# IC Compiler II Error Messages

Version V-2023.12, December 2023

SYNOPSYS®

Feedback

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# 1

# IC Compiler II Error Messages

This document describes the error messages supported by the IC Compiler II tool.

## 3DCODE

### 3DCODE-001

(error) %s.

**Description**

The input 3DCODE file(s) may contain syntax error.

**What Next**

Read the error message to identify and correct the syntax error in the input 3DCODE file(s).

### 3DCODE-002

(error) Python execute file (%s) error. Details are in the *.debug.log file.

**Description**

An error occurred while executing a Python command. Detailed information is available in the *.debug.log file.

**What Next**

Get the detailed information about the error and contact your AE for help.

### 3DCODE-003

(error) Syntax error: %s.

**Description**

The input 3DCODE file(s) contains syntax error. Please check the error message.

Feedback

**What Next**

Read the error message to identify and correct the syntax error in the input 3DCODE file(s).

---

## 3DCODE-004

(error) Data error: %s.

**Description**

The input 3DCODE file(s) contains incorrect data value. Please check with the error message.

**What Next**

Read the error message to identify and correct the data error in the input 3DCODE file(s).

---

## 3DCODE-005

(warning) Loop connection is found for block instance: %s in %s.

**Description**

There is loop connection in specified .3dBuilder file, thus z level of all the block instances will be set to zero.

**What Next**

Please check the source file to find the loop connection and remove it.

---

## 3DCODE-006

(error) Invalid library name '%s' specified in '%s' command option of read_3dcode.

**Description**

The library name specified is invalid because it contains spaces.

**What Next**

Ensure that there are no spaces in library name specified.

---

## 3DCODE-007

(error) %s.

Feedback

**Description**

The specified library name exists in directory. read_3dcode creates library path under ./refLibs/ directory. The specified library name from 3DCODE files already exists.

**What Next**

Please check the existing directory. Back up or delete it, then run read_3dcode again.

---

## 3DCODE-008

(error) The specified library path '%s' doesn't exist.

**Description**

The library name specified by -library contains path, but the path doesn't exist.

**What Next**

Ensure the library path exist.

---

## 3DCODE-009

(warning) BlockDesign '%s' has different bump thickness (%s and %s) under one region.

**Description**

Bumps belong to one region have different bump thickness.

**What Next**

Check the related bump thickness values are expected or not.

---

## 3DCODE-010

(warning) The bumps between block instance '%s' and '%s' are with inconsistent thickness %s and %s.

**Description**

A connection is detected by write_3dcode since their bump regions are overlapped.
But they have different bump thickness.write_3dcode chooses the top thickness as the connection thickness.

**What Next**

Check the related bump thickness values are expected or not.

Feedback

# 3DIC

## 3DIC-001

(error) Incorrect command %s option usage. %s.

**Description**

Incorrect command option usage.

**What Next**

See command man page for correct option usage.

## 3DIC-002

(Information) net %s already connects %s %s. net is ignored.

**Description**

Command *assign_3d_interchip_nets* will assign physical contact for logical net. This message tells user that certain net is already assigned so it is skipped in automatic optimal assignment.

**What Next**

User should double confirm whether the existed assignment status for this net is as expected. If not, please remove existed assignment of the net and run command again.

**See Also**

- assign_3d_interchip_nets

## 3DIC-003

(error) %s array %s already exists, please use another %s array name to create %s array.

**Description**

Errors are found during the processing of the command options.

**What Next**

Please check the manpage of the command to correct any errors related to option settings.

Feedback

## 3DIC-004

(error) The minCutSpacing value in tech file is %s, and specified TSV pitch (%s %s) violates minimal cut spacing rule.

**Description**

The spacing between two TSVs is less than minCutSpacing that defined in the tech file.

**What Next**

Check TSV array pitch value.

## 3DIC-005

(info) Current design %s is a 3DIC design.

**Description**

Current design type is 3DIC.

**What Next**

See command man page for correct command usage.

## 3DIC-006

(Error) Found multiple chip instances refer to same design %s. MIB is not supported.

**Description**

MIB is not supported in command assign_3d_interchip_nets.

**What Next**

Check the design again.

## 3DIC-007

(Error) -pins is only valid when source chip is hard macro.

**Description**

If -pin option is sepcified, then the source chip should be hard macro.

**What Next**

Please make sure that the source chip is hard macro before -pin is specified.

Feedback

## 3DIC-012

(Error) %s pair %s's matching type %s is different from %s's matching type %s.

**Description**

In 3DIC flow, matching type should be consistent on any physical path or logical path. Otherwise, inter-chip nets assignment will fail.

**What Next**

Set proper matching type on port or pins by set_matching_type or propagate_3d_matching_type.

## 3DIC-013

(Error) Logically connected bumps or drivers on net %s have different matching types.

**Description**

In 3DIC flow, matching type should be consistent on any physical path or logical path. Otherwise, inter-chip nets assignment will fail.

**What Next**

Set proper matching type on cells, terminals or pins by set_matching_type or propagate_3d_matching_type.

## 3DIC-014

(Error) The design_type of the die "%s" is not valid.

**Description**

In 3DIC design, the valid design_type of a die should be one of: module, macro, analysis, die, bridge, interposer, substrate.

**What Next**

Set proper design_type for the die.

## 3DIC-030

(warning) Z value must be an integer. Rounding value to %d.

**Description**

Z value stands for chip stack z value and must be an integer.

Feedback

**What Next**

Specify an integer to z value will be better.

---

## 3DIC-031

(error) No chip coordinates specified. Use -location {x y z}.

**Description**

3-D coordinate is used to describe the location of a chip in design.

**What Next**

Please input coordinate like -location {x y z}.

---

## 3DIC-032

(error) The specified scaling_factor value is not in range (0, 1].

**Description**

The scaling_factor value should be greater than 0 and no greater than 1.

**What Next**

Please specify a value greater than 0 and no greater than 1.

---

## 3DIC-033

(warning) Set attribute of chip %s, other chips will be ignored.

**Description**

This command set a chip each time.

**What Next**

Please input a chip name with -chip.

---

## 3DIC-034

(error) Z value must be no less than 0.

**Description**

Z value must be no less than 0.

Feedback

**What Next**

Use set_3d_chip_placement to set chip's z value.

---

## 3DIC-035

(warning) Block '%s' will not be saved. Library '%s' is in read mode.

**Description**

The specified block will not be saved because its library is open in read mode.

**What Next**

Open the specific library and block in edit mode and try again.

---

## 3DIC-036

(warning) Block '%s' will not be saved because it is in read mode.

**Description**

The specified block will not be saved because it is open in read mode.

**What Next**

Open the block in edit mode and try again.

---

## 3DIC-037

(error) Specified via def %s is not a TSV via def.

**Description**

The via def specified is not a TSV via def.

**What Next**

Specify TSV via def.

---

## 3DIC-038

(error) Specified %s is not a TSV.

**Description**

The specified option value is not a TSV.

Feedback

**What Next**

Specify TSV.

---

## 3DIC-039

(error) %s.

**Description**

Errors are found during the processing of the command options.

**What Next**

Please check the manpage of the command to correct any errors related to option settings.

---

## 3DIC-040

(error) Command should run on top level and current design should be a legal 3DIC design.

**Description**

Current design is not a legal 3DIC design or command is not running on top level.

**What Next**

Please check the current design.

---

## 3DIC-041

(error) Less than two chips in the design.

**Description**

3DIC design should contains more than one chip.

**What Next**

Add more chips into the design.

---

## 3DIC-042

(error) The chip %s does not have a Z value.

**Description**

Chip should be assigned a Z value to describe the location in design.

Feedback

**What Next**

Assign a Z value to chip by command set_cell_location.

---

## 3DIC-043

(error) Chip %s is not editable.

**Description**

3DIC commands should be executed on editable chip.

**What Next**

Make sure the corresponding chip is editable.

---

## 3DIC-044

(error) Command should be run on top level and %s for chip %s should be set.

**Description**

%s should be set first for chip %s or this command is not running on top level.

**What Next**

Use set_3d_chip_placement to set chip's attribute.

---

## 3DIC-045

(warning) Specified pin %s is not connected to an existing net.

**Description**

Specified pin is not connected to an existing net.

**What Next**

Check command option value.

---

## 3DIC-046

(error) Specified TSV %s is not belong to current design.

**Description**

Errors are found during the processing of the command options.

Feedback

**What Next**

Please check the manpage of the command to correct any errors related to option settings.

---

## 3DIC-047

(warning) Specified pin %s is not I/O driver pin, bump pin or pin of standard cell.

**Description**

Specified pin is not I/O driver pin, bump pin or pin of standard cell.

**What Next**

Check command option value.

---

## 3DIC-048

(error) No specified matching type will be propagated.

**Description**

The specified matching types is not existed in source chip or these matching types contain no object.

**What Next**

Please check the design.

---

## 3DIC-049

(warning) No object has matching type %s, it will be ignored.

**Description**

No object in matching type %s, there is no way to propagate this matching type.

**What Next**

Add objects to matching type by using command add_to_matching_type.

---

## 3DIC-050

(warning) Matching type %s does not exist in any source chip and will be ignored.

**Description**

%s in -matching_type list does not exist in any source chip or the design.

Feedback

**What Next**

This matching type will be ignored.

---

## 3DIC-051

(error) No matching type in the '-matching_types {list}' exist in the source chip.

**Description**

There isn't any matching type with the name in '-matching_type {list}' existed in source chip.

**What Next**

Please check the option -matching_types.

---

## 3DIC-052

(Error) No valid matching type defined in design or chips specified by '-from'.

**Description**

No valid matching type defined in the design or chips specified by '-from'.

**What Next**

Please add matching types in source chip and do propagation.

---

## 3DIC-053

(error) Conflicting matching types %s and %s found on source chip %s. Related objects: %s.

**Description**

There are more than one matching type in a physical or logical connection on the source chip, it will cause error when propagate matching type from source chip to target chip.

**What Next**

Please check the design.

---

## 3DIC-054

(error) Matching types %s in target chip corresponding to the matching type %s in source chip during propagation. Related objects: %s.

Feedback

**Description**

This messages appears because one of scenario below happens: 1. There are more than one matching types in target chip corresponding to the same matching type in source chip; 2. The corresponding matching type in source chip and target chip has different matching type name.

**What Next**

Please check the design. Using -force may help to solve this conflict.

---

## 3DIC-055

(error) Objects do not have physical contact or logical connection with the other objects of the same matching type %s and were not propagated.

**Description**

Objects do not have physical contact or logical connection with the other objects of the same matching type and were not propagate.

**What Next**

Please check the design. Using -force may help to solve this situation.

---

## 3DIC-056

(warning) Conflicting matching types %s and %s found on source chip %s. Related objects: %s.

**Description**

There are more than one matching type in a physical or logical connection on the source chip, it will cause error when propagate matching type from source chip to target chip.

**What Next**

Please check the design.

---

## 3DIC-057

(warning) Matching types %s in target chip corresponding to the matching type %s in source chip during propagation. Related objects: %s.

**Description**

This messages appears because one of scenario below happens: 1. There are more than one matching types in target chip corresponding to the same matching type in source chip;

Feedback

2. The corresponding matching type in source chip and target chip has different matching type name.

**What Next**

Please check the design.

---

## 3DIC-058

(warning) Objects do not have physical contact or logical connection with the other objects of the same matching type %s and were not propagated.

**Description**

Objects do not have physical contact or logical connection with the other objects of the same matching type and were not propagate.

**What Next**

Please check the design.

---

## 3DIC-059

(warning) %s is specified, %s will be ignored.

**Description**

%s has high priority than %s.

**What Next**

Specify one of these two options.

---

## 3DIC-060

(error) The objects with same matching type name %s in source chip and target chip have no connections.

**Description**

The objects with same matching in two chips must have connection with each other.

**What Next**

Rename or remove matching type %s in target chip. Or use -force to solve this conflict.

Feedback

## 3DIC-061

(warning) The objects with same matching type name %s in source chip and target chip have no connections.

**Description**

The objects with same matching in two chips must have connection with each other.

**What Next**

Please check the design.

## 3DIC-062

(error) Specified pins are not connected to the same existing net.

**Description**

Specified pins connected to multiple nets.

**What Next**

Check command option value.

## 3DIC-063

(error) %s %s does not physically connect to the %s on the adjacent side of the vertical neighboring chip, the reason is '%s'.

**Description**

Found single bump or bond pad on the chip. These single bump and bond pad cannot be used to construct physical connections.

**What Next**

Adjust the location of chips or remove the single bump/bond pad or use create_3d_mirror_bumps to create bump pair/bond pad pair.

## 3DIC-064

(error) Non-aligned %s pair {%s, %s}, exceed tolerance of %s.

**Description**

%s %s and %s are physically contact but non-aligned, exceed tolerance of %s.

Feedback

**What Next**

Adjust bump or bond pad location or enlarge the alignment threshold of two bumps' center by set plan.3dic.bump_alignment_threshold.

---

## 3DIC-065

(error) %s {%s, %s} overlap with each other in a single chip.

**Description**

%s {%s, %s} overlap with each other in a single chip.

**What Next**

Adjust bump or bond pad location.

---

## 3DIC-066

(error) %s %s contacts to more than one %s {%s} of the neighboring die.

**Description**

%s %s contacts to more than one %s {%s} of the neighboring die.

**What Next**

Adjust bump location.

---

## 3DIC-067

(warning) No bump pair or bond pad pair on the adjacent sides between two vertical neighboring and overlap chips: {%s} and {%s}.

**Description**

No bump pair or bond pad pair between two vertical neighboring and overlap chips, then it cannot build any legal inter-chip connections between the two chips.

**What Next**

Check the design about below issues: 1. check the orientation of adjacent chips. 2. check there are valid bumps or bond pads on the adjacent sides between two vertical neighboring chips, the valid bumps must have metal shape on the off-die layer and its design type is flip_chip_pad. The valid bond pads also must have metal shape on the off-die layer and the attribute of 'is_bond_pad' is true.

Feedback

## 3DIC-068

(info) Specified pins are not connected to net, creating a default net %s to connect the pins.

**Description**

Specified pins are not connected to net, creating a default net to connect the pins.

**What Next**

Check command option value.

## 3DIC-069

(info) Creating a default net %s to connect the TSV.

**Description**

Creating a default net to connect the TSV.

**What Next**

Check command option value.

## 3DIC-070

(info) BPConstraint is created on TSV port %s and the destination pin is %s.

**Description**

BPConstraint is created on TSV and the destination pin.

**What Next**

Check BPConstraint.

## 3DIC-071

(error) No bump pair or bond pad pair for the top-level logical connection %s.

**Description**

Each logical connection should correspond to a bump pair or bond pad pair.

**What Next**

Make sure the top-level logic connection is correct.

Feedback

## 3DIC-072

(warning) %s %s and %s are contacted, but they have no net.

**Description**

If two bumps or bond pads are center-aligned, one bump-pin or bond pad should be connected to a top-level net, otherwise, the bump or bond pad in other side cannot be propagated with net connection. If two or one of these two bumps is dummy, the issue is legal and this violation will not be reported.

**What Next**

Make sure the top-level connection is correct.

## 3DIC-073

(error) Center-aligned %s pair {%s, %s} does not connect to the same net.

**Description**

If two bumps or bond pads are center-aligned, but the bump-pins' or bond pads' connected driver-pins are in different top-level nets, the logical net propagation will skip the scenario due to such critical error. Also, if one bump or bond pad is connected to top-level net while another has no top level net connection, the error message should also be reported. If one bump or bond pad is dummy, the other bump or bond pad is connected to top-level power net, this issue is legal and this violation will not be reported.

**What Next**

Make sure the top-level connection is correct.

## 3DIC-074

(info) TSV %s port %s is connected to net %s.

**Description**

Connect TSV port to net.

**What Next**

Complete assign tsv.

## 3DIC-075

(info) Pin %s is connected to net %s.

Feedback

**Description**

Connect pin to net.

**What Next**

Complete assign tsv.

---

## 3DIC-076

(error) The Z value of chips are not continuous.

**Description**

The Z value of chips are not continuous.

**What Next**

Modify the Z value of chips by command set_3d_chip_placement or set_cell_location.

---

## 3DIC-077

(error) All chips in the design have the same Z value.

**Description**

In 3D design, chips that are stacked should have different Z value.

**What Next**

Modify the Z value of chips by command set_3d_chip_placement or set_cell_location.

---

## 3DIC-078

(error) The Z value of the chips do not start from 0.

**Description**

In 3D design, chip with a Z value equals to 0 is seen as base chip, so it is required to have one chip with Z value equal to 0 in design.

**What Next**

Modify the Z value of chips by command set_3d_chip_placement or set_cell_location.

---

## 3DIC-079

(warning) %s.

Feedback

**Description**

Warnings are found during the processing of the command options.

**What Next**

Please check the manpage of the command to correct any errors related to option settings.

---

## 3DIC-080

(error) The chips {%s, %s} with the same Z value are overlapping.

**Description**

In 3D design, the chips with the same Z value are in the same layer and should not overlapped.

**What Next**

Modify the attributes of chips by command set_3d_chip_placement or set_cell_location.

---

## 3DIC-081

(error) Vertical neighboring chips {%s, %s} are partially overlapped rather than the large one completely covers the small one.

**Description**

For two neighboring chips, if they overlapped on virtual top level, the larger chip should contain the other chip.

**What Next**

Modify the attributes of chips by command set_3d_chip_placement or set_cell_location.

---

## 3DIC-082

(info) BPConstraint is created on TSV port %s and the destination pin is TSV %s's pin %s.

**Description**

BPConstraint is created on TSV and TSV pin.

**What Next**

Check BPConstraint.

Feedback

## 3DIC-083

(error) Originally TSV %s connected to net %s, and can not be connected to net %s.

**Description**

Can not connect TSV to net.

**What Next**

Disconnect TSV original net and reconnect it.

## 3DIC-084

(error) Found distance %s between the TSV %s and the boundary of design %s on %s direction less than %s.

**Description**

You receive this message because the spacing between TSV and chip's boundary is less then tsvKeepoutSpacing that defined in the tech file.

**What Next**

Check and move TSV's location.

## 3DIC-085

(error) The TSV %s is not inside the boundary of design %s.

**Description**

You receive this message because some TSV are not in chip's boundary.

**What Next**

Check and move TSV's location.

## 3DIC-086

(error) found distance %s between the TSV %s and the TSV %s on %s direction less than %s.

**Description**

You receive this message because the spacing between two TSV is less than minCutSpacing that defined in the tech file.

Feedback

**What Next**

Check and move TSV's location.

---

## 3DIC-087

(error) The bump cluster %s in design %s has more than 1 connected nets.

**Description**

You receive this message because some bump cluster in the design has more than 1 connected nets.

**What Next**

Check the related bump cluster.

---

## 3DIC-088

(error) The physical contacts between %s and %s utilize clusters that are physically inconsistent.

**Description**

If one bump in a physical contact pair belongs to a cluster, the other bump in this physical contact pair should also belong to another cluster.

**What Next**

Check the design to find where the inconsistency comes from and fix this issue manually.

---

## 3DIC-089

(error) Chip %s with %s to package does not have Z = 0.

**Description**

Chip with net to package should has Z = 0.

**What Next**

Check the settings of chip or re-assign the related nets.

---

## 3DIC-090

(error) Net %s on the top level has %s pin(s).

Feedback

**Description**

Nets on top level should link two and only two pins.

**What Next**

Check the design.

---

## 3DIC-091

(error) Net %s on the top level between bottom-most die and package does not have a connected bump or bond pad.

**Description**

The net on top level between bottom-most die and package should have a connected bump.

**What Next**

Check the design about below issues: 1. check the orientation of adjacent chips. 2. check there are valid bumps or bond pads on the adjacent sides between two vertical neighboring chips, the valid bumps must have metal shape on the off-die layer and its design type is flip_chip_pad. The valid bond pads also must have metal shape on the off-die layer and the attribute of 'is_bond_pad' is true. If need to create bumps or bond pads between bottom-most die and package, use command create_bump_array or create_bond_pad_array to create.

---

## 3DIC-092

(Error) The chip '%s' and the chip '%s' have spacing violation(s). Minimum spacing required: %s, actual spacing: %s.

**Description**

In ICC 3DIC top level, user can set a "minimum spacing" rule for any two chips of same vertical level. The rule will be honored by 3D chip placement commands or check commands to report violations.

**What Next**

Double confirm the reported violation is as expected. If yes, correct the 3D chip placement in top level design. If not, re-set the rule to proper value.

Feedback

## 3DIC-093

(Error) Chip '%s' and chip '%s' have enclosure violation. minimum enclosure required:%s, real enclosure:%s.

**Description**

In ICC 3DIC top level, user could set a "minimum enclosure" rule for any two vertically adjacent chips. The rule will be honored by 3D chip placement commands or check commands to report violations.

**What Next**

Double confirm the reported violation is as expected. If yes, correct the 3D chip placement in top level design. If not, re-set the rule to proper value.

## 3DIC-094

(error) Net %s on top level does not connect the adjacent overlapped chip.

**Description**

You receive this message because the net %s is physically infeasible.

**What Next**

Check the design.

## 3DIC-095

(error) The TSV %s is overlapping with the TSV %s.

**Description**

You receive this message because two TSV are overlapping.

**What Next**

Check and move TSV's location.

## 3DIC-096

(warning) Cannot create TSV at coordinate %s.

**Description**

The TSV cannot be created in this position.

Feedback

**What Next**

Please check TSV array.

---

## 3DIC-099

(error) Pattern cell: %s is not valid for bump cluster pattern cell.

**Description**

Pattern cell specified must meet certain pre-requisites to be valid: (a) tool does not support hierarchical pattern cells; (b) tool needs at least one bump cell to connect bump_cluster net.

**What Next**

Fix pattern cell for error situation specified in message, or specify a different pattern cell.

---

## 3DIC-100

(Error) There is no corresponding bump on the mirroring face of source chip %s.

**Description**

In command *create_3d_mirror_bumps*, user should guarantee that there are one more bumps on the mirroring face of source chip, otherwise, the error message would be reported.

**What Next**

Check the situation in source chip to assure that there exists bump on the mirroring face.

---

## 3DIC-101

(Error) Cannot find library cell %s.

**Description**

The command *create_3d_mirror_bumps* should specify correct library cell.

**What Next**

Please specify the correct library cell.

---

## 3DIC-102

(Error) Bump %s in source chip %s is overlapped with bump %s in target chip %s.

Feedback

**Description**

Before running the command *create_3d_mirror_bumps*, the user should make sure that there is no overlapped bump between source chip and target chip .

**What Next**

Use *create_3d_mirror_bumps -force* or remove the overlapped bumps in target chip.

---

## 3DIC-103

(Info) Totally mirrored %d bumps, please use 'save_block' to retain the modification, otherwise, the data will be lost.

**Description**

Report the result *create_3d_mirror_bumps*.

**What Next**

Use *save_block* to save data.

---

## 3DIC-104

(Error) Chip %s is not in design view or outline view

**Description**

The command *create_3d_mirror_bumps* can only support the operation in design view or outline view chip, if the chip is in frame view, timing view or abstract view, the command does not support the scenarios.

**What Next**

Please specify the chip with correct view.

---

## 3DIC-105

(Error) Chip %s is not aligned with chip %s

**Description**

During the execution of command *create_3d_mirror_bumps*, the source chip and the target chip should be touched with together.

**What Next**

Make sure that the chip is intersected with the other chip.

Feedback

## 3DIC-106

(Warning) Bump %s is skipped because it is not within corresponding mirrored chip %s 's die-boundary.

**Description**

During the execution of command *create_3d_mirror_bumps*, the bump on source chip should be inside the boundary of target chip.

**What Next**

Make sure that the bump is inside the die boundary of target chip.

## 3DIC-107

(Error) The source chip %s and target chip %s 's differential z value is not 1.

**Description**

In command create_3d_mirror_bumps, the source chip and target chip 's differential z value should be 1.

**What Next**

Make sure the different z value of source chip and target chip is 1.

## 3DIC-108

(Info) Please use 'save_block -hierarchical' to retain the modification, otherwise, the data will be lost.

**Description**

The IC Compiler use save_block -hierarchical to retain the modification in multiple physical hierarchy.

**What Next**

N/A

## 3DIC-109

(Warning) Bond pad %s is skipped because it is not within corresponding mirrored chip %s 's die-boundary.

Feedback

**Description**

During the execution of command *create_3d_mirror_bumps*, the bond pad on source chip should be inside the boundary of target chip.

**What Next**

Make sure that the bond pad is inside the die boundary of target chip.

---

## 3DIC-110

(Error) No correct net in top level has been defined or specified.

**Description**

In command *propagate_3d_connections* and *disconnect_3d_bumps*, users can specify the corresponding nets in the command option, however, if there is no correct net in design, the error message will be reported.

**What Next**

Check the top-level netlists.

---

## 3DIC-111

(Error) Floating port without net connection.

**Description**

In command *propagate_3d_connections*, the port should be connected with the nets, otherwise, the nearby bump cannot be propagated.

**What Next**

Check the situation in block to make sure that the port is connected with nets.

---

## 3DIC-112

(Information) There are no nets in the current block.

**Description**

It is a critical information in 3DIC commands, which there are no nets in top-level.

**What Next**

Check the top-level netlists.

Feedback

## 3DIC-113

(Warning) Chip %s is not editable, thus the change inside chip cannot be saved.

**Description**

3DIC commands should be executed on editable chip. This warning message is reported due to several reasons, e.g., the reference block of the chip is read-only, the top design is read-only or the reference library cannot be edited.

**What Next**

Make sure the corresponding chip is editable.

## 3DIC-114

(Error) Chip %s is not design view or outline view.

**Description**

3DIC commands can be only executed on the chip with design view or outline view if this chip is not a hard macro.

**What Next**

Make sure the corresponding chip is design view or outline view.

## 3DIC-115

(Error) The net %s has two unaligned bumps %s and %s connected, this net is illegal.

**Description**

In command *propagate_3d_connections*, if a net has two bumps in different chips connected, the bump should be center-aligned to keep logical and physical consistency.

**What Next**

Make sure the two bumps on the same nets are logically and physically consistent.

## 3DIC-116

(Warning) Although the bump %s and the bump %s are contacted, they have no net to propagate, skipped.

Feedback

**Description**

In command *propagate_3d_connections*, if two bumps are center-aligned, one bump-pin should be connected to a top-level net, otherwise, the bump in other side cannot be propagated with net connection.

**What Next**

Make sure the top-level connection is correct.

---

## 3DIC-117

(Error) Although the bump %s and the bump %s are center-aligned but their associated logical ports %s and %s are connected to different nets %s and %s.

**Description**

In command *propagate_3d_connections*, if two bumps are center-aligned, but the bump-pins are connected to different ports with different nets, it will cause shorts.

**What Next**

Make sure the top-level connection is correct.

---

## 3DIC-118

(Warning) Although the bump %s and the bump %s are contacted, they have no net to propagate, skipped.

**Description**

In command *propagate_3d_connections*, if two bumps are center-aligned, but with no logical connection, it is a critical error.

**What Next**

Make sure the top-level connection is correct.

---

## 3DIC-119

(Error) Derive net on %s %s failed.

**Description**

In command *propagate_3d_connections* failed to propagate logical connections for some bumps or bond pads.

Feedback

**What Next**

N/A

---

## 3DIC-120

(Error) Remove net on bump %s failed.

**Description**

In command *disconnect_3d_bumps* failed to remove logical connections for some bump pins.

**What Next**

N/A

---

## 3DIC-121

(Info) Propagate %d nets.

**Description**

The command *propagate_3d_connections* result report.

**What Next**

N/A

---

## 3DIC-122

(Info) Totally %d bump pins are disconnected from nets.

**Description**

The command *disconnect_3d_bumps* result report.

**What Next**

N/A

---

## 3DIC-123

(Error) The aligned %s %s and %s have different matching type.

Feedback

**Description**

The command *assign_3d_interchip_nets* will choose aligned bump pair or bond pad pair which flip chip drivers can be assigned, the bump pair or bond pad pair should have same matching type, otherwise, the error message will be reported.

**What Next**

Change one bump's matching type.

---

## 3DIC-124

(Error) The connected driver pin %s and %s have different matching type.

**Description**

The command *assign_3d_interchip_nets* will choose two connected driver pins in different chip to assign to bumps, the driver pins should have same matching type, otherwise, the error message will be reported.

**What Next**

Change one driver pin's matching type.

---

## 3DIC-125

(Warning) Interchip net assignment between %s and drivers cannot be implemented because the number of %s pair is %d while driver pair number is %d.

**Description**

The command *assign_3d_interchip_nets* should guarantee that bump or bond pad number is not less than driver number in same matching type group. Also, both driver pair number and bump pair or bond pair number should be at least one.

**What Next**

N/A

---

## 3DIC-126

(Error) There is no bump on net %s.

**Description**

The command *propagate_3d_connections -nets -check_only* will check whether specified nets has bump pin for connection, if no bump pin is connected, the error message will be reported.

Feedback

**What Next**

Connect the net with at least one bump pin.

---

## 3DIC-127

(Error) The net %s has less than two driver pins.

**Description**

The command *propagate_3d_connections -nets* will check whether specified net must have at least two driver pins, if not, the error message will be reported.

**What Next**

N/A

---

## 3DIC-128

(Error) The net %s has one bump %s 's pin connected, the bump has not corresponding center-aligned bump for logical connection propagation.

**Description**

The command *propagate_3d_connections -nets* will check whether the bump on specified net has another center-aligned bump to be a pair.

**What Next**

Check the physical position of the bumps

---

## 3DIC-129

(Warning) The net %s has the bump-pin connected to the driver-pin, it will be skipped in inter-chip net assignment.

**Description**

The command *assign_3d_interchip_nets* will check whether the net has the bump pin connected on it, if there is bump pin on the net, this net will be skipped in the assignment afterwards.

**What Next**

Check the logical connection of the net

Feedback

## 3DIC-130

(Warning) The matching type '%s' group have no corresponding driver-pin pair for inter-chip net assignment.

**Description**

The command *assign_3d_interchip_nets* can do net assignment for different matching type, however, if there is no driver-pin pair, the warning message will be reported.

**What Next**

Make sure there is driver-pin in this matching type group.

## 3DIC-131

(Warning) The matching type '%s' group have no corresponding bump-pin pair for inter-chip net assignment.

**Description**

The command *assign_3d_interchip_nets* can do net assignment for different matching type, however, if there is no bump-pin pair, the warning message will be reported.

**What Next**

Make sure there is bump-pin in this matching type group.

## 3DIC-132

(Warning) Matching type %s does not exist in the child blocks of top design and will be ignored.

**Description**

%s in -matching_type list does not exist in the child blocks of top design.

**What Next**

This matching type will be ignored.

## 3DIC-133

(error) No matching type defined in child blocks of top design.

**Description**

No matching type defined in child blocks of top design.

Feedback

**What Next**

Please add matching types in child blocks as chips in the top design.

---

## 3DIC-134

(Error) There is no corresponding bond pad on the mirroring face of source chip %s.

**Description**

In command *create_3d_mirror_bumps*, user should guarantee that there are one more bond pads on the mirroring face of source chip, otherwise, the error message would be reported.

**What Next**

Check the situation in source chip to assure that there exists bond pad on the mirroring face.

---

## 3DIC-135

(Error) Bond pad %s in source chip %s is overlapped with bond pad %s in target chip %s.

**Description**

Before running the command *create_3d_mirror_bumps*, the user should make sure that there is no overlapped bond pad between source chip and target chip .

**What Next**

Use *create_3d_mirror_bumps -force* or remove the overlapped bond pads in target chip.

---

## 3DIC-136

(Info) Created %d bond pads on instance: %s, layer %s by mirroring from instance: %s, layer %s.

**Description**

Mirroring bond pads from source instance to target instance.

**What Next**

Assigning and checking 3DIC interchip nets.

---

## 3DIC-137

(warning) Layer %s is not bond pad layer.

Feedback

**Description**

Layer should be bond pad layer to create bond pads.

**What Next**

Specify bond pad layer which should be off-die layer.

---

## 3DIC-138

(Info) Created %d bumps and %d bump clusters on instance %s by mirroring from instance %s.

**Description**

Mirroring bumps and bump clusters from source instance to target instance.

**What Next**

Assigning and checking 3DIC interchip nets.

---

## 3DIC-139

(error) Collection specified for pattern cell contains %d entries, it must contain exactly one entry.

**Description**

User must specify a unique specification for pattern cell. If more than one entry matches given specification, user should update specification to match only the correct pattern cell for bump cluster creation.

**What Next**

Please check the name of pattern cell as well as list of reference libraries in design setup.

---

## 3DIC-140

(error) Cannot find pattern cell, or if a collection is specified, collection is empty.

**Description**

Tool cannot create bump_cluster because given pattern cell is not found in either design or reference library.

**What Next**

Please check the name of pattern cell as well as list of reference libraries in design setup.

Feedback

## 3DIC-141

(error) Pattern cell: %s is not valid for bump_cluster creation. Reason: %s

**Description**

Pattern cell specified for create_bump_cluster command must meet certain pre-requisites to be valid for bump_cluster creation. Some of the reasons, pattern cell could be invalid is: (a) pattern cell contains hierachy - tool doe snot support hierarchical pattern cells (b) pattern cell does not contain atleast one bump cell - tool needs atleast one bump cell to connect bump_cluster net.

**What Next**

Fix pattern cell for error situation specified in message, or specify a different pattern cell.

## 3DIC-142

(error) Bump cluster with name: %s is already existing in design.

**Description**

If user specifies name for a bump cluster, it must not conflict with another existing bump cluster.

**What Next**

Provide a new non-conflicting name for creating bump cluster

## 3DIC-143

(error) Bump cluster %s overlaps with existing %s. Aborting bump cluster creation.

**Description**

Tool does not create overlapping bump clusters (unless -force option is used). It should be noted that overlap is computed based on bounding box of each bump cluster.

**What Next**

If overlap is intentional and will not impact bump cluster functionality, user can use -force option. Otherwise, user must check -coordinates and orientation for all overlapping bump clusters.

## 3DIC-144

(warning) Bump cluster %s overlaps with existing %s. Continuing with bump cluster creation as -force option is used.

Feedback

**Description**

Tool does not create overlapping bump clusters unless -force option is used. Use of -force indicates taht overlap is intentional and will not impact bump cluster functionality. It should be noted that overlap is computed based on bounding box of each bump cluster.

**What Next**

User must ensure overlap is intentional and will not impact bump cluster functionality.

---

## 3DIC-145

(warning) Net: %s exists in design, bump cluster cells will connect to this net.

**Description**

If user-specified net is already existing in design, all bump_cluster cells will connect to this net. This may cause unexpected connectivity to other terminals on the net.

**What Next**

User should ensure that specified net is supposed to exist and any resulting connectivity to the bump cluster cells is expected

---

## 3DIC-146

(warning) Object: %s inside pattern cell: %s is unsupported design type: %s ; Ignoring for bump cluster creation.

**Description**

Bump cluster creation from given pattern cell only supports certain type of design objects as valid objects. It ignores specified design object even though presence of such object in pattern cell does not invalidate pattern cell.

**What Next**

Review warning messages to see if ignored design objects are relavent to bump cluster functionality

---

## 3DIC-147

(warning) No bump clusters matched specified input, returning empty string.

Feedback

**Description**

None of bump cluster names matched the input pattern name or specified -of_objects did not belong to any bump cluster. This condition is not considered error, and command returns empty string. However, user should check if no match was expected.

**What Next**

If user expected command to match one or more bump clusters, check input to the command, or name / content of all bump clusters in the design.

---

## 3DIC-148

(error) command cannot find any bump cluster for given name, or collection is empty.

**Description**

Command did not find any bump cluster with given name, if name was provided as text. Command performs standard wildcard expansion if necessary. If collection was provided, collection was empty.

**What Next**

Check the name or collection creation command for any spelling errors. You may use get_bump_cluster_name command to see list of all bump clusters.

---

## 3DIC-149

(warning) command did not match any object for obj_type: %s.

**Description**

Command did not find any object for given type in given bump cluster(s).

**What Next**

User may check pattern cell used for such bump cluster creation if given object exist in pattern cell. Other possibility is subsequent edit of bump_cluster contents. It should be noted that TSV instances are returned with -tsvs option, and not with either -bumps or -vias option.

---

## 3DIC-150

(error) Bump Cluster creation failed.

Feedback

**Description**

This message indicates that create_bump_cluster command failed due to one or more previously reported errors.

**What Next**

User should inspect all error (and warning) messages reported by create_bump_cluster command. Also check the manpage of the command to correct any errors related to option settings.

---

## 3DIC-151

(error) %s.

**Description**

This message describes specific situations that may cause create or get commands related to bump_cluster to fail.

**What Next**

User should inspect all error (and warning) messages reported by create_bump_cluster command. Also check the manpage of the command to correct any errors related to option settings.

---

## 3DIC-152

(warning) %s.

**Description**

This message describes specific situations that may cause bump cluster create or get commands to generate unexpected output.

**What Next**

User should inspect all messages reported by create_bump_cluster command. Also check the manpage of the command to correct any warnings related to option settings.

---

## 3DIC-153

(error) Z value is greater than %d.

**Description**

The specified z value can not exceed the given value. It must be equal to or greater than 0 and equal to or less than the given max value.

Feedback

**What Next**

Specify a new z value less than or equal to the given max value.

---

## 3DIC-154

(warning) The user specified scaling factor will be ignored since the value of %s has been defined in TLU+ file.

**Description**

User can specify the scaling factor only if it is not set in TLU+ file. If there is scaling factor in TLU+ file, the user specified scaling factor will be ignored.

**What Next**

Use the TLU+ file based scaling factor instead of specifying a new one.

---

## 3DIC-155

(error) There is no off-die layer on the die %s that physical contact the die %s.

**Description**

The source die and target die should have physical contact off-die layers. Generally, the off-die layer is the most top metal layer with mask_name as "metal#" and the most bottom metal layer with mask_name as "bmetal#". For example, if the back-side of a die faces to another die, then generally, this die should have back-side off-die layer with mask_name as "bmetal#", # here stands for a number. Furthermore, user can use "set_3d_multilevel_stacking" to set off-die layer of a die.

**What Next**

Please check the layer settings in the technology file, or use "set_3d_multilevel_stacking" to set off-die layer of a die.

---

## 3DIC-156

(Info) Created %d bump region patterns and %d bump regions with %d pseudo bumps on instance %s by mirroring from instance %s.

**Description**

Mirroring bump regions from source instance to target instance.

**What Next**

Commit pseudo bump regions to create real bumps or bond pads.

Feedback

## 3DIC-157

(Warning) bump region %s is skipped because it is not within corresponding mirrored chip %s 's die-boundary.

**Description**

During the execution of command *create_3d_mirror_bumps*, the bump region on source chip should be overlappled with the boundary of target chip.

**What Next**

Make sure that the bump region is overlapped with the die boundary of target chip.

## 3DIC-158

(Warning) Chip %s is not editable, thus the '%s' setting on chip cannot be saved.

**Description**

3DIC commands should be executed on editable chip. This warning message is reported due to the reference block of the chip is read-only.

**What Next**

Make sure the corresponding chip is editable.

## 3DIC-160

(error) Collection specified for -from option contains object of unsupported type: %s, object name: %s

**Description**

derive_3d_interface error message description is TBD

**What Next**

User should inspect all error and warning messages reported by derive_3d_interface command. Also check the manpage of the command to correct any errors related to option settings.

## 3DIC-161

(error) Collection specified for -from option contains objects of multiple types; Prior type: %s, Conflicting type: %s at index: %d.

Feedback

**Description**

derive_3d_interface error message description is TBD

**What Next**

User should inspect all error and warning messages reported by derive_3d_interface command. Also check the manpage of the command to correct any errors related to option settings.

---

## 3DIC-162

(error) Collection specified for -to_object_ref option contains object of invalid type: %s; Allowed types are: TSV_ViaDef, BumpCell_Design, BondPad_Def or BondPad_Layer.

**Description**

derive_3d_interface error message description is TBD

**What Next**

User should inspect all error and warning messages reported by derive_3d_interface command. Also check the manpage of the command to correct any errors related to option settings.

---

## 3DIC-163

(error) Total %d derived object will lie outside block boundary. First occurence is given below:

**Description**

derive_3d_interface error message description is TBD

**What Next**

User should inspect all error and warning messages reported by derive_3d_interface command. Also check the manpage of the command to correct any errors related to option settings.

---

## 3DIC-164

(error) Collection specified for -from option is empty. It must contain at least one object.

**Description**

derive_3d_interface error message description is TBD

Feedback

**What Next**

User should inspect all error and warning messages reported by derive_3d_interface command. Also check the manpage of the command to correct any errors related to option settings.

---

## 3DIC-165

(error) Collection specified for -to_object_ref option is empty. It must contain at least one object.

**Description**

derive_3d_interface error message description is TBD

**What Next**

User should inspect all error and warning messages reported by derive_3d_interface command. Also check the manpage of the command to correct any errors related to option settings.

---

## 3DIC-166

(error) Collection specified for -to_object_ref reference cell/layer contains %d entries, it must contain exactly one entry.

**Description**

derive_3d_interface error message description is TBD

**What Next**

User should inspect all error and warning messages reported by derive_3d_interface command. Also check the manpage of the command to correct any errors related to option settings.

---

## 3DIC-167

(error) Value specified for -coord_spec contains %d tokens, it must contain exactly 2 tokens.

**Description**

derive_3d_interface error message description is TBD

Feedback

**What Next**

User should inspect all error and warning messages reported by derive_3d_interface command. Also check the manpage of the command to correct any errors related to option settings.

---

## 3DIC-168

(error) Value specified for -coord_spec contains invalid token(s). It must be one of c|bl|br|tl|tr.

**Description**

derive_3d_interface error message description is TBD

**What Next**

User should inspect all error and warning messages reported by derive_3d_interface command. Also check the manpage of the command to correct any errors related to option settings.

---

## 3DIC-169

(error) Value specified for -connect must be one of true|false

**Description**

derive_3d_interface error message description is TBD

**What Next**

User should inspect all error and warning messages reported by derive_3d_interface command. Also check the manpage of the command to correct any errors related to option settings.

---

## 3DIC-170

(error) Command aborting due to input errors. No objects were created. Reason: %s

**Description**

derive_3d_interface error message description is TBD

**What Next**

User should inspect all error and warning messages reported by derive_3d_interface command. Also check the manpage of the command to correct any errors related to option settings.

Feedback

## 3DIC-171

(error) Command must specify -bond_pad_width/-bond_pad_height if to_object_ref is BondPad type.

**Description**

derive_3d_interface error message description is TBD

**What Next**

User should inspect all error and warning messages reported by derive_3d_interface command. Also check the manpage of the command to correct any errors related to option settings.

## 3DIC-172

(error) Command must specify positive value for -bond_pad_width and -bond_pad_height.

**Description**

derive_3d_interface error message description is TBD

**What Next**

User should inspect all error and warning messages reported by derive_3d_interface command. Also check the manpage of the command to correct any errors related to option settings.

## 3DIC-173

(warning) Command will ignore specified -bond_pad_width/-bond_pad_height because to_object_ref is not of BondPad type.

**Description**

derive_3d_interface error message description is TBD

**What Next**

User should inspect all error and warning messages reported by derive_3d_interface command. Also check the manpage of the command to correct any errors related to option settings.

## 3DIC-174

(warning) Created Objects cannot use _index1_index2 naming because %s.

Feedback

**Description**

If the object name to be copied have naming style like <string>_<index1>_<index2>, derive_3d_interface will name the derived objects with <name_prefix>_<index1>_<index2>. Otherwise, the command will name the object with <name_prefix>_<string>. And a postfix _<number> will be added when there is a name conflict.

**What Next**

User should inspect all error and warning messages reported by derive_3d_interface command. Also check the manpage of the command to correct any errors related to option settings.

---

## 3DIC-175

(warning) BlkInst: %s is connected to multiple nets: i) %s , and ii) %s ; First will be used for connecting derived object.

**Description**

derive_3d_interface error message description is TBD

**What Next**

User should inspect all error and warning messages reported by derive_3d_interface command. Also check the manpage of the command to correct any errors related to option settings.

---

## 3DIC-176

(error) No bump needs any RV landing via.

**Description**

There is not a single bump needs any RV landing.

**What Next**

Please check your design and the pattern file. Only bumps on the front side of the die can have RV landings. All bumps in the exclusion section of the pattern file are excluded. In addition, only bumps on the top metal layer are chosen by default. You can try to specify viaDef of the RV landing in the pattern file.

---

## 3DIC-177

(error) The following line in pattern file %s: %s.

Feedback

**Description**

The description of the pattern file is not clear.

**What Next**

Please check your pattern file. All lines must follow the pattern file format.

---

## 3DIC-178

(error) %s.

**Description**

The bump via insertion fails.

**What Next**

Please change the pattern file based on the error information.

---

## 3DIC-179

(error) distance value missing in line %s.

**Description**

No distance value is found in this line of the pattern file.

**What Next**

Please change the pattern file based on the error information.

---

## 3DIC-180

(error) Bump cell that specified by '-ref_cell %s' has no terminal on the metal layer %s.

**Description**

To form contact bump pair, the mirrored bump that specified by '-ref_cell %s’ should have terminals on the metal layer %s.

**What Next**

Please specify another lib_cell for "-ref_cell" or specify another orientation for the target chip by usign cmd "set_cell_location".

Feedback

## 3DIC-181

(error) via %s cannot be on level %d since it requires more routing layers than that defined by the current technology.

**Description**

User pattern file requires more layers than what is defined.

**What Next**

Please check the technology of the design or change the pattern file.

## 3DIC-182

(warning) Command will ignore specified -bond_pad_width/-bond_pad_height because to_object_ref is BondPad_Def type.

**Description**

derive_3d_interface error message description is TBD

**What Next**

User should inspect all error and warning messages reported by derive_3d_interface command. Also check the manpage of the command to correct any errors related to option settings.

## 3DIC-190

(Error) No correct %s in the top level has been defined or specified.

**Description**

In command *derive_3d_connections*, users can specify the corresponding nets and ports in the top level in the command option, otherwise, the error message will be reported.

**What Next**

Check the top-level netlists.

## 3DIC-191

(Error) No valid die has been defined or specified.

Feedback

**Description**

In command *derive_3d_connections*, users should specify the correct dies in the command option, otherwise, the error message will be reported.

**What Next**

Check the dies in the current design.

---

## 3DIC-192

(Warning) %s is not in the top level, skip it.

**Description**

In command *derive_3d_connections*, users can specify the corresponding nets and ports in the top level in the command option, otherwise, the warning message will be reported.

**What Next**

Check the top-level netlists.

---

## 3DIC-193

(Warning) Pin %s is connected to the top net %s which doesn't connect to the port %s, skip it.

**Description**

If a pin is not connected to the port with the same name, then it should not be connected to any net in the top level.

**What Next**

Check the top-level netlists.

---

## 3DIC-194

(Warning) Net %s is connected to the top net %s, skip it.

**Description**

If a net in the die is not connected to the top net with the same name, then it should not be connected to any net in the top level.

**What Next**

Check the top-level netlists.

Feedback

## 3DIC-195

(Error) More than one library %s have block with name %s.

**Description**

More then one library have the block with the required name, it's hard for the command to decide in which library the block need to be updated.

**What Next**

Check the design.

## 3DIC-196

(Error) No library has block %s.

**Description**

No valid library or no specified library has the block with name as %s.

**What Next**

Check the design.

## 3DIC-197

(Error) Can't find the port(s) '%s' that specified by '%s' from the instances that specified by '%s'.

**Description**

The port specified by '-input' or '-output' option should exist on the specified instances.

**What Next**

Check the design.

## 3DIC-198

(Warning) Command '%s' is deprecated, please use the command '%s'.

**Description**

The old command is deprecated and replaced with another command with same functions.

**What Next**

NA.

Feedback

## 3DIC-200

(info) %s.

**Description**

3DIC Information message

**What Next**

Not applicable

## 3DIC-201

(error) 3DIC top-level chip placement is not correct. Use check_3d_design -chip_placement to identify and resolve all issues.

**Description**

If 3DIC top-level design has chip placement issues, these should be debug using check_3d_design -chip_placement command, and all reported errors must be fixed before proceeding.

**What Next**

Use check_3d_design -chip_placement to identify and resolve all issues.

## 3DIC-202

(warning) Via for bump %s cannot be placed in net bounding box due to user location constraints.

**Description**

User has provided location constraints for the via. The location constraint, if honored, results in the placement of via outside of the bounding box of the corresponding net.

**What Next**

Please check the user constraints and make sure it is correct. In addition, please check the via location. Change location constraints in the pattern file is needed.

## 3DIC-203

(warning) Via for bump %s has %s spacing violation due to user location constraints.

Feedback

**Description**

User has provided the exact location for the via. Although the tool places the via at the location, there is at least one design rule violation. The via may be too close to or overlap with bumps, vias, pre-routes, or blockages.

**What Next**

Please check the user constraints and make sure it is correct. Alternatively, user can change the design to fix the design rule violation.

---

## 3DIC-204

(error) because %s, the following line in pattern file cannot be processed: %s.

**Description**

The specified the line in pattern file contains incorrect information.

**What Next**

Please delete or correct the line.

---

## 3DIC-205

(warning) only single via will be inserted by default, stacked via spacing from pattern file is ignored.

**Description**

Since user does not specify any stacked via information nor stack level, the tool will insert single vias not stacked vias. Therefore, the stacked via spacing information in the pattern file is ignored.

**What Next**

If stacked vias are needed, please specify additional information in the pattern file. For example, user can use stack_level keyword.

---

## 3DIC-206

(warning) stacked via spacing larger than stack level %d is ignored.

**Description**

If stacked via spacing cannot be applied, it will be ignored. For instance, if stacked level is 1, you will have two levels of vias. If you specify the via spacing constraint for the third level or more, the information is ignored.

Feedback

**What Next**

Please correct any mistake in the pattern file related to via configuration.

---

## 3DIC-207

(warning) via_def in config section is choose for level %d.

**Description**

if both configuration section and drc section specify via_def for the same level, the via_def in configuration section is used.

**What Next**

Please fix the pattern file if needed.

---

## 3DIC-208

(error) stacked vias creation failure for net %s.

**Description**

Errors are found during the via placement.Although the top level via is placed successfully, one or many stacked via are not placed. Please check the result.

**What Next**

Examine all vias on the net. Perform manual editing of the design if needed.

---

## 3DIC-209

(warning) ignore %d top routing layers because via %s is on level %d.

**Description**

User pattern file specifies the via level. Based on this specification, some routing layers are ignored. Note that only bumps on the top layer according to the pattern file are considered. It is very likely that the pattern file is incorrect.

**What Next**

Please check the technology of the design or change the pattern file.

---

## 3DIC-210

(error) via cannot be inserted for bump %s.

Feedback

**Description**

The tool fails to find a legal location to place a via for a bump. There can be various reasons for this to occur. There might be routes or blockages that prevent the placement of a via. There might be bumps or other vias that prevent the placement of a via. The bounding box of the bump net might be too small.

**What Next**

Please check your design. User can provide the exact location for the via using the location constraint in pattern file.

---

## 3DIC-211

(warning) multiple stack level constraints are given and the largest value will be used

**Description**

There are multiple stack level constraints in the pattern file. The tool will use the largest value.

**What Next**

Change pattern file if needed.

---

## 3DIC-212

(warning) extra via_def beyond stack level in config section of the pattern file is ignored.

**Description**

There are more via_def given than the value defined by stack level.

**What Next**

Change pattern file if needed.

---

## 3DIC-213

(warning) via for net %s cannot be placed on pre-routes.

**Description**

User has selected "flat_net" mode for the via insertion. However, the tool cannot place the via on any RDL-layer pre-routes of the net. There can be several reason for this situation. There might not be any pre-routes in the design. The net might not be connected to any other pins except for bump pins on RDL layer. The tool will try to place the via close to a bump of the net instead.

Feedback

**What Next**

Please check the via created. If user wants to place the via on pre-routes, please add routes on RDL layers either manually or using a RDL router.

---

## 3DIC-214

(error) The following line does not describe a single %s cell: %s.

**Description**

The tool expects the name of a single bump cell or a single reference cell in the given portion of the pattern file.

**What Next**

Please correct the pattern file.

---

## 3DIC-215

(warning) some bump is not placed on min-grid.

**Description**

When bumps are not on min-grid, escaping vias cannot be placed next to bumps. The tool will use an enlarge boundary box of the corresponding bump to place the via.

**What Next**

Check the bump placement. If needed, adjust the relative position constraint of escaping vias.

---

## 3DIC-216

(warning) Pattern cell does not contain any cell of type: flip_chip_pad.

**Description**

Pattern cell specified for create_bump_cluster command may not be valid for bump_cluster creation because pattern cell does not contain at least one bump cell.

**What Next**

Fix pattern cell for warning situation specified in message, or specify a different pattern cell.

Feedback

## 3DIC-220

(error) bump pair {%s %s} have different matching types {%s %s}.

**Description**

There are more than one matching type in a physical contact pair, it will cause error when propagate matching type from source chip to target chip.

**What Next**

Please check the design. Using -force may help to solve this conflict.

## 3DIC-221

(warning) bump pair {%s %s} have different matching types {%s %s}.

**Description**

There are more than one matching type in a physical contact pair.

**What Next**

Please check the design.

## 3DIC-222

(error) Matching types {%s} in target chip %s corresponding to the matching types {%s} in source chip during propagation.

**Description**

This messages appears because one of scenario below happens: 1. There are more than one matching types in target chip corresponding to the same matching type in source chip; 2. The corresponding matching type in source chip and target chip has different matching type name.

**What Next**

Please check the design. Using -force may help to solve this conflict.

## 3DIC-223

(warning) Matching types {%s} in target chip %s corresponding to the matching types {%s} in source chip during propagation.

Feedback

**Description**

This messages appears because one of scenario below happens: 1. There are more than one matching types in target chip corresponding to the same matching type in source chip; 2. The corresponding matching type in source chip and target chip has different matching type name.

**What Next**

Please check the design.

---

## 3DIC-224

(error) routing on layer %s cannot go around bump escaping vias.

**Description**

Bump escaping vias are aligned to their corresponding routing. The routing of signal nets between the lower and upper layer of a via must go around the via if the via is close. In current 3D routing where tracks are evenly spaced, the track of the via cannot be the same as that of the net which needs to go around the via. Otherwise, there is no room for the net to go round.

**What Next**

Please set the stub_alignment_type option to staggered_before or staggered_after in the pattern file.

---

## 3DIC-225

(warning) exclusion section is ignored when bump_array_box is used in single_hbm style.

**Description**

If user uses the bump_array_box to define a 3D channel, the exclusion section in the pattern file is ignored. The option excluded_bump_in_box should be used to exclude any bumps inside the bump array boxes.

**What Next**

Use excluded_bump_in_box if needed. Remove the exclusion section from the pattern file.

---

## 3DIC-226

(warning) The reference_cell and bump constraints in the inclusion or exclusion section are ignored when single_hbm style is chosen.

Feedback

**Description**

When the single_bump style is chosen, the channel components cannot be described using the reference_cell and bump constraints in the inclusion or exclusion section. The channel can be defined using net constraints. It is recommended to use bump_array_box constraint in the exception section for 3D channel definition.

**What Next**

User bump_array_box for 3D channel definition. The following keywords are supported for hbm channel definition: net in inclusion section, float_bump, 2pin_through_net, bump_array_box, and excluded_bump_in_box in the exception section.

---

## 3DIC-227

(error) The middle_turn_offset is too large to fit in the middle section of the 3D channel.

**Description**

The value given is too large.

**What Next**

Reduce the value. If the value cannot be reduced, please try to route the middle section of the channel using a different pattern.

---

## 3DIC-228

(error) bump array does not match the pattern of an 3D channel: %s.

**Description**

An 3D channel usually contains two bump arrays, on the two ends of the channel, respectively. There are same number of bumps in either bump array. The bump count in a bump array is usually an even number. All bumps must be described in the pattern file although some bumps might not need routing. Moreover, bumps form an array of either inline or staggered pattern on either side of the channel.

**What Next**

Modify the pattern file to fully describe the 3D channel. User can use net or bump description.

---

## 3DIC-229

(error) 3D channel route planning cannot handle nets connected to three or more bumps.

Feedback

**Description**

During 3D channel routing, only two pin nets are routed.

**What Next**

Modify the netlist and/or pattern file so that the target design meet an 3D channel description.

---

## 3DIC-230

(error) Net specified by 2pin_through_net has incorrect connectivity or a net is partially selected.

**Description**

A through net must connect a single bump at the front side of the die and a single bump at the back side of the die. If a net has other connectivity, it cannot be a through net. Another reason that this message is issued is that a net is partially selected for the 3D channel. User must select the entire net to proceed.

**What Next**

Check all through nets. Modify the netlist and/or pattern file to correct the connectivity information.

---

## 3DIC-231

(warning) routing stubs snapped to the min-grid.

**Description**

Routing shapes must be placed on the min-grid. The stub locations specified by user using boundary_stub_corners is not on the mid-grid. The tool moves the stubs to the nearest mid-grid locations.

**What Next**

Check the routing result. If needed, change the pattern file.

---

## 3DIC-232

(error) via dimension is too wide for track alignment.

**Description**

In 3D routing, the pitch of wire tracks is the sum of wire width and wire spacing. The wire pitch must be large enough so that bump connection vias can be placed on a wire without

Feedback

introducing routing rule vioation between the via and neighoring wire track. Normally, via width or height should not be larger than twice of wire pitch minus wire width, for vertical or horizontal channels, respectively.

**What Next**

Choose a different via configuration to reduce via dimension or increase wire pitch.

---

## 3DIC-233

(error) bump %s cannot be excluded since it is connected to a two-bump net inside the 3D channel.

**Description**

All nets connected to two and only two bumps inside the 3D channel must be routed. Any bump connected to such nets cannot be excluded using float_bump.

**What Next**

Modify the netlist by removing the bump from its net or change the pattern file. The netlist and pattern file must be consistent.

---

## 3DIC-234

(error) middle section turn direction is ambiguous, please %s mid-track routing pitch.

**Description**

The middle section routing cannot be completed. Please adjust midtrack_spacing and/or midtrack_width.

**What Next**

Change midtrack_spacing and/or midtrack_width.

---

## 3DIC-235

(error) middle section routes cannot be created: %s.

**Description**

The middle section routing cannot be completed. Please check the dimension of the specified fraction. In particular, the length of the fraction must be no less than the width of its neighbor fraction, which is equal to the routing pitch times N-1, where N is the number of tracks, including tracks with no route. This error only occurs for Manhattan style middle section routing.

Feedback

**What Next**

Modify the middle section constraint midtrack_lengths to change the fraction lengths or change fraction routing pitch using midtrack_turn_spacing, midtrack_turn_width, midtrack_spacing, midtrack_width.

---

## 3DIC-236

(error) %s.

**Description**

In the current routing mode, the wire width must be less than the wire spacing in order to derive a DRC violation free routing result.

**What Next**

Modify the wire_spacing and/or wire_width in the pattern file.

---

## 3DIC-237

(error) fail to assign nets to routing stubs.

**Description**

The tool cannot assign all nets to routing tracks inside the 3D channel. It is likely that the channel routing capacity, which is determined by boundary_stub_range, is insufficient.

**What Next**

Increase the channel capacity by changing the values after boundary_stub_range in the pattern file.

---

## 3DIC-238

(error) the number of nets is %d, which is larger than the maximal track count %d.

**Description**

The 3D channel contains more nets than the routing capacity. It is impossible to route all nets.

**What Next**

Modify the pattern file or the netlist.

Feedback

## 3DIC-239

(error) Routing layer %s is invalid.

**Description**

The routing layer cannot be used. User may use the incorrect routing layer name.

**What Next**

Please check the routing layer names.

## 3DIC-240

(error) cannot derive valid shielding layers for middle section.

**Description**

Normally, the shielding layers should not be the same as layers used for signal routing. For an 3D channel, two layers are used for signal routing. The two layers below the corresponding signal routing layers are candidated for middle section shielding.

**What Next**

Please check the signal routing layers. The two layers cannot be adjacent.

## 3DIC-241

(error) cannot route all %d nets with space only enough for %d tracks.

**Description**

There is not enough space to route all nets in the bump array regions of an 3D channel.

**What Next**

There are multiple ways to increase the channel capacity. User can reduce routing width and/or spacing. User can reduce the size of bump access vias. If user uses boundary_stub_range in the pattern file, its parameters can be changed to increase the route area.

## 3DIC-242

(error) cannot route all nets within the specified range.

**Description**

There is not enough space to route all nets within the user specified range.

Feedback

**What Next**

Please increase the boundary_stub_range in the pattern file.

---

## 3DIC-243

(error) not enough routing stubs are available for net assignment: have %d but expect %d.

**Description**

There is not enough space to route all nets within the user specified range.

**What Next**

Please increase the boundary_stub_range in the pattern file. User can try without using boundary_stub_range in order to find out how wide the channel need to be.

---

## 3DIC-244

(error) fail to assign nets to boundary stubs on layer %s.

**Description**

There is not enough routing stubs for all nets on the specified layer.

**What Next**

Please increase the boundary_stub_range in the pattern file. User can try without using boundary_stub_range in order to find out how wide the channel need to be.

---

## 3DIC-245

(warning) bump access via dimension is equal to or larger than half of track pitch.

**Description**

Wider vias occupy more space. Therefore it is more difficult to route around them. In the extreme case, routing violations might not be completed avoided.

**What Next**

Please check the routing result to search for any routing violations.

---

## 3DIC-246

(error) bump access via dimension is too large.

Feedback

**Description**

Wider vias occupy more space. The tool cannot find a routable solution.

**What Next**

Please increase the space between adjacent tracks or use narrow vias.

---

## 3DIC-247

(error) failed to create stacked vias: %s.

**Description**

The tool is not able to create stacked vias for signal routing due to the given reason.

**What Next**

Please fix the specified issue.

---

## 3DIC-248

(error) fail to assign tracks in bump array regions.

**Description**

The tool is not able to assign nets to routing tracks in the bump array regions. It is probably due to insuffcient tracks or tracks are not aligned to their corresponding bumps. Note that bump track locations are defined by boundary_stub_corners and are limited by boundary_stub_range.

**What Next**

Please increase the channel width defined by boundary_stub_range, or move the track ranges also defined by boundary_stub_range, or adjust bump-net assignment. User might also change the track locations by changing boundary_stub_corners and/or boundary_stub_range.

---

## 3DIC-249

(error) some new stub overlaps with bump-access vias.

**Description**

The tool is able to proceed. However, a routing stub overlaps with a bump-access via. The result is not routing rule clean.

Feedback

**What Next**

Please adjust pattern file paramenters to resolve overlap violations.

---

## 3DIC-250

(warning) Only %d routing layers are available.

**Description**

The tool is able to proceed. However, lack of routing layers might lead to bad routing and/or shielding result.

**What Next**

Please check the technology file and see whether any layer information is missing.

---

## 3DIC-251

(error) Cannot handle net %s with %d bumps.

**Description**

The tool can only route nets with two bumps.

**What Next**

Please check the netlist. If the netlist is correct, disconnect the bumps from the specified net and re-connect them after 3D routing.

---

## 3DIC-252

(error) bump %s and bump %s are not assigned to the same net.

**Description**

In an 3D design, there are two bump arrays at the two ends of the channel, respectively. Bumps in the two arrays are connected by 2-pin nets. For each pair of bumps connected by a net, they must be placed at the same relative locations inside the corresponding arrays.

**What Next**

Adjust the bump-to-net assignment or replace bumps.

---

## 3DIC-253

(error) fail to place access via for bump %s

Feedback

**Description**

Tool cannot find a feasible location to place a bump access via.

**What Next**

Reduce the via dimension described in the pattern file.

---

## 3DIC-254

(warning) bump %s ignored in low design hierarchy.

**Description**

Tool only processes top level bumps.

**What Next**

Please pop-up bumps in side any block.

---

## 3DIC-255

(error) %s are placed at different layers.

**Description**

all micro-bumps must be on the same metal layers. All C4 bumps must be on the same layer.

**What Next**

Check bumps or C4s in the design. Remove nets from the command line option if needed.

---

## 3DIC-256

(error) the chip specified by -to should be block with design view.

**Description**

The target chip should be block with design view.

**What Next**

Please check the design.

---

## 3DIC-257

(error) %s is destructible, it contacts with %s %s.

Feedback

**Description**

The object that contacts with destructible probe pad or probe pad cell should also be probe pad or probe pad cell and both have the same top level connection. The "destrutible" object is defined by the attribute "is_destrutible".

**What Next**

Please check the design.

---

## 3DIC-260

(error) The pg net %s doesn't connect to the package.

**Description**

The package supply power to the pg net in the design, the pg net should connect to the port on the package.

**What Next**

Please check the design.

---

## 3DIC-261

(error) The pg net %s is not continuous between package and %s.

**Description**

The package supply power to the pg net in the design and the pg net should always connect to the vertical adjacent dies. The the Z value of the dies that connect to the same PG net should be continuous.

**What Next**

Please check the design.

---

## 3DIC-262

(error) %s %s does not physically connect to the %s on the adjacent side of the vertical neighboring chip, the reason is '%s'.

**Description**

Found dangling dummy bump or bond pad on the chip.

Feedback

**What Next**

Adjust the location of chips or remove the dangling bump/bond pad or use create_3d_mirror_bumps to create bump pair/bond pad pair.

---

## 3DIC-263

(error) Non-aligned %s pair {%s, %s}, exceed tolerance of %s.

**Description**

%s %s and %s are physically contact but non-aligned, exceed tolerance of %s.

**What Next**

Adjust bump or bond pad location or enlarge the alignment threshold of two bumps' center by set plan.3dic.bump_alignment_threshold.

---

## 3DIC-264

(error) %s %s contacts to more than one %s {%s} of the neighboring die.

**Description**

%s %s contacts to more than one %s {%s} of the neighboring die.

**What Next**

Adjust bump location.

---

## 3DIC-265

(error) %s {%s, %s} overlap with each other in a single chip.

**Description**

%s {%s, %s} overlap with each other in a single chip.

**What Next**

Adjust bump or bond pad location.

---

## 3DIC-266

(warning) %s %s is not covered by the %s neighboring chip.

**Description**

Found dangling bump or bond pad is not covered by the vertical neighboring chip .

Feedback

**What Next**

Adjust the location of chips or remove the dangling bump or bond pad if needed.

---

## 3DIC-267

(warning) %s %s is not covered by the %s neighboring chip.

**Description**

Found dangling dummy bump or bond pad is not covered by the vertical neighboring chip .

**What Next**

Adjust the location of chips or remove the dangling dummy bump or bond pad if needed.

---

## 3DIC-268

(error) The dummy %s %s connect to no-dummy cell(s) %s in the same die.

**Description**

The dummy bump or bond pad connect to no-dummy cells in the same die which is not allowed.

**What Next**

Check the net connections of the dummy bump.

---

## 3DIC-269

(error) The dummy %s %s connect to more than one no-dummy cells which are in different dies.

**Description**

The dummy bump or bond pad connect to more than one no-dummy cells, this dummy bump or bond pad may be used for the net connections which is not allowed.

**What Next**

Check the net connections of the dummy bump.

---

## 3DIC-270

(error) The %s spacing between the chip {%s} and {%s} is %s which violate the constraint. The valid range is {%s, %s}.

Feedback

**Description**

In 3DIC design, user can set some spacing constraint for any two chips of same vertical level by setting app options: "plan.3dic.chip_vertical_spacing" or "plan.3dic.chip_horizontal_spacing", these constraints will be honored by check commands to report violations.

**What Next**

Double confirm the reported violation is as expected. If yes, correct the 3D chip placement in top level design. If not, re-set the constraint to proper value.

---

## 3DIC-271

(error) The %s misalignment between the chip {%s} and {%s} is %s which violate the constraint. The valid range is {%s, %s}.

**Description**

In 3DIC design, user can set some alignment constraint for any two chips of same vertical level by setting app options: "plan.3dic.chip_vertical_misalignment" or "plan.3dic.chip_horizontal_misalignment", these constraints will be honored by check commands to report violations.

**What Next**

Double confirm the reported violation is as expected. If yes, correct the 3D chip placement in top level design. If not, re-set the constraint to proper value.

---

## 3DIC-272

(error) The enclosure spacing between the chip {%s} and {%s} is %s which violate the constraint. The valid range is {%s, %s}.

**Description**

In 3DIC design, user can set enclosure constraint for two vertical neighboring chips by setting app option: "plan.3dic.chip_enclosure", this constraint will be honored by check commands to report violations.

**What Next**

Double confirm the reported violation is as expected. If yes, correct the 3D chip placement in top level design. If not, re-set the constraint to proper value.

Feedback

## 3DIC-273

(error) The %s %s %s is in the forbidden zone.

**Description**

In 3DIC design, user can set forbidden zone for connected or unconnected bump by setting app options: "plan.3dic.forbidden_spacing_of_unconnected_bump" or "plan.3dic.forbidden_spacing_of_connected_bump", this constraint will be honored by check commands to report violations.

**What Next**

Double confirm the reported violation is as expected. If yes, correct the bump placement in the chips. If not, re-set the constraint to proper value.

## 3DIC-274

(error) The %s %s %s is in the forbidden zone.

**Description**

In 3DIC design, user can set forbidden zone for connected or unconnected bump by setting app options: "plan.3dic.forbidden_spacing_of_unconnected_bump" or "plan.3dic.forbidden_spacing_of_connected_bump", this constraint will be honored by check commands to report violations.

**What Next**

Double confirm the reported violation is as expected. If yes, correct the bump placement in the chips. If not, re-set the constraint to proper value.

## 3DIC-275

(error) The %s of bump cluster %s is larger than the constraint.

**Description**

In 3DIC design, user can set app option: "plan.3dic.max_bump_cluster_size" to make sure all bump clusters are within the specified {width height}, this constraint will be honored by check commands to report violations.

**What Next**

Double confirm the reported violation is as expected. If yes, re-generate the bump clusters. If not, re-set the constraint to proper value.

Feedback

## 3DIC-276

(error) Fail to create routing guidance file: %s.

**Description**

Cannot open the guidance file. It could happen when you do not have write permission in current working directory.

**What Next**

Please make sure that the guidance file can be created.

## 3DIC-277

(error) Fail to create routing guidance file for middle section routing.

**Description**

This error occurs when simple angled routing in the middle section fails due to insufficient routing resource.

**What Next**

Please check the design and adjust wire_spacing, wire_width, boundary_stub_gaps, or the number of routing layers in the pattern file.

## 3DIC-278

(error) Fail to create routing guidance file for vias to C4 bump connection.

**Description**

This error occurs due to improper C4 bump arrangement or wrong net-to-C4 assignment. For example, two C4 bumps are too close, or there are two or more nets assigned to the same C4 bump.

**What Next**

Please check and adjust the C4 bump arrangement.

## 3DIC-279

(error) Route planning failed for metal layer %s due to track bump alignment.

Feedback

**Description**

For two layer routing, tracks can be aligned in three different types (inline, staggered_before, staggered_after) specified by stub_alignment_type in the pattern file. This error occurs when the wire width and spacing cannot satisfy the specified alignment type.

**What Next**

Please adjust the settings related to wire width and spacing in the pattern file: wire_width, midtrack_turn_width, midtrack_width, wire_width_left_array, wire_width_lower_array, wire_width_right_array, wire_width_upper_array, boundary_stub_staggered_offset, wire_spacing, midtrack_spacing, midtrack_turn_spacing, boundary_stub_gaps, boundary_stub_gaps_high, boundary_stub_gaps_low, boundary_stub_gaps_middle.

---

## 3DIC-280

(error) Fail to place escaping vias to access bumps for lower routing layer.

**Description**

Fail to place escaping vias to access bumps in lower routing layer. This error may result from insufficient routing resource in bump array or multiple nets assigned to the same bump.

**What Next**

Please check if the routing channel is sufficient to route the number of target nets with given routing pitch, considering the space for escaping via placement or there are more than one net assigned to the same bump.

---

## 3DIC-281

(error) %s.

**Description**

Stubs are small routing segments to guide the detailed router to complete 3D routing. Boundary stubs are placed between bump array and middle section. The tool may fail to create stubs when the offset of stubs is too large or when the channel width is too small.

**What Next**

Please decrease boundary_stub_gaps. Adjust routing pitch using command option -wire_width or -wire_spacing. Alternatively, adjust wire_spacing or wire_width in pattern file.

Feedback

## 3DIC-282

(error) Error found when loading via_def and metal layers: %s.

**Description**

This error occurs when loading via_def and metal layers due to the specifed issue.

**What Next**

Please fix the specified issue.

## 3DIC-283

(error) Fail to route differential net pair.

**Description**

This error occurs when routing differential net pair failed because there is no enough space for differential net pair gap.

**What Next**

Please adjust differential_net_spacing in the pattern file.

## 3DIC-284

(error) Fail to place vias according to given location constraints.

**Description**

This error occurs because the given via locations are too close or the via location constraint data are not complete.

**What Next**

Please check and adjust the given via location constraints.

## 3DIC-285

(error) %s.

**Description**

This error occurs when the constraint value in the pattern file is invalid.

Feedback

**What Next**

Please check and correct the constraint value in the pattern file. See more explanation for pattern file constraints by using the command list_3d_routing_constraints.

---

## 3DIC-286

(error) Loading %s failed.

**Description**

This error occurs when loading the specified data from the tech file or the design failed.

**What Next**

Please check the tech file and the design.

---

## 3DIC-287

(error) Only 2-pin nets can be processed: %s.

**Description**

We expect to route the nets that connect bump-to-bump or bond-pad-to-bond-pad in 3D routing.

**What Next**

Please check if you have non-2-pin net.

---

## 3DIC-288

(error) Bump information error: %s.

**Description**

uBump layer should be on the top of C4 bump layer and PG net layer. A ubump should aligned inside the corresponding C4 bump. A C4 bump can contain at most one ubump. Floating bumps cannot have net connection. Missing lib-cell for bumps.

**What Next**

Please fix the specified issue. Check the layer configuration in the tech file. Adjust the positions of ubump and C4 bump. Remove nets on the floating bumps. Check the missing lib-cell for bumps in the cell library file.

Feedback

## 3DIC-289

(warning) Preroutes or vias exist in bump array regions.

**Description**

There exist preroutes or preplaced vias in bump array regions, which may reduce routability. Routing can still be successful. User needs to check whether these components are there due to mistakes.

**What Next**

Please remove unintended existing preroutes or vias in bump array regions.

## 3DIC-290

(error) Some bumps are not in pairs.

**Description**

In 3D routing, each 2-pin net should connect a pair of bumps. This error occurs when some bumps are not in pair for a net.

**What Next**

Please check and correct the bump-to-net assignment.

## 3DIC-291

(error) Total number of open nets = %d.

**Description**

This error message shows the number of open nets after auto 3D or die-to-die routing.

**What Next**

Please check the open nets in the design. Try to adjust wire_width, wire_spacing, or other constraints regarding wire spacing/width in the pattern file.

## 3DIC-292

(error) Non-uniform bumps detection error in narrow bump gap case.

**Description**

This error occurs when non-uniform bump dimension is found in the narrow bump gap case.

Feedback

**What Next**

Please increase bump gap in bump arrays.

---

## 3DIC-293

(error) %s.

**Description**

There exist two bumps that are supposed to be assigned to the same net but actually mismatched.

**What Next**

Please check and adjust the bump-to-net assignment. Make sure that a 2-pin net can only be assigned to exactly two bumps.

---

## 3DIC-294

(error) Narrow bump gaps are found in %s 3D bump array(s).

**Description**

In 3D routing, the gap between two bumps in the bump array(s) should keep a minimum distance for escaping routing. However, the bump gaps in %s 3D bump array are found to be too narrow.

**What Next**

Please check and increase the bump gaps in the 3D bump array(s).

---

## 3DIC-295

(error) Incorrect bump pairing specification.

**Description**

This error occurs when the bump pairing specification is incorrect.

**What Next**

NA.

---

## 3DIC-296

(error) Incorrect channel orientation.

Feedback

**Description**

This error occurs when the channel orientation is incorrect for the placement of bump arrays.

**What Next**

Please adjust channel_direction in the pattern file.

---

## 3DIC-297

(error) Incorrect routing layers specified: %s.

**Description**

The specified routing layer is incorrect or cannot be used.

**What Next**

Please specify valid routing layers in the tech file.

---

## 3DIC-298

(error) Incorrect stacked vias discovered.

**Description**

This error occurs when a bump is connected to multiple stacked vias or a stacked via is not connected to a correct bump.

**What Next**

Please check and correct the stacked vias connected to lower bumps.

---

## 3DIC-299

(error) Inclusion and exclusion sections cannot exist at the same time.

**Description**

When processing the RV pattern file for RV placement, the inclusion and exclusion sections exist at the same time.

**What Next**

Please check and adjust the inclusion and exclusion sections in the RV pattern file.

Feedback

## 3DIC-300

(warning) port duplication cannot be applied to power net %s.

**Description**

power nets specified by -nets or -ports are ignored by command.

**What Next**

No action is needed.

## 3DIC-301

(error) no valid net is specified for port duplication.

**Description**

User specified nets are all power nets which cannot be processed.

**What Next**

Please choose the correct nets or use no option.

## 3DIC-302

(warning) die %s is not hierarchical.

**Description**

If a die is a leaf cell, no new feedthrough can be created and no new port can be created on it.

**What Next**

Please change the design data if the die needs to be modified.

## 3DIC-303

(error) net %s cannot be processed since die %s is not hierarchical.

**Description**

Non-hierarchical dice cannot be changed. As a result, the specified net cannot be processed.

**What Next**

Please change the design data if the die needs to be modified.

Feedback

## 3DIC-304

(error) lower half of the %s bump array is not aligned to higher half.

**Description**

According to 3D design standard, the bumps at either end of an 3D channel must form a single array. The tool has detected misalignment of the bumps.

**What Next**

Please correct the mis-placed bumps.

## 3DIC-305

(error) lower half of the %s bump array has different number of rows in comparison to higher half.

**Description**

According to 3D design standard, the bumps at either end of an 3D channel must form a single array. The tool has detected misalignment of the bumps.

**What Next**

Please correct the mis-placed bumps.

## 3DIC-306

(error) bump pairing information does not match bump placement.

**Description**

Bump row and/or column index based on current bump placement does not match bump pairing information. It is likely that the bump are not placed correctly. It is also possible that bump pairing information is incorrect. Please examine the pairing information and make sure that all bumps in the pairs exist in the bump array.

**What Next**

Please correct the mis-placed bumps or modify the bump pairing information.

## 3DIC-307

(error) bump at {%d %d} already appears in a pair.

Feedback

**Description**

A bump can only appear in a single pair in bump_index_pair constraints at most.

**What Next**

Please correct the bump_index_pair constraints.

---

## 3DIC-308

(error) net %s is skipped since it is not on top level.

**Description**

Nets specified by -nets must be on top level.

**What Next**

Please change the net name.

---

## 3DIC-309

(warning) net %s is skipped since it belongs to the same flat net of net %s.

**Description**

The net is skipped because it belongs to the same flat net of another net which is already processed.

**What Next**

Please check the result netlist.

---

## 3DIC-310

(error) feedthrough port creation failed for net %s.

**Description**

The net needs some new feedthrough ports which cannot be created.

**What Next**

Please process the net manually.

---

## 3DIC-311

(warning) only %d tracks of routes can be placed on layer %s.

Feedback

**Description**

The C4 bump is not big enough to contain the number of tracks specified by the user. Only a small number of routes are added.

**What Next**

Please change the pattern file.

---

## 3DIC-312

(error) bump %s interferes with a via of net %s.

**Description**

Although a via is placed, there is a bump of other net too close. The min-spacing is violated.

**What Next**

Please adjust the pattern file or netlist to resolve the violation. For example, rdl_via_offset can be increased. User can specify a smaller spacing using rdl_route_spacing in the pattern file.

---

## 3DIC-313

(error) routing detour placed with routing rule violation.

**Description**

Routing cannot be completed because the routing detour around bump-accessing vias are too close to other routes.

**What Next**

Use detour_spacing in the pattern file to adjust the detour routing location. Check tech file to be sure its correctness.

---

## 3DIC-314

(error) stub range %s micron is not enough to hold %d net tracks on the %s side of the 3d channel.

**Description**

The stub range constraints define a narrow corridor which is not wide enough to route all nets of the 3D channel.

Feedback

**What Next**

Please check and modify the constraints in the pattern file. For example, user can increase the stub range of reduce the routing width and/or spacing.

---

## 3DIC-315

(error) No bumps are selected.

**Description**

The tool cannot find any bump to process.

**What Next**

Please check the command options.

---

## 3DIC-316

(error) layer information is missing for PG net %s.

**Description**

The tool cannot find layer information of the given PG net.

**What Next**

Please check the command options.

---

## 3DIC-317

(error) technology file is missing.

**Description**

The tool cannot find technology file, which is required to collect information such as layers.

**What Next**

Please check and fix the design.

---

## 3DIC-318

(error) %s.

**Description**

The tool find information that is unexpected and cannot be used.

Feedback

**What Next**

Please check and fix the command-line option values.

---

## 3DIC-319

(error) cannot reach pin layer %s of bump %s.

**Description**

The layer description is insufficient.

**What Next**

Please add more layers.

---

## 3DIC-320

(error) margin plus length is larger than the bump dimension for %s.

**Description**

The margin value and/or wire length is too large.

**What Next**

Please reduce margins and/or wire lengths.

---

## 3DIC-321

(error) grill span exceeds bump dimension for %s.

**Description**

The grill dimension is too large.

**What Next**

Please reduce wire width, wire spacing, margins, and/or number of routes.

---

## 3DIC-322

(error) cannot connect to pg net for bump %s.

**Description**

The grill cannot be connect to existing pg net.

Feedback

**What Next**

Please check whether the bump is placed under existing pg routes.

---

## 3DIC-323

(error) grill broken for bump %s.

**Description**

Either routes or vias cannot be added for a layer for the specified bump.

**What Next**

Please check the input parameters and modify.

---

## 3DIC-324

(warning) bump %s is connected to a different net, skipping.

**Description**

The bump is connected to a different net.

**What Next**

Please check the netlist.

---

## 3DIC-325

(error) There are multiple tech files {%s} in chip %s.

**Description**

Multiple tech files are used inside a single die, this is invalid. This should only be allowed on the top-level of the 3DIC flow. It should not be allowed at the die-level and below.

**What Next**

Check the tech file setting in the chip.

---

## 3DIC-326

(warning) Total %d %s {%s} were snapped to match the grid in target chip %s.

Feedback

**Description**

The bump cell/bond pad/bump cluster must placed to the grid correctly, if not, the tool will snap the location of target bump to grid. If the snapped behavior is not expected, please add "-no_snap" in command "create_3d_mirror_bumps" to prevent the snap.

**What Next**

NA.

---

## 3DIC-327

(error) The dummy %s %s contacts to the %s %s which is not dummy.

**Description**

The bump/bond_pad marked as is_dummy true must be in contact with any bump/bond_pad marked as is_dummy true.

**What Next**

Check the 'is_dummy' attribute of bumps or bond pads.

---

## 3DIC-328

(Warning) unable to set location of %d port(s) %s: %s.

**Description**

These ports cannot be set location because the is_pad attribute of related pin in bump cell is false.

**What Next**

please check the lib cell of connected bump.

---

## 3DIC-329

(Warning) the pin %s cannot be set the location to %s, the reason is '%s'.

**Description**

these pins cannot be set the location to the specified port because of the reason message describes.

**What Next**

please check the lib cell of connected bump.

Feedback

## 3DIC-330

(error) these bumps that are mirrored by %s %s and %s %s will be overlapped in the die %s.

**Description**

During the process of mirroring bumps, these new created bumps may be overlapped with each other, this behavior violates DRC rules and cannot be accepted. If this situation is expected by customer, please add the "-force" command option.

**What Next**

Please check the location of source bumps.

## 3DIC-331

(error) the layout for %s probe bumps {%s} is not vertices of a square with edges parallel to die edges which is violated the rule %s in the chip %s.

**Description**

NA.

**What Next**

Please adjust the location of probe bumps.

## 3DIC-332

(error) the number of %s probe bumps is %s which is %s than the constraint <%s>, the valid range is {%s %s} in the chip %s.

**Description**

The rule value of single probe bumps is "NUM_PROBE". The rule value of quad or group probe bumps is "NUM_QPROBE".

**What Next**

Please adjust the number of probe bumps.

## 3DIC-333

(error) the number of %s bumps that connect to %s is %s which is %s than the constraint <%s>, the valid range is {%s %s} in the chip %s.

Feedback

**Description**

The rule value of single probe bumps is "NUM_PROBE_POWER" or "NUM_GPROBE_POWER".

**What Next**

Please adjust the number of probe bumps.

---

## 3DIC-334

(error) the pitch between %s probe bumps {%s} and {%s} is %s which is %s than the constraint <%s>, the valid range is {%s %s} in the chip %s.

**Description**

The rule value of quad or group probe bumps is "PITCH_GPROBE_GPROBE".

**What Next**

Please adjust the location of probe bumps.

---

## 3DIC-335

(error) find the dangling %s probe bumps {%s} that violate the constraint <%s>, the valid range is {%s %s} in the chip %s.

**Description**

The pitch rule value of quad or group probe bumps is "PITCH_GPROBE_GPROBE". or the pitch rule value of quad or group probe bumps is "PITCH_GPROBE_BUMP".

**What Next**

Please adjust the location of probe bumps.

---

## 3DIC-336

(error) the %s pitch between %s probe bump %s and single probe bump %s is %s which is %s than the constraint <%s> setting, the valid range is {%s %s} in the chip %s.

**Description**

The pitch rule value is "PITCH_GPROBE_PROBE_ORTHOGONAL".

**What Next**

Please adjust the location of probe bumps.

Feedback

## 3DIC-337

(error) the pitch between %s probe bumps {%s} and single probe bump %s is %s which is %s than the constraint <%s> setting, the valid range is {%s %s} in the chip %s.

**Description**

The pitch rule value is "PITCH_GPROBE_BUMP";

**What Next**

Please adjust the location of probe bumps.

## 3DIC-338

(error) find the dangling single probe bump %s that violate the constraint <%s>, the valid range is {%s %s} in the chip %s.

**Description**

The pitch rule value of single probe bump is "PITCH_PROBE_BUMP_SQUARE". The pitch rule value of single probe bump is "PITCH_PROBE_DUMMYE". The pitch rule value of single probe bump is "PITCH_PROBE_BUMP".

**What Next**

Please adjust the location of probe bumps.

## 3DIC-339

(error) the pitch between single probe bump %s and %s in the true-dummy region is %s which is %s than the constraint <%s> setting, the valid range is {%s %s} in the chip %s.

**Description**

The rule value of single probe bump is "PITCH_PROBE_DUMMYREGION".

**What Next**

Please adjust the location of probe bumps.

## 3DIC-340

(error) the pitch between single probe bump %s and %s in the non-dummy region where the single probe bumps are in %s pattern is %s which is %s than the constraint <%s> setting, the valid range is {%s %s} in the chip %s.

Feedback

**Description**

The rule value of single probe bump is "PITCH_PROBE_SQUARE" or "PTICH_PROBE".

**What Next**

Please adjust the location of probe bumps.

---

## 3DIC-341

(error) the pitch between single probe bump %s and unprobed %s-dummy bump %s where the single probe bumps are in %s pattern is %s which is %s than the constraint < %s> setting, the valid range is {%s %s} in the chip %s.

**Description**

The rule value of single probe bump is "PITCH_PROBE_DUMMY" or "PITCH_PROBE_BUMP" or "PITCH_PROBE_BUMP_SQUARE".

**What Next**

Please adjust the location of probe bumps.

---

## 3DIC-343

(Information) No valid anchor pins are found, so these top nets between chip %s and chip %s will be skipped.

**Description**

Anchor pins are connected to the top nets in the die, these pins are used to determine the location of port and which bump pair should be used to assign the corresponding top nets.

**What Next**

Check the connectivity between neighboring chips.

---

## 3DIC-344

(Information) Collect %lu out of %lu nets between chip %s and %s for assignment.

**Description**

Collect top nets for net assignment between neighboring chips. These top nets that were already assigned to bumps or didn't connect to anchor pins are skippped.

**What Next**

NA.

Feedback

## 3DIC-345

(Information) %lu total nets are found between chip %s and chip %s. Bumps are fully assigned, no nets will be assigned.

**Description**

Nets between specified dies already have bumps assigned, no assignment is needed. Command will not re-assign existing assigned bumps and nets.

**What Next**

NA.

## 3DIC-346

(error) bump resource is not enough%s. Net count is %d while bump count is %d.

**Description**

The number of top nets that connect to anchor pins is larger than the number of bumps with the same matching type.

**What Next**

check the matching type setting of bumps.

## 3DIC-347

(warning) missing feedthrough port for top net %s in the chip %s.

**Description**

When propagate/assign top nets, the tool needs to punch new ports to construct the complete connections.

**What Next**

Use command propagate_3d_connections or assign_3d_interchip_nets to construct complete connections.

## 3DIC-348

(warning) port %s connect to %d terminals, exceed the limitation %d, skip to set location for this port.

Feedback

**Description**

If the port connects to too many terminals, the tool will skip to update the port location in order to reduce runtime. The default max number of terminals is 200.

**What Next**

Adjust the max number of terminals by app option 'plan.3dic.max_terminal_number_for_port_setting'.

---

## 3DIC-349

(Information) create new top net %s from top net %s.

**Description**

create the new top net by splitting the existed top net in order to connect the adjacent chip.

**What Next**

NA.

---

## 3DIC-350

(Information) No DFT issues found.

**Description**

This message tells you that the command found no DFT issues.

**What Next**

No further action in terms of DFT is required.

---

## 3DIC-351

(error) There are more than one type signal of type %s specified those signals are %s.

**Description**

There are more than one type signal of type specified in the message marked as specified TAP type.

**What Next**

Look at design and figure which one of the ports marked of specified type is a duplicate and remove it.

Feedback

## 3DIC-352

(error) Die %s is missing %d %s signals of types: %s

**Description**

The two ports mentioned in the error message are not correctly connected in the top level netlist.

**What Next**

Check the design for one of the two issues. The DFT marking on the specified port was incorrectly set. The logical connection to the specified port is not correct in the top level netlist.

## 3DIC-353

(error) Port %s of type %s is connected to %s of type %s should be connected to type %s.

**Description**

The ports mentioned in the error message is not connected a port of the correct type.

**What Next**

Check the design for one of the two issues. The DFT marking on the specified port was incorrectly set. The logical connection to the specified port is not correct in the top level netlist.

## 3DIC-354

(warning) %s, skipping.

**Description**

Tool cannot add enough via arrays for the specified bump.

**What Next**

Please check whether there are enough routes on the right layer near the bump.

## 3DIC-355

(warning) unexpected sub-channel dimension and/or count: %d sub-channels discovered.

Feedback

**Description**

Tool finds that the number of sub-channels is different from that in 3D standard. An 3D channel should have 37 sub-channels. Tool will route all sub-channels discovered.

**What Next**

Please check whether the bump arrays are correct.

---

## 3DIC-356

(warning) layer %s skipped since it is not below target layer %s.

**Description**

Tool skips a user-specified layer during create_vertical_connection run.

**What Next**

Please check whether the result is acceptable. Adjust target layer or layer list if needed.

---

## 3DIC-357

(warning) skipping non-bump cell %s.

**Description**

Tool skips a cell because it is not a bump.

**What Next**

Please check cell types.

---

## 3DIC-358

(warning) skipping cell %s since it is not on top level.

**Description**

Tool can only process cell on top level.

**What Next**

Please check physical hierarchy of cells.

---

## 3DIC-359

(error) cannot perform shielding since there is no PG net to use.

Feedback

**Description**

Tool cannot perform shielding for 3D channels because there is no PG net in the design or all PG nets do not have routes to tie to.

**What Next**

Please create either power or gound net. If users select a tie option, please route the PG net first so that the shielding tie can be made.

---

## 3DIC-360

(error) %s on net %s is not connected to correct TAP.

**Description**

The ports mentioned in the error message is missing a connection to a port of the correct type.

**What Next**

Add the marking that is missing on the specified net.

---

## 3DIC-361

(error) %s incorrectly connected to %s on net %s.

**Description**

The ports mentioned in the error message is not connected a port of the correct type.

**What Next**

Check the design for one of the two issues. First, make sure that the DFT marking on the specified port was set. Second, make sure that the logical connection to the specified port is correct in the top level netlist.

---

## 3DIC-362

(error) The direction of port %s is set to %s, it should be set to %s.

**Description**

The direction on the mentioned port is not set to the correct direction.

**What Next**

Correct the direction of the port in the netlist.

Feedback

## 3DIC-363

(warning) Ignoring Bump %s at location (%.4f,%.4f) due to overlap with another bump

**Description**

This message tell you that when the tool finds another bump in the same location as a previous bump that bump is ignored.

**What Next**

Modify your design so that bumps are not overlapping.

## 3DIC-364

(warning) Command should be run from the virtual 3DIC top. Bond pads will be ignored.

**Description**

This message tells you that you ran the command from the die instead of the virtual 3DIC top and that bond pads will be ignored.

**What Next**

Run the command form the virtual 3DIC top instead of the die.

## 3DIC-365

(Information) Auto-generating tolerance values as no tolerances were specified.

**Description**

This message tells you that the command is auto-generating tolerance values.

**What Next**

If the auto-generated tolerance is too small or large rerun the command specifing a better tolerance value.

## 3DIC-366

(Information) Using auto-generated tolerances values: x=%f, y=%f.

**Description**

This message tells you that values the command is using for auto-generated tolerance values.

Feedback

**What Next**

If the auto-generated tolerance is too small or large rerun the command specifing a better tolerance value.

---

## 3DIC-367

(Warning) The terminal %s on the %s side does not have 'is_bond_pad' attribute.

**Description**

The terminal %s on the %s side will not be seen as a valid bond_pad since it does not have 'is_bond_pad' attribute.

**What Next**

If this terminal should be a bond_pad, please use command "set_attribute" to set 'is_bond_pad' attribute as true.

---

## 3DIC-368

(Information) Value specified for -x_tolerence is too small, adjusting from %f to %f.

**Description**

This message tells you that the value you specified for -x_tolerance is too low and the tool has adjusted it larger value for you.

**What Next**

Check the value specified for -x_tolerance and make sure that it correct.

---

## 3DIC-369

(Information) Value specified for -y_tolerence is too small, adjusting from %f to %f.

**Description**

This message tells you that the value you specified for -y_tolerance is too low and the tool has adjusted it larger value for you.

**What Next**

Check the value specified for -y_tolerance and make sure that it correct.

Feedback

## 3DIC-370

(error) Object %s is not a port, leaf pin, hierarchical pin or net and therefore cannot be used with -3d_objects option.

**Description**

Only objects that are ports, leaf pins, hierarchical pins or nets can be specified with the -3d_objects option.

**What Next**

Remove the objects that do not meet the criteria from the -3d_objects option.

## 3DIC-371

(warning) Ignoring unconnected bump %s

**Description**

This message informs you that tool has found an signal bump that is not connected to a verilog port and omitted it from the fault lists because it is not connected to a net.

**What Next**

Connect the specified bump to a verilog port.

## 3DIC-372

(error) Bump-access via has DRC violations which cannot be fixed.

**Description**

This error occurs when bump-access via DRC violation is found due to insufficient room, and the violation cannot be removed.

**What Next**

Please set bump_via_shift_allowed or adjust rdl_via_spacing in the pattern file.

## 3DIC-373

(error) Incorrect stub-net layer assignment for switching routing layer in the middle section.

**Description**

A 2-pin net should access the bumps in two bump arrays from different layers. This error occurs when the layer assignment is incorrect.

Feedback

**What Next**

NA.

---

## 3DIC-374

(error) Cannot route nets on higher routing layer.

**Description**

This error occurs when routing nets on higher routing layer due to congestion.

**What Next**

Please adjust wire_spacing, wire_width, boundary_stub_gaps, for higher routing layer in the pattern file.

---

## 3DIC-375

(error) %s.

**Description**

This error occurs when middle section routing failed due to congestion.

**What Next**

Please adjust wire_spacing, wire_width, boundary_stub_gaps, or the number of routing layes in the pattern file.

---

## 3DIC-376

(error) No enough room to place bump-access vias inside C4 bump %s.

**Description**

This tool fails to place bump-access vias inside C4 bump due to not enough room in the bump array.

**What Next**

Please adjust the bump placement or adjust rdl_via_offset and rdl_via_spacing in the pattern file.

---

## 3DIC-377

(error) Cannot place stubs in the middle section for nets connected to C4 bumps from %s layer.

Feedback

**Description**

The tool fails to plan stubs in the middle section for nets connected to C4 bumps due to routing congestion.

**What Next**

Please adjust wire_spacing or wire_width in the pattern file.

---

## 3DIC-378

(error) Cannot place stacked vias for %s metal layer routing.

**Description**

This tool cannot create stacked vias from lower/higher metal layer to bumps. This error occurs when there exists bump overlap or via_def is not found.

**What Next**

Please fix the specified issue.

---

## 3DIC-379

(error) Cannot find complete bump pairs for all 2-pin nets.

**Description**

Each 2-pin net should be assigned to a bump pair. However, some 2-pin nets are not assigned to a bump pair. This error occurs when the bump numbers in bump arrays are enough for all 2-pin nets.

**What Next**

Please check if the number of bumps in each bump array is enough for all 2-pin nets.

---

## 3DIC-380

(error) C4 bump overlap is found for cell %s.

**Description**

This error occurs when there exists C4 bump overlap for some C4 bump cells.

**What Next**

Please check and remove the C4 bump overlap for the specified cell.

Feedback

## 3DIC-381

(error) Cannot sort odd number of bump columns.

**Description**

For 3D routing, the number of bump columns in a routing channel should be even.

**What Next**

Please check and correct the bump design for 3D routing.

## 3DIC-382

(error) Cannot use the same metal layer to route both upper and lower bump arrays.

**Description**

This tool cannot route both upper and lower bump arrays by using the same metal layer.

**What Next**

Please check and correct the specified metal layers in the option (-routing_layers) for upper and lower bump arrays in 2-layer routing mode.

## 3DIC-383

(error) Cannot create via in the middle section: net %s.

**Description**

Vias should be created in the middle section for switching layer. This errors occurs when via creation failed in the middle section due to routing congestion.

**What Next**

NA.

## 3DIC-384

(error) Cannot support %s.

**Description**

This tool currently cannot support the specified function.

**What Next**

Please fix the specified issue in the design, pattern file, or tech file.

Feedback

## 3DIC-385

(error) Error found in middle section: %s.

**Description**

This error occurs when the routing resource in the middle section is insufficient. It could be caused by large middle_turn_offset, midtrack_turn_spacing, midtrack_turn_width, or small midtrack_lengths.

**What Next**

Please adjust middle_turn_offset, midtrack_turn_spacing, midtrack_turn_width, or midtrack_lengths in the pattern file.

## 3DIC-386

(error) Fail to fix bump-to-net assignment, %s.

**Description**

This error occurs when creating the bump-to-net assignment failed and cannot be fixed due to the specifed issue.

**What Next**

Please check the bump and net design to fix the specified issue.

## 3DIC-387

(error) Error found in via_def: %s.

**Description**

This error occurs when an error is found in via_def. It could be missing via library for specific layers or bumps.

**What Next**

Please fix the specified issue in via_def.

## 3DIC-388

(error) Non-uniform bump dimension within a single bump array.

Feedback

**Description**

The dimension of bumps (width and height) in a single bump array should be the same. This tool cannot support bump array with non-unifrom bump dimension.

**What Next**

Please check and adjust bump dimensions in the design.

---

## 3DIC-389

(error) Netlist checking fails.

**Description**

This error occurs when the nets in two bump arrays are mismatched. Each 2-pin net should consist of two bumps in each bump array.

**What Next**

Please check the netlist and the bump-to-net assignment.

---

## 3DIC-390

(error) Error found during bump array analysis.

**Description**

This error occurs when the bump array analysis fails due to wrong bump placement. The tool detects unexpected bump array patterns.

**What Next**

Please correct the bump array accordingly.

---

## 3DIC-391

(warning) Cross-boundary net found: %s

**Description**

A net will be routed only if all bumps of the net are inside the specified boundary. The tool skips all cross-boundary nets in routing.

**What Next**

Please check whether the net needs to be routed or not. If the net needs to be routed, please adjust the command line option -channel_boundaries to include the net.

Feedback

## 3DIC-392

(warning) Congestion analysis fails: %s.

**Description**

The congestion analysis is to check if the wire width/spacing values can be used for nets to be routed through bump areas. If the wire width/spacing values are too large, the capacity in the bump area will be insufficient. Thus, routing will definitely fail.

**What Next**

Please adjust -wire_width or -wire_spacing in the command line. Alternatively, adjust wire_width or wire_spacing in the routing pattern file.

## 3DIC-393

(error) Net %s has a large bump/pin alignment offset %f (required <= %f).

**Description**

For die-to-die routing, the bump/pin alignment offset of a net should be less than a threshold value. A large bump/pin alginment offset would cause routing failure. User can specify the threshold value by setting the app option 3dic.route.die2die_net_alignment_max_offset.

**What Next**

Please check and adjust the bump-net assignment to decrease the bump/pin alignment offset.

## 3DIC-394

(error) Net %s crosses subchannel %d and subchannel %d.

**Description**

In 3D routing, two bumps of a net should be in the same subchannel. A cross-subchannel net is invalid and will cause routing failure.

**What Next**

Please check and correct the bump-net design.

## 3DIC-395

(warning) %d is an invalid maximum number of die-to-die routing iterations.

Feedback

**Description**

User specifies an invalid value for the maximum number of die-to-die routing iterations, so the tool is using one iteration instead. The maximum number of die-to-die routing iterations should be a positive integer. User can modify the value by the app-option 3dic.route.die2die_max_routing_iterations.

**What Next**

Please check and correct the value of the app-option 3dic.route.die2die_max_routing_iterations.

---

## 3DIC-396

(warning) The net ordering in sub-channel %u is not sequential. The nonsequential nets in the sub-channel are:%s.

**Description**

The net assignments of both bump arrays are inconsistent in an 3D sub-channel. That is, the net ordering is not squential and twisted on the bump arrays, which causes net crossing in the sub-channel. In an 3D sub-channel, inconsistent bump-net assignments (or net crossing) may cause routing quality degradation. However, some local net crossing may not significantly impact routing quality. Therefore, if the tool successfully completes the routing, please just ignore this warning.

**What Next**

Please adjust the net assignments of both bump arrays to remove net crossing in the sub-channel.

---

## 3DIC-397

(warning) The bump signal assignment is not sequential for bump %s (net: %s) and bump %s (net: %s) in subchannel %u.

**Description**

The net assignments of bumps are inconsistent on the bump arrays. That is, the net ordering is not squential and twisted on the bump arrays, which causes net crossing in a subchannel or some nets even cross subchannels. In an 3D subchannel, inconsistent bump-net assignments (or net crossing) may cause routing quality degradation. Please check the bump net assignments to avoid route planning issue or a suboptimal routing quality. However, some local net crossing may not significantly impact routing quality. Therefore, if the tool successfully completes the routing, please just ignore this warning. User can also add excluded nets into exclusion section of pattern file so that the tool will

Feedback

ignore the nets in bump order check. Besides, the information of wire length variation can also help to check if there is a bump order issue.

**What Next**

Please adjust the net assignments of both bump arrays to remove net crossing in the subchannel.

---

## 3DIC-398

(warning) Bump cells are placed with multiple %s pitches on %s.

**Description**

The X/Y pitches of bumps are inconsistent on left (lower) or right (upper) bump arrays. This mighit cause route planning issue or a suboptimal routing quality. However, if it is intended by users, please just ignore this warning.

**What Next**

Please check the bump pitch if it is intended.

---

## 3DIC-399

(warning) The bump arrays use multiple reference cells.

**Description**

The reference cells of bumps are inconsistent in the 3D channel. If this is intended by users, please just ignore this warning.

**What Next**

Please check the bump arrays and reference cells if they are as expected.

---

## 3DIC-400

(Information) Simulation concluded successfully with no errors, and log: %s.

**Description**

This message tells you that simulation concluded successfully with no errors.

**What Next**

Check the log file for details of the simulation.

Feedback

## 3DIC-401

(Information) Running external tool %s using path %s

**Description**

This message tells you that what version of the external tool is being used.

**What Next**

Check to make sure that the correct version is being used.

## 3DIC-402

(Information) Differential pair setting only specified on transmitter: %s but not on receiver: %s even though they share net %s.

**Description**

The message tells you that differential pair setting was only applied on the transmitter.

**What Next**

Preferably, fix the specification of differential pair so that is correct. If this is not fixed the command will still run correctly.

## 3DIC-403

(Information) Differential pair setting only specified on receiver: %s but not on transmitter: %s even though they share net %s.

**Description**

The message tells you that differential pair setting was only applied on the receiver.

**What Next**

Preferably, fix the specification of differential pair so that is correct. If this is not fixed the command will still run correctly.

## 3DIC-404

(Information) Found executable %s for %s.

**Description**

This message tells you that the given executable was found in the specified location.

Feedback

**What Next**

Check the path to make sure it is the correct version.

---

## 3DIC-405

(Warning) Object %s was specified as part of a differential pair but not specified as either a transmitter or receiver.

**Description**

The message tells you that the object specified was not specified as either a transmitter or receiver.

**What Next**

Check and make sure that specified object name is correct, if it is correct make sure it is listed in either the -receivers or -transmitters option.

---

## 3DIC-408

(Information) Extraction concluded successfully with no errors, log %s.

**Description**

This message tells you that extraction concluded successfully with no errors.

**What Next**

Check the log file for details of the extraction.

---

## 3DIC-409

(Error) Extraction was not successful, log %s.

**Description**

This message tells you that extraction had errors and did not conclude successfully.

**What Next**

Check the log file for details of the errors in the extraction.

---

## 3DIC-410

(Error) Unable to find the executable for %s.

Feedback

**Description**

This message tells you that the specified executable was not found.

**What Next**

Setup the path for the specified executable.

---

## 3DIC-411

(error) %s.

**Description**

The tool encountered the error specified.

**What Next**

Please check and fix the command-line options specified to the command.

---

## 3DIC-412

(error) Incorrect command %s option usage. %s.

**Description**

Incorrect command option usage.

**What Next**

See command man page for correct option usage.

---

## 3DIC-413

(Error) Simulation was not successful, log: %s

**Description**

This message tells you that simulation had errors and did not conclude successfully.

**What Next**

Check the log file for details of the errors in the simulation.

---

## 3DIC-414

(Error) Reading Directory is not supported. File %s is a directory.

Feedback

**Description**

The input file is a directory.

**What Next**

Change the input that specifies a file instead of directory.

---

## 3DIC-415

(Error) Different differential pair setting on port %s and %s even though they share net %s.

**Description**

The message tells you that specified ports have a different differential pair settings even though both ports share a net.

**What Next**

Fix the specification of differential pair so that is correct.

---

## 3DIC-416

(Error) Unable to get licence for external tool %s.

**Description**

The message tells you that command was unable to get licence for external tool specified.

**What Next**

Check your licensing settings, including environmental setting to make sure they are correct.

---

## 3DIC-417

(Error) File %s is not supported as a model file.

**Description**

The message tells you that specified file is not a supported model file.

**What Next**

First, check the file and make sure it is correct file and file type. The correct file types end with .s2p and .s4p. If it is the correct type, make sure that the file is not corrupted.

Feedback

## 3DIC-418

(warning) differential net routing cannot be implemented.

**Description**

The differential nets specified by the user cannot be routed in the required routing style. Thus, they are treated as normal nets.

**What Next**

Differential net routing is only supported for 2-layer 3D routing. There must be and only be two pairs of differential signals. In addition, bumps of a pair of differential nets cannot be selected arbitrarily. They need to be close to each other. Moreover, the two pairs of nets must be in different bump regions so that they can routed on different layers with the same routing foot-print. Please check the routing result and adjust bump-to-net assignment if needed.

## 3DIC-419

(error) current max_net_count_per_layer (%d) is not large enough to complete 3D routing: needs to be at least %d.

**Description**

The maximal net count per layer constraint in the pattern file is not sufficient to complete the 3D routing.

**What Next**

User can either increase max_net_count_per_layer or add more routing layers.

## 3DIC-420

(error) Cannot derive layer assignment for nets.

**Description**

The tool is not able to derive layer assignment for the 3D netlist.

**What Next**

User can either increase max_net_count_per_layer or change the bump-to-net assignment. It is recommended that the bumps of a single net should be at the same relative locations of the two corresponding bump arrays.

If user uses keyword target_net_count_for_layer, it is important to check whether the values are consistent with the bump-to-net assignment. Specifically, the tool requires all

Feedback

bumps of the same column to be assigned to the same layer. The column orientation is perpendicular to the channel orientation. In addition, the columns of bumps assigned to a layer must be consecutive. Please check whether the values of target_net_count_for_layer are possible to achieve or not.

---

## 3DIC-421

(error) Stubs at the %s bump array boundary must have a consistent pitch when bump-access vias are abutted to micro-bumps.

**Description**

When bump-access vias are abutted to micro-bumps, the stubs at either bump array boundaries must have the same widths and pitch, except for differential signal nets, which can have different spacing.

**What Next**

Please change boundary_stub_gaps_high and/or boundary_stub_gaps_low if bump-to-via abutment is required. Otherwise, please consider choosing non-abutted routing mode.

---

## 3DIC-422

(warning) differential nets cannot be routed in the overlapping pattern since inline stub pattern is not selected.

**Description**

The differential nets specified by the user cannot be routed in the required routing style because inline stub pattern is not selected.

**What Next**

Please select inline for stub_alignment_type.

---

## 3DIC-423

(error) bump %s does not abut its bump-access via.

**Description**

The tool fails to place a bump and its via with perfect abutment.

**What Next**

Please adjust stub locations and/or spacings to fix the abutment error.

Feedback

## 3DIC-424

(warning) max_net_count_per_layer in pattern file is igored for single or dual layer 3D routing.

**Description**

User has chosen either single layer or dual layer 3D routing. The constraint max_net_count_per_layer is not supported. It is supported only for routing with 3 or more layers. It is also supported for 2-layer routing with hard_spacing in the pattern file.

**What Next**

Please fix any error in the pattern file.

## 3DIC-425

(warning) modified middle subchannel routing skipped.

**Description**

The 3D subchannel identifical find inconsistency between middle subchannels on the two sides of the 3D channel. The routing of middle channels is skipped.

**What Next**

Please route the middle channel separately or use "keep_inconsistent_mid_subchannel"

## 3DIC-426

(warning) Expanding routing region beyond bump array area.

**Description**

The bump array area contains vias and/or pre-routes which prevent routing from going through. The tool has to create detours beyond bump array area. As a result, DRC violations might occur.

**What Next**

Check the routing result. Remove pre-existing vias and/or preroutes if needed before running 3D routing.

## 3DIC-427

(error) Tool fails to route due to existing blockages.

Feedback

**Description**

There are vias and/or pre-routes in the bump array area, which prevent routing from going through. As the result, there is no enough space for all the routing tracks.

**What Next**

Check the routing result. Remove pre-existing vias and/or preroutes before running 3D routing.

---

## 3DIC-430

(Error) The value of the app option %s is required for thermal model construction.

**Description**

The value of this app option is required but it has no default value.

**What Next**

Set the corresponding app option value.

---

## 3DIC-431

(Error) The height of the dies %s is not set in the app option %s.

**Description**

The height of the dies is not specified in the app option.

**What Next**

Provide the height of the dies in the app option in unit of um, where die is of libraryName:designName format.

---

## 3DIC-432

(Error) The height of the dies %s from the app option %s needs to be greater than 0.

**Description**

This occurs when the specified height value of the dies is not positive.

**What Next**

Specify a positive value for the height of the dies in unit of um.

Feedback

## 3DIC-433

(Error) No power map file specified for any of the cell instances.

**Description**

This occurs when no power map file is specified for cell instances.

**What Next**

Specify power map for at least one cell instance.

## 3DIC-434

(Error) Could not find the power %s for the cell instance %s from the app option %s.

**Description**

This occurs when the power for the corresponding cell instance is not found in the given path.

**What Next**

Correct the path of power for the given cell instance in the corresponding app option.

## 3DIC-435

(Error) Failed to find the interposer of %s.

**Description**

The design does not have interposer.

**What Next**

Check the design.

## 3DIC-436

(Error) The cell %s with 0 z_offset has no C4 bump.

**Description**

This occurs when the cell with 0 z_offset does not have C4 bump.

**What Next**

Check the design.

Feedback

## 3DIC-437

(Error) The die %s has no micro-bumps.

**Description**

The die has no micro-bumps.

**What Next**

Check the design.

## 3DIC-438

(Error) Cannot find the must-keys %s of the app option %s.

**Description**

Cannot continue because some required attributes are missing from the given prototype.

**What Next**

Set the missing attributes in the given prototype, where the formats of some attributes are as follows: size: {LL_X, LL_Y, UR_X, UR_Y} # Origin is at its center-center layers: {Layer Name, Thickness, Type, Metal Composition Percentage}, where Type has possible values: 0 is Signal, 1 is Dielectric, and 2 is Power molding: {MOLD Name, LL_X, LL_Y, UR_X, UR_Y, Thickness} solderball: {X Pitch, Columns, Y Pitch, Rows}

## 3DIC-439

(Warning) Failed to evaluate the command %s.

**Description**

This warning message occurs when executing a command.

**What Next**

None.

## 3DIC-440

(Error) Error when running %s, check the log file '%s' for details.

**Description**

This error message occurs when executing the executable.

Feedback

**What Next**

Check the log file for errors/warnings for hints.

---

## 3DIC-441

(Error) Error when running %s.

**Description**

This error message occurs when executing the executable.

**What Next**

None.

---

## 3DIC-442

(Information) %s executed successfully.

**Description**

The executable executed successfully.

**What Next**

None.

---

## 3DIC-443

(Information) Starting to analyze thermal...

**Description**

Starting to analyze thermal with the given design and settings.

**What Next**

Started the thermal analysis.

---

## 3DIC-444

(Information) Extracting TSV, C4 bumps, and micro-bumps...

**Description**

Extracting TSV, C4 bumps, and micro-bumps from the design.

Feedback

**What Next**

The TSV, C4 bumps, and micro-bumps are extracted from the design.

---

## 3DIC-445

(Information) Finished the extraction of TSV, C4 bumps, and micro-bumps.

**Description**

The TSV, C4 bumps, and micro-bumps are all extracted successfully.

**What Next**

Generating the thermal script after finishing the extraction of TSV, C4 bumps, and micro-bumps.

---

## 3DIC-446

(Information) Generating the thermal script...

**Description**

Generating the thermal script for thermal analysis.

**What Next**

Generated the thermal script.

---

## 3DIC-447

(Information) Generated the thermal script successfully.

**Description**

Generated the thermal script for successfully.

**What Next**

Starting thermal analysis.

---

## 3DIC-448

(Information) Updating thermal database...

**Description**

Completed thermal analysis. Starting to update the thermal result in thermal database.

Feedback

**What Next**

The thermal result is generated in thermal database.

---

## 3DIC-449

(Information) Deleting old thermal database %s.

**Description**

Try to delete existing thermal database and create a new one.

**What Next**

Check if you can delete the old thermal database.

---

## 3DIC-450

(Error) Failed to remove old thermal database files.

**Description**

Cannot delete the old thermal result files in the thermal database. You may not have permission to access the files or the some of files are opened by other processes.

**What Next**

Check the permission and the opened files of the directory and try again.

---

## 3DIC-451

(Information) Created the thermal database %s.

**Description**

Created the thermal database for thermal result.

**What Next**

Make sure the thermal database can be populated with the thermal result files.

---

## 3DIC-452

(Information) Updated the thermal database successfully. The thermal result is %s.

**Description**

Show the thermal result after updating the thermal database.

Feedback

**What Next**

The thermal result can be opened by 'open_thermal_result' or automatically loaded by 'analyze_3d_thermal'.

---

## 3DIC-453

(Information) Finished thermal analysis successfully.

**Description**

Finished thermal analysis successfully.

**What Next**

None.

---

## 3DIC-454

(Information) Starting data preparation for thermal analysis...

**Description**

Generating the thermal script and input data for the analysis.

**What Next**

Generated the thermal script and input data.

---

## 3DIC-455

(Information) Finished the data preparation for thermal analysis.

**Description**

Generated the thermal script and input data for the analysis.

**What Next**

None.

---

## 3DIC-456

(Error) The cell instance %s has non-zero z_offset with no supporting cell instance below it.

**Description**

The cell instance has non-zero z_offset, but has no supporting cell instance below.

Feedback

**What Next**

Check if the supporting cell instances are available in the design.

---

## 3DIC-457

(Warning) %s is an invalid value for %s, so %s is used instead.

**Description**

The value is not set properly.

**What Next**

Modify the value if necessary.

---

## 3DIC-458

(Error) %s is an invalid value for %s.

**Description**

This occurs when the value is not valid.

**What Next**

Correct the value.

---

## 3DIC-459

(Information) The analysis is powered by Ansys RedHawk-SC Electrothermal, and the required version is 2023R2.3.

**Description**

The analysis is powered by Ansys RedHawk-SC Electrothermal, and the required version is 2023R2.3. There could be some unexpected issues if unmatched version is used.

**What Next**

None

---

## 3DIC-460

(info) StarRC channel extraction clock net settings cleared.

**Description**

This option tells you that StarRC channel extraction clock net settings have cleared.

Feedback

**What Next**

Make sure this was intended.

---

## 3DIC-461

(info) StarRC channel extraction inductance settings cleared.

**Description**

This option tells you that StarRC channel extraction inductance settings have cleared.

**What Next**

Make sure this was intended.

---

## 3DIC-462

(Error) The instance %s can not be found.

**Description**

The instance can not be found in the block.

**What Next**

Check whether the instance is in the block or not.

---

## 3DIC-463

(Error) The point %s is outside of %s's boundary bbox.

**Description**

The point is outside of the boundary bbox.

**What Next**

Choose a point inside the boundary bbox.

---

## 3DIC-464

(Error) No %s Profile file result.

**Description**

The Profile file result was not generated.

Feedback

**What Next**

Check if there is something wrong at previous step.

---

## 3DIC-465

(Error) Unable to find %s's thermal file %s of static thermal analysis.

**Description**

Unable to find the thermal file of static thermal analysis.

**What Next**

Check whether the thermal file exists or not.

---

## 3DIC-466

(Error) The Profile %s already exists.

**Description**

The Profile already exists in memory.

**What Next**

Specify a unique name.

---

## 3DIC-467

(Error) The Profile %s does not exist.

**Description**

The Profile does not exist in memory.

**What Next**

Check whether the Profile exists in memory or disk or not. You can also create a Power Profile by *create_power_profile or create_rail_power_profile. Run thermal analysis to obtain Thermal Profile or Probe Profile.*

---

## 3DIC-468

(Error) The format of the value %s of %s is invalid for the %s data type.

Feedback

**Description**

The value is invalid; we can not convert it to the certain data type. The data types include string, double, ndmBBox, and ndmCoord.

**What Next**

Check the format of the input value.

---

## 3DIC-469

(Error) The value of -kind option does not match the file extension.

**Description**

The value of -kind option does not match the file extension. We only support "spr" and "mhs" currently.

**What Next**

Use "spr" and "mhs" as the file extension instead.

---

## 3DIC-470

(error) mirror bump %s will cause miss alignment with bumps in chip %s.

**Description**

When 3DIC structure has MIB(Multiply Instantiated Blocks), create mirror bumps from one block to one MIB block, this operation will cause miss alignment between bumps in another MIB chip.

**What Next**

Check MIB die location and source bump location.

---

## 3DIC-471

(warning) mirror bump %s cannot find corresponding bump in chip %s.

**Description**

When 3DIC structure has MIB(Multiply Instantiated Blocks), create mirror bumps from one block to one MIB block, this operation will cause dangling bump in another MIB chip.

**What Next**

Check MIB die location and source bump location.

Feedback

## 3DIC-472

(error) propagate nets {%s} will change the golden netlist.

**Description**

When 3DIC structure has MIB(Multiply Instantiated Blocks), propagate nets will change the golden netlist.

**What Next**

Check nets connection.

## 3DIC-473

(error) assign nets {%s} will change the golden netlist.

**Description**

When 3DIC structure has MIB(Multiply Instantiated Blocks), assign nets will change the golden netlist.

**What Next**

Check nets connection.

## 3DIC-474

(warning) found short violation in 3D routing result.

**Description**

Tool detects overlapping violation between pre-routes and 3D routes. Normally, 3D routing is performed before power routing and other signal routing. Therefore, there should not be pre-routes in the 3D channel. If there are such pre-routes, customer should use pattern file to make tool avoid them.

**What Next**

Remove or modify preroutes to resolve overlapping violations. Alternatively, user can change the pattern file to re-generate 3D routing result.

## 3DIC-475

(warning) pre-existing route shape %s overlaps with 3D routes.

Feedback

**Description**

Tool detects a route shape which overlaps 3D routes.

**What Next**

Remove or modify the route. Alternatively, user can change the pattern file to re-generate 3D routing result to avoid the route shape.

---

## 3DIC-476

(warning) pre-existing via %s overlaps with 3D routes.

**Description**

Tool detects a via which overlaps 3D routes.

**What Next**

Remove or modify the via. Alternatively, user can change the pattern file to re-generate 3D routing result to avoid the via.

---

## 3DIC-477

(warning) net %s is skipped during routing.

**Description**

Tool skips the specified net since the net does not cross the 3D channel.

**What Next**

Route the net after auto-3D routing.

---

## 3DIC-478

(error) middle track offset does not fit the %s switch offset: minimal adjusting distance is %s.

**Description**

The middle track offset must be increased or the upper edge switch offset must be decreased by the given amount if the violation is on the upper side so that the middle section can be routed. The middle track offset must be decreased or the lower edge switch offset must be decreased by the given amount if the violation is on the lower side so that the middle section can be routed.

Feedback

**What Next**

Please adjust the mid_routing_switch_direction_gaps.

---

## 3DIC-479

(error) The middle channel is too narrow for single-waterfall routing pattern.

**Description**

The offset of bump arrays is large. As a result, it is impossible to route the middle channel section using the single-waterfall routing pattern.

**What Next**

Set mid_routing_switch_direction to true or adjust bump array locations to reduce offset.

---

## 3DIC-480

(error) Pattern dimensions are illegal, please check the pattern setting.

**Description**

In pattern's setting, 'dx' and 'dy' must be not negative, other dimensions must be not zero or negative.

**What Next**

NA.

---

## 3DIC-481

(error) When create objects by lib cell in '-lib_cell_selection' option, these objects will overlap in %s direction.

**Description**

The near objects that created by lib cell will overlap because the distance between the near objects is less than pattern's pitch.

**What Next**

check the '-lib_cell_selection' option or '-pattern' setting.

---

## 3DIC-482

(error) The command cannot support multiple dies scenario.

Feedback

**Description**

The command cannot support multiple dies scenario.

**What Next**

check '-core', '-design_boundary', '-voltage_area', '-block' or '-boundary' option setting.

---

## 3DIC-483

(error) Cannot find any region to create objects.

**Description**

From command options, it's impossible to get any region to create objects.

**What Next**

check '-core', '-design_boundary', '-voltage_area', '-block' or '-boundary' option setting.

---

## 3DIC-484

(error) Assign net failed, the hier of object %s is not same as the hier of net %s.

**Description**

The net cannot be assigned to the object that create by the command.

**What Next**

check '-net_assignment' option setting.

---

## 3DIC-485

(error) The region %s that plan to create objects is overlapped because of Multipy Instance Blocks.

**Description**

Overlapped region will cause incorrect objects placement.

**What Next**

check '-boundary' option setting.

---

## 3DIC-486

(warning) The boundary of the existing region '%s' overlaps with the boundary of the newly created region boundary {%s}.

Feedback

**Description**

The overlapped region boundary may lead to the failure to create a new region.

**What Next**

NA

---

## 3DIC-487

(warning) Net assignment is not supported in 3DIC top Level.

**Description**

In top level, net assignment is not supported, '-net_assignment' option will be ignored.

**What Next**

NA

---

## 3DIC-488

(warning) Region '%s' is existed, %d associated objects were removed, and the region will be reset.

**Description**

When users run 'create_3d_region_objects' or 'copy_3d_region_objects', and the target region is existed , the region will be reset after running these commands.

**What Next**

NA

---

## 3DIC-489

(warning) Pattern '%s' is existed, and the pattern will be reset.

**Description**

When users run 'create_3d_object_pattern', and the pattern is existed , the pattern will be reset after running the command.

**What Next**

NA

Feedback

## 3DIC-490

(error) All objects cannot be created and placed correctly in the #%u ring with boundary {%s}.

**Description**

When users use ring pattern to create new region and objects, the specified ring cannot create and place new objects correctly.

**What Next**

NA

## 3DIC-491

(warning) Link and link_location cannot be copyed from region '%s' to new region.

**Description**

NA.

**What Next**

See command 'create_3d_region_objects'

## 3DIC-492

(error) '-aligned_top_level' option only supports the scenario when command runs in top level and sets '-boundary' option.

**Description**

The command can support to keep new objects aligned only when set '-boundary' option in top level.

**What Next**

NA.

## 3DIC-493

(error) When command sets '-boundary' option on the top level, the '-target_die' option must be specified at the same time.

**Description**

NA.

Feedback

**What Next**

NA.

---

## 3DIC-494

(error) The region '%s' is %s set back side , it's conflict with the layer set by bond pad reference '%s'.

**Description**

When the region is set to back side, the layer set by bond pad reference must be in the back side. When the region is not set to back side, the layer set by bond pad reference also must be not in the back side.

**What Next**

Check the setting of bond pad layer.

---

## 3DIC-495

(error) The name of bond pad reference '%s' cannot be the same as the name of any lib cell or bond pad def.

**Description**

The bond pad reference name must be unique, it cannot have the same name with any lib cell or bond pad def.

**What Next**

Set a unique bond pad reference name.

---

## 3DIC-496

(warning) Bond pad reference '%s' is existed, and the bond pad reference will be reset.

**Description**

When users run 'create_bond_pad_reference', and the reference is existed, the reference will be reset after running the command.

**What Next**

NA.

Feedback

## 3DIC-497

(error) Cannot find any valid region to commit objects.

**Description**

From command options, it's impossible to get any valid region to commit objects. No valid or specified regions are existed in current block.

**What Next**

Check if none of regions in "region_names" option is existed in top level; Check if there are no regions in current block when "region_names" option is not specified.

## 3DIC-498

(warning) The associated edit group with region '%s' is existed, %d associated objects were removed, the region will be re-committed.

**Description**

If the region has been committed, the existed committed objects will be removed and the new objects will be re-committed.

**What Next**

NA.

## 3DIC-499

(warning) There is no %s in the die %s.

**Description**

There is no %s in the die %s.

**What Next**

Create the bumps in the die.

## 3DIC-500

(warning) There is no valid <%s> for current 3dic design.

**Description**

There is no valid <%s> setting can match current 3dic design. For <FP_PATTERN>, it means the floorplan of current design doesn't match any pattern; For <SOC_TYPE>, it

Feedback

means the height/width/length/diagonal of 'SOC' can't statisfy all of the settings in any <SOC_TYPE> section.

**What Next**

Adjust the floorplan by 'set_cell_location' or adjust the boundary of the die by 'set_block_boundary'.

---

## 3DIC-501

(error) The %s between %s and %s is %s which is %s than the constraint <%s> setting, the valid range is {%s %s}.

**Description**

The spacing between the bump and its parent die violate the constraint.

**What Next**

Adjust the bump's location.

---

## 3DIC-502

(error) The %s of %s is %s which is %s than the constraint <%s> setting, the valid range is {%s %s}.

**Description**

The shape of bump violates the constraint.

**What Next**

Check the bumps in the design.

---

## 3DIC-503

(error) Find dangling %s %s that violate the constraint <%s> setting, the valid range is {%s %s}.

**Description**

The bump is too far away with other bumps that it will be seen as a dangling bump.

**What Next**

Adjust the bump's location.

Feedback

## 3DIC-504

(error) The %s between %s and %s is %s which is %s than the constraint <%s> setting, the valid range is {%s %s}.

**Description**

The relative location of two bumps violates the constraint.

**What Next**

Adjust the bump's location.

## 3DIC-505

(error) The bump(s) reside in the %s corner triangular zone of %s violate the constraint <%s> setting, the wedge length is {%s}.

**Description**

The constraint specifies a triangular zone at the corner of a wafer-level fanout box where the specify rank (corner-most) bump (electrically unconnected) must reside.

**What Next**

Adjust the bump's location.

## 3DIC-506

(error) %s abutt or overlap with %s which violate the constraint <%s> setting.

**Description**

The bumps are abutted or overlpped.

**What Next**

Adjust the bump's location.

## 3DIC-507

(error) The %s between %s and %s is %s which is %s than the constraint <%s> setting, the valid range is {%s %s}.%s

**Description**

The relative location or area_ratio between two dies violates the constraint.

Feedback

**What Next**

Adjust the boundary or location of the die.

---

## 3DIC-508

(error) The %s of %s is %s which is %s than the constraint <%s> setting, the valid range is {%s %s}.%s

**Description**

The shape of die violates the constraint.

**What Next**

Adjust the boundary of the die.

---

## 3DIC-509

(error) %s.

**Description**

Errors are found during the processing of the command options.

**What Next**

Please check the manpage of the command to correct any errors related to option settings.

---

## 3DIC-510

(warning) %s.

**Description**

Errors are found during the processing of the command options.

**What Next**

Please check the manpage of the command to correct any errors related to option settings.

---

## 3DIC-511

(warning) The bump %s of net %s is ignored for port creation as it is not within the user specified bounding box.

Feedback

**Description**

The bump is ignored as it is not fully contained within the user specified bounding box.

**What Next**

Please check your bounding box.

---

## 3DIC-512

(warning) The bump %s of reference net %s is ignored for port creation as it is not within the user specified bounding box.

**Description**

The bump is ignored as it is not fully contained within the user specified bounding box.

**What Next**

Please check your bounding box.

---

## 3DIC-513

(error) The reference net %s has zero bumps inside the user defined bounding box.

**Description**

The bump is ignored as it is not fully contained within the user specified bounding box.

**What Next**

Please check your bounding box.

---

## 3DIC-514

(warning) %s.

**Description**

The port name defined for the net does not correspond to a bump and cannot be used for simulate_signal_integrity.

**What Next**

Please check the manpage of the command to correct any errors related to option settings.

Feedback

## 3DIC-515

(warning) Object of type %s is not supported in extract_channel_parasitics right now. Objects of this type will not be written to the EDB.

**Description**

The aforementioned object is not supported and will not be written to the EDB.

**What Next**

Please check the object type.

## 3DIC-516

(error) The %s between %s and %s is %s which is %s than the constraint <%s> setting, the valid range is {%s %s}.

**Description**

The area_ratio of toal area of bumps in specific region between area of specific region violates the constraint.

**What Next**

Adjust the location of the bumps.

## 3DIC-517

(error) Find dangling bump %s that violate the constraint <%s> & <%s> setting, the valid <%s> is {%s %s} and the valid <%s> is {%s %s}.

**Description**

The bump is too far away with other bumps that it will be seen as a dangling bump.

**What Next**

Adjust the bump's location.

## 3DIC-518

(warning) Detected incomplete net %s

**Description**

Detected an incomplete net on a 3DIC design. This might be caused by bump mis-alignment or bad net assignment on pins.

Feedback

A top-level inter-die net is incomplete if any one of its logical ports is unreachable from another.

**What Next**

Please refer to the error browser for more information.

---

## 3DIC-519

(warning) Detected short violations for nets %s on bumps %s.

**Description**

Detected a die to die short violation on bumps. This might be caused by bump mis-alignment or bad net assignment on pins.

**What Next**

Please refer to the error browser for more information.

---

## 3DIC-520

(warning) Detected short violations for nets %s at pin %s.

**Description**

Detected a short violation caused by bad net assignment at pins.

---

## 3DIC-521

(warning) Ignore invalid cell %s which is not a bump or TSV cell or the reference cell is not in the "-include_ref_cells" list.

**Description**

Only 3DIC related objects, including bumps, TSVs, special included reference cells and routes are the copying target of this command, other objects are ignored and not copied.

**What Next**

Please check the input.

---

## 3DIC-522

(warning) Invalid block net input for '%s' option.

Feedback

**Description**

The input net is not a valid net.

**What Next**

Please check if the input net is valid.

---

## 3DIC-523

(warning) Net conflict found between '%s' and '%s' when doing net tracing.

**Description**

Net conflict is found when trying to propagate the net from one shape to other touched objects.

**What Next**

Please check the conflict and correct the net assignment or remove the objects.

---

## 3DIC-524

(warning) Copied object %s is outside of design boundary.

**Description**

One of the copied objects is outside of design boundary.

**What Next**

Please check the -to or -delta input option to correct the copy destination.

---

## 3DIC-525

(Info) Objects already exist in destination bounding box.

**Description**

There are already objects in the destination bounding box. This situation might be expected for copying objects to connect exsiting objects. But it can also be an error due to wrong destination given.

**What Next**

Please check if the copy result is expected.

Feedback

## 3DIC-526

(warning) No valid objects to copy.

**Description**

Only 3DIC related objects, including bumps, TSVs, and routes are the copying target of this command, other objects are ignored and not copied.

**What Next**

Please check the input.

## 3DIC-527

(warning) control_channel_bus_width ignore since control channel cannot be identified.

**Description**

The control channel is the middle subchannel. Since there are even number of subchannels, no control channel can be identified. The control_channel_bus_width constraint is ignored.

**What Next**

Please check the design.

## 3DIC-528

(error) channel length needs to be increased by %s.

**Description**

The channel length is too short. 3D routing cannot be completed.

**What Next**

Please increase the channel length or reduce routing pitch.

## 3DIC-529

(warning) Ignore invalid object %s which is not a cell or shape.

**Description**

Only 3DIC related objects, including bumps, TSVs, and routes are the copying target of this command, other objects are ignored and not copied.

Feedback

**What Next**

Please check the input.

---

## 3DIC-531

(error) When commit objects by adjacent design/bond_pad_reference '%s' and '%s', these objects will overlap because the adjacent distance between these two designs is less than the '%s' of pattern '%s'.

**Description**

The adjacent objects that created by design collection will overlap because the distance between the adjacent objects is less than pattern's pitch.

SH WHAT NEXT If this phenomenon is which you expect, please add the '-force' option in command "commit_pseudo_bumps". check the '-bumps', '-probe_bumps' or '-dummy_bumps' option setting.

---

## 3DIC-532

(error) Total %s pseudo bumps {%s} are not within its '%s' in chip '%s'.

**Description**

Pseudo bump must be within its region. Pseudo bump must be within its die boundary.

SH WHAT NEXT Adjust the boundary setting of bump region.

---

## 3DIC-533

(error) Total %s pseudo bump pairs {%s} overlap in chip '%s'.

**Description**

Two pseudo bumps does not overlap.

SH WHAT NEXT Adjust the pitch or stagger pitch setting of bump region pattern.

---

## 3DIC-534

(error) Pseudo bump '%s' is dummy and it still assigns net '%s' in chip '%s'.

**Description**

Dummy bump cannot be assigned to a net.

SH WHAT NEXT Check the net attribute setting of pseudo bump.

Feedback

## 3DIC-535

(warning) connector spacing is reduced in current subchannel.

**Description**

The route spacing in the connector region is reduced.

**What Next**

Please increase channel length.

## 3DIC-536

(error) cannot apply default 2-layer routing due to bump-net assignments.

**Description**

In the default 2-layer routing mode, the bump array is divided evenly. Half of the bumps are routed using a single layer. It requires that no net can span across the two halves of bump arrays. If there is one or more nets whose bumps are in different halves of bump arrays, the routing cannot be performed.

**What Next**

Consider routing the channel in a single layer with subchannel_layer_list in the pattern file. Or check the bump to net assignment and perform modification. User can also use create_interposer_routeplan to route the channel separately, with non-default bump array boxes defined by bump_array_box in the pattern file.

## 3DIC-537

(warning) Bump lib cell %s has more than 1 port.

**Description**

Bumps should only have a single port. The tool detects a bump lib cell with more than 1 port.

**What Next**

Please check the bump lib cell.

## 3DIC-538

(Error) The configuration file is not specified for the app option %s.

Feedback

**Description**

The occurs when the .gds file is specified for 3dic.thermal.pkg_file and/or 3dic.thermal.pcb_file, and the corresponding configuration file is not set.

**What Next**

Set the corresponding configuration file in the app option.

---

## 3DIC-539

(Error) Could not find the file %s from the app option %s because the file does not exist.

**Description**

This occurs when the file is not found in the given path.

**What Next**

Provide correct filepath in the corresponding app option.

---

## 3DIC-540

(Error) tracks are too close to abutted vias.

**Description**

Although the tool places bump access vias, these vias are too close to tracks and routing cannot be completed without DRCs.

**What Next**

Please check constraints. Use smaller spacing values with keyword rdl_route_spacing

---

## 3DIC-541

(error) %s is outside of the boundary of die %s which violates the constraint <%s>.

**Description**

The bump is not fully inside the die boundary.

**What Next**

Please adjust the bump location.

---

## 3DIC-542

(error) %s overlaps with %s which violate the constraint <%s> setting.

Feedback

**Description**

The tag cell overlaps with bump.

**What Next**

Adjust the tag cell or bump's location.

---

## 3DIC-543

(error) The %s between %s and %s is %s which is %s than the constraint <%s> setting, the valid range is {%s %s}.

**Description**

The relative location of tag cell and bump violates the constraint.

**What Next**

Adjust the bump or tag cell's location.

---

## 3DIC-544

(error) The %s between %s and %s is %s which is %s than the constraint <%s> setting, the valid range is {%s %s}.

**Description**

The relative location of bump and tag cell violates the constraint.

**What Next**

Adjust the location of bump or tag cell.

---

## 3DIC-545

(error) The %s between %s and %s is %s which is %s than the constraint <%s> setting, the valid range is {%s %s}.

**Description**

The spacing between the tag cell and its parent die violate the constraint.

**What Next**

Adjust the tag cell's location.

Feedback

## 3DIC-546

(error) %s is outside of the boundary of die %s which violates the constraint <%s>.

**Description**

The tag cell is not fully inside the die boundary.

**What Next**

Please adjust the tag cell's location.

## 3DIC-547

(error) The %s between %s and %s is %s which is %s than the constraint <%s> setting, the valid range is {%s %s}.

**Description**

The area_ratio of toal area of tag cells in specific region between area of specific region violates the constraint.

**What Next**

Adjust the location of some tag cells or remove/add some tag cells.

## 3DIC-548

(warning) failed to set preferred_pin for port %s.

**Description**

The command "commit_pseudo_bumps" cannot set preferred pins for the assigned port from corresponding pseudo bump.

**What Next**

Please check the specified lib cell and port.

## 3DIC-549

(warning) The bump region '%s' is at the top level and it cannot be committed.

**Description**

The command "commit_pseudo_bumps" cannot support the bump region at the top level to be commited. Please create bump region in dies.

Feedback

**What Next**

Use command "create_bump_region" in dies.

---

## 3DIC-550

(warning) The off-die tsv layer is not frontSide port layer or backSide port layer, the layer will be ignored.

**Description**

The first five characters of frontSide port layer's mask name must be 'metal'; The first six characters of backSide port layer's mask name must be 'bmetal'; The specified off-die layer must be frontSide port layer or backSide port layer, or the off-die layer will be ignored. The command will use the current frontSide port layer and backSide port layer to report informatton of feedThrough TSV.

**What Next**

NA.

---

## 3DIC-551

(warning) These 'is_dummy' and 'is_probe_pad' attributes of pseudo bump '%s' are true at the same time, the pseudo bump cannot be commited.

**Description**

NA.

**What Next**

NA.

---

## 3DIC-552

(warning) apply user spacing of layer %s even it is smaller than that in the tech-file, which is %s microns.

**Description**

The min-spacing specified by user is smaller than that in the tech-file.

**What Next**

Please verify the tech-file and/or user spacing for the specified layer.

Feedback

## 3DIC-553

(error) cannot find '%s' lib cell specified in plan.3dic.pattern_marker_lib_cell_names. PRS checking will be skipped.

**Description**

Both two lib cells specified in app option plan.3dic.pattern_marker_lib_cell_names must be existed.

**What Next**

check if the reported lib cell is existed.

## 3DIC-554

(warning) source bump cell '%s' has no pin.

**Description**

The source bump has no pin, and attribute "is_dummy" is not set true.

**What Next**

NA.

## 3DIC-555

(error) the probe bumps cluster {%s} is invalid in chip %s.

**Description**

Probe bump cluster must be exactly 2x2 bumps, all 4 identical bumps shape, size, layer, perfect equal square spacing, all 4 connected to the same net.

**What Next**

NA.

## 3DIC-556

(error) the distance between quad probe bumps {%s} and {%s} is less than min distance %s in chip %s.

Feedback

**Description**

The min distance between two quad probe bumps is set by app option "plan.3dic.min_probe_bumps_quad_spacing" or "plan.3dic.qprobe_min_pitch". If the two app options are specified, the command will use the max value to check the violations.

**What Next**

NA.

---

## 3DIC-557

(error) bump %s in outermost edge must be dummy or assign to VSS/VDD net in chip %s.

**Description**

The outermost bump along the edge must be dummy or the bump assigned to VSS/VDD net.

**What Next**

NA.

---

## 3DIC-558

(error) there is no unique bump pattern in chip %s.

**Description**

One bump pattern in chip corner must be unique.

**What Next**

NA.

---

## 3DIC-559

(error) the distance of the intradie bump %s in orientation %s is less than min distance %s in chip %s.

**Description**

The min distance of intradie bump is set by app option "plan.3dic.min_bump_zone_street_keepout_margin".

**What Next**

NA.

Feedback

## 3DIC-560

(error) there are no two opposite corners which have PRS cells with specified margins in chip %s, the reason is '%s'.

**Description**

Two opposite corners which have PRS cells in each chip. These PRS cells must satisfy conditions as below: 1.The die contains exactly two unique PRS markers specified by the user. 2.The two cells must be placed such that two edges are flush with the EDM (within a tolerance - specify by app options,default is zero) 3.The two cells must be placed in opposite corners.

**What Next**

check the number of PRS Cells or location.

## 3DIC-561

(error) the %s length of %s's plate hole is less than %s from rv rule in %s chip.

**Description**

The rv rules are set by app options, these rules limit the width/height length of via's plate hole.

**What Next**

check the width/height of via's plate hole.

## 3DIC-562

(error) the spacing distance between %s and %s is less than %s from rv rule in %s chip.

**Description**

The rv rules are set by app options, these rules limit the distance between two objects.

**What Next**

check the spacing distance between two objects.

## 3DIC-563

(warning) the dimension of the die is out of industry standard.

Feedback

**Description**

This sanity check ensure the dimension is within industry standard.

**What Next**

Either check the dimension or apply "estimate_3d_thermal -force" with caution in simulation result.

---

## 3DIC-564

(Warning) The size of %s is out of industry standard.

**Description**

This sanity check by command *estimate_3d_thermal* ensure the dimension is within industry standard.

**What Next**

Either check the dimension or apply *estimate_3d_thermal –force* to proceed simulation with caution.

---

## 3DIC-565

(error) File format error in block power file '%s'.

**Description**

Please check your file and correct the part that does not match the format.

Block power file format is as follows:

```
region_currents = [
    {
        'name':'A',
        'coord':RealRect(0.000000,0.000000,12000.000000,12000.000000),
        'method':'Method',
        'current':{Net('VDD'):1. 0e-3,Net('VSS'):1.5e-03}
    }
]
```

where the 'coord' and 'current' input are the necessary information. Please keep the format of having each information and the brackets in a single line.

**What Next**

Correct the file input.

Feedback

## 3DIC-566

(error) In option '%s', '%s' is not a valid die instance name.

**Description**

Please provide a valid die instance name for the -block_power_files, -exculde_regions, -via_def or -lib_cell option. The die instance name should be one of the 'get_cells' result collection of the top die.

**What Next**

Correct the option input.

## 3DIC-567

(error) In file '%s', '%s' is not a valid %s input. %s.

**Description**

Please check your file and correct the part that does not match the format.

Block power file format is as follows:

```
region_currents = [
    {
        'name':'A',
        'coord':RealRect(0.000000,0.000000,12000.000000,12000.000000),
        'method':'Method',
        'current':{Net('VDD'):1. 0e-3,Net('VSS'):1.5e-03}
    }
]
```

where the 'coord' and 'current' input are the necessary information.

**What Next**

Correct the file format.

## 3DIC-568

(error) In block power files associated with die inst '%s', there are overlapped regions which is not supported. Overlapped regions: %s and %s.

**Description**

Currently, multiple current regions in one die is supported when the regions are disjoint. Overlapped region are not supported.

Feedback

**What Next**

Please correct your input.

---

## 3DIC-569

(error) Wrong input for option '-block_power_files'. Correct format {{die_instance_name1 file_name1} {die_instance_name2 file_name2} ...}.

**Description**

The input format for '-block_power_files' option is {{die_instance_name1 file_name1} {die_instance_name2 file_name2} ...}.

**What Next**

Please correct your input.

---

## 3DIC-570

(warning) In option '-block_power_files', die '%s' and '%s' refer to the same reference design. Input '%s' is ignored.

**Description**

For multiply instantiated module (MIM) or block (MIB), the block power data should be the same. Ignore the second input to avoid conflict.

**What Next**

Check the result and correct the option input if needed.

---

## 3DIC-571

(warning) Net %s specified from '-pg_nets' option %s. Skip it.

**Description**

The net input from '-pg_nets' option either does not exist in the target creation die or is not a PG nets. The net is skipped during PG TSV creation.

**What Next**

Please check the option input.

---

## 3DIC-572

(info) No PG nets are specified by option -pg_nets. Use all PG nets as target nets.

Feedback

**Description**

When the option -pg_nets is not specified or all the nets specified by -pg_nets options are invalid, then all PG nets of the target creation die which has power density information is used to create PG TSVs.

**What Next**

Check the result.

---

## 3DIC-573

(warning) There are already %d TSVs on die '%s' which are not taken into calculation. New created TSV might have DRC violations with them.

**Description**

This command does not support incremental TSV creation. The existing TSVs are not taken into density calculation, thus the final TSV number might not meet the required power density needs. Moreover, newly created TSV might have DRC violation with the pre-exist ones.

**What Next**

Please check the result and if the result is not correct, remove all TSVs and rerun the command.

---

## 3DIC-574

(warning) the committed editgroup '%s' cannot be assigned to the port '%s'.

**Description**

When the corresponding pseudo bump was assigned to one port, but the committed editgroup cannot support to be assigned to any port.

**What Next**

NA.

---

## 3DIC-575

(warning) there are more than one ports: %s on the net '%s' which assigned on pseudo bump '%s'.

Feedback

**Description**

Multiple ports are connected to the assigned net, the tool cannot decide which port can be used to create terminal.

**What Next**

Check the net attribute setting of pseudo bump.

---

## 3DIC-576

(warning) the pseudo bump '%s' was set net '%s' and port '%s' at the same time and it will be committed to a bond pad, the corresponding assigned net will be ignored.

**Description**

The committed bond pad will be assigned to the port which connected to the specified net. If the assigned port is also specified, the assigned port will be conficted. During this scenario, the specified net will be ignored.

**What Next**

NA.

---

## 3DIC-577

(error) the bump cell pin '%s' cannot be setted preferred pin to port '%s'.

**Description**

The bump cell pin which can be setted to preferred pin to one port must satisfy below conditions: 1. The pin in bump cell must be IO Pad. 2. The specified port cannot have any regular shapes.

**What Next**

NA.

---

## 3DIC-578

(error) the net which connected to the port '%s' is '%s', it's not same with the net '%s' which assigned to pseudo bump '%s'. it will cause unexpected behavior that one net will be removed.

**Description**

Two nets connect to one bump cell, it will caused one net will be removed. If this scenario is expected, please add '-force' option.

Feedback

**What Next**

NA.

---

## 3DIC-579

(warning) there are more than one ports: %s on the net '%s' which assigned on pseudo bump '%s' in chip '%s'.

**Description**

NA.

**What Next**

Check the net attribute setting of pseudo bump.

---

## 3DIC-580

(error) connet bond pad '%s' to net '%s' failed, target net '%s' is dangling, it is not connected to a valid port. Cannot create bond pad terminal without an associated port.

**Description**

If committed bond pad needs to connected to the specified net, the specified net must connect to a valid port.

**What Next**

NA.

---

## 3DIC-581

(error) committed %s '%s' from pseudo bump '%s' using design/bond pad reference '%s' in bump region '%s' cannot be assigned to port '%s'.

**Description**

Please check 1. If the specified port has any regular shapes. 2. If the attribute 'is_pad' of the pin in the lib cell is false.

**What Next**

Please add '-force' option in command.

Feedback

## 3DIC-582

(error) committed %s '%s' from pseudo %s '%s' using %s '%s' in bump region '%s' cannot be connected to net '%s'.

**Description**

If the committed object type is 'bump cell', please check if there is any valid pin from specified design. If the committed object type if 'bond pad' , please check if there is any valid port connected to the speccified net.

**What Next**

NA.

## 3DIC-583

(warning) Need %d TSV to support the current setting for this region, and the region is too small to create these TSV without DRC violation. Skip the creation for this region.

**Description**

The distance between TSVs need to be larger than minimum spacing and pitch defined in technology file. According to the input current value from the block_power_files and tsv_conversion_factor, the number of TSV needed to support the current setting is too many to keep enough distance between each other. In order to avoid long runtime for creating TSV with DRC violations, the TSV creation for this region is skipped.

**What Next**

Check the input and correct it.

## 3DIC-584

(error) In option '-exclude_regions', '%s' is not a valid bbox list.

**Description**

Please provide a valid bbox list with the associate die instance name. The input format of '-exclude_regions' option is {{die_inst_name1 {{{box1_llx box1_lly} {box1_urx box1_ury}} {{box2_llx box2_lly} {box2_urx box2_ury}}}} {die_inst_name2 {{{box3_llx box3_lly} {box3_urx box3_ury}}}}}. If the bbox list comes from other tcl command, such as [get_attribute [get_cells $myCell] bbox], please use double quote instead of the most outer curly brackets. Example: "{die_inst_name1 {[get_attribute [get_cells $myCell] bbox]}} {die_inst_name2 {[get_attribute [get_cells $myCell2] bbox]}}"

Feedback

**What Next**

Correct the option input.

---

## 3DIC-585

(error) In option '%s', the input is not a valid list of %s.

**Description**

Correct format for -via_def is {{die_instance_name1 via_def_name1} {die_instance_name2 via_def_name2} ...}. Correct format for -lib_cell is {{die_instance_name1 tsv_cell_name1} {die_instance_name2 tsv_cell_name2} ...}.

**What Next**

Correct the option input.

---

## 3DIC-586

(warning) Die %s has "face-down" orientation FN with no adjacent die at a higher stacking level. TSV creation is skipped.

**Description**

A die which has "face-down" orientation and no adjacent die at a higher stacking level does not need TSV, since there is no adjacent die or package to connect.

**What Next**

Check your design.

---

## 3DIC-587

(warning) Die %s has no TSV layer defined in technology file. TSV creation is skipped.

**Description**

The die has no TSV layer (mask name is tsv). Cannot create TSV for this die.

**What Next**

Check your design.

Feedback

## 3DIC-588

(warning) Need %d TSV to support the current setting for net %s in this region, and the PG stripes are too short to create these TSV without DRC violation. Skip the creation for the net in this region.

**Description**

The distance between TSVs need to be larger than minimum spacing and pitch defined in technology file. According to the input current value from the block_power_files and tsv_conversion_factor, the number of TSV needed to support the current setting for the net is too many to keep enough distance between each other along the PG stripe. In order to avoid long runtime for creating TSV with DRC violations, the TSV creation for this net in this region is skipped.

**What Next**

Create more PG stripes for the net in the region or correct the settings.

## 3DIC-589

(error) the deisgn type of cell '%s' is not flip_chip_pad but it's on the off-die layer in '%s' die.

**Description**

The design type of cell must be flip_chip_pad when it's on the off-die layer.

**What Next**

Please contact foundry or bump cell library provider.

## 3DIC-590

(error) the bump '%s' is not on the off-die layer in '%s' die.

**Description**

The bump must be on the off-die layer.

**What Next**

Please contact foundry or bump cell library provider.

## 3DIC-591

(warning) there is no bumps on the %s layer in '%s' die.

Feedback

**Description**

NA.

**What Next**

NA.

---

## 3DIC-592

(error) the pin '%s' is not %s but %s in '%s' die.

**Description**

The bond pad must be on off-die layer and set bond pad attribute.

**What Next**

NA.

---

## 3DIC-593

(error) tsv cell '%s' has no shape on 'tsv' layer in '%s' die.

**Description**

Tsv cell must have one or more shapes on 'tsv' layer.

**What Next**

Please contact foundry or tsv cell library provider.

---

## 3DIC-594

(error) tsv cell '%s' has no shape on %s layer in '%s' die.

**Description**

Tsv cell must have one or more shapes on front layer whose mask_name is 'metal*' and back layer whose mask_name is 'bmetal*'.

**What Next**

Please contact foundry or tsv cell library provider.

---

## 3DIC-595

(error) the frame view of lib cell '%s' is missing.

Feedback

**Description**

The frame view of lib cell will be used during 3DIC Designing Planning flwo, it's missing will cause unexpected error.

**What Next**

Please contact foundry or tsv cell library provider.

---

## 3DIC-596

(error) the lib cell of 'flip_chip_pad' type '%s' is not on the off-die layer.

**Description**

Lib cell of 'flip_chip_pad' type must be on the off-die layer.

**What Next**

Please contact foundry or tsv cell library provider.

---

## 3DIC-597

(error) the lib cell of 'tsv' type '%s' has no shape on layers '%s'.

**Description**

Lib cell of 'tsv' type must have one or more shapes on 'tsv', 'metal*' and 'bmetal*' layers at the same time.

**What Next**

Please contact foundry or tsv cell library provider.

---

## 3DIC-598

(error) the layer setting of bump cell '%s' is different with its parent block in '%s' die.

**Description**

The layer setting of bump cell is different with its parent block, because the order of layer setting is different or some layers setting is missing.

**What Next**

Please contact foundry or tsv cell library provider.

Feedback

## 3DIC-599

(error) the attribtue 'is_pad' of pin '%s' in the lib cell %s is false, it will cause the port assignment failure.

**Description**

The pin can be set to the preferred_pin of the port, the attribte 'is_pad' of the pin must be true.

**What Next**

Please contact foundry or bump cell library provider.

## 3DIC-600

(error) the pitch of single probe bumps %s in the non-dummy region is less than value %s where the single probe bumps are in %s pattern in chip '%s'.

**Description**

The rule value of single probe bump in square pattern is set by app option 'plan.3dic.sprobe_min_pitch_square'; The rule value of single probe bump in other pattern is set by app option 'plan.3dic.sprobe_min_pitch_other';

**What Next**

Please adjust the placement of bumps.

## 3DIC-601

(error) the pitch of single probe bumps %s in the true-dummy region is less than value %s in chip '%s'.

**Description**

The rule value of single probe bump in other pattern is set by app option 'plan.3dic.sprobe_min_pitch_true_dummy_region'.

**What Next**

NA.

## 3DIC-602

(error) dummy region area.

Feedback

**Description**

The rule value of single probe bump in other pattern is set by app option 'plan.3dic.sprobe_dummy_region_area'.

**What Next**

NA.

---

## 3DIC-603

(error) the pitch between single probe bump '%s' and unprobed %s-dummy bump '%s' is less than value %s where the single probe bumps are in %s pattern in chip '%s'.

**Description**

The rule value of the pitch between single probe bump and unprobed non-dummy bump in square pattern is set by app option 'plan.3dic.sprobe_min_pitch_non_dummy_ubump_square'; The rule value of the pitch between single probe bump and unprobed non-dummy bump in other pattern is set by app option 'plan.3dic.sprobe_min_pitch_non_dummy_ubump_other'; The rule value of the pitch between single probe bump and unprobed true-dummy bump is set by app option 'plan.3dic.sprobe_min_pitch_true_dummy_ubump';

**What Next**

Please adjust the placement of bumps.

---

## 3DIC-604

(error) the layout for quad probe bump '%s' is not vertices of a square with edges parallel to die edges in chip '%s'.

**Description**

NA.

**What Next**

Please adjust the placement of bumps.

---

## 3DIC-605

(error) the pitch between quad bump '%s' and bump '%s' is less than value %s in chip '%s'.

**Description**

The rule value is set by app option 'plan.3dic.qprobe_pitch_ubump';

Feedback

**What Next**

Please adjust the placement of bumps.

---

## 3DIC-606

(error) the pitch between quad bumps %s and %s is less than value %s in chip '%s'.

**Description**

The bump quad must be square. The rule value of is set by app option 'plan.3dic.qprobe_min_pitch'.

**What Next**

Please adjust the placement of bumps.

---

## 3DIC-607

(error) the pitch between quad probe bump '%s' and single probe bump '%s' is less than value %s at %s direction in chip '%s'.

**Description**

The rule value is set by app option 'plan.3dic.qprobe_min_orthogonal_pitch_sprobe';

**What Next**

Please adjust the placement of bumps.

---

## 3DIC-608

(error) the number of probe bumps is %s, it's more than value %s in chip '%s'.

**Description**

The rule value is set by app option 'plan.3dic.probe_num_max';

**What Next**

Please adjust the number of probe bumps.

---

## 3DIC-609

(error) the number of probe bumps that connect to %s is %s, it's less than value %s in chip '%s'.

Feedback

**Description**

The rule value is set by app option 'plan.3dic.probe_num_min_discrete_power_ground';

**What Next**

Please adjust the number of probe bumps.

---

## 3DIC-610

(error) the number of quad probe bumps is %s, it's less than value %s in chip '%s'.

**Description**

The quad probe bump is the probe bump with square pattern. The rule value is set by app option 'plan.3dic.qprobe_num_min'.

**What Next**

Please adjust the number of quad probe bumps.

---

## 3DIC-611

(warning) No PG stripes of net %s found in this region. Skip the creation of %d TSVs for this net.

**Description**

With the '-pg_align_layer' option, the TSVs are created on the PG stripes of the nets. If there are no PG stripes in the region, the TSV creation for this net is skipped.

**What Next**

Create more PG stripes for the net in the region.

---

## 3DIC-612

(warning) Cannot find layer '%s' in tech file of die '%s'

**Description**

The layer name specified in '-pg_align_layer' option cannot be found in the technology file of the associated die.

**What Next**

Correct the layer name or the die name in the '-pg_align_layer' option.

Feedback

## 3DIC-613

(info) Found current information of net '%s' in upper die '%s'. Add TSV to supply the current for upper die.

**Description**

If net is found in upper die by connectivity tracing or name matching and the current information is defined in associated block power file, then enough TSVs are created to support the current for upper die.

## 3DIC-614

(info) No current information of net '%s' found in upper die '%s'. Create TSV to supply current for this die only.

**Description**

If no corresponding net is found in upper die by connectivity tracing or name matching, or there is no current information defined in associated block power file, then TSVs are created only to support the current for the creation target die.

## 3DIC-615

(warning) Cannot snap TSV from (%.3f %.3f) to the nearest TSV grid (%3.f %3.f).

**Description**

If there are already another TSV on the nearst TSV gird, then snaping the TSV to the TSV grid causes violation. Skip the snapping in such case.

**What Next**

Undo the creation anf try to create TSV again with smaller TSV pitch.

## 3DIC-616

(error) %s.

**Description**

Some via might overlap with preroutes.

**What Next**

Please perform DRC checking and fix any errors.

Feedback

## 3DIC-617

(error) bump %s has insufficient RDL routing.

**Description**

RDL routing connected to the given bump does not meet the min-area rule.

**What Next**

Please check the routing and modify it using GUI if needed.

## 3DIC-618

(error) %s %.3f is smaller than %s %.3f.

**Description**

The -delta value cannot be smaller than the TSV pitch defined in techfile. Otherwise, there will be spacing violation between created TSV. The -offset value cannot be smaller than the TSV width/height. Otherwise, the created TSV will overlap with the creation boundary.

**What Next**

Correct the command input.

## 3DIC-619

(warning) Cannot honor %s %.3f setting. Use internal calculated vaule to create TSV.

**Description**

The -delta and -offset settings are soft constraints. If TSV arrary cannot be created with these values, command uses internal calculated vaule to create TSVs.

**What Next**

Adjust the option input.

## 3DIC-622

(error) the design type of lib cell '%s' is not flip_chip_pad but it has pin or shape on the off-die layer.

**Description**

The design type of lib cell must be flip_chip_pad when it has pin or shape on the off-die layer.

Feedback

**What Next**

Please contact foundry or bump cell library provider.

---

## 3DIC-623

(error) commit pseudo %s failed. %d objects have been deleted.%s

**Description**

Error happened during commit pseudo bumps or tsvs. Please check the bump region setting.

**What Next**

NA.

---

## 3DIC-624

(warning) The bump %s is connected to multiple probe pads %s.

**Description**

The bump can only connect to one probe pad.

**What Next**

Please check the 'connected_bumps' attribute setting.

---

## 3DIC-627

(warning) backside bump %s is not assigned to TSV, it will be ignored for net assignment.

**Description**

Bumps in the backside must be assigned to TSV, otherwise it will not be used to net assignment. TSV is used to construct the connection between frontside and backside bumps.

**What Next**

NA.

---

## 3DIC-628

(warning) It is obsolete to use manufacturing shape to represent %s of die %s, please use 'create_keepout_margin -type %s' to recreate %s object.

Feedback

**Description**

It is obsolete to use manufacturing shape to represent seal_ring or assembly_die of a die.

**What Next**

please use 'create_keepout_margin -type seal_ring / assembly_die' to recreate object.

---

## 3DIC-629

(error) the bump %s with attribute 'is_direct_probe' as true and 'is_probe_pad' as false is invalid in chip %s.

**Description**

NA.

**What Next**

Please check the attribute setting.

---

## 3DIC-630

(warning) The %s of reference lib cell or via def '%s' is not equal with the diameter of pseudo tsv def '%s' in pseudo tsv '%s'.

**Description**

NA.

**What Next**

NA.

---

## 3DIC-631

(error) These tsvs that are mapped with pseudo tsv '%s' of def '%s' and using lib cell or via def '%s' to committed will be overlapped.

**Description**

The committed tsvs will be overlapped, because the sum of spacing or diameter in pseudo tsv def is less than the width or height of reference lib cell or via def, this phemomenon is illegal.

**WHAT NE**

Check the pseudo tsv def setting or adjust reference lib cell or via def.

Feedback

## 3DIC-640

(warning) no 3DBlox %s.

**Description**

No 3DBlox %s in current design.

**What Next**

Please check the 3DBlox settings.

## 3DIC-641

(error) the region %s is not entirely inside die %s.

**Description**

A region must be entirely inside a die.

**What Next**

Please check the 3DBlox settings.

## 3DIC-642

(error) the region %s overlaps with region %s in the die %s.

**Description**

Two regions on the same side of a die must not overlap. Regions of same die on different sides have no such restriction.

**What Next**

Please check the 3DBlox settings.

## 3DIC-643

(error) the region %s and %s are overlaped but not covered with each other.

**Description**

If two regions belong to different chiplets, check that either the first region completely contains the second region or vice versa in the XY dimension.

**What Next**

Please check the 3DBlox settings.

Feedback

## 3DIC-644

(error) the connection %s connects two regions %s and %s which are from the same die %s.

**Description**

Two regions of a connection must be from different dies.

**What Next**

Please check the 3DBlox settings.

## 3DIC-645

(error) there is connection %s that from %s to %s, connection %s that has opposite direction is prohibited.

**Description**

If there is a connection from die_A to die_B, then connection from die_B to die_A is prohibited. It will lead to inconsistency of the top-bottom relationship.

**What Next**

Please check the 3DBlox settings.

## 3DIC-646

(error) Based on connection %s, the connection %s is invalid, it will cause the inconsistency of the top-bottom relationship.

**Description**

Two connections with the same bottom region must have the same Z-elevation, otherwise it will cause the inconsistency of the top-bottom relationship.

**What Next**

Please check the 3DBlox settings.

## 3DIC-647

(error) The die %s is isolated from the spcified connectivity.

**Description**

The die %s is totally isolated from the connectivity specified in 3DBlox input.

Feedback

**What Next**

Please check the 3DBlox settings.

---

## 3DIC-648

(error) the path assertion %s has less than 2 regions.

**Description**

The path assertion should contain more than 2 regions.

**What Next**

Please check the 3DBlox settings.

---

## 3DIC-649

(error) Assert path %s. Assert result: FAIL.

**Description**

There is no valid path to satisfy the path assertion %s.

**What Next**

Please check the 3DBlox settings.

---

## 3DIC-650

(error) The connection %s is invalid. The top region %s is on the chiplet instance %s and the bottom region %s is on the chiplet instance %s, but chiplet instance %s is not on the top of %s.

**Description**

In a connection, the top die should be on the top of the bottom die.

**What Next**

Please check the 3DBlox settings.

---

## 3DIC-651

(error) the connection %s connects two regions whose parent die %s and %s are not neighboring dies.

Feedback

**Description**

A connection should connect two regions in the neighboring dies.

**What Next**

Please check the 3DBlox settings.

---

## 3DIC-652

(error) The connection %s is invalid, the %s region %s is on the %s side of face %s chiplet instance %s.

**Description**

In a connection, the bottom region should be on the front side of face up chiplet instance or the back side of face down chiplet instance, the top region should be on the back side of face up chiplet instance or the front side of face down chiplet instance.

**What Next**

Please check the 3DBlox settings.

---

## 3DIC-653

(warning) Basic Connectivity Check failed, skip %s Check.

**Description**

Basic Connectivity Check failed, the top-bottom relationship of all chiplets are inconsistent, skip %s Check.

**What Next**

Please check the 3DBlox settings.

---

## 3DIC-654

(error) the connection %s connects two regions %s and %s which are not physically connected.

**Description**

A connection should connect two regions which are physically overlapped.

**What Next**

Please check the 3DBlox settings.

Feedback

## 3DIC-655

(error) the bump %s is not on the %s manufacturing grid. The grid is %s.

**Description**

All bumps in 3D design must be placed on the manufacturing grid.

**What Next**

Please check the 3DBlox settings.

## 3DIC-656

(error) based on connection %s, the connection %s will cause extra cycle.

**Description**

Two connections with the same start must have the same Z-elevation, otherwise it will cause extra cycle.

**What Next**

Please check the 3DBlox settings.

## 3DIC-657

(error) Assert connection %s, the calculated stacking height(z) of die %s is %s, the stacking height(z) value in 3DBlox is %s. Assert result: FAIL.

**Description**

If two regions in connection are both on the die border, then it should satisfy the stacking hieght check "conn.top.z = conn.bot.z + conn.bot.thickness + conn.thickness".

**What Next**

Please check the 3DBlox settings.

## 3DIC-658

(error) For ChipletDef %s, assert levelness of region %s with layer %s. Assert result: FAIL. %s

Feedback

**Description**

For the layer definition in the region, it should satisfy: 1. each region has a layer definition. 2. the layer definition matches the side definition. 3. the bumps in the bmap haves shape on the layer.

**What Next**

Please check the 3DBlox settings.

---

## 3DIC-659

(error) The connections %s form a loop.

**Description**

The loop connections will lead to inconsistency of the top-bottom relationship.

**What Next**

Please check the 3DBlox settings.

---

## 3DIC-660

(error) Current design is not a valid 3DIC design. All chiplet instance's z_level are 0.

**Description**

A valid 3DIC design should have dies with different z_level. If the design is read in by "read_3dblox", please check the 'Connection' setting in the 3DBlox file.

**What Next**

Please check the 3DBlox settings.

---

## 3DIC-661

(error) Multiple %s ports %s are connected to the probe bump %s.

**Description**

The probe bump can only connect to one signal/PG port.

**What Next**

Please check the connection.

Feedback

## 3DIC-662

(warning) The port %s is connected to multiple probe pads %s.

**Description**

The port can only connect to one probe pad.

**What Next**

Please check the connection.

## 3DIC-663

(error) the pitch between probe %s %s and %s is %s, it's less than the minimum value %s in chip %s.

**Description**

The rule value of the pitch between two probe bumps is set by app option 'plan.3dic.min_3dblox_probe_pitch'; Probe bump: is_probe_pad is true, is_direct_probe is true. Probe pad: is_probe_pad is true, is_direct_probe is false.

**What Next**

Please adjust the placement of probe bumps.

## 3DIC-664

(error) the top net %s does not have both in and out direction ports.

**Description**

The top net cannot have the same in/out direction.

**What Next**

Please check the connection.

## 3DIC-665

(error) the bump %s cannot be covered by bump region %s in chip %s.

**Description**

The bump must be covered by the bump region which has the same name with the bump's associated edit group.

Feedback

**What Next**

Please adjust the placement of bumps.

---

## 3DIC-666

(error) %s %s does not physically connect to the %s on the adjacent side of the vertical neighboring chip.

**Description**

Found single bump or bond pad on the chip. These single bump and bond pad cannot be used to construct physical connections.

**What Next**

Adjust the location of chips or remove the single bump/bond pad or use create_3d_mirror_bumps to create bump pair/bond pad pair.

---

## 3DIC-667

(warning) the bump %s cannot find any associated bump region in chip %s.

**Description**

The bump must be associate with one bump region.

**What Next**

Please add the bump to the correct bump region.

---

## 3DIC-668

(warning) the bump %s in bbox {%s} will be overlapped with %s %s, skip create it.

**Description**

The shape in the created bump cell will be overlapped with the existed shape/blockage/ stiching zone. it violates drc rules.

**What Next**

Check the command option setting.

---

## 3DIC-669

(warning) the %s %s which is committed with %s will be overlapped with %s %s, skip commit it.

Feedback

**Description**

The shape in the committed cell/bond pad will be overlapped with the existed shape/ blockage/stiching zone. it violates drc rules.

**What Next**

Check the bump region setting.

---

## 3DIC-670

(Error) Cannot find the result '%s' in the thermal database.

**Description**

The specified result name cannot be found in the current thermal database.

**What Next**

Run analyze_3d_thermal with the specified result name first.

---

## 3DIC-671

(Error) Cannot find the data of %s thermal analysis with given result name '%s'.

**Description**

The given type of thermal analysis result cannot be found in the current result.

**What Next**

Run analyze_3d_thermal with the specified result name and analysis type.

---

## 3DIC-672

(Information) Finished thermal report generation successfully.

**Description**

The reports of the thermal analysis has been generated, containing summary report and detailed report.

**What Next**

Refer to the path ${THERMAL_DATABASE}/${RESULT}/${ANALYSIS_TYPE} for the reports.

Feedback

## 3DIC-673

(Error) Invalid value '%s' specified for design '%s' of app option '3dic.thermal.power_depth_ratio'.

**Description**

The value is not allowed in the app option '3dic.thermal.power_depth_ratio'.

**What Next**

The value of power depth ratio can only be: - Floating point from 0.0 to 100.0 - "substrate"

## 3DIC-674

(Warning) '%s' value '%s' is ignored in the command set_thermal_grid.

**Description**

The specified region does not intersect with the boundary of the instance.

**What Next**

Make sure the specified region intersects with the boundary of the instance.

## 3DIC-675

(Error) Failed to generate the thermal report.

**Description**

The reports of the thermal analysis cannot be generated properly.

**What Next**

Refer to the path ${THERMAL_DATABASE}/${RESULT}/${ANALYSIS_TYPE} for the reasons.

## 3DIC-676

(Error) Cannot find the stage '%s' in the thermal database.

**Description**

The specified stage name cannot be found in the current thermal database.

**What Next**

Run analyze_3d_thermal with the transient analysis first.

Feedback

## 3DIC-679

(warning) the bond pad %s in bbox {%s} will be overlapped with %s %s, skip create it.

**Description**

The shape in the created bond pad will be overlapped with the existed shape/blockage/ stiching zone. it violates drc rules.

**What Next**

Check the command option setting.

## 3DIC-680

(error) %s in the chip %s %s correctly cover %s in the chip %s, the reason is '%s'.

**Description**

The check will be done when the app option plan.3dic.probe_cover_dummy_bump is true, and plan.3dic.probe_cover_dummy_bump_size can set the column and row, default is 2*2.

**What Next**

Please check the bump location.

## 3DIC-681

(error) The %s between %s and %s is %s which is %s than the constraint <%s> setting, the valid range is {%s %s}.

**Description**

The enclosure spacing between the bump shapes in different layers violate the constraint.

**What Next**

Please contact foundry or bump cell library provider.

## 3DIC-682

(error) The %s between %s and %s is %s which is %s than the constraint <%s> setting, the valid range is {%s %s}.

**Description**

The area_ratio of toal area of utag objects in specific region between area of specific region violates the constraint.

Feedback

**What Next**

Adjust the location of some utag objects or remove/add some utag objects.

---

## 3DIC-683

(error) The %s between %s and %s is %s which is %s than the constraint <%s> setting, the valid range is {%s %s}.

**Description**

The area_count of tag objects in specific region between area of specific region violates the constraint.

**What Next**

Adjust the location of some tag objects or remove/add some tag objects.

---

## 3DIC-684

(error) The direct probe bump %s cannot have any associated bumps in attribute connected_bumps.

**Description**

For a direct probe bump probing, its connected_bumps should be empty, and it can’t connect to a probe pad.

**What Next**

NA.

---

## 3DIC-685

(warning) The probe pad %s has no values in attribute connected_bumps.

**Description**

The connected_bumps for every probe pad instance is supposed to have value.

**What Next**

NA.

---

## 3DIC-686

(error) The attribute 'connected_bumps' of %s %s and bump %s do not contains each other.

Feedback

**Description**

If the value of probe bump/pad pInst1's connected_bumps is bInst1, then the value of bInst1's connected_bumps must include pInst1.

**What Next**

NA.

---

## 3DIC-687

(error) The %s %s and its connected_bump %s are not logically connected.

**Description**

If the bump is set to the attribute connected_bump of one probe bump/pad, these bumps must have the same logical connection.

**What Next**

NA.

---

## 3DIC-688

(error) The %s %s and its connected_bump %s are not at the same side of chip %s.

**Description**

The bump and pad associated can only be at the same side of chip.

**What Next**

NA.

---

## 3DIC-689

(warning) The %s %s cannot set connected_bumps with %s %s because it's %s.

**Description**

pInst: attribute is_probe_pad is true, is_direct_probe is false; bInst: attribute is_probe_pad and is_direct_probe is both true or false; pInst only have bInst in connected_bump, bInst only have pInst in connected_bump.

**What Next**

NA.

Feedback

## 3DIC-695

(warning) the new mirrored %s from source %s %s in bbox {%s} will be overlapped with %s %s, skip create it.

**Description**

The shape in the mirrored bump or bond pad will be overlapped with the existed shape/ blockage/stiching zone. it violates drc rules.

**What Next**

NA.

## 3DIC-700

(error) net %s and net %s cross different subchannels.

**Description**

The tool partitions the routing tasks into subchannels. Therefore, all nets must be within the same subchannel. The tool finds nets crossing subchannels, causing the tool to error out.

**What Next**

Adjust bump-to-net assignment.

## 3DIC-701

(error) %d nets cannot be selected for routing.

**Description**

One or more nets cannot be routed since there is no enough space within the specified region. Namely, there are too many nets in the routing region. The tool calculated the maximal number of nets that can be routed, which is less than the total net count.

**What Next**

Add more routing layers or reduce routing pitches. It is also possible to resolve this issue by adjusting the routing region.

## 3DIC-702

(error) either pitch in x or y, %s or %s, is smaller than the size in x or y of %s, %s or %s, and thus they are in conflict.

Feedback

**Description**

Pitches from users' input should be smaller than the die size or larger than the size of bump/tsv lib cell. If the pitches are larger than the die size, no single bump/tsv can be created. If the pitches are smaller than the size of bump/tsv lib cell, each bump/tsv would overlap with another bump/tsv as invalid condition.

**What Next**

Please adjust and ensure the both pitches are larger than the size of bump/tsv lib cell as well as smaller than the die size.

---

## 3DIC-703

(warning) Detected short violation for nets %s on bumps/bond_pads (%s) and tap point (%s).

**Description**

Taps are created on top of bumps/bondPads of different nets.

---

## 3DIC-704

(Warning) Detected floating taps %s on net %s.

**Description**

Taps are not created on any bump/bond_pads.

---

## 3DIC-705

(warning) Detected redundant %s taps on %s..

**Description**

More than one tap are added on bumps/bond_pads.

---

## 3DIC-706

(warning) Detected missing tap vioation on net %s.

**Description**

There's no tap created on PG net. Please use create_taps command to create tap points.

Feedback

## 3DIC-707

(Information) The lengths of %s in x and y are %s mm and %s mm, respectively.

**Description**

The lengths in x and y of an rectangular object like PCB or package are shown in the unit of millimeter (mm) before sanity check in command estimate_3d_thermal.

**What Next**

Please check if the input of lengths is provided or within common industry standard.

## 3DIC-708

(Information) The ratio of size of pcb over package is %s.

**Description**

The ratio of sizes of PCB to package are shown before sanity check in command estimate_3d_thermal. Given a pcb is commonly larger than a package, the ratio is the maximum dimension of the PCB to the minimum dimension of the package.

**What Next**

Please check if the ratio is within common industry standard.

## 3DIC-709

(Information) The lengths of %s in x and y are %s um and %s um.

**Description**

The length in x and y of a die are shown before sanity check in command estimate_3d_thermal. The sanity check is to verify if the length is within common industry standard

**What Next**

Please check if the lengths within common industry standard.

## 3DIC-710

(Information) The ratio of die sizes is %s.

Feedback

**Description**

The ratio of sizes of dies is shown before sanity check in command estimate_3d_thermal. Given multiple dies are in an analysis, the ratio is the maximum dimension among the dies to the minimum dimension among the dies.

**What Next**

Please check if the ratio and length are within common industry standard.

---

## 3DIC-711

(Information) The height of %s is %s um.

**Description**

1.) The height of a bump array is shown before sanity check in command estimate_3d_thermal. If there are multiple bump arrys, the heights of all bump arrays are listed, respectively.

2.) The height of a die is shown before sanity check in command estimate_3d_thermal. If there are multiple dies, the heights of all dies are listed, respectively.

**What Next**

1.) Please check if the height of each bump array is within common industry standard.

2.) Please check if the height of each die is within common industry standard.

---

## 3DIC-712

(Warning) PG tap %s of die %s cannot form external electrical connection because the die is not located at the lowest Z-level.

**Description**

Taps are not created in the correct die. Please correct.

**What Next**

report_taps

---

## 3DIC-713

(Warning) PG tap %s of a die %s on metal layer %s cannot form external electrical connection because it is not the outer-most metal layer.

Feedback

**Description**

Taps are not created in the correct layer of bottom die. Please correct.

**What Next**

report_taps

---

## 3DIC-714

(Error) Syntax error found in 3dic.rail.expanded_pg_pattern.

**Description**

There is syntax error in the expanded PG pattern string.

**What Next**

Check 3dic.rail.expanded_pg_pattern man-page.

---

## 3DIC-715

(error) Net %s cannot be connected to bumps during die-to-die bus routing.

**Description**

Congestion causes open nets.

**What Next**

Adjust bumps of the net so that they are aligned along the channel direction. If bumps cannot be modified, fix routing manually.

---

## 3DIC-716

(error) Bumps of different types are not allowed in each region of die-to-die routing.

**Description**

Bumps of each region must be of the same type during die-to-die routing.

**What Next**

Adjust bump lib cells if needed. If bumps cannot be modified, consider disconnecting the nets of the bumps. Apply different routing methods to these nets after die-to-die routing.

---

## 3DIC-717

(error) cannot derive pin bbox for bump %s.

Feedback

**Description**

It is assumed that a bump should have one and only one pin on the bump layer. The tool either finds no pin or multiple pins on the bump layer.

**What Next**

Please check the bump and fix its reference if needed. User can define the bump layer using via_def in the pattern file.

---

## 3DIC-718

(warning) Only the last %s section is considered.

.SH DESCRIPTION

There are multiple inclusion and/or exclusion sections in the pattern file. The tool only considers the last one.

**What Next**

Please merge the multiple inclusion and/or exclusion sections in the pattern file.

---

## 3DIC-719

(error) The shielding option cannot be selected in multithreaded mode in current version of the tool.

**Description**

The shielding option cannot be selected in multithreaded mode in current version of the tool.

**What Next**

Please either use single threaded mode or removing the shielding option. Alternatively, please find newer version of the tool, which support multi-threaded shielding.

---

## 3DIC-720

(Warning) Template setting/values for %s are brought in via command set_thermal template.

**Description**

The template setting/values is brought in and may replace any setting before set_thermal template.

Feedback

**What Next**

If customized inputs are needed, Please be aware of using set_app_options after command set_thermal template.

---

## 3DIC-721

(Warning) skipping %s template input.

**Description**

Template input is skipped.

**What Next**

Please be aware of using set_app_options for customized inputs.

---

## 3DIC-730

(error) The current shielding pattern is not supported: %s.

**Description**

The shielding pattern specified by the user is not supported during 3D routing.

**What Next**

Please use create_rdl_shields to perform the shielding task after 3D routing is completed.

---

## 3DIC-731

(warning) some %s is in both inclusion and exclusion sections of the pattern file.

**Description**

Same net or bump is found in both inclusion and exclusion sections of the pattern file.

**What Next**

Net or bump in both inclusion and exclusion sections will not be processed. Please adjust the pattern file if needed.

---

## 3DIC-732

(warning) shielding constraints are ignored.

Feedback

**Description**

Shielding is impossible due to the lack of shielding net. So shielding related constraints in the pattern file are ignored.

**What Next**

Please check the netlist and add power or ground nets.

---

## 3DIC-739

(Error) The power can not be negative.

**Description**

The power must be non-negative

**What Next**

Specify non-negative power value

---

## 3DIC-740

(Error) The region is invalid.

**Description**

The region is invalid for the bounding box.

**What Next**

Check the coordinate of the bounding box.

---

## 3DIC-741

(Error) Duplicate probes.

**Description**

There is another probe having the same name.

**What Next**

Give the probe another name.

---

## 3DIC-742

(Error) Invalid Thermal Profile file format.

Feedback

**Description**

The Profile file is invalid. Its kind must be "Thermal Profile".

**What Next**

Use the Thermal Profile file.

---

## 3DIC-743

(Error) Invalid Probe Profile file format.

**Description**

The Profile file is invalid. Its kind must be "Probe Profile".

**What Next**

Use the Probe Profile file.

---

## 3DIC-744

(Warning) User defined %s in 3dic.thermal.heat_spreader is too small. %s is used instead.

**Description**

The specified size of heatspreader is ignored because it is does not cover the bbox of the topmost dies.

**What Next**

Make sure the specified value is large enough to cover the bbox of the topmost dies.

---

## 3DIC-748

(Error) The value of the duration time of stage %s must be greater than 0.

**Description**

The value of the stage duration time must be greater than 0.

**What Next**

Assign the duration time with a non-negative value instead.

---

## 3DIC-749

(Error) The step time is longer than the duration time in stage %s.

Feedback

**Description**

Since the step time is the temporal resolution of the output result, the step time should be shorter than or equal to the duration time in the stage.

**What Next**

Assign a value for the step time, which should be smaller than or equal to the duration time.

---

## 3DIC-750

(Error) No die for thermal analysis.

**Description**

After filtering the dies, there is no valid one for thermal analysis.

**What Next**

Please check the design type of the cells first, then use the *get_thermal_cells* command with *-ignore* option to see what dies are ignored and use the *set_thermal_cells* command to specify the filter for ignoring the thermal dies.

---

## 3DIC-751

(Error) Unable to find Power Profile of transient thermal analysis.

**Description**

The Power Profile of transient thermal analysis does not exist in the thermal database.

**What Next**

None

---

## 3DIC-752

(Error) The probe name %s has been added already.

**Description**

The probe name has been added already. It cannot have two probes named the same.

**What Next**

Use another name for the probe.

Feedback

## 3DIC-753

(Error) Unable to find %s's Probe Profile file %s.

**Description**

Unable to find the Probe Profile file.

**What Next**

Check whether the probe file exists or not.

## 3DIC-754

(Error) Unable to find thermal profile of transient thermal analysis.

**Description**

The thermal profile of transient thermal analysis does not exist in the thermal database.

**What Next**

Transient thermal analysis does not generate the thermal profile. Use Probe Profile to view the transient result instead.

## 3DIC-755

(Warning) Layer will be ignored in mhs format.

**Description**

The mhs format does not store the date layer-wisely.

**What Next**

Check the data is not layer-wise.

## 3DIC-756

(Error) Fail to parse the design bbox from %s.

**Description**

The input should be four double numbers.

**What Next**

Follow the input format to specify the region of the bbox.

Feedback

## 3DIC-757

(Warning) Fail to set fixed temperature boundary condition '%s'.

**Description**

Fail to set fixed temperature boundary condition because it cannot be applied on the prototype package and prototype PCB now.

**What Next**

Check app options 3dic.thermal.package_file and 3dic.thermal.pcb_file if the prototype package and prototype PCB are used.

## 3DIC-758

(Information) The design box of profile %s is %s that mismatches the design boundary box %s.

**Description**

The design box of the disk profile file is different from the design boundary box of the targeted design.

**What Next**

Use center-center alignment to map the data of the disk profile to the targeted design.

## 3DIC-759

(Warning) The design box size of profile %s mismatches the size of design boundary box.

**Description**

The design box size of the disk profile file is different from the design boundary box size of the targeted design.

**What Next**

Use center-center alignment to map the data of the disk profile to the targeted design.

## 3DIC-760

(Error) The stacking from %s to %s is invalid.

**Description**

The z-offset of the above die is greater than the z-offset of the below die.

Feedback

**What Next**

Check z-offset of the dies.

---

## 3DIC-761

(Error) The height of the containing die %s is smaller than the sinking die %s.

**Description**

The height of the containing die is higher than the sinking die.

**What Next**

Check the z-offsets and z-span of the containing die and the sinking die.

---

## 3DIC-762

(Error) %s's number of z-span heights is %d, but the z-span is %d.

**Description**

Inconsistent number of z-span heights.

**What Next**

Check the die heights.

---

## 3DIC-763

(Error) No power map file specified for any of the die instances in transient flow.

**Description**

No power map file is specified for die instance in transient analysis.

**What Next**

Use *set_thermal_transient_sequence* and *set_thermal_transient_stage* to specify the power map file.

---

## 3DIC-764

(Error) Either PCB or Package has to be specified.

**Description**

At least one of PCB or Package needs to be specified.

Feedback

**What Next**

Specify the PCB or package using the app options.

---

## 3DIC-765

(Error) The fusion power file %s is allowed only in fusion thermal engine.

**Description**

The fusion power file is only supported for fusion thermal engine.

**What Next**

Use other kind of power map.

---

## 3DIC-766

(Error) The power file %s is not supported.

**Description**

The power file is not supported.

**What Next**

Use other kind of power map.

---

## 3DIC-767

(Warning) The lid mode is not supported in current setting. The plate heatsink is used instead.

**Description**

The lid heatsink can be specified only when package size is available and a valid void space can be found.

**What Next**

None

---

## 3DIC-768

(Error) There should be at least 1 transient stage.

**Description**

At least one transient stage shoud be specified.

Feedback

**What Next**

Use *set_thermal_transient_sequencefP to add transient stages.*

---

## 3DIC-769

(Information) For details, please refer to below file(s):%s.\nPlease note that some files might not be generated if any error occurs during analysis.

**Description**

Detailed information are provided in the listed file(s), but some files might not be generated if any error occurs during analysis.

**What Next**

Check the files for details if needed.

---

## 3DIC-770

(Error) %s is an invalid value for %s attribute in app option %s.

**Description**

This occurs when the value is not valid.

**What Next**

Correct the value.

---

## 3DIC-771

(error) In option '%s', '%s' is not a valid 3DIC die instance name.

**Description**

Please provide a valid die instance name for the -inst_name.

**What Next**

Correct the option input.

---

## 3DIC-772

(Warning) There is no Power Profile to be removed.

**Description**

There is no Power Profile to be removed.

Feedback

**What Next**

Check the Power Profile exists or not.

---

## 3DIC-773

(Warning) There is no Thermal Profile to be removed.

**Description**

There is no Thermal Profile to be removed.

**What Next**

Check the Thermal Profile exists or not.

---

## 3DIC-774

(warning) Subchannel %d is placed too close to other subchannels.

**Description**

The minimal bus width is the routing pitch times the number of nets to be routed on the same layer. If two subchannels are placed too close, the bus routing of the two subchannels can overlap, leading to open nets or routing rule violations.

**What Next**

Please adjust the bump placement or use smaller routing width and/or spacing.

---

## 3DIC-775

(Error) Thermal model files of the instance %s are incomplete in the thermal database.

**Description**

There are top, middle, and bottom thermal model files for each instance per simulation run. Some of the files are missing in the thermal database.

**What Next**

Check the log file under the current working directory.

---

## 3DIC-776

(Warning) Thermal model files of the instance %s are missing in the thermal database.

Feedback

**Description**

All thermal model files of the instance are not presented in the thermal database. This may happen because the instance is ignored in the thermal analysis.

**What Next**

None

---

## 3DIC-777

(Error) The value %s of %s is not supported by built-in thermal engine.

**Description**

The specified value is not supported by built-in thermal engine.

**What Next**

Specify a supportd value.

---

## 3DIC-778

(warning) Wire length of net %s is 0.

**Description**

This error occurs when users do the check on an unrouted design or there exists an open net.

**What Next**

Please do the check on a routed design and check if there is an open net.

---

## 3DIC-779

(error) %s.

**Description**

This error occurs when users do not specify the channel_boundaries so that all 3D channels are treated as one 3D channel. which may cause some bumps cannot be identified which subchannels they belong to. Therefore, a subchannel may have odd number of bumps, which is not allowed.

**What Next**

Please specify the 3D channel boundaries by -channel_boundaries.

Feedback

## 3DIC-780

(warning) Non-two-bump net %s exists.

**Description**

This error occurs when there is a non-two-bump net. It will be skipped for the current check. The non-two-bump net may be a design issue or it is in control channel. If the net is in control channel, please ignore it.

**What Next**

Please check the design.

## 3DIC-781

(warning) The option %s is obsolete. Please use %s instead.

**Description**

The given option is obsolete and will be removed from the future release. Please change to use other options or a new command instead, which has similar functionality.

**What Next**

Check the command man page to see the correct options to use.

## 3DIC-782

(Warning) %s connection check error.

**Description**

The server connection is failed.

**What Next**

Please check whether the license of the Ansys tool is associated with the server that can not be connected.

## 3DIC-783

(Warning) %s.

**Description**

There is no specified license of Ansys tool.

Feedback

**What Next**

Please check whether the license of the Ansys tool exists.

---

## 3DIC-784

(Warning) %s.

**Description**

The available licenses issued is occupied.

**What Next**

Please check whether the license of the Ansys tool is occupied.

---

## 3DIC-785

(Error) Erratic net pattern detected. Cannot identify the net pattern.

**Description**

The net pattern of the followed objects cannot be identified to horizontal or vertical. All of the followed objects touching the same stripe must be fully inside the stripe unless the stripe width is smaller than the object size.

**What Next**

Please check the direction of the net pattern of the followed objects. If the direction is feasible but the pattern cannot be identified, it could be due to the absence of stripe width on the objects' layer.

The stripe width will be determined based on the following priorities: 1. The stripe width on the followed objects' layer specified in the -pg_constraints. 2. The existing stripe width on the followed objects' layer. 3. One of the followed objects' size.

---

## 3DIC-786

(warning) shielding constraints cannot be honored for layer %s.

**Description**

Either width or spacing or both of shielding cannot be satisfied. Please adjust the values. The shielding constraints must fit the routing width and spacing constraints.

**What Next**

The tool will try to complete the routing and/or shielding. But the shielding rate will be low since the constraints cannot be honored. Please adjust the values if needed.

Feedback

## 3DIC-787

(Error) %s.

**Description**

The license is expired.

**What Next**

Please check whether the license of the Ansys tool is expired.

## 3DIC-788

(Warning) %s.

**Description**

The specified file was not generated successfully.

**What Next**

Please check whether the file is generated successfully.

## 3DIC-789

(Error) %s.

**Description**

Cannot find the RedHawk-SC(ET) binary.

**What Next**

Please check whether the binary of RedHawk-SC(ET) exists.

## 3DIC-790

(Error) %s.

**Description**

Cannot detect the RedHawk-SC(ET) version.

**What Next**

Please check whether the version shows up by adding flag -v to RedHawk-SC(ET) binary.

Feedback

## 3DIC-791

(Warning) %s.

**Description**

There is layer inconsistence between Ansys and Synopsys techfile.

**What Next**

Please check the layer order between Ansys and Synopsys techfile.

## 3DIC-792

(Error) %s.

**Description**

There is a net cannot be found in the die or there is no port can be found on a net.

**What Next**

Please check the specified net name in -nets or the port setup on the net.

## 3DIC-793

(Error) %s.

**Description**

Fail to copy the library(block).

**What Next**

Please check whether the copied library(block) name exists.

## 3DIC-794

(Error) %s.

**Description**

Fail to add the copied library in ref_libs.

**What Next**

Please check whether the path of the copied library exists.

Feedback

## 3DIC-795

(Warning) %s.

**Description**

There is no instance information in the specified file.

**What Next**

Please check whether the file is generated successfully with instance information.

## 3DIC-796

(Error) %s.

**Description**

The specified variable is not set.

**What Next**

Please check whether the specified variable is set.

## 3DIC-797

(Error) %s.

**Description**

Library coverage check is failed.

**What Next**

Please check whether the library information for Ansys tool is complete.

## 3DIC-798

(Error) No tsv def found in the design.

**Description**

Cannot find any TSV def in the design to create TSV instance.

**What Next**

Please check the TSV def is valid or net.

Feedback

## 3DIC-799

(Error) %s.

**Description**

There is no license exists or all licenses are expired.

**What Next**

Please check whether the license of the Ansys tool exists or all the license of the Ansys tool is expired.

## 3DIC-800

(Error) %s.

**Description**

Fail to execute command analyze_rail.

**What Next**

Please check whether the command analyze_rail can be used.

## 3DIC-801

(Error) %s.

**Description**

Fail to open(remove) the copied library(block).

**What Next**

Please check whether the copied library(block) cannot be opened(removed).

## 3DIC-802

(Error) %s.

**Description**

Fail to set the reference of the specified die.

**What Next**

Please check whether the specified die cannot change the reference library.

Feedback

## 3DIC-803

(Warning) %s.

**Description**

Fail to parse Synopsys or Ansys techfile.

**What Next**

Please check whether the Synopsys techfile or the Ansys techfile exists or the csv file has the correct format.

## 3DIC-804

(Warning) No flyline is created on net: %s in die instance: %s. There is no bump connected to the corresponding port.\n

**Description**

Failed to create flyline since there is no bump connected to the corresponding port.

**What Next**

Please check the port of the net and create the corresponding bump.

## 3DIC-805

(Warning) No flyline is created on net: %s in die instance: %s. There is only one bump connected to the corresponding port.\n

**Description**

Failed to create flyline since there is only one bump connected to the corresponding port.

**What Next**

Please create more than one bump connected to this port.

## 3DIC-806

(Warning) No flyline is created on net: %s in die instance: %s. Bumps connected to the corresponding port do not have valid direction.\n

**Description**

Failed to create flyline since the bumps connected to the corresponding port do not have valid direction.

Feedback

**What Next**

Please check the direction of the bumps connected to the corresponding port.

---

## 3DIC-807

(Warning) %s.

**Description**

The specified check of Redhawk-SC(ET) setup is failed.

**What Next**

Please check whether the specified check is failed.

---

## 3DIC-808

(Error) Pseudo objects and physical objects have conflicts in die inst %s.

**Description**

A pseudo bump (TSV) and a physical bump (TSV) are overlapped. Command create_*_template_obejcts would not create objects in this case.

**What Next**

Please remove overlaps or make two objects be connected to the same net and at the same center location.

---

## 3DIC-809

(Error) The pseudo %s %s and the phyiscal %s %s are overlapped and at different center locations.

**Description**

A pseudo bump (TSV) and a physical bump (TSV) are overlapped. Command create_*_template_obejcts would not create objects in this case.

**What Next**

Please remove overlaps or make two objects be connected to the same net and at the same center location.

---

## 3DIC-810

(Error) The lib_cell bump %s in -lib_bump_cell cannot be found on layer %s.

Feedback

**Description**

The specified lib_cell bump cannot be found in the library or don't contain any pin on the layer corresponding to the specified side.

**What Next**

Please check the name, side and the pin's layer of the specified bump.

---

## 3DIC-811

(Error) Net %s is trying to be specified with lib_cell bump %s in -lib_bump_cell, but the net has been specified with the lib_cell bump %s.

**Description**

The net has been already specified with a bump and can only be specified with one bump.

**What Next**

Please don't specify multiple bumps with the same net, or check if the net is wrongly specified to a used net.

---

## 3DIC-812

(Error) Cannot specify lib_cell bump %s as default in -lib_bump_cell. The default lib_cell bump %s has been specified.

**Description**

There is only one default lib_cell bump can be specified in -lib_bump_cell.

**What Next**

Please don't specify multiple bumps as default, or check if the pg net is missed.

---

## 3DIC-813

(Error) TSV %s has no %s.

**Description**

The TSV's def or library is incomplete.

**What Next**

Please check the TSV's def or library.

Feedback

## 3DIC-814

(Error) No lib_cell bump found on layer %s.

**Description**

Cannot find any lib_cell bump which has any pin on the layer corresponding to the specified side.

**What Next**

Please check if there is at least a feasible lib_cell bump in the library.

## 3DIC-815

(warning) %d existing shapes in 3D channel.

**Description**

This occurs when there are pre-existing shapes in 3D channels. It may block the subchannels and cause routing quality degradation due to detour. However, some shapes may not significantly impact routing quality. Therefore, if the tool successfully completes the routing, please just ignore this warning.

**What Next**

Please check the design.

## 3DIC-816

(warning) There is no channel marking result in 3D channel.

**Description**

This occurs when there is no channel marking in 3D channel. Therefore, users cannot use any tcl command with the subchannel index attribute, since the attribute is set to -1 by default. Channel marking will be done automatically during 3D channel routing. However, the post-route report can still be generated. Please just ignore this warning if users do not need to use the attribute. Note: The subchannel index attribute of all non-3D channel nets will be set to -1 by default.

**What Next**

Please use the 3D channel routing command then use the report command.

## 3DIC-817

(warning) %s.

Feedback

**Description**

This occurs when there is no pattern file specified in the option, or the pattern file cannot be saved into the attribute. Therefore, users cannot use any tcl command with the pattern file attribute, However, if users do not need to use the attribute, please just ignore this warning.

**What Next**

Please check the command line and the pattern file.

---

## 3DIC-818

(warning) %s.

**Description**

This occurs when there is a value that does not meet the constraints of tech file, or some settings are not set correctly. However, some may not significantly impact the routing quality.

**What Next**

Please check the command line, the pattern file, and the design to see if they are as expected.

---

## 3DIC-819

(error) %s.

**Description**

This occurs when bumps are not aligned in bump region, which may cause the channel identification or routing fail. Bumps should be aligned along the channel direction.

**What Next**

Please check the bump arrays.

---

## 3DIC-820

(Warning) Cannnot find any pseudo bump in the specified region, please use create_bump_region_pattern and create_bump_region to create pseudo bumps first.

**Description**

There is no pseudo bump to be read when -read_object_type pseudo|all is specified in the command create_rail_template_objects.

Feedback

**What Next**

Please use create_bump_region_pattern and create_bump_region to create pseudo bumps first.

---

## 3DIC-821

(Warning) Cannnot find any pseudo TSV in the specified region, please use create_pseudo_tsv_def, create_bump_region_pattern and create_bump_region with -tsv_pattern to create pseudo TSVs first.

**Description**

There is no pseudo TSV to be read when -read_object_type pseudo|all is specified in the command create_rail_template_objects.

**What Next**

Please use create_pseudo_tsv_def, create_bump_region_pattern and create_bump_region to create pseudo TSVs first.

---

## 3DIC-822

(error) Complex bump %s exists in 3D channels.

**Description**

This occurs when there are complex bumps in the 3D channels, which means that some bumps are connected to multiple nets. Currently, the tool does not support complex bumps with multiple pins.

**What Next**

Please check the bump arrays.

---

## 3DIC-823

(Error) Pixel model %s already exists.

**Description**

The pixel model already exists.

**What Next**

Specify a different pixel model name.

Feedback

## 3DIC-824

(Error) Unable to find pixel model %s.

**Description**

The pixel model does not exists.

**What Next**

Specify a valid pixel model name.

## 3DIC-825

(Error) Unable to remove pixel model %s because it is referenced by voltage model %s.

**Description**

The pixel model is referenced by the voltage model and cannot be removed.

**What Next**

Remove the voltage models that are referencing the pixel model.

## 3DIC-826

(Error) The region is not a rectilinear polygon.

**Description**

The voltage model must have a rectilinear region.

**What Next**

Specify a rectilinear polygon.

## 3DIC-827

(Error) The voltage model region is not enclosed by %s's boundary bbox.

**Description**

The voltage model region must be enclosed by die boundary bbox.

**What Next**

Specify a rectilinear polyrect inside the die boundary bbox.

Feedback

## 3DIC-828

(Error) Voltage model %s already exists.

**Description**

The voltage model already exists.

**What Next**

Specify an unique voltage model name.

## 3DIC-829

(Error) Unable to find voltage model %s.

**Description**

The voltage model does not exists.

**What Next**

Specify a valid voltage model name.

## 3DIC-830

(error) %s.

**Description**

This occurs when there is a value that does not meet the constraints of tech file, or some settings are not set correctly. The issues will cause routing to fail or design rule violations.

**What Next**

Please check the command line, the pattern file, and the design to see if they are as expected.

## 3DIC-831

(Error) Voltage area %s is overlapped with other voltage area that shares the same net.

**Description**

Overlapping voltage area of the same die can not share the same net.

**What Next**

Remove the overlapping voltage area.

Feedback

## 3DIC-832

(error) adjusting via_def and metal layers failed for layer %s.

**Description**

This error occurs when via_def and metal layer cannot be adjusted for the specified top layer name. Please check whether the name is correct.

**What Next**

Please check the tech file and specified layer name.

## 3DIC-833

(error) The constraint {%s} is ignored.

**Description**

This occurs when the constraints in pattern files, which should be specified in a particular section, are not added correctly. It may lead to the channel identification or routing fail.

**What Next**

Please check the constraints in pattern file and specify them in a correct section.

## 3DIC-834

(info) Merging subchannel %u and %u due to cross-subchannel net %s.

**Description**

This occurs when there is a cross-subchannel net originally and the subchannels are merging due to the cross-subchannel net. Therefore, the net will no longer cross subchannels.

**What Next**

Please check if the merge is as expected.

## 3DIC-835

(error) Net %s is still crossing subchannels since subchannel %u and %u cannot be merged as %s.

Feedback

**Description**

This occurs when the tool tries to merge the subchannels due to a cross-subchannel net. However, due to the given reason, the merge fails and the cross-subchannel net still exists.

**What Next**

Please check the bump arrays and the corresponding nets.

---

## 3DIC-836

(Warning) Fail to get the custom pixel model resistance from %s. All resistances will set to default value 0.

**Description**

The spice file can not find the subckt name, or the file format is incorrect.

**What Next**

Check the spice file.

---

## 3DIC-837

(error) Fail to route using side-RV mode because %s.

**Description**

Customer choose top_layer_no_via and via_under_bump in the pattern file. As a result, the route will place routing tracks on the side of each bump column. However, the routing cannot be completed successfully due to the given reason.

**What Next**

Correct the input netlist or choose a different routing mode.

---

## 3DIC-838

(warning) Some sideband nets are assigned to bumps of different columns.

**Description**

The two bumps of a net should be assigned to bumps of the same column for easy routing. If bumps of different columns are chosen, net flylines might crossover each other, leading to open nets during routing.

Feedback

**What Next**

Check the routing result. If open nets are found, please consider adjusting net-bump assignments.

---

## 3DIC-839

(error) %s.

**Description**

This occurs when lower-level vias do not meet the design rules. Although the tool may still finish the routing, there will be design rule violations. Please adjust the dimension of lower-level vias.

**What Next**

Please check the lower-level vias and adjust them if needed.

---

## 3DIC-840

(error) no pre-routes are allowed in the channel for side-via routing mode.

**Description**

If side-via routing mode is selected, there cannot be any pre-routes in the channel.

**What Next**

Please check the design and remove any pre-routes. If the pre-routes cannot be removed, try to use another routing mode.

---

## 3DIC-841

(Error) The detected net pattern is different from the specified direction and the preferred routing direction of the layer %s.

**Description**

The direction of the followed objects' net pattern is different from the direction of the layer specified in *set_3d_template -pg_constraints* and the preferred routing direction of the specified layer in the tech file.

**What Next**

Please check the direction of the followed objects' net pattern and *-coordinates*/*-coordinate_system*. If *-coordinates* is set too large, it may lead to the wrong results of net pattern detection.

Feedback

# ABS

## ABS-007

(Information) Block %s is a design view. No constraint comparison will be run.

**Description**

The block specified is linked as a full design view. When checking the constraint consistency between abstracts and top-level, this block will be skipped.

**What Next**

This is an informational message. No action needed.

## ABS-008

(information) Did not find any mismatched constraints between abstract '%s' (reference : '%s') and top level.

**Description**

This message is issued when the constraint consistancy between the abstract and top-level is checked and no mismatches are found.

**What Next**

No action is necessary.

## ABS-009

(error) Found %d mismatched constraints between abstract '%s' (reference : '%s') and top level.

**Description**

This message is issued when the constraint consistancy between the abstract and top-level is checked and some mismatches are found.

**What Next**

Update the top and/or the block constraints to ensure that both the top-level and the block designs have the correct constraints. If the block constraints were changed re-create the abstract.

Feedback

## ABS-034

(Error) Can't split UPF because it's not loaded yet.

### Description

UPF has not been loaded in the design. Hence split_constraints -upf_only cannot proceed.

### What Next

Load the UPF and try the command again.

### See Also

- load_upf
- source

## ABS-100

(warning) Abstract view '%s' is not being saved. Library '%s' is in read mode.

### Description

The specified block view is not saved because its library is open in read mode.

### What Next

Use "save_lib -as" to save the current library as some other library. You can then use that other library to access the block.

```
prompt> save_lib -as <new_library_name>
```

### See Also

- open_block
- open_lib

## ABS-101

(error) Cannot merge pin guide/blockage %s which is associated with objects other than ports and nets.

### Description

You get this message when the tool tries to merge data from the outline view to the design view, or from the abstract view to the design view, or from the design view to the abstract view.

Feedback

This message indicates that there is a pin guide or pin blockage in the outline view, or the abstract view, or the design view that is associated with objects of a type that is not port or net.

The tool does not support merging this type of pin guide or pin blockage between views. As a result, the corresponding pin guide or pin blockage created in the design or abstract view might have incomplete data. This might cause the pin guide or pin blockage affect objects that they are not intended to affect. This might also cause the pin guide or pin blockage to take no effect on the objects they are intended to affect.

**What Next**

If the pin guide or pin blockage is no longer needed, please delete it. If it is still needed, please delete it and recreate it in the design view after the data is merged or in the abstract view after it is created.

**See Also**

- create_pin_guide
- create_pin_blockage

---

## ABS-110

(Error) Some objects in the collection used for the -include_objects command option are not valid in the current block.

**Description**

When the create_abstract command is run with the -include_objects option, all the objects in the collection given to that option need to be ports, pins or nets that are valid in the current block. This error message is issued if some of the objects in the collection are invalid, or belong to a different block (or a different view).

**What Next**

Examine the collection used and remove/add objects to ensure that all objects in the collection are valid ports, pins or nets in the current block.

**See Also**

- create_abstract
- filter_collection
- remove_from_collection
- add_to_collection

Feedback

## ABS-115

(Warning) The 'none' option value for -timing_level option of create_abstract command is deprecated and will be removed in a future release.

### Description

The option value 'none' for the command option -timing_level of create_abstract command is deprecated. It will be removed in the coming release.

## ABS-125

(error) The abstract view was not created from this version of design view. Changes from abstract view cannot be merged back to design view. Aborting merging changes.

### Description

The present abstract view was not created from this version of the associated design view. Under such conditions, changes cannot be merged from abstract view to design view.

### What Next

Make sure the abstract view is alwasy associated with the design view, from which it was created.

### See Also

- create_abstract

## ABS-130

(error) cell '%s' has no reference module.

### Description

This message is issued during abstract merging when there is a cell in abstract view that does not have reference module.

### What Next

This error happens when the specified cell has no reference module. This could happen if the reference library is not set up properly.

Please check reference library settings and make sure the reference library for the cell is set properly.

Feedback

## ABS-150

(Warning) Property changes detected for routing rule '%s'. User need to manually update the routing rule.

**Description**

This warning message comes when property changes are detected for some routing rule. ECO changelist will not update these properties. User need to manually review for the property changes and update the routing rule accordingly.

**What Next**

If the property changes are needed, then manually update the routing rule.

## ABS-152

(Information) Eco change list files with header description '%s' will be written in directory '%s'.

**Description**

This information message specifies the location where eco files will be written for abstracts. At the specified location, sub-directories will be created for individual blocks. Each block specific directory will contain a 'eco.tcl' file which would have the eco changes written in tcl format.

If the eco write out is happening in the same location multiple times, each sub-block specific 'eco.tcl' will be appended with new eco changelist.

The 'eco.tcl' files will contain a header description and the users will be informed about this header in this information message. If the eco write out is happening in the same location multiple times, each 'eco.tcl' will contain multiple headers.

**What Next**

This is an informational message. No action needed.

## ABS-187

(Error) Invalid value %f for option -min_segment_percent

**Description**

You are getting this message because you specified a value that is not between 0.0 and 0.5 for the option -min_segment_percent.

Feedback

The value given with the -min_segment_percent option is a percentage of the clock period. The tool computes the minimum budget for the budget segments by multiplying the value specified here with the clock period.

**What Next**

Please specify a value that is between 0 and 0.5.

---

## ABS-188

(Error) Command option '%s' is not supported when app option '%s' is set to %s.

**Description**

The given option of this command is not supported with the given setting of the app option.

**What Next**

Please change the setting of your app option, or do not use this command option.

---

## ABS-189

(Error) Module '%s' has been specified twice.

**Description**

You have specified the given module name to a -design_subblocks, -hier_abstract_blocks, or -shell_subblocks option. But this name has appeared twice in one list, or is in two separate lists. Each name can only appear once.

**What Next**

Please remove the redundant name.

---

## ABS-190

(Error) Module '%s' is not a valid budget block.

**Description**

You have specified the given module name to a -design_subblocks, -hier_abstract_blocks, or -shell_subblocks option. But the module name does not correspond to any of the instances you are trying to budget.

You need to specify a module that corresponds to one of the instances you supplied with the "set_budget_options -add" command. For example, if you specify module "CPU", an instance of CPU (perhaps called my_cpu_inst) my appear like:

set_budget_options -add my_CPU_inst

Feedback

### What Next

Make sure that the module you specify corresponds to one of the instances you supplied with the "set_budget_options -add" command. You can add new instances to consider by calling set_budget_options.

---

## ABS-191

(Error) Can't find installation of Library Compiler%s. Budget shell cannot be created.

### Description

When creating a budget shell, it is necessary to invoke Synopsys Library Compiler. And often, the Library Compiler executable is installed in another directory and not with ICC2. This error indicates that icc2_shell was unable to locate your Library Compiler installation.

### What Next

Please set the SYNOPSYS_LC_ROOT environment varible to point to the directory where Library Compiler is installed. For example, if you set

setenv SYNOPSYS_LC_ROOT /remote/depot/lc

then the icc2_shell will expect to find a library compiler executable at /remote/depot/lc/bin/lc_shell. It will also expect to find an executable under an machine-specific directory. For example, if you are using linux64, then you should see this executable:

/remote/depot/lc/linux64/lc/lc_shell_exec

If you do not see this executable, then please check your installation.

You can set the SYNOPSYS_LC_ROOT environment variable in your ICC2 tcl script, if you wish. But it is often more convenient to set the environment variable in your .cshrc or at the unix prompt. If you set the environment variable in unix, you must set it before starting your icc2_shell program.

---

## ABS-192

(Warning) At budget pin %s, exception of type '%s %s' is overridden by '%s %s' because precedence has changed.

### Description

In SDC, there is a "precedence" that determines what happens when two timing exceptions overlap on the same timing path. Please refer to the man page for set_multicycle_path to see details on exception precedence.

Feedback

When you create a budget for a lower-level block, the precedence for an exception may change. For example, at the top level, you might have the following exception:

set_multicycle_path 2 -from reg1/CP

When you create a budget, this constraint changes at the block level:

set_multicycle_path 2 -from virtual_clock -through block_port

The budgeted exception has lower precedence and has the potential to be dominated by a different constraint in your block. This can cause the constraint to change.

When budgeting detects that your budgeted top-level exception has been overridden by a block exception, this warning is printed.

**What Next**

No action is required on your part because the budget will adapt to compensate for the change in exception. You will not get an overconstrained budget.

If you want to get rid of this message, you can do one of the following:

1. Suppress the message by setting:

```
set_app_options -name plan.budget.report_exception_precedence \\
       -value false
```

2. Change your constraints so they are consistent. For example, if you have a path across a block boundary with two exceptions:

```
set_multicycle_path 3 -from reg1/CP
set_multicycle_path 2 -to block/reg2/D
```

   then you will trigger this warning. If you use "set_multicycle_path 2" for both constraints, there will be no warning, because the constraint remains the same.

---

## ABS-193

(Warning) The current scenario must have setup analysis activated. Timing analysis disabled.

**Description**

This command analyzes the setup times in the current scenario. Either the current_scenario has not been specified or the current_scenario does not have setup analysis activated.

The analysis will continue using only connectivity and no timing.

Feedback

**What Next**

Set your current scenario to an appropriate choice using "current_scenario", or activate setup timing with "set_scenario_status".

---

## ABS-194

(warning) Clipped %d %s values by an average of %s.

**Description**

You have set the app option plan.budget.clip_unoptimized_values to true. As a result, whenever the budgeter sees a delay or capacitance that seems unreasonably high, it will choose a lower value to use in its analysis.

This message gives you a general idea of how much values have been reduced.

**What Next**

Please see the man page for plan.budget.clip_unoptimized_values for more details. In general, you should only be using this app option in early phases of design, when timing accuracy is less important.

---

## ABS-195

(warning) Budget shell for corner %s will be given default PVT values.

**Description**

In order to properly use budget shells, you should assign each of your top-level design's corners "process", "voltage", and "temperature" (PVT) values.

No values were assigned for the given corner, so the internal processing has assigned default values for you. These values are not really used, and will not effect the outcome of budget shell timing. But they may appear confusing if you see them in some reports.

**What Next**

To get rid of this warning, fill in PVT values for all of your corners. This assignment can be done either with the set_operating_conditions command or with the set_process, set_voltage, and set_temperature commands.

---

## ABS-196

(error) Unable to generate db files for the budget shells: %s

Feedback

**Description**

Something has gone wrong with processing. The budget shells could not be processed.

**What Next**

If you have ruled out problems with your network, such as running out of disk space, please contact Synopsys and report this problem.

---

## ABS-197

(error) Cannot generate budgets shells because of %s in your installation.

**Description**

Generation of budget shells requires auxiliary files in the installation root. If you get this message, it means that your installation is incomplete.

**What Next**

Ensure that the software is installed correctly and all files in the installation are present.

---

## ABS-198

(warning) The -hold option is no longer needed. Setting plan.budget.write_hold_budgets to true.

**Description**

The -hold options are no longer needed in order to generate hold budgets. They are now ignored. Hold constraints will automatically be generated by the write_budgets command whenever the plan.budget.write_hold_budgets app option is set to true. The app option has been set for you.

**What Next**

Please modify your script. Remove the -hold option from compute_budget_constraints and set_pin_budget_constraints. Instead, add this to your script:

set_app_options -name plan.budget.write_hold_budgets -value true

---

## ABS-199

(error) Can't process leaf instance '%s'. Skipping.

**Description**

Split_constraints and write_budget can only be performed on hierarchical instances. You have asked to process a leaf instance.

Feedback

**What Next**

Double check the results of design linking. Perhaps a problem there caused the hierarchy to not be loaded. If this really is a leaf instance, you should remove it from your list of budgeted blocks.

---

## ABS-200

(error) Pin '%s' is not on a budgeted block. Cannot set budget.

**Description**

Pin budgets can only be set on budgeted blocks.

**What Next**

Use add_budget_block to select a block for budgeting.

---

## ABS-201

(error) Aborting after %d errors.

**Description**

The command only executed until the specified number of errors occurred.

**What Next**

Fix the specified errors. For the set_budget_options command, consider using the -all flag.

---

## ABS-202

(error) Budget includes a shell for cell '%s', but the cell is not in the design.

**Description**

At the time write_budgets was run, the -shell_subblocks option was applied to a cell of this name in the design. In order to use this budget, the given cell must still exist in the design. It does not.

Because of the problem, the constraints have not been loaded.

**What Next**

If you expected this cell to exist, you should check to see if your design has been properly linked. If this cell is no longer important, you need to use write_budgets to generate a new budget based on the current design.

Feedback

## ABS-203

(error) Can't apply budget because it assumes that cell '%s' is an abstract.

**Description**

At the time write_budgets was run, the hierarchy corresponding to the given cell was treated as an abstract. This means that constraints were generated for the interface of the cell, but not for the internal paths.

There are two possible reasons why write_budgets may have written constraints for an abstract rather than a full design.

1. You failed to supply the proper -design_subblocks option to write_budgets.
2. At the time your budget was created, the given cell was represented as an abstract, not a full design.

In your current design, one of the following problems has occurred:

1. The cell is missing. Constraints in the budget cannot be applied.
2. The cell is has been replaced by a budget shell or ETM. Constraints in the budget cannot be applied.
3. The cell corresponds to a full design netlist, not an abstracted version. Constraints in the budget are incomplete.

Because of the problem, the constraints have not been loaded.

**What Next**

If you expected the cell to be an abstract, you should check to see if your design has been properly linked. If you now want the cell to be something other than an abstract, you need to use write_budgets to generate a new budget based on the current design. Make sure that you specify the correct -design_subblocks option.

## ABS-204

(error) '%s' has not been declared as a budget block.

**Description**

Budgeting can only act on instances that have been added to the budget block list. Also, the instance must point to a logical or physical hierarchy. It is not legal to budget a leaf cell.

**What Next**

Use "set_budget_options -add_blocks" to add your instance to the budget list. Make sure that the instance you are adding corresponds to a hierarchy.

Feedback

## ABS-205

(error) Can't open '%s' for writing.

**Description**

The file could not be opened. The budget has not been written.

**What Next**

Check unix permissions on the target directory and file.

## ABS-206

(warning) When writing multiple budgets, the file_name must contain `%%d'

**Description**

When writing multiple budgets, you need to specify %d somewhere in the file name. The %d characters will automatically be replaced by the name of the budgeted design. If the design has multiple constraint modes, you should also use %m, which will be replaced with the mode name. Failure to use %d would cause multiple budgets to dump to the same file and overwrite each other.

**What Next**

Use %d in at least one place in your file name specification. You can also use %m for the constraint mode of the block.

## ABS-207

(error) Cannot find -source pin for generated clock '%s' in block '%s'. Master clock does not propagate to the block boundary. Output will be incorrect.

**Description**

When budgeting a generated clock within a lower level block, you must conform to the following two rules:

rule 1

Generated clocks should be "expanded" in the full design to resolve their master clock. If the generated clock is not expanded before budgeting, it is not be possible to expand it in the budgeted block.

rule 2

If a master clock source is outside of the block and the generated clock source is inside, the master must propagate to the block boundary.

Feedback

If you received this message, rule 2 has been violated, and it is not possible to generate a legal definition for the generated clock -- because the master clock source is not available. The output SDC file will not be usable until the problem is corrected.

You should look in your program output directly before this error message. You will see information that is useful in figuring out why your master clock did not propagate to the block boundary.

**What Next**

This problem is very serious and it should be fixed. Certainly, you should not start post-CTS optimization with the bad generated clock. However, if you want to continue with prects optimization you can set:

```
set_app_options -name plan.budget.allow_disconnected_gclocks -value true
```

This will generate a budget that will let you continue. Please refer to the app option man page for details.

To correct problems with rule 1, make sure that a master clock is listed for every generated clock shown by the report_clocks command.

You can correct the problem with rule 2 by ensuring that either the master clock propagates to the block boundary or the generated clock is declared at or outside of the boundary.

For example -- consider this verilog for top and block:

```
module top;
    PLL   my_pll(.CLKOUT(clk));
    DFF   reg1(.CLK(clk), .Q(divclk));
    block b1(.clkin(divclk));
    endmodule

    module block(clkin);
    input clkin;
    BUF clkbuf(.A(clkin), .Y(clkin2));
    DFF reg2(.CLK(clkin2));
    endmodule
```

An error will result while budgeting block "b1" if you define your top level SDC as:

```
create_clock -name clock my_pll/CLKOUT
    create_generated_clock -name gclock \\
            -source my_pll/CLKOUT b1/clkbuf/Y
```

This is because clock "clock" propagates only to reg1 and not the block boundary.

Feedback

You can fix the problem in 3 ways:

fix 1

Declare your generated clock at the block boundary:

```
create_clock -name clock my_pll/CLKOUT
    create_generated_clock -name gclock \\
            -source my_pll/CLKOUT b1/clkin
```

fix 2

Declare your generated clock before the block boundary:

```
create_clock -name clock my_pll/CLKOUT
    create_generated_clock -name gclock \\
            -source my_pll/CLKOUT reg1/Q
```

fix 3

Declare an intermediate generated clock at or before of the block boundary, and then make the intermediate clock act as the master for your generated clock.

```
create_clock -name clock my_pll/CLKOUT
    create_generated_clock -name intermediate \\
            -source my_pll/CLKOUT reg1/Q
    create_generated_clock -name gclock \\
            -source reg1/Q b1/clkbuf/Y
```

---

## ABS-208

(warning) Clock '%s' crosses boundary of block '%s' in both inverted and non-inverted sense.

**Description**

A clock at the boundary of a block can only be characterized in one of inverted and non-inverted sense at one time. In this case, two separate clock definitions are needed. This can sometimes be a problem. For example, a generated clock may depend on the clock in this warning message. Since a generated clock may only have one "master", the generated clock will only depend on the positive sense of the clock, not the negative.

**What Next**

To avoid this warning, allow only one sense of the clock to cross the block boundary.

---

## ABS-209

(warning) Generated clock '%s' has sources both inside and outside block '%s'.

Feedback

**Description**

In order to properly time a generated clock, the "master" clock must be visible in the current context. This means that generated clock sources which are outside of the block, but propagate to the block boundary must be converted into regular (not generated) clocks. They will no longer be part of the generated clock with sources inside of your block. Also, if your generated clock propagates out of the block and then back in, the re-entering part of the clock will be converted to a regular clock.

**What Next**

To avoid this warning, declare generated clock sources either all inside or all outside of your block. Additionally, you should avoid having generated clocks that propagate outside of the block and then re-enter.

---

## ABS-210

(warning) Cannot propagate master of generated clock '%s' to input '%s' of block '%s'.

**Description**

In your design, the given pin hierarchical pin lies between the given generated clock source and the generated clock's master clock "source pin". However, it is not possible to trace directly from the master clock to the hierarchical pin. This may have occurred for a few different reasons:

+ There is a register between the master clock and the hierarchical pin.

+ Clock propagation from master clock to generated clock is tracing through registers that you don't intend. This may or may not be a problem.

**What Next**

This warning is not a problem unless all of the paths from a master clock to the block boundary of a generated clock are untracable. In this case, you will also receive an ABS-207 error. Please refer to the manual page of ABS-207 for details.

---

## ABS-211

(warning) Propagating clock '%s' to boundary of block '%s'.

**Description**

In your top level design, the given clock does not usually propagate to the boundary of the given block. This is probably because some generated clock's source sits between the given clock and the block boundary.

Feedback

For the purpose of budgeting, the given clock will appear on the boundary of the given block. This is necessary because the given clock is the master of a generated clock within your block.

**What Next**

If you don't want the given clock to appear at the boundary of your block, you must change the master of the generated clock that is inside of your block. The generated clock inside your block should only depend on clocks which propagate to your block boundary.

---

## ABS-212

(warning) Pin '%s' is missing budget specifications.

**Description**

When creating budgets, you must use the set_pin_budget_constraints command to specify how each interface pin of a block is to be budgeted. The given pin has no specification or has a subset of paths that have no specification.

**What Next**

If you want the pin to receive a budget, specify a budget with set_pin_budget_constraints. If you want the pin to receive no budget, you can use "set_pin_budget_constraints -none" to explicitly select no budget. This warning message will then be supressed.

You can use "report_budget -pin" to see more details about how your current budget specification has been applied to paths through this pin.

---

## ABS-213

(warning) %d pin(s) are missing budget specifications.

**Description**

When creating budgets, you must use the set_pin_budget_constraints command to specify how each interface pin of a block is to be budgeted. The given number of pins have no specification.

**What Next**

If you want the pin to receive a budget, specify a budget with set_pin_budget_constraints. If you want the pin to receive no budget, you can use "set_pin_budget_constraints -none" to explicitly select no budget. This warning message will then be suppressed.

Use "report_budget -warning_pins" or use the -verbose flag of the write_budgets command to see a list of all of the pins with no budget specification.

Feedback

You can use "report_budget -pin" to see more details about how your current budget specification has been applied to paths through a pin.

## ABS-214

(warning) Pin '%s' has no timing paths. No budget created for this pin.

### Description

You have asked for a budget on the given pin, but there are no valid timing paths through the pin. This may be caused by missing clock or datapath connections. It might also be caused by set_false_path specifications. It is expected that non-datapath pins (such as clocks) will have no timing paths.

### What Next

You should use timing constraints to create a path through the given pin. Alternatively, you can use the black-box-timing set of commands to add a "virtual" timing path to your block. For example:

```
set_blackbox_clock_port CLK
    create_blackbox_delay -rise_from CLK -to dataOut -value 1.0
```

Once the timing path has been added, you can create a budget for it.

## ABS-215

(warning) %d pin(s) have no timing paths. No budget created for these pins.

### Description

You have asked for a budget on the given number of pins, but there are no valid timing paths through the pins. This may be caused by missing clock or datapath connections. It might also be caused by set_false_path specifications. It is expected that non-datapath pins (such as clocks) will have no timing paths.

### What Next

Use "report_budget -warning" or use the -verbose option of write_budgets to see a list of pins with no timing path.

You should use timing constraints to create a path through the given pin. Alternatively, you can use the black-box-timing set of commands to add a "virtual" timing path to your block. For example:

```
set_blackbox_clock_port CLK
    create_blackbox_delay -rise_from CLK -to data_out -value 1.0
```

Feedback

Once the timing path has been added, you can create a budget for it. A budget cannot be created for pins with no timing path.

---

## ABS-216

(information) Forcing clock %s (%c) for budgeting at pin %s

**Description**

You have used the -force option of the set_pin_budget command to declare a clock which does not yet exist in your real circuit and/or constraints. The time budget will be created as if the clock actually exists.

**What Next**

No action is necessary.

---

## ABS-217

(information) %d pin(s) in the budget have forced clocks.

**Description**

You have used the -force option of the set_pin_budget command to declare a clock which does not yet exist in your real circuit and/or constraints. The time budget will be created as if the clock actually exists.

**What Next**

For more details, you can recreate your budget using the -verbose option.

---

## ABS-218

(error) %s is not a %spin of a budgeted block.

**Description**

You can only query pins of a block that has been added to the budget with "set_budget_options -add_block".

**What Next**

Add the block with set_budget_options. Or perhaps you have tried to access an input/output pin with specified direction output/input.

Feedback

## ABS-219

(error) The same pin (%s) cannot be used as both -from pin and -to pin in the set_segment_budget_constraints command.

**Description**

This message indicates that in the set_segment_budget_constraints command the same pin was used as both a -from and a -to pin

It is required that the -from and -to pins are two different pins that are adjacent along a timing path.

**What Next**

Correct the pins used in the set_segment_budget_constraints command to indicate a correct budget segment.

## ABS-220

(warning) There are %d pin constraints that appear impossible to meet.

**Description**

Some of the timing paths in your design have constraints that appear too tight to meet. There are several possible causes, which should be fixed if possible. The causes include: Paths with negative slack and only non-optimizable (fixed delay) logic. Very bad clock skews Missing or inconsistent clock_latency declarations Errors in the SDC specification Missing clock_group, false_path or multicycle_path declarations Extremely large fanout nets which have not been declared ideal Excessively large set_input_delay or set_output_delay in the top-level SDC Paths leading to macros with very large setup time, where the macro's clock has not been adjusted. Failure to use estimated timing on an unoptimized netlist.

**What Next**

The automatic budgeter has isolated the paths that appear impossible and budgeted them separately. But the budgets are probably not achievable. You should fix the timing problem on these paths and run the automatic budgeter again to get a high-quality budget.

You can use "report_budget -warning_pins" to see a list of pins which seem to have bad constraints. You can use "report_budget -pins" to see an overview of the bad budget timing. You can use "report_timing -through" to see a detailed timing path.

## ABS-221

(warning) No Timing information found for lib_cell %s.

Feedback

**Description**

The specified cell does not have timing information. This means that timing paths will not propagate through this cell. And constraints that apply to such paths may be lost.

**What Next**

It is OK to get this warning for physical-only cells like antenna diodes. If you see this warning issued for standard logic cells or timed macros, you should check to see if your library has been properly built.

---

## ABS-222

(Information) Retaining interface logic of lower level sub-blocks.

**Description**

This message is generated during abstract creation. It indicates that the interface to lower-level blocks will be retained in the abstract to facilitate MPH budgeting.

**What Next**

This is an informational message. No action needed.

---

## ABS-223

(warning) Clock '%s' has unknown period

**Description**

The waveform of the given clock could not be determined during timing analysis, so the clock has been given an arbitrary period in the output constraints. The clock does not propagate to the current block so the arbitrary period will not adversely affect your timing.

**What Next**

Check the warnings that occur during regular timing analysis. It is likely that you have a generated clock with no master clock driving it. If you fix or remove the generated clock, this error message should go away.

---

## ABS-224

(error) Clock '%s' has unknown period. The resulting constraints may be incorrect.

**Description**

The waveform of the given clock could not be determined during timing analysis. So the clock has been turned into a virtual clock (with no sources) and has been given an

Feedback

arbitrary period in the output constraints. This clock is defined in or propagates to the edge of the current block. In your original design, the paths relating to this clock cannot be timed. In your budgeting design, the timing paths will not even be defined.

**What Next**

Check the warnings that occur during regular timing analysis. It is likely that you have a generated clock with no master clock driving it. If you fix or remove the generated clock, this error message should go away.

---

## ABS-225

(warning) The created abstract has no active %s corners for mode %s. Timing will be omitted.

**Description**

When you create an abstract, timing information is recorded for each corner that is active when the abstract is created. If you have active setup corners, long timing paths will be recorded. If you have active hold corners, short timing paths will be recorded. If you have no active corners, then the abstract will contain no timing paths. The abstract will not be useful for timing analysis.

**What Next**

Use the set_scenario_status command to make sure that you have active setup and hold corners set before calling create_abstract. Otherwise, the abstract will not be useful for the corner of interest.

---

## ABS-226

(error) You must have at least one active corner for abstraction.

**Description**

When an abstract is created, the current timing of one of more modes is needed to so that cells may be selected to be in the abstract. Currently, none of the declared operating modes has timing activated in any corner.

**What Next**

Use set_scenario_status to turn on setup and/or hold timing in at least one mode. Or you can use the -physical_only flag of create abstract is you really don't care about timing.

Feedback

## ABS-227

(information) Information for mode %s is not available in abstract view of design %s for block %s.

**Description**

Data for the specified mode doesn't exist on the specified abstract view or cannot be opened to read.

**What Next**

If the top mode should use data from a different block mode, please use the set_block_to_top_map command to set it up. If the block should have this mode, please make sure the abstract view was created successfully and check the read permission of the design.

## ABS-228

(warning) Port '%s' of design '%s' is driven by both zero and one.

**Description**

When creating budgets, top-level constant values usually cause a set_case_analysis statement to be set for input ports of the budgeted block.

But in this case, the block design has been multiply instantiated. And some of the connections to the given port have a constant zero and others have a constant one. As a result, no set_case_analysis has been applied for this port.

**What Next**

If you want a set_case_analysis statement to appear in the budget, you must make sure that all instances of the design are driven by the same constant.

## ABS-229

(warning) %d port(s) of design '%s' are driven by both zero and one.

**Description**

When creating budgets, top-level constant values usually cause a set_case_analysis statement to be set for input ports of the budgeted block.

But in this case, the block design has been multiply instantiated. And some of the connections to the ports have constant zero and others have constant one. As a result, no set_case_analysis has been applied for the ports.

Feedback

**What Next**

If you want a set_case_analysis statement to appear in the budget, you must make sure that all instances of the design are driven by the same constant.

Use "report_budget -warning" or use the -verbose flag of the write_budgets command to see a list of all of the ports driven by both one and zero.

---

## ABS-230

(warning) Port '%s' of design '%s' is driven by both constants and non-constants.

**Description**

When creating budgets, top-level constant values usually cause a set_case_analysis statement to be set for input ports of the budgeted block.

But in this case, the block design has been multiply instantiated. And some of the connections to the given port have a constant and others don't. As a result, no set_case_analysis has been applied for this port.

**What Next**

If you want a set_case_analysis statement to appear in the budget, you must make sure that all instances of the design are driven by the same constant.

---

## ABS-231

(warning) %d port(s) of design '%s' are driven by both constants and non-constants.

**Description**

When creating budgets, top-level constant values usually cause a set_case_analysis statement to be set for input ports of the budgeted block.

But in this case, the block design has been multiply instantiated. And some of the connections to the ports have constant and others don't. As a result, no set_case_analysis has been applied for the ports.

**What Next**

If you want a set_case_analysis statement to appear in the budget, you must make sure that all instances of the design are driven by the same constant.

Use "report_budget -warning" or use the -verbose flag of the write_budgets command to see a list of all of the ports driven by both constants and non-constants.

Feedback

## ABS-232

(Error) Budget block "%s" has no linked design.

**Description**

The given block currently is not linked to any design. Since the block cannot be timed, budgets cannot be created.

**What Next**

Please make sure that the block is correctly linked, and then try creating the budget again.

## ABS-233

(Error) Cannot mix pin budget constraints with budget pin controls.

**Description**

It is not supported to mix budget pin constraints like -from_percent or -to_delay with budget pin controls like -same_as_pin or -frozen.

**What Next**

Please use two separate set_pin_budget_constraints commands, one with a budget constraint and one with budget controls.

## ABS-234

(warning) Design %s.%s is not abstract view.

**Description**

The specified design is not of abstract view. The command only takes abstract view design.

**What Next**

Please check the view type of the specified design.

## ABS-235

(Warning) Corner '%s' is not currently defined.

**Description**

You have created a constraint for a corner that is not currently defined. This is OK. You can choose to define the corner later. Otherwise the constraint will be ignored.

Feedback

**What Next**

If you has typed the wrong name for your intended corner, you can simply ignore the constraint from this command and retype the command with the correct corner name.

---

## ABS-236

(Error) Budget boundary '%s' is not defined.

**Description**

In order to apply a budget boundary constraint, you must first define the constraint using the set_budget_boundary command.

**What Next**

Use the set_budget_boundary command to create the constraint.

---

## ABS-237

(Warning) Top-level mode '%s' has no setup corners for budgeting.

**Description**

In order to generate a budget associated with a top-level mode, it must be possible to trace delays through timing paths. Because your mode has no corners with setup enabled, no set_input_delay or set_output_delay can be generated for setup timing. Only virtual clocks and exceptions will be generated.

**What Next**

If you want budgets for setup, please enable setup timing for at least one corner in the mode.

---

## ABS-238

(Warning) No budget information is available for %s '%s'.

**Description**

The budget information you are reading was created without information for the given mode or corner. No budget will be applied for this mode or corner. Any old budget information which may have been applied will be retained.

**What Next**

Consider regenerating the budget information. Make sure that the desired mode or corner is defined at the top level.

Feedback

## ABS-239

(Warning) Budget information could not be applied for %s '%s'.

**Description**

Budget information is available for the given mode or corner, but your current design does not have this mode or corner defined.

**What Next**

In order to apply a budget to a block, you must first have constraints for the block defined. The budget data is merely an overlay on top of the defined constraints. It defines new constraints at the block boundary.

## ABS-240

(Information) Budget for this design will be based on estimated_corner delays.

**Description**

By default, automatic budgeting creates constraints based on the worst delays across all active timing corners. But currently, the automatic budgeter for this design is in "estimate_timing mode" and will base the budget only on delays in the estimated_corner. This is desirable when some or all of your design is unoptimized.

**What Next**

The default value for the app option "plan.budget.estimate_timing_mode" is "auto". When "auto" is selected, estimate_timing budgeting will be used whenever a estimated_corner is defined.

You can force regular budgeting by either removing the estimated_corner or by setting.

set_app_option -name plan.budget.estimate_timing_mode -value false

## ABS-241

(Information) Restoring the reg2reg timing paths across sub-blocks previously ignored by command set_timing_paths_disabled_blocks.

**Description**

Budgeting restores the reg2reg timing paths across sub-blocks by running remove_timing_paths_disabled_blocks internally.

Feedback

### What Next

No action is required. The block internal timing paths can be disabled again after budgeting by running the command

set_timing_paths_disabled_blocks -all_sub_blocks

---

## ABS-242

(Warning) The estimated_corner is not initialized for mode '%s'. Estimated timing budget not generated for this mode.

### Description

You have asked for "estimate_timing mode" budgets in this design, but the "estimated_corner" for this timing mode does not exist or is not active. No set_input_delay or set_output_delay can be generated and old budgets will be kept. Only virtual clocks and exceptions will be updated.

Note that the estimated_corner is only created for setup analysis and not hold, and estimate_timing-based analysis only works for setup_delay budgeting. So if the given mode has no setup scenarios, then it does not make sense to use estimate_timing for budgeting on the mode.

### What Next

You can create the estimated_corner by using the "estimate_timing" command. Or you can choose not to use the estimated corner by using the "-no_estimate_timing" option of this command. You can also turn off estimate_timing mode budgeting for this design by using one of the following:

set_app_option -name plan.budget.estimate_timing_mode -value false

set_app_option -name plan.budget.estimate_timing_mode -value auto

---

## ABS-244

(warning) Cannot propagate master of generated clock '%s' to input '%s' of block '%s'. Consider using "create_generated_clock -combinational".

### Description

In your design, the given pin hierarchical pin lies between the given generated clock source and the generated clock's master clock "source pin". However, it is not possible to trace directly from the master clock to the hierarchical pin. This probably occurred because you have not used the -combinational option of create_generated_clock.

Feedback

When you do not use the -combinational flag, clock propagation from master clock to generated clock may be tracing through registers that you don't intend. The tracing through registers triggers this warning. You might want to consider using the -combinational option of create_generated_clock when you use "-divide_by 1".

**What Next**

This warning is not a problem unless all of the paths from a master clock to the block boundary of a generated clock are untraceable. In this case, you will also receive an ABS-207 error. Please refer to the manual page of ABS-207 for details.

You should consider using the -combinational option with create_generated_clock. That option will make this warning go away.

---

## ABS-245

(error) Cannot overwrite the existing abstract view. Changes pending merge from abstract view to design view.

**Description**

You get this message when running create_abstract command. The error message is issued because there are changes in the abstract view of the design that has not been merged back to the design view yet. The tool cannot create a new abstract view to overwrite the old abstract view from an out-of-date design view.

**What Next**

You should run merge_abstract first.

If you are sure that you want to abandon the data in the current abstract view and create a new abstract view from the out-of-date design view, you can consider using remove_blocks to delete the existing abstract view from both in-memory and on-disk. Then, you can proceed with create_abstract.

---

## ABS-246

(error) cell %s has no reference module.

**Description**

You get this message when running create_abstract command. The error message is issued because, in order to have a leaf cell in the abstract view, it has to have reference module.

Feedback

**What Next**

This error happens when the specified cell has no reference module. This could happen if the reference library is not set up properly.

Please check reference library settings and make sure the reference library for the cell is set properly, then run create_abstract again.

If you are running RTL Floorplanning flow, please run *estimate_design* command to generate physical views for unmapped logic.

---

## ABS-247

(Information) %s abstract for block %s:%s has been saved to disk.

**Description**

The given abstract is now written to disk and can be linked by other blocks. You will not be allowed to save read-only abstracts later.

**What Next**

You might see future information messages that the read-only abstract cannot be saved. This is OK, because it was saved in this step.

---

## ABS-248

(Warning) Design %s.%s is of version %u.0 which is not compatible with current version.

**Description**

The specified abstract view design contains annotation data that is of an older version that is no longer supported. This abstract view can no longer correctly represent the corresponding design view with the executable your are using.

**What Next**

If there is any modification on the abstract view that you need to merge back to the design view, please do so by calling merge_abstract. Then, use the current executable and run create_abstract to create a new version of the abstract view.

If you want to discard the existing abstract view, please use remove_abstract command to remove it. Then, use the current executable and run create_abstract to create a new version of the abstract view.

---

## ABS-249

(Warning) The design '%s' has no abstract view.

Feedback

**Description**

There is no abstract view, so merge_abstract cannot be performed.

**What Next**

Abstracts are create by the create_abstract command. If you ran create_abstract, perhaps you also called remove_abstract by accident.

It is also possible that this design was created by using copy_block or "save_block -as". Neither of these commands will copy original design's abstract.

---

## ABS-250

(warning) Missing timing constraints. Abstract may be incorrect.

**Description**

When an abstract is created, the current timing of one of more modes is needed to so that cells may be selected to be in the abstract. None of your modes have declared clocks, and it is possible that the improper timing paths are being represented in the abstract. A top-level timing report that uses this abstract may not show the worst timing path.

The requirements for your block constraints are different, depending on whether you are using "compact" abstracts or "full" abstract.

Full abstract are bigger, but they require less complete constraints at the block level. Before making a full abstract, the block should have the following defined:

- All clocks and all generated clocks

Compact abstracts are significantly smaller than full abstracts. This will give you a significant memory and CPU advantage when processing a top-level design. Before making a compact abstract, the block should have the following defined:

- All clocks and all generated clocks
- All set_case_analysis and set_disable_timing
- All exception constraints (set_false_path, set_multicycle_path) that affect timing paths that enter or leave the block.

**What Next**

If your block really has no clocks in it, then everything is OK. Otherwise, you should make sure to load your block SDC constraints before creating an abstract.

Feedback

## ABS-251

(warning) This abstract cannot be used for timing analysis.

### Description

The created abstract is not intended to be used for timing operations such as report_timing or optimization. If you instantiate this abstract at the top level, your timing information may be misleading or incorrect. In particular, the delay values might be incorrect and the worst timing path might not be represented.

### What Next

Create a new abstract with timing information.

## ABS-252

(information) Input port '%s' has high fanout. Some fanouts filtered from the abstract.

### Description

"Full" abstracts contain nearly all of the boundary (donut) logic that can be found in the original design. But some of the boundary logic is filtered out in order to keep the abstract size reasonable. In particular, logic connected only to high-fanout networks is filtered. Usually, the filtered logic includes clock networks, scan enable, and reset. For these signals, only the essential part of the logic is kept, instead of all of the logic.

### What Next

You can use the "abstract.high_fanout_limit" app option to adjust the filtering behavior.

## ABS-253

(warning) Top-level clock '%s' cannot be mapped to a block-level clock. Assuming the block and top use the same clock name.

### Description

It is possible that clocks defined in your blocks have names that do not match connected top-level clock names. For example, a top level clock named "SYS_CLK" may be connected to a block-level clock named "CLK". The *write_budgets* command understands that names may not match, and will automatically derive the mapping between top-level and block-level clocks. This allows *write_budgets* to generate the proper constraints for the block.

This warning informs you that the given top-level clock connects to the block, but write_budgets cannot figure out which block-level clock is connected. write_budgets

Feedback

assume that the block-level clock name matches the top-level clock name. If this is a bad assumption, then the constraints generated by write_budgets will be incorrect.

**What Next**

Make sure that your block has constraints specified, and then use set_block_to_top_map to specify the clock association.

---

## ABS-254

(warning) %d clocks top-level clocks(s) cannot be mapped to block-level clocks. Assuming the block and top use the same clock name.

**Description**

It is possible that clocks defined in your blocks have names that do not match connected top-level clock names. For example, a top level clock named "SYS_CLK" may be connected to a block-level clock named "CLK". The *write_budgets* command understands that names may not match, and will automatically derive the mapping between top-level and block-level clocks. This allows *write_budgets* to generate the proper constraints for the block.

This warning informs you that some of the clocks in this block could not be mapped to top level clocks. write_budgets assume that the block-level clock name matches the top-level clock name. If this is a bad assumption, then the constraints generated by write_budgets will be incorrect.

**What Next**

Make sure that your block has constraints specified, and then use set_block_to_top_map to specify the clock association.

Use the -verbose option of write_budgets to see a list of clocks with no name mapping.

---

## ABS-255

(warning) No clocks defined for mode '%s'. Block constraints should be loaded prior to applying budget.

**Description**

By default, the write_budgets command only writes constraints that apply to a subblock boundary. The output of write_budgets is not a complete set of constraints for a block unless you use the -full_budget_blocks option. Rather, the output is intended to be

Feedback

"layered" on top of your existing block constraints. So, in order to use the budget file from write_budgets, you need to perform the following steps:

- Load your block
- Apply your regular block constraints
- At the top level, obtain budget constraints automatically by using compute_budget_constraints or manually using budget option commands.
- Write budget SDC using write_budgets
- Source the output from write_budgets into your block

**What Next**

If your block really should have no clocks or registers in it, then you are OK. Otherwise, you need to follow the procedure above or use the -full_budget_blocks option of write_budgets.

If you do not have regular block constraints, consider using the split_constraints command to derive block-level constraints based on chip-level constraints.

---

## ABS-256

(error) This command must be run on a 'design' view. Your current block is a '%s' view.

**Description**

The command was not designed to run on the current view.

**What Next**

Open the 'design' view of your block. If you only have an 'outline' view, you need to run the expand_outline command.

---

## ABS-257

(error) No block specified.

**Description**

You are seeing this message because you ran *create_abstract* or *merge_abstract* with option *-host_options* but did not specify the blocks you want to create or merge abstract view for.

Feedback

**What Next**

Please use option *-blocks* to specify the blocks you want the tool to create or merge abstract view for. Or, use option *-all_blocks* if you want all child blocks' abstract view to be created or merged.

---

## ABS-258

(error) Cell %s does not belong to root design %s.

**Description**

You are seeing this message because you ran *create_abstract* or *merge_abstract* with option *-blocks* but the block cells do not belong to the same root design.

When the specified cells do not belong to the same root design, it is not possible for the tool to figure out block dependencies. In order to run the blocks in the correct order, the tool has to know their dependencies of each other and maybe dependencies to other designs.

**What Next**

Please specify block cells that belong to the same root design.

---

## ABS-259

(error) Wire parasitics are too low. Can't buffer properly.

**Description**

The tool has detected a problem with the parasitic settings in your design. With the current values, it appears as if wires will have no delay and should never be buffered. The current parasitic settings, TLU+ or tech file information is probably wrong.

**What Next**

Please check the timing and parasitic settings of your design.

---

## ABS-260

(error) Cannot find reference design for cell %s.

**Description**

This message indicates that the design is not fully linked. The specified physical block cell has no reference design.

Feedback

**What Next**

Please check your reference library settings to make sure that the design is linked successfully.

---

## ABS-261

(error) Cannot open file %s to %s.

**Description**

This message indicates that the tool cannot open the mentioned file for the specified operation.

**What Next**

Please check file and directory permissions.

---

## ABS-262

(error) Syntax error in line %u of file %s.

**Description**

This message indicates that there is a syntax error in the block constraint mapping file.

The correct syntax for each line of the block constraint mapping file is: <block_design_name> <constraint_type> <file_name> where, *constraint_type* is one of UPF, SDC, DEF, BUDGET, CLKNET, PG_CONSTRAINT, COMPILE_PG, ETM_UPF, CTS_CONSTRAINT, BTM, FLOORPLAN.

**What Next**

Please check the content of the block constraint mapping file and fix any syntax error.

---

## ABS-263

(error) Cannot find block with design name %s.

**Description**

This message indicates that there is no physical block in current design with the specified reference design name.

**What Next**

Please check the content of the block constraint mapping file and the reference design name of the physical block.

Feedback

## ABS-264

(warning) Design %s for block %s has no design view.

**Description**

This message indicates that there is no design view for the specified design.

You are calling an operation that requires the design has design view. Since the block has no design view, the block is skipped.

**What Next**

Please check if there should be a design view for the specified design.

## ABS-265

(error) Timing is not available for this budget.

**Description**

It is not possible to budget or report on timing in the current mode. This is probably caused by missing process corners for the mode.

Possibly you have asked for a estimated_corner budget when the estimate_timing command has not been run. If you see a ABS-240 message, please read the man page for that message.

**What Next**

Make sure that the current mode has a corner defined. Make sure you have an estimated_corner when your budget is looking at estimate_timing delays.

## ABS-266

(warning) Abstract design %s.%s cannot be used for timing analysis.

**Description**

The specified abstract is not intended to be used for timing operations such as report_timing or optimization. By instantiating this abstract at the top level, your timing information may be misleading or incorrect. In particular, the delay values might be incorrect and the worst timing path might not be represented.

**What Next**

If you want to look at timing related to the instantiation of this abstract, create a new abstract with timing information.

Feedback

## ABS-267

(error) No abstract design to remove.

**Description**

Either the specified abstract design is not present in the library or there are no abstract designs for remove_abstract to work on.

**What Next**

Please check the name of the abstract design to be removed and if there are any abstract designs to be removed.

## ABS-268

(Warning) Top-level mode '%s' has no hold corners for budgeting.

**Description**

In order to generate a budget associated with a top-level mode, it must be possible to trace delays through timing paths. Because your mode has no corners with hold enabled, no set_input_delay or set_output_delay can be generated for hold timing. Only virtual clocks and exceptions will be generated.

**What Next**

If you want budgets for hold, please enable hold timing for at least one corner in the mode.

## ABS-269

(Warning) The current mode has no defined clocks. Skipping timing evaluation for path analysis.

**Description**

Flyline analysis for paths can only use timing information if your design has clocks defined. Analysis will continue for path counts, but not for timing.

**What Next**

Provide timing constraints (including clocks) prior to running path analysis.

## ABS-270

(Information) Flyline analysis for paths %s include timing information.

Feedback

### Description

This message tells you whether or not timing analysis will included when performing flyline analysis for paths. If you have received a ABS-269 message, please refer to that message for details. Otherwise, the behavior is controlled by the plan.path_analysis.enable_timing app option. Set the option to true to analyze timing, or false to save runtime during analysis.

### What Next

Set the app option appropriately if it isn't already, and then click the "reload" button in the GUI. To set the app option use:

set_app_option -name plan.path_analysis.enable_timing -value false

or

set_app_option -name plan.path_analysis.enable_timing -value true

---

## ABS-271

(Warning) There is no constraint mapping file for the design.

### Description

This message tells you that the constraint mapping file is missing for the current design.

The constraint mapping file tells the tool where the constraints (for example, DEF, UPF, SDC, BUDGET, etc.) are for the current design as well as each sub-block reference designs. The tool picks up the information and loads the necessary constraints to each block and top level at different stages during the design planning flow when it creates the sub-blocks' abstract view.

If you started the design with a full design SDC and UPF file and ran *split_constraint*, the command generates a mapfile with the UPF and SDC information for the top level and each sub-block.

*write_budget* writes out a mapfile with BUDGET constraints for each sub-block.

User can write the mapfile manually if they have the constraint files for the top and blocks.

To learn more about the constraint mapping file, please refer to the man page for command *set_constraint_mapping_file*.

### What Next

If you don't have the constraints yet and want to continue without the constraints, you can ignore this warning.

Feedback

If you have the constraints but forgot to set the mapping file, please set the mapping file and re-create the abstract views.

---

## ABS-272

(Warning) Block signal pins (e.g. %s) tied directly to power/ground net '%s' will not be budgeted.

### Description

The given net has been identified as power or ground and also crosses a budgeted block boundary. The signal cannot be timed, so block crossings associated with this net will not get a budget. The proper set_case_analysis *will* still be in the budget.

### What Next

If you want this signal to be considered despite its constant value, please isolate it from the rails by using a buffer or a tie cell.

---

## ABS-273

(Error) The current scenario must have setup analysis activated.

### Description

This command analyzes the setup times in the current scenario. So you must activate setup timing in the current scenario to continue.

### What Next

Set your current scenario to an appropriate choice using "current_scenario", or activate setup timing with "set_scenario_status".

---

## ABS-274

(Error) One or more block(s) failed. Please look for more detailed error message(s) earlier in the log.

### Description

The command tried to do some operations on the child blocks and some block(s) failed. There should be script and log file for each block. The location of these files should be stated in the current log file before this message, "*all scripts and log files are under ...*". There should also be *CDPL-XXX* or *ABS-296* error messages before this message to indicate which block(s) failed.

Feedback

**What Next**

Please check the log for the failed block(s) to find more information about the reason behind the failed job. If there are no log files for the blocks, then this most likely means the distributed processing mechanism failed. Check the cdpl_log/*.err file. This may give more insight into the distributed processing failure. You can then use the check_host_options command to see if your host option setting is properly set up.

Check the time stamp when looking at the log files. Because log files are not removed by default, you may accidentally look at an old log file, not the one from the current run.

If the log file for a failed block seems to end before finishing the block's script but without any error message, it is most likely that the job crashed. You should be able to verify the crash by running the tcl script for the block. The tcl script for the block is located under the same directory where the block's log file is.

---

## ABS-275

(Error) Instance %s (module %s) must not be an abstract.

**Description**

When creating budgets for multiple levels of hierarchy, only the leaf levels of hierarchy may be represented as abstracts. The top level hierarchy and intermediate level hierarchies must be full "design" views, that contain the full netlist.

**What Next**

Change the given hierarchy to a full design view, or you can remove nested hierarchies from the list of budgeted modules.

---

## ABS-276

(Error) Instance %s (module %s) may not be an abstract when plan.budget.all_design_subblocks is set to "true".

**Description**

When the app option plan.budget.all_design_subblocks is set to "true" you may not use abstract blocks in your hierarchy. Please refer to the man page of plan.budget.all_design_subblocks for further details.

**What Next**

You can change to full designs using change_abstract, or you can reset plan.budget.all_design_subblocks option using the set_app_options command.

Feedback

## ABS-277

(Error) Your library must have technology information in order to use budget shells.

**Description**

In order to make a budget shell, the technology information (the tech file) must be known.

**What Next**

Include a library with technology information in the reference libraries.

## ABS-278

(Error) Instance %s (module %s) cannot be made into a budget shell.

**Description**

In order to make a budget shell for an instance, the instance must be associated with an independently linkable design hierarchy. The instance may not be a leaf cell, a simple logic hierarchy, or unresolved.

**What Next**

Make sure your instance links properly to an independent block.

## ABS-279

(Information) Abstract view of design %s is not changed since its creation. No merge needed.

**Description**

Upon loading a design view or when *merge_abstract* is called, the tool checks whether the abstract view (if one exists) has been changed since it was created. You are getting this message because the abstract view of the specified design has not been changed.

**What Next**

This is an informational message. No action needed.

## ABS-280

(Information) Abstract view of design %s is read-only. No merge needed.

Feedback

### Description

Upon loading a design view or when *merge_abstract* is called, the tool checks whether there is anything in the abstract view that need to be merged to the design view. You are getting this message because the abstract view of the specified design is read-only.

### What Next

This is an informational message. No action needed.

---

## ABS-281

(Information) Abstract view of design %s is not changed since last merge. No merge needed.

### Description

When switching the current design or when *merge_abstract* is called, the tool checks whether there is anything in the abstract view that need to be merged to the design view. You are getting this message because the abstract view of the specified design has not been changed since the last merge.

### What Next

This is an informational message. No action needed.

---

## ABS-282

(Information) Abstract view of design %s is changed since last merge.

### Description

When switching the current design or when *merge_abstract* is called, the tool checks whether there is anything in the abstract view that need to be merged to the design view. You are getting this message because the abstract view of the specified design has changed since the last merge which indicates that another merge is needed.

### What Next

This is an informational message. No action needed.

---

## ABS-283

(Error) Design view of design %s is changed since last merge. Cannot merge new abstract changes to the design view.

Feedback

### Description

When switching the current design or when *merge_abstract* is called, the tool checks whether there is anything in the abstract view that need to be merged to the design view.

You are getting this message because both the design view and the abstract view of the specified design has changed which makes it impossible for the tool to perform view merge.

### What Next

You need to decide whether to keep the content of the design view and discard the abstract view or keep the content of the abstract view discard the changes in the design view.

If you choose to keep the content of the design view, please call remove_abstract then continue with your flow.

If you choose to keep the content of the abstract view, please close and purge the design view and open it again. To close and purge the design view, you can call the command *close_block -purge -force*.

---

## ABS-284

(Error) No blocks have been specified for splitting/budgeting. Nothing done.

### Description

Before you run split_constraints or write_budgets, you must declare which instances you want to operate on. This is done with the -add_block option of the set_budget_options command. No instances have been declared, or you are trying to operate on instances that have not been declared.

### What Next

Make sure you declare all budgeted blocks with the -add_block option of the set_budget_options command.

---

## ABS-285

(Error) None of the shell_subblocks listed are budget instances.

### Description

When you use the shell_subblocks option, you must specify the module name of one of the instances that you are budgeting. The given names do not match nay of the instance modules.

Feedback

### What Next

Please give the module name of one of the budgeted instances, or do not use the shell_subblocks option.

---

## ABS-286

(error) Cannot find -source pin for generated clock '%s' in block '%s'. Master clock does not propagate to the generated clock. Output will be incorrect.

### Description

When budgeting a generated clock within a lower level block, you must conform to the following two rules:

rule 1

Generated clocks should be "expanded" in the full design to resolve their master clock. If the generated clock is not expanded before budgeting, it is not be possible to expand it in the budgeted block.

rule 2

If a master clock source is outside of the block and the generated clock source is inside, the master must propagate to the block boundary.

If you received this message, there is no path that leads from the master clock to the generated clock source. This means that it is impossible to determine the correct point for the master clock to enter your block. It also means that it is not possible to generate a legal definition for the generated clock. The output SDC file will not be usable until the problem is corrected.

### What Next

This problem is very serious and it should be fixed. Certainly, you should not start post-CTS optimization with the bad generated clock. However, if you want to continue with prects optimization you can set:

```
set_app_options -name plan.budget.allow_disconnected_gclocks -value true
```

This will generate a budget that will let you continue. Please refer to the app option man page for details.

You need to make sure that there is a legal path between the master clock and the generated clock. You should also make sure that the clock path is not stopped by set_disable_timing or set_clock_sense. Also pay attention to situations where clock gating, clock muxing, or case analysis may be switching off the path that crosses between the master clock and the generated clock.

Feedback

Note that when the master clock is cut off by clock gating or clock muxing, the latency to your generated clock will not be propagated. So you could consider removing the generated clock completely, since the timing is not correct in any case.

If you don't care about the path from the master clock to the generated clock (because you don't care about tracing the clock latency between the master and generate clocks), then consider changing your generated clock in a regular clock.

---

## ABS-287

(information) Using latencies from corner '%s' as the default for other corners.

**Description**

When setting budget latency targets, each of the currently defined corners is analyzed and assigned appropriate values. For any new corner and for the estimated_corner, a default value will be set that is copied from the given corner.

**What Next**

No action is necessary.

---

## ABS-288

(warning) Html output may be too big for your browser. Consider using -html_dir.

**Description**

The -html option of the report_budget command writes a single large file. The files produced by -html can be very large and most browsers do no perform well on large files.

**What Next**

If you encounter problems due to a large report file, try using the -html_dir option.

The -html_dir option splits the report across many files in the specified directory to allow easier browsing and to avoid overwhelming the browser with one large file. The command creates the index.html file in the specified directory to contain a table of contents that references the individual files.

---

## ABS-289

(error) You must define budget latencies before setup delays in mode '%s'.

Feedback

**Description**

In order for budgeting to work properly, you must define budget target latencies for your clock. The relative values of the target latencies will indicate to the budgeter if you want purposeful clock skew or OCV in you blocks.

Your current circuit either has propagated clocks or user defined set_clock_latency values, indicating that there can be clock skew or OCV intended in your circuit. But no budget target latencies have been set.

**What Next**

You can manually set budget target latencies by using the set_latency_budget_constraints command. If you want to set all target latencies to zero, use:

set_latency_budget_constraints -early_latency 0 -late_latency 0 -default

Alternatively, you can automatically set up budget target latencies based on actual latencies in your current circuit. This is done by using:

compute_budget_constraints -latency_targets actual -balance true

Please refer to the respective command man pages for more details.

---

## ABS-290

(error) Cannot do %s on design %s because it is not saved after rename.

**Description**

You are getting this message because you are trying to do an operation that would trigger the tool to automatically save the design. But this design was renamed from another design and there is no user-specified save after the design was renamed. A partial automatic save design on a design that is potentially a part of a set of designs that was renamed would corrupt the database and cause a severe error down the flow. That is why this operation is not allowed.

**What Next**

If you have not called rename_block command explicitly, there might be some commands that you call from the top level to operate on child blocks that would do rename internally to avoid committing any changes to disk without user calling save command.

If you want to continue with the operation, you can do a *save_lib -all* to save the designs first, then continue.

---

## ABS-291

(error) Cannot create abstract view for block %s because %s.

Feedback

### Description

You are getting this message because you are trying to run create_abstract on a block that it is not allowed for the specified reason.

### What Next

Here are the possible reasons you might see from the message and what you can do next.

*plan.flow.design_view_only is true*: Use *report_app_options* or *get_app_option_value* to check the app option setting on the specified block. The tool is not allowed to create abstract view for a block with this app option set to true.

*It is instantiated in another block with abstract view*: This indicates that this block is instantiated in another block which has an abstract view. The tool can only create abstract view at one level of physical hierarchy. If an ancestor block has abstract view, the descedent blocks are not allowed to have abstract view.

*It contains a partition*: This block has no abstract view, yet it has a child physical hierarchy which is represented in the form of a partition. This situation should not happen. Please review the flow.

*Has lower level block that can be abstract view*: The tool creates abstract view for the lowest level of physical hierarchy that is allowed. You get this error either because there is one or more lower level blocks in the specified block can be abstract view, or because one or more lower level blocks already has an abstract view. If a lower level blocks already has an abstract view, you need to remove the abstract view. If none of the lower level blocks has abstract view, that means one or more lower level blocks has plan.flow.design_view_only setting to false, so the abstract view should be created at a lower level, not this block level. If you want to create abstract view for this block, please set the app option plan.flow.design_view_only to true and remove any existing abstract view for all the lower level blocks.

*It has no write permission on disk*: You have no write permission for the on-disk data for this block. Since creating abstract for a block means writing data to the block's on-disk database, if you do not have disk write permission, the tool cannot create abstract for the block.

*The tool cannot find the block*: If the message provides a name for the block that it complains about, check whether the block is instantiated anywhere in the current block or its child blocks. If so, whether it is linked properly.

---

## ABS-292

(warning) Cannot create abstract view for block %s because %s.

Feedback

### Description

You are getting this message because you are trying to run a command that is attempting to create abstract on a block that it is not allowed for the specified reason.

### What Next

Here are the possible reasons you might see from the message and what you can do next.

*abstract.allow_all_level_abstract is false*: Use *report_app_options* or *get_app_option_value* to check the app option setting on the specified block. The tool is not allowed to create abstract view for a block with this app option set to false.

*It is instantiated in another block with abstract view*: This indicates that this block is instantiated in another block which has an abstract view. The tool can only create abstract view at one level of physical hierarchy. If an ancestor block has abstract view, the descedent blocks are not allowed to have abstract view.

*It contains a partition*: This block has no abstract view, yet it has a child physical hierarchy which is represented in the form of a partition. This situation should not happen. Please review the flow.

*Has lower level block that can be abstract view*: The tool creates abstract view for the lowest level of physical hierarchy that is allowed. Since there is one or more lower level blocks in the specified block can be abstract view, the abstract view should be created at a lower level, not this block level. If you want to create abstract view for this block, please set the app option abstract.allow_all_level_abstract to false for all the lower level blocks.

*It has no write permission on disk*: You have no write permission for the on-disk data for this block. Since creating abstract for a block means writing data to the block's on-disk database, if you do not have disk write permission, the tool cannot create abstract for the block.

*The tool cannot find the block*: If the message provides a name for the block that it complains about, check whether the block is instantiated anywhere in the current block or its child blocks. If so, whether it is linked properly.

---

## ABS-293

(Error) Cannot operate on current design %s.outline, because it is already expanded to design view.

### Description

You are getting this message because you are trying to run a command on outline view of a design, but the design view of the same design already exists because *expand_outline* was called previously.

Feedback

**What Next**

If you intend to open the design view, please close the current design and use *open_block* to load the design view and continue.

If you intend to discard the existing design view, please use *remove_block* to explicitly delete it. Please also delete the abstract view and/or frame view of the same design, if they exist.

---

## ABS-294

(warning) Cannot find -source pin for generated clock '%s' in block '%s'. Master clock does not propagate to the block boundary. Generated clock will be converted to a regular clock.

**Description**

You have set the app_option plan.budget.allow_disconnected_gclocks to true, which has caused you to receive this "warning" message instead of an ABS-207 "error" message.

The generated clock in your budget has been changed into a regular clock, which means that clock latency will no longer be propagated to this clock. This will be a serious problem for post-CTS optimization.

**What Next**

Please refer to the man page for ABS-207 for details on how to address this problem. You should fix this problem and then reset the plan.budget.allow_disconnected_gclocks app_option back to false.

---

## ABS-295

(warning) Cannot find -source pin for generated clock '%s' in block '%s'. Master clock does not propagate to the generated clock. Generated clock will be converted to a regular clock.

**Description**

You have set the app_option plan.budget.allow_disconnected_gclocks to true, which has caused you to receive this "warning" message instead of an ABS-286 "error" message.

The generated clock in your budget has been changed into a regular clock, which means that clock latency will no longer be propagated to this clock. This will be a serious problem for post-CTS optimization.

**What Next**

Please refer to the man page for ABS-286 for details on how to address this problem. You should fix this problem and then reset the plan.budget.allow_disconnected_gclocks app_option back to false.

Feedback

## ABS-296

(error) Block %s failed.

**Description**

You get this error message because you are running a command that does block level jobs from top level, and the job for the specified block failed.

**What Next**

Please check the log file for the specified block to find out what exactly failed.

To find the location of the log file for the block, search the main log file before this message for "*all scripts and log files are under ...*". The log file for the block can be found in a sub-directory there with the same name as the block.

## ABS-297

(error) Could not find any instantiation for the given block(s).

**Description**

You get this error message because you are running a command that does block level jobs from top level for a specified set of blocks. But the tool is unable to find any instantiation of the given blocks in the current block.

**What Next**

Please check the blocks you specified and their view names.

Please check the reference block attributes of a cell to get the library name, reference block name and reference view name.

## ABS-298

(error) Current design is read-only. Cannot save the modified design and launch jobs for child blocks.

**Description**

You get this error message because you are running a command that does block level jobs from top level. In preparation to start running the child blocks, the tool saves and closes the top level. But the tool is unable to do so because the top level design is modified and the top level design is read-only.

Feedback

### What Next

Please check the write permission of the top level design on disk. If the design has no disk write permission, you can use *save_block -as* to save the design to another design and continue with the operation.

If the design has disk write permission, please check if there is an editable abstract view for the current design. If there is, you can either call *remove_abstract* to remove the abstract view and make the current design editable, or call *reopen_block -edit* to promote the current design to edit mode, and then continue with the flow.

---

## ABS-299

(warning) Placement abstract views are always editable abstracts, option -read_only is ignored.

### Description

You get this error message because you are running a command that tries to create placement abstract view with option *-read_only* specified for command *create_abstract*.

A placement abstract view is always an editable abstract view. An editable abstract view doesn't necessarily mean that the top level operations are allowed to modify this block. An editable abstract view just means that the design view of the block is locked and that the abstract view has the latest data of the design.

### What Next

If you want the block to be read-only for commands issued from the top level where this block is instantiated, you can set_editability to restrict modifications to this block. Another option is to remove the write unix write permission for the disk directory where this block is stored.

---

## ABS-300

(error) Block level clock '%s' at pin '%s' maps to multiple top level clocks for mode '%s'.

### Description

There are multiple top level clocks with the correct waveform that propagate to the given pin. It is not possible to automatically determine which top level clock you wish to associate with your top level clock.

### What Next

You need to set the association manually using the set_block_to_top_map command.

Feedback

## ABS-301

(error) Block level clock '%s' at pin '%s' does not match any top level clocks for mode '%s'.

**Description**

The clocks that propagate to the given pin of your block boundary have a waveform that is different than what the clock expects.

**What Next**

You must either supply the correct clock at the top level or change the clock definition in your block. You might want to consider declaring another operating mode for your block which has the appropriate clock frequency.

## ABS-302

(warning) Block level clock '%s' at pin '%s' has no top level clocks that correspond to it in mode '%s'.

**Description**

There are no top-level clocks that propagate to the source pin of your block's clock.

**What Next**

You can force a top-level clock source to be declared at you block boundary by using the -map option of the set_block_to_top_map command.

## ABS-303

(error) Block level clock '%s' in cell '%s' maps to top level clock '%s' and '%s' for mode '%s'.\n",

**Description**

The clock in your block has multiple sources, but the sources are connected to different clocks at the top level.

**What Next**

You should fix your top level clocks to supplies both block sources the same way. Alternatively, you can declare two separate clocks in your block.

## ABS-305

(information) %s clock '%s' created in cell '%s' for mode '%s'.

Feedback

**Description**

It was not possible to connect your block clock to a clock at the top level. A top-level clock source has been created to correspond to your block-level clock source. Timing analysis will use this clock source, but it is not necessarily electrically correct.

**What Next**

You should check if this clock is electrically correct. If it's not, consider removing it later in the design flow.

---

## ABS-306

(error) Clock '%s' in cell '%s' has a waveform that doesn't match top level clock '%s' in mode '%s'.

**Description**

The two named clocks cannot be connected together because their waveforms are different. A separate unconnected clock will be created for use in the block.

**What Next**

You should fix the waveforms to be the same and then manually reassociate the clocks using set_block_to_top_map. Alternatively, you can force the clock to be associated, even though their waveforms mismatch, by using set_block_to_top_map.

---

## ABS-307

(error) No constraint data found for instance '%s'. Mapping not applied.

**Description**

It is not possible to error check the setting of set_block_to_top_map for this instance. Either the instance does not represent independently linkable design hierarchy or the instance has no valid constraints. You must load your block constraints and save the design before using set_block_to_top_map at the top level.

**What Next**

Make sure your instance points to a constrained design.

If you wish, you can use the -path option of set_block_to_top_map to skip error checking. Please refer to the man page for details.

---

## ABS-308

(error) The '%s' specification must have %d values.

Feedback

**Description**

The given command line option is not formatted correctly. For example, a mode option with 2 values should be formatted as:

-mode {block_mode top_mode}

**What Next**

Refer to the man page for more details.

---

## ABS-309

(error) The top level %s '%s' does not exist.

**Description**

The mapping cannot be completed because the given object was not defined at the top level.

**What Next**

Make sure you map from/to existing objects. Use report_block_to_top_map to see a list of objects that exist.

---

## ABS-310

(error) Top level clock '%s' in mode '%s' does not connect to block level clock %s source in block %s.

**Description**

The top level clock does not connect to the block level clock as it should. For block clock sources on the block boundary, the top level clock should propagate to that boundary. For block clock source within the block, the top level clock source must be on the same pin.

Another possible problem is that your current block abstraction does not contain the block clock's source.

**What Next**

You can force the top level clock and block level clock to be related by using set_block_to_top_map command. This will confirm your intention to subsequent budgeting and abstraction commands. Timing constraints created by the budgeter will work at the block level, but you may encounter problems when you assemble your final blocks at the top level.

Feedback

## ABS-311

(error) The %s '%s' does not exist in block '%s'.

**Description**

The mapping cannot be completed because the given object was not defined in the block.

**What Next**

Make sure you map from/to existing objects. Use report_block_to_top_map to see a list of objects that exist.

## ABS-312

(error) Cannot find %s in block '%s' corresponds to %s '%s' at top level.

**Description**

The mode/corner mapping between top level and the specified block is not available.

**What Next**

Make sure your modes/corners are properly set up at the top level as well as inside the block. By default, block modes/corners are associated with top modes/corners if they have the same name. If your mode/corner names have different names at the top level and at the block level, please use set_block_to_top_map command to set up the desired mapping.

## ABS-313

(error) Cannot open file to write. Abstract annotations cannot be saved.

**Description**

The tool cannot open the file to save abstract data. So some important data with the abstract view will not be saved.

**What Next**

Please check free disk space and write permission of the design library. Make sure you have the write permission to create abstract view for the design, and there is enough free disk space.

## ABS-314

(error) Abstract %s (ref: %s) does not have information for %s. Activate %s at block level and re-create abstract.

Feedback

**Description**

Relevant data for the specified mode/corner and early/late is not available for the specified block abstract.

**What Next**

If there is mismatch between top and block mode/corner names, please use the set_block_to_top_map command to set up the mapping.

The message is issued for a particular mode/corner if a certain optimization type (setup/hold) is only enabled at top-level but not at the block level. You can do *report_abstract* at the top level or *report_scenario* in the block's design view to find out. Please recreate the abstracts by enabling the needed optimization type in block-level.

---

## ABS-315

(warning) Subdesign %s (instance %s) has no constraints

**Description**

Block to top mapping cannot be performed on this block because it appears to have no local constraints. This is OK if you don't intend to create budgets for or promote constraints from this block.

**What Next**

If you want to budget this block or promote constraints from this block, you need to apply SDC constraints to it. This is done by using open_block or current_design to make the subdesign the current design. Then you can use create_mode, create_corner, and create_clock on the subdesign.

It is possible that you only have SDC constraints for a top-level block, and not your subdesign. In this case, consider using the "split_constraints" command to generate SDC constraints for your lower-level block.

---

## ABS-316

(information) Mapped %d clocks. %d clocks could not be resolved.

**Description**

This is just a summary of set_block_to_top -auto. If you used the -report_only option, then no mappings have actually been changed.

Feedback

### What Next

If some clocks have not been mapped, you might want to run "set_block_to_top -auto" again to see if more clocks will propagate. Or you could use the -promote option and select "local" or "all".

---

## ABS-317

(warning) Cannot resolve clock latencies for non-block '%s' (instance '%s').

### Description

You have asked to budget a logical hierarchy which is not a block in the physical hierarchy, so the budgeter is unable to determine what constraints will be applied to the logical hierarchy if it is optimized independently.

The budgeter will assume that this logical hierarchy will not be subject to clock tree synthesis or to the actions of the compute_clock_latency command, since those commands require a physical hierarchy. So the budget will constrain the logic hierarchy as if it has zero clock latency.

### What Next

If possible, you should be budgeting and optimizing physical hierarchies. If you really want to budget and optimize this logic hierarchy that has non-zero clock latencies, then you will need to manually account for clock latency at the boundaries of the logic hierarchy.

---

## ABS-318

(warning) No block latencies available for block '%s' (instance '%s') mode '%s'.

### Description

There is no mode information stored for the given block. So none of the top level clocks can be mapped to block level clocks, and clock latencies cannot be adjusted. The budgeter will assume the clock latencies in the block are zero.

Please refer to the man page for ABS-320 for more details.

### What Next

Perhaps you have specified an incorrect mode mapping with the set_block_to_top_map command. Or perhaps you have neglected to specify the given mode in the block. Use report_block_to_top_map to get more details.

Feedback

## ABS-319

(warning) No block latencies available for '%s' (instance '%s') mode '%s' corner '%s'.

**Description**

There is no corner information stored for the given block. So none of the top level clocks can be mapped to block level clocks, and clock latencies cannot be adjusted. The budgeter will assume the clock latencies in the block are zero.

**What Next**

Perhaps you have specified an incorrect corner mapping with the set_block_to_top_map command. Or perhaps you have neglected to specify the given corner in the block. Use report_block_to_top_map to get more details.

## ABS-320

(warning) For block '%s' (instance '%s'), %d top-level clock(s) could not be mapped to block-level clocks.

**Description**

The top-level design defines clocks which overlap the given lower-level block. But the block has not declared clocks in the equivalent places. Because of this, there is no way to determine the clock latency of these clocks in the block. This means that the latency adjustment of the write_budgets command cannot be correctly performed. The budgeter will assume the clock latencies in the block are zero.

You should refer to the man page of the write_budgets command for more details concerning clock latency adjustment.

**What Next**

You need to define the block's clocks in SDC input that is loaded at the block level. If you have done this, then perhaps you have specified an incorrect mode or corner mapping with the set_block_to_top_map command. Or perhaps you have neglected to specify the clocks in the block.

If your budgeted block will not be implemented as a separate physical hierarchy, then you do not need to worry that latency has not been adjusted. If this message refers to a block that is not budgeted, then you do not need to worry.

Use report_block_to_top_map and "report_budget -latency" to get more details.

Feedback

## ABS-321

(error) Multiply instantiated instance '%s' must have the same adjust_latency setting as instance '%s' (%s versus %s).

**Description**

The set_budget_options command allows you to set different "adjust_latency" settings for different block instances. If a block is multiply instantiated, then the setting must be the same for all instances of the block.

**What Next**

Please use the set_budget_options command to set all multiply instantiated instances to have the same "adjust_latency" setting.

## ABS-322

(warning) Cannot find %s '%s' in block '%s' that corresponds to top-level %s '%s'. Clock mapping cannot be performed.

**Description**

The given mode is not defined in the given block. This means that there will be no mapping for any clocks in this mode and block. Please refer to the man page for ABS-320 to learn more about clock mapping.

**What Next**

Please refer to the man page for ABS-320 to learn more about clock mapping, and what you might need to do next.

## ABS-323

(warning) Propagated clock latencies in block '%s' have not been updated for mode '%s' corner '%s'.

**Description**

In the given block, the clocks have been set to propagated, but the update_clock_latency command has never been run. Because of this the actual clock latencies in the block have never been recorded, and budgeting cannot properly adjust the latencies in the budget. Budgeting will assume the latencies in the block are zero.

**What Next**

Load each of your blocks and run "update_clock_latency -real_only". Then save the block database to disk. This will cause current latency each of your block's clocks to be

Feedback

recorded. After the block latencies have been recorded, budgeting at the top level can adjust for them.

## ABS-324

(warning) Information for %s is missing in abstract view of design %s for block %s.

**Description**

You are getting this message because you are doing timing operations involving estimated_corner. The specified data is not available in the abstract view for the specified block.

This could happen if *estimated_corner* did not exist inside the block when *create_abstract* was executed and *-estimate_timing* option was not specified for *create_abstract*.

**What Next**

If the abstract view was created before any timing optimization was done on the block design, this could be a serious problem that affect top level *estimate_timing* quality.

If timing optimization was performed on the block design before *create_abstract*, this warning can be ignored as the situation would not cause any issue for *estimate_timing*.

## ABS-325

(warning) Reference %s is unbound.

**Description**

You are getting this message because you are attempting to *change_abstract* for cell instances of the specified reference when the cells are unbound.

This could happen if you have missing reference library.

**What Next**

Check whether the specified reference design was correctly created. If it is supposed to be from a reference library, find out which library the specified reference design is located, check if the library is a reference library of the current library.

## ABS-326

(error) Can't change reference '%s' because library '%s' is not in the necessary ref_lib lists.

Feedback

**Description**

When using change_abstract, references can only be changed if the target block is inside one of the libraries listed in the ref_lib list of the current design. Additionally, if the block is instantiated at lower levels of hierarchy, those ref lib lists must also include the library.

No changes will be made for any reference.

**What Next**

Make sure that the target library is the library you want. If it is, you need to add the library to your ref_lib list by using the set_ref_libs command.

---

## ABS-327

(error) Can't change reference '%s' because block '%s' does not exist.

**Description**

The block you have specified does not exist.

No changes will be made for any reference.

**What Next**

Make sure that this is the block that you want. You might want to adjust your parameters to change_abstract if you wanted a different lib, block, label, or view.

---

## ABS-328

(error) The ports of reference %s do not match block %s.

**Description**

By default, change_abstract requires that the ports of the newly inserted block be identical to the original ports. The change cannot be performed.

**What Next**

You can use the -force option of change_abstract to force the change to be made. Be careful, because connections and constraints to the reference will be dropped if ports are missing.

---

## ABS-329

(error) Nested blocks found: cell %s (reference: %s) is an ancestor of cell %s (reference: %s).

Feedback

**Description**

You are getting this message because change_abstract or change_view is called on two blocks that have ancestor-descendant relationship.

When this happens, it is unclear to the tool whether the descendant block in the ancestor block's current view should be changed or whether the descendant block in the ancestor block's new changed-to view should be updated.

**What Next**

Please use separate change_abstract or change_view calls for the ancestor block and the descendant block.

---

## ABS-330

(warning) The constraint map contains no information for the %s constraints for block %s.

**Description**

You are getting this message because the tool cannot find the specified information in the block constraint map in the current block.

This might cause missing constraints for some blocks down the flow.

**What Next**

Please use *report_constraint_mapping_file* to check the current block constraint map. You can use the *set_constraint_mapping_file* command to add information to the block constraint map.

---

## ABS-331

(warning) Could not find the Verilog file %s for module %s of outline view block %s.

**Description**

You are getting this message because the tool cannot find the Verilog file that would be needed to do *expand_outline* for the specified block.

This might cause error when you do *expand_outline* on the specified block.

**What Next**

Please find the Verilog file that was originally used by *read_verilog_outline* and make sure that it is where it was.

Feedback

## ABS-332

(warning) The design has no %s.

**Description**

You are getting this message because the tool cannot find a timing corner named *estimated_corner* or a timing scenario associated with *estimated_corner*.

If the netlist is not optimized and you do timing budgeting on this design, you might get bad timing budgets.

**What Next**

Check your flow and see if you need to run *estimate_timing* first and then use the result of *estimate_timing* to guide timing budgeting to get a more reasonable timing budget for your blocks.

## ABS-333

(error) %s is not one of: DEF, UPF, ETM_UPF, SDC, BUDGET, CLKNET, PG_CONSTRAINT, COMPILE_PG, CTS_CONSTRAINT, BTM, FLOORPLAN.

**Description**

The constraint type name is not valid. For constraint type names, only DEF, UPF, ETM_UPF, SDC, BUDGET, CLKNET, PG_CONSTRAINT, COMPILE_PG, CTS_CONSTRAINT, BTM, FLOORPLAN are valid.

**What Next**

Please check the specified constraint type name to make sure that it is one of: DEF, UPF, ETM_UPF, SDC, BUDGET, CLKNET, PG_CONSTRAINT, COMPILE_PG, CTS_CONSTRAINT, BTM, FLOORPLAN.

## ABS-334

(error) Option -remove_all should not be used together with option -remove_blocks or option -remove_types.

**Description**

Command update_constraint_mapping_file has three options: -remove_all, -remove_blocks and -remove_types. When -remove_all is used, all constraint entries for all blocks in the map will be removed, so it could not be used together with -remove_blocks or -remove_types.

Feedback

**What Next**

Please check the options of command update_constraint_mapping_file. Do not use -remove_all and -remove_blocks, or -remove_all and -remove_types at the same time.

---

## ABS-335

(error) Must specify at least one option for command update_constraint_mapping_file.

**Description**

Command update_constraint_mapping_file has three options: -remove_all, -remove_blocks, and -remove_types. This command could not be used if no option is specified.

**What Next**

Please check whether command update_constraint_mapping_file is used without any option. Specify at least one option for this command.

---

## ABS-340

(error) Abstract %s (ref: %s) does not have enough parasitic data. Activate needed scenarios at block level and re-create abstracts.

**Description**

Enough parasitic data is not available for the specified block abstract. At Block level, please recreate abstract after activating all scenarios which are needed for TOP level flow.

**What Next**

Please make sure the abstract view was created successfully and check the read permission of the design.

If there is mismatch between top and block mode/corner names, please use the set_block_to_top_map command to set up the mapping.

---

## ABS-345

(warning) Parasitic data for the following corner(s) is not present in the abstract %s (ref: %s). Tool will use minimal R,C values for them.

**Description**

At Top level, enough parasitic data is not available for the specific corner(s) of the block abstract. As a result, tool will use very small R,C values for them. User can choose to either stay with these values or fix the corner related mismatch between top and abstracts.

Feedback

### What Next

Please make sure the abstract view was created successfully and check the read permission of the design.

If all this looks good, then follow one of the ways to fix the mismatch. One way is to use set_block_to_top_map command to set up the mapping between extra corner at TOP and an existing corner in block. Another way is to reacreate abstract after creating extra corners/scenarios which are needed for TOP level flow. For such situation, try to match parasitic parameters between block and TOP before re-creating abstracts. report_parasitic_parameters can be used for that purpose.

### See Also

- report_parasitic_parameters
- set_block_to_top_map

---

## ABS-350

(error) Abstract created with '-placement' option cannot be used in top-level implementation.

### Description

The abstract created with no timing information cannot be used in top-level implementation flow. The value for '-target_use' option of the *create_abstract* command is implementation. However, the other options provided conflicts with the targeted use.

### What Next

Check and correct the options of the *create_abstract* command. If the abstract is intended to be used for top-level implementation, create an abstract with timing information. Else, if the abstract is intended to be used for planning, specify planning as the value for target_use option.

### See Also

- create_abstract

---

## ABS-351

(error) Application option 'abstract.allow_all_level_abstract' is not enabled. Failed to create abstract on %s which has lower level block(s) that can be abstract view.

Feedback

### Description

The application option 'abstract.allow_all_level_abstract' is set to false. If abstracts need to be created on designs that are already linked with lower level physical hiearchies, this application option needs to be set to true.

### What Next

Check and correct the value of application option 'abstract.allow_all_level_abstract'

### See Also

- abstract.allow_all_level_abstract

---

## ABS-352

(warning) Application option 'abstract.allow_all_level_abstract' is not enabled. Failed to create abstract on %s which has lower level block(s) that can be abstract view.

### Description

The application option 'abstract.allow_all_level_abstract' is set to false. If abstracts need to be created on designs that are already linked with lower level physical hiearchies, this application option needs to be set to true.

### What Next

Check and correct the value of application option 'abstract.allow_all_level_abstract'

### See Also

- abstract.allow_all_level_abstract

---

## ABS-353

(error) No valid block(s) specified for abstract creation.

### Description

The block(s) specified for option *'-blocks'* are invalid. No such block(s) are found. The abstract creation has failed.

### What Next

Check and correct the blocks specified for the *'-blocks'* option of *create_abstract* command. Specify list of valid block designs for which abstract has to be created.

Feedback

**See Also**

- create_abstract

---

## ABS-354

(error) The leaf cell %s is unmapped. Abstract cannot be created for the current design.

**Description**

The specified leaf cell in the current design is unmapped. Abstracts can be created on a design only when all the leaf cells are mapped. So, abstract creation on the current design will be aborted.

**What Next**

Please go through the appropriate synthesis flow to avoid any unmapped netlist.

**See Also**

- create_abstract

---

## ABS-355

(error) Current design has %d unmapped leaf cell instance(s). Abstract cannot be created for the current design.

**Description**

Some leaf cell(s) in the current design are unmapped. Abstracts can be created on a design only when all the leaf cells are mapped. So, abstract creation on the current design will be aborted.

**What Next**

Look out for ABS-354 messages that report the unmapped cells. ABS-354 message limit may need to be increased to view complete list of unmapped cells. Please go through the appropriate synthesis flow to avoid any unmapped netlist.

**See Also**

- create_abstract
- ABS-354
- set_message_info

Feedback

## ABS-356

(error) Creation of abstract with target use DCNXT is not allowed in default IC Compiler II shell.

**Description**

Abstract targeted for use in DCNXT cannot be directly created in IC Compiler II shell.

**What Next**

Use *import_ndm_block* in DCNXT to create and use IC Compiler II abstract in DCNXT.

## ABS-400

(error) Top level area '%s' already exists.

**Description**

You must give a unique name to each top-level area.

**What Next**

Remove the existing area with the same name or use another name.

## ABS-401

(error) Can't add bound '%s' to the top-level area.

**Description**

You are only allowed to add move bounds, macro instances, and pad instances to the top level area. Other types of bounds and instances are not allowed.

**What Next**

If you are trying to add standard cell instances, you must first put them in a move bound. Then you can add the move bound.

## ABS-402

(error) Top level area has not been declared.

**Description**

You must give a unique name to each top-level area.

Feedback

**What Next**

Add the area by using the -add_top_area option.

---

## ABS-403

(error) Instance '%s' is not a hierarchical block.

**Description**

This option only works on instances of hierarchical blocks. It does not work on hard macros or standard cells.

**What Next**

Please use these options only on design hierarchy instances.

---

## ABS-404

(warning) Can't add %d standard cells to the top-level area.

**Description**

You are only allowed to add move bounds, macro instances, and pad instances to the top level area.

**What Next**

If you are trying to add standard cell instances, you must first put them in a move bound. Then you can add the move bound.

---

## ABS-405

(error) Can't add %d ports to the top-level area.

**Description**

You are only allowed to add placed ports to the top level area. These ports were not placed.

**What Next**

Ports may be placed by reading a DEF file or by running place_pins at the chip level and pushing down pin locations.

---

## ABS-406

(error) Tool currently does not support SI timing with abstracts.

Feedback

**Description**

The tool currently does not support signal integrity timing on top-level designs linked with abstracts. The tool proceeds with this error but signal integrity timing might not be accurate.

**What Next**

Set the *time.si_enable_analysis* application option to *false* and proceed with the non-signal-integrity flow.

---

## ABS-407

(warning) Skipping coupling capacitance of %s net %s of block %s which has no para annotation.

**Description**

The net does not have base parasitics annotated to it in the abstract. Hence the coupling capacitance of the net is also not saved to the abstract.

**What Next**

Check the nature of the net. It may be a dangling net in abstract, ideal net or a high fanout net.

---

## ABS-408

(warning) Net %s%s does not have expected parasitic information.

**Description**

The net does not have parasitics annotated to it in the abstract.

**What Next**

Check if the net has parasitics present in design view of the block. Check the nature of the net. It may be a dangling net in abstract.

---

## ABS-409a

(warning) %d eligible %snet(s) do not have parasitics%s.%s

**Description**

The given number of nets do not have base parasitics annotated in the abstract. For a timing abstract, nets eligible for parasitic annotation include all the non-HFN and non-

Feedback

ideal nets of the abstract. For an abstract with estimated timing, nets eligible for parasitic annotation include only the non-HFN and non-ideal boundary nets of the abstract.

The eligible nets without parasitic annotation will be estimated in top-level and if SI is enabled,the coupling capacitance of these nets will also be affected. This can lead to some timing inaccuracy.

The HFN and ideal nets without parasitic annotation can be ignored. These nets may impact timing only if the nature of the net changes when the abstract gets linked to a higher level.

**What Next**

If this message is issued during create_abstract, check if the parasitics are present in design view of the block using *report_parasitics* or *write_parasitics* command. If this message is issued during timing update in top-level, look out for ABS-409a message in the abstract creation log of the corresponding block. If parasitics are missing during abstract creation itself, check for parasitics in the block level, otherwise look out for ABS-411 messages in the top-level timing update which indicate failure in loading of parasitics from the abstract.

**See Also**

- ABS-409
- ABS-411
- report_parasitics
- write_parasitics

---

## ABS-409b

(information) %d eligible %snet(s) do not have parasitics%s.%s

**Description**

The given number of nets do not have base parasitics annotated in the abstract. For a timing abstract, nets eligible for parasitic annotation include all the non-HFN and non-ideal nets of the abstract. For an abstract with estimated timing, nets eligible for parasitic annotation include only the non-HFN and non-ideal boundary nets of the abstract.

The eligible nets without parasitic annotation will be estimated in top-level and if SI is enabled,the coupling capacitance of these nets will also be affected. This can lead to some timing inaccuracy.

The HFN and ideal nets without parasitic annotation can be ignored. These nets may impact timing only if the nature of the net changes when the abstract gets linked to a higher level.

Feedback

In optimizable abstracts, it is expected to have some nets missing parasitics and they can be ignored.

**What Next**

If this message is issued during create_abstract, check if the parasitics are present in design view of the block using *report_parasitics* or *write_parasitics* command. If this message is issued during timing update in top-level, look out for ABS-409a message in the abstract creation log of the corresponding block. If parasitics are missing during abstract creation itself, check for parasitics in the block level, otherwise look out for ABS-411 messages in the top-level timing update which indicate failure in loading of parasitics from the abstract.

**See Also**

- ABS-409
- ABS-411
- report_parasitics
- write_parasitics

---

## ABS-409

(information) %d nets (%.2f%% of eligible nets) of %s%s%s%s have parasitics.

**Description**

The given number of nets have base parasitics annotated in the abstract. For a timing abstract, nets eligible for parasitic annotation include all the non-HFN and non-ideal nets of the abstract. For an abstract with estimated timing, nets eligible for parasitic annotation include only the non-HFN and non-ideal boundary nets of the abstract.

If less than 100% eligible nets have parasitics, the eligible nets without parasitic annotation will be estimated in top-level and if SI is enabled, the coupling capacitance of these nets will also be affected. This can lead to some timing inaccuracy.

**What Next**

If this message is issued during create_abstract and reports less than 100% eligible nets with parasitics, check if the parasitics are present in design view of the block using *report_parasitics* or *write_parasitics* command. If this message is issued during timing update in top-level and reports less than 100% eligible nets with parasitics, look out for ABS-409 message in the abstract creation log of the corresponding block. If parasitics are missing during abstract creation itself, check for parasitics in the block level, otherwise look out for ABS-411 messages in the top-level timing update which indicate failure in loading of parasitics from the abstract.

Feedback

## ABS-410

(information) %d nets of %s.abstract%s have load/delay annotation for corner %s (HFN: %d, Ideal: %d, others: %d).

### Description

The given number of nets have load/delay value annotated to them in the abstract. These load/delay values will be used in top-level delay calculation.

### What Next

This is an informational message. No action needed.

## ABS-411

(warning) Failed to load parasitics for net %s. Coupling cap with %d nets are affected. %s

### Description

The base parasitics of the net failed to load from the abstract. Hence the coupling capacitance of the net is also not loaded from the abstract. This may lead to timing inaccuracy.

### What Next

Check the nature of the net and its connections. It may be a dangling net in abstract.

### See Also

- ABS-409
- ABS-409a

## ABS-412

(information) Optimizable net %s%s does not have expected parasitic information and will be estimated in top-level.

### Description

The optimizable net does not have parasitics annotated to it in the abstract. This is an expected behavior and the net will be estimated in top-level.

### What Next

This is an information message and no action needs to be taken.

Feedback

## ABS-413

(warning) Extractor is not initailized. No parasitics will be saved to %s.abstract.

### Description

Extractor is not initialized for this design. Hence no net prarasitics is available for annotation onto the nets of the abstract. The parasitics file annotated to the abstract will be empty. The nets of the abstract will be estimated in top-level. This can lead to timing inaccuracy. If SI is enabled in top-level,the coupling capacitance of these nets will also be affected.

### What Next

Check for TIM-102 message that indicates that the extractor is not initialized. Fix the reason for extractor not being initialized. If TIM-102 is not present, it might have happened that the extractor got uninitialized by an issue in tool.

### See Also

- TIM-102
- ABS-408
- ABS-409
- ABS-409a
- report_parasitics
- write_parasitics

## ABS-414

(warning) Failed to load coupling caps for %s (victim) - %s (aggressor) pair from abstract instance %s.

### Description

Coupling caps could not be loaded for a victim-aggressor pair because of the presence of invalid nodes (para node ID is greater than the maximum node ID in the net's base parasitics). Correlation issues might exist due of missing of coupling cap information.

## ABS-415

(warning) Some coupling caps couldn't be loaded for %s (victim) - %s (aggressor) pair from abstract instance %s.

Feedback

### Description

Some coupling caps could not be loaded for a victim-aggressor pair because of the presence of invalid nodes (para node ID is greater than the maximum node ID in the net's base parasitics). Correlation issues might exist due of missing of some coupling cap information.

---

## ABS-416

(warning) Zero Interconnect Delay is used. So no parasitic data will be saved to %s.abstract.

### Description

Zero Interconnect Delay calculation style is used, so no parasitic data will be saved to the block abstracts.

### What Next

If parasitics are required to be saved with the abstract, set any other valid delay calculation style which is required to generate non-zero delay along with valid interconnect parasitics and then call *create_abstract*.

### See Also

- time.delay_calculation_style

---

## ABS-417

(warning) Zero Interconnect Delay had been used during abstract creation. %s

### Description

Zero Interconnect Delay calculation style is used during creation of block abstract and hence no parasitics data was saved for those blocks. So parasitics data cannot be loaded for such blocks.

### What Next

If parasitics need to be loaded from abstract, set any other valid delay calculation style which is required to generate non-zero delay along with valid interconnect parasitics and then call *create_abstract*.

### See Also

- ABS-416
- time.delay_calculation_style

Feedback

## ABS-444

(error) Disconnecting pin/port %s from partially retained net %s in %s.abstract. This change may not merge back into %s.design.

**Description**

Pins/Ports connected to partially retained nets in abstract should never be disconnected. Such changes may not be merged back.

## ABS-445

(error) Modifying connection of dangling pin/port %s in %s.abstract. This change will merge back into %s.design; but there can be netlist issues.

**Description**

Pins/Ports that are dangling in abstract due to abstraction should never be modified. Such changes will be merged back; but there can be netlist issues.

## ABS-446

(error) Found logical hierarchy mismatch between views of design %s: %s.

**Description**

This message indicates that the information merging from one view of a design to another view of the same design detected mismatch in the logical hierarchy of the two views of the same design.

It is required that the logical hierarchy stay the same between different views of the same design in order to perform merge.

**What Next**

Find out what caused the logical hierarchies to be different.

If you are encountering this error during *expand_outline*, check if any setting for verilog loading is changed after *read_verilog_outline* and before *expand_outline*.

If you are encountering this error during merge from abstract view to the design view, please review the flow after the last *create_abstract* operation for this design to find which step changed the logical hierarchy in the abstract view.

Feedback

## ABS-447

(error) Block %s has no outline view or design view. This should never happen. Please check your flow to see when the database is broken.

**Description**

Any block should have at least a design view or an outline view. If a block is found to have neither of these two views, something must have gone wrong earlier in the flow before the command which issued this error message.

**What Next**

Go back in the flow to the time when the design view or the outline view is created, trace it step by step to find out when it disappeared.

## ABS-448

(error) The physical hierarchy in the %s view and the %s view of design %s:%s do not match.

**Description**

This message indicates that the information merging from one view of a design to another view of the same design detected mismatch in the physical hierarchy of the two views of the same design.

It is required that the physical hierarchy stay the same between different views of the same design in order to perform merge.

**What Next**

Find out what caused the physical hierarchies to be different.

Please check if there is any link error.

If you are encountering this error during merge from abstract view to the design view, please also review the flow after the last *create_abstract* operation for this design to find which step could have changed the physical hierarchy in the abstract view.

## ABS-449

(error) Cell %s is a physical block in %s view but not in %s view.

Feedback

**Description**

This message indicates that the information merging from one view of a design to another view of the same design detected mismatch in the physical hierarchy of the two views of the same design.

It is required that the physical hierarchy stay the same between different views of the same design in order to perform merge.

**What Next**

Find out what caused the physical hierarchies to be different.

Please check if there is any link error.

If you are encountering this error during merge from abstract view to the design view, please also review the flow after the last *create_abstract* operation for this design to find which step could have changed the physical hierarchy in the abstract view.

---

## ABS-450

(information) Instance %s with reference block %s has the correct abstract view type, no update needed.

**Description**

This message indicates that the specified block has an existing abstract view with the correct type that the current command requires.

Since the existing abstract view type is correct, there is no need to create a new abstract view for the block. So the command will skip abstract creation for this block.

**What Next**

If you want to force the tool to re-create abstract view for the block, there are two choices.

One is to open the block's design view and call *create_abstract*.

The other is, at the top level, use the *-force_recreate* option with the *create_abstract* command and *-all_blocks* or *-blocks* option.

---

## ABS-451

(warning) Reference change for physical hierarchy instance %s cannot be merged after linking. Use change_abstract to change the reference to %s.

Feedback

### Description

This message indicates that the reference for the specified physical hierarchy instance in the abstract view is different from the design view. This change has not been merged since the design view is already in a linked state. The tool proceeds with the original reference in the design view and this can lead to unexpected results.

### What Next

Check and correct any usage of non-recommended abstract related app options that can affect merge_abstract functionality. Look out for the block references to which the design has linked with. Manually perform the reference change using *change_abstract* as needed.

### See Also

- change_abstract
- change_view
- set_reference

---

## ABS-452

(warning) Cannot find block with design name %s.

### Description

This message indicates that there is no block in current design with the specified reference design name.

### What Next

Please check the content of the block constraint mapping file and the reference design name of the block.

---

## ABS-480

(warning) Boundary change detected for editable abstract block '%s'. Command will skip this block.

### Description

Command 'update_block_views' works only for readonly abstract blocks.

### What Next

Check the abstract creation step for this block. Recreate a readonly abstract.

Feedback

**See Also**

- create_abstract

---

## ABS-501

(error) False path constraint mismatch between block and top level.

**Description**

A false path constraint referencing a pin in the interface of an abstract is mismatched. The constraint exists either in the block or at top-level but not in both.

**What Next**

Check the constraints both at the top level and in the block and modify them to ensure that the correct constraints appear in both designs. If the block constraints were changed, create the abstract again using the create_abstract command.

---

## ABS-502

(error) Multicycle path constraint mismatch between block and top level.

**Description**

A multicycle path constraint referencing a pin in the interface of an abstract is mismatched. The constraint exists either in the block or at top-level but not in both.

**What Next**

Check the constraints both at the top level and in the block and modify them to ensure that the correct constraints appear in both designs. If the block constraints were changed, create the abstract again using the create_abstract command.

---

## ABS-503

(error) set_min_delay/set_max_delay constraint mismatch between block and top level.

**Description**

A set_min_delay or a set_max_delay constraint referencing a pin in the interface of an abstract is mismatched. The constraint exists either in the block or at top-level but not in both.

Feedback

**What Next**

Check the constraints both at the top level and in the block and modify them to ensure that the correct constraints appear in both designs. If the block constraints were changed, create the abstract again using the create_abstract command.

---

## ABS-504

(error) Data sense constraint mismatch between block and top level.

**Description**

A set_sense -type data constraint referencing a pin in the interface of an abstract is mismatched. The constraint exists either in the block or at top-level but not in both.

**What Next**

Check the constraints both at the top level and in the block and modify them to ensure that the correct constraints appear in both designs. If the block constraints were changed, create the abstract again using the create_abstract command.

---

## ABS-505

(error) Clock source mismatch between block and top level.

**Description**

A clock source defined on a pin in the interface of an abstract is mismatched. The clock source exists either in the block or at top-level but not in both.

**What Next**

Check the constraints both at the top level and in the block and modify them to ensure that the correct constraints appear in both designs. If the block constraints were changed, create the abstract again using the create_abstract command.

---

## ABS-506

(error) Generated clock source mismatch between block and top level.

**Description**

A generated clock source defined on a pin in the interface of an abstract is mismatched. The generated clock source exists either in the block or at top-level but not in both.

Feedback

**What Next**

Check the constraints both at the top level and in the block and modify them to ensure that the correct constraints appear in both designs. If the block constraints were changed, create the abstract again using the create_abstract command.

---

## ABS-507

(error) Case analysis mismatch between block and top level.

**Description**

A set_case_analysis constraint defined on a pin in the interface of an abstract is mismatched. The constraint exists either in the block or at top-level but not in both.

**What Next**

Check the constraints both at the top level and in the block and modify them to ensure that the correct constraints appear in both designs. If the block constraints were changed, create the abstract again using the create_abstract command.

---

## ABS-508

(error) Disable timing mismatch between block and top level.

**Description**

A set_disable_timing constraint defined in the interface of an abstract is mismatched. The constraint exists either in the block or at top-level but not in both.

**What Next**

Check the constraints both at the top level and in the block and modify them to ensure that the correct constraints appear in both designs. If the block constraints were changed, create the abstract again using the create_abstract command.

---

## ABS-509

(error) CTS delay point mismatch between block and top level.

**Description**

A CTS delay point constraint defined in the interface of an abstract (using the command set_clock_balance_points) is mismatched. The constraint exists either in the block or at top-level but not in both.

Feedback

**What Next**

Check the constraints both at the top level and in the block and modify them to ensure that the correct constraints appear in both designs. If the block constraints were changed, create the abstract again using the create_abstract command.

---

## ABS-510

(error) CTS ignore point mismatch between block and top level.

**Description**

A CTS ignore point constraint defined in the interface of an abstract (using the command set_clock_balance_points -consider_for_balancing false) is mismatched. The constraint exists either in the block or at top-level but not in both.

**What Next**

Check the constraints both at the top level and in the block and modify them to ensure that the correct constraints appear in both designs. If the block constraints were changed, create the abstract again using the create_abstract command.

---

## ABS-511

(error) Clock sense constraint mismatch between block and top level.

**Description**

A set_sense -type clock constraint referencing a pin in the interface of an abstract is mismatched. The constraint exists either in the block or at top-level but not in both.

**What Next**

Check the constraints both at the top level and in the block and modify them to ensure that the correct constraints appear in both designs. If the block constraints were changed, create the abstract again using the create_abstract command.

---

## ABS-512

(error) Timing derate mismatch between block and top level.

**Description**

A timing derate constraint defined in the interface of an abstract is mismatched. The constraint exists either in the block or at top-level but not in both.

Feedback

### What Next

Check the constraints both at the top level and in the block and modify them to ensure that the correct constraints appear in both designs. If the block constraints were changed, create the abstract again using the create_abstract command.

---

## ABS-514

(warning) Abstract %s (ref: %s) does not have information for %s.

### Description

Relevant data for the specified mode/corner and early/late is not available for the specified block abstract.

### What Next

If you are getting this message for late (setup), please contact Synopsys.

If you are getting this message for early (hold), please check whether there are data-to-data checks in the design or timing window analysis during crosstalk analysis enabled only in top-level which internally enables the hold analysis. Activate hold at the block level and re-create the abstract view.

Data-to-data check can come from two places: non-sequential check arc of lib cell and SDC set_data_check constraint.

To find which library cell has data check, use:

foreach_in_collection lib [get_libs *] { report_lib -timing_arcs $lib }

Then search for *nonseq_* in the output. Those would be the non-sequential check arcs on the lib cells. But non-sequential check arc does not necessary mean that the instantiation is data-data check if the reference pin is a clock pin. It is only when signals reaching both pins are data signals when instantiated, the non-sequential check arc is data check.

User have two choices to resolve this warning message caused by data-data check. And the decision is based on whether data-data check can be ignored at top-level.

If data check can be ignored, then, remove set_data_check constraints if the data-data check is from SDC, or set the following app option if it is from library: *set_app_options name time.enable_non_sequential_checks value false*

If data check cannot be ignored, please go to the block level, activate hold for all the scenarios needed by the top level, and re-create the abstract view.

Check if timing window analysis during crosstalk analysis is enabled in top-level by reporting the value of application option 'time.enable_si_timing_windows'. If it is necessary but enabled only in top-level, enable it in block-level also and re-create the abstract view.

Feedback

## ABS-515

(error) Temperature mismatch between block and top level.

**Description**

A set_temperature constraint defined in the interface of an abstract is mismatched. The constraint exists either in the block or at top-level but not in both.

**What Next**

Check the constraints both at the top level and in the block and modify them to ensure that the correct constraints appear in both designs. If the block constraints were changed, create the abstract again using the create_abstract command.

## ABS-516

(error) Process label mismatch between block and top level.

**Description**

A process label constraint defined in the interface of an abstract is mismatched. The constraint exists either in the block or at top-level but not in both.

**What Next**

Check the constraints both at the top level and in the block and modify them to ensure that the correct constraints appear in both designs. If the block constraints were changed, create the abstract again using the create_abstract command.

## ABS-517

(error) Process number mismatch between block and top level.

**Description**

A process number constraint defined in the interface of an abstract is mismatched. The constraint exists either in the block or at top-level but not in both.

**What Next**

Check the constraints both at the top level and in the block and modify them to ensure that the correct constraints appear in both designs. If the block constraints were changed, create the abstract again using the create_abstract command.

## ABS-518

(error) Clock latency mismatch between block and top level.

Feedback

**Description**

A clock latency constraint defined in the interface of an abstract is mismatched. The constraint exists either in the block or at top-level but not in both.

**What Next**

Check the constraints both at the top level and in the block and modify them to ensure that the correct constraints appear in both designs. If the block constraints were changed, create the abstract again using the create_abstract command.

---

## ABS-519

(error) Clock gating check mismatch between block and top level.

**Description**

A clock gating check constraint defined in the interface of an abstract is mismatched. The constraint exists either in the block or at top-level but not in both.

**What Next**

Check the constraints both at the top level and in the block and modify them to ensure that the correct constraints appear in both designs. If the block constraints were changed, create the abstract again using the create_abstract command.

---

## ABS-520

(error) Input delay mismatch between block and top level.

**Description**

An input delay constraint defined in the interface of an abstract is mismatched. The constraint exists either in the block or at top-level but not in both.

**What Next**

Check the constraints both at the top level and in the block and modify them to ensure that the correct constraints appear in both designs. If the block constraints were changed, create the abstract again using the create_abstract command.

---

## ABS-521

(error) Output delay mismatch between block and top level.

Feedback

**Description**

An output delay constraint defined in the interface of an abstract is mismatched. The constraint exists either in the block or at top-level but not in both.

**What Next**

Check the constraints both at the top level and in the block and modify them to ensure that the correct constraints appear in both designs. If the block constraints were changed, create the abstract again using the create_abstract command.

---

## ABS-522

(error) Max time borrow mismatch between block and top level.

**Description**

An set_max_time_borrow constraint defined in the interface of an abstract is mismatched. The constraint exists either in the block or at top-level but not in both.

**What Next**

Check the constraints both at the top level and in the block and modify them to ensure that the correct constraints appear in both designs. If the block constraints were changed, create the abstract again using the create_abstract command.

---

## ABS-523

(error) Data check mismatch between block and top level.

**Description**

An set_data_check constraint defined in the interface of an abstract is mismatched. The constraint exists either in the block or at top-level but not in both.

**What Next**

Check the constraints both at the top level and in the block and modify them to ensure that the correct constraints appear in both designs. If the block constraints were changed, create the abstract again using the create_abstract command.

---

## ABS-524

(error) Annotated delay mismatch between block and top level.

Feedback

**Description**

An set_annotated_delay constraint defined in the interface of an abstract is mismatched. The constraint exists either in the block or at top-level but not in both.

**What Next**

Check the constraints both at the top level and in the block and modify them to ensure that the correct constraints appear in both designs. If the block constraints were changed, create the abstract again using the create_abstract command.

---

## ABS-525

(error) Annotated transition mismatch between block and top level.

**Description**

An set_annotated_transition constraint defined in the interface of an abstract is mismatched. The constraint exists either in the block or at top-level but not in both.

**What Next**

Check the constraints both at the top level and in the block and modify them to ensure that the correct constraints appear in both designs. If the block constraints were changed, create the abstract again using the create_abstract command.

---

## ABS-526

(error) Latch loop breaker mismatch between block and top level.

**Description**

An set_latch_loop_breaker constraint defined in the interface of an abstract is mismatched. The constraint exists either in the block or at top-level but not in both.

**What Next**

Check the constraints both at the top level and in the block and modify them to ensure that the correct constraints appear in both designs. If the block constraints were changed, create the abstract again using the create_abstract command.

---

## ABS-527

(error) Cell mode mismatch between block and top level.

Feedback

**Description**

An set_cell_mode constraint defined in the interface of an abstract is mismatched. The constraint exists either in the block or at top-level but not in both.

**What Next**

Check the constraints both at the top level and in the block and modify them to ensure that the correct constraints appear in both designs. If the block constraints were changed, create the abstract again using the create_abstract command.

---

## ABS-528

(error) AOCVM coefficient mismatch between block and top level.

**Description**

A set_aocvm_coefficient constraint defined in the interface of an abstract is mismatched. The constraint exists either in the block or at top-level but not in both.

**What Next**

Check the constraints both at the top level and in the block and modify them to ensure that the correct constraints appear in both designs. If the block constraints were changed, create the abstract again using the create_abstract command.

---

## ABS-530

(Information) Including all high-fanout interface nets in the abstract.

**Description**

This message is generated during abstract creation. It indicates that all high-fanout interface nets will be traced and included in the abstract.

This behavior can be turned off using: set_app_options -name abstract.full_interface_unfiltered -value false

**What Next**

This is an informational message. No action needed.

---

## ABS-531

(Information) Preparing fault-in information for all the cells in the block

Feedback

### Description

This message is generated during abstract creation. It indicates that for every cell in the block fault-in information will be computed and stored. The default behavior is to compute and store fault-in information only for cells that are on the interface of the block.

This behavior can be turned off using: set_app_options -name abstract.full_interface_unfiltered -value false

### What Next

This is an informational message. No action needed.

---

## ABS-532

(error) No timing constraints found for mode '%s'. Clock longest and shortest paths will not be traced.

### Description

A specific timing mode is defined, but it doesn't have complete timing constraints associated with it. The longest and shortest clock paths for this mode cannot be computed and retained.

### What Next

Make sure that timing constraints are defined for the mode and that there is an active scenario associated with it.

---

## ABS-549

(warning) Application option '%s' will be made block-scoped in the upcoming %s release.

### Description

The specified application option will be changed from global-scoped to block-scoped and persistant in the upcoming release. Block-scoped application options will work for only the particular block for which it is set against.

### What Next

Correct scripts in the upcoming release to ensure the application option is set for appropriate block.

### See Also

- set_app_options

Feedback

## ABS-550

(info) The default value of app-option %s will be changed to %s in %s.

**Description**

There is a change in default value.

**What Next**

No action necessary.

## ABS-551

(error) The valid views are "design" and "abstract".

**Description**

The valid views of the value specified are design and abstract.

**What Next**

Need to change the input view specified.

## ABS-552

(warning) Missing leaf driver and/or loads in -include_objects for global net %s. No connections will be retained.

**Description**

For any net to be included, at least one of the load and a driver pin/port needs to be provided with -include_objects option.

**What Next**

Check -include_objects option to make sure a leaf driver and a leaf load are specified for each net to be included.

**See Also**

- create_abstract

## ABS-553

(information) Keeping objects with timing constraints in the abstract.

Feedback

### Description

This message indicates that objects that have timing constraints defined on them will be kept in the placement abstract.

This could cause a substantial increase in the size of the abstract, in cases where the design has many constrained objects.

### What Next

If you want to avoid retaining objects with timing constraints in the placement abstract you can set the app option *abstract.keep_constrained_objects* to false.

set_app_options -name abstract.keep_constrained_objects -value true

### See Also

- create_abstract
- abstract.keep_constrained_objects

---

## ABS-554

(warning) Deletion of hierarchical cell %s will not be merged to design view.

### Description

Hierarchical cell deletion is not supported in abstract view. The cell will get deleted in abstract view but will not be merged back to design view.

### See Also

- merge_abstract

---

## ABS-555

(Error) Clock Trunk Planning needs to be run before using the -latency_targets estimated option in the compute_budget_constraints command.

### Description

When the *-latency_targets estimated* option is used in the *compute_budget_constraints* command, estimated clock latencies are used. Estimated clock latencies are queried from the "clock trunk planning" tool, so in order for the option to work correctly, clock trunk planning needs to run first. This error message indicates that the *-latency_targets estimated* was used without running clock trunk planning.

Feedback

**What Next**

Either run clock trunk planning, or use *-latency_targets actual*

**See Also**

- compute_budget_constraints

---

## ABS-556

(Information) Using estimated clock latencies from clock trunk planning.

**Description**

This message is generated during budgeting, when the option *-latency_targets estimated* is used. It indicates that the estimated clock latencies from clock trunk planning will be used.

**What Next**

This is an informational message. No action needed.

**See Also**

- compute_budget_constraints
- synthesize_clock_trunks

---

## ABS-560

(Information) Using a min segment percent budget value of %f.

**Description**

This message is generated during budgeting, when the option *-min_segment_percent* has been set to a positive number. It indicates that the budgeting computation will try to give to each non-fixed budget segment at least the percentage of the clock cycle that has been specified by the option.

Other constraints as well as zero-slack budgeting will be also taken into account, so the precise specified percentage might not be reached.

**What Next**

This is an informational message. No action needed.

Feedback

**See Also**

- set_budget_options
- compute_budget_constraints

---

## ABS-570

(error) Cannot specify option '-format' without also specifying option '-shell_subblocks'.

**Description**

The write_budgets command was used with the option *-format*, but without the option *-shell_subblocks*. The option *-format* affects only the creation of budget shells, so the option should only be used when budgets shells are created.

**What Next**

Please specify the blocks for which budget shells are needed, using the option *-shell_subblocks*.

**See Also**

- write_budgets

---

## ABS-571

(error) Cannot specify '-format' other than 'icc2' and 'dc'.

**Description**

In the *write_budgets* command the option *-format* was used with a value other than *icc2* or *dc*. Only the above two formats are supported by *write_budgets*.

**What Next**

Please specify the tool you need to create budget shells for by using either *icc2* or *dc* for the *-format* option.

**See Also**

- write_budgets

---

## ABS-572

(Error) Can't find installation of Library Manager %s. LEF file cannot be created.

Feedback

### Description

When creating a LEF file for the ICC2 -> DC flow, it is necessary to invoke Synopsys Library Manager. And sometimes, the Library Manager executable is installed in another directory and not with ICC2. This error indicates that icc2_shell was unable to locate your Library Manager installation.

### What Next

Please set the SYNOPSYS_LM_ROOT environment varible to point to the directory where Library Manager is installed. For example, if you set

setenv SYNOPSYS_LM_ROOT /remote/depot/lm

then the icc2_shell will expect to find a Library Manager executable at /remote/depot/lm/bin/lm_shell. It will also expect to find an executable under an machine-specific directory. For example, if you are using linux64, then you should see this executable:

/remote/depot/lm/linux64/lm/icc2_lm_exec

If you do not see this executable, then please check your installation.

You can set the SYNOPSYS_LM_ROOT environment variable in your ICC2 tcl script, if you wish. But it is often more convenient to set the environment variable in your .cshrc or at the unix prompt. If you set the environment variable in unix, you must set it before starting your icc2_shell program.

---

## ABS-573

(Information) Creating LEF with %s.

### Description

This message is generated during budgeting, when the options *-shell_subblocks* and *-format dc* have been used. Budgeting will generate LEF files to be used by DC, by running the script indicated in the message.

### What Next

This is an informational message. No action needed.

### See Also

- write_budgets

---

## ABS-574

(Error) Can't find installation of Milkyway executable%s. Milkyway libraries cannot be created.

Feedback

### Description

When creating Milkyway libraries for the ICC2 -> DC flow, it is necessary to invoke Synopsys Milkyway. This error indicates that icc2_shell was unable to locate your Milkyway installation.

### What Next

Please set the SYNOPSYS_MW_ROOT environment varible to point to the directory where Milkyway is installed. For example, if you set

setenv SYNOPSYS_MW_ROOT /remote/depot/mw

then the icc2_shell will expect to find a Milkyway executable at that location, under a machine specific directory. For example, if you are using linux64, then you should see this executable:

/remote/depot/mw/bin/linux64/Milkyway

If you do not see this executable, then please check your installation.

You can set the SYNOPSYS_MW_ROOT environment variable in your ICC2 tcl script, if you wish. But it is often more convenient to set the environment variable in your .cshrc or at the unix prompt. If you set the environment variable in unix, you must set it before starting your icc2_shell program.

---

## ABS-575

(Information) Creating Milkyway libraries with %s.

### Description

This message is generated during budgeting, when the options *-shell_subblocks* and *-format dc* have been used. Budgeting will generate Milkyway libraries to be used by DC, by running the script indicated in the message.

### What Next

This is an informational message. No action needed.

### See Also

- write_budgets

---

## ABS-576

(Error) Failed to create %s. Cannot complete file generation for the ICC2->DC budget shell flow.

Feedback

### Description

Budgeting tried to create all the files needed for the ICC2 -> DC budget shell flow but was not successful.

### What Next

Please check that the output directory given in the command *write_budgets* has the correct write permisions and that there is enough disk space. Also check that the environment variables *SYNOPSYS_LM_ROOT* and *SYNOPSYS_MW_ROOT* have been set to point to the correct executable location.

### See Also

- write_budgets

---

## ABS-577

(Information) Writing DC design loading wrapper script at %s.

### Description

This message is generated during budgeting, when the options *-shell_subblocks* and *-format dc* have been used. Budgeting will generate a wrapper script to help load the design in DC and save it in the indicated location.

### What Next

This is an informational message. No action needed.

### See Also

- write_budgets

---

## ABS-578

(Error) Can't find installation of Library Compiler%s given by SYNOPSYS_LC_ROOT. Budget shell cannot be created.

### Description

When creating a budget shell, it is necessary to invoke Synopsys Library Compiler. And often, the Library Compiler executable is installed in another directory and not with ICC2. This error indicates that icc2_shell was unable to locate your Library Compiler installation. Specifically, the variable SYNOPSYS_LC_ROOT was set to point to a location that did not contain an installation of Library Compiler.

Feedback

**What Next**

Please set the SYNOPSYS_LC_ROOT environment varible to point to the directory where Library Compiler is installed. For example, if you set

setenv SYNOPSYS_LC_ROOT /remote/depot/lc

then the icc2_shell will expect to find a library compiler executable at /remote/depot/lc/bin/lc_shell. It will also expect to find an executable under an machine-specific directory. For example, if you are using linux64, then you should see this executable:

/remote/depot/lc/linux64/lc/lc_shell_exec

If you do not see this executable, then please check your installation.

You can set the SYNOPSYS_LC_ROOT environment variable in your ICC2 tcl script, if you wish. But it is often more convenient to set the environment variable in your .cshrc or at the unix prompt. If you set the environment variable in unix, you must set it before starting your icc2_shell program.

---

## ABS-579

(Error) Can't find installation of Library Compiler. Please set SYNOPSYS_LC_ROOT.

**Description**

When creating a budget shell, it is necessary to invoke Synopsys Library Compiler. And often, the Library Compiler executable is installed in another directory and not with ICC2. This error indicates that icc2_shell was unable to locate your Library Compiler installation.

**What Next**

Please set the SYNOPSYS_LC_ROOT environment varible to point to the directory where Library Compiler is installed. For example, if you set

setenv SYNOPSYS_LC_ROOT /remote/depot/lc

then the icc2_shell will expect to find a library compiler executable at /remote/depot/lc/bin/lc_shell. It will also expect to find an executable under an machine-specific directory. For example, if you are using linux64, then you should see this executable:

/remote/depot/lc/linux64/lc/lc_shell_exec

If you do not see this executable, then please check your installation.

You can set the SYNOPSYS_LC_ROOT environment variable in your ICC2 tcl script, if you wish. But it is often more convenient to set the environment variable in your .cshrc or at the unix prompt. If you set the environment variable in unix, you must set it before starting your icc2_shell program.

Feedback

## ABS-580

(Error) Can't find installation of Library Manager%s as given by SYNOPSYS_LM_ROOT. LEF file cannot be created.

### Description

When creating a LEF file for the ICC2 -> DC flow, it is necessary to invoke Synopsys Library Manager. And sometimes, the Library Manager executable is installed in another directory and not with ICC2. This error indicates that icc2_shell was unable to locate your Library Manager installation. Specifically, the variable SYNOPSYS_LM_ROOT was set to point to a location that did not contain an installation of Library Manager.

### What Next

Please set the SYNOPSYS_LM_ROOT environment varible to point to the directory where Library Manager is installed. For example, if you set

setenv SYNOPSYS_LM_ROOT /remote/depot/lm

then the icc2_shell will expect to find a Library Manager executable at /remote/depot/lm/bin/lm_shell. It will also expect to find an executable under an machine-specific directory. For example, if you are using linux64, then you should see this executable:

/remote/depot/lm/linux64/lm/icc2_lm_exec

If you do not see this executable, then please check your installation.

You can set the SYNOPSYS_LM_ROOT environment variable in your ICC2 tcl script, if you wish. But it is often more convenient to set the environment variable in your .cshrc or at the unix prompt. If you set the environment variable in unix, you must set it before starting your icc2_shell program.

## ABS-581

(Error) Can't find installation of Library Manager. Please set SYNOPSYS_LM_ROOT.

### Description

When creating a LEF file for the ICC2 -> DC flow, it is necessary to invoke Synopsys Library Manager. And sometimes, the Library Manager executable is installed in another directory and not with ICC2. This error indicates that icc2_shell was unable to locate your Library Manager installation.

### What Next

Please set the SYNOPSYS_LM_ROOT environment varible to point to the directory where Library Manager is installed. For example, if you set

Feedback

setenv SYNOPSYS_LM_ROOT /remote/depot/lm

then the icc2_shell will expect to find a Library Manager executable at /remote/depot/lm/bin/lm_shell. It will also expect to find an executable under an machine-specific directory. For example, if you are using linux64, then you should see this executable:

/remote/depot/lm/linux64/lm/icc2_lm_exec

If you do not see this executable, then please check your installation.

You can set the SYNOPSYS_LM_ROOT environment variable in your ICC2 tcl script, if you wish. But it is often more convenient to set the environment variable in your .cshrc or at the unix prompt. If you set the environment variable in unix, you must set it before starting your icc2_shell program.

---

## ABS-582

(Error) Can't find installation of Milkyway executable%s given by SYNOPSYS_MW_ROOT. Milkyway libraries cannot be created.

**Description**

When creating Milkyway libraries for the ICC2 -> DC flow, it is necessary to invoke Synopsys Milkyway. This error indicates that icc2_shell was unable to locate your Milkyway installation. Specifically, the variable SYNOPSYS_MW_ROOT was set to point to a location that did not contain an installation of Milkyway.

**What Next**

Please set the SYNOPSYS_MW_ROOT environment varible to point to the directory where Milkyway is installed. For example, if you set

setenv SYNOPSYS_MW_ROOT /remote/depot/mw

then the icc2_shell will expect to find a Milkyway executable at that location, under a machine specific directory. For example, if you are using linux64, then you should see this executable:

/remote/depot/mw/bin/linux64/Milkyway

If you do not see this executable, then please check your installation.

You can set the SYNOPSYS_MW_ROOT environment variable in your ICC2 tcl script, if you wish. But it is often more convenient to set the environment variable in your .cshrc or at the unix prompt. If you set the environment variable in unix, you must set it before starting your icc2_shell program.

Feedback

## ABS-583

(Error) Can't find installation of Milkyway executable. Please set SYNOPSYS_MW_ROOT.

### Description

When creating Milkyway libraries for the ICC2 -> DC flow, it is necessary to invoke Synopsys Milkyway. This error indicates that icc2_shell was unable to locate your Milkyway installation.

### What Next

Please set the SYNOPSYS_MW_ROOT environment varible to point to the directory where Milkyway is installed. For example, if you set

setenv SYNOPSYS_MW_ROOT /remote/depot/mw

then the icc2_shell will expect to find a Milkyway executable at that location, under a machine specific directory. For example, if you are using linux64, then you should see this executable:

/remote/depot/mw/bin/linux64/Milkyway

If you do not see this executable, then please check your installation.

You can set the SYNOPSYS_MW_ROOT environment variable in your ICC2 tcl script, if you wish. But it is often more convenient to set the environment variable in your .cshrc or at the unix prompt. If you set the environment variable in unix, you must set it before starting your icc2_shell program.

## ABS-584

(warning) clocks associated with instance %s for the block %s will not be considered for create_clock

### Description

In case of multi instantiated block ,the split_constraint flow will take the first instance defined in set_budget_options as the reference for clock.If different clocks say clk and clk2 (create_clock at top at different port) reach instances of the block (inst1 and inst2) then first instance (inst1) of the block which has clk reaching it will be used to write the block level clock (create_clock -name {clk} [get_ports {clk}]) The other clk2 clock will be defined as virtual clock at the block level.

### What Next

if you want to change the clock defined in the block level according to the instance 2 then define that instance prior in the list of set_budget_options.

Feedback

## ABS-585

(warning) Pin '%s' of design '%s' is driven by both zero and one.

**Description**

When creating budgets, top-level constant values usually cause a set_case_analysis statement to be set for input ports of the budgeted block.

But in this case, the block design has been multiply instantiated. And some of the connections to the given port have a constant zero and others have a constant one. As a result, no set_case_analysis has been applied for this port.

**What Next**

If you want a set_case_analysis statement to appear in the budget, you must make sure that all instances of the design are driven by the same constant.

## ABS-586

(warning) set_case_analysis constraint is missing for '%s' instance.

**Description**

When creating budgets, top-level constant values usually cause a set_case_analysis statement to be set for input ports of the budgeted block.

But in this case, the block design has been multiply instantiated. And some of the MIM instances are not considered for set_case_analysis.

**What Next**

If you want a set_case_analysis statement to appear in the budget, you must make sure that all instances of the design are included for set_case_analysis and are driven by the same constant.

## ABS-587

(warning) The value provided is out of range. Using %d as the default percentage value for computation.

**Description**

The percent value cannot be greater than 100 or lesser than zero.

**What Next**

Need to change the percent value. Entered value is either greater than 100 or lesser than zero.

Feedback

## ABS-588

(warning) Options -internal_percent and -logic_depth are specified together. Calculating the delay on the basis of logic_depth.

**Description**

Options -internal_percent and -logic_depth cannot be specified together.

**What Next**

Need to specify either -internal_percent or -logic_depth option for split_constraints command.

## ABS-589

(warning) Option %s should be specified with two percentage values, an input delay percentage and an output delay percentage.

**Description**

This option should be specified with two percentage values, the input delay percentage and the output delay percentage.

**What Next**

Need to specify two percentage values with this option for split_constraints command.

## ABS-590

(error) Block level virtual clock '%s' cannot be mapped top level clocks for mode '%s'. Relevant exceptions are dropped.

**Description**

There are no top level clocks with the correct waveform that propagate to the I/O delay pin set wrt to this virtual clock.

**What Next**

You need to set the association manually using the set_block_to_top_map command.

## ABS-591

(Error) Option %s should not be specified without option(s) %s.

Feedback

**Description**

The first option should not be specified without the second option(s).

**What Next**

Need to specify the first option together with the second option(s) for split_constraints command.

---

## ABS-592

(Warning) Abstract for the block %s cannot be created since the block or it's parent block is read only.

**Description**

You get this error message because you are creating abstract for a block that is read-only or it's parent block is a read only block.

**What Next**

Please check the write permission of the block/parent block on disk. If the design has no disk write permission, you can use *save_block -as* to save the design to another design and continue with the operation.

If the design has disk write permission, please check if the block and it's parent block have their editability set as true. This can be done using *set_editability* command.

---

## ABS-593

(Warning) The setting %s will cause negative constraints. Please refer to the file "segment_budget_reset.tcl" for segment budget constraints removal.

**Description**

When the write_budgets command is used with -check_segment_constraints option, any segment budgets given by the user utilizing the set_segment_budget_constraint commands are reviewed while writing out the budgets. If any of them are too large, causing negative *set_input_delay* or *set_output_delay* constraints, this warning is generated.

Also, a command that can be used to remove this user constraint is generated and written out in the file segment_budget_reset.tcl in the budget output directory.

**What Next**

Please examine the file segment_budget_reset.tcl in the budget output directory and use the commands in the file to remove the segment budgets that are not necessary.

Feedback

## ABS-594

(Warning) Budget margins set using the %s option are not supported for hold budgeting.

**Description**

The command *set_budget_margins* can be used to add margins to budgets computed for specific blocks. The tool supports defining the margin as a percentage of the allowable delay on the budgeted path, using the options *-launch_percent* and *-capture_percent*. This capability works only for setup budgeting. Margins defined in this way will not affect the budgets in hold budgeting.

This warning is given when a percent-based margin for hold budgeting is considered, to notify the user that it will not be taken into account.

**What Next**

Please use *-setup* together with the *-launch_percent* and *-capture_percent* options of the *set_budget_margins* command to avoid getting this warning.

## ABS-596

(warning) Option %s is specified with more than two percentage values. Only the first two values are taken into consideration.

**Description**

This option should be specified with only two percentage values, the input delay percentage and the output delay percentage. When more than two values are specified with this option then only the first two values would be considered.

**What Next**

Need to specify only two percentage values with this option for split_constraints command.

## ABS-597

(Error) Constraint promotion may only be performed on immediate (direct children) blocks. Block '%s' is not a direct child of top.

**Description**

The given block is not a direct child of the top. Constraint promotion can only be performed on immediate blocks which are direct children of the top.

**What Next**

Need to specify a block which is a direct child of the top for promotion.

Feedback

## ABS-598

(error) Cannot open Block clock Mapping file %s .

### Description

This message indicates that the tool cannot open the mentioned file for the specified operation.

### What Next

Please check file location and directory permissions.

## ABS-600

(Warning) Input/output delay constraints for budget pin %s create an exception precedence problem. Block-level timing through the pin will not correlate with top-level timing.

### Description

When creating constraints for the pins of the block, budgeting checks if the constraint will change the precedence of exceptions associated with the pin. If the input/output delay written out would change the exception precedence and make the timing more pessimistic, the input/output delay constraint is dropped by default.

The user can change the default behavior by setting the app option *plan.budget.output_bad_precedence_delay* to true. When this app option is true and an input/output delay that creates pessimistic timing is written out, this warning is printed.

### What Next

If you want to get rid of this message, you can do one of the following:

1. Change the value of the app option *plan.budget.output_bad_precedence_delay* to false.

   ```
   set_app_options -name plan.budget.output_bad_precedence_delay \\
          -value false
   ```

2. Change your constraints so they are consistent. For example, if you have a path across a block boundary with two exceptions:

   ```
   set_multicycle_path 3 -from reg1/CP
   set_multicycle_path 2 -to block/reg2/D
   ```

   then you will trigger this warning. If you use "set_multicycle_path 2" for both constraints, there will be no warning, because the constraint remains the same.

Feedback

## ABS-601

(Error) In mode '%s', the node is invalid at connection source '%s' of the block clock '%s' for the instance '%s'.

**Description**

The node corresponding to the block clock's connection source is invalid. Top clock points won't be written out.

**What Next**

Please review the block and top designs, block clock's connection source.

## ABS-602

(Information) Instance : '%s' Mode : '%s' Possible top clock port for block clock '%s' create_clock -name <fill> -period <period> -waveform <fill> [get_ports '%s']

**Description**

This is a probable top clock port for the unmapped block clock.

**What Next**

Please review the top clock points and create clocks if required.

## ABS-603

(Information) Instance : '%s' Mode : '%s' Possible top generated clock point for block clock '%s' create_generated_clock -name <fill> -source '%s' -divide_by <fill> [get_pins '%s']

**Description**

This is a probable top generated clock point for the unmapped block clock.

**What Next**

Please review the top clock points and create clocks if required.

## ABS-610

(warning) Cannot optimize budgets for pin '%s'.

**Description**

When creating budgets, a cost computation is used to decide the optimal value for the budgets allocated to both sides of each pin.

Feedback

But in this case, the cost function could not be computed accurately and the budgets for some paths through the pin might not be optimal.

**What Next**

Check the timing through the indicated pin to verify that it is properly constrained.

---

## ABS-611

(Information) Restoring the reg2reg timing paths across lower level sub-blocks previously ignored by command set_timing_paths_disabled_blocks.

**Description**

Budgeting restores the reg2reg timing paths across lower level sub-blocks by running set_app_options -list {time.ignore_sub_block_internal_paths_use_common_ancestor false} internally

**What Next**

No action is required. The block internal timing paths that cross lower level blocks can be disabled again after budgeting by running the command

set_app_options -list {time.ignore_sub_block_internal_paths_use_common_ancestor true}

---

## ABS-612

(Information) time.ignore_sub_block_internal_paths is set to true.

**Description**

The design has time.ignore_sub_block_internal_paths set to true while running budgeting.

**What Next**

No action is required. If reseting the value of the app option is needed, the command

set_app_options -name time.ignore_sub_block_internal_paths -value false

can be used.

---

## ABS-613

(Warning) budgeting with time.ignore_sub_block_internal_paths set to true is not recomended for MPH designs.

Feedback

**Description**

While running budgeting commands, it was detected that the app option time.ignore_sub_block_internal_paths is true and the design has multiple levels of physical hierarchy. This combination is not recomended, because setting time.ignore_sub_block_internal_paths to true might hide some interface paths of lower level blocks.

**What Next**

You can change the app option value using the command:

set_app_options -name time.ignore_sub_block_internal_paths -value false

and run budgeting again.

---

## ABS-614

(Error) No budget segments found '%s'.

**Description**

While running the report_budget -segment_delay command, no segments were specified by the options used (-from, -to or -segment). No report can be generated.

**What Next**

Review the options used in the command to ensure that at least one segment is specified.

---

## ABS-700

(information) Top-level clock '%s' mapped with block level clock '%s' in mode '%s' for block instance '%s'. The list of other top-level clocks reaching the same block-pin include: %s.

**Description**

There are multiple top level clocks that propagate to the given pin. If one of the top-level clocks has same name as block clock, mapping happens with that top-level clock. The rest of the top-level clocks reaching the block pin will not be mapped to the block-level clock.

---

## ABS-701

(warning) Top-level clock '%s' mapped with block level clock '%s' in mode '%s' for block instance '%s'. The list of other top-level clocks reaching the same block-pin include: %s.

Feedback

**Description**

There are multiple top level clocks that propagate to the given pin. If one of the top-level clocks has same name as block clock, mapping happens with that top-level clock. The rest of the top-level clocks reaching the block pin will not be mapped to the block-level clock.

---

## ABS-702

(warning) In mode %s, multiple block clocks are defined for block pin %s. The list of clocks: %s.

**Description**

Multiple block clocks are defined on the same block pin to which a top-level clock is propagated. The tool maps all the block clocks on the block pin to the top clock irrespective of the name.

**What Next**

All the block level clocks defined on the block pin are listed. User can set the constraints manually.

---

## ABS-703

(Error) Top-level %s '%s' could not be mapped to any block %s for block instance '%s'.

**Description**

Top-level mode/corner is not mapped to a block-level mode/corner. Promotion will not happen for these top-level objects.

**What Next**

Use set_block_to_top_map to set the mapping.

---

## ABS-704

(Error) Block-level clock '%s' could not be mapped to any top-level clock in mode '%s' for block instance '%s'.

**Description**

Block-level clock is not mapped to a top-level clock. Promotion of constraints from this block clock will not happen.

**What Next**

Use set_block_to_top_map to set the mapping.

Feedback

## ABS-705

(warning) Top-level clock '%s' is not mapped to any block-level clock in any mode.

**Description**

Given top-level clock is not mapped to a block-level clock in any mode. Promotion will not happen for this top-level clock.

**What Next**

Use set_block_to_top_map to set the mapping.

## ABS-706

(warning) Skipped reason reporting for %d %s objects.

**Description**

Reason(s) for inclusion to abstract is stored only for abstract and non-hierarchical netlists. If reason is requested for a non-abstract or hierarchical object, the object is skipped.

**What Next**

Plese provide only abstract and non-hierarchical objects for reporting reasons.

## ABS-707

(Error) The option '%s' is obsoleted and will be removed in subsequent release. %s

**Description**

The given command option is obsoleted. Please see the man-page of the command for migration assistance.

**What Next**

Please change the setting of your app option, or do not use this command option.

## ABS-708

(Error) The option '%s' is obsoleted and will be removed in subsequent release. Since, the app option "plan.budget.estimate_timing_mode" and the option '%s' are conflicting to each other, compute_budget_constraints command will error out. Please set app-option correctly to continue.

Feedback

**Description**

The command option and the app option "plan.budget.estimate_timing_mode" value are conflicting. Please see the man-page of the command for migration assistance.

**What Next**

Please check the command option and the app option "plan.budget.estimate_timing_mode" value.

---

## ABS-709

(warning) The multi_clock_mapping has been enabled and '%s' flow is disabled under N-1 clock mapping setting.

**Description**

The promote_constraints command supports N-1 clock mapping, where N top clocks are mapped to a single block top. However the budgetting flow does not support the sdcs generated from N-1 promotion flow.

**What Next**

Use report_block_to_top to check the mappings.

---

## ABS-710

(Error) Instance %s is not a budget block.

**Description**

The command set_budget_options -add_master_blocks was used to define the master of an MIB family for split_constraints and budgeting, but one of the blocks in the command is not a budget block.

**What Next**

Please use the command set_budget_options -add_blocks to add the missing block to the budgeted blocks. Or check the command option and remove any blocks that are not defined as budget blocks.

---

## ABS-711

(Information) Instance %s is already defined as a primary block.

Feedback

**Description**

The command set_budget_options -add_primary_blocks was used to define the primary of an MIB family for split_constraints and budgeting, but one of the blocks in the command has alreadt be defined as a primary block.

**What Next**

This is only an information message, no further action is needed.

---

## ABS-712

(Error) Instance %s with reference %s cannot be a primary block, a primary instance for this reference has already been defined.

**Description**

The command set_budget_options -add_primary_blocks was used to define the primary of an MIB family for split_constraints and budgeting, but one of the blocks in the command belongs to an MIB family that has already a primary defined.

**What Next**

In order to define a new primary for this MIB family, please use the command set_budget_options -remove_primary_blocks first, to remove the already defined primary block.

---

## ABS-713

(Error) Two instances with the same reference %s are defined as primary.

**Description**

The command set_budget_options -add_primary_blocks was used to define the primary of an MIB family for split_constraints and budgeting, but two of the blocks in the command belong to the same MIB family.

**What Next**

Please check the instances in the command and select only one from each MIB family to be the primary instance.

---

## ABS-714

(Error) Instance %s is not a primary block.

Feedback

**Description**

The command set_budget_options -remove_primary_blocks was used to remove a primary block definition of an MIB family for split_constraints and budgeting, but the specified instance was not previously defined as a primary.

**What Next**

Please check the instances in the command and and use only the instances that have been defined as a primary instance.

---

## ABS-715

(Information) Selecting primary instance %s.

**Description**

The command split_constraints chose the indicated instance to be the primary for an MIB family.

**What Next**

This is only an information message, no further action is needed. If another instance is prefered as the primary, please set it explicitly, by using the command *set_budget_options -add_primary_blocks [instance]* before running the split_constraints command.

---

## ABS-716

(Information) Disabling the physical mode of abstract creation since the flow is non-physical.

**Description**

The flow is non-physical and hence switching to non-physical mode of abstract creation.

**What Next**

This is an informational message. No action needed.

---

## ABS-720

(Information) split_constraints - %s.

**Description**

The command split_constraints is currently performing the operation indicated.

Feedback

**What Next**

This is only an information message, no further action is needed.

---

## ABS-721

(Information) split_constraints - %s.

**Description**

The command split_constraints is currently writing out mode constraints for the indicated block or for the top-level

**What Next**

This is only an information message, no further action is needed.

---

## ABS-722

(Information) split_constraints - %s.

**Description**

The command split_constraints is currently writing out corner constraints for the indicated block or for the top-level.

**What Next**

This is only an information message, no further action is needed.

---

## ABS-723

(Information) split_constraints - %s.

**Description**

The command split_constraints is currently writing out scenario constraints for the indicated block or for the top-level

**What Next**

This is only an information message, no further action is needed.

---

## ABS-724

(Information) split_constraints - %s.

Feedback

**Description**

The command split_constraints is currently processing the specified type of constraints and writing the result to the output files.

**What Next**

This is only an information message, no further action is needed.

### ABS-725

(information) clocks associated with instance %s for the block %s will be considered for create_clock

**Description**

In case of multi instantiated block ,the split_constraint flow will take the first instance defined in set_budget_options as the reference for clock.If different clocks say clk and clk2 (create_clock at top at different port) reach instances of the block (inst1 and inst2) then the first instance (inst1) of the block which has clk reaching it will be used to write the block level clock (create_clock -name {clk} [get_ports {clk}]) The other clk2 clock will be defined as virtual clock at the block level.

**What Next**

If you want to change the clock defined in the block level and use the clocks associated with instance 2, then define that instance prior in the list of set_budget_options.

## ACG

### ACG-001

(error) %s.

**Description**

Analog constraint groups related error message. The message is self explanatory allowing proper actions to be taken.

## ADES

### ADES-021

(info) Running user-defined checker : %s.

Feedback

**Description**

This status message is printed during *check_constraints*, while running user-defined checkers. It shows the currently running user-defined checker.

# AFP

## AFP-1000

(warn) Warning: %s

**Description**

The printed constraint value is not a valid value.

**See Also**

- report_auto_floorplan_constraints

## AFP-2000

(error) No floorplan and floorplan initialization is disabled.

**Description**

You have no design floorplan and have the app option to create die turned off.

**What Next**

Please set the app option compile.auto_floorplan.initialize to true or auto

## AFP-2001

(error) Some macros are macros unplaced and automatic macro placement turned off.

**Description**

You have unplaced macros and have the app option to place macros turned off.

**What Next**

Please set the app option compile.auto_floorplan.place_hard_macros to true or auto

## AFP-2002

(error) Some pins are unplaced and automatic pin placement turned off.

Feedback

**Description**

You have some uplaced pins in the floorplan and have the app option to place ports/pins to false.

**What Next**

Please set the app option compile.auto_floorplan.place_pins to true or auto

---

## AFP-2003

(error) Some io pads are unplaced and automatic IO placement turned off.

**Description**

You have some uplaced pads in the floorplan and have the app option to place IO's to false.

**What Next**

Please set the app option compile.auto_floorplan.place_ios to true or auto

---

## AFP-2004

(info) Removing placement from placed pins

**Description**

Based on your usage the existing pin placement is removed.

---

## AFP-2005

(info) Removing placement info from Macros

**Description**

Based on your usage the existing macro placement is removed.

---

## AFP-2007

(info) Some macros have placement honoring them for macro placement

**Description**

Based on your usage the existing macro placement is kept.

Feedback

## AFP-2025

(info) Detected floorplan rule violations. Use check_floorplan_rules more details.

### Description

Autofloorplan detected floorplan rule violations. Use check_floorplan_rules -all to get more details on the violations. report_floorplan_rules can provide more details on existing floorplan rules.

### What Next

fix_floorplan_rules can help fix violations in the floorplan.

### See Also

- check_floorplan_rules
- fix_floorplan_rules
- report_floorplan_rules

## AFP-2026

(info) Sourcing track script (%s) after creating floorplan

### Description

Specified script is sourced after creating the floorplan.

### See Also

- report_auto_floorplan_constraints
- set_auto_floorplan_constraints

## AFP-2027

(warning) Auto floorplan constraint specified track script '%s' doesn't exist. Ignoring the option.

### Description

Specified track script either doesn't exist or not readable.

Feedback

**See Also**

- report_auto_floorplan_constraints
- set_auto_floorplan_constraints

---

## AFP-2028

(warning) Track script is not specified. Creating default tracks.

**Description**

set_auto_floorplan_constraints -track_script is not specified. Autofllorplan will create default tracks with default offset and pitch based on technology file information. The created default tracks may not match the advanced node technology and library specific track requirements.

**What Next**

Specify track script to create technology library compatible tracks

**See Also**

- report_auto_floorplan_constraints
- set_auto_floorplan_constraints

---

## AFP-2098

(warning) Auto-Floorplan is disabled in RTS incremental run.

**Description**

Auto-Floorplan is disabled in incremental synthesis. Unplaced macros could be present.

**What Next**

Please fix standard cells/macros from initial synthesis and use create_placement -floorplan to place newly added macros

---

## AFP-2099

(warning) Unplaced macros detected, either provide location post incremental run or run initial run with changed RTL.

**Description**

Auto-Floorplan is disabled in incremental synthesis. Unplaced macros are detected.

Feedback

**What Next**

Please either provide location post incremental run or run initial run with changed RTL.

---

## AFP-4000

(error) Error: %s

**Description**

The specified constraint value is not a valid value.

**See Also**

- set_auto_floorplan_constraints

---

# AIF

---

## AIF-001

(error) Design view name '%s' is not supported by %s. View name should be 'design'.

**Description**

Only design view is supported in AIF read/write.

---

## AIF-002

(error) Incomplete/incorrect physical data : %s.

**Description**

Physical data might not have been read in or the data is incorrect like no bounding box, invalid pin shape, etc. Output error message provides more details.

**What Next**

open_block or report_attribute commands can be executed before read_aif to clear the error which is caused by physical data haven't been read.

---

## AIF-003

(warn) Incomplete/incorrect physical data : %s.

Feedback

**Description**

Physical data might not have been read in or the data is incorrect like no bounding box for bump instances. These bump instances will be skipped.

---

## AIF-004

(error) Missing connectivity : %s.

**Description**

Net or port is missing or the connectivity between them or the association with a cell is missing. Output error message provides more details.

---

## AIF-005

(warn) Missing connectivity : %s.

**Description**

Port is missing or there are no connections for bump instance. Output warning message provides more details.

---

## AIF-006

(information) There are no bump cell instances to be put in AIF.

**Description**

No bump cell instances are present in the design.

---

## AIF-007

(warn) Input AIF '%s' was generated with -hierarchy. It will not be read in.

**Description**

AIF generated with -hierarchy option is not read back in a design.

---

## AIF-008

(error) Syntax error : %s.

**Description**

Refers to syntax errors in input AIF like fields missing, wrong values, wrong types, etc. or in input string -pad_to_ref_list. Output error message provides more details.

Feedback

## AIF-009

(information) Number of entries is 2 in value {%s} given with -pad_to_ref_list. Assuming orientation to be N.

**Description**

Pattern given in -pad_to_ref_list switch must be of the form {{str1 str2 str3} {...} ...}. str1 is reference bump cell in AIF, str2 is target reference bump cell and str3 is target orientation and is optional. Default target orientation is N.

## AIF-010

(error) Reference library error : %s.

**Description**

Reference library cell is not found, is not of bump cell type or the instance is unbound.

## AIF-011

(error) Can't create bump cell instance '%s'.

**Description**

Bump cell instance couldn't be created.

## AIF-012

(warn) Bound library bump cell '%s' for instance '%s' doesn't match reference library bump cell '%s' given in AIF.

**Description**

Bump instance is bound to a different reference library bump cell than what is given in the input AIF.

## AIF-013

(warn) Cannot find parent cell for string '%s' given in AIF.

**Description**

Check the parent cell is available or not for the string given in the input AIF. Only when the parnt cell is available, the string-specific object is legal.

Feedback

# ALCP

## ALCP-001

(Warning) You did not select all lib cells from some variant groups, so the variant pass rate reported by analyze_lib_cell_placement may be inaccurate.

### Description

The lib cells that you passed to *analyze_lib_cell_placement* partially covered one or more variant groups. This might lead to false variant group pass rate output. Any line in the ALCP report which has ** next to the variant pass rate indicates a lib cell that is part of an incomplete group.

### What Next

Pass all lib cells in a variant group to *analyze_lib_cell_placement* to get correct variant pass rate.

## ALCP-002

(Warning) lib cell %s failed at {%g %g}:%s

### Description

When using *analyze_lib_cell_placement* with the *-verbose* option, a lib cell failed to be legalizable in any of the tested orientations. The lib cell name, the origin of the test site, and the reason are given. The reason may be a single rule that applies to all orientations, like "Basic Rules" or "pin_color_align", or it may be a string of reasons, one per orientation, like "R0 Basic Rules; MX legal_orient;" and so on.

### What Next

In addition to the message, the application wrote a Tcl script file per lib cell with failures which when sourced, annotates the GUI with shapes to represent each failure. You can use this feature to further analyze the failures.

## ALCP-003

(Warning) No legal site row for reference %s/%s

### Description

When using *analyze_lib_cell_placement*, you passed in a lib cell (reference) which requires a specific site row (from the *site_name* attribute on the lib cell). There are no site rows in the design matching the requirement.

Feedback

**What Next**

If you are expecting to use this lib cell in your design, you need to ensure that the floorplan has site rows for it.

---

## ALCP-004

(Warning) No legal sites for reference %s in region {{%g %g} {%g %g}}

**Description**

When using *analyze_lib_cell_placement*, you passed in a lib cell (reference) which requires a specific site row (from the *site_name* attribute on the lib cell). There are site rows in the design matching the requirement, but you have most likely used *-region* to narrow the focus for the command, and the area you selected does not have any of the required sites.

**What Next**

For the given reference, use a different region, or none.

---

## ALCP-005

(Warning) No frame view found for design %s

**Description**

When using *analyze_lib_cell_placement*, you passed in a lib cell (reference) which does not have a frame view. The lib cell is ignored.

**What Next**

Only pass lib cells with frame views to this command.

---

## ALCP-006

(Warning) No library cells are available to analyze

**Description**

When using *analyze_lib_cell_placement*, all of the lib cells you passed in were unable to be analyzed for various reasons like no legal site rows, no legal sites, and so on.

**What Next**

Choose a different set of lib cells, or use a wider region

Feedback

## ALCP-007

(Information) The "pass_rate" attribute has been created on %d lib cells processed by analyze_lib_cell_placement. The attribute is not persistent

**Description**

When *analyze_lib_cell_placement* completes, it sets a "pass_rate" attribute on each lib cell which was analyzed. The attribute lasts throughout the session, but it is not saved.

# APPOPT

## APPOPT-001

(Error) Value %s for %s is not in range: must be %s.

**Description**

The specified numeric value is not in the range. Ranges are either within a minimum and maximum value, greater than or equal to a minimum value, or less or equal to a maximum value. This message text indicates violation of range constraint.

**What Next**

Enter an appropriate value for the app-option. In case of typed_value for physical quantites, please qualify the numeric value with SI prefix and unit. Example: 10nm is the typed_value for length that represents the physical quantity 1.0e-8 meter, where "10" is the numeric value, 'n' is the SI prefix for "nano" and 'm' is the symbol of SI unit of length meter. Similarly, 34.56kOhm is the typed_value for resistance that represents the physical quantity 3.456e4 Ohm, where "34.56" is the numeric value, 'k' is the SI prefix for "kilo" and "Ohm" is the symbol of SI unit of resistance Ohm.

## APPOPT-002

(Error) Value %s for %s is not valid. Specify one of: %s.

**Description**

The specified string value is not one of the valid strings defined for the app-option. This message text indicates the allowable values.

**What Next**

Enter an appropriate value for the app-option.

Feedback

### APPOPT-003

(Information) Value %s for %s is ignored. Reason: %s.

**Description**

The specified string value is ignored and has no effect because of the reason is that is mentioned. This message is issued when the functionality you are trying to get with the setting is already available with another feature.

**What Next**

Should you need this setting to take effect, see if there is any other way to achieve this based on the reasoning mentioned in the message.

### APPOPT-004

(Information) The two application options "design.disable_auto_link" and "gui.auto_link_blocks" are linked together, which means changing the value of one will update the value of the other.

**Description**

The specific application option has been made deprecated and will hence be removed in a future release. The man page of the particular application option will have details about the required migration activity.

**What Next**

See the man page for this application option for migration assistance.

## APS

### APS-001

(error) No scenario with setup timing found

**Description**

The current design does not contain any valid scenario with setup timing. To run estimate_timing atleats one one such scenario much exist.

**What Next**

Create a scenario with proper setup timing.

Feedback

**See Also**

- create_scenario
- create_mode
- report_scenarios

---

## APS-002

(error) Cannot find current design and/or its hierarchy

**Description**

The tool cannot find any valid design and/or hierarchy. To run estimate_timing a valid design and/or hierarchy must exist.

**What Next**

Load a valid design.

---

## APS-003

(error) Cannot find hierachical root of current design

**Description**

The tool cannot find the hierarchical root for the current design.

**What Next**

Load a valid design that has a proper hierarchical root.

---

## APS-004

(error) Cannot find timing library for current design

**Description**

The tool cannot find any timing library in the libraries loaded.

**What Next**

Make sure the libraries in the design contains atleast one timing library.

Feedback

## APS-005

(error) Cannot create estimated_corner because current_corner is the default corner without user settings.

### Description

The tool has detected only one corner which is also the default one. Since the default corner does not have any manual setting, estimate_timing cannot be run on this corner.

### What Next

Create a corner containig manual settings and this will override the default corner.

### See Also

- create_corner

## APS-006

(error) Cannot find block design for current module

### Description

The tool cannot find any blocks in the current module.

### What Next

Create a block inside the current module to run estimate_timing.

## APS-007

(Information) Top %d layers are not good for layer optimization.

### Description

User will see this message because the tool finds the average unit length delay of user selected top layers is equal to or greater than average unit length delay of all layers. So there will be no benefit to do layer promotion in estimate_timing on this design even if layer promotion is on.

### What Next

User does not need to take any action. If user still wants layer promotion happens in estimate_timing, please try to set another promoted layers count.

Feedback

**See Also**

- estimate_timing
- plan.estimate_timing.optimize_layers
- plan.estimate_timing.optimize_layers_count

---

## APS-008

(error) No scenario enabled for setup analysis

**Description**

To run estimate_timing, there must exist a scenario enabled for setup analysis.

**What Next**

Create a setup scenario, then rerun estimate_timing.

**See Also**

- create_scenario
- set_scenario_status
- report_scenarios

---

## APS-009

(error) A user defined estimated_corner already exists

**Description**

To run estimate_timing, the user defined estimated_corner should be deleted.

**What Next**

Delete estimated_corner, then rerun estimate_timing.

**See Also**

- create_scenario
- set_scenario_status
- report_scenarios

Feedback

## APS-010

(warning) %s will be made obsolete and removed from future releases starting with the %s release.

**Description**

This is a warning message indicating the specified command, option or variable is in the process of being obsoleted and will not be supported in future releases.

**What Next**

Please consult Synopsys Application Consultant for more details about the obsolescence, and whether there are alternative approaches or replacements for this feature being obsoleted.

## APS-011

(warning) Tcl variable %s has become obsolete since the %s release.

**Description**

This is a warning message indicating the specified command, option or variable is obsolete.

**What Next**

Please consult Synopsys Application Consultant for more details about the obsolescence, and whether there are alternative approaches or replacements for this feature being obsoleted.

## APS-012

(warning) The %s annotation to net %s, pin %s failed

**Description**

Annotation of optimized net/cell delay value to the specified net/cell failed.

**What Next**

Annotation may have failed due to scenario specific setup/hold mismatch.

## APS-013

(warning) The cell delay annotation for arc %s > %s %s failed.

Feedback

**Description**

Annotation of optimized cell delay value to the specified cell failed.

**What Next**

Annotation may have failed due to scenario specific setup/hold mismatch.

---

## APS-014

(Warning) Existing congestion map is not detected. estimate_timing is generating it.

**Description**

Virtual layer promotion in estimate_timing relies on congestion map to measure whether net could be promoted to higher metal layer or not. By default, estimate_timing will generate the congestion map from scratch. However, user could control estimate_timing to re-use existing congestion map. This message just reminds user that there is no existing congestion map in current design so estimate_timing will still generate it anyway.

**What Next**

Nothing.

**See Also**

- estimate_timing

---

## APS-015

(Warning) estimate_timing rule %s on net %s does not exist.

**Description**

estimate_timing honors layer or buffer specific rule on nets. Tool will check the attribute estimate_timing_net_rule set by the user. If the attribute exists, command will look for the rule by the attribute value in current design. If the rule is not defined, user will see this message and command will treat the net like there is no rule on it.

**What Next**

User needs to confirm whether rule is expected on the net. If not, remove estimate_timing_net_rule attribute, else re-define the rule by set_net_estimation_rule.

Feedback

**See Also**

- estimate_timing
- set_net_estimation_rule
- net_attributes

---

## APS-016

(warning) Current scenario is not enabled for setup analysis. Updating current scenario to %s which has setup analysis enabled.

**Description**

To run estimate_timing, there must exist a scenario enabled for setup analysis.

**What Next**

Tool will update current scenario to a setup scenario. User does not need to do anything

**See Also**

- create_scenario
- set_scenario_status
- report_scenarios

---

## APS-017

(Warning) net %s is skipped for layer promotion since target layer for promotion is out of the range of net's layer constraint.

**Description**

estimate_timing layer promotion honors net specific layer constraints. Layer constraints could be from routing rule or estimate_timing_net_rule attribute on the net. Only when layer promotion's layer range is within net's layer constraint, the net become a candidate for layer promotion. Otherwise, the net will not be considered for layer promotion.

**What Next**

User needs to double confirm whether layer constraints on the net is expected and the layer range for layer promotion is expected. If no, adjust layer constraints or layer range and run estimate_timing again.

Feedback

**See Also**

- estimate_timing
- set_net_estimation_rule
- net_attributes
- set_routing_rule
- plan.estimate_timing.optimize_layers
- plan.estimate_timing.optimize_layers_count

---

## APS-018

(information) Using user specified %s %s layer for chain buffer model creation.

**Description**

Using user specified layer for chain buffer model creation

**What Next**

This message is to inform user input is taken or not

---

## APS-019

(information) Using user specified %s buffer library cell for chain buffer model creation.

**Description**

Using user specified buffer lib cell for chain buffer model creation

**What Next**

This message is to inform user input is taken or not

---

## APS-020

(error) Something is wrong and optimization cannot start

**Description**

Something is wrong and optimization cannot start. Please check previous messages.

**What Next**

check previous messages to know the actual reason

Feedback

## APS-021

(error) Current scenario should be setup and active for create_ml_data -mode postOpt

**Description**

To run create_ml_data -mode postOpt, the current scenario must be enabled for setup analysis.

**What Next**

Create a setup scenario and set it to current scenario, then rerun create_ml_data -mode postOpt.

**See Also**

- create_scenario
- set_scenario_status
- report_scenarios

## APS-022

(warning) Pin %s is skipped from estimate_timing since it has an overlap pin %s.

**Description**

This message indicates that the pins with overlap pin(s) are skipped during estimate_timing. This message will appear if and only if route support feature in estimate_timing is enabled through the app option plan.estimate_timing.honor_routes.

**What Next**

Make Sure pins doesn't overlap.

## APS-101

(error) Block %s does not have valid %s.

**Description**

To run estimate_timing, the block should have valid corner, mode and setup scenario.

**What Next**

Please check the timing constraint setup for the current block.

Feedback

**See Also**

- create_corner
- create_mode
- create_scenario
- estimate_timing

---

## APS-102

(warning) Block %s with placement abstract is compromising the check.

**Description**

The specified block is currently referencing to placement abstract view. There is not enough information to perform the necessary checks before *estimate_timing* for blocks referencing to placement abstract views..

**What Next**

No action is needed from the user.

**See Also**

- create_abstract

---

## APS-103

(error) Instance %s is referencing to %s view.

**Description**

In order to run estimate_timing, a child block should reference to either abstract view or design view.

**What Next**

Please check your flow to see why the block is not referencing to abstract view or design view.If the block is still referencing to outline view, make sure that DEF with the full block placement information is properly loaded or set in the constraint mapping file, before proceeding to estimate_timing.

**See Also**

- create_abstract

Feedback

## APS-104

(error) Cannot find %s in block '%s' corresponds to %s '%s' at top level.

### Description

The mode/corner of scenario mapping between top level and the specified block is not available.

### What Next

Make sure your modes/corners of scenario are properly set up at the top level as well as inside the block. By default, block modes/corners are associated with top modes/corners if they have the same name. If your mode/corner names have different names at the top level and at the block level, please use set_block_to_top_map command to set up the desired mapping.

## APS-200

(Error) Annotation file of previous timing estimation is not found in incremental timing estimation mode.

### Description

This error message will be reported when user want run incremental timing estimation, however, there is no annotation file generated by previous full chip timing estimation.

### What Next

Run full chip timing estimation.

### See Also

- create_scenario
- create_mode
- estimate_timing

## APS-201

(Warning) Skip because net %s is not a signal net, also, this net is not a signal net or a clock net when doing clock tree synthesis timing estimation.

### Description

During incremental timing estimation, user must guarantee that specified net is signal net while specified pin should be connected to signal net. For clock tree synthesis timing

Feedback

estimation mode, user must guarantee that specified net is signal net or clock net. Otherwise, the warning message would be reported.

**What Next**

No action is needed from the user.

**See Also**

- estimate_timing

---

## APS-202

(Warning) Layer optimization is automatically disabled with incremental timing estimation.

**Description**

Incremental timing estimation does not support layer promotion strategy currently.

**What Next**

Run incremental timing estimation without layer optimization.

**See Also**

- estimate_timing

---

## APS-203

(Warning) The reference design of block instance %s is not in supported view.

**Description**

Only design view, vipo abstract view is supported for incremental timing estimation.

**What Next**

Check the view of reference block for corresponding instance.

**See Also**

- estimate_timing

---

## APS-204

(Warining) Library cells with inverter/buffer types are useless due to zero input capacitance. For example, Library cell %s has zero input capacitance.

Feedback

### Description

This particular library has buffer or inverter cells with zero input capacitance, and cannot be used for estimate_timing.

### What Next

Check the library to provide enough usefull library buffer or inverter cells.

### See Also

- estimate_timing

---

## APS-205

(warning) The block %s is a read-only block. This may affect the results of estimate_timing.

### Description

The design containts read-only blocks. The estimate_timing optimizes only boundary nets of these read-only blocks. The delay of nets and cells inside the read-only blokcs will not be estimated. This leads to non-optimimal results.

### What Next

Check why these blocks are read-only and if possible make them editable to get optimal results.

---

## APS-250

(error) A corner named estimated_corner is already exist

### Description

To run estimate_timing with "force_user_corner", the corner named estimated_corner should be deleted.

### What Next

Delete estimated_corner, then rerun estimate_timing.

### See Also

- create_scenario
- set_scenario_status
- report_scenarios

Feedback

## APS-251

(error) Cannot run estimate_timing because current scenario is not enabled for setup analysis

**Description**

To run estimate_timing, the current scenario should be enabled for setup analysis.

**What Next**

Enable setup analysis for current scenario or set the current scenario properly, then rerun estimate_timing.

**See Also**

- create_scenario
- set_scenario_status
- report_scenarios

## APS-252

(information) Changing the corner %s back to normal corner from estimated_corner

**Description**

Set the estimated_corner back tonormal corner.

**What Next**

Set the estimated_corner back tonormal corner.

**See Also**

- create_scenario
- set_scenario_status
- report_scenarios

## APS-253

(Warning) The design already has an estimated_corner and one/many user corner/corners is/are also estimated corner.

Feedback

### Description

When estimate_timing is ran with force_user_corner, it converts user corner to estimated corner. The command reset_et_corner reverts the user corner back to normal corner. If design already has a corner named estimated_corner and user tries to run reset_estimated_corner, this message identifies such situation.

### What Next

Please check why user corner is estimated corner and a separate estimated_corner present in the design. Please delete estimated_corner and rerun estimate_timing with/without force_user_corner based on user requirement.

### See Also

- estimate_timing

---

## APS-254

(error) One/more blocks are in abstract view, cannot run estimate_timing with plan.estimate_timing.force_user_corner option true.

### Description

One/more blocks are in abstract view, Cannot run estimate_timing with plan.estimate_timing.force_user_corner option true.

### What Next

Turn off plan.estimate_timing.force_user_corner option and rerun estimate_timing.

---

## APS-255

(Information) The app option plan.estimate_timing.force_user_corner is true, so the command estimate_timing uses current corner to annotate estimated value.

### Description

As the app option plan.estimate_timing.force_user_corner is true, the command estimate_timing will not create estimated_corner separately. Instead, it will use current user setup corner to annoate estimated values.

### What Next

Informational message printed by the command estimate_timing.

Feedback

**See Also**

- estimate_timing

## APS-260

(error) MV initialization Errors, can not run estimate_timing.

**Description**

MV initialization Errors, can not run estimate_timing. Please turn off "plan.estimate_timing.check_et_errors" and turn on "plan.estimate_timing.ignore_mv_errors" to continue. Alternatively, please fix the MV errors to continue.

**What Next**

Please turn off "plan.estimate_timing.check_et_errors" and turn on "plan.estimate_timing.ignore_mv_errors" option and rerun estimate_timing.

## APS-261

(error) estimate_timing errors out as zero interconnect delay is set for "time.delay_calculation_style".

**Description**

zero interconnect model is set for "time.delay_calculation_style", can not run estimate_timing. Please change it or turn on the app option "plan.estimate_timing.ignore_extractor" to continue.

**What Next**

Please change "time.delay_calculation_style" to non-zero interconnect model or turn on the app option "plan.estimate_timing.ignore_extractor" and rerun estimate_timing.

# ASDP

## ASDP-1000

(information) Auto SDP events will generate for the command(s) %s.

**Description**

Auto SDP events will generate for the commands which was specified with the app option "shell.common.auto_sdp_commands".

Feedback

## ASDP-1001

(information) Auto SDP stack_tracer frequency is set to %s.

**Description**

Auto SDP is set with the frequency which was specified with the app option "shell.common.auto_sdp_stack_trace_frequency".

## ASDP-1002

(information) Auto SDP crte timeperiod is set to %s.

**Description**

Auto SDP is set with the crte timeperiod which was specified with the app option "shell.common.auto_sdp_crte_timeperiod".

## ASDP-1003

(information) Removing old \'sdp\' directory.

**Description**

Auto SDP is removing the existing sdp directory before creating the new sdp directory.

## ASDP-1004

(information) Auto SDP has stopped.

**Description**

Auto SDP has stopped, which was unset using the app option "shell.common.auto_sdp" (or) the shell has exit from the current state.

## ASDP-1005

(information) SDP directory is compressed via tar and gzip.

**Description**

SDP directory is compressed via tar and gzip.

## ASDP-1006

(information) Auto SDP started for the command %s.

Feedback

**Description**

Auto SDP has started for the command which was specified with the app option "shell.common.auto_sdp_commands".

# ATR

## ATR-010

(warning) Unable to %s attribute '%s' for %s %s

**Description**

The attribute you are referencing cannot be set or removed by the user using set_attribute or remove_attributes, respectively. There might be other commands which allows you to set or remove the attribute.

**What Next**

Consult the documentation to determine if it is possible to set or remove this attribute.

## ATR-011

(warning) User attributes cannot be defined for '%s' objects

**Description**

The specified object class does not allow user attributes.

**What Next**

Make use of existing attributes.

## ATR-012

(warning) Attribute '%s' is already defined in class '%s'

**Description**

While defining a new user attribute, you specified an attribute name that is already defined. There is no mechanism to change an attribute definition.

**What Next**

Make use of the existing attribute or use another name for the new attribute.

Feedback

## ATR-013

(Error) Attribute '%s' is already defined as %s for another class.

**Description**

While defining a new user attribute, you specified an attribute name which is in use for another object class, but the data type which you specified was not the same as for the other object class. An attribute must have the same data type for all object classes for which it is defined.

**What Next**

Determine which is the correct data type, and re-define the attribute for each affected class.

## ATR-014

(warning) Persistent user attributes cannot be defined for '%s' objects

**Description**

The specified object class does not allow persistent user attributes.

**What Next**

Use a non-persistent attribute.

## ATR-015

(warning) Attributes of type '%s' cannot be persistent.

**Description**

The specified attribute type cannot be used for persistent user attributes.

**What Next**

Use a non-persistent attribute, or a type (integer, Boolean, float, double, or string) that can be persistent.

## ATR-016

(Error) Attribute '%s' is not user-defined for class(es): '%s'. Cannot undefine it.

Feedback

**Description**

The attribute you are referencing is an application attribute for some( or all) classes. Application attributes can't be undefine using undefine_user_attribute.

**What Next**

Remove the classes from class-list for which given attribute is an application attribute. And then try to undefine the attribute.

---

## ATR-017

(info) Attribute '%s' has been undefined from class(es): '%s'.

**Description**

The class-list contains some(or all) classes who do not support user defined attributes. The message is giving information about the classes from which attribute has been undefined.

**What Next**

This is informational message only. To turn it off, use quiet option of the command.

---

## ATR-018

(Error) Attribute '%s' has been set for object(s) of following class(es): '%s'. Cannot undefine it.

**Description**

The attribute you are undefining has been set for some objects of the class(es) given in the class list. The command cann't undefine the attribute successfully because it would cause loss of attribute information.

**What Next**

To undefine the attribute from such class(es) use '-force' option of the command.

---

## ATR-019

(Error) %s does not support more than one attribute name at a time.

**Description**

The command will take one attribute name at once. Please provide attribute name without spaces as the command support only one attribute name at a time.

Feedback

### ATR-020

(warning) %s does not support '%s' objects.

**Description**

The specified object class does not supported by the command.

## ATTR

### ATTR-1

(warning) Attribute '%s' has not been defined for the %s object class.

**Description**

The attribute you are referencing is not defined for the object class you are using. It is possible that the attribute is not defined at all.

Application attributes are all defined at runtime by the application. You can create user-defined attributes at any time.

**What Next**

Verify that the attribute name is spelled correctly. If this is a user-defined attribute, ensure that you have defined the attribute for all appropriate classes.

### ATTR-2

(warning) Attribute '%s' is not user-defined for %ss; can't %s it

**Description**

The attribute you are referencing is an application attribute. These cannot be set or removed by the user using set_user_attribute or remove_user_attribute, respectively. There might be other commands which allows you to set or remove the attribute.

**What Next**

Consult the documentation to determine if it is possible to set or remove this attribute.

### ATTR-3

(warning) Attribute '%s' does not exist on %s %s

Feedback

**Description**

The attribute you are trying to get is not found on the object, or is out-of-date. If the attribute is out-of-date, a timing update may establish its value. There is a way to suppress this message if you desire. For example, if this is in a loop or in a procedure (-quiet?)

---

## ATTR-4

(warning) Value '%s' is not valid for '%s' on %ss

**Description**

The value you are trying to set on the attribute cannot be converted to the data type defined for the attribute. For example, if the attribute is defined as "float", setting the attribute to "true" is not valid.

**What Next**

Enter an appropriate value for the attribute.

---

## ATTR-5

(warning) Value '%s' for '%s' is not in range (%s)

**Description**

The numeric value you are trying to set on the attribute is not in the range specified for the attribute. Ranges are either within a minimum and maximum value, greater than or equal to a minimum value, or less or equal to a maximum value. The message text indicates the violated constraint.

**What Next**

Enter an appropriate value for the attribute.

---

## ATTR-6

(warning) Value '%s' for '%s' is not valid. Specify one of: %s

**Description**

The string value you are trying to set on the attribute is not one of the valid strings defined for the attribute. The message text indicates the allowable values.

**What Next**

Enter an appropriate value for the attribute.

Feedback

## ATTR-7

(information) Inferred definition of the '%s' attribute for the '%s' class because it is imported for the '%s' class.

### Description

You used the *define_user_attribute* command to define a design or port attribute, and asked for it to be imported from .db files. Because design attributes are inherited onto cells (an instance of a design) and port attributes are inherited onto pins (an instance of a port), the attribute you defined for a design or port must also be defined for a cell or pin, respectively. The tool defines the attribute for you if you have not already done so, and issues this message.

### What Next

If you want to remove this message, add the appropriate class to the list of classes in your *define_user_attribute* command, or add a dedicated *define_user_attribute* command earlier in the script to define the attribute for the cell or pin class.

### See Also

- define_user_attribute

## ATTR-8

(error) Illegal min/max layer constraint.

### Description

This error occurs when the min layer constraint is on a higher layer than the max layer constraint. This constraint is ignored for the net.

### What Next

Please redefine the min/max layer constraint.

## ATTR-9

(warning) Net '%s' already had '%s' constraint set to '%s', which will be overwritten.

### Description

A min/max layer constraint has already been applied to this net. The new constraint will overwrite it.

Feedback

## ATTR-10

(warning) The min layer constraint is on a higher layer than the max layer constraint.

**Description**

Please redefine the min/max layer constraint.

## ATTR-11

(information) The returned value for %s '%s' is the design specific attribute override defined within the current block '%s'.

**Description**

This command returned the design specific attribute value from the current block, instead of the actual value from the library. Within the block, the design specific attribute value overrides the library value. It was set previously by the *set_attribute* command, because library settings may not be overwritten. This override may be removed by the *remove_attributes* command.

## ATTR-12

(information) The design specific attribute override for %s '%s' is set in the current block '%s', because the actual library setting may not be overwritten.

**Description**

This command set the design specific attribute value in the current block, instead of the actual value in the library, because library settings may not be overwritten.

Within the block, the design specific attribute value overrides the library value, and will be persistently stored in the block when saved. This attribute value is returned by the *get_attribute* command, and may be be removed by the *remove_attributes* command.

## ATTR-13

(information) The design specific attribute override for %s '%s' is removed from the current block '%s'.

**Description**

This command removed the design specific attribute value from the current block. Within the block, the design specific attribute value overrides the library value. It was set previously by the *set_attribute* command, and was returned by the *get_attribute* command.

Feedback

Now that the design specific attribute value is removed, the *get_attribute* command will return the library value.

---

## ATTR-14

(information) Attribute %s will be obsoleted, please use %s instead.

**Description**

This message informs you that the attribute specified in the information message is becoming obsolete. You are advised to use the specified replacement attribute instead.

**What Next**

Use the replacement attribute instead.

---

## ATTR-15

(error) Cannot set design specific attribute override for %s '%s' because current block is not defined.

**Description**

Library settings cannot be directly modified within this tool. However you can override the library setting within the scope of a design. In order to do so, there must be a current block in which to store the override.

**What Next**

Open or create the current block and then call *set_attributes* to set the override. Or use Library Manager to directly modify the library setting.

---

## ATTR-18

(error) Cannot set the attribute because %s.

**Description**

The attribute is settable only if the current mismatch config is "prototype".

**See Also**

- get_current_mismatch_config
- set_current_mismatch_config

Feedback

## ATTR-19

(error) Unable to set the lib pin "direction" attribute because %s.

**Description**

There are some restrictions for lib_pin "direction" attribute set.

The lib_pin object "direction" attribute can only be set when the following listed conditions are satisfied:

(1) under lm_shell (2) the current mismatch config is "prototype". (3) the lib_pin is not used in logic function. (4) no timing delay arc on the lib_pin. (5) the lib_pin is not bus port.

**See Also**

- get_current_mismatch_config
- set_current_mismatch_config

## ATTR-20

(warning) setting dont_touch attribute on unmapped %s '%s' is not supported.

**Description**

If set_dont_touch is applied on cells, designs or modules, the objects have to be mapped to accept the dont_touch attribute.

## ATTR-21

(warning) Cannot set size_only on hierarchical cell '%s'.

**Description**

Only leaf cells can accept the size_only attribute.

## ATTR-22

(information) Duplicate '%s' attribute value to '%s' view.

**Description**

When set indicated attribute, the value will be duplicated to all views to keep it consistent between different views.

Feedback

## ATTR-23

(error) Cannot set lib-cell attribute %s in ICC2 shell.

### Description

It is not possible to change the named lib-cell attribute from inside ICC2 shell. There are three types of lib-cell attributes. Some cannot be changed after the lib-cell library is built. Others cannot be changed outside of the ICC2 Library Manager because the checks which make sure the value is changed properly only exist within the Library Manager. Still others can be changed from inside the ICC2 shell, but only when the app option design.enable_lib_cell_editing is set to "mutable".

### What Next

If the value must be changed, return to the original ICC2 Library Manager setup used to create the library and change the value before the workspace is checked and committed.

## ATTR-24

(error) Cannot set unsafe lib-cell attribute %s in ICC2 shell.

### Description

It is not possible to change the named lib-cell attribute from inside ICC2 shell. There are three types of lib-cell attributes. Some cannot be changed after the lib-cell library is built. Others cannot be changed outside of the ICC2 Library Manager because the checks which make sure the value is changed properly only exist within the Library Manager. Still others can be changed from inside the ICC2 shell, but only when the app option design.enable_lib_cell_editing is set to "mutable".

### What Next

If the value must be changed, return to the ICC2 Library Manager and use the edit flow to change this value. Optionally, you can return to the original setup used to create the library and change the value before the workspace is checked and committed.

## ATTR-25

(error) Cannot set mutable lib-cell attribute %s in ICC2 shell.

### Description

It is not currently possible to change the named lib-cell attribute from inside ICC2 shell. There are three types of lib-cell attributes. Some cannot be changed after the lib-cell library is built. Others cannot be changed outside of the ICC2 Library Manager because the checks which make sure the value is changed properly only exist within the Library

Feedback

Manager. Still others can be changed from inside the ICC2 shell, but only when the app option design.enable_lib_cell_editing is set to "mutable".

**What Next**

If the value must be changed, set the app option design.enable_lib_cell_editing to "mutable". Optionally, you can return to the Library Manager and use the edit flow to change this value. Additionally, you can return to the original setup used to create the library and change the value before the workspace is checked and committed.

---

## ATTR-26

(information) Setting unsafe lib-cell attribute %s in ICC2 shell.

**Description**

The lib-cell attribute was changed inside ICC2 shell. It is considered an "unsafe" attribute in that the checks which make sure the value is changed properly only exist within the Library Manager. This means that changing the value to an improper one may lead to improper results or even a crash inside ICC2.

**What Next**

If you would like to have the changed value checked, you can choose to return to the Library Manager and use the edit flow. Under these circumstances the new value will be checked and validated as consistent.

---

## ATTR-27

(warning) There's no user set dont_touch attribute on %s '%s'.

**Description**

Only user set dont_touch attribute can be overwritten by user command set_dont_touch false.

---

## ATTR-28

(warning) Cannot set size_only on unmapped DesignWare cell '%s'.

**Description**

User can not set size_only attribute on unmapped DesignWare. Only leaf cells can accept the size_only attribute.

Feedback

## ATTR-29

(error) Cannot set attribute '%s' on cell '%s' which is not from top-level of 3d design.

**Description**

The specified attribute can be set only on cells of top-level 3d design.

## ATTR-30

(error) Cannot set package_type on block '%s' since it is not a top-level 3D IC design.

**Description**

The package_type attribute can only be set on a top-level 3D IC design. In other words, the design_type of the current_design must be "3dic".

## ATTR-31

(error) Design types 'rtl', '3dic', 'bridge', 'interposer', 'substrate', 'package'and 'analysis' are read-only and cannot be changed.

**Description**

The design_type settings 'rtl', '3dic', 'bridge', 'interposer', 'substrate', 'package' and 'analysis' reflect underlying characteristics of the design and cannot be changed.

## ATTR-32

(warning) Attribute '%s' does not exist on %s of type %s

**Description**

The attribute you are trying to get or set is not found on the object of given type.

# ATTRDEF

## ATTRDEF-001

(error) This visual (%s) does not exist.

**Description**

This visual does not exist. Please check the visual name.

Feedback

## ATTRDEF-002

(error) Internal error. Not valid type set %s.

**Description**

Type set is not valid.

## ATTRDEF-003

(error) Not valid class name.

**Description**

Design object class is not valid. Please, specify one of the following: %s

## ATTRDEF-004

(error) (error) Incompatible type and subtype values.

**Description**

Subtype can be specified only if type is string.

## ATTRDEF-005

(error) (error) Type must be specified.

**Description**

Type must be specified if subtype is presented.

## ATTRDEF-006

(error) Incompatible type and format values.

**Description**

Format can be specified only if type is int or double.

## ATTRDEF-007

(error) Type must be specified.

**Description**

Type must be specified if format is presented.

Feedback

## ATTRDEF-008

(error) Attribute %s not found.

**Description**

Specified attribute not found.

## ATTRDEF-009

(error) Internal error. Can't create attribute %s.

**Description**

Specified attribute can not be created.

## ATTRDEF-010

(error) Attribute %s already exists.

**Description**

Specified attribute already exists and can not be created.

## ATTRDEF-011

(error) Attribute %s is not user-defined.

**Description**

Specified attribute is not user-defined and not allowed for this command.

## ATTRDEF-012

(error) Internal error. Removing of attribute %s failed.

**Description**

Removing of attribute failed.

## ATTRDEF-013

(error) Internal error. Error getting list of attdefs for %s.

**Description**

Getting list of attdefs failed.

Feedback

## ATTRDEF-014

(error) Internal error. Error getting list of object types.

**Description**

Getting list of object types failed.

## ATTRDEF-015

(error) Error. Option -name valid with -class only

**Description**

Option -name valid with -class only.

## ATTRDEF-016

(error) Attribute group %s already exist.

**Description**

Specified attribute group already exist and can not be created.

## ATTRDEF-017

(error) Attribute group %s not found.

**Description**

Specified attribute group not found.

## ATTRDEF-018

(error) Internal error. Removing of attribute group %s failed.

**Description**

Removing of attribute group failed.

## ATTRDEF-019

(error) Internal error. Removing all attribute groups of class %s failed.

**Description**

Removing of attribute groups failed.

Feedback

## ATTRDEF-020

(error) Error. Options -add, -delete or -move are exclusive with -attr_list.

**Description**

Options -add, -delete or -move are exclusive with -attr_list.

## ATTRDEF-021

(error) Error. Attribute name -attr is required.

**Description**

Attribute name -attr is required.

## ATTRDEF-022

(error) Error. Attribute list or single attribute is required.

**Description**

Attribute list or single attribute with add/delete/move option is required.

## ATTRDEF-023

(error) Error. Option -anchor should be specified with -move.

**Description**

Option -anchor should be specified with -move after/before options.

## ATTRDEF-024

(error) Error. Either -add, -delete or -move should be specified.

**Description**

Either -add, -delete or -move should be specified.

## ATTRDEF-025

(error) Internal error. Error getting list of attribute groups for class %s failed.

**Description**

Getting list of attribute groups failed.

Feedback

## ATTRDEF-026

(error) Attribute %s is application defined.

**Description**

Specified attribute is aplication defined and can not be deleted.

## ATTRDEF-027

(error) Attribute %s is user-defined.

**Description**

Specified attribute is user-defined. Options -width, -show|-hide, and -show_infotip|-hide_infotip are valid for this type of attribute.

## ATTRDEF-028

(error) Content of attribute group %s is system defined.

**Description**

Specified attribute group is system defined and cannot be deleted or modified.

# AUTOREAD

## AUTOREAD-100

(information) Adding '%s'.

**Description**

This information message occurs when the autoread feature finds an HDL file that will be used in future *analyze* commands, if needed, or a directory that contains HDL sources.

**What Next**

This is an information-only message. No action is required.

However, you might want to analyze the HDL file.

## AUTOREAD-101

(information) Removing from autoread database '%s'.

Feedback

### Description

This information message occurs when autoread determines that the specified file or directory is no longer needed or that it was not specified in the current autoread command.

The reason for the file or directory deletion from the autoread database may be the following:

1. The source is no longer provided in the list of sources of the command line.
2. The source belonged to a directory, and the directory or the source is no longer available.
3. There is no HDL language bound to the source file extension, therefore, it does not need to be analyzed later.

The file or directory is removed logically only from the internal autoread database, but it is not removed from the disk.

### What Next

This is an information-only message. No action is required.

---

## AUTOREAD-102

(information) Source %s ignored by autoread due to filtering.

### Description

This information message occurs when the autoread feature finds an HDL source file or directory that appears in the exclude list you provided.

The file is not read for direct analyze.

### What Next

This is an information-only message. No action is required.

However, if the file requires being analyzed directly, remove the file from the list provided by the *-exclude* option in the *analyze -autoread* or *read_file -autoread* command.

---

## AUTOREAD-103

(information) HDL language of file %s is not defined.

### Description

This information message occurs when the specified file does not match with any file extension known to be related to HDL, such as VHDL or Verilog.

Feedback

### What Next

This is an information-only message. No action is required.

However, if the result is not what you intended, and the file is known to belong to HDL, add its extension to the *hdlin_autoread_verilog_extensions* or *hdlin_autoread_vhdl_extensions* variable.

---

## AUTOREAD-104

(information) Recursion disabled for directory %s.

### Description

This information message occurs when the source directory cannot be scanned for new HDL sources because recursion is disabled for autoread.

By default, when autoread scans the contents of a directory, it does not recursively scan any subdirectory found.

### What Next

This is an information-only message. No action is required.

However, if you want to enable recursion, use the *-recurse* argument in the *analyze -autoread* or the *read_file -autoread* command.

---

## AUTOREAD-105

(warning) Adding missing directory to search_path - '%s'.

### Description

This warning message occurs when a source directory is passed to autoread through the command line, but the directory is not available in the *search_path* system variable.

To avoid future errors, autoread added the new directory to the previously defined *search_path*, appending it to the end.

### What Next

This only a warning message. No action is required.

If the directory added is always used, add it manually to the *search_path* system variable.

### See Also

- search_path

Feedback

## AUTOREAD-106

(information) Setting new value for search_path - '{ %s }'.

**Description**

This information message occurs when autoread changes the system variable *search_path* to the value shown in the message.

The change in the search path ensures that all files will be reachable for later analyze processes.

**What Next**

This is an information-only message. No action is required.

A change in the search path by autoread means that it did not find the required files using the original *search_path* value.

You can set the *search_path* variable to the desired value before calling autoread to prevent autoread from setting a new value.

**See Also**

- search_path

## AUTOREAD-107

(warning) Defining design library '%s' at directory '%s'.

**Description**

This warning message occurs when autoread detects that a design library is being used by a design, but the design has not been mapped to a UNIX directory.

Autoread is adding a mapping pointing to the *LIB_NAME*_autoread directory.

**What Next**

This is only a warning message. No action is required.

However, you can prevent this warning message when you manually configure the design library mapping by setting the mapping in your script using the *define_design_lib* command.

## AUTOREAD-108

(warning) Using user-excluded file '%s'.

Feedback

**Description**

This warning message occurs when autoread detects that a file is needed or is being used by the HDL languages, but the file was previously marked as excluded.

A typical scenario is a Verilog `include macro that forces the inclusion and usage of a file, but that file is marked as excluded. As the excluded file is being requested by the HDL language, the autoread feature cannot block its usage.

**What Next**

This is only a warning message. No action is required.

However, if the result is not what you intended, check if the file really needs to be excluded for a proper analyze and elaborate flow. Modify the `include macro or the HDL expression that forces the file inclusion.

---

## AUTOREAD-109

(warning) Using HDL language defined via -format for file '%s'.

**Description**

This warning message occurs when a file does not have a valid extension, and the command-line format (*-format* option) was defined.

Autoread will use the HDL language defined by -format, even if it cannot be deduced from the file's extension.

**What Next**

This is only a warning message. No action is required.

However, if the file does not belong to the HDL language defined by *-format*, analyze errors will occur later.

Add the file extension to the corresponding file extension variables with the *hdlin_autoread_verilog_extensions* variable.

---

## AUTOREAD-200

(error) Source %s not found.

**Description**

This error message occurs when the source provided for autoread was not found in the search path.

Feedback

The source might come from the command line or it might be mentioned in another HDL source as an included file.

**What Next**

Add the file path to the *search_path* variable and run the command again.

---

## AUTOREAD-201

(error) Error accessing to source %s.

**Description**

This error message occurs when the autoread feature finds an HDL source that cannot be properly accessed.

**What Next**

Check that the source file or directory exists and that you have the proper file system permissions to read it.

---

## AUTOREAD-202

(error) No HDL source files found.

**Description**

You receive this error message because the autoread feature is unable to detect or find valid HDL source files with which to work.

**What Next**

Check that the source files are provided in command-line arguments.

Check that directories provided at the command line contain valid HDL source files.

Check if you need to provide the *-recurse* option to allow autoread to scan subdirectories where the HDL source files may be located.

Check that the language extensions variables are properly set for the file extensions you are using.

Check that you are not incorrectly excluding valid HDL sources.

---

## AUTOREAD-203

(error) Design library '%s' is not writeable.

Feedback

### Description

This error message occurs because the autoread feature was unable to write to the design library.

### What Next

Check that the mapped directory exists and that it has the correct permissions for writing.

---

## AUTOREAD-204

(warning) Broken link %s.

### Description

This error message occurs when a source found in a directory is a broken symbolic link or it is not reachable.

### What Next

This is only a warning message. No action is required.

However, if the result is not what you intended, you can do the following:

- If the source is a symbolic link, verify that the link is valid.

- Check for file permissions.

---

## AUTOREAD-205

(error) Conflicting language extension mapping '%s'.

### Description

This error message occurs when the autoread language extension mapping receives a file extension that is already in use for a different language.

For example, this error occurs if you add extension ".v" to SystemVerilog extensions specified with the *hdlin_autoread_sverilog_extensions* variable, while the same extension is also defined in Verilog extensions with the *hdlin_autoread_verilog_extensions* variable.

### What Next

Check the content of the *hdlin_autoread_verilog_extensions*, *hdlin_autoread_vhdl_extensions*, and *hdlin_autoread_systemverilog_extensions* system variables and ensure every extension appears only on a single variable.

Feedback

## AUTOREAD-300

(information) HDL source file %s is out-of-date.

**Description**

This information message occurs when the autoread feature detects that a file is new or has changed since the last time it was reviewed.

**What Next**

This is an information-only message. No action is required.

However, the out-of-date status may trigger future *analyze* commands over the file or their related dependencies.

## AUTOREAD-301

(information) Dependencies changed. File '%s' has to be analyzed.

**Description**

This information message occurs when the autoread feature detects that the file is up-to-date, but because of dependencies on other files that are out of date, the file must be marked as out of date.

**What Next**

This is an information-only message. No action is required.

However, be aware that the out of date status might trigger future *analyze* commands over the file or related dependencies.

## AUTOREAD-302

(information) Source file %s marked as up to date because it does not require analyze.

**Description**

This information message occurs when the autoread feature determines that the specified file is out-of-date, but the file does not require the *analyze* command to be invoked.

This message might occur because other HDL sources depend on this file. Since the file is included, it will be read when the other HDL sources run their own *analyze* command.

**What Next**

This is an information-only message. No action is required.

Feedback

However, the out-of-date status might trigger future *analyze* commands over the file-related dependencies.

## AUTOREAD-303

(information) Scanning file %s.

**Description**

This information message advises you that the autoread feature is reading the HDL source file and detecting the language abstractions defined, declared, and used in the file that lead to dependencies from other source files.

**What Next**

This is an information-only message. No action is required.

After all files are read and their abstractions found, the autoread algorithm is able to calculate the HDL sources' dependencies, and analyze the sources in the correct order to prevent *analyze* command errors.

## AUTOREAD-304

(warning) %s:%d: error at or near '%s' prevents file scanning.

**Description**

This warning message occurs when the autoread feature of *read_file* or *analyze* commands tried to read the content of a RTL file to detect the possible HDL abstraction declared on it, but possible syntax errors prevented the process from completing.

This message might be issued if an HDL file depends on a file that should be read together with additional sources. For example, in Verilog when a file uses a macro while a different Verilog file defines it.

This messages does not stops the autoread flow for continuing with *analyze* (and *elaborate*) stages.

After an AUTOREAD-304 warning *autoread* will stop reading the reported file at the line reported. If there are language constructs that generates file dependencies after the reported line then those construct will not be visible for dependency analysis, and the file could be not included for *analyze*/*elaborate* flow, leading to an incomplete design building.

This message is not a replacement of the *analyze* command internal analysis for syntax and semantic errors in the RTL source, therefore the absence of this message for a RTL file is not a guarantee that the file will not have syntax/semantic errors during *analyze* stage.

Feedback

Note that AUTOREAD-304 message is issued before the dependency analysis stage of *autoread*, and the offending file may not be analyzed later or being part of the designs elaborated. Therefore if you want to debug all your RTL sources having every RTL source file analyzed, you must use the *analyze* command with *-autoread* option but without using *-top* option. Doing that will force *autoread* to analyze every non-excluded RTL file, and then every possible syntax/semantic error will be properly reported by *analyze*.

**What Next**

Review the reported file and line number to check if a typo or evident error triggers the warning. If so, fix the RTL file.

If the file is not intended for being read or it is not required for the top-down elaboration the user needs, the file could be excluded using the *-exclude* command line option to prevent this warning message from being issued.

---

## AUTOREAD-305

(information) HDL language of file %s not supported for autoread scanning.

**Description**

This information message occurs because the autoread feature does not support the HDL language of the specified file for scanning the internal HDL abstractions that it contains.

**What Next**

This is an information-only message. No action is required.

---

## AUTOREAD-320

(information) Generic dependency: '%s' depends on '%s'.

**Description**

This information message occurs when the autoread feature detects that the first file contains HDL abstractions that depend on the content of the second file.

The file dependency helps autoread to determine the proper file ordering to analyze all sources without ordering errors.

**What Next**

This is an information-only message. No action is required.

However, depending on the file changes, some of the files may require running the *analyze* and/or *elaborate* command on the contained designs.

Feedback

## AUTOREAD-321

(information) Analyze dependency: '%s' depends on '%s'.

**Description**

This information message occurs when the autoread feature detects that the first file contains HDL abstractions that depend on the content of the second file in a way that the second file must be successfully analyzed prior the analysis of the first file.

The file dependency helps autoread to determine the proper file ordering to analyze all sources with no errors caused by incorrect ordering.

If the second file is changed and requires a new *analyze* command, the first file will also require analysis. This will be detected and performed automatically by autoread.

**What Next**

This is an information-only message. No action is required.

However, depending on whether any of the files change, some of the files may require the *analyze* and/or *elaborate* commands be run on the contained designs.

## AUTOREAD-322

(information) Elaborate dependency: '%s' depends on '%s'.

**Description**

This information message occurs when the autoread feature detects that the first file contains HDL abstractions that depend on the content of the second file in a way that the second file must be successfully analyzed before elaborating any design defined in the first file.

The file dependency helps autoread to determine the proper file ordering to analyze all sources without ordering errors.

This dependency requires that the second file be analyzed for a proper elaboration of the first file, but the analyze order is not critical between the two files.

**What Next**

This is an information-only message. No action is required.

However, depending on the file changes, some of the files may require running the *analyze* and/or *elaborate* command on the contained designs.

Feedback

## AUTOREAD-323

(information) Include dependency: '%s' depends on '%s'.

**Description**

This information message occurs when the autoread feature detects that the first file contains HDL abstractions that directly include the contents of the second file (like `include in Verilog language).

The file dependency helps autoread to determine the proper file ordering to analyze all sources without ordering errors.

The second file may not require direct analysis if it is directly included from all sources where it is needed.

**What Next**

This is an information-only message. No action is required.

However, depending on the file changes, some of the files may require running the *analyze* and/or *elaborate* command on the contained designs.

## AUTOREAD-324

(information) Compilation unit dependency: '%s' depends on '%s'.

**Description**

This information message occurs when the autoread feature detects that the first file contains HDL abstractions that depend on the content of the second file in a way that the second file must analyze in the same compilation unit of the first file in order for the analysis to succeed.

The file dependency helps autoread to determine the proper file ordering to analyze all sources without ordering errors.

If the second file is changed and requires a new *analyze* command, the first file will also require analysis. This is detected and performed automatically by autoread.

**What Next**

This is an information-only message. No action is required.

However, depending on the file changes, some of the files may require running the *analyze* and/or *elaborate* command on the contained designs.

Feedback

## AUTOREAD-330

(error) Multiple definitions found for %s.

**Description**

This error message occurs when the autoread feature detects multiple definitions of a language element such as Verilog modules, VHDL architectures, and so on. The autoread feature cannot safely determine the correct one to choose as the right dependency.

**What Next**

Depending on the nature of the duplicated definitions, you can do any of the following:

- Rename one of the element's definitions and its valid usages inside of the HDL sources.
- Avoid adding one of the sources that defines the duplicated element.
- Exclude the undesired source that defines the duplicated element using the *-exclude* option of *analyze* or *read_file* commands.
- For Verilog macros and SVerilog $UNIT definition the AUTOREAD-330 error could be solved using `include for the file that contains the right definitions in the RTL file that requires them, resolving the definition ambiguity.

## AUTOREAD-331

(error) No design found for '%s'.

**Description**

This error message occurs when the autoread feature tries to determine the source file where a named design is defined, but the element (Verilog module, VHDL entity, and so on) is not found inside the autoread database.

**What Next**

Check the following:

- Check that the file that defines the design element exists and that it is being added to the *autoread* command properly.
- Check if there are parsing errors while reading the file that prevents reading it properly.
- Check that there are no typographical errors in the design name you are providing through the command-line *-top* argument.

Feedback

## AUTOREAD-332

(warning) Unsolvable dependency loop found.

### Description

This warning message occurs when the autoread feature detects a dependency loop inside the list of files and directories provided.

The loop could be unsolvable if every file requires that a different file must be analyzed first to succeed at its own *analyze* command call, but there is no way of analyzing one of the files as head of the dependency chain due the dependency loop.

This kind of loop might produce errors at analyze time.

### What Next

This is only a warning message. No action is required.

However, if Presto generates an error due the loop, fix the error using the information provided by Presto.

## AUTOREAD-333

(error) Ambiguous dependency resolution for symbol '%s'.

### Description

This error message occurs when the autoread feature detects multiple valid definitions of a language element such as SystemVerilog typedef, localparams, or function, but it cannot safely determine the correct one to choose as the right dependency because all are valid definitions.

For example, this error is issued when a definition declared at $unit scope is also available as a wildcard import from a SV Package:

```
== File package.sv ==
        package P;
                localparam VAL = 1;
        endpackage

== File unit.sv ==
        localparam VAL = 2;

== File module.sv ==
        import P::*;
        module #(parameter LOCAL_VAL = VAL) top();
        // ...
====================
```

Feedback

Given the above RTL files, the VAL symbol could be valid used from the package P definition; therefore, autoread could behave as if there were two consecutive calls to the *analyze* command:

```
prompt> analyze -f sverilog package.sv

prompt> analyze -f sverilog module.sv
```

Or, as an another valid alternative, autoread could analyze the module.sv file as a single compile unit with the unit.sv file; therefore, VAL will be used from the $unit scope VAL localparam, behaving as this call to analyze:

```
prompt> analyze -f sverilog { unit.sv module.sv }
```

As there is no way autoread could determine your intentions, autoread cannot safely select the right alternative for defining the VAL symbol.

**What Next**

Depending on the nature of the duplicated definitions, you can do any of the following:

- Rename one of the element's definitions and its valid usages inside of the HDL sources to avoid the symbol clash.
- Avoid adding one of the sources that defines the duplicated element.
- Move one of the sources that defines the duplicated element to the exclude list.
- Include the definitions of $unit scope directly using`the include directive to disambiguate the symbol to be used.
- Explicitly import the symbol from the package to disambiguate the symbol to be used.

---

## AUTOREAD-334

(warning) Cannot determine dependencies for %s.

**Description**

This warning message occurs because the autoread feature is unable to determine the real dependency for the element listed.

This message could be issued if some designs will be linked later to precompiled designs.

The message could warn about missing files that define the elements mentioned and that must be added to the autoread command-line call to allow the tool find them.

**What Next**

This is only a warning message. No action is required.

Feedback

However, be aware that if the files are not provided properly, black-box cells might be created later, and link errors could occur.

## AUTOREAD-335

(error) Detected potential side effect of conditional compilation directive '%s'.

**Description**

This error message is issued because the autoread feature determines several HDL source files need to be analyzed together, but joining them would alter how the RTL of the files is interpreted because there is a side effect of a conditional compilation directive.

Error arises when a (S)Verilog file contains an `ifdef/`ifndef conditional compilation directive that could be altered by a dependency file that contains a `define/`undef entry for the same identifier. Adding the dependency file or altering the analyze order could alter the RTL being analyzed by HDL Compiler.

Allowing the analyze/elaborate flow to continue under that circumstances may lead to bad logic or unexpected synthesis outcomes. Therefore autoread stops the flow for safety reasons.

For example, given this two files:

== File top.sv == localparam P = EXT_P;

`ifndef SYN_DISABLED module bottom (); endmodule `endif ==========

== File define.sv == localparam EXT_P = 64; `define SYN_DISABLED ==========

When autoread first read the files, identifier SYN_DISABLED is not defined, therefore autoread dependency analysis will detect file top.sv declares design bottom, but also requires an external symbol EXT_P. Later, when dependencies are resolved, autoread will determine that file "define.sv" that declares the localparam EXT_P symbol must be analyzed together with the first file to succeed analyze. But doing that will make the macro SYN_DISABLED visible to file "top.sv", altering the RTL being analyzed.

**What Next**

Fix the RTL code to explicitly include the file that defines the macro, forcing the RTL to be always the same no mather what the read order is.

If the files are not required for top design elaboration, exclude them.

## AUTOREAD-400

(information) Invoking analyze command for file %s.

Feedback

**Description**

This information message occurs when the autoread feature detects that the file is required to be analyzed by Presto.

**What Next**

This is an information-only message. No action is required.

If the analyze command succeeds, the file is marked as up to date, and no other analyze commands should be needed until the file or its dependencies change.

---

## AUTOREAD-401

(error) Cannot analyze file %s because the HDL language is unknown.

**Description**

This error message occurs when the autoread feature determines that the file needs to be analyzed by Presto, but the HDL language of the file is unknown.

**What Next**

Check that the file extension is registered with a known HDL language, such as VHDL or Verilog. Use the *hdlin_autoread_vhdl_extensions* or *hdlin_autoread_verilog_extensions* variable, or explicitly define the file format using the *-format* argument of the appropriate command.

---

## AUTOREAD-402

(warning) Analyze command for file %s did not succeed.

**Description**

This warning message occurs because after the autoread feature determines that the file must be analyzed by Presto, the *analyze* command returns an error status on the specified file.

The error during *analyze* prevents the designs from being elaborated.

**What Next**

This is only a warning message. No action is required.

However, if the result is not what you intended, you can check the *analyze* command output to identify the HDL errors in the source files. After making the necessary corrections, run the command again.

Feedback

## AUTOREAD-500

(error) Autoread internal error.

### Description

This error message occurs if the Autoread feature encounters an internal error and cannot proceed. The command shell is not affected by this failure, so your work session can continue without loss of data. Internal errors in the Autoread feature are usually related to one or two constructs that can be recoded to work around the problem until Synopsys releases a new compiler free of this defect.

The Autoread feature of Presto HDL Compiler contains several internal checks on its operation designed to catch miscompilation problems as close as possible to their source. Internal errors prevent the production of incorrect results.

### What Next

If you encounter an internal error when using the most recent release from Synopsys, submit a test case that reproduces the problem to the Synopsys Support Center by using Enter A Call at http://solvnet.synopsys.com/EnterACall.

For information about the latest software releases, go to the Synopsys SolvNet Release Library at http://solvnet.synopsys.com/ReleaseLibrary.

For information about creating, packaging, and sending a test case, go to http://www.synopsys.com/testcase.

## AUTOREAD-600

(error) The source file for the following VHDL unit was not found: '%s'.

### Description

This error message occurs when the autoread feature cannot find a source file containing the selected VHDL design unit.

### What Next

Resolve the error by doing the following:

1. Check that the file that defines the design unit really exists and has been added to the autoread command properly.
2. Check if there are parsing errors while reading the file that prevents reading it properly.

Feedback

## AUTOREAD-602

(error) Unit %s must be analyzed in libraries %s and %s. This is not yet supported.

**Description**

Autoread infers that a unit must go in library L if it appears as L.unit in a use clause or an explicit reference inside the source code. Other constraints include all units in the same file must go in the same library, the primary unit (for example the entity) must go in the same library as its secondary units (the architecture), and so on.

Some designs may appear conflicting, such as L1.unit and L2.unit. This requires that the unit be analyzed twice, once in library L1 and once in library L2. This is currently not supported by autoread.

**What Next**

Resolve the error by doing the following;

1) Check if the unit appears referenced simultaneously as L1.unit and L2.unit.

2) Check if 2 units inferred to be in different libraries belong to the same file.

3) Check if the entity and architectures have been inferred to be in different libraries.

## AUTOREAD-603

(error) The following VHDL unit is not supported: '%s'.

**Description**

This error message occurs when autoread finds that the top requested design is a configuration. This feature is not currently supported by autoread.

**What Next**

Avoid configurations when using autoread.

## AUTOREAD-604

(warning) The following VHDL unit is not supported: '%s'.

**Description**

This warning message occurs when a reference is done on a VHDL configuration. This feature is not currently supported by autoread. The analysis will fail if this configuration is required by the design.

Feedback

### What Next

Avoid configurations when using autoread.

---

## AUTOREAD-2100

(error) The program cannot find all initialization files.

### Description

You receive this message because the program did not find all files that are necessary for proper initialization. The Synopsys software installation is incomplete or corrupted.

### What Next

Reinstall the software to fix the problem. You can continue working with the current installation, although some functionality is not available.

---

## AUTOREAD-2101

(error) %s

### Description

You receive this message because an error occurs when the *acs_read_hdl* command is scanning through the source files to detect interfile dependencies. It has found either a syntax error or an unsupported construct.

### What Next

Fix the syntax error or work around the unsupported construct. Another way to avoid this error message is to run the *acs_read_hdl* command with the *-no_dependency_check* option. This omits dependency checking so you should specify all source files in the proper order. For Verilog source, this means that files included by any other file should not appear in the *hdl_source* list. VHDL files defining any library objects should be specified before any files using those objects.

---

## AUTOREAD-2102

(error) Internal error: The '%s' command is not available.

### Description

You receive this message because an internal error occurred during the initialization phase. The described command could not be registered with the shell. It is not available. This error should not occur under normal circumstances.

Feedback

**What Next**

Reinstall the software to fix the problem. You can continue working with the current installation, although the described functionality is not available.

---

## AUTOREAD-2103

(error) Internal error: Some functions are not available.

**Description**

You receive this message because an internal error occurred during the initialization phase. An ACS command could not be registered with the shell. Some ACS functionality will not be available. This error should not occur under normal circumstances.

**What Next**

Reinstall the software to fix the problem. You can continue working with the current installation, although you should not use any AUTOREAD commands.

---

## AUTOREAD-2104

(error) The above libraries are not readable.

**Description**

You receive this message because the specified design libraries are used in the design, but are not currently mapped to a valid directory name.

**What Next**

Use the *define_design_lib* command to create a new design library. The directory to which the library is mapped must exist and should have read permissions set. To analyze HDL source files into a specific library, the directory to which the library is mapped should also have write permissions set.

---

## AUTOREAD-2105

(error) Cannot access file '%s'.

**Description**

You receive this message because an attempt failed to retrieve information about the specified file from the operating system.

Feedback

### What Next

In most cases this error should not cause any problems. However, if the specified file is an include file, it may be analyzed and cause analyzer errors. If you run the command again, the error should not reoccur.

---

## AUTOREAD-2106

(information) The file '%s' appears more than once in the source file list. Ignoring all further occurrences.

### Description

You receive this message because you specified an HDL source file more than once in the source file list of the *acs_read_hdl* command. It is analyzed only once.

---

## AUTOREAD-2107

(information) All use statements of the form 'use <LIB>.all' are ignored during dependency checking.

### Description

The *acs_read_hdl* command attempts to detect interfile dependencies to analyze files in the correct order. You receive this message because the above statement creates a dependency that is too general to be of any use. Thus it is ignored.

### What Next

If the *acs_read_hdl* command cannot resolve all dependencies and analysis of the source code fails, it may help if you replace 'use <lib>.all' statements with more-specific 'use <lib>.<package>' statements in the HDL source code.

---

## AUTOREAD-2108

(error) Cyclic dependencies are detected in the above source files.

### Description

The *acs_read_hdl* command attempts to detect interfile dependencies to analyze source files in the correct order. Definitions of library objects must be analyzed before the object is used.

You receive this message because this is not possible for all library objects with the current source files. The above list shows the cyclic dependency in the files.

Feedback

### What Next

Change your source files and move object definitions so that the files can be analyzed correctly. The above list gives hints about the files and library objects that are involved in the cyclic dependency. There are some implicit dependencies, such as a package body that depends on its package definition or an architecture that depends on its entity. Use of a package implies a dependency on the package body.

---

## AUTOREAD-2109

(error) Library object(s) are multidefined.

### Description

The list above this error message shows all library objects that are defined in more than one file and the corresponding filenames. You receive this message because it is not clear which definition to use.

### What Next

Modify the source files, deleting all definitions except one for each multidefined object, or remove the source files containing the additional definitions from the source file list.

---

## AUTOREAD-2110

(error) Multiple architectures are defined for one entity.

### Description

The list above this error message shows all files defining architectures for each entity with multiple architectures. You receive this message because it is not clear which architecture to use.

### What Next

Modify the source files, deleting all architecture definitions except one for each entity, or remove the source files containing the extra architecture definitions from the source file list.

---

## AUTOREAD-2111

(error) You cannot specify -recursive when using the search_path as a Verilog source file list.

Feedback

### Description

You receive this message because using the *-recursive* option in conjunction with the *search_path* variable can cause unexpected behavior. For the command to recursively look for source code files in directories, you cannot use the *search_path* variable to specify the same directories.

### What Next

Use the *acs_hdl_source* variable or the *-hdl_source* option to specify the source directories (using the *-recursive* option, if needed), and include the names of all include directories in the *search_path* variable.

---

## AUTOREAD-2112

(error) Cannot detect source language of file '%s'.

### Description

This error message occurs when the specified file exists in the HDL source list, but has none of the language-specific extensions. The *acs_read_hdl* command cannot determine the source language of the file, because the *-format* option is not set.

### What Next

Do either of the following:

- Specify the *acs_read_hdl* command with the *-format* option. A side effect of this is that it analyzes only Verilog or VHDL files (with an extension from the corresponding *acs_verilog_extensions* or *acs_vhdl_extensions* variable).
- Add the extension of the identified file to one (or both) of the *acs_verilog_extensions* and *acs_vhdl_extensions* variables to indicate the source language. This causes all files with this extension to be analyzed, not only the one specified above.

---

## AUTOREAD-2113

(error) File '%s' is specified twice in the HDL source files list.

### Description

You receive this message because you explicitly specified the mentioned file twice in the HDL source files list for the *acs_read_hdl* command. If you have symbolic links in your source directories, different paths may specify the same file.

### What Next

Remove the specified filename from the HDL source files list.

Feedback

## AUTOREAD-2114

(warning) No source files are found for '%s'.

**Description**

You receive this message because you specified an item in the HDL source files list that could not be expanded to any filename. If the specified item is a file, that file was not found; if it is a directory, that directory does not contain any source files.

**What Next**

You can ignore the warning if all of your source files have been analyzed. Remove the specified HDL source list item from the HDL source files list to eliminate this warning. If the *acs_read_hdl* command has found source files matching the expression, check the syntax of that item, and check the *acs_exclude_extensions*, *acs_exclude_list*, *acs_verilog_extensions*, and *acs_vhdl_extensions* Tcl variables.

## AUTOREAD-2115

(error) No input files specified.

**Description**

You receive this message because you specified an empty HDL source file list.

**What Next**

Set the *acs_hdl_source* Tcl variable to a non-empty list of file and directory names or specify that list by using the *-hdl_source* option at the command line.

## AUTOREAD-2116

(error) No top-level module name specified.

**Description**

You receive this message because the command requires the name of the top-level module to elaborate the design.

**What Next**

Specify the top-level module name at the command line. You can also set the *-no_elaborate* option, which prevents the *acs_read_hdl* command from performing elaboration.

Feedback

## AUTOREAD-2117

(error) An error or unsupported construct is detected during the Verilog include file detection phase.

### Description

You receive this message because an error or unsupported construct was detected during the Verilog include file detection phase. It is also possible that all specified source files are include files so no files are left to be analyzed.

### What Next

Correct all syntax errors or work around unsupported constructs, and rerun the command. Another workaround for this error message is to use the *acs_read_hdl* command with the *-no_dependency_check* option. This prevents the command from detecting Verilog include files; however, you have to manually remove all include files from the HDL source files list.

## AUTOREAD-2118

(warning) Mixed language designs are not automatically elaborated.

### Description

You receive this message because the tool has implicitly set the *-no_elaborate* flag, since it cannot elaborate a design that has Verilog and VHDL source code parts.

### What Next

After the command finishes you must manually do the elaboration. Execute the *elaborate* command separately for each top-level module of each language, doing the design top-level module last.

## AUTOREAD-2119

(information) Autoread is analyzing Verilog file '%s'.

### Description

This information message occurs when the *acs_read_hdl* command is analyzing a Verilog file.

### What Next

This is an information-only message. No action is required.

If you do not want to see these messages, remove the *-verbose* option from the command line.

Feedback

## AUTOREAD-2120

(information) Autoread is analyzing VHDL file '%s' into library '%s'.

**Description**

This information message occurs when the *acs_read_hdl* command is running.

**What Next**

This is an information-only message. No action is required.

If you prefer not to view these messages, remove the *-verbose* option from the command line.

## AUTOREAD-2121

(error) No files have been analyzed.

**Description**

You receive this message because either you have not specified an HDL source files list or the *acs_read_hdl* command has not found any source files in the specified locations.

**What Next**

Include all directories containing source files in the HDL source files list, and properly set the exclude and extensions variables.

## AUTOREAD-2122

(warning) Analyzer ended with errors.

**Description**

You receive this message because analyzer ended for one or more files with errors. You used the *-ignore_analyze_errors* option at the command line, so this message informs you only that one or more errors occurred and does not describe the errors.

**What Next**

Ignore this message or fix the problems that generated the errors and rerun the command.

## AUTOREAD-2123

AUTOREAD is elaborating module '%s'.

Feedback

**Description**

You receive this message because the *acs_read_hdl* command is describing its activity.

**What Next**

No action is necessary. The result is informative. Remove the-*verbose* option from the command line if you prefer not to view these messages,

---

## AUTOREAD-2124

(error) More than one module with name '%s' found.

**Description**

You receive this message because you specified a top-level module name that is not unique. Another design has the same name in memory. The current design cannot be set. The GTECH result of the *acs_error_user* command is still in memory.

**What Next**

Ensure that the *current_design* variable is set to the top-level module of your design.

---

## AUTOREAD-2125

(warning) The command line argument '%s' overrides the Tcl variable '%s'.

**Description**

You receive this message because you have assigned a value to the specified Tcl variable and you are providing an option at the command line that specifies the same thing. This warning just informs you that the *acs_read_hdl* command ignores the value of the Tcl variable.

**What Next**

Nothing to do, if this doesn't surprise you.

---

## AUTOREAD-2126

(error) Invalid file or directory name: '%s' is ignored.

**Description**

You receive this message because you specified an invalid item in a file list. The invalid expression is specified in the message. The invalid item is ignored.

Feedback

**What Next**

This error message may be caused by an unknown user (in a "~<user>" expression) or an inaccessable current working directory (if you specified "."). You do not have to take any action unless you require the ignored expression as part of your specification of the source code location.

---

## AUTOREAD-2127

(error) Invalid extension: '%s' is Ignored.

**Description**

You receive this message because you used an invalid extension in one of the extensions lists. The invalid extension is specified in the message and is ignored. The slash character (/) and backslash character (\\) are not allowed in extensions.

**What Next**

Remove the invalid extension from the list. If you actually have a source code file with a slash or backslash character in the extension you must rename it.

---

## AUTOREAD-2128

(information) AUTOREAD is starting Verilog include file detection.

**Description**

You receive this message because the AUTOREAD *acs_read_hdl* command is reporting its current activity: starting Verilog include file detection.

---

## AUTOREAD-2129

(information) AUTOREAD is star VHDL dependency extraction.

**Description**

You receive this message because the AUTOREAD *acs_read_hdl* command is reporting its current activity: starting VHDL dependency extraction.

---

## AUTOREAD-2130

(information) AUTOREAD is starting Verilog analysis.

Feedback

**Description**

You receive this message because the AUTOREAD *acs_read_hdl* command is reporting its current activity: starting Verilog analysis.

---

## AUTOREAD-2131

(information) AUTOREAD is starting VHDL analysis.

**Description**

You receive this message because the AUTOREAD *acs_read_hdl* command is reporting its current activity: starting VHDL analysis.

---

## AUTOREAD-2132

(error) DC-Expert license for AUTOREAD is not enabled.

**Description**

You receive this message because the license feature DC-Expert is not available. Every AUTOREAD command checks it.

---

## AUTOREAD-2133

(error) No additional DC-Expert licenses are available.

**Description**

You receive this message because all your licenses are already checked out. The command you executed failed to retrieve one.

Each dc_shell using an AUTOREAD command checks out one license of the DC-Expert feature. Other commands might also be using this license feature.

**What Next**

Do one of the following, and then try again to execute the command:

- Quit any other dc_shell that uses DC-Expert licenses.
- Wait until any other dc_shell that uses DC-Expert licenses is done.
- Obtain more licenses.

---

## AUTOREAD-2134

(information) Starting Design Compiler Design Budgeting.

Feedback

**Description**

You receive this message to inform you that Design Compiler (DC) is reporting its current activity: starting Design Budgeting.

**What Next**

This is an informational message only. No action is required on your part.

---

## AUTOREAD-2135

(information) Starting PrimeTime Design Budgeting.

**Description**

You receive this message because PrimeTime is reporting its current activity: starting Design Budgeting.

---

## AUTOREAD-2136

(error) Variable '%s' must be a value between 0 and 100.

**Description**

You receive this message because the variable value is not between 0 and 100.

---

## AUTOREAD-2137

(warning) The current design contains less than %d levels of hierarchy.

**Description**

You receive this message because you used the *-level* option to specify where to set the compile partitions on the design, but the number of hierarchy levels in the design is less than the specified value.

**What Next**

Take either of the following actions:

- Specify a different hierarchy level (a lower number) that contains instances with the *-level* option.
- Use a different option to specify where to set compile partitions. For detailed information, see the *set_compile_partitions* man page.

Feedback

## AUTOREAD-2138

(error) Unresolved Multiple Instances.

### Description

You receive this message because there are multiply-instantiated modules in the design and the settings of the *MasterInstance* attribute are not correct.

### What Next

See the *MasterInstance* man page or the *AUTOREAD, User Guide* for more detailed information. If the directions you find there for resolving this problem are not acceptable, take one of the following actions:

- Uniquify the design.
- Set the *MasterInstance* attribute on exactly one instance of each multiply-instantiated design.
- Set the *dont_touch* attribute on the multiply-instantiated design.
- Set the *dont_touch* attribute on all but one instance of the multiply-instantiated design.

### See Also

- set_attribute
- uniquify

## AUTOREAD-2139

(error) Design '%s' is not part of the current design.

### Description

You receive this message because the command works only with modules that are linked to the *current design*. The *current_design* command is set to a design that, within its hierarchy, does not instantiate the specified design.

### What Next

Perform one of the following tasks:

- Call the command with a design that is part of the *current design*.
- Set the *current_design* command to a parent module of the specified design.
- Set the *current_design* command to the design itself.

Feedback

**See Also**

- current_design

---

## AUTOREAD-2140

(warning) MasterInstance of module '%s' is not found in design '%s' children. Randomly choosing '%s' instance instead.

**Description**

You receive this message because you specified the *-master_only* command line option, but one of the following conditions applies to the instance that has the *MasterInstance* attribute set:

- It is not a child of the specified design.
- It is not in the design subtree (if the *-hierarchy* option was set).
- The master instance does not qualify with respect to the other options (for example like *-dt_only* or *-ndt_only*).

The command returns an instance representing the module, but it is not the master instance.

**What Next**

Do one of the following:

- Specify a design that has the master instance of the specified module in its subtree
- Change the other command line options so that the master instance is not filtered out.
- Change the *MasterInstance* attribute settings.

**See Also**

- report_attribute
- set_attribute

---

## AUTOREAD-2141

(warning) Multiple master instances found for design '%s': '%s' and '%s'. Randomly choosing '%s' instance.

Feedback

### Description

You receive this message because you specified the *-master_only* command line option, and the command found two instances of the same design that have the *MasterInstance* attribute set. This compromises the concept of the master instance, which is that you can have only one.

### What Next

Remove the *MasterInstance* attribute from one of the specified instances.

### See Also

- report_attribute
- set_attribute

---

## AUTOREAD-2142

(error) Multiple designs are found matching the '%s' name.

### Description

You receive this message because you specified a name that is not unique.

### What Next

When designs with identical names are loaded into memory, the filename is added to the design name to make it unique. You can use the *find* and *get_designs* commands to determine the unique name of the design that you want to specify.

### See Also

- get_designs

---

## AUTOREAD-2143

(error) No parent partition found in the current design for design '%s'.

### Description

You receive this message because you need to set your compile partitions in order for this command to work properly, and the current_design has to be set to a design that has partitions in its subhierarchy (or is a partition itself).

### What Next

Set the compile partitions on the current design, or set the *current design* to a design (preferably the top level module) that has partitions in its design subtree.

Feedback

### See Also

- current_design

---

## AUTOREAD-2144

(error) Multiply-instantiated design '%s' is not a compile partition.

### Description

You receive this message because the specified design is a parent module in the logical hierarchy of the module for which you want to find the parent partition. It is impossible to find the parent partition as it might be different for the different instances. The design is multiply-instantiated, but is not marked as partition. This should not happen if the compile partitions are correctly set.

### What Next

Correctly set the compile partitions on the current design. This entails either of the following:

- The design must be made unique by using the *uniquify* command.
- All multiply-instantiated designs and the top level module must be partitions, and an instance must be picked as master instance.

### See Also

- current_design
- uniquify

---

## AUTOREAD-2145

(error) Directory %s, which should contain the mapped design, does not exist.

### Description

You receive this message because the directory which should contain the mapped designs does not exist. This is either the default directory (pass0) or a directory you specified by using the *-mapped* and *-type* options of the *acs_merge_design* command. The program cannot merge the designs in the process of being updated into the mapped designs.

### What Next

Use the *acs_merge_design* command with the *-mapped* option at the command line to specify a directory for the mapped designs.

Feedback

## AUTOREAD-2146

(error) No unique top design '%s' after merge from unmapped directory '%s'.

### Description

You receive this message because after the merge from the directory specified by the *-unmapped* option (default: elab/db) to the design specified by the *-mapped* option (default: pass0) there does not exist a unique top design. Either there is no top design at all because the unmapped directory does not contain this design, or there are several top designs with the same name because there are several db files in the unmapped directory containing the top design.

### What Next

Ensure that in the unmapped directory there exists a complete unique design database containing exactly one version of the top design.

## AUTOREAD-2147

(warning) Previous partition '%s' does not have\n \ta corresponding design after the merge.

### Description

You receive this message because the design specified by the *-mapped* option of *acs_merge_design* command contained a partition whose associated design does no longer exists in the design hierarchy after merging the updated designs. Thus, the previous partition will no longer be present in the updated design. This can happen if your design updates contains changes to the design hierarchy, e.g. if you have ungrouped or flattened some design within the updated design part.

### What Next

No action is necessary if this is what you expected. However, after major changes to the design hierarchy it is recommended to re-partition the design and map it from scratch using *acs_compile_design*.

## AUTOREAD-2148

(error) The top design '%s' does not\n \tlink after merging updates from directory '%s' specified\n \tby the -unmapped option (elab/db by default).

Feedback

**Description**

You receive this message because the design does not link completely after the *acs_merge_design* command has merged in the updated designs from the directory specified by the *-unmapped* option (elab/db by default).

**What Next**

Ensure that the specified unmapped design directory contains a complete design database for the top design which is consistent with the mapped design specified by option '-mapped' for acs_merge_design.

---

## AUTOREAD-2149

(warning) Designs are merged into mapped designs \n \tin default mapped path %s.

**Description**

You receive this message because you did not use the *-mapped* option of the *acs_merge_design* command in the command line to provide a directory for the mapped designs. This warning message informs you that the program is using the default directory for the mapped designs.

**What Next**

No action is necessary, if this is what you expected.

---

## AUTOREAD-2150

(warning) Using unmapped designs in default directory elab/db.

**Description**

You receive this message because you did not use the *-unmapped* option of the *acs_merge_design* command in the command line to provide a directory for the unmapped designs and designs being merged. This warning message informs you that the program is using the default directory for the unmapped designs.

**What Next**

No action is necessary, if this is what you expected.

---

## AUTOREAD-2151

(warning) MasterInstance of the '%s' module is not found. \n \tRandomly choosing the '%s' instance instead.

Feedback

### Description

You receive this message because you specified the *-master_only* command line option, but the instance that has the *MasterInstance* attribute set is not found for the specified module.

### What Next

Perform one of the following tasks:

- Specify a module that has a master instance.
- Change the *MasterInstance* attribute settings on one of the instances of the specified module.

### See Also

- report_attribute
- set_attribute

---

## AUTOREAD-2152

(warning) Report for design %s might not be valid.

### Description

You receive this message because the compile flow is a bottom-up compile. At the end of the compile on the top level design, the *compile* command is run with the *-top* option, which might cause changes in subpartitions. Thus, reports generated immediately after the compile of the subpartitions might be invalid after running after *compile -top*.

### What Next

You can either choose not to use custom report scripts on subpartitions, or choose to ignore the warning message, but consider that the subpartition reports you have might be invalid.

---

## AUTOREAD-2153

(warning) No area estimation numbers are available yet.

### Description

You receive this message because the *report_partitions* command has been executed before any area estimation has been done. Area estimation is performed only by the "auto partitioning" command, which is the *-auto* option of the *set_compile_partitions* command. At this time the report of the design partitions shows only zeros in the percentage column.

Feedback

### What Next

No action is necessary, if this is what you expected. However, if it is necessary to get useful percentage numbers, run the *set_compile_partitions* command with the *-auto* option.

---

## AUTOREAD-2154

(error) Cannot open file '%s' for reading.

### Description

You receive this message from the *acs_read_hdl* command because the tool is unable to open (or read data or text from) the named file in the message.

### What Next

Verify the existence and correct spelling of the file you specified and that you have read privileges to this file and all directories leading to it. Also check to confirm that you spelled correctly the path to the file. Make any necessary corrections. Then run the *acs_read_hdl* command again.

---

## AUTOREAD-2155

(information) Expanding the search_path with '%s'.

### Description

You receive this message from the *acs_read_hdl* command to let you know that, because AUTOREAD detects that HDL Analyzer might not be able to locate some of the HDL files you listed in the *search_path* variable, AUTOREAD is appending one or more paths to the variable.

If you specified the *-recurse* option of the *acs_read_hdl* command, the tool might see more Verilog include files than the *analyze* command sees, because the *analyze* command does not search recursively the directories of the files listed in the *search_path* variable.

This automatic expansion of the search path eliminates your having to enter manually into the *search_path* variable all paths to Verilog include files.

### What Next

This message is informational only and requires no action on your part.

### See Also

- search_path

Feedback

## AUTOREAD-2156

(information) Found %d include files in the search_path matching '%s'.

### Description

You receive this message from the *acs_read_hdl* command to inform you that AUTOREAD has encountered an include Verilog preprocessor directive that gives the file name of the include file but does not specify the path to the include file.

After searching all directories in the order in which they are listed in the *search_path* variable, the tool has found more than one file with the same name as that given in the preprocessor directive. AUTOREAD is assuming that the first matching file is the one you want to include.

### What Next

This message is informational only and requires no action on your part. However, because the Design Compiler *analyze* and *read_verilog* commands will show the same behavior, and because the behavior of these commands is always implementation- and release-dependent, it is no doubt wise to identify the correct path information and add it to the preprocessor directive. Then run the *acs_read_hdl* command again.

### See Also

- search_path

## AUTOREAD-2157

(warning) Could not locate include file %s, included by file %s.

### Description

You receive this message from the *acs_read_hdl* command because Automated Chip Synthesis (ACS) encountered an include Verilog preprocessor statement that does not specify a path to the include file.

ACS first tries to locate the file by searching all the directories listed in the *search_path* variable. If this search is not successful, ACS then searches all of the directories of all files listed in the *-hdl_source* option (if defined) of *acs_read_hdl*.

If you also specified the *-recurse* option of *acs_read_hdl*, ACS searches further all of the subdirectories of all the files listed in the *-hdl_source* option of *acs_read_hdl*.

The *-recurse* option of *acs_read_hdl* does not search all subdirectories listed in the *search_path* variable, however. (In this respect, the command mimics the behavior of the *analyze* and *read_verilog* commands.)

Feedback

After searching all of these directories, ACS did not find the missing include file.

**What Next**

This warning message alerts you to a possible conflict, which, if not resolved, might prevent a later *analyze* command being able to find the include file, resulting in an error.

To ensure a smooth process here, it is no doubt preferable to locate and identify the correct include file and add its path to the statement or to the *search_path* variable. Then run the *acs_read_hdl* command again.

**See Also**

- search_path

---

## AUTOREAD-2158

(warning) Found more than one matching include file %s, wanted by file %s.

**Description**

You receive this message from the *acs_read_hdl* command because AUTOREAD encountered an include Verilog preprocessor statement that does not specify a path to the include file.

ACS first tries to locate the file by searching all the directories listed in the *search_path* variable. If this search is not successful, AUTOREAD then searches all of the directories of all files listed in the *-hdl_source* option (if defined) of *acs_read_hdl*.

If you also specified the *-recurse* option of *acs_read_hdl*, AUTOREAD searches further all of the subdirectories of all the files listed in the *-hdl_source* option of *acs_read_hdl*.

The *-recurse* option of *acs_read_hdl* does not search all subdirectories listed in the *search_path* variable, however. (In this respect, the command mimics the behavior of the *analyze* and *read_verilog* commands.)

Although AUTOREAD could not find the include file in the directories listed in the *search_path* variable, it did locate, by searching all directories (and possibly subdirectories) listed in the *-hdl_source* option of *acs_read_hdl*, more than one include file with the same name. (The implicit assumption is that the *search_path* variable presents an ordered list of directories, but this assumption does not apply to the list of directories in the *-hdl_source* option.)

**What Next**

This warning message alerts you to a possible conflict, which, if not resolved, might prevent a later *analyze* command being able to find the include file, resulting in an error.

Feedback

To ensure a smooth process here, it is no doubt preferable to identify the correct include file and add its path to the *search_path* variable. Then run the *acs_read_hdl* command again.

**See Also**

- search_path

---

## AUTOREAD-2159

(warning) Multiple Master Instances found for design '%s'. Using instance '%s'.

**Description**

You receive this message because you've specified multiple Master Instances for a multiply instantiated design. You've also specified the '-force' option with the 'set_compile_partitions' command, causing the command to pick the alphabetically smallest instance as Master Instance now.

**What Next**

There is nothing to do, if the instance mentioned in the warning is the instance you want to use as Master Instance for this design. If you want to specify a different instance as Master Instance you would need to remove the *MasterInstance* attribute on all but the desired instance of this design.

See the *MasterInstance* man page or the *AUTOREAD, User Guide* for more detailed information.

**See Also**

- set_attribute
- uniquify

---

## AUTOREAD-2160

(warning) No qualifying designs are found in level '%d'.

**Description**

You receive this message because you used the *-level* option to specify where to set the compile partitions on the design, but no design at this hierarchical level qualified to become a partition. Only standard hierarchical cells can be treated as partitions. A cell will not be treated as partition if it is not hierarchical, dont touched, or a DesignWare part.

Feedback

**What Next**

Take either of the following actions:

- Specify a different hierarchy level (a lower number) that contains instances with the *-level* option.
- Use a different option to specify where to set compile partitions. For detailed information, see the *set_compile_partitions* man page.

---

## AUTOREAD-2161

(warning) Executable '%s' does not exist.

**Description**

You receive this message because the specified executable could not be found. This is often due to a wrong setting of the AUTOREAD variable *acs_dc_exec*. If one of these variables is set, ACS will use the specified executable instead of the default ones.

**What Next**

Take either of the following actions:

- Check the ACS variables listed above for typographical errors.
- Use absolute path descriptions instead of relative ones.

---

## AUTOREAD-2162

(warning) You do not have execute permissions for file '%s'.

**Description**

You receive this message because you do not have the right permissions to execute the specified file.

**What Next**

Take either of the following actions:

- Try to set the execute-permission (e.g., with 'chmod' on UNIX platforms).
- Ask your system administrator for help.

---

## AUTOREAD-2163

(information) Starting Design Compiler Design Budgeting.

Feedback

### Description

This message indicates that AUTOREAD is using Design Compiler for slack allocation (budgeting).

AUTOREAD distinguishes between RTL budgeting and gate-level budgeting. RTL budgeting is performed, e.g., by the *acs_compile_design* command and utilizes Design Compiler's built-in design budgeting facilities.

Gate-level budgeting is performed, e.g., by the *acs_recompile_design* and *acs_refine_design* command. ACS offers three different methods for gate-level slack allocation: Design Compiler budgeting, PrimeTime budgeting, and budgeting via a user budgeting script. Design Compiler budgeting is the recommended way for design budgeting and selected by default. PrimeTime budgeting is deprecated and currently kept for backwards compatibility.

### What Next

If you wish to use a custom budgeting script for gate-level budgeting, please place a script named *budget.scr* into the directory defined by the *user_budget_script* file type.

---

## AUTOREAD-2164

(information) Starting PrimeTime Design Budgeting.

### Description

This message indicates that AUTOREAD is using PrimeTime for slack allocation (budgeting).

AUTOREAD distinguishes between RTL budgeting and gate-level budgeting. RTL budgeting is performed, e.g., by the *acs_compile_design* command and utilizes Design Compiler's built-in design budgeting facilities.

Gate-level budgeting is performed, e.g., by the *acs_recompile_design* and *acs_refine_design* command. AUTOREAD offers three different methods for gate-level slack allocation: Design Compiler budgeting, PrimeTime budgeting, and budgeting via a user budgeting script. Design Compiler budgeting is the recommended way for design budgeting and selected by default. PrimeTime budgeting is deprecated and currently kept for backwards compatibility.

### What Next

If you wish to use a custom budgeting script for gate-level budgeting, please place a script named *budget.scr* into the directory defined by the *user_budget_script* file type.

Feedback

## AUTOREAD-2165

(information) Using custom budgeting script '%s'.

**Description**

This message indicates that AUTOREAD is using a custom busgeting script for slack allocation (budgeting).

AUTOREAD distinguishes between RTL budgeting and gate-level budgeting. RTL budgeting is performed, e.g., by the *acs_compile_design* command and utilizes Design Compiler's built-in design budgeting facilities.

Gate-level budgeting is performed, e.g., by the *acs_recompile_design* and *acs_refine_design* command. AUTOREAD offers three different methods for gate-level slack allocation: Design Compiler budgeting, PrimeTime budgeting, and budgeting via a user budgeting script. Design Compiler budgeting is the recommended way for design budgeting and selected by default. PrimeTime budgeting is deprecated and currently kept for backwards compatibility.

**What Next**

If you do not wish to use a custom busgeting script, please remove file *budget.scr* from the directory defined by the *user_budget_script* file type.

## AUTOREAD-2166

(information) Compile design '%s' in '%s'.

**Description**

This message is issued by *acs_compile_design* and indicates the source where the specified design is loaded from.

## AUTOREAD-2167

(information) Compile design in memory.

**Description**

This message is issued by *acs_compile_design* and indicates that ACS is going to compile the in-memory design.

## AUTOREAD-2168

(information) Re-compile design '%s' in '%s'.

Feedback

**Description**

This message is issued by *acs_recompile_design* and indicates the source where the specified design is loaded from.

---

## AUTOREAD-2169

(information) Re-compile design in memory.

**Description**

This message is issued by *acs_recompile_design* and indicates that ACS is going to re-compile the in-memory design.

---

## AUTOREAD-2170

(information) Refine design '%s' in '%s'.

**Description**

This message is issued by *acs_refine_design* and indicates the source where the specified design is loaded from.

---

## AUTOREAD-2171

(information) Refine design in memory.

**Description**

This message is issued by *acs_refine_design* and indicates that AUTOREAD is going to refine the in-memory design.

---

## AUTOREAD-2172

(information) Reading partition '%s' from '%s'.

**Description**

This message is issued whenever a partition is read into memory.

---

## AUTOREAD-2173

(information) Writing partition '%s' to '%s'.

**Description**

This message is issued whenever a partition is written to disc.

Feedback

## AUTOREAD-2174

(information) Deriving budgets from gate-level design in '%s'.

### Description

This message is issued by *acs_recompile_design* and indicates the source of the gate-level design that is utilized for slack allocation (budgeting). Note that in *acs_recompile_design*, the design used for budgeting is different than the design to be compiled.

## AUTOREAD-2175

(information) Removing designs from memory.

### Description

This message indicates that all in-memory designs are discarded. Design deletion takes place at the end of each invocation of *acs_compile_design*, *acs_recompile_design*, and *acs_refine_design* in order to avoid confusion between the pre-compiled and the post-compiled designs. If designs were not deleted, a possible confusion could occur, e.g., if *acs_refine_design* is called right after *acs_compile_design*. The refine command would compile the design in memory which is unchanged and not the result of *acs_compile_design*. The automatic design deletion ensures than *acs_refine_design* does not find any design in memory and therefore reads the design, by default, from the destination directory of *acs_compile_design*.

## AUTOREAD-2176

(information) Using user override procedure for command '%s'.

### Description

This message indicates that AUTOREAD has found a user override procedure for the specified command. This procedure will be executed instead of the default one. User override procedures can be defined for the commands *acs_compile_design* (procedure *acs_compile_design_user*), *acs_recompile_design* (procedure *acs_recompile_design_user*), and *acs_refine_design* (procedure *acs_refine_design_user*). The easiest way to set up a user override procedure is to customize one of the predefined *.template* files *acs_compile_design.template*, *acs_recompile_design.template*, and *acs_refine_design.template*.

### What Next

If you do not wish to override the default behaviour, delete the procedures *acs_compile_design_user*, *acs_recompile_design_user*, and *acs_refine_design_user*.

Feedback

## AUTOREAD-2177

(Error) ILM design '%s' is a subdesign of ILM design '%s'.

### Description

You receive this message because you have nested ILMs in your design. This is not supported yet. ILMs are detected by testing the *is_interface_model* or the *useILM* attribute. The first indicates that a design is an ILM, the latter tells AUTOREAD to create an ILM for the design. In your current design the mentioned designs are ILMs (or are tagged to become ILMs), but the first is instantiated in the logical hierarchy of the other. This is currently not allowed.

### What Next

Make sure you don't have nested ILMs in your design and you don't set the *useILM* attribute on any design that is instantiated in the logical hierarchy of another ILM (or a design with the "useILM" attribute set). You can either remove the "useILM" attribute from the specified designs, or make sure that the full design representation of the designs is linked in, instead of the ILM representation.

### See Also

- set_attribute

## AUTOREAD-2178

(Error) Partition '%s' is inside the ILM design '%s'.

### Description

You receive this message because you have marked the specified design as AUTOREAD partition, but it is instantiated in the logical hierarchy of the ILM design. Part of the ILM design is or will be cut out which may lead to unexpected constraints for the partition. This is why partitions inside ILMs are not allowed.

### What Next

Make sure you only created ILMs for designs marked as partitions or designs below partitions. Also, don't set the *useILM* attribute on a design above an AUTOREAD partition.

### See Also

- set_attribute

Feedback

## AUTOREAD-2179

(Error) ILM design '%s' contains an unresolved reference '%s'.

### Description

You receive this message because you have marked the specified design with the *useILM* attribute to become an ILM, but it instantiates a design that couldn't be resolved by the linker.

### What Next

Make sure your *search_path* and *link_library* are set correctly (see the man page of the *link* command).

### See Also

- set_attribute

## AUTOREAD-2180

(Error) ILM design '%s' detected in full design.

### Description

You receive this message because you are trying to compile a design in memory with *acs_refine_design* and this design contains an ILM. The AUTOREAD flow needs the full design representation (in memory or in the source pass) in order to run a useful ILM based flow. The ILM in your design will be considered as full design representation for this partition (i.e. the "is_interface_logic" attribute will be removed). The post_compile db file created by acs_refine_design will contain only the functionality contained in the mentioned ILM design.

## AUTOREAD-2182

(warning) Multiple instances found for design '%s'. Using instance '%s'.

### Description

You receive this message because the design you specified is multiply instantiated and cannot be uniquely identified with a single instance in the design. Therefore, the command you invoked has selected an arbitrary cell instance to proceed.

Feedback

### What Next

There is nothing to do, if the instance mentioned in the warning is the one you want to use as instance for this design. If the wrong instance has been chosen, you might want to uniquify the design in order to eliminate multiple instances in your design.

### See Also

- uniquify

---

## AUTOREAD-2183

(error) Option '%s' can be used only inside of another tool.

### Description

This error occurs because because you have used an option that is not supported in this tool. For example, physical AUTOREAD rtl2pg and gates2pg flow can be applied only inside a different tool than the one you are currently using.

---

## AUTOREAD-2184

(error) Option '%s' is not comatible with the update option.

### Description

You receive this message because you have specified an option that cannot be combined with the 'update' feature. E.g., the physical AUTOREAD flows cannot be used inside an update flow.

---

## AUTOREAD-2185

(error) Option '%s' is not comatible with physical AUTOREAD flows.

### Description

You receive this message because you have specified an option that cannot be combined with the physical AUTOREAD flows.

---

## AUTOREAD-2186

(warning) Option '-ignore_analyze_errors' of command 'acs_read_hdl' is no longer supported.

Feedback

### Description

This warning message occurs when the specified option is no longer supported.

Even with this option set, the *analyze* command used by *acs_read_hdl* internally halts upon encountering an error and forgets all data obtained during that call to *analyze*. As *acs_read_hdl* always tries to implicitly call *analyze* with as many files as possible, the result of this option is not exactly what you would expect.

Preceding internal calls to *analyze* and following calls to *analyze* are still executed. Therefore, you can still use this option for a trial analysis.

### What Next

This is only a warning message. No action is required.

---

## AUTOREAD-2187

(warning) false dont_touch attribute is detected on object '%s'. It may be lost or AUTOREAD may not be able to complete successfully.

### Description

This message indicates that there is dont_touch being set to false in your design. Right now, AUTOREAD cannot guarantee that AUTOREAD could complete successfully if your design has false dont_touch attribute and the object who has false dont_touch is a multiply instantiated design or is an instance of a multiply instantiated design. Or, the false dont_touch attribute is not be hornored if the object who has false dont_touch is a singly instantiated non-partition design or its instance, and this false dont_touch attribute may cause long runtime on the upper level partitions' compile (object who has false dont_touch attribute may be re-compiled during upper level partitions' compile). Or, the false dont_touch attribute may be lost after compile by using AUTOREAD if the object who has false dont_touch is a singly instantiated partition design or its instance.

### What Next

You need to optimize/compile the design (who or whose instance has false dont_touch) independently, and remove the false dont_touch attribute on your design before AUTOREAD to make sure that AUTOREAD could run through successfully. It is not recommended to apply false dont_touch attribute on your design when you use AUTOREAD.

---

## AUTOREAD-2188

(error) '%s' is not an autoread-supported attribute.

Feedback

### Description

This error message occurs when the specified attribute is not currently supported by autoread.

### What Next

To view the complete list of attributes supported by autoread, their valid values and their default values, see the *acs_set_attribute* man page.

---

## AUTOREAD-2190

(error) Top module does not link.

### Description

This error message occurs when the the *link* command issued within the AUTOREAD auto update flow fails. The failure occurs when the GTECH library or the target Technology library is omitted or incorrectly specified.

### What Next

Ensure that the *search_path* is set correctly and that all of the source directories are specified in the *-hdl_source* option of the *acs_read_hdl* command.

### See Also

- current_design

---

## AUTOREAD-2191

(error) Corrupt elab_times table.

### Description

This error message occurs when the design's elab_times table is corrupted.

The elab_times table is an ASCII table created during the AUTOREAD auto update flow to keep track of the update times of the designs in the design hierarchy. The default location for the table is elab/db.

### What Next

To continue, delete the design's elab_time table. Then regenerate the table using the *acs_read_hdl* command with the *-auto_update* or *-update* option.

Feedback

## AUTOREAD-2192

(warning) Cell '%s' does not have reference design.

### Description

This message is printed because AUTOREAD could not locate referencing design for the cell.

### What Next

Take either of the following actions:

- Make sure design is properly linked.
- Make sure all design units are read properly and there are no syntax errors.

## AUTOREAD-2193

(error) Valid target library for cell '%s' could not be found.

### Description

AUTOREAD issues this error when valid target library for cell could not be found. This can happen when libraries which are characterized for specified operating condition are not found. AUTOREAD also issues this message when target library subset and library set characterized for operating conditions have no common elements.

### What Next

- Make sure operating conditions and target library subset is properly specified.

### See Also

- set_target_library_subset
- set_operating_conditions

## AUTOREAD-2194

(error) Attribute '%s' is no longer supported by AUTOREAD.

### Description

AUTOREAD issues this message when attempting to set an obsolete AUTOREAD attribute. AUTOREAD attributes become obsolete when their corresponding DC commands become obsolete.

The list of obsolete attributes and the corresponding DC commands is given below.

Feedback

```
Attribute Name               Design Compiler Command
-------------------------------------------------------------------
HasArithmetic                transform_csa
PartitionDP                  partition_dp
```

**What Next**

Remove the obsolete attribute from the compilation scripts and re-run AUTOREAD.

## AUTOREAD-2195

(warning) Attribute '%s' set on partition '%s' is obsolete and will be ignored.

**Description**

AUTOREAD prints this message when it finds an obsolete attribute on a partition. Please refer to the AUTOREAD Users Guide or the acs_set_attribute man page for a list of supported attributes.

**What Next**

Remove the obsolete attribute from the partition and run AUTOREAD again.

# BBT

## BBT-001

(information) Port %s already has %s setting, overwriting.

**Description**

This message indicates that the specified port already has its load or drive information set by a previous call of set_blackbox_port_load or set_blackbox_port_drive. The more recent call to set_blackbox_port_load or set_blackbox_port_drive is overwriting the previous setting.

**What Next**

No action is needed if this intended.

## BBT-201

(warning) Port %s does not belong to the current block.

Feedback

### Description

The command only takes port objects that are ports of the current block. The specified port is not a port of the current block. It is ignored by the command.

### What Next

If the current block is the block you are trying to create BBT for, no action is needed.

---

## BBT-202

(warning) Port %s is a feedthrough port.

### Description

You are getting this message because you attempted to specify blackbox timing information for a feedthrough port created by the tool.

A feedthrough port created by the tool is either connected directly to another non-feedthrough port of the design and is going to share the same timing characteristics as the other port, or connected to an existing path going through the design in which case the connected netlist is going to be used to time through the port. Either case, user cannot specify blackbox timing on a feedthrough port. The specified timing information for this port is ignored.

### What Next

No action is needed.

---

## BBT-203

(warning) Failed to find a default %s lib cell.

### Description

You are getting this message because you attempted to create timing model for a blackbox design, and the tool could not find certain type of library cell to use in order to construct the model. This may cause the model to be incomplete.

### What Next

Please note that the tool cannot use any MV cell, such as level shifter, isolation cell, or always-on cell, for the purpose of building the blackbox timing model. It is also not allowed to use any lib cell marked as don't use.

Please check reference libraries to make sure that there is useable lib cell of the specified type.

Feedback

## BBT-204

(warning) Clock %s does not belong to the current block.

**Description**

The command only takes clock objects that are clocks of the current block. The specified clock is not a clock of the current block. It is ignored by the command.

**What Next**

If the current block is the block you are trying to create BBT for, no action is needed.

## BBT-301

(error) Value %f for option %s is invalid.

**Description**

The command option only accepts positive number.

**What Next**

Please correct the value.

## BBT-302

(error) Cannot specify %s without -type.

**Description**

The specified option can only be specified when option -type is specified. Specifying the option without -type is meaningless.

**What Next**

Please correct the command and re-issue.

## BBT-303

(error) Must specify one of %s.

**Description**

Must specify one of the specified options. Not specifying any of them is meaningless.

**What Next**

Please correct the command and re-issue.

Feedback

## BBT-304

(error) Value %d for option %s is invalid.

**Description**

The command option only accepts positive number.

**What Next**

Please correct the value.

## BBT-305

(error) Design %s already has BBT defined.

**Description**

You are getting this message because you are calling a BBT (blackbox timing) command for the design which already has BBT defined.

**What Next**

If you are trying to add information to the existing BBT, please use *write_blackbox_timing_script <file_name>* to write out a script containing commands that can re-create the existing BBT, then edit the script to add the things you want to add. After you finish editing the script, please do a remove_blackbox_timing then source the script.

## BBT-306

(error) Value %s for option -edge is invalid.

**Description**

Valid values for the option are *rise* and *fall*.

**What Next**

Please correct the value.

## BBT-307

(error) Cannot find blackbox timing data for non-PG port %s.

**Description**

This message indicates that the tool cannot find the specified non-PG port in the blackbox timing model which is being defined through a set of tcl commands.

Feedback

If you are getting this message but find the port exists on the current block, there are two possibile reasons:

1. BBT (Blackbox Timing) does not allow user to specify any constraint for an output/inout port if the port already has a driver in the existing netlist, or an input/inout port if the port already has load in the existing netlist. If the specified port falls into this category, you cannot create blackbox timing constraint on this port.
2. The first blackbox timing model command issued on the current block would take a snapshot of the ports of the block. All commands that define the blackbox timing for the current block are expected to be issued in one shot without any netlist modification during the process. Please check your scripts to make sure that no netlist change has been done after issuing the first blackbox timing model command.

**What Next**

Check if you specify the name correctly. If it is correct, check if the port has existing connection in the netlist. If so, do not use this port in BBT. If you really need to use this port in BBT, see if you can disconnect it from its existing driver/load. Or, if it is a port that you created after issing the first blackbox timing command, please change your flow to add the port before issuing any blackbox timing command, or do a *remove_blackbox_timing* after adding the port and re-issue all the blackbox timing commands.

---

## BBT-308

(error) Module %s is not a %s.

**Description**

The specified module is not of the right type. The command needs a library cell which is a buffer or inverter.

**What Next**

Correct the command and re-issue.

---

## BBT-309

(error) Cannot find %s type %s.

**Description**

The tool cannot find the load or drive type with the specified name. It is possible that it was never created or was not created successfully.

To create a load type, use the command *create_blackbox_load_type*. To create a drive type, use the command *create_blackbox_drive_type*.

Feedback

**What Next**

Please check to make sure that the specified load or drive type is created successfuly.

---

## BBT-310

(error) Cannot get mode and corner information.

**Description**

Mode and corner information are needed for creating BBT, even for a single mode single corner design.

You are getting this error message because the tool cannot find mode or corner information.

**What Next**

Please check if design is loaded and linked successfully.

---

## BBT-311

(error) Port %s is not a clock input port for BBT.

**Description**

You are getting this message because you are calling *create_blackbox_clock_network_delay* or *create_blackbox_constraint -from* for the specified port. But the port is not a clock input port.

Only clock input port can have network delay information or be the from port of a constraint path.

**What Next**

Please check if the specified port is an input port and a clock typed port or have run set_blackbox_clock_port on it.

---

## BBT-312

(error) Port %s is not input or inout type.

**Description**

Only input or inout type ports can have load setting, or be the from port of a delay path, or be a from/to port of a constraint path.

Feedback

**What Next**

Please check the type of the specified port.

---

## BBT-313

(error) Port %s is not output or inout type.

**Description**

Only output or inout type ports can have drive setting or be the to port of a delay path.

**What Next**

Please check the type of the specified port.

---

## BBT-314

(error) Port %s already has a driver.

**Description**

BBT (Blackbox Timing) does not allow user to specify any constraint for an output/inout port if the port already has a driver in the existing netlist, or an input/inout port if the port already has load in the existing netlist.

**What Next**

Please do not use this port in BBT. If you really need to use this port in BBT, see if you can disconnect it from its existing driver/load.

---

## BBT-315

(error) Port %s already has a driver.

**Description**

BBT (Blackbox Timing) does not allow user to specify any output port constraint for an inout port if the port already has a driver in the existing netlist.

**What Next**

Please do not specify any constraint that is intended for an output port on this inout port in BBT. If you really need to use this port, see if you can disconnect it from its existing driver.

---

## BBT-316

(error) Load cap value %f is smaller than cap value %f of lib pin %s/%s.

Feedback

**Description**

You are getting this message because you specified a load capacitance value for a port of the design and the value is too small.

The tool uses the smallest buffer library cell's input pin capacitance as a guideline to check whether the load cap user specifies is reasonable.

**What Next**

Please specify a value larger than the lib pin's cap indicated in the message.

If you want to use the input pin cap of a specific buffer or inverter cell, please use *create_blackbox_load_type* and *set_blackbox_port_load -type*.

---

## BBT-317

(error) Invalid value %f for option -value in presence of -clock.

**Description**

You are getting this message because you specified a value that is not between 0.0 and 1.0 for option *-value* while you also specified option *-clock*.

When option *-clock* is specified, the meaning of *-value* changes from the absolute delay value to the percentage number of the clock period. The tool multiply the value of *-value* with the period of the specified clock to derive the absolute delay value.

**What Next**

Please specify a value that is larger than 0 and no larger than 1. Or, if you want to specify the absolute delay value, please remove the *-clock* option.

---

## BBT-318

(error) Current design is not design view.

**Description**

You are getting this message because you are trying to specify blackbox timing information for the current design which is not design view.

Blackbox timing can only be specified when the current design is design view.

**What Next**

Please open the design view of the blackbox design to continue.

If you only have the outline view of the design and do not have the design view yet, please do *expand_outline* to get the design view.

Feedback

## BBT-319

(error) Cannot find blackbox timing data for clock %s.

**Description**

This message indicates that the tool cannot find the specified clock in the blackbox timing model which is being defined through a set of tcl commands.

If you are getting this message but find the clock exists on the current block, there are two possibile reasons:

1. BBT (Blackbox Timing) does not allow user to specify any constraint for an output/inout port if the port already has a driver in the existing netlist, or an input/inout port if the port already has load in the existing netlist. If the specified port falls into this category, you cannot create blackbox timing constraint on this port.
2. The first blackbox timing model command issued on the current block would take a snapshot of the ports of the block. All commands that define the blackbox timing for the current block are expected to be issued in one shot without any netlist modification during the process. Please check your scripts to make sure that no netlist change has been done after issuing the first blackbox timing model command.

**What Next**

Check if you specify the name correctly. If it is correct, check if the port has existing connection in the netlist. If so, do not use this port in BBT. If you really need to use this port in BBT, see if you can disconnect it from its existing driver/load. Or, if it is a port that you created after issing the first blackbox timing command, please change your flow to add the port before issuing any blackbox timing command, or do a *remove_blackbox_timing* after adding the port and re-issue all the blackbox timing commands.

## BBT-320

(error) You cannot specify an output port '%s' to be a generated clock master source.

**Description**

A generated clock master can be an input or inout port or a pin.

**What Next**

Determine an appropriate master source for the generated clock.

## BBT-321

(error) Generated clock name %s has been used by other clocks.

Feedback

**Description**

When creating blackbox generated clock using command create_blackbox_generated_clock, the name specified for the generated clock has been used by other clocks.

**What Next**

Please change the name of the blackbox generated clock.

---

## BBT-322

(error) Blackbox generated clock has no master clock source pin/port.

**Description**

Blackbox generated clock must have a master clock pin/port. User could use command create_blackbox_generated_clock -source to specify a master clock source pin/port for the generated clock. If it is not specified, the master clock specified for this generated clock must have a source pin/port.

**What Next**

Please specify a master source clock pin/port for the blackbox generated clock.

---

## BBT-323

(warning) In-memory blackbox timing model for design %s is removed because %s.

**Description**

In blackbox timing model, if user create or remove a mode, corner or port, all the in-memory blackbox timing model information will be deleted. If the blackbox timing model is already committed to disk, the on-disk model is not affected.

**What Next**

Reset the blackbox timing model information if needed.

---

## BBT-501

(error) Incremental blackbox creation failed for designs: %s.

**Description**

Incremental blackbox could not be created for designs in the message. The tool will continue using the blackbox it was using prior to creating this incremental abstract.

Feedback

**What Next**

Check the blackbox script provided in set_bbt_override.

# BLOX

## BLOX-001

(error) %s.

**Description**

The input 3DBlox file(s) may contain syntax error.

**What Next**

Please follow the error message to find if there is syntax error in the input 3DBlox file(s).

## BLOX-002

(error) Python execute file (%s) error, please find detail in file *.debug.log.

**Description**

There occurs error while executing python command. Detail info is dumped in file *.debug.log.

**What Next**

Please get the detail error info and contact AE for help.

## BLOX-003

(error) Syntax error: %s.

**Description**

The input 3DBlox file(s) contains syntax error. Please check the error message.

**What Next**

Please follow the error message to find if there is syntax error in the input 3DBlox file(s).

## BLOX-004

(error) Data error: %s.

Feedback

**Description**

The input 3DBlox file(s) contains wrong data value. Please check with the error message.

**What Next**

Please follow the error message to find if there is data error in the input 3DBlox file(s).

---

## BLOX-005

(warning) Loop connection is found for chiplets: %s in %s.

**Description**

There is loop connection in specified .3dbx file, thus z level of all the chip insts will be set to zero.

**What Next**

Please check the source file to find the loop connection and remove it.

---

## BLOX-006

(error) Invalid library name '%s' specified in '%s' command option of read_3dblox.

**Description**

The library name specified is invalid because it contains spaces.

**What Next**

Ensure that there are no spaces in library name specified.

---

## BLOX-007

(error) %s.

**Description**

The specified library name exists in directory. read_3dblox creates library path under ./refLibs/ directory. The specified library name from 3dblox files already exists.

**What Next**

Please check the existing directory. Back up or delete it, then run read_3dblox again.

---

## BLOX-008

(error) The specified library path '%s' doesn't exist.

Feedback

**Description**

The library name specified by -library contains path, but the path doesn't exist.

**What Next**

Ensure the library path exist.

---

## BLOX-009

(warning) ChipletDef '%s' has different bump thickness (%s and %s) under one region.

**Description**

Bumps belong to one region have different bump thickness.

**What Next**

Check the related bump thickness values are expected or not.

---

## BLOX-010

(warning) The bumps between chiplet instance '%s' and '%s' are with inconsistent thickness %s and %s.

**Description**

A connection is detected by write_3dblox since their bump regions are overlapped. But they have different bump thickness.write_3dblox chooses the top thickness as the connection thickness.

**What Next**

Check the related bump thickness values are expected or not.

---

## BLOX-011

(error) 3Dblox version 1.0 does not support -mode feasibility.

**Description**

3Dblox version 1.0 does not support -mode feasibility, please specify -version upper than 1.0

**What Next**

Please do not use -version 1.0 with -mode feasibility.

Feedback

# BO

## BO-2001

(Information) Setting global option with individual option violates mutual Exclusivity.

**Description**

This message invokes when set_boundary_optimization values are set with auto | none | all along with any of one of more below mentioned options. -constant_propagation, -unloaded_propagation, -equal_opposite_propagation, -phase_inversion. When set_boundary_optimization is set to false | true then there is no issue of mutual exclusivity so as to maintain the backward compatibity. Example set_boundary_optimization [object_list] -false -constant_propagation false | NO ERROR set_boundary_optimization [object_list] -true -constant_propagation false -unloaded_propagation false | NO ERROR set_boundary_optimization [object_list] -auto -constant_propagation false | ERROR set_boundary_optimization [object_list] -none -constant_propagation false -unloaded_propagation false | ERROR

set_boundary_optimization [object_list] -none | NO ERROR set_boundary_optimization [object_list] -constant_propagation false | NO ERROR

## BO-2002

(Information) Global Option Values are not set with auto | none | all.

**Description**

This message invokes when set_boundary_optimization values are not set with auto | none | all. User shall only use any of the above described values while setting set_boundary_optimization command.

## BO-2004

(Information) Please update your script to include the new Boundary Optimization UI. Use all, auto, none instead of true or false.

**Description**

This warning message is invoked when either true of false is used with set_boundary_optimization. Please update your script to include new UI (all, none or auto option with set_boundary_optimization command).

Feedback

## BO-2005

(Error) Boundary optimization settings on %s are inherited from instance %s, please use 'remove_boundary_optimization' on instance.

**Description**

This error message is invoked when remove_boundary_optimization command is executed for pins which have inhereted boundary optimization restriction from their instance.

## BO-2006

(Error) Boundary optimization settings on %s are inherited from module %s, please use 'remove_boundary_optimization' on module.

**Description**

This error message is invoked when remove_boundary_optimization command is executed for pins which have inhereted boundary optimization restriction from their module.

# CAS

## CAS-0001

(Error) Net net has conflicting user case analysis values on load pins.

**Description**

Multiple conflicting case analysis values are specified on load pins of the net. This is a situation that cannot occur in normal circuit operation and may result in incorrect analysis. The specified case values are propagated forward for each conflicting load pin, even though they disagree with one another.

A condition that is not possible in circuit operation will be analyzed. This can result in poor analysis coverage or in the analysis of false timing paths.

**ONE POSSIBLE SCENARIO**

Consider a scenario where a single hierarchical net is driven by a single driver, and connects to two input pins. A set_case_analysis of 0 is set on one of the input pins and a set_case_analysis of 1 is set on the other. Because these two pins are connected to the same net, this condition is not physically possible, and a CAS_0001 rule violation will be issued.

Feedback

**What Next**

The "Violation Details" section will give you the names of the net and its load pins, along with their case values. To find the drivers of the net use the report_net -connections command. For the example scenario, the command can be issued as follows:

rtl_shell> report_net -connections [get_net net1]

**********
Report : net -connections
**********
Connections for net 'net1':
Driver Pins Type
----------
----------
driver/Z Output Pin (AN2)

Load Pins Type
---------- ----------
load2/A Input Pin (IV)
load1/A Input Pin (IV)

The report contains the driver pins for the net as well as its load pins.

**FIX SUGGESTION**

To resolve the issue, move the case value setting to the driver of the net with the appropriate value for this scenario.

For the example scenario, the command can be issued as follows:

set_case_analysis 1 driver/Z remove_case_analysis load1/A remove_case_analysis load2/A

All load pins of the net will have the case value of 1.

set_case_analysis report_net remove_case_analysis

# CATEGORY

## CATEGORY-001

(error) cannot create category rule '%s' since the previously created rule is not built-in.

**Description**

This error message occurs when an attempt is made to create a category rule with the -builtin option even though the previously created rule is not built-in.

**What Next**

Rerun gui_create_category_rule without the -builtin option.

**See Also**

- gui_create_category_rule

Feedback

## CATEGORY-002

(error) cannot create category rule with name '%s' since a rule with that name already exists.

### Description

This error message occurs when an attempt is made to create a category rule with a rule name that is already in use.

### What Next

Rerun gui_create_category_rule with a -name option value which is not a rule name that is already in use.

### See Also

- gui_create_category_rule

## CATEGORY-003

(error) for category rule '%s' cannot add subrule '%s' since no such rule exists.

### Description

This error message occurs when gui_create_category_rule -subrules references a rule which does not already exist.

### What Next

One possibility is that the subrule name was mis-spelled when passed to -subrules. In that case rerun gui_create_category_rule but passing the correctly spelled rule name to -subrules. Another possibility is that the rule name passed to -subrules was correctly spelled but does not yet exist. In that case, create a new rule with that name (using gui_create_category_rule -name) and then rerun the gui_create_category_rule -subrules command that failed.

### See Also

- gui_create_category_rule

## CATEGORY-004

(error) cannot add subrule '%s' since that subrule has already been added to the category rule '%s'.

Feedback

### Description

This error message occurs when gui_create_category_rule -subrules references the same subrule (using the same rule name) more than once in the subrules list.

### What Next

Rerun gui_create_category_rule -subrule but removing duplicate references to the rule name that was listed in the subrules more than once.

### See Also

- gui_create_category_rule

---

## CATEGORY-005

(error) -rule_names or -names contains invalid category rule name '%s'.

### Description

This error message occurs when a command is run which accepts a list of category rule names via -rule_names or -names and that list includes a category rule name which does not exist.

### What Next

Rerun the command with corrected spelling for the rule name (passed to -rule_names or -names) which was mis-spelled.

### See Also

- gui_list_category_rules

---

## CATEGORY-011

(error) cannot create category rule '%s' since it has an error in the category specification string at position %d.

### Description

This error message occurs when gui_create_category_rule is run with a -category option value which has a syntax error.

### What Next

Fix the syntax error in the -category option string value and rerun gui_create_category_rule.

Feedback

**See Also**

- gui_create_category_rule

---

## CATEGORY-012

(error) cannot create category rule '%s' since it has an error in the filter specification string.

**Description**

This error message occurs when gui_create_category_rule is run with a -filter option value which has a syntax error.

**What Next**

Fix the syntax error in the -filter option string value and rerun gui_create_category_rule.

**See Also**

- gui_create_category_rule

---

## CATEGORY-021

(error) cannot remove category rule '%s' since it is in use by one or more existing categories.

**Description**

This error message occurs when gui_remove_category_rule is run to remove a rule that one or more existing categories depends on.

**What Next**

Fix the issue by removing the affected categories and rerun gui_remove_category_rule.

---

## CATEGORY-031

(error) cannot evaluate expression since no attribute '%s' exists for '%s'.

**Description**

This error message occurs when -category or -filter specifications reference an attribute which does not exist for an object that the category rule is executed for.

**What Next**

Revise the -category or -filter specification to use an attribute which exists for the object in question.

Feedback

**See Also**

- gui_create_category_rule

---

## CATEGORY-032

(error) cannot evaluate expression since attribute '%s' is a collection.

**Description**

This error message occurs when -category or -filter specifications reference an attribute whose value is a collection containing multiple objects.

**What Next**

Referencing a collection containing multiple objects is currently not supported in -category or -filter specifications. Use a different attribute instead.

**See Also**

- gui_create_category_rule

---

## CATEGORY-033

(error) cannot evaluate expression since attribute '%s' of type '%s' has sub-attributes.

**Description**

This error message occurs when -category or -filter specifications reference an attribute whose value is a collection containing a single object.

**What Next**

Consider using "dot notation" in -category or -filter specification to access an attribute of the object in the collection. For example, instead of -category <startpoint> try -category <startpoint.full_name>. Use list_attributes to discover the attributes supported for the type mentioned in the error message.

**See Also**

- gui_create_category_rule

---

## CATEGORY-034

(error) cannot evaluate filter expression since attribute '%s' of type '%s' of data type '%s' is not of data type '%s'.

Feedback

### Description

This error message occurs when evaluating a filter expression where an attribute for an object that the category rule is executed for doesn't match a required type in the expression.

### What Next

Revise the -filter specification so that the type for the attribute matches what is required.

### See Also

- gui_create_category_rule

---

## CATEGORY-035

(error) cannot evaluate filter expression because of an operand type mismatch.

### Description

This error message occurs when evaluating a filter expression where a operand type mismatch has occured.

### What Next

Revise the -filter specification so that the operands match in type.

### See Also

- gui_create_category_rule

---

## CATEGORY-036

(error) cannot evaluate filter expression because operator '%s' has a type mismatch.

### Description

This error message occurs when evaluating a filter expression where a operator type mismatch has occured.

### What Next

Revise the -filter specification so that the operators match in type.

### See Also

- gui_create_category_rule

Feedback

## CATEGORY-040

(error) cannot create category tree with name '%s' since a category tree with that name already exists.

### Description

This error message occurs when an attempt is made to create a category tree with a category tree name that is already in use.

### What Next

Rerun gui_create_category_tree with a -name option value which is not a category tree name that is already in use.

### See Also

- gui_create_category_tree

## CATEGORY-052

(error) No current category tree found.

### Description

This error message occurs when the -tree option is omitted when calling one of the category commands, and there is no current category tree. With these category commands, you can pass an explicit -tree option value to specify the category tree to be operated on. If that optional option is omitted, an attempt will be made to use the current category tree. This error happens if there is no current category tree (which can only happen if no category trees exist).

### What Next

Create at least one category tree (so that there will be a current category tree) and rerun the command which generated the error.

## CATEGORY-053

(error) the categorization script file '%s' already exists.

### Description

This error message occurs when the gui_write_category command is run with a -file option value which identifies a file which already exists, and the -overwrite option was not specified.

Feedback

### What Next

Rerun the gui_write_category command specifying the -overwrite option to overwrite the existing file. Or else specify a file which does not yet exist via the -file option.

---

## CATEGORY-054

(error) the categorization script file '%s' could not be opened for write.

### Description

This error message occurs when the gui_write_category command is run with a -file option value which identifies the file to be written, and the file cannot be opened for write.

### What Next

Investigate parent directory permissions to figure out why the file could not be opened for write. Fix any such permissions problems and rerun the gui_write_category command.

---

## CATEGORY-055

(error) cannot create new subcategories of a hierarchical category which already has subcategories that were created via a separator character

### Description

The categorization engine can create category hierarchy as a result of interpreting hierarchical separator characters embedded in the string value produced by a category rule's category specification. This error message occurs when the gui_create_category command is used to try to add additional subcategories to a category which already has one or more subcategories that were created by interpreting a separator character.

### What Next

Choose a different category to pass to the -parent_category option of gui_create_category.

### See Also

- gui_create_category_rule

---

## CATEGORY-056

(error) cannot remove subcategories of a hierarchical category which has subcategories that were created via a separator character

Feedback

### Description

The categorization engine can create category hierarchy as a result of interpreting hierarchical separator characters embedded in the string value produced by a category rule's category specification. This error message occurs when the gui_remove_category command is used to try to remove subcategories of a category which has one or more subcategories that were created by interpreting a separator character.

### What Next

Choose a different category to pass to the -parent_category option of gui_remove_category.

### See Also

- gui_create_category_rule

## CATEGORY-091

(warning) cell '%s' has no block mark.

### Description

This warning message occurs when a command is run to get or remove the block mark of a cell instance which has no block mark.

### What Next

Rerun the command but specify a cell instance for which a block mark has been previously set.

### See Also

- gui_get_cell_block_marks
- gui_remove_cell_block_marks

# CCD

## CCD-001

(info) Skip %s, reason '%s'.

### Description

This information prints the detailed reason why an object is being skipped by CCD. Objects can be clock sinks or intermediate pins in the clock network.

Following are the possible reasons that can cause CCD to skip an object:

**Sink is boundary**

The sink is in the fanin/fanout of I/O ports. So this is considered as a boundary flop. *ccd.optimize_boundary_timing* is set as false and hence boundary flops cannot be skewed by CCD.

**Sink is a fixed balance pin**

The sink has the attribute *cts_fixed_balance_pin* set to true and *ccd.respect_cts_fixed_balance_pins* application option is enabled.

**Pin is not in the balance portion of the clock tree**

Pin is in the ignore portion of the clock tree and CCD will not skew this pin. Use *report_clock_qor -type structure* and *report_clock_balance_points* to confirm that the ignore exception settings are user intended.

**Pin is on dont_touch net**

Clock net connected to the pin is marked as dont_touch. Use *report_dont_touch* on the clock net to check if the dont_touch setting is user intended.

**Pin is on read_only net**

Clock net connected to this pin is marked as read_only. This could be a sub-block not being editable.

**Pin is in abstract or ETM**

Pins in abstracts or ETMS lack proper timing/power models and cannot be power shaped.

**What Next**

See *CCD-002* for the total number of skipped objects for each reason. Review the above details and ensure that all the settings are reasonable and user intended.

---

## CCD-002

(info) There were %d objects skipped by CCD due to the reason '%s'.

**Description**

This information prints the total number of objects skipped by CCD for each reason.

**What Next**

See details of which objects were skipped and their reasons in *CCD-001*.

Feedback

## CCD-003

(info) Using default fall-back value %s for %s: Please set app option %s with desired value.

**Description**

This message indicates that some options have not been set and a fall-back default is used instead.

**What Next**

Set the recommended option to the desired value.

## CCD-004

(info) DPS ignoring effective resistance - %s

**Description**

This message indicates that DPS is ignoring the use of effective resistance feedback. Optimization will continue without it.

**What Next**

Please check that rail results are loaded.

## CCD-005

(error) An error prevented DPS optimization. %s

**Description**

Something in the design or the configuration prevented the successful run of a Dynamic Power Shaping optimization.

**What Next**

It may be necessary to alter the design, configuration or application options to complete the information required for the DPS optimization.

## CCD-006

(warning) No %s scenarios for DPS optimization.

Feedback

**Description**

The DPS optimization step did not find any active setup or hold enabled scenarios. This may result in uncontrollable violations of setup or hold timing.

**What Next**

Add setup or hold scenarios in the place_opt stage to make sure both setup and hold timing is constrained in the DPS optimization.

---

## CCD-007

(error) Design partitioning not possible. %s

**Description**

DPS did not create clock sink partitions which makes it impossible to start a DPS optimization. This could be due to design traits such as abstracts or ETMs, clock nets being marked dont_touch, \ sinks being set as prefixed balance points or ccd.dps.powershape_* app options limiting the number of sinks that can be skewed. CCD-001 and CCD-002 messages in the log may tell what types of settings affected the partitioning.

**What Next**

Change the design traits or app options to allow sinks skewing which will allow creation of sink partitions for optimization.

---

## CCD-008

(warning) Use of property %s is deprecated in use cases.

**Description**

A property which is deprecated has been applied to a DPS use case.

**What Next**

See the man page for ccd.dps.use_cases for a list of valid properties.

---

## CCD-009

(information) %d clock pins are clocked by multiple clocks: %s. Resolved by %s.

Feedback

**Description**

When doing the DPS power simulation clock pins were encountered which were clocked by multiple active clocks. The simulation will pick one of them, if they are identical it takes any, otherwise it picks the fastest of the active clocks.

**What Next**

If you want the simulation to use a different clock you can specify the active clocks in the use case. See the man page for ccd.dps.use_cases for more detail.

---

## CCD-010

(information) %d input ports are constrained by multiple clocks: %s. Resolved by %s.

**Description**

When doing the DPS power simulation input ports were encountered which were constrained by multiple active clocks. The simulation will pick one of them, if they are identical it takes any, otherwise it picks the fastest of the active clocks.

**What Next**

If you want the simulation to use a different clock you can specify the active clocks in the use case. See the man page for ccd.dps.use_cases for more detail.

---

## CCD-011

(Warning) Relocation detects : %s %s

**Description**

The warning shows tool detects multiple driver or pin count of the net is less than 2.

**What Next**

Please check this net in the design.

---

## CCD-012

(error) Errors in supply net connectivity found during DPS initialization. %s

**Description**

Dynamic Power Shaping needs connected supply nets to be able to build a valid power model

Feedback

**What Next**

Make sure the UPF definition is complete. The command 'connect_pg_net -automatic' can be used to check the number of instance connections on each supply net and if there are supply net connections which cannot be resolved.

---

## CCD-014

(info) %s.

**Description**

This message prints the detailed information while executing report_ccd_visualization.

---

## CCD-015

(error) %s.

**Description**

This message prints the detailed error while executing report_ccd_visualization.

---

## CCD-016

(warning) %s.

**Description**

This message prints the detailed warning while executing report_ccd_visualization.

---

## CCD-017

(Warning) Clock buffer chain is longer than 500. From pin %s to sink %s (pin slack: %f, clock: %s)

**Description**

The warning shows unexpected too long clock buffer chain.

**What Next**

Please check the timing on the path in the design.

---

## CCD-018

(error) Invalid CCD reference corner. Corner %s %s

Feedback

**Description**

User specified reference corner via ccd.reference_corner app option, but the corner was not recognized or lacks any timing scenarios.

**What Next**

Fix the reference corner name or make sure there are active timing scenario for that corner.

---

## CCD-019

(warning) CCD reference corner is %s, but previous ccd run used reference corner %s

**Description**

There are different reference corners used for CCD runs in different flow stages. This may lead to mixed offsets with different resolutions as well as occasional mis-interpretation of prepone and postpone limits.

**What Next**

Make sure the available corners are the same throughout the flow, or specify a reference corner to use for all stages in the ccd.reference_corner app option.

---

## CCD-020

(Warning) Compute Useful Skew is running in mixed mode (found %d ideal and %d propagated clocks).

**Description**

This warning indicates that the design contains both ideal and propagated clocks. This will not necessarily work well in the optimization and may result in worse QoR due to many clock sinks being set to fixed-latency.

**What Next**

Please check the clock definitions to make sure all clocks are ideal (for pre-CTS CCD) or all clocks are propagated (for CTO CCD).

---

## CCD-021

(Error) Invalid or missing FSDB file '%s'.

Feedback

**Description**

The provided FSDB file either does not exist or is invalid. The DPS power modelling is unable to continue.

**What Next**

Please check the path is correct.

---

## CCD-022

(Warning) Sink '%s' not found in FSDB file.

**Description**

A sink pin of the design cannot be found in the provided FSDB file. DPS will fallback to use vectorless modelling for the sink. Too many warnings of this kind indicates that the FSDB file is unsuitable for driving the power modelling.

**What Next**

Please make sure that the FSDB file is correct.

---

## CCD-023

(Warning) Pin '%s' not found in FSDB file.

**Description**

A pin of the design cannot be found in the provided FSDB file. DPS will fallback to use vectorless modelling for the corresponding sink. Too many warnings of this kind indicates that the FSDB file is unsuitable for driving the power modelling.

**What Next**

Please make sure that the FSDB file is correct.

---

## CCD-024

(Warning) A total of %d sinks, pins or ports not found in FSDB. %d%% of sinks being randomized.

**Description**

A sumary message indicating how well the provided FSDB matches the netlist. A too high percentage reported indicates that the FSDB does not match the netlist , and the results of the power simulation will not correlate well with the original simulation.

Feedback

**What Next**

Please make sure that the FSDB file is correct.

---

## CCD-025

(error) CCD (Concurrent Clock and Data) optimization failed due to %s.

**Description**

The CCD optimization experienced an error and was not able to complete the optimization. This could be caused by insufficient design data or invalid app options.

**What Next**

Check the error messages and fix the related settings.

---

## CCD-026

(info) Endpoint %s has reached %s limit defined by %s (constraint: %.3f, offset: %.3f)

**Description**

This messages reports information relating the endpoint with the constraint limit which is either reached or is one resolution away from being reached by the skew profile from CUS.

---

## CCD-027

(Information) Skip CCD clock DRC fixing as there is no DRC violation or all DRC violations are not fixable.

**Description**

CCD will not fix the DRC violation on clock since there is no DRC violation or all DRC violations are not fixable.

**What Next**

Please check if there are valid DRC constraints on design/clock/pin. Please check if there are dont_touch attributes on DRC violation pins/nets.

---

## CCD-028

(info) Sink '%s' is marked as boundary.

Feedback

**Description**

This message indicates that the named object is identified as a boundary sink. It will only be output if boundary sinks are treated differently than non-boundary sinks.

**What Next**

Review the reported sinks and ensure that all the settings are reasonable and user intended.

---

## CCD-029

(info) LP solver could not fully converge. Returned result is '%.2f' percent away from the real optimal.

**Description**

This message indicates that the mathematical solver which performed the optimization could not reach the optimal result.

**What Next**

If the user is not satisfied with the returned QoR then they should file a testcase to the CCD team to check if a better result was attainable.

---

## CCD-100

(info) There are %d pins with cts_fixed_balance_pin attribute set to true, which could limit skewing during CCD.

**Description**

When cts_fixed_balance_pin attribute is set on a pin and ccd.respect_cts_fixed_balance_pins is set to true or upstream, CCD will no skew this pin.

---

## CCD-101

(Warning) %s has inconsistent balance points.

**Description**

The pin has inconsistent balance points set on it. For example, balance points are set on some clocks/corners but missing on other clocks/corners. CCD result can be jeopardized because of this.

Feedback

### What Next

Check balance point setting on this pin and make sure it is consistent across clocks and corners.

---

## CCD-102

(Warning) CCD is overwriting exclude exception on %s.

### Description

This pin is skewed by CCD and CCD is applying a balance point on it, which will overwrite user exclude exception.

### What Next

Check the exclude exception on the pin.

---

## CCD-103

(Info) Useful skew copied/scaled %d balance points and %d clock latencies.

### Description

CCD will copy and scale CCD generated balance points and latencies to new clocks and scenarios which are not presented in earlier CCD calls. This will make sure useful skew information is consistent across clocks and scenarios.

---

## CCD-104

(Warning) Unable to apply ccd_max_prepone attribute with value %.3f in order to replace estimated latency value on protected arc to pin %s

### Description

The ccd.skew_opt_max_prepone_override_latency_estimation app option is set to true and CCD therefore attempted to replace the prepone constraint calculated by minimum latency estimation by the value of the ccd_max_prepone attribute. It cannot do so because the arc to the pin is protected, for instance by a dont_touch constraint on the driving net.

---

## CCD-105

(Info) Generated %d seed macro offsets using strategy "%s".

Feedback

**Description**

CCD has adjusted the useful skew in macros according to the selected strategy. The balance points and latencies in macros may change during this step.

# CCGUI

## CCGUI-001

(Error) Custom Compiler failed to open. %s

**Description**

Custom Compiler failed to open.

**What Next**

Custom Compiler failed to open, please look at the previous error messages to understand the reasons for failure.

# CDPL

## CDPL-001

(error) Block ref / task %s, script %s: Run failed to complete.

**Description**

This errors occurs when running the given script on the given block reference (or task not associated with any block) in parallel. This run failed to complete. There are a number of reasons why this could happen. Possibly, the tool could have crashed or the network may have failed.

**What Next**

Look at the log file of the block reference / task run which is in the command specified <work directory>/<block reference> to see why the run failed. If nothing appears to have gone wrong, it could have been a network problem and you can try to rerun the command. If the run failed, then debug it as any normal failure.

You can also run serially to see if that goes through. If not, then there is a script related problem rather than a distributed processing related problem.

Feedback

## CDPL-002

(error) Block ref / task %s, script %s: Script errored out.

**Description**

This errors occurs when running the given script on the given block reference (or task not associated with a block) in parallel. This run failed to complete because one of the commands in the script errored out.

**What Next**

Look at the log file of the block reference / task run which is in the command specified <work directory>/<block reference> or <work directory>/<task name> to see what command errored out. Fix the script to avoid this problem.

It may help to run serially with only the smallest / fastest of the failing blocks / tasks so as to quickly find a fix.

## CDPL-003

(error) Block ref / task %s, script %s: Internal error.

**Description**

This errors occurs when running the given script on the given block reference (or task not associated with any block) in parallel. This run failed to complete because of an unexpected internal error.

**What Next**

Look at the log file of the block reference / task run which is in the command specified <work directory>/<block reference> or <work directory>/<task reference> to see what command errored out. You can try to change the script to avoid this problem.

You can run serially to avoid this issue. If only a few blocks are failing but others are ok, then you can run only the failing ones serially.

Report this issue to Synopsys.

## CDPL-004

(error) Host option %s has not specified any host names.

**Description**

This errors occurs when you are using rsh or ssh for distributed processing but have not allocated any hosts (ie machines) to be used.

Feedback

### What Next

Look at man page of set_host_options to see how to properly set up the host names to be used.

---

## CDPL-005

(error) Distributed run failed. error code: %s.

### Description

This error means that the distributed run failed. There are a number of possibilities for why this might happen. Common issues include lack of compute farm access, lack of network access, or incorrect host_options settings. Another possibility is if you set a timeout, it is possible that the timeout limit was reached.

### What Next

Check that the host_options used is set up properly. Use the check_host_options command for this. If check_host_options command is not successful, then any distributed processing in ICC2 will not work. Try your submit command (from set_host_option -submit_command) at the unix prompt with a trivial job like "xterm". If it doesn't work at the unix prompt, then you should work with your farm administrator on how to get your farm access properly set up.

If you are using a timeout (set_host_option -timeout), it is possible that the timeout limit was reached. Try with a longer timeout.

Another thing is that your work directory must be on a network disk (i.e., not /tmp). That disk must be accessible from all hosts in distributed run because that disk will be used to save log files, etc.. If you did not set the work directory to be a network disk then change this.

There are a couple workarounds, first you can run sequentially rather than distributed, but this could lead to long wall clock run times. Second, if you are having compute farm setup issues, then switch to use rsh or sh in set_host_options -submit_command. This has advantage that it does not need much setup so you can still run distributed while farm issue is being resolved.

---

## CDPL-006

(warning) Characters [] used in -name option were substituted with ().

Feedback

**Description**

This warning is issued to indicate that the characters [ and/or ] were used in the -name option to the *run_blokc_scripts* command. For this option, [ and ] are reserved characters and they were automatically replaced with ( or ).

**What Next**

Pass a value to the -name option that does not contain [ or ].

---

## CDPL-007

Error: CDPL connection to parent process was lost. Exiting.

**Description**

The network connection to the parent process was lost. The child process can no longer communicate with the parent, and any results produced will be lost. To avoid tying up resources, the child process will exit.

This can occur if the distributed command on the parent process is interrupted due to an error in one of the child processes. This can also occur if the user pressed Ctrl-C during read-only jobs, or because the main process was terminated.

**What Next**

Determine why the parent process stopped communicating and resolve the problem.

If this behavior is not desired, disable the distributed process watchdog timer before launching distributed jobs.

set_app_options -name plan.distributed_run.watchdog_timer_enabled -value false

---

## CDPL-008

Error: Distributed processing version mismatch. Main version: %s, worker version %s

**Description**

The worker started as part of a distributed run was not running the same version of the tool as the main process. This will not happen if your distributed (farm) environment is set up the same as that of the main machine. Ensure that the same version of the executable is available from the absolute/real path the main session was started from.

---

## CDPL-009

(error) Cannot reschedule a task whose dependent task didnot complete successfully.

Feedback

### Description

This error means that the task could not be rescheduled. The reason is that the task is dependent on other tasks which were not all successfully completed.

### What Next

Check why the task and the dependent tasks didnot complete successfully.

---

## CDPL-100

(warning) Waiting to start %d distributed jobs.

### Description

This warning indicates that there are a certain number of distributed jobs that have been requested but not been started yet. There are 2 main reasons for this. The first is that your compute farm is busy and thus the distributed job is pending in the farm queue. This is normal and you just need to wait. One thing that could help is to remove any lower priority jobs that you may have running that you can run later. The second is that there is a problem with your farm resource request. For example if you request a resource that is not available on the farm, this request can be pending forever. Also if you request a resource that is incompatible with the tool, then the distributed job will wait forever. Note that if there is at least 1 distributed job running then progress is being made on the run, but it will be slower than if all the distributed jobs are running.

### What Next

Check the farm queue using a command like qstat (SGE) or bjobs (LSF) that is appropriate for your compute farm. You will either see your job pending in queue or you will not see it (to avoid confusion with other runs, only run one distributed job at this time).

1. If you see your job is pending, then check that the resources you requested can be satified by the queue. If so, then wait. If not, then fix the resource request made in *set_host_options*.
2. If you don't see your job in the queue, then it likely failed to run due to a machine and/or operating system incompatibility. In this case change the *set_host_options* resource requirement to ensure all the machines you get are compatible with this version of executable. Note that it is common for a farm to have machines with different operating systems installed. The tool may be incompatible with older versions of operating system.

---

## CDPL-101

(warning) Only able to obtain license for %d distributed processes. %d distributed processes were requested.

Feedback

### Description

This warning indicates that there are a certain number of distributed jobs which will not be actively running because there were not enough licenses for them. When one actively running processes finishes, then a waiting process can start actively running.

### What Next

There is nothing to do but note that your run may take longer to complete because the maximum parallelism is not used. You may see that some processes are not progressing because they are waiting for a license to proceed.

---

## CDPL-102

(warning) Did not recognize protocol setting: %s.

### Description

This warning indicates that you have set a communication protocol that that the tool does not understand. The tool understands protocols like rsh, ssh, lsf, sge, and sh.

### What Next

If the distributed job is running properly, then you can ignore this message. If distributed job is not running, then you may need to set one of the above listed protocols depending on what type of distributed processiong you are using. For example, if you are using an SGE farm, then you can set the sge protocol.

---

# CELLEM

---

## CELLEM-025

(error) Scenario '%s' is not active for cell EM analysis.

### Description

A scenario must be active for cell EM analysis before the analysis can be performed. Either the scenario is not active at all, or its cell_em setting is set false.

### What Next

To perform cell EM analysis, please set the scenario's status to include cell EM.

### See Also

- set_scenario_status

Feedback

# CGR

## CGR-0001

(Error) Clocks generated from the same master clock master_clock are declared as asynchronous.

### Description

Clocks which are generated from the same master clock cannot be asynchronous with each other. Paths between the clocks declared as asynchronous will not be analyzed. In addition, for crosstalk delay, infinite windows analysis is performed on nets in aggressor/victim relationships between the clocks. This can result in overly conservative signal integrity analysis.

### ONE POSSIBLE SCENARIO

Consider a scenario where two generated clocks with the same master clock are declared asynchronous:

create_clock -period 4 clk1 create_generated_clock -source clk1 -divide_by 2 clk2 create_generated_clock -source clk1 -divide_by 4 clk3 set_clock_groups -asynchronous -group {clk2} -group {clk3}

The two generated clocks, clk2 and clk3, cannot be asynchronous because they both have the same master clock, clk1, and a CGR_0001 rule violation will be issued.

### What Next

The "Violation Details" section will give you the names of the clocks which are generated from the same master clock, but declared as asynchronous. You can click on the clock names to see the relevant clock definition, and the location of the problematic clock group definition is also provided.

### FIX SUGGESTION

Consider adjusting the clock grouping relationship of the two clocks. In the preceding example, the asynchronous relationship can be changed by removing the set_clock_groups command:

set_clock_groups -asynchronous -group {clk2} -group {clk3}

Alternatively, you might want to change the master/generated clock relationship such that the reported clocks are not generated from the same source.

### See Also

set_clock_groups create_clock create_generated_clock report_clock report_clock_crossing remove_clock_groups remove_clock remove_generated_clock

Feedback

## CGR-0002

(Error) Clock clock1 is generated directly or indirectly from the clock clock2. They should not be declared as asynchronous.

### Description

Clocks which are generated from one another cannot be asynchronous. Paths between the clocks declared as asynchronous will not be analyzed. In addition, for crosstalk delay, infinite windows analysis is performed on nets in aggressor/victim relationships between the clocks. This may result in overly conservative signal integrity analysis.

### ONE POSSIBLE SCENARIO

Consider a scenario where a generated clock and its master clock are declared asynchronous:

create_clock -period 4 clk1 create_generated_clock -source clk1 -divide_by 2 clk2 set_clock_groups -asynchronous -group {clk1} -group {clk2}

The clock, clk2, is a generated clock with master clock, clk1. Because clk2 is generated from clk1, they cannot be asynchronous to each other, and a CGR_0002 rule violation will be issued.

### What Next

The "Violation Details" section will give you the names of the generated clock and the master clock that are declared as asynchronous. You can click on the clock names to see the relevant clock definition, and the location of the problematic clock group definition is also provided.

### FIX SUGGESTION

To resolve this issue, consider adjusting the clock grouping relationship of the two clocks. In the preceding above, the asynchronous relationship can be changed by removing the set_clock_groups command:

set_clock_groups -asynchronous -group {clk1} -group {clk2}

Alternatively, you might want to change the master/generated clock relationship of the reported clocks.

### See Also

set_clock_groups create_clock create_generated_clock report_clock -groups report_clock_crossing remove_clock_groups remove_clock remove_generated_clock

Feedback

## CGR-0003

(Error) Clock clock1 and clock clock2 are defined asynchronous with each other. There are clocks generated directly or indirectly from these clocks which are not defined as asynchronous.

### Description

If two clocks are asynchronous, then all the clocks generated from these clocks should be asynchronous with each other. A generated clock does not inherit the asynchronous properties of its master clock, so it should be declared asynchronous to clocks that are asynchronous with its master clock. For clocks not declared asynchronous, paths between the clocks will be analyzed, possibly resulting in timing violations being reported for false paths. In addition, for crosstalk delay, the timing window relationship will be preserved on nets in aggressor/victim relationships between the clocks. This may result in optimistic signal integrity analysis.

### ONE POSSIBLE SCENARIO

Consider a scenario where two master clocks are declared asynchronous, while no asynchronous relationship is specified between other clock pairs:

create_clock -period 4 clk1 create_clock -period 4 clk2 create_generated_clock -source clk1 -divide_by 2 clk3 create_generated_clock -source clk2 -divide_by 2 clk4 set_clock_groups -asynchronous -group {clk1} -group {clk2}

Two master clocks, clk1 and clk2 are asynchronous to each other. The two generated clocks clk3 and clk4 should also be asynchronous because this is the relationship between their master clocks. Since the generated clocks clk3 and clk4 do not have an asynchronous relationship, a CGR_0003 rule violation will be issued.

### What Next

The "Violation Details" section will give you the names of the clock pairs that are missing asynchronous clock declarations. You can click on the clock names to see the relevant clock definition, and the location of the problematic clock group definition is also provided.

### FIX SUGGESTION

To resolve this issue, consider declaring the clock pairs as asynchronous. In the preceding example, the correct asynchronous relationship can be declared by changing the set_clock_groups command as follows:

set_clock_groups -asynchronous -group {clk1 clk3} \ -group {clk2 clk4}

Alternatively, you can change the master/generated clock relationship of the reported clocks.

### See Also

Feedback

set_clock_groups create_clock create_generated_clock report_clock -groups report_clock_crossing remove_clock_groups remove_clock remove_generated_clock

---

## CGR-0004

(Error) Clock clock1 is missing an asynchronous definition with at least one clock in each of the following asynchronous clock pairs.

### Description

Clocks which are not generated directly or indirectly from either of the two asynchronous clocks should have an asynchronous relationship with at least one of them. A clock cannot be synchronous with two clocks that are not synchronous to each other. For clocks not declared asynchronous, paths between the clocks will be analyzed, possibly resulting in timing violations being reported for false paths. In addition, for crosstalk delay, the timing window relationship will be preserved on nets in aggressor/victim relationships between the clocks. This may result in optimistic signal integrity analysis.

### ONE POSSIBLE SCENARIO

Consider a scenario where two clocks are declared asynchronous. Another clock that is not generated from either of the two clocks is not declared as asynchronous to either one:

create_clock -period 4 clk1 create_clock -period 4 clk2 create_clock -period 4 clk3 set_clock_group -asynchronous -group {clk1} -group {clk2}

The clocks, clk1 and clk2 are asynchronous to each other. The third clock, clk3, should have an asynchronous relationship to either clk1 or clk2. Because clk3 does not have an asynchronous relationship with either clk1 or clk2, a CGR_0004 rule violation will be issued.

### What Next

The "Violation Details" section will give you the names of the clocks pairs which are declared asynchronous. The reported clock should be declared asynchronous to at least one clock in each asynchronous clock pair. You can click on the clock names to see the relevant clock definition, and the location of the problematic clock group definition is also provided.

### FIX SUGGESTION

To resolve this issue, consider declaring the reported clock as asynchronous to at least one clock in each asynchronous clock pair.

### See Also

set_clock_groups create_clock create_generated_clock report_clock -groups report_clock_crossing remove_clock_groups remove_clock remove_generated_clock

Feedback

## CGR-0005

(Error) Clock clock1 is generated directly or indirectly from the clock clock2. They should not be declared as physically exclusive.

### Description

Clocks which are generated from one another cannot be physically exclusive. Paths between the clocks declared as physically exclusive will not be analyzed. An incomplete timing analysis may be performed on the design. In addition, for crosstalk delay, it is assumed that there is no possible crosstalk interaction on nets in aggressor/victim relationships between the clocks. This may result in optimistic signal integrity analysis.

### ONE POSSIBLE SCENARIO

Consider a scenario where a generated clock and its master clock are declared physically exclusive:

create_clock -period 4 clk1 create_generated_clock -source clk1 -divide_by 2 clk2 set_clock_group -physically_exclusive -group {clk1} -group {clk2}

The generated clock, clk2, has clock, clk1, as its master. The two clocks cannot be physically exclusive because they are physically related to each other, and a CGR_0005 rule violation will be issued.

The "Violation Details" section will give you the names of the related clocks that are declared as physically exclusive. You can click on the clock names to see the relevant clock definition, and the location of the problematic clock group definition is also provided.

### FIX SUGGESTION

To resolve this issue, consider adjusting the clock grouping relationship of the two clocks. In the example above, this can be achieved by removing the following command:

set_clock_group -physically_exclusive -group {clk1} -group {clk2}

Alternatively, you might want to change the master/generated clock relationship of the reported clocks.

### See Also

set_clock_groups create_clock create_generated_clock report_clock -groups report_clock_crossing remove_clock_groups remove_clock remove_generated_clock

## CGR-0006

(Warning) Clock clock1 and clock clock2 are defined physically exclusive with each other. There are clocks generated from these clocks which are not defined as physically exclusive.

Feedback

### Description

If two clocks are physically exclusive, then all the clocks generated from these clocks should be physically exclusive with each other. A generated clock does not inherit the physically exclusive properties of its master clock, so it should be declared physically exclusive to clocks that have this relationship to its master clock. For clocks not declared physically exclusive, paths between the clocks will be analyzed, possibly resulting in timing violations due to false paths. In addition, for crosstalk delay, the timing window relationships will be preserved on nets in aggressor/victim relationships between the clocks. This may result in pessimistic signal integrity analysis because physically exclusive clocks assume that there is no possible crosstalk interaction on nets with this relationship.

### ONE POSSIBLE SCENARIO

Consider a scenario where two master clocks are declared physically exclusive, while no physically exclusive relationship is specified between clocks derived from the master clocks:

create_clock -period 4 clk1 create_clock -period 4 clk2 create_generated_clock -source clk1 -divide_by 2 clk3 create_generated_clock -source clk2 -divide_by 4 clk4 set_clock_groups -physically_exclusive -group {clk1} -group {clk2}

Two master clocks, clk1 and clk2, are physically exclusive. The two generated clocks, clk3 and clk4, should also be physically exclusive with each other because this is the relationship between their master clocks. Since the generated clocks clk3 and clk4 do not have a physically exclusive relationship, a CGR_0006 violation will be issued.

### What Next

The "Violation Details" section will give you the names of the clocks pairs which should have been declared physically exclusive. You can click on the clock names to see the relevant clock definition, and the location of the problematic clock group definition is also provided.

### FIX SUGGESTION

To resolve this issue, consider declaring the clock pairs as physically exclusive. In the example above, the correct physically exclusive relationship can be declared by changing the set_clock_groups command as follows:

set_clock_groups -physically_exclusive -group {clk1 clk3} \ -group {clk2 clk4}

Alternatively, you might change the master/generated clock relationship of the reported clocks.

### See Also

set_clock_groups create_clock create_generated_clock report_clock -groups report_clock_crossing remove_clock_groups remove_clock remove_generated_clock

Feedback

## CGR-0007

(Warning) Multiple clocks or a clock with either set_input_delay or set_output_delay are defined at pin/port pin_port. In both cases, (the defined multiple clocks or a defined clock with the reference clock), these clocks should be defined as exclusive/asynchronous.

**Description**

Clocks originating from the same source should be set as physically exclusive. The tools will conservatively analyze all potential coupling conditions between the clocks that are not declared asynchronous or physically exclusive. This may result in poor crosstalk analysis performance and over-conservative analysis result.

**ONE POSSIBLE SCENARIO**

Consider a scenario where two clocks are created on the same port, but no physically exclusive relationship is specified:

create_clock -per 4 -name clk1 clk1 create_clock -per 4 -name clk1_dup -add clk1

Consider another scenario where a clock is defined at a port and a set_input_delay is also specified, but no physically exclusive relationship is specified between the defined clock and the reference clock:

create_clock -per 5 -name jtag_clk jtag_clk set_output_delay 1.4 -clock mclk { jtag_clk }

**What Next**

The "Violation Details" section will give you the names of the clock pairs which should be declared as physically exclusive.

**FIX SUGGESTION**

To resolve this issue, consider declaring the clock given pairs as physically exclusive. In the previous examples, this can be achieved with the following command:

set_clock_group -physically_exclusive -group {clk1} -group {clk1_dup}

**See Also**

report_clock_crossing create_clock set_clock_groups remove_clock_groups

## CGR-0008

(Warning) Multiple clocks defined at pin/port pin_port have false path set between them. For accurate signal integrity results use the set_clock_groups command.

Multiple clocks with false path relationships have been defined at the same source. It is recommended that a clock group relationship be defined in place of the individual false path exceptions. Defining a clock group relationship will disable the timing paths between

Feedback

clocks while allowing more accurate delay calculations in SI mode with PrimeTime. This violation can result in less accurate SI analysis results. Modeling the clocks interactions with set_clock_groups will allow the following nuances in SI delay calculations:

Two clocks defined to be logically exclusive of each other have no logical timing paths between them. PrimeTime does not check the logical timing between the clocks, but PrimeTime SI still checks for possible crosstalk interaction between them.

Two clocks defined to be physically exclusive of each other have no logical timing paths between them, and furthermore, are considered physically isolated from each other. PrimeTime does not check the logical timing between the clocks and PrimeTime SI assumes no possible crosstalk interaction between them.

Two clocks defined to be asynchronous with each other have no timing relationship at all. PrimeTime does not check the logical timing between the clocks, but PrimeTime SI still checks for possible crosstalk interaction between them, using infinite arrival windows.

**ONE POSSIBLE SCENARIO**

Consider a scenario where two clocks are created on the same port, and set_false_path relationships are specified between the two clocks:

create_clock -per 4 -name clk1 clk1 create_clock -per 4 -name clk1_dup -add clk1 set_false_path -from clk1 -to clk1_dup set_false_path -from clk1_dup -to clk1

**What Next**

The "Violation Details" section will give you the names of the clocks that are created on the same port with set_false_path relationships specified.

**FIX SUGGESTION**

To resolve this issue, consider using set_clock_groups to replace set_false_path between clocks. Below is an example of the command to achieve this:

set_clock_groups -physically_exclusive -group {clk1} -group {clk1_dup}

**See Also**

report_clock_crossing set_clock_groups remove_clock_groups

---

## CGR-0009

(Error) Clocks clock1 and clock2 have valid timing paths between them, but their common base period is more than the limit times the clock period.

**Description**

These two clocks should be mutually exclusive even though a valid timing path exists between them. The clocks can be either asynchronous, logically exclusive, or physically

Feedback

exclusive. A CGR_0009 rule violation is triggered when one of the following conditions is met:

The common base period of those two clocks is greater than the *max_period_multiplier* property multiplied by the period of the larger of the two clock periods.

The *min_period_multiplier* property multiplied by the period of the smaller of the two clock periods.

The common base period is the least common multiple between the two clock periods.

If you change the *skip_clocks_with_MCP* property, the tool will not issue CGR_0009 rule violations if there are clock-to-clock multicycle paths between the clocks. Use the set_rule_property command to change the value of properties.

For clocks not defined as asynchronous in the set_clock_groups command, the timing paths between the two clocks will be analyzed, and that can result in large timing violations caused by false paths.

**ONE POSSIBLE SCENARIO**

Consider a scenario in which two functional mode constraints are merged into a single set of constraints. The first functional mode has clocks VCLK1, CLK1, and CLK2. The second functional mode has clocks VCLK1_test, CLK1_test, and CLK2_test. The two virtual clocks VCLK1 and VCLK1_test are missing from the clock group definition.

```
create_clock -name VCLK1 -period 3.1

create_clock -name VCLK1_test -period 11.07

create_clock -name CLK1 -period 3.1 [get_ports cport1] -add

create_clock -name CLK2 -period 6.2 [get_ports cport2] -add

create_clock -name CLK1_test -period 11.07 [get_ports cport1] -add

create_clock -name CLK2_test -period 22.14 [get_ports cport2] -add

set_clock_groups -group {CLK1 CLK2} -group \ {CLK1_test CLK2_test}
-physically_exclusive

set_input_delay 1.23 -clock VCLK1 -add [get_port data1]

set_input_delay 4.23 -clock VCLK1_test -add [get_port data1]
```

Because the clock group for VCLK1 and VCLK1_test has not been defined, CGR_0009 rule violations will be issued:

CGR_0009: Clocks VCLK1 and CLK1_test have valid timing paths between them, but their common base period is more than the limit times the clock period.

Feedback

CGR_0009: Clocks CLK1 and VCLK1_test have valid timing paths between them, but their common base period is more than the limit times the clock period.

**What Next**

The "Violation Details" section identifies the clocks that should be declared asynchronous and the associated clock periods. You can review the clock names for the relevant clock definitions or review the clock periods using the report_clock command.

**FIX SUGGESTION**

To resolve the scenario issue described previously, add the clocks, VCLK1 and VCLK1_test, to separate clock groups and define the clock pair as asynchronous, as shown in the following command:

set_clock_groups -group {CLK1 CLK2 VCLK1} \ -group {CLK1_test CLK2_test VCLK1_test} -asynchronous

*Rule Property*: *max_period_multiplier*

A CGR_0009 rule violation is issued when the base period between the two clocks is greater than this property multiplied by the larger of the two clock periods. Increasing or decreasing the value of this property decreases or increases the number of CGR_0009 violations. The default for this parameter is 101.

Note that the base period expansion is also affected by the *min_period_multiplier* property. The violation is issued only if the common base period exceeds the smaller of the following two values:

max_period_multiplier times larger clock period

min_period_multiplier times smaller clock period

*Rule Property*: *min_period_multiplier*

A CGR_0009 rule violation is issued when the base period between the two clocks is greater than this property multiplied by the smaller of the two clock periods. Increasing or decreasing the value of this property decreases or increases the number of CGR_0009 violations. The default for this parameter is 1000.

Note that the base period expansion is also affected by the *max_period_multiplier* property. The *min_period_multiplier* value should be set to a number greater than or equal to the *max_period_multiplier* value. If not, the *max_period_multiplier* value is used as the multiplication factor for smaller periods as well. The violation is issued only if the common base period exceeds the smaller of the following two values:

max_period_multiplier times larger clock period

min_period_ multiplier times smaller clock period

*Rule Property*: *skip_clocks_with_MCP*

Feedback

By default, a violation is issued for all pairs of interacting clocks that satisfy the period checks described previously. However, if there are MCPs between the two clocks, they might not cause timing violations even if the clocks are not defined as asynchronous or physically exclusive. You can avoid issuing violations in this situation by enabling the *skip_clocks_with_MCP* rule property. Note:

It is recommended that the clocks be declared asynchronous or physically exclusive. This ensures that the paths between the two clocks are not timed and also takes care of signal integrity effects.

*Rule Property*: *skip_clocks_with_max_delay*

By default, a violation is issued for all pairs of interacting clocks that satisfy the period checks described previously. However, if there are min/max delays between the two clocks, they might not cause timing violations even if the clocks are not defined as asynchronous or physically exclusive. You can avoid issuing violations in this situation by enabling the *skip_clocks_with_max_delay* rule property.

*Rule Property*: *skip_clocks_with_allow_paths*

By default, a violation is issued for all pairs of interacting clocks that satisfy the period checks described previously. However, if there is allow_paths set between the two clocks, they might not cause timing violations even if the clocks are not defined as asynchronous. You can avoid issuing violations in this situation by enabling the *skip_clocks_with_allow_paths* rule property.

**See Also**

set_clock_groups report_clock

---

## CGR-0010

(Error) Clocks clock1 and clock2 might be interacting, but their common base period is more than the limit times the clock period.

Status: Disabled by default

**Description**

The two clocks are likely to be mutually exclusive but are not specified as such. The clocks can be asynchronous or physically exclusive. Note that this rule is an extension of the CGR_0009 rule where all pairs of clocks that are not explicitly defined as asynchronous or physically exclusive are considered for base period expansion. This is because even if valid timing paths do not exist between a pair of clocks, they might still require base clock expansion for signal integrity analysis, therefore this rule is a part of the PrimeTime SI ruleset. This rule is disabled by default. A common base period is the least common

Feedback

multiple between the two clock periods. This violation is issued if the common base period is larger than the smallest of the following two values:

the *max_period_multiplier* property multiplied by the period of the larger of the two clock periods

the *min_period_multiplier* property multiplied by the period of the smaller of the two clock periods

The value of the properties can be changed with the set_rule_property command.

For clocks not defined as asynchronous in the set_clock_groups command, the timing paths between the two clocks will be analyzed, and that can result in large timing violations due to false paths. Also, for signal integrity analysis, the base period expansion might be clipped to a large limit and can lead to large runtimes. For clocks not defined as physically exclusive in the set_clock_groups command, the signal integrity analysis will still be considered which can lead to pessimistic timing results causing timing violations.

**ONE POSSIBLE SCENARIO**

Consider a scenario in which two functional mode constraints are merged into a single set of constraints. The first functional mode has clocks VCLK1, CLK1, and CLK2. The second functional mode has clocks VCLK1_test, CLK1_test, and CLK2_test. The two virtual clocks VCLK1 and VCLK1_test are missing from the clock group definition.

create_clock -name VCLK1 -period 3.1

create_clock -name VCLK1_test -period 11.07

create_clock -name CLK1 -period 3.1 [get_ports cport1] -add

create_clock -name CLK2 -period 6.2 [get_ports cport2] -add

create_clock -name CLK1_test -period 11.07 [get_ports cport1] -add

create_clock -name CLK2_test -period 22.14 [get_ports cport2] -add

set_clock_groups -group {CLK1 CLK2} -group \ {CLK1_test CLK2_test} -physically_exclusive

set_input_delay 1.23 -clock VCLK1 -add [get_port data1]

set_input_delay 4.23 -clock VCLK1_test -add [get_port data1]

Because the clock group for VCLK1 and VCLK1_test is not defined, CGR_0010 rule violations are issued:

CGR_0010: Clocks VCLK1 and CLK1_test might be interacting, but their common base period is more than the limit times the clock period.

Feedback

CGR_0010: Clocks CLK1 and VCLK1_test might be interacting, but their common base period is more than the limit times the clock period.

**What Next**

The "Violation Details" section identifies the clocks that should be declared asynchronous along with their periods. You can review the clock name or names for the relevant clock definition or review the clock periods using the report_clock command.

**FIX SUGGESTION**

To resolve the scenario issue described previously, add the clocks, VCLK1 and VCLK1_test, to separate clock groups and define the clock pair as physically exclusive. The following example syntax shows the clock definition using the set_clock_groups command.

set_clock_groups -group {CLK1 CLK2 VCLK1} \ -group {CLK1_test CLK2_test VCLK1_test} \ -physically_exclusive

The following clock group definition also eliminates the CGR_0010 violation, but it can lead to pessimistic signal integrity results and should not be used.

set_clock_groups -group {CLK1 CLK2 VCLK1} \ -group {CLK1_test CLK2_test VCLK1_test} -asynchronous

*Rule Property*: *max_period_multiplier*

A CGR_0010 violation is issued when the base period between the two clocks is greater than this property multiplied by the larger of the two clock periods. Increasing or decreasing the value of this property will decrease or increase the number of CGR_0010 rule violations. The default for this parameter is 101.

Note that the base period expansion is also affected by the *min_period_multiplier* property. The violation is issued only if the common base period exceeds the smaller of the following two values:

max_period_multiplier times larger clock period

min_period_multiplier times smaller clock period

*Rule Property*: *min_period_multiplier*

A CGR_0010 rule violation is issued when the base period between the two clocks is greater than this property multiplied by the smaller of the two clock periods. Increasing or decreasing the value of this property will decrease or increase the number of CGR_0010 violations. The default for this parameter is 1000.

Note that the base period expansion is also affected by the *max_period_multiplier* property. The *min_period_multiplier* value should be set to a number greater than or equal to the *max_period_multiplier* value. If not, the *max_period_multiplier* value is used as

Feedback

the multiplication factor for the smaller period as well. The violation is issued only if the common base period exceeds the smaller of the following two values:

max_period_multiplier times larger clock period

min_period_ multiplier times smaller clock period

*Rule Property*: *skip_clocks_with_min_max_delay*

By default, a violation is issued for all pairs of interacting clocks that satisfy the period checks described previously. However, if there are min/max delays between the two clocks, they might not cause timing violations even if the clocks are not defined as asynchronous or physically exclusive. You can avoid issuing violations in this situation by enabling the *skip_clocks_with_min_max_delay* rule property.

# CGRP

## CGRP-001

(warn) %s.

**Description**

Cell group consistency related warning messages.

## CGRP-002

(error) %s.

**Description**

Cell group creation related error messages.

# CGRR

## CGRR-0001

(info) Register pin %s has been identified as constant by the redundancy removal engine and may be removed as a constant register in later optimization steps.

**Description**

This message is issued when Fusion Compiler's redundancy removal engine identifies an input pin of a register as redundant.

Feedback

Redundancy removal engine identifies register as constant, but does not remove it. The task of removing constant registers is done by register removal engine which is executed later in the compile flow. The register identified as constant may or may not be removed by the register removal engine. For example, such register will not be removed if: (1) User has marked it as dont_touch (2) User has disabled removal of constant registers using compile.seqmap.remove_constant_registers. (3) User has disabled removal of unloaded registers using compile.seqmap.remove_unloaded_registers.

Fusion compiler applies checks (2) and (3) when a register is found to be constant as well as unloaded.

If a register is identified as constant but could not be removed, the registers in its fanout may not get removed if boundary restrictions prevent propagation of constants in the forward direction. Details can be found in man page of set_boundary_optimization.

Command report_transformed_registers can be used to find if a register is removed or preserved as constant.

**See Also**

- report_transformed_registers

---

## CGRR-0002

(info) Register pin %s has been identified as unloaded by the redundancy removal engine and may be removed as an unloaded register in later optimization steps.

**Description**

This message is issued when Fusion Compiler's redundancy removal engine identifies an output pin of a register as unloaded.

Redundancy removal engine identifies register as unloaded, but does not remove it. The task of removing constant registers is done by the register removal engine which is executed later in the compile flow. The register identified as unloaded may or may not be removed by the register removal engine. For example, such register will not be removed if: (1) User has marked it as dont_touch (2) User has disabled removal of constant registers using compile.seqmap.remove_constant_registers. (3) User has disabled removal of unloaded registers using compile.seqmap.remove_unloaded_registers.

Fusion compiler applies checks (2) and (3) when a register is found to be constant as well as unloaded.

If register is identified as unloaded but could not be removed, the registers in its fanin may not get removed if boundary restrictions prevent propagation of unloaded in the backward direction. Details can be found in man page of set_boundary_optimization.

Feedback

Command report_transformed_registers can be used to find if a register is removed or preserved as unloaded.

**See Also**

- report_transformed_registers

# CGT

## CGT-2500

(Warning) Class %s of object %s is not supported by -object command option.

**Description**

The command *set_clock_gate_style* was called with -object option and the existing design, module or instance is not hierarchical.

**What Next**

Check that the objects used for *set_clock_gate_style* are hierarchical.

## CGT-2501

(Warning) There were problems when extracting hierarchies from the object %s.

**Description**

The command *set_clock_gate_style* was called with -object option an object in the design is not hierarchical.

**What Next**

Check that the objects used for *set_clock_gate_style* are hierarchical.

## CGT-2700

(Error) %s was given for more than one option.

**Description**

The command *set_clock_gating_objects* was called with the same object for more than one option, which is not allowed.

Feedback

**What Next**

Check that the objects used for *set_clock_gating_objects* are only present in one of the options.

---

## CGT-2701

(Error) Cannot specify -enable_source without -force or -include

**Description**

The command *set_clock_gating_objects* was called with the option *-enable_source* but without the options *-force* or *-include*.

**What Next**

Check that the options *-force* or *-include* are used when *-enable_source* is used.

---

## CGT-2702

(Error) The provided collection does not contain valid elements. No attribute will be set.

**Description**

The command *set_clock_gating_objects* was called with no valid elements. Valid elements are sequential cells, hierarchies, power domains, and modules.

**What Next**

Check that the objects used for *set_clock_gating_objects* are valid elements.

---

## CGT-2703

(Error) Value for option '%s' must be %s.

**Description**

If the command *set_clock_gate_latency* is called with the option *-stage* the value must be non-negative and the zero is allowed. If the command *set_clock_gate_latency* is called with the option *-levels* the value must be greater than zero.

**What Next**

Check if the *set_clock_gate_latency* command does not have incorrect values.

---

## CGT-2704

(Error) Range %u-%s is overlapping with range %u-%s.

Feedback

**Description**

During the command *set_clock_gate_latency* fanout ranges cannot overlap.

**What Next**

Check the *set_clock_gate_latency command* settings and set ranges that do not overlap.

---

## CGT-2705

(Error) Lower limit for range cannot be zero.

**Description**

In the fanout range the lower limit value cannot be zero.

**What Next**

Check the *set_clock_gate_latency* command and ensure the lower limit is greater than zero.

---

## CGT-2706

(Error) Lower limit %u for range is higher than its upper limit %u.

**Description**

In the command *set_clock_gate_latency* the lower limit value must be lower than its upper limit value.

**What Next**

Check the *set_clock_gate_latency* command settings and change the incorrect value.

---

## CGT-2707

(Error) There are no specified fan-out ranges for -fanout_latency option.

**Description**

The option *-fanout_latency* of the *set_clock_gate_latency* command must be assigned a range of values.

**What Next**

Check the *set_clock_gate_latency* command settings and specify a range for *-fanout_latency* option.

Feedback

## CGT-2708

(Error) Specified delay value for %s %u cannot be negative

**Description**

Range values of the *-fanout_latency* option cannot be negative.

**What Next**

Check the range values of the *-fanout_latency* option and correct them.

## CGT-2709

(Error) For stage 0, only -fanout_latency {{1-inf value}} is accepted.

**Description**

When the option *-stage 0* is specified, the only accepted range is {1-inf}.

**What Next**

Check that only one range at {1-inf} was specified for stage 0 in the *set_clock_gate_latency* command.

## CGT-2710

(Error) Fan-out ranges do not cover the count of %s.

**Description**

The ranges for the option *-fanout_latency* must cover all values in the range {1-inf}.

**What Next**

Check that the range values cover {1-inf} in the *set_clock_gate_latency* command.

## CGT-2711

(Error) Invalid syntax for -fanout_latency option

**Description**

The *fanout_latency* option has an incorrect syntax.

**What Next**

Check and correct the syntax in the *set_clock_gate_latency* command.

Feedback

## CGT-2712

(Error) No clocks found on current the design.

**Description**

When no clock is defined execution will be aborted.

**What Next**

Check the *create_clock* command and create a clock. Verify a clock exists with the *get_clocks* command.

## CGT-2713

(Error) There are no clock gates in the given object_list.

**Description**

No clock gates were found in the argument of *set_clock_gate_transformations*.

**What Next**

Check that the collection of cells passed to *set_clock_gate_transformations* has valid clock gate cells.

## CGT-2714

(Error) There are no clock gates in the given object_list.

**Description**

The *report_clock_gating_enable_condition* command must receive a valid list of clock gates.

**What Next**

Pass a valid list of clock gates in the *report_clock_gating_enable_condition* command.

## CGT-2715

(Error) The provided collection does not contain valid elements. No setting would be stored.

**Description**

The command *set_self_gating_objects* was called with no valid elements. Valid elements are sequential cells, hierarchies, power domains, and modules.

Feedback

### What Next

Check that the objects used for *set_self_gating_objects* are valid elements.

### See Also

- set_self_gating_objects

---

## CGT-2716

(Error) Option -tree_type must be used together with either -include or -force.

### Description

The option *-tree_type* was specified for the *set_self_gating_objects* command, but neither *-include* nor *-force* was specified.

### What Next

Specify one of the *-include* or *-force* options.

### See Also

- set_self_gating_objects
- report_self_gating_objects

---

## CGT-2717

(Error) At least one of the options -include, -exclude, -force, -clear, or -reset must be used.

### Description

The *set_self_gating_objects* command was called with no valid option.

### What Next

Use at least one of the *-include*, *-exclude*, *-force*, *-clear*, or *-reset* options.

### See Also

- set_self_gating_objects
- report_self_gating_objects

---

## CGT-2718

(Error) Must specify the -fanout_latency option with the -stage option.

Feedback

**Description**

The command *set_clock_gate_latency* was called with the *-fanout_latency* option and without the *-stage* option, which is not allowed. The option *-fanout_latency* must be used with the *-stage* option.

**What Next**

Check that the option *-fanout_latency* is used with the *-stage* option.

---

## CGT-2719

(Error) Cannot use -level and -stages in the same session.

**Description**

The option of using *-level* and *-stages* in the same session is currently not supported.

**What Next**

Use either *-level* or *-stages* in the *set_clock_gate_latency* command, not both.

---

## CGT-2720

(Error) Cannot combine -include with -enable_source none.

**Description**

The command *set_clock_gating_objects* was called with the option *-enable_source none* and the option *-include*, which is not allowed. The option *-enable_source none* is only allowed with the option *-force*.

**What Next**

Check that the option *-include* is not used with *-enable_source none*.

---

## CGT-2721

(Error) Must specify %s with either the -gated or -gating_elements options.

**Description**

The command *report_clock_gating* was called with either *-origin* or *-type* option and without either the *-gated* option or *-gating_elements* option, which is not allowed. The options *-origin* and *-type* must be used with either the *-gated* option or *-gating_elements* option.

Feedback

### What Next

Check that the options *-origin* or *-type* are used with either the *-gated* or *-gating_elements* option.

### See Also

- report_clock_gating

---

## CGT-2722

(Error) The value specified for -max_bitwidth is smaller than that for -min_bitwidth.

### Description

The max bitwidth value specified for the *set_self_gating_options* command must be equal to or higher than the min bitwidth value.

### What Next

Specify a max bitwidth value that is higher than or equal to the min bitwidth value.

### See Also

- set_self_gating_options
- report_self_gating_options

---

## CGT-2730

(Warning) Overwriting previous clock gating tree option setting on source pin %s of Clock %s.

### Description

The clock gating tree option command is applied to clocks with the same driver more than once. The previous setting will be overwritten.

### What Next

No action is required. For more details check the /fBreport_clock_gating_tree_option command.

---

## CGT-2731

(Warning) Clock %s is invalid. Only clocks with valid clock sources are supported.

Feedback

**Description**

The *set_clock_gating_tree_options* command is applied to clocks with invalid clock sources. Virtual clocks with no clock sources or clocks created on inout ports are not supported.

**What Next**

Check the *create_clock* command settings.

---

## CGT-3000

(Warning) Scan-stitched clock gate %s was removed. %s.

**Description**

Clock Gating optimization ungated the fanout of a pre-existing clock gate whose SE pin was connected to a non-constant signal.

**What Next**

If the removal of the clock gate is expected, Formality directive *set_constant*, for the given port and constant or for the clock gate's SE pin must be used during Formality verification to validate this transformation.

If the removal of this clock gate is not expected, directive *set_clock_gate_transformations* must be used during Synthesis to prevent this transformation.

---

## CGT-3001

(Warning) There are missing latency values specification. Use *report_clock_gate_latency* for more information.

**Description**

There are (latency) values that were not specified or written for the clock gate cells.

**What Next**

Check that stages for all clock gates were specified and use *report_clock_gate_latency* for more information.

---

## CGT-3002

(Warning) There are latency value conflicts on cells. Use report_clock_gate_latency for more information.

Feedback

**Description**

Two consecutive clock gates are presenting conflicting latency values.

**What Next**

For more details check the /fBreport_clock_gate_latency command.

---

## CGT-3003

(Warning) This option %s will be ignored during clock gating insertion/optimizations.

**Description**

The given legacy option is not supported by the tool and will be ignored.

**What Next**

Remove the legacy options since it will have no effect.

---

## CGT-3004

(Warning) For hier %s, given min/max settings (%u/%u), some clock gates might not meet max fanout or minimum bitwidth.

**Description**

The tool cannot ensure that both the minimum bitwidth and maximum fanout will be honored after splitting a large clock gating bank if the minimum bitwidth is greater than half of the maximum fanout.

**What Next**

No action required.

**See Also**

- set_fanin_sequential_clock_gating_options

---

## CGT-3005

(Warning) No valid reference latency value for clock gate %s. Using default value of 0. See documentation of clockgate.ignore_latency_on_pin for more information.

Feedback

### Description

Verify the status of variable *clockgate.ignore_latency_on_pin*. If the variable is set to true, the reference used to compute the latency value in stage 1 is no longer valid and zero is used as a reference.

### What Next

Check the *set_clock_latency* and *set_clock_gate_latency* command settings. Check the variable *clockgate.ignore_latency_on_pin*.

---

## CGT-3006

(Warning) The cell %s has two clocks on pin %s and pin %s. Must have only one clock defined.

### Description

A combinational cell having a clock defined in two input pins is not a valid clock gate.

### What Next

Check that only one clock has been defined for the cell.

### See Also

- create_clock
- get_clock_gates
- report_clock_gating

---

## CGT-3009

(Warning) The register %s is scan stitched and it will not be replaced.

### Description

If a register is a scan stitched, the combinational cell that drives it will not be replaced.

### What Next

Can run *replace_clock_gates before compile command.*

### See Also

- report_clock_gating

Feedback

## CGT-3010

(Warning) The combinational cell %s has more than two input and it will not be replaced.

### Description

The given combinatorial cell has an incorrect number of pins..

### What Next

Verify that all the combinational cells to be replaced have a correct number of pins.

### See Also

- report_clock_gating

## CGT-3011

(Warning) No clocks found on the current design.

### Description

No clocks are found in the current design, a clock must be defined before executing the command *replace_clock_gates*.

### What Next

Check that a clock is defined or create it using *create_clock* command.

### See Also

- create_clock
- report_clock_gating

## CGT-3012

(Warning) The design is already placed and no replacement will be done.

### Description

If the design has already been placed, no replacement will be made for any combinational cell.

### What Next

Verify is the design is already placed.

Feedback

### See Also

- report_clock_gating

---

## CGT-3013

(Warning) The combinational cell %s has edge conflict with the gated registers and it will not be replaced.

### Description

If a combinational cell has edge conflict with any of its gated registers, it will not be replaced.

### What Next

Check the edge sense of the combinational cell and the registers.

### See Also

- report_clock_gating

---

## CGT-3300

(Warning) There is no integrated clock-gating cell in the technology library that satisfies the specified style and operating condition for module %s. No %s insertion will be done in this module.

### Description

No valid integrated clock gating (ICG) library cells can be found for the clock gate style in effect for this module. Clock gating cannot work.

### What Next

Check your library for the existence of the ICG library cells you want to use. Check whether these library cells have the "CTS" purpose. Check whether these library cells do not have the dont_touch attribute and whether the target library subset for this module may exclude these library cells specifically. Check whether the set_clock_gate_style settings in effect for this module match the style of these library cells. The check_clock_gate_library_cell_availability command can give a detailed report on clock gate library cell availability issues.

---

## CGT-3700

(Warning) Some of the elements in the object_list are not clock gating candidates and will be ignored. Attributes will only be set on valid objects.

Feedback

**Description**

The command *set_clock_gating_objects* was called with instances that are not sequential registers, which will not be gated anyways.

**What Next**

No action required.

---

## CGT-4000

(Information) Clock gate %s cannot be collapsed to honor the maximum number of levels.

**Description**

The clock gate cannot be collapsed to meet the constrain set by *compile.clockgate.max_number_of_levels".*

**What Next**

No action required.

---

## CGT-4001

(Information) Collapsing clock gates %s with %s to honor the maximum number of levels.

**Description**

The clock gates are being collapsed to meet the constrain set by *compile.clockgate.max_number_of_levels".*

**What Next**

No action required.

---

## CGT-4002

(Information) Removing clock gate %s by ungating to honor the maximum number of levels.

**Description**

The clock gate is being removed by ungating the gated registers to meet the constrain set by *compile.clockgate.max_number_of_levels".*

**What Next**

No action required.

Feedback

## CGT-4003

(Information) Clock gate %s cannot be removed to honor the maximum number of levels.

**Description**

The clock gate cannot be removed by ungating the gated registers to meet the constrain set by *compile.clockgate.max_number_of_levels".*

**What Next**

No action required.

## CGT-4005

(Information) The automatic clock gate timing flow is activated. Clock latencies set on clock gate cells will be automatically adjusted by an estimate for latency.

**Description**

The automatic clock gate timing flow is activated. Clock latencies set on clock gate cells will be automatically adjusted by an estimate for latency. Clock gate latencies derived from the *set_lock_gate_latency* command will not be used.

**What Next**

To preserve annotated clock latencies from set_clock_latency, set the *dont_estimate_clock_latency* attribute on the clock pin of the clock gating cell. E.g. *set_attribute [get_pins {my_CG/CK}] dont_estimate_clock_latency true*

## CGT-4309

(Information) Activity driven multi level clock gating is enabled.

**Description**

In the activity-driven mode, *Fusion Compiler* considers the switching activities on a clock gate when it performs level expansion. The application option *compile.clockgate.enable_level_expansion* controls whether *compile_fusion* executes clock-gate level expansion in activity-driven mode.

**What Next**

No action required.

Feedback

## CGT-4314

(Information) Clock gate max-fanout specification is not recommended when timing driven clock gate splitting is enabled.

**Description**

In clock gating flow, *Fusion Compiler* performs a timing-driven clock gate splitting. In this mode, not specifying max-fanout setting might yield better Qor. The application option *compile.clockgate.timing_based_split_in_cg_timing_flow* controls whether *compile_fusion* executes timing-driven clock gate splitting.

**What Next**

No action required.

## CGT-5007

(Information) A clock gate was removed which had a connection to pin %s please set to constant %u for verification.

**Description**

A scan-stitched clock gate has been removed by the tool. When removing a clock gate, the enable logic driving the enable pin of the clock gate is transferred to the driven registers while the logic driving the test enable pin of clock gate is dropped. This is because the scan enable logic is used only to bypass the clock gate for DFT purposes and is not needed for functional correctness. Hence for correct functional formal verification, the test enable pin of the clock gate must be set to a non-controlling value on the reference side.

If the reported clock gate needs to be preserved instead, use *set_clock_gate_transformations* command.

**What Next**

Constrain the clock gate test enable pin to a non-controlling constant for verification purposes. Alternatively, the clock gate can be retained using *set_clock_gate_transformations* command.

# CHF

## CHF-001

(error) For command %s option %s: You can only specify one from the list of valid option values: %s.

Feedback

**Description**

For the named command option, only one valid value can be specified.

**What Next**

See command man page for valid option value.

---

## CHF-002

(error) Incorrect command %s option usage. %s.

**Description**

Incorrect command option usage.

**What Next**

See command man page for correct option usage.

---

## CHF-003

(error) %s: Incorrect library cell type specified.

**Description**

Incorrect library cell type specified. Library cell must be one of these sub-types: std cell, physical only or filler.

**What Next**

Please specify library cell with correct library cell type.

---

## CHF-004

(Warning) %s: Reach limit of %d on this message. No more of this message will be printed.

**Description**

The preset limit on number of this message is reach. No more of this message will be shown.

**What Next**

Please check related message for further instruction.

Feedback

## CHF-005

(Warning) cell %s not multiple of row height.

**Description**

This cell is not multiple of row height of filler cells used. This maybe because it is on a unit tile differ from filler cell unit tile. It may also because it is not snap to row/tile.

**What Next**

Please check if this cell is snap to row/tile.

## CHF-006

(Error) filler %s has site %s but filler %s has site %s.

**Description**

All fillers specified in -lib_cells must have same site def

**What Next**

Use correct fillers.

## CHF-007

(Error) %s.

**Description**

create_stdcell_fillers can't continue because of described problem.

**What Next**

Please make necessary correction as explained.

## CHF-008

(Warning) %s.

**Description**

You receive this warning message because your fillers are not specified from largest to smallest. Fillers are inserted in the specified order. If they are not listed from largest to smallest, tool may not be able to fill all gaps.

Feedback

**What Next**

Please check your filler list and make necessary correction.

---

## CHF-009

(Warning) %s.

**Description**

You receive this warning message because without those size fillers, gaps may be left unfilled to prevent leaving gaps smaller than smallest sized cells.

**What Next**

Please check your filler list and make necessary correction.

---

## CHF-010

(Error) filler ref cell %s are not marked filler type.

**Description**

You receive this error message because you have -rules { post_route_auto_delete} turn on but specified filler ref cell is not marked filler type.

**What Next**

Please check your filler ref cell type setting.

---

## CHF-011

(Warning) filler ref cell %s not marked filler type.

**Description**

You received this warning message because your filler ref cell is not marked filler type.

**What Next**

Please check your filler ref cell type.

---

## CHF-012

(Error) Fillers siteDef is %s and there is no placeable row area with same site def.

Feedback

**Description**

You receive this error message because there is no placeable row area for fillers. This can be because all the area is occupied or because rows have different siteDef as fillers

**What Next**

Please check your fillers or design and make necessary corrections.

---

## CHF-013

(Error) Filler %s has no site def set.

**Description**

You receive this error message because this filler ref lib cell's site def is not set.

**What Next**

Please set filler ref cell site def.

---

## CHF-014

(Warning) option -continue_on_error turned on automatically.

**Description**

You receive this warning message because you did not specify -continue_on_error and -rules { no_1x}, and doesn't provide size 1x fillers. Tool therefore turn on -continue_on_error automatically so it will skip unfillable 1x gaps and continue to fill the rest. This may lead to unfillable gaps.

**What Next**

Please include fillers of all needed sizes

---

## CHF-015

(Warning) Center cell %s will be skipped since its cell type is not supported.

**Description**

Incorrect center cell type specified. Center cell must be one of these sub-types: std cell, filler, well tap, end cap, diode or feed through.

**What Next**

Please specify center cell with correct library cell type.

Feedback

## CHF-016

(Error) Site %s of lib cell %s is not compatible with site %s of lib cell %s.

**Description**

All lib cells specified in -lib_cells must have same site def.

**What Next**

Use correct lib cells.

## CHF-017

(Error) Height of center lib cell %s is not multiple integer of the height of %s filler lib cell %s.

**Description**

The height of center lib cells specified must be multiple integer of the height of its corresponding left or right filler lib cells.

**What Next**

Use correct lib cells.

## CHF-018

(Information) Total %d instances of %s filler %s inserted for center cell %s.

**Description**

This message tells you how many left or right filler cells were inserted into the design by the command.

**What Next**

Compare the number of inserted left or right filler cells to what you anticipated.

## CHF-019

(Information) Total %d left right filler cells inserted successfully.

**Description**

This message tells you how many left right filler cells were inserted into the design by the command.

Feedback

**What Next**

Compare the number of inserted left right filler cells to what you anticipated.

---

## CHF-020

(error) Vertical rows are not supported.

**Description**

This command works for designs with only horizontal rows.

**What Next**

Use this command for designs with only horizontal rows.

---

## CHF-021

(info) Boundary cell placement rules are not set.

**Description**

This message indicates no boundary cell placement rules are set by set_boundary_cell_rules.

**What Next**

Use set_boundary_cell_rules to set boundary cell placement rules before running this command.

---

## CHF-022

(Information) Total %d %s rules found in design %s.

**Description**

This message tells you how many rules were set by set_boundary_cell_rules command for this design.

**What Next**

Compare the number of rules to what you anticipated.

---

## CHF-023

(warning) %s rule %s will not be honored by %s.

Feedback

**Description**

This message tells you the rule is not supported for this command.

**What Next**

NA

---

## CHF-024

(Information) No %s violation.

**Description**

This message tells you no violation for this rule.

**What Next**

NA

---

## CHF-025

(error) There is a gap in {{%s, %s} {%s, %s}}.

**Description**

This message indicates that the tool has detected a continuity violation.

**What Next**

Insert an appropriate boundary cell in the gap.

---

## CHF-026

(error) Illegal %s cell %s in {{%s, %s} {%s, %s}}.

**Description**

This message indicates that the tool has detected a corner and boundary cell violation.

**What Next**

Change the illegal corner or boundary cell to appropriate one.

---

## CHF-027

(error) Illegal orientation for cell %s in {{%s, %s} {%s, %s}}.

Feedback

**Description**

This message indicates that the tool has detected an orientation violation.

**What Next**

Change the cell's orientation according to the orientation rule.

---

## CHF-028

(error) Illegal %s abutment cell %s in {{%s, %s} {%s, %s}}.

**Description**

This message indicates that the tool has detected an abutment cell violation.

**What Next**

Change the illegal abutment cell to appropriate one.

---

## CHF-029

(error) There are %d even boundary cells and %d odd boundary cells in vertical edge {{%s, %s} {%s, %s}}.

**Description**

This message indicates that the tool has detected a consistent parity violation.

**What Next**

Remove the inappropriate parity type boundary cells and insert new ones to make all the boundary cells along this vertical edge have the same parity.

---

## CHF-030

(error) Boundary cell parity violation in {{%s, %s} {%s, %s}}, parity in left is %s and parity in right is %s.

**Description**

This message indicates that the tool has detected a boundary cell parity violation.

**What Next**

Change the boundary cells' parity type in left or right side according to the boundary cell parity rule.

Feedback

## CHF-031

(error) %s segment parity violation in {{%s, %s} {%s, %s}}, %s is %s number (%d) %s sites.

**Description**

This message indicates that the tool has detected a segment parity violation.

**What Next**

Insert hard placement blockages in the left/right side or top/bottom of this segment to meet the segment parity rule.

## CHF-032

(error) %s %s violation in {{%s, %s} {%s, %s}}, %s is %s.

**Description**

This message indicates that the tool has detected a short edge or segment violation.

**What Next**

Check the value for the min horizontal jog or min row width to ensure that it is correct. If it is correct modify your floorplan to extend the edge or segment in the error message so that it is longer than the min value.

## CHF-033

(error) No valid segment rule type is specified.

**Description**

This message indicates no segment rule type is specified by the app option plan.flow.segment_rule.

**What Next**

Set the app option plan.flow.segment_rule before running this command.

## CHF-034

Rows with margin(gap or space) are not supported.

Feedback

**Description**

Do not support gaps between site rows (y_margin) or gaps between sites in a row(x_margin).

**What Next**

Check margin value(s). It must be 0.

---

## CHF-035

(error) -smallest_cell_size must be an integer between 1 and 11.

**Description**

You received this error message because you specified invalid value for option -smallest_cell_size

**What Next**

Please see man page for option usage.

---

## CHF-036

(error) inconsistant option usage between -rules {no_1x} and -smallest_cell_size.

**Description**

You received this error message because you specified both -smallest_cell_size and -rules {no_1x}. Please only use -rules {no_1x} if your smallest cell size is 2x. Otherwise, only use -smallest_cell_size.

**What Next**

Please use only one of the option. See man page for more details.

---

## CHF-037

(error) Left boundary cell or right boundary cell is not specified.

**Description**

This message indicates no left boundary cell or right boundary cell is specified by set_boundary_cell_rules.

**What Next**

Use set_boundary_cell_rules to set left boundary cell and right boundary cell before running this command.

Feedback

## CHF-038

(error) %s of the %s separation (%.4f, %.4f), (%.4f, %.4f) is smaller than the allowed minimum value.

**Description**

This message indicates that the tool has detected a short separation violation.

**What Next**

Check the value for the min horizontal separation or vertical separation to ensure that it is correct. If it is correct modify your floorplan to extend the separation in the error message so that it is longer than the min value.

## CHF-039

(Warning) Multi-threading is forced off because multi-height fillers specified.

**Description**

You enabled multi-threading but currently multi-threading is not supported when multi-height fillers specified.

**What Next**

Please turn off multi-threading when multi-height fillers specified.

## CHF-040

(Warning) No advanced rule in design. Multi-threading will not provide noticeable runtime improvement.

**Description**

You received this message because there is no advanced rules( min-vth, TPO, or OD rule) in design and you enabled multi-threading. When no advanced rules, multi-threading will not provide noticeable runtime improvement.

**What Next**

Please turn off multi-threading when there is no advanced rule.

## CHF-041

(error) The area specified by user is not inside block boundary.

Feedback

**Description**

The bbox specified by user should be inside block boundary.

**What Next**

Specify a correct value.

---

## CHF-042

(warning) The area specified should be inside block boundary, the tool will automatically adjusted it.

**Description**

During *create_tap_meshes*, the *-bbox* specified by users should be inside block boundary, otherwise, the tool will adjust it.

**What Next**

N/A

---

## CHF-043

(error) %s: Incorrect library cell type specified for left/right cell.

**Description**

Incorrect library cell type specified. Library cell must be one of these sub-types: std cell, filler, well tap or feed through.

**What Next**

Please specify library cell with correct library cell type.

---

## CHF-044

(warning) Perimeter constraint routing blockage can not be created on design block %s, and only metal created in preferred direction.

**Description**

Perimeter constraint routing blockage can not be created on block.

**What Next**

Please check the block design.

Feedback

## CHF-045

(error) Target filler cell and custom cells have different height or width.

**Description**

You see this error message because you specified target filler cell and custom cells of different size.

**What Next**

Use the custom cells with the same size of target filler cell instead.

## CHF-046

(error) Max vertical constraint violation between cell %s and %s.

**Description**

You see this error message because you specified -check_only to check max vertical constraint violation.

**What Next**

Remove -check_only or use other method to fix the max vertical constraint violation.

## CHF-047

(error) No rules are specified. Please set place.legalize.enable_2D_query_placer_rules.

**Description**

You see this error message because you turned off auto 2D rule setting but specified no rules by place.legalize.enable_2D_query_placer_rules.

**What Next**

Use app option place.legalize.enable_2D_query_placer_rules to specify some rules.

## CHF-048

(warning) There is metal shape %s on layer %s overlaps bbox %s in block %s.

**Description**

This is check_only mode for command derive_perimeter_constraint_objects.

Feedback

**What Next**

Creating perimeter constraint objects.

---

## CHF-049

(Warning) %s.

**Description**

Please set the correct command options.

**What Next**

Please check command options usage.

---

## CHF-050

(warning) There is route guide %s on layer %s overlaps bbox %s in block %s.

**Description**

This is check_only mode for command derive_perimeter_constraint_objects and derive_metal_cut_routing_guides.

**What Next**

Creating perimeter constraint or metal cut route guide objects.

---

## CHF-051

(warning) There is no max pattern perimeter constraint route guide on layer %s in bbox %s in block %s.

**Description**

This is check_only mode for command derive_perimeter_constraint_objects.

**What Next**

Creating perimeter constraint objects.

---

## CHF-052

(warning) Layer '%s' is not a via cut layer.

**Description**

The specified layer is not a via cut layer. The layer must be a via cut layer.

Feedback

**What Next**

Check the technology layer information.

---

## CHF-053

(warning) There is no perimeter constraint metal shape on layer %s in bbox %s in block %s.

**Description**

This is check_only mode for command derive_perimeter_constraint_objects.

**What Next**

Creating perimeter constraint objects.

---

## CHF-054

(warning) The option %s should be used with boundary cell rule -at_va_boundary.

**Description**

The given option should be used with boundary cell rule -at_va_boundary.

**What Next**

Check the command man page to see correct options to use.

---

## CHF-055

(error) Found %s boundary cell %s in {{%s, %s} {%s, %s}}.

**Description**

This message indicates that the tool has detected a redundant extra boundary cell.

**What Next**

Check redundant extra boundary cell.

---

## CHF-056

(warning) The option %s is obsolete, please use %s instead.

**Description**

The given option is obsolete and will be removed from the future release. Please change to use the new option or the new command instead, which has similar functionality.

Feedback

**What Next**

Check the command man page to see correct options to use.

---

## CHF-057

(Error) Option %s's value should not be a negative value.

**Description**

Option value should not be a negative value.

**What Next**

Please set the correct option value.

---

## CHF-058

(error) %s.

**Description**

Errors are found during the processing of the command options.

**What Next**

Please check the manpage of the command to correct any errors related to option settings.

---

## CHF-059

(warning) No preferred routing direction defined for routing layer %s.

**Description**

There is no preferred routing direction defined for a routing layer. The checking is performed for all routing layers if max/min routing layers are not defined. If max/min routing layers are defined, the tool only check layers between max and min layers.

**What Next**

Please set preferred routing layer direction in order for the tool to proceed

Example: set_attribute [get_layers M1] routing_direction horizontal

set_ignored_layers -max_routing_layer METAL -min_routing_layer METAL3

Feedback

## CHF-060

(warning) Perimeter constraint max pattern nonpreferred direction route guide's width is too large for routing layer %s.

**Description**

Max pattern nonpreferred direction route guide's width is too large for perimeter constraint on routing layer.

**What Next**

Please set the correct constraint value of routing guides.

## CHF-061

(Warning) Center cell %s is specified more than one time.

**Description**

Center cell %s is specified more than one time, the latest setting will overwrite previous settings.

**What Next**

Please check -lib_cells settings.

## CHF-062

(error) Creating perimeter constraint objects group failed on design block %s.

**Description**

Errors are found during the processing of perimeter constraint objects group creating on block.

**What Next**

Please check the block design.

## CHF-063

(info) Perimeter constraint objects group %s is created on design block %s.

**Description**

Perimeter constraint objects group is created on block.

Feedback

**What Next**

Please check the block design.

---

## CHF-064

(error) Lib cell %s has site def %s that cannot be found in the site rows of current deign.

**Description**

Incorrect lib cells specified because its site def cannot be found in the site rows of current design.

**What Next**

Please specify correct lib cells.

---

## CHF-065

(error) Please specify option %s's value %s with %s for design block %s.

**Description**

Perimeter constraint objects creating need correct command options setting.

**What Next**

Please check the command options setting.

---

## CHF-066

(warning) There is route blockage %s on layer %s overlaps bbox %s in block %s.

**Description**

This is check_only mode for command derive_perimeter_constraint_objects.

**What Next**

Creating perimeter constraint objects.

---

## CHF-067

(warning) Perimeter constraint short metal is ended in max pattern routing guide on routing layer '%s'.

**Description**

The specified short metal length is too small for the routing layer.

Feedback

**What Next**

Check the short metal length and spacing from boundary.

---

## CHF-068

(warning) Layer '%s' is not a routing layer.

**Description**

The specified layer is not a routing layer. The layer must be a routing layer.

**What Next**

Check the technology layer information.

---

## CHF-069

(warning) Perimeter constraint routing blockage can not be created on design block %s, and only max pattern routing guide created.

**Description**

Perimeter constraint routing blockage can not be created on block.

**What Next**

Please check the block design.

---

## CHF-070

(Error) %s.

**Description**

The specific valid region is too small to place tap cell.

**What Next**

Please check the -bbox settings or other options.

---

## CHF-071

(Error) no edge is detected in the rectangular region prescribed.

**Description**

The region specified by -bbox should touch at least one valid object.

Feedback

**What Next**

Please check the -bbox settings.

---

## CHF-072

(Error) The -skip_fixed_cells option is not supported by the rule specified with -replacement_rule.

**Description**

The -skip_fixed_cells option only supports rules mentioned in the command man page.

**What Next**

Check the man page of -skip_fixed_cells option and specify the supported replacement rule.

---

## CHF-073

(Warning) Smallest filler provided is greater than specified in option place.rules.min_vt_filler_size

**Description**

The smallest fillers specified in "create_stdcell_fillers -lib_cells" is larger than what is specified in app option place.rules.min_vt_filler_size. This may cause unfillable gaps.

**What Next**

Please include missing size fillers in create_stdcell_fillers -lib_cells.

---

## CHF-074

(Warning) User overwrite prefer same VT heuristics

**Description**

User uses app option chf.create_stdcell_fillers.follow_orders to overwrite prefer same VT heuristics. This will increase run time and the increase may be significant.

**What Next**

Only use app option chf.create_stdcell_fillers.follow_orders when absolute necessary. An alternative is use -type_utilization.

Feedback

## CHF-075

(Error) Filler insertion encounters unfillable gaps. Please run check_legality before filler insertion to ensure design is fillable.

**Description**

create_stdcell_fillers encounters unfillable gaps. Please run check_legality before filler insertion to ensure design is fillable.

**What Next**

Please run check_legality before filler insertion to ensure design is fillable.

## CHF-076

(Error) Incomplete/Invalid cell group (variants) defined for lib cell %s.

**Description**

You receive this error because specified lib cell has incomplete/invalid cell group setup.

**What Next**

Please check your cell group (variant) setup for the specified lib cell.

## CHF-077

(error) Same object cannot appear in both option %s and option %s.

**Description**

Incorrect command option usage.

**What Next**

See command man page for correct option usage.

## CHF-078

(warning) There are lib cells in -lib_cells but not in -type_utilization. Option -fill_remaining is not specified.

**Description**

If there are lib cells in -lib_cells but not in -type_utilization, those lib cells can be used by -fill_remaining option. If -fill_remaining is not specified, please check if the option -fill_remaining is missed or it is intentional to specify more lib cells in -lib_cells option.

Feedback

**What Next**

Check if the option -fill_remaining is missed or it is intentional to specify more lib cells in -lib_cells option.

---

## CHF-079

(error) Option %s and %s cannot be specified together.

**Description**

User specified options or value are mutually exclusive.

**What Next**

Please check man page for command option usage.

---

## CHF-080

(Warning) VT leakage order not defined. Command continue without considering VT leakage order.

**Description**

This warning occurs when both -leakage_vt_order option and app option place.legalize.boundary_leakage_info_files not used or tool not able to retrieve valid vt leakage order from them, so create_stdcell_fillers command will run without considering vt leakage order.

**What Next**

Specify -leakage_vt_order option or use app option place.legalize.boundary_leakage_info_files.

---

## CHF-081

(Warning) app option place.legalize.boundary_leakage_info_files specified. -leakage_vt_order option will be ignored.

**Description**

This warning occurs when both -leakage_vt_order option and app option place.legalize.boundary_leakage_info_files is set, so create_stdcell_fillers command will use leakage side files and ignore -leakage_vt_order option.

Feedback

**What Next**

Only use one of -leakage_vt_order option or app option place.legalize.boundary_leakage_info_files.

---

## CHF-082

(Warning) cannot derive vt leakage order from files.

**Description**

This warning occurs when either vt leakage order files not specified, or incorrect value specified in those files. Tool will use orders from -leakage_vt_order option if specified.

**What Next**

Please specify correct vt leakage files, or check the data in the files.

---

## CHF-085

(Information) set_tap_boundary_wall_cell_rules command is executed and it will overwrite all rules defined previously.

**Description**

This message remind user that set_tap_boundary_wall_cell_rules has top priority and it will overwrite all rules defined previously.

**What Next**

User could review if current rules are expected.

---

## CHF-090

(error) Boundary cell %s contains %s layer outside of cell boundary.

**Description**

Boundary cell %s contains %s layer outside of cell boundary.

**What Next**

Please check to see if the legal cells have been specified.

---

## CHF-100

(Information) Total %d %s inserted successfully into %s.

Feedback

**Description**

This message tells you how many cells were inserted into the design by the command.

**What Next**

Compare the number of inserted cells to what you anticipated.

---

## CHF-101

(Error) Cannot place tap cell at location (%f, %f).

**Description**

This error occurs the when the tool cannot place a tap cell in specified location, which is needed to ensure that tap distances are met.

**What Next**

Check the location specified for blockages or pre-existing tap cells that are preventing the tap cell from being inserted.

---

## CHF-102

(Error) Cannot instantiate current_design within itself.

**Description**

The *insert_tap_cell* command was used to try to create a tap cell in the current design that referenced the current design. This would create a cycle in the design hierarchy, which is not allowed.

**What Next**

Choose a different *reference* for the *insert_tap_cell* command.

---

## CHF-103

(Error) Library specified in -lib_cell was not found.

**Description**

The *insert_tap_cell* cannot find the library specified in the *-lib_cell* option.

**What Next**

Check and make sure that the specified library in the -lib_cell option exists and is spelled correctly.

Feedback

## CHF-104

(Error) Cell specified in -lib_cell was not found.

**Description**

The *insert_tap_cell* cannot find the cell specified in the *-lib_cell* option.

**What Next**

Check and make sure that the specified cell in the -lib_cell option exists and is spelled correctly.

## CHF-105

(Error) No '/' in -lib_cell option to seperate library and cell name.

**Description**

The *-lib_cell* option of the *insert_tap_cell* does not contain a '/' to seperate between the library and the cell.

**What Next**

Add '/' to the *-lib_cell* option to separate between the library and the cell in between the library and cell names.

## CHF-106

(Error) Net %s is not a %s net.

**Description**

The *-connect_power_name* and *-connect_ground_name* options of the *insert_tap_cell* command can only accept power and ground nets respectively.

**What Next**

Change the net specified by the option to a net that is a power or ground net.

## CHF-107

(Error) Failed to connect net %s to tap cell %s.

**Description**

The net specified to the either *-connect_power_name* and *-connect_ground_name* options of the *insert_tap_cell* command could not be connected to newly a created tap cell.

Feedback

**What Next**

Check the net specified to the option and make sure that it is valid and is a power or ground net.

---

## CHF-108

(Error) Failed to insert %s in block named %s.

**Description**

The command could not insert cells into a block.

**What Next**

Site rows are needed in the block to insert cells. If the block is a soft-macro, check and make sure that site rows have been pushed down from the top level.

---

## CHF-109

(Error) %s are not supported.

**Description**

The feature listed in the error message is not supported.

**What Next**

Remove the feature list in the error message.

---

## CHF-110

(Error) Block %s is read-only.

**Description**

The command cannot insert into the block because the block is read-only.

**What Next**

Change the block so that it is writable.

---

## CHF-111

(Error) No placement site with correct height found for cell %s.

Feedback

**Description**

The command could not find a placement site definition in the block that is compatible with the height of the cell that is to be inserted.

**What Next**

First, check and make sure you that cell you are trying to insert is the correct one. Then check to make sure that all the values including height are correctly set in the site definition for the cell you are trying to insert. Finally, if necessary create site rows with correct height for the cell that is to be inserted.

---

## CHF-112

(Error) Specified lib cells do not have the same site definition.

**Description**

The lib cells you specified do not have the same site definition. When doing insertion all library cells specified must have the same site definition.

**What Next**

Check and make sure you that cells you are trying to insert are the correct ones and the dimensions of these cells are compatible with each other.

---

## CHF-113

(Error) Voltage area %s is not associated with any power domain..

**Description**

In order for cells to be inserted into the correct hierarchy, non-default voltage areas must be associated with power domains.

**What Next**

Check your upf specifications and make sure non-default voltage areas are related to power domains.

---

## CHF-114

(Error) Tap distance is smaller than the width of a single cell placement site.

**Description**

The tap distance specified is smaller than the width of the cell placement site being used for insertion.

Feedback

**What Next**

Increase the specified tap distance so that it is larger than the width of a cell placement site.

---

## CHF-115

(Warning) Cannot place %s at location (%f, %f) because %s.

**Description**

The cell cannot be placed in the location due to the specified violation.

**What Next**

Check your library setup to make sure it is correct. For example if you have a cell that has an orientation that is not legal for the row it is placed and causing a violation. Make sure that the site that the cell is being created in allows symmetry on the x-axis.

---

## CHF-116

(Warning) Failed to fix violation for cell %s.

**Description**

The specified cell is in violation of a rule, but we failed to fix it by swapping equivalent cells in its place.

**What Next**

Make sure that correct substitutes are specified for this cell, or move it where it will not violate rules.

---

## CHF-117

(Error) Cell %s is in both %s and %s list.

**Description**

The specified cell is in two conflicting lists.

**What Next**

Remove cell from one of the lists.

---

## CHF-118

(Warning) Unable to fix max constraint length violation between cell %s and %s.

Feedback

**Description**

The command failed to resolve the violation between the two specified cells.

**What Next**

Consider moving or replacing some of the cells between the two.

---

## CHF-119

(error) Illegal fin type for cell %s in {{%s, %s} {%s, %s}}.

**Description**

This message indicates that the tool has detected a fin type violation.

**What Next**

Change the cell's LibCell according to the fin type rule.

---

## CHF-120

(error) Site master "%s" does not have Y-Symmetry but the design contains Y-mirrored site rows.

**Description**

This message indicates that the tool has detected that the site master being used does not have Y-Symmetry but the block that insertion is being performed in contains site rows that either have a MY or R180 orientation. This is a problem because there is no legal way that cells can inserted to meet the cell symmetry restrictions and the match the orientation of the row.

**What Next**

Either add Y-Symmetry to the site master or change the orientation of the rows from R180/MY to R0/MX.

---

## CHF-121

(Error) The height of library cell %s does not match the height of the other site definitions.

**Description**

The height of the specified library cell does not match the height of the other site definitions being used. This will happen for example if you specify cells that have different heights for different library cell options of create_boundary_cells.

Feedback

### What Next

First, check to see if the correct cells have been specified. Then check if all of the cells specified to the command are the same height. If it is correct then check the site definition of the specified cell to make sure they are all correct.

---

## CHF-122

(error) Line segment (%.4f, %.4f), (%.4f, %.4f) is smaller than the allowed minimum jog.

### Description

This message indicates that the tool has detected a line segment that is smaller than the allowed the minimum jog.

### What Next

Check the value for the minimum jog to ensure that it is correct. If it is correct modify your floorplan to extend the line segment specified in the error message so that it is longer than the minimum jog.

---

## CHF-123

(Warning) Overwriting value %s.

### Description

The tool is warning you that you are overwriting a value that you have already set.

### What Next

Please check the parameters in the command you just issued and make sure they are correct.

---

## CHF-124

(warning) Row (%.4f, %.4f, %.4f, %.4f) is not wide enough to accommodate non-abutted tap cells.

### Description

This message indicates that the tool has detected a row that is not wide enough to accommodate non-abutted tap cells.

Feedback

**What Next**

Before inserting tap cells, insert placement blockages in the rows that are not wide enough to accommodate non-abutted tap cells or run the command without the -no_abutment option.

---

## CHF-125

(warning) No 1x or 2x + 3x wide cells provided for top or bottom option.

**Description**

You receive this warning message because you have not specified either a cell that is equal in width to the unit site or a pair of cells that are twice and three times the width of the unit tile for the bottom and top boundary cell options.

**What Next**

Add cells to the bottom or top boundary cell list so that you mee the requirement of having a either 1x or a pair of 2x and 3x cells for both top and bottom boundary cells.

---

## CHF-126

(Error) The lower left corner of the boundary of library cell %s is not at (0,0).

**Description**

The lower left corner of the boundary of the specified library cell is not set to (0,0). Only library cells where the lower left corner of the boundary is set to (0,0) can be used with this command.

**What Next**

Go back to the LEF used to generate the library cell and set the lower left corner of the boundary to (0,0).

---

## CHF-127

(Warning) The design already has boundary cells of type %s.

**Description**

The design already has boundary cells of type %s.

**What Next**

Remove existing boundary cells and then re-run the command.

Feedback

## CHF-128

(warning) The design has advanced rules. Using create_stdcell_fillers is recommended over create_vtcell_fillers.

**Description**

This message indicates that the tool has detected that the design has advanced rules such as VT or CNOD.

**What Next**

Change your script to use create_stdcell_fillers instead of create_vtcell_fillers.

## CHF-129

(Error) Cannot terminate tap column with tap cap at location (%f, %f).

**Description**

The specified tap column cannot be terminated with a tap cap. This occurs when there is column consisting of one single tap cell that needs to be placed in a location that has a placement obstruction directly above and below it.

**What Next**

Change the floorplan so that it does not produce a column that consists of one single tap cell that has a placement obstruction directly above and below it.

## CHF-130

(warning) Skipping routing layer %s because routing direction is not set for that layer.

**Description**

This message indicates that the routing direction is not set on the given routing layer and that boundary routing blockages will not be created for that layer.

**What Next**

Set routing direction for the layer using the set_attribute command. For example, to set the routing direction for metal layer 3 to horizontal, issue the following command: set_attribute [get_layers M3] routing_direction horizontal

## CHF-131

(Error) Cannot find correct vt filler to fill gap at (%f, %f).

Feedback

**Description**

This error occurs the when the tool cannot find a correct vt filler cell to fill the gap at specified location.

**What Next**

Check the vt table to make sure that you have specified all the vt filler cells supplied by the library vendor for all the different vt types. Also check to make sure there are not gaps that are not less wide than the smallest filler vt filler cell.

---

## CHF-132

(Information) Inserted %d cell %s successfully in region %s.

**Description**

This message tells you how many cells were inserted into the region by the command.

**What Next**

Compare the number of inserted cells to what you anticipated.

---

## CHF-133

(Information) Dropped %d cell %s in region %s due to overlapping with blockages, macros, keepout regions and other existing cells.

**Description**

This message tells you how many cells were not inserted into the region due to blockages, macros, keepout regions and other existing cells by the command.

**What Next**

Compare the number of dropped cells to what you anticipated.

---

## CHF-134

(Information) Tap cells will be shifted to legal location due to %s.

**Description**

This message tells that tap cells may be shifted to accommodate certain rules.

Feedback

**What Next**

Check what options you specified to the command to make sure that you intended to turn those rules on. Make sure to include a tolerance between what you specify to the distance option and the actual tap rule DRC distance.

---

## CHF-135

(warning) The create_boundary_cells command will be deprecated and replaced with the compile_boundary_cells.

**Description**

Migrate to the compile/check_boundary_cells commands flow.

---

## CHF-136

(warning) Row (%.4f, %.4f, %.4f, %.4f) is not wide enough to accommodate %s.

**Description**

This message indicates that the tool has detected a row that is not wide enough to accommodate a tap cell insertion feature that requires shifting the location of the tap cells.

**What Next**

Before inserting tap cells, insert placement blockages in the rows that are not wide enough to accommodate shifting tap cells or turn-off the feature listed in the warning message.

---

## CHF-137

(warning) The chipfinishing.standard_cell_region_horizontal_shrink_factor and chipfinishing.standard_cell_region_vertical_shrink_factor app option will be deprecated and replaced with the derive_standard_cell_region_routing_guides command options: -standard_cell_region_vertical_shrink_factor and -standard_cell_region_horizontal_shrink_factor.

**Description**

The chipfinishing.standard_cell_region_horizontal_shrink_factor and chipfinishing.standard_cell_region_vertical_shrink_factor app option will be deprecated.

**What Next**

Change your script to replace the use of chipfinishing.standard_cell_region_horizontal_shrink_factor and chipfinishing.standard_cell_region_vertical_shrink_factor

Feedback

to use the -standard_cell_region_vertical_shrink_factor and -standard_cell_region_horizontal_shrink_factor options of the derive_standard_cell_region_routing_guides command.

---

## CHF-138

(Warning) Same cell specified for boundary cell (%s) as for boundary tap cell (%s)

**Description**

This warning occurs when you have specified the same cell for bottom or top boundary cell as you do for a boundary tap cell.

**What Next**

Modify your script so that you do not specify any cells for boundary tap cells.

---

## CHF-139

(Warning) Floorplan does not align correctly with double height boundary cell power rails at (%.4f, %.4f, %.4f, %.4f).

**Description**

This warning occurs when you are using double height boundary cells and the Y locations of placement blockages, macros and voltage area edges are not snapped to the correct row site row.

For example if you have double height cells where the middle power rail is a VDD rail the top of row of all the place-able row must have a row orientation of MX and the bottom row of all place-able rows must have a row orientation of R0. On the contrary, if the middle power rail is a GND rail the top of row of all the place-able row must have a row orientation of R0 and the bottom row of all place-able rows must have a row orientation of MX.

**What Next**

Modify the placement blockages, macros and voltage areas of the floorplan so that the top and bottom rows are of the correct orientation.

---

## CHF-140

(Warning) Not enough horizontal clearance at (%f, %f, %f, %f).

Feedback

**Description**

This warning occurs the when there is not enough horizontal clearance to insert boundary cells with the targets that where specified to the command. the tool cannot find a correct vt filler cell to fill the gap at specified location.

**What Next**

Modify the floorplan so that there is enough horizontal clearance in the area specified in the warning.

---

## CHF-141

(Warning) Not enough vertical clearance at (%f, %f, %f, %f).

**Description**

This warning occurs the when there is not enough vertical clearance to insert boundary cells with the targets that where specified to the command. the tool cannot find a correct vt filler cell to fill the gap at specified location.

**What Next**

Modify the floorplan so that there is enough vertical clearance in the area specified in the warning.

---

## CHF-142

(Information) Both core and voltages ares were specifed as boundary insertion targets, core area target will be ignored.

**Description**

This message tells you that you specified both core and voltages ares as targets to the boundary insertion command. If voltage areas are specified there is no need specify core area.

**What Next**

Update your script so that only voltage areas are specified as targets to boundary cell insertion.

---

## CHF-148

(Error) Library cell name %s is specified twice in the -multiple_tap_cell option.

Feedback

**Description**

This message lets you know that you specified the same library cell twice in the -multiple_tap_cell option.

**What Next**

Check over the library cells specified to the -multiple_tap_cell option remove one of them. If they library options have different parameters please take care to remove the one that has incorrect parameters.

---

## CHF-149

(Warning) Variants found for library cell %s, but variant awareness is turned off.

**Description**

This message lets you know that the library cell that you specified has variants, but variant awareness is not turned on. This gives unexpected results and might cause cells to be placed in locations where they are not legal due to variant rules.

**What Next**

Turn on the enable_variant_aware application option as follows:

set_app_option -name place.legalize.enable_variant_aware -value true

---

## CHF-150

(Information) Total %d blockages inserted successfully into %s.

**Description**

This message tells you how many blockages were inserted into the design by the command.

**What Next**

Compare the number of inserted blockages to what you anticipated.

---

## CHF-160

(Information) Total %d route guides inserted successfully into %s.

**Description**

This message tells you how many route guides were inserted into the design by the command.

Feedback

**What Next**

Compare the number of inserted route guides to what you anticipated.

---

## CHF-161

(Information) No areas without tap coverage found.

**Description**

This message tells you that the command found no place-able areas in the design that were missing tap coverage.

**What Next**

Tap insertion should now be complete and no further action in tap cell insertion is required.

---

## CHF-162

(Information) Found %d areas without tap cell coverage. Tap cell insertion will be performed in the areas using the specified pattern.

**Description**

This message tells you the number areas in the design that were missing tap coverage.

**What Next**

After insertion you should check the design to make sure that the final tap cell placement is what you anticipate.

---

## CHF-163

(Information) Found %d areas without tap cell coverage. The areas will be added to the error browser.

**Description**

This message tells you the number areas in the design that were missing tap coverage and that they have been added to the error browser.

**What Next**

After seeing this message you should open up the error browser and inspect and add tap cell coverage to the areas highlighted in the error browser.

Feedback

## CHF-164

(Information) Found %d areas with abutment violations.

**Description**

This message tells you the number areas in the design that have abutment violations.

**What Next**

After insertion you should check the design to make sure that the final tap cell placement is what you anticipate.

## CHF-165

(Error) Found DRC errors in check.

**Description**

This message tells you that DRC errors were found during checking.

**What Next**

Inspect the error browser and investigated the errors.

## CHF-166

(Warning) Tap cell insertion was executed without the -no_abutment options on a design that contains vertical abutment rules.

**Description**

This message tells you that you ran the tap cell insertion command without using vertical abutment options in a design that contains vertical abutment rules.

**What Next**

Consult with the design rules for tap cell insertion and add one more of the following no abutment options as as appropriate: -no_abutment, -no_abutment_cells, -no_abutment_horizontal_spacing and/or -no_abutment_corner_spacing.

## CHF-168

(Information) Tap cells with a VT that cannot abut boundary cells detected, no 0x mode will be turned on.

Feedback

**Description**

This message tells you that the command has detected tap cells that have a VT that cannot abut boundary cells. As a result of this the command turns on no 0x mode, which means that tap cells will not be placed at the beginning or end of place-able site rows.

**What Next**

Run check_legality after tap cell insertion to insure that there are no violations.

---

## CHF-169

(Warning) App option place.legalize.ignore_keepout_margins_against_filler is not supported in advanced legalizer filler flow, use -ignore_standard_cell_keepout_margins instead.

**Description**

You set the place.legalize.ignore_keepout_margins_against_filler app option to true.

**What Next**

Change your script so that instead of setting the place.legalize.ignore_keepout_margins_against_filler app option to true you specify the -ignore_standard_cell_keepout_margins option to the create_stdcell_fillers command.

---

## CHF-170

(Warning) Voltage area %s does not have hierarchy associated with it. Cells inserted in this voltage may be associated with DEFAULT_VA instead of the former voltage area.

**Description**

In order for cells to be inserted and associated into those voltage areas the voltage areas need to have a hierarchy. This warning lets you know if there is a voltage area in your design that do not have associated hierarchy.

**What Next**

Create a module for the hierarchy using create_module and then create a cell into that hierarchy using the create_cell command.

---

## CHF-171

(Error) Leakage vt order violation at %s. Cell %s can be replaced by lib cell %s's instance.

Feedback

**Description**

There is a leakage vt order violation at the reported location. Using the reported lib cell's instance to replace the reported cell can reduce the leakage power.

**What Next**

Fix the violation to reduce the leakage power.

---

## CHF-172

(Information) Attribute %s for library cell %s has been specified both on the command line and in PRF, the value specified on the command line will be used.

**Description**

This message tells you that the given attribute has been specified both on the command line and in PRF and that the value specified on the command line will be used.

**What Next**

Check whether you need to specify the attribute on the command line since it is already specified in PRF and that the values are correct.

---

## CHF-173

(Error) The placeable region {%s} is too %s that not larger than ('-x_offset' + '-x_offset') or ('-left_offset' + '-right_offset'), falied to derive objects in this area.

**Description**

The placeable area is derived according to the settings, if the settings is incorrect, the placeable area will be too narrow or short to place objects.

**What Next**

Please adjust the floorplan or change the command options' value.

---

## CHF-199

(warning) The usage of option -segment_parity in boundary cell insertion is not recommended for advanced technology process due to inefficient area usage. Recommend the floorplan checking and fixing flow using check_floorplan_rules and fix_floorplan_rules commands.

**Description**

This message suggests using the floorplan checking and fixing flow.

Feedback

**What Next**

Check floorplan rules.

---

## CHF-200

(Information) Starting %s insertion into %s using site master "%s".

**Description**

This message tells you that insertion has started and which site definition is being used for insertion.

**What Next**

Check the site definition to make sure it matches your expectation.

---

## CHF-201

(Information) Replaced cell %s with unconstrained cells.

**Description**

This message indicates that the specified constrained cell has been replaced with one or more unconstrained cells.

**What Next**

Check that the new cells match your expectation.

---

## CHF-210

(warning) The MIM capacitor cell whose bbox is {%s} is intersected with %s %s, the creation is skipped.

**Description**

The new created MIM capacitor cannot be intersected with these objects whose type is specified by command option "-no_overlap".

**What Next**

NA.

---

## CHF-250

(Information) Total %d %s successfully removed from %s.

Feedback

**Description**

This message tells you how many cells were removed from the design by the command.

**What Next**

Compare the number of removed cells to what you anticipated.

---

## CHF-300

(error) Different replacement rules cannot be run at the same time.

**Description**

This error message occurs when you specify different rule option sets in one replace_fillers_by_rules command. For example, you cannot run max vertical constraint rule (related to -max_constraint_length option) and abutment rulea(related to -replace_abutment option) in one command.

**What Next**

Run one type of replace_fillers_by_rules at a time. See man page for details.

---

## CHF-301

(error) options %s must be used together.

**Description**

You receive this error message because you didn't specify all the required options for this command.

**What Next**

Please check command man page for detail option usage.

---

## CHF-302

(error) Invalid syntax in -replace_abutment option.

**Description**

This error message occurs when replace_fillers_by_rules -replace_abutment option syntax incorrect. The correct syntax is -replace_abutment { {targetRef1 customRef1} {targetRef2 customRef2} ...}

**What Next**

Please check your option syntax.

Feedback

## CHF-303

(error) Unable to find ref cell %s.

**Description**

This error message occurs when command replace_fillers_by_rules cannot find the ref cell you specified

**What Next**

Please check your script and make sure the ref cell exists.

## CHF-304

(error) Invalid syntax in -tap_cells option.

**Description**

This error message occurs when replace_fillers_by_rules -tap_cells option syntax incorrect. If -left_tap/-right_tap/-both_tap options are used, only specify the tap ref cell to be swapped out in -tap_cells.

If -left_tap/-right_tap/-both_tap aren't used, list the tap cell to be swapped out first, followed by substitute tap cells. For example, -tap_cells {swap_out_tap, substitute_tap1, substitute_tap2, ...}

**What Next**

Please check your option syntax. See man page for more details.

## CHF-305

(error) Option %s must be specified.

**Description**

This error message occurs when indicated option isn't specified.

**What Next**

Please specify requested option. See man page for more details.

## CHF-306

(error) Invalid %s option usgae: %s.

Feedback

**Description**

This error message occurs of indicated option usage error

**What Next**

Please check your option syntax. See man page for more details.

---

## CHF-307

(error) Filler ref cell %s is specified twice

**Description**

This error message occurs when a filler ref cell is used in replace_fillers_by_rules -replace_lib_cells option twice.

**What Next**

Please check your command option usage. See man page for more details.

---

## CHF-308

(error) options %s cannot be used together.

**Description**

You receive this error message because you specified options that are mutually exclusive.

**What Next**

Please check command man page for detail option usage.

---

## CHF-309

(error) duplicate entry %s in -refill_table option.

**Description**

You receive this error message because you specified the indicated ref lib cell twice in -refill_table option.

**What Next**

Please check command man page for detail option usage.

---

## CHF-310

(error) Ref cell %s are specified as target ref cell and replacement ref cell.

Feedback

**Description**

You receive this error message because you specified the indicated ref lib cell as both the target ref cell and replacement ref cell .

**What Next**

Please check command man page for detail option usage.

---

## CHF-311

(error) Command does not support multi-height tap cell.

**Description**

You receive this error message because command does not support multi-height tap cell .

---

## CHF-312

(Warning) Cell %s has end to end orientation violation.

**Description**

You can get this warning when running the replace_fillers_by_rules command with the -check_only option, when the command detects a violation it prints this message.

**What Next**

Fix the violation either manually or be using the replace_fillers_by_rules command without the -check_only option.

---

## CHF-313

(Error) lib cell %s is specified in more than one option.

**Description**

You can get this error when running the replace_fillers_by_rules command with a lib cell specified in more than one option.

**What Next**

Check man page for correct option usage and correct the script.

---

## CHF-314

(Error) Specified filler cells have different heights.

Feedback

**Description**

You can get this error when running the replace_fillers_by_rules -rule small_filler_stacking command, and specifed fillers in -small_fillers and/or -replacement_fillers of different heights. This command only support fillers of same height and siteDef.

**What Next**

Check man page for correct option usage and correct the script.

---

## CHF-315

(Error) specified small filler %s cannot be greater than 2 site width.

**Description**

You can get this error when running the replace_fillers_by_rules -rule small_filler_stacking command, and specifed fillers in -small_fillers that are greater than 2 site width

**What Next**

Check man page for correct option usage and correct the script.

---

## CHF-316

(Error) specified replacement filler %s must be greater or equal to 3 site width.

**Description**

You can get this error when running the replace_fillers_by_rules -rule small_filler_stacking command, and specifed fillers in -replacement_fillers that are less than 3 site width

**What Next**

Check man page for correct option usage and correct the script.

---

## CHF-400

(warning) more than 2500 windows.

**Description**

You received this warning message because based on the specified window size, there will be more than 2500 windows

**What Next**

Check if you use the correct window size

Feedback

## CHF-401

(error) window size greater than core area's.

**Description**

You received this error message because specified window height or width is greater than the core area's, and that you didn't use option -include_small_windows.

**What Next**

Check if you use the correct window size or use option -include_small_windows

## CHF-402

(error) TCD reference cell is not of type macro cell.

**Description**

You received this error message because frontend TCD reference cell must be macro cell type.

**What Next**

Check your frontend TCD reference cell cell type

## CHF-403

(error) design does not have FinFet grid.

**Description**

You received this error message because you used -snap_to_fin_grid but design doesn't have fin grid.

**What Next**

Check your design or option usage.

## CHF-404

(error) backend TCD reference cell is not of type cover cell.

**Description**

You received this error message because backend TCD reference cell must be cover cell type.

Feedback

**What Next**

Check your backend TCD reference cell cell type

---

## CHF-405

(error) specified insertion bbox is outside die area.

**Description**

You received this error message because specified -bbox is outside die area

**What Next**

Check value specified for -bbox option.

---

## CHF-406

(error) incompatible values specified for -tcd_spacing and -stack_with_backend

**Description**

You received this error message because you specified -tcd_spacing for 2 backend tcd layers, but also specified them in -stack_with_backend that they are stackable

**What Next**

Check value specified for -tcd_spacing and -stack_with_backend option.

---

## CHF-407

(error) option %s must be greater than 0

**Description**

You received this error message because values for this option must be greater than 0

**What Next**

Check value specified for the option and man page for additional details.

---

## CHF-502

(error) ICOVL reference cell is not of type macro cell.

**Description**

You received this error message because ICOVL reference cell must be macro cell type.

Feedback

**What Next**

Check your icovl reference cell cell type

---

## CHF-503

(error) Unable to create placement blockage for icovl cell %s because placement blockage of same name already exists.

**Description**

You received this error message because tool tried to create placement blockage for named ICOVL cell but identically named placement blockage already exists. This maybe caused by creating and removing ICOVL cells without removing corresponding placement blockages.

**What Next**

Remove existing ICOVL placement blockages.

---

## CHF-510

(error) Lib cell %s has no vt layer or contains vt layer not in option -leakage_vt_order.

**Description**

You received this error message because tool failed to get vt layer of the lib cell or the vt layer tool got was not specified in the option -leakage_vt_order.

**What Next**

Remove the lib cell or add the vt layer of the lib cell in the option -leakage_vt_order.

---

## CHF-511

(error) specified fller cell %s has unit width %d, whcih is smaller than user defined smallest filler size of %d .

**Description**

You received this error message because you specified a filler cell with unit site width less than what is specified in app option place.rules.min_vt_filler_size.

**What Next**

Remove the offending filler cell from create_stdcell_fillers command or update app option place.rules.min_vt_filler_size setting.

Feedback

## CHF-601

(error) No layer density rule %s exists.

**Description**

You received this error message because tool there is no layer density rule of this name.

**What Next**

Check if you specified correct rule name or define the rule first.

## CHF-602

(warning) For window (%f, %f)-(%f, %f): Impossible to satisfy layer %s upper bound density requirement of %f. After replacing all filler cells with min density cells, the density does not go below %f.

**Description**

You received this error message because after replacing all filler cells with min density cells, it still exceeds max density limit.

**What Next**

Check if you specified correct density requirement or adjust cell placement in window.

## CHF-603

(warning) For window (%f, %f)-(%f, %f): Impossible to satisfy layer %s lower bound density requirement of %f. After replacing all filler cells with max density cells, the density does not exceed %f.

**Description**

You received this error message because after replacing all filler cells with max density cells, it is still below the min density limit.

**What Next**

Check if you specified correct density requirement or adjust cell placement in window.

## CHF-604

(error) Exceed limit of %d windows that are impossible to satisfy density bounds. Command execution terminated.

Feedback

**Description**

You received this error message because the number of windows that are impossible to satisfy density bounds exceeds the set limit. You can change the limit using option -max_dead_window.

**What Next**

Check density rule, cell placement, or adjust max impossible to fix window limit.

---

## CHF-605

(warning) Window (%f, %f)-(%f, %f) is ignored because it has no filler cells.

**Description**

You received this warning message because there is no filler in this window. Therefore no layer density fix can be performed.

---

## CHF-606

(warning) The boundary tap distance has been reduced by one site unit to %s to have an even site length because the -force_even_tile_rows option is used.

**Description**

If the -force_even_tile_rows option is set, then all boundary tap cells will be placed at an even distance location. If an odd tap distance is specified, the distance will be reduced by one site unit to make it even.

**What Next**

Please check boundary tap distance or -force_even_tile_rows option.

---

## CHF-607

(warning) The boundary tap distance has been reduced by one site unit to %s to have an even site length because the -segment_parity option is set as horizontal_even.

**Description**

If the -segment_parity option is set as horizontal_even, then all boundary tap cells will be placed at an even distance location. If an odd tap distance is specified, the distance will be reduced by one site unit to make it even.

**What Next**

Please check boundary tap distance or -segment_parity option.

Feedback

## CHF-608

(warning) The %s has been reduced by one site unit to an even-site %s because -force_even_sites is used.

**Description**

If the -force_even_sites option is specified, then all cells must be placed at even-site locations. If the X-coordinate location provided by the user is not at even-site location, the coordinate location will be rounded down to an even-site location. If the X distance provided by the user is not at even length (in unit of sites), the distance will be rounded down by one site width to an even length. If the X-coordinate location and/or distance provided by the user is already even, the values are accepted without further adjustments.

**What Next**

Please check -x_offset option or -x_pitch option.

## CHF-700

(error) No %s replacement availables.

**Description**

You received this error message because no valid replacement filler of specified type provided for command "replace_fillers_by_rules -replacement_rule half_row_adjacency".

**What Next**

please check that you provided correct replacement filler. See man page for details

## CHF-701

(error) MIMCap reference cell is not of type cover cell.

**Description**

You received this error message because MIMCap reference cell must be cover cell type.

**What Next**

Check your MIMCap reference cell's cell type

## CHF-702

(error) The stub can be specified only when the metal_preferred option is provided.

Feedback

**Description**

Add the metal_preferred option if you need use the stub option.

**What Next**

Check the specifications of your options.

---

## CHF-703

(error) Option -stub's %s is invalid. A valid range is: %s.

**Description**

An invalid value is found for the stub option.

**What Next**

Please check the values of the stub options and use values in the specified range.

---

## CHF-704

(warning) None of MIM capacitor cell is inserted because of the mismatch between PG nets and PG pins of MIM capacitor.

**Description**

VDD and VSS net in the specified boundary can’t cover the VDD and VSS pin of MIM capacitor together.

**What Next**

Please check the PG nets and PG pins of the MIM capacitor in specified layer.

---

## CHF-705

(error) total %d empty-space blocks are found in current design.

**Description**

This error message summarizes total number of empty-space blocks in current design.

**What Next**

Check if the existance of these empty-space blocks is proper. If yes, you can use suppress_message to ignore this error.

Feedback

## CHF-706

(error) Design %s has inconsistent tap distance defined in PRF.

**Description**

All designs should have same default tap distance defined in PRF. This error message indicates there's inconsistent setting.

**What Next**

Check all design and set same tap distance.

## CHF-707

(Warning) Unable to fill gap at {%s %s}.

**Description**

You received this warning message because cannot fill gap at {%s %s} because legal check fail.

**What Next**

Check filler cell can insert {%s %s} coordinate with legal rules.

## CHF-708

(Information) %s .

**Description**

This message tells you verbose information about creating MIM Capacitors.

**What Next**

Please check your information message.

## CHF-710

(Warning) Unable to insert %s filler for center cell %s.

**Description**

You received this warning message because tool cannot legally insert the left/right filler for indicated center cell .

Feedback

**What Next**

Check if design is legal before create_left_right_filler_cells.

---

## CHF-711

(Warning) There are no %s site rows in the design but site_name of lib cell %s is %s.

**Description**

Incorrect lib cell specified because its site def cannot be found in the site rows of current design.

**What Next**

Please specify correct lib cells.

---

## CHF-712

(Error) There is no frame view for lib cell design %s.

**Description**

The lib cell without frame view cannot be used in set_boundary_cell_rules.

**What Next**

Make sure that each lib cell has frame view.

---

## CHF-715

(warning) Unable to connect some pins of '%s' at location {%s} to PG net '%s'.

**Description**

NA.

**What Next**

NA.

---

# CHK

---

## CHK-001

(Error ) The option '%s' is not specified.

Feedback

**Description**

The Option is not specified.

**What Next**

Specify the option in the correct format. Please see the manpage of the command for the correct type of the option.

---

## CHK-002

(warning) The checkpoint action '%s' has a previous definition that will be overwritten.

**Description**

The specified action has a previously definition which will be overwritten.

**What Next**

Please verfiy if the overwriting of the action definition is intended.

---

## CHK-003

(Error) Cannot remove checkpoint action '%s'. No action with that name is defined.

**Description**

The specified action is not found in the checkpoint system. The chechpoint action cannot be removed

**What Next**

The specified named action object is not found in the checkpoint system. You can list the defined action names using "get_checkpoint_data -list_actions".

---

## CHK-004

(Error ) One of the options '%s' must be always be specified.

**Description**

The specified Options for the command are mutually exclusive but atleast one of them must be always be specified.

**What Next**

Please specify one of the specified options.

Feedback

## CHK-005

(warning) The nested checkpoint '%s' will be ignored.

**Description**

The checkpointing system ignores the nested checkpoint and only the outermost checkpoint is ignored.

**What Next**

Please verify the named checkpoint.

## CHK-006

(warning) Checkpoint name '%s' was previously used. Changing to unique name '%s'.

**Description**

The checkpoint name been specified has been defined previously. The checkpointing system will generate a new name for the specified checkpoint.

**What Next**

Please verify the named checkpoint.

## CHK-007

(warning) The checkpoint report '%s' has a previous definition that will be overwritten.

**Description**

The specified report has a previously definition which will be overwritten.

**What Next**

Please verfiy if the overwriting of the report definition is intended.

## CHK-008

(Error) Cannot remove checkpoint report '%s'. No report with that name is defined.

**Description**

The specified report is not found in the checkpoint system. The chechpoint report cannot be removed

Feedback

**What Next**

The specified named report object is not found in the checkpoint system. You can list the defined report names names using "get_checkpoint_data -list_reports".

---

## CHK-009

(information) Running action '%s' before checkpoint '%s'.

**Description**

The specified action is running before the given checkpoint is executed.

**What Next**

Please verfiy if the running of the action at the specified position is intended.

---

## CHK-010

(information) Running action '%s' after checkpoint '%s'.

**Description**

The specified action is running after the given checkpoint is executed.

**What Next**

Please verfiy if the running of the action at the specified position is intended.

---

## CHK-011

(information) Running action '%s' as a replacement for the original contents of checkpoint '%s'.

**Description**

The specified action is running as a replacement for the given checkpoint.

**What Next**

Please verfiy if the running of the action at the specified position is intended.

---

## CHK-012

(information) Running this sequence of reports before checkpoint '%s': '%s'.

**Description**

The specified reports is running before the given checkpoint is executed.

Feedback

**What Next**

Please verfiy if the running of the reports at the specified position are intended.

---

## CHK-013

(information) Running this sequence of reports after checkpoint '%s': '%s'.

**Description**

The specified reports is running after the given checkpoint is executed.

**What Next**

Please verfiy if the running of the reports at the specified position are intended.

---

## CHK-014

(warning) The command will not return checkpoint information because it is run outside of a checkpoint.

**Description**

The command "get_current_checkpoint" will not return checkpoint information because it is been run outside of checkpoint.

**What Next**

Please verify if the command "get_current_checkpoint" is intended to be run outside of checkpoint.

---

## CHK-015

(information) Checkpoint configuration file checkpoint.config.tcl is found in the run directory and will be sourced.

**Description**

Checkpoint configuration file "checkpoint.config.tcl" is found in the run directory and will be sourced.

**What Next**

Please verfiy if the configuration file is intended to be sourced.

---

## CHK-016

(warning) Checkpoint '%s' has not yet been executed.

Feedback

**Description**

The given Checkpoint has not yet been executed.

**What Next**

Please verfiy if a different checkpoint name is to be provided.

---

## CHK-017

(Error ) The action name '%s' has Special Characters that are not allowed.

**Description**

Special characters are not allowed in action name of checkpoint system. The special characters not allowed are 'space', 'carriage return', 'line break', '$', '', '[', ']', '{', '}', '(', ')', '#', ' ;', '\\', ' !'.

**What Next**

Please verify the action name provided.

---

## CHK-018

(Error ) The report name '%s' has Special Characters that are not allowed.

**Description**

Special characters are not allowed in report name of checkpoint system. The special characters not allowed are 'space', 'carriage return', 'line break', '$', '', '[', ']', '{', '}', '(', ')', '#', ' ;', '\\', ' !'.

**What Next**

Please verify the report name provided.

---

## CHK-019

(Error ) The checkpoint name '%s' has Special Characters that are not allowed.

**Description**

Special characters are not allowed in checkpoint name of checkpoint system. The special characters not allowed are 'space', 'carriage return', 'line break', '$', '', '[', ']', '{', '}', '(', ')', '#', ' ;', '\\', ' !'.

**What Next**

Please verify the checkpoint name provided.

Feedback

## CHK-020

(warning) Action '%s' has not yet been defined.

**Description**

The given Checkpoint action has not yet been defined.

**What Next**

Please verfiy if a different checkpoint action is to be provided.

## CHK-021

(warning) Report '%s' has not yet been defined.

**Description**

The given Checkpoint report has not yet been defined.

**What Next**

Please verfiy if a different checkpoint report is to be provided.

## CHK-022

(warning) Checkpoint system is disabled by 'set_checkpoint_options -active false'. eval_checkpoint will evaluate the enclosed Tcl commands but will not run any actions or reports.

**Description**

The checkpoint system is disabled. The command 'eval_checkpoint' will not run the '-before/-after' reports and '-before/-re[lace/-after' actions. The checkpoint will just evaluate the enclosed TCL commands.

**What Next**

Please verfiy if a the checkpoint system is to be enabled.

## CHK-023

(information) Starting 'eval_checkpoint %s ...'.

**Description**

The specified checkpoint and the associated actions and reports are preparing to get executed.

Feedback

**What Next**

Please verfiy if the running of the specified checkpoint is intended.

---

## CHK-024

(information) Ending 'eval_checkpoint %s ...'.

**Description**

The specified checkpoint and the associated actions and reports have been successfuly executed.

**What Next**

Please verfiy if the running of the specified checkpoint is intended.

---

## CHK-025

(Error) Checkpoint name '%s' does not exist.

**Description**

The specified checkpoint doesn't exists.

**What Next**

Please verfiy if the checkpoints that has been executed.

---

# CHUNB

---

## CHUNB-001

(Error) Library cell '%s' missing. '%s' and '%d' other cell instance(s) which reference this lib cell are unbound.

**Description**

This message appears when a leaf cell's reference is missing.

---

## CHUNB-002

(Error) Cell instance '%s' does not have any reference and is unbound.

**Description**

This message appears when a leaf cell is unbound.

Feedback

## CHUNB-003

(Error) Site definition '%s' is missing. '%s' and '%d' other site row(s) which reference this site definition are unbound.

**Description**

This message appears when a site row is unbound. Please check if the site definition is removed.

## CHUNB-004

(Error) Site row '%s' does not have any site definition and is unbound.

**Description**

This message appears when a site row is unbound.

## CHUNB-005

(Error) Site definition '%s' is missing. '%s' and '%d' other site array(s) which reference this site definiton are unbound.

**Description**

This message appears when a site array is unbound. Please check if the site definition is removed.

## CHUNB-006

(Error) Site array '%s' does not have any site definition and is unbound.

**Description**

This message appears when a site array is unbound.

## CHUNB-007

(Error) Via definition '%s' is missing. '%s'and '%d' other via(s) which reference this via definition are unbound.

**Description**

This message appears when a via is unbound. Please check if the via definition is removed.

Feedback

## CHUNB-008

(Error) Via '%s' does not have any via definition and is unbound.

**Description**

This message appears when a via is unbound.

## CHUNB-009

(Error) Shape '%s' has unbound layer '%s'.

**Description**

This message appears when a shape has unbound layer.

## CHUNB-010

(Error) Track '%s' has unbound layer '%s'.

**Description**

This message appears when a track has unbound layer.

## CHUNB-011

(Error) Layer '%s' is unbound layer.

**Description**

This message appears when a layer is unbound layer.

## CHUNB-012

(Error) Pin guide '%s' has unbound layer '%s'.

**Description**

This message appears when a Pin guide has unbound layer.

## CHUNB-013

(Error) Pin blockage '%s' has unbound layer '%s'.

**Description**

This message appears when a Pin blockage has unbound layer.

Feedback

### CHUNB-014

(Error) Routing guide '%s' has unbound layer '%s'.

**Description**

This message appears when a Routing guide has unbound layer.

### CHUNB-015

(Error) Routing corridor shape '%s' has unbound layer '%s'.

**Description**

This message appears when a Routing corridor shape has unbound layer.

### CHUNB-016

(Error) %s '%s' has unbound layer(%s) '%s'.

**Description**

This message appears when a Via defs has unbound layer.

## CL

### CL-001

(error) Cannot find current design and/or its hierarchy

**Description**

The tool cannot find any valid design and/or hierarchy. Please load a valid design.

**What Next**

Load a valid design.

### CL-002

(error) Clone count is less than 1.

**Description**

For clone_logic command, the value of "-clone_count" should NOT less than 1.

Feedback

**What Next**

Set an valid option value for clone count.

---

## CL-003

(error) Can only clone one driver cell at a time.

**Description**

The clone_logic command can only clone one driver cell at one time.

**What Next**

Set an valid option value for -driver_cell option.

---

## CL-004

(error) Driver cell must be leaf sequential cell.

**Description**

Must specify leaf sequential cell to option "-driver_cell".

**What Next**

Set an valid value for -driver_cell option.

---

## CL-005

(error) Can NOT clone don't touch cell %s.

**Description**

Don't touch cell can not be cloned or removed. User needs to double check the cell attributes.

**What Next**

Double check cell attribute.

---

## CL-006

(error) Can NOT clone fixed cell %s.

**Description**

Fixed cell can not be cloned or removed. User needs to double check the cell attributes.

Feedback

**What Next**

Double check cell attribute.

---

## CL-007

(error) Can NOT clone RP group cell %s.

**Description**

RP group cell can not be cloned or removed. User needs to double check the cell attributes and RP group.

**What Next**

Double check cell attribute and RP group.

---

## CL-008

(error) Can NOT clone bound cell %s.

**Description**

A cell in bound can not be cloned or removed. User needs to double check the cell attributes and bound.

**What Next**

Double check cell attribute and bound.

---

## CL-009

(error) Can NOT clone MV cell %s.

**Description**

MV cell (level-shifter, isolation cell) can not be cloned or removed.

**What Next**

Double check specified driver cell.

---

## CL-010

(error) Can NOT clone don't touch net %s.

**Description**

Don't touch net can not be cloned or removed.

Feedback

**What Next**

Double check net attribute

---

## CL-011

(error) Failed to cluster endpoints into %d clusters, invalid endpoints location or number.

**Description**

The command failed to cluster endpoints into several clusters.

**What Next**

Double check the value of option "-clone_count". Make sure the endpoint number is larger than clone cluster count, make sure endpoints are placed.

---

## CL-012

(error) Can not clone un-placed cell %s.

**Description**

Cell must be placed before clone_logic is performed. User needs to double check the placement result.

**What Next**

Double check cell placement result.

---

## CL-013

(error) Can not clone driver with combinational loop.

**Description**

The command can not clone logic with combinational loop, which means the driver can not be one of the endpoints.

**What Next**

User needs to double check the net-list logic.

---

## CL-014

(error) Can not clone cell %s within top level block.

Feedback

**Description**

The command can not clone cells inside top level block.

**What Next**

User needs to double check the net-list logic.

# CLE

## CLE-01

(information) Command line editor mode is set to %s successfully.

**Description**

This information message confirms that the command line editor mode is set to the mode you specified. Use command *set_cle_options* to set the line editing mode to either vi or emacs.

**What Next**

This is an information message only. No action is required.

## CLE-02

(error) Command line editor mode cannot be set to %s. Proceeding with %s mode.

**Description**

This error message occurs when you attempt to set the line editor mode to an invalid value. The value can be either set to *vi* or *emacs*. If you attempt to set the edit mode to an invalid value, then the tool uses either the existing edit mode, if the mode is set, or the default emacs mode.

**What Next**

Use the *set_cle_options* command to specify a valid mode.

## CLE-03

(warning) Command line editor is already in %s mode.

**Description**

This warning message occurs when you specify the same value for the command line editor mode that is currently in use by the editor.

Feedback

**What Next**

This is a warning message only. No action is required.

However, if the result is not what you intended, use the *set_cle_options* command to set the line editing mode to either vi or emacs.

---

## CLE-04

(warning) Variable sh_enable_line_editing can be set only in .synopsys_dc.setup file.

**Description**

This warning message occurs when you attempt to enable command line editing by setting the *sh_enable_line_editing* variable in the shell rather than in the .synopsys_dc.setup file.

**What Next**

This is a warning message only. No action is required.

However, you can enable command line editing by setting the *sh_enable_line_editing* variable to true in the .synopsys_dc.setup file.

Refer to the *Design Compiler Command-Line Interface Guide* for more information.

---

## CLE-05

(error) Command line editor is not enabled.

**Description**

This error message occurs when command line editing is not active.

**What Next**

Set the *sh_enable_line_editing* variable to true in the .synopsys_dc.setup file to enable command line editing.

Refer to the *Design Compiler Command-Line Interface Guide* for more information.

---

## CLE-06

(information) %s is currently in %s editing mode.

**Description**

This information message displays the current editing mode.

Feedback

**What Next**

This is only an informational message. No action is required.

However, you can use the *set_cle_options* command to set the line editing mode to either vi or emacs.

---

## CLE-07

(information) Terminal beep is %s.

**Description**

This information message displays the current terminal beep mode.

**What Next**

This is only an informational message. No action is required.

However, you can use the *set_cle_options* command to set the terminal beep to either on or off.

---

## CLE-08

(error) Terminal beep mode value can be either on or off.

**Description**

This message occurs when you attempt to set the line editor beep mode to an invalid value. The value can be either *on* or *off*. If you attempt to set the beep mode to an invalid value, then the tool uses the existing beep mode.

**What Next**

Use the *set_cle_options* command to specify a valid beep mode.

---

## CLE-09

(warning) -defaults option will override other options.

**Description**

-defaults option will override the other options of *set_cle_options* command.

**What Next**

This is only a warning message. No action is required.

Feedback

### CLE-10

(information) Term was not able to be set up using %s . Using "xterm" by default instead.

**Description**

This warning message occurs in certain OS when the term cannot be set up under its default term name.

## CLFR

### CLFR-001

(information) Read antenna property CLF file into '%s' library '%s'.

**Description**

This message reports the name of the library to accomodate CLF data.

**What Next**

No action required.

### CLFR-002

(error) Library '%s' is not a physical lib_cell library.

**Description**

The library specified for read CLF must be a physical lib_cell library.

**What Next**

Choose a physical lib_cell library to read CLF in.

### CLFR-003

(error) Invaild library name '%s' for option -library.

**Description**

There is no library with the name specified.

**What Next**

Please check the library name.

Feedback

## CLFR-004

(error) The library has no technology information.

**Description**

The library was created without a technology file. The library must have technology information to correctly read the CLF file.

**What Next**

Load a technology file into the library using the *read_tech_file* command.

## CLFR-005

(error) Cannot find CLF file '%s'.

**Description**

The file, specified in the read_clf_antenna_properties command, cannot be read.

**What Next**

Check the existence of the file in the search_path using the *which* command.

## CLFR-006

(error) Failed to read CLF file '%s'.

**Description**

reading CLF file failed.

**What Next**

Check CLF file syntax and content.

## CLFR-007

(error) Failed to find cell '%s' with '%s' view in lib '%s'.

**What Next**

Check the existence of the cell in the lib.

## CLFR-008

(error) Failed to find lib_pin '%s' in the lib_cell '%s'.

Feedback

**What Next**

Check the existence of the lib_pin in the lib_cell.

---

## CLFR-009

(error) '%s' at the line '%d' of file '%s'.

**What Next**

Check the syntax.

---

## CLFR-010

(error) Cannnot read CLF file to workspace library.

**Description**

The library specified for read CLF cannot be the workspace library.

**What Next**

Choose a library name other than the workspace name to read CLF in.

---

## CLFR-011

(error) Library name '%s' cannot be same as workspace name.

**Description**

The library name specified for read CLF cannot be the workspace name.

**What Next**

Choose a library name other than the workspace name to read CLF in.

---

## CLFR-012

(warning) Invalid hierarchy antenna properties specified on the layer "%s" for port "%s" of cell "%s".

**Description**

You got this message because the number of the hierarchy antenna property on the layer is not equal to 7. The syntax of scheme command "defineHierAntennaProp" is as follows. (defineHierAntennaProp "cellName" "portName" '( ("Metal1_layerName" m1_gate_size m1_mode1_area m1_mode2_ratio m1_mode3_area m1_mode4_area m1_mode5_ratio m1_mode6_area) )) These hierarchy antenna properties will be ignored.

Feedback

**What Next**

Check the number of the hierarchy antenna property specified on the layer.

---

## CLFR-013

(error) Layer '%s' does not exist in the library.

**Description**

You got this message because the indicated layer was not found in the library. All the antenna properties defined on this missing layer will be ignored.

**What Next**

Manually add the missing layer into the technology file, load the technology file, and then read the CLF file..

---

# CLFW

---

## CLFW-001

(warning) The layer information of the antenna property "%s" "%s" for port "%s" is missing.

**Description**

The antenna property will be ignored when the layer information is missing.

---

## CLFW-002

(error) The library "%s" does not contain any cell with "%s" view.

**Description**

write_clf_antenna_properties always dumps antenna properties from frame view.

**What Next**

Check whether there is any cell with frame view.

---

## CLFW-003

(error) The library has no technology information.

Feedback

### Description

The library was created without a technology file. The library must have technology information to correctly write the CLF file.

### What Next

Load a technology file into the library using the *read_tech_file* command.

# CLK

## CLK-0001

(warning) Clock clock has count sources on inout ports

### Description

The circuit behavior of an inout port can be unpredictable if it has a clock signal driving it. If the port is functioning as an output while the clock is driving, bus contention occurs. If the clock is a three-state condition, any registers in the fanout of the clock can lose state. If the inout port is operating only in input mode for this scenario, there is no risk. It is also possible that the clock is defined on the incorrect port, or the port is defined with the wrong direction.

### ONE POSSIBLE SCENARIO

Consider the scenario where a clock is defined on an inout port:

create_clock bidirClock -period 2

Since the port is bidirectional (or inout), and a clock has been defined on it, a CLK_0001 rule violation will be issued.

### What Next

Verify that the circuit is safe according to your design rules. Synopsys tools assume that the clock is valid. For the example scenario, you can verify the direction of the port with the get_attribute command:

get_attribute [get_port bidirClock] direction

### FIX SUGGESTION

There are several possibilities for resolving this rule violation. If the inout port is only operating in input mode for the scenario of interest, no change may be needed. If the port has the wrong direction, the Verilog module related to the port can be updated. Last, if the clock is defined on the incorrect port, it can be moved to the correct port for your design.

Feedback

For the example scenario, based on knowledge of the design, the bidirClock port only acts as an input port for the design in this mode, so no change is needed.

**See Also**

create_clock get_attribute

---

## CLK-0002

(Warning) Generated clock clock has count sources on input or inout ports.

**Description**

When creating a generated clock on an input or inout port, the generated clock will only derive its waveform characteristics from the master clock. Delay is derived from delay information at the port itself. Additionally, creating a generated clock to drive an inout port can cause unpredictable circuit behavior. If the port is functioning as an output while the clock is driving, bus contention occurs. If the clock is a three-state condition, any registers in the fanout of the clock can lose state. In post-layout conditions, a source latency for the generated clock cannot be calculated from the master clock. This may lead to inaccuracies in the timing checks related to the generated clock.

**ONE POSSIBLE SCENARIO**

Consider the scenario where a generated clock is defined on an input port: create_clock CLK1 -period 2 create_generated_clock -source CLK1 -name gen_clk [get_ports CLK2]

Since the generated clock was defined on input port CLK2, a CLK_0002 rule violation will be issued.

**What Next**

Verify that the circuit is safe according to your design rules. Synopsys tools assume that the generated clock is valid. For the example scenario, you can examine the generated clock definition by clicking on the name of the generated clock in the ‘Violation Details’. You can use the report_port command to verify the direction of the port specified in the "Violation Details":

rtl_shell> report_port [get_ports CLK2]

Pin Cap Wire Min Max Min Port Dir rise/fall rise/fall rise/fall rise/fall Attributes
---------- CLK2 in 0.00/0.00 0.00/0.00 0.00/0.00 0.00/00

The report shows that CLK2 is an input port, so the generated clock has been defined on an input port.

**FIX SUGGESTION**

Feedback

There are several possibilities for resolving this rule violation. If the waveform characteristics are the only thing required from the master clock, and the delay characteristics of the port are to be used for the generated clock, then no change may be needed.

If the port is an inout port and has the wrong direction, the Verilog module related to the port can be updated.

Last, if the generated clock is defined incorrectly on a port, it can be moved to the correct pin for your design. For the example scenario, based on knowledge of the design, the generated clock was incorrectly defined on the port and the constraint was changed to an internal pin of the design:

create_generated_clock –source CLK1 –name gen_clk [get_pins ff1/Q]

**See Also**

create_clock create_generated_clock report_port get_attribute

---

## CLK-0003

(Error) Generated clock clock is not expanded because it has no clock reaching its master source pin.

**Description**

The pin or port given for the -source option of this generated clock has no clock, therefore, no master clock can be determined for it. A generated clock without a master clock has no waveform, and it is not propagated forward to register clock pins. The portion of the circuit that should be clocked by this generated clock is untested, resulting in incomplete timing analysis on the design.

**ONE POSSIBLE SCENARIO**

This example scenario has a clock divider, div2. The constraints have a generated clock definition at the output, div2/Q, with the div2/CP clock pin, as the -source pin. No -master_clock option has been given. A CLK_0003 rule violation will be issued if no clock propagates to the div2/CP pin. (A CLK_0028 violation will be issued if there are multiple clocks at div2/CP.) In this example scenario, there is no clock at the div2/CP pin because of the set_disable_timing constraint at the U1/Z pin.

**What Next**

Determine why there is no clock at the -source pin or if an incorrect -source pin was specified in the create_generated_clock command. If the -source pin is correct, then the clock may be missing because:

The master clock has not been defined.

Feedback

The master clock is blocked from the -source pin by another constraint.

**FIX SUGGESTION**

Add the missing master clock, remove the constraints blocking the correct master clock, or move the -source pin to the correct location.

For the preceding example scenario, the set_disable_timing constraint can be removed by issuing the command:

remove_disable_timing u1/Z

Or the set_disable_timing command can be removed from the constraint file.

*Corresponding PrimeTime Error Message*

*Error Message: PTE-023*

*Severity:* Warning

*Description*

The generated clock '%s' has not been expanded. Create or activate its master clock.

**See Also**

create_generated_clock

---

## CLK-0004

(Warning) Mismatch between generated clock definitions at pin_or_port and potential master clocks.

**Description**

At the given pin or port location, there should be a generated clock (specified with the -add and -master_clock options) for each clock signal meeting the following requirements:

Present on the master sources

Propagating through source latency paths to this pin or port

The violation details section lists the master clocks with missing generated clock statements at the reported pin or port location.

This rule looks for potential masters as follows:

All the pins provided through the -source option of all generated clocks on the violated pin or port.

Any clock blocked at the violated pin or port by the generated clock definitions.

Feedback

Any register clock pin in the source latency paths of the existing generated clocks and in the fanout of the master clocks of those generated clocks.

Any generated clocks defined in the source latency path of the target generated clock with the same master clock. This portion of the rule check is controlled by the warn_on_generated_clocks_of_same_master rule property.

The register clock pins in the fanout of the given pin will be missing some of the clock waveforms that are physically possible for the circuit. This may result in a circuit that has not been completely verified and potentially create unbalanced clock trees during clock tree synthesis.

**ONE POSSIBLE SCENARIO**

Consider a clock divider, div2, with two clocks, CLK1 and CLK2 propagating to the clock pin of the clock divider. Only one generated clock is created on the output of the clock divider with CLK1 as its master clock. Since two clocks are actually propagating to this location (CLK1 and CLK2), a CLK_0004 rule violation will be issued for the div2/Q pin. The “Violation Details” section reports that potential CLK2 master clock exists at div2/CP, but it is not used when creating a generated clock at div2/Q.

**What Next**

Examine the "Violation Details" section to find which generated clocks exist at the violated pin or port.

**FIX SUGGESTION**

Add the missing generated clocks to the pin or port as shown here:

```
create_generated_clock DIV2/q -add \ -name gclk2 \ -source [get_port clk2] \
-master_clock clk2 \ -divide_by 2
```

**See Also**

create_generated_clock

---

## CLK-0005

(Info) Virtual clock clock has the same name as a design object.

**Description**

A virtual clock was defined with the same name as a port, pin, or net, and might not be what was intended. This kind of virtual clock, if intended to be an actual clock in the design, can lead to unconstrained paths in the design and incorrect constraints. Some constraints related to the virtual clock can be ignored and result in unrealistic timing violations.

Feedback

### ONE POSSIBLE SCENARIO

Consider a simple scenario where the clock is mistakenly defined as a virtual clock.

create_clock -name CLK1 -period 2 create_clock -name CLK2 -period 1 [get_ports CLK2] set_multicycle_path 2 -from [get_clocks CLK1] -to [get_clocks CLK2]

The ‘CLK1’ clock is a virtual clock that is not associated with any clock source port or pin. In this case, the following warning message is also issued:

Warning: Creating virtual clock named 'CLK1' with no sources. (UIC-021)

The design is unconstrained because there is no clock signal associated with the ‘CLK1’ port. In addition, the multicycle path exception from ‘CLK1’ to ‘CLK2’ is ignored because the ‘CLK1’ virtual clock is not associated with the ‘CLK1’ clock port.

### FIX SUGGESTION

Change the virtual clock to a non-virtual clock or change the virtual clock name such that the clock name differs from an existing port, pin or net name in the design.

In the example scenario, the virtual clock can be made into a non-virtual clock as follows:

create_clock -name CLK1 -period 1 [get_ports CLK1] create_clock -name CLK2 -period 2 [get_ports CLK2] set_multicycle_path 2 -from [get_clocks CLK1] -to [get_clocks CLK2]

Now the design is properly constrained and the multicycle path exception will be applied correctly.

### See Also

create_clock

---

## CLK-0006

(Warning) Clock clock source source has a logic constant or case value.

### Description

A clock source has been defined at a port or pin that has a constant value due to a logic constant or set_case_analysis constraint. The constant value will suppress the propagation of the clock waveform. A constant value at a clock source will deactivate the clock, and no timing analysis will be performed for the portion of design that is affected by the clock.

### ONE POSSIBLE SCENARIO

Consider a simple scenario involving a generated clock at a register output. The generated clock ‘GCLK1’ is defined on the output pin of a register, and this pin’s value is set to 0 by

Feedback

set_case_analysis. This constant suppresses the propagation of the generated clock and deactivates timing analysis of the downstream logic affected by ‘GCLK1’.

create_clock -name CLK -period 2 [get_ports CLK] create_generated_clock -name GCLK1 -source [get_ports CLK] -divide_by 2 [get_pins ff1/Q] set_case_analysis 0 [get_pins ff1/Q]

Because the generated clock is defined at a pin with a constant value, a CLK_0006 rule violation will be issued.

**What Next**

The violation message gives the name of the clock and the clock source pin. To further identify the root cause of the constant arriving at the source pin, use the report_case_details command. This command can show how logic constants and user case values propagate to the clock source pin.

For the example scenario, the command can be issued as follows:

rtl_shell> report_case_details -to [get_pins ff1/Q]

As can be seen in the following report excerpt, the clock source pin has a constant value of 0 due to a user-specified set_case_analysis.

Properties Value Pin/Port ---------- user case 0 ff1/Q

**FIX SUGGESTION**

If the intent is not to suppress the clock signal, remove the logic constant or the user-specified case value that caused the clock source pin to assume a constant value.

In the example scenario, this can be done by simply removing the set_case_analysis constraint:

create_clock -name CLK -period 2 [get_ports CLK] create_generated_clock -name GCLK1 -source [get_ports CLK] -divide_by 2 [get_pins ff1/Q] set_case_analysis 0 [get_pins ff1/Q]

This will allow generated clock ‘GCLK1’ to propagate.

**See Also**

create_generated_clock create_clock set_case_analysis report_case_details

---

## CLK-0007

(Error) Generated clock clock master source source has a logic constant or case value.

**Description**

The master source (a port or a pin) of the reported generated clock has a constant value due to logic or set_case_analysis constraint. The constant value will suppress the propagation of the generated clock waveform.

Feedback

Constant value at the master clock source will deactivate the generated clock and no timing analysis will be performed for the portion of design that is affected by the generated clock. This could result in an incompletely timed design.

**ONE POSSIBLE SCENARIO**

Consider a simple scenario involving two generated clocks. The first generated clock‘s (‘GCLK1’) source pin is defined at the output of a register whose value is set to 0 by set_case_analysis. The second generated clock (‘GCLK2’) has its master clock source pin defined at the output of the first register. This constant master clock source suppresses the clock propagation and de-activates the timing analysis of the downstream logic affected by the generated clock ‘GCLK2’.

create_clock -name CLK -period 2 [get_ports CLK] create_generated_clock -name GCLK1 -source [get_ports CLK] -divide_by 2 [get_pins ff1/Q] create_generated_clock -name GCLK2 -source [get_pins ff1/Q] -divide_by 1 [get_pins inv2/Z] set_case_analysis 0 [get_pins ff1/Q]

**What Next**

The violation message gives the name of the clock and its master clock source pin. To identify further the root cause of the constant arriving at the master clock source pin, use the report_case_details command. This command shows how logic constants and user case values propagate to the master clock source pin.

For the example scenario, the command can be issued as follows:

rtl_shell> report_case_details ff1/Q

As can be seen in the report excerpt below, the clock source pin has a constant value of 0 due to user case.

Properties Value Pin/Port ---------- user case 0 ff1/Q

To analyze how the constant value propagates to the source pin of the generated clock ‘GCLK2’, supply the generated clock source pin in the -to option of the report_case_details command:

rtl_shell> report_case_details -to inv2/Z

Properties Value Pin/Port ---------- from user case 0 inv2/Z

Case fanin report: Verbose Source Trace for pin/port inv2/Z: Path number: 1 Path Ref # Value Properties Pin/Port ---------- 0 F()=A' inv2/Z 1 inv2/A 1 F()=A' inv1/Z 0 inv1/A 0 user case ff1/Q

**FIX SUGGESTION**

If the intent is not to suppress the generated clock signal, remove the logic constant or the user case that caused the master clock source pin to assume a constant value.

Feedback

In the example scenario, this can be done by simply removing the set_case_analysis constraint:

create_clock -name CLK -period 2 [get_ports CLK] create_generated_clock -name GCLK1 -source [get_ports CLK] -divide_by 2 [get_pins ff1/Q] create_generated_clock -name GCLK2 -source [get_pins ff1/Q] -divide_by 1 [get_pins inv2/Z] #set_case_analysis 0 [get_pins ff1/Q]

**See Also**

create_generated_clock create_clock set_case_analysis report_case_details

---

## CLK-0008

(Warning) Generated clock clock has paths from the sources of master clock master_clock to generated clock sources with differing sequential depth.

**Description**

There are several possible latency paths reaching the generated clock. Not all the paths cross the same number of sequential elements and not all the paths may be needed for the latency calculation. The clock latency calculation may become too pessimistic. The generated clock may need to be modified to remove the pessimism. If this violation is not addressed, the timing analysis results may be too pessimistic since the timing analysis tool has to account for every path. This can lead to unnecessary increased effort in optimization and timing closure.

**ONE POSSIBLE SCENARIO**

Consider a scenario where there are two latency paths from the master clock to a generated clock. One path involves a sequential depth of one and the other path is combinational. In this example, the design intent is to propagate clock through the combinational path and use the sequential path as the signal pin of clock gating logic. However, clock latency calculation involves both latency paths and the “worst” path is chosen for timing analysis. It is likely that the sequential path is chosen for hold latency calculation because a sequential path delay tends to be larger than a combinational path delay. This may lead to undesirable hold violations associated with the generated clock.
*Figure 1: Two Latency Paths From Master Clock to Generated Clock*

create_clock -name sys_clk -period 4 [ get_ports CLK ] create_generated_clock -name gclk -divide_by 1 -source [get_ports CLK] [get_pins and1/Z]

**What Next**

The violation message gives the name of the generated clock and the Violation Details section provides the location where the differing sequential depth branches merge.

**FIX SUGGESTION**

Feedback

Different scenarios may require different approaches. The basic idea is to ensure that the latency path for the generated clock includes only the actual clock propagation path from the master clock to the generated clock. This may be done by using one of the following techniques:

moving the generated clock source

creating intermediate generated clocks

using the -stop_propagation option with the set_clock_sense command

using the -combinational option with the create_generated_clock command

For the example scenario, this can be done by changing the generated clock “gclk” to “combinational”. This means that the latency path for “gclk” will include only combinational paths where the master clock propagates to the generated clock and no paths involving sequential elements.

The modification can be done as follows:

create_clock -name sys_clk -period 4 [ get_ports CLK ] create_generated_clock -name gclk -combinational -source [get_ports CLK] [get_pins and1/A]

**See Also**

create_generated_clock create_clock set_clock_sense

---

## CLK-0009

(Error) Generated clock clock is not expanded because master clock master_clock does not reach its master source pin.

**Description**

The source pin of this generated clock does not belong to the clock network of the given generated clock’s master clock. The sense of the master clock at the -source pin is required to compute the waveform of a generated clock. No waveform will be created and, as a consequence, the generated clock cannot be propagated forward to register clock pins. The portion of the circuit that should be clocked by this generated clock is untested. This can result in potentially incorrect timing if the generated clock is supposed to be propagated to registers.

**ONE POSSIBLE SCENARIO**

In this example scenario, two clocks are propagating to some clock divider structure. In order to guarantee the proper timing analysis, generated clocks are created on the output of the clock dividers as seen in the following sample script.

Feedback

create_clock [get_port A] -name clk1 -period 2 create_clock [get_port B] -name clk2 -period 3 create_generated_clock -name gclk1 -divide_by 2 div2/Q \ -source [get_port B] -add -master_clock clk1 create_generated_clock -name gclk2 -divide_by 2 div2/Q \ -source [get_port A] -add -master_clock clk2

However, some mistakes were introduced in those constraints, resulting in CLK_0009 rule violations:

Generated clock 'gclk1' with 'clk1' as its master 'clk1' but refers to port B as its source. This is incorrect; the source should be port A.

Similarly, generated clocks 'gclk2' should refer to port B as its source pin rather than port A.

**What Next**

Check that the master_clock option is correct with respect to the source pin.

Check that the source pin is correct with respect to the master clock.

You can use the report_clock command to achieve this as demonstrated below.

rtl_shell> report_clock ****************************************** Report : clock Design : CLK_0009 Scenario: default Version: … Date : … ******************************************

Attributes: p - Propagated clock G - Generated clock I - Inactive clock

Clock Period Waveform Attrs Sources ---------- clk1 2.00 {0 1} {A} clk2 3.00 {0 1.5} {B} gclk1 G {div2/Q} gclk2 G {div2/Q}

Generated Master Generated Master Waveform Clock Source Clock Modification ---------- gclk1 B div2/Q clk1 div(2) gclk2 A div2/Q clk2 div(2)

This reports shows the source discrepancy. Port “A” is the source of clock “clk1”, but port “B” is the source of generated “gclk1”.

If the master clock and the source pin seem correct, investigate why the master_clock doesn’t propagate to the source pin. Most likely, another constraint or a design structure is preventing proper clock sensitization:

The master clock could be blocked from the source pin by another constraint.

The master clock could be blocked by an unresolved cell in the clock network. When there appears to be no path and you expect one, an unresolved cell in the path, or a cell with an incomplete model is a possibility. Use the all_fanout command to look for where the path is disconnected.

rtl_shell> all_fanout -from [get_ports A] {"A", "b2/A", "b2/Z"}

**FIX SUGGESTION**

Feedback

Change the generated clock to correct the master_clock and -source pin/port or remove constraints that are blocking the master clock from the -source pin/port. For the example scenario above, fix the -source values:

create_clock [get_port A] -name clk1 -period 2 create_clock [get_port B] -name clk2 -period 3 create_generated_clock -name gclk1 -divide_by 2 div2/Q \ -source [get_port A] -add -master_clock clk1 create_generated_clock -name gclk2 -divide_by 2 div2/Q \ -source [get_port B] -add -master_clock clk2

*Corresponding PrimeTime Error Message*

*Error Message: PTE-023*

*Severity:* Warning

*Description*

The generated clock '%s' has not been expanded. Create or activate its master clock.

**See Also**

create_generated_clock report_clock

---

## CLK-0010

(Error) Generated clock clock is not expanded because its master clock master_clock is not expanded.

*Status:* Disabled by default

**Description**

A generated clock cannot be activated; the specified master_clock is not expanded. The portion of the circuit that should be clocked by this generated clock is unconstrained. This is likely to result in incorrect timing for the design.

**ONE POSSIBLE SCENARIO**

Consider the following scenario involving two cascading generated clocks. The second generated clock “d4” specifies the first generated clock “d2” as its master clock. However, no clock has been created at the port clk1, and thus the generated clock “d2” cannot be activated. This results in a CLK_0003 violation being flagged on “d2” and no waveform can be computed for generated clock “d2”. This CLK_0003 violation in turn creates a CLK_0010 violation for “d4” since a valid waveform would need to be present on “d2” to compute “d4”.

create_generated_clock -name d2 -divide_by 2 div2/Q -source div2/CP create_generated_clock -name d4 -divide_by 2 div4/Q \ -source div2/Q -master_clock d2 -add

Feedback

**What Next**

Investigate and fix the violation CLK_0003 or CLK_0009 related to the master clock. When this is done, the CLK_0010 rule violation should be addressed as a side effect.

**FIX SUGGESTION**

See the violation for the master clock. (CLK_0003 or CLK_0009). For the example scenario above, fix the *-source* values:

create_clock clk1 -period 2

*Corresponding PrimeTime Error Message*

*Error Message: PTE-023*

*Severity:* Warning

*Description*

The generated clock '%s' has not been expanded. Create or activate its master clock.

**See Also**

create_generated_clock report_clock CLK_0003 CLK_0009

---

## CLK-0011

(Error) Generated clock clock is not expanded because potential master clock master_clock is not expanded.

**Description**

A generated clock cannot be activated if only unexpanded master clocks drive the master source. Fixing the issue with the unexpanded master clocks resolves this violation.
This rule differs from the CLK_0010 rule because the -master_clock option for the create_generated_clock command was not used, therefore the unexpanded master clock appears as a potential master clock. The portion of the circuit clocked by this generated clock is unconstrained. No timing analysis will be performed for the portion of the circuit that is associated with this generated clock.

**ONE POSSIBLE SCENARIO**

Consider the following scenario involving two cascaded generated clocks. The generated clock d4 specifies the generated clock d2 as its master clock. The generated clock d2 cannot be activated because no clock has been created at the port clk1. This results in a CLK_0003 violation being issued for d2. A CLK_0011 rule violation will be issued for d4 because its definition does not contain the -master_clock option, and no waveform can be computed for it due to the error at d2.

Feedback

create_generated_clock -name d2 -divide_by 2 [get_pins div2/Q] \ -source [get_pins div2/CP] create_generated_clock -name d4 -divide_by 2 [get_pins div4/Q] \ -source [get_pins div4/CP]

**What Next**

Investigate the CLK_0003, CLK_0009, CLK_0010, or CLK_0011 rule violation related to the master clock.

**FIX SUGGESTION**

Fix the CLK_0003, CLK_0009, CLK_0010, or CLK_0011 rule violation related to the master clock. For the preceding example scenario, specify the *-source* values for the potential master clock:

create_clock [get ports clk1] -period 2

*Corresponding PrimeTime Error Message*

*Error Message: PTE-023*

*Severity:* Warning

*Description*

The generated clock '%s' has not been expanded. Create or activate its master clock.

**See Also**

create_generated_clock create_clock report_clock CLK_0003 CLK_0009 CLK_0010

---

## CLK-0012

(Warning) Generated clock clock has non-unate sense for master clock master_clock at its master source.

The waveform of a generated clock is dependent on the sense at its master source. When the sense is non-unate, only the positive sense of the master clock is used. If both clock senses need to be considered, create additional generated clocks to avoid incorrect or pessimistic timing analysis. The choice of clock sense can be made explicit by adding the set_clock_sense command at the non-unate location.

If this violation is not addressed, the resulting timing analysis may be incorrect or pessimistic depending on how the generated clock interacts with the master clock. Incorrect results may be obtained if the expected clock sense is negative but only the positive clock sense propagates. Incorrect clock sense may lead to a half cycle shift in clock waveforms. Pessimistic result may arise if there are interactions with the negative sense of the master clock and the positive sense of the generated clock. Pessimistic timing can lead to increased optimization and timing closure efforts.

Feedback

**ONE POSSIBLE SCENARIO**

Consider a scenario involving two generated clocks. The second generated clock *GCLK2* is essentially derived from the first generated clock *GCLK1*. The path from *GCLK1* source pin to G*CLK2* source pin involves a MUX selector pin to MUX output pin path.

Depending on the logic values at the inputs of the MUX, the clock sense may become positive or negative through the MUX. As a result, both positive and negative sense of *GCLK1* is available at the master source pin of *GCLK2* but only the positive sense will be used for the waveform and latency calculation of *GCLK1*. When timing results are analyzed at register ‘ff3/D’, the following cases are possible due to the interaction between *GCLK1* and *GCLK2:*

Launching Clock Sense (GCLK2) Capturing Clock Sense (GCLK1)

Case 1 Positive Negative

Case 2 Positive Positive

Case 3 Negative Negative

Case 4 Negative Positive

Due to the way *GCLK2* is defined, case 3 and case 4 will not be considered. As a result, an incorrect timing result may be obtained if a negative clock sense propagation is expected (case 3). As can be seen from the waveform diagram below, the positive sense waveform of *GCLK2* differs from the negative sense waveform by a half a cycle.

At the same time, a pessimistic timing may arise since case 1 may not be possible due to the mutually exclusive nature of the MUX: Z = A S’ + B S

A B Clock Sense at mux1/Z 0 0 Clock is inactive due to 0 0 1 Positive 1 0 Negative 1 1 Clock is inactive due to 1

create_clock -name CLK -period 2 [get_ports CLK] create_generated_clock -name GCLK1 [get_pins inv2/Z] \ -source [get_ports CLK] -combinational create_generated_clock -name GCLK2 [get_pins ff1/Q] \ -source [get_pins mux1/Z] -divide_by 2 set_propagated_clock [all_clocks]

**What Next**

The Violation Details section provides the names of the generated clock and the non-unate master clock as well as the master source pin. There are multiple ways to analyze the problem. One way is to use the report_clock command to view the clock waveform for *GCLK2* to see if the clock edges are correct. As can be seen from the report below, only the positive clock sense is propagated for *GCLK2*.

Clock Period Waveform Attrs Sources ---------- CLK 2.00 {0 1} p {CLK} GCLK1 2.00 {0 1} p, G {inv2/Z} GCLK2 4.00 {0 2} p, G {ff1/Q}

Feedback

Another way to analyze this problem is to use the clocks_with_sense attribute. This attribute provides, for each pin or port, information on clock and its sense for all its associated clocks. The clocks_with_sense attribute can be queried as follows:

rtl_shell> get_attribute [get_pins mux1/Z] clocks_with_sense { { "GCLK1" "negative" } { "GCLK1""positive" } } rtl_shell> get_attribute [get_pins ff2/CP] clocks_with_sense { { "GCLK2" "positive" } }

The attribute shows that there are both positive and negative senses of the master clock GCKL1 at the master source pin of GCLK2 but only the positive sense of GCLK2 propagates to register ‘ff2’.

**FIX SUGGESTION**

If both the positive clock sense and the negative clock sense need to be considered, create additional generated clocks to fix incorrect clock waveforms or remove unrealistic pessimism.

For the example scenario, this can be done as follows:

create_clock -name CLK -period 2 [get_ports CLK] create_generated_clock -name GCLK1_POS [get_pins mux1/Z] \ -source [get_ports CLK] \ -combinational -add -master_clock CLK create_generated_clock -name GCLK1_NEG [get_pins mux1/Z] \ -source [get_ports CLK] \ -combinational -add -master_clock CLK -invert create_generated_clock -name GCLK2_POS [get_pins ff1/Q] \ -source [get_pins mux1/Z] \ -divide_by 2 -add -master_clock GCLK1_POS create_generated_clock -name GCLK2_NEG [get_pins ff1/Q] \ -source [get_pins mux1/Z] \ -divide_by 2 -add -master_clock GCLK1_NEG set_clock_groups -logically_exclusive -group {GCLK1_POS GCLK2_POS} \ -group {GCLK1_NEG GCLK2_NEG} set_propagated_clock [all_clocks]

Note that the positive sense and negative sense of both generated clocks are declared to be mutually exclusive.

If it is known that only one clock sense is possible, the pessimism can be reduced even further by using the set_clock_sense command. In the example scenario, if the input conditions A =1 and B = 0 can never happen, then the negative clock sense can be eliminated by using

set_clock_sense -positive mux1/Z

Then, the resulting timing analysis will not include any violations at register ‘ff3/D’ that are due to a launching clock associated with the negative sense of GCLK1 and a capturing clock with the negative sense of GCLK2.

**See Also**

create_generated_clock create_clock get_attribute set_clock_sense report_clock

Feedback

## CLK-0013

(Error) Generated clock clock ignores multiple rising edges of master clock master_clock due to a user-defined duty cycle in the generated clock definition.

### Description

A generated clock specified with the -multiply_by option and a user-defined duty cycle, creates its waveform using the first rising edge and the period of the master clock. The master clock has multiple rising edges defined in its period, though the additional edges are ignored by this generated clock.

The generated clock waveform may be incorrect, especially for the falling edge, since this is defined by the create_generated_clock command using the -duty_cycle option. Timing analysis will be performed, but the results are based on an incorrect generated clock because the master clock has multiple rising edges.

### ONE POSSIBLE SCENARIO

Consider the two frequency divider clocks produced by phase lock loop (PLL) circuits. Both of the generated clocks *GCLK1* and *GCLK2* are defined using the -multiply_by 2 option with a master clock that has multiple rise and fall edges within a single period. *GCLK1* has an additional duty cycle constraint of 40%.

create_clock -name CLK -period 10 -waveform {2 4 6 8} \ [get_ports CLK] create_generated_clock -name GCLK1 -source [get_ports CLK] \ -multiply_by 2 -duty_cycle 40 [get_pins pll1/CLK_OUT] create_generated_clock -name GCLK2 -source [get_ports CLK] \ -multiply_by 2 [get_pins pll2/CLK_OUT]

The master clock CLK has multiple rising edges at time units 2 and 6. The generated clock, GCLK1, will only use the first rising edge to create its waveform due to the -duty_cycle option constraint in its specification. GCLK2 does not use the -duty_cycle option but is a -multiply_by 2 clock derived from the master clock CLK and has the same number of rising or falling edges.

### What Next

The “Violation Details” section provides the waveforms details for the generated clock and its master clock.

Master Clock Period: 10.0000 Waveform: {2.0000 4.0000 6.0000 8.0000} Generated Clock Period: 5.0000 Waveform: {1.0000 3.0000}

Check the master clock and the generated clock waveforms for correctness. The waveforms for clocks in the design can be obtained by using the report_clock command. For the example scenario, the command can be issued as follows for the clocks of interest:

rtl_shell> *report_clock [get_clocks {CLK GCLK1 GCLK2}]*

Feedback

Attributes: p - Propagated clock G - Generated clock I - Inactive clock

Clock Period Waveform Attrs Sources ---------- CLK 10.00 {2 4 6 8} {CLK} GCLK1 5.00 {1 3} G {pll1/CLK_OUT} GCLK2 5.00 {1 2 3 4} G {pll2/CLK_OUT}

Generated Master Generated Master Waveform Clock Source Clock Modification ---------- GCLK1 CLK pll1/CLK_OUT CLK div(2) GCLK2 CLK pll2/CLK_OUT CLK div(2)

**FIX SUGGESTION**

If the generated clock waveform needs to be reshaped with the duty cycle option, consider changing the master clock waveform definition to have a single rising edge and a single falling edge. This will affect the shape of the master clock, and it may also affect the clock waveform of other generated clocks that use the same master clock. In the example scenario, this can be done by removing the last two edges from the master clock waveform:

create_clock -name CLK -period 10 -waveform {2 4} [get_ports CLK]

Alternatively, if the generated clock does not need a modified duty cycle, the -duty_cycle option can be removed from the generated clock definition.

**See Also**

create_generated_clock create_clock report_clock

---

## CLK-0014

(Info) Clock clock is created on a pin instead of a port.

**Description**

A clock has been defined on an internal a pin. Any clock defined on a pin will ignore incoming slews and latencies up to that pin which can impact timing accuracy. This rule will not be triggered if the pin has no incoming timing arcs.

The presence of a CLK_0014 rule violation is preceded by a UIC-019 message when reading the SDC constraints.

Ignoring incoming slews or latencies on clock paths may lead to optimistic timing results resulting in slower chip operation or even failure.

**ONE POSSIBLE SCENARIO**

Consider a simple scenario where the clock is defined at pin ‘inv2/Z’ that lies on the path from the clock port ‘CLK’ to the register. Since the clock source is at the inverter output pin ‘inv2/Z’, any clock slews or latencies from the port ‘CLK’ to the pin ‘inv2/Z’ will be ignored

Feedback

during clock latency calculation. This may lead to optimistic timing results on the flip-flop and might cause the chip to fail.

create_clock -name CLK1 -period 2 [get_pins inv2/Z]

**What Next**

The violation message gives the name of the clock and the Violation Details section provides the location of the clock source pin.

**FIX SUGGESTION**

There are two possible approaches to address this violation. One approach is to move the clock definition directly to the input port. In the example scenario, this can be done by:

create_clock -name CLK1 -period 2 [get_ports CLK]

Another approach is to create a generated clock as the output of the inverter ‘inv2’. A generated clock will automatically inherit the latency and slew data from its master clock. In the example scenario, this can be done by first creating a master clock at the port and then driving a generated clock at ‘inv2/Z’ from the master clock:

create_clock -name CLK1 -period 2 [get_ports CLK] create_generated_clock -name GCLK1 [get_pins inv2/Z] \ -combinational -source [get_ports CLK]

**See Also**

create_generated_clock create_clock

---

## CLK-0015

(Warning) Clock clock is created on a hierarchical pin.

**Description**

Defining a clock on a hierarchical pin breaks the net parasitics in a way that reduces the accuracy of latency calculation. Because a hierarchical pin does not have a set physical location, the latency calculation will be arbitrary. Creating clocks on hierarchical pins will result in potentially incorrect latency calculation and, as a consequence, incorrect timing analysis.

**ONE POSSIBLE SCENARIO**

The following clock definition shows that a clock source has been defined on a hierarchical pin.

create_clock -period 10 -name hier_clock b1/CLK

The physical location at which this hierarchical pin exists is arbitrary. In such a scenario CLK_0015 rule violation is given.

Feedback

### What Next

The all_fanin command can be used to find out which driving cell leaf pins correspond to this hierarchical pin. In the above example this command would be:

all_fanin -to b1/CLK -flat -pin_levels 1 {"b1/CLK", "CLK"}

### FIX SUGGESTION

Consider moving this clock definition to the driving leaf pins. The location of the cell leaf pin will now be known and the latency will be computed in an accurate fashion.

### See Also

create_clock all_fanin

---

## CLK-0016

(Error) Generated clock clock has no source latency path from its master clock master_clock.

### Description

There is no valid path from the source of the master clock to the source of the generated clock due to:

no topological path (possibly due to an unresolved reference or a black box)

disabled logic

the path is blocked by another clock source

a set_clock_sense-stop command in the path to the source of the generated clock

The "Violation Details" section provides reasons for the lack of a source latency path.

The generated clock will not be expanded, and registers clocked by the generated clock will not be constrained. Timing analysis will not occur for paths related to the registers, and the timing results will be incomplete for the design.

### ONE POSSIBLE SCENARIO

Consider the following scenario where the path from the master clock to the generated clock is blocked by a set_case_analysis command on the select pin of a MUX. The case analysis specified on the select pin of the MUX allows clock *CLK2* to propagate, but does not allow clock *CLK1* to propagate. The generated clock *GCLK* specifies clock *CLK1* as its master clock.

create_clock -name CLK1 -period 1 [get_ports CLK1] create_clock -name CLK2 -period 2 [get_ports CLK2] create_generated_clock -name GCLK -divide_by 2 \ -source [get_pins ff1/CP] -master_clock CLK1 \ -add [get_pins ff1/Q] set_case_analysis 1 [get_ports SEL]

Feedback

Since the case analysis on the select pin of the MUX propagates the clock *CLK2*, there is no latency path from *CLK1* to *GCLK*, and a *CLK_0016* rule violation will be issued.

**What Next**

Examine the "Violation Details" section for reasons the latency path is missing between the master clock source and the source of the generated clock.

If the reason reported is no topological path from the master clock to the generated clock, check if there are any unresolved references (black boxes) in the design. Black boxes are flagged with LNK-805 warning message of the form:

Warning: Unable to resolve reference to ‘BBOX’ in ‘top’. (LNK-805)

The reason for the missing clock latency path might be because it is blocked, disabled, or the clock is not propagating.

**FIX SUGGESTION**

If black boxes are blocking the clock paths, remove them by reading in additional library or design files. If the clock path is disabled/blocked, remove the disabling condition, or redefine the generated clock so that the path from the master clock to the generated clock is not disabled/blocked. For the example scenario, the set_case_analysis command can be removed:

create_clock -name CLK1 -period 1 [get_ports CLK1] create_clock -name CLK2 -period 2 [get_ports CLK2] create_generated_clock -name GCLK -divide_by 2 \ -source [get_pins ff1/CP] -master_clock CLK1 \ -add [get_pins ff1/Q] #set_case_analysis 1 [get_ports SEL]

Alternatively, the generated clock *GCLK* can be derived from clock *CLK2* instead of *CLK1*:

create_clock -name CLK1 -period 1 [get_ports CLK1] create_clock -name CLK2 -period 2 [get_ports CLK2] create_generated_clock -name GCLK -divide_by 2 \ -source [get_pins ff1/CP] -master_clock CLK2 \ -add [get_pins ff1/Q] set_case_analysis 1 [get_ports SEL]

*Corresponding PrimeTime Error Message*

*Error Message: PTE-075*

*Severity:* Warning

*Description*

Generated clock '%s' has no path to its master clock.

**See Also**

create_generated_clock create_clock report_disable_timing set_clock_sense

Feedback

## CLK-0017

(Error) The master source of generated clock clock is not in the fanin of the generated clock source pin.

### Description

The master source of the generated clock, as given by the -source option to the create_generated_clock command, is not in the fanin cone of the generated clock source pin. The generated clock will still propagate, but there will be issues with the latency of the generated clock. In addition, the sense of the master clock used to create the generated clock waveform may be affected. This rule differs from CLK_0016 in that a topological path from the master clock start point to the generated clock source pin exists and is not blocked or disabled. If the incorrect sense of master clock is used to create the generated clock waveform, the generated clock may have the wrong waveform propagated, and it will have zero clock latency. Zero clock latency may cause pessimistic timing results that may lead to increased optimization and timing closure effort. Incorrect generated clock waveforms will produce inaccurate timing results because the launching or capturing clock edges are wrong.

### ONE POSSIBLE SCENARIO

Consider a scenario involving multiple paths from the master clock source (start) point. The generated clock *GCLK1* is defined with the -source option set to the inverter output pin *inv1/Z*, even though the inverter *inv1* is not in the fanin cone of register *ff1*where the source pin of generated clock *GCK1* is specified. However, a topological path from the master clock start point *CLK* to the generated clock source pin *ff1/Q* does exist and is not blocked or disabled.

create_clock -name CLK -period 2 [get_ports CLK] create_generated_clock -name GCLK1 [get_pins ff1/Q] \ -divide_by 2 -source [get_pins inv1/Z] \ -master_clock CLK -add set_propagated_clock [all_clocks]

There are two problems with this scenario:

The resulting clock waveform is wrong. The generated clock waveform is neither a fully inverted nor a fully non-inverted divide-by-2 waveform of the original master clock waveform. As shown in the diagram below, the rising edge of the master clock is at time 0 and the falling edge at time 1. The rising edge of the generated clock will be at time 1 and the falling edge at time 3, instead of 0 and 2, or 2 and 4.

Zero source clock latency will be assumed for the generated clock so the clock network delay from ff1/D to ff1/Q will not be included in timing check at ff3/D. This will lead to pessimistic setup timing results at ff3/D.

For this scenario, the PrimeTime tool issues a *UITE-461* error message to alert you of this situation. However, not all violations of this rule causes a *UITE-461* error message in PrimeTime. *Figure 2: Master Clock and Generated Clock Waveforms*

Feedback

### What Next

Use the report_clock command to compare the master clock and generated clock waveforms to make sure that the generated clock waveforms are correct. For the example scenario, the command and its output are as follows:

rtl_shell> report_clock Attributes: p - Propagated clock G - Generated clock I - Inactive clock

Clock Period Waveform Attrs Sources ---------- CLK 2.00 {0 1} p {CLK} GCLK1 4.00 {1 3} p, G {ff1/Q}

The report output shows that the generated clock has a rising edge at time 1 and a falling edge at time 3.

In addition, look for other violations related to the generated clock involving edge relations. In the example scenario, a violation of CLK_0020 will also be reported:

Generated clock GCLK1 has edge relationships with its master clock CLK that cannot be satisfied. Only paths with 'rise_triggered' sense exist from the master clock to source pin ff1/Q. A 'fall_triggered' sense is expected.

### FIX SUGGESTION ONE

Consider changing the *–source* pin or port of the generated clock so that it lies in the fanin cone of the generated clock source pin. In the example scenario, the -source option can be changed to *CLK* from *inv1/Z*.

create_clock -name CLK -period 2 [get_ports CLK] create_generated_clock -name GCLK1 [get_pins ff1/Q] \ -divide_by 2 -source [get_ports CLK] \ -master_clock CLK -add set_propagated_clock [all_clocks]

The generated clock waveform now has a rising edge at 0 and a falling edge at 2. See *GCLK1 (FIX1)* in the timing diagram.

### FIX SUGGESTION TWO

Alternatively, the generated clock can be inverted without changing the -source option. This will also remove the current and CLK_0020 violations in the example scenario.

create_clock -name CLK -period 2 [get_ports CLK] create_generated_clock -name GCLK1 [get_pins ff1/Q] \ -divide_by 2 -source [get_pins inv1/Z] \ -master_clock CLK -add -invert set_propagated_clock [all_clocks]

The generated clock waveform will have a rising edge at 2 and a falling edge at 4. See *GCLK1 (FIX2)* in the timing diagram.

### See Also

create_generated_clock create_clock report_clock

Feedback

## CLK-0018

(Warning) Clock clock has non-unate logic in the clock network.

### Description

There is non-unate logic in the clock network. The locations where the clock network becomes non-unate are given in the details of this warning. Both the positive and negative sense of the clock will be propagated. If both clock senses are possible, but not simultaneously, consider creating generated clocks that are mutually exclusive to remove unrealistic pessimism. If only one clock sense can propagate, consider adding a set_clock_sense constraint to remove the pessimism. If this violation is not addressed, the resulting timing analysis may be pessimistic because both the positive sense and negative sense of the clock will be assumed to propagate simultaneously. In practice, only one clock sense may propagate, or both clock senses may propagate, but not simultaneously. Pessimistic timing results can lead to increased effort in optimization and timing closure.

### ONE POSSIBLE SCENARIO

Consider a scenario where a MUX has both a clock signal and an inverted clock signal as inputs. If the mode of operation of the circuit is not specified by setting a case analysis value on the select pin of the MUX, both the positive and negative sense of the clock CLK will be propagated.

create_clock -name CLK -period 2 [get_ports CLK]

Timing analysis assumes that both positive and negative clock senses can propagate simultaneously as shown in the following waveform diagram.

Timing violations will be reported at register ‘ff2/D’ that are due to a launching clock associated with the positive sense of ‘CLK’ and a capturing clock with the negative sense of ‘CLK’ and vice versa.

### What Next

The “Violation Details” section provides the name of the clock and the non-unate pins. There are several ways to analyze the root cause of the non-unateness.

Another way to analyze the root cause of non-unate clocks is to use the clocks_with_sense attribute. This attribute provides, for each pin or port, information on a clock and its sense for all clocks associated with the object. The clock_with_sense attribute can be queried as follows:

rtl_shell> get_attribute [get_pins mux1/A] clocks_with_sense \ get_attribute [get_pins mux1/A] clocks_with_sense \ { { "CLK" "positive" } }

rtl_shell> get_attribute [get_pins inv1/Z] clocks_with_sense \ get_attribute [get_pins inv1/Z] clocks_with_sense \ { { "CLK" "negative" } } rtl_shell> get_attribute [get_pins mux1/

Feedback

Z] clocks_with_sense \ get_attribute [get_pins mux1/Z] clocks_with_sense \ { { "GCLK" "negative" } { "GCLK" "positive" } }

**FIX SUGGESTION**

If the positive clock sense and the negative clock sense cannot propagate simultaneously, create two mutually exclusive generated clocks at the non-unate pins to remove the unrealistic pessimism. If only one clock sense can propagate, consider adding a set_clock_sense constraint to remove the pessimism, or a set_case_analysis constraint to propagate the desired sense of the clock for the mode of operation.

For the example scenario, case analysis can be specified on the MUX select pin to propagate the positive edge clock:

set_case_analysis 0 [get_pins mux1/S]

Now, only one sense of clock ‘CLK’ will arrive at the register clock pins ‘ff1/CP’ and ‘ff2/CP’.

*Corresponding PrimeTime Error Message*

*Error Message:PTE-070*

*Severity:* Warning

*Description*

A non-unate path in clock network detected. Propagating both inverting and noninverting senses of clock '%s' from pin '%s'.

**See Also**

create_generated_clock create_clock get_attribute set_clock_sense set_case_analysis

---

## CLK-0019

(Warning) Design has both propagated and ideal non-virtual clocks.

**Description**

The design has some propagated clocks and some ideal clocks. For pre-layout analysis, ideal clocks are usually appropriate. For post-layout analysis, propagated clocks should be used. It is not common for some clocks to be pre-layout and some to be post-layout. Note that virtual clocks are ignored for this check and so are the clocks defined at output ports. Ideal clocks in post-layout analysis will lead to incorrect timing results because only specified clock latencies are considered. The clock latencies given with set_clock_latency may be larger or smaller than the actual latencies. Incorrect timing results may lead to slower chip operation or even failure. Propagated clocks in pre-layout analysis may add

Feedback

unintended pessimism or optimism to timing results because the clock network will not have been implemented with proper drive strength and skew.

**ONE POSSIBLE SCENARIO**

Consider a post-layout design where all the clocks need to be propagated. In the constraint file, there is a set_propagated_clock command that is applied to all the clocks.

create_clock -name CLK1 -period 10 [get_ports CLK1] create_clock -name CLK2 -period 20 [get_ports CLK2] set_propagated_clock [all_clocks] create_clock -name CLK3 -period 30 [get_ports CLK3]

However, the clock definition for ‘CLK3’ appears after the set_propagated_clock command, so ‘CLK3’ becomes ideal instead of propagated.

**What Next**

The "Violation Details" section provides a list of ideal clocks and a list of propagated clocks. Consider making either one of the clock lists empty by changing clocks from ideal to propagated or vice versa. Note that virtual clocks and clocks defined at output ports are not included in these lists.

**FIX SUGGESTION**

For post-layout analysis, declare the ideal clocks shown in the Violation Details section as propagated clocks. For the example scenario, this can be done by moving the set_propagated_clock command after the last clock definition as in:

create_clock -name CLK1 -period 10 [get_ports CLK1] create_clock -name CLK2 -period 20 [get_ports CLK2] create_clock -name CLK3 -period 30 [get_ports CLK3] set_propagated_clock [all_clocks]

For pre-layout analysis, consider changing the propagated clocks to ideal clocks to remove unintended pessimism. This can be done by removing set_propagated_clock commands.

**See Also**

create_generated_clock create_clock set_propagated_clock report_clock

---

## CLK-0020

(Error) Generated clock clock has edge relationships with its master clock master_clock that cannot be satisfied. Only paths with path_sensesense exist from the master clock to source pin source_pin. An expected_sensesense is expected.

Feedback

### Description

A generated clock definition implies certain relationships between the master clock source and the generated clock source. For example, a simple -divide_by 2 generated clock implies that a rising master clock edge can cause both a rising and falling generated clock edge (see the following figure). *Figure 1: Relation Between a Clock and its divide_by_2 Generated Clock*

A generated clock definition specifies how the master clock waveform, as it exists at the -source pin (from the create_generated_clock command), is altered.

If the *-source* pin given is one of the master clock sources, then the waveform modifier (-divide_by, -multiply_by, -edges, -combinational, -invert) specifies how the master clock waveform is modified to become the generated clock waveform.

If the *-source* pin is a clock network pin, the master clock waveform may be inverted at that pin. In the inverted case, that generated clock modifier specifies how the inverted master clock waveform is altered to become the generated clock waveform.

The CLK_0020 rule violation looks at the source latency paths of the generated clock and determines whether the specified relationship to its master exists. If the relationship cannot be proven, a CLK_0020 rule violation is triggered for that particular generated clock. In addition, a zero propagated latency is used for the edges of the generated clock where no source latency path is found.

This rule violation correlates to the PrimeTime message *UITE-461*.

The generated clock waveform might be incorrect, the needed source latency paths might be disabled for some reason, and the propagated latency might not be able to be computed. These problems can lead to incorrect or unexpected timing results.

### ONE POSSIBLE SCENARIO

Consider a scenario where an inverted clock reaches the input of a multiplexer. A generated clock (with a -combinational modifier) is added to the output of the MUX. The clock port (CLK) is defined as the -source pin of that generated clock, as shown here:

create_clock -name clk1 -period 2 [get_ports CLK] create_generated_clock -name g1 mux/Z -source [get_ports CLK] \ -combinational -master_clock clk1 -add

These constraints specify that the generated clock, g1, has the same waveform as its master clock, clk1. The tool expects that there is a positive unate path from the port CLK to the output of the mux. This means that a rising edge on CLK should produce a rising edge on g1 and a falling edge on CLK should produce a falling edge on g1. However,

Feedback

when constraint consistency does the clock analysis, it concludes that only negative unate paths exist. This will trigger a CLK_0020 rule violation as follows:

CLK_0020: Generated clock g1 has edge relationships with its master clock that cannot be satisfied. Only paths with negative sense exist from the master clock to source pin mux/Z. A positive sense is expected.

**What Next**

The relationship between the generated clock and its master clock can be seen using the report_clock command. In the preceding example, the report shows that the waveform of *g1* is the same as the waveform of the master clock, *clk1*.

rtl_shell> report_clock {clk1 g1}

****************************************** Report : clock Design : CLK_0020 Scenario: default Version: ... Date : ... ******************************************

Attributes: p - Propagated clock G - Generated clock I - Inactive clock

Clock Period Waveform Attrs Sources
---------- clk1 2.00 {0 1} {CLK} g1 2.00 {0 1} G {mux/Z}

Generated Master Generated Master Waveform Clock Source Source Clock Modification
---------- g1 CLK mux/Z clk1 div(1)

**FIX SUGGESTION**

This violation usually indicates some errors specifying the generated clocks. You should alter the generated clock definition so the waveform matches the existing source latency paths.

In the preceding example, this can be done by either:

Adding the -invert option to the create_generated_clock command

create_generated_clock -name g1 mux/Z -source [get_ports CLK] \ -invert -combinational -master_clock clk1 -add

Changing the -source to a pin after the master clock is inverted

create_generated_clock -name g1 mux/Z -source [get_pins mux/A] \ -combinational -master_clock clk1 -add

The first corrected create_generated_clock command will lead to the following waveform as seen by the report_clock command. Note that the g1 waveform is inverted. The other generated clock statement will produce the same waveform.

rtl_shell> report_clock {clk1 g1}

Feedback

********** Report : clock Design : CLK_0020 Scenario: default Version: ... Date : ... **********

Attributes: p - Propagated clock G - Generated clock I - Inactive clock

Clock Period Waveform Attrs Sources ---------- clk1 2.00 {0 1} {CLK} g1 2.00 {1 2} G {mux/Z}

Generated Master Generated Master Waveform Clock Source Source Clock Modification ---------- g1 CLK mux/Z clk1 div(1), combinational, inv

Note: In the case of complex clock generation circuits, clock modifiers such as *divide_by*, *invert*, or *multiply_by* might not be enough to correctly specify your generated clocks. In these cases, the *-edge* option should be used.

*Corresponding PrimeTime Error Message*

*Error Message:UITE-461*

*Severity:* Error

*Description*

Generated clock '%s' '%s' is not satisfiable%s

**See Also**

create_generated_clock report_clock

---

## CLK-0021

(Info) Clock clock is not used in this scenario.

**Description**

A clock defined in the constraints is not used in any scenario. A clock is classified as unused if all the following conditions are met:

no register clock pins receive the clock’s signal

no input or output delay is defined relative to the clock

no generated clock has the clock specified as its master clock

the clock does not propagate to an output/inout port

Unused clocks might be a symptom of an incorrectly or incompletely constrained design. This will also have a memory or runtime adverse effect in a static timing analysis tool since the clock still has to be accounted for.

Feedback

### ONE POSSIBLE SCENARIO

One possible scenario demonstrating this violation involves black boxes or unresolved references associated with a clock path. Such black boxes prevent the clock signal from propagating to register clock pins or output/inout ports and are flagged with LNK-805 warning message of the form:

Warning: Unable to resolve reference to 'BBOX' in 'top'. (LNK-805)

### FIX SUGGESTION

Check if there are any black boxes that are blocking the clock paths. It is a good idea to remove all the black boxes by reading in additional library or design files. A cell is considered a black box if a definition for the cell is not found in any of the Verilog or library technology files.

The clock might be blocked from all registers with case settings or other constraints. If it is correct in this scenario for the clock not to be used, the clocks can be safely removed as they do not affect the function of the design.

### See Also

create_generated_clock create_clock set_input_delay set_output_delay

---

## CLK-0023

(Warning) There are count clocks on source source with the same period and waveform.

### Description

More than one clock is defined on the same source, with identical periods and waveforms. This can result in increased runtime and memory usage. Consider combining the clocks unless they are needed for other constraint reasons such as clock latency pessimism removal. By default, this violation will be issued for generated clocks only if they have identical periods, waveforms and master clocks. If the *exclude_different_primary_masters* rule property is set to *false*, then generated clocks with identical periods and waveforms will be reported to have this violation even if their master clocks are different.

If this violation is not addressed, the timing analysis runtime and memory usage may increase without any reduction in clock latency pessimism. Failure to remove clock latency pessimism may lead to unrealistic setup or hold timing violations.

### ONE POSSIBLE SCENARIO A

Consider a scenario where there are two generated clocks with the same generated clock source but with different master clock source pins. The master clock source pin for ‘gclkA’ is specified at ‘mux1/A’, and the master source pin for ‘gclkB’ at ‘mux1/B’. However, for the purpose of clock latency calculation, the clock latency path begins from the point where

Feedback

the master clock originates, not where the master source pin is specified in the generated clock. Thus, generated clocks ‘gclkA’ and ‘gclkB’ have

identical clock periods

identical clock waveforms

identical master clocks

identical clock latency paths that start from the port ‘CLK’ and end at pin ‘mux2/A’

create_clock -name sysclk -period 4 [get_ports CLK] create_generated_clock -name gclkA -divide_by 1 \ -source [get_pins mux1/A] -master sysclk \ [get_pins mux2/A] -add create_generated_clock -name gclkB -divide_by 1 \ -source [get_pins mux1/B] -master sysclk \ [get_pins mux2/A] -add

**ONE POSSIBLE SCENARIO B**

Another possible scenario involves multiple clock MUXs controlled by the same select line. Since some clock latency paths are mutually exclusive, the pessimism in latency calculation can be removed by declaring some clocks as logically or physically exclusive. Consider two clock MUXs controlled by one common select line as follows:

create_clock -name CLK1 -period 1 [get_ports CLK1] create_clock -name CLK2 -period 2 [get_ports CLK2] create_generated_clock [get_pins mux1/Z] -name CLK1_SHORT \ -source [get_pins mux1/A] \ -divide_by 1 -add -master_clock [get_clocks CLK1] create_generated_clock [get_pins mux1/Z] -name CLK1_LONG \ -source [get_pins inv11/A] -divide_by 1 -add -master_clock [get_clocks CLK1] create_generated_clock [get_pins mux2/Z] -name CLK2_SHORT \ -source [get_pins mux2/A] -divide_by 1 -add -master_clock [get_clocks CLK2] create_generated_clock [get_pins mux2/Z] -name CLK2_LONG \ -source [get_pins inv21/A] -divide_by 1 -add -master_clock [get_clocks CLK2] set_propagated_clock [all_clocks] set_clock_groups -logically_exclusive \ -group {CLK1_SHORT CLK2_SHORT} \ -group {CLK1_LONG CLK2_LONG}

Two generated clocks are created at the output of each clock MUX. The long and short versions of each clock are declared as mutually exclusive by using set_clock_groups. However, both the long and the short versions of the generated clocks have identical periods, waveforms, master clocks and latencies because the paths from the master clock source pin to the generated clock source pin are identical. As a result CLK_0023 violation is issued and no pessimism in clock latency is removed despite an increase in runtime and memory usage.

**What Next**

The "Violation Detail" section provides the name of the clock that is being used as data as well as the clock source locations for the clock.

**FIX SUGGESTION**

Feedback

It is best to remove unnecessary generated clocks unless they are needed for purposes like removing pessimism in clock latency calculation. For example scenario A, the extra generated clock ‘gclkB’ can be removed without any impact on timing results.

For example scenario B, the duplicate generated clocks need to be changed to unique generated clocks to remove clock latency pessimism. This can be done by moving the generated clock source pins to the input pins of clock MUXs. This will create two unique generated clocks from each master clock such that the short (long) latency path traverses the path from the clock port to pin ‘A’ (‘B’) of the clock MUX. Now that correct latencies are associated with each generated clock, there will be fewer setup or hold violations to fix. The generated clock modification can be done as follows:

create_clock -name CLK1 -period 1 [get_ports CLK1] create_clock -name CLK2 -period 2 [get_ports CLK2] create_generated_clock [get_pins mux1/A] -name CLK1_SHORT \ -source [get_ports CLK1] \ -divide_by 1 -add -master_clock [get_clocks CLK1] create_generated_clock [get_pins mux1/B] -name CLK1_LONG \ -source [get_ports CLK1] -divide_by 1 \ -add -master_clock [get_clocks CLK1] create_generated_clock [get_pins mux2/A] -name CLK2_SHORT \ -source [get_ports CLK2] \ -divide_by 1 -add -master_clock [get_clocks CLK2] create_generated_clock [get_pins mux2/B] -name CLK2_LONG \ -source [get_ports CLK2] -divide_by 1 \ -add -master_clock [get_clocks CLK2] set_propagated_clock [all_clocks] set_clock_groups -logically_exclusive \ -group {CLK1_SHORT CLK2_SHORT} \ -group {CLK1_LONG CLK2_LONG}

Rule Property: exclude_different_primary_masters

By default, the exclude_different_primary_masters rule property is set to true. The rule property can be set to false as follows:

set_rule_property exclude_different_primary_masters false CLK_0023

If the rule property is set to *false*, more CLK_0023 violations will be reported because generated clocks with identical periods and waveforms will be reported to have this violation even if their master clocks are different.

**See Also**

create_generated_clock create_clock set_clock_groups set_propagated_clock

---

## CLK-0024

(Warning) Register Clock pin pin has count clocks.

*Status:* Disabled by default

**Description**

This rule is disabled by default because it can give a large number of violations that the user may not be concerned about. It can be used to investigate further a PRF_0001 rule

Feedback

violation. The number of clocks allowed at a register clock pin without triggering a violation is set with the *max_clocks_per_register* property on this rule.

The PRF_0001 rule gives at most one violation. It gives the number of registers that exceed the *max_clocks_per_register* property of the PRF_0001 rule. The performance of many tools is negatively impacted by having many clocks propagating to a single clock pin.

There is a risk that the timing analysis will run slower and take more memory for a set of constraints with many clocks propagating to the same registers. There is also a risk that the design is not constrained for the functional modes as expected. The large number of clocks could also be due to an error in the case analysis settings for clock definitions in the design.

**ONE POSSIBLE SCENARIO**

You have a set of constraints that is meant to represent one functional mode for a design. It is expected that no register has more than one clock for this set of constraints. If any register has more than one clock, then for this example design there are other constraints missing that select the correct clock for this mode. To check the number of clocks reaching clock pin registers in the design, you would include the following commands in your run script:

enable_rule CLK_0024 set_rule_property max_clocks_per_register 1 CLK_0024

During check_constraints, a CLK_0024 violation will be issued if any clock pin has more than one clock.

**What Next**

The “Violation Details” section provides the names of the clocks at the register that has more than the maximum number of clocks.

**FIX SUGGESTION**

If you are using this rule to check that only one clock reaches each register, then you need to add additional case analysis values such that only one clock propagates through the design’s various clock selector logic. For the example scenario, the violation can be resolved by deciding to propagate only the positive sense clock CLK1. This can be achieved with the following command:

set_case_analysis 0 [get_ports SEL]

If you are using this rule to determine whether merging several functional modes has created a large number of registers with more than max_clocks_per_register, you might want to turn off this rule and just use the PRF_0001 rule.

*Rule Property: max_clocks_per_register*

Feedback

The default of the rule property, max_clocks_per_register, can be queried using the following command:

get_rule_property max_clocks_per_register CLK_0024

It can be set to a different value using the set_rule_property command:

set_rule_property max_clocks_per_register 10 CLK_0024

The number of violations reported by this rule is dependent on the setting of this property. If it is set to a large limit, then pins below that limit will not be reported as violations.

**See Also**

PRF_0001 get_rule_property set_rule_property check_constraints disable_rule enable_rule

---

## CLK-0026

(Warning) Clock clock is used as data. One or more sources of the clock fans out to a register data pin or to a constrained primary output or inout port.

**Description**

The clock is used as data in the design. This can cause multiple problems during optimization, clock tree synthesis, and testing. Clock signals have a high fanout. When they are used as data this can cause large transition time violations in data paths which cannot be fixed by optimization tools. Also, if the data paths are not identified the clock tree synthesis runtime and quality can be adversely impacted. When the clock signals drive the data pin of registers, a race condition is created between the data pin and the clock pin of the registers. The race conditions may lead to the creation of poor test patterns or low test coverage. Clocks used as data may cause large transition time violations which cannot be fixed during optimization; this can increase runtime or degrade quality during clock tree synthesis. It can also result in poor test pattern generation or low test coverage for design testing.

**ONE POSSIBLE SCENARIO**

Consider a scenario where a clock signal drives register data pins and register clock pins. At register ‘ff2’ the clock signal ‘CLK’ appears as data at the data pin and as clock at the clock pin. *Figure 1: Clock Signal Driving Register Data Pins and Register Clock Pins*

create_clock -name CLK -period 1 [get_ports CLK] set_output_delay -clock CLK 0.5 [all_outputs]

**What Next**

The "Violation Details" section provides the name of the clock that is being used as data and the clock source locations.

Feedback

### FIX SUGGESTION

Check if the clock needs to be used as both data and clock. If possible, modify the design so the clock is not used as data. If the clock must be used as both clock and data, consider using clock isolation logic so the data paths and clock paths are separated. A simple clock isolation solution using a two-phase shifted clock divider is shown in the following figure:

### See Also

create_clock

---

## CLK-0027

(Error) At pin, there is a conflicting set_clock_sense for clock.

### Description

There is a set_clock_sense constraint at the given pin and the requested clock sense does not exist at the pin for the given clock. This may be due to a conflicting set_clock_sense constraint on a previous pin, or it may be because the only clock paths to that pin are not of the sense specified. The set_clock_sense constraint for the pin will be ignored because the requested clock sense does not exist. The timing analysis performed on the design will not use the specified clock sense. This could affect the timing results for the design.

### ONE POSSIBLE SCENARIO

Consider a scenario where a clock signal propagates from a select pin to an output pin of a MUX. The clock sense may become positive or negative through the MUX, based on the logic values at the inputs of the MUX.

Port SEL value Clock Sense at mux1/Z 0 Positive 1 Negative

create_clock -name CLK -period 2 [get_ports CLK] set_case_analysis 0 [get_ports SEL] set_clock_sense -negative mux1/Z

### What Next

The “Violation Details” section provides the name of the clock and the location of the set_clock_sense setting. It will also show the requested sense and the available sense of the clock.

### FIX SUGGESTION

Change the constraints so that the conflicting clock sense is resolved.

For the example scenario, it is not necessary to select the clock sense if the case value is used. The set_clock_sense command can be removed from the golden constraint file to resolve the violation.

Feedback

### See Also

set_clock_sense

---

## CLK-0028

(Warning) The master clock for generated clock clock is ambiguous. There are multiple clocks on -source pin master_source.

### Description

The generated clock has been created without a -master_clock or -add options. The -source pin has more than one potential master clock leading to an ambiguous definition. Constraint consistency will, by default, pick the master clock with the least numbers of generated clocks as the primary master. If the potential master clocks tie for the number of levels, then the first created master clock will be chosen. Because the clock definition is ambiguous, there is a non negligible level of unknown associated with that clock definition. Not all tools in the flow may pick the same master clock and, even if they do, the constraints may be altered and the master clocks order changed in the constraint file changing the master clock for this generated clock. The waveform computed for this generated clock is not dependable and can potentially lead to incorrect timing analysis.

### ONE POSSIBLE SCENARIO

In the example below, a generated clock *CLK_gate* has been added at the output of a clock gating cell. The generated clock does not have a *-master_clock* specified, and the *-source* port has two clocks defined on it. This leads to an ambiguous clock definition. Constraint consistency will select the first clock created at the port, *CLK_SCAN* as the master clock of generated clock *CLK_gate*. This example also has an additional generated clock (properly specified) at the same location with the -master_clock and -add options.

create_clock -name CLK_SCAN -period 5 [get_port CLK] -add create_clock -name CLK_NORM -period 2.3 [get_port CLK] -add create_generated_clock -name CLK_gate -combinational \ -source [get_port CLK] ck_gate/Z create_generated_clock -name CLK_gate2 -combinational \ -source [get_port CLK] ck_gate/Z \ -master_clock CLK_NORM -add

### What Next

You can use the report_clock command to see the master clock and waveform for the generated clock.

For this example scenario, the command would be:

rtl_shell> report_clock

Attributes: p - Propagated clock G - Generated clock I - Inactive clock

Feedback

Clock Period Waveform Attrs Sources
---------- CLK_NORM 2.30 {0 1.15} p {CLK} CLK_SCAN 5.00 {0 2.5} p {CLK} CLK_gate 5.00 {0 2.5} p, G {ck_gate/Z} CLK_gate2 2.30 {0 1.15} p, G {ck_gate/Z}

Generated Master Generated Master Waveform Clock Source Clock Modification
---------- CLK_gate CLK ck_gate/Z CLK_SCAN div(1), combinational CLK_gate2 CLK ck_gate/Z CLK_NORM div(1), combinational

**FIX SUGGESTION**

Remove the ambiguity when creating the generated clock by using the -master clock and -add options to the create_generated_clock. This will allow a correct and dependable computation of the clock waveforms.

For the preceding example scenario:

create_generated_clock -name CLK_gate -combinational \ -source [get_port CLK] ck_gate/Z \ -master_clock CLK_SCAN -add

*Corresponding PrimeTime Error Message*

*Error Message:PTE-004*

*Severity:* Warning

*Description*

The master clock for generated clock clock is ambiguous. There are multiple clocks on the -source pin master_source.

**See Also**

create_generated_clock report_clock

---

## CLK-0029

(Warning) The last edge shift value for generated clock clock is not equal to the first edge shift.

**Description**

The generated clock is specified with the -edges and -edge_shift options. The last edge specifies when the generated waveform repeats. Normally, the shift value for this last edge should be the same as the first shift value, resulting in a generated clock period equal to the difference between the last edge and the first edge. Specifying a different shift for the last edge may result in an edge relationship between the master clock and

Feedback

generated clock that changes from period to period. Clock generation circuitry is typically not designed to generate clocks of this type.

**ONE POSSIBLE SCENARIO**

Consider a scenario where a generated clock is created using edges and the timing of the edges is modified with the -edge_shift option. The last edge shift value is *1.1* time units and the first edge shift value is *0.2* time units in the generated clock definition:

create_clock -period 10 [get_ports CLK] create_generated_clock -source CLK ff1/Q -name genc \ -edges {1 1 3} -edge_shift {0.2 0.6 1.1}

Because the first and last edge shift values are different, the tool issues a CLK_0029 rule violation.

**What Next**

The violation message gives you the name of the generated clock that has different first and last edge shift values.

Examine the edge relationships between the clocks to verify whether or not the edge relationship between the master clock and generated clock remains the same between periods. For the example scenario, the first two periods of the master clock and generated clock are shown as follows:

For the first period, the time difference between the rising edge of the master clock and generated clock is 0.2 time units because (0.2 – 0 = 0.2). For the second period, the time difference between the rising edge of the master clock and generated clock is 1.1 time units because (11.1 – 10 = 1.1). The difference in the edge shift values causes the relationship between the rising and falling edges of the master clock and generated clock to change with each period of the clock.

**FIX SUGGESTION**

If the edge relationship between the master clock and generated clock remains the same between periods, the violation can be waived. If the edge relationship varies, consider modifying the *-edge_shift* specification so that the first and last edge shift values match.

**See Also**

create_generated_clock create_clock

---

## CLK-0031

(Info) The waveform for clock clock has more than two edges.

Feedback

### Description

Most waveforms have a single pair of rise and fall edges. Some tools do not support waveforms with four or more edges. There may be design tools that do not support clock definitions with more than two edges in their specification. All Galaxy tools support such a clock definition.

### ONE POSSIBLE SCENARIO

Consider a clock definition with more than one pair of rise and fall edges.

create_clock -period 10 CLK -waveform {0 1 3 4}

All Galaxy tools support more than two edges in the waveform specification. When a clock has more than two edges in its definition, the tool issues a CLK_0031 rule violation.

### What Next

Check to see if clock specifications of this type are supported by all of the design tools in your flow. If they are, this violation can be waived.

### See Also

create_clock

---

## CLK-0032

(Error) Non-combinational generated clock clock has a source identical to its master source object.

### Description

The reported generated clock has the same source as its master source. Since the generated clock is not combinational, this is an incorrect statement. The constraint describes a condition that is not physically possible in a real design. This will most likely result in incorrect timing analysis since the clock relationship is erroneous.

### ONE POSSIBLE SCENARIO

Consider a scenario where a generated clock is declared as divide-by-2 from its master source, also acting as its source pin:

create_generated_clock -name GCLK -source ff1/CP -divide_by 2 ff1/CP

### What Next

The “Violation Details” reports the pin acting as both the source and master source pin for the generated clock.

### FIX SUGGESTION

Feedback

There are two different approaches to solving this issue depending where the error was made in the generated clock definition:

You can declare the generated clock to have a combinational relationship with the master clock

You can move the master source pin such that the latency path satisfies the master/generated clock relationship

For example, in the preceding scenario, changing the generated clock declaration to the follow would solve the issue:

create_generated_clock -name GCLK -source ff1/CP -combinational

**See Also**

create_generated_clock

---

## CLK-0033

(Error) Clock clock has ideal latency during post-layout analysis.

*Status:* Disabled by default

**Description**

All clocks should be propagated during post-layout analysis. Virtual clocks and clocks defined on output ports are excluded from this check. Ideal clocks in post-layout analysis may lead to inaccurate timing results because clock latencies are not considered. This can lead to missed timing violations in the design.

**ONE POSSIBLE SCENARIO**

Consider a post-layout design where it is desired that all clocks should be propagated. In the constraint file, there is a set_propagated_clock command that is applied to all the clocks.

create_clock -name CLK1 -period 1 [get_ports CLK1] set_propagated_clock [all_clocks]
create_clock -name CLK2 -period 2 [get_ports CLK2]

However, the clock definition for *CLK2* appears after the set_propagated_clock command, so *CLK2* does not get propagated. As a result, a *CLK_0033* violation will be issued by the tool for *CLK2*.

**What Next**

The violation message will include the name of the clock that is not propagated. Virtual clocks and clocks defined on output ports are not included in this check.

Feedback

You can also obtain a report of the clocks in your design and whether they are propagated with the report_clock command. For the example scenario, the report can be obtained as follows:

rtl_shell> report_clock

****************************************** Report : clock Design : ... Scenario: ... Version: ... Date : ... ******************************************

Attributes: p - Propagated clock G - Generated clock I - Inactive clock

Clock Period Waveform Attrs Sources ---------- CLK1 1.00 {0 0.5} p {CLK1} CLK2 2.00 {0 1} {CLK2}

The report shows that *CLK1* is a propagated clock, while *CLK2* is an ideal clock.

**FIX SUGGESTION**

For post-layout analysis, determine if the specified clock should be made propagated and include it in the collection of clocks to be propagated. For the example scenario, this can be done by moving the set_propagated_clock command in the constraint file so that it occurs after the last clock definition:

create_clock -name CLK1 -period 1 [get_ports CLK1] create_clock -name CLK2 -period 2 [get_ports CLK2] set_propagated_clock [all_clocks]

**See Also**

set_propagated_clock create_clock create_generated_clock report_clock

---

## CLK-0034

(Error) Clock clock has propagated latency during pre-layout analysis.

*Status:* Disabled by default

**Description**

All clocks should be ideal during pre-layout analysis. Virtual clocks and clocks defined on output ports are excluded from this check. Propagated clocks in pre-layout analysis will have their latency calculated with cell delays only, since no detailed parasitics are present in the design for interconnect delay. This may lead to inaccurate timing results.

**ONE POSSIBLE SCENARIO**

Consider a pre-layout design with two clocks, CLK1 and CLK2. In the constraint file, there is a set_propagated_clock command that is applied to clock *CLK2*.

create_clock -name CLK1 -period 1 [get_ports CLK1] create_clock -name CLK2 -period 2 [get_ports CLK2] set_propagated_clock [get_clocks CLK2]

Feedback

As a result, *CLK1* will be ideal, but *CLK2* is propagated. When this rule is enabled, a CLK_0034 violation will be issued for *CLK2* because it is propagated.

**What Next**

The violation message will include the name of the clock that is propagated. Virtual clocks and clocks defined on output ports are not included in this check.

You can also obtain a report of the clocks in your design and whether they are propagated with the report_clock command. For the example scenario, the report can be obtained as follows:

rtl_shell> report_clock

****************************************** Report : clock Design : ... Scenario: ... Version: ... Date : ... ******************************************

Attributes: p - Propagated clock G - Generated clock I - Inactive clock

Clock Period Waveform Attrs Sources ---------- CLK1 1.00 {0 0.5} {CLK1} CLK2 2.00 {0 1} p {CLK2}

The report shows that *CLK2* is a propagated clock, while *CLK1* is an ideal clock.

**FIX SUGGESTION**

For pre-layout analysis, consider removing the propagated clock declaration and replacing it with the set_clock_latency command. This command can be removed later when post-layout data is available for the design. For the example scenario, the set_propagated_clock command can be removed and replaced with a set_clock_latency command.

create_clock -name CLK1 -period 1 [get_ports CLK1] create_clock -name CLK2 -period 2 [get_ports CLK2] set_clock_latency 2.36[get_clocks CLK2]

**See Also**

set_propagated_clock remove_propagated_clock set_clock_latency create_clock create_generated_clock report_clock

---

## CLK-0035

(Warning) No clock-gating check inferred for the instance: cell clock pin: clk_pin enable pin: enable_pin lib cell: lib_cell clk name: clk_name.

**Description**

The enable pin specified converges with a clock network, but no clock gating check can be inferred. The library cell logic does not create a *high* or *low* clock gating check, and there

Feedback

is no set_clock_gating_check-high or -low or set_disable_clock_gating_check constraint at this location.

Constraint consistency can infer the clock gating check level (high or low) if the gate logic can be determined to be equivalent to logic AND or logic OR. Otherwise, the you need to specify the check level to enable clock gating checks on the cell. The clock gating checks will not be created because the clock gating check level cannot be inferred. Any timing violations related to the clock gating signal cannot be detected and will not be reported.

**ONE POSSIBLE SCENARIO**

Consider a scenario shown in the following figure, where a clock is created on the clock port *M1*, and a clock gating timing check cannot be inferred on the *MUX21H* cell since it does not have the logic AND or logic OR:

create_clock -period 5 [get_ports M1] -name m1

**What Next**

The violation message provides the name of the cell where a clock gating check cannot be inferred, along with the corresponding clock pin, enable pin, and library cell.

**FIX SUGGESTION**

To enable the clock gating check on this cell, use the set_clock_gating_check command with the -high or -low option to specify the check level information. If clock gating checks are not required, you can remove this violation by disabling the clock gating checks for this cell with the set_disable_clock_gating_check command.

**See Also**

set_clock_gating_check set_disable_clock_gating_check report_clock_gating_check

---

## CLK-0036

(Error) Generated clock clock (file:gclk_flr) has incorrect waveform inversion with its master clock master_clock(file: mclk_flr) that cannot be satisfied.

**Description**

There are edge triggered cells in the paths between the master clock and the generated clock that might involve waveform inversion. The specified clock waveform cannot be computed under the current conditions. It is likely that your generated clock definition has either:

a missing or an incorrect -invert option

a missing or an incorrect inverting -edges option

Feedback

This violation will not be issued for generated clocks that already have CLK_0020 or CLK_0008 violations.

If the generated clock definition incorrectly specifies waveform inversion, all timing paths affected by the generated clock can receive the wrong waveform. This can affect timing results of those paths and can negatively affect runtime and accuracy during timing optimization and timing closure. When performing this analysis, constraint consistency looks at inversions created with the -invert switch as well as inversions created by the -edges definitions.

**ONE POSSIBLE SCENARIO**

create_clock -period 10 -name clk1 [get_ports clk1] create_generated_clock -divide_by 2 -name clk2 \ -source [get_ports clk1] \ -master [get_clocks clk1] \ [get_pinsFGclock/Q] -add -invert

In the preceding example, the generated clock, clk2, is defined with a waveform inversion on the create_clock command. However, this doesn't match the connectivity in the actual design (the waveform should not be inverted). This results in constraint consistency issuing a CLK_0036 rule violation.

**What Next**

The Violation Details section shows whether the -invert option is missing or incorrectly specified. For the preceding example, it shows the following information:

Generated clock 'clk2' is specified with the -invert option but inversion is not expected.

Check the generated clock definition and the design and redefine the generated clock with the -invert option added or removed. In the example scenario previously described, the -invert option should be removed to match the clk2 waveform to the design.

**See Also**

report_clock CLK_0020 CLK_0008

---

## CLK-0037

(Warning) There are other clock sources in the clock latency path from the sources of the master clock master_clock to the source of generated clock clock.

*Status:* Enabled by default

**Description**

There are clock sources that launch clocks different from the current generated clock or the master of current generated clock in the paths between the master clock and the generated clock. You might want to redefine the generated clock in terms of the clocks that are present in these paths instead of the current master clock. When clock latency

Feedback

paths from a master clock source to a generated clock source contain pins which are the sources of other clocks, static timing tools might perform traversals involving both sequential logic and other clock sources, causing analysis complications (such as missing inferred clock gating checks) or increased runtime.

**ONE POSSIBLE SCENARIO**

Consider the clock generation structure described in the schematic below and the following clock definitions. The clock paths are highlighted in orange in the schematic:

create_clock -name master_clk -period 10 -waveform [list 0 5] [get_ports CLK]

create_generated_clock -add -name gclk1 -master master_clk -source [get_portsCLK] -divide_by 2 [get_pinsFF1/Q]

create_generated_clock -add -name gclk2 -master master_clk -source [get_ports CLK] -divide_by 4 [get_pinsFF2/Q]

In this example, both generated clocks gclk1 and gclk2 are specified with respect to the same master clock master_clk. However, the generated clock, gclk1, is created between the clocks master_clk and gclk2. As a consequence, gclk2 should be expressed in terms of gclk1 instead of in terms of master_clk. Constraint consistency issues a CLK_0037 violation whenever this situation is detected in the design.

**What Next**

The Violation Details section displays additional information related to the reported generated clock source latency. Most importantly, it identifies clock sources in the clock source latency path. In the preceding example, the violation details show the following message:

The following clock source pins occur in the source latency paths between the master clock (master_clk) and generated clock (gclk2):

* FF1/Qp

The Debugging Help section gives the command that can be used to further identify the cone of logic between the master clock and the reported generated clock source pin.

**FIX SUGGESTION**

The recommended practice is to define the generated clocks in the design with respect to the intermediate clock source present in the source latency paths instead of the current master clock. In the previous example, the constraints can be rewritten as follows:

create_clock -name master_clk -period 10 -waveform [list 0 5] [get_ports CLK]
create_generated_clock -add -name gclk1 -master master_clk -source [get_ports CLK] -divide_by 2 [get_pins FF1/Q] create_generated_clock -add -name gclk2 -master gclk1 -source [get_pins FF1/Q] -divide_by 2 [get_pins FF2/Q]

Feedback

**See Also**

report_clock CLK_0008

---

## CLK-0038

(Warning) Detects generated clocks with a sequential source network spanning block boundaries.

*Status:* Disabled by default

**Description**

The generated clock can be defined inside a block, whereas the source latency network of the generated clock could be crossing the block boundary where the generated clock is defined. This kind of scenario can lead to problems during the hierarchical analysis.

This rule uses the *set_hierarchical_config* command to define the blocks and the instances in the current design. This block information is used by this rule to identify the blocks that are of interest to the user.

set_hierarchical_config -block block_name -instances instances list

The source latency numbers for the generated clock might be different in the block-level analysis and the top-level analysis. This can lead to inaccurate results in hierarchical designs.

**What Next**

The Violation Details section displays additional information related to the reported generated clock source latency. Most importantly, it identifies clock sources, the master source, and the master clock for the reported generated clock.

**FIX SUGGESTION**

The recommended practice is to define generated clocks within the same block as the master clock.

**See Also**

check_constraints set_hierarchical_config

---

## CLK-0040

(Warning) Detects clock definition conflicts on the same net.

*Status:* Enabled by default

Feedback

### Description

This rules detects the clock definition conflicts on the same net. This is most useful when you have a hierarchical design, and a clock definition at the block level and another at the top level. When performing a flat analysis, the clocks might be defined with different definitions on the same net. This rule helps in identifying such problems. It is not physically possible to have different clock waveforms in the same net.

### What Next

The Violation details section displays additional details about the clocks with conflicting definitions. It shows all of the details, such as where the clock is defined, its master clock, and the SDC browser link. Most importantly, it lists all of the net segments in a hierarchical design that can lead to the conflict.

### FIX SUGGESTION

Different scenarios might require different approaches. The basic idea is to ensure that there are no conflicting clock definitions on the same hierarchical net segments.

### See Also

check_constraints

---

## CLK-0042

(Error) The user-applied case analysis overlaps with clock propagation.

### Description

This violation occurs when clock propagation overlaps with user-applied case analysis. There is a risk that an unwanted case will stop the clock propagation and might lead to a missing path.

### ONE POSSIBLE SCENARIO

create_clock -name ck -period 10 -waveform {0 5} [get_ports ck] set_case_analysis [get_pins b0/z]

### What Next

Check if the case value is wanted. Remove the case value if it is an incorrect set.

### FIX SUGGESTION

Use the remove_case_analysis command on the port/pin to remove the case value.

### See Also

set_case_analysis remove_case_analysis

Feedback

## CLK-0043

(Warning) Creation of a clock clock blocks the propagation of clocks reaching to the specified point.

### Description

This violation is given when the creation of clock on a port/pin is blocking the propagation of clocks reaching until that point. The clock created on the propagating path of other clocks will lead to all the other clocks stop propagation. If this is unintended it might lead to incorrect clock definition across the design.

### ONE POSSIBLE SCENARIO

create_clock -name ck -period 10 -waveform {0 5} [get_ports ck] create_clock -name ck1 -period 20 -waveform {0 10} [get_pins b0/z]

In this scenario, clock ck1 is created on the path of clock ck. It should use create_generated_clock -master_clock ck -source [get_pins b0/a] -divided_by 2 -name ck1 to define the clock.

### What Next

Check if this is wanted and change the clock definition accordingly.

### FIX SUGGESTION

If the propagation of any blocked clock has to continue, then create a generated clock for all such clocks.

### See Also

create_clock create_generated_clock

# CMD

## CMD-001

(error) Cannot specify '%s' with '%s'.

### Description

The listed command options are exclusive. Only one of them can be specified.

### What Next

Look at the manpage for this command for more information on command options.

Feedback

## CMD-002

(error) Value for '%s' cannot be negative

**Description**

The value for this option must be greater than or equal to zero.

**What Next**

Enter the command again with a valid option value.

## CMD-003

(error) Cannot specify %s without %s.

**Description**

One command option requires another.

**What Next**

Refer to the manual page for this command for detailed information on valid options.

## CMD-004

(error) Must specify one of these options: %s.

**Description**

This command requires that one of the options in the list is specified.

**What Next**

Refer to the manual page for this command for detailed information on valid options.

## CMD-005

(error) unknown command '%s'

**Description**

The command is not recognized.

**What Next**

Look for a typographical error in the command. If it is correct, make sure that the program you are running supports the command, or you have the license to use the command.

Feedback

## CMD-006

(error) ambiguous command '%s' matched %d commands:\n \t(%s)

**Description**

The command does not have sufficient characters to distinguish it from other commands. The first three commands which match the abbreviation are listed. To see them all, use the help as follows: if the abbreviation is cmd, type 'help cmd*'. This lists all commands that begin with 'cmd'.

**What Next**

Type enough characters so the command is unambiguous.

## CMD-007

(error) Required argument '%s' was not found

**Description**

The listed argument to the command might not be omitted.

**What Next**

Supply the required argument.

## CMD-008

(error) value not specified for option '%s'

**Description**

The listed argument requires a value (that is, it is not a boolean option), and none were supplied.

**What Next**

Supply a value for the argument.

## CMD-009

(error) value '%s' for option '%s' not of type '%s'

**Description**

The value given for the listed argument is not the correct type. For example, if 'abc' is given for an integer option, this error occurs.

Feedback

**What Next**

Supply a compatible value for the argument.

---

## CMD-010

(error) unknown option '%s'

**Description**

The option is not recognized.

**What Next**

If this is not a simple mistake, retype the command with by the -help option. This lists all of the possible options.

---

## CMD-011

(error) ambiguous option '%s'

**Description**

The option does not have sufficient characters to distinguish it from other options.

**What Next**

Type enough characters so that the option is unambiguous.

---

## CMD-012

(error) extra positional option '%s'

**Description**

The command expects some positional arguments and has already received enough. It might also be the case that this was intended as a dash option and is misspelled.

**What Next**

Verify that the option given is not a misspelled dash option. If a list is provided directly instead of as a variable, ensure it is enclosed by curly braces or double quotation marks. Use -help with the command to verify which arguments are already given.

---

## CMD-013

(error) %s\n \tUse error_info for more info.

Feedback

**Description**

A script or complex command failed and there is a stack trace for the failure. The trace points out the source files and loops where the error occurred. The error_info command is used to display this stack.

**What Next**

Fix the error indicated by error_info.

---

## CMD-014

(error) Invalid %s value '%s' in list.

**Description**

A list argument is expected to be a common type (like integer or float) and one or more elements cannot be converted to that format.

**What Next**

Fix the offending list element.

---

## CMD-015

(error) could not open %s file

**Description**

A script or an output redirect file cannot be opened.

**What Next**

Verify that the file exists or that you have write access to the directory. Write access depends on the file type.

---

## CMD-016

(error) could not close %s file

**Description**

A script or an output redirect file cannot be closed.

---

## CMD-017

(warning) duplicate option '%s' ignored.

Feedback

**Description**

The given option is already issued. This command uses the first value of the option, and subsequent values are ignored.

**What Next**

Make sure this is the option you want to use. If so, decide which value you want and verify that you get the correct one.

---

## CMD-018

(warning) duplicate option '%s' overrides previous value.

**Description**

The given option has already been issued. This command uses the last value of the option, and previous values are ignored.

**What Next**

Make sure that this is the option you want to use. If so, decide which value you want and make sure that you get the correct one.

---

## CMD-019

(error) value '%s' for option '%s' not in range (%s).

**Description**

The value given for the listed argument is not in the allowable range. For example, if 4 is given for an integer option, which has a range of 1 to 3, this error occurs.

**What Next**

Supply a compatible value for the argument.

---

## CMD-020

(error) unknown OR extra positional option '%s'

**Description**

The dash option is not recognized. Further, all positional arguments have already been received. This is most likely a misspelled dash option.

Feedback

**What Next**

Check to see if the option is misspelled. Look at the entire command, as other options may have misled the interpreter.

---

## CMD-021

(warning) invoked %s outside of a loop

**Description**

The listed control command (break or continue) was used outside of the context of control structure (such as foreach, while, and so on).

**What Next**

Look for a loop that ends prematurely or for a misspelled control word.

---

## CMD-022

(warning) Can't create alias named '%s' - %s%s.

**Description**

An attempt was create an alias with an invalid name. Invalid names include those which match an existing command or procedure, and those which can be converted to a decimal, hexadecimal, or octal number.

**What Next**

Choose another name. Use 'help' and 'alias' (with no arguments) to see what names are in use.

---

## CMD-023

(error) Alias loop: %s

**Description**

You have aliases that refer to one another.

**What Next**

Use the alias command to look at the aliases listed in the diagnostic. Remove the loop and re-execute the command.

Feedback

## CMD-024

(error) can't %s "%s": %s

**Description**

You attempted an operation on a variable which failed. You may have tried to read a non-existent variable (set var). Or, you may have tried to unset a non-existent or application-owned variable. The text of the message will indicate which operation failed.

**What Next**

Verify that the variable exists with the printvar command. If it's not a user variable, you cannot remove (unset) it.

## CMD-025

(error) No manual entry for '%s'

**Description**

The topic for which you requested man pages does not exist.

**What Next**

Verify that the topic is spelled correctly.

## CMD-026

(error) %s required for the '%s' argument.

**Description**

The command is incomplete as entered. The specified argument requires a valid object or list of objects.

**What Next**

Enter the command with valid values for all arguments.

## CMD-027

(error) couldn't change working directory to '%s'

**Description**

The directory which you specified to the cd command is not valid.

Feedback

**What Next**

Verify that the directory is spelled correctly.

---

## CMD-028

(error) couldn't get working directory name

**Description**

The pwd command was unable to access the current directory. It is most likely the case that the directory which you are in no longer exists,

**What Next**

Use the cd command to get into an existing directory.

---

## CMD-029

(warning) no aliases matched '%s'

**Description**

You specified a pattern to the unalias command, and there are no aliases which match that pattern.

**What Next**

There is no adverse effect of this action. However, check the spelling of the arguments to unalias to ensure that you removed all of the aliases which you wanted to remove.

---

## CMD-030

(warning) File '%s' was not found in search path.

**Description**

The 'which' command evaluated an filename argument and the file was not found.

**What Next**

No adverse effect on the result of the command, but check spelling, etc.

---

## CMD-031

(error) value '%s' for option '%s' is not valid. Specify one of:\n \t%s

Feedback

**Description**

The value given for the listed argument is not one of the limited allowable strings. This messages lists all of the appropriate values.

**What Next**

Supply a compatible value for the argument.

---

## CMD-032

(warning) command '%s' requires some options.

**Description**

No options were given for the command, yet some are required.

**What Next**

Supply appropriate arguments.

---

## CMD-033

(error) cannot source the current log file.

**Description**

An attempt to source the log file of the currently running interpreter is not allowed. It would cause the tool to infinitely loop.

**What Next**

Copy the part of the log to be a source for another file, then source that file instead.

---

## CMD-035

(error) Value for %s cannot be larger than the %s value.

**Description**

Some commands work in pairs, specifying a maximum and minimum value. The minimum value should be less than the maximum value. For example, never specify a *min_capacitance* which is larger than the *max_capacitance* for the same design or port.

**What Next**

Remove the old value or use a different value.

Feedback

## CMD-036

(error) Value for list '%s' must have %s elements.

**Description**

The value given for the list argument does not have the correct number of elements. Some commands have list arguments which require either a specific number or an even number of elements. The message will indicate which it is.

**What Next**

Supply a correct number of elements in the list. If the list is provided directly instead of as a variable, ensure it is enclosed by curly races or double quotation marks.

## CMD-037

(error) value '%s' for option '%s' is invalid: must be %s.

**Description**

The value given for the listed argument is greater than or less than the allowable limit. For example, if 4 is given for an integer option, which is required to be less than or equal to 3, this error occurs.

**What Next**

Supply a compatible value for the argument.

## CMD-038

(information) The '%s' option for %s is unsupported.%s

**Description**

The option which you specified is not currently supported.

## CMD-039

(information) The '%s' variable is unsupported.%s

**Description**

The variable which you specified is not supported.

**What Next**

If a replacement variable is specified, use it instead of this one.

Feedback

## CMD-040

(information) No %s matched '%s'.

### Description

In command or variable search functions (help and printvar), you specified a pattern that did not match any variables or commands.

Note that printvar cannot find a specific array element; it can only find the entire array by name.

### What Next

Try using more wildcards (* or ?) in your search pattern.

## CMD-041

(information) Defining new variable '%s'.

### Description

This message is issued when a variable is set for the first time.

When combined with the *printvar* command, this message can be used to isolate spelling errors in system (application) variables. However, like many debugging features, this has significant CPU cost. Therefore, the feature should only be used interactively or when developing scripts.

This feature is enabled by setting the *sh_new_variable_message* variable to true. When combined with a true value for variables *sh_new_variable_message_in_script* or *sh_new_variable_message_in_proc*, this setting causes a warning message (CMD-042) to be issued, which indicates that the performance of scripts (or Tcl procedures) will be adversely affected. To enable the feature in Tcl procedures, set the *sh_new_variable_message_in_proc* variable to true. To enable the feature in Tcl scripts, set the *sh_new_variable_message_in_script* variable to true.

In the following example, the user has misspelled the variable *sh_continue_on_error* by making it plural. With this feature, debugging is simplified.

```
prompt> set sh_continue_on_errors true
Information: Defining new variable 'sh_continue_on_errors' (CMD-041)
true
prompt> printvar sh*
sh_arch               = "sparcOS5"
sh_continue_on_error = "false"
sh_continue_on_errors = "true"
sh_enable_page_mode  = "false"
sh_new_variable_message = "true"
```

Feedback

```
sh_new_variable_message_in_proc = "false"
sh_product_version   = ""
sh_source_uses_search_path = "false"
prompt> unset sh_continue_on_errors
prompt> set sh_continue_on_error true
true

Application variables are always defined, so if this message appears,
a new user-defined variable has been created.
```

### What Next

If attempting to set an application variable, use *printvar* with wildcards to get the correct spelling for the variable.

### See Also

- printvar
- sh_new_variable_message
- sh_new_variable_message_in_proc
- sh_new_variable_message_in_script
- CMD-042

---

## CMD-042

(warning) Enabled new variable message tracing -\n \tTcl scripting optimization disabled.

### Description

This message is issued when you enable new variable tracing for Tcl scripts or procedures. That occurs when you set the variable *sh_new_variable_message* to TRUE, and when you set the variables *sh_new_variable_message_in_proc* or *sh_new_variable_message_in_script* to TRUE. It warns you that the performance of the application will be negatively mpacted because this feature is costly in CPU time when enabled.

This feature is intended for debugging, and should only be used interactively or when developing scripts. It should not be used in a main flow.

### What Next

Set one or more of the variables to FALSE unless you are debugging a script.

Feedback

**See Also**

- sh_new_variable_message
- sh_new_variable_message_in_proc
- sh_new_variable_message_in_script

---

## CMD-050

(error) Unknown procedure '%s'.

**Description**

The procedure name argument to *define_proc_attributes* is not a procedure.

**What Next**

Verify that the argument is correct.

---

## CMD-051

(error) Procedure '%s' cannot be modified.

**Description**

The procedure that you passed to *define_proc_attributes* is a permanent procedure that cannot be modified.

**What Next**

The procedure might be part of the application, in which case it was correctly defined with *-permanent*. If it is not part of the application, it is possible that it was erroneously defined with *-permanent*.

---

## CMD-052

(error) Unknown command group '%s'

**Description**

The command group referenced does not exist. For example, using the *-command_group* option with the *define_proc_attributes* command, and passing in a non-existent command group will raise this error.

**What Next**

Verify that the correct command group name is being used.

Feedback

## CMD-053

(warning) The body of procedure '%s' is protected

**Description**

You attempted to examine the body of a procedure using *info body*. That procedure was protected by the writer so that it's body cannot be displayed.

**What Next**

No action required.

## CMD-060

(error) Syntax error in argument definition %d for proc '%s'.

**Description**

Using the *-define_args* option for *define_proc_attributes*, there is some kind of syntax error, for example, an improperly formatted list.

**What Next**

Use *error_info* to narrow the problem, then reenter the command.

## CMD-061

(error) Need at least 2 fields in argument definition %d for proc '%s'.

**Description**

Using the *-define_args* option for *define_proc_attributes*, an argument definition had insufficient arguments. At least 2 are required: the argument name and the option help text.

**What Next**

Reenter the argument definition with the correct number of fields.

## CMD-062

(error) Unknown %s '%s' in argument definition %d (%s) for proc '%s'.

**Description**

Using the *-define_args* option for *define_proc_attributes*, either a data type or attribute is invalid.

Feedback

The allowable data types are string, boolean, int, float, and list. The allowable attributes are required and optional.

**What Next**

Correct the invalid data and reenter the command.

---

## CMD-063

(error) Illegal name '%s' for Boolean argument definition %d for\n \t proc '%s': must begin with '-'.

**Description**

Using the *-define_args* option for *define_proc_attributes*, you attempted to create a Boolean argument with a name not preceded by a '-'. Boolean arguments require a leading '-'.

**What Next**

Correct the argument name and reenter the command.

---

## CMD-064

(warning) Value help ignored for Boolean option %s\n \tin argument definition %d for proc '%s'.

**Description**

Using the *-define_args* option for *define_proc_attributes*, you tried to add value help for a Boolean argument. Boolean arguments cannot have the value help.

**What Next**

Remove the value help field for boolean arguments and reenter the command.

---

## CMD-065

(error) Can't specify both 'optional' and 'required' in argument\n \tdefinition %d (%s) for procedure '%s'

**Description**

This message indicates an attempt to specify conflicting flag values as part of the definition of a procedure argument within the *define_proc_attributes* command.

**What Next**

Decide whether the argument is optional or required, and remove the opposite flag.

Feedback

## CMD-066

(error) Must specify a value for attribute 'values' when using '%s'\noption type as in option %d (%s) for procedure '%s'

### Description

This message is issued by the *define_proc_attributes* command when you attempt to define an argument whose value must be one of a set of pre-defined strings (the one_of_string data type), without specifying the set of valid strings.

### What Next

If the value type really needs to be one_of_string, pass the values in as a list within the attributes list (i.e. {values {a b c}}).

## CMD-067

(error) Invalid attribute specification for attribute '%s'\n \t(%s)\n \tin option %d (%s) for procedure '%s'

### Description

This message is issued by the *define_proc_attributes* command. It indicates an incorrect attempt at specifying an attribute for a procedure argument. The reason for the error is included in the message.

### What Next

Fix the syntax of the command and try again.

## CMD-068

(error) Could not find procedure '%s'. Arguments can't be parsed.

### Description

This message indicates an attempt to use the parse_proc_arguments command from within a procedure which has not been defined using define_proc_attributes.

### What Next

Define the procedure's arguments using define_proc_attributes and try again.

## CMD-069

(error) Could not set '%s(%s)' while parsing arguments in '%s'.

Feedback

### Description

This message indicates that the *parse_proc_arguments* command was not able to set the specified Tcl array variable to hold the value of a command option.

### What Next

This typically indicates that the variable was read-only. Use a different variable and try again.

---

## CMD-070

(error) %s can only be called from within a procedure

### Description

This message indicates an attempt to use the given command from the interpreter command line. Calls to this command are only supported from within a Tcl procedure.

### What Next

Create a procedure and call the command from within the scope of the procedure body.

---

## CMD-080

(warning) Command '%s' is disabled.

### Description

Although part of the application, the listed command is not currently enabled. The sh_disabled_is_error variable controls the severity of this message. When true, a Tcl error is issued for this command. When false, the command is a no-op.

### What Next

Look at the user documentation to determine how various commands are enabled and disabled.

### See Also

- sh_disabled_is_error

---

## CMD-081

(information) script '%s'\n \tstopped at line %d due to %s.

Feedback

### Description

The execution of a script was terminated. This message tells you which script stopped, the line number where it stopped, and why it stopped.

If the *sh_continue_on_error* variable is false (the default), any Tcl error, either syntax or semantic, stops the script. If *sh_continue_on_error* is false and the *sh_script_stop_severity* variable is W or E, messages of that severity or higher stop the script.

If the *sh_continue_on_error* variable is true, the *sh_script_stop_severity* variable is ignored and the script continues even if there are errors or warnings.

### What Next

Use the information in this message to identify and correct the source of errors and warnings. Then reexecute the script.

### See Also

- sh_continue_on_error
- sh_script_stop_severity

---

## CMD-082

(information) %s occurred at or before line %d in\n \tscript '%s'.

### Description

You receive this message if an error or warning occurs while a script is executing, and the variable *sh_source_emits_line_numbers* is set to E or W. A setting of E causes this message to be issued only if an error occurs, while for a setting of W, this message is issued for both warnings and errors. This message tells you the error or warning and the line and script in which it occurred.

The setting of the *sh_script_stop_severity* variable affects the output of the CMD-082 message. If *sh_script_stop_severity* is set to E, the script stops executing if an error occurs; for a setting of W, the script stops executing if a warning or error occurs. In both cases, message CMD-081 is issued, and takes precedence over CMD-082.

### What Next

Use the information in this message to identify and correct the source of errors and warnings. Then reexecute the script.

Feedback

### See Also

- sh_script_stop_severity
- sh_source_emits_line_numbers
- CMD-081

## CMD-085

(warning) Renaming %s %s cause %s commands which use it to fail.

### Description

You receive this message if you rename a command which is not a user-defined Tcl procedure. Renaming commands can be dangerous. Parts of the application are written in Tcl, and if you rename a command that the application is using, it is possible that those parts of the application will not function.

The only true use for *rename* is to wrap a command. For example:

```
shell> rename command1 command1_orig
shell>  \\
  proc command1 {args} {
    # ...
    eval command1_orig $args
    # ...
  }
```

If you use *rename* in this way, it is more likely that application will continue to function correctly. Still, use *rename* with extreme care and at your own risk.

### What Next

Consider using *alias*, Tcl procedures, or a private namespace before using *rename*.

### See Also

- rename

## CMD-086

(error) Could not find command '%s'.

### Description

This message indicates that the command name entered does not exist and therefore operation on associated command mode could not be performed.

Feedback

**What Next**

Check to make sure command name is typed correctly.

---

## CMD-087

(error) The command requires either a command name or a command mode name.

**Description**

This command requires either a command name or a command mode name to be specified.

**What Next**

Enter *set_current_command_mode* with the *-command* option flag followed by a command name or the *-mode* option flag followed by a command mode name. These options are mutually exclusive.

---

## CMD-088

(error) Could not find command mode '%s'.

**Description**

This message indicates that the command mode name entered does not exist and therefore could not be set as the current mode.

**What Next**

Check to make sure command mode name is typed correctly. *get_command_modes -all* lists all defined command mode names.

---

## CMD-089

(error) Initialization of command '%s' failed.

**Description**

This message indicates that a failure occurred during initialization and the specified command could not be evaluated.

---

## CMD-090

(error) Initialization of command mode '%s' failed.

Feedback

**Description**

This message indicates that a failure occurred during initialization of the command mode and the specified command mode could not be made current.

---

## CMD-100

(warning) Detected use of obsolete/unsupported feature. The following\n \twill not be available in a future release of the application:\n \t%s. Use %s instead

**Description**

You have used a feature which is no longer supported by the application, and the feature is planned to be removed at some future date. The supported method is given in the message.

**What Next**

Update your command usage as indicated.

---

## CMD-101

(error) Failed to set value of option %s for command %s.

**Description**

A run of a command such as set_command_option_value failed to set the default or current value of an option. The command option may not have been enabled for value-tracking or a conversion error may have occurred when attempting to set the option value.

**What Next**

It may be necessary to enable the option for value-tracking.

---

## CMD-102

(error) No such positional option %d for command %s.

**Description**

An attempt was made to find the positional option of the command at the given position. No such positional option was found. Either the given command has no positional options, or the given position is "out of range". Note that positional options are numbered 0, 1, 2, ... (N-1) where N is the number of postional options of the command.

**What Next**

Retry the operation using a positional option position that is "in range" for the command.

Feedback

## CMD-103

(error) A Severe error has occurred. \n \tTo ensure that the script does not continue, the value of sh_continue_on_error\n \thas been overridden to be false. Your script is being interrupted.\n \tTo see the Tcl call stack for the part of your script which generated\n \tthe Severe error use the error_info command.

### Description

A Severe error has occurred during a command execution. To ensure that the script does not continue, the value of sh_continue_on_error has been overridden to be false. Your script is being interrupted. To see the Tcl call stack for the part of your script which generated the Severe error use the error_info command.

### What Next

For details on the Severe error please look in your log file. You can also run man on the Severe error id to learn more about the error. Study the Severe error and try to fix the error in your script.

## CMD-104

(error) Variable '%s' is not an application variable. Using Tcl global variable.

### Description

The specified variable is not declared as an application variable (not returned by get_app_var -list). This message is only generated when the application variable sh_allow_tcl_with_set_app_var is true.

Please see the manpages for get_app_var and set_app_var for additional details.

### What Next

Make sure you are using the correct variable name.

## CMD-105

(warning) Option '%s' is deprecated, use '%s' instead.

### Description

This option is deprecated, you should use a different option for this command feature. The code has automatically used a compatible option setting, but in the future the old option may be removed, so you should update your scripts.

Feedback

**What Next**

Update your script to use the new option.

---

## CMD-106

(warning) Option '%s' for command '%s' is obsolete. See the command's man page for alternatives.

**Description**

This option is no longer supported, specifying it has no effect.

**What Next**

Update your script

---

## CMD-107

(error) Not enough values specified for option '%s', requires %d values found %d.

**Description**

The listed option requires that the specified number of values and not enough values were supplied.

**What Next**

Supply all the required values for the option.

---

## CMD-108

(warning) Command %s is obsolete. See the command's man page for alternatives.

**Description**

This command is obsolete. Calling it has no effect. Please see product documentation for alternatives.

**What Next**

Update your script as needed.

---

## CMD-109

(warning) Command %s is deprecated. See the command's man page for alternatives.

Feedback

**Description**

This command is deprecated. Please see product documentation for alternatives.

**What Next**

Update your script as needed.

---

## CMD-110

(warning) Option '%s' for command '%s' is deprecated. See the command's man page for alternatives.

**Description**

This option is deprecated, you should use a different option for this command feature.

**What Next**

Update your script to use the new option.

---

## CMD-111

(error) The write operation failed: %s.

**Description**

The write operation failed for the reason shown.

---

## CMD-112

(error) duplicate option '%s'

**Description**

Duplicate options are disallowed for this command option.

**What Next**

Provide a single entry for this option.

---

## CMD-113

(error) problem running user derived attribute command for %s attribute %s:\n \t%s

**Description**

Duplicate options are disallowed for this command option.

Feedback

**What Next**

Provide a single entry for this option.

---

## CMD-114

(severe) Command raised a C++ exception:\n %s

**Description**

A command raised a C++ exception. This is a tool bug and should be reported to Synopsys. This script will terminate execution and you may try to conintue by saving or attempting some other operation. Other issues are likely.

**What Next**

Report the issue to Synopsys.

---

## CMD-999

(severe) A Severe error has occurred during testing.

**Description**

A Severe error has occurred during testing. This should never happen in production.

---

# CMP

---

## CMP-009

(Warning) The custom clock mapping is found to be incomplete.

**Description**

The option is provided for custom clock mapping. However, it is found that the clock mapping is incomplete.

**What Next**

No action is needed if this is intentional. Otherwise, update the script for custom clock mapping.

---

## CMP-010

(Error) Conflicting clock mapping detected at '%s'; previous mapping for this clock defined at '%s'.

Feedback

**Description**

Conflicting entries are found in the custom clock mapping for the given clock. It is likely that you need to redefine a clock as a multi-source clock instead of as two different clocks.

**What Next**

Custom clock mapping for this clock will be discarded. Check your clocks and the mapping scripts.

---

## CMP-011

(waring) Duplicated clock mapping detected at '%s'; previous mapping for this clock defined at '%s'.

**Description**

Duplicate entries are found for same clock in the custom clock mapping for the given clock.

**What Next**

Consider removing the duplicated entry in the script.

---

# CNL

---

## CNL-0001

(Warning) Inconsistent clock network latency values on object_type object.

**Description**

The clock network latency values are inconsistent. Each min value should be smaller than the corresponding max value. By definition, min/max values are used for minimum/maximum operating conditions. It is inconsistent with the intent of having the different analysis modes to have the max clock network latency to be less than the min clock network latency. This could lead to incorrect delay calculation and analysis.

**ONE POSSIBLE SCENARIO**

Consider a scenario where the clock network latency is set on a clock *CLK* as follows:

set_clock_latency -rise -min 11 [get_clock CLK] set_clock_latency -fall -min 2 [get_clock CLK] set_clock_latency -rise -max 3 [get_clock CLK] set_clock_latency -fall -max 4 [get_clock CLK]

Feedback

The *-rise -min* value is larger than the *-rise-max* value which is inconsistent. It should be the smallest value. For this situation, the tool will issue a CNL_0001 rule violation.

**What Next**

The “Violation Details” section of the rule violation shows a table for the source latency values defined at the object. For the scenario described previously, the table reported is shown as follows:

*Min Max*

*Rise* 11.0000 2.0000

*Fall* 3.0000 4.0000

You can also view the related constraint definitions in the Constraint Browser by clicking on the links in the “Network latency specified at:” section of the “Violation Details.”

**FIX SUGGESTION**

Changing the constraints to have consistent min/max clock network latency values will fix this violation. For the example scenario, changing the -rise -min value to be less than the -rise -max value will fix the violation.

set_clock_latency -rise -min 1 [get_clock CLK]

---

## CNL-0002

(Warning) Input/output delays are defined relative to clock clock. This clock has incomplete network latency values required for those external delays even though its source ports/pins have more complete network latencies defined on them.

**Description**

Since the network latency values for the clock are incomplete, no network latency will be applied for paths with from/to points which have input or output delay relative to this clock. The network latency set on the clock sources is not applied on these paths. This could result in incorrect timing analysis. Network latency should be applied to the clock object.

Clock network latency values applied on a clock source pin or port will not be used when delays need to be calculated on paths that have ports with external delay settings on them.

Only latency values applied on clock objects are used for path delay calculations, and if these are not specified or are incomplete, zero values are assumed. This could affect both setup and hold timing calculations if it is assumed that the clock network latency values specified on the clock source pin/port will be used instead.

**ONE POSSIBLE SCENARIO**

Feedback

Consider a scenario, where a clock is defined on an input port:

create_clock -period 10 [get_ports CLK]

Clock network latencies are defined on both the clock object and the clock source port.

set_clock_latency -min 3 [get_clock CLK] set_clock_latency 4 [get_ports CLK]

An input delay is specified for a port relative to this clock:

set_input_delay 5 -clock CLK -add_delay [get_ports INPUT]

The input delay set on the INPUT port has been defined with respect to clock CLK, but since no network latency is defined on the clock object for the max condition, the clock network latency for maximum delay calculations will be zero. The network latency of 4 applied on the CLK port object will not be used. For this situation, a CNL_0002 rule violation is issued to warn the user that the clock network latency applied on the port object will not be used for clocks specified in external delay definitions.

The “Violation Details” section shows a table for the network latency values: one for the clock and one for its sources. For the scenario described above, the tables reported in the “Violation Details” section would be:

Network Latency Values on the Clock

*Min Max*

*Rise* 3.0000 --

*Fall* 3.0000 --

Network Latency Values on Source ‘CLK’

*Min Max*

*Rise* 4.0000 4.0000

*Fall* 4.0000 4.0000

You can also view the related constraint definitions in the Constraint Browser by clicking on the link in the “Network latency specified at:” section associated with each table in the “Violation Details.”

**FIX SUGGESTION**

Modify the constraints so that a complete set of clock network latency values is specified. For the example scenario, adding network latency for the clock object max condition as follows will fix the violation:

set_clock_latency -max 4 [get_clock CLK]

Feedback

If an incomplete network latency definition is the desired analysis for the from/to ports having external delays with respect to this clock, then no fix is required.

---

## CNL-0003

(Warning) Clock network latency is specified on object_type object, but that object is not in any clock network.

**Description**

Clock network latency should only be specified on clocks, pins, or ports in a clock network.

The clock network latency specified for an object that is not in a clock network will be ignored. This could affect both setup and hold timing of the design.

**What Next**

The “Violation Details” section provides a link with the file name and line number to the clock network latency specification.

This violation can be caused by a pin or port that may be part of a clock network, but another constraint is blocking the clock. Use the *potential_clocks* attribute to check if the pin or port could be in a clock network, but was blocked by the set_case_analysis or set_disable_timing command specification or some other constraint.

For the example scenario, the value of the potential_clocks attribute is as follows:

rtl_shell> get_attribute [get_pins inv/Z] potential_clocks {"clk"}

If the object is not in a potential clock network, then there may be an error in specifying the pin, or there is a missing clock definition. An error in the pin specification could be caused if the wrong pin is specified, or if a wildcard is used in the set_clock_latency command, and it returns more pins than expected.

**FIX SUGGESTION**

If the pin or port is in a potential clock network, check if the blockage is correct for this scenario. If it is correct, the violation can be waived or the set_clock_latency command can be altered to remove the object associated with the violation.

If the pin or port is not part of a clock network, the latency specification should be moved to other pins or ports that are part of the clock network so that it is not ignored.

---

## CNL-0004

(Warning) Clock network latency is specified on object_type object with respect to clock clock, but that object is not in the clock's network.

Feedback

### Description

Clock network latency should only be specified on pins or ports which are part of the specified clock network.

The clock network latency specified for an object that is not in a clock's network will be ignored. This could affect both setup and hold timing of the design.

### ONE POSSIBLE SCENARIO

Consider the scenario below where a design contains two clocks. The clock network latency specification is specified on pin *ff1/CP* with respect to clock *clk2* instead of *clk1*. Since the *ff1/CP* pin is not in the *clk2* network, this clock network latency specification is ignored, and the tool will issue a CNL_0004 rule violation.

create_clock -period 10 [get_clocks clk1] create_clock -period 10 [get_clocks clk2] set_clock_latency 1.3 [get_pins ff1/CP] -clock clk2

### What Next

The "Violation Details" section provides a link with the file name and line number to the clock network latency specification.

This violation can be due to an incorrect clock that is specified with the *set_clock_latency* command. Query the clocks attribute on the pin or port to determine which clocks reach the pin or port. If a different set of clocks reaches this pin or port than the one specified in the constraint, then check to see if one of those clocks should be defined in the clock network latency command.

For the example scenario, the clocks attribute can be queried as follows:

rtl_shell> get_attribute [get_pins ff1/CP] clocks {"clk1"}

If the contents of the clocks attribute are not the expected ones, then the clock of interest could potentially be blocked from reaching the pin or port. The potential_clocks attribute associated with the object will help determine which clocks could reach the pin or port. For the example scenario, the attribute can be queried as follows:

rtl_shell> get_attribute [get_pins ff1/CP] potential_clocks Warning: Attribute 'potential_clocks' does not exist on pin 'ff1/CP' (ATTR-3)

For the example scenario, the clock clk1 arrives at the ff1/CP pin, but there are no other potential clocks for that pin. Further debug can be completed as described below to identify if there are other potential clocks that exist on the object or if the expected clock is not in the clocks or potential_clocks attributes query. Resolution of the violation is discussed in the "Fix Suggestion" section below.

### FIX SUGGESTION

Feedback

Check to see if the clock name in the -clock option of the network latency is correctly specified. In the example scenario, the network latency needs to be specified relative to the clock clk1.

set_clock_latency 1.3 [get_pins ff1/CP] -clock clk1

If the pin or port is in a potential clock network, see whether the blockage is correct for this scenario. If it is correct, the violation can be waived, or the set_clock_latency command can be altered to remove the object associated with the violation.

---

## CNL-0005

(Warning) All register clock pins in the fanout of pin have no clock latency for clock clock.

**Description**

During pre-layout analysis, all clocks should have an ideal clock network latency.

In pre-layout designs, path delays are not finalized. The propagated clock network latency may not reflect the real latency from the placement, clock-tree synthesis, and routing stages. Missing a realistic latency at the register clock pins may lead to inaccurate timing results.

**ONE POSSIBLE SCENARIO**

Consider the scenario shown in Figure 1, where clock signals reach the register clock pin U2/CP.

If the following constraints are applied,

create_clock -period 10 [get_clocks clk1] create_clock -period 5 [get_clocks clk2] set_clock_latency -clock clk1 2.0 [get_pins U1/Z]

no clock network latency is defined at the register clock pin with respect to the clock clk2. This results in a CNL_0005 rule violation indicating that no clock latency has been defined for the fanout of port clk2.

**What Next**

The violation message reports the name of the clock and the pin where the ideal clock network latency can be set. In addition, the More Info section of the message lists up to 20 register clock pins affected by the missing clock network latency.

**FIX SUGGESTION**

Specify a realistic clock network latency value at the given pin or at the fanout of the pin with respect to the reported clock. For the scenario above, specify the clock network latency at the output of the multiplexer:

set_clock_latency -clock clk2 1.2 [get_pins U1/Z]

Feedback

# CORR

## CORR-001

(information) The reference %s version used for correlation check is '%s'.

**Description**

This information message advises you to use compatible image version for correlation checker.

**What Next**

Please use a compatible version of image to avoid potential problems.

It is recommended that you specify correct image version with command *set_consistency_settings_options*.

**See Also**

- check_consistency_settings

## CORR-002

(error) The command '%s' is obselete, please use command '%s'.

**Description**

This information message advises you to use new change for old command functionality.

**What Next**

Please use a new command to accomplish your work.

**See Also**

- check_consistency_settings

## CORR-003

(information) The command will be applied to current scenario only.

**Description**

This information message indicates the command will be applied to current scenario only.

Feedback

### What Next

No need action.

### See Also

- check_consistency_settings

---

## CORR-401

(warning) Starting with the Q-2019.12 release the default behaviour of *analyze_timing_correlation* with respect to compatibility settings will change from 'enabled' to 'disabled'

### Description

If you do not specify either the -disable_compatibility_settings switch or the -enable_compatibility_settings switch, the default behaviour of *analyze_timing_correlation* will be to disable compatibility settings. This will mean that the tool setup for correlation will use the current design settings rather than changing options to the recommended settings for correlation. This may lead to worse timing correlation than if the recommended settings were used.

### What Next

Explicitly specify either one of the *-enable_compatibility_settings* or *-disable_compatiblity_settings* switches.

---

## CORR-800

(Error) The %s %s setting should be set to %s.

### Description

The specified PrimeTime setting is not set to the recommended value for PrimeTime and IC Compiler correlation consistency.

Please see the manpages for specified PrimeTime setting for additional details.

### What Next

Change the specified PrimeTime setting to recommended value.

### See Also

- check_consistency_settings

Feedback

## CORR-801

(Error) For consistency checking, it is recommended to disable fast analysis mode in PrimeTime.

### Description

This message indicates a correlation issue with PrimeTime setting. *sh_fast_analysis_mode_enabled*.

### What Next

To resolve this issue, you are reocmmended to disable *sh_fast_analysis_mode_enabled* in PrimeTime.

### See Also

- check_consistency_settings

## CORR-802

(Error) The %s %s setting should be set to %s.

### Description

The specified setting in image is not set to the recommended value for correlation consistency

Please see the manpages for specified setting for additional details.

### What Next

Change the specified setting to recommended value.

### See Also

- check_consistency_settings

## CORR-803

(Error) The %s setting is inconsistent between %s and %s. The %s value is %s; the %s value is %s.

### Description

The specified setting is inconsistent between both two tools, and one of the settings is not set to the recommended value for correlation consistency.

Please see the manpages for specified setting for additional details.

**What Next**

Change the specified setting to recommended value in both tools.

**See Also**

- check_consistency_settings

---

## CORR-804

(information) The %s setting in both %s and %s are set to '%s'; The recommendation is for both tools to set to '%s'.

**Description**

The specified setting are both not set to the recommended value for correlation consistency.

Please see the manpages for specified setting for additional details.

**What Next**

Change the specified setting to recommended value in both tools.

**See Also**

- check_consistency_settings

---

## CORR-805

(Error) For post-route delay calculation mode, Arnoldi should be used.

**Description**

Arnoldi is not specified for post-route delay calculation mode, as recommended for PrimeTime and IC Compiler correlation consistency.

Please see the manpages for IC Compiler command set_delay_calculation_options for additional details.

**What Next**

Change post-route delay calculation mode to Arnoldi in IC Compiler .

**See Also**

- check_consistency_settings

Feedback

## CORR-806

(Error) For post-route Arnoldi delay calculation effort, high effort should be used.

**Description**

High effort is not specified for post-route Arnoldi delay calculation effort, as recommended for PrimeTime and IC Compiler correlation consistency.

Please see the manpages for IC Compiler command set_delay_calculation_options for additional details.

**What Next**

Change post route Arnoldi delay calculation effort to high in IC Compiler.

**See Also**

- check_consistency_settings

## CORR-807

(Error) SI setting should be enabled in %s.

**Description**

SI, signal integrity, is not enabled in specific tool. It's recommended to enable it for correlation consistency.

**What Next**

Enable SI in tool.

**See Also**

- check_consistency_settings

## CORR-808

(Error) SI setting should be enabled in %s.

**Description**

SI, signal integrity, is not enabled in specific tool. It's recommended to enable it for correlation consistency.

Please see the manpages for command set_si_options for additional details.

Feedback

**What Next**

Enable SI in tool.

**See Also**

- check_consistency_settings

---

## CORR-809

(information) The %s setting is %s in both %s and %s. For consistency checking, it's recommended to %s this feature in both tools.

**Description**

The specified setting is not set to the recommended value for both tools correlation consistency.

Please see the manpages for specified setting for additional details.

**What Next**

Change the specified setting to recommended value in both tools.

**See Also**

- check_consistency_settings

---

## CORR-810

(information) The %s setting does not have same recommended value between %s and %s. %s has recommended value '%s', %s has recommended value '%s'.

**Description**

The recommended values are not same between tools.

**What Next**

Make sure the recommended values settings are correct in both tools.

**See Also**

- check_consistency_settings

---

## CORR-812

(Error) The %s %s setting should be set to %s.

Feedback

**Description**

The specified setting in image is not set to the recommended value for correlation consistency

Please see the manpages for specified setting for additional details.

**What Next**

Change the specified setting to recommended value.

**See Also**

- check_consistency_settings

---

## CORR-813

(Error) SI setting should be enabled in %s.

**Description**

SI, signal integrity, is not enabled in specific tool. It's recommended to enable it for correlation consistency.

**What Next**

Enable SI in tool.

**See Also**

- check_consistency_settings

---

## CORR-814

(Error) %s does not support the same functionality as %s %s%s.

**Description**

The tool does not support the same functionality like the other tool.

Please see the manpages for specified setting for additional details.

**What Next**

Change the specified setting to recommended value in both tools.

**See Also**

- check_consistency_settings

Feedback

## CORR-815

(Error) The %s setting is inconsistent between %s and %s. The %s value is %s; the %s value is %s.

### Description

The specified setting is inconsistent between both two tools, and one of the settings is not set to the recommended value for correlation consistency.

Please see the manpages for specified setting for additional details.

### What Next

Change the specified setting to recommended value in both tools.

### See Also

- check_consistency_settings

## CORR-816

(Warning) The %s %s setting %s.

### Description

The setting does not meet the correlation checking rules. Please correct related setting per warning message.

### What Next

Please correct related setting per warning message.

### See Also

- check_consistency_settings

## CORR-817

(Warning) %s.

### Description

The message shows some issues with PPSlave.

### What Next

Please correct related setting per warning message.

Feedback

**See Also**

- check_consistency_settings

# CSL

## CSL-0001

(Warning) Inconsistent source latency values on object_type object.

**Description**

The source latency values defined on the reported object are inconsistent. In order to correctly specify the latency on this object, each “early” value should be smaller than its corresponding “late” value and each “min” value should be smaller than its corresponding “max” value. By definition early/late and min/max values are used for minimum/maximum operating conditions. Having the late/max values being less than the early/min does not allow for proper analysis of the different operating corners of the design. If left as is, this could lead to incorrect delay calculation and analysis.

**ONE POSSIBLE SCENARIO**

Consider a scenario where the clock source latency is set on a clock port ‘CLK’ as follows:

set_clock_latency -source -early -rise -min 11 CLK set_clock_latency -source -early -fall -min 2 CLK set_clock_latency -source -early -rise -max 3 CLK set_clock_latency -source -early -fall -max 4 CLK

set_clock_latency -source -late -rise -min 6 CLK set_clock_latency -source -late -fall -min 7 CLK set_clock_latency -source -late -rise -max 8 CLK set_clock_latency -source -late -fall -max 9 CLK

The -early -min -rise value is larger than the other rise values. This is most likely not the expected behavior for this design and should be investigated. The combination of -early and -min should probably have the smallest latency here.

Constraint consistency will flag a CSL_0001 rule violation to bring the problem to your attention.

**What Next**

The Violation Detail section displays a table containing the various source latency values defined on the reported object. In the above scenario, this is what that table would look like:

*EARLY LATE*

*Min Rise* 11.0000 6.0000

Feedback

*Min Fall* 2.0000 7.0000

*Max Rise* 3.0000 8.0000

*Max Fall* 4.0000 9.0000

The various places in your SDC file where those values are defined are also reported in this section. One can easily find out from the table where the inconsistency is and then use the SDC hyperlinks to navigate to the exact SDC command where the latency is defined.

**FIX SUGGESTION**

Using realistic and consistent data for min/max and early/late latency values will address this problem.

In the preceding example scenario, changing the *-early -rise -min* value to be less than the other set of rise values would fix the problem.

set_clock_latency -source -early -rise -min 1 CLK

---

## CSL-0002

(Warning) Input or output delays are defined relative to clock clock. This clock has incomplete source latency values required for those external delays even though its source ports or pins have more complete source latencies defined on them.

**Description**

Some input and output delays are relative to a clock with incompletely specified source latency. Because of this, paths starting or ending at ports relative to this clock will not have source latency either. The source latency data specified on the clock source (for example the port where a clock is specified) will not be carried over to those paths. This could create incorrect timing results during static timing analysis.

Only latency values applied on clock objects are used for path delay calculations and if these are not specified or are incompletely specified, zero delay values are assumed. The clock source latency values applied on a clock source pin/port will not be used when delays need to be calculated on paths which have ports with external delay settings on them.

This could lead to optimistic delay calculations if the user assumed that the clock source (pin or port) latency values will be used in place of the clock object latency values.

**ONE POSSIBLE SCENARIO**

Consider the following scenario where a clock is defined on an input port

create_clock –period 10 CLK

Feedback

Clock source latencies are defined on both the clock object and the clock source port.

set_clock_latency –source –min 3 [get_clock CLK] set_clock_latency –source 4 [get_port CLK]

An input delay is specified to a port relative to this clock:

set_input_delay 4 –clock CLK –add IN

The input delay set on the *IN* port has been defined with respect to clock *CLK*, but since no source latency is defined on the clock object for the max condition, the clock source latency for maximum delay calculations will be zero. The source latency applied on the port object CLK (with a value of 4) will not be used in place of the clock object source latency here.

Constraint consistency will flag a CSL_0002 violation in this case to notify you that incomplete or missing latency values on a clock object could be problematic.

**What Next**

The “Violation Detail” section displays a table containing the source latency values both for the clock and its sources. In the preceding scenario, the table would look like this:

Source Latency Values on the Clock

*EARLY LATE*

*Min Rise* 3.0000 3.0000

*Min Fall* 3.0000 3.0000

*Max Rise* -- --

*Max Fall* -- --

Source Latency Values on Source ‘CLK’

*EARLY LATE*

*Min Rise* 4.0000 4.0000

*Min Fall* 4.0000 4.0000

*Max Rise* 4.0000 4.0000

*Max Fall* 4.0000 4.0000

**FIX SUGGESTION**

It is recommended that clock latency be completely defined. This would help STA tools computing accurate timing reports.

Feedback

In the example scenario presented above, one easy fix would be to add a -max latency value on the clock object as seen below:

set_clock_latency -source –max 4 [get_clock CLK]

SH "SEE ALSO" set_clock_latency set_input_delay set_output_delay

---

## CSL-0003

(Warning) Clock source latency is specified on object_type object, but that object is not a clock source.

**DSECRIPTION**

Source latency should only be specified on clocks or on clock source pins or ports.

Even though the set_clock_latency command itself does not allow latency values to be set on a non clock source pin or port, if a clock is removed or redefined to a new location, its latency values still persist at the outdated clock source pin or port. This is usually not the designer’s intention when moving clocks.

When a clock is redefined to another clock source in the design, the clock latency values set on its original clock source (pin or port) do not get copied over to the new source location. The clock source latency is ignored and has no value.

This is a situation which frequently arises when as a result of merging different constraint scripts together.

**ONE POSSIBLE SCENARIO**

Consider the following scenario where you move a clock from source port CLK to source port CL. In the original clock definition, a source latency value is specified for the clock.

create_clock -period 10 -name clk1 [get_port CLK] set_clock_latency -source -max 3 [get_port CLK]

One can report the current clocks and latency values specified in the design using the report_clock and report_clock-skew commands. Here are the current values:

rtl_shell> report_clocks

Clock Period Waveform Attrs Sources
---------- clk1 10.00 {0 5} {CLK}

You can see in the report above that the clock “clk1” is defined on port “CLK”.

rtl_shell> report_clocks -skew

Min Condition Source Latency Max Condition Source Latency
---------- Object

Feedback

Early_r Early_f Late_r Late_f Early_r Early_f Late_r Late_f Rel_clk
---------- CLK 3.00 3.00 3.00 3.00 - - - - --

You can see from the report above that clock source latency values exist on port “CLK”, which is also the source of clock “clk1”. Now if one moves the clock “clk1” definition to a different port, using the command below, the clock latency data will not be moved along.

create_clock -period 10 -name clk1 [get_port CL]

rtl_shell> report_clock

Clock Period Waveform Attrs Sources ---------- clk1 10.00 {0 5} {CL}

The source of clock “clk1” has been move from port “CLK” to pot “CL”.

rtl_shell> report_clock -skew

Min Condition Source Latency Max Condition Source Latency
---
Object Early_r Early_f Late_r Late_f Early_r Early_f Late_r Late_f Rel_clk
---------- CLK 3.00 3.00 3.00 3.00 - - - - --

From the preceding report, however, the clock latency is still specified on port “CLK”. No new latency values are available on port “CL”.

**What Next**

The designer should examine the script and make sure the clock redefinition was intentional. If this is the case, any latency specification should also be manually transferred to the new clock source and the old latency values should be removed.

**FIX SUGGESTION**

Remove unnecessary latency values and reapply the old values to the new clock source.

In the preceding example scenario, adding the source latency at port ‘CL’ addresses the problem.

set_clock_latency -source -max 3 [get_clock CL]

In order to remove the violation, the unnecessary clock source latency value (which was specified on port CLK) should be removed. For the example scenario the command would look like this:

remove_clock_latency -source [get_port CLK]

Feedback

## CSL-0004

(Warning) Generated clock clock master clock master_clock is not propagated and has zero or incomplete source latency values.

### Description

The reported clock's master is not propagated. It is expected that realistic source latency values be provided to represent the delay to the clock source. Constraint consistency could not find those latency values. Since no source latency values are specified, timing analysis tools will assume a zero delay value. This might result in incorrect timing analysis.

### ONE POSSIBLE SCENARIO

Consider a scenario where a generated clock gen_clock is defined with respect to a master_clock.

create_clock -period 10 master_clock create_generated_clock -name gen_clock \ -source master_clock gen/Q -divide_by 2

If the master_clock is defined as ideal and not propagated, the generated clock *gen_clock* will have zero source latency values. Constraint consistency flags a CSL_0004 in such a case to warn you that there are potentially missing latency data on those clocks.

### What Next

The "Violation Details" section reports whether the clock latency values are undefined, have a value of zero, or are incomplete.

### FIX SUGGESTION

There are two different possible ways to address this issue depending on the situation. For the example scenario described above, here are the two different approaches to solve this problem:

The master clock was supposed to be propagated but somehow the command was never issued. In this case, propagating the clock will address the issue and the generated clock will inherit the proper latency values.

set_propagated_clock master_clock

The master clock is ideal. In this case, some realistic latency values should be provided in order to ensure correct timing analysis.

set_clock_latency -source 3 [get_clock gen_clock]

Feedback

## CSL-0005

(Warning) Generated clock clock has source latency values less than that of master clock master_clock when the clock edge relationships are considered.

### Description

The source latency of a generated clock is automatically computed when all the various clocks in the design are propagated. When you specify the source latency values for the generated clock, those values should be larger than the source latency of the corresponding master clock. The latency is created by taking into account the edge relationship between the master and the generated clocks.

A generated clock source is further away from the master clock source and therefore has more latency than its master clock. Specifying smaller source latency values than the master clock leads to overly optimistic results.

### ONE POSSIBLE SCENARIO

Consider a scenario where a generated clock ‘gen_clock’ is defined with respect to a master_clock. The generated clock has a divide_by 2 relationship with respect to its master without sense inversion.

create_clock -period 10 -name master_clock create_generated_clock -name gen_clock -source master_clock gen/Q -divide_by 2

Clock source latencies are manually defined as follows for both the master clock and its related generated clock:

set_clock_latency -source -rise 4 [get_clocks master_clock] set_clock_latency -source -fall 1 [get_clocks master_clock] set_clock_latency -source -rise 2 [get_clocks gen_clock] set_clock_latency -source -fall 2 [get_clocks gen_clock]

When looking at the edge relationship between the master clock and its related generated clock, as shown in the following figure, you can see the rising edges of the master clock triggers both the rising and the falling edges of the generated clock.

As consequence, on the generated clock side, the latency (for both rise and fall) should be at least as large as the “rising edge” latency of the master (four in the preceding figure). If the latency values for the generated clock are less, a CSL_0005 violation is issued.

### What Next

The “Violation Details” section reports the various latency data in a table for both the master and the generated clocks.

Master Clock Source Latency

*EARLY LATE*

Feedback

*Min Rise* 4.0000 4.0000

*Min Fall* 1.0000 1.0000

*Max Rise* 4.0000 4.0000

*Max Fall* 1.0000 1.0000

Generate Clock Source Latency

*EARLY LATE*

*Min Rise* 2.0000 2.0000

*Min Fall* 2.0000 2.0000

*Max Rise* 2.0000 2.0000

*Max Fall* 2.0000 2.0000

**FIX SUGGESTION**

To achieve the correct source latency for the generated clock the edge relationships between both the master and the generated clocks should be considered.

In the preceding example, the source latency computation for the generated clock is at least four units (the master clock latency) for both the rise and fall data. In general, the source latency computation of the generated clock is the total of the master clock source latency and any latency between the master clock and the generated clock source; this includes all gates within the path. As an example, the following value would solve the CSL_0005 rule violation:

set_clock_latency -source -rise 5 [get_clock gen_clock] set_clock_latency -source -fall 5 [get_clock gen_clock]

---

## CSL-0006

(Warning) Clock source latency is specified on object_type object with respect to clock clock, but that object is not in the clock's network.

**Description**

The source latency specification is incorrect, the port or pin reported using in the set_clock_latency command is not part of the clock network specified through the -clock option.

If the referenced object is a valid clock source, the CSL_0006 rule verifies that the source latency is specified on an object belonging to the clock network of -clock.

For constraint consistency, the set_clock_latency command generates a UIC-002 error message if the referenced object is not a proper clock source.

Feedback

Since the object where the source latency is specified does not belong to the related clock network, the source latency value will be ignored. This could result in incorrect clock network delay calculations and overly optimistic results.

**ONE POSSIBLE SCENARIO**

Consider a scenario shown in Figure 1, with two different clock networks CLK1 and CLK2. Each of those clock networks originates respectively from ports PORT_CLK1 and PORT_CLK2.

create_clock -period 10 [get_ports PORT_CLK1] -name CLK1 create_clock -period 10 [get_ports PORT_CLK2] -name CLK2 set_clock_latency 2 -source [get_ports PORT_CLK1] -clock CLK2

The set_clock_latency command specifies the source latency for clock network CLK2 at port CLK1. Since CLK1 is not in the clock network CLK2, this latency will be ignored. A CSL_0006 violation is issued as a warning.

**What Next**

The “Violation Details” section reports the file name and the line number where the set_clock_latency command was issued.

**FIX SUGGESTION**

The source latency specification needs to be revised so the clock and its relative clock source match.

In the preceding scenario, if the requirement is to set the clock latency on the port PORT_CLK1 with respect to the clock CLK1, the revised specification would be:

set_clock_latency -source 5 [get_ports PORT_CLK1] -clock CLK1

---

## CSL-0007

(Warning) Source latency defined on pin/port pin_port will overwrite the clock source latency for clock clock.

**Description**

The clock source latency should be specified on either the clock or the clock definition point, but not both at one time. Doing so results in overwriting the clock source latency of the clock and could create incorrect timing results during static timing analysis.

Since the source latency value is changed, it can potentially lead to unexpected and incorrect analysis results, if ignored.

**ONE POSSIBLE SCENARIO**

Feedback

Consider the following scenario where a clock is defined on an input port:

create_clock -name clk1 -period 10 -waveform { 0 5 } [get_ports {clkp}]

A clock source latency is defined on the clock object:

set_clock_latency -source -min 55 -dynamic 11 [get_clocks {clk1}]

set_clock_latency -source -max 55 -dynamic 11 [get_clocks {clk1}]

A clock source latency is also defined on the clock definition point of clk1:

set_clock_latency -source -clock clk1 1000 [get_ports clkp] -early

set_clock_latency -source -clock clk1 2000 [get_ports clkp] -late

This clock source latency on the clkp port will overwrite the source latency of clk1. PrimeTime constraint consistency flags a CSL_0007 in such a case to warn the designer that the source latency of the clock object has been changed so that any unexpected results can be avoided.

**What Next**

The “Violation Details” section displays the clock for which the source latency has been overwritten, along with the clock definition.

**FIX SUGGESTION**

You should check the source latencies applied on the clock definition point and remove the incorrect ones.

In the example scenario presented previously, one easy fix is to remove the source latency applied on the port clkp as follows:

remove_clock_latency -source -clock clk1 [get_ports {clkp}]

# CSTR

## CSTR-001

(warning) The %s analysis mode is not supported; using on_chip_variation\nanalysis mode instead.

**Description**

The bc_wc or single analysis modes are not supported in this tool.

Feedback

**What Next**

Ensure your scripts are written using modes and corners appropriate for on-chip-variation analysis.

---

## CSTR-002

(error) Mode '%s' already exists.

**Description**

The mode has already been created. Each mode must have a unique name.

**What Next**

Use a different name for each mode.

---

## CSTR-003

(warning) The -clock option cannot be specified for clock '%s'. Ignoring option for this clock.

**Description**

This option exists to specify clock objects to be associated with the network latency that is placed on pin/port objects. So providing this option with clock already in the object_list is irrelevant and the execution of the command proceeds as if -clock was not given.

**What Next**

Check if the -clock option is redundant or incorrect.

---

## CSTR-004

(error) '%s' and '%s' are already defined as '%s' not allowing paths.

**Description**

You receive this error message because the -allow_paths defined by asynchronous clock groups is conflict with false path set by either the asynchronous, physically exclusive or logically exclusive clock groups for the same clock pair.

**What Next**

Use the report_clock with -groups option to check what clock groups have been set. To remove the existing clock groups, use the remove_clock_groups command.

Feedback

**See Also**

- set_clock_groups
- remove_clock_groups
- report_clock

---

## CSTR-005

(error) '%s' and '%s' are already defined as -allow_paths in asynchronous clock groups.

**Description**

You receive this error message because the clock pair is already defined as asynchronous clock relationships which allow paths between these two clocks.

**What Next**

Use the report_clock with -groups option to check what clock groups have been set. To remove the existing clock groups, use the remove_clock_groups command.

**See Also**

- set_clock_groups
- remove_clock_groups
- report_clock

---

## CSTR-006

(warning) '%s' %s existing false paths.

**Description**

The set_clock_groups command won't analyze the paths between exclusive and asynchronous clocks. Previous manually defined false paths between these exclusive and asynchronous clocks will be removed by set_clock_groups during remove_clock_groups command.

**See Also**

- set_clock_groups
- remove_clock_groups

Feedback

- report_exceptions
- set_false_path

## CSTR-007

(Information) From and to clocks are exclusive or asynchronous clocks.

**Description**

The set_clock_groups command won't analyze the paths between exclusive and asynchronous clocks. Therefore, the set_false_path between these exclusive or asynchronous clocks is a redundant command.

**What Next**

Use the remove_clock_groups command to remove existing clock groups. Use the report_exceptions command to see the existing false paths.

**See Also**

- reset_path
- remove_clock_groups
- report_exceptions
- set_false_path

## CSTR-008

(warning) Attempting to remove a clock gating check that was not previously set.

**Description**

Removal of clock gating check on an object is valid only if a clock gating check was set before on that object.

**What Next**

This command will be ignored. Please check the spelling of the object for the *remove_clock_gating_check* command.

## CSTR-009

(error) Cannot remove internal path group '%s'.

Feedback

**Description**

You cannot use *remove_path_group* to remove internal path groups (such as the default group).

**What Next**

Reenter the command without the internal path group name.

---

## CSTR-010

(warning) Path group '%s' has no paths; it will not affect optimization.

**Description**

The *group_path* command can be used to create or modify path groups. You can use *group_path -name clock -weight 5* to change the weight of an existing group. This warning is to let you know when you have created a path group that has no paths. This path group will not affect optimization unless paths are added to it by additional *group_path* commands.

**What Next**

You should use *group_path -name clock -to clock* if you wish to include all paths to object "clock" in the path group. Use *report_path_group* to see the path groups that are currently defined.

---

## CSTR-011

(warning) The '%s' command is not supported in this program. The command will be ignored.

**Description**

The command is not supported in this program. The command will be ignored.

**What Next**

Check if the program version is correct.

---

## CSTR-012

(Error) Cannot specify -clock_path or -data_path with objects other than clocks.

**Description**

The command options, -clock_path and -data_path, are provided only for clock objects. These option are not allowed with other object types.

Feedback

**What Next**

Check if the objects passed to the command are clock objects.

---

## CSTR-013

(warning) Removed %s '%s' had a constraint. The constraint is no longer applied.

**Description**

The given database object was removed, but it had a constraint that referred to it. The constraint for the object has now been removed.

**What Next**

If your object was subsequently replaced, you may want to consider reapplying your constraint.

If many constraints were removed and your constraints were from an sdc or tcl file, you can reapply all of constraints by using: "reset_design" followed by resourcing your constraints.

---

## CSTR-014

(error) Cell '%s' has no %s port named '%s'. Pin '%s' cannot be set the same.

**Description**

In order to set the the budget for to pins to be the same, those pins must have the same direction and must be pins of the same hierarchical cell in the netlist. The port you named cannot be found on the cell you have named.

**What Next**

Perhaps the port is the wrong direction, or perhaps you used the full hierarchical pin name instead of the simple port name. The syntax is:

set_budget_pin_options -same_as_pin IN[0] U1/U2/U3/IN[*]

---

## CSTR-015

(error) Corner '%s' already exists.

**Description**

The corner has already been created. Each corner must have a unique name.

Feedback

**What Next**

Use a different name for each corner.

---

## CSTR-016

(error) Scenario '%s' already exists.

**Description**

The scenario has already been created. Each scenario must have a unique name.

**What Next**

Use a different name for each scenario. Note that scenarios without a user-given name are auto-named, and it is possible for a user-given name to be the same as the auto-generated name of a different scenario.

---

## CSTR-017

(information) Full design constraints for the block %s need to be read when changed from a more abstracted view to a less abstracted view.

**Description**

When switching from a more abstracted view to a less abstracted view, timing constraints are not automatically filled in for the core of the block. Only constraints which cross the boundary are guaranteed to be preserved. For example, if you initially had a abstract view, the abstract holds no constraints for the core of a block. When you switch to a design view, it will be necessary to fill in constraints for the core.

**What Next**

Reread your block constraints from a previouly saved SDC file.

---

## CSTR-018

(information) The set_clock_balance_points constraint on block output port %s is not promoted.

**Description**

The 'promote_constraints -cts' command will not promote set_clock_balance_points constraints on output block ports. Such constraints are usually used to allow block-level CTS in the presence of clock outputs. They should not be necessary at the toplevel.

Feedback

**What Next**

If the promotion of the constraint is really required, it needs to be re-applied manually while the toplevel design is current.

---

## CSTR-019

(warning) Removed %s '%s' is the last one of one path exception's -from/-through/-to set. Removing it will cause the whole path exception being dropped.

**Description**

The given database object is removed, but it is the last object in one path exception's -from/-through/-to set, so removing this object will cause the whole path exception being dropped. The path exception including set_min/max_delay, set_false_path, set_multicycle_path, set_path_margin, and group_path.

**What Next**

If your object is subsequently replaced, you may want to re-apply your constraint.

If many constraints are removed and the constraints are from an sdc or tcl file, you can re-apply all of constraints by using "reset_design", followed by sourcing the constraints.

---

## CSTR-020

(information) '%u' cells set to '%s'.

**Description**

The given number of cells have had their size_only or dont_touch status set or cleared.

---

## CSTR-021

(warning) set_load on nets is not supported.

**Description**

For script compatibility the *set_load* command will accept nets, but the command will have no effect.

---

## CSTR-022

(warning) The -library option of set_driving_cell is not supported.

Feedback

### Description

The *-library* option of the *set_driving_cell* command is not supported. Driving_cell references will be resolved using the same *ref_libs* mechanism used for linking the design. The library name given by the *-library* option will be stored and written out by *report_port*, *write_script*, and *write_sdc*, but it will have no other effect on the tool operation.

### See Also

- set_driving_cell
- report_port

---

## CSTR-025

(error) There is already a scenario for mode %s and corner %s.

### Description

A scenario already exists for this mode and corner. There can be only one scenario for any mode-corner pair.

### What Next

Identify the existing scenario for this mode and corner, and either use it, or delete it and re-create a new one.

---

## CSTR-026

(warning) Scenario %s has no analysis types to activate.

### Description

The set_scenario_status command has been given the "-active true" option, which is meant to enable the actual setup, hold, power, and DRC analysis settings. However, none of these settings are turned on, and thus the "-active true" option will have no immediate effect.

### What Next

Turn on some of the actual analysis settings, with the -setup, -hold, etc. options.

---

## CSTR-027

(warning) Scenario %s is inactive, so the specified analysis will not actually be performed.

Feedback

**Description**

The set_scenario_status command has turned on one of the setup, hold, power, or DRC analysis settings. However, the scenario is not marked as active, and thus no analysis will be done for it.

**What Next**

Make the scenario active with the "-active true" option.

---

## CSTR-028

(warning) Constraint on clock was not %s for following scenario(s): %s

**Description**

The command was passed list of scenarios or modes but no clock object specified for the one or more scenarios. This warning is generated if a scenario, default or specified using -modes/-scenarios options, has no clock objects specified. This can happend if a clock object from mode1 is specified with no other clocks in set_max_capacitance/ set_max_transition and remove_max_transition/remove_max_capacitance commands and passed with scenarios of mode1 and mode2. The warning will report all scenarios of mode2 for which constraint was not set.

For example, consider 2 scenarios s1 and s2 with modes m1 and m2 and corners c1 and c2, respectively. Following will generate this warning message for scenario s2 because no clock object from mode m2 was specified.

all_scenarios

{s1 s2}

current_scenario s1 // <-- current_mode is m1

set_max_transition 1 [get_clocks *] -scenarios [all_scenarios]

*get_clocks* will return clocks from mode m1 only and there is no constraint set in scenario s2 for any clock from mode m2.

**What Next**

Make sure the scenarios specified are related to the modes of the clock objects passed.

---

## CSTR-029

(warning) %s -comment has newline and/or '#' and is replaced by ' '.

Feedback

### Description

The specified constraint command -comment line has either newline and/or '#' which is illegal and is replaced by white space ' '.

### What Next

The illegal characters is replaced automatically by white spaces ' '

---

## CSTR-030

(error) Operating condition of name %s not found in any library.

### Description

The given operating condition does not exist in any reference library.

### What Next

Use the *report_lib* command with each reference library to determine the correct operating condition name.

### See Also

- report_lib

---

## CSTR-031

(error) Found multiple versions of the %s operating condition in the %s reference library.

### Description

The specified reference library contains multiple versions of the specified operating condition. This can happen when a reference library is built from multiple logic libraries, and two or more of them have an operating condition with the same name.

### What Next

Use the *-lib* option to specify one of the logic libraries that was used to build the reference library. Use the *report_lib* command to understand what operating conditions the reference library contains.

### See Also

- report_lib

Feedback

## CSTR-032

(error) Multiple versions of operating condition %s found among the design's reference libraries.

### Description

Multiple versions of the given operating condition exist among the reference libraries of the design.

### What Next

Use the *-lib* option to specify a particular library or a particular .db file that was used to build the libraries. Use the *report_lib* command with the reference libraries to understand what operating conditions they contain.

### See Also

- report_lib

## CSTR-033

(error) %s is not the name of any library, and does not match any .db file used to build any library.

### Description

The given name does not identify any reference library, and it also does not identify any .db file used to build any library.

### What Next

Use the *get_libs* command to determine which reference libraries are loaded. Use the *report_lib* command to determine which .db files were used to build each reference library.

### See Also

- get_libs
- report_lib

## CSTR-034

(warning) Multiple references to .db library %s found among the design's reference libraries.

Feedback

### Description

The given .db library has apparently been used to build more than one of the design's reference libraries, and multiple copies of its operating conditions have been found.

### What Next

Use the *report_lib* command on all reference libraries to understand what .db files they were built from, and what operating conditions they contain.

### See Also

- report_lib

---

## CSTR-035

(error) Cannot specify -db with objects other than lib_pins.

### Description

The option,-db, is used to specify .db library or file name for lib_pin objects. This error is generated when it is used with objects other than lib_pins.

### What Next

Check the object list and remove objects that are not lib_pins.

### See Also

- set_max_transition
- set_max_capacitance
- set_min_capacitance
- report_lib

---

## CSTR-036

(error) Specified .db library %s not found for the lib_pin.

### Description

The given .db library was not found in library pane for the lib_pin.

### What Next

Use the *report_lib* command on reference libraries to understand what .db files were used.

Feedback

### See Also

- report_lib
- set_max_transition
- set_max_capacitance

---

## CSTR-037

(error) The .db library name %s matches multiple libraries. Please specify unique .db library name.

### Description

The given .db library name doesn't specify a unique .db library file name. The command expects a unique .db name. If multiple .db libraries have same local names then use full path name of the .db file.

### What Next

Use the *report_lib* command on relevent reference libraries to understand what .db files they were built from and what pvt and pane they contribute to.

### See Also

- report_lib
- set_max_transition
- set_max_capacitance
- set_min_capacitance

---

## CSTR-038

(information) Automatically promoting voltage %.2f for supply net '%s' in corner %s from the block constraints of '%s'.

### Description

The given supply net corresponds to a voltage region that is isolated from the top level. And there is no set_voltage command in the top-level constraints that assigns a voltage for the region.

The same net was assigned a voltage in the block-level constraints. That value will be used for top-level timing and optimization.

Feedback

### What Next

If you wish to use a different voltage value, you should assign the voltage directly in the top-level constraints.

### See Also

- set_voltage

---

## CSTR-039

(warning) Process label %s is not used by the reference library %s.

### Description

The given process label is not used by the given library, so specifying it will have no effect.

### What Next

Use the *report_lib* command with the reference library to determine which process labels it uses.

### See Also

- report_lib

---

## CSTR-040

(warning) Process label %s is not used by any reference library.

### Description

The given process label is not used by any library, so specifying it will have no effect.

### What Next

Use the *report_lib* command with the design's reference libraries to determine which process labels they use.

### See Also

- report_lib

---

## CSTR-041

(error) Process labels may not contain embedded whitespace.

Feedback

### Description

The given process label contains embedded whitespace, which is not allowed.

### What Next

Use the *report_lib* command with the design's reference libraries to determine which process labels they use.

### See Also

- set_process_label

---

## CSTR-042

(error) %s may not contain embedded whitespace.

### Description

The given identifier contains embedded whitespace, which is not allowed.

### What Next

Ensure that identifiers such as object names are a single word, without any embedded spaces, tabs, or newlines.

---

## CSTR-050

(warning) Cannot set %s on hierarchical pin, '%s'.

### Description

THis warning message is generated when a hierarchial pin is passed to set_max_transition or set_max_capacitance commands. The commands support constraints only on leaf pins. Hierarchical pins are ignored with this warning. This usually happens when a collection of pins, with a mix of leaf and hierarchical pins, is passed to the commands.

### What Next

User can just ignore the warning message or filter out all the hierarchical pins from the collection passed to the comamand for clean log file.

### See Also

- set_max_transition
- set_max_capacitance

Feedback

## CSTR-051

(Error) Can't find valid HyperScale context data under given path: %s

**Description**

The HyperScale context data can't be found in the path given -to *read_hyperscale_context*, with *-path* option.

With a HyperScale directory generated by PrimeTime, you can specify the top directory, or TOP/CONTEXT, or TOP/CONTEXT/BLOCK_NAME, or TOP/CONTEXT/BLOCK_NAME/HyperScale_Default.

**What Next**

Make sure the given path for HyperScale context is correct.

## CSTR-052

(Information) The mode '%s' has been loaded with other scenarios, skipping the mode data loading for scenario '%s'.

**Description**

The HyperScale context has already been loaded for the mode, with some other scenarios. In MCMM with HyperScale, we expect the HyperScale contexts have exact same mode related data for the scenarios of the same mode. So the mode data is loaded only once for the first HyperScale scenario context of that mode, and loading is skipped for following scenarios.

**What Next**

No action needed.

## CSTR-053

(Warning) In mode '%s', HyperScale clock '%s' can't be found in block level SDC.

**Description**

When HyperScale timing context is generated, the clock name used in the timing context will be the top level clock name, which could be different from the name used in the block level implementation (usually constraints generated by budgeting tool).

**What Next**

To correct this, before *update_timing*, use *set_clock_map* to provide a name mapping between top and block level clocks, so tool can resolve the clock names correctly.

Feedback

## CSTR-054

(Warning) In mode %s, clock (%s)'s properties are different in different HyperScale timing contexts for the same mode.

**Description**

The HyperScale timing contexts carries mode specific information like clock definitions. When multiple scenarios of the same mode have HyperScale timing context loaded with *read_hyperscale_context*, we expect the mode related information from those timing contexts to be consistent.

**What Next**

To correct this, make sure when the scenario timing contexts is generated in PrimeTime, they are actually using constraints from same mode.

## CSTR-055

(Information) HyperScale context exists for scenario '%s', will be overwritten.

**Description**

The given scenario already has HyperScale context loaded. This *read_hyperscale_context* will overwrite existing context.

**What Next**

No action needed.

## CSTR-056

(Warning) ignore *set_dont_override* on the virtual clock: '%s'.

**Description**

The virtual clock is not modeled in HyperScale. We can't apply option *setDontOverride* on virtual clocks.

**What Next**

No action needed.

## CSTR-057

(Warning) In mode %s, clock (%s)'s properties are different from those captured in HyperScale timing contexts.

Feedback

### Description

The HyperScale timing contexts carries mode specific information like clock definitions. There is mismatch between the SDC clock definition and HyperScale timing context.

### What Next

To correct this, make sure when the scenario timing contexts is generated in PrimeTime, they are actually using constraints from same mode. You can also modify SDC clock definition to make them consistent.

### See Also

- create_clock
- create_generated_clock

---

## CSTR-058

(Information) No HyperScale context exists for scenario %s.

### Description

The given scenario doesn't have HyperScale context loaded so nothing is removed with this *remove_hyperscale_context*.

HyperScale context can be loaded with *read_hyperscale_context*.

### What Next

No action needed.

---

## CSTR-059

(Warning) '%s' has inconsistent mode data in scenario '%s'.

### Description

In MCMM HyperScale, it's expected that the HyperScale contexts for different scenarios should have exact the same mode specific information, like clock, exception, disable timing, case value, etc.

The warning message indicate that for the mentioned pin or clock, inconsistent mode data is detected in the mentioned scenario, compared with the scenario of the same mode which is previously loaded.

Feedback

**What Next**

Check HyperScale contexts for the scenarios of the same mode, make sure they have same mode-specific settings.

---

## CSTR-060

(Information) In mode '%s', HyperScale clock '%s' is mapped to block level SDC clock '%s', based on '%s'.

**Description**

When HyperScale timing context is generated, the clock used in the timing context will be the top level clock, it's mapped to block level SDC clock during block level timing analysis, either based on *set_clock_map* or auto matching based on clock properties like period, waveform, source pins, propagated status, etc.

**What Next**

This is informative, no actions needed.

---

## CSTR-061

(Information) HyperScale data for mode '%s' is cleared after set_clock_map.

**Description**

The clock mapping between HyperScale top-level clock and SDC block-level clock will change the clock resolution in HyperScale data loading, so the loaded HyperScale data is cleared and needs to be reloaded.

**What Next**

No action needed.

---

## CSTR-062

(Warning) In mode '%s', HyperScale exception is dropped because it's invalid in block level.

**Description**

If the objects (clock or netlist objects) referred by the HyperScale exception captured in top-level are missing, exceptions could become invalid and will be dropped.

Feedback

**What Next**

Ensure that the netlist and constraints are consistent between HyperScale generation and block level, especially clock definition. *set_clock_map* would be needed to resolve missing matching clock objects.

---

## CSTR-063

(Error) Can't find top-level clock named '%s' in loaded HyperScale context for mode '%s'.

**Description**

The given top-level clock name in *set_clock_map* can't be found in the HyperScale context.

**What Next**

Make sure the name is correctly specified.

---

## CSTR-064

(Information) In mode '%s', SDC clock '%s' is mapped to top level HyperScale clock '%s', based on '%s'.

**Description**

When HyperScale timing context is generated, the clock used in the timing context will be the top level clock, it's mapped to block level SDC clock during block level timing analysis, either based on *set_clock_map* or auto matching based on clock properties like period, waveform, source pins, propagated status, etc.

**What Next**

This is informative, no actions needed.

---

## CSTR-065

(Error) Ignore user set clock mapping for clock '%s' in mode '%s', because the top level clock '%s' doesn't exist in context.

**Description**

This message is issued when HyperScale context is loaded, so that the user set clock mapping can be checked to see whether the clocks exist in the context. If not, the clock map setting for that clcok will be ignored.

Feedback

### What Next

Check the clock name used in *set_clock_map -parent_clock parent_clock_name*, make sure the clock does exist in the context.

---

## CSTR-066

(warning) No corresponding clock for top-level clock '%s' in mode '%s' with period %s and waveform %s.

### Description

In HyperScale analysis, clocks from block level have to be mapped to clocks defined at top level in order to properly interpret the timing data and constraints. If this mapping fails for any clock, the warning is issued.

### What Next

To see the details of failed clock mapping, please use *report_clock -map*. To specify correct clock mapping, use *set_clock_map*.

### See Also

- report_clock

---

## CSTR-068

(Information) HyperScale data for %s '%s' is cleared.

### Description

When HyperScale related settings are changed, the loaded HyperScale data is cleared, next timing update will reload those data if needed.

### What Next

No action needed.

---

## CSTR-069

(Warning) %s context data at port '%s' for pin '%s' because it is missing or not on the same net.

### Description

This message is issued during context override at the HyperScale block level analysis, where there is a netlist mismatch found on the port net between current block design and

Feedback

HyperScale top level context data. A best-effort context override will be applied to the related port.

**What Next**

No action required. To ensure accurate context override, please use consistent block and top netlist.

---

## CSTR-070

(Warning) The pin '%s' is no longer connected to port '%s'. Its context value will be applied to pin '%s' instead.

**Description**

At block, ECO fixing (such as buffer insertion and removal) on I/O ports can sometimes cause the pin that context was initially captured on to no longer exist. In this case, the context value will be applied on the new load pin of the port as informed in the warning message.

**What Next**

Check whether the netlist change is intended.

---

## CSTR-071

(Warning) Ignore context data at port '%s' because it's missing.

**Description**

This message is issued during context override at the HyperScale block level analysis, where there is a netlist mismatch found on the port net between current block design and HyperScale top level context data. The context data is ignored because the port can't be found.

**What Next**

No action required. To ensure accurate context override, please use consistent block and top netlist.

---

## CSTR-072

(Warning) Netlist object (pin/port) '%s' referred in HyperScale context is not in the design.

**Description**

The given object was in the design at the time the HyperScale context was captured, but the object is gone now. The constraint refering to the object will be dropped.

Feedback

If the object is no longer important, then your constraints are OK.

**What Next**

If needed, regenerate the HyperScale context with consistent netlist.

---

## CSTR-081

(warning) Missing or unusable driving_cell '%s'.

**Description**

The specified library cell could not be found, or is not suitable for use as a driving_cell. This may happen if no library in the ref_lib list contains the cell, if the specified output or input pins do not exist, if no pins were specified and the cell has multiple output pins, or if no suitable cell arcs exist to provide the drive capability.

**What Next**

If the driving cell requires a library that has not been loaded and included in the ref_lib list, the ref_lib list should be changed with *set_ref_libs* to include that library. Otherwise, check the command options for errors in cell name or pin names.

---

## CSTR-100

(warning) Propagated clock %s is written as ideal and source %s as propagated causing another clock %s to become propagated in scripts.

**Description**

This warning is generated by write_sdc and write_script for formats that cannot support io path latency for propagated clocks. In such cases, propagated clocks are written out as ideal and their sources as propagated.

To generate timing comparable script for io paths, the clock sources are generated as propgated and the clocks are generated as ideal. But, when a propagated clock source is also a source for another ideal clock then the tools can not be compared without first changing the ideal clock to propagated.

**What Next**

The generated script can be used to compare timing for clocks only after the reported ideal clock is converted to propagated in this tool.

Alternatively, convert all the clocks from the reported source to propagated or all the clocks to ideal and re-generate the scripts.

Feedback

## CSTR-101

(warning) Ideal clock %s has zero network latency but non-zero io path latency.

**Description**

This warning is generated by write_sdc and write_script when the tool generates network latency for io paths only. In general, network latency is same as io path network latency for ideal clocks but remove_propagated_clock on a propagted clock can change a clock to ideal with zero network latency and non-zero io path latency. To generate timing comparable script for io paths, the clock or the sources should be changed to propgated.

**What Next**

The generated script can be used to compare timing only after the reported ideal clock is converted to propagated or the clock sources made propagated in this tool. For other tools, the clock sources should be made propagated.

Alternatively, convert the reported clock or the clock sources to propagated in this tool and generate the scripts again. If the intent is to use the clock as ideal then set network latency for the clock using set_clock_latency command and re-generate the scripts.

## CSTR-102

(Error) Failed to create physical context, '%s.'

**Description**

Command failed to create design physical context. Check if the object already exist or incomplete command arguments specified.

**What Next**

If you want to modify the existing physical context then remove and recreate.

## CSTR-103

(Error) Can't find physical context, '%s.'

**Description**

Command failed because no pysical context was found by name in the design. Use report command to find existing design physical contexts.

**What Next**

Correct physical context name or use different command option.

Feedback

## CSTR-104

(Error) Physical context '%s' exists for process label '%s.'

### Description

This error is generated when trying to create multiple physical contexts for a process label. Only one physical context is allowed for a process label in the design and a valid physical context already exists for the process label.

### What Next

Create physical context for different label or remove existing physical context first.

## CSTR-105

(warning): Constraint loading found unsupported physical context format, '%c' and skipped.

### Description

Constraint loading has skipped unsupported format found in the current version. Valid formats are: M - physical contexts and G - physical context groups

### What Next

Use the correct version of the tool or recreate data. Ignore if physical context are not needed.

### See Also

- open_block

## CSTR-106

(Error): Physical context group could not be set. Reason: %s.

### Description

Physical context group was not set becauase of one of the following reasons. Reason:

1. Physical context overlap: One or more physical contexts are already used by another group in the corner. 2. Less than two in a group: Specified less than two valid physical contexts in a group specified. Minimum of two contexts are required.

### What Next

Fix the issue by removing overlap or specify minimum of two contexts.

Feedback

**See Also**

- open_block

---

## CSTR-108

(Warning): Physical context group(s) removed for corner %s.\n %s

**Description**

Physical context group(s) removed for the corner becauase of one or more of the physical contexts were removed by remove_physical_context command.

**What Next**

Set process physical context groups for the corner again using remaining or new physical contexts.

---

## CSTR-400

(error) Loading of constraint configuration failed.

**Description**

An error occured while reading information about the saved design constraints. It is possible that the database is corrupted and the constraints are now incorrect.

**What Next**

You should consider clearing your constraints with reset_design, and then reloading the constraints from your original tcl or sdc file.

---

## CSTR-401

(error) Hierarchy instance '%s' was in the netlist when constraints were saved, and is not in the netlist now.

**Description**

The configuration of the netlist included the given instance when the design was saved. The configuration has changed now -- most likely because design linking failed. Old constraints that refered to the missing instance will be dropped.

**What Next**

If your subblock is no longer needed, then your constraints are still OK for the netlist that still remains. You will need to take action to restore the dropped constraints if the missing hierarchy is ever restored.

Feedback

If the missing hierarchy should not be missing, then a more serious problem has occured. You should fix the hierarchy and reload the constraints.

---

## CSTR-402

(error) Errors have occurred during constraint loading.

**Description**

There is a mismatch between the constraints on the disk and the configuration of the netlist in memory. So it is likely that the constraints on disk are not complete.

**What Next**

You can use "load_constraints -force" to ignore this error and read the constraints as they are. Or you can try "load_constraints -promote" to try to fill in missing top-level constraints with saved constraints from the lower level blocks.

---

## CSTR-403

(warning) For hierarchy instance '%s', constraints were stored for design '%s:%s', but the current design is '%s:%s'.

**Description**

This warning occurs when you design configuration has change since the time that the design was last saved.

**What Next**

If this message was unexpected, you should check your ref_lib specifications to see if they are correct.

---

## CSTR-404

(warning) Hierarchy instance '%s' was added to the netlist since constraints were saved.

**Description**

The saved constraints for this design do not cover the given new instance hierarchy.

**What Next**

You should consider adding constraints for the new hierarchy.

Feedback

## CSTR-405

(warning) Hierarchy instance '%s' changed from view '%s' to view '%s' since constraints were saved.

**Description**

The saved constraints for this design might not cover the given new hierarchy view.

**What Next**

You should consider adding constraints for the new hierarchy.

## CSTR-406

(warning) Hierarchy instance '%s' changed from view 'design' to view '%s'.

**Description**

The new hierarchy instance has fewer objects than the original. Your constraints are still OK for the objects that remain, but the constraints that pointed to the missing instances will now be dropped.

**What Next**

You will need to take action to restore the dropped constraints if the missing hierarchy is ever restored. The constraints can be reapplied from the top.

## CSTR-407

(warning) Constrained %s '%s' is not in the design.

**Description**

The given object was in the design at the time the constraints were saved, but the object is gone now. The constraint refering to the object will be dropped.

If the object is no longer important, then your constraints are OK.

**What Next**

If the constraint refering to the object was important, you can apply new constraints.

## CSTR-408

(warning) Constraint promotion for instance '%s' cannot be performed because top-level %s '%s' is not mapped to a block %s.

Feedback

**Description**

Before you can promote constraints from a lower level block, you must provide a mapping between block and top constraints. The given object is not mapped.

**What Next**

Use set_block_to_top_map to set the mapping.

---

## CSTR-409

(warning) Constraint promotion for instance '%s' cannot be performed because %d block-level clock(s) are not mapped to a top-level clock in mode '%s'.

**Description**

Before you can promote constraints from a lower level block, you must provide a mapping between block and top constraints. The given clock is not mapped.

**What Next**

Use set_block_to_top_map to set the mapping. Use report_block_to_top_map to see which clocks need to be mapped.

---

## CSTR-410

(information) Promoting constraints for instance '%s'.

**Description**

Constraints are being promoted from the lower-level instance to the top-level design. This promotion may have been triggered automatically when the configuration of your design hierarchy changed, and constraints needed to be filled in.

---

## CSTR-411

(Error) Promoting constraints for instance '%s' failed.

**Description**

An error occured while promoting constraints from the lower-level instance to the top-level design. The constraints have not been promoted or are only partially promoted.

**What Next**

Check the error messages that preceded this one for more details. One common reason for promotion to fail is that the modes, clocks and/or corners of the block could not be mapped to their counterparts at the top level.

Feedback

## CSTR-412

(Error) %s '%s' is not an independently linkable, hierarchical block.

**Description**

This operation can only be done on hierarchical instances, where the instance or reference points to a block in a seperate design. If the instance is not hierarchical, or the instance is only logical hierarchy within a single design, this operation cannot be performed.

**What Next**

You could consider using commit_block to break out this hierarchy into another design. That is to say -- you could make a new physical hierarchy.

If the instance should already point to a hierachical block, then double check to make sure that your design has successfully linked. Perhaps a reference lib is missing.

## CSTR-413

(Information) No map information for %s '%s'.

**Description**

No hierarchy mapping information has been recorded for the given object. The block clocks must be mapped before constraint promotion can be done.

**What Next**

You can set map information with the set_block_to_top_map command.

## CSTR-414

(warning) Block currently has no constraints defined. Assuming block and top clock names are the same.

**Description**

It is possible that clocks defined in your blocks have names that do not match connected top-level clock names. For example, a top level clock named "SYS_CLK" may be connected to a block-level clock named "CLK". The *write_budget* command understands that names may not match, and will automatically derive the mapping between top-level and block-level clocks. This allows *write_budget* to generate the proper constraints for the block.

In order for write_budget to properly map clock names, it is necessary for constraints to already be loaded for your block. Since your block currently has no defined constraints,

Feedback

name mapping cannot be performed. Write_budget will assume that your block-level and top-level clock names match. If this is a bad assumption, then the constraints generated by write_budget will be incorrect.

**What Next**

Load your regular block constraints and call write_budget again. This will avoid this warning message, and ensure that clock names are properly mapped between block and top.

---

## CSTR-415

(Warning) %s '%s' is not an independently linkable, hierarchical block. Timing constraints may be incorrect.

**Description**

The change_abstract command attempts to preserve timing constraints which are applied to a hierarchical block views such as "design" and "abstract". When you switch between these two views, constraints at the boundaries of the blocks will be properly maintained.

You have switched from a non-hierarchical block to a hierarchical "abstract" or "design" view. The constraints that were applied to your original non-hierarchical view are probably not suitable for the new hierarchical view that you have inserted. So your timing constraints are now probably incorrect.

**What Next**

You probably need to reread your design constraints from your original top-level SDC before timing reports can be reliable.

---

## CSTR-416

(Error) You cannot specify a clock edge without a clock.

**Description**

When constraining with a clock edge, you must specify a clock or clock group. For example, *::rise and *::fall are not supported.

**What Next**

If you want to specify a clock edge, please also specify a specific clock or clock group.

---

## CSTR-417

(Error) You may not specify clock groups based on generated clock '%s'.

Feedback

**Description**

You have specified a clock group like my_gclock+. Clock groups can only be specifed on non-generate clocks. The group will implicitly include all generated clocks that transitively depend on the regular clock.

**What Next**

If you want to specify a clock edge, please also specify a non-generated clock.

---

## CSTR-418

(Error) Clock groups cannot be specified for specific blocks or clock pins.

**Description**

You have supplied a clock group specification similar to clk+:block or clk+:block:rise. This is not supported. You may only use clock groups in cases where no block or pin is specified. The following specifications are acceptable:

clk+ clk+::rise clk+::fall.

**What Next**

Please use clock groups without blocks or pins.

---

## CSTR-419

(Error) value %s for option '%s' is invalid: must be '%s'.

**Description**

Multi-Input Switching (MIS) coefficients can only be applied to combinational cell with more than 2 input pins, while the given lib cell is either a sequential one or with less than 2 input pins. You can use lib cell's attribute 'is_combinational == true' to filter the sequential lib cells.

**What Next**

Supply a compatible value for the argument.

---

## CSTR-420

(error) Found invalid objects during constraint loading.

**Description**

There is a mismatch between the constraints on the disk and the configuration of the netlist in memory. Some netlist objects may have been deleted and updated constraints

Feedback

is not saved to disk. The error line and column(s) numbers printed refer to the constraint saved on disk. The constraint command with missing objects skipped is printed.

**What Next**

You can compare the constraint command printed to the original and locate what is missing. In the case missing or no arc, please check design linking errors. You can also consult CSTR-402.

---

## CSTR-430

(information) Promoting clock_data for instance '%s'.

**Description**

Clock data is being promoted from the lower-level instance to the top-level design.

---

## CSTR-431

(Error) Promoting clock_data for instance '%s' failed.

**Description**

An error occured while promoting clock-data from the lower-level instance to the top-level design. The data has not been promoted or is only partially promoted.

**What Next**

Check the error messages that preceded this one for more details.

---

## CSTR-435

(warning) For instance '%s', %d block-level clock(s) are not mapped to a top-level clock in mode '%s'.

**Description**

In order to promote clock balance points from a lower level block through promote_clock_data command, you must provide a mapping between block and top constraints. The given clock is not mapped. Since clocks are not mapped, command will only promote clock-independent balance points.

**What Next**

Use set_block_to_top_map to set the mapping. Use report_block_to_top_map to see which clocks need to be mapped.

Feedback

### See Also

- promote_clock_data
- set_block_to_top_map

---

## CSTR-438

(warning) Clock Mapping Failed. Promoting clock independent balance points (if any) for instance '%s'.

### Description

While promoting clock balance points from a lower level block through promote_clock_data command, clock mapping has failed between block and top. Since clocks are not mapped, command will only promote clock-independent balance points.

### What Next

In order to promote clock balance points also, use set_block_to_top_map to set the mapping. Use report_block_to_top_map to see which clocks need to be mapped.

### See Also

- promote_clock_data
- set_block_to_top_map

---

## CSTR-439

(Warning) In mode '%s' case value conflict at pin '%s'. Block value of '%s' will be used.

### Description

A case value is different from the value given to set_case_analysis at this pin for this mode. The case value set at Top will be ignored and the value set at Block will be used.

### What Next

Check that your constraint files are meant to be used for this mode.

---

## CSTR-440

(information) Applied size_only to '%s' but %s permitted.

Feedback

**Description**

User size_only restriction is applied to the specified cell but constant or unloaded cell gobbling, or constant propagation through the cell is permitted.

---

## CSTR-441

(warning) For instance '%s', based on connectivity and IO delay block-level virtual clock(s) are mapped to a top-level clock in mode '%s'. User to verify.

**Description**

The IO delays associated with the virtual clocks are used to map the block level virtual clocks.

**What Next**

Use set_block_to_top_map to set the mapping. If user is aware of which clocks need to be mapped.

---

## CSTR-442

(Information) In auto POCV scaling, voltage settings in OCVM tables of lib cell are ignored.

**Description**

When application option *time.ocvm_auto_pocv_scaling* is turned on and object_type is lib cell in POCV side file, the voltage settings are ignored. The exact matching and scaling for coefficient and distance derate are performed based on the panes specified in the object_spec.

**What Next**

This is informative, no actions needed.

---

## CSTR-443

(warning) Dropping clock_groups '%s' during promotion because one of the group has only virtual clocks which are either unmapped or driving the ports in the mode '%s'.

**Description**

You receive this warning message because one of the clock groups has only virtual clocks which are driving the ports.

Feedback

### What Next

Use the report_clock with -groups option to check what clock groups have been set. To remove the existing clock groups, use the remove_clock_groups command.

### See Also

- set_clock_groups
- remove_clock_groups
- report_clock

---

## CSTR-444

(warning) %s and %s in clock group %s will trigger clock exclusivity outside the block in the mode '%s'.

### Description

You receive this warning message because the clock pair will trigger clock-exclusivity outside the block.

### What Next

Use the report_clock with -groups option to check what clock groups have been set. To remove the existing clock groups, use the remove_clock_groups command.

### See Also

- set_clock_groups
- remove_clock_groups
- report_clock

---

## CSTR-445

(warning) exclusivity between %s and %s has been established in clock group %s due to promotion in the mode '%s'.

### Description

You receive this warning message because exclusivity between the clock pair has been established due to promotion.

Feedback

### What Next

Use the report_clock with -groups option to check what clock groups have been set. To remove the existing clock groups, use the remove_clock_groups command.

### See Also

- set_clock_groups
- remove_clock_groups
- report_clock

---

## CSTR-446

(warning) %s Top has clock group between user-mapped clocks %s and %s but block does not have a corresponding clock group in the mode '%s'.

### Description

You receive this warning message because the user-mapped clock pair has clock group at the top, whereas block does not have a corresponding clock group.

### What Next

Use the report_clock with -groups option to check what clock groups have been set. To remove the existing clock groups, use the remove_clock_groups command.

### See Also

- set_clock_groups
- remove_clock_groups
- report_clock

---

## CSTR-447

(warning) %s Conflicting clock group type between clocks: %s, %s and their master/top level clocks in the mode '%s'.

### Description

You receive this warning message because the clock pair has a conflicting clock group type with respect to their master/top level clocks.

Feedback

### What Next

Use the report_clock with -groups option to check what clock groups have been set. To remove the existing clock groups, use the remove_clock_groups command.

### See Also

- set_clock_groups
- remove_clock_groups
- report_clock

---

## CSTR-448

(error) %s Same clock group exists with a different clock group type in the mode '%s'.

### Description

You receive this error message because the exactly same clock group exists with a different clock group type.

### What Next

Use the report_clock with -groups option to check what clock groups have been set. To remove the existing clock groups, use the remove_clock_groups command.

### See Also

- set_clock_groups
- remove_clock_groups
- report_clock

---

## CSTR-449

(Information) In mode %s, dropping %s latency for block clock %s because it is mapped to existing top clock %s.

### Description

During promote_constraints the source latency set on block clock is dropped if its mapped to existing top clock as it will be over ridding the actual top clock latency.

Feedback

## CSTR-450

(Information) In mode %s, dropping %s latency for block clock %s because it is mapped to existing top clock %s.

**Description**

During promote_constraints the network latency set on block clock is dropped if its mapped to existing top clock as it will be over ridding the actual top clock latency.

## CSTR-451

(Information) In mode %s, promoting clock uncertainty from block clock %s to %s even though its master clock also has uncertainty.

**Description**

During promote_constraints uncertainity set on block clock is promoted if the mapped top clock is a generated clock.

## CSTR-452

(Information) In mode %s, dropping clock uncertainty from block clock %s because it is mapped to existing top clock %s.

**Description**

During promote_constraints the uncertainity set on block clock is dropped if it is mapped to the existing top clock to avoid the top clock uncertainity being overridden.

## CSTR-453

(warning) Overwriting top level derates of cell %s with block level derates in the corner '%s'.

**Description**

You receive this warning message because top level derates of the cell is overwritten with block level derates.

**What Next**

Use report_timing and report_timing_derate to check what derates have been set. To reset derate factors, use the reset_timing_derate command.

Feedback

**See Also**

- report_timing
- report_timing_derate
- reset_timing_derate

---

## CSTR-454

(warning) Skipping promotion of block level derates of lib cell %s in the corner '%s'.

**Description**

You receive this warning message because block level derates of the lib cell is skipped during promotion.

**What Next**

Use report_timing and report_timing_derate to check what derates have been set. To reset derate factors, use the reset_timing_derate command.

**See Also**

- report_timing
- report_timing_derate
- reset_timing_derate

---

## CSTR-455

(Information) In mode %s, dropping input/output delays specified on port %s, as it is estimated in the absence of top.

**Description**

During promote_constraints we promote the block level constraints to top, while we drop the input/output delays set on ports.

---

## CSTR-456

(Information) In mode %s, dropping clock transition for block clock %s because master clock %s is propagated.

Feedback

**Description**

During promote_constraints if master clock is propagated for a mode then clock transition for block clock is dropped for that particular mode.

**What Next**

Stop Master Clock Propagation to the top for a mode to avoid dropping of block clock transition for that mode during constraints promotion.

---

## CSTR-457

(Information) In mode %s, dropping source latency for block clock %s because master clock %s is propagated.

**Description**

In promote_constraints we are dropping source latency along with dynamic latency, as after promotion there is no need to estimate the source latency if the master clock is propagated from top.

---

## CSTR-458

(Warning) In mode %s dropping set_clock_gating_check on %s. The object should not be hierarchical-cell.

**Description**

During promote_constraints if the instance is a hierarchical cell then set_clock_gating_check is dropped for that mode.

**What Next**

Use report_clock_gating to check what clock gatings have been set. To set clock gating check, use the set_clock_gating_check command.

**See Also**

- report_clock_gating
- set_clock_gating_check

---

## CSTR-459

(warning) Dropping set_latency_adjustment_options during promotion because block clock '%s' is a virtual clock which is driving a port in the mode '%s'.

Feedback

### Description

You receive this warning message because the clock referred in the *-clock_to_update* option in *set_latency_adjustment_options* is a virtual clock and driving a port in the block.

### See Also

- set_latency_adjustment_options

---

## CSTR-460

(warning) Dropping group_path '%s' during promotion because one of the group has only virtual clocks which are driving the ports in the mode '%s'.

### Description

You receive this warning message because one of the path groups mapped to top has only virtual clocks which are driving the ports.

### What Next

Use the report_path_groups to check what path groups have been set. To remove the existing clock groups, use the remove_path_groups command.

---

## CSTR-461

(Information) In mode %s, dropping clock uncertainty from block clock %s because it is an unmapped clock.

### Description

During promote_constraints the uncertainity set on a block clock which is unmapped is dropped as the block clock itself is dropped.

---

## CSTR-462

(Information) In mode %s, dropping %s latency for block clock %s because it is an unmapped clock.

### Description

During promote_constraints the source or the network latency set on a block clock which is unmapped is dropped as the block clock itself is dropped.

Feedback

## CSTR-463

(Information) In scenario %s, dropping %s %s for block clock %s because it is an unused clock.

### Description

During promote_constraints, the max_transition / max_capacitance set on a block clock which is unused is dropped as the block clock itself is dropped.

### What Next

To remove the existing max_transition, use the remove_max_transition command. To remove the existing max_capacitance, use the remove_max_capacitance command.

## CSTR-464

(warning) '%s' exception is different between the block clock '%s' and the top clock '%s' in the mode '%s'.

### Description

You receive this warning message because one of the exceptions is different between the top clock and the block clock. Multiple contrasting exceptions are being promoted for the same clock.

### What Next

Use the command report_exceptions to check what exceptions have been specified.

## CSTR-465

(Warning) In mode '%s', removed %s clock from '%s' design during promote_constraints, since this clock is mapped by the user which is defined on the internal pin of the promoted block.

### Description

During promote_constraints, the top clock cannot be mapped with a block clock, when it is created on an internal pin[a level deeper inside the block].

### What Next

Define the top clock on an hierarchial pin or on the top design and then map with right clock using set_block_to_top_map. Also see report_clocks, promote_constraints, set_block_to_top_map.

Feedback

## CSTR-466

(Information) Dropping clock latency during promotion for block clock '%s' because it is a virtual clock which is driving a port in the mode '%s'.

**Description**

You receive this message because the block clock has delays set on port and promoting this shall affect the mapped existing top clock latency.

## CSTR-467

(Information) Dropping clock uncertainity during promotion for block clock '%s' because it is a virtual clock which is driving a port in the mode '%s'.

**Description**

You receive this message because the block clock here has delays set on port and promoting this shall affect the mapped existing top clock uncertainity.

## CSTR-468

(warning) The '%s' exception at '%s', exists on top and hence dropping in the mode '%s'.

**Description**

You receive this warning message because one of the exceptions already exists on top. One of the reasons could be that during split_constraints if the exception is set from top pin to a block pin then while splitting the exception is moved to block interface.

**What Next**

Use the command report_exceptions to check what exceptions have been specified.

## CSTR-469

(error) Exception is not set because no through objects could be found .

**Description**

The object list of the exception path is empty at the from, to, or through position. This might happen if the specified object in the command line is a net that does not have any global driver or is a cell with no output pins .

**What Next**

Verify that this is the intended behavior.

Feedback

## CSTR-470

(warning) Clock path max transition constraints for output pins for clock '%s' are skipped as not supported for sdc and script with pt/dc format.

**Description**

Clock path max transition constraints for output pins of clocks are not supported for sdc and scripts for format other than icc/fc.

**What Next**

If want to set maximum transition on clock output pins for these formats set it on specific lib pins .

## CSTR-471

(Warning) -'%s' option for set_sense -type clock command is not written out with write_sdc as read_sdc does not this option for this command.

**Description**

New version of sdc will support this option for this command.

**What Next**

Checkout other options for this command.

# CTP

## CTP-001

(warning) No phase delay could be computed for anchor point %s

**Description**

During anchor point creation, a pin or port was deemed a candidate for becoming an anchor point. However, no phase delay could be computed for this point. The likely reason is that the clock tree is not or not fully implemented, so that no reasonable delays could be computed.

**What Next**

Check the logfile for messages pertaining to clock tree synthesis, to see what problems were reported.

Feedback

## CTP-002

(error) Pin/port %s has the wrong direction.

**Description**

The specified pin or port has the wrong direction.

**What Next**

Specify a pin or port of the correct direction.

## CTP-003

(error) No physical design data.

**Description**

There was no physical design data. Most likely this is due to the design flow not yet ready for clock trunk planning.

**What Next**

Perform floorplanning first.

## CTP-004

(error) Core area is not defined.

**Description**

The design core area is not defined.

**What Next**

This is a general setup issue. Check that the flow has been run succesfully so far, in particular check if floorplanning has been performed.

## CTP-005

(information) For mode %s, port %s: %s

**Description**

Clock trunk planning has derived a temporary set_driving_cell constraint for the given mode and port.

Feedback

**What Next**

This message is informational only. If the given set_driving_cell constrain is not desired, an explicit one can be given using the CTS set_block_constraint file.

---

## CTP-006

(information) Pushed-down cell %s has been removed after being used to guide pin placement.

**Description**

During push_down_clock_trunks, guide buffers are pushed down, then used to guide pin placmeent, and finally removed. This message is to inform the user of the last step.

**What Next**

No action required. This message is informational only.

---

## CTP-007

(error) No valid libcells with CTS purpose found for cell %s in block %s.

**Description**

During clock trunk planning, an attempt was made to push down clock cells into a block. However, there are no valid libcells with CTS purpose configured for the block.

**What Next**

Either configure some libcells with CTS purpose for the block or close the block for feedthrough.

---

## CTP-008

(error) No %s libcells found for cell %s in block %s.

**Description**

During clock trunk planning, an attempt was made to push down clock cells into a block. However, there are no valid libcells with the correctl polarity (biffer/inverter) with CTS purpose configured for the block.

**What Next**

Configure some buffer or inverter libcells with CTS purpose for the block.

Feedback

## CTP-009

(error) Cannot replace existing repeater %s in block %s.

**Description**

The existing cell could not be replaced because it appears not to be a repeater, even though its libcell indicates it is.

**What Next**

This may indicate a library setup issue. Please check the libcell. We expect a repeater to have a single input and a single putput.

## CTP-010

(error) Attempt to make %s a clock tuning point, which is not the input pin of a buffer.

**Description**

Clock tuning points can only be placed at the input pin of a buffer.

**What Next**

You can first insert a buffer at the desired place in the netlist, then mark its input as a clock tuning point.

## CTP-011

(information) CTP constraints %s are set for block %s.

**Description**

constraints set by the command set_ctp_constraints.

**What Next**

This message is informational only. It prints constraints set by the command set_ctp_constraints.

## CTP-012

(Error) Problem found in offset value. %s.

**Description**

Problem found in offset value provided with set_ctp_constraints command. Ignoring these values.

Feedback

**What Next**

Check the offset values and apply set_ctp_constraints again with correct offset value.

---

## CTP-013

(Error) Length of the side %u of Block %s is less than offset %u provided. Ignoring the CTP constraints for the block %s.

**Description**

Problem found in offset value provided with set_ctp_constraints command. Ignoring these values.

**What Next**

Check the offset values and apply set_ctp_constraints again with correct offset value.

---

## CTP-014

(Error) Valid edges are not found for %s based on applied CTP constraints. Ignoring CTP constraints set on this block.

**Description**

Problem found in CTP constraints set by the user. Ignoring these values.

**What Next**

Check the CTP constraints and apply set_ctp_constraints again with correct value.

---

## CTP-015

(Information) CTP pin constraints will happen based on following constraints.

**Description**

This is an information message to print CTP pin constraints".

**What Next**

Check for other CTP messages.

---

## CTP-038

(error) Number of endpoints too large: actual number is %d , limit is %d.

Feedback

**Description**

The number of endpoints which have been automatically identified for clock trunk planning is too large.

**What Next**

The recommended approach is to define endpoints manually using set_clock_trunk_end_point. Alternatively, the limit can be increased by using app_option "plan.clock_trunk.endpoint_limit".

---

## CTP-039

(error) Different supply nets on %s and %s.

**Description**

Two sinks that were to be buffered together have a different supply net.

**What Next**

Consider removing one of the sinks from the set of sinks to be buffered.

---

## CTP-040

(error) Sink %s is not a physical pin or port.

**Description**

The sink is not a physical pin or port. A logical pin or port cannot be buffered.

**What Next**

Please specify only physical sinks.

---

## CTP-041

(error) Pin/port %s is unconnected.

**Description**

A sink to be buffered was unconnected, so there is no net to buffer.

**What Next**

Please specify only connected pins or ports.

Feedback

## CTP-042

(error) Sinks are deriven by multiple physical nets: %s and %s

**Description**

The sinks are driven by multiple physical nets, so they cannot be buffered by a single buffer.

**What Next**

Please specify only sinks attached to a single physical net.

## CTP-043

(error) Sinks are in different physical hierarchies.

**Description**

The sinks are in different physical hierarchies. This command can only buffer sinks in the same physical hierarchy.

**What Next**

Please specify sinks in the same physical hierarchy.

## CTP-044

(error) Multiple-driven nets are not supported: %s and %s

**Description**

The net to be buffered is multiple-driven. This is not supported.

**What Next**

If a buffer is really desired here it should probably be inserted at the Verilog level.

## CTP-045

(error) Net %s to be buffered has no driver.

**Description**

The net to be buffered has no driver. This may indicate a problem in the netlist.

**What Next**

Check the netlist if an undriven clock net is really intended.

Feedback

## CTP-046

(error) No buffer libcell found.

**Description**

No buffer could be inserted since the library contains no buffer libcell. This is probably a library setup error.

**What Next**

Check if the standard cell library is complete.

## CTP-047

(error) No sinks were specified.

**Description**

No sinks were specified, while at least a single sink is required in order to put a buffer in front of it.

**What Next**

Please specify at least one sink.

## CTP-048

(error) No physical net to buffer found.

**Description**

A physical net to buffer could not be found.

**What Next**

Please check if all specified sinks are driven by a single physical net.

## CTP-049

(error) No logical hierarchy can be found to place the buffer in.

**Description**

No logical hierarchy can be found to place the buffer in. This may indicate a software bug.

**What Next**

Please contact Synopsys with this problem.

Feedback

## CTP-050

(error) No buffer could be inserted.

**Description**

No buffer could be inserted. This may indicate a software bug.

**What Next**

Please contact Synopsys with this problem.

## CTP-051

(error) Expected a single input and a single output pin on the buffer.

**Description**

A library cell which purports to be a buffer was found which does not have a single input and a single output pin. This may indicate a library setup problem.

**What Next**

Please check your standard cell library.

## CTP-052

(error) Could not hook up new buffer to %s.

**Description**

The new buffer could not be properly connected. This may indicate a software bug.

**What Next**

Please contact Synopsys with this problem.

## CTP-053

(error) No physical design data available.

**Description**

No physical design data is available. Probably the state of the flow is too early to perform clock trunk planning.

**What Next**

Proceed further in the flow before attempting clock trunk planning.

Feedback

## CTP-054

(error) No allowed buffers found for clock %s and supply %s. Cannot perform timing estimation.

**Description**

No buffers could be found that may be used to buffer the clock under consideration. This may indicate a library setup problem.

**What Next**

Check your library setup that there are some buffers which can be used to buffer the clock tree.

## CTP-055

(error) No technology library could be found.

**Description**

No technology library could be found. This may indicate a library setup problem.

**What Next**

Check your library setup.

## CTP-056

(error) Cannot obtain any allowed routing layers for clock tree synthesis.

**Description**

The tool cannot obtain any allowed routing layers for clock tree synthesis. This may indicate a setup problem.

**What Next**

Check that there are some layers which may be used by clock tree synthesis.

## CTP-057

(error) No unit length delay found for nets powered by supply net %s.

**Description**

No unit-length delay could be estimated.

Feedback

**What Next**

Pleas check earlier warnings for a setup problem.

---

## CTP-058

(error) Cannot create timer.

**Description**

The timer could not be created.

**What Next**

Please try report_timing and check the problems it reports.

---

## CTP-059

(error) Unexpected problem with path group creation.

**Description**

There was an unexpected problem with path group creation. This may indicate a software bug.

**What Next**

Please contact Synopsys with this problem.

---

## CTP-060

(error) Syntax error in driving cell specification. %s.

**Description**

The format of driving cell specified through app_option "plan.clock_trunk.default_driving_cell" is wrong.

**What Next**

Please refer to the man page for app_option "plan.clock_trunk.default_driving_cell".

---

## CTP-061

(information) For mode %s: %s

Feedback

**Description**

Driving cell specification applicable to a given port for a given mode and specified either through app_option plan.clock_trunk.default_driving_cell or automatically chosen by the tool for Clock trunk planning.

**What Next**

This message is informational only. This is applicable in case set_driving_cell constraint is missing in CTS_CONSTRAINT file for a given mode.

---

## CTP-062

(error) Incorrect value %s specified for option '-sort_by' of report_clock_trunk_qor.

**Description**

Option '-sort_by' of report_clock_trunk_qor can only accept either "wns" or "wns_ocv" as sorting criteria.

**What Next**

Specify correct sorting criteria (either "wns" or "wns_ocv") with option '-sort_by' before running report_clock_trunk_qor.

---

## CTP-063

(information) Block clock pin %s is not placed.

**Description**

It is recommended to place block clock pins before running CTP.

**What Next**

Place clock pins for all blocks before running CTP.

---

## CTP-064

(warning) CTS_CONSTRAINT file specification for block %s is missing from set_constraint_mapping_file.

**Description**

CTP does not consider clock tree constraints for a particular block if its CTS_CONSTRAINT file specification is missing from set_constraint_mapping_file.

Feedback

**What Next**

Specify the CTS_CONSTRAINT file for all blocks before running CTP.

---

## CTP-065

(warning) CTS_CONSTRAINT file %s specified in set_constraint_mapping_file for block %s is empty.

**Description**

CTS_CONSTRAINT file specified in set_constraint_mapping_file should not be empty.

**What Next**

Specify the CTS_CONSTRAINT file for all blocks before running CTP.

---

## CTP-066

(warning) Latency estimation cannot be done at pin %s for clock %s.

**Description**

This may be because no path exists between clock source and pin. Look for GRF-014 messages in the logfile which is one of the potential causes of this issue.

**What Next**

Ensure that a valid path exists between clock source and the pin. Eliminate the reason for GRF-014 message in logfile if it exists.

---

## CTP-100

(Warning) Unable to determine side constraint %s. Command will proceed with pin placement.

**Description**

During topological constraint file generation, CTP engine was unable to find a valid side constraint.

**What Next**

Check the log file for CTP related related warning/error messages leading upto this point.

---

## CTP-101

(Error) Loop detected %s for pin '%s' for clock source pin '%s'.

Feedback

**Description**

Loop detected in the topology for the said pin.

**What Next**

Check the log file for CTP related related warning/error messages leading upto this point.

---

## CTP-105

(Error) Loop detected in the topology tree%s.

**Description**

The topology tree generated by CTP has a loop.

**What Next**

Check the log file for CTP related related warning/error messages leading upto this point.

---

## CTP-107

(Error) Unable to determine the location for pin '%s' for clock source pin '%s'.

**Description**

CTP was unable to find proper location for the said pin.

**What Next**

Check the log file for CTP related related warning/error messages leading upto this point.

---

## CTP-108

(Warning) Placement for port '%s' is not valid. This port will be ignored during CTP.

**Description**

CTP engine will ignore output ports if it is not placed.

**What Next**

Check the non-placed output ports present in the design.

---

## CTP-109

(Error) Pin constraint writer detected an issue for pin '%s'. Pin placement will be skipped.

Feedback

**Description**

CTP was unable to find or create the said pin.

**What Next**

Check the log file for CTP related related warning/error messages leading upto this point.

---

## CTP-110

(Error) Logical loop detected. Output pin '%s' is driving input pin '%s' of the same physical hierarchy instance '%s'.

**Description**

There is a logical loop as detailed in the message. These type of logical loops are not supported.

**What Next**

Modify the design to remove such logical loops.

---

## CTP-121

(Error) Consistency checks failed for clock source pin '%s'. Pin placement will be skipped for this.

**Description**

CTP ran into consistency issue for the said clock source pin. Subsequent pin placement step will be skipped for this clock net.

**What Next**

Check the log file for CTP related related warning/error messages leading upto this point.

---

## CTP-125

(information) None of the clocks could be mapped to block-level clocks for block '%s'.

**Description**

No block level clocks of the specfied block could be mapped to any top level clocks. Command will proceed for all the clocks in that block.

**What Next**

Check the clock mapping for the blocks.

Feedback

## CTP-126

(warning) None of the specified clocks could be mapped to block-level clocks for block '%s'.

**Description**

None of the block level clocks of the specfied block could be mapped to any of the specified top level clocks. Command will skip this block.

**What Next**

Check the clock mapping for the blocks.

## CTP-151

(Warning) detected %s pin %s on the read-only block %s, %s.

**Description**

During tree generation, CTP engine will ignore pins beyond first pin of read only block.

**What Next**

Check the read only blocks present in the design.

## CTP-152

(Warning) The port %s is not driving any load. This port will ignored.

**Description**

During tree construction, CTP engine will ignore ports if it is not driving any load.

**What Next**

Check the ports with no load present in the design.

## CTP-153

(Error) Not able to create a Feedthrough pin/port %s on read-only block %s.

**Description**

During Feedthrough pin/port construction, CTP will not create any Feedthrough pin/port on a read only block.

Feedback

**What Next**

Check the connection to find out why feedthrough pin/port is required on a read only block.

---

## CTP-180

(Error) Topology constraint file '%s' is missing.

**Description**

The topological constraint file was could not be found.

**What Next**

Check the log file for CTP related related warning/error messages leading upto this point.

---

## CTP-181

(Error) Topology constraint file '%s' is empty.

**Description**

The topological constraint file was found to be empty.

**What Next**

Check the log file for CTP related related warning/error messages leading upto this point.

---

## CTP-200

(information) Setting app option '%s' to '%s' for CTP %s flow.

**Description**

Tool has detected that app option 'top_level.optimize_subblocks' and/or 'cts.common.enable_subblock_optimization' has not been set to proper value for the specified flow. Command will set the app options internally to proper values for the specfied flow.

**What Next**

Check the intention of using present values of the app options 'top_level.optimize_subblocks' and/or 'cts.common.enable_subblock_optimization'.

---

## CTP-201

(Warning) Ignoring '%s' argument for CTP %s flow.

Feedback

**Description**

The arguments '-from' and '-to' for command 'synthesize_clock_trunks' is applicable only in case of clock trunk planning in THO mode. Command will ignore these arguments in any other mode.

**What Next**

Check the intention of using the '-from' and '-to' arguments.

---

## CTP-202

(Error) All the sub-blocks are found to be readonly. Command will not proceed.

**Description**

Tool detected that all the sub-blocks are readonly. Command will not proceed further.

**What Next**

Check the setup. Check for other CTP messages leading up to this point.

---

## CTP-203

(Warning) Found editable abstract instance ‘%s’ without optimization context. Command will treat it as readonly.

**Description**

Tool detected the said block instance as linked with abstract without optimization context. However it has been marked as editable. Command is treating it as readonly.

**What Next**

Check the setup. Either mark the said block instance as non-editable or run "synthesize_clock_trunk_setup_hier_context -init" to setup the instances.

---

## CTP-204

(Warning) Found editable abstract instance '%s'. But abstract editing is not enabled. Command will treat it as readonly.

**Description**

Tool detected that abstract editing is not enabled for command "synthesize_clock_trunks". But there are some sub-blocks linked with abstracts that have been marked as editable. Command is treating it as readonly.

Feedback

**What Next**

Check the setup. Either abstract editing needs to be enabled or blocks linked with abstract need to set as non editable.

---

## CTP-205

(Information) Block instance '%s' will be treated as %s.

**Description**

This is an information message specifying editability of blocks for command "synthesize_clock_trunks".

**What Next**

Check for other CTP messages.

---

## CTP-206

(Waring) Setup issue detected while checking for sub-block editability. The checks will be skipped.

**Description**

Tool detected setup issue while performing the editability checks on sub-blocks. The checks will not be performed.

**What Next**

Check the setup. Check for other CTP messages leading up to this point.

---

## CTP-207

(Information) The MPH flag is true, Tool will ignore app option "plan.clock_trunk.write_separate_files".

**Description**

This is an information message specifying Tool will ignore the app option "plan.clock_trunk.write_separate_files" as MPH flag is set.

**What Next**

Check the MPH flag. Check for other CTP messages leading up to this point.

Feedback

## CTP-208

(Information) %s

**Description**

This is an information message.

**What Next**

This is an information message.

## CTP-209

(Error) Incrmental CTP can not run on MPH design.

**Description**

Incrmental CTP can not run on MPH design.

**What Next**

Check the design and run full CTP instead of Incremental CTP.

## CTP-210

(Error) Detect no full CTP has been run before. To run Incrmental CTP previous full CTP has to be run.

**Description**

To run Incrmental CTP previous full CTP has to be run.

**What Next**

To run Incrmental CTP previous full CTP has to be run.

## CTP-307

(Error) CTP sanity check has failed. Run check_design_for_clock_trunk_planning -clock command for detailed errors.

**Description**

This message indicates that the sanity checks have failed at top level. The issues may cause the CTP flow to be incorrect or fail at some point. It is recommended to correct the reported errors before proceeding with the CTP flow. Run check_design_for_clock_trunk_planning -clock <clocks> command to see detailed errors.

Feedback

### What Next

Run check_design_for_clock_trunk_planning -clock command to see detailed errors. If app-option top_level.continue_flow_on_check_hier_design_errors is set to true, command will continue even if sanity checks fail and CTP-308 error message will be given.

### See Also

- check_design_for_clock_trunk_planning
- top_level.continue_flow_on_check_hier_design_errors

---

## CTP-308

(Error) CTP sanity check has failed but the command will continue. Run check_design_for_clock_trunk_planning -clock command for details.

### Description

This message is issued because the sanity checks have failed at top level and the app-option top_level.continue_flow_on_check_hier_design_errors is set to true. The issues may cause the CTP flow to be incorrect or fail at some point. It is recommended to correct the reported errors before proceeding with the CTP flow. Run check_design_for_clock_trunk_planning -clock <clocks> command to see detailed errors.

### What Next

Run check_design_for_clock_trunk_planning -clock command to see detailed errors.

### See Also

- check_design_for_clock_trunk_planning
- top_level.continue_flow_on_check_hier_design_errors

---

## CTP-400

(Error) Load site %s is not reachable, will be ignored.

### Description

During CTP if a load is not reachable this error will be issued.

### What Next

Check if the any of user constraints or the arrangement of block/hardmacro/readonly-blocks is making this load unreachable.

Feedback

### CTP-401

(Error) With given MIB-set, clock trunk is not possible. Loads: %s.

**Description**

During CTP if among MIBs, an agreeable placement of pin can not be found, this error is issued.

**What Next**

Check if the any of user constraints or the arrangement of block/hardmacro/readonly-blocks is making MIB-set unreachable with agrreable placement of pin.

## CTS

### CTS-001

(Warning) '%s' pin/port not a clock source, ignored.

**Description**

The pin/port specified is not a source of a clock, and is thus ignoered.

### CTS-002

(Warning) There are active CTS scenarios with no clock definition.

**Description**

There are timing scenarios enabled for CTS that do not have any clock definitions. CTS will work on any active scenario that is enabled for setup or hold fixing with the set_scenario_status command.

**What Next**

Ensure that all of the active CTS scenarios have at least one clock defined. Control the CTS scenarios by using the *set_scenario_status* command. New clocks are defined using the SDC commands create_clock and create_generated_clock.

### CTS-003

(Warning) Skipping source pin '%s' of clock '%s' (mode '%s') because there is no valid clock net for CTS to work on.

Feedback

**Description**

When a clock is defined on an input pin or internal pin of a clock gate, CTS will iterate through all the output pins of the clock gate to find root clock nets. This message indicates that no output net of the clock gate is found.

**What Next**

Please check the clock definition. CTS will exclude this clock source pin because there is no net to work on.

---

## CTS-004

(Warning) There are generated clocks that cannot be reached by their master clock.

**Description**

A generated clock is logically derived from its master clock, as specified in the create_generated_clock constraint. For a master clock to derive a generated clock, it should be possible to trace a path from the master clock to the generated clock root pin in the netlist. This message indicates that CTS failed to find such a path. As a result, CTS will not balance the generated clock with the master clock. The generated clock will be balanced independently.

**What Next**

Examine your netlist and timing constraints to understand why the generated clock could not be reached by the master clock. You can proceed with CTS despite this issue, but the generated clock will not be balanced with the master clock. To balance them together, the netlist or timing constraints must be modified so that the master clock propagates to the generated clock's root pin.

---

## CTS-005

(Warning) Generated clocks defined on bidirectional pins

**Description**

A generated clock definition is present on a bidirectional pin. The timer may have problems correctly handling this constraint, which can affect CTS balancing of this generated clock domain, or of the generated clock with its master clock.

**What Next**

Use remove_generated_clock to remove the existing generated clock definition. Then use create_generated_clock to redefine the clock on another pin that is not bidirectional.

Feedback

**See Also**

- create_generated_clock
- remove_generated_clock

---

## CTS-006

(Warning) Balancing conflicts exist between different clocks

**Description**

The partially overlapping clock structure shared by two clocks makes it impossible to balance their delays to produce low clock skew. The structure contains common sinks for both clocks, but it also has at least one pin that is a sink for only one of the clocks, but is an intermediate pin in the clock tree for the other clock.

The most common reason for this is a situation where a generated clock is defined on the output of a divider flop for a particular master clock, but there are multiple clocks propagating to the divider flop clock pin. As a result, the flop is a divider for one master clock, and the tree continues to downstream sinks. For the other master clock, the divider flop is instead a sink.

**What Next**

Check the clock structure, and modify the timing constraints to avoid a situation where one clock arrives at a sink but the other clock continues through it. If the structure is defined as intended and cannot be modified, then understand that the clocks cannot be balanced. Instead, consider modifying the interclock delay balancing constraints with the create_clock_balance_group command, so that these two master clocks are not balanced together.

---

## CTS-007

(Warning) Failed to specify any clock buffers or inverters for CTS

**Description**

CTS requires at least one buffer or inverter library cell to build the clock trees.

**What Next**

Use 'set_lib_cell_purpose -include cts $lib_cell_list', where $lib_cell_list includes at least one buffer or inverter lib_cell.

**See Also**

- set_lib_cell_purpose

Feedback

## CTS-008

(Warning) Clock reference cells have dont_touch or dont_use

**DESCRIPTION"**

If a lib_cell has a dont_touch or dont_use attribute, then its instances in the design cannot be modified. This also prevents CTS from inserting new instances of that lib_cell, or resizing instantiations of that cell. This message indicates that one or more clock references are dont_touch or dont_use.

**What Next**

Modify the available clock references by using 'set_lib_cell_purpose -include cts $lib_cell_list', where $lib_cell_list includes at least one lib_cell that is not dont_touch. Alternately, 'set_dont_touch $lib_cell false' can be used to clear dont_touch attributes on clock references in the library.

**See Also**

- set_lib_cell_purpose
- set_dont_touch
- get_attribute

## CTS-009

(Warning) Cell instances in the clock tree have multiple conditional delay arcs between the same pins

**Description**

This message indicates that a CTS reference cell has more than one timing arc.

**What Next**

Please check the CTS reference settings.

**See Also**

- set_lib_cell_purpose

## CTS-010

(Information) Source pin '%s' of clock '%s' has a clock tree balance point defined on it.

Feedback

### Description

The source pin of this clock is defined as clock tree balance point. CTS will fix design constraints on this clock tree, but will not try to balance its sinks.

### What Next

Check the clock to see if it is what you expect.

### ALSO SEE

`set_clock_tree_balance_point (2), remove_clock_tree_balance_point (2)`

---

## CTS-011

(Information) Source pin '%s' of clock '%s' has ExplicitIgnore pin defined on it.

### Description

The source pin of a clock is defined as an ExplicitIgnore pin.

---

## CTS-012

(Warning) Nets in the clock network have a dont_touch constraint

### Description

Nets in the clock network have been marked with a dont_touch attribute. dont_touch on a net prevents sizing, optimization, and replacement of all attached cells. It also prevents buffering on the net. dont_touch can limit CTS optimization, which can cause worse latency, skew, or area in the clock network.

### What Next

Examine the reported nets to ensure that the dont_touch attributes have been applied with good reason. In particular, dont_touch on high-fanout or other large nets can cause latency problems after CTS, as that large net will not be buffered. dont_touch attributes can be removed with the set_dont_touch, set_dont_touch_network, or mark_clock_tree commands. set_dont_touch is normally used for setting and clearing dont_touch on individual nets or cell instances. set_dont_touch_network is used to apply dont_touch to an entire fanout from a clock or pin. Note that set_dont_touch_network will not trace from a master clock through to a generated clock. mark_clock_tree can be used also to apply or clear dont_touch. Unlike set_dont_touch_network, mark_clock_tree will trace through generated clocks.

Feedback

**See Also**

- set_dont_touch
- set_dont_touch_network

---

## CTS-013

(Warning) Cells in the clock network have a dont_touch constraint

**Description**

Cells in the clock network have been marked with a dont_touch attribute. dont_touch on a cell prevents sizing, optimization, and replacement during CTS. dont_touch can limit CTS optimization, which can cause worse latency, skew, or area in the clock network.

**What Next**

Examine the reported cells to ensure that the dont_touch attributes have been applied with good reason. dont_touch attributes can be removed with the set_dont_touch, set_dont_touch_network, or mark_clock_tree commands. set_dont_touch is normally used for setting and clearing dont_touch on individual nets or cell instances. set_dont_touch_network is used to apply dont_touch to an entire fanout from a clock or pin. Note that set_dont_touch_network will not trace from a master clock through to a generated clock. mark_clock_tree can be used also to apply or clear dont_touch. Unlike set_dont_touch_network, mark_clock_tree will trace through generated clocks.

**See Also**

- set_dont_touch
- set_dont_touch_network

---

## CTS-014

(Warning) There is a balance conflict at pin '%s' (corner '%s') : %s %s.

**Description**

This message indicates that balance point(stop, float) values are conflict in different mode.

**What Next**

Check the balance point settings. Make sure they are consistent across different mode.

---

## CTS-015

(Warning) set_max_delay or set_min_delay constraints are defined in the clock network

Feedback

### Description

This message indicates that there is max/min delay set on clock. CTS will ignore this constraint.

### What Next

Check the max/min delay settings.

### See Also

- set_max_delay
- set_min_delay

---

## CTS-016

(Warning) There is Max_delay or Min_delay set on pin '%s'.

### Description

This message indicates that there is max/min delay set on a clock pin. CTS will ignore this constraint.

### What Next

Check the max/min delay settings.

### See Also

- set_max_delay
- set_min_delay

---

## CTS-017

(Warning) clock gate '%s' is dont touch.

---

## CTS-018

(Warning) clock net '%s' is dont touch.

---

## CTS-019

(Warning) Clocks propagate to output ports

Feedback

### Description

One or more clocks in the design propagate to output ports. This is allowable, and could be due to a feedthrough clock in a block, or clocks generated inside a block that need to propagate to the top-level. Default CTS behavior is dependent on the constraints on the output port.

If the output port is a valid timing endpoint, due to a set_output_delay, set_max_delay, set_min_delay, or set_data_check constraint to that output port, then the port is considered a data endpoint. It will not be balanced as a clock sink, and any nets in the clock tree fanning out only to data pins will not be synthesized by CTS.

If the output port is unconstrained, it will be treated as an ignore pin and will not be balanced. A DRC clean clock tree will still be built to this port.

If the user explicitly puts a stop pin or float pin constraint on this port using set_clock_balance_points, then it will be treated as a sink, and will be balanced appropriately during CTS.

### What Next

Check each port that is reported, and see how it will be handled based on the constraints on that port. It may be treated as a data sink, a clock ignore point, or a clock balance point. If the port is not constrained to be handled as expected, then modify the constraints on the port. set_output_delay can force the port to be a data endpoint. set_clock_balance_points can be used to set it as a balance point. set_clock_ignore_points can be used to set it as an ignore point.

### See Also

- set_clock_balance_points
- set_output_delay
- set_max_delay
- set_min_delay
- set_data_check

---

## CTS-020

(Warning) Corner '%s' has different RC value for early and late. early: %f %f %f %f and late: %f %f %f %f

### Description

The message indicates that there are different TLU+ for early and late for the same corner.

Feedback

## CTS-021

(Warning) -max and -min are set differently at input port '%s' of clock '%s'. max: %f, %f; min: %f, %f.

### Description

The message indicates that there are different -max and -min settings on clock input ports.

### What Next

Check the -max and -min settings in set_input_transition.

### See Also

- set_input_transition

## CTS-022

(Error) clock '%s' is a generated clock. This command does not take a generated clock as input.

### Description

This command does not take a generated clock as input.

### What Next

Check the master clock of the specified generated clock, and use it as input.

## CTS-023

(Error) No clock routing rule matched specified pattern.

### Description

The pattern which you specified did not match any clock routing rule.

### What Next

Check the pattern to see if it is what you expected.

### See Also

- set_clock_routing_rules

Feedback

## CTS-024

(Warning) Skip the balance group '%s' because its mode '%s' has no associated scenario with setup or hold actviated.

### Description

The clocks of the clock balance group are in the mode which has no associated scenario with setup or hold activated. Therefore, the program skips to balance this group.

### What Next

Check the clock balance group and scenario status to see if it is what you expected.

### See Also

- report_clock_balance_groups

## CTS-025

(Warning) Gate '%s' is not sizable.

### Description

The gate may have dont_touch attribute, or may be in a read only block, thus sizing was not performed.

## CTS-026

(Error) The '-objects' list mixes clocks and skew groups.

### Description

To prevent confusion the object list has to be all clocks or all skew groups. If you want to balance a skew group with a clock which has no skew group defined on it, you can specify the default skew group of this clock. The name of the default skew group is 'default_<clock name>', for example, the default skew group of clock 'clk1' is 'default_clk1'.

### What Next

Check clocks and skew groups to see if it is what you expected.

### See Also

- create_clock_balance_group

Feedback

## CTS-027

(Error) Clock '%s' is in both -offset_from and -offset_to clock list.

### Description

The clock which you specified is in the lists of both -offset_from and -offset_to options. This caused a conflicting settings.

### What Next

Check the clock to see if it is what you expected.

### See Also

- create_clock_balance_group

## CTS-028

(Warning) Skipped delay insertion on driver pin '%s' because its net cannot be buffered. CTS also failed to find its downstream nets to buffer without degrading the balance of this sub-tree.

### Description

CTS failed to insert a delay buffer here. This net may have dont_touch attribute or may be in a read only block.

### What Next

Check the clock to see if it is what you expected.

## CTS-029

(Warning) Replace net '%s' with '%s' in clock routing rule.

### Description

CTS automatically replaces the user specified net with the net connecting the driver pin in the net based clock routing rule, because a hierarchical net may be lost after buffer insertion.

### What Next

Check the result to see if it is what you expect.

### ALSO SEE

set_clock_routing_rules (2)

Feedback

## CTS-030

(Error) Clock '%s' exists in the other clock balance group '%s'. To combine a clock into an existing group, remove the existing group and create a new group.

### Description

The clock which you specified already exists in the other clock balance group. Since a clock cannot be found in multiple groups, you need to consolidate the new group and existing groups. If you intend to combine a new clock into an existing group, remove the existing group and re-create a new group which includes the new clock and the other clocks in the existing group.

### What Next

Consolidate the group which you specified and existing groups to make a new one.

### See Also

- create_clock_balance_group

## CTS-031

(Error) The minimum layer constraint is either equal to the maximum layer constraint, or on a higher layer than the maximum layer constraint.

### Description

The minimum layer constraint has to be on a lower layer than the maximum layer constrait. Please redefine the min/max layer constraint.

## CTS-032

(Warning) There are fixed routing shapes on the net '%s', so the dont_touch flag is set on this net to prevent buffering.

### Description

Since the net has fixed routing shapes, the clock synthesis tool will prevent buffering on this net.

### What Next

Check the result to see if it is what you expect.

Feedback

## CTS-033

(Warning) Both logical nets '%s' and '%s' have been set with net based routing rules, but they correspond to the same physical net. Therefore, the setting on the latter net is ignored and removed.

### Description

These two logical nets correspond to the same physical net, so the setting on one net will be honored and the other will be ignored.

### What Next

Check the result to see if it is what you expect.

### ALSO SEE

set_clock_routing_rules (2)

## CTS-034

(Warning) The specicified reference cell '%s' for isolate port '%s' either has dont_touch flag or does not have the cts purpose, so this setting will not be honored.

### Description

The specified reference cell either has dont_touch flag or does not have the cts purpose. The clock synthesis tool will ignore this setting and choose a cts purpose reference.

### What Next

If you still prefer this reference cell to isolate a clock port, please use *set_lib_cell_purpose* command to add cts purpose for the reference cell. If this cell has dont_touch flag, you need to remove the flag.

### ALSO SEE

```
set_isolate_ports (2)
set_lib_cell_purpose (2)
set_dont_touch (2)
```

## CTS-035

(Warning) The internal clock tree structure is inconsistent on net '%s'.

Feedback

**Description**

The clock tree synthesis tool finds unexpected connections on this net. This problem is usually caused by the preserved hierarchical ports or dont_touch flag on some segments of this net. This internal problem may damage the clock skew and DRC.

**What Next**

This is just a warning message. It is better to remove constraints on its hierarchical pins and dont_touch on the net.

**ALSO SEE**

---

## CTS-036

(Warning) No clock was specified or found in all active scenarios.

**Description**

CTS will work on any active scenario with setup or hold timing analysis enabled. There was no clock in those scenarios.

**What Next**

Check clock and analysis settings in all scenarios. Modify the scenarios settings by *set_scenario_status* command.

---

## CTS-037

(info) CTS QoR %s: GlobalSkew = %s; ID = %s;%s%s NetsWithDRC = %d; %s; ClockBufCount = %d; ClockBufArea = %s; ClockCellArea = %s; ClockWireLen = %s; Clock = %s; Mode = %s; Corner = %s; ClockRoot = %s.

**Description**

Clock Qor information.

---

## CTS-038

(Warning) No clock routing rule is specified.

**Description**

There is no clock routing rule used in the clock tree synthesis.

**What Next**

Check the result to see if it is what you expect.

Feedback

**ALSO SEE**

set_clock_routing_rules (2)

---

## CTS-039

(info) Clock '%s' mode '%s' root '%s' is removed from skew and latency optimization.

**Description**

This message indicates that a clock is removed from skew and latency optimization, because optimization engine finds out that it is identical with another clock. For the clocks that are identical, optimization engine will pick only one of them to process. The optimization of the remaining clocks is implicit.

---

## CTS-040

(info) Balance point auto scaling set on term '%s' clock '%s' corner '%s' early rise %s early fall %s late rise %s late fall %s.

**Description**

This message indicates that a balance point is set by CTS automatically. CTS will detect missing balance points that are necessary for clock balancing purpose and set them according to the user balance point settings.

**ALSO SEE**

```
set_clock_balance_points (2)
report_clock_balance_points (2)
```

---

## CTS-041

(Warning) Gate '%s' is not sizable because of %s dont_touch.

**Description**

The gate has dont_touch attribute, thus sizing was not performed.

---

## CTS-042

(Warning) Gate '%s' is not sizable because of dont_touch_network on %s.

**Description**

The gate has dont_touch_network attribute, thus sizing was not performed.

Feedback

## CTS-043

(Warning) Gate '%s' is not sizable because of fixed placement.

**Description**

The gate is fixed placement, thus sizing was not performed.

## CTS-044

(Warning) Gate '%s' is not sizable because of read-only physical block.

**Description**

The gate is inside a read-only physical block, thus sizing was not performed.

## CTS-045

(Warning) Gate '%s' is not sizable because of its lib_cell specified by users.

**Description**

The lib_cell of the gate was specified by users, thus sizing was not performed.

## CTS-046

(Warning) Net %s has pin %s with invalid location: ignored.

**Description**

Load pin on the net doesnt't have valid location and would not be buffered.

## CTS-047

(Error) Driver %s is not placed. buffering will not happen for this net.

**Description**

Driver pin of the net is not placed. Net would be skipped for buffering.

## CTS-048

(Warning) clock '%s' is a virtual clock. This command does not take a virtual clock as input.

**Description**

This command does not accept a virtual clock, so it has been ignored.

Feedback

## CTS-049

(Error) Failed to disconnect pin '%s' from net '%s' in routing update.

**Description**

The interactive router failed to disconnect a pin from its net.

**What Next**

Please check the pin in the reference library. The most common situation is this pin does not have a physical shape in the reference library.

## CTS-050

(Error) Failed to disconnect pins from net '%s' in routing update.

**Description**

The interactive router failed to disconnect pins from its net.

**What Next**

Please check pins on this net. The most common situation is that some pins do not have a physical shape in the reference library.

## CTS-051

(Warning) clock '%s' has been set. This command will overwrite its settings.

**Description**

This command will overwrite the clock settings.

## CTS-052

(Warning) The specified lib cell '%s' either has dont_touch attribute or no cts purpose, so this setting will not be honored.

**Description**

The specified lib cell either has dont_touch attribute or no cts purpose. This will prevent clock tree synthesis tools to use it.

Feedback

**What Next**

If you would like to use this lib cell for clock tree synthesis, please apply *set_lib_cell_purpose* command to add cts purpose, and *set_dont_touch* command to clear dont_touch attribute.

**ALSO SEE**

```
set_lib_cell_purpose (2)
set_dont_touch (2)
```

---

## CTS-053

(Warning) The net '%s' will not be buffered because it is a pad net.

**Description**

This net is a pad net, so it will not be buffered.

---

## CTS-054

(Warning) The net '%s' will not be buffered because it is in dont_touch_network.

**Description**

This net is in dont_touch_network, so it will not be buffered.

---

## CTS-055

(Warning) The net '%s' will not be buffered because it is a dont_touch net.

**Description**

This net is a dont_touch net, so it will not be buffered.

---

## CTS-056

(Warning) The net '%s' will not be buffered because it is in a read only block.

**Description**

This net is in a read only block, so it will not be buffered.

---

## CTS-057

(Warning) The net '%s' will not be buffered because it connects a special cell.

Feedback

### Description

This net connects a special cell, such as port isolate buffer, so it will not be buffered.

---

## CTS-058

(Warning) The net '%s' has user's specified %s routing rule, so it will not be overwritten by CTS.

### Description

In clock synthesis stage, CTS will apply clock routing rules on all clock nets. Here clock routing rules are the rules specified by the *set_clock_routing_rules* command. However, this process will not overwrite existed rules already set on nets by *set_routing_rule* command. Since this net has a routing rule already, it will not be overwritten by CTS.

### What Next

If this behavior does not meet user's need, user can clear routing rules before clock tree synthesis.

### See Also

- set_routing_rule
- set_clock_routing_rules

---

## CTS-059

(Warning) There is no clock tree balance point on user specified primary corner '%s', but this settings are shown on other corners.

### Description

In the gate by gate clock tree synthesis stage CTS will choose a primary corner and balance all loads based on this corner, so it is better to have all balance points on this corner. Now the user specified primary corner does not have any setting, but the other corner has. This inconsistency will affect the result of clock synthesis.

### What Next

Please check the primary corner and balance point settings.

### See Also

- cts.compile.primary_corner

Feedback

## CTS-060

(Warning) No layer constraint is specified in this set_clock_routing_rules command.

### Description

Since no '-max_routing_layer' or '-min_routing_layer' option is specified, the clock synthesis tool will use all routing layers available in the design for net type specified in this *set_clock_routing_rules* command.

### What Next

If you do not want the tool to use all routing layers for specified net type, set desired routing layers with min_routing_layer and max_routing_layer options.

### ALSO SEE

```
set_clock_routing_rules(2)
remove_clock_routing_rules(2)
```

## CTS-061

(Error) Clock %s has zero period.

### Description

Since the synthesize_clock_tree depends on the clock period for optimizing clock trees, a zero-period clock may cause unpredictable result.

### What Next

Check the clock definition and make sure the clock's period is not zero

### ALSO SEE

```
create_clock(2)
report_clocks(2)
```

## CTS-062

(Warning) Exception '%s' is applied on %s '%s' %s.

### Description

Port / Net may have restrictions applied to it. The type of the restriction is reported, as well as name of the port / net.It will also print the reason of the exception.

Feedback

## CTS-063

(Warning) The set_clock_routing_rules has incomplete min/max_routing_layer constraints.

### Description

The user only set min_routing_layer or max_routing_layer, but not both min_routing_layer and max_routing_layer. In a lot of cases, QoR result of synthesize_clock_tree with/ without complete min_max_routing_layer can be significantly different. If both min/max routing layers are desired, please make sure to provide both the -min_routing_layer and -max_routing_layer options. Even if users use two set_clock_routing_rules commands to set the min_routing_layer and max_routing_layer seperately, we still recommend users to set them in the same set_clock_routing_rules commands to avoid confusion.

## CTS-064

(Error) Skip to work on clock_balance_group '%s' because it is impossible to balance its members together.

### Description

The program skips this balance group because some members in this group cannot be balanced with others.

### What Next

Please check each member in this group, and find the ones which cannot be balanced with others. Then remove this balance group and create a new one without those bad members.

### See Also

- create_clock_balance_group
- remove_clock_balance_groups

## CTS-065

(Warning) trying to balance a very large downstream delay of '%f' at driver '%s' is restricted to balance only upto '%f'.

### Description

The downstream of this driver pin has load with very large and possiblly unreasonable delay that is needed to be balanced against other loads with much smaller delay. This will result in a large number of buffers being inserted. CTS will only balance it to a pre-defined upper limit to prevent excessive large latency and buffer count/area.

Feedback

### What Next

Please check the balancing requirements of this driver's load pins, and make sure that this is what user want. Usually it is due to some really large drc violation downstream, unreasonally large balance point, etc.

### See Also

- set_clock_balance_points

---

## CTS-066

(Warning) The net '%s' will not be buffered because it has multiple drivers.

### Description

This net has multiple drivers, so it will not be buffered.

---

## CTS-067

(Error) Failed to initialize clocks.

### Description

The program fails to initialize clocks, so it will not continue.

### What Next

Here are the main reasons to fail clock initialization. 1. There is no available buffer/inverter lib cell. CTS will use those lib cells with "cts" purpose and no "dont_touch" flag. Please check the reference lib cell settings. 2. Failed to characterize corners in the designs.

### See Also

- set_lib_cell_purpose
- set_dont_touch

---

## CTS-068

(Warning) MV violation on pin $$s on clock $$s. The connected inst and net will not be considered for buffering and optimization.

### Description

A design object was found to be in violation of MV rules at the start of clock tree synthesis, and will not be buffered or optimized.

Feedback

### What Next

Check for MV violations using check_mv_design before starting clock tree synthesis.

### See Also

- check_mv_design

---

## CTS-069

(Error) No available %s in the clock reference list. The app_option setting 'cts.compile.repeater_selection %s' will be ignored.

### Description

User provided repeater_selection setting does not match with the clock reference list.

### What Next

Modify the clock reference list to include repeaters of the specified type.

---

## CTS-070

(Information) Multi-threaded CTO is disabled for VR mode, when minimum-pulse-width optimization is enabled, or when wire_em is enabled.

### Description

When global routing (GR) is available and enabled in the flow, GR-based multi-threaded CTO will be used; otherwise single-threaded optimization will occur with virtual routing (VR). If cts.optimize.minimum_pulse_width or cts.optimize.wire_em_aware is enabled (both default false), multi-threaded CTO will also be disabled due to compatibility and will fall back to single-threaded CTO.

### What Next

If GR is available in the flow but disabled by app option, cts.optimize.minimum_pulse_width is enabled, or cts.optimize.wire_em_aware is enabled, please review your settings.

---

## CTS-071

(Warning) There is a huge DRC transition violation on the net driven by %s, which is on longest path of clock %s. This is causing very large clock latency. The clock tree optimization is disabled to prevent from unnecessary delay insertion to match this very large clock latency.

Feedback

**Description**

Some large drc violations (especially large transition violations) might be present at the CTO skew/latency optimization step. There are several possible reasons, like some kind of user dont-touch settings, or simply the gate-by-gate build clock tree step wont be able to fix all the drc. When this happens, the clock latency can be a very large value since large transition values often have large stage delays and these nets with often appear on the longest path. In this case, if CTO skew/latency optimization proceed with normal balancing, it will have to add huge amount of delay buffers and often cause very large runtime. This is not necessary, user need to check either the constraint of the nets with large drc, or check if there is anything wrong in the gate-by-gate step.

**What Next**

Check user constraints on the terms, or check the gate-by-gate step on the terms/net pointed out.

---

## CTS-072

(Warning) There is a very large arc delay between term %s and term %s for clock %s on corner %s

**Description**

Some very large delay is on the longest path and take majority of the longest path, at a certain mode/corner. This often means either there is a very large drc violation or a lib arc being used for that scenario itself has very large delay. The clock will be ignored for clock tree optimization to prevent unnecessary balancing.

**What Next**

Check user constraints on the terms/arcs, or check the gate-by-gate step on the terms/net pointed out.

---

## CTS-080

(Warning) Exception '%s' is applied %d times %s.

**Description**

Buffer tree has restrictions applied to it. The type of the restriction is reported, as well as number of the occurences of that restriction for this clock tree net.

---

## CTS-101

(Information) %s will work on the following scenarios.

Feedback

**Description**

Lists the scenarios that command will work on.

---

## CTS-102

(Information) %s will work on the following clocks.

**Description**

Lists the clocks that command will work on.

---

## CTS-103

(Information) '%s' is identified as primary corner for initial clock tree building.

**Description**

The *synthesize_clock_trees* command will pick a primary corner in the initial clock tree building stage. By default the worst delay corner will be selected as primary corner.

---

## CTS-104

(Information) The run time for %s is %s, cpu time is %s.

**Description**

Prints the run time information for a step.

---

## CTS-105

(Information) All clock objects will be converted from ideal to propagated clock during CTS.

**Description**

CTS will change the clocks in active scenarios to be propagated.

**See Also**

- set_propagated_clock

---

## CTS-106

(Information) Relocated the clock cell '%s' from %s to %s.

Feedback

**Description**

This clock cell is relocated by CTS to improve the clock latency.

---

## CTS-107

(Information) %s will work on all clocks in active scenarios, including %d master clocks and %d generated clocks.

**Description**

Print the number of clocks for command to work on.

---

## CTS-108

(Information) There are %d skew groups in the mode '%s'.

**Description**

Print the number of skew groups in the specified mode for CTS to work on.

---

## CTS-109

(Information) Clock %s had a previously specified reference setting, which will be overwritten.

**Description**

Specified clock had a previously specified reference setting, which will be overwritten with latest setting.

**ALSO SEE**

`set_clock_tree_reference_subset (2)`

---

## CTS-110

(Information) %d references enabled for CTS.

**Description**

Prints the count of refrences enabled for LEQ cell using command derive_clock_cell_references.

---

## CTS-111

(Information) %d references are already enabled for CTS.

Feedback

**Description**

Prints the count of refrences which are already enabled for LEQ cell using command derive_clock_cell_references

---

## CTS-112

(Information) %d LEQ cells found but not enabled due to user set exclude purpose.

**Description**

Prints the count of LEQ cell refrences which are not enabled as user has set exclude purpose of lib cell using command derive_clock_cell_references.

---

## CTS-113

(Information) %d LEQ cells found but not enabled due to dont_touch on lib_cell.

**Description**

Prints the count of LEQ cell refrences which are not enabled due to dont_touch attribute using command derive_clock_cell_references.

---

## CTS-114

(Information) %d LEQ cells found but not enabled due to dont_use on lib_cell.

**Description**

Prints the count of LEQ cell refrences which are not enabled due to dont_use attribute using command derive_clock_cell_references.

---

## CTS-115

(Warning) Failed to legalize cell '%s'.

**Description**

CTS cannot to legalize this cell. The failure could be caused by various reasons, such as congested floorplan, blockages, or large clock cell spacing rules.

**What Next**

Check the floorplan for high utilization or blockages. If a clock cell spacing rule is specified, try to remove or modify this rule.

Feedback

## CTS-116

(Warning) Mode '%s' is not active. set_clock_tree_options -copy_exception_across_modes will be ignored.

**Description**

The "-from_mode" and "-to_mode" modes specified in the "set_clock_tree_options -copy_exception_across_modes" command are required to be active. Otherwise, there will be no exception copying and this option is ignored.

**What Next**

Make sure that the scenarios of the "-from_mode" and "-to_mode" are active before invoking synthesize_clock_trees.

## CTS-117

(Error) There is no active scenario with setup or hold timing analysis enabled.

**Description**

CTS will work on an active scenario with setup or hold timing analysis enabled. Since CTS does not find such a scenario, it will not do anything.

**What Next**

Check the analysis settings in all scenarios. Modify the scenarios settings by *set_scenario_status* command.

## CTS-118

(Error) The core area of this design is invalid.

**Description**

Since the core area of the design is invalid, it is not suitable to run clock synthesis.

**What Next**

Check design and fix the block.

## CTS-119

(Error) The current design has been routed, so it is not appropriate to run this command.

Feedback

**Description**

This command works on a pre-routed design. It exits because this design has been routed.

**What Next**

Check the design and the flow.

---

## CTS-120

(Warning) The balance_point phase delay values '%s' set on mode '%s' and at pin '%s' is too large and will be removed.

**Description**

The phase delay value of the balance_point (considered for balancing) is much more than the tool's internal upperLimit (currently set as 20ns). This may cause synthesize_clock_tree to either hang or have bad results. The balance_point exception will be removed.

**What Next**

Make sure that the balance point phase delay values are correctly set to a reasonable value.

---

## CTS-121

(Information) Firewall ICG %s sized to %s.

**Description**

This information shows the firewall ICG has been sized for power in gate-by-gate synthesis stage.

---

## CTS-122

(Information) %d firewall ICGs are found in the design, %d of them have been sized.

**Description**

This information prints the summary of firewall ICG sizing for power in gate-by-gate synthesis stage.

---

## CTS-123

(Warning) Skipped buffering for net %s as buffer models are not available in all domains.

Feedback

**Description**

Net cannot be buffered if buffer models are not available for the net in all voltage domains.

---

## CTS-124

(info) Total %d ICGs are not considered in merging for reasons: don't touch - %d; fixed - %d; other reasons - %d.

**Description**

This information prints the summary of ICGs not considered for merging because of restrictions.

---

## CTS-125

(info) %s: reason - %s

**Description**

This information prints the detailed restrictions of an ICG not considered for merging.

---

## CTS-126

(info) Total %d ICGs are unique in the design.

**Description**

This information prints the summary of unique ICGs for merging in design.

---

## CTS-127

(info) %s: reason - unique

**Description**

This information prints unique ICG for merging in design.

---

## CTS-128

(info) Information: Transitive-fanout-based Root NDR feature is turned ON.

**Description**

This information notifies user that the transitive-fanout-based Root NDR feature is on. User has set_clock_tree_options -root_ndr_fanout_limit set to a positive integer value.

Feedback

## CTS-129

(info) Information: There are %d pre-existing gates having transitive fanout greater than user-defined -root_ndr_fanout_limit %d

**Description**

This information notifies user that with transitive-fanout-based Root NDR feature on, the statistics of the pre-existing gates with transitive fanout greater than user-defined threshold.

## CTS-130

(Information) Clock {%s} with %d user-defined skew group%s has a target latency applied to its default skew group. The latenc%s of the user skew group%s may be altered to achieve the target on the default skew group.

**Description**

This is a notice that, when a target latency is applied to a clock with user skew group(s), the latencies of the user skew groups may not necessarily be preserved if the default skew group needs to be delayed to achieve the target latency.

## CTS-131

(Information) Clock {%s} with %d user-defined skew group%s has a target latency applied to its default skew group. The latenc%s of the user skew group%s will be preserved, but more buffers may need to be added and global skew may not be preserved in achieving the target latency.

**Description**

This is a notice that, when a target latency is applied to a clock with user skew group(s) and the user has requested the user skew group latencies to be preserved, more buffer chains may need to be added to achieve the target latency on the default skew group without affecting the user skew groups. As this may result in parallel buffer chains downstream from the clock root, the global skew may not be preserved during target delay insertion.

## CTS-132

(Information) To achieve the target latency on the default skew group of clock {%s} while preserving the latencies of user skew groups, %d parallel repeater chains will be inserted.

Feedback

**Description**

This is an informational message quantifying the effect of preserving the user skew group latencies of the clock in order to meet the target latency on the default skew group.

---

## CTS-133

(Warning) Skew group {%s} contains %ld non-boundary sink%s. The target latency constraint on this skew group will be ignored.

**Description**

A boundary skew group must only contain boundary sinks for the target latency constraint to be effective.

**What Next**

Check the skew group definition and remove the non-boundary sinks as listed in the report.

---

## CTS-134

(Warning) Unable to characterize library cell '%s' in corner '%s'. Please check your CTS reference cells.

**Description**

An unexpected or out-of-range value in the library cell definition has caused a calculation error while characterizing the delay model.

**What Next**

Check the definition of library cells used for CTS purpose.

---

## CTS-135

(Warning) A guide buffer of type '%s' could not be inserted on the net driven by '%s'.

**Description**

CTS inserts guide buffers for various reasons in order to separate a subtree of the clock network for the purpose of clock tree synthesis and optimization. No such guide buffer could be inserted on any of the hierarchies of this net, and the resulting lack of separation may have unintended consequences for clock tree synthesis and/or optimization.

**What Next**

Check the bufferability of this net.

Feedback

## CTS-136

(Warning) A guide buffer of type '%s' could not be inserted in front of %ld of %ld load(s) on the net driven by '%s'.

### Description

CTS inserts guide buffers for various reasons in order to separate a subtree of the clock network for the purpose of clock tree synthesis and optimization. No such guide buffer could be inserted on any of the hierarchies of this net, and the resulting lack of separation may have unintended consequences for clock tree synthesis and/or optimization.

### What Next

Check the bufferability of this net.

## CTS-137

(Warning) Delay insertion on driver pin '%s' skipped as the estimated number of repeater levels required to achieve the target delay exceeds the threshold.

### Description

CTS attempted to insert a chain of repeaters on this driver pin to achieve a certain target delay for skew reduction, but achieving such a target delay may result in long runtime and/or high power and area use.

### What Next

If it is expected to have to insert such a large number of repeaters to reduce skew within the clock tree, this checking may be disabled with *set_app_options -list {cts.common.unlimit_delay_insertion_stages true}*.

### See Also

- cts.common.unlimit_delay_insertion_stages

## CTS-138

(Warning) No buffer is inserted after root '%s' in %s.

### Description

CTS could not add buffers to balance the clock tree at this root with other clocks in the same balance group or roots within a multi-source clock.

Feedback

### What Next

Check constraints, available reference cells, and physical layout on the root net.

---

## CTS-139

(Information) CTO is relaxing latency of clock/skewgroups '%s' to benefit skew of clocks/skewgroups '%s'.

### Description

CTO is entering a mode of relaxing the latency of some clocks in order to benefit skew of some other clocks. This happens when the shortest path of the clocks to benefit skew are overlapping with longest/near-longest path of the clocks at least partially, and when there are no other places on the shortest path of the skew-benefit clocks to insert delay except the common path which share with the longest path of the clocks to relax latency.

### What Next

Please check the clock structure, especially those settings that is preventing delay insertion on shortest path of the skew-benefit clocks. If the settings are correct/expected, the relax of latency is expected.

---

## CTS-140

(Warning) The user-set target skew '%f' for clock '%s' is too tight (< '%f' of latency) and may cause too much latency degradation for the victim clock when advanced-skew-balance is trying to improve skew toward target skew.

### Description

CTO is entering advanced-skew-balance mode (a mode of relaxing the latency of some clocks in order to benefit skew of some other clocks). The target skew is being used to achieve better skew with the latency of some other clock increase tolerated. But the user-set target skew is too tight comparing to a certian percentage of its latency. Since CTO will try to improve skew towards user target skew, it may increase the other victim clock's latency by too much.

### What Next

Please check the target skew setting for the clock, keep in mind that working on skew too hard will increase latency of another clock.

Feedback

## CTS-141

(Information) Latency Bottleneck Path for clock '%s' and skew group '%s' in mode '%s' for clock root '%s':

**Description**

Reports the latency critical path for every clock and skew group in each mode for the primary corner.

## CTS-142

(Warning) To achieve the target latency for %s, %ld parallel repeater chains will be inserted.

**Description**

This warning message shows multiple parallel repeater chains will be added to cover the specific skew group without affecting the other sinks. It may heavily increase the repeater count.

**What Next**

Redefine the skew group or refine the clock tree structure to make all the sinks of that skew group in a separate branch.

## CTS-200

(Error) An unexpected error has occurred during CTS. Please report this information to Synopsys.

**Description**

An unexpected error has occurred during CTS.

**What Next**

The results of this command following this error may be unexpected. Please report this information to Synopsys.

## CTS-500

(error) Ignoring option set name '%s' as it hasn't been defined for '%s'.

Feedback

**Description**

The specified option set name has either not been defined or is not defined for the current application. Use the command 'set_multisource_clock_subtree_options' to define a set of options for subtree synthesis.

---

## CTS-501

(error) No option set exists for '%s'. Please use the '%s' command to define a set of options.

**Description**

The executed command requires that at least one option set with driver objects has been defined using the specified command.

---

## CTS-502

(info) option '%s' is ignored when option '%s' is specified.

**Description**

The printed options should not be used together. This is not an error but the first mentioned option is not considered by the command.

---

## CTS-503

(error) Net '%s' has no driver.

**Description**

The specified net cannot be used to determine a clock subtree driver. No port with direction 'in' or cell output pin is connected to the specified net.

---

## CTS-504

(error) Only driver pins are allowed. Pin '%s' is ignored.

**Description**

The specified cell pin is not of direction 'out'. Only output pins are allowed as driver objects.

---

## CTS-505

(error) Only driver ports are allowed. Port '%s' is ignored.

Feedback

**Description**

The specified port is not of direction 'in'. Only input ports are allowed as driver objects.

---

## CTS-506

(error) The -max_total_wire_delay option requires the -corners option.

**Description**

A corner or a list of corners is required in case the option -max_total_wire_delay is applied. The maximum total wire delay of a clock path from driver object to a sink can only be specified per corner.

---

## CTS-507

(error) Maximum total wire delay must be larger than 0.

**Description**

The maximum total wire delay is a delay value constraint that has to be positive.

---

## CTS-508

(error) The -corners option requires the -max_total_wire_delay option.

**Description**

A corner specification is only required if the -max_total_wire_delay option has been specified. The total maximum wire delay requires a corner.

---

## CTS-509

(error) '%s' is not a valid group name.

**Description**

The specified group name has not been defined using the 'set_multisource_clock_subtree_options' command.

---

## CTS-510

(warning) Clock '%s' not found in group '%s'.

Feedback

**Description**

The specified clock was not defined in the specified group using the 'set_multisource_clock_subtree_options' command.

---

## CTS-511

(warning) No maximum total wire delay defined for corner '%s' in group '%s'.

**Description**

There was no total maximum wire delay defined for the specified corner using the 'set_multisource_clock_subtree_options' command.

---

## CTS-512

(warning) No drivers defined for group '%s'.%s

**Description**

There are no driver objects specified using the 'set_multisource_clock_subtree_options' command for the specified group.

---

## CTS-513

(error) Wrong order of flow steps of options -from and -to.

**Description**

The flow step as specified by option -from has to be equal to or comes before the flow step as specified by option -to. The flow step order is: merge, optimize, route_clock.

---

## CTS-514

(warning) Couldn't create a unique %s name with prefix '%s' and suffix '%s'.

**Description**

The tool was unable to create a database object of the specified type with the given prefix and suffix settings. A reason could be that too many objects of the given type had to be created.

---

## CTS-515

(warning) %s Couldn't delete cell '%s'.

Feedback

**Description**

The tool was unable to delete a cell. Please check attributes on the cell or the hierarchy the cell belongs to.

---

## CTS-516

(warning) Failed to disconnect pin '%s'.

**Description**

The tool was unable to disconnect a cell pin from its net.. Please check attributes on the cell or the net the pin is connected to.

---

## CTS-517

(warning) Clock '%s' in mode '%s' is not valid in any timed scenario. The clock is skipped.

**Description**

There is no active timed scenario (hold or setup) that refers to the mode of the clock. As a consequence, no timing infromation for the clock is available.

**What Next**

Create and/or activate a scenario with the mode of the clock.

---

## CTS-518

(warning) No clock found for driver '%s'. This driver is skipped.

**Description**

A driver object was defined for a group using the 'set_multisource_clock_subtree_options' command which is not part of a clock path, i.e. no clock is arriving at that object. The driver object is ignored by synthesis.

---

## CTS-519

(warning) No driver for sink '%s'.

**Description**

While traversing a clock subtree to prepare synthesis, a clock sink was reached which is not reachable from a driver object.

Feedback

## CTS-520

(warning) Cannot resize lib cell '%s' because no context-valid lib cells are available.

**Description**

No library cell class nor a logically equivalent library cell was found for sizing.

## CTS-521

(error) The option set name '%s' is already used by %s.

**Description**

Subtree synthesis and tap assignment cannot use the same option set name.

**What Next**

Please use the -name option and select a different name.

## CTS-522

(warning) Failed to resize cell '%s' from '%s' to '%s'.

**Description**

The tool was unable to change the library cell of the specified cell to the specified library cell. Please check attributes on the given objects that might prevent resizing.

## CTS-523

(error) Pin '%s' is part of subtree '%s' but a driver of a subtree as well.

**Description**

A driver object that has been specified by the 'set_multisource_clock_subtree_options' command is in the fanout of another driver object. A nesting of clock subtrees is not allowed.

## CTS-524

(info) Selecting clock '%s' in scenario '%s' for driver '%s'.

**Description**

The specified clock arrives at the specified driver object. Constraints for that clock in the indicated scenario will be used to synthesize the subtree starting at the driver object.

Feedback

## CTS-525

(info) A maximum total wire delay of %s has been defined for this driver.

**Description**

A maximum total wire delay has been specified for a clock subtree using the 'set_multisource_clock_subtree_options' command.

## CTS-526

(info) %s '%s' is not splittable since %s.

**Description**

The object with the given path name cannot be split. This message prints the reason for it.

## CTS-527

(info) Found clock cell '%s' without a sub-tree behind it. Will treat this cell as a sink of the tree.

**Description**

A non-sequential cell in the clock subtree has no fanout. The cell is treated as a sink during subtree synthesis.

## CTS-528

(info) Resized %d cells to a different library cell.

**Description**

To prepare the synthesis of clock subtrees all non-sink clock cells are resized to a certain drive strength.

## CTS-529

(warning) Cannot move cell '%s' to (%s).

**Description**

The tool failed to place a clock cell to the desired location. Reasons could be cell congestion or blockages.

Feedback

## CTS-530

(warning) Large displacement of %s while moving cell '%s'. Placed location: {%s} target location: {%s}.

**Description**

The tool failed to move a clock cell to the desired location. Reasons could be cell congestion or blockages.

## CTS-531

(error) Couldn't connect pin '%s' of a splitted cell.

**Description**

The tool failed to connect a new cell pin. Please check attributes on the cell and the corresponding hierarchy levels.

## CTS-532

(error) No tree found for node '%s'.

**Description**

While splitting clock cells to assign them to preroute structures, the tool was unable to find a matching subtree for a given node. This indicates a synthesis problem.

## CTS-533

(warning) Total wire delay violation on path through cell '%s' which cannot be split. Total wire delay: %s.

**Description**

A maximum total wire delay violation was detected on a clock path through the specified cell. However, the cell could not be split to reduce the total wire delay.

## CTS-534

(warning) Latency violation at cell '%s' which cannot be split to fix the violation. Latency %s.

Feedback

**Description**

A latency violation was detected on the specified cell. However, the cell could not be split to reduce the latency.

---

## CTS-535

(warning) Failed to split cell '%s'. This might degrade QoR.

**Description**

The tool was unable to split the specified cell. A reason could be that cells in the direct fanout are placed at identical locations.

---

## CTS-536

(warning) DRC violation at output '%s' which cannot be split. Max trans: %s max cap: %s fanout: %d.

**Description**

A DRC violation (max transition, max capacitance, or max fanout) was detected on the specified cell output. However, the cell could not be split to reduce or fix the violation.

---

## CTS-537

(warning) Failed to move sink '%s' to nearest cluster at (%s).

**Description**

The tool was unable to reassign a clock sink to a different driver pin. This might degrade QoR.

---

## CTS-538

(warning) Failed to delete a dangling tree node at output '%s' because it still has still loads.

**Description**

The tool was unable to remove a cell which is supposed to not drive any sinks anymore. The connected pins are printed out.

---

## CTS-539

(warning) Unable to connect loads to output '%s'.

Feedback

**Description**

The tool was unable to reconnect clock cell inputs to the specified output. The input pins are printed out.

---

## CTS-540

(info) %s violation detected at cell '%s' (latency: %s, wire RC: %s).

**Description**

The tool has detected a violation on a cell which will be split.

---

## CTS-541

(info) Created new cell '%s' with a fanout of %d. tran: %g latency: %g.

**Description**

The tool has created a new cell by splitting.

---

## CTS-542

(info) Created new cell '%s' with a fanout of %d.

**Description**

The tool has created a new cell by splitting.

---

## CTS-543

(info) Preroute splitting created new cell '%s' with a fanout of %d.

**Description**

The tool has created a new cell by splitting for preroute shapes.

---

## CTS-544

(warning) Cannot extract preroutes: no net connected to driver '%s'.

**Description**

No net preroute shapes could be extracted as the clock subtree driver has no net connected to it.

Feedback

## CTS-545

(warning) Cannot extract preroutes: no physical net attached to net '%s'.

**Description**

No net preroute shapes could be extracted as the net connected to the clock subtree driver has no physical information.

## CTS-546

(error) Could not identify any multi-source subtree drivers.

**Description**

The 'synthesize_multisource_clock_subtrees' command could not find any groups to synthesize. Please use the 'set_multisource_clock_subtree_options' command to define a group.

## CTS-547

(error) Failed to mark clock trees.

**Description**

The clock tree marking step failed.

## CTS-548

(error) Failed to legalize placement.

**Description**

The final legalization step failed.

## CTS-549

(error) Failed to route clock nets.

**Description**

The final detail routing of the clock nets failed.

## CTS-550

(info) Successfully %s.

Feedback

**Description**

The given step finished successfully after synthesizing the clock trees.

---

## CTS-551

(warning) Merge: Failed to move first-level cell '%s' from root '%s' to root '%s'.

**Description**

The command 'synthesize_multisource_clock_subtrees' was unable to reconnect a first-level clock cell to a closer driver object.

---

## CTS-552

(warning) Cannot rename cell '%s' while merging to '%s'.

**Description**

The command 'synthesize_multisource_clock_subtrees' could not change the cell name of a merged cell according to the merge naming rule using prefix, suffix, and name concatenation.

---

## CTS-553

(info) Merge: Moved first-level cell '%s' from root '%s' to root '%s'.

**Description**

The command 'synthesize_multisource_clock_subtrees' reconnected a first-level clock cell to a closer driver object.

---

## CTS-554

(error) Check failed after %s.

**Description**

This message indicates a severe problem. The synthesis might have created a netlist which is not equivalent with the original netlist. Check previous warning and error messages for failures.

---

## CTS-555

(error) Failed to create subtree '%s'.

Feedback

**Description**

The command 'synthesize_multisource_clock_subtrees' was unable to create a tree representation of the given driver object.

---

## CTS-556

(error) Splitting %s failed for subtree '%s'.

**Description**

The command 'synthesize_multisource_clock_subtrees' was unable to perform the described task for the subtree of the given driver object. Please check for earlier error and warning messages or enable verbose output.

---

## CTS-557

(error) Snapping of first-level cells failed.

**Description**

The command 'synthesize_multisource_clock_subtrees' was unable to snap the first-level clock cells close to the preroutes of the subtree driver.

---

## CTS-558

(error) Switching clocks to propagated mode of the trees failed.

**Description**

The command 'synthesize_multisource_clock_subtrees' was unable to switch the relevant clocks to propagated mode. This might result in wrong timing results in the clock tree.

---

## CTS-559

(warning) Failed to snap first level cell '%s' to %s.

**Description**

The command 'synthesize_multisource_clock_subtrees' was unable place the specified cell close to a preroute shape. Reasons might be cell congestion or placement blockages.

---

## CTS-560

(warning) Preconditioning of the clock trees failed.

Feedback

**Description**

Some setup steps failed while executing the command 'synthesize_multisource_clock_subtrees'. Please check earlier warning or error messages or enable verbose output.

---

## CTS-561

(warning) Clock subtrees of clock '%s' at root pin '%s' have no considered sinks. Optimization will be skipped.

**Description**

The clock subtrees of the clock do not have considered sinks. This means that there are no sinks, or that none of the sinks is considered for balancing. As a consequence no latency nor skew optimization will be performend.

**What Next**

Check the netlist to see whether any sinks are connected to the mentioned root pin. Check the 'set_clock_balance_point -consider_for_balance false' constraints to make sure that not all sinks are ignored for balancing.

---

## CTS-562

(warning) Adding multiple clocks to a group '%s' is not recommended. This might degrade QoR or lead to other problems during synthesis. For example, constraints referring to these clocks are not adapted when moving first-level clock cells from a clock source of one clock to a clock source of a different clock.

**Description**

It is recommended that you use multiple groups if you want to synthesize a set of clocks. Each group can have several clock sources of the same clock.

---

## CTS-563

(error) Design %s has no technology data.

**Description**

There is no technology data associated with the design.

---

## CTS-564

(error) Design %s has no physical data.

Feedback

**Description**

There is no floorplan defined for the design.

---

## CTS-565

(error) Design %s has invalid global-route cells.

**Description**

The design is not yet setup properly for running global routing.

---

## CTS-566

(error) Detected reconvergence in the clock tree at pin '%s'. This is not supported.

**Description**

There is a reconvergent clock structure at the reported pin. This command does not support subtrees with such clock structures.

---

## CTS-567

(error) There are design constraints that refer to hierarchical clock pin '%s'. This is not supported.

**Description**

There are design constraints that refer to the reported hierarchical pin. This command does not support subtrees with such hierarchical pins.

---

## CTS-568

(error) Multiple clocks used for drivers in group '%s'. Please restrict the number of clocks to one when defining the group with 'set_multisource_clock_subtree_options'.

**Description**

The tool was unable to select a single clock for all subtree drivers of the specified group. It is recommended that you use multiple groups if you want to synthesize a set of clocks. Each group can have several clock sources of the same clock.

---

## CTS-569

(info) Optimization %s %d cell%s to %s.

Feedback

**Description**

This message summarizes netlist changes for subtree optimization.

---

## CTS-570

(error) Sink alignment failed.

**Description**

This message indicates that the sink alignment step failed. Check previous warning and error messages.

---

## CTS-571

(warning) Routing failed for net '%s': %s.

**Description**

This message explains why the given net cannot be routed.

---

## CTS-572

(error) No options have been defined for the specified clock%s.

**Description**

The -clocks option specifies clocks for which no option set has been defined.

---

## CTS-573

(info) Using target latency %s for clock '%s' at corner '%s'.

**Description**

A target latency value has been specified using the set_clock_tree_options command for the given clock.

---

## CTS-574

(info) Skew mode enabled. Using an estimated target latency %s for clock '%s' at corner '%s'.

Feedback

**Description**

No target latency value has been specified using the set_clock_tree_options command for the given clock. The synthesis command will use an estimated target latency to perform clock cell splitting.

---

## CTS-575

(info) Skew mode is enabled for all clocks.

**Description**

A target latency value has not been specified using the set_clock_tree_options command for all the clocks. The synthesis command will use an estimated target latency to perform clock cell splitting. You can increase the verbosity to see the estimated target latency per clock driver.

---

## CTS-576

(Error) Port '%s' is unplaced. This command requires a placed design.

**Description**

This command requires that the ports driving the clock subtrees, are placed.

**What Next**

Check that the reported port has terminals.

---

## CTS-577

(Error) Cell '%s' is unplaced. This command requires a placed design.

**Description**

This command requires that the cells connected to the clock subtrees, are placed.

**What Next**

Check the placement status of the reported cell.

---

## CTS-578

(warning) %s arrive at the driver '%s'. The cell of that driver is skipped.

Feedback

**Description**

The user has restricted the command to a set of clocks. However, none of the specified clocks propagate through the specified driver. The driver object is ignored by synthesis.

---

## CTS-579

(warning) The clock '%s' of group '%s' doesn't arrive at the driver '%s'. The cell of that driver is skipped.

**Description**

The user has specified a clock of a synthesis group. However, that specified clock doesn't propagate through the specified driver. The driver object is ignored by synthesis.

---

## CTS-580

(error) No groups specified. Please use the 'set_multisource_clock_subtree_options' command to define at least one group.

**Description**

The executed command requires that at least one group has been defined using the 'set_multisource_clock_subtree_options' command.

---

## CTS-581

(warning) %s {%s} is outside of track area.

**Description**

The requested strap shape cannot be created because there are no routing tracks at the strap location.

**What Next**

No routing tracks could be identified at or near the requested location that satisfy the -margins setting. Make sure that there are routing tracks in the requested layer at the requested location.

---

## CTS-582

(info) Created %d %s (shifted %d tracks%s).

Feedback

**Description**

Straps were created. The message contains information about the number of straps, the shift with respect to the requested location and about the shielding of the straps.

**What Next**

No action is required. This is an informative message.

---

## CTS-583

(info) Found %d %s shielding shapes for biasing.

**Description**

Informative message about the number of shielding shapes that were found for biasing the straps.

---

## CTS-584

(warning) Cannot create %s side of ring.

**Description**

The indicated side of the ring cannot be created.

**What Next**

Check for blockages at the location where the ring should be created.

---

## CTS-585

(error) No spine shapes found%s.

**Description**

The tools was unable to detect spine shapes that can be used for stripe connection to the net.

**What Next**

Make sure that there are spine shapes to connect the stripes to.

---

## CTS-586

(info) Spine {%s} of net %s has influence area {%s}.

Feedback

**Description**

Reports the influence area of a detected spine. Stripes in that area will be connected to this spine.

---

## CTS-587

(info) Created ring with bounding box {%s}.

**Description**

Reports the bounding box of the created ring around the mesh.

---

## CTS-588

(error) Could not create ring shapes.

**Description**

The ring around the mesh could not be created.

**What Next**

Check for blockages at the locations where the ring should be created.

---

## CTS-589

(warning) Could not create the %s-ends part.

**Description**

The ends part as requested by the -create_ends option could not be implemented.

**What Next**

Check for blockages at the locations where the ends should be created.

---

## CTS-590

(warning) No %s grid defined.

**Description**

No grid definition for the named direction was specified. As a consequence, no regular stripes in that direction will be created.

Feedback

**What Next**

If this is the intention, no action is required. Otherwise, please specify a grid using -grid {start stop step}

---

## CTS-591

(warning) Margin is set to 0 for %s straps. Grid point should be exactly on routing tracks.

**Description**

When the margin is 0, straps will only be created if the specified grids are exactly on routing tracks.

**What Next**

Specify a non-zero margin using the -margin option to allow snapping of the grids to routing tracks.

---

## CTS-592

(warning) Skipped creation of %s strap {%s} because it is not inside the boundary.

**Description**

The strap was not created because it is outside the area for strap creation. The area for strap creation is the core area of the block. If an explicit boundary was specified using the *-boundary* option, then the intersection of that boundary and the core area is considered as the area for strap creation.

**What Next**

Enlarge the boundary or change the strap grid definition.

---

## CTS-593

(warning) Could not create %s strap {%s} because %s.

**Description**

The strap was not created. There can be multiple reasons. 1. The strap would cause a design rule violation (spacing, overlap, ..) with existing shapes. 2. The strap would overlap with user specified keepouts (using the -keepouts option). 3. The strap does not connect to shapes in the orthogonal direction because it does not overlap with any of those orthogonal shapes, or it is impossible to connect to the orthogonal shape with a via.

Feedback

**What Next**

Inspect the layout for other shapes in the same layer that may cause DRCs. Check the position of keepouts and see whether the requested strap would overlap with a keepout region. If the -allow_splitting option was set to false, you don't allow partial straps to be created. In that case, straps that do not span the full width or height of the bounding box will be skipped. Use the -allow_floating option if you do not require straps to connect to shapes in the orthogonal layer. Use app option cts.common.verbose to increase the verbosity such that more information about the exact cause of the problem is printed.

---

## CTS-594

(info) Requested creation of %d %s stripes.

**Description**

Reports the requested number of stripes in the named direction based on the grid definition.

---

## CTS-595

(warning) %d %s grid lines were skipped because %s.

**Description**

Reports the reason why straps at certain grid lines were not created.

**What Next**

Increase the value of cts.multisource.verbose to increase the verbosity level of the command.

---

## CTS-596

(info) Created %d %s straps on %d grid lines.

**Description**

Reports the number of straps that were created in the named direction. It also reports how many grid lines are addressed by these straps.

---

## CTS-597

(info) Created %d end-straps on %d grid lines.

Feedback

**Description**

Informative message on the number of end straps (induced by the -create_ends options) that were created.

---

## CTS-598

(info) For influence area {%s} the specified length is too large for the specified backoff value.

**Description**

Both -length and -backoff values were specified. The length could not be realized without violating the backoff value. The backoff is respected, and the length was decreased.

**What Next**

In general, this is not a problem. Both -length and -backoff define the final length of the stripes and the length will be adjusted if the backoff value is violated.

---

## CTS-599

(warning) Physical net for %s was not found. This net is skipped.

**Description**

During clearing of straps, the named physical net could not be found.

---

## CTS-600

(info) Removed %d shapes from net %s.

**Description**

During clearing of straps, the number of shapes that is removed from the named net is reported.

---

## CTS-601

(info) Creating ring.

**Description**

Informative message that ring creation has started.

Feedback

## CTS-602

(error) %s%s%s%s%s

**Description**

There is an error in the argument list of the create_clock_straps command. The additional information should be self explanatory.

**What Next**

Correct the error in the argument list. See the manpage of create_clock_straps for more information about argument syntax and semantics.

## CTS-603

(error) Could not create straps.

**Description**

The command failed because no straps were created.

**What Next**

Earlier errors should indicate what the cause of the failure is. Please scroll up in the log to check for earlier errors.

## CTS-604

(info) %s layer %s will be used in non-prefered direction.

**Description**

For the named layer, straps in the non-prefered direction will be created. This should only happen if the user has requested this.

**What Next**

No further action required. This is just an informative messages.

## CTS-605

(info) No -boundary specified. Using core area {%s}.

**Description**

The user did not specify a -boundary for strap creation. As a consequence, straps will be created for the full core area of the block.

Feedback

**What Next**

No further action required. This is just an informative messages.

---

## CTS-606

(info) Using double spacing (%s) for backoff.

**Description**

The user did not specify a -backoff for strap creation. By default, double width-dependent minimum spacing is used as a separation of stripes that are connected to different spines.

**What Next**

See the manual page of create_clock_straps for the semantics of -backoff.

---

## CTS-607

(info) Found %s net for shielding: %s.

**Description**

The named power or ground net was identified as a net that can be used for shielding of the created straps.

**What Next**

No further action required. This is just an informative messages.

---

## CTS-608

(error) %s%s%s%s%s

**Description**

There is an error in the argument list of the create_clock_drivers command. The additional information should be self explanatory.

**What Next**

Correct the error in the argument list. See the manpage of create_clock_drivers for more information about argument syntax and semantics.

---

## CTS-609

(warning) Failed cell placement prevents further splitting of that cell and its clones.

Feedback

**Description**

The tool failed to place a clock cell to the desired location. No additional splitting will happen to that cell and its clones.

**What Next**

Check for limited area in voltage areas or move bounds. Placement blockages might also cause over-utilization.

---

## CTS-610

(warning) The option '%s' is deprecated and will be removed in future. Please use option '%s' instead.

**Description**

The given option is deprecated. Please use the replacement option as given in this message.

---

## CTS-611

(error) Multi-output clock cell '%s' detected. This is only supported if at most one output is connected to a net.

**Description**

Cells with multiple connected outputs are currently not supported by the executed command. Only one of the outputs is allowed to be connected to a net.

**What Next**

If multiple output pins are connected to nets but only one of them has loads, then remove the net without any loads.

---

## CTS-612

(warning) Some cells are not allowed to be %s. Reasons are:

**Description**

The listed constraints prevent the command from merging or splitting some cells.

**What Next**

Set the application option cts.multisource.verbose to 8 to get list of cells with a reason why it cannot be split or merged with CTS-613 messages. Check whether the constraints are intended and remove them to enable cell merging and splitting.

Feedback

## CTS-613

(warning) Cell '%s' is not %s since the cell %s.

### Description

The message provides an indication of why a cell cannot be split or merge. Following are the reasons that prevent clock cell splitting and merging:

```
1. Cell has physical_status: locked, legalize_only, or fixed
2. Cell has user_size_only or dont_touch
3. Constraints on path between its driver and the loads of this cell:
    a. A don't-touch net between driver and a load
    b. Path crosses a physical hierarchy
    c. Path crosses a power domain boundary and port punching on it is
 not allowed
    d. A logical read-only hierarchy (like a black box)
    e. A hierarchy with frozen ports: Either due to set_freeze_ports or
 due to
       timing constraints on the hierarchical pin
```

### What Next

Please check net don't touch settings or constraints on the logical and physical hierarchies.

## CTS-614

(error) The %s '%s' is ignored as it is already defined as a driver object for %s.

### Description

The given pin or port has already been specified as a driver object for a different purpose.

### What Next

Either correct the pin or port name or remove ther option setting that revers to this driver object.

## CTS-615

(Error) Net %s cannot be routed by route_clock_straps, because %s.

### Description

The route_clock_straps command can only be used to route clock nets with clock straps. The given net does not satisfy one of these criteria, as explained in the message.

Feedback

## CTS-616

(Error) The %s routing step failed.

**Description**

The given routing step of the route_clock_straps command failed for one or more of the given nets. See output earlier in the log for more details.

## CTS-617

(Error) %s routing cannot route net %s, because %s.

**Description**

Routing for given topology cannot be done, for the reason explained in the message.

## CTS-618

(warning) Large legalization distance %s for cell %s. It was moved from preferred location {%s} to legalized location {%s}.

**Description**

The cell was moved over a large distance during legalization. No -max_displacement option was given, so the cell was inserted at the legalized location. This may point to high utilization, or blocked area near the preferred location.

**What Next**

Check the floorplan for high utilization or blockages. Consider to specify a -max_displacement to avoid that cells get inserted far away from the preferred location.

## CTS-619

(warning) Option -convert_ideal_latencies is deprecated. It will not be supported in future releases. Please use derive_clock_balance_points for ideal latency conversion.

**Description**

The automatic conversion of ideal latencies to set_clock_balance_points constraints is not supported as an option anymore. There is a separate command *derive_clock_balance_points* that is doing this conversion. Use the *-reference_latency 0.0* option to get the same behavior as -convert_ideal_latencies.

**What Next**

Adapt your script to change

Feedback

```
synthesize_multisource_clock_subtrees -convert_ideal_latencies
```

to

```
derive_clock_balance_points -reference_latency 0.0
    synthesize_multisource_clock_subtrees
```

## CTS-620

(warning) Below lib cells will not be used because these lib cells %s%s:

**Description**

The specified lib cells will not be used during clock driver insertion for the reason as mentioned in the message.

**What Next**

Check whether the mentioned reason is expected. If not, use commands set_lib_cell_purpose, set_dont_touch or set_dont_use to change the lib cell attributes as required. If the reason is expected, you may consider to not pass the offending lib cells to the create_clock_drivers command. That will avoid the message.

## CTS-621

(error) Below lib cells are not buffers%s:

**Description**

The specified lib cells can not be used during clock driver insertion because these are not buffer lib cells.

**What Next**

The create_clock_drivers command only accepts buffers as lib cells. If other cells need to be inserted, you need to insert a template cell in the netlist and use the -template option to indicate that this template cell should be cloned.

## CTS-622

(error) A mix of buffers and inverters is specified in the -lib_cells option%s.

**Description**

The polarity of all cells in the list of lib_cells should be the same.

**What Next**

Remove either the inverters or the buffers from the list of lib_cells.

Feedback

## CTS-623

(error) No usable lib cells found, aborting clock driver insertion%s.

**Description**

None of the specified lib_cells can be used by create_clock_drivers. The reasons why certain lib_cells could not be used were listed in earlier messages.

**What Next**

Make sure to specify at least one lib_cell that can be used for clock driver insertion.

## CTS-624

(error) %s Both -template and -lib_cells defined%s. This is not supported.

**Description**

For each level, the create_clock_drivers command can either insert buffers/inverters, or clone an existing template cell to create the drivers. It cannot do both.

## CTS-625

(error) Pin '%s' is not connected to the proper net. This pin is skipped.

**Description**

All load pins should be connected to the same physical net. The mentioned pin is connected to a different physical net than the others.

**What Next**

Make sure that all specified load pins are connected to the same physcial net.

## CTS-626

(error) Net driver pin/port could not be identified%s.

**Description**

The net of the specified load pins does not have a driver. The create_clock_drivers command requires that the net of the load pins is driven.

**What Next**

Make sure that a driver of the load pins exists in the netlist.

Feedback

## CTS-627

(error) Driver of net '%s' is a port. This is not supported with -template%s.

**Description**

The -template option instructs the create_clock_drivers command to clone the existing driver cell of the load net. If the load net is driven by a port, there is no template cell that can be cloned.

**What Next**

Make sure that the load net is driven by a lib_cell when using -template.

## CTS-628

(error) Load net '%s' is multi-driven. This is not supported with -template%s.

**Description**

The -template option instructs the create_clock_drivers command to clone the existing driver cell of the load net. If there are multiple driving cells, the driver cannot be identified.

**What Next**

Make sure that the load net is driven by a single lib_cell when using -template.

## CTS-629

(Warning) Restricting the %s value to %s, to ensure proper comb routing.

**Description**

The given fishbone (sub-)span is restricted to the value reported in the message, to avoid that in the resulting fishbone routing, the distance from the pins to the fishbone topology exceeds the maximum comb routing distance.

## CTS-630

(Warning) Ignoring stripe {{%.3f %.3f} {%.3f %.3f}} on layer %s because it does not follow the preferred routing direction.

**Description**

The stripe shape indicated in the message is ignored by fishbone routing, because it does not follow the preferred routing direction of the layer. Comb routing can still connect to the stripe.

Feedback

## CTS-631

(Error) Option %s can only be used with a fishbone topology.

**Description**

The given option of the route_clock_straps command only applies to the fishbone topology and cannot be used with -topology comb.

## CTS-632

(Warning) Failed to route a fishbone finger to the following pins: %s. These pins will still get connected to a clock stripe by comb routing.

**Description**

The route_clock_straps command failed to route a fishbone finger to the reported pins. These pins will still be connected to the nearest clock stripe by comb routing, but that can require a longer comb route.

## CTS-633

(Warning) Found: %d dont-touch net%s, %d power-domain boundary pin%s, %d physical hierarchy pin%s, and %d frozen hierarchy pin%s limiting clock cell merge and split.

**Description**

In case there is a dont-touch net, a power domain boundary, a physical hierarchy pin, or a frozen hierarchy pin between a driver and its fanout, then the driver cell cannot be split or merged together with an equivalent cell. You can set the application option cts.multisource.verbose to 8 to see a full list of pins that are affected by these settings.

**What Next**

Check that any reported don't touch net or frozen pin is required. If not, clear the corresponding setting and rerun the command.

## CTS-634

(Error) Detected cyclic dependency while calculating the level of cell '%s'.

**Description**

Before performing splitting to fix DRC violations, all cells get a level assigned. Cells with highest level (counted from the driver object) are split first. Levelizing cells requires that there are no cycles in the clock path. This message indicates that a cycle has been found around the given cell.

Feedback

**What Next**

If the cycle is inside the enable logic paths, then try to use a lower tracing level number.

---

## CTS-635

(error)MV aware global clock tree construction is not possible, because %s.

**Description**

The analysis of the problem instances showed that global clock tree construction is not possible due to limitations in the supported configurations. Either change the location of the global tree sinks, or modify the area where to place the clock drivers.

---

## CTS-636

(error)MV analysis indicates that below required buffering configuration does not lead to valid supply information.

**Description**

During the global clock tree construction a required buffering configuration was encountered for which the multi voltage infrastrcuture indicated that no valid setup could be found.

**What Next**

To find more details use command *check_bufferability* and pass the shown configuration. If possible correct the setup and re-run the command.

---

## CTS-637

(error) Stop processing because no valid buffering solution was found for net '%s'.

**Description**

The command *synthesize_multisource_global_clock_trees* evalutes several solutions to buffer a given net. Many restrictions, foremost multi-volatge related ones, are limiting or excluding possible solutions.

**What Next**

To find more details use command *check_bufferability* for the failing net. If possible correct the setup and re-run the command.

Feedback

## CTS-638

(warning) Dropping %s rail %s '%s' because %s.

**Description**

The command *synthesize_multisource_global_clock_trees* expects that either all user specified repeaters are of version single or dual rail, or for each repeater type there is a pair of single and dual rail version specified which are from their repective electrical behavior (input capacitance, internal deay and load dependend delay) exchangable. This is necessary to obtain a balanced global tree. Please note that the pairing is based on the specified order of library cells passed with option *-lib_cells* of command *synthesize_multisource_global_clock_trees*.

**What Next**

Check the set of library cells passed with option *-lib_cells* into command *synthesize_multisource_global_clock_trees* and adopt it if necessary to match the above specified requirements.

## CTS-639

(error) Stop processing because no buffering solution was found at level '%d'.

**Description**

The command *synthesize_multisource_global_clock_trees* evalutes several solutions to buffer a given net. Many restrictions, foremost multi-volatge related ones, are limiting or excluding possible solutions.

**What Next**

To find more details, see warning *CTS-640*. If possible choose different lib-cells and re-run the command.

## CTS-640

(warning) There was no meaningful pairing of single/dual rail buffers to drive.

**Description**

Not able to find a pair of single/dual rail buffers to drive a meaningful distance.

## CTS-641

(Information) Selecting via ladder %s for pin %s.

Feedback

### Description

The command *synthesize_multisource_global_clock_trees* applies suitable via ladder constraints on output and input pins of inserted buffers considering user input as well as library constraints.

---

## CTS-642

(Error) The timer anchor driver '%s' of the driver object '%s' is not on the clock network of clock '%s'.

### Description

In the MSCTS flow, a mesh net can be modeled by defining annotated delay and transition values for all mesh drivers along with the mesh net and mesh loads. For a multi-driven net the timer disables all but one mesh driver internally, the driver with timing arc retained is termed as anchor driver. Ensure there are no user defined disabled timing arcs for the mesh drivers and let tool derive the same since a mismatch can result in a case where no clock reaches the subtree driver object.

### What Next

Check with the 'report_disable_timing' command whether the input to output arc of the timer anchor is enbled.

---

## CTS-643

(Warning) Could not place cell %s at location (%f, %f) due to existing pre-routes. Removing the temporary placement blockages for the remaining flow and trying to place the cell again. This could lead to high run time at routing stage and could also leave some nets unrouted in the worst case.

### Description

Metal shapes in preferred routing layers are converted to placement blockages before placing the cells. However, if it is not possible to place one or more cells with the existing blockages, the placement blockages are removed and the cells are placed. This is equivalent to the behaviour of app option cts.multisource.enable_pin_accessibility_for_global_clock_trees being set to false. This could lead to high run time at routing stage and could also leave some nets unrouted if the pins of some of the cells are inaccessible due to pre-routes.

---

## CTS-644

(Error) No active scenario with setup or hold enabled found for clock '%s' in mode '%s' for option set '%s'.

Feedback

**Description**

Each option set for MSCTS has a clock defined for a given mode. It is required that a scenario exists with that mode which is active and has either setup or hold analysis defined. This message indicates that no such scenario is found for the named option set.

**What Next**

Check with the 'report_multisource_clock_subtree_options' respectively the 'report_multisource_clock_tap_options' command whether the specified clock is defined for the right mode. Use the 'report_scenarios' command to check that the desired scenario is active and activated for stup or hold analysis.

---

## CTS-645

(info) Reconvergence detected for clock '%s' at pin '%s' through input pins %s.

**Description**

Multiple paths of a clock are reconverging through different input pins of clock cell.

---

## CTS-646

(info) Clocks %s overlap at pin '%s' through input pins %s.

**Description**

Paths of different clocks are overlapping through different input pins of a clock cell.

---

## CTS-650

(error) %s%s%s%s%s

**Description**

There is an error in the argument list of the synthesize_multisource_global_clock_trees command. The additional information should be self explanatory.

**What Next**

Correct the error in the argument list. See the manpage of synthesize_multisource_global_clock_trees for more information about argument syntax and semantics.

---

## CTS-651

(error) Skipping library cell '%s' because it is neither a buffer nor an inverter.

Feedback

**Description**

The specified library cell has to be of type buffer or inverter.

**What Next**

Check the option '-repeaters' and remove all library cells which are neither buffer nor inverter.

---

## CTS-652

(error) Skipping repeater '%s' because flag 'dont_touch' is set to true.

**Description**

The specified library cell of type repeater is marked 'dont_touch' and hence is not allowed to be used.

**What Next**

If the repeater has to be used use command 'set_dont_touch' to overwrite the flag. Otherwise check the option '-repeaters' and remove all library cells with mark 'dont_touch'.

---

## CTS-653

(error) Skipping library cell '%s' because it does not belong to a reference library of the current design.

**Description**

The specified library cell belongs to a library which has not been used for linking the current design.

**What Next**

Check the option '-repeaters' and remove all library cells which are not belonging to the list of linked libraries.

---

## CTS-654

(error) No legal repeater remain from option '-repeaters'.

**Description**

The checking of all library cells specified with option '-repeaters' end in an empty list of usable repeaters.

Feedback

**What Next**

Check the option '-repeaters' and specify a list of library cells which are legal repeaters.

---

## CTS-655

(error) No physical net found for net '%s' which should be synthesized.

**Description**

No physical net information has been found for given logical net.

**What Next**

Check the consistency of the database.

---

## CTS-656

(error) No %s routing layer specified for net '%s'.

**Description**

For each net to be synthesized and/or routed a min and max routing layer must be found to guide the router where to implement the physical layout of the net.

**What Next**

Use command 'set_routing_rule' or 'set_clock_routing_rules' to explicitly defined a lowest or highest routing layer to be used for that net.

---

## CTS-657

(error) The min/max layer specification for net '%s' ' does not include a %s routing layer.

**Description**

The specified min/max routing layers for the given net have to cover a horizontal and a vertical routing layer.

**What Next**

Check the specification of command 'set_routing_rule' and/or 'set_clock-routing_rules' for that net. If neither is given, check if the highest two routing layers of the technolgy are orthogonal to each other.

Feedback

## CTS-658

(error) The min/max layer specification for net '%s' has to form an adjacent set of horizontal/vertical routing layers."

**Description**

The sepcified min/max routing layer pair for the given net has to be an adjacent layer pair.

**What Next**

Check the specification of command 'set_routing_rule' and/or 'set_clock-routing_rules' for that net.

## CTS-659

(info) Using routing H/V layer pair '%s'/'%s' for net '%s'.

**Description**

Reports the used horizontal an vertical routing layer for the given net.

## CTS-660

(error) Stop processing because the set of loads of net '%s' are not distributed on a regular and uniform grid.

**Description**

The command 'synthesize_multisource_global_clock_trees' assumes that the loads for which a global tree should be synthesized are distributed in a regular and uniform grid.

**What Next**

Inspect the location of the loads of the net and adjust them such that they are distributed in a uniform and regular grid.

## CTS-661

(info) Using the following DRC values for net '%s' :%s

**Description**

Reports the design rules constraints used during synthesize of the given net.

Feedback

## CTS-662

(info) Using routing H/V layer pair '%s'/'%s' for all the nets in logica level %d.

**Description**

Reports the horizontal an vertical routing layer used for the routing of all nets at the given logical level counted from the specified roots.

## CTS-663

(error) The processing of the route-only flow stopped because %s%s%s.

**Description**

The input data for the route-only flow does not match the required specification.

**What Next**

Check and correct the database and re-run the command.

## CTS-664

(error) The pre-processing of the clock data failed.

**Description**

Some setup in the general clock data is invalid.

**What Next**

Check the logfile for other warnings and errors.

## CTS-665

(error) Stop processing of command '%s' because %s.

**Description**

The input data for the command '%s' does not match the required specification.

**What Next**

Check and correct the database and re-run the command.

## CTS-666

(error) Cannot split template cell '%s' because the cell %s%s.

Feedback

**Description**

The command was unable to split (clone) the existing template cell as requested by the -template option because of the mentioned reason.

**What Next**

Remove the reason that prevents the template cell from being split.

---

## CTS-667

(warning) Cannot insert driver '%s' at location {%s} because %s.

**Description**

The command was unable to insert a driver at the specified location because of the reason mentioned in the message.

**What Next**

Investigate whether the mentioned reason is valid and whether the dropping of the driver is acceptable.

---

## CTS-668

(error) No drivers were inserted%s because MV violations would be created.

**Description**

No drivers were inserted, because a multi-voltage legal solution could not be found.

**What Next**

Set the application option cts.common.verbose to 1 (or higher) to see more information about the reason(s).

---

## CTS-669

(error) No drivers could be inserted in power domain '%s'.

**Description**

The command was unable to insert any drivers in the mentioned power domain, while drivers are required in that power domain.

**What Next**

Investigate the reason why the drivers could not be inserted. Earlier messages in the log may help in finding the reason(s).

Feedback

## CTS-670

(error) Cannot identify driver of net '%s'.

**Description**

The named net has no driver.

**What Next**

Check the netlist.

## CTS-671

(error) Cannot set output net name to '%s'.

**Description**

You specified a specific net name for the shorted output net of the drivers. The command was unable to set the name of the net to the specified string.

## CTS-672

(warning) Shorted output net has been renamed from '%s' to '%s' to avoid a name conflict.

**Description**

You specified a specific net name for the shorted output net of the drivers. This name is already present in the netlist, and a suffix has been added to avoid a name conflict.

## CTS-673

(warning) Clock '%s' in mode '%s' is skipped because %s.

**Description**

The named clock is not considered for balance point derivation because of the mentioned reason.

## CTS-674

(warning) Skipped balance point '%s' because clock '%s' is not used as a clock by the timer.

Feedback

### Description

The named balance point is not affected because the clock at the balance point is not used as a clock by the timer. The propagation of the set_clock_latency constraints is only done for the part of the clock network that fans out to pins that need a clock for timing analysis.

### What Next

Check that the delay value of the balance point is in line with the set_clock_latency constraints because derive_clock_balance_points will not affect this (manual) delay value.

---

## CTS-675

(info) Cell statistics%s after %s of driver object set '%s': clock cell count: %d (ICGs %d) clock cell area: %s (ICGs %s).

### Description

This message prints the number of non-sink clock cells and their area after each step during tap assignment.

---

## CTS-676

(warning) The -allow_buffering option will not be respected for the tree rooted at '%s' because there are still high-fanout drivers in the tree (maximum fanout is %d).

### Description

The *-allow_buffering* option of the *set_multisource_clock_subtree_options* command allows for buffer insertion during skew optimization of structural multisource clock sub-trees. These skew buffers will be used to slow down fast clock paths. In order to do effective skew optimization, the clock tree should have been constructed by splitting existing cells in the netlist. This splitting should break up high fanout nets into smaller fanout nets to eliminate DRC (transition) violations. When the command detects high fanout nets during the optimization step, it will turn off skew-buffer insertion because the timing of high fanout nets is not reliable and skew optimization would be ineffective.

### What Next

When this message occurs, there is a setup problem. Make sure that there are enough cells in the netlist such that splitting can resolve DRC issues. Check the value of the maximum allowed transitions (set_max_transition) and make sure the value is tight enough. Make sure that no dont_touch, size_only or other reason prevents the existing cells from being split.

Feedback

## CTS-677

(error) No driver cells were inserted%s. Buffers at all other levels are being removed.

**Description**

No driver cells were inserted. This is an error situation. Earlier messages indicate the reason why driver cells could not be inserted. For the command to complete successfuly, at each level, at least one driver should be inserted.

**What Next**

Correct the configuration such that, at each level, at least one driver is inserted.

## CTS-678

(info) Multisource Global Clock Tree Metrics: Skew = %s; ID = %s; NetsWithDRC = %s; Worst Trans/Cap violation = %s/%s; ClockBufArea = %s; Clock = %s; Mode = %s; Corner = %s; GlobalTreeRoot = %s.

**Description**

Multisource global clock tree Qor information.

## CTS-679

(error) Pin '%s' is not on the clock network%s.

**Description**

The specified pin is not on the clock network. This pin is either the driver pin or a load pin of a net that is addessed by create_clock_drivers. The create_clock_drivers command can only operate on objects that are on the clock network.

**What Next**

Make sure all clock definitions are correct.

## CTS-680

(error) The area spanned by the leaves of net '%s' is too small to be handled by global clock tree synthesis.

**Description**

The specified problem instance is too small to be handled by the global clock tree synthesis command.

Feedback

**What Next**

Either spread the sinks over a larger area, or use smaller repeater and thinner wiring.

---

## CTS-681

(error) No load terminals could be identified.

**Description**

Either the specified net of the -loads option does not have any load pins, or the set of load pins as specified by the -loads option is empty.

**What Next**

Check the specified net for load pins.

---

## CTS-682

(error) No drivers were inserted%s because port punching is not allowed at power domain crossing hierarchies.

**Description**

No drivers were inserted, because ports would need to be punched at power domain crossing hierarchies. This is not allowed. An example situation is a power domain with multiple existing entry ports. In that case, the command cannot make a choice on what driver to connect to what entry port. Since it is not allowed to create a dedicated entry port per driver, or attach all drivers to a single entry port, we do not insert any drivers at all and we consider this an error situation.

**What Next**

Set the application option cts.common.verbose to 1 (or higher) to see more information about the reason(s).

---

## CTS-683

(warning) QoR report disabled for clock %s in mode %s at root %s because it is in ideal mode.

**Description**

QoR report is only possible if clock is in propagted mode.

**What Next**

Set clock into propagted mode and re-run the command.

Feedback

## CTS-684

(error) Different clocks arrive at multi-source drivers '%s' and '%s'.

**Description**

Since the command may move clock structures from one multi-source driver to another, it is required that the clocks at each driver pin/port are identical.

## CTS-685

(error) None of the repeaters are able to drive just a sink of net '%s' without creating DRC violations.

**Description**

All repeaters passed with command option -lib_cells to command synthesize_multisource_global_clock_trees are not able to drive just one sink of the net to synthesize without creating DRC violations.

**What Next**

Relax the max capacitance and/or max transition time limits for the net or clock and re-run the command.

## CTS-686

(warning) Since command option %s is set, command option %s is ignored.

**Description**

The two options are orthognal to each other.

## CTS-687

(Error) Timing characterization failed for lib_cell %s.

**Description**

The timing behavior of the reported lib_cell cannot be characterized, while this lib_cell is available for clock synthesis

**What Next**

Review the delay calculation for the lib_cell to check whether there is any missing data that causes this issue. If delay calculation cannot be performed, exclude cts from the purpose of the lib_cell to indicate that the lib_cell is not available for clock synthesis.

Feedback

## CTS-688

(Error) Both a horizontal and a vertical routing layer are required when sub-finger routing is enabled.

**Description**

When using the -fishbone_sub_span and -fishbone_layers options of the route_clock_straps command together, it is required that both horizontal and vertical routing layers are specified.

## CTS-689

(Warning) Skipping cell '%s' marked as global tree buffer because library cell '%s' is not identified as a repeater.

**Description**

It is expected that every cell in the design marked as a global tree repeater cell is instantiated from a library cell which can be identfied as a buffer or inverter.

## CTS-690

(error) %s%s%s%s%s

**Description**

There is an error in the argument list of the remove_multisource_global_clock_trees command. The additional information should be self explanatory.

**What Next**

Correct the error in the argument list. See the manpage of remove_multisource_global_clock_trees for more information about argument syntax and semantics.

## CTS-691

(info) %s %d cells are marked as global tree repeaters out of which %d are matching the prefix '%s'. In total %d repetears are removed (%d %s, %d %s).

## CTS-692

(error) %s%s%s%s%s

Feedback

**Description**

There is an error in the argument list of the remove_clock_drivers command. The additional information should be self explanatory.

**What Next**

Correct the error in the argument list. See the manpage of remove_clock_drivers for more information about argument syntax and semantics.

---

## CTS-693

(error) %s.

**Description**

There is an error in the expected input for command remove_clock_drivers. The additional information should be self explanatory.

**What Next**

Correct the input and re-run the command.

---

## CTS-694

(warning) %s.

**Description**

There is an inconsistency in the input for command remove_clock_drivers. The additional information should be self explanatory.

**What Next**

Although the command finished consider to repair the input and re-run the command.

---

## CTS-695

(info) %s

**Description**

Information about the result of calling command remove_clock_drivers.

---

## CTS-696

(info) %s

Feedback

### Description

General information about the initialization, setup and processing of MS-CTS commands.

---

## CTS-697

(error) Placement of %s '%s' failed, because no legal location was found (target = (%f, %f)).

### Description

This typically happens if the library cells specified to command *synthesize_multisource_global_clock_trees* with option *-lib_cells* are larger than the space remaining after converting the metal shapes in the prefered routing layers into placement blockages.

### What Next

To solve the issue either user smaller library cells, or disable the app option cts.multisource.enable_pin_accessibility_for_global_clock_trees.

---

## CTS-698

(error) MV aware global clock tree construction failed, because %s.

### Description

Instantiation of a cell in a multi-voltage setup for multi-source global clock tree construction failed. There are several types of reasons why such a failure happen.

If the message reads "*none of the provided library cells is valid for voltage area*" the multivoltage setup requires a type of repeater (dual or single rail) for a given location and hence voltage area which is not listed in the set of library cells passed with the command option *-lib_cells*.

If the reasons reads "*none of the specified driver locations belong to the power domain*" all locations where clock drivers should be placed belong to a voltage area, and hence power domain, which are different to the one of the provided template cell. Since the template cell cannot be moved into a different power domain without loosing its original power domain information the command *remove_clock_drivers* is not able to re-create the original state. Such a change is not allowed. As a consequence it is required that the user has to specify at least one clock driver location which falls into the voltage area of the template cell.

### What Next

Based on the explanation above either check the library cells listed in the command option *-lib_cells or adopt the targeted driver locations such that at least on of them fall into the voltage area of the template cell, then re-run the command.*

Feedback

## CTS-699

(error)MV aware global clock tree construction is not possible, because %s.

**Description**

The analysis of the problem instances showed that global clock tree construction is not possible without port punching on power domain boundaries. This could be resolved by enabling the app option *opt.common.allow_physical_feedthrough*.

**What Next**

Set the app option *opt.common.allow_physical_feedthrough* to true and re-run the command.

## CTS-700

(error) Tap assignment requires that option -num_taps is used on option set '%s' and matches the number of driver objects.

**Description**

In the current version of the tool tap assignment requires that the number of requested tap drivers matches the number of pins and ports given by option -driver_objects.

## CTS-701

(error) The number of tap drivers does not match the number of specified taps using option -num_taps on option set '%s'.

**Description**

In the current version of the tool tap assignment requires that the number of requested tap drivers matches the number of pins and ports given by option -driver_objects.

## CTS-702

(warning) Warning: The tap drivers should get explicitly defined on option set '%s'.

**Description**

In the current version of the tool tap assignment requires that the option -driver_objects specifies all tap drivers. The implicit definition of tap drivers by considering the clock sources will be disabled in future.

**What Next**

Make sure that the option -driver_objects specifies all tap driver pins or ports.

Feedback

## CTS-704

(warning) The -clock option will become a required option in future releases. Please specify a clock using the -clock option.

**Description**

Previous versions of the tool didn't require the -clock option. The clock specification has become a requirement now to resolve any ambiguity.

## CTS-705

(warning) The strategy cannot be set using this command. The option -strategy is deprecated and will be removed in future releases.

**Description**

Previous versions of the tool allowed the specification of a strategy for a group. This has changed as separate commands for structural synthesis and tap assignment (regular clock tree synthesis) have been introduced.

## CTS-706

(error) Sink group '%s' already exists.

**Description**

A sink group with the given name already exists.

**What Next**

Either remove the existing sink group using the 'remove_multisource_clock_sink_groups' command or specify a different sink group name using the -name option.

## CTS-707

(error) Sink group '%s' doesn't exist.

**Description**

A sink group with the given name doesn't exist.

**What Next**

Use the 'report_multisource_clock_sink_groups' command to list the existing sink groups.

Feedback

## CTS-708

(info) Removed %d sink group%s.

**Description**

Successfully removed the given amount of sink groups.

## CTS-709

(info) No sink groups defined.

**Description**

There have been no sink groups using the 'create_multisource_clock_sink_group' command created.

## CTS-710

(warning) No valid clock sinks specified.

**Description**

There were no valid clock sinks specified using the -sinks option.

## CTS-711

(info) Fanout of tap driver '%s' of option set '%s' with %d endpoint%s will get distributed.

**Description**

This message indicates how man clock tree sinks have been identified for tap assignment for the given option set and its tap drivers.

## CTS-712

(error) Couldn't find input of tap cell '%s' that is on the clock path of clock '%s'.

**Description**

None of the input pins of the tap driver cell is on the clock network of the specified clock.

**What Next**

Check the netlist and the timing constraints to make sure that a clock input pin can be detected.

Feedback

## CTS-713

(warning) No tap cell found for option set '%s' whose fanout can be distributed.

**Description**

After the merging operation none of the tap cells of the given option set had a clock network connected to it.

**What Next**

Make sure that at least one tap driver drives a clock tree with clock sinks for the given clock of the option set.

## CTS-714

(warning) Reducing the number of subtrees to %d for option set '%s'.

**Description**

The given number of tap drivers as given by option -num_taps cannot be synthesized. Check for earlier warnings why some tap drivers could not be used.

**What Next**

Check the defined tap drivers of the option set. Verify that they can be reached by the given clock.

## CTS-715

(error) Tap assignment requires that all tap driver cells have a fixed location. Cell '%s' is not marked as fixed.

**Description**

The given tap driver doesn't have a fixed placement location as currently required by tap assignment..

**What Next**

Make sure to use the 'set_placement_status' command to mark the tap cells as fixed.

## CTS-716

(warning) Tap driver '%s' has multiple exclusive sink groups assigned.

Feedback

**Description**

The specified tap driver has multiple sink groups assigned which are of type exclusive. This means that an exclusive assignment is not possible.

**What Next**

Use the 'report_multisource_clock_sink_groups' command to check the definition of the existing sink groups.

---

## CTS-717

(info) Tap driver '%s' has the sink group '%s' assigned.

**Description**

The specified tap driver has the specified sink group assigned using the 'create_multisource_clock_sink_group' command.

---

## CTS-718

(warning) Additional sinks are added to exclusive sink group driver '%s' because the pin '%s' is not splittable%s. Enable message CTS-725 to see them all.

**Description**

An exclusive sink assignment as defined for the given sink group is not possible due to constraints put on the clock tree. Exclusive sink assignment requires that clock cells can be split to achieve the assignment goal. In this case the specified pin and its entire fanout has to be assigned to the tap driver due to the reason given in the message.

**What Next**

Check any constraints like size-only, dont-touch, or frozen ports whether splitting can be enabled for the given pin. Also balance point definitions in front of sink-group sinks might cause this warning.

---

## CTS-719

(warning) Pin '%s' of sink group '%s' is not reachable from any of the existing driver objects. This sink is not assigned to any tap driver.

**Description**

Not all of the sinks of the given sink group can be assigned to the tap driver of that group. The clock might not reach the given sink.

Feedback

**What Next**

Check the netlist and the clock related constraints that might prevent a path from the tap driver to the sink.

---

## CTS-720

(warning) Cannot use %d tap%s as only %d pin%s to connect to %s available.

**Description**

The netlist doesn't offer as much splitting opportunities as there are tap drivers defined. Constraints might restrict the amount of splitting.

**What Next**

Check the netlist and the constraints that might prevent splitting.

---

## CTS-721

(info) Removing %d option set%s.

**Description**

The given number of option sets for tap assignment are removed.

---

## CTS-722

(warning) Non-propagated upstream clock '%s' detected.

**Description**

There is a non-propagated clock in the upstream clock network.

**What Next**

Switch the upstream clocks to propagated mode. See the documentation of message CTS-723 for more information.

---

## CTS-723

(warning) Detected ideal clock(s) in the upstream clock network. Global skew balancing accross the multisource subtrees will be skipped.

**Description**

Some part of the upstream clock network, i.e. the part of the clock network between the clock source and the multisource drivers, is still in ideal clock mode. Typically this means

Feedback

that a (master) clock in the upstream clock network was not switched to propagated mode using the set_propagated_clock command. As long as parts of the upstream clock network are ideal, it makes no sense to balance the subtrees because the upstream delays are unknown. Subtree balancing could even be counter productive in that case.

**What Next**

Switch the upstream clocks to propagated mode. If you want to model certain delays and/or transitions, please use set_annotated_delay and set_annotated transition.

---

## CTS-724

(error) Tap driver '%s' has a multi-voltage violation on its connected net.

**Description**

Tap assignment requires that the tap drivers are MV clean.

**What Next**

Please resolve the MV violation first before executing tap assignment.

---

## CTS-725

(warning) Additional sink '%s' added to exclusive sink group driver '%s'.

**Description**

This message reports all clock sinks that get added to an exclusive sink group tap driver but which are not part of that sink group. Reasons are constraints put on the clock tree. Exclusive sink assignment requires that clock cells can be split to achieve the assignment goal. In this case the specified pin and its entire fanout has to be assigned to the tap driver due to the reason given in the message.

**What Next**

Check any constraints like size-only, dont-touch, or frozen ports whether splitting can be enabled for the given pin. Also balance point definitions in front of sink-group sinks might cause this warning.

---

## CTS-726

(error) Tap cell '%s' doesn't have a valid library cell class.

**Description**

Tap assignment requires equivalent lib cell class for all tap cells. The lib-cells need to have the same function_id attribute value. If this attribute is not set, the tool will consider the

Feedback

user_function_class lib-cell attribute. In case this attribute is also not set, then the tap cells need to have the same lib-cells to be considered equivalent.

**What Next**

Please check the library cell type of the tap drivers. Make sure that either the function_id or the user_function_class attributes do match.

---

## CTS-727

(error) Tap cells '%s' and '%s' don't have the same library cell class.

**Description**

Tap assignment requires equivalent lib cell class for all tap cells.

**What Next**

Please check the lib cell subset attribute 'subset_name' and the library cell type of the two tap cells.

---

## CTS-728

(Error) Conflict detected. User mode %s has been included in another different scale_across_modes group already.

**Description**

The user specifed mode has been grouped with other modes in another existing group before, which is either from a previous command in the same session or from the database. The existing group will be kept and user input is being ignored.

**What Next**

Check the existing scale_across_modes settings and either remove the existing settings of scale_across_modes, or discard the current attempt of change.

---

## CTS-729

(Error) Regular multi-source clock group '%s' does not have any loads connected to the drivers.

**Description**

A regular multi-source clock group was defined using the set_multisource_clock_tap_options command. The drivers of this group, however, do not drive any load pins. This is an error.

Feedback

**What Next**

Check the loading of the driver objects of the multi-source clock group as configured by set_multisource_clock_tap_options.

---

## CTS-730

(info) Removed %d %s strap(s) because these do not connect to shapes in the orthogonal direction and floating shapes are not allowed.

**Description**

When floating shapes are not allowed, all straps that do not connect to shapes in the orthogonal direction are removed.

---

## CTS-731

(error) All %s straps were removed because these do not connect to shapes in the orthogonal direction and floating shapes are not allowed.

**Description**

When floating shapes are not allowed, all shapes that do not connect to shapes in the orthogonal direction are removed. In this case, no shapes in the given direction are remaining, and that is flagged as an error.

**What Next**

If you do allow floating shapes, please use option *-allow_floating true*.

---

## CTS-732

(Warning) Hierarchical pin '%s' is ignored. Only leaf input pins or ports are accepted.

**Description**

The -loads option of the split_clock_cells command accepts a list of collections of leaf cell pins and ports. Hierarchical pins are not allowed.

---

## CTS-733

(Warning) The collection of option '-loads' at position %d is empty.

**Description**

The -loads option of the split_clock_cells command accepts a list of non-empty collections of leaf cell pins and ports.

Feedback

## CTS-734

(Error) No valid pins nor ports have been specified for option '-loads'.

**Description**

The -loads option of the split_clock_cells command accepts a list of non-empty collections of leaf cell pins and ports.

## CTS-735

(Error) Load '%s' is driven by port '%s' which cannot be cloned.

**Description**

All pins or ports that are specified with option -loads of the split_clock_cells command have to by driven by a leaf cell which can be split. Ports as drivers are not allowed.

## CTS-736

(Error) Load '%s' is not connected to a driver pin.

**Description**

All pins or ports that are specified with option -loads of the split_clock_cells command have to by driven by a leaf cell which can be split.

## CTS-737

(Error) Load '%s' is driven by '%s' which is different than the driver of other loads in the collections. Skip splitting for the collection with '%s'.

**Description**

Loads that are connected to different drivers have been specified in same collection to option -loads of the split_clock_cells command.

**What Next**

Make sure that all load pins and ports specified in a collection are driven by the same pin.

## CTS-738

(Warning) Expecting list of pins or ports for option '%s'.

Feedback

**Description**

The -loads option of the split_clock_cells command accepts a list of collections of leaf cell pins and ports.

---

## CTS-739

(information) %d connected components were removed because these do not connect to the connected component %s.

**Description**

When the -keep_connected_component option is set, shapes of connected components that are not selected to be kept are removed.

**What Next**

Check whether the connected component that survives is the intended one.

---

## CTS-740

(Error) Driver object '%s' of option set '%s' overlaps with other driver objects.

**Description**

The bounding box of a driver object cell or port is overlapping with one of the other driver objects of the same multisource option setting. This typically indicates a setup problem.

**What Next**

Make sure that all driver objects are legally placed with repsect to each other.

---

## CTS-741

(Warning) Sink pin '%s' cannot be assigned to closest tap driver '%s' due to MV considerations.

**Description**

The specified sink pin needs to be driven by a tap driver which is further way than the closest tap driver as the closest tap driver is not allowed to drive the first-level cell of that sink.

---

## CTS-742

(Error) No first-level pin found that can be connected to the tap driver '%s'.

Feedback

**Description**

The specified tap driver doesn't have any first-level pin that can be connected to it.

---

## CTS-743

(info) %s

**Description**

The tool has reordered an ICG with a buffer.

---

## CTS-744

(Error) Logic level balancing constraints not meeting for the subtree driver '%s'.

**Description**

The logic level balancing step failed because no possible configuration/buffer combinations found fit for meeting the constraints.

The reason could be one of the following : - Don't touch applied on net which prevents buffer cells to be inserted. - No valid buffer cell found for insertion. - No matching buffer cell found for the matching hierarchy for power domain. - Target level settings are too strict to satisfy.

**What Next**

Please review the log file for probable cause like buffer not available, don't touch net etc.

---

## CTS-745

(warning) ICG '%s' is not reordered because %s.

**Description**

The message provides an indication of why an ICG cannot be reordered. Following are the reasons that prevent ICG reordering with the buffer:

```
1. ICG cannot be reordered in case the ICG and the buffer driving it:
    a. Are marked as dont_touch
    b. Are marked fixed
    c. A don't-touch net between ICG and buffer
    d. Have an MV violation
    e. Are in different voltage areas
```

Feedback

**What Next**

Please Check whether the constraints are intended and remove them to enable ICG reordering.

---

## CTS-746

(Warning) No buffers available to buffer the net '%s' for logic level balancing.

**Description**

The logic level balancing step couldn't insert buffers on the specified net possibly because (a) no buffer cells available in the design or (b) no matching buffer found for the corresponding hierarchy of net. If buffer cells exists but not available for this particular net or hierarchy then it will be automatically tried on other nets and solution will be found.

---

## CTS-747

(Warning) '%s' net is don't touch net, cannot be buffered for logic level balancing.

**Description**

The logic level balancing step couldn't insert buffer on the specified net because the net is marked as don't touch. It will now try other nets for buffering in order to meet the level balancing constraints. If a solution is possible then it will be tried by inserting some extra buffer cells on other un-constrained nets.

---

## CTS-748

(Warning) The level balancing setting of group '%s' is smaller than the existing number of levels(%d). Only balancing subtrees with a level below %d for balancing the subtree levels.

**Description**

The target level constraint set with "set_multisource_clock_subtree_options" or "set_multisource_clock_subtree_constraints" command is too strict and can't be honoured.

**What Next**

Please reivew the target level constraint set with "set_multisource_clock_subtree_options" or "set_multisource_clock_subtree_constraints"

Feedback

## CTS-749

(info) No buffer found driving only a single ICG as immediate load, hence no ICG re-ordering performed pre DRC Fixing stage.

**Description**

During ICG re-ordering if no buffer found driving only a single ICG as immediate load then pre DRC fixing stage,ICG re-ordering will not be performed.

## CTS-750

(info) MSCTS QoR %s : GlobalSkew = %s; ID=%s; Median Latency=%s; Std.dev Latency=%s; Deviation from Target Latency=%s; MaxSubtreeLevels=%s; NetsWithDRC = %d; Worst Trans/Cap violation = %s/%s; ClockCellCout=%s; ClockCellArea =%s; ClockBufArea = %s; Clock = %s; Mode = %s; Corner = %s;

**Description**

Multi Source Clock Qor information.

## CTS-751

(Warning) Subtree synthesis is leaving %d first-level net%s as ideal after optimization.

**Description**

Multisource structural subtree synthesis is marking nets connected to the driver objects as timing ideal during the first phases of optimization if these nets have preroutes that are considered for snapping the first-level clock cells. If the app option 'cts.multisource.subtree_routing_mode' is set to 'none', i.e. virtual routing is used for parastics estimation, then these nets will stay in ideal mode after the optimization finishes. Multi-driven nets will always stay in ideal mode as their timing should be modeled by transition and delay annotations using 'analyze_subcircuit'.

**What Next**

Consider setting the app option 'cts.multisource.subtree_routing_mode' to 'global' or 'fishbone' to get more accurate parasitics and timing estimations during optimization.

## CTS-752

(warning) DRC violation at driver object '%s' which might not get fixed by subsequent splitting or sizing operations. Max trans: %s (limit: %s) max cap: %s (limit: %s) fanout: %d (limit: %d).

Feedback

**Description**

A DRC violation (max transition, max capacitance, or max fanout) was detected on the specified driver object which is most likely not fixable by subsequent optimization steps.This can be due to one of the following reasons:

1. DRC or latency constraints are too tight. Tight constraints might result in too many first-level cells that were created during the DRC splitting phase or the initial load of the driver object was already too high.
2. If the net driven by subtree driver is root net or mesh net then it has not been modeled correctly with set_annotated_delay and set_annotated_transition.
3. Other design issues. Example: no-placeable area close to pre-route shapes on the net driven by subtree driver for snapping.

QoR is not guaranteed with a large DRC violation on the subtree driver or its loads.

**What Next**

Please check the root cause of DRC on subtree driver and resolve the same:

1. Ensure DRC constraints for max-fanout, max-cap, and max-transition are reasonable.
2. If the net driven by a subtree driver is a root or mesh net with high fanout and capacitance, then model the net with set_annotated_delay and set_annotated_transition for command to see reasonable transition and delay in the clock tree.
3. Ensure pre-route shapes on nets driven by subtree drivers are reasonable and cells can be snapped under them without creating DRCs.

---

## CTS-753

(Error) Partial Stage Constraints Definition Found : '%s'.

**Description**

S-MSCTS require complete stage constraint definition to work in Structural Mode.

Following constraints must be defined to work in Structural Mode. - Min & Max Cell Delay for all the cells used in the clock tree. - For cells with subset_name attribture defined; use command : set_multisource_clock_subtree_constraints -subset_names - For cells without subset_name attribture defined; use command : set_multisource_clock_subtree_constraints -undefined_subset_name - Max RC Delay for all the cell levels used in the clock tree. - use command : set_multisource_clock_subtree_constraints -net_level 2 -max_rc_delay 10

Feedback

The reason for this error could be one of the following :

**What Next**

Report existing constratins with report_multisource_clock_subtree_constraints Please review the clock subtree for missing constriants on the cells or nets.

---

## CTS-754

(info) Running in Structural Mode; since valid stage constraints are set.

**Description**

This message indicates that the S-MSCTS is running in "Structural Mode" which is primarily used to control min/max cell delay and stage RC delay.

---

## CTS-755

(info) Stage Constraint Violators : %s : Max Cell Delay Violators = %s(%s), Min Cell Delay Violators = %s(%s), RC Delay Violators = %s(%s), Clock = %s; Mode = %s; Corner = %s; ClockRoot = %s.

**Description**

Multi Source Clock Stage Constraints Violator information.

---

## CTS-756

(info) Running multisource tap assignment.

**Description**

This message indicates that the multisource tap assignment is running.

---

## CTS-757

(info) Running multisource incremental tap assignment.

**Description**

This message indicates that the multisource tap assignment is running with incremental mode.

---

## CTS-758

(Warning) Could not find legal location for the context instance %s

Feedback

**Description**

This message indicates that unable to find legal location while adding buffer during clustering.

---

## CTS-759

(Warning) No %s routing layer specified for net '%s'.

**Description**

Min-max layer constraints must be specified for clock net to enable correct routing topology.

**What Next**

Use command ‘set_clock_routing_rule’ to define the correct NDR rule along with min-max routing layers to be used for routing.

---

## CTS-760

(Warning) The MSCTS clock tree input '%s' doesn't have an annotated transition time.

**Description**

This message indicates that the given input to MSCTS doesn't have any transition time annotation given. This might lead to sub-optimal timing QoR due to pessimistic transition times.

**What Next**

Please use the 'set_annotated_transition' command to annotate reasonable treansition times in the scenario applied by MSCTS.

---

## CTS-761

(Warning) The MSCTS clock tree input '%s' doesn't have a valid transition time even though some annotations on this input exist.

**Description**

This message indicates that the given input to MSCTS doesn't have a valid transition time annotation given. This might lead to sub-optimal timing QoR due to pessimistic transition times.

Feedback

**What Next**

Please use the 'set_annotated_transition' command to annotate reasonable treansition times in the scenario applied by MSCTS.

---

## CTS-762

(Warning) The MSCTS clock tree input '%s' has a max transition violation. Annotated transition time: %s limit: %s.

**Description**

This message indicates that the given input to MSCTS has a transition time annotation which is violating a given max transition constraint. This might lead to sub-optimal timing QoR due to pessimistic transition times.

**What Next**

Either reduce the transition time annotation or relax the constraint, if appropriate.

---

## CTS-763

(Warning) The MSCTS clock tree input '%s' doesn't have a valid transition constraint defined.

**Description**

This message indicates there is no max transition constraint that applies to the given MSCTS input. Hence, the tool cannot check whether the annotated transition time on this input is reasonable.

**What Next**

Pleae check whether a set_max_transition for the entire design or this input would be required.

---

## CTS-764

(info) Ignoring %s for all subtree driver objects due to app option 'cts.multisource.ignore_drc_on_subtree_driver'.

**Description**

This message indicates that some DRC constraints (max capacitance and/or max-fanout) are ignored on the driver objects due to the user defined app option setting.

Feedback

## CTS-765

(Warning):Sizing pre-exsiting cell %s with reference module %s.

### Description

Sizing pre-exsiting cell whose reference is not in CTS purpose list with references from CTS reference list during synthesize_clock_trees command.

## CTS-766

(Error) No valid clock cells to split.

### Description

The message shows that none of the cells provided for splitting are valid cells. Following are the reasons that prevent clock cell splitting and merging:

```
1. Cell has physical_status: locked, legalize_only, or fixed
2. Cell has user_size_only or dont_touch
3. Constraints on path between its driver and the loads of this cell:
    a. A don't-touch net between driver and a load
    b. Path crosses a physical hierarchy
    c. Path crosses a power domain boundary and port punching on it is
 not allowed
    d. A logical read-only hierarchy (like a black box)
    e. A hierarchy with frozen ports: Either due to set_freeze_ports or
 due to
       timing constraints on the hierarchical pin
```

### What Next

Please check net don't touch settings or constraints on the logical and physical hierarchies.

## CTS-767

(warning) H-Tree NDR rule '%s' with layers %s-%s, specified using set_regular_multisource_clock_tree_options is ignored because a net specific NDR rule '%s' with layers %s-%s already exists on clock net '%s'.

### Description

The H-Tree NDR rule specified using the option -htree_routing_rule could not be applied to H-Tree nets because a net specific NDR rule already exists on the clock net.

Feedback

**What Next**

Remove the net specific NDR rules already existing on the clock net to apply the H-Tree NDR rule.

---

## CTS-768

(Error) Command synthesize_regular_multisource_clock_trees is not supported for Multiple Physical Hierachy Designs.

**Description**

H-tree creation is not supported in Multiple Physical Hierarchy Designs.

**What Next**

Turn off the app option 'cts.multisource.enable_mlph_flow' and re-run to ignore the physical hierachy blocks.

---

## CTS-769

(Information) relaxing user dont_touch/size_only limit on cell '%s' for splitting.

**Description**

The message provides an information that the cell is considered for splitting during tap assignment relaxing the user_dont_touch/user_size_only constraints. This is since the cell is specified with set_multisource_clock_tap_options -relax_split_restrictions_for_cells.

---

## CTS-770

(warning) Latency target to intermediate pin %s(%s) is higher than the latency target to %s(%s) in its fanout; hence ignoring the target to intermediate pin.

**Description**

Reports the conflict between the intrmediate latency set by user at the non-sink pin and the target latency(with balance point offset) at the sink pin.

---

## CTS-771

(warning) Cannot merge %s (delay_offset: %s) and %s (delay_offset: %s) due to different delay_offsets.

Feedback

**Description**

Merging of cells is not allowed if the pins of 2 merging instances have different intermediate latency offset.

---

## CTS-772

(info)Found %d intermediate pins with latency targets for group %s and will be honored by subtree synthesis.

**Description**

Intrmediate latency constraints are found and will be hounered by the subtree synthesis.

---

## CTS-773

(error) '%s' is not a valid driver. It will be ignored.

**Description**

The specified driver object is not valid. Below are valid driver objects - input port - output pin - net driven by an output pin

**What Next**

Correct the error in the argument list. See the manpage of set_multisource_clock_tap_options or set_multisource_clock_subtree_options for more details of valid driver_objects.

---

## CTS-774

(error) %s required at location (%f, %f) in voltage area '%s'.

---

## CTS-775

(warning) '%s'

**Description**

The buffer could not be inserted at a location suitable to build H-Tree. Hence it is being relocated to nearest valid location which could have a QoR impact.

**What Next**

Review the floorplan constraints at the ideal location and ensure that there are no placement blockages, unsuitable voltage areas at that location.

Feedback

## CTS-776

(warning) Logical net looping encountered during traversal of net: '%s'. Ignoring this net for backward traversal.

**Description**

Tool encountered a case of looping while backward traversion on the gvien net. The net may not be intended for the clocks. If such a net is found then some of the MSCTS netlist operations like merging, splitting may not work.

**What Next**

Review the net and fix the loop structure.

## CTS-777

(warning)Template cell %s has not been used and will be deleted.

**Description**

The specified template cell could not be used during buffer insertion as there are no locations falling in the template cell hierarchy. Hence the template cell would be deleted.

**What Next**

Check why no locations have been specified for buffer insertion in the template cell hierarchy. If you do not want the cell to be deleted, remove if from the create_clock_drivers command input.

## CTS-778

(info) Running initial multisource subtree synthesis.

**Description**

This message indicates that initial multisource subtree synthesis is running.

## CTS-779

(info) Running multisource subtree synthesis for optimization.

**Description**

This message indicates that the multisource subtree synthesis is running for latency based optimization.

Feedback

## CTS-780

(warning) Skipping driver '%s' as its net is multi-driven.

**Description**

This warning message indicates that we have a multi-driven net from the given driver, so ignoring this tap driver from tap assignment.

## CTS-781

(info) Running multisource subtree synthesis for preprocess and merge.

**Description**

This message indicates that the multisource subtree synthesis preprocess and merge is running.

## CTS-782

(info) Skipping SMSCTS %s step as optimization has already been done.

**Description**

This message indicates that final_place SMSCTS steps will be skipped as subtree optimizations has already been done.

## CTS-783

(Warning) Found more than %d percent (%d) of clock nets are detail-routed, which is unexpected before starting S-MSCTS refine.

**Description**

S-MSCTS refine is not expected to see detail-routed clock-nets. This may lead to inconsistencies of timing number after refine due to DR <-> GR correlation.

## CTS-784

(Warning) Found stripe shapes in both horizontal and vertical directions on output nets of driver objects, this is not supported. Ensure stripe shapes in any one direction only on all driver objects for subtree synthesis.

Feedback

### Description

Currently S-MSCTS supports either horizontal or vertical shape preroutes for subtree driver output net. Having both dir preroute is not a recommended flow for now.

---

## CTS-785

(warning) %s is unsplit because it %s.

### Description

The message provides an indication of why a cell cannot be split. Following are the reasons that prevent clock cell splitting:

```
1. Cell has physical_status: locked, legalize_only, or fixed
2. Cell has user_size_only or dont_touch
3. Constraints on path between its driver and the loads of this cell:
    a. A don't-touch net between driver and a load
    b. Path crosses a physical hierarchy
    c. Path crosses a power domain boundary and port punching on it is
 not allowed
    d. A logical read-only hierarchy (like a black box)
    e. A hierarchy with frozen ports: Either due to set_freeze_ports or
 due to
       timing constraints on the hierarchical pin
```

### What Next

Please check net don't touch settings or constraints on the logical and physical hierarchies.

---

## CTS-786

(warning) Pin %s is not a boundary sink and will be ignored from the boundary skew group %s.

### Description

The message provides an indication of a pin will be ignored from the boundary skew group since it's not a real boundary sink.

### What Next

Update the setting and remove the pin from the boundary skew group.

---

## CTS-787

(Information) Driver pin %s will not be splitted since it is on reconvergent path.

Feedback

### Description

The message shows the information that pins on reconvergent clock path will not be splitted.

---

## CTS-788

(Error) %s is unsplit because it %s.

### Description

The message provides an indication of why a cell cannot be split. Following are the reasons that prevent clock cell splitting:

```
1. Cell has physical_status: locked, legalize_only, or fixed
2. Cell has user_size_only or dont_touch
3. Constraints on path between its driver and the loads of this cell:
    a. A don't-touch net between driver and a load
    b. Path crosses a physical hierarchy
    c. Path crosses a power domain boundary and port punching on it is
 not allowed
    d. A logical read-only hierarchy (like a black box)
    e. A hierarchy with frozen ports: Either due to set_freeze_ports or
 due to
       timing constraints on the hierarchical pin
```

### What Next

Please check net don't touch settings or constraints on the logical and physical hierarchies.

---

## CTS-789

(Error) Secondary cost attribute %s is not defined for following lib cells: %s. Define user secondary cost annotation for these lib cells.

### Description

An error to be produced by SMSCTS when there are cells of a given lib cell class with missing user secondary cost annotations. For example: Consider lib cell class ‘ABC’ which contains lib cells “D1/D2/D4/D6/D8/…” etc..

If one of the lib cell i.e. ‘D2’ in the lib cell class ‘ABC’ doesn’t have user secondary cost defined (but at least one other cell in the same lib cell class does) then produce an error and stop command execution.

### What Next

Check the missing user secondary annotation for given lib cell class.

Feedback

## CTS-790

(info) MSCTS Power QoR %s : InputPinCap = %s; Internal Power = %s; Switching Power = %s; Total Dynamic Power = %s; Clock = %s; Mode = %s; Corner = %s;

**Description**

Multi Source Clock Power Qor information.

## CTS-791

(warning) Found instance %s with 'placed' status. Manually modified clock cells must be legalized and marked as 'application_fixed' before next clock step.

**Description**

Route step expects the cell status as 'application_fixed'. For manually modified clock cells, they must be legalized and marked as 'application_fixed' to avoid any legalization issues lat later step during subtree synthesis.

## CTS-800

(Warning) '%s' is not a clock net, ignored.

**Description**

The net specified is not a clock net, and is thus ignored.

## CTS-801

(Warning) Net %s cannot be routed by route_fishbone, because %s.

**Description**

The fishbone command can only be used to route clock nets appropriate status. The given net does not satisfy one of these criteria, as explained in the message.

## CTS-802

(Error) No valid net to route.

**Description**

The net(s) specified are not valid for fishbone routing.

Feedback

## CTS-803

(Warning) Could not legalize net %s.

**Description**

Wire or Via shapes on the net could not be legalized.

## CTS-804

(Warning) Net %s, cannot legally route %s sub strap at %.3f, estimated length %.3f (%s to %s).

**Description**

Sub straps could not be routed successfully due to legalization problem which could be due to DRC not being clean. This could be due to routing blockages, availability of vias etc.

## CTS-805

(Information) Net %s, routed %s sub strap at %.3f, estimated length %.3f (%s to %s).

**Description**

A sub strap was routed successfully at given location.

## CTS-806

(info) Driver '%s' corresponding to specified location {%s} will not drive any loads.

**Description**

Clock drivers at specified location will be left dangling and will not connect to any loads.

## CTS-900

(Warning) Clock routing rules are outside of allowable layers

**Description**

Routing rules for nets are defined by using the *create_routing_rule* command, and then applied to clock nets by using the *set_routing_rule* or *set_clock_routing_rules* command. The *set_routing_rule* and *set_clock_routing_rules* commands have *-min_routing_layer* and *-max_routing_layer* options to restrict nets with that routing rule to those routing layers. There is also a global minimum and maximum routing layer for the design, which is set

Feedback

by the *-min_routing_layer* and *-max_routing_layer* options of the *set_ignored_layers command.*

This warning is issued when a routing rule defined on a clock net uses layers outside of the allowable range for the design. For such cases, it is not possible to route on all layers specified in the net routing rule. This can lead to different RC delay characteristics than expected on that clock net, or could contribute to routing congestion on certain layers.

**What Next**

No change is needed if it is acceptable to route on a more limited set of layers than what was specified with the *set_routing_rule* or *set_clock_routing_rules* command. Otherwise, modify the routing layer ranges specified by the *set_routing_rule*, *set_clock_routing_rules*, or *set_ignored_layers* commands to resolve the routing layer conflict.

**See Also**

- create_routing_rule
- set_ignored_layers
- set_routing_rule
- set_clock_routing_rules

---

## CTS-901

(Warning) Clock nets have MV violations

**Description**

Clock nets should be free of MV violations before started CTS. This check reports cases where clock nets already have MV violations.

**What Next**

Use check_mv_design to further investigation the reasons for the MV violation. Depending on the violation, it may be necessary to modify the UPF constraints, insert special cells like level shifters or isolation cells, or take other corrective action.

**See Also**

- check_mv_design

---

## CTS-902

(Warning) No AON (always-on) buffers or inverters available for CTS

Feedback

### Description

Always-on cells are often required on MV designs, typically when a signal needs to remain active in a voltage domain that can be shut-down. This check indicates that none of the specified clock buffer or inverter references can be used as an always-on cell.

### What Next

Use 'set_lib_cell_purpose -include cts $lib_cell_list', where $lib_cell_list includes at least one always-on buffer or inverter

### See Also

- set_lib_cell_purpose

---

## CTS-903

(Warning) Cells instantiated in the clock network are not in the clock reference list

### Description

CTS can only size to lib_cells that are specified in the clock reference list, constrained using 'set_lib_cell_purpose -include cts $lib_cell_list'. This message indicates that some cells instantiated in the clock network have reference lib_cells that are not in the clock reference list. If there are logically equivalent lib_cells in the reference list, then these instances can still be resized to those logically equivalent lib_cells, if necessary. There is no guarantee that the cell will be resized, and once resized it cannot be sized back to the original lib_cell that was not in the reference list.

If there are no logically equivalent cells in the reference list, then this cell will not be resized by CTS, and a CTS-904 message will also be issued by check_clock_trees, indicating that no logically equivalent cells are available.

### What Next

No action is necessary if the behavior described above is okay. If resizing of the cell is desired, ensure that logically equivalent lib_cells are available using the 'set_lib_cell_purpose -include cts $lib_cell_list' command. Note that there is no guarantee that the cell will be resized, and so it may remain in the clock network with its original reference cell which was not supplied in the clock reference list. If you want to guarantee that this reference cell does not exist in the clock network after CTS completes, then you must manually associate it to a new lib_cell using the change_link command before running CTS.

Feedback

**See Also**

- set_lib_cell_purpose
- change_link

---

## CTS-904

(Warning) Some clock reference cells have no LEQ cell specified for resizing

**Description**

CTS cannot resize a cell in the clock network unless there are logically equivalent lib_cells specified in the clock reference list. This message indicates that there are cells instantiated in the clock network that have no logically equivalent lib_cells in the reference list, so no sizing can be done.

**What Next**

If the cell does not need to be sized, or has optimization constraints like dont_touch that prevent sizing, then no action is necessary. Otherwise, use 'set_lib_cell_purpose -include cts $lib_cell_list', where $lib_cell_list includes logically equivalent lib_cells to the ones reported by this check. This will ensure that the cell can be resized during CTS.

**See Also**

- set_lib_cell_purpose

---

## CTS-905

(Warning) There are clocks with no sinks

**Description**

Normally a clock is expected to propagate to a set of sinks for clock construction and balancing. Alternately, a clock may not propagate directly to any sinks, but rather is only used to generate other clocks. This check flags cases that violate this rule.

For the purposes of this check, a sink can either be a standard clock sink like a flip-flop or macro clock input, or it can be a clock balance point defined by the set_clock_balance_points command. For example, if a clock propagates to 100 sinks but all were set as ignore points, those would still be considered sinks for this check, even though report_clock_qor would not report them as sinks. Also, if a clock has no sinks but does generate another clock, it will not be flagged by this check.

Feedback

### What Next

Typically a clock should always arrive at sinks, or generate another clock. Clocks flagged by this check normally are prevented from arriving at their sinks by a timing constraint, like set_case_analysis blocking clock propagation through a mux or ICG, or set_disable_timing preventing clock propagation through a timing arc. Check the netlist to ensure that the clock should arrive at some sinks. If it does, then check the timing constraints to see what constraint could be preventing propagation of this clock through its network. The reporting commands report_case_analysis and report_disable_timing can assist in debugging these timing constraints.

### See Also

- create_clock
- create_generated_clock
- set_case_analysis
- set_disable_timing
- set_clock_balance_points
- report_case_analysis
- report_disable_timing

---

## CTS-906

(Warning) There are sinks with no clock

### Description

Clock sinks exist in the design that do not receive a clock signal. As a result, CTS will not be able to build a clock tree to these sinks.

For the purposes of this check, a sink can either be a standard clock sink like a flip-flop or macro clock input, or it can be a clock balance point defined by the set_clock_balance_points command. For example, if a clock propagates to 100 sinks but all were set as ignore points, those would still be considered sinks for this check, even though report_clock_qor would not report them as sinks.

### What Next

Typically a sink should always receive a clock. When it doesn't, it's likely caused by a timing constraint blocking clock propagation, or by an incorrectly specified set_clock_balance_points constraint.

Feedback

Timing constraints like set_case_analysis and set_disable_timing can block clock propagation through a timing arc. You can use report_case_analyis and report_disable_timing to help debug whether either of these constraints are causing a clock to be blocked from propagating to its sinks.

If a set_clock_balance_points constraint is set on a pin in the datapath, then a clock would not propagate to that balance point. This could also trigger this check.

**See Also**

- set_case_analysis
- set_disable_timing
- set_clock_balance_points
- report_case_analysis
- report_disable_timing

---

## CTS-907

(Warning) There are disabled timing arcs in the clock network

**Description**

This message indicates that there are timing arcs in the clock network that have been disabled. The clock cannot propagate through a disabled arc, which can affect which sinks a clock can propagate to, and as a result this has an impact on clock construction and balancing.

Typically a disabled timing arc would come from either an explicit set_disable_timing constraint, or indirectly from set_case_analysis propagating a constant value to a cell in the clock network. For example, an ICG enable pin may have case analysis that disables the clock from propagating through the ICG.

**What Next**

Check the set_disable_timing and set_case_analysis constraints in your design. Clear any of these constraints that are preventing clocks from correctly propagating through the clock tree. The commands report_disable_timing and report_case_analysis can help in debugging these issues.

**See Also**

- set_case_analysis
- set_disable_timing

Feedback

- report_case_analysis
- report_disable_timing

## CTS-908

(Warning) Large phase delay in abstracted sub-blocks

**Description**

Phase delay indicates the downstream skew from a paricular node in the clock tree. For the case of an abstract model like an ILM or ETM, it's possible that there is large phase delay at the clock pin, even when that pin is a leaf-level sink of the clock tree. This message indicates that a large phase delay is present on some hierarchical cells. This can hurt skew balancing, since CTS cannot modify the contents of that sub-block, it can do nothing to reduce that large phase delay.

**What Next**

Reexamine your abstracted sub-block and determine why it has large phase delay. It may be expected due to actual skew inside the block. If there is large skew in the sub-block, it should be addressed during implementation of that sub-block.

To ignore the large phase delay on the sub-block, use set_clock_balance_points to force new downstream delay values on that block's clock pin.

**See Also**

- create_abstract
- set_clock_balance_points

## CTS-909

(Warning) set_load constraints detected in the clock tree

**Description**

set_load constraints are used to force a capacitive load value on a port or net in the design. Typically, set_load constraints should only be used for output ports. A set_load constraint on an input clock port or a clock net will force a load value onto that net, even as it is being synthesized. This can cause convergence problems where inserting repeaters does not reduce the load of the net. It can also lead to unfixable max_cap violations in the clock tree. This check looks for any set_load constraints set on a clock input port or a clock net. It will ignore set_load constraints on output ports attached to the clock network.

Feedback

### What Next

Examine the nets or ports reported by this check, and verify that the set_load constraint should have been applied. Typically these constraints should only be used during place-and-route on output ports, but in some cases they may be used to override dirty library data, like incorrect pin load values.

Pay special attention to set_load constraints that exceed the max_cap constraint being applied on the clock net. These will cause unfixable cap DRCs on the clock network.

### See Also

- set_load

---

## CTS-910

(Warning) Balance point constraints are defined downstream of another balance point or ignore point constraint

### Description

set_clock_balance_points is used to set balance points in the clock tree to direct clock skew balancing. set_clock_ignore_points is used to define pins in the clock tree that should be ignored for skew balancing. Any of these constraints, when applied at an intermediate node of the clock tree, cause the downstream clock tree to be ignored for balancing purposes (a DRC clean clock tree is still built downstream). As a result of this behavior, if a balance point constraint is applied downstream of another balance point or ignore point constraint, it will have no effect on CTS.

This message indicates that some clock balance point constraints will be ignored due to upstream balance point or ignore point constraints.

### What Next

This behavior may not be a problem, but it is reported to make the user aware that some constraints will be ignored. If the downstream balance point must be honored, then the upstream balance or ignore point constraint should be removed. Balance point constraints can be removed using remove_clock_balance_points. Ignore point constraints can be removed using remove_clock_ignore_points.

### See Also

- set_clock_balance_points
- remove_clock_balance_points
- report_clock_balance_points

Feedback

## CTS-911

(Warning) Clock pins downstream of a balance point or ignore point have been added to a skew group

### Description

When a balance point is defined with the *set_clock_balance_points* command, any downstream sinks or balance points are not balanced. Likewise, when an ignore point is defined with the *set_clock_balance_points -consider_for_balancing false* command, any downstream sinks or balance points are not balanced. Balancing stops at the first encountered balance point or ignore point, and the entire downstream clock tree is synthesized DRC-free but not balanced.

This message detects the situation where a pin added to a skew group by the *create_clock_skew_group* command is downstream of a balance point. In this case, even though the pin is added to a skew group, it is not balanced.

### What Next

If the pin that was added to a skew group should be skew balanced, reconsider the balance point or ignore point definition upstream. You can clear it by using the *remove_clock_balance_points* command.

If the pin that was added to a skew group does not need to be balanced, no change in constraints is required. A DRC-free clock tree will be built to that pin downstream of the balance point or ignore point.

### See Also

- set_clock_balance_points
- remove_clock_balance_points
- report_clock_balance_points
- create_clock_skew_group

## CTS-912

(Warning) set_load constraints on output clock ports exceed the max capacitance limit

### Description

set_load is used to explicitly constrain a capacitive load value on a port or net. This check looks for set_load constraints on output clock ports, and reports cases where the set_load constraint exceeds the capacitance limit on the clock port. In these cases, an unfixable DRC violation will exist on the clock port.

Feedback

### What Next

Either the set_load constraint or the capacitance limits should be changed on the port. Capacitance limits can come from the .lib or .db description of the reference cell, or from the set_max_capacitance SDC constraint. CTS honors both capacitance limits.

### See Also

- set_load
- set_max_capacitance

---

## CTS-913

(Warning) Explicit ignore points have been added to a skew group, and will not be balanced

### Description

set_clock_ignore_points is used to define pins in the clock tree that should be ignored for skew balancing. create_clock_skew_group is used to add clock pins into a separate skew group from the default skew group for that clock. This message reports clock pins that are defined as explicit ignore points but also have been added to a skew group. In this case, one constraint indicates to not balance the pin, and the other constraint indicates that the pin should be balanced as part of a user defined skew group. Ignore point constraints take precedence over skew group constraints, so these pins will not be balanced.

### What Next

Check the pins that are reported and consider whether the pin should be balanced as part of the skew group or not. If it should be balanced, then the ignore point constraint should be cleared with remove_clock_ignore_points.

### See Also

- create_clock_skew_group

---

## CTS-914

(Warning) set_input_transition on clock ports exceeds the max transition limit

### Description

set_input_transition is used to constrain a transition time on an input port. This check looks for set_input_transition constraints on clock input ports, and reports cases where the constraint exceeds the max transition limit on the clock port. In these cases, an unfixable DRC violation will exist.

Feedback

### What Next

Either the set_input_transition constraint or the max transition limits should be changed on the port. Transition limits can come from the .lib or .db description of reference cells, or from the set_max_transition SDC constraint. CTS honors both transition limits.

### See Also

- set_input_transition
- set_max_transition

---

## CTS-915

(Warning) Excessively small max capacitance constraints in the clock network

### Description

Max capacitance constraints have been detected on the clock network that are small enough to adversely affect clock QOR. Excessive buffering triggered by the max cap constraints could hurt clock area, latency, or both. The max capacitance constraints could be coming from the .lib / .db description of the clock tree reference cells, or could be coming from set_max_capacitance constraints applied on ports, clocks, or the design.

### What Next

Examine the max capacitance constraints reported by this command and relax them if possible. Otherwise, the constraints will be honored as they are, and CTS QOR can be affected.

### See Also

- set_max_capacitance

---

## CTS-916

(Warning) Excessively small max transition constraints in the clock network

### Description

Max transition constraints have been detected on the clock network that are small enough to adversely affect clock QOR. Excessive buffering triggered by the max trans constraints could hurt clock area, latency, or both. The max transition constraints could be coming from the .lib / .db description of the clock tree reference cells, or could be coming from set_max_transition constraints applied on ports, clocks, or the design.

Feedback

### What Next

Examine the max transition constraints reported by this command and relax them if possible. Otherwise, the constraints will be honored as they are, and CTS QOR can be affected.

### See Also

- set_max_transition

---

## CTS-917

(Warning) Implicit ignore points have been added to a skew group, and will be balanced

### Description

Implicit ignore points are constraints automatically derived by CTS and propagated in the clock tree. Certain pins in the clock tree are not balanced, including explicit ignore pins constrained by the set_clock_ignore_points command, as well as data pins attached to the clock network (such as flip-flop D pins). Implicit ignore constraints are propagated up the clock tree from these ignore pins, so that the whole clock fanout leading only to ignore pins is implicitly ignored. This check reports the case that the user has constrained an implicit ignore pin to be part of a skew group for balancing in that group. A skew group constraint has higher precedence than an implicit ignore constraint, so the pin will be skew balanced as part of the skew group.

### What Next

Normally implicit ignore constraints only propagate through the parts of the clock network that are not expected to be balanced, because no downstream pin needs to be balanced. As such, it is unexpected that an implicit ignore should be balanced as part of a skew group, which is the reason for this check. You can use report_clock_qor -type structure, or the Abstract Clock Graph feature in the GUI, to investigate and debug the propagation of implicit ignore constraints in the clock network.

If the implicit ignore point constraints are expected and the pin should be balanced as part of the skew group, then no further action is required. If after investigation you determine the pin should not be balanced at all, then redefine the skew group using the create_clock_skew_group command to exclude this pin. If the pin is not expected to be implicit ignore but should be balanced as part of the skew group, no action may be necessary. The pin will be balanced in the skew group. But, you can remove downstream explicit ignore constraints that may be causing the implicit ignore propagation, using the remove_clock_ignore_points command. Note that this command can be used only to remove explicit ignore constraints, not implicit ignores, which are automatically derived and propagated.

Feedback

**See Also**

- create_clock_skew_group
- report_clock_qor

---

## CTS-918

(Warning) Voltage area blocked for buffering.

**Description**

Volatage area are blocked for buffering through app option "opt.common.blocked_vas". CTS will not abe able to add buffer in the blocked VAS. This is just a warning message for user for reporting the blocked VAS

**What Next**

To remove the blocked VAS, reset the value of app option "opt.common.blocked_vas" to empty string.

---

## CTS-919

(Error) Core area of block %s is unbound.

**Description**

The core area is unbound. This is possibly due to removed or invalid site def information.

**What Next**

Recreate the block information with valid site def information and re-run the command.

---

## CTS-920

(Warning) There are %d instances connected to routes in the design. No optimization will be performed on the routed nets and instances connected to them.

**Description**

CTO THO flow currently works in VR mode only. If there are global or detail routed nets at the top level or block level, no optimization will be run on them

**What Next**

Delete the global and detail routes and run again.

Feedback

## CTS-921

(Information) Design is being run in THO mode. Global router is being turned off, all timing computations will happen in virtual route mode.

**Description**

synthesize_clock_trees will use virtual router in THO mode. Global router will be disabled internally. No additional app options or setting need to be provided for doing this.

## CTS-922

(error) %s%s%s%s%s

**Description**

There is an error in the argument list of the set_regular_multisource_clock_tree_options command. The additional information should be self explanatory.

**What Next**

Correct the error in the argument list. See the manpage of set_regular_multisource_clock_tree_options for more information about argument syntax and semantics.

## CTS-923

(error) %s%s%s%s%s

**Description**

There is an error in the argument list of the synthesize_regular_multisource_clock_trees command. The additional information should be self explanatory.

**What Next**

Correct the error in the argument list. See the manpage of synthesize_regular_multisource_clock_trees for more information about argument syntax and semantics.

## CTS-924

(warning) Below lib cells will not be used because these lib cells %s%s:

**Description**

The specified lib cells will not be used during clock driver insertion for the reason as mentioned in the message.

Feedback

**What Next**

Check whether the mentioned reason is expected. If not, use commands set_lib_cell_purpose, set_dont_touch or set_dont_use to change the lib cell attributes as required. If the reason is expected, you may consider to not pass the offending lib cells to the set_regular_multisource_clock_tree_options command. That will avoid the message.

---

## CTS-925

(error) Below lib cells are not buffers%s:

**Description**

The specified lib cells can not be used during clock driver insertion because these are not buffer lib cells.

**What Next**

The set_regular_multisource_clock_tree_options command only accepts buffers as tap lib cells. If other cells need to be inserted, you need to insert a template cell in the netlist and use the -tap_template_cells option to indicate that this template cell should be cloned.

---

## CTS-926

(error) Redhawk cannot be initialized for voltage drop optimization:

**Description**

Voltage drop option is turned on but Redhawk data is not available.

**What Next**

Please make sure Redhawk resistance calculation is run before invoking clock_opt with voltage drop option turned on.

---

## CTS-927

(info) Voltage drop aware optimization is turned on:

**Description**

Grid resistance based voltage drop aware option is turned on in clock_opt.

---

## CTS-928

(information) No option set exists for '%s'. Please use the '%s' command to define a set of options.

Feedback

**Description**

The executed command requires that at least one option set with driver objects has been defined using the specified command.

---

## CTS-929

(Warning) Layers` '%s' of clock strap do not match the top layer of via-ladder '%s'. Via-ladders would not be inserted.

**Description**

The via ladder would not be inserted if layers of clock strap do not match the top layer of via-ladder.

---

## CTS-930

(Warning) Skipping RC delay calculations because of missing RC Tree for net '%s'. Continuing.

**Description**

The RC delay calculations are being skipped because of missing RC Tree for the net. The reasons could be failed routing or failed extraction of the net.

---

## CTS-931

(Information) H tree checking for the set of tap drivers failed due to tap locations.

**Description**

In auto tap synthesis for H tree topology, tap drivers are under feasibilty check for H tree implementation. This message tells the tap drivers as a set failed the checking due to their locations. If none of the tap drivers are inserted in the end, please check the floor plan and may need to give boundary to guide the tap synthesis.

---

## CTS-932

(Information) H tree checking in tap driver insertion failed for the set of tap drivers due to some reasons other than tap locations.

**Description**

In auto tap synthesis for H tree topology, tap drivers are under feasibilty check for H tree implementation. This message tells the tap drivers as a set failed the checking for the reasons other than the tap locations.

Feedback

## CTS-934

(info) Dynamic voltage drop ccd is turned on:

**Description**

Dynamic voltage drop ccd is turned on in ccd.

## CTS-935

(error) Redhawk cannot be initialized for ccd dynamic voltage drop optimization:

**Description**

Voltage drop option is turned on but Redhawk multi grid data is not available.

**What Next**

Please make sure Redhawk dynamic voltage calculation is run before invoking ccd with voltage drop option turned on.

## CTS-936

(Warning) Skipping RC delay calculations because of net '%s' is global routed. Continuing.

**Description**

The RC delay calculations are being skipped because the design is global routed and not detailed routed. If the design is global routed, the calculations are not accurate because the design is not DRC clean

## CTS-938

Warning: NDR spacing '%f' is greater than %s grid step size '%f'. Some straps may be skipped.

**Description**

In presence of defined ndr, some straps may be skipped when the ndr spacing value is higher.

## CTS-939

(warning) %s%s%s%s%s

Feedback

**Description**

This is a warning message for the synthesize_multisource_global_clock_trees command. The additional information should be self explanatory.

**What Next**

Check the warning and correct if needed. See the manpage of synthesize_multisource_global_clock_trees for more information about argument syntax and semantics.

---

## CTS-940

(error) %s%s%s%s%s

**Description**

There is an error in the argument list of the set_multisource_global_clock_tree_options command. The additional information should be self explanatory.

**What Next**

Correct the error in the argument list. See the manpage of set_multisource_global_clock_tree_options for more information about argument syntax and semantics.

---

## CTS-941

(info) Removed %d group%s.

**Description**

Successfully removed the given amount of groups.

---

## CTS-942

(info) No groups defined.

**Description**

There have been no groups created using the 'set_multisource_global_clock_tree_options' command.

---

## CTS-944

(error)Global clock tree synthesis failed in spine construction, because %s.

Feedback

**Description**

Global clock tree spine construction is not possible due to reasons as given.

---

## CTS-945

Warning: No via-ladder inserted with top layer matching clock strap layers '%s', hence skipping zroute connection.

**Description**

No via-ladder inserted with top layer matching clock strap layers M8/M9, hence skipping further zroute connectioni.

---

## CTS-946

Warning: Excluding '%s' from the non-exclusive bounds.

**Description**

CTS/CTO does not support non-exclusive hier move bounds. If non-exclusive hier move bound is found during buffer insertion in that hier, than CTS/CTO will ignore the bound constraint

---

## CTS-947

(Error) No delay arc between '%s' and '%s.'

**Description**

There is no cell arc between the specified two pins.

**What Next**

Specify the input pin and output pin of the same cell.

---

## CTS-948

(Error) No delay arc between '%s' and '%s.'

**Description**

There is no delay arc between the specified two pins.

**What Next**

Specify the driver and sink of the same net.

Feedback

## CTS-949

(Information) Tap synthesis inserted single level of inverter, subsequent global clock tree building step (htree_synthesis) will correct the polarity.

**Description**

Given -invert and inverter as lib_cells from create_clock_drivers, or given inverter lib cells for taps and H tree from auto tap sunthesis, tap drivers will be inverting the logic and H tree will invert again to make polarity correct.

## CTS-950

(Information) Htree synthesis adding odd number of inverter levels to accommodate single inverter level from tap synthesis and address the clock tree polarity.

**Description**

Given -invert and inverter as lib_cells from synthesize_multisource-global_clock_trees or given inverter lib cells for taps and H tree from auto tap sunthesis, tap drivers will be inverting the logic and H tree will invert again to make polarity correct.

## CTS-951

(Warning) Htree synthesis is not under invert mode but taps are inverted. Poloarity of the clock tree maybe wrong.

**Description**

When taps are inverting the logic, H tree is expected to invert the logic again to make polarity correct. Use -invert and inverter as lib_cells when calling synthesize_multisource-global_clock_trees.

## CTS-952

(error) %s.

**Description**

The basic sanity checks of HTree are failing as specified.

## CTS-953

(Warning) %s

Feedback

### Description

This indicates a warning scenario in H-tree building flow.

---

## CTS-954

(info) Clock cell relocation is restricted to cells without data connection because app option cts.compile.enable_cell_relocation is set to "auto" and clock gate latency aware placement is enabled.

### Description

Since the timing-driven placer has placed the clock gates with consideration of clock latency, it is advised not to relocate the clock gates during CTS to improve clock metrics. Moving the clock gate cells may disturb clock gate enable path timing that was optimized by the placer. The *auto* value of cts.compile.enable_cell_relocation imposes this recommendation

Clock gate latency aware placement is enabled by setting place.coarse.clock_gate_latency_aware to "true", or by setting place.coarse.clock_gate_latency_aware to "auto" and enabling clock gate latency estimation at the flow level by seting one of app option place_opt|refine_opt|compile.flow.estimate_clock_gate_latency to "true". To have a consistent flow, it is recommended to set the *.flow.estimate_clock_gate_latency app options for all used mega commands to the same value.

### What Next

If you want clock gates to be relocated anyway, you can set app option cts.compile.enable_cell_relocation to *all*.

### See Also

- cts.compile.enable_cell_relocation
- place.coarse.clock_gate_latency_aware
- place_opt.flow.estimate_clock_gate_latency

---

## CTS-955

(warning) App option cts.compile.enable_cell_relocation is not set to "auto" while clock gate latency aware placement is enabled. Consider to set it to "auto".

### Description

Since the timing-driven placer has placed the clock gates with consideration of clock latency, it is advised not to relocate the clock gates during CTS to improve clock metrics.

Feedback

Setting cts.compile.enable_cell_relocation to e.g. "all" or "leaf_only" will make CTS move clock gates that are connected to signal nets and that may disturb the clock gate enable path timing. It is recommended to set cts.compile.enable_cell_relocation to "auto" when clock gate latency aware placement is enabled.

Clock gate latency aware placement is enabled by setting place.coarse.clock_gate_latency_aware to "true", or by setting place.coarse.clock_gate_latency_aware to "auto" and enabling clock gate latency estimation at the flow level by setting one of app options place_opt|refine_opt|compile.flow.estimate_clock_gate_latency to "true". To have a consistent flow, it is recommended to set the *.flow.estimate_clock_gate_latency app options for all used mega commands to the same value.

**What Next**

Consider to set cts.compile.enable_cell_relocation to "auto".

**See Also**

- cts.compile.enable_cell_relocation
- place.coarse.clock_gate_latency_aware
- place_opt.flow.estimate_clock_gate_latency

---

## CTS-956

(warning) Please use -largest / -smallest / -all switches with -show_verbose_paths / -show_paths to report the clock paths.

**Description**

-largest / -smallest / -all switches have to add with -show_verbose_paths / -show_paths to report the clock paths.

**What Next**

Add -largest / -smallest / -all switches to report the clock paths.

---

## CTS-957

(Warning) Clock drivers will not be snapped as %s.

**Description**

Clock driver could not be snapped due to the reason specified.

Feedback

**What Next**

Review the message and correct the setup.

---

## CTS-958

(warning) Cannot snap driver '%s' at location {%s} because %s.

**Description**

The command was unable to snap a clock driver at the specified location because of the reason mentioned in the message.

**What Next**

Investigate whether the mentioned reason is valid and whether the dropping of the driver is acceptable.

---

## CTS-959

(warning) Could not get phase delay of pin '%s'

**Description**

The phase delay of the pin was not found.

---

## CTS-960

(error) There are no repeater lib cell references opened for CTS.

**Description**

There are no repeater lib cell references opened for CTS. Command cannot proceed.

**What Next**

Correct the error and rerun the command.

---

## CTS-961

(error) None of the repeater lib cell references opened for CTS are usable.

**Description**

The is no repeater lib cell references opened for CTS. Command cannot proceed

**What Next**

Correct the error and rerun the command.

Feedback

## CTS-962

(warning) "%s"

**Description**

Lib cells are unusable as per the provided reason.

**What Next**

Check if not using those lib-cells are intentional.

## CTS-963

(warning) "%s constraint is set to %f which is too tight".

**Description**

Constraints are too tight. Tight constraints might result in unexpected long runtime.

**What Next**

Ensure DRC constraints for max-transition, max-cap, max-net-length, max-fanout are reasonable.

## CTS-964

(Information) Successfully relocated %d leaf ICGs out of %d total leaf ICGs for dynamic power improvement.

**Description**

This message indicates how many leaf ICGs are successfully relocated out of total leaf ICGs.

## CTS-965

(Information) Total %d leaf ICGs are not relocated for reasons: fixed - %d, legalize_only - %d, Zero_input_toggle_rate - %d, Toggle_Ratio - %d, Wirelength_Check - %d, Other_Reason - %d

**Description**

This message indicates how many leaf ICGs are not relocated and provides a reason wise break up.

Feedback

## CTS-966

(Information) %s

**Description**

This message indicates ICG gate name which is not relocated during Activity Driven Clock Gate Relocation and print the corresponding reason.

## CTS-967

(Error) %s is sink for generated clock %s but pass through for master clock.

**Description**

A master clocks tree's non sink pin is a sink pin for the generated clock. This can cause issues with skew balancing on endpoints beyond this pin.

**What Next**

Check generated clock definitions.

## CTS-968

(Error) Corner %s has no valid parasitic.

**Description**

Could not find valid parasitic corner, usually because constraint corner has no parasitic tech setup. RC extraction can not be performed.

**What Next**

Check constraint corner's parasitic tech setup in run script. Run report_parasitic_parameters to make sure all corners are correctly set.

## CTS-969

(Warning) All leaf ICGs have 0 input toggle rate. Please verify if power scenarios are activated.

**Description**

Since all leaf ICGs have 0 toggle rate on input net, no ICG will be moved. Please check if power scenarios are enabled .

Feedback

## CTS-970

(Information) Successfully moved %d ICGs (including %d reparented) out of %d total candidate ICGs for dynamic power improvement.

**Description**

This message indicates how many ICGs are successfully moved (including reparented) out of total candidate ICGs.

## CTS-971

(error) The net '%s' has a %s structure. Disconnecting the net '%s' from the hierarchical term %s so that CTS can still buffer the net.

**Description**

This error message will triggered when there is a reconvergent path or a cycle on hier net.

**What Next**

One of the hier pin will be disconnected from the net such that CTS can still buffer the net.

## CTS-972

(error) Subtree synthesis failed to get reference cell for %d output ports

**Description**

Tool failed to get a reference cell based on the number of output loads.

**What Next**

Please check the script to make sure complete set of splitter library cells are available for CTS usage.

## CTS-973

(info) The value of option cts.compile.enable_cell_relocation has been overridden to "leaf_only" to support latency-driven placement.

**Description**

If option place.coarse.latency_driven_placement is true, placement will attempt to move clock gating cells to improve clock latency. To prevent clock gate relocation from altering the locations of these cells, clock gate relocation will be done in leaf_only mode. Only leaf-level clock gates will be moved by clock gate relocation. This overrides option cts.compile.enable_cell_relocation.

Feedback

Latency-driven placement happens at the place_opt flow stage. Clock gate relocation happens at the clock_opt flow stage.

**What Next**

To use clock gate relocation mode other than "leaf_only", it is necessary to disable latency-driven placement. Set option place.coarse.latency_driven placement to false during place_opt and clock_opt flow stages. Clock latency may degrade as a consequence of this.

**See Also**

- cts.compile.enable_cell_relocation

---

## CTS-974

(warning) Mode '%s' doesn't have any valid clock in any timed scenario. The mode is skipped.

**Description**

The clock definition and constraints are not set properly for all the modes. Hence, no clock information is available for the mode.

**What Next**

Set clock constraints on all the modes.

---

## CTS-975

(info) Fanout %s from clock gate %s is moved to clock gate %s

**What Next**

None

---

## CTS-976

(info) Clock gate %s is removed

**What Next**

None

---

## CTS-977

(info) Clock gate %s is renamed to %s

Feedback

**What Next**

None

---

## CTS-978

(info) Clock gate %s is cloned from clock gate %s

**What Next**

None

---

## CTS-979

(info) %d fanout(s) from clock gate %s are moved to clock gate %s

**What Next**

None

---

## CTS-980

Warning: Clock net ‘%s’ routed with user net length constraint of ‘%f’ has DRC violation. (tran/cap cost : $trancost/$capcost)

**Description**

This clock net has length limit constraint. To meet contraint, routed wire length may have been impacted. The constraint value can be adjusted to fix violation

**What Next**

adjust constraint value

---

## CTS-981

(info) Annotating zero clock path delay till subtree driver '%s'.

**Description**

This message indicates we are annotating zero delay till subtree driver.

---

## CTS-982

(info) Removing internally annotated clock path delay till subtree drivers.

Feedback

**Description**

This message indicates we are removing annotated clock path delay till subtree drivers.

---

## CTS-983

(Warning) %s

**Description**

At least one tap driver must be inserted in each uncommitted hierarchy. Ensure that an appropriate tap configuration is used which can cover all uncommitted hierarchy boundaries.

---

## CTS-984

(Warning) Enable logic tracing of latch with or gate driver is set to true for enable logic cloning, split_enable_max_logic_levels/split_enable_max_flop_levels will be ignored.

**Description**

Tracing of OR gate will ONLY be supported with latch. If max_logic_level or max_flop_level is given, it will be ignored while enable logic tracing.

---

## CTS-985

(error) Section : '%s' is not confined in one Uncommitted Hierarchy

**Description**

There is an error in the argument list of the set_regular_multisource_clock_tree_options command, Section is not confied in one Uncommitted Hierachy. The additional information should be self explanatory.

**What Next**

Correct the error in the argument list. Every section must be confined in one Uncommitted hierarchy. See the manpage of set_regular_multisource_clock_tree_options for more information about argument syntax and semantics.

---

## CTS-986

(Warning) DRC disabled for clock nets.

**Description**

DRC is disabled for special nets through *set_auto_disable_drc_nets* command.

Feedback

**What Next**

To enable drcs on clock network do: *set_auto_disable_drc_nets -on_clock_network false* command.

---

## CTS-987

(Warning) Instance '%s' is not placed on a legal location, moving it to a legal location.

**Description**

The instance is not placed on legal location, moving the instance to a legal location..

---

## CTS-988

(Error) No load is detected for %s. Tap synthesis cannot be done for this %s.

**Description**

No load pins of the net being synthesized are inside the section boundary. Please check if the correct net is specified for synthesis or the boundary for the section is correct.

**What Next**

If the input net is correct, this section can be deleted. Rerun after correcting the input.

---

## CTS-989

(Error) Uncommitted hierarchy for %s cannot be determined.

**Description**

Some loads are not part of any uncommitted hierarchy. For tap synthesis to work correctly, loads of the root net should belong to one of the specified hierarchy.

**What Next**

See if the loads are belonging to top hierarchy. In such case, modify input such that top becomes one of the uncommitted hierarchy.

---

## CTS-990

(warning) %s%s%s%s%s

**Description**

This is a warning message for the synthesize_regular_multisource_clock_trees command. The additional information should be self explanatory.

Feedback

**What Next**

Check the warning and correct if needed. See the manpage of synthesize_regular_multisource_clock_trees for more information about argument syntax and semantics.

---

### CTS-991

(warning) Could not initialize power scenario

**Description**

Power scenario couldnt be initialized.

---

## DA

---

### DA-001

(error) Layer %s density %s for rule %s %s.

**Description**

This message is to report a layer density violation.

**What Next**

Fix it.

---

### DA-002

(error) There are %d violations of rule %s.

**Description**

This message is to report number of layer density violations of a rule.

**What Next**

Fix them.

---

### DA-003

(info) There are %d violations.

**Description**

This message is to report number of layer density violations.

Feedback

**What Next**

Fix them.

---

## DA-004

(error) Layer %s area of lib cell %s is not set.

**Description**

This message is to report an error that layer areas of lib cell are not complete. If the layer is defined by other layers. This message means at least one of these layers lack layer area setting of this lib cell. The reported lib cell may be a rpGroup if its layer areas are set but not complete.

**What Next**

Complete all layer areas then run this command again. If this is not a layer defined by other layers, please check whether its density of the reported cell is set by set_auxiliary_layer_area. If this is a layer defined by other layers, please check whether densities of all these layers of the reported cell is set by set_auxiliary_layer_area.

---

## DA-005

(error) Fail to get valid bbox of %s %s.

**Description**

bbox of the reported object is not valid. So this object cannot be supported by this command.

**What Next**

Make bbox of this object valid and run this command again.

---

## DA-006

(error) %s appears more than one time in %s

**Description**

The reported item or object appears more then one time. This is not supported.

**What Next**

Check whether this is a typo and run this command again.

Feedback

## DA-007

(error) Same %s %s cannot appears in %s and %s.

**Description**

The reported item or object appears multiple times conflictly. This is not supported.

**What Next**

Check whether this is a typo and run this command again.

## DA-008

(error) %s %s alreayd exists.

**Description**

The reported item or object to create already exists. This is not supported.

**What Next**

Check whether this is a typo and run this command again.

## DA-009

(error) Layer %s is not defined.

**Description**

When reference a layer in -layers of define_auxiliary_layer or -layer of set_auxiliary_layer_density_rule, this layer must be defined. Here defined means appearing in -name of define_auxiliary_layer or -layer_areas / -layer_densities of set_auxiliary_layer_area.

**What Next**

Run this command again with defined layer.

## DA-010

(error) Cannot set layer density of layer %s that is defined by define_auxiliary_layer.

**Description**

Density of layer defined by define_auxiliary_layer is computed from other layers. So cannot set its layer density.

Feedback

**What Next**

Set density of layers that are used to compute this layer.

---

## DA-011

(error) Cannot define layer %s by other layers. Densities of this layer are already set by set_auxiliary_layer_area

**Description**

Density of layer defined by define_auxiliary_layer is computed from other layers. So if its layer densities cannot be explicitly set.

**What Next**

Define with another layer name.

---

## DA-012

(Warning) No %s are defined %s.

**Description**

This message reports the things this command must based on are not defined.

**What Next**

Define them then run this command again.

---

## DA-013

(Error) The existing layer area and shapes information, which are set by the library, cannot be overwritten.

**Description**

This message reports the layer area and shapes information setting by -read_prf cannot be overwritten by the other options.

**What Next**

Avoid using -layer_area, -layer_shapes and -layer_density if the layer information has been set by -read_prf.

Feedback

# DB

## DB-1

(error) File is not a DB file.

**Description**

The identified file was not in the DB file format.

**What Next**

Check to see what format the file is in, and read it with the appropriate command. For example, edif files can be read into *dc_shell* with the *read -format edif <filename>* command.

## DB-3

(warning) Can't locate file '%s'.

**Description**

This warning message will be printed out the first time if we can not resolve a link to a given file. In the absense of the given file, there will be some unresolved references.

**What Next**

Check the link_library and the search_path variables and set their value accordingly.

## DB-4

(error) Open gen_state (0x%x) -- %s object '%s' (0x%x)\n attach '%s' id = %d

## DB-5

(error) Limit exceeded: Cannot create attribute %s for class %s, so program cannot continue.

**Description**

This message appears when the internal limit of 32k attributes per database object is exceeded. Once this limit is exceeded, work in progress cannot be saved and the program cannot continue.

Feedback

**What Next**

Check the scripts to see if an excessive number of *create_attribute* commands are run with unique attribute names. Try to reuse names where possible. If many scripts are run in one large session, attempt to break up the single long run into a number smaller runs, saving your work each time.

---

## DB-6

(error) Limit exceeded: Cannot create attach %s for class %s, so program cannot continue.

**Description**

This message appears when the internal limit of 32k types of attaches per database object is exceeded. Once this limit is exceeded, work in progress cannot be saved and the program cannot continue.

**What Next**

If many scripts are run in one large session, attempt to break up the single long run into a number smaller runs, saving your work each time.

---

# DBPQ

---

## DBPQ-1

(error) Failed to export library data into parquet file: '%s'. Check the disk space and run again.

**Description**

This message occurs when one or more of the following exist: 1) Data written into parquet don't meet schema requirement 2) The writing process failed

**What Next**

Check the data written from the library. Check the memory to confirm whether there is enough space to generate the corresponding parquet file.

---

## DBPQ-2

(error) Unable to find a valid db in the specific path: '%s'.

Feedback

### Description

This message occurs when the following exist: There is no valid db library under the given path.

The following is an example message:

```
Error: Unable to find a valid db in the specific path: '/.../test.db'.
 (DBPQ-2)
```

### What Next

Please check the path to the db library and library name.

---

## DBPQ-3

(error) Failed to create parquet dir: '%s'.

### Description

This message occurs when one or more of the following exist: 1) There is one existing directory with the same name 2) Fail to give written permission

The following is an example message:

```
Error: Failed to create parquet dir: './test'. (DBPQ-3)
```

### What Next

Check if the permission allows you to creat this new dir. Check if the dir already existed. If yes, create a new directory with another name, or careful clean the existing one.

---

## DBPQ-4

(error) Failed to aquire db data.

### Description

This message occurs when the following exist: 1) Failed to create db database 2) Failed to create db root

The following is an example message:

```
Error: Failed to aquire db data.(DBPQ-4)
```

### What Next

Check if the library is valid and good to use.

Feedback

## DBPQ-5

(error) Failed to read db root

**Description**

This message occurs when the following exist: the library has syntax/logic error

The following is an example message:

```
Error: Failed to read db root (DBPQ-5)
```

**What Next**

Check if the library is valid and good to use.

## DBPQ-6

(warning) No parquet generated since no relevant data is extracted from library.

**Description**

This message occurs when one or more of the following exist: 1) The library doesn't have timing group 2) The library has specific timing table format which haven't been supported yet

**What Next**

Check if the library is valid and good to use. If the library is good, please report to Library Compiler Team (Synopsys) the library has been tested with. There might be some special nldm/ocv table format which haven't been supported yet.

## DBPQ-7

(error) An unexpected error occurred while writing into parquet.

**What Next**

If this message occurs continually, please report to Library Compiler Team (Synopsys) the library has been tested with.

## DBPQ-8

(warning) Up to 2 power type voltage_maps are supported. Loaded library has %d voltage_maps.

Feedback

**Description**

This message occurs when the following exist: In the loading library there are more than two voltage_maps referenced by the cell-level pg_pin group existing. In this case only the first two voltage_maps will be extracted and exported into parquet file.

# DCFP

## DCFP-010

(Error) No floorplan exists for block %s.

**Description**

This error occurs when the most basic floorplan information, like boundary and core, are not defined for the block.

**What Next**

Check if the floorplan information has been read in correctly for the design. In case no floorplan has been read in, run the initialize_floorplan command to create a new floorplan.

## DCFP-011

(Error) Boundary bbox is not defined for the block %s.

**Description**

This error occurs when the boundary is not defined for the block or the boundary is invalid.

**What Next**

Check if the DIEAREA definition in your input DEF file is valid. Read in a DEF with the correct DIEAREA defined or run initialize_floorplan command to create a new floorplan.

## DCFP-012

(Error) Block %s has invalid core area.

**Description**

This error occurs when no core area is defined for the block or the core area defined is invalid.

Feedback

**What Next**

Check if the ROW definitions in your input DEF file are valid. Use create_site_array command to create site rows or run initialize_floorplan command to create a new floorplan.

---

## DCFP-020

(Error) Unit tile is missing in reference library.

**Description**

Unit tile is not found in the physical library. This message indicates one of three errors:

- No physical library is accessible.

- There is no site definition in any physical library.

- There are sites defined in the library, but none of them is a unit tile.

**What Next**

Add unit site definition in the technology portion of the physical library, and regenerate the library.

---

## DCFP-021

(Error) Site type %s height %s is not an integer multiple of site type %s height %s.

**Description**

This error occurs because the abnormal site height is not an integer multiple of the normal site height. Detail placement requires that the abnormal site height is an integer multiple of the normal site height.

**What Next**

Check the physical library make sure that the multiheight site is an integer multiple of the normal site height.

---

## DCFP-030

(Error) Tracks not defined for layer %s.

**Description**

This message indicates that the tracks are not defined on the layer %s. The tracks are necessary for routing on this layer.

Feedback

**What Next**

Use create_track command to create the tracks on the specific metal layers.

---

## DCFP-031

(Warning) Track %s doesn't cross core area.

**Description**

This message indicates that the track %s doesn't cross core area. These tracks will not be considered by the tool during optimization and estimation.

**What Next**

Check the track definition and make sure the tracks cover the core area of the block.

---

## DCFP-040

(Error) Macro cell %s is not fixed.

**Description**

This message indicates that the macro cell %s either doesn't have a location or has a location but is not fixed.

**What Next**

If the macro doesnt have a location, use create_placement command to place it. If the macro is placed, then fix the macros.

---

## DCFP-041

(Error) No location specified for fixed macro cell %s.

**Description**

Macro cell %s is fixed but its location is invalid.

---

## DCFP-042

(Warning) Fixed macro %s is out of the core area.

**Description**

This message indicates that the fixed macro %s is outside the core area.

Feedback

**What Next**

Move this cell into the core area manually.

---

## DCFP-043

(Warning) Fixed cell %s is overlapping with blockages at (%s, %s).

**Description**

This message is indicates that the fixed cell instance is overlapping with a blockage.

**What Next**

Manually move the fixed cell outside the blockage or run create_placement command.

---

## DCFP-044

(Error) Macro cell %s is overlapping other macro cells at (%s, %s).

**Description**

This message is indicates that this macro cell and one other neighbour macro cell are overlapping each other.

**What Next**

Manually move the cells to ensure that they don't overlap.

---

## DCFP-050

(Error) Site row %s overlaps with other site rows of same site definition at (%s, %s).

**Description**

Site rows with same site definition are not allowed to overlap.

**What Next**

Check your floorplan to find overlapping rows. Modify your DEF file to eliminate the overlap.

---

## DCFP-051

(Error) Site array has rows and columns specified at the same time.

Feedback

**Description**

You receive this error message because the site array you defined in the input DEF contains both horizontal and vertical arrays. Only one orientation of arrays can be specified in the site array, horizontal rows or vertical columns, but not both.

**What Next**

Modify the ROW definitions in the input DEF.

---

## DCFP-052

(Error) Site rows with site definition %s are misaligned.

**Description**

All the site rows with the same site site definition need to be aligned such that there is no offset between sites vertically.

**What Next**

Change the floorplan so that the rows align properly. Ensure that the distance between x-origins of the different rows is zero modulo the site width.

---

## DCFP-053

(Error) Row %s has a non-zero extra spacing of %s.

**Description**

Non-Zero extra spacing in rows is not supported.

**What Next**

Set row_extra_spacing to zero

---

## DCFP-060

(Warning) Move bound %s partially or completely overlaps with other move bound %s.

**Description**

This message indicates that two or more bounds overlap with each other. This may lead to invalid placement of cells.

**What Next**

Make sure that the bounds don't overlap each other.

Feedback

## DCFP-061

(Info) Non-exclusive movebound %s is completely covered by blockages.

**Description**

You receive this information because a movebound is completely covered by a blockage. When this occurs, the placer ignores the bound.

**What Next**

Move the movebounds to ensure that they do not conflict with the blockages.

## DCFP-062

(Warning) Placement blockage %s lie partially or completely outside the bounds of the block.

**Description**

You receive this warning message because the dimensions or locations of the specified blockages cause them to lie outside the block. Therefore, these blockages in your design are located off the chip.

**What Next**

Correct the core area to include all the blockages within the block. Or correct the locations or dimensions of the placement blockage to fit within the block.

## DCFP-063

(Error) Keepout margin for cell %s is larger than the core area.

**Description**

This error indicates that the keepout area corresponding to the specified cell to be greater than the core area. This will result in no placeable cell area in the block.

**What Next**

Ensure that the keepout value is more reasonable or increase the core area to accommodate the keepout area.

## DCFP-064

(Error) Exclusive move bounds %s and %s partially or completely overlap with each other.

Feedback

**Description**

This error occurs when two exclusive bounds overlap each other partially or completely. This may lead to insufficient space to place the cells constrained by the bounds as intersection area of two exclusive bounds is unusable.

**What Next**

Make sure that the bounds don't overlap each other.

---

## DCFP-065

(Warning) Keepout margin of cell %s lies partially or completely outside the bounds of the core.

**Description**

This message indicates that the keepout margin of the cell are not withon the core or boundary of the block.

**What Next**

Check the DEF file and ensure that the given keepout margin values are within the core area.

---

## DCFP-066

(Error) Exclusive movebound %s is completely covered by blockages.

**Description**

Exclusive movebound %s is completely covered by a blockage. This bound will be ignored.

**What Next**

Move the movebounds to ensure that they do not conflict with the blockages.

---

## DCFP-070

(Error) IO Pad %s doesn't lie between die bounding box and core bounding box.

**Description**

The placement for the specified pad or I/O is illegal.

Feedback

## DCFP-071

(Warning) IO Pad %s overlaps with other pads at (%s, %s).

## DCFP-080

(Error) No preferred routing direction defined for layer %s.

**Description**

This error indicates that there is no routing direction defined for a routing layer. The checking is performed for all routing layers if max/min routing layers are not defined.

If max/min routing layers are defined, the tool only check layers between max and min layers.

**What Next**

Please set the routing_direction attribute on the layer.

## DCFP-081

(Warning) Consecutive metal layers have the same preferred routing direction.

**Description**

This warning message occurs when two or more consecutive metal layers have the same prefferred routing direction.

**What Next**

Set the routing_direction attribute on the layer.

# DCHK

## DCHK-001

(warning) In design '%s', input pin '%s' of leaf cell '%s' is not connected to any net.

**Description**

This message appears when a leaf cell has an unconnected input pin.

Feedback

## DCHK-002

(warning) In design '%s', input pin '%s' of leaf cell '%s' is connected to undriven net '%s'.

**Description**

This message appears when leaf cell input pin has a connected net, but the net has no driver.

## DCHK-003

(warning) In design '%s', input pin '%s' of hierarchical cell '%s' has one or more internal loads, but is not being driven.

**Description**

This message appears when a hierachical cell input pin has internal load, but no driver.

## DCHK-004

(warning) In design '%s', input pin '%s' of hierarchical cell '%s' has no internal loads and is not being driven.

**Description**

This message appears when hierarchical cell input pin has no internal load and no driver.

## DCHK-005

(warning) In design '%s', the pin '%s' on submodule '%s' is connected to logic '%s'.

**Description**

This warning is issued when the *check_netlist* finds an instance of a hierarchical design (for example, a sub-module) that has an input connected to a logic constant. This warning is issued to verify that this is a desired connection on the submodule.

## DCHK-006

(warning) In design '%s', the same net is connected to more than one pin on submodule '%s'.

Feedback

**Description**

This warning is issued when the command *check_netlist* finds an instance of a hierarchical design (for example, a sub-module) that has more than one input connected to the same net. This warning is issued to verify that these are desired connections on the submodule.

---

## DCHK-007

(warning) In design '%s', cell '%s' has no output pins.

**Description**

This warning message indicates that the command *check_netlist* has found a cell with no output pins.

---

## DCHK-008

(warning) In design '%s', cell '%s' does not drive any nets.

**Description**

This warning alerts you that the output(s) of a cell is not connected to any load nets. This usually indicates that a design has not been correctly specified.

---

## DCHK-009

(warning) In design '%s', '%s' net '%s' driven by %s '%s' has no loads.

**Description**

This warning message occurs when a net is driven by an output pin (or pins) but has no load pins connected to it. This usually indicates that a design is not correctly specified.

---

## DCHK-010

(warning) In design '%s', '%s' net '%s' has no drivers.

**Description**

This warning message occurs when there is a net that is not driven by any source pins.

---

## DCHK-011

(information) In design '%s', '%s' net '%s' has multiple drivers.

Feedback

**Description**

This warning indicates that the *check_netlist* has found a net with multiple source pins.

**What Next**

Make sure that you intended to have multiple drivers on the given net. Errors in specifying a design could lead to this message in a situation where wired logic is not intended.

---

## DCHK-012

(error) In design '%s', '%s' net '%s' is driven by both logic 0 and logic 1.

**Description**

This error is issued when the *check_netlist* command encounters a shorted power ground net, which is connected to both logic 0 and logic 1.

---

## DCHK-013

(warning) In design '%s', multiply-driven '%s' net '%s' is driven by constant '%s'.

**Description**

A multiply-driven net has a constant as one of its drivers. This is likely to be a design error. If you intended to drive the net with a pullup/pulldown, instantiate a pullup/pulldown cell in your design instead.

---

## DCHK-014

(warning) In design '%s', '%s' net '%s' has a single tri-state driver '%s'.

**Description**

This warning message is printed when the *check_netlist* command detects that a net is driven by a single tri-state driver.

---

## DCHK-015

(warning) In design '%s', three-state '%s' net '%s' has non three-state driver '%s'.

**Description**

Synopsys tools classify a net as a three-state net if it is driven by at least one pin that has three-state attribute. Normally, if there are multiple drivers on such nets, it is assumed that all driving pins should be three-state drivers, for correct operation of the three-state

Feedback

bus. This warning message indicates a situation where at least one non-three-state driver appears on a three-state net.

## DCHK-016

(warning) In design '%s', port '%s' is not connected to any nets.

**Description**

This warning alerts you that a port in a design is not connected to any nets. This usually indicates that a design has not been correctly specified.

## DCHK-017

(warning) In design '%s', input port '%s' is unloaded.

**Description**

The *check_netlist* command issues this warning when it finds an unloaded input port on a design. This means that the given input port does not connect to any logic inside the design.

## DCHK-018

(warning) In design '%s', input port '%s' drives wired logic.

**Description**

This warning message occurs when the *check_netlist* command encounters an input port connected to a net that has multiple drivers. The input port (which acts as a driver on the net) is part of a wired-logic gate.

**What Next**

Make sure that you want multiple drivers on the net driven by the indicated port. Errors in specifying a design can lead to this message in a situation where wired logic is not intended. For example, if the direction of an output port is accidentally specified as input, then this message could occur.

## DCHK-019

(information) In design '%s', input port '%s' is connected directly to output port '%s'.

**Description**

This message alerts you to a situation where an input port in a design is directly connected to an output port.

Feedback

## DCHK-020

(warning) In design '%s', output port '%s' is connected directly to '%s'.

**Description**

This warning alerts you to a situation where an output port in a design is connected directly to Logic 1 or Logic 0.

## DCHK-021

(warning) In design '%s', net connected to output port '%s' is not driven.

**Description**

This warning message occurs when the *check_netlist* command finds the specified output port for the design is not driven by any signal.

## DCHK-022

(warning) In design '%s', output port '%s' is driven from the outside.

**Description**

This error message occurs when the *check_netlist* command finds the specified output port for the design is driven from the outside.

## DCHK-023

(warning) In design '%s', output port '%s' is connected directly to output port '%s'.

**Description**

This warning alerts you to a situation where an output port in a design is connected directly to another output port.

## DCHK-024

(warning) In design '%s', port '%s' is connected to a zero width bus.

**Description**

This messages indicates that the specified port is part of a zero width bus.

Feedback

## DCHK-025

(warning) In design '%s', port '%s' is not connected to any net. However the net with same name '%s' is also present in the design.

### Description

This messages indicates that the specified port is not connected to any net. However there is a net with the same name as that of port present in the design. You may want to check the connections of this port.

## DCHK-026

(warning) In design '%s', port '%s' is connected to net '%s' which does not have same name.

### Description

This messages indicates that the specified port is connected to net which does not have the same name. This is a violation of the verilog naming rules. Typically it happens when a logical hierarchy has an input port directly driving an output port, or a net that is driving multiple output ports directly. If nothing is done to resolve these warnings, then any Verilog written out will include assign statements to resolve the naming rule violations.

### What Next

To avoid assign statements in Verilog, Fusion Compiler offers the set_fix_multiple_port_nets command to resolve this early during the compile_fusion command. The -feedthroughs option enables insertion of buffers between logical hierarchy input and output ports. The -outputs option enables insertion of buffers on nets driving multiple output ports, such that each net only drives one output port after the buffers are inserted.

The Fusion Compiler and ICC II backend flow also offers the opt.port.eliminate_verilog_assign app option, which when set true, enables optimization steps in the compile_fusion, place_opt, and clock_opt commands to insert buffers to resolve these violations.

## DCHK-027

(warning) In design '%s', port '%s' is connected to net '%s' which does not have same name. However a net '%s' with same name as port exists in design.

Feedback

### Description

This messages indicates that the specified port is connected to a net with a different name. However a net with same name as the port also exists in the design. Hence, you may want to check the connections of the port as this is in violation of verilog naming rules.

---

## DCHK-028

(warning) The %s design contains an unused reference, %s.

### Description

This warning indicates that you design contains a dangling "reference" with the specified name. In Synopsys terminology, "reference" refers to the actual component for a given instantiation or cell. For example, if your cell library contains an inverter, INV, and you instanatiate that inverter in your design as cell13 and cell23, the INV library cell is the "reference" for each of these instantiations (cells). Sometimes, during the process of building or modifying a design, a reference remains, even though all cells that refer to it have been removed or changed. This should not impair the functionality of the design or the operation of Synopsys tools.

### What Next

A large number of dangling references in your design can increase the memory size of the design. Sometimes they can be removed by writing the entire design out and then reading it back in.

---

## DCHK-029

(warning) %s does not have any output ports.

### Description

This warning message indicates a design/cell without output pins. Unless such designs are protected through the use of the dont_touch command, the compile command will remove instances of such designs, since they have no functionality.

### What Next

This warning probably appeared because of an error in specifying a component in your design, or a component in your technology library. Check the design/library to find the problem, and correct the description of the given cell/design.

---

## DCHK-030

(Error) No floorplan exists for block '%s'.

Feedback

**Description**

This error occurs when the most basic floorplan information, like boundary and core, are not defined for the block.

**What Next**

Check if the floorplan information has been read in correctly for the design. In case no floorplan has been read in, run the initialize_floorplan command to create a new floorplan.

---

## DCHK-031

(Error) Boundary bbox is not defined for the block '%s'.

**Description**

This error occurs when the boundary is not defined for the block or the boundary is invalid.

**What Next**

Check if the DIEAREA definition in your input DEF file is valid. Read in a DEF with the correct DIEAREA defined or run initialize_floorplan command to create a new floorplan.

---

## DCHK-032

(Error) Block '%s' has invalid core area.

**Description**

This error occurs when no core area is defined for the block or the core area defined is invalid.

**What Next**

Check if the ROW definitions in your input DEF file are valid. Use create_site_array command to create site rows or run initialize_floorplan command to create a new floorplan.

---

## DCHK-033

(warning) In design '%s' net '%s' has a connection class violation.

**Description**

This warning message indicates that *check_netlist* has found a net that does not meet the connection class requirements described in the technology library. These connection class requirements ensure that all pins on a net can legally be connected. All pins connected together on a net must share at least one connection class in common. Each pin's class

Feedback

is defined in the technology library description for the pin, through the connection_class attribute.

**What Next**

Determine whether the design you are checking was (1) created manually as a netlist, or (2) created by Design Compiler (possibly from a high-level description) or DFT Compiler. If (1) is true, investigate the violation to be sure that you have specified the net correctly in your design. If (2) is true, examine the illegal net(s) created by the tool. It might not always be possible for the given net to be legalized. For example, a net might connect though hierarchy to pins whose connection classes cannot be matched. Or, other design rules, such as fanout restrictions or transition time restrictions might have made legalization for connection rules impossible. Also, check your technology library to determine whether there are level shifting components available to convert between the given connect classes.

---

## DCHK-034

(error) In design '%s', input port '%s' is connected to a multiply-driven net and the drivers include a hierarchical output pin.

**Description**

This error message occurs when the *check_netlist* command encounters an input port connected to a net that has multiple drivers and one or more of the drivers is a hierarchical output pin.

**What Next**

Make sure that you intend to have multiple drivers on the net driven by the indicated port.

Check that design ports are specified correctly. For example, if the direction of an output port is accidentally specified as input, then this message occurs.

Check the design to ensure that this is the functionality that you originally intended.

If you want to skip this checking, set the *da.check_netlist.allow_multiply_driven_nets_by_inputs_and_outputs* app option to true.

---

## DCHK-035

(error) In design '%s', net '%s' is a wire loop.

**Description**

This error message occurs when the *check_netlist* command encounters a wire loop, which is a timing loop with no cells in it.

Feedback

**What Next**

Make sure that you break the wire loop. Wire loops might cause further problems later.

If you want to skip this checking, set the *da.check_netlist.check_for_wire_loop* app option to false.

---

## DCHK-036

(error) In design '%s', leaf cell output pin '%s' is connected to constant net '%s'.

**Description**

This error message occurs when the *check_netlist* command encounters an cell output pin attempting to drive a constant net. This likely reflects a design error.

**What Next**

Please modify the design, so the output pin is not connected to a constant.

---

## DCHK-037

(warning) In design '%s', %s port '%s' is not being used in accordance with its stated direction.

**Description**

This warning message occurs when a port is not being used in accordance with its stated direction. An example is when a module assigns a value to an input port. These situations can cause bad logic.

**What Next**

This is only a warning message. No action is required.

However, if the result is not what you intended, search the source RTL for the offending assignment and correct it.

---

## DCHK-038

(information) In current design '%s', module '%s' is instantiated '%s' times.

**Description**

This information message indicates that a single module is instantiated more than once below the current design.

Feedback

## DCHK-039

(warning) In design '%s', cell '%s' is unmapped.

**Description**

This message appears when a cell in the design is found to be unmapped.

**What Next**

If the design has not undergone synthesis, this warning can be ignored. Otherwise, you should examine the cell that is reported to see why it is still unmapped.

## DCHK-040

(Error) Unit tile is missing in reference library.

**Description**

Unit tile is not found in the physical library. This message indicates one of three errors:

- No physical library is accessible.

- There is no site definition in any physical library.

- There are sites defined in the library, but none of them is a unit tile.

**What Next**

Add unit site definition in the technology portion of the physical library, and regenerate the library.

## DCHK-041

(Error) Site type '%s' height '%s' is not an integer multiple of site type '%s' height '%s'.

**Description**

This error occurs because the abnormal site height is not an integer multiple of the normal site height. Detail placement requires that the abnormal site height is an integer multiple of the normal site height.

**What Next**

Check the physical library make sure that the multiheight site is an integer multiple of the normal site height.

Feedback

## DCHK-042

(information) Use the related options for more information about warnings.

### Description

This message indicates that a terse version of informational messages or warnings on a design has just been displayed.

### What Next

Use the corresponding options to obtain more information about the design.

## DCHK-043

(error) In design '%s', cell '%s' does not have a reference.

### Description

This error describes a problem in the database representation for your design. The error means that a particular instantiation of a component (or "cell" in Synopsys terms) does not indicate which type of component it actually is. In Synopsys terminology "reference" describes the part of a design representation that points to the actual component for a given instantiation or cell. If you have an inverter (INV) in your technology library, and you've instantiated that inverter in your design as cell13 and cell23, then each of the instantiations (cells) should own a pointer (reference) to the actual component (in this case INV). This error indicates that, for some reason, the reference does not exist on the cell.

### What Next

Make sure that your design specifies the type of the given cell. If you specified a cell without a type, fix the problem by changing your design description. Otherwise this error could indicate that there is an internal problem in the Synopsys software that needs to be reported to the Synopsys Hotline.

## DCHK-044

(Information) Module '%s' does not contain any cells or nets.

### Description

Please double check the empty module.

## DCHK-045

(warning) In design '%s', the output pin '%s' of hierarchical cell '%s' has internal drivers, but has no loads.

Feedback

**Description**

This warning is issued when the *check_netlist* finds a floating output pin, which does not connect to any load nets.

---

## DCHK-047

(Warning) Macro cell '%s' is not placed.

**Description**

Macro cell %s location is invalid.

---

## DCHK-048

(Warning) Port '%s' is not placed.

**Description**

Port %s location is invalid.

---

## DCHK-050

(Error) Tracks not defined for layer '%s'.

**Description**

This message indicates that the tracks are not defined on the layer %s. The tracks are necessary for routing on this layer.

**What Next**

Use create_track command to create the tracks on the specific metal layers.

---

## DCHK-051

(Warning) Track '%s' on layer '%s' does not cross core area.

**Description**

This message indicates that the track %s doesn't cross core area. These tracks will not be considered by the tool during optimization and estimation.

**What Next**

Check the track definition and make sure the tracks cover the core area of the block.

Feedback

## DCHK-052

(warning) Block instance %s and %s are not aligned on cell row. There is cell row %s that orientation is %s end in location (%s, %s) at block instance %s's boundary edge %s, while there is no aligned cell row%s at block instance %s's boundary edge %s.

### Description

Cell rows in two block instances do not align. This will cause potential DRC errors in implementation.

### What Next

Check the cell rows in both block instances.

## DCHK-053

(warning) Block instance %s and current block %s are not aligned on cell row and wire track. There is cell row %s that orientation is %s end in location (%s, %s) at block instance %s's boundary edge %s in current block, while there is no aligned cell row%s at boundary edge %s in block instance %s.

### Description

Cell rows in two block instances do not align. This will cause potential DRC errors in implementation.

### What Next

Check the cell rows in both block instances.

## DCHK-054

(warning) Block instance %s and current block %s are not aligned on cell row. There is cell row %s that orientation is %s end in location (%s, %s) at boundary edge %s in block instance %s, while there is no aligned cell row%s at block instance %s's boundary edge %s in current block.

### Description

Cell rows in two block instances do not align. This will cause potential DRC errors in implementation.

### What Next

Check the cell rows in both block instances.

Feedback

## DCHK-055

(warning) Block instance %s and %s are not aligned on wire track. There is track %s%s %s on layer %s end in location (%s, %s) at block instance %s's boundary edge %s, while there is no same track%s at block instance %s's boundary edge %s.

### Description

Wire tracks in two block instances do not align. This will cause potential DRC errors in implementation.

### What Next

Check the wire tracks in both block instances.

## DCHK-056

(warning) Block instance %s and current block %s are not aligned on wire track. There is track %s%s%s on layer %s end in location (%s, %s) at boundary edge %s in block instance %s, while there is no same track%s at block instance %s's boundary edge %s in current block.

### Description

Wire tracks in two block instances do not align. This will cause potential DRC errors in implementation.

### What Next

Check the wire tracks in both block instances.

## DCHK-057

(warning) Block instance %s and current block %s are not aligned on wire track. There is track %s%s%s on layer %s end in location (%s, %s) at block instance %s's boundary edge %s in current block, while there is no same track%s at boundary edge %s in block instance %s.

### Description

Wire tracks in two block instances do not align. This will cause potential DRC errors in implementation.

### What Next

Check the wire tracks in both block instances.

Feedback

## DCHK-060

(Error) Macro cell '%s' is not fixed.

**Description**

This message indicates that the macro cell %s either doesn't have a location or has a location but is not fixed.

**What Next**

If the macro doesnt have a location, use create_placement command to place it. If the macro is placed, then fix the macros.

## DCHK-061

(Error) No location specified for fixed macro cell '%s'.

**Description**

Macro cell %s is fixed but its location is invalid.

## DCHK-062

(Warning) Fixed macro %s is out of the core area.

**Description**

This message indicates that the fixed macro %s is outside the core area.

**What Next**

Move this cell into the core area manually.

## DCHK-063

(Warning) Fixed cell %s is overlapping with blockages.

**Description**

This message is indicates that the fixed cell instance is overlapping with a blockage.

**What Next**

Manually move the fixed cell outside the blockage or run create_placement command.

Feedback

## DCHK-064

(Error) Macro cell %s is overlapping other macro cells.

**Description**

This message is indicates that this macro cell and one other neighbour macro cell are overlapping each other.

**What Next**

Manually move the cells to ensure that they don't overlap.

## DCHK-070

(Error) Site row %s overlaps with other site rows of same site definition.

**Description**

Site rows with same site definition are not allowed to overlap.

**What Next**

Check your floorplan to find overlapping rows. Modify your DEF file to eliminate the overlap.

## DCHK-071

(Error) Site array has rows and columns specified at the same time.

**Description**

You receive this error message because the site array you defined in the input DEF contains both horizontal and vertical arrays. Only one orientation of arrays can be specified in the site array, horizontal rows or vertical columns, but not both.

**What Next**

Modify the ROW definitions in the input DEF.

## DCHK-072

(Error) Site rows with site definition %s are misaligned.

**Description**

All the site rows with the same site site definition need to be aligned such that there is no offset between sites vertically.

Feedback

**What Next**

Change the floorplan so that the rows align properly. Ensure that the distance between x-origins of the different rows is zero modulo the site width.

---

## DCHK-073

(Error) Row '%s' has a non-zero extra spacing of '%s'.

**Description**

Non-Zero extra spacing in rows is not supported.

**What Next**

Set row_extra_spacing to zero

---

## DCHK-080

(Warning) Move bound '%s' overlaps with movebound '%s'.

**Description**

This message indicates that two or more bounds overlap with each other. This may lead to invalid placement of cells.

**What Next**

Make sure that the bounds don't overlap each other.

---

## DCHK-081

(Info) Non-exclusive movebound '%s' is completely covered by blockages.

**Description**

You receive this information because a movebound is completely covered by a blockage. When this occurs, the placer ignores the bound.

**What Next**

Move the movebounds to ensure that they do not conflict with the blockages.

---

## DCHK-082

(Warning) Placement blockage '%s' lie partially or completely outside the bounds of the block.

Feedback

**Description**

You receive this warning message because the dimensions or locations of the specified blockages cause them to lie outside the block. Therefore, these blockages in your design are located off the chip.

**What Next**

Correct the core area to include all the blockages within the block. Or correct the locations or dimensions of the placement blockage to fit within the block.

---

## DCHK-083

(Error) Keepout margin for cell '%s' is larger than the core area.

**Description**

This error indicates that the keepout area corresponding to the specified cell to be greater than the core area. This will result in no placeable cell area in the block.

**What Next**

Ensure that the keepout value is more reasonable or increase the core area to accommodate the keepout area.

---

## DCHK-084

(Error) Exclusive move bounds %s and %s partially or completely overlap with each other.

**Description**

This error occurs when two exclusive bounds overlap each other partially or completely. This may lead to insufficient space to place the cells constrained by the bounds as intersection area of two exclusive bounds is unusable.

**What Next**

Make sure that the bounds don't overlap each other.

---

## DCHK-085

(Warning) Keepout margin of cell %s lies partially or completely outside the bounds of the core.

**Description**

This message indicates that the keepout margin of the cell are not withon the core or boundary of the block.

Feedback

**What Next**

Check the DEF file and ensure that the given keepout margin values are within the core area.

---

## DCHK-086

(Error) Exclusive movebound %s is completely covered by blockages.

**Description**

Exclusive movebound %s is completely covered by a blockage. This bound will be ignored.

**What Next**

Move the movebounds to ensure that they do not conflict with the blockages.

---

## DCHK-090

(Error) IO Pad %s doesn't lie between die bounding box and core bounding box.

**Description**

The placement for the specified pad or I/O is illegal.

---

## DCHK-091

(Warning) IO Pad %s overlaps with other pads.

---

## DCHK-100

(Error) No preferred routing direction defined for layer %s.

**Description**

This error indicates that there is no routing direction defined for a routing layer. The checking is performed for all routing layers if max/min routing layers are not defined.

If max/min routing layers are defined, the tool only check layers between max and min layers.

**What Next**

Please set the routing_direction attribute on the layer.

Feedback

## DCHK-101

(Warning) Consecutive metal layers '%s' and '%s' have the same preferred routing direction.

**Description**

This warning message occurs when two or more consecutive metal layers have the same prefferred routing direction.

**What Next**

Set the routing_direction attribute on the layer.

## DCHK-102

(Error) Shape '%s' of port '%s' is of unexpected type '%s'.

**Description**

This message appears when a shape of port is of an unexpected type other than point, rectangle or polygon.

## DCHK-103

(Warning) The orientation of cell instance '%s' does not match any legal orientations.

**Description**

This message appears when a cell instance's orienation is not one of the legal orientation of the library cell.

## DCHK-104

(Information) The layer '%s' does not contain any PG shapes.

**Description**

This message appears when a layer doesn't have on PG shapes on it.

## DCHK-105

(Warning) The spacing of layer '%s' is greater than the difference of the pitch and width of the layer.

Feedback

**Description**

This message appears when a layer's spacing is greater than the difference of pitch and width.

---

## DCHK-106

(Error) Layer '%s' is not defined in the tech file.

**Description**

This message appears when a layer is not present in the tech file. Please check the tech file.

---

## DCHK-107

(Warning) The utilization of '%s':'%s' is above '%s'.

**Description**

This message appears when a movebound/groupbound object's utilization is above a certain value.

---

# DCX

---

## DCX-0003

(Information) Performing timing-driven clock gate split.

**Description**

The timing-driven clock gate split is being executed.

**What Next**

If that is not what you intended, set the application option *compile.clockgate.enable_timing_driven_clock_gate_split* to false.

---

# DCXS

---

## DCXS-0001

(error) Sub-Ckt build failed. No scenarios found during sub-ckt creation.

Feedback

**Description**

XForm sub-ckt build failed because no scenarios are found.

**What Next**

Client should add valid scenarios before calling XForm.

# DDB

## DDB-1

(warning) Cell %s not added to design %s because a cell with that name already exists.

## DDB-2

(warning) Net %s not added to design %s because a net with that name already exists.

**Description**

A net with the specified name already exists.

**What Next**

Reissue the command with a new net name. Use the *report_net* command to find a list of all the nets currently present in the design.

## DDB-3

(warning) Consistency problem: port %s is not owned by any design or reference.

**Description**

The specified port is not owned by any design.

**What Next**

Check the consistency of your design database.

## DDB-4

(warning) Consistency problem: a pin is not owned by any cell.

**Description**

The specified pin is not owned by any design.

Feedback

**What Next**

Check the consistency of your design database.

---

## DDB-5

(warning) Consistency problem: cell %s is not owned by any design.

---

## DDB-7

(warning) Consistency problem: net %s is not owned by any design.

---

## DDB-8

(warning) Consistency problem: a pin was found without a corresponding port.

---

## DDB-9

(warning) Consistency problem: cell %s was found without a corresponding reference.

---

## DDB-10

(warning) Consistency problem detected in design %s\n%s

---

## DDB-11

(warning) Internal error\n%s

---

## DDB-12

(warning) Removed duplicate cell '%s'.

---

## DDB-13

(warning) The name of net '%s' in design '%s' can't be changed \n\ \tto the name of both ports '%s' and '%s' to which it's connected.

---

## DDB-14

(warning) The net '%s' in design '%s' is connected to both \n \tports '%s' and '%s'.

Feedback

### Description

Two ports can only be connected to a net if they are inout ports or if they drive wired logic. Connecting two input ports to a net can limit the optimization that can be performed by Design Compiler on the net.

### What Next

Disconnect one of the ports from the net using the *disconnect_net* command.

---

## DDB-21

(error) Conflict between logic 0 and 1. Can't %s.

### Description

The two specified ports were set previously as logic 0 and 1. They might not have been directly set, but could have been implicitly set by a sequence of *set_equal* and *set_opposite* commands.

### What Next

If the ports were directly set as logic 0 and 1, the next steps are as follows:

```
Use report_attributes commands on the two ports. They will have the
driven_by_logic_zero and driven_by_logic_one attributes set.

Remove one of these two attributes, one of which is incorrect,
using the remove_attribute command

Reissue the original command.
```

If the ports were implicitly set as logic 0 and 1 through a sequesce of *set_equal* and *set_opposite* commands, the next step is as follows:

```
Use the reset_design command to remove the setting of 0 and 1 on the
ports. this command removes all other constraints on the design and
 should
be used with caution.
```

---

## DDB-22

(error) Can't set equal ports opposite in design '%s': '%s' '%s'.

### Description

The two ports were set as equal ports previously. They might not have been directly set, but could have been implicitly set by a sequence of *set_equal* and *set_opposite* commands.

Feedback

### What Next

The two specified ports were set previously as equal ports. The only way to revert back this setting is to reset the design using the *reset_design* command. This command removes all other constraints on the design and should be used with caution.

---

## DDB-23

(error) Can't set opposite ports equal in design '%s': '%s' '%s'.

### Description

The two ports were set as opposite ports previously. They might not have been directly set, but could have been implicitly set by a sequence of *set_equal* and *set_opposite* commands.

### What Next

The two specified ports were previously set as opposite ports. The only way to revert back this setting is to reset the design using the *reset_design* command. This command removes all other constraints on the design and should be used with caution.

---

## DDB-24

(warning) Overwriting design file '%s/%s'.

### Description

Overwriting the specified version of the design with more recent version.

### What Next

Warning only. No action is required.

---

## DDB-27

(error) '%s' value must be positive.

### Description

A negative value was entered. Constraint values must be positive (or 0).

### What Next

Please enter a non-negative value.

Feedback

## DDB-28

(error) '%s' cannot be set on %s pin '%s'.

**Description**

The specified constraint cannot be set on a pin of the specified direction. The constraint command will be ignored.

**What Next**

Please enter a correct constraint command.

## DDB-29

(error) '%s' cannot be set on %s port '%s'.

**Description**

The specified constraint cannot be set on a port of the specified direction. The constraint command will be ignored.

**What Next**

Please enter a correct constraint command.

## DDB-30

(error) Can't specify output port '%s' as a path startpoint.

**Description**

Output ports are not valid as path startpoints. There are no timing paths from such ports.

**What Next**

Examine the design to determine the correct startpoint for the path.

## DDB-31

(error) Can't specify input port '%s' as a path endpoint.

**Description**

Input ports are not valid as path endpoints. There are no timing paths to such ports.

**What Next**

Examine the design to determine the correct endpoint for the path.

Feedback

## DDB-32

(error) Can't specify hierarchical cell '%s' as a path '%s'.

**Description**

Hierarchical cell names are not valid as path startpoints or endpoints.

**What Next**

Use a clock, port, pin (leaf or hierarchical), or cell (leaf only) as the path startpoint or endpoint.

## DDB-33

(error) Pin '%s' does not have a library hold time.

**Description**

None.

**What Next**

Check the target library.

## DDB-34

(error) %s '%s' is in design '%s', but %s '%s' is in design '%s'.

**Description**

None.

**What Next**

None.

## DDB-35

(error) '%s' does not exist in library '%s'.

**Description**

The listed key in the error message does not exist in the library.

**What Next**

Check the key file.

Feedback

## DDB-38

(error) Can't open security file '%s' for protected library '%s'.

**Description**

Cannot open security file.

**What Next**

Check the security file.

## DDB-39

(error) Bad security key in file '%s' for library '%s'.

**Description**

You are trying to invoked *read_lib* on a protected library.

**What Next**

If a nodelocked library, check and correct these attributes: key_file, key_seed, and key_bit. If a network licensing library, check and correct these attributes: key_feature, key_version, and key_seed. Then reinvode the *read_lib* command.

## DDB-40

(error) Can't read protected library '%s'.

**Description**

You are trying to invoked *read_lib* on a protected library.

**What Next**

If a nodelocked library, check and correct these attributes: key_file, key_seed, and key_bit. If a network licensing library, check and correct these attributes: key_feature, key_version, and key_seed.

## DDB-41

(error) Incomplete library protection attributes.

## DDB-43

(warning) Could not create attribute '%s' for %s objects.

Feedback

## DDB-44

(error) Can't read unprotected library '%s'.

## DDB-45

(error) A design with name '%s' already exists in the same\n \tdesign file as design '%s'.

## DDB-46

(error) A reference with name '%s' already exists in design '%s'.

## DDB-47

(error) A cell with name '%s' already exists in design '%s'.

## DDB-48

(warning) Creating port '%s' on design '%s' with direction unknown.

## DDB-51

(warning) In the value of %s, \n \tthe characters separating the \tthose separating each

## DDB-52

(error) The value of %s \n \tmust include one

## DDB-53

(error) The value of %s \n \tmust include one

**Description**

The message is printed out when the current command found invalid value in the indicated variable.

**What Next**

Check the value of the given variable and provide the valid value.

Feedback

## DDB-54

(error) The value of %s \n \tmust include only one

**Description**

The error message is printed out when the current command found invalid value in the indicated variable.

**What Next**

Provide the valid value for this variable.

## DDB-55

(error) The value of %s \n \tmust include only one

**Description**

The error message is printed out when the current command found invalid value for the indicated variable name included in the message.

**What Next**

Provide the valid value for this variable.

## DDB-56

(error) In the value of %s, \n \tthere are no characters separating %s.

**Description**

This error message is printed out when the current command found that there are no character separators.

**What Next**

Provide the valid value for the command.

## DDB-57

(error) In the value of %s, \n \tthe %s of the characters separating \n \t%s must not be%s a digit.

**Description**

This error message is printed out when the current command found digit in the character separators.

Feedback

**What Next**

Provide the valid value for the command.

---

## DDB-58

(warning) In the value of %s, \n \tthere are no characters separating the \tbe ambiguous).

**Description**

If no characters separate the array name from the member number, the bus names will be ambiguous for arrays whose names end in a digit. For example, with *bus_naming_style* set to "%s%d", the name of the third member of array "A1" and the thirteenth element of array "A" would both have the name "A13".

**What Next**

Most likely you should change the value of the bus naming style to contain some characters between the "%s" and "%d". See the help page for *bus_naming_style* for details.

---

## DDB-60

(error) Could not find library pin for pin '%s'.

**Description**

The back-annotation failed because the pin does not have a corresponding pin in the link library.

**What Next**

Verify that the design is fully linked with the 'link' command. Verify that all link and target library search paths are valid.

---

## DDB-66

(warning) Removing group '%s'.

**Description**

The path group is being removed because the operation being performed has removed the last path from the group.

**What Next**

Be aware that any subsequent attempts to use this path group will fail because the group has been deleted. If you think the group should not have been deleted, identify at least

Feedback

one path you think should still be in the group and trace back to find which action deleted that path.

## DDB-67

(warning) Removing %s from group '%s'.

## DDB-68

(warning) Removing external delay related to clock %s.

**Description**

When a clock source is deleted from the design using the 'remove_clock' command, all input and output delay values that were specified relative to that clock are also deleted from the design. For example, consider the following script:

create_clock -name CLK -period 10 set_input_delay 2 -clock CLK all_inputs()
set_output_delay 2 -clock CLK all_outputs()

remove_clock CLK

The 'remove_clock' command will also have the effect of removing the input and output delay values that were specified relative to CLK.

**What Next**

No action is required, this is merely an informational warning.

## DDB-70

(error) None of the selected cells were grouped.

**Description**

When using the 'group' command on an hdl design that has not been compiled yet, it is possible that the cells you specified to be grouped cannot be grouped because they need to stay with thier neighboors until the design has gone through resource sharing in compile.

**What Next**

Use the 'group' command after the design has been compiled.

## DDB-71

(error) Design '%s' requires one of the following licenses: '%s'.

Feedback

**Description**

The specified design is licensed and requires one of the listed licenses to be available. An error has occurred because none of the licenses could be obtained.

**What Next**

Verify that your site has at least one of the specified licenses. If it does not, then one must be purchased. If your site does have one of the licenses, try again when the license is available.

---

## DDB-72

(information) Added key list '%s' to design '%s'.

**Description**

Indicates that the listed set of licenses have been associated with the specified design. Accessing the design will require that one of the listed licenses can be successfully checked out.

**What Next**

No action is required since this is just an information message.

---

## DDB-73

(warning) License '%s' contains the illegal character '%c'.\n\tIt was ignored.

**Description**

One of the licenses associated with a design contains an invalid character. That license will be dropped from the list of licenses which can be used to access the design.

**What Next**

If a design is associated with multiple licenses, it may be possible to access the design via a different, valid license. In any case the license name should be updated so that it does not contain any invalid characters. Invalid characters include: ',', '}', '{', '*', '\'.

---

## DDB-74

(warning) Design '%s' inherited license information from design '%s'.

Feedback

### Description

All of the licenses associated with one design have now been associated with another design. This typically happens when a design is ungrouped. In that situation, the parent design will inherit all licensing information from the ungrouped design.

### What Next

Although this is classified as a warning, it is essentially just an information message. No action is required.

---

## DDB-75

(warning) Design '%s' is being converted to a limited design.

### Description

This message indicates that the indicated design has been converted into a limited design. A limited design is a design which can be compiled and analyzed, but whose contents may not be examined or written out.

A design gets converted into a limited design in two situations. In the first case you have only an evaluation license for the design (obtained by manually setting synlib_disable_limited_licenses = "FALSE"), and the design will remain limited because the intent of an evaluation license is for the design data to be restricted. In the second case the design was derived from a DesignWare part and has not yet been run through the *compile* command. In this latter case the design remains limited only until after it has been compiled, at which point the limitation is removed. The intent of this latter limitation is to protect the technology- independent structure of the design.

### What Next

If you have only an evaluation license for the limited design, then the only way to gain access to the internals of that design is through the use of a full license for that part; if a full license is subsequently obtained then the first *compile* command will remove the 'limited design' restrictions. If the design has already been compiled, it is sufficient to run the *get_license* command to convert the limited design back to a regular, unrestricted format.

If you already have a full license for the limited design, then after the first *compile* command you can expect the design to be converted back to a regular, unrestricted format.

---

## DDB-76

(error) Cannot load design '%s'.

Feedback

**Description**

The named design cannot be loaded into Design Compiler's internal data structures. This can happen if certain synthetic parts cannot be found.

**What Next**

Attempt to link the design with the command *link -all*. If this fails, check that the software is correctly installed.

---

## DDB-77

(error) License '%s' is a Synopsys internal key and should\n \tnot have a seed associated with it.

**Description**

When license names are set on a design, third party keys must have a seed associated with them. It is an error, however, for Synopsys internal keys to have seeds.

**What Next**

When creating your own license names, make sure that they do not clash with the Synopsys internal license names.

---

## DDB-78

(error) No seed provided for the third-party license '%s'.

**Description**

When adding a third-party license, you must specify both a license name and a non-zero seed. This prevents third party vendors from creating keys for each other's licenses.

**What Next**

Specify the license name as *name/seed*. For example: "MY_LIC/1234"

---

## DDB-79

(error) The license '%s' has an invalid seed associated with it.

**Description**

This error occurs when the *set_design_license command* is used to store a third party license and a seed, and the seed that is specified is not consistent with the seed that was used to create the key.

Feedback

### What Next

Either generate a new key with the correct seed, or change the *set_design_license* command line so that the seeds are consistent.

---

## DDB-80

(error) The seed '%s' specified for license '%s' is\n\tnot a valid 32-bit integer.

### Description

Licenses can be specified in the from '<license>/<seed>' where <seed> is supposed to be a valid integer which can be represented in 32 bits. In this case the <seed> specified was either not an integer, or was too large to be represented in 32 bits.

### What Next

The <seed> associated with the license needs to be corrected such that it is a valid integer which can be stored in at most 32 bits.

---

## DDB-81

(warning) Unable to find specified driving_cell for port '%s'.

### Description

The driving_cell attributes indicate that a port should inherit its drive capability from a certain library cell. This error means that the tool was unable to locate a matching library cell or pin on that library cell. This may happen if the link_library does not contain the library for that cell, or if the cell name or pin name was incorrect. The driving cell information can be seen using *report_port -drive*. It may have been set by either *set_driving_cell* or *characterize*.

### What Next

If the driving cell requires a library that has not been identified in the *link_library*, the *link_library* should be changed to include that library. Otherwise, check the information for errors in cell_name, library name, or pin names using *report_port -drive -only port_name.*

---

## DDB-84

(error) Only ports of the same direction can be grouped.

---

## DDB-85

(error) Objects must be either all ports or all nets.

Feedback

**Description**

This error message is issued if a mixture of objects of different types is given as input to a command that requires a homogenous set of ports or nets as input.

**What Next**

Re-issue the command specifying only ports or only nets as input.

---

## DDB-86

(error) Bus name '%s' conflicts with existing names.

**Description**

This error message is issued when an attempt is made to create a bus with a name that conflicts with the name of an existing bus.

**What Next**

Re-issue the command with a non-conflicting name for the bus.

---

## DDB-87

(error) All objects must be from the same design.

**Description**

This error message is issued if a command that requires a set of objects belonging to the same design is invoked with objects that belong to different designs.

**What Next**

Re-issue the command with a set of objects that belong to the same design.

---

## DDB-88

(error) At least one of the port objects specified is already a member of a bus.

**Description**

This error message is displayed if an attempt is made to insert a port that belongs to a bus into a new bus.

**What Next**

Re-issue the command with ports that do not belong to an existing bus.

Feedback

## DDB-89

(error) Type name '%s' conflicts with existing type.

**Description**

This error message is issued if an attempt to create a type for a new bus object is made with a type name that is already used for another bus object with a different width.

**What Next**

Change the name of the new bus type.

## DDB-90

(error) specified range is different from number of objects to be bussed.

## DDB-91

(error) only designs or references can have busses created in them.

## DDB-92

(error) Cannot load design '%s' for an HDL embedded command.

**Description**

The named design cannot be loaded for the current command embedded in an HDL file. The current embedded command is not legal within an embedded script. See the HDL Compiler Manual for details.

**What Next**

Change the embedded script so that it does not use the offending command.

## DDB-95

(warning) Unable to find net instance '%s' in design '%s'.

**Description**

Back-annotation, such as a set_load or set_resistance value, was stored on a net instance within the design, but that instance can no longer be found. This can occur if a lower-level design containing the net instance was modified but the top-level design was unaware of this change. Commands which could modify lower level designs include *ungroup*, *change_names*, and *compile*.

Feedback

### What Next

Perform *ungroup* and *change_names* at the top level so that the top level design will have a chance to update its back-annotation records. Use *characterize* to move annotation to a subdesign before running *compile* or *reoptimize_design* on that subdesign.

---

## DDB-100

(warning) Unable to find minimum version of library cell '%s/%s' in library '%s'.

### Description

The *set_min_library* command has been used to indicate that timing data for minimum analysis must use a particular library, but this library cell cannot be found in that minimum library. Design Compiler will use the maximum version of the cell for both maximum and minimum analysis in this case.

### What Next

Check that the *set_min_library* command specified the correct library for minimum analysis. See if the indicated library cell was accidentally left out of the minimum analysis library.

---

## DDB-101

(warning) Unable to find minimum version of library pin '%s/%s' in library '%s'.

### Description

The *set_min_library* command has been used to indicate that timing data for minimum analysis must use a particular library, but the indicated library cell has conflicting pin descriptions between the maximum and minimum libraries. In this case, a pin exists on the library cell in the maximum library, but that pin does not exist in the minimum library. Design Compiler will use the maximum version of the library pin for both maximum and minimum analysis in this case.

### What Next

Check that the *set_min_library* command specified the correct library for minimum analysis. See if the indicated library pin was accidentally left out of the minimum analysis library.

---

## DDB-102

(warning) Conflicting timing arc descriptions between maximum library '%s' and minimum library '%s' to pin '%s/%s'.

Feedback

### Description

The *set_min_library* command has been used to indicate that timing data for minimum analysis must use a particular library, but the indicated library cell has conflicting timing arc descriptions between the maximum and minimum libraries. In this case, a timing arc exists on the library cell in the maximum library, but a corresponding timing arc does not exist in the minimum library. Design Compiler will use the maximum version of the timing arc for both maximum and minimum analysis in this case.

In order to consider timing arcs as compatible between minimum and maximum libraries, they must have the same sense (for example, positive unate, or clear), they must have the same SDF condition, and there must be the same number of arcs of each type between a pair of pins.

### What Next

Check that the *set_min_library* command specified the correct library for minimum analysis. See if timing arcs to the indicated library pin were accidentally left out of the minimum analysis library.

---

## DDB-103

(warning) Pin %s of lib cell %s exists in maximum library %s but not in minimum library %s. Assuming min delay for pin to be same as max delay.

### Description

The *set_min_library* command has been used to indicate that timing data for minimum analysis must use a particular library, but the indicated library cell has some pins in the max library which don't exist in the minimum libraries. Design Compiler will use the maximum version of the timing arc for both maximum and minimum analysis in this case.

In order to be compatible between minimum and maximum libraries, the library cells must have the same pins and same timing arcs (same sense (for example, positive unate, or clear), they must have the same SDF condition), and there must be the same number of arcs of each type between a pair of pins.

### What Next

Check that the *set_min_library* command specified the correct library for minimum analysis. See if the specified pins of the specified library cell were accidentally left out of the minimum analysis library.

---

## DDB-105

(warning) Design '%s' requires one of the following licenses: '%s'. \n\tWaiting for license to become available, press <ctrl>-C to terminate.

Feedback

### Description

The specified design is licensed and requires one of the listed licenses to be available. None of the licenses could be obtained. Design Compiler will wait for one of the licenses to become available and then continue.

### What Next

Verify that your site has at least one of the specified licenses. If it does not, then one must be purchased. If you wish to terminate the command instead of waiting for the license to become available, set synlib_wait_for_design_license = {}

---

## DDB-107

(warning) Deleted or recreated %d internal pin(s) on cell '%s'. All attributes and attaches on it(them) are lost.

### Description

The *access_internal_pins* variable controls creation and deletion of internal pins and user access to them. Because of the source of internal pins, they are created or deleted during link time, depending on the setting of this variable. You can use the *find* and *get_pins* commands to show internal pins, if such pins exist. Certain timing commands can also set constraints on internal pins.

### What Next

If this is what you intended to do, no action is necessary.

### See Also

- get_pins

---

# DDC

---

## DDC-1

(error) Unable to open DDC file '%s' for writing.

### Description

*dc_shell* encountered an I/O error when it attempted to open the specified DDC file for writing.

Feedback

**What Next**

Check that the file name is correct, that the directory exists and is writable, and that the filesystem is not full. If the target file already exists it must be writable in order for dc_shell to overwrite it.

---

## DDC-2

(error) Unable to open file '%s' for reading.

**Description**

The tool encountered an error when it attempted to open the specified DDC file for reading.

**What Next**

Check that the path to the file is correct and that the file is readable by the current user. Also verify that the file was written in DDC format.

---

## DDC-3

(error) DDC internal write error in design '%s'.

**Description**

*dc_shell* encountered an internal error while attempting to write the specified design. No DDC file was produced.

**What Next**

Please contact the Synopsys Support Center for assistance.

---

## DDC-4

(error) DDC internal read error in file '%s'.

**Description**

*dc_shell* encountered an internal error while attempting to read the specified file. No designs were read.

**What Next**

Please contact the Synopsys Support Center for assistance.

Feedback

## DDC-5

(error) Design data is corrupt in DDC file.

**Description**

The file being read has been modified or corrupted since it was originally written by *dc_shell*.

**What Next**

The DDC file must be re-created from the original design source. Note that DDC files cannot be edited by the user.

## DDC-6

(error) DDC file version is not compatible. The DDC file was written with\n dc_shell version %s dated %s.

**Description**

The file being read was written with an incompatible version of of the .ddc format that cannot be read with this release of the tool. The .ddc format is updated from time to time to support new features or functionality, and sometimes the new format cannot be read with older tool versions.

**What Next**

Use the current version of the tool to read the .ddc file.

## DDC-7

(error) File is not in DDC format.

**Description**

*dc_shell* has determined that the file being read is not a DDC format file.

**What Next**

Verify that the correct format option has been provided to the read_file command, and that the file path is correct.

## DDC-8

(error) Attempt to write duplicate design name '%s' to DDC file.

Feedback

### Description

*dc_shell* detected an attempt to write multiple designs with the same name to a DDC file, which is not allowed by the DDC format.

### What Next

Check the list of designs provided to the write command. It is possible, although unusual, for duplicate design names to have been read from different files, such as multiple .db files. If the duplicate design names must be preserved, they must be written to separate DDC files.

---

## DDC-9

(warning) The DDC format cannot be used to store physical data. Any physical\n information that is present will not be written to the DDC file.

### Description

You receive this warning message when you write out a DDC file that contains physical data. The DDC file format is not capable of representing physical data. Only the logical representation will be written to the DDC file.

### What Next

It is considered best practice to use the Milkyway storage format for designs that contain physical information. You can use the *write_milkyway* command to write the design to a Milkyway database.

---

## DDC-10

(error) Scenario '%s' does not exist.

### Description

The specified scenario, which was provided with the -scenarios option to the *write* command, does not exist.

### What Next

Check that the scenario name is correct, and that any scenarios which are to be written to the DDC file have been created.

### See Also

- create_scenario

Feedback

## DDC-11

(error) DDC file contains packed command syntax that cannot be processed by\n this version of dc_shell. Please use a current version of dc_shell\n to read this file.

### Description

The current DDC file contains embedded commands (constraints) which were written by a newer version of *dc_shell* and utilize a syntax which cannot be parsed by the current executable.

### What Next

Use the same (or newer) version of *dc_shell* to read the file as was used to write it.

## DDC-12

(warning) DDC file was written with a newer version of dc_shell.\n Some embedded commands may be ignored.

### Description

You are trying to read a DDC file which was written by a newer version of *dc_shell*, and the DDC file contains some packed commands (constraints) which are not recognized by the older version of *dc_shell*. Normally this would result in a DDC-6 error; however, if the variable ddc_allow_unknown_packed_commands is set to "true" then *dc_shell* will attempt to read the file, but any unrecognized packed commands will be ignored. (A DDC-13 warning message will be printed when each such command is first encountered.)

### What Next

To avoid losing potentially important information, always read the DDC file with the same (or newer) version of *dc_shell* as was used to write it.

### See Also

- DDC-13

## DDC-13

(warning) Ignoring unknown packed command #%d: %s

### Description

*dc_shell* attempted to unpack an embedded command which it does not recognize. This can happen when a DDC (or Milkyway) file which was written with a newer version of *dc_shell* is read with an older version of the product, while the variable

Feedback

ddc_allow_unknown_packed_commands is set to "true". The unrecognized command is simply discarded.

**What Next**

To avoid losing potentially important information, always read the DDC file with the same (or newer) version of *dc_shell* as was used to write it.

---

## DDC-14

(information) This file contains data for the following %d scenarios:\n %s

**Description**

The DDC or Milkyway reader prints this message if the file being read contains any scenario-specific constraint data. The message will report the name of each scenario for which the file contains data. If the scenario was inactive when the file was written, an asterisk (*) appears after the scenario name.

**What Next**

This is an informational message. No action is required.

**See Also**

- create_scenario
- current_scenario
- all_scenarios

---

## DDC-15

(warning) Ignoring %s attribute %s on %s object(s):\n Attribute type conflicts with existing attributes.

**Description**

An attribute contained in the DDC file conflicts in type with with an attribute that is already registered with the tool. The conflicting attribute is ignored by the DDC reader.

An attribute saved in a DDC file must have the same type as an existing attribute of the same name on the same netlist object class (e.g., string attribute on design objects).

This message is most likely seen when reading in DDC files written by the X-2005.09 release or earlier. Attribute conflicts from more-recent files are silently discarded.

Feedback

### What Next

This message could appear if there happens to be a user-defined attribute that has the same name (but different type) as a built-in attribute that was added in a later version of the tool. It is good practice to use a prefix for user-defined attributes, such as your company's name, that is unlikely to be used by Synopsys in the future.

### See Also

- set_attribute
- report_attribute

---

## DDC-16

(error) ddc file contains no scenario-specific data. Cannot specify\n -scenarios option for this file.

### Description

The *-scenarios* option to the *read_file* command can only be used to read files that contain scenario-specific constraint data.

### What Next

See *read_file*(2) for more information on controlling which scenarios' constraints are read from the ddc file.

---

## DDC-17

(error) ddc file contains no data for any of the requested scenarios

### Description

The specified ddc file does not contain any constraint data for any of the scenarios specified to the *-scenarios* option to the *read_file* command.

### What Next

Check the list of scenario names supplied to the *read_file -scenarios* option. See *read_file*(2) for more information on controlling which scenarios' constraints are read from the ddc file.

---

## DDC-18

(error) DDC file contains no data for any of the requested active scenarios

Feedback

### Description

The specified ddc file does not contain any constraint data for any of the scenarios specified to the *-active_scenarios* option to the *read_file* command.

### What Next

Check the list of scenario names supplied to the *read_file -active_scenarios* option. See *read_file*(2) for more information on controlling which scenarios' constraints are read from the ddc file.

---

## DDC-19

(warning) ddc file contains no data for the following requested scenarios:\n %s

### Description

The specified ddc file does not contain any constraint data for one or more of the scenarios specified to the *-scenarios* option to the *read_file* command. The file will be read and the list of scenarios to be read in will be restricted according to the remaining scenarios specified to the *-scenarios* option.

### What Next

Check the list of scenario names supplied to the *read_file -scenarios* option. See *read_file*(2) for more information on controlling which scenarios' constraints are read from the ddc file.

---

## DDC-20

(warning) This .ddc file contains special features and may only be read by\n\t tools that support the required features.

### Description

This warning message occurs when you write a .ddc file that is "feature locked." It is a reminder that the tool that is writing the file has flagged the file as requiring specific features in order for it to be read, and the file might not be readable by other tools or by the same tool if the required features have not been enabled.

### What Next

This is only a warning message. No action is required.

### See Also

- DDC-21

Feedback

## DDC-21

(error) The feature used to generate this .ddc file is not supported by\n\t this tool or is not enabled in the current session.

### Description

This error message occurs when you attempt to read a .ddc file that was "feature locked" by the tool that wrote it. The current session does not have, or has not enabled the specific features required to read the file. This usually happens when *dc_shell* writes the file when it is in a special mode and the resulting files cannot be read by all tools that consume the .ddc format.

### What Next

Review any other messages that were printed to understand which tool and features are required to read the file.

### See Also

- DDC-20

## DDC-22

(error) Cannot read design '%s' as a block abstraction because the file does not contain block abstraction information.

### Description

This error occurs when the *set_top_implementation_options* command has been run to specify that the design be read as a block abstraction, but no block abstraction information was saved in the .ddc file.

### What Next

Make sure you run the *create_block_abstraction* command immediately before writing the .ddc file. Also, verify that the correct design name is specified by the *set_top_implementation_options* command.

## DDC-23

(error) Cannot read design '%s' as a block abstraction because the specified interface features are not present in the file.

Feedback

### Description

This error occurs when the *set_top_implementation_options -load_logic compact_interface* command is used to specify that the named design be read as a block abstraction with a compact_interface, but the block abstraction has been generated with an older version of the tool that does not support the annotation of compact_interface logic.

### What Next

Regenerate the block abstraction with the latest version of the tool in order to obtain support for compact_interface markings. Alternatively, the block abstraction can be loaded as a block abstraction with a full_interface.

---

## DDC-24

(error) Output file %s is the source file for the block abstraction.

### Description

This error occurs when the specified output file is the source .ddc file for the block abstraction that is being written out. Design Compiler does not allow you to overwrite the original .ddc file because each design's core (non-interface) logic is retrieved from that file and merged with the block abstraction's interface logic in order to write out the complete netlist for the design.

### What Next

Specify an output file that is not the source of any block abstractions that are currently in memory.

---

## DDC-25

(error) Block abstraction source file %s appears to have been modified since the design was read.

### Description

This error occurs when the design that is being written out is a block abstraction and the .ddc file that it was originally read from has since been modified. The .ddc files containing block abstractions must not be modified until after their blocks have been written back out because the core (non-interface) logic is retrieved from the original files.

### What Next

Do not modify the source files for block abstractions if the designs are still being used.

Feedback

## DDC-26

(error) Cannot open block abstraction source file %s.

### Description

This error occurs when the design that is being written out is a block abstraction and the .ddc file that it was originally read from cannot be opened. The file might have been deleted or renamed, or there might be a permissions problem. The .ddc files containing block abstractions must remain available until their blocks have been written back out because the core (non-interface) logic is retrieved from the original files.

### What Next

Make sure the original .ddc files remain available as long as the block abstractions are being used.

## DDC-27

(information) Reading %s block abstraction for design '%s' and its hierarchy.

### Description

This message occurs when the specified design hierarchy is being loaded as a block abstraction. Only the specified type of interface logic (full or compact interface) is being loaded, not the entire design netlist. This behavior is controlled by the *set_top_implementation_options* command.

### What Next

This is an informational message. No action is required.

## DDC-28

(error) Block abstraction interface logic has been modified in an unsupported manner. Cannot write .ddc file.

### Description

This message occurs when the interface logic of a block abstraction has been modified in a way that the *write_file* command does not expect. This prevents the merging of the interface logic in memory with the core logic from the original .ddc file. This could be caused by manual modifications you have made to the netlist.

### What Next

Do not make manual modifications to block abstractions. Any such changes should be made prior to running the *create_block_abstraction* command.

Feedback

## DDC-29

(error) You must specify the output file name when writing a block abstraction.

**Description**

This message occurs if you are writing a block abstraction to a .ddc file and you have not specified the *-output* option with the *write_file* command. In this case, "block abstraction" refers to either the full design that has been processed by the *create_block_abstraction* command or the block abstraction (interface logic only) as specified by the *set_top_implementation_options* command.

Specifying the *-output* option causes Design Compiler to write all designs to a single file, which is required for block abstraction hierarchies.

**What Next**

Specify an output file when writing block abstractions to .ddc files.

## DDC-30

(warning) Forcing the -hierarchy option because a block abstraction is being written.

**Description**

This message occurs if you are writing a block abstraction to a .ddc file and you have not specified the *-hierarchy* option with the *write_file* command. In this case, "block abstraction" refers to either the full design that has been processed by the *create_block_abstraction* command or the block abstraction (interface logic only) as specified by the *set_top_implementation_options* command.

Design Compiler requires that block abstractions be written hierarchically, starting from the top design of the block abstraction. Therefore, the *-hierarchy* option is inferred if it has not been explicitly specified.

**What Next**

Specify the *-hierarchy* option when writing block abstractions to .ddc files.

## DDC-31

(error) You must specify only the top design of the block abstraction hierarchy.

**Description**

This message occurs if you are writing a block abstraction to a .ddc file and you have specified more than one design name with the *write_file* command or the specified design is not the top design of the block abstraction hierarchy. In this case,

Feedback

"block abstraction" refers to either the full design that has been processed by the *create_block_abstraction* command or the block abstraction (interface logic only) as specified by the *set_top_implementation_options* command.

A block abstraction's entire hierarchy must be written to a single .ddc file. The only design name that can be specified with the *write_file* command is the top design of the block abstraction. However, the design name can be omitted if that design is the current design. The *-hierarchy* and *-output* options are mandatory when writing block abstractions.

**What Next**

Specify only the top design of the block abstraction when writing .ddc files.

---

## DDC-32

(information) Merging block abstraction with non-interface netlist data from %s.

**Description**

This message occurs when writing out a design that was loaded as a block abstraction as specified by the *set_top_implementation_options* command. Since the block abstraction's interface logic might have been modified after it was loaded into memory, the interface logic in memory is merged with the core (non-interface) logic from the original .ddc file, and a new .ddc file is written out.

**What Next**

This is an informational message. No action is required.

---

## DDC-33

(information) Block abstraction information will not be preserved. If this\n information is required, you must run the create_block_abstraction\n command prior to writing the design.

**Description**

This message occurs when you read a .ddc file that contains block abstraction information that was created by the *create_block_abstraction* command. If the *set_top_implementation_options* command was not used to tell Design Compiler to read the file as a block abstraction (to read the interface logic only), then the design is read in its entirety. This message is a reminder that the block abstraction information is not automatically preserved if the design is written to a new .ddc file. If you want to preserve the block abstraction's information in a new file, you must run the *create_block_abstraction command immediately before writing it out.*

Feedback

**What Next**

This is an informational message. No action is required.

---

## DDC-34

(error) Cannot write IC Compiler block abstraction %s in ddc format.

**Description**

This message occurs when you attempt to write a block abstraction that was created by IC Compiler to a .ddc file. Only block abstractions created by Design Compiler can be written in the ddc format.

---

## DDC-35

(error) Reading of ILM designs is no longer supported. Cannot read %s.

**Description**

This message occurs when you try to read ILM designs in ddc format. ILM designs are no longer supported for use in Design Compiler.

**What Next**

Migrate to the use of block abstractions instead of ILMs for loading the blocks at the top level.

---

# DDCR

---

## DDCR-001

(error) Design '%s' in library '%s' is currently opened; reading DDC into an opened design is not allowed.

**Description**

The DDC file cannot be read into a design that is currently opened in memory.

**What Next**

Close the currently opened design and then run *read_ddc* again.

---

## DDCR-002

(information) Reading DDC into new design '%s' in library '%s'.

Feedback

**Description**

A new design has been created by *read_ddc*. The DDC file is read into the new design.

---

## DDCR-003

(information) Reading DDC into existing design '%s' in library '%s'; the existing design contents has been truncated.

**Description**

The DDC file is read into the specified existing design. Incremental DDC updates are not allowed. So *read_ddc* opens the design in truncate mode, discards the existing contents, and then reads the new DDC file into the design.

**What Next**

When saving truncated designs, use the *save_block -as* option to avoid overwriting your existing design on disk.

---

# DDF

---

## DDF-001

(error) Failed to start a design dash fusion server on {%s:%s}.

**Description**

You receive this error message when it failed to start a design dash fusion server on the address you set. It is very likely the address has already been used or the address cannot be assigned.

**What Next**

Please check if the address is legal at first. It the address is legal, please check if the address has already been used. Using stop() in resttcl package to shut down the old server before start a new one on the same address.

---

## DDF-002

(error) Detected invalid python.

**Description**

You receive this error message when the python binary specified by "-path" option is invalid.

Feedback

**What Next**

Please check the python binary and update the "-path" option accordingly.

---

## DDF-003

(error) Don't support the python version below 3.2.3 .

**Description**

You receive this error message when the version python binary specified by "-path" option is not supported.

**What Next**

Please check the python binary and update the "-path" option with a python binray higher than version 3.2.3 .

---

## DDF-004

(error) %s.

**Description**

You receive this error message when exception are caught during the execution of your command.

**What Next**

Please follow the exception error messgage to fix.

---

## DDF-005

(error) Fail to get the object class for object '%s'.

**Description**

The tool cannot get the object class for this object. This will stop the tool to get the feature value.

**What Next**

Double check the specified objects.

---

## DDF-006

(error) The input objects have different object classes.

Feedback

**Description**

The command cannot support the mixed objects with different object classes.

**What Next**

Correct the specified objects.

---

## DDF-007

(error) The value of option %s is empty.

**Description**

The command cannot accpet the empty option. Please double check it.

**What Next**

Correct the empty option.

---

## DDF-008

(warning) The name pair {%s %s} will be ignored since it is not included in the specified features.

**Description**

The labels will be ignored since it is not specifed by the option -features.

---

## DDF-009

(error) The specified new feature name "%s" given by -labels has conflict with -features.

**Description**

You receive this error message when the new feature name in -labels is a duplicate of one feature in -features.

**What Next**

Please provide correct name pairs.

---

## DDF-010

(error) A table named "%s" already exists.

**Description**

Cannot create two tables with the same name.

Feedback

**What Next**

Please provide another valid table name.

---

## DDF-011

(error) The table "%s" does not exist.

**Description**

Cannot update nonexistent table.

**What Next**

Please create the table before updating it.

---

## DDF-012

(error) The feature '%s' does not exist.

**Description**

You receive this error message when the object class does not have the specified feature(attribute).

**What Next**

Please check your input for -features.

---

## DDF-013

(warning) Failed to get feature "%s" for %d of %d objects. Default value will be used for the missing value.

**Description**

The tool fails to get the feature value, so the defaul value nan will be applied.

**What Next**

Check if the feature is specified correctly.

---

## DDF-014

(warning) Failed to get feature "%s" for all objects. This feature will be ignored.

**Description**

The tool cannot get the feature value for all the objects, so this feature will be skipped.

Feedback

**What Next**

Double check your feature name.

---

## DDF-015

(warning) Failed to get any feature for %d of %d objects. These objects will be ignored.

**Description**

If none of the feature value can be extracted for the object, it will be skipped.

**What Next**

Double check the specified objects.

---

## DDF-016

(error) The attribute type of feature "%s" is %s, which is not supported.

**Description**

The tool cannot support this type of attribute.

**What Next**

Consider using -function to derive this feature.

---

## DDF-017

(error) The specified data sizes of %s and %s are not equal.

**Description**

The sizes are inconsistent in these two options.

**What Next**

Please specify equal number of elements for these two options.

---

## DDF-018

(info) Feature "%s" is duplicated. It will only be extracted once.

**Description**

You receive this error message when there are duplicates in -features.

Feedback

**What Next**

Please remove duplicates in -features. If you really want to keep duplicates, you can use update_feature_table with -labels to rename a feature.

---

## DDF-019

(info) Feature %s is a duplicate of index. It will be ignored.

**Description**

You receive this error message when the feature in -features is a duplicate of index.

**What Next**

Please remove this feature from -features. If you really want to keep duplicates, you can use -labels to rename this feature.

---

## DDF-020

(error) Operation failed because: %s

**Description**

You receive this error message when the command failed for some reason.

**What Next**

Please correct your inputs according to the shown message.

---

## DDF-021

(error) Failed to get index. %s

**Description**

You receive this error message when the tool failed to extract the index.

**What Next**

Please correct your inputs according to the shown message.

---

## DDF-022

(error) Invalid objects. Make sure -objects is a valid collection.

**Description**

You receive this error message when the input for -objects is not a valid collection.

Feedback

**What Next**

Please use valid collection for -objects.

---

## DDF-023

(error) Failed to execute tcl proc %s. %s

**Description**

You receive this error message when the tool failed to execute the proc given by -function or -index_function.

**What Next**

Please correct the input for -function or -index_function according to the shown message.

---

## DDF-024

(warning) %d rows are skipped due to duplicated index. The latter duplicated row will overwrite the former. Duplicated indexes: %s

**Description**

You receive this error message when the input for -objects has duplicates.

**What Next**

You can remove duplicates from the collection of -objects.

---

## DDF-025

(error) Cannot parse feature value. %s

**Description**

You receive this error message when the tool failed to convert the feature value to expected data type.

**What Next**

Check if the type given by -type if correct.

---

## DDF-026

(error) Cannot use database. %s

Feedback

**Description**

You receive this error message when the tool failed to use database.

**What Next**

Check if the database is opened correctly.

---

## DDF-040

(error) %s

**Description**

You receive this error message when the specified database does not exist.

**What Next**

Please use an existing database.

---

## DDF-041

(error) %s

**Description**

Cannot open multiple databases.

**What Next**

Please close current open database first.

---

## DDF-042

(error) %s

**Description**

You receive this error message when you try to close database while there is no open database.

**What Next**

Use open_ml_db to open an database before closing it.

---

## DDF-043

(error) %s

Feedback

**Description**

Cannot execute the opration without an open database.

**What Next**

Please open a database first.

---

## DDF-044

(error) %s

**Description**

You receive this error message when you try to remove the current open database.

**What Next**

Please close current open database before removing it.

---

## DDF-045

(error) %s

**Description**

Cannot open different types of databases.

**What Next**

Please close previously opened database first.

---

## DDF-046

(error) %s

**Description**

You receive this error message when you try to create a record while this record already exists.

**What Next**

Please use another record name.

---

## DDF-047

(error) %s

Feedback

**Description**

You receive this error message when you specify a nonexistent record.

**What Next**

Please specify an existing record.

---

## DDF-048

(error) %s

**Description**

You receive this error message when the specified record is not a hier record.

**What Next**

Please specify a hier record.

---

## DDF-049

(error) Environment variable CERBERUS_HOME is not set. Either use this environment variable or -server option to specify the server path.

**Description**

You receive this error message when you try to use cassandra datastore without specifying server path.

**What Next**

Please set environment variable CERBERUS_HOME or use -server to specify the server path.

---

## DDF-050

(error) Database operation failed. %s

**Description**

You receive this error message when the operation failed for some reason.

**What Next**

Please correct your input according to the shown message.

Feedback

## DDF-117

(Error) Unable to create directory in %s. Please use -working_dir to specify a working directory.

**Description**

You receive this error message when directory cannot be created.

**What Next**

Please use -working_dir to change working directory.

## DDF-118

(Info) The background python process is terminating.

**Description**

Please note that the background python process will be terminated

## DDF-119

(Error) The Python screen output option can only be specified after the Python code run.

**Description**

In command run_python, the -screen_output can be specified alone without any other options, or specified with -file or -cmd option.

**What Next**

Correct the usage of -screen_output option for run_python.

## DDF-120

(error) The Python variable '%s' does not exist.

**Description**

The tool fails to get the Python vaiable, please double check.

## DDF-123

(error) Python file %s is not found.

Feedback

**Description**

You receive this error message when Python file cannot be found.

**What Next**

Provide correct path to Python file.

---

## DDF-125

(error) Host file %s is not found.

**Description**

You receive this error message when host file cannot be found.

**What Next**

Please check the path of host file.

---

## DDF-126

(error) The host name should be provided for RSH.

**Description**

For RSH protocol, the correct host name is needed.

**What Next**

Please provide the correct host name.

---

## DDF-128

(Error) Directory %s is not writable. Please use -working_dir to specify a working directory.

**Description**

You receive this error message when directory is not writable.

**What Next**

Please use -working_dir to change working directory.

---

## DDF-132

(error) There are some exceptions in python commands, except for 'SystemExit'.

Feedback

**Description**

The python commands throw out an error or exceptions, which will not include the base exceptions 'SystemExit'.

**What Next**

Please check the python commands.

---

## DDF-134

(error) Failed to start background Python process.

**Description**

Background Python process failed to start.

**What Next**

Please check if Python executable is available.

---

# DEFR

---

## DEFR-001

(error) Cannot find def file '%s'.

**Description**

The file, specified in the read_def command, cannot be read.

**What Next**

Check the existence of the file in the search_path using the *which* command.

---

## DEFR-002

(error) read def '%s' failed.

**Description**

read def file failed.

**What Next**

Check def file syntax and content.

Feedback

## DEFR-003

(error) Cannot create %s '%s'.

**Description**

The specified object cannot be created in the design.

**What Next**

Check the object definition in the DEF file. Make sure that an object with the same name does not already exist in the design.

## DEFR-004

(error) Cannot find %s '%s'.

**Description**

The specified object cannot be found in the database.

**What Next**

Check that the specified object exists in the design, if it is a design object. Or check that the specified object exists in the library, if it is a library or technology object.

## DEFR-005

(error) Port '%s' cannot be connected to net '%s', because it is already connected to net '%s'.

**Description**

The specified net cannot be reconnected to the net specified in the DEF netlist. It is already connected to another net. The DEF reader is not allowed to change port connections. It can only connect unconnected ports and reconnect ports from tie-hi/lo to power/ground nets. x

**What Next**

Check the Verilog and DEF netlists and makes sure they are consistent with each other.

## DEFR-006

(error) Pin '%s' of instance '%s' cannot be connected to net '%s', because it is already connected to net '%s'.

Feedback

**Description**

The specified net cannot be reconnected to the net specified in the DEF netlist. It is already connected to another net. The DEF reader is not allowed to change pin connections. It can only connect unconnected pins and reconnect pins from tie-hi/lo to power/ground nets. x

**What Next**

Check the Verilog and DEF netlists and makes sure they are consistent with each other.

---

## DEFR-007

(error) memory allocation error.

**Description**

There is an memory allocation or reallocation error when reading def file.

**What Next**

Check available machine memory and if memory is enough to hold the design.

---

## DEFR-008

(error) error of BUSBITCHARS '%s'.

**Description**

There is an syntax error BUSBITCHARS in the reading DEF file.

**What Next**

Check syntax BUSBITCHARS in the def file.

---

## DEFR-009

(error) unsupported or wrong UNITS.

**Description**

The UNITS in def file is wrong or not supported.

**What Next**

Check "UNITS DISTANCE MICRONS dbuPerMicron" in def file.

Feedback

## DEFR-010

(error) Failed to create net wiring for net '%s'.

**Description**

An error occurred while creating the net wiring (i.e., shapes) of the specified net.

**What Next**

Check the net wiring of the specified net in the DEF file.

## DEFR-011

(error) Pin '%s' has an invalid DIRECTION.

**Description**

The specified pin has no DIRECTION or an invalid DIRECTION.

**What Next**

Check the DIRECTION of the pin in the DEF file.

## DEFR-012

(error) Net %s has an invalid net wiring extension or STYLE.

**Description**

The specified net has an invalid wiring extension or STYLE.

**What Next**

Check the net wiring extension and STYLE of the net in the DEF file.

## DEFR-013

(error) Site '%s' does not support %s rows with site orientation '%s'.

**Description**

The specified site cannot be oriented in the direction specified by the row and site orientations.

The site symmetry must support the rotation and mirror specified by the row site orientation. Sites do not support rotations and mirrors that would swap the width and height dimensions.

Feedback

**What Next**

Check the LEF site definition and make sure the specified symmetry supports the necessary row site orientation.

---

## DEFR-014

(warning) Context design name '%s' does not match the DEF DESIGN name '%s'.

**Description**

The context design name should match the design name specified in the DEF file's DESIGN statement.

The context design is specified by the -design option. If the -design option is not specified, the current design is assumed to be the context design.

**What Next**

If the context design and DEF DESIGN names must match, specify the correct design using the -design option.

---

## DEFR-015

(warning) Ignoring unsupported %s.

**Description**

The specified DEF construct is not supported and will be ignored. When read_def is complete, a summary shows the total number of ignored constructs.

---

## DEFR-016

(information) Reading %s.

**Description**

The specified DEF construct is being read.

When read_def is complete, a summary shows the actual number of processed objects out of the total number of read constructs. Processed objects are those that are annotated or verified with the DEF information.

---

## DEFR-017

(warning) Skipping unplaced component '%s'.

Feedback

**Description**

The specified component is skipped because it has no placement information to annotate to the design.

---

## DEFR-018

(warning) Skipping physical-only pin '%s'.

**Description**

The specified pin is skipped, because it is physical-only. There is no corresponding logical pin.

---

## DEFR-019

(warning) Skipping unrouted net '%s'.

**Description**

The specified net is skipped, because it has no routing information to annotate to the design.

---

## DEFR-020

(warning) Skipping regionless group '%s'.

**Description**

The specified group is skipped, because it has no region. Only groups with physical regions are supported.

---

## DEFR-021

(error) Cannot find pin '%s' of instance '%s'.

**Description**

The specified pin cannot be found in the design. It cannot be connected to the net indicated in the DEF netlist.

**What Next**

Check that the pin exists on the specified cell instance.

Feedback

## DEFR-022

(error) Cannot find non-default rule '%s' of net '%s'.

**Description**

The specified non-default rule does not exist and cannot be assigned to the specified net. The net wiring shapes cannot be created without the non-default rule defintiion.

**What Next**

Add the non-default rule to the DEF file or create the rule using the *create_routing_rule* command, and then read the DEF file.

## DEFR-023

(warning) Cannot find net '%s' in hierarchy '%s' based on the DEF net name '%s'.

**Description**

The specified net cannot be found in the design. Its net wiring will still be created, but will be left unconnected to any net.

**What Next**

Check that the net exists in the design. Check that the DEF net name is escaped correctly. If the DEF net name contains escaped hierarchical characters, the hierarchy is extracted from the DEF net name. The DEF net name is stripped of the escape characters and looked up in the database. Sometimes a hierarchical character is mistakenly escaped or not escaped which alters the hierarchy in which the net is located.

## DEFR-024

(error) The instance '%s' of reference design '%s' is not physical.

**Description**

The specified cell instance is not physical and will be skipped. The instance exists only in the logical hierarchy. All DEF components must be physical.

**What Next**

Check the DEF file's component section and make sure the specified instance references the correct design.

## DEFR-025

(warning) %s '%s' already exists in the %s.

Feedback

**Description**

The specified object already existed and will not be created again during *read_def*.

Only one via definition, with the specified name, can exist in the design. This applies to both DEF fixed and generated vias. The name must be unique among all fixed and generated vias in the design.

Only one fixed via definition, with the specified name, can exist in the library and design. The name must be unique among all fixed vias in the library and design.

Only one pin (top-level port shape), with the specified name, can exist in the design.

**What Next**

If the existing object is correct, do nothing. If the existing object should be replaced by the DEF definition, remove the existing object or rename the DEF file's via, and then run *read_def*.

---

## DEFR-026

(error) Cannot create %s '%s'; %s.

**Description**

The specified object cannot be created in the design for the specified reason.

**What Next**

If the layer does not exist, define the layer in the technology or correct the layer name in the DEF input file.

If the geometry is invalid, correct the geometry definition in the DEF input file.

If the via cut pattern is invalid, correct the via cut pattern in the DEF input file.

---

## DEFR-027

(error) Failed to read into unsupported current design view '%s'.

**Description**

The DEF reader supports only the design and outline views.

**What Next**

Switch the current design to one of the supported views.

Feedback

## DEFR-028

(warning) Created cell '%s' has non-PG cross hierarchy net connections.

### Description

The specified cell was created by the *read_def* command. However the DEF specifies signal/clock net connections to this cell across logical hierarchy. These cross hierarchy connections are ignored, because *read_def* does not perform signal/clock port punching on hierarchical boundaries.

### What Next

These non-PG cross hierarchy net connections must be specified in the input Verilog file or manually using the *connect_net* command.

## DEFR-029

(warning) Cannot find cell '%s'.

### Description

The specified cell instance cannot be found in the design. The DEF reader does not create any cell instances. It only annotates data onto existing cell instances, which are usually created by the Verilog reader.

### What Next

Check your input Verilog file and make sure it contains the specified cell instance. Or you may force *read_def* to create the missing cell instances using the *-allow_cell_creation* option. Note that *read_def* connects cell instances to PG nets, but not to signal nets. Signal connections are expected to come through Verilog.

## DEFR-030

(error) Cannot find reference design '%s'.

### Description

The specified design cannot be found in the database, and the input DEF file contains an instance of this design. The DEF reader will skip this instance, and no data will be annotated to the design for this instance.

### What Next

Check your libraries and read the appropriate input files to create the missing reference design.

Feedback

## DEFR-031

(information) Converted site definition '%s' to '%s'.

**Description**

The indicated row site definition name was changed to the new name, as specified in the *-convert_sites* option. All rows associated with site definition will be associated with the newly named site definition.

**What Next**

Verify that you have specified the correct site definition names.

## DEFR-032

(warning) Pin '%s' is specified with direction '%s' that does not match existing direction '%s'.

**Description**

The specified pin has a DEF-specified direction that differs from the current pin direction. The pin's direction will not be changed.

**What Next**

Check your input DEF file and your current design, and make sure you are reading the correct DEF file.

## DEFR-033

(warning) Pin '%s' is specified with connection to net '%s' that does not match existing connection to net '%s'.

**Description**

The specified pin has a DEF-specified net connection that differs from the current net connection. The pin's net connection will not be changed.

**What Next**

Check your input DEF file and your current design, and make sure you are reading the correct DEF file.

## DEFR-034

(error) Scan chain '%s' has an unspecified %s.

Feedback

**Description**

The scan chain is ignored and omitted from the design for the indicated reason.

**What Next**

Check the DEF file's scan chain definition and make sure it is defined correctly.

---

## DEFR-035

(error) Library '%s' is missing technology information.

**Description**

The technology information is missing in the specified library. This information is necessary in order to read the DEF file and create physical design data.

**What Next**

Load a technology file into the library using the *read_tech_file* command.

---

## DEFR-036

(error) %s '%s' cannot be set as the %s of scan chain '%s' because it is already a part of scan chain '%s'.

**Description**

In the DEF file, the port or pin is specified as the starting or ending point of a scan chain. However it is currently either a starting point, ending point, or in the middle of another scan chain.

**What Next**

Correct the invalid scan chain data, then read the corrected DEF file.

---

## DEFR-037

(error) Scan chains are invalid; removing all scan chain data from the design.

**Description**

One or more scan chains were defined with incorrect data. This invalidates all scan chains in the design. All scan chains will be removed from the design.

**What Next**

Correct the invalid scan chain data, then read the corrected DEF file.

Feedback

## DEFR-038

(error) The instance '%s' is not in the current design.

**Description**

The specified cell instance is within a lower level of the physical hierarchy. It is in another design. This command may only annotate data to the current design.

**What Next**

Check the DEF file. The DEF file should contain data for one and only one design.

## DEFR-039

(warning) Setting sequential scan cell '%s' bit length to 1; ignoring the DEF specified bit length of %d.

**Description**

All sequential cells must have a minimum bit length of 1. The DEF specified bit length was overridden with the default bit length 1.

**What Next**

Verify that the default bit length 1 is correct for this sequential cell. If not, correct the input DEF file to reflect the correct bit length, which should be greater or equal to 1.

## DEFR-040

(warning) Setting non-sequential scan cell '%s' bit length to 0; ignoring the DEF specified bit length of %d.

**Description**

All non-sequential (i.e., combinational) cells must have a bit length of 0. The DEF specified bit length was overridden with the bit length 0.

**What Next**

Verify that the bit length 0 is correct for this combinational cell.

## DEFR-041

(warning) Via definition '%s' specifies a via rule name '%s' that references a %s technology via definition. A design via definition will be created.

Feedback

### Description

The DEF generated via definition specifies a via rule name that does not correspond to a valid technology via definition. The technology via definition must meet these requirements:

1) If the via rule name is DEFAULT, the default via definition must exist in the library technology for the specified cut layer. 2) If the via rule name is anything other than DEFAULT, the via definition must exist in the library technology having the same name as the specified via rule name. 3) The via definition must be a simple via definition, and not a custom via definition.

A new design simple via definition will be created. This via definition has design scope only, and is not visible outside of this design nor in the library technology. Instances of this new via definition will be created for all references in the DEF NETS and SPECIALNETS sections.

### What Next

If the new design via definition is not desired, verify that your design library is using the correct technology file, and that the DEF via definition specifies a valid via rule. The technology file must contain the via rule referenced by the DEF generated via definition. Reload the correct technology file and recreate your design.

---

## DEFR-042

(warning) Via definition '%s' specifies a via rule name '%s' that references a technology via definition with a different %s. A design via definition will be created.

### Description

The DEF generated via definition specifies a via rule name that corresponds to a technology via definition that has different layers, cut size, cut spacing, enclosure dimensions, or origin offset values.

A new design simple via definition will be created. This via definition has design scope only, and is not visible outside of this design nor in the library technology. Instances of this new via definition will be created for all references in the DEF NETS and SPECIALNETS sections.

### What Next

To avoid creating a new design via definition, the DEF and technology via definition must have matching layers, cut size, cut spacing, enclosure dimensions, and origin offset values. If the new design via definition is not desired, verify that your design library is using the correct technology file, and that the DEF via definition matches the technology via definition. Then reload the correct technology file and recreate your design.

Feedback

## DEFR-043

(warning) Failed to create the blockage with '+ COMPONENT %s' as cell keepout margins because %s; creating %s instead.

### Description

The indicated DEF blockage cannot be created as cell keepout margins for the indicated reason. It will be created as a placement blockage, losing its association with the specified cell.

DEF blockages defined with the "+ COMPONENT compName" statement must be created as cell keepout margins, in order to create a placement blockage around a specific cell. It may be created as a cell keepout margin if and only if the blockage is defined as a single geometry which extends out uniformly from the cell's boundary on each side (i.e., there is one margin for all boundary left-edges, one margin for all boundary bottom-edges, one margin for all boundary right-edges, and one margin for all boundary top-edges).

### What Next

Verify that the created placement blockage is the desired representation for the DEF blockage.

## DEFR-044

(warning) Fixed via definition '%s' has been converted into a generated via definition, because it has more than %d cuts.

### Description

The indicated DEF fixed via definition has too many cuts which leads to runtime degradation of downstream applications. It has been converted into a DEF generated via definition. This DEF generated via definition is either mapped to a matching library technology simple (parameterized) via definition, or it is created as a design simple via definition.

Note that once converted, it remains as a simple via definition. The write_def command does not unconvert it back into a DEF fixed via definition.

### What Next

Verify the results.

## DEFR-045

(warning) Invalid mask value %d for %s.

Feedback

**Description**

The specified shape has an invalid mask information. Following situations are checked: (1) Rect, polygon, track or via metal shape has a number of mask more than three. (2) The mask number of track is more than the number_of_masks of the routing layer used by the track. (3) The number_of_masks of the routing layer used by track is less than two when the track is specified which mask for double or triple patterning lithography to use.

**What Next**

Please fix the mask number of the shape and try command again.

---

## DEFR-046

(error) Failed to read in the layers in COMPONENTMASKSHIFT section.

**Description**

The specified layers are invalid. Following situations are checked: 1.The specified layer must be either a two or three mask layer. 2.The order of the layers must be increasing from the highest layer down to the lowest layer in the LEF layer order.

**What Next**

Please check the number_of_masks and the order of the layers and correct the invalid layer data, then read the corrected DEF file.

---

## DEFR-047

(warning) Invalid masks for '%s'.

**Description**

The specified masks are invalid because there is no layer specified in COMPONENTMASKSHIFT section, so read_def will ignore the masks.

**What Next**

Please specify the shifting layers, then read the corrected DEF file.

---

## DEFR-048

(error) Invalid mask values for the cut-shapes of via '%s'.

Feedback

**Description**

The tool issues this error when it has partially colored cuts for one via. Because the cut-shapes must all be colored or none of them colored for a fixed via made up of RECT/ POLYGON statements.

**What Next**

Reset the mask numbers for the cut-shapes of via then read the DEF file again.

---

## DEFR-049

(warning) Invalid mask values for the cut layer of via '%s'.

**Description**

The specified via has invalid mask information. Following situations are checked: (1) The mask value of via cut-shape is greater than 15. (2) The mask value of via cut-shape is greater than the number_of_masks of the cut layer. (3) The mask values of all via cut-shapes are neither alternating(e.g.123123...) nor uniform(e.g.111...). In above cases, all mask values for the specified via will be ignored.

**What Next**

Please check the number_of_masks of the cut layer and the mask values of via cut-shapes, and correct the invalid color data then read the DEF file again.

---

## DEFR-050

(error) Via definition '%s' has shapes on invalid layer '%s'.

**Description**

The indicated via definition has shapes on an invalid layer. All via shapes must be on an interconnect, local_interconnect, via_cut, or local_via_cut layer.

**What Next**

Check the layer_type attribute of the indicated layer. Correct the DEF input file and then read it again.

---

## DEFR-051

(error) Failed to delete '%s' '%s'.

**Description**

The tool was unable to delete this physical data.

Feedback

**What Next**

Check its attributes.

---

## DEFR-052

(warning) Invalid mask shift value '%d' for the compontent '%s'.

**Description**

You received this message because of the specified cell instance has invalid mask shift value. DEF shiftLayerMasks value must meet the following conditions, otherwise it will be ignored. 1.It must be 0 or 1 for two-mask layers, and 0, 1, 3, 4 or 5 for three-mask layers. 2.It must be 3, 4 or 5 when the shifted layer has fixed mask 1, 2 or 3.

**What Next**

Please check the DEF shiftLayerMasks value and the shift layer, and correct the invalid data then read the DEF file again.

---

## DEFR-053

(warning) Fail to perform the consistency check for the mask shift value '%d' of compontent '%s'.

**Description**

You received this message because the corresponding fixed mask in technology file is not 1, 2 or 3 when DEF shiftLayerMasks value is 3, 4 or 5. In this case, the mask shift value for the specified cell instance will be ignored.

**What Next**

Please correct the invalid DEF shiftLayerMasks value, and then read the DEF file again.

---

## DEFR-054

(warning) The cut layer of via '%s' is not a multi-mask layer.

**Description**

This warning message occurs when the cut layer is a single-mask layer and the mask pattern of the via definition is alternate. In this case, the mask of the cuts will be pre-defined as a uniform pattern, alternating pattern is only allowed on multi-mask layer.

Feedback

**What Next**

Please check if the attribute number_of_masks of layer is what you want, if not, please correct the attribute and read the DEF file again, otherwise, please check where the input DEF is from.

---

## DEFR-055

(warning) Invalid trim metal shape on the layer "%s".

**Description**

Trim metal shape will be ignored if it is not defined in the FILLS section.

**What Next**

Please correct the section and then read the DEF file again.

---

## DEFR-056

(warning) Layer "%s" has wrong-way default width.

**Description**

This warning occurs when layer has wrong-way default width and the DEF version is not 5.8. In this case, default width will apply to signal routing in both the preferred and non-preferred directions.

**What Next**

Please use DEF 5.8 if you want to honor wrong-way default width.

---

## DEFR-057

(warning) In the input DEF file, the geometry for STYLE %d is invalid.

**Description**

According to DEF syntax, a style polygon consists of two to eight points.

**What Next**

Fix input DEF.

---

## DEFR-058

(warning) In the input DEF file, the styleNum value '%d' in "STYLE styleNum" is invalid.

Feedback

**Description**

This warning message occurs in the following two situations. 1. The styleNum is less than 0. 2. The styleNum is only specified in NETS/SPECIALNETS section but not in STYLES section. The invalid styleNum will be ignored.

**What Next**

Fix input DEF styleNum value.

---

## DEFR-059

(warning) The reference design of instance '%s' is inconsistent between database and DEF file.

**Description**

This warning message occurs when the instance has existed in database but its reference design is inconsistent between database and DEF file. The DEF reader does not update the reference design of an existing instance.

**What Next**

Please check the attribute ref_name of the cell instance to make sure the DEF is consistent with the reference library.

---

## DEFR-060

(warning) Existing terminal '%s' of port '%s' has been replaced.

**Description**

The specified terminal (i.e., DEF pin) already exists in the design and has been replaced with the new terminal definition in the DEF PINS section. Only one terminal, with the specified name, can exist in the design.

**What Next**

Verify that the new terminal, defined in the DEF PINS section, is correct.

---

## DEFR-061

(warning) Multiple via def matches found for via ladder layer %s for via ladder at location ('%d', '%d').

**Description**

Multiple via rules are found for one via ladder, we do not restore the viaRule for this ladder.

Feedback

## DEFR-062

(warning) No via_def found for vias on via ladder layer '%s'.

**Description**

No via_def found for this via ladder. Vias in this via ladder will be created as custom vias. These vias will not have is_via_ladder bit set on them.

## DEFR-063

(warning) Via rule '%s' references undefined cut name '%s'. Hence this via rule is not a candidate to be considered as a correct match to the via ladder.

**Description**

Via rule references undefined cut name in cutNameTbl. Hence this via rule is not a candidate to be considered as a correct match to the via ladder.

## DEFR-064

(warning) Failed to set DEF %s property on %s '%s', because %s.

**Description**

The DEF property could not be set on the specified object. Either the property has not been defined, or it is defined with a different data type.

**What Next**

Make sure the property definition exists and is defined with the correct data type.

## DEFR-065

(warning) Cannot find %s '%s'.

**Description**

The specified object cannot be found in the database.

**What Next**

Check that the specified object exists in the design, if it is a design object. Or check that the specified object exists in the library, if it is a library or technology object.

Feedback

## DEFR-066

(error) Failed to create a scan clain element with scan-in pin '%s' and scan-out pin '%s' in scan chain '%s', because %s.

### Description

The specified scan chain element is invalid. A pair of scan-in and scan-out pins may appear as an element in at most one scan chain. Duplicate elements are not allowed.

### What Next

Remove the invalid element from the scan chain and then read the Scan DEF.

## DEFR-067

(error) Fail to do the consistentcy check on the attribute '%s' of the reference block for cell '%s'.

### Description

This message occurs when the mask shift information of the cell in DEF file is inconsistent with the reference block of the cell.

```
Take the following DEF file as an example, there is a cell U210 which
references NAND3X1_LVT, this message will be trigged when the reference
block meets any of the following conditions.
1. The attribute "is_mask_shiftable" is false.
2. The attribute "mask_shift_layers" does not exist or the values
   does not contain M7 or M6.

DESIGN TEST ;
COMPONENTMASKSHIFT M7 M6 M3 M2 M1 ;
COMPONENTS 1 ;
 - U210 NAND3X1_LVT + UNPLACED + MASKSHIFT 11000 ;
END COMPONENTS
END DESIGN
```

### What Next

Check the attribute of the reference block and make sure the specified instance references the correct design.

## DEFR-068

(information) Cells referencing design '%s' will use design label '%s'.

Feedback

**Description**

When searching for reference designs, read_def chooses the reference design with the label as indicated by the label switch list if any. Otherwise read_def chooses the reference design with the same label as the top design. If none is found, then read_def chooses the reference design with the default label. If no default label exists, then read_def chooses the reference design with the smallest label name and prints this information message.

---

## DEFR-069

(warning) Unable to convert RECT ( %d %d %d %d ) to a path shape, because %s.

**Description**

The application option, file.def.convert_rects_to_paths, is set true, so *read_def* attempted to create the indicated DEF RECT as a path shape. However *read_def* was unable to do so for the indicated reason. This specific DEF RECT will be created as a rectangle shape in the same manner as when the application option is false.

**What Next**

If the application option was untentionally set to true, set it back to false to avoid this warning. If the DEF RECT must be converted into a path shape, fix the problem indicated in the warning message. If the problem is a length precision limitation, try adjusting the tech length precision or library scale factor.

---

## DEFR-070

(error) TRACK statement: '%s' in PROPERTYDEFINITIONS section, DESIGN TRACKPROPERTIES statement could not be parsed.

**Description**

Parsing failed.

**What Next**

Check the DESIGN TRACKPROPERTIES statement in PROPERTYDEFINITIONS.

---

## DEFR-071

(warning) Skipping invalid scan chain %s; %s.

**Description**

The scan chain in DEF is ignored for the indicated reason.

Feedback

**What Next**

Check the specified scan chain and make sure it is defined correctly.

---

## DEFR-072

(error) DEF fill purpose map file error at line %u - %s

**Description**

The fill purpose map file has some syntax error.

**What Next**

Refer to the file.def.fill_purpose_map_per_layer man page.

**See Also**

- file.def.fill_purpose_map
- write_def

---

## DEFR-073

(error) No via_def '%s' found for via ladder on net '%s'.

**Description**

No via_def found for this via ladder. Via defs for via ladder should be defined in VIAS section of the DEF file.

---

## DEFR-079

(warning) Found mismatch on %s and layers for color check.

**Description**

This warning message occurs when there are some colored shapes/tracks on uni-pattern layers or some uncolored shapes/tracks on multi-pattern layers in the database. The check is controlled by the app option file.def.check_mask_constraints, value "loose". Colored shapes on uni-pattern layer will be converted to colorless shapes. Rest other shapes will be read as usual by read_def.

**What Next**

Please use file.def.check_mask_constraints app option with value "none" to read all these shapes.

Feedback

### DEFR-080

(error) Found mismatch on %s and layers for color check.

**Description**

This error message occurs when there is mismatch on shapes/tracks and layers for color check in the database, the check is controlled by the app option file.def.check_mask_constraints, value "strict".

**What Next**

Please use file.def.check_mask_constraints app option with value "none" to read all these shapes.

## DEFW

### DEFW-001

(warning) The specified units-per-micron value of %d is less than the database length precision of %d.

**Description**

The specified units-per-micron is specified with the *-units* option. The value should be greater or equal to the database technology length precision. If it is less, the length values may be truncated in the output DEF file.

**What Next**

Specify a larger units value or specify none to use the technology length precision.

### DEFW-002

(warning) The DEF syntax does not support multiple spacings for non-default rule %s layer %s.

**Description**

The DEF non-default rule syntax supports only one spacing per layer. The specified rule has multiple spacings, along with weights and effort levels, on the specified layer. In these cases, the output DEF file will only include the first spacing of the layer.

**What Next**

To export the complete set of non-default rules, use the *report_routing_rules -output* command to write out the rules to a script.

Feedback

## DEFW-003

(warning) Skipping invalid scan chain %s; %s.

**Description**

The scan chain is omitted from the DEF output file for the indicated reason.

**What Next**

Check the specified scan chain and make sure it is defined correctly.

## DEFW-004

(error) Failed to write from unsupported current design view '%s'.

**Description**

The DEF writer supports only the design view and outline view.

**What Next**

Switch the current design to the supported view.

## DEFW-005

(warning) Cell '%s' has multiple keepouts; only the %s keepout is written.

**Description**

The DEF writer can only output one hard or soft keepout. If there are multiple keepouts, the DEF writer will output the hard keepout only if it is defined, otherwise the soft keepout is output if it is defined.

**What Next**

Use write_floorplan to dump all the keepouts of a cell.

## DEFW-006

(information) Creating dummy net '%s' for unconnected port '%s'.

**Description**

There are unconnected ports in the design. For each unconnected port, write_def generates a dummy net and connects the port to it. The dummy net will appear in the PINS and regular NETS sections. The dummy net is not actually created and connected in the design; it is merely a placeholder in the DEF output file.

Feedback

### What Next

Verify whether the port should be connected in the netlist. Connect them to avoid seeing this message and dummy nets in the output DEF file.

---

## DEFW-007

(error) Library '%s' is missing technology information.

### Description

The technology information is missing in the specified library. This information is necessary in order to write the DEF file.

### What Next

Load a technology file into the library using the *read_tech_file* command.

---

## DEFW-008

(error) The specified %s '%s' cannot be written; %s.

### Description

The object specified in the *-objects* option cannot be written out to the DEF file for indicated reason. All objects must be in the current block and must be either a physical cell, net, or port; non-reserved routing_blockage; hard, soft, or partial placement_blockage; or move bound.

### What Next

Remove the invalid object from the *-objects* option, and call *write_def* again.

---

## DEFW-009

(information) DEF %s section is filtered by the -objects option list; %d %s written to this section.

### Description

The indicated objects, specified in the *-objects* option list, are written to the indicated section in the DEF file, but all others are omitted.

### What Next

If all objects must be written to the indicated section, do not use the *-objects* option.

Feedback

## DEFW-010

(warning) %d %s ignored in the -objects option list; %s.

### Description

The indicated number of objects, specified in the *-objects* option list, are not written out to the DEF file. Objects are omitted if the corresponding data type is not specified in the *-include* option value or is specified in the *-exclude* option value. Unrouted nets (i.e., non-PG nets having no connections or shapes) are always omitted.

### What Next

If the indicated objects must be written, explicitly include the object type in the *-include* option or do not specify the object type in the *-exclude* option.

## DEFW-011

(warning) Unsupported mask '%s' found for %s '%s'.

### Description

The mask information for specified object is not supported by DEF or is invalid. Following situations are checked:

- The value same_mask is not supported by DEF, if a mask value is same_mask, it will be treat as no_mask. - For mask pattern of track, DEF only support uniform and alternating mask pattern. The uniform mask pattern means all masks on this track are same. The alternating mask pattern means the masks on this track is alternative, for example 'mask_one:mask_two' or 'mask_two:mask_one', given that the layer of the track has number of masks as 2. - If a mask is not for a via or via definition, it shall less than or equal to 3. If the mask is great than 3, it will still be written out, however, this warning message will be printed. - If a mask is for cut layer of a via or via definition, it shall less than or equal to 9. If the mask is great than 9, the whole via mask will be ignored. - If a mask is for upper or lower layer of a via or via definition, it shall less than or equal to 3. If the mask is great than 3, but less or equal to 9, it will still be written out, however, this warning message will be printed. If the mask is great than 9, the whole via mask will be ignored.

### What Next

If the mask of the object is not correct, correct it and try the command again.

## DEFW-012

(information) Converted site definition '%s' to '%s'.

Feedback

**Description**

The indicated row site definition name was changed to the new name, as specified in the *-convert_sites* option. All rows associated with site definition will be associated with the newly named site definition.

**What Next**

Verify that you have specified the correct site definition names.

---

## DEFW-013

(warning) Design '%s' contains net fills which cannot be expressed using DEF 5.7 syntax.

**Description**

The design contain net fills which can only be expressed using DEF 5.8 or newer syntax.

**What Next**

Run write_def with the "-version 5.8" option.

---

## DEFW-014

(warning) 'routing_rules' is not in the include_list for option -include.

**Description**

There are non-default-routing rules are used in the NETS section, but the NONDEFAULTRULES section are not output. When reading the output DEF file, the width of net shapes might be changed.

**What Next**

Please check if the non-default-routing rules are needed in the DEF file, if so, please add 'routing_rules' to the include_list.

---

## DEFW-015

(warning) Technology information is missing for layer number '%d'.

**Description**

The technology information for the specified layer number is missing. If the database has the shapes on such layer, the shapes will be ignored.

Feedback

### What Next

Add the missing layer information by loading a technology file using the *read_tech_file* command if you want to write out the shapes.

---

## DEFW-016

(error) Via definition '%s' referenced by routing rule '%s' is undefined.

### Description

The specified via definition is not defined in the library technology or design. It is omitted from the DEF VIAS and NONDEFAULTRULES sections.

### What Next

Create the missing via definition or remove it from the routing rule.

---

## DEFW-018

(warning) Layer "%s" has wrong-way default width.

### Description

This warning occurs when layer has wrong-way default width and user does not use -version 5.8 for write_def. In this case, net wiring will be written out to NETS section if its width is equal to the default width, so the net wiring with wrong-way default width may be written out to the SPECIALNETS section unless the wrong-way default width is equal to the default width.

### What Next

Please use option -version 5.8 if you want to honor wrong-way default width.

---

## DEFW-019

(warning) Technology information is missing for purpose number '%d' of layer number '%d'.

### Description

The technology information for the specified layer purpose is missing; the shape is written into the DEF file on the base layer.

To exclude these shapes from the DEF file,

- Define the layerDataType for this layer and purpose in the technology file with nonMask=1.
- Use the *-exclude non_mask_purpose* option with the *write_def* command.

Feedback

### What Next

Add the missing layer purpose information to the technology file and load the modified technology file by using the *read_tech_file* command.

---

## DEFW-020

(error) DEF version 5.7 does not support '%d' units per micron.

### Description

The DEF version 5.7 file cannot be written out with the specified units per micron value.

### What Next

The valid values of units per micron for DEF version 5.7 are 100, 200, 1000, 2000, 10000 and 20000.

### See Also

- create_lib
- write_def
- write_lef
- write_gds

---

## DEFW-021

(error) Tech precision value is used as units per micron and DEF version 5.7 does not support '%d' unitsPerMicron.

### Description

The DEF version 5.7 file cannot be written out with current units per micron value.

### What Next

The valid values of units per micron for DEF version 5.7 are 100, 200, 1000, 2000, 10000 and 20000. Please use one of the above valid values with -units option or use a tech file with valid precision value.

### See Also

- create_lib
- write_def

Feedback

- write_lef
- write_gds

---

## DEFW-022

(error) Routing rules must be included when the application option 'file.def.non_default_width_wiring_to_net' is true.

### Description

Routing rules are necessary because write_def needs to generate nondefault rules in DEF, when the the application option 'file.def.non_default_width_wiring_to_net' is set to true. This application option causes write_def to output nondefault width wiring to the DEF NETS section. In order to do so correctly, write_def needs to generate dummy nondefault rules with the wiring width in the DEF NONDEFAULTRULES section. These rules are referenced by the DEF NETS section to define the nondefault width wiring. If the rules are missing, the DEF is incomplete and read_def cannot process the DEF NETS section.

### What Next

Either include routing rules when calling write_def or set the application option'file.def.non_default_width_wiring_to_net' to false.

---

## DEFW-023

(warning) Cannot output shield net for net '%s', because it is shielded by unnetted shapes.

### Description

The specified net is shielded by shapes that are not owned by any net. In order to express the net's shield net relationship with the DEF "+ SHIELDNET shieldNetName" statement in the output, the shield shapes must be owned by a net (e.g., a ground net).

### What Next

Find all unnetted shield shapes (i.e., shapes with shape_use "shield_route") and set their net owner attibute.

### See Also

- write_def

---

## DEFW-024

(warning) 'cut_metal' is not in the include_list for option -include.

Feedback

**Description**

There are cut metals in the design and 'fills' or 'specialnets' is listed in the -include, but cut metal and connect below cut metal will be not included in DEF file as you didn't specify 'cut_metal' in -include option.

**What Next**

Please check if cut metal shapes or connect below cut metal shapes are needed in DEF file, if yes, please add 'cut_metal' to the include_list.

---

## DEFW-025

(warning) The user specified view '%s' for flatten lib-cell '%s' is not available.

**Description**

The user specified view for lib-cells is not available for this lib-cell design. The flatten shapes for the current lib-cell design will not be written to the DEF file.

**What Next**

The user may need to change the view to be written to DEF file for the lib-cell design using option "-lib_cell_view".

---

## DEFW-026

(warning) Found uncolored multi-patterning shapes in the database which won't be output.

**Description**

This warning message occurs when there are some uncolored multi-patterning shapes in the database the check is controlled by the app option file.def.ignore_uncolored_shapes or the syntax forbidNoMaskConstraint in technology file. write_def will continue to write out all other shapes except the uncolored ones.

**What Next**

Please use command check_shapes -uncolored to find out all these shapes and color them, or turn off the app option mentioned above to output uncolored shapes. Note, if the syntax forbidNoMaskConstraint is 1, the value of the app option will be ignored.

---

## DEFW-027

(warning) Found viaLadder path '%s' that does not connect with any vias. This path will be written as normal path in DEF file.

Feedback

### Description

This warning message occurs when there is viaLadder path that does not connect with any vias. write_def will write out this path as as normal path in DEF file.

### What Next

Please check the missing via in this viaLadder.

---

## DEFW-028

(warning) Failed to find path(s) that overlap with leading wire of '%s' in the via ladder. This leading wire from (%d, %d) to (%d, %d) is ignored in DEF file.

### Description

Leading wire is generated on the fly during DEF writer, which should overlap with path(s) in via ladder.

This message occurs when there is no path in the via ladder that overlaps with the leading wire of this DEF NR_STACK. The leading wire of this NR_STACK is ignored in the DEF file.

### What Next

Please check why there is no path in the via ladder that overlaps with the leading wire.

---

## DEFW-029

(error) Invalid Via PCell Design named %s given in -cell_as_via option.

### Description

To be printed as via, Via PCell design must exist already and be of design_type "feedthrough". It must contain shapes in three different adjacent layers, which like in via, must be via_cut layer between two interconnect metal layers.

### What Next

Please enter valid existing Via PCell design name in -cell_as_via option.

### See Also

- get_attribute
- write_def

Feedback

## DEFW-030

(warning) Net shape '%s' shields multiple nets. Writing the first shielded net only.

### Description

The given shape shields more than one nets. The DEF SPECIALNETS wiring syntax "+ SHIELD shieldNetName" supports shielding one net only. The DEF Writer writes out this syntax, only for the first net shielded by the net shape.

### What Next

Use 'add_shield_association' and 'remove_shield_association' commands to correct the shield association between the specified net shape and shielded net.

### See Also

- remove_shield_association
- add_shield_association

## DEFW-031

(error) write_def failed because: '%s'.

### Description

This error message occurs when there is some unrecoverable error in the current design which leads to write_def failure.

### What Next

Please check the design and rectify it.

## DEFW-078

(warning) Found uncolored shapes on multi-patterning layers in the database which won't be output.

### Description

This warning message occurs when there are some uncolored shapes on multi-patterning layers in database. The check is controlled by the app option file.def.check_mask_constraints, value "loose". write_def will continue to write out all other shapes except the uncolored ones.

Feedback

**What Next**

Please use command check_shapes to find out all these shapes and color them, or turn off the app option mentioned above to output uncolored shapes.

## DEFW-079

(warning) Found mismatch on %s and layers for color check.

**Description**

This warning message occurs when there are some colored shapes/tracks on uni-pattern layers or some uncolored shapes/tracks on multi-pattern layers in the database. The check is controlled by the app option file.def.check_mask_constraints, value "loose" or "strict". When app option value is "loose", colorless shape on multi-pattern layer will be dropped. When the app option value is "strict", the colored shapes in uni-pattern layers are written out as colorless shapes. Rest other shapes are written out as usual by write_def.

**What Next**

Please use command check_shapes to find out all these shapes.

## DEFW-080

(error) Found mismatch on %s and layers for color check.

**Description**

This error message occurs when there is mismatch on shapes/tracks and layers for color check in the database, the check is controlled by the app option file.def.check_mask_constraints, value "strict".

**What Next**

Please use command check_shapes to find out all these shapes and fix them.

# DES

## DES-0001

(Warning) Register clock pin pin has no clock signal propagating to it.

**Description**

The clock pin does not have a clock signal propagating to it. The design is not fully constrained. A register clock pin is any pin that is a reference to a timing check or has a

Feedback

sequential timing arc from it. Special cases of pins that are not clocked are identified in rule DES_0003.

The design is not fully constrained. Setup or hold checks cannot be properly performed at the register, and the paths to and from the register are not constrained. As a result, timing checks for the paths related to the register cannot be performed.

**ONE POSSIBLE SCENARIO**

Consider a scenario where a scan register is clocked by either the functional clock CLK or the scan clock TCLK through a MUX. A case analysis value is specified to select the MUX input of the functional clock CLK; however, the clock specification for CLK is missing from the constraints. As a result, no clock signal reaches the clock pin of the scan register ff1/CP. Since ff1 is not clocked, a DES_0001 rule violation will be issued by the tool.

create_clock -period 30 [get_ports TCLK] set_case_analysis 0 [get_ports TE]

**What Next**

The violation message identifies the clock pin that does not have a clock reaching it.

The “Debugging Help” section provides context specific suggestions for investigating the violation to determine why the clock pin does not have a clock signal reaching it.

There are several possible causes for this violation. Some common possibilities are that the clock is blocked from reaching the pin due to a constraint blocking it, that a clock definition is missing, or that a generated clock is not expanded.

Use the potential_clocks attribute to see if the pin is in a clock network, but was blocked by a set_case_analysis, set_disable_timing, or some other constraint.

For the example scenario, the attribute can be queried as follows:

rtl_shell> get_attribute [get_pins ff1/CP] \ potential_clocks {"TCLK"}

If the potential_clocks attribute returns a clock, then use the analyze_clock_networks command with the -traverse_disabled option to find which constraint blocked the clock from reaching the pin. For the example scenario, the command can be issued as follows:

If a generated clock is not expanded, review the violations in the CLK_0003, CLK_0009, CLK_0010, or CLK_0011 rules for the reasons the generated clock is not expanded.

**FIX SUGGESTION**

For most cases, constraints might be missing for the design or contain an error if,

a clock signal is blocked from reaching the register clock pin, consider adjusting the constraints so the clock can propagate to the clock pin

the register clock pin is not in any clock’s potential clock network, evaluate whether a clock definition is missing from the constraints

Feedback

a generated clock is not being expanded, consider adjusting the constraints so that the clock can be expanded and propagate to the clock pin

For the example scenario, a clock definition for the clock, CLK, needs to be added to the constraints for the functional mode of the design:

create_clock -period 10 [get_ports CLK] create_clock -period 30 [get_ports TCLK] set_case_analysis 0 [get_ports TE]

---

## DES-001

(error) Current block is not defined.

**Description**

The current block is not defined. Many commands require that the current block is set.

**What Next**

Use *open_block* and *link_block* to open and link a block.

---

## DES-002

(error) Cannot find %s '%s' in design '%s'

**Description**

The specified object cannot be found in the given design. This is sometimes seen while reading SDF and parasitics files. In those cases, it could indicate a file which is out of sync with the design.

**What Next**

If reading SDF or parasitics, verify that the file matches the design.

---

## DES-003

(error) '%s' cannot be used on %s %s '%s'.

**Description**

Certain commands are valid only for input or output objects.

**What Next**

Enter the command with a valid list of objects.

Feedback

## DES-004

(error) Cannot find block '%s'.

**Description**

There is no block with that name is in memory.

**What Next**

Read in the block or reenter the command with a different name.

## DES-005

(error) Cannot set current instance to leaf cell '%s'.

**Description**

The current instance must be a hierarchical cell.

## DES-006

(error) Cannot find pin '%s' on cell '%s'.

**Description**

The pin does not exist on the specified cell.

**What Next**

Use *query_objects [select_pin -of_object [select_cell cell_name]] to list pin names on the cell.*

## DES-007

(warning) '%s' is not a valid object type.

**Description**

When -from or -to option is used with set_disable_timing or remove_disable_timing , the object list can only be a cell or a lib cell. This warning is generated for objects of type port and pin.

**What Next**

Don't use -from or -to option to disable pins or ports.

Feedback

## DES-008

(error) Invalid block name '%s'%s.

**Description**

Block name cannot contain ":", ".", "/", or empty spaces. The name of a block eventually becomes the directory name. So it cannot contain the "/" character or empty spaces. The ":" character is used to delimit library and block names, and the "." character is used to delimit block and view names. So they cannot be in a block name either.

**What Next**

Remove invalid characters from block name.

## DES-009

(error) Cannot find pin '%s' on library cell '%s'.

**Description**

The pin does not exist on the specified library cell.

**What Next**

Enter the command again with a valid object name.

## DES-010

(error) Cannot find %s '%s'.

**Description**

The specified object cannot be found.

**What Next**

Enter the command again with a valid object name.

## DES-011

(error) %s failed. block '%s' is in read mode.

**Description**

The specified operation failed because the block is in read mode.

Feedback

**What Next**

re-open the block in edit mode and try again.

---

## DES-012

(error) Cannot use '%s' command on %s '%s'.

**Description**

The command works only on ports of a specific direction.

---

## DES-013

(information) Creating block '%s' in library '%s'.

**Description**

This informational message shows the block.view being created in the library

**What Next**

Showing the block and library name to help understand when a block is being created.

---

## DES-014

(Error) Cannot copy block to self.

**Description**

The source and destination block are the same. Copying a block to itself is not allowed.

**What Next**

Change the destination block name and try again.

---

## DES-015

(Error) Cannot move block to self.

**Description**

The source and destination block are the same. Moving a block to itself is not allowed.

**What Next**

Change the destination block name and try again.

Feedback

## DES-016

(error) Via_def '%s' already exists in the given block or tech.

**Description**

A via_def with the specified name already exists. Two via_defs cannot have the same name in the same block or tech.

**What Next**

Specify a different name for the via_def or try the command again with the -force option to override and replace the existing via_def.

## DES-017

(Error) %s failed. Block '%s' is not open.

**Description**

The specified operation failed because the block has not been opened.

**What Next**

Open the block and try again.

## DES-018

(Error) %s failed. Block '%s' in library '%s' is modified.

**Description**

The specified operation failed because the block contains modified and un-saved data.

**What Next**

Try again with the -force option.

## DES-019

(information) Re-initializing block '%s' in library '%s'.

**Description**

This informational message shows the block.view being re-initialized in the library

Feedback

### What Next

Showing the block and library name to help understand when a block is being re-initialized.

---

## DES-020

(warning) Block '%s' is not being saved. Library '%s' is in read mode.

### Description

The specified block view is not saved because its library is open in read mode.

### What Next

Open the specific library and block in edit mode and try again.

```
prompt> open_block <block_name> -edit
```

### See Also

- open_block
- open_lib

---

## DES-021

(Information) Incrementing open_count of block '%s' to %d.

### Description

The block is already in memory. Each open_block command on a block will increase a block's open_count by one.

---

## DES-022

(Information) Decrementing open_count of block '%s' to %d.

### Description

The block is still in memory. Each close_block command will decrease a block's open_count by one. The block is removed from memory when its open_count drops to zero.

---

## DES-023

(Error) 'reopen_block %s' failed. Block '%s' in library '%s' hasn't been saved to disk.

Feedback

### Description

Using the "-read" or "-edit" option of reopen_block command discard the run-time changes and revert the block to its previously saved state. When the block hasn't been saved, there is no saved state to revert to. If what you really want is to re-initialize the block again, use the "-new" option.

### What Next

Try reopening the block in new mode if this is what you really want:

```
prompt> reopen_block <block_name> -new
```

### See Also

- save_block
- open_block

---

## DES-024

(error) Cannot create block. Block '%s' already exists in library '%s'.

### Description

create_block failed because the specified block already exists.

### What Next

Use a different block name, or use the -force option to override existing block.

---

## DES-025

(information) Overwriting block '%s' in library '%s'.

### Description

The block is being overwritten. It will be unbound from all currently opened blocks that reference it, and be replaced with a new block.

### What Next

Showing the block and library name to help understand when a block is being overwritten.

---

## DES-026

(error) Block '%s' does not have a top module.

Feedback

**Description**

This error message occurs when the command specifies a block that does not have a top module.

**What Next**

Re-create the block from verilog.

---

## DES-027

(information) Renaming '%s' to '%s' and saving.

**Description**

The block is being renamed to the specified name in memory nd saved to disk.

**What Next**

Showing the block and library name to help understand when a block is being renamed and saved.

---

## DES-028

(information) Saving '%s' to '%s'.

**Description**

The block is being saved to the named destination on disk.

**What Next**

Showing the block and library name to help understand when a block is being saved.

---

## DES-029

(error) %s failed. Block '%s' is in a read mode library.

**Description**

The specified operation failed because the block is in a read mode library.

**What Next**

re-open the library in edit mode and try again, or set the block.no_open_edit_in_read_only_libs app option to false. Then the library open_mode will be auto promoted when a block is opened in edit mode.

```
prompt> set_app_options -global { block.no_open_edit_in_read_only_libs
 false }
```

Feedback

### See Also

- get_app_option_value
- open_block
- reopen_block

---

## DES-030

(Information) Promoting open mode of the %s design library from read to edit.

### Description

The design library is already open in read mode. By default, when you open a block in the library in edit or new mode, the tool promotes the library's open mode from read to edit.

### What Next

To disable library open mode promotion, set the *design.edit_read_only_libs* application option to *false* before running the *open_block* command.

```
prompt> set_app_options -name design.edit_read_only_libs -value false
```

### See Also

- get_app_option_value
- open_block
- reopen_block
- set_app_options
- design.edit_read_only_libs

---

## DES-031

(warning) Block '%s' is not being saved because it is in read mode.

### Description

The specified block view is not saved because it is open in read mode.

### What Next

Open the block in edit mode and try again.

```
prompt> open_block <block_name> -edit
```

Feedback

**See Also**

- open_block
- open_lib

---

## DES-032

(Error) The '%s' command failed because '%s' is a library cell.

**Description**

The specified command failed because the source block is a library cell. Library cell blocks cannot be individually saved, copied, moved, or removed in icc2_shell. They can only be modified in the library manager shell, lm_shell.

**What Next**

Use the library manager shell to make changes to this library.

---

## DES-033

(Warning) Opening design '%s' in read mode because it is a library cell.

**Description**

Library cells cannot be modified in icc2_shell. They can be modified only in the library manager shell, lm_shell.

**What Next**

Use the library manager shell to make changes to this library cell.

---

## DES-034

(error) The reopen_block command failed because '%s' is a timing view library cell.

**Description**

The *reopen_block* command failed because the source block is a timing view library cell. Timing view library cells cannot be opened in icc2_shell. They can be modified only in the library manager shell, lm_shell.

**What Next**

Use the library manager shell to make changes to this library.

Feedback

## DES-035

(error) 'save_block -hierarchical -as' failed. You must specify a design label different from the source.

**Description**

save_block -hierarchical -as failed because the source and destination label are the same. When saving a hierarchy as a different name, the user must specify a design label that is different from the source design. That label will be used to save the lower-level reference blocks in the hierarchy.

## DES-036

(error) 'save_block -hierarchical -as' failed. Reference library '%s' is read-only.

**Description**

save_block -hierarchical -as requires all non-lib-cell reference libraries to be editable, so that a labeled version of the reference block can be created.

**What Next**

Open the library and its reference libraries in edit mode with

open_lib <lib_name> -edit -ref_libs_for_edit

## DES-037

(error) 'save_block -hierarchical -as' failed. Destination view is different from source.

**Description**

save_block -hierarchical -as can only be used to save a block and its sub-blocks to the same view. Saving to a different view is not allowed.

**What Next**

Remove the view specification from the -as argument and try again:

save_block -hierarchical -as <designName>/<labelName>

## DES-038

(error) 'save_block -as' failed. '%s' is a sub-block of '%s'.

Feedback

**Description**

save_block -as failed because you are trying to save a block into one of its sub-block. This will cause cyclic/self referencing and is not allowed.

---

## DES-039

(error) Both label '%s' and '%s' of sub-block '%s' are referenced.

**Description**

It is an error to save or rename a block hierarchically into a different label if the block references more than one label of of the same sub-block.

---

## DES-040

(error) 'rename_block -hierarchical' failed. You must specify a design label different from the source.

**Description**

rename_block -hierarchical failed because the source and destination label are the same. When renaming a hierarchy to a different name, the user must specify a design label that is different from the source design. That label will be used to rename the lower-level reference blocks in the hierarchy.

---

## DES-041

(error) 'rename_block -hierarchical' failed. Reference library '%s' is read-only.

**Description**

rename_block -hierarchical requires all non-lib-cell reference libraries to be editable, so that a labeled version of the reference block can be created.

**What Next**

Open the library and its reference libraries in edit mode with

open_lib <lib_name> -edit -ref_libs_for_edit

---

## DES-042

(error) 'rename_block -hierarchical' failed. Destination view is different from source.

Feedback

**Description**

rename_block -hierarchical can only be used to rename a block and its sub-blocks to the same view. Renaming to a different view is not allowed.

**What Next**

Remove the view specification from the -as argument and try again:

rename_block -hierarchical -to <designName>/<labelName>

---

## DES-043

(error) Layer '%s' is not a via cut layer.

**Description**

The specified layer is not a via cut layer. The layer must be of type VIA_CUT or LOCAL_VIA_CUT.

**What Next**

Check the technology layer information.

---

## DES-044

(warning) The x and y values of '%s' should not be less than the corresponding minimum pitch values of %d and %d specified by the given via_def.

**Description**

The specified via_def has a minimum cut pitch that is larger than the cut pitch specified for this array via. The minimum x-pitch is the sum of the via_def's *cut_width* and *min_cut_x_spacing* (or *min_cut_spacing* if *min_cut_x_spacing* is not set) attributes. The minimum y-pitch is the sum of the via_def's *cut_height* and *min_cut_y_spacing* (or *min_cut_spacing* if *min_cut_y_spacing* is not set) attributes.

**What Next**

Specify pitch values greater than or equal to the via_def's minimum pitch values as indicated in the warning message.

---

## DES-045

(error) The row and column values of '%s' cannot be less than the corresponding minimum numbers of rows and columns {%d %d} per via defined on the cut layer specified by the given via_def.

Feedback

**Description**

A via_def has been specified on which the cut layer has a minimum number of rows or columns of cuts per via that is larger than the value specified for the cuts of this array via.

**What Next**

Check the number of rows and columns of cuts of this array via and provide values greater than or equal to the corresponding values on the cut layer specified in the via_def.

---

## DES-046

(error) Min cut spacing cannot be specified for a single-cut via_def.

**Description**

A via_def for an array via is created by specifying min_rows, min_columns or both. Otherwise a via_def for a single-cut via is created, in which case min cut spacing cannot be specified.

**What Next**

To create a via_def for an array via, check the number of rows and columns of cuts. At least one of them must be greater than 1. To create a single-cut via_def, do not set the min cut spacing.

---

## DES-047

(error) create_via_def requires one of -cut_layer, -shapes, or -tsv_keepout_spacing. Specify -cut_layer to create a simple via_def, -shapes to create a custom via_def, or -tsv_keepout_spacing to create a through-silicon via_def.

**Description**

A via_def for a simple via is created by specifying -cut_layer. A via_def for a custom via is created by specifying a collection of shapes with -shapes. A via_def for a through-silicon via is created by specifying -tsv_keepout_spacing. Exactly one of these three options must be provided.

**What Next**

To create a via_def for an simple via, specify a cut layer with -cut_layer. To create a via_def for a custom via, specify a collection of shapes with -shapes. To create a via_def for a through-silicon via, specify a keepout distance with -tsv_keepout_spacing.

Feedback

## DES-048

(error) Layer '%s' is not in the specified tech or tech referenced by the specified design or library.

**Description**

A via_def cannot reference a cut layer in an unrelated tech database. For a tech via_def, the cut layer of the via_def must be in the same tech database. For a design via_def, the cut layer must be in the tech referenced by the library containing the design.

**What Next**

When referencing a layer by name, check that the layer exists in the correct tech database. When passing a layer object, set the current_lib to the correct library before getting the layer with 'get_layers' to ensure that you have a layer object from the correct tech database.

## DES-049

(Error) Cannot rename block to self.

**Description**

The source and destination block are the same. Renaming a block to itself is not allowed.

**What Next**

Change the destination block name and try again.

## DES-050

(error) Rail name '%s' does not exist.

**Description**

You tried to use a non existent rail name, for example, in trying to set the rail_name attribute on a PG pin.

**What Next**

Specify a different rail name.

## DES-051

(error) Port '%s' does not exist.

Feedback

**Description**

You tried to use a non existent port name.

**What Next**

Specify a different port name.

---

## DES-052

(error) Port '%s' is not a %s port.

**Description**

You tried to use a port of an incorrect type. For example, an attribute might require a power port, but you used a signal port.

**What Next**

Specify a different port name.

---

## DES-053

(error) Cannot attach a %s port to a %s rail

**Description**

You tried to use a port and rail combination that is incompatible. For example, if you tried to set the *rail_name* attribute for a power pin to refer to a ground rail, this message would be issued.

**What Next**

Check the port and rail selections.

---

## DES-054

(error) Can't change '%s'- %s

**Description**

You tried to change an attribute which cannot be changed for the reason given. For example, you might have tried to change a port_type attribute on a port from signal to power, but that port is used in the logic function of the cell so it cannot be changed.

---

## DES-055

(information) Renaming '%s' to '%s'.

Feedback

**Description**

The block is being renamed in memory to the named destination.

**What Next**

Showing the block and library name to help understand when a block is being renamed.

---

## DES-056

(error) Block '%s' cannot be renamed. It is referenced by '%s'.

**Description**

The block cannot be renamed because it or one of its views is referenced by another block in memory. You can rename the entire design hierarchy from the top block but renaming individual sub-blocks at the lower level is not allowed.

**What Next**

Use copy_block to make a copy of the block.

---

## DES-057

(error) 'rename_block' failed. '%s' is a sub-block of '%s'.

**Description**

rename_block failed because you are trying to rename a block into one of its sub-block. This will cause cyclic/self referencing and is not allowed.

---

## DES-058

(error) Layer '%s' does not have a metal layer %s it in the layer stack. It cannot be the cut layer of a simple via_def.

**Description**

create_via_def failed because the specified cut layer does not have a layer of type INTERCONNECT or LOCAL_INTERCONNNECT both above and below it in the layer stack. A simple via_def must have metal above and below its cut layer.

**What Next**

Choose another cut layer or modify your tech to add the missing metal layer definition.

Feedback

## DES-059

(error) 'save_block -label' failed. You must specify a design label without the characters `:', `/', or `.'.

**Description**

save_block -label failed because the label specified is not legal. Because of the syntax used for specifying a block (lib:block/userLabel.view), the lib, block, view, and userLabel names must not include the characters `:', `/', or `.'.

**What Next**

Choose a user label name that meets the criteria above and repeat the save_block command.

## DES-060

(error) Current block is a %s and is not supported by this command.

**Description**

The current block is a type that is not supported by the current command.

**What Next**

Use the *current_block* command to switch to a block that is supported by this command.

## DES-061

(error) Topology %s '%s' already exists in the plan '%s'.

**Description**

A topology object with the specified name already exists in the plan.

**What Next**

Specify a different name, or remove the existing topology object. Alternatively, you may omit the name option, and the system will automatically select a unique name.

## DES-062

(error) Topology repeater '%s' already exists on the edge '%s'.

**Description**

A topology repeater with the specified name already exists on the edge.

Feedback

**What Next**

Specify a different topology repeater name, or use the remove_topology_repeaters command to remove the existing topology repeater. Alternatively, you may omit the name option, and the system will automatically select a unique name for the topology repeater.

---

## DES-063

(error) Topology plan '%s' already exists in the block.

**Description**

A topology plan with the specified name already exists in the block.

**What Next**

Specify a different topology plan name, or use the remove_topology_plans command to remove the existing topology plan. Alternatively, you may omit the name option, and the system will automatically select a unique name for the topology plan.

---

## DES-064

(error) Topology node '%s' already exists in the plan '%s'.

**Description**

A topology node with the specified name already exists in the plan.

**What Next**

Specify a different topology node name, or use the remove_topology_nodes command to remove the existing topology node. Alternatively, you may omit the name option, and the system will automatically select a unique name for the topology node.

---

## DES-065

(error) Cell array pattern '%s' already exists.

**Description**

A cell array pattern with the specified name already exists in the specified block, or the current block if the block is unspecified.

**What Next**

Specify a different cell array pattern name, or use the remove_cell_array_patterns command to remove the existing cell array pattern. Alternatively, you may omit the name option, and the system will automatically select a unique name for the cell array pattern.

Feedback

## DES-078

(Error) Variable '%s' cannot be set after define_scaling_lib_group.

**Description**

This variable needs to be set before define_scaling_lib_group.

## DES-089

(error) Terminal '%s' already exists in block '%s'.

**Description**

A terminal with the specified name already exists in the block. In a block, each terminal must have a unique name.

**What Next**

Specify a different name that is not already used by a terminal in this block, or do not specify a name and a unique name will be automatically generated. This restriction can be controlled by the design.enforce_terminal_name_uniqueness app option.

## DES-090

(error) Block '%s' is not linked.

**Description**

This command requires that the current block is linked.

**What Next**

Use the *link_block* command to link the current block before using this command.

## DES-091

(error) Block '%s' is not opened.

**Description**

The specified block must be opened before using this command.

**What Next**

Use the *open_block* command to open the block before using this command.

Feedback

## DES-092

(error) '%s' is not the top module in current block '%s'.

**Description**

When using absolute path syntax in current_instance, the module name must be the top module name of the current block

**What Next**

Please use the top module name in the command. For example, *current_instance /top/inst1*

If the instance belongs to a different block, please use *current_block* command to change the current block before running *current_instance*

## DES-100

(error) Bound '%s' already exists in the design.

**Description**

A bound with the specified name already exists in the design.

**What Next**

Specify a different bound name, or use the remove_bounds command to remove the existing bound.

## DES-101

(error) %s '%s' cannot be assigned to the %s bound; it is already assigned to %s bound '%s'.

**Description**

The specified object is already assigned to another bound.

Ports may be assigned to at most one bound. A port cannot be reassigned to another bound without first unassigning it from its current bound.

Cells may be assigned to at most one move bound and one group bound. A cell cannot be reassigned to another move bound without first unassigning it from its current move bound. Likewise a cell cannot be reassigned to another group bound without first unassigning it from its current group bound. Cells may be

Feedback

**What Next**

Use the remove_from_bound command to unassign the object from its current bound, and then call this command again.

---

## DES-102

(error) The bound region %s is not rectilinear.

**Description**

The specified bound region is not rectilinear. Each region must be a rectilinear polygon or rectangle.

**What Next**

Correct the bound region.

---

## DES-103

(warning) %s '%s' cannot be unassigned from the bound; %s.

**Description**

The specified object cannot be unassigned from the bound. Objects may only be unassigned from bounds to which it is assigned.

Note that while a cell may be in a bound, it might not have been assigned to one.This occurs if the cell's parent hierarchical cell (or any of its ancestors in the same block) are assigned to a bound. This command only removes cells that were assigned explicitly to a bound.

**What Next**

Use the remove_from_bound command to unassign the object from its currently assigned bound if any. For a cell, unassign the cell's parent hierarchical cell from its bound if any.

---

## DES-104

(error) Ports cannot be assigned to this type of bound.

**Description**

This error comes from either create_bound or add_to_bound command. It implies that the existing or new bound being created, is limited to having only specific type of cells in it.These bounds are not allowed to have ports.

Feedback

### What Next

If the error occurs in create_bound command, you can either change the options used in creating the bound, such that it can have ports or remove the ports from the object list to the new bound. If the error occurs in add_to_bound command, you would need to choose a different bound for the port.

---

## DES-105

(error) The cell '%s' cannot be assigned to this type of bound.

### Description

This error comes from either create_bound or add_to_bound command. It implies that the existing or new bound being created is limited to having only specific types of cells in it. The specified cells cannot be added to the bound. For example, module bounds can only contain module boundary cells, which are not ancestors or descendants of any existing cell in the bound.

### What Next

If the error occurs in create_bound command, you can either change the options used in creating the bound, such that it can have cells of this type or remove the cell from the object list to the new bound. If the error occurs in add_to_bound command, you would need to choose a different bound for the cell.

---

## DES-106

(error) %s '%s' cannot be assigned to the bound; it is in a different block than the bound.

### Description

The specified object is in a different block than the bound. The bound and its containing cells and ports must be in the same physical block.

### What Next

Specify cells and ports that are in the same block as the bound.

---

## DES-110

(error) Non-default rule '%s' already exists in the design.

### Description

A non-default routing rule already exists with the specified name in the design.

Feedback

**What Next**

Remove the existing rule and then define the rule.

---

## DES-111

(error) Layer '%s' does not exist.

**Description**

The specified layer does not exist.

**What Next**

Check the technology layer information.

---

## DES-112

(error) Layer '%s' is not a routing layer.

**Description**

The specified layer is not a routing layer. The layer must be a routing layer.

**What Next**

Check the technology layer information.

---

## DES-113

(error) The number of %s values does not match the number of spacing values on layer '%s'.

**Description**

The number of spacing weight and effort values must match the number of spacing values specified for the layer in the non-default rule.

**What Next**

Check the number of spacing, weight, and effort values.

---

## DES-114

(error) Layer spacing weights must be specified on layer '%s'.

Feedback

**Description**

If multiple spacing values are specified on a layer, spacing weight values must also be specified on that layer.

**What Next**

Specify the spacing weight values.

---

## DES-115

(error) Spacing value %s is specified multiple times on layer '%s'.

**Description**

The indicated spacing value is specified more than once on the layer. Each spacing value may be specified at most once per layer per routing rule.

**What Next**

Check and correct the spacing values on the indicated layer.

---

## DES-120

(error) Io guide '%s' already exists in the design.

**Description**

An io guide with the specified name already exists in the design.

**What Next**

Specify a different io guide name, or use the remove_io_guides command to remove the existing io guide.

---

## DES-121

(error) Cell '%s' is already in io guide '%s'.

**Description**

The specified cell is already in another io guide. Cells cannot be moved from one io guide to another io guide.

**What Next**

Use the remove_from_io_guide command to remove the cell from the current io guide, and then call this command again.

Feedback

## DES-122

(error) '%s' is not a pad cell.

**Description**

The specified object is not a pad cell. Only pad cells can be in an io guide.

**What Next**

Check the get_attribute command to verify the cell is a pad cell.

## DES-123

(error) Invalid line specification: %s

**Description**

The specified line for the io_guide is invalid because of the given reason.

**What Next**

Specify a different line for the io_guide

## DES-130

(error) Io ring '%s' already exists in the design.

**Description**

An io ring with the specified name already exists in the design.

**What Next**

Specify a different io ring name, or use the remove_io_rings command to remove the existing io ring.

## DES-131

(error) Io guide '%s' is already in io ring '%s'.

**Description**

The specified io guide is already in another io ring. Io Guides cannot be moved from one io ring to another io ring.

Feedback

**What Next**

Use the remove_from_io_ring command to remove the io guide from the current io ring, and then call this command again.

---

## DES-132

(error) Invalid offset specification for io_ring

**Description**

The specified offset for the io_ring is invalid because it will create an io_ring with zero width and length.

**What Next**

Specify a different io_ring offset

---

## DES-133

(error) Invalid bbox specification for io_ring

**Description**

The specified bbox for the io_ring is invalid because it is outside the die area.

**What Next**

Specify a bbox with the die area.

---

## DES-134

(error) Cannot find pad cell '%s' in io guide '%s'.

**Description**

The specified io guide does not contain this pad cell.

**What Next**

Verify the pad cell is in the io guide and try again.

---

## DES-135

(error) Cannot find io ring '%s' in design '%s'.

**Description**

The specified io ring cannot be found in the design.

Feedback

**What Next**

Verify the io ring is valid and try again.

---

## DES-136

(error) The specified region is not rectilinear. The region must be a rectilinear polygon or rectangle.

**Description**

The specified region is not rectilinear.

**What Next**

Specify a rectilinear region and try this command again.

---

## DES-137

(error) Invalid region. Region is outside the die area.

**Description**

The specified region is outside the die area.

**What Next**

Specify a region inside the die area and try this command again.

---

## DES-138

(error) An x and y guard_band must be specified for each voltage_area_shape.

**Description**

Guard_band specifications are missing for the voltage_area_shape.

**What Next**

Specify a valid x and y guard_band and try this command again.

---

## DES-140

(error) Term '%s' is already connected to net '%s'.

**Description**

The term is already connected to the indicated net, and cannot be connected to the user-specified net unless it is disconnected first.

Feedback

**What Next**

Disconnect the term from the indicated net, and then connect it to the desired net.

---

## DES-141

(error) Cannot connect term '%s' across hierarchy to net '%s'.

**Description**

The indicated term and net are in different hierarchies and cannot be connected.

**What Next**

If the indicated term and net must be connected, create the appropriate nets and ports in the intermediate hierarchies, and then connect the term to the corresponding net in its local hierarchy.

---

## DES-142

(error) Term '%s' cannot be connected to net '%s'.

**Description**

The indicated term cannot be connected to the net.

**What Next**

Check the term and net connections.

---

## DES-143

(error) Cannot connect %s %s pin '%s' to signal net '%s'.

**Description**

The indicated PG pin has either an input or inout direction, and cannot be driven by a signal net.

**What Next**

Check the net and pin types, and check the pin direction.

---

## DES-144

(error) Cannot connect signal output pin '%s' to %s net '%s'.

Feedback

**Description**

The indicated signal pin has an output direction, and cannot drive a PG net.

**What Next**

Check the net and pin types, and check the pin direction.

---

## DES-145

(error) Term '%s' is currently connected to net '%s'.

**Description**

The term is currently connected to the indicated net, and cannot be disconnected from the user-specified net.

**What Next**

Disconnect the term from the indicated net.

---

## DES-146

(error) Cannot disconnect term '%s' across hierarchy from net '%s'.

**Description**

The indicated term and net are in different hierarchies and cannot be disconnected.

**What Next**

If the indicated term and net must be disconnected, disconnect the pin from the corresponding net in its local hierarchy.

---

## DES-147

(error) Term '%s' cannot be disconnected from net '%s'.

**Description**

The indicated term cannot be disconnected from the net.

**What Next**

Check the term and net connections.

---

## DES-149

(information) Block '%s' cannot be auto-linked.

Feedback

**Description**

A summary message indicating that the auto link process is not allowed for your block.

**What Next**

Explicitly link the block using the set_top_module command

---

## DES-150

(error) Net '%s' already exists%s.

**Description**

A net with the specified name already exists. Two nets cannot have the same name in the same cell.

**What Next**

Specify a different name for the net.

---

## DES-151

(error) Unable to create net '%s'; %s.

**Description**

The net cannot be created for the specified reason.

**What Next**

If the message indicates an invalid instance path, check the name specified in the option *net_names* and make sure the name includes the correct path to the new net.

If the message indicates that the cell instance is a leaf cell, check the name specified in the option *net_names* and make sure the name includes a path to the parent hierarchical cell of the new net. Nets cannot be added to leaf cells.

---

## DES-152

(error) Port '%s' already exists%s.

**Description**

A port with the specified name already exists. Two ports cannot have the same name on the same cell.

Feedback

**What Next**

Specify a different name for the port.

---

## DES-153

(error) Unable to create port '%s'; %s.

**Description**

The port cannot be created for the specified reason.

**What Next**

If the message indicates an invalid instance path, check the name specified in the option *port_names* and make sure the name includes the correct path to the new port.

If the message indicates that the cell instance is a leaf cell, check the name specified in the option *port_names* and make sure the name includes a path to the parent hierarchical cell of the new port. Ports cannot be added to leaf cells.

---

## DES-154

(error) Pin '%s' already exists%s.

**Description**

A pin with the specified name already exists. Two pins cannot have the same name on a cell.

**What Next**

Specify a different name for the pin.

---

## DES-155

(error) Unable to create pin '%s'; %s.

**Description**

The pin cannot be created for the specified reason.

**What Next**

If the message indicates an invalid instance path, check the name specified in the option *pin_names* and make sure the name includes the correct path to the new pin.

Feedback

If the message indicates that the cell instance is a leaf cell, check the name specified in the option *pin_names* and make sure the name includes a path to the parent hierarchical cell of the new pin. Pins cannot be added to leaf cells.

---

## DES-160

(warning) Cannot set keepout on non-physical %s '%s'.

**Description**

Keepout margins may only be set on physical designs and cells. The indicated design or cell is not physical and will be skipped.

**What Next**

Check the input object list and verify that you have specified the correct physical designs and cells.

---

## DES-161

(warning) Cannot set inner keepout on leaf cell '%s'.

**Description**

Inner keepout margins may only be set on physical designs. Inner keepout margins may not be set on leaf cells. The indicated cell is a leaf cell and will be skipped.

This has no bearing on the outer keepout. Leaf cells may have an outer keepout. If an outer keepout is specified, the cell's outer keepout will still be set.

**What Next**

Check the input object list and verify that you have specified designs rather than leaf cell instances.

---

## DES-162

(warning) Cannot derive pin-based keepout on %s '%s'.

**Description**

The pin-based outer keepout margins may only be derived for macro cells. They cannot be set on designs, partitions, or other leaf cells. The indicated cell is not a macro cell.

**What Next**

Check the input object list and verify that you have specified the correct macro cells.

Feedback

## DES-163

(error) Minimum routing layer and layer preferred routing directions are not set on the design.

**Description**

The pin-based keepout cannot be derived, because the minimum routing layer and layer preferred routing directions are not set on the design. One of these must be set in order to obtain a valid pitch value, which is used to derive the pin-based keepout margins.

**What Next**

Either set the minimum routing layer using the *set_ignored_layers* command or set the layer preferred routing directions using the *set_attribute* command on each routing layer.

## DES-164

(error) Layers are not specified for route blockage keepout.

**Description**

Layers must be specified when setting a route blockage keepout.

**What Next**

Use the option -layer to specified layers.

## DES-165

(warning) Only hard or soft type keepout margin can be set on leaf cell '%s'.

**Description**

Only hard or soft type keepout margin can be set on leaf cells.The indicated a invalid type keepout margin has been specified to set on a leaf cell and will be skipped.

**What Next**

Check the input keepout margin type and verify that you have specified the correct keepout margin type on leaf cells.

## DES-166

(warning) Layers can only be specified for route blockage keepout,layers will be ignored.

Feedback

**Description**

Layers will be ignored when keepout type is not route_blockage.

**What Next**

Do not use the option -layer to specified layers.

---

## DES-167

(warning) Can't set outer keepout for current design, setting outer keepout for current design '%s' is ignored.

**Description**

Current design will be skipped when set outer keepout for it.

**What Next**

Do not set outer keepout for current design.

---

## DES-170

(error) Pin blockage '%s' already exists in the design.

**Description**

A pin blockage with the specified name already exists in the design.

**What Next**

Specify a different pin blockage name, or use the remove_pin_blockages command to remove the existing pin blockage first.

---

## DES-171

(error) The pin blockage cannot be assigned %s; %s.

**Description**

The pin blockage cannot be assigned the specified object type for the indicated reason.

The pin blockage may only be assigned a homogeneous object list, either a list of pins, nets, or ports. If the pin blockage is already assigned an object of one type, it may not be assigned an object of another type.

**What Next**

Correct the problem with object list.

Feedback

## DES-172

(error) Pin guide '%s' already exists in the design.

**Description**

A pin guide with the specified name already exists in the design.

**What Next**

Specify a different pin guide name, or use the remove_pin_guides command to remove the existing pin guide first.

## DES-173

(error) Cell '%s' is not a macro or partition cell.

**Description**

The indicated cell is not a macro or partition cell. Only macro and partition cells may be set as the parents of a pin guide.

**What Next**

Specify a list of macro and partition cells for the parents option.

## DES-174

(error) The pin guide cannot be assigned %s; %s.

**Description**

The pin guide cannot be assigned the specified object type for the indicated reason.

The pin guide may only be assigned a homogeneous object list, either a list of pins, nets, or ports. If the pin blockage is already assigned an object of one type, it may not be assigned an object of another type.

Furthermore, if the pin guide's parent is the current design, it may only be assigned net and port objects. If the pin guide's parents are cells, it may only be assigned net and pin objects.

**What Next**

Correct the problem with object list.

Feedback

## DES-175

(error) The boundary is not rectilinear.

**Description**

The specified boundary is not rectilinear. The boundary must be a rectilinear polygon or rectangle.

**What Next**

Correct the boundary coordinate list.

## DES-176

(error) The object list is not homogeneous.

**Description**

The specified object list contains objects of different types. The object list may only contain objects of one type.

**What Next**

Specify objects of one type only.

## DES-177

(error) %s name cannot begin with the system-reserved prefix '%s'.

**Description**

A name cannot begin with a system-reserved prefix. This prefix is used by the system for unnamed objects of the indicated type.

**What Next**

Specify a different name that does not begin with the indicated system-reserved prefix.

## DES-178

(warning) %s '%s' is not physical.

**Description**

The indicated object is not physical. Only physical pins, nets, and ports may be added to a pin blockage or pin guide. This object is skipped and not added.

Feedback

**What Next**

Check that the correct physical objects are added.

---

## DES-179

(error) View '%s' of cell %s does not exist.

**Description**

The cell does not have indicated view.

**What Next**

Check that the correct view type is specified.

---

## DES-180

(error) Cannot change view to cell '%s'.

**Description**

The view of specified instance cannot be changed.

**What Next**

Check if the indicated view exists.

---

## DES-181

(error) Unable to change view on hierarchical cell '%s'.

**Description**

This command does not accept hierarchical cell.

**What Next**

Only specify leaf cells for this command.

---

## DES-182

(error) '%s' is not a legal view name.

**Description**

This command does not support the specified view type.

Feedback

**What Next**

Correct the view name.

---

## DES-183

(error) Edit group '%s' already exists in the design.

**Description**

An edit group with the specified name already exists in the design.

**What Next**

Specify a different edit group name, or use the remove_edit_groups command to remove the existing edit group. Alternatively, you may omit the name option, and the system will automatically select a unique name for the edit group.

---

## DES-184

(error) '%s' is not a valid design.

**Description**

The specified object is not a top module. A design must also be a top module.

**What Next**

Specify a top module.

---

## DES-185

(error) Addition of object '%s' to edit group '%s' forms a membership cycle.

**Description**

Adding the specified object to the specified edit group results in a membership cycle, such that the object ultimately both contains and is contained by one of its member objects, which is not permitted.

**What Next**

Use the remove_from_edit_group command to remove another object in the membership cycle, and then call this command again.

---

## DES-186

(error) Object '%s' is already in edit group '%s'.

Feedback

**Description**

The specified object is already in an edit group. Objects cannot be moved from one edit group to another edit group.

**What Next**

Use the remove_from_edit_group command to remove the object from the current edit group, and then call this command again.

---

## DES-187

(error) Object '%s' cannot be added to edit group '%s'.

**Description**

The specified object's type is not supported for inclusion in edit groups, so it cannot be added to the edit group.

**What Next**

Select a different object to add to the edit group.

---

## DES-188

(error) Unable to create pin '%s' in top module.

**Description**

A pin cannot be created in the top module of the design.

**What Next**

Only ports can be created in the top module of the design. Use the command *create_port* to create a new port in the top module of the design.

---

## DES-189

(error) Routing guide '%s' already exists in the design.

**Description**

A Routing guide with the specified name already exists in the design.

**What Next**

Specify a different Routing guide name, or use the remove_Routing_guide command to remove the existing Routing guide first.

Feedback

## DES-190

(error) Terminal '%s' already exists on port '%s'.

**Description**

A terminal with the specified name already exists on the port. On a port, each terminal must have a unique name.

**What Next**

Specify a different name that is not already used by a terminal on this port, or do not specify a name and a unique name will be automatically generated.

## DES-191

(error) At least one net name must be specified.

**Description**

One or more net names must be specified when calling the create_net command.

**What Next**

Specify one or more non-empty names.

## DES-192

(error) %s '%s' is in a different Physical hierarchy.

**Description**

The specified object to be added/removed is in a different physical hierarchy than the container object.

## DES-193

(error) Object '%s' cannot be removed from edit group '%s'.

**Description**

The specified object is not in the specified edit group, so it cannot be removed from it.

**What Next**

Select a different object to remove from the edit group.

Feedback

## DES-194

(Information) Opening sub-blocks in read-only mode.

### Description

Sub-blocks are being opened in read-only mode. The -ref_libs_for_edit option is overridden because the design.sub_blocks_for_read app option value is true.

### See Also

- open_block
- design.sub_blocks_for_read

## DES-195

(error) Topology edge '%s' already exists in the design.

### Description

An topology edge with the specified name already exists in the design.

### What Next

Specify a different topology edge name, or use the remove_topology_edges command to remove the existing topology edge. Alternatively, you may omit the name option, and the system will automatically select a unique name for the topology edge.

## DES-196

(error) Topology node '%s' already exists in the design.

### Description

An topology node with the specified name already exists in the design.

### What Next

Specify a different topology node name, or use the remove_topology_nodes command to remove the existing topology node. Alternatively, you may omit the name option, and the system will automatically select a unique name for the topology node.

## DES-197

(Information) The -ref_libs_for_edit option is overridden.

Feedback

**Description**

The -ref_libs_for_edit option is overridden because the library is in read mode and the design.edit_read_only_libs option is false. The block and its sub-blocks are opened in read mode.

**See Also**

- open_block
- design.edit_read_only_libs

---

## DES-198

(error) %s '%s' is in a different block than the %s '%s'.

**Description**

The specified object to be added/removed is in a different block than the container object.

---

## DES-199

(error) The pin guide's parent cells must be in the same physical hierarchy.

**Description**

The pin guide may only be assigned parent cells which reside in the same physical hierarchy as the pin guide itself.

**What Next**

Correct the problem with the parent cell list.

---

## DES-200

(error) Unable to create %s in leaf cell %s.

**Description**

The object cannot be created in a leaf cell.

**What Next**

Provide the correct cell for object creation.

---

## DES-201

(error) Can't save design. Design '%s' is not linked.

Feedback

**Description**

save_block requires this design to be linked.

**What Next**

Explicitly link the design prior to saving it

---

## DES-202

(warning) Not assigning %s '%s' to the %s bound because it is already assigned to %s bound '%s'.

**Description**

The specified object is already assigned to another bound.

Ports may be assigned to at most one bound. A port cannot be reassigned to another bound without first unassigning it from its current bound.

Cells may be assigned to at most one move bound and one group bound. A cell cannot be reassigned to another move bound without first unassigning it from its current move bound. Likewise a cell cannot be reassigned to another group bound without first unassigning it from its current group bound.

**What Next**

Use the remove_from_bound command to unassign the object from its current bound, and then call this command again.

---

## DES-203

(information) Block '%s' cannot be linked with link_block command, please use set_top_module.

**Description**

A summary message indicating that the auto link process is not allowed for your block, and set_top_module must be used intead of link_block.

**What Next**

Explicitly link the block using the set_top_module command

---

## DES-210

(error) Unable to create pin_bus '%s'; %s.

Feedback

**Description**

The pin_bus cannot be created for the specified reason.

**What Next**

If the message indicates an invalid instance path, check the name specified in the option cell_bus and make sure the name includes the correct path to the correct cell_bus.

---

## DES-211

(error) Cannot create pin bus on cell bus %s whose members are leaf cells.

**Description**

The pin bus cannot be created on cell bus whose members are leaf cells

**What Next**

Provide the correct cell bus for pin bus creation.

---

## DES-219

(error) cannot create net. Bus net with same name '%s' already exists%s.

**Description**

A bus net with the specified name already exists.

**What Next**

Specify a different name for the net.

---

## DES-220

(error) Block '%s' failed to properly link during 'set_top_module'.

**Description**

A message indicating that the command had problems processing your block. Please prior review output for pertinent error messages.

**What Next**

Explicitly link the block using the 'set_top_module' command after correcting the reported problems.

Feedback

## DES-221

(error) Command cannot use RTL block %s.

**Description**

A message indicating that the command cannot use an RTL block but the implicitly or explicitly-specified block is an RTL block.

**What Next**

Provide a design-type block for the command.

## DES-222

(error) Command cannot use analysis block %s.

**Description**

A message indicating that the command cannot use an analysis block, which is a block that has been created or changed using approximate techniques for the purposes of analysis and isn't suitable for use later on in the flow, and the implicitly or explicitly-specified block is an RTL block.

**What Next**

Provide a normal design block for the command.

## DES-223

(Warning) pre_extract command has not produced pre-extracted block from the original block %s.

**Description**

A message indicating that the command has not produced a new pre-extracted block from the original block, because pre-extraction proecess had not needed to producenew data different from the original.

**What Next**

Extraction tools can run on the original block.

## DES-224

(Error) Command aborted because not able to find technology in design %s.

Feedback

**Description**

A message indicating that the command has aborted because no technology is found in the given design (library).

**What Next**

Make sure the design (library) has access to technology, usually set-up during design set-up stages.

---

## DES-225

(error) Block '%s' is not linked.

**Description**

This command requires that the current block is linked. The block was not linked because auto-linking was disabled.

**What Next**

To enable auto-linking, set the app-option design.disable_auto_link to false. Alternatively use the *link_block* command to link the current block before using this command.

```
prompt> set_app_option -name design.disable_auto_link -value false
```

---

## DES-226

(error) Layer '%s' doesnot have a mask order in the tech layer stack. Mandatory to specify the lower and upper routing tech layers.

**Description**

A via_def must reference a cut layer in the tech layer stack. Othwise, please specify the lower and upper routing layers also.

**What Next**

Please specify the lower and upper routing tech layers.

---

## DES-227

(error) Command is not allowed to run on 3DIC design %s.

**Description**

The current command cannot run on the block, as 3DIC design types are not supported for this command.

Feedback

**What Next**

Check the design type and run the command on a non-3DIC design.

# DFT

## DFT-002

(Information) Scan chain %s is not in a partition. Only reordering is possible.

**Description**

A scan chain exists in the netlist without an associated partition. Such scan chains cannot be re-partitioned, only re-ordered.

## DFT-003

(Error) The design scandef is incorrect because %s.

**Description**

The design scandef is incorrect due to the described reason like buffer or inverter existence in floating list, or multiple occurence of the same scan pin, or MV related cell existence as scan cell.

**What Next**

Check the design scandef to solve the reported issue, and reload the correct scandef into the design to continue.

## DFT-004

(Error) Inconsistency exists between the design scandef and the netlist because %s.

**Description**

The design scandef is inconsistent with the design netlist due to the reported issue like missing scan pin in netlist versus scandef, or net connection inconsistency between scandef and netlist.

**What Next**

If the message is seen in check_scan_chain before place_opt or create_placement, please check the design to correct the reported issue, reload the revised scandef if necessary. If the message is seen only at the end of optimize_dft, please contact synopsys for the issue.

Feedback

## DFT-005

(warning) The scan chain %s is abnormal and may introduce logic error in later flow because %s.

**Description**

The reported scan chain is abnormal and may introduce logic error in later flow due to the described issue like buf/inv existence within OREDERED list driving both scan and data loads, or buf/inv as stop point driving both scan and data loads.

**What Next**

Check the design scandef to see whether the described issue in the reported scan chain is intended.

## DFT-006

(warning) The scan chain %s is abnormal because %s.

**Description**

The reported scan chain is abnormal due to the described issue. This abnormality may limit the dft optimization efficiency.

**What Next**

Check the design scandef to see whether the described issue in the reported scan chain is intended.

## DFT-007

(warning) The scan reordering on scan chain %s is skipped because %s.

**Description**

The scan reordering on scan chain %s is skipped due to the described reason.

**What Next**

Please check whether the described reason is caused by design issue. If yes, please correct the problem and redo dft optimization.

## DFT-008

DFT optimization is terminated due to the engine defect.

Feedback

**Description**

DFT optimization is terminated due to the engine defect.

**What Next**

Please contact synopsys engineer for the solution.

---

## DFT-009

(warning) DFT repartition is skipped on PARTITION label %s group due to the failed scan chain %s.

**Description**

DFT repartition is skipped on the named PARTITION label group due to the existence of failed scan chain.

**What Next**

Please check the displayed error message DFT-003 or DFT-004 to correct the scandef error, and reload the correct scandef to continue.

---

## DFT-010

(warning) DFT reordering is skipped for the failed scan chain %s.

**Description**

DFT reordering is skipped for the named failed scan chain.

**What Next**

Please check the displayed error message DFT-003 or DFT-004 to correct the scandef error, and reload the correct scandef to continue.

---

## DFT-011

(information) NO SCANCHAIN defined in SCANDEF.

**Description**

optimize_dft does nothinig or check_scan_chain cannot perform since there is no scan chain defined in the design.

**What Next**

use read_def to read in the SCANDEF with SCANCHAIN info

Feedback

## DFT-012

(information) scan repartition is disabled.

**Description**

scan repartition is disabled, which may impact the total scan wirelength reduction efficiency.

**What Next**

remove the app option which disables scan repartition.

## DFT-013

(Error) %s pin is set on the %s pin - %s.

**Description**

This error is shown when the scan-in pin is set on an output pin of scan cell or the scan-out pin is set on the input pin of scan cell.

Here the scan-in/out pin is the pin of scan element. Scan-in pin should be an input pin of cell and scan-out pin should be an output pin.

**What Next**

1. Checks whether the scandef is correct. Maybe a wrong pin is specified. 2. It is possible that scandef is correct but the pin direction is wrong. User has to correct the db.

## DFT-014

(warnning) optimize_dft is skipped in spg flow due to unplaced scan cell.

**Description**

optimize_dft is skipped in spg flow due to unplaced scan cell.

**What Next**

please ensure all scan cells being placed if optimize_dft is expected to be employed.

## DFT-015

optimize_dft is terminated due to unplaced scan cell.

**Description**

optimize_dft is terminated due to unplaced scan cell in the design.

Feedback

**What Next**

use legalize_placement beforehand to ensure all the scan cells being placed.

---

## DFT-016

(warnning) net %s is with timing exception constraint and will be preserved during dft optimization.

**Description**

The described net is with timing exception constraint and its connection will be preserved during dft optimization.

**What Next**

please check the net to see whether the timing exception on this net is intended.

---

## DFT-017

(warning) mv cell(s) exist in scan chain %s, this scan chain will be split to perform further optimization.

**Description**

Isolation cell or level shifter exists in the scan chain, this scan chain will be split into multiple valid ones to perform further dft optimization.

**What Next**

Check the scan def to ensure there is no mv cell inserted in newly generated scan def and go on other dft optimization steps.

---

## DFT-018

(warning) mv cell(s) locate inside an ordered list of scan chain %s, this ordered list will be break up during scan def repairment.

**Description**

Isolation cell or level shifter exists in an order list of the scan chain, this scan chain will be split into multiple valid ones so that the original ordered list will be break up.

**What Next**

Check the scan def to ensure there is no mv cell inserted in newly generated scan def and go on other dft optimization steps.

Feedback

## DFT-019

(warning) New scan chain %s is inserted into scan def.

**Description**

A new generated scan chain is inserted into scan def.

**What Next**

Check the scan def to ensure validity of new inserted scan chain and go on other dft optimization steps.

## DFT-020

(warning) Scan chain %s is deleted from scan def.

**Description**

An origianl scan chain is deleted from scan def because of mv cell inserted and this scan chain is split.

**What Next**

Check the scan def to ensure validity of new inserted scan chain and go on other dft optimization steps.

## DFT-021

(Error) Unqualified instance besides sequential cells exists in scandef as scan element, skip DFT optimization

**Description**

An non-squential cell found as scan element in scan chain, this is not an expeceted behaviour so DFT optimization will be skipped.

**What Next**

Check the location of non-squential cell inside scan chain, and re-design scan chains if necessary.

## DFT-022

(warning) new flip-flop %s replaced original flip-flop %s in scandef

Feedback

**Description**

When one flip-flop have multiple loads, this flip-flop will be cloned and assigned to each load path seprately, and a new cloned flip-flop will replace original one in as scan element in scan path.

**What Next**

Go on further DFT optimization steps.

---

## DFT-023

(warning) The start term of scan chain: %s is revised.

**Description**

Isolation cell or level shifter exists between start term and first element in scan chian, the start term of this scan chain will be reset.

**What Next**

Report this chain to see the new start term of this scan chain.

---

## DFT-024

(warning) The end term of scan chain: %s is revised.

**Description**

Isolation cell or level shifter exists between end term and last element in scan chian, the end term of this scan chain will be reset.

**What Next**

Report this chain to see the new end term of this scan chain.

---

## DFT-025

(Error) optimize_dft introduces new failed scan chain, now the failed scan chain count is %d versus the number %d before optimize_dft.

**Description**

The message occurs when optimize_dft introduces new failed scan chain.

**What Next**

Please disable optimize_dft call with the command 'set_optimize_dft_options' or contact ynopsys engineer for the solution.

Feedback

## DFT-026

(information) There exists %s in scan chain %s, and MV aware optimize_dft will be employed.

### Description

The tool found mv cell(s) or net(s) cross different voltage areas, mv aware dft will be automatically turned on to ensure no mv vialotions introduced in DFT optimization.

### What Next

check_mv_design after dft optimization ends.

## DFT-027

(information) optimize_dft is skipped in spg flow since the majority scan nets are shared by data path or the chip is too long and narrow.

### Description

optimize_dft is skipped in spg flow since the majority scan nets are shared by data path or the chip is too long and narrow. For such designs, optimize_dft call within place_opt spg flow helps little but may bring negative impact, so it is skipped.

### What Next

If the user insists on the use of dft optimization, he/she can call standalone optimize_dft command in the flow.

## DFT-028

(Warning) The timing scenario manager could not be started and optimize_dft is being skipped.

### Description

optimize_dft is being skipped because of a problem with the timing scenario manager.

### What Next

If this message is seen in the flow, check for extraction or timer messages in the log file.

### See Also

- check_timing

Feedback

## DFT-029

(Warning) Partition %s is overconstrained. It has %d bits of total stub chain capacity and %d bits of stub chain used capacity.

**Description**

The indicated partition has overfull stub chains with maxbits set so that the number of bits it contains exceeds its capacity. DFT repartitioning will be skipped.

**What Next**

If this message is seen in the flow, check for chains with maxbits on the given partition.

## DFT-1001

(Error) CTL Model could not be read

**Description**

CTL Model could not be read

**What Next**

Please Review DFT Specifications

## DFT-1002

(Error) No DFT valid partition from %s

**Description**

No DFT valid partition from %s

**What Next**

Please Review DFT Specifications

## DFT-1003

(Error) Test Mode %s is not defined

**Description**

Test Mode %s is not defined

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1004

(Error) Total number of '%d' DFT signal(s) of Type '%s' required%s. '%d' DFT signal(s) of type '%s' specified.

**Description**

Total number of '%d' DFT signal(s) of Type '%s' required%s. '%d' DFT signal(s) of type '%s' specified.

**What Next**

Please Review DFT Specifications

## DFT-1005

(Error) Port %s is not defined

**Description**

Port %s is not defined

**What Next**

Please Review DFT Specifications

## DFT-1006

(Error) -Timing is specified with either -view option wrongly or -view option as spec

**Description**

-Timing is specified with either -view option wrongly or -view option as spec

**What Next**

Please Review DFT Specifications

## DFT-1007

(Error) Multiple pins cannot be specified for %s '%s'.

**Description**

Multiple pins cannot be specified for %s '%s'.

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1009

(Error) Test Mode %s is not defined

**Description**

Test Mode %s is not defined

**What Next**

Please Review DFT Specifications

## DFT-1010

(Error) The '-timing' option requires two values

**Description**

The '-timing' option requires two values

**What Next**

Please Review DFT Specifications

## DFT-1011

(Error) The specified '-type' option is not supported.

**Description**

You receive this error message if you specify the wrong DFT signal type by set_dft_signal command.

Error: The specified '-type' option is not supported. (DFT-1011)

**What Next**

Define the correct DFT signal type.

## DFT-1012

(Error) Direction of '-hookup_pin' pin is incorrect.

**Description**

Tool throw this error message if it sees invalid direction of hookup_pin of signal type. Direction of hookup_pin should be input for signal type Reset, RefClock, MasterClock, ScanSlaveClock, TestMode, PllBypass, PllReset, CodecCntrl, Constant, ScanMasterClock for tck, ScanEnable, ScanDataIn, TestControl for wso.

Feedback

Direction of hookup_pin should be output for signal type ScanDataOut.

If there is any other combination defined by the user, Tool will throw error.

Error: Direction of '-hookup_pin' pin is incorrect. (DFT-1012)

**What Next**

Define the correct direction of hookup_pin of signal type.

---

## DFT-1013

(Error) Usage type of '-usage' pin is incorrect

**Description**

Usage type of '-usage' pin is incorrect

**What Next**

Please Review DFT Specifications

---

## DFT-1016

(Error) Either Internal pins flow or Clock Controller should be enabled

**Description**

Either Internal pins flow or Clock Controller should be enabled

**What Next**

Please Review DFT Specifications

---

## DFT-1018

(Error) View name of '-view' is incorrect

**Description**

View name of '-view' is incorrect

**What Next**

Please Review DFT Specifications

---

## DFT-1019

(Error) Direction of '-associated_internal_clocks' pin '%s' is incorrect

Feedback

**Description**

Direction of '-associated_internal_clocks' pin '%s' is incorrect

**What Next**

Please Review DFT Specifications

---

## DFT-1020

(Error) Could not find the specified DFT location name

**Description**

Could not find the specified DFT location name

**What Next**

Please Review DFT Specifications

---

## DFT-1022

(Error) There is no pin named %s on the instance %s.

**Description**

There is no pin named %s on the instance %s.

**What Next**

Please Review DFT Specifications

---

## DFT-1023

(Error) The %s pin on the instance %s is not an input pin.

**Description**

The %s pin on the instance %s is not an input pin.

**What Next**

Please Review DFT Specifications

---

## DFT-1024

(Error) %s is not a valid control signal; only ScanEnable and TestMode are valid.

Feedback

**Description**

%s is not a valid control signal; only ScanEnable and TestMode are valid.

**What Next**

Please Review DFT Specifications

---

## DFT-1026

(Error) '%s' is not a valid test mode for %s command.

**Description**

'%s' is not a valid test mode for %s command.

**What Next**

Please Review DFT Specifications

---

## DFT-1027

(Error) Unsuccessful DFT Insertion

**Description**

Unsuccessful DFT Insertion

**What Next**

Please Review DFT Specifications

---

## DFT-1028

(Error) The command requires -test_mode specification or current test_mode set.

**Description**

The command requires -test_mode specification or current test_mode set.

**What Next**

Please Review DFT Specifications

---

## DFT-1029

(Error) Gate level flow on designs with unmapped cells is not supported. First found unmapped cell is %s.

Feedback

**Description**

Gate level flow on designs with unmapped cells is not supported. First found unmapped cell is %s.

**What Next**

Please Review DFT Specifications

---

## DFT-1030

(Error) The -period option can only be used for signals of type RefClock

**Description**

The -period option can only be used for signals of type RefClock

**What Next**

Please Review DFT Specifications

---

## DFT-1031

(Error) The -period value should be greater than 0

**Description**

The -period value should be greater than 0

**What Next**

Please Review DFT Specifications

---

## DFT-1032

(Error) Incorrect value '%s' specified for option '-type' in command 'create_dft_netlist'.

**Description**

The Error message is shown when incorrect value is specified for option '-type' option.

**What Next**

1. Correct value for option '-type' is 'extest'.

Feedback

## DFT-1033

(Error) An integer greater than 0 should be specified for option '-debug' in command 'create_dft_netlist'.

**Description**

The Error message is shown when incorrect value is specified for option '-debug'.

**What Next**

1. Value must be greater than 0 and less than 5.

## DFT-1034

(Error) %s is not available.

**Description**

This error message is issued when neither External Test nor Internal Scan modes are found in the DFT specifications during Shell Netlist flow. Shell Netlist is aborted under such circumstances.

**What Next**

Please define either External Test or Internal Scan Mode in the script before DFT insertion and shell netlist.

## DFT-1035

(Error) "%s file can't be opened.

**Description**

This error message is issued when create_dft_netlist -debug x (x>=1) is unable to create a debug report by the name of shell_netlist.rpt. This report prints out logs of shell netlist run.

**What Next**

Please retry the command with -debug (1|2|3).

## DFT-1036

(Error) DFT has already been inserted

**Description**

DFT has already been inserted

Feedback

**What Next**

Please Review DFT Specifications

---

## DFT-1037

(Error) Name '%s' is already in use by scan skew group specification

**Description**

Name '%s' is already in use by scan skew group specification

**What Next**

Please Review DFT Specifications

---

## DFT-1038

(Error) Invalid element '%s'. set_scan_path -ordered/-head/-tail does not support %s.

**Description**

Invalid element '%s'. set_scan_path -ordered/-head/-tail does not support %s.

**What Next**

Please Review DFT Specifications

---

## DFT-1039

(Error) set_scan_path with %s .

**Description**

You receive this error message in the following scenario : 1) When interface option is defined with other class apart from 'bsd' | 'ieee_1838_3dcr' | 'ieee_1838_dwr'. 2) When exact_length option is defined with other class apart from 'bsd' | 'ieee_1687'. 3) When 'bsd' | 'ieee_1687' class option is defined and exact_length | ordered_elements is not defined. 4) When -init_data is defined with class bsd | ieee_1687. 5) When length of init_data and exact_length do not match. 6) When length of reset_state and exact_length do not match. 7) When class occ is defined with -input_wrapper_cells_only. 8) When class occ is defined with -output_wrapper_cells_only. 9) When class spc is defined with -input_wrapper_cells_only. 10) When class spc is defined with -output_wrapper_cells_only. 11) When class scan is defined with -input_wrapper_cells_only. 12) When class scan is defined with -output_wrapper_cells_only. 13) When class occ is defined with -cell_set. 14) When class occ is defined with -ordered_elements | -head_elements | -tail_elements 15) When

Feedback

class wrapper is defined with -input_wrapper_cells_only | -output_wrapper_cells_only and -include_elements | -ordered_elements/-head_elements | -tail_elements'. 16) When -include_elements/-head_elements/-ordered_elements/-tail_element is defined with -cell_set.

Error : set_scan_path with '-interface' is supported for '-class bsd | ieee_1838_3dcr | ieee_1838_dwr'. (DFT-1039) Error : set_scan_path with '-exact_length' is supported for '-class bsd | ieee_1687'. (DFT-1039) Error : set_scan_path with '-class bsd | ieee_1687' is supported when -exact_length is positive and order_element list is not empty. (DFT-1039) Error : set_scan_path with -init_data is supported for '-class bsd | ieee_1687'. (DFT-1039) Error : set_scan_path with -init_data is supported when length of init_data is same as exact_length. (DFT-1039) Error : set_scan_path with -reset_state is supported when length of reset_state is same as exact_length. (DFT-1039) Error : set_scan_path with '-class occ' and '-input_wrapper_cells_only' not supported. (DFT-1039) Error : set_scan_path with '-class occ' and '-output_wrapper_cells_only' not supported. (DFT-1039) Error : set_scan_path with '-class spc' and '-input_wrapper_cells_only' not supported. (DFT-1039) Error : set_scan_path with '-class spc' and '-output_wrapper_cells_only' not supported. (DFT-1039) Error : set_scan_path with '-class scan' and '-input_wrapper_cells_only' not supported. (DFT-1039) Error : set_scan_path with '-class scan' and '-output_wrapper_cells_only' not supported. (DFT-1039) Error : set_scan_path with '-class occ' and '-cell_set' not supported. (DFT-1039) Error : set_scan_path with '-class occ' and '-ordered_elements/-head_elements/-tail_elements' not supported. (DFT-1039) Error : set_scan_path with '-class wrapper', '-input_wrapper_cells_only/-output_wrapper_cells_only' and '-include_elements/-ordered_elements/-head_elements/-tail_elements'. (DFT-1039) Error : set_scan_path with '-include_elements/-head_elements/-ordered_elements/-tail_elements' and '-cell_set' not supported. (DFT-1039)

**What Next**

Define interface option when class is bsd | ieee_1838_3dcr | ieee_1838_dwr. Define exact_length option when class is bsd | ieee_1687. Define exact_length | ordered_elements when class is bsd | ieee_1687 and exact_length should be positive Define init_data for class bsd | ieee_1687. Define init_data | reset_state of same length as exact_length. Do not define class occ with input_wrapper_cells_only. Do not define class occ with output_wrapper_cells_only. Do not define class spc with output_wrapper_cells_only. Do not define class spc with output_wrapper_cells_only. Do not define class scan with input_wrapper_cells_only. Do not define class scan with output_wrapper_cells_only. Do not define class occ with -cell_set. Do not define class occ with -ordered_elements | -head_elements | -tail_elements. Do not define class wrapper with -input_wrapper_cells_only | -output_wrapper_cells_only and -include_elements | -ordered_elements | -head_elements | -tail_elements. Do not define -include_elements | -head_elements | -ordered_elements | -tail_element with -cell_set.

Feedback

## DFT-1043

(Error) Could not find '%s' in the model

**Description**

Could not find '%s' in the model

**What Next**

Please Review DFT Specifications

## DFT-1044

(Warning) The signal '%s' must be specified as '%s'

**Description**

This warning message reports that the port specified in set_scan_path -scan_enable|-scan_data_in|-scan_data_out was not declared with the correct type using the set_dft_signal command.

**What Next**

Verify the port definition in the set_dft_signal command. In set_scan_path specification, for -scan_data_in the port specified must be declared as ScanDataIn, for -scan_data_out it must be declared as ScanDataOut and for -scan_enable it must be declared as ScanEnable.

## DFT-1045

(Error) The specified class '%s' is not supported.

**Description**

The specified class '%s' is not supported.

**What Next**

Please Review DFT Specifications

## DFT-1046

(Error) Could not find %s named '%s'

**Description**

Could not find %s named '%s'

Feedback

**What Next**

Please Review DFT Specifications

---

## DFT-1047

(Error) Only 'set_scan_path -class scan|wrapper|occ|spc|bsd|ieee_1838_3dcr|ieee_1838_dwr|ieee_1687' is currently supported.

**Description**

Only 'set_scan_path -class scan|wrapper|occ|spc|bsd|ieee_1838_3dcr|ieee_1838_dwr|ieee_1687' is currently supported.

**What Next**

Please Review DFT Specifications

---

## DFT-1048

(Error) The controller index for '%s' exceeds available chain count '%d'

**Description**

The controller index for '%s' exceeds available chain count '%d'

**What Next**

Please Review DFT Specifications

---

## DFT-1050

(information) Missing '-test_mode_port' specification.

**Description**

While using the command "set_dft_clock_controller" the -test_mode_port switch must be defined.

**What Next**

Please use -test_mode_port switch in the "set_dft_clock_controller" command.

---

## DFT-1051

(Error) %s not of usage scan or wrp_if

Feedback

**Description**

%s not of usage scan or wrp_if

**What Next**

Please Review DFT Specifications

---

## DFT-1052

(Error) Could not find %s named '%s'

**Description**

Could not find %s named '%s'

**What Next**

Please Review DFT Specifications

---

## DFT-1053

(Error) %s is not defined as a Test Mode

**Description**

%s is not defined as a Test Mode

**What Next**

Please Review DFT Specifications

---

## DFT-1054

(Error) Test Mode specified is not defined

**Description**

Test Mode specified is not defined

**What Next**

Please Review DFT Specifications

---

## DFT-1055

(Error) Value entered is invalid

Feedback

**Description**

Value entered is invalid

**What Next**

Please Review DFT Specifications

---

## DFT-1056

(Error) Mode already been defined

**Description**

Mode already been defined

**What Next**

Please Review DFT Specifications

---

## DFT-1057

(Error) %s not a valid usage type

**Description**

%s not a valid usage type

**What Next**

Please Review DFT Specifications

---

## DFT-1058

(Error) %s is not defined as a TestMode port

**Description**

%s is not defined as a TestMode port

**What Next**

Please Review DFT Specifications

---

## DFT-1059

(Error) The value %s for the port is incorrect

Feedback

**Description**

The value %s for the port is incorrect

**What Next**

Please Review DFT Specifications

---

## DFT-1060

(Error) Encoding specifications for modes %s and %s are identical

**Description**

Encoding specifications for modes %s and %s are identical

**What Next**

Please Review DFT Specifications

---

## DFT-1061

(Error) %s is a tool reserved mode name for %s mode with usage %s only

**Description**

%s is a tool reserved mode name for %s mode with usage %s only

**What Next**

Please Review DFT Specifications

---

## DFT-1062

(Error) Partition %s has already been defined

**Description**

Partition %s has already been defined

**What Next**

Please Review DFT Specifications

---

## DFT-1063

(Error) Path %s includes elements from previously defined partition

Feedback

**Description**

Path %s includes elements from previously defined partition

**What Next**

Please Review DFT Specifications

---

## DFT-1067

(Error) %s is not a valid clock. Use set_dft_signal to define it as a valid clock.

**Description**

%s is not a valid clock. Use set_dft_signal to define it as a valid clock.

**What Next**

Please Review DFT Specifications

---

## DFT-1068

(Error) Specified serilaizer IP outputs and inputs should be equal

**Description**

Specified serilaizer IP outputs and inputs should be equal

**What Next**

Please Review DFT Specifications

---

## DFT-1069

(Error) %s and %s cannot both be specified.

**Description**

This error is issued when two uncompatible options of set_scan_group are specified.

**What Next**

Please check and update set_scan_group specification.

---

## DFT-1070

(Error) -connect_to %s is not a pin/cell/hierarchy/clock and cannot be connected as specified

Feedback

**Description**

-connect_to %s is not a pin/cell/hierarchy/clock and cannot be connected as specified

**What Next**

Please Review DFT Specifications

---

## DFT-1071

(Error) Invalid specification for scan path '%s': %s is not allowed.

**Description**

This message is issued when some cells of set_scan_path specification are not compatible. For example,

Error: Invalid specification for scan path 'PATH1': rising to falling clock edge is not allowed. (DFT-1071)

**What Next**

Check set_scan_path specification.

---

## DFT-1076

(Error) %s cannot be a port or a cell in the design.

**Description**

This error is issued when incorrect name is specified.

**What Next**

Please check and update set_scan_group specification.

---

## DFT-1077

(Error) Scan synthesis cannot run because %s.

**Description**

This message is issued when scan synthesis is missing a critical prerequisite and cannot continue executing without it.

This can happen if scan enable pipelining is enabled via set_scan_configuration, but corresponding LOSPipelineEnable signals are not provided.

Feedback

**What Next**

The error message describes the exact reason why scan synthesis stopped executing. If the reason refers to the DFT specification, review it to ensure it is complete and consistent.

---

## DFT-1078

(Warning) Physical location for %s not found.

**Description**

There are cells or its pins that do not report a valid physical location on the design's layout.

This message can arise when performing DFT Scan Synthesis operations like during commands *preview_dft* or *insert_dft*.

Cells needs to be properly placed before calling DFT SCAN insertion, as the default status after *compile_fusion -to logic_opto* stage was completed before initializing DFT SCAN insertion, when coarse placement should have been applied to all elements of the design.

Lack of physical location during the DFT SCAN insertion stage might lead to sub-optimal SCAN chain architecting or excesive wire-length in SCAN connections because incomplete information of placement for elements in the design might produce bad choices for SCAN interconnection in the internal algorithms.

**What Next**

Review the design setup and run logs to identify previous errors in design setup or compile_fusion execution that prevented all cells being properly placed.

---

## DFT-1079

(Error) Could not find the specified DFT location name '%s'

**Description**

Could not find the specified DFT location name '%s'

**What Next**

Please Review DFT Specifications

---

## DFT-1080

(Error) Architecting of chains in mode '%s' failed. Please run 'set_scan_configuration -chain_count'.

Feedback

**Description**

This Error message will be displayed when chain count is not specified through set_scan_configuration command.

---

## DFT-1081

(Error) %s should be specified for %s

**Description**

%s should be specified for %s

**What Next**

Please Review DFT Specifications

---

## DFT-1083

(Error) Core Wrapping - Shared wrapper flow in netlist is enabled and number of short wrapper chains are not set.

**Description**

The Error message is shown when shared wrapper flow is enabled and number of short wrapper chains in compression mode are not specified

**What Next**

See Information DFT-3143 to fix this error.

Specify wrapper chain count of short chains mode (compression_mode|wrp_if) as following and run again set_wrapper_configuration -chain_count N -test_mode (compression_mode|wrp_if)

---

## DFT-1085

(warning) UPF check failed for the path (%s) -> (%s). Netlist connection not UPF legal.

**Description**

This message is issued when during the SCAN stitching process, where connection from ScanEnable, TestMode, ScanInput, ScanOutput or similar signals, or between SCAN cells (SCAN cell ScanOutput to next SCAN chain cell ScanInput) has been detected by MultiVoltage (MV) APIs as invalid. This means it can introuce a MV violation.

Feedback

### What Next

Use command *check_mv_design* to determine what is the MV problem in the connection reported by this message. With that information, please validate the UPF setup is valid, complete and enought for defining all UPF rules required for connecting that particular driver and load combination. If there is a UPF setup issue or deficiency, please update UPF setup.

### See Also

- check_mv_design

---

## DFT-1086

(Information) Test DRC reports %d violating cells.

### Description

This message reports the exact amount of cells that were marked as violated by DFT DRC and TetraMAX. These are ignored by scan synthesis.

### What Next

Review your DFT specification if the number of violated cells is unexpected.

---

## DFT-1090

(Error) %s has to be specified with %s as its value.

### Description

This error is issued when specified option is not specified with proper value.

### What Next

Please check and update set_scan_group specification.

---

## DFT-1091

(Error) App options %s and %s are not compatible.

### Description

This message is issued when some app options are not compatible. For example,

Error: App options enable_general_multimode_support 'false' and occ_spc_chain_arch_flat_flow 'true' are not compatible. (DFT-1091)

Feedback

**What Next**

Check which combinations of app options are supported. Update the settings accordingly.

---

## DFT-1092

(Error) Invalid parameters: %s

**Description**

Invalid parameters: %s

**What Next**

Please Review DFT Specifications

---

## DFT-1094

(Error) %s should %s.

**Description**

This error is issued when specified option has value of inccorect type.

**What Next**

Please check and update set_scan_group specification.

---

## DFT-1095

(warning) Scan chain element %s has no power information

**Description**

This message warns user that some element in the design has incomplete, invalid or inconsistent UPF information related to Power Domains (PD) or Voltage Supply (VS) data.

DFT SCAN Synthesis requires precise PD and/or VS information for preparing DFT SCAN domains of cells that can be safely put into SCAN chains without creating UPV or MultiVoltage issues, and without requiring the insertion of unneeded MultiVoltage cells like Level Shifter, Isolation cells, etc. If the previous information is not available for a cell, SCAN architecting decisions might be hampered or erroneous regarding UPF and MV criteria, introducing MV violations.

**What Next**

Verify that libraries are properly loaded and UPF data & setup has beeing loaded successfully.

Feedback

**See Also**

- check_mv_design

---

## DFT-1096

(Error) %s is already specified as an equivalent clock

**Description**

%s is already specified as an equivalent clock

**What Next**

Please Review DFT Specifications

---

## DFT-1097

(Error) %s option is only supported with type shared_only of set_boundary_cell. Ignoring %s option.

**Description**

add_loopback_logic or add_toggle_logic is supported only with type shared_only for set_boundary_cell command.

---

## DFT-1098

(Error) -exclude option is only valid with -ports option and not -instance

**Description**

-exclude option is only valid with -ports option and not -instance

**What Next**

Please Review DFT Specifications

---

## DFT-1100

(Error) %s.

**Description**

This is a generic scan architecting error message. You receive this error message if it is not possible to build scan chains according to provided set_scan_configuration settings.

Feedback

**Messages:**

Scan partitioning trying to overflow chain %i, current chain len: %i, max_length for the given group of elements: %i.

**What Next**

Please review set_scan_configuration settings and set_scan_compression_configuration settings if scan compression is used.

---

## DFT-1101

(Error) dft.insertion_post_logic_opto disabled, cannot run insert_dft in this stage

**Description**

dft.insertion_post_logic_opto disabled, cannot run insert_dft in this stage

**What Next**

Please Review DFT Specifications

---

## DFT-1102

(Error) '%s' is not a valid Test Mode

**Description**

'%s' is not a valid Test Mode

**What Next**

Please Review DFT Specifications

---

## DFT-1106

(Warning) DFT insertion is not supported in FC extended RTL shell. Hence ignoring DFT configurations.

**Description**

DFT insertion is not supported in FC extended RTL shell.Hence DFT related configurations will be ignored.

---

## DFT-1107

(Warning) Skipping scan synthesis: %s

Feedback

**Description**

DFT SCAN insertion (*insert_dft* command) cannot be executed and will be skipped.

Usual reasons are: 1) The design was already DFT SCAN inserted.

This might happen if a previous *insert_dft* command was succesfully executed. In that case, incremental DFT SCAN insertion is both not supported and not required.

2) Unexisting or insuficient DFT setup is detected.

DFT SCAN insertion requires the definition of all required DFT signals, correct setup of SCAN chain length or SCAN chain quantity, etc. If one or more of those required setup data are not found, DFT SCAN insertion cannot proceed.

**What Next**

Check if the command *insert_dft* was not previously executed for the design, and if is not, check that DFT Specifications are complete.

---

## DFT-1108

(Warning) %s

**Description**

This is a generic warning message for preview_dft reporting.

By default, after running Scan Synthesis from compile_fusion or insert_dft commands, a 'scan_dft.rpt' file with the scan chains report from preview_dft will be created automatically in the working directory. To specify a different file name for the report, use the *dft.scan.scan_chain_report* application option. DFT-1108 message will be printed if there was an issue with the file defined for automatic reporting.

**Messages:**

"Filename %s already exists. Report will overwrite it."

"Can't open file %s for writing. Report will be skipped."

**What Next**

If the file defined for reporting could not be opened for writing, please verify whether the path specified in the file name is valid, and whether the path and the file have valid writing permissions.

---

## DFT-1109

(Error) Unable to find signal definition for %s.

Feedback

**Description**

This message is issued when ScanEnable or TestMode signals are not found. For example,

Error: Unable to find signal definition for ScanEnable or TestMode. (DFT-1109)

**What Next**

Check signals specification

---

## DFT-1112

(Error) Required OCC existing flow signal option: '%s'

**Description**

Required OCC existing flow signal option: '%s'

**What Next**

Please Review DFT Specifications

---

## DFT-1113

(Error) The clock controller bit list should be a multiple of three

**Description**

The clock controller bit list should be a multiple of three

**What Next**

Please Review DFT Specifications

---

## DFT-1114

(Error) Could not find the occ control bit pin '%s' from the current design

**Description**

Could not find the occ control bit pin '%s' from the current design

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1115

(Error) The active state value of the OCC signal control bit should be either '0' or '1'

**Description**

The active state value of the OCC signal control bit should be either '0' or '1'

**What Next**

Please Review DFT Specifications

## DFT-1116

(Error) Could not find the occ control bit cycle value from '%s'

**Description**

Could not find the occ control bit cycle value from '%s'

**What Next**

Please Review DFT Specifications

## DFT-1117

(Error) The specified '-internal_clocks' option is incorrect - .%s

**Description**

The specified '-internal_clocks' option is incorrect - .%s

**What Next**

Please Review DFT Specifications

## DFT-1121

(Error) Unknown location type '%s' specified.

**Description**

Unknown location type '%s' specified.

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1122

(Error) Design '%s' has no test model

**Description**

Design '%s' has no test model

**What Next**

Please Review DFT Specifications

## DFT-1123

(Error) Could not find the specified DFT location name '%s'

**Description**

Could not find the specified DFT location name '%s'

**What Next**

Please Review DFT Specifications

## DFT-1128

(Warning) %s.

**Description**

This is a generic warning issued when set_scan_group -access_pins settings are not compatible.

**Messages:**

Sequential cell %s is connected to the access pin %s of a serially routed chain. Cell will be excluded from scan synthesis - applies to set_scan_group virtual case (without -include_elements). In such setting the cell will be automatically excluded.

Sequential cell %s is on the access pin %s of a serially routed chain. Cell will be excluded from scan synthesis. - applies to set_scan_group virtual case (without -include_elements). In such setting the cell will be automatically excluded.

BypassSe can be specified only for -class bypass. Ignoring BypassSe spec.

**What Next**

Please review set_scan_group settings.

Feedback

## DFT-1129

(Error) Other DFT Ips can not be grouped with AIT, SERVER and BSD

**Description**

Other DFT Ips can not be grouped with AIT, SERVER and BSD

**What Next**

Please Review DFT Specifications

## DFT-1130

(Error) Signal of data type 'Reset' is not valid for -connect_to option

**Description**

Signal of data type 'Reset' is not valid for -connect_to option

**What Next**

Please Review DFT Specifications

## DFT-1131

(Error) Pin %s direction is not valid for -connect_to option.

**Description**

Pin %s direction is not valid for -connect_to option.

**What Next**

Please Review DFT Specifications

## DFT-1132

(Error) Multiple signal types cannot be specified for hookup port '%s'.

**Description**

Multiple signal types cannot be specified for hookup port '%s'.

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1134

(Error) For '-timing' option, rise time and fall time cannot be same

**Description**

For '-timing' option, rise time and fall time cannot be same

**What Next**

Please Review DFT Specifications

## DFT-1135

(Error) Test design rule checking reported FATAL violations.

**Description**

Message shows that dft_drc command in under the hood tetramax script has failed. The message will only be reported when enhaced dft drc reporting is enabled.

SH "SEE ALSO"

```
dft_drc
set_dft_drc_configuration
```

## DFT-1139

(Error) DFT DRC cannot performed due to too many open files.

**Description**

dft_drc command internally instantiate a process to invoke tetramax atpg tool. To perform the design rule checks on the design. This process has a maximum limit of 1024 for file descriptors. If number exeeds 1024,then dft_drc exits pre-maturely.

## DFT-1140

(Error) Cannot specify shared -%s_shared_pipeline_stages without enabling hierarchical or shared SI/SO flow.

**Description**

Cannot specify shared -%s_shared_pipeline_stages without enabling hierarchical or shared SI/SO flow.

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1142

(Error) Must specify 'true' or 'false' for the %s_scan_flop option

**Description**

Must specify 'true' or 'false' for the %s_scan_flop option

**What Next**

Please Review DFT Specifications

## DFT-1143

(Error) Must specify 'true' or 'false' for the enable_bypass option

**Description**

Must specify 'true' or 'false' for the enable_bypass option

**What Next**

Please Review DFT Specifications

## DFT-1144

(Error) Since the enable_bypass option is set to true, serializer mode %s cannot be encoded to all zeros

**Description**

Since the enable_bypass option is set to true, serializer mode %s cannot be encoded to all zeros

**What Next**

Please Review DFT Specifications

## DFT-1145

(Error) A serializer clock must be provided to enable hierarchical serialization

**Description**

A serializer clock must be provided to enable hierarchical serialization

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1147

(Error) DFT DRC violations cannot be reported. dft_drc was not run for %s mode.

**Description**

DFT DRC violations cannot be reported. dft_drc was not run for %s mode.

**What Next**

Please Review DFT Specifications

## DFT-1151

(Error) To group AIT, SERVER and TAP, new hierarchy should be specified insideTOP

**Description**

To group AIT, SERVER and TAP, new hierarchy should be specified insideTOP

**What Next**

Please Review DFT Specifications

## DFT-1154

(Error) The '-connect_to clock' option is not supported for '%s'.

**Description**

The '-connect_to clock' option is not supported for '%s'.

**What Next**

Please Review DFT Specifications

## DFT-1159

(Error) Design-state not compatible with requested starting point '%s'. If you are attempting to run the pre-compile flow please refer to man page of this message for further information.

**Description**

You receive this error message when tool is not able to find proper license settings.
If pre-compile flow is intended then below app option should be set after setting the current_design in the flow: set_app_options -name dft.insertion_pre_compile_fusion -value true

Feedback

If in-compile or post-compile flow is intended then 'create_test_protocol', 'preview_dft' and 'insert_dft' should be run after 'compile_fusion -to logic_opto' or 'compile' step.

---

## DFT-1169

(Error) Specify positive value for option %s.

**Description**

Option's value must be >= 1

**What Next**

Please Review DFT Specifications

---

## DFT-1170

(Error) Number of %s must be >= 1

**Description**

Number of %s must be >= 1

**What Next**

Please Review DFT Specifications

---

## DFT-1172

(Error) Could not find CTL model for design '%s'.

**Description**

Could not find CTL model for design '%s'.

**What Next**

Please Review DFT Specifications

---

## DFT-1176

(Error) No cells matched for %s option

**Description**

No cells matched for %s option

Feedback

**What Next**

Please Review DFT Specifications

---

## DFT-1177

(Error) Number of usage for mode should not be more than two, discarding test mode specification.

**Description**

Number of usage for mode should not be more than two, discarding test mode specification.

**What Next**

Please Review DFT Specifications

---

## DFT-1178

(Error) Currently only 'scan_compression', 'streaming_compression' and 'wrp_of' usage types are supported for multi usage specification, discarding test mode specification.

**Description**

Currently only 'scan_compression', 'streaming_compression' and 'wrp_of' usage types are supported for multi usage specification, discarding test mode specification.

**What Next**

Please Review DFT Specifications

---

## DFT-1179

(Error) Required argument '-usage' was not found, discarding test mode specification.

**Description**

Required argument '-usage' was not found, discarding test mode specification.

**What Next**

Please Review DFT Specifications

---

## DFT-1180

(Error) Required environment not found

Feedback

**Description**

Required environment not found

**What Next**

Please Review DFT Specifications

---

## DFT-1181

(Error) RP group %s not found

**Description**

RP group %s not found

**What Next**

Please Review DFT Specifications

---

## DFT-1188

(Error) wrp_disable option is only supported with app option dft.test_wrp_custom_wrp_cell_disable_sig_support enabled. Ignoring -wrp_disable option

**Description**

You receive this error message when you specifiy -wrp_disable option without app option 'test_wrp_custom_wrp_cell_disable_sig_support' being ON

---

## DFT-1189

(Error) %s option is only supported with type WC_Q1 | WC_Q0 ignoring %s

**Description**

You receive this error message when you do not specify -type WC_Q1 or WC_Q0 with the specified option

---

## DFT-1190

(Error) No wrp_disable signal defined for port/pin specified with '-wrp_disable' option

**Description**

You receive this error message when port/pin spcecified in -wrp_disable option does not have a dft signal of -type wrp_disable.

Feedback

## DFT-1193

(Error) Building model for design rules checking failed.

**Description**

This build error occurs when hookup pin is not available in netlist when driver/load of the hookup pin doesn't exist and when library cells are modeled incorrectly and TetraMAX cannot parse it properly due to incorrect modeling.

**What Next**

user needs to apply a restriction to prevent driver/load from being optimized away or need to check if ndm library is generated by latest version of library compiler, at least later than IC Compiler II O-2018.06-SP5.

## DFT-1196

(Error) Invalid class specified, supported values : wrapper, bsd

**Description**

Invalid class specified, supported values : wrapper, bsd

**What Next**

Please Review DFT Specifications

## DFT-1197

(Error) Invalid cell type specified, supported values : WC_D1 | shared_only | WC_OBSERVE | WC_Q0 | WC_Q1 | WC_Q0_AO | WC_Q1_AO | BC_1 | BC_2 | BC_4 | BC_7 | AC_1 | AC_2 | AC_7 | AC_SELU | AC_SELX | BC_1_pinmux | BC_7_pinmux | none

**Description**

Invalid cell type specified, supported values : WC_D1 | shared_only | WC_OBSERVE | WC_Q0 | WC_Q1 | WC_Q0_AO | WC_Q1_AO | BC_1 | BC_2 | BC_4 | BC_7 | AC_1 | AC_2 | AC_7 | AC_SELU | AC_SELX | BC_1_pinmux | BC_7_pinmux | none

**What Next**

Please Review DFT Specifications

## DFT-1198

(Error) Empty port and instance list

Feedback

**Description**

Empty port and instance list

**What Next**

Please Review DFT Specifications

---

## DFT-1199

(Error) An integer greater than or equal to 0 should be specified

**Description**

An integer greater than or equal to 0 should be specified

**What Next**

Please Review DFT Specifications

---

## DFT-1205

(Error) Incorrect input parameters %s is excluded by user. Will be ignored

**Description**

When instance given in the set_dft_clock_gating_pin is excluded by set_dft_clock_gating_configuration -exclude_elements.DFT-1205 message will be printed.

**What Next**

Remove the instance from the set_dft_clock_gating_configuration -exclude_cells

---

## DFT-1208

(Error) Value %i for set_test_assume not in range (0 to 1)

**Description**

Value %i for set_test_assume not in range (0 to 1)

**What Next**

Please Review DFT Specifications

---

## DFT-1209

(Error) Could not find pin %s in design

Feedback

**Description**

Could not find pin %s in design

**What Next**

Please Review DFT Specifications

---

## DFT-1210

(Error) Incorrect value '%s' specified for option '%s'

**Description**

Incorrect value '%s' specified for option '%s'

**What Next**

Please Review DFT Specifications

---

## DFT-1213

(Error) Please use either all top level port or TDR pins as TestModes. Mixing of port and TDR pins are not allowed as TestMode type.

**Description**

You receive this error message if you define port and tdr_pin both as TestMode type.

Error: Please use either all top level port or TDR pins as TestModes. Mixing of port and TDR pins are not allowed as TestMode type (DFT-1213)

**What Next**

Please define either all ports or TDR pins as TestMode type.

---

## DFT-1216

(Error) '%s' is not a valid Test Mode

**Description**

'%s' is not a valid Test Mode

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1217

(Error) '%s' should be 'true' or 'false'

**Description**

'%s' should be 'true' or 'false'

**What Next**

Please Review DFT Specifications

## DFT-1218

(Error) Q-Gating is disabled on FM-ECO mode

**Description**

Q-Gating is disabled on FM-ECO mode

**What Next**

Please Review DFT Specifications

## DFT-1219

(Error) '%s' cannot be %s

**Description**

'%s' cannot be %s

**What Next**

Please Review DFT Specifications

## DFT-1220

(Error) Unspecified option '%s'

**Description**

Unspecified option '%s'

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1221

(Error) Unrecognized dft ip design type '%s' specified with the option '%s'

**Description**

Unrecognized dft ip design type '%s' specified with the option '%s'

**What Next**

Please Review DFT Specifications

## DFT-1223

(Error) Specified %s name '%s' with option '%s' is not found

**Description**

Specified %s name '%s' with option '%s' is not found

**What Next**

Please Review DFT Specifications

## DFT-1224

(Error) Specified Dft IP design '%s' doesn't have a pin by name '%s'

**Description**

Specified Dft IP design '%s' doesn't have a pin by name '%s'

**What Next**

Please Review DFT Specifications

## DFT-1225

(Error) Mandatory Dft IP design pin of type '%s' is not specified

**Description**

Mandatory Dft IP design pin of type '%s' is not specified

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1227

(Error) Unrecognised design type!, Expected design type format as $DesignType$

**Description**

Unrecognised design type!, Expected design type format as $DesignType$

**What Next**

Please Review DFT Specifications

## DFT-1230

(Error) An integer greater than or equal to 0 should be specified.

**Description**

While using the option -wir_width or -wby_width of the command "set_ieee_1500_configuration" the value specified should be a positive integer

**What Next**

Specify a positive integer

## DFT-1231

(Error) Invalid signaltype '%s' specified for design type '%s'

**Description**

Invalid signaltype '%s' specified for design type '%s'

**What Next**

Please Review DFT Specifications

## DFT-1232

(Error) Invalid parameter '%s' specified for design type '%s'

**Description**

Invalid parameter '%s' specified for design type '%s'

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1233

(Error) %s is not a valid pin_value

**Description**

%s is not a valid pin_value

**What Next**

Please Review DFT Specifications

## DFT-1234

(ERROR) The interface list is parsed as key-value pair of interface name and portname, please provide correct number of arguments.

**Description**

The Provided Interface List Count is Invalid.

**What Next**

Please Provide Proper Interface List Count. Please Refer Below Specification for Interface List: ---------- | Interface Name | Attribute | |----------| | i_ait_clk | Input | |----------| | i_ref_clk | Input | |----------| | i_ait_rstn | Input | |----------| | i_timeout_cnt_strap | Input | |----------| | i_mode | Input | |----------| | i_test_start | Input | |----------| | i_test_enable | Input | |----------| | i_data_valid | Input | |----------| | i_abort | Input | |----------| | i_mem_start_addr | Input | |----------| | i_mem_data_in | Input | |----------| | i_ait21500_wrck | Input | |----------| | i_i15002ait_wso | Input | |----------| | i_reset_counter | Input | |----------| | o_mem_rd_enable | Output | |----------| | o_mem_rd_addr | Output | |----------| | wso_compare_status | Output | |----------| | status_ready | Output | |----------| | max_compare_wso_done | Output | |----------| | o_status | Output | ---------- Allowed -interface [list <interface-name> <portname> <interface-name> <portname> <interface-name> <portname> ....]

Not Allowed -interface [list <interface-name> <portname> <interface-name> <interface-name> <portname> ....]

Feedback

## DFT-1235

(ERROR) The parameter list is parsed as key-value pair of parameter name and width, please provide correct number of arguments.

**Description**

The Provided Parameter List Count is Invalid. One or Many Parameter(s) are missing.

**What Next**

Please Provide Proper Parameters Count as Mentioned in Below Table:

---------- | Name of Parameter | ---------- | MEM_DATA_WIDTH | ---------- | MEM_ADDR_WIDTH | ---------- | TIMEOUT_STRAP_WIDTH | ---------- | PROP_DLY_CNT_VAL | ---------- | WRSTN_TOGGLE_PULSE_COUNT | ---------- | MAX_COMPARE_VALUE | ----------

Allowed -parameter [list <parameter-name> <parameter-value> <parameter-name> <parameter-value> <parameter-name> <parameter-value> ....]

Not Allowed -parameter [list <parameter-name> <parameter-value> <parameter-name> <parameter-name> <parameter-value> ....]

## DFT-1236

(ERROR) The provided interface type "%s" is incorrect. List of supported interface types: %s

**Description**

The Provided Interface Keyword Name is Invalid.

**What Next**

Please Review the Defined Interface Keyword Name(s) and Correct Them. Please use Below Table for Reference. ---------- | Interface Name | Attribute | |----------| | i_ait_clk | Input | |----------| | i_ref_clk | Input | |----------| | i_ait_rstn | Input | |----------| | i_timeout_cnt_strap | Input | |----------| | i_mode | Input | |----------| | i_test_start | Input | |----------| | i_test_enable | Input | |----------| | i_data_valid | Input | |----------| | i_abort | Input | |----------| | i_mem_start_addr | Input | |----------| | i_mem_data_in | Input | |----------| | i_ait21500_wrck | Input | |----------| | i_i15002ait_wso | Input | |----------| | i_reset_counter | Input | |----------| | o_mem_rd_enable | Output | |----------|

Feedback

| o_mem_rd_addr | Output | |----------| | wso_compare_status | Output | |----------| | status_ready | Output | |----------| | max_compare_wso_done | Output | |----------| | o_status | Output | ----------

---

## DFT-1237

(ERROR) The provided parameter type "%s" is incorrect. List of supported paramaters: %s

**Description**

The Provided Parameter Name(s) is Wrong.

**What Next**

Please Define Correct Parameter Name(s) as Mentioned in Below Table:

---------- | Name of Parameter | Default Value | Minimum Value | Maximum Value | ---------- | MEM_DATA_WIDTH | 64 | 8 | 64 | ---------- | MEM_ADDR_WIDTH | 16 | 8 | 64 or Till Memory supports | ---------- | TIMEOUT_STRAP_WIDTH | 4 | 4 | 8 | ---------- | PROP_DLY_CNT_VAL | 10 | 10 | 255 | ---------- | WRSTN_TOGGLE_PULSE_COUNT | 5 | 5 | 7 | ---------- | MAX_COMPARE_VALUE | 8 | 8 | 65535 | ----------

---

## DFT-1238

(ERROR) The provided port name "%s" is incorrect. Please provide a valid port name.

**Description**

The Provided Port Name(s) is Wrong.

**What Next**

Please Provide Proper Port Name(s).

---------- | Interface Name | Attribute | |----------| | i_ait_clk | Input | |----------| | i_ref_clk | Input |

Feedback

|----------| | i_ait_rstn | Input | |----------| | i_timeout_cnt_strap | Input | |----------| | i_mode | Input | |----------| | i_test_start | Input | |----------| | i_test_enable | Input | |----------| | i_data_valid | Input | |----------| | i_abort | Input | |----------| | i_mem_start_addr | Input | |----------| | i_mem_data_in | Input | |----------| | i_ait21500_wrck | Input | |----------| | i_i15002ait_wso | Input | |----------| | i_reset_counter | Input | |----------| | o_mem_rd_enable | Output | |----------| | o_mem_rd_addr | Output | |----------| | wso_compare_status | Output | |----------| | status_ready | Output | |----------| | max_compare_wso_done | Output | |----------| | o_status | Output | ----------

## DFT-1239

(information) The provided parameter width "%d" for "%s" is invalid. Width for "%s" can range from "%d" to "%d" and be incremented only in steps of "%d".

**Description**

The Provided Parameter Width is Invalid.

**What Next**

Please Define Proper Width for Parameters as Mentioned in Below Table:

---------- | Name of Parameter | Default Value | Minimum Value | Maximum Value | ---------- | MEM_DATA_WIDTH | 64 | 8 | 64 | ---------- | MEM_ADDR_WIDTH | 16 | 8 | 64 or Till Memory supports | ---------- | TIMEOUT_STRAP_WIDTH | 4 | 4 | 8 | ---------- | PROP_DLY_CNT_VAL | 10 | 10 | 255 | ---------- | WRSTN_TOGGLE_PULSE_COUNT | 5 | 5 | 7 | ---------- | MAX_COMPARE_VALUE | 8 | 8 | 65535 | ----------

## DFT-1240

(Error) Could not find the specified DFT location name

Feedback

**Description**

Could not find the specified DFT location name

**What Next**

Please Review DFT Specifications

---

## DFT-1241

(Error) Unsuccessful DFT RTL Generation

**Description**

Unsuccessful DFT RTL Generation

**What Next**

Please Review DFT Specifications

---

## DFT-1242

(Error) Missing current design

**Description**

Missing current design

**What Next**

Please Review DFT Specifications

---

## DFT-1243

(Error) Design '%s' has no test model

**Description**

Design '%s' has no test model

**What Next**

Please Review DFT Specifications

---

## DFT-1246

(Error) Please use set_dft_configuration -dft_access enable to enable set_dft_access_element -type ait option.

Feedback

**Description**

Please use set_dft_configuration -dft_access enable to enable set_dft_access_element -type ait option.

**What Next**

Please Review DFT Specifications

---

## DFT-1247

(Error) Please use set_dft_configuration -dft_access/-mbist enable to enable set_dft_access_element command.

**Description**

Please use set_dft_configuration -dft_access/-mbist enable to enable set_dft_access_element command.

**What Next**

Please Review DFT Specifications

---

## DFT-1248

(Error) Wrong type provided. Valid values are top_server/sub_server/shadow_server/ait/1838_stap/processor/mmb_processor/cam_processor/wrapper/cam_wrapper/ams_wrapper/subchip

**Description**

Wrong type provided. Valid values are top_server/sub_server/shadow_server/ait/1838_stap/processor/mmb_processor/cam_processor/wrapper/cam_wrapper/ams_wrapper/subchip

**What Next**

Please Review DFT Specifications

---

## DFT-1250

(Error) Inputs are not fully shared for shared codec SI with dedicated SO.

**Description**

Inputs are not fully shared for shared codec SI with dedicated SO.

Feedback

**What Next**

Please Review DFT Specifications

---

## DFT-1251

(Error) Incorrect polarity specified for signal '%s' acceptable polarity : '%s'

**Description**

Incorrect polarity specified for signal '%s' acceptable polarity : '%s'

**What Next**

Please Review DFT Specifications

---

## DFT-1252

(Error) Codecs outputs sizes must be identical for shared codec SI with dedicated SO.

**Description**

Codecs outputs sizes must be identical for shared codec SI with dedicated SO.

**What Next**

Please Review DFT Specifications

---

## DFT-1254

(Error) Nothing to do here. Use set_dft_configuration -mbist/-dft_access enable to use 'plan_dft' command.

**Description**

The 'plan_dft' command is used when SMS network is present. SMS network is inserted when mbist or dft_access client is enabled.

**What Next**

User need to enable either mbist or dft_access or both to insert SMS Network. For example, 'set_dft_configuration -mbist enable' OR 'set_dft_configuration -dft_access enable'

---

## DFT-1255

(Error) Invalid format specified. Valid values are ctl/icl/pdl/ict/wks/hdl/masis

Feedback

**Description**

Invalid format specified. Valid values are ctl/icl/pdl/ict/wks/hdl/masis

**What Next**

Please Review DFT Specifications

---

## DFT-1256

(Error) Option "-design" is mandatory for icl/pdl/ict/wks/hdl/masis formats

**Description**

Option "-design" is mandatory for icl/pdl/ict/wks/hdl/masis formats

**What Next**

Please Review DFT Specifications

---

## DFT-1257

(Error) File %s does not exist or unreadable

**Description**

File %s does not exist or unreadable

**What Next**

Please Review DFT Specifications

---

## DFT-1258

(Error) Please use set_dft_configuration -dft_access/-mbist/-ieee1687 enable to enable set_dft_ring command.

**Description**

Please use set_dft_configuration -dft_access/-mbist/-ieee1687 enable to enable set_dft_ring command.

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1260

(Error) The provided parameter list does not contain the mandatory parameters %s. List of mandatory parameters: %s

**Description**

The provided parameter list does not contain the mandatory parameters %s. List of mandatory parameters: %s

**What Next**

Please Review DFT Specifications

## DFT-1261

(Error) The provided interface list does not contain the mandatory interfaces %s. List of mandatory interfaces: %s

**Description**

The provided interface list does not contain the mandatory interfaces %s. List of mandatory interfaces: %s

**What Next**

Please Review DFT Specifications

## DFT-1263

(Error) set_scan_group functionality with lockup_exists and serial_routed as FALSE option is not supported. Scan group specification will be discarded.

**Description**

set_scan_group supports -lockup_exists just with -serial_routed true option.

**What Next**

Please Review DFT Specifications

## DFT-1264

(Error) The specified class '%s' is not supported.

**Description**

The specified class '%s' is not supported.

Feedback

**What Next**

Please Review DFT Specifications

---

## DFT-1265

(Error) Expected width of "%s" being connected to "%s" is "[%d:%d]", but found it "[%d:%d]" in design.

**Description**

Expected width of "%s" being connected to "%s" is "[%d:%d]", but found it "[%d:%d]" in design.

**What Next**

Please Review DFT Specifications

---

## DFT-1270

(ERROR) The interface list is parsed as key-value pair of interface name and portname <interfaceName, portName>, please provide even number of arguments in the interface list. List of mandatory interface keywords %s.

**Description**

The interface list has key-value pair of interface name and portname <interfaceName, portName>. Therefore, it cannot have odd number of arguments.

**What Next**

Please provide even number of arguments in the interface list.

---

## DFT-1276

(Error) %s is not found in the netlist.

**Description**

This message is issued when cell, pin or port from scan specification is not found in netlist. For example,

Error: SO0 is not found in the netlist. (DFT-1276)

**What Next**

Check signals specification

Feedback

## DFT-1277

(Error) Could not find valid scan group element. It should be either ndmInstance or ctl segment.

**Description**

set_scan_group has incorrect -include specification: the elements are not identified either as cells or ctl segments.

**What Next**

Please Review DFT Specifications

## DFT-1278

(Error) Please use set_dft_configuration -hsio enable option.

**Description**

Please use set_dft_configuration -hsio enable option.

**What Next**

Please Review DFT Specifications

## DFT-1282

(Error) HSIO Configuration is empty.

**Description**

HSIO Configuration is empty.

**What Next**

Please Review DFT Specifications

## DFT-1286

(Error) Path %s has scan group elements declared in discontinuous order.

**Description**

Path %s has scan group elements declared in discontinuous order.

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1292

(Error) Requested %s scan chains in test mode %s, but only %s chains are available in the cores targeted by test mode %s.

**Description**

Requested %s scan chains in test mode %s, but only %s chains are available in the cores targeted by test mode %s.

**What Next**

Please Review DFT Specifications

## DFT-1294

(Error) %s PinMap package doesn't exist

**Description**

%s PinMap package doesn't exist

**What Next**

Please Review DFT Specifications

## DFT-1297

(Error) Multiple (%d) design ports map to the '%s' package pin.

**Description**

Multiple (%d) design ports map to the '%s' package pin.

**What Next**

Please Review DFT Specifications

## DFT-1299

(Error) Option '%s' is only allowed for 'default_partition'.

**Description**

Option '%s' is only allowed for 'default_partition'.

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1300

(Error) %s pin %s can not be a %s.

**Description**

set_scan_group -access_pin of specified type has wrong direction.

**What Next**

Please Review DFT Specifications

## DFT-1302

(Error) -occ_clock_weights option accepts only list of weights and clocks. Weights can take only positive integers or always_on or always_off.

**Description**

-occ_clock_weights option accepts only list of weights and clocks. Weights can take only positive integers or always_on or always_off.

**What Next**

Please Review DFT Specifications

## DFT-1303

(Error) -xtolerance high or -xtolerance extended is expected for XLBIST. BIST without xtolerance is not supported.

**Description**

BIST is mainly differentiated from normal logicbist by using the '-xtolerance high or -xtolerance extended' option of the "set_logicbist_configuration" command.

**What Next**

Please use '-xtolerance high or -xtolerance extended' switch in the "set_logicbist_configuration" command while using XLBIST configuration.

## DFT-1304

(ERROR) %s value is not supported for option %s. This option is ignored.

Feedback

**Description**

This error message is shown, when specified values of options do not match predefined enums. In this case the setting of the option is ignored.

**What Next**

Please review set_scan_group settings

---

## DFT-1306

(Error) Invalid clocks provided as argument to -occ_clock_weights

**Description**

Invalid clocks provided as argument to -occ_clock_weights

**What Next**

Please Review DFT Specifications

---

## DFT-1307

(Error) -reset_weights option accepts only list of weights and clocks. Weights can take only positive integers or or always_off.

**Description**

-reset_weights option accepts only list of weights and clocks. Weights can take only positive integers or or always_off.

**What Next**

Please Review DFT Specifications

---

## DFT-1308

(Error) Number of chainCounts in the list specified by -internal_chains_per_misr must be <= the number of outputs specified (-outputs).

**Description**

Number of chainCounts in the list specified by -internal_chains_per_misr must be <= the number of outputs specified (-outputs).

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1309

(Error) Sum of chainCounts in the list of -internal_chains_per_misr option should match the -chain_count value

**Description**

Sum of chainCounts in the list of -internal_chains_per_misr option should match the -chain_count value

**What Next**

Please Review DFT Specifications

## DFT-1312

(Error) -scan_en option is mandatory with type WC_Q0/WC_Q1/WC_Q0_AO/WC_Q1_AO cell

**Description**

-scan_en option is mandatory with type WC_Q0/WC_Q1/WC_Q0_AO/WC_Q1_AO cell

**What Next**

Please Review DFT Specifications

## DFT-1313

(Error) No atpg_shift signal defined for port/pin specified with '-scan_en' option

**Description**

No atpg_shift signal defined for port/pin specified with '-scan_en' option

**What Next**

Please Review DFT Specifications

## DFT-1314

(Error) Number of misrLengths in the list specified by -misr_lengths must be equal to the number of chainCounts specified by -internal_chains_per_misr.

**Description**

Number of misrLengths in the list specified by -misr_lengths must be equal to the number of chainCounts specified by -internal_chains_per_misr.

Feedback

**What Next**

Please Review DFT Specifications

---

## DFT-1316

(Error) Boundary Cell Types AC_1, AC_2, AC_7, AC_SELU, AC_SELX can't be used for ports which are not AC ports.

**Description**

Boundary Cell Types AC_1, AC_2, AC_7, AC_SELU, AC_SELX can't be used for ports which are not AC ports.

**What Next**

Please Review DFT Specifications

---

## DFT-1322

(Error) Failed to create heterogeneous collection

**Description**

Failed to create heterogeneous collection

**What Next**

Please Review DFT Specifications

---

## DFT-1328

(Error) %s '-signal' %s found for %s custom BC cell.

**Description**

You receive this error message if 1. When needed signal is not specified using interface option of define_dft_design 2. When extra signal specified for custom BC cell

Error: Extra signal update_clk found for "BC_4" custom BC cell. (DFT-1328) Error: The signal update_clk not found for "BC_2" custom BC cell. (DFT-1328)

**What Next**

Define the correct DFT signal for specified custom BC cell.

Feedback

## DFT-1329

(Error) Could not find name in set_dft_access_element in 3DIC.

**Description**

Could not find name in set_dft_access_element in 3DIC.

**What Next**

Please Review DFT Specifications

## DFT-1339

(Error) Number of scan inputs shared across CODECs should be atleast %d.

**Description**

Number of scan inputs shared across CODECs should be atleast %d.

**What Next**

Please Review DFT Specifications

## DFT-1340

(Error) An integer greater than or equal to 0 should be specified

**Description**

An integer greater than or equal to 0 should be specified

**What Next**

Please Review DFT Specifications

## DFT-1341

(Error) An integer greater than or equal to 0 should be specified

**Description**

An integer greater than or equal to 0 should be specified

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1343

(Error) Incorrect port for security_lock

**Description**

Incorrect port for security_lock

**What Next**

Please Review DFT Specifications

## DFT-1344

(Error) Incorrect type for dft_signal

**Description**

Incorrect type for dft_signal

**What Next**

Please Review DFT Specifications

## DFT-1350

(Error) Option "-instance" is supported for "top_server", "sub_server", "processor", "mmb_processor", "cam_processor", "ams_wrapper" and "subchip" type only

**Description**

Option "-instance" is supported for "top_server", "sub_server", "processor", "mmb_processor", "cam_processor", "ams_wrapper" and "subchip" type only

**What Next**

Please Review DFT Specifications

## DFT-1351

(Error) Option "-module" is supported for "top_server", "sub_server", "processor", "mmb_processor", "cam_processor", "wrapper", "cam_wrapper", "ams_wrapper" and "subchip" type only

Feedback

**Description**

Option "-module" is supported for "top_server", "sub_server", "processor", "mmb_processor", "cam_processor", "wrapper", "cam_wrapper", "ams_wrapper" and "subchip" type only

**What Next**

Please Review DFT Specifications

---

## DFT-1352

(Error) Option "-parameter" is not supported for "%s" type

**Description**

Option "-parameter" is not supported for "%s" type

**What Next**

Please Review DFT Specifications

---

## DFT-1353

(Error) Option "-instance"/"-module" is mandatory for "%s" type

**Description**

Option "-instance"/"-module" is mandatory for "%s" type

**What Next**

Please Review DFT Specifications

---

## DFT-1354

(Error) Option "-location" is supported for "top_server", "sub_server", "processor", "mmb_processor", "cam_processor", "wrapper" and "ams_wrapper" type only

**Description**

Option "-location" is supported for "top_server", "sub_server", "processor", "mmb_processor", "cam_processor", "wrapper" and "ams_wrapper" type only

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1355

(Error) Option "-module" is mandatory with option "-instance" for "processor", "mmb_processor", "cam_processor" and "ams_wrapper" type

**Description**

Option "-module" is mandatory with option "-instance" for "processor", "mmb_processor", "cam_processor" and "ams_wrapper" type

**What Next**

Please Review DFT Specifications

## DFT-1356

(Error) Option "-server_select_instruction" is only supported for "top_server" type

**Description**

Option "-server_select_instruction" is only supported for "top_server" type

**What Next**

Please Review DFT Specifications

## DFT-1357

(Error) Module '%s' is not a valid subchip at current level

**Description**

Module '%s' is not a valid subchip at current level

**What Next**

Please Review DFT Specifications

## DFT-1358

(Error) No compiler found for component '%s'. Read compiler libraries before this command

**Description**

You receive this error message if no compiler is found for the type specified in set_dft_access_element command. Each type must have a associated compiler as some sanity checks use the compiler information.

Feedback

**What Next**

User must make sure they are reading the compilers using 'set_mbist_configuration/ set_dft_access_configuration -compiler_libraries' command and the path contain the compiler for specified type.

---

## DFT-1359

(Error) LogicBIST Configuration is empty.

**Description**

LogicBIST Configuration is empty.

**What Next**

Please Review DFT Specifications

---

## DFT-1360

(Error) Parameter '%s' not found in compiler '%s'

**Description**

Parameter '%s' not found in compiler '%s'

**What Next**

Please Review DFT Specifications

---

## DFT-1361

(Error) Invalid hierarchical name '%s'. Use '%c' as hierarchy separator

**Description**

Invalid hierarchical name '%s'. Use '%c' as hierarchy separator

**What Next**

Please Review DFT Specifications

---

## DFT-1362

(Error) Cannot use "-type processor/mmb_processor/cam_processor/wrapper/ cam_wrapper/ams_wrapper/top_server/sub_server/shadow_server/subchip" with "name" positional argument

Feedback

**Description**

Cannot use "-type processor/mmb_processor/cam_processor/wrapper/cam_wrapper/ams_wrapper/top_server/sub_server/shadow_server/subchip" with "name" positional argument

**What Next**

Please Review DFT Specifications

---

## DFT-1363

(Error) Option "-interface" is supported for "top_server", "sub_server", "processor" , "mmb_processor" , "cam_processor", "ams_wrapper", "subchip" type only

**Description**

Option "-interface" is supported for "top_server", "sub_server", "processor" , "mmb_processor" , "cam_processor", "ams_wrapper", "subchip" type only

**What Next**

Please Review DFT Specifications

---

## DFT-1364

(Error) Unable to find UPF legal hierarchy for %s

**Description**

You receive this error message when tool is unable to find a UPF legal hierarchy for new cell insertion

---

## DFT-1365

(Error) Width of pin '%s' (W: %d) does not match with width of connection '%s' (W: %d)

**Description**

Width of pin '%s' (W: %d) does not match with width of connection '%s' (W: %d)

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1366

(Error) Incorrect input for option '-algorithm'. Expecting -default/-hardwired/-soft_programmable <name1> <name2> <name3>...

**Description**

Incorrect input for option '-algorithm'. Expecting -default/-hardwired/-soft_programmable <name1> <name2> <name3>...

**What Next**

Please Review DFT Specifications

## DFT-1368

(Error) Module definition of instance "%s" not found.

**Description**

Module definition of instance "%s" not found.

**What Next**

Please Review DFT Specifications

## DFT-1371

(Error) Incorrect algorithm type %s is specified. Valid types are -default/-hardwired/-soft_programmable

**Description**

Incorrect algorithm type %s is specified. Valid types are -default/-hardwired/-soft_programmable

**What Next**

Please Review DFT Specifications

## DFT-1372

(Error) '%s' algorithm not provided

**Description**

'%s' algorithm not provided

Feedback

**What Next**

Please Review DFT Specifications

---

## DFT-1373

(Error) Max size(%lu) of selected algorithms is greaten than user specifed tbox size(%lu)

**Description**

Max size(%lu) of selected algorithms is greaten than user specifed tbox size(%lu)

**What Next**

Please Review DFT Specifications

---

## DFT-1374

(Error) Option "-algorithm" is supported for "processor", "mmb_processor" and "cam_processor" type only

**Description**

Option "-algorithm" is supported for "processor", "mmb_processor" and "cam_processor" type only

**What Next**

Please Review DFT Specifications

---

## DFT-1375

(Error) Custom algorithm must be enabled to select algorithm for processors

**Description**

Custom algorithm must be enabled to select algorithm for processors

**What Next**

Please Review DFT Specifications

---

## DFT-1376

(Error) Cannot select soft programmable algorithm('%s') as defualt algorithm because parameter 'talgo_prog_enable' is disabled

Feedback

**Description**

Cannot select soft programmable algorithm('%s') as defualt algorithm because parameter 'talgo_prog_enable' is disabled

**What Next**

Please Review DFT Specifications

---

## DFT-1377

(Error) Cannot select 'TSR' as defualt algorithm because parameter 'treg_enable' is disabled

**Description**

Cannot select 'TSR' as defualt algorithm because parameter 'treg_enable' is disabled

**What Next**

Please Review DFT Specifications

---

## DFT-1378

(Error) Parameter 'treg_enable' cannot be disabled if parameter 'talgo_prog_enable' is enabled

**Description**

Parameter 'treg_enable' cannot be disabled if parameter 'talgo_prog_enable' is enabled

**What Next**

Please Review DFT Specifications

---

## DFT-1379

(Error) Default algorithm %s not found in hardwired or soft programmable algorithms

**Description**

Default algorithm %s not found in hardwired or soft programmable algorithms

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1380

(Error) The input memory file %s is empty or does not exist

**Description**

The input memory file %s is empty or does not exist

**What Next**

Please Review DFT Specifications

## DFT-1381

(Error) -input_memory or -input_icl option specifed without -memory option

**Description**

-input_memory option specifed without -memory option

**What Next**

Please Review DFT Specifications

## DFT-1383

(Error) Could not find valid scan group exclude element. It should be either ndmInstance or ctl segment.

**Description**

set_scan_group has incorrect -exclude specification: the elements are not identified either as cells or ctl segments.

**What Next**

Please Review DFT Specifications

## DFT-1384

(Error) Enable '%s' for using test_mode %s

**Description**

Enable '%s' for using test_mode %s

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1385

(Error) Multiple arguments provided with "%s" option

**Description**

Multiple arguments provided with "%s" option

**What Next**

Please Review DFT Specifications

## DFT-1390

(Error) Algorithm '%s' is present in both 'hardwired' and 'soft_programmable' algorithm lists

**Description**

Algorithm '%s' is present in both 'hardwired' and 'soft_programmable' algorithm lists

**What Next**

Please Review DFT Specifications

## DFT-1391

(Error) Algorithms '%s' is(are) not allowed in default and soft programmable algorithms

**Description**

Algorithms '%s' is(are) not allowed in default and soft programmable algorithms

**What Next**

Please Review DFT Specifications

## DFT-1395

(Error) Invalid bit/range('%s') for port/pin '%s' of design/instance '%s'

**Description**

Invalid bit/range('%s') for port/pin '%s' of design/instance '%s'

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1397

(Error) Port/Pin '%s' not found in design/instance '%s'

**Description**

Port/Pin '%s' not found in design/instance '%s'

**What Next**

Please Review DFT Specifications

## DFT-1398

(Error) Endianness mismatch for port/pin '%s' of design/instance '%s'

**Description**

Endianness mismatch for port/pin '%s' of design/instance '%s'

**What Next**

Please Review DFT Specifications

## DFT-1399

(Error) Port '%s' not found on component '%s'

**Description**

Port '%s' not found on component '%s'

**What Next**

Please Review DFT Specifications

## DFT-1401

(Error) Ac ports list doesn't exist

**Description**

Ac ports list doesn't exist

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1403

(Error) Linkage ports list doesn't exist

**Description**

Linkage ports list doesn't exist

**What Next**

Please Review DFT Specifications

## DFT-1405

(Information) Accepted %s dft signal specification in partition '%s' for %s

**Description**

DFT signal specification has been successfully accepted for the given partition and mode.

## DFT-1406

(Error) %s is not a pin/cell/hierarchy/clock and cannot be excluded

**Description**

%s is not a pin/cell/hierarchy/clock and cannot be excluded

**What Next**

Please Review DFT Specifications

## DFT-1414

(Error) Only codecs with identical number of outputs is supported in 'test_enable_codec_sharing' flow

**Description**

Only codecs with identical number of outputs is supported in 'test_enable_codec_sharing' flow

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1415

(Error) Only codecs with identical number of inputs is supported in 'test_enable_codec_sharing' flow

**Description**

Only codecs with identical number of inputs is supported in 'test_enable_codec_sharing' flow

**What Next**

Please Review DFT Specifications

## DFT-1416

(Error) DFTMax Codec test mode must have maximum number of %s in 'test_enable_codec_sharing' flow

**Description**

DFTMax Codec test mode must have maximum number of %s in 'test_enable_codec_sharing' flow

**What Next**

Please Review DFT Specifications

## DFT-1422

(Error) Cannot open STIL file:'%s'.

**Description**

Cannot open STIL file:'%s'.

**What Next**

Please Review DFT Specifications

## DFT-1423

(Error) No 'core_module_name' and 'pattern_exec' found for label '%s'.

**Description**

No 'core_module_name' and 'pattern_exec' found for label '%s'.

Feedback

**What Next**

Please Review DFT Specifications

---

## DFT-1424

(Error) No module with name '%s' found

**Description**

No module with name '%s' found

**What Next**

Please Review DFT Specifications

---

## DFT-1426

(Error) File %s does not exist or unreadable.

**Description**

File %s does not exist or unreadable.

**What Next**

Please Review DFT Specifications

---

## DFT-1430

(Error) Unable to open '%s' file.

**Description**

Unable to open '%s' file.

**What Next**

Please Review DFT Specifications

---

## DFT-1433

(Error) Unable to collect server compiler from compiler libraries '%s'.

**Description**

Unable to collect server compiler from compiler libraries '%s'.

Feedback

**What Next**

Please Review DFT Specifications

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## DFT-1434

(Error) Unable to collect wrapper compiler from compiler library '%s'.

**Description**

Unable to collect wrapper compiler from compiler library '%s'.

**What Next**

Please Review DFT Specifications

---

## DFT-1437

(Error) Permission denied. Cannot access ICL file '%s'.

**Description**

Permission denied. Cannot access ICL file '%s'.

**What Next**

Please Review DFT Specifications

---

## DFT-1440

(Error) '%s' signal not found for design '%s'.

**Description**

'%s' signal not found for design '%s'.

**What Next**

Please Review DFT Specifications

---

## DFT-1446

(Error) Corresponding PDL file not specified for ICL file '%s'

**Description**

Corresponding PDL file not specified for ICL file '%s'

Feedback

**What Next**

Please Review DFT Specifications

---

## DFT-1448

(Error) Core instance '%s' not found in design.

**Description**

Core instance to module mapping not found during port_patterns/write_test_protocol command.

**What Next**

Please look that correct icl/pdl files are provided in 'port_patterns' or 'write_test_protocol' commands.

SH "SEE ALSO"

```
port_patterns
write_test_protocol
```

---

## DFT-1451

(Error) Directory '%s' for core specific PDLs not found.

**Description**

Directory '%s' for core specific PDLs not found.

**What Next**

Please Review DFT Specifications

---

## DFT-1452

(Error) No 'pattern_file' found for label '%s'.

**Description**

No 'pattern_file' found for label '%s'.

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1453

(Error) Unable to extract pattern exec from STIL file '%s'.

**Description**

Unable to extract pattern exec from STIL file '%s'.

**What Next**

Please Review DFT Specifications

## DFT-1454

(Error) Permission denied. Cannot access STIL file '%s'.

**Description**

Permission denied. Cannot access STIL file '%s'.

**What Next**

Please Review DFT Specifications

## DFT-1455

(Error) Verbose option value is invalid. Valid values are 'on' and 'off'

**Description**

Verbose option value is invalid. Valid values are 'on' and 'off'

**What Next**

Please Review DFT Specifications

## DFT-1457

(Error) No group names specified for '-merge' option.

**Description**

No group names specified for '-merge' option.

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1458

(Error) The file '%s' has not been generated by Testmax. This cannot be used as a generated config file for 'port_patterns' command.

**Description**

The file '%s' has not been generated by Testmax. This cannot be used as a generated config file for 'port_patterns' command.

**What Next**

Please Review DFT Specifications

## DFT-1463

(Error) Template file '%s' does not exist or unreadable.

**Description**

Template file '%s' does not exist or unreadable.

**What Next**

Please Review DFT Specifications

## DFT-1464

(Error) Directory '%s' does not exist.

**Description**

Directory '%s' does not exist.

**What Next**

Please Review DFT Specifications

## DFT-1465

(Error) SPF file '%s' does not exist or unreadable.

**Description**

SPF file '%s' does not exist or unreadable.

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1466

(Error) Invalid value '%s' specified for '-occ_only' option.

**Description**

Invalid value '%s' specified for '-occ_only' option.

**What Next**

Please Review DFT Specifications

## DFT-1467

(Error) Invalid value '%s' specified for '-occ_bypass_only' option.

**Description**

Invalid value '%s' specified for '-occ_bypass_only' option.

**What Next**

Please Review DFT Specifications

## DFT-1468

(Error) Config file '%s' does not exist or unreadable.

**Description**

Config file '%s' does not exist or unreadable.

**What Next**

Please Review DFT Specifications

## DFT-1469

(Error) The file '%s' specified for '-generated_config' is not valid.

**Description**

The file '%s' specified for '-generated_config' is not valid.

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1470

(Error) No configuration found for group '%s'.

**Description**

No configuration found for group '%s'.

**What Next**

Please Review DFT Specifications

## DFT-1471

(Error) Use command 'set_dft_access_configuration -group <group_name> -cores <core_1 core_2 ... core_N>' to specify cores group.

**Description**

Use command 'set_dft_access_configuration -group <group_name> -cores <core_1 core_2 ... core_N>' to specify cores group.

**What Next**

Please Review DFT Specifications

## DFT-1476

(Error) Permission denied! Could not create Stilgen config file '%s'.

**Description**

Permission denied! Could not create Stilgen config file '%s'.

**What Next**

Please Review DFT Specifications

## DFT-1482

(Error) Verilog list file '%s' does not exist or unreadable.

**Description**

Verilog list file '%s' does not exist or unreadable.

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1483

(Error) System Verilog list file '%s' does not exist or unreadable.

**Description**

System Verilog list file '%s' does not exist or unreadable.

**What Next**

Please Review DFT Specifications

## DFT-1489

(Error) Migm run failed. %s.

**Description**

You receive this error message when masis generation failed during 'port_patterns' or 'write_test_protocol' commands.

**What Next**

Please look that correct icl/pdl files are provided in 'port_patterns' or 'write_test_protocol' commands.

SH "SEE ALSO"

```
port_patterns
write_test_protocol
```

## DFT-1490

(Error) YA Script generation failed. %s.

**Description**

You receive this error message when you do not provide valid input to 'port_patterns' and 'write_test_protocol' commands.

**What Next**

Please use correct inputs to 'port_patterns' and/or 'write_test_protocol'.

SH "SEE ALSO"

```
port_patterns
write_test_protocol
```

Feedback

## DFT-1494

(Error) Hier file '%s' does not exist or unreadable.

**Description**

Hier file '%s' does not exist or unreadable.

**What Next**

Please Review DFT Specifications

## DFT-1495

(Error) Masis files directory '%s' does not exist or unreadable.

**Description**

Masis files directory '%s' does not exist or unreadable.

**What Next**

Please Review DFT Specifications

## DFT-1496

(Error) Permission denied. Can't open script file '%s' for writing.

**Description**

Permission denied. Can't open script file '%s' for writing.

**What Next**

Please Review DFT Specifications

## DFT-1499

(Error) Unable to update simulation models in '%s' file.

**Description**

Unable to update simulation models in '%s' file.

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1500

(Error) -pins options is only supported with -type none

**Description**

-pins options is only supported with -type none

**What Next**

Please Review DFT Specifications

## DFT-1501

(Error) Option '-ded_hier_path' is missing. Ignoring '-ded_hier_wrp_clock' option.

**Description**

Option '-ded_hier_path' is missing. Ignoring '-ded_hier_wrp_clock' option.

**What Next**

Please Review DFT Specifications

## DFT-1502

(Error) -ded_hier_wrp_clock '%s' is not defined as wrapper clock. Ignoring 'ded_hier_wrp_clock' option.

**Description**

-ded_hier_wrp_clock '%s' is not defined as wrapper clock. Ignoring 'ded_hier_wrp_clock' option.

**What Next**

Please Review DFT Specifications

## DFT-1503

(Error) '-class bsd' and '-type AC_*/BC_*' must come together.

**Description**

'-class bsd' and '-type AC_*/BC_*' must come together.

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1504

(Error) Option '%s' is supported only for class '%s'.

**Description**

Option '%s' is supported only for class '%s'.

**What Next**

Please Review DFT Specifications

## DFT-1505

(Error) Incorrect BSR type '%s' specified for function '%s'.

**Description**

Incorrect BSR type '%s' specified for function '%s'.

**What Next**

Please Review DFT Specifications

## DFT-1506

(Error) Invalid ports specified for function '%s', Please verify ports directions.

**Description**

Invalid ports specified for function '%s', Please verify ports directions.

**What Next**

Please Review DFT Specifications

## DFT-1507

(Error) Invalid function type '%s'.

**Description**

Invalid function type '%s'.

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1508

(Error) Option '%s' is not enabled. Ignoring '%s' option.

**Description**

Option '%s' is not enabled. Ignoring '%s' option.

**What Next**

Please Review DFT Specifications

## DFT-1509

(Error) Option '-share' is applicable only for Control/Ac Select Boundary Cells.

**Description**

Option '-share' is applicable only for Control/Ac Select Boundary Cells.

**What Next**

Please Review DFT Specifications

## DFT-1510

(Error) %s signal missing. Cannot perform clock masking

**Description**

You receive this warning message in Dft core wrapper insertion when tool is trying to find signal to perform clock masking. This error is for 2 types of signals 1) core_cell_clock_control 2) wrp_cell_clock_control

## DFT-1511

(Error) Specified streaming compressed chain count %d is less than the minimum value of %d for %d input(s)/output(s)

**Description**

Specified streaming compressed chain count %d is less than the minimum value of %d for %d input(s)/output(s)

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1512

(Error) Specified or Computed number of Shift Power Chain length is less than 2

**Description**

Specified or Computed number of Shift Power Chain length is less than 2

**What Next**

Please Review DFT Specifications

## DFT-1515

(Error) OCC Chain Count is greater than total number of Cycles

**Description**

OCC Chain Count is greater than total number of Cycles

**What Next**

Please Review DFT Specifications

## DFT-1516

(Error) A [set_scan_path -class occ/spc] is required if DFT Shift Power Control feature has been enabled.

**Description**

A [set_scan_path -class occ/spc] is required if DFT Shift Power Control feature has been enabled.

**What Next**

Please Review DFT Specifications

## DFT-1517

(Error) [set_scan_compression_configuration -virtual_chain_count ] is not supported for DFTMax Ultra.

**Description**

[set_scan_compression_configuration -virtual_chain_count ] is not supported for DFTMax Ultra.

Feedback

**What Next**

Please Review DFT Specifications

---

## DFT-1518

(Error) Inconsistent xtol mode between the top level and core level, xtol mode at both top level and core level should either be default or high.

**Description**

Inconsistent xtol mode between the top level and core level, xtol mode at both top level and core level should either be default or high.

**What Next**

Please Review DFT Specifications

---

## DFT-1519

(Error) Minimum -chain_count is 10 for Streaming Compression

**Description**

Minimum -chain_count is 10 for Streaming Compression

**What Next**

Please Review DFT Specifications

---

## DFT-1521

(Error) Found empty string when processing: %s

**Description**

Empty string received when trying to create a category or a message. or when calling options -(de)activate_list. A string was mandatory

**What Next**

Please add your category or message with a valid string or review your call to enable_dft_debug -activate_list/-deactivate_list, use a valid string

---

## DFT-1522

(Error) Message %s couldn't be %s

Feedback

**Description**

(De)activation of message failed due to invalid tag format

**What Next**

Please review you message tag.

---

## DFT-1524

(Error) Syntax Error: Invalid category format -- %s

**Description**

Syntax error related to the category format

**What Next**

Please review the syntax of the category you are trying to (de)activate or create

---

## DFT-1536

(Error) The options -chain and -cell cannot be specified at the same time.

**Description**

The options -chain and -cell cannot be specified at the same time.

**What Next**

Please Review DFT Specifications

---

## DFT-1537

(Error) Specified rule %s is not supported in %s

**Description**

Specified rule %s is not supported in %s

**What Next**

Please Review DFT Specifications

---

## DFT-1538

(Error) %s is not a valid mode name. Mode name can only contain alphanumeric characters (a-z, A-Z, 0-9) and the underscore (_).

Feedback

**Description**

%s is not a valid mode name. Mode name can only contain alphanumeric characters (a-z, A-Z, 0-9) and the underscore (_).

**What Next**

Please Review DFT Specifications

---

## DFT-1539

(Error) Specified test mode is empty

**Description**

Specified test mode is empty

**What Next**

Please Review DFT Specifications

---

## DFT-1540

(Error) Missing TestMode port and/or encoding value in option -encoding

**Description**

Missing TestMode port and/or encoding value in option -encoding

**What Next**

Please Review DFT Specifications

---

## DFT-1541

(Error) Mode %s not found.

**Description**

You receive this error message because specified mode is not found.

---

## DFT-1542

(Error) Command set_streaming_compression_configuration is not supported.

**Description**

Command set_streaming_compression_configuration is not supported.

Feedback

**What Next**

Please Review DFT Specifications

---

## DFT-1543

(Error) Invalid Timing specification [%f, %f] or timing is not specified for the clock signal.

**Description**

Invalid Timing specification [%f, %f] or timing is not specified for the clock signal.

**What Next**

Please Review DFT Specifications

---

## DFT-1544

(Error) Invalid option value '%s' in %s

**Description**

Invalid option value '%s' in %s

**What Next**

Please Review DFT Specifications

---

## DFT-1545

(Error) Number of Test Mode signals defined in option -encoding of define_test_mode %d is not sufficient to encode %d Test Modes.

**Description**

Number of Test Mode signals defined in option -encoding of define_test_mode %d is not sufficient to encode %d Test Modes.

**What Next**

Please Review DFT Specifications

---

## DFT-1546

(Error) Test mode '%s' is not defined

**Description**

Test mode '%s' is not defined

Feedback

**What Next**

Please Review DFT Specifications

---

## DFT-1548

(Error) Test mode '%s' is not defined

**Description**

Test mode '%s' is not defined

**What Next**

Please Review DFT Specifications

---

## DFT-1549

(Error) No TestMode pin found with name '%s'.

**Description**

No TestMode pin found with name '%s'.

**What Next**

Please Review DFT Specifications

---

## DFT-1550

(Error) Specified rule %s is not supported in %s

**Description**

Specified rule %s is not supported in %s

**What Next**

Please Review DFT Specifications

---

## DFT-1551

(Error) One switch out of error, allow or ignore should be active in a single invocation.

**Description**

One switch out of error, allow or ignore should be active in a single invocation.

Feedback

**What Next**

Please Review DFT Specifications

---

## DFT-1552

(Error) Invalid TDR pin usage : '%s'

**Description**

Invalid TDR pin usage : '%s'

**What Next**

Please Review DFT Specifications

---

## DFT-1554

(Error) Unsupported signal type '%s' specified for TDR pin : '%s'

**Description**

Unsupported signal type '%s' specified for TDR pin : '%s'

**What Next**

Please Review DFT Specifications

---

## DFT-1555

(Error) No hookup pin specified for TDR pin '%s' as Testdata

**Description**

No hookup pin specified for TDR pin '%s' as Testdata

**What Next**

Please Review DFT Specifications

---

## DFT-1556

(Error) -tdr_pin and -hookup_pin options are mutually exclusive except for type 'TestData'

**Description**

-tdr_pin and -hookup_pin options are mutually exclusive except for type 'TestData'

Feedback

**What Next**

Please Review DFT Specifications

---

## DFT-1557

(Error) '-test_mode_port' switch doesn't accept collection it expects name of pin/port. Either input pin is invalid or get_object_name should be used with collection to get pin/port name.

**Description**

'-test_mode_port' switch doesn't accept collection it expects name of pin/port. Either input pin is invalid or get_object_name should be used with collection to get pin/port name.

**What Next**

Please Review DFT Specifications

---

## DFT-1560

(Error) configuration ports should be of type TestControl

**Description**

configuration ports should be of type TestControl

**What Next**

Please Review DFT Specifications

---

## DFT-1561

(Error) Tdr pin type for usage 'sms_dm' | 'shs_dm' must be 'TestControl'.

**Description**

Tdr pin type for usage 'sms_dm' | 'shs_dm' must be 'TestControl'.

**What Next**

Please Review DFT Specifications

---

## DFT-1562

(Error) Invalid pin index/name specified for tdr pin '%s'. Correct specification : TDR[msb:lsb] or TDR[index] or TDR

Feedback

**Description**

Invalid pin index/name specified for tdr pin '%s'. Correct specification : TDR[msb:lsb] or TDR[index] or TDR

**What Next**

Please Review DFT Specifications

---

## DFT-1563

(Error) Pin length of load pin '%s' doesn't match to TDR pin length

**Description**

Pin length of load pin '%s' doesn't match to TDR pin length

**What Next**

Please Review DFT Specifications

---

## DFT-1565

(Error) Invalid -shift_power_disable pin : %s

**Description**

This error message is issued during *set_scan_compression_configuration* if spc_disable pin specified using the *-shift_power_disable* option is Invalid.

For example,

Error: Invalid -shift_power_disable pin : SPC_DISABLE (DFT-1565)

**What Next**

Please specify a valid spc_disable pin in the *set_scan_compression_configuration* command to resolve this Error.

---

## DFT-1567

(Error) The specified port %s in option %s is not an %s port.

**Description**

The specified port %s in option %s is not an %s port.

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1568

(Error) Feed through connection for output %s already exists in model or user specification.

**Description**

Feed through connection for output %s already exists in model or user specification.

**What Next**

Please Review DFT Specifications

## DFT-1569

(Error) Mode '%s' is not defined

**Description**

Mode '%s' is not defined

**What Next**

Please Review DFT Specifications

## DFT-1570

(Error) Missing TestMode port and/or encoding value in option -encoding

**Description**

Missing TestMode port and/or encoding value in option -encoding

**What Next**

Please Review DFT Specifications

## DFT-1571

(Error) %s is not defined as a TestMode port

**Description**

%s is not defined as a TestMode port

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1572

(Error) The value %s for the port is incorrect

**Description**

The value %s for the port is incorrect

**What Next**

Please Review DFT Specifications

## DFT-1573

(Error) Pin length of load pin '%s' when processing indx '%d' exceeded TDR bus length.

**Description**

This error message is issued when TDR bus length is less than the pin length of the load pins.

Error: Pin length of load pin 'dm1' when processing indx '35' exceeded TDR bus length (DFT-1573)

## DFT-1574

(Error) Found more than one ISO/LS cell for input port %s. Cannot skip ISO/LS cell for the port.

**Description**

This error message is issued during UPF analysis for adding dedicated wrapper when more than a single ISO/LS is connected to a net which itself is connected to an input port. This happens when the app option "test_soc_core_wrap_skip_iso_cells" is set to true.

For example,

Error: Found more than one ISO/LS cell for input port i_rd. Cannot skip ISO/LS cell for the port. (DFT-1574)

**What Next**

Tool does not handle this scenario as of now. This will be supported in the near future to skip all ISO/LS cells and add DW cell for all those paths.

Feedback

## DFT-1575

(Error) "Options '-clock_mixing', '-voltage_mixing' and '-power_domain_mixing' are allowed with app option 'dft.scan.allow_wrapper_chain_clock_mixing_setting' only."

**Description**

This error message is issued when user try to specify the options '-clock_mixing', '-voltage_mixing' and '-power_domain_mixing' with 'set_wrapper_configuration' without enabling the app option 'dft.scan.allow_wrapper_chain_clock_mixing_setting'.

**What Next**

Please retry the options with enabling the app option 'dft.scan.allow_wrapper_chain_clock_mixing_setting'.

## DFT-1579

(Error) Option '%s' is not supported if option -type is not user specified.

**Description**

You receive this error message when -type option is not user specified and user has specified option %s in set_boundary_cell command. If option -type is specified the error will be removed.

## DFT-1581

(Error) Invalid class name, only class pmm is allowed.

**Description**

Valid value for -class is pmm. No other value is allowed.

**What Next**

Please Review DFT Specifications

## DFT-1582

(Error) Slack bottom limit cannot be greater than slack top limit.

**Description**

Slack bottom limit cannot be greater than slack top limit.

Feedback

**What Next**

Please Review DFT Specifications

---

## DFT-1583

(Error) Either '-clock_buf_ref_name' or '-clock_inv_ref_name' must be specified.

**Description**

Either '-clock_buf_ref_name' or '-clock_inv_ref_name' must be specified.

**What Next**

Please Review DFT Specifications

---

## DFT-1584

(Error) Invalid stitch methodology, should be either coverage or slack.

**Description**

Invalid stitch methodology, should be either coverage or slack.

**What Next**

Please Review DFT Specifications

---

## DFT-1585

(Error) Value of option '-clock_latency_percentage' should be a positive integer.

**Description**

'-clock_latency_percentage' value should be greater than 0.

**What Next**

Please Review DFT Specifications

---

## DFT-1586

(Warning) More than one modules specified in switch '%s'. Only first module will be considered.

**Description**

Among the list of module names provided, only first module will be considered for processing.

Feedback

**What Next**

Please Review DFT Specifications

---

## DFT-1587

(Warning) More than one port/pin specified in switch '%s'. Only first port/pin will be considered.

**Description**

Among the list of port/pins provided in the switch, only first one will be considered for processing.

**What Next**

Please Review DFT Specifications

---

## DFT-1588

(Error) Invalid values provided in '-stage'. Valid stitching stages are 'postcts' and 'postroute'.

**Description**

Invalid values provided in '-stage'. Valid stitching stages are 'postcts' and 'postroute'.

**What Next**

Please Review DFT Specifications

---

## DFT-1589

(Error) Invalid values provided in '-stitching_methodology'. Valid stitching method are 'full', 'restitch', and 'none'.

**Description**

Invalid values provided in '-stitching_methodology'. Valid stitching method are 'full', 'restitch', and 'none'.

**What Next**

Please Review DFT Specifications

---

## DFT-1592

(Warning) Invalid option value '%s' in set_slm_stitching_strategy '%s'.

Feedback

**Description**

Please look into 'man set_slm_stitching_strategy' output in fc_shell to know about valid values of each option.

---

## DFT-1600

(Error) Scan path name should not start with digit.

**Description**

Scan path name should not start with digit.

**What Next**

Please Review DFT Specifications

---

## DFT-1650

(Error) The %s name provided "%s" is invalid

**Description**

The %s name provided "%s" is invalid

**What Next**

Please Review DFT Specifications

---

## DFT-1651

(Error) Multiple file formats specifed with -format option

**Description**

Multiple file formats specifed with -format option

**What Next**

Please Review DFT Specifications

---

## DFT-1652

(Error) Architecting plan not available. generate_dft either failed or not run

**Description**

Architecting plan not available. generate_dft either failed or not run

Feedback

**What Next**

Please Review DFT Specifications

---

## DFT-1653

(warning) No port specified, specify port or port list

**Description**

You receive this warning message when no port is specified for test point.

---

## DFT-1654

(Error) Could not find '%s' in the model

**Description**

You receive this error message when you specify wrong signal in '-control_signal' field of 'set_testability_configuration' command.

**What Next**

Please use correct signal name in '-control_signal' field of 'set_testability_configuration' command.

---

## DFT-1655

(Error) The signal '%s' must be specified as '%s'

**Description**

You receive this error message when you specify wrong signal type with 'set_testability_configuration' command.

**What Next**

Please use correct signal type with 'set_testability_configuration' command.

---

## DFT-1656

(Error) The ['%s'] field value range is '%d' to '%d'.

**Description**

You receive this error message when you specify a value that is out of allowed range for a field like '-test_points_per_scan_cell', '-max_test_points', '-max_test_point_registers' and '-max_test_point_register_percent' in 'set_testability_configuration' command.

Feedback

**What Next**

Please use a value within the allowed range.

---

## DFT-1657

(Error) The '%s' value range is '0' to '%d'.

**Description**

You receive this error message when you specify a value that is out of allowed range for field '-target_test_coverage' in 'set_testability_configuration' command.

**What Next**

Please use a value within the allowed range.

---

## DFT-1658

(Error) Incorrect -effort switch value: '%s'. Please specify: 'low', 'high' or 'medium'

**Description**

You receive this error message when you specify a value that is not allowed for field '-effort' in 'set_testability_configuration' command.

**What Next**

Please use an appropriate value for '-effort' field.

---

## DFT-1659

(Error) Incorrect -target switch value: '%s'. Please specify: atpg_conflict | auto_self_gating | multicycle_paths | random_resistant | self_gating | shadow_wrapper | untestable_logic | user | x_blocking

**Description**

You receive this error message when you specify a value that is not allowed for field '-target' in 'set_testability_configuration' command.

**What Next**

Please specify: atpg_conflict | auto_self_gating | multicycle_paths | random_resistant | self_gating | shadow_wrapper | untestable_logic | user | x_blocking in '-target' field of 'set_testability_configuration' command

Feedback

## DFT-1660

(Error) Incorrect -target usage for '%s' switch. Please specify '-target %s'

**Description**

You receive this error message when you specify an option that is not supported with specified target.

**What Next**

Please specify the appropriate target as suggested by the message.

## DFT-1661

(Error) Target %s is required to enable %s

**Description**

You receive this error message when you specify target auto self gating in set_testability_configuration but do not have self gating optimization enabled.

## DFT-1662

(Error) The signal '%s' must be specified as '%s'

**Description**

You receive this error message when you specify a wrong signal type for a field in 'set_test_point_element' command.

**What Next**

Please use correct signal type with 'set_test_point_element' command.

## DFT-1663

(Error) Invalid value for '-type'. It should be control_0 | control_1 | observe | force_0 | force_1 | force_01

**Description**

You receive this error message when you specify a value that is not allowed for '-type' field in 'set_test_point_element' command.

**What Next**

Use an allowed value for '-type' field in 'set_test_point_element' command.

Feedback

## DFT-1664

(Error) Options %s cannot be specified together

**Description**

You receive this error message when you specify options that are not supported together in the 'set_testability_configuration' command

For example: 1) Options test_point_type and target_test_coverage cannot be specified together 2) Options max_test_points, max_test_point_registers, and max_test_point_register_percent cannot be specified together

## DFT-1665

(Error) Options '%s' must be specified together with %s

**Description**

You receive this error message when you do not specify options that must be specified together for a field in 'set_testability_configuration' command

For example: 1) Option only_from_file should be specified together with option test_point_file 2) Option effort should be specified together with option target random_resistant 3) Option isolate_elements should be specified together with option target shadow_wrapper

## DFT-1666

(Error) Invalid command syntax. Please provide at least one option.

**Description**

You receive this error message when you specify no option with 'set_testability_configuration' command.

**What Next**

Please provide at least one option with 'set_testability_configuration' command.

## DFT-1667

(Error) Invalid value for '-type'. It should be control_0 | control_1 | observe | force_0 | force_1 | force_01

**Description**

You receive this error message if you use invalid value for test point '-type' field

Feedback

## DFT-1668

(Error) '%s' option not supported with partition

**Description**

You receive this error message when you specify an option that is not supported in partition

**What Next**

Use an appropriate option that is supported in partition. Refer to the man page of 'set_testability_configuration' to check which options can be specified on a per-partition basis

## DFT-1669

(Error) Ignoring test point '%s' specified in wrong partition '%s'. Specify under '%s' or set the application option 'dft.tp_enable_partition_search' to true

**Description**

You receive this error message when the test point is ignored because it is specified in the wrong partition. This could happen because you may have used a set_test_point_element to insert the test point on a hierarchy from a DFT partition different from the current_dft_partition

**What Next**

Specify the test point in the correct partition or set the application option 'dft.tp_enable_partition_search' to true.

## DFT-1670

(Error) Unable to remove user defined test point '%s'. The test point is either not defined in partition '%s' or already removed.

**Description**

You receive this error message when the tool is not able to remove user defined test point.

**What Next**

You can check if the user defined test point exists in the defined partition.

Feedback

## DFT-1671

(Error) Ignoring test point '%s' specified in wrong partition '%s'. Specify under '%s' or set the application option 'dft.tp_enable_partition_search' to true.

### Description

You receive this error message when the test point is ignored because it is specified in wrong partition.

### What Next

Specify the test point in correct partition or set the application option 'dft.tp_enable_partition_search' to true.

## DFT-1672

(Error) Ignoring the command 'set_testability_configuration' as client is not enabled. Use 'set_dft_configuration -testability enable' to enable the client

### Description

You receive this error message if you use the 'set_testability_configuration' command without having specified 'set_dft_configuration -testability enable' previously

### What Next

You can add 'set_dft_configuration -testability enable' prior to the current command

## DFT-1673

(Error) Ignoring the command 'set_test_point_element' as client is not enabled. Use 'set_dft_configuration -testability enable' to enable the client.

### Description

You receive this error message if you use the set_test_point_element command without having specified set_dft_configuration -testability enable previously

### What Next

Add set_dft_configuration -testability enable prior to the current command

## DFT-1674

(Error) Ignoring the command 'run_test_point_analysis' as client is not enabled. Use 'set_dft_configuration -testability enable' to enable the client.

Feedback

**Description**

You receive this error message if you use the run_test_point_analysis command without having specified set_dft_configuration -testability enable previously

**What Next**

Add set_dft_configuration -testability enable prior to the current command

---

## DFT-1675

(Error) Ignoring the command 'run_test_point_analysis' as testability configuration is not defined. Use 'set_testability_configuration' to specify configuration.

**Description**

You receive this error message if you use the run_test_point_analysis command without having specified set_testability_configuration previously

**What Next**

Add set_testability_configuration configuration prior to the current command

---

## DFT-1676

(Error) Ignoring the command 'remove_test_point_element' as client is not enabled. Use 'set_dft_configuration -testability enable' to enable the client.

**Description**

You receive this error message if you use the 'remove_test_point_element' command without having specified 'set_dft_configuration -testability enable' previously.

**What Next**

You can add 'set_dft_configuration -testability enable' prior to the current command.

---

## DFT-1677

(Error) Test point report generation failed (DFT-3705).

**Description**

You receive this error message when the test point report generation is failed.

---

## DFT-1678

(Error) Value specified for option %s should be within %d and %d.

Feedback

**Description**

You receive this error message when you specify a value that is out of range for a field like '-test_points_per_scan_cell' in 'set_test_point_element' command.

**What Next**

Use a value within the allowed range.

---

## DFT-1679

(Error) No port specified, specify port or port list

**Description**

You receive this error message when no port is specified for test point.

---

## DFT-1680

(Error) Failed to insert %s test point at %s

**Description**

You receive this error message when the tool is unable to insert a control or a force_01 test point at the specified location

---

## DFT-1681

(Error) Missing control signal for test point target %s

**Description**

You receive this error message when control signal is missing for the specified test point target.

**What Next**

Check that you have specified a correct control signal for the specified test point target.

---

## DFT-1682

(Error) Missing clock signal and local clock hookup pin specification for user defined test points for partition %s

**Description**

You receive this error message when clock signal and local clock specification are missing for user defined test points.

Feedback

## DFT-1684

(Error) Unable to open the test point file %s

**Description**

You receive this error message when the tool is unable to open test point file at the specified location.

**What Next**

Check the existence of the file at the specified location.

## DFT-1685

(Error) Value %s for option '-test_point_types' is not valid. Please specify one of: control_0, control_1, control, observe, force_01

**Description**

You receive this error message when you specify a value for field '-test_point_types' that is not valid. Please specify one of: control_0, control_1, control, observe, force_01.

**What Next**

Correct the test point types.

## DFT-1686

(Error) Missing target specification for test point insertion for partition %s

**Description**

You receive this error message when the target specification is missing for test point insertion for the specified partition

**What Next**

Check the test point target specification for the partition.

## DFT-1687

(Error) Missing specifications for Max test point registers for one or more targets for partition %s

Feedback

**Description**

You receive this error message when max test point registers specification is missing for one or more targets for the specified partition.

**What Next**

Check the max test point registers specification for the test point target for the specified partition.

---

## DFT-1688

(Error) Tool cannot locate successful SpyGlass batch run to load GUI.

**Description**

You receive this error message when tool is not able to locate successful SpyGlass batch run to load GUI.

**What Next**

Check that you have a valid Spyglass project file generated by Fusion Compiler test point analysis step.

---

## DFT-1689

(Error) Tool cannot locate existing SpyGlass project directory.

**Description**

You receive this error message when tool is not able to locate existing SpyGlass project directory.

**What Next**

Check you have a valid SpyGlass project directory generated by Fusion Compiler test point analysis step.

---

## DFT-1690

(Error) Missing current design.

**Description**

You receive this error message when current design is not specified.

**What Next**

Check you have specified the correct design.

Feedback

## DFT-1691

(Error) Elaborate Failed. Make sure module %s is read properly.

**Description**

You receive this error message when the module specified using define_dft_design is not properly analyzed.

**What Next**

Check the module you have specified as DFT design is elaborated and built correctly.

## DFT-1692

(Error) DFT IP %s is unavailable. Make sure it is analyzed post compile.

**Description**

You receive this error message during insert_dft when tool is unable to find the user defined IP since it would have been optimized in logic opto hence try to analyze it post compile command.

## DFT-1693

(error) Vectored name '%s' of a bus port specified in interface. Specify pin name '%s' only

**Description**

You receive this error message when no a vectored pin is specified in the interface list

## DFT-1694

(error) Specified file '%s' doesn't exist or can't be read

**Description**

You receive this error message when specified file doesn't exist or can't be read.

## DFT-1695

(Error) Clock %s cannot be specified together in %s and %s

**Description**

Clock %s cannot be specified together in %s and %s

Feedback

**What Next**

Please Review DFT Specifications

---

## DFT-1696

(Error) Clock %s cannot be specified together with -clock_signal/-allowed_clock_signal and -disallowed_clock_signals

**Description**

You receive this error message when clock gating enable signal is missing for the specified test point target.

---

## DFT-1700

(Error) No test point test mode control has been specified.

**Description**

You receive this error message because no test point test mode control has been specified.

---

## DFT-1703

(Error) Incorrect combination for synchronized clocks was specified.

**Description**

Incorrect combination for synchronized clocks was specified.

**What Next**

Please Review DFT Specifications

---

## DFT-1704

(Error) Required argument '-pllclocks' was not found.

**Description**

Required argument '-pllclocks' was not found.

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1705

Error : You must specify either '%s' or '%s' option.

**Description**

This information is shown when it is mandatory to define one of the option and no option is defined.

For example,

write_test_protocol -out des_unit.comp1.spf Error: You must specify either '-pattern_exec' or '-test_mode' option. (DFT-1705)

write_test_protocol -target {{u1 ./des_unit_withSIB_merged.pdl} {u2 ./des_unit_withSIB_merged.pdl} {twox_top ./twox_top_merged.pdl}} -output ./twox_top.scan.spf Error: You must specify either '-pattern_exec' or '-test_mode' option. (DFT-1705)

**What Next**

Please Review DFT Specifications

## DFT-1707

(Error) Test mode '%s' is not defined

**Description**

Test mode '%s' is not defined

**What Next**

Please Review DFT Specifications

## DFT-1715

(Error) Unsuccessful chain update: %s

**Description**

You receive this error message if an issue occurred when processing a update_stub_chain command.

**What Next**

Verify that the options provided on the update_stub_chain command are correct. The -start_point option must be a driver pin while the -stop_point option must be a load pin, and both pins must be present in the stub chain provided by the -chain option.

Feedback

## DFT-1716

(Error) Unsuccessful chain connect: %s

**Description**

You receive this error message if an issue occurred when processing a connect_element_to_stub_chain command.

**What Next**

Verify that the options provided on the connect_element_to_stub_chain command are correct. The -floating_scanin option must be a load pin while the -floating_scanout option must be a driver pin, and both pins must be part of the same cell or sub-block.

## DFT-1717

(Error) Unsuccessful chain disconnect: %s

**Description**

You receive this error message if an issue occurred when processing a disconnect_element_from_stub_chain command.

**What Next**

Verify that the options provided on the disconnect_element_from_stub_chain command are correct. The -scanin option must be a load pin and must be from a stub chain element of the current ScanDEF.

## DFT-1718

(Error) Cannot add element with scan-in '%s' to stub chain '%s' since it's already present on SCANDEF

**Description**

You receive this error message when trying to add a new stub chain element through connect_element_to_stub_chain command from a cell that is already in the ScanDEF.

**What Next**

Verify that the cell specified by the -floating_scanin and -floating_scanout options is not already present in one of the existing stub chains of the current ScanDEF.

Feedback

## DFT-1719

(Error) Cannot remove element with scan-in '%s' from stub chain since it's not present on SCANDEF

### Description

You receive this error message when trying to remove the stub chain element of a cell that is not in the ScanDEF through disconnect_element_from_stub_chain command.

### What Next

Verify that the cell specified by the -scanin option is present in one of the existing stub chains of the current ScanDEF.

## DFT-1720

(Error) Pin '%s' cannot be used for SCANDEF purposes since it does not have physical information

### Description

You receive this error message when trying to add a new stub chain element through connect_element_to_stub_chain command from a cell does not have physical information.

### What Next

Verify that the cell specified by the -floating_scanin and -floating_scanout options do have physical information as it is required to be added to the ScanDEF.

## DFT-1721

(Error) Mismatch on provided (%d) and computed (%d) bit-length for segment with scan-in pin '%s'

### Description

You receive this error message when the bit length specified in the -bit_len option of connect_element_to_stub_chain command does not match the bit length of the multibit cell specified by the -floating_scanin and -floating_scanout options.

### What Next

Verify that the -bit_len value is correct for the given -floating_scanin and -floating_scanout options. If the -floating_scanin and -floating_scanout options point to a multi-bit cell with a reference library cell, the tool will compute the bit length automatically so the -bit_length option is optional.

Feedback

If the value provided by the -bit_len option its correct, there could be an issue with the design library or the library cell of the specified scan pins.

For single-bit cells, macros and sub-blocks, the tool will rely on the value provided by the -bit_len option. If the -bit_len option is not provided in those cases, the tool will consider the new stub element as bit length one.

---

## DFT-1751

(Error) Invalid '%s' file specified for design '%s'

**Description**

Invalid '%s' file specified for design '%s'

**What Next**

Please Review DFT Specifications

---

## DFT-1752

(Error) No instance found for design '%s'. Ignoring specified test model

**Description**

No instance found for design '%s'. Ignoring specified test model

**What Next**

Please Review DFT Specifications

---

## DFT-1753

(Error) Sub-server -> sub-server architecure not allowed due to known limitations

**Description**

Sub-server -> sub-server architecure not allowed due to known limitations

**What Next**

Please Review DFT Specifications

---

## DFT-1754

(Error) Test model '.ict.zip' missing for design '%s'

Feedback

**Description**

Test model '.ict.zip' missing for design '%s'

**What Next**

Please Review DFT Specifications

---

## DFT-1755

(Error) DFT_ACCESS client is enabled and test models (wks/ict/icl/pdl/masis) are not provided.

**Description**

DFT_ACCESS client is used to enable hierarchical flow where the lower level blocks are integrated at the current level. Hence the user must specify the test models (wks/ict/icl/pdl/masis) for the lower level blocks.

**What Next**

Specify any one of the test models (wks/ict/icl/pdl/masis) using the 'read_test_model' command for the lower level.

---

## DFT-1757

(Error) Sub-server -> shadow-server architecure not allowed due to known limitations

**Description**

Sub-server -> shadow-server architecure not allowed due to known limitations

**What Next**

Please Review DFT Specifications

---

## DFT-1758

(Error) Functional clock pin not present for the memory module

**Description**

The 'plan_dft' command provides clock information on memory clock pins which is used during memory grouping in the next step. Hence it is mandatory to have functional clock pins in MASIS files when spyglass is invoked for memory clock report generation.

Feedback

**What Next**

User can review and provide the functional clock pin in MASIS files if it is missing. If functional clock pins are skipped intentionally from MASIS files then user must run 'plan_dft -single_clock_grouping on' command which will skip the memory clock report generation and consider all memories under single clock domain.

---

## DFT-1760

(Error) Invalid compiler type. Valid types are - %s

**Description**

Invalid compiler type. Valid types are - %s

**What Next**

Please Review DFT Specifications

---

## DFT-1761

(Warning) No instance found for design '%s'. Ignoring specified icl/pdl/masis test model

**Description**

No instance found for design '%s' in input rtl. Ignoring specified icl/pdl/masis test model

**What Next**

Please Review input rtl or read the icl/pdl/masis testmodel after plan dft stage.

---

## DFT-1765

(Error) Incorrect compiler name '%s' specified for compiler type '%s'.

**Description**

Incorrect compiler name '%s' specified for compiler type '%s'.

**What Next**

Please Review DFT Specifications

---

## DFT-1770

(Error) Invalid input. Expecting list of pair (<compiler_type>,<compiler_name>).

Feedback

**Description**

Invalid input. Expecting list of pair (<compiler_type>,<compiler_name>).

**What Next**

Please Review DFT Specifications

---

## DFT-1771

(Error) Required argument '-interface' was not found.

**Description**

Required argument '-interface' was not found.

**What Next**

Please Review DFT Specifications

---

## DFT-1772

(Error) Error: Missing current design. Information: Skipping set_testmax_configuration command option check.

**Description**

Error: Missing current design. Information: Skipping set_testmax_configuration command option check.

**What Next**

Please Review DFT Specifications

---

## DFT-1773

(Error) Unrecognized option '%s' passed. For correct command info use 'validate_dft -help'.

**Description**

Unrecognized option '%s' passed. For correct command info use 'validate_dft -help'.

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1778

(Error) %s signal '%s' from design '%s' does not have %s clock information in test model.

**Description**

The Error message is shown when ScanDataIns/ScanDataOuts have missing capture/launch clock information in the CTL.

**What Next**

1. Please add the required information in the CTL being read. Otherwise, CTL core segments cannot be integrated/stitched into scan chains.

## DFT-1791

(Error) There is no DFT IP specification to be reported.

**Description**

You receive this error message if you do not specify any IP/DFT configuration(ieee_1687/Bsd/ieee_1500/Hsio).

Error: There is no DFT IP specification to be reported (DFT-1791)

**What Next**

Define the correct IP Specification by set_dft_configuration command.

## DFT-1794

(Error) Insufficient top interface pins for supporting serdes upstream { %d %d } with topInterface of {SI: %d, SO: %d}

**Description**

This error message reports that the size of upstream of serdes is greater than the size of the top interface (upstream interface) pins available in streaming fabric.

**What Next**

Verify that the size of upstream in serdes defined with -upstream {n_si n_so} is less than the size of sfip upstream interface.

## DFT-1800

(Error) Test DRC failed.

Feedback

**Description**

This message is issued, when internal call to test DRC fails.

**What Next**

Please run dft_drc and check errors.

---

## DFT-1801

(Error) %s is not available.

**Description**

This message is issued when part of the DFT specification is unexpectedly missing or incomplete. For example,

Error: LOS PipelineScan enable on partition default_partition is not available. (DFT-1801)

Potential causes can be:

- Pipeline scan enable has been enabled, but LOSPipelineEnable signal specification is missing for a partition.
- ScanDataIn and ScanDataOut pins cannot be found or are mismatched (specified more of one type than the other)
- Elements specified via set_scan_path cannot be found in the netlist.

**What Next**

The error message describes the part of the DFT specification that is inconsistent. Review it to ensure all necessary signals were specified and accepted by the tool.

Pay special attention to DFT partitions, as it is easy to accidentally specify signals that only apply to one partition, which can leave the other partitions missing necessary signal specifications.

Verify that pins and cells in the DFT specification still exist in the netlist by the time insert_dft occurs. Optimizations occurring between specification and DFT insertion can result in the removal of elements. In these cases you can try setting dont_touch on them.

---

## DFT-1802

(Error) %s is not supported.

Feedback

**Description**

This message is issued when part of the DFT specification is not supported. For example,

Error: Multiple pad cells on SO are not supported. (DFT-1802)

**What Next**

The error message describes the part of the DFT specification that is inconsistent. Review it to ensure all necessary signals were specified and accepted by the tool.

---

## DFT-1803

(Error) Scan path %s has incompatible elements: %s.

**Description**

This message is issued when some cells of set_scan_path specification are not compatible. For example,

Error: Scan path PATH0 has incompatible elements: different power domains. (DFT-1803)

**What Next**

Check set_scan_path specification.

---

## DFT-1804

(Error) %s is not found in the netlist.

**Description**

This message is issued when part of the DFT specification is not found in the netlist. For example,

Error: SO0 is not found in the netlist. (DFT-1804)

**What Next**

The error message describes the part of the DFT specification that is inconsistent. Review it to ensure all necessary signals were specified and accepted by the tool.

---

## DFT-1805

(Error) %s.

Feedback

**Description**

This message is issued when part of the DFT SCAN Synthesis can not find prerequisite data. For example,

Error: The design hierarchy is not accessible. (DFT-1805) Error: The design's test model is not accessible. (DFT-1805)

**What Next**

The error message describes the part of the DFT specification that is inconsistent. Review it to ensure all necessary signals were specified and accepted by the tool.

---

## DFT-1811

(Error) %s is not defined as a partition

**Description**

%s is not defined as a partition

**What Next**

Please Review DFT Specifications

---

## DFT-1816

(Error) %s

**Description**

This is a generic scan architecting error message. You receive this error message if it is not possible to build scan chains according to provided set_scan_configuration settings.

**Messages:**

%i zero-size chain(s) - Please check set_scan_configuration settings or if the design has enough valid scan cells for scan stitching.

max_length violation. max_length should be at least %i - Scan architecting finished constructing chains but couldn't follow set_scan_configurantion -max_length setting. Please check set_scan_* settings.

Requested 0 chain from scan synthesis. Settings error - Scan architecting received request to build 0 chains. Please check set_scan_* settings.

set_scan_path -class scan/occ must have equal number of scan chains (%i USCAN) groups for %i available chains - Some user defined scan paths are empty. Please check if elements of set_scan_path are valid.

Feedback

Sufficient number of chains not available to architect internal scan cells - Scan architecting couldn't build requested number of scan chains. Please check clock or voltage mixing settings or provide enough number of scan chains.

Sufficient number of chains not available to architect wrapper/internal scan cells - Scan architecting couldn't build requested number of scan chains. Please check clock or voltage mixing settings or provide enough number of scan chains.

Requirement of chain_count cannot be satisfied with Given number of ScanDataIn & ScanDataOut ports - There is a mismatch between requested number of chains and provided SI/SO ports. Please check set_scan* settings.

Requirement of chain_count cannot be satisfied. Minimum chain_count to be %i - Scan architecting couldn't build scan chains successfully with given scan specification. Please check clock or voltage mixing settings or provide enough number of scan chains.

Given scanDataIn & ScanDataOut ports are not sufficient. Minimum ports to be %i - Please match chain_count settings with the count of ScanDataIn/ScanDataOut signals or provide additional ScanDataIn/ScanDataOut signals.

Requirement of set_scan_configuration can not be satisfied with given number of ports. Required %i, available %i SI/SO port pairs. Mode is %s, partition is %s. - ScanDataIn/ ScanDataOut signals are defined less than required for scan architecting.

One of -chain_count or -max_length should be specified - At least one setting chain_count or max_length should be specified. Please update set_scan_configuration.

No scan chain end-points found' - ScanDataIn / ScanDataOut signals are not found. Please add necessary signals.

Requirement of chain_count specification cannot be satisfied. Minimum chain_count to be %i - Scan architecting needs more chains to meet clock / power requirements. Please update set_scan_confituration -chain_count.

Failed to balance %s chains. - Scan architecting couldn't balance chains. Please check clock / power mixing settings.

**What Next**

Please review set_scan_configuration settings and set_scan_compression_configuration settings if scan compression is used.

---

## DFT-1817

(Error) %s

Feedback

**Description**

This is a generic set_scan_path related scan architecting error message. You receive this error message if it is not possible to build scan chains according to provided set_scan_path settings.

**Messages:**

There are no enough chains to allocate existing-occ

There are no enough chains to allocate internal set_scan_path

Not enough chains available for architecting in partition %s, mode %s. Problematic clock is edge is %s.

**What Next**

Please review set_scan_path settings.

---

## DFT-1818

(Error) The design does not have scan chains

**Description**

This message is issued when there are no valid scan chains defined for the design. For example,

Error: The design does not have scan chains (DFT-1818)

**What Next**

The error message shows fundamental error in scan specification. Please check scan specifications.

---

## DFT-1819

(Error) The design does not have scan cells

**Description**

This message is issued when there are no valid scan cells to process in the design. For example,

Error: The design does not have scan cells (DFT-1819)

**What Next**

The error message shows fundamental error in the design. Please check design netlist and scan specifications.

Feedback

## DFT-1820

(Error) %s

**Description**

This is a generic wrapper scan architecting error message. You receive this error message if it is not possible to build scan chains according to provided set_wrapper_configuration settings.

**Messages:**

Cannot meet set_wrapper_configuration -chain_count. Minimum chain_count should be %i. Mode is %s, partition is %s.

Can't distribute wrapper chains for %s clock.

**What Next**

Please review set_wrapper_configuration settings.

## DFT-1821

(Error) Subblock mixes clock edges while top block does not allow mixing. (partition %s)

**Description**

This message is issued when one of subblocks has mixed clock edges, while DFT specification does not allow mixing clock edges. For example,

Error: Subblock mixes clock edges while top block does not allow mixing. (partition default_partition) (DFT-1821)

**What Next**

The error message shows that DFT clock mixing specification is not compatible with subblocks.

Check subblock generation script and disable edge mixing or update set_scan_configuration -clock_mixing settings for the top block.

## DFT-1822

(Error) Subblock mixes clocks while top block does not allow mixing. (partition %s)

Feedback

**Description**

This message is issued when one of subblocks has mixed clockd, while DFT specification does not allow mixing clocks. For example,

Error: Subblock mixes clocks while top block does not allow mixing. (partition default_partition) (DFT-1822)

**What Next**

The error message shows that DFT clock mixing specification is not compatible with subblocks.

Check subblock generation script and disable clock mixing or update set_scan_configuration -clock_mixing settings for the top block.

---

## DFT-1823

(Error) There are no scan cells to process in mode %s, partition %s.

**Description**

This message is issued when there are no valid scan cells to process in specific mode and partition. For example,

Error: There are no scan cells to process in mode Internal_scan, partition is default_partition. (DFT-1823)

**What Next**

The error message shows fundamental error in the design. Please check design netlist and scan specifications.

---

## DFT-1824

(Error) Number of clock edge crossing (%s) scan segments exceeds requested chain count (%s). Since only one clock edge crossing is allowed per chain, cannot honor specification. Problematic mode is %s, partition is %s.

**Description**

This message is issued when number of segments with crossing clock edges exceeds scan chain count requested. In such setting it is not possible to honor specification. For example,

Error: Number of clock edge crossing (5) scan segments exceeds requested chain count (4). Since only one clock edge crossing is allowed per chain, cannot honor specification. Problematic mode is Internal_scan, partition is default_partition. (DFT-1824)

Feedback

**What Next**

The error message shows that DFT specification can not be honored. Please update chain_count.

---

## DFT-1825

(Error) No clock/test_mode signal(via ‘set_dft_signal’ or user_opt_file) is specified. Please specify clock/test_mode signals through the “set_dft_signal” command in order to achieve the ‘check_dft’ command purpose.

**Description**

The 'check_dft' command is required to have clock/reset or testmode signals in order to perform scan analysis.

**What Next**

Users need to define clock/reset or testmode signals through the 'set_dft_signal' command

---

## DFT-1830

(Error) %s

**Description**

This is a generic set_scan_group processing error message. You receive this error message if it is not possible to build scan group according to the spec.

**Messages:**

Scan group's %s clock was not identified correctly. Group skipped.

**What Next**

Please review set_scan_group settings.

---

## DFT-1831

(Error) %s

**Description**

This is a generic set_scan_group settings error message. You receive this error message if settings of scan group are not compatible with other set_scan* commands

Feedback

**Messages:**

Scan group's %s clock %s is not specified correctly as clock in the test model. Group skipped.

Scan group's %s clock %s is set equivalent to %s signal. This signal is not specified correctly as clock in the test model. Group skipped.

**What Next**

Please review set_scan_group settings.

---

## DFT-1832

(Error) Ignoring test model of '%s' for scan architecting as it has missing information. Found while processing signal '%s' from core segment '%s' and test mode '%s'

**Description**

This error message is shown when signal information is not found in the corresponding test model file. In this case test model will be ignored for scan stitching purposes. For example,

Error: Ignoring test model of 'shift_r1' for scan architecting as it has missing information. Found while processing signal 'CaptureClock' from core segment 'chain2' and test mode 'Internal_scan'

**What Next**

Please check the test model file using the information printed by this message as a starting point.

---

## DFT-1836

(Error) Must specify 'true' or 'false' for the %s option

**Description**

Must specify 'true' or 'false' for the %s option

**What Next**

Please Review DFT Specifications

---

## DFT-1837

(Error) Required '%d' %s, but only '%d' DFT signal(s) of type '%s' are specified.

Feedback

**Description**

Required '%d' %s, but only '%d' DFT signal(s) of type '%s' are specified.

**What Next**

Please Review DFT Specifications

---

## DFT-1838

(Error) Minimum '%d' DFT signal(s) of type '%s' are required for testing core(s), but only '%d' parallel %s are specified in streaming fabric specification.

**Description**

Minimum '%d' DFT signal(s) of type '%s' are required for testing core(s), but only '%d' parallel %s are specified in streaming fabric specification.

**What Next**

Please Review DFT Specifications

---

## DFT-1840

(Error) The specified fabric clock %s is one of the excluded clocks.

**Description**

The specified fabric clock %s is one of the excluded clocks.

**What Next**

Please Review DFT Specifications

---

## DFT-1841

(Error) The specified value for option %s is invalid.

**Description**

The specified value for option %s is invalid.

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1842

(Error) Option %s is required to be defined together with option %s for fabric clock pre-shift pulse suppress.

**Description**

Option %s is required to be defined together with option %s for fabric clock pre-shift pulse suppress.

**What Next**

Please Review DFT Specifications

## DFT-1843

(Error) Missing pipe stage value, %s must end with a positive integer

**Description**

Missing pipe stage value, %s must end with a positive integer

**What Next**

Please Review DFT Specifications

## DFT-1844

(Error) The specified value %s is not a valid instance name for %s option.

**Description**

The specified value %s is not a valid instance name for %s option.

**What Next**

Please Review DFT Specifications

## DFT-1846

(Error) Ignoring command 'set_dft_streaming_fabric_configuration' as client is not enabled. Use 'set_dft_configuration -streaming_fabric enable' to enable the client.

**Description**

Ignoring command 'set_dft_streaming_fabric_configuration' as client is not enabled. Use 'set_dft_configuration -streaming_fabric enable' to enable the client.

Feedback

**What Next**

Please Review DFT Specifications

---

## DFT-1851

(Error) The option value for %s is not sufficient, total number of option values must be even.

**Description**

The option value for %s is not sufficient, total number of option values must be even.

**What Next**

Please Review DFT Specifications

---

## DFT-1852

(Error) The specified %s value for option -name does not match with any streaming fabric element name.

**Description**

The specified %s value for option -name does not match with any streaming fabric element name.

**What Next**

Please Review DFT Specifications

---

## DFT-1853

(Error) set_scan_group -class bypass doesn't support negative clocks in spec. Ignoring clocks spec for scan group %s. Problematic clock is %s.

**Description**

This error message is sent when set_scan_group -class bypass has -bypass_head_clock or -bypass_tail_clock negative clock specification. In this case both -bypass_head_clock and -bypass_tail_clock specifications will be ignored. For example,

Error: set_scan_group -class bypass doesn't support negative clocks in spec. Ignoring clocks spec for scan group BYPASS_GROUP. Problematic clock is tail. (DFT-1853)

**What Next**

Please check clock specifications.

Feedback

## DFT-1854

(Error) The specified value for option %s does not match with any name defined with set_dft_streaming_fabric.

**Description**

The specified value for option %s does not match with any name defined with set_dft_streaming_fabric.

**What Next**

Please Review DFT Specifications

## DFT-1855

(Error) The specified %s value for option -name does not match with any streaming fabric element name.

**Description**

The specified %s value for option -name does not match with any streaming fabric element name.

**What Next**

Please Review DFT Specifications

## DFT-1856

(Error) The specified configuration requires streaming fabric to have %d number of signal of type %s, (specified %d).

**Description**

The specified configuration requires streaming fabric to have %d number of signal of type %s, (specified %d).

**What Next**

Please Review DFT Specifications

## DFT-1857

(Error) The specified %s value for option name does not match with any name defined with set_dft_streaming_fabric_interface.

Feedback

**Description**

The specified %s value for option name does not match with any name defined with set_dft_streaming_fabric_interface.

**What Next**

Please Review DFT Specifications

---

## DFT-1861

(Error) %s is not defined as %s in set_dft_signal

**Description**

This error shows that input specified to the option -test_data is not of type test_data. Please define it as -type test_data using set_dft_signal.

**What Next**

Please specify correct port with option test_data in set_autofix_configuration

SH "SEE ALSO"

```
set_autofix_configuration
set_dft_signal
```

---

## DFT-1866

(Error) Ignoring command 'set_dft_streaming_fabric' as client is not enabled. Use 'set_dft_configuration -streaming_fabric enable' to enable the client.

**Description**

Ignoring command 'set_dft_streaming_fabric' as client is not enabled. Use 'set_dft_configuration -streaming_fabric enable' to enable the client.

**What Next**

Please Review DFT Specifications

---

## DFT-1867

(Error) Ignoring command 'set_dft_streaming_fabric_element' as streaming fabric client is not enabled. Use 'set_dft_configuration -streaming_fabric enable' to enable the client.

Feedback

**Description**

Ignoring command 'set_dft_streaming_fabric_element' as streaming fabric client is not enabled. Use 'set_dft_configuration -streaming_fabric enable' to enable the client.

**What Next**

Please Review DFT Specifications

---

## DFT-1870

(Error) Wrong port direction, The Bypass signal %s must have direction %s.

**Description**

Wrong port direction, The Bypass signal %s must have direction %s.

**What Next**

Please Review DFT Specifications

---

## DFT-1871

(Error) Port %s is not of type %s

**Description**

Port %s is not of type %s

**What Next**

Please Review DFT Specifications

---

## DFT-1873

(Error) Failed to generate correct streaming fabric IP or model. Fix reported error to proceed.

**Description**

Failed to generate correct streaming fabric IP or model. Fix reported error to proceed.

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1875

(Error) scandump mode is not specified for bypass variation2. Stitching bypass groups as usual scan groups.

**Description**

This error message is sent when bypass variation2 is specified but scandump is not provided.

**What Next**

Please add scandump mode specification via set_app_option –name dft.scan.scandump_mode_for_bypass.

## DFT-1876

(Error) bypass with global clock is enabled, but global clock is not specified. Ignoring the option, stitching bypass groups without global clock.

**Description**

This error message is sent when bypass variation2 with global clock is enabled, but global clock was not specified.

**What Next**

Please add global clock specification.

## DFT-1877

(Error) bypass with global clock is enabled, but bypass variation2 is not enabled. Ignoring the option.

**Description**

This error message is sent when bypass variation2 with global clock is enabled, but bypass variation itself was not enabled.

**What Next**

Please enable bypass variation2 via set_app_option –name dft.scan.scan_segment_bypass_variation2.

## DFT-1878

(Error) Regex for chain names is not correct. The command is discarded. Please check regex.

Feedback

**Description**

Correct regex expression for chain names is required for add_chain_bypass command. The error shows that provided regex expression was not correct.

**What Next**

Please update regex expression in add_chain_bypass comman specification.

---

## DFT-1879

(Error) Failed to attach segment %s to fabric.

**Description**

Failed to attach segment %s to fabric.

**What Next**

Please Review DFT streaming fabric Specifications

---

## DFT-1882

(Error) Validate model %s segment %s failed.

**Description**

Validate model %s segment %s failed.

**What Next**

Please Review DFT streaming fabric Specifications

---

## DFT-1883

(Error) SFIP does not have socket modes.

**Description**

SFIP does not have socket modes.

**What Next**

Please Review DFT streaming fabric Specifications

---

## DFT-1884

(Error) Fail to create %s segment for %s

Feedback

**Description**

Fail to create %s segment for %s

**What Next**

Please Review DFT streaming fabric Specifications

---

## DFT-1889

(Error) Pin specified in -hookup_pin is inside abstract design of type ctl/bam/etm. Specified pin can't be used for specifying dft signal.

**Description**

The Error message is shown when pin specified under '-hookup_pin' switch in 'set_dft_signal' is inside any abstract design(ctl/bam/etm).

**What Next**

1. Specify pin which is not inside abstract design.

---

## DFT-1890

(Error) "

**Description**

"

**What Next**

Please specify mandatory arguments '-algo_name' and '-algo_value' for create_mbist_algorithm command.

---

## DFT-1892

(Error) the controller clock chain %s : %d has been traced in multiple scan paths. \n

**Description**

In non multimode flow, same occ controller cannot be specified in multiple set_scan_path. Please enable multimode flow <set_app_options -name dft.scan.enable_general_multimode_support -value true> and enable the spc/occ flow <set_app_options -name dft.scan.occ_spc_chain_arch_flat_flow -value true> to trace occ ctrl in multiple scan path

Feedback

**What Next**

Please Review DFT Specifications

---

## DFT-1893

(Error) A [set_scan_path -class occ/spc] is required if DFT Shift Power Control segment %s is present , please add set_scan_path for %s\n

**Description**

Must specify 'set_scan_path' for the %s , please add set_scan_path for %s\n

**What Next**

Please Review DFT Specifications

---

## DFT-1894

(Error) %s instance is not present , %s can not be specified inside %s hierarchy, Use 'set_ungroup [get_cells %s] false'

**Description**

%s instance is not present, %s can not be specified inside % hierarchy , Use 'set_ungroup [get_cells %s] false'

**What Next**

Please Review DFT Specifications

---

## DFT-1895

(Error) Unable to instantiate the DFT IP %s in %s hierarchy, due to %s instance does not exist.

**Description**

Unable to instantiate the DFT IP %s in %s hierarchy, due to %s instance does not exist.

**What Next**

Please Review DFT Specifications

---

## DFT-1896

(Error) Failed to instantiate: %s --- Ref module: %s.

Feedback

**Description**

Failed to instantiate: %s --- Ref module: %s.

**What Next**

Please Review DFT Specifications

---

## DFT-1897

(Error) Invalid Scan Path '%s' element already given in '%s' chain for same set of SI/SO and test mode.

**Description**

Invalid Scan Path '%s' element already given in '%s' chain for same set of SI/SO and test mode.

**What Next**

Please Review DFT Specifications

---

## DFT-1901

(Error) Ignoring the command 'define_test_mode -usage scan_compression' as client is not enabled. Use 'set_dft_configuration -scan_compression enable' to enable the client.

**Description**

You receive this error message if you use the define_test_mode -usage scan_compression command without having specified set_dft_configuration -scan_compression enable previously

**What Next**

Add set_dft_configuration -scan_compression enable prior to the current command

---

## DFT-1902

(Error) Ignoring the command 'define_test_mode -usage logicbist' as client is not enabled. Use 'set_dft_configuration -logicbist enable' to enable the client.

**Description**

You receive this error message if you use the define_test_mode -usage logicbist command without having specified set_dft_configuration -logicbist enable previously

Feedback

**What Next**

Add set_dft_configuration -logicbist enable prior to the current command

---

## DFT-1903

(Error) Ignoring the command 'define_test_mode -usage wrp_if' as client is not enabled. Use 'set_dft_configuration -wrapper enable' to enable the client.

**Description**

You receive this error message if you use the define_test_mode -usage wrp_if command without having specified set_dft_configuration -wrapper enable previously

**What Next**

Add set_dft_configuration -wrapper enable prior to the current command

---

## DFT-1904

(Error) Ignoring the command 'define_test_mode -usage wrp_of' as client is not enabled. Use 'set_dft_configuration -wrapper enable' to enable the client.

**Description**

You receive this error message if you use the define_test_mode -usage wrp_of command without having specified set_dft_configuration -wrapper enable previously

**What Next**

Add set_dft_configuration -wrapper enable prior to the current command

---

## DFT-1905

(Error) Ignoring the command 'set_dft_clock_controller' as client is not enabled. Use 'set_dft_configuration -clock_controller enable' to enable the client.

**Description**

Ignoring the command 'set_dft_clock_controller' as client is not enabled. Use 'set_dft_configuration -clock_controller enable' to enable the client.

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1906

(Error) Ignoring command 'set_scan_compression_configuration' as client is not enabled. Use 'set_dft_configuration -scan_compression enable' to enable the client.

**Description**

Ignoring command 'set_scan_compression_configuration' as client is not enabled. Use 'set_dft_configuration -scan_compression enable' to enable the client.

**What Next**

Please Review DFT Specifications

## DFT-1907

(Error) Ignoring the command 'set_ieee_1500_configuration' as client is not enabled. Use 'set_dft_configuration -ieee_1500 enable' to enable the client.

**Description**

You receive this error message if you use the set_ieee_1500_configuration without having specified set_dft_configuration -ieee_1500 enable previously

**What Next**

Add set_dft_configuration -ieee_1500 enable prior to the current command

## DFT-1908

(Error) Ignoring the command 'set_logicbist_configuration' as client is not enabled. Use 'set_dft_configuration -logicbist enable' to enable the client.

**Description**

You receive this error message if you use the set_logicbist_configuration without having specified set_dft_configuration -logicbist enable previously

**What Next**

Add set_dft_configuration -logicbist enable prior to the current command

## DFT-1909

(Error) Ignoring the command 'set_pipeline_configuration' as client is not enabled. Use 'set_dft_configuration -pipeline_scan_data enable' to enable the client.

Feedback

**Description**

You receive this error message if you use the set_pipeline_scan_data_configuration without having specified set_dft_configuration -pipeline_scan_data enable previously

**What Next**

Add set_dft_configuration -pipeline_scan_data enable prior to the current command

---

## DFT-1910

(Error) Ignoring command 'set_scan_configuration' as client is not enabled. Use 'set_dft_configuration -scan enable' to enable the client.

**Description**

Ignoring command 'set_scan_configuration' as client is not enabled. Use 'set_dft_configuration -scan enable' to enable the client.

**What Next**

Please Review DFT Specifications

---

## DFT-1912

(Error) Ignoring the command 'set_wrapper_configuration' as client Wrapper is not enabled. Use 'set_dft_configuration -wrapper enable' to enable the client.

**Description**

Ignoring the command 'set_wrapper_configuration' as client Wrapper is not enabled. Use 'set_dft_configuration -wrapper enable' to enable the client.

**What Next**

Please Review DFT Specifications

---

## DFT-1913

(Error) Ignoring the command 'set_test_point_element' as client is not enabled. Use 'set_dft_configuration -testability enable' to enable the client.

**Description**

Ignoring the command 'set_test_point_element' as client is not enabled. Use 'set_dft_configuration -testability enable' to enable the client.

Feedback

**What Next**

Please Review DFT Specifications

---

## DFT-1919

(Warning) Option '%s' will be ignored as 'dft_access' client is enabled. Use 'set_dft_access_configuration command to specify this option

**Description**

You receive this error message if mbist and dft_access clients are enabled and common options in set_mbist_configuration and set_dft_access_configurations commands are specified through set_mbist_configuration command

**What Next**

When mbist and dft_access clients are enabled then all common option must be specified through set_dft_access_configuration command only

---

## DFT-1920

(Error) Ignoring the command '%s' as client is not enabled. Use 'set_dft_configuration -mbist enable' to enable the client (mismatch).

**Description**

Ignoring the command '%s' as client is not enabled. Use 'set_dft_configuration -mbist enable' to enable the client (mismatch).

**What Next**

Please Review DFT Specifications

---

## DFT-1921

(Error) Ignoring the command '%s' as client is not enabled. Use 'set_dft_configuration -dft_access enable' to enable the client (mismatch).

**Description**

Ignoring the command '%s' as client is not enabled. Use 'set_dft_configuration -dft_access enable' to enable the client (mismatch).

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1922

(Error) Ignoring the command. 'set_bsd_configuration' as client is not enabled. Use 'set_dft_configuration -bsd' to enable the client (mismatch).

**Description**

Ignoring the command. 'set_bsd_configuration' as client is not enabled. Use 'set_dft_configuration -bsd' to enable the client (mismatch).

**What Next**

Please Review DFT Specifications

## DFT-1923

(Error) Specified IR width %d is not within 2-63 inclusive, ir_width specification discarded.

**Description**

Specified IR width %d is not within 2-63 inclusive, ir_width specification discarded.

**What Next**

Please Review DFT Specifications

## DFT-1924

(Error) Ignoring the command. 'set_bsd_configuration -boundary_register enable' option not supported, if 'set_dft_configuration -bsd tap_only' is used

**Description**

Ignoring the command. 'set_bsd_configuration -boundary_register enable' option not supported, if 'set_dft_configuration -bsd tap_only' is used

**What Next**

Please Review DFT Specifications

## DFT-1925

(Error) Incorrect value '%s' specified for option '-load_gui'

**Description**

Incorrect value '%s' specified for option '-load_gui'

Feedback

**What Next**

Please Review DFT Specifications

---

## DFT-1926

(Error) Incorrect value '%s' specified for option '-load_gui'

**Description**

Incorrect value '%s' specified for option '-load_gui'

**What Next**

Please Review DFT Specifications

---

## DFT-1927

(Error) Incorrect value '%s' specified for option '-verbose'

**Description**

Incorrect value '%s' specified for option '-verbose'

**What Next**

Please Review DFT Specifications

---

## DFT-1928

(Error) Incorrect value '%s' specified for option '%s'

**Description**

Incorrect value '%s' specified for option '%s'

**What Next**

Please Review DFT Specifications

---

## DFT-1929

(Error) An integer greater than or equal to %s should be specified

**Description**

An integer greater than or equal to %s should be specified

Feedback

**What Next**

Please Review DFT Specifications

---

## DFT-1930

(Error) Ignoring the command 'set_ieee_1687_configuration' as client is not enabled. Use 'set_dft_configuration -ieee_1687 enable' to enable the client.

**Description**

Ignoring the command 'set_ieee_1687_configuration' as client is not enabled. Use 'set_dft_configuration -ieee_1687 enable' to enable the client.

**What Next**

Please Review DFT Specifications

---

## DFT-1931

(Error) Option "-name" is mandatory. Specify 'set_dft_ops -name <mode name>'

**Description**

Option "-name" is mandatory. Specify 'set_dft_ops -name <mode name>'

**What Next**

Please Review DFT Specifications

---

## DFT-1932

(Error) File %s does not exist or unreadable. Provide valid file path

**Description**

File %s does not exist or unreadable. Provide valid file path

**What Next**

Please Review DFT Specifications

---

## DFT-1933

(Error) Empty command argument passed

**Description**

Empty command argument passed

Feedback

**What Next**

Please Review DFT Specifications

---

## DFT-1934

(Error) "-name and -value" must be specified before "-pins".

**Description**

"-name and -value" must be specified before "-pins".

**What Next**

Please Review DFT Specifications

---

## DFT-1935

(Error) '%s' is not available in mode list. First define '%s' mode using 'set_dft_ops -name <mode name>.

**Description**

'%s' is not available in mode list. First define '%s' mode using 'set_dft_ops -name <mode name>.

**What Next**

Please Review DFT Specifications

---

## DFT-1939

(Error) Incorrect value '%s' specified for option '%s'

**Description**

Incorrect value '%s' specified for option '%s'

**What Next**

Please Review DFT Specifications

---

## DFT-1940

(Error) An internal error occured during plan_dft. AIT configuration not set.

**Description**

An internal error occured during plan_dft. AIT configuration not set.

Feedback

**What Next**

Please Review DFT Specifications

---

## DFT-1942

(Error) License consistency check(client enablers not consistent with internal app options) issue. Check man page for more details

**Description**

TMM use some internal app options to store the status of client enablers. When these app options are not consistent with client enablers i.e. one of them is "on" and other is "off" then TMM throws this error. This could happen in below cases. 1) User has disabled the internal app option unkowningly. 2) User is using same session to run multiple runs and all the clients(which are enabled in first session) are not disabled explicity in the second session.

**What Next**

User can check the app options settings or they can use different TMM session to run different runs

---

## DFT-1947

(Error) Compression is disabled. Please run [set_dft_configuration -scan_compression enable] to enable compression.

**Description**

Compression is disabled. Please run [set_dft_configuration -scan_compression enable] to enable compression.

**What Next**

Please Review DFT Specifications

---

## DFT-1948

(Error) DFT Wrapper configuration issue.

**Description**

DFT Wrapper configuration issue occurs mostly when there is mismatch in wrapping configuration. For example: a) licence checkout for DFT Test core integration support is disabled with app option "_shared_core_wrapping_enable_" b) set_boundary_cell -add_shared_wrapper_cell_only true

Feedback

**What Next**

set_boundary_cell set_app_option

---

## DFT-1949

(Error) Test Mode %s is not defined

**Description**

Test Mode %s is not defined

**What Next**

Please Review DFT Specifications

---

## DFT-1950

(Error) Ignoring scan configuration for mode %s, since no scan configuration was present for intest mode.

**Description**

Ignoring scan configuration for mode %s, since no scan configuration was present for intest mode.

**What Next**

Please Review DFT Specifications

---

## DFT-1951

(Error) Cound not find %s signal from user specification.

**Description**

Cound not find %s signal from user specification.

**What Next**

Please Review DFT Specifications

---

## DFT-1952

(Error) Mandatory IEEE 1149.1 test port %s missing.

**Description**

Mandatory IEEE 1149.1 test port %s missing.

Feedback

**What Next**

Please Review DFT Specifications

---

## DFT-1954

(Error) Invalid value '%s' specified for option '%s'. Valid values are '%s'"

**Description**

Incorrect value '%s' specified for option '%s'. Accepted values are '%s'

**What Next**

Please Review DFT Specifications

---

## DFT-1961

(Error) %s is not a valid directory path

**Description**

%s is not a valid directory path

**What Next**

Please Review DFT Specifications

---

## DFT-1962

(Error) Design read failed for file %s

**Description**

Design read failed for file %s

**What Next**

Please Review DFT Specifications

---

## DFT-1963

(Error) No dft signal found on %s

**Description**

This error shows that signal specificed in option -test_data or -control_signal of set_autofix_configuration is not a valid dft signal.

Feedback

**What Next**

Please specify correct signal with option -test_data or -control_signal in set_autofix_configuration

SH "SEE ALSO"

`set_autofix_configuration`

---

## DFT-1964

(Error) Signal %s provided for option %s should be defined as -type TestMode -usage autofix or -type ScanEnable.

**Description**

This error shows that signal specificed with option %s is not of type TestMode with usage autofix or of type ScanEnable that is ScanOnly in its set_dft_signal specification. Please use the signal which has -type %s in his set_dft_signal specifiation.

**What Next**

Please specify correct signal with option %s in set_autofix_configuration

SH "SEE ALSO"

`set_autofix_configuration`

---

## DFT-1965

(Error) No signal specified for option %s

**Description**

This error shows that no signal is specified for option -test_data or -control_signal in set_autofix_configuration.

**What Next**

Please specify correct signal with option -test_data or -control_signal in set_autofix_configuration

SH "SEE ALSO"

`set_autofix_configuration`

---

## DFT-1971

(Error) Incorrect value '%s' specified for option '-load_gui'

Feedback

**Description**

Incorrect value '%s' specified for option '-load_gui'

**What Next**

Please Review DFT Specifications

---

## DFT-1972

(Error) Incorrect value '%s' specified for option '-extract_dft_clocks'

**Description**

Incorrect value '%s' specified for option '-extract_dft_clocks'

**What Next**

Please Review DFT Specifications

---

## DFT-1977

(Error) Empty command argument passed

**Description**

Empty command argument passed

**What Next**

Please Review DFT Specifications

---

## DFT-1978

(Error) "-class" must be specified. Please specify a class type using "set_dft_ring -class <IEEE1500/IEEE1687/SMS>"

**Description**

"-class" must be specified. Please specify a class type using "set_dft_ring -class <IEEE1500/IEEE1687/SMS>"

**What Next**

Please Review DFT Specifications

---

## DFT-1979

(Error) '%s' option is supported for 'set_dft_ring' for IEEE1687 only."

Feedback

**Description**

'%s' option is supported for 'set_dft_ring' for IEEE1687 only."

**What Next**

Please Review DFT Specifications

---

## DFT-1980

(Error) Invalid value '%s' for option '-sib' specified, allowed options are enable | disable

**Description**

Invalid value '%s' for option '-sib' specified, allowed options are enable | disable

**What Next**

Please Review DFT Specifications

---

## DFT-1981

(Error) Invalid class type '%s' is specified. Please specify a valid class type (IEEE1500/IEEE1687/SMS)

**Description**

Invalid class type '%s' is specified. Please specify a valid class type (IEEE1500/IEEE1687/SMS)

**What Next**

Please Review DFT Specifications

---

## DFT-1982

(Error) Invalid ring class '%s' is specified. Please specify a valid ring class sms|ieee1500|ieee1687

**Description**

Invalid ring class '%s' is specified. Please specify a valid ring class sms|ieee1500|ieee1687

**What Next**

Please Review DFT Specifications

Feedback

## DFT-1983

(Error) Ring name '%s' is not found in populated ring information. Please specify a valid ring name

**Description**

Ring name '%s' is not found in populated ring information. Please specify a valid ring name

**What Next**

Please Review DFT Specifications

## DFT-1984

(Error) Module name '%s' is not found in the design. Please specify a valid module name

**Description**

Module name '%s' is not found in the design. Please specify a valid module name

**What Next**

Please Review DFT Specifications

## DFT-1987

(Error) IEEE 1500 IP configuration issue.

**Description**

IEEE 1500 IP configuration issue.

**What Next**

Please Review DFT Specifications

## DFT-1990

(error) SCANDEF based Optimization failed.

**Description**

This message is issued if an DFT Optimization call (*optimize_dft* or equivalent command) could not finish is task successfully.

The DFT SCANDEF Optimization can perform repartioning or reordering of SCANDEF chains elements.

Feedback

### What Next

Check for previous issued Warning or Error messages on DFT tag that details the problem detected during the optimization.

### See Also

- optimize_dft

---

## DFT-1991

(warning) SCANDEF based Optimization skipped: No SCANDEF chains found.

### Description

This message is issued when SCANDEF based Optimization (like *optimize_dft* or similar commands) was attempted but no SCANDEF chain data was found. The SCANDEF chain data is basic for performing this kind of optimizations.

### What Next

Check if SCANDEF data should be available, for example if *insert_dft* was previously executed succesfully, or if SCANDEF data has been loaded from setup using *read_def* command.

### See Also

- optimize_dft
- read_def

---

## DFT-1992

(info) SCANDEF based Optimization skipped: No SCANDEF chains found.

### Description

This message is issued when SCANDEF based Optimization (like *optimize_dft* or similar commands) was attempted but no SCANDEF chain data was found. The SCANDEF chain data is basic for performing this kind of optimizations.

### What Next

Check if SCANDEF data should be available, for example if *insert_dft* was previously executed succesfully, or if SCANDEF data has been loaded from setup using *read_def* command.

Feedback

**See Also**

- optimize_dft
- read_def

---

## DFT-1994

(Error) Chain %s %s has not valid driving logic.

**Description**

This message is issued when provided DFT IP stump chains ends are not connected. This shows problem in scan compression logic creation. For example,

Error: Chain SO DES_UNIT_WRAP/i_dft_control/scan_chain_end[4] has not valid driving logic. (DFT-1994)

**What Next**

Please check DFT compression logic insertions.

---

## DFT-1995

(Warning) Chain %s %s has not valid driving logic.

**Description**

This message is issued when provided DFT IP stump chains ends are not connected. This shows problem in scan compression logic creation. For example,

Warning: Chain SO DES_UNIT_WRAP/i_dft_control/scan_chain_end[4] has not valid driving logic. (DFT-1995)

**What Next**

Please check DFT compression logic insertions.

---

## DFT-1996

(Error) %s

**Description**

This is a generic DFT-IP stump term related error message. You receive this error message if provided DFT-IP stump term data is not sufficient to built chains.

Feedback

**Messages:**

Received 0 channels from DFT IP config. Can't continue.

**What Next**

Please review set_scan_compression_configuration and set_scan_configuration settings.

---

## DFT-1997

(Error) %s

**Description**

This is a generic DFT scan chain ends error in preinserted flows. You receive this error message if scan chain ends data from previously run insert_dft command is not available.

**What Next**

Please review DFT flow.

---

## DFT-1998

(Error) %s is a only mode name allowed with usage %s

**Description**

%s is a only mode name allowed with usage %s

**What Next**

Please Review DFT Specifications

---

## DFT-1999

(Error) Partial encoding exist. Define encoding for all modes or do not define encoding for any mode

**Description**

Partial encoding exist. Define encoding for all modes or do not define encoding for any mode

**What Next**

Please Review DFT Specifications

Feedback

## DFT-2000

(Warning) CTL Model read for design %s.

**Description**

CTL model has been successfully read for the given design.

## DFT-2003

(Warning) There is no DFT IP specification to be reported.

**Description**

No DFT IP specifications are found. As a result no DFT IPs have been inserted in the design. DFT IPs are inserted if one of the following flows are enabled (a) compression (b) OCC (c) wrapper or (d) test points.

## DFT-2007

(Warning) Object %s is already in an %s hierarchy

**Description**

This message indicates that one of the included or excluded object is present in a hierarchy that has been already included or excluded in another specification.

## DFT-2008

(Warning) Inconsistency in DFT Specification Data. Failed to read it.

**Description**

An inconsistency has been detected in DFT specification while saving it for persistency or while loading the save data in current session. This may happen due to mismatch in specification data from one release to another or an error in current DFT specification data.

**What Next**

1. Regenerate the saved block with the current version of the tool 2. As a temporary workaround, disable the persistency check using the following environment variable. This way the check is ignored and the flow is allowed to proceed further. setenv dft_tdm_spec_persistency false

Feedback

## DFT-2009

(Error) %s pin '%s' is not defined as dft signal.

**Description**

%s pin '%s' is not defined as dft signal.

**What Next**

Please Review DFT Specifications

## DFT-2012

(Error) A previous specification for scan cell set '%s' exists

**Description**

A previous specification for scan cell set '%s' exists

**What Next**

Please Review DFT Specifications

## DFT-2013

(Error) Test modes must be defined before being specified in set_scan_path command.

**Description**

Test modes must be defined before being specified in set_scan_path command.

**What Next**

Please Review DFT Specifications

## DFT-2014

(Error) A previous specification for %s '%s' exists

**Description**

A previous specification for %s '%s' exists

**What Next**

Please Review DFT Specifications

Feedback

## DFT-2015

(Error) Could not find chain %s in core %s.

**Description**

Could not find chain %s in core %s.

**What Next**

Please Review DFT Specifications

## DFT-2021

(Error) Unsuccessful DFT Architect

**Description**

Unsuccessful DFT Architect

**What Next**

Please Review DFT Specifications

## DFT-2022

(Warning) ICG cell feeding scan cells for multiple (%d) partitions. ICG %s, partitions %s.

**Description**

Integrated Clock Gating (ICG) cells are required to have their TE (Test Enable) pins connected to ScanEnable or TestMode signals during stages of *dft_drc*, *preview_dft* and/or *insert_dft* commands for allowing the SCAN clocks pass through the ICG cells in a transparent mode, else the SCAN clock data can not reach the DFT SCAN cells, producing a SCAN data blockage.

The ScanEnable or TestMode DFT signals can be defined per DFT partition (see command *define_dft_partition and set_dft_signal for details). Given that, different SCAN cells can be associated to different DFT partitions, and therefore they need to be handled by different ScanEnable and/or TestMode signals defined for its specific partition.*

For setting up all ICG TE connection properly, the ICG TE pins needs to be connected to ScanEnable or TestMode signals in sync with the SCAN cells they are providing clock to.

If this message is issued, it means the ICG reported is driving the clock of SCAN cells that belong to different DFT partitions, and there might be clashes or conflicts about what would be the correct ScanEnable or TestMode signal to use for that ICG cell, if DFT partition uses different signals ports/hookup pins.

Feedback

**What Next**

Review the *dft_drc* reports after DFT SCAN insertion to identify potential SCAN chain blockages (S1 errors) that might be related to this scenario.

As ICG insertion happens before DFT SCAN Synthesis and it is not aware to DFT setup, or DFT setup might not even exist at the time ICG cells are inserted, there is no way to prevent this problem.

Consider using ECO operation to bypass the clock signal for SCAN cells that might get blocked because its driving ICG is connected to other DFT partition signal. Remember that the ICG is still driving some SCAN cells for that partition so not all SCAN cells on that ICG clock tree are subject to a potential issue.

---

## DFT-2023

(Information) Using enable data from partition %s for ICG enable %s

**Description**

The message explicitly informs which partition's signals will be used to drive the TestEnable signal of the ICG cell.

This information is important when more an ICG cells is driving SCAN cells from different DFT partitions, and therefore there might be more than one candidate TestMode or ScanEnable signals valid to drive the ICG TestEnable pin.

**What Next**

Review the *dft_drc* reports after DFT SCAN insertion to identify potential SCAN chain blockages (S1 errors) that might be related to this scenario.

As ICG insertion happens before DFT SCAN Synthesis and it is not aware to DFT setup, or DFT setup might not even exist at the time ICG cells are inserted, there is no way to prevent this problem.

Consider using ECO operation to bypass the clock signal for SCAN cells that might get blocked because its driving ICG is connected to other DFT partition signal. Remember that the ICG is still driving some SCAN cells for that partition so not all SCAN cells on that ICG clock tree are subject to a potential issue.

---

## DFT-2025

(Warning) Unable to update the target hierarchy for terminal lockup insertion of dual rail cell %s: %s.

Feedback

### Description

This warning message is thrown by DFT scan Synthesis during terminal lockup latch insertion when the dft.scan.mv_aware_terminal_lockup_placement application option is enabled.

If a terminal lockup is about to be inserted for a scan cell whose clock pin is fed by a secondary voltage supply and the dft.scan.mv_aware_terminal_lockup_placement application option is enabled, the tool will try to insert the terminal lockup latch in the hierarchy of the clock source instead, preventing the creation of a LS cell to fix the MV violation in the clock net of the terminal lockup latch. If the tool was unable to insert the lockup latch in the hierarchy of the clock source, this warning message will be thrown with the name of the affected scan cell and the issue that occurred, and the terminal lockup latch insertion will be skipped for that cell.

### What Next

Verify that the voltage supplies of the clock source and their containing hierarchy are compatible. If the clock source is a port or a hierarchical pin, try creating a buffer and defining it as hookup pin of the clock signal using set_dft_signal command.

---

## DFT-2032

(Error) Register name cannot be empty for user defined instruction.

### Description

Register name cannot be empty for user defined instruction.

### What Next

Please Review DFT Specifications

---

## DFT-2033

(Error) Register name 'BOUNDARY' or 'BYPASS' cannot be specified for user defined instruction.

### Description

Register name 'BOUNDARY' or 'BYPASS' cannot be specified for user defined instruction.

### What Next

Please Review DFT Specifications

Feedback

## DFT-2034

(Error) Register name 'BOUNDARY' cannot be specified for BYPASS, HIGHZ, CLAMP, IDCODE, and USERCODE instruction.

**Description**

Register name 'BOUNDARY' cannot be specified for BYPASS, HIGHZ, CLAMP, IDCODE, and USERCODE instruction.

**What Next**

Please Review DFT Specifications

## DFT-2035

(Error) Register name 'BYPASS' cannot be specified for EXTEST, INTEST, SAMPLE, PRELOAD, IDCODE, and USERCODE instruction.

**Description**

Register name 'BYPASS' cannot be specified for EXTEST, INTEST, SAMPLE, PRELOAD, IDCODE, and USERCODE instruction.

**What Next**

Please Review DFT Specifications

## DFT-2036

(Error) Specified opcode '%s' is not binary.

**Description**

Specified opcode '%s' is not binary.

**What Next**

Please Review DFT Specifications

## DFT-2037

(Error) Cellname entered in option -instance is not present in the design '%s'

**Description**

Cellname entered in option -instance is not present in the design '%s'

Feedback

**What Next**

Please Review DFT Specifications

---

## DFT-2038

(Error) Invalid user defined instruction name.

**Description**

Invalid user defined instruction name.

**What Next**

Please Review DFT Specifications

---

## DFT-2039

(Error) 'BYPASS' instruction should have all 1's in opcode.

**Description**

'BYPASS' instruction should have all 1's in opcode.

**What Next**

Please Review DFT Specifications

---

## DFT-2040

(Warning) DFT clock not found for CTL clock pin in DFT DRC clock table.

**Description**

DFT relies on TetraMAX to determine accurate clock domain information for scan cells in a design. This data is referred to as the DFT DRC clock table.

When a clock pin belonging to a CTL cell is found in the fanout of a scan clock, this clock table is consulted to obtain accurate clock domain information. If this pin is not found in that table, the tool issues this warning message. This means the tool could assign the wrong clock domain to this CTL, which can result in scan chains that violate user-provided DFT specifications. If this warning is issued for a CTL chain's clock, that chain may not end up stitched during Scan Synthesis to prevent future DFT DRC issues.

**What Next**

The message contains the name of the specific CTL clock pin instance that caused the problem. Verify the CTL models of the affected cells to confirm whether there's a problem with its clock pin specifications.

Feedback

Otherwise, it could be indicative of a bug in DFT DRC or TetraMAX.

## DFT-2041

(Warning) Port '%s' is connected to no I/O Registers and %d clock-gating cells through enable pin and %d CTL cell pins; dedicated wrapper cell will be added to the port due to missing I/O registers

**Description**

This warning message is issued when tool is inserting a dedicated wrapper cell to port which is connected to no I/O Registers and clock-gating cells through enable pin and CTL cell pins

For example,

Warning: Port 'in3' is connected to no I/O Registers and 5 clock-gating cells through enable pin and 3 CTL cell pins; dedicated wrapper cell will be added to the port due to missing I/O registers (DFT-2041)

The combinations of commands that can trigger this warning message are:

- *set_wrapper_configuration*

## DFT-2042

(Error) Currently '-view exist' is not supported for BSD.

**Description**

Currently '-view exist' is not supported for BSD.

**What Next**

Please Review DFT Specifications

## DFT-2043

(Error) Invalid View specified: %s

**Description**

Invalid View specified: %s

**What Next**

Please Review DFT Specifications

Feedback

## DFT-2044

(Error) %s instruction is not defined.

**Description**

%s instruction is not defined.

**What Next**

Please Review DFT Specifications

## DFT-2045

(Error) Opcode '%s' already exists for other instruction, bsd instruction specification discarded.

**Description**

Opcode '%s' already exists for other instruction, bsd instruction specification discarded.

**What Next**

Please Review DFT Specifications

## DFT-2047

(Warning) Port '%s' is connected to no I/O Registers and %d clock-gating cells through enable pin; dedicated wrapper cell will be added to the port due to missing I/O registers

**Description**

This warning message is issued when tool is inserting a dedicated wrapper cell to port which is connected to no I/O Registers and clock-gating cells through enable pin

For example,

Warning: Port 'in3' is connected to no I/O Registers and 5 clock-gating cells through enable pin; dedicated wrapper cell will be added to the port due to missing I/O registers (DFT-2047)

The combinations of commands that can trigger this warning message are:

- *set_wrapper_configuration*

Feedback

## DFT-2048

(Error) '-test_mode_port' option doesn't accept collection it expects name of pin/port. Either input pin is invalid or get_object_name should be used with collection to get pin/port name.

**Description**

'-test_mode_port' option doesn't accept collection it expects name of pin/port. Either input pin is invalid or get_object_name should be used with collection to get pin/port name.

**What Next**

Please Review DFT Specifications

## DFT-2049

(Error) No TestMode pin found with name '%s'.

**Description**

No TestMode pin found with name '%s'.

**What Next**

Please Review DFT Specifications

## DFT-2052

(Warning) %s

**Description**

The list of objects specified for the command is either empty or the listed objects are not found in design or are invalid. For Example, Warning: object_list is empty (DFT-2052)

## DFT-2058

(Error) Must provide a top-level wrapper clock

**Description**

Must provide a top-level wrapper clock

**What Next**

Please Review DFT Specifications

Feedback

## DFT-2059

(Error) Invalid signature flop specification '%s'

**Description**

Invalid signature flop specification '%s'

**What Next**

Please Review DFT Specifications

## DFT-2060

(Error) Discarding test mode '%s' as test mode of type '%s' has already been defined.

**Description**

Discarding test mode '%s' as test mode of type '%s' has already been defined.

**What Next**

Please Review DFT Specifications

## DFT-2061

(Warning) Dedicated wrapper has not been added for port '%s' for any load, as no powersafe location has been found for it.

**Description**

This warning message is issued when tool is unable to find any hierarchy on the connected net of the port which is safe to insert from UPF point of view

For example,

Warning: Dedicated wrapper has not been added for port 'i_rd' for any load, as no powersafe location has been found for it. (DFT-2061)

The combinations of commands that can trigger this warning message are:

- *set_wrapper_configuration -add_wrapper_cells_to_power_domain enable.*
- *define_dft_partition <> -wrapper enable*

**What Next**

Change design to make sure there is powersafe location where dedicated wrapper cells can be inserted.

Feedback

## DFT-2062

(Warning) Dedicated wrapper has been added for port %s with fixable violation.

**Description**

This warning message is issued when a dedicated wrapper cell has been added to port at a location which can lead to a UPF violation which can be fixed by insertion of MV cells.

For example,

Warning: Dedicated wrapper has been added for port i_rd with fixable violation. (DFT-2062)

The combinations of commands that can trigger this warning message are:

- *set_wrapper_configuration -add_wrapper_cells_to_power_domain enable.*
- *define_dft_partition <> -wrapper enable*

**What Next**

Check violation and try to fix that violation depending on type of violation.

## DFT-2063

(Warning) Dedicated wrapper has not been added for load %s of port '%s' as no powersafe location has been found for it.

**Description**

This warning message is issued when tool is unable to find any hierarchy on the connected net of the port which is safe to insert from UPF point of view for a specific flat load of the port.

For example,

Warning: Dedicated wrapper has not been added for load ABC of port 'i_rd' as no powersafe location has been found for it. (DFT-2063)

The combinations of commands that can trigger this warning message are:

- *set_wrapper_configuration -add_wrapper_cells_to_power_domain enable.*
- *define_dft_partition <> -wrapper enable*

**What Next**

Change design to make sure there is powersafe location where dedicated wrapper cells can be inserted.

Feedback

## DFT-2064

(Warning) Dedicated wrapper cell cannot be inserted to output port '%s' as there is a feedthrough path to input port, which is excluded from wrapping

**Description**

This warning message is issued when tool is inserting a dedicated wrapper cell to an output port which is connected to input port via feedthrough path, which is excluded from wrapping.

This is to show that output port wasn't wrapped with dedicated wrapper cell in order to avoid xvalue propogation in lbist combinational path.

Please note that this message has precedence over all preceding messages showing that a dedicated wrapper cell will be added to the port. That is, if this message is shown along other messages for the same port, a dedicated wrapper cell will not be added to the port.

For example,

Warning: Dedicated wrapper cell cannot be inserted to output port 'g3_out3[0]' as there is a feedthrough path to input port, which is excluded from wrapping (DFT-2064)

The combinations of commands that can trigger this warning message are:

- *set_wrapper_configuration*
- *define_dft_partition <> -wrapper enable*

**What Next**

Change design to make sure there is powersafe location where dedicated wrapper cells can be inserted.

## DFT-2065

(Warning) A dedicated wrapper cell is added to input port '%s' as there is a feedthrough path to output port, which is wrapped with a dedicated wrapper cell

**Description**

This warning message is issued when tool is inserting a dedicated wrapper cell to an input port which is connected to output port via feedthrough path, which is wrapped with dedicaated wrapper cell.

This is to avoid xvalue propogation to output port which is connected to input port via feedthrough path. A dedicated wrapper cell is added to input port to avoid x value propogation.

Feedback

For example,

Warning: A dedicated wrapper cell is added to input port 'g1_in2[2]' as there is a feedthrough path to output port , which is wrapped with a dedicated wrapper cell (DFT-2065)

The combinations of commands that can trigger this warning message are:

- *set_wrapper_configuration*
- *define_dft_partition <> -wrapper enable*

**What Next**

Change design to make sure there is powersafe location where dedicated wrapper cells can be inserted.

---

## DFT-2066

(Error) An integer greater than or equal to 0 should be specified

**Description**

An integer greater than or equal to 0 should be specified

**What Next**

Please Review DFT Specifications

---

## DFT-2067

(Warning) Port '%s' is connected to no I/O Registers and %d CTL cell pins; dedicated wrapper cell will be added to the port due to missing I/O registers

**Description**

This warning message is issued when tool is inserting a dedicated wrapper cell to port which is connected to no I/O Registers and CTL cell pins

For example,

Warning: Port 'in3' is connected to no I/O Registers and 3 CTL cell pins; dedicated wrapper cell will be added to the port due to missing I/O registers (DFT-2067)

The combinations of commands that can trigger this warning message are:

- *set_wrapper_configuration*

Feedback

## DFT-2068

(Warning) Port '%s' is connected to %d clock-gating cells through enable pin which exceeds clock-gating cell threshold %d; dedicated wrapper cell will be added to the port

**Description**

This warning message is issued when tool is inserting a dedicated wrapper cell to port which is connected to clock-gating cells through enable pin which exceeds clock-gating cell threshold

For example,

Warning: Port 'in3' is connected to 5 clock-gating cells through enable pin which exceeds clock-gating cell threshold 3; dedicated wrapper cell will be added to the port (DFT-2068)

The combinations of commands that can trigger this warning message are:

- *set_wrapper_configuration*

## DFT-2069

(Warning) Port '%s' is connected to %d CTL cell pins which exceed CTL cell threshold %d; dedicated wrapper cell will be added to the port

**Description**

This warning message is issued when tool is inserting a dedicated wrapper cell to port which is connected to CTL cell pins which exceeds CTL cell threshold

For example,

Warning: Port 'in3' is connected to 5 CTL cell pins which exceeds CTL cell threshold 3; dedicated wrapper cell will be added to the port (DFT-2069)

The combinations of commands that can trigger this warning message are:

- *set_wrapper_configuration*

## DFT-2070

(Error) Only one hierarchy is allowed with '-include' option while '-wrapper' option is enabled

**Description**

Only one hierarchy is allowed with '-include' option while '-wrapper' option is enabled

Feedback

**What Next**

Please Review DFT Specifications

---

## DFT-2071

(Error) %s is not a valid partition name. Partition name can only contain alphanumeric characters (a-z, A-Z, 0-9) and the underscore (_).

**Description**

%s is not a valid partition name. Partition name can only contain alphanumeric characters (a-z, A-Z, 0-9) and the underscore (_).

**What Next**

Please Review DFT Specifications

---

## DFT-2072

Warning: Expected value on pin '%s' %s, actual value is %s

**Description**

The dft_drc command issues this warning message while validating a test model instance connectivity. The warning may be serious. It indicates that at least one constant pin (with the Constant data type) or test-mode pins on the DFT-modeled instance is not set to its expected logical value, as a result of the instance integration.

**What Next**

This is only a warning message. No action is required.

However, since this warning may be serious, you can look for the accompanying TEST-3083 information message(s) for the details on which pins are involved.

---

## DFT-2075

(Warning) pin %s is inside a BAM instance. Hence ignoring it from user specification

**Description**

Pin present inside bam instance would be ingored from user specification.

---

## DFT-2076

(Error) Mode '%s' not found

Feedback

**Description**

Mode '%s' not found

**What Next**

Please Review DFT Specifications

---

## DFT-2077

(Error) Option '%s' not found

**Description**

Option '%s' not found

**What Next**

Please Review DFT Specifications

---

## DFT-2085

(Warning) Failed to load test information from database. Refer to man page for more details.

**Description**

An issue has been detected in CTL restoration while loading the data which is saved in earlier session. This may happen due to bug in CTL.

**What Next**

1. If the ndmLib is compiled with older tool version, then try recompiling with current verion before using it. 2. If issue still persists, please report issue to Synopsys R&D.

---

## DFT-2087

(Error) Option '%s' is not supported %s.

**Description**

Option '%s' is not supported %s.

**What Next**

Please Review DFT Specifications

Feedback

## DFT-2088

(Error) Option '%s' is supported %s.

**Description**

Option '%s' is supported %s.

**What Next**

Please Review DFT Specifications

## DFT-2090

(Warning) Overwriting specification for scan group %s.

**Description**

scan group with such name already exists.

**What Next**

Please Review DFT Specifications. Specify scan groups with unique names.

## DFT-2094

(Error) Direction of '-hierarchical_pin' pin is incorrect

**Description**

Direction of '-hierarchical_pin' pin is incorrect

**What Next**

Please Review DFT Specifications

## DFT-2095

(Error) Unable to get valid driver for the hierarchical pin.

**Description**

Unable to get valid driver for the hierarchical pin.

**What Next**

Please Review DFT Specifications

Feedback

## DFT-2098

(Error) '-streaming true' should be specified for streaming compression mode. No '-streaming' or '-streaming false' should be specified for scan compression mode.

**Description**

'-streaming true' should be specified for streaming compression mode. No '-streaming' or '-streaming false' should be specified for scan compression mode.

**What Next**

Please Review DFT Specifications

## DFT-2099

(Error) cell %s specified with '%s' doesn't exist

**Description**

cell %s specified with '%s' doesn't exist

**What Next**

Please Review DFT Specifications

## DFT-2100

(Error) %s is not a valid hookup pin.

**Description**

%s is not a valid hookup pin.

**What Next**

Please Review DFT Specifications

## DFT-2101

(Error) Specified hook-up pin is not a valid hookup pin.

**Description**

Specified hook-up pin is not a valid hookup pin.

**What Next**

Please Review DFT Specifications

Feedback

## DFT-2104

(Warning) 'dft.test_split_wrp_cell' is only supported in UPF aware dedicated wrapper insertion flow. App-option will be ignored.

**Description**

The app-option 'dft.test_split_wrp_cell' can only be supported for UPF flow as it requires extra UPF checks for split logic placement. If its a case of non-upf flow than this app-option won't be honored and regular flow would be triggered for wrapper cell insertion.

**What Next**

To honor the app-option please enable upf flow by the following command set_wrapper_configuration -add_wrapper_cells_to_power_domains enable

## DFT-2105

(Error) Invalid TDR pin bit index for '%s' check supported register length

**Description**

You receive this error message when you specify a value that is out of allowed range for option '-exact_length' in 'set_scan_path' command.

set_dft_signal -view spec -tdr_pin TDR[6] -type TestMode Error: Invalid TDR pin bit index for 'TDR' check supported register length (DFT-2105)

**What Next**

Please use a value within the allowed range.

## DFT-2106

(Error) Invalid TDR register name. '%s' register is not defined.

**Description**

You receive this error message if you use the register pin in the 'set_dft_signal' command without specifying the register by 'set_scan_path' command.

set_dft_signal -view spec -tdr_pin temp_TDR[0] -type TestMode Error: Invalid TDR register name. 'temp_TDR' register is not defined (DFT-2106).

**What Next**

Add set_scan_path temp_TDR -class ieee_1687 -exact_length 3 -reset_state 000 -init_data 000 prior to the current command.

Feedback

## DFT-2107

(Error) Error while generating SPF during port pattern API call

**Description**

Error while generating SPF during port pattern API call

**What Next**

Please Review DFT Specifications

## DFT-2108

(Error) Bus type '%s' '%s' is not supported

**Description**

Bus type '%s' '%s' is not supported

**What Next**

Please Review DFT Specifications

## DFT-2109

(Error) Bsd Interface Signal of type '%s' is missing in '%s' command, it is mandatory to specify this signal.

**Description**

Bsd Interface Signal of type '%s' is missing in define_dft_design/set_dft_signal, it is mandatory to specify this signal.

**What Next**

Please Review define_dft_design Specification.

## DFT-2110

(Warning) Unsuccessful Q-Gating insertion on partition %s: %s

**Description**

This message is issued during Q-Gating insertion of Scan Synthesis when invalid Q-Gating setup is found. For example:

Warning: Unsuccessful Q-Gating insertion on partition PART1: missing or invalid Q-Gating enable signal (DFT-2110)

Feedback

### What Next

Verify that the values from set_scan_suppress_toggling options are properly defined following set_scan_suppress_toggling man page. For -min_slack option, the max value is defined by dft.scan.q_gating_min_slack_max_value app option, so the value in set_scan_suppress_toggling command should be under that range. Also verify that a Q-Gating signal, defined by set_dft_signal -usage scan_toggle, is specified on each DFT partition with Q-Gating setup or in the default partition instead.

---

## DFT-2120

(Warning) Cannot find valid allowed hierarchy for DFT insertion. Using %s

### Description

You receive this warning message if you use commands *set_allow_new_cells* or *create_voltage_area_rule* for limiting the creation of new cells on certain hierarchies, and *insert_dft* requires to insert a DFT cell at any of those restricted hierarchies but cannot locate an alternative allowed hierarchies (like parent hierarchies) available for such insertion.

In the scenario described above, the tool warns that the user intent of limiting cell insertion/creation on the reported hierarchy cannot be honored and one or more DFT cells will be added in the restricted hierarchy.

### What Next

Check if the restricted hierarchy really requires to be in that state, or check if parent hierarchies of the restricted hierarchy can be set as allowed for insertion of new cell as an alternative placeholder.

---

## DFT-2121

(Warning) Autofix ignored for %s violation because set_dft_configuration %s not enabled.

### Description

This warning shows that the option %s is not enabled in set_dft_configuration before executing the autofix feature to fix %s violation in the design.

### What Next

Please enable %s option in set_dft_configuration before executing insert_dft -autofix.

SH "SEE ALSO"

```
set_autofix_configuration
set_dft_configuration
set_dft_signal
```

Feedback

## DFT-2122

(Warning) Ignoring %s because scan cells are not allowed in %s mode.

**Description**

This warning shows that the option %s is not supported for the test mode %s due to scan cells not allowed in this mode. Hence, will be ignored while executing the set_wrapper_configuration command.

SH "SEE ALSO"

`set_wrapper_configuration`

## DFT-2126

(Error) Can not honor %s specification for %s

**Description**

This is error message is thrown when mixing retention and non-retention flops is not allowed (via set_scan_configuration ), but set_scan_group / set_scan_path specification contains both retention and non retention flops. Thus, set_scan_configuration and set_scan_group / set_scan_path specs are not compatible and the error is thrown.

**Messages:**

Error: Can not honor set_scan_group specification for scan group "SSG_ERROR" since it includes a mix of retention and non-retention registers, and retention cell mixing is turned off by the user. (DFT-2126)

Error: Can not honor set_scan_path specification for scan path "SSP_ERROR1" since it includes a mix of retention and non-retention registers, and retention cell mixing is turned off by the user. (DFT-2126)

**What Next**

Please review set_scan_configuration / set_scan_path / set_scan_group settings.

## DFT-2130

(Warning) Cannot enable routing distance analysis. No connected leaf driver found

**Description**

Some optimizations like SCAN chain architecting with blockage awareness requires to initialize the routing engine with at least one leaf cell driver pin as reference.

Feedback

In too small designs, like reduced test cases, it might not be possible to find a valid leaf cell driver pin connected to a valid net.

If that is the case, blockage aware optimizations might not be initialized and they will continue in non-blockage aware mode.

**What Next**

Check that the design is big enough to have valid SCAN cells with proper driver pins connected to valid nets before starting *preview_dft* or *insert_dft* commands. If it is too small to find them, this warning message can be simply ignored.

---

## DFT-2131

(Warning) New hold mux architecture flow is only supported with in-compile flow

**Description**

You receive this warning message when you try to use new hold mux architecture with pre-compile flow enabled.

**What Next**

Disable the pre-compile flow and use breakpoint flow while using new hold mux architecture to avoid this warning.

---

## DFT-2147

(Error) Register name '%s' cannot be specified for IDCODE and USERCODE instruction.

**Description**

Register name '%s' cannot be specified for IDCODE and USERCODE instruction.

**What Next**

Please Review DFT Specifications

---

## DFT-2148

(Error) Invalid argument in the capture value list. '%s' is not a valid design pin, sized binary or a sized hexadecimal constant.

**Description**

Invalid argument in the capture value list. '%s' is not a valid design pin, sized binary or a sized hexadecimal constant.

Feedback

**What Next**

Please Review DFT Specifications

---

## DFT-2149

(Error) Capture Pattern '%s' already exists for other instruction, bsd instruction specification discarded.

**Description**

Capture Pattern '%s' already exists for other instruction, bsd instruction specification discarded.

**What Next**

Please Review DFT Specifications

---

## DFT-2151

(Error) Specified capture pattern '%s' should be less than or equal to 32 bits binary.

**Description**

Specified capture pattern '%s' should be less than or equal to 32 bits binary.

**What Next**

Please Review DFT Specifications

---

## DFT-2152

(Error) Invalid bsd init data - '%s'.

**Description**

Invalid bsd init data - '%s'.

**What Next**

Please Review DFT Specifications

---

## DFT-2153

(Error) Core %s specified with '%s' for group sharing doesn't contain compressor

**Description**

Core %s specified with '%s' for group sharing doesn't contain compressor

Feedback

**What Next**

Please Review DFT Specifications

---

## DFT-2154

(Error) Partition %s is included in previously defined shared group

**Description**

Partition %s is included in previously defined shared group

**What Next**

Please Review DFT Specifications

---

## DFT-2155

(Error) no instance or reference can be attached to partition %s

**Description**

no instance or reference can be attached to partition %s

**What Next**

Please Review DFT Specifications

---

## DFT-2156

(Error) Invalid or missing %s specified for standard instructions : %s

**Description**

Invalid or missing %s specified for standard instructions : %s

**What Next**

Please Review DFT Specifications

---

## DFT-2157

(Error) value '%f' for option '-time' not in range (0.001 to 1e+06).

**Description**

value '%f' for option '-time' not in range (0.001 to 1e+06).

Feedback

**What Next**

Please Review DFT Specifications

---

## DFT-2158

(Error) Wrong value '%s' given for option '%s'.

**Description**

Wrong value '%s' given for option '%s'.

**What Next**

Please Review DFT Specifications

---

## DFT-2159

(Error) '%s' option is supported for %s instruction.

**Description**

'%s' option is supported for %s instruction.

**What Next**

Please Review DFT Specifications

---

## DFT-2160

(Error) It is mandatory to specify '-clock_cycles' option with '-time' option for EXTEST_TRAIN."

**Description**

It is mandatory to specify '-clock_cycles' option with '-time' option for EXTEST_TRAIN."

**What Next**

Please Review DFT Specifications

---

## DFT-2161

(Error) 'Clock cycle' can be specified for RUNBIST, INTEST, EXTEST_PULSE, EXTEST_TRAIN instruction.

Feedback

**Description**

'Clock cycle' can be specified for RUNBIST, INTEST, EXTEST_PULSE, EXTEST_TRAIN instruction.

**What Next**

Please Review DFT Specifications

---

## DFT-2162

(Error) Wrong value given for option '%s'.

**Description**

Wrong value given for option '%s'.

**What Next**

Please Review DFT Specifications

---

## DFT-2163

(Error) Options -time and -clock_cycles are mutually exclusive for EXTEST_PULSE instruction."

**Description**

Options -time and -clock_cycles are mutually exclusive for EXTEST_PULSE instruction."

**What Next**

Please Review DFT Specifications

---

## DFT-2165

(Warning) The core wrapper client is not enabled.

**Description**

You receive this message because you have issued a command to configure a DFT client, but that client has not been enabled by the set_dft_configuration command. For some clients, the specification is still accepted and stored, but will not take effect unless the client is enabled before DFT insertion. For other clients, the specification is discarded.

**What Next**

Please Review DFT Specifications. Enable the client by using the set_dft_configuration(2) command along with set_wrapper_configuration(2)/set_boundary_cell(2) commands.

Feedback

## DFT-2169

(Error) Option '%s' is not allowed with '-reuse_threshold' option.

**Description**

Option '%s' is not allowed with '-reuse_threshold' option.

**What Next**

Please Review DFT Specifications

## DFT-2170

(Error) Option '%s' is not specified. Options '-input_reuse_threshold' and '-output_reuse_threshold' are mandatory options.

**Description**

Option '%s' is not specified. Options '-input_reuse_threshold' and '-output_reuse_threshold' are mandatory options.

**What Next**

Please Review DFT Specifications

## DFT-2171

(Error) Options '-input_reuse_threshold', '-output_reuse_threshold' and '-input_fanin_reuse_threshold' are allowed with app option 'dft.test_core_wrap_enable_separate_reuse_threshold' only.

**Description**

Options '-input_reuse_threshold', '-output_reuse_threshold' and '-input_fanin_reuse_threshold' are allowed with app option 'dft.test_core_wrap_enable_separate_reuse_threshold' only.

**What Next**

Please Review DFT Specifications

## DFT-2174

(Info) Port '%s' is excluded from wrapping as cell '%s' connected to it is excluded from wrapping due to user specification.

Feedback

**Description**

Port '%s' is excluded from wrapping as cell '%s' connected to it is excluded from wrapping due to user specification.

**What Next**

Please Review DFT Specifications

---

## DFT-2175

(Info) Port '%s' is excluded from dedicated wrapper insertion as cell '%s' connected to it is excluded from wrapping due to shared only user specification.

**Description**

Port '%s' is excluded from dedicated wrapper insertion as cell '%s' connected to it is excluded from wrapping due to shared only user specification.

**What Next**

Please Review DFT Specifications

---

## DFT-2176

(Warning) Port '%s' is connected to cell '%s', which is represented by a CTL model. However, dedicated wrapper cell is not added due to shared only user specification for the port.

**Description**

Port '%s' is connected to cell '%s', which is represented by a CTL model. However, dedicated wrapper cell is not added due to shared only user specification for the port.

**What Next**

Please Review DFT Specifications

---

## DFT-2177

(Warning) Port '%s' is connected to cell '%s', which is represented by a BAM model. However, dedicated wrapper cell is not added due to shared only user specification for the port.

**Description**

Port '%s' is connected to cell '%s', which is represented by a BAM model. However, dedicated wrapper cell is not added due to shared only user specification for the port.

Feedback

**What Next**

Please Review DFT Specifications

---

## DFT-2178

(Warning) Port '%s' is connected to Clock Gating cell '%s'. However, dedicated wrapper cell is not added due to shared only user specification for the port.

**Description**

Port '%s' is connected to Clock Gating cell '%s'. However, dedicated wrapper cell is not added due to shared only user specification for the port.

**What Next**

Please Review DFT Specifications

---

## DFT-2179

(Warning) Port '%s' is connected to cell '%s', which is included in user scan chain. However, dedicated wrapper cell is not added due to shared only user specification for the port.

**Description**

Port '%s' is connected to cell '%s', which is included in user scan chain. However, dedicated wrapper cell is not added due to shared only user specification for the port.

**What Next**

Please Review DFT Specifications

---

## DFT-2180

(Warning) Port '%s' is connected to a Multi-Bit IO register '%s'. However, dedicated wrapper cell is not added due to shared only user specification for the port.

**Description**

Port '%s' is connected to a Multi-Bit IO register '%s'. However, dedicated wrapper cell is not added due to shared only user specification for the port.

**What Next**

Please Review DFT Specifications

Feedback

## DFT-2181

(Info) Port '%s' has reuse threshold value as -1. However, dedicated wrapper cell is not added due to shared only user specification for the port.

### Description

Port '%s' has reuse threshold value as -1. However, dedicated wrapper cell is not added due to shared only user specification for the port.

### What Next

Please Review DFT Specifications

## DFT-2182

(Warning) Port '%s' is connected to output port having dedicated wrapper cell via feedthrough path. However, dedicated wrapper cell is not added due to shared only user specification for the port.

### Description

Port '%s' is connected to output port having dedicated wrapper cell via feedthrough path. However, dedicated wrapper cell is not added due to shared only user specification for the port.

### What Next

Please Review DFT Specifications

## DFT-2184

(Info) Port '%s' has no I/O Registers associated with it; not adding dedicated wrapper cells for this port.

### Description

Port '%s' has no I/O Registers associated with it; not adding dedicated wrapper cells for this port.

### What Next

Please Review DFT Specifications

Feedback

## DFT-2185

(Warning) Port '%s' has I/O registers associated with it which have combinational logic depth %d greater than the specified depth threshold %d. However, dedicated wrapper cell is not added due to user specification for the port.

**Description**

Port '%s' has I/O registers associated with it which have combinational logic depth %d greater than the specified depth threshold %d. However, dedicated wrapper cell is not added due to user specification for the port.

**What Next**

Please Review DFT Specifications

## DFT-2186

(Warning) Port '%s' is connected to %d input registers and %d output registers, which exceeds %s reuse threshold %d. However, dedicated wrapper cell is not added due to shared only user specification for the port.

**Description**

Port '%s' is connected to %d input registers and %d output registers, which exceeds %s reuse threshold %d. However, dedicated wrapper cell is not added due to shared only user specification for the port.

**What Next**

Please Review DFT Specifications

## DFT-2187

(Warning) Port '%s' is connected to %d input registers, which exceeds %s reuse threshold %d. However, dedicated wrapper cell is not added due to shared only user specification for the port.

**Description**

Port '%s' is connected to %d input registers, which exceeds %s reuse threshold %d. However, dedicated wrapper cell is not added due to shared only user specification for the port.

**What Next**

Please Review DFT Specifications

Feedback

## DFT-2188

(warning) %s Warning %s

**Description**

This message is issued when off-by-default SCAN SQL event database is enabled, and some isues are detecting during database operations.

Typical messages includes:

- "Connection warning Connection with database fail"
- "Query warning <SQL QUERY ERROR MESSAGE>"
- "Pragma warning <SQL PRAGMA ERROR MESSAGE>"

**What Next**

This feature for SCAN SQL event database is experimental and unreleased, so its operation and debugging requires R&D analysis. Please contact DFT PE team for further support diagnosing what the root cause issue is.

## DFT-2189

(Warning) Port '%s' is connected to %d output registers, which exceeds %s reuse threshold %d. However, dedicated wrapper cell is not added due to shared only user specification for the port.

**Description**

Port '%s' is connected to %d output registers, which exceeds %s reuse threshold %d. However, dedicated wrapper cell is not added due to shared only user specification for the port.

**What Next**

Please Review DFT Specifications

## DFT-2192

(Warning) In-place ports does not support power aware dedicated wrapper insertion for area aware optimise flow.

**Description**

In-place ports does not support power aware dedicated wrapper insertion for area aware optimise flow.

Feedback

**What Next**

Please Review DFT Specifications

---

## DFT-2194

(Error) Dedicated Wrapper cell for %s port %s will introduce new non fixable MV violation, not inserting Dedicated Wrapper cell to the port.

**Description**

Dedicated Wrapper cell for %s port %s will introduce new non fixable MV violation, not inserting Dedicated Wrapper cell to the port.

**What Next**

Please Review DFT Specifications

---

## DFT-2195

(Error) Unable to find location for Dedicated Wrapper cell for %s port %s, not inserting Dedicated Wrapper cell to the port.

**Description**

Unable to find location for Dedicated Wrapper cell for %s port %s, not inserting Dedicated Wrapper cell to the port.

**What Next**

Please Review DFT Specifications

---

## DFT-2196

(Warning) No valid UPF location found for DW cell for port %s. Dedicated wrapper cell is not added for this port.

**Description**

No valid UPF location found for DW cell for port %s. Dedicated wrapper cell is not added for this port.

**What Next**

Please Review DFT Specifications

Feedback

## DFT-2198

(Info) Port %s has BSD cell inserted; not adding wrapper cell for this pin.

**Description**

Port %s has BSD cell inserted; not adding wrapper cell for this pin.

**What Next**

Please Review DFT Specifications

## DFT-2199

(Warning) Pin %s is an isolation control pin; not adding wrapper cell for this pin.

**Description**

Pin %s is an isolation control pin; not adding wrapper cell for this pin.

**What Next**

Please Review DFT Specifications

## DFT-2248

(Error) %s capture value '%s' cannot be represented in the specified '%d' bits.

**Description**

%s capture value '%s' cannot be represented in the specified '%d' bits.

**What Next**

Please Review DFT Specifications

## DFT-2291

(Warning) Ignoring instance '%s' because it is located inside a block abstraction model.

**Description**

Any hierarchy specified inside the block abstraction model is being ignored.

**What Next**

Remove the hierarchy specified inside block abstraction model to avoid this warning.

Feedback

## DFT-2300

(Warning) No PMU instance found using the given pattern '%s' in set_slm_stitching_strategy -pmm_instance_name_pattern option. Please specify correct pattern name in regular expression format for e.g. path_monitor_unit.*_loop1.

**Description**

This msg comes if no PMU instance is found with the given regular expression in set_slm_stitching_strategy -pmm_instance_name_pattern option. Please specify correct pattern name in regular expression format for e.g. path_monitor_unit.*_loop1.

## DFT-2323

(Warning) Option value '%s' is deprecated. Please use '%s' .

**Description**

You receive this warning message if you specify the deprecated class option by set_dft_ring command.

Warning: Option value 'ieee_1687' is deprecated. Please use 'ieee1687'. (DFT-2323)

**What Next**

Define correct class type for set_dft_ring command.

## DFT-2520

(Warning) Threshold check failed for the port '%s', no dedicated wrapper cell is added to the port as the port is traced to Dft Signal pin.

**Description**

You receive this warning message in Dft core wrapper insertion when threshold check fails for a port, but the tool is not adding a dedicated wrapper cell as the port is traced to Dft Signal internal pin in shared wrapper analysis.

When app option dft.test_core_wrap_ignore_dft_signal_fanout_logic is set to true, core wrapper insertion will not add dedicated wrapper cell to a port when threshold check fails for the port, but the port is traced to Constant/TestMode/ScanEnable Dft signal internal pin in shared wrapper analysis.

## DFT-2521

(Error) Specified in-place wrapper cell %s is not found in current design, ignoring the cell.

Feedback

**Description**

You receive this error message in Dft core wrapper insertion when a cell to be in-place wrapped is not found in current design. User specifies cells to be in-place wrapped using the app option dft.test_core_wrap_in_place_wrp_cell_names.

When a specified cell is not found in current design, tool ignores the cell.

Specify correct cell names with the app option to avoid the error.

---

## DFT-2522

(Warning) Specified reuse threshold value %s for in-place wrapper cell %s is not valid, ignorning the specified reuse threshold value.");

**Description**

You receive this warning message in Dft core wrapper insertion when incorrect reuse threshold value is specified for a in-place wrapper cell. Allowed reuse threshold values for in-place wrapper cells are >= -1. User specifies cells to be in-place wrapped using the app option dft.test_core_wrap_in_place_wrp_cell_names.

Tool uses parent partition's reuse threshold value in partitions flow and current design reuse threshold value in non partitions flow as reuse threshold value for the specified in-place wrapper cell when incorrect reuse threshold is specified for the cell.

Specify correct reuse threshold value with the app option to avoid the warning.

---

## DFT-2530

(Warning) Ignoring dedicated wrapper cell insertion on clock pin %s path as app option test_core_wrap_analysis_avoid_dwc_insertion_clk_path is enabled

**Description**

App option test_core_wrap_analysis_avoid_dwc_insertion_clk_path is on so avoid DWC insertion on clk path and also during fanin analysis via clock pin of wrapper cells

**What Next**

Please review the warning to check pins which are not wrapped due to this violation or disable app option if tool should not check this condition and continue adding dedicated wrapper cell on clock path

---

## DFT-2552

(Error) Couldn't find '%s' pin on wrapper cell '%s'

Feedback

**Description**

You receive this warning message in Dft core wrapper insertion when tool is trying to connect wrapper cell to pin of a port. This error is when tool couldn't find pin for connections.

---

## DFT-2600

(Error) The pattern list is parsed as key-value pair of portname and associated bit value, please provide correct number of arguments.

**Description**

The pattern list is parsed as key-value pair of portname and associated bit value, please provide correct number of arguments.

**What Next**

Please Review DFT Specifications

---

## DFT-2601

(Error) The provided bit value is invalid, please provide valid binary value 0 or 1

**Description**

The provided bit value is invalid, please provide valid binary value 0 or 1

**What Next**

Please Review DFT Specifications

---

## DFT-2602

(Error) Invalid option for remove_bsd_Compliance. Supported options are {-all, -name "pattern name"}

**Description**

Invalid option for remove_bsd_Compliance. Supported options are {-all, -name "pattern name"}

**What Next**

Please Review DFT Specifications

Feedback

## DFT-2603

(Error) BSD Compliance list is empty

**Description**

BSD Compliance list is empty

**What Next**

Please Review DFT Specifications

## DFT-2605

(Error) One or more of the provided ports are invalid, please provide valid port names

**Description**

One or more of the provided ports are invalid, please provide valid port names

**What Next**

Please Review DFT Specifications

## DFT-2606

(Error) Compliance ports can only be input or inout, please provide valid port direction for "%s"

**Description**

Compliance ports can only be input or inout, please provide valid port direction for "%s"

**What Next**

Please Review DFT Specifications

## DFT-2650

(Warning) Tool cannot find dominant clock %s. Default clock %s shall be used

**Description**

You receive this warning message when tool is not able to find specified dominant clock and specified default clock shall be used.

Feedback

## DFT-2651

(Warning) Tool cannot find local clock hookup pin %s

**Description**

You receive this warning message when tool is not able to find the specified local clock hookup pin.

## DFT-2652

(Warning) %s.

**Description**

You receive this warning message when you specify an invalid object name for a field in 'set_testability_configuration' command.

## DFT-2653

(Warning) Ignoring test point specification for '%s'. %s

**Description**

You receive this warning message when the test point specifications are ignored because a. Test point inside reference modules are not supported b. new cells are not allowed in current test point location

## DFT-2654

(Warning) %s.

**Description**

You receive this warning message when you specify an invalid instance name for a field in 'set_test_point_element' command.

## DFT-2656

(Warning) Rejected test point %s '%s'. %s

**Description**

You receive this warning message when a test point is rejected at a location for reason like max test point sharing limit reached for registers, hierarchical pin is not connected,

Feedback

test point is reported inside reference module, pin is not connected, test point insertion will result in MV violations, net is a constant net, test point already exists at the location etc.

---

## DFT-2657

(Warning) Tool couldn't find signal %s in design

**Description**

You receive this warning message when the specified signal is not found in the design

---

## DFT-2659

(Warning) Supply net difference between test point location %s and %s %s, causing MV violation.

**Description**

You will receive this warning message when there is mismatch in power of test point and selected clock/control signal.

---

## DFT-2661

(Warning) Make sure observe pin is a vector with size equivalent to minimum test point sharing ratio

**Description**

You receive this warning when you define tp_observe/tp_observe_control type design using define_dft_design

---

## DFT-2662

(Warning) Rejected duplicate test point '%s'.

**Description**

You will receive this warning message when test point location specified is duplication of exisiting specification.

---

## DFT-2690

(warning) No debug license detected, file %s contains activated debug logs

Feedback

**Description**

No debug license detected, file will be activated by default an it will contain all activated debug logs

---

## DFT-2691

(Warning) Debug file is already opened. Using debug file %s

**Description**

DFT debug command *enable_dft_debug* requires to define an explicit debug file. But if other previous debug commands like *enable_debug* or *enable_compile_debug* already opened a debug file, their debug file stream is reused for DFT debug log data too.

**WHAT'S NEXT**

Take note of the log file path and name effectively used, so the file can be easily retrieved and analyzed later when required.

---

## DFT-2702

(Error) Incorrect value '%s' specified for option '%s'

**Description**

Incorrect value '%s' specified for option '%s'

**What Next**

Please Review DFT Specifications

---

## DFT-2704

(Info) No license to enable security logic insertion

**Description**

This information message is shown during security insertion if security licence is not enable.

---

## DFT-2705

(Error) Incorrect value '%s' specified for option '%s'

**Description**

Incorrect value '%s' specified for option '%s'

Feedback

**What Next**

Please Review DFT Specifications

---

## DFT-2707

(Error) An integer greater than or equal to 0 should be specified

**Description**

An integer greater than or equal to 0 should be specified

**What Next**

Please Review DFT Specifications

---

## DFT-2708

(Error) Incorrect value '%s' specified for option '%s'

**Description**

Incorrect value '%s' specified for option '%s'

**What Next**

Please Review DFT Specifications

---

## DFT-2712

(Error) Incorrect value '%s' specified for option '%s'

**Description**

Incorrect value '%s' specified for option '%s'

**What Next**

Please Review DFT Specifications

---

## DFT-2713

(Error) Incorrect value '%s' specified for option '%s'

**Description**

Incorrect value '%s' specified for option '%s'

Feedback

**What Next**

Please Review DFT Specifications

---

## DFT-2721

(Warning) Tool couldn't find signal %s in design

**Description**

You receive this warning message when the specified signal is not found in the design

---

## DFT-2722

(Error) Incorrect pattern for set_access_control

**Description**

Incorrect pattern for set_access_control

**What Next**

Please Review DFT Specifications

---

## DFT-2724

(Error) '%s' file not found

**Description**

'%s' file not found

**What Next**

Please Review DFT Specifications

---

## DFT-2757

(Warning) Unable to fork new processes due to resource issues. Please try on bigger compute machine

**Description**

You receive this warning message when current compute machine is not sufficient(does not have enough memory) to run tool on current design. This could lead to missing some importtant steps in the flow which results in error or undefined behavior or incorrect results.

Feedback

### What Next

User can try to run the design on different/larger(memory) compute machine.

---

## DFT-2759

(Warning) Environment variable '%s' will be depricated in future release. Please use '%s' app option instead of this environment variable

### Description

Users get this messsage when they use environment varibale to enable specific flow or behavior for which an app option already exists. Even with this message the flow will run fine and this is just a heads up to user to update their scripts as the environment varibale will be removed in future releases.

### What Next

Users are expected to update their scripts to remove environment varibale and replace it with app option.

---

## DFT-2821

(Warning) Subblock mixes clock edges while top block does not allow mixing. (partition %s)

### Description

This message is issued when one of subblocks has mixed clock edges, while DFT specification does not allow mixing clock edges. For example,

Warning: Subblock mixes clock edges while top block does not allow mixing. (partition default_partition) (DFT-2821)

### What Next

The warning message shows that DFT clock mixing specification is not compatible with subblocks.

Check subblock generation script and disable edge mixing or update set_scan_configuration -clock_mixing settings for the top block.

---

## DFT-2822

(Warning) Subblock mixes clocks while top block does not allow mixing. (partition %s)

Feedback

**Description**

This message is issued when one of subblocks has mixed clockd, while DFT specification does not allow mixing clocks. For example,

Warning: Subblock mixes clocks while top block does not allow mixing. (partition default_partition) (DFT-2822)

**What Next**

The warning message shows that DFT clock mixing specification is not compatible with subblocks.

Check subblock generation script and disable clock mixing or update set_scan_configuration -clock_mixing settings for the top block.

---

## DFT-2831

(Warning) Invalid cell for ScanDEF generation of serial routed group found in -include_elements specification (%s). ScanDEF generation of group %s will be skipped

**Description**

This message is issued during Scan Synthesis when an invalid cell was defined in set_scan_group -include_elements specification if used with -serial_routed true. For example:

Warning: Invalid cell for ScanDEF generation of serial routed group found in -include_elements specification (core_cell). ScanDEF generation of group scan_group1 will be skipped (DFT-2831)

**What Next**

This warning shows that some of the cells defined in the -include_elements option of a scan group defined with -serial_routed true where found invalid.

For serial routed groups, only leaf cells are allowed in the -include_elements specification. Verify that all the cells defined in -include_elements specification are correctly written and don't include hierarchical cells or CTL designs.

---

## DFT-2832

(Warning) Skipping serial routed group %s from ScanDEF (unable to trace internal elements through access data pins)

Feedback

**Description**

This message is issued during ScanDEF generation step of Scan Synthesis when the tool couldn't generate a ScanDEF ordered segment for a pre-stitched scan group because the internal trace of the group failed. For example:

Warning: Skipping serial routed group scan_group1 from ScanDEF (unable to trace internal elements through access data pins) (DFT-2832)

**What Next**

This warning shows that the internal tracing of the reported serial routed scan group was not possible and it will be excluded from the ScanDEF. Verify that the internal elements that connect the ScanDataIn and ScanDataOut are properly connected and, if the group was defined using the -include_elements option, that all the internal elements were specified and in the correct order.

---

## DFT-2902

(Warning) 'check_dft' command should be run to get an early estimate of the coverage and scan readiness of the design.

**Description**

It is recommended to run 'check_dft' to get the coverage and scan readiness of the design before actual IP insertion.

---

## DFT-2905

(Error) "-type '1839_stap' without enabling -std {ieee1838_2019} is not allowed."

**Description**

This error message is issued when -type '1838_stap' is used without enabling -std {ieee1838_2019}.

Error: -type '1839_stap' without enabling -std {ieee1838_2019} is not allowed.(DFT-2905)

**What Next**

Enable -std {ieee1838_2019} using set_bsd_configuration.

---

## DFT-2907

(Error) "-type '1838_stap' is not allowed when 1838 stap ports are defined using -view exist."

Feedback

**Description**

This error message is issued when any of the 1838 stap port are defined using option -view exist.

Error: -type '1838_stap' is not allowed when 1838 stap ports are defined using -view exist. (DFT-2907)

**What Next**

Don't define any of the 1838 stap port using option -view exist.

---

## DFT-2910

(Error) Not able to build PRPG. Consider increasing the PRPG width.

**Description**

This error message will be displayed when PRPG width is small, Increase the PRPG width to build PRPG.

---

## DFT-2911

(Warning) the 'setup_rtl_flow' command is not specified. Specify this command before the 'create_port, create_bus_port, create_connection' command. Otherwise, testmax_advisor can lead to FATAL at runtime

**Description**

This warning will be displayed only in the RTL shell if the users does not specify the 'setup_rtl_flow' command. This command is required to observe pseudo signal notification from NDM

---

## DFT-2912

(Warning) the 'Enabling external clock in few/all partitions, default partition will be considered for OCC Scan chain'

**Description**

this will be be due to Enabling external clock in few/all partitions, default partition will be considered for OCC Scan chain. if in few/all partitions, external clocks is enabled with pipeline, default partition is always considered for enabling external clock dummy OCC chain

Feedback

## DFT-2913

(Warning) the 'External clocks should be enabled in all partitions else leads to K41 DRC violation

**Description**

This warning will be displayed if the external clocks are not enabled in all partitions

## DFT-2921

(Error) Instance '%s' not found in design '%s'

**Description**

Instance '%s' not found in design '%s'

**What Next**

Please Review DFT Specifications

## DFT-2922

(Error) File path '%s' not found or not accessible

**Description**

File path '%s' not found or not accessible

**What Next**

Please Review DFT Specifications

## DFT-2923

(Error) No memories found in design. Aborting plan_dft command...

**Description**

No memories found in design. Aborting plan_dft command...

**What Next**

Please Review DFT Specifications

## DFT-2928

(Error) No corresponding PDL test models specified (for specified ICL test models).

Feedback

**Description**

No corresponding PDL test models specified (for specified ICL test models).

**What Next**

Please Review DFT Specifications

---

## DFT-2929

(Error) No corresponding ICL test models specified (for specified PDL test models).

**Description**

No corresponding ICL test models specified (for specified PDL test models).

**What Next**

Please Review DFT Specifications

---

## DFT-2932

(Error) Please enable top_server/shadow_server command when IEEE_1687 and dft_access both are enable.

**Description**

You receive this error message when IEEE_1687 and dft_access both are enable and server is not used.

Error: Please enable top_server/shadow_server command when IEEE_1687 and dft_access both are enable. (DFT-2932)

**What Next**

Please enable server.

---

## DFT-2933

(Warning) Current version does not match with recommended version '%s' for '%s'."

**Description**

This warning message will be displayed when selected SMS/SHS compiler vesrion does not match with recommended compiler version. User must use recommended compiler versions to avoid unforeseeable issues. The message will be displayed when compilers information is populated.

Feedback

## DFT-2935

(Error) Hier SIDO-MIMC case missing local_codec_drc_cycle specification in '%s'.

**Description**

This error message will be displayed when local_codec_drc_cycle specification is missing in Hier SIDO-MIMC case.

## DFT-2951

(Error) Current version '%s' of '%s' is earlier than recommended version '%s'. Aborting execution

**Description**

This error message will be displayed when version of underlying tools SpyGlass, EMBEDIT is earlier than the recommended version. User must stick to recommended version of tools to avoid unforeseeable issues. The error will be reported on fist use of SpyGlass/EMBEDIT modules.

## DFT-2952

(Error) Current version '%s' of '%s' is greater than recommended version '%s'. Aborting execution

**Description**

This error message will be displayed when version of underlying tools SpyGlass, EMBEDIT is greater than the recommended version. User must stick to recommended version of tools to avoid unforeseeable issues. The error will be reported on fist use of SpyGlass/EMBEDIT modules.

## DFT-2953

(Error) Synopsys spyglass module not loaded or corrupted. Aborting execution

**Description**

This error message will be displayed when no SpyGlass module is found in TMM installtion area(testmax_advisor link) and SPYGLASS_HOME is not set. The error is also reported when specified SpyGlass is corrupt(incomplete installation). The error will be reported on fist use of SpyGlass module.

Feedback

## DFT-2954

(Error) Synopsys embedit module not loaded. Aborting execution

**Description**

This error message will be displayed when no EMBEDIT module is found in TMM installtion area(embedit directory) and EMBEIDT_HOME is not set and EMBEDIT binaries is not in PATH variable. The error will be reported on fist use of EMBEDIT module.

## DFT-2955

(Error) Synopsys vcs/vcsmx module not loaded or VCS_HOME is not set. Aborting execution

**Description**

This error message will be displayed when no VCS module is loaded and VCS_HOME is not set. The error will be reported on fist use of VCS module.

## DFT-2956

(Error) "%s tck time (%f.00) must be less than strobe_time (%f.00) (specified with test_bsd_default_strobe App option)."

**Description**

You receive this error message when you specify a strobe time which is less than TCK fall and rise time.

set_dft_signal -view exist -type TCK -port tck -timing [list 45 55] set_app_options -as_user_default -name dft.test_bsd_default_strobe -value 40 //strobe time less than tck timing

write_testbench -o bsd_rtl_patterns -test_mode all_dft_protocol -type {all} Error: Rising tck time (45.000000.00) must be less than strobe_time (40.000000.00) (specified with test_bsd_default_strobe App option). (DFT-2956) Error: Falling tck time (45.000000.00) must be less than strobe_time (40.000000.00) (specified with test_bsd_default_strobe App option). (DFT-2956)

**What Next**

Please use a strobe time more than the TCK rise and fall time.

Feedback

## DFT-2961

(Warning) %s option is not supported in %s shell for set_boundary_cell. Ignoring this option.

**Description**

Following option of set_boundary_cell is not supported in the designated shell. It is supported only in TestMax Manager shell.

## DFT-2978

(Warning) Specified interface merge type %s is invalid. Using default interface merge type mux.

**Description**

This warning message reports that the specified streaming fabric interface merge type for ScanDataOut is invalid, and the default merging type mux will be used.

**What Next**

Verify the merge type definition in the set_dft_streaming_fabric_interface command. Only xor and mux merge tyes are accepted.

## DFT-2989

(Error) Invalid specification with -dedicated_chains_for option.

**Description**

Describes any configuration error while specifying *set_scan_configuration* -dedicated_chains_for.

The error message contains a description of the problem. Common causes can be:

- Tried to specify this option for a specific test mode or partition.
- Specified an unknown or unsupported scan type.

**What Next**

Review usage of *set_scan_configuration* -dedicated_chains_for

## DFT-2999

(Information) splitting chain %s to %d sub-chains, dropping %d elements to maximize number of optimizable elements.

Feedback

### Description

Stub chains which have MV crossings or non transparent elements are split during dft optimization, and elements might be dropped. Stub chains can be reconfigured during dft optimization to improve QoR. If a stub chain has MV crossings (a change in voltage area, power domain, or power supply), which can occur if no MV information was available during dft synthesis, the stub chain is split where MV crossings are found, to avoid incompatible connections. If stub chain splitting would produce a chain that is not optimizable (for example, no cells are available to be promoted to START and STOP for the new stub chain), then those elements are dropped from the scandef considered in dft optimization. The dropped elements remain scan stitched.

### What Next

If this message is seen in the flow, verify that the UPF setup has been properly sourced before dft synthesis.

### See Also

- report_stub_chains

---

## DFT-3001

(Information) Writing protocol for test mode %s in %s.

### Description

Tool has started writing test protocol for the given test mode. The tool will create a .spf file.

---

## DFT-3003

(Error) Could not find hookup pin %s.

### Description

Could not find hookup pin %s.

### What Next

Please Review DFT Specifications

---

## DFT-3004

(Error) DFT Design not defined for RTL PAD %s.

### Description

DFT Design not defined for RTL PAD %s.

Feedback

### What Next

Please Review DFT Specifications

---

## DFT-3005

(Information) Logical scan chain information and reporting not available after P&R backend commands.

### Description

You receive this Information message if you used DFT Scan reporting commands that use logical Scan chain information after a FC or ICC2 backend command was called.

The complete logical DFT scan chain data needed for reporting and similar commands is available after DFT was inserted (*insert_dft*), during the frontend flow steps like *compile_fusion* and it is kept persistent during NDM based *save_block*, *open_block* or alike operations.

During optimizations changes on logical DFT scan chains are tracked and the data is kept up to date, but once any backend command is called (FC or ICC2) it is not possible to maintain the information up to date, and therefore the data is reset, and commands depending on it cannot report anymore, so if DFT logical scan chain reporting is needed, then it should be obtained before any backend command is being executed, e.g. after *insert_dft* or *compile_fusion*.

Frontend commands affected by this, are e.g.: *report_scan_path*, *report_dft* (with *-scan_path* option and/or Scan Path Architecture section), *get_scan_cell_names*, *get_scan_chain_names* and *get_scan_chains*.

Nevertheless, all DFT stub chain (ScanDEF) based reports will continue to work as expected because they are based on independent scan stub chain (ScanDEF) data.

### What Next

If DFT logical scan chain reporting is needed, then it should be obtained before any backend command is being executed, e.g. directly after *insert_dft* or *compile_fusion*, but user can continue using all DFT stub chain (ScanDEF) based reporting commands.

### See Also

- check_scan_chain

---

## DFT-3006

(Error) Couldn't find Top Interface signal for select signal %s

Feedback

**Description**

Couldn't find Top Interface signal for select signal %s

**What Next**

Please Review DFT Specifications

---

## DFT-3008

(Error) '-multi_configurations' should have list size in the range [1,3]

**Description**

'-multi_configurations' should have list size in the range [1,3]

**What Next**

Please Review DFT Specifications

---

## DFT-3009

(Error) '-configuration_ports' should have list size in the range [1,2]

**Description**

'-configuration_ports' should have list size in the range [1,2]

**What Next**

Please Review DFT Specifications

---

## DFT-3010

(Error) Wrong DFTMAX parameters specified, chain_count/max_length value is either too high or too low, Specify correct value

**Description**

The message occurs when user specifies chain_count or max_length value too high or too low

**What Next**

Please specify correct DFTMAX parameters (chain_count/max_length value).

Feedback

## DFT-3023

(Error) Invalid value for option '-mode' specified, allowed options are any|shift| capturenDiscarding DFT,SDC and Formality constraints generation.

**Description**

Invalid value for option '-mode' specified, allowed options are any|shift|capturenDiscarding DFT,SDC and Formality constraints generation.

**What Next**

Please Review DFT Specifications

## DFT-3024

(Error) Invalid value for option '-occ' specified, allowed options are any|bypass| activenDiscarding DFT,SDC and Formality constraints generation.

**Description**

Invalid value for option '-occ' specified, allowed options are any|bypass|activenDiscarding DFT,SDC and Formality constraints generation.

**What Next**

Please Review DFT Specifications

## DFT-3025

(Error) Invalid value for option '-constant' specified, allowed options are defined| undefinednDiscarding DFT,SDC and Formality constraints generation.

**Description**

Invalid value for option '-constant' specified, allowed options are defined| undefinednDiscarding DFT,SDC and Formality constraints generation.

**What Next**

Please Review DFT Specifications

## DFT-3026

(Error) Invalid value for option '-test_mode' specified, allowed options are none| definednDiscarding DFT,SDC and formality constraints generation

Feedback

**Description**

Invalid value for option '-test_mode' specified, allowed options are none| definednDiscarding DFT,SDC and formality constraints generation

**What Next**

Please Review DFT Specifications

---

## DFT-3027

(Error) Invalid value for option '-input_delay' specified, allowed options are enable| disablenDiscarding DFT,SDC and formality constraints generation.

**Description**

Invalid value for option '-input_delay' specified, allowed options are enable| disablenDiscarding DFT,SDC and formality constraints generation.

**What Next**

Please Review DFT Specifications

---

## DFT-3028

(Error) Invalid value for option '-output_delay' specified, allowed options are enable| disablenDiscarding DFT,SDC and formality constraints generation.

**Description**

Invalid value for option '-output_delay' specified, allowed options are enable| disablenDiscarding DFT,SDC and formality constraints generation.

**What Next**

Please Review DFT Specifications

---

## DFT-3029

(Error) Required argument '-output' was not found

**Description**

Required argument '-output' was not found

**What Next**

Please Review DFT Specifications

Feedback

## DFT-3030

(Error) Required argument '-sms_outdir' was not found

**Description**

Required argument '-sms_outdir' was not found

**What Next**

Please Review DFT Specifications

## DFT-3034

(Error) Invalid value for option '-occ' specified, allowed options are any|bypass|activenDiscarding Timing constraints generation.

**Description**

Invalid value for option '-occ' specified, allowed options are any|bypass|activenDiscarding Timing constraints generation.

**What Next**

Please Review DFT Specifications

## DFT-3035

(Error) Invalid value for option '-constant' specified, allowed options are defined|undefinednDiscarding Timing constraints generation.

**Description**

Invalid value for option '-constant' specified, allowed options are defined|undefinednDiscarding Timing constraints generation.

**What Next**

Please Review DFT Specifications

## DFT-3036

(Error) Invalid value for option '-test_mode' specified, allowed options are none|definednDiscarding Timing constraints generation

Feedback

**Description**

Invalid value for option '-test_mode' specified, allowed options are none| definednDiscarding Timing constraints generation

**What Next**

Please Review DFT Specifications

---

## DFT-3037

(Error) Invalid value for option '-input_delay' specified, allowed options are enable| disablenDiscarding Timing constraints generation.

**Description**

Invalid value for option '-input_delay' specified, allowed options are enable| disablenDiscarding Timing constraints generation.

**What Next**

Please Review DFT Specifications

---

## DFT-3038

(Error) Invalid value for option '-output_delay' specified, allowed options are enable| disablenDiscarding Timing constraints generation.

**Description**

Invalid value for option '-output_delay' specified, allowed options are enable| disablenDiscarding Timing constraints generation.

**What Next**

Please Review DFT Specifications

---

## DFT-3039

(Error) Required argument '-output' was not found

**Description**

Required argument '-output' was not found

**What Next**

Please Review DFT Specifications

Feedback

## DFT-3040

(Error) Permission denied! Can't open user connection file

**Description**

Permission denied! Can't open user connection file

**What Next**

Please Review DFT Specifications

## DFT-3041

(Error) Invalid value for option '-mode' specified, allowed options are any|shift| capturenDiscarding Timing constraints generation.

**Description**

Invalid value for option '-mode' specified, allowed options are any|shift|capturenDiscarding Timing constraints generation.

**What Next**

Please Review DFT Specifications

## DFT-3042

(Error) No tdmEnvironment found. Skipping insertCoreWrappingLogic.

**Description**

No tdmEnvironment found. Skipping insertCoreWrappingLogic.

**What Next**

Please Review DFT Specifications

## DFT-3047

(Error) Design %s has no pattern_exec called '%s'.

**Description**

You receive this error message if you specify incorrect value to pattern_exec option of write_test_protocol command.

Error: Design subsystem0 has no pattern_exec called 'scan_occ_bypass' (DFT-3047)

Feedback

**What Next**

Define the correct DFT signal type.

---

## DFT-3049

(Error) %s is not defined by set_scan_path.

**Description**

%s is not defined by set_scan_path.

**What Next**

Please Review DFT Specifications

---

## DFT-3053

(Error) Required argument '-sms_outdir' was not found

**Description**

Required argument '-sms_outdir' was not found

**What Next**

Please Review DFT Specifications

---

## DFT-3054

(Error) Pin length of load pin '%s' doesn't match to TDR pin length.

**Description**

Pin length of load pin '%s' doesn't match to TDR pin length.

**What Next**

Please Review DFT Specifications

---

## DFT-3055

(Error) '%s' option is supported for 'set_scan_path %s %s' only."

**Description**

'%s' option is supported for 'set_scan_path %s %s' only."

Feedback

**What Next**

Please Review DFT Specifications

---

## DFT-3056

(Error) Invalid value '%s' for option '-sib' specified, allowed options are enable | disable

**Description**

Invalid value '%s' for option '-sib' specified, allowed options are enable | disable

**What Next**

Please Review DFT Specifications

---

## DFT-3057

(Error) Options '-register' and '-instruction' are mutually exclusive.

**Description**

Options '-register' and '-instruction' are mutually exclusive.

**What Next**

Please Review DFT Specifications

---

## DFT-3058

(Error) Cannot run DRC on unmapped netlist

**Description**

Cannot run DRC on unmapped netlist

**What Next**

Please Review DFT Specifications

---

## DFT-3059

(Information) Skipping DRC. No spec changes since last DRC Run.

**Description**

Tool has detected that since the last DRC run, there has been no changes in the DFT specifications and the last DRC results are still valid. Hence, the DRC run has been skipped.

Feedback

## DFT-3060

(Error) Missing isolation control. Please specify isolation control signal name

**Description**

Missing isolation control. Please specify isolation control signal name

**What Next**

Please Review DFT Specifications

## DFT-3061

(Error) No isolation control found with specified name %s. Please specify isolation control signal with set_dft_signal -type IsolationControl -port IsoCtrl

**Description**

No isolation control found with specified name %s. Please specify isolation control signal with set_dft_signal -type IsolationControl -port IsoCtrl

**What Next**

Please Review DFT Specifications

## DFT-3062

(Error) Port '%s' specified for safe val 1 and 0 both. Please provide correct specification

**Description**

Port '%s' specified for safe val 1 and 0 both. Please provide correct specification

**What Next**

Please Review DFT Specifications

## DFT-3063

(Error) %s is not defined by set_scan_path/set_dft_ring.

**Description**

You receive this error message if you use the element in ordered element list without specifying the element by set_scan_path command or by set_dft_ring command.

Error: RING is not defined by set_scan_path/set_dft_ring. (DFT-3063)

Feedback

**What Next**

If element is ring, define it by set_dft_ring. If it is register, define it by set_scan_path.

---

## DFT-3066

(ERROR) "%s" interface keyword specified with "%s" option.

**Description**

The following Interface name should NOT be specified with "-ieee_1500_interface_only enable" option:

---------- | Interface Name | Attribute | |----------| | i_mode | Input | |----------| | i_test_start | Input | |----------| | i_test_enable | Input | |----------| | i_timeout_cnt_strap | Input | |----------| | i_mem_start_addr | Input | |----------| | i_reset_counter | Input | ----------

The following Interface name should be specified with "-memory_write enable" option only:

---------- | Interface Name | Attribute | |----------| | o_mem_wr_en | Output | |----------| | o_mem_wr_data | Output | ----------

**What Next**

Please provide correct option or remove these keywords from the "set_dft_access_element -interface" command(s).

---

## DFT-3067

(Error) PDL file name '%s' doesn't exist or unreadable.

**Description**

PDL file name '%s' doesn't exist or unreadable.

**What Next**

Please Review DFT Specifications

---

## DFT-3068

(Error) %s"%s" driving %s "%s" ports

**Description**

%s"%s" driving %s "%s" ports

Feedback

**What Next**

Please Review DFT Specifications

---

## DFT-3069

(Error) select port '%s' is not defined as dft signal.

**Description**

select port '%s' is not defined as dft signal.

**What Next**

Please Review DFT Specifications

---

## DFT-3070

(Error) No TDR Bus port found with name '%s' in existing DFT IP .

**Description**

No TDR Bus port found with name '%s' in existing DFT IP .

**What Next**

Please Review DFT Specifications

---

## DFT-3071

(Error) Input design doesn't have any %s instance with name '%s'.

**Description**

Input design doesn't have any %s instance with name '%s'.

**What Next**

Please Review DFT Specifications

---

## DFT-3072

(Error) '-include' option is mandatory for class 'bsd' with view existing.

**Description**

'-include' option is mandatory for class 'bsd' with view existing.

Feedback

**What Next**

Please Review DFT Specifications

---

## DFT-3074

(Error) Unable to instantiate wrapper cell %s due to allow_new_cells disabled on %s hierarchy. Use set_dft_location/set_boundary_cell to change wrapper cell location.

**Description**

New cell creation on %s hierarchy has been disabled through set_allow_new_cells command. So dedicated wrapper cell can not be created. User need to change the location of wrapper cell instantiation using set_dft_location/set_boundary_cell.

SH "SEE ALSO"

```
set_dft_location
set_boundary_cell
set_allow_new_cells
```

---

## DFT-3076

(Error) Unable to instantiate DFTTop IP %s in %s hierarchy, due to allow_new_cells disabled in the hierarchy. Use set_dft_location to change DFTIP location.

**Description**

Unable to instantiate DFTTop IP %s in %s hierarchy, due to allow_new_cells disabled in the hierarchy. Use set_dft_location to change DFTIP location.

**What Next**

Please Review DFT Specifications

---

## DFT-3079

(Error) Masis file name '%s' doesn't exist or unreadable.

**Description**

Masis file name '%s' doesn't exist or unreadable.

**What Next**

Please Review DFT Specifications

Feedback

## DFT-3080

(Error) AIT Configuration is empty.

**Description**

AIT Configuration is empty.

**What Next**

Please Review DFT Specifications

## DFT-3081

(Error) 3Dic Configuration is empty.

**Description**

3Dic Configuration is empty.

**What Next**

Please Review DFT Specifications

## DFT-3082

(Error) DFT Access Configuration is empty.

**Description**

DFT Access Configuration is empty.

**What Next**

Please Review DFT Specifications

## DFT-3083

(Information) Constant pins of DFT-modeled instance %s(%s) are incorrectly controlled in all_dft mode.

**Description**

The message shows that testmode/constant signal polarity of core and the top level differs.

**What Next**

This is an information-only message. No action is required.

Feedback

## DFT-3085

Information: Stopping core wrapping analysis at cell %s for core %s.

**Description**

Stopping core wrapper analysis at cell as it crosses the boundary of partition currently being processed.

**What Next**

This is an information-only message. No action is required.

For example,

Information: Stopping core wrapping analysis at cell gtcgpinf_vccgt_wrap1 for core "gtcgpinf_vccinf_wrap1". (DFT-3085)

The combinations of commands that can trigger this warning message are:

- *set_wrapper_configuration -add_wrapper_cells_to_power_domain enable.*
- *define_dft_partition <> -wrapper enable*

## DFT-3086

(Error) Unable to insert AON custom wrapper cell for port '%s' as no lib cell specified for AON dual rail mux. App option 'dft.test_ao_dedicated_wrp_mux_lib_cell' is not set. Ignoring port for wrapper analysis.

**Description**

Ignoring port for AON custom wrapper insertion. This could be because of the app option 'dft.test_ao_dedicated_wrp_mux_lib_cell' is not set. The app option sets the value for AON dual rail mux library cell name which is required for AON custom wrapper insertion.

**What Next**

1. Set the app option 'dft.test_ao_dedicated_wrp_mux_lib_cell' value.

## DFT-3087

(Warning) Could not find clock control signals on top IP. Cannot mask clock gating cells of partition %s.

**Description**

This warning message is issued when DFT Top IP does not have gate cells control pins for specified partition for wrapper gate cells feature

Feedback

For example,

Warning: Could not find clock control signals on top IP. Cannot mask clock gating cells of partition default_partition (DFT-3087)

The commands that can trigger this warning message are:

- *insert_dft*
- *preview_dft*
- *set_wrapper_configuration*

**What Next**

Specify gate cells control signals wrp_cell_clk_ctrl and core_cell_clk_ctrl using set_dft_signal command for the partition missing these signals

---

## DFT-3088

(Warning) Unable to find IP in partition %s, falling back to default_partition IP signals.

**Description**

This warning message is issued when DFT Top IP is not present in specified partition for wrapper gate cells feature

For example,

Warning: Unable to find IP in partition default_partition, falling back to default_partition IP signals (DFT-3088)

The commands that can trigger this warning message are:

- *insert_dft*
- *preview_dft*
- *set_wrapper_configuration*

**What Next**

Specify gate cells control signals wrp_cell_clk_ctrl and core_cell_clk_ctrl using set_dft_signal command for the partition missing these signals

---

## DFT-3089

(Error) Could not find clock control signals in design. Cannot perform clock masking.

Feedback

**Description**

This error message is issued when tool is unable to find gate cell control signal via user specification or DFTTop IP to perform clock masking for one or all DFT partitions

For example,

Error: Could not find clock control signals in design. Cannot perform clock masking (DFT-3089)

The commands that can trigger this error message are:

- *insert_dft*
- *preview_dft*
- *set_wrapper_configuration*

**What Next**

Specify gate cells signals wrp_cell_clk_ctrl and core_cell_clk_ctrl for gate cells functionality or disable gate_cells functionality via set_wrapper_configuration -gate_cells none option

---

## DFT-3090

(Information) Skipping Q-Gating insertion on partition %s: %s

**Description**

This message report that Q-Gating insertion will be skipped on a certain DFT partition. For example:

Information: Skipping Q-Gating insertion on partition PART1: Invalid Q-Gating configuration (DFT-3090)

**What Next**

Verify that the values from set_scan_suppress_toggling options are properly defined following set_scan_suppress_toggling man page. For -min_slack option, the max value is defined by dft.scan.q_gating_min_slack_max_value app option, so the value in set_scan_suppress_toggling command should be under that range. Also verify that a Q-Gating signal, defined by set_dft_signal -usage scan_toggle, is specified on each DFT partition with Q-Gating setup or in the default partition instead.

---

## DFT-3091

(Warning) No gate cells signals found for partition %s. Clock gating cells in partition %s wont be masked

Feedback

**Description**

This warning message is issued when control signals (wrp_cell_clk_ctrl and core_cell_clk_ctrl) are missing for wrapper gate cells feature

For example,

Warning: No gate cells signals found for partition default_partition. Clock gating cells in partition default_partition wont be masked (DFT-3091)

The commands that can trigger this warning message are:

- *insert_dft*
- *preview_dft*
- *set_wrapper_configuration*

**What Next**

Specify gate cells control signals wrp_cell_clk_ctrl and core_cell_clk_ctrl using set_dft_signal command for the partition missing these signals

---

## DFT-3092

(Error) Leaf level hierachy %s is not allowed with '-include' option while '-wrapper' option is enabled.

**Description**

This error message is issued when user tries to give a leaf level hierarchy as a soft core partition via -wrapper enable option

For example,

Error: Leaf level hierachy de_a/o_rrdy_reg is not allowed with '-include' option while '-wrapper' option is enabled (DFT-3092)

The commands that can trigger this error message are:

- *define_dft_partition*

**What Next**

Check the ICG module name provided for new gate cell addition is a valid ICG cell

---

## DFT-3093

(Error) %s pin not found for ICG reference cell module '%s'. New CG cells wont be added.

Feedback

**Description**

This error message is issued when tool is unable to find relevant pins on clock gating cell specified for set_wrapper_configuration gate_cell feature via dft.test_icg_p_ref_for_dft app option

For example,

Error: Clock input pin not found for ICG reference cell module 'CGLPXT' (DFT-3093)

The commands that can trigger this error message are:

- *insert_dft*
- *preview_dft*
- *set_wrapper_configuration*

**What Next**

Check the ICG module name provided for new gate cell addition is a valid ICG cell

---

## DFT-3094

(Error) Test mode encoding for test mode '%s' is same as test mode '%s'.

**Description**

This error message is issued when tool encounters same encoding values values for two user define test modes

For example,

Error: Test mode encoding for test mode 'test_mode1' is same as test mode 'test_mode2' (DFT-3094)

The commands that can trigger this error message are:

- *insert_dft*
- *preview_dft*
- *set_test_mode_encoding*

**What Next**

Review and correct TestMode signal encoding for the defined test modes

---

## DFT-3095

(Warning) set_scan_path -class wrapper is not supported without dedicated SI/SO signals.

Feedback

**Description**

This warning message is issued when a set_scan_path -class wrapper specification is encountered without ScanDataIn and ScanDataOut signal. This is not supported in tool.

For example,

Warning: set_scan_path -class wrapper is not supported without dedicated SI/SO signals (DFT-3095)

The commands that can trigger this warning message are:

- *insert_dft*
- *preview_dft*

**What Next**

Remove the set_scan_path -class wrapper specification or provide ScanDataIn and ScanDataOut for those set_scan_paths in order to provide set_scan_path specification and rerun. Otherwise tool will ignore the specification and move on further with insert_dft

---

## DFT-3096

(Error) Failed to create DFT Top IP for partition %s.

**Description**

This error message is issued tool is unable to create a valid DFT Top IP module for a partition

For example,

Error: Failed to create DFT Top IP for partition default_partition (DFT-3096)

The commands that can trigger this error message are:

- *insert_dft*
- *preview_dft*

---

## DFT-3097

(Information) Accepted test mode encoding specification in partition %s for %s mode

**Description**

This information message is printed when set_test_mode_encoding command is accepted.

Feedback

For example,

Information: Accepted test mode encoding specification in partition "part_A" for wrp_if_c mode (DFT-3097)

The commands that can trigger this error message are:

- *set_test_mode_encoding*

---

## DFT-3098

(Information) Port '%s' is excluded from wrapping due to user specification

**Description**

This information is shown when port is excluded from wrapping due to user specification which is set by command set_boundary_cell -type none.

For example,

Information: Port 'out3' is excluded from wrapping due to user specification (DFT-3098)

The commands that can trigger this error message are:

- *insert_dft*
- *preview_dft*
- *set_boundary_cell -type none*

**What Next**

Check wrapping status of port which is mentioned above. Port should not be wrapped.

---

## DFT-3099

(Information) Port '%s' is excluded from wrapping as pin connected to it is excluded from wrapping due to user specification

**Description**

This information is shown when port is excluded from wrapping as it is connected to pin which is excluded from wrapping by command set_boundary_cell -type none.

For example,

Information: Port 'out3' is excluded from wrapping as pin connected to it is excluded from wrapping due to user specification (DFT-3099)

Feedback

The commands that can trigger this error message are:

- *insert_dft*
- *preview_dft*
- *set_boundary_cell -type none*

**What Next**

Check wrapping status of port which is mentioned above. Port should not be wrapped.

---

## DFT-3100

(Information) DFT Scan insertion was disabled by %s

**Description**

This message is issued when there are external reasons for *insert_dft* command to skip performing DFT SCAN insertion over a design.

For example, this message will be issued if user disables DFT SCAN insertion using application option dft.scan.insertion_disabled (default false).

**What Next**

If the behavior is undesired, review the DFT & design setup to remove the commands or options that enabled the skipping of DFT SCAN insertion.

---

## DFT-3105

(Information) Port '%s' is connected to hierarchical cell '%s' having user specification; dedicated wrapper cell will be added to the port.

**Description**

This information is shown when a port is connected to hierarchical cell having set_boundary_cell -type WC_D1 specification; dedicated wrapper cell will be added to the port

For example,

Information: Port 'in1' is connected to hierarchical cell 'in1' having user specification; dedicated wrapper cell will be added to the port. (DFT-3105)

Feedback

The commands that can trigger this error message are:

- *insert_dft*
- *preview_dft*
- *set_boundary_cell -type WC_D1*

**What Next**

Check wrapping status of port which is mentioned above. Port should not be wrapped.

---

## DFT-3106

(Error) Misr Load value should be greater than number of outputs in the CODEC to consider.

**Description**

Misr Load value should be greater than number of outputs in the CODEC to consider.

**What Next**

Please Review DFT Specifications

---

## DFT-3107

(Error) Number of MISRs should always be factor of scanouts per CODEC for correct concatenation.

**Description**

Number of MISRs should always be factor of scanouts per CODEC for correct concatenation.

**What Next**

Please Review DFT Specifications

---

## DFT-3108

(Error) -shared_outputs value should be same across all partitions.

**Description**

-shared_outputs value should be same across all partitions.

**What Next**

Please Review DFT Specifications

Feedback

## DFT-3109

(Information) Port '%s' is connected to cell '%s', which is represented by a CTL model; dedicated wrapper cell will be added to the port.

**Description**

This information is shown when a port is connected to cell having represented by CTL model; dedicated wrapper cell will be added to the port

For example,

Information: Port 'in1' is connected to cell 'de_a', which is represented by a CTL model; dedicated wrapper cell will be added to the port. (DFT-3109)

The commands that can trigger this error message are:

- *insert_dft*
- *preview_dft*

**What Next**

Check wrapper report. Port mentioned above should have ded wrapper cell.

## DFT-3110

(Information) Port '%s' is connected to cell '%s', which is represented by a BAM model; dedicated wrapper cell will be added to the port.

**Description**

This information is shown when a port is connected to cell having represented by BAM model; dedicated wrapper cell will be added to the port

For example,

Information: Port 'in1' is connected to cell 'de_a', which is represented by a BAM model; dedicated wrapper cell will be added to the port. (DFT-3110)

The commands that can trigger this error message are:

- *insert_dft*
- *preview_dft*

**What Next**

Check wrapper report. Port mentioned above should have ded wrapper cell.

Feedback

## DFT-3111

(Information) Port '%s' is connected to clock-gating cell '%s'; dedicated wrapper cell will be added to the port.

**Description**

This information is shown when a port is connected clock-gating cell; dedicated wrapper cell will be added to the port

For example,

Information: Port 'in1' is connected to clock-gating cell 'clock_gate1'; dedicated wrapper cell will be added to the port. (DFT-3111)

The commands that can trigger this error message are:

- *insert_dft*
- *preview_dft*

**What Next**

Check wrapper report. Port mentioned above should have ded wrapper cell.

## DFT-3112

(Information) Port '%s' is connected to cell '%s', which is included in a user scan chain; dedicated wrapper cell will be added to the port.

**Description**

This information is shown when a port is connected cell which is included in user scan chain; dedicated wrapper cell will be added to the port

For example,

Information: Port 'out3' is connected to cell 'de_a', which is included in a user scan chain; dedicated wrapper cell will be added to the port. (DFT-3112)

The commands that can trigger this error message are:

- *insert_dft*
- *preview_dft*

**What Next**

Check wrapper report. Port mentioned above should have ded wrapper cell.

Feedback

## DFT-3113

(Information) Port '%s' is connected to Multi-Bit IO register '%s'; dedicated wrapper cell will be added to the port.

**Description**

This information is shown when a port is connected to multi-bit IO register; dedicated wrapper cell will be added to the port

For example,

Information: Port 'out3' is connected to Multi-Bit IO register 'i_reg'; dedicated wrapper cell will be added to the port. (DFT-3113)

The commands that can trigger this error message are:

- *insert_dft*
- *preview_dft*

**What Next**

Check wrapper report. Port mentioned above should have ded wrapper cell.

## DFT-3114

(Information) Port '%s' has a dedicated wrapper cell user specification; dedicated wrapper cell will be added to the port.

**Description**

This information is shown when a port has dedicated wrapper cell specification; dedicated wrapper cell will be added to the port

For example,

Information: Port 'out3' has a dedicated wrapper cell user specification; dedicated wrapper cell will be added to the port. (DFT-3114)

The commands that can trigger this error message are:

- *insert_dft*
- *preview_dft*

**What Next**

Check wrapper report. Port mentioned above should have ded wrapper cell.

Feedback

## DFT-3115

(Information) I/O registers associated with port '%s' have combinational logic depth %d greater than the specified depth threshold %d; dedicated wrapper cell will be added to the port.

### Description

This information is shown I/O registers associated with port have combimational logic depth greater than specified depth threshold dedicated wrapper cell will be added to the port.

For example,

Information: I/O registers associated with port 'out3' have combinational logic depth 4 greater than the specified depth threshold 2; dedicated wrapper cell will be added to the port. (DFT-3115)

The commands that can trigger this error message are:

- *insert_dft*
- *preview_dft*

### What Next

Check wrapper report. Port mentioned above should have ded wrapper cell.

## DFT-3116

(Information) Port '%s' is connected to %d input registers and %d output registers, which exceeds %sreuse threshold %d; dedicated wrapper cell will be added to the port.

### Description

This information is shown when port is connected to input and output registers whose count exceed reuse threshold value; dedicated wrapper cell will be added to the port.

For example,

Information: Port 'out3' is connected to 4 input registers and 5 output registers, which exceeds input reuse threshold 3; dedicated wrapper cell will be added to the port. (DFT-3116)

The commands that can trigger this error message are:

- *insert_dft*
- *preview_dft*

Feedback

**What Next**

Check wrapper report. Port mentioned above should have ded wrapper cell.

---

## DFT-3117

(Error) Size of the list specified by -misr_custom_names must be equal to the number of MISRs specified by set_logicbist_configuraion.

**Description**

Size of the list specified by -misr_custom_names must be equal to the number of MISRs specified by set_logicbist_configuraion.

**What Next**

Please Review DFT Specifications

---

## DFT-3118

(Information) Port '%s' is excluded from wrapping due to %s DFT signal specification

**Description**

This information is shown when port is excluded from wrapping due to DFT signal specification

For example,

Information: Port 'out3' is excluded from wrapping due to Reset DFT signal specification (DFT-3118)

The commands that can trigger this error message are:

- *insert_dft*
- *preview_dft*
- *set_boundary_cell*

**What Next**

Check wrapper report. Port mentioned above should not be wrapped.

---

## DFT-3120

(Information) Soft core port '%s' is excluded from core wrapping as it is connected to port/pin '%s' having DFT signal specification

Feedback

**Description**

This information is shown when soft core port is excluded from wrapping as it is connected to port/pin which has DFT signal specification

For example,

Information: Soft core port 'in3[0]' is excluded from core wrapping as it is connected to port/pin 'out3[0]' having DFT signal specification (DFT-3120)

The commands that can trigger this error message are:

- *insert_dft*
- *preview_dft*

**What Next**

Check wrapper report. Port mentioned above should have ded wrapper cell.

---

## DFT-3121

(Information) Port '%s' is connected to %d input registers, which exceeds %sreuse threshold %d; dedicated wrapper cell will be added to the port.

**Description**

This information is shown when port is connected to input registers whose count exceed reuse threshold value; dedicated wrapper cell will be added to the port.

For example,

Information: Port 'out3' is connected to 4 input registers, which exceeds input reuse threshold 2; dedicated wrapper cell will be added to the port. (DFT-3121)

The commands that can trigger this error message are:

- *insert_dft*
- *preview_dft*

**What Next**

Check wrapper report. Port mentioned above should have ded wrapper cell.

---

## DFT-3122

(Information) Port '%s' is connected to %d output registers, which exceeds %sreuse threshold %d; dedicated wrapper cell will be added to the port.

Feedback

**Description**

This information is shown when port is connected to output registers whose count exceed reuse threshold value; dedicated wrapper cell will be added to the port.

For example,

Information: Port 'out3' is connected to 5 output registers, which exceeds output reuse threshold 2; dedicated wrapper cell will be added to the port. (DFT-3122)

The commands that can trigger this error message are:

- *insert_dft*
- *preview_dft*

**What Next**

Check wrapper report. Port mentioned above should have ded wrapper cell.

---

## DFT-3127

(Information) Adding dedicated wrapper cell on %s branch for port '%s' on pin '%s'

**Description**

This information is shown when

For example,

Information: Adding dedicated wrapper cell on out branch for port 'out3' on pin 'out3[0]' (DFT-3127)

The commands that can trigger this error message are:

- *insert_dft*
- *preview_dft*

**What Next**

Check wrapper report. Port mentioned above should have ded wrapper cell.

---

## DFT-3128

(Error) Insufficient bus length %d of TDR register '%s' specified for bist_cdr, expected length : %d

**Description**

Insufficient bus length %d of TDR register '%s' specified for bist_cdr, expected length : %d

Feedback

**What Next**

Please Review DFT Specifications

---

## DFT-3129

(Error) Invalid register width %d has been specified for %s in set_logicbist_configuraion.

**Description**

Invalid register width %d has been specified for %s in set_logicbist_configuraion.

**What Next**

Please Review DFT Specifications

---

## DFT-3136

(Information) value missing for %s , please provide input in {%s:%s} format.

**Description**

This information is shown ipunt/value specified in the wrong format

For example,

Information: set_pipeline_scan_enable_configuration -clock_based_fanout_limit {losp 12}

The commands that can trigger this error message are:

- *set_pipeline_scan_enable_configuration*

**What Next**

Check the command , there is not value specified for mention input.

---

## DFT-3137

(Info) Adding dedicated wrapper cell for internal pin '%s' because of user specification

**Description**

This information message is issued when tool is inserting a dedicated wrapper cell to internal_pin because of user specification. set_boundary_cell -internal_pins [list of pins]

Example,

set_boundary_cell -internal_pins { dd_d/o_tval} -type WC_D1

Feedback

## DFT-3138

(Info) A dedicated wrapper will be added to port '%s' because dedicated wrapper insertion flow is enabled

**Description**

This information message is shown when reuse threshold is set to -1. In this scenerio every port will have a dedicate wrapper cell.

Example,

set_wrapper_configuration -reuse_threshold -1

## DFT-3139

(Information) Modified wrapper IP will be instantiated in top hierarchy as user didn't specified the location for partial wrapper cell in split wrapper cell flow.

**Description**

This information is shown when user donot specifies the location for partial wrapper cell in split wrapper cell flow. This information will only occur if the app-option 'dft.test_split_wrp_cell' is enabled without specifying the location of DFT_CX hierarchy. Hence, as a fall-back mechanism the modified wrapper IP would be instantiated in the top hierarchy.

**What Next**

Please specify the location for partial wrapper cell in split wrapper cell flow using the following commands:

set_dft_location set_boundary_cell -location

## DFT-3140

(Info) For port '%s', not adding Dedicated Wrapper cell to loads '%s' due to non fixable MV violations

**Description**

This information message is shown in upf aware flow when tool could not find the location to insert dedication wrapper cell without introducing non fixable MV violations.

## DFT-3141

(Information) Option %s is obsolete, use %s option inside parameters or map_parameters.

Feedback

**Description**

This information is shown when option is not part of latest tool , its obsolete and should be used with parameter list option

For example,

Information: set_watermark_configuration -location de_a

The commands that can trigger this error message are:

- *set_watermark_configuration*

**What Next**

Check the command , use option inside parameterlist option.

---

## DFT-3142

(Information) %s Reg length updated with new value = %d.

**Description**

This information is shown when exact_length in set_scan_path is different than specified in security configuration

For example,

Information: set_scan_path KEY -class ieee_1687 -exact_length 10
set_logic_lock_configuration -exact_length 32 -level gate -exec_name TRLL generate_dft

The commands that can trigger this error message are:

- *generate_dft*

**What Next**

Check the command , Reg length will be updated .

---

## DFT-3143

(Information) In shared wrapper flow user needs to specify short wrapper chains using set_wrapper_configuration in scan compression mode.

**Description**

The information message is shown when shared wrapper flow is enabled and number of short wrapper chains are not specified. This information message helps fix DFT-1083 error.

Feedback

For example,

Information: In shared wrapper flow user needs to specify short wrapper chains using set_wrapper_configuration in scan compression mode. (DFT-3043)

The combinations of commands that can trigger this warning message are:

- *set_wrapper_configuration* • *generate_dft*

**What Next**

Specify wrapper chain count of short chains mode (compression_mode|wrp_if) as following and run again set_wrapper_configuration -chain_count N -test_mode (compression_mode|wrp_if)

set_wrapper_configuration generate_dft

---

## DFT-3144

(Information) fanout_limit for %s should be positive integer.

**Description**

This information is shown when fanout_limit is specified with negative value.

For example,

Information: set_pipeline_scan_enable_configuration -clock_based_fanout_limit {losp:-12}

The commands that can trigger this error message are:

- *set_pipeline_scan_enable_configuration*

**What Next**

Check the command , please specified the positive value.

---

## DFT-3145

(Information) no specified value of clock_based_fanout_limit for %s, honoring global setting....

**Description**

This information is shown when clock_based_fanout_limit is specified with zero value

For example,

Information: set_pipeline_scan_enable_configuration -clock_based_fanout_limit {losp:0}

Feedback

The commands that can trigger this warning message are:

- *set_pipeline_scan_enable_configuration*

**What Next**

Warnning : global fanout_limit will be consider.

---

## DFT-3146

(Information) clock_based_fanout_limit value for %s is overwritten with %d.

**Description**

This information is shown clock_based_fanout_limit value is specified more than once for a port/hookup pin

For example,

Information: set_pipeline_scan_enable_configuration -clock_based_fanout_limit {losp:12} set_pipeline_scan_enable_configuration -clock_based_fanout_limit {losp:20}

The commands that can trigger this warning message are:

- *set_pipeline_scan_enable_configuration*

**What Next**

Check the command , clock_based_fanout_limit value is overridden by new value.

---

## DFT-3147

(Information) Please regenerate the library %s with library manager version newer than O-2018.06-SP3.

**Description**

You receive this information message when library manager version used to generate ndm Libraries is O-2018.06-SP3 or older.

---

## DFT-3148

(Information) Violating all segments of cell instance %s(%s)

**Description**

Message displays that the CTL cell has failed the test model connectivity checks. Hence all the segments of the CTL cell will not be considered for scan stitching during insert_dft.

Feedback

**What Next**

To skip test model connectivity checks for all the CTL cells in the design. Please do set_dft_drc_configuration -test_validate_test_model_connectivity false

---

## DFT-3150

(Error) No aon_enable signal defined for port/pin specified with '-aon_enable' option

**Description**

No aon_enable signal defined for port/pin specified with '-aon_enable' option

**What Next**

Please Review DFT Specifications

---

## DFT-3151

(Error) -aon_enable option is mandatory with type WC_Q0_AO/WC_Q1_AO cell.

**Description**

Specifying '-aon_enable' option in set_boundary_cell command is mandatory for cell type WC_Q0_AO/WC_Q1_AO. The option '-aon_enable' accepts aon_enable signal type port/pin.

**What Next**

1. Use option '-aon_enable' in set_boundary_cell command.

---

## DFT-3153

(Info) Port '%s' is connected to latch '%s'.Dedicated wrapper cell will be added due to app option test_core_wrap_add_dw_for_ports_connected_to_latch is enabled

**Description**

This information message is issued when tool is inserting a dedicated wrapper cell to port if port is connected to a latch and app option test_core_wrap_add_dw_for_ports_connected_to_latch is enabled.

---

## DFT-3154

(Information) New hold mux architecture support is enabled.

Feedback

**Description**

You recieve this information message when the app option for the new hold mux architecture is enabled and pre-compile flow is not used.

---

## DFT-3155

(Information) Skipping wrapping port '%s' as the number of dedicated wrapper cells %d is exceeding the user specified dedicated wrapper cell threshold per port %d.\n

**Description**

Port '%s' is not wrapped since it exceeds max DWC per port threshold set by user.

**What Next**

set_wrapper_configuration -max_dwc_per_port

---

## DFT-3160

(Warning) CG cell %s requires TE stitching but design is already CG stitched.

**Description**

Integrated Clock Gating (ICG) cells need to have their TE pins connected to the correct ScanEnable or TestMode signals, task that needs to be performed during *insert_dft* command.

There are special flows like the usage of command *connect_clock_gate_test_enables* that can perform the TE pin ICG connections on advance given the proper DFT Specification is provided. If that's the case, and as ICG TE pin connection is expected to be done once per design/flow, a later call to *insert_dft* command is not expected to connect again the TE pins of ICGs.

But if in previous scenario, during *insert_dft* command some ICG cells are detected like requiring to have their TE pins connected to the DFT signals, this warning message will be issued to alert user that this TE pin unconnected cells can lead to DFT SCAN insertion problems.

**What Next**

If this message is issued, check that the ICG is manually connected to the correct TestMode or ScanEnable signal before executing DFT Scan insertion (*insert_dft* command).

Feedback

## DFT-3179

(Error) "Wrong partial range specified for the following interface port %s"

### Description

You receive this error message when you specify a range for bus port or pin that is not in correct range or syntax for option '-interface' in 'set_dft_hsio_element' command.

set_dft_hsio_element \ interfcae [list \ o_SO {{sub_inst/i_SO1 sub_inst/i_SO2[8:]}} \ // wrong syntax, missing lsb ] Error: Wrong partial range specified for the following interface port 'sub_inst/i_SO2' (DFT-3179)

set_dft_hsio_element \ interfcae [list \ o_SO {{sub_inst/i_SO1 sub_inst/i_SO2[4:8]}} \ //lsb is greater than msb ] Error: Wrong partial range specified for the following interface port 'sub_inst/i_SO2' (DFT-3179)

### What Next

Please use a value in the correct range. Check proper syntax to pass partial range to avoid error.

## DFT-3180

(Information) %s port '%s' is connected to an %s port via combinational path having 0 I/O registers. Dedicated wrappers will be added to CG/CTL branches.

### Description

This information is shown when an input/output port is connected to an output/input port via combinational path and has 0 I/O registers. Additionally, dedicated wrappers are to be added on CG/CTL branches.

For example,

Information: Input port 'sr_u2/in13' is connected to an Output port via combinational path having 0 I/O registers. Dedicated wrappers will be added to CG/CTL branches. (DFT-3180)

The commands that can trigger this message are:

- *insert_dft*
- *preview_dft*

### What Next

Check wrapper report. Port should have one or more dedicated wrappers on CG/CTL branches.

Feedback

## DFT-3219

(Information) Starting test protocol creation.

**Description**

Tool has started creation of test protocol.

## DFT-3248

(Error) Provided value '%s' of app-option dft_cloned_cg_hookup_pin is not per expected format. Expected format is {SourcePin:DestPin}

**Description**

App option dft_cloned_cg_hookup_pin value for specifying cloned clock gating cell source pin is not as per expected format which is {SourcePin:DestPin}. For e.g. - C_reg/pin1:A_reg/pin2

**What Next**

Please provide correct value of app_option dft_cloned_cg_hookup_pin as per expected format

## DFT-3265

(Information) Identified system/test clock port %s (%5.1f,%5.1f).

**Description**

Tool has identified the given port as a system or test clock with specified rise and fall times.

## DFT-3266

(Information) Identified active low asynchronous control port "%s".

**Description**

Tool has identified the given port as an asynchronous control with specified active state.

## DFT-3267

(Information) Skipping ScanDEF-based optimization due to user option.

Feedback

**Description**

You receive this information message when the tool is going to skip ScanDEF-based optimization during compile_fusion due to user option. This can be triggered if the user sets the 'dft.scan.scandef_based_opto' application option to false.

**What Next**

Enable ScanDEF-based optimization setting 'dft.scan.scandef_based_opto' to true.

---

## DFT-3500

(Information) endpoint clk pin '%s' is not connected to ICG rather connected to top level port, ignoring it.

**Description**

If an endpoint(register) clk pin is connected to top level port then it's clkgate can't be determined and PMM stitching needs clkgate information. So this endpoint will not be considered for stitching in *slm_stitch* command.

---

## DFT-3501

(Information) %s.

**Description**

It prints status message like which step is currently running and status message after the step is done.

---

## DFT-3502

(Information) PMU '%s' is not connected to any endpoint because of '%s'.

**Description**

If there are no available endpoint is within the PMU boundary area then this message might be printed. In case of slack based connection methodology this message might be printed if PMU search range endpoints clock gates have less endpoints than set_slm_stitching_strategy -num_eps_per_icg option value.

---

## DFT-3551

(Warning) Current version '%s' of '%s' is earlier than recommended version '%s'

Feedback

**Description**

This warning message will be displayed when version of underlying tools SpyGlass, EMBEDIT is earlier than the recommended version. By default the severity of this message is error(DFT-2951) but user can downgrade the severity to warning, by using environment variable(setenv TESTMAX_SKIP_RELEASE_COMPATIBILITY_CHECKS true), and move ahead in specical cases under R&D guidence.

---

## DFT-3552

(Warning) Current version '%s' of '%s' is greater than recommended version '%s'

**Description**

This warning message will be displayed when version of underlying tools SpyGlass, EMBEDIT is greater than the recommended version. By default the severity of this message is error(DFT-2952) but user can downgrade the severity to warning, by using environment variable(setenv TESTMAX_SKIP_RELEASE_COMPATIBILITY_CHECKS true), and move ahead in specical cases under R&D guidence.

---

## DFT-3553

(Warning) Recommended version not specified for '%s'. Skipping version check

**Description**

This warning message will be displayed when TMM is not able to find recommended version of SpyGlass and EMBEDIT for version checking. The version checking will be skipped in this case.

---

## DFT-3554

(Warning) Not able to extract current version for '%s'. Skipping version check

**Description**

This warning message will be displayed when TMM is not able to extract version of SpyGlass and EMBEDIT for recommended version checking. The version checking will be skipped in this case.

---

## DFT-3555

(Warning) Overriding '%s' using '%s' = '%s'

Feedback

**Description**

This warning message will be displayed when user set SpyGlass and EMBEDIT modules using envormnet varibale SPYGLASS_HOME and EMBEDIT_HOME.

---

## DFT-3556

(Error) Requested more %s than total %s stages. %s must be less than total %s stages.

**Description**

This Error message will be displayed when user sets IduCodecDepth value greater than the pipeline stage value.

---

## DFT-3599

(Error) Invalid value '%s' for option '%s' specified, allowed options are n'b1, n'b0 or binary value.

**Description**

You receive this error message if you specify the wrong value for reset_state and init_data of TDR.

Error: Invalid value '' for option '-init_data' specified, allowed options are n'b1, n'b0 or binary value (DFT-3599).

**What Next**

Define the correct value for reset_state/init_data of TDR.

---

## DFT-3650

(Information) Test point file generated at: %s

**Description**

You receive this information message when the test point file is generated at the specified location.

---

## DFT-3651

(Information) Test point architect completed.

**Description**

You receive this information message when the test point architect is completed.

Feedback

## DFT-3652

(Information) Test point insertion completed.

**Description**

You receive this information message when the test point insertion is completed.

## DFT-3653

(Information) Unable to open file : %s

**Description**

You receive this information message when the tool is not able to open a file mentioned in the message.

**What Next**

Check if working directory has write permission and space available.

## DFT-3654

(Information) Test point architect failed.

**Description**

You receive this information message when the test point architect is failed.

## DFT-3656

(Information) Test point report generation successful.

**Description**

You receive this information message when the test point report is successfully generated.

## DFT-3657

(Information) Test point '%s' is specified in partition '%s' but considering it under '%s' since partition search is enabled.

**Description**

You receive this information message when the test point is specified in one partition but tool considers it under another partition because partition search is enabled.

Feedback

## DFT-3658

(Information) Test point '%s' is specified in partition '%s' but considering it under '%s' since partition search is enabled.

**Description**

You receive this information message when the test point is specified in one partition but tool considers it under another partition because partition search is enabled.

## DFT-3660

(Information) Skipping test point insertion for target %s

**Description**

You receive this information message when the test point insertion is skipped for the specified target.

## DFT-3661

(Information) No test points shall be inserted for partition %s, as no target has been enabled.

**Description**

You receive this information message when the no test point will be inserted by the tool because of missing target specifications for that partition.

## DFT-3662

(Information) Max test point register specifications for targets for partition %s shall be ignored

**Description**

You receive this information message when the max test point register specifications will be ignored for the specified target.

## DFT-3663

(Information) Loading Spyglass GUI using existing project directory '%s'.

**Description**

This information message contains the path to SpyGlass project directory

Feedback

## DFT-3664

(Information) Exiting Spyglass GUI.

**Description**

You receive this information message when the SpyGlass GUI exits.

## DFT-3665

(Information) Opening a pipe %s to SpyGlass.

**Description**

You receive this information message when SpyGlass execution is going to begin

## DFT-3666

(Information) SpyGlass setup files are created at %s

**Description**

This information message contains the path to SpyGlass setup files.

## DFT-3667

(Information) Skipping test point analysis.

**Description**

You receive this information message when the tool is going to skip test point analysis. Please check the test point commands.

## DFT-3668

(Information) Using %s as DFT test point IP

**Description**

You receive this information message when the tool is unable to find the user defined module in the library and uses default controller IP.

## DFT-3669

(info )Additional port name(s) of module %s not defined is(are) : %s.

Feedback

**Description**

You receive this info message when there are additional ports in the module but not/ cannot define in the interface list. Just an FYI for the user.

---

## DFT-3670

(info )Changing test point from '%s' (%s) to '%s' (%s) due to legal move at higher hierarchy.

**Description**

You receive this info message when there is there is test point relocation to another functionally equivalent location in upper hierarchy.

---

## DFT-3671

(info ) 'control_01' type test points will be implented as 'force_01' test points.

**Description**

You receive this info message when control_01 type test points are specified through set_test_point_element. We implement control_01 to force_01 test points through set_test_point_element to minimize area as control_01 test points use two registers.

---

## DFT-3672

(Info) Using '%s' (%s) as clock-gating enable signal for test point target '%s' for partition '%s'

**Description**

You receive this info message when a signal is used for clock-gating enable for test point target and partition.

---

## DFT-3700

(Information) Picking compilers from path '%s'

**Description**

This information message contains the path to compilers picked from TMM installtion area.These compilers will be picked automatically if user does not specify compilers path in TMM TCL script.

Feedback

## DFT-3712

(Information) Stub chain '%s' was removed from the ScanDEF as it does not have remaining stub elements. Dropping %d bit(s) on %d element(s)

**Description**

This information message provide details about a stub chain being dropped after a successful update_stub_chain command.

## DFT-3713

(Information) Updating START point of stub chain '%s' to '%s'. Dropping %d bit(s) on %d element(s)

**Description**

This information message provide details about the updated START point after a successful update_stub_chain command.

## DFT-3714

(Information) Updating STOP point of stub chain '%s' to '%s'. Dropping %d bit(s) on %d element(s)

**Description**

This information message provide details about the updated STOP point after a successful update_stub_chain command.

## DFT-3715

(Information) Splitting stub chain '%s' into sub-chains '%s' and '%s'. Dropping %d bit(s) on %d element(s)

**Description**

This information message provide details about the stub chain split after a successful update_stub_chain command.

## DFT-3716

(Information) Adding element to floating section of stub chain '%s'. It has %d bit(s) and scan-in pin '%s'

Feedback

**Description**

This information message provide details about the added stub chain element after a successful connect_element_to_stub_chain command.

---

## DFT-3717

(Information) Removing element from stub chain '%s'. It has %d bit(s) and scan-in pin '%s'

**Description**

This information message provide details about the removed stub chain element after a successful disconnect_element_from_stub_chain command.

---

## DFT-3718

(Information) Updating test point clock to %s

**Description**

You receive this information message when the tool is updating test point clock.

---

## DFT-3719

(Error) Test model '.ict.zip' missing of component '%s' for design '%s'

**Description**

Test model '.ict.zip' missing of component '%s' for design '%s'

**What Next**

Please Review DFT Specifications

---

## DFT-3802

(Warning) %s.

**Description**

This warning message is issued when an invalid cell specification is passed to some DFT commands that take a list of cells as a parameter.

There are multiple reasons that can cause this warning to be issued, so the exact description depends on the logic of each particular command. The message provides an explanation of what parameter was considered invalid and why. For example,

Warning: Skipping u_des_unit/ctl/ctmi_1027: is not a sequential cell. (DFT-3802)

Feedback

The commands that can trigger this warning message are:

- *define_scan_cell_set*
- *set_scan_configuration*
- *set_scan_element*
- *set_scan_group*
- *set_scan_path*
- *set_scan_skew_group*

**What Next**

Verify the specifications being passed to these DFT commands. The warning message names the exact cell that couldn't be accepted and the reason for rejection.

---

## DFT-3888

(Information) DFT Scan Event SQL Observer -- SQL Report

**Description**

This message is used to print reports for two different scenarios:

1) PSE clusters results (DEBUG). Available if debug message DFT-5958 is enabled.

2) If experimental SQL SCAN statistics are enabled. Internal R&D feature. Not released.

**What Next**

Disable the enabling conditions if these reports are no longer required.

---

## DFT-3889

(Error) Parameter '%s' cannot be active low for non-differential pad types, in flow to add wrapper cells on pads

**Description**

When wrapper configuration is enabled along with specifications for pads, the following pad pins data_in, data_out and port cannot be active low.

**What Next**

Please Review define_dft_design specifications and set data_in, data_out and port to active high.

Feedback

## DFT-3890

(Information) Ignoring -add_wrapper_cells_to_power_domains option during wrapper analysis in initial_map

**Description**

When wrapper configuration is enabled with -add_wrapper_cells_to_power_domains, it will not be honoured during initial_map stage.

**What Next**

Wrapper configuration with -add_wrapper_cells_to_power_domains is honoured only during DFT insertion phase.

## DFT-4000

(Error) Total no of tdr bits defined are not enough to define all the testmodes.

**Description**

This error message is issued when defined number of TDR bits are not enough to define all the testmodes.

Error: Total no of tdr bits defined are not enough to define all the testmodes. (DFT-4000)

**What Next**

Increase the TDR bits according to the testmodes defined.

## DFT-4001

(Warning) Could not find any instance of ctl design '%s'

**Description**

When there is not matching ctl instance for design provided in read_test_model cmd.

**What Next**

Please provide matching ctl instance.

## DFT-4002

(Error) Given partition %s has not been defined

Feedback

**Description**

When an undefined partition is listed in the design provided in get_total_flop_count command.

**What Next**

Please provide valid partition name in the command.

---

## DFT-4010

(Error) Test Mode specified is not defined

**Description**

Test Mode specified is not defined

**What Next**

Please Review DFT Specifications

---

## DFT-4011

(Error) %s is not defined as a Test Mode

**Description**

%s is not defined as a Test Mode

**What Next**

Please Review DFT Specifications

---

## DFT-4020

(Error) Unable to open file '%s' with information for constraint-based wrapper analysis.

**Description**

This message is shown when constraint-based wrapper analysis is enabled but the wrapper constraint file was not found. The wrapper constraint file is generated automatically by the run_test_point_analysis command prior to DFT insertion. If you have a wrapper constraint file with a non-default name or you want to change the default value, you can specify it by setting the application option dft.test_core_wrap_constraints_file_name

**What Next**

1. check run_test_point_analysis output.

Feedback

## DFT-4021

(Warning) Pin '%s' is not found in the design for constraint-based wrapper analysis.

### Description

Tool is unable find the pin present in spyglass generated wrapper constraints file during run_test_point_analysis.

### What Next

1. check if pin exists in the design. 2. check the app run_test_point_analysis output.

## DFT-4022

(Warning) Ignoring in-place wrapping for instance '%s' present inside a block abstraction model hierarchy.

### Description

You receive this error message in Dft core wrapper insertion when a cell to be in-place wrapped is located inside a block abstraction model. User specifies cells to be in-place wrapped using the app option dft.test_core_wrap_in_place_wrp_cell_names.

When a specified cell is located inside a block abstraction model, tool ignores the cell.

Specify correct cell names with the app option to avoid the message.

## DFT-4025

(Warning) Since dft.test_wrp_power_domain_aware_dw_wrp_clk is enabled, user needs to provide a set of wrapper clocks and disable use_system_clock_for_dedicated_wrp_cells. Disabling power aware clock selection for dedicated core wrapper cells

### Description

When user enable power aware clock selection for dedicated wrapper cells, there has to be a set of wrapper clock in the system specified via "set_dft_signal -type wrp_clock", to perform power aware clock selection. If this not the case power aware clock selection will be disabled. Also for power aware clock selection use_system_clock_for_dedicated_wrp_cells should be disable in set_wrapper_configuration command.

SH "SEE ALSO"

```
set_wrapper_configuration(1)
dft.test_wrp_power_domain_aware_dw_wrp_clk(2)
```

Feedback

## DFT-4030

(Error) Unable to insert AON custom wrapper cell for port '%s' as no AON dual rail mux libcell found with name %s. Ignoring port for wrapper analysis.

**Description**

Ignoring port for AON custom wrapper insertion. This could be because tool is not able to find AON dual rail mux library cell module which is required for AON custom wrapper insertion.

**What Next**

1. check the app option 'dft.test_ao_dedicated_wrp_mux_lib_cell' value.

## DFT-4031

(Error) Failed to write %s to file %s.

**Description**

Failed to write wrapper IP. This may occur because of bad wrapper IP logic.

**What Next**

1. Recheck the wrapper IP logic.

## DFT-4033

(Warning) Overriding '%s' wrapper cell type to %s wrapper cell as per user specification against type %s wrapper cell identified by wrapper analysis.

**Description**

This message is shown when there is a type mismatch between user specified wrapper cell and wrapper analysis determined cell type. In such cases we honor user specifications and issue the warning. Because of user forced type for wrapper cells, the wrapping status of all ports associated with such wrapper cells might be affected.

**What Next**

1. check set_wrapper_configuration.

## DFT-4091

(Error) Invalid value for option '-unified' specified, allowed options are enable|disablenDiscarding Timing constraints generation.

Feedback

**Description**

Invalid value for option '-unified' specified, allowed options are enable|disablenDiscarding Timing constraints generation.

**What Next**

Please Review DFT Specifications

---

## DFT-4851

(info) ICG TE stitching verbose report

**Description**

This message can be enabled by user configuration to print extra information about the ICG TE stitching results, using set_msg command:

set_msg DFT-4851 -level information

**What Next**

Set the message back to verbose or delete the original set_msg call if this report is no longer required.

---

## DFT-4860

(info) Scan chain post processing verbose messages

**Description**

This message can be enabled by user configuration to print extra information about extra optimizations performed over SCAN chains after they have been architected.

It can be enabled by the following command.

set_msg DFT-4860 -level information

**What Next**

Set the message back to verbose or delete the original set_msg call if this report or information is no longer required.

---

## DFT-4861

(info) DFT Scan Synthesis clock gathering stats.

Feedback

**Description**

This message can be enabled by user configuration to print extra statistical information about the SCAN clock gathering process, i.e. the stage where all DFT clock signals are traversed to locate and identify every SCAN cell driven by the corresponding clock.

It can be enabled by the following command.

set_msg DFT-4861 -level information

**What Next**

Set the message back to verbose or delete the original set_msg call if this report or information is no longer required.

---

## DFT-4895

(info) PSE reclustering summary report

**Description**

This message can be enabled by user configuration to print extra information about the Pipeline Scan Enable (PSE) reclustering process, showing the different compatible PSE groups and the amount of loads assigned to them.

It can be enabled by the following command.

set_msg DFT-4895 -level information

**What Next**

Set the message back to verbose or delete the original set_msg call if this report or information is no longer required.

---

## DFT-5041

(Error) Incorrect value '%s' specified for option '%s'

**Description**

Incorrect value '%s' specified for option '%s'

**What Next**

Please Review DFT Specifications

Feedback

## DFT-5050

(Error) audit_report_option is disabled in set_testmax_configuration command, ignoring insert_ded_wrp_in_netlist option.

**Description**

audit_report_option is disabled in set_testmax_configuration command, ignoring insert_ded_wrp_in_netlist option.

**What Next**

Please Review DFT Specifications

## DFT-8991

(Error) Failed in reading attachment data for design %s and design library %s.

**Description**

Inconsistency in DFT Specification Data. Failed to read it. This may occur because of the inconsistency in the design attachment data.

**What Next**

1. Regenerate the saved block with the current version of the tool 2. As a temporary workaround, disable the persistency check using the following environment variable. This way the check is ignored and the flow is allowed to proceed further. setenv dft_tdm_spec_persistency false

## DFT-8993

(Error) Incorrect -target specification, either empty or invalid list provided. Correct specification : -target [list [list inst pdl] [list inst pdl]] / -target {{inst pdl} {inst pdl}}.

**Description**

This error message is issued when target pair is empty/invalid.

Error: Incorrect -target specification, either empty or invalid list provided. Correct specification : -target [list [list inst pdl] [list inst pdl]] / -target {{inst pdl} {inst pdl}}. (DFT-8993)

**What Next**

Define valid target pair

Feedback

### DFT-8994

(Error) "PDL file passed as the target doesn't exist."

**Description**

This error message is sent when during write_test_protocol command

For example,

Error: "PDL file passed as the target doesn't exist." (DFT-8994)

**What Next**

Please check for correct PDL file name to be passed with target under write_test_protocol command.

## DFTSCHD

### DFTSCHD-001

(warning) No schematic support for DFT violation (%s).

**Description**

Check the man page for %s DRC violation.

## DIO

### DIO-001

(error) Incorrect value specified for option %s.

**Description**

The input option value may not meet the requirements.

**What Next**

Please follow the command man page to input option value that matches the format.

### DIO-002

(error) The option %s is mutually exclusive with option %s.

Feedback

**Description**

The mutually exclusive options are specified.

**What Next**

Please correct the options so that do not violate any mutual exclusion requirement.

---

## DIO-003

(error) Python execute file (%s) error, please find detail in file *.debug.log.

**Description**

There occurs error while executing python command. Detail info is dumped in file *.debug.log.

**What Next**

Please get the detail error info and contact AE for help.

---

## DIO-004

(error) Could not create instance %s since it is outside block boundary (%s line %u).

**Description**

This error can occur while executing read_design_io command. It means that the object with the specified name could not be created as it will extend outside the boundary of the current block. Hence the entry at the specified CSV file and line number will be ignored.

**What Next**

Please check the CSV file and modify the X_origin and Y_origin on the particular line number so that the object lies inside the current block's boundary.

---

## DIO-005

(warning) Mismatch in object types of reference cell and instance (%s line %u).

**Description**

This warning message is issued when the bump type or object type of the reference is different from the type inferred from the instance. For example, if the instance name is specified under C4_inst column, but the ref cell specified under Reference_name is of type ubump, it is considered as a mismatch.

Feedback

**What Next**

Please check if instance name is written at right column or value of reference name is correct.

---

## DIO-006

(error) Could not find bond_pad_def with name %s.

**Description**

No bond_pad_def with the given name was found in the current design.

**What Next**

Please check that a correct reference name for is specified for the bond pad object. Also ensure that the bond_pad_def with the specified name is already existing in the current design.

---

## DIO-007

(warning) Option -cell_origin_type is ineffective when origin for the specified reference cell is its center.

**Description**

The -cell_origin_type option depends on the reference cell - whether its origin {0 0} is at the lower left of the ref cell, or at the center. This option is effective only for ref cells with origin at lower left.

**What Next**

Please check that the reference cell with correct origin type is specified if -cell_origin_type option needs to be effective.

---

## DIO-008

(error) The net name in cell(s) %s is invalid.

**Description**

This message will be triggered if 1. xlsx cell content only consists of symbols, e.g., %, $,',etc. 2. there are some spaces in the middle of cell content, e.g., aa bb

**What Next**

Please check the content in the cell and correct it.

Feedback

## DIO-009

(information) Assuming via_def %s is to be used to create TSV instance %s (%s line %u).

**Description**

This message indicates that no ref cell with the specified name was found, however a via def with the same name is available. Hence the tool will use the via def with the specified name for creating the TSV instance.

**What Next**

Confirm that ref cell with the specified name is indeed not available, and it is ok to use the TSV def for creating the TSV.

## DIO-010

(warning) Could not set preferred pins for port %s.

**Description**

This warning message indicates that preferred_pin attribute could not be set for the specified port. This could be due to various reasons. For example, the die side of the preferred pins is not consistent or could not be determined. Second possibility is that the preferred pin is not a pad.

**What Next**

Check whether the correct set of bump cells are specified corresponding to the given port, and that the relevant attributes are set correctly.

## DIO-011

(information) Incorrect value '%s' of %s on line %u as it should have been '%s' as per pseudo_bump data.

**Description**

This message indicates that the specified attribute value provided is not correct as per pseudo_bump data.

**What Next**

Please provide the correct value as per pseudo_bump data.

## DIO-012

(warning) The chiplet instance '%s' do contain probe pads or direct probes.

Feedback

**Description**

The chiplet name i.e., instance name, provided in the command line do not conatin probe pads or direct probes.

**What Next**

Please check the instance name provided in the command line has probe pads or direct pads.

---

## DIO-013

(warning) The chiplet instance '%s' do not exist in the design.

**Description**

The chiplet name i.e., instance name, provided in the command line do not exist in the design or provided instance name is not a valid chiplet.

**What Next**

Please check the instance name provided in the command line is valid or not.

---

## DIO-014

(error) Failed to write probe card for the chiplet instance '%s' for the region '%s' and the side is '%s'.

**Description**

Failed to write probe card spec for the chiplet i.e., instance name, provided in the command line

**What Next**

Please get the detail error info and contact AE for help.

---

## DIO-015

(warning) No chiplet/die found in the design.

**Description**

No chiplet/die present in the design to write the bumps count in to csv file.

**What Next**

Please make sure that the design has valid chiplet(s)/die(s).

Feedback

## DIO-016

(warning) No bumps found in the design.

**Description**

No bumps found in any of the chiplet(s)/die(s) of the design to write bumps count in to csv file.

**What Next**

Please make sure that the design has bumps to write a csv file.

## DIO-017

(Information) No bump found in the region '%s' of the chiplet '%s'.

**Description**

No bump found in the one of the regions of the chiplet instance.

## DIO-018

(Error) No port named '%s' found on top-level design (%s line %u).

**Description**

The port with the given name on the specified line does not exist on the current top-level design.

**What Next**

Check whether the port name in CSV file is correct, or use invoke the comamnd with *-create_ports* option so that the missing port will be automatically created in the design.

# DM

## DM-100

(error) Unable to create port '%s' because port creation is not allowed on design '%s.'

**Description**

Port creation is disabled in the design. This is usually done to prevent "port-punching" in a design.

Feedback

**What Next**

Change the value of the *is_port_punching_ok* attribute on the design if appropriate to allow modification of ports.

---

## DM-101

(error) Unable to delete port '%s' because port deletion is not allowed on design '%s.'

**Description**

Port deletion is disabled in the design. This is usually done to prevent "port-punching" in a design.

**What Next**

Change the value of the *is_port_punching_ok* attribute on the design if appropriate to allow modification of ports.

---

# DMM

---

## DMM-010

(Warning) Switching between mismatch configs is not recommended.

**Description**

It is not recommended to switch between configs because there might be some design mismatches already repaired in previous config. And those mismatches will not be in sync with current config anymore. There is currently no provision to re-mitigate or invalidate those already mitigated objects.

**What Next**

Make sure not to change current config if some mismatches have already been repaired.

---

## DMM-011

(Warning) Changing current repair(handler) for a mismatch type in between a flow is not recommended.

**Description**

It is not recommended to change current repair(handler) for a mismatch type in between a flow because there might be some design mismatches already repaired with previous

Feedback

handler. And those mismatches will not be in sync with current repair anymore. There is currently no provision to re-mitigate or invalidate those already mitigated objects.

**What Next**

Make sure not to change current repair if some mismatches have already been repaired.

---

## DMM-012

(Warning) Changing action for a mismatch type in between a flow is not recommended.

**Description**

It is not recommended to change action for a mismatch type in between a flow because there might be some design mismatches already repaired with previous action. And those mismatches will not be in sync with current action anymore. There is currently no provision to re-mitigate or invalidate those already mitigated objects.

**What Next**

Make sure not to change action if some mismatches have already been repaired.

---

## DMM-013

(Information) No mismatch exists on ref library '%s'.

**Description**

The specified reference library is found with no mismatch information.

---

## DMM-014

(error) Action '%s' is not allowed for '%s' mismatch type.

**Description**

This action type is not in the allowed list of actions for this mismatch type.

**What Next**

Use report_mismatch_config output to check which action types are allowed for this mismatch type.

---

## DMM-015

(error) Mismatch config '%s' do not exists.

Feedback

**Description**

The provided mismatch config do not exists. Check and use existing configs.

**What Next**

Use report_mismatch_configs output to check which mismatch configs are available to use.

---

## DMM-016

(error) Mismatch config '%s' already exists.

**Description**

The provided mismatch config already exists.

**What Next**

Use report_mismatch_configs to see available configs to use.

---

## DMM-017

(error) Config '%s' does not exists.

**Description**

The named config does not exists. Please use '-all' option to see the existing configs and use one of them.

**What Next**

Use report_mismatch_configs to see available configs to use.

---

## DMM-021

(Warning) Over-riding current design mismatch config '%s' by block-specific config '%s'.

**Description**

The design mismatch config settings are stored on a block when it is created. When it is opened or made the current block, these are restored. If they are different from the session's current config settings, the block's settings over-ride the session's settings and come into effect.

**What Next**

Use 'get_current_mismatch_config' and 'report_mismatch_config -verbose' to check what config settings came into effect. for this mismatch type.d

Feedback

## DMM-022

(Information) Restoring current design mismatch config value to '%s'.

**Description**

The block-specific design mismatch config settings are active when the block is open and is the current block. When the block is closed, these block-specific settings are removed and are replaced by the session's config settings that were in effect before this block was opened. This message indicates that the mismatch config setting has been restored to the previous value.

**What Next**

Use 'get_current_mismatch_config' and 'report_mismatch_config -verbose' to check what config settings came into effect.

## DMM-030

(error) '%s' is an empty rtl module.

**Description**

This denotes that an empty rtl module is read-in. Configure DMM mismatch empty_logic_module to convert it into macro.

**What Next**

Use report_mismatch_config output to check which action types are allowed for this mismatch type.

## DMM-040

(information) Resizing cell ‘%s’.

**Description**

When creating a macro type blackbox on an empty module mismatch mitigated using the create_macro strategy, tool will resize the macro without creating a new blackbox.

**What Next**

This is an information message no action is needed.

## DMM-041

(warning) Mitigated empty module cannot be moved to specified library or renamed.

Feedback

### Description

When creating a macro type block-box on an empty module mismatch mitigated using the create_macro strategy -library & -name options will be ignored.

### What Next

Review your design mismatch configuration and set the correct strategy action for the empty_logic_module, to disable the mitigation.

### See Also

- report_design_mismatch
- report_mismatch_configs

---

## DMM-042

(warning) Mitigated empty module cannot be move to library ‘%s’

### Description

When creating a macro type blackbox on an empty module mismatch mitigated using the create_macro strategy the -name option will be ignored.

### What Next

Review your design mismatch configuration and set the correct strategy action for the empty_logic_module, to disable the mitigation

### See Also

- report_design_mismatch
- report_mismatch_configs

---

## DMM-100

(Info) Information: Mismatch type %s is detected on object type %s at object %s. There are total %d objects of this mismatch type on this design. Out of these %d objects are repaired using %s repair strategy.

### Description

The mismatch type %mmtype is found to exists on %objtype %objname .

Feedback

The following table lists more information of the mismatch type:

Table:

```
Field   Information
  Mismatch Object Name   %mmobjname
  Object Name   %objname
  Object Type   %objtype
  Repaired State   %repairedstate
  Repair Strategy   %mmhandler
```

## DMM-101

(Info) Information: Mismatch type %s is detected on object type %s at object %s. There are total %d objects of this mismatch type on this design. Out of these %d objects are repaired using %s repair strategy.

**Description**

The mismatch type %mmtype is found to exists on %objtype %objname .

The following table lists more information of the mismatch type:

Table:

```
Field   Information
  Mismatch Object Name   %mmobjname
  Object Name   %objname
  Object Type   %objtype
  Repaired State   %repairedstate
  Repair Strategy   %mmhandler
```

## DMM-102

(Info) Information: Mismatch type %s is detected on object type %s at object %s. There are total %d objects of this mismatch type on this design. Out of these %d objects are repaired using %s repair strategy.

**Description**

The mismatch type %mmtype is found to exists on %objtype %objname .

The following table lists more information of the mismatch type:

Table:

```
Field   Information
  Mismatch Object Name   %mmobjname
  Object Name   %objname
  Object Type   %objtype
```

Feedback

```
  Repaired State    %repairedstate
  Repair Strategy    %mmhandler
```

## DMM-103

(Info) Information: Mismatch type %s is detected on object type %s at object %s. There are total %d objects of this mismatch type on this design. Out of these %d objects are repaired using %s repair strategy.

**Description**

The mismatch type %mmtype is found to exists on %objtype %objname .

The following table lists more information of the mismatch type:

Table:

```
Field    Information
  Mismatch Object Name    %mmobjname
  Object Name    %objname
  Object Type    %objtype
  Repaired State    %repairedstate
  Repair Strategy    %mmhandler
```

## DMM-104

(Info) Information: Mismatch type %s is detected on object type %s at object %s. There are total %d objects of this mismatch type on this design. Out of these %d objects are repaired using %s repair strategy.

**Description**

The mismatch type %mmtype is found to exists on %objtype %objname .

The following table lists more information of the mismatch type:

Table:

```
Field    Information
  Mismatch Object Name    %mmobjname
  Object Name    %objname
  Object Type    %objtype
  Repaired State    %repairedstate
  Repair Strategy    %mmhandler
```

Feedback

## DMM-105

(Info) Information: Mismatch type %s is detected on object type %s at object %s. There are total %d objects of this mismatch type on this design. Out of these %d objects are repaired using %s repair strategy.

**Description**

The mismatch type %mmtype is found to exists on %objtype %objname .

The following table lists more information of the mismatch type:

Table:

```
Field   Information
  Mismatch Object Name   %mmobjname
  Object Name   %objname
  Object Type   %objtype
  Repaired State   %repairedstate
  Repair Strategy   %mmhandler
```

## DMM-106

(Info) Information: Mismatch type %s is detected on object type %s at object %s. There are total %d objects of this mismatch type on this design. Out of these %d objects are repaired using %s repair strategy.

**Description**

The mismatch type %mmtype is found to exists on %objtype %objname .

The following table lists more information of the mismatch type:

Table:

```
Field   Information
  Mismatch Object Name   %mmobjname
  Object Name   %objname
  Object Type   %objtype
  Repaired State   %repairedstate
  Repair Strategy   %mmhandler
```

## DMM-107

(Info) Information: Mismatch type %s is detected on object type %s at object %s. There are total %d objects of this mismatch type existing on this design. Out of these %d objects are repaired using %s repair strategy.

Feedback

**Description**

The mismatch type %mmtype is found to exists on %objtype %objname .

The following table lists more information of the mismatch type:

Table:

```
Field   Information
  Mismatch Object Name   %mmobjname
  Object Name   %objname
  Object Type   %objtype
  Repaired State   %repairedstate
  Repair Strategy   %mmhandler
  Related Instance Count   %instcount
```

## DMM-108

(Info) Information: Mismatch type %s is detected on object type %s at object %s. There are total %d objects of this mismatch type on this design. Out of these %d objects are repaired using %s repair strategy.

**Description**

The mismatch type %mmtype is found to exists on %objtype %objname .

The following table lists more information of the mismatch type:

Table:

```
Field   Information
  Mismatch Object Name   %mmobjname
  Object Name   %objname
  Object Type   %objtype
  Repaired State   %repairedstate
  Repair Strategy   %mmhandler
  Related Instance Count   %instcount
```

## DMM-109

(Info) Information: Mismatch type %s is detected on object type %s at object %s. There are total %d objects of this mismatch type on this design. Out of these %d objects are repaired using %s repair strategy.

**Description**

The mismatch type %mmtype is found to exists on %objtype %objname .

Feedback

The following table lists more information of the mismatch type:

Table:

```
Field   Information
  Mismatch Object Name   %mmobjname
  Object Name   %objname
  Object Type   %objtype
  Repaired State   %repairedstate
  Repair Strategy   %mmhandler
  Related Instance Count   %instcount
```

## DMM-110

(Info) Information: Mismatch type %s is detected on object type %s at object %s. There are total %d objects of this mismatch type on this design. Out of these %d objects are repaired using %s repair strategy.

**Description**

The mismatch type %mmtype is found to exists on %objtype %objname .

The following table lists more information of the mismatch type:

Table:

```
Field   Information
  Mismatch Object Name   %mmobjname
  Object Name   %objname
  Object Type   %objtype
  Repaired State   %repairedstate
  Repair Strategy   %mmhandler
  Related Instance Count   %instcount
```

## DMM-111

(Info) Information: Mismatch type %s is detected on object type %s at object %s. There are total %d objects of this mismatch type on this design. Out of these %d objects are repaired using %s repair strategy.

**Description**

The mismatch type %mmtype is found to exists on %objtype %objname .

The following table lists more information of the mismatch type:

Table:

```
Field   Information
  Mismatch Object Name   %mmobjname
  Object Name   %objname
```

Feedback

```
  Object Type   %objtype
  Repaired State   %repairedstate
  Repair Strategy   %mmhandler
  Related Instance Count   %instcount
```

## DMM-112

(Info) Information: Mismatch type %s is detected on object type %s at object %s. There are total %d objects of this mismatch type on this design. Out of these %d objects are repaired using %s repair strategy.

**Description**

The mismatch type %mmtype is found to exists on %objtype %objname .

The following table lists more information of the mismatch type:

Table:

```
Field   Information
  Mismatch Object Name   %mmobjname
  Object Name   %objname
  Object Type   %objtype
  Repaired State   %repairedstate
  Repair Strategy   %mmhandler
```

## DMM-113

(Info) Information: Mismatch type %s is detected on object type %s at object %s. There are total %d objects of this mismatch type on this design. Out of these %d objects are repaired using %s repair strategy.

**Description**

The mismatch type %mmtype is found to exists on %objtype %objname .

The following table lists more information of the mismatch type:

Table:

```
Field   Information
  Mismatch Object Name   %mmobjname
  Object Name   %objname
  Object Type   %objtype
  Repaired State   %repairedstate
  Repair Strategy   %mmhandler
```

Feedback

## DMM-114

(Info) Information: Mismatch type %s is detected on object type %s at object %s. There are total %d objects of this mismatch type on this design. Out of these %d objects are repaired using %s repair strategy.

### Description

The mismatch type %mmtype is found to exists on %objtype %objname .

The following table lists more information of the mismatch type:

Table:

```
Field   Information
  Mismatch Object Name   %mmobjname
  Object Name   %objname
  Object Type   %objtype
  Repaired State   %repairedstate
  Repair Strategy   %mmhandler
```

## DMM-115

(Warning) Auto deriving '%s' routing direction for layer '%s'.

### Description

There is no layer routing direction provided by user and also not available in technology file. So auto deriving the routing direction for the specified layers.

## DMM-116

(information) Total %d mismatches are found on block '%s'.

### Description

If there exists any inconsistency between design and reference data, they are regarded as a mismatch in the flow. These mismatches can be related to a missing reference, port name case, missing routing direction on a metal layer etc.

When such a mismatch is detected, it is collected and recorded on design. These mismatches are captured during specific flows like linking and routability checks.

### What Next

Check output of report_design_mismatch to find more details about mismatches.

Feedback

### See Also

- report_design_mismatch

---

## DMM-117

(warning) Reporting design mismatch(es) or creating or setting design mismatch configuration after early data check has been enabled.

### Description

User is reporting on design mismatch(es) or is creating or setting design mismatch configuration after the early data check (EDC) system has been enabled. The design mismatch commands and reports are still accepted by the tool after enabling the EDC system. However, the design mismatch manager turned inactive and will no longer check, record, or repair design mismatches since these are now checked and handled by the EDC system. It is still possible to report the mismatches that have been recorded by the design mismatch manager before the EDC system got enabled.

### What Next

The warning can be ignored when reporting design mismatches that got recorded before enabling EDC. When creating or configuring design mismatch configuration, consider using the EDC system instead by using related EDC checks with matching policies and strategies. The EDC system is enabled by 'set_early_data_check_policy' or by 'set_qor_strategy -mode early_design' command. Please refer to below commands related to the EDC system.

set_early_data_check_policy

```
report_early_data_checks
get_early_data_check_records
```

---

## DMM-121

(warning) Repair skipped for '%s' for object '%s' as it would cause loss of UPF data.

### Description

Repaired skipped for design mismatch as it would cause UPF data loss as part of converting empty logic module to macro.

### What Next

If you do not want to repair then change action to 'ignore'. If still repair is needed then enable design mismatch to repair before loading upf.

Feedback

# DPAP

## DPAP-001

(Information) Tool calculated threshold pin count of the partitions is '%d'.

### Description

When pin balancing of partitions is enabled during partitioning, the tool internally calculates a threshold value of number of pins in the partition.

### What Next

This is an information message. No action is required.

### See Also

- explore_logic_hierarchy

## DPAP-002

(warning) The value of plan.auto_group.group_npins_max, set by the user is significantly small. This may result in large number of partitions and runtime cost.

### Description

User set threshold value of pin count of partitions is small.

### What Next

Set a higher value for threshold pin count.

### See Also

- explore_logic_hierarchy

## DPAP-003

(Information) %s. Skipping pin balancing of partitions. %s

### Description

Pin Balancing of partitions is skipped.

### What Next

This is an information message. No action is required.

Feedback

**See Also**

- explore_logic_hierarchy

---

## DPAP-004

(Information) %s. The second priority is implicitly assumed.

**Description**

The second priority in the priority order of constraints in partitioning is implicitly assumed.

**What Next**

This is an information message. No action is required.

**See Also**

- explore_logic_hierarchy

---

## DPAP-005

(Information) %s Running %s auto-partition %s.

**Description**

The above flavor of partitioner will be run.

**What Next**

This is an information message. No action is required.

**See Also**

- explore_logic_hierarchy

---

## DPAP-006

(Information) %s. %s.

**Description**

Skipping the user given invalid values in 'plan.auto_group.results_grading_criteria'. Using default values in case none of values are valid.

**What Next**

Check app-option man page for valid values.

Feedback

**See Also**

- explore_logic_hierarchy

---

## DPAP-010

(Information) The priority order of constraints used in auto-partitioning is : %s %s.

**Description**

The priority order of partitioning constraints is as specified in the message.

**What Next**

This is an information message. No action is required.

**See Also**

- explore_logic_hierarchy

---

## DPAP-101

(error) Max cell count of a partition has either not been specified or invalid value has been specified for auto-partitioning. Specify by using app-option 'plan.auto_group.group_ncells_max'. Exiting Command.

**Description**

The specified command option requires a valid value of maximum cell count allowed in a partition to function correctly. Specify by using app-option 'plan.auto_group.group_ncells_max'.

**What Next**

Specify a valid value of maximum cell count allowed in a partition by using app-option 'plan.auto_group.group_ncells_max'.

**See Also**

- setup_design_for_auto_partition
- explore_logic_hierarchy

---

## DPAP-102

(warning) Value of app-option '%s' has been changed after the setup_design_for_auto_partition command. Ignoring the changed app-option value.

Feedback

**Description**

The above app-option need to be set before setup_design_for_auto_partition command is run. The setting should remain the same throughout auto-partition.

**What Next**

Set the above app-option before setup_design_for_auto_partition command is run.

**See Also**

- setup_design_for_auto_partition
- explore_logic_hierarchy

---

## DPAP-103

(warning) Virtual Groups have changed since setup_design_for_auto_partition command. Ignoring the changes in virtual groups.

**Description**

The virtual group definitions should remain the same through auto-partition setup and run.

**What Next**

Do not make any virtual group changes between setup and run of auto-partition.

**See Also**

- setup_design_for_auto_partition
- explore_logic_hierarchy

---

## DPAP-104

(warning) Partition candidate hierarchy '%s' specified by the user is ignored because of user specified constraints %s.

**Description**

Partition candidate specified are ignored as candidates for partition due to user specified constraints on the chierarchy, its parent or any child hierarchy.

**What Next**

Remove the user constraints.

Feedback

**See Also**

- setup_design_for_auto_partition
- explore_logic_hierarchy

---

## DPAP-105

(warning) Leaf cell count of the partition candidate hierarchy '%s' specified by user is greator than the maximum cell count of a partition specified by the user through the app-option 'plan.auto_group.group_ncells_max'.

**Description**

Leaf cell count of the partition candidate specified by the user is more than the maximum cell count of a partition. This will violate the partition max cell count.

**What Next**

Increase the maximum cell count of a partition or choose a smaller partition candidate.

**See Also**

- setup_design_for_auto_partition
- explore_logic_hierarchy

---

## DPAP-201

(Warning) %s. Ignoring the module.

**Description**

The module is not an MIM in the design and will be ignored in mim handling during auto-partitioning.

**What Next**

Please correct the reference module name in the command option -enable_mim of setup_design_for_auto_partition command.

**See Also**

- setup_design_for_auto_partition
- explore_logic_hierarchy

Feedback

## DPAP-202

(Information) %s (Module: %s) is/are not identified as MIM. (%s)

### Description

The module instances in the message are not considered MIMs in auto-partitioning. The reason is stated in the message.

### What Next

This is an information message. No action is required.

### See Also

- setup_design_for_auto_partition
- explore_logic_hierarchy

## DPAP-203

(Information) %s.

### Description

The instance(s) are not considered as MIM instances. These instances are either user specified existing partitions/physical blocks or parent/child hierarchies of the user specified existing partition/physical blocks.

### What Next

This is an information message. No action is required.

### See Also

- setup_design_for_auto_partition
- explore_logic_hierarchy

## DPAP-204

(Information) User specified hierarchical instance %s is not treated as a partition as %s.

### Description

The hierarchical instance is not treated as a partition because of the reason specified in the message.

Feedback

### What Next

This is an information message. No action is required.

### See Also

- setup_design_for_auto_partition
- explore_logic_hierarchy

---

## DPAP-205

(Information) %s.

### Description

The hierarchical instance is not treated as a partition because of the reason specified in the message.

### What Next

This is an information message. No action is required.

---

## DPAP-206

(Information) %s.

### Description

The Design has option set value as true/false.

### What Next

This is an information message. No action is required.

---

## DPAP-207

(Information) Continuing auto partition process with file: %s .

### Description

Continuing the auto partition process with user provided file/best file.

---

## DPAP-208

Error : %s .

Feedback

**Description**

The auto_partition/set_auto_partition_constraints flow stops with message.

---

## DPAP-209

(Information) %s .

**Description**

The set_auto_partition_constraints/auto_partition_design command process continues with message.

---

## DPAP-210

(warning) %s .

**Description**

The set_auto_partition_constraints command throws warning for wrong value of option.

---

## DPAP-211

(Information) %s (Module: %s) identified as MIM.

**Description**

The module instance is considered as an MIM instance during auto-partitioning. A separate partition will be created for the module instance.

**What Next**

This is an information message. No action is required.

**See Also**

- setup_design_for_auto_partition
- explore_logic_hierarchy

---

## DPAP-212

(Information) Resetting %s old values set by user.

**Description**

Reset the old value set by user.

Feedback

### What Next

This is an information message. No action is required.

---

## DPAP-220

(Warning) Some hierarchies in the existing virtual groups are dont-touch. Setting value of the option '-honor_existing_vitual_groups' to false for this run.

### Description

Existing virtual groups contain dont-touch hierarchies. This may cause issues in using the auto-partition solution.

### What Next

Do not create existing virtual-groups with dont-touch hierarchies inside them.

---

## DPAP-231

(Information) Instance '%s' is treated as an existing partition. All its parent and child hierarchies will not be considered for partitioning by the tool.

### Description

The instance, its parent and child hierarchies will not be considered for partitioning by the tool.

### What Next

This is an information message. No action is required.

### See Also

- setup_design_for_auto_partition
- explore_logic_hierarchy

---

## DPAP-250

(Information) No hierarchies in the design have been identified as candidates for partition. Exiting Auto-Partition.

### Description

Either there are no hierarchies in the design or the hierarchies in the design have been constrained so that they are not identified as candidates for partitioning the design. Exiting Auto-Partition.

Feedback

### What Next

This is an information message. No action is required.

### See Also

- auto_partition_design
- set_auto_partition_constraints

# DPBUS

## DPBUS-001

(error) No bus plan exists.

### Description

You used a command that needs a bus plan to have been created, but no bus plan was found.

### What Next

Create a bus plan using create_busplans.

## DPBUS-002

(error) Constraint %s not supported.

### Description

There are a limited set of constraints supported by the tool. The supported constraints are <, <=, >, >= and =.

### What Next

Change to use a supported constraint.

## DPBUS-003

(error) Bus %s cannot be constrained to itself.

### Description

The same bus is specified in both the *-from* and in the *-to* bus sets. This means a bus should be contrained to itself. This is not supported.

Feedback

**What Next**

Change *-from* or *-to* to ensure the same bus does not appear in both sets.

---

## DPBUS-004

(error) Buses %s and %s cannot be constrained to be in a cycle.

**Description**

Adding the requested bus constraint would cause a constraint cycle involving the two buses reported. This is not supported.

**What Next**

Change *-from* or *-to* to ensure the constraints do not form a cycle.

---

## DPBUS-005

(error) Layer %s not recognized.

**Description**

You have specified a layer which the tool does not recognize as a layer.

**What Next**

Use *get_layers* to see list of layers.

---

## DPBUS-006

(error) Buffer %s is not recognized.

**Description**

You have specified a buffer which the tool does not recognize.

**What Next**

This command wants the ref_module_name of a buffer. Find an example buffer cell in the design and get the attribute ref_module_name. Use the ref_module_name for this command.

---

## DPBUS-007

(error) Buffer %s is not usable.

Feedback

**Description**

You have specified a buffer which the tool cannot use. This is because the specified buffer is marked as "dont use".

**What Next**

Look at the "dont use" attribute on the lib cell. For example, *get_attribute -name dont_use [get_lib_cells */bufferName]*. To set attribute to false, for example use *set_attribute -name dont_use -value false [get_lib_cells */bufferName]*. An alternate way is to use *set_lib_cell_purpose*.

---

## DPBUS-008

(error) Cannot specify %s without %s or %s

**Description**

You used a command option that requires another option to also be specified.

**What Next**

Refer to the manual page for this command for detailed information on valid options.

---

## DPBUS-010

(error) Bus %s is part of a fanin bus %s. It cannot be removed.

**Description**

You have specified to remove a bus which is part of a fanin bus. To remove this bus you need to remove the fanin bus instead.

**What Next**

Use *remove_busplans -fanin_bus <fanin_bus_name>* to remove the fanin bus.

---

## DPBUS-011

(error) Bus %s is not a starting bus for fanin bus %s. It cannot be the current bus.

**Description**

You have specified an incorrect bus to be the current bus for a fanin bus. Only buses at the start of a fanin bus can be the current bus. The current bus represents the bus that timing analysis will be done for the fanin bus. To uniquely identify all the paths in the fanin bus only the starting buses can be made into the current bus.

Feedback

**What Next**

Use *get_attribute -name start_buses [get_busplans -fanin_bus <fanin_bus_name>]* to see all the starting buses on <fanin_bus_name>. Pick one of these buses to be the current bus.

---

## DPBUS-012

(error) Bus %s is not found.

**Description**

You have specified a bus which does not exist.

**What Next**

Use *report_busplans* * to see all the buses.

---

## DPBUS-013

(error) No fanin bus gates found.

**Description**

You have specified certain gates to be used to create a fanin bus, but those gates were not found.

**What Next**

Make sure the gates exist in design.

---

## DPBUS-014

(error) Fanin bus gate %s has attribute busplan_trace_through.

**Description**

You specified a fanin bus gate which has the busplan_trace_through attribute set. This is not allowed as the fanin bus gate has special treatment which conflicts with busplan_trace_through setting.

**What Next**

Remove the busplan_trace_through attribute from this cell.

---

## DPBUS-015

(error) Could not create the sub-buses needed for fanin bus %s

Feedback

**Description**

A fanin bus is composed of normal fanout buses. These buses are called sub-buses of the fanin bus. For some reason, the sub-buses were not able to be created.

**What Next**

There could be a number of reasons why this happens. First check that the fanin gates are properly specified so as to create a fanin bus. Second check that none of the nets are already used by some other bus.

Another thing to try is to manually create the sub-buses to see where the problem occurs. This can be done by adding the fanin gates to the start_end_cells by *create_busplans -add_start_end_cells*, then use *create_busplans -from -to* to create each sub-bus.

---

## DPBUS-016

(error) Found sub-buses with branches for fanin bus %s.

**Description**

A fanin bus is composed of normal fanout buses. These fanout buses are called sub-buses of the fanin bus. A current limitation is that the sub-buses cannot have branches. Another way to look at this is that the tool does not support a bus with both fanin and fanout nodes.

**What Next**

Decide whether to have a fanin bus or a fanout bus.

---

## DPBUS-017

(error) Could not create the fanin bus %s.

**Description**

This is a generic message indicating the fanin bus creation failed. There should be more detailed information in previous messages.

**What Next**

There could be a number of reasons why this happens. First check that the fanin gates are properly specified so as to create a fanin bus. Second check that none of the nets are already used by some other bus.

Another thing to try is to manually create the sub-buses to see where the problem occurs. This can be done by adding the fanin gates to the start_end_cells by *create_busplans -add_start_end_cells*, then use *create_busplans -from -to* to create each sub-bus.

Feedback

## DPBUS-018

(error) Bus(es) {%s} have branches. This command does not support branching buses.

**Description**

You used a command that only supports 2 pin buses. The given buses have branches and thus are not a 2 pin buses. These buses are skipped.

**What Next**

Plan these buses using the manual controls.

## DPBUS-019

(error) The name %s has already been used for %s.

**Description**

The name given has already been used and this command requires a unique name.

**What Next**

Choose a different name for this object.

## DPBUS-020

(error) Bus(es) {%s} have feedthroughs. This command does not support buses with feedthroughs.

**Description**

You used a command that only supports top level buses. The given buses have feedthroughs and thus are not a top level buses. These buses are skipped.

**What Next**

Plan these buses using the manual controls.

## DPBUS-021

(error) Bus(es) {%s} have fixed elements. This command does not support buses with fixed elements.

**Description**

You used a command that does not support fixed elements. Buses with fixed elements are skipped.

Feedback

**What Next**

Remove the fixed attribute on bus elements or plan these buses using the manual controls.

---

## DPBUS-022

(error) Cannot add elements before the starting element or after the ending element.

**Description**

The starting and ending elements cannot be changed so it is not allowed to add elements before the starting element or after the ending element.

**What Next**

Do not attempt to add elements before the starting element or after the ending element.

---

## DPBUS-023

(error) Bus(es) {%s} does not have proper layers.

**Description**

You used a command that requires each bus to have proper layers. Layer for a bus are given in the associated net estimation rule. The bus needs to have correct horizontal as well as vertical layer assigned in the net estimation rule.

To determine the rule associated with a bus use *get_attribute -name rule [get_busplans <busname>]*

To see the horizontal and vertical layers associated with a rule use *report_net_estimation_rule <rulename>*

To change a rule use *set_net_estimation_rule -parameter layer -horizontal_value <horLayer> -vertical_value <verLayer> <rulename>*

**What Next**

Change the rule so that the horizontal and vertical layers are properly set.

---

## DPBUS-024

(error) Bus(es) {%s} has unassigned pins.

**Description**

You used a command that requires all pins on all the nets in the bus to have pins assigned. To assign pins use *place_pins* command

Feedback

**What Next**

Place all the pins then try again.

---

## DPBUS-025

(error) Bus %s did not route because %s.

**Description**

The specified bus failed to route due to an internal issue as stated in message.

**What Next**

Plan this bus using auto-plan or manually.

---

## DPBUS-026

(error) Rule %s: specified layers %s and %s are not adjacent. Adjacent layers or user-provided register / buffer spacing is required.

**Description**

The layers specified for the indicated net estimation rule are not adjacent. In this situation, the net estimation timing engine cannot compute register / buffer spacing based on indirect attributes (layer, clock, scenario, etc). Non-adjacent layers are permitted, but only if the user directly specifies user-computed register / buffer spacing values.

**What Next**

Modify the net estimation rule settings to use 2 adjacent layers.

Or

Directly specify register / buffer spacing values for the net estimation rule.

**See Also**

- set_net_estimation_rule
- report_net_estimation_rules

---

## DPBUS-027

(error) Option %s is mutally exclusive with these options %s.

**Description**

The listed command option is mutually exclusive with all other options listed.

Feedback

**What Next**

Look at manpage for this command for more information on command options.

---

## DPBUS-028

(error) %s is not a physical cell.

**Description**

The listed cell is logical cell only. The command requires a physical cell.

**What Next**

Change input cell to a physical cell.

---

## DPBUS-029

(error) Can't find frame view for reference design %s.

**Description**

The listed reference design does not have a frame view. The command requires a frame view to get boundary and area information.

**What Next**

Check that the reference library for this design is linked and available.

---

## DPBUS-030

(error) Balance does not support buses with repeated MIBs.

**Description**

The balancing feature has a limition that it does not support buses with repeated MIBs. A repeated MIB bus is a bus that goes through 2 or more instances of the same MIB using the same pins in each case.

**What Next**

Balancing must be done manually for this type of bus.

---

## DPBUS-050

(error) Error in XML file: %s

Feedback

### Description

An error was detected in the specified XML file while executing the command load_busplans. Examine the detailed error message and attempt to resolve the problem with the file before calling load_busplans again.

### See Also

- load_busplans

---

## DPBUS-051

(error) Error creating busplan %s

### Description

An error was detected while creating a busplan. This may be due to an incorrectly specified XML input file passed to load_busplans. Examine the output messages and try to correct the problem.

### See Also

- load_busplans

---

## DPBUS-052

(error) Error saving XML file %s

### Description

An error was detected while writing an XML file from report_busplans, write_busplans, or write_net_estimation_rules.

### What Next

Examine the output message(s) and try to correct the problem.

### See Also

- write_busplans
- report_busplans
- write_net_estimation_rules

Feedback

## DPBUS-053

(error) General list processing error: %s

### Description

An error was detected while processing an XML file passed to load_busplans. Verify the format of your input XML file.

### See Also

- load_busplans

## DPBUS-054

(error) Bus % does not have any ports.

### Description

This command only works on buses with top level ports. The given bus has no top level ports.

### What Next

Only apply command on buses with top level ports.

## DPBUS-055

(error) Copy busplan %s to %s failed because %s.

### Description

The copy failed. The most common reason is that the structure of the 2 bus section is not the same. There must be a 1-1 mapping between the non-virtual objects in the 2 bus section. Non-virtual objects means pins, ports, and registers elements.

### What Next

These two buses sections cannot be copied. Find a matching section to copy instead.

## DPBUS-056

(error) Error in XML file: %s

Feedback

### Description

An error was detected in the specified XML file while executing the command read_net_estimation_rules. Examine the detailed error message and attempt to resolve the problem with the file before calling read_net_estimation_rules again.

### See Also

- read_net_estimation_rules

---

## DPBUS-057

(warning) Busplan %s is missing element(s) %s that were found in the save file.

### Description

While executing the *load_busplans* command, the indicated busplan did not contain all the non-virtual elements that were expected, based on the file being loaded. For example: busplan1 was created at some point in time and had actual (not virtual) registers as part of its elements and the busplan definition was saved to a file. Later, an attempt to load/recreate busplan1 into a different session is made, where 1 or more of those registers do not exist. This warning will be printed to indicate the missing elements and their type.

Note: the busplan will exist with the same start and end points as before, but the intermediate elements will be different, and an undetermined number of virtual elements were re-created. Depending on whether the virtual elements were present before, between or after the missing elements, some, none or all of them may have been re-created.

### What Next

Remove the busplan using the *remove_busplans* command and use *load_busplans -only_create ...* to recreate the busplan without virtual elements. Alternatively, use *load_busplans -only_create -replace_existing virtual elements (if any).*

### See Also

- load_busplans
- remove_busplans
- start_busplan_gui
- write_busplans

---

## DPBUS-058

(Warning) Busplan %s has new elements (%s). Use merge to restore virtual elements.

Feedback

### Description

The *load_busplans* command encountered a netlist that is different from the original netlist used to create the save file. Specifically, there are new elements as detailed in the error message. The new elements prevent a successful re-creating of previous virtual elements, so the busplan was not restored the same way it was saved.

### What Next

Use GUI-based merge to resolve the differences between the saved version of the busplan and the current busplan.

### See Also

- load_busplans
- start_busplan_gui
- write_busplans

---

## DPBUS-100

(error) Failed to set net estimation rule %s parameter %s

### Description

Setting the net estimation rule parameter failed. The most likely cause is that the value specified for the parameter is not valid. For example if the parameter is layer, then the specified value must be a routing layer.

### What Next

Fix the value to a valid setting.

### See Also

- DPBUS-101

---

## DPBUS-101

(error) Error creating net estimation rule %s.

### Description

The net estimation rule was not created due to an error reported just before this error. This is most likely due to an error in trying to set an invalid value for a rule that does not yet exist. If the rule already exists, *DPBUS-100* will be issued instead.

Feedback

### What Next

Fix the parameter value to be valid.

### See Also

- DPBUS-100

---

## DPBUS-102

(warning) Using top-level corner %s for net estimation rule %s.

### Description

The net estimation rule is defined in a sub-block, but the corner referenced by the net estimation rule was not defined in the same sub-block. However, a corner with the same name was found to be defined in the top-level block (current design) and is being used instead.

### What Next

According to the recommended methodology, the corner referenced by the net estimation rule would also be defined in the sub-block and be defined the same way as the corner in the top block. However, depending on how the current blocks and associated constraints were defined, this may be awkward. Accordingly, this situation does not require any corrective action, but the warning is issued so that corrective action can be taken, if desired.

### See Also

- create_corner
- report_net_estimation_rules
- set_net_estimation_rule

---

## DPBUS-103

(Error) Corner %s defined in net estimation rule %s was not found.

### Description

The corner referred to by the net estimation rule could not be found. If the net estimation rule is defined in a sub-block, both the sub-block and the top-level block (current design) were searched to try to locate a defined corner by that name.

Feedback

### What Next

Define a corner with the referenced name in the block that the net estimation rule is defined in, or use a different, already existing corner for the net estmation rule.

### See Also

- create_corner
- report_net_estimation_rules
- set_net_estimation_rule

---

## DPBUS-200

(warning) bus %s skipped for spreading due to invalid timing.

### Description

The command *route_busplans -spread* checks to ensure a busplan has valid timing before attempting to spread it. Invalid timing is the result of incomplete data in the net estimation rule associated with the busplan. In order to spread this busplan, ensure the net estimation rule associated with the busplan is properly configured.

### See Also

- route_busplans
- set_net_estimation_rule

---

## DPBUS-201

(warning) bus %s skipped for spreading due to negative slack.

### Description

The command *route_busplans -spread* checks to ensure a busplan has positive slack before attempting to spread it and the named busplan had negative slack when attempting to spread it. This can be resolved by either modifying the busplan to have positive slack, such as by adding virtual registers or using a different net estimation rule, or by using the *-skip_wns_check-fP option of route_busplans -spread (or setting the check box under "Settings" in the Pipeline Register Planning GUI).*

### See Also

- modify_busplan
- route_busplans

Feedback

- set_net_estimation_rule
- start_busplan_gui

---

## DPBUS-202

(warning) bus %s skipped for spreading due to branches.

**Description**

The command *route_busplans -spread* currently cannot handle spreading of busplans with branches in channels. Such busplans are skipped.

**See Also**

- route_busplans

---

## DPBUS-203

(warning) bus %s skipped for spreading - not rectilinear.

**Description**

The command *route_busplans -spread* cannot handle spreading of busplans with non-rectilinear sections. The *-no_plan_busplans* option was specified to the *route_busplans* command, so this busplans was skipped without attempting to resolve the non-rectilinear section(s).

**See Also**

- route_busplans

---

## DPBUS-204

(warning) bus %s skipped for spreading - not rectilinear.

**Description**

The command *route_busplans -spread* cannot handle spreading of busplans with non-rectilinear sections. An attempt was made to perform an "Auto-plan" (*route_busplans -quick*) operation on the busplan, but the result still contained one or more non-rectilinear sections. This busplan was skipped for spreading.

**See Also**

- route_busplans

Feedback

## DPBUS-205

(Warning) No busplans found suitable for spreading.

### Description

The command *route_busplans -spread* was invoked with one or more busplans passed to it. After checking the busplans, no busplans suitable for spreading were found. No spreading was performed.

### See Also

- route_busplans

## DPBUS-500

(warning) From pin %s is dropped from consideration.

### Description

There are several reasons why a from pin can be filtered out. First, the from pin must be on a start_end_cell or the from pin must be a top level port. By default all blocks and hard macros are considered to be start_end_cells but any physical cell can be added by create_busplans -add_start_end_cells. Second, the from pin must be a net driver pin. Third, the tracing done from the from pin must reach a pin on a start_end_cell (or top level port). If using the -to option, then the from pin must reach a to pin.

### What Next

Most likely need to use -add_start_end_cells option to include the cell the from pin is on.

## DPBUS-501

(warning) To pin %s is not on any start_end_cell and is dropped from consideration.

### Description

It is required that to pins must be on a start_end_cell (specified by create_busplans -start_end_cells) or to pins must be a top level port. This is not the case for the given to pin.

### What Next

Most likely need to set the -start_end_cells to include the cell the to pin is on.

Feedback

## DPBUS-502

(warning) All to pin %s have been dropped.

**Description**

It is required that to pins must be on a start_end_cell (specified by create_busplans -start_end_cells) or to pins must be a top level port. All the specified to pins have been dropped by above criteria so no buses can be created.

**What Next**

Most likely need to set the -start_end_cells to include the cells the to pins are on.

## DPBUS-503

(warning) To pin %s is not on a bus starting from any from pin.

**Description**

A bus trace starting from the from pins did not reach the given to pin. Thus there is no bus going to this to pin. Note that bus tracing does not go through arbitrary combinational logic. It only traces through repeaters, registers, latches, level shifers, and isolation cells.

**What Next**

If this is unexpected, one thing to try is to see what buses can be found using create_busplan -auto. This will show what, if any, buses can be found that end in the to pins.

## DPBUS-504

(warning) Movebound %s is overlapping block and/or placement blockages.

**Description**

When creating a movebound for top level registers, the tool attempts to create the movebound such that it will not overlap top level blocks or top level soft/hard placment blockages. Sometimes this fails and then the movebound will overlap top level blocks or placement blockages. This can often happen if the top level register is placed on top of a block instead of in a top level channel.

**What Next**

Investigate the given movebound and manual adjust it if needed.

Feedback

## DPBUS-505

(warning) Cannot remove default net estimation rule %s.

**Description**

The default net estimation rule cannot be removed as certain pipeline register planning functionalities depend on this. Command will proceed without removing this rule.

**What Next**

Do not attempt to remove the default net estimation rule.

## DPBUS-506

(warning) Cannot remove net estimation rule %s. It is used by busplan %s.

**Description**

The given net estimation rule is being used by 1 or more busplans, one of which is given. To remove a net estimation rule, it must not be used by any busplan.

**What Next**

Change all the busplans using the given net estimation rule to use a different rule. Can use modify_busplan -change_rule for this.

## DPBUS-508

(warning) On bus plan %s, cannot find net estimation rule %s. The default net estimation rule %s will be used instead.

**Description**

The given net estimation rule could not be found. Most likely it was deleted via the command *remove_net_estimation_rules*. Since the rule could not be found, the default rule will be used in its place.

**What Next**

It is recommended to fix this warning. To fix this, use *modify_busplan* to change the rule on the bus plan to an existing net estimation rule. You can also create a new net estimation rule using *set_net_estimation_rule*.

## DPBUS-509

(warning) Cell %s has been deleted, busplan %s correspondingly is deleted.

Feedback

**Description**

The given cell has been removed. That cell is part of the given busplan. The busplan is no longer valid so it has been removed.

**What Next**

Consider if this cell removal is correct.

---

## DPBUS-510

(error) Cannot rename bus plan %s to %s - the name %s is already in use.

**Description**

The new bus plan name is already being used by another bus. To rename a bus plan, the new name must not be used by any existing bus.

**What Next**

Choose a different new name for the bus plan rename operation.

---

## DPBUS-511

(warning) Bus %s is part of fanin bus %s and has been removed without removing the corresponding fanin bus.

**Description**

A sub-bus of a fanin bus has been removed without removing the corresponding faning bus. This causes the fanin bus to be inconsistent and could lead to incorrect results.

**What Next**

Remove the fanin bus before removing the bus.

---

## DPBUS-512

(warning) Cell instance %s is already part of bus %s, so it cannot be a start end cell instance.

**Description**

The given cell is already a part of the given bus. Cells which are part of existing buses cannot be added to the start end cell instance list. This is because it would cause an inconsistency between the start cell instances and the existing busplans.

Feedback

If the given cell instance is a hierarchical cell instance, then this means that there is cell in given bus which is a desendant of that hierarchical cell instance. This is also considered as being part of bus and has same issue as described above.

**What Next**

If it is desired that the given cell be a start end cell instance, then remove the given bus first.

---

## DPBUS-513

(warning) Movebound %s is overlapping movebound %s.

**Description**

When creating the script for creating movebounds for pipeline registers, the tool may create movebounds which overlap. If the movebounds are hard, then this may create a placement over-utilization issue.

**What Next**

Investigate the given movebound and manual adjust it if needed and recreate the movebound script. Alternately make the movebounds soft by editing the movebound script.

---

## DPBUS-514

(warning) Unable to determine bus %s %s budget.

**Description**

The launch or capture budget was not determined. This is most likely because there is a problem with the timing setup. Another reason is that there are no input/output delays at the ports of the bus.

**What Next**

Try using report_timing to see if is any issue with the timing setup. Check if there are input/output delays on the ports of the bus.

---

## DPBUS-515

(warning) Unable to determine layers for bus %s. All layers will be used.

Feedback

### Description

There is some problem with the layer specification for the given bus. Most likely the issue is that here is no horizontal or no vertical layer specified. The layer specification is given by the net estimation rule associated with the given bus.

### What Next

If you are ok to use all layers, then nothing needs to be done. If you want to pick only specific layers, then determine the net estimation rule associated with the bus. Check the layer specification for that rule. The rule must include a horizontal and a vertical layer specification. Certain commands treat the horizontal and vertical layers as a range of layers rather than 2 independent layers. In this case the range of layers must include at least 1 horizontal and 1 vertical layer.

---

## DPBUS-516

(warning) Bus(es) %s are not planned and so were ignored.

### Description

This command requires that buses are planned to be considered. Planned buses are buses with valid timing (in busplanning GUI bus will have green color). For most accurate results, the topology of the bus should also be legal.

### What Next

If you are ok to ignore these buses, then nothing needs to be done. If you want these buses to be considered, then plan the buses by setting a net estimation rule with valid timing (one way is to set the register spacing on the net estimation rule), adding the suggested registers, and creating a topology for the bus(es).

---

## DPBUS-517

(warning) The bus elements %s and %s are not MIB compliant.

### Description

The given bus elements correspond to 2 sets of MIB feedthrough ports, each of which is part of a bus (ie a bus element). These are called MIB bus feedthrough port sets.

For MIB compliance it is required that every MIB bus feedthrough port set are either unrelated or one is a subset of the other or they are a subset of a parent MIB bus.

For example suppose there are 3 MIBS, B1, B2, and B3. If there is a feedthrough bus element corresponding to the ports B1/data[0..10] and there is a feedthrough bus element corresponding to the ports B2/data[5..10]. Then in this case these 2 MIB bus port sets are compliant because B2 is subset of B1. If there is also a feedthrough bus element

Feedback

corresponding to the ports B3/data[1..6], that is also compliant because B2 and B3 are subsets of B1.

On the other hand if B1 did not exist, then B2 port set and B3 port set are not MIB compliant because one port sets is not a subset of the other.

A second requirement is that the fanout inside the MIB is the same. For example if the bus1 includes the path, port B1/data[0] to B1/out1[0] and B1/out2[0], then if bus2 only includes the path, port B1/data[0] to B1/out1[0], then this is not compliant.

**What Next**

Manually recreate the buses to ensure all the MIB bus feedthrough port sets match.

---

## DPBUS-518

(warning) Port %s is in %s, but is not in %s.

**Description**

This indicates half of an MIB compliance issue between 2 buses. There should be two such messages. One indicating port in first bus, but not in second bus. The other indicating port in second bus, but not in first bus. These 2 messages indicate these 2 buses are incompatible with each other.

For MIB compliance it is needed that one of these buses feedthrough port set is a subset of the other.

**What Next**

Manually recreate the buses to ensure the MIB bus feedthrough port sets of one bus is subset of the other.

---

## DPBUS-519

(warning) The bus elements %s and %s are not MIB %s compliant.

**Description**

For 2 buses to be MIB compliant it is necessary for the structure of the bus inside the MIB to be the same. It is also necessary for the placement and fixed attribute of the bus elements inside the MIBs to be the same with respect to the MIB. Finally it is necessary for the net estimation rule to be the same and the clock be the same.

This warning indicates that either the structure of the bus inside the MIBs is not the same or the (fixed) location of the bus elements is not the same or the rule or clock is not the same. The structure is related to the number and type of bus elements in the MIB.

Feedback

### What Next

Decide which bus has the desired structure, location, rule, or clock. Use the *copy_busplan* command to copy the structure and locations from the desired to the other. Set the rule to be the same using *modify_busplan -change_rule*. Finally if the clock is the issue, check the SDC to see which clock is the correct one to use.

---

## DPBUS-520

(warning) Found that removal of bus element %s may cause MIB compliance issue.

### Description

This warning is because you have requested an operation that would require the removal of a bus element that is the parent bus of an MIB bus group. To maintain that all busplans are MIB compliant, this is not allowed. The command will proceed without the removal of this bus element.

### What Next

Manually create the busplans in such a way that removal is not needed. This can be done by creating the bus with the given bus element first.

---

## DPBUS-521

(info) Creating busplan %s requires busplan %s to be created as well.

### Description

In order to maintain MIB compliance, the parent MIB busplan of the busplan currently being processed must be created first. Since this busplan does not already exist, it is being created before the child MIB busplan, even though it is not part of the list of busplans to be created based on the parameters specified to load_busplans.

### See Also

- load_busplans
- report_busplans

---

## DPBUS-522

(Error) Busplan %s requires MIB parent busplan %s, which could not be created.

Feedback

### Description

In order to maintain MIB compliance, the MIB parent busplan of the busplan currently being loaded needs to exist. However, it does not already exist, and its definition does not exist in the current XML file. Creation of the current busplan is canceled.

### What Next

Create the MIB parent busplan first and retry the current operation, load a different XML file, or use the *-ignore_MIB_compliance* option to load_busplans.

### See Also

- load_busplans
- report_busplans

---

## DPBUS-523

(warning) Busplans %s and %s have a required MIB parent loop. The dependency %s to %s is dropped in order to continue.

### Description

In order to maintain MIB compliance, the MIB parent busplan of the busplan currently being loaded needs to exist. However, it does not already exist, and trying to create it requires the current busplan - a loop. In order to try to continue, this depdendency will be dropped.

### What Next

This situation should not occur if the saved file was created by *write_busplans*, but it may occur due to manual editing of the .xml file. If the resulting busplans are not correct, verify that the .xml file was correctly constructed.

### See Also

- load_busplans
- write_busplans

---

## DPBUS-800

(info) Did not find any blocks or hard macros to be start_end cells. You may have to add cells using -add_start_end_cells.

Feedback

**Description**

By default, blocks, IOs and hard macros are considered to be objects from which busplans can be created. For example, there could be a busplan created between two blocks, B1 and B2. For example, there could be a busplan created between IOs and block B1.

This message is indicating that the tool could not find any blocks or hard macros in the design. Thus the only busplans that could be found would be between IOs.

**What Next**

If there are no blocks or hard macros in the design then there is nothing to do. If there are blocks or hard macros in the design then use the -add_start_end_cells option to add those blocks or hard macros into consideration if desired. You can see the list of start end cells using *report_busplans -start_end_cells*

---

## DPBUS-801

(info) While tracing %s, unable to trace through gate %s.

**Description**

To identify buses, the netlist is traced. The tracing will only go through buffers and inverters. During tracing from the given pin, the given gate was found which tracing was not able to go through so no bus can be created with the given pin.

**What Next**

This bus cannot be created at this time.

---

# DPCHK

---

## DPCHK-001

(Error) boundary point (%s %s) of site row '%s' is not on FinFET grid.

**Description**

You received this message because the FinFET grid checker reported a detail violation of site row.

**What Next**

Fix the violation and run FinFET grid checker again.

Feedback

**See Also**

- check_finfet_grid

---

## DPCHK-002

(Error) boundary point (%s %s) of %s '%s' is not on FinFET grid.

**Description**

You received this message because the FinFET grid checker reported a detail violation of the cell.

**What Next**

Fix the violation and run FinFET grid checker again.

**See Also**

- check_finfet_grid

---

## DPCHK-003

(Error) boundary point (%s %s) of site array '%s' is not on FinFET grid.

**Description**

You received this message because the FinFET grid checker reported a detail violation of site array.

**What Next**

Fix the violation and run FinFET grid checker again.

**See Also**

- check_finfet_grid

---

## DPCHK-004

(Error) boundary point (%s %s) of design '%s' is not on FinFET grid.

**Description**

You received this message because the FinFET grid checker reported a detail violation of design boundary.

Feedback

### What Next

Fix the violation and run FinFET grid checker again.

### See Also

- check_finfet_grid

---

## DPCHK-005

(Error) core area point (%s %s) of design '%s' is not on FinFET grid.

### Description

You received this message because the FinFET grid checker reported a detail violation of design core area.

### What Next

Fix the violation and run FinFET grid checker again.

### See Also

- check_finfet_grid

---

## DPCHK-006

(Error) height (%s) and width (%s) of cell reference '%s' do not match FinFET grid.

### Description

You received this message because the FinFET grid checker found the boundary points of this cell reference could not snap to FinFET grid perfectly no matter what the orientation of its instance is.

### What Next

Do not use this cell reference before its boundary is modified to conform to FinFET grid.

### See Also

- check_finfet_grid

---

## DPCHK-007

(Error) height (%s) and width (%s) of site '%s' do not match FinFET grid.

Feedback

### Description

You received this message because the FinFET grid checker found the boundary points of this site could not snap to FinFET grid perfectly no matter what the orientation this site is with.

### What Next

Do not use this site before its boundary is modified to conform to FinFET grid.

### See Also

- check_finfet_grid

---

## DPCHK-008

(Warning) Design '%s' does not have FinFET grid.

### Description

You received this message because FinFET grid was not detected on the design.

### What Next

If it is expected, just ignore this message. Otherwise, please double check the FinFET grid definition in tech file.

### See Also

- report_grids

---

## DPCHK-009

(error) Found %d %s point(s) violated to FinFET grid in block %s.

### Description

You received this message because some object points not on FinFET grid.

### What Next

If it is expected, just ignore this message. Otherwise, please double check the FinFET grid definition in tech file.

### See Also

- report_grids

Feedback

## DPCHK-010

(error) found region {%s} from {%s} to {%s} violate rule %s %s: %s.

### Description

You received this message because the gap/space between two objects violated the rule.

### What Next

If it is expected, just ignore this message. Otherwise, please check the loacation of related objectss.

### See Also

- set_floorplan_spacing_rules
- set_floorplan_enclosure_rules
- set_floorplan_halo_rules
- report_floorplan_rules

## DPCHK-011

(error) found distance %f from corner {%s} on {%s} to corner {%s} on {%s} violate rule %s: %s.

### Description

You received this message because the distance between two objects' specified corner violated the rule.

### What Next

If it is expected, just ignore this message. Otherwise, please check the location of related objects.

### See Also

- set_floorplan_enclosure_rules
- report_floorplan_rules

## DPCHK-012

(error) found %s rule %s violated between {%s} and {%s}%s.

Feedback

### Description

You received this message because for halo/enclosure rules, from-object can't fully cover to-object; For spacing rules, two objects are overlapping with each other.

### What Next

If it is expected, just ignore this message. Otherwise, please check the loacation of related objects.

### See Also

- set_floorplan_enclosure_rules
- set_floorplan_halo_rules
- set_floorplan_spacing_rules
- report_floorplan_rules

---

## DPCHK-013

(error) found %s type width of region {%s} of {%s} violate rule %s %s: %s.

### Description

You received this message because the width of some segments of object violated the rule.

### What Next

If it is expected, just ignore this message. Otherwise, please check the shape of related objectss.

### See Also

- set_floorplan_width_rules
- report_floorplan_rules

---

## DPCHK-014

(error) found area of region {%s} of {%s} violate rule %s: %s.

### Description

You received this message because the area of some regions of the object violated the rule.

Feedback

### What Next

If it is expected, just ignore this message. Otherwise, please check the shape of related objects.

### See Also

- set_floorplan_area_rules
- report_floorplan_rules

---

## DPCHK-015

(warning) for rule %s, there is no %s in the current design.

### Description

You received this message because for rule %s, there is no %s can be found in the current design.

### What Next

Please check the design or related rule settings.

### See Also

- set_floorplan_area_rules
- report_floorplan_rules

---

## DPCHK-016

(error) for rule %s, there is no %s in the current design.

### Description

You received this message because for rule %s, there is no %s can be found in the current design.

### What Next

Please check the design or related rule settings.

### See Also

- set_floorplan_area_rules
- report_floorplan_rules

Feedback

## DPCHK-017

(error) found track %s on location {%s} violate track constraint %s.

**Description**

You received this message because some track violated track constraints.

**What Next**

If it is expected, just ignore this message. Otherwise, please check the location of tracks.

**See Also**

- set_track_constraint
- remove_track_constraint

## DPCHK-018

(warning) found %s in current design that violated rule %s.

**Description**

You received this message because %s can't be palaced in current design according to constrain %s.

**What Next**

Please check the design or related rule settings.

**See Also**

- set_floorplan_area_rules
- report_floorplan_rules

## DPCHK-020

(error) port %s has more than one terminal which are on different side of chip.

**Description**

You received this message because port %s can't have backside and frontside terminals simultaneously.

**What Next**

Please check the design.

Feedback

**See Also**

- check_design

---

## DPCHK-021

(error) Poly rule violation in {%s}, width is %d tiles.

**Description**

The length of this place-able rectangle is less than the poly rule threshold and no exception applies to this rectangle.

**What Next**

Adjust the position of the related hard macros, placement blockages, soft macros and voltage areas.

---

## DPCHK-022

(error) %s in %s overlaps with %s in %s.

**Description**

This message indicates that the tool has detected an overlapping violation.

**What Next**

Change the locations of overlapping liner cells or padding cells.

---

## DPCHK-023

(error) %s in %s is dangling.

**Description**

This message indicates that the tool has detected a dangling violation.

**What Next**

Change the locations of the dangling liner cell or padding cell.

---

## DPCHK-024

(error) Illegal spacing between %s in %s and %s in %s.

Feedback

**Description**

This message indicates that the tool has detected a spacing violation.

**What Next**

Change the locations of relevant liner cells or padding cells.

---

## DPCHK-025

(warning) boundary point (%s %s) of %s '%s' is not on FinFET grid.

**Description**

You received this message because the FinFET grid checker reported a detail violation of the cell.

**What Next**

Fix the violation and run FinFET grid checker again.

**See Also**

- check_finfet_grid

---

## DPCHK-026

(warning) height (%s) and width (%s) of cell reference '%s' do not match FinFET grid.

**Description**

You received this message because the FinFET grid checker found the boundary points of this cell reference could not snap to FinFET grid perfectly no matter what the orientation of its instance is.

**What Next**

Do not use this cell reference before its boundary is modified to conform to FinFET grid.

**See Also**

- check_finfet_grid

---

## DPCHK-027

(warning) Found %d %s point(s) violated to FinFET grid in block %s.

Feedback

**Description**

You received this message because some object points not on FinFET grid.

**What Next**

If it is expected, just ignore this message. Otherwise, please double check the FinFET grid definition in tech file.

**See Also**

- report_grids

---

## DPCHK-028

(error) the %s %s is %s which violate the density rule %s: %s.

**Description**

You received this message because the density(area_ratio or amount) of some objects violated the rule.

**What Next**

Please check the design or related rule settings.

**See Also**

- set_floorplan_density_rules
- report_floorplan_rules

---

## DPCHK-029

(error) the %s %s is %s which violate %s: %s.

**Description**

You received this message because the density of some objects violated the density rules. These density rules are from technology file.

**What Next**

Please check the design.

**See Also**

- read_tech_file

Feedback

## DPCHK-100

(warning) Option %s of the rule %s may not necessarily be fixed

### Description

When removing violations of spacing/enclosure/location rules, some rule value options are not targeted by the fix_floorplan_rules command and may not be fixed, even though they are checked by check_floorplan_rules. In particular, the command does not bring macros closer to each other or to the boundary. So, -max option is ignored, as well as anything that behaves like -max. For example, if there is a valid_range, increasing the distance to at least the lower bound of the range is supported, but decreasing the distance to the upper bound of the range is not, i.e. since the valid range {A B} behaves like -min A -max B, only -min A is supported. Valid list and forbidden range are treated similarly; also note that valid list values must be on grid in order to be satified.

### What Next

If this is expected, just ignore this message. Otherwise, please check the rule setting.

### See Also

- fix_floorplan_rules
- set_floorplan_spacing_rules
- set_floorplan_enclosure_rules
- set_floorplan_location_rules
- report_floorplan_rules

# DPI

## DPI-001

(error) total unplaced IO width is larger than total unused IO guide length

### Description

There is not enough IO area for placing all IO cells No physical design is available.

### What Next

Please use create_io_guide or create_io_ring to increase IO area.

Feedback

## DPI-002

(error) Only -width or -height was specified.

**Description**

The options -width and -height can be either both specified or both not specified.

**What Next**

Please specify -width and -height options together.

## DPI-003

(error) Design boundary does not exist.

**Description**

The design does not have any boundary defined. Therefore the -keep_boundary will not work.

**What Next**

Please remove the -keep_boundary option from this command.

**See Also**

- initialize_floorplan

## DPI-004

(error) zero or negative height was specified.

**Description**

The specified height needs to be greater than 0.

**What Next**

Please specify positve height value.

## DPI-005

(error) zero or negative chip-to-site was specified.

**Description**

The specified chip-to-site needs to be greater than 0.

Feedback

**What Next**

Please specify positve chip-to-site value.

---

## DPI-006

(error) The specified utilization is equal to or less than 0.01.

**Description**

The specified utilization needs to be greater than 0.01.

**What Next**

Please specify a value greater than 0.01.

---

## DPI-007

(error) The specified aspect ratio is less than 0.01.

**Description**

The specified aspect ratio needs to be greater than 0.01.

**What Next**

Please specify a value greater than 0.01.

---

## DPI-008

(warning) the specified core area is too small.

**Description**

The specified core area is too small, may need specify larger area to contain all the cells in the design.

**What Next**

Please specify larger core dimensions.

---

## DPI-009

(error) Unable to create core placement area.

**Description**

The tool is unable to create core placement area with computed boundaries.

Feedback

## DPI-010

(error) Unable to create die area.

**Description**

The tool is unable to create die area with computed boundaries.

## DPI-011

(error) no boundaries defined for cell %s.

**Description**

Boundaries of all cells need to be defined in library.

**What Next**

Please check correspinding .NDM library or LEF/DEF

## DPI-012

(error) the IO %s has been assigned two IO guides.

**Description**

Same IO has been assigned to two different IO guides.

**What Next**

Please remove the IO from one of the IO guide.

## DPI-013

(error) unable to write the IO placement file %s.

**Description**

The IO placement file is not able to be opend for write.

**What Next**

Please check if there is enough disk space for the file. Also, please make sure the sub-directory exists.

## DPI-014

(error) unable to read the IO placement file %s.

Feedback

**Description**

The IO placement file is not able to be opend for read.

**What Next**

Please check if the file is readable.

---

## DPI-015

(error) unable to find the IO guide %s from IO placement file.

**Description**

The IO guide specified in IO placement file does not exist.

**What Next**

Please create the IO guide before specifying it in the IO placement file.

---

## DPI-016

(error) unable to find the IO pad %s from IO placement file.

**Description**

The IO pad specified in IO placement file does not exist.

**What Next**

Please check if the IO pad is in the database.

---

## DPI-017

(error) There is no IO guide created.

**Description**

IO guide and/or ring are required for IO placement.

**What Next**

Please use create_io_guide or create_io_ring to create IO guide and/or IO ring.

---

## DPI-018

(error) unable to find cornerpad %s.

Feedback

**Description**

The specified master does not exist.

**What Next**

Please check the specified master name.

---

## DPI-019

(error) unable to place IO %s in IO guide %s.

**Description**

To perform IO placement successfully, the width of IO guide needs to be larger than the summation of IO width

**What Next**

Please either increase the width of IO guide or remove IO pads from the IO guide.

---

## DPI-020

(error) Invalid core side dimension for the given shape.

**Description**

When creating floorplan by using core side dimension length, the number of core side dimensions must match the given shape. For rectangular, 2 sides must be specified; for L shape, 4 sides must be specified; for T and U shapes, 6 sides must be specified.

**What Next**

Please specify the correct number of side dimensions.

---

## DPI-021

(error) invalid core offset for the given shape.

**Description**

When specifying the offset between core and die, user can specify one core to die offset value which will be applied to all sides of the core shape, or specify a list of offset values which will be applied to individual side of the core shape. However,if user specify multiple offset values, the number of offsets must match the given shape. For rectangular, 4 offset values must be specified; for L shape, 6 offset values must be specified; for T and U shapes, 8 offset values must be specified. If -coincident option is set to *false*, only 4 offset values should be specified.

Feedback

**What Next**

Please specify the correct number of core offset.

---

## DPI-022

(error) the core area is too large to fit in the existing die boundary.

**Description**

This error occurs when user specifies the -keep_boundary option in the *initialize_floorplan* command and the core area is too big to fit into the existing die boundary or the core side offset is too large.

**What Next**

Please modify one or several of the constraints of the core area such as utilization ratio, shape, dimension and core side offsets.

---

## DPI-023

(error) invalid %s shape definition.

**Description**

The %s shape based on the specified dimension is not a valid shape. It either has intersecting points or not a regular polygon.

**What Next**

Please check the specified dimension.

---

## DPI-024

(error) zero or negative dimension is specified.

**Description**

The specified dimension needs to be greater than 0.

**What Next**

Please specify positve dimension value.

---

## DPI-025

(error) negative core-to-die offset is specified.

Feedback

**Description**

The specified core-to-die offset needs to be no less than 0.

**What Next**

Please specify positve offset value.

---

## DPI-026

(error) cannot find site def.

**Description**

The site def information defined in technology file is required to create floorplan.

**What Next**

Please update the technology file.

---

## DPI-027

(error) the row core ratio %f is smaller than the core utilization %f.

**Description**

There is not enough core area to honor the row-core space. Row core ratio must be greater than or equal to the core utilization.

**What Next**

Please specify a smaller row core ratio or increase the core area.

---

## DPI-028

(error) invalid core-side ratio %f.

**Description**

The core side ratio is too large or too small such that one of the core area side exceeds the maximal allowable dimension.

**What Next**

Please specify a reasonable core side ratio.

---

## DPI-029

(error) unable to choose a tile from technology file.

Feedback

**Description**

There are multiple tiles defined in the technology file. However, user did not specify which tile to use for floorplan creation.

**What Next**

Please specify the -tile option in *initialize_floorplan* command.

---

## DPI-030

(error) Initialize floorplan failed because no technology information is available in the library.

**Description**

The library is missing technology information. This command requires the technology information.

**What Next**

Please check your library.

---

## DPI-031

(error) Pathline loading failed.

**Description**

This command cannot load pathline information. The database is corrupted.

**What Next**

Please erase the existing pathlines and then create your IO placement result using the recommend flow.

---

## DPI-032

(error) Bump assignment cannot be completed due to the lack of accessible bumps.

**Description**

The tool is unable to find enough bumps to assign to IO pad pins. Specifically, bumps assigned to any pad pins will eventually be connected those pins through RDL routes. During IO placement, the routability is checked using pathline routing. If the space between bumps are too small to route rdl pathlines to connect bumps to pad pins, no bump assignment will be performed for the corresponding IO pads.

Feedback

**What Next**

Please check the pathlines in IO placement result to identify the region where pathlines are not rerouted. If possible, users can adjust the bump locations and/or pad locations to reduce congestion. Users can also use matching types to guide bump assignment.

---

## DPI-033

(error) Net %s connecting I/O bumps and pads is connected to cell instance %s of another type.

**Description**

Nets between bumps and pad should not be connected to other on-chip devices such as standard cells. The design netlist is wrong if such nets exist.

**What Next**

Check the design netlist and fix any netlist errors.

---

## DPI-034

(error) Remove bump %s to pad %s assignment due to netlist.

**Description**

User assignment cannot have conflicts with current netlist. If a bump is connected to a pad driver through a net, this bump cannot be assigned to other pads.

**What Next**

Please check the design netlist and fix any netlist errors. The specified constraint will be removed since it cannot be satisfied.

---

## DPI-035

(error) Currently, only one-to-one mapping between pad and bump is supported.

**Description**

Currently, IO placement can only support one-to-one mapping between pad and bump.

**What Next**

Please check the IO placement result and make modification if needed.

Feedback

## DPI-036

(error) bump %s to pad %s assignment will not be considered during io placement.

**Description**

The specified bump to pad assignment will not be honored.

**What Next**

If the assignment is needed, please modified the netlist so that the specified bump and pad are connected by a net, which will not be connected to other bumps or pads.

## DPI-037

(error) pad drivers overflow in IO guide %s.

**Description**

Pads are placed beyond the specified IO guide.

**What Next**

Please reassign pads to the specified IO guide or change the IO guide constraints. Alternatively, you can increase the IO guide length. Make sure that the IO guide is wide enough and then perform io placement again.

## DPI-038

(error) run out of bumps in guide %s.

**Description**

Some pads in the specified IO guides are not connected to any bumps.

**What Next**

Please reassign pads to the specified IO guide or adjust the bumps in the design. Perform io placement again after any adjustment.

## DPI-039

(error) cannot assign bump %s to pad %s since it is connected to pad %s.

**Description**

A bump cannot be connected to multiple pads.

Feedback

**What Next**

Choose a different bump to pad assignment.

---

## DPI-040

(error) cannot assign pad %s to bump %s since it is connected to bump %s.

**Description**

A bump cannot be connected to multiple pads.

**What Next**

Choose a different bump to pad assignment.

---

## DPI-041

(error) conflicted assignments are removed.

**Description**

Bump-to-pad assignments that are conflict to netlists are removed.

**What Next**

Choose a different bump to pad assignment if necessary.

---

## DPI-042

(error) bump %s to pad %s assignment cannot be satisfied according to the pathline routing result.

**Description**

The pathline routing creates a linear order for all bumps of the corresponding IO guide. The signal IO constraints create a linear order for all pads in the same IO guide. The position of the specified bump in the bump order and the position of the specified pad make it impossible to connect the bump and pad without introducing crossovers of rdl routes.

**What Next**

Change the IO signal constraints or bump-pad assignment or both. Note that the bump-pad assignment may be due to the netlist.

Feedback

## DPI-043

(error) bump %s to pad %s assignment and bump %s to pad %s assignment cannot be satisfied at the same time according to the pathline routing result.

### Description

The pathline routing creates a linear order for all bumps of the corresponding IO guide. The signal IO constraints create a linear order for all pads in the same IO guide. The order of the two specified bumps and the order of the two specified pads are different. It is impossible to connect the bumps and pads, respectively, without introducing crossovers of rdl routes.

### What Next

Change the IO signal constraints or bump-pad assignment or both. Note that the bump-pad assignment may be due to the netlist.

## DPI-044

(error) pad %s and pad %s violate the power spacing constraint of lib cell %s in IO guide %s.

### Description

The specified power spacing constraint is violated.

### What Next

Try to use ratio based constraints instead of spacing based constraints if possible. Try to change the existing signal IO constraints and/or bump-pad assignments.

## DPI-045

(error) pad %s and pad %s violate the power ratio constraint of lib cell %s in IO guide %s.

### Description

The specified power ratio constraint is violated.

### What Next

Try to change the existing signal IO constraints and/or bump-pad assignments.

## DPI-046

(error) power offset constraint of lib cell %s is violated in IO guide %s.

Feedback

**Description**

The specified power offset constraint is violated.

**What Next**

Try to change the existing signal IO constraints and/or bump-pad assignments. Check if the length of the IO guide can be increased for addition power pads.

---

## DPI-047

(error) net %s connect multiple bumps or multiple pads.

**Description**

Bump-pad assignment among multiple bumps and pad is not supported.

**What Next**

Enhancement will be added in later releases.

---

## DPI-048

(error) cannot assign pad cells to IO guides.

**Description**

place_io fails to assign pad cells to IO guides. There are two possible reasons. First, the total length of the selected IO guides is too short to hold all IO pad cells. Second, there are not enough usable bumps.

**What Next**

If the selected guide length is too short, you can increase the IO guide length or select more IO guides. If there are too few usable bumps, add more bumps.

---

## DPI-049

(error) Cannot perform bump and pad assignment for the I/O guides.

**Description**

The tool cannot identify the sets of bumps and pads for all I/O guides.

Feedback

**What Next**

You can try the following methods to resolve this issue:

- Adjust the I/O guide length.
- Select different I/O guides.
- Manually assign pads to I/O guides.
- Modify the number and locations of bumps.

---

## DPI-050

(error) No enough bumps are selected.

**Description**

Users do not specify enough bumps.

**What Next**

Please create additional bumps using create_bump_array command or select additional existing bumps.

---

## DPI-051

(error) Failed to store I/O signal constraints in the database.

**Description**

Errors occured during the storage of I/O signal constraint data. This could happend because the command could not identify the I/O guide of the constraint or the constraint data could be corrupted.

**What Next**

Try removing all existing I/O constraints and resetting them.

---

## DPI-052

(error) Failed to store or change bump assignment data in the database.

**Description**

Errors occurred during the storage of bump assignment data. Check the log file for previous error messages that might indicate why this occurred. This issue could also occur due to a corrupted design database.

Feedback

**What Next**

Try removing all existing bump assignments and resetting them.

---

## DPI-053

(error) Errors found on existing bumps.

**Description**

The tool found data errors on existing bumps. The design database might be corrupted.

**What Next**

Reload the design data.

---

## DPI-054

(error) Errors found during loading of design data

**Description**

The *place_io* command failed because it could not load design data, including bumps, pads, and I/O guides.

**What Next**

Make sure that the design data is loaded correctly.

---

## DPI-055

(warning) lines of IO guide %s and IO guide %s crossover.

**Description**

The lines of the specifies two IO guides cross over each other.

**What Next**

Try to move IO guides to avoid IO guide crossovers. Otherwise, pad cells within IO guide might overlap.

---

## DPI-056

(error) Error during netlist modification: %s.

**Description**

place_io failed to change the netlist according to io placement result.

Feedback

**What Next**

This is an obsolete message.

---

## DPI-057

(error) Anchor bump is missing in bump array %s.

**Description**

The lower left bump is the anchor bump of a given bump array. The anchor bump cannot be found. It is impossible for place_io to access bump array location.

**What Next**

The bumps in a bump array cannot be removed individually. Please recreate the bump array by removing the incomplete array and create a new array.

---

## DPI-058

(error) pad assignment and constraint conflict discovered.

**Description**

A pad is assigned to an IO guide according to its signal io constraints but to another guide according to pad assignment. Or a pad given in signal IO constraints cannot be found in the design.

**What Next**

Change either the signal io constraints or pad-guide assignment.

---

## DPI-059

(error) pad %s in signal constraint cannot be found in the design.

**Description**

A pad cell name is described in the signal io constraint but cannot be found in current design. This cell may be removed accidentally.

**What Next**

If the pad is deleted by mistake, load the correct netlist. Otherwise, change the signal io constraint before performing io placement.

Feedback

## DPI-060

(error) pad %s in IO guide %s is constrainted in IO guide %s.

**Description**

A pad is assigned to an IO guide according to its signal io constraints but to another guide according to pad assignment.

**What Next**

Change either the signal io constraints or pad-guide assignment.

## DPI-061

(error) pad %s cannot be placed correctly since its bump %s is not selected.

**Description**

place_io find the specified pad and bump are assigned to each other. Although the pad is selected in the current run, the bump is not selected. As a result, the specified pad cannot be placed correctly.

**What Next**

Select the specified bump, e.g., using -bump_array or -all options.

## DPI-062

(error) Not enough bumps: %s.

**Description**

Since pads need to be connected to bumps, there must be enough bumps being selected during IO placement.

**What Next**

Add or select additional bumps. Please check the manpage of place_io.

## DPI-063

(error) Design data error is detected during initial I/O guide and bump loading.

Feedback

**Description**

The *place_io* command must load the design data before performing I/O placement. This error occurs when the command finds errors during the data loading and cannot continue with I/O placement.

**What Next**

Check the log file for error and warning messages that occur before this message. Update the design data to correct any errors and reload the design data.

---

## DPI-064

(error) overlapping bumps exist.

**Description**

IO placement cannot be performed due to bump overlaps.

**What Next**

Search for earlier messages to find out which bumps overlap. Remove the bump overlap by moving or deleting bumps.

---

## DPI-065

(error) Internal error: %s.

**Description**

The *place_io* command encountered unexpected data.

---

## DPI-066

(error) fail to create new power pad for IO guide %s.

**Description**

place_io cannot insert new power pad for the given IO guide to meet the power io constraints. There are several potential reasons. The command may not know how to connect the new pad to the rest of the netlist. The IO guide may be too short to hold the new pad. There may not be enough bumps for the IO guide.

**What Next**

Look for additional error messages for the reason of this error and then fix it. You can also change the power io constraints.

Feedback

## DPI-067

(error) pathline crossing detected.

**Description**

The command found pathline crossings.

**What Next**

This is an internal error.

## DPI-068

(error) cannot route bump %s.

**Description**

Due to congestion, place_io cannot route pathlines from the specified bump to its pad cell.

**What Next**

Adjust the bump location distribution. If the bump is assigned to its pad by users, consider change the bump assignment.

## DPI-069

(error) bump assignment cannot be satisfied due to missing bump %s.

**Description**

The specified bump is not assigned to the IO guide which its pad is assigned to. As a result, the pad may not be placed appropriately.

**What Next**

Adjust the bump location, bump assignment, or the pad assignment.

## DPI-070

(error) bump %s assignment cannot be satisfied due to missing pad.

**Description**

The bump-pad assignment of the specified bump cannot be satisfied because the corresponding pad cell is not assigned to the I/O guide of the bump.

Feedback

**What Next**

Increase the I/O guide length.

---

## DPI-071

(error) bump %s assignment cannot be satisfied due to mismatching pads.

**Description**

The bump-pad assignment of the specified bump cannot be satisfied because different pads are assigned to the bump.

**What Next**

Check the bump assignment and fix any incorrect assignments. Currently, only one-to-one mapping is allowed between bump and pad.

---

## DPI-073

(error) Bump cell %s has more than one port.

**Description**

The tool supports only passive bump cells with a single port connection.

**What Next**

Change the type of bump cells used in the design.

---

## DPI-074

(error) Cannot identify a pin on pad %s to connect to the RDL net.

**Description**

A bump is assigned to the specified pad during bump assignment. However, the tool cannot identify a pin of the pad to be connected to the bump through an RDL net. As a result, the bump is not connected to any pin of the pad.

**What Next**

Check the pad to see if its RDL pins are labeled correctly. You can also create a matching type to select the pin.

---

## DPI-075

(error) bump %s has no connection for rdl nets.

Feedback

**Description**

The specified bump has no ports that can be connected to a rdl net.

**What Next**

Please change the bump cell.

---

## DPI-076

(error) cannot revert netlist modification on cell %s.

**Description**

The netlist is modified by a previous I/O placement command. The modification is not reverted but is used in the current I/O placement run. The modification is related to the specified cell instance.

---

## DPI-077

(warning) bump %s and bump %s have different dimensions.

**Description**

The specified bumps have different bounding boxes.

**What Next**

This message is issued once even if there are many bumps that have different dimensions. The reasons for bumps to have different bounding boxes can vary. Sometimes, bumps are of different reference cells. Sometimes bumps might have different orientations. Please check all bumps and make sure that they are correctly placed.

When bumps have different dimensions, place_io will combine various bounding boxes to create a single bounding box and use it during IO placement. Specifically, the width and height of the generated bounding box are equal to the largest width and height of all bump bounding boxes, respectively.

If the bump dimension variation is small, no additional action is needed. Otherwise, users can improve the IO placement result by running place_io multiple times. Specifically, if chip design area can be partitioned into areas that contain bumps of the same/similar sizes, users can perform IO placement one area at a time. Such an approach requires users to partition pad cells accordingly, however.

---

## DPI-078

(error) confusing netlist modification information ignored: %s.

Feedback

**Description**

The specified netlist modification cannot be reversed.

**What Next**

This may be caused by netlist modification after place_io. Please check the netlist for correctness.

---

## DPI-079

(error) Cannot store netlist modification information.

**Description**

The tool stores the netlist modifications performed during I/O placement so that subsequent I/O placement runs can revert the changes. However, some modifications cannot be stored during the current I/O placement run.

**What Next**

You can ignore this error if you do not plan run I/O placement multiple times.

**See Also**

- place_io

---

## DPI-080

(error) IO guides cannot hold pads %s.

**Description**

IO guides must be long enough to hold all IO pads assigned plus any pads and/or spacing specified by IO constraints.

**What Next**

Please change IO guide dimensions, pads assigned to guides, and/or IO guide constraints. Users can also check the orientations of pad reference cells. The orientation of reference cell should be the one as if the pad cell is placed along a bottom IO guide. If the reference cell orientation is not, users should set the reference_orientation attribute of the ref_block of the pad. The value is that, when the reference cell is placed with such an orientation, the cell is in the correct orientation placed along a bottom guide.

---

## DPI-081

(error) The option honor_pad_limit supports only rectangle shapes, not L, T, or U shape.

Feedback

**Description**

The option honor_pad_limit supports only rectangle shapes, not L, T, or U shape.

**What Next**

Please specify rectangle shape using -shape R, or remove the option -honor_pad_limit.

---

## DPI-082

(error) Cannot load matching types.

**Description**

IO placement cannot load matching types.

**What Next**

Please check all matching type and reset data if needed.

---

## DPI-083

(error) found IO pad overlap.

**Description**

IO placement finds pad overlap among pad cells. It might be due to wrong user constraints. For example, users might set fixed locations for two pads that overlap each other. Another reason of pad overlap is that some pads do not have lib-cell origins equal to (0,0). These pads are placed at locations shifted by their lib-cell origins, resulting in pad overlap.

**What Next**

Please correct or remove pad constraints. Please check the place_io messages for pad lib-cells with origins not equal to (0, 0). Change the origins to (0, 0) or insert space to avoid pad overlap.

---

## DPI-084

(error) The total width of pads selected is larger than the space of IO guides selected.

**Description**

Since the IO pads are placed within IO guides, the total available space within the selected IO guides must be equal to or larger than the total pads to be placed. Note that IO guide space is needed for pads in the design but not assigned to IO guides yet.

Feedback

**What Next**

Please select additional IO guides or enlarge guide dimensions. Alternatively, users can select few pads.

---

## DPI-085

(error) Break cell location cannot be determined by the two guides specified.

**Description**

This message is issued when users specify two IO guides when inserting break cells. However, the two IO guides are not sufficient for the command to derive the location of insertion.

The location of the break cells is the crossing point of extended lines of IO guides when the two guides are perpendicular to each other. When the two IO guides are in parallel, the two guide must have the same orientation and their extended lines must overlap. In addition, there must exist a gap between the two IO guides. The break cells will be placed at the end of the first guide if the first guide is in front of the second guide. If the first guide is after the second guide, the break cells are placed before the first guide. When none of the aforementioned conditions is satisfied, the break cell locations cannot be derived.

**What Next**

Please verify that the two IO guides specified are correct.

---

## DPI-086

(error) Fail to create a new break cell instance.

**Description**

The new break cell instance cannot be created.

**What Next**

Please check the reference cell name of the break cell.

---

## DPI-087

(error) The order of fixed pad %s and %s is conflicting with the IO signal constraints.

**Description**

The order of the specified pads in signal IO constraints does not match that derived from their locations. Since the pads have fixed locations, the order constraint cannot be satisfied.

Feedback

**What Next**

Please remove or change either order constraints or fixed location constraints or both.

---

## DPI-088

(error) The order constraint of IO guide %s cannot be satisfied due to fixed pads.

**Description**

The order constraints and fixed locations constraints of the specified IO guide cannot be honored at the same time.

**What Next**

Please remove or change either order constraints or fixed location constraints or both.

---

## DPI-089

(error) In IO guide %s, filler cells cannot cover the gap at %s with length %s.

**Description**

There still exist gaps after filler cell insertion in the specified IO guide. This can be caused by the fact that the gap is narrower than the smallest filler cell and only non-overlapping filler cells are specified. The starting coordinate and length of the gap are given. The gap extending direction is clock-wise, which is the same as that of the IO guide.

**What Next**

Please use overlapping filler cells. Alternatively, users can modify the pad placement to change the widths of gaps for filler cells. Users should check the orientation of filler cells and/or lib-cells. If the filler cell orientation is not set correctly, filler cells will not be placed properly, which can result in unfilled gaps.

---

## DPI-090

(error) IO guide %s overflow information: %s.

**Description**

This message explains why the specified IO guide cannot hold all its pads.

**What Next**

Users can increase IO guide lengths or assign pads to IO guides. Users can also check the orientations of pad reference cells. The orientation of reference cell should be the one as if the pad cell is placed along a bottom IO guide. If the reference cell orientation is not,

Feedback

users should set the reference_orientation attribute of the ref_block of the pad. The value is that, when the reference cell is placed with such an orientation, the cell is in the correct orientation placed along a bottom guide.

## DPI-091

(error) cannot create core area.

### Description

The core area cannot be created.

## DPI-092

(error) IO pad %s cannot fit in any IO guide.

### Description

There is no enough room in any IO guide to hold the specified pad. Sometimes, it is not enough that the total length of IO guides is equal to or a little longer than the total length of IO pads because a pad must be placed entirely within a single guide. For example, two IO guides of 10-unit long is not enough for a pad of 15-unit long. Note that only one pad name is reported. There might be other pads. When this error occurs, place_io command will still assign this pad to an IO guide and place the pad after the endpoint of the IO guide. The overflow amount is minimized for all IO guides.

### What Next

Users can increase IO guide lengths. If users create IO guides using the create_io_ring command, a larger IO ring is needed. To increase the ring dimension, users might need to re-run the initialize_floorplan command. It is recommended that the -honor_pad_limit option is used with the initialize_floorplan command if the die area is limited by IO pads.

## DPI-093

(error) Cannot perform pad assignment for the I/O guides.

### Description

The tool cannot identify the sets of pads for all I/O guides. For most cases, this error is caused by the fact that there is no enough IO guide space for all IO pads. Specifically, pads must be placed in IO guides. If guides are not long enough, pads cannot be placed properly. Instead, pads are placed outside of IO guides. Users can check each IO guide to find out which one contains more pads than its capacity.

Feedback

**What Next**

You can try the following methods to resolve this issue:

- Adjust the I/O guide length.
- Select different I/O guides.
- Manually assign pads to I/O guides.

One common case that results in this error is that the chip area is too small for all IO pads to be placed in a single IO ring. To resolve the problem, users can run the initialize_floorplan command with the option -honor_pad_limit. The chip area will be adjusted so that a single IO ring can hold all IO pads in the design.

---

## DPI-094

(error) Cells %s and %s overlap.

**Description**

The tool has identified a pair of overlapping cells.

**What Next**

Legalize the cell placement, if needed.

---

## DPI-095

(error) Pad %s cannot be placed in IO guide %s.

**Description**

Users assign the specified pad to the specified IO guide through pad assignment file. However, the guide is not large enough to hold the pad.

**What Next**

Remove other pads assigned in the guide or modify the pad assignment file. In addition, users can change the dimension of the IO guide.

---

## DPI-096

(error) power constraints cannot be satisfied.

Feedback

**Description**

Power constraints specified by users cannot be satisfied. Specifically, power pads might be added to the netlist in order to satisfy the power constraints. However, there is no enough IO guide capacity for the additional power pads.

**What Next**

Please modify power constraints or increase the IO guide length.

---

## DPI-097

(Error) The width '%f' of site def '%s' doesn't match the litho-grid-pitch '%f'.

**Description**

This message was issued because the width of the Site def to be used is not multiple integer of litho-grid-pitch.

**What Next**

Change the tile-definition in technology-file, so that it matches the min-grid(litho-grid).

---

## DPI-098

(Error) The height '%f' of site def '%s' doesn't match the litho-grid-pitch '%f'.

**Description**

This message was issued because the height of the Site def to be used is not multiple integer of litho-grid-pitch.

**What Next**

Change the tile definition in technology-file, so that it matches the min-grid(litho-grid).

---

## DPI-099

(Warning) The length '%f' of '%s' is not an integral multiple of the litho-grid-pitch '%f'.

**Description**

This message was issued because the length is not an integral multiple of litho-grid's pitch.

**What Next**

May need to change the length, so that it should be integral multiple of litho-grid-pitch.

Feedback

## DPI-100

(Warning) The block-boundary's coordinate '(%f, %f)' does not align with the FinFET grid.

**Description**

This message was issued because the user uses "initialize_floorplan -keep_boundary" to derive the block-boundary whose coordinates don't align with the FinFET grid.

**What Next**

Use "initialize_floorplan" without "-keep_boundary" specified.

## DPI-101

(Warning) The block-boundary's coordinate '(%f, %f)' does not align with the litho-grid.

**Description**

This message was issued because the user uses "initialize_floorplan -keep_boundary" to derive the block-boundary whose coordinates don't align with the litho-grid.

**What Next**

Use "initialize_floorplan" without "-keep_boundary" specified.

## DPI-102

(Warning) The origin '(%f, %f)' of physical-block %s does not align with the FinFET grid.

**Description**

This message was issued because the user uses "initialize_floorplan -keep_block_placement" to keep the placement of physical-blocks during floorplan process. However, the origin of the physical-block does not align with the FinFET grid

**What Next**

Redo "shape_blocks" to place the physical-blocks on FinFET grid.

## DPI-103

(Warning) The origin '(%f, %f)' of physical-block %s does not align with the litho-grid.

Feedback

**Description**

This message was issued because the user uses "initialize_floorplan -keep_block_placement" to keep the placement of physical-blocks during floorplan process. However, the origin of the physical-block does not align with the litho-grid

**What Next**

Redo "shape_blocks" to place the physical-blocks on litho-grid.

---

## DPI-104

(Warning) The origin '(%f, %f)' of macro %s does not align with the FinFET grid.

**Description**

This message was issued because the user uses "initialize_floorplan -keep_macro_placement" to keep the placement of macros during floorplan process. However, the origin of the macro does not align with the FinFET grid

**What Next**

Redo "create_placement -floorplan" to place the macros on FinFET grid.

---

## DPI-105

(Warning) The origin '(%f, %f)' of macro %s does not align with the litho-grid.

**Description**

This message was issued because the user uses "initialize_floorplan -keep_macro_placement" to keep the placement of macros during floorplan process. However, the origin of the macro does not align with the litho-grid

**What Next**

Redo "create_placement -floorplan" to place the macros on litho-grid.

---

## DPI-106

(Warning) The origin '(%f, %f)' of pad %s does not align with the FinFET grid.

**Description**

This message was issued because the user uses "initialize_floorplan -keep_io_placement" to keep the placement of io-pad during floorplan process. However, the origin of the pad does not align with the FinFET grid

Feedback

**What Next**

Redo "place_io" to place the pads on FinFET grid.

---

## DPI-107

(Warning) The origin '(%f, %f)' of pad %s does not align with the litho-grid.

**Description**

This message was issued because the user uses "initialize_floorplan -keep_io_placement" to keep the placement of io-pad during floorplan process. However, the origin of the pad does not align with the litho-grid

**What Next**

Redo "place_io" to place the pads on litho-grid.

---

## DPI-108

(error) bump %s is in block %s, not on top.

**Description**

Command place_io can only work if all bumps are on top level.

**What Next**

Try to pop the bump up to the top level.

---

## DPI-109

(error) pad %s is in block %s, not on top, and cannot be placed.

**Description**

Command place_io can only place pads that are on top level.

**What Next**

Try to pop the pad up to the top level or set the pad status to fixed.

---

## DPI-110

(error) unable to get technology data.

**Description**

The tool is not able to get technology data from main library.

Feedback

**What Next**

Please check if the main library contain technology data.

---

## DPI-111

(error) the total cell area is larger than the limit the tool can support.

**Description**

The total cell area is larger than the limit the tool can support.

**What Next**

Please reduce the total cell area.

---

## DPI-112

(error) %s in matching type %s is not the right type.

**Description**

Only pads, bumps, flip-chip drivers, and their pins and terminals can be added to matching types for IO placement.

**What Next**

Remove the object from the matching type.

---

## DPI-113

(error) some pads are placed outside of IO guides.

**Description**

place_io fails to place all pads within IO guides. It often happened when IO guides are very close to each other. As a result, one pad can occupy space in multiple IO guides so that there is less space in guides to place pads.

**What Next**

Please check the in the GUI to find out which pads are placed outside of IO guides based on previous error messages. User can either move IO guide far apart or assign pads to other guides using add_to_io_guide and remove_from_io_guide commands.

---

## DPI-114

(error) pad %s is placed outside of IO guide %s.

Feedback

**Description**

There is no enough room to place the specified pad in its guide. The guide might be selected by user or the tool. The guide is long enough to hold its pads. However, because pad in other guides occupy some space in this guide, there is no sufficient space left. This problem often occurs for designs with multiple IO rings.

**What Next**

User can adjust the locations of IO guides so that they are far away from each other or assign pads to guides using IO constraints or add_to_io_guide command.

---

## DPI-115

(error) pad %s is placed outside of IO guide.

**Description**

The tool is not able to place the specified pad in a guide. This often occurs for designs with multiple IO rings. Although the total length of IO rings is sufficient to hold all pads, because some pads are large and occupy multiple IO guide, the effective IO guide length is not enough.

**What Next**

User can adjust the IO guide locations and/or dimensions. User can also assign pads to IO guides. Contact Synopsys R&D if the problem cannot be resolved.

---

## DPI-116

(warning) Signal pin %s is overlapped with signal pin %s.

**Description**

There are signal pins overlapped after snapping them to wire tracks.

**What Next**

User can adjust signal pins location.

---

## DPI-117

(warning) Signal pin %s color is not same with track color.

**Description**

There is signal pin color is not same with wire track color.

Feedback

**What Next**

User can adjust signal pins location.

---

## DPI-118

(warning) The option %s will be obsolete in future release, please use %s instead.

**Description**

The given option will be obsolete in future release. Please change to use the new option or the new command instead.

**What Next**

Check the command man page to see correct options to use.

---

## DPI-120

(Warning) Command option '-core_rate_ratio' should not be set together with app option 'plan.flow.segment_rule'.

**Description**

Spacing between site rows is not support in the scenario when setting 'plan.flow.segment_rule'.

**What Next**

Check the design and specify the proper command.

---

## DPI-121

(Warning) %s'%s' outside of chip boundary.

**Description**

The chip boundary is no longer encapsulating objects that specified to be kept when specifying -keep_all or -keep_xxx_palcament.

**What Next**

Check the design.

---

## DPI-122

(warning) Invalid bump-padpin assignment.

Feedback

**Description**

The current existing assignment of bumps and padpins is invalid as there are multiple bumps and multiple pins/terminals in this assignment. So this assignment will be skipped when writing out matching types.

**What Next**

Please check the current existing assignment.

---

## DPI-123

(error) reference of cell %s cannot be found.

**Description**

The tool is not able to identify the reference cell of the specified cell instance. It cannot continue due to lack of information.

**What Next**

Please check and make sure no design data is missing. Reload the entire design before running IO placement.

---

## DPI-124

(warning) %s of pad %s will not have bump assignment.

**Description**

There is no available bump for the specified pad pin. These can be due to two reasons. First, the pin is part of a matching type. There are no enough bumps in the matching type for all pins, based on the uniquify number. Second, the pin is connected to a logic net. By default, each bump on the net is assigned to a unique pin on the same net. If there are more pins on the net, some of the pins will not have bump assigned to them.

**What Next**

If the pin belongs to a matching type, users can add more bumps to the matching type or change the uniquify number. If the pin is connected to a net, users can connect more bumps to the net or create a matching type for the pin with uniquify number larger than 1.

---

## DPI-125

(error) current top block is not defined.

Feedback

**Description**

Current top block is not defined. Top block must be defined to be able to load matching types or bump assginments.

**What Next**

Please check if top block is defined.

---

## DPI-126

(Warning) The '%s' of macro %s with value %f does not match litho grid's '%s' with value %f

**Description**

This warning message is reported because the hard macro 's width or height violates the litho grid on its potential allowed orientations.

**What Next**

Redo the macro 's library preparation or technology file setting.

---

## DPI-127

(Warning) The '%s' of macro %s with value %f does not match finfet grid's '%s' with value %f

**Description**

This warning message is reported because the hard macro 's width or height violates the FinFET grid on its potential allowed orientations.

**What Next**

Redo the macro 's library preparation or technology file setting. Also, user can restrict macro 's allowable orientation to avoid violating the FinFET grid.

---

## DPI-128

(warning) signal pad %s cannot satisfy min pitch because spacing is constrained by signal io constraints.

**Description**

The tool is not able to move this signal pad to satisfy min pitch because of the spacing set in signal io constraints.

Feedback

**What Next**

Please check and modify the signal io constraints if min pitch needs to be satisfied.

---

## DPI-129

(warning) IO pad %s cannot be moved to satisfy min pitch because the pad position is fixed.

**Description**

The tool is not able to move this I/O pad to satisfy min pitch because the pad position is fixed.

**What Next**

Please check and modify the status of the pad if min pitch needs to be satisfied.

---

## DPI-130

(warning) bump assignment for bump %s is lost due to netlist change.

**Description**

The bump had connections to pads after place_io. But the assignment is lost due to netlist change.

**What Next**

Please check the modifications on netlist.

---

## DPI-131

(error) Cannot handle some matching types.

**Description**

There are some matching types that cannot be processed.

**What Next**

Please look for matching types previously reported and delete or modify them.

---

## DPI-132

(error) rdl net %s has multiple ports on block %s.

Feedback

**Description**

No feedthrough is allowed on RDL nets. It means that a net can only have at most one port on a block instance.

**What Next**

User should modify the netlist to remove all feedthrough ports.

---

## DPI-133

(warning) cannot place some IO pads in blocks.

**Description**

The tool cannot place pads in blocks when there is no IO guide physically inside of the block instance.

**What Next**

Set status of all pads in blocks as fixed or use -bump_assignment_only option. Or add IO guides in the blocks. Note that IO guides can be created either on top or at block level.

---

## DPI-134

(error) io guide %s crosses the boundary of %s.

**Description**

No IO guide can cross the boundary of physical hierarchy.

**What Next**

Modify the IO guide or block boundary or both.

---

## DPI-135

(error) pad %s has incorrect assignment constraint in pad assignment file.

**Description**

The logic hierarchy of the specified pad is not the same as the physical hierarchy of the IO guide which the pad is assigned to.

**What Next**

Modify the IO guide or pad assignment file.

Feedback

## DPI-136

(error) guide %s in block %s has pad assignment from block %s.

**Description**

The specified guide is in a block but contains pads outside of the block.

**What Next**

Move the IO guide to the location consistent with the pads assigned to it. Or remove pads from the guides. The pads can be part of the signal constraint of the guide.

## DPI-137

(error) guide %s and guide %s match exactly in different MIB instances but have different power constraints.

**Description**

The specified guides are in different MIB instances. They overlap exactly within the reference block. However, they have different power constraints. Such a scenario is not permitted.

**What Next**

Modify the IO guides and/or power constraints.

## DPI-138

(error) guide %s and guide %s across each other in the same reference block but are not equal.

**Description**

The specified guides are either in the same instance or different instances of the same reference block. They are not identical but overlap each other. Such a scenario is not permitted.

**What Next**

Modify the IO guides so that they will not overlap.

## DPI-139

(error) block %s contains IO cells but is not placed.

Feedback

**Description**

All blocks with IO cells must be placed before running IO placement or bump assignment. Note that, if there are multiple physical hierarchy, all blocks containing a block with IO cells must also be placed.

**What Next**

Set the status of the specified block and all blocks containing the block, if any, to placed.

---

## DPI-140

(warning) Cannot satisfy no space constraints of IO guide %s as first pad %s is fixed.

**Description**

Due to the nature of the constraints or the netlist the no space constraints for an IO guide could not be satisfied.

**What Next**

Please verify existing signal constraints or the netlist.

---

## DPI-141

(warning) Cannot satisfy no space constraints of IO guide %s as last pad %s is fixed.

**Description**

Due to the nature of the constraints or the netlist the no space constraints for an IO guide could not be satisfied.

**What Next**

Please verify existing signal constraints or the netlist.

---

## DPI-142

(error) guide %s and guide %s match exactly in different MIB instances but have different signal constraints.

**Description**

The specified guides are in different MIB instances. They overlap exactly within the reference block. However, they have different signal constraints. Such a scenario is not permitted.

Feedback

**What Next**

Modify the IO guides and/or signal constraints.

---

## DPI-143

(error) bumps in bump cluster %s are in different matching types.

**Description**

The bumps in the same bump cluster must be connected together. They are not distinguishable. Therefore, you cannot assign those bumps to difference matching types.

**What Next**

Please remove or modify those matching types. You can have at most one matching type. The matching type can contain at most 1 bump in each bump cluster.

---

## DPI-144

(error) multiple bumps in bump cluster %s are in matching type %s.

**Description**

The bumps in the same bump cluster must be connected together. They are not distinguishable. Therefore, you cannot have two or more bumps described in a matching type.

**What Next**

Please remove extra bumps from the matching type. Adjust uniquify number if needed.

---

## DPI-145

(error) bump %s is in matching type %s and %s.

**Description**

The bump or its pin or its terminal is in more than one matching type. For example, the bump instance is in a matching type. The pin of the same bump is in another matching type. Since a bump has a single pin, the pin and the bump are the same object, which cannot be in two or more matching types.

**What Next**

Please change the matching types. A bump or its pin or its terminal can only be in a single matching type at most.

Feedback

## DPI-146

(error) terminal %s and its %s are in matching type %s and %s, respectively.

**Description**

The terminal is in more than one matching type. Specifically, the owner pad or owner pin of a terminal is in a different matching type from that of the terminal.

**What Next**

Please change the matching types.

## DPI-147

(error) pin %s and its pad are in matching type %s and %s, respectively.

**Description**

The pin is in more than one matching type. Specifically, the owner pad is in a different matching type from that of the pin.

**What Next**

Please change the matching types.

## DPI-148

(error) fail to connect power pad %s according to power constraint.

**Description**

The newly created cell is not connected to existing netlist according to its power constraints.

**What Next**

Please check the power constraint and netlist. Connect the pins of the pad to nets manually if necessary.

## DPI-149

(error) connectivty of power constraint is incorrect since port %s is not on power reference cell %s.

Feedback

**Description**

The pin connectivity information in the database does not match the reference cell. In particular, the connection related to the given pin cannot be honored since the port cannot be found on the reference cell.

**What Next**

Please check the power constraint and design library. Check all the ports of the reference cell. Re-set the power constraints if needed.

---

## DPI-151

(warning) Although '%s' is specified, the %s is still adjusted.

**Description**

The origin block_boundary or specified block_boundary has conflict with floorplan rule or does not snap to the finfet grid. Although '-keep_boundary' or '-control_type die -boundary' is specified, the block_boundary is still adjusted.

**What Next**

Please check if the adjusted block_boundary is expected. If not, please try some of below methods: (1) disable the finfet grid: set_app_options -name finfet.ignore_grid.std_cell -value true (2) remove the floorplan rules: remove_floorplan_rules (3) set following two DCM options: set_early_data_check_policy -check plan.floorplan.core_offset_conflict_enclosure_rules -policy tolerate set_early_data_check_policy -check plan.floorplan.core_area_length_conflict_width_rules -policy tolerate

---

## DPI-152

(Warning) Although '%s' is specified, the block_boundary is still adjusted.

**Description**

If user specify one of options '-side_ratio, -side_length, -shape, -boundary, -core_utilization, -macro_utilization', command will calculate core_area first and derive block_boundary from core_area, then the block_boundary may be changed although '-keep_boundary' is specified.

**What Next**

Please check if the adjusted block_boundary is expected. If not, please try to not specify '-keep_boundary' with any of options '-side_ratio, -side_length, -shape, -boundary, -core_utilization, -macro_utilization'.

Feedback

## DPI-201

(error) Boundary is illegal.

**Description**

The arguments(points) for -boundary can't form a legal boundary.

**What Next**

Please specify the correct values of points for -boundary.

**See Also**

- initialize_floorplan

## DPI-202

(error) Can't create %s with -core_offset values.

**Description**

Can't create %s with -core_offset values, it may self-intersect with settings.

**What Next**

Please specify the correct values of core offset.

**See Also**

- initialize_floorplan

## DPI-203

(error) Can't create core area.

**Description**

Can't create core area with the settings, the value of die boudary side ratio are conflict with the core offset values, these arguments can't form a legal core area.

**What Next**

Please specify the correct values of core offset or die boundary side ratio.

**See Also**

- initialize_floorplan

Feedback

## DPI-204

(warning) this option %s is not valid when you specify the -boundary option.

**Description**

*initialize_floorplan* command only honors the option -boundary when the option %s is used. In other words, the specified boundary is used to create the floorplan.

**What Next**

Please check if you specify the correct options to *initialize_floorplan* command.

## DPI-501

(warning) no boundaries defined for cell %s.

**Description**

Boundaries of all cells need to be defined in library.

**What Next**

Please check correspinding .NDM library or LEF/DEF

## DPI-502

(warning) The length of IO guide %s is less than total assigned pad width.

**Description**

The length of IO guide is not enough for assigned IOs.

**What Next**

Please increase the length of the IO guide, or remove some IOs from the IO guide.

## DPI-503

(warning) this option -core_utilization/-macro_utilization is only valid when you specify the -side_ratio option.

**Description**

*initialize_floorplan* command only honors the option -core_utilization/-macro_utilization when the option -side_ratio is used. Otherwise, the specified core or macro utilization is not used when creating the floorplan.

Feedback

**What Next**

Please check if you specify the correct options to *initialize_floorplan* command.

---

## DPI-504

(warning) only %d core side dimension is needed for %s shape core area.

**Description**

*initialize_floorplan* only need %d core side dimensions to define a %s shape core area.

**What Next**

The rest of core side dimension values in this option will be ignored.

---

## DPI-505

(warning) pad cell %s is placed away from its connection point.

**Description**

The specified pad cell might not be connected to its bump by RDL routes.

**What Next**

Please check the pathline connecting to the specified pad and adjust bump locations or pad constraints.

---

## DPI-506

(warning) Half of minimal spacing is used to create rdl pathlines for IO guide %s.

**Description**

The pathline spacing is equal to half of the default spacing since the default spacing cannot be satisfied.

**What Next**

Dual layer rdl routing might be needed.

---

## DPI-507

(warning) skipping IO guide %s since it contains no driver cells.

**Description**

The specified IO guide contains no IO drivers.

Feedback

**What Next**

Please assign drivers to the IO guide or use -all option. Note the if you use the -all option, place_io command will try to assign drivers based on bump distribution. So it is possible that some IO guides are not used for driver assignment.

---

## DPI-508

(warning) bump %s is far away from its driver.

**Description**

The specified bump is far away from the pad it is assigned to. The corresponding RDL routing might contain violations.

**What Next**

Please consider changing the bump-pad assignment or the pad-guide assignment or both.

---

## DPI-509

(warning) net %s is connected to multiple ports.

**Description**

Multiple top level terminals are connected to a single net. As a result, during IO placement, these ports might be assigned to a single bump.

**What Next**

Request for enhancement if you need to specify the bump for each port. Otherwise, you can ignore this message.

---

## DPI-510

(warning) There exists IO guide bbox overlap.

**Description**

IO guide boxes are defined by the IO guide lines and pad cells placed within IO guides. When IO guide boxes overlap with each other, the pad cells placed within the IO guides may overlap. To avoid pad overlaps, certain regions of IO guides may not be used, which reduce the capacity of IO guides.

**What Next**

Please check IO placement result for pad cell overlaps.

Feedback

## DPI-511

(warning) bump %s cannot be used for any selected IO guide.

**Description**

A bump can only be used for an IO guide if it is placed on the accessible side of the guide. The bump specified cannot be used for any IO guide selected for io placement.

**What Next**

If the given bump must be used, please move the bump to a new location. Otherwise, ignore this warning.

## DPI-512

(warning) bump placed beyond %s bound of IO guide %s.

**Description**

The range of bump location is usually within the length of IO guide. When the bumps are placed beyond the IO guide length, pathlines overlap might occur.

**What Next**

Please check the pathline routing result.

## DPI-513

(warning) pad cell %s has no pins for rdl connection and is treated as filler cells during IO placement.

**Description**

The specified cell is labeled as a pad cell. However, it has no pin so that it cannot be connected to any net. It will be placed as a filler cell.

**What Next**

Please check whether this cell is labeled or designed correctly.

## DPI-514

(warning) due to limited bump count, some pad pins are not connected to bumps.

Feedback

**Description**

During IO placement, pad pins with RDL connection property that are not connected to any bumps are connected to bumps without any net connections. However, since there are no enough bumps available, some or all of those pad pins are not connected to bumps.

**What Next**

Please check the netlist. If pad pins are not labeled correctly, fix the labeling. Otherwise, add additional bumps to the design.

---

## DPI-515

(warning) pin shape %s selected in matching type %s has no BUMP class label.

**Description**

The specified pin shape is selected in a matching type. Therefore, it will be connected to RDL nets. However, this pin shape does not have BUMP class label.

**What Next**

Please verify pin shape names in the matching type. Mark the pin shape to BUMP class if needed.

---

## DPI-516

(warning) The pad pins with BUMP class labels in matching type %s are more than what are expected.

**Description**

The expected number of pad pins with BUMP class labels in a matching type can be derived based on the associated bump count and uniquify number. The actual pin count is more than the expected count. As a result, some pad pins will not be connected to bumps.

**What Next**

Please change the bump count or uniquify number. Alternatively, users can change the pad pin labels.

---

## DPI-517

(warning) Pad pin %s without BUMP class label will be connected to bump %s.

Feedback

**Description**

In General, only pad pins with BUMP class labels will be connected to bumps. The specified pad pin does not have BUMP class label. However, a two pin net connect a bump to this pin. Therefore, the IO placement will assign the bump to the pad pin.

**What Next**

Please set BUMP class label to the pin or remove the two-pin net.

---

## DPI-518

(warning) There are not enough bumps on net %s for pad pins.

**Description**

When users define a uniquify number on a net connecting bumps and pad pins, the number of bumps on the net needs to be sufficient in order to honor the uniquify number.

**What Next**

Please check the net. Users can add more bumps to the net or reduce pad pins with BUMP class label. Users can also set uniquify number for the bumps and pad pins.

---

## DPI-519

(warning) There are no pad pins that will be connected to bumps.

**Description**

To be connected to a bump and RDL interconnect, a pad pin must have BUMP class label or is connected to a bump through a two-pin net. The place_io command cannot find any such pins. No bumps will be connected to any pads.

**What Next**

Please set pad pin labels correctly.

---

## DPI-520

(warning) No pad pins have BUMP class labels.

**Description**

During lib-cell preparation, pads should have their pins labeled as BUMP class so that place_io can connect these pins to RDL nets. However, no pad pins with BUMP class labels are found in the design.

Feedback

**What Next**

Please set pad pin labels correctly.

---

## DPI-521

(warning) The pitch format cannot be used for IO guide %s since its pads are not placed uniformly.

**Description**

Only when the locations of pads are placed with equal spacing, the pitch option can be used. When pitch option cannot be used, the order_only format is used instead. Note that pitch is difference from spacing. Pitch is the distance between the starting points of consecutive pads whereas the spacing is the gap between adjacent pads. To see the offset of the starting point of each pad, use the option fixed.

**What Next**

Please try to use other options or adjust pad locations.

---

## DPI-522

(warning) The spacing format cannot be used for IO guide %s since its pads have overlaps.

**Description**

When pad overlap exists, the spacing value cannot be computed since there is no non-negative gap between pads. The order_only format is used instead.

**What Next**

Please try to use other options or adjust pad locations.

---

## DPI-523

(warning) The design contains pads of type %s which have no RDL connection pins.

**Description**

The pads of the specified type have no pins with BUMP class labels. They might not be connected to bumps.

**What Next**

Please check the pin labeling of the lib-cell.

Feedback

## DPI-524

(warning) Relax spacing rule to generate pathlines.

**Description**

The default spacing rule or selected spacing rule cannot be used to generate pathlines due to congestion. A relaxed rule is used.

**What Next**

Please use RDL router to verify the routability of the IO pad placement.

## DPI-525

(warning) Object %s of matching type %s is ignored during IO placement.

**Description**

The specified object in the matching type is invalid. It might not exist anymore. IO placement will not consider it.

**What Next**

Please check the matching type.

## DPI-526

(warning) some pads are skipped and not written into the constraint file.

**Description**

A pad is only written into the signal IO constraint file when it is placed correctly within an IO guide. If there is no guide in the design or pad cells are not placed in any IO guide, these pads are not included in the signal constraints.

**What Next**

Please check the IO guides in the design or adjust IO pad locations.

## DPI-527

(warning) reference cell %s is treated as a filler.

**Description**

The specified reference cell is labeled as a pad cell. However, it will be treated as a filler cell.

Feedback

**What Next**

Please check whether this reference cell is labeled or designed correctly.

---

## DPI-528

(warning) No pad is written into the constraint file since there is no IO guide in the design.

**Description**

A pad is only written into the signal IO constraint file when it is placed correctly within an IO guide. There is no guide in the design. Therefore, nothing is written into the constraint file.

**What Next**

Please create IO guides.

---

## DPI-529

(warning) No pad is written into the constraint file since pads are not placed in IO guides.

**Description**

A pad is only written into the signal IO constraint file when it is placed correctly within an IO guide. If a pad is not placed with the right edge aligning any IO guide, it will not be reported as part of the signal constraint.

**What Next**

Please modify IO guides or adjust pad locations.

---

## DPI-530

(warning) pad %s is skipped and not written into the constraint file.

**Description**

A pad is only written into the signal IO constraint file when it is placed correctly within an IO guide. This means that the pad must be entirely within the IO guide and in the right orientation. Otherwise, the pad is not included in the signal constraints.

**What Next**

Please check the location and orientation of the pad.

Feedback

## DPI-531

(warning) Non-IO cell %s is detected during IO placement.

**Description**

Command place_io only considers IO pads and do not consider non-IO cells. If non-IO cells exist, they will not be placed and might overlap with IO pads.

**What Next**

Please check if the specified cell is an IO macro. If yes, please label it as IO pad.

## DPI-532

(warning) IO pad %s of reference %s in constraint file will have different orientation if placed by place_io.

.SH DESCRIPTION

The specified IO pad is assigned in an IO guide in the constraint file. However, its current orientation does not match that of the IO guide. If this IO constraint file is used during a place_io run, the pad will be placed in a different orientation.

**What Next**

Please check the specified pad. If users want to keep its orientation, its reference orientation must be changed. Users can also modify the constraint file if the pad should not be assigned to the IO guide.

## DPI-533

(warning) Fixed IO pad %s will not be placed in any IO guide.

**Description**

The specified IO pad is fixed. Therefore, it will not be moved during IO placement. However, it is not currently placed in any IO guides. It will remain at its location, not in any IO guide.

**What Next**

Please check the specified pad. If users want to place it into IO guides, please change its attribute.

## DPI-534

(warning) Fixed IO pad %s will not be moved according to users constraint.

Feedback

**Description**

The specified IO pad is fixed. However, it also has a fixed location constraint from users. The IO placement will ignore the user constraint.

**What Next**

Please check the specified pad. If users want it moved, please remove the fixed attribute.

---

## DPI-535

(warning) Fixed IO pad %s will be reassigned from guide %s to guide %s.

**Description**

The specified IO pad is fixed. The guide which it is assigned to is different from the guide which it is placed in. Therefore, the pad is reassigned to the new guide.

**What Next**

Please check the specified pad. If users do not want it reassigned, please remove the fix location constraint.

---

## DPI-536

(warning) matching type %s is removed since it has conflicts with net %s in the netlist.

**Description**

The user defined matching type cannot have conflicts with netlist connectivity. For example, if a user matching type only contain a fraction of components connected by a net, it is considered incomplete. Users either have to remove components of the net or add the rest of the components of the net. This matching type is ignored during IO placement.

**What Next**

Users need to modify the specified matching type.

---

## DPI-537

(warning) pad cell %s is removed from signal IO constraints of guide %s.

**Description**

The specified pad cell is fixed at a location different from the one in the constraint or outside of the IO guide. Therefore the pad is removed from the signal IO constraint.

Feedback

### What Next

Users need to modify the signal IO constraint or change the fixed status of the cell.

---

## DPI-538

(warning) pad cell %s is not placed in guides.

### Description

The specified pad cell is fixed at a location not aligned properly to its IO guide. Because the pad is close to or overlap with an IO guide, a fraction of the IO guide is reserved for the pad.

### What Next

Users need to check if the pad location is indeed correct.

---

## DPI-539

(warning) pad cell %s is assigned to guide %s, violating pad assignment requirement.

### Description

The specified pad cell has been assigned to an IO guide already, either by explicit pad assignment or signal IO guide constraints. This guide is not one of the guides specified in the pad assignment file. The pad assignment file will not be honored.

### What Next

Users need to check if the pad assignment is acceptable. If yes, modify the pad assignment file. If not, remove the pad assignment or modify signal IO constraints.

---

## DPI-540

(warning) pad cell %s is not connected to any bumps.

### Description

During IO placement, new power pads might be added to the netlist to satisfy power IO constraints. There might not be enough bumps for these power pads. As a result, these power pads are not connected to any bumps.

### What Next

Please verify existing power constraints. If powers pads are needed, please rerun place_io. Use matching types to specify bumps for those pads if needed.

Feedback

## DPI-541

(warning) %s of cell %s is not multiple of %s

**Description**

The dimension of an IO pad cell is not multiple of grid size. As a result, pads will not be snapped to grid points. Note that both width and height of any pad must be multiple of vertical and horizontal grid size since pads might be rotated when placed.

**What Next**

Please check grid definition or change IO pad lib-cells.

## DPI-542

(warning) finfet grid will not be honored during IO placement.

**Description**

The finfet grid cannot be honored during IO placement since some IO cells do not have proper dimensions. The litho grid will be used instead.

**What Next**

Please check finfet grid settings or change IO pad lib-cells.

## DPI-543

(warning) guide %s is skipped since it is entirely inside of guide %s.

**Description**

If an IO guide is completely inside of another IO guide, placing pads in both guides will lead to pad overlap. As a result, the smaller guide is ignored by place_io.

**What Next**

Please check all existing IO guides and remove wrong IO guides as needed.

## DPI-544

(warning) matching type %s is modified since it has conflicts with net %s in the netlist.

**Description**

when a net connects a pad pin in a matching type and a bump outside matching type, this net is said to have a conflict with the matching type. When a conflict exist, the matching

Feedback

type is modified. Specifically, the pin inside the matching type is removed from the matching type. Similarly, when a net connects a bump inside a matching type and a pad pin outside matching type, the bump is also removed from the matching type.

**What Next**

Users need to modify the specified matching type or the netlist.

---

## DPI-545

(warning) pad %s is in block %s, not on top.

**Description**

Command place_io cannot perform netlist changes on pads within blocks. Users need to connect pins of such pads to bump separately, if necessary.

**What Next**

If no netlist modification is needed, you can ignore this warning. Otherwise, try to pop the pad up to the top level or make the netlist change before running place_io.

---

## DPI-546

(warning) %s in matching type %s cannot be skipped since it is connected to a bump.

**Description**

Normally, if a matching type does not contain any bump, the RDL pins in the matching type will not be assigned to bumps. However, if a pin has already been connected to a bump through a two-pin net in the netlist, such a connection will remain as is.

**What Next**

If the connection is correct, user should remove the pin from the matching type. Otherwise, disconnect the pin from the net.

---

## DPI-547

(warning) bump %s in matching type %s cannot be skipped since it is connected to an IO pad pin.

**Description**

Normally, if a matching type only contains bumps, none of the bumps will be assigned to IO pads. However, if a bump has already been connected to an IO pad pin through a two-pin net in the netlist, such a connection will remain as is.

Feedback

### What Next

If the connection is correct, user should remove the bump from the matching type. Otherwise, disconnect the bump from the net.

---

## DPI-548

(warning) matching type %s is modified with components connected to net %s removed.

### Description

The user defined matching type contains components, e.g., bumps or IO pad pins, already being connected to the specified net. Since the net connectivity will be honores separately, such components are removed from the user defined matching type. IO placement will honor the components remaining in the matching type, if anything exist.

### What Next

Users need to modify the specified matching type. In particular, it is recommended to create a matching type for components that are connected by a single net, not multiple nets.

---

## DPI-549

(warning) %s is removed from matching type %s since it is not labeled as a valid IO component.

### Description

Only cells whose design types are pad or flip_chip_driver will be processed as IO cells. If a matching type contains non-IO cell instances, pins of non-IO cell instances, and/or terminals of non-IO cell instances, all such components will be removed from the matching type. Moreover, even a cell is a pad or flip_chip_driver, it is still considered an invalid IO component if it does not have any RDL pin, because no bump assignment is possible for a pad without a single RDL pin.

When a component is removed from a matching type, this matching type will not be honored fully. Only the remaining components, if any, will be honored.

### What Next

Users need to modify the specified matching type or the design_type of cell instances. If the cell is excluded because it does not have a RDL pin, users need to set the bump class labels of terminals of the cell. Alternatively, users can explicitly add pins or terminals into the matching type.

Feedback

## DPI-550

(error) matching type %s does not contain enough bumps.

### Description

A matching type should contain a minimal number of bumps, which is derived by the total number of pad pins and the uniquify number. If there are not enough bumps, the matching type cannot be honored fully.

### What Next

Users need to modify the specified matching type. For example, more bumps can be added to the matching type. Users can also change the number of pad pins to be assigned to a single bump by changing the uniquify number of the matching type. For instance, when the uniquify number of the matching type is set to 2, each bump in the matching type can be assigned to 2 pad pins at most. Note that the uniquify number is applied to all bumps in the matching type. If users only want to assign multiple pad pins to a specific bump in the matching type, they need to create a separate matching type only for that bump with the corresponding uniquify number.

## DPI-551

(warning) pad %s will be placed since it is assigned to guide %s.

### Description

Normally, if user specifies pad-only matching types in a non-flip-chip design, place_io will only place pads in the matching types. However, when a pad has already be assigned to a user selected guide or part of the signal IO constraint of a user selected guide, it will be placed even it is not in any matching type.

### What Next

User needs to modify the matching types, signal IO constraints, or pad-to-guide assignment. Specifically, if the pad needs to be placed, add it to a matching type. If the pad does not need to be placed, remove it from the corresponding IO guide or IO guide constraint.

## DPI-552

(warning) pad %s will not be placed since it is assigned to guide %s, which is not selected.

Feedback

**Description**

In this IO placement run, user selects a subset of IO guides in the command line. All pads that are assigned to IO guides not in the set will not be placed even they are in the selected matching types.

**What Next**

User needs to modify the matching types, signal IO constraints, or pad-to-guide assignment. Specifically, if the pad needs to be placed, add its IO guide to the guide list in the command line or change its guide assignment. If the pad does not need to be placed, remove it from the corresponding matching type.

---

## DPI-553

(error) unplaced bumps found.

**Description**

Bumps must be placed before running place_io. The unplaced bumps will ignored and not be used for bump assignment.

**What Next**

User must places bumps first. If bumps are at the correct locations, change bump status attributes from unplaced to correct values such as placed or fixed.

---

## DPI-554

(warning) Object %s is unplaced and will be ignored for overlap checking.

**Description**

The specified object has not been placed yet. It will not be considered for overlap checking.

**What Next**

Please place the specified object.

---

## DPI-555

(error) Object %s is not assigned to any IO pad pin.

**Description**

The specified object is not assigned to any pad pin.

Feedback

**What Next**

Please assign the specified object to an IO pad pin.

---

## DPI-556

(warning) pad %s will not be processed since it is in matching type %s without any bump.

**Description**

If a pad is in a matching type without any bumps, the pad will not be processed. No bump will be assigned to any RDL pin of this pad. This pad will not be moved.

**What Next**

Modify matching types if the pad needs to be processed during IO design.

---

## DPI-557

(warning) pad %s is fixed at a position outside of die area.

**Description**

The pad is fixed. It is not entirely within the die area.

**What Next**

Adjust the pad location or die area.

---

## DPI-558

(warning) Object %s is not an IO pad.

**Description**

The object specified for unplaced checking is not an IO pad.

**What Next**

Specify IO pads for unplaced checking..

---

## DPI-559

(warning) Cell %s will not be processed because %s.

**Description**

The cell will be ignored during IO design procedure.

Feedback

**What Next**

Check the reason and modify the design if needed.

---

## DPI-560

(warning) Matching type %s is not processed because it is not chosen.

**Description**

The matching type is not selected by the user. As a result, none of its components will be processed during IO placement.

**What Next**

If users want to process the matching type, included it in the command line. Do not use -matching_types option if all matching types need to be processed.

---

## DPI-561

(warning) %d cells of IO reference cell %s, for example %s, have %s not being multiple of %s

**Description**

The dimension of one or more IO pad cells is not multiple of grid size. As a result, pads will not be snapped to grid points. Note that both width and height of any pad must be multiple of vertical and horizontal grid size since pads might be rotated when placed. Users can ignore this warning if pads do not need to be aligned to the grid.

**What Next**

Please check grid definition or change IO pad lib-cells.

---

## DPI-562

(warning) Deleting filler cell %s which is a part of signal IO constraints on IO guide %s.

**Description**

The user has decided to delete a filler cell which is a part of an existing signal IO constraint.

**What Next**

The user should modify or delete the constraint if he chooses to delete a cell which is a part of a constraint.

Feedback

## DPI-563

(warning) Bump %s is inside a block instance and is ignored.

**Description**

The user has created a bump which is inside a block instance and hence it will not be processed.

**What Next**

The user should delete the particular bump instance from inside the block.

## DPI-564

(warning) Matching type %s contains component in a block.

**Description**

Currently, only bump assignment support multiple physical hierarchy. The matching type cannot be processed.

**What Next**

The user should delete the matching type. Or the bump_assignment_only option should be used. Please use report_matching_types command to check the matching type.

## DPI-565

(warning) Bumps exist in a block.

**Description**

The tool can only process bumps on top level. Other bumps will be ignored.

**What Next**

To perform bump assignment for bumps in blocks, open the block as the top design.

## DPI-566

(warning) pad pins in blocks are ignored since they are assigned to block-level bumps.

**Description**

The tool will excludes pad pins in blocks that are already connected to block level bumps.

Feedback

**What Next**

Verify the connectivity between block level bumps and block level pad pins.

---

## DPI-567

(warning) matching types in block %s are ignored.

**Description**

The tool will only honor matching types on top level.

**What Next**

Remove matching types in reference blocks.

---

## DPI-568

(warning) io guide %s in block %s is ignored during IO placement.

**Description**

No MPH support is available for IO pad placement. The block level IO guide is ignored.

**What Next**

Remove IO guides in in reference blocks.

---

## DPI-569

(warning) pad %s is not on top level and cannot be placed.

**Description**

No MPH support is available for IO pad placement. Some or all pins of the pad may not have bump assignment.

**What Next**

Place the pad at the right location and set it status to fixed.

---

## DPI-570

(warning) matching type %s is not properly assigned.

**Description**

Matching type does not have enough bumps assigned to meet user requirement.

Feedback

**What Next**

Assign the required bumps to pads or change the matching type specification.

---

## DPI-571

(warning) Newly created corner cell %s overlaps with existing pads and corners.

**Description**

The newly created corner cell overlaps with existing corner and pads in the design.

**What Next**

Move the objects to remove overlap.

---

## DPI-572

(warning) Matching type %s does not have enough bumps to satisfy uniquify number.

**Description**

The matching type has insufficient bumps.

**What Next**

Modify the matching type.

---

## DPI-574

(warning) Matching type %s and matching type %s have conflicting components.

**Description**

These matching types have components which are similar to each other.

**What Next**

Modify the matching types.

---

## DPI-575

(warning) Matching type %s.

**Description**

These matching types have components which are similar to each other.

Feedback

**What Next**

Modify the matching types.

---

## DPI-576

(warning) Matching type %s has bumps in blocks.

**Description**

The matching type has bump(s) inside blocks.

**What Next**

Modify the matching types.

---

## DPI-577

(warning) The %s.

**Description**

The named pad pins are counterparts of each other but belong to matching types with different uniquify number.

**What Next**

Modify the matching types so that these counterparts belong to the same matching type or different matching types with same uniquify number.

---

## DPI-578

(warning) matching type %s will be modified with components connected to net %s removed.

**Description**

The user defined matching type contains components, e.g., bumps or IO pad pins, already being connected to the specified net. Since the net connectivity will be honored separately, such components are removed from the user defined matching type. IO placement will honor the components remaining in the matching type, if anything exist.

**What Next**

Users need to modify the specified matching type. In particular, it is recommended to create a matching type for components that are connected by a single net, not multiple nets.

Feedback

## DPI-579

(warning) matching type %s will be removed since it has conflicts with net %s in the netlist.

**Description**

The user defined matching type cannot have conflicts with netlist connectivity. For example, if a user matching type only contain a fraction of components connected by a net, it is considered incomplete. Users either have to remove components of the net or add the rest of the components of the net. This matching type is ignored during IO placement.

**What Next**

Users need to modify the specified matching type.

## DPI-580

(warning) matching type %s will be modified since it has conflicts with net %s in the netlist.

**Description**

when a net connects a pad pin in a matching type and a bump outside matching type, this net is said to have a conflict with the matching type. When a conflict exist, the matching type is modified. Specifically, the pin inside the matching type is removed from the matching type. Similarly, when a net connects a bump inside a matching type and a pad pin outside matching type, the bump is also removed from the matching type.

**What Next**

Users need to modify the specified matching type or the netlist.

## DPI-581

(error) IO guide %s and IO guide %s are of the same side and overlap with each other.

**Description**

These two guides have the same direction and overlap with each other. Please modify.

**What Next**

Please modify the length of these two guides so that they don't overlap.

## DPI-582

(error) New location of specified hierarchical cell %s does not lie within the parent block instance boundary.

Feedback

**Description**

The new location of the specified hierarchical cell does not lie within the block boundary. Please modify.

**What Next**

Please modify the location so that the new location is within the block boundary.

---

## DPI-583

(warning) Specified cell %s overlaps with multiple hierarchical instances.

**Description**

The new location of the specified hierarchical cell overlaps with multiple hierarchical instances.

**What Next**

Please modify the location in order to avoid overlap.

---

## DPI-584

(warning) New location specified will cause the corner cell to overlap with multiple hierarchical instances.

**Description**

The new location of the specified hierarchical cell overlaps with multiple hierarchical instances.

**What Next**

Please modify the location in order to avoid overlap or use -force.

---

## DPI-585

(Error) Specified lib cell names have different heights.

**Description**

The different specified lib cell names have different heights.

**What Next**

Please modify the selection of lib cell names so that they have the same height.

Feedback

## DPI-586

(Error) Filler cells cannot be inserted on IO guide %s between offset %s and %s as that area overlaps with multiple hierarchical instances.

**Description**

Filler cells cannot be inserted in the specifed area as it overlaps with multiple hierarchical instances.

**What Next**

Please use -force to specify the parent block instance for the filler cells to be inserted.

## DPI-587

(warning) New location of specified hierarchical cell %s does not lie within the parent block instance boundary.

**Description**

The new location of the specified hierarchical cell does not lie within the block boundary. In case of MIBs all MIB instances of the modified cell will be moved correspondingly to the new location.

**What Next**

Please modify the location so that the new location is within the block boundary.

## DPI-588

(warning) New location for break cell module %s overlaps with multiple hierarchical instances and hierarchical instance %s is chosen as the parent instance.

**Description**

New location specifies with multiple instances.

**What Next**

Please modify so that overlap does not occur.

## DPI-589

(warning) pad %s for reference block %s is skipped and not written into the constraint file.

Feedback

**Description**

A pad is only written into the signal IO constraint file when it is placed correctly within an IO guide. This means that the pad must be entirely within the IO guide and in the right orientation. Otherwise, the pad is not included in the signal constraints.

**What Next**

Please check the location and orientation of the pad.

---

## DPI-590

(warning) io guide %s crosses the boundary of %s.

**Description**

IO guide crosses the boundary of physical hierarchy.

**What Next**

Modify the IO guide or block boundary or both.

---

## DPI-591

(error) Block %s and block %s overlap.

**Description**

User defined blocks overlap with each other.

**What Next**

Please modify the blocks to prevent overlap.

---

## DPI-592

(Warning) New cell location overlaps with multiple block instances, but it is still being created as user has specified option -force.

**Description**

New cell location overlaps with multiple block instance but cell is created.

**What Next**

Please check the cell location.

Feedback

## DPI-593

(Error) New cell location does not overlap with the pathname specified by -force.

**Description**

New cell location does not overlap with the pathname specified by -force.

**What Next**

Please check the -force option

## DPI-594

(Error) Filler cells cannot be inserted on IO guide %s between offset %s and %s as that block instance specified does not overlap with that area.

**Description**

Filler cells cannot be inserted in the specifed area as block instance specified by -force does not overlap with that area.

**What Next**

Please use -force to specify the parent block instance among the overlapping block instances in that area for the filler cells to be inserted.

## DPI-595

(warning) Filler cells inserted on IO guide %s between offset %s and %s eventhough it overlaps with multiple block instances as -force is specified.

**Description**

Filler cells inserted in the specifed area as block instance is specified by -force.

**What Next**

Please validate the new area is correct.

## DPI-596

(warning) cell %s is connected to a different net during place_io.

Feedback

**Description**

The netlist is modified. The specified cell is disconnected from its original net and connected to a new net according to bump assignment result. This can occur when two nets are merged to form a single net.

**What Next**

Please check the connectivity of the specified cell. Use LVS if needed.

---

## DPI-597

(error) IO cell %s has different owner from IO guide %s.

**Description**

Specified cell in constraint has different owner compared to IO guide.

**What Next**

Modify the IO constraint.

---

## DPI-598

(error) Fixed pad %s in IO guide %s is constrainted in IO guide %s.

**Description**

A fixed pad is in one IO guide and is constrained to another guide.

**What Next**

Change either the signal io constraints or pad-guide assignment or the status of the pad.

---

## DPI-599

(Warning) Cells %s %s and %s %s overlap.

**Description**

The tool has identified a pair of overlapping cells. These two cells cannot be moved by the tool. Therefore, the overlap cannot be resolved. The reason why both cells cannot be moved can vary. First, a cell might be fixed. Second, a cell's location might be specified by signal IO constraints. Thirdly, users may exclude the cell from placement using a matching type with no bump. Fourthly, the pad may be inside of a block which does not contain any IO guides.

Feedback

**What Next**

Check the cell placement if it meets desired requirements.

---

## DPI-600

(Error) Column %s occurs more than once. Initial occurence was at index %s.

**Description**

A column in CSV file occurs more than once.

**What Next**

Please check the file.

---

## DPI-601

(Error) Compulsary column %s does not exist in CSV file.

**Description**

A column which is compulsary does not exist in file

**What Next**

Please check the file.

---

## DPI-602

(Error) In line number %s of CSV file the number of entries in row (%s) does not match number of columns specified (%s).

**Description**

Number of entries does not match number of columns for specified line number.

**What Next**

Please check the file.

---

## DPI-603

(Error) In line number %s of CSV file distance specified for %s is incorrect.

**Description**

Distance in specified line number is not a number.

Feedback

**What Next**

Please check the file.

---

## DPI-604

(Error) In line number %s of CSV file the pad name for column %s is empty.

**Description**

Pad name in specified line number is empty.

**What Next**

Please check the file.

---

## DPI-605

(Warning) Ignoring line number %s of map file as number of entries in map file %s should be exactly 2.

**Description**

Bad specificaltion for map file in specified line number.

**What Next**

Please check the file.

---

## DPI-606

(Error) In line number %s standard column %s is specified in the first column of map file %s.

**Description**

Incorrect syntax in the map file.

**What Next**

Please check the file.

---

## DPI-607

(Warning) In line number %s specified standard column %s is not recognised in map file %s.

Feedback

**Description**

Incorrect syntax in the map file.

**What Next**

Please check the file.

---

## DPI-608

(Error) Standard column %s named as custom column %s has occured previously. Previous occurence was at %s.

**Description**

A column in CSV file occurs more than once.

**What Next**

Please check the file.

---

## DPI-609

(warning) Specified cell %s on top is moved to a location where it overlaps with hierarchical instance(s).

**Description**

The new location of the specified hierarchical cell overlaps with hierarchical instance(s).

**What Next**

Please check the locaton if it is correct.

---

## DPI-610

(warning) For reference block %s no pad is written in the constraint file as there is no IO guide in the particular reference block.

**Description**

A pad is only written into the signal IO constraint file when it is placed correctly within an IO guide. There is no guide in the reference block. Therefore, nothing is written into the constraint file.

**What Next**

Please create IO guides.

Feedback

## DPI-611

(warning) No pad is written into the constraint file since pads are not placed in IO guides for reference cell %s.

**Description**

A pad is only written into the signal IO constraint file when it is placed correctly within an IO guide. If a pad is not placed with the right edge aligning any IO guide, it will not be reported as part of the signal constraint.

**What Next**

Please modify IO guides or adjust pad locations.

## DPI-612

(warning) No pad is written in the constraint file as there is no IO guide in reference blocks.

**Description**

A pad is only written into the signal IO constraint file when it is placed correctly within an IO guide. There is no guide in the reference blocks. Therefore, nothing is written into the constraint file.

**What Next**

Please create IO guides.

## DPI-613

(warning) layer information is missing.

**Description**

Due to inconsistency of technology file update, some physical objects have unbounded layers.

**What Next**

Please correct technology information in the database. User can run report_unbound to get additional information

## DPI-614

(warning) bump assignment information cannot be stored %s.

Feedback

**Description**

Due to inconsistency of technology file update, some physical objects have unbounded layers. As a result, bump assignment information cannot be stored properly.

**What Next**

Please correct technology information in the database. User can run report_unbound to get additional information

---

## DPI-615

(warning) io guide %s has %d power constraints, which is more than 10.

**Description**

Although ICC 2 does not have limitation on how many power constraints an IO guide can have, it is rare that a guide can have more than 10 power constraints. It is possible that user does not describe the power constraints correctly.

**What Next**

Please use report_power_io_constraints to check whether all constraints are correct. Make any correction if needed.

---

## DPI-616

(warning) In power constraints specified for IO guide %s, for reference cell %s, the spacing value of %s is too less.

**Description**

For the above mentioned power constraint the spacing value is very low and is probably incorrect.

**What Next**

Please use report_power_io_constraints to check whether all constraints are correct. Make any correction if needed.

---

## DPI-617

(warning) No task is specified.

**Description**

The command has no task to perform.

Feedback

**What Next**

Please select *-report* option to report bump placement information. Please select *-min_spacing* option to check bump placement.

---

## DPI-700

(error) invalid Definition of app_option plan.flow.site_def_mapping.

**Description**

The specified app_option plan.flow.site_def_mapping is not valid.

**What Next**

Please specify the valid app_option plan.flow.site_def_mapping.

---

## DPI-701

(warning) definition of app_option plan.flow.site_def_mapping is dependent on -hybrid_site_def option

**Description**

The app_option plan.flow.site_def_mapping is dependent on command option -hybrid_site_def for command initialize_floorplan.

**What Next**

Please specify -hybrid_site_defs option for command initialize_floorplan.

---

## DPI-702

(error) invalid specified -last_row_site_fef_index option.

**Description**

The range of specified command option -last_row_site_fef_index should set between 0 to the last index of hybrid site def list.

**What Next**

Please specify the valid command option -last_row_site_fef_index.

---

## DPI-703

(warning) If -last_row_site_def_index option is defined, vertical parity (basic row account should be even or odd) cannot be guaranteed.

Feedback

**Description**

If -last_row_site_fef_index option is defined, vertical parity (basic row account should be even or odd) cannot be guaranteed.

**What Next**

Please skip setting plan.flow.segment_rule if -last_row_site_fef_index option is defined.

---

## DPI-704

(error) Failed because of the invalid hybrid site_rows creation.

**Description**

Failed for generating hybrid site_rows.

**What Next**

Please check the existing pre-defined site_defs and command option for command initialize_floorplan.

---

## DPI-705

(error) Failed because of the invalid multi-height site_rows creation.

**Description**

Failed for generating multi-height site_rows.

**What Next**

Please check the existing pre-defined site_defs, app_option plan.flow.site_def_mapping and command option for command initialize_floorplan.

---

## DPI-706

(error) invalid specified multiHeight site_def mapping: the sum of mapped site_def height should be equal to specidied multi-height site_def %s.

**Description**

The sum of mapped site_def height should be equal to specidied multi-height site_def.

**What Next**

Please specify the valid app_option plan.flow.site_def_mapping.

Feedback

## DPI-707

(error) invalid specification for overlapping site_def %s, failed for mapping the overlapping site_defs.

**Description**

The specified overlapping site_def cannot be mapped in specified row_pattern.

**What Next**

Please check the validation of row_pattern specification in the Physical Rule File (PRF).

## DPI-708

(error) invalid specified hybrid or multi-height site_def %s.

**Description**

The specified hybrid or multi_height site_def cannot be found in the existing site_defs.

**What Next**

Please check the existing site_defs and the specified app_option plan.flow.site_def_mapping, as well as the -hybrid_site_defs option for command initialize_floorplan.

## DPI-709

(error) the site_width for the site_defs which specified in site_defs under row_pattern should be equal.

**Description**

The specified site_defs under row_pattern should have equal site_def width.

**What Next**

Please check the validation of row_pattern specification in the Physical Rule File (PRF).

## DPI-710

(error) invalid specified multi-height site_def.

**Description**

The specified multi_height site_def is not valid.

Feedback

**What Next**

Please check the existing site_defs and the specified app_option plan.flow.site_def_mapping, as well as the -hybrid_site_defs option for command initialize_floorplan.

---

## DPI-711

(error) when specifying the -site_def_mapping, duplicated multi-height site_def is not allowed.

**Description**

The specified multi-height site_def cannot be duplicated.

**What Next**

Please check the specified app_option plan.flow.site_def_mapping.

---

## DPI-712

(error) multiple site_defs should be specified for -hybrid_site_def option.

**Description**

The number of specified hybrid_site_defs should be larger than one.

**What Next**

Please check the -hybrid_site_defs option for command initialize_floorplan.

---

## DPI-713

(warning) segment_rule for vertical parity (row counts) won't be honored if -hybrid_site_def is specified.

**Description**

plan.flow.segment_rule won't be honored if -hybrid_site_def is specified.

**What Next**

Please unset the valid plan.flow.segment_rule for command initialize_floorplan.

---

## DPI-714

(warning) User specified side_length %s will be automatically tuned to %s due to the requirement of %s.

Feedback

### Description

The user specified side_length will be automatically tuned due to some requirement in the command initialize_floorplan.

### What Next

Please check the user specified side_length option for command initialize_floorplan.

---

## DPI-715

(warning) The core boundary won't be automatically tuned for meeting the requirement of last_row_site_index specified in PRF.

### Description

If the command option "-keep_boundary" and command option "-row_pattern" is specified at same time for command initialize_floorplan, the core boundary won't be automatically tuned for meeting the requirement of last_row_site_index specified in PRF. Neverthess, if the core boundary already satisfied the requirement of last_row_site_index specified in PRF, initialize_floorplan -keep_boundary won't change the boundary.

### What Next

Please use command option -side_length instead.

---

## DPI-801

(error) Can not create %s track.

### Description

The checking is performed for uniform and non_uniform track creation in floorplan generation. Wire track pattern should be specified correctly before initialize_floorplan.

### What Next

Please set correct wire track pattern for the tool to proceed.

### See Also

- initialize_floorplan

---

## DPI-802

(warning) io guide %s is too close to a block $s.

Feedback

**Description**

When pads are placed along such an io guide, they may be partially inside some other blocks.

**What Next**

Please check the pad placement result. Adjust the guide location and or dimension if needed. The location of blocks can also be adjusted to avoid this issue.

---

## DPI-804

(error) Initialize_floorplan can not calculate the core area because both core area and standard cell area are 0.

**Description**

This error message is delivered because both core area and standard cell area are 0 so that initialize_floorplan cannot be conducted.

**What Next**

Please use initialize_floorplan -boundary instead.

---

## DPI-805

(warning) Power pad %s with reference cell %s has different connectivity from existing power constraint.

**Description**

The existing power pad has same reference cell as exisitng power constraint but different connectivity information.

**What Next**

Please check the existing power pads and the power constraints.

---

## DPI-806

(warning) Optional column %s does not exist in CSV file.

**Description**

A column which is optional does not exist in file

**What Next**

Please check the file and add the optional column if it is required.

Feedback

## DPI-807

(error) The %s.

**Description**

The pin of the specified filler cell can be connected to two possible nets.

**What Next**

Modify the netlist connection.

## DPI-808

(warning) Pin %s cannot be connected to a net.

**Description**

The filler pin cannot be connected to a net by abutment.

**What Next**

Please check the abutment in case the filler pin needs to be connected.

## DPI-809

(error) %s.

**Description**

This error message is dispalyed because there was a problem while parsing the provided patterns.

**What Next**

Please use the manpage to correct errors.

## DPI-810

(error) No patterns found for current design.

**Description**

No patterns were found to parse.

**What Next**

Please check the pattern definition file.

Feedback

## DPI-811

(error) Unable to create directory %s.

**Description**

Unable to create specified directory.

**What Next**

Please check if write permission is provided.

## DPI-812

(error) No pad pins in design match pad pins specified in patterns.

**Description**

Unable to find any pad pins in design.

**What Next**

Please check the pin assignment names in the pattern file.

## DPI-813

(error) Error encountered while parsing design data.

**Description**

Unable to parse design data.

**What Next**

Please check the design data.

## DPI-814

(error) Too many objects specified. Exceeds the number of colours available.

**Description**

Please specify less patterns.

**What Next**

Please specify less patterns.

Feedback

## DPI-815

(error) Cannot create images for pattern %s.

**Description**

Unable to create an image for specified pattern.

**What Next**

Please check earlier error messgaes and pattern specification to resolve the issue.

## DPI-816

(error) Cannot parse directory %s.

**Description**

Unable to parse directory specified.

**What Next**

Please check permissions and existence for specified directory.

## DPI-817

(error) Cannot find any training models to process.

**Description**

No training models specified.

**What Next**

Please check if the placeio directory has training models.

## DPI-818

(error) Cannot parse any model checkpoint files in %s.

**Description**

No checkpoint files found.

**What Next**

Please check if the checkpoint files are readable.

Feedback

## DPI-819

(error) Cannot find scaling factor information in %s.

**Description**

No scaling factor was found.

**What Next**

Please check if the file is present.

## DPI-820

(error) Cannot find any matching objects during detection.

**Description**

No matching objects found.

**What Next**

Please check objects present in design match iwth pattern specification file.

## DPI-821

(warning) Cannot find any pattern with name %s in current design.

**Description**

No pattern with specified name found in current design.

**What Next**

Please check pattern specification for the model specified and current design.

## DPI-822

(warning) Cannot find any instance of pattern %s in design.

**Description**

No instance of specified pattern found in the design.

**What Next**

Please check if any particular instance of specified pattern exists in design.

Feedback

## DPI-823

(warning) Cannot process ML data for pattern detection.

**Description**

Cannot process ML data for given design.

**What Next**

Please check log file for errors/warning for hints.

## DPI-824

(warning) The app option 'plan.floorplan.enable_user_core_offset' has been set as true. The actual core_offset will be strictly as users specified. The core area and die boundary will not be snapped to the site_def and grid.

**Description**

If users use the 'plan.floorplan.enable_user_core_offset' to assign the actual core_offset, core area and die boundary will not be snapped to the site_def and grid, which may bring about gaps between them.

**What Next**

Please make sure the app option is what you need.

## DPI-826

(warning) %s.

**Description**

Warnings are found during creating liner cells in current design.

**What Next**

Please check the design related to the warnings.

## DPI-827

(error) %s.

**Description**

Errors are found during creating liner cells in current design.

Feedback

**What Next**

Please check the design related to the errors.

---

## DPI-828

(warning) Not enough usable ports available to create terminals for bump shapes.

**Description**

There are not enough ports available to create terminals for the bump shapes which is necessary for RDL routing.

**What Next**

Please review the netlist and make necessary port connections to be able to perform RDL routing.

---

## DPI-829

(error) Cannot create a terminal for bump shape %s.

**Description**

Unable to create a terminal for specified bump shape.

**What Next**

Please check if the net connected to the bump shape is connected to a port.

---

## DPI-830

(warning) Cannot consider shape %s for place_io.

**Description**

place_io can only process shapes which are polygons or rectangles or trapezoids.

**What Next**

Please check if the shape is a polygon or a rectangle or a trapezoid.

---

## DPI-831

(warning) Conflict: %s, Related Core Offset Number: %u, Solution: %s.

Feedback

**Description**

There are conflicts between two or more floorplan enclosure rules. Conflicts show which floorplan rules have conflicts. Related core offset number shows which core offset conflicts with floorplan rules. Solution shows what the command has implemented to fix the conflict.

**What Next**

Check the floorplan enclosure rules and initializing floorplan options.

---

## DPI-832

(warning) Conflict: %s, Related Core Side Number: %u, Solution: %s.

**Description**

There are conflicts between two or more floorplan width rules. Conflicts show which floorplan rules have conflicts. Related core side number shows which core area side conflicts with floorplan rules. Solution shows what the command has implemented to fix the conflict.

**What Next**

Check the floorplan width rules and initializing floorplan options.

---

## DPI-833

(error) Width rule: %s conflicts with site %s.

**Description**

There are conflicts beetween width rules and site height or width.

**What Next**

Check the floorplan width rules and site width or height setting.

---

## DPI-834

(warning) The repaired result cannot satisfy width rules completely.

**Description**

When users use option '-boundary' to initialize floorplan, the 'initialize_floorplan' command cannot ensure the repaired result satisfy width rules completely.

When users set width rules for block boundary, the 'initialize_floorplan' command cannot ensure the repaired result satisfy width rules completely.

Feedback

**What Next**

Please use the command 'check_floorplan_rules' to check and fix_floorplan_rules to repair again.

---

## DPI-835

(error) There are conflicts between enclosure and width rules in direction %splease check floorplan rules.

**Description**

There are conflicts beetween enclosure and width rules in some directions. The command cannot create core area because of these conflicts.

**What Next**

Please check the step and offset options in enclosure and width rules.

---

## DPI-836

(warning) Cannot find suitable die side%d length value to satisfy floorplan block boundary width rules.

**Description**

Cannot find one suitable die side value to satisfy block boudnary width rules, and correspongding core area and core offset can satisfy std cell width rules and enclosure rules at the same time.

**What Next**

Please use the command 'check_floorplan_rules' to check and fix_floorplan_rules to repair again.

---

## DPI-837

(warning) Core area cannot accommodate the max macro size, please check side length/ ratio setting.

**Description**

When initialize floorplan by 'side_length' or side_ratio' option, the core area cannot accommodate the max macro size. This scenario may cause some bad results in next design planning flow.

Feedback

**What Next**

Please check 'side_length' or side_ratio' option setting.

---

## DPI-838

(error) Boundary points %s are unaligned, please check '-boundary' option setting.

**Description**

When initialize floorplan by '-boundary' option, the die or core area cannot be created to a poly rectangle shape. This scenario may cause some bad results in next design planning flow.

**What Next**

Please check '-boundary' option setting.

---

## DPI-839

(warning) %s is not a valid lib-cell name.

**Description**

The tool cannot open the given lib-cell. It will try to place pads without the given lib_cell. If the lib-cell is linked during IO placement, you can ignore this message.

**What Next**

Please try the full name or correct the name.

---

## DPI-840

(error) Some IO guides miss N5 rule settings.

**Description**

Some IO guides do not have N5 rules. The tool cannot continue.

**What Next**

Please set N5 rules on all IO guides or remove N5 rules from all IO guides.

---

## DPI-841

(error) some pads are not assigned to any IO guide.

Feedback

**Description**

During IO placement with N5 rules, all pads must be assigned to IO guides.

**What Next**

Please assign pads to IO guides using add_to_io_guide command or using IO guide constraints.

---

## DPI-842

(error) guide %s does not have complete order constraints.

**Description**

During IO placement with N5 rules, each IO guide must have an order signal constraint consisting of all pads on the guide.

**What Next**

Please set the signal contraint for the given guide.

---

## DPI-843

(error) cannot derive pad locations for guide %s with N5 rules.

**Description**

Only certain length values can have feasible pad placement solution when N5 rules are honored. The length of the given guide must be modified. Otherwise, no solution is possible.

**What Next**

Please modify guide length, pad-guide assignment, and/or N5 rules.

---

## DPI-844

(error) derived pad locations are incorrect for guide %s with N5 rules.

**Description**

Although tool derives the pad locations for the given guide, the derived values do not pass the internal checking. It is likely caused by incorrect design data.

**What Next**

Please contact tool supporting engineers. Modify guide length, pad-guide assignment, and/or N5 rules if possible.

Feedback

## DPI-845

(error) no possible pad placement solution for guide, please adjust guide length by %s.

**Description**

Only certain length values can have feasible pad placement solution when N5 rules are honored. The length of the given guide must be modified. Otherwise, no solution is possible. The recommended adjustment amount is provided.

**What Next**

Please modify guide length, pad-guide assignment, and/or N5 rules.

## DPI-846

(error) the block boundary cannot be non-rectilinear when -keep_boundary is set.

**Description**

The block boundary should be rectilinear when -keep_boundary is set.

**What Next**

Please specify a rectilinear boundary.

## DPI-847

(error) failed to create the ipc boundary ring.

**Description**

The ipc boundry ring cannot be created.

**What Next**

Please check the values of options of initialize_floorplan command.

# DPP

## DPP-001

(error) Failed to get the physical design.

**Description**

No physical design is available. This design does not have a core area or a die boundary.

Feedback

**What Next**

initialize_floorplan can be used to create the physical design. It is also possible to read it from DEF using read_def.

---

## DPP-002

(error) %s

**Description**

The inputs of the command option did not pass the sanity check.

**What Next**

Please refer to the specific message related to this error and make the changes in the option.

---

## DPP-003

(error) %s

**Description**

Failed to load the design from database.

**What Next**

When loading the design and building its internal data structure, the tool performs a number of sanity checks. Before this error message is reported, it should report the detailed information about what sanity check is failed. Please refer to those specific information to find the reason.

---

## DPP-004

(error) Failed to load the row and site structures.

**Description**

Shaping only supports certain types of row structures. This error message indicates that the input design has the row structure which is not supported by current shape_blocks.

**What Next**

Before reporting this error message, the tool will also report the specific row structure used in the design which is causing the problem. Please create the alternative rows which is supported by shape_blocks.

Feedback

## DPP-005

(error) Failed to get the big enough placeable area for %s.

**Description**

Found placeable cells/blocks to be placed but no sufficient placeable areas.

**What Next**

Please check if there are extra blockages.

## DPP-006

(error) failed to create the initial shape for MIB %s

**Description**

When shaping the design with MIBs, the tool needs to find the legal MIB shape so that it can fit all macros inside the MIB. If this is failed, the tool will stop proceeding.

**What Next**

Check if the design utilization is normal and also if macros inside MIBs have very restrict allowed orientations.

## DPP-007

(error) Incorrect shaping constraints.%s

**Description**

Failed in checking the constraints for shaping.

**What Next**

Please refer to the above checking messages for more information. Also check shape_blocks man page for shaping constraint syntax and examples.

## DPP-008

(error) Failed when loading the routing information. %s

**Description**

Could not get the correct routing information for channel estimation.

Feedback

**What Next**

Please refer to the reporting individual message to help resolve the problem. It is possible that you set the ignore layers incorrectly or perhaps need to set the ignore layer. There needs to be at least 1 routing layer in the horizonal direction and at least 1 routing layer in the vertical direction.

---

## DPP-009

(error) There are unplaced cells in the design.

**Description**

report_placement requires that all cells in the design are placed or fixed. This message indicates that one or more cells are not placed.

**What Next**

Please please check the design to make sure it is properly placed.

---

## DPP-010

(error) Fail to open the specified file %s.

**Description**

Option "-constraint_file file_name" provides the input file name. This error message indicates that the specified input file cannot be opened for reading.

**What Next**

Please check if file name is correct, or if file is in the correct path, or if the file has the read permission.

---

## DPP-011

(error) Minimum required channel size specified in option -min_channel_size is too big.

**Description**

The minumum channel size is larger than half of the minimum(core_width, core_height), which is apparently too big. This normally indicates a wrong input.

**What Next**

Please note that the option unit is in microns. Also please note that the bigger -min_channel_size is specified, the larger each block utilization will be.

Feedback

## DPP-012

(error) Channel size or keepout margin is too big.

**Description**

The channel sizes or keepout margins are utilizing too much of the available placement area, causing the utilization to exceed 100%.

**What Next**

Please use option "set_shaping_options -min_channel_size" to reduce the channel size to get an acceptable utilization. If the channel size has already been minimized but there is still a utilization problem, try reducing the keepout margins.

## DPP-013

(error) Target utilization is too small.

**Description**

The target utilization (shaping_target_utilization) specified in the constraint file is too small, which is making the total utilization over 100%.

**What Next**

Please increase the target utilization so as to get an acceptable utilization or increase the die size.

## DPP-014

(error) Block %s is read only.

**Description**

This command requires that blocks be editable. This is likely because the block will be changed by this command.

**What Next**

Make block editable by open_block -edit. Be careful to make the appropriate block view (e.g., design view or abstract view) editable.

## DPP-015

(error) Edit group %s contains objects in different physical hierarchies.

Feedback

**Description**

This command only supports edit groups in a single physical hierarchy. The command has detect an edit group that violates this rule and so cannot proceed.

**What Next**

Please remove the problematic edit group, or remove the objects that are in different physical hierarchies from the edit group.

---

## DPP-016

(error) Edit group %s contains unsupported placeable objects

**Description**

This command only supports certain types of edit groups. It only supports edit groups with hard macros in it or edit groups that have no placeable objects. This edit group violates this limitation.

**What Next**

Please remove the problematic edit group, or the unsupported objects within the edit group.

---

## DPP-017

(warning) Block instance %s is not on block grid.

**Description**

You get this message because the specified block instance's snap-point location is not on its associated block grid.

**What Next**

Please check the location of the block instance. You can use snap_cells_to_block_grid to move the cell to be on grid.

---

## DPP-018

(warning) The current orientation of block instance %s is not allowed by its associated block grid.

**Description**

You get this message because the specified block instance's orientation is not among the allowed orientations of its associated block grid.

Feedback

**What Next**

Please check the orientation of the block instance. Either correct the cell's orientation, or change the block grid.

---

## DPP-019

(error) Edit group %s is not supported.

**Description**

This command only supports edit groups with hard macros in the same physical hierarchy, voltage area, and/or move bound. Any edit groups containing cell instances which violates this limitation, are not supported. For example an edit group with a hard macro in movebound M1 and also a hard macro in movebound M2 is not supported. Another example is an edit group with a hard macro and an IO cell.

**What Next**

Remove the edit groups which are not supported, or remove the unsupported objects from the edit group.

---

## DPP-020

(error) Macro placement failed.

**Description**

Macro placement failed to run. There are a number of reasons why this may happen. The most common is that utilization of the design or a block in the design is over 100 percent. This could happen if the sliver size is set too high or a placement blockage is covering up a large area of the design.

**What Next**

Check any previous messages for reason for failure and fix the issue.

---

## DPP-021

(error) Placement for standard cells failed.

**Description**

Standard cell placement failed to run. There are a number of reasons why this may happen. The most common is that utilization of the design or a block in the design is over 100 percent.

Feedback

**What Next**

Check any previous messages for reason for failure and fix the issue.

---

## DPP-022

(warning) Block instance %s is out of the core area of design block %s.

**Description**

Block instance is out of the core area of design block.

**What Next**

Check block placement.

---

## DPP-023

(error) Block %s overlaps with %s.

**Description**

The two specified objects are placed overlapping each other. This command requires that there is no overlap to start with.

**What Next**

Check the design to make sure they are properly placed without overlap.

---

## DPP-024

(error) Block %s has no site rows.

**Description**

The specified block has no site rows and thus no core area defined. This command requires that blocks have valid site rows or site array.

**What Next**

Create valid site rows for block. Can do this by pushing down top level site rows using push_down_objects from the parent (ie top level) block or by using create_site_row inside the block.

For example, from the parent block which already has site rows do:

push_down_objects [get_site_rows]

Feedback

## DPP-025

(error) Could not find global routing congestion map.

**Description**

You got this error from incremental congestion-driven macro placement because the tool could not find the global route congestion map that is needed by the incremental congestiven-driven macro placement.

**What Next**

Run global routing using command create_channel_congestion_map or route_global to get the global routing congestion map.

If the design has physical hierarchies underneath and you want to run congestion-driven macro placement for the macros inside the child blocks as well, please run virtual flat global route to generate congestion map for the whole chip. Please see man page of create_channel_congestion_map or route_global for how to get congestion map for the whole chip.

## DPP-026

(error) No blocks or voltage areas are found. Nothing can be done for shape_blocks.

**Description**

The command shape_blocks requires that the design have blocks/VAs defined. If there are no blocks/VAs in the design and shape_blocks is running, it is most likely that the create block step has a problem or UPF is missing.

**What Next**

Create the blocks and/or load UPF before using this command.

## DPP-027

(error) Block %s is not a design view, and should be fixed before placement.

**Description**

If a block is not a design view, it must be fixed before placement.

**What Next**

Fix the block or change the block to be a design view.

Feedback

## DPP-028

(error) Block %s should have a valid shape before placement.

**Description**

It is required that each block hava a valid shape before running placement.

**What Next**

Create shapes for blocks using shape_blocks.

## DPP-029

(warning) The specified option %s value is too large for block %s. This might result in undesired results from report_placement.

**Description**

The specified option value is too large.

**What Next**

Set the correct option value.

## DPP-031

(error) %d blocks have the fixed shapes, which are over half of total %d blocks.

**Description**

If the design has too many blocks with fixed shapes, shape_blocks cannot perform well.

**What Next**

Reduce the number of blocks with fixed shapes so that the number is less than half of the total blocks. Another way is to mark the blocks with fixed shape as fixed (i.e., not placeable).

## DPP-032

(error) Available effective shape for macros of %s.

**Description**

The macros will not fit in this region or the region is too big to be handled by the tool. Macro placement will skip this region but continue for any other regions.

Feedback

**What Next**

Check the shape of this region and the relevant blockages in it. Also check for the presence of fixed objects that block area for macro placement.

If the shape is too large, check if the shape is correct. If the design is really that big, please consider changing the internal units. See *create_workspace -scale_factor* and *create_lib -scale_factor* for more information.

---

## DPP-033

(error) Sanity check for the grouping constraint failed.

**Description**

shape_blocks only support limited types of grouping constraints.

**What Next**

Remove the groups which are not supported based on the detailed messages reported.

---

## DPP-034

(error) Utilization is too high after considering the channel size and keepout margins.

**Description**

The adjusted design utilization after considering the channel size and the keepout margins is too high to be used in the later stage of the flow.

**What Next**

Please reduce the channel size and/or the block keepout margins.

---

## DPP-035

(error) Design %s has no core area.

**Description**

The design has no core area defined. This command requires that there be a valid core area.

**What Next**

Create a valid core area by reading it in via a DEF file or using initialize_floorplan command.

Feedback

If it is a block level design in a hierarchical design context, push_down_objects can be used to push the site rows and/or site arrays from the top level into the block.

---

## DPP-036

(error) Overlaps between move bounds or between move bound and physical block are found.

**Description**

The tool does not support partially overlapped move bounds.

**What Next**

Create move bounds without partial overlaps.

---

## DPP-037

(error) Macro packing failed for %s.

**Description**

Macro packing did not find a legal macro placement for this region. Please check that the macros can fit in the shape.

**What Next**

If the macros look like they should fit into the region, it is possible that the default keepouts are too large. You can turn off creation of default keepouts by setting the app option plan.place.auto_generate_blockages to false. However, this is applied to entire design being placed. If you want a more focused change, then put a manual hard_macro blockage on the macros in the region using *create_keepout_margin -type hard_macro*.

Another solution is to increase the size of the region.

---

## DPP-038

(error) Block %s has no valid boundary.

**Description**

The specified block has no boundary defined. This command requires that blocks have a valid boundary.

Feedback

**What Next**

Create a valid boundary for the block. Can do this by pushing down top level rows using push_down_objects or by using initialize_floorplan inside the block. For example:

push_down_objects [get_site_rows]

---

## DPP-039

(error) Top level design has no valid boundary.

**Description**

The top level design has no valid boundary defined. This command requires that there be a valid boundary.

**What Next**

Create a valid boundary by reading it in via a DEF file or using initialize_floorplan command.

---

## DPP-040

(error) Sanity check for the nested_va constraint failed.

**Description**

shape_blocks encountered an issue while parsing or applying the nested_va constraints.

**What Next**

Check existence of voltage areas named in the nested_va constraints, and check that there are no loops defined among voltage areas named in multiple constraints.

---

## DPP-041

(error) Can not run congestion driven due to macro overlaps.

**Description**

Can not run congestion driven placement due to macro overlaps. Congestion driven placement requires legal (no macro overlaps, macros inside core) macro placement.

**What Next**

Remove macro overlaps and make sure they are placed inside core. A new congestion map may need to be generated to reflect the new macro locations. To display the list of violations, use *report_placement -hard_macro_overlap -verbose high* .

Feedback

## DPP-042

(error) Macro %s is out of the placement area.

**Description**

Macro is not completely inside the placement area.

**What Next**

Make sure they are placed inside core. To display the list of violations, use *report_placement -hard_macro_overlap -verbose high* .

## DPP-043

(error) Macro %s overlaps another macro or a blockage.

**Description**

Macro overlaps another macro or a placement blockage.

**What Next**

Remove macro overlaps. To display the list of violations, use *report_placement -hard_macro_overlap -verbose high* .

## DPP-044

(error) There were issues with grid assignment. See previous warnings for details.

**Description**

The tool failed to properly match the grids to macros. According to the value of plan.macro.grid_error_behavior app option, the command will stop.

**What Next**

Check whether user choices for the grids were accurate. Previous warnings should indicate particular grid assignment issues. If you want to continue with the placement anyway, change the value of plan.macro.grid_error_behavior app option.

## DPP-045

(warning) Unused block shaping constraint (possibly overwritten by subsequent constraints, or no applicable regions found):\n%s

Feedback

**Description**

This block shaping constraint will not be taken into account. There are two situations in which this can occur. If a subsequent constraint of the same type applies to the same object (block, voltage area, move bound) the last defined constraint is considered, and previous definition will be ignored. Alternatively, this warning is also issued if a constraint cannot be applied to a suitable region. As an example, take a utilization constraint defined on a group that only contains block instances. Because utilization constraints don't apply to block instances, it will not be used. In this case, the utilization constraint should have been specified on the respective reference blocks.

**What Next**

Verify whether this constraint is allowed to be overwritten. If not, move the definition of this constraint to a later stage, such that it will not be overwritten by others. Also verify whether the constraint is applied to the proper type of object. Some constraints must be specified on the reference block instead of on a block instance.

---

## DPP-046

(error) Failed to place due to over utilization.

**Description**

Placement has failed because the block utilization is over 100%. The most common reason for this is that some part of the row structure is missing. Placement respects the row stucture.

**What Next**

Take a look at the rows in the block. If some part is missing recreate the rows.

---

## DPP-047

(error) Poly rule threshold is not redefined as a positive integer.

**Description**

Poly rule threshold must be positive integer when checking poly rule violations.

**What Next**

Redefine the poly rule threshold as a positive integer.

---

## DPP-048

(error) Poly rule violation in {{%s, %s} {%s, %s}}, width is %d

Feedback

**Description**

The length of this place-able rectangle is less than the poly rule threshold and no exception applies to this rectangle.

**What Next**

Adjust the position of the related hard macros, placement blockages, soft macros and voltage areas.

---

## DPP-049

(error) There are more than one type of unit tiles in the design.

**Description**

Poly rule checker can only check designs containing one type of unit tiles.

**What Next**

Delete the redundant types of unit tiles to make the design containing one type of unit tiles.

---

## DPP-050

(error) Vertical rows are not supported.

**Description**

Poly rule checker can check designs with only horizontal rows.

**What Next**

Use the poly rule checker for designs with only horizontal rows.

---

## DPP-051

(error) Cannot use -use_existing_placement because too many cells are not placed.

**Description**

The option -use_existing_placement uses the existing placement in the design to determine the module boundary shapes. If the cells are not placed then this option will not work.

**What Next**

First place the cells in the design then you can use the -use_existing_placement option. Use create_placement to place cells in the design.

Feedback

**See Also**

- create_placement

---

## DPP-052

(error) Macro %s has an illegal orientation.

**Description**

Macro orientation is not in the allowed set.

**What Next**

Change the macro orientation to one of the allowed set. The allowed set can be found using *report_macro_constraints -allowed_orientations*.

---

## DPP-053

(info) Trace mode is set to %s.

**Description**

The trace mode controls the way connectivity is extracted from the netlist. It can be set via plan.place.trace_mode.

**What Next**

You can check whether the displayed information matches the intended one.

---

## DPP-054

(info) Default error view %s is created in GUI error browser.

**Description**

Default error view is created in error browser.

**What Next**

Check error browser.

---

## DPP-055

(error) Detecting block instances of design %s using distinct views: %s.

Feedback

**Description**

Block shaping and macro placement require multiple instances of a block to use the same view.

**What Next**

Change the view of the reported block instances such that they only use a single view per reference block.

---

## DPP-056

(error) Macro orientation %s is not legal for macro cell %s, and macro relative location constraint is not derived since the macro is unplaced.

**Description**

The checking is performed for target macro orientation. The given orientation conflicts with the set of legal orientations for the given cell. Legal orientations are gotten from the reference library of the cell or set by command set_macro_options.

**What Next**

Please set correct option value for the tool to proceed.

---

## DPP-057

(info) %s.

**Description**

Command information report.

**What Next**

n/a.

---

## DPP-101

(error) Found above illegal shapes for placement.

**Description**

If the shape of block (or voltage area or move bound) is completely outside the boundary of its parent, it usually indicates some problems before placement.

**What Next**

Please move the related shape into its parent's boundary.

Feedback

## DPP-103

(error) %s overlaps with %s.

**Description**

Found overlaps between blocks and/or voltage areas after shape_blocks.

**What Next**

This should not happen when shape_blocks succeeds. Please check if it runs into the existing limitations in the log file; otherwise please report the problem.

## DPP-104

(error) Found the above situation which is not supported by incremental shaping.

**Description**

shape_blocks has some limitations when applying incremental shaping.

**What Next**

Before the tool removes the limitations, please manually shape the blocks.

## DPP-105

(error) MIB instance %s is fixed, but the MIB instance %s is not.

**Description**

It is required that either all MIB instances of the same master are fixed or all are not fixed for incremental shaping.

**What Next**

Please either unfix all MIBs or fix all MIBs of the same master.

## DPP-107

(error) Hierarchy %s to be expanded has no connectivity.

**Description**

The given hierarchy does not have any connectivity. The likely reason is that you are using an outline view, and the connectivity of this hierarchy was not loaded.

Feedback

Outline views are created using *read_verilog_outline*. The outline view has only a limited amount of connectivity to save on memory. The amount of netlist to load is controlled via options on *read_verilog_outline*.

**What Next**

If you have outline view and want to load the connectivity for the current hierarchy, look at the man page for *read_verilog_outline* and use options to load more of the netlist.

If you want to expand this hierarchy boundary without any connectivity, please use the option *-force* of command *explore_logic_hierarchy*.

---

## DPP-108

(error) Found invalid move bounds.

**Description**

If a move bound is completely blocked by a physical block, it is an invalid move bound. If the boundary of a move bound of a block is outside the block's boundary, it is invalid. Also if the element of a move bound is the same element of voltage area, it is invalid for placement.

**What Next**

Remove invalid move bounds or change the boundaries of invalid move bounds to make them valid.

---

## DPP-109

(error) Found fixed macros which are not inside the top level block boundary.

**Description**

Any fixed macro inside a physical block or voltage area should be placed inside the top level block (or chip) boundary so that shaping can create the legal shape to cover them.

**What Next**

Either unfix macros or move fixed macros inside the chip boundary.

---

## DPP-111

(error) Found the fixed block or voltage area which is not inside the chip boundary.

Feedback

**Description**

It is not allowed that the shape of fixed physical block or fixed voltage area is not completely inside the chip boundary.

**What Next**

Change the fixed shape so that it is completely inside the chip boundary.

---

## DPP-112

(error) Found overlaps between blocks.

**Description**

Any overlap between blocks is not allowed. This is mainly caused by the tool limitations.

**What Next**

Please fix the overlaps before moving to the next step, and also report the issue.

---

## DPP-113

(error) Found unsupported move bounds in shaping.

**Description**

Shaping is checking if the defined move bound is supported. When it detects the unsupported move bound, it will issue the detailed information for each related move bounds. Please make the change accordingly, otherwise those unshaped move bounds will cause the problem in the flow.

**What Next**

Remove the unsupported move bounds before shaping.

---

## DPP-114

(error) %s is not inside the core area.

**Description**

Incremental shaping requires that all blocks and voltage areas are inside the core area.

**What Next**

Move the corresponding blocks and voltage areas into the core area before applying incremental shaping.

Feedback

## DPP-115

(error) Found fixed shape blocks or voltage areas which are over utilized.

**Description**

When fixed shape blocks are over utilized, it will cause the problem in placement and optimization.

**What Next**

Change the shapes so that they are not over-utilized.

## DPP-116

(error) Found fixed shape block or voltage area whose input shape cannot be aligned to the boundary grid.

**Description**

When the width or height of the input fixed shape is not divisible by the boundary grid's X-step or Y-step, it is impossible to align this shape to the grid.

**What Next**

Change the input fixed shapes so that they can be aligned to the boundary grid.

## DPP-117

(error) shape_blocks failed to generate the valid solution.

**Description**

shape_blocks failed to find the shapes which can fit all macros and meet the utilization target.

**What Next**

There are many reasons for shape_blocks to fail. If the design has MIBs, using the constraint file to set MIB as the rigid blocks may help. If the design has many rigid blocks, fix some blocks if their locations are known to reduce the number of rigid blocks. If the design has large or many macros, fix the blocks which are macro dominant may help.

## DPP-118

(error) No more than %d physical blocks, voltage areas, and move bounds are supported. There are curretly %d.

Feedback

**Description**

There are too many physical blocks, voltage areas or move bounds in the design.

**What Next**

The pysical blocks, voltage areas, and move bounds are not intened to be used at a very fine scale, but are intended to contain larger groups of cells. Please constrain the cells in larger groups.

---

## DPP-119

(error) App option values %s and %s are mutally exclusive.

**Description**

The stated app option values are incompatible, so the command cannot proceed.

**What Next**

Please change the value for one of the options to a compatible one.

---

## DPP-120

(error) %s.

**Description**

To run placement or shaping, it is required that at least one site row has been created based on the valid site def.

**What Next**

Please create the site row with valid site def.

**See Also**

- create_site_def
- get_site_defs
- create_site_array
- create_site_row

---

## DPP-121

(Error) There is no active scenario with setup analysis activated for design %s.

Feedback

**Description**

The command expects to be able to analyze the setup times in the current scenario, or at least some scenario. If no active scenario has setup analysis activated, that can not happen. So you must activate setup timing in at least one scenario to continue.

**What Next**

Activate setup timing with "set_scenario_status".

---

## DPP-122

(Error) Block %s is nested inside module boundary %s

**Description**

Option "-place" does not support the nesting between block and module boundary.

**What Next**

To keep the physical block for placement, the module boundary has to be expanded such that the physical block is not nested inside any module boundary.

---

## DPP-123

(Error) No valid input is found %s

**Description**

The command will first check whether the input is valid or not. If it identifies the input which is not supported (for example, try to expand a physical block), it will issue a message and ignore this input. If all inputs are ignored, the tool will error out.

**What Next**

Please refer to the reported message before this error and make the changes in the option.

---

## DPP-124

(Error) Cannot find the corresponding %s for %s

**Description**

The tool could not find the valid module boundary (or bound) to continue the operation.

Feedback

**What Next**

Please check if there are other messages before this error. If not, please report this problem.

---

## DPP-125

(Error) Both center and distributed block placement types for block %s

**Description**

Specified block has some instances with the *block_placement_type* attribute equal to *center*, and some equal to *distributed*. For a given block, only one of those two values is allowed.

**What Next**

Check the values of this attribute for instances of the specified block, and change them appropriately.

---

## DPP-126

(warning) Block instance %s is out of the die area of design block %s.

**Description**

Block instance is out of the die area of design block.

**What Next**

Check block placement.

---

## DPP-127

(Error) Design is unmapped.

**Description**

Unmapped designs are not supported in macro placement. You can force the tool to try anyway, by setting the app option plan.macro.allow_unmapped_design to true, but, in that case, the result, or even that the tool will not have a serious problem, cannot be guaranteed.

**What Next**

Map the design.

Feedback

## DPP-128

(Error) Command-line option %s of create_placement -floorplan is not compatible with a shell design.

**Description**

The command-line options mentioned above is not compatible with shell designs.

**What Next**

Remove the command-line option.

## DPP-129

(Error) Style all_islands is compatible only with shell designs.

**Description**

The app option plan.macro.style has value all_islands, which is only compatible with shell designs.

**What Next**

Change the value of this app option to an appropriate one.

## DPP-201

(warning) No placeable block is found. Quit shaping.

**Description**

Did not find any placeable block.

**What Next**

Please make sure that blocks are created and not fixed in order to shape them.

## DPP-202

(warning) Ignored user input (%s %s).

**Description**

Wrong user input is identified and ignored.

**What Next**

Please correct the input based the detailed messages.

Feedback

## DPP-203

(warning) Cells in block %s are not placed.

### Description

You are getting message when running *report_placement -hierarchical*. When cells inside a block are not placed, report_placement will not report wire length or check violations inside this block.

### What Next

Please make sure that cells are placed, or run *report_placement* without the option *-hierarchical*.

## DPP-204

(warning) There are unplaced cells in the design.

### Description

report_placement only considers cells in the design that are placed or fixed. This message indicates that one or more cells are not placed or fixed. These cells will be ignored by this command. For example, the wirelength given by this report will be less than if all the cells were placed.

### What Next

If expectation is that design is placed, check the design to make sure all cells are properly marked with the placed attribute.

## DPP-205

(warning) Edit group %s is ignored

### Description

This command only supports edit groups with hard macros in the same physical hierarchy, voltage area, and/or move bound. Any edit group which violates this limitation will be ignored. Another reason for an edit group to be ignored is if it is a nested edit group. This command only considers the top most edit group. However note that if the top most edit group is supported, all nested edit groups will automatically be supported.

### What Next

The edit group objects relative locations may change after this command is run. You may want to check on this.

Feedback

## DPP-206

(warning) %s %s %s %.1f

**Description**

User specified -min_channel_size is not achieved.

**What Next**

One of the most common reasons is in order to fit macros inside each shape such that the shape cannot be reduced to get the required channel size.

If there is only one or two channels missing the required channel size, reducing the target utilization of the macro dominated block may make it easier to place macros inside the proposed shape, thus enabling to shrink the shape for the required channel size.

If many channels are missing the required channel size, please check if there are many macro dominated blocks and if some small macro dominated blocks can be merged into a big one; or if min_channel_size is too big to be achieved.

## DPP-207

(warning) Top level connections to the ports without locations will be ignored by placement.

**Description**

Ports are not placed before applying shape_blocks. This will impact the placement quality and thus impact the shaping results.

**What Next**

Please place ports before shaping.

## DPP-208

(warning) Block %s has an invalid boundary.

**Description**

The given block's boundary is not valid and so the block will be ignored. This could be because the block was not created properly, or the block boundary was somehow set to an incorrect value.

**What Next**

Take a look at the block and see why its boundary is invalid.

Feedback

## DPP-209

(warning) Merge of %s failed on cell %s.

**Description**

A floorplan merge between two different views was performed, likely during create_abstract, merge_abstract, or expand_outline. During this merge, the given information was dropped.

**What Next**

If this information is important you will need to reapply it.

## DPP-210

(info) Repelling group bound %s is ignored by macro placement.

**Description**

The macro placer detected a repelling group bound. These bounds are not considered during macro placement.

**What Next**

There's nothing that needs to be done.

## DPP-211

(warning) top level cells are not loaded.

**Description**

If the design is the outline view, top level connectivity is missing, which impacts the shaping QoR.

**What Next**

Run expand_outline to load the top level cells.

## DPP-212

(warning) Macro group location planning had difficulties for %s.

Feedback

### Description

The planning phase of macro placement had some difficulties finding a good solution. The macro legalizer will create a legal solution (if possible) but you might want to check the macro placement quality for the given region.

### What Next

Continue with the normal flow, but check the macro placement QoR.

---

## DPP-213

(Information) Created macro group %s.

### Description

This message gives the name of a group of macros which will be placed together.

### What Next

This is an informational message only. No action is required.

### See Also

- create_placement

---

## DPP-214

(Error) The group_bound %s contains fixed cells (e.g.: %s). Macro placement only supports group_bounds containing floating cells.

### Description

This message is shown for group_bounds that do not pass the macro placer's signin check.

### What Next

Modify the contents of the group_bound such that the group_bound only contains floating cells.

### See Also

- create_bound
- create_placement

Feedback

## DPP-215

(warning) Block %s: core area is (partially) outside boundary area

### Description

This message indicates that for the given block, the core area is not completely inside the block boundary. This may create problems for placement.

### What Next

This likely means that the block core area is not set up correctly. It can be fixed by using initialize_floorplan on the block or by push_down_objects of site rows from the top level.

## DPP-216

(Warning) The group bound %s was created with a nondefault value for %s. Macro placement will ignore this option.

### Description

This message is shown for group bounds that have unsupported options.

### What Next

Do not specify the option mentioned in the warning when creating the group bound. Macro placement supports effort levels low and medium, and does not depend on the dimension of a group bound. The contents of medium effort group bounds are kept together during macro placement, without forcing a particular location or shape as move bound or edit groups do. You can use a low effort group bound on higher level hierarchies, to tell macro placement to keep these hierarchies physically separate if possible.

### See Also

- create_bound
- create_placement

## DPP-217

(Warning) The -congestion option is not yet supported for macro placement in designs with physical hierarchy.

### Description

This design has physical hierarchy (sub-blocks), which is currently not supported for the congestion aware macro placement. Macro placement continues without congestion awareness. The standard cell placement will still use the -congestion flag.

Feedback

### What Next

Ignore the message until support has been added.

### See Also

- create_placement

---

## DPP-218

(warning) Block %s does not have a view suitable for placement.

### Description

You have asked to run a global placement inside given block, but the given block has a view that is incompatible with this. To run global placement, blocks need to have design view or a placement compatible abstract view. If you do not want a global placement, but want a flat placement instead, then use the command set_editability to control what blocks to place.

Placement will proceed without placing inside this block.

### What Next

Change to use a design view or placement compatible abstract for block or use set_editability to control what blocks to place.

---

## DPP-219

(warning) Hard macro %s is a spare cell.

### Description

The given hard macro is marked as a spare cell. A spare cell is a cell that has no connectivity and typically is a standard cell. It is unusual to have a hard macro marked as a spare cell and if many hard macros are marked as spare cells, then hard macro placement QoR will suffer because of lack of connectivity to drive placement.

### What Next

Check to see if this hard macro is really meant to be unconnected. If the hard macro is inside an abstract view, and you determine that the abstraction process caused the hard macro to be unconnected then switch to use design view instead.

---

## DPP-220

(warning) Potential over-utilization on %s %s.

Feedback

### Description

Based on a quick rough calculation of utilization, it appears that the given VA or block is over utilized. Since this calculation is rough, only a warning is issued. Later, a more accurate computation is done, and at that time if there is an issue, an error will be issued.

### What Next

If no error is issued, then there is nothing to do. If an error is issued, then check the block or VA to see why it is over utilized. It is possible the shape is too small.

---

## DPP-221

(warning) The -congestion_effort option only effects the standard cell congestion reduction, which is not enabled in this run.

### Description

The *-congestion_effort* option to *create_placement -floorplan* sets the effort level for the standard cell congestion reduction. The standard cell congestion reduction is only enabled if the *-congestion* option is used and the application option *plan.place.congestion_driven_mode* is set to *std_cell* or *both*.

In this run, the standard cell congestion reduction is not enabled, so the *-congestion_effort* option is ignored.

### What Next

Enable standard cell congestion reduction or leave out the *-congestion_effort* option.

### See Also

- create_placement
- plan.place.congestion_driven_mode

---

## DPP-222

(error) Option %s and app option %s are mutally exclusive.

### Description

The stated command option and app option settings are not supported together, so the command cannot proceed.

### What Next

Please change the option or app option to be compatible ones.

Feedback

## DPP-230

(warning) Can not run incremental placement due to unplaced cells, switching to non-incremental.

### Description

When running *-create_placement -floorplan -incremental*, it is expected that the macro cells are already placed. If they are not, *-incremental* is ignored and the cells are placed in a standard way.

### What Next

Make sure the cells are placed before using *-incremental*.

## DPP-231

(warning) Can not run incremental placement due to too many macro overlaps, switching to non-incremental mode.

### Description

When running *-create_placement -floorplan -incremental*, it is expected that the macro cells are inside core and if they overlap, they do it lightly. Heavy overlaps or too many cells outside their block will result in a switch to non-incremental placement.

### What Next

Make sure the macros do not overlap too much before using *-incremental*.

## DPP-232

(warning) Unable to parse app_option placement_file_mode: %s.

### Description

The app_option placement_file_mode accepts a mode, which is one of "none", "read", "write", optionally followed by a file name. For example: set_app_options -list {plan.macro.placement_file_mode "write preferred_macro_locations.tcl"} will instruct "create_placement -floorplan" to output the resulting macro locations to the file called "preferred_macro_locations.tcl".

### What Next

Verify that app_option placement_file_mode adheres to the proper format.

Feedback

## DPP-233

(error) Found multiple regions for voltage area %s (with element %s).

**Description**

When a voltage area has multiple regions which are not fixed, shape_blocks will ignore this voltage area.

**What Next**

Please fix the voltage area regions before running shape_blocks.

## DPP-234

(warning) Hierarchy %s has no connectivity.

**Description**

The given hierarchy does not have any connectivity. If this is expected, then there is no issue. However if it was expected to have connectivity, the likely reason is that you are using an outline view, and the connectivity of this hierarchy was not loaded.

Outline views are created using *read_verilog_outline*. The outline view has only a limited amount of connectivity to save on memory. The amount of netlist to load is controlled via options on *read_verilog_outline*.

**What Next**

If you have outline view, look at the man page for *read_verilog_outline* and use options to load more of the netlist.

## DPP-235

(warning) Did not find any buffer or inverter whose reference library cell site matches the block site.

**Description**

All the cells in the reference library were checked to find buffers and inverters. The tool was unable to find at least one buffer and one inverter whose site name matches with the block's site name. This is not a problem for placement, so this is only a warning. However this implies there is a problem either in the reference library setup or the block row site creation. Later, during virtual IPO, this will be a problem.

Feedback

### What Next

The problem is either with library prep or with the block sites created by *initialize_floorplan*. Change one or other so that site names match.

---

## DPP-236

(warning) Auto blockages in block/VA %s not created.

### Description

The most common reason for this messages is that the given block or voltage area (VA) has a very high utilization after auto blockages are created. This may lead to placement over utilization errors. Because of this, the tool did not create any auto blockages for this block or VA. The upper limit of this utilization is controlled using app option *plan.place.max_utilization_for_auto_blockages*.

A second possible reason for this message is that the VA has no hard macros. Auto blockages are not created in VAs that have no hard macros.

### What Next

If this is unexpected (ie the current utilization is low and there are not many channels to be blocked by auto blockages), the reason could be because one or more of below app options is set too high. Please see man pages for each of these app options to get more details on their functionality. App options: plan.place.max_utilization_for_auto_blockages, place.floorplan.sliver_size, plan.place.auto_generate_hard_blockage_channel_width, plan.place.auto_generate_soft_blockage_channel_width.

If one or more of these app options is set, try reducing the setting. You can recreate auto blockages with command *derive_placement_blockages*.

To force creation of blockages even if utilization is high use *derive_placement_blockages -force*.

---

## DPP-237

(warning) Macro %s does not have pins on one side, so it will not be considered for sandwich logic.

### Description

Sandwich logic is considered only for macros with signal pins only on one side. Pins of the two macros belonging to the sandwich will be placed facing each other.

### What Next

Do not use macros with pins on two sides for sandwich logic.

Feedback

## DPP-238

(warning) Macros %s and %s do not have the same shape, so they will not be considered as a pair for sandwich logic.

**Description**

Sandwich logic is considered only for pairs of macros with same shape.

**What Next**

Do not use pairs of macros with different shape for sandwich logic.

## DPP-239

(warning) Only floating standard cells can be marked as sandwich logic. Offending cell is %s.

**Description**

Only floating standard cells can be marked as sandwich logic for placement between a related macro pair.

**What Next**

Only include cells that are not marked fixed and have model type "standard cell".

## DPP-240

(warning) %s.

**Description**

Missing connectivity for macro placement.

**What Next**

Recreate the placement abstracts with the same settings for connectivity extraction that are currently set for macro placement.

## DPP-241

(warning) Channel %s of %s is not a multiple of %s grid pitch.

**Description**

Channel width or height is not a multiple of the grid pitch. This is likely to make it impossible to align macros to grid.

Feedback

**What Next**

Make sure channel widths and heights are multiples of the grid pitch.

---

## DPP-242

(warning) Macro %s is assigned grid %s, but there is no grid with that name.

**Description**

Specified macro was assigned a grid using set_macro_constraint command, but that grid does not exist. This setting will be ignored and macro will be assigned default grid, which is FinFET grid, if one exists, else litho grid.

**What Next**

Check if the grid name is misspelled or a grid with that name is deleted.

---

## DPP-243

(warning) -align_pins_to_track grid constraint on macro %s for orientation %s ignored as align point could not be determined.

**Description**

The tool failed to determine align point in order to snap pins to tracks. This is most likely because there are not enough pins (less than half) that can be simultaneously aligned to tracks, or perhaps there are pin shapes on multiple layers with the same routing direction. This setting will be ignored and macro will be assigned default grid, which is FinFET grid, if one exists, else litho grid.

**What Next**

Create a user grid that matches the tracks that you want to align pins to, and provide location of one of the pins (in master coordinates) to use as the align point.

---

## DPP-244

(warning) Can not color match all pins for macro %s. %s

**Description**

The tool failed to color match all pins to tracks. The specific reason is provided in the message. The tool will try to match as many pins as possible.

Feedback

**What Next**

Check whether the coloring for macro and tracks match properly. Check whether macro orientation precludes color matching, and if so, disallow such orientation using set_macro_constraints -allowed_orientations.

---

## DPP-245

(warning) There were issues with grid assignment. The command will stop after macro placement.

**Description**

The tool failed to properly match the grids to macros. According to the value of plan.macro.grid_error_behavior app option, the command will stop after macro placement, if possible.

**What Next**

Check whether user choices for the grids were accurate. Previous warnings should indicate particular grid assignment issues. If you want to continue with the placement anyway, change the value of plan.macro.grid_error_behavior app option.

---

## DPP-246

(warning) Cannot snap the fixed boundary object %s to the FinFet grid.

**Description**

When any dimension of the fixed block boundary (from shaping constraints) is not divisible by the FinFet grid, shaping cannot align all boundary edges to the FinFet grid.

**What Next**

It is better to create the shape so that its dimensions (i.e., edge lengths) are divisible by the FinFet grid.

---

## DPP-247

(warning) %s

**Description**

When the input pg strategy is invalid, compile_pg cannot create the pseudo PG shapes for shaping. Then shape_blocks will ignore the option -pg_strategy.

Feedback

**What Next**

Run compile_pg command to find why it is failing (e.g., missing nets) and provide the valid pg strategy.

---

## DPP-248

(warning) %s%s%s%s

**Description**

When the input is an existing module boundary, physical block or move bound, it will be ignored for the specified exploration.

**What Next**

Check if the input is wrong. Otherwise this warning can be ignored as the module boundary or move bound is already existing.

---

## DPP-249

(error) %s.

**Description**

Error during Importing the connections from an attribute for macro placement.

**What Next**

Check the design with the same settings for connectivity extraction that are currently set for macro placement.

---

## DPP-250

(warning) Found fixed shape blocks or voltage areas which are over utilized.

**Description**

When fixed shape blocks are over utilized, it will cause the problem in placement and optimization.

**What Next**

Change the shapes so that they are not over-utilized.

---

## DPP-251

(warning) Found master %s with edges that are neither horizontal nor vertical.

Feedback

**Description**

The command expects only cells whose boundary is a Manhattan polygon (i.e. polygon containing only horizontal and vertical edges). If the boundary is not such a polygon, the command will use the smallest Manhattan polygon containing the boundary instead of the actual boundary.

**What Next**

Check if the boundary is correct, or if it is supposed to be a Manhattan polygon.

---

## DPP-252

(warning) %s %s

**Description**

Could not find the virtual hierarchy which matches the specified input name. So the corresponding input is ignored.

**What Next**

Check if the input names are correct. With *set_app_options -list { plan.place.hierarchy_by_name true}*, get_cells cannot be used. Instead the virtual hierarchy names are used as inputs.

---

## DPP-253

(warning) %s%s%s%s

**Description**

When the input is not the expected module boundary or move bound, the specified exploration cannot be applied.

**What Next**

Check if the input is wrong. Otherwise follow the level based -expand/-collapse (or -virtual_group/-virtual_ungroup) to get the expected module boundary or move bound.

---

## DPP-254

(warning) %s is inside the physical block %s, which is ignored for exploration

**Description**

When the input is inside any committed block, the specified exploration cannot be applied.

Feedback

### What Next

To explore module boundary or move bound inside any committed block, first set the current design to the committed block and then apply the exploration options.

---

## DPP-255

(warning) Found two tracks on the same layer %s and with the same direction (%s), but different steps

### Description

There are multiple tracks on the same layer and the same direction, but with different steps. This adversely affects the ability to snap macro pins onto tracks (using *create_placement -floorplan* command), as well as to check whether they are snapped properly (using *report_placement -user_grid* command).

### What Next

Delete unnecessary tracks.

---

## DPP-256

(warning) Bound %s cannot be used for exploration because of content change(s).

### Description

When cells or ports are added to a bound or removed from a bound which is created by explore_logic_hierarchy, the bound cannot be used for exploration any more.

### What Next

Do not change the bound contents if the bound is to be explored using explore_logic_hierarchy.

---

## DPP-257

(warning) When plan.macro.auto_buffer_channels is set to true, plan.macro.%s will be ignored.

### Description

This message shows up when plan.macro.auto_buffer_channels is set to true, and some of plan.macro.max_buffer_stack_height, plan.macro.buffer_channel_height, plan.macro.max_buffer_stack_width, and plan.macro.buffer_channel_width are set to non-zero values. Those non-zero values will be ignored, as the coresponding values will now be automatically calculated.

Feedback

**What Next**

Make sure you wanted the values automatically calculated.

---

## DPP-258

(warning) No rule named %s, using default rule instead.

**Description**

The rule specified using plan.macro.buffer_rule_name does not exist. The tool will use the default rule instead.

**What Next**

Check the name of the rule.

---

## DPP-259

(warning) No buffer available for rule %s, using the default rule's buffer.

**Description**

The rule specified using plan.macro.buffer_rule_name does not have a buffer specified. The tool will use the default rule's buffer size instead.

**What Next**

Check the contents of the specified rule.

---

## DPP-260

(warning) No buffer available for the default rule, turning off automatic channel buffering.

**Description**

The default rule does not have a buffer specified; hence the tool cannot properly determine the buffer channel sizes and will turn off automatic channel buffering.

**What Next**

Check the contents of the default rule.

---

## DPP-261

(warning) No buffer spacings available for rule %s, using the default rule's values.

Feedback

**Description**

The rule specified using plan.macro.buffer_rule_name does not have buffer spacings. The tool will use the default rule's buffer spacing instead.

**What Next**

Check the contents of the specified rule.

---

## DPP-262

(warning) No buffer spacings available for the default rule, turning off automatic channel buffering.

**Description**

The default rule does not have buffer spacings; hence the tool cannot properly determine the buffer to buffer distances and will turn off automatic channel buffering.

**What Next**

Check the contents of the default rule.

---

## DPP-263

(info) %s

**Description**

Information about override and auto_select controlling the BB-star hierarchy selection algorithm.

**What Next**

If an error is reported, adjust the specified hierarchy names accordingly.

---

## DPP-264

(warning) Edit group %s marked flexible, but treated as rigid.

**Description**

The edit group (macro array) has is_flexible attribute set to true, but was not treated as flexible. In order to be treated as flexible, we expect clearly defined rectangular set of rows and columns of macros. This issue happens if the macros are not organized relatively neatly to be able to easily deduce rows/columns, or if the array is rectilinear.

Feedback

**What Next**

Make sure the macros are arranged in a way that rows and columns are easily deduced; using GUI macro array tool will ensure that. Additionally, make sure that it is a rectangular macro array; currently, the tool does not support flexible rectilinear macro arrays.

---

## DPP-265

(warning) Style %s is not compatible with a shell design, using style %s instead.

**Description**

The app option plan.macro.style has a value that is incompatible with a shell design, so the tool will treat it as though the another value is specified.

**What Next**

Next time use a compatible value.

---

## DPP-266

(warning) Value %s of app option %s is not compatible with a shell design, using the default value %s instead.

**Description**

Some app options values are not compatible with shell design, so they are reset to a compatible value in order for the command to proceed.

**What Next**

Next time use a compatible value.

---

## DPP-401

(info) cell %s is outside of placement area

**Description**

The specified cell is placed outside of its parent physical hierarchy's placement area.

**What Next**

Please check the design to make sure it is properly placed.

---

## DPP-403

(info) cell %s overlaps block %s

Feedback

**Description**

The specified cell is overlapping with the specified block.

**What Next**

Please check the design to make sure it is properly placed.

---

## DPP-405

(info) %s and %s overlap

**Description**

The two specified objects are placed overlapping each other. The objects could be hard macro cells or one could be a hard macro blockage. For a hard macro cell, overlapping includes any user defined hard macro keepout (create_keepout_margin -type hard_macro).

**What Next**

Please check the design to make sure it is properly placed.

---

## DPP-406

(info) %s is not on user grid (%s).

**Description**

The reported object is not on user grid. To see what user grids are defined for this object use *report_macro_constraints*.

**What Next**

Check the object to make sure it is properly placed and manually snap it to grid if desired. If the user grid is incorrect for this object use *set_macro_constraints -alignment_grid* to change it.

---

## DPP-407

(info) cell %s is not placed.

**Description**

One or more cells in design are not placed. In particular, the specified cell is not placed. This may have implications based on what command is being executed. For example, not placed cells may be ignored.

Feedback

**What Next**

This is an informational message.

---

## DPP-408

(info) %s and %s overlap

**Description**

The two specified physical hierarchy cells are placed overlapping each other.

**What Next**

Please check the design to make sure they are properly placed.

---

## DPP-409

(info) %s and %s have macro spacing rule violation

**Description**

The two specified objects have macro spacing rule violation each other. The objects could be hard macro cells. The macro spacing rule is set by app option plan.macro.spacing_rule_widths and plan.macro.spacing_rule_heights.

**What Next**

Please check the design to make sure it is properly placed.

---

## DPP-410

(info) Nothing is expanded because no sub-hierarchy size is more than %.1f%% of current hierarchy (%s) size.

**Description**

This message informs the user that either the input hierarchy has no sub-hierarchies or each sub-hierarchy is too small with respect to the threshold.

**What Next**

You can ignore this message if this is ok. Otherwise, adjust threshold using "explore_logic_hierarchy -threshold" if small sub-hierarchies are to be explored.

---

## DPP-411

(info) Block %s violates core area.

Feedback

**Description**

The specified block is not completely inside the core area. Typically, blocks should be placed inside the core area.

**What Next**

Check the design to make sure the block is placed correctly.

---

## DPP-412

(info) Block %s utilization differs from target by %.2f percentage points

**Description**

The actual utilization of the specified block does not match the target utilization by the amount specified. For example, if the target utilization is 60% and the current utilization is 65% then the difference is 5 percentage points.

**What Next**

It is normal for the actual block utilization to not exactly match the target utilization, but if the difference is large, the block shape may need to be adjusted.

---

## DPP-413

(info) Placing macro %s on preferred location {%s}.

**Description**

The location constrained through set_macro_constraints -preferred_location has been applied.

---

## DPP-414

(info) Ignoring preferred location of macro %s because it is fixed.

**Description**

The location constrained through set_macro_constraints -preferred_location has been ignored because the macro has a fixed location.

**What Next**

If the macro should be placed at the preferred location, mark it placeable. If the fixed location is correct, remove the preferred location constraint for this macro to avoid this message.

Feedback

## DPP-415

(info) Hierarchy bound %s has no connectivity.

**Description**

The given hierarchy bound does not have any connectivity. If this is expected, then there is no issue. However if it was expected to have connectivity, the likely reason is that you are using an outline view, and the connectivity of this hierarchy bound was not loaded.

Outline views are created using *read_verilog_outline*. The outline view has only a limited amount of connectivity to save on memory. The amount of netlist to load is controlled via options on *read_verilog_outline*.

**What Next**

If you have outline view, look at the man page for *read_verilog_outline* and use options to load more of the netlist.

## DPP-416

(info) All hard macros are fixed, no hard macro placement is done.

**Description**

You have requested the -floorplan option to create_placement which includes hard macro (HM) placement. However the tool has determined that all the HMs are fixed and thus there are no HMs to be placed.

It this is expected, then no action is required.

**What Next**

If this is not expected, then you need to determine why all the HMs are fixed and unfix the HMs that you want the tool to place. To determine the placement status of a HM use *get_attribute -name status [get_cells <cell_name>]*. Change all the HMs you want to place to status "placed" using the *set_attribute* command.

Another reason that all the macros are fixed could be that the app option place.coarse.fix_hard_macros is set to true (this is not the default). Use *report_app_options place.** to see values of all placement app options. If you dont want macros to be fixed, then use *set_app_options -list {place.coarse.fix_hard_macros false}*.

## DPP-417

(info) max_density auto set to %.2f for placement.

Feedback

**Description**

The app option plan.place.auto_max_density is set to true. Because of this, the tool will automatically turn on max_density control during placement of a block that has subblocks in it, if it determines that utilization is low. That means standard cells will be allowed to be placed in a clumped fashion, thus minimizing wirelength but potentially creating congestion.

**What Next**

If you do not want max_density to be turned on automatically, then turn off this app option using *set_app_options* command.

---

## DPP-418

(info) Macro %s is assigned %sgrid%s.

**Description**

This message displays grid information for a macro.

**What Next**

You can check whether the displayed information matches the intended one.

---

## DPP-419

(info) checking for fixed hard macro %s is skipped.

**Description**

This message shows the tool will skip checking density gradient violations for this fixed hard macro since user has specified '-ignore_fixed' option.

**What Next**

You can check whether the displayed information matches the intended one.

---

## DPP-420

(warning) checking for hard macro %s is skipped since it is overlapped with other hard macros.

**Description**

The tool find the hard macro is overlapped with other hard macros, so it will skip checking density gradient violations for this hard macro.

Feedback

**What Next**

Move those overlapped hard macros, then run report_placement -hard_macro_overlap and report_placement -physical_hierarchy_violations to make sure there is no placement violation in current design.

---

## DPP-421

(info) checking for small hard macro %s is skipped.

**Description**

This message shows the tool will skip checking density gradient violations for this small hard macro since both the width and height are less than the specified min macro size.

**What Next**

You can check whether the displayed information matches the intended one.

---

## DPP-422

(info) checking for small hard macro cluster %s is skipped.

**Description**

This message shows the tool will skip checking density gradient violations for this small hard macro cluster since both the width and height are less than the specified min macro size.

**What Next**

You can check whether the displayed information matches the intended one.

---

## DPP-425

(error) There exist relative location placement constraints loop:

**Description**

The macro or macro array relative location placement constraints are invalid when there exist constraint loop. For example, macro a anchor to b, macro b anchor to c, macro c anchor to a to construct a loop. The relavent macros or macro arrays are listed below this message. Since there is error in the constraint setting, the relative location constraints for these macros or macro arrays will be ignored by placement.

Feedback

**What Next**

Please review the macro relative location constraints for the macros or macro arrays listed below this message. You probably need to remove one or more of the constraints.

---

## DPP-426

(error) There exist bad relative location constraints path with root anchor object has no fixed location:

**Description**

If a macro is anchored by other macros while it does not anchor to other objects, it should be fixed, otherwise, the error message will be reported. The relavent macros or macro arrays are listed below this message. Since they don't eventually anchor to a fixed object, the relative location constraints for these macros will be ignored by placement.

**What Next**

Please review the relative location constraints on the macros or macro arrays listed below this message. If you think some constraints should not be there, please use *remove_macro_relative_locations* to remove them. It is possible that you missed setting a constraint on a macro to anchor it to a fixed object. If so, please use *set_macro_relative_location* to add the constraint. Also, user can mark the macro which has no relative location placement constraints but is anchored by other macros as "fixed".

---

## DPP-427

(Warning) Target object %s and anchor object %s are not in the same physical hierarchy. Skip it!

**Description**

The checking is performed for target and anchor object. The two objects should be in the same physical hierarchy.

**What Next**

Please set correct option value for the tool to proceed.

---

## DPP-428

(error) Insufficient space for packing contents. Detailed information: %s

Feedback

**Description**

The block shaper found that it has insufficient room for fitting the contents within the boundary of a block or voltage area. Contents of such a region may be (a combination of) macros, nested blocks or nested voltage areas.

**What Next**

Check macros, predefined (nested) block boundaries, keepouts, channel constraints, which influence the available and required space.

---

## DPP-429

(warning) Potentially insufficient space for packing contents. Detailed information: %s

**Description**

The block shaper may have difficulty finding a solution, because it might have insufficient room for fitting the contents within the boundary of a block or voltage area. Contents of such a region may be (a combination of) macros, nested blocks or nested voltage areas.

**What Next**

Check macros, predefined (nested) block boundaries, keepouts, channel constraints, which influence the available and required space.

---

## DPP-430

(error) Detected utilization issues. Detailed information: %s

**Description**

The block shaper found that it has insufficient room for fitting the contents within the boundary of a block or voltage area. Contents of such a region may be (a combination of) macros, nested blocks or nested voltage areas.

**What Next**

Check constraints that influence the target utilization, like channel size constraints, target area/utilization constraints and keepouts. Relax them to make more space available.

---

## DPP-431

(warning) The design block %s shape has changed.

Feedback

**Description**

The macro or macro array relative location constraints that derived from command derive_macro_relative_location do not guarantee to support the design block that has been changed the shape, such as from R floorplan to T floorplan.

**What Next**

Please review the design block floorplan shape.

---

## DPP-432

(warning) The placed macro or macro array %s is outside the core area boundary of block %s.

**Description**

The relative location placement place macro or macro array outside the block core area boundary.

**What Next**

Please review the relative location constraint or place the object inside block core area.

---

## DPP-433

(warning) There is relative location constraints both on macro array %s and macro %s that the macro array contains. Macro placement take the constraint on macro array.

**Description**

The relative location constraints on macro array and macro have conflict, and macro placement take the constraint on macro array.

**What Next**

Please review the relative location constraint for macro array and macros.

---

## DPP-434

(warning) There is relative location constraints on macro array %s's macro %s. Macro placement take the macro's constraint as the constraint on macro array.

**Description**

Macro placement take the macro's constraint as the constraint on macro array to resolve the conflict between the constraints of macro array and its macros.

Feedback

**What Next**

Please review the relative location constraint for macro array and macros.

---

## DPP-435

(info) Total utilization difference from target in design %s: %.2lf percentage points.

**Description**

The total utilization difference will be count at the top level.

**What Next**

N/A

---

## DPP-436

(error) Macro cell %s is associated with voltage area %s, but is not placed inside its effective voltage area region.

**Description**

Macro cell has its corresponding voltage area, during placement, it should be in right place, otherwise, the error message will be reported.

**What Next**

Please check the design to make sure it is properly placed.

---

## DPP-437

(error) Macro cell %s does not belong to voltage area %s, but the cell is placed inside the unrelated voltage area.

**Description**

Macro cell has its corresponding voltage area, during placement, it should be in right place, otherwise, the error message will be reported.

**What Next**

Please check the design to make sure it is properly placed.

---

## DPP-438

(error) Too many rigid blocks detected in %s.

Feedback

**Description**

This design contains too many rigid subblocks which is beyond the current capabilities of shape_blocks.

**What Next**

Define a rigid boundary for fewer subblocks, or enlarge the boundary of this design to provide more space for placing the rigid subblocks. Verify that padding by channel constraints and inner and outer keepouts don't take up too much space between the subblocks. Also take into account space taken up for achieving block grid alignment. Alternatively, consider placing the blocks manually.

---

## DPP-439

(error) Black box %s 's target utilization or target area or target boundary area is not valid.

**Description**

The black box should have a legal target area and target boundary area, if not, it should have a legal target utilization value which is bigger than 0 and not bigger than 1.

**What Next**

Check the settings of black box's target area and target boundary area, the value should be set, also, if these two values have not been set, the target utilization should be a valid value.

---

## DPP-441

(Warning) Found no block voltage area or power domain for shaping

**Description**

There must be at least one block, voltage area or power domain for shaping

**What Next**

If this is intended to be a hierarchical design and the user wants to run *shape_blocks* to place child blocks, please use *commit_block* command to create the child blocks.

If there are voltage areas which needs the placement of tool, user can use *create_voltage_area*, *shape_blocks* would create voltage area automatically for any power domain that does not have voltage areas. *create_power_domain* will create corresponding power domains.

Feedback

## DPP-442

(Warning) UPF data is invalid.

**Description**

During shaping and placement, valid UPF data is needed.

**What Next**

Please use *check_mv_design* to check the validation of MV setting.

## DPP-443

(Warning) App option plan.place.hierarchy_by_name is off.

**Description**

If the design has no logical hierarchy, there may be potential long run time issue for placement. If the logical hierarchy for the design is flat, it could result in long run time for / fBcreate_placement -floorplan/fp. To avoid the long placement run time, you can set app option "plan.place.hierarchy_by_name" to true

**What Next**

If the design is flat, user can set app option "plan.place.hierarchy_by_name" to true to reduce the run time.

## DPP-444

(Info) Macro %s belongs to RP group.

**Description**

In IC Compiler II, if the user wants to place a set of hard macros together and align them in a particular way, we recommend using the macro array tool to create edit group for the set of hard macros. However, sometimes the user needs to mix hard macros and standard cells in one RP Group, then *create_placement* will handle it during placement stage. For floorplan stage, the user wants to know which macro belongs to RP group and which has no RP ownership.

**What Next**

If the macro belongs to RP group, user must fix it after *create_placement -floorplan*.

## DPP-445

(Info) The number of illegal pin shapes of %s : %d.

Feedback

**Description**

For a legal pin shape, it must meet the following two requirements at the same time: (1) color attribute matched between pin shape and track should be matched, (2) the track center should be aligned with pin shape. If there are illegal pin shape, the report will be generated.

**What Next**

Redo the macro cell library preparation.

---

## DPP-446

(warning) Fixed content is hard to slice for %s

**Description**

The block shaper may have difficulty finding a solution, because it has trouble slicing the fixed contents within the boundary of a block or voltage area.

**What Next**

Check the locations of fixed macros or fixed nested blocks and rearrange them such that content belonging to the same nested block, voltage area or move bound is easier to cover by a rectangle or L-shape. Avoid situations that require more complex shapes (e.g. due to detouring around content of siblings).

---

## DPP-447

(Info) Cannot create child module boundary for %s because there is a module boundary in parent hierarchy %s

**Description**

By default, the command will not create the nested module boundary. So the input logic hierarchy is ignored.

**What Next**

Use *set_app_options -name plan.explore.allow_nested -value true* to force the tool to create nested module boundary.

---

## DPP-448

(Info) Reserved edge space for %d pins and/or ports

Feedback

**Description**

This message describes how much space is going to be kept free of macros along block/movebound/voltage area.

**What Next**

If this value is unexpectedly too low or even 0, check whether the tracks are reaching block boundary; otherwise, the command may interpret that as no space available for pins, and therefore would not attempt to leave any space free of macros. Also note that the command separately handles fixed/placed pins, so they do not count towards this.

---

## DPP-449

(Warning) Cannot group macros of block %s by hierarchy

**Description**

Cannot group macros for the block based on hierarchy either because: 1. This block contains a sub-block which crosses different logical hierarchies, or 2. This block is a sub-block itself which crosses different logical hierarchies.

**What Next**

Check the movebound or voltage area setting to see whether any of them crosses different logical hierarchies.

---

## DPP-450

(info) %s

**Description**

Block shaping has detected overutilization, and has found block instances referring to uneditable blocks. The current boundary of such blocks cannot be shaped, and might be the cause of overutilization.

**What Next**

Use set_editability to make the referred blocks editability, thus enabling shape_blocks to modify their boundary.

---

## DPP-451

(warning) Block shaping will load shaping constraints from Tcl, not from the file "%s".

Feedback

**Description**

Block shaping has been configured to import shaping constraints from Tcl, and therefore ignores the constraints provided by shape_blocks' option "-constraint_file %s".

**What Next**

Use "set_app_options -name plan.shaping.import_tcl_shaping_constraints -value false" to ignore Tcl shaping constraints and load shaping constraints from file.

---

## DPP-452

(warning) Ignoring detected shaping constraints in Tcl, because app_option plan.shaping.import_tcl_shaping_constraints is set to false.

**Description**

Block shaping can import shaping constraints from file or from Tcl, which is controlled by app_option plan.shaping.import_tcl_shaping_constraints. Use "set_app_options -name plan.shaping.import_tcl_shaping_constraints -value true" to have "shape_blocks" import shaping constraints from Tcl.

**What Next**

Use "set_app_options -name plan.shaping.import_tcl_shaping_constraints -value true" to enable importing Tcl shaping constraints during block shaping.

---

## DPP-453

(info) Imported shaping constraints from Tcl (%d constraints, %d channels and %d groups). For details, use app_option plan.shaping.report_import_constraints.

**Description**

This message gives an overview of the shaping constraints from Tcl that have been imported by "shape_blocks".

**What Next**

Use "set_app_options -name plan.shaping.report_import_constraints -value log" to get a detailed report about the imported Tcl shaping constraints during block shaping.

---

## DPP-454

(info) Exported (converted) shaping constraints to Tcl (%d constraints, %d channels and %d groups). For details, use app_option plan.shaping.report_export_constraints.

Feedback

**Description**

This message gives an overview of the shaping constraints that were converted to Tcl by "shape_blocks -constraint_file <name>".

**What Next**

Use "set_app_options -name plan.shaping.report_export_constraints -value log" to report the converted constraints during block shaping in the log. Use "set_app_options -name plan.shaping.report_export_constraints -value file" to write out a Tcl file containing the converted constraints.

---

## DPP-455

(error) %s

**Description**

Failed exporting (converting) shaping constraints.

**What Next**

Please refer to the above message for more information. Also check shape_blocks man page for shaping constraint syntax and examples.

---

## DPP-456

(error) The mutually exclusive app_options plan.shaping.export_tcl_shaping_constraints and plan.shaping.import_tcl_shaping_constraints are both enabled.

**Description**

The app_options plan.shaping.import_tcl_shaping_constraints and plan.shaping.export_tcl_shaping_constraints are mutually exclusive.

**What Next**

To import shaping constraints from tcl, only enable plan.shaping.import_tcl_shaping_constraints. To export (convert) shaping from a file to tcl, only enable plan.shaping.export_tcl_shaping_constraints.

---

## DPP-457

(error) Options %s and %s are mutually exclusive.

**Description**

Found mutually exclusive options.

Feedback

**What Next**

Drop one of the options and rerun the command.

---

## DPP-458

(error) Macros of macro group %s are not in same placeable area.

**Description**

All macros of a macro group must belong to same placeable area, i.e, a physical bloc, a movebound or a voltage area.

**What Next**

Remove macros which belong to a different placeable area and rerun the command.

---

## DPP-459

(error) Macros of macro array %s are not in same placeable area.

**Description**

All macros of a macro array must belong to same placeable area, i.e, a physical bloc, a movebound or a voltage area.

**What Next**

Remove macros which belong to a different placeable area and rerun the command.

---

## DPP-460

(error) Macro %s and macro group %s don't belong to same placeable area.

**Description**

All macros of a macro array must belong to same placeable area, i.e, a physical bloc, a movebound or a voltage area.

**What Next**

You cannot add the macro to this group.

---

## DPP-461

(warning) macro group with name %s already exists.

Feedback

**Description**

Macro group name must be unique in its block.

**What Next**

You cannot create two macro groups with same name.

---

## DPP-462

(warning) Macros don't belong to same placeable area.

**Description**

Macros of a macro group must belong to same placeable area, i.e., a physical block, a move bound, or a votage area.

**What Next**

You cannot create macro group with these macros.

---

## DPP-463

(warning) Cell %s is not a hard macro. Ignored.

**Description**

The given cell is not a hard macro cell.

**What Next**

This cell will be ignored.

---

## DPP-464

(error) Group %s has no macros, can not set boundary of this type.

**Description**

The given macro group has no macros, so the shape can not be auto-determined.

**What Next**

Add macros to the group, or change type to contour and add boundary list of points.

---

## DPP-465

(error) There is no such macro group.

Feedback

**Description**

The specified object is not a macro group.

**What Next**

Make sure the specified object is a macro group.

---

## DPP-466

(error) If -type is contour, boundary must be provided.

**Description**

The boundary list of points was not provided, but it is required when -type is contour.

**What Next**

Provide the list of points, or change type to something other than contour.

---

## DPP-467

(error) If -type is not contour, boundary can not be provided.

**Description**

When -type is not contour, the tool will automatically calculate the boundary, so the list of boundary points can not be provided.

**What Next**

Remove the list of points, or change type to contour.

---

## DPP-468

(error) Specified shape can have only horizontal and vertical edges.

**Description**

The specified shape contains edges that are neither horizontal nor vertical. This is not allowed.

**What Next**

Change the shape so that each edge is horizontal or vertical.

---

## DPP-469

(warning) Macro groups %s and %s don't match and cannot be cloned..

Feedback

**Description**

Macro groups must match each other to be cloned.

**What Next**

Use et_macro_group_packing_clone_candidates command to get clone candidates.

---

## DPP-470

(warning) Cell %s is ignored because %s.

**Description**

The cell is not valid.

**What Next**

No futher action is required.

---

## DPP-478

(warning) Usage of repelling group bounds%s with more than %d cores is still experimental.

**Description**

Repelling group bounds of two cores is supported for DCLS (Dual Core Lock Step). The use of more than two cores within one repelling group bound is still experimental.

**What Next**

The use of more than two cores may have an impact on QoR.

---

## DPP-479

(error) Block instance %s is out of the die area of design block %s.

**Description**

Block instance is out of the die area of design block.

**What Next**

Check block placement.

---

## DPP-480

(warning) Macro %s will be removed from movebound %s and added to macro group %s.

Feedback

**Description**

Macro group is as sub type of movebound. One macro can only belong to one movebound at one time. If a macro belonging to a movebound is added to a macro group, it will be removed from orignal movebound.

**What Next**

None.

---

## DPP-481

(warning) Macro %s will be moved from macro group %s to a new macro group %s.

**Description**

A macro can only belong to one macro group at one time. If it is added to a new macro group, it will be removed from old macro group.

**What Next**

None.

---

## DPP-482

(info) All macors are fixed, no macro group will be created.

**Description**

All macros are fixed, the command will not create any macro group.

**What Next**

None.

---

# DPPA

---

## DPPA-001

(warning) Failed to %s %s %s

**Description**

place_pins failed to place or legalize (in legalization mode) this pin.

Feedback

### What Next

This may be due to pin constraints setting error or internal data error. Please check pin constraint settings. In legalization mode, a legal slot cannot be found for the pin and it is not legalized as a result. This can be due to insufficient slot/track resources or pin constraint settings.

### See Also

- create_pin_blockage
- create_pin_guide
- create_routing_corridor
- create_topological_constraint
- read_pin_constraints
- set_block_pin_constraints
- set_bundle_pin_constraints
- set_individual_pin_constraints

---

## DPPA-002

(Error) Failed to place some of the block pins.

### Description

place_pins failed to place some pins.

### What Next

This may be due to the fact that there are no enough slots available to assign all the pins according to the pin constraints. Please check the pin constraint setting and make sure that there are enough slots available to place pins.

### See Also

- create_pin_blockage
- create_pin_guide
- create_routing_corridor
- create_topological_constraint
- read_pin_constraints

Feedback

- set_block_pin_constraints
- set_bundle_pin_constraints
- set_individual_pin_constraints

## DPPA-003

(warning) Not enough slots for %s %s of net %s on abutting edges between blocks %s and %s. Ignoring pin spacing constraints...

### Description

place_pins failed to place this pin while honoring pin spacing constraints.

### What Next

This may be due to the fact that there are no enough slot available to assign all the abutted pins according to the pin constraints. Please check the pin constraint setting and make sure that there are enough slots available to place abutted pins.

### See Also

- create_pin_blockage
- create_pin_guide
- create_routing_corridor
- create_topological_constraint
- read_pin_constraints
- set_block_pin_constraints
- set_bundle_pin_constraints
- set_individual_pin_constraints

## DPPA-004

(Warning) Not enough slots available for pin placement. Ignore pin spacing constraints.

### Description

place_pins failed to place some pins while honoring pin spacing constraints. place_pins will relax spacing constraints and place those pins.

Feedback

### What Next

This may be due to the fact that there are no enough tracks available to assign all the pins according to the pin constraints. Please check the pin constraint setting and make sure that there are enough tracks available to place pins.

### See Also

- create_pin_blockage
- create_pin_guide
- create_routing_corridor
- create_topological_constraint
- read_pin_constraints
- set_block_pin_constraints
- set_bundle_pin_constraints
- set_individual_pin_constraints

---

## DPPA-005

(Error) There are no valid blocks defined.

### Description

place_pins doesn't find any blocks or the blocks don't have valid boundaries.

### What Next

Command place_pins places block pins but it could not find any valid blocks. Please define block or set valid boundary for the block before perform place_pins.

### See Also

- commit_block
- initialize_floorplan

---

## DPPA-006

(Warning) There are unrecognized ports or pins on net %s.

### Description

The net reported is connected to a port or pin that may be corrupted.

Feedback

### What Next

This may be due to a bad netlist or an internal data error. Please check the net and make sure it is correct.

### See Also

- check_design
- report_nets

---

## DPPA-007

(Warning) Net %s contains a hierarchical loop.

### Description

A hierarchical loop is detected for the net reported.

### What Next

Please check the netlist connection use report_nets and make sure it is correct.

### See Also

- check_design
- report_nets

---

## DPPA-008

(Warning) There are unrecognized nets connected to net %s.

### Description

The net reported is not correctly connected to its upper hierarchy or lower hierarchy.

### What Next

This may be due to a bad netlist or an internal data error. Please check the netlist and input data and make sure it is correct.

### See Also

- check_design
- report_nets

Feedback

## DPPA-009

(Warning) There are routing errors on net %s.

**Description**

The net reported is not routed correctly.

**What Next**

This may be due to a bad cell placement. Please check that all cells connected to the net are placed within or outside of the right corresponding boundaries using command report_placement -physical_hierarchy_violations.

**See Also**

- report_placement

## DPPA-010

(Warning) There are routing errors on net %s.

**Description**

The net reported is not routed correctly.

**What Next**

This may be due to a bad cell placement. Please check that all cells connected to the net are placed within or outside of the right corresponding boundaries using command report_placement -physical_hierarchy_violations.

**See Also**

- report_placement

## DPPA-011

(Warning) Net %s is not routed.

**Description**

The net reported is not routed.

**What Next**

Please check the routing of the net. Make sure the net is routed.

Feedback

**See Also**

- place_pins
- route_global
- route_group

---

## DPPA-012

(Warning) The driver of the net %s is not connected to the route.

**Description**

The net reported is not fully routed.

**What Next**

Please check the routing of the net. Make sure the net is fully routed.

**See Also**

- place_pins
- route_global
- route_group

---

## DPPA-013

(Warning) There are floating ports or pins on net %s.

**Description**

The global router tries to connect a port or pin on the reported net. However, the port or pin is not hierarchically connected to the net.

**What Next**

This may be due to a bad netlist or an internal data error. Please check the routing of the net; netlist and the input data

**See Also**

- place_pins
- route_global
- route_group

Feedback

## DPPA-014

(Warning) There are errors when analyzing the routing on net %s.

**Description**

Netlist analysis failed for the indicated net.

**What Next**

Please check the net and report this error.

**See Also**

- place_pins
- route_global
- route_group

## DPPA-015

(Warning) There is a mismatch between the routing and the hierarchical netlist of net %s.

**Description**

The physical routing connects objects that are not hierarchically connected to the net reported.

**What Next**

This may be due to a bad netlist, bad placement or an internal data error. Please check the routing of the net and the netlist. Or use command report_placement -hierarchical to check if there is any hierarchical error in placement.

**See Also**

- check_design
- place_pins
- report_placement
- route_global
- route_group

Feedback

## DPPA-016

(Warning) Floating routes found on net %s.

**Description**

There exist routes that do not connect to any objects. This may be because of bad netlist, incorrect placement or other errors.

**What Next**

Please check the routing of the net and the netlist. Please check the placement using command report_placement -hierarchical.

**See Also**

- check_design
- place_pins
- report_placement
- route_global
- route_group

## DPPA-017

(Warning) Failed to derive feedthrough port direction for net %s.

**Description**

The tool cannot decide the feedthrough port direction on the specified net.

**What Next**

The feedthrough port direction is not set correctly during pin placement or the port direction was manually edited but not set correctly.

**See Also**

- place_pins

## DPPA-018

(Warning) Incomplete feedthrough creation on net %s.

Feedback

**Description**

Only a fraction of feedthrough ports are created on the specified net.

**What Next**

There is a mismatch between the routing and the netlist such that the ports or pins the net connects to does not match the physical routing. Please check the routing of the net and the netlist.

**See Also**

- place_pins
- route_global

---

## DPPA-019

(Warning) Cannot find the driver of net %s.

**Description**

The driver of the net cannot be identified. The feedthrough ports created on this net will be inout direction.

**What Next**

Please check the routing of the corresponding net and make sure inout feedthroughs are allowed.

**See Also**

- place_pins
- route_global

---

## DPPA-020

(Warning) Net %s contains routing loops.

**Description**

The routing of the indicated net is incorrect because it contains loops. The feedthrough creation on this net is skipped.

**What Next**

Please check and fix the routing of the net by removing the loop.

Feedback

**See Also**

- place_pins
- route_global

---

## DPPA-021

(Warning) Cannot find the driver of net %s.

**Description**

The driver of the net cannot be identified. The feedthrough ports created on this net will be inout direction.

**What Next**

Please check the routing of the corresponding net and make sure inout feedthroughs are allowed.

**See Also**

- place_pins

---

## DPPA-022

(Warning) The driver of net %s connects to an input port.

**Description**

The driver of the reported net is connected to an input terminal. Any feedthrough ports created on this net will be inout ports.

**What Next**

Check the net and make sure that inout feedthrough port is allowed.

**See Also**

- place_pins
- route_global

---

## DPPA-023

(Warning) Cannot place pins or create feedthroughs based on the global routing for net %s.

Feedback

### Description

The net reported is not fully routed or the routing does not follow the design hierarchy. Possible reasons are: there are missing terminals or pin shapes on this net, there are terminals or pin shapes on a non-routing layer, or there are unplaced cell that this net is connected to. Feedthrough creation is skipped for this net. Pins on this net will be placed based on connectivity rather than global routing results.

### What Next

Please fix the routing problem.

### See Also

- route_global
- route_group

---

## DPPA-024

(Warning) Found corrupted ports or pins.

### Description

There are corrupted ports or pins in the design.

### What Next

Please check the netlist. Possible examples of corrupted ports or pins can be unrecognized data of one or more ports or pins from the database. Make sure the database is saved correctly from previous session and not modified manually.

### See Also

- check_design

---

## DPPA-025

(Error) Internal error found on design.

### Description

Current operation cannot proceed due to unexpected data in feedthrough creation step.

### What Next

Please check the log to see if there are other error or warning messages right before this one. Fixing those errors may resolve this error.

Feedback

## DPPA-026

(warning) pin guide or route corridor is conflict with pin blockage or routing blockage or side constraint or layer constraint or topological constraint for %s %s, the pin guide or route corridor constraint will be ignored on this pin

### Description

There is a conflict beween pin guide or route corridor and the pin constraints as reported by this message. The pin guide or route corridor is therefore ignored.

### What Next

This may be due to the exclusive pin guide, pin blockage, routing blockage, side constraint, layer constraint or topological constraint conflicts with the pin guide or route corridor. Please fix the conflicting constraints.

### See Also

- create_pin_blockage
- create_pin_guide
- create_routing_corridor
- create_topological_constraint
- read_pin_constraints
- set_individual_pin_constraints
- set_block_pin_constraints
- set_bundle_pin_constraints

## DPPA-027

(warning) %s %s connected to net %s which connects to more than 500 ports on blocks

### Description

There are too many block pins connected to the same net. This is more likely a user error as too many block pins connect to the same net will usually lead to low quality of result of pin placement.

### What Next

This may be due to tie high or tie low net that is not created properly. Please check original net list or check how the tie nets are created.

Feedback

**See Also**

- check_mv_design
- connect_pg_net

---

## DPPA-028

(Warning) No physical data defined.

**Description**

The tool cannot find the physical data from the database. Please check the design input.

**What Next**

Please check the design input and make sure that the physical data exists before running this command.

**See Also**

- write_pin_constraints

---

## DPPA-029

(Warning) The %s %s is in conflict with side or layer pin constraint on block %s.

**Description**

A constraint conflict is detected. The pin guide or routing corridor's overlapping portition with a block is on a side which is excluded from the block or individual pin constraints. Or the block or individual pin has a layer constraint that does not overlap with pin guide or routing corridor's layers. If the conflict exists during pin placement, pin placement will honor pin guide or routing corridor based constraint.

**What Next**

Please check the settings of the pin guide or routing corridor and the conflicting block or individual pin constraints and remove the conflict.

---

## DPPA-030

(Warning) There is a conflict between constraints of %s %s and %s %s which are connected together.

Feedback

### Description

The two connected block pins have pin constraints either from pin guide or from individual pin constraints on abutted edges but they do not overlap. As such, the pin placement and global route cannot find a valid location and layer to honor both pin constraints.

### What Next

Please check the constraints settings and remove the conflict.

### See Also

- create_pin_blockage
- create_pin_guide
- create_routing_corridor
- create_topological_constraint
- read_pin_constraints
- set_individual_pin_constraints
- set_block_pin_constraints
- set_bundle_pin_constraints

---

## DPPA-031

(warning) net %s on block %s has forbidden feedthrough implied by a pin guide

### Description

This is probably because you have feedthroug implied by a pin guide on a net, however, the block cell or the net has feedthrough not allowed. This is feedthrough constraint conflict and feedthrough constraint from block cell or net will be ignored and pin guide will be honored.

### What Next

Please remove the feedthrough conflict and try again.

---

## DPPA-032

(warning) found pin guide and corner keepout constraint conflict for %s %s, all the pin guide constraint will be honored for this port

Feedback

**Description**

place_pins found pin guide corner keepout contraint conflict

**What Next**

Please re-create pin guide or reset corner keepout constraints so that they are NOT conflicting with each other

---

## DPPA-033

(warning) for the net %s, pin guide will not be honored because the number of intersections between the pin guide and block %s is larger than the number of top level connections for that net.

**Description**

For each pin guide, if you have multiple intersection with a block, it implies feedthrough guide. In order for a feedthrough guide to work properly, the number of intersections between this pin guide and a block should be smaller or equal than number of top level connections for that net.

**What Next**

Please re-create pin guide so that it meets this requirement.

---

## DPPA-034

(warning) net %s has pin guide which has only one intersection between pin guide and block cell %s while this net doesn't connect to this block cell, pin guide will be ignored

**Description**

Since it has only one intersection between pin guide and block cell and the net doesn't connect to this block cell, the pin guide is considered NOT created properly since there is NO pin associated with this pin guide.

**What Next**

Please re-create pin guide so that it meets this requirement.

---

## DPPA-035

(warning) Pin Guide %s is applied to the current design. Ignoring pin guide %s during push-down operation.

Feedback

**Description**

Since the pin guide's parent is the current design, push down operation will not push down this pin guide

**What Next**

Please re-create pin guide so that it meets this requirement.

---

## DPPA-036

(warning) pin guide %s has no intersection with its parent cell %s. Ignoring pin guide %s.

**Description**

The specified pin guide is outside of the bounding box of its parent cell. Push down or pin placement command will ignore this pin guide.

**What Next**

Please re-create pin guide so that it meets this requirement.

---

## DPPA-037

(warning) The %s associated with pin guide %s have no connection to the parent cell %s. Ignore the pin guide %s during push-down operation

**Description**

The associated physical pins or nets of the specified pin guide have no connection to the pin guide's parent cell.

**What Next**

Please re-create pin guide so that it meets this requirement.

---

## DPPA-038

(warning) No %s are popped up since command pop_up_objects does not support %s objects.

**Description**

The warning message was issued because site rows or tracks are in the object list for pop up, but these objects are not supported for command pop_up_objects

**What Next**

Please check the man page for command pop_up_objects.

Feedback

## DPPA-039

(warning) The pin %s %s associates with physical pins. Ignore this pin %s for pop up operation.

**Description**

During pin guide or pin blockage pop up, command pop_up_objects only handles the pin guides and pin blockages that associate with ports or nets.

**What Next**

Please re-create pin guide or pin blockage so that it meets this requirement.

## DPPA-040

(warning) The nets associated with pin %s %s either have no port connection or are feedthrough nets. Ignore this pin %s during pop up operation

**Description**

The associated nets of the specified pin guide or pin blockage have no port connection, or the associated nets are feedthrough nets. Command pop_up_objects only pops up pin guides or pin blockages that associate with ports or associate with non-feedthrough nets that have port connections.

**What Next**

Please re-create pin guide or pin blockage so that it meets this requirement.

## DPPA-041

(warning) %s objects will not be removed from block level since only partial MIB instances are selected for objects pop-up.

**Description**

The warning message is issued during popping up objects from MIB designs. Because user selects partial MIB instances for objects pop-up and sets block action to "remove". During pop-up, the block action "remove" is ignored and objects inside block will not be removed

**What Next**

Please read man page of pop_up_objects for details.

Feedback

## DPPA-042

(error) block cell or top level design %s doesn't have valid boundary or the boundary is not a rectilinear shape

**DESCRIPTION Command cannot find valid boundary for the block cell**

or top level design. This might be due to the un-setted boundary or the boundary is not a rectilinear boundary for a block cell of top level design. Global route and pin placement will skip such block cell or top level design.

**What Next**

Please set valid boundary for the block cell or top level design.

## DPPA-043

(error) only one of -pins, -nets, -ports, -bundles, -cells and -self options can be specified

**Description**

Command only accept one of the -pins, -nets, -ports, -bundles, -cells and -self options.

**What Next**

Please select only one of the -pins, -nets, -ports, -bundles, -cells and -self options.

**See Also**

- place_pins

## DPPA-044

(Warning) %s objects will not be pushed down due to %s mis-alignment among MIB instances.

**Description**

In case of mis-alignment, the default behavior of command push_down_objects is to issue warning message and do not push down the objects. User can set "-ignore_misalignment" option to true via command set_push_down_object_options.

**What Next**

Please fix the mis-aligned objects or set "-ignore_misalignment" to true using command set_push_down_object_options.

Feedback

## DPPA-045

(Warning) Pin locations are not updated because block boundary is incorrect.

**Description**

The warning message is issued because user streching the block boundary to an illegal shape.

**What Next**

Please change the boundary to the correct shape.

## DPPA-046

(Warning) Pin overlap occured.

**Description**

The warning message is issued because pins cannot be legally placed due to the block shape being changed and no enough slots.

**What Next**

Please re-shape block or run place_pins.

## DPPA-047

(Error) cannot open file %s

**Description**

The error message is issued because the file cannot be opened with the given file name. This could be there is no direcotry for the name. For instance, if you specify "dir/file_name" for the file name, but there is no "dir" under current directory, this error will be issued.

**What Next**

Please check if the file name with directory is correct.

## DPPA-048

(warning) The pin guide %s has parents. Ignore this pin guide for pop up operation.

**Description**

During pin guide pop up, command pop_up_objects only handles the pin guides that associate with ports or nets.

Feedback

### What Next

Please re-create pin guide or pin blockage so that it meets this requirement.

---

## DPPA-049

(Warning) Cannot find valid %s %s at line %d.

### Description

The command cannot find valid object when parsing pin constraints file. The object type could be cell, pin, port, net or bundle.

### What Next

Check if the object name and object is valid

### See Also

- read_pin_constraints

---

## DPPA-050

(warning) find invalid number %s at line %d

### Description

The command expect a number but cannot find it when parsing pin constraints file. This could happen when the command try to get a side number for specified "side" constraint. Or when try to get a pin offset location from "offset" constraints. Or when try to get a starting location or ending location for pin location range or pin order within a location range with "start" or "end" locations.

### What Next

Check if the specified number is valid or not.

### See Also

- read_pin_constraints

---

## DPPA-051

(warning) find invalid layer name %s at line %d

Feedback

### Description

The command expect a layer name but cannot find valid one when parsing pin constraints file. This could happen when the command try to get layer names for "layer" constraints.

### What Next

Check if the specified layer name is valid or not.

### See Also

- read_pin_constraints

---

## DPPA-052

(warning) no feedthrough constraint is specified at line %d

### Description

The command expect a feedthrough constraint statement (Feedthrough or NoFeedthrough) when parsing pin constraint file but could not find the constraint statement.

### What Next

Check if the feedthrough constraint statement is missing.

### See Also

- read_pin_constraints

---

## DPPA-053

(warning) no block cell is specified at line %d

### Description

The command expect a block cell name when parsing pin constraint file but could not find the valid block cell name

### What Next

Check if the cell name missing or incorrect.

### See Also

- read_pin_constraints

Feedback

## DPPA-054

(warning) invalid constraints specified at line %d

### Description

The command expect a valid pin constraints (layer, side, offset, spacing, order, pin range...) specified, but cannot parse such constraints when reading pin constraints file.

### What Next

Check if the specified constraint is valid or not at specified line.

### See Also

- read_pin_constraints

## DPPA-055

(warning) invalid %s constraint is specified

### Description

The command expect failed to parse a valid pin constraints (layer, side, offset, spacing, order, pin location range...).

### What Next

Check if the specified constraint is valid or not.

### See Also

- read_pin_constraints

## DPPA-056

(warning) side number %d is larger than number of sides for block %s, side constraint will be ignored.

### Description

A side number is specified for side or exclude_side constraints, but this number is larger than the total number of side on selected block.

### What Next

Correct the side number then reset the constraint again.

Feedback

**See Also**

- read_pin_constraints

---

## DPPA-057

(Warning) The mixed feedthrough ports of net %s inside cell instance %s all have OUT direction. Skip feedthrough removal on this net.

**Description**

The specified net is a mixed feedthrough net without any IN or INOUT feedthrough ports. Buffers may have been inserted into this net. Please check the man page to see if buffered feedthrough removal is supported

**What Next**

Check the man page of remove_feedthroughs on the support and usage of buffered feedthrough removal.

---

## DPPA-058

(warning) Found %s constraint assoicated with %s %s conflict with non-default routing rule on net %s, the non-default routing rule will be honored.

**Description**

There is a conflict between individual pin constraint and non-default routing rule, pin placement will honor the non-default routing rule when place pins.

**What Next**

Please check pin constraints on associated pins or nets and non-default routing rules

---

## DPPA-059

(warning) invalid key word "%s" is specified at line %d

**Description**

The command expect failed to parse a valid pin constraints key word (layer, side, offset, spacing, order, pin location range, cell, pin, port...).

**What Next**

Check if the specified constraint is valid or not.

Feedback

**See Also**

- read_pin_constraints

---

## DPPA-060

(warning) %s %s's offset constraint is longer than length of side of block %s.

**Description**

The offset constraint set on the pin or net or bundle is longer than block's side length.

**What Next**

Check if the specified constraint is valid or not.

**See Also**

- read_pin_constraints

---

## DPPA-061

(warning) unrecognized pin constraints statement detected in line %d.

**Description**

The command expects pin constraints statement, but cannot find a valid one which ends with ";".

**What Next**

Check if the pin constraints specified correctly.

**See Also**

- read_pin_constraints

---

## DPPA-062

(warning) invalid pin constraints at line %d.

**Description**

The command expect a valid pin constraints, but cannot parse the pin constraints statement.

**What Next**

Check if the pin constraints specified correctly.

Feedback

### See Also

- read_pin_constraints

---

## DPPA-063

(Error) MIB instance misalignment prevents the push-down of wire tracks. push_down operation will terminate.

### Description

The command encountered a situation where one or more wire tracks were found to be aligned differently for different MIB instances, and the user did not instruct the command to ignore misalignment of MIB instances. In this situation, the command will not push down the wire tracks and will announce an error and prematurely exit.

### What Next

Correct the MIB misalignment, or call the command with the option to ignore MIB misalignment.

---

## DPPA-064

(Error) Partial MIB selection encountered within the specified collection of block instances. Command will stop.

### Description

The command encountered a collection of block instances that included one or more MIB block instances, and at least one MIB reference block was instantiated only one time. If an MIB instance of a particular MIB reference block is specified, the command must perform MIB alignment checks. These checks require at least two MIB instances of each MIB block reference.

### What Next

Correct the specified collection of MIB block instances on the command line.

---

## DPPA-065

(Warning) MIB instance %s was singly specified within it's MIB group. MIB alignment checks will not be performed for this block instance.

### Description

The command encountered a situation where the command detected that a single MIB instance from among its MIB group was specified for push-down. When this occurs,

Feedback

MIB alignment checks cannot be made because at least 2 are required to perform this alignment check. The result is that the alignment check is not run for this particular instance. This may result in problems later if other MIB instances are not correctly aligned with this instance. Specification of singly-selected MIB block instances is discouraged.

**What Next**

Make sure that this MIB instance is correctly aligned with it's siblings.

---

## DPPA-066

(Warning) The following command-line-specified MIB block instances of the MIB reference block %s were specified partially (not all instances of the MIB group were included): \n%s \n MIB alignment checks will be run for this MIB group, but only on the partially specified MIB block instances\n\n.

**Description**

The command encountered a situation where the command detected that some, but not all, of MIB group of instances were specified on the command line. MIB alignment checks will only be made for those instances that were specified on the command line. This means that there is the potential for any omitted instance of the MIB group to be misaligned with the specified MIB instances.

**What Next**

Make sure that any non-specified MIB instance are correctly aligned with the ones that were specified.

---

## DPPA-067

(Error) push_down_objects does not explicitly support wire tracks. They are indirectly and automatically pushed down with cell rows.

**Description**

The command encountered a collection of wire tracks passed to the command. push_down_objects does not explicitly support wire tracks. Wire tracks are automatically created in a block when cell rows are specified for push-down.

**What Next**

Do not include a collection of wire tracks when calling push_down_objects. Instead, push down cell rows, which will automatically push down wire tracks.

Feedback

## DPPA-068

(Error) No physical block defined in the design.

**Description**

The command did not find any physical blocks in the design.

**What Next**

Make sure you have physical blocks before calling this command.

## DPPA-069

(Error) Failed to create and connect port-connected child net %s into block %s

**Description**

The command failed during push-down of cells to make child level connections

**What Next**

This is an internal error. Contact Synopsys.

## DPPA-070

(Error) Failed to create and connect tapped port-connected child net %s into block %s

**Description**

The command failed during push-down of cells to make child level connections on an existing child net

**What Next**

This is an internal error. Contact Synopsys.

## DPPA-071

(Warning) multi-tap cell connections not yet supported (net %s into block %s)

**Description**

The command encountered a cell that drives a block on multiple existing taps. This scenario is not yet supported by push_down_objects.

**What Next**

This is an internal warning. Contact Synopsys.

Feedback

## DPPA-072

(Warning) cannot find end statement for %s

### Description

The command expect an end constraint statement but couldn't find one. This may due to missing semi-column or incorrect statement. For topological constraint, please specify "END TOPOLOGICAL MAP;". For feedthrough control, please specify "END FEEDTHROUGH CONTROL;". For physical pin constraints, please specify "END PHYSICAL PIN CONSTRAINTS;". For pin spacing control, please specify "END PIN SPACING CONTROL;".

### What Next

Check to see if the end constraint statement correct.

### See Also

- read_pin_constraints

## DPPA-073

(Warning) Option "-use_existing_routing" is specified , however at least some of global routing is not present in the database.

### Description

When option "-use_existing_routing" specified, the tool expect the design has been routed already. If the nets are not routed , pin locations will not be determined based on routing , and will likely be non-optimal. It is recommended to re-run place_pins without using "-use_existing_routing" option.

### What Next

Check to see if the end constraint statement correct.

## DPPA-074

(Warning) Unrecognized pin constraints syntax at line %d

### Description

When parsing pin constraints file, the tool expect a valid pin constraints statement but cannot find one. Please use one of the following statement: START TOPOLOGICAL MAP; END TOPOLOGICAL MAP; START FEEDTHROUGH CONTROL; END

Feedback

FEEDTHROUGH CONTROL; START PHYSICAL PIN CONSTRAINTS; END PHYSICAL PIN CONSTRAINTS; START PIN SPACING CONTROL; END PIN SPACING CONTROL;

**What Next**

Check to see if the pin constraints statement correct or not

**See Also**

- read_pin_constraints

---

## DPPA-075

(Warning) find curly bracked mis-match pin constraints statement at line %d

**Description**

The tool expect pin constraint statement with matched curly bracket. This is probably because user input error or typo in the statement.

**What Next**

Check to see if the pin constraints statement correct or not

**See Also**

- read_pin_constraints

---

## DPPA-076

(Warning) push_down_objects has pushed cells into a MIB reference block %s for block instance %s. No other MIB block instances of this ref block will be considered for pushing down cell instances during this call to push_down_objects.

**Description**

The push_down_objects only allows pushing cells into multiply-instantiated block instance for one instance. This affects all other instances internally and for their port interfaces. Any other instances specified or encountered which have cells to be pushed down will be disregarded for push-down of cells during this command call

**What Next**

Be sure to only attempt to push cells into a single instance of a group of multiply-instantiated blocks (MIBs).

Feedback

## DPPA-077

(Warning) Pin location coordinate for %s %s is outside of block cell boundary and will be ignored.

### Description

If pin location is not inside block boundary, pin placement will ignore this constraint. If the pin location is specified inside block boundary and if the pin is not an off-edge pin, pin placement will snap the location to closest edge and wire track

### What Next

Please check pin location coordinate and make it inside block boundary

### See Also

- set_individual_pin_constraints

## DPPA-078

(Warning) A routing segment of block internal net %s with bounding box (in child) {{%s %s} {%s %s}} extends outside of the destination block boundary.

### Description

A net which is a block-internal net has a wire segment which extends outside of the boundary of a block that a top level cell instance is being pushed into.

### What Next

Modify your routes so that all block-owned net paths, contacts, and contact arrays are within the block boundaries. Or re-route this net in the child so that the routing is contained within the child block boundary.

## DPPA-079

(Warning) A contact of block internal net %s with bounding box (in child) {{%s %s}{%s %s}} extends outside of the destination block boundary.

### Description

A net which is a block-internal net has a contact which extends outside of the boundary of a block that a top level cell instance is being pushed into.

Feedback

### What Next

Modify your routes so that all block-owned net paths, contacts, and contact arrays are within the block boundaries. Or re-route this net in the child so that the routing is contained within the child block boundary.

---

## DPPA-080

(Warning) Find current routing topology does not match with topological constraints for net %s

### Description

The tool find that there is a mis-match between current routing results with topological constraints. This could because the topological constraints are conflict with other pin constraints or the topological constraints have errors in it

### What Next

Check the topological constraints and other pin constraints to see if they are conflict.

### See Also

- read_pin_constraints

---

## DPPA-081

(Warning) net %s violates topological constraints by not having a direct connection between %s %s and %s %s

### Description

The tool cannot honor topological constraints which has direct connection between two objects. This could happen due to bad constraints (topological constraints conflict with other pin constraints or routing layer direction is conflict with side direction) or due to bad design data (unplaced cells/pins/ports or block pins are placed and fixed outside of block boundary)

### What Next

Check the topological constraints and other pin constraints to see if they conflict, check design data to see if there are unplaced objects assoicated with this topological constraint or placed/fixed block pins outside owner block's boundary

### See Also

- read_pin_constraints

Feedback

## DPPA-082

(Warning) allowed routing layers range is different from pin constraint layer on cell %s, common layer range will be used

### Description

The tool detects that the minimum and maximum routing layers are different from pin constraint layers, the tool will use common layers for pin placement purpose

### What Next

Check the minimum and maximum routing layer and also pin constraint layers for this cell and see if this is intended

### See Also

- read_pin_constraints
- set_individual_pin_constraints
- set_block_pin_constraints

## DPPA-083

(Error) there is no common layer range between allowed routing layers and pin constraint layer on cell %s

### Description

The tool detects that theres is no overlap layer range between minimum and maximum allowed routing layers and pin constraint layers, the tool will not proceed until there are common routing layers between those two constraints

### What Next

Remove the conflict between the minimum and maximum allowed routing layer and pin constraint layers for this cell

### See Also

- read_pin_constraints
- set_individual_pin_constraints
- set_block_pin_constraints

Feedback

## DPPA-084

(Error) %s does not yet support the %s object type.

**Description**

The command was passed an object type that is not yet supported by the command.

**What Next**

Do not include this object type when calling this command.

## DPPA-085

(Warning) layer and sides topological constraints are conflict for net %s

**Description**

The command detect that there is no preferred layer when using the side specified from topological constraint, topological constraint will be ignored

**What Next**

Please check the conflict from layer constraint and side constraint

**See Also**

- read_pin_constraints
- create_channel_congestion_map

## DPPA-086

(Error) Pin %s has off-edge constraint, but there is no location or pin guide constraint associated with this pin.

**Description**

The command expects a pin guide constraint or a pin location constraint for this pin but cannot find those constraints. For off-edge pin placement, user has to specify either a pin guide or a pin location constraint.

**What Next**

Please add pin guide or pin location constraint for this pin.

Feedback

**See Also**

- read_pin_constraints
- set_individual_pin_constraints
- set_block_pin_constraints
- create_pin_guide

---

## DPPA-087

(Warning) push_down_objects has encountered cells specified to be pushed into MIB reference block %s for block instance %s, but the reference block had already had cells pushed into it through a previously processed MIB instance. No other MIB block instances of this ref block are considered for pushing down cell instances during this call to push_down_objects.

**Description**

The push_down_objects only allows pushing cells into multiply-instantiated block instances for one instance. This affects all other instances internally and for their port interfaces. Any other instances specified or encountered which have cells to be pushed down will be disregarded for push-down of cells during this command call.

**What Next**

Be sure to only attempt to push cells into a single instance of a group of multiply-instantiated blocks (MIBs).

---

## DPPA-088

(Error) Failed to punch ports for net %s and pin %s.

**Description**

The push_down_objects attempted to punch ports to a connection in another hierarchical level and failed.

**What Next**

Report this problem

---

## DPPA-089

(Warning) %d pins on block %s have been placed with relaxed %s constraint(s).

Feedback

**Description**

The command relaxed some constraints in order to place those pins. For the pin spacing constraint, it might not be accurate as the pins that already placed without relaxing spacing may still have spacing violation if later placed pins violated pin-to-pin spacing. Please use check_pin_placement to get the pin list which has pin constraint violations.

**What Next**

Please use check_pin_placement to get more information on pins that violate pin constraint.

**See Also**

- read_pin_constraints
- set_individual_pin_constraints
- set_block_pin_constraints
- check_pin_placement

---

## DPPA-090

(Warning) %d pins on block %s failed to place.

**Description**

The command failed to place pins. This may be because the pin constraint conflict, or no enough wire tracks available.

**What Next**

Please check pin constraints and also available wire tracks.

**See Also**

- read_pin_constraints
- set_individual_pin_constraints
- set_block_pin_constraints

---

## DPPA-091

(Warning) Cannot create feedthrough routing for topological constrains on net %s

Feedback

**Description**

The command detects that the topological constraint specifies the routing topology to connect to a block which doesn't have original connection to the net. However place_pins cannot find other pins that can be used to create pure feedthrough routing on this block. This may be because the topological constraint specified incorrectly.

**What Next**

Please check topological constraints

**See Also**

- read_pin_constraints

---

## DPPA-092

(Error) Failed when trying to re-use exising tap-connected portName %s.

**Description**

The push_down_objects command attempted to trace a path across pushed-down buffers in an attempt to re-use a tap connection (existing connection to the block) name, and experienced a problem.

**What Next**

Report this problem

---

## DPPA-093

(Warning) Timing library is not loaded. Unable to determine if reference module %s is a buffer or inverter.

**Description**

The push_down_objects command, when attempting to traverse buffer chains for tap-connected port name re-use, discovered that the timing library was not loaded, so it cannot determine which instances are buffers or inverters.

**What Next**

Please change your library set-up to include timing libraries.

---

## DPPA-094

(Warning) Layer and sides pin constraints are conflict for pin %s.

Feedback

### Description

The command detect that there is no preferred allowed layer when using the side specified from pin constraint. If user intends to honor both side and layer constraints even the layer is not a preferred layer direction, user should set location and layer hard constraints in set_block_pin_constraints. If there is no hard constraints, the layer constraint will be ignored.

### What Next

Please check the conflict from layer constraint and side constraint.

### See Also

- read_pin_constraints
- set_individual_pin_constraints
- set_block_pin_constraints

---

## DPPA-095

(Warning) Pin %s has off edge pin constraint that is conflict with bundle pin constraint, bundle pin constraint will be ignored for this pin.

### Description

The command detects that the pin has off edge pin constraint and also bundle pin constraint. When user set bundle pin constraints to be placed together, the tool expect to place all the bundle pins together on block edges. When there is off edge constraint for a bundle pin, the tool will honor off edge pin constraint and place rest of bundle pins together.

### What Next

Please check if the off edge pin constraint and bundle pin constraints are set correctly.

### See Also

- read_pin_constraints
- set_individual_pin_constraints
- set_bundle_pin_constraints

Feedback

## DPPA-096

(Warning) found %s on side %d of block %s on layer %s and there is no enough wire tracks available on this side

### Description

The command detects that the number of corner keepout wire tracks is larger than available tracks on block cell edge or corner keepout distance is larger than the length of block cell edge, pins will not be placed on this edge if there are wiretracks available on other edges

### What Next

Please check if the corner keep out pin constraints are setting correctly

### See Also

- set_block_pin_constraints

## DPPA-097

(warning) Find %s constraints assoicated with block %s conflict with non-default routing rule on net %s, non-default routing rule will be honored.

### Description

place_pins find block pin constraints and non-default routing rule conflict, pin placement will honor non-default routing rule and ignore block pin constraints when place pins

### What Next

Please check block pin constraints and and non-default routing rule

### See Also

- read_pin_constraints
- set_individual_pin_constraints
- set_block_pin_constraints
- create_routing_rule

## DPPA-098

(Error) While traversing route shapes from the driver for net %s, not all shapes were reached.

Feedback

**Description**

Route tracing code failed to reach all routing shapes that it expected to reach. This is either a problem with the routing, or this may be an internal problem with the route tracing algorithm.

**What Next**

Report this problem

---

## DPPA-099

(Error) Failed to process/analyze routing for net %s, skipping this net.

**Description**

The command attempted to process/analyze routing for this net, but failed. This is probably due either to some routing problem (discontinuous routing outside of block instance boundaries or routing loop for example) or possibly an internal error.

---

## DPPA-100

(Error) Failed to process net %s for feedthrough creation. Will skip this net.

**Description**

The command attempted to consider this net for feedthrough creation, but failed. This could happen if the routing doesn't follow hierarchical connection on physical blocks. Or there is a routing loop that going through physical block area and the tool cannot determine the feedthrough pins direction

**What Next**

Please check if the net is routed correctly.

---

## DPPA-101

(Error) Failed to update subnets after feedthrough creation for net %s. Skipping this net.

**Description**

The command attempted to update route subnets after feedthrough creation, but failed.

**What Next**

Report this problem

Feedback

## DPPA-102

(Error) Failed to push down subnets after feedthrough creation for net %s. Skipping this net.

**Description**

The command attempted to push down net routing shapes for subnets of the routing graph, but failed.

**What Next**

Report this problem

## DPPA-103

(Warning) Unexpected net driver pinShapeType %s type: %d

**Description**

The command encountered a net driver shape that it did not expect.

**What Next**

Report this problem

## DPPA-104

(Error) When tracing routing from driver shapes, the command failed to find a driver shape to traverse from for net %s.

**Description**

The command encountered a route tracing problem when attempting to trace the route from a driver shape.

**What Next**

Report this problem

## DPPA-105

(Warning) Failed to find a net driver shape in a block that has an output port on the net for top-level net %s.

Feedback

**Description**

The command was searching for a driver shape within a block because the block has an output port on the top-level net.

**What Next**

Report this problem

---

## DPPA-106

(Error) Failed to find a crossing point where a path crosses a block boundary. Path: %s Block: %s Net: %s

**Description**

The command was processing a path crossing of a block boundary, but failed to correctly identify a specific coordinate for the center of the path as it crossed the block.

**What Next**

Report this problem

---

## DPPA-107

(Error) Failed to find an existing logical connection for entry crossing of subNet %d for blkNet %s.

**Description**

The command was processing a path crossing of a block boundary for a net that has a pre-existing logical connection to the block, but failed to conclusively associate that port with the block subnet.

**What Next**

Report this problem

---

## DPPA-108

(Error) net %s has collinear route segments or vias. Skipping net. Please re-route the net.

**Description**

The command was processing signal net for pushing down routing into blocks. It discovered that some routing of the net overlaps blocks to be pushed into in a collinear direction. This is not supported in signal routing push-down.

Feedback

### What Next

Please re-route the net so that it does not have collinear overlaps with a block that is being pushed into.

---

## DPPA-109

(Warning) The %s of the bundle pin constraint applied on pin %s conflicts with the %s of the previously applied %s pin constraint. The %s ignored.

### Description

The command detects a conflict between bundle pin constraints applied on the specified pin or port. The most possible reason is that the net of the pin or port is included in more than one bundles and these bundles have conflict constraint settings.

### What Next

Please check the conflict and reset the constraint accordingly.

### See Also

- create_bundle
- set_bundle_pin_constraints

---

## DPPA-110

(Warning) net %s has incomplete routing. Skipping net. Please completely route the net for signal route push-down into blocks.

### Description

The command was processing signal net for pushing down routing into blocks with feedthroughs allowed. It discovered that the net was not fully routed. This is not supported in signal routing push-down.

### What Next

Please fully route the net so that it can be properly analyzed for push-down into blocks.

---

## DPPA-111

(Warning) global bundle pin range or order constraints doesn't have associated edge

Feedback

**Description**

The command detects that the bundle pin range or order constraints don't have the associated edge. For the command to honor those constraints, user needs to specify the associated edge.

**What Next**

Please check and reset the constraint accordingly.

**See Also**

- set_bundle_pin_constraints
- set_individual_pin_constraints
- read_pin_constraints

---

## DPPA-112

(Error) Net %s has logical vs. physical inconsistency in nonFeedthrough physical crossings. Skipping net. Please re-route the net.

**Description**

The command detected that physical route crossings of one or more blocks by a net was not consistent with the logical connectivity of the net. This net needs to be re-routed in such a was that the physical is consistent with logical. There should be at least as many physical crossings as the logical connections to the block

**What Next**

Please re-route the net.

---

## DPPA-113

(Warning) Block sides pin constraints conflict with pin %s.

**Description**

The command detected that there are side constraints on the pin that conflict with block side constraint. The constraints on individual pins, nets or bundles will override block pin constraints.

**What Next**

Please check the conflict from block side constraint and individual pins/nets/bundles side constraint.

Feedback

**See Also**

- read_pin_constraints
- set_block_pin_constraints
- set_bundle_pin_constraints
- set_individual_pin_constraints

---

## DPPA-114

(Error) place_pins and design planning global route cannot be used with clock routing topology routing mode

**Description**

The command detected that clock routing topology routing mode is turned on while it cannot be used together with place_pins or design planning global route since design planning global route is used to determine the pin locations for pin placement purpose.

**What Next**

Please reset the app option route.common.clock_topology's value to "normal"

---

## DPPA-115

(Error) Skipping propagation of shape %s of from-cell net %s because it would overlap with or touch %s of net %s in cell %s.

**Description**

The push_down_objects command or the pop_up_objects command, because the option to do overlap checking was on, detected that a net shape in the "from" cell, if propagated to the "to" cell, would overlap with existing routing. If overlap checking was indicated, the command infers that the existence of overlap should be considered an error, and the shape is skipped for propagation push-down.

**What Next**

Reroute the offending pre-routed net.

---

## DPPA-116

(Error) Skipping propagation of shape %s of from-cell net %s because it would overlap exactly %s of net %s in cell %s.

Feedback

**Description**

The push_down_objects command, because the option to do overlap checking was on, detected that a net shape in the top cell, if pushed down, would overlap exactly with existing routing. If the -routing_overlap_check is specified as true, then the command infers that the existence of overlap should be considered an error, and the shape is skipped for push-down.

The push_down_objects command or the pop_up_objects command, because the option to do overlap checking was on, detected that a net shape in the "from" cell, if propagated to the "to" cell, would overlap exactly with existing routing. If overlap checking was indicated, the command infers that the existence of overlap should be considered an error, and the shape is skipped for propagation push-down.

**What Next**

Reroute the offending pre-routed net.

---

## DPPA-117

(Warning) Pin %s and pin %s are connected to nets that belong to same bundle, however, both pins have pin offset/location constraints. The constraint on pin %s will be honored.

**Description**

In bundle pin placement, the tool will try to place all the pins together when their connected nets belong to same bundle. However, if multiple pins also have pin offset or location constraints on the same bundle, it is likely that the pin constraints will conflict with each other. When this happens, the tool will pick one of the pin location/offset constraint and ignore others.

**What Next**

Please remove pin location/offset constraints if they are more than one set on the pins that connected to nets belong to same bundle.

---

## DPPA-118

(error) Cell %s already exists.

**Description**

The *push_up_objects* command was used to try to create a cell in the current cell from a cell in a child block, but a cell with that name already exists in the current cell.

**What Next**

Report this problem.

Feedback

## DPPA-119

(Warning) Skipping creation of %s in block %s from the push-down of top net %s because it would overlap with or touch %s of net %s in the same block.

### Description

The push_down_objects command, because the option to do overlap checking was on, detected that the creation of a pin in a block, if created, would overlap with existing routing or pins in the block. If overlap checking was indicated, the command infers that the existence of overlap should be considered an error, and the pin is prohibited during routing push-down.

### What Next

Reroute the offending pre-routed net.

## DPPA-120

(Warning) Net %s has top-level discontinuity.

### Description

The check_nets_for_push_down command detected a pre-routed signal net that has top-level discontinuity.

### What Next

Reroute the offending pre-routed net.

## DPPA-121

(Warning) Pin %s has connections that are not same for all abutted mulitple instantiated blocks, this pin maybe placed shorted with other pins.

### Description

The connection to block cells are not same for all MIB instances. For instance, there are two MIB groups in a design with mib1_inst1 and mib1_inst2 belong to same MIB group while mib2_inst1 and mib2_inst2 belong to another MIB group. Pin A on mib1_inst1 connected to pin B on mib2_inst1 and pin A on mib1_inst2 connected to pin C on mib2_inst2. Instance mib1_inst1 is abutted with mib2_inst1 while mib2_inst2 is abutted with mib2_inst2. This can cause shorted pins situation when pins need to be placed on abutted edge as the pin B and pin C on mib2_inst1 or mib2_inst2 need to be placed together in order to create abutted connection to pin A to avoid single pin connection.

Feedback

### What Next

Check to see if the logic connection and floorplan is correct.

### See Also

- check_mib_for_pin_placement

---

## DPPA-122

(Error) check_nets_for_push_down detected exactly redundant paths. These can cause route traversal problems for the push_down_object command, and need to be reduced to single, non-redundant paths.

### Description

The push_down_objects command when pushing down signal nets needs to be able to traverse from the driver to all loads on the net. When paths exist which are perfectly redundant, this confuses the traversal and push-down algorithm, and usually indicates some error in the flow, so it is reported as an error.

### What Next

Examine the reported paths and delete the redundant ones.

---

## DPPA-123

(Warning) tap-connected P/G net %s has no child P/G nets in block instance %s (only tie high/low signal nets). Skipping block-level connections of this net.

### Description

The push_down_objects command when pushing down cells detected a scenario where a top-level cell(s) were connected to a P/G net which also connected to signal ports of the block that the cells were being pushed into. But the command detected that no block-level P/G net exists. This usually indicates a problem with the UPF setup.

### What Next

Please make sure the P/G has been defined correctly for the block that the cells are being pushed into.

---

## DPPA-124

(Warning) tap-connected P/G net %s has multiple P/G nets in block instance %s. Using the first one in the list.

Feedback

**Description**

The push_down_objects command when pushing down cells detected a scenario where a top-level cell(s) were connected to a P/G net which had multiple tap connections to the block that the cells were being pushed into. The command chose one of the ports for getting a child-level P/G net.

**What Next**

Please make sure that the results are what you wish.

---

## DPPA-125

(Error) No physical block defined in the design.

**Description**

A hierarchical design planning command detected that there are no physical blocks in the design. Hierarchical design planning commands require physical blocks upon which to perform hierarchical operations.

**What Next**

Please make sure that the design contains physical blocks before attempting hierarchical design planning commands.

---

## DPPA-126

(Error) Could not find a driver for net %s

**Description**

A hierarchical design planning command could not find a driver for a net when it was trying to make net connections

**What Next**

Please check the net.

---

## DPPA-127

(Error) No input tap found for connecting net %s for push-down cells.

**Description**

push_down_objects, when pushing down cells, encountered a net connected to the pushed-down cells that was previously connected to the block into which the cells were pushed. It was determined that the cells being pushed into the block included a driver on

Feedback

this tap-connected net meaning that it needed to find a pre-existing input connection to the block, and it could not find one.

**What Next**

Please check the net.

---

## DPPA-128

(Error) No input or output tap found for connecting net %s for push-down cells.

**Description**

push_down_objects, when pushing down cells, encountered a net connected to the pushed-down cells that was previously connected to the block into which the cells were pushed. It was determined that the cells being pushed into the block contained no driver on this tap-connected net meaning that it needed to find a pre-existing input or output connection to the block, and it could not find one.

**What Next**

Please check the net.

---

## DPPA-129

(warning) no target object in topological constraint pair in line %d.

**Description**

The command expects source and target topological constraint pair, but cannot find target object.

**What Next**

Check if the topological constraints specified correctly.

---

## DPPA-130

(Warning) child cell inst %s is outside the boundary of block design %s

**Description**

The command detected that a child cell inst connected to a net considered for push-down is outside of the block boundary. This can cause problems during push-down of the route.

**What Next**

Please fix the placement issue.

Feedback

## DPPA-131

(Error) net %s has block-level cell connections that include cells that are not placed inside the respective cell boundary. Skipping net. Please fix the cell placement issue.

### Description

The command detected that a net considered for push-down into a block contained child-level cell inst connections which included one or more cell instances that were not inside the block boundary. This can cause problems for the push-down of that net.

### What Next

Please fix the placement issue.

## DPPA-132

(Error) same-net MIB pin direction conflict detected. 2 boundary crossings occurring at the same location in the reference block (%s) but different directions. Topcell locations Location 1: {%s %s} Location 2: {%s %s} layer: %d

### Description

The command detected that a net considered for push-down into blocks crossed multiple instances of a reference block at the same location on the reference blocks, but in different directions. This usually results from one of the blocks being in a different orientation. For instance, abutted MIB instances cannot allow a pre-route crossing the abutted edge.

### What Next

Please fix the placement issue.

## DPPA-133

(Error) Net %s has a pin direction conflict on a MIB block instance. %s

### Description

The command detected that a net considered for push-down into blocks crossed multiple instances of a reference block at the same location on the reference blocks, but in different directions. This usually results from one of the blocks being in a different orientation. For instance, abutted MIB instances cannot allow a pre-route crossing the abutted edge.

### What Next

Please fix the placement issue.

Feedback

## DPPA-134

(Information) Deleted obsoleted port %s from reference block %s.

### Description

The command detected that a tap connection on a block has now become obsolete because all cells on the related net have been pushed down into the block. This is just an informational note, which by default will occur for 10 examples of obsoleted ports per call to the command. If you would wish to see more, then increase the message limit for the DPPA-134 information code. If you would prefer that these obsolete ports be retained, then use the block_action "retain_obsolete_ports" for object_type cells in the set_push_down_object_options command.

### What Next

Nothing needed. Just informational.

## DPPA-135

(Info) Obsoleted port(s) have been detected as a result of pushing cells into blocks. This means that they are no longer necessary on the block as all of the cells on the net have been pushed into the block for the given net. If you would prefer to retain these obsoleted ports, use the set_push_down_object_options command, and the block_action "retain_obsolete_ports" to suppress the deletion of these ports.

### Description

The command detected that one or more tap connections on a block has now become obsolete because all cells on the related net have been pushed down into the block. This is just an informational note and does not require any action. If you would prefer that these obsolete ports be retained, then use the block_action "retain_obsolete_ports" for object_type cells in the set_push_down_object_options command.

### What Next

Nothing needed. Just informational.

## DPPA-136

(Info) Filtered redundant %s (bbox: {{%s %s}{%s %s}} layer: %d) from MIB reference design %s for child net %s (when pushing down top-level net %s)

### Description

The command detected that, when pushing down shapes into reference blocks of MIB instances, that some redundant shapes were present. This can occur when different

Feedback

instances of MIB blocks have routing pushed into them. The command determines that it is legal for them to overlap, and if so, any shapes that are completely redundant (removable), it removes them.

**What Next**

Nothing needed. Just informational.

---

## DPPA-137

(Error) Failed to delete redundant %s (bbox: {{%d %d}{%d %d}} layer: %d) from MIB reference design %s for child net %s (when pushing down top-level net %s)

**Description**

The command detected that, when pushing down shapes into reference blocks of MIB instances, that some redundant shapes were present. This can occur when different instances of MIB blocks have routing pushed into them. The command determines that it is legal for them to overlap, and if so, any shapes that are completely redundant (removable), it removes them. In this case, the attempt to delete the redundant shape failed.

**What Next**

Report this. In the mean time, try to re-route the top-level routing to avoid this conflict.

---

## DPPA-138

(Warning) Failed to initialize uhSplitCntr object. UPF may not be properly loaded. New feedthrough ports won't get UPF MV constraints placed upon them.

**Description**

The command attempted to create and initialize a uhSplitCntr object for the purpose of later adding UPF constraints to new feedthrough ports created by either the place_pins command or the push_down_objects command (for signal net feedthroughs). It likely results from UPF not loaded properly.

**What Next**

Make sure that UPF is properly loaded.

---

## DPPA-139

(Warning) Failed to add MV constraints to new feedthrough ports on blkInst %s (ref: %s) %s

Feedback

### Description

The command attempted to add UPF MV constraints to newly added feedthrough ports.

### What Next

Make sure that UPF is properly loaded.

---

## DPPA-140

(Error) Net driver %s seems to be not connected to any routing.

### Description

The net driver for a net considered for push-down is a top-level driver pin that does not seem to be connected to any routing.

### What Next

Make sure top-level portions of routed nets is connected to top-level pins/ports.

---

## DPPA-141

(Warning) The error view specified with the -error_view option uses an invalid file extension (%s); using the file extension "err" instead.

### Description

The tool issues this warning when the error view specified with the *-error_view* option has an invalid file extension.

The command requires a file extension of "err" for the error view. If you specify a different file extension, the tool keeps the error view name, but replaces the file extension with "err".

Note that you can also specify the error view name without an extension. In this case, the tool adds the "err" extension, but does not issue a warning.

### What Next

To avoid this warning message, either use the "err" file extension or no file extension when you specify the error view name.

### See Also

- check_boundary_cells
- check_floorplan_rules
- check_io_placement

Feedback

- check_pin_placement
- report_block_shaping
- report_placement

## DPPA-142

(Warning) Preferred routing direction conflict detected between existing direction %s and new direction %s for layer %s in block %s. Skipping preferred routing direction propagation for this block and layer.

### Description

When propagating preferred routing directions from one block to another, the command encountered a layer in the destination block that had an existing preferred routing direction that was different from the direction in the same layer in the source block. In this situation, the layer in this block is skipped for setting the preferred routing direction.

### What Next

Make sure that the new preferred routing directions are what you want in the destination block.

### See Also

- push_down_objects

## DPPA-143

(Warning) %s on %s is less than technology rule. This constraint is ignored.

### Description

The user defined pin width or length constraint, or the non default routing rule width associated with net on the pin, is less than the minimum value defined by the technology rule. As a result, the user defined constraint is ignored for pin placement or pin check. Note that pin creation uses the maximum of the minimum width and default width specified in technology file.

### What Next

Please double check the technology rule to make sure that user defined width or length constraint, or the non default routing rule, does not violate those defined in technology rule.

Feedback

**See Also**

- set_bundle_pin_constraints
- set_individual_pin_constraints
- set_routing_rule

---

## DPPA-144

(Error) MIB feed crossing conflict detected in reference block %s at reference coord: {%d %d} layerNumber: %d

**Description**

During checking of push-down of signal routing, top-level net routing required a feedthrough net in a Multiply-Instantiated-Block (MIB), but another crossing of this MIB produced a crossing previously that conflicts with this subsequent crossing.

**What Next**

Make sure that your MIB-crossing routes for this reference block do not conflict with other crossings of other instance of this reference block.

---

## DPPA-145

(Warning) When checking MIB feedthrough crossing conflicts, a conflict was detected for a MIB subnet from top net %s over reference block %s.

**Description**

During checking of push-down of signal routing, top-level net routing required a feedthrough net in a Multiply-Instantiated-Block (MIB), but another crossing of this MIB produced a crossing previously that conflicts with this subsequent crossing. This is not necessarily a problem, as long as the crossing that do exist in common match between instances of the reference block.

**What Next**

Make sure that your MIB-crossing routes for this top-level net do not conflict with other crossings of other instance of this reference block, or if they do, that it's just a matter of one instance having fewer crossings than another instance.

---

## DPPA-146

(Error) creation of pin in reference block %s with bbox {{%d %d} {%d %d}} on layer %d would cause this pin shorted with an existing pin. Skipping creation of this pin.

Feedback

**Description**

During pushing down of signal routing and during terminal creation on a block, it was detected that a pin location conflicted with an existing pin. This could happen when routing over MIB instances conflict.

**What Next**

Make sure that your MIB-crossing routes for this top-level net do not conflict with other crossings of other instance of this reference block.

---

## DPPA-147

(Error) MIB multiple child subnets conflict with each other in reference block %s from one ->top-net %s crossing instances in different ways. childNet1: %s childNet2: %s

**Description**

During block-level routing creation, it was detected that Multiply-Instantiated-Blocks (MIBS) were encountere and routing crossing a particular instance of a reference block would cause conflicts with routing that was previously created for a previous instance crossing.

**What Next**

Make sure that your MIB-crossing routes for this top-level net do not conflict with other crossings of other instance of this reference block.

---

## DPPA-148

(Error) Net %s has a MIB feedthrough conflict in the number of feed ports on a block. %s

**Description**

During push_down_objects, MIB's were encountered which involved multiple crossings of MIB instances resulting in block-level subnets that are in conflict with subnets from other routes pushed down into another instance of the MIB reference block.

**What Next**

Make sure that your MIB-crossing routes for this top-level net do not conflict with other crossings of other instance of this reference block.

---

## DPPA-149

(Error) Net %s has a MIB feedthrough conflict where matched feedthroughs' shapes don't match. %s

Feedback

### Description

During push_down_objects, MIB's were encountered which involved multiple crossings of MIB instances resulting in block-level subnets that are in conflict with subnets from other routes pushed down into another instance of the MIB reference block.

### What Next

Make sure that your MIB-crossing routes for this top-level net do not conflict with other crossings of other instance of this reference block.

---

## DPPA-150

(Warning) MIB feed crossing conflict detected in reference block %s at reference coord: {%s %s} layerNumber: %d current net: %s current instance: %s coord in parent: {%s %s}

### Description

During push_down_objects, MIB's were encountered which involved multiple crossings of MIB instances resulting in block-level subnets that are in conflict with subnets from other routes pushed down into another instance of the MIB reference block. This is only a warning, as the mismatch may simply be that a crossing of a boundary occurred for one instance, but not the other. If all other crossings are consistent, it is probably okay.

### What Next

Make sure that your MIB-crossing routes for this top-level net do not conflict with other crossings of other instance of this reference block, or that the re-used feedthrout is correct.

---

## DPPA-151

(Error) Only matched %d out of %d crossings of two MIB feed subnets that have the same entry crossing point on reference block %s. (block Instance: %s entry crossing point {%s %s}

### Description

During push_down_objects, MIB's were encountered which involved multiple crossings of MIB instances resulting in block-level subnets that are in conflict with subnets from other routes pushed down into another instance of the MIB reference block.

### What Next

Make sure that your MIB-crossing routes for this top-level net do not conflict with other crossings of other instance of this reference block.

Feedback

## DPPA-152

(Error) Pin %s is shorted with %s %s.

**Description**

The tool detects that when placing block pins based on their connected top terminals or when place synthesized pins, they are shorted with other top level terminals or other block pins.

**What Next**

Make sure that the top terminals are placed properly and the pin location constraints is set properly when place synthesized pins before *place_pins*.

## DPPA-153

(Warning) A no-net collinear %s, %s, was detected during push-down. Due to the collinear nature of this %s, it will be created in the block as a copy, and retained in the parent block.

**Description**

During push-down of no-net routing, a path or via was determined to be collinear. Since push-down operation never splits a path laterally, or a via (in any way), the command just propagated a copy to the block and left the original in the parent level.

**What Next**

If you want a no-net element to be pushed down fully, be sure that it does not straddle a block boundary.

## DPPA-154

(Warning) A parent-net-to-block-net mapping problem has been detected for net %s and reference block %s. Skipping parent-to-child net mapping for this net and reference block.

**Description**

During push-down of routing, where a net has an existing connection to a block, the command builds a map between the parent net and the block net. If the parent net and the block net are not of the same type (P/G or signal), this is a mapping problem and is skipped if the parent type is P/G and the block type is signal.

**What Next**

Make sure that you have correctly run connect_pg_net and created P/G nets in the block correctly.

Feedback

## DPPA-155

(Warning) The push_down_objects command detected a top-level P/G net to block-level signal net conflict for block %s. This usually indicates improper UPF and P/G connectivity prior to calling push_down_objects. Top level P/G net %s block-level signal net: %s

### Description

During push-down of routing, where a net has an existing connection to a block, the command builds a map between the parent net and the block net. If the parent net and the block net are not of the same type (P/G or signal), this is a mapping problem and is skipped if the parent type is P/G and the block type is signal.

### What Next

Make sure that you have correctly run connect_pg_net and created P/G nets in the block correctly.

## DPPA-156

(Warning) A parent-net-to-block-net mapping problem has been detected for net %s and reference block %s. A top-level P/G net has a connection to a block instance pin (%s) but no block-level P/G net previously existed. A new block-level net %s was created.

### Description

During push-down of routing, where a net has an existing connection to a block, the command builds a map between the parent net and the block net. If the parent net is a P/G net and a P/G port exists on the block, but no corresponding P/G net exists in the block, one is created.

### What Next

Make sure that you have correctly run connect_pg_net and created P/G nets in the block correctly.

## DPPA-157

(Warning) top terminal %s partially overlaps with block %s

### Description

The tool detects that a top terminal partially overlaps with a block. If this top terminal is connected to a block pin on the same block, then the block pin can be placed outside its block boundary.

Feedback

**What Next**

Make sure that the top terminals are placed properly before run *place_pins*.

---

## DPPA-158

(Warning) move_bound %s is not completely enclosed by the hierBoundary for module instance %s.

**Description**

During the command commit_blocks, a move_bound was encountered which overlapped a hierBoundary or partition, but only partially.

**What Next**

Make sure that if a move_bound is is not fully inside a hierBoundary or partition prior to running commit_block, that it is what you intend.

---

## DPPA-159

(Warning) move_bound %s is completely outside of hierBoundary for module instance %s, but it has members that belong to that module instance.

**Description**

During the command commit_blocks, a move_bound was encountered which did not overlap a hierBoundary or partition, but had cell members that belonged to that module instance.

**What Next**

Make sure that if a move_bound is not fully inside a hierBoundary or partition prior to running commit_block, and if the move_bound belongs to the module instance, that this is what you intended.

---

## DPPA-160

(Warning) pin %s are placed %s, router may not be able to connect to such pin

**Description**

The tool detects that a block pin is placed on the edge which is abutted with another block or with top design, or the pin is placed totally inside block. When this happens, the router might not be able to route to such a pin. This normally happens in multiple instantiated blocks (MIB) when the top level terminal is placed inside one of the MIB block boundary and the pin needs to be placed right below this terminal.

Feedback

**What Next**

Make sure that the top terminals are placed properly before *place_pins*.

---

## DPPA-161

(Error) Could not find %s %s in destination block %s. Won't add that %s to new move bound propagated during hierarchical DP operation.

**Description**

During a hierarchical DP command, when adding cells to a new move bound, a cell could not be found by name in the destination block.

**What Next**

Report this.

---

## DPPA-162

(Error) Failed to sort hierarchical pins based upon location for hierSubNet index %d of top net %s.

**Description**

During the push-down of signal net routing, an attempt to sort hierarchical pins based upon their locations on a block failed.

**What Next**

Report this.

---

## DPPA-163

(Warning) move_bound %s is not completely exclusive but overlaps the hierBoundary for module instance %s. It will be skipped from propagation.

**Description**

During the push-down of move bounds during commit_block, a move bound was detected which partially overlapped a hierBoundary but the cells in the bound were not exclusive to the module.

**What Next**

Make sure this is what was intended.

Feedback

## DPPA-164

(Warning) move_bound %s is partially exclusive and does not overlap the hierBoundary for module instance %s. It will be skipped from propagation.

**Description**

During the push-down of move bounds during commit_block, a move bound was detected which was only partially exclusive and did not overlap a hierBoundary of the module being committed.

**What Next**

Make sure this is what was intended.

## DPPA-165

(Warning) Shape %s is non-orthogonal. Skipping this shape.

**Description**

During the push-down/pop-up of routing, a non-orthogonal shape, %s, was detected, which is not supported.

**What Next**

Do not attempt to push down or pop up shapes that are non-orthogonal.

## DPPA-166

(Warning) Net %s has non-orthogonal shapes. Nets with non-orthogonal shapes are not supported by push-down/pop-up DP operations.

**Description**

During the push-down/pop-up of routing, a non-orthogonal shape, %s, was detected on a net, which is not supported.

**What Next**

Do not attempt to push down or pop up nets that have shapes that are non-orthogonal.

## DPPA-167

(Error) Pin %s is shorted with pin %s.

Feedback

**Description**

Pin placement detects that the pin currently placed is shorted with another pin. This might be because the pins created are based on top level objects (terminals or cells) but the top level objects are placed shorted with each other or they are placed at same location on top of different MIB instances which have same reference block

**What Next**

Please check top level objects (terminals or cells) and remove the shorted pins

---

## DPPA-168

(Error) Failed to create new pin in block %s by name %s

**Description**

An attempt to create a pin in a block failed during push-down or pop-up of objects

**What Next**

Report this

---

## DPPA-169

(Warning) The command encountered a non-feedthrough dangling route over block instance %s for net %s

**Description**

During signal route push-down for feedthrough creation, a section of routing was found which dangles over a block with no pre-existing logical connection, and the routing does not feed all the way through the block.

**What Next**

Please check your routing for this net.

---

## DPPA-170

(Error) The pop_up_object command did not receive any objects to pop up.

**Description**

The pop_up_objects command needs to have objects passed to it to pop up. None were detected.

Feedback

### What Next

Make sure that the collection that you have passed to the command contain valid objects to pop up.

---

## DPPA-171

(Warning) The topological segment from %s to %s of %s %s is not starting from or ending at a physical block. This segment has no effect and is ignored.

### Description

Neither the source nor the target in the topological constraint pair is on block instance.

### What Next

Make sure that the constraint pair specified correctly

### See Also

- read_pin_constraints

---

## DPPA-172

(Warning) Instance-based feedthrough allowance was detected for net %s, where some, but not all instances allowed feedthroughs. The only supported scenario is for all instances to allow feedthroughs. Since one instance was found to allow feedthroughs, the command will allow feedthroughs on all instances. If this is not what you want, then remove all instance-based pin constraints which allow feeds on some but not other instances.

### Description

The push_down_objects command detected a block-based pin constraint that allowed feedthroughs on the block instance, but also found one or more block-based constraints that disallowed feedthrough creation. The user control of feedthrough creation/noncreation currently is only offered on a whole net basis, on or off, for all blocks that the routing crosses. The command will honor the instance-based pin constraint allowing feedthroughs for all blocks that it crosses, and disregard the other, conflicting, pin constraints disallowing feedthrough creation.

### What Next

Please set pin constraints according to this restriction.

Feedback

## DPPA-173

(Warning) Routing shapes for net %s in block %s (for block instance %s) will not be popped up because it appears to be an internal net with no connections to any net outside the block.

**Description**

The pop_up_objects command encountered a specified net for pop-up which was not connected to any net outside the block that the net exists in. The pop_up_objects command only supports pop-up of routing for nets that are connected to nets outside the block.

**What Next**

Please only attempt to pop-up routing for nets which are connected to nets outside the block.

## DPPA-174

(Error) Pin mask constraint conflict between abutted pins detected.

**Description**

The mask constraints on abutted pins conflict with each other.

**What Next**

Change the mask constraints on the abutted pins to resolve the conflict.

## DPPA-175

(Error) Pin size violation detected.

**Description**

The actual pin size does not honor the value defined by individual pin constraints, or derived from net or bus constraints, or the default pin length to avoid mask constraint violation.

**What Next**

Ensure that constraints are set correctly, validate incorrect pin size in the error-browser GUI and check the log of pin-placement for any related warning or error messages.

## DPPA-176

(Error) Pin spacing violation detected.

Feedback

**Description**

The actual pin spacing does not honor the value defined by individual pin constraints, or derived from net or bus constraints, or the non default routing rule.

**What Next**

Ensure that constraints are set correctly, validate incorrect pin spacing in the error-browser GUI and check the log of pin-placement for any related warning or error messages.

---

## DPPA-177

(Error) Signal pin stacking violation detected.

**Description**

Two signal pins are overlapping on different layers. It is regarded as a violation if the pin placement constraints do not allow signal pin to signal pin overlapping.

**What Next**

Ensure that constraints are set correctly, validate incorrect stacking in the error-browser GUI and check the log of pin-placement for any related warning or error messages.

---

## DPPA-178

(Error) Signal pin to P/G stacking violation detected.

**Description**

Signal pin is overlapping with power or ground straps. It is regarded as a violation if the pin placement constraints do not allow signal pin to power or ground strap overlapping.

Ensure that constraints are set correctly, validate incorrect stacking in the error-browser GUI and check the log of pin-placement for any related warning or error messages.

---

## DPPA-179

(Error) Collinear overlap of shape %s with block instance %s for signal net %s detected. Skipping this shape for push-down.

**Description**

During push-down of signal net routing, it was determined that one or more shapes overlap collinearly with a block boundary, which is not allowed during signal route push-down. The shapes were not pushed down, and were left in the top block.

Feedback

**What Next**

Correct the signal routing.

---

## DPPA-180

(Error) The net %s has wire stubs in combination with multiple crossings of a blkInst that has wire-stubs in it. This would lead to potential ambiguity in how the wire-stubs should be traversed, and is not supported during routing push-down. Skipping this net.

**Description**

During push-down of signal net routing, it was determined that wire-stubs were in use in one or more block instances, and that the routing crossed a block instance that had wire-stubs on more than one occasion. This can lead to potentially incorrect traversal of wire stubs because the command makes an assumption that all routing in a block that has wire stubs should all belong to the same subnet in the block. And this may very probably not be what the user intended if the routing crosses multiple times.

**What Next**

Correct the wire-stub signal routing.

---

## DPPA-181

(Error) Wire-Stubs detected along with multiple crossings of block instance %s

**Description**

During push-down of signal net routing, it was determined that wire-stubs were in use in a block instances, and that the routing crossed that block instance that had wire-stubs on more than one occasion. This can lead to potentially incorrect traversal of wire stubs because the command makes an assumption that all routing in a block that has wire stubs should all belong to the same subnet in the block. And this may very probably not be what the user intended if the routing crosses multiple times.

**What Next**

Correct the wire-stub signal routing.

---

## DPPA-182

(Error) Failed to re-connect other affected MIB connections when pushing cells into block instance %s.

Feedback

**Description**

During push-down of cells into blocks, some connections on an MIB instance required using a new port (because the original port could not be reused). This requires finding the same port on other MIB instances and switching their connections to the newly-created port. This attempt to re-connect the MIB "other" port instances apparently failed.

**What Next**

Contact Synopsys.

---

## DPPA-183

(Error) Driver for net %s inside block %s is not placed inside the block boundary.

**Description**

During push-down of signal routing into blocks, a driver inside a block was found, but the cell instance was not placed inside the block boundary. For signal route push-down with feedthrough creation, it is necessary that cells be placed inside the block boundary.

**What Next**

Place the cells inside the block before running push-down of signal routing with feedthroughs.

---

## DPPA-184

(Error) Block %s via definition %s conflicts with top-cell via definition of the same name. Nets with routing in this block specified for pop-up which contain this via type will be skipped for pop-up.

**Description**

During pop-up of signal routing, a via definition conflict between a via type in the block and a via type in the top cell was discovered before the actual pop-up operation began. When this occurs, then for every block that has this conflict with the top block, nets specified for pop up will be checked before popping up the net, and if a net contains a via type that has been detected as having a conflict with the top, the net will be skipped for pop-up.

**What Next**

Correct the via name in either the top block or the child block, and try popping up the net again.

Feedback

## DPPA-185

(Error) Net %s in block %s contains a via type %s which has a conflict with a via type in the top block of the same name. Skipping this net for pop-up.

### Description

During pop-up of signal routing, a via definition conflict between a via type in the block and a via type in the top cell was discovered before the actual pop-up operation began. When this occurs, then for every block that has this conflict with the top block, nets specified for pop up will be checked before popping up the net, and if a net contains a via type that has been detected as having a conflict with the top, the net will be skipped for pop-up.

### What Next

Correct the via name in either the top block or the child block, and try popping up the net again.

## DPPA-186

(Error) Ambiguous wire-stubs detected for net %s over block instance %s.

### Description

During push-down of signal routing for feedthrough creation, routing was detected to involve wire-stubs (routing that does not touch pins inside a block). Push-down of signal routing for feedthrough creation can tolerate wire-stubs based upon the assumption that the command makes that all wire-stubs over a specific block instance should internally be considered physically connected, or stated alternately, as part of the same block-level subnet. Traversal for this net has resulted in the command detecting that wire-stubs within a block instance belong to multiple block-level subnets, which violates the assumption that all wire-stubs in an instance should be considered connected physically. If multiple subnets with wire-stubs within a block are implied by the routing, this means that there is ambiguity in how the command should treat these wire-stubs, and it is very likely that the result may not be what the user expects.

Such a scenario may likely happen if wire-stubs are used for an instance where routing leaves a block instance and feeds back upon the same instance (perhaps an existing output drives another port on the same instance) where a wire stub is used for both crossings of the boundary.

### What Next

Make sure that the routing for this net is non-ambiguous, particularly if the instance has multiple existing port connections to the net.

Feedback

## DPPA-187

(Warning) cell %s only partially overlaps block %s. Note that push-down of associated routing will very likely fail or produce incorrect results.

### Description

During push-down of cells into blocks, a cell was detected to overlap a block only partially. This is inadvisable, as the portion that exists outside of the block boundary will not be seen in the top cell. Note, also, that if any top-level routing associated with a pin of a pushed-down cell where the pin remains outside of the block boundary, the routing will likely not be able to be pushed into the block because the routing might not ever cross the block boundary.

### What Next

Make sure that all cells specified for push-down into a block are fully inside the block instance boundary before specifying them for push-down.

## DPPA-188

(Warning) cell %s specified for push-down overlaps more than one block. The command will only push this cell into one block, determined by the first block processed to receive cells. Note that push-down of associated routing will very likely fail or produce incorrect results.

### Description

During push-down of cells into blocks, a cell was detected to overlap more than one block instance. This is a very inadvisable scenario, and should be corrected by the user. Note, also, that if any top-level routing associated with a pin of a pushed-down cell where the pin remains outside of the block boundary, and inside the adjacent block, the routing will likely not be able to be pushed into the block because the result may be unpredictable and unsatisfactory to the user.

### What Next

Make sure that all cells specified for push-down into a block are fully inside one block instance boundary before specifying them for push-down.

## DPPA-189

(Warning) cell %s specified for push-down overlaps more than one block. The command will only push this cell into block %s .

Feedback

**Description**

During push-down of cells into blocks, a cell was detected to overlap more than one block instance. This is a very inadvisable scenario, and should be corrected by the user. But regardless, the command pushed down the cell in to a block that was chosen based upon which block is overlapped the most by the cell to be pushed down. If the area amounts of overlap are equal for multiple blocks, then the command chooses the block whose boundary contains the origin of the cell that overlaps multiple blocks. If that cell's origin does not fall within a block instance that can be pushed into, then the command chooses the block instance whose origin is closest to the cell's origin.

**What Next**

Make sure that all cells specified for push-down into a block are fully inside one block instance boundary before specifying them for push-down.

---

## DPPA-190

(warning) physical pin constraints can only be set on block cell instances or block pins, the constraints set on top level ports or standard-cell pins will be ignored for topological constraints at line %d

**Description**

The command failed to parse a valid topological constraint because a physical constraint (side, layer or offset) is set on a standard-cell pin or top level port. The tool can only set such (physical) constraints on block cell instances or block pins, but not on standard-cell pins or top level ports. Constraints set on top level ports will be ignored when runing place_pins from within this block, however if place_pins is run from parent of current block, then physical constraints set on top level terminals will be honored.

**What Next**

Remove the specified constraint

---

## DPPA-191

(Error) Asymmetric MIB Port connections detected.

**Description**

The MIB port has asymmetric connections across its block instances. This can result in sub-optimal pin-placement.

---

## DPPA-192

(Error) Pin and track mask constraint mismatch detected.

Feedback

**Description**

The mask constraint on the specified pin does not match the mask constraint on the track.

**What Next**

Ensure that constraints are set correctly, validate mismatch from the error-report from check_pin_placement check the log of place_pins for any related warning or error messages.

**See Also**

- create_track
- propagate_pin_mask_constraint
- read_def

---

## DPPA-193

(Error) Net %s has a dangling piece of wire not over a block. Skipping this net.

**Description**

The command detected that a piece of routing would be left dangling outside a block that the routing feeds through. The push_down_objects command does not allow a feedthrough creation where one side dangles over the top cell.

**What Next**

Please correct your routing for this net and try to push-down again.

---

## DPPA-194

(Error) net %s experienced a traversal problem and not all shapes on the net were visited. %s %s

**Description**

The command detected that some parts of the net (either pins or terminals or route shapes) did not get reached during traversal of the net to determine how feedthroughs should be created.

**What Next**

Please correct your routing for this net and try to push-down again.

Feedback

## DPPA-195

(Error) Failed to reach all original connections for net %s%s

**Description**

The command detected that some parts of the net (either pins or terminals or route shapes) did not get reached during traversal of the net to determine how feedthroughs should be created.

**What Next**

Please correct your routing for this net and try to push-down again.

## DPPA-196

(Error) The net %s violates feedthrough constraints on cell %s with feedthrough ports : %s

**Description**

The check_feedthrough command detected that the specified net violates feedthrough constraints on the specified block. The violating feedthrough ports are also listed.

**What Next**

Ensure that the constraints were read/set correctly with no errors or warnings.

## DPPA-197

(Error) The net %s violates topological constraints by not having a direct connection between cells %s and %s.

**Description**

The check_feedthrough command detected that the specified net violates its associated topological constraints by not having a direct connection between the two listed blocks.

**What Next**

Ensure that the topological constraints are correctly specified in the pin-constraint file and read-in correctly without any errors or warnings. Also check the logfile for warnings during pin-placement.

## DPPA-198

(Error) The net %s violates topological constraints by not having a direct connection between cell %s and instance pin %s.

Feedback

**Description**

The check_feedthrough command detected that the specified net violates its associated topological constraints by not having a direct connection between the block and instance pin listed.

**What Next**

Ensure that the topological constraints are correctly specified in the pin-constraint file and read-in correctly without any errors or warnings. Also check the logfile for warnings during pin-placement.

---

## DPPA-199

(Error) The net %s violates topological constraints by not directly connecting the cell %s and top level port %s.

**Description**

The check_feedthrough command detected that the specified net violates its associated topological constraints by not having a direct connection between the block and the top level port listed.

**What Next**

Ensure that the topological constraints are correctly specified in the pin-constraint file and read-in correctly without any errors or warnings. Also check the logfile for warnings during pin-placement.

---

## DPPA-200

(Error) The net %s violates topological constraints by not directly connecting the instance pins %s and %s.

**Description**

The check_feedthrough command detected that the specified net violates topological constraints by not having a direct connection between the instance pins that are listed.

**What Next**

Ensure that the topological constraints are correctly specified in the pin-constraint file and read-in correctly without any errors or warnings. Also check the logfile for warnings during pin-placement.

Feedback

## DPPA-201

(Error) The net %s violates topological constraints by not directly connecting the instance pin %s and top level port %s.

### Description

The check_feedthrough command detected that the specified net violates topological constraints by not having a direct connection between the instance pin and top level port listed.

### What Next

Ensure that the topological constraints are correctly specified in the pin-constraint file and read-in correctly without any errors or warnings. Also check the logfile for warnings during pin-placement.

## DPPA-202

(Error) The net %s violates topological constraints by not directly connecting top level port %s and top level port %s.

### Description

The check_feedthrough command detected that the specified net violates topological constraints by not having a direct connection between the listed top level ports.

### What Next

Ensure that the topological constraints are correctly specified in the pin-constraint file and read-in correctly without any errors or warnings. Also check the logfile for warnings during pin-placement.

## DPPA-203

(Error) Port %s on MIB Reference Module %s is a potentially swapped port.

### Description

The check_mib_for_pin_placement command detected that the specified port on the MIB reference module is potentially a swapped port. That is it does not connect to the same MIB pin across its MIB instances.

### What Next

check to see if such connection is reasonable for pin placement

Feedback

## DPPA-204

(Error) Port %s on MIB Reference Module %s connects to top level terminals which are not all at the same location.

**Description**

The check_mib_for_pin_placement command detected that the specified port on the MIB reference module connects to different top-level terminals (across the MIB instances) that do not all have the same location (relative to the MIB instance).

**What Next**

check to see if such connection is reasonable for pin placement

## DPPA-205

(Error) Port %s on MIB Reference Module %s has asymmetric connections.

**Description**

The check_mib_for_pin_placement command detected that the specified port on the MIB reference module has asymmetric. That is it does not connect to the same set of pins across its MIB instances and may cause problems during pin-placement.

**What Next**

check to see if such connection is reasonable for pin placement

## DPPA-206

(Warning) Total %d top level terminal violations detected in MIB instances.

**Description**

The check_mib_for_pin_placement command detected top level terminal violations. These can cause problems during pin-placement.

**What Next**

check to see if such connection is reasonable for pin placement

## DPPA-207

(Warning) Skipping net %d for alignment checking, because it has unplaced objects.

Feedback

**Description**

The specified net is being skipped for alignment checking since the -exclude_unplaced_objects option is set to true.

**What Next**

Ensure that the net has placed connections to allowe for alignment computation or ignore such nets for alignment.

---

## DPPA-208

(Warning) track %s is not within the boundary of block %s

**Description**

Tool detects that the track is defined outside of the boundary of a block instance

**What Next**

Please re-define the tracks so that they are inside the block boundary

---

## DPPA-209

(Error) Only site rows which belongs to a site array was passed into the push_down_object collection as stand-alone site rows. No actual standalone site rows were detected. Site rows that belong to site arrays are never explicitly pushed down. Instead, they are pushed down as part of a site array. This may have been unintentional and the result of the use of get_site_rows, which includes site-array-owned rows.

**Description**

The push_down_objects command detected that a site row which belongs to a site array was specified to be pushed down like a standalone site row. This is incorrect procedure. Only standalone site rows should be included in the object collection as site rows. If you want to push down a site array (which will include the site rows), then specify the site arrays using get_site_arrays. This may have occurred because get_site_rows now includes site-array-owned rows. You should first decide whether you even have standalone site rows in your design. If not, but you do have site arrays, you should just push down site arrays, instead of site rows. If you find you still must push down rows separately, you must deliberately apply a filter to get only the standalone rows. This can be done using the following: push_down_objects [get_site_rows -filter "is_derived==false"] .

**What Next**

Please correct your push_down_objects command line to push down site arrays instead of attempting to push down individual site rows belonging to a site array. Or, apply the "is_derived" filter to the get_site_rows command.

Feedback

## DPPA-210

(Error) Voltage Area %s is a primary VA. Only non-primary VA's (charging stations) are currently supported by push_down_objects. Skipping it for push-down.

**Description**

The push_down_objects command detected that a voltage area was specified for push-down. But the voltage area was a primary voltage area. Currently, the only type of voltage areas supported by push_down_objects is a charging_station voltage area.

**What Next**

Please only specify charging station voltage areas when calling push_down_objects.

## DPPA-211

(Error) This command does not support paths that are neither 1) Manhattan-style nor 2) 45-degree. Skipping shape %s.

**Description**

The push_down_objects/pop_up_objects command detected that a path was passed to the command which is diagonal, but not a 45 degree angle. Currently, these commands only support push/pop of paths that are either horizontal, vertical or 45-degree diagonal.

**What Next**

Please only specify paths to push_down_objects or pop_up_objects that are either 1) horizontal, 2) vertical, or 3) 45-degree diagonal.

## DPPA-212

(Error) This command only supports 45-degree paths that are for RDL nets. Skipping shape %s.

**Description**

The push_down_objects/pop_up_objects command detected that a path was passed to the command which is 45-degree diagonal, but had no net associated with it. Currently, these commands only support push/pop of 45-degree paths that are part of a net that is an RDL net.

**What Next**

Please only specify 45-degree paths to push_down_objects or pop_up_objects belong to RDL nets.

Feedback

## DPPA-213

(Error) Pin %s on cell %s is shorted with pin %s on cell %s.

**Description**

The specified pins are shorted with each other, that is they are on the same layer and touch or overlap with each other.

**What Next**

Confirm in GUI/Error-Browser that the pins are indeed shorted. Move/edit pin locations as appropriate.

place_pins

## DPPA-214

(Error) Port %s on cell %s has no pin.

**Description**

The specified port does not have a physical pin created.

**What Next**

Use GUI/Error-Browser to confirm that physical pins are missing.

## DPPA-215

(Error) A mixture of site row types was passed into the push_down_objects object collection as site-rows. This includes 1) standalone site rows and 2) site-array-owned site rows. The push_down_object command only pushes down standalone site row, and will ignore the site-array-owned rows that were passed. If you want to push down rows that belong to site arrays, you need to include site arrays in the collection (using get_site_arrays).

**Description**

The push_down_objects command detected that, along with standalone site rows, a site row which belongs to a site array was specified to be pushed down like a standalone site row. If you want to push down a site array (which will include the site rows), then specify the site arrays using get_site_arrays. This may have occurred because get_site_rows now includes site-array-owned rows. If you find you need to push down standalone site rows, and you don't want to see this error message, you must deliberately apply a filter to get only the standalone rows. This can be done using the following: push_down_objects [get_site_rows -filter "is_derived==false"] .

Feedback

**What Next**

Please correct your push_down_objects command line to apply the "is_derived" filter to the get_site_rows command.

---

## DPPA-216

(warning) Net %s is assigned to a pin guide, but appears to be covered by a previous pin guide. The pin guide %s is ignored.

**Description**

Pin guides should uniquely describe a range or set of ranges for pins. All pin guides must be obeyed; the router will not choose between alternate pin guides. If it appears that a net has already been fully covered by prior pin guides, and further pin guide will be discarded.

**What Next**

Please make sure that there are not duplicate or alternate pin guides for a single net.

---

## DPPA-217

(Warning) Path %s is neither manhattan style or 45 degree. This all-angle path is not supported for push-down or pop-up.

**Description**

A push or pop objects operation encountered a path that is "all-angle" (neither manhattan, nor 45-degree). Such a path is unsupported in this tool, and will not be pushed or popped.

**What Next**

Try to find out how such a path got introduced into your design. It is an illegal path.

---

## DPPA-218

(Error) Net %s was determined to have one or more paths that overlap a block edge collinearly for block instance %s. This overlap scenario is unsupported by push_down_objects for signal net routing.

**Description**

A push_down_objects for signal routing encountered a path that is "collinear" with a block edge. While this scenario is allowable for P/G nets, it is not allowed for signal routing. A "collinear" overlap is a wire that "straddles" a block edge in a way that is not perpendicular to the edge. Also, any via that only partially overlaps a block edge is also considered to be "collinear". These wire and via "collinear" scenarios are not supported in signal routes

Feedback

because they cannot be split in a way that part is put into the block and the other part is left at the top, or in an abutted block.

**What Next**

Re-route the specified net so that no section of the routing overlaps a block edge collinearly (in parallel with).

---

## DPPA-219

(Warning) Net %s has global routing elements. Push-down/pop-up operations only support detailed routing elements. Skipping this net.

**Description**

A push_down_objects or pop_up_objects operation detected a net that has global routing elements. These operations only support routing which is of type "detail" routing. This net will be skipped.

**What Next**

Make sure the net that you are attempting to push-down or pop-up has detail routing.

---

## DPPA-220

(Warning) P/G routing objects will not be pushed down into block instance %s (or any block instance) due to P/G strap mis-alignment among MIB instances. (This can be overridden with pg_routing push-down option -ignore_misalignment using set_push_down_object_options.)

**Description**

When mis-alignment of P/G straps over MIB instance is encoutered, the default behavior of command push_down_objects is to issue a warning message and to not push down the P/G routing objects in any block. The user can over-ride this by setting the "-ignore_misalignment true" option to the command "set_push_down_object_options -object_type pg_routing".

**What Next**

Please fix the mis-aligned objects or set "-ignore_misalignment" to true using command set_push_down_object_options.

---

## DPPA-221

(Warning) pin %s on cell %s will be ignored for spacing check due to a mismatch in routing layer direction and pin orientation.

Feedback

**Description**

The specified pin has a mismatch in routing layer direction and pin/edge direction. That is the pin is on a vertical routing layer but the pin is horizontal (placed on a vertical edge) or vice-versa.

**What Next**

Make sure that the routing direction is specified correctly for all layers.

---

## DPPA-222

(Warning) The direction of track %s in block %s does not match its layer (%s) routing direction.

**Description**

The specified track's direction (horizontal/vertical) does not match that of its routing layer's direction.

**What Next**

Make sure that the routing direction is specified correctly for all layers and for this track.

---

## DPPA-223

(Warning) Inconsistent routing direction. The layer %s has no tracks in its specified routing direction on block %s.

**Description**

The specified layer has no tracks defined for this cell that match its routing direction (horizontal/vertical).

**What Next**

Make sure that the routing direction is specified correctly for all layers and tracks.

---

## DPPA-224

(Error) Net %s specified for push-down into block instance %s is completely contained with the block instance boundary, but has no logical connection to it. This is not a supported net push-down scenario.

**Description**

The push_down_objects command detected a net that was specified for push-down into blocks that was an interconnection net of cell instances where all the cells overlapped the

Feedback

interior of the block instance boundary. In this case, we do not allow the net push-down, but require instead that the cells be pushed down into the block. If the cells are pushed down, all nets which interconnect between these cells exclusively will automatically be pushed into the block, along with the routing.

**What Next**

Push the cells which connect to this net into the block instead of trying to push the net that interconnect the cells.

---

## DPPA-225

(Error) NDR and track mask constraint mismatch detected.

**Description**

The pin has a non default routing rule that requires a particular mask constraint, which conflicts with the mask constraint on the track.

**What Next**

Check the non defualt routing rule and the track's mask constraints to make sure there are no conflicts.

**See Also**

- create_track
- create_routing_rule
- set_routing_rule

---

## DPPA-226

(Error) NDR and pin mask constraint mismatch detected.

**Description**

The pin has a non default routing rule that requires a particular mask constraint, which conflicts with the mask constraint on the pin.

**What Next**

Check the non default routing rule and the pin's mask constraints to make sure there are no conflicts.

Feedback

**See Also**

- create_routing_rule
- set_routing_rule

---

## DPPA-227

(Warning) Location constraint (%.3f %.3f) for %s %s is outside of block boundary.

**Description**

The pin location constraint is outside of the block boundary. In general, block pin should always be inside block boundary. This constraint will be ignored by the tool.

**What Next**

Please reset pin location constraint

**See Also**

- set_individual_pin_constraints
- read_pin_constraints

---

## DPPA-228

(Warning) MIB abutted pins %s and %s are shorted together.

**Description**

When synthesizing abutted pin connections, the push_down_objects command detected abutted MIB pins which resulted from push-down of routing into another MIB instance.

**What Next**

Examine your placement and/or recreate pre-routes.

push_down_objects(2)

---

## DPPA-229

(Warning) MIB abutted pin %s and terminal %s are stacked, but both are unconnected. Expected that the terminal would be connected to a net.

**Description**

When synthesizing abutted pin connections, the push_down_objects command detected an MIB feedthrough pin which touches a top terminal, but both are unconnected.

Feedback

**What Next**

Examine your placement and/or recreate pre-routes.

push_down_objects (2)

---

## DPPA-230

(Warning) MIB abutted pins %s and top terminal %s are shorted together.

**Description**

When synthesizing abutted pin connections, the push_down_objects command detected an MIB feedthrough pin which touches a top terminal, but both are connected to different nets.

**What Next**

Examine your placement and/or recreate pre-routes.

push_down_objects (2)

---

## DPPA-231

(Warning) Found incompatible abutted MIB pin pair (%s : %s)

**Description**

When synthesizing abutted pin connections, the push_down_objects command detected an MIB feedthrough pin which touches another MIB pin but the pin is inpcompatible with automatic MIB abutted pin stitching.

**What Next**

Examine your placement and/or recreate pre-routes.

push_down_objects (2)

---

## DPPA-232

(warning) Relax non default rule based mask constraints to place pin %s.

**Description**

The pin has non default rule based mask constraints that are incompatible with the track mask constraints settings. The tool has to relax the non default rule based mask constraints to place this pin.

Feedback

### What Next

Check the non default routing rule to make sure the mask constraints are compatible with the track mask constraints settings.

### See Also

- create_routing_rule
- set_routing_rule

---

## DPPA-233

(Warning) Site array %s in block instance %s is unaligned with it's existing counterpart in the parent block. Skipping propagation of this site array.

### Description

When uncommitting a block with site arrays, a site arrays was determined to already exist in the parent block, but one in the child would be unaligned with the one in the parent if propagated to the parent.

### What Next

Check the placement of your block or the offset of the site array within the block and re-run uncommit_block.

### See Also

- uncommit_block

---

## DPPA-234

(Warning) No %s found for parent net %s of popped net %s.

### Description

When popping up a net that was a feedthrough, the command attempted to determine the best parent net to inherit the split net connections, and the reconnect all connections to the chosen parent net. But in doing so, the command failed to either find a driver or other loads on the chosen parent net.

### What Next

This is a system error. Contact your AC.

Feedback

**See Also**

- pop_up_objects

---

## DPPA-235

(Warning) Split net %s is not empty. Will not delete this net.

**Description**

When popping up a net that was a feedthrough, the command attempted to exchange connections from a split net to an inheriting net and then delete the net, but after swapping connections and shapes, apparently the net still was not empty.

**What Next**

This is a system error. Contact your AC.

**See Also**

- pop_up_objects

---

## DPPA-236

(Warning) Non-MIB pin %s/%s would short with MIB feedthrough single pin %s

**Description**

When synthesizing abutted pins for single pins on MIB blocks, it was discovered that a short would occur if the pin were synthesized in the adjacent block.

**What Next**

Please check your placement and pre-routes.

**See Also**

- pop_up_objects

---

## DPPA-237

(Warning) Detected %s blockages in the top with more than %d instances with the same name %s. count = %d\nThese blockages, when propagated, will not be propagated using uniquified names.

Feedback

### Description

When propagating blockages in push_down_objects or pop_up_objects, the command detected blockages passed into the command that have the same name. Normally, these commands try to generate the propagated blockages using unique names. But if the number of redundant blockages with the same name is quite large, the confirmation of uniqueness of a name can be quite costly performance-wise. So when name redundancy exceeds a certain threshold, the command concludes that the redundant use of the name is intentional, and that the user would not want uniquification of blockage names.

### What Next

Please be sure that you intended to pass blockages that use redundant names.

### See Also

- push_down_objects
- pop_up_objects

---

## DPPA-238

(Error) Failed to reconnect split net connections when popping up feedthrough routing for net %s in block instance %s

### Description

When popping up feedthrough routing from blocks in pop_up_objects, the command attempted to re-combine split nets that were previously connected to feedthrough ports of a block, but encountered a problem.

### What Next

This is an internal error. Please report this.

### See Also

- push_down_objects
- pop_up_objects

---

## DPPA-239

(Error) failed to prepare outputport-connected conn-driven split net for consolitation for net %s.

Feedback

### Description

When popping up feedthrough routing from blocks in pop_up_objects, the command attempted to re-combine a port-connected split net that was previously connected to feedthrough ports of a block, but encountered a problem.

### What Next

This is an internal error. Please report this.

### See Also

- push_down_objects
- pop_up_objects

---

## DPPA-240

(Error) Shorted non-MIB pin %s/%s (abuts with MIB feedthrough pin %s)

### Description

When synthesizing MIB abutted pins (because the app option plan.pins.synthesize_abutted_pins was specified), a pin on a MIB instance was found to short with another pin with which it was not logically connected, and the abutment did not match the source abutted pin's type.

### What Next

Correct your existing pin placements or modify the routint pushed into MIB instances before pushing down.

### See Also

- push_down_objects

---

## DPPA-241

(Warning) Wire-stub crossing into blkInst %s at location {%d %d} on net %s which has multiple original inputs. Resolving randomly and assigning this crossing to existing input block pin %s .

### Description

During the push-down of routing that involves "wire-stub"'s crossing into a block instance, the command determined that multiple input connections exist for this block instance, and since the stubs don't physically touch a block-level pin or terminal, this introduces an ambiguity scenario regarding how to assign such input crossings. In this context, the

Feedback

command decides to automatically resolve the ambiguity randomly by choosing the 1st unassigned input connection to this crossing. The same is done for the other input wire stub crossing for this block instance and net.

**What Next**

Confirm that the randomly selected mapping of a wire-stub crossing to a pre-existing input connection is acceptable.

**See Also**

- push_down_objects

---

## DPPA-242

(Warning) Push_down_objects randomly chose one of a group of existing block-level port-connected nets that were part of the same net push-down to assign the feedthrough output ports to for block instance %s.

**Description**

During the push-down of routing that involves "wire-stub"'s crossing into a block instance, the command determined that multiple input connections exist for this block instance, as well as one or more output feedthrough crossings. The command randomly chose one of the existing block-level nets to assign the feedthrough output ports to.

**What Next**

Confirm that the randomly selected mapping of a wire-stub output crossing to a pre-existing input connection is acceptable.

**See Also**

- push_down_objects

---

## DPPA-243

(Warning) Skipping net for feedthrough removal due to MPH connections %s.

**Description**

remove_feedthroughs does not support feedthrough removal on nets that span Multiple Levels of Physical Hierarhcy. So this net is being skipped for feedthrough removal. User can set into lower levels of hierarchy and remove feedthroughs there before removing at the top (or higher level).

Feedback

### What Next

Confirm that the net has MPH connections and remove feedthroughs on lower levels of physical hierarhcy first.

### See Also

- report_feedthroughs

---

## DPPA-244

(warning) Cannot parse side or layer constraints at line %d

### Description

The command cannot parse side or layer constraints pin constraint file. This could happen due to a syntax error. For instance, when specifying side constraints, the command can parse side constraints {sides {1 2 3}} or {sides 1 2 3} but not {sides {1 2} 3} or {sides {{1 2} 3}}

### What Next

Check if the side or layer constraints are specified correctly

---

## DPPA-245

(Warning) Removing feedthrough ports of net %s due to MIB connections.

### Description

The specified net was not passed using the -nets option, but connects to MIB instances whose feedthrough ports are being removed due to feedthrough removal on a (different) user specified net. If feedthrough ports are not removed on this net, it would lead to dangling/unconnected nets.

### What Next

If feedthrough removal was not desired on this net, refine the set of nets specified on the command prompt to not include nets that connect to MIB instances.

---

## DPPA-246

(Warning) Inconsistent routing direction. Layer %s has no tracks in its specified routing direction on block %s

Feedback

**Description**

The specified layer has no tracks in the routing direction that is set on it for this particular block. This can be due to incorrectly set routing or track direction.

**What Next**

Confirm that the track directions and layer routing directions are set correctly.

---

## DPPA-247

(Info) Block pin %s lies directly under and is connected to a top level terminal.

**Description**

The specified port is directly under and connected to a top level terminal, hence no individual or block pin layer constraints are checked this port. The check_pin_placement command ensures that this pin is on the same layer as the corresponding top level terminal.

**What Next**

Nothing needed. Just informational.

---

## DPPA-248

(Warning) Failed to initialize uhSplitCntr object for charging stations. UPF may not be properly loaded. Will not be able to push-down charging stations.

**Description**

The command attempted to create and initialize a uhSplitCntr object for the purpose of checking for or pushing down charging stations (special voltage areas) It likely results from UPF not loaded properly.

**What Next**

Make sure that UPF is properly loaded.

---

## DPPA-249

(Error) Failed to update pushed repeater cells with UPF information.

**Description**

The push_down_objects command, during push-down of charging stations, and associated repeater cells, attempted to associate the pushed-down repeater cells with UPF information of their associated voltage area, but a problem occurred.

Feedback

**What Next**

Contact your Synopsys representative.

---

## DPPA-250

(Warning) Could not locate hierarchical repeater instance %s during charging station push-down.

**Description**

The push_down_objects command, during push-down of charging stations, and associated repeater cells, failed to find a repeater cell in a hierarchical table.

**What Next**

Contact your Synopsys representative.

---

## DPPA-251

(Error) Failed to create child-level charging station %s in block %s.

**Description**

The push_down_objects command, during push-down of charging stations, failed to find a child-level charging station (voltage area).

**What Next**

Contact your Synopsys representative.

---

## DPPA-252

(Error) Charging station %s is not appropriate for pushing down into block instance %s. Reason: %s

**Description**

The push_down_objects command, during push-down of charging stations, determined that the reference block of the instance did not meet necessary UPF qualifications for pushing down the charging station. The reason should be specified in the error message.

**What Next**

Depending upon the reason reported, the user may want to 1) check the original UPF, 2) check current design MV status (using report_power_domains, or check_mv_design, etc.), or 3) check if any warning message can be found in the split_constraints log output.

Feedback

## DPPA-253

(warning) No wire track is defined for block %s

### Description

The tool detects that there is no track defined in a block. If no track is defined for all the blocks including top level block, pin placement will not be able to place any pins.

### What Next

Please define wire tracks and try again.

### See Also

- create_track

## DPPA-254

(Warning) Cell instance %s belongs to a non-primary voltage area %s and is not a single-rail repeater cell. This cell cannot be pushed down into any block.

### Description

The push_down_objects command, during push-down of cells, determined that a cell instance belonged to a non-primary voltage area (also referred to as a gas station), and that cell instance is not a single-rail repeater cell. Currently, we do not support push-down of any cells that belong to a gas station unless it is a single-rail repeater cell. And then, in that case, such single-rail repeater cells can only be pushed down by pushing down the parent gas station which contains the repeater cell.

### What Next

Only attempt stand-alone cell push-down for cells that belong to a primary voltage area.

## DPPA-255

(Warning) Cell instance %s belongs to a non-primary voltage area %s and is a single-rail repeater cell. This cell can only be pushed down into a block by pushing down the owning voltage area (also referred to as gas station), during which the member repeater cells will be pushed down along with the gas station.

### Description

When pushing down cells, the *push_down_objects* command determined that a cell instance belonged to a non-primary voltage area (also referred to as a gas station), and that cell instance is a single-rail repeater cell. The tool does not support push down of

Feedback

any cells that belong to a gas station unless it is a single-rail repeater cell. Such single-rail repeater cells can be pushed down only by pushing down the parent gas station that contains the repeater cell.

**What Next**

Instead of attempting to push down gas station repeater cells standalone, push down the owning voltage area (gas station).

---

## DPPA-256

(Error) Cell instance %s is a dual-rail cell instance. This is currently only supported by push_down_objects if the cells have no UPF strategy and if you use the -allow_multi_rail_cells option to the set_push_down_object_options.

**Description**

The push_down_objects command, during push-down of cells, determined that a cell instance was a dual-rail cell. Push_down_objects currently does not support the MV requirements of dual-rail cells if they have a UPF strategy, and so this instance will be skipped. This can be over-ridden by using the set_push_down_object_options option, "-allow_multi_rail_cells true", but only if the cell instance has no UPF strategy.

**What Next**

Use the "set_push_down_object_options -object_type cells -allow_multi_rail_cells true" option to push down multi-rail cells with no UPF strategy.

**See Also**

- set_push_down_object_options

---

## DPPA-257

(Info) No feedthroughs are found in the design. Feedthrough shadow ECO script creation is skipped.

**Description**

The feedthrough shadow ECO script is to be used for re-creating the feedthroughs from push_down_objects and/or place_pins. However, the design does not have feedthroughs or the call of push_down_objects and/or place_pins with the -write_shadow_eco option does not create any feedthroughs. Therefore, feedthrough shadow ECO script creation is skipped and no scripts are created.

**What Next**

Nothing needed. Just informational.

Feedback

## DPPA-258

(Warning) Specified width/length for layer %s is less than the layer minimum width/length %s

### Description

The user defined pin width or length constraint, or the non default routing rule width associated with net on the pin, is less than the minimum value defined by the technology rule. As a result, the user defined constraint is ignored for pin placement or pin check.

### What Next

Please double check the technology rule to make sure that user defined width or length constraint, or the non default routing rule, does not violate those defined in technology rule.

## DPPA-259

(Warning) Specified %s for layer %s is less than the layer minimum %s %s

### Description

The user defined pin width or length constraint associated with net on the pin, is less than the minimum value defined by the technology rule for this particular net. As a result, the user defined constraint is ignored for pin placement or pin check.

### What Next

Please double check the technology rule to make sure that user defined width or length constraint, does not violate those defined in technology rule.

## DPPA-260

(Error) Check_objects_for_push_down currently only supports object types: nets and charging stations.

### Description

Currently, the check_objects_for_push_down only supports 2 types: 1) Nets and 2) charging stations (gas stations or non-primary voltage areas).

### What Next

Only attempt to call this command for Nets or Charging Stations (but not both types at the same time).

Feedback

## DPPA-261

(Error) Check_objects_for_push_down received no valid net (with routing) or charging station objects.

**Description**

The check_objects_for_push_down command did not receive any objects of type "net" or "charging station" (the 2 object types currently supported by the command).

**What Next**

Please specify a valid net or charging station for the command to check for push_down. (But one type at a time, net or charging station).

## DPPA-262

(Error) Check_objects_for_push_down can only be called for one object type at a time (currently either nets or charging stations, but not both).

**Description**

Because the check_objects_for_push_down returns a collection of objects found to violate some rule or cause a problem with push_down, it will not attempt to mix object types in the collection returned. For this reason, it is necessary to restrict the use of this command by allowing only one object type to be checked at a time. The user is not allow to mix object types in the same call to this command. Thus, the collection which may be returned by the command will be specific to the type specified to the command.

**What Next**

Only call the command for one object type at a time per call. (Either net or charging station.)

## DPPA-263

(Error) Could not intialize UPF for charging station push-down or check.

**Description**

During push_down_objects or check_objects_for_push_down for charging stations, the command could not initialize UPF. This usually means that UPF is not loaded or complete for this design.

**What Next**

Make sure that the UPF is correctly defined/loaded for this design.

Feedback

## DPPA-264

(Error) Shapes of charging station %s only partially overlap block instance %s.

**Description**

During push_down_objects or check_objects_for_push_down for charging stations, the command discovered that a charging station shape overlapped a block instance, but was not fully enclosed within the block instance boundary. This is not supported by push_down_objects for charging stations.

**What Next**

Make sure that the charging station voltage area shapes are fully enclosed by a block instance before attempting to push the charging station down into the block.

## DPPA-265

(Error) Failed to create charging station %s in block %s for blkInst %s

**Description**

During push_down_objects or check_objects_for_push_down for charging stations, the command failed to create a charging station in the deepest block that the charging station overlapped.

**What Next**

This may be a tool problem. Report this.

## DPPA-266

(Error) No technology defined for current design

**Description**

The tool cannot find technology assoicated with current design.

**What Next**

Please setup technology before running pin placement or global route related commands

## DPPA-267

(Warning) Is necessary to extend a pin geometry outside the block boundary to meet DRC requirement. Block %s port: %s pin box: {{%s %s}{%s %s}} layer: %s

Feedback

**Description**

During push-down of routing, a route overlap with a block resulted in a pin geometry that did not meet DRC rule requirements for minimum width. This is usually due to a rare case of the path only very slightly overlapping the block. In this case, the command is forced to extend the pin outside the boundary a very small amount, because extending the pin inside the boundary may result in a spacing violation.

**What Next**

Nothing required.

---

## DPPA-268

(Warning) Didn't find any enabled planning block for %s.

**Description**

The tool couldn't find an enabled block to run the specified command. This could happen if user uses set_editability to disable all the physical blocks from design planning or user forgets to define physical block in the design. This could also happen if there is no writable physical blocks (blocks are read only)

**What Next**

Check that set_editability are set correctly and/or check if there is any physical block in the design or check if all the blocks are writable.

---

## DPPA-269

(Error) Place_pins cannot be used with crosstalk driven mode in global route.

**Description**

The command detected that crosstalk driven mode is turned on while it cannot be used together with place_pins since place_pins uses global route to determine the pin locations.

**What Next**

Please reset the app option route.global.crosstalk_driven to "false"

---

## DPPA-270

(Error) Cannot load in technology of layer %s using its layer number.

Feedback

**Description**

The command cannot load the layer due to a failed query of the layer's technology using its layer number. This may happen due to user error or data corruption in the database.

**What Next**

Use check_design command for a sanity check.

---

## DPPA-271

(Warning) Place_pins and design planning global route cannot be used with single connection to pin routing mode. This single connection option will be ignored.

**Description**

The command detected that single connection to pin routing mode is turned on while it cannot be used together with place_pins or design planning global route since design planning global route is used to determine the pin locations for pin placement purpose.

**What Next**

Please reset the app option route.common.single_connection_to_pins to "off"

---

## DPPA-272

(Warning) Net %s connects to unplaced instance %s.

**Description**

The check_pin_placement detour check detected that this net connects to unplaced instances. Pins on such instances cannot be used for detour computation and will be ignored for detour computation. This can cause large detour values for such nets.

**What Next**

Ensure such instances are placed to allow for meaningful detour computation or ignore these nets for detour if applicable.

---

## DPPA-273

(Error) Push_down_objects detected that, during the push-down of the net %s into block instance %s, a feedthrough chosen for re-use in the block (based upon how the routing crosses the block) is already connected on this instance (to nets %s). Skipping this net for push-down.

Feedback

**Description**

The command detected that it could re-use an existing feedthrough based upon a pre-existing feedthrough in the block, but that the feedthrough ports of the block feedthrough are already connected to nets in the top. This is probably due to redundant top-level pre-routes of different nets that are exactly the same.

**What Next**

Please examine and correct the pre-routes over this block instance and make sure that they are not in conflict with each other.

---

## DPPA-274

(Warning) push_down_objects detected that, during the push-down of the charging station %s into block instance %s, a voltage area already existed by that name. The name that was created (%s) was modified to produce a unique name.

**Description**

The command detected that voltage area already existed in the ultimate destination block, and so the new voltage area of the charging station was created with a uniquified name based upon the original name.

**What Next**

Nothing required.

---

## DPPA-275

(Warning) A parent-net-to-block-net mapping problem has been detected for net %s and reference block %s. A top-level CLOCK net has a connection to a block instance pin (%s) but no block-level CLOCK net previously existed. A new block-level net %s was created.

**Description**

During push-down of routing, where a net has an existing connection to a block, the command builds a map between the parent net and the block net. If the parent net is a CLOCK net and a port exists on the block connected to that net, but no corresponding CLOCK net exists in the block, one is created.

**What Next**

Make sure that this is the behavior that you expect, or make sure that you have created clock nets in the block correctly.

Feedback

## DPPA-276

(Error) unexpected failure to locate a feedthrough net in block instance %s when updating feedthrough information for feedthroughs created during the push-down of net %s.

**Description**

During push-down of routing, some unexpected failure occurred in feedthrough creation in one of the block instances that the routing passed over.

**What Next**

Please check the routing for this net and report this problem.

## DPPA-277

(Warning) Net %s does not overlap all originally-connected blocks on the net. Block Instances that have no routing overlap: {%s}

**Description**

During push-down of routing with feedthroughs allowed, the command determined that not all originally-connected block instances had routing overlapping the block instance.

**What Next**

Please check the routing for this net and, if needed, fix the routing so that the command can traverse routing into every orgiginaly-connected block instance.

## DPPA-278

(Warning) Encountered more wire stub crossings into block instance %s than original inputs to the block instance for net %s. The push_down_objects command will generate feedthrough connections since feedthroughs are allowed for this net, but you may wish to make sure this result is what is intended, as this involves ambiguity concerning wire-stub crossings into the same block.

**Description**

During push-down of routing with feedthroughs allowed, the command encountered wire-stubs crossings from a driver to a receiving block in a scenario where more stubs existed thatn the number of inputs into the block. In this scenario, the command will create separate feedthrough split nets from the driver block to the receiving block. The stubs that overlap the receiving block (beyond the stubs consumed by the original inputs) will result in separate dangling subnets in that block. If this is not what you intended, then please consider turning of feedthrough allowance for this net and re-run push-down. The result

Feedback

with feedthroughs turned off will be EEQ pins for one of the original input pins into the block.

**What Next**

Please check the routing for this net and make sure the results are what you desire.

---

## DPPA-279

(Error) path %s crosses blkInst %s boundary at 45 degrees but crossing is unsupported.

**Description**

During RDL push-down of routing, an unsupported 45-degree crossing of a block boundary was encountered that is not supported.

**What Next**

Please check the routing for this net and make sure the routing is what was intended.

---

## DPPA-280

(Error) Failed to connect new synthesized port %s to tie-low net %s.

**Description**

During abutted pin synthesis during push-down of routing, the command failed to create a tie-low net in one of the abutted blocks.

**What Next**

Report this issue.

---

## DPPA-281

(Error) %s segment of path %s overlaps block inst %s in a collinear direction. Segment bbox: {{%s %s}{%s %s}} block edge: {{%s %s}{%s %s}}

**Description**

The command detected routing that overlaps a block collinearly (overlaps in a parallel fashion). The command does not support such a routing scenario.

**What Next**

Check the routing for this path and correct it before attempting to push it down again.

Feedback

## DPPA-282

(Warning) Charging Station (%s) shape was trimmed to fit inside destination block %s boundary.

**Description**

The push_down_objects command detected a charging station shape which did not fit completely inside the destination block boundary. It was trimmed to fit inside the block.

**What Next**

Make sure this is what was expected. If not, then you may wish to modify the original top-level charging station boundary to fit inside the final destination lower-level block.

## DPPA-283

(Warning) %s is not on the same physical level with %s. The instance will be skipped for writing topological constraints.

**Description**

The current instance is not one the same physical level with driver. Therefore, the instance cannot be recognized by ID in topological constraints. Skip the instance and continue building topological record.

**What Next**

Skip the current instance and continue building topological record.

## DPPA-284

(Warning) Instance %s %s. This instance will be ignored.

**Description**

The specified instance is is not a valid physical block instance that can be pushed down or popped up.

**What Next**

Make sure that you passed the cells that you intended into the command.

## DPPA-285

(Error) Destination block %s does not contain tech via definition %s. Skipping creation of via in destination block (source via %s). This may indicate a design flow problem.

Feedback

### Description

During creation of a via in a block (during push_down_objects or pop_up_objects), it was determined that the via was a tech via, meaning it is defined in a tech file. The destination block was in a different library thant the source block, and that libray tech did not contain the via definition. This likely indicates a design flow problem. Perhaps a bottom-up flow using DEF files, in which the DEF file for the block did not contain the via definition.

### What Next

If you are using a bottom-up DEF flow, make sure that the DEF file for blocks contains via definitions for all vias you wish to push into the block from the top.

---

## DPPA-286

(Warning) Cannot parse keyword %s at line %d in pin constraints file %s. This line is ignored.

### Description

The command cannot parse the specified keyword on the line indicated in the pin constraints file. Make sure the keyword and its required argument(s) are properly enclosed by a pair of curly braces. For example, {BUNDLE bundle1}, {SIDES {1 2 3}}, {CELL U1}.

### What Next

Check if the keyword and its required argument(s) are correctly specified and properly enclosed by a pair of curly braces.

### See Also

- read_pin_constraints

---

## DPPA-287

(Warning) Some nets were pushed down. However, one or more nets failed to push down. Please check all Warning and Error messages.

### Description

During push-down of routing, some nets pushed down without problems. But some nets encountered problems, and were not pushed down. The command will continue to push down net routing even if errors are detected on some. Please carefully check the log for Error message and fix problems described in the error messages.

### What Next

Check the log file for errors.

Feedback

## DPPA-288

(warning) net %s connects to %d ports on blocks

### Description

There are too many block pins connected to the same net. This is more likely a user error as too many block pins connect to the same net will usually lead to low quality of result of pin placement.

### What Next

This may be due to tie high or tie low net that is not created properly. Please check original net list or check how the tie nets are created.

### See Also

- check_mv_design
- connect_pg_net

## DPPA-289

(Warning) The following blocks are disabled for planning (no pin or feedthrough will be created on those blocks):%s.

### Description

Report the blocks that are disabled for planning in the current design.

Here, "disabled" can mean any one of the following: - The block is not editable - The block is in a library that is read-only on disc - The block is opened in another session

### What Next

Try setting the block as editable by using the *set_editability* command.

### See Also

- set_editability

## DPPA-290

(Warning) Did not push down cell rows (site arrays) or wire tracks into any block instance because of MIB misalignment.

Feedback

**Description**

The command has detected that the user has chosen to consider all block instances for cell row (site array) / wire track push-down. But the command has also detected that some MIB instances are misaligned with respect to each other. In this case, cell row (site array) and wire track push-down is prevented for all block instances (MIB and non-MIB instances). If you want to force push-down into all instances, consider setting the session option, -ignore_misalignment:

set_push_down_object_options -object_type pg_routing -ignore_misalignment true

Alternatively, you may force push-down of cell rows (site arrays) and wire tracks by using the -cells option of the push_down_objects command, and specify only non-MIB instances. If only non-MIB instances are included in the -cells collection, then the command disregards the MIB misalignment and allows the requested push-down operation.

**What Next**

Fix the misalignment issue with MIB's, or choose one of the option mentioned above.

---

## DPPA-291

(Warning) No slots found for pin creation on edge %d

**Description**

The command has detected that there were no slots for pin placement/creation on the specified edge. This is most likely due to lack of any tracks overlapping that particular edge, and can happen during pin-update due to wire-tracks not being appropriately pushed down.

In such a situation the tool will attempt to place/move pins to nearby edges. It is recommended to run place-pins or place-pins -incr after block reshaping.

**What Next**

Ensure that tracks are defined correctly either through the GUI or command line. Also run place-pins after reshaping is complete.

---

## DPPA-292

(Info) No edges changed for cell %s

**Description**

The pin-update code detected no change in edges for the specified cell. Hence pin-placement is kept unchanged.

Feedback

**What Next**

Nothing needed. Just informational.

---

## DPPA-293

(Error) Two cells overlap the same MIB reference block %s at the same reference location in the reference cell, but are different cell types. Cell1: %s Cell1 type: %s Cell2: %s Cell2 type: %s

**Description**

During push-down of cells over MIB instances, two cells were found to conflict in their location over their respective MIB block instances. In this case, the types of the cells were different. If multiple cells overlap an instance of an MIB reference block, they must be placed exactly by using the same orientation and the same reference block.

**What Next**

Fix the placement of cells overlapping the MIB instances.

---

## DPPA-294

(Error) Two cells overlap the same MIB reference block %s at the same reference location in the reference cell, but have different reference orientations. Cell1: %s Cell1 ori: %s Cell2: %s Cell2 ori: %s

**Description**

During push-down of cells over MIB instances, two cells were found to conflict in their location over their respective MIB block instances. In this case, the difference was in the orientation of the cells. If multiple cells are overlapping an instance of a MIB reference block, then they must be placed exactly by using the same orientation and the same reference block.

**What Next**

Fix the placement of cells overlapping the MIB instances.

---

## DPPA-295

(Error) Two cells overlap the same MIB reference block %s at the same reference location in the reference cell, but are not identically overlapping. Cell1: %s Cell1 bbox: {{%s %s} {%s %s}} Cell2: %s Cell2 bbox: {{%s %s}{%s %s}}

Feedback

**Description**

During push-down of cells over MIB instances, two cells were found to conflict in their location over their respective MIB block instances. In this case, the difference was in the bounding box of the cells. If multiple cells are overlapping an instance of a MIB reference block, then they must be placed exactly by using the same orientation and the same reference block.

**What Next**

Fix the placement of cells overlapping MIB instances.

---

## DPPA-296

(Warning) Net %s is logically a bidirectional (INOUT) net, but physically it is unidirectional. Feedthroughs will be created because the net direction can be determined.

**Description**

Feedthrough creation on INOUT nets has been disabled. The net in question is logically an INOUT net, but has been implemented in the physical netlist as unidirectional. Because the direction is known the feedthrough port directions can be properly set and a feedthrough net will be created.

**What Next**

If the desire is to not have a feedthrough net on this net, it must be excluded using set_individual_pin_constraints.

---

## DPPA-297

(Error) Push down of cells over MIB instance %s failed. Skipping push down for all instances of MIB block reference %s.

**Description**

During push-down of cells over MIB instances, a problem occurred when pushing cells into an MIB instance.

**What Next**

Check and ensure that the cell placement over all instances of this MIB block is correct.

---

## DPPA-298

(Error) Attempt to create pin %s at location where a pin %s alreaddy exists.

Feedback

**Description**

During abutted pin synthesize, it was detected that a pin about to be created would overlap with a previously created abutted pin.

**What Next**

Please report this problem.

---

## DPPA-299

(Warning) Parent signal net to child clock net mapping detected for parent net %s to child net %s in block %s for block instance %s. Allowing this mapping, but you may wish to check if this was intended.

**Description**

During push_down_objects for routing, a parent-to-child mapping process detected a net type mis-match between the parent net and a net in a child block that the top-level net was connected by a port on the child block. The command allowed this mapping to occur even though the parent net was of type SIGNAL and the child net was of type CLOCK. If this is not what you intended, you may wish to investigate this mis-match.

**What Next**

Make sure this net type mis-match is what you intended.

---

## DPPA-300

(Error) Chains over MIB instances that have at least one identical cell overlap, are not identical for all MIB instance of the same reference block %s.

**Description**

During push-down of cells over MIB instances of the type reported, the push_down_objects command detected that there were at least some identical cell overlap over at least some of the instances. When identical cell overlap is detected, and involves a connected change of more than one cell, it is required that this chain be replicated identically over all MIB instances of the same type for any instances that have any identical cell overlap.

**What Next**

Make sure that your cell chain that has some identical overlap is identical for each MIB instance that you want to re-use a chain within.

Feedback

## DPPA-301

(Error) Problem with identical cell overlap over MIB insances (for reference type %s). Skipping pushdown into block instances of this type.

### Description

During push-down of cells over MIB instances of the type reported, the push_down_objects command detected that there were at least some identical cell overlap over at least some of the instances. When identical cell overlap is detected, and involves a connected change of more than one cell, it is required that this chain be replicated identically over all MIB instances of the same type for any instances that have any identical cell overlap.

### What Next

Make sure that your cell chain that has some identical overlap is identical for each MIB instance that you want to re-use a chain within.

## DPPA-302

(Warning) Cell instance %s is a Multi-Rail cell with no associated strategy. It will be allowed for push-down, but the user should be aware that this cell, and any related ports created during push-down of this cell will NOT be automatically updated with UPF constraints. It is the responsibility of the user to perform UPF updating of this cell and related ports.

### Description

During push_down_objects for cells, an option to allow Multi-Rail cells with no MV strategy to be pushed down was enabled, and a cell that qualified for this exemption was detected and pushed down. In this scenario, the command does NOT automatically update the new new cell with MV constraints, nor any new ports created associated with the push-down of this cell, so it is the responsibility of the user to do this UPF updating.

### What Next

Nothing.

## DPPA-303

(Error) Cell instance %s is a Multi-Rail cell %s. The push_down_objects command does not support such a cell for push-down. This cell will not be included for push-down.

Feedback

**Description**

During push_down_objects for cells, an option to allow Multi-Rail cells with no MV strategy to be pushed down was enabled, and a cell that was not qualified for this exemption was detected and prevented from pushing down.

**What Next**

Make sure that you used the option to allow Multi-Rail cell push-down with no MV strategy, that you only pass cell which qualify for this option.

---

## DPPA-304

(Error) Cell instance %s is an MV Power-Switch cell. The push_down_objects command does not support such a cell for push-down. This cell will not be included for push-down.

**Description**

During push_down_objects for cells, a MV power switch cell was included in cells for push-down. This is because the insert_power_switch should have been used to directly add power switch cells into the block already.

**What Next**

Don't attempt to push-down top-level power switches.

---

## DPPA-305

(Error) Physical block instance %s does not have a valid physical boundary. This instance will be skipped for push/pop consideration.

**Description**

During push_down_objects block instances were examined for suitability for push-down (to receive objects). This block instance was either specified for push-down, or it was among all blocks for push-down. It was determined that this block did not have a valid physical boundary, without which it is not possible to perform push-down operations.

**What Next**

Check why you seem to have physical blocks, but some don't have valid physical boundaries.

---

## DPPA-306

(Error) Problem experienced in loading block boundaries.

Feedback

**Description**

During push_down_objects block instances were examined for suitability for push-down (to receive objects). During that process one or more blocks were encountered which had some problem.

**What Next**

Make sure all your physical blocks in your design are valid.

---

## DPPA-307

(Error) No valid blocks to be pushed into from top.

**Description**

During push_down_objects block instances were examined for suitability for push-down (to receive objects). During that process, the command concluded that there were no valid physical blocks to push into.

**What Next**

Make sure you have at least one valid physical block before calling push_down_objects.

---

## DPPA-308

(Information) %d %s created for %s.

**Description**

This message reports the number of mask constraint routing blockages or placement blockages created for a given block. If the reported number is 0, it is most likely there are no mask constraints defined.

**See Also**

- create_mask_constraint_routing_blockages

---

## DPPA-309

(Warning) There are(is) %d terminal(s) on %s with invalid layer number. The first a few are(is) %s.

**Description**

The indiciated terminals have invalid layer number. Please check the design to make sure they are created correctly.

Feedback

**See Also**

- create_mask_constraint_routing_blockages
- create_shape

---

## DPPA-310

(Warning) The top_action

**Description**

During push_down_objects for cells, the top_action "keep" was specified. The push_down_objects command only supports push-down of cells that are physical-only, and ignores any other cells passed into the command.

**What Next**

Nothing required.

---

## DPPA-311

(Warning) Top_action was

**Description**

During push_down_objects for cells, the top_action "keep" was specified. The push_down_objects command only supports push-down of cells that are physical-only, and ignores any other cells passed into the command. In this case, during inspection of cells passed into the command, the command found no cells which were physical-only.

**What Next**

Make sure to include only cells which are physical-only when using the cells object type top_action "keep"

---

## DPPA-312

(Warning) Skipped some non-physical-only cells during push-down of cells.

**Description**

During push_down_objects for cells, the top_action "keep" was specified. The push_down_objects command only supports push-down of cells that are physical-only, and ignores any other cells passed into the command. In this case, apparently the collection of cells passed into the command for push-down included non-physical-only cells. These were disregarded during this cell push-down session.

**What Next**

Make sure to include only cells which are physical-only when using the cells object type top_action "keep"

---

## DPPA-313

(Warning) The location of block pin %s is illegal. The block pin will be skipped for writing out existing physical and topological constraints.

**Description**

The location of the current pin is illegal. This might be because that the pin is placed outside of owner blocks's boundary, which is not supposed to happen. The pin will be skipped with this warning printed when writing out existing physical or topological constraints.

**What Next**

Check the pin placement result of this pin.

---

## DPPA-314

(Warning) Warning output directory %s already exists, over-writing existing data.

**Description**

The check-pin-placement command detected that the output directory specified already exists. The command will over-write any existing previously created files/reports in this direcotry.

**What Next**

Ensure the directory name is specified as expected.

---

## DPPA-315

(Error) Failed to push down site array into block %s (blkInst: %s

**Description**

During push-down of site arrays into a block, the command failed to successfully create a site array in the destination block. Please look for other messages describing unexpected scenarios.

**What Next**

Report this problem.

Feedback

## DPPA-316

(Warning) Destination ndm library %s has no tech information. Propagation of physical elements to the new block is not possible. Disregard this message if physical information in the new block is not required at this point.

**Description**

During commit_blocks (and during create_black_box) it was detected that a destination block did not possess technology file information. This information is necessary in order to do any real physical propagation of objects from a parent block to a new block. This can happen if the destination block is targeted to be placed into a separate ndm library which has been created without any technology file information.

**What Next**

Please make sure that you have created a destination library with technology file information. If physical information is not required at this point in the flow, then this message may be disregarded.

## DPPA-317

(Error) Destination ndm library %s has no tech information. Cannot propagate preferred routing directions.

**Description**

During propagation of preferred routing directions during wire track push-down into a destination block, the command determined that the destination block's ndm library had no technology information. This is a serious missing item.

**What Next**

Please make sure that you have created a destination library with technology file information.

## DPPA-318

(Error) Cannot open file %s . Skipping output file creation.

**Description**

Check Pin Placement could not open the specified file for writing the results of the check. As a result the command will revert back to displaying results on the command prompt.

**What Next**

Make sure read/write permissions are set appropriately.

Feedback

## DPPA-319

(Warning) Block %s has multiple views across all its instances.

### Description

As the instances of the same block have different views, the change made to one view will not sync to the other.

## DPPA-320

(Warning) Block connection %s has multiple physical pins (EQ pins - %d pins). Skipping this connection for Abutted Pin Synthesis.

### Description

During Abutted Pin Synthesis as part of push_down_objects for net routing , the command encountered some connections on block instances that had multiple physical pins.
These are referred to as EQ pins. Currently, Abutted Pin Synthesis does not support the synthesis of connections that have multiple physical pins in different parts of the block.

### What Next

Nothing, except just be aware that such pins are currently not supported.

## DPPA-321

(Information) The following blocks do not have associated technology files:%s.

### Description

Report the blocks that do not have associated technology files. If a block does not have a technology file with it, the tool cannot know physical data needed, e.g. minimum width allowed on a certain layer. Therefore, any operations or checks that are related to physical data cannot be performed.

### What Next

Check that the technology file is correctly set.

### See Also

- create_lib
- set_ref_libs

Feedback

## DPPA-322

(Error) Only one of -pins, -nets, -ports and objects can be specified.

### Description

Command only accepts one of the -pins, -nets, -ports and objects arguments. All of these options can be used to limit the checking to the specified set of pins/nets/ports/instances respectively.

### What Next

Please select only one of the -pins, -nets, -ports, objects arguments.

### See Also

- place_pins

## DPPA-323

(Error) %s definition does not exist for %s %s.

### Description

During Design Planning command, the command encountered a via or a row site which does not have a definition. This can happen if the user has removed the via or site definition using TCL commands. If the user has done this, it renders all vias or site rows which reference that definition invalid.

### What Next

Make sure your tech file contains the definitions for all vias and cell sites.

## DPPA-324

(Warning) Detected nets inside blocks that are of ABSTRACT views. The first few such nets are: %s.

### Description

During estimating wire length, the tool discovered some nets inside blocks that are of ABSTRACT views. Because ABSTRACT view does not contain the complete netlist, the estimated wire length of the corresponding net segments inside the blocks with ABSTRACT views may not be accurate.

### What Next

Change the blocks of ABSTRACT view to DESIGN view.

Feedback

**See Also**

- get_estimated_wirelength
- change_view

---

## DPPA-325

(Warning) Detected pins or ports without physical pin shapes. The first few such pins or ports are: %s.

**Description**

During estimating wire length, the tool discovered some pins or ports that do not have physical pin shapes. Therefore, the wire length estimation for the net segments that connect to those pins or ports will be ignored.

**What Next**

Check if the reference libraries are correctly linked, check if all block pins are properly placed.

**See Also**

- get_estimated_wirelength
- place_pins

---

## DPPA-326

(Warning) Detected pins or ports with multiple physical pin shapes. The first few such pins or ports are: %s.

**Description**

During estimating wire length, the tool discovered some pins or ports that have multiple physical pin shapes. The wire length estimation for the net segments that connect to those pins or ports will randomly choose one of the pin shapes.

**What Next**

Check if the multiple pin shapes with the same pin or port is expected.

**See Also**

- get_estimated_wirelength
- place_pins

Feedback

## DPPA-327

(Error) Block pin %s (shape %s) does not overlap the block it belongs to.

### Description

During push_down_objects for signal routing with feedthroughs, the command detected a driving pin of a physical block which did not overlap the block it belongs to. This may indicate a problem in the place_pins command.

### What Next

Check why such a pin exists.

### See Also

- place_pins

## DPPA-328

(Warning) Offset/location constraints ignored by place_pins for fixed pin %s.

### Description

The command detected offset/location constraints set on the specified pin and the pins physical status is fixed. Because such pins cannot be moved by place_pins, this constraint is ignored.

### What Next

Ensure that the pin is marked fixed correctly as expected, also check location/offset constraints to ensure correct intent.

## DPPA-329

(Information) The pin shape of %s is created on the same layer underneath the existing pin shape or terminal of %s because the latter is completely inside the boundary of %s and is connected to the former.

### Description

When a block port is connected to an existing pin that is completely inside the corresponding block boundary, the pin shape of the block port will be exactly the same as that existing pin, i.e. its pin shape, layer, and position.

### What Next

Check if the connected existing pin is created or placed correctly beforehand.

Feedback

**See Also**

- place_pins
- set_individual_pin_constraints

---

## DPPA-330

(Error) Failed to clear timing in top block %s.

**Description**

The command required that existing timing data be cleared for a block before proceeding, but the attempt to clear timing for that block failed.

---

## DPPA-331

(Error) Skip remove feedthroughs due to %s.

**Description**

Remove feedthroughs could not proceed due to the specified reason.

**What Next**

Address the issues mentioned in the previous warning/error messages prior to run remove_feedthroughs.

---

## DPPA-332

(Warning) Parent clock net to child signal net mapping detected for parent net %s to child net %s in block %s for block instance %s. Allowing this mapping, but you may wish to check if this was intended.

**Description**

During push_down_objects for routing, a parent-to-child mapping process detected a net type mis-match between the parent net and a net in a child block that the top-level net was connected by a port on the child block. The command allowed this mapping to occur even though the parent net was of type CLOCK and the child net was of type SIGNAL. If this is not what you intended, you may wish to investigate this mis-match.

**What Next**

Make sure this net type mis-match is what you intended.

Feedback

## DPPA-333

(Information) Identified % instance %s.

### Description

The check pin placement command identified an instance to be of the specified type. This is used to alert the user for non-typical instances that are being checked for pin violations.

### What Next

Ensure such instances are intended for checking pin placement.

### See Also

- report_cells

## DPPA-334

(Warning) The command %s does not support to use options %s together. Option %s is ignored.

### Description

The specified command line options of the corresponding design planning command cannot be used together.

### What Next

Remove the ignored command line option and try again.

## DPPA-335

(Warning) %s of value %.3f on pin %s is ignored due to its abutment to a fixed or existing pin.

### Description

The pin mentioned has a constrained width by individual pin constraint, bundle pin constraint or non default routing rule. But it also abuts to a fixed pin or an existing pin, whose width does not match the constrined width. A pair of abutted pins should have matched pin width. Therefore, the constrained width is ignored.

### What Next

Remove the constrined width or set it to match the pin it abuts to.

Feedback

**See Also**

- set_individual_pin_constraints
- set_bundle_pin_constraints

---

## DPPA-336

(Error) net %s connects to one or more cells which are found to overlap a block that is a target to be pushed into. Please make sure to include these cells for push-down, as the net cannot be pushed down without the cell.

**Description**

The push_down_objects command, when considering pushing down routing for a net into blocks, discovered that the net had pin connections to cells which are overlapping physical blocks that are eligible for pushing into. The push_down_objects command requires that in such a scenario, that the cell be pushed down before, in combination with the net that was specified for push-down. The command cannot push down a net that is connected to some leaf cell on top of a block that the routing overlaps.

**What Next**

Include all connected leaf cells for push-down if they overlap blocks that the routing may be pushed into.

---

## DPPA-337

(Error) Physical block connection %s in reference block %s has no layer. Skip this pin for abutted pin synthesis.

**Description**

The push_down_objects command, when examining new pins on MIB instances from the push-down of routing, encountered a pin that has no shape or layer.

**What Next**

Check if the technology file is correctly set.

---

## DPPA-338

(Information) Using %d threads for pin placement.

**Description**

This message reports the number of threads used in multi-threaded pin placement.

Feedback

### What Next

No action is required.

### See Also

- set_host_options

---

## DPPA-339

(Warning) %s %s from block being uncommitted, %s, could not be propagated upward.

### Description

During uncommit_block, there was a shape that could not be popped-up. This may be a via that exists in the block, but has no via definition that exists in the technology of the block that the via is being attempted to pop up to.

### What Next

Check if the technology file is correctly set.

---

## DPPA-340

(Warning) %s %s from block being uncommitted, %s, could not be propagated upward because a via_def in the parent block exists by the same name (%s), but is defined differently. Skipping this via during uncommit_block route pop-up.

### Description

During uncommit_block, via could not be popped up to a parent block because the parent block already had a via definition of the type of the via being popped up from a child block. But in this case, the definitions were found to be different. In this case, a warning is given and the via is skipped.

### What Next

Make sure that your blocks, if from different ICC2 libraries, have consistent via definitions.

---

## DPPA-341

(Warning) Net %s does not connect to cell %s. The constraint at line %d is skipped. Please consider using topological constraints.

### Description

The net and cell are not connected. The constraint is skipped. This case should be constrainted with topological constraints instead of physical pin constraints.

Feedback

### What Next

Please consider using topological constraints to constrain this case.

### See Also

- read_pin_constraints
- create_topological_constraint

---

## DPPA-342

(Warning) Command %s found that the pin %s has a non-defined block port. This may be the result of a linking problem.

### Description

The command encountered a pin that it can't find the block port information. This may be the result of a linking problem. The command will skip the net that contains this pin.

### What Next

Make sure that your environment set-up includes all libraries that the contains the blocks in this design.

---

## DPPA-343

(Error) Could not find the driver of the net %s. Skip this net.

### Description

During push_down_objects for net routing, the command encountered a problem when examining pins of the net while searching for the driver. This may be the result of a linking problem. The command will skip the net that contains this pin when this happens.

### What Next

Make sure that your environment set-up includes all libraries that the contains the blocks in this design.

---

## DPPA-344

(Warning) Pin geometry extended as a result of the pin_meet_fatwire_rule option. Block %s port: %s pin box: {{%s %s}{%s %s}} layer: %s

Feedback

**Description**

During push-down of routing, a route overlap with a block resulted in a pin geometry that was sized according to the pin_meet_fatwire_rule option. In this case, the command may extend the pin outside the boundary by some amount, because extending the pin inside the boundary may result in a spacing violation.

**What Next**

Nothing required.

---

## DPPA-345

(Error) Abutted pin synthesis failed.

**Description**

During push-down of routing, thus user has specified that the command should synthesize pins and nets for pins which abut after push_down. This process failed for some reason.

**What Next**

Report this.

---

## DPPA-346

(Information) Identified port %s with multiple pins.

**Description**

The check pin placement command identified a port with multiple pins. This message is used to inform users of non-typical ports/pins detected by the command. Such ports with multiple physical pins are usually power/ground or hard-macro pins.

**What Next**

Ensure such pins are intended for checking pin placement.

---

## DPPA-347

(Information) Option %s is ignored by the %s check.

**Description**

The specified option is ignored by the given check as part of the check_pin_placement command. This is expected since certain checks are not intended to honor specified options (like -cell_type, -pin_type etc.). The check will still run as expected while ignoring the specified option. Refer to the man-page for further details.

Feedback

**What Next**

Ensure that options are used as intended.

---

## DPPA-348

(Information) block cell %s is not placed.

**Description**

The tool detected that some of the block cells are not placed. Please fix the placement issue and try again

**What Next**

Ensure that the reported block cells are placed

**See Also**

- report_placement

---

## DPPA-349

(Warning) Net %s in bundle %s does not connect to cell %s. The constraint is skipped. Please consider using topological constraints.

**Description**

The net and cell are not connected. The constraint is skipped. This case should be constrainted with topological constraints instead of physical pin constraints.

**What Next**

Please consider using topological constraints to constrain this case.

**See Also**

- read_pin_constraints
- create_topological_constraint

---

## DPPA-350

(Warning) Net %s in bundle %s does not connect to cell %s. The constraint is skipped. Line number is %d. Please consider using topological constraints.

Feedback

### Description

The net and cell are not connected. The constraint is skipped. This case should be constrainted with topological constraints instead of physical pin constraints.

### What Next

Please consider using topological constraints to constrain this case.

### See Also

- read_pin_constraints
- create_topological_constraint

---

## DPPA-351

(Warning) Topological constraint in line %d is not associated with any net and will be ignored.

### Description

The topological constraints specified are not associated with any net.

### What Next

Check if topological constraints are specified correctly and the check could start from the last net before the given line number.

### See Also

- read_pin_constraints

---

## DPPA-352

(Warning) %s does not have the corresponding layer data of layer number %d.

### Description

The indiciated object has a layer number that does not have the corresponding layer data describing its technology rules and other properties. Make sure the design is correctly linked with the right technology files or libraries.

### What Next

Make sure that your environment setup includes all technology files or libraries.

Feedback

## DPPA-353

(Warning) Filtering MV cell %s from push-down. MV cells are not supported for push-down.

### Description

The push_down_objects command encountered an MV cell instances when examining cells passed into the command for push-down. Currently, such cells are not supported by the push_down_objects command.

### What Next

Avoid attempting to push down MV cells.

## DPPA-354

(Error) Unsupported tech file. Default width %s less than minimum width %s for layer %s .

### Description

Command detected that an unsupported tech file value is used, please fix the tech file with consistent values. Place-pins will error out when this mismatch is detected, the pin checker will issue this message and use the minimum width for size checking and ignore the default width. It is recommended to fix the tech-file and proceed for either command.

### What Next

Fix and re-run command.

## DPPA-355

(Warning) A P/G wire for P/G net %s that is wider than it's length was found to be running collinear to a block instance that it overlaps. Creating pin like a normal pin created from perpendicular overlap.

### Description

During push_down_objects, for a P/G net wire, a wire was determined to overlap a block instance collinearly (the direction of the length of the wire is the same as the direction of the edge of the instance). But in this rare case, the width of the wire is greater than the length of the wire, making the wire look like the opposite type (vertical looks like horizontal, horizontal looks like vertical). Normally, P/G collinear routing results in a pin that is the full shape of the collinear wire that runs collinear with the block edge, but in this rare case, the pin will be created like a normal perpendicular overlapping wire.

Feedback

**What Next**

Make sure the result is what you want.

---

## DPPA-356

(Warning) A P/G wire for P/G net %s that is wider than it's length (based upon start and stop endpoints) was found to be running collinear to a block instance that it overlaps. Creating pin like a normal pin created from perpendicular overlap.

**Description**

During push_down_objects, for a P/G net wire, a wire was determined to overlap a block instance collinearly (the direction of the length of the wire is the same as the direction of the edge of the instance). But in this rare case, the width of the wire is greater than the length of the wire, making the wire look like the opposite type (vertical looks like horizontal, horizontal looks like vertical). Normally, P/G collinear routing results in a pin that is the full shape of the collinear wire that runs collinear with the block edge, but in this rare case, the pin will be created like a normal perpendicular overlapping wire.

**What Next**

Make sure the result is what you want.

---

## DPPA-357

(Warning) Command %s can't find a parent hierarchy for logical instance %s . This may be the result of a linking problem.

**Description**

During a hierarchical command, the command failed to find a parent hierarchy for a logical instance. Sometimes, when there is a problem with linking, the ICC2 database does not get correctly updated to include this parent hierarchy information in association with logical instances.

**What Next**

Make sure that your design includes all necessary reference libraries in the library search path.

---

## DPPA-358

(Warning) Command %s can't find a logical instance from a physical instance %s .

Feedback

**Description**

During a hierarchical command, the command failed to find a logical instance from a physical instance. Sometimes, when there is a problem with linking, the ICC2 database does not get correctly updated to include this physical-to-logical mapping in association with instances.

**What Next**

Make sure that your design includes all necessary reference libraries in the library search path.

---

## DPPA-359

(Warning) Command %s can't find a physical parent instance of a physical pin %s .

**Description**

During a hierarchical command, the command failed to find a physical parent instance from a physical pin. Sometimes, when there is a problem with linking, the ICC2 database does not get correctly updated to include this physical pin to its owning block instance mapping in association with pins.

**What Next**

Make sure that your design includes all necessary reference libraries in the library search path.

---

## DPPA-360

(Warning) Command %s can't find a logical hierarchy for reference block %s .

**Description**

During a hierarchical command, the command failed to find a logical hierarchy for a physical reference block. Sometimes, when there is a problem with linking, the ICC2 database may not get correctly updated to include this logical hierachy to physical reference block mapping.

**What Next**

Make sure that your design includes all necessary reference libraries in the library search path.

---

## DPPA-361

(Warning) Command %s can't find a logical hierarchy for physical block instance %s .

Feedback

**Description**

During a hierarchical command, the command failed to find a logical hierarchy for a physical block instance. Sometimes, when there is a problem with linking, the ICC2 database may not get correctly updated to include this physical instance to logical hierachy mapping.

**What Next**

Make sure that your design includes all necessary reference libraries in the library search path.

---

## DPPA-362

(Error) The command %s detected that some blocks that are eligible for pushing-into (including any block instances specified using the -cells option) do not have tech-file information. The command cannot continue as necessary data may be missing in these blocks and yield bad/wrong results. Blocks without tech file info: %s.

**Description**

During a push_down_objects or pop_up_objects command, the command checked each eligible block for push-down or pop-up, to determine whether necessar tech-file information was presend, and it discovered that some eligible blocks were lacking this information. Since it would be dangerous to continue, the command exited with an error status.

**What Next**

Make sure that each of the blocks in the design have correct tech-file information.

---

## DPPA-363

(Warning) Pin %s with location range constraints has been placed with relaxed spacing.

**Description**

During pin placement, the tool detects that it has to relax spacing constraints in order to place this pin which has location range constraints. The location range constraints maybe set through topological constraints, individual pin constraints or from pin constrain file

**What Next**

Make sure that there are enough room to place this pin within the defined location range constraints

Feedback

**See Also**

- set_individual_pin_constraints
- read_pin_constraints

---

## DPPA-364

(Error) Hierarchical physical block instance %s is not overlapped by top-level routing for net %s. Skip this net.

**Description**

During push_down_objects, for a signal net specified for feedthrough creation where it crosses blocks in ways that require a feedthrough, it was determined that the top-level routing did not reach every block instance that it the net is logically connected to. For feedthrough creation to work at all physical levels, it is required that all physical block instances have routing which at least touches all block that the flat physical net is connected to logically.

**What Next**

Make sure the top-level routing for the net reaches all connected blocks at all levels that are enabled for push-down.

---

## DPPA-365

(Warning) Net %s is connected to mib %s. The related feedthrough ports on all instances of %s will be removed.

**Description**

Feedthrough removal is performanced on blocks. All instances of a same block will be affected.

---

## DPPA-366

(Warning) Ignoring topological constraint on net %s due to %s.

**Description**

During pin placement, the tool detected a topological constraint that could not be honored for the specified reason. The constraint will be ignored.

**What Next**

Correct constraint and re-run command.

Feedback

## DPPA-367

(Warning) Failed to get physical hierarchical block instance for logical instance %s.

**Description**

During the current command, the command tried to generate a physical hierarchical block instance based upon a logical hierarchical instance of a physical block, but failed to generate it correctly. This may be due to a failure in linking or a reference library issue.

**What Next**

Make sure your design can link properly and that your reference libraries are complete.

## DPPA-368

(Warning) The command encountered an invalid hierarchical block port associated with pin %s.

**Description**

During the current command, the command tried to retrieve a hierarchical block port from a pin but the hierarchical block port was invalid. This may be the result of a linking problem.

**What Next**

Make sure your design can link properly and that your reference libraries are complete.

## DPPA-369

(Warning) The command failed to get a logical pin from a logical port for port %s.

**Description**

During the current command, the command tried to retrieve a logical pin for a logical port of a block, but the logical pin was invalid. This may be the result of a linking problem.

**What Next**

Make sure your design can link properly and that your reference libraries are complete.

## DPPA-370

(Warning) The command failed to get a physical pin from a logical pin for logical pin %s.

Feedback

**Description**

During the current command, the command tried to retrieve a physical pin from a logical pin of a block, but the physical pin was invalid. This may be the result of a linking problem.

**What Next**

Make sure your design can link properly and that your reference libraries are complete.

---

## DPPA-371

(Warning) The command failed to get a logical pin from a physical pin for physical pin %s.

**Description**

During the current command, the command tried to retrieve a logical pin from a physical pin of a block, but the logical pin was invalid. This may be the result of a linking problem.

**What Next**

Make sure your design can link properly and that your reference libraries are complete.

---

## DPPA-372

(Warning) The command failed to get a logical port from a physical port for physical port %s.

**Description**

During the current command, the command tried to retrieve a logical port from a physical port of a block, but the logical port was invalid. This may be the result of a linking problem.

**What Next**

Make sure your design can link properly and that your reference libraries are complete.

---

## DPPA-373

(Warning) The command failed to get a physical block port from a physical pin for physical pin %s.

**Description**

During the current command, the command tried to retrieve a physical block port from a physical pin of a block, but the physical port was invalid. This may be the result of a linking problem.

Feedback

**What Next**

Make sure your design can link properly and that your reference libraries are complete.

---

## DPPA-374

(Warning) Command can't find a physical instance from a logical instance %s.

**Description**

During a hierarchical command, the command failed to find a physical instance from a logical instance. Sometimes, when there is a problem with linking, the ICC2 database does not get correctly updated to include this logical-to-physical mapping in association with instances.

**What Next**

Make sure that your design includes all necessary reference libraries in the library search path.

---

## DPPA-375

(Warning) Command can't find a parent block from a physical block instance %s.

**Description**

During a hierarchical command, the command failed to find a parent physical block from a physical block instance. Sometimes, when there is a problem with linking, the ICC2 database does not get correctly updated to include this logical-to-physical mapping in association with instances.

**What Next**

Make sure that your design includes all necessary reference libraries in the library search path.

---

## DPPA-376

(Warning) Command can't find a reference block from a physical block instance %s.

**Description**

During a hierarchical command, the command failed to find a reference block from a physical block instance. Sometimes, when there is a problem with linking, the ICC2 database does not get correctly updated to include this logical-to-physical mapping in association with instances.

Feedback

### What Next

Make sure that your design includes all necessary reference libraries in the library search path.

---

## DPPA-377

(Warning) Block instance being uncommitted, %s, has a tech file that is incompatible with the tech file of the parent block, %s. Routing, if it exists, cannot be safely popped up for this instance. Skipping routing pop-up for this instance.

### Description

During uncommit_blocks, the tool detected that a block instance being uncommitted had a tech file that was found to be incompatible with the parent block of the block instance being uncommitted. This can cause problems during routing pop-up, so routing pop-up for this instance is skipped.

### What Next

Make sure that tech files of blocks to be uncommitted are compatible with their parent blocks' tech files.

---

## DPPA-378

(Warning) Cannot specify port for bundle topological constraints at line %d

### Description

Port is not valid for topological constraints of bundle.

### What Next

Use cell keyword to specify instances for topological constraints of bundle.

### See Also

- read_pin_constraints

---

## DPPA-379

(Warning) Pin %s is placed with relaxed %s.

### Description

During pin placement, the tool detects that it has to relax one or more constraints as indicated in order to place this pin. The specified constraints maybe set through

Feedback

topological constraints, individual pin constraints, bundle pin constraints, block pin constraints, incremental pin constraints, or from a pin constrain file.

**What Next**

Make sure that there are enough room to place this pin within the specified constraints.

**See Also**

- set_individual_pin_constraints
- set_bundle_pin_constraints
- set_block_pin_constraints
- read_pin_constraints

## DPPA-380

(Error) Site array %s already exists in block %s with normal shadow status. Skipping push-down of this site array in that block.

**Description**

During push_down_objects, a site arrays was specified to be pushed into blocks that it overlapped. The command detected that a site array already exists in the block with the same name, and the existing site array has a normal shadow status. In this case, the command will not delete the existing site array and replace it with a pushed-down one. Push-down of the site array from the parent block will be skipped.

**What Next**

Delete the site array manually in the block before attempting to push-down a site array from above.

## DPPA-381

(Error) Site array %s already exists in block %s but has a different site def than the one being pushed down. Skipping push-down of this site array in that block.

**Description**

During push_down_objects, a site arrays was specified to be pushed into blocks that it overlapped. The command detected that a site array already exists in the block with the same name, but the existing site array has different site definition. In this case, the command will not delete the existing site array and replace it with a pushed-down one. Push-down of the site array from the parent block will be skipped.

Feedback

**What Next**

Delete the site array manually in the block before attempting to push-down a site array from above.

---

## DPPA-382

(Error) Problem encountered processing routing push-down for net %s and wire %s and block instance %s.

**Description**

During push_down_objects, non-feedthrough routing push-down encountered a wire which failed to push-down properly into a block instance. This is likely an internal problem.

**What Next**

Report this.

---

## DPPA-383

(Warning) pin %s of hard macro block %s was specified for pop-up, but the pin is not P/G. Skipping this block pin.

**Description**

During pop_up_objects, a hard macro pin was specified for pop-up as a top-level terminal. But the pin specified was not a P/G pin. Currently, pop_up_objects only allows pop-up of pins from hard macros if the pin is P/G.

**What Next**

Please only attempt to pop-up hard macro pins that are P/G pins, and that are connected to a net in the top cell that is port-connected.

---

## DPPA-384

(Warning) Hard macro pin %s is not connected to any net. Skipping this hard macro pin connection for pop-up.

**Description**

During pop_up_objects, a hard macro pin was specified for pop-up as a top-level terminal. But the pin specified was not connected to a net. The pop_up_objects command only allows pop-up of pins that are connected to a net, and the net must be connected to a top-level port.

Feedback

### What Next

Please only attempt to pop-up hard macro pins that are P/G pins, and that are connected to a net in the top cell that is port-connected.

---

## DPPA-385

(Warning) Hard macro pin %s is connected to a net %s, but the net is not connected to a top-level port. Skipping this hard macro pin connection for pop-up.

### Description

During pop_up_objects, a hard macro pin was specified for pop-up as a top-level terminal. But the pin specified was connected to a net, but the net was not connected to a top-level port. The pop_up_objects command requires that the net that the hard macro pin is connected to is also connected to a top-level port so that the command can create a terminal for that port.

### What Next

Please only attempt to pop-up hard macro pins that are P/G pins, and that are connected to a net in the top cell that is port-connected.

---

## DPPA-386

(Warning) pin %s is not on any routing layer

### Description

During pin placement, the tool detects that the pin shape is not on any routing layers. In synthesize abutted pin mode, the tool will skip synthesizing abutted pins for this pins

### What Next

Make sure that the pin shape is on routing layer

---

## DPPA-387

(Warning) The command failed to pre-process a pre-existing feedthrough net, %s in reference block %s, in preparation for allowing that block feedthrough net to be re-used in subsequent feedthrough nets which may cross the same MIB block the same way. This feedthrough net will not be able to be reused.

### Description

During push_down_objects, the command was preparing all possible pre-existing feedthrough nets which feed through blocks. During this process, the command was

Feedback

unable to identify the coordinate and layer of the point in the block where the routing touches the block boundary. As a result, the command will not be able to re-use this pre-existing feedthrough when processing similar crossings of block instances of this same block type.

**What Next**

This is an unexpected scenario. Report this issue.

---

## DPPA-388

(Error) Failed to make top-level reconnections after push-down of cells into blocks.

**Description**

During push_down_objects, the command pushed down specified cells into blocks, but failed to successfully re-connect connections on the blocks that cells were pushed into.

**What Next**

This is an unexpected scenario. Report this issue.

---

## DPPA-389

(Warning) Block instance %s is not DP-enabled for modification.

**Description**

The command attempted to pre-open blocks and encountered a block instance(s) that is not DP-enabled for modification. This block instance will not be modified in this command.

**What Next**

Make sure you expect this block to not be modified.

---

## DPPA-390

(Error) Failed to reopen block %s for edit.

**Description**

The command attempted to pre-open blocks and encountered a block instance(s) that could not re-opened for EDIT. This block instance will not be modified in this command.

**What Next**

Do not include this block for hierarchical design planning operations.

Feedback

## DPPA-391

(Warning) Failed to create RDL pin shapes for net %s.

**Description**

During push_down_objects, the command pushed down an RDL net.

**What Next**

This is an unexpected scenario. Report this issue.

## DPPA-392

(Warning) Block net %s already is associated with an existing routing corridor (%s). Skip this net for current corridor %s.

**Description**

During push-down of a routing corridor, the command considered associating a net with a routing corridor in the block, but the command determined that the net was already associated with an existing routing corridor. In this case, the command will not associae this net with a new routing corridor.

**What Next**

Make sure this is what you expect.

## DPPA-393

(Warning) Failed to set %s %s on new voltage area %s.

**Description**

During push-down of a charging station, the command failed to set one of several charging station attributes from the charging statino in the parent.

**What Next**

Make sure that your UPF is correct.

## DPPA-394

(Warning) The new child-level charging station %s name was changed from the pushed-down charging station name %s due to a name conflict in the destination block.

Feedback

**Description**

During push-down of a charging station, the command encountered a charging station that already existed with the name of the station being pushed down. In this case, the command creates a unique name for the new charging station.

**What Next**

Make sure that this is what you expect.

---

## DPPA-395

(Warning) Failed to set new voltage area attributes for charging station %s.

**Description**

During push-down of a charging station, the command failed to successfully set all attributes on the new charging station.

**What Next**

Make sure that your UPF is correct.

---

## DPPA-396

(Warning) Overlap detected with pre-existing routing in one of the blocks the net %s crosses. Skip this net.

**Description**

During push-down of routing, the command detected a conflict with a pre-existing wire in a block that the net crosses. The command will skip the push-down of the net into the block.

**What Next**

Make sure that you expect this.

---

## DPPA-397

(Error) Failed to re-connect pins for net %s after cell push down.

**Description**

During push-down of cells, the command failed to re-connect a net affected by the cell push-down in the top cell.

**What Next**

Report this.

Feedback

## DPPA-398

(Error) Failed when checking for existing input tap-connections which are driven by cells respectively which ultimately are driven by one outside driver. Existing input tap connection name %s.

**Description**

The push_down_objects command attempted to trace a path across pushed-down buffers in an attempt to collect existing input tap connections which can be traced through multiple paths back to an outside driver.

**What Next**

Report this problem

## DPPA-399

(Error) Failed when processing existing input tap-connections which are driven by cells respectively which ultimately are driven by one outside driver. Existing input tap connection names %s.

**Description**

The push_down_objects command attempted to process a cell push-down connection scenario where multiple input tap connections which can be traced through multiple paths back to an outside driver, but encountered a failure.

**What Next**

Report this problem

## DPPA-400

(Error) Failed when tracing back and existing tap connection %s to the original driving net %s.

**Description**

The push_down_objects command attempted to trace back from an existing input tap connection across buffers to the original driving net, but failed.

**What Next**

Report this problem

Feedback

## DPPA-401

(Warning) No instances of hard macros found in the current block. Skipping hard macro P/G pin pop-up.

**Description**

The pop_up_objects command detected hard macro terminals specified for pop-up, but there were no instances of any hard macros found in the top block. Currently, pop_up_objects does not support MPH pop-up (specifying objects in blocks multiple levels below the current block.

**What Next**

Make sure that you are currently in a block that contains instances of the hard macros that you wish to have P/G pin popped up from.

## DPPA-402

(Warning) No instances of hard macros of type specified for pop-up (%s) found in the current block. Skipping hard macro P/G pin pop-up.

**Description**

The pop_up_objects command detected hard macro terminals specified for pop-up, but there were no instances of that specific hard macro type found in the top block. Currently, pop_up_objects does not support MPH pop-up (specifying objects in blocks multiple levels below the current block.

**What Next**

Make sure that you are currently in a block that contains instances of the hard macros that you wish to have P/G pin popped up from.

## DPPA-403

(Information) Block pin constraints %s are set for %s.

**Description**

Show information about which block pin constraints are set on which blocks.

## DPPA-404

(Information) Block pin constraints of %s are reset to default.

Feedback

**Description**

Show information about which block pin constraints on blocks are removed (reset to default).

---

## DPPA-405

(Information) %d individual pin constraints are removed from %s.

**Description**

Show information about individual pin constraints are removed from blocks.

---

## DPPA-406

(Information) %d bundle pin constraints are removed from %s.

**Description**

Show information about bundle pin constraints are removed from blocks and bundles.

---

## DPPA-407

(Information) %d topological constraints are removed from %s.

**Description**

Show information about topological constraints are removed from blocks.

---

## DPPA-408

(Information) %s pin constraints associated with %s are removed from block %s.

**Description**

Show information about individual pin constraints are removed from blocks.

---

## DPPA-409

(Warning) Net %s is connected to mib %s. The related feedthrough ports on all instances of %s will be removed.

**Description**

Feedthrough removal is performanced on blocks. All instances of a same block will be affected.

Feedback

## DPPA-410

(Warning) Synthesizing abutted pin creation is disabled under multithreading mode of command place_pins.

### Description

Synthesizing abutted pin creation is currently not supported under multithreading mode of command *place_pins*. To enable synthesizing abutted pin creation, please run *place_pins* in single threading mode.

### What Next

plan.pins.synthesize_abutted_pins (3) plan.pins.multi_threading (3)

## DPPA-411

(Warning) Failed to pop block net shapes for net %s from blkInst %s.

### Description

The current command attempted to pop up shape associated with a net in a block, but the attempt encountered a problem.

### What Next

Report this problem

## DPPA-412

(Warning) Ambiguous parent-to-child mapping for top-level net %s and multiple child-level nets in block instance %s. Chosen mapping is to net %s (chosen randomly).

### Description

The push_down_objects command, when trying to map a parent-net to child net for a particular instance found that an instance was connected to the parent net for multiple pins on the same block instance. This creates an ambiguous mapping scenario, and the command must choose one. It chooses the first pin encountered which has a child net and uses that net as the child net to assign the new child-level net shapes to.

### What Next

Make sure that the chosen child net is acceptable.

Feedback

## DPPA-413

(Error) Net %s has fixed pin(s) that do not align with routing. Pin(s): %s. Skipping this net for push-down.

**Description**

The check_objects_for_push_down or the push_down_objects command, when pre-analyzing signal-route push-down, found that one or more fixed pins exist in some physical level which are not connected to any routing or is not aligned with routing being pushed down. This is considered an error and the net will be rejected for push-down until the pin is removed or made non-fixed, or the routing is changed to reach the fixed pin.

**What Next**

Fix the issue related to this/these pin(s) and re-try push-down.

## DPPA-414

(Warning) Skip unplaced pin %s in feedthrough length measurement.

**Description**

Exclude the unplaced pins as measuring the feedthrough net length.

## DPPA-415

(Warning) Instance %s is not a physical block. Skip this instance for topological constraints.

**Description**

Topological constraints does not support non-physical instances. If non-physical instances are specified, the instance will be skipped.

## DPPA-416

(Information) non-fixed pin %s of net %s is overlapped by a route shape but the route shape does not overlap the block instance %s. This non-fixed pin will be preserved because it is necessary for connectivity.

**Description**

The push_down_objects command, when running a pre-check for routing completeness, determined that a logically connected block instance boundary was not overlapped by routing, but it was determined by the checker that a non-fixed pin of the block instance was overlapped by routing because the pin sticks outside of the block instance boundary.

Feedback

Normally, the command will delete pre-existing non-fixed pins on the net, but in this case, the pin is necessary to complete connectivity, so it will not be removed.

**What Next**

This is just information letting you know that a pin was not deleted. There should be no action necessary on your part..

---

## DPPA-417

(Warning) The %s allowed layers on %s %s are in conflict with block pin constraints allowed layers or min/max routing layers.

**Description**

The individual pin constraints, bundle pin constraints, topological constraints allowed layers or the NDR allowed layers of the indicated pin or port do not have common layers with the block allowed layers on the parent block or the top design. Or they are not within the min and max routing layers of the top design. The individual pin constraints, bundle pin constraints or topological constraints allowed layers will be honored.

**What Next**

Check constraints settings to remove the conflict. Or if the tool behavior is expected it is safe to ignore this warning.

**See Also**

- create_topological_constraint
- read_pin_constraints
- set_block_pin_constraints
- set_bundle_pin_constraints
- set_individual_pin_constraints
- set_ignored_layers

---

## DPPA-418

(Warning) Failed to find pin boundary-touching point for reference port %s on reference block %s.

**Description**

When collecting existing pin boundary-touching points for a reference block and port, the command failed to find a valid touching point.

Feedback

**What Next**

Report this

---

## DPPA-419

(Error) Multiple-input wire-stub ambiguity detected with wire-stub overlap of net %s with block instance %s .

**Description**

The command detected that some of the net routing was in the form of wire stubs that overlap a block instance that is multiply-connected to the net, with apparent existence of stubs intended for feedthrough outputs. This is an ambiguous scenario, as the command cannot know which input should be associated with which output feedthrough pin.

**What Next**

Please re-design your wire stubs over this instance to remove ambiguity. A good way to do this is to use full connectivity from input to output.

---

## DPPA-420

(Error) Net %s has wire-stub ambiguity involving some connected instances that are multi-input connections. Skipping this net.

**Description**

The command detected that some of the net routing was in the form of wire stubs that overlap a block instance that is multiply-connected to the net, with apparent existence of stubs intended for feedthrough outputs. This is an ambiguous scenario, as the command cannot know which input should be associated with which output feedthrough pin.

**What Next**

Please re-design your wire stubs over all instances that have been reported to have ambiguity, to remove ambiguity. A good way to do this is to use full connectivity from input to output over a multiply-connected instance.

---

## DPPA-421

(Information) Deleting pre-existing Non-Fixed pin %s for port %s of block %s.

**Description**

The command detected that some non-fixed pre-existing physical pins existed, which get deleted automatically, to be replaced with physical pins based upon the physical overlap

Feedback

of the routing being pushed down. Any pre-existing physical pin which are FIXED are honored and left where they are.

**What Next**

This is just an information message. No action is required of the user.

---

## DPPA-422

(Warning) Ultimate leaf cell driver, %s, has no routing connected to it, and the driver is placed on top of a physical block and this cell has not been pushed into that block, and has no existing connection to that block. No further push-down will be done in block %s for net %s.

**Description**

The command detected that during push-down of routing, an ultimate driving leaf cell at a physical level below the ultimate top block exists which is placed on top of a block with no connection to that block. Push-down will not push routing further down into that block because the driving leaf cell has not been pushed into that block.

**What Next**

This is just an warning message. Make sure the results are what you desire.

---

## DPPA-423

(Information) Skipping net %s because it does not appear to have detailed routing.

**Description**

The push_down_objects command detected a net that has no detailed routing shapes. The command only pushes down nets that have detailed routing (wires and vias). This is only an informational message, as the user may pass all nets into the command, but only some nets may have routing.

**What Next**

Make sure the net that you are attempting to push-down has detailed routing if you specifically expected this net to be pushed down.

---

## DPPA-424

(Warning) Creating new P/G net %s in block %s because block instance %s is not connected to the P/G net that a pushed cell was connected to in the top cell.

Feedback

### Description

The push_down_objects command detected a situation during cell push-down where the cell being pushed down was connected to a P/G net in the top, but the block the cell was pushed into was not connected to that P/G net. In this situation, the command creates a new P/G net using the name of the net that the pushed cell was connected to and appending the "__PUSHDOWN" name tag. This is a warning because this is an indicator that the recommended flow was not followed where the user applies P/G connections to the to block prior to pushing down cells into the block. It is strongly advised that the user deliberately creates the proper P/G nets in the block and run connect_pg_net in the block and in the top cell prior to pushing cells into the block, as connect_pg_net is the proper way to create P/G in a block rather than relying upon push_down_objects to create the power and ground

### What Next

Make sure that the P/G net created by push_down_objects is correctly applying P/G in the block, but it is better if the recommended flow of creating P/G with special P/G commands is followed.

---

## DPPA-425

(Warning) Creating new P/G port %s in block %s because block instance %s is not connected to the P/G net that a pushed cell was connected to in the top cell.

### Description

The push_down_objects command detected a situation during cell push-down where the cell being pushed down was connected to a P/G net in the top, but the block the cell was pushed into was not connected to that P/G net. In this situation, the command creates a new P/G port using the name of the net that the pushed cell was connected to and appending the "__PUSHDOWN" name tag. This is a warning because this is an indicator that the recommended flow was not followed where the user applies P/G connections to the to block prior to pushing down cells into the block. It is strongly advised that the user deliberately creates the proper P/G nets in the block and run connect_pg_net in the block and in the top cell prior to pushing cells into the block, as connect_pg_net is the proper way to create P/G in a block rather than relying upon push_down_objects to create the power and ground

### What Next

Make sure that the P/G port created by push_down_objects is correctly applying P/G in the block, but it is better if the recommended flow of creating P/G with special P/G commands is followed.

Feedback

## DPPA-426

(Warning) Duplicated constraints are found in topological constraints on line %d.

**Description**

Duplicated constraints are found inside the topological constraints specified in the given line. The duplicated constraints are skipped.

**What Next**

Please remove the duplicated constraints.

## DPPA-427

(Error) MIB feed crossing conflict detected in reference block %s at reference coord: {%s %s} layerNumber: %d Conflicting block instance: %s (input crossing point: {%s %s} top-cell coordinates, {%s %s} reference block coordinates) Reference block instance: %s (input crossing point {%s %s} top-cell coordinates, {%s %s} reference block coordinates).

**Description**

During checking of push-down of signal routing, top-level net routing required a feedthrough net in a Multiply-Instantiated-Block (MIB), but another crossing of this MIB produced an input crossing previously that conflicts with this subsequent input crossing.

**What Next**

Make sure that your MIB-crossing routes for this reference block do not conflict with other crossings of other instance of this reference block.

## DPPA-428

(Error) MIB feedthrough crossing direction conflict detected in reference block %s at reference coord: {%s %s} layerNumber: %d Conflicting block instance: %s ( crossing dir: %s crossing point: {%s %s} top-cell coordinates). Other block instance: %s ( crossing dir: %s crossing point {%s %s} top-cell coordinates).

**Description**

During checking of push-down of signal routing, top-level net routing required a feedthrough net in a Multiply-Instantiated-Block (MIB), but another crossing of this MIB block (involving a previous instance) produced a different crossing direction (in or out) at the same location in the reference block. Feedthrough nets have an input crossing and one or more output crossings, at specific coordinates and layers. When these occur on MIB block instances, the command must check to see if a previously created feedthrough can be re-used. When doing this, it compares the prospective feedthrough (the one that it

Feedback

is trying to see if it can match an already-created feedthrough), with existing feedthroughs, and for each crossing of the current block, it compares direction, and location and layer. In this case, it matched location and layer of an existing feedthrough crossing, but the direction was not the same. This feedthrough cannot be re-used, and in fact represents an error.

**What Next**

Make sure that your MIB-crossing routes for this reference block do not conflict with other crossings of other instance of this reference block.

---

## DPPA-429

(Warning) The sides/layer/spacing constraints on a block are conflict with topological constraints on net %s.

**Description**

The command detected that there are topological constraints on a net that are conflict with side/layer/spacing constraints on a block. The topological constraints will override block's side/layer/spacing constraints.

**What Next**

Please resolve the conflict between topological constraints and block's side/layer/spacing constraints

**See Also**

- read_pin_constraints
- set_block_pin_constraints
- report_topological_constraints

---

## DPPA-430

(Error) Net %s has sub-level routing which does not overlap a receiving block instance in a top-level instance which has multiple connections to the net being pushed down.

**Description**

The command detected, after analyzing the routing for this net, that inside of a block which has multiple input connections to the pushed net, that one or more of the child nets does not overlap all of the logically-connected bocks inside the multiply-connected block.

Feedback

**What Next**

Please fix the routing inside the multiply-connected block instance to reflect the child net logical connectivity.

---

## DPPA-431

(Error) Instance %s is not overlapped by its local routing for net %s.

**Description**

The command detected, after analyzing the routing for this net, that inside of a block which has multiple input connections to the pushed net, that one or more of the child nets does not overlap all of the logically-connected bocks inside the multiply-connected block.

**What Next**

Please fix the routing inside the multiply-connected block instance to reflect the child net logical connectivity.

---

## DPPA-432

(Error) Short(s) was detected during pre-examination of the routing of net %s. This net cannot be pushed down.

**Description**

The command detected, after analyzing the top-level routing for this net, that some of the routing shorted with one or more existing pins that were not connected to the full flat net. This may have occurred in physical levels below the top.

**What Next**

Please fix the routing that shorts with other pins.

---

## DPPA-433

(Error) Top level routing of net %s shorts with pin %s at location {%s,%s} layer number %d.

**Description**

The command detected, after analyzing the top-level routing for this net, that some of the routing shorted with one or more existing pins that were not connected to the full flat net. This may have occurred in physical levels below the top.

Feedback

### What Next

Please fix the routing that shorts with other pins.

---

## DPPA-434

(Warning) Routing for top-level net %s does not overlap a connected block %s, but this will not prevent push-down into the parent block of the non-overlapped instance.

### Description

The push_down_objects command, when running a pre-check for routing completeness, determined that a logically connected block instance was not overlapped by routing, but it was determined by the checker that this would not cause a problem for push-down into the instance's parent block, this non-overlap instance will not prevent the push-down of the net. It will result in no pin created on the non-overlapped instance.

### What Next

Make sure that this is acceptable. You may need to further route this lower level net to connect to the block instance.

---

## DPPA-435

(Warning) Power or ground net %s is detected in bundle %s as bit %u. The bundle pin constraint on this net is ignored.

### Description

A power or ground net is included in a bundle. The bundle pin constraint defined on this bundle, or the global defined bundle pin constraint, is ignored by the tool for this particular net. In general, power or ground net is excluded by the tool during place_pins, check_pin_placement, or check_mib_for_pin_placement.

### What Next

Check the bundle definition and make sure the power or ground nets are excluded from the bundles.

### See Also

- create_bundle

---

## DPPA-436

(Warning) The %s object %s of topological segment of %s %s is not on the same physical level as %s %s. This segment has no effect and is ignored.

Feedback

### Description

The starting object or ending object do not belong to the same parent level as specified, which is illegal for topological constraint segments.

### What Next

Make sure that the constraint pair specified correctly.

### See Also

- read_pin_constraints

---

## DPPA-437

(Error) Net %s in block %s has vias that lack a definition in the database.

### Description

During push-down of routing, a via was detected which has no definition in the database. The net which contains the via was skipped for push-down. This is probably the result of a library setup problem.

### What Next

Fix the via definition issue and re-try push down of this net.

---

## DPPA-438

(Warning) Skipping power or ground net %s. Power or ground nets are excluded for %s.

---

## DPPA-439

(Warning) Potential conflicting topological side constraint detected. The fixed pin %s on net %s is off-edge, %s boundary of %s. The fixed pin will be ignored in global route and pin placement.

### Description

In a segment of topological side constraint, if an element is a fixed pin and is off-edge, the fixed pin will be skipped. New pin will be created on block boundary based on the topological constraints.

Feedback

## DPPA-440

(Warning) Potential conflicting topological constraints detected. Topological segment of net %s connects %s with %s, but the fixed pin %s is not on the facing edge to %s.

**Description**

In topological constraints, the two elements of a segment are not on facing edges and connection of one element is a fixed pin.

## DPPA-441

(Warning) Conflicting block pin constraints detected. Routing direction of allowed layers %s incompatible with %s sides %s of block %s.

**Description**

The allowed sides and allowed layers set in block pin constraints are incompatible.

**What Next**

Please modify the conflicts in block constraints.

## DPPA-442

(Warning) Fixed pin %s might cause newly-placed abutted pin in cell %s to overlap with blockage %s that is close to the abutted edge %d of cell %s.

**Description**

For two abutted blocks, if there is a fixed pin in one block, the tool might place a new pin on the abutted edge of the other block, which could result in overlap with blockage that is close to the abutted edge.

**What Next**

Please review the pin or routing blockages and if the blockages are incorrectly created, either remove or re-create them so that they don't overlap or abut to the fixed pins in the design and also cause the abutted pin to be placed to be placed inside the blockage.

If the blockages are to be retained as it is, then either remove the terminals associated with the reported fixed pins or unfix the terminals (set physical_status to 'unrestricted') so that place_pins can work on the pin to find a legal location for it outside the blockage.

Feedback

**See Also**

- report_pin_blockages
- remove_pin_blockages
- remove_from_pin_blockage
- remove_routing_blockages

---

## DPPA-443

(Warning) Port %s is not a top level port. Please use keyword PIN.

**Description**

Keyword port can only be used to specify a top level port. Otherwise, please use pin.

---

## DPPA-444

(Warning) net %s connected to P/G or tie net, push down will skip pushing down this net and no feedthroughs will be created

**Description**

The push_down_objects command detected a situation during net push-down where the net being pushed down was connected to a P/G net or tie net segment. The PG net or tie net segment can be either in top or inside child block. Push down will skip processing this net and no feedthrough will be created for this net as well.

**What Next**

Make sure that the netlist generated properly.

---

## DPPA-445

(Warning) net %s connected to top level port %s which has multiple terminals. Push down will skip this nets for feedthrough creation.

**Description**

The push_down_objects command detected that the tool trying to push down a net and create feedthroughts. But the tool finds that there is a top level port on this net which has multiple terminals. Push down doesn't support creating feedthroughs for multiple terminals from same port.

Feedback

### What Next

Remove multiple terminals and try again

---

## DPPA-446

(Warning) The marker layer shape %s in hardmarco %s will not be popped up, because the block_action is not "keep".

### Description

The marker layer in the hardmacros can be popped up only when the block_action is "keep".

### See Also

- set_pop_up_object_options

---

## DPPA-447

(Warning) A valid number must be provided for %s.

### Description

The offset range or layer range must have a number equal or larger than zero.

### What Next

Please set a valid number for offset or layer range.

---

## DPPA-448

(Warning) Cannot find a non-reserved wire track whose track width is compatible to pin %s. Use the most closely matched track instead.

### Description

The pin indicated in the warning has width constrained from individual pin constraints, bundle pin constraints, or non-default routing rule. But none of the wire track available for pin placement has the compatible track width. The tool chooses the most closely matched track instead.

### See Also

- place_pins
- create_track

Feedback

## DPPA-449

(Warning) track %s in block %s is ignored for pin placement.

**Description**

The specified track is ignored for pin creation, due to incorrect track specification. As a result no pins can be created centered on this track. If there are many such tracks, then it can limit resources for pin placement.

**What Next**

Please check track defition and re-define tracks as needed.

## DPPA-450

(Warning) Fixed pin %s might cause newly-placed abutted pin to overlap with pre-existing pin %s that is close to the abutted edge %d of cell %s.

**Description**

For two abutted blocks, if there is a fixed pin in one block, the tool might place a new pin on the abutted edge of the other block, which could result in overlap with pre-existing pin that is close to the abutted edge.

## DPPA-451

(Warning) Signal net %s has feedthrough disallowed, push down will not creating feedthrough for this net

**Description**

The push_down_objects command finds that when pushing down a none-PG type net, the feedthorugh setting is disallowed. Push down will push down the net without creating feedthroughs

**What Next**

If user intends to create feedthroughts, please use set_individual_pin_constraints command to set feedthrough allowed for this net

## DPPA-452

(Warning) Net %s is tied to power or ground net. The individual pin constraint is applied to the connected %s %s instead%s.

Feedback

### Description

The signal net that is to apply the individual pin constraint is tied to power or ground net and physically is a power or ground net. Individual pin constraints cannot be applied to power or ground net. Therefore, the tool is applying the same constraint to the applicable pins or ports instead.

### What Next

Check if the signal net tied to power or ground is user's intention.

### See Also

- set_individual_pin_constraints
- connect_pg_net

---

## DPPA-453

(Error) Loop back topological constraints start from and end at cell %s detected without indicating pin names. This constraint is applied on net %s.

### Description

Because the topological path both start and end at the reported cell, user must indicate which pin it starts from and which pin it ends at. Note that such topological constraint can be applied either on the reported net, or on the bundle which contains the reported net.

### What Next

Check the topological constraints.

---

## DPPA-454

(Warning) Fixed pin %s in cell %s is overlapping with blockage %s.

### Description

This warning is reported when a terminal which has physical-status as 'Fixed' is found to abut or overlap with a routing blockage or an associated pin blockage and the pin layer is same as the layers associated with the blockage.

### What Next

If the blockages are incorrectly created, either remove or re-create them so that they don't overlap or abut to the fixed pins in the design.

Feedback

If the blockages are to be retained as it is, then either remove the terminals associated with the reported pins or unfix the terminals (set physical_status to 'unrestricted') so that place_pins can work on the pin to find a legal location for it outside the blockage.

**See Also**

- report_pin_blockages
- remove_pin_blockages
- remove_from_pin_blockage
- remove_routing_blockages

---

## DPPA-455

(Error) Fixed pin %s on cell %s is shorted with pin %s on cell %s.

**Description**

The specified pins are shorted with each other, that is they are on the same layer and touch or overlap with each other.

**What Next**

Confirm in GUI/Error-Browser that the pins are indeed shorted. Move/edit pin locations as appropriate.

---

## DPPA-456

(Information) Skipping pin %s for pin legalization.

**Description**

The place_pins command detected that the specified pin/port does not have any shape and is hence skipped for legalization. The legalization mode of place_pins is intended to be used to legalize pre-existing pin placement.

**What Next**

Use place_pins in regular mode (that is without -legalize) to create pin placement for such pins.

---

## DPPA-457

(Warning) Pin %s has incomplete routing.

Feedback

**Description**

The specified pin/port has routing, but its routing is incomplete and hence does not connect to another pin.

**What Next**

Check if the routing is missing.

---

## DPPA-458

(Warning) The fixed pin %s on net %s is off-edge, %s boundary of %s. The fixed pin will be ignored in global route and pin placement.

**Description**

The tool detects that there is a fixed pins not placed within owner block boundary, the fixed pin will be skipped. New pin will be created on block boundary.

---

## DPPA-459

(Warning) Synthesizing abutted pin creation is disabled under legalize mode of command place_pins.

**Description**

Synthesizing abutted pin creation is not supported when using the *-legalize* option of the command *place_pins*. This option is hence ignored during pin legalization. To enable synthesizing abutted pin creation, please run full *place_pins* without the legalize option.

**What Next**

plan.pins.synthesize_abutted_pins (3)

---

## DPPA-460

(Warning) Edge %d on block %s is not aligned with the manufacturing grid.

**Description**

The specified edge is not aligned with the manufacturing grid. This can cause pins to also not align with the grid and create potential routability issues.

**What Next**

Ensure that the design/floorplan setup is valid.

Feedback

## DPPA-461

(Warning) Cannot find compatible mask constraint for pin %s. Use center track mask constraint instead.

### Description

The tool cannot find a compatible mask constraint to set on the specified pin. It then uses the center track's mask constraint instead, which may cause mask constraint conflict with surrounding tracks.

## DPPA-462

(Warning) Detected invalid topological constraint. Cell %s and cell %s cannot be connected directly. This constraint is applied on net %s.

### Description

Topological constraints have indicated direct connection between two cells. However, in the given floorplann, these two cells cannot be connected directly.

## DPPA-463

(Warning) Detected infeasible connection of net %s, between cell %s and cell %s. There are feedthrough-unallowed or un-editable cell(s) separate cell %s and cell %s.

### Description

In the given floorplan, two cells can be connected only through some other cells. However, these other cells are not allowed for feedthroughs, or marked as un-editable.

### What Next

Review the block pin constraints for all the blocks that are between the cells reported in the message and modify them to enable feedthroughs through the required blocks. Also, review the hierarchical edit control set on the blocks that separate the reported cells.

### See Also

- report_block_pin_constraints
- report_editability

## DPPA-464

(Warning) Detected inconsistent blockage across MIBs. Blockage %s for cell %s does not have a counterpart for cell %s.

Feedback

**Description**

Inconsistent blockage across MIBs could cause routing/pin-assignment difficulties.

**What Next**

Verify if the blockages that are present on only a subset of the MIB instances (that belong to the same block reference group) are required. Remove them if they are not required before running place_pins.

In the scenario where the blockages are required, it is not necessary to replicate similar blockage across all the MIB instances. Pin placement would be able to detect and tolerate such inconsistencies and will place the pins with relaxed constraint (spacing, offset, layer) if necessary.

---

## DPPA-465

(Warning) Detected range/offset overlap between net/bundle/pins constraints. Net/bundle/pins %s overlaps with net/bundle/pins %s.

**Description**

Range/offset overlap between net/bundle/pins constraints could cause routing/pin-assignment difficulties.

**What Next**

If it is an offset overlap, please re-define the pin constraint to avoid offset violations/pin alignment issues after pin placement.

In case of a range overlap, if the overlapping range has enough tracks to place all the bundle pins within the specified range then you could ignore this warning. If not, please re-define the pin constraint to avoid offset violations/pin alignment issues after pin placement.

---

## DPPA-466

(Warning) Detected fixed pin %s associated with user constraints.

**Description**

Fixed pin will not be touched in pin assignment. So the user constraints on these pins will be ignored.

---

## DPPA-467

(Warning) Detected range/offset overlap between net/bundle/pins constraint and existing fixed pin. Offset/range constraint of net/bundle/pins %s overlaps with fixed pin %s.

Feedback

**Description**

Range/offset overlap between net/bundle/pins constraint and existing fixed pin could cause routing/pin-assignment difficulties.

**What Next**

If it is an offset overlap, please re-define the pin constraint to avoid offset violations/pin alignment issues after pin placement.

In case of a range overlap, if the overlapping range has enough tracks to place all the bundle pins within the specified range then you could ignore this warning. If not, please re-define the pin constraint to avoid offset violations/pin alignment issues after pin placement.

---

## DPPA-468

(Warning) Detected invalid topological constraint. The sides specified for cell %s and cell %s cannot be connected directly. This constraint is applied on net %s.

**Description**

Topological constraints have indicated direct connection between two cells on specified sides. However, in the given floorplann, the specified sides cannot be connected directly.

---

## DPPA-469

(Warning) Detected bad topological constraint. The sides specified for cell %s and cell %s are opposite to the facing sides. This constraint is applied on net %s.

**Description**

Topological constraints have indicated very unfavorable side connection for two cells, as the specified sides are opposite to the facing sides of the two cells.

---

## DPPA-470

(Warning) User provided window length is too small.

**Description**

The user provided value for window length to be used for the pin density map is too small. The default value will be used instead for pin density computation. A small window length leads to a very large number of windows/buckets and can adversely affect runtime.

**What Next**

Use a value greater than or equal to the default/minimum window length value.

Feedback

## DPPA-471

(Information) Using window length %s microns for pin density computation.

**Description**

The specified value is being used for pin density computation. Each edge of a physical block will be divided into sub-sections of the specified size for demand/capacity computation. Refer to man-page for further details.

**What Next**

This is just an information message. No action is required of the user.

## DPPA-472

(Information) Port instance %s is associated with pin blockage/guide %s, but this port is NOT on block.

**Description**

The port reported is not on a physical block.

## DPPA-473

(Warning) Detect block edge with no available tracks: side %d in cell %s does not have tracks on layer %s.

**Description**

The edge reported did not have intersecting track on the reported layer. This will cause pin assignment to skip such block edges for pin placement and may also have a significant impact on pin assignment QoR.

## DPPA-474

(Warning) Detected a net included in multiple bundles: net %s in block %s is included in the following bundles: %s.

**Description**

The net reported is included in multiple bundles. These bundles may have conflicting constraints for that net.

**What Next**

Review the bundle definitions and conflicting bundle pin constraints. Resolve any unintentional flow errors and bad pin constraints.

Feedback

## DPPA-475

(Warning) Detected dimension mismatch between the boundary and bbox, for block %s in %s view. At least one side the block bbox cannot be touched by the block boundary.

### Description

This can happen when physical objects of a block is outside of the boundary. This mismatch can cause incorrect result in global-routing/pin-assignment.

### What Next

Move the objects into the block boundary, or change the block boundary to enclose the objects.

### See Also

- move_objects
- resize_objects
- reshape_objects

## DPPA-476

(Warning) Detected inconsistent boundary between the %s view and frame view of the block %s.

### Description

This boundary inconsistency can lead to incorrect results of global-routing/pin-assigment.

### What Next

Either remove the frame view of this block from the design library or re-create the frame view for the block.

### See Also

- remove_blocks
- create_frame

## DPPA-477

(Warning) Detected fixed pin outside of block boundary: fixed pin %s is outside of the boundary of cell %s.

Feedback

### Description

*place_pins* will retain this out-of-boundary fixed pin shape while create an additional pin shape on the block boundary based on the applicable pin constraints. The redundant out-of-boundary pin shape can trigger off-edge and off-track violations during pin assignment quality checks.

### What Next

To avoid such redundant pin shapes, user can remove such terminals before running pin placement.

### See Also

- remove_terminals

---

## DPPA-478

(Warning) Detected conflict in layer constraint between pin spacing control and block pin constraints specified for block %s, on side %u. Layer %s %s allowed in block pin constraint, but %s allowed in pin spacing control. Layer %s will be ignored when pin is placed on side %u of block %s.

### Description

Layer constraint conflict has been detected between the block pin constraints and pin spacing control section defined in the pin constraint file read in using *read_pin_constraints*. Hence the layer mentioned in the warning will not be used for pin placement on the highlighted edge of the block. Such conflicts can arise due to the following reasons:

(1)Pin spacing control section does not specify spacing for a metal layer which is included in the list of allowed routing layers for a block in block pin constraints.

Note: For those edges specified in the pin spacing control section, those layers that are NOT listed are NOT allowed for pin placement.

(2)Pin spacing control section specifies spacing for a metal layer which is not included in the list of allowed routing layers for a block in block pin constraints.

Note: If the layers listed for an edge of the block in pin spacing control are NOT allowed for the corresponding reference blocks in block pin constraints, then those metal layers are NOT allowed for pin placement.

Such incomplete constraints might impact pin assignment QoR and also cause runtime issues because of less resources available for placing the pins. Please refer to the man page of *read_pin_constraints* for more details on pin spacing control section and its applications.

Feedback

### What Next

Report the allowed layers and pin spacing control constraints using *report_block_pin_constraints*, review the conflict and resolve them accordingly. In the pin spacing control section, specify spacing for all the layers that are to be considered for pin placement.

### See Also

- read_pin_constraints
- set_block_pin_constraints
- report_block_pin_constraints
- remove_block_pin_constraints

---

## DPPA-479

(Warning) Detected invalid topological constraint. Port/pin %s cannot reach its target on cell %s as specified in topological constraint. This constraint is applied on net %s.

### Description

Topological constraints have indicated direct connection between a port/pin cand its target. However, in the given floorplann, they cannot be connected directly.

### What Next

Verify if the port/pin and cell objects specified in the topology map can actually have a direct connection in the given abutted style floorplan. It is possible that an intermediate block cell exists between the two objects and the routing topology should pass through it, therefore not allowing a direct connection between the port/pin and cell.

Also, review the side/layer/offset values specified in the topological constraint and verify if they allow for a successful direct connection between the port/pin and cell objects in the given floorplan.

### See Also

- report_topological_constraints
- remove_topological_constraints

---

## DPPA-480

(Warning) Detected infeasible connection between port/pin %s and its target on cell %s. There are feedthrough-unallowed or un-editable cell(s) separate them.

Feedback

### Description

In the given floorplan, the port/pin reported can only connect to its target through some other cells. However, these other cells are not allowed for feedthroughs, or marked as un-editable.

### What Next

Review the block pin constraints for all the blocks that are between the objects reported in the message and modify them to enable feedthroughs through the required blocks. Also, review the hierarchical edit control set on the blocks that separate the reported cells.

### See Also

- report_block_pin_constraints
- report_editability

---

## DPPA-481

(Warning) Detected conflicting %s constraint. The allowed layer is incompatible with the allowed sides. The constraint is applied on net/pin/bundle %s.

### Description

The command detected that there is no preferred allowed layer compatible with the sides specified in pin constraint. Layer constraint will be ignored or relaxed while placing the pins.

### What Next

Resolve the conflict between the side and layer values specified in pin constraint.

### See Also

- report_individual_pin_constraints
- report_bundle_pin_constraints
- report_pin_constraints
- write_pin_constraints

---

## DPPA-482

(Warning) Detected conflicting bundle constraint. The allowed layer is incompatible with the allowed sides. The constraint is applied on all the bundles %s.

Feedback

**See Also**

- set_bundle_pin_constraints
- report_bundle_pin_constraints

## DPPA-483

(Error) Cell %s will not be moved down into cell %s, since it only partially overlaps with cell %s.

## DPPA-484

(Warning) Detected bad topology plan %s. Cell %s is specified as both start-point and end-point of a path, but the names of the pins are not specified for the start-point and/or the end-point.

**See Also**

- create_topology_plan
- create_topology_node
- create_topology_edge

## DPPA-485

(Warning) Topology node %s is in a loop-back path.

**See Also**

- create_topology_plan
- create_topology_node
- create_topology_edge

## DPPA-486

(Warning) Instantiation pairs of %s, %s and %s, %s have homogeneous logical connections but are associated with heterogeneous physical placement. The related pairs of pins are %s, %s and %s, %s. The related topology plan(s) are %s.

Feedback

### Description

There are homogeneous logical connections between the instantiation pairs of the reported two blocks, however, these pairs have heterogeneous physical placement. For example, block A has two instantiations A_0, A_1, block B has two instantiations B_0, B_1. Logical connection is as, A_0/out-->B_0/in, A_1/out-->B_1/in. The physical placement is as, A_0, B_0 both have orientation of R0; on the other hand,A_1 has orientation of R0, however, B_1 has other orientation (e.g MX). This mismatch between logical connection and physical placement can cause difficulties in pin assignment.

### What Next

It is likely that the heterogeneous physical placement here is a mistake. Please consider make the physical placement also homogeneous.

---

## DPPA-487

(Warning) MIB topology node inconsistency. %s

### See Also

- create_topology_node
- remove_topology_nodes
- report_topology_plans

---

## DPPA-488

(Error) Inconsistency between the actual netlist connectivity and the connectivity of topology edge %s.

### See Also

- get_topology_edges
- report_topology_plans

---

## DPPA-490

(Error) Topology edge %s should have repeater(s)%s, according to net estimation rule of topology plan %s.

### Description

The topology repeaters specified in the reported topology edge failed to honor the corresponding net estimation rule.

Feedback

**See Also**

- get_topology_repeaters
- report_topology_plans

## DPPA-491

(Warning) Topology edge %s of block %s intersects topology edge %s of block %s. This intersection happens inside block %s. The intersection between topology edges can cause difficulty for buffer/repeater insertion.

**Description**

The intersection between topology edges can cause difficulty for buffer/repeater insertion, since the prefered buffer/repeater zones for these two topology edges may both fall in the intersection part.

**See Also**

- report_topology_plans
- create_topology_edge

## DPPA-492

(Warning) Instantiation pairs of %s, %s and %s, %s have heterogeneous logical connections but are associated with homogeneous physical placement. The related nets are %s and %s. The related topology plan(s) are %s.

**Description**

There are heterogeneous logical connections between the instantiation pairs of the reported two blocks, however, these pairs have homogeneous physical placement. For example, block A has two instantiations A_0, A_1, block B has two instantiations B_0, B_1. Logical connection is as, {A_0/out[0],A_0/out[1]}-->{B_0/in[0],B_0/in[1]}, {A_1/out[0],A_1/out[1]}-->{B_1/in[1],B_1/in[0]}. The physical placement is as, A_0, B_0 both have orientation of R0; on the other hand, A_1 and B_1 also both have orientation of R0. This mismatch between logical connection and physical placement can cause difficulties in pin assignment.

**What Next**

It is likely that the homogeneous physical placement here is a mistake. Please consider make the physical placement also heterogeneous.

Feedback

## DPPA-493

(Warning) Detected inconsistent topology edges over MIBs. Topology edge %s and topology edge %s are indicating different implementations for the same net(s) in block %s.

### Description

The reported topology edges have inconsistent either shape layers, or topology repeaters, or net estimation rules.

### See Also

- create_topology_edge
- remove_topology_edges
- report_topology_plans

## DPPA-494

(Information) The pin order specified in topology node %s is inconsistent with the net order specifed in the bundle associated with the topology plan.

### See Also

- create_topology_node

## DPPA-495

(Warning) The bundle associated with topology plan %s is too wide. Extra repeater(s) may be needed to compensate the timing difference among the bundle bits.

### Description

When the bundle is too wide, extra register(s)/repeater(s) may be needed to balance the timing difference between the LSB and MSB, in case of turn. While these extra register(s)/repeater(s) may cause difficulty in the implementation of the topology plan.

### See Also

- create_topology_plan

## DPPA-496

(Error) The topology node %s has pins or ports placed on layer %s, but this layer is not included in the Net Estimation Rule (NER) that is associated with this node.

Feedback

**Description**

The objects assoicated with the topology node should placed only on the NER layers.

---

## DPPA-497

(Error) Topology edge %s of block %s is too close to topology edge %s of block %s. The collision happens in block %s.

**Description**

There may be difficulty in routing topology edges that are too close to each other.

**See Also**

- report_topology_plans
- create_topology_edge

---

## DPPA-498

(Error) Topology edge %s of block %s is too close to the boundary of cell %s.

**Description**

There may be difficulty in routing topology edges that are too close to cell boundary.

**See Also**

- report_topology_plans
- create_topology_edge

---

## DPPA-499

(Error) Topology edge %s of block %s overlaps cell %s, however, the reported edge is not allowed to create feedthroughs or flyover on the reported cell.

**See Also**

- report_topology_plans
- create_topology_edge

---

## DPPA-500

(Information) %s.

Feedback

**Description**

Generic information. For debug purpose. Do not use in normal message.

---

## DPPA-501

(Warning) MIB pins %s and %s connect to same pins on other abutted MIB instances, but the abutted MIB instances have different relative placement.

**Description**

The place_pins or design planning global route command detected that same MIB pins from different MIB instances connecting other abutted MIB instances with same pins. There is no soultion for this placement/connections to place pins on abutted edges. Global router will skip routing the nets connecting to those MIB pins. Pin placement will ignore single pin checking when place those pins

**What Next**

check to see if such connection and placement are reasonable for pin placement

---

## DPPA-502

(Warning) Constraints modified on an uneditable(read-only) block %s and it will be ignored.

**Description**

This message is issued when a read-only block is being modified.

**What Next**

check to see if this block is reasonable to be modified

---

## DPPA-503

(warning) value 'off' for option '-bundle_order' is not valid. Specify one of: ordered, increasing, decreasing, equal-distance, equal_distance

**Description**

Command only accept one of the ordered, increasing, decreasing, equal-distance,equal_distance for option 'bundle_order.

**What Next**

Command only accept one of the ordered, increasing, decreasing, equal-distance,equal_distance for option 'bundle_order.

Feedback

## DPPA-504

(Warning) The user specified nets do not connect to any valid pins.

**Description**

The user provided nets to the check_pin_placement command do not connect to any valid pins. This is most often due to an incorrect/incompatible -cell_type option used with the command. For example if the nets only connect to hard-macro pins but the -cell_type is the default value (BLOCK). This can lead to effectively no pins being checked.

Refer to the man page of check_pin_placement for more details.

**What Next**

Check to see what the correct -cell_type should be used.

## DPPA-505

(Warning) net %s has routing issue: routing causes short between pin %s and %s

**Description**

Command check_objects_for_push_down detects that a net has pre-route that overlap with a MIB instance. When creating pin based on such pre-route on MIB instance, this pin or same pins on other MIB instances will be shorted with other pin(s)

**What Next**

Correct routing issues

## DPPA-506

(Warning) track %s has invalid defintion.

**Description**

Tool detects that the track has an invalid definition. Most likely this is due to incorrect bounding box or high/low coordinates. The track will be ignored.

**What Next**

Please re-define the tracks so that they are valid or remove invalid tracks.

## DPPA-507

(Warning) Track width for track %s is less than layer's minimum width, default width will be used instead.

Feedback

### Description

This message is issued when a user-defined track width is less than the layer's minimum width. In such scenarios, the maximum of the default width and min-width rule defined by the technology file will be used during pin-creation and pin-checking.

### What Next

check to see if the track definition is correct.

---

## DPPA-508

(Warning) Inputs specified for command options -pins, -ports, -cells or -self will be ignored for writing topological constraints.

### Description

When -pins, -ports, -cells or -self command options are specified together with -topological_map and -from_existing_pins, the tool will ignore the values passed to those options while writing out topological map constraints based on existing pin locations.

### What Next

Please specify either net(s) or bundle(s) objects to print their routing topology map to the pin constraint file.

---

## DPPA-509

(Warning) Detected mismatch between origin and bounding bbox lower left coordinates for block %s.

### Description

Origin co-ordinates of the block is not the same as it's bottom left corner co-ordinates of the bounding bbox. This can happen when you resize the left-most or bottom most edge of the block and the origin will no longer be at the bottom left corner of the bounding bbox.

This mismatch can cause tool to print incorrect location co-ordinates based on existing pin placement since the location co-ordinates are written with respect to the block's origin.

If these location constraints are to be re-used in a new version of the design (which now has an origin == bounding_box.ll) to replicate golden pin placement, then read_pin_constraints will discard such invalid location constraints and thus replication of pin placement will not be possible.

### What Next

Reset the block origin to match it's bounding bbox's lower left co-ordinates before writing out pin constraints.

Feedback

## DPPA-510

(Warning) Pin offset/range or start/end specified for side %d of cell %s is outside the block boundary

### Description

Offset/range or start/end values specified for pins/port lies outside the block boundary and hence will be ignored.

### What Next

Input a valid offset/range or start/end value which is within the limits of the associated block edge's perimeter.

## DPPA-511

(Warning) Edge %d on block %s is too close to its parent block edge.

### Description

Tool detects that the specified edge is too close to its parent block edge. Most likely this is due to incorrect floorplan setup. The gap between those edges is less than min-depth on at least one layer. The place_pins command will treat the two edges as abutted, this can cause parent block pins to extend into the child block boundary. It is expected that block edges should either be abutted or at least min-depth/width apart if not abutted.

### What Next

Please fix floorplan issues.

### See Also

- report_block_shaping
- report_placement

## DPPA-512

(Warning) Detected conflicting constraints between %s and %s in line %d.

### Description

When there are conflicting constraints specified in the pin constraint file, the last constraint takes priority when we read in the pin con- straints file.

### What Next

Please review and resolve the conflict between the constraints in the pin-constraint file.

Feedback

### See Also

- read_pin_constraints

---

## DPPA-513

(Warning) Could not find any information in the database regarding how the shadow objects are created. The option -latest_rounds of command write_shadow_eco is ignored.

### Description

The shadow objects in the database could be created by old version of ICC-2. Or they could be created manually or by using shadow eco script created by write_shadow_eco. In either of the cases, they miss certain attribute that the option -latest_rounds of write_shadow_eco relies on.

### What Next

If the option -latest_rounds must be used, then please remove the feedthroughs and recreate them using the latest version of ICC-2 and then do write_shadow_eco again.

### See Also

- remove_feedthroughs
- place_pins
- push_down_objects

---

## DPPA-514

(Warning) There are no valid values specified by option -latest_rounds of command write_shadow_eco. This option is ignored.

### Description

None of the shadow objects in the database were created within the ranges of steps specified by the option -latest_rounds of command write_shadow_eco. It is most likely that the shadow objects in the database were created by N steps place_pins or push_down_objects, but the specified values by option -latest_rounds are all larger than N.

### What Next

Please specify valid values.

Feedback

**See Also**

- place_pins
- push_down_objects

---

## DPPA-515

(Warning) No routing layers found in the design. The command cannot continue.

**Description**

No routing layers were found in the design. They either have not been defined or have been removed. The command cannot complete as a result.

**What Next**

Please specify valid routing layers.

**See Also**

- check_design

---

## DPPA-516

(Information) Loading in %s span spacing rules on layer %s.

**Description**

The command detected the specified span spacing rules and will be loaded into the tool as a technology layer related spacing constraint.

**What Next**

This is just an information message. No action is required of the user.

---

## DPPA-517

(Warning) Found neither the mask1/mask2 spacing rules or same color span rules on layer %s.

**Description**

Tool expects to find exactly one of the per mask based span spacing rules or the generic same color span spacing rules on double patterning layers.

**What Next**

Please check the technology file for specified layer.

Feedback

## DPPA-518

(Warning) Inconsistent entries in the %s spacing rules on layer %s.

**Description**

Tool found that spacing related tables for the specified rule were of in-consistent sizes. This can lead to unexpected spacing constraints during pin creation or checking.

**What Next**

Please check the technology file for specified layer.

## DPPA-519

(Warning) Layer %s has both color based and non color based span spacing rules.

**Description**

Tool expects to find only one of color based span rules and non color based span rules on any given layer. The non color based rules will be ignored.

**What Next**

Please check the technology file for specified layer.

## DPPA-520

(Warning) Run the location check to validate location constraints on ports.

**Description**

The tool detected at least one port with a location constraint. Prior versions of the tool would check these constraints under the -offset switch for on-edge ports. The new option -location should now be used to check location constraints of both on and off edge ports.

**What Next**

Use the -location switch for such ports.

## DPPA-521

(Warning) Hard constraint off is ignored due to other values are specified at the same time.

**Description**

The value spacing, location, layer, length or width is specified together with value off when setting hard constraint in command *set_block_pin_constraints*. The value off is ignored.

Feedback

### What Next

Do not specify off when other hard constraint values are used.

### See Also

- set_block_pin_constraints

---

## DPPA-522

(Warning) Location constraint of %s %s is snapped to edge (side=%u).

### Description

The location constraint specified by user is snapped to the nearest edge. If the intention is to create an off-edge pin by location constraint, please use -off_edge option at the same time.

### What Next

Update the location constraint with -off_edge option is necessary.

### See Also

- set_individual_pin_constraints

---

## DPPA-523

(Warning) %s constraint applied on net %s will be ignored.

### Description

Location/off_edge constraints are not allowed on nets. Therefore, such constraints specified by the user for a net will be ignored.

### What Next

Instead, apply the location/off-edge constraint on the pin that connects to the corresponding net.

### See Also

- set_individual_pin_constraints

Feedback

## DPPA-524

(error) Pin location coordinate for %s %s is outside of block cell boundary and will be ignored.

### Description

If pin location is not inside block boundary, the set_individual_pin_constraints will ignore all setting and error out.

### What Next

Please check pin location coordinate and make it inside block boundary

### See Also

- set_individual_pin_constraints

## DPPA-525

(Warning) Edge %d on block %s is not aligned with the manufacturing grid.

### Description

Boundary co-ordinates of the reported block instance are not compatible with the length precision/scale factor defined in the technology file. Hence the reported block edge is not aligned with the manufacturing grid. The length of all edges of the block must be evenly divisible by the manufacturing grid precision defined in the technology file as shownbelow.

Technology { <snip> unitLengthName = "micron" <snip> gridResolution = 1 lengthPrecision = 2000 <snip> } TF length precision = 1 um / 2000 = 0.0005 um = manufacturing grid

### What Next

Verify the block boundary co-ordinates. Update the block boundary to make all edges be aligned with the manufacturing grid.

### See Also

- set_block_boundary

## DPPA-526

(Warning) Exclude-side constraint set for %s %s includes all the sides of it's parent block.

Feedback

**Description**

All sides of the block are disallowed for pin placement. This could have bad consequences on place_pins outcome and resulting QoR.

**What Next**

Update the exclude sides constraints.

**See Also**

- remove_individual_pin_constraints

---

## DPPA-527

(Warning) Pin-Guide(s) %s on block cell %s on side %d are overlap with routing blockages or pin blockages.

**Description**

The specified block has pin-guides that are overlap with routing blockages or pin blockages. This conflict can lead to sub-optimal pin-placement results or slower runtime as the tool will try to honor pin guide and avoid blockages at same time.

**What Next**

Update the pin-guide/pin blockage/routing blockages constraints.

---

## DPPA-528

(Warning) Entry on row %d of rule %s on layer %s is ignored.

**Description**

The specified entry in the specified rule or table is ignored. This most likely happens because the entry is not correctly formed as expected by the tool.

**What Next**

Ensure the specified rule has correct input.

---

## DPPA-529

(Warning) Loop back connection is detected on net %s with cell(s)%s. These objects are not pushed down into cell %s.

Feedback

**Description**

During push down, loop back net connection is detected in block. The command will not push down anything into that block.

**What Next**

Avoid to push down such unsupported connection.

---

## DPPA-530

(Error) The net %s has no wire-stubs representing the feedthrough connection. Skipping this net.

**Description**

During push-down of signal net routing, it was determined that no wire-stubs represent the required feedthrough.

**What Next**

Correct the wire-stub signal routing.

---

## DPPA-531

(Warning) No -pins, -ports, -nets option specified, constraints will not set for any object.

**Description**

During setting individual pin constraints, must have object specified for setting. Otherwise, the command will not set any constraints into DB.

**What Next**

Specifed object for setting constraints.

---

## DPPA-532

(Error) There is no netlist connection between the %s pin and the %s pin. The command will not add feedthroughs.

**Description**

During specifying the -from and -to pins, please make sure the netlist connection is present before executing the command.

**What Next**

Please run the command on the pins which are connected.

Feedback

## DPPA-533

(warning) cell %s is ignored as its parent cell is not enabled.

**Description**

The cell has been ignored

**What Next**

Check the editability of its parent block

## DPPA-534

(warning) MIB block %s is not aligned on power/ground strap. Found misaligned %s strap %s on metal layer %s at bbox %s.

**Description**

The MIB instance has misaligned strap.

**What Next**

Move or re-shape the MIB blocks with correct dimension and offsets, or update P/G mesh

## DPPA-535

(warning) Terminal(s) will not be created for %d pin(s) that do not have applicable region based pin constraints.

**Description**

The command *create_terminals_for_pins* only creates terminals for pins or ports that have preferred region based constraints. The region based constraint can be just the side constraints or with additional offset or range constraints. Or if the pins or ports already have existing non-fixed terminals.

**What Next**

Please apply applicable region based pin constraints.

## DPPA-536

(warning) Failed to create bundle pin objects. Cannot meet both bundle pin spacing and range constraints.

Feedback

**Description**

The command attempted to create pin objects for bundle pin, however with given bundle pin constraints, there is no enough space meet all requirments.

**What Next**

Modify bundle pin constraints

---

## DPPA-537

(warning) Bundle pins %s to %s contain one or more fixed terminals. Pin creation is skipped.

**Description**

The specified bundle pins have one or more pre-existing fixed pins, i.e. terminals. This prevents pin creation for them.

**What Next**

Remove the pre-exising fixed pins.

---

## DPPA-538

(warning) Logical %s %s has one or more existing fixed terminals. Pin creation is skipped.

**Description**

The specified logical port or pin already has pre-existing fixed pins, i.e. terminals. This prevents pin creation for it.

**What Next**

Remove the pre-exising fixed pins.

---

## DPPA-539

(warning) Cannot derive preferred pin location(s) from pin constraint(s) or existing physical terminal(s) of %s. Pin creation is skipped.

**Description**

The specified port(s) or pin(s) do not have applicable location based pin constraints or existing physical terminals for command *create_terminals_for_pins* to derive an initial pin location.

Feedback

### What Next

Apply suitable pin constraints with side, offset, or range.

---

## DPPA-540

(warning) Pin creation is skipped on block %s because it is read-only.

### Description

The specified is in read-only status because user purposely set it so, or the database is accessed by other session at the same time and is locked. Pin creation is therefore skipped on it.

### What Next

Remove the read-only restriction on the database and re-run *create_terminals_for_pins*.

---

## DPPA-541

(Error) Detected overlapping topology nodes on a same layer. The pins or ports on topology node %s are placed interleavingly with topology node %s.

### See Also

- report_topology_plans

---

## DPPA-542

(Information) Pre-existing pin %s is placed but not fixed/locked. However the newly created topology node will still anchor on it.

### See Also

- create_default_topology_plans

---

## DPPA-543

(Warning) Topology node %s contains or, abuts and connects to fixed pins or ports that are placed on layer %s, but this layer is not included in the Net Estimation Rule (NER) associated with this node.

Feedback

**See Also**

- create_topology_plan
- create_topology_edge

## DPPA-544

(Warning) Bundle %s does not have consistent routing topologies among each of its nets. Ports %s to %s will not be treated as bundle ports.

**Description**

The specified bundle has inconsistent routing topologies and the tool is not able to keep all bits of the pins together. Therefore they are not treated as bundle pins any more.

**What Next**

Check if there are conflicting bundle pin constraints, individual pin constraints, topological pin constraints or other routing related constraints are applied.

**See Also**

- set_bundle_pin_constraints
- set_individual_pin_constraints
- set_block_pin_constraints
- create_topological_constraint
- create_routing_corridor

## DPPA-545

(Warning) The pins or ports associated with topology node %s is inconsistent with the nets in the bundle associated with the topology plan.%s

**What Next**

Investigate the connectivity of the objects listed in the warning for incomplete or wrong connections.

**See Also**

- report_topology_plans

Feedback

## DPPA-546

(Warning) Skip topology edge %s because it does not have valid start or end nodes.

### See Also

- create_topology_edge
- report_topology_plans

## DPPA-547

(Error) Invalid buffer/register module specified for topology edge %s. Skip repeater cell insertions.

### See Also

- create_topology_edge
- create_topology_plan
- set_net_estimation_rule

## DPPA-548

(Warning) Detected orphan topology node %s.

### Description

The reported topology node either connects to no topology edge, or it is a virtual node which connects to only one topology edge.

### See Also

- create_topology_edge
- create_topology_node

## DPPA-549

(Error) Bundle %s has inconsistent number of existing %s across the bits.

### Description

The topology plan associated with the reported bundle will have incorrect topology repeater settings after *optimize_topology_plans*.

**See Also**

- create_topology_plan
- optimize_topology_plans

## DPPA-550

(Error) The route shape(s) for topology edge path %s deviated from the shape layer description.

## DPPA-551

(Error) Topology repeater %s does not have any physical repeater(s).

## DPPA-552

(Error) The physical repeater(s) associated with topology repeater %s are too far away from the previous or the next stage of repeater(s). This distance exceeds the spacing specified in Net Estimation Rule.

**See Also**

- create_topology_plan
- create_topology_edge

## DPPA-553

(Warning) Topology node %s owns pin(s) or port(s) that are not placed together. This can make the topology plan difficult to implement.

**See Also**

- create_topology_plan
- create_topology_node

## DPPA-554

(Warning) Topology edge %s has invalid net estimation rule %s.

Feedback

**See Also**

- create_topology_plan
- create_topology_edge
- set_net_estimation_rule

---

## DPPA-555

(Error) Topology repeater %s is too close to a corner on topology edge %s.

**Description**

The reported topology repeater is infeasible to implement.

**See Also**

- create_topology_repeater
- optimize_topology_plans

---

## DPPA-556

(Error) Topology repeater %s overlaps topology repeater %s.

**Description**

The reported topology repeaters are infeasible to implement.

**See Also**

- create_topology_repeater
- optimize_topology_plans

---

## DPPA-557

(Error) Topology repeater %s overlaps cell or blockage %s.

**Description**

The reported topology repeater is infeasible to implement.

**See Also**

- create_topology_repeater
- optimize_topology_plans

## DPPA-558

(Information) Identified ETM instance %s.

**Description**

The ETM instance is identified while loading the design and is treated as a hard macro.

**What Next**

This message is for information only. No action from user is needed.

**See Also**

- check_pin_placement

## DPPA-559

(Warning) Interleaving bundle pin constraints are too tight for %s to %s on bundle %s. Ignored.

**Description**

The interleaving bundle pin constraints applied on the specified pins of specified bundle are too tight to honor. The tool ignores them and treats the bundle pins as not interleaving.

**What Next**

Review the interleaving bundle pin constraints and relax them or create feasible interleaving bundle pin constraints.

**See Also**

- set_bundle_pin_constraints

## DPPA-560

(Warning) Bundle pin constraints are too tight for %s to %s on bundle %s. Ignored.

Feedback

### Description

The bundle pin constraints applied on the specified pins of specified bundle are too tight to honor. The tool ignores them and treats the bundle pins as normal block pins.

### What Next

Review the bundle pin constraints and relax them or create feasible bundle pin constraints.

### See Also

- set_bundle_pin_constraints

---

## DPPA-561

(Error) Not fully-optimized topology plans (%lu plans): %s.

### Description

The reported topology plans can be one of the following cases:

- The reported plan have node(s) whose constraint type is edge. However, *optimize_topology_plans* should have associated these node(s) with block pins.
- The reported plan have edge(s) without shape_layers. However, *optimize_topology_plans* should have created shape_layers for these edge(s).

### See Also

- optimize_topology_plans

---

## DPPA-562

(Error) Bad shape layers for alignable topology edge %s.

### Description

The start and end nodes of the reported topology edge have pins on a same layer, residing on alignable locations. Therefore, the reported topology edge SHOULD only include that layer.

### See Also

- optimize_topology_plans

Feedback

## DPPA-563

(Error) The topology node %s inconsistent with adjacent shape layer(s). The reported node has pin(s)/port(s) placed on layer %s, but the adjacent shape layer is on %s.

### See Also

- optimize_topology_plans

## DPPA-564

(Error) Topology plan %s has incorrect register stages.

### Description

The number of expected register stages can be specified as the following attributes on the topology plan: *numSeqRepeaters*, *sameNumSeqRepeaters*, or *seqRepeaters*.

### See Also

- optimize_topology_plans
- set_attribute

## DPPA-565

(Error) Topology plan versus netlist connectivity mismatch. %s

### Description

This will be infeasible for optimize_topology_plans.

### See Also

- create_topology_edge
- create_topology_node
- optimize_topology_plans

## DPPA-567

(Warning) Feedthrough pins with alignment violation: %s

### Description

The intra block feedthough pins are unaligned.

Feedback

### What Next

Ensure that alignment tolerance distance is set properly. Its default value is 0.

---

## DPPA-568

(Warning) Failed to update topology plan %s.

### Description

Please check topology plan, it could be routing result not valid, topology plan created with invalid node or edge, etc.

### See Also

- report_topology_plans

---

## DPPA-569

(warning) Edit group %s can not be pushed into cell %s as partially overlap.

### Description

The edit group partially overlaps the cell, cannot be pushed down

### What Next

Double check the location of the edit group.

---

## DPPA-570

(warning) Edit group %s has unsupported members %s to be pushed down.

### Description

The edit group has unsupported members.

### What Next

Double check the edit group members.

---

## DPPA-571

(warning) Edit group %s is not aligned with %s.

### Description

The edit group is not aligned with another edit group.

Feedback

**What Next**

Check for the alignment of these edit groups.

---

## DPPA-572

(warning) The only supported %s action is %s, please make sure the %s action of %s set as %s.

**Description**

The specified top/block action is unsupported.

**What Next**

Specify supported top/block actions.

---

## DPPA-573

(Warning) Feedthrough pins aligned, but on different metal layers: %s

**Description**

The intra block feedthough pins are aligned, but their layers mismatch.

**What Next**

Ensure that alignment tolerance distance is set properly. Its default value is 0.

---

## DPPA-574

(Warning) Skip bundle %s as it does not contain supernet(s).

**Description**

Currently only the supernets bundle is supported.

---

## DPPA-575

(Error) Narrow channel violation detected: Pin %s on edge %d. Missing connected pin on edge %d of cell %s

**Description**

Single pin across narrow channel.

Feedback

## DPPA-576

(Error) Narrow channel violation detected: Pin %s on edge %d. Unaligned with pin %s on edge %d

**Description**

The pins across a narrow channel are unaligned.

## DPPA-577

(Error) Narrow channel violation detected: Pin %s on edge %d. Connecting to multi-fanout net %s

**Description**

The number of pins across a narrow channel should be 2.

## DPPA-578

(Error) Narrow channel violation detected: Pin %s on edge %d. Not connecting to pin %s on edge %d

**Description**

The pins across a narrow channel are not connected.

## DPPA-579

(Error) Narrow channel violation detected: Pin %s on edge %d (on layer M%d). Layer-Mismatch with pin %s on edge %d(on layer M%d)

**Description**

The pins across a narrow channel are on different layers.

## DPPA-580

(Warning) pin %s from bundle %s has been placed at different edge from global routing's location.

**Description**

The bundle pins are not placed at the locations passed from global router. This may happen when there is no enough routing resource to place all the bundle pins together at routing location.

Feedback

### What Next

Check the routing location to see if there are enough routing resource for the bundle nets to route through. If the resource is not enough, you may need to use bundle pin constraints, topological pin constraints, pin/routing blockages or routing corridor to guide the routing to go through desired topologies.

### See Also

- check_pin_placement

---

## DPPA-581

(warning) new segment of original path %s is too short, the end cap is adjusted to fit its length.

### Description

The new segment is too short.

### What Next

Ensure the cut segment is long enough.

---

## DPPA-582

(Warning) Insufficient layers specified for bundle %s. Topology edge %s need to use both horizontal and vertical layers.

---

## DPPA-583

(Error) Invalid topology node %s.%s

---

## DPPA-584

(Warning) Contradictive net: pin %s tries to extend check scope to include net %s, but the net is explicitly excluded.

---

## DPPA-585

(Warning) Conflicting layer constraints between bundle pin constraint %s and Net Estimation Rule (NER) of topology plan %s. The NER will have higher priority.

Feedback

## DPPA-586

(Warning) Bundle %s cannot be folded.

## DPPA-587

(Error) Invalid topology node %s. The reported node has constraint type of origin, but has objects or constrained_objects.

## DPPA-588

(Warning) Skip topology edge %s because its shape layer(s) is crossing current block boundary.

### See Also

- create_topology_edge
- report_topology_plans

## DPPA-589

(Error) Bad sequential_repeaters constraint for topology plan %s.

### Description

The reported topology plan either has invalid syntax for *sequential_repeaters*, or the value specified in *sequential_repeaters* forms a loop with other topology plan(s).

### See Also

- create_topology_plan

## DPPA-590

(Warning) The min/max routing layer specified for net %s inconsist with %s.

### See Also

- set_routing_rule
- read_pin_constraints
- set_block_pin_constraints

Feedback

## DPPA-591

(Warning) Topology plan cannot have '/' in the name. All the '/' will be replaced with '_'.

**See Also**

- create_topology_plan

## DPPA-592

(Error) Physical repeater %s does not associate with any topology repeater.

## DPPA-593

(warning) Bump region %s with pattern anchor block is not supported.

**Description**

Bump region with pattern anchor block is not supported.

**What Next**

Check the attribute.

## DPPA-594

(warning) There is a misalignment region %s in block %s, can not push down the bump region.

**Description**

There is a confliction between new child bump regions.

**What Next**

Check the alignment.

## DPPA-595

(warning) Grid %s is misaligned in block %s, can not push down the grid.

**Description**

There is a confliction between new child grid among MIB cells.

Feedback

**What Next**

Check the alignment.

## DPPA-596

(Error) Overbooked bump region %s. The signal number of the topology plan(s) is too large for the reported bump region.

**What Next**

Please check the bump count and the signal number of the topology plan(s).

**See Also**

- report_attributes

## DPPA-597

(Error) Topology edge %s of block %s is too close to the boundary of bump region %s.

**Description**

There may be difficulty in routing topology edges that are too close to bump region boundary.

**See Also**

- create_topology_edge

## DPPA-598

(Error) Infeasible topology node %s. The reported node maps to a fixed pin %s, but they are on different edges of cell %s.

**Description**

This will be infeasible for optimize_topology_plans.

## DPPA-599

(Warning) There is more than 1 pin connecting to %s.

## DPPA-600

(Warning) The routing informations of net %s are not enough to build a netlist.

Feedback

## DPPA-601

(Warning) Cannot decide the direction of routings for net %s, switch to non-feedthrough mode.

## DPPA-602

(Warning) There is no solution for net %s.

## DPPA-603

(Warning) The routings of net %s has multiple solutions.

## DPPA-604

(Warning) There are multiple solutions for pure internal shapes of net %s.

## DPPA-605

(Warning) Cannot find top level connectivity of %s, skip to map it.

## DPPA-606

(Warning) Skip object %s as its orientation is not supported to push down/pop up.

## DPPA-607

(Warning) Minimum area is too large or minimum length is too small on layer %s, this can cause large minimum pin length in pin placement.

**Description**

Command detected that the min area is too large or min length is too small for a layer. This will result in large minimum pin length for pin placement and can cause pin placement failure. This is likely a setup error, please check to see if there is an issue in the tech file definition.

**What Next**

Fix and re-run command.

Feedback

## DPPA-608

(Warning) The specified spine edge has invalid shape layers. Tool will not create fishbone connection for topology plan %s.

**What Next**

Fix and re-run command.

## DPPA-609

(warning) Feedthroughs of pattern name %s in block %s cannot construct a bus with same dimension to original bus.

## DPPA-610

(warning) Cannot find a unified new bus name for bus %s, skip matching related net-port names

## DPPA-611

(Warning) The specified cell %s has no hierarchical boundary.

## DPPA-612

(Warning) Multiple references are specified for module %s.

## DPPA-613

(Warning) The specified plan has invalid NER layers. Tool will not create fishbone connection for topology plan %s.

**What Next**

Fix NER and re-run command.

## DPPA-614

(error) Specified pin(s) and layer(s) have mismatched directions.

**Description**

Specified pin(s) and layer(s) should have same directions: horizontal or vertical.

Feedback

**What Next**

NA

---

## DPPA-615

(warning) Skip %d unplaced pins.

**Description**

Only apply to placed pins.

**What Next**

NA

---

## DPPA-616

(warning) Term %s is invalid to derive (%s)

**Description**

The term name should meet some conditions.

**What Next**

NA

---

## DPPA-617

(warning) Topology plan %s has node with constraint type tap, cannot proceses this plan.

**Description**

The node with constraint type tap is unsupported.

**What Next**

NA

---

## DPPA-618

(warning) Site rows that belong to site arrays cannot be explicitly popped. Instead, they can only be popped as part of a site array.

**Description**

Site rows belongs to a site array cannot be popped stand alone.

Feedback

**What Next**

NA

## DPPA-619

(Warning) Name collision happens: original name %s, new name %s.

**Description**

Two objects should use different names.

**What Next**

NA

# DPTOPO

## DPTOPO-026

(warning) Rule %s: specified layers %s and %s are not adjacent. User-provided register / buffer spacing is required.

**Description**

The layers specified for the indicated net estimation rule are not adjacent. In this situation, the net estimation timing engine cannot compute register / buffer spacing based on indirect attributes (layer, clock, scenario, etc). Non-adjacent layers are permitted, but only if the user directly specifies user-computed register / buffer spacing values.

**What Next**

Modify the net estimation rule settings to use 2 adjacent layers.

Or

Directly specify register / buffer spacing values for the net estimation rule.

**See Also**

- set_net_estimation_rule
- report_net_estimation_rules

## DPTOPO-027

(error) Net Estimation Rule %s: corner %s has no parasitic parameters set.

Feedback

### Description

The corner assigned to the indicated net estimation rule has no parasitic parameters set - early, late or both parameters are missing. Without this information, this net estimation rule cannot be used to obtain signal velocity for topology plans using the *get_topology_timing* command.

estimate_topology_timing will try to use net estimation rules defined on previous edges (or the main topology plan) to see if those have valid settings. If successful, DPTOPO-030 will be be issued to indicate the situation.

### What Next

Modify the corner parasitic parameter settings using the *set_parasitic_parameters* command.

### See Also

- set_parasitic_parameters

---

## DPTOPO-028

(error) Error timing topology plan %s from %s to %s.

### Description

There was an error estimating timing between the indicated elements of the named topology plan. An earlier error message should indicate the specific problem.

### What Next

Look for errors before this message and attempt to rectify them.

---

## DPTOPO-029

(Error) Rule %s has invalid corner setting (or none)

### Description

The corner referred to by the net estimation rule could not be found, or it was not set.

### What Next

Define a valid corner for the given net estmation rule.

Feedback

**See Also**

- create_corner
- report_net_estimation_rules
- set_net_estimation_rule

---

## DPTOPO-030

(Info) Net Estimation Rule %s was used instead of current rule %s

**Description**

The currently-in-effect net estimation rule for a given part of the topology plan was not properly initialized. Either the corner was not set, or parasitics were not properly initialized, etc. Look for previous errors/warnings.

By searching backwards along the topology plan (towards the start), a propoerly initialized rule was found, and this was used instead to estimate timing for the *estimate_topology_timing* command.

This situation may occur after *characterize_topology_plan* has pushed plans into child blocks and created reference nodes, but not all constraints and/or parasitics for the child block(s) have been properly set up. This is done as a convenience for the user to proceed even in the presence of incomplete data.

**What Next**

Ensure the rule labeled as "current" (the second name given in the error message) is properly configured, with a valid corner, valid parasitics, etc.

**See Also**

- create_corner
- set_net_estimation_rule
- set_parasitic_parameters

---

## DPTOPO-031

(error) Unsupported object type

**Description**

The topology node is associated with unsupported objects in its objects or constrained_objects attribute.

Feedback

### What Next

Update the topology plan or topology node to associate it with supported objects, such as hierarchical block pins or cells.

### See Also

- create_topology_plan
- create_topology_node

---

## DPTOPO-032

(error) Error Find loop in the topology plan %s, can not estimate delay for the topology plan.

### Description

There is a loop in the topology plan. So, estimate_topology_timing can not compute delay for the plan. Please fix the topology plan.

### What Next

Look for the loop and attempt to fix it.

---

## DPTOPO-033

(error) Error Unable to find -%s node, can not estimate delay for the topology plan.

### Description

The user specified node is not found. So, estimate_topology_timing can not continue. Please specify correct node for reporting.

### What Next

Specify correct node for reporting.

---

## DPTOPO-034

(error) Unable to find a path with specified to/from/through topology nodes/repeaters.

### Description

Unable to find a path with specified to/from/through topology nodes. Please provide correct to/from/through topology nodes/repeaters.

Feedback

**What Next**

Please provide correct to/from/through topology nodes/repeaters.

---

## DPTOPO-035

(error) Error Unsupported object type is found in %s list %s.

**Description**

The user specified node or repeater is not found. So, estimate_topology_timing can not continue. Please specify correct node or repeater for reporting.

**What Next**

Specify correct node for reporting.

---

## DPTOPO-036

(error) Error - No valid timing path found with given to/from/through options.

**Description**

Any timing path passes through user specified nodes or repeaters is not found. So, estimate_topology_timing can not report. Please specify correct nodes or repeaters for reporting.

**What Next**

Specify correct nodes/repeaters for reporting.

---

## DPTOPO-050

(Info) Topology Plan %s (referenced by plan %s) was added to the list of reported plans.

**Description**

While creating the report for the *write_topology_report* command, a topology plan in list of plans to be reported contained a topology node that referenced a topology plan not already in list of plans to be reported. To ensure completeness, the referenced plan was added to the report.

**What Next**

To disable this automatic adding of referenced plans, add the *-no_referenced_plans* option to the *write_topology_report* command.

Feedback

**See Also**

- report_topology_plans
- write_topology_report

---

## DPTOPO-100

(Error) No sources available for reporting.

**Description**

Calling the command *report_topology_project* with all 4 options: -no_rules, -no_supernets, -no_bundles and -no_plans will result in nothing to report.

**What Next**

Retry the command and remove at least one of the *-no_* options.

---

## DPTOPO-101

(Info) Reading XML file %s.

**Description**

Informational message printed during the running of read_topology_project.

---

## DPTOPO-102

(Error) Error reading XML datafile %s.

**Description**

An internal error occurred during the reading of a Topology Project XML file.

**What Next**

Verify that the file exists and has not been modified and was produced by a compatible version of write_topology_project.

---

## DPTOPO-103

(Error) %s is a directory, not a valid XML filename.

**Description**

A directory was passed to the -filename argument of read_topology_project. This option expects a valid Topology Project XML file, not a directory.

Feedback

**What Next**

Either use the -directory argument, if this directory is a valid XML directory created with write_topology_project, or pass in a valid XML filename.

---

## DPTOPO-104

(Errro) file %s does not exist.

**Description**

The file passed to read_topology_project does not exist.

**What Next**

Verify that the filename is correct and is available for reading by the current user.

---

## DPTOPO-105

(Information) Found %lu %s entries.

**Description**

Informational message printed by the command read_topology_project.

---

## DPTOPO-106

(Error) %s is a file, not a valid XML Topology Project directory.

**Description**

A filename was passed to the -directory argument of read_topology_project. This option expects a valid directory, not a file.

**What Next**

Either use the -filename argument, if this directory is a valid XML file created with write_topology_project, or pass in a valid XML directory name.

---

## DPTOPO-201

(warning) Topology plan %s has a non-Manhattan segment %s. Using Manhattan approximation.

Feedback

### Description

The *estimate_topology_timing* command detected a non-Manhattan wire / connection (not horizontal or vertical) as part of the indicated topology plan. The wire / connection will be converted to two separate wires, one vertical, one horizontal, and estimation will continue.

### What Next

For more accurate results, ensure that the plan has Manhattan routing / connections.

### See Also

- estimate_topology_timing

---

## DPTOPO-202

(error) Register cell setting '--ideal--' in net estimation rule %s is incompatible with estimate_topology_timing.

### Description

The command *estimate_topology_timing* requires a real lib cell assigned to both the buffer and register settings of the associated net estimation rule. The command cannot proceed to estimate timing with a setting of --ideal-- for the register.

### What Next

Set the register setting of the net estimation rule to a real register lib cell.

### See Also

- report_net_estimation_rules
- set_net_estimation_rule

---

## DPTOPO-203

(error) %s value set for net estimation rule %s is invalid.

### Description

The command *estimate_topology_timing* requires a real lib cell assigned to both the buffer and register settings of the associated net estimation rule. The rule either had an invalid register or buffer setting that was unable to be resolved to a valid lib cell.

### What Next

Set the register and/or buffer setting of the net estimation rule to a real lib cell.

### See Also

- report_net_estimation_rules
- set_net_estimation_rule

---

## DPTOPO-204

(error) Could not find a valid %s pin in lib cell %s of net estimation rule %s.

### Description

A valid pin of the specified type could not be found in the lib cell assigned to the net estimation rule. The *estimate_topology_timing* command requires a net estimation rule to have valid lib cells set for register and buffer settings. Both the register and buffer cells needs to have valid in and out pins. As well, the register needs to have a valid clk pin.

### What Next

Assign lib cells to the register and/or buffer settings for the net estimation rule that correspond to valid lib cells, with valid pins identified in the library.

### See Also

- report_net_estimation_rules
- set_net_estimation_rule

---

## DPTOPO-205

(error) Segment %s object %s has no nets.

### Description

In order to use the *-routing real* option for the command *estimate_topology_timing*, real nets with route shapes must exist in the netlist and be associated to the topology plan.

The specified objects were part of the plan, but no nets were associated with them which could be used to look up layer shape data.

### What Next

Ensure that all the elements of the topology plan are routed.

### See Also

- optimize_topology_plans

Feedback

## DPTOPO-206

(error) Cannot find net connectivity between %s (objects %s) and %s (object %s).

**Description**

In order to estimate timing for topology plans using *-routing real*, each element of the topology plan must be connected by real nets that have routing shapes.

**What Next**

Ensure that the objects are connected in the netlist.

## DPTOPO-207

(error) Cannot determine tech layer for the route shape %s.

**Description**

There was an error determining the tech layer information from the given route shape. *estimate_topology_timing* cannot continue with the option *-routing real* in this situation.

**What Next**

Ensure that the routing shapes and tech layer information is correct.

## DPTOPO-208

(error) Segment %s has %d bits, but sibling %s has %d bits.

**Description**

*estimate_topology_timing* with option *routing real* requires that sibling branches have the same number of bits in a branching topology plan.

**What Next**

Ensure that all the bits of the topology plan are present at every branch.

## DPTOPO-209

(error) Invalid net estimation rule for edge %s.

**Description**

The net estimation rule for the topology edge could not be determined.

Feedback

### What Next

Ensure a valid net estimation rule is set for the topology edge and/or the topology plan.

### See Also

- create_topology_edge
- create_topology_plan

---

## DPTOPO-210

(warning) Instance %s (associated with topology repeater %s) is not placed - only virtual cell timing is possible.

### Description

The indicated cell is not placed. In order to use the *-cell real* option of *estimate_topology_timing*, all cells associated with topology repeaters must be placed in order to determine the distance of the cell from the origin of the repeater.

### What Next

Ensure all cells associated with topology repeaters are placed, or use the *-cell virtual* option (or let *estimate_topology_timing* automatically select that option).

### See Also

- estimate_topology_timing

---

## DPTOPO-211

(error) topology plan %s cannot be timed with real routing.

### Description

The status of the topology plan, its associated objects and the netlist prevent *estimate_topology_timing* from using real routing for calculating the delays of the topology plan.

### What Next

See earlier errors/warnings for information about this situation.

---

## DPTOPO-212

(warning) Edge %s, bit %d, net %s has no routing shapes - cannot time using real routing.

Feedback

### Description

The indicated net, associated with the indicated bit of the indicated topology edge did not have any routing shapes. In order for *estimate_topology_timing* to estimate delays for topology plans using the option *-routing real*, actual routing shapes are required for each net associated with the plan.

### What Next

Ensure all nets associated with the topology plan are routed, or use virtual route timing.

### See Also

- estimate_topology_timing

---

## DPTOPO-213

(warning) Edge %s, bit %d has no nets with route shapes

### Description

The indicated bit of the indicated topology edge did not have any nets with routing shapes. In order for *estimate_topology_timing* to estimate delays for topology plans using the option *-routing real*, actual routing shapes are required for each net associated with the plan.

### What Next

Ensure all nets associated with the topology plan are routed, or use virtual route timing.

### See Also

- estimate_topology_timing

---

## DPTOPO-214

(warning) Edge %s has no nets

### Description

The indicated topology edge did not have any nets associated with it. In order for *estimate_topology_timing* to estimate delays for topology plans using the option *-routing real*, actual nets with routing shapes are required for each bit in the plan.

### What Next

Ensure all edges in the topology plan have real nets associated with them, or use virtual route timing.

Feedback

**See Also**

- estimate_topology_timing

---

## DPTOPO-215

(warning) No prior node with netlist objects found for node %s.

**Description**

In order to use the *-cell real* option of the command *estimate_topology_timing*, each repeater must have actual cells in the netlist associated with it, and they must match the object count of the previous node (ignoring virtual/tap nodes).

**What Next**

Ensure that the topology plan has netlist objects associated consistently with nodes/ repeaters.

---

## DPTOPO-216

(warning) Folded edge %u has %lu objects, but master edge %s has %lu objects.

**Description**

Folded edges must contain the same number of repeater cell references as the master edge in order to use the *-cell real* option of *estimate_topology_timing* command.

**What Next**

Ensure all repeaters / edges have the same number of associated repeater objects and it matches the node object count, or use virtual cell computations.

---

## DPTOPO-217

(warning) Repeater %u of edge %s has 0 objects.

**Description**

Repeaters must have an associated netlist cell for each bit of the topology plan in order to use the *-cell real* option of *estimate_topology_timing* command.

**What Next**

Ensure all repeaters have the required number of assoicated netlist cells.

Feedback

## DPTOPO-218

(warning) Repeater %u of edge %s has %u objects, but the prior node has %u objects.

**Description**

Repeaters must have an associated netlist cell for each bit of the topology plan in order to use the *-cell real* option of *estimate_topology_timing* command. This cell count must match the associated object count of the prior topology node (excluding virtual/tap nodes).

**What Next**

Ensure all repeaters have the required number of assoicated netlist cells and that this count matches the object count of the prior topology node.

## DPTOPO-219

(error) Topology element %s (edge %s) has %d bits, but %s has %d bits.

**Description**

*estimate_topology_timing* with option *routing real* requires that all topology objects on an edge have the same number of associated objects/nets/bits.

**What Next**

Ensure that all topology objects on the edge have properly associated objects.

## DPTOPO-220

(error) topology plan %s cannot be timed with real cell instances.

**Description**

The status of the topology plan, its associated objects and the netlist prevent *estimate_topology_timing* from using real cell instances for calculating the delays of the topology plan.

**What Next**

Verify that every topology repeater in the plan has one cell instance per net associated with it.

Earlier errors/warnings may also give additional information.

## DPTOPO-221

(warning) Cannot determine net(s) for bit %d of edge %s.

Feedback

### Description

*estimate_topology_timing* was unable to find any nets along the given edge for the given bit. This may be because the topology plan is unfinished and does not yet match up to the actual netlist. For example, if the topology plan expects a feedthrough edge, but the actual feedthroughs have not yet been created, or are not part of the supernet(s) of the plan.

### What Next

Ensure that the plan is not unintentionally inconsistent with the netlist.

### See Also

- estimate_topology_timing

---

## DPTOPO-225

(Error) Routing found %d bits, cell objects have %d bits

### Description

During topology plan netlist analysis, *estimate_topology_timing* found a difference in the number of bits between nets associated with the plan and cell/pin/port objects associated with nodes/repeaters.

### What Next

Correct the discrepancy before calling estimate_topology_timing again.

### See Also

- estimate_topology_timing
- report_topology_plans

---

## DPTOPO-226

(Error) Cannot time fully implemented plan %s in virtual cell mode

### Description

An attempt was made to call *estimate_topology_timing* with the options -routing real -cells virtual when the netlist is a fully implemented plan, which has route shapes on each net. This situation is incompatible with real routing, virtual cell mode. Either real routing, real cell mode or virtual routing, virtual cell mode can be used instead.

### What Next

Use -routing virtual -cell virtual, or -routing real -cell real options.

Feedback

**See Also**

- estimate_topology_timing

---

## DPTOPO-227

(Information) Estimating Delay for timing plan %s in 3dIC.

**Description**

*estimate_topology_timing* estimates the timing for the indicated topology plan defined in a 3DIC design.

**What Next**

If timing estimation is acceptable, nothing needs to be done.

**See Also**

- estimate_topology_timing

---

## DPTOPO-228

(error) Topology node %s does not associate with any object. The command estimate_topology_timing can not estimate the delay.

**Description**

The topology node of the topology plan does not associate with any object. The command *estimate_topology_timing* can not estimate delays of the topology plan.

**What Next**

Verify that topology node in the plan and associate object to it.

Earlier errors/warnings may also give additional information.

---

## DPTOPO-229

(Information) Estimating Delay for timing plan %s using ns_per_mm of net estimation rule.

**Description**

*estimate_topology_timing* estimates the timing for the indicated topology plan using ns_per_mm of net estimation rule.

**What Next**

If timing estimation is acceptable, nothing needs to be done.

Feedback

**See Also**

- estimate_topology_timing

---

## DPTOPO-230

(Error) Cut shapes for path segment %s to %s are inconsistent.

**Description**

*estimate_topology_timing* can operate in different timing modes, based on user preference or based on automatic selection. One such mode is *-routing real* (or real routing selected by the auto-detection algorithm) and virtual cell mode (again, user specified or auto-selected). In this mode, since there are not real standard cells associated with some or all of the topology repeaters, the center (origin) of the virtual topology repeater object is used as the location of the repeater.

An attempt is made to create a "cut line" to find the closest route shape that approaches the origin of the virtual repeater. This cut line is first drawn through the origin, perpendicular to the edge direction at the point where the repeater is. If, after a certain search distance, no route shape is found a bounding box search centered on the origin is used instead.

If the route shapes created for a given plan deviate significantly from the virtual plan locations (i.e. the topology object locations), then it may happen the route shapes found for two adjacent topology plan objects are "out of order". That is, repeater2 comes after repeater1 on a given plan, but the cut shape algorithm finds a route shape for repeater2 that is closer to the start pin/port of the topology plan (in route shape traversal order) than the route shape found for repeater1. In this case, the algorithm to find the shapes belonging to the topology plan subset repeater1 to repeater2 cannot be determined, and in fact, points to a serious issue with the route shapes - it would not be sensible to continue.

Depending on user paramters, if the "auto" setting for routing is in effect, then *estimate_topology_timing* will select the virtual routing option instead and re-estimate the timing of the plan using that mode instead. If the user has specified *-routing real*, an error is generated instead.

**What Next**

Improve either the virtual locations of the topology plan objects or the path of the route shapes associated with the nets/supernets of the plan such that they correlate better.

Feedback

### See Also

- create_topology_plan
- estimate_topology_timing
- set_net_estimation_rule

---

## DPTOPO-231

(Information) Unable to estimate delay for plan %s with %s - trying with %s.

### Description

*estimate_topology_timing* attempted to estimate the timing for the indicated topology plan using the indicated setting (either real routing or real cells) and unable to continue. As the setting was in "auto" mode (i.e. not user set), the virtual version of the setting (virtual routing or virtual cells) will be used instead, and the plan timing will be estimated with the virtual setting instead.

### What Next

If timing estimation in virtual mode is acceptable, nothing needs to be done.

Otherwise, see the log for prior errors indicating the problem that caused the switch to virtual mode and attempt to correct them.

### See Also

- estimate_topology_timing

---

## DPTOPO-232

(warning) Corner attached to the current design is a default corner without user settings.

### Description

The corner attached to current design is a default corner without user settings. So, the command *estimate_topology_timing* will use "ns_per_mm" of net estimation rule to estimate delay.

### What Next

The command *estimate_topology_timing* will use "ns_per_mm" of net estimation rule to estimate delay. The user may create a corner containig manual settings and this will override the default corner.

Feedback

**See Also**

- report_net_estimation_rules
- set_net_estimation_rule

---

## DPTOPO-233

(Information) Topology %s is a zero netlist TIP, so ns_per_mm of net estimation rule will be used to estimate delay.

**Description**

The given TIP is a zero netlist TIP. The command *estimate_topology_timing* will use ns_per_mm of net estimation rule to estimate its delay.

**What Next**

If timing estimation is acceptable, nothing needs to be done.

**See Also**

- estimate_topology_timing

---

## DPTOPO-234

(error) Unable to find a valid ns_per_mm of the nest estimation rule %s.

**Description**

The net estimation rule does not have a valid ns_per_mm. Thus, the command estimate_topology_timing can not estimate delay of the topology plan.

**What Next**

Verify that net estimation rule.

Earlier errors/warnings may also give additional information.

---

## DPTOPO-250

(Warning) Cannot determine associated pins on topology node %s. Topology plan merging is skipped on its edge.

**Description**

Merging topology plans, i.e., to match real repeater cells in the netlist to the corresponding topology repeaters in the topology plans, needs the related topology nodes with correct

Feedback

*objects* attribute. This normally means the topology node of a block should have corresponding block pins set as its *objects* attribute. The indicated topology node does not have this attribute set correctly.

**What Next**

Please set correct *objects* attribute on the specified topology node.

**See Also**

- create_topology_node
- set_attribute

---

## DPTOPO-251

(Warning) Netlist tracing backward from pin %s does not match topology plan tracing backward from topology node %s. Merging is skipped on its edge.

**Description**

The tool detects a mismatch between the netlist and the topology plan on the specified pin and the specified topology node. Merging topology plans, i.e., to match real repeater cells in the netlist to the corresponding topology repeaters in the topology plans, will be skipped for the topology repeaters of the specified topology edges.

**What Next**

Please fix the mismatch between the netlist and the topology plan.

**See Also**

- create_topology_plan
- create_topology_node
- create_topology_edge

---

## DPTOPO-252

(Warning) Cannot find matching topology repeater for flop %s.

**Description**

The netlist constains a flip-flop as indicated that does not have a matching sequential type topology repeater from the topology plan.

Feedback

### What Next

Please add a matching sequential type topology repeater in the topology plan, or remove the extra flip-flop from the netlist.

### See Also

- create_topology_plan
- create_topology_node
- create_topology_edge
- create_topology_repeater

---

## DPTOPO-253

(Warning) Cannot find matching flop for topology repeater %s.

### Description

The topology plans contains a sequential type topology repeater as indicated that does not have a matching flop-flop from the netlist.

### What Next

Please add a matching flip-flop in the netlist, or remove the extra sequential type topology repeater from the topology plan.

### See Also

- create_topology_plan
- create_topology_node
- create_topology_edge
- create_topology_repeater

---

## DPTOPO-254

(Warning) Cannot find matching topology repeater for buffer cell %s.

### Description

The netlist constains a buffer cell as indicated that does not have a matching repeater type topology repeater from the topology plan.

Feedback

### What Next

Please add a matching repeater type topology repeater in the topology plan, or remove the extra buffer cell from the netlist.

### See Also

- create_topology_plan
- create_topology_node
- create_topology_edge
- create_topology_repeater

---

## DPTOPO-255

(Warning) Cannot find matching buffer cell for topology repeater %s.

### Description

The topology plans contains a repeater type topology repeater as indicated that does not have a matching buffer cell from the netlist.

### What Next

Please add a matching buffer cell in the netlist, or remove the extra repeater type topology repeater from the topology plan.

### See Also

- create_topology_plan
- create_topology_node
- create_topology_edge
- create_topology_repeater

---

## DPTOPO-256

(Warning) Cannot find a path ending at %s %s of topology node %s that is compatible to topology plan %s. Supernet exception extraction is skipped on this net.

### Description

The specified pin or port on the topology node is not reachable from any of the pins of the start topology node. In most cases it indicates that the netlist is incompatible to

Feedback

the topology plan. Or the tool cannot automatically trace and identify a pair of supernet exception pins.

**What Next**

Please examine the netlist and make sure it is compatible to the topology plan definition. Or manually add supernet exception on the corresponding net of the indicated pin or port.

**See Also**

- create_topology_plan
- create_topology_node
- create_topology_edge
- create_topology_repeater
- set_supernet_exceptions

---

## DPTOPO-257

(Error) Topology node %s is not on the boundaries of its associated block. The command *merge_topology_plans* cannot proceed.

**Description**

The command *merge_topology_plans* requires the topology nodes must be defined on the block boundaries, block pins, top ports, move bounds, or of the origin type. The indicated topology node does not meet the requirements.

**What Next**

Fix the issues reported on the topology nodes. In most cases, it means the input topology plan does not come from the results of the command *optimize_topology_plans*.

**See Also**

- create_topology_plan
- create_topology_node
- create_topology_edge
- optimize_topology_plans

---

## DPTOPO-258

(Warning) Cannot determine the associated nets of topology plan %s.

Feedback

### Description

The tool cannot determine the associated nets from the indicated topology plan. It will be ignored in the corresponding operation.

### What Next

Please update the topology plan to associated it with correct object type. The supported object type is bundle of supernets.

### See Also

- create_topology_plan
- create_topology_node
- create_topology_edge
- optimize_topology_plans

---

## DPTOPO-259

(Warning) The net %s associated to topology plan %s is dangling.

### Description

The indicated net is a dangling net. It will be ignored in the corresponding operation.

### What Next

Please update the bundle or topology plan definition to exclude the dangling net or connect the dangling net to correct driver and loads.

### See Also

- create_topology_plan
- create_topology_node
- create_topology_edge
- create_bundle

---

## DPTOPO-260

(Error) The associated objects of topology node %s have inconsistent object types.

Feedback

### Description

The indicated topology node has different types of objects associated to it, which is not supported by command *merge_topology_plans*.

### What Next

Please update the topology node's associated objects so that they are of the same object types.

### See Also

- create_topology_plan
- create_topology_node
- create_topology_edge
- create_bundle

---

## DPTOPO-261

(Error) The associated constrained_objects of topology node %s have inconsistent object types.

### Description

The indicated topology node has different types of constrained_objects associated to it, which is not supported by command *merge_topology_plans*.

### What Next

Please update the topology node's associated constrained_objects so that they are of the same object types.

### See Also

- create_topology_plan
- create_topology_node
- create_topology_edge
- create_bundle

---

## DPTOPO-262

(Warning) Mismatch between topology node %s and pin %s.

Feedback

**Description**

By topology plan tracing and netlist tracing, the indicated topology node should have the indicated pin as part of its associated objects. But the actual objects or constrained_objects attribute of the topology node conflicts to the indicated pin. The command *merge_topology_plans* cannot merge the netlist to the topology plan properly.

**What Next**

Please check the predecessors and successors of the indicated topology node and pin and fix the mismatch.

**See Also**

- create_topology_plan
- create_topology_node
- create_topology_edge
- optimize_topology_plans

---

## DPTOPO-263

(Warning) Cannot find a valid topology node in topology plan %s.

**Description**

The topology plan does not have any valid topology nodes.

**What Next**

Please create valid topology nodes and edges for the topology plan.

**See Also**

- create_topology_plan
- create_topology_node
- create_topology_edge
- optimize_topology_plans

---

## DPTOPO-290

(Warning) Plan %s could not be written out - skipping.

Feedback

### Description

The indicated topology plan could not be processed by *write_topology_plans* as it constains invalid topology nodes.

### What Next

Please correct invalid topology nodes and re-invoke *write_topology_plans*.

### See Also

- create_topology_node
- write_topology_plans

---

## DPTOPO-300

(error) Plan %s could not be timed - skipping.

### Description

The indicated topology plan could not be processed by *estimate_topology_timing*.

### What Next

Look at earlier error/warning messages to determine the nature of the problem, then re-invoke *estimate_topology_timing*

---

## DPTOPO-301

(Information) The worst delay was %s from topology plan %s.

### Description

This is an informational messages. When the detail option of *estimate_topology_timing* is set to > 0, and more than one topology plan is timed, the worst plan name is reported.

### See Also

- estimate_topology_timing

---

## DPTOPO-302

(Error) No valid netlist object found for bit %d, end node %s, plan %s.

Feedback

### Description

When attempting to trace the netlist from end to start, the tracing logic could not determine a valid netlist object for the indicated bit index of the indicated topology node that belongs to the indicated topology plan.

This may be because no (or not enough) pin/port objects were set for the "objects" attribute of the topology node, or because the objects associated with the node are not connected to the supernet(s)/bundle(s) associated with the overall topology plan.

When this condition is encountered, netlist tracing analysis cannot be performed successfully, and the current topology analysis command stops with an error.

### What Next

Correct the object attribute of the indicated topology node, such that the correct number of pin/port objects are associated with the node and that the associated objects are associated with the supernet(s)/bundle(s) set in the overall topology plan object attribute.

### See Also

- get_attribute
- get_topology_nodes
- get_topology_plans
- report_topology_plans
- set_attribute

---

## DPTOPO-303

(Error) Unable to initialize netlist tracing for topology plan %s.

### Description

There was an error in the topology plan specification and/or netlist that prevents successful initialization of the topology plan / netlist tracing data, which is used to connect netlist objects to topology plan objects.

The *estimate_topology_timing* command cannot continue.

### What Next

Look for previous errors to describe the specific issue and resolve them and re-run *estimate_topology_timing*.

**See Also**

- DPTOPO-302
- get_attribute
- get_topology_nodes
- get_topology_plans
- report_topology_plans

## DPTOPO-350

(info) Finished processing %lu topology plans - created %d new supernet exceptions

**Description**

Informational message printed by the command *merge_topology_plans* to indicate how many topology plans were processed and how many new supernet exceptions were created.

**See Also**

- merge_topology_plans

## DPTOPO-351

(info) Finished processing %lu topology plans - associated %d std cells arrays with repeaters.

**Description**

Informational message printed by the command *merge_topology_plans* to indicate how many topology plans were processed and how many std cells arrays were associated with topology repeaters that previously had no objects associated with them.

**See Also**

- merge_topology_plans

# DPUI

## DPUI-001

(error) Cannot find or read file %s for option %s.

Feedback

**Description**

This option requires that the given file exist and be readable.

**What Next**

Check that the file exists and you have read permission.

---

## DPUI-002

(error) Unable to create directory %s.

**Description**

This command requires to have certain directories with write permission to put log files and other data. The given directory was not able to be created.

**What Next**

Check that you have permission to create and write files in given directory. If not, specify a different directory to the command.

---

## DPUI-003

(error) One of these options %s, %s is required.

**Description**

The command requires that you specify one of the listed options.

**What Next**

Please refer to the man page of the command to see why one of these options is required.

---

## DPUI-004

(error) Option %s requires one of %s.

**Description**

The option requires that you specify one of the listed possibilities.

**What Next**

Please refer to the man page of the command for more details.

---

## DPUI-005

(error) Error parsing option %s.

Feedback

**Description**

There has been an error when processing the specified option. This likely means that an unexpected input has been given to that option.

**What Next**

Look in log file for further errors or warnings regarding this option. Please refer to the man page of the command for more details on what input is required for the specified option.

---

## DPUI-006

(error) This command is only for a distributed processing worker.

**Description**

The requested command is intended only for an executable which is run as a distributed processing worker. This executable is not a distributed processing worker so this command is invalid.

**What Next**

To learn more about using distributed processing see set_host_options command.

---

## DPUI-007

(error) Error parsing option -var_list.

**Description**

There has been an error when processing the option -var_list. The recommended format for this option is a list of lists with one variable alternating with its corresponding value. For example, if you want to have a variable "a" with value "1" and variable "b" with value "hello world" then use -var_list {{a} {1} {b} {hello world}}. Other formats may also work but above is recommended format.

**What Next**

Try using above format for this option.

---

## DPUI-008

(error) Error parsing option -var_list, variable %s not accepted.

**Description**

There has been an error when processing the option -var_list. The listed variable has a space in it which is not allowed.

Feedback

The recommended format for this option is a list of lists with one variable alternating with its corresponding value. For example, if you want to have a variable "a" with value "1" and variable "b" with value "hello world" then use -var_list {{a} {1} {b} {hello world}}. Other formats may also work but above is recommended format.

**What Next**

Do not use spaces in variables. Try using above format for this option.

---

## DPUI-009

(error) Internal error: %s

**Description**

There has been an unexpected internal error.

**What Next**

Report this error to Synopsys.

---

## DPUI-010

(Error) Incorrect %s for option %s.

**Description**

This option requires certain values, but the given value is not the option asks for.

**What Next**

Check the command man page to see correct usages for option.

---

## DPUI-011

(error) %s is only valid for %s.

**Description**

The request option is only valid for certain object type.

**What Next**

Check the command man page to see correct usages for option.

---

## DPUI-012

(error) Reference lib %s is open. It must be closed to proceed.

Feedback

**Description**

To run a block in distributed mode, the block reference lib must be closed. This is to avoid database contention issues.

**What Next**

Close the reference lib of all affected blocks using close_lib. Be aware that a lib can be opened more than once.

---

## DPUI-013

(error) Options %s and %s are mutually exclusive.

**Description**

The listed command options are mutually exclusive. Only one of them can be specified.

**What Next**

Look at manpage for this command for more information on command options.

---

## DPUI-014

(error) The corner height of IO ring is less than the height of IO driver cell %s.

**Description**

The corner height of an IO ring should be at least as large as the height of IO driver cells within the IO ring.

**What Next**

Increase the corner height or remove the IO driver out of the IO ring.

---

## DPUI-015

(error) There are unsaved modified nlibs. These need to be discarded or saved. To bypass this check use -force option.

**Description**

While running in distributed mode, all nlibs will be closed, including the current (ie top level) nlib. To ensure no data is lost, the command checks to see if there are any modified nlibs. If so, then the command errors out for you to fix this issue.

Sometimes you are ok to discard the unsaved nlibs. In that case use the option -force to bypass this check.

Feedback

### What Next

Save all the modified nlibs using save_lib -all. If this doesn't work, then use the option -force.

---

## DPUI-016

(error) Block '%s' must be expanded before this operation.

### Description

This command requires that the current block is defined and is a fully expanded design view. Currently, your design is either missing, or it has not yet been expanded into a full netlist.

In the "high-capacity" flow for design preparation, the tool first loads an "outline view" of your design. This outline view contains a small fraction of the total design data, and it is intended to be used during hierarchy exploration and partitioning. The advantage of using an outline view is that you can work with a smaller representation of your design without the need to load tens of millions of instances into computer memory.

Once you have partitioned your design into reasonable sized blocks, you should call the expand_ouline command on each block to load the full netlist for that block. After the netlist is loaded, you can continue with this operation.

### What Next

There are several ways you might proceed after receiving this error.

If your design is reasonable small (a few million instances), you can immediately call the *expand_outline* command to obtain your full netlist. In the future, you might want to consider using the "flat design" flow instead of the "high capacity" flow in the task assistant. The "flat flow" will immediately read your full netlist, without requiring you to call *expand_outline*. If you are reading your design using a tcl script, you should consider using the *read_verilog* command instead of the *read_verilog_outline* command.

If your design is large, and has already been partitioned using the *commit_block* command, then you are ready to call the *expand_outline* command to obtain your full netlist.

If your design is large and has not yet been partitioned into hierarchical blocks, you should do that now. You should look at "hierarchy preparation" in the task assistant to aid you. Or you can just call *commit_block* if you already know what you want. After you have committed your blocks, you can call *expand_outline* to get a full netlist.

Feedback

## DPUI-017

(error) Block '%s' must be expanded before this operation.

**Description**

This command requires that the current block is a fully expanded design view. Your design has not yet been expanded into a full netlist.

In the "high-capacity" flow for design preparation, the tool first loads an "outline view" of your design. This outline view contains a small fraction of the total design data, and it is intended to be used during hierarchy exploration and partitioning. The advantage of using an outline view is that you can work with a smaller representation of your design without the need to load tens of millions of instances into computer memory.

Once you have partitioned your design into reasonable sized blocks, you should call the expand_ouline command on each block to load the full netlist for that block. After the netlist is loaded, you can continue with this operation.

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If your design is large, and has already been partitioned using the *commit_block* command, then you are ready to call the *expand_outline* command to obtain your full netlist.

If your design is large and has not yet been partitioned into hierarchical blocks, you should do that now. You should look at "hierarchy preparation" in the task assistant to aid you. Or you can just call *commit_block* if you already know what you want. After you have committed your blocks, you can call *expand_outline* to get a full netlist.

## DPUI-018

(error) Block '%s' is not a design view.

**Description**

This command requires that all the blocks be design views.

Feedback

There are three types of design views with different types of netlist abstraction, which is needed to increase tool capacity. A normal design view has the full netlist, an outline view has a limited netlist used for hierarchy/design exploration, and an abstract view has a special netlist for when this block is used as a sub-block in another block.

**What Next**

Use *change_view* command to change the block to the design view.

---

## DPUI-019

(error) Block '%s' must be expanded before this operation.

**Description**

This command requires that the given block is a fully expanded design view. Your design has not yet been expanded into a full netlist.

In the "high-capacity" flow for design preparation, the tool first loads an "outline view" of your design. This outline view contains a small fraction of the total design data, and it is intended to be used during hierarchy exploration and partitioning. The advantage of using an outline view is that you can work with a smaller representation of your design without the need to load tens of millions of instances into computer memory.

Once you have partitioned your design into reasonable sized blocks, you should call the expand_ouline command on each block to load the full netlist for that block. After the netlist is loaded, you can continue with this operation.

**What Next**

There are several ways you might proceed after receiving this error.

If your design is reasonable small (a few million instances), you can immediately call the *expand_outline* command to obtain your full netlist. In the future, you might want to consider using the "flat design" flow instead of the "high capacity" flow in the task assistant. The "flat flow" will immediately read your full netlist, without requiring you to call *expand_outline*. If you are reading your design using a tcl script, you should consider using the *read_verilog* command instead of the *read_verilog_outline* command.

If your design is large, and has already been partitioned using the *commit_block* command, then you are ready to call the *expand_outline* command to obtain your full netlist.

If your design is large and has not yet been partitioned into hierarchical blocks, you should do that now. You should look at "hierarchy preparation" in the task assistant to aid you. Or you can just call *commit_block* if you already know what you want. After you have committed your blocks, you can call *expand_outline* to get a full netlist.

Feedback

## DPUI-020

(error) Block '%s' is an abstract view. Please set constraints from its parent block where the abstract view is instantiated.

**Description**

This command requires that the current block cannot be an abstract view.

**What Next**

Since you have an abstract view of this block, you must also have a design view for this block (abstract views are created using *create_abstract* from design views). Use *open_block* to open the design view for this block to proceed.

## DPUI-021

(error) '%s' is not a pad cell.

**Description**

The command expects a pad cell. However, the specified object is not a pad cell.

**What Next**

Change the string to a valid pad cell name.

## DPUI-022

(error) '%s' is not a bump cell.

**Description**

The command expects a bump cell. However, the specified object is not a bump cell.

**What Next**

Change the string to a valid bump cell name.

## DPUI-023

(error) '%s' cannot be processed by set_bump_assignment.

**Description**

The assignment option of set_bump_assignment expect users to give one or more pairs of names. Each pair must contain two and only two names. The first one must be a pad cell name. The second must be a bump cell name. Please check and modify the string

Feedback

so that it contains one pad cell name followed by a bump cell name. The two names are separated by whitespaces only. If more than one pair is specified, each pair must be placed in curly brackets. All pairs are placed within a pair of outer curly brackets.

**What Next**

Check and modify the string.

---

## DPUI-024

(error) bump cell '%s' appears more than once.

**Description**

set_bump_assignment currently can only assign one pad cell to one bump cell.

**What Next**

Please check the information after the assignment option and make sure the bump cell only appears once.

---

## DPUI-025

(error) pad cell '%s' appears more than once.

**Description**

set_bump_assignment currently can only assign one pad cell to one bump cell.

**What Next**

Please check the information after the assignment option and make sure the pad cell only appears once.

---

## DPUI-026

(error) IO guide list cannot be processed correctly.

**Description**

Incorrect description of IO guide list is encountered.

**What Next**

The tool cannot understand values after -io_guide option. Please correct the values.

Feedback

## DPUI-027

(error) Incorrect netlist is encountered

**Description**

Errors are discovered when the netlist is loaded.

**What Next**

Please check the design database.

## DPUI-028

(error) Both flip-chip and non-flip-chip IO driver cells are found in the design.

**Description**

The design can have either flip-chip IO driver cells or non-flip-chip IO driver cells, but not both.

**What Next**

Please check the cell library and choose the right IO driver cells according to the design style. If the design is a flip-chip design, use only flip-chip IO driver cells. Otherwise, use non-flip-chip IO driver cells.

## DPUI-029

(error) No IO driver cell is found in the design.

**Description**

There is not a single IO driver cell in the design. IO placement cannot be performed.

**What Next**

Fix the design netlist.

## DPUI-030

(warning) %s is not a bump array.

**Description**

The specified string is not the name of a bump array

Feedback

**What Next**

Correct the name string or remove it from the bump array list.

---

## DPUI-031

(warning) bump array list cannot be processed correctly.

**Description**

Incorrect description of bump array list is encountered.

**What Next**

The tool cannot understand values after -bump_array option. Please correct the values.

---

## DPUI-032

(error) No bumps are specified in flipchip design.

**Description**

Current design contains flip-chip IO driver cells. However, no bump is specified for IO placement.

**What Next**

If the design is non-flip-chip, change the IO driver cells to non-flip-chip pads. If the design is flip-chip, use -bump_array option to specify target bump cells or use the -all option so that place_io command will search for bumps not in bump arrays. If there is no bump in the design, add bumps using command create_bump_array

---

## DPUI-033

(error) No target IO guides can be found.

**Description**

The command place_io will place IO driver cells in target IO guides. Currently, there IO guides are missing.

**What Next**

If you specified the target IO guides using -io_guide option, check the names of specified guides and make sure that the names are correct. If you did not specify target IO guide names, make sure that there are IO guides in the design. IO guides are created with the create_io_guide command. Alternatively, you can create IO guides using the create_io_ring command. Each IO ring is made of four IO guides.

Feedback

## DPUI-034

(error) cannot create new power cell of type %s in IO guide %s.

**Description**

The command place_io cannot create new power driver cells to satisfy power constraints.

**What Next**

Please check the power constraints to make sure that the power driver cell type is correct.

## DPUI-035

(error) power constraints of IO guide %s cannot be satisfied.

**Description**

The power constraint requirement set by the user cannot be met.

**What Next**

Please modify the IO signal constraint and/or IO power constraint of the specified IO guide.

## DPUI-036

(error) IO guide %s overflows.

**Description**

The total width of pads and spacing assigned to the specified IO guide exceeds its length.

**What Next**

Please modify the pad assignment or io constraints of the IO guide. Users can also check the orientations of pad reference cells. The orientation of reference cell should be the one as if the pad cell is placed along a bottom IO guide. If the reference cell orientation is not, users should set the reference_orientation attribute of the ref_block of the pad. The value is that, when the reference cell is placed with such an orientation, the cell is in the correct orientation placed along a bottom guide.

## DPUI-037

(error) The GUI window is not opened.

Feedback

**Description**

The GUI window is closed. The GUI window needs to be open in order for the current command to continue.

**What Next**

Please open the GUI window using command gui_start.

---

## DPUI-038

(error) %s is not an IO guide.

**Description**

The specified string is not the name of an IO guide.

**What Next**

Correct the name string or remove it from the list.

---

## DPUI-039

(error) Cannot store power constraint for I/O guide %s.

**Description**

The specified power constraints cannot be stored in the database.

**What Next**

The database is corrupted. You must re-create the design.

---

## DPUI-040

(error) bump arrays do not contain any bumps.

**Description**

No bump cell instance is found for the bump arrays in the design.

**What Next**

Delete the existing bump arrays and recreate new ones.

---

## DPUI-041

(error) Unsupported orientation, %s.

Feedback

**Description**

The given orientation is not using legal syntax. This command only takes orientations in the form R0, R90, R180, R270, MX, MXR90, MY, MYR90.

**What Next**

Use above syntax for orientations.

---

## DPUI-042

(error) Block '%s' does not have a design view.

**Description**

This command requires that all the blocks have a design view.

**What Next**

Create design views for all the blocks. Use expand_outline to create a design view from an outline view.

---

## DPUI-043

(error) %s is not a valid lib-cell name for a power pad.

**Description**

A lib cell name is expected.

**What Next**

Please check whether the lib cell name is correct and/or the lib cell is in the library.

---

## DPUI-044

(error) This following line cannot be processed: %s.

**Description**

The string specified is not in the right format.

**What Next**

Please revise the specified string using the correct format according to the manpage of this command.

Feedback

## DPUI-045

(error) cell type %s is not described in -reference_cell option.

**Description**

All pad cell types must be included in the -reference_cell option. The specified lib cell is not.

**What Next**

Please check whether the lib cell name is correct. If it is, add it to the list after the -reference_cell option.

## DPUI-046

(error) Design '%s' is a timing abstract view.

**Description**

This command requires that the current design cannot be a timing abstract view. You may want to be using a non-timing abstract view or a design view.

**What Next**

Change to use non-timing abstract view or design view of this design.

## DPUI-047

(error) string "%s" is skipped.

**Description**

Signal IO constraints must be described using the right formats. The string fraction specified in the error message is not in the right format. Therefore, it is removed from the constraint.

**What Next**

Please change the constraint description. Read the man page for the description of the right formats. Note that all elements must be placed in curly brackets.

## DPUI-048

(error) string "%s" cannot be processed.

Feedback

**Description**

The specified string in signal IO constraints is not in the right format and cannot be processed correctly.

**What Next**

Please change the constraint description. Read the man page for the description of the right formats.

---

## DPUI-049

(error) pad %s is assigned to IO guide %s.

**Description**

The specified pad cell is assigned to a different guide from the guide of the signal constraint. This pad cell is ignored.

**What Next**

Please remove the pad from the constraint or its current IO guide. You can use report_signal_io_constraints to check the exact constraint stored in the data based.

---

## DPUI-050

(error) pad %s is already in constraint.

**Description**

A pad cell can only be in an order-based signal io constraint once. The specified pad is in the constraint more than once. The second or later appearance is ignored.

**What Next**

Correct the signal io constraint. You can use report_signal_io_constraints to check the exact constraint stored in the data based for the IO guide.

---

## DPUI-051

(error) pitch value must be positive.

**Description**

The pitch value in a signal IO constraint must be positive and no less than the widths of respective pad cells.

Feedback

**What Next**

Correct the pitch value.

---

## DPUI-052

(error) pad cell %s is wider than its pitch value.

**Description**

Pitch value must be as large as the widths of respective pad cells. Otherwise the pad cells may overlap with each other.

**What Next**

Correct the pitch value or remove pad cells from those share the pitch value.

---

## DPUI-053

(error) negative spacing value is changed to zero.

**Description**

Spacing values must be positive. Any negative spacing values are changed to zero.

**What Next**

Correct the spacing values if needed.

---

## DPUI-054

(error) %s is not a valid spacing value.

**Description**

The command expects a non-negative spacing value but cannot find it.

**What Next**

Change the string to a valid spacing value.

---

## DPUI-055

(error) later absolute location value is used for pad %s.

**Description**

The command finds more than one absolute location value setting for the specific pad cells. The last one will be used.

Feedback

**What Next**

Modify the signal io constraint if needed.

---

## DPUI-056

(error) missing pad or location information.

**Description**

The command expects both a pad cell and a location. However, one or both is missing.

**What Next**

Please correct the wrong signal constraint description.

---

## DPUI-057

(error) The cell '%s' does not have any boundary.

**Description**

This command requires that all the cells have a boundary. The given cell does not have a boundary. That could be because the cell is a black box without any size, or it could be that the cell has no view (eg design view or abstract view).

**What Next**

Create the design view for the cell to give it a boundary. If you are running RTL Floorplanning flow, please run *compile* command to generate physical views for unmapped logic.

---

## DPUI-058

(error) %s.

**Description**

Errors are found during the processing of the command options.

**What Next**

Please check the manpage of the command to correct any errors related to option settings.

---

## DPUI-059

(error) Error found during file processing: %s.

Feedback

**Description**

Command cannot parse the constraint file successfully.

**What Next**

Please check the constraint file and fix any errors. Read manpage for correct data format.

---

## DPUI-060

(error) existing cell %s prevents the creation of bump array %s.

**Description**

The bumps within a bump array will have their names created based on the bump array name. Currently, there exists a cell whose name is identical to the name of a bump to be created.

**What Next**

Change the bump array name or rename the existing cell.

---

## DPUI-061

(error) bump array creation failed: %s.

**Description**

The bump array cannot be created due to the given error. There can be several potential reasons. For instance, a bump array of the same name may alreasy exist. The bounding box of the bump array may be too small to hold a single bump cell. In addition, the design data may be corrupted.

**What Next**

Please check the command option values and perform the command again.

---

## DPUI-062

(error) Module %s cannot be used because it has no ports.

**Description**

A power pad must have at least one feasible port. For pads that are not connected to bumps, any port is a feasible port. For pads that are connected to bumps, a feasible port must be labeled with an RDL connection flag.

Feedback

**What Next**

Change the power reference cell if it does not have a feasible port.

---

## DPUI-063

(error) The library %s is read only.

**Description**

The *create_physical_hierarchy* command needs to modify the library %s. If the library is read only, the *create_physical_hierarchy* command cannot be completed.

**What Next**

Please change the write permission of the library.

---

## DPUI-064

(warning) The commit_cell command is being deprecated and will be removed on 02/07/2013, to be replaced with the command commit_block. Please begin to migrate your scripts to use commit_block instead of commit_cell.

**Description**

The *commit_cell* command is being replaced with the new command *commit_block*.

**What Next**

Please change your script soon to switch to the new command name.

---

## DPUI-065

(warning) The uncommit_cell command is being deprecated and will soon be removed on 02/07/2013, to be replaced with the command uncommit_block. Please begin to migrate your scripts to use uncommit_block instead of uncommit_cell.

**Description**

The *uncommit_cell* command is being replaced with the new command *uncommit_block*.

**What Next**

Please change your script soon to switch to the new command name.

---

## DPUI-066

(error) the -file and -incremental options cannot be used together.

Feedback

**Description**

The two specified options cannot be used concurrently in command place_io.

**What Next**

Change the options.

---

## DPUI-067

(Error) push_down_objects does not support the top-action "keep" for object type %s.

**Description**

This option currently only accepts the top-action "remove" for the specified object_type.

**What Next**

Please do not attempt to use the top_action "keep" when specifying this object_type.

---

## DPUI-068

(Error) uncommit_block only accepts the -type options boundary and module.

**Description**

The -type option for uncommit_block only accepts "boundary" and "module".

**What Next**

Please do not attempt to use any other hierarchy types with this command.

---

## DPUI-069

(error) Unable to remove directory %s.

**Description**

This command attempted to remove the given directory and all its contents but was not able to do so.

**What Next**

Check that you have permission to remove all the files in the directory. Remove the directory manually.

---

## DPUI-070

(error) Cannot load rule %s.

Feedback

**Description**

The specified routing rule cannot be loaded.

**What Next**

Check whether the name of the rule is spelled correctly.

---

## DPUI-071

(error) Found blocks which cannot be edited.

**Description**

When a block has a design view and an abstract view, only one of the two can be changed (ie is editable), the other view cannot be changed (ie is non-editable). This message appears when you try to change the non-editable view.

For example, commands like shape_blocks and place_pins cannot work on the non-editable view.

**What Next**

Change the block view you are using to the editable view for the block using change_abstract. Alternately, you can use merge_abstract to put the design view and abstract view in sync and use create_abstract to recreate the abstract view. By default the abstract view is editable. If you want the design view to be editable, then use create_abstract -read_only.

---

## DPUI-072

(error) Found blocks and/or fixed hierBoundaries which are outside the core area.

**Description**

Option -place in command explore_logic_hierarchy requires that all blocks and hierBoundaries with the fixed intent have their boundaries inside the core area.

**What Next**

Move blocks and the hierBoundaries with fixed intent into the core area.

---

## DPUI-073

(error) Library %s:%s.%s that contains block design %s is READ-ONLY, but the -remove_design option was specified. This design cannot be uncommitted with removal of the design! Skipping this design.

Feedback

**Description**

The command has detected that a block specified for uncommit belongs to a library that is not opened for EDIT (READ-ONLY). Uncommit_block requires that the block's design library be opened for EDIT before it will uncommit the block with removal of the design.

**What Next**

Make sure you have opened the relevant libraries for EDIT before running uncommit_block, or run uncommit_block without specifying the -remove_design option.

---

## DPUI-074

(error) Block design %s is READ-ONLY, but uncommit_block was called using the -remove_design option. This design cannot be uncommitted with design removal if the design is READ-ONLY! Skipping this design.

**Description**

The command has detected that a block specified for uncommit is READ-ONLY, and the -remove_design option was also specified. Uncommit_block requires that the block's design be in EDIT mode before it will uncommit the block and remove the design.

**What Next**

Make sure that the design is not in READ_ONLY mode before running uncommit_block with design removal. Or call uncommit_block without the -remove_design option.

---

## DPUI-075

(error) Option -matching_type and option -all are mutual exclusive.

**Description**

Option -matching_type and option -all cannot be used concurrently.

**What Next**

If all pads and bumps are considered, do not use -matching_type. If a fraction of bumps and/or pads are considered, do nto use -all.

---

## DPUI-076

(error) IO constraints cannot be written as requested.

Feedback

**Description**

The specified options cannot be used to generate IO constraints. The locations of IO drivers in certain IO guides make it impossible to write IO constraints in the required format.

**What Next**

Please check the warning messages to find out the exact reason of errors.

---

## DPUI-077

(error) Corner pad is wrongly specified in signal IO constraints.

**Description**

Corner pad can only be the last pad in signal IO constraints. The specified corner pad is not the last one.

**What Next**

Remove the corner cell from the signal IO constraint or place it as the last cell in the IO guide constraint.

---

## DPUI-078

(warning) %s is not a valid lib-cell name for a filler pad.

**Description**

A lib cell name of a filler is expected.

**What Next**

Please check whether the lib cell name is correct and/or the lib cell is in the library.

---

## DPUI-079

(error) Block %s cannot be opened for edit because %s.

**Description**

The given block cannot be opened for edit. There are several possible reasons for this. First, the parent block of the given block may be not open for edit. In this case, you must open the parent block for edit. Second, only one of design/abstract view can be editable. When you have a block with a design view and an abstract view, only one of these two can be editable. This is necessary to maintain database consistency. The view you are using currently is the one which is not editable. Please refer to the man page of create_abstract

Feedback

for more details on this. Third, failed to open library for edit. This means you will need to open the libraries for edit manually using open_lib -ref_libs_for_edit. Fourth, failed to open design for edit. This means you will need to open the design for edit manually using open_block -edit.

**What Next**

Depending on what error you see, different things need to be done as described above.

---

## DPUI-080

(error) the extension bounding box is incorrect.

**Description**

The bounding box must be within the design area.

**What Next**

Please check and fix the values of the bounding box.

---

## DPUI-081

(error) No filler cell is specified.

**Description**

Users must specify at least one filler reference cell.

**What Next**

Please use -reference_cells option or -overlap_cells option or both.

---

## DPUI-082

(error) Cannot specify cells from different physical hierarchies to be in one macro array.

**Description**

You are getting this message because cells belong to different physical hierarchies are specified in one create_macro_array command call.

The tool requires that all cell in a macro array belong to the same physical hierarchy.

**What Next**

Please put cells of different physical hierarchies into different macro arrays.

Feedback

## DPUI-083

(error) There are non-fixed macros in design. Use the -floorplan option to place macros or fix all macros.

### Description

You are getting this message because the current design has non-fixed macros. If you want to place these macros then you need to use the -floorplan option; this option invokes the macro placement engine. If you already have a macro placement and want to fix all the macros, set the place.fix_hard_macros application option to true.

### What Next

Specify the -floorplan option to place the macros, or fix the macros by setting the place.fix_hard_macros application option to true.

## DPUI-084

(error) Could not parse preferred location: %s.

### Description

The preferred location could not be parsed. The expected format is a coordinate in the form of {x y}, where x and y are numbers ranging from 0 to 1. The coordinate expresses the preferred location inside the block to which the macro(s) belong. Coordinate {0 0} is corresponds to the lower left corner of that block's bounding box, while {1 1} corresponds to the upper right corner.

### What Next

Check the value set for the -preferred_location option.

## DPUI-085

(error) Unable to write to file %s.

### Description

The file specified by the -file option could not be opened or written to.

### What Next

Check the file name and write permissions.

## DPUI-086

(error) Block %s has an unknown view type, "%s" view type is expected.

Feedback

**Description**

The current block has a view type that is unrecognized by this command. The expected type for this command is given.

**What Next**

Use *open_block* to open up the block with the appropriate view type.

---

## DPUI-087

(error) Options %s requires option %s.

**Description**

The first options requires that the second option be used. Some possible reasons are that the first option does not make sense without the second option or that the first option will have no effect without the second option.

**What Next**

Look at manpage for this command for more information on command options.

---

## DPUI-088

(error) No preferred routing direction defined for routing layer %s

**Description**

There is no preferred routing direction defined for a routing layer. The checking is performed for all routing layers if max/min routing layers are not defined. If max/min routing layers are defined, the tool only check layers between max and min layers.

**What Next**

Please set preferred routing layer direction in order for the tool to proceed

Example: set_attribute [get_layers M1] routing_direction horizontal

set_ignored_layers -max_routing_layer METAL -min_routing_layer METAL3

---

## DPUI-089

(Error) No instance found of design %s

**Description**

The uncommit_block command did not find an instance of the specified design anywhere in the physical hierarchy of the current design.

Feedback

**What Next**

Please make sure that you specified the correct design name to be uncommitted.

---

## DPUI-090

(Error) User specified dimension is off finfet grid: %s

**Description**

The object location or size is not on the finfet grid. As a result, the object cannot be created. Specifically, the starting and end points of IO guides must be on the grid points. The four corners of IO rings must be on grid. Any spacing constraints must be multiples of finfet pitches.

**What Next**

Please modify input parameters as suggested by the message. For IO rings, users need to adjust different parameters if the bounding box of the ring is not on grid. If the boundary box is derived using an offset, the offset needs to be multiples of both horizontal and vertical pitch of finfet grid. If the bounding box is derived based on the chip boundary, the chip bound must be on grid. The corner height of IO rings should be multiples of both horizontal and vertical pitch of finfet grid. If the corner height is derived automatically from the IO pad dimensions, users need to provide a valid value explicitly.

Users can disable the finfet grid using set_app_options. Even there is no finfet grid, the placement of any component still needs to honor litho grid. If users still receive this message after disabling finfet grid, users need to adjust the parameters according to litho grid.

---

## DPUI-091

(Error) User specified dimension is off litho grid: %s

**Description**

The object location or size is not on the litho grid. As a result, the object cannot be created or the constraints cannot be accepted.

**What Next**

Please modify input parameters as suggested by the message.

---

## DPUI-092

(Error) Module %s appears to be a sparse outline module. Commit_block does not support sparse outline modules. This module will be skipped.

Feedback

### Description

A "sparse" outline module is one that is imported by read_verilog_outline and has no ports when unexpanded, but does have outline ports if expanded. Please read the man page for read_verilog_outline for information regarding "sparse" outlines.

### What Next

Please do not specify sparse outline modules for commit_block.

### See Also

- read_verilog_outline

---

## DPUI-093

(Error) the specified gradient tolerance is out of range.

### Description

This error message occurs when the specified gradient tolerance is greater than or less than the allowable limit. The gradient tolerance value must be >= 0 and <= 1.

### What Next

Supply a compatible gradient tolerance value or use the default value.

---

## DPUI-094

(Error) white space density is not specified for layer %s.

### Description

Value for the option '-white_space_density' must be layer name and density pairs.

### What Next

Add a density value after the layer name or remove the unpaired layer name.

---

## DPUI-095

(Error) the specified white space density is out of range.

### Description

This error message occurs when the specified white space density is greater than or less than the allowable limit. The density value must be >= 0 and <= 1.

Feedback

**What Next**

Supply a compatible density value or use the default value.

---

## DPUI-096

(Error) edge zone width is not specified for layer %s.

**Description**

Value for the option '-edge_zone_width' must be layer name and inner width pairs.

**What Next**

Add an inner width value after the layer name or remove the unpaired layer name.

---

## DPUI-097

(Error) the specified edge zone width is out of range.

**Description**

This error message occurs when the specified edge zone width is less than the allowable limit. The edge zone width value must be > 0.

**What Next**

Supply a compatible edge zone width value or use the default value.

---

## DPUI-098

(Error) outer zone width is not specified for layer %s.

**Description**

Value for the option '-outer_zone_width' must be layer name and outer width pairs.

**What Next**

Add an outer width value after the layer name or remove the unpaired layer name.

---

## DPUI-099

(Error) the specified outer zone width is out of range.

**Description**

This error message occurs when the specified outer zone width is less than the allowable limit. The outer zone width value must be > 0.

Feedback

**What Next**

Supply a compatible outer zone width value or use the default value.

---

## DPUI-100

(Error) gradient tolerance is not specified for layer %s.

**Description**

Value for the option '-gradient_tolerance' must be layer name and tolerance value pairs.

**What Next**

Add a tolerance value after the layer name or remove the unpaired layer name.

---

## DPUI-101

(error) Block actions %s and %s are mutually exclusive.

**Description**

The listed block actions are mutually exclusive. Only one of them can be specified.

**What Next**

Look at manpage for this command for more information on block_actions.

---

## DPUI-102

(error) Unsupported orientation, %s.

**Description**

The given orientation is not using legal syntax. This command only takes orientations in the form all, R0 (includes R0, R180, MX, MY) or R90 (includes R90, R270, MXR90, MYR90).

**What Next**

Use above syntax for orientations.

---

## DPUI-103

(error) No layer with the routing direction %s

**Description**

Both horizontal and vertical layers need to be existed

Feedback

**What Next**

Please set routing layer direction correctly in order for the tool to proceed

Example: set_attribute [get_layers M1] routing_direction horizontal

---

## DPUI-104

(error) uncommit_block candidate %s has a default site array but the parent block %s does not. Skipping this block for uncommit.

**Description**

While uncommitting blocks, the uncommit_block command detected that a block to be uncommitted possessed a default site array, while its parent did not. This is a scenario that is not supported in uncommit_block.

**What Next**

Make sure that if you are uncommitting a block that has a default site array, that the parent block also has a default site array that is compatible with the block being uncommitted (same site definition, and rows aligned, etc.).

---

## DPUI-105

(error) uncommit_block candidate %s has a default site array that is incompatible with a parent block %s. Skipping this block for uncommit.

**Description**

While uncommitting blocks, the uncommit_block command detected that a block to be uncommitted possessed a default site array that was in some way incompatible with the parent block (site definition, misaligned, etc).

**What Next**

Make sure that if you are uncommitting a block that has a default site array, that the parent block also has a default site array that is compatible with the block being uncommitted (same site definition, and rows aligned, etc.).

---

## DPUI-106

(error) Invalid constraint type %s. Ignored.

Feedback

**Description**

The type name you specified with the *-type* option is invalid. Valid type names include: PRECHECK, DEF, UPF, SDC, BUDGET, CLKNET, CTS_CONSTRAINT, BTM, FLOORPLAN.

**What Next**

Check the command line and specify the correct name of the type of constraints you want to load.

If you want to load more than one types of constraints, please use multiple *-type* options, one *-type* option for one type.

---

## DPUI-107

(error) No constraint mapping file is available for design %s.

**Description**

You get this message because you are trying to run *load_block_constraints* to load block level constraints for child blocks in the current design.

The tool needs the constraint mapping file to get the block level constraints in order to load into child blocks. Without the constraint mapping file, the command cannot do anything.

**What Next**

Please use *set_constraint_mapping_file* to load the constraint mapping file.

---

## DPUI-108

(error) Could not find a module instance of type %s in the calculated lowest physical block %s. Skipping this module type.

**Description**

In multi-physical level block analysis for commit_blocks, the preparation steps determined that a module instance existed in a block that was determined to be the lowest physical block that contained an instance of that type, but a later step failed to find an instance of the module type in that block.

**What Next**

This is likely a tool problem. Report this problem.

Feedback

## DPUI-109

(Warning) Block instances are bound to an abstract view, %s, but uncommit_block does not support uncommitting an abstract view! Skipping instances of this type

### Description

The commit_block encountered a scenario where a block type specified for uncommit existed as abstract bound references. Uncommit_block does not support the uncommitting of instances bound as abstracts. It is necessary to call change_abstract and change the view for these instances to design view so that uncommit can be performed on these instances.

### What Next

Use change_abstract to change the view type to design for instances of this type.

### See Also

- change_abstract

## DPUI-110

(error) No valid cells in cell list for -cell option.

### Description

There is no valid macros in the provided cell list.

### What Next

Please provide valid macro names in the cell list for -cell option.

## DPUI-111

(error) Invalid order type %s. Valid values are top_down or bottom_up.

### Description

The order type you specified with the *-run_order* option is invalid. Valid types include: top_down, bottom_up.

### What Next

Check the command line and specify the correct order type you want to run the blocks.

Feedback

## DPUI-112

(error) Unknown offset type for -offset_type option.

**Description**

Cannot recognize the specified value in -offset_type option.

**What Next**

Please choose between "fixed", "auto" and "scalable" for this option.

## DPUI-113

(error) Found nested module boundaries.

**Description**

Option -place in command explore_logic_hierarchy does not support nested module boundaries.

**What Next**

Remove the module boundaries so that there are no nested module boundaries.

## DPUI-114

(error) Invalid corner value for -anchor_corner option.

**Description**

The value for -anchor_corner option should be either among "bl", "tl", "tr" and "br" if block BBox is to be used for deriving the relative locations, or a positive numerical value no larger than the total corner count of the current design block.

**What Next**

Please provide a valid corner value for this command.

## DPUI-115

(Error) Block instance %s (reference block %s) cannot be uncommitted.

**Description**

During uncommit_block, the command determined that a block could not be uncommitted. A possible cause could be that the port interface of the block instance does not match the design master for the reference block.

Feedback

**What Next**

A workaround for this problem might be to close and then reload the design and then re-call uncommit_block.

---

## DPUI-116

(Warning) Timing constraints purged in block %s prior to uncommitting block instances in that block.

**Description**

During uncommit_block, the command removed timing constraints in a block that contained a block instance to be uncommitted. This is necessary because the major re-structuring of the owning block as a result of uncommitting block instances results in many constraints becoming obsolete or invalid.

**What Next**

Nothing is required except that constraints may need to be reloaded in the owning block after uncommit completes.

---

## DPUI-117

(Error) push_down_objects only supports the top_action (keep) for routing_corridors.

**Description**

This push_down_objects command currently only accepts the top_action "keep" for routing_corridor objects.

**What Next**

Please do not attempt to use any top_action other than "keep" when specifying routing_corridor objects.

---

## DPUI-118

(Error) push_down_objects only supports the block_action (copy) for routing_corridors.

**Description**

This push_down_objects command currently only accepts the block_action "copy" for routing_corridor objects.

Feedback

**What Next**

Please do not attempt to use any block_action other than "copy" when specifying routing_corridor objects.

---

## DPUI-119

(Warning) the top_action (keep) in push_down_objects for cells can only work for physical-only cells and ignore the rest.

**Description**

This push_down_objects command currently only accepts the top_action "keep" for pushing down physical-only cells. Any non-physical-only cells passed into the command will be ignored during this push-down session for cell push-down.

**What Next**

Just be aware that if the user uses the top_action "keep" in conjunction with cell push-down, that it will only push-down physical-only cells and ignore the rest.

---

## DPUI-120

(Warning) No technology information exists for target library %s! All committed blocks will have no physical information!

**Description**

During the command, commit_blocks, it was discovered that the target directory for the new committed block as no technology information. Without this technology information, the block can have no physical information in it. This can happen if the target library has been created without passing a technology file.

**What Next**

Be sure that any target library for a committed block was created with technology file information. If no physical information is needed at this stage, then this warning can disregarded.

---

## DPUI-121

(Error) This design does not have any hierarchical (uncommittable) block instances.

Feedback

**Description**

During the command, uncommit_blocks, it was discovered that the design did not contain any hierarchical physical blocks. All blocks to be uncommitted must have the "is_hierarchical" attribute set to true in order to be uncommittable.

**What Next**

Be sure that the block you wish to uncommit is truly a physical block that bears the "is_hierarchical" attribute.

---

## DPUI-122

(error) Invalid value %s. Valid values are globals, namespaces and procs.

**Description**

The setting entry you specified with the *-pass_settings* option is invalid. Valid values are one or more of: globals, namespaces and procs.

**What Next**

Check the command line and specify the correct setting value you want to use.

---

## DPUI-123

(Error) Failed to create voltage area %s.

**Description**

During the command, uncommit_blocks, when popping up block-level voltage areas to the top from blocks, the command failed to create the new voltage area in the top.

**What Next**

Make sure the the voltage areas in the blocks are correct, and if so, report this.

---

## DPUI-124

(Error) Failed to create Voltage Area shape with bbox {{%s %s}{%s %s}} for Voltage Area %s

**Description**

During the command, uncommit_blocks, when popping up block-level voltage areas to the top from blocks, the command failed to add a new voltage area shape for a voltage area that exists in the top cell.

Feedback

**What Next**

Make sure the the voltage area shapes in the blocks are correct, and if so, report this.

---

## DPUI-125

(Warning) Some blocks to be uncommitted are not supported for automatic UPF merging!

**Description**

During the command, uncommit_blocks, pre-uncommit analysis determined that some of the blocks specified for uncommit are not supported for automatic UPF merging. These blocks will be uncommitted, but they won't have voltage areas created for them and they won't have their UPF information merged with the UPF in the top.

**What Next**

Make sure the the UPF is correct for this design.

---

## DPUI-126

(Warning) Failed to successfully update UPF associations after uncommit!

**Description**

During the command, uncommit_blocks, post-uncommit analysis determined that some of the blocks uncommitted failed in automatic UPF merging.

**What Next**

Make sure the the UPF is correct for this design.

---

## DPUI-127

(Warning) The commit_block command has determined that the module instance %s does not have an associated module boundary that the command could use to determine an origin and/or shape for the committed module instance. Child member cells within the committed reference block will be unplaced. And the resulting block instance will also be unplaced in the top cell.

**Description**

During the command commit_blocks, the command determined that a module instance specified to be committed did not have a module boundary that the command could use to determine a resulting origin and shape of the committed block. Withouth this information, the command will leave child leaf cells unplaced in the reference block and the block boundary will be a square whose area is calculated based upon criteria such as number of

Feedback

cells instances, and hard macros and other physical blocks. In this case, the resulting top-cell block instance will also be unplaced.

If you want more deterministic commit_blocks results, please consider creating a module boundary for this module instance that the command can use to provide cell placement within the block, and shaping and placement of the top-level block instance.

**What Next**

Make sure that the results are ok based upon your flow. If not, then add a module boundary for this module instance.

**See Also**

- explore_logic_hierarchy
- set_cell_hierarchy_type

---

## DPUI-128

(Error) parent block %s has no existing rows or default site array!

**Description**

During the command uncommit_blocks, the command found that the parent block did not have any existing rows or a default site array. This is a problem because uncommit_blocks will not pop-up a default site array because it expects the parent block to already have a default site array.

**What Next**

Make sure that the parent block has rows or a default site array before uncommitting blocks.

---

## DPUI-129

(Error) parent block %s and uncommitting block %s have different site definitions!

**Description**

During the command uncommit_blocks, the command found that the parent block has a default site array and the block being uncommitted has a default site array, but their site definitions don't match.

**What Next**

Make sure that the parent block and the block being uncommitted have the same site definitions.

Feedback

## DPUI-130

(Error) Parent block %s has a different default row direction (%s) from uncommitted instance default default site array row direction (%s) for instance %s! Skipping propagation of this block default site array!

**Description**

During the command uncommit_blocks, the command found that the parent block has a default site array and the block being uncommitted has a default site array, but their row directions are different.

**What Next**

Make sure that the parent block and the block being uncommitted have the same row directions.

## DPUI-131

(Error) Parent block %s and uncommitting block %s have different default site array alternate_rows_flipped definitions!

**Description**

During the command uncommit_blocks, the command found that the parent block has a default site array and the block being uncommitted has a default site array, but their alternate_rows_flipped definitions are different.

**What Next**

Make sure that the parent block and the block being uncommitted have the same alternate_rows_flipped definititions.

## DPUI-132

(Error) Parent block %s and uncommitting block %s have different default site array row or site spacings!

**Description**

During the command uncommit_blocks, the command found that the parent block has a default site array and the block being uncommitted has a default site array, but their row and/or site spacings are different.

**What Next**

Make sure that the parent block and the block being uncommitted have the same row/site spacings.

Feedback

## DPUI-133

(error) Block (Design with physical block) with name %s already exists in current library which prevents commit_block from committing the module to that block. You may confirm this existence using the list_blocks command.

**Description**

The commit_block command detected that the current library already has a block with the specified name, therefore a new block with the same name can not be created and commit_block cannot be allowed to continue for this module. One way that this can happen is if the block is first committed, then uncommitted (without the -remove_design option) and then an attempt is made to run commit_block on the module name again.

**What Next**

Please specify a name which does not conflict with existing block names.

## DPUI-134

(Warning) The list of module names to be committed is empty. No blocks to commit.

**Description**

The commit_block command detected that the list of module names passed into the command to be committed was empty. Therefore there is nothing for the command to do.

**What Next**

Please specify one or more module names when calling commit_block. ~

## DPUI-135

(error) Block (Design with physical block) with name %s already exists in current library which prevents commit_block from committing the module to that block. You may confirm this existence using the list_blocks command.

**Description**

The commit_block command detected that the current library already has a block with the specified name, therefore a new block with the same name can not be created and commit_block cannot be allowed to continue for this module. One way that this can happen is if the block is first committed, then uncommitted (without the -remove_design option) and then an attempt is made to run commit_block on the module name again.

**What Next**

Please specify a name which does not conflict with existing block names. ~

Feedback

## DPUI-136

(error) Cannot find bump lib cell %s in current library.

**Description**

Tool cannot find user specified bump reference cell in opened libraries. Or the specified reference cell is not bump.

**What Next**

Users need to check whether the bump lib cell name is spelled correctly. Moreover, users can check whether the library of the bump lib cell is opened or not. Use open_lib to open the library if needed. Users can use get_lib_cell command to check whether the lib cell exists. Users may need to specify the library name when using get_lib_cell. If the lib cell exists, users need to check whether its design type is flip_chip_pad. Correct the design type if needed.

## DPUI-137

(error) Bump lib cell %s is not prepared correctly.

**Description**

Tool can find the specified bump lib cell. However, the lib cell data is corrupted.

**What Next**

User needs to recreate the bump lib cell and set all attributes correctly. Please use get_lib_cell command to check whether the lib cell exists. Users may need to specify the library name when using get_lib_cell.

## DPUI-138

(error) Current block is not editable for design planning.

**Description**

This command requires that the current block's editability is enabled for design planning.

**What Next**

Use *set_editability* command to change the current block's editability to enable it for design planning.

## DPUI-139

(Error) Option %s's value should be a positive value.

Feedback

**Description**

Option value should be a positive value.

**What Next**

Please set the correct option value.

---

## DPUI-140

(Error) Option %s's value should not be a negative value.

**Description**

Option value should not be a negative value.

**What Next**

Please set the correct option value.

---

## DPUI-141

(warning) Layer '%s' is not a routing layer.

**Description**

The specified layer is not a routing layer. The layer must be a routing layer.

**What Next**

Check the technology layer information.

---

## DPUI-142

(warning) No preferred routing direction defined for routing layer %s.

**Description**

There is no preferred routing direction defined for a routing layer. The checking is performed for all routing layers if max/min routing layers are not defined. If max/min routing layers are defined, the tool only check layers between max and min layers.

**What Next**

Please set preferred routing layer direction in order for the tool to proceed

Example: set_attribute [get_layers M1] routing_direction horizontal

set_ignored_layers -max_routing_layer METAL -min_routing_layer METAL3

Feedback

## DPUI-143

(error) %s can only work with %s.

**Description**

The first option value can only work with the second value.

**What Next**

Specify both two values to make it work.

## DPUI-174

(error) Unknown offset type for -offset_type option.

**Description**

Cannot recognize the specified value in -offset_type option.

**What Next**

Please choose between "fixed" and "scalable" for this option.

## DPUI-200

(error) Tracer data file for design %s incompatible with current settings: %s

**Description**

The given abstract design name was part of the *-include* option given to *compute_dff_connections*, but the stored tracer data file for the abstract design (if it exists) was incompatible with the current tracer settings. To force a re-trace of the design, add the design to the *-retrace_blocks* option of *compute_dff_connections*.

**What Next**

See the given incompatiblity reasons and change DFF tracer settings (and/or filter settings) to remove the incompatiblity.

Alternatively, add the design to the *-retrace_blocks* option of *compute_dff_connections* to force a re-trace with the current settings.

## DPUI-201

(error) Design %s has no abstract version: cannot be specified for %s.

Feedback

**Description**

The design passed to the indicated option of *compute_dff_connections* is not an abstract view design, and no abstract view version of it exists.

**See Also**

report_abstracts(2) create_abstracts(2)

---

## DPUI-202

(warning) DFF Tracer Filter file entries refer to MIB instance(s): %s

**Description**

While reading DFF Tracer Filter entries from a file using the *create_dff_trace_filters* command, one or more block references in the filter file referred to MIB (Multiply-Instantiated Block) instance names, rather than the MIB reference design name.

This could lead to unpredictable results during the DFF Tracer Filtering process: some pins could be filtered if the MIB instance is in design view, but then not be filtered at a different point in time if switched to abstract view.

**What Next**

If this was not intended, remove the filter patterns associated with the MIB instance name as block name (refer to the warning message for a list of these block names) and re-add them using the MIB reference design name as block name instead.

**See Also**

create_dff_trace_filters(2) write_dff_trace_filters(2) remove_dff_trace_filters(2) compute_dff_connections(2)

---

## DPUI-203

(error) compute_dff_connections cannot proceed when working design stack size > 1.

**Description**

An attempt was made to invoke *compute_dff_connections* with a working design stack size > 1. The DFF tracer can only be invoked when the working design stack size is equal to 1 (i.e. working with the top design).

**What Next**

Return back to the top of the working design stack with the command *set_working_design -pop -level 0*

Feedback

### See Also

set_working_design(2) set_working_design_stack(2) get_working_design_stack(2)

---

## DPUI-204

(error) Block %s is read only. This command needs blocks to be editable.

### Description

This command needs to save and modify the given block which is currently read only.

### What Next

Open the block as editable. One way to do this is when you open up the top level design use open_lib -ref_libs_for_edit option.

### See Also

- set_editability

---

## DPUI-205

(error) No valid DFF results available.

### Description

The *report_dff_connections* command was called without calling *compute_dff_connections* first. This command needs valid DFF results.

### What Next

Call the *compute_dff_connections* command first.

### See Also

- compute_dff_connections

---

## DPUI-206

(error) %s requested - only %s available.

### Description

The *report_dff_connections* command was called with the -max_reg or -max_gate option, requesting a value larger than available in the trace data.

Feedback

### What Next

Call the *compute_dff_connections* command first.

### See Also

- compute_dff_connections

---

## DPUI-207

(error) No valid DFF results available.

### Description

The *get_dff_connections* command was called without calling *compute_dff_connections* first. This command needs valid DFF results.

### What Next

Call the *compute_dff_connections* command first.

### See Also

- compute_dff_connections

---

## DPUI-208

(error) %s requested - only %s available.

### Description

The *get_dff_connections* command was called with the -max_reg or -max_gate option, requesting a value larger than available in the trace data. Either adjust the value of the option to match what is available from the current trace data, or re-run *compute_dff_connections* with appropriate parameters.

### What Next

Call the *compute_dff_connections* command first.

### See Also

- compute_dff_connections

---

## DPUI-209

(warning) %s is skipped because it is read_only.

Feedback

### Description

The indicated block was included as of a distributed processing operation. However, the block was marked as read_only, and the operation requires the block to be editable. The block was skipped for this operation.

### What Next

If you wanted the block to be processed in read_only mode, issue a command that can work with read_only blocks. Alternatively, modify the editability of the block using the set_editability *command.*

### See Also

- report_editability
- run_block_script
- set_editability

---

## DPUI-210

(error) Found reference block %s can not be edited.

### Description

This message appears when you try to edit the reference block in non-editable view.

### What Next

Change the block view you are using to the editable view for the reference block.

---

## DPUI-211

(error) Specified block instance is not defined.

### Description

The specified block instance is not defined.

### What Next

Check the specified block instance name.

---

## DPUI-250

(warning) User threshold is below 0.90. DFF connection data is unlikely to be accurate.

Feedback

### Description

This warning is issued if the user issues the load_dff_connections command and sets the -threshold *option below 0.90. The threshold option is use to allow loading of DFF connection files to succeed, even if there are some errors. This allows re-using of outdated connection files if some parts of the design have changed a small amount.*

It is not advised to rely on the connection results if the threshold is set below 0.90, but is allowed, if the user is willing to accept any incorrect information.

### What Next

Proceed with caution, or create a new DFF connection trace

---

## DPUI-251

(Error) Loading DFF Connections canceled - success rate %0.6f is below required threshold: %0.6f.

### Description

This error is issued if the user issues the load_dff_connections command and the required error threshold is not met. The default threshold is 1.0 (100% matching), but may be set as low as 0.50 (50% matching). If the calculated error rate while loading the file is such that (1 - error rate) is less than the threshold setting, this error message is issued and loading of the dff connections fails.

### What Next

Reduce the required threshold using the -threshold option to load_dff_connections, or create a new trace using create_dff_connections

---

## DPUI-252

(Warning) Loading DFF Connections - errors detected, but threshold %0.6f is above required threshold: %0.6f.

### Description

This warning is issued if the user issues the read_dff_connections command and errors are detected while loading, but the required error threshold is met. Loading succeeds.

### See Also

- write_dff_connections

Feedback

## DPUI-253

(Error) No DFF Connection files found matching specified criteria.

### Description

This message indicates that no DFF connection files were found that matched the user criteria. For *load_dff_connections*, this means using *-list* or *-report* options along with the *-tag* option. For *remove_dff_connections*, this means using wildcards with one or both of the *-filename* or *-tag* options.

### What Next

Check the values *-filename* and/or *-tag* settings.

## DPUI-254

(Error) DFF Connection file error: %s.

### Description

There was an error processing one or more DFF Connection file(s). Please see the detailed message for further details.

### See Also

- remove_dff_connection_files

## DPUI-255

(Error) Design contains unmapped instances. DFF tracing not supported.

### Description

The DFF feature does not support unmapped instances. The DFF tracer cannot readily identify unmapped instances as buffers or registers. Tracing a deisgn without buffers or registers causes the internal graph to essentially map the entire design, which causes high memory and CPU usage.

### What Next

Ensure all instances are mapped before trying to use DFF features. Or use the -override option to compute_dff_connections to allow it to proceed.

## DPUI-256

(Error) No bounds found in the design.

Feedback

**Description**

This command works on bounds to compute the connections.

**What Next**

Ensure that there are bounds in the design. Use command explore_logic_hierarchy to create bounds.

---

## DPUI-257

(Error) DFF Connections reading failed.

**Description**

DFF connections reading failed due to unrecoverable errors. This can happen because of intermediate netlist changes.

**What Next**

Use command compute_dff_connections to recompute the DFF connections.

**See Also**

- write_dff_connections

---

## DPUI-258

(Error) Incompatible DFF data found ... exiting without reading the data.

**Description**

DFF connections reading failed due to version mismatch.

**What Next**

Recreate the DFF data with command write_dff_connections.

**See Also**

- write_dff_connections

---

## DPUI-300

(warning) Use %s as the output script file name.

Feedback

**Description**

The user does not provide an output file name or the provided value is a directory. The tool then uses default output file name instead.

**What Next**

If the default script file name is all right, user can safely ignore this warning. Otherwise, please append the desired script file name to the output path name.

---

## DPUI-301

(warning) %s is not a directory. Use %s as the output directory instead.

**Description**

In block-based mode *write_shadow_eco*, the option *-output* is expecting a directory name. If it is provided with an existing file name instead, the tool will use the file's directory as the output directory and the file name is ignored.

**What Next**

User can safely ignore this warning if the tool's behavior is what user expects.

---

## DPUI-302

(warning) Option %s requires option %s.

**Description**

The first option requires the second option to be specified. The first option will be ignored for this command execution.

**What Next**

Can safely ignore this warning if first option is not desired.

---

## DPUI-303

(warning) There are %d invalid macros in -cell option.

**Description**

The cell list for -cell option should consist of only hard macro cells.

**What Next**

Please check for misspellings in the cell list.

Feedback

## DPUI-304

(Error) %s is a regular file.

**Description**

The changelist output directory option requires an empty directory name or a new directory name as its value.

**What Next**

Use an empty directory name or a new directory name as the changelist output directory.

## DPUI-305

(Error) Changelist output directory %s is not empty.

**Description**

The changelist output directory option requires an empty directory name or a new directory name as its value.

**What Next**

Use an empty directory name or a new directory name as the changelist output directory.

## DPUI-306

(Warning) No valid blocks are found. Option -output_dangling_removal_blocks is ignored.

**Description**

The list of blocks provided by user does not contain a valid block name.

**What Next**

Review the design and provide the correct list of blocks.

## DPUI-307

(Error) The valid attribute value for attribute %s can only be %s.

**Description**

The attribute value is incorrect.

**What Next**

Use correct attribute value.

Feedback

## DPUI-308

(Error) The attribute %s cannot be modified by command %s.

**Description**

The attribute cannot be set the this command.

**What Next**

Use correct command to set the attribute value.

## DPUI-309

(error) Invalid bounding box for IO ring.

**Description**

The tool is not able to derive a valid bounding box for the IO ring.

**What Next**

Please check the design and technology. The boundary of the die must be valid.

## DPUI-501

(error) From object '%s' must be an input port or an output pin.

**Description**

The starting point for a bus can only be an input port of your block or an output pin of a hierarchy or macro.

**What Next**

Please specify the correct ports or pins.

## DPUI-502

(error) From pin '%s' is not an output of a specified start cell.

**Description**

When you create buses, the start of the bus must come either from a block port or from an output pin of one of the declared "start cells". Start cells are declared by using the -start_end_cells option of the create_busplans command.

Feedback

### What Next

Please list the cell of this this pin when you specify -start_end_cells to the create_busplans command. You need only specify -start_end_cells once. You do not need to repeat the option on every call to create_busplans.

---

## DPUI-503

(error) Could not determine a bus from the given pins.

### Description

When you create buses, the bus must start and end at cells that were declared by the -start_end_cells option of the create_busplans command. In between the cells, the bus must only consist of the following types of elements: buffers, inverters, level shifters, isolation cells, flops, and hierarchy pins. If there are other elements along the bus's path (e.g. multiplexors) the bus cannot be connected.

### What Next

Make sure that you have listed the cell at the end of your intended bus when you specify -start_end_cells to the create_busplans command. You need only specify -start_end_cells once. You do not need to repeat the option on every call to create_busplans.

If there are unspported elements in the middle of your path, consider breaking the bus into multiple buses -- one before the element and one after.

---

## DPUI-504

(warning) Could not find path for %d bits of the bus.

### Description

When you create buses, the bus must start and end at cells that were declared by the -start_end_cells option of the create_busplans command. In between the cells, the bus must only consist of the following types of elements: buffers, inverters, level shifters, isolation cells, flops, and hierarchy pins. If there are other elements along the bus's path (e.g. multiplexors) the bus cannot be connected.

Some of the bits of the bus had no path from the start of the bus to an end cell.

### What Next

This may not be a serious problem. Perhaps some of the bits of your bus are simply not connected. That is OK. The bits have been dropped from your bus.

Feedback

Make sure that you have listed the cell at the end of your intended bus when you specify -start_end_cells to the create_busplans command. You need only specify -start_end_cells once. You do not need to repeat the option on every call to create_busplans.

If there are unspported elements in the middle of your path, consider breaking the bus into multiple buses -- one before the element and one after.

---

## DPUI-505

(error) Cannot reorder element '%s' because it would cause a loop

**Description**

When you reorder elements of a bus the fanouts of the reordered item stay attached to that item. So you cannot order an element A after element B if element B is currently after A. It would result in a loop.

**What Next**

Make sure that element B is not after element A when you attempt to put element A after B.

---

## DPUI-506

(error) Cannot reorder element '%s'. This will cause group order violation.

**Description**

When you reorder elements of a bus, you cannot move certain objects arbitrarily. For example, a register inside a block cannot move past the pin on the block. Similarly, a pin on a block cannot move past a top level register. This type of violation is happening for this element.

**What Next**

Determine what type of violation is happening and determine if you can achieve result in a different way.

---

## DPUI-507

(error) All bits of a bus must start from same hierarchy.

**Description**

When you are creating a busplan, all of the pins specified for -from must be on the same logical hierarchy or leaf cell. Or you must specify all top-level ports.

Feedback

**What Next**

If you are trying to make a bus that starts from many sibling cells, consider starting the bus at the common logical hierarchy that contains the leaf cells.

---

## DPUI-508

(error) Cannot create a new bus at pin %s because bus %s is already using that pin.

**Description**

When you create buses, each pin in the design can only belong to one bus. The given pin is already part of another bus.

**What Next**

If you use the -force option, the pin (and all fanouts of the pin) will be removed from the existing bus and added to your new bus.

---

## DPUI-509

(error) Busplan %s already exists.

**Description**

You already have a busplan with the given name. You are not allowed to have two busplans with the same name.

**What Next**

You can choose a different name or remove the existing plan with remove_busplans.

---

## DPUI-510

(error) Element '%s' of busplan '%s' already exists.

**Description**

You already have a busplan element with the given name. You are not allowed to have two elements with the same name.

**What Next**

You can choose a different name or remove the existing element with modify_busplan -remove.

Feedback

## DPUI-511

(error) Cannot calculate register / buffer spacing for net estimation rule %s.

### Description

In order to analyze register / buffer placement, the net estimation timing engine requires the setting of some attributes. Without this information, indirect computation of these values cannot be done.

### What Next

There are several ways to give the necessary information. Choose any of the following:

- Set up your timing scenario complete with parasitic parameters. And also make sure that a valid timing path passes through your bus. (Use report_timing through the bus source to see if the timing path exists.)
- Set up your timing scenario complete with parasitic parameters. And also set the "clock" or "sdc_stage_delay" parameter in the net_estimation_rule.
- Set the "ns_per_mm" parameter in your net_estimation_rule. And also make sure that a valid timing path passes through your bus.
- Set the "ns_per_mm" parameter in your busplan net_estimation_rule. And also set the "clock" or "sdc_stage_delay" parameter in the net_estimation_rule.
- Set the "register_spacing" and "buffer_spacing" parameters in the net_estimation_rule with the set_net_estimation_rule command.

### See Also

- set_net_estimation_rule

## DPUI-512

(warning) Cannot process block_budget parameters for busplan_bus '%s' (net_estimation_rule %s).

### Description

In order to completely process budgeted delays for a bus, it must be possible to compute the propagation speed of the bus. Without this information, the tool cannot figure out how much wire length to reserve for block_budget parameters.

The launch_block_budget and capture_block_budget parameters have been ignored.

Feedback

### What Next

There are a several ways to give the necessary information. Choose any of the following:

- Set up your timing scenario complete with parasitic parameters.
- Set the "ns_per_mm" parameter in your net_estimation_rule.
- Make sure that a valid timing path passes through your bus. (Use report_timing through the bus source to see if the timing path exists.)
- Set the "clock" or "sdc_stage_delay" parameter in the net_estimation_rule.

---

## DPUI-513

(error) The option %s requires objects of type %s. No such objects found.

### Description

The specified option requires objects of certain type. No objects of the correct type were specified.

### What Next

Please specify objects of the correct type.

---

## DPUI-514

(error) All bits of a bus must have the same clock period, found different clock periods on terms %s and %s.

### Description

When you are creating a busplan, all of the pins specified for -from must be have the same clock period (or not have a clock period).

### What Next

If you dont care about this, then you can use remove_sdc to remove all the constraints. Then all the clocks will be the same. Another way is to set the constraints such that all the pins specified from -from have same clock. Can use report_timing -through <pin> to see associated clock.

---

## DPUI-515

(warning) The bus %s has specified %s constraint but the net %s has a % constraint which is not consistent.

Feedback

### Description

The specified bus has a constraint which is not consistent with the given net which is part of the bus. Buses are associated with a net estimation rule which specifies certain constraints. For example the net estimation rule of a bus might specify that it is to be routed on layer M3 and M4. If a net in the bus is specified to be routed on M5 and M6 (with min_layer, max_layer constraints), then this contradicts the net estimation rule and may lead to problems when implementing this bus.

### What Next

Consider if the inconsistency will create a problem during implemenation. In above example, if M5/M6 has vastly different timing characteristics then M3/M4 that would be a problem. If so, then can change the net estimation rule to be consistent with the net constraint. Alternately can change the net constraint to be consistent with the net estimation rule.

---

## DPUI-516

(warning) Cannot compute register spacing for net estimation rule %s.

### Description

In order to perform register placement for topology plans, the net estimation timing engine requires the setting of some attributes. Without this information, indirect computation of these values cannot be done.

If register spacing cannot be computed due to missing information, this warning will be issued, and the internal spacing value of 1 meter will be used (essentially, no registers) in order to allow buffer-only flows to succeed.

### What Next

There are several ways to give the necessary information. Choose any of the following:

- Set up your timing scenario complete with parasitic parameters. And also set the "clock" or "sdc_stage_delay" parameter in the net_estimation_rule.
- Set the "ns_per_mm" parameter in your topology plan net_estimation_rule. And also set the "clock" or "sdc_stage_delay" parameter in the net_estimation_rule.
- Set the "register_spacing" parameter directly in the net_estimation_rule with the set_net_estimation_rule command.

### See Also

- optimize_topology_plans
- set_net_estimation_rule

Feedback

## DPUI-517

(warning) Cannot compute buffer spacing for net estimation rule %s.

### Description

In order to perform buffer placement for topology plans, the net estimation timing engine requires the setting of some attributes. Without this information, indirect computation of these values cannot be done.

If buffer spacing cannot be computed due to missing information, this warning will be issued, and the internal spacing value of 1 meter will be used (essentially, no registers) in order to allow register-only flows to succeed.

### What Next

There are several ways to give the necessary information. Choose any of the following:

- Set up your timing scenario complete with parasitic parameters. And also set the "clock" or "sdc_stage_delay" parameter in the net_estimation_rule.
- Set the "ns_per_mm" parameter in your topology plan net_estimation_rule. And also set the "clock" or "sdc_stage_delay" parameter in the net_estimation_rule.
- Set the "buffer_spacing" parameter directly in the net_estimation_rule with the set_net_estimation_rule command.

### See Also

- optimize_topology_plans
- set_net_estimation_rule

## DPUI-518

(Error) Net Estimation Rule parameter %s cannot be negative.

### Description

Some numeric (float) parameters for net estimation rules can only have values >= 0.0. These are: utilization, xtalk_fraction, ff_per_mm, ohms_per_mm, buffer_spacing, delay_derate, ns_per_mm, std_stage_delay and register_spacing.

### What Next

Assign a value >= 0.0 for this parameter.

Feedback

## DPUI-520

(error) Net '%s' is not part of a recognizable bus.

**Description**

When you create buses, the start of the bus must come either from a block port or from an output pin of one of the declared "start cells". for the create_budget_busplan command, the start cells are the budget cells as declared by set_budget_options.

The net you specify must be connected to a legal bus start, with only buffers in between.

**What Next**

Please declare the cell where you want the path to start as a start cell. If you believe you have done this, make sure that the net you have specifed is on a simple buffered path from the start cell or block port.

## DPUI-521

(warning) Cannot calulate ns_per_mm for busplan net_estimation_rule '%s'.

**Description**

In order to use busplans for segment budgets, it must be possible to calculate the ns_per_mm parameter for the busplan net_estimation_rule. This can be done manually be using the set_net_estimation_rule command.

If you want the value to be calculated automatically, you need to make sure that your corners and parasitic paramters are fully defined. This will allow extraction tools to characterize your library.

**What Next**

There are a few ways to give the necessary information.

- Set up your timing scenario complete with parasitic parameters.
- Set the "ns_per_mm" parameter in your busplan net_estimation_rule.

Please refer to the set_net_estimation_rule man page for more details.

## DPUI-522

(error) Cannot reorder element '%s'.

Feedback

**Description**

Reordering the given element will cause an illegal busplan. Most likely the illegality is that the previous element of the given element will be left as an end of the busplan. Ends of busplans can only be top level ports or pins on start end cells.

**What Next**

Try to change the busplan in a different way so that each change results in a legal busplan.

---

## DPUI-523

(error) Cell specified for parameter %s is not a physical block.

**Description**

The specified parameter requires a physical block (design) cell. The passed value does not refer to a physical block.

**What Next**

Specify a physical block for the parameter.

---

## DPUI-600

(warning) The option %s of command %s is deprecated. It will be completely removed in release %s. Please use %s instead.

**Description**

The option indicated is deprecated and should not be used in future.

**What Next**

Check the command man page to see correct command options.

---

## DPUI-601

(warning) The object %s is not appropriate for option %s.

**Description**

This option requires objects of a certain type, but the given object is not that type. This object will be ignored.

**What Next**

Check the command man page to see correct objects types for option.

Feedback

## DPUI-602

(warning) The variable %s failed to set to desired value %s.

**Description**

The tool was unable to set the given variable to the given value. Run will continue without this setting.

**What Next**

Try setting the variable to the value manually. Try renaming variable.

## DPUI-603

(warning) Duplicate reference module %s detected in option %s. Instance %s ignored.

**Description**

This command option requires unique reference modules. In other words, you can specify only 1 cell instance from an MIB (multiply instantiated block) group. You have specified 2 or more cell instances with the stated reference module. The stated instance will be ignored.

**What Next**

Specify only 1 cell instance in an MIB group. It does not matter which one.

## DPUI-604

(warning) The block %s is missing its reference lib.

**Description**

The specified block does not have a reference lib. This is likely because the reference lib was not added to design or it was previously closed. This block will be skipped.

**What Next**

Open the reference lib for block (open_lib) and/or add the reference lib (set_ref_lib) and rerun command.

## DPUI-605

(warning) IO guides have overlaps.

Feedback

**Description**

Two or more IO guides ovelap with each other.

**What Next**

Check the location and size of each IO guide in the design. Modify IO guides to remove overlaps.

---

## DPUI-606

(warning) treat non-flip-chip pad drivers in design as flip-chip pads since bumps are detected in the design.

**Description**

The IO driver cells in the design are of non-flip-chip style. However, they will be treated as flip-chip IO driver cells since the design has bump cells.

**What Next**

If the design is a flip-chip design, change IO driver cells to flip-chip driver cells. If the design is non-flip-chip, remove all bumps.

---

## DPUI-607

(warning) find IO driver cells with design_type attribute as pad and flip_chip_driver and treat all driver cells as flip-chip pads since bumps are detected in the design.

**Description**

Various types of IO drivers exist in the design. They will be treated as flip-chip IO driver cells since the design has bump cells.

**What Next**

Check all IO driver cells in the design. If they are flip-chip IO drivers, ignore this warning or change their design_type to flip_chip_driver.

---

## DPUI-608

(warning) find IO driver cells with design_type attribute as pad and flip_chip_driver and treat all driver cells as non-flip-chip pads since there is no bump found in the design.

**Description**

Various types of IO drivers exist in the design. They will be treated as non-flip-chip IO driver cells since the design has no bump.

Feedback

**What Next**

If the design is a flip-chip design but you want to place all pads without bumps, ignore this warning. Otherwise, create bumps before performing io placement. If the design is a non-flip-chip design, check all IO driver cells in the design. If they are non-flip-chip pads, ignore this warning or change their design_type to pad.

---

## DPUI-609

(warning) flip-chip pads are placed without bumps.

**Description**

Because there is no bump in the design, flip-chip pads are placed as non-flip-chip pads without consideration of bump-to-pad routing.

**What Next**

If the design is not a flip-chip design, check the pad cells. If the design is a flip-chip design and you want the io placer to consider bump-to-pad routing, please add bumps to the design. Otherwise, ignore this warning.

---

## DPUI-610

(warning) The orientation %s is not legal for cell %s.

**Description**

The given orientation conflicts with the set of legal orientations for the given cell. Legal orientations are gotten from the reference library of the cell. If the cell is a hard macro, then the given orientation will be ignored. If the cell is an IO, then the given orientation will be allowed and thus legal orientations could be violated.

**What Next**

Check if the orientation setting is correct and/or if the reference library orientations are correct.

---

## DPUI-611

(warning) The following string is ignored: %s.

**Description**

The command ignores the specified string since it has received enough information.

Feedback

**What Next**

Please check whether the command line is correct.

---

## DPUI-612

(warning) The string %s is not a pad cell and is ignored.

**Description**

The command ignores the specified string, which is not a pad cell name as it should be.

**What Next**

Please specify the right pad cell names. Only pad cell names can be accepted at the current location of signal IO constraint.

---

## DPUI-613

(warning) extra power reference cells ignored.

**Description**

At most two power reference cells can be specified in the power constraint of each IO guide. Additional reference cells are ignored.

**What Next**

Contact Synopsys if you need to specify more than two power reference cells. You can manually place power pads in IO guides.

---

## DPUI-614

(warning) bump overlap exists.

**Description**

There are overlapping bumps cells in the design. The overlap must be removed before IO placement

**What Next**

Move bumps or bump arrays to eliminate bump overlap.

---

## DPUI-615

(error) Object %s not in the top-level physical hierarchy of the current design.

Feedback

**Description**

The object should be in the top-level physical hierarchy of the current design. This command cannot manipulate constraints on objects in sub-blocks.

**What Next**

Change the current_design to the physical hierarchy directly enclosing the object.

---

## DPUI-616

(error) No constraints have been specified.

**Description**

The function of this command is to apply some pin placement constraints to certain objects, but no constraints were specified.

**What Next**

Please specify some constraints.

---

## DPUI-617

(warning) Block %s passed to option %s was not referenced by the current design. Block will be skipped.

**Description**

This command option requires blocks which are referenced by the curernt design. Either a version of a block with a label or view different from the one instantiated in the current design was specified, or the default expansion of block base names resulted in more than one version of a block being passed to this option.

This warning will NOT be generated if multiple versions of a given block were passed, and at least one version was referenced by the current design.

**What Next**

If you passed a specific block/label.view name, then ensure that it is referenced by the current design. If you used a block base name (such as block1), then ensure a version of this block is referenced in the current design.

---

## DPUI-619

(warning) %s.

Feedback

**Description**

Errors are found during the processing of the command options. If there is no other error, the command will still be completed.

**What Next**

Please check the result of the command. Correct any errors related to option settings.

---

## DPUI-620

(warning) Could not derive a preferred location for %s.

**Description**

The given macro's location could not be converted into a preferred location, either because it did not have a proper location or because the block it belongs to has no proper boundary assigned.

**What Next**

Place the macro and ensure that the block it belongs to has a boundary assigned.

---

## DPUI-621

(warning) Failed to identify the host option submit command %s, assuming it is a custom command.

**Description**

This command uses distributed processing and you have specified a host option which uses the given submit command. There are certain host option submit commands which are natively supported, such as bsub, qsub, sh, rsh, and ssh. The tool does not recognize your submit command as being one of these natively supported submit commands and thus it will treat it as a custom submit command.

**What Next**

Typically nothing needs to be done. However if the tool is unable to handle your custom submit command try using one of the supported submit commands listed above. Alternately, you can supply the correct protocol using set_host_options -submit_protocol <protocol>

---

## DPUI-622

(warning) Specified coordinate(s) %s is adjusted to %s to be on FinFET grid.

Feedback

**Description**

The specified coordinate(s) are adjusted to be on FinFET grid.

**What Next**

No action is needed if you want the coordinate(s) to be on FinFET grid.

If you don't want the tool to adjust the coordinate(s) to FinFET grid, please turn off the function by setting the following: *set_app_options -global {finfet.enable_grid false}*

Set the app option to true to restore the function.

---

## DPUI-623

(error) Maximum (%d) monitor gui instances reached. PIDs of instances: %d, %d. Use -kill option to terminate.

**Description**

This session has already launched the maximum number (2) of distributed processing Monitor Gui instances.

**What Next**

In order to launch a new instance, you must either manually close 1 or more existing instances, or use the -kill option to the *run_monitor_gui* command.

**See Also**

- run_monitor_gui

---

## DPUI-624

(error) The grid %s does not exist.

**Description**

An alignment grid that does not exist has been specified for a macro (or a set of macros).

**What Next**

Define the alignment grid and rerun the command.

---

## DPUI-625

(warning) The option %s is obsolete, please use %s instead.

Feedback

**Description**

The given option is obsolete and will be removed from the future release. Please change to use the new option or the new command instead, which has similar functionality.

**What Next**

Check the command man page to see correct options to use.

---

## DPUI-626

(error) The defined alignment point is outside macro %s.

**Description**

The alignment point defined by the command set_macro_constraints -alignment_point {x y} is outside the macro.

**What Next**

Please use an alignment point inside the macro.

---

## DPUI-627

(warning) Net %s does not connect to cell %s. The constraint is skipped. Please consider using topological constraints.

**Description**

The net and cell are not connected. The constraint is skipped. This case should be constrainted with topological constraints instead of physical pin constraints.

**What Next**

Please consider using topological constraints to constrain this case.

---

## DPUI-628

(warning) %s pitch of finfet grid is invalid. Litho grid pitch will be used.

**Description**

You are getting this message because the pitch of the finfet grid in the specified dimension is zero or smaller than zero. This is likely because it is defined so in the tech file.

When invalid finfet grid pitch is detected, design planning operations will use the litho grid pitch for the specified dimension.

Feedback

**What Next**

If you are ok with the tool using litho grid pitch, no action is needed.

If you want a different valid pitch to be used for the finfet grid, please check the tech file and make sure that the finfet grid is defined correctly.

---

## DPUI-629

(error) The option %s is not appropriate for %s.

**Description**

The request option is only valid for certain object type.

**What Next**

Check the command man page to see correct usages for option.

---

## DPUI-650

(error) There was a %s conflict between option %s and previous options. Previous setting: %s. Conflicting setting: %s.

**Description**

There was a conflict in the indicated setting between multiple host options. When multiple host options are specified for one command, they must agree on target applicatoin, working directory and max_cores settings. It an error to specify host options that have conflicting values for these settings.

**What Next**

Change your host options to agree using the set_host_options command, or choose a different list of host options.

**See Also**

set_host_options(2) report_host_options(2)

---

## DPUI-801

(error) Option value for %s should be %s.

**Description**

The checking is performed for command option. Option value specified should be correct value.

Feedback

**What Next**

Please set correct option value for the tool to proceed.

---

## DPUI-802

(error) Cannot find site def.

**Description**

The checking is performed for -site_def option. Option value is not specified for -site_def.

**What Next**

Please set correct site def name for the tool to proceed.

---

## DPUI-803

(warning) Option -site_def is not specified. Using default unit tile %s.

**Description**

The checking is performed for -site_def option. Option value is not specified for -site_def and use default unit tile.

---

## DPUI-804

(error) Value %s for option %s must be %s %s %s.

**Description**

The checking is performed for command option. Option value specified must be greater than or less than an allowable limit.

**What Next**

Please set correct option value for the tool to proceed.

---

## DPUI-805

(warning) Option -mode is not specified. Using default uniform mode.

**Description**

The checking is performed for -mode option. Option value is not specified for -mode and use default uniform mode.

Feedback

## DPUI-806

(warning) Option value for -direction should be same as prefered direction %s.

**Description**

The checking is performed for -direction option. Option value specified for -direction should be same as prefered direction.

**What Next**

Please set correct direction for the tool to proceed.

## DPUI-807

(error) Offset value in option -coord cannot exceed site def %s %s.

**Description**

The checking is performed for -coord option. Option value specified for -coord cannot exceed site def height or width.

**What Next**

Please set correct direction for the tool to proceed.

## DPUI-809

(warning) Direction is not specified. Using the layer preferred direction.

**Description**

You receive this warning because the direction is not specified in the command invoked, so the tool uses the layer preferred direction.

**What Next**

No action is required on your part unless you want to specify a direction. You can examine the design to see the layer preferred direction and the man page for the command that caused the warning for detailed information on specifying a direction value.

## DPUI-810

(error) Option -space is only used for uniform mode.

Feedback

**Description**

The checking is performed for -space option. Option value specified for -space is only used for uniform mode.

**What Next**

Please set correct option mode for the tool to proceed.

---

## DPUI-811

(warning) Space is not specified. Using pitch.

**Description**

You receive this warning because the space between the tracks is not provided. The tool uses the default value, which is the pitch value provided by the library.

**What Next**

No action is required on your part. You can see the man page for the command that caused the warning for detailed information on specifying optional and required parameters.

---

## DPUI-812

(error) Couldn't get the pitch value of the layer %s from the libray.

**Description**

You receive this error because the specified layer's pitch value cannot be fetched from the edit library.

**What Next**

Please check the technology information of the edit library.

---

## DPUI-813

(warning) Coordinate value is not specified. Using coordinate that is half of the space distance inside of the die area.

**Description**

You receive this warning because the value of the start of the first track is not specified. The tool attempts to use half of the space.

Feedback

**What Next**

No action is required on your part. You can see the man page for the command that caused the warning for detailed information on specifying optional and required parameters.

---

## DPUI-814

(error) Option value for -coord should be specified in non_uniform mode.

**Description**

The checking is performed for -coord option. Option value specified for -coord should be used for non_uniform mode.

**What Next**

Please set correct option mode for the tool to proceed.

---

## DPUI-817

(error) List size of option %s should be %s in %s mode.

**Description**

The checking is performed for option value consistency.

**What Next**

Please set correct option value for the tool to proceed.

---

## DPUI-820

(error) Target object %s should not be fixed.

**Description**

The checking is performed for target macro object. The target hard macro specified should not be fixed.

**What Next**

Please set correct option value for the tool to proceed.

Feedback

**See Also**

- set_macro_relative_location
- remove_macro_relative_location
- report_macro_relative_location

---

## DPUI-821

(error) Target orientation %s is not legal for macro cell %s.

**Description**

The checking is performed for target macro orientation. The given orientation conflicts with the set of legal orientations for the given cell. Legal orientations are gotten from the reference library of the cell or set by command set_macro_options.

**What Next**

Please set correct option value for the tool to proceed.

---

## DPUI-822

(error) Target object and anchor object are not in the same design.

**Description**

The checking is performed for target and anchor object. The two objects should in the same design.

**What Next**

Please set correct option value for the tool to proceed.

---

## DPUI-823

(Warning) Target orientation using default legal value %s for macro cell %s.

**Description**

The checking is performed for target macro orientation. Default legal orientation are gotten from the reference library of the cell or set by command set_macro_options.

Feedback

**See Also**

- set_macro_relative_location
- remove_macro_relative_location
- report_macro_relative_location

---

## DPUI-824

(Error) No legal target orientation found for macro %s.

**Description**

The checking is performed for target macro orientation. Default legal orientation are gotten from the reference library of the cell or set by command set_macro_options.

**What Next**

Please set the correct legal orientation for macro.

**See Also**

- set_macro_relative_location
- remove_macro_relative_location
- report_macro_relative_location

---

## DPUI-825

(Error) No hard macros, macro arrays or blocks found.

**Description**

The checking is performed for hard macros, macro arrays or blocks specified.

**What Next**

Please set correct hard macro, macro array edit group or block list.

---

## DPUI-826

(Error) Can not set std cell as anchor object.

**Description**

The checking is performed for anchor object. The anchor object should be IO pad cell, hard macro, block, port, pin.

Feedback

**What Next**

Please set the correct anchor object.

---

## DPUI-827

(Error) Anchored object corner should be bl, tl, tr, br for rectangle or a positive integer number from 1 to n for each corner of a shape

**Description**

The checking is performed for anchored object corner. The anchored object corner should be bl, tl, tr, br for rectangle or 1, 2, 3, 4, 5... for each corner of a shape. A rectilinear corner number is a positive integer that starts at one, and increments by one as you proceed around each edge in the shape. Given any rectilinear shape, the bottom corner of the left-most edge is the starting corner (corner number 1). You cannot specify a value of 0 for this option.

**What Next**

Please set the correct anchored object corner.

---

## DPUI-828

(Error) Scalable offset value should be in [-1 1]

**Description**

The checking is performed for scalable offset. The scalable offset value should be in [-1 1].

**What Next**

Please set the correct offset value.

---

## DPUI-829

(Error) Scale edge number should be a positive integer number from 1 to n.

**Description**

The checking is performed for scale edge number. The scale edge number should be 1, 2, 3... for each edge. A edge number is a positive integer that starts at 1, and increments by one as you proceed around each edge in the shape. Given any rectilinear shape, the lower leftmost vertical edge is the starting edge (edge number 1).

**What Next**

Please set the correct edge number.

Feedback

## DPUI-830

(Warning) Scale edge only takes effect with scalable offset type.

**Description**

The checking is performed for scale edge option. The scale edge takes effect with scalable offset type.

**What Next**

Please set the correct offset type.

## DPUI-831

(warning) The only supported orientation for macro array/group is R0.

**Description**

The given orientation is not using legal syntax for relative location constraint on macro array/group. This command only takes orientation for macro array/group in the form R0.

**What Next**

Use above syntax for orientations.

## DPUI-832

(Error) The target macro can not anchor to its macro array.

**Description**

When a macro was set to anchor its parent macro array, this setting is illegal.

**What Next**

Set the correct anchor object.

## DPUI-833

(warning) Target object %s is fixed.

**Description**

The checking is performed for target macro object. If the target macro is still fixed by the time macro placement is run, this constraint could not be honored.

Feedback

**What Next**

Please check the target object fixed attribute for the tool to proceed.

---

## DPUI-834

(error) Cannot load rule %s.

**Description**

The specified floorplan rule cannot be loaded.

**What Next**

Check whether the name of the rule is correct.

---

## DPUI-835

(error) Cannot load object type %s.

**Description**

The specified object type cannot be loaded.

**What Next**

Check whether the name of the object type is correct.

---

## DPUI-836

(error) all %s that specified are invalid.

**Description**

All the specified arguments are invalid.

**What Next**

Check whether the name of the arguments are correct.

---

## DPUI-837

(warning) Specified object %s is fixed.

**Description**

Specified object is fixed.

Feedback

**What Next**

Please check the specified object fixed attribute.

---

## DPUI-838

(warning) some pins remain dangling for power constraints.

.SH DESCRIPTION

The power constraint only describes pad pin connectivity for some pins. The other pins will remain unconnected.

**What Next**

Please describe pin connection for all pins if needed.

---

## DPUI-839

(warning) There are %d invalid macros in -cell option.

**Description**

The cell list for -cell option should consist of only macro cells.

**What Next**

Please check the specified cell list.

---

## DPUI-840

(Error) Targeted object corner should be bl, tl, tr, br for rectangle or a positive integer number from 1 to n for each corner of a shape

**Description**

The checking is performed for targeted object corner. The targeted object corner should be bl, tl, tr, br for rectangle or 1, 2, 3, 4, 5... for each corner of a shape. A rectilinear corner number is a positive integer that starts at one, and increments by one as you proceed around each edge in the shape. Given any rectilinear shape, the bottom corner of the left-most edge is the starting corner (corner number 1). You cannot specify a value of 0 for this option.

**What Next**

Please set the correct targeted object corner.

Feedback

## DPUI-841

(Error) Specified soft macro scale factor value is not reasonable.

**Description**

The checking is performed for soft macro scale factor.

**What Next**

Please set the correct option value.

## DPUI-842

(error) Option %s should be used if option %s is not specified.

**Description**

The two options shouldn't be not specified together.

**What Next**

Check the command options.

## DPUI-843

(error) Soft macro %s's relative location constraint is not derived since the soft macro is unplaced.

**Description**

Deriving relative location constraints for soft macros when some of them are unplaced will result in invalid constraints and hence the constraint derivation will be skipped.

**What Next**

Check the unplaced soft macros.

## DPUI-901

(information) Instance %s is referencing to %s view%s.

**Description**

You are getting this message because the command you called requires a block instance to reference to certain view types, and the specified instance is using a view that is not allowed by the command.

Feedback

**What Next**

The tool will switch the block instance to use an allowed view. If the allowed view is not available, the tool will create it for you.

---

## DPUI-902

(information) Elapsed time for %s excluding pending time: %s.

**Description**

You are getting this message because the command you called can use distributed processing. To accurately measure the elapsed time when using distributed processing it is necessary to remove time spent pending on the queue (ie the job is waiting to be run and not actually running). This message indicates the elapsed time after pending time is removed or in other words the elapsed time considering the longest running distributed job with no pending time.

**What Next**

No action is required.

---

## DPUI-903

(information) CPU time for %s : %s.

**Description**

You are getting this message because the command you called can use distributed processing. To accurately measure the CPU time when using distributed processing it is necessary to only consider the maximum CPU time of all the jobs running in parallel.

The reported CPU time includes the CPU time spent on the parent process as well as the maximum CPU time spent for distributed jobs.

**What Next**

No action is required.

---

## DPUI-904

(information) Peak memory usage for %s : %lu MB.

**Description**

You are getting this message because the command you called can use distributed processing. To accurately measure the peak memory when using distributed processing

Feedback

it is necessary to consider the peak memory of the parent process as well as the peak memory of all the distributed jobs.

The reported memory is the largest peak memory of all the distributed jobs as well as the parent process.

**What Next**

No action is required.

---

## DPUI-905

(Error) No valid hard placement placement blockages or macros specified

**Description**

This error message occurs when you use the command "replicate_route_blockages" without macros or placement blockages specified. Or user specified invalid objects like soft placement blockage or standard cells.

**What Next**

```
User must specifiy valid placement blockages or macros
```

---

## DPUI-906

(Error) Not a valid routing blockage layer.

**Description**

This error message occurs when you specify a metal layer which is not valid routing blockage layer to create routing blockage.

**What Next**

Specify a valid routing blockage layer.

---

## DPUI-907

(Warning) Cell %s is not a hard macro, skip it.

**Description**

This warning message occurs when you specify a object is not hard macro, but you use the command option "replicate_route_blockages -macros".

**What Next**

Specify a valid macro cell.

Feedback

## DPUI-908

(Error) Module with name '%s' already exists in parent design '%s'.

**Description**

This error message occurs when uncommit_block is trying to create a module with name (that is top module name of physical design for which uncommit_block is called) and module with same name already exists in parent design.

**What Next**

User has to remove existing module.

## DPUI-910

(warning) Set %s first, %s will be ignored.

**Description**

This error message occurs when you specify two conflict options simultaneously.

**What Next**

Specify one of these options.

## DPUI-911

(information) The constraint mapping file does not have the specified type(s) of constraint for block %s.

**Description**

You are getting this message from *load_block_constraints* command because the tool does not find any entry in the constraint mapping file for the specified block that contains information for the type(s) of constraint you want to load. So the command would not run this block to load any constraint.

**What Next**

If this is expected, no action is needed. If you think that the specified block should have the type(s) of constraint to load, please check the constraint mapping file. You can check the content of the constraint mapping file by calling command *report_constraint_mapping_file*.

Feedback

### See Also

- set_constraint_mapping_file
- report_constraint_mapping_file

---

## DPUI-912

(error) Design %s has unresolved reference %s. Command %s will be skipped for design %s.

### Description

Design that has unresolved reference(s) will not be uncommited.

---

## DPUI-913

(error) Design %s will be skipped for command %s. It is uneditable in the context of current top design.

### What Next

The editability of the design can be set by *set_editability* command.

### See Also

- set_editability
- report_editability

---

## DPUI-914

(Warning) Arguments passed to %s will be ignored since -from_existing_pins option was not specified concurrently

### Description

You are getting this message because write_pin_constraints command will only honor the list of arguments specified when -from_existing_pins option is used simultaneously. If -from_existing_pins is not specified, the command writes out all the pin constraints associated with the specified objects stored in design database and ignores the arguments. Please refer to the man page description of write_pin_constraints for more details.

### What Next

If this is expected, no action is needed.

Feedback

**See Also**

- write_pin_constraints

---

## DPUI-915

(warning) Keyword %s is specified as %s.

**Description**

Keyword for power constraints is wrongly spelt.

**What Next**

Please check the spelling of the keyowrds.

---

## DPUI-916

(warning) %s.

**Description**

Warnings are found during the processing of the command options.

**What Next**

Please check the manpage of the command to correct any errors related to option settings.

---

## DPUI-917

(warning) For guide %s and reference cell %s existing power constraints already have connectivity information defined.

**Description**

For the above mentioned IO guide and reference cell there are already exisitng power constraints with connectivity information.

**What Next**

Please check the constraints.

---

## DPUI-918

(error) Error found during file processing: %s.

Feedback

**Description**

Command cannot parse the pattern description file successfully.

**What Next**

Please check the description file and fix any errors. Read manpage for correct data format.

---

## DPUI-919

(error) Keyword %s is not found for pattern %s.

**Description**

Particular pattern is missing a keyword.

**What Next**

Please check if all parameters of specified pattern is present.

---

## DPUI-920

(error) Not support derive_origin type %s.

**Description**

Support derive_origin type: auto, bl, br, tl, tr.

**What Next**

Please check if derive_origin type is supported.

---

## DPUI-921

(error) Incorrect %s in %s option.

**Description**

The argument provided to the command is invalid.

**What Next**

Check the arguments to the command and try again.

---

## DPUI-922

(error) Routing directions for layer '%s' are missing.

Feedback

**Description**

It is not permitted that layers' routing directions are not provided by user and also not available in technology file.

**What Next**

Check the layers' routing directions and set these unknown directions.

---

## DPUI-924

(warning) The routing directions for layer '%s' are missing. Auto deriving '%s' routing direction for layer '%s'.

**Description**

There exist some routing layers whose routing directions are not provided by user and also not available in technology file. Therefore, the routing directions of all routing layers are automatically derived from bottom to top. These existing routing directions will be overridden.

**What Next**

Check the layers' routing directions and set these unknown directions by yourself.

---

## DPUI-930

(Warning) Option '-reuse_processes' is deprecated and will be removed in a future release. Please use the app option 'plan.distributed_run.reuse_rbs_workers' instead.

**Description**

Option '-reuse_processes' is deprecated. It will be removed in the coming release.

---

## DPUI-931

(error) Floorplan rules: %s, conflict with initialization floorplan's %s .

**Description**

When DCM policy is 'Error'. The Message shows which floorplan rules conflict with initialization floorplan's core offset or core area length. The command 'initialize_floorplan' will report error and return.

**What Next**

NA

Feedback

## DPUI-932

(warning) Floorplan rules: %s, conflict with initialization floorplan's %s .

**Description**

When DCM policy is 'Tolerate'. The Message shows which floorplan rules conflict with initialization floorplan's core offset or core area length. The command 'initialize_floorplan' will continue to run and use the core offset or core area length set by users as the final value.

**What Next**

NA

## DPUI-933

(Error) Design with Physical Blocks are not supported in Auto Partitioning

**Description**

Auto-partitioning does not take into consideration any physical block.

**What Next**

NA

## DPUI-934

(Error) Invalid value for app option %s

**Description**

App option must be set to a non-zero value

**What Next**

Please set the correct app option value. NA

## DPUI-935

(Error) Invalid value for app options %s and %s for a design with total cell count %d %s

**Description**

App options value are too small to accomodate all the cells in the design

Feedback

**What Next**

Please set the correct app option value. NA

---

## DPUI-936

(error) Either app_option %s or %s need to be specified for conducting the auto_partition

**Description**

Either of App option must be set to a non-zero value

**What Next**

Please set the correct app option value. NA

---

## DPUI-937

(error) Do nothing if only one pin is specified.

**Description**

Need specify more than one pin.

**What Next**

NA

---

## DPUI-938

(error) Can't find tech constraints for current design.

**Description**

Both of lib and version must be set correctly.

**What Next**

Please set the correct technology configuration.

---

## DPUI-940

(error) Can't get technology version, please use set_floorplan_tech_constraint to set.

**Description**

Technology version must be set.

Feedback

**What Next**

Please set the correct technology configuration.

---

## DPUI-941

(error) Can't get technology node, please use set_technology.

**Description**

technology node is unset.

**What Next**

Please set the correct technology node.

---

# DPWF

---

## DPWF-201

(Warning) Found %s. This information will not be transfered to IC Compiler.

**Description**

The *write_floorplan* command found data that is not supported in IC Compiler. This data will not be transferred to IC Compiler.

**What Next**

Verify that this data is not needed while working in IC Compiler and proceed with the flow.

---

## DPWF-202

(warning) %d objects ignored, as their types are not supported.

**Description**

-objects option only supports cells, pins, blockages, voltage areas, bounds, edit groups, and corridors. Objects of other types will be ignored.

**What Next**

Specify only objects of supported types.

---

## DPWF-203

(warning) Both macros and cells values are specified; macros will be ignored.

Feedback

**Description**

*-include* and *-exclude* options support both *macros* and *cells* values; when they are both specified, *cells* is used as more general.

**What Next**

Specify *cells* when you want both macros and standard cells; use *macros* for macro cells only.

---

## DPWF-204

(warning) Voltage area %s is not associated with any power domain, so it will be ignored.

**Description**

Voltage areas must be associated with power domains, or they will not be output by write_floorplan.

**What Next**

Check your upf specifications and make sure voltage areas are related to power domains.

---

## DPWF-205

(warning) Option %s is obsolete, please use %s instead..

**Description**

This message informs you that the option specified in the information message will be obsolete. You are advised to use the specified replacement option instead.

**What Next**

Use the replacement command instead.

---

## DPWF-206

(Warning) %s overrides setting for %s.

**Description**

The two options may have a potential conflict.

**What Next**

Pleases review the input options based on demand and solve the conflict.

Feedback

### DPWF-207

(warning) Shape Pattern won't support icc format in command write_floorplan.

**Description**

This message informs you that shape pattern in the design won't support icc format in command write_floorplan.

**What Next**

Use icc2 format instead.

## DPX

### DPX-001

(Information) Distributed multi-processing workers are started successfully after %s.

**Description**

Workers specified by *dpx_host_options* have started successfully.

### DPX-002

(Information) Distributed multi-processing flow status: %s %s.

**Description**

It shows the current status of distributed multi-processing flow.

### DPX-003

(Error) Distributed multi-processing task %d fails because of %s.

**Description**

Distributed multi-processing task fails because of the described error. Please check logs under work_dir/debug_log or work_dir/dpx_log for details. Frequent cause of this error includes crash in workers, dpx_host_options are not valid, jobs are killed because of timeout.

Feedback

## DPX-004

(Error) Failed to launch distributed multi-processing workers because dpx_host_options is not specified or -num_process value in dpx_host_options is not positive.

**Description**

Distributed multi-processing workers are specified by

```
set_host_options -name dpx_host_options -num_process ....
```

When they are not specified, *start_dpx_workers* fails.

**What Next**

```
start_dpx_workers(2)
```

## DPX-005

(Information) Distributed multi-processing flow finishes successfully %s.

**Description**

It shows the current status of distributed multi-processing flow.

## DPX-006

(Error) Distributed multi-processing flow fails %s.

**Description**

Distributed multi-processing flow fails. The detailed information can be found in previous error messages. To debug this, please check dpx_log and debug_log under work_dir.

## DPX-007

(Warning) Failed to launch distributed multi-processing workers after %s.

**Description**

Distributed multi-processing workers are specified by

```
set_host_options -name dpx_host_options -num_process ....
```

When they are not specified correctly, *start_dpx_workers* fails. One possible cause is that the timeout setting in dpx_host_options is too short. Please check logs in dpx_debug directory under work directory specified by start_dpx_workers command for details. Non-distributed flow is performed instead.

Feedback

### What Next

```
start_dpx_workers(2)
```

---

## DPX-008

(Information) Requesting distributed workers...

### Description

*start_dpx_workers* uses information specified by dpx_host_options to request workers.

---

## DPX-009

(Information) All distributed workers stopped.

### Description

*stop_dpx_workers* stops all distributed workers.

---

## DPX-010

(Error) Failed to get distributed workers because they are not reserved yet.

### Description

It is required to use *start_dpx_workers* to acquire all workers before *get_dpx_workers* is used.

### What Next

```
start_dpx_workers(2)
get_dpx_workers(2)
```

---

## DPX-011

(Error) Failed to get information of distributed workers.

### Description

Workers are not responding to query request. Please check workers' status.

### What Next

```
start_dpx_workers(2)
get_dpx_workers(2)
```

Feedback

## DPX-012

(Error) DPX is not enabled.

### Description

App option dpx.enable has to be set to true before start_dpx_workers command.

### What Next

```
start_dpx_workers(2)
get_dpx_workers(2)
```

## DPX-013

(Warning) Synchronization stage %d (%s) takes %lu seconds.

### Description

Synchronization between workers may take long time because of unbalanced work loads in workers.

## DPX-014

(Warning) Number of cores (%d) specified in submit_command of dpx_host_options is less than number of cores (%d) specified by -max_cores; runtime will degrade.

### Description

Number of cores used by DPX workers is equal to min(number of cores specified in -max_cores in dpx_host_options or global host option in master, cores specified in submit_command of dpx_host_options). For example,

set_host_options -name dpx_host_options -num_processes 2 -max_cores 4 -submit_command "qsub -pe mt 2"

Then every worker uses min(4, 2) = 2 cores. The reason why number of cores specifed in -max_cores in dpx_host_options is not always used is that farm/grid admin might kill job when CPU overutilization happens.

Our recommended usage is that the number of cores specified in submit_command should be equal or greater than the number of cores specifed in -max_cores.

### What Next

Please correct the core specification in either -max_cores or submit_command.

Feedback

## DPX-015

(Error) Task %d fails because of error in worker scripts.

**Description**

The given task fails because of errors in worker scripts.

**What Next**

Please check workers' log files under work_dir/debug_dir for script errors.

## DPX-016

(Error) %s; and please check in workers' logs under %s. Details:

**Description**

Workers crashed.

**What Next**

Please check workers' log files for stacktraces.

## DPX-017

(Error) %s; please check in workers' logs under %s. There might not be enough memory or CPU cores reserved for workers. Details:

**Description**

One of possible reasons is that there is not enough memory or CPU cores reserved for workers, and farm admin kills these jobs that over-utilize resources.

**What Next**

Please make sure that there is enough resource reserved for every worker (e.g., check dpx_host_options).

## DPX-018

(Warning) Either time.estimate_all_corners or extract.extract_all_corners is set to false, which causes some runtime improvement features in DPX to be turned off.

**Description**

When time.estimate_all_corners or extract.extract_all_corners is set to false, certain runtime improvement features in DPX cannot be turned on.

Feedback

**What Next**

Please revert values of time.estimate_all_corners and extract.extract_all_corners to their default values.

# DWS

## DWS-0000

(error) Can't find module to implement part %s

**Description**

Datapath generation was unable to find an implementation module for the named synthetic part.

## DWS-0001

(error) Unable to initialize synthetic library

**Description**

Synthetic part management was unable to start.

## DWS-0002

(error) User design contains synthetic operator that is not available in synthetic library: %s

**Description**

The design being compiled is using a synthetic operator that has been marked as disabled in the synthetic library.

The most likely cause is that the synthetic operator requires a license that was not available to the current compilation.

## DWS-0003

(error) User design contains synthetic module that is not available in synthetic library: %s

**Description**

The design being compiled is using a synthetic module that has been marked as disabled in the synthetic library.

Feedback

The most likely cause is that the synthetic module requires a license that was not available to the current compilation.

---

## DWS-0004

(error) No legal implementation found for %s

**Description**

The synthetic library does not contain an implementation that is legal for the current design context.

**What Next**

Check the parameter values used in that instance. Check that the library can support an instance using those parameters.

---

## DWS-0005

(error) Implementation failed for %s,%s

**Description**

No implementation in the synthetic library was usable for the part in the named design context.

**What Next**

Check that the library has an implementation for the part. If you are using a Synopsys library, this is an internal error. Contact Synopsys.

---

## DWS-0006

(warning) Found module name conflict during DW implementation: %s

**Description**

Unexpected conflicting module names were found during the compilation of the named part.

**What Next**

Module names are required to be unique. Rename conflicting module names.

---

## DWS-0007

(warning) Unable to rename module "%s" to "%s"

Feedback

**Description**

Synthetic part management was unable to rename a module. The most likely cause is a conflicting module name.

**What Next**

Module names are required to be unique. Rename conflicting module names.

If renaming module doesn't resolve the issue, contact Synopsys.

---

## DWS-0008

(warning) Illegal DW module suffix: "%s"

**Description**

Synthetic part management found an unexpected module name suffix. This may trigger later issues. It is not a serious issue in the local design context.

**What Next**

This is an internal error. Contact Synopsys.

---

## DWS-0009

(warning) QoR might be affected because can't get license for %s

**Description**

One or more licensed implementations for the part couldn't be used. An existing un-licensed implementation was used. That implementation might affect final QoR of the design.

**What Next**

Check that needed licenses are available.

---

## DWS-0010

(warning) %s '%s' not found in synthetic library.

**Description**

A command referencing a synthetic library was unable to find the named object.

Feedback

**What Next**

Check that the library definition is correct. If this is a Synopsys synthetic library, contact Synopsys.

---

## DWS-0011

(warning) Pruned implementation '%s' because its dependency is pruned

**Description**

An implementation object was pruned from the synthetic library due to a dependent object being pruned.

---

## DWS-0012

(warning) Pruned implementation '%s' of '%s' due to dont_use setting

**Description**

An implementation of a part was pruned from the synthetic library due to a dont_use command.

---

## DWS-0013

(warning) Pruned all basic implementations of '%s' due to dont_use setting

**Description**

All basic implementation of a synthetic library object were pruned due to dont_use commands.

---

## DWS-0014

(warning) Synthetic libraries have component dependency issue

**Description**

A dependency issue is preventing a complete synthetic library analysis.

**What Next**

Check that synthetic library are defined correctly. If the compilation uses a Synopsys synthetic library, contact Synopsys.

Feedback

## DWS-0015

(error) Implementation failed for %s,%s

**Description**

This component is only available in total power mode.

**What Next**

Enable total power mode for this compilation.

## DWS-0016

(error) %s

**Description**

Unable to read the required synthetic library

## DWS-0017

(error) %s

**Description**

User specified universal module name prefix for DesignWare is not valid.

## DWS-0100

(fatal) Pseudocell function %s doesn't have an implementation in the reference library. Check library dont_use settings.

**Description**

The datapath generators tried to use a library cell function that is not available in the libraries used for this compilation.

**What Next**

Check library cell dont_use settings. If that doesn't resolve the issue, this is an internal error. Contact Synopsys.

## DWS-0101

(error) No library cell resources are available to compile %s,%s.\n Check library dont_use settings.

Feedback

**Description**

DesignWare target library characterization didn't find any usable library cells. The most likely cause is incorrect library "dont_use" cell markings.

**What Next**

Ensure that "dont_use" setting in the current compilation haven't marked needed cells as "dont_use".

If that doesn't resolve the issue, this is an internal error. Contact Synopsys.

---

## DWS-0102

(error) No reference libraries were found; unable to generate synthetic parts

**Description**

The target libraries used for the current compilation do not have any usable cells for datapath generation. Check library dont_use settings.

**What Next**

Ensure that the current compilation contains the correct target libraries.

---

## DWS-0103

(error) Insufficient library cells to generate synthetic parts. Need either one of nand2/nor2 or an inverter and one of and/or

**Description**

Datapath generation needs a minimum subset of target library cells. This subset consists of either a nand2 or a nor2 gate, or an inverter plus either an and2 or or2 gate.

**What Next**

Ensure that the libraries used in the compilation contain the minimum required cell subset.

---

## DWS-0104

(error) Unexpected license problem during compile is found on synthetic part: %s

**Description**

The design being compiled is using a synthetic part but the license check on this part failed.

Feedback

All required license should have been checked out before compile. This error indicates some unexpected license problem is detected during synthetic part synthesis.

---

## DWS-0200

(error) Invalid implementation '%s' for cell '%s'.

**Description**

The command tried to set a module or implementation that isn't valid for the named cell.

**What Next**

The most likely cause for this issue is an incorrect synthetic library. This may be an internal error. Contact Synopsys.

---

## DWS-0201

(error) The cell '%s' is not a synthetic cell, so set_implementation was not applied to it.

**Description**

The set_implementation command can only be used on synthetic part cells.

---

## DWS-0202

(error) The cell '%s' is neither a synthetic cell nor a hierarchical cell, so %s was not applied to it.

**Description**

The command can only be used for synthetic part leaf cells, or for hierarchical cells that contain synthetic parts.

---

## DWS-0204

(information) Option '%s' may not take any effect depending on optimization opportunity and lib cell availability

**Description**

The named command option may not have any effect. The option depends on how the cell is optimized during compilation, and on the availability of certain target library cells.

Feedback

## DWS-0205

(warning) Conflicting datapath architecture options between '%s' and -mult_arch on '%s'. The '%s' option setting is ignored.

**Description**

Multiplier architecture generation options set on the named instance conflict. The named option will be ignored.

## DWS-0206

(warning) Conflicting datapath architecture options between '%s' and -booth_encoding on '%s'. The '%s' option setting is ignored

**Description**

Booth encoding generation options set on the named instance conflict. The named option will be ignored.

## DWS-0207

(warning) Cells '%s' that are extracted into a single block, in design '%s', have different datapath architecture options. All cells in this OP_OP cell are optimized using the architecture option settings '%s'

**Description**

Cells that are extracted into a single datapath block must use the same datapath architecture options.

## DWS-0208

(warning) The '%s' option setting could be ignored if total power mode is not on

**Description**

The named datapath architecture option can only be used in the total power mode. If total power mode is not on, the datapath generation will not be affected by ths option.

## DWS-0209

(warning) The option settings on '%s' has changed, either overwriting or appending.

Feedback

**Description**

The command has changed existing optimization options previously set on the named instance.

---

## DWS-0210

(warning) Datapath generation options '%s' can't be applied due to missing library cell

**Description**

The named datapath architecture option can only be used if the target library contains needed library cells.

An example is if the compilation tries to force the usage of Booth multiplier cells, and the libraries used in the compilation don't have Booth multiplier cells.

---

## DWS-0211

(warning) The forced implementation %s set by set_implementation was ignored for cell %s

**Description**

A set_implementation command was used to force the use of the named implementation. The specified implementation can't be used for this cell.

---

## DWS-0212

(error) Can not find synthetic cell matching '%s'.

**Description**

The set_synlib_dont_use command can only be used on synthetic part cells.

---

## DWS-0213

(error) Specified library '%s' is not a synthetic library.

**Description**

The synthetic library is expected but a non-synthetic library is given. The tool can't proceed.

Feedback

## DWS-0214

(error) Can't find specified synthetic library to report. Either the library is not loaded, or the name is mispelled.

**Description**

The tool can't find any match in loaded libraries with given name.

**What Next**

Check spelling to make sure it matches one of the loaded synthetic libraries. If the library is not loaded, use *open_lib* command to load it.

## DWS-0215

(warning) Area information not found for cell %s

**Description**

The area information for the cell is missing.

## DWS-0216

(information) Added key list '%s' to design '%s'.

**Description**

Indicates that the listed set of licenses have been associated with the specified design. Accessing the design will require that one of the listed licenses can be successfully checked out.

**What Next**

No action is required since this is just an information message.

## DWS-0217

(warning) Detected unbound synthetic operator '%s' on instance '%s'. This is unexpected.

**Description**

DesignWare synthesis engine found unbound synthetic operator during compile. It indicates a potenial flow problem in the tool.

**What Next**

Contact Synopsys.

Feedback

## DWS-0218

(warning) Cells '%s' that are extracted into a single block, in design '%s', have different datapath gating options. The OP_OP cell will be using the gating option settings '%s'.

**Description**

Cells that are extracted into a single datapath block must use the same datapath gating options.

## DWS-0219

(error) The %s implementation is only available in total power mode. To enable total power mode use set_qor_strategy -stage synthesis -metric total_power

**Description**

The command allows to read libraries only when total_power mode is enabled.

**What Next**

Low power DW implementations are enabled only in total power mode. This may be an internal error. Contact Synopsys.

## DWS-0220

(error) Cannot use generic library cell '%s' for synthesis of Synchronizers

**Description**

The library cells to be used to map Synchronizer designs have to be technology library cells. They cannot be generic library (GTECH) cells. The specified library cell will not be used for the synthesis of Synchronizers.

## DWS-0221

(error) Cannot use non-sequential library cell '%s' for synthesis of Synchronizers

**Description**

The library cells to be used to map Synchronizer designs have to be sequential technology library cells. They cannot be combinational (or any other kind of non-sequenital) library cells. The specified library cell will not be used for the synthesis of Synchronizers.

Feedback

## DWS-0222

(warning) Cannot use library cell '%s' with dont_use attribute set for synthesis of Synchronizers

**Description**

The library cells marked as "dont use" - i.e dont_use attribute set to be true will not to be used to map Synchronizer designs because the dont_use attribute takes precedence over the list passed to set_syncff_library. The specified library cell will not be used for the synthesis of Synchronizers.

## DWS-0223

(Error) architecture %s cannot be specified with this command.

**Description**

The circuit of pparch/apparch is built dynamically based on constraints. This architecture can not be specified with set_implementation.

## DWS-0225

(error) Cell name or cell list is missing.

**Description**

Rounding positions can only be set on cells, not globally.

## DWS-0226

(error) Internal rounding can not be set on '%s' because it is not of multiplier type.

**Description**

Internal rounding can only be used for multiplier-based synthetic or DW component cells.

## DWS-0227

(error) Incorrect value for rounding or internal rounding position, must be >= 0 and round_pos >= round_pos_int.

**Description**

Rounding position must be larger or equal to internal rounding position. Both positions must be >= 0.

Feedback

## DWS-0300

(error) No implementation is available for module %s

**Description**

The datapath generation system can't implement the named module. The most likely cause is a missing implementation in the synthetic library.

**What Next**

This is an internal error. Contact Synopsys.

## DWS-0301

(error) No implementation is available for operator %s

**Description**

The datapath generation system can't implement the named operator.

**What Next**

This is an internal error. Contact Synopsys.

## DWS-0302

(error) Datapath generation failed for %s

**Description**

Datapath generation failed for the named cell.

**What Next**

This is an internal error. Contact Synopsys.

## DWS-0303

(error) Datapath generation function is not supported: %s

**Description**

Datapath generation tried to use a function that is not yet implemented.

**What Next**

This is an internal error. Contact Synopsys.

Feedback

## DWS-0304

(error) Blocks with POP multipliers as a single node cannot be implemented: %s

**Description**

The graph describing a datapath block has an inconsistency that prevents datapath generation from continuing.

**What Next**

This is an internal error. Contact Synopsys.

## DWS-0305

(error) DFG hierarchical node type not supported: %d

**Description**

The graph describing a datapath block has an inconsistency that prevents datapath generation from continuing.

**What Next**

This is an internal error. Contact Synopsys.

## DWS-0306

(error) Unexpected DW node in graph

**Description**

The graph describing a datapath block has an inconsistency that prevents datapath generation from continuing.

**What Next**

This is an internal error. Contact Synopsys.

## DWS-0307

(error) Unexpected node type in graph: %d

**Description**

The graph describing a datapath block has an inconsistency that prevents datapath generation from continuing.

Feedback

### What Next

This is an internal error. Contact Synopsys.

---

## DWS-0310

(warning) Hybrid DesignWare library '%s' is already in use and can't be modified!

### Description

The hybrid DesignWare library has been successfully set up in current session, and user is trying to change that setting, which is not allowed.

### What Next

```
dw.hybrid.synopsys_root(3)
dw.hybrid.library_path(3)
dw.hybrid.library_version(3)
```

---

## DWS-0311

(error) Failed to setup hybrid DesignWare library '%s'.

### Description

The hybrid DesignWare library can't be successfully set up. There could be multiple reasons for this failure, such as file permission issue, concurrent lock issue, or simply because user specified illegal data.

### What Next

```
DWS-0313(n)
dw.hybrid.synopsys_root(3)
dw.hybrid.library_path(3)
dw.hybrid.library_version(3)
```

---

## DWS-0312

(error) The specified value '%s' for variable '%s' is not valid.

### Description

The specified value is not valid to use hybrid DesignWare library. Please check your input.

### What Next

```
DWS-0313(n)
dw.hybrid.synopsys_root(3)
dw.hybrid.library_path(3)
dw.hybrid.library_version(3)
```

Feedback

## DWS-0313

(error) %s

### Description

This is a generic error message in hybrid DesignWare library setup. It indicates the hybrid DesignWare library setup is failed. Please follow the instructions in the message and fix the problem.

### What Next

```
dw.hybrid.synopsys_root(3)
dw.hybrid.library_path(3)
dw.hybrid.library_version(3)
```

## DWS-0314

(information) Hybrid DesignWare library "%s" is in use.

### Description

This information message indicates that the synthetic libraries being used are from hybrid DesignWare library root, instead of the one from current image of the tool.

### What Next

```
dw.hybrid.synopsys_root(3)
dw.hybrid.library_path(3)
dw.hybrid.library_version(3)
```

## DWS-0315

(warning) The design "%s" contains synthetic operator or instance that was previously compiled with a different DesignWare version "%s".

### Description

The tool found that the specified design was compiled with a different DesignWare version than current. It may cause problem in datapath synthesis for some components. User should avoid using different versions of the tool for datapath synthesis in one design.

## DWS-0316

(error) It's too late to activate hybrid DesignWare library in this session.

Feedback

**Description**

User tried to use hybrid DesignWare library feature but it's too late to do so in current session.

The hybrid DesignWare library feature must be activated before any synthetic library reading commands are called, such as *elaborate*, *analyze*, *set_top_module*, *compile_fusion* etc.

**What Next**

For more details, refer to the DesignWare reference manual.

---

## DWS-0400

(error) Implementation of instance %s failed in module %s

**Description**

The implementation of the named instance, contained within the named module of a hierarchical static part failed.

**What Next**

This is an internal error. Contact Synopsys.

---

## DWS-0401

(error) Elaboration of module %s:%s in %s using parameters %s failed

**Description**

RTL elaboration failed on the named synthetic library module and implementation, using the named set of parameters, failed.

**What Next**

The likely cause is an inconsistency between RTL sources used in a synthetic library, and the synthetic library. If this is for a Synopsys library, contact Synopsys.

---

## DWS-0402

(error) Can't find module to implement part %s

**Description**

Hierarchical static part implementation was unable to find the needed module to implement the named contained synthetic part.

Feedback

**What Next**

This is an internal error. Contact Synopsys.

---

## DWS-0403

(error) Can't find specified module. Probably it has been ungrouped. Please try again with -no_autoungroup option.

**Description**

Hierarchical static part implementation was unable to find the needed module to implement the named contained synthetic part.

**What Next**

This is an internal error. Contact Synopsys.

---

## DWS-0500

(information) Couldn't locate model: %s and label: %s, training ML model

**Description**

specify the correct path to trained ml model and label.

**What Next**

For internal use, contact Synopsys.

---

## DWS-0501

(error) unable to read files %s and %s, using default flow

**Description**

trained ml model and label files not found.

**What Next**

For internal use, contact Synopsys.

---

## DWS-0502

(error) couldn't locate data file %s for training, exiting ML flow

**Description**

starting ML prediction

Feedback

**What Next**

For internal use, contact Synopsys.

---

## DWS-0503

(error) ML flow failed

**Description**

ML flow failed.

**What Next**

For internal use, contact Synopsys.

---

# ECO

---

## ECO-001

(error) Lib '%s' of reference lib cell '%s' is not a reference lib of lib '%s'.

**Description**

You receive this error message when you attempt to use a reference lib cell. However, its library is not part of the reference libary list on a library.

**What Next**

Use set_ref_libs to add the library.

---

## ECO-002

(Warning) Name of net %s is not same with the port(s) it is connected.

**Description**

Name of a net is not same with the port(s) it is connected.

**What Next**

Manually change the name of either the net or a port later.

---

## ECO-003

(error) Unable to find hierarchy %s; %s.

Feedback

**Description**

Unable to find the hierarchy.

**What Next**

Make sure the hierarchy exist.

---

## ECO-004

(Error) library cell '%s' is not a buffer.

**Description**

You recieve this error message, if the library cell matching the specified name is not a buffer library cell from the target library.

**What Next**

Check to make sure that the libray is set correctly and use the buffer.

**See Also**

- report_lib

---

## ECO-005

(Error) '%s' is a driving pin/port.

**Description**

This error message occurs if it is a driving pin/port.

**What Next**

specify load pin/port.

---

## ECO-006

(Error) Unable to find the hierarchy '%s'.

**Description**

This error message occurs if the name of hieararcy can not be found.

**What Next**

specify correct hierarchy.

Feedback

## ECO-007

(Error) All loads are not under the same connection.

**Description**

This error message occurs if any loads are connected to the different connections.

**What Next**

specify loads with the same connection

## ECO-008

(error) '%s' is not a driver pin/port.

**Description**

The specified pin/port is not a driver.

**What Next**

Specified the driver pin/port.

## ECO-009

(error) '%s' is not a load pin/port.

**Description**

The specified pin/port is not a load.

**What Next**

Specified the load pin/port.

## ECO-010

(error) pin_port:'%s' does not have no location

**Description**

There are no location for the pin or port. The location is required for this command.

**What Next**

Set the location

Feedback

## ECO-011

(warning) Reference change of instance '%s' from '%s' to '%s' is ignored.

**Description**

The reference of this instance is soft macro and this kind of reference change cannot be supported by change_link currently, so ignore it.

## ECO-012

(warning) Can't set location for cell instance "%s".

**Description**

This message is shown when the tool can't set the location of cell instance.

## ECO-013

(Error) Target module "%s" doesn't exist in golden.

**Description**

This message is shown by eco_netlist when the module in -compare_target_modules doesn't exist in golden block.

**What Next**

Check the target modules name and ensure it exists in golden block.

## ECO-014

(Error) The reference %s of instance %s in %s block cannot be resolved or is not in target module list.

**Description**

This message is shown by eco_netlist when the reference of module instance can not be resolved or is not in target module list when doing partial diff.

**What Next**

Make sure that all instances in both working and golden block can be resolve. If you are doing partial netlist diff and the reported reference module get changed or is not in working, make sure it exists in golden and is included in target module list.

Feedback

## ECO-015

(Warning) Not support adding buffer in primary Voltage Areas '%s' whose associating power domains have no hierarchy cells.

**Description**

This message is shown when there is a voltage area whose is_primary is true but all of its associating power domains have no hierarchy cells (e.g. all elements of power domain are leaf cells). If the net to buffer crosses such primary voltage area, the VA will be treated as a blocked region that no buffers will be added inside.

**What Next**

User can try to change such primary VA as a gas-station then run this buffering command again.

## ECO-016

(Error) Cell %s has no original information recorded.

**Description**

This message is shown by revert_cell_sizing when the cell is not resized by size_cell or has been reverted before.

**What Next**

Please remove it from the input list and run command again.

## ECO-017

(Info) Input and output routes are not merged after removing buffer %s.

**Description**

This message is shown by remove_buffers when the input and output routes of the removed buffer are not merged due to the following reasons.

- the routes are not aligned.
- the routes are not at same layer.
- the removed buffer has eco_legalized change status.

**What Next**

You should run route command to make the route complete.

Feedback

## ECO-018

(error) The design %s has no site row.

### Description

You receive this error message when the design has no site row, and the following commands does not support designs without site row: *add_spare_cells*.

### What Next

Create site rows before using these commands.

## ECO-019

(error) There is no enough placeable area to add new spare cells.

### Description

You receive this error message when there is no enough space in target placeable regions. The reason may be too many cell number specified or the placeable area of the specified regions is small.

When calculating 'placeable area', the existing cells, blockages and exclusive bounds will be considered.

### What Next

Specify less number of cells or specify larger placeable regions. If you use the option '-repetitive_window', specify larger window size.

## ECO-020

(warning) Skip to spread spare cells belonging to the target %s (%s) since there is no enough available space.

### Description

There is no enough available space in the target region to spread spare cells. The command skips to do spreading for the spare cells belonging to this target region.

### What Next

Check the utilization of target region.

### See Also

- spread_spare_cells

Feedback

## ECO-021

(error) %s.

## ECO-022

(error) Failed to get voltage area for cell (%s).

**Description**

Cannot get voltage area for the given cell.

**What Next**

Check the voltage area of the cell.

**See Also**

- spread_spare_cells
- get_voltage_areas

## ECO-023

(error) Failed to get library cell for cell (%s).

**Description**

Cannot get library cell for the given cell.

**What Next**

Check the library cell of the cell.

**See Also**

- spread_spare_cells
- get_lib_cells

## ECO-024

(error) The number of valid locations (%d) does not match with the number of cell to be spread (%d).

**Description**

The number of valid locations is not equal to the number of cell to be spread.

Feedback

**See Also**

- spread_spare_cells

---

## ECO-025

(Info) Routes are re-connected after removing buffer %s.

**Description**

This message is shown by remove_buffers when the input and output routes of removed buffer are re-connected.

**What Next**

No need to re-route to keep the route complete after removing buffer.

---

## ECO-026

(Error) The specified pin or port (%s) has already been connected to other virtual connection.

**Description**

This error only happens when the specified pin or port has already been connected to other virtual connection.

**What Next**

Please re-specify the pin or port to be connected.

---

## ECO-027

(Error) The specified virtual connection does not exist.

**Description**

This error only happens when the specified virtual connection does not exist.

**What Next**

Please specify the valid virtual connection name.

---

## ECO-028

(Error) The given pin or port (%s) is not connected to the specified virtual connection (%s).

Feedback

**Description**

This error only happens when the specified pin or port is not connected to the specified virtual connection.

**What Next**

Please input valid pin or port.

---

## ECO-029

(Error) The virtual connection (%s) already exists.

**Description**

This error only happens when the specified virtual connection already exists.

**What Next**

Please specify the new virtual connection name.

---

## ECO-030

(error) There is no unmapped eco cells to be placed

**Description**

There is no eco cells to be placed

**What Next**

Please check the eco cell's attributes: "is_fs_eco_add" as TRUE and "eco_change_status" is one of {create_cell, add_buffer, add_buffer_on_route, change_link, size_cell}.

**See Also**

- place_freeze_silicon

---

## ECO-031

(error) Cell %s is not mappable.

**Description**

This command only supports mapping eco cell added by freeze silicon eco and still not mapped. It must be leaf cell (attribute "is_hierarchical" is FALSE) and have attributes: "is_fs_eco_add" as TRUE and "eco_change_status" as one of {create_cell, add_buffer, add_buffer_on_route, change_link, size_cell}.

Feedback

### What Next

Specify correct eco cell and rerun the command.

### See Also

- place_freeze_silicon
- map_freeze_silicon

---

## ECO-032

(Error) Some eco cells has no corresponding spare cells to swap.

### Description

ECO lib cells and the lib cells of spare cells may have psc_type_id or bitmap, or neither. This command will map eco cells to spare cells based on these lib cell attributes. For eco cells whose lib cells having psc_type_id, they can be swapped with PSC fillers whose lib cells also having psc_type_id. This attribute is manually set by user for traditional PSC mapping. For eco cells whose lib cells having bitmap, they can be swapped with PSC fillers whose lib cells also having bitmap. This attribute is automatically set by derive_programmable_spare_cell_mapping_rules for advanced PSC mapping. For eco cells whose lib cells without psc_type_id and bitmap, they can be only swapped with non-PSC spare cells of same library cell name. This is logical spare cell based mapping. This error message occurs when 1. there is any eco cell whose lib cell has psc_type_id but there is no PSC filler whose lib cell has psc_type_id. There are three possible reasons: 1) psc_type_id is set on ECO lib cells but not set on PSC filler lib cells 2) PSC filler cell instances of these lib cells are not created by create_stdcell_fillers 3) attribute is_spare_cell is not set on PSC filler cell instances. 2. there is any eco cell whose lib cell has bitmap but there is no PSC filler whose lib cell has bitmap. There are three possible reasons: derive_programmable_spare_cell_mapping_rules has been applied on ECO lib cells but not on PSC filler lib cells. 2) PSC filler cell instances of these lib cells are not created by create_stdcell_fillers 3) attribute is_spare_cell is not set on PSC filler cell instances. 3. there is any eco cell with neither psc_type_id nor bitmap, but the design has no non-PSC spare cell.

### What Next

Firstly, please confirm which kind of mapping you are going to do. 1) logical spare cell based mapping 2) traditional PSC mapping 3) advanced PSC mapping Secondly, please read the user guide of corresponding mapping and make sure the flow is all correct.

Feedback

**See Also**

- eco_netlist
- create_cell
- create_stdcell_fillers
- check_freeze_silicon
- place_freeze_silicon
- derive_programmable_spare_cell_mapping_rules

---

## ECO-033

(Error) Can't find any unmapped eco cells in the design.

**Description**

This error message occurs when there are no any unmapped cell which it must have attributes: "is_fs_eco_add_cell" as TRUE and "eco_change_status" is one of {create_cell, add_buffer, add_buffer_on_route, change_link, size_cell}.

**What Next**

Use eco_netlist -freeze_silicon to get unmapped cells or set_attribute..

**See Also**

- eco_netlist
- set_attribute

---

## ECO-034

(Error) No spare cell found under the region: %s %s in the design.

**Description**

This error message occurs when there is no any spare cell under a region in the design.

**What Next**

add the spare cell by either design import, verilog eco, netlist editing or add_spare_cells.

---

## ECO-035

(Error) No %s spare cell found under the region: %s %s in the design.

Feedback

**Description**

This error message occurs when there is no any spare cell of a particular lib cell under a region in the design.

**What Next**

add the spare cell by either design import, verilog eco, netlist editing or add_spare_cells.

---

## ECO-036

(Error) %d %s spare cell is less than %d %s ecoCell under the region:%s %s.

**Description**

This error message occurs when there is no enough spare cell of a particular lib cell under a region in the design.

**What Next**

add enough spare cells by either design import, verilog eco, netlist editing or add_spare_cells.

---

## ECO-037

(Error) Option "%s" doesn't support collection input.

**Description**

This message is shown when the command option can't receive collection input.

**What Next**

Remove the collection input and try again.

---

## ECO-038

(Warning) Skip the gas station voltage Area %s.

**Description**

This Warning message occurs when the command found a voltage area which is gas station. And the gas station voltage area is not supported yet by this command.

**What Next**

None

Feedback

## ECO-039

(Error) No voltage area overlaps with the specified boundary.

**Description**

This error message occurs when there is no voltage area overlaps with the specified boundary.

**What Next**

Specify a new boundary that have voltage areas overlap with it.

## ECO-040

(Error) No voltage area supported in the specified voltage area list.

**Description**

This error message occurs when there is no voltage area supported in the specified voltage area list. This command does not support gas station.

**What Next**

Specify supported voltage areas.

## ECO-041

(Error) Failed to get the placeable area for the hierarchical cell %s.

**Description**

This error message occurs when the command failed to get the placealbe area for the specified hierarhcical cell to place new cells. This may happend because there is no voltage area or bound associated with the hierarchical cell.

**What Next**

Associate the hierarchical cell to a power domain and its voltage areas or specify another hierarchical cell.

## ECO-042

(error) The current design (%s) is not same as the design (%s) with virtual connection.

Feedback

**Description**

This error message occurs when the current design is not same as the design with virtual connection.

**What Next**

User may switch to the design with virtual connection and clean up the virtual connection. And then switch to the working design and build new virtual connection for this design.

---

## ECO-043

(error) No virtual connection matched the specified objects.

**Description**

This error message occurs when no virtual connection matched the specified objects.

---

## ECO-044

(error) Cannot map eco cell %s to spare cell %s, which is in different voltage area.

**Description**

Eco cell must be mapped to spare cell in same voltage area.

**What Next**

Specify spare cell in same voltage area and rerun the command.

**See Also**

- map_freeze_silicon

---

## ECO-045

(error) Cannot map eco cell %s to spare cell %s, whose reference lib cell is not compatible.

**Description**

Eco cell must be mapped to spare cell of a compatible lib cell, which means either:

1. Eco cell and spare cell are of same reference lib cell. In this scenario, user can specify only one eco cell and one spare cell in the mapping pair.
2. Spare cell is programmable spare cell (PSC) filler, whose PSC type is same as the PSC type of the eco cell. In this scenario, user can specify one or multiple adjacent

Feedback

PSC fillers with the same PSC type of eco cell(s) in the mapping pair. 3. Spare cell is programmable spare cell (PSC) filler and PSC mapping rule is defined. In this scenario, user can specify one or multiple PSC fillers in the mapping pair.

**What Next**

Specify a compatible spare cell and rerun the command.

**See Also**

- map_freeze_silicon

---

## ECO-046

(error) Cannot map multi eco cells %s to spare cell %s

**Description**

Cannot map multi eco cells to one spare cell, unless the spare cell is a programable spare cell (PSC) filler.

**What Next**

Map different eco cells to different spare cells, or specify a PSC filler.

**See Also**

- map_freeze_silicon

---

## ECO-047

(error) Cannot map eco cells %s to multiple spare cells %s

**Description**

Cannot map one eco cell to multiple spare cells. Note: this command also does not support merging programmable spare cell (PSC) fillers. So user cannot specify multiple PSC fillers to map one eco cell as well.

**What Next**

Specify one spare cell and rerun the command.

**See Also**

- map_freeze_silicon

Feedback

## ECO-048

(error) Strategy %s is invalid for %s, whose orientation is %s.

### Description

Strategy specified must match orientation of the spare cell. E.g. for a R90 spare cell, pack_top is valid while pack_right is invalid.

### What Next

Specify correct strategy or do not specify strategy (use default strategy). Then rerun the command.

### See Also

- map_freeze_silicon

## ECO-049

(error) Strategy %s is invalid for %s, which are not is_placed.

### Description

Such strategy is designed for eco cells which are already put near expected locations. Attribute is_placed of such eco cells must be true.

### What Next

Please move eco cells to expected locations before using this strategy. If they are already in expected locations but is_placed is not true , please try to set phsycial_status attribute as placed. Or you could specify other strategies or do not specify strategy (use default strategy). Then rerun the command.

### See Also

- map_freeze_silicon

## ECO-050

(error) Size of PSC filler %s that is specified to map eco cells %s is not enough.

### Description

Size of PSC filler must be enough for all eco cells mapping to it. When mapping rule is defined, this message means no enough PSC fillers including compatible (including splitting/merging) PSC Types. Space of PSC filler under PG straps cannot be used if eco

Feedback

cells are not allowed to use. When closest rule is used, it is possible that firstly placed eco cells will split PSC fillers into fragments that make rest eco cells fail to place.

**What Next**

Select larger PSC filler or map rest eco cells to other PSC fillers. Then rerun the command.

**See Also**

- map_freeze_silicon

---

## ECO-051

(error) Unused white space (length %s, psc_type_id %s) at location {%s} cannot be recovered to be programmable spare cell(s).

**Description**

You receive this error message when there is no valid programmable spare cell (PSC) filler lib cell found for recovering the unused white space left by PSC cells after PSC mapping or PSC recycling.

**What Next**

If this message is issued when the option *-lib_cells_for_filler_recovery* is specified, please specify small enough PSC filler lib cells with the same PSC type in the command option. Otherwise, add reference libraries which contain small enough PSC filler lib cells with the same PSC type.

**See Also**

- place_freeze_silicon
- map_freeze_silicon
- recycle_programmable_spare_cells

---

## ECO-052

(error) Feasibility check of programmable spare cells mapping failed.

**Description**

You receive this error message when the programmable spare cells (PSC) are not enough for the mapping of eco cells with PSC type.

Feedback

### What Next

Run command "check_freeze_silicon" to get more detailed information of the PSC feasibility check failure and add enough PSC fillers.

### See Also

- place_freeze_silicon
- check_freeze_silicon

---

## ECO-053

(info) %d cells have been recycled to be PSC fillers successfully.

### Description

This info shows how many cells with attribute 'psc_filler_is_used = true' have been recycled.

---

## ECO-054

(error) Mapping engine failed to find enough nearby programmable spare cells (PSC) for mapping %d eco cells with psc_type_id %d under voltage area %s.

### Description

You receive this error message when the nearby programmable spare cells (PSC) are not enough for mapping eco cells under the voltage area. Some programmable spare cells may be far awary from the target eco cells.

### What Next

Consider re-designing the ECO to use only the nearby available PSC resources.

### See Also

- place_freeze_silicon

---

## ECO-055

(error) Failed to place %d eco cells to the matched programmable spare cells (PSC), psc_type_id %d and voltage area %s.

### Description

You receive this error message when command failed to place the eco cells to the matched programmable spare cells.

Feedback

**See Also**

- place_freeze_silicon

---

## ECO-056

(info) Totally %d cells have not been mapped due to the programmable spare cell recovery issue or the other issues reported in previous command log.

**Description**

This info shows how many eco cells have not been mapped to programmable spare cells (PSC) due to the issue that there is no valid PSC filler library cell to fill the unused white space of PSC cells or the other issues reported by ECO-069.

**What Next**

For ECO-051: please refer to the error message ECO-051 printed by command and prepare valid PSC filler library cells in command option or in the reference libraries. Then rerun command with these eco cells again.

For ECO-069: please check the failure reason and rerun command with these eco cells after fixing the issue.

You can filter the mapped cells by the attribute eco_change_status and then pass the left cells to the comand again. For mapped eco cells, the eco_change_status is set to eco_legalized.

**See Also**

- place_freeze_silicon

---

## ECO-057

(info) %d cells have not been recycled to be PSC fillers due to the recovery issue.

**Description**

This info shows how many cells have not been recycled to be PSC fillers due to the issue that there is no valid PSC filler library cell to fill and recover the white space after recycling.

**What Next**

Please reference the error message ECO-051 printed by command and prepare valid PSC filler library cells in command option or in the reference libraries. Then rerun command with these cells again.

Feedback

**See Also**

- recycle_programmable_spare_cells

---

## ECO-058

(info) %d eco cells without PSC type (programmable spare cell) have not been mapped successfully.

**Description**

This info shows how many eco cells have not been mapped successfully for non-PSC type.

**See Also**

- place_freeze_silicon

---

## ECO-059

(error) Could not find any valid spare cell for user input buffering location (%s, %s, layer %d).

**Description**

You receive this error message when there is no valid spare cell for the user input buffering location.

**See Also**

- add_buffer_on_route

---

## ECO-060

(Error) There is no cell to be connected in the specified cells.

**Description**

You receive this error message when you attempt to use connect_freeze_silicon_tie_cells to connect cells with tie connection to tie cells but there is no valid cell to be connected to tie cell in the specified cells. The valid cell must have pins connected to tie net, and should be placed. The cells inside the sub block are not supported.

**What Next**

Specify placed cells with tie connection in top block.

Feedback

**See Also**

- connect_freeze_silicon_tie_cells

---

## ECO-061

(Warning) There are %s cells ignored in the specified cells.

**Description**

You receive this error message when you attempt to use connect_freeze_silicon_tie_cells to connect cells with tie connection to tie cells, and there are some invalid cells in the specified cells, these cells will be skipped and other cells will be connected to tie cells. A valid cell must have pins connected to tie net, and should be placed. The cells inside the sub block are not supported.

**What Next**

Nothing.

**See Also**

- connect_freeze_silicon_tie_cells

---

## ECO-062

(Error) No enough tie %s cells in the voltage area %s.

**Description**

You receive this error message when you attempt to use connect_freeze_silicon_tie_cells to connect cells with tie connection to tie cells, but there are not enough tie cells in the voltage area.

**What Next**

Add more tie cells if possible or specify fewer cells to be connected.

**See Also**

- connect_freeze_silicon_tie_cells

---

## ECO-063

(Error) No enough tie %s cells surrounding the target cells in the voltage area %s

Feedback

### Description

You receive this error message when you attempt to use connect_freeze_silicon_tie_cells to connect cells with tie connection to tie cells but there are not enough tie cells surrounding the target cells in the voltage area. This may also be caused by max_fanout or max_wire_length constrains.

### What Next

Broaden the max_fanout limitation to allow a tie cell to drive more loads. If you specify the option max_wire_length, also broaden this limitation.

### See Also

- connect_freeze_silicon_tie_cells

---

## ECO-064

(warning) The boundary {%s} of cell instance "%s" is not set or invalid.

### Description

The boundary of the target cell instance is not set or invalid and this would affect the correctness of command. Please make sure the cell boundary is set correctly.

### What Next

Check the boundary of target cell instance and set it correctly.

### See Also

- check_freeze_silicon
- place_freeze_silicon
- map_freeze_silicon

---

## ECO-065

(warning) Fail to set name of net {%s} to match names of ports that connects this net.

### Description

The name of a net should be same with the ports that it connects. This message is shown when tool fail to achieve this.

Feedback

### What Next

Check ports that connects this net by all_connected. Check whether there is already another net whose name is same with the port that connects this net.

---

## ECO-066

(Error) Eco legalization failed due to physical unfriendly cells.

### Description

You receive this error message when you attempt to use place_eco_cells to do minimum physical impact legalization with the option max_displacement_threshold, but failed because there are physical unfriendly cells. Here, "physical unfriendly cells" mean cells that have displacement larger than the max_displacement_threshold value.

Minimum physical impact legalization is the combination of free site only legalization and incremental(push non-eco cell) legalization. The engine will first try to do free site only legalization for target cells, and do incremental legalization for cells that have displacement larger the the *displacement_threshold*. But if there are cells have displacement larger the the *max_displacement_threshold*, the command will fail. And the physical unfriendly cells will be stored in the collection named epl_max_displacement_cells.

### What Next

You can raise the value for the option *max_displacement_threshold*, or you can decide to reject the ECO changes for physical-unfriendly cells in collection epl_max_displacement_cells, and redo minimum physical impact legalization.

### See Also

- place_eco_cells

---

## ECO-067

(warning) The layout change of the desgin %s has not been saved and will be lost after closed.

### Description

You receive this error message when the design is closed before saving layout change. The layout editing recording for the design has been started by "record_layout_editing -start", but has not been saved by calling "record_layout_editing -stop" before closing the design. After that the design is closed, the recorded data will be lost.

Feedback

## ECO-068

(warning) The previous recording of layout editing has not been saved and will be lost.

**Description**

You receive this error message when you start new layout recording before saving previous layout recording. when the new layout recording starts, the previous recording will be lost.

## ECO-069

(error) No available layout editing need to be saved for the current design.

**Description**

You receive this error message when you call "record_layout_editing -stop" to save the layout change but actually there is no recording that has been started by "recrod_layout_editing -start".

## ECO-070

(error) Failed to open the file %s for write to output the change file%s.

**Description**

You receive this error message when you specified an invalid file name for "record_layout_editing -output" or "record_layout_editing -undo_output".

**What Next**

Specify a valid file name. previous recording will be lost.

## ECO-071

(Error) The pin or port (%s) is under the first common branch of route from the specified loads.

**Description**

If the pin or port is not part of specified loads and is uner the first common branch of route from the specified load, the buffer will not be able to added to drive the specific loads.

**What Next**

Please re-specify the pin or port to buffer the specific loads.

Feedback

## ECO-072

(error) Could not find a compatible spare cell within the specified maximum distance %s for cell %s.

### Description

You receive this warning message when there is no valid compatible spare cell found within the specified maximum distance for the eco cell.

### What Next

Increase the maximum distance to spare cell and rerun command again.

### See Also

- size_cell
- add_buffer_on_route

## ECO-073

(error) Cell sizing failed for cell %s due to the spare cell aware checking.

### Description

You receive this warning message when failed to do cell sizing due to the spare cell aware checking not passed.

### What Next

Increase the maximum distance to spare cell and rerun command again.

### See Also

- size_cell

## ECO-074

(error) Block partition between working design and golden design is inconsistent for %s.

### Description

This message is shown by eco_netlist when -by_block is used. The working design and golden design have block partition difference and the name of the different block is shown in the message.

Feedback

### What Next

Make sure the block patition is consistent between working design and golden design.

### See Also

- eco_netlist

---

## ECO-075

(warning) In the netlist change of instance %s, the target reference %s is not in the reference library list of working library %s.

### Description

The target reference of this netlist change is not in the reference library list of working library. When source back this change, it would encounter the issue that the reference could not be found.

### What Next

Before source back the change list generated by eco_netlist, please add the library of this reference to the reference library list.

### See Also

- eco_netlist

---

## ECO-076

(error) In %s, two different sub blocks ("%s" and "%s") have the same top module name "%s";

### Description

This message is shown by eco_netlist when -cross_physical_hierarchy is used. The working (or golden) have two sub blocks which have the same top module name. Currently eco_netlist does not support this kind scenario.

### What Next

Make sure in working (or golden) there are no two sub blocks which have the same top module name.

### See Also

- eco_netlist

Feedback

## ECO-077

(info) %d buffers (ideally %d) have been added on net %s.

### Description

This info shows how many buffers have been added on the net considering the spare cell aware. Some place of net may have no buffer added for there is no compatible spare cell within the max searching distance.

### See Also

- add_buffer_on_route

## ECO-079

(warning) Programmable spare cell %s (psc_type_id %s) is skipped due to PG stripe overlap on %s side in layer { %s}, which is not allowed to overlap with PG stripe on the specified side of layers.

### Description

You receive this warning message when detect programmable spare cell (PSC) which has PG stripe overlap but this kind of PSC has been set with the mapping rule that it is not allowded to overlap with PG stripe on some side of layers. So this kind of PSC cells will be ignored and skipped during the PSC mapping.

### What Next

Check the mapping rules and make sure the PG overlap status of programmable spare cell is consistent with the mapping rule set.

### See Also

- place_freeze_silicon
- report_programmable_spare_cell_mapping_rule

## ECO-080

(error) Cell (%s) is not placed, it should be placed when -map_spare_cells_only is used.

### Description

The target eco cells should be placed when the option -map_spare_cells_only is used.

Feedback

### What Next

For unplaced cells, run place_freeze_silicon -no_spare_cell_swapping first to do coarse placement. Or move the unplaced cell to target location and set its attribute physical_status as placed.

### See Also

- place_freeze_silicon
- set_attribute

---

## ECO-081

(warning) The supply net for cell %s is not updated, because %s.

### Description

This message is shown when the tool will not update supply net for a cell.

If "in gas-station but non-single rail" is reported as the reason, it means the cell has more than one power or ground pin, which is not supported to update supply net in gas station.

### What Next

If "in gas-station but non-single rail" is reported, specify single-rail reference when creating new cell in gas station.

---

## ECO-082

(Warning) no driver found for cell %s. Fail to update supply for eco buffer tree includes this cell.

### Description

This Warning message occurs if no driver found for the cell when updating supply nets for cells.

### What Next

Fix the connections for the cell.

### See Also

- eco_update_supply_net

Feedback

## ECO-083

(warning) Total %u cells specified by -cells are skipped to update supply nets.

### Description

This message is shown if some cells are skipped to updated supply nets when using "eco_update_supply_net -cells". This command only updates buffers in the top block that is added by timing eco commands. And for other type cells, the command will skip them and not update the supply nets.

### What Next

Specify supported cells.

### See Also

- eco_update_supply_net

## ECO-084

(info) No eco timing cell need to update supply net.

### Description

This information shows if no buffer need to update supply net. When "-cells" is specified, the message is issued if the supply nets for all supported buffers has already been updated in the specified cells; otherwise, the message is issued when the supply nets for all supported buffers has already been updated in the top block.

### What Next

Nothing

## ECO-085

(warning) Failed to update supply nets for the buffer tree driven by %s.

### Description

This message occurs when there is any mv violation during updating supply nets for the buffer tree. For example, the command failed to annotate supply on the buffer tree, or some buffers/inverters in the tree are invalid, either the lib cell type is wrong, or they cannot be even put there (logically and physically).

The cells that failed to update supply net will be added to the collection $ecoUpdateSupplyFailedCells.

Feedback

**What Next**

- Use *fix_mv_design -buffer* to fix any MV legality issues such as Power domain-Voltage area mismatch.
- Use *add_buffer_on_route -select_mv_buffers* to enable MV legality checks when inserting buffer.
- Please run *check_bufferability* to report information for supply assignment failure and additional details.
- Check if appropriate supplies are available in buffered power domain.

**See Also**

- eco_update_supply_net
- check_bufferability
- fix_mv_design
- add_buffer_on_route

---

## ECO-086

(warning) Failed to update supply net for cells in collection $%s.

**Description**

This message occurs when there is any cell that failed to update supply net due to mv violation. And the cells that failed to update supply net will be added to the collection $ecoUpdateSupplyFailedCells.

**See Also**

- eco_update_supply_net

---

## ECO-087

(error) Failed to update supply nets for all target cells.

**Description**

This message occurs when failed to update supply nets for all target cells due to mv violation. Here "target cells" means the cells specified by the option "-cells" or all timing cell cells that have no supply nets set yet. And the cells that failed to update supply nets will be added to the collection $ecoUpdateSupplyFailedCells.

Feedback

**See Also**

- eco_update_supply_net

## ECO-088

(info) Total %u cells have updated supply nets.

**Description**

This information shows how many cells are updated to set supply nets.

**What Next**

Nothing

## ECO-089

(Warning) no load found for cell %s. Fail to update supply for eco buffer tree includes this cell.

**Description**

This Warning message occurs if no load found for the cell when updating supply nets for cells.

**What Next**

Fix the connections for the cell.

**See Also**

- eco_update_supply_net

## ECO-090

(warning) Totally %s programmable spare cells are skipped due to the PG overlap rule violation under %s. The detail number is ( %s).

**Description**

You receive this warning message when detect programmable spare cells (PSC) which have PG overlap but no such kind of mapping rule is specified. This kind of PSC cells will be ignored and skipped during the PSC mapping.

Feedback

### What Next

Check the mapping rules and make sure the PG overlap status of programmable spare cell is consistent with the mapping rule set.

### See Also

- place_freeze_silicon
- report_programmable_spare_cell_mapping_rule

---

## ECO-091

(error) The programmable spare cells %s in one mapping pair are not adjacent and they cannot be merged together for mapping eco cells.

### Description

When specifiy multiple programmable spare cells for one mapping pair, these programmable spare cells must be adjacent and can be merged together for the mapping.

### What Next

Specify correct adjacent programmable spare cells for this mapping pair.

### See Also

- map_freeze_silicon

---

## ECO-092

(error) Programmable spare cells clustered within the bbox %s.

### Description

You receive this error message when the programmable spare cells clustered within the bbox so that many eco cells may have conflicts when they are mapped to the nearby spare cells. The mapping engine assumes the spare cells are distributed evenly.

### What Next

Spread spare cells more evenly or add more spare cells outside the bbox.

### See Also

- place_freeze_silicon

Feedback

## ECO-093

(info) Total %u target cells are found.

**Description**

This information shows how many target cells of this command. Definition of target cells is described in command man page.

## ECO-094

(warning) %s is not a target of this command, because %s.

**Description**

This object is not target of this command.

**What Next**

If this object is not expected to process, nothing need to do. If the reported reason means an issue, please fix it and re-run the command. Otherwise, please find commands support this object to process this object. Please see also man page of this command to find out what are its target objects.

## ECO-095

(warning) Meet issue at %s, because %s.

**Description**

Tool meets an issue at the reported object because the reported reason.

**What Next**

If this is not a influential problem, just ignore it. If this can be fixed, please fix it and re-run the command. Otherwise, please find commands support this object to process this command.

## ECO-096

(error) Failed to create bus guides for %d bus bundles: { %s }.

**Description**

You receive this error message when command failed to create bus guides for some of the input bus bundles.

Feedback

**What Next**

Please check the failed reason and setting, make sure there is enough resource to create the bus guides for these bus bundles.

---

## ECO-097

(error) Failed to set bus guide connection of %s for reason: %s.

**Description**

You receive this error message when command failed to set bus guide connection for the listed reason.

**What Next**

Check the reason listed and adjust the input options.

---

## ECO-098

(error) Cannot set bundle %s with partial preference, it has not be set with complete bus guide preference before.

**Description**

You receive this error message when try to set partial preference for the bundle listed which has not be set with complete bus guide preference before.

**What Next**

Please set complete guide preference for this bundle first.

---

## ECO-099

(error) Failed to do GCR resource setting for %s.

**Description**

You receive this error message when failed to do GCR resource setting for the bundle listed.

**What Next**

Please check the detail failed error messages.

---

## ECO-100

(Warning) Signoff ECO signatures were removed.

Feedback

### Description

You receive this warning message when you attempt to make ECO changes for design with signoff ECO signatures. If the design is not under recording mode, the design's signoff ECO signatures will be cleaned up with any ECO changes.

### What Next

Double check if the ECO changes are expected without enable signoff ECO change recording.

### See Also

- record_signoff_eco_changes

---

## ECO-101

(Warning) Singoff ECO logical recording is terminated due to functional ECO change.

### Description

You receive this warning message if you attempt to make functional ECO changes when you are recording the signoff ECO changes.

### What Next

Avoid to make functional ECO change when recording signoff ECO changes.

### See Also

- record_signoff_eco_changes

---

## ECO-102

(Error) Signature check between design and tcl file failed.

### Description

This check will be triggered by tcl command *record_signoff_eco_changes -start -input <tcl file>*. The tool will check the signoff ECO signature consistency between the tcl file and the current design. If the signatures are not consistent with each other, or there is no signature inside design or tcl file, the check will fail. The signoff eco changes recording is not started. And also the eco TCL file is not sourced. The signature check provide one extra layer protection to make sure the PT-ECO tcl file applied to the right design.

### What Next

Choose tcl file with consistent signatures as input.

Feedback

**See Also**

- record_signoff_eco_changes

---

## ECO-103

(Error) No signoff eco signature ininitalized in current design.

**Description**

This check will be triggered by tcl command *record_signoff_eco_changes -start -input <tcl file>*. The tool will check the signoff ECO signature consistency between the tcl file and the current design. If the signatures are not consistent with each other, or there is no signature inside design or tcl file, the check will fail. The signoff eco changes recording is not started. And also the eco TCL file is not sourced. The signature check provide one extra layer protection to make sure the PT-ECO tcl file applied to the right design.

**What Next**

Use *record_signoff_eco_changes -init* to initialize the signoff eco signature.

**See Also**

- record_signoff_eco_changes

---

## ECO-104

(error) Failed to get bus guide preference data of bundle %s for reason: %s.

**Description**

You receive this error message when failed to get some bus guide preference data for the listed reason.

**What Next**

Please check the reason listed.

---

## ECO-105

(error) There is one issue found when do GCR resource setting for bundle %s: %s.

**Description**

You receive this error message when failed to do GCR resource setting for the reason listed.

Feedback

**What Next**

Please check the failed reason.

---

## ECO-106

(warning) Bundle %s has no bus guide preference.

**Description**

You receive this warning message when try to remove the bus guide preference for the bundle which has not been set with bus guide preference.

---

## ECO-107

(error) There is one issue found when execute GCR operation: %s.

**Description**

You receive this error message when failed to execute GCR operation for the reason listed.

**What Next**

Please check the failed reason.

---

## ECO-108

(error) There is one issue found when route bus guides: %s.

**Description**

You receive this error message when failed to route bus guides for the reason listed.

**What Next**

Please check the failed reason.

---

## ECO-109

(warning) For bundle %s, the %s of layer %s is smaller than the minimal valid value. Use the minimal valid value %s instead.

**Description**

You receive this warning message when the specified setting of layer is smaller than the minimal valid value.

Feedback

## ECO-110

(error) Failed to initialize the bus buffer arrays for reason: %s.

**Description**

You receive this error message when the initialization of bus buffer arrays failed for the reason listed.

**What Next**

Please check the failed reason listed.

## ECO-111

(error) Failed to create bus buffer arrays in database for %s.

**Description**

You receive this error message when command failed to create bus buffer arrays in data base for some reason.

**What Next**

Please check the failed error messages printed previous.

## ECO-112

(error) Failed to %s for reason: %s.

**Description**

You receive this error message when command failed to do the operation in database for the reason listed

**What Next**

Please check the failed reason listed.

## ECO-113

(error) Failed to commit the changes of %s.

**Description**

You receive this error message when command failed to commit the changes of specified object or operation in database.

Feedback

**What Next**

Please check the failed reason;

---

## ECO-114

(Warning) Not support the route shape %s which is %s.

**Description**

Only horizontal and vertical route shapes are supported.

**What Next**

Change the route shape to horizontal and vertical route shapes.

---

## ECO-115

(Error) Found %s with bus guides %s.

**Description**

Only one net of bus to be processed by this command must have valid bus guides. Valid bus guides are route shapes created by route_bus_guides without -exploration option.

**What Next**

Run route_bus_guides to create valid bus guides before this command.

---

## ECO-116

(error) The inverter number from the driver to load are not even.

**Description**

This error occurs when tool detects that it will insert non-even inverters for the path from the driver to the load or tool cannot make this number even.

**What Next**

Please check the bufferable route shapes and use proper options to make it possible to insert even number of inverters from the driver to loads.

---

## ECO-117

(error) Cannot remove pattern %s, since it is referenced by %s.

Feedback

**Description**

The pattern cannot be removed because it is referenced by other objects.

**What Next**

Remove objects referencing this pattern before removing the pattern.

---

## ECO-118

(warning) The target location %s of clock bit net %s has conflit with the target location %s of one other clock bit net %s. Will skip adjusting the location of this clock bit net %s.

**Description**

You receive this warning message when trying to connect two clock bit nets to the same buffer of buffer array.

**What Next**

Please change the clock bit location to avoid conflict.

---

## ECO-119

(warning) Failed to derive the bus guide capacity. Set used capcity to be default value.

**Description**

You receive this warning message when command failed to calcluate the bus guide capacity from route width and buffer array pattern size.

---

## ECO-120

(warning) Set bus trunk width of bundle %s to be smaller than the minimal valid value. So the routes created are just for exploration and cannot be used to add bus buffer arrays.

**Description**

You receive this warning message when command sets small bus trunk width. Then the routes created are just for exploration and cannot be used to add bus buffer arrays.

---

## ECO-121

(error) Failed to do cell array snapping (cell location %s).

**Description**

You receive this error message when command failed to do cell array snapping.

Feedback

## ECO-122

(error) Failed to %s for reason: %s.

**Description**

You receive this error message when command failed to do the operation in database for the reason listed

**What Next**

Please check the failed reason listed.

## ECO-123

(warning) The attribute 'is_spare_cell' of programmble spare cell '%s' has not been set with true.

**Description**

You receive this warning message when command detects a programmable spare cell but its 'is_spare_cell' attribute is false not true. Then it will not be treated as a spare cell by ECO freeze-silicon commands.

**What Next**

Please set this cell's 'is_spare_cell' attribute to be true if it is a programmable spare cell.

## ECO-124

(warning) The attribute 'is_fs_eco_add' of ECO cell '%s' (eco_change_status is '%s') has not been set with true. It may be created by editing command not in Freeze-Silicon mode.

**Description**

You receive this warning message when command detects an ECO cell in Freeze-Silicon flow but its 'is_fs_eco_add' attribute is false not true. This ECO cell may be created by editing command not in Freeze-Silicon mode. Then it will not be treated as ECO cell by ECO Freeze-Silicon commands.

**What Next**

Please set this cell's 'is_fs_eco_add' attribute to be true if it is an ECO cell created in Freeze-Silicon flow. In Freeze-Silicon flow, the editing commands should be used in Freeze-Silicon mode.

Feedback

## ECO-125

(warning) Skip to connect pg net for the cell %s since no pg net found in its hierarchy.

### Description

This message is shown when there is no pg net in the hierarchy of the cell. After spare cell mapping, the engine tries to connect pg net for the eco cell, but failed since there is no existing pg net in the hierarchy of the cell. The engine only do pg connection for existing pg net, and do not create any pg net for connection.

### What Next

Do pg connection after netlist change or create needed pg nets before spare cell mapping.

### See Also

- place_freeze_silicon
- map_freeze_silicon
- connect_pg_net

## ECO-126

(Error) Eco legalization failed for free site only.

### Description

You receive this error message when you attempt to use place_eco_cells to do free site only legalization, but failed because there is no enough free sites to place cells. With free site only legalization, legalized cells will be treated as fixed and cannot be moved, so there may be high fragmentation caused by many fixed cells. And this situation will make it difficult for the legalizer to pack in the cells and can result in high displacements and long runtimes. Both *free_site_only* mode and *minimum_physical_impact* will call free site only legalization.

### What Next

Try to use *allow_move_other_cells* legalization mode by option *legalize_mode*.

### See Also

- place_eco_cells

## ECO-127

(Error) Not support derive terminal shape for %s %s.

Feedback

**Description**

This command can only derive terminal shape for block pin.

**What Next**

Create terminal shape manually. Then run this command again.

---

## ECO-128

(Error) Failed to derive terminal shape for %s, because %s.

**Description**

This command can only derive terminal shape for a block pin based on a unique obvious wire end inside its block. And this command can only derive terminal shape of one block pin for each block.

**What Next**

Modify wires inside that block to have a unique obvious wire end or create terminal shape manually. Then run this command again.

---

## ECO-129

(Info) Not support %s %s of net %s, it will be ignored.

**Description**

This command can only derive connection based on detail routes (manhattan path) of the specified net. Other shapes will be kept but ignored during deriving.

**What Next**

If you want this command use this shape, please convert it to a manhattan path and re-run this command.

---

## ECO-130

(Error) Failed to create %s %s.

**Description**

Tool meets an issue when creating an object.

**What Next**

Check why the object cannot be created. For example, user can try to create this object manually and see whether there is an error.

Feedback

## ECO-131

(error) Meet issue at %s, because %s.

**Description**

Tool meets an issue at the reported object because the reported reason.

**What Next**

If this is not a influential problem, just ignore it. If this can be fixed, please fix it and re-run the command. Otherwise, please find commands support this object to process this command.

## ECO-132

(Error) Failed to connect disjoint routes for %s, because %s.

**Description**

This command failed to connect disjoint routes.

**What Next**

Read the reason reported by this command. Connect not supported routes manually. Then run this command again.

## ECO-133

(error) Fail to get reference lib cell of '%s'.

**Description**

You receive this error message when you attempt to get the reference lib cell of the instance but failed.

## ECO-134

(Error) All spare cells under the region: %s %s in the design have trial mapped eco cell, so no available spare cell for %d input non-trial mapped eco cells.

**Description**

This error message occurs when there is no any available spare cell under a region in the design. The spare cells are not available since they have been trial mapped to eco cells before or set as is_eco_fs_dont_use true.

Feedback

**What Next**

add the spare cell by either design import, verilog eco, netlist editing or add_spare_cells.

---

## ECO-135

(Error) All spare cells of lib cell %s under the region: %s %s in the design have trial mapped eco cell, so no available spare cell for %d input non-trial mapped eco cells.

**Description**

This error message occurs when there is no any available spare cell of a particular lib cell under a region in the design. The spare cells are not available since they have been trial mapped to eco cells before or set as is_eco_fs_dont_use true.

**What Next**

add the spare cell by either design import, verilog eco, netlist editing or add_spare_cells.

---

## ECO-136

(Error) Total %d %s eco cells under the region:%s %s, %d of them have trial mapped spare cell, but other %d eco cells have no enough available spare cell (total %d).

**Description**

This error message occurs when there is no enough spare cell of a particular lib cell under a region in the design.

**What Next**

add the spare cell by either design import, verilog eco, netlist editing or add_spare_cells.

---

## ECO-137

(Error) The eco_change_status attribute of cell %s is not size_cell.

**Description**

This message is shown by revert_cell_sizing when the cell has no eco_change_status attribute or its value is not size_cell.

**What Next**

Please remove it from the input list and run command again.

Feedback

## ECO-138

(warning) Under voltage area %s, the width %s of %s with same reference lib cell %s (psc_type_id is %s) is not a multiple of the smallest lib cell width %s with the same psc_type_id. This would cause unrecoverable white space left after eco cell mapping.

### Description

You receive this warning message when command detects the width of some cells is not a multiple of the smallest lib cell width which would be used for the unused white space recovery. This would cause some errors (such as ECO-051: unrecoverable white space) during the eco cell mapping in command place_freeze_silicon or map_freeze_silicon.

### What Next

Please try to add a programmable spare cell (PSC) filler lib cell which has the same psc_type_id and a small with. The width of all the eco cells and programmable spare cell with same psc_type_id can be divisible by this small width.

## ECO-140

(Error) Failed to create logic equivalence change list for cell '%s' because its library cell '%s' has invalid function string.

### Description

This message is shown by create_freeze_silicon_leq_change_list when library cell of the cell to map has function string that can not be parsed successfully.

### What Next

Please check function string of the library cell.

### See Also

- create_freeze_silicon_leq_change_list

## ECO-141

(error) Failed to create logic equivalence change list for cell '%s' because we cannot get function string for its library cell '%s'.

### Description

This message is shown by create_freeze_silicon_leq_change_list when it's failed to get function string of the library cell.

Feedback

**What Next**

Please check function string of the library cell.

**See Also**

- create_freeze_silicon_leq_change_list

---

## ECO-142

(error) Failed to create logic equivalence change list for cell '%s' because its logic gate level exceeds max limitation.

**Description**

The maximum logic gate level create_freeze_silicon_leq_change_list supports is 2. We don't create logic equivalence change list for the cell whose logic gate level is greater than 2.

**See Also**

- create_freeze_silicon_leq_change_list

---

## ECO-143

(error) Failed to create logic equivalence change list for cell '%s' because it has multiple output pins.

**Description**

Multiple output pin is not supported by create_freeze_silicon_leq_change_list. The command will skip ECO cells that have more than one output pin.

**See Also**

- create_freeze_silicon_leq_change_list

---

## ECO-144

(error) Spare cell (%d) in the design is less than ECO cell (%d).

**Description**

Number of spare cell is not enough to perform logical equivalence change for all ECO cells in the design.

Feedback

**What Next**

Revert ECO changes.

---

## ECO-145

(Error) Failed to create logic equivalence change list for cell '%s' in max logic gate level.

**Description**

The maximum logic gate level create_freeze_silicon_leq_change_list supports is 2. This message means either spare cells is not enough to create logic equivalence change list for the cell or spare cell is enough but logic gate level of the replacement is greater than 2.

**See Also**

- create_freeze_silicon_leq_change_list

---

## ECO-146

(error) The programmable spare cells %s in one mapping pair are not of same direction (horizontal or vertical)

**Description**

When specifiy multiple programmable spare cells for one mapping pair, these programmable spare cells must be of same direction.

**What Next**

Specify correct programmable spare cells for this mapping pair.

**See Also**

- map_freeze_silicon

---

## ECO-147

(error) The lib cell %s with psc_type_id %s has a different height %s compared with the other same psc_type_id lib cells (height is %s).

**Description**

The command detects the same psc_type_id lib cells have different height sizes and this is not allowed. All the same psc_type_id lib cells shoud have the same height size.

Feedback

### What Next

Please set correct psc_type_id for the lib cells (the same type lib cells must have the same height size). If there already are PSC mapping rules set for this type, please clear all the mapping rules and set again. Finally re-try this command.

### See Also

- set_attribute
- remove_programmable_spare_cell_mapping_rule
- set_programmable_spare_cell_mapping_rule

---

## ECO-151

(warning) There is no available spare cell around the buffering location (%s, layer %d) within the maximum distance %s.

### Description

This message is shown when the tool could not find any available spare cell around the target buffering location within the maximum distance. And this would cause the subsequent spare cell mapping to be difficult to find a proper and nearby spare cell for mapping this new buffer eco cell in freeze-silicon flow.

### What Next

Please be careful about the spare cell mapping of this new buffer eco cell in the subsequent flow. Or undo the buffering of this location and choose a new buffering location which has available spare cell around.

### See Also

- add_buffer_on_route

---

## ECO-152

(warning) Skip net %s to find driver or load pins for reason: %s.

### Description

You receive this warning message when command failed the listed reason.

### What Next

Check the reason listed and specify the only driver or load for the net by the command set_bus_guide_connection.

Feedback

## ECO-153

(error) Failed to find bundles for bound %s.

**Description**

You receive this error message when command failed the listed reason.

**What Next**

Create bundle of the nets connected the cells attached to the bounds. For a bound, the command place_pipeline_bounds needs two bundles, one for input, and one for output.

**See Also**

- create_bundle

## ECO-154

(error) Failed to find driver pin or load pin for bundle %s.

**Description**

You receive this error message when command failed the listed reason.

**What Next**

Check the net connection of bundle and make sure the nets have driver and loads, and the owner cell of driver or load pins are placed.

## ECO-155

(error) %s.

**Description**

You receive this error message when command failed the listed reason.

## ECO-156

(error) Cell %s of the pin %s is not placed!

**Description**

You receive this error message when command failed the listed reason.

**What Next**

Make sure the cell is placed.

Feedback

## ECO-157

(error) The current design (%s) is not same as the design (%s) with pipeline bound info.

**Description**

This error message occurs when the current design is not same as the design with pipeline bound info.

**What Next**

User may switch to the design with pipeline bound info and clean up memory data with the command remove_pipeline_bounds_info. And then switch to the working design and place pipeline bounds with the command place_pipeline_bounds for this design.

## ECO-158

(error) Failed to find pipeline bound path created by place_pipeline_bounds from input %s.

**Description**

You receive this error message when command failed the listed reason.

**What Next**

Check the input.

## ECO-159

(error) Failed to %s for %d pipeline bound paths.

**Description**

You receive this error message when command failed to create bus guides or place bounds for some of pipeline bound paths.

**What Next**

Please check the failed reason, make sure there is enough resource to create the bus guides.

## ECO-160

(error) One move bound should connect to two bundles, one for input and one for output. But the bound %s is not, bundle number: input (%d), output(%d).

Feedback

**Description**

You receive this error message when command failed the listed reason.

**What Next**

For a bound, the command place_pipeline_bounds needs two bundles, one for input, and one for output. Check the bundle connection for cells attached to the bound.

---

## ECO-161

(error) The bundle %s connects more than two bounds.

**Description**

You receive this error message when command failed the listed reason.

**What Next**

The bundle should connect two bounds. Check the bundle connection to bounds.

---

## ECO-162

(error) Unused white space (length %s, height %s) at location {%s} cannot be recovered to be programmable spare cell(s).

**Description**

You receive this error message when there is no valid programmable spare cell (PSC) filler lib cell found for recovering the unused white space left by PSC cells after PSC mapping or PSC recycling.

**What Next**

If this message is issued when the option *-lib_cells_for_filler_recovery* is specified, please specify small enough and compatible PSC filler lib cells in the command option. Otherwise, add reference libraries which contain small enough and compatible PSC filler lib cells and they can be used to recover the empty space.

**See Also**

- place_freeze_silicon
- map_freeze_silicon
- recycle_programmable_spare_cells

Feedback

## ECO-163

(error) Mapping engine failed to find enough matched programmable spare cells (PSC) for mapping %d eco cells under voltage area %s.

### Description

You receive this error message when the matched programmable spare cells (PSC) are not enough for mapping eco cells under the voltage area. The programmable spare cells are less than target eco cells or some programmable spare cells are not mappable for the target eco cells.

### What Next

Consider re-designing the ECO to make sure there are enough available PSC resources.

### See Also

- place_freeze_silicon

## ECO-164

(error) Failed to place %d eco cells to the matched programmable spare cells (PSC) under voltage area %s.

### Description

You receive this error message when command failed to place the eco cells to the matched programmable spare cells.

### See Also

- place_freeze_silicon

## ECO-165

(error) Failed to create bus guides for %d pipeline bound paths, the driver bundles are: { %s }.

### Description

You receive this error message when command failed to create bus guides for some of the pipeline bound paths. The driver bundle is the first bundle on a pipeline bound path.

### What Next

Please check the failed reason and setting, make sure there is enough resource to create the bus guides for these pipeline bound paths.

Feedback

## ECO-166

(error) The inner length of %s is greater than even distance: %s.

**Description**

You receive this error message when the inner length is greater than even distance..

**What Next**

Please check the attribute inner_length of driver bundle or load bundle. And set the proper value.

**See Also**

- get_attribute

## ECO-167

(error) There are multiple %s as load for bound %s.

**Description**

You receive this error message when command failed the listed reason.

**What Next**

For pipeline path, one bound should only have one load, either bound or pin.

## ECO-168

(error) Failed to find load for bound %s.

**Description**

You receive this error message when command failed the listed reason.

**What Next**

For pipeline path, one bound should only have one load, either bound or pin.

## ECO-169

(error) %d eco cells (%s) are unmapped due to: %s.

**Description**

You receive this error message when detect eco cells are unmapped due to some reason.

Feedback

### What Next

Get the unmapped eco cells by the listed lib cells and check the failure reason.You can rerun place_freeze_silicon for these eco cells after fixing the issues.

### See Also

- place_freeze_silicon

---

## ECO-170

(warning) Skip to create_bus_guides for %d bounds { %s }.

### Description

You receive this error message when command failed to find valid bundles for bounds. Such bounds are skipped to create_bus_guides.

### What Next

For a bound, the command route_bus_guides needs two bundles, one for input, and one for output. Check the bundle connection for cells attached to the bound.

---

## ECO-171

(warning) The width %s of library cell %s (psc_type_id is %d) is not a multiple of the smallest lib cell width %s with the same psc_type_id or compatible psc_type_id. This would cause unrecoverable white space left after eco cell mapping.

### Description

You receive this warning message when command detects the width of one library cell with PSC type is not a multiple of the smallest library cell width with the same PSC type. This would cause unexpected issues during the PSC mapping, such as ECO-051: unrecoverable white space.

### What Next

Please try to add a small programmable spare cell (PSC) filler lib cell which has the same psc_type_id or compatible psc_type_id. The width of all the eco cells and programmable spare cell with same psc_type_id or compatible psc_type_id can be divisible by this small width.

Feedback

## ECO-172

(warning) The orientation (%s) set by mapping rule for PSC type id %d is none of the allowable orientation (%s) of library cell %s. Use the orientation set by mapping rule for this library cell instead.

### Description

You receive this warning message when command detects the orientation set by mapping rule is none of the allowable orientations of the library cell. Command will use the orientations set by maping rule for this library cell instead.

### What Next

Please make sure the orientation set by mapping rule is correct.

### See Also

- set_programmable_spare_cell_mapping_rule

## ECO-173

(error) The window size %s is smaller than the library cell size.

### Description

You receive this error message when the specified repetitive window size for add_spare_cells is smaller than the size of library cells.

### What Next

Specify window size which is larger than the max size of the specified library cells.

## ECO-174

(info) Total %d pipeline bound paths based on the input are skipped since the status is not qualified to apply the command.

### Description

You receive this error message when the path status is not qualified to apply the command.

### What Next

Please check the input to figure out the skipped paths with report_pipeline_bound_paths and find the reason.

Feedback

## ECO-175

(error) Failed to derive PSC mapping rules for all the %d input library cells.

### Description

You receive this error message when all the input lib cells are failed to derive PSC mapping rule. Please check the previous failure messages.

### What Next

Please check the previous failure messages and fix the issues reported.

### See Also

- derive_programmable_spare_cell_mapping_rules

## ECO-176

(info) Totally %d %s lib cells are failed to derive PSC mapping rules due to some issues reported in previous command log.

### Description

This info shows how many ECO/PSC library cells are failed to derive PSC mapping rules. Please check the previous command error messages.

### What Next

Please check the previous command error messages and fix the issues reported.

### See Also

- derive_programmable_spare_cell_mapping_rules

## ECO-177

(error) The width %s of library cell %s is not a multiple of the unit width %s and this would cause unexpected PSC mapping issues. Skipped the rule deriving for this library cell.

### Description

You receive this error message when command detects the width of one library cell is not a multiple of the unit width. The unit width is specified by command option or it is the smallest lib cell widthj in the input list. This would cause unexpected issues during the PSC mapping, such as ECO-051: unrecoverable white space.

Feedback

### What Next

Please try to specify a proper unit width or add a small programmable spare cell (PSC) filler lib cell to the input list. The width of all the input lib cells can be divisible by this small width.

### See Also

- derive_programmable_spare_cell_mapping_rules

---

## ECO-178

(error) Failed to check the physical information for library cell %s.

### Description

You receive this error message when command failed to check the physical information for this library cell which may have no physical data or have some error in the physical data, such as no frame view nor design view.

So the rule deriving for this library cell is failed.

### What Next

Please double check whether the specified library cell has physical information and it is valid for deriving PSC mapping rules.

### See Also

- derive_programmable_spare_cell_mapping_rules

---

## ECO-179

(Warning) Failed to derive voltage area at {%s}.

### Description

You receive this error message when command failed to derive voltage area at the reported location. This may cause tool fail in some voltage area specific features, e.g add_buffer_on_route may fail to use correct lib cell or to insert into correct hierarchy.

### What Next

Check the voltage area of reported location

---

## ECO-180

(warning) No valid bufferable locations can be found for the %s.

Feedback

**Description**

You receive this warning message when command failed to find any valid bufferable location for the specified bus.

It may be caused by: 1. The bus guide path is too short for buffering comparing with the repeat distance. 2. Could not find bufferable location along bus guide path, such as the bus guide path is too close to the blockages/macros.

**What Next**

Please re-specify suitable option value or modify the bus guide path if you still want to buffer the bus by this command.

---

## ECO-181

(error) Failed to derive PSC mapping rules for the following reason: %s.

**Description**

You receive this error message when command failed to derive the PSC mapping rule for the specified reason.

**What Next**

Please check the failed reason and fix the issues reported.

**See Also**

- derive_programmable_spare_cell_mapping_rules
- place_freeze_silicon

---

## ECO-182

(warning) There is no route on first net of driver bundle: %s.

**Description**

You receive this warning message when there is no route on first net of driver bundle. It will cause issue when you run the commmand place_pipeline_bounds for this pipeline bound path.

When you run route_bus_guides -bounds -by_route_file, the route is from the input file, however, the owner net of route for the pipeline bound path is not the first net of driver bundle. In long-hop flow, the command route_bus_guides keeps route with the first net of driver bundle and the command place_pipeline_bounds honors the route from first net of driver bundle to do bound placement.

Feedback

**What Next**

Please make sure the the owner net of route is first net of driver bundle for each pipeline bound path and run route_bus_guides again. You can change the owner net of route by modifying the input route file directly or by set_attribute and then dump out the route to file by write_routes.

```
For example,
set_attribute [get_shapes PATH*] owner [get_nets bus_12_7_net[0]]
set_attribute [get_vias VIA*] owner [get_nets bus_12_7_net[0]]
```

---

## ECO-183

(warning) The related pipeline bound path info will be clear since %s will be removed.

**Description**

You receive this warning message when command will remove bound or bundle. The related pipeline bound path info will be clear to avoid tool crash due to dirty data.

**What Next**

Please run route_bus_guides -bounds to re-generate pipeline bound path if needed.

---

## ECO-184

(info) Complete to check derived bound connection and pipeline bound paths before routing bus guide.

**Description**

You receive this message when route_bus_guides -check_only is used.

**What Next**

Please check the information and fix the issue if there is any. Otherwise, the command route_bus_guides will skip to build pipeline bound path and route for invalid bounds. After all issues are fixed, please run route_bus_guides without -check_only to create bus guides.

---

## ECO-185

(warning) bundle %s has no bus buffer array added, so skip the array writing for it.

**Description**

You receive this message when command finds the target bundle has no bus buffer array added, so this bundle will be ignored and skipped.

Feedback

### What Next

please make sure the target bundle has bus buffer array added, otherwise it will be ignored and skipped.

---

## ECO-186

(warning) There is no location to satisfy %s constraint for eco cell %s.

### Description

You receive this message when command cannot find the location to satisfy constraint.

### What Next

Please check design utilization and specified density threshold for density constraint. Please check the voltage area of eco cell for voltage area constraint.

---

## ECO-187

(warning) failed to find a proper reference lib cell (%s) for swapping the reference of ECO cell %s, which would have pin track alignment violation after PSC mapping.

### Description

You receive this warning message when there is no proper reference lib cell found for swapping the reference of ECO cell and this ECO cell would have pin track alignment violation after PSC mapping.

Normally the reference lib cell of ECO cell has a cell group (sharing timing data) defined and one proper lib cell in cell group can be selected to swap with the current reference of ECO cell considering the pin track alignment rule.

### What Next

Please check the cell group of ECO cell's reference lib cell and select a proper lib cell in cell group considering the pin track alignment rule. Finally use change_link command to change the ECO cell's reference to be the proper one.

### See Also

- place_freeze_silicon
- change_link

---

## ECO-188

(warning) failed to swap the reference lib cell from '%s' to '%s' on ECO cell %s.

Feedback

**Description**

You receive this warning message when command failed to swap the reference lib cell of ECO cell, which would have pin track alignment violation after PSC mapping.

Normally when the ECO cell has pin track alignment violation after PSC mapping, command would swap the reference lib cell of ECO cell to be proper one, which is selected from the same cell group.

**What Next**

Please check the ECO cell's reference lib cell and the new target reference. Make sure they are compatible for swapping. Finally use change_link command to change the ECO cell's reference lib cell.

**See Also**

- place_freeze_silicon
- change_link

---

## ECO-189

(error) The '%s' design '%s' has unmapped cells.

**Description**

The command "eco_netlist" should work on synthesised design.

You receive this error message when command finds that the golden or working design has unmapped cells.

**What Next**

The design should undergo synthesis. Make sure both the working and golden design are synthesised.

---

## ECO-190

(warning) %s for %d eco cells, the cells are {%s}

**Description**

You receive this warning message when failed to do reverting.

**What Next**

Please check the failed reason.

Feedback

**See Also**

- revert_eco_changes

---

## ECO-191

(warning) Additionally revert one eco cell (%s) due to overlap with input cell

**Description**

You receive this warning message when revert_eco_changes revert non input cell additionally.

**What Next**

After some input cell is reverted, if it is overlapped with some cell with eco_change_status eco_legalized, the command revert_eco_changes will do reclusively reverting for those overlapping eco cells until there is no overlapping eco cell.

**See Also**

- revert_eco_changes

---

## ECO-192

(warning) %s for %d eco cells, the cells are {%s}

**Description**

You receive this warning message when failed to do reverting.

**What Next**

Please check the failed reason.

**See Also**

- revert_eco_changes

---

## ECO-193

(warning) Do not support vertical design.

**Description**

You receive this warning message when the design is vertical.

Feedback

**See Also**

- report_eco_physical_changes

---

## ECO-194

(warning) Found non-Eco cells that are not legalized.

**Description**

You receive this warning message when there are non-Eco cells that not legalized in the design. And these cells are added to the collection with name ecoNonLegalCellsForSpaceCheck.

**See Also**

- report_cell_feasible_space

---

## ECO-195

(warning) Cutline {{%s} {%s}} is not vertical nor horizontal. Skipped.

**Description**

You receive this warning message when the cutline is not vertical nor horizontal. It will be skipped.

**See Also**

- add_group_repeaters

---

## ECO-196

(warning) Unable to find location to insert repeater groups.

**Description**

You receive this warning message when there is no valid locations based on user specified rule.

**See Also**

- add_group_repeaters

Feedback

## ECO-198

(error) Route has large open gap (> %s um) or multiple gaps.

### Description

You receive this warning message when route of the net(s) has large open gap, or multiple gaps.

### See Also

- add_group_repeaters

## ECO-199

(warning) failed to process the incomplete route segment %s of net %s for the reason: %s.

### Description

Command failed to process the target route for the specified reason. This route segment would be dangling or incomplete. If it is an incomplete route, this would cause the buffering failed.

### What Next

If buffering failed, please check the failure reason of this message and optimize the command options or net routes. If buffering successed, please ignore this warning message.

### See Also

- add_group_repeaters
- add_buffer_on_route

## ECO-200

(warning) failed to process the incomplete routes of net %s.

### Description

Command failed to process the incomplete routes of specified net. The buffering for this net would be failed.

### What Next

Please check the related error or warning messages printed for this net and follow the guidance of these messages.

Feedback

**See Also**

- add_group_repeaters
- add_buffer_on_route

---

## ECO-201

(error) Failed to create diode cell for load %s: %s.

**Description**

You receive this error message when command failed to create the diode cell for the specified load. Please check the failure reason listed.

**What Next**

Please check the listed failure reason of this load.

---

## ECO-202

(warning) Failed to place the %d diode cells to the proper location.

**Description**

You receive this warning message when command failed to place new created diode cells to the proper location. And these diode cells will be placed at the initial location (nearby or overlap with the target loads).

**What Next**

Please place these diode cells to the proper location again.

---

## ECO-203

(warning) Failed to place the diode cell %s of load %s to the proper location.

**Description**

You receive this warning message when command failed to place the created diode cell to the proper location. And this diode cell will be placed at the initial location (overlap with the target load).

**What Next**

Please place this diode cell to the proper location again.

Feedback

## ECO-204

(info) No valid location to insert repeater groups within 20 um of location {%s}.

### Description

You receive this warning message when there is no valid locations to place repeater group with the specified cutline.

### See Also

- add_group_repeaters

## ECO-205

(warning) Failed to find enough place for route cutting on path %s around coordinate %s.

### Description

You receive this warning message when failed to find enough place for route cutting in command add_group_repeaters when honor option -layer_cutting_spacing.

### What Next

Please reset layer cutting spacing.

### See Also

- add_group_repeaters

## ECO-206

(error) %s.

### Description

You receive this error message when command failed the listed reason.

## ECO-207

(error) Failed to derive long hop guidances for %d bundles {%s}.

### Description

You receive this error message when command failed to derive long hop guidances for some bundles.

Feedback

**What Next**

Please check the failed reason.

---

## ECO-208

(error) Failed to open file %s for write.

**Description**

You receive this error message when command failed to open file for write.

**What Next**

Please check the failed reason.

---

## ECO-209

(error) %s for bundle %s.

**Description**

You receive this error message when command failed the listed reason.

---

## ECO-210

(error) %s.

**Description**

You receive this error message when command failed the listed reason.

---

## ECO-211

(error) Driver list is empty or load list is empty or driver list and load list have different size.

**Description**

You receive this error message when cannot get drivers or loads from UTC or drivers and loads got from UTC have different size.

**What Next**

Check the start and the end node of the topology edge

Feedback

## ECO-212

(error) The pattern (%s) of the topology repeater (%s) not found.

**Description**

You receive this error message when the pattern of a topology repeater is not existed.

**What Next**

Check the pattern of the topology repeater

## ECO-213

(error) The lib cell (%s) of the pattern (%s) not found.

**Description**

You receive this error message when the lib cell of a cell array pattern is not existed.

**What Next**

Check the lib cell of the cell array pattern

## ECO-214

(error) The object of the topology node belonging to the topology edge (%s) is neither a pin nor a port.

**Description**

You receive this error message when an object of a topology node is neither a pin nor a port.

**What Next**

Check the objects field of the topology node.

## ECO-215

(error) The object of the topology node belonging to the topology edge (%s) is neither a driver nor a load.

**Description**

You receive this error message when an object of a topology node is neither a driver nor a load.

Feedback

### What Next

Check the objects field of the topology node.

---

## ECO-216

(error) Failed to set mask_shift attribute for cell %s.

### Description

This message is shown when tool failed to set mask_shift attribute for the specified cell which needs update the color shifting setting.

### What Next

Check whether the "mask_shift" attribute of this cell is setable or not.

### See Also

- place_freeze_silicon

---

## ECO-217

(error) Failed to find color shifting layer by name %s.

### Description

This message is shown when tool failed to find color shifting layer by name set via the app option "eco.freeze_silicon.color_shift_layers".

### What Next

Please check the layer name saved in the app option "eco.freeze_silicon.color_shift_layers" and set with correct color shifting layer names.

### See Also

- place_freeze_silicon

---

## ECO-218

(error) There are %d eco cells whose power intention need to be updated.

### Description

You receive this error message when there are some eco cells whose power intention need to be updated before the command place_eco_cells or place_freeze_silicon. These cells are added by "add_buffer_on_route -allow_physical_feedthrough_buffer", but their

Feedback

power intention are not updated by the command set_eco_power_intention. So the place command may place them to wrong locations if their power intention are not correct. These cells are saved in the collection $ecoAborPhysicalFeedthroughCells.

**What Next**

Run set_eco_power_intention for the cells in the collection $ecoAborPhysicalFeedthroughCells, and then run the place command again.

---

## ECO-219

(error) The cell %s is not associated with any topology repeaters.

**Description**

You receive this error message when a cell is on the netlist but not associated with any topology repeaters.

**What Next**

Associate the cell with a proper topology repeater before run place_interconnect_repeaters.

---

## ECO-220

(warning) Failed to connect to route for %d object: { %s }.

**Description**

You receive this warning message when command failed to connect to route for driver or load.

**What Next**

Please check the pin shape of object and route shapes. The object as driver or load can be port, macro pin and placed cell. For details please find message ECO-320.

**See Also**

- place_group_repeaters

---

## ECO-221

(Error) Failed to link %s block: %s.

Feedback

### Description

This message is shown by eco_netlist when failed to link the target block and this would cause wrong diff result.

### What Next

Check the link error messages printed and resolve the link issue before rerun.

### See Also

- eco_netlist

---

## ECO-222

(warning) net %s still drives %d loads which is larger than the user specified maximum fanout number %d.

### Description

You receive this warning message when command failed to add enough buffers for some reason (such as the expected location is not bufferable or the load number of some route segments is already equal or smaller than the max fanout number).

### What Next

Please check the net route segments or other messages printed.

### See Also

- split_fanout

---

## ECO-223

(Error) size_cell is skipped due to dont_use attribute on lib cell %s.

### Description

This message is shown when new library cell specified for size_cell has dont_use attribute.

### What Next

Check lib cell status and choose another lib cell for size_cell.

Feedback

## ECO-224

(Error) The option -repeater_groups cannot be used when the app option eco.placement.eco_enable_pipeline_register_placer is false.

### Description

This message is shown when the option -repeater_groups is used but the app option eco.placement.eco_enable_pipeline_register_placer is false.

### What Next

Please set_app_options -name eco.placement.eco_enable_pipeline_register_placer -value true if you want to use -repeater_groups.

### See Also

- place_group_repeaters

## ECO-225

(Error) Some nets (%ld out of %ld) do not have a net group ID. Find the collection of nets without group ID in %s.

### Description

This message is shown when add_gorup_repeaters -honor_user_group_id option is used, but some nets do not have net group ID.

### What Next

Run create_group_repeaters_guidance to create net group IDs, or set net attribute eco_net_group_id manually.

### See Also

- create_group_repeaters_guidance

## ECO-226

(Error) Failed to place cell %s\n.

### Description

This message is shown when implement_topology_repeaters fails to place a cell.

Feedback

### What Next

Check if the design has incomplete routes. If the design has incomplete routes, set app option eco.placement.itr_max_incomplete_route_distance.

---

## ECO-227

(warning) Failed to place %d repeaters for %d drivers in collection (%s), and failed cells are in collection (%s).

### Description

You receive this warning message when command failed to place some repeaters.

### What Next

Please check the failed repeaters by get_cells. Note that the driver can be port, macro pin and placed cell. So please check the failed drivers by query_objects, get_ports, get_pins and get_cells.

### See Also

- place_group_repeaters

---

## ECO-228

(warning) Some repeaters from path driver (%s) have same repeater group id. The cells belonging to the path are {%s}.

### Description

You receive this warning message when command skipped to place some repeaters.

### What Next

Please check the repeater group id for the cells and make sure the group id of cells on one path are different.

### See Also

- place_group_repeaters
- report_repeater_groups
- set_repeater_group

Feedback

## ECO-229

(warning) Failed to find intersection of cutline %s and route for cell {%s}.

**Description**

You receive this warning message when command failed to find intersection of cutline and route with specific cell.

**See Also**

- place_group_repeaters

## ECO-230

(error) Lib cell %s names of all registers in the same topology plan must be the same.

**Description**

You receive this error message when different lib cell input/output names of registers are found in the same topology plan.

## ECO-231

(error) The repeater group (%d) %s.

**Description**

You receive this error message when the repeater group has no cutline or invalid cutline.

**What Next**

Please set correct cutline for this group by set_repeater_group.

**See Also**

- place_group_repeaters
- set_repeater_group

## ECO-232

(error) Cutline {{%s} {%s}} is invalid.

**Description**

You receive this warning message when the cutline is not vertical nor horizontal. The valid format is {{x1 y} {x2 y}} or {{x y1} {x y2}}.

Feedback

### What Next

Please set correct cutline.

### See Also

- set_repeater_group

---

## ECO-233

(error) The cells in collection (%s) are invalid since they have no target input pin name (%s) and output pin name (%s).

### Description

You receive this warning message when the cell has no target pin name.

### What Next

Please double check the cells in collection. You can specify input pin name and output pin name by the option -lib_cell_input and -lib_cell_output if needed, or filter out invalid cells from input. The default value for -lib_cell_input is D. The default value for -lib_cell_output is Q.

### See Also

- place_group_repeaters
- set_repeater_group

---

## ECO-234

(warning) There is no repeater to be placed for the path driver (%s).

### Description

You receive this warning message when there is no repeater to be placed for the path driver.

### See Also

- place_group_repeaters

---

## ECO-235

(warning) There is no route on driver net (%s).

Feedback

**Description**

You receive this warning message when there is no route on driver net.

**What Next**

Please make sure there is route on driver net.

**See Also**

- place_group_repeaters

---

## ECO-236

(warning) There is no target load for path driver (%s).

**Description**

You receive this warning message when there is no target load for the path.

**What Next**

Please make sure there is target load (i.e. port, macro pin, placed cell) for the path.

**See Also**

- place_group_repeaters

---

## ECO-237

(warning) Some cell to be placed has no load for path driver (%s).

**Description**

You receive this warning message when there is some cell to be placed as end fanout since the cell has no load.

**What Next**

Please make sure the cell to be placed has target load (i.e. port, macro pin, placed cell).

**See Also**

- place_group_repeaters

---

## ECO-238

(warning) Failed to update power intention for %d new added buffers/inverters.

Feedback

### Description

You receive this warning message when command failed to update the power intention for the new added buffers or inverters.

### What Next

Please check the new added buffers/inverters and update the powner intention by manullay calling set_eco_powner_intention command.

### See Also

- add_buffer_on_route
- set_eco_power_intention

---

## ECO-239

(info) %s.

### See Also

- place_group_repeaters

---

## ECO-240

(error) The option -cells cannot be used when the app option eco.placement.eco_enable_virtual_connection is true.

### Description

You receive this warning message when the option -cells is used with the app option eco.placement.eco_enable_virtual_connection as true.

### What Next

You can set the app option eco.placement.eco_enable_virtual_connection as false if you need to specify -cells. Or you can specify -repeater_groups if you need to use virtual connection.

### See Also

- place_group_repeaters
- set_repeater_group

Feedback

## ECO-241

(warning) The distance based placement does not support multi-fanout path driver (%s).

### Description

You receive this warning message when the command applies distance based placement for multi-fanout path driver.

### What Next

Please apply cutline based placement for the path driver. You need to specify repeater group by set_repeater_group first. And then use place_group_repeaters -repeater_groups to apply cutline based placement.

### See Also

- place_group_repeaters
- set_repeater_group

## ECO-242

(warning) The cell (%s) has no driver or load (number of driver: %d, number of load: %d) based on connection from lib pin names (%s).

### Description

You receive this warning message when the command cannot identify driver or load of the cell.

### What Next

Please check the cell connectivity based on real connection from the specified pin names. Please make sure the cell has both driver and load.

### See Also

- place_group_repeaters
- set_repeater_group
- report_repeater_groups

## ECO-243

(error) The cells in collection (%s) are invalid since they have no driver or load.

Feedback

**Description**

You receive this error message when some cells to be placed have no driver or load.

**What Next**

Please check ECO-242 for such cells. Make sure the cell to be placed has both driver and load. Or remove such cell from input.

**See Also**

- place_group_repeaters
- set_repeater_group

---

## ECO-244

(warning) The %s %s has no shape.

**Description**

You receive this warning message when the top port or macro pin has no shape.

**What Next**

Please make sure the top port or macro pin has shape.

**See Also**

- place_group_repeaters

---

## ECO-245

(error) Failed to flip ECO cell %s with orientation %s at placed location.

**Description**

This message is shown when tool failed to flip target ECO cell with speicified orientation. The ECO cell needs to be set with legal orientation at the location.

**What Next**

Check whether the orientation of this ECO cell is legal or not. If current orientation is not legal, may need manually set a legal orientation instead.

**See Also**

- place_freeze_silicon

Feedback

## ECO-246

(warning) Skip the path driver (%s) since the driver or load for the path has no shape.

### Description

You receive this warning message when the driver or load has no shape for path driver. This path driver will be skipped.

### What Next

Please double check ECO-244 and make sure the top port or macro pin has shape.

### See Also

- place_group_repeaters

## ECO-247

(error) Failed to add buffer for %d (out of %d) nets: { %s }

### Description

You receive this error message when command failed to add buffer for some input nets which are listed in the error message.

### What Next

Please double check the other reported error messages to find out the root cause.

### See Also

- add_buffer_on_route

## ECO-248

(error) Bundle %s: Some nets have different routing topology or not routed. Nets in the bundle are grouped into %ld groups based on their routes. Net group IDs are %s. Use net attribute eco_net_group_id to query net groups (e.g. get_nets -filter eco_net_group_id==0).

### Description

You receive this error message when the nets of specified bundle are not routed together.

### What Next

Please correct bundle routing, or use -nets option.

Feedback

## ECO-249

(error) Error of UTC transition for %s:\n%s

**Description**

You receive this error message when UTC tansition fails to provide some necessary information for placement or insertion.

**What Next**

Please double check you design based on the error informatin first

## ECO-250

(error) %d of %lu nets don't have route: { %s}

**Description**

You receive this error message when some, but not all, nets don't have route.

**What Next**

Please double check your design to see if you attach routes to all nets as expected.

**See Also**

- route_custom
- create_trunk

## ECO-251

(info) Place group repeaters successfully.

**Description**

You receive this info message when all input cells are placed successfully.

**See Also**

- place_group_repeaters

## ECO-252

(warning) Failed to place repeaters for repeater path which driver is %s.

Feedback

### Description

You receive this warning message when the command failed to place repeaters for one repeater path.

### See Also

- place_group_repeaters

---

## ECO-253

(warning) The cutline information %s for cell %s is not on valid route.

### Description

You receive this warning message when the cutline for cell is not on valid route.

### What Next

Please check the cutline and route information related to the cell.

### See Also

- place_group_repeaters
- set_repeater_group

---

## ECO-254

(warning) Failed to cut route for cell %s.

### Description

You receive this warning message when the command failed to cut route for cell.

### See Also

- place_group_repeaters

---

## ECO-255

(warning) gap (%s) between %s (%s) and nearest route is larger than placing distance (%s).

### Description

You receive this warning message when the route to nearest pin distance is larger than placing distance for driver or load.

Feedback

**See Also**

- place_group_repeaters

---

## ECO-256

(warning) The %s distance (%s) specified by command option is shorter than the gap (%s) between pin and route for %s (%s).

**Description**

You receive this warning message when the -first_distance or -last_distance specified is shorter than the gap between pin and route for driver or load.

**What Next**

Please check the option -first_distance, -last_distance and the route completeness.

**See Also**

- place_group_repeaters

---

## ECO-257

(warning) Cannot find cutline information for cell %s.

**Description**

You receive this warning message when the command cannot find cutline information for cell.

**What Next**

Please check repeater group info of cell.

**See Also**

- place_group_repeaters
- set_repeater_group
- report_repeater_groups

---

## ECO-258

(info) The app option about virtual connection is turned on. Please make sure the cells in repeater group keep in order.

Feedback

### Description

You receive this info message when the app option eco.placement.eco_enable_virtual_connection is true and user specifies place_group_repeaters -repeater_groups.

### See Also

- place_group_repeaters

---

## ECO-259

(error) The number (%d) of cells from repeater group (%d) is not consistent with the number (%d) of drivers from driver group (%s).

### Description

You receive this warning message when the number of cells is not consistent with the number of drivers.

### What Next

Please check the repeater group and make sure the number of drivers from driver group and cells from repeater group are same.

### See Also

- place_group_repeaters
- set_repeater_group
- report_repeater_groups

---

## ECO-260

(warning) The number of cells from repeater group (%d) is not consistent with the number of loads from load group (%s).

### Description

You receive this warning message when the number of cells is not consistent with the number of loads.

### What Next

Please check the repeater group and make sure the number of loads from load group and cells from repeater group are same.

Feedback

**See Also**

- place_group_repeaters
- set_repeater_group
- report_repeater_groups

---

## ECO-261

(warning) The number of path loads is less than the number of cells in group (%d), so skip these path loads.

**Description**

You receive this warning message when the number of path loads is less than the number of cells in group.

**What Next**

Please check the repeater group and make sure the number of path loads is an integral multiple of the number of cells in group.

**See Also**

- place_group_repeaters
- set_repeater_group
- report_repeater_groups

---

## ECO-262

(warning) The number of path loads is not an integral multiple of the number of cells in group (%d), so use path loads in order.

**Description**

You receive this warning message when the number of path loads is not an integral multiple of the number of cells in group.

**See Also**

- place_group_repeaters
- set_repeater_group
- report_repeater_groups

Feedback

## ECO-263

(warning) Net %s is unbufferable in the voltage area region {%s} (VA shape %s) of voltage area %s, skip the buffering in this region.

### Description

You receive this warning message when the command detects the target voltage area region is unbufferable for the net to be buffered. Command will skip the buffering in this region for the net.

### What Next

Please check the MV information of this region for the net to be buffered.

### See Also

- add_buffer_on_route

## ECO-265

(warning) The pin shape {%s} of %s (%s) is projected to route %s {%s}.

### Description

You receive this warning message when the route is far away from driver or loads. The tool will project the closest pin shape to the route.

### See Also

- place_group_repeaters

## ECO-266

(Warning) The cells in collection %s which are not placed before cannot be unplaced by unplace_group_repeaters.

### Description

This message is shown the cells which are not placed before cannot be unplaced. Please note that only the cells added by add_group_repeaters or placed by place_group_repeaters can be unplaced by the command unplace_group_repeaters.

### What Next

Please double check the cell status.

Feedback

**See Also**

- place_group_repeaters
- add_group_repeaters

## ECO-267

(Error) Input and output routes are not merged after unplacing repeater %s since the routes are not aligned or not at the same layer.

**Description**

This message is shown by unplace_group_repeaters when the input and output routes of the unplaced repeaters are not merged due to the following reasons.

- the routes are not aligned.
- the routes are not at the same layer.

**What Next**

You should check the routes of the driver net and load net of the unplaced repeater.

## ECO-268

(Error) The cells in collection (%s) are invalid since they are placed before. Please apply the command unplace_group_repeaters to unplace them.

**Description**

This message is shown when the cells which are placed before cannot be placed again, where the cells are either added by add_group_repeaters or placed by placed_group_repeaters.

**What Next**

Please double check the cell status. The placed cells should be unplaced by unplace_group_repeaters first.

**See Also**

- place_group_repeaters
- add_group_repeaters

Feedback

## ECO-269

(error) Unable to check the current reference of cell '%s': %s.

### Description

You receive this error message when command is unable to check the current reference of cell is legal or not for the reported reason. Please check the reported reason and other printed messages for more details.

### What Next

Please check the reported reason. If root cause is that command is unable to identify the design technology, please check whether there are some wrong ECO app option settings for the variant aware support.

### See Also

- eco_change_legal_reference

## ECO-270

(error) Unable to find swappable variant reference for cell '%s': %s.

### Description

You receive this error message when command found the current reference of cell is illegal but unable to find a swappable variant reference for the reported reason.

### What Next

Please check the variant cell group definition for the specified cell's reference library cell.

### See Also

- eco_change_legal_reference

## ECO-271

(error) Candidate reference '%s' and reference (%s) of cell (%s) are not in the same variant cell group.

### Description

You receive this error message when command detects compatible issue between thecandidate reference and the current reference of cell. They do not belong to the same variant cell group.

Feedback

### What Next

Please check whether the current reference and the candidate reference are in the same variant cell group or not.

### See Also

- eco_change_legal_reference

---

## ECO-272

(Error) The mixed input cells are not supported, which includes the cells added by add_group_repeaters and the cells placed by place_group_repeaters.

### Description

This message is shown that the command unplace_group_repeaters cannot unplace the cells added by add_group_repeaters and the cells placed by place_group_repeaters together.

### What Next

Please double check the input cells.

### See Also

- place_group_repeaters
- add_group_repeaters

---

## ECO-273

(Error) There are no placed cells to be unplaced. Please double check the cell status.

### Description

Please note that only the cells added by add_group_repeaters or placed by place_group_repeaters can be unplaced by the command unplace_group_repeaters.

### What Next

Please double check the cell status.

### See Also

- place_group_repeaters
- add_group_repeaters

Feedback

## ECO-274

(Warning) You are deleting cell %s with eco_repeater_group_id %d.

**Description**

You receive this warning message when you are trying to delete a cell belonging to a repeater group.

**What Next**

Please double check if it is necessary to delete the cell. This may cause the following action failed.

**See Also**

- set_repeater_group
- report_repeater_groups
- place_group_repeaters

## ECO-275

(Error) Failed to traverse cutlines. The nets may have too many blockages.

**Description**

Cannot find locations to insert cell groups.

**What Next**

Remove blockages or find cutlines and use -repeater_group option.

**See Also**

- place_group_repeaters
- add_group_repeaters

## ECO-276

(Error) Unable to find free area for repeater on net (%s) at location (%s) within range {%s}.

**Description**

You recieve this message when there is not enough free space for repeater.

Feedback

### What Next

Check if the range is enough for repeater groups.

### See Also

- place_group_repeaters
- add_group_repeaters

---

## ECO-277

(Warning) You are trying to override cells of repeater group %d. Please use -override if you have to override cells.

### Description

You receive this warning message when you try to override cells without using -override optoin.

### What Next

Please make sure you have to override cells of the repeater group. Please use -override to override cells.

---

## ECO-278

(Error) Get 0 transparent cells from all supernets.

### Description

You receive this error message when no transparent cells are found from all supernets passed in.

### What Next

Please double check the definition of the supernets.

### See Also

- set_supernet_exceptions
- create_supernet

---

## ECO-279

(Error) %s has multiple(%lu) inputs.

Feedback

**Description**

You receive this error message because create_repeater_groups doesn't support multiple fan-in cases.

**What Next**

Please make sure all cells and loads have only one input.

---

## ECO-280

(warning) Unable to find any nearby route to connect %s. Extend the maximum searching distance to be %s.

**Description**

You receive this warning message when command is unable to find any nearby route to connect the reported driver/load with current maximum searching distance setting.

Command will automatically extend the searching distance to be specified value and try to search the nearby routes for connecting driver/load again.

**What Next**

If command still could not find any nearby route to connect driver/load, please check whether the net routes is too far awary from the specified driver/load.

**See Also**

- place_group_repeaters

---

## ECO-281

(info) Distance between repeaters has been adjusted to spread out evenly.

**Description**

You receive this warning message when the specified repeater distance cannot be met without some short distance near the load.

**What Next**

Use -disable_distance_adjustment to avoid automatic spreading of repeater groups.

**See Also**

- add_group_repeaters

Feedback

## ECO-282

(error) Cutline in repeater group is not a valid location {%s %s} (layer %s). It could be near a bus turn, crossing, or blocked by macro or placement blockages.

### Description

You receive this warning message when the specified cutline in a repeater group is near a bus turn, crossing, or blocked by macro or placement blockages.

### What Next

Move cutline away from corner and crossing. Check overlapped macro or placement blockages.

### See Also

- place_group_repeaters

## ECO-283

(warning) Skip the load group (%d) for group (%d) since the load group is not input group and the cell in group is unplaced.

### Description

You receive this warning message when the load group is skipped since it is not user-input group and the cell in group is unplaced.

### See Also

- place_group_repeaters

## ECO-284

(error) The repeater group (%d) %s.

### Description

You receive this error message when the repeater group has neither driver group nor path driver, or has neither load group nor path load.

### What Next

The repeater group should have driver group (or path drivers), and load group (or path loads) when virtual connection is on. You need to make the group virtual connection complete by set_repeater_group.

Feedback

**See Also**

- set_repeater_group
- place_group_repeaters

---

## ECO-285

(error) Bad cutline order for group (%d) and its driver group (%d).

**Description**

You receive this error message when the cutline order is not reasonable.

**What Next**

Please double check the group and its driver group to make sure the group relation is reasonable and as your expectation.

This is just a sanity check and there could be false alarm. Here is how the tool check. For example, if from the driver side, the group connectivity is g1 -> g2 -> g3 -> g4. For one group (such as g3), we will calculate the Manhattan's distance (dist1) between the center of cutline of a group (g3) and its driver group (g2), and the distance (dist2) between the center of cutline of a group (g3) and driver group of its driver group (g1). If dist1 > dist2, this message will be issued.

If you think it is false alarm, you can use the app option below to bypass the order check so that place_group_repeaters can continue. set_app_options -name eco.placement.eco_enable_cutline_order_check -value false

**See Also**

- set_repeater_group
- place_group_repeaters

---

## ECO-286

(warning) Bad cutline order for group (%d) and its driver group (%d).

**Description**

You receive this warning message when the cutline order is not reasonable.

**What Next**

Please double check the group and its driver group to make sure the group relation is reasonable and as your expectation.

Feedback

**See Also**

- set_repeater_group
- place_group_repeaters

---

## ECO-287

(error) Net %s has no drivers or loads.

**Description**

You receive this error message when the net doesn't connect to any drivers or loads.

**What Next**

Please double check the net's connection.

**See Also**

- connect_net

---

## ECO-288

(error) Net %s has multiple drivers.

**Description**

You receive this error message when the net connects to multiple drivers. derive_bundles only support one-driver nets.

**What Next**

Please double check the net's connection.

---

## ECO-289

(warning) The repeater group (%d) %s.

**Description**

You receive this warning message when the repeater group has no info above.

**See Also**

- get_repeater_group_info
- set_repeater_group

Feedback

## ECO-290

(error) You are trying to create a new repeater group without specifying any cells.

**Description**

You receive this error message when you are trying to create a new repeater group without specifying any cells. You must specify cells when you create a new repeater group. A repeater group's cell list cannot be empty.

**What Next**

Please speciify cells when you create a new repeater group.

## ECO-291

(Error) The repeater group (group id: %d) you are trying to delete doesn't exist.

**Description**

You receive this error message when you are trying to delete an inexistent repeater group.

**See Also**

- set_repeater_group
- report_repeater_groups

## ECO-292

(warning) The number of %s is less than the number of cells in group (%d), so skip to set attribute about neighbors for cells in this group.

**Description**

You receive this warning message when the number of path drivers / path loads is less than the number of cells in group.

**What Next**

Please check the repeater group and make sure the number of path drivers / path loads is an integral multiple of the number of cells in group.

Feedback

**See Also**

- place_group_repeaters
- set_repeater_group
- report_repeater_groups

---

## ECO-293

(warning) The number of %s is not an integral multiple of the number of cells in group (%d), so use %s in order to set cell attribute about neighbors.

**Description**

You receive this warning message when the number of path drivers / path loads is not an integral multiple of the number of cells in group.

**See Also**

- place_group_repeaters
- set_repeater_group
- report_repeater_groups

---

## ECO-294

(warning) The number (%d) of cells from repeater group (%d) is not consistent with the number (%d) of drivers from driver group (%s), skip to set cell attribute about neighbors.

**Description**

You receive this warning message when the number of cells is not consistent with the number of drivers.

**What Next**

Please check the repeater group and make sure the number of drivers from driver group and cells from repeater group are same.

**See Also**

- place_group_repeaters
- set_repeater_group
- report_repeater_groups

Feedback

## ECO-295

(error) The path line from point (%s) to point (%s) is not vertical or horizontal.

**Description**

You receive this error message when two adjacent turning points cannot make a vertical or horizontal path line.

**What Next**

Please check if you provide correct turning point list in correct order.

## ECO-296

(error) Location specified is not valid {%s}. It could be near a bus turn, crossing, or blocked by macro or placement blockages.

**Description**

You receive this warning message when the specified location is near a bus turn, crossing, or blocked by macro or placement blockages.

**What Next**

Move cutline away from corner and crossing. Check overlapped macro or placement blockages.

**See Also**

- place_group_repeaters

## ECO-297

(error) Failed to remove previously created route guidance objects of bundle net.

**Description**

You receive this error message when command failed to remove the previously created route guidance objects.

**What Next**

Please check whether the route guidances objects are editable or not.

## ECO-298

(error) Failed to load route guidances from bundle %s: %s.

Feedback

**Description**

You receive this error message when command failed to load previously created route guidance objects for the specified reason.

**What Next**

Please check whether the bundle has route guidance objects or not. And also check whgether there are dangling route guidance objects or not.

---

## ECO-299

(error) Failed to commit individual bit route creation for bundle %s: %s.

**Description**

You receive this error message when command failed to commit the individual bit route creation for the specified reason.

**What Next**

Please check the reported reason.

---

## ECO-300

(error) Failed to commit route guidance creation for bundle %s: Unable to create wire objects.

**Description**

You receive this error message when command failed to commit route guidance creation for the specified reason.

**What Next**

Please check the reported reason.

---

## ECO-301

(error) Unable to derive the load group location for bundle %s: %s.

**Description**

You receive this error message when command is unable to derive the leaf load group location for the specified reason.

**What Next**

Please check the reported failure reason and update the bundle path settings.

Feedback

## ECO-302

(warning) The load %s of load group %d has not been set with physical location. Ignore it when derive the load group location of bundle %s.

**Description**

You receive this warning message when command detects the leaf load of one load group has not been set with physical location. It would be caused by the bundle path settings or the bunlde load found is a hierarchical object.

Command will ignore this load when derive the load group location.

**What Next**

Please check the specified load and bundle path settings.

## ECO-303

(error) You are not allowed to set the driver group id to %d because the repeater group %d already has path drivers. Please use -driver_group_id $id -clear path_drivers together to remove path drivers from the current repeater group and set a driver group id it.

**Description**

You receive this error message when you are trying to set a driver group id to the current repeater group while the current repeater group has already been set path drivers. This action is not allowed because a driver group id and path drivers are mutually exclusive of a repeater group.

## ECO-304

(error) You are not allowed to set path drivers to the repeater group %d because it already has a driver group. Please use -path_drivers $drivers -clear driver_group_id together to remove the driver group id from the current repeater group and set path drivers to it.

**Description**

You receive this error message when you are trying to set the path drivers to the current repeater group which has already been set a driver group id. This action is not allowed because a driver group id and path drivers are mutually exclusive of a repeater group.

## ECO-305

(error) The size of the new cell list (%u) does not match the size of the current cell list (%u).

Feedback

### Description

You receive this error message when you are trying to override the cell list of the current repeater group, but the size of the new cell list is not the same as the size of the current cell list of the repeater group.

### What Next

Please input a cell list with the same size of the current cell list when you are trying to override the current cell list of the repeater group.

---

## ECO-306

(error) For repeater group (%d), the number (%d) of objects from driver group (%s) is not consistent with the number (%d) of output nets.

### Description

You receive this error message when the number of objects is not consistent with the number of output nets of objects.

### What Next

Please double check the input options -lib_cell_input and -lib_cell_output. The default value is D and Q. The tool will find net connection based on pin name specified by the two options. Please check the output net of objects from driver group and make sure the number of objects from driver group and output nets of objects are same.

### See Also

- place_group_repeaters
- set_repeater_group
- report_repeater_groups

---

## ECO-307

(error) Failed to get output net of %s (%s).

### Description

You receive this error message when the tool cannot get the output net of object.

### What Next

Please double check the input options -lib_cell_input and -lib_cell_output. The default value is D and Q. The tool will find net connection based on pin name specified by the two options. Please check the output net of object based on the specified lib pin name.

Feedback

**See Also**

- place_group_repeaters
- set_repeater_group
- report_repeater_groups

---

## ECO-308

(error) Failed to create repeater groups for %s.

**Description**

You receive this error message when the command failed to create repeater groups.

**What Next**

Please check the related error messages.

**See Also**

- place_group_repeaters
- set_repeater_group
- report_repeater_groups

---

## ECO-309

(warning) The options -lib_cell_input and -lib_cell_output are not specified, so the attribute pin_pair for each cell in this group is set based on default value D and Q.

**Description**

You receive this warning message when the command options -lib_cell_input and -lib_cell_output are not specified.

**What Next**

The pin_pair set on repeater is used to: 1. Find real connection of cell based on specified pin names when set_repeater_group -check_connection. 2. Find net connected to specified pin name to assign the split route when place_group_repeaters -repeater_groups.

Feedback

**See Also**

- set_repeater_group
- get_attribute

---

## ECO-310

(warning) The cell (%s) has no driver or load (number of driver: %d, number of load: %d).

**Description**

You receive this warning message when the command cannot identify driver or load of the cell.

**What Next**

Please make sure the cell has both driver and load. Please check the cell connectivity based on user-defined virtual connection. Check if the repeater group that the cell belongs to has driver group or path driver or path load. If missing, you can use set_repeater_group -path_drivers or -driver_group_id to specify drivers of this group, while use -path_loads to specify path loads. Noth that -path_drivers is only for the first group on the path and -path_loads is only for the last group on the path.

**See Also**

- place_group_repeaters
- set_repeater_group
- report_repeater_groups

---

## ECO-311

(error) Failed to find a track pattern for layer %s.

**Description**

You receive this error message when input track patterns does not cover a certain layer which is used by at least an input topology edge.

**What Next**

Please input track patterns to cover all layers used by input topology edges.

---

## ECO-312

(error) Provided more than one track pattern for layer %s.

Feedback

**Description**

You receive this error message when more than one track pattern for a certain layer has been found.

**What Next**

Please input only one track pattern for each layer.

---

## ECO-313

(error) Layer %s is neither horizontal nor vertical.

**Description**

You receive this error message when the layer in shape_layers of the topology edge is neither horizontal nor vertical. slot_topology_edges only support horizontal or vertical layers.

**What Next**

Please only use horizontal or vertical layers to create topology edges.

**See Also**

- create_topology_edge

---

## ECO-314

(error) Failed to find track patterns on the topology edge %s.

**Description**

You receive this error message when no track patterns are found on the topology edge.

**What Next**

Please set user track patterns to topology edges before slotting.

---

## ECO-315

(warning) Failed to place the %d buffer cells to the proper location.

**Description**

You receive this warning message when command failed to place new created buffer cells to the proper location. And these buffer cells will be placed at the initial location (nearby or overlap with the target cells).

Feedback

### What Next

Please place these buffer cells to the proper location again.

### See Also

- add_buffer

---

## ECO-316

(warning) Failed to place the buffer cell %s of terminal %s to the proper location.

### Description

You receive this warning message when command failed to place the created buffer cell to the proper location. And this buffer cell will be placed at the initial location (overlap with the target cell).

### What Next

Please place this buffer cell to the proper location again.

### See Also

- add_buffer

---

## ECO-317

(error) The topology repeater at {%s} does not project to any route%s.

### Description

You receive this error message when the command failed to find a line to intersects with routes on the layer and at the location where the topology repeater is.

### What Next

Please make sure there are routes on the layer and at the location where the topology repeater is.

---

## ECO-318

(warning) The cell (%s) is not supported since it is not macro cell or sequential cell or combo cell or repeater.

### Description

You receive this warning message when the command failed to get valid label for cell.

Feedback

### What Next

Please check the cell attribute. The command create_interconnect_groups only supports macro cell, sequential cell, combo cell and repeater based on backward path tracing from input loads. For valid cell, the label is like below. S is for sequential cell. R is for repeater. C is for combo gate. M is for hard or soft macro.

### See Also

- create_interconnect_groups

---

## ECO-319

(error) %s.

### Description

You receive this error message when command failed the listed reason.

### See Also

- set_repeater_group

---

## ECO-320

(warning) Failed to find any nearby route to connect %s in search region %s.

### Description

You receive this warning message when command is unable to find any nearby route to connect the reported driver/load in search region.

### What Next

Please check whether the net routes are too far awary from the specified driver/load. You can use the option -max_distance_for_incomplete_route to control the search distance.

### See Also

- place_group_repeaters

---

## ECO-322

(error) The specified load pins are not driven by the driver pin or net.

Feedback

### Description

You receive this error message when you use option -same_set_connection_only, while the specified load pins are not driven by the specified driver pin or net. If -same_net_connection_only is not specified. This message will only be an information message.

### What Next

Please specify load pins that are driven by the driver pin or net.

---

## ECO-323

(error) Unable to connect load %s to driver %s, route is incomplete.

### Description

You receive this error message when command is unable to find route to connect the reported load to the driver and the route is incomplete.

### What Next

Please make sure there is route to connect the driver and load.

### See Also

- place_group_repeaters
- add_group_repeaters

---

## ECO-324

(error) Failed to find any nearby route to connect %s in search region %s.

### Description

You receive this error message when command is unable to find any nearby route to connect the reported driver/load in search region.

### What Next

Please check whether the net routes are too far awary from the specified driver/load.

### See Also

- place_group_repeaters
- add_group_repeaters

Feedback

## ECO-325

(error) In the search region %s of %s, there is more than 1 open route of the same net.

### Description

You receive this error message when command finds more than 1 open route of the same net in the search region to connect the reported driver or load.

### What Next

Please make sure there is only 1 open route of the same net in the search region of the driver or load.

### See Also

- place_group_repeaters
- add_group_repeaters

## ECO-326

(error) Unable to connect %s and %s, via is not created correctly.

### Description

You receive this error message when command is unable to connect specified route, via, pin or port because they are created in different layers.

### What Next

Please make sure that the specified route, via, pin or port is created in the same layer.

### See Also

- place_group_repeaters
- add_group_repeaters

## ECO-328

(error) The net %s does not have a driver.

### Description

You receive this error message when an input net does not have a driver. Each input net should have one and only one driver.

Feedback

### What Next

Please make sure each input net has a driver.

### See Also

- connect_net

---

## ECO-329

(error) The net %s does not have loads.

### Description

You receive this error message when an input net does not have loads. Each input net should have at least one load.

### What Next

Please make sure each input net has loads.

### See Also

- connect_net

---

## ECO-330

(error) The net %s has multiple (%lu) drivers.

### Description

You receive this error message when an input net has more than one driver. Each input net should have only one driver.

### What Next

Please make sure each input net has one and only one driver.

### See Also

- disconnect_net

---

## ECO-331

(error) The cell %s of the pin %s is not placed.

Feedback

### Description

You receive this error message when a cell of a driver or load pin is not placed. When a pin on a standard cell, the cell should be placed.

### What Next

Please set the cell of a pin to be placed.

### See Also

- set_attribute

---

## ECO-332

(error) The %s %s does not have shapes.

### Description

You receive this error message when a pin or port of a net has no shapes.

### What Next

Please make sure each net pin or port have shapes.

### See Also

- create_shape
- set_object_shape

---

## ECO-333

(error) There are multiple (%d) drivers (%s) for load %s.

### Description

You receive this error message when an input supernet has more than one driver.

### What Next

Please make sure each input supernet has one and only one driver.

### See Also

- disconnect_net

Feedback

## ECO-334

(error) The cell %s cannot find the pin with the pin name %s.

**Description**

You receive this error message when a cell does not have the pin with a specific pin name.

**What Next**

Please use cells with library cells that have a pin with the pin name above.

**See Also**

- get_lib_cells
- create_cell

## ECO-335

(error) The cell %s does not have a library cell.

**Description**

You receive this error message when an input cell does not have a library cell.

**What Next**

Please make sure each cell has a library cell.

**See Also**

- get_lib_cells
- create_cell

## ECO-336

(error) There are no routes on the net %s.

**Description**

You receive this error message when an input net has no routes.

**What Next**

Please make sure each net is routed.

Feedback

### See Also

- route_custom
- create_trunk

---

## ECO-337

(Error) Some nets (%ld out of %ld) do not have a net group ID or the net group ID is not specified by option %s. Find the collection of nets in %s.

### Description

This message is shown when add_group_repeaters -first_distance_of_net_groups or -cutlines_of_net_groups is used, but some nets do not have net group ID, or some nets have net group ID but ID is not specified by the option.

### What Next

If the net does not have a net group ID, please run create_group_repeaters_guidance to create net group IDs, or set net attribute eco_net_group_id manually. If the net has net group ID, please specify the pair of group ID and first distance by the option -first_distance_of_net_group, or specify cutlines for each group ID by the option -cutlines_of_net_groups.

### See Also

- create_group_repeaters_guidance

---

## ECO-338

(error) Unable to find all nets for topology repeater at %s (total nets: %d, found nets: %d)

### Description

You receive this warning message when not all nets can be found by projecting the topology repeater location.

### What Next

Check and correct your topology plan.

### See Also

- implement_topology_repeaters

Feedback

## ECO-339

(error) One or more specified cutlines are not valid!

**Description**

You receive this warning message when the specified cutline(s) are not valid.

**What Next**

Please check cutline locations to make sure they intersect with the routes.

**See Also**

- add_group_repeaters

## ECO-340

(Error) The input net group ID %d is unused. No input net has such net group id.

**Description**

This message is shown when add_group_repeaters -first_distance_of_net_groups or -cutlines_of_net_groups is used, but the input net group ID is unused.

**What Next**

Remove such net group ID from input.

**See Also**

- add_group_repeaters

## ECO-341

(Error) The lib cell of cell (%s) cannot be found in -pin_mapping.

**Description**

This message is shown when create_repeater_groups -multi_bit -pin_mapping is used, but the lib cell of input cell is not specified in -pin_mapping.

**What Next**

Specify the lib cell of input cell in -pin_mapping.

**See Also**

- create_repeater_groups

Feedback

## ECO-342

(warning) Smart net grouping is on by defaul now. The option -smart_net_grouping is deprecated. Please use these options to control net groups if needed: -group_spacing, ignore_layers, and -exclude_pin__port.

### Description

This message is shown when -smart_net_grouping option is used in add_group_repeaters.

### What Next

Smart net grouping is on by default.

### See Also

- place_group_repeaters

## ECO-343

(error) Failed to get the location of net %s for add_buffer with -respect_voltage_areas option. The net will be ignored.

### Description

You receive this error message when failing to get the location of input pin, port or driver of net for add_buffer with -respect_voltage_areas option.

### What Next

Add buffer with -respect_voltage_areas option on leaf pin, leaf port or net with leaf driver.

### See Also

- add_buffer

## ECO-344

(error) Unsupported multiple fanout net %s for add_buffer with -respect_voltage_areas option. And there are lost loads or additional loads due to hier net change. The net will be ignored.

### Description

You receive this error message when trying to add buffer on unsupported multiple fanout net using add_buffer with -respect_voltage_areas option.

Feedback

**See Also**

- add_buffer

---

## ECO-345

(error) Failed to get voltage area according to the location (%s) of specified pin, port or driver of net %s. And the net will be ignored.

**Description**

You receive this error message when using add_buffer with -respect_voltage_areas option, which is based on the voltage area at the location of specified pin, port or driver of net. Since there is no voltage area at the location, voltage area aware add_buffer is not supported. And the net will be ignored.

**What Next**

Add buffer with -respect_voltage_areas option on pin, port or net in voltage area.

**See Also**

- add_buffer

---

## ECO-346

(error) Empty load detected. It might be due to unequal fanouts in the bus. It is not supported.

**Description**

You receive this error message when repeater paths do not have the same number of loads.

---

## ECO-347

(info) Unequal fanout loads detected for topology repeater %s. Treat it as a single-load path.

**Description**

You receive this error message when repeater paths do not have the same number of loads.

Feedback

## ECO-348

(warning) The location of net %s for add_buffer with -respect_voltage_areas option is outside of core area. Buffer's location will be updated from (%s) to (%s).

### Description

You receive this warning message when the location for add_buffer with -respect_voltage_areas option is outside of core area.

### What Next

Buffer's location will be projected into the core area.

### See Also

- add_buffer

## ECO-349

(error) Module net is not supported for add_buffer with -respect_voltage_areas option. And the net %s will be ignored.

### Description

You receive this error message when trying to add buffer on module pin, port or net with -respect_voltage_areas option.

### What Next

Add buffer with -respect_voltage_areas option on hierarchy pin, port or net.

### See Also

- add_buffer

## ECO-350

(error) Cannot find correct hier net for add_buffer with with -respect_voltage_areas option. And the net %s will be ignored.

### Description

You receive this error message when using add_buffer with -respect_voltage_areas option, but the correct hier net cannot be found base on buffer's location.

Feedback

### What Next

Add buffer with -respect_voltage_areas option on pin, port or net that correct hier net can be found base on buffer's location.

### See Also

- add_buffer

---

## ECO-351

(error) Net %s should have one ultimate leaf driver.

### Description

You receive this error message when using add_buffer with -respect_voltage_areas option to a flat net without leaf driver.

### What Next

Add buffer with -respect_voltage_areas option on a flat net with leaf driver.

### See Also

- add_buffer

---

## ECO-352

(Warning) There are %d cells in total with eco_change_status attribute eco_legalized. These cells will not be moved during place_eco_cells.

### Description

You receive this warning message when you try to issue command place_eco_cells with command option -legalize_mode allow_move_other_cells.

### What Next

If you would like to move these cells during place_eco_cells, please change these cells eco_change_status attribute to be eco_placed.

### See Also

- place_eco_cells

Feedback

## ECO-353

(error) Cannot specify block pin or net with block pin as driver for add_buffer with with -respect_voltage_areas option. And the net %s with block pin %s will be ignored.

### Description

You receive this error message when using add_buffer with -respect_voltage_areas option, and inputing block pin or net with block pin as driver.

### What Next

Add buffer with -respect_voltage_areas option on non-block pin.

### See Also

- add_buffer

## ECO-354

(Error) Cannot find valid hierarchy given input net %s.

### Description

You receive this warning message when you try to issue command add_buffer_on_route with command option -honor_net_hierarchy.

### What Next

Please make sure given net is in correct hierarchy.

### See Also

- add_buffer_on_route

## ECO-355

(Error) %s has no connected freeze layer objects on the highest frozen layer %s.

### Description

You receive this warning message when you try to disconnect net from the pin/port in freeze layer mode.

### See Also

- disconnect_net

Feedback

## ECO-356

(Error) One or more connected freeze layer objects on %s is user route or locked route.

### Description

You receive this warning message when you try to disconnect net from the pin/port in freeze layer mode.

### What Next

Check routes and vias connected to the pin/port to make sure they are not user route or locked.

### See Also

- disconnect_net
- get_shapes
- get_vias

## ECO-357

(Info) There are %ld shapes buried under frozen layers: %s

### Description

Reports freeze layer routes and vias that cannot be reconnected.

### See Also

- disconnect_net
- get_shapes
- get_vias

## ECO-358

(Error) %s.

### Description

You receive this error message when the sum of distances at branch exceeds the value of -last_distance.

Feedback

### What Next

Please check the route length of each route branch and change -last_distance. Make sure the value is smaller than each route branch or bigger than the total length of all route branches.

### See Also

- add_buffer_on_route

---

## ECO-359

(error) failed to process the route segments of net %s for the reason: %s.

### Description

Command failed to process the routes for the specified reason which would cause the buffering failed.

### What Next

Please check the failure reason of this message and optimize the command options or net routes. Please check the route topology and increase max search distance.

### See Also

- add_group_repeaters
- add_buffer_on_route

---

## ECO-360

(Error) Inconsistentcy between power domain %s of specified hierarchy %s and voltage area %s.

### Description

When specify both command option -honor_net_hierarchy and -respect_voltage_areas, power domain of specified hierarchy may not match with voltage area.

### What Next

Please try only use -honor_net_hierarchy or -respect_voltage_areas, or check if the power domain of specified net hierarchy matches with voltage area.

### See Also

- add_buffer_on_route

Feedback

## ECO-361

(Error) MV buffer selection is failed for net %s. No buffer is inserted.

### Description

You receive this error message when some specified/computed locations are not bufferable.

### What Next

Modify locations or change distance rule. Check bufferability using command check_bufferability.

### See Also

- check_bufferability

## ECO-362

(Error) Bus %d (net count = %d) is not fully connected.\n%s

### Description

The routes of these nets are not fully connected. The route traversal failed.

### What Next

Check if the routes are connected correctly. Instead of using detail route, try create_trunk to route nets before add_group_repeaters.

### See Also

- create_trunk
- create_trunk_pin_to_pin

## ECO-363

(Warning) Bus %d (net count = %d) has no location to insert.\n%s

### Description

The routes of these nets do not have location to insert.

### What Next

Check for blockages, macros, route corners, and existing placed repeaters that may block the area for insertion.

Feedback

## ECO-364

(Info) set app option %s to be %s.

**Description**

This message shows the value of one app option.

**What Next**

None.

## ECO-365

(Info) %s color shifting feature is auto-enabled in ECO PSC freeze-silicon flow.

**Description**

This message shows the mode of color shifting feature in ECO PSC freeze-silicon flow.

**What Next**

None.

## ECO-366

(error) unit width %s is not integral multiple of %s %s.

**Description**

In advanced PSC mapping of place_freeze_silicon, unit width is a base unit of PSC filler split and merge. By default, unit width is same as the smallest width of the lib cells processed by command derive_programmable_spare_cell_mapping_rules. It could be overwritten by command option -unit_width. To ensure advanced PSC mapping succeeds, unit width should follow 3 rules: 1. It should be integral multiple of site width. 2. It should be integral multiple of pitch. 3. It should be integral multiple of the width of one cycle of track pattern. You receive this error message when any of these rules are broken.

**What Next**

Please try to specify a proper unit width by derive_programmable_spare_cell_mapping_rules -unit_width width to meet above 3 conditions. Since the default value is derived from the smallest lib cell width, you may also need to double check if the lib cells passed to this command are all needed, and filter unexpected lib cells.

Feedback

**See Also**

- derive_programmable_spare_cell_mapping_rules
- report_site_defs
- report_tracks

---

## ECO-367

(error) First track offset is not specified and has no default value.

**Description**

For advanced PSC mapping, place_freeze_silicon needs to check color and track alignment for ECO cells and refilled PSC fillers based on the first track offset. You receive this error message when offset value is not specified and has no default value.

**What Next**

Please query the first track offset by report_tracks and set it by app option ecolm.freeze_silicon.psc_track_offset.

**See Also**

- derive_programmable_spare_cell_mapping_rules
- report_tracks

---

## ECO-368

(error) Input first track offset value %s is a negative value.

**Description**

For advanced PSC mapping, place_freeze_silicon needs to check color and track alignment for ECO cells and refilled PSC fillers based on the first track offset. It should be greater than or equal to 0. You receive this error message when you input a negative first track offset.

**What Next**

Please query the first track offset by report_tracks and set it correctly by app option ecolm.freeze_silicon.psc_track_offset.

Feedback

**See Also**

- derive_programmable_spare_cell_mapping_rules
- place_freeze_silicon
- report_tracks

---

## ECO-369

(warning) The specified first track offset %s is different from default value %s.

**Description**

For advanced PSC mapping, place_freeze_silicon needs to check color and track alignment for ECO cells and refilled PSC fillers based on the first track offset. If it is not set correctly, PSC mapping may cause legality violations. You receive this error message when you input a different first track offset from the default value and the default value will be overwritten.

**What Next**

Please query the first track offset by report_tracks and check if it is the same as the value of app option ecolm.freeze_silicon.psc_track_offset. If not, please correct your setting. Otherwise, you can ignore this message.

**See Also**

- derive_programmable_spare_cell_mapping_rules
- report_tracks

---

## ECO-370

(warning) PG layer is not specified. Use default layer %s.

**Description**

For advanced PSC mapping of place_freeze_silicon, PG layer must be set correctly for PG DRC check. You receive this warning message when PG layer is not specified. The command will use the default PG layer.

**What Next**

Please double check if the default PG layer is appropriate for this design. If not, please change it by command derive_programmable_spare_cell_mapping_rules -pg_layer $layer_name.

Feedback

**See Also**

- derive_programmable_spare_cell_mapping_rules
- place_freeze_silicon

---

## ECO-371

(error) Could not find default PG layer %s, please set correct layer by command option.

**Description**

For advanced PSC mapping of place_freeze_silicon, PG layer must be set correctly for PG DRC check. You receive this error message when default PG layer could not be found.

**What Next**

Please set PG layer by command derive_programmable_spare_cell_mapping_rules -pg_layer layer.

**See Also**

- derive_programmable_spare_cell_mapping_rules
- place_freeze_silicon

---

## ECO-372

(warning) Input PG layer %s is different from default layer %s.

**Description**

For advanced PSC mapping of place_freeze_silicon, PG layer must be set correctly for PG DRC check. You receive this warning message when you input a different PG layer from the default layer. Your input PG layer will overwrite the default layer.

**What Next**

Please double check if the input PG layer is appropriate for this design. If not, please change it by command derive_programmable_spare_cell_mapping_rules -pg_layer $layer_name.

**See Also**

- derive_programmable_spare_cell_mapping_rules

Feedback

## ECO-373

(warning) First track offset is not manually set, use default offset %s.

### Description

For advanced PSC mapping, place_freeze_silicon needs to check color and track alignment for ECO cells and refilled PSC fillers based on the first track offset. If it is not set correctly, PSC mapping may cause legality violations. You receive this warning message when the first track offset is not manually set. The command will try to use the default offset according to technology node.

### What Next

Please query the first track offset by report_tracks. If the default offset is different from it, please set the first track offset correctly by app option ecolm.freeze_silicon.psc_track_offset. Otherwise, you can ignore this message.

### See Also

- derive_programmable_spare_cell_mapping_rules
- place_freeze_silicon
- report_tracks

## ECO-374

(warning) The tech node of this design is not set.

### Description

You receive this warning message when design tech node is not set. Without setting tech node layer, PG checking layer will use default value, which may be different from real layer.

### What Next

You can set tech node by command set_technology.

### See Also

- derive_programmable_spare_cell_mapping_rules
- set_technology

Feedback

## ECO-375

(information) Do %s.

### Description

You receive this information message when you execute command place_freeze_silicon. The message indicates what freeze silicon ECO flow is being done. Logical spare cell mapping is based on reference lib cell. An eco cell can only be mapped with a logical spare cell with same reference lib cell name. Traditional PSC mapping is based on psc_type_id. An eco cell can be mapped to one or multiple abutted PSC fillers with same psc_type_id if total width of PSC fillers is larger than the eco cell. Advanced PSC mapping, which is for advanced nodes, will check not only cell size but also many advanced rules. Before place_freeze_silicon, you need to run derive_programmable_spare_cell_mapping_rules for both ECO cells and PSC fillers.

### See Also

- add_spare_cells
- create_stdcell_fillers
- derive_programmable_spare_cell_mapping_rules

## ECO-376

(error) No spare cell has same psc_group_id %d with eco cell.

### Description

You receive this error message when you do advanced PSC mapping. Lib cell attribute psc_group_id is for handling different VT/Channel. If the lib cell of an eco cell has psc_group_id, it can only be mapped to PSC filler spare cells whose lib cells have the same psc_group_id. If the lib cell of an eco cell has no psc_group_id, it can only be mapped to PSC filler spare cells whose lib cells do not have psc_group_id as well.

### What Next

Please check the psc_group_id setting for ECO lib cells and PSC filler lib cells.
You also need to check the ECO lib cells and PSC filler lib cells passed to derive_programmable_spare_cell_mapping_rules. Make sure ECO cells have compatible PSC fillers in terms of psc_group_id.

### See Also

- derive_programmable_spare_cell_mapping_rules

Feedback

## ECO-377

(warning) Skip mapping eco cell %s since its eco_change_status is eco_legalized.

**Description**

You receive this warning message when you execute command place_freeze_silicon. Eco cells will be skipped to do mapping if the eco_change_status attributes are eco_legalized.

**What Next**

Please set the attribute eco_change_status of eco cell to create_cell/add_buffer/change_link/size_cell/eco_placed/add_buffer_on_route if you want to map this eco cell.

## ECO-378

(error) No spare cell is spread successfully since there is no enough available space.

**Description**

The command skips to do spreading for all the spare cells belonging to each target region since every target region has no enough available space.

**What Next**

Check the utilization of all target regions.

**See Also**

- spread_spare_cells

## ECO-379

(Error) Failed to elevate net %s due to %s\n

**Description**

The net cannot be elevated.

**What Next**

Check the net.

## ECO-380

(Error) Frontside FIB insertion failed during %s\n

Feedback

**Description**

Frontside FIBs insertion failed.

---

## ECO-381

(Error) Port %s is not connected to net!\n

**Description**

Elevated port is not connected to any net.

---

## ECO-382

(Error) Port %s is not connected to signal elevation buffer or pin!\n

**Description**

The port is not a valid elevation port.

---

## ECO-383

(Error) Failed to eval tcl command for %s\n

**Description**

Command failed to execute.

---

## ECO-384

(Error) Port %s terminal creation failed due to pin %s does not have shape!\n

**Description**

Pin does not have a shape which caused port terminal creation failure.

**What Next**

Check the pin.

---

## ECO-385

(Warning) Net %s is skipped due to %s.

**Description**

The net is skipped in signal elevation.

Feedback

**What Next**

Check the net.

# ECOUI

## ECOUI-001

(error) Could not find parent cell for '%s'.

**Description**

You receive this error message when you attempt to create either a cell, net or port in an instance which does not already. This command will not create any new hierarchy except the new leaf object which is specified in the command.

**What Next**

Review the man page for the creation command being used, and try using current_instance to move to the instance within which you wish to create the new object. If you need to create levels of hierarchy, use the create_cell command to create each level one at a time.

## ECOUI-002

(error) Can't remove port '%s'; it's part of a bus.

**Description**

You receive this error message when you attempt to remove bus port.

**What Next**

choose the port without bus.

## ECOUI-003

(error) The pin '%s' connects to a PG/Tie net.

**Description**

remove_buffers doesn't support removing the buffer which connect to PG/Tie net

**What Next**

Please remove the buffer manually.

Feedback

## ECOUI-004

(error) Failed to create golden design from the Verilog file "%s".

**Description**

The specified Verilog file may have some issues and it could not be loaded to be the golden design. Use read_verilog command to check whether the Verilog file can be read successfully or not.

For complete netlist diff, sometime this issue would be caused by the multiple top modules in verilog file. Please check the top module of input verilog file.

**What Next**

Use read_verilog command to check whether the Verilog file content is correct or has some issues.

For complete netlist diff, if the input verilog file has multiple top modules, user can use option "-top_module" to sepcify a correct top module for eco_netlist.

**See Also**

- eco_netlist

## ECOUI-005

(error) Type "%s" is not valid.

**Description**

The command supports all blockage types same as create_placement_blockage.

**What Next**

Please refer to the blockage type defined in create_placement_blockage.

**See Also**

- add_spare_cells
- spread_spare_cells
- create_placement_blockage

## ECOUI-006

(Error) net '%s' should have one driver pin

Feedback

### Description

This error message occurs when the operating net has more than one driver pin or the net has no valid driver pin.

### What Next

*remove_buffers* and *add_buffer* cannot operate on nets having multiple driver pins.

### See Also

- remove_buffers
- add_buffer

---

## ECOUI-007

(Error) Error in argument '%s'. '%s'.

### Description

You recieved this error message because an option in the command has an incorrect value. Either the tool could not find the expected value for the argument or the number of arguments is invalid.

### What Next

Clarify the proper use of the option, see the man page of the command.

---

## ECOUI-008

(error) Unable to create %s '%s'. A %s already exists%s.

### Description

The specified operation failed because an object with the given name already exists.

### What Next

Choose a different name for this new object.

---

## ECOUI-009

(error) The app option eco.freeze_silicon.spreader_spacing (%f) is not valid.

### Description

The app option should be greater than or equal to 0. The unit is micron.

Feedback

### What Next

Please specify the valid value.

### See Also

- add_spare_cells
- spread_spare_cells

---

## ECOUI-010

(Error) library cell '%s' is not a buffer or an inverter.

### Description

You recieve this error message, if the library cell matching the specified name is not a buffer or inverter library cell from the target library.

### What Next

Check to make sure that the libray is set correctly and use the buffer or inverter.

### See Also

- add_buffer
- report_lib

---

## ECOUI-011

(Error) cell '%s' is not a buffer or an inverter.

### Description

You recieve this error message because the specified cell is not a buffer or an inverter.

### What Next

Select the cell that its library cell is a buffer or inverter.

### See Also

- add_buffer
- report_lib

Feedback

## ECOUI-012

(error) Unable to create %s '%s'. Measure side '%s' cannot be used with first buffer at '%s'.

**Description**

If creating pattern for horizontal route, first buffer must be at 'bottom' or 'top' and specify measure side 'left' or 'right'. If creating pattern for vertical route, must specify first buffer at 'left' or 'right' and specify measure side 'bottom' or 'top'. Other combination are not supported.

**What Next**

Specify supported combination.

## ECOUI-013

(Error) Error in argument '%s'. '%s'.

**Description**

You recieved this error message because an option in the command has an incorrect value. Either the tool could not find the expected value for the argument or the number of arguments is invalid.

**What Next**

Clarify the proper use of the option, see the man page of the command.

## ECOUI-014

(error) There are eco cells created by spare cell awared add_buffer_on_route or size_cell in design, which have psc_type_id but have not been completely trial mapped or real swapped.

**Description**

The command does not allow user to do place_freeze_silicon -no_spare_cell_swapping or -map_spare_cells_only unless all eco cells with psc_type_id created by spare cell awared add_buffer_on_route or size_cell are completely trial mapped or real swapped.

**What Next**

Please find out all eco cells with eco_change_status as add_buffer_on_route or size_cell and the attribute fs_psc_is_spread_in_slot as false. And apply place_freeze_silicon -no_spare_cell_swapping or -map_spare_cells_only on such cells first.

Feedback

**See Also**

- place_freeze_silicon
- get_attribute

---

## ECOUI-015

(error) The %s view is not supported by the command %s.

**Description**

You receive this error message when you attempt to run the command in the view which is not supported. This command has not supported this view yet.

**What Next**

Run this command in other views.

---

## ECOUI-016

(error) The input bus buffer array pattern %s for option %s is not square or rectangle array type and it is not supported by the command %s.

**Description**

You receive this error message when you specify a non square nor rectangle array type bus buffer array pattern for the option, but this command can only support square or rectangle array type pattern input.

**What Next**

Run this command with square or rectangle array type bus buffer array pattern.

---

## ECOUI-017

(error) Object in bundle %s is not net type, please specify net type bundles for option %s.

**Description**

You receive this error message when the input bundles contain objects which are not net type.

**What Next**

Specify the net type bundles for option and rerun the command again.

Feedback

## ECOUI-018

(error) Failed to set bus guide connection for %s.

**Description**

You receive this error message when command failed to set bus guide connection for the specified driver or some nets.

**What Next**

Check the previous error messages and find out why failed to set the bus guide connection for these nets. Then specify the correct options and run again.

## ECOUI-019

(error) input object %s of %s is not %s.

**Description**

You receive this error message when specify wrong type object for the option.

**What Next**

Specify the correct type object for the option.

## ECOUI-020

(Error) net '%s' does not have any loads.

**Description**

This error message occurs when the operating net has no load pin connected to it.

**What Next**

*add_buffer* will not buffer unconnected nets.

**See Also**

- add_buffer

## ECOUI-021

(Error) The lib cell %s is not tie-%s lib cell for the option %s.

Feedback

### Description

This error message occurs when there is invalid lib cell specified for the lib cell option. The specified lib cells must be tie-high lib cell for the option *-tie_high_lib_cell*. And the specified lib cells must be tie-low lib cell for the option *-tie_low_lib_cell*.

### What Next

Specify the correct lib cells.

### See Also

- connect_freeze_silicon_tie_cells

---

## ECOUI-022

(Error) The library cell %s is not of %s type.

### Description

This error message occurs when there is invalid library cell specified for the lib cell option. The specified lib cells must be the type as required by the option.

### What Next

Check the design_type of library cells and specify the library cells of the required type.

### See Also

- place_eco_cells

---

## ECOUI-023

(error) Pin %s has no connected net.

### Description

You receive this error message when the specified pin has no connected net and command cannot be executed successfully.

### What Next

Select valid pin for the option and run again.

---

## ECOUI-024

(error) Bundle %s has no net object.

Feedback

**Description**

You receive this error message when the specified bundle has no net object.

**What Next**

Specify valid bundles which must have net objects for the command option.

---

## ECOUI-025

(error) Driver pin %s and load pin %s do not belong to the same net.

**Description**

You receive this error message when the pins belong to different nets.

**What Next**

Specify correct pins for the options which must belong to the same net.

---

## ECOUI-026

(error) Failed to set bus guide preferences for %s bus bundles {%s}.

**Description**

You receive this error message when command failed to set bus guide preferences for the bundles listed.

**What Next**

Check the failure reason of the bundles listed.

---

## ECOUI-027

(error) Failed to add bus buffer arrays for %s.

**Description**

You receive this error message when command failed to add bus buffer arrays for the bundle listed.

**What Next**

Check the failure reason of the bundle listed.

---

## ECOUI-028

(error) Check multiple loads of %s failed for reason: %s.

Feedback

**Description**

You receive this error message when command failed to do multiple loads checking.

**What Next**

Check the failed reason listed.

---

## ECOUI-029

(error) Fail to get unit site info.

**Description**

Fail to get unit site info for this command.

**What Next**

Please check if there is site def and site row in design.

**See Also**

- get_site_defs
- get_site_rows

---

## ECOUI-030

(error) Check command option %s failed for reason: %s

**Description**

You receive this error message when command failed to check the specified command option.

**What Next**

Check the failed reason listed.

---

## ECOUI-031

(error) Fail to get via def.

**Description**

Fail to get necessary via def for this command.

**What Next**

Create via def or not remove via def.

Feedback

## ECOUI-033

(error) input cell %s already has a parent cell group %s.

### Description

You receive this error message when the input cell already has a parent cell group and command could not create cell group for this cell.

## ECOUI-034

(Error) Library cell '%s' is an inverter.

### Description

You recieve this message, if you have specified an inverter library cell for *buffer_lib_cell* option without specifying the *-inverter_pair* option.

### What Next

a) specify a library buffer cell.

b) specify -inverter_pair option along with inverter library cell.

### See Also

- add_buffer
- report_lib

## ECOUI-035

(error) There still are %s eco cells with psc_type_id created by spare cell awared add_buffer_on_route or size_cell not in the target list of command.

### Description

The command must do place_freeze_silicon -no_spare_cell_swapping or -map_spare_cells_only for all the eco cells which have psc_type_id and were created by spare cell awared add_buffer_one_route and size_cell at the same time.

### What Next

Please find out all eco cells with eco_change_status as add_buffer_on_route or size_cell and the attribute fs_psc_is_spread_in_slot as false. Then apply place_freeze_silicon -no_spare_cell_swapping or -map_spare_cells_only on such cells first.

Feedback

**See Also**

- place_freeze_silicon
- get_attribute

---

## ECOUI-036

(error) value for list '%s' cannot have %d elements.

**Description**

The value given for the list argument does not have the correct number of elements. Some commands have list arguments which require either a specific number or an even number of elements. The message will indicate which it is.

**What Next**

Supply a correct number of elements in the list.

**See Also**

- add_buffer

---

## ECOUI-037

(error) Could not find the block by name "%s" for option "%s".

**Description**

The specified block does not exist or the block name format is not correct. Please check whether the block exists. If the block exists, then check the option man page and use the correct block name format again.

**What Next**

Provide the block with correct name format for option.

**See Also**

- eco_netlist

---

## ECOUI-038

(error) Currently the spare cell aware add_buffer_on_route does not support %s.

Feedback

**Description**

Currently the specified option cannot be used in spare cell aware add_buffer_on_route.

**What Next**

Remove the specified option or disable spare cell aware feature by two methods: (1) Use option "-not_spare_cell_aware" to disable the spare cell aware feature in this command. (2) Use app option "eco.freeze_silicon.not_spare_cell_aware" to disable the spare cell aware feature for both add_buffer_one_route and size_cell in the whole freeze-silicon flow.

When command is not spare cell aware, please be careful about the spare cell mapping result of these new eco cells in the subsequent flow.

**See Also**

- add_buffer_on_route

---

## ECOUI-039

(error) Option "%s" can only be used in freeze-silicon mode.

**Description**

The specified option can only be used in freeze-silicon mode.

**What Next**

Remove the specified option or enable freeze-silicon mode.

**See Also**

- size_cell
- add_buffer_on_route

---

## ECOUI-040

(error) Failed to get the specified voltage areas.

**Description**

The value given for the voltage area list does not have the correct voltage area.

**What Next**

Supply correct voltage areas in the list.

Feedback

**See Also**

- spread_spare_cells

---

## ECOUI-041

(error) Cell %s is not spare cell.

**Description**

The given cell is not spare cell.

**What Next**

Specify a spare cell, or set the attribute "is_spare_cell" if you are sure the given cell is a spare cell. Then rerun the command.

**See Also**

- spread_spare_cells
- map_freeze_silicon

---

## ECOUI-042

(error) The target %s (%s) is outside the user specified boundary.

**Description**

The target region to spread spare cells is not within the user specified boundary.

**What Next**

Refine the boundary with the option -boundary to cover the target region. Or remove the spare cells belong to this target region from the given cell list. Or remove the target voltage area from the given voltage area list.

**See Also**

- spread_spare_cells

---

## ECOUI-043

(error) No placeable area for target %s (%s).

**Description**

There is no placeable area in the target region.

Feedback

**What Next**

Check the effective regions of target region.

**See Also**

- spread_spare_cells

---

## ECOUI-044

(error) There is no spare cell to be spread.

**Description**

There is no spare cell to be spread.

**What Next**

Please check the spare cells in top design or in any voltage area.

**See Also**

- spread_spare_cells

---

## ECOUI-045

(error) There is no eco cell to be placed.

**Description**

There is no eco cell to be spread.

**What Next**

Please specify eco cells to be placed.

**See Also**

- place_freeze_silicon

---

## ECOUI-046

(error) The spare cell aware feature is turned on, but there are completely trial mapped eco cells which have psc_type_id but have not been real swapped.

Feedback

**Description**

When the command runs with spare cell aware feature, it is not allowed that there are completely trial mapped eco cells (which have psc_type_id) in design.

**What Next**

Please find out all eco cells with the attribute fs_psc_is_spread_in_slot as true and apply place_freeze_silicon -map_spare_cells_only on such cells first.

Or turn off the spare cell awared feature by specifying option "-not_spare_cell_aware" for the command.

**See Also**

- add_buffer_on_route
- size_cell
- place_freeze_silicon
- get_attribute

---

## ECOUI-047

(error) Command failed to process %s. No change will be applied on it.

**Description**

Command failed when processing object and will not apply any change on it.

**What Next**

Fix the problem and re-run the command.

---

## ECOUI-048

(error) There are programmable spare cell mapping rules set and the option %s cannot be used with the mapping rules together.

**Description**

The command does not allow user to do place_freeze_silicon -no_spare_cell_swapping or -write_map_file when there are programmable spare cell (PSC) mapping rules set. Currently the trial mapping feature cannot be used with the PSC mapping rules together.

**What Next**

Please turn off this option or remove all the existing PSC mapping rules, then rerun place_freeze_silicon.

Feedback

**See Also**

- place_freeze_silicon
- report_programmable_spare_cell_mapping_rule
- remove_programmable_spare_cell_mapping_rule

## ECOUI-049

(error) In the input list of option "%s", there are two PSC types %u and %u have the same lib cell height.

**Description**

You receive this error message when you attempt to set two PSC types in the input list of this option, but they have the same lib cell height or there is no lib cell has been set with these PSC types.

**What Next**

Make sure there are lib cells have been set with these two PSC types and the input PSC types of this option must have different lib cell height.

**See Also**

- set_programmable_spare_cell_mapping_rule
- set_attribute

## ECOUI-050

(error) Invalid name for the option '%s'.

**Description**

You receive this error message when you specify an invalid name for the option. This command option requires a simple name without the hierarchy separator character.

**What Next**

Specify a name without the hierarchy separator character for this option.

## ECOUI-051

(error) Cannot instantiate current_design within itself.

Feedback

**Description**

The *add_spare_cells* command was used to try to create a cell in the current design that referenced the current design. This would create a cycle in the design hierarchy, which is not allowed.

**What Next**

Choose a different *reference* for the command.

---

## ECOUI-052

(error) Design '%s' is not in the reference library list.

**Description**

The referenced design must belong to the current library containing the current design, or a reference library of the current library.

**What Next**

Either choose a different library for the reference design, or add the desired library to the current design's library reference library list. refer to NDMUI-079 for more detail about design and its libarary

---

## ECOUI-053

(error) Invalid reference library cells for the option '%s'.

**Description**

You receive this error message when you specify invalid reference library cells for the option.

**What Next**

Specify valid reference library cells for this option.

---

## ECOUI-054

(error) Invalid instance numbers for the option '%s'.

**Description**

You receive this error message when you specify invalid instance numbers for the option.

**What Next**

Specify integers larger than one as the instance numbers for this option.

Feedback

## ECOUI-055

(error) Only logical hierarchical cell in the top block is valid for the option '%s'.

**Description**

You receive this error message when you specify an invalid cell name or the specified cell is not a logical hierarchical cell in top block for the option.

**What Next**

Specify logical hierarchical cell in top block for this option.

## ECOUI-056

(error) Only voltage areas in the top block is valid for the option '%s'.

**Description**

You receive this error message when you specify invalid voltage area names or the specified voltage areas are not in the top block for the option.

**What Next**

Specify voltage areas in the top block for this option.

## ECOUI-057

(error) The specified library cell "%s" is programmable spare cell (PSC) type, currently the spare cell aware feature of command does not support it.

**Description**

You receive this error message when you specify PSC type library cell for spare cell aware feature.

**What Next**

Use option "-not_spare_cell_aware" to turn off the spare cell aware feature or use non-PSC type library cell instead.

**See Also**

- size_cell
- add_buffer_on_route

Feedback

## ECOUI-058

(error) The PSC type %u and %u in the compatible type list have different lib cell height.

### Description

You receive this error message when you attempt to set compatible rule between two PSC types by using set_programmable_spare_cell_mapping_rule, but their lib cells have different height. The compatible PSC types must have the same lib cell height.

### What Next

Make sure the compatible types must have the same lib cell height, if you want to set compatible relationship between these PSC types.

### See Also

- set_programmable_spare_cell_mapping_rule

## ECOUI-059

(error) The lib cell height %s of %s type %u is not smaller than the height %s of target PSC type %u.

### Description

You receive this error message when you attempt to set split or merge rule for the target PSC type by using set_programmable_spare_cell_mapping_rule, but the lib cell height of split or merge type is not smaller than the target PSC type.

### What Next

Make sure the lib cell height of split or merge type is smaller than the height of target PSC type, if you want to set split or merge relationship for them.

### See Also

- set_programmable_spare_cell_mapping_rule

## ECOUI-060

(error) The pg stripe layers of type %u and type %u are not consistent.

### Description

You receive this error message when you attempt to set relationship between two types by using set_programmable_spare_cell_mapping_rule, but their pg stripe layer constrain are not consistent. When using the command set_programmable_spare_cell_mapping_rule

Feedback

with option "-compatible", "-multi_height_merge" or "-multi_height_split", the pg stripe layers of the types must be consistent.

**What Next**

Change the pg stripe layer constrain of the types to be consistent if you really want to set relationship between these types.

**See Also**

- set_programmable_spare_cell_mapping_rule

---

## ECOUI-061

(warning) Only support right side currently, ignore the value.

**Description**

You receive this error message when you attempt to specify the value "left" for the option "-side" of set_programmable_spare_cell_mapping_rule. Currently, this option only support "right".

**What Next**

Specify "right" for the option "-side" or just remove this option.

**See Also**

- set_programmable_spare_cell_mapping_rule

---

## ECOUI-062

(warning) The value %d for option '-size' is ignored since not support yet.

**Description**

You receive this error message when you attempt to specify the value for the option "-size" of set_programmable_spare_cell_mapping_rule. Currently, this option is not supported yet.

**What Next**

Just remove this option.

**See Also**

- set_programmable_spare_cell_mapping_rule

Feedback

## ECOUI-063

(error) Must specify at least 2 types for option '-compatible'.

### Description

You receive this error message when you attempt to specify less than 2 types for the option "-compatible" of the command set_programmable_spare_cell_mapping_rule.

### What Next

Specify more types for the option "-compatible".

### See Also

- set_programmable_spare_cell_mapping_rule

## ECOUI-064

(error) No enough information for creating mapping rule.

### Description

You receive this error message when you attempt to create psc mapping rule by using the command set_programmable_spare_cell_mapping_rule with the option "-psc_type_id" but missing other informations. When user specify the option "-psc_type_id", at least one of the following options need to be specified together: "-dont_overlap_pg_stripe_layer", "-partial_overlap_pg_stripe_layer", "-multi_height_merge", "-multi_height_split".

### What Next

Specify at least one of the above four options with "-psc_type_id".

### See Also

- set_programmable_spare_cell_mapping_rule

## ECOUI-065

(warning) No psc mapping rule is set for the type %u.

### Description

You receive this error message when you attempt to remove or report rules for a type which has no rule set.

Feedback

### What Next

Specify the type that has set rule.

### See Also

- remove_programmable_spare_cell_mapping_rule
- report_programmable_spare_cell_mapping_rule

---

## ECOUI-066

(error) Same PSC type %u in options '%s' and '%s'.

### Description

You receive this error message when you attempt to specify same PSC type in the two options of the command set_programmable_spare_cell_mapping_rule. These two options require different PSC types.

### What Next

Specify different types for the two options.

### See Also

- set_programmable_spare_cell_mapping_rule

---

## ECOUI-067

(error) The specified layers of type %u are not consistent with its convertable types.

### Description

You receive this error message when you attempt to specify inconsistent PG stripe layers using the command set_programmable_spare_cell_mapping_rule. The specified type has convertable types and some of these type are PG stripe aware for some layers, but the specified layers are not consistent with there layers.

### What Next

User report_programmable_spare_cell_mapping_rule to get the detal layers information and specify the consistent layers. If you really want to use the specified layers, please remove other related rules first.

Feedback

**See Also**

- set_programmable_spare_cell_mapping_rule
- report_programmable_spare_cell_mapping_rule

## ECOUI-068

(warning) The "%s" is not supported by the spare cell aware feature of command. The spare cell aware feature will be disabled automatically.

**Description**

You receive this warning message when you specify one option for the spare cell aware feature of command which is not supported. So the spare cell aware feature will be disabled automatically.

**What Next**

To avoid this warning message, use option "-not_spare_cell_aware" to disable the spare cell aware feature in this command or use app option "eco.freeze_silicon.not_spare_cell_aware" to disable the spare cell aware feature for both add_buffer_one_route and size_cell in the whole freeze-silicon flow.

Since the command is not spare cell aware, please be careful about the spare cell mapping result of these new eco cells in the subsequent flow.

**See Also**

- add_buffer_on_route

## ECOUI-069

(error) Window size should be greater than 0.

**Description**

You receive this error message when specify invalid window size for the option.

**What Next**

Specify the correct window size for the option.

## ECOUI-070

(error) There is no bound to be placed.

Feedback

**Description**

There is no bound to be placed.

**What Next**

Please check the command input.

---

## ECOUI-071

(error) The input value inner_length_ratio is not valid.

**Description**

The inner_length_ratio should be greater than or equal to 0, and less than 1.

**What Next**

Please input the valid value.

**See Also**

- set_attribute

---

## ECOUI-072

(warning) There are programmable spare cell mapping rules set, which are not supported by the spare cell aware feature of command. The spare cell aware feature will be disabled automatically.

**Description**

You receive this warning message when you set programmable spare cell (PSC) mapping rules previously and the input lib cells of command are also PSC type. Since the spare cell aware feature of command does not support these PSC mapping rules, it will be disabled automatically.

**What Next**

To avoid this warning message, use option "-not_spare_cell_aware" to disable the spare cell aware feature in this command or use app option "eco.freeze_silicon.not_spare_cell_aware" to disable the spare cell aware feature for both add_buffer_one_route and size_cell in the whole freeze-silicon flow.

**See Also**

- add_buffer_on_route
- size_cell

Feedback

## ECOUI-073

(error) The bitmap string '%s' is not valid.

### Description

The specified bitmap string is not valid for the option "-bit_map". And the valid bitmap should follow the rules: 1. the bitmap string can only have charactor '0', '1' and space. 2. The type bits of the contiguous base should have the different value. 3. Space will be internally filtered before bit map checking. 4. The size of bitmap string should be multiple of the specified base length.

### What Next

Please check the bitmap and specify the valid value.

### See Also

- place_freeze_silicon

## ECOUI-074

(error) The input value inner_length is not valid.

### Description

The inner_length should be greater than or equal to 0.

### What Next

Please input the valid value.

### See Also

- set_attribute

## ECOUI-075

(error) No pipeline bound path info matched the specified objects.

### Description

There is no bound path info matched the specified objects.

### What Next

Please check the input.

Feedback

## ECOUI-076

(error) Bound %s is not move bound.

**Description**

The command only supports move bound.

**What Next**

Please check the input.

## ECOUI-077

(error) The site row %s does not belong to the site array %s.

**Description**

You receive this error message when the specified site row does not belong to the specified site array. The site row specified in the option '-allowable_site_rows' should belong to the site array specified in the option '-site_array'.

**What Next**

Please check the site row name and site array name to make sure they are consistent.

**See Also**

- set_programmable_spare_cell_mapping_rule

## ECOUI-078

(error) Invalid value for option '-allowable_site_rows'.

**Description**

You receive this error message when the specified value for the option '-allowable_site_rows' is not valid. The valid format is {row_name incre_num} and incre_num should be greater than 0.

**What Next**

Please specify the right value for the option '-allowable_site_rows'.

**See Also**

- set_programmable_spare_cell_mapping_rule

Feedback

## ECOUI-079

(error) No site row matched the specified name %s.

### Description

You receive this error message when the specified site row name is not valid and no site row can be found to match the name in the option '-allowable_site_rows'.

### What Next

Please check the site row name and specify the correct site row name.

### See Also

- set_programmable_spare_cell_mapping_rule

## ECOUI-080

(error) No valid library cell found for the option '%s'.

### Description

You receive this error message when you specify invalid value for the option and command could not find the target library cells with this specified option value.

### What Next

Specify valid value for this option.

## ECOUI-081

(warning) Some of the input bundles belong to the paths of the input bounds and these bundles will be ingored: { %s }.

### Description

Some of the input bundles are long to the paths of the input bounds. Since they are duplicated with the bounds, these bundles will be ingored when do the bus guide routing.

### What Next

Please check the input bundles whether belong to the paths of the input bounds.

## ECOUI-082

(Error) Must specify at least one of these options '%s'.

Feedback

**Description**

User must specify at least one of these options.

**What Next**

Add at least one of the specified options to command and run again.

---

## ECOUI-083

(Error) There are %d capacity values specified for option "-capacity", but which needs %d capacity value(s) input.

**Description**

User must input correct number of capacity values. For "-bounds" and "-bundles", When only of them is specified, the "-capacity" option can only support one capacity value input. But when both of "-bounds" and "-bundles" are specfiied, user needs input two capacity values for the "-capacity" option, especially the first capacity value is set for "-bounds" and the second capacity value is set for "-bounds".

**What Next**

Please input correct number of capacity values for "-capacity" option and rerun command again.

---

## ECOUI-084

(Error) The location list is not specified correctly: %s.

**Description**

User must input correct location list for option, like {x1 y1 x2 y2 ...}. The value must be valid coordinate.

**What Next**

Please input correct location list for option and rerun again.

---

## ECOUI-085

(Error) Lib cell %s has %d %s, %s.

**Description**

This command can only support lib cell with single input pin and single output pin. If the lib cell has multiple input/output pins, must specify only one explicitly.

Feedback

**What Next**

Specify one input/output pin and re-run the command.

---

## ECOUI-086

(warning) Skip %d input cells to revert, the cells are {%s}

**Description**

Some input cells are skipped to revert since the attribute eco_change_status is not size_cell/add_buffer/add_buffer_on_route.

---

## ECOUI-088

(error) The %s pin is not specified correctly for %s cell %s.

**Description**

This option -lib_cell_input/-lib_cell_output is not correctly specified.

**What Next**

Please provide the specified pins of the cell.

---

## ECOUI-089

(error) Invalid value for option '%s', you can specify '%s' for this option.

**Description**

You receive this error message when you specify an invalid name for the option. This command option requires only two pattern.

**What Next**

Correct and re-run.

---

## ECOUI-090

(warning) Option '%s' is only effective in single-load nets or bundles.

**Description**

You get the warning when using this option in multi-load nets or bundles.

**What Next**

The command continues but the result may or may not honor the option.

Feedback

## ECOUI-091

(error) Option '%s' is not allowed in multi-load nets or bundles.

**Description**

You get the warning when using this option in multi-load nets or bundles.

**What Next**

Do no use the option with the specified nets or bundles, or use this option with single-load nets or bundles.

## ECOUI-092

(error) Cannot change_link on cell: '%s' because the design '%s' not compatible with current lib_cell '%s'.

**Description**

The *change_link* command was unable to change the lib_cell used for the specified cell. The new lib_cell either has a different port interface, or a different function from the current lib_cell.

The *change_link* command cannot be used to change the functionality or the connectivity of the net list.

**What Next**

Try using a different design when change_link for this cell.

## ECOUI-093

(error) Cannot change_link on cell '%s', the new reference design %s has a different port interface.

**Description**

The *change_link* command was unable to set the cell to the new reference the given design. The new reference design has a different port interface from the current design.

The *change_link* command cannot be used to change the connectivity of the net list.

**What Next**

Try using a different lib_cell when sizing this cell.

Feedback

## ECOUI-094

(error) Cannot change_link on cell '%s', it is an unresolved reference to '%s'.

**Description**

The *change_link* command was unable to set the cell to reference the new design. The cell's current reference is unresolved or unbound.

The *change_link* command cannot be used to change the connectivity of the net list.

**What Next**

Use the *link_design -force* command to resolve the cell's reference. Then use *change_link* to change the reference.

## ECOUI-095

(error) Value of option %s must be set to %s when %s .

**Description**

This message shows that some option must be set to certain value in certain condition.

**What Next**

Fix the command line.

## ECOUI-096

(error) Value of option %s must be %s the value of option %s.

**Description**

This message shows that some option must be in certain condition compared to the other option.

**What Next**

Fix the command line.

## ECOUI-097

(error) Please start GUI before using -preview option in add_group_repeaters.

**Description**

This message shows that preview option is used without starting GUI.

Feedback

**What Next**

Use command start_gui before using -preview option in add_group_repeaters.

---

## ECOUI-098

(error) There does not exist the %s pin named %s in cell %s.

**Description**

This option -lib_cell_input/-lib_cell_output is not correctly specified.

**What Next**

Please provide the correct pins of the cell.

---

## ECOUI-099

(error) Module with name '%s' not found in the library.

**Description**

A module with the specified name must already exist in the library. for the command to work properly.

**What Next**

Specify a design which already exists in the library.

---

## ECOUI-100

(error) Fail to find the core area.

**Description**

The The core area is not found in the design.

**What Next**

Specify a design with the core area.

---

## ECOUI-101

(error) Bidirectional pin/port:%s cannot be buffered.

**Description**

Bidirectional pin/port can not be buffered.

Feedback

### What Next

Choose the non-bidirectional pin to buffer.

---

## ECOUI-102

(error) The inverter %s is not paired.

### Description

This error occurs when tool detects that the inverters to be removed are not paired.

### What Next

Please check the inverters specified to remove are paired.

---

## ECOUI-103

(Warning) Not relinking cell '%s'.

### Description

This warning message occurs when the above cell is already linked with the specified target library cell.

### What Next

This is only a warning message.

However, the result is not what is expected, specify the reference library cell different from the current one.

### See Also

- size_cell

---

## ECOUI-104

(Error) No buffer(s) or inverter-pair(s) connected to net '%s'.

### Description

This error message occurs when the *remove_buffers* command is not able to find any buffer or inverter-pair for the given net.

### What Next

Report the buffer tree for the net using *report_cell* and *report_net* commands and see if there are any buffer or inverter-pairs associated with it.

Feedback

**See Also**

- remove_buffers

---

## ECOUI-105

(error) Can not change link for hierarchical instance %s.

**Description**

The change_link command can only work on leaf level instances. The reference of hierrachical instances can not be changed using change_link.

**What Next**

Please provide correct inputs to the command

---

## ECOUI-106

(Warning) Not relinking cell '%s'.

**Description**

This warning message occurs when the above cell is already linked with the specified target library cell.

**What Next**

This is only a warning message.

However, the result is not what is expected, specify the reference library cell different from the current one.

**See Also**

- size_cell
- change_link

---

## ECOUI-107

(Error) Hierarchy Deliminator is detected for '%s' for option: -%s.

**Description**

This error message occurs if the input name contains hiearchy delimiantor.

Feedback

**What Next**

specify the name without hiearchy deliminator.

**See Also**

- add_buffer

---

## ECOUI-108

(Error) No net connection for %s.

**Description**

This error message occurs if pin or port is not connected to the net.

**What Next**

specify the name with net connection.

**See Also**

- add_buffer

---

## ECOUI-109

(Error) The pin '%s' is PG.

**Description**

This error message occurs if PG pin is used

**What Next**

specify pin of the signal type.

**See Also**

- add_buffer

---

## ECOUI-110

(error) Input lib cell '%s' is not PSC filer.

**Description**

You receive this error message when you specify Non-PSC filler library cell for the option.

Feedback

**What Next**

Specify valid PSC filler reference library cells for this option. Make sure the library cells have correct attribute of psc_type_id.

---

## ECOUI-111

(error) Input cell '%s' in option '%s' cannot be recycled.

**Description**

This command only supports recycling spare cell which comes from PSC filler. It must have attribute "psc_filler_is_used" TRUE.

**What Next**

Specify valid cells for this option, you can check whether the cells have attribute 'psc_filler_is_used' set with true value or not.

---

## ECOUI-112

(error) There is no valid cell specified in option '%s'.

**Description**

You receive this error message when you use the option with no valid cell specified.

**What Next**

Specify valid cells for this option or run command without this option.

---

## ECOUI-113

(warning) This command does not check the polarity of the signal. Please ensure the inverters provided are in pair.

**Description**

Pay attention that this command will not check the polarity of the signal.

**What Next**

Please ensure that the number of inverters is even and they are in pair.

---

## ECOUI-114

(error) Failed to add diode cells for %s loads {%s}.

Feedback

**Description**

You receive this error message when command failed to add diode cells for the load listed.

**What Next**

Check the failure reason of the loads listed.

---

## ECOUI-115

(error) Input %s for option '%s' is invalid: %s.

**Description**

The input value for the list option is invalid. Please check the specified reason.

**What Next**

Please input correct value for the listed option and try again.

---

## ECOUI-116

(error) The block "%s" of "%s" is uneditable.

**Description**

This message shows that the hierarchical edit control of the specified block is false.

**What Next**

Check the editability of the related block.

---

## ECOUI-118

(Warning) %ld nets have no wire.

**Description**

You receive this error message when the net has no wire. These nets will be skipped. Use -verbose option to list net names.

**What Next**

Please check the route status of the net.

---

## ECOUI-119

(Warning) %ld nets have no repeater inserted.

Feedback

**Description**

You receive this error message after the add_group_repeaters command, there is no new repeater in some nets. Use -verbose option to list net names.

**What Next**

Please check the net that fails to insert the repeaters.

---

## ECOUI-120

(error) Failed to initialize Python.

**Description**

You receive this error message when something is wrong in Python.

**What Next**

Run it again.

---

## ECOUI-122

(Warning) Skip signal polarity checking for inverting repeaters.

**Description**

add_group_repeaters will not check polarity when inserting repeater groups.

---

## ECOUI-124

(error) The topology plan or the topology repeater (%s) does not belong to the current block and it is not supported by the command %s.

**Description**

You receive this error message when you try to run the command on a topology plan or a topology repeater not belonging to the current block.

**What Next**

Run this command with topology plans or topology repeaters belonging to the current block.

---

## ECOUI-130

(Error) The current design is not linked.

Feedback

**Description**

This error message occurs when the current design is not linked.

**What Next**

Please link the design first.

---

## ECOUI-131

(Warning) No repeater is added due to %s.

**Description**

You receive this error message after the add_group_repeaters command, there is no new repeater for the net group.

**What Next**

Please check the nets and the insertion rule that failed to insert the repeaters.

---

## ECOUI-135

(error) The provided shapes are invalid in layer %s.

**Description**

The tool cannot recognize the shapes information. Please double check.

**What Next**

Please correct the geometry list for each layer.

---

## ECOUI-136

(error) The repeater group (%d) is invalid.

**Description**

The tool cannot recognize the input group id.

**What Next**

Please double check if the input group id exists by report_repeater_groups.

Feedback

**See Also**

- place_group_repeaters
- set_repeater_group
- report_repeater_groups

---

## ECOUI-137

(error) Incorrect value format for option '%s', which should be '%s'.

**Description**

The value given for the list argument does not have the correct number of elements. Some commands have list arguments which require either a specific number or an even number of elements. The message will indicate which it is.

**What Next**

Please see the man page of the command and input a correct number of elements in the list.

---

## ECOUI-138

(error) Incorrect value found in option '%s': %s.

**Description**

The value given for this option is not correct for the specified reason.

**What Next**

Please check the specified error reason and input a correct value.

---

## ECOUI-139

(error) The lib cell of cell (%s) is not same as value (%s) by -lib_cell.

**Description**

The value given for this option is not correct for the specified reason.

**What Next**

The lib cell of cells by -cells should be same as value by -lib_cell.

Feedback

**See Also**

- add_eco_repeater

---

## ECOUI-140

(error) The lib cell %s doesn't have lib pin %s.

**Description**

The value given for this option is not correct for the specified reason.

**What Next**

The lib pin name specified by -pin_mapping should exist on the lib cell by -lib_cell.

**See Also**

- add_eco_repeater

---

## ECOUI-141

(error) The direction of lib pin %s of lib cell %s is not expected direction %s.

**Description**

The value given for this option is not correct for the specified reason.

**What Next**

The format of -pin_mapping should be { {inp1 outp1} {inp2 outp2}...}.

**See Also**

- add_eco_repeater

---

## ECOUI-142

(error) could not buffer for the input location %s, because %s.

**Description**

You receive this error message when command detects some error for the input buffering location.

**What Next**

Please check the failed reason reported. Then rerun with correct buffering location or other options again.

Feedback

**See Also**

- add_buffer_on_route

---

## ECOUI-143

(error) There is no reference lib cell defined in the option %s for voltage area %s.

**Description**

You receive this error message when command detects the target voltage area (VA) has no reference libaray cell defined in the specified option. This would cause MV violation when command adds buffer(s) in this VA and the default used reference library cell is not legal for this VA.

**What Next**

Please check the target voltage area and define proper reference library cells in the specified option for it.

**See Also**

- add_buffer_on_route

---

## ECOUI-144

(error) No voltage area region is bufferable for target net %s when honor MV-aware.

**Description**

You receive this error message when there is no bufferable voltage area region for target net considering MV-aware.

**What Next**

Please check the target net and its crossed voltage areas.

**See Also**

- add_buffer_on_route

---

## ECOUI-145

(error) Option '%s' cannot be used in freeze-silicon mode unless option '-not_spare_cell_aware' is turned on.

Feedback

**Description**

In freeze-silicon mode, command will enable spare cell aware feature by default.But the specified option has conflict with spare cell aware feature which needs voltage-area aware, so this option cannot be used in freeze-silicon mode.

If you still want to use this option in freeze-silicon mode, you can disable the spare cell aware by option '-not_spare_cell_aware". But be aware that the buffers added may not have corresponding spare cells nearby and would be moved far away from routes by place_freeze_silicon command followed.

**What Next**

Considering the case situation, not use this option or turn on "-not_spare_cell_aware" to allow using this option in freeze-silicon mode.

**See Also**

- add_buffer_on_route

---

## ECOUI-146

(Info) Do not treat lower layer routes as blockages.

**Description**

add_group_repeaters/place_group_repeaters will ignore lower layer routes and do not treat them as blockages.

---

## ECOUI-147

(Info) Disable automatic repeater distance adjustment to spread repeaters evenly.

**Description**

add_group_repeaters will not adjust distance between repeaters to avoid short distance to load.

---

## ECOUI-149

(error) The load number of {%s} is %d which is not equal to bunlde net number %d.

**Description**

You receive this error message when the load number in one sub path is not equal to the bundle net number.

Feedback

**What Next**

Please specify the load collection group with correct object number.

---

## ECOUI-150

(error) Input path point list of option '%s' have %d sub paths, but the load group number is only %d.

**Description**

You receive this error message when total load group number is not equal to sub path number. For multiple fanout scenario, each sub path should have the corresponding load collection group input.

**What Next**

Please specify correct load groups.

---

## ECOUI-151

(error) Cannot specify -clear when trying to create a new repeater group.

**Description**

You receive this error message when you are trying to clear path drivers, path loads or the driver group id while creating a new repeater group

**What Next**

Please specify -clear only for an existing repeater group

---

## ECOUI-152

(error) There is no base region on layer %s, in this case, you cannot create a user track pattern that track type of first track is S.

**Description**

You receive this error message when you try to create a user track pattern that track type of first track is S, on a layer without base region.

**What Next**

Please add base region on the layer before create the track pattern.

Feedback

## ECOUI-153

(error) There is base region %s on layer %s already, only 1 base region is allowed on a layer.

**Description**

You receive this error message when you try to add base region on a layer that contains base region already.

**What Next**

Please remove the existing base region on the layer before add a new base region.

## ECOUI-154

(error) There is %s %s on layer %s already, same name for base region or user track pattern is not allowed.

**Description**

You receive this error message when the name of the base region or the user track pattern that you try to create is exiting.

**What Next**

Please create the base region or user track pattern with a different name.

## ECOUI-155

(error) On layer %s, the direction of %s %s is %s, which is incompatible with the direction of current %s.

**Description**

You receive this error message when the direction of the base region or the user track pattern that you try to create is incompatible with the direction of existing base region or user track pattern on the layer.

**What Next**

Please create the base region or user track pattern with a compatible direction.

## ECOUI-156

(error) On layer %s, the first track's type of user track pattern %s is S, then user track pattern's offset, which is %s, must be larger than the offset of base region %s, which is %s.

Feedback

**Description**

You receive this error message when the offset of user track pattern that the first track's type is S, is not larger than the offset of the base region.

**What Next**

Please create the base region or user track pattern with a compatible offset.

---

## ECOUI-157

(error) On layer %s, the first track's type of user track pattern %s is P or G, then user track pattern's offset, which is %s, must be equal to the offset of base region %s, which is %s, plus zero or one or multiple times of base region's length, which is %s.

**Description**

You receive this error message when the offset of user track pattern that first track's type is P or G, is not equal to base region's offset plus zero or one or multiple times of base region's length.

**What Next**

Please create the base region or user track pattern with a compatible offset.

---

## ECOUI-158

(error) On layer %s, width of track pattern %s, which is %s, must be equal to one or multiple times of the length of base region %s, which is %s.

**Description**

You receive this error message when the width of user track pattern is not equal to one or multiple times of base region's length.

**What Next**

Please create the base region or user track pattern with a compatible width.

---

## ECOUI-159

(error) Failed to process input supernet bundle %s for the reason: %s.

**Description**

You receive this error message when command failed to process input supernet bundle. The supernet of bundle must be in current block only and does not contain any net segment in sub blocks.

Feedback

### What Next

Please check the failed reason and resolve it.

---

## ECOUI-160

(error) The track pattern (%s) is invalid.

### Description

The tool cannot recognize the input track pattern.

### What Next

Please double check if the input track pattern exists by report_user_track_pattern.

---

## ECOUI-162

(error) The lib cell (%s) specified by -lib_cell is not multi-bit buffer or inverter.

### Description

The input lib cell is not valid multi-bit buffer or inverter.

### What Next

The function id of multi-bit buffer is "a1.0(0)_a1.0(1)". The function id of multi-bit inverter is "Ia1.0(0)_Ia1.0(1)". You can query the function id by get_attribute [get_lib_cells $libCell] function_id. Please input the valid lib cell when -multi_bit is on.

### See Also

- add_eco_repeater
- get_attribute

---

## ECOUI-164

(warning) The value %s for option %s is larger than 20, in this case, the result will be unreliable.

### Description

You receive this warning message when specified value for option "-max_distance_for_incomplete_route" is larger than 20, in this case, the result will be unreliable.

Feedback

### What Next

Specify "-max_distance_for_incomplete_route" option with value smaller than 20 when needed.

### See Also

- add_buffer_on_route
- add_group_repeaters
- place_group_repeaters

---

## ECOUI-165

(warning) The value %s for app option eco.place.itr_max_incomplete_route_distance is larger than 20, in this case, the result will be unreliable.

### Description

You receive this warning message when specified value for app option eco.place.itr_max_incomplete_route_distance is larger than 20, in this case, the result will be unreliable.

### What Next

Specify eco.place.itr_max_incomplete_route_distance app option with value smaller than 20 when needed.

### See Also

- implement_topology_repeaters

---

## ECOUI-167

(Error) Dual-rail buffer is required when inserting buffers in secondary PG regions only.

---

## ECOUI-168

(Error) Option %s cannot be specified with -secondary_pg_region.

---

## ECOUI-185

(error) The PG net or the tie net is not allowed for this command.

Feedback

**Description**

The PG net or the tie net is not allowed for this command.

**What Next**

Run the command with correct data

---

## ECOUI-205

(error) Cell %s is not supported, because it is not a leaf level cell.

**Description**

This command can only work on leaf level cells.

**What Next**

Please provide correct inputs to the command

---

## ECOUI-206

(info) %s is specified more than one times

**Description**

The reported object is specified more than one times in option.

---

## ECOUI-207

(error) Both %s and %s has same output net.

**Description**

This command dose not support multiple driving situation.

**What Next**

Please provide correct inputs to the command

---

## ECOUI-208

(error) %s is to remove as first inverter of a pair, but not all of its loads are to remove.

**Description**

An inverter is to remove as first inverter of a pair, but not all of its loads are to remove. Such situation will cause rest loads be driven by inverted signal.

Feedback

**What Next**

Please provide correct inputs to the command

---

## ECOUI-209

(error) Not support buffer '%s' because it dose not has 2 pins.

**Description**

The command can only support buffer that have 2 pins. Buffers with more or less than 2 pins is not supported by this command.

The command doesn't know how to process the connections for cells that have other than 2 pins.

**What Next**

Manually using basic editing commands.

---

## ECOUI-210

(error) Unable to create %s '%s'; %s.

**Description**

The object cannot be created for the specified reason.

---

## ECOUI-211

(error) The net is not specified correctly.

**Description**

This command require that if the specified net exists, it should connecte with the cell. If the pin is specified, it should also connected with this pin.

**What Next**

Check the input of command.

---

## ECOUI-212

(error) This command does not support %s.

**Description**

This command not support this usage or situation.

Feedback

**What Next**

Use this command as its man page.

---

## ECOUI-213

(error) %s does not support %s.

**Description**

This command not support this usage or situation.

**What Next**

Change the situation or use another command.

---

## ECOUI-214

(warning) Timer is now invalid.

**Description**

Timer cannot be incrementally updated and is now invalid. It may take some time to rebuild the timer for the next command which requires it.

---

## ECOUI-215

(error) Line %s in %s is in error. This line will be skipped.

**Description**

This line with error will be skipped. Other lines in the file still will be executed.

**What Next**

Correct this line. Then re-run it through shell or as a line of a new file.

---

## ECOUI-216

(info) %s run in freeze-silicon mode.

**Description**

Detail behavior of freeze-silicon mode can be found in manpage of the command.

Feedback

## ECOUI-217

(error) %s can only run in freeze-silicon mode.

**Description**

This command must be run in freeze-silicon mode.

**What Next**

Set app option design.eco_freeze_silicon_mode true

## ECOUI-218

(error) Cell instances of same cell %s are with different %s in different hierarchy of same module.

**Description**

This command does not support this situation in module based editing

**What Next**

Make cell instances of same cell be same before using module based editing. Or you can use instance based editing instead.

## ECOUI-219

(error) Syntax error: too many fields are specified at line %s.

**Description**

Too many fields are specified at this line. Please read this command's man page for the right format of file.

**What Next**

Correct this line. Then re-run it through shell or as a line of a new file.

## ECOUI-220

(error) %s specified at line %s could not be found in design.

**Description**

Objects specified could not be found in design.

Feedback

**What Next**

Correct this line. Then re-run it through shell or as a line of a new file.

---

## ECOUI-221

(error) Invalid value %s for field %s specified at line %s.

**Description**

This value for this field is invalid. Please view this command's manpage to see the right format of file.

**What Next**

Correct this line. Then re-run it through shell or as a line of a new file.

---

## ECOUI-222

(error) Syntax error: required filed %s is not specified at line %s.

**Description**

The command could not find this required field at this line. Please read this command's man page for the right format of file.

**What Next**

Correct this line. Then re-run it through shell or as a line of a new file.

---

## ECOUI-223

(error) Syntax error: missing close-brace '}' at line %s.

**Description**

Braces must be balanced. Please read this command's man page for the right format of file.

**What Next**

Correct this line. Then re-run it through shell or as a line of a new file.

---

## ECOUI-224

(error) %s is specified more than one times %s, each %s can only be specified one time.

Feedback

**Description**

This object can only be specified one time.

**What Next**

Remove duplicated specified objects and rerun the command.

---

## ECOUI-225

(info) %s of %s specified mappings succeed%s.

**Description**

This info shows how many mappings succeed. Failed lines will be listed after this message, unless all lines are failed. It is a summary. Details of failed lines have been printed in front of this message.

**What Next**

Correct the failed mappings. Then re-run them through shell or as a line of a new file.

---

## ECOUI-226

(error) %s is not a programmable spare cell (PSC) filler.

**Description**

A programmable spare cell (PSC) filler must be with psc_type_id and its design type must be filler.

**What Next**

Correct and re-run.

---

## ECOUI-227

(error) Value of option %s must be set to %s when using option %s .

**Description**

This message shows that some option must be used with other option with certain values.

**What Next**

Fix the command line.

Feedback

## ECOUI-228

(error) Option -no_spare_cell_swapping and -map_spare_cells_only cannot be used in PSC (Programmable Spare Cells) eco flow (also known as gate array flow).

### Description

This error message occurs when there are eco cells with PSC type to be mapped while option -no_spare_cell_swapping or -map_spare_cells_only is turned on.

### What Next

Turn off -no_spare_cell_swapping and -map_spare_cells_only and run place_freeze_silicon again.

## ECOUI-230

(error) The attribute fs_dont_use of cell %s is true.

### Description

The spare cell with the attribute fs_dont_use as true cannot be used when mapping.

### What Next

Set the attribute fs_dont_use as false and do mapping again.

### See Also

- set_attribute
- map_freeze_silicon
- place_freeze_silicon

## ECOUI-231

(Error) Inverter cell '%s' cannot be used with option -on_route or -max_fanout.

### Description

Inverter cell cannot be used when -on_route or -max_fanout option is specified.

### What Next

Use a buffer cell instead, or remove -on_route or -max_fanout option.

Feedback

## ECOUI-232

(Error) Failed to turn on eco.placement.enable_new_advanced_filler_removal with advanced legalizer

**Description**

When advanced legalizer is enabled, place_eco_cells should have turned on app option eco.placement.enable_new_advanced_filler_removal automatically. Somehow it failed to set this app option true successfully. Hence, this error message is output.

**What Next**

Please double check if the app option place.legalize.enable_advanced_legalizer is set true, then turn on this feature manually as follows. set_app_options -name eco.placement.enable_new_advanced_filler_removal -value true

# EDC

## EDC-001

(Info) Policy for early data check '%s' is '%s'%s.

**Description**

This message informs the user about the configured policy for a failing early data check. Early data checks will fail if the data is incomplete, inconsistent or overconstrained. The policy defines what action should be taken when such a check fails. Possible policies are 'error', 'tolerate' and 'repair'. When the data is still dirty, checks can be configured as 'repair' or 'tolerate', while in the later stages, close to tape-out, the policy will typically be configured as 'error'.

**What Next**

Use command 'set_early_data_check_policy' to modify the policy (error, tolerate, repair) for specific checks.

## EDC-002

(Info) No further messages for early data check '%s' will be generated.

**Description**

This message informs the user that the number of early data check messages that were generated for this check has reached the tool-imposed maximum. No further messages for this check will be issued.

Feedback

**What Next**

Use command 'report_early_data_checks' or 'get_early_data_check_records' to inspect the records that were created for this check.

---

## EDC-003

(Info) No further records for early data check '%s' will be created.

**Description**

This message informs the user that the number of early data check records that were created for this check has reached the tool-imposed maximum. No further records for this check will be created. As a consequence, the early data check records that are in the data based do not reflect all failing objects.

**What Next**

Use command 'report_early_data_checks' or 'get_early_data_check_records' to inspect the records that were created for this check.

---

## EDC-004

(warning) Repair skipped for '%s' for object '%s' as it would cause loss of UPF data.

**Description**

Repaired skipped for early data check as it would cause UPF data loss as part of converting empty logic module to macro.

**What Next**

If you do not want to repair then change policy to 'tolerate'. If still repair is needed then enable early data check repair strategy before loading upf.

---

# ELAB

---

## ELAB-1

(error) %s Different registers '%s' and '%s' exist in the same always/process block. Relative placement register bank will not be supported.

Feedback

**Description**

This error message occurs when different registers exist in the same always/process block. The relative placement register bank coding style requires each always/process block to have only one register array.

**What Next**

Rewrite the RTL code to meet the coding style requirement and run the command again.

---

## ELAB-2

(error) %s Relative placement register bank group '%s' does not contain a latch/FlipFlop.

**Description**

This error message occurs when the relative placement register bank group does not contain a latch or flip-flop.

**What Next**

Remove the relative placement directive for the given always/process block.

---

## ELAB-3

(warning) %s Extraneous argument for system function/task %s ignored.

**Description**

If you provide more arguments than the compiler requires when you make a system function or task call, the compiler ignores the extraneous arguments.

**What Next**

This is a warning message only and requires no action on your part.

---

## ELAB-5

(warning) %s Encountered extra 'dc_script_begin' pragma.

**Description**

The *read* command or *elaborate* command issues this warning code when an extra 'dc_script_begin' is encountered in the vhdl design.

For e.g. -- pragma dc_script_begin -- set_dont_touch find (net, N1) false -- pragma dc_script_end -- pragma dc_script_begin <<-- extra 'dc_script_begin' pragma.

Feedback

**What Next**

Locate the extra 'dc_script_begin' in the vhdl design at the specified line number. Then either provide corresponding 'dc_script_end' pragma or remove it, as intended.

---

## ELAB-6

(warning) %s Encountered extra 'dc_script_end' pragma.

**Description**

The *read* command or *elaborate* command issues this warning code when an extra 'dc_script_end' is encountered in the vhdl design.

For e.g. -- pragma dc_script_begin -- set_dont_touch find (net, N1) false -- pragma dc_script_end -- pragma dc_script_end <<-- extra 'dc_script_end' pragma.

**What Next**

Locate the extra 'dc_script_end' in the vhdl design at the specified line number. Then either provide corresponding 'dc_script_begin' pragma or remove it, as intended.

---

## ELAB-7

(error) %s Disable Presto features in Verification Friendly DC Mode

**Description**

If you use multiple dimension array in Verication Friendly DC Mode, this error message will happen.

**What Next**

Rewrite RTL source codes or remove special Presto directives.

---

## ELAB-8

(warning) %s Disable Presto features in verification friendly mode.

**Description**

This warning message occurs when you use the enum directive and the label directive in verification friendly Design Compiler mode.

**What Next**

This is only a warning message. No action is required.

Feedback

However, if the result is not what you intended, rewrite the RTL source codes or remove the special Presto directives.

---

## ELAB-9

(warning) %s Manual resource sharing pragmas will be ignored.

**Description**

This warning message occurs when the *hdlin_disable_manual_resource_sharing* variable is set to true. All of the manual resource sharing pragmas are ignored. The default value of the variable is false.

**What Next**

This is only a warning message. No action is required.

However, if the result is not what you intended, you can use manual resource sharing by setting the *hdlin_disable_manual_resource_sharing* variable to false.

---

## ELAB-012

(information) %s '%s' has non-canonical array bounds [%d:%d];\n\tusing C-style bounds [%d] would align with the external port names

**Description**

This information message issues when you

- declare an interface (or modport) array as a formal port of your design; and
- its bounds do *not* run from [0:*length*-1], (or in C-style: [*length*]); and
- the module's (or its hierarchical parent's) hdlin_interface_port_ABI was set to an odd number (low-order bit = 1).

If your view of this design remains exclusively inside SystemVerilog, the choice of bounds or direction, when used consistently, makes no difference. But if you require a Verilog-/netlist-level interaction with this design (e.g., to add constraint or power annotations, or to link with a Verilog testbench), you may find non-canonical bounds to be confusing and to increase the risk of human error.

**HOW AND WHY CANONICAL INDEX SETS ARE USED (an ABI oddity)**

Feedback

External port names exist in their own namespace/scope; they do not change how your SV indexing scheme works within the module's scope. External port names only determine how other designs link to your design and how you address the ports of your design cell in later stages of the work flow.

SystemVerilog elaboration introduces separate netlist-level ports for your interface/modport array's contents. Each element of the array adds another copy of its modport list into this design's port list. When by-name port connection syntax is used, list position no longer suffices to identify them; so each "[element].content" port needs a unique external port name, including its element index (or offset). In SV, a top design is *not obliged* to pass an interface array whose bounds are an exact match - only its length(s along corresponding dimensions) must be equal. This is the same rule as applies to bused data ports: widths must match; bounds needn't be identical.

Allowing for the fact that, during a top-down link, the reference cell's port names are unchangeable "givens", there are just three ways to align array element numbering across a named port connection:

- by writing the RTL to manually match the bounds on both sides of an interface array port connection;
- by letting the compiler choose a matching set of "canonical" bounds (hdlin_interface_port_ABI bit[0]=1, the default); or
- by requesting custom-elaboration of the down design for each different combination of array bounds needed by a caller. When hdlin_interface_port_ABI bit[0] is set to 0, these customized elaborations will align with each reference cell's interface array port indices by duplicating the down module's design and inventing a matching set of external port names for each unique instantiation. To *minimize distinct elaborations* (and avoid functional duplication of designs), all the reference cells to any design should agree on a single interface array numbering scheme via one of the first two options.

Note that when part-selects are used to connect a slice of an interface array to an array port, the slice's bounds are always canonical. This limits the flexibility of the manual-matching method and means that (on average) interface arrays declared with canonical (i.e. C-style) bounds need the minimum amount of net-name-translation across port connections.

**What Next**

This is an information-only message. No action is required. Some users may turn off bit[0] of hdlin_interface_port_ABI.

Feedback

**Should you change your RTL to use [0:length-1] indexing?**

If part-selections from an interface array are connected to down design ports and you need a high level of concordance between the internal and external names of ports, we strongly recommend that you make the RTL change (carefully and thoroughly!). Trying to keep an index set reversal in mind throughout the design cycle is asking for trouble. If you can adjust your indexing expressions to work with [0:*length*-1], you'll find that the netlist names correspond to your RTL-source references without translation; one source of later confusion will be eliminated.

**Should you just switch the ABI setting?**

If your interface arrays always use precisely the same bounds and you never need part selection syntax, you will find that setting hdlin_interface_port_ABI to 2 aligns your netlisted port indices with your non-canonical RTL indices. The setting must be applied when the SV source of the interface is analyzed or read and it affects each interface separately.

When refering to port or pin names of interface arrays in constraint annotations, SAIF, UPF, or other netlist-level parts of the flow, always use the indexing convention from the top design's reference cells. Notice that by-name connections may invent (formal port) names whose indices differ from the actual nets being connected. These are the names controlled by the ABI switch setting of the interface.

**See Also**

- suppress_message

---

## ELAB-21

(error) %s Unable to add '%s' to column %d and row %d of group '%s' because that location is already occupied.

**Description**

This error message occurs when the specified location is already occupied by other relative placement group.

**What Next**

Change the location for the given group.

---

## ELAB-22

(warning) %s Columns %d and rows %d of relative placement group '%s' are larger than the specified columns %d and rows %d. The actual size will be used when creating the relative placement group.

Feedback

**Description**

This error message occurs when the size of the relative placement group is larger than the specified size.

**What Next**

Change the size for the specified group and run the command again.

---

## ELAB-23

(error) %s Wrong format of relative placement directive '%s'.

**Description**

This error message occurs when the format of the specified relative placement directive is incorrect.

**What Next**

Change the format of the relative placement directive.

---

## ELAB-24

(error) %s switch hdlin_enable_relative_placement has an incorrect value of '%s'.

**Description**

This error message occurs when the *hdlin_enable_relative_placement* variable has an invalid value. The valid values are *none*, *mux*, *rb*, *mux;rb*, *rb;mux*.

**What Next**

Change the value of *hdlin_enable_relative_placement* to one of the values shown above.

---

## ELAB-28

(warning) %s SCALARED keywords are ignored.

**Description**

You receive this warning to let you know that for synthesis purposes, the compiler ignores the scalared keyword.

**What Next**

This is a warning only and requires no action on your part.

Feedback

## ELAB-30

(error) %s Relative placement group name '%s' does not match end group name '%s'.

**Description**

This error message occurs when the group name specified in the rp_group compiler directive is different from the group name specified in the rp_endgroup compiler directive.

**What Next**

Change the group name in rp_group and rp_endgroup, so the group name is the same in both compiler directives.

## ELAB-31

(warning) %s Relative placement register bank group '%s' does not exist.

**Description**

This warning message occurs when the specified relative placement register bank group does not exist.

**What Next**

This is only a warning message. No action is required.

However, to avoid this warning message, do not place this relative placement group in the hierarchy relative placement group.

## ELAB-32

(warning) %s cell '%s' in relative placement group '%s' has negative row/column '%d'/'%d'.

**Description**

This warning message occurs when the cell in the relative placement group has a negative row/column. The tool does not accept a negative row/column.

**What Next**

This is only a warning message. No action is required.

However, if the result is not what you intended, fix the negative row/column and run the command again.

Feedback

## ELAB-33

(warning) %s SV Assertions are ignored for synthesis since %s is not set to true.

### Description

This warning message issues once for each module elaboration where the Presto HDL Compiler encounters a labelled assertion or assertion severity system task call that must be ignored due to the tcl environment variable setting cited in the message. Because SystemVerilog Assertions (SVA) allow synthesis' optimizers to recognize more "don't care" conditions than otherwise specified by the RTL, they are not applied by default but must be enabled - both as a product feature and also by certifying that individual assertions have proven to be unreachable prior to synthesizing this RTL.

### What Next

This is only a warning message. No action is required.

However, if you want the construct to be either exploited or ignored with no warnings, you must set *hdlin_enable_assertions* to true and also establish its assertion label (with lexical scope) as an element of the *confirmed_SVA()* environment array giving this a true or false value.

## ELAB-34

(warning) %s The configuration '%s' needs to be analyzed; module or interface '%s' will be elaborated but not configured

### Description

This warning message issues when the design of a module or interface should be configured, but the configuration description was either: not analyzed, not readable, or is missing from the indicated design library. Earlier warnings often may give more information.

This generally happens during down-stream linking when an unresolved reference requires a configured design. Configuration instructions are suffixed to the parameter string of some unresolved references when the elaboration of the parent design is configured.

### What Next

This is only a warning message. No action is required, but the result, if it can even be produced, will not embody the indicated configuration.

Check that the configuration's design library has been defined and is found at its specified -path in the file system. The analyzed intermediate file of a Verilog "config" must be accessible during elaboration of every design it might affect.

Feedback

## ELAB-35

(warning) %s %s %s already has a specific %s; others are ignored

### Description

This warning message issues in a Verilog configuration when an *instance* or *cell* clause is repeated (either literally, or effectively) with the same type of expansion clause - either a *liblist* or a *use*. Since only one can be implemented, the first, most-specific, occurence wins and other specifications for the same cell or instance are ignored.

### What Next

This is only a warning message. It is not issued when a more-specific selection clause overrides a less-specific one, only when multiple similarly-specific selections apply to one configuration choice. Review the cited configuration rule and remove duplicates.

## ELAB-36

(error) %s Missing analyzed description of '%s.%s:config'.

### Description

This error message issues once in each module which uses the cited hierarchical configuration when that config description is not analyzed (yet) by a compatible Verilog compiler, or the analyzed config cannot be found or read.

Without regard to whether the intended design cell is already loaded in memory or must still be elaborated sometime in the future, the reference you're constucting here (to that configured component, from this module) must include full instructions for the linker to find and elaborate it. The key library and template name information this requires only exists in the down-configuration description.

### What Next

This message is usually preceded by a "Missing File" warning during the attempt to obey a hierarchical configuration specification. If you mean to heed that warning and you arrange for the .pvl file to be found and read, you must re-link or re-elaborate the current (configured) module. The file containing the config description may need to be analyzed into the expected library using a compatible S/Verilog compiler, or the library identifier may need to be defined to the right file system path, or the file or library-directory must grant access permission for you to read the file.

## ELAB-65

(warning) %s Always block that has both a timing control statement as well as embedded event (@) expression is not supported by synthesis.

Feedback

**Description**

You receive this warning when an always block has both a timing control statement and an embedded event (@) expression, a situation that Synthesis does not support.

**What Next**

Change the always block description by removing either the timing control statement or the embedded event (@) expression. Then invoke the compiler again.

---

## ELAB-66

(warning) %s The event depends on both edge and nonedge expressions, which synthesis does not support.

**Description**

You receive this warning message because you have assigned to an event both edge and nonedge expressions, which Synthesis does not support.

**What Next**

Modify your description so that the event depends on either an edge expression or a nonedge expression, but not both.

---

## ELAB-67

(warning) %s Only simple variables are allowed in the sensitivity list for synthesis.

**Description**

You receive this warning because you have not included simple variables in the sensitivity list. Synthesis supports only simple variables in the sensitivity list.

**What Next**

Use simple variables in the sensitivity list. Then invoke the compiler again.

---

## ELAB-68

(warning) %s The event depends on two edges of the same variable, which synthesis does not support.

**Description**

You receive this warning because you have assigned to an event two edges of the same variable, which synthesis does not support.

Feedback

**What Next**

Modify your description so that the event depends on only one edge of the variable. Then invoke the compiler again.

---

## ELAB-69

(warning) %s In the event expression with 'posedge' or 'negedge' qualifier, synthesis allows only simple variables.

**Description**

You receive this warning to let you know that synthesis allows only simple variables in an event expression with a 'posedge' or 'negedge' qualifier. If the event expression in your description contains no simple variables, you receive this warning.

**What Next**

Modify your description to include only simple variables in the event expression.

---

## ELAB-90

(error) %s Always block that has both a timing control statement as well as embedded event (@) expression is not supported by synthesis.

**Description**

You receive this error message when an always block has both a timing control statement and embedded event (@) expression, which synthesis does not support.

**What Next**

Change the always block description by removing either the timing control statement or the embedded event (@) expression. Then invoke the compiler again.

---

## ELAB-91

(error) %s The event depends on both edge and nonedge expressions, which synthesis does not support.

**Description**

You receive this error message because you have assigned to an event both edge and nonedge expressions, which Synthesis does not support.

Feedback

**What Next**

Modify your description so that the event depends on either an edge expression or a nonedge expression, but not both.

---

## ELAB-92

(warning) %s Only simple variables are allowed in the sensitivity list for synthesis.

**Description**

You receive this warning message because you have not included simple variables in the sensitivity list. Synthesis supports only simple variables in the sensitivity list.

**What Next**

Use simple variables in the sensitivity list. Then invoke the compiler again.

---

## ELAB-93

(error) %s Events that depend on two edges of the same variable are not supported by synthesis.

**Description**

You receive this error message because you have put in the event expression both 'posedge' and 'negedge' of the same variable. Synthesis supports only one edge of a variable in the event expression.

**What Next**

Review your design description and decide which edge you want to use. Then invoke the *elaborate* command again.

---

## ELAB-94

(error) %s In the event expression with 'posedge' or 'negedge' qualifier, synthesis allows only simple variables.

**Description**

You receive this error message because you did not specify simple variables only under a 'posedge' or 'negedge' qualifier in the event expression. Synthesis supports only simple variables under those qualifiers.

Feedback

### What Next

Review your design description and make the appropriate changes, using only simple variables under 'posedge' or 'negedge'. Then invoke the compiler again.

---

## ELAB-95

(error) %s Unsupported expressions appear in the sensitivity list.

### Description

You receive this error message because unsupported expressions appear in the sensitivity list with posedge or negedge. Only simple variables or variables indexed by a constant or a constant expression as an offset are allowed in the sensitivity list with posedge or negedge, by synthesis policy. For example:

```
parameter p=5;
 always @(posedge a[p+2])
```

### What Next

Change the expression following posedge or negedge so it conforms to the above synthesis policy.

---

## ELAB-100

(warning) %s Variable '%s' is a legacy of HDL Compiler. For Presto HDL compiler, please use '%s' instead.

### Description

This is an old HDLC variable.

---

## ELAB-101

(error) %s %s in constant function is illegal.

### Description

Illegal action inside constant function.

---

## ELAB-102

(warning) %s %s in the constant function will be ignored.

Feedback

### Description

This warning message advises you that the action inside the constant function will be ignored.

### What Next

This is only a warning message. No action is required.

---

## ELAB-103

(error) %s Try to get value of parameter before the declaration of the parameter.

### Description

Try to get value of parameter before the declaration of the parameter!.

---

## ELAB-104

(error) %s Symbol '%s' is not included in modport.

### Description

This error issues where a port connection, whose type is <interface>.<modport>, is used to access contents (the cited symbol name) of an interface instance that has not been properly connected by the actual modport. The interface declaration did not declare this symbol as accessible via the connected modport. The cited symbol need not appear explicitly on the cited source line; it may be a symbol accessed from the interface scope via a global variable reference within some interface function or task called directly or indirectly on the cited source line.

To appreciate why this is undesireable during logic synthesis, please recognize that the interface content is typically a net or cell instantiated remotely by the parent design who must then connect it to an instance of this design. Remote signals are conveyed into or out of the present module context via physical port connections. The modport in question selects a subset of the interface's contents and presents them each with a definite port direction and in a definite positional order. This is what allows a physically coherent linkage to be established across the hierarchical boundary.

### What Next

Interface-port connections built (for simulation purposes) around function calls that implement a "data abstraction" must be rendered synthesizable by declaring a suitable modport that gives port directions to the entire range of communication signals the module will finally need.

Additional signals needed by the down-design clients of an interface must be added explicitly to the modport(s) being connected to those instances.

Feedback

### See Also

- ELAB-398

---

## ELAB-105

(error) %s The name '%s' is not a modport of the interface instance actually provided on port '%s'.

### Description

This issues during the elaboration of an module or interface one of whose formal ports requires a modport with the specified name to be supplied as the actual port of the instantiation. The actual interface provided on the indicated port did not satisfy that expectation. Either an interface instance was provided with no such modport defined, or a modport of a different name was passed by mistake.

### What Next

Check the spelling of the modport name vs the modports declared by the interface type of the instance being provided on this port.

---

## ELAB-106

(warning) %s A symbol named '%s' is already defined differently in the receiving scope.

### Description

This warning message occurs in SystemVerilog contexts when a problem arises merging two copies of a well-known scope. It indicates that two or more reuses of a common definition are not in perfect agreement about its authorship or meaning.

SystemVerilog modules and interfaces share access to global "compilation unit" scopes such as $unit (formerly known as $root) and to SV package scopes. When they are separately analyzed, each module's or interface's .pvl file, and each package's .pvk file holds a replica of selected entities within these scopes. Package imports, interface instantiations, and type parameterizations expect to share these common definitions. As the separate parts are recombined, the SystemVerilog compiler applies a "trust-but-verify" protocol before accepting the equivalence of a foreign definition to its local homograph. If the declaration, scope, and provenance (such as source code location) of the two symbols match up perfectly, they can become identified with each other. When this protocol fails, compilation proceeds by accepting both homographs.

The message is only a warning, because both definitions survive after the merger and continue to support all their existing referents. References to the foreign definition that arrive alongside it continue to refer to the foreign definition. References to the name in the receiving scope are bound to its lexically local definition. However, future mergers

Feedback

of the receiving scope with any others can cause multiple repeats of the warning. Also, the existence of two meanings for one name might make some compiler output ambiguous. The presence of type homographs can result in strict datatype mismatches and unexpected results from type comparison operations.

**What Next**

This is only a warning message. No action is required.

However, if the result is not what you intended, locate the relevant declaration(s) of this name. If the name is not one you declared (for example, it looks like it might be a compiler temporary), it usually belongs to a struct or enum type declaration that you might know by one or more typedef'ed names. The internal name for a struct or enum type becomes visible in instantiation reports when the type is passed as a type parameter; experiments using that feature can help to locate these.

The location information in this report might point either to the semantic action that merges scopes, or to one of the homograph declarations. Merge actions can occur deeply nested in the actual receiving scope, so be sure to consider the name's definition in all parent scopes of the reported line number. If the declaration sites are not all within the source file currently being analyzed or read, consider how they come to be included in this compilation. There must be only one lexical occurrence of the declaration of any strong data type. Function, task, and named block declarations must be reproduced verbatim, and ideally should be acquired by file inclusion or package reference.

The problem may be related to `include file revisions, to `include'ing one file into several scopes, or to an attempt to pass a type parameter or refer to a package between incompatible compilations of the same application.

Check that files `included by both compilation units are the same version of the same file, and that they use the same basename for the file, even if the directory paths differ. Some matching criteria are sensitive to line location and token sequencing. Note that macro expansion outcomes can differ due to the `define namespace of separate analyze commands, or the prevailing `defines at different points of inclusion. One file, analyzed with different releases of the shell tool, may be incompatible with itself; although analyses done on distinct hosts or even operating systems with the same release level of dc_shell should be interchangeable. ELAB-106 is extra sensitive to many subtle kinds of miscoordination between separately compiled components. If you are integrating subprojects for the first time, you will need to establish uniform ways of preparing any infrastructure they share.

**See Also**

- VER-21
- VER-930
- VER-934

## ELAB-107

(error) %s Wrong slice size .

**Description**

Wrong slice size.

## ELAB-108

(error) %s Attempt to operate %s on null string.

**Description**

Attempt to operate on null string.

## ELAB-109

(error) %s Type query specifies nonexistent dimension.

**Description**

Type query specifies nonexistent dimension.

## ELAB-110

(error) %s Invalid hierarchical modport port.

**Description**

Invalid hierarchical modport port.

## ELAB-111

(error) %s Unsupported operation on double: %s.

**Description**

Unsupported operation on double.

## ELAB-112

(error) %s Multiple sequential assignment.

**Description**

Multiple sequential assignment.

Feedback

## ELAB-113

(error) %s Non-standard synchronous coding style not supported.

**Description**

Non-standard synchronous coding style not supported.

## ELAB-114

(warning) %s Variable %s has an invalid value: %s.

## ELAB-115

(error) %s RTL Netlist found.

**Description**

RTL Netlist found.

## ELAB-116

(warning) %s Variable %s is not supported in Incremental Mode.

## ELAB-117

(error) %s Array of GTECH_* instances is not supported yet.

**Description**

Array of GTECH_* instances is not supported yet.

## ELAB-118

(error) %s Unsupported resource : %s .

## ELAB-119

(error) %s Array length should be positive integer.

**Description**

This message issues because you have provided a zero or negative value when specifying an array length.

Feedback

### Examples

When you try to convert an integer to an array type in VHDL, you might write

```
...
   CONV_UNSIGNED(A, 0)
   ...
```

which requests an empty vector of bits - the zero here is not correct. Or in SystemVerilog you might use a C-style declaration to declare

```
byte lcd_text[N];
   ...
```

which can provoke this error if N = -42.

### What Next

Correct the values of the array length in your design.

---

## ELAB-120

(warning) %s %s does not exist, resource declaration will be ignored.

### Description

This warning message will happen when the operator does not have the operator label in resouce declaration, or the opearator is optimized away. Then the resource declaration will be ignored.

### What Next

If the user wants to use resouce declaration, please add the operator label for the operator.

---

## ELAB-123

(information) %s Port .%s()'s array bounds [%d:%d] were translated to [%d:%d] by renaming at the module boundary

### Description

This information message issues when you

declare

its

the choice of bounds or direction, makes no difference (only the length along each axis is important). But if your workflow uses a Verilog-/netlist-level interaction with this design

Feedback

(e.g., to add constraint or power annotations, or to link with a Verilog testbench), you may need to address this design's ports using the translated indices that were taken from the actual port expression.

**What Next**

This is an information-only message. No action is required. Some users may turn on bit[0] of hdlin_interface_port_ABI, or alter the formal port's array bounds.

**See Also**

- suppress_message
- ELAB-012

---

## ELAB-130

(warning) %s The initial value for signal '%s' is not supported for synthesis. Presto ignores it.

**Description**

You receive this warning message if the signal is assigned an initial value when it is declared. Presto VHDL ignores initial values for signals. If the only assignment to a signal is from the initial value declaration, the signal will be treated as an undriven net for synthesis. Any nets that remain undriven will be tied to zero during optimization. This will cause a simulation-synthesis mismatch if the initial value is not zero.

**What Next**

You may ignore this warning, if the declaration initialization is not the only assignment to this signal. However, if the declaration initialization is the only assignment to this signal, then either a second assignment to the signal should be added, or a constant should be used instead. Leaving the declaration initialization as the only the assignment to the signal will likely cause a simulation-synthesis mismatch.

---

## ELAB-160

(error) %s Redeclaration of port '%s' is not allowed.

**Description**

You receive this message because, during *analyze* or *read* command activity, the Presto HDL Compiler has detected that the same port is declared twice inside the same function, task, or module. Following is an example that illustrates this type of error:

```
function f;
   input a;
```

Feedback

```
  input a;      // redeclaration of already declared port
   ...
endfunction;
```

### What Next

Remove or rename as appropriate one of these ports.

---

## ELAB-161

(error) %s Forml port .%s()'s width [%d] does not match the corresponding actual axis' width [%d]

### Description

This message issues when you declare an interface (or modport) array as a formal port of your design and its shape does *not* conform with the shape of the actual port_expression at one of its instantiation sites.

### What Next

This message prevents elaboration from resolving a reference cell of the parent design being linked. The HDL source of either the instantiating design or the design's definition must be adjusted so that the interface/modport array shapes of the formal and actual ports conform when connected. Consider using parameters in the formal array bounds that can be overridden to match each instantiation site. Re-analyze any edited source before re-elaborating or re-linking.

### See Also

- ELAB-012

---

## ELAB-162

(warning) %s Formal port .%s()'s width [%d] does not match the corresponding actual axis' width [%d]

### Description

This DC Explorer message issues when you declare an interface (or modport) array as a formal port of your design and its shape does *not* conform with the shape of the actual port_expression at one of its instantiation sites. This elaboration and/or linkage has been allowed to proceed by forming random connections unrelated to your intended logic. It is not guaranteed to reach completion and its results, if any, are not suitable for tape-out.

Feedback

### What Next

The HDL source of either the instantiating design or the design's definition must be adjusted so that the interface/modport array shapes of the formal and actual ports conform when connected. Consider using parameters in the formal array bounds that can be overridden to match each instantiation site. Re-analyze any edited source before re-elaborating or re-linking.

### See Also

- ELAB-123
- ELAB-161

---

## ELAB-163

(error) %s %s port .%s() has more dimensions than the corresponding %s port

### Description

This message issues when

you

declares

the

minus

does

### What Next

This message prevents elaboration from resolving a reference cell or cell array of the parent design being linked. The HDL source of either the instantiating design or the design's definition must be adjusted so that the interface/modport array shapes of the formal and actual ports conform when connected. Check that the shape of the array of instances conforms to the leading axes of this port if the port will be distributed across instance array elements. Consider using parameters in the formal array bounds that can be overridden to match each instantiation site. Re-analyze any edited source before re-elaborating or re-linking.

### See Also

- ELAB-209

Feedback

## ELAB-164

(warning) %s %s port .%s() has more dimensions than the corresponding %s port

### Description

This DC Explorer serious warning message issues when

you

declares

the

minus

does

If the actual array has more dimensions, the formal will only connect with the leading element of each of the excess trailing axes. If the formal has more dimensions, something happens which is too horrible to describe; fix this quickly before you have to explain it...

### What Next

This message tries not to prevent elaboration from resolving a reference cell or cell array of the parent design being linked, it does this by a technically unsound process called "making stuff up", said to be the elixir of life for DC Explorer, and yet it signals corruption of the design result. The HDL source of either the instantiating design or the design's definition must be adjusted so that the interface/modport array shapes of the formal and actual ports conform when connected. Check that the shape of the array of instances conforms to the leading axes of this port if the port will be distributed across instance array elements. Consider using parameters in the formal array bounds that can be overridden to match each instantiation site. Re-analyze any edited source before re-elaborating or re-linking. Discard any results incorporating the design which produced this warning.

### See Also

- ELAB-209
- ELAB-163

## ELAB-165

(warning) 'default' is used as HDL library and it will be replaced by the value of 'hdlin.hdl_library.default_name'.

Feedback

**Description**

When user specifies HDL library name as default, this message will come out.

---

## ELAB-166

(error) '%s' is a reserved library name.

**Description**

The specified library name is a reserved name, and cannot be used to define a new HDL library name. Please choose a different name to resolve this issue.

---

## ELAB-182

(error) %s Task call %s should not have parentheses unless it has a parameter.

**Description**

You receive this error message because the task declaration you wrote has an empty parameter list.

The following example shows a task declaration that would generate this error:

```
task foo();
   ...
   endtask
```

The correct task declaration is

```
task foo;
   ...
   endtask
```

**What Next**

Rewrite the task declaration as shown in the example, removing the parentheses.

---

## ELAB-189

(error) Current NDM library is not defined.

**Description**

This error message issues when current NDM library is not defined. For example:

```
analyze -f verilog top.v
elaborate top
```

Feedback

Presto will create default template library to save the analyzed result. Presto does not create current NDM library.

**What Next**

Create NDM library before analyzing. For example:

```
create_lib my_work
analyze -f verilog top.v
elaborate top
```

## ELAB-190

(warning) %s Constructs from STD.TEXTIO are not supported for synthesis, they will be ignored. %s

**Description**

You receive this warning from the *read* command or the *elaborate* command, either of which issues this message when constructs from STD.TEXTIO package are in use. The details of the error appear in the message.

**What Next**

To avoid receiving this warning message in the future, remove the constructs from STD.TEXTIO package. This warning message can be safely ignored.

## ELAB-191

(error) %s Constructs from STD.TEXTIO package are not supported. %s

**Description**

You receive this error from the *read* command or the *elaborate* command, either of which issues this message when constructs from STD.TEXTIO package are in use. The details of the error appear in the message.

**What Next**

To avoid receiving this error message in the future, remove the constructs from STD.TEXTIO package.

## ELAB-192

(warning) %s There are nonblocking assignments inside TASK that set output/inout port '%s'; this may introduce simulation mismatch.

Feedback

### Description

This warning message occurs when TASK has nonblocking assignments that set the output or inout ports. This may introduce simulation mismatch, for example:

```
module ShiftR (Out, In);
output Out;
input In;
reg Out;
always@(In)
     tShift(Out, In );
task tShift(output Q, input D);
      Q <= D ;
endtask
endmodule
```

### What Next

This is only a warning message. No action is required.

However, to avoid receiving this warning message, do not set the output or inout port of TASK using a nonblocking assignment. You can change the output or inout port as a global variable. The above example can be rewritten as follows:

```
module ShiftR (Out, In);
output Out;
input In;
reg Out;
always@(In)
     tShift(In);
task tShift(input D);
      Q <= D ;
endtask
endmodule
```

---

## ELAB-193

(information) Elaborating HDL template %s:%s %s.

### Description

This informational message issues when a new elaboration of an HDL template commences indirectly via the link step of another command (typically an *elaborate* or *link* of some parent design). The message indicates the logical library and template names of the new module or entity and gives other details of the circumstances that require the subsequent elaboration effort.

Feedback

### What Next

This is only an information message. Next comes any log output produced by the elaboration work described. Consider the details given in the ELAB-193 message as an audit trail of the flow leading to any upcoming messages.

---

## ELAB-194

(error) elaborate cannot write designs to a template library: %s.

### Description

This error message issues when command *elaborate* attempts to write a design to a template library. For example:

```
create_lib my_work_lib
analyze -f verilog -library top_lib top.v
elaborate -library top_lib top
```

"top_lib" is a template library, elaborate can't write a design into it.

### What Next

Either use -ref_libs instead, for example, elaborate -ref_libs top_lib top. Or do not specifiy library, such as: elaborate top.

---

## ELAB-195

(warning) %s Missing design or template for '%s' to glob .* port-names;\n\t... ignoring .* due to 'link_allow_design_mismatch'.

### Description

This DC Explorer warning message issues when elaboration of a .* (wildcard globbing) port connection requires either DC Link to find loaded instances of the down design, or Presto HDL Compiler to find an analyzed intermediate (template) file. The spelling of the down design's port names is needed to link them with the corresponding signals in the design being elaborated. Absent any information about the down design, an unresolved (black box) reference has been produced with no port connections corresponding to your .* globbing request. Note that this will usually be bad logic; it has been synthesized only so that Design Exploration may proceed.

### What Next

Ensure that the module or interface is analyzed before next running the elaborate command. If elaboration is proceeding bottom-up, pre-load the .ddc designs that will be linkage candidates for the current instantiation reference. All designs produced under this

Feedback

warning contain bad logic and must be re-elaborated to achieve standard SystemVerilog semantics.

### See Also

- ELAB-197

---

## ELAB-196

(warning) %s Connecting local definition '%s' to non-optional formal port '%s' of '%s';\n \tNon-standard extension 'link_allow_pin_name_synonym' permits forming this bad logic

### Description

This warning message issues when DC Explorer allows a ".*" port connection between a non-optional formal port and an actual definition with a synonymous spelling if no identically declared name was found in the instantiating design. Note that such connection requests cannot trigger a package wildcard import, nor declare implicit wires, nor can they reach elements of $unit not used elsewhere in the instantiating design.

### What Next

Ensure that the (down design ) module or interface that you've analyzed or loaded is well-matched to the current version of the design you're elaborating. If elaboration is proceeding bottom-up, with only .ddc files, remove obsolete designs that still expect a port no longer provided by this instantiation site. Consider explicitly naming each port in this connection, .* entails some risk of miscoordination that can be easily caught by a (redundant) copy of the intended port list. In all cases, the design must be re-elaborated to produce verifiable standard logic.

### See Also

- ELAB-198

---

## ELAB-197

(error) %s No module or interface '%s' has been loaded and/or analyzed.

### Description

This error message issues when elaboration of a .* (wildcard) port connection requires either DC Link to find loaded instances of the down design or Presto HDL Compiler to find an analyzed intermediate file. Information about the down design's port names is needed to establish connections to the corresponding signals in the design being elaborated.

Feedback

### What Next

Ensure that the module or interface is analyzed before running the command. If elaboration is proceeding bottom-up, pre-load the .ddc designs that will be linkage candidates for the current instantiation reference.

---

## ELAB-198

(error) %s No local definition matches the formal port '%s' of '%s'

### Description

This error message issues when a ".*" port connection to the cited formal port cannot be completed due to lack of an identically declared name in the instantiating design. Note that such connection requests cannot trigger a package wildcard import, nor declare implicit wires, nor can they reach elements of $unit not used elsewhere in the instantiating design.

### What Next

Ensure that the (down design ) module or interface that you've analyzed or loaded is well-matched to the current version of the design you're elaborating. If elaboration is proceeding bottom-up, with only .ddc files, remove obsolete designs that still expect a port no longer provided by this instantiation site. Consider explicitly naming each port in this connection, .* entails some risk of miscoordination that can be easily caught by a (redundant) copy of the intended port list.

---

## ELAB-199

(warning) %s No local definition matches the non-optional formal port '%s' of '%s';\n \t'link_allow_design_mismatch' is a non-standard extension which treats this port as optional

### Description

This DC Explorer warning message issues when a ".*" port connection to the cited, non-optional, formal port is given access to that port's default value or is disconnected due to the lack of an identically declared name in the instantiating design. This violates the IEEE-1800-2009 SystemVerilog standard; it is done here at your request for exploratory purposes only. Connections formed this way are not made by standard-conforming products.

Note that ".*" connection requests cannot trigger a package wildcard import, nor declare implicit wires, nor can they reach elements of $unit not used elsewhere in the instantiating design.

Feedback

### What Next

Ensure that the (down design) module or interface that you've analyzed or loaded is well-matched to the current version of the design you're elaborating. If elaboration is proceeding bottom-up, with only .ddc files, remove obsolete designs that still expect a port no longer provided by this instantiation site. Consider explicitly naming each port in this connection, .* entails some risk of miscoordination that can be easily caught by a (redundant) copy of the intended port list. In all cases, the design in which the warning was issued must be re-elaborated to produce verifiable standard logic.

### See Also

- ELAB-198

---

## ELAB-201

(warning) %s Unconnected output '%s' in function call tied to ground.

### Description

Presto has found that the named output is not driven by any logic for this instance of function call. Sometimes this represents an issue in your design and sometimes it doesn't. You should review your source code to ensure that this will not translate into incorrect outputs.

Presto has grounded the logic that may depend on these function outputs.

An example of this situation happens in the following code:

```
procedure elab_201_example(in1: in std_logic;
                           out1: out std_logic;
                           undriven_out : out std_logic_vector)
begin
    out1 := '0';
end procedure elab_201_example;
```

---

## ELAB-202

(warning) %s In IEEE function CONV_INTEGER, the unsigned input signal contains more than 31 bits.

### Description

You receive this warning message when the unsigned input of the CONV_INTEGER function contains more than 31 bits, which can cause overflow when converted to integer type.

Feedback

### What Next

This is only a warning message. No action is required.

However, if the result is not what you intended, modify your VHDL source code to resolve this issue.

---

## ELAB-207

(error) %s Malformed specify block in Verilog source code.

### Description

You receive this error message because your Verilog source code contains an invalid use of a specify block.

The following example, containing two specify blocks in the same module (an invalid instance), illustrates this type of error.

```
module m;
      ...
      specify
       ...
      endspecify
      ...
      specify
       ...
      endspecify
      ...
   endmodule;
```

### What Next

Modify your Verilog source code to remove the inavlid usage.

---

## ELAB-208

(error) %s Instance array's %d%s axis has length %d, but the corresponding axis of its actual port '%s' has length %d.

### Description

This error message occurs when the (leading) dimensions of an actual interface array port do not conform to the (trailing) dimensions of the instance array over which that port connection should be distributed.

Distribution of array ports with unpacked dimensions begins at the first instance dimension whose length matches the leading dimension of the array port. Subsequent contiguous dimensions of both arrays must match each other.

Feedback

### What Next

If the mismatch has occurred because a coincidental length match between leading dimensions which should instead replicate the array port, switch to using a generate loop to cause this replication. Otherwise, modify your SystemVerilog source code to make distributed unpacked dimensions match.

---

## ELAB-209

(error) %s Instance array has %d dimensions, only %d of which match(es) its actual interface array port '%s'.

### Description

This error message occurs on a source line where an instance array has more unpacked dimensions than one of its interface array ports, but where the axes whose lengths match do not include the last (fastest varying) dimension(s) of the instance array.

Distribution of array ports with unpacked dimensions begins at the first instance dimension whose length matches the leading dimension of the array port. Subsequent contiguous dimensions of both arrays must match each other.

### What Next

If the mismatch has occurred because a coincidental length match between leading dimensions which should instead replicate the array port, switch to using a generate loop to cause this replication. Otherwise, modify your SystemVerilog source code to make the unpacked dimensions of a distributed port match those of the array instance.

---

## ELAB-210

(Error) %s integer overflow occurs.

### Description

You receive this error message because there is an integer overflow in an arithmetic operation.

For example, the following code causes the error message because the operation 2**32 overflows in the range of integers.

```
module M (a,d);
input [31:0] a;
output [31:0] d;
reg [31:0] d;

always begin
  d = 2**32 - a;
  $display ("d = %d\n", d);
```

Feedback

```
end

endmodule
```

Another possible integer overflow case is introduced by verilog shifter. By verilog language standard, the right operand of a shifter is always treated as an unsigned number.
The constant '-1' (32'b11111111111111111111111111111111) in the following example is interpreted as unsigned number, which causes the integer overflow.

```
module m (input [1:0] in1, output [1:0] out1);
assign out1 = (in1 << -1);
endmodule
```

### What Next

Check the operand of the operator, on the line indicated by the error message, to ensure that the result is within integer range, or use a Verilog number vector to represent a large value.

---

## ELAB-211

(Error) %s Conversion to bit-stream types larger than 64 bits is not supported.

### Description

You receive this error message because there is an unsupported conversion from real type expression to bit-stream type with size of larger than 64 bits.

For example, the following code causes the error message because there is casting from real type expression to bit-stream type with size of 71 (>64) bits.

```
typedef logic [70:0] t;

module M (output logic [31:0][1:0]);
 int i;
 always_comb
   begin
     i =3;
     o2 =  t' (i + real'(1) );
   end
endmodule
```

---

## ELAB-229

(error) %s 0-repeat concat not supported in generate expressions.

### Description

This error message occurs because an expression in generate block contains a 0-repeat concat operation that Presto HDL Compiler does not support.

Feedback

### What Next

Rewrite your source code to remove the occurance of 0-repeat concats and try again.

---

## ELAB-230

(error) %s Multiple blocks have been named '%s' in a scope.

### Description

You get this error when two (or more) blocks with the same name are declared in the same local scope.

### What Next

Change the relevant code to conform to the IEEE Std 1364.

---

## ELAB-231

(error) %s %s declaration redeclares symbol '%s'.

### Description

You see this message because you have assigned a function or task declaration to a symbol that has already been defined.

In the following example, which demonstrates such a situation, the symbol *f* is declared twice: first as a register and then as a function:

```
reg f;          // initial declaration of 'f'

   function f;     // illegal redeclaration of 'f'
   ...
   endfunction
```

The example elicits the VER-106 message.

### What Next

Rename either the function declaration or the other declaration and all corresponding uses of the symbol.

---

## ELAB-233

(error) %s Unsupported %d levels of cross module reference when declaring hierarchical reference '%s'.

Feedback

### Description

You receive this error message because unsupported levels of cross module reference appear when declaring hierarchical reference. For example:

```
assign a.b.c = i_1;
```

### What Next

Reduce cross-module references by adding explicit ports to carry signals between modules.

---

## ELAB-255

(error) %s Internal error in %s at line %d.

### Description

You receive this error message because an internal compiler error has occurred, related to the indicated position in the source code.

The nature of this error prevents availability of more specific information about it.

### What Next

Please file a star against Presto with your testcase if it's possible.

---

## ELAB-256

(error) %s %s call violation: %s.

### Description

You receive this error because the RTL power constructs were used incorrectly. The RTL power constructs include power, isolate and retain calls. The reason can be incorrect number of parameters, unsupported parameter types, width mismatch or the wrong place to call the functions.

### What Next

For details on how to use RTL power constructs, please refer to the Power Compiler User Guide, chapter "Implementing Multivoltage Designs Using RTL Isolation and Power Constructs", and section "Multivoltage Elements: $isolate and $power".

---

## ELAB-275

(error) %s You must specify a named block with infer_mux when used outside a case statement.

Feedback

### Description

You receive this error message because you have used the Synopsys *infer_mux* pragma outside of a *case* statement without specifying the label of a named block on which MUX inference is to be applied.

### What Next

Using *infer_mux*, specify the label of the named block on which you want to apply MUX inference. Then invoke the compiler again.

---

## ELAB-276

(warning) %s All symbols and listed symbols are designated as %s set/reset in block '%s'; using the all designation.

### Description

The warning occurs because you have specified incompatible conditions, which the compiler has to resolve:

- You have explicitly marked a symbol as a set or reset signal in a list of symbols annotated with the Synopsys set/reset pragma.
- You have implicitly marked the symbol as a set or reset signal using the Synopsys pragma that applies set/reset inferencing to all symbols.

Under these circumstances, the compiler uses set/reset inferencing on all symbols (not just the listed symbols), and this warning informs you of that fact.

### What Next

This warning is for your information and requires no action on your part.

---

## ELAB-277

(warning) %s More than one multibit pragma specified for symbol '%s'; using '%s'.

### Description

You receive this warning message from the *elaborate* or *read* command if there are two multibit pragmas on one variable. The message informs you about which pragma is being used.

### What Next

Remove the pragma you do not want to use, and reexecute the command.

Feedback

## ELAB-278

(warning) %s The pragma '%s' for object '%s' will be ignored because the object could not be found.

### Description

You receive this warning because you tried to place a pragma on an object that does not exist.

### What Next

Inspect your design and make sure it is correct and that there are no errors in the spelling of objects and other components.

## ELAB-279

(warning) %s More than one fsm_complete pragma specified for symbol '%s'.

### Description

You receive this warning message from the *elaborate* or *read* command if there are two fsm_complete pragmas on one variable. The message informs you about which pragma is being used.

### What Next

Remove the pragma you do not want to use, and reexecute the command.

## ELAB-280

(warning) %s Signal on port %d renamed from '%s' to '%s' in '%s'.

### Description

You receive this warning when the name of a signal on a port, such as on the second port in the following example,

```
module test(x, \\x[0] );
        input [0:0] x;
        output \\x[0] ;
        assign \\x[0] = x[0];
     endmodule
```

is different in the output netlist than it would usually be, because of a name conflict, such as a conflict with the name of a component of a bus.

In this example, the output signal on the second port will be renamed from "x[0]" to "x[0]1".

Feedback

## ELAB-281

(error) %s Default branch isn't the last branch in the case statement.

**Description**

You receive this error message because you have specified a default branch that is not the last branch in the *case* statement, which is must be.

**What Next**

Review your design and make sure the default branch you specify is the last branch in the *case* statement.

## ELAB-282

(warning) %s Parameter range specification is only meaningful to synthesis, not simulation, which might result in different behavior.

**Description**

Providing a range (size) for a parameter is only used during synthesis. You might see different behavior between synthesis and simulation if you have explicitly sized parameters.

**What Next**

Review your design description and make appropriate changes.

## ELAB-283

(warning) %s Function '%s' is mapped to module '%s' but body is not empty and will be ignored.

**Description**

You receive this warning because you specified a function whose body contains code. When you use the *map_to_module* pragma, because calls to that function are mapped to the specified module, the body of the function is not used. This warning lets you know that your design has such a function and its body contains code and will be ignored.

**What Next**

This warning is informational only and requires no action on your part.

## ELAB-284

(error) %s Return port name '%s' conflicts with names of function input parameters.

Feedback

### Description

You receive this error message because a function description in your design contains an output port that has the same name as one of the input ports, an invalid condition. This occurred in connection with your use of the *map_to_module* or *map_to_operator* pragmas. Following is an example of a description that would elicit this error message. The *return_port_name* is specified as *o* and so is one of the input ports.

```
function f;
          // synopsys map_to_module ADDER
          // synopsys return_port_name "o"
        input o;
        input p;
endfunction
```

### What Next

Change the name of the port whose name duplicates the name of another port. Then invoke the command again.

---

## ELAB-285

(error) %s Conversion function on formal port is not supported.

### Description

You receive this error message because conversion function is called with formal port as argument while port mapping, which is not supported.

```
begin
        U1 : TEST_E
            port map ( Cone(A) => Ctwo(TA), Z => TZ );
```

### What Next

Remove the conversion functions on port formals and try again

---

## ELAB-286

(error) %s Type conversion on formal port is not supported.

### Description

You receive this error message because port formals have Type conversion on them, while port mapping which is not supported by Presto.

### What Next

Remove the type conversion on formal port and try again.

Feedback

## ELAB-287

(error) %s Incorrect specification of port formals.

**Description**

You receive this error message because a syntax error is detected on port formals during port mapping.

**What Next**

Please check the correct syntax of specifying port formals during port mapping and try again.

## ELAB-288

(error) %s synthesis_on and synthesis_off directives are not used in pair within current design unit.

**Description**

You receive this error message because either synthesis_on or synthesis_off is missed within current design unit.

**What Next**

Please follow the error message to complete the synthesis_off/on pair and try again.

## ELAB-289

(error) %s A synthesis_off or translate_off object has been used in an expression.

**Description**

You receive this error message because a synthesis_off or translate_off object has been used in an expression.

**What Next**

Please follow the error message to check if the synthesis_off/on translate_on/off pragma has been put at a wrong place.

## ELAB-290

(error) %s The synthesis_on/synthesis_off or translate_on/translate_off attribute is inconsistent on arg #%d.

Feedback

### Description

You receive this warning message because the synthesis_on/synthesis_off or translate_on/translate_off is inconsistent between the formal and actual argument.

### What Next

Please follow the warning message to check if the synthesis_off/on translate_on/off pragma is put at a wrong place or is missing.

---

## ELAB-291

(error) %s Received the reserved word 'all' while expecting signal names in the sensitivity list.

### Description

You receive this error message because the reserved word "all", instead of a signal name, appeared in the sensitivity list.

### What Next

Use of the reserved word "all" in sensitivity list is a VHDL 2008 only feature. If this usage is intended, please set hdlin.vhdl.standard to 2008.

---

## ELAB-292

(warning) %s '%s' is being read, but does not appear in the sensitivity list of the block.

### Description

You should not receive this warning by default. If you do receive this warning, it may or may not be accurate; please evaluate your RTL code accordingly. If you wish to remove this warning from your DC sessions, please contact Synopsys support.

You receive this warning message if not all the signals that are read appear in the sensitivity list of the block. This may result in simulation-synthesis mismatches.

Consider the following example:

For Verilog:

```
always @(in1)
   myout = in1 | in2 ;
```

For VHDL:

```
entity e is
   port(in1, in2: in bit; myout: out bit);
end;
```

Feedback

```
architecture a of e is
begin
   process(in1) begin
      myout <= in1 or in2;
   end process;
end;
```

This example will result in an OR gate whose inputs are *in1* and *in2*, and whose output is *myout*. The output of the OR gate will potentially change whenever *in1* or *in2* changes. However, in simulations of the original description, *myout* will change only when *in1* changes, because *in1* is the only signal to appear in the sensitivity list of the block.

**What Next**

Add the signal specified in the warning to the sensitivity list of the block.

---

## ELAB-293

(warning) %s Variable %s is read before being assigned; the synthesized result may not match simulations.

**Description**

You receive this warning message because a variable that is local to a function or task was read before being assigned. When the *hdlin_infer_function_local_latches* variable is set to *false*, the Presto HDL Compiler does not preserve variable values across separate calls to a task or function, unlike a simulator. Any bits of the variable that are not assigned before being used are given the value 0. No latches are inferred for variables local to a subprogram.

When the *hdlin_infer_function_local_latches* variable is set to *true*, the compiler matches the simulator's behavior. This may result in additional latches in your design.

**What Next**

If you intended to use a value set during a previous call, set *hdlin_infer_function_local_latches* to *true* in order to preserve the value across calls.

If you did not intend to use a value across calls, check your design to verify that you assigned all local variables before using them.

---

## ELAB-294

(warning) %s Floating pin '%s' connected to ground.

Feedback

### Description

You receive this warning message when a previously unconnected pin has been connected to ground. This message reminds you to check whether the behavior resulting from connecting the pin to ground conforms to your expectations.

### What Next

Verify that connecting the specified pin to ground results in the desired behavior.

---

## ELAB-295

(error) %s Number of actual arguments in task or function call does not match the number of formals.

### Description

You receive this error message because the number of actual arguments in a function or task call does not match the number of formal arguments expected by the corresponding function or task.

### What Next

Identify the offending parameters and add or remove them accordingly. See the function or task definition to determine which parameters are required.

---

## ELAB-296

(error) %s Index bounds '%s' for vector '%s' are wrong.

### Description

You receive this error message because the bounds of a vector declaration are not an integer type or sized parameter value is overflow. The following example shows the compiler attempting to use a real number as a vector bound.

```
...
   reg [3.0:0] x;      // error: "3.0" is not a valid array bound
   ...
```

The following example shows the compiler attempting to override sized parameter, the new value is overflow for the given size.

```
module test (a, b, c);
   parameter WIDTH = 8;
   input [WIDTH-1:0] a, b;
   output [WIDTH-1:0] c;
   bottom  #(.W(WIDTH)) U1 (a, b ,c ); // W has 3 bits, but 8 need 4
 bits.
   endmodule
```

Feedback

```
module bottom (a, b, c);
parameter [2:0] W = 4;
input [W-1:0] a, b;
output [W-1: 0] c;
assign c = a + b;
endmodule
```

**What Next**

Modify each part select so that it uses only integer bounds. In the example, you would replace "3.0" with "3". Modify sized parameter definiton or new parameter value.

---

## ELAB-297

(error) %s '%s' is a task and not a function.

**Description**

You receive this error message because a function call is attempting to enable a task. A function call can only enable functions. The symbol a function call refers to must be defined by a function declaration.

A function returns a value; thus, a function call must be used as an expression.

**What Next**

Check that you did not attempt to enable a task. If you did not, either change the function call to a task, or rewrite the task to a function.

---

## ELAB-298

(error) %s Array index out of bounds %s.

**Description**

You receive this error message because the index used to access the array is not within the range of valid indices defined by the array declaration.

In the following example, the valid array indices must be in the range 0:1 because of the way the array is declared. Attempting to access location 2 outside the valid range, as in the following example, results in this error.

```
reg [1:0] a;
   ...
   a[2] = 1'b1;         // error: index out of bounds
```

Feedback

### What Next

Modify the array index so that it is within the valid index range given by the array declaration.

---

## ELAB-299

(error) %s Array index out of bounds %s.

### Description

You receive this error message because the index used to access the array is not within the range of valid indices defined by the array declaration. In the following example, the array index must be within the range 0:1 because of the way the array is declared. Attempting to access location 2, which is outside the valid range, results in this error message.

```
reg [1:0] a;
   ...
   x = a[2];         // error: index out of bounds
```

### What Next

Modify the array index so that it is within the valid index range given by the array declaration.

---

## ELAB-300

(error) %s Cannot test variable '%s' because it was not in the event expression or with wrong polarity.

### Description

You receive this error message when the outermost if statement in the always block tests some other variable that does not appear in a *posedge* or *negedge* expression or with wrong polarity. When an always block's sensitivity list contains more than one *posedge* or *negedge* expression, the Presto HDL Compiler assumes that one expression represents the clock, and any remaining expressions represent asynchronous sets or resets. For each set or reset condition, the always block should contain an if statement that tests it.

### What Next

Modify your rtl to meet the above requirements. See the *HDL Compiler for Verilog Reference Manual* or the *VHDL Compiler Reference Manual* for information on how the compiler infers flip-flops.

Feedback

## ELAB-301

(warning) %s Only constant-valued subscripts are checked in the sensitivity list; other subscripted expressions are ignored.

### Description

You should not receive this warning by default. If you do receive this warning, it may or may not be accurate; please evaluate your RTL code accordingly. If you wish to remove this warning from your DC sessions, please contact Synopsys support.

You receive this warning message because Presto HDL Compiler does not perform sensitivity list checking on variable subscripts.

Sensitivity list checking verifies that all signals read by a combinational always block are in its sensitivity list.

### What Next

Remove the variable subscript from the sensitivity list.

## ELAB-302

(error) %s The statements in this 'always' block are outside the scope of the synthesis policy. Only an 'if' statement is allowed at the top level in this always block.

### Description

You receive this error message when the statement at the top level of the always block is not an if statement.

### What Next

See the *HDL Compiler for Verilog Reference Manual* and the *VHDL Compiler Reference Manual* for ways to infer flip-flops and latches from always blocks with an if statement.

## ELAB-303

(error) %s The expression in the reset condition of the 'if' statement in this 'always' block can only be a simple identifier or its negation.

### Description

You receive this error message when the reset condition of the *if* statement is not a simple identifier or a negation of the simple identifier. Presto HDL Compiler only supports an identifier or its negation as the reset condition.

Feedback

### What Next

Modify the reset condition so that it is either a simple identifier or a negation of the simple identifier. The following example shows how to modify the RTL code.

Original RTL code:

```
if (~reset || jt_cbits[6]) ec_mode <= 1'b0;
        else ec_mode <= 1'b1;
```

Modified RTL code:

```
if (~reset) ec_mode <= 1'b0;
        else if (jt_cbits[6]) ec_mode <= 1'b0;
        else ec_mode <= 1'b1;
```

---

## ELAB-304

(warning) %s Case or if statement, or ?: operator has an infer_mux attribute and a default branch or incomplete mapping. This might cause nonoptimal logic if a MUX is inferred. %s.

### Description

This warning message occurs when MUX inference has been requested for a case statement, if statement, or ?: operator.

You receive this warning when a case statement does not have all of the values of the case variable explicitly enumerated. Instead, a default branch is used (explicitly or implicitly) to represent several of them together as shown in the following example:

```
case(c) // synopsys infer_mux
  2'b00: out = in1 ;
  2'b11: out = in2;
  default: out = in2;
endcase
```

A MUX_OP is not the optimal hardware for this representation. You will not receive this warning if you delete the default branch and explicitly add labels for 2'b10 and 2'b01.

You also receive this warning when an if statement has a final else branch, or the control variable is not compared against all possible values, as in the following example:

```
if (sel == 2'b00) // synopsys infer_mux
  dout <= a;
else if (sel == 2'b11)
  dout <= b;
else
  dout <= in2;
```

Feedback

You will not receive the warning if you change this code as shown below:

```
if (sel == 2'b00) // synopsys infer_mux
  dout <= a;
else if (sel == 2'b11)
  dout <= b;
else if (sel == 2'b10)
  dout <= in2;
else if (sel == 2'b01)
  dout <= in2;
```

In the current version this message always occurs for the ?: operator. Change this operator to a case statement to avoid the warning message.

You may receive many ELAB-304 warning messages when you set the *hdlin_infer_mux* variable to *all* because this is equivalent to using the infer_mux pragma or attribute in any case or if statement, or ?: operator.

**What Next**

To avoid this warning message, remove the infer_mux pragma, or recode by a case or if statement where all possible values are enumerated.

---

## ELAB-305

(error) %s Clock %s used as data.

**Description**

You receive this error message because Presto HDL Compiler does not allow clock signals to be used for purposes other than clocking.

**What Next**

Remove the unsupported code from your design.

---

## ELAB-306

(error) %s Illegal use of tristate value.

**Description**

You receive this error message when a tristate value is used in an expression in a way that Presto does not support, as in the following example.

```
out = in & 1'bz
```

This error is also issued when a variable assigned to a tristate value is used in an expression, as in the following example.

Feedback

```
if(c) tmp = 1'bz;
else tmp = in1 ;

out = tmp & in2
```

**What Next**

Remove the unsupported portion of code.

---

## ELAB-307

(error) %s Case labels are not parallel in %s at line %d. A MUX_OP cannot be inferred.

**Description**

You receive this error message because Presto HDL Compiler does not infer a MUX_OP for the case statements that are not parallel. A case is parallel when all its branches are mutually exclusive.

**What Next**

Remove the "infer_mux" pragma from this case statement, or if the branches really will be mutually exclusive, prepend a "unique" qualifier to the case statement in SystemVerilog or add a "parallel_case" pragma in traditional Verilog.

---

## ELAB-308

(warning) %s Mismatch between simulation and synthesis may occur due to three-state value.

**Description**

You receive this warning message when the design contains a variable that is conditionally assigned to a three-state value, and the variable is read in the same always block in which it is conditionally assigned. Reading such a variable in the HDL source code generates logic that uses the output of a three-state buffer, which may cause a mismatch between simulations and synthesis.

In the following example, *tmp* is assigned to a three-state value and assigned to out when both *c* and *in1* are false.

```
if(c)
      tmp = 1'bz ;
   else if (in1)
      tmp = in2 ;

   out = tmp ;
```

Feedback

### What Next

Modify the code so that the design does not read a variable that is not conditionally assigned to *z*.

---

## ELAB-309

(information) %s Cannot infer a MUX_OP in %s at line %d.

### Description

The tool issues this message if a MUX_OP cannot be inferred for an if or case statement.

### What Next

Contact Synopsys support.

---

## ELAB-310

(warning) %s Comparison against '?', 'x', or 'z' values is always false. It may cause simulation/synthesis mismatch.

### Description

You receive this warning when any operand in a comparison expression has *x or z values. The result of such expressions are always determined to be false, and the corresponding branch is never executed.*

In the following example, 'then-branch' of the if-conditional is never executed.

```
if(2'b0x) begin
      out = in1;       /* Will never fire */
   end
   else begin
      out = in2;       /* Will always fire */
   end
```

In the following example, the case branch condition that contains *x* or *z* values are never reached.

```
case(c)
     1'bx : out = in1 ;  /* Will never fire */
     1'b0 : out = in2 ;
   endcase
```

### What Next

Rewrite the design to eliminate the occurance of *x* or *z* from conditional expressions, or convert the case into a casex or casez statement.

Feedback

## ELAB-311

(warning) %s DEFAULT branch of CASE statement cannot be reached.

### Description

You receive this warning message when you execute the *read* or *elaborate* command and your design contains a case statement in which the default branch can never be executed. This warning is issued when the compiler detects that at least one of the other branches will always be executed.

```
case(c)
     1'b1 : out = in1 ;
     1'b0 : out = in2 ;
     default: out = in2 ;
   endcase
```

### What Next

Verify that your case statement is written as you intended. If so, delete the default branch.

## ELAB-312

(warning) %s Out of bounds bit select %s.

### Description

You receive this warning message when you apply a subscript to a bit vector, and the bit vector attempts to access a bit that is out of range of the vector.

### What Next

Modify the subscript so that it attempts to access a bit that is within range of the vector.

## ELAB-313

(error) %s Repetition multiplier in a concatenation is not constant expression.

### Description

You receive this error message because you have specified a repetition multiplier in a concatenation that is not a constant expression, which is not allowed. The repetition multiplier in a concatenation must be a constant. For example, {i{x}} is not allowed if *i* is a varying expression.

### What Next

Review your design and make appropriate corrections.

Feedback

## ELAB-314

(warning) %s Case branch %s is unreachable.

### Description

You receive this warning message when one or more branches in a case statement are unreachable due to stronger branch conditions in earlier branches.

For example, in the following Verilog code the first branch of the case statement is always taken. As a result, all subsequent case branches are unreachable.

```
module m (a, b, y);
 input a, b;
 output reg y;

 reg t0, t1;

 always begin
  case (1'b1)
    1'b1: y = a;
    b: y = ~a;
    default: y = b;
  endcase
 end

endmodule // m
```

### What Next

Dead case branches might indicate a design error. Examine the case statement and verify that the unreachable branches are intentional.

## ELAB-315

(error) %s A statement in your design description tried to read a three-state value.

### Description

You receive this error message because a statement in your source code tried unsuccessfully to read a *Z* value. Synthesis does not support *Z* values.

### What Next

Rewrite your description to change or eliminate the value or values in question.

Feedback

## ELAB-316

(error) %s A statement in your design description tried to read an 'x', '?', or three-state value.

### Description

You receive this error message because a statement in your source code tried unsuccessfully to read an *x*, *?*, or *Z* value. Synthesis does not support these values.

### What Next

Rewrite your description to change or eliminate the value or values in question.

## ELAB-317

(error) %s A statement in your design description tried to read an 'x', '?', or three-state value in a case expression.

### Description

You receive this error message because a statement in your source code tried unsuccessfully to read an *x*, *?*, or *Z* value in a case expression. Synthesis does not support these values in a case expression.

### What Next

Rewrite your description to change or eliminate the value or values in question.

## ELAB-318

(error) %s Unable to read from modfile %s: %s

### Description

You receive this error message because your intermediate file could not be read, a situation that might be caused by restricted directory permission or by disk corruption.

### What Next

Change the directory permission to allow access. Then invoke the *analyze* command again to analyze the corrupted module file and attain a modfile that is not corrupted.

## ELAB-319

(information) %s Reading module %s from file %s.

Feedback

**Description**

Reading module from the corresponding .pvl file.

**What Next**

No action is necessary.

---

## ELAB-320

(warning) %s File %s not found, or does not contain a usable description of %s.

**Description**

This warning issues when the intermediate files (.mr, .pvl, .syn, etc.) corresponding to the given module, interface, checker, or config could not be found or, if present, were not usable for (this stage of) elaboration. It may be optional in this context, but probably should be provided.

**What Next**

This is only a warning. To avoid it, reanalyze the RTL description source (with the same DC release as issued this message). For warnings arising through .* wildcard port connections, loading the referenced design will also satisfy this requirement.

---

## ELAB-321

(error) %s Number of ports on reference design '%s' is inconsistent with specified instance '%s'.

**Description**

You receive this error message because the number of ports on reference design is inconsistent with specified instance.

**What Next**

Rewrite your description to make the number of ports on reference design is consistent with specified instance.

---

## ELAB-322

(error) %s Width of port '%s' on reference design '%s' is inconsistent with specified instance '%s'.

Feedback

**Description**

You receive this error message because the width of port on reference design is inconsistent with specified instance.

**What Next**

Rewrite your description to make the width of port on reference design is consistent with specified instance.

---

## ELAB-323

(error) %s Too many ports found on gate instantiation

**Description**

Too many ports found on gate instantiation

**What Next**

Check that the correct number of ports has been specified on the gate instantiation.

---

## ELAB-324

(error) %s Too many ports passed to instance %s

**Description**

Too many ports passed to instance

**What Next**

Check that the correct number of ports has been passed to the instance listed above.

---

## ELAB-325

(error) %s Unknown port %s in instance %s in module %s

**Description**

Unknown port in instance in module

**What Next**

Check that the offending port listed above has been specified correctly.

---

## ELAB-326

(error) %s module data file %s revision mismatch: Expected %d, got %d

Feedback

**Description**

module data file revision mismatch.

**What Next**

analyze the rtl file again using new version.

---

## ELAB-327

(error) %s Cannot resolve pin %s on cell %s:%s.

**Description**

Cannot resolve pin on cell.

**What Next**

modify rtl source to make sure the module instanitation agree with submodule declaration.

---

## ELAB-328

(error) %s Replication constant contains 'x' or 'z'.

**Description**

The repeat count of this replication operator is a constant expression that contains 'x' or 'z'. It is unusable for elaborate-time replication of the data or assignment pattern. The following example will trigger this error:

```
x = { 3'bx11 { y } };
```

**What Next**

Replace the replication constant by a constant expression resolvable to 0s and 1s. Or, if this constant was meant to produce 'x' or 'z' data values in the result, insert a comma after the constant so that it parses as a list element and not as a replication operator.

---

## ELAB-329

(warning) %s A 'disable' statement for block '%s' appears outside the block and is being ignored.

Feedback

### Description

You receive this warning message because a disable statement appears outside of the block it attempts to disable, as in the following example:

```
always begin
      begin : block_a
         y = 1'b1;
      end
      disable block_a;
   end
```

A disable statement must appear with the block to be disabled; otherwise, the disable statement has no effect and is ignored.

### What Next

Examine your code to verify the block you want to disable, and revise the code so that the disable statement appears inside the block to be disabled.

---

## ELAB-330

(error) %s Attempt to disable block '%s' out of function scope is not supported.

### Description

You receive this error message when a disable statement inside a function refers to a block that is either outside the function or that does not contain the disable statement. The Presto HDL Compiler for Verilog supports only disable statements that refer to a block inside the current function.

### What Next

Change the design so that all disable statements are within the block they disable, and that all disable statements refer to blocks within the same function or task.

---

## ELAB-331

(error) %s Attempt to disable block '%s' out of task scope is not supported.

### Description

You receive this error message when a disable statement inside a task refers to a block that is either outside the task or that does not contain the disable statement. The Presto HDL Compiler of Verilog supports only disable statements that refer to a block inside the current task.

Feedback

### What Next

Change the design so that all disable statements are within the block they disable, and that all disable statements refer to blocks within the same function or task.

---

## ELAB-332

(warning) %s Concatenations cannot have unsized numbers; assuming 32 bits.

### Description

You receive this warning message when a concatenation contains an unsized constant. The compiler cannot calculate the correct size of the concatentation. The size is assumed to be 32 bits.

This construction is disallowed by the IEEE Verilog standard, but is permitted by the compiler for backward compatibility.

The following two concatenations will trigger this warning:

```
x = { 0, a };
  z = { b, 'hf, c };
```

### What Next

Avoid this style. Use an explicit size on all constants within concatenations.

---

## ELAB-333

(warning) %s This SV Assertion is ignored for synthesis since %s is not set to true.

### Description

This warning message issues when the Presto HDL Compiler finds a labelled assertion or assertion severity system task call whose usage is not confirmed by the tcl environment variable setting cited in the message. Because SystemVerilog Assertions (SVA) can lead synthesis to using a larger set of "don't care" conditions during optimization, you must separately certify that these assertions have proven to be unreachable prior to synthesizing this RTL.

### What Next

This is only a warning message. No action is required.

However, if you do not want the construct to be ignored, you must set hdlin_enable_assertions to true and also set an element of the confirmed_SVA environment array to true whose element name identifies the label and lexical scope of the applicable assertion. The most specific form of its element name is used in the warning message when hdlin_enable_assertions is already true.

Feedback

The recommended way to confirm assertions is to assemble a separate tcl file to source into your dc_shell before elaborating the design elements which use SVA. This same file should then be sourced into fm_shell when formally verifying the results of this elaboration.

## ELAB-334

(error) %s Part-select direction does not match declared direction of symbol '%s' (declared [%d:%d], part-select is [%d:%d]).

### Description

You receive this error message when a part-select statement on a variable conflicts with the variable's declaration. If the variable is declared with the left bound greater than the right bound, then the part-select must also use a left bound greater than a right bound, and vice versa.

For example, the following code will trigger this error because the part-select 'x[1:2]' does not match the declared direction of 'x'.

```
module test(x, y);
        input  [2:0] x;
        output [1:0] y;

    assign y = x[1:2];

    endmodule
```

### What Next

Correct the part-select statement so that the bounds are compatible with the variable's declaration.

## ELAB-335

(warning) %s 'Case' statement is full and has only one nontrivial branch; it will be inlined.

### Description

You receive this warning message because a case statement in the design either is marked as a full case by the user, or is proved to be a full case by the compiler, and has only one useful branch. Because in a the full case, at least one branch must execute, the single branch of the case statement will always be executed. The compiler has detected this situation and will skip generating logic to test the case condition, executing the code in the body as if it were not surrounded by the case statement.

Feedback

Most commonly, this message arises when case statements have one normal branch, and a default branch that only assigns variables to 'bx. For example:

```
module test(a,y);

       input  a;
       output y;
       reg        y;

       always
          case (a)
            1'b1: y = 1'b1;
            default: y = 1'bx;
          endcase
    endmodule
```

In this case, the presence of the default branch means that the case statement is full (that is, one branch will always be taken.) Then, Because the default branch only assigns variables to 1'bx, it can be optimized away, and the resulting single-branch case can be simplified to unconditionally assign y = 1'b1.

In the next example, the // synopsys full_case directive tells the compiler that all possible values of *x* are handled in the case statement. However, there is only one branch. The compiler concludes that *a* must always be 1. Thus, the branch will always be executed.

```
module test(a,y);

       input  a;
       output y;
       reg        y;

       always
          case (a) // synopsys full_case
            1'b1: y = 1'b1;
          endcase
    endmodule
```

This second example is an uncommon coding style. The compiler issues this warning to remind you to verify that the case statement is as you intended it.

**What Next**

Verify that the case statement implements the behavior you intended.

---

## ELAB-336

(error) %s Presto does not support processes with multiple event statements.

Feedback

### Description

You receive this error message because Presto Compiler does not support multiple event statements in the same process.

### What Next

Change the source code to remove the use of multiple event in the same process.

---

## ELAB-337

(warning) %s Assignment to supply0/1 variable '%s' is ignored.

### Description

You receive this warning message if you attempt to make an assignment to a variable declared as 'supply0' or 'supply1'. These variables always have values of 0 and 1, respectively; any assignments you attempt are ignored.

### What Next

Remove the specified assignment, then reexecute the command.

---

## ELAB-338

(warning) %s Module '%s' contains a supply%d variable '%s'. Replacing with wire driven by a continuous assignment to %d.

### Description

You receive this warning message when a supply0 or supply1 variable appears in your design.

### What Next

This is an informational message only. No action is required on your part.

---

## ELAB-339

(warning) %s Symbol '%s', declared as an enum, may be assigned non-enum values.

### Description

A variable declared using the 'synopsys enum' pragma may be assigned to a value that is not a member of the enum. An enumerated type declaration represents an assertion by the user that a variable can only be assigned a certain set of values; if the design assigns a value that is not part of the user-declared enum, results may be unpredictable.

Feedback

The compiler is not always able to determine conclusively when an expression will evaluate to a value listed in the enum. For example, the compiler cannot determine the possible values of input ports. Another case occurs when a symbol declared as an enum is assigned the result of an arithmetic or logical expression that cannot be evaluated to a constant at compile time.

In the following example, the expression 'current_state + 2'b01' will trigger this error. By examining the code, you can determine that the expression evaluates to a value that is contained in the enumerated type, except when 'current_state' is equal to 'last_state'. Since it is only executed when 'current_state' is not 'last_state', it would be safe to leave this code as is.

```
module test (clock);

      input clock;

      parameter [1:0] /* synopsys enum states */ first_state = 2'b00,
        second_state = 2'b01, last_state = 2'b10;

      reg [1:0] /* synopsys enum states */ current_state, next_state;

      always @(posedge clock) begin
         if (current_state == last_state)
           next_state = first_state;
         else
           next_state = current_state + 2'b01;
      end
    endmodule
```

However, if the 'always' block were replaced with:

```
assign next_state = current_state + 2'b01;  // result is 2'b11 when
                                                 // current_state ==
 last_state
```

then 'next_state' could be assigned 2'b11. This contradicts the user-declared enum, and could have unpredictable results and cause simulation/synthesis mismatches.

**What Next**

If you receive this warning, make certain that the symbol is ONLY assigned a values that are listed in the enumerated type declaration.

If the variable is assigned to the value of an input port, make sure that the port will only be assigned to values within the enum. If that is the case, declare the port as a member of the same enumerated type using the 'synopsys enum' pragma.

If the variable is assigned to an expression result, make sure that expression can never evaluate to a non-enum value.

Feedback

If you cannot determine that the value assigned will be a member of the declared enumerated type, you must remove the enum declaration from the variable.

## ELAB-340

(warning) %s Enumerated type '%s' is not compatible with type of symbol '%s'.

### Description

You receive this error message because the specified enumerated type is not compatible with the specified symbol type. This error occurs at elaboration when parameters in the size specifications are resolved.

In the following example, the variable *current_state* has bounds of [0:SIZE], which resolves to [0:3], but the enumerated type states has bounds of [0:SIZE2], which resolves to [0:2].

```
parameter SIZE = 3;
   parameter SIZE2 = 2;

   parameter [0:SIZE] /* synopsys enum states */ state1 = 4'd0, state2 =
 4'd1;
   reg [0:SIZE2] /* synopsys enum states */ current_state;
```

### What Next

If it is a typo, modify the design so that the symbol and the enumerated type share the same type.

If user intends to let the symbol and the enumerated type have different type, the user have to make sure that the synthesis behavior is what the user wants.

## ELAB-341

(error) %s The number of enum encodings given does not match the number of literals.

### Description

You receive this error message when, in an enumerated type declaration, there is a different number of literals and encodings.

### What Next

Fix the design so that the number of encodings and literals is the same.

## ELAB-342

(error) %s Expression involving %s operator cannot have a real operand.

Feedback

### Description

You receive this error message because the specified operator does not support real operands. Real operands are not supported for synthesis.

### What Next

If your design is intended for synthesis only, remove real operands from the design.

---

## ELAB-343

(error) %s Source file for '%s' was not analyzed by this release of the compiler; re-analyze it.

### Description

Your *read, analyze,* or *elaborate* command requires data about the specified entity, module, or package that was stored by a previous *analyze* command. A file of such data was found, but it had been produced by an incompatible version of the Presto HDL compiler. The file cannot be read in its current version.

### What Next

Reanalyze all design source files needed for this compilation using *this release* of Design Compiler - on any platform. Or adjust your working directory settings to use the environment that contains the properly analyzed sources.

When the analyzed data files are in synch with the compiler release you are using, reexecute the failed command.

---

## ELAB-344

(error) %s File '%s' does not contain an analyzed design.

### Description

You receive this error message when you execute the *read* or *elaborate* command and specify a file that does not contain an analyzed design. The compiler was expecting a file that is an analyzed design. The Presto HDL Compiler stores pre-analyzed designs in disk files.

### What Next

Re-analyze the module or entity you are elaborating or reading and specify a file that contains an analyzed design.

Feedback

## ELAB-345

(error) %s Symbol '%s' is a function that is being called as a task.

**Description**

You receive this error message when a function in your design is being called as a task. A task call is a stand-alone statement, whereas a function call generates a value.

**What Next**

Modify the design so that the function is called only in places where an expression is expected.

## ELAB-346

(error) %s Variable '%s' must be non-negative.

**Description**

You receive this error message when a variable that controls the Presto HDL Compiler has a negative value. Only non-negative values are supported.

**What Next**

Identify the variable that has a negative value and change it to a non-negative value. See the manual page of the variable for valid values.

## ELAB-347

(warning) %s Division by 0 evaluates to 'x'.

**Description**

You receive this warning message because an expression contains a division by 0, which evaluates to 'x' in Verilog.

**What Next**

Verify that you intended to write an expression with 0 as the divisor.

## ELAB-348

(error) %s Incorrect datatype for variable '%s'.

Feedback

### Description

You receive this error message when a variable that controls the behavior of the Presto HDL Compiler has a value that is the wrong datatype. This occurs when a non-numeric value is assigned to a variable that should contain a string, or when a variable that should be *true* or *false* has a numeric value, or contains another string.

### What Next

Use the correct datatype in interactive use, and correct any scripts that set the variable. To determine what type of value the variable should have, see the manual page of the appropriate variable.

---

## ELAB-349

(warning) %s Potential simulation-synthesis mismatch if index exceeds size of array '%s'.

### Description

You receive this warning message if the bit-width of the index expression is larger than the bit-width of the array bounds, or if an array has bounds that cannot be expressed as [0 : 2^n-1] or vice versa.

An array in the program is indexed by an expression that can evaluate to an out-of-bounds index, and the compiler cannot determine if the index will evaluate to a legal subscript.

If the index does evaluate to an out-of-bounds value, the synthesized result may not match Verilog simulations. Assigning to, or reading from, an out-of-bounds location may have unpredictable results.

For example, in the following Verilog module, a register is indexed by the input port 'addr'. If 'addr' ever has the value 3'd7, then the register will be accessed out-of-bounds, but the correct logic can be generated.

```
module test(addr, data, write);
  input [2:0] addr;
  input data, write;
  reg [6:0] mem;

always @(addr or data or write)
  if (write)
    mem[addr] = data;

endmodule
```

For example, in the following Verilog module, a register is indexed by the input port 'addr'. If 'addr' ever is more than the value 4'd6, then the register will be accessed out-of-bounds, and the bad logic will be generated. When 'addr' ever has the value 4'd8, it will access mem[0], not mem[8].

Feedback

```
module test(addr, data, write);
  input [3:0] addr;
  input data, write;
  reg [6:0] mem;

always @(addr or data or write)
  if (write)
    mem[addr] = data;

endmodule
```

### What Next

Verify that the index expression cannot evaluate to an out-of-bounds value.

---

## ELAB-350

(error) %s Can't resolve constant expression for symbol '%s'.

### Description

In certain circumstances, the Presto HDL compiler may not evaluate constant expressions with operands larger than 32 bits. To work around this problem, pre-compute such expressions in your design, or use HDL Compiler to elaborate that design. This problem will be fixed in an upcoming release.

### What Next

Either pre-compute the value, or use HDL Compiler on this design.

---

## ELAB-351

(warning) %s Comparison against 'unknown' value is always false; may cause simulation/ synthesis mismatch.

### Description

You receive this message because the result of a comparison, by definition, can produce only one result. In the following example, the body of the first IF statement always executes, and the body of the second IF statement never executes. All comparisons against the uninitialized value *U* fail because the result of an equality comparison against *U* is always false, and the result of a disequality comparison against *U* is always true.

```
IF(A2 /= "U") THEN
      ONE_2 <= '1';              -- SHOULD ALWAYS FIRE
    END IF;

    IF(A2 = "U") THEN
      ZERO_2 <= '1';             -- SHOULD NEVER FIRE
    END IF;
```

Feedback

### What Next

Ensure that you intended to compare against *U*; and, if not, correct the comparison.

---

## ELAB-352

(error) %s Operator '%s' not found in synthetic library.

### Description

You receive this message because the operator defined by a *map_to_operator* attribute is not described in any of the synthetic libraries in the *synthetic_library* variable (and is not found in *standard.sldb*).

### What Next

To use the synthetic library operator, ensure that it can be found in the files defined by the *synthetic_library* variable.

---

## ELAB-353

(error) %s Width of the actual at the port '%s' does not match. %s

### Description

You receive this message because a temporary signal/variable does not have a proper bit width.

The following example shows that a *temp_out* signal connected to a *Y* port is too wide or too narrow when represented in hardware.

```
architecture A of E is
component C1
   port (A : integer range 0 to 15;
         Y : integer range 0 to 15);
end component;
signal temp_in : integer range 0 to 15;
signal temp_out : integer range 0 to 31;
begin
U1: C1 port map (A => temp_in, Y => temp_out);
end A;
```

### What Next

Define a temporary signal/variable with the proper bit width. The following example is a modification of the one above to provide a proper bit width for the *temp_out* signal.

```
architecture A of E is
component C1
   port (A : integer range 0 to 15;
```

Feedback

```
         Y : integer range 0 to 15);
end component;
signal temp_in : integer range 0 to 15;
signal temp_out : integer range 0 to 31;
signal temp : integer range 0 to 15;
begin
temp_out <= temp;
U1: C1 port map (A => temp_in, Y => temp);
end A;
```

## ELAB-354

(error) %s Cannot read architecture file %s: %s.

### Description

You receive this message because during a read or elaborate command, the Presto HDL compiler cannot open or read a file describing the source language (and, for VHDL, the architecture) of the module or entity being elaborated.

You also receive this message if the module or entity has not yet been analyzed or if the module or entity name is misspelled.

### What Next

Ensure that the module or entity being elaborated has already been analyzed, and that the entity or module name is correctly specified.

## ELAB-355

(information) %s Elaborating module '%s'.

### Description

You automatically receive this message for each module elaborated during a read or elaborate command.

### What Next

No action is necessary.

## ELAB-356

(error) %s No modules exist in the analyzed file.

Feedback

**Description**

You receive this message because the Presto HDL compiler finds no modules or entities in the file being read from the *read* command using a *verilog* or *vhdl* format. The error is reported by the *read* command.

**What Next**

Ensure that the file being read contains at least one module or entity.

---

## ELAB-357

(error) %s Module '%s' cannot be found for elaboration.

**Description**

You receive this message because the Presto HDL compiler cannot find the module to be elaborated. The error is reported by the *elaborate* command.

**What Next**

Ensure that the module to be elaborated has been analyzed.

---

## ELAB-358

(error) %s Mismatching number of connections in module/component instantiation.

**Description**

The number of port connections of the module/component instantiation does not match the number of ports in the module/component declaration. Either you specified too many or not enough connections to the module/component.

**What Next**

Check against the module/component declarations to find out which ports are unconnected or whether you specified too many connections.

---

## ELAB-359

(error) %s The procedure/task %s cannot be mapped to %s because it has an inout port (%s).

**Description**

HDL Compiler issues this error when, during the activity of the *read* command or *elaborate* command, the compiler encounters a procedure or task that has a map-to-entity or map-

Feedback

to-operator or map_to_module directive and also has an inout port or signal port. HDL Compiler does not support inout or signal ports on mapped procedures.

**What Next**

Remove the inout port from the mapped function or procedure, and issue the command again.

---

## ELAB-360

(error) %s The REAL data type is not supported for synthesis.

**Description**

HDL Compiler does not support the use of the REAL data type when synthesizing VHDL.

**What Next**

Rewrite your design so that it does not use the REAL data type.

---

## ELAB-361

(error) %s Unable to open standard synthetic libraries; check for correct installation.

**Description**

The Presto HDL Compiler could not open the standard synthetic library file, standard.sldb, or the standard GTECH library, gtech.db. This probably indicates that Design Compiler is not installed correctly.

**What Next**

Verify the successful completion of the the installation of Design Compiler.

---

## ELAB-362

(error) %s Cannot find port '%s' on synthetic operator '%s'.

**Description**

This error message occurs when the Presto HDL Compiler cannot find a port it expected to find on a synthetic operator. This usually indicates that a function that uses the map_to_operator pragma has incorrect parameter names for the synthetic operator to which it is mapped.

**What Next**

Verify that any functions mapped to synthetic operators use the correct port (parameter) names.

Feedback

## ELAB-363

(error) %s Incorrect value for hdlin_infer_mux: '%s'; allowed values are 'all', 'default', or 'none'.

### Description

This error message occurs when an incorrect value is set for the *hdlin_infer_mux* variable. Allowed values for *all*, *none*, and *default*.

When *all* is the value, HDL Compiler attempts to infer a MUX_OP for a signal or variable assigned in a case statement.

When *none* is the value, HDL Compiler does not attempt to infer any MUX_OPs for a Verilog or VHDL design.

When *default* (the default value) is the value, HDL Compiler attempts to infer a MUX_OP for a signal or variable assigned in a case statement or if statement, if the statement is either marked with the *infer_mux* directive, or is in a process associated with the *infer_mux* attribute or directive. A MUX_OP cannot be inferred if it violates the *hdlin_mux_size_limit* variable.

For details, see the *HDL Compiler for Verilog User Guide* or the *HDL Compiler for VHDL User Guide*.

To determine the current value of this variable, use the *printvar hdlin_infer_mux* command. For a list of HDL variables and their current values, use *print_variable_group hdl*.

### What Next

Set *hdlin_infer_mux* to a value of *all*, *default*, or *none*.

## ELAB-364

(warning) %s Replication constant {%d{...}} is non-standard.

### Description

You get this message when elaborating a Verilog-1995 or Verilog-2001 module and a zero replication constant is encountered and the application option *hdlin.verilog.v2005_replication_semantics* has been disabled.

A replication constant in a Verilog or SystemVerilog concatenation, such as {P{expression}}, can be a positive number or 0. In early versions of Verilog before 2005, it was non-standard for it to be 0, but most tools would implement {0{expression}} as 1'b0 instead of issuing an error. Verilog-2005 implements {0{expression}} as empty, but permits such empty replication results only within an otherwise non-empty concatenation.

Feedback

In Fusion Compiler that old behavior on zero replication constants in Verilog-1995 and Verilog-2001 can still be obtained by overriding

```
set_app_options -name hdlin.verilog.v2005_replication_semantics -value
 false
```

The setting of hdlin.verilog.standard at the time the Verilog source is analyzed determines whether the old implementation of the replication operator can be used when elaborating zero replications. If the module was analyzed as Verilog-2005, then overriding the above variable has no effect.

**What Next**

Make sure you intend to use a zero replication constant, and, if so, make sure you intend to use the old behavior. A risk of doing so is that it might be a mismatch with your simulator, because most tools use the Verilog-2005 semantics for this by default. If you don't want the old behavior, find where the application variable has been disabled in your script and undo it. If you need consultation about migrating legacy codes to current standards, contact Synopsys.

---

## ELAB-365

(warning) %s Net '%s' or a directly connected net is driven by more than one process or block.

**Description**

You receive this warning when a 'reg' variable is driven by more than one always block or a 'wire' variable is driven by more than one continuous assignment and you have also set the variable *hdlin_prohibit_nontri_multiple_drivers* to *false*.

*Note*: This warning is issued because an invalid design could result.

In simulation, if a 'reg' variable is driven by more than one always block, the definition, in effect, depends on which block executed most recently. In synthesis, since all blocks are concurrently executing at all times, this is not the case, and invalid designs can result if the drivers are simply shorted together. However, when the *hdlin_prohibit_nontri_multiple_drivers* variable is set to *false*, multiple drivers of 'reg' variables are permitted.

A multiple-driven 'wire' variable is prohibited by the Verilog standard. Although some simulators permit this behavior, HDL Compiler issues the ELAB-366 error if it occurs. However, when *hdlin_prohibit_nontri_multiple_drivers* is set to *false*, multiple drivers are permitted. *Note* that this variable setting might cause the generation of invalid designs.

The compiler permits multiple drivers on nets declared as 'tri', although it issues a warning if all the drivers are not three-state devices.

Feedback

### What Next

An invalid design might result when this warning is issued. Multiple driver nets are only generated because the *hdlin_prohibit_nontri_multiple_drivers* variable is set to *false*.

Consider modifying your design, to avoid driving a variable from more than one always block or continuous assignment.

---

## ELAB-366

(error) %s Net '%s' or a directly connected net is driven by more than one source, and not all drivers are three-state.

### Description

You receive this error message when a 'reg' variable is driven by more than one always block or a 'wire' variable is driven by more than one continuous assignment or input port and not all of the drivers are three-state.

In simulation, if a 'reg' variable is driven by more than one always block, the definition, in effect, depends on which block executed most recently. In synthesis, since all blocks are concurrently executing at all times, this behavior is not possible, and invalid designs can result if the drivers are simply shorted together. Therefore, HDL Compiler issues an error if any bits of any variable are driven by more than one process.

A multiply-driven net declared as 'wire' is not supported in synthesis and Presto HDL Compiler issues the ELAB-366 error if it occurs. If you want the compiler to permit such behavior, set the *hdlin_prohibit_nontri_multiple_drivers* variable to *false*. *Note* that this variable setting might cause the generation of invalid designs.

The compiler permits multiple drivers on nets declared as 'tri', although it issues a warning if all the drivers are not three-state devices.

### COMMON ISSUES

Sharing the same loop index variable between processes is a common source of this error. At the end of each process or always block, the loop index variable is driven by the last value it had during the loop.

```
integer i;
  always begin
    if (c)
      for (i = 0; i < N; i = i + 1)
        x[i] = ~x[i];
  end
  always begin
    if (c)
      for (i = 0; i < N; i = i + 1)
        y[i] = ~y[i];
  end
```

Feedback

To avoid this situation, use different variables in the different always blocks or declare the loop index variables locally, as in the following example.

```
always begin: x_block
    integer i;
    if (c)
      for (i = 0; i < N; i = i + 1)
        x[i] = ~x[i];
  end
  always begin: y_block
    integer i;
    if (c)
      for (i = 0; i < N; i = i + 1)
        y[i] = ~y[i];
  end
```

In some cases the compiler can determine that the variable *i* is never read before being assigned in the block, and in those cases the error is not issued.

Previous versions of HDL Compiler treated loop variables as a special case and did not allow them to be read after the loop. However, the Presto HDL Compiler treats loop variables as full-fledged program variables, which can lead to this situation.

**What Next**

Modify your design to avoid driving a variable from more than one always block or continuous assignment. Or if all drivers of the variable are three-state devices, change the declaration of the variable to 'tri', which permits multiple drivers.

Another alternative is to set *hdlin_prohibit_nontri_multiple_drivers* to *false*. *Note* that this setting might allow the generation of invalid designs to be generated. For that reason, it is not recommended.

---

## ELAB-367

(error) %s Clock expression must be one bit wide.

**Description**

An expression you use as a clock signal must be one bit wide. If you use an expression that is wider than one bit, you receive this error message. In Verilog, you specify the clock using an event statement, such as "@(posedge clock)". In VHDL, you specify a clock event using "if (clock'event and clock = '1')".

**What Next**

Modify your design so that all clock expressions are one bit wide.

Feedback

## ELAB-368

(error) %s Net '%s', or a directly connected net, is driven by more than one source, and at least one source is a constant net.

### Description

You receive this error when a 'reg' variable is driven by more than one always block, or a 'wire' variable is driven by more than one continuous assignment or input port, and at least one of the drivers is a constant. This is an error because an invalid design results when the output of a gate is connected directly to the Logic0 or Logic1 nets.

In simulation, if a 'reg' variable is driven by more than one always block, the definition, in effect, depends on which of the blocks executed most recently. In synthesis, since all blocks are concurrently executing at all times, this behavior is not possible, and invalid designs can result if the drivers are simply shorted together. Therefore, HDL Compiler issues an error if any bits of any variable are driven by more than one process.

A multiply-driven 'wire' variable is prohibited by the Verilog standard. Although some simulators permit this behavior, HDL Compiler issues the ELAB-366 or ELAB-368 error in this situation. If you wish to permit this behavior, you may set *hdlin_prohibit_nontri_multiple_drivers* to *false*. *Note* that this variable setting may cause the generation of invalid designs.

Ordinarily, multiple drivers are permitted on nets declared as 'tri'; however, constant nets are an exception to this rule because the implementation would result in an invalid design.

### COMMON ISSUES

A common source of this error message is sharing the same loop index variable between processes. At the end of each process or always block, the loop index variable is driven by the last value it had during the loop.

```
integer i;
  always begin
    if (c)
      for (i = 0; i < N; i = i + 1)
        x[i] = ~x[i];
  end
  always begin
    if (c)
      for (i = 0; i < N; i = i + 1)
        y[i] = ~y[i];
  end
```

To avoid this situation, either use different variables in the different always blocks, or declare the loop index variables locally, as in the following example.

```
always begin
    integer i;
```

Feedback

```
    if (c)
      for (i = 0; i < N; i = i + 1)
        x[i] = ~x[i];
  end
  always begin
    integer i;
    if (c)
      for (i = 0; i < N; i = i + 1)
        y[i] = ~y[i];
  end
```

In some cases the compiler can determine that the variable *i* is never read before being assigned in the block, and in those cases the error is not issued.

Previous versions of HDL Compiler treated loop variables as a special case, and did not allow them to be read after the loop; however, the Presto HDL Compiler treats loop variables as full-fledged program variables, which can lead to this situation.

**What Next**

Modify your design to avoid driving a variable from more than one always block or continuous assignment.

When none of the drivers is a constant net, another alternative is to set *hdlin_prohibit_nontri_multiple_drivers* to *false*. However, when one of the drivers is a constant net, an error is always generated, because the result would be an invalid design.

---

## ELAB-369

(error) %s The width of port %s on instance %s of design %s is inconsistent with other instantiations of the same design.

**Description**

You receive this error message when a module is instantiated in your design more than once, but the width of the ports is inconsistent among all of the instances.

In the following example, the module named *foo* is instantiated twice, but the ports named *a* and *b* have different widths, causing the error message.

```
module E (a, b, out1, out2);
input [2:0] a;
input [3:0] b;
output [3:0] out1, out2;

foo U1 (a, out1);
foo U2 (b, out2);
endmodule
```

Feedback

The instance for which this error is reported may not be the actual instance that uses the wrong width. Since the design is not yet linked, the compiler cannot check which of the instances is correct and can only report the inconsistency.

In the example above, the error is reported for instance U2, because instance U1 has already been seen and uses a different width. As the first in the list, the module named *foo* might require a 3-bit port, a 4-bit port, or another size. Errors that affect all instances equally are checked during linking instead of during elaboration.

**What Next**

Correct the width of the ports so that they are consistent, and then invoke the compiler again.

---

## ELAB-370

(warning) %s No MUX_OP inferred for the case because it might lose the benefit of resource sharing.

**Description**

You receive this warning message because the compiler did not infer the MUX_OP you specified for the case statement, because the case statement would lose the benefit of resource sharing.

**What Next**

This is a only a warning and requires no action on your part.

---

## ELAB-371

(warning) %s Pragmas has been removed in compiling.

**Description**

The tool issues this warning when pragmas are removed in the specified location during compilation, usually in rewriting intermediate language code.

For example, when inferring carryin for the following statement, one of the two adders will be removed, along with the label pragma associated with that adder.

```
a = a1 + //synopsys label adder1
    a2 + //synopsys label adder2
    cin;
```

**What Next**

No action is required on your part. However, if you want to avoid the removal of pragmas, change your Verilog code.

Feedback

## ELAB-372

(warning) %s Unsupported use of user-encoded enumeration type as array index.

### Description

This warning message occurs because synthesis does not support user-encoded enumeration types used as an array index. A user-encoded enumeration type is one that has an associated ENUM_ENCODING attribute.

During synthesis, enumeration values are implemented by choosing a binary encoding. VHDL requires that the meaning of an enumeration value used as an index is the position of the enumeration value in the enumeration type definition. Since the encoded value and the position are usually different values, correct synthesis requires the insertion of inefficient decoding logic.

### Examples

The following declaration of A1 is not supported because the index type ENUM1 is a user-encoded enumerated type:

```
...
attribute ENUM_ENCODING : string;
attribute ENUM_ENCODING : string;
type ENUM1 is (M1, M2, M3, M4);
attribute ENUM_ENCODING of
    ENUM1 : type is "0010 1000 1001 1111";

type A1 is array (ENUM1 range <>) of BOOLEAN;
 ...
```

### What Next

This is only a warning message. No action is required.

However, if the result is not what you intended, rewrite the source code so that it does not use arrays with user-encoded enumeration types as an index. The tool allows the use of the enumeration type with default encoding as an array index type.

## ELAB-373

(warning) %s Potential overflow which may cause synthesis and simulation mismatch in operator '%s'.

### Description

This warning message occurs when the tool detects a potential overflow in an operator (such as exponent or abs) of the expression.

Feedback

For example, in the following expression if *tmp* is less than zero, there will be an integer overflow.

```
2 ** (31 - tmp)
```

In the following expression, if *tmp* is the minimum integer, there will be an integer overflow.

```
abs(tmp)
```

**What Next**

This is a warning message only. No action is required on your part. However, it is good practice to remove the overflow exponent or abs from the expression.

---

## ELAB-374

(information) %s Inferred unloaded sequential element for %s.

**Description**

You receive this message when the Presto HDL Compiler infers a sequential element that is not required for proper functioning of the design being elaborated, but whose inference was requested by the user.

An RTL-level design description in Verilog or VHDL may contain variables that ordinarily would create a sequential cell (either because they are assigned under a clock edge, or because they are conditionally assigned) but whose values are never read, or whose values are not used in computing any output of the design. Because the their values are not used (that is, they have no loads, or no path to the the outputs), by default HDL Compiler (Presto) will not create sequential cells for these variables.

In some circumstances designers wish to retain these sequential cells. Unloaded sequential cells can be inferred by two methods. First, if hdlin_preserve_sequential is TRUE, then a sequential cell will be inferred for any variable that is conditionally assigned or assigned under a clock, regardless of whether its value is used or not. Second, if hdlin_preserve_sequential is not set, but some variables are marked with the "preserve_sequential" pragma, only those variables will receive sequential cells even if they have no loads.

**What Next**

Examine the set of variables for which unloaded sequential cells are inferred.

If you do not want any unloaded sequential cells, make sure that hdlin_preserve_sequential is "none" and no pragmas appear in your HDL source files.

If you do intend to infer some unloaded sequential cells, but some unwanted cells also appear, and the variable hdlin_preserve_sequential is not "none", consider using the pragma method for finer control over the inference.

Feedback

Otherwise, no action is required on your part.

## ELAB-375

(error) %s Could not resolve hierarchical reference '%s'.

**Description**

You receive this message when the object referred to by a hierarchical identifier could not be found.

**What Next**

Find the object that the hierarchical identifier was meant to refer to and fix the hierarchical name to correctly refer to this object.

## ELAB-376

(Error) %s elaborating interface '%s'.

**Description**

This error message occurs if you attempt to elaborate a SystemVerilog interface, because that is not supported in synthesis.

**What Next**

Refer to the SystemVerilog User Guide for recommended use models.

## ELAB-377

(warning) %s The resolution function '%s' is not marked with a resolution method directive. It is being ignored.

**Description**

The tool issues this warning message if any function that is used as a resolution function is not marked with one of the following resolution method directives:

```
-- synopsys resolution_method wired_and
-- synopsys resolution_method wired_or
-- synopsys resolution_method wired_three_state
```

The tool does not support arbitrary resolution functions. Only *wired_and*, *wired_or*, and *three_state* functions are allowed. The tool requires that all resolution functions are marked with one of the resolution method directives indicating the kind of resolution being performed.

Feedback

This resolution function will be ignored.

**What Next**

This is a warning message only. No action is required on your part. However, if you do not want the resolution function to be ignored, add one of the following resolution method directives to the resolution function:

```
-- synopsys resolution_method wired_and
-- synopsys resolution_method wired_or
-- synopsys resolution_method wired_three_state
```

---

## ELAB-378

(error) %s The content '%s' is not defined in interface %s's instance '%s'.

**Description**

This error message issues when you attempt to hierarchically access the content of the interface via its instance name, but the content is not defined in the interface. The following example contains two causes of this error message:

```
interface simple_bus;
  logic a;
  modport mp( output a );
endinterface;

module top(input i, output logic o);
   bottom     forward(  sb.mp ); // A modport isn't
 hierarchically-accessible content
  simple_bus  sb();              // Adds the instance's modport to this
 scope.
   assign o =       i & sb.b;    // Error: "b" is not defined in the
 interface
   bottom     backward( sb.mp ); // OK to refer to a known-instance's
 modport
endmodule
```

All forward references to an instance are presumed to be "hierarchical"; they may only refer to content found in some upcoming instance. When you connect an interface instance, or one of its modports to an interface port, the interface's instance must be instantiated before it can be connected to the down-design. Such interface.modport references are not "hierarchical", even though they use the same "." notation; the modport is not "content". By the same token, a port expression that is a forward reference to the whole interface instance will be read as an implicit wire, and the instance name will thus be defined-already come the instantiation.

Feedback

### What Next

Access the correct interface content or else add the content definition into the cited interface declaration, reanalyze the interface definition, and re-elaborate the module that instantiates it. If, instead, you are trying to connect an interface port: relocate the interface instantiation statement prior to this reference within in the same scope.

---

## ELAB-379

(Error) %s value of double type is not supported in synthesis.

### Description

You receive this error message because you are attempting to synthesize a real, float, or double data type, which synthesis does not support.

### What Next

Do not use real, float, or double data types for synthesis. Correct the data type and run the command again.

---

## ELAB-380

(Error) %s name of port %s is inconsistent with module definition or other instances.

### Description

You receive this error message because the specified port name is inconsistent with module definition or other module instances having the named port connection. One of the ports is an interface or modport instance in the module instantiation.

In the following example, the foo module is instantiated with a named port connection. It uses the named ports a, o, and ifc, where ifc is an interface instance. However, a is not in the port list of the foo module, so the error message occurs.

```
interface ifc;
logic x,y;
endinterface

module E (a, out1);
input [2:0] a;
output [3:0] out1;
iface ifc();
foo U1 (.a(a), .o(out1), .ifc(ifc));
endmodule

module foo(input [2:0] i, output [3:0] o, iface ifc);
 ...
endmodule
```

Feedback

### What Next

Correct the names of the ports so that they are consistent, and then invoke the compiler again.

---

## ELAB-381

(error) %s The formal %s port '%s' is connected to an incompatible %s.

### Description

This error message issues during the specialization of a module or interface that is done to accomodate its interface ports. The interface type name or modport name of the incoming interface instance (if any) is not compatible with the declaration of the corresponding (cited) formal port. Here are some sample cases of this mismatch:

- Connecting an interface port or instance to a wire or variable port
- Connecting a wire or variable to an interface/modport port
- Connecting an interface port or instance instantiated from one template to an interface port declared with a different interface template name.
- Connecting one modport of an interface instance to a different modport (by name)

In the following example, the module named foo is instantiated with the ports named a, out1, and ifc. The ifc port is an interface instance that is connected to the output port named o2. However, the o2 port is not a compatible interface port. This causes the error message.

```
interface iface;
  logic x,y;
  modport MP(input x, output y);
endinterface

module E (a, out1);
  input [2:0] a;
  output [3:0] out1;
  iface ifc();
  foo U1 (ifc.MP);
  ...
endmodule

module foo(output o2);
 ...
endmodule
```

Another example that results in the error message is when you attempt to connect a modport of an interface instance to a different modport port.

Feedback

```
interface iface;
  wire x,y;
  modport MP1(input x, output y);
  modport MP2(input x, output y);
endinterface

module E (a, out1);
  input [2:0] a;
  output [3:0] out1;
  iface ifc();
  foo U1 (ifc.MP1);
 ...
endmodule

module foo(iface.MP2 ifc);
  ...
endmodule
```

**What Next**

The immediately preceding HDL-193 ("Building ...") indicates one design's reference cells which are incompatible with the cited port. Correct the ports in the module instantiation and/or the module definition so that they agree on what is being connected. Re-analyze the changed RTL and re-link the design.

---

## ELAB-382

(error) %s Module %s was not elaborated because it contains the generic interface as port.

**Description**

You receive this error message because you are elaborating a module with generic interface as port.

**What Next**

Specify the interface in elaboration or elaborate the upper designs containing instantiations of this module.

---

## ELAB-383

(error) %s Too many elements in assignment pattern expression list

**Description**

The elements listed in an aggregate expression must be in one-to-one correspondence with the members of the structure type or the elements of the array type into which the aggregate expression is being assigned, aggregated, or cast.

Feedback

### Examples

In the following example, structures of type *shoe* have just two members. Since *shoe* is not declared to be a *packed* structure, lists enclosed within curly braces that are assigned to *shoe*s are interpreted according to the rules for aggregate expressions. Thus, lists longer than than two elements are rejected, even if they would be valid as (packed array) concatenations.

```
typedef struct {
  logic [3:0]heel;
  logic [3:0]toe;
} shoe;

shoe left, right;
shortint tongue;

left = { right.toe, 4'b1111, tongue };      // Shoe is too small
right= { 1'b1, left.heel[2:0], left.toe };  // Not a Verilog concat
```

### What Next

Check for a mismatch between the structure declaration and its uses, due to incompatible versions of two source files.

If concatenation was intended where an unpacked aggregate expression is now parsed, consider changing the structure declaration to *packed*, or regrouping the elements to correspond to the structure members as declared.

---

## ELAB-384

(error) %s Too few elements in assignment pattern expression list

### Description

The elements listed in an aggregate expression must be in one-to-one correspondence with the members of the structure type or the elements of the array type into which the aggregate expression is being assigned, aggregated, or cast.

### Examples

In the following example, structures of type *shoe* have just two members. Since *shoe* is not declared to be a *packed* structure, lists enclosed within curly braces that are assigned to *shoe*s are interpreted according to the rules for aggregate expressions. Thus, lists with less than two elements are rejected, even if they would be valid as (packed array) concatenations.

```
typedef struct {
  bit [3:0] sole;
  bit [3:0] tongue;
} shoe;
```

Feedback

```
shoe left, right;
byte heel;

left = { right.sole }; // Shoe is too big
right= { heel };       // Not a Verilog concat
```

**What Next**

Check for a mismatch between the structure declaration and its uses, due to incompatible versions of two source files.

If concatenation was intended where an aggregate expression is now parsed, check that the structure declaration is *packed*, or try decomposing and regrouping the listed elements to correspond to the structure members as declared.

---

## ELAB-385

(Error) %s A design template for '%s' cannot be found.

**Description**

This error message occurs because the SystemVerilog compiler cannot find an analyzed template needed to inline an instance of a module or interface. Interface and module instantiations are syntactically indistinguishable. When an instance_name forms the root of a hierarchical reference (XMR), or appears as the actual port of another instance, it is presumed to be a templated design. Its content will be copied in-line into the instantiating design.

Unknown identifiers used in place of port directions or user-defined data types are assumed to be interface types; so this error can issue for certain cases that previously would have been cited as syntax errors. Similarly unknown identifiers used as the roots of cross-module-references (XMRs) were previously reported as syntax errors, but now are assumed to be instances of templated modules or interfaces intended to be fully inlined so as to expose their hierarchically referenced content inside the parent design.

**What Next**

Ensure that the interface or module has been analyzed and the intermediate files are in the working directory or the library search path, and then re-elaborate the design which failed.

Check the identifier for typographical errors, or for mistaken usage. For example, the actual interface instance name should not be used as the interface type name when declaring a formal port to receive it. Instead, use the interface type identifier from which that instance was instantiated in the parent design. Conversely the interface or module name should not be used as the root of a hierarchical reference into one of its instances; use the instance name instead.

Feedback

## ELAB-386

(error) %s Name clash on net '%s'.

### Description

You receive this error message because your two names in your design cause a name clash in the netlist. The name clash can be between to same named objects or two similar named objects in conjunction with a naming style.

Eg. when choosing net naming style "%n_%d" the following declarations can create a name clash:

```
// net naming style set to "%s_%d"

  // nameclash regarding r_0:
  reg r_0;
  reg [0:3] r;  // will result in r_0, r_1, r_2 and r_3
```

### What Next

Ensure that all object names in your design are unamibiguous, considering the naming style you are using.

## ELAB-387

(error) %s The '%s' attribute is supported only if that attribute is used in conformance with the style described in the Synopsys manual for the VHDL Compiler.

### Description

You receive this error message when an attribute usage is not in conformance with the style described in the Synopsys manual for the VHDL Compiler.

You might have used the attribute as the RHS expression of blocking assignment, non-blocking assignment, or, in the return expression of function, etc.

### What Next

See the *HDL Compiler for VHDL Reference Manual* for ways to use the attributes.

## ELAB-388

(warning) %s X and/or Z bits occur in an actual parameter to the '%s' design template.

### Description

This warning message occurs when you create an instance of a parameterized subdesign using a constant value that contains X or Z bits.

Feedback

In classical synthesis, these bits are zeroed to form ordinary integer or string constants. However, in the current operating mode, the full four-state parameter value is used as written when the subdesign is elaborated. If the subdesign uses it as a casex or casez statement label, there may be a mismatch between synthesis and simulation results.

Don't care bits in parameters do not indicate wildcard matching within the design libraries.

**What Next**

Convert the casex item into a literal or convert the casex statement into a case statement.

For more information, see the *-parameters* option on the *elaborate* command man page.

---

## ELAB-389

(warning) %s X and Z bits in parameters of subdesign '%s' are set to zero.

**Description**

You are creating an instance of a parameterized subdesign using a constant value which contains x or z bits. In classical synthesis coding styles, parameters are integer or string constants. In the current operating mode, 4-state parameter values are being cast to 2-state values (by setting x and z bits to zero) before the subdesign is elaborated. If the subdesign uses this parameter as a casex or casez statement label, there may be a mismatch between synthesis and simulation results.

Don't-care bits in parameters do not indicate wildcard matching within the design libraries.

**What Next**

Try to convert the casex item into a literal or convert the casex statement into a case statement.

---

## ELAB-390

(warning) %s Variable or function call in casex item is a potential simulation/synthesis mismatch.

**Description**

You receive this warning message when a variable or function call is used in a casex item. Even if an x bit is assigned to a variable or returned by a function call, the value of the case item will not match both 0 and 1, as it would for an x bit in a literal in a casex item. For example:

```
b = 2'b1x ;
   casex (a)
     b : ... // in this case, 2'b1x only matches 2'b10
   endcase
```

Feedback

A full example:

```
module m( o1 , o2 );
output o1, o2 ;
reg o1, o2 ;

  function f ;
  input [1:0] R ;
  casex (2'b11)
    R : f = 1'b0 ;
    default : f = 1'b1 ;
  endcase
  endfunction

  always begin
   o1 = f(2'b1x) ;
  end

  reg [1:0] R;
  always begin
   R = 2'b1x;
   casex (2'b11)
     R : o2 = 1'b0 ;
     default: o2 = 1'b1 ;
   endcase
  end

endmodule
```

**What Next**

Convert the casex item into a literal or convert the casex statement into a case statement.

---

## ELAB-391

(warning) %s Variable %s may be read before being assigned; the synthesized result may not match simulations.

**Description**

You receive this warning message because a variable that is local to a function or task may be read before being assigned. When the *hdlin_infer_function_local_latches* variable is set to *false*, the Presto HDL Compiler does not preserve variable values across separate calls to a task or function, unlike a simulator. Any bits of the variable that are not assigned before being used are given the value 0. No latches are inferred for variables local to a subprogram.

When the *hdlin_infer_function_local_latches* variable is set to *true*, the compiler matches the simulator's behavior. This may result in additional latches in your design.

Feedback

**What Next**

If you intended to use a value set during a previous call, set *hdlin_infer_function_local_latches* to *true* in order to preserve the value across calls.

If you did not intend to use a value across calls, check your design to verify that you assigned all local variables before using them.

---

## ELAB-392

(warning) %s Value of function %s may be used before being assigned; the synthesized result may not match simulations.

**Description**

You should not receive this warning by default. If you do receive this warning, it may or may not be accurate; please evaluate your RTL code accordingly. If you wish to remove this warning from your DC sessions, please contact Synopsys support.

A Verilog function definition implicitly declares a variable, internal to the function, with the same name as the function. You receive this warning message when the function may return without the variable having been assigned or when the variable may be read by the function body before the variable is assigned.

When the *hdlin_infer_function_local_latches* variable is set to *false*, the Presto HDL Compiler does not preserve variable values across separate calls to a task or function, unlike a simulator. Any bits of the variable that are not assigned before being used are given the value 0. No latches are inferred for variables local to a subprogram.

When the variable *hdlin_infer_function_local_latches* is set to *true*, the compiler matches the simulator's behavior. This may result in additional latches in your design.

**What Next**

If you intended to use a return value set during a previous call, set *hdlin_infer_function_local_latches* to *true* in order to preserve the value across calls.

If you did not intend to use a return value across calls, check your design to verify that a meaningful value is returned by the function.

---

## ELAB-393

(warning) %s Value of function '%s' is used before being assigned; the synthesized result may not match simulations.

Feedback

### Description

A Verilog function definition implicitly declares a variable, internal to the function, with the same name as the function. You receive this warning message when the function returns without the variable having been assigned or when the variable is read by the function body before the variable is assigned.

When the variable *hdlin_infer_function_local_latches* is set to *false*, the Presto HDL Compiler does not preserve variable values across separate calls to a task or function, unlike a simulator. Any bits of the variable that are not assigned before being used will be given the value 0. No latches will be inferred for variables local to a subprogram.

When the variable *hdlin_infer_function_local_latches* is set to *true*, the compiler will match the simulator's behavior. This might result in additional latches in your design.

### What Next

If you intended to use a return value set during a previous call, set *hdlin_infer_function_local_latches* to *true* in order to preserve the value across calls.

If you did not intend to use a return value across calls, check your design to verify that a meaningful value is returned by the function.

---

## ELAB-394

(error) %s Specified resoure sharing (label %s) is not realizable. There is conflict among the operations.

### Description

You receive this error message when the specified resource sharing is not realizable. For example:

module ppf (cond, cond2, a,b, c, d, z, y); input cond, cond2; input [3:0] a, b, c, d; output [3:0] z, y; reg [3:0] z, y;

always @(a or b or c or d) begin : b1 /* synopsys resource r0 : ops = "a0 a1", map_to_module = "DW01_add", implementation = "rpl", add_ops="false";*/

if (cond) begin z = a + /* synopsys label a0 */ b; y = c + /* synopsys label a1 */ d; end else z = a + c; end

endmodule

Resoure r0 can not be shared by a0 and a1.

### What Next

Modify resource sharing pragma, make sure the specified resources can be shared.

Feedback

## ELAB-395

(warning) %s Latch inferred in design %s read with 'hdlin_check_no_latch'

**Description**

The variable hdlin_check_no_latch directs the read command to ensure no latch is inferred in the HDL file.

**What Next**

Check whether the value of hdlin_check_no_latch is correct or HDL file contains error hence latches are inferred unexpectedly.

## ELAB-396

(warning) %s Floating net '%s' connected to ground.

**Description**

You receive this warning message when a previously unconnected net has been connected to ground. This message reminds you to check whether the behavior resulting from connecting the net to ground conforms to your expectations.

**What Next**

Verify that connecting the specified net to ground results in the desired behavior.

## ELAB-397

(error) %s The module or interface '%s' needs to be analyzed.

**Description**

This error message occurs when the Presto HDL Compiler fails to read the intermediate file after analyzing.

**What Next**

Ensure that the module or interface is analyzed before running the command.

## ELAB-398

(error) %s The content '%s' is not provided via interface %s's modport %s, which is connected to '%s'.

Feedback

### Description

This error message occurs when attempting to access the content of an interface's modport through a port connection, but the content is not provided by the modport you received. The interface either never defined content with that name, or the actual modport that was connected did not list it among the accessible names. The following example results in this error message:

```
interface simple_bus;
logic a,b,c;
modport M(b,c);
endinterface;

module middle(i, o, sb);
input i;
output o;
simple_bus.M sb;
assign o = i & sb.a;   // a is not accessible via this interface's M
 modport

endmodule
```

### What Next

Correct the interface content access or add the content definition in the corresponding interface and run the command again.

### See Also

- ELAB-104

---

## ELAB-399

(error) %s Type query %s on undefined expression or type.

### Description

You receive this error message because the tool detected an error when querying on an undefined expression or type.

In the example below, the argument of $bits is undefined, which causes the error message to be issued.

```
module top(i, o);
input [$bits(t)-1:0] i;
output [$bits(t)-1:0] o;
endmodule
```

Feedback

### What Next

Correct the query argument and run the command again.

---

## ELAB-400

(error) %s Type query %s requires an array reference as the first argument.

### Description

This error message occurs when you attempt an array query on a non-array type.

In the following example, the $left argument does not have an array type, so the error message is issued.

```
typedef logic t;

module top(i, o);
input [$left(t)-1:0] i;
output [$left(t)-1:0] o;
endmodule
```

### What Next

Correct the query argument and run the command again.

---

## ELAB-401

(warning) %s Comparisons to don't-care or to unknown literals are always treated as being false. This may cause simulation to disagree with synthesis.

### Description

This warning message occurs because comparisons in hardware are inherently different than in the simulator. Digital hardware comparators can only distinguish between ones and zeros. As a result, it does not make sense to synthesize comparisons to three-state, don't-care, or unknown literals. Since signals are assumed to carry a value of one or zero, equality tests to non-one or zero values always return false in synthesis.

### What Next

This is only a warning message. No action is required.

If the comparison was used only for simulation purposes you do not need to modify your HDL. For example, if you want to print a message when an input port is in the high-impedance state, it is acceptable to compare its value to z.

If you are using the comparison to affect the state of your circuit, you should consider recoding your HDL. For example, if you need a four-state state machine, do not declare a

Feedback

single-bit variable and use 0, 1, x, and z as your states. In hardware, each bit only holds two useful states: 1 and 0.

## ELAB-402

(information) %s Unsigned 0-extension of a signed quantity

### Description

This informational message issues because arithmetic used in your design implicitly used an unfortunate, but standardized, difference between Verilog 95 and many other languages, including ANSI C/C++. In IEEE-1800-2009 Verilog/SystemVerilog and all its predecessor standards, a signed quantity used in an unsigned context becomes unsigned, and any bit-width extension it may require is done via zero-extension, not sign-extension (i.e.the sign bit does not fanout to the extension bits) as other languages and common sense would have it.

ELAB-402 is not reported if $unsigned() or unsigned'() casts are used to explicitly discard signedness from an operand, if the leading bit of the operand is already 0, or if no extension is necessary for such an operand. It is reported if there appears to be a possibility that zero-extension bits of a signed operand will be used by some downstream operation or storage location. It issues in addition to the VER-318 warning about the same coercion to indicate that the arithmetic outcome has probably changed from what a C programmer would have gotten.

### What Next

This is only an informational message. No action is required.

Verilog context-determined typing is tricky and, in our opinion, poorly documented in the IEEE standards. Simply inserting casts may not be the proper response to this information message. Consider other ways to decompose the cited expression, or case comparison, that do not rely upon implicit type coercion.

### See Also

- VER-318

## ELAB-403

(warning) %s Presto uses shift to implement signed division.

### Description

You receive this warning message when Presto HDL Compiler uses shift to implement signed division. The result does not match the simulation result, but it is the same behavior as the HDL Compiler and Formality.

Feedback

If you want to match simulation behavior, set the *hdlin_signed_division_use_shift* variable to *false*, which is the default value.

**What Next**

This is only a warning message. No action is required.

However, you can perform the verification and change the value of the *hdlin_signed_division_use_shift* variable to *false* to match the simulation result.

---

## ELAB-404

(error) %s Unpacked type used as index.

**Description**

You receive this error message when an unpacked type is used as an index for a subscription or a vector part-select.

**What Next**

Change the type of index expression or cast it to packed types and run the command again.

---

## ELAB-405

(warning) %s Net %s or a directly connected net may be driven by more than one process or block.

**Description**

You receive this warning message when a reg variable is driven by more than one always block, or a wire variable could be driven by more than one continuous assignment, and you have also set the *hdlin_prohibit_nontri_multiple_drivers* variable to *false*.

This warning message is issued because an invalid design could result.

In the following example, the assignment in the module could cause multidriven issues for wire t.

```
module M(input i, output o);
wire t = i;
assign t = o;
 ...
endmodule
```

The compiler permits multiple drivers on nets declared as *tri*, but it issues a warning if one of the drivers is not a three-state device.

Feedback

### What Next

This is only a warning message.

An invalid design may result when this warning is issued. Multiple driver nets are only generated because the *hdlin_prohibit_nontri_multiple_drivers* variable is set to *false*.

Consider modifying the design to avoid driving a variable from more than one always block or continuous assignment.

---

## ELAB-406

(warning) %s The binding of %s is ignored.

### Description

You receive this message from the *elaborate* or *read* command when the compiler eliminates binding of the specified signal. The bindings are usually made by assignments. The reason for the elimination is that there is no reference to the specified signal before the next binding.

For example:

```
always @* begin
  t = 1;
  t = 2;
end
```

The binding of the first assignment to t is overwritten by the second assignment. The message warns you that the first assignment is eliminated.

The following is another example:

```
module M (input i, output o);
reg t;
always @* begin
  t = 1;
end
endmodule
```

The binding of t is not used anywhere else, so you are warned when it is eliminated.

### What Next

This is only a warning message.

However, if the elimination is unexpected, modify the code and run the tool again.

Feedback

## ELAB-407

(error) %s All edge events in a single process must depend on the same signal.

**Description**

This error message occurs when a process contains more than one event wait (in VHDL), or when an always block contains more than one edge expression (in Verilog or SystemVerilog). All of the waits must be for an edge on the same signal.

**What Next**

Recode the design so that all event waits depend on the same variable.

## ELAB-408

(error) %s Only some paths through the process contain event waits; waits must occur on all or no paths through a process.

**Description**

This error message occurs when a process contains any event wait (in VHDL), or when an always block contains any edge expression (in Verilog or SystemVerilog). All paths through that block must reach at least one event wait. It must be impossible to execute the process or block without triggering a wait.

**What Next**

Recode the design so that all paths reach an event wait.

## ELAB-409

(error) %s Edge expression being tested is not a simple edge.

**Description**

This error message occurs when a process contains any event wait (in VHDL), or when an always block contains any edge expression (in Verilog or SystemVerilog). The event the system is waiting for must be a simple edge expression.

**What Next**

Recode the design so that all event waits are for simple edge expressions.

## ELAB-410

(error) %s Processes with both 'wait' statements and 'if' statements that test events are not currently supported.

Feedback

### Description

This error message occurs when a process contains an event wait, and an if statement in the same process contains an event as part of the condition. This process type is not supported by Presto.

### What Next

Recode the design so that it contains only event waits or an if statement that tests an event condition.

---

## ELAB-411

(error) %s Processes or always blocks with multiple events are disallowed when hdlin_allow_multiple_waits is false.

### Description

This error message occurs when the Presto HDL Compiler encounters a process containing more than one event wait when reading or elaborating a Verilog or VHDL design.

Presto supports the read and elaborate processes, except when the *hdlin_allow_multiple_waits* variable is set to false. The default value is true.

The most common reason for setting *hdlin_allow_multiple_waits* to false is as part of a range of settings that ensure that designs can be elaborated by the Formality RTL readers.

### What Next

Either set *hdlin_allow_multiple_waits* to true, or recode the design so that it contains only a single event wait or an if statement that tests an event condition.

---

## ELAB-412

(error) %s Wait or event statements that do not depend on edge events are not supported.

### Description

This error message occurs when the design being read or elaborated contains a process with a wait statement that does not depend on an edge event, or an always block that does not depend on a posedge or negedge expression.

This error also occurs when SystemVerilog designs contain special-form always blocks, such as always_comb or always_ff, and also contain a sensitivity list. Designs containing any of these statements are not supported.

Feedback

### What Next

Recode the design so that it contains only event waits.

---

## ELAB-413

(error) %s Processes containing more than one 'if' statement that tests an event expression are not supported.

### Description

This error message occurs because a process that contains more than one if statement that tests an event expression is not supported.

### What Next

Recode the design so that each process contains no more than one if statement that tests an event expression in each process.

---

## ELAB-414

(warning) %s Ignoring %s - too complex for synthesis to utilize.

### Description

This warning message issues when the Presto HDL Compiler finds a labelled assertion or assertion severity system task call whose logical structure lies outside the current synthesis subset.

- Generally, the asserted expression should compare only one variable or signal to a compile-time constant, or else
- It should make a $onehot or $onehot0 claim about a collection of signal wires.
- It must not have any statements in its action_block.
- It should use bitwise NOT (~) of the complete argument, rather than reduction NOT (!) of each bit when forming a $onehot() argument to assert a one-cold relationship.

  Consult the SystemVerilog User Guide for the exact definition of the subset currently supported by this release.

### What Next

This is only a warning message. No action is required.

However, if you do not want the construct to be ignored, you must simplify it, perhaps into several separate relations.

Feedback

## ELAB-415

(error) %s Processes containing nested 'if' statement that tests an event expression are not supported.

**Description**

This error message occurs because a process that contains nested if statement that tests an event expression is not supported.

**What Next**

Recode the design so that processes contain no nested if statement that tests an event expression.

## ELAB-420

(warning) %s Invalid symbol '%s' or expression in a signal name list of a pragma or assertion. It is being ignored.

**Description**

This warning message occurs when the Presto HDL Compiler finds an invalid symbol or expression in the list of signal names submitted to a Verilog or VHDL compiler pragma or in a SystemVerilog assertion. This can be caused by one of the following:

- The type of the signal or expression is invalid for the pragma or asserted condition.
- If the signal or its expression is an array, its index or bounds should be a constant expression. No function call is allowed, including a constant function call.
- The type of the signal or the expression accessing it cannot involve a union type.
- For one_hot and one_cold pragmas or $onehot and $onehot0 system functions, the signal name list can only contain signals that are ports.

**What Next**

This is only a warning message. No action is required.

However, if you do not want the symbol and/or its expression to be ignored in this context, correct the symbol name and/or its expression and analyze the RTL again.

## ELAB-421

(warning) %s The value of type %s assigned to the '%s' attribute will be converted to %s

Feedback

### Description

This warning issues when the built-in, declared, or implied type for this attribute's data differs from the type of expression in this setting. Sensible arithmetic conversions occur, possibly involving numeric formatting. Packed data types are treated as arithmetic bit vectors.

### What Next

This is only a warning message. No action is required, but the final attribute value should be checked for range overflow or sign change issues.

To repair this warning, the declaration of the attribute and this usage must be brought into type-agreement. For DC's built-in attributes, your value setting must change. For attributes you've added via (* attr=value *) syntax, all value expressions for an attribute (across all modules presented to the compiler in a single read command) must have data types compatible within the small set of possible attribute types. Unpacked array dimensions need not be the same for these purposes.

---

## ELAB-422

(warning) %s Unsupported expression type for '%s' attribute; this setting is ignored

### Description

This warning issues when no attribute type exists to represent the value provided, or no conversion exists from the given expression type to the previously known type of this attribute.

### What Next

This is only a warning message; no action is required. Review Verilog/SystemVerilog type representations and the attribute value expectations of any cooperating tools or DC commands. Choose simpler data types in order to lower the complexity of data format conversion efforts.

To repair this warning, the declaration of the attribute and this usage must be brought into type-agreement. For DC's built-in attributes, your value setting must change. For attributes you've added via (* attr=value *) syntax, all value expressions for an attribute (across all modules presented to the compiler in a single read command) must have data types compatible within the small set of possible attribute types. Unpacked array dimensions need not be the same for these purposes.

---

## ELAB-425

(error) %s Found expression without known width in context that requires one

Feedback

### Description

You receive this error when you use an expression that cannot have a width or whose width cannot be determined in a context that requires an expression with a well-defined width.

An example from SystemVerilog:

```
interface IFC;
    logic a, b;
    assign b = a;
    modport mp(output a, input b);
  endinterface

  module test(output out, input in);
    IFC ifc();
    bot bot(ifc.mp, out, in);
  endmodule

  module bot (IFC.mp ifc, output out, input in);
    assign out = ifc; // Error -- intended ifc.b, not just ifc
    assign ifc.a = in;
  endmodule
```

### What Next

Make sure that you have not written an expression without a well-defined width.

---

## ELAB-433

(warning) %s This SV Assertion is treated as confirmed for synthesis although %s has not been set to true.

### Description

This warning message issues when the Presto HDL Compiler finds a labelled assertion or assertion severity system task call whose usage is neither confirmed nor disconfirmed by the tcl environment variable setting cited in the message, but a blanket confirmation of all such assertions has been made and hdlin_enable_assertions is true. Because SystemVerilog Assertions (SVA) can lead synthesis to using a larger set of "don't care" conditions during optimization, you should separately certify that this assertion has proven to be unreachable prior to a final synthesis of this RTL.

### What Next

This is only a warning message. No action is required.

If you ultimately do want this construct to be used in a final synthesis, you should set an element of the confirmed_SVA environment array to true whose element name identifies

Feedback

the label and lexical scope of the applicable assertion. The most specific form of its element name is used in this warning message.

The recommended way to confirm assertions is to assemble a separate tcl file to source into your dc_shell before elaborating the design elements which use SVA. This same file should then be sourced into fm_shell when formally verifying the results of this elaboration.

---

## ELAB-434

(warning) %s The assertion %s would currently be too complex for synthesis to utilize.

**Description**

This warning message issues when the Presto HDL Compiler finds a labelled assertion or assertion severity system task call whose logical structure lies outside the current synthesis subset when a blanket confirmation has been issued.

- Generally, the asserted expression should compare only one variable or signal to a compile-time constant, or else
- It should make a $onehot or $onehot0 claim about a collection of signal wires.
- It must not have any statements in its action_block.
- It should use bitwise NOT (~) of the complete argument, rather than reduction NOT (!) of each bit when forming a $onehot() argument to assert a one-cold relationship.

  Consult the SystemVerilog User Guide for the exact definition of the subset currently supported by this release.

**What Next**

This is only a warning message. No action is required.

However, if you do not want the construct to be ignored, you must simplify it, perhaps into several separate relations.

---

## ELAB-463

(warning) %s Incorrect value for hdlin_infer_mux: '%s'; allowed values are 'all', 'default', or 'none'. Using 'default'.

**Description**

This warning message occurs when an incorrect value is set for the *hdlin_infer_mux* variable. Allowed values for *all*, *none*, and *default*.

When *all* is the value, HDL Compiler attempts to infer a MUX_OP for a signal or variable assigned in a case statement.

Feedback

When *none* is the value, HDL Compiler does not attempt to infer any MUX_OPs for a Verilog or VHDL design.

When *default* (the default value) is the value, HDL Compiler attempts to infer a MUX_OP for a signal or variable assigned in a case statement or if statement, if the statement is either marked with the *infer_mux* directive, or is in a process associated with the *infer_mux* attribute or directive. A MUX_OP cannot be inferred if it violates the *hdlin_mux_size_limit* variable.

For details, see the *HDL Compiler for Verilog User Guide* or the *HDL Compiler for VHDL User Guide*.

To determine the current value of this variable, use the *printvar hdlin_infer_mux* command. For a list of HDL variables and their current values, use *print_variable_group hdl*.

**What Next**

Set *hdlin_infer_mux* to a value of *all*, *default*, or *none*.

---

## ELAB-468

(warning) %s Port '%s' of design '%s' matches a local name whose type differs from this structured instance;\n\tit cannot be distributed; so it will be treated as '.%s'

**Description**

This warning issues when a structure of instances has used .* port name globbing to form a connection with a variable (or net) whose type does not match the structure of instances'. Since ports are only distributed one-bit-per structural instance, this port will have to be replicated. If that is the intended connection, it should be specified as suggested in the message text.

In the following example, the module named *foo* is twice instantiated with the 3-bit wide shape of 'a', but the port named *clk* is just one bit wide (doesn't match a's shape) which provokes the warning message in the case where the connection is formed by port globbing.

```
module foo(input clk, a,b, output out );
 ...
endmodule
module E (clk, a, b, out);
  input clk;
  input [2:0] a;
  input [2:0] b;
  output [2:0] out;

  foo WARN<$type(a)> (.*);       // warning that clk isn't 3-bits wide
  foo GOOD<$type(a)> (.clk, .*); // explicit fanout of 1-bit clk
endmodule
```

Feedback

### What Next

Correct the type of the cited variable, so it will (implicitly) distribute over the instances; rename objects so they connect better when port globbing (or pin synonyms) are in use; or explicitly connect this port without <distribution> brackets to indicate that it should be replicated.

### See Also

- remove_pin_name_synonym
- report_pin_name_synonym
- set_pin_name_synonym

---

## ELAB-469

(warning) %s The width of port %s on instance %s of design %s is inconsistent with other instantiations of the same design.

### Description

You receive this warning message when a module is instantiated in your design more than once, but the width of the ports is inconsistent among all of the instances.

In the following example, the module named *foo* is instantiated twice, but the ports named *a* and *b* have different widths, causing the warning message.

```
module E (a, b, out1, out2);
input [2:0] a;
input [3:0] b;
output [3:0] out1, out2;

foo U1 (a, out1);
foo U2 (b, out2);
endmodule
```

The instance for which this warning is reported may not be the actual instance that uses the wrong width. Since the design is not yet linked, the compiler cannot check which of the instances is correct and can only report the inconsistency.

In the example above, the warning is reported for instance U2, because instance U1 has already been seen and uses a different width. As the first in the list, the module named *foo* might require a 3-bit port, a 4-bit port, or another size. Errors that affect all instances equally are checked during linking instead of during elaboration.

### What Next

Correct the width of the ports so that they are consistent, and then invoke the compiler again.

Feedback

## ELAB-490

(error) %s Variable or function call in casex item is a potential simulation/synthesis mismatch.

### Description

You receive this error message in verification friendly mode when a variable or function call is used in a casex item. Even if an x bit is assigned to a variable or returned by a function call, the value of the case item will not match both 0 and 1, as it would for an x bit in a literal in a casex item. For example:

```
b = 2'b1x ;
   casex (a)
     b : ... // in this case, 2'b1x only matches 2'b10
   endcase
```

A full example:

```
module m( o1 , o2 );
output o1, o2 ;
reg o1, o2 ;

  function f ;
  input [1:0] R ;
  casex (2'b11)
    R : f = 1'b0 ;
    default : f = 1'b1 ;
  endcase
  endfunction

  always begin
   o1 = f(2'b1x) ;
  end

  reg [1:0] R;
  always begin
   R = 2'b1x;
   casex (2'b11)
     R : o2 = 1'b0 ;
     default: o2 = 1'b1 ;
   endcase
  end

endmodule
```

### What Next

Convert the casex item into a literal or convert the casex statement into a case statement.

Feedback

## ELAB-491

(warning) %s Embedded configuration statements are not supported for synthesis. They are ignored.

### Description

This warning is issued when the design on which the *elaborate* or *read* command is invoked includes embedded configuration statement. Embedded configuration statements are not supported, and is ignored.

### What Next

Rewrite the design using configuration blocks and try again.

## ELAB-492

(warning) %s Direct configuration instantiations are not supported for synthesis. They are ignored and treated as direct entity instantiations.

### Description

This warning message occurs when the design on which the *elaborate* or *read* command is invoked includes a direct configuration instantiation. These configuration instantiation statements are not supported, and are ignored; that is, they will be treated as direct entity instantiations.

### What Next

This is only a warning message. No action is required.

However, if the result is not what you intended, rewrite the design using a standalone nested configuration and a component to be configured.

## ELAB-495

(error) %s Variable or function call in case-inside item is a potential simulation/synthesis mismatch.

### Description

You receive this error message in when a variable or function call is used in a case-inside item. Even if an x bit is assigned to a variable or returned by a function call, the value of the case-inside item will not match both 0 and 1, as it would for an x bit in a literal in a case-inside item. For example:

```
b = 2'b1x ;
   case (a) inside
```

Feedback

```
    b : ... // in this case, 2'b1x only matches 2'b10
  endcase
```

A full example:

```
module m( o1 , o2 );
output o1, o2 ;
reg o1, o2 ;

  function f ;
  input [1:0] R ;
  case (2'b11) inside
    R : f = 1'b0 ;
    default : f = 1'b1 ;
  endcase
  endfunction

  always begin
   o1 = f(2'b1x) ;
  end

  reg [1:0] R;
  always begin
   R = 2'b1x;
   case (2'b11) inside
     R : o2 = 1'b0 ;
     default: o2 = 1'b1 ;
   endcase
  end

endmodule
```

**What Next**

Convert the case-inside item into a literal or convert the case-inside statement into a case statement.

---

## ELAB-496

(warning) %s Case statement can not be optimized using boolean optimization, because its branch number(%d) is greater than the max number of branches. You can change the public variable hdlin_optimize_pla_max_branch setting for using boolean optimization, but runtime and QoR may be hit.

**Description**

This warning message occurs when the number of branches of a case statement is greater than the max number of branches which is set by the public variable hdlin_optimize_pla_max_branch. This makes case statement optimization not use boolean optimization.

Feedback

**What Next**

This is only a warning message. No action is required.

However, if the result is not what you intended and you want to try boolean optimization, you can increase the setting value for the public variable hdlin_optimize_pla_max_branch. This may increase runtime, the worse case may hang elaboration; QoR also is impacted.

If the case statement has a lot of dont_care values, increasing the max number of branches threshold and using boolean optimization, another story may happen.

---

## ELAB-500

(warning) %s No Design Compiler license, Presto optimization: %s can not be enabled.

**Description**

Presto advanced optimizations need check Design Compiler (DC) license. If no DC license, the optimizations will not be enabled and this warning message will be generated.

**What Next**

Get DC license.

---

## ELAB-501

(error) %s defparam assignment for %s was ignored.

**Description**

Defparam assignment to task, function or block parameters are ignored for synthesis.

**What Next**

Recode the design using named parameter assignments and try again.

---

## ELAB-502

(warning) %s Label '%s' does not refer to a unique operation.

**Description**

When pragma label is added into RTL file, if the pragma label can't refer a unique operation, this warning message will happen.

**What Next**

Go to RTL source file and check whether the label is correct or not.

Feedback

## ELAB-510

(information) %s Presto preserved an unloaded net '%s'.

**Description**

You receive this information message when Presto HDL Compiler preserves a net that may not be driving any loads.

**What Next**

Check your design for possible misconfiguration of port directions or port connections and try again. You may ignore this information if the connections are part of the required design specifications.

## ELAB-600

(error) %s Pre-analyzed files for module '%s' not found.

**Description**

You receive this message because the Presto HDL compiler cannot find the pre-analyzed files for the given module during elaboration.

**What Next**

Ensure that the module to be elaborated has been analyzed with the variable hdlin_sv_ieee_assignment_patterns set to a value of 2 or higher.

## ELAB-601

(error) %s %s is not supported for %s.

**Description**

The combination of array of instances and assignment patterns on ports on a module instantiation is not supported for synthesis.

**What Next**

Please recode to remove either the array of instances, or the assignment pattern on the ports construct and try again.

## ELAB-602

(error) %s %s is not supported for %s.

Feedback

### Description

This error is issued for unsupported features in Unified Compile flow.

---

## ELAB-747

(warning) %s The body of module '%s' is being discarded, because the module name is %s.

### Description

This warning message issues when an instance of a module with a non-empty body and its name in overridden hdlin.elaborate.black_box or not in overridden hdlin.elaborate.black_box_all_excepted is encountered during elaboration. The body will be discarded, except for port declarations.

### What Next

If the body of this module should not be ignored, remove the module name from hdlin.elaborate.black_box or add it to hdlin.elaborate.black_box_all_except.

---

## ELAB-749

(Error) %s The module '%s' is %s.

### Description

This error message issues when an instance of a module with a non-empty body is encountered during elaboration and its name is in the values of both application options hdlin.elaborate.black_box and hdlin.elaborate.black_box_all_except.

### What Next

If the body of this module should be discarded during elaboration, remove the module name from hdlin.elaborate.black_box_all_except. If it shouldn't be discarded during elaboration, remove the module name from hdlin.elaborate.black_box.

---

## ELAB-781

(warning) %s Formal %s, named '%s', misconnects with %s.

### Description

This warning message issues during the specialization of a module or interface that is done to accomodate its interface ports. The interface type name or the modport name of the incoming interface instance is not compatible with the declaration of the corresponding (cited) formal port. For the purposes of Design Exploration, the formal port is being

Feedback

synthesized to match the reference cell's expectations, but is left unconnected within the present module. This is Bad Logic in that it appears to permit:

- Connecting an interface port or instance to a wire or variable port
- Connecting a wire or variable to an interface/modport port
- Connecting an interface port or instance instantiated from one template to an interface port declared with a different interface template name.
- Connecting one modport of an interface instance to a different modport (by name)

### What Next

The immediately preceding HDL-193 ("Building ...") indicates one design's reference cells which are incompatible with the cited port. Correct the ports in the module instantiation and/or the module definition so that they agree on what is being connected. Re-analyze the changed RTL and re-link the design.

### See Also

- ELAB-381

---

## ELAB-800

(error) %s The loop variable is not initialized to a constant.

### Description

You receive this error message when your loop variable is not initialized to a constant, as required by Presto synthesis. The following example results in the error message.

```
for(i = non_constant ; i< 64 ; i = i+1)
 begin
   loop_body
 end
```

### What Next

Rewrite the loop so that the loop variable is initialized to a constant. The previous example can be rewritten as follows if non_constant is positive:

```
for (j = 0; j<64; j = j+1)
 begin
   i = j + non_constant;
   if ( i < 64 )
      loop_body
 end
```

Feedback

## ELAB-801

(warning) %s BDD result may not be complete.

### Description

You receive this error message because the BDD memory limit has been reached. The BDD result may not be complete, so the QOR (Quality of Results) may be worse than expected. The default BDD memory limit is 6M.

### What Next

Use the hidden variable *hdlin_bdd_memory_limit* to increase the memory limit.

## ELAB-802

(warning) %s The port default value in entity declaration for port '%s' is not supported. Presto ignores it.

### Description

You receive this warning message because port default values in entity declaration are ignored by Presto VHDL.

In the following example, the port default values '1' and 'Z' for the ports A and Z will be ignored:

```
entity test is
  port (A: in std_logic := '1';
        EN: in std_logic;
        Z: out std_logic := 'Z'
       );
end;
```

### What Next

Verify if it implements the behavior you intended after ignoring the port default values, and modify your code if needed.

## ELAB-803

(warning) %s There is a name conflict between the instance '%s' and an existing cell. The name of the instance is being renamed to '%s'.

Feedback

### Description

This warning message occurs when you specify an instance name of a module instantiation that has already been used by another existing cell. The message may be caused by the following circumstances:

- There is a name conflict between your instance name and another instance name generated in FOR-GENERATE statement. In this case, if the original instance name in the FOR-GENERATE statement is *mod_name*, the tool may generate a series of instance names based on it in the form: *mod_name*, *mod_name_1*, *mod_name_2*, etc.
- There is a name conflict between your instance name and another instance name in your code.

### What Next

If you want the instance name to be renamed to the new name indicated in the warning message, no action is required on your part. However, if you want to keep the instance name unchanged, you can resolve the conflict by modifying the name of the conflicting instance name in your code, and then run the tool again.

---

## ELAB-804

(error) %s error found for parameters value near token '%s'.

### Description

This error message occurs when there is an error in the parameter value list given by the module or component instantiation or by the *elaborate* command.

Presto VHDL adds support for the following types:

```
integer
bit
bit_vector
std_ulogic
std_ulogic_vector
std_logic
std_logic_vector
signed
unsigned
```

Presto Verilog adds support for parameters with Verilog numbers and strings.

Presto SVerilog adds support for parameters with Verilog numbers, strings, and type parameters.

The parameter specification is a list of values separated by commas, based on parameter order (for example, "3,4,5") or parameter names (for example, "N=>8,M=>16").

Feedback

### What Next

Correct the error in the parameter value list and run the command again. See the man page for the *elaborate* command for information on the accepted parameter list syntax.

---

## ELAB-805

(error) %s The value '%s' is out of the range of the parameter that it overrides.

### Description

This error message occurs because a generic parameter in an entity instantiation has been overridden with a value that is out of the declared range of that parameter.

### What Next

Specify a value inside the declared range or modify the entity to accept a larger range. Correcting this issue is critical because in most cases, out of range values would be truncated, thus inferring non-attended logic.

---

## ELAB-810

(warning) %s The name of the register or instance '%s' will be changed to its hierarchial form name, e.g. from '%s' to '%s', in next release. Similar changes may only warn once.

### Description

You should not receive this warning by default. If you do receive this warning, it may or may not be accurate; please evaluate your RTL code accordingly. If you wish to remove this warning from your DC sessions, please contact Synopsys support.

You receive this warning because the mentioned register or instance is embadded in one or more hierarchically nested block(s), but is named with a flat name without the hierarchical path attached. The name will be changed into its hierarchical form with the hierarchial path attached in next release.

### What Next

If you are satisfied with the flat name, no further action on your part is required.

If you want to use the hierarchical naming style for register or instance in this release, you can set the *hdlin_enable_hier_naming* variable (default value is *false*) to *true*.

---

## ELAB-820

(error) %s This use of clock'event specification not supported

Feedback

### Description

This error occurs if you use more than one *if clock'event* expression in a process or if you embed the *clock'event* expression inside a subprogram or loop.

The following example shows a positive- and negative-edge trigger in the same process.

```
entity test is
  port(clock, a: in bit; z1, z2: out bit);
end;

architecture a of test is
begin
  process (clock)
  begin
    if(clock'event and clock = '1') then
      z1 <= a;
    end if;

    if(clock'event and clock = '0') then -- ERROR
      z2 <= a;
    end if;
  end process;
end;
```

### What Next

Restructure your code to have a single "if clock'event" construct at the start of the process.

The previous example can be fixed by splitting the process into two processes.

```
entity test is
  port(clock, a: in bit; z1, z2: out bit);
end;

architecture a of test is
begin
  process (clock)
  begin
    if(clock'event and clock = '1') then
      z1 <= a;
    end if;
  end process;

  process (clock)
  begin
    if(clock'event and clock = '0') then
      z2 <= a;
    end if;
  end process;
end;
```

Feedback

## ELAB-832

(warning) %s Undriven register '%s' is connected to primary output '%s'.

### Description

This warning message occurs when Presto determines that the listed register is not driven in the source code. The cause might be an issue in your design. Review your source code to ensure that this will not translate to incorrect outputs.

Normally, Presto grounds undriven registers, but the listed register will not be grounded because it is connected to a primary output.

An example of this situation happens in the following code:

```
module example(input [1:0] x, input y, output reg [1:0] z);
  always @* begin
    z[2*x]= y;
  end
endmodule
```

For z[1], Presto normally creates the register z_reg[1], but it can never be assigned because 2*x will never be 1. Therefore, z_reg[1] is an undriven register and it is not created. The output z[1] is also undriven.

### What Next

Modify your code to drive the listed register properly.

## ELAB-833

(warning) %s Register '%s' is undriven.

### Description

This warning message occurs when Presto determines that the listed register is not driven in the source code. The cause might be an issue in your design. Review your source code to ensure that this will not translate to incorrect outputs.

An example of this situation happens in the following code:

```
module example(input [1:0] x, input y, output reg z);
  reg [1:0] t;
  always @* begin
    t[2*x]= y;
    z= &t;
  end
endmodule
```

Feedback

For t[1], Presto normally creates the register t_reg[1], but it can never be assigned because 2*x will never be 1. Therefore, t_reg[1] is an undriven register and it is not created.

Note that Presto grounds automatically undriven registers that are not connected to primary outputs. In the example Presto handles t[1] as constant 0 because it is not directly connected to output z. The output z is 0.

For undriven registers connected to primary outputs, Presto issues a warning with the ELAB-832 message.

**What Next**

Modify your code to drive the listed register properly.

**See Also**

- ELAB-832

---

## ELAB-834

(error) %s The type of port '%s' is an unconstrained multidimensional array.

**Description**

This error message occurs because HDL Compiler does not support multidimensional arrays.

**What Next**

Specify the full dimensions of the array and run the command again.

---

## ELAB-840

(error) %s Invalid null range operand in expression.

**Description**

This error message occurs because one of the operands of a VHDL expression is a null range. Null range can only be legal in concat or repeat concat operations.

**What Next**

Fix your VHDL source to avoid null ranges in expressions.

---

## ELAB-900

(error) %s Loop exceeded maximum iteration limit.

Feedback

### Description

You receive this error message when your loop runs on forever, or is designed to run some number of times greater than the iteration limit. When the compiler processes loops, the loop condition must fail at some point.

### What Next

Set the *hdlin.elaborate.maximum_while_loop_iterations* variable to a higher number.

---

## ELAB-901

(error) %s Function call stack exceeded maximum depth.

### Description

This error message occurs when the depth of the function call stack exceeds the maximum depth currently supported by Presto to prevent runaway function call chains.

The most probable cause of this error is that your design has an infinite recursion.

Note that this message may be shown for recursive functions that finish in simulation, but are infinite in synthesis. One example of infinite recursion, and thus unsupported, is this design:

```
module fact (input [3:0] n, output [40:0] z);

  function [40:0] f;
  input [3:0] k;
  begin
    if (k==0)
      f = 1;
    else
      f = k * f(k-1);
  end
  endfunction

  assign z = f(n);
endmodule
```

This example will not finish in synthesis because the terminating condition depends on a port and therefore is never a proper constant.

### What Next

Check your design for an infinite recursion and then run the command again.

---

## ELAB-902

(warning) %s In the call to '%s', the '%%%c' format specifier is not supported.

Feedback

### Description

You receive this warning message when you execute the *elaborate* or the *read* command with a format specifier that is not supported. The following specifiers are supported, and the rest are ignored:

`%b, %d, %h, %c, %o, %s.`

### What Next

Reexecute the *elaborate* or *read* command with a format specifier that is supported.

---

## ELAB-903

(warning) %s Insufficient pins supplied to array of modules for instance '%s' of design '%s'.

### Description

You receive this warning message when the wires you specify for a given port do not meet the following requirements:

1) Wires must match the width of the port on the module.

2) The wires must be an integer multiple of the width of the port.

### What Next

Respecify the wires so that they meet the above requirements.

---

## ELAB-904

(error) %s When instantiating '%s' array of modules, the implementation design '%s' must be available.

### Description

You receive this error message when you execute the *elaborate* or *read* command and specify an array of modules that contain a black box.

Because of the strict nature of the port connections for instantiating an array of modules, the design to be instantiated cannot be a black box; you must provide the complete design.

### What Next

Verify that the implementation design is available. Also check to see that there is not a typo in the name of the module to be instantiated.

Feedback

## ELAB-906

(error) %s Too many parameters passed to module instance %s.

**Description**

You receive this message because the list of parameter override values exceeds the number of parameters declared in the module. This is a list of parameter override values that can be supplied when instantiating a module.

**What Next**

Ensure that the number of parameters passed to a module instance matches the number of parameters declared in the module.

## ELAB-908

(error) %s Exponentiation is only supported if the base is a power of 2 or the exponent is 2.

**Description**

You receive this error message because one of these situations exist. Either

1. The base of an exponentiation expression is variable and the exponent is not 2.

or

1. The exponent is not 2.

**What Next**

Correct the expression so that its base is constant or its exponent is not 2.

## ELAB-909

(warning) %s Case statement is not a full case.

**Description**

You receive this warning message because you used the "// synopsys full_case" directive for this case statement and you set the *hdlin_check_user_full_case* variable to *true*. However, the case statement is not a full case statement.

**What Next**

No action is necessary on your part. However, you can verify that the case statement implements the behavior you intended, try to make the case statement full, or set the *hdlin_check_user_full_case* variable to *false*.

Feedback

## ELAB-910

(warning) %s Case statement is not a parallel case.

**Description**

You receive this warning message because you used the "// synopsys parallel_case" directive for this case statement and you set the *hdlin_check_user_parallel_case* variable to *true*. However, this case statement is not parallel. A case is parallel when all of its branches are mutually exclusive.

**What Next**

Try to make the case statement parallel, or set the *hdlin_check_user_parallel_case* variable to *false*.

## ELAB-911

(error) %s Can not find return port name of type function '%s'.

**Description**

You receive this error message because this type function does not have the required return port name.

**What Next**

Define a return port name for the type function by using -- pragma *return_port_name*.

## ELAB-912

(error) %s Negative exponent.

**Description**

You receive this error message when the tool detects a negative exponent.

The exponent might be entered directly or might be the result of an expression. For example, the expression (2**(M-N)) has a negative exponent when both M and N are constant, and N is larger than M.

**What Next**

Remove the negative exponent from the expression.

## ELAB-913

(error) %s Divide by zero.

Feedback

### Description

You receive this error message when the tool detects division by a constant value, which is zero in a VHDL file.

The zero might be entered directly or it might be the result of an expression. For example, the expression (WIDTH/(M-N)) divides by zero if both M and N are equal constants.

### What Next

Remove the division by zero from the expression.

---

## ELAB-914

(error) %s Direction ('to' or 'downto') does not agree with 'first' and 'last' values in range.

### Description

You receive this error message because you specified a direction that conflicts with the position in the range.

- Use the keyword *downto* when the first value in the range is greater than the last value.
- Use the keyword *to* when the first value in the range is less than the last value.

### What Next

Rewrite your descriptions using *downto* and *to* as described above. In the following example, the direction is *downto*, because 15 is greater than 0.

```
VHDL:
   signal my_bus : bit_vector ( 15 to 0 );
```

---

## ELAB-915

(error) %s Expected value but was supplied unknown

### Description

You receive this error message because a case statement contains "unknown" instead of a value.

For example, the tool issues the message if an enumeration literal with a U encoding is supplied to a case statement.

### What Next

Change the case choice to avoid an unknown value.

Feedback

## ELAB-916

(error) %s Can not find definition for %s .

### Description

You receive this error message if the declared function, operator, or procedure cannot be found in the VHDL libraries or packages.

### What Next

Include the correct library or package.

## ELAB-917

(error) %s Type of port '%s' is unconstrained.

### Description

This error message issues when you elaborate an entity (in which the given port's data_type is unconstrained) either as a topmost design or for an instantiation that does not connect it to a constrained-type data object. This can happen when you use read_vhdl, where every entity is elaborated separately (as a topmost design) prior to linking. Some actual port connection must provide the constraining type information.

### What Next

Use analyze and elaborate instead of read_vhdl. Only elaborate entities whose ports' data_types are constrained. Add constrained bounds to this port's type declaration.

## ELAB-918

(warning) %s Physical types are not supported. Presto ignores it.

### Description

You receive this warning message because physical types are not supported by Presto VHDL. Presto ignores physical types.

### What Next

No action is required on your part. Use caution when physical types are used.

## ELAB-919

(error) %s Unsupported range type used in array declaration.

Feedback

**Description**

You receive this error message because the range type you specified in the array declaration is not supported. For example, std_logic is not supported as the range of an array.

**What Next**

Rewrite the array declaration using a supported range type.

---

## ELAB-920

(warning) %s OPS entry '%s' is already included in another resource. Entry ignored.

**Description**

You receive this warning message because during *elaborate* or *read* command activity, the compiler detected a label that appears in the ops list of more than one resource, which is invalid.

The following example shows the invalid entry:

```
synopsys resource r0 : ops = "op1 op2";    //op2 appears twice
synopsys resource r1 : ops = "op2 op3";
```

**What Next**

Correct your code so that each label appears in the ops list of only one resource.

---

## ELAB-921

(error) %s Built-in %s currently is not supported yet.

**Description**

The built-in used is not supported yet. Changes to VHDL sources may be needed.

**What Next**

Change your VHDL source file to avoid using the built-in.

---

## ELAB-922

(error) %s Constant value required.

**Description**

This error message occurs because an expression in the indicated line of your RTL description does not evaluate to a constant value, as required by the language.

Feedback

In the example below, the expression in the generate case statement is required to evaluate to a constant value at elaboration time. The use of an input port in that expression violates this requirement.

```
module m (a);
    input a;

    generate
      case (a)         // error: not constant
      ...
    endgenerate
  endmodule
```

**What Next**

Change your source file to use a constant value and run the command again.

---

## ELAB-923

(error) %s Potential glitch on net '%s': A signal/variable can't be synchronously and asynchronously assigned within the same process.

**Description**

This error is issued when the design on which the *elaborate* or *read* command is invoked is not supported for synthesis because there is a potential glitch on a net. This is caused by the order in which multiple sequential statements that assign to the same net are executed.

In the following example, the if statements are executed sequentially.

```
architecture RTL of E is
begin
  process (CLK, RST)
  begin
    if (CLK'event and CLK ='1') then
      Q <= D;
    end if;
    if (RST = '1') then
      Q <= '0';
    end if;
  end process;
end RTL;
```

In the event that the conditions on both if statements evaluate to TRUE, Q will momentarily be assigned the value of D before being assigned '0', resulting in a possible glitch.

To avoid this situation, if a D flip-flop with an asynchronous reset is desired, the rtl can be rewritten as shown.

Feedback

```
architecture RTL of E is
begin
  process (CLK, RST)
  begin
    if (RESET = '1') then
      Q <= '0';
    elsif (CLK'event and CLK ='1') then
      Q <= D;
    end if;
  end process;
end RTL;
```

**What Next**

Modify the design to avoid a potential glitch on a net and ensure that the design accurately specifies the desired functionality. Invoke the compiler again.

---

## ELAB-924

(warning) %s Signal assignment delays are not supported for synthesis. They are ignored.

**Description**

This warning is issued when a signal assignment statement in the design on which the *elaborate* or *read* command is invoked includes delay information. The delay is not supported and is ignored.

**What Next**

No modification is required. Ensure the delay that is ignored is not integral to the functionality of the design.

---

## ELAB-925

(error) %s Attribute %s is not supported for synthesis.

**Description**

This error is issued when the compiler encounters an attribute that is not supported for synthesis in the design on which the *elaborate* or *read* command is invoked.

**What Next**

Remove the attribute that is causing the error and rerun the compiler.

---

## ELAB-926

(error) %s %s types are not supported for synthesis.

Feedback

### Description

This error is issued when the compiler encounters a physical, access, or file type in the design on which the *elaborate* or *read* command is invoked. These types are not supported for synthesis.

### What Next

Avoid using unsupported types in the design.

---

## ELAB-927

(error) %s cannot find architecture %s of entity %s in library %s.

### Description

This error message occurs when the compiler is unable to find a specific entity or architecture pair in a library.

### What Next

Ensure that all entity and architecture names are specified correctly. Remove any special characters that may be problematic.

---

## ELAB-928

(error) %s An if statement containing a clock event has an illegal else branch and is not supported for synthesis.

### Description

This error occurs when a value is assigned on the "else" part of an "if clock'event".

An example of RTL that is not synthesizable because an assignment does not depend on a clock edge is shown:

```
entity test is
  port(clock, a, b: in bit; z: out bit);
end;

architecture a of test is
begin
  process (clock, a, b)
  begin
    if(clock'event and clock = '1') then
      z <= a;
    else
      z <= b; -- ERROR
    end if;
```

Feedback

```
  end process;
end;
```

The following example shows RTL that is not synthesizable because assignments occur on multiple clock edges:

```
entity test is
  port(clock, a, b: in bit; z: out bit);
end;

architecture a of test is
begin
  process (clock, a, b)
  begin
    if(clock'event and clock = '1') then
      z <= a;
    elsif(clock'event and clock = '0') then
      z <= b; -- ERROR
    end if;
  end process;
end;
```

A variable can receive both asynchronous and synchronous assignments, but the asynchronous assignment must take precedence. That is, there must be a Boolean condition that enables the asynchronous part and disables the synchronous part.

**What Next**

Use an "if" statement to ensure that either that the asynchronous or synchronous assignments fire in any execution of the process. Make sure that the condition controlling this "if" (the asynchronous reset condition) is level sensitive, not edge sensitive.

---

## ELAB-929

(error) %s Illegal use of unpacked type %s.

**Description**

This error message occurs because the RTL code uses an unpacked type in a location where unpacked types are illegal, as shown in the following example.

```
struct {int i;} ctrl;   // unpacked structure
  ...
  y = ctrl ? a:b;  // error: control expression not allowed to be of
 unpacked type
```

**What Next**

Change the unpacked type to a packed type and run the command again.

Feedback

## ELAB-930

(error) %s Incompatible types %s.

### Description

This error message occurs because the language standard requires the equivalence of two types in the indicated context, but the actual provided types are not equivalent, as shown in the example below.

```
// two non equivalent structures
  struct {int i;} a;
  struct {char c;} b;
  ...
  y = c ? a : b;  // error: types of both conditional branches must be
 equivalent
```

The precise rules for type compatibility are given in section 5.8 of the *SystemVerilog LRM*. Incompatibility can arise due to differences in vector widths, packed versus unpacked dimensions, unpacked structures that come from different declaration sites, or other inconsistencies between two data types where they are required to correspond.

### What Next

The end of the error message indicates the kind of construction that requires the type equivalence, which is usually an assignment, an aggregate construction, or a relational operator.

Identify the part of the cited line that corresponds to the indicated language feature. Determine the source of the two types that are involved. Correct the data accesses or their type declarations so that equivalent types are used and run the command again.

## ELAB-931

(warning) %s implicit truncation caused by range overflow.

### Description

This warning message occurs when a value is truncated by the CONV_SIGNED or CONV_UNSIGNED function, because the value overflows the range specified by SIZE.

### What Next

This is a warning message only. No action is required on your part. However, be aware that some information may be lost because of the truncation.

Feedback

## ELAB-932

(Error) %s %s type as range type is not supported.

**Description**

You receive this error message because the range type you specified in the array declaration or loop parameter is not supported. For example, std_logic/bit/boolean is not supported as the range of an array. or std_logic/bit/boolean is not supported as the discrete range in the loop parameter.

**What Next**

Rewrite the array declaration or loop parameter range using a supported range type.

## ELAB-933

(warning) %s REM operator is synthesized differently in this version of HDL Compiler.

**Description**

This warning message occurs when a REM operator with a potentially negative first operand is encountered.

Previous versions of HDL Compiler synthesized the REM operation as a MOD operation, resulting in a potential synthesis and simulation mismatch.

The Presto engine within HDL Compiler handles a REM operation with a negative first operand correctly and does not translate it as a MOD operation. Therefore, Presto HDL Compiler does not produce the same result as previous versions of HDL Compiler.

**What Next**

This is a warning message only. No action is required on your part.

If a REM operation is intended, be aware that the synthesis results may differ from that of older versions of HDL Compiler. However, if a MOD operation is intended, modify the source code to use a MOD operator.

To suppress further instances of this warning, add ELAB-933 to the list of error codes specified by the suppress_errors variable.

## ELAB-934

(error) %s Architecture '%s' of design '%s' cannot be found.

**Description**

This error occurs when the specified architecture for a design cannot be found.

Feedback

**What Next**

Ensure the desired architecture exists and has been specified correctly.

---

## ELAB-935

(warning) %s the direction (to or downto) does not agree with the values first and last in the range.

**Description**

This warning message occurs when you specify a direction that conflicts with the position in the range.

- Use the keyword *downto* when the first value in the range is greater than the last value.
- Use the keyword *to* when the first value in the range is less than the last value.

**What Next**

This is a warning message only. No action is required.

Ignore the warning message if the code is unreachable. However, if the code does not reflect the design you intended, make the necessary corrections and rerun the command.

---

## ELAB-936

(warning) %s Attribute %s is not supported for synthesis and will be ignored.

**Description**

This warning is issued when the compiler encounters an attribute that is not supported in the design on which the *elaborate* or *read* command is invoked.

**What Next**

No user action is required. However, be aware that the attribute will be ignored.

---

## ELAB-937

(error) %s Constant expression too complex.

**Description**

You are receiving this error message because in a context requiring a constant expression you are using an expression that is not constant or too complex to be handeled by the compiler. This can be a restriction of the language or the compiler.

Feedback

Example:

```
module (in);
  input [0:3] in;
  parameter p = 2;

  generate
    if (in[p])    // error: in[p] is not a constant expression
      ...
  end generate
```

**What Next**

Simplify the constant expression so the compiler can determine the value at compile time.

---

## ELAB-938

(error) %s An order based parameter was specified after a name based parameter for the module instance %s.

**Description**

This error happens when the argument to the elaborate command includes a combination of order-based and name-based parameter values, and all the order-based parameters are not specified before name-based parameters.

**What Next**

Modify the arguments to the elaborate command so that all order-based parameters occur before name-based parameters.

---

## ELAB-939

(error) %s The system function '%s' is not supported for synthesis.

**Description**

This error message occurs when a system function called in the design is not implemented for synthesis.

**What Next**

Remove the specified system function call in your design and reread or reanalyze it.

---

## ELAB-940

(warning) %s real types are ignored.

Feedback

**Description**

This warning message occurs when a real type is encountered in the design. Real types are not supported by the compiler and are ignored.

**What Next**

This is a warning message only. No action is required.

However, be aware the portion of the design that involves real types will not be implemented.

---

## ELAB-941

(error) %s attribute %s of prefix %s cannot be found.

**Description**

This error message occurs when the attribute specification of the named attribute in the attribute reference is nonexistent or is defined in an invalid scope.

**What Next**

Ensure that the attribute specification of the desired attribute is defined in the current scope and then rerun the command.

---

## ELAB-942

(Error) %s %s of physical types are not supported in synthesis.

**Description**

You receive this error message because physical types are not supported by Presto VHDL as procedure or function parameters, entity or block ports types, etc.

**What Next**

Remove the procedure or function parameters, or the entity or block ports that are assuming physical types and rerun the command.

---

## ELAB-943

(warning) %s Generic %s does not have a default value.

**Description**

You receive this warning message because the generic default value is not specified.

Feedback

### What Next

Set the default value for the given generic and run the command again.

---

## ELAB-944

(warning) %s Non-standard coding style used to implement a sequential element.

### Description

You receive this warning message when the design contains a non-standard coding style to describe a sequential element. Non-standard coding styles can lead to problems during simulation and may not be supported by other tools.

The following example contains a non-standard coding style to infer a sequential element with asynchronous reset. Two separate if statements are used to describe the synchronous part and the asynchronous reset. This can cause a problem during simulation.

```
entity e is
 port (d: in bit;
       q: out bit;
       clk: in bit;
       r: in bit
      );
end;

architecture a of e is
 signal t: bit;
begin

 p: process (clk) is
 begin
  if (clk'event and clk = '1') then
    t <= d;
  end if;
  if (r = '1') then
    t <= '0';
  end if;
 end process;

 q <= t;

end;
```

The next example shows a standard way to describe the sequential element:

```
entity e is
 port (d: in bit;
       q: out bit;
       clk: in bit;
```

Feedback

```
       r: in bit
      );
end;

architecture a of e is
 signal t: bit;
begin

 p: process (clk) is
 begin
  if (r = '1') then
    t <= '0';
  else if (clk'event and clk = '1') then
    t <= d;
  end if;
 end process;

 q <= t;

end;
```

**What Next**

This is only a warning message. No action is required.

If you are willing to accept potential simulation problems, you can leave the code unchanged.

If you want to avoid possible simulation problems, refer to the *HDL Compiler (Presto Verilog) Reference Manual* or *HDL Compiler (Presto VHDL) Reference Manual* for the recommended coding style.

---

## ELAB-945

(error) %s Attribute not supported with index value different from '1'.

**Description**

You receive this error because the VHDL attribute is used with an index value different from '1'. The attribute is only supported for index value '1'.

**What Next**

Rewrite your VHDL code so it doesn't use the attribute with an index value different from '1'.

---

## ELAB-946

(error) %s Assignment to %s overflows declared bounds.

Feedback

### Description

This error message occurs when a variable is assigned a value whose size is larger than the declared bounds of the variable.

### What Next

Fix the offending assignment and try again.

---

## ELAB-949

(error) %s The content '%s' cannot be selected out of interface or modport array '%s'.

### Description

This error message occurs when the dot (.) operator tries to select content across all elements of an interface or modport array. The type and data access methods of IEEE-1800 do not support this as of the 2012 edition.

In the actual port list of a down-instance, a dot like this *can specify a modport* across an entire array of interfaces, but only if the interface port identifier being modified does not *already* have a modport specified. That can come from either its formal port declaration, or from the actual interface instance connected to it.

### Examples

```
interface simple_bus;
  logic a,b,c;
  modport M(b,c);
endinterface;

module middle(i, o, sb);
  input i;
  output o;
  simple_bus.M sb[3];
  bottom B1( sb.M     // sb's modport M has already been specified.
           , sb.b );  // Discontinuous slice of array sb[0:2].b
endmodule
```

### What Next

If this is a redundant modport specification, omit it. If you are attempting to scatter/gather distributed content, generate a for loop of continuous assignments from/to a temporary variable to make the scatter/gather activity explicit.

### See Also

- ELAB-398

Feedback

## ELAB-950

(error) %s out of bounds index [%d] into scope array '%s'.

### Description

This error message occurs when the index value specified is not within the declared range of the scope array. For example:

```
generate
  genvar i;
  for (i=3; i>0; i=i-1) begin : forlbl
    always @(posedge clk) begin : alwlbl
        reg tmp;
        o[i] = tmp;
        forlbl[i+1].alwlbl.tmp = b[i] & a[i];  // Error on accessing
 forlbl[4]
    end
  end
 endgenerate
```

### What Next

Rewrite your design to change or eliminate the illegal access and run the command again.

## ELAB-951

(error) %s Indexed access to %s not allowed.

### Description

This error message occurs when an indexed access is made into a scope that does not support indexed operation.

### What Next

Rewrite your design to change or eliminate the illegal access and try again.

## ELAB-952

(error) %s The genvar '%s' has been assigned the value '%s' repeatedly.

### Description

This error message occurs when the same *genvar* variable is assigned more than one time. According to the Verilog LRM this is illegal. Only one assignment to each *genvar* variable is allowed.

Feedback

### What Next

Rewrite the Verilog RTL code so that there is only one assignment to each *genvar* variable.

---

## ELAB-953

(error) %s Array of scopes '%s' cannot be referenced without an index.

### Description

This error message occurs when your Verilog RTL code contains a hierarchical reference to an object inside a generate loop scope. The hierarchical reference must specify which iteration of the for loop it refers to. This is accomplished by subscripting the for loop scope name with the value of the *genvar* variable specific to this loop iteration, as shown in the following example:

```
module m (y);
  output [0:3] y;
  genvar i;

  generate
    for (i=0; i<4; i=i+1) begin : blk
      wire t;
    end
  endgenerate

  assign y = blk[1].t;    // subscript 'blk' with genvar value '1'
endmodule
```

### What Next

Rewrite your Verilog RTL code so that the hierarchical name uses subscripts to access specific generate loop iterations.

---

## ELAB-954

(warning) %s Nonstandard generate block outside of generate conditional or generate loop.

### Description

This warning message occurs because the Verilog standard no longer permits generate blocks that do not belong to the branches of a generate conditional or a generate loop.

Presto Verilog still handles singleton generate blocks for backward compatibility, but other tools may not.

Feedback

The following example shows a singleton generate block that causes this warning message:

```
module m ();

   begin : blk    // deprecated singleton generate block
   end

  end
```

**What Next**

This is only a warning message.

You can eliminate this warning message by using a generate block that is associated with a generate conditional or a generate loop.

---

## ELAB-955

(warning) %s State register '%s': changing state encoding from gray to binary.

**Description**

This warning message indicates that the requested FSM encoding style "gray" cannot be implemented because the number of states is not a power of 2. In this case the encoding is changed to binary.

**What Next**

This is only a warning message. The state machine will still be implemented, but with binary encoding.

---

## ELAB-956

(Warning) %s Unknown FSM encoding '%s' requested.

**Description**

You are receiving this warning message because you set the Presto option "hdlin_force_fsm_encoding" to an unknown value. The permitted values are: one_hot, binary, gray, none.

In case of an unknown value, the value 'none' is assumed.

**What Next**

Change the value of the option "hdlin_force_fsm_encoding" to one of the permitted values.

Feedback

## ELAB-957

(Error) %s Generate construct not inside generate region.

**Description**

You are receiving this error message because the Verilog standard requires that generate constructs must occur inside a generate region (generate regions are indicated by the keywords 'generate...endgenerate'). The generate construct at the indicated source location is not inside a generate region.

**What Next**

Add a generate region, by surrounding generate constructs with the keywords 'generate...endgenerate'.

## ELAB-958

(Error) %s Generate loop requires named begin block.

**Description**

You are receiving this error message because the Verilog 2001 standard requires that loop generate constructs contain a named begin block. Your RTL either does not contain a begin block or the begin block is not named. Please note that the 2005 update of the Verilog standard all

**What Next**

Add a named block to the loop generate construct.

## ELAB-959

(error) %s Illegal value assigned to a genvar variable.

**Description**

This error message occurs when any bit of of a genvar variable is set to an X or Z. Genvar values with X or Z are not supported in Synthesis.

**What Next**

Rewrite the Verilog RTL code such that genvar variables are assigned legal values and try again.

## ELAB-960

(error) %s Constant value not supported in given context.

Feedback

**Description**

This error message occurs when any bit of a constant value is an X or Z, and the context in which it is used does not allow for such values in the constant.

**What Next**

Rewrite the Verilog RTL code such that the constant value does not evalutate to bits with X or Z, and try again.

---

## ELAB-961

(error) %s Constant expression ( %s ) not supported in given context.

**Description**

This error message occurs the reported operation is not supported in evaluation context.

**What Next**

Rewrite the Verilog RTL code such that the reported operation does not happen in the given context, and try again.

---

## ELAB-968

(error) %s The width=%s of the indexed part-select is less than 1.

**Description**

You receive this message because the width of the indexed part-select is not at least 1.

**What Next**

Modify code to resolve this issue.

---

## ELAB-971

(error) %s Assignment of NULL waveforms not supported.

**Description**

You receive this error when you assign a null waveform to a signal. This is not supported by the tool.

**What Next**

Make sure you do not have any such statements in your code. Assign a proper (non-null) value to the signal.

Feedback

## ELAB-972

(error) %s Construct %s.%s not supported

**Description**

This error is seen if the package/Unit definition is not present.

**What Next**

In case of Package definition missing you must replace the code that makes use of this package by one that uses packages that have already been defined.

## ELAB-973

(error) %s Only selected types are supported for generics.

**Description**

Currently only the following types are supported for generics:

Integer, std_logic_vector, std_logic, bit, bit_vector.

**What Next**

Change the source code to remove the usage of the generics types that are not supported.

## ELAB-974

(warning) %s Netlist for always_comb block contains a latch%s.

**Description**

This warning message occurs during *read* or *elaborate* command activity to advise you that a latch is inferred for an always_comb block.

**What Next**

This is only a warning message. You can eliminate this warning message by following the instructions below.

If combinational logic is intended, you can rewrite the always_comb block.

If combinational logic is not intended, you can remove the always_comb keyword.

## ELAB-975

(warning) %s Netlist for always_latch block does not contain a latch.

Feedback

**Description**

This warning message occurs during *read* or *elaborate* command activity to advise you that the netlist for an always_latch block does not contain a latch. This may be because the always_latch block represents combinational logic or because all latches could be optimized away.

**What Next**

This is only a warning message. You can eliminate this warning message by following the instructions below.

If latched logic is intended, determine whether the body of the always_latch block represents latched logic. If it does not, then a design problem in the always_latch block has been detected. However, if the always_latch block does represent latched logic, then a coding simplification opportunity may have been detected.

If latched logic is not intended, you can replace the always_latch keyword.

---

## ELAB-976

(warning) %s Netlist for always_ff block does not contain a flip-flop.

**Description**

This warning message occurs during *read* or *elaborate* command activity to advise you that the netlist for an always_ff block does not contain a flip-flop. This may be because the always_ff block does not actually represent sequential logic or because all of the flip-flops could be optimized away.

**What Next**

This is only a warning message. You can eliminate this warning message by following the instructions below.

If sequential logic is intended, determine whether the body of the always_ff block represents sequential logic. If not, then a design problem in the always_ff block has been detected. However, if the always_ff block does represent sequential logic, then a coding simplification opportunity may have been detected.

If sequential logic is not intended, you can replace the always_latch keyword.

---

## ELAB-977

(warning) %s Netlist for always_latch block contains a flip-flop.

Feedback

**Description**

This warning message occurs during *read* or *elaborate* command activity to advise you that a flip-flop is inferred for an always_latch block.

**What Next**

This is only a warning message. You can eliminate this warning message by following the instructions below.

If latched logic is intended, you can rewrite the always_latch block.

If latched logic is not intended, you can replace the always_latch keyword. Complete your changes and run the command again.

---

## ELAB-978

(warning) %s Netlist for always_ff block contains a latch.

**Description**

This warning message occurs during *read* or *elaborate* command activity to advise you that a latch is inferred for an always_ff block.

**What Next**

This is only a warning message. You can eliminate this warning message by following the instructions below.

If sequential logic is intended, you can rewrite the always_ff block.

If sequential logic is not intended, you can replace the always_ff keyword.

---

## ELAB-979

(warning) %s Netlist for always_comb block contains a flip-flop.

**Description**

This warning message occurs during *read* or *elaborate* command activity to advise you that a flip-flop is inferred for an always_comb block.

**What Next**

This is only a warning message. You can eliminate the warning message by following the instructions below.

If combinational logic is intended, you can rewrite the always_comb block.

If combinational logic is not intended, you can remove the always_comb keyword.

Feedback

## ELAB-980

(error) %s Real type signal is not supported.

### Description

This error message occurs because the Presto HDL Compiler does not support the real type signal. Currently, only the real type constant is supported.

### What Next

Declare a real type constant instead of the real type signal, and run the command again.

## ELAB-981

(error) %s Real type variable is not supported.

### Description

This error message occurs because the Presto HDL Compiler does not support the real type variable. Currently, only the real type constant is supported.

### What Next

Declare a real type constant instead of the real type variable, and run the command again.

## ELAB-982

(warning) %s Netlist for always_comb block is empty.

### Description

This warning message occurs during *read* or *elaborate* command activity to advise you that no logic is inferred for an always_comb block, because none of the signals driven from that block are needed to compute the outputs of the module. For example:

```
module test(input  logic in,
              output logic out);
  logic tmp;
  assign out = in ;
  always_comb begin
      tmp = in;
  end
  endmodule
```

Feedback

### What Next

If combinational logic is intended, determine whether the body of the always_comb block represents combinational logic. If not, then a design problem in the always_comb block has been detected.

However, if the always_comb block does represent combinational logic, then either the signals driven by this block are not connected as intended or a coding simplification opportunity has been detected and the always_comb block can be removed.

---

## ELAB-983

(warning) %s Netlist for always_latch block is empty.

### Description

This warning message occurs during *read* or *elaborate* command activity to advise you that no logic is inferred for an always_latch block, because none of the signals driven from that block are needed to compute the outputs of the module. For example:

```
module test(input  logic clk,
            input  logic in,
            output logic out);
  logic tmp;
  always_latch begin
    if(clk)
      tmp <= in;
  end
  endmodule
```

### What Next

If latched logic is intended, determine whether the body of the always_latch block represents latched logic. If not, then a design problem in the always_latch block has been detected.

However, if the always_latch block does represent latched logic, then either the signals driven by this block are not connected as intended or a coding simplification opportunity has been detected and the always_latch block can be removed.

---

## ELAB-984

(warning) %s Netlist for always_ff block is empty.

Feedback

### Description

This warning message occurs during *read* or *elaborate* command activity to advise you that no logic is inferred for an always_ff block, because none of the signals driven from that block are needed to compute the outputs of the module. For example:

```
module test(input  logic clk,
              input  logic in,
              output logic out);
  logic tmp;
  always_ff @(posedge clk) begin
      tmp <= in;
  end
  endmodule
```

### What Next

If sequential logic is intended, determine whether the body of the always_ff block represents sequential logic. If not, then a design problem in the always_ff block has been detected.

However, if the always_ff block does represent sequential logic, then either the signals driven by this block are not connected as intended or a coding simplification opportunity has been detected and the always_ff block can be removed.

---

## ELAB-985

(warning) %s Netlist for always block is empty.

### Description

This warning message occurs during *read* or *elaborate* command activity to advise you that no logic is inferred for a SystemVerilog always block with an implicit or explicit sensitivity list, because none of the signals driven from that block are needed to compute the outputs of the module. For example:

```
module test(input  logic in,
              output logic out);
  logic tmp;
  assign out = in ;
  always @* begin
      tmp = in;
  end
  endmodule
```

### What Next

Either the signals driven by this block are not connected as intended or a coding simplification opportunity has been detected and the always block can be removed.

Feedback

## ELAB-989

(error) %s Packed union members must be of the same size.

### Description

You receive this error message because a packed union has been declared with members of different sizes. The *Language Reference Manual* (LRM) specifies that "A packed union shall contain members that must be packed structures, or packed arrays or integer data types all of the same size".

For example:

```
module test #(parameter OFFSET_BITSIZE = 16,
                parameter PAGE_BITSIZE = 16);
    union packed {
      int numeric;
      struct packed {
        bit [OFFSET_BITSIZE-1:0] offset;
        bit [PAGE_BITSIZE-1:0] page;
      } vaddress;
    } one_union;
  endmodule
```

If the module test is going to be instantiated with parameter values other than the defaults, they must add 32, in order for the bit width of the packed structure vaddress to be of the same size than the integer numeric (32).

### What Next

Modify your source file so that all union members have the same bit width.

## ELAB-990

(error) %s Expression width of .%s(<distributed port>) does not equal the number of instances here.

### Description

You receive this message because a distributed port expression, such as .p(<expr>), does not have the same number of bits as the type model declared for this structured instantiation. In the error message, portnames are given if available; "p#" positional indices are used otherwise. Note that all ports found by a ".*" request must connect via the port distribution (not the port replication) rule for structured instances.

### What Next

Modify code so that the given distributed port expression has the same number of bits as the declared type of the structure of instances. All signals that replicate (i.e. fanout) to

Feedback

all instances must connect without angle brackets (<>), or .* globbing. Replication is not usually appropriate for output direction ports, as the instances will then become multiple drivers of the replicated net.

---

## ELAB-991

(warning) %s The type of .%s(<distributed port expression>) does not match the declared <type> of this structured instance.

### Description

You receive this message because a distributed port expression, such as .p(<expr>) in a structure of instances, does not have the declared type for the structure of instances. For example, the type declared for the structure of instances might be a packed structure, while the type of the distributed port expression might be a simple bit-vector type of the same length.

### What Next

If the expression bit width of each distributed port matches the number of instances, they will be distributed anyway. However, as a precaution, you should examine the RTL code to make sure that the distributed port expression's type is actually intended to align with the instance structure in a bit-streamed order. If so, consider using an explicit cast of the port expression to clarify that this is what you intend.

---

## ELAB-992

(error) %s The lhs width=%d does not match the rhs width=%d of the assignment statement.

### Description

You receive this message because the assignment statement has a width mismatch problem.

### What Next

Modify code to resolve this issue.

---

## ELAB-993

(warning) %s Synthesis/simulation mismatch for multiplier.

### Description

You receive this error message because the multiplier is optimized assuming no result truncation. Verilog/VHDL semantics require result truncation to be taken into account. In

Feedback

most cases assuming no result truncation results result in the intended behaviour and better QOR. If the design relies on multiplier truncation, this message indicates bad logic. When you see this message, there may be a synthesis/simulation mismatch.

Strict Verilog/VHDL semantics can be forced by setting the following option:

```
set hdlin_idxopt_optimize_mul false
```

This will resolve the synthesis/simulation mismatch, but can result in QOR degradation.

**What Next**

Modify code to reolve this issue.

---

## ELAB-994

(warning) %s Found an unknown pragma '%s'; it will be ignored.

**Description**

You receive this warning message from the *elaborate* or *read* command, if there is an unknown pragma. The unknown pragma will be ignored.

**What Next**

Check the spelling of the unknown pragma and correct it if necessary; otherwise consider to remove the unknown pragma.

---

## ELAB-995

(error) %s Real to integer conversion is not supported.

**Description**

This error message occurs because the Presto HDL Compiler does not support the real to integer conversion currently.

**What Next**

Modify the code to remove the error source.

---

## ELAB-996

(error) %s WAIT statement inside FOR loop is not supported.

**Description**

This error message occurs because the Presto HDL Compiler does not support WAIT statements inside FOR loops.

Feedback

**What Next**

Modify the code to remove the error source.

---

## ELAB-997

(error) %s Multidimensional part-select is not supported.

**Description**

This error message occurs because the Presto HDL Compiler does not support part-select operators (+: and -:) with multidimensional arrays.

**What Next**

Modify the code to remove the error source.

---

## ELAB-998

(error) %s The rhs width does not match the lhs width of the logical operator statement.

**Description**

You receive this message because the logical operator statement has a width mismatch problem.

**What Next**

Modify code to resolve this issue.

---

## ELAB-999

(error) %s Width of actual parameter does not match the formal parameter '%s'.

**Description**

You received this error message because width of the actual parameter does not match the mismatch the actual parameter, given by module/component instantiation or by elaborate command.

**What Next**

Modify formal or actual parameter type to match the width.

---

## ELAB-1000

(error) %s Overflow in a conversion of a real type number: %s.

Feedback

**Description**

You receive this message because there is an overflow in a conversion of a real type number.

**What Next**

Modify code to resolve this issue.

---

## ELAB-1001

(error) %s Overflow in a real type number or calculation.

**Description**

You receive this message because there is an overflow in a real type number or calculation. The number/result is greater than REAL'HIGH = 1.0e+38 or smaller than REAL'LOW = -1.0e+38.

**What Next**

Modify code to resolve this issue.

---

## ELAB-1002

(warning) %s Real types with ranges are not supported and will be ignored.

**Description**

This warning message occurs when there is a real type with a range defined. This is not supported. A real type without ranges will be used.

**What Next**

This is a warning message only. No action is required.

However, be aware that the portion of the design that involves real types with ranges could behave differently.

---

## ELAB-1003

(error) %s The operator (%s) specified by pragma map_to_operator does not define type function and make the width big (%d) for VHDL design.

**Description**

This error message occurs when the operator that is specified by the map_to_operator pragma does not define the type function. This makes the width of the operator exceed the supported range for VHDL design.

Feedback

**What Next**

Define the type function for the operator when the map_to_operator pragma is used.

---

## ELAB-1004

(error) %s In VHDL, element width=%d does not match width=%d of the element subtype.

**Description**

You receive this message because VHDL element has a width mismatch problem. This can happen in assigment statement, array aggregate, or etc.

**What Next**

Modify code to resolve this issue.

---

## ELAB-1010

(error) %s Complex usage of multiple events statements is not supported.

**Description**

This error message occurs because the Presto HDL Compiler does not support complex multiple events in an verilog always block or a VHDL process.

**What Next**

Modify the code to have only one event per always or process.

---

## ELAB-1011

(warning) %s Comparison operator always returns '%s' when comparing values with different widths.

**Description**

This warning message occurs when the tool detects that there is a comparison with values of different width which evaluates always as unequal.

This can result in never reached statements within a comparison branch.

**What Next**

Resolve the potential problem.

Feedback

## ELAB-1012

(error) %s The width (%d %s) is too big.

**Description**

This error message occurs when the tool detects a width that exceeds the supported range.

**What Next**

Decrease the used width and run the command again.

## ELAB-1013

(error) %s PORT/GENERIC declaration in BLOCK header is not supported.

**Description**

Declaration and use of generics and ports in a block header is not supported.

**What Next**

Remove port and generic declarations from the block header and adapt the block statements if necessary.

## ELAB-1014

(warning) %s The lhs width=%s does not match the rhs width=%s of the comparison.

**Description**

This warning message occurs when the tool detects that there is a comparison with values of different widths.

**What Next**

Correct the widths so that they are the same and run the command again.

## ELAB-1015

(warning) %s Ignoring SystemVerilog checker '%s' instantiation: '%s'.

**Description**

This warning message occurs when the SystemVerilog *checker* construct and its instantiations are ignored.

Feedback

### What Next

This is only a warning message. No action is required.

---

## ELAB-1016

(error) %s Port name '%s' is not declared by this module or interface.

### Description

This error message occurs when a port name used in a module or interface instantiation or used in an *elaborate -parameter* command argument is not declared as a port by that module or interface.

In the following example, the foo module is instantiated with a named port connection. It uses the named ports a, o, and state. However, state is not in the port list definition of the foo module, so the error message occurs.

```
module E (a, out1, state);
input [2:0] a;
output [3:0] out1;
output state;
foo U1 (.a(a), .o(out1), .state(state));
endmodule

module foo(input [2:0] i, output [3:0] o, output status);
 ...
endmodule
```

### What Next

Correct the names of the ports so that they are consistent with the module definition, and then invoke the compiler again.

### See Also

- set_pin_name_synonym

---

## ELAB-1017

(information) %s Verification priority is set to '%s' for module '%s'.

### Description

This information message indicates that the named module is synthesized with verification priority.

### What Next

This is an information-only message. No action is required.

Feedback

See the *set_verification_priority* command man page for details.

## ELAB-1018

(warning) %s Out of range comparison is always %s.

**Description**

This warning message occurs when the tool detects that there is a comparison with a constant value that is out of the range of the other compared type.

This can result in never or always reached statements within a comparison branch.

**What Next**

Resolve the potential problem.

## ELAB-1019

(error) did not find pre-elaborated design : %s.

**Description**

This error message occurs when the tool detects that pre elaborated design is missing.

This can cause crash in tool.

## ELAB-1060

(error) %s Symbol '%s' results in multiple objects.

**Description**

The named symbol results in creation of objects that conficts with each other. The most likely cause of this error is importing of symbols from interfaces or classes into a module. During the import phase, homographs are created that in most cases can be automatically resolved, however in some cases such as this, automatic resolution is not possible.

**What Next**

Look for ELAB-106 warnings for the symbol in question, and rename or resolve the homographs.

**See Also**

- ELAB-106

Feedback

## ELAB-1090

(error) %s HDL Compiler has run out of memory.

**Description**

This error message occurs when HDL Compiler runs out of memory while trying to process the design. This can happen when you try to read or elaborate a design that is too large to be compiled with less than 4 GB.

**What Next**

Try reading or elaborating the design on a 64bit version of the tool.

## ELAB-1093

(error) The elaboration process reaches its limits while building the design %s.

**Description**

This error message occurs when HDL Compiler runs into a process limit when building a design module. This can happen if a design module is too large due to a large number of RTL constructs, such as always blocks and generate loops.

**What Next**

Try recoding the RTL design or break the design up into smaller modules.

## ELAB-1094

(error) %s State register '%s': has attribute FSM_COMPLETE which value is TRUE, but it defines unreachable states.

**Description**

This error message indicates that the requested FSM has attribute: FSM_COMPLETE, the attribute value is TRUE. FSM defines unreachable states.

**What Next**

First solution: FSM adds default state

Second solution: FSM defines all of the states.

Third solution: FSM remove attribute FSM_COMPLETE

Fourth solution: FSM set attribute value as FALSE.

Feedback

## ELAB-1095

(error) %s State register '%s' specifies Hamming2 encoding, its default state can't be uniquely determined, a strict RTL coding style must be enforced.

### Description

This error message indicates that When Hamming2 encoding is specified, a strict RTL coding style must be enforced for Presto to uniquely determine the default state and pass it to compile. No conditional statements are allowed in "default:" branch of case statement in Verilog or "when others" branch in VHDL, only simple assigments are allowed.

For example:

```
type state_values is (s0, s1, s2, s3, s4);
  signal state, next_state: state_values;

  when others =>
      out1 <= "000";
      next_state <= s3; // default state is s3
```

Complex "when others"/"default" case:

```
type state_values is (s0, s1, s2, s3, s4);
  signal state, next_state: state_values;

  when others =>
      if (in1)
         next_state <= s3;
          // default state can't be uniquely determined. Presto will
 issue an error
      else
         next_state <= s1;
```

### What Next

Use simple assigments when Hamming2 encoding is specified.

## ELAB-1096

(error) %s failsafe FSM pragma '%s' exists, switch hdlin_failsafe_fsm is not enabled.

### Description

You receive this error message if failsafe FSM pragma is used, the switch hdlin_failsafe_fsm is not enabled.

### What Next

Turn on the switch: hdlin_failsafe_fsm.

Feedback

## ELAB-1097

(warning) %s When switch hdlin_failsafe_fsm is enabled, no failsafe FSM pragma is used.

**Description**

You receive this warning message when switch hdlin_failsafe_fsm is enabled, no failsafe FSM pragma is used.

**What Next**

Turn off the switch: hdlin_failsafe_fsm.

## ELAB-1098

(warn) %s Pragma fsm_syn_encoding '%s' is defined multiple time, the latest value '%s' will override the previous value '%s'.

**Description**

This warning message indicates that pragma fsm_syn_encoding is defined multipe time, the latest value will override the previous value.

**What Next**

Check the RTL source codes to remove the redundant definitons.

## ELAB-1099

(warn) %s Pragma fsm_syn_encoding is obsolete, please use SSF(Synopsys Safe Format) way to implement it.

**Description**

This warning message indicates that pragma fsm_syn_encoding in the RTL file does not do anything, please use SSF way to implement it.

**What Next**

Check the RTL source codes to remove the pragma.

## ELAB-1500

(information) Running Presto elaboration using a maximum of %d cores.

Feedback

### Description

This inforamtion message occurs during *read* or *elaborate* command activity to advise you that the tool is running in multicore mode. You can specify the maximum number of CPU cores to be used with the *set_host_options* command.

### What Next

This is an information message only. No action is required.

### See Also

- set_host_options

---

## ELAB-1800

(information) Filelist %s regenerated as %s.

### Description

This inforamtion message occurs during *write_active_rtl_file_list* command when used with the option *-prune_filelist*.

### What Next

This is an information message only. No action is required.

---

## ELAB-1991

(warning) %s The concat with unpacked operand(s) is detected in instance port expression; make sure it can match the corresponding submodule port declaration in RTL.

### Description

You receive this message because a concat with unpacked operand(s) is detected in instance port expression, such as *.A({a, b[3:2], c})* in the following sample test case:

```
module top (
    input  logic a,
    input  logic b[3:2],
    input  logic c,
    output logic d
);

sub sub(.A({a, b[3:2], c}), .B(d));

endmodule

module sub(
    input  logic A[3:0],
```

Feedback

```
    output logic B
);
  ...
endmodule
```

where, the *b[3:2]* in the port expression *{a, b[3:2], c}* is an unpacked array.

**What Next**

If the bit width of the port expression matches the corresponding port declaration in the submodule, it will be connected anyway. However, as a precaution, you should examine the RTL code to make sure the connection is as expected. For example, in the corresponding port declaration, the port should also be declared as an unpacked array, instead of a packed array such as an integer or bit-vector.

---

## ELAB-2000

(error) %s Incorrect value for hdlin_infer_ram: '%s'; possible values are 'all', 'default', or 'none'.

**Description**

Allowed values for the *hdlin_infer_ram* variable are *all*, *none*, and *default*. When *all* is the value, HDL Compiler attempts to infer a RAM_OP for a 2-dimensional register. When *none* (the default value) is the value, HDL Compiler does not attempt to infer any RAM_OPs for a Verilog or VHDL design.

When *default* is the value, HDL Compiler attempts to infer a RAM_OP for a 2-dimensional register, only if the register is marked with the *infer_ram* directive,

For details, see the *HDL Compiler for Verilog Reference Manual* or the *VHDL Compiler Reference Manual*.

To determine the current value of this variable, type *list hdlin_infer_ram*. For a list of hdl variables and their current values, type *list -variables hdl*.

**What Next**

Set *hdlin_infer_ram* to one of the values *all*, *default*, or *none*.

---

## ELAB-2001

(warning) %s Ignored infer_ram pragma value '%s' on register '%s'.

**Description**

You receive this message because, during *elaborate* or *read* command activity, the compiler has detected invalid synopsys infer_ram pragma value on the memory element.

Feedback

Currently supported values are none, auto, distrubuted and no_rw_check. These values specify the implementation of memory.

For details, see the *HDL Compiler for Verilog Reference Manual* or the *VHDL Compiler Reference Manual*.

---

## ELAB-2002

(warning) %s The design '%s' has no parameters. The parameters '%s' has been ignored.

### Description

You receive this message because, during *elaborate* or *read* command activity, the compiler has detected that cell has be instantiated with the parameter. But the down design to which this cell is linked does not take any parameters.

---

## ELAB-2004

(warning) %s Signal %s has %s attribute, but is %s. A possible mismatch of attribute and behavior.

### Description

You receive this warning message because one_hot signal is used in an active low situation while one_hot attribute indicates the signal should be active high, or a one_cold signal is used in an active high situation while one_cold attribute indicates the signal should be active low.

This warning reminds the designer to keep the attribute of the signal consistent with the behavior of the signal.

### What Next

This is a only a warning and requires no action on your part.

---

## ELAB-2008

(information) %s Complex logic will not be considered for set/reset inference.

### Description

You receive this information when, during *elaborate* or *read* command activity, some possible set/reset logics have been excluded from consideration for set/reset inference, because they were estimated to be too complex, for example, because they were driven by another instance.

Feedback

If the hdlin.elaborate.infer_complex_set_reset is set to TRUE, if the control logic is complex and that drives the data pin of register to zero, the control logic gets moved to the reset line;and any control logic that drives the data pin to one, gets moved to the set line. And if hdlin.elaborate.infer_complex_set_reset is set to FALSE (default), no complex control logic gets moved to the set/reset lines and it will stay on the data line.

Usually excluding these complex possibilities leads both to better quality of result and to better runtime. Hence this message is not a warning, but only informational. Occasionally, however, the designer knows reasons why it would be good for the compiler to consider additional, more complex, possibilities. When this is so, it can be communicated to the compiler by overriding to true the default value of the hdlin.elaborate.infer_complex_set_reset variable.

---

## ELAB-2010

(error) %s Size mismatch in the assignment.

**Description**

This error message occurs when right hand side of the assignment is not the same as left hand size.

**What Next**

Fix the offending assignment and try again.

---

## ELAB-2015

(warning) %s illegal tristate driver '%s' on the wired types. may cause simulation/synthesis mismatch.

**Description**

You receive this message because Output of the tristate is used to drive more than one wired net. These nets may be driven directly by the tristate, or might be driven indirectly through another wired net.

**What Next**

Check the assignments on wired type, to fix the error

---

## ELAB-2016

(error) %s Found illegal tristate driver/drivers on the wired types.

Feedback

### Description

You receive this message because Output of the tristate is used to drive more than one wired net. These nets may be driven directly by the tristate, or might be driven indirectly through another wired net. To overide this error message set hdlin_error_illegal_driver_on_wired_type to FALSE. Overring this error might cause

### What Next

To overide this error message set hdlin_error_illegal_driver_on_wired_type to FALSE. Overring this error might cause simulation/synthesis mismatch.

---

## ELAB-2017

(warning) %s Ignoring 'infer_onehot_mux' pragma because case not marked %s.

### Description

You receive this message because the case statement marked as "infer_onehot_mux" is not marked as "full_case" or "parallel_case". Both pragmas "full_case" and "parallel_case" are required for the "infer_onehot_mux" pragma to be honored by the compiler.

### What Next

Make sure the case statement is really "full" and "parallel", then add the "full_case" and "parallel_case" pragmas to the "infer_onehot_mux" pragma.

---

## ELAB-2018

(error) %s Conflicting pragmas '%s' and '%s' cannot be applied simultaneously.

### Description

You receive this message because you applied to conflicting pragmas simultaneously. The pragmas are either conflicting or the tool does not support the application of both pragmas.

### What Next

Keep the pragma that was intended to be used and remove the conflicting other pragma.

---

## ELAB-2020

(error) %s %s.

### Description

This error message occurs when Presto VHDL detects the following semantic error when running the *analyze* command.

Feedback

**What Next**

Fix the error and reanalyze this file.

---

## ELAB-2024

(warning) %s Clock %s used as data.

**Description**

You receive this warning message because Presto HDL Compiler detected clock signals used for purposes other than clocking.

**What Next**

Modify the design to avoid using clock for purposes other than clocking.

---

## ELAB-2025

(error) %s Call to impure function '%s' is not supported.

**Description**

You receive this error message because Presto VHDL does not support calls to Impure functions.

---

## ELAB-2026

(warning) %s Impure function %s is used in your design.

**Description**

This warning message occurs because HDL Compiler detected an impure function in your design. Impure functions cause side effects that can lead to nondeterministic behavior, and can cause synthesis simulation mismatches.

**WHAT NEXT.**

This is only a warning message. No action is required.

However, to avoid this message, check your design for the given impure function.

---

## ELAB-2028

(error) %s Default value of the sized parameter '%s' has been overridden with an expression whose self-determined type is not as wide as the parameter.

Feedback

### Description

You receive this warning message when the default value of a sized parameter '%s' has been overridden with an expression whose self-determined type is not as wide as the parameter.

In synthesis the expression used in a parameter override is evaluated in a self-determined context, so in the following example, parameter P in the instantiation "bot1" is overridden with an expression that is evaluated with 1-bit width precision.

But in simulation, because the width of the sized parameter P in the module BOT is 2, the expression would be evaluated with 2-bit width precision. As in this example, the difference in precision can lead to different results between simulation and synthesis.

```
`define E 1'b1 + 1'b1
module TOP (output test_bit);
  wire [1:0] w1, w2;
  assign test_bit = (w1 == w2);
  BOT#(.P(`E)) bot1(w1);
  BOT bot2(w2);
endmodule

module BOT #(parameter [1:0] P = `E) (output [1:0] o);
  assign o = P;
endmodule
```

### What Next

Either reduce the size of the parameter or override the parameter with an expression whose self-determined type is as least as wide as the parameter.

---

## ELAB-2029

(warning) %s Default value of the sized parameter '%s' has been overridden with an expression whose self-determined type is not as wide as the parameter.

### Description

You receive this warning message when the default value of a sized parameter '%s' has been overridden with an expression whose self-determined type is not as wide as the parameter.

In synthesis the expression used in a parameter override is evaluated in a self-determined context, so in the following example, parameter P in the instantiation "bot1" is overridden with an expression that is evaluated with 1-bit width precision.

But in simulation, because the width of the sized parameter P in the module BOT is 2, the expression would be evaluated with 2-bit width precision. As in this example, the difference in precision can lead to different results between simulation and synthesis.

Feedback

```
`define E 1'b1 + 1'b1
module TOP (output test_bit);
  wire [1:0] w1, w2;
  assign test_bit = (w1 == w2);
  BOT#(.P(`E)) bot1(w1);
  BOT bot2(w2);
endmodule

module BOT #(parameter [1:0] P = `E) (output [1:0] o);
  assign o = P;
endmodule
```

### What Next

Either reduce the size of the parameter or override the parameter with an expression whose self-determined type is as least as wide as the parameter.

---

## ELAB-2030

(warning) %s DC runtime and Memory usage might increase significantly with (hdlin_mux_size_limit = %d) > 32.

### Description

Adding the infer_mux pragma in RTL and setting hdlin_mux_size_limit to a value greater than 32 will impact DC runtime and memory usage significantly.

### What Next

Make sure you really need to use hdlin_mux_size_limit > 32.

---

## ELAB-2031

(information) %s Pruning inferred sequential element for %s.

### Description

You receive this message when HDL Compiler detects that a sequential element that has alrady been inferred is not actually required for proper functioning of the design being elaborated, and the inference has not been explicitly requested by the user, so the sequential element is removed.

An RTL-level design description in Verilog or VHDL may contain variables for which a sequential cell would ordinarily be inferred (either because they are assigned under a clock edge, or because they are conditionally assigned) but whose values are never read, or whose values are not used in computing any output of the design. Because their values are not used (that is, they have no loads, or no path to the outputs), by default HDL Compiler will attempt not to infer sequential cells for these variables.

Feedback

But if it does infer them, it will then try to prune them before the end of elaborate.

In some circumstances designers wish to retain these sequential cells. Unloaded sequential cells can be inferred by two methods. First, if hdlin_preserve_sequential is "all", then a sequential cell will be inferred for any variable that is conditionally assigned or assigned under a clock, regardless of whether its value is used or not. Second, if hdlin_preserve_sequential is not set, but some variables are marked with the "preserve_sequential" pragma, only those variables will receive sequential cells even if they have no loads.

**What Next**

Examine the set of variables for which unloaded sequential cells are inferred.

If you do intend to infer some of unloaded sequential cells, try setting the the variable hdlin_preserve_sequential to "ff", "latch", or "ff+latch". Or consider using the pragma method for finer control over the inference.

Otherwise, no action is required on your part.

---

## ELAB-2032

(error) %s Generic %s does not have a default value.

**Description**

You receive this error message because the generic default value is not specified.

**What Next**

Set the default value for the given generic and run the command again.

---

## ELAB-2040

(error) %s Processes containing both positive edge and negative edge event expressions are not supported.

**Description**

This error occurs if you use both positive-edge and negative-edge triggers in multiple clock'event statements in the same process.

An example is shown below.

```
entity test is
  port(clock, a: in bit; z1, z2: out bit);
end;

architecture a of test is
begin
```

Feedback

```
  process (clock)
  begin
    if(clock'event and clock = '1') then
      z1 <= a;
    end if;

    if(clock'event and clock = '0') then -- ERROR
      z2 <= a;
    end if;
  end process;
end;
```

## ELAB-2041

(error) %s Module or interface %s was not elaborated because no default or override values were provided for parameter(s) '%s'.

### Description

This error occurs when default values for the parameters are omitted in the module definition and no overriding paramter values are specified during the module instantiation.

### What Next

Explicitly provide values for the parameters reported in the message at all instantiations of the module.

## ELAB-2050

(error) %s $fatal(%s) output: %s

### Description

User $fatal elaboration system task message. 'finish_number' can be one of 0, 1 or 2 and will be printed as $fatal(finish_number) in the message, but the tool does no further processing with finish_number. After outputting message, the tool will stop elaboration of the current module.

## ELAB-2051

(error) %s $error output: %s

### Description

User $error elaboration system task message.

Feedback

## ELAB-2052

(warning) %s $warning output: %s

**Description**

User $warning elaboration system task message.

## ELAB-2053

(info) %s $info output: %s

**Description**

User $info elaboration system task message.

## ELAB-2054

(error) Cannot explicitly elaborate '%s'; it has the same name as a DesignWare entity.

**Description**

User elaboration of Synopsys DesignWare entities is not permitted.

## ELAB-2057

(info) Loop optimization is performed.

**Description**

CDFG loop optimization is performed on some loops in the module.

## ELAB-2079

(warning) %s Default value of the parameter '%s' has been overridden with an expression that will be a simulation-synthesis mismatch if the parameter is sized to '%d' bits or more.

**Description**

You receive this warning message when the default value of a parameter has been overridden with an expression that will be a simulation-synthesis mismatch if the parameter is sized at or beyond the size reported in the message.

In synthesis the expression used in a parameter override is evaluated in a self-determined context, so in the following example, parameter P in the instantiation "bot1" is overridden with an expression that is evaluated with 1-bit width precision.

Feedback

But in simulation, because the width of the sized parameter P in the module BOT is 2, the expression would be evaluated with 2-bit width precision. As in this example, the difference in precision can lead to different results between simulation and synthesis.

```
`define E 1'b1 + 1'b1
module TOP (output test_bit);
  wire [1:0] w1, w2;
  assign test_bit = (w1 == w2);
  BOT#(.P(`E)) bot1(w1);
  BOT bot2(w2);
endmodule

module BOT #(parameter [1:0] P = `E) (output [1:0] o);
  assign o = P;
endmodule
```

### What Next

Assign the expression to a local parameter of the expected size and override the parameter with the local parameter instead.

```
`define E 1'b1 + 1'b1
module TOP (output test_bit);
  wire [1:0] w1, w2;
  assign test_bit = (w1 == w2);
  localparam [1:0] P_val = `E;
  BOT#(.P(P_val)) bot1(w1);
  BOT bot2(w2);
endmodule

module BOT #(parameter [1:0] P = `E) (output [1:0] o);
  assign o = P;
endmodule
```

---

## ELAB-2080

(error) %s Default value of the parameter '%s' has been overridden with an expression that will be a simulation-synthesis mismatch if the parameter is sized to '%d' bits or more.

### Description

You receive this error message when the default value of a parameter has been overridden with an expression that will be a simulation-synthesis mismatch if the parameter is sized at or beyond the size reported in the message.

In synthesis the expression used in a parameter override is evaluated in a self-determined context, so in the following example, parameter P in the instantiation "bot1" is overridden with an expression that is evaluated with 1-bit width precision.

Feedback

But in simulation, because the width of the sized parameter P in the module BOT is 2, the expression would be evaluated with 2-bit width precision. As in this example, the difference in precision can lead to different results between simulation and synthesis.

```
`define E 1'b1 + 1'b1
module TOP (output test_bit);
  wire [1:0] w1, w2;
  assign test_bit = (w1 == w2);
  BOT#(.P(`E)) bot1(w1);
  BOT bot2(w2);
endmodule

module BOT #(parameter [1:0] P = `E) (output [1:0] o);
  assign o = P;
endmodule
```

### What Next

Assign the expression to a local parameter of the expected size and override the parameter with the local parameter instead.

```
`define E 1'b1 + 1'b1
module TOP (output test_bit);
  wire [1:0] w1, w2;
  assign test_bit = (w1 == w2);
  localparam [1:0] P_val = `E;
  BOT#(.P(P_val)) bot1(w1);
  BOT bot2(w2);
endmodule

module BOT #(parameter [1:0] P = `E) (output [1:0] o);
  assign o = P;
endmodule
```

---

## ELAB-2081

(error) %s Module ' %s ' is found in both enable_sync_set_reset and enable_sync_set_reset_all_except. Conflicting assignment.

### Description

This error message issues when the module in the values of both application options hdlin.elaborate.enable_sync_set_reset and hdlin.elaborate.enable_sync_set_reset_all_except.

### What Next

To enable FLIP FLOP synchronous set/reset inference on the module, remove the module name from hdlin.elaborate.enable_sync_set_reset_all_except. To disable FLIP FLOP synchronous set/reset inference on the module, remove the module name from hdlin.elaborate.enable_sync_set_reset.

Feedback

## ELAB-2082

(error) enable_sync_set_reset and enable_sync_set_reset_all_except cannot be used when always_ff_sync_set_reset is set to true.

**Description**

You receive this warning enable_sync_set_reset and enable_sync_set_reset_all_except are used when always_ff_sync_set_reset is set to true.

Scenario 1: To disable FLIP FLOP synchronous set/reset inference on all module, set always_ff_sync_set_reset to FALSE.

Scenario 2: To enable FLIP FLOP synchronous set/reset inference on all module, set always_ff_sync_set_reset to TRUE.

Scenario 3: To use per module FLIP FLOP synchronous set/reset inference, set always_ff_sync_set_reset to FALSE and list the modules using enable_sync_set_reset/ enable_sync_set_reset_all_except.

**What Next**

To use module level FF SYNC SET/RESET inference, set always_ff_sync_set_reset to false and use enable_sync_set_reset/enable_sync_set_reset_all_except.

## ELAB-2083

(error) %s first argument of $pow should be or evaluate to a integer constant '2' or a real constant '2.0'.

**Description**

$pow accepts only '2' or '2.0' as base. Parameter substitution is also accepted. For example: $pow(2,3);

localparam P = 2.0 $pow(P,3);

**What Next**

Specify base as '2' or '2.0'.

## ELAB-2084

(warning) %s For svf checksum calculation , the original file %s not found

**Description**

elaborate generates svf file for formality ECO flow and as a part of that includes checksum for the RTL files. However if the RTL file is moved since it was analyzed or analyze

Feedback

happened in different directory, in some cases elaborate may not be able to find that file for checksum calculation.

# EMS

## EMS-001

(Error) No database specified to save.

**Description**

There is no database specified nor current EMS database available. No database to be saved.

**What Next**

Either use -database flag or set_current_ems_database command to specify a database to be saved.

## EMS-002

(Error) File '%s' already exists. Use -overwrite flag to force writing of messages database.

**Description**

The specified filename already exists.

**What Next**

Either to specify a new filename or use -overwrite flag to force writing of messages database.

## EMS-003

(Error) File '%s' does not exist.

**Description**

The specified filename does not exist .

**What Next**

To check or provide a valid filename.

Feedback

## EMS-004

(Error) Illegal EMS filename. Filename needs to have suffix of '.ems'.

**Description**

The specified filename does not meet EMS database naming convention.

**What Next**

Specify a new filename with suffix of '.ems'.

## EMS-005

(Error) No database specified. No current EMS DB, no database will be closed

**Description**

Neither database specified nor current EMS DB exists.

**What Next**

Specify a database collection after 'close_ems_database' or set_current_ems_database for the database to be closed.

## EMS-006

(Error) No db or command option is specified.

**Description**

When 'set_current_ems_database' no db or command option is specified.

**What Next**

Specify a database or command option .

## EMS-007

(Error) No current DB exists.

**Description**

When'set_current_ems_database -reset', but no current DB exists.

**What Next**

No action needed. No need to reset since no current DB exists.

Feedback

## EMS-008

(Error) '%s" is not current DB, cannot be reset.

**Description**

When'set_current_ems_database -reset db', db is not current DB cannot reset.

**What Next**

check and provide current DB name or remove db list. By default -reset will reset current DB.

## EMS-009

(Error) No EMS Current DB, no message will be get.

**Description**

When'get_ems_message', no current DB exists.

**What Next**

To make sure current DB exists so get_ems_message returns its message.

## EMS-010

(error) %s

**Description**

The specified error happend during command action.

**What Next**

Check the specified error before continue.

## EMS-011

(warning) %s

**Description**

The specified warning happend during command action.

**What Next**

Check the specified warning before continue.

Feedback

## EMS-012

(info) %s

**Description**

The command action returned the specified information message.

**What Next**

n/a.

## EMS-013

(info) %s successfully

**Description**

This message tell command progress, the specified step completed successfully.

**What Next**

n/a.

## EMS-014

(Error) Cannot specify '-of_objects' with '-database'.

**Description**

When'get_ems_messages' specifies '-of_objects' with '-database'.

**What Next**

To psecify with -of_objects.

## EMS-015

(Error) message parameter '%s' not found!

**Description**

When open_ems_database, the paramter stored in the database is not currently defined by the rule. To GUI display this database will encounter ambiguous result.

**What Next**

To make sure the rule currently defined is consistent with the database stored.

Feedback

## EMS-016

(Error) parameter '%s' was not found in rule '%s'.

**Description**

When open_ems_database, the parameter currently defined with the rule doesnt exist in the database. To GUI display this database will encounter ambiguous result.

**What Next**

To make sure the rule currently defined is consistent with the database stored.

## EMS-017

(Error) parameter '%s' was used, but not set in message of type '%s'.

**Description**

The parameter defined in the rule but its correspondent instance doesnt exist. This will cause derive_ems_message failed.

**What Next**

To provide needed instance for the rule when you want to generate its message.

## EMS-018

(Error) no rule name entered.

**Description**

When 'create_ems_rule' the required string object of the option -name is missing.

**What Next**

To provide the required information in order to complete the command.

## EMS-019

(Error) 'create_ems_rule' cannot create non-USER rules.

**Description**

When 'create_ems_rule' trying to create a rule which not having the prefix of 'USER', this error message will be issued. 'create_ems_rule' commands only creates 'USER-xxx' rules. 'USER' is selected and limited for end users.

Feedback

**What Next**

Please change your rule name to have prefix of 'USER' if you want to create non-Synopsys rule. All Synopsys internal rules need to be registered and defined under 'nwtn/doc/EMS'.

---

## EMS-020

(Error) no rule message entered.

**Description**

When 'create_ems_rule' no string object provided for option of '-message'.

**What Next**

To provide a string of message for the rule.

---

## EMS-021

(Error) no parameter name or type entered.

**Description**

When'create_ems_rule', either object of -parameter_name or -parameter_type was not provided. For multiple entries use curly bracket {} or double quote "". No bracket or quote needed for single entry.

**What Next**

To provide complete and consistent information for parameter name and type in order to complete the command.

---

## EMS-022

(Error) illegal 'severity' entry- need to be 'info', 'warn' or 'err'.

**Description**

When 'create_ems_rule' a 'string' object provided for -severity is an illegal entry. The severity for EMS ruless is either 'information', 'warning' or 'error'.

**What Next**

To make sure legal entry entered for -severity. The entry is case insensitive and accepts shorter wording such as 'warn', 'info' or , 'err'.

Feedback

## EMS-023

(Error) illegal 'parameter_type' entry - need to be S, I, F or D.

**Description**

When 'create_ems_rule' the 'string' object entered for -parameter_type is illegal. The entry should be either 'string' 'integer', 'float' or 'double'.

**What Next**

To make sure the information is provided correctly. The entry is case insensitive. It also accepts one alphabet entry such as : 'I', 'S', 'd', 'f'.

## EMS-024

(Error) parameter_name and parameter_type entered inconsistently.

**Description**

When 'create_ems_rule' the number of parameter_name entered inconsistently with the number of parameter_type entered, this error message will be issued.

**What Next**

Check if the name or type of any parameter is missing. Provide complete info and complete the command.

## EMS-025

(Error) rule create failed!.

**Description**

Both rule and message need to be created successfully, otherwise 'create_ems_rule' will be failed .

**What Next**

To follow error messages provided to fix failure.

## EMS-026

(Warning) rule '%s' already exists.

Feedback

**Description**

When 'create_ems_rule' trying to create a rule which already defined, this warning message will be issued. No rule will be created.

**What Next**

No need to redefine the issue, it's already exist.

---

## EMS-027

(Warning) rule '%s' already exists but defined differently.

**Description**

When 'create_ems_rule' trying to create a rule which name already exist. However it was defined differently.

**What Next**

To use different rule ID in case you want to complete rule creation. To avoid confusing, no duplication of rule name is allowed.

---

## EMS-028

(Error) rule '%s' does not exist.

**Description**

When'derive_ems_message' or 'open_ems_database' trying to generate or assemble a rule which does not currenly exist. 'derive_ems_message' or 'open_ems_database' command will fail.

**What Next**

To create a rule which is consistent with the message tried to generate/assemble.

---

## EMS-029

(Error) no instance enetered.

**Description**

When 'derive_ems_message', no instance entered. Instance entered through a nameless option, just follows after the rule name. For example derive_ems_message -rule USER-001 {pin1 12}

Feedback

**What Next**

To provide instance(s) which needed for derive_ems_message.

---

## EMS-030

(Error) instance '%s' is not a '%s' number .

**Description**

When 'derive_ems_message' a 'string' instance was mistakenly provided for an integer or float/dobule parameter.

**What Next**

To check and make sure the parameter type is consistent with the instance provided. For a 'string' parameter, if an integer or float/double instance was entered, no error message will be issued. Still users need to check the consistency between parameter type and its correspondent instance provided.

---

## EMS-031

(Warning) '-name' option needs to be used with '-current' option .

**Description**

When 'create_ems_database' if users specify database name by using 'name' option need to use '-current' otpion. The intention is to make this specified database as users current one and ready for writing message to it.

**What Next**

To add '-current' to continue.

---

## EMS-032

(Error) file '%s" already exists. It cannot be created.

**Description**

When 'create_ems_database' if users specify a database name which already a file exists, create_ems_database will return with this Error message.

**What Next**

To specify a different name or delete the file with the same name under working directory to continue.

Feedback

## EMS-033

(Error) Filename "%s" is invalid.

**Description**

Only alpha-numeric([a-z] [A-Z] [0-9]) and underscore("_") characters are allowed. Character "." is allowed only as part of file-extension ".ems".

**What Next**

Specify a different name for the file which meets the above specified constraints.

## EMS-034

(Warning) Overwriting the already existing file "%s".

**Description**

If a file by the same name already exists, it will be overwritten.

**What Next**

If above is not the intended behavior, then provide a different filename.

## EMS-035

(Error) "%s" is not a valid check name.

**Description**

Check provided is not a pre-defined check or an user-defined check.

**What Next**

Run the command "get_design_checks" to obtain the collection of all valid check names.

## EMS-036

(Warning) No current EMS database present.

**Description**

There is no EMS database which is "current" or active.

**What Next**

Use one of the commands "open_ems_database", "set_current_ems_database" or "create_ems_database" to make an EMS database "current".

Feedback

## EMS-037

(Warning) No current EMS database present, ignoring "-merge" option.

### Description

There is no EMS database which is "current" or active. Hence cannot merge the newly opened EMS database with current EMS database.

### What Next

If needed, firstly, use one of the commands "open_ems_database", "set_current_ems_database" or "create_ems_database" to make an EMS database "current". Then open the EMS database which needs merging from the current EMS database.

## EMS-038

(Warning) Skipping merge since current EMS database and EMS database opened are same.

### Description

The current EMS database and the EMS database opened are same. So its redundant to merge them. Hence the merge operation is skipped and the only the EMS database is opened.

### What Next

Set the current EMS database to the right EMS database which you want to merge with opened database.

## EMS-039

(Error) More than one EMS database name is provided.

### Description

"-name" option of report_ems_database command expects only one EMS database name. If more than one EMS database is provided then it is invalid.

### What Next

Provide only one EMS database name. Make sure there are no whitespaces in the EMS database name provided.

Feedback

## EMS-040

(Warning) EMS database "%s" already exists, over-writing it.

**Description**

The command executed creates an EMS database even if one exists by the same name(over-writes the existing EMS database).

**What Next**

Use "-ems_database" to provide a new EMS database name if you do not want to overwrite the existing EMS database.

## EMS-100

(Error) Name of EMS rule provided is empty.

**Description**

EMS rule name should be a non-empty string. The provided name does not contain any characters or only comprises of whitespace characters.

**What Next**

Provide a valid EMS rule name.

## EMS-101

(Error) Name of EMS rule %s, does not obey regular expression [A-Z]+[-][0-9]+

**Description**

EMS rule name provided is invalid. It should contain only upper-case alphabets follwed by a hyphen character and then followed by numerals.

**What Next**

Provide a valid EMS rule name.

## EMS-102

(Error) An EMS rule by name %s, already exists.

**Description**

An EMS rule already exists by the given name and hence cannot create the rule.

Feedback

**What Next**

Provide a different name for the EMS rule to be created.

---

## EMS-103

(Error) Severity of EMS rule provided is empty.

**Description**

The severity provided for EMS rule should be non-empty and should be one of valid severity strings.

**What Next**

Provide a valid severity string.

---

## EMS-104

(Error) Severity of EMS rule provided is invalid.

**Description**

The severity provided for EMS rule should be one of valid severity strings.

**What Next**

Provide a valid severity string.

---

## EMS-105

(Error) Message-template of EMS rule provided is empty.

**Description**

The message-template for EMS rule should be a non-empty string. It may or may not contain EMS parameters.

**What Next**

Provide a valid message-template string.

---

## EMS-106

(Error) Unescaped but stray percent(%) character is found.

Feedback

**Description**

Percent character is used to indicate the start of EMS parameter or can be a literal. If a literal then it needs to be escaped with one more percent character. But in this case it does not fall into any of these categories.

**What Next**

Correct the message-template either by providing a parameter name after percent character or by escaping the percent character.

---

## EMS-107

(Error) Invalid parameter name; it can contain only alphanumeric or underscore characters.

**Description**

EMS parameter name can contain only alphanumeric and underscore characters. string "tcl" is a reserved word and it also cannot be used as a parameter name.

**What Next**

Provide a valid parameter name

---

## EMS-108

(Error) "tcl" is a reserved word; it cannot be a parameter name.

**Description**

EMS parameter name can contain only alphanumeric and underscore characters. But string "tcl" is a reserved word and it cannot be used as a parameter name.

**What Next**

Provide a valid parameter name

---

## EMS-109

(Error) Nested tcl{} is not allowed.

**Description**

One cannot specify construct tcl{} inside a tcl{} construct. Nesting is not allowed.

Feedback

**What Next**

Remove the nested tcl{} construct and try saving results of tcl commands into variables and then using them in message-template.

---

## EMS-110

(Error) Failed to parse parameter in message-template of EMS rule.

**Description**

EMS has failed to parse the parameter in the message-template. The reason could be invalid parameter name or corrupt message -template string.

**What Next**

Try to remove parameters one at a time from message-template to identify the culprit parameter.

---

## EMS-111

(Error) Construct tcl{} does not contain any tcl commands inside its curly-braces.

**Description**

Construct tcl{} has empty content or contains only whitespace characters, which is invalid.

**What Next**

Provide a tcl command inside the tcl{} construct

---

## EMS-112

(Warning) There are EMS parameters with undefined type; assuming string type for them.

**Description**

By default EMS parameter names are assigned string type.

**What Next**

If you do not want the parameter to have string type, use edit_ems_rule command to re-define the type of parameter.

---

## EMS-113

(Error) Value for parameters option is empty.

Feedback

**Description**

Empty string provided for option parameters.

**What Next**

Do not use parameters option if you do not have any parameters or provide a valid definition of parameter properties

---

## EMS-114

(Error) Properties for one or more EMS parameters are not specified.

**Description**

Properties for EMS parameters are not defined.

**What Next**

Provide definition for properties of EMS parameters

---

## EMS-115

(Error) One or more parameter properties are not specified, using format name:value.

**Description**

Properties for EMS parameters are not defined using correct format.

**What Next**

Provide definition for properties of EMS parameters using correct syntax.

---

## EMS-116

(Error) Invalid value %s, provided for type-property of parameter.

**Description**

Value provided for type-property of parameter is not valid. It should be one of int, float, double or string.

**What Next**

Provide a valid value for type-property of parameter.

---

## EMS-117

(Error) Invalid parameter property %s.

Feedback

**Description**

The property name provided for EMS parameter is not valid.

**What Next**

Provide a valid value for type-property of parameter which is either name or command or type.

---

## EMS-118

(Error) Name-property of EMS parameter is missing which is mandatory.

**Description**

Name property of EMS parameter is mandatory. It is needed to associate the property definitions with the right parameter.

**What Next**

Provide the name-property along with other property definitions for EMS parameter

---

## EMS-119

(Warning) Parameter %s is not used in message-template; ignoring it.

**Description**

Parameter properties are defined but it is not used in the message template of EMS rule.

**What Next**

Check if the parameter is missed from message-template by oversight.

---

## EMS-120

(Error) Failed to create user-defined rule %s.

**Description**

EMS could not create the user-defined rule.

**What Next**

Verify whether the parameter names are correct.

---

## EMS-121

(Error) Built-in rule %s cannot be edited.

Feedback

**Description**

Only user-defined rules, which have not been frozen, can be edited.

**What Next**

Create a user-defined rule to meet the requirements.

---

## EMS-122

(Error) EMS rule %s which already has an associated EMS message, cannot be edited.

**Description**

Only user-defined rules, which have not been frozen, can be edited.

**What Next**

Create a new user-defined rule to meet the requirements.

---

## EMS-123

(Error) Rule by name %s does not exist; so cannot edit it.

**Description**

Only user-defined rules which are already created can be edited.

**What Next**

Create the user-defined rule first and then edit it, if needed.

---

## EMS-124

(Error) Options -pattern and -all are mutually exclusive.

**Description**

Only one of options -pattern and -all can be used at a time.

**What Next**

Use either option -pattern or -all.

---

## EMS-125

(Error) Failed to report user-defined rules.

Feedback

**Description**

Reporting of user-defined rules failed.

---

## EMS-126

(Error) Failed to get parameters for rule %s.

**Description**

While reporting the user-defined rule/s, parameter information could not be retrieved.

---

## EMS-127

(Error) Failed to get parameter %s for rule %s.

**Description**

While reporting the user-defined rule/s, parameter information could not be retrieved.

---

## EMS-128

(Error) Filename provided is empty.

**Description**

Filename provided is empty.

---

## EMS-129

(Error) Failed to open file %s.

**Description**

Failed to open the given file.

---

## EMS-130

(Error) Failed to write user-defined rules.

**Description**

Failed to write user-defined rules.

Feedback

## EMS-131

(Error) Rule %s is either invalid or disabled.

**Description**

Rule is either invalid or disabled.

## EMS-132

(Error) Failed to create EMS message.

**Description**

Failed to create EMS message.

## EMS-133

(Error) Option parameters does not contain valid information.

**Description**

Option parameters does not contain valid information.

## EMS-134

(Error) Parameter option should be provided with parameter_name:parameter_value pairs.

**Description**

Parameter name and value for the parameters should be provided.

## EMS-135

(Error) Failed to add parameter %s to EMS message of rule %s.

**Description**

Parameter value information could not be added to EMS message.

## EMS-136

(Error) Rule by name %s does not exist.

**Description**

Given rule name does not exist.

Feedback

## EMS-137

(Error) Failed to create EMS message for built-in rule.

**Description**

Only messages for user-defined rules can be created using command create_ems_message.

## EMS-138

(Error) Expecting %s type for parameter %s.

**Description**

There is a mismatch in type of parameter provided during EMS rule creation/edit and during EMS message creation.

## EMS-139

(Error) Failed to edit rule %s.

**Description**

Rule provided could not be edited.

## EMS-140

(Error) Setting 'severity' attribute for user-defined EMS rule which is frozen, is not allowed.

**Description**

If EMS rule is defined by user using 'create_ems_rule' command then such rule becomes user-defined rule. If an EMS message is instantiated/created for this rule then that EMS rule is considered 'frozen'. 'severity' attribute can be set for user-defined rules which are not frozen. Valid values for 'severity' attribute are "error", "warn", "warning", "info" and "information". These values can be in upper-case or lower-case or a mix of both.

## EMS-141

(Error) Setting 'severity' attribute for built-in EMS rule is not allowed.

**Description**

If EMS rule is defined by the tool then such rule is considered built-in EMS rule. For such EMS rules, setting 'severity' attribute is not allowed.

Feedback

## EMS-142

(Warning) Failed to remove EMS rules.

**Description**

If an EMS rule cannot be removed the there can be multiple reasons. The EMS rule could be built-in or EMS is not enabled or the given EMS rule does not exist.

## EMS-143

(Error) Failed to report EMS rules.

**Description**

If an EMS rule cannot be reported then the EMS rule might be disabled or it might not exist.

## EMS-144

(Warning) Failed to remove EMS rule %s since it has EMS messages instantiated.

**Description**

The EMS rule could not be removed since there are EMS messages already created for the EMS rule. Removal of such EMS rules will make associated EMS messages invalid and might lead to poblems reading the EMS databases containing such messages.

## EMS-145

(Warning) Using -force option to remove EMS rules which might have EMS messages instantiated.

**Description**

If -force option is used then EMS rules which have EMS messages are removed. This might result in failure to open EMS databases on disk which have the EMS messages corresponding to removed EMS rules. It is recommended to avoid using -force option.

## EMS-146

(Warning) EMS rule by name %s already exists; Ignoring the new definition.

Feedback

**Description**

While opening the EMS database if an EMS rule definition is encountered which has the same name as already existing one in memory then the newly encountered EMS rule is ignored. Any messages instantiated for this ignored rule present in the EMS database will also be ignored.

---

## EMS-147

(Error) Failed to open EMS database %s

**Description**

EMS database could not be opened since it is corrupted or has outdated or invalid data.

---

## EMS-148

(Error) Since '%s' is a built-in check name, it cannot be used to define a new check.

**Description**

A built-in check by same name as provided already exists. It cannot be re-used. Please use a different check name to define a new check.

---

## EMS-149

(Warning) Option '-reset' is made obsolete and it will be ignored.

**Description**

Since user is not allowed to reuse built-in check names while creating new checks, the -reset option loses its meaning and hence has become defunct. Currently this option is ignored. Please change your scripts to remove this option.

---

## EMS-150

(Warning) EMS rule by name %s already exists; Ignoring the new definition.

**Description**

While opening the EMS database if an EMS rule definition is encountered which has the same name as already existing one in memory then the newly encountered EMS rule is ignored. Any messages instantiated for this ignored rule present in the EMS database will also be ignored.

Feedback

# EMX

## EMX-0001

(info) Early Mux Extraction engine extracted '%lu' SelectOps in module '%s'.

### Description

This message signifies how many SelectOps were extracted in a given module in a design by the Early Mux Extraction engine. This engine looks at mux structures represented with AND/OR gates in the netlist and builds SelectOps from those structures. The 'report_mux_stats' command gives additional information about the number of such extracted SelectOps in the whole design.

# EOPT

## EOPT-001

(Error) Please use M-2017.06-SP3-1 and beyond PrimeTime executable.

### Description

This message indicates that you are using an old version PrimeTime.

### What Next

User needs to specify a new version PrimeTime.

### See Also

- set_pt_options

## EOPT-002

(Error) PrimeTime executable does not exist or file specified is not an executable: %s.

### Description

This message indicates that you specify a bad PrimeTime exectuable.

### What Next

User needs to specify the right PrimeTime executable.

Feedback

**See Also**

- set_pt_options

---

## EOPT-003

(Error) pre link script %s does not exist

**Description**

This message indicates that you give a bad pre link script file.

**What Next**

User needs to specify a good pre link script file.

**See Also**

- set_pt_options

---

## EOPT-004

(Error) post link script %s does not exist

**Description**

This message indicates that you give a bad post link script file.

**What Next**

User needs to specify a good post link script file.

**See Also**

- set_pt_options

---

## EOPT-005

(Error) no write permission on working directory.

**Description**

This message indicates that you give a working directory that you have no permission.

**What Next**

User needs to specify a working directory with write permission.

Feedback

**See Also**

- set_pt_options

---

## EOPT-006

(Error) Invalid optimization type: %s.

**Description**

This message indicates that you give a bad fixing type in eco_opt.

**What Next**

User needs to specify the available ECO fixing types.

**See Also**

- eco_opt

---

## EOPT-007

(Error) eco_script file %s does not exist

**Description**

This message indicates that you give a user file that does not exist.

**What Next**

User needs to specify an valid file name.

**See Also**

- eco_opt

---

## EOPT-008

(Error) : Please use set_starrc_options command to set config file in order to run StarRC in ECO fusion. For other PrimeTime based analysis without StarRC, please set app_option extract.starrc_mode to false.

**Description**

This message indicates that you need to run a command to set up StarRC settings.

**What Next**

User needs to set up the StarRC config file.

Feedback

**See Also**

- set_starrc_options
- eco_opt
- check_pt_qor

---

## EOPT-009

(Error) : The -punch_port option works only for the max_transition, DRC and noise fixing type.

**Description**

This message indicates that you used -punch_port option for fixing types other than max_transition, DRC and noise.

**What Next**

User needs to use this option along with max_transition DRC or noise fixing.

**See Also**

- set_starrc_options
- eco_opt
- check_pt_qor

---

## EOPT-010

(Error) : '%s' is not a valid option for the '%s' type.

**Description**

This message indicates that you run check_pt_qor -type with invalid option(s).

**What Next**

User needs to use the right options to run check_pt_qor.

**See Also**

- set_starrc_options
- eco_opt
- check_pt_qor

Feedback

## EOPT-011

(Error) : '%d' errors encountered.

### Description

This message indicates that errors are encountered during changelist implentation.

### What Next

User can examine the changelist output txt file for more details.

### See Also

- eco_opt
- check_pt_qor

## EOPT-012

(warning) : The following hold buffers have been automatically selected for hold-fixing: '%s' .

### Description

Tool has identified the list of buffers used for hold_fixing in addition to what is defined with set_lib_cell_purpose.

### What Next

User can override the tool generated list with eco_opt -hold_fix_buffers.

### See Also

- eco_opt

# EPL

## EPL-001

(Error) Cell %s (%d) is not placed!

### Description

This error only happens when the non eco cell connected to eco cell is not placed.

Feedback

**See Also**

- check_legality

---

## EPL-002

(Warning) There is no ECO cell to be placed.

**Description**

There is no ECO cell to be placed. Maybe all input cells are unsupported cell if -cells is used.

---

## EPL-003

(Warning) %d Cells are not supported and have been skipped silently!

**Description**

This warning means there are %d unsupported cells in input cell list or unplaced cells. And they have been filtered out silently.

In general, user do not count the following cells as ECO cells. hierarchical cell, fixed cell, macro cell, filler cell, io_pad cell, spare cell, physical only cell, cell without signal nets connection. (The cells only connect to tie nets are more like spare cells.)

**See Also**

- get_attribute

---

## EPL-004

(Warning) The connection of net (%s) is ignored for exceeding the fanout limit (%d).

**Description**

This warning means the net connection is ignored since its fanout exceeds the maximum fanout.

**What Next**

The default fanout limit is 100. You can use -max_fanout to specify the fanout limit as you need.

Feedback

## EPL-005

(Info) The current maximum fanout to ignore high fanout net connection of ECO cells is %d.

### Description

This message shows the current maximum fanout value.

### What Next

The default fanout limit is 100. You can use -max_fanout to specify the fanout limit as you need.

## EPL-006

(Warning) %d Cells have no connectivity and have been ignored silently!

### Description

This warning means %d cells in input cell list or unplaced cells have no connectivity or its all connections are ignored due to the option -max_fanout or -ignore_pin_connection. They have been filtered out silently and will not be placed.

## EPL-007

(Error) The reference pin name %s is invalid!

### Description

This error only happens when the reference pin name is invalid.

### What Next

Please specify the valid reference pin name for -ignore_pin_connection.

### See Also

- get_pins

## EPL-008

(error) Design has no site rows.

### Description

There are no site rows defined for the design. Site rows are required for this command.

Feedback

**What Next**

Define the site rows.

---

## EPL-009

(error) Cells in relative placement groups are not provided together.

**Description**

This message occurs when the cells in the relative placement groups are not provided together for legalization. The relative placement group need to be moved as a whole unit and cells in the group should not be legalized individually.

**What Next**

Please provide all cells of the relative placement groups for legalization. If you really want to legalize individual cells only, then please remove the relative placement group.

---

## EPL-010

(error) Skip to legalize cells in relative placement group %s.

**Description**

This message occurs when the cells in the relative placement group are not provided together for legalization. The relative placement group need to be moved as a whole unit and cells in the group should not be legalized individually.

**What Next**

Please provide all cells of the relative placement group for legalization. If you really want to legalize individual cells only, then please remove the relative placement group.

---

## EPL-011

(error) The ECO cell (%s) is a filler cell of the specified reference by option -remove_filler_references.

**Description**

This message occurs when any eco cell is a filler cell of the specified reference by option -remove_filler_references. It may lead to unexpected result (i.e. crash) since the eco cell may be removed when legalization since the reference is specified by option -remove_filler_references.

Feedback

**What Next**

Please remove the eco cell from input or remove the reference in option -remove_filler_references.

---

## EPL-020

(Error) There is unplaced eco cell in the given list when the option -legalize_only is used with -cells.

**Description**

When the option -legalize_only is used with -cells, it expects all the eco cells have been already placed. The option allows only doing legalization.

**What Next**

Use place_eco_cells -unplace to place the eco cells.

**See Also**

- get_attribute

---

## EPL-021

(warning) Skip to legalize the cell (%s) since it is not placed.

**Description**

When the option -legalize_only is used with -eco_changed_cells, the command will skip the cells that is not placed (attribute is_placed == false). The command should not legalize the unplaced cells.

**What Next**

Use place_eco_cells -unplaced(-cells) to place and legalize the cell.

**See Also**

- get_attribute

---

## EPL-022

(Warning) There is no ECO cell to be legalized.

**Description**

There is no ECO cell to be legalized.

Feedback

### What Next

Please check the given cells. Maybe all input cells are unsupported cell.

---

## EPL-023

(Error) Cannot find user specified net weight.

### Description

When the option -user_specified_net_weight is used, the tool expects user already set net weight for individual net before place_eco_cells.

### What Next

Please use the command set_place_eco_cells_net_weight to set net weight for individual net as user-defined net weight priority.

### See Also

- set_eco_placement_net_weight

---

## EPL-024

(Warning) %d Cells are not supported for legalization and have been skipped silently!

### Description

This warning means there are %d unsupported cells in input cell list for legalization. And they have been skipped silently.

In general, we do not support the following cells to do eco legalization. hierarchical cell, fixed cell, macro cell, pad cell.

### See Also

- get_attribute
- legalize_placement

---

## EPL-025

(Warning) Cell %s is not in top level design and has been skipped silently!

### Description

This warning means the given cell is in sub block and has been skipped silently. The command only works for cells in top level design.

Feedback

### EPL-026

(Error) Cannot find user specified net weight data in memory.

**Description**

This message occurs when there is no user specified net weight data in memory.

**What Next**

Please use the command set_place_eco_cells_net_weight to set net weight for individual net as user-defined net weight priority.

**See Also**

- set_eco_placement_net_weight

## ERRDM

### ERRDM-001

(error) Failed to open error data which is not for the current design.

**Description**

An attempt was made to open an error data which is not associated with the current design.

**What Next**

Select an error data file which is associated with the current design.

### ERRDM-002

(error) Failed to open error data which does not exist.

**Description**

An attempt was made to open an error data file which does not exist.

**What Next**

Select an existing error data file which is associated with the current design.

### ERRDM-003

(error) Could not create the error data file.

Feedback

**Description**

The error data file could not be created.

**What Next**

Check for inadequate write permissions on the disk or a possible collision with existing filenames.

---

## ERRDM-004

(error) Invalid file format.

**Description**

An invalid file format was specified for the error data file.

**What Next**

Provide a valid file format.

---

## ERRDM-005

(error) Corrupt error data.

**Description**

A syntax error was encountered while reading a corrupt error data file.

---

## ERRDM-006

(error) The error data has unsaved changes.

**Description**

An attempt was made to close an error data file which has unsaved changes.

**What Next**

Use the -force option to force closure without saving changes, discarding any pending changes. For exported error data files, you may also use the -save option to first save the unsaved changes. For standard error data attached to a design block, first save the block with save_block, then close the error data.

---

## ERRDM-007

(error) The error data %s is open.

Feedback

**Description**

An attempt was made to remove an error data file which has not been closed.

**What Next**

Close the error data before removing it.

---

## ERRDM-008

(error) An error type with the given name already exists.

**Description**

An attempt was made to create an error type with an existing name. Error type names must be unique.

**What Next**

Specify a unique name for the new error type.

---

## ERRDM-009

(error) The error class was missing or was invalid.

**Description**

An attempt was made to create an error type without specifying a valid error class.

**What Next**

Specify a valid error class.

---

## ERRDM-010

(error) Invalid error type.

**Description**

An attempt was made to create an error for an invalid error type.

**What Next**

Specify a valid error type.

---

## ERRDM-011

(error) Invalid error specification.

Feedback

**Description**

An attempt was made to create an error with invalid parameters.

**What Next**

Specify a valid parameters.

---

## ERRDM-012

(error) An error data file with the given name already exists.

**Description**

An attempt was made to create an error data file with an existing name. Error file names must be unique.

**What Next**

Specify a unique name for the new error data file.

---

## ERRDM-013

(error) The given error data file is read-only.

**Description**

An attempt was made to edit an error data file which was open in read-only mode.

**What Next**

Open the error data file in read-write mode.

---

## ERRDM-014

(error) For a top level 3DIC design, you cannot create drc errors with layers.

**Description**

create_drc_error command cannot be used to specify layers for a top level 3DIC design.

---

## ERRDM-032

(error) For a top level 3DIC design, you cannot create drc errors with layers.

**Description**

create_drc_error_shapes command cannot be used to specify layers for a top level 3DIC design.

Feedback

## ERRDM-033

(error) At least one of options '-polygons', '-polylines', '-points' or '-endpoints' for create_drc_error_shapes should be specified.

**Description**

create_drc_error_shapes command requires at least one of options: '-polygons', '-polylines', '-points' or '-endpoints' to be specified.

## ERRDM-034

(warning) There must be two coordinates given for each endpoint shape.

**Description**

There must be two coordinates given for each endpoint shape.

## ERRDM-035

(error) ERRDM_LOCK_FAIL_ACTION `%s' not one of: `off', `complain', or `die'. Turning off.

**Description**

This is an internal error. Please report to Synopsys.

**What Next**

Internal error, please report to Synopsys.

## ERRDM-036

(info) lock check failed!.

**Description**

This is an internal message. Please report to Synopsys.

**What Next**

Internal message, please report to Synopsys.

## ERRDM-037

(error) %s cannot be saved independent of the associated block.

Feedback

**Description**

Error data that are attached to a block cannot be saved independent of saving the block.

**What Next**

To save changes to the given error data, call save_block to save the block. Alternatively, you may export the error data to a file using the command write_drc_error_data.

---

## ERRDM-038

(warning) option '-readonly' and '-readwrite' will be ignored when opening error data which is attached with current block.

**Description**

option '-readonly' and '-readwrite' will be ignored when opening error data which is attached with current block.

---

## ERRDM-039

(error) Use '-endpoints' option with two point coordinates to specify two endpoints for an open error.

**Description**

Open error accepts only '-endpoints' option with two point coordinates to describe open error shape.

---

## ERRDM-040

(error) Saving to the given version %s is not supported. %s

**Description**

Provide a version string in the supported range.

---

## ERRDM-041

(error) The drc error data schema version string %s is invalid.

**Description**

Provide a properly formed version string, <major version number>.<minor version number>, "1.6", for example.

Feedback

## ERRDM-042

(error) The design schema version string %s is invalid.

**Description**

Provide a properly formed version string, <major version number>.<minor version number>, "1.090", for example.

## ERRDM-043

(error) The error data %s is closed.

**Description**

An attempt was made to edit or access data of an error data file which has not been opened.

**What Next**

Open the error data before editing or accessing its types and violations.

**See Also**

- open_drc_error_data

## ERRDM-044

(error) Unopened error data %s cannot be saved.

**Description**

An attempt was made to save an error data file which has not been opened.

**What Next**

Check the error data file input to make sure you specified the correct one.

**See Also**

- open_drc_error_data

## ERRDM-045

(information) Incrementing open_count of error data '%s'.

Feedback

### Description

open_drc_error_data was called for an already opened error data. A reference count was added to the error data. Use close_drc_error_data to decrement the reference count. When the reference count drops to zero, the error data will be closed.

### See Also

- close_drc_error_data

# EX

## EX-78

(Warning) Two Overlapping rectangles with different cap_levels and same picture levels Layer %s overlaps with %s Suggested Fix in Transistor Level: Use Boolean Not in runset to remove the overlap r1=%s r2=%s

### Description

Detected short between two rectagles with the same picture level and different capcap_level. This may cause accuracy issues.

### What Next

Use boolean operation NOT in runset to remove the overlap.

## EX-80

(Information) Can not find via model matching with the via: (%d %d), (%d %d) on level %d. Via coverage checking is not performed on the via.

### Description

Missing via model information.

### What Next

Check via coverage model in the runset.

## EX-81

(Information) Via coverage code is not found for model index %d via: (%d %d), (%d %d) on level %d

Feedback

**Description**

Via coverage code is not present in the model for given via.

**What Next**

Check via coverage model.

---

## EX-82

(Warning) Probe text id %d at (%.4f %.4f) is not on any shapes on layer %s, ignored.

**Description**

Probe text is not on any shapes on that layer, ignored.

**What Next**

Adjust probe coordinates.

---

## EX-83

(Warning) Text not touching any polygons %d

**Description**

Text doesn't touch any polygon.

**What Next**

Adjust text coordinates.

---

## EX-85

(Warning) The user defined subckt in %s will be skipped because its corresponding cell %s is not available in HN.

**Description**

StarRC will not merge subckt model of cell into final netlist as the cell is not used.

**What Next**

Check if the cell name used in 2D_BUMP_SUBCKT_FILE is correct.

---

## EX-86

(Warning) Instance "%s" port number is not 1.

Feedback

**Description**

Instance of BUMP cell has more than 1 port. The final netlist will have problem after StarRC replacing the instance with its subckt.

**What Next**

Check if the BUMP cell has correct port definition in input data.

---

## EX-87

(Warning) Cannot found bump Port of instance: %s

**Description**

StarRC failed to find port of the BUMP instance for subckt replacement.

**What Next**

Contact the Synopsys Support Center.

---

## EX-94

(Fatal) INTERNAL Error: Detected negative resistance: %g between nodes %ld and %ld in net %d.

**Description**

An unexpected result in an internal sanity check.

**What Next**

Contact the Synopsys Support Center.

---

## EX-96

(Fatal) Reluctance matrix is too large to invert; suggest to use NETLIST_INDUCTANCE: NO

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

Feedback

## EX-97

(Warning) Extrapolation on RPSQ_VS_WIDTH_AND_SPACING is used, result might be inaccurate.

**Description**

StarRC invoked extrapolation functionality when RPSQ_VS_WIDTH_AND_SPACING table look up is out of bound and EXTRAPOLATE_YES is set.

**What Next**

Contact the Synopsys Support Center.

## EX-98

(Warning) Extrapolation on RPSQ_VS_WIDTH_AND_SPACING is enabled, but only one value in the table, boundary value is used. result might be inaccurate.

**Description**

StarRC invoked extrapolation functionality when RPSQ_VS_WIDTH_AND_SPACING table look up is out of bound and EXTRAPOLATE_YES is set.

**What Next**

Contact the Synopsys Support Center.

## EX-113

(Warning) there are still %d opens not reported for net %s!

**Description**

There are opens not report for some nets.

**What Next**

Double check opens.sum file to make sure the opens are as expected or rerun StarRC. Contact the Synopsys Support Center if the error message persists.

## EX-114

(Fatal) Incompatible command provided: 3D_IC and an undocumented option

Feedback

**Description**

Two incompatible StarRC commands have been provided. One of them is 3D_IC, the other is an undocumented command.

**What Next**

Check that your StarRC comand file does not contain any undocummented commands that is not really needed. Contact the Synopsys Support Center if the error message persists.

---

## EX-115

(Warning) The maximum number of regions for calculating effective density must be less than 6. Skipping extra regions.

**Description**

The DENSITY_BOX_WEIGHTING_FACTOR command in the nxtgrd file contains more than 5 density box entries. In this case, StarRC extraction only considers the first five entries for computing density.

**What Next**

Update the DENSITY_BOX_WEIGHTING_FACTOR commands in the nxtgrd file so that the number of density box entries is 5 or less.

---

## EX-116

(Warning) The requested size of the region for effective density calculation is too large: %g um. Setting it to the maximum limit of: %g um.

**Description**

The DENSITY_BOX_WEIGHTING_FACTOR command in the nxtgrd file contains a density box size larger than 500 microns. In this case, StarRC extraction will limit the density box size to a maximum of 500 microns.

**What Next**

Update the DENSITY_BOX_WEIGHTING_FACTOR commands in the nxtgrd file so that the density box size is 500 microns or less.

---

## EX-117

(Warning) The requested size of the region for effective density calculation is too small: %g um. Setting it to the minimum limit of: %g um

Feedback

**Description**

The DENSITY_BOX_WEIGHTING_FACTOR command in the nxtgrd file contains a density box size smaller than 6 microns. In this case, StarRC extraction will limit the density box size to a minimum of 6 microns.

**What Next**

Update the DENSITY_BOX_WEIGHTING_FACTOR commands in the nxtgrd file so that the density box size is 6 microns or more.

---

## EX-126

(Fatal) Secondary corner process variation tables are not consistent with primary corner.

**Description**

Multi-corner nxtgrd files are not compatible with each other.

**What Next**

Please make sure that nxtgrd files from foundry are all valid and original.

---

## EX-127

(Fatal) Inconsistent thickness variation tables detected. All corner ITFs must have same resistive and capactive thickness variation tables specified for a given metal layer,

**Description**

Different types of resistive or capacitive thickness variation tables are found in multi-corner nxtgrd files. These tables can have different values but should all have the same type.

**What Next**

Please check with foundry and make sure that the multi-corner nxtgrd files are valid and original.

---

## EX-128

(Fatal) corrupt grd file, SPLNCV information incomplete

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

Feedback

## EX-129

(Fatal) corrupt grd file, SPLNCV information incomplete

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

## EX-130

(Fatal) corrupt grd file, PLLAYERMAP information incomplete

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

## EX-131

(Fatal) Nxtgrd file inconsistency, technology information incomplete

**Description**

Temperature coefficients from nxtgrd file are not valid.

**What Next**

Please check with foundry and make sure that the nxtgrd file is valid and original.

## EX-132

(Fatal) xTractor found an inconsistency in reading the data from the TCAD_GRD_FILE. Please use the correct xTractor version compatible with the provided TCAD_GRD_FILE.

**Description**

The nxtgrd file is not compatible for the current StarRC.

**What Next**

Please check with foundry and make sure that the nxtgrd file is valid and original.

Feedback

## EX-133

(Warning) IGNORE_GATE_CHANNEL_CAPACITANCE forced to %s to ensure consistency with NXTGRD

**Description**

IGNORE_GATE_CHANNEL_CAPACITANCE is reset to be compatible with the nxtgrd file.

**What Next**

Please check with foundry and make sure that the nxtgrd file is valid and original.

## EX-135

(Information) Reading binary grd file %s ...

**Description**

Reading binary nxtgrd file.

**What Next**

Please refer to StarRC manual.

## EX-136

(Information) Reading ascii grd file %s ...

**Description**

Reading ascii nxtgrd file.

**What Next**

Please refer to StarRC manual.

## EX-138

(Fatal) Unknown sensitivity type %s in GRD file

**Description**

Sensitivity type must be one of the following: T, THICKNESS, W, WIDTH, RPV, RHO, ER. Other sensitivity types are not supported.

Feedback

**What Next**

Please make sure that the nxtgrd files are all valid, original and consistent. Please refer to StarRC manual on SENSITIVITY and VARIATION_PARAMETER.

---

## EX-139

(Fatal) temperature_sensitivity:yes is not supported for different global temperatures for different layers %f %f

**Description**

Temperature sensitivity is not supported for a technology with multiple global temperatures on different layers.

**What Next**

Please make sure that the nxtgrd file is valid and original. Please make sure that StarRC commands are compatible with the nxtgrd file. Please refer to StarRC manual on TEMPERATURE_SENSITIVITY.

---

## EX-141

(Warning) No global temperature is found in grd, will use default temperature 25 degrees

**Description**

Default global temperature 25 Celsius degrees is used since there is no defined global temperature in nxtgrd file.

**What Next**

Please refer to StarRC manual on GLOBAL_TEMPERATURE.

---

## EX-143

(Fatal) nxtgrd file is corrupted

**Description**

The nxtgrd file may be corrupted.

**What Next**

Please make sure that the nxtgrd file is valid and original. Please make sure that you have enough disk space.

Feedback

## EX-144

(Fatal) inconsistent layer information in nxtgrd file

**Description**

The nxtgrd file may be corrupted with inconsistent layer information.

**What Next**

Please make sure that the nxtgrd file is valid and original. Please make sure that you have enough disk space.

## EX-150

(Fatal) Inconsistent bottom thickness information

**Description**

The nxtgrd file may be corrupted with inconsistent bottom thickness variation information.

**What Next**

Please make sure that the nxtgrd file is valid and original. Please make sure that you have enough disk space.

## EX-153

(Warning) Version mismatch between extractor engine and .nxtgrd (tcad) file for THICKNESS_VS_DENSITY effect. Skipping this effect, you have to re-run the process characterization step using the new format for the THICKNESS_VS_DENSITY command with a newer grdgenxo. The re-characterization process is extremely fast since new tables are processed only.

**Description**

The version of the nxtgrd file is too old. Process effects defined in THICKNESS_VS_DENSITY statements are ignored.

**What Next**

Please make sure that the nxtgrd file is valid, original and up-to-date. Please refer to StarRC manual on THICKNESS_VS_DENSITY.

## EX-154

(Warning) ETCH_VS_WIDTH_AND_SPACING is incorrect, please check your grd file

Feedback

**Description**

The nxtgrd file may be corrupted with inconsistent ETCH_VS_WIDTH_AND_SPACING statements.

**What Next**

Please make sure that the nxtgrd file is valid and original. Please refer to StarRC manual on ETCH_VS_WIDTH_AND_SPACING.

---

## EX-155

(Warning) ETCH_VS_WIDTH_AND_SPACING does not work with resistance only extraction

**Description**

ETCH_VS_WIDTH_AND_SPACING statements in the nxtgrd file are not compatible with resistance only extraction.

**What Next**

Please refer to StarRC manual on EXTRACTION and ETCH_VS_WIDTH_AND_SPACING.

---

## EX-156

(Warning) in grd file, CEVWSE line is incorrect, please check your grd file

**Description**

The nxtgrd file may be corrupted with inconsistent CAPACITIVE_ONLY ETCH_VS_WIDTH_AND_SPACING statements.

**What Next**

Please make sure that the nxtgrd file is valid and original.

---

## EX-157

(Warning) ETCH_VS_CONTACT_AND_GATE_SPACINGS in ITF is empty or incorrect. please check your grd file

**Description**

The nxtgrd file may be corrupted with inconsistent ETCH_VS_CONTACT_AND_GATE_SPACING statements.

Feedback

**What Next**

Please make sure that the nxtgrd file is valid and original.

---

## EX-158

(Warning) ETCH_VS_WIDTH_AND_LENGTH table in ITF is empty or incorrect. please check your grd file

**Description**

The nxtgrd file may be corrupted with inconsistent ETCH_VS_WIDTH_AND_LENGTH statements.

**What Next**

Please make sure that the nxtgrd file is valid and original.

---

## EX-159

(Warning) GATE_TO_DIFFUSION_CAP table in ITF is empty or incorrect, please check your grd file

**Description**

The nxtgrd file may be corrupted with inconsistent GATE_TO_DIFFUSION_CAP statements.

**What Next**

Please make sure that the nxtgrd file is valid and original.

---

## EX-161

(Warning) RAISED_DIFFUSION_TO_GATE_SMIN_TABLE is corrupted, please check your ITF or grd file

**Description**

The nxtgrd file may be corrupted with inconsistent RAISED_DIFFUSION_TO_GATE_SMIN_TABLE statements.

**What Next**

Please make sure that the nxtgrd file is valid and original.

Feedback

## EX-162

(Warning) Width and spacing dependent thickness feature does not work with resistance only extraction

**Description**

Process effects such as width and/or spacing dependent thickness variations are not supported in resistance only extraction.

**What Next**

Please refer to StarRC manual on EXTRACTION and all ITF statements associated with thickness variations.

## EX-163

(Warning) Cannot define THICKNESS_VS_WIDTH_AND_SPACING together with command option use_polygon_thickness, ignore the THICKNESS_VS_WIDTH_AND_SPACING.

**Description**

When polygon thickness data are defined together with THICKNESS_VS_WIDTH_AND_SPACING statements for the same layer, the former takes precedence and the latter is ignored.

**What Next**

Please make sure that the polygon thickness data are indeed the preferred thickness variation process effects.

## EX-164

(Warning) density dependent thickness feature does not work with resistance only extraction

**Description**

Process effects such as density based thickness variations are not supported for resistance only extraction.

**What Next**

Please refer to StarRC manual on EXTRACTION and all density based thickness variation statements in the ITF file.

Feedback

## EX-165

(Fatal) corrupt grd file, via temperature derating information incomplete

**Description**

The nxtgrd file is corrupted with inconsistent via temperature derating data.

**What Next**

Please make sure that the nxtgrd file is valid and original.

## EX-166

(Fatal) corrupt grd file, via capactive only etch information incomplete

**Description**

The nxtgrd file may be corrupted with inconsistent via capacitive only etch statements.

**What Next**

Please make sure that the nxtgrd file is valid and original.

## EX-167

(Fatal) corrupt grd file, %s line is inconsistent with VIA_FRT line.

**Description**

The nxtgrd file may be corrupted with inconsistent via resistance data.

**What Next**

Please make sure that the nxtgrd file is valid and original.

## EX-176

(Fatal) corrupt grd file, CVASKIP line is inconsistent with VIA_FRT line.

**Description**

The nxtgrd file may be corrupted with inconsistent CRT statements.

**What Next**

Please make sure that the nxtgrd file is valid and original.

Feedback

## EX-179

(Fatal) SEARCH_DISTANCE line is corrupted, please check grd file.

**Description**

The nxtgrd file may be corrupted with inconsistent layer based search distance.

**What Next**

Please make sure that the nxtgrd file is valid and original.

## EX-180

(Fatal) SEARCH_DISTANCE for CONDUCTOR %s in grd file must be positive.

**Description**

The nxtgrd file may be corrupted because there are negative search distance specified in ITF file.

**What Next**

Please make sure that the nxtgrd file is valid and original.

## EX-181

(Fatal) SEARCH_DISTANCE for CONDUCTOR %s in grd file is too large. It must be no greater than 10um and no greater than 25 * (WMIN + SMIN).

**Description**

The nxtgrd file may be corrupted because the search distance in ITF file may be too large.

**What Next**

Please make sure that the nxtgrd file is valid and original.

## EX-183

(Warning) Operating frequency smaller than minimum frequency in TCAD file.

**Description**

The operating frequency specified in StarRC command file is smaller than the minimum frequency from the nxtgrd file.

Feedback

**What Next**

Please consider changing the operating frequency according to the range specified in the nxtgrd file.

---

## EX-184

(Warning) Operating frequency larger than maximum frequency in TCAD file.

**Description**

The operating frequency specified in StarRC command file is larger than the minimum frequency from the nxtgrd file.

**What Next**

Please consider changing the operating frequency according to the range specified in the nxtgrd file.

---

## EX-185

(Fatal) Invalid operating frequency for reluctance or it is not specified!

**Description**

Operating frequency for reluctance extraction is either empty or invalid.

**What Next**

Please specify a valid operating frequency for reluctance extraction. Please refer to StarRC manual on reluctance extraction.

---

## EX-186

(Warning) The grd file doesn't contain SCT models, but (MULTIGATE_MODELS: yes) or (DEVICE_MODEL_LEVEL: 2).

**Description**

The nxtgrd file contains FinFET or other multigate features but no SCT models.

**What Next**

Please contact foundry for an up-to-date nxtgrd file.

---

## EX-187

(Warning) Port %s is changed to upper-case.

Feedback

**Description**

The ports in SPICE_SUBCKT_FILE are defined in lower case while CALIBRE_LAYOUT_CASE_SENSITIVE is set to NO, changing the port names to upper-case in case of wrong connection.

**What Next**

No action is needed.

---

## EX-188

(Warning) Failed to evaluate schematic property %s in instance %s, %s.

**Description**

StarRC trys to copy schematic properties to OA view when OA_PROP_ANNOTATION_VIEW is set. This warning means the property is using pPar/iPar expression, and StarRC cannot evaluate it. The reason could be: 1. the property defined with pPar/iPar cannot be found in the design. 2. there is incorrect mathematical expression in the property expression.

**What Next**

Check if the schematic property is correctly defined.

---

## EX-199

(Fatal) TCAD_GRD_FILE is not compatible with the xTractor

**Description**

The TCAD_GRD_FILE is incompatible with this version of the xTractor.

**What Next**

Regenerate TCAD_GRD_FILE with a compatible version of grdgenxo.

---

## EX-201

(Fatal) INTERNAL Error: unknown layer has been referenced: %s

**Description**

The layer is not found in internal data base.

**What Next**

Contact the Synopsys Support Center.

Feedback

## EX-202

(Warning) INTERNAL Warning: unknown layer has been referenced: %s

**Description**

The layer is not found in internal data base.

**What Next**

Contact the Synopsys Support Center.

## EX-204

(Warning) INTERNAL Warning: via top_height (%g) < bottom_height (%g) on db layer (%s)

**Description**

An unexpected result in an internal sanity check.

**What Next**

Contact the Synopsys Support Center.

## EX-205

(Info) INTERNAL Info: via layer %s has multiple via cap clone levels %d and %d mapped to same pic level %d, removed extra via cap clone level

**Description**

An information is issued in an internal sanity check.

**What Next**

Contact the Synopsys Support Center.

## EX-207

(Fatal) INTERNAL Error: Two layers with different picture levels have been connected in the layer_info file. first layer: %s second layer: %s

**Description**

An internal connectivity check has failed.

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## EX-208

(Fatal) INTERNAL Error: via layer %s for via_to has to be within [%d, %zd]

**Description**

An internal connectivity check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## EX-209

(Fatal) INTERNAL Error: via layer %s for via_from has to be within [%d, %zd]

**Description**

An internal connectivity check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## EX-210

(Fatal) INTERNAL Error: via_from and via_to are set to the same layer %s for via layer %s

**Description**

An internal connectivity check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## EX-211

(Fatal) INTERNAL Error: via layer %s for both via_from and via_to layers are vias. This is not allowed.

**Description**

An internal connectivity check has failed.

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## EX-212

(Fatal) INTERNAL Error: via_to %s and via_from %s for via layer %s are conneected together. This is not allowed.

**Description**

An internal connectivity check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## EX-214

(Fatal) INTERNAL Error: inconsistancy in layer_info file for layer: %s. The following three fields should all_or_none be set VIA_FROM , VIA_TO , VIA_CONDUCTIVITY_PER_MICRONS2

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## EX-215

(Warning) INTERNAL Warning: Conductivity is unset for layer %s; set it to 1.0

**Description**

An unexpected result in an internal sanity check.

**What Next**

Contact the Synopsys Support Center.

---

## EX-216

(Warning) INTERNAL Warning: Conductivity is unset for via layer %s. Also cann't find any other layer connected to the via layer, set conductivity to 1.0.

Feedback

**Description**

An unexpected result in an internal conductivity check.

**What Next**

Contact the Synopsys Support Center.

---

## EX-219

(Fatal)INTERNAL_Error: invalid layer in grd_layer_info: %s

**Description**

Invalid layer during parsing an internal file "grd_layer_info".

**What Next**

Contact the Synopsys Support Center.

---

## EX-221

(Fatal) Can not find resistor between %ld and %ld for net %d in corner %d! make sure REDUCTION is set to TOPOLOGICAL.

**Description**

A resistance cannot be found. This may be caused by setting inproper REDUCTION value in the command.

**What Next**

Set REDUCTION to TOPOLOGICAL and rerun. Contact the Synopsys Support Center if the error message persists.

---

## EX-222

(Fatal) can not find node %ld or %ld for net %d in corner %d! make sure REDUCTION is set to TOPOLOGICAL.

**Description**

A resistance cannot be found. This might be caused by setting inproper REDUCTION value in the command.

**What Next**

Set REDUCTION to TOPOLOGICAL and rerun. Contact the Synopsys Support Center if the error message persists.

Feedback

## EX-224

(Information) **** MAIN CIRCUIT STATISTICS ****

**Description**

Not available

**What Next**

Nothing is required at your side at the moment.

## EX-225

(Information) **** POWER CIRCUIT STATISTICS ****

**Description**

Not available.

**What Next**

Nothing is required at your side at the moment.

## EX-226

(Fatal) Desired number of corners of %u exceeds maximum number of allowed corners (%u).

**Description**

You receive this message because the total corners which are seleceted to netlist out exceed maximum number of allowed corners.

**What Next**

Reduce selected corners to make sure they don't exceed the maximum number of allowed corners.

## EX-227

(Fatal) failed to attach via_coverage file.

**Description**

via_coverage file cannot be attached.

Feedback

### What Next

Check if via_coverage file exists and rerun. Contact the Synopsys Support Center if the error message persists.

---

## EX-230

(Warning) device model linking enabled in xTractor but not in grdgenxo

### Description

Many device models have been generated in grdgenxo for equivalent layouts. SCT models for equivalent layouts may have slightly different caps due to field solver statistical noise. Device model linking is now applied in xTractor so only one of the equivalent model group will be loaded. That may cause a small version-to-version change in xTractor capacitances, corresponding to SCT field solver stattistical noise.

### What Next

You can avoid this information message by generating nxtgrd with -link_device_models grdgenxo option. This option will speed up nxtgrd generation by avoiding to simulate equivalent layouts. If using Quickcap for SCT generation, the LINKED_TO statements in ITF must be consitent with the QTF.

---

## EX-231

(Severe) No aspect ratio defined on layer '%s' to DISABLE_ASPECT_RATIO

### Description

DISABLE_ASPECT_RATIO is used to disable the ART difined on an ITF layer. Please check the ART for LINE_END_EXTENSION_TABLE is correctly defined in TCAD_GRD_FILE.

### What Next

You can check option DISABLE_ASPECT_RATIO to avoid this information.

---

## EX-256

(Warning) Line continuation is not supported in SPEF

### Description

In SPEF, each line presents a item, such as, a coupling capacitance, a resistance, a pin, etc ... and multiple items cannot be written in the same line.

Feedback

**What Next**

Nothing is required at your side at the moment.

---

## EX-257

(Warning) Line continuation is not supported in SPEF

**Description**

In SPEF, each line presents a item, such as, a coupling capacitance, a resistance, a pin, etc ... and multiple items cannot be written in the same line.

**What Next**

Nothing is required at your side at the moment.

---

## EX-261

(Fatal) A dynamic linking error occured: %s

**Description**

One of the dynamic libraries required cannot be found or linked.

**What Next**

Check your StarRC installation and that your environment is set properly.

---

## EX-262

(Fatal) Invalid netlist output. OA is only supported on linux and amd64 platforms.

**Description**

The OA flow is not supported on the current architecture.

**What Next**

Try rerunning on a supported architecture.

---

## EX-263

(Fatal) PRL output is supported only for SUN5, SUN64 and IA32

**Description**

PRL output is not supported in the current architecture

Feedback

**What Next**

Try rerunning on a supported architecture.

---

## EX-266

(Warning) Extraction was done %s only. Netlisting is requested resistance only. Please verify.

**Description**

Extraction and netlisting were done with different modes, for instance, extraction was done with capacitance-only while netlisting was done with resistance-only.

**What Next**

Nothing is requied at your side at the moment.

---

## EX-267

(Warning) Extraction was done %s only. Netlisting is requested capacitance only. Please verify.

**Description**

Extraction and netlisting were done with different modes, for instance, extraction was done with resistance-only while netlisting was done with capacitance-only.

**What Next**

Nothing is requied at your side at the moment.

---

## EX-269

(Warning) Only one corner is selected, NETLIST_SMC_FORMULAR is not needed

**Description**

NETLIST_SMC_FORMULAR is helpful when multiple corners are selected and it is not neccessary to be specified when one corner is selected.

**What Next**

Nothing is required at your side at the moment.

---

## EX-270

(Fatal) Could not find %s in LD_LIBRARY_PATH.

Feedback

**Description**

One of the dynamic libraries required cannot be found.

**What Next**

Check your StarRC installation and that your environment is set properly.

---

## EX-272

(Warning) NETLIST_POWER_MEM_OPT will be ignored because the number of nets (%ld) exceeds the limit (%d).

**Description**

When the number of nets exceeds a limit, NELIST_POWER_MEM_OPT is ignored.

**What Next**

Nothing is required at your side at the moment.

---

## EX-275

(Fatal) multi-corner netlist is only supported for NETLIST_FORMAT: STAR, NETNAME, SPF, SPEF.

**Description**

multi-corner netlist is only supported for netlist formats including STAR, NETNAME, SPF, SPEF.

**What Next**

Try with a format that supports Multi-Corner Netlisting. Contact the Synopsys Support Center if the error message persists.

---

## EX-277

(Fatal) Could not load symbols from OA.

**Description**

The OA library could not be found or it is corrupted.

**What Next**

Check your StarRC installation and that your environment is set properly. Contact the Synopsys Support Center if the error message persists.

Feedback

## EX-278

(Fatal) Please specify input xout using -i option

**Description**

Input for netlisting could not be found.

**What Next**

Try running StarRC again. Contact the Synopsys Support Center if the error message persists.

## EX-279

(Fatal) Please specify output xout using -o option

**Description**

Output of netlisting could not be found

**What Next**

Try running StarRC again from a clean directory. Contact the Synopsys Support Center if the error message persists.

## EX-281

(Fatal) NETLIST_TOTALCAP_THRESHOLD cannot be less than 0.

**Description**

NETLIST_TOTALCAP_THRESHOLD cannot be set to a negative value.

**What Next**

Set NETLIST_TOTALCAP_THRESHOLD to a positive value and rerun. Contact the Synopsys Support Center if the error message persists.

## EX-293

(Fatal) INTERNAL Error: Unable to parse line %d of coupling report file %s.

**Description**

An intermediate file generated by StarRC is corrupted.

Feedback

**What Next**

A corrupt file may be produced for a variety of reasons such as the disk being full or more than one job running concurrently in the same run directory. Try running StarRC one more time.

Contact the Synopsys Support Center if the error message persists.

---

## EX-296

(Fatal) I/O error while writing to file %s

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## EX-304

(Information) Parsing option_file: %s

**Description**

xTractor has started parsing its option file.

**What Next**

This message is for informational purposes only. No further action is needed.

---

## EX-306

(Fatal) INTERNAL Error: num_same_text_id for text id:%d has been previously assigned as %d but is being reset to %d

**Description**

An unexpected result in an internal sanity check during circuit node creation.

**What Next**

Contact the Synopsys Support Center.

---

## EX-314

(Fatal) Data inconsistency, please rerun StarRC

Feedback

### Description

An intermediate file produced by StarRC is corrupted which caused data inconsistency. A corrupt file may be produced for a variety of reasons such as the disk being full, or more than one job running concurrently in the same run directory, or running a newer version of StarRC on data produced by an older version.

### What Next

Try running StarRC again from a clean directory. Contact the Synopsys Support Center if the error message persists.

---

## EX-315

(Fatal) Invalid option MODE:%d

### Description

Setting mode or license is not valid. This may be caused by the license being invalid.

### What Next

Make sure the license is valid. Contact the Synopsys Support Center if the error message persists.

---

## EX-318

(Fatal) File %s can't be opened.

### Description

An intermediate file produced by StarRC is corrupted. A corrupt file may be produced for a variety of reasons such as the disk being full, or more than one job running concurrently in the same run directory, or running a newer version of StarRC on data produced by an older version.

### What Next

Try running StarRC again from a clean directory. Contact the Synopsys Support Center if the error message persists.

---

## EX-320

(Fatal) INTERNAL Error: intermediate file %s is corrupted.

Feedback

**Description**

An intermediate file produced by StarRC is corrupted. A corrupt file may be produced for a variety of reasons such as the disk being full, or more than one job running concurrently in the same run directory, or running a newer version of StarRC on data produced by an older version.

**What Next**

Try running StarRC again from a clean directory. Contact the Synopsys Support Center if the error message persists.

---

## EX-321

(Fatal) Internal error: netId %d is greater than the total number of nets %d

**Description**

An intermediate file produced by StarRC is corrupted. A corrupt file may be produced for a variety of reasons such as the disk being full, or more than one job running concurrently in the same run directory, or running a newer version of StarRC on data produced by an older version.

**What Next**

Try running StarRC again from a clean directory. Contact the Synopsys Support Center if the error message persists.

---

## EX-324

(Fatal) No reluctance data in tcad_file or tcad_file corrupted!

**Description**

TCAD_GRD_FILE does not contain reluctance data.

**What Next**

TCAD_GRD_FILE must be regenerated using ITF containing reluctance data. Contact the Synopsys Support Center for instructions on obtaining a compatible grd file.

---

## EX-325

(Fatal) Unknown reluctance model

**Description**

Unknown reluctance model is found in TCAD_GRD_FILE.

Feedback

**What Next**

Regenerate TCAD_GRD_FILE with a compatible version of grdgenxo. Contact the Synopsys Support Center for further information.

---

## EX-326

(Fatal) negative Self-K in model %d\t possible TCAD_GRD_FILE file mismatch

**Description**

Self-K values are required by xTractor to be non-negative. TCAD_GRD_FILE file may be incompatible with this version of the xTractor.

**What Next**

Contact the Synopsys Support Center to get more information on generating a compatible TCAD_GRD_FILE.

---

## EX-327

(Information) Invalid RV via array id %d is accessed for net %s

**Description**

Invalid RV via array id is accessed for the given net.

**What Next**

No action is required on your side.

---

## EX-329

(Fatal) XOUT_K_AUXILIARY file corrupted!

**Description**

An intermediate file produced by StarRC is corrupted. A corrupt file may be produced for a variety of reasons such as the disk being full, or more than one job running concurrently in the same run directory, or running a newer version of StarRC on data produced by an older version.

**What Next**

Try running StarRC again from a clean directory. Contact the Synopsys Support Center if the error message persists.

Feedback

## EX-334

(Fatal) Can not have NETLIST_PRINT_CC_TWICE: YES for NETLIST_FORMAT: SPF

**Description**

SPF prints coupling capacitance in either victim net or aggressive net depending on their order in the netlist file. Setting NETLIST_PRINT_CC_TWICE: YES with NETLIST_FORMAT: SPF, thus, is not allowed.

**What Next**

Try to remove NETLIST_PRINT_CC_TWICE: YES and rerun. Contact the Synopsys Support Center if the error message still persists.

## EX-338

(Fatal) Inconsistent corner information provided ...

**Description**

The number of valid nxtgrd files provided is not identical to the number of process corners specified in the StarRC command file.

**What Next**

Please check the validity and consistency of your statements in StarRC command file. Please also make sure that nxtgrd files from foundry are all valid and original.

## EX-339

(Fatal) Temperature statements in nxtgrd File are inconsistent.

**Description**

Statements related to TEMPERATURE_DERATION in the nxtgrd file are not consistent.

**What Next**

Please make sure that nxtgrd files from foundry are all valid and original.

## EX-340

(Fatal) CRT statements in nxtgrd file are inconsistent.

**Description**

CRT statements and temperature statements in nxtgrd file are not consistent. .

Feedback

**What Next**

Please make sure that nxtgrd files from foundry are all valid and original.

---

## EX-341

(Warning) The thickness variation defined in the technology file is ignored. The polygon thickness variation defined in the THICKNESS_VARIATION_FILE is used instead.

**Description**

When polygon thickness variation data defined in THICKNESS_VARIATION_FILE exist together with the thickness variation data from ITF file, the former takes precedence.

**What Next**

Please make sure that using polygon thickness variation data from THICKNESS_VARIATION_FILE is indeed intended.

---

## EX-343

(Fatal) ild thickness variations defined but pbtv data and thickness_vs_density_and_width are missing in level %d

**Description**

Nxtgrd file is corrupt because it contains ILD thickness variation data but no PBTV data or thickness_vs_density_and_width data.

**What Next**

Please make sure that nxtgrd files from foundry are all valid and original.

---

## EX-348

(Fatal) Cap level greater than max cap level.

**Description**

New ITF file pointed by RES_UPDATE_FILE may contain more conductor layers than the nxtgrd file. This is not allowed when using RES_UPDATE_FILE.

**What Next**

Please make sure that both nxtgrd file and the ITF file used by RES_UPDATE_FILE are valid, original and consistent. Please refer to StarRC manual on RES_UPDATE_FILE command for more information.

Feedback

## EX-349

(Fatal) Cannot update RPSQ_VS_SI_WIDTH_AND_LENGTH for layer %s which is not a GATE layer.

**Description**

RPSQ_VS_SI_WIDTH_AND_LENGTH can only be used with gate poly and those conductor layers with LAYER_TYPE = GATE. It is not allowed to use RPSQ_VS_SI_WIDTH_AND_LENGTH with other conductor layers without LAYER_TYPE = GATE.

**What Next**

Please make sure that both nxtgrd file and the ITF file used by RES_UPDATE_FILE are valid, original and consistent. Please refer to StarRC manual on RES_UPDATE_FILE command and ITF statement RPSQ_VS_SI_WIDTH_AND_LENGTH for more information.

## EX-350

(Fatal) Cannot update RPSQ_VS_SI_WIDTH_AND_LENGTH for layer %s because there is no FIELD POLY layer.

**Description**

RPSQ_VS_SI_WIDTH_AND_LENGTH can only be used on gate conductors with the presence of field poly layers in the technology files. If there are no field poly (LAYER_TYPE = FIELD_POLY) in the technology files, RPSQ_VS_SI_WIDTH_AND_LENGTH can not be used on any layer even if the layer is a gate conductor (LAYER_TYPE = GATE).

**What Next**

Please make sure that both nxtgrd file and the ITF file used by RES_UPDATE_FILE are valid, original and consistent. Please refer to StarRC manual on RES_UPDATE_FILE command and ITF statement RPSQ_VS_SI_WIDTH_AND_LENGTH for more information.

## EX-352

(Fatal) INTERNAL Error: xTractor tech_file -r value out of range: %d

**Description**

An unexpected result in an internal sanity check.

**What Next**

Contact the Synopsys Support Center.

Feedback

## EX-356

(Information) SMIN: Location BBox=(%g,%g),(%g,%g) Net=%d,%d Layer=%d,%d r1= %s %d %d r2= %s %d %d

**Description**

Spacing values between these polygons and their neighbors are much smaller than minimum spacing. This message lists the location of the violating polygon.

**What Next**

Please check if the small spacing values in the layout are indeed desired.

## EX-357

(Warning) 3DIC: cannot read subckt file %s, StarRC model will be used.

**Description**

StarRC could not find subckt file provided by user. The default model will be used to replace TSV in netlist.

**What Next**

Check if 3D_IC_SUBCKT_FILE is set correctly.

## EX-358

(Fatal) INTERNAL Error: Format in file TsvRC is not correct.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## EX-359

(Warning) The user defined subckt in %s will be skipped because its corresponding layer %s is not TSV or bump via.

**Description**

Cell name of subckt is not layer name of TSV or bump via so the model is ignored.

Feedback

**What Next**

Check if the cell name in 3D_IC_SUBCKT_FILE is correct.

---

## EX-361

(Fatal) INTERNAL Error: The Grd file was not declared in the tech file.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## EX-362

(Fatal) INTERNAL Error: The HN database was not declared in the tech file.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## EX-363

(Fatal) INTERNAL Error: Could not find tsv cell %s in data base.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## EX-365

(Fatal) 3DIC: syntax error in subckt file %s line %d:%s

**Description**

Model defined in 3D_IC_SUBCKT_FILE has problem so StarRC could not use it.

Feedback

**What Next**

Check the content defined in 3D_IC_SUBCKT_FILE.

---

## EX-366

(Fatal) 3DIC: subckt file %s did not specify DB layer.

**Description**

The cell name in 3D_IC_SUBCKT_FILE needs to be TSV layer name.

**What Next**

Check the cell name used in 3D_IC_SUBCKT_FILE.

---

## EX-367

(Fatal) 3DIC: subckt file %s specified %d res and %d cap. StarRC only supports 1Res, or 2Res, or 2Res+1Cg models.

**Description**

StarRC only support some pre-defined models. Others are not acceptable.

**What Next**

Check the model defined in 3D_IC_SUBCKT_FILE.

---

## EX-368

(Fatal) Unsupported format for TSV as cell. Please run -clean using SPEF or SPF.

**Description**

In TSV-as-cell flow, StarRC only support SPEF/SPF netlist. Other formats are not acceptable.

**What Next**

Reset NETLIST_TYPE:[SPEF/SPF] in StarRC command file.

---

## EX-375

(Warning) TSV instance:%s connected to 2 different nets

Feedback

**Description**

2 TSV terminals connect to different nets at front and back side. This could be short problem in netlist.

**What Next**

Check TSV connection in data input.

---

## EX-377

(Warning) TSV cell %s two pins connected to the same node

**Description**

The two terminals of TSV cell connect to same node.

**What Next**

Information only; no action is needed.

---

## EX-388

(Fatal) Multicontact via on layer %s (%d) is detected.

**Description**

Detected multicontact via is not supported for certain layers.

**What Next**

Please set AUTO_RUNSET: YES and restart StarXtract

---

## EX-390

(Fatal) For the layer %s option RPV_VS_AREA or RPV_VS_WIDTH_AND_LENGTH is not supported when multiple vias defined between the same pair of conductors.

**Description**

Option RPV_VS_AREA or RPV_VS_WIDTH_AND_LENGTH is not supported when multiple vias defined between the same pair of conductors.

**What Next**

Modify ITF definitions and regenerate nxtgrd.

Feedback

## EX-391

(Fatal) For the layer %s option RPV_VS_AREA can't be used for a trench contact array.

**Description**

Option RPV_VS_AREA can't be used for a trench contact array.

**What Next**

Modify ITF definitions and regenerate nxtgrd.

## EX-392

(Fatal) For the layer %s option RPV_VS_WIDTH_AND_LENGTH or RPV_VS_SI_WIDTH_AND_LENGTH can't be used for a via array.

**Description**

Option RPV_VS_WIDTH_AND_LENGTH or RPV_VS_SI_WIDTH_AND_LENGTH can't be used for a via array.

**What Next**

Modify ITF definitions and regenerate nxtgrd.

## EX-393

(Fatal) For the layer %s option RPV_VS_AREA must be specified for a via array when other via statement are not defined in ITF.

**Description**

Option RPV_VS_AREA must be specified for a via array when other via statement are not defined in ITF.

**What Next**

Modify ITF definitions and regenerate nxtgrd.

## EX-394

(Fatal) Fake via length is zero.

**Description**

Fake via length is zero.

Feedback

**What Next**

No action is required on your side.

---

## EX-395

(Fatal) Layer %s is not a TRENCH CONTACT FAKE VIA. Option RPV_VS_WIDTH_AND_LENGTH or RPV_VS_SI_WIDTH_AND_LENGTH can only be specified for TRENCH CONTACT FAKE VIA layers.

**Description**

Option RPV_VS_WIDTH_AND_LENGTH or RPV_VS_SI_WIDTH_AND_LENGTH can only be specified for TRENCH CONTACT FAKE VIA layer.

**What Next**

Modify ITF definitions and regenerate nxtgrd.

---

## EX-397

(Fatal) For the layer %s option RPV_VS_AREA or RPV_VS_WIDTH_AND_LENGTH or RPV_VS_SI_WIDTH_AND_LENGTH or RPV_VS_COVERAGE must be specified when other via statements are not defined in ITF.

**Description**

Option RPV_VS_AREA or RPV_VS_WIDTH_AND_LENGTH or RPV_VS_SI_WIDTH_AND_LENGTH or RPV_VS_COVERAGE must be specified when other via statements are not defined in ITF.

**What Next**

Modify ITF definitions and regenerate nxtgrd.

---

## EX-398

(Warning) Found mismatch between bridge via array and bridge via stucture, via resistor may not be correct for the bridge via (%d %d), (%d %d) on level %d.

**Description**

Found mismatch between input and actual information about bridge via.

**What Next**

Check the layout and runset data to avoid missmatch.

Feedback

## EX-410

(Fatal) Probe text on gate affects the accuracy of mos_gate_delta_resistance.

**Description**

Probe text on gate affects the accuracy of mos_gate_delta_resistance.

**What Next**

Remove the probe and rerun. Contact the Synopsys Support Center if the error message persists.

## EX-411

(Warning) Unable to convert to Delta configuration for gate %s %s located at(%f,%f) in cell %s.

**Description**

StarRC is unable to convert to Delta configuration for the particular gate

**What Next**

Nothing is required at your side at the moment if you are aware of this. Contact the Synopsys Support Center if you expect StarRC to convert the Delta configuration.

## EX-412

(Warning) Find contact on gate %s %s located at (%f, %f) in cell %s while converting to delta network.

**Description**

StarRC find contact on gate and is unable to convert to Delta configuration for the particular gate

**What Next**

Nothing is required at your side at the moment if you are aware of this. Contact the Synopsys Support Center if you expect StarRC to convert the Delta configuration.

## EX-414

(Warning) found open net(s) in the design. Check opens.sum file.

Feedback

**Description**

One or more open nets are found in the design and output in opens.sum file.

**What Next**

Nothing is required at your side at the moment.

---

## EX-415

(Warning) NETLIST_REMOVE_LOOPS will break a resistive loops in nets. Total pin-to-pin resistances will not be preserved, and REDUCTION_MAX_DELAY_ERROR may be violated. Use EXTRACT_LESS_LOOPS:YES (default) instead.

**Description**

Using NETLIST_REMOVE_LOOPS optoin will break a resistive loops in nets. Thus, the total pin-to-pin resistances may not be preserved, and REDUCTION_MAX_DELAY_ERROR may be violated. Use EXTRACT_LESS_LOOPS:YES (default) instead.

**What Next**

Nothing is required at your side at the moment.

---

## EX-417

(Fatal) Failed to copy %s to netlist %s

**Description**

An intermediate file produced by StarRC is corrupted. A corrupt file may be produced for a variety of reasons such as the disk being full, or more than one job running concurrently in the same run directory, or running a newer version of StarRC on data produced by an older version.

**What Next**

Try running StarRC again from a clean directory. Contact the Synopsys Support Center if the error message persists.

---

## EX-420

(Fatal) Design contains too many internal nodes. Consider using the SPF or SPEF formats to output the netlist or extract a reduced number of nets.

Feedback

**Description**

Design contains too many internal nodes. Consider using the SPF or SPEF formats to output the netlist or extract a reduced number of nets.

**What Next**

Nothing is required at your side at the moment.

---

## EX-432

(Warning) Missing model for Dieletric Thickness versus width and spacing in NXTGRD

**Description**

The NXTGRD does not contain a capacitance model table that is supposed to be present. This may have impact on capacitance accuracy. Note that if the ITF contains: 1. SIDE_DAMAGE_VS_WIDTH_AND_SPACING or SW_T_VS_WIDTH_AND_SPACING 2. Directionnal ETCH_VS_WIDTH_AND_SPACING 3. LINE_END_EXTENSION_TABLE

**What Next**

Regenerate the NXTGRD using a more recent grdgenxo version or contact your foundry to get a newer NXTGRD. If the warning persists, please contact Synopsys.

---

## EX-441

(Information) Parsing nxtgrd file: %s ...

**Description**

This message indicates the nxtgrd file specified by TCAD_GRD_FILE in StarRC command file is being parsed.

**What Next**

This is an informational message only. No action is required on your part.

---

## EX-443

(Fatal) Failed to parse nxtgrd file: %s

**Description**

The nxtgrd file specified by TCAD_GRD_FILE in StarRC command file has not been parsed sucessfully.

Feedback

**What Next**

Check TCAD_GRD_FILE in StarRC command file. Make sure that the nxtgrd file is not corrupted or obsolete. Also, check that the StarXtract version is at least as recent as the grdgenxo version used to generate the nxtgrd file.

---

## EX-444

(Information) Successfully parsed nxtgrd file: %s

**Description**

Not available.

**What Next**

This is an informational message only. No action is required on your part.

---

## EX-502

(Information) Messages about shorts/smin/vias violations are obsolete and will be removed in a future release. Please refer to shorts_all.sum, smin.sum and vias.sum files for durable report process.

**Description**

Detailled info messages about shorts, smin violations and via violations written in log files are now obsolete.

**What Next**

Contact the Synopsys Support Center.

---

## EX-503

(Information) SHORT: Two different nets TOUCH each other; ignoring the touch. BBox=(%.5f,%.5f),(%.5f,%.5f) Net=%d,%d Layer=%d,%d Inst=%d,%d r1= %s %d %d type1=0x%llx r2= %s %d %d type2=0x%llx

**Description**

Detected short between two polygons from different nets. This may cause accuracy issues.

**What Next**

Modify the layout to avoid shorts between polygons from two different nets.

Feedback

## EX-504

(Information) Two different texts touching each other; text_id1=%s text_id2=%s

**Description**

Detected short between two polygons with different text ids. This may cause accuracy issues.

**What Next**

Modify the layout to avoid shorts between polygons with different test ids.

## EX-505

(Information) FILL_SHORT: A short is found between floating metal fill polygon and design data. BBox=(%.5f,%.5f),(%.5f,%.5f) Net=%d,%d Layer=%d,%d r1=%s %d %d type1=0x%llx r2=%s %d %d type2=0x%llx

**Description**

Detected short between metal fill and design data. This may cause accuracy issues.

**What Next**

Modify the layout to avoid shorts between metal fills and design data.

## EX-506

(Information) UDDR: R %d Instance Port (%g,%g) Contact (%g,%g) %g $p_c=%d $sn_gn=%d $d_gc=%g $d_gb=%g $et_cd=%g $eb_cd=%g $w_g=%g $w_n=%g $w_d=%g $model=%s

**Description**

Detected the extraction of a User Defined Diffusion Resistance.

**What Next**

Check the correctness of the resistance and the parameters in the summary/uddr.sum file. This ouput is for debug purpose.

## EX-507

(Information) UDDR: R %d Instance Port (%g,%g) Contact (%g,%g) %g $p_c=%d $d_gc=%g $d_gb=%g $w_g=%g $w_n=%g $w_d=%g $Cenc=%g $model=%s

Feedback

**Description**

Detected the extraction of a User Defined Diffusion Resistance in the case of a bend case.

**What Next**

Check the correctness of the resistance and the parameters in the summary/uddr.sum file. This ouput is for debug purpose.

---

## EX-600

(Warning) The %d%s variation parameter specified in the ITF for %s produces greater capacitive sensitivity than the recommended %.3g%% of nominal capacitance in %.3g%% of related capacitance tables. This potentially affects VX extraction accuracy. Reduce the variation by %.3g%% for better accuracy. Statistics of the capacitive deviation: Mean = %.3g%% Standard Deviation = %.3g%%

**Description**

The variation parameters provided for sensitivity analysis allow too much variation for accurate capacitance extraction.

**What Next**

Limit the impact from variation parameters within the recommended values. In addition, ensure that the nxtgrd file is a valid and original file obtained from the foundry.

Refer to the User Guide for this StarRC version to obtain more information about using the SENSITIVITY and VARIATION_PARAMETERS commands.

---

## EX-603

(Warning) corrupt grd file, nominal_temperature is not specified.

**Description**

The nxtgrd file may be corrupted because CRT statements are specified without nominal temperature.

**What Next**

Please make sure that the nxtgrd file is valid and original. Please refer to StarRC manual on all CRT statements.

---

## EX-635

(Fatal) Unrecognized keyword %s found while parsing the device model in the nxtgrd file.

Feedback

**Description**

The nxtgrd file might be corrupted with inconsistent device model statements.

**What Next**

Ensure that the nxtgrd file is a valid and original file obtained from the foundry. If the problem persists, contact the Synopsys Support Center.

---

## EX-660

(Fatal) Failed to parse the nxtgrd file.

**Description**

The nxtgrd file specified by TCAD_GRD_FILE in StarRC command file has not been parsed sucessfully.

**What Next**

Check TCAD_GRD_FILE in StarRC command file. Make sure that the nxtgrd file is not corrupted or obsolete. Also, check that the StarXtract version is at least as recent as the grdgenxo version used to generate the nxtgrd file.

---

## EX-666

(Fatal)INTERNAL_ERROR: unknown layer has been referenced: %s

**Description**

The db layer is not found in an internal layer info table.

**What Next**

Contact the Synopsys Support Center.

---

## EX-670

(Warning) 3DIC: tsv model '%s' is not defined in 3D_IC_SUBCKT_FILE. The default model will be used to replace TSV in netlist.

**Description**

StarRC could not find tsv from user's subckt file. The default model will be used to replace TSV in netlist.

**What Next**

Check if 3D_IC_SUBCKT_FILE is set correctly.

Feedback

## EX-702

(Warning) SMIN: Found less than %sminimum spacing. SMIN: %g, found: %g

**Description**

Spacing values between neighbor polygons are found to be much smaller than minimum spacing. This message lists spacing value and the minimum spacing on this layer.

**What Next**

Please check if the small spacing values in the layout are indeed desired.

## EX-703

(Warning) Subckt device %s is declared more than once.

**Description**

TSV model has duplicated device. That could result to problem in final netlist.

**What Next**

Check the model defined in 3D_IC_SUBCKT_MODEL.

## EX-712

(Information) No trench contact shape is found for the fake via clone c1=%s c2=%s

**Description**

No trench contact shape is found for the fake via clone.

**What Next**

No action is required on your side.

## EX-714

(Information) VIA: Detected via with one or no connections: BBox=(%g,%g),(%g,%g)um Net=%d Layer=%d

**Description**

Detected via with one or no connections.

**What Next**

Correct via connections in the layout.

Feedback

## EX-716

(Warning) In the generated SPF file there are %d nets which have been netlisted in more than one part ("MULTI-PART" nets). Multi-part nets are not resistively connected; hence are open. You can find the multi-part net names below.%s

**Description**

In the generated SPEF file there might be nets that have been netlisted in more than one part ("MULTI-PART" nets). Multi-part nets are not resistively connected; hence are open.

**What Next**

Nothing is required at your side at the moment.

## EX-718

(Fatal) Cannot open TCAD_GRD_FILE: %s

**Description**

xTractor was not able to read the TCAD_GRD_FILE.

**What Next**

Check to make sure that TCAD_GRD_FILE exists and that the xTractor has read permissions for it.

## EX-719

(Fatal) TCAD_GRD_FILE %s is corrupted. The number of layers %d is outside possible range.

**Description**

Invalid layer count found in GRD file.

**What Next**

Check to make sure that GRD file is correct.

## EX-720

(Fatal) Invalid conductor layer in TCAD_GRD_FILE

**Description**

xTractor found invalid conductor layer in TCAD_GRD_FILE.

Feedback

**What Next**

Check to make sure that GRD file is compatible with this version of xTractor. Contact the Synopsys Support Center to get more information on generating a compatible GRD file.

---

## EX-721

(Fatal) Invalid layer %s in RC_SCALING_FILE

**Description**

xTractor found invalid layer in RC_SCALING_FILE.

**What Next**

Check to make sure that the input file is compatible with the GRD file.

---

## EX-724

(Information) Initializing extraction ...

**Description**

xTractor has started initializing extraction data structures.

**What Next**

This message is for informational purposes only. No further action is needed.

---

## EX-738

(Information) Parsed A tcad of %d levels

**Description**

xTracor is reporting the number of levels in the tcad file.

**What Next**

This message is for informational purposes only. No further action is required by the user.

---

## EX-739

(Warning) Cannot initialize error file!

**Description**

Not available.

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## EX-741

(Fatal) Zero input and Zero Output; Exiting ...

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## EX-742

(Warning) In the region x(%d, %d) y(%d, %d), overlapping polygons are found on two layers, either between layer %s and layer %s or between via layers connecting them. Please check design or LVS runset to remove the overlap.

**Description**

Detected possible short between two polygons. This may cause accuracy issues.

**What Next**

Locate the overlapping polygons first. If there is indeed overlap, use boolean operation NOT in runset to remove the overlap.

---

## EX-743

(Warning) Number of shorted nets reported is higher than the SHORTS LIMIT specified.

**Description**

To get a report of all violations, return StarRC with SHORTS_LIMIT set to 2147483647 in the StarRC command file. Having StarRC report an excessively large number of violations can significantly increase the overall runtime.

**What Next**

Use SHORTS_LIMIT to increase shorted nets limit in the command file.

Feedback

## EX-744

(Warning) Can't find coverage option for via width '%f' length '%f' X(%f - %f) Y(%f - %f) on layer '%s'. Please check VIA_COVERAGE_OPTION_FILE or VIA_COVERAGE in StarRC command file, or VIA_COVERAGE in nxtgrd file.

### Description

Can't find coverage option for via on the layer.

### What Next

Check VIA_COVERAGE_OPTION_FILE or VIA_COVERAGE in StarRC command file, or VIA_COVERAGE in nxtgrd file.

## EX-745

(Warning) Can't find coverage option in cmd or VIA_COVERAGE in nxtgrd for via on layer '%d'

### Description

Can't find coverage option in cmd or VIA_COVERAGE in nxtgrd for this via layer.

### What Next

Information only; no action is needed.

## EX-746

(Fatal) The GPD is not compatible to SPEF netlist format since NETILST_REMOVE_DANGLING_BRANCHES or NETLIST_MERGE_SHORTED_PORTS or NETLIST_MINRES_THRESHOLD or REDUCTION: HIGH was used when generating this GPD. Command NETILST_REMOVE_DANGLING_BRANCHES and NETLIST_MERGE_SHORTED_PORTS are not supported for SPEF format. Command NETLIST_MINRES_THRESHOLD and REDUCTION: HIGH support both SPF and SPEF, but the GPDs for SPF format and SPEF format are incompatible. If you used NETLIST_FORMAT: SPF (or didn't specify NETLIST_FORMAT) along with NETLIST_MINRES_THRESHOLD or REDUCTION: HIGH, the generated GPD is incompatible to SPEF format. You can only convert SPF netlist from this GPD.

### Description

Command NETILST_REMOVE_DANGLING_BRANCHES and NETLIST_MERGE_SHORTED_PORTS are not supported for SPEF format. If you use either of these 2 commands and use NETLIST_FORMAT: SPEF, StarRC will error out; If you use either of these 2 commands and use NETLIST_FORMAT: SPF (or didn't specify

Feedback

NETLIST_FORMAT), GPD will be generated but the generated GPD is not compatible to SPEF format and you can't convert SPEF netlist from this GPD. You can only convert SPF netlist from this GPD. Command NETLIST_MINRES_THRESHOLD and REDUCTION: HIGH support both SPF and SPEF, but the GPDs for SPF format and SPEF format are incompatible. If you use either of these 2 commands and use NETLIST_FORMAT: SPF (or didn't specify NETLIST_FORMAT), the generated GPD is incompatible to SPEF format and you can't convert SPEF netlist from this GPD. You can only convert SPF netlist from this GPD. However, if you use either of these 2 commands and use NETLIST_FORMAT: SPEF, you can get the SPEF netlist.

**What Next**

If you used NETILST_REMOVE_DANGLING_BRANCHES or NETLIST_MERGE_SHORTED_PORTS, remove them and re-generate the GPD.
If you used NETLIST_MINRES_THRESHOLD or REDUCTION: HIGH, you can set NETLIST_FORMAT: SPEF to generate the SPEF netlist, or remove the 2 commands and re-generate the GPD if you want to generate a GPD which is compatible to both SPF and SPEF formats.

---

## EX-747

(Warning) Number of fill shorts reported is higher than the FILL SHORTS LIMIT specified.

**Description**

To get a report of all violations, return StarRC with FILL_SHORTS_LIMIT set to 2147483647 in the StarRC command file. Having StarRC report an excessively large number of violations can significantly increase the overall runtime.

**What Next**

Use FILL_SHORTS_LIMIT to increase shorted nets limit in the command file.

---

## EX-748

(Fatal) INTERNAL Error: The input to temperature sensitivity flow is invalid. Rerun StarRC.

**Description**

The input to temperature sensitivity flow is corrupted.

**What Next**

Please rerun StarRC to solve the issue.

Feedback

## EX-749

(Warning) GATE_TO_DIFFUSION_CAP_VS_DIFFUSION_LENGTH table in ITF is empty or incorrect, please check your grd file

**Description**

The nxtgrd file may be corrupted with inconsistent GATE_TO_DIFFUSION_CAP_VS_DIFFUSION_LENGTH statements.

**What Next**

Please make sure that the nxtgrd file is valid and original.

## EX-750

(Information) Parsing RC scaling file: %s

**Description**

xTractor has started parsing RC scaling file.

**What Next**

This message is for informational purposes only. No further action is needed.

## EX-1977

(Warning) Parameter mag for '%s' '%s' '%s' is not of type float.

**Description**

Type for parameter "mag" is not 'oacFloatParamType'.

**What Next**

Make sure that the type for parameter "mag" is 'oacFloatParamType'.

## EX-1978

(Fatal) Detected N-1 mapping in PROPMAP of device '%s' :'%s' conflicts with earlier entry '%s'.

**Description**

When adding property mapping, the user-defined property type conflicts with property type from dspf.

Feedback

### What Next

Make sure that the user-defined property type is the same with property type from dspf.

---

## EX-1979

(Warning) Detected multiple definitions for cell '%s' in skip cell mapping files. The latest definition overrides all previous definitions.

### Description

StarRC detects multiple definitions for cells listed in files specified by the OA_SKIPCELL_MAPPING_FILE or OA_DEVICE_MAPPING_FILE command. The latest definition overrides all previous definitions.

### What Next

Make sure that the definitions are correct in the skip cell mapping files.

---

## EX-1980

(Warning) Detected multiple definitions for model '%s' in the device mapping file. The latest definition overrides all previous definitions.

### Description

StarRC detects multiple definitions for a model in the device mapping file specified by the OA_DEVICE_MAPPING_FILE command. The latest definition overrides all previous definitions.

### What Next

Make sure that the definitions are correct in device mapping file.

---

## EX-1983

(Fatal) Library '%s' is not defined in the OA_LIB_DEF file.

### Description

In the OpenAccess flow, all libraries referenced by the following commands must be defined in the OA_LIB_DEF file:

OA_DEVICE_MAPPING_FILE OA_LAYER_MAPPING_FILE OA_LIB_NAME OA_SKIPCELL_MAPPING_FILE OA_PORT_ANNOTATION_VIEW OA_PROPERTY_ANNOTATION_VIEW

Feedback

**What Next**

Make sure that all libraries are defined in the OA_LIB_DEF file.

---

## EX-1984

(Fatal) Cell '%s' is not defined in library '%s'.

**Description**

StarRC has determined that the cell specified by the OA_CELL_NAME command is not defined in the library specified by the OA_LIB_NAME command.

**What Next**

Make sure that the library and cell are correctly defined.

---

## EX-1985

(Fatal) '%s'

**Description**

StarRC encountered this OpenAccess error message when trying to open the design.

**What Next**

Ensure that the design database is correct. If the error persists, contact the Synopsys Support Center.

---

## EX-1986

(Fatal) '%s'

**Description**

StarRC encountered this OpenAccess error message when trying to open the design.

**What Next**

Ensure that the design database is correct. If the error persists, contact the Synopsys Support Center.

---

## EX-1991

(Warning) Pin count mismatch for instance '%s' of device '%s'. '%s' has '%d' pins. '%s' has '%d' pins. Device will remain unconnected.

Feedback

**Description**

You receive this message because of the mismatch between number of device ports and number of ports in the symbol view. To correctly connect the device, the number of device ports should be equal to number of ports in the symbol view. If number of device ports is not equal to number of ports in the symbol view, spice_subckt_file can be used to match number of ports in the symbol view. The device is invalid and unconnected.

**What Next**

Check why the device has different number of device ports and number of ports in the symbol view. If the number of device ports can be matched by spice_subckt_file, you can define the ports you want to connect in this device in spice_subckt_file.

---

## EX-1992

(Warning) Pin count mismatch: '%s'

**Description**

In OA flow, StarRC trys to match pins in skip cell with symbol view before using device mapping file, there's a mismatch between the number of instance pins and the number of its master cell pins.

**What Next**

Make sure the number of the skip cell instance pins is correct, or you can use device mapping file to match the pins.

---

## EX-1993

(Warning) '%s'layer '%s' undefined in technology library.

**Description**

The layer defined in layer_mapping_file is not defined in technology library.

**What Next**

Make sure that the layer is defined in technology library.

---

## EX-1994

(Warning) In layer mapping file line '%d' purpose '%s' undefined in technology library.

**Description**

The purpose defined in layer_mapping_file is not defined in technology library.

Feedback

**What Next**

Make sure that the purpose is defined in technology library.

---

## EX-1995

(Warning) Flylines for resistors will not be generated: please define valid *pres layer-purpose in layer mapping file, or define the default lpp (y0, drawing) in the technology library.

**Description**

When adding pres flyline layer-purpose pair(lpp), the layer-purpose pair is not corecttly defined in layer_mapping_file.

**What Next**

Define valid *pres layer-purpose in layer mapping file, or define the default lpp (y0, drawing) in the technology library.

---

## EX-1996

(Warning) Flylines for capacitors will not be generated: please define valid *pcap layer-purpose in layer mapping file, or define the default lpp (y1, drawing) in the technology library.

**Description**

When adding pcap flyline layer-purpose pair(lpp), the layer-purpose pair is not corecttly defined in layer_mapping_file.

**What Next**

Define valid *pcap layer-purpose in layer mapping file, or define the default lpp (y1, drawing) in the technology library.

---

## EX-1997

(Warning) Parse line: '%d' in layer_mapping_file.

**Description**

Syntax error(s) seen while parsing layer_mapping_file. This can lead to some layers not getting mapped as expected.

**What Next**

Make sure that the layer_mapping_file does not have syntax errors.

Feedback

## EX-1998

(Warning) Only one corner is selected, NETLIST_SMC_FORMULAR is ignored

**Description**

You receive this message because you use NETLIST_SMC_FORMULAR while specify only one corner. In a simultaneous multicorner flow, you can use the NETLIST_SMC_FORMULA command to specify that the output should be a single netlist containing RC values written as formulas that use the corner names as variables. If you only specify one corner name in SELECTED_CORNERS, the NETLIST_SMC_FORMULA will be ignored.

**What Next**

Check if your NETLIST_SMC_FORMULAR and SELECTED_CORNERS settings are correct.

## EX-2000

(Warning) Physical shape and presistor of design layer '%s' ('%d') is not available in parasitic view as there is no corresponding LPP in OA_LAYER_MAPPING_FILE.

**Description**

Physical shape and presistor of design layer is not available in parasitic view as there is no corresponding LPP in OA_LAYER_MAPPING_FILE.

**What Next**

Make sure that the LPP for the specified layer is correctly defined in OA_LAYER_MAPPING_FILE.

## EX-2001

(Fatal) lib '%s' is not defined in OA_LIB_DEF file.

**Description**

StarRC failed to open an OA view.

**What Next**

Make sure the OA library is correctly defined in OA_LIB_DEF file.

## EX-2002

(Fatal) '%s' is not defined in '%s'.

Feedback

**Description**

In OA flow, cell is not defined in lib.

**What Next**

Make sure that the cell is defined in lib.

---

## EX-2003

(Warning) Create view %s/%s/%s failed, waiting to re-try. Remaining: %d s.

**Description**

Create OA view failed. It might be locked by other session, or you do not have the permission. StarRC is waiting to re-try.

**What Next**

Please close other process which opens this view. Or wait for specified time.

---

## EX-2005

(Fatal) '%s'

**Description**

There is a OpenAccess Exception, when StarRC is trying to open the design.

**What Next**

Contact the Synopsys Support Center.

---

## EX-2006

(Fatal) '%s'

**Description**

There is an OpenAccess Exception, when StarRC is trying to open the design.

**What Next**

Contact the Synopsys Support Center.

---

## EX-2007

(Fatal) Can not create design '%s'/'%s'/'%s'

Feedback

**Description**

StarRC failed to generate symbol view for skip cell.

**What Next**

Check if you have the permission to write.

---

## EX-2008

(Fatal) lib '%s' is not defined in '%s'

**Description**

In OA flow, lib is not defined in OA_LIB_DEF file.

**What Next**

Make sure that the lib is defined in OA_LIB_DEF file.

---

## EX-2009

(Fatal) Cannot create or overwrite the view '%s'/'%s'/'%s', It might be locked by another session, or you do not have the permission.

**Description**

In OA flow, an attempt to create or overwrite the OA view failed. It may be locked by another session, or you do not have the permission to write.

**What Next**

Make sure that the OA view is not locked by another session, and you do have the permission to write.

---

## EX-2010

(Fatal) Cannot open tech database for library '%s'

**Description**

In OA flow, cannot open the technology library.

**What Next**

Make sure that the technology lib exists and you have the permission to read.

Feedback

## EX-2012

(Warning) '%s': '%s' is ignored due to illegal format.

**Description**

In OA flow, a syntax error was encountered when reading StarRC commands OA_PORT_ANNOTATION_VIEW or OA_PROPERTY_ANNOTATION_VIEW.

**What Next**

Make sure that the OA_PORT_ANNOTATION_VIEW or OA_PROPERTY_ANNOTATION_VIEW is correctly defined.

## EX-2013

(Warning) Cannot access the view '%s': '%s'

**Description**

The OpenAccess view cannot be opened.

**What Next**

Make sure that the OpenAccess view exists, is defined in cds.lib, and you have permission to read it.

## EX-2014

(Warning) Some instances cannot get schematic properties, please check oa_property_annotation.sum

**Description**

In OA flow, some instances cannot get schematic properties because they are not xref-ed.

**What Next**

Contact the Synopsys Support Center.

## EX-2015

(Warning) The program failed to create the OA view because of '%d': '%s'

**Description**

There is an OpenAccess Exception, when StarRC is trying to create the OA view.

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## EX-2016

(Fatal) '%d': '%s'

**Description**

There is an OpenAccess Exception, when StarRC is trying to create the OA view.

**What Next**

Contact the Synopsys Support Center.

---

## EX-2017

(Warning) warning: '%s'

**Description**

StarRC is unable to read the OA_LIB_DEF file.

**What Next**

Please make sure the OA_LIB_DEF file exists and you have the read permission.

---

## EX-2018

(Warning) syntax issue in '%s': '%s'

**Description**

There is an OpenAccess exception, when StarRC tries to open OA_LIB_DEF file.

**What Next**

Contact the Synopsys Support Center.

---

## EX-2019

(Warning) '%s' load warning: '%s'

**Description**

StarRC ran into a syntax error while reading the OA_LIB_DEF file.

Feedback

**What Next**

Please make sure the OA_LIB_DEF file does not have syntax errors.

---

## EX-2020

(Fatal) '%s'

**Description**

StarRC found a fatal error.

**What Next**

Contact the Synopsys Support Center.

---

## EX-2021

(Fatal) '%s'

**Description**

There is an OpenAccess exception, when StarRC tries to get mapping information for device modeling.

**What Next**

Make sure the device mapping information is correctly defined in OA_DEVICE_MAPPING_FILE or OA_SKIPCELL_MAPPING_FILE.

---

## EX-2054

(Fatal) extraction output appears to be corrupted or truncated.

**Description**

StarRC's internal circuit representation seems corrupted.

**What Next**

Check that the extraction stages completed successfully. Contact the Synopsys Support Center for further assistance.

---

## EX-2058

(Fatal) unable to determine offset for file %s, operating system error status: %s.

Feedback

**Description**

StarRC is unable to determine the offset of the specified file because of problems with the runtime environment.

**What Next**

Probable cause is due to unsupported OS. Please make sure to use OS that are supported by StarRC. If the OS is supported and issue is not clear, please contact the Synopsys Support Center.

---

## EX-2059

(ERROR) The program failed to create the OA view because of: %s

**Description**

There is an exception, when StarRC tries to get mapping information for device modeling.

**What Next**

Make sure the device mapping information is correctly defined in OA_DEVICE_MAPPING_FILE or OA_SKIPCELL_MAPPING_FILE, and layer mapping information is correctly defined in OA_LAYER_MAPPING_FILE.

---

## EX-2065

(Fatal) Unable to read data from file %s, error status %s

**Description**

StarRC is unable to read data from a file because of issues in the runtime environment.

**What Next**

Carry out remedial actions as suggested by the operating system error status. Contact your system administrator if necessary.

---

## EX-2082

(Warning) OPERATING_TEMPERATURE is ignored from corner file because TEMPERATURE_SENSITIVITY is set to NO.

**Description**

Because OPERATING_TEMPERATURE and TEMPERATURE_SENSITIVITY: NO are mutually exclusive the OPERATING_TEMPERATURE will be ignored.

Feedback

**What Next**

No action is required.

---

## EX-2085

(Fatal) The corner name %s specified in the NETLIST_CORNER_NAMES option cannot be found in the file specified by NETLIST_CORNER_FILE.

**Description**

StarRC detects inconsistency between the NETLIST_CORNER_NAMES option and the corners defined in the file specified by the NETLIST_CORNER_FILE option.

**What Next**

Ensure that all of the NETLIST_CORNER_NAMES are present in the NETLIST_CORNER_FILE.

---

## EX-2101

(Fatal) Invalid entry in command %s: %s

**Description**

Invalid entry detected for a command in xTractor options file.

**What Next**

Contact Synopsys Support Center.

---

## EX-2110

(Warning) Routing layer direction for layer %s cannot be determined. VMF will be disabled for that layer.

**Description**

The routing layer direction for layer %s cannot be determined, therefore VMF flow will be disabled for that layer.

**What Next**

If this is intentional, there is no action to take. If you want VMF to be enabled for that layer, please specify the layer direction with the direction statements in the VMF parameter file.

Feedback

## EX-2112

(Warning) Failed to openLibs because of: '%d': '%s'

**Description**

There is an OpenAccess exception, when StarRC tries to open the library.

**What Next**

Contact the Synopsys Support Center.

## EX-2114

(Warning) GATE_TO_DIFFUSION_CHANNEL_CAP table is corrupted, please check ITF or NXTGRD file

**Description**

The nxtgrd file may be corrupted with inconsistent GATE_TO_DIFFUSION_CHANNEL_CAP statements.

**What Next**

Please make sure that the nxtgrd file is valid and original.

## EX-2116

(Fatal) INTERNAL Error: problem while processing internal circuit representation.

**Description**

A internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## EX-2117

(Warning) INTERNAL Warning: Node %d_%ld is connected to port %ld but is also shorted to port %ld.

**Description**

A non-critical internal consistency check has failed. This warning message is printed for use by the Synopsys Support Center in case a program error occurs during StarRC's execution.

Feedback

**What Next**

Contact the Synopsys Support Center if StarRC terminates abnormally or other issues are found.

---

## EX-2118

(Warning) INTERNAL Warning: Node %d_%ld is connected to port %ld and also to port %ld

**Description**

A non-critical internal consistency check has failed. This warning message is printed for use by the Synopsys Support Center in case a program error occurs during StarRC's execution.

**What Next**

Contact the Synopsys Support Center if StarRC terminates abnormally or other issues are found.

---

## EX-2120

(Warning) INTERNAL Warning: Node %d_%ld is shorted to port %ld but is also shorted to port %ld

**Description**

A non-critical internal consistency check has failed. This warning message is printed for use by the Synopsys Support Center in case a program error occurs during StarRC's execution.

**What Next**

Contact the Synopsys Support Center if StarRC terminates abnormally or other issues are found.

---

## EX-2121

(Warning) INTERNAL Warning: Node %d_%ld is shorted to port %ld but is also connected to port %ld.

**Description**

A non-critical internal consistency check has failed. This warning message is printed for use by the Synopsys Support Center in case a program error occurs during StarRC's execution.

Feedback

**What Next**

Contact the Synopsys Support Center if StarRC terminates abnormally or other issues are found.

---

## EX-2125

(Warning) In NXTGRD backdoor, the fourth column in some EVWSB line is incorrect. It should be same as the first EVWS table on the same layer if it is the first sub EVWS table, or 0 if not.

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## EX-2126

(Warning) In NXTGRD file, some sub EVWS table is missing. Please use newest version of grdgenxo to regenerate the NXTGRD file.

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## EX-2127

(Warning) In NXTGRD backdoor, the fourth column in some CEVWSB line is incorrect. It should be same as the first EVWS table on the same layer if it is the first sub EVWS table, or 0 if not.

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

Feedback

## EX-2128

(Warning) In NXTGRD file, some sub CEVWS table is missing. Please use newest version of grdgenxo to regenerate the NXTGRD file.

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

## EX-2129

(Fatal) SMC via_coverage corrupted: RPV table size is different cross corners.

**Description**

When using corner-dependent VIA_COVERAGE_OPTION_FILE in SMC flow, each RPV table should have consistent number of entries across corners.

**What Next**

Correct the input files for VIA_COVERAGE_OPTION_FILE and restart.

## EX-2130

(Fatal) Only temperature_sensitivity is supported for CCP sensitivity flow.

**Description**

StarRC-CCP sensitivity extraction flow only supports temperature variations (TEMPERATURE_SENSITIVITY: YES) but not other variations.

**What Next**

Change the setup and restart. If more variations are required, please use SIMULTANEOUS_MULTI_CORNER flow.

## EX-2131

(Fatal) Failed to read global temperature used by CCP engine for temperature sensitivity calculation.

**Description**

Failed to retrieve the global temperature value.

Feedback

**What Next**

First check whether the nominal temperature or global temperature is defined in nxtgrd file. Or contact the Synopsys Support Center.

---

## EX-2132

(Fatal) Global temperature mismatch: %.1f in ITF and %.1f in CCP.

**Description**

There is a discrepancy on the global temperature values used by CCP and StarRC.

**What Next**

Check the consistency of nominal temperature and global temperature in the nxtgrd file. Or contact the Synopsys Support Center.

---

## EX-2133

(Information) ANALOG_SYMMETRIC_NET forced to YES for this process

**Description**

ANALOG_SYMMETRIC_NET option is forced to YES for this process.

**What Next**

Please check StarRC manual for the definite of ANALOG_SYMMETRIC_NET.

---

## EX-2140

(Information) Detected large percentage (%g%%) ECO nets changed.

**Description**

Large percentage of ECO nets have changed. This may affect speed of ECO extraction.

**What Next**

Please refer to section on ECO extraction in the StarRC user guide.

---

## EX-2141

(Warning) Unexpected number (%d) of polygons in in layer %d in window: (%d,%d) (%d, %d)

Feedback

**Description**

Detected a polygon that is surrounded by the window overlaps with too many other polygons. The coordinates of the window is in database units.

**What Next**

Check if the large number of overlaps are reasonable.

---

## EX-2142

(Warning) The grd file contains SCT models, but they are ignored (MULTIGATE_MODELS:NO)

**Description**

Although the nxtgrd file contains SCT models, they are ignored because MULTIGATE_MODELS is set to NO.

**What Next**

Please refer to StarRC manual on MULTIGATE_MODELS.

---

## EX-2144

(Fatal) Characterization Initialize Failed

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## EX-2145

(Fatal) Characterization Failed

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

Feedback

## EX-2147

(Warning) FLOATING_GATE_TO_DIFFUSION_CONTACT_CAP table in ITF is empty or incorrect, please check your grd file.

**Description**

The nxtgrd file may be corrupted with inconsistent FLOATING_GATE_TO_DIFFUSION_CONTACT_CAP statements.

**What Next**

Please make sure that the nxtgrd file is valid and original.

## EX-2148

(Warning) SPACER_ER_VS_WIDTH_AND_SPACING table in ITF is empty or incorrect, please check your grd file.

**Description**

The nxtgrd file may be corrupted with inconsistent SPACER_ER_VS_WIDTH_AND_SPACING statements.

**What Next**

Please make sure that the nxtgrd file is valid and original.

## EX-2149

(Fatal) FLOATING_GATE_TO_DIFFUSION_CONTACT_CAP table has too few entries for GATE_POLYGON_LENGTH. Please check your NXTGRD file.

**Description**

Statements related to FLOAT_GATE conductors in your nxtgrd file does not have enough entries for GATE_POLYGON_LENGTH. This may lead to errors in capacitance extraction.

**What Next**

Please make sure that nxtgrd files from foundry are all valid and original.

## EX-2150

(Fatal) The values in FLOATING_GATE_TO_DIFFUSION_CONTACT_CAP table are invalid. Please check your NXTGRD file.

Feedback

**Description**

Statements related to FLOAT_GATE conductor capacitance values are not valid.

**What Next**

Please make sure that nxtgrd files from foundry are all valid and original.

---

## EX-2151

(Warning) FinFET width and spacing violates design rules described at TCAD_GRD_FILE and it will lead to parasitic capacitance inaccuracy

**Description**

Width and spacing values on FinFET structures are not compatible with foundry design rules.

**What Next**

Please check FinFET layout to make sure that the design satisfies foundry rules.

---

## EX-2155

(Fatal) %s is incompatible with this version of StarRC.

**Description**

The internal circuit representation is not compatible with this version of StarRC. This might be caused by using -cleanN to skip extration with a run generated by an older version of StarRC.

**What Next**

Please rerun StarXtract to generate compatible xout files. For further assistance, please contact the Synopsys Support Center.

---

## EX-2156

(Warning) Inconsistency between StarRC and CCP extraction output files is detected on: %s.

**Description**

The extraction output files from StarRC and CCP are not compatible to be merged.

Feedback

### What Next

Check whether the extraction setups in star command file and CCP_EXTRACTION_FILE are correct. If the error does not go away, contact the Synopsys Support Center.

---

## EX-2159

(Fatal) While loading multi-corner nxtgrd files, found and inconsistent number of capacitance tables

### Description

StarRC is loading many nxtgrd files for multi-corner flow. Some capacitance tables are present in some nxtgrd files but not in another nxtgrd file.

### What Next

Ensure that the nxtgrd files have been generated using identical grdgenxo versions and identical options. If the error persists, contact the Synopsys Support Center.

---

## EX-2160

(Fatal) The process TSMC N20 V1.0 is no longer supported in StarRC versions J-2014.12 or later.

### Description

The nxtgrd file from TSMC N20 V1.0 is no longer supported in StarRC versions J-2014.12 or later.

### What Next

Please contact the foundry for an up-to-date version of nxtgrd files.

---

## EX-2161

(Fatal) INTERNAL Error: intermediate file '%s' does not exist.

### Description

An intermediate file expected to be produced by StarRC does not exist.

### What Next

It is possible that a file written by a process on one host may not be immediately visible on another host if the disk or network is busy. Try rerunning StarRC on a lightly loaded disk or network.

Contact the Synopsys Support Center if this problem persists.

Feedback

## EX-2162

(Fatal) Unable to open gzip file '%s' for writing: %s

**Description**

StarRC failed to open a gzip file for writing.

**What Next**

Make sure that you have permission to write to this directory, and that the file system is not full.

## EX-2163

(Fatal) Unable to close gzip file '%s'

**Description**

StarRC was not able to close a gzip file.

**What Next**

Check to see if the file system is full.

## EX-2164

(Fatal) Attempt to write to file '%s' failed.

**Description**

StarRC failed to write to a file.

**What Next**

Contact the Synopsys Support Center.

## EX-2165

(Fatal) Unable to write %ld bytes to gzip file '%s': %s

**Description**

StarRC failed to write to a gzip file.

**What Next**

Make sure that you have permission to write to this directory, and that the file system is not full.

Feedback

## EX-2168

(Fatal) Unable to flush buffer for file '%s': %s

**Description**

Unable to complete writing to a file.

**What Next**

Make sure that the file system is not full.

## EX-2169

(Fatal) unable to flush buffer for file '%s' after %d attempts: %s

**Description**

Unable to complete writing to a file after multiple attempts.

**What Next**

Make sure that the file system is not full.

## EX-2170

(Fatal) Unable to write %ld bytes to gzip file '%s': %s

**Description**

Failed to write to a gzip file.

**What Next**

Make sure that the file system is not full.

## EX-2171

(Warning) Found short between clock net %d and net %d; inductance not extracted for region of BBox BBox=(%g,%g),(%g,%g) layer %s

**Description**

A short found in clock net inductance extraction. Inductance calculation is skipped for the affected polygon. Accuracy of inductance for this net may be adversely affected.

**What Next**

Contact Synopsys Support Center.

Feedback

## EX-2172

(Warning) Cannot find a capacitance table that shall be in the nxtgrd file; that may cause cap underestimation.

### Description

You receive this message when StarXtract cannot retrieve a capacitance table that shall be in the nxtgrd file. This is generally due to an incorrect ITF description.

### What Next

This is only a warning message. No action is required. However, if the accuracy is not good and the ITF file is available, check any warning or error message when launching the following comand:

grdgenxo -parse <ITF>

If the issue can be fixed, relaunch the nxtgrd file generation. In other cases, contact the Synopsys Support Center.

## EX-2175

(Fatal) Fail to create file '%s' to merge shorted ports.

### Description

An intermediate file cannot created. This may be produced for a variety of reasons such as the disk being full, or more than one job running concurrently in the same run directory and one of them holding the file.

### What Next

Try running StarRC again from a clean directory. Contact the Synopsys Support Center if the error message persists.

## EX-2176

(Fatal) Detected too many shorts (%lld) in this partition. BBox=(%g,%g),(%g,%g)

### Description

The number of polygon shorts exceeds the number of polygons in this partition. The coordinates of the bounding box are in microns.

Feedback

### What Next

Check the design and remove the shorts. The excessive shorts could be due to many unplaced cells in the design. To ignore the error and proceed, use STOP_EXTRACTION_ON_NUMEROUS_SHORTS:NO.

---

## EX-2180

(Warning) Clock net polygon has no shielding polygon on a power net within %f microns. Coordinates: (%f,%f) (%f,%f), layer %d; for inductance extraction purpuses, StarRC will assume a shielding distance of %f microns";

### Description

Clock net inductance extraction expects shielding wires on power or ground nets within a maximum distance. The absence of such shielding may affect the accuracy of the inductance extractor.

### What Next

Check the design to make sure clock net polygons are shielded. Read the section on clock net inductance extraction in the StarRC user manual.

---

## EX-2190

(Fatal) Cannot output temperature coefficients of grouped corners (the selected corners concatenated with ':').

### Description

The grouped corners are the selected corners which are concatenated with ':'. StarRC cannot output temperature coefficients (TC1/TC2) for the grouped corners.

### What Next

Remove the grouped corners or separate the corners with ' '.

---

## EX-2191

(Warning) Multi-corner temperature coefficients output does not support command "NETLIST_CORNER_FILE" and command "NETLIST_CORNER_NAMES".

### Description

Multi-corner temperature coefficients output does not support command "NETLIST_CORNER_FILE" and command "NETLIST_CORNER_NAMES".

Feedback

**What Next**

Use SELECTED_CORNER to select corners to output temperature coefficients.

---

## EX-2192

(Warning) Using GLOBAL_TEMP in corner definition to output multi-corner TC1/TC2 will be obseleted in 2016.06.

**Description**

Please use SIMULTANEOUS_MULTI_CORNER: YES and TEMPERATURE_SENSITIVITY: YES to output multi-corner TC1/TC2.

---

## EX-2193

(Warning) GATE_TO_DIFFUSION_ADJUSTMENT_CAP table is corrupted, please check your ITF or grd file

**Description**

The nxtgrd file may be corrupted with inconsistent GATE_TO_DIFFUSION_ADJUSTMENT_CAP statements.

**What Next**

Please make sure that the nxtgrd file is valid and original.

---

## EX-2194

(Warning) Multiple connections on both upper and lower layer is found for via: (%d %d), (%d %d) on level %d.

**Description**

Multiple connections on both upper and lower layer is not expected for any via. In this case, it is not treated as a bridge via, but only one via resistor is extracted from center to center.

**What Next**

Modify the layout.

---

## EX-2195

(Warning) Multiple connections on both horizontal and vertical direction of a via are found at : (%d %d), (%d %d) on level %d.

Feedback

**Description**

A via with multiple connections on one direction on upper or lower layer is considered as a bridge via. Multiple connections on both horizontal and vertical direction might cause open.

**What Next**

Modify the layout.

---

## EX-2196

(Warning) Via center is not covered by any metal at: (%f %f), (%f %f)um on layer %s.

**Description**

Via node is created at the center of the via. If the center is not covered by any metal, there will be a shorting resistor connecting via node and the node on the metal.

**What Next**

Please check the original layout with -Display to confirm if the via misalignmane is reported correctly. If not, please check if via merging is performed (MERGE_VIAS_IN_ARRAY in command file or MAX_VIA_ARRRAY_LENGTH/SPACING). If root cause is still not clear, please contact Synopsys Support Center.

---

## EX-2197

(Warning) The center of via and one leg overlapping area is not covered by any metal at: (%d %d), (%d %d)nm on level %d.

**Description**

Via node is created at the center of the overlapping area of the via and one leg. If the center is not covered by any metal, there will be a shorting resistor connecting via node and the node on a nearby metal.

**What Next**

Nothing

---

## EX-2198

(Warning) Ignoring unsupported vector: (%g %g), (%g %g) on level %d.

**Description**

Current vector file supports only horizontal or vertical vectors.

Feedback

**What Next**

Fix file content.

---

## EX-2199

(Fatal) Internal error on CUT_END_EXTENSION_TABLE for conductor layer %s. nxtgrds are broken.

**Description**

Some or all of the nxtgrds are broken. Regenerate all the nxtgrds and re-run.

**What Next**

If the issue still exist after nxtgrd regeneration, contact the Synopsys Support Center.

---

## EX-2200

(Warning) The grd file contains %s models, but they are ignored

**Description**

Although the nxtgrd file contains models described in the message, they are ignored because of customor setting.

**What Next**

Please refer to StarRC manual on how to turn on the models.

---

## EX-2201

(Warning) Tall via layer %s and %s have couplings, but there is no PAIR_TO defined for them in ITF. Caps between these two layers may be extracted with inaccuracy.

**Description**

The two tall via layers described in the warning message have coupling caps in design. However, there is no PAIR_TO defined for them in TALL_VIA_CONFIG of either one of these two layers in ITF. Coupling cap accuracy is not good for these two layers.

**What Next**

Add the reported layers in PAIR_TO section of TALL_VIA_CONFIG.

Feedback

## EX-2202

(Warning) Instance port via center is not covered by any metal at: (%d %d), (%d %d)nm on level %d.

**Description**

Via node is created at the center of the via. If the center is not covered by any metal, there will be a shorting resistor connecting via node and the node on the metal.

**What Next**

Nothing

## EX-2203

(Warning) '%s'

**Description**

StarRC detect inconsist layers in ignore cap behavior.

**What Next**

Overlapping polygons' ignore cap pair behavior inconsistency is detected, which may introduce accuracy issues, user need to fix overlapping layer definition.

## EX-2204

(Warning) GATE_TO_GROUND_ADJUSTMENT_CAP table is corrupted, please check your ITF or grd file

**Description**

The nxtgrd file may be corrupted with inconsistent GATE_TO_GROUND_ADJUSTMENT_CAP statements.

**What Next**

Please make sure that the nxtgrd file is valid and original.

## EX-2205

(Warning) Thermal parameter file(s) defined by user, has failed to open, please check thermal parameter file location etc. Turns off thermal aware extraction.

Feedback

**Description**

Thermal parameter file(s) defined by user, has failed to open, please check thermal parameter file location etc. Turns off thermal aware extraction.

**What Next**

Please check thermal parameter file(s) defined in either SMC command file or user command file.

---

## EX-3005

(Fatal) For the layer %s option RPV_VS_COVERAGE can't be used for a via array.

**Description**

Option RPV_VS_COVERAGE can't be used for a via array.

**What Next**

Modify ITF definitions and regenerate nxtgrd.

---

## EX-3006

(Warning) Excessive number (%d) of overlapping polygons encountred in the input database at location: (%d,%d) DB layer %d.

**Description**

Detected a polygon that overlaps with significant number of other polygons.

**What Next**

Remove overlaps in the input database.

---

## EX-3007

(Warning) Cannot open EM_PARAM_MAPPING_FILE: %s.

**Description**

xTractor was not able to read the EM_PARAM_MAPPING_FILE.

**What Next**

Check to make sure that EM_PARAM_MAPPING_FILE exists and that the xTractor has read permissions for it.

Feedback

## EX-3008

(Warning) EM_PARAM_MAPPING_FILE syntax: %s.

**Description**

EM_PARAM_MAPPING_FILE has syntax error. The line with syntax error is ignored.

**What Next**

Check to make sure that EM_PARAM_MAPPING_FILE has correct syntax.

## EX-3009

(Warning) Model for TSV layer found multiple times in 3D_IC_SUBCKT_FILE specification. Selecting first model found in file %s, ignoring subsequent model from file %s.

**Description**

This warning is seen whenever the 3D IC model for a TSV layer is specified more than once in the input file(s) specified with the 3D_IC_SUBCKT_FILE option. StarRC selects the first model it reads from the user-specified file(s) and disregards all others, even if they appear in the same input file.

**What Next**

Remove extraneous model definitions and specify only one 3D IC model for each TSV layer.

## EX-3010

(Warning) Density grid is enlarged due to huge input layout size and small number of polygons. It may introduce errors in density calculation.

**Description**

When the number of polygon is small, the layout is extracted in a single partition. To prevent memory from being exhausted, density grid is enlarged when layout area is too big. It may introduce error in density value calculation.

**What Next**

Reduce the layout size to 10mm X 10mm for N10 and above or 2mm X 2mm for N7 and below.

Feedback

## EX-3011

(Fatal) StarRC command MOS_GATE_NON_NEGATIVE_DELTA_RESISTANCE is not supported when ITF file has VERTICAL_RESISTANCE_VS_SI_WIDTH_AND_LENGTH table.

**Description**

StarRC command MOS_GATE_NON_NEGATIVE_DELTA_RESISTANCE is not supported when ITF file has VERTICAL_RESISTANCE_VS_SI_WIDTH_AND_LENGTH table.

**What Next**

Remove MOS_GATE_NON_NEGATIVE_DELTA_RESISTANCE from the StarRC command file.

## EX-3012

(Fatal) The maximum VIA_INDEX in layer_info file (%d) is bigger than that in nxtgrd file (%d), due to different version of StarXtract and grngenxo.

**Description**

Using different version of StarXtract and grngenxo may cause different number of via layers in layer_info file and nxtgrd file, which may cause program crash.

**What Next**

Regenerate nxtgrd file with newer version grdgenxo.

## EX-3013

(Warning) %d violation are present for net %d!

**Description**

One or more of the following netlist integrity violation are present for this net. 1. NodeID Bound - some nodes are beyond nodeCount for the net. 2. Holes in node Id - For res nodes, we should have all the nodes from 0 to nodeCount -1 but we have some of the missing from the list. 3. Negative parasitics - some cap/res values are negative. 4. Netlist Connectivity (open count) - there are open nets present in the final netlist.

**What Next**

These could be design problems. Ensure that your setting is correct. If problme still persists, contact StarRC.

Feedback

## EX-3014

(Warning) IC Compiler II InDesign StarRC's %s option is invoked. It is not meant for final signoff flow because aggregate scaling may not represent the actual process corner characteristics of a technology node.

**Description**

StarRC scales parasitic values according to provided coefficients.

**What Next**

Please make sure this option is not used during sign-off.

## EX-3015

(Warning) IC Compiler II InDesign StarRC's %s option is invoked.

**Description**

StarRC provides the details of the parasitics of the routed sections of the specified nets while ICC2 native extraction estimates the parasitics for the open parts of the network.

**What Next**

Please make sure this option is not used during sign-off.

## EX-3016

(Warning) Trench contact polygon at (%d %d %d %d) on layer %d abut two other polygons with trench contact extension.

**Description**

StarRC only support trench contact extension from one neighbor.

## EX-3017

(Fatal) Aborting extraction on core %d because of an error during extraction on another core.

**Description**

If an error occurs during extraction on one core, extraction on all other cores terminates with this error message.

Feedback

### What Next

See the summary file for information about the nature of the error on the first failing core.

---

## EX-3018

(Warning) Resistor width smaller than WMIN are snapped to WMIN.

### Description

Resistor width smaller than WMIN are snapped to WMIN in the netlist tail comments. Please refer to snap_resistor_width file in the star directory for pairs of nodes of the resistance with width snapped.

### What Next

Nothing is required at your side at the moment.

---

## EX-3019

(Information) [STAR_MAIN_SUMMARY] Extracting %s with bounding box left: %.2f bottom: %.2f right: %.2f top: %.2f

### Description

The displayed coordinates represent values for the lower-left and upper-right corners of the layout being extracted by a particular extraction job in a StarRC run. The ID of the partition being extracted is also listed if StarRC has internally divided the layout into multiple partitions.

### What Next

This is an informational message only. Remove the PRINT_EXTRACTION_BBOX option from the StarRC command file or set its value to NO to prevent this message from appearing in the summary file.

---

## EX-3020

(Information) Extrapolation on RPV_VS_AREA table for via connecting a TRENCH_CONTACT, FIELD_POLY or GATE layer at bottom is handled differently in TSMC N5 and below.

### Description

For via connecting a TRENCH_CONTACT, FIELD_POLY or GATE layer at bottom, extrapolation on RPV_VS_AREA table is handled differently in TSMC N5 and below comparing to other nodes. For TSMC N5 and below, the tool determines the conductance

Feedback

of the via with the smallest or biggest area entry and uses that value to calculate the via resistance for any vias with smaller or bigger areas. For other nodes, if via area is outside the bounds of area entry in the RPV_VS_AREA table, the tool uses the RPV value according to the smallest or biggest area.

**What Next**

This is an informational message only. If you do not want to extrapolate RPV_VS_AREA table on a perticular layer, you can specify EXTRAPOLATE_NO on that table.

---

## EX-3021

(Warning) Process %d for extracting partition %d exited after receiving signal %d

**Description**

Sometimes a StarRC run exits without an error messages in the summary file if the processes used for extraction are delivered an uncaught signal. This message is output to help identify the root cause of such failures.

---

## EX-3022

(Fatal) RCAPI_VERSION RCAPI_MODEL_VERSION or RCAPI_PARAM_FILE is missing in the nxtgrd file.

**Description**

RCAPI_VERSION RCAPI_MODEL_VERSION and RCAPI_PARAM_FILE must be specified together with RCAPI_LIBRARY_NAME in the ITF header and stored in nxtgrd file.

**What Next**

specify RCAPI_VERSION RCAPI_MODEL_VERSION and RCAPI_PARAM_FILE together with RCAPI_LIBRARY_NAME in the ITF header and regenerate nxtgrd file.

---

## EX-3023

(Warning) There are two nets with different net id has the same netname net1: %s net_id: %ld, net2: %s net_id: %ld. if they are different actually, please set CASE_SENSITIVE: YES.

**Description**

if case_sensitive:no, we think net1 and NET2 are the same.

Feedback

**What Next**

Set CASE_SENSITIVE:YES

---

## EX-3024

(Fatal) TMIRC initialization failed! Please check if PARAM_FILE specified in %s exist, and if proper version (%s) of library is specified by TMIRC_LIBRARY_FILE.

**Description**

TMIRC initialization function failed. Possible reasons include not limited to: missing file or wrong path of files in the TMIRC configure file specified by TMIRC_CONFIG_FILE; wrong version of library specified by TMIRC_LIBRARY_FILE.

**What Next**

Make sure to use proper version of TMIRC_LIBRARY_FILE is used, all files specified in TMIRC_CONFIG_FILE exist. If problem persist, please contact TSMC for TMIRC initialize issue.

---

## EX-3025

(Fatal) TMIRC gate resistance extraction function failed.

**Description**

StarRC calls a function in TMIRC libary to extract resistance on gate. The function failed.

**What Next**

contact Synopsys for support.

---

## EX-3026

(Fatal) TMIRC capacitance extraction function failed.

**Description**

StarRC calls a function in TMIRC libary to extract capacitance around gate and diffusion. The function failed.

**What Next**

contact Synopsys for support.

Feedback

## EX-3027

(Fatal) TMIRC failed. %s.

**Description**

There is an issue in TMIRC library.

**What Next**

get updated library from foundry.

## EX-3028

(Fatal) Failed to load shared object: %s

**Description**

There is an issue loaing shared object.

**What Next**

Contact the Synopsys Support Center.

## EX-3030

(Fatal) corrupted TCNR simulation result file.

**Description**

TCNR simulation result file should include three columns: instancer_name TCNR_source and TCNR_drain. The TCNR simulation result of this test case is corrupted.

**What Next**

Contact Synopsys Inc.

## EX-3031

(Fatal) The number of tcnr value in TCNR simulation result is wrong. Please check if proper version of hspice is specified in TCNR_SIMULATION_SETUP_FILE.

**Description**

hspice version Q-2020.03-SP1-2 or later is needed to support TCNR SMC.

**What Next**

set hspice version Q-2020.03-SP1-2 or later in TCNR_SIMULATION_SETUP_FILE.

Feedback

## EX-3032

(Warning) The RPSQ is set to 0.001 on superconductive on layer %s for TCNR extraction.

**Description**

When TCNR is on, diffusion layer needs to be conductive to be extracted.

**What Next**

Check RPSQ setting in nxtgrd and mapping files. Set RPSQ on diffusion layer not smaller than 0.001.

## EX-3033

(Fatal) Unexpected connection between trench contact and RVTV at (%f %f), (%f %f).

**Description**

All trench contact should connect RVTV from one side of RVTV.

**What Next**

Check the layout.

## EX-3034

(Fatal) There is no trench contact connection on RVTV layer at (%d %d), (%d %d).

**Description**

Every RVTV shape need to have 1 and more trench contact connections.

**What Next**

Check the layout.

## EX-3035

(Warning) no resistor was added between the bulk layer and a pin of net '%s' for NETLIST_BULK_TO_PIN: yes because the net did not have any pins.

**Description**

The NETLIST_BULK_TO_PIN option can only work on nets that have pins. The option has no effect for nets that do not have any pins.

Feedback

**What Next**

Check your layout for possible errors. Remove NETLIST_BULK_TO_PIN: yes from the StarRC command file to prevent this warning from being generated.

---

## EX-3036

(Fatal) INTERNAL Error: ML model file '%s' does not exist.

**Description**

ML model is expected to be provided with StarRC installation.

**What Next**

Contact the Synopsys Support Center if this problem persists.

---

## EX-3037

(Fatal) 3DIC: syntax error when parsing 3C RLC model.

**Description**

StarRC only support the specific 3C RLC models.

**What Next**

Check the model defined in 3D_IC_SUBCKT_FILE.

---

## EX-3039

(Fatal) Density vaule of Conductor Layer '%s' in density layers file '%s' is not between 0 and 1

**Description**

Density value should be in 0~1

**What Next**

Check the density value defined in density layers file.

---

## EX-3040

(Fatal) Conductor Layer '%s' in density layers file '%s' is not DB layer.

**Description**

StarRC only support DB layer in density layers file.

Feedback

**What Next**

Check the conductor layer defined in density layers file.

---

## EX-3041

(Warning) Missing metal on layer %s for virtual via at (%d %d %d %d)

**Description**

Virtual via needs to be covered by metals on both top and bottom sides.

**What Next**

Check your layout for possible errors.

---

## EX-3045

(Fatal) corrupt grd file, duplicate mask pair in %s table.

**Description**

One mask pair can only be specified once in the same table in the ITF file. Duplicate mask pair is found during parsing nxtgrd file in extraction.

**What Next**

Contact foundry for a regenerated nxtgrd file.

---

## EX-3046

(Fatal) corrupt grd file, missing mask pair in %s table.

**Description**

All mask pairs must be specified the same table if NUMBER_OF_MASKS is specified in the ITF file. Missing mask pair is found during parsing nxtgrd file in extraction.

**What Next**

Contact foundry for a regenerated nxtgrd file.

---

# EXC

---

## EXC-0001

(Error) An exception has conflicting rise_or_fall and to_rise_or_fall options specified.

Feedback

### Description

The specified endpoint cannot simultaneously be both rising and falling. The -rise/-fall options refer to a rising/falling value at the end point. As a result, the -rise_to option conflicts with the -fall option and vice versa. The exception will be ignored. The exception will be ignored because of ambiguity due to the conflicting options. Ignored exceptions might result in undesired analysis of the design, resulting in larger optimization or timing closure effort. Multicycle paths in the design might be analyzed with the wrong timing, or timing violations might be reported for false paths. Maximum or minimum path delays will be checked against the library specified setup or hold time of the capturing clock.

### ONE POSSIBLE SCENARIO

Consider a scenario where a constraint is specified with a conflicting -rise and -fall_to option:

set_multicycle_path 2 -rise -from [get_pins ff1/CP] \ -fall_to [get_pins ff2/D]

This exception has conflicting values specified for the ff2/D endpoint. The -fall_to option specifies a falling edge at this point, and the -rise option specifies a rising edge at this point. Because both transitions cannot occur simultaneously, the exception will be ignored, and an EXC_0001 rule violation will be issued.

### What Next

The "Violation Details" section in the GUI contains the design objects in the exception and the location of the constraint conflicts.

### FIX SUGGESTION

Revise the exception to remove the inconsistency. If the exception is not revised, it will be ignored. For the following example, the exception could be applied to both transitions at the endpoint, the -rise/-fall or -rise_to/-fall_to options need to be removed.

The following example shows possible ways to resolve the conflict:

If the required analysis at the endpoint is for a rising edge, it can be rewritten in one of two ways:

set_multicycle_path 2 -rise -from [get_pins ff1/CP] \ -to [get_pins ff2/D]

Or

set_multicycle_path 2 -from [get_pins ff1/CP] \ -rise_to [get_pins ff2/D]

Similarly, if the required analysis at the endpoint is for a falling edge, it can be rewritten in one of two ways:

set_multicycle_path 2 -fall -from [get_pins ff1/CP] \ -to [get_pins ff2/D]

Or

Feedback

```
set_multicycle_path 2 -from [get_pins ff1/CP] \ -fall_to [get_pins ff2/D]
```

**See Also**

set_false_path set_multicycle_path set_min_delay set_max_delay

---

## EXC-0002

(Warning) An exception has both valid and invalid startpoints and endpoints specified.

**Description**

The *-from* objects are not valid timing path startpoints, and the *-to* objects are not valid timing path endpoints.

Valid startpoints are:

Input or bidirectional ports

Register clock pins

Transparent latch D-pins

Internal pins that have a *set_input_delay* constraint set

Internal pins that have a set_min_delay or set_max_delay constraint set with the *-from*, *-rise_from*, or *-fall_from* option

A clock source

Valid endpoints are:

Output or bidirectional ports

Pins constrained by a setup/hold/recovery/removal timing check including:

Register data pins

Register preset and clear pins

Clock gating enable pins

Internal pins that have a set_output_delay constraint set

Internal pins that have a set_min_delay/set_max_delay constraint set with the -to, -rise_to, or -fall_to option

The "Violation Details" section of the GUI shows the -from or -to objects that are invalid. The reasons are as follows:

Non-startpoint – The pin is internal and is not a timing path startpoint

Non-endpoint – The pin is internal and is not a timing path endpoint

Feedback

Endpoint – The pin or port is used as a -from object and is a timing path endpoint

Startpoint – The pin or port is used as a -to object and is a timing path startpoint

Case disabled – The pin or port has been disabled by set_case_analysis

Check disabled – Timing check at the pin or port has been disabled

Sequential disabled – A sequential arc at the timing path startpoint is disabled

Node disabled – A valid endpoint or startpoint is disabled

Hierarchical pin – The pin is on a hierarchical boundary

Not a start cell – The cell used as a -from object does not have any valid timing path startpoints

Not an end cell – The cell used as a -to object does not have any valid timing path endpoints

Only false path (set_false_path) and multicycle path (set_multicycle_path) exceptions are included in this check.

The invalid startpoints and endpoints will be ignored, and the exception will take effect only on the valid timing points. The result is poor analysis or relaxed timing requirements being applied to the design.

**ONE POSSIBLE SCENARIO**

Consider a scenario where wildcards are used to specify that all paths that end at register pins with the name ‘D’ in a design are false:

set_false_path -to [get_pins */D]

The name ‘D’ could also be a pin name for other gates in the library that may not be valid timing path endpoints. In this scenario, the false path exception is not applied to ‘gate/D’ because this is a combinational cell, and it is not a valid endpoint. An EXC_0002 rule violation will be issued.

**FIX SUGGESTION**

Revise the exceptions to specify only valid startpoints and endpoints.

Exceptions containing wildcard may lead to unintended false paths or multicycle paths being applied to the design. For example, false paths originally intended only for D-type flip-flops may be applied more broadly when new sequential elements are added to the design.

For the preceding example scenario, the false path can be specified as:

set_false_path -from ff1/D

Feedback

**See Also**

set_false_path set_multicycle_path

---

## EXC-0003

(Warning) All -from objects for an exception are not valid path startpoints.

**Description**

All of the *-from* objects are not valid path startpoints. This exception will be ignored.

Valid startpoints are:

Input or bidirectional ports

Register clock pins

Transparent latch D-pins

Internal pins that have a set_input_delay constraint set

Internal pins that have a set_min_delay/set_max_delay constraint set with the -from, -rise_from, or -fall_from option

A clock source

The "Violation Details" section in the GUI shows the -from objects that are not valid. The reasons are as follows:

Non-startpoint – The pin is internal and is not a timing path startpoint

Endpoint – The pin or port is used as a -from object and is a timing path endpoint

Case disabled – The pin or port has been disabled by set_case_analysis

Check disabled – Timing check at the pin or port has been disabled

Sequential disabled – A sequential arc at the timing path startpoint is disabled

Node disabled – A valid startpoint is disabled

Hierarchical pin – The pin is on a hierarchical boundary

Non-start cell – The cell used as a -from object does not have any valid timing path startpoints

Only false path (set_false_path) and multicycle path (set_multicycle_path) exceptions are included in this check.

The exception will be completely ignored. Ignored exceptions can lead to larger optimization or timing closure effort.

Feedback

**ONE POSSIBLE SCENARIO**

Consider the following command used to disable a path between two registers:

set_false_path -from [get_pins ff1/Q] -to [get_pins ff2/D]

Even though the -to objects are correct, this exception is entirely ignored because the -from object is invalid, making the path specification ambiguous. The EXC_0003 rule violation will be issued.

**FIX SUGGESTION**

Revise the exceptions to specify only valid startpoints.

For the example, set the false path to specify the clock pin of the register as the startpoint:

set_false_path –from [get_pins ff1/CP] –to [get_pins ff1/D]

**See Also**

set_false_path set_multicycle_path

---

## EXC-0004

(Warning) All -to objects for an exception are invalid timing path endpoints.

**Description**

All of the *-to* objects are not valid path endpoints. This exception will be ignored.

Valid endpoints are

- Output or bidirectional ports
- Pins constrained by a setup/hold/recovery/removal timing check including:
- Register data pins
- Register preset and clear pins
- Clock gating enable pins
- Internal pins that have a set_output_delay constraint set
- Internal pins that have a set_min_delay/set_max_delay constraint set with the -to, -rise_to, or -fall_to option

Feedback

The "Violation Details" section in the GUI shows the -to objects that are invalid. The reasons are as follows:

- Non-endpoint – The pin is internal and is not a timing path endpoint
- Startpoint – The pin or port is used as a -to object but is a timing path startpoint
- Case disabled – The pin or port has been disabled by set_case_analysis
- Check disabled – The timing check at the pin or port has been disabled
- Sequential disabled – A sequential arc at the timing path startpoint is disabled
- Node disabled – A valid endpoint is disabled
- Hierarchical pin – The pin is on a hierarchical boundary
- Not an end cell – The cell is used as a -to object and does not have any valid timing path endpoints

Only false path (set_false_path) and multi-cycle path (set_multicycle_path) exceptions are included in this check.

The exception will be ignored, resulting in timing violations due to false paths or stricter timing requirements. Ignored exceptions can lead to larger optimization or timing closure effort.

**ONE POSSIBLE SCENARIO**

Consider a scenario where a path is disabled between two registers with the constraint:

create_clock -period 10 [get_ports CLK] set_false_path -from [get_pins ff1/CP] -to [get_pins inv/Z]

Even though the -from object is correct, this exception will be ignored because the -to object is invalid, which makes the path specification ambiguous. As a result, the EXC_0004 rule violation will be issued.

**ANOTHER POSSIBLE SCENARIO**

Consider a scenario where the correct startpoint and endpoint are specified for disabling a path between two registers with the following constraints:

create_clock -period 10 [get_ports CLK] set_case_analysis 0 [get_pins ff_end2/CP] set_false_path -from [get_pins ff_start1/CP] -to [get_pins ff_end2/D]

Even though both the -from and the -to objects are correct, this exception will be ignored. The reason is that the timing checks on the -to object are disabled by a set_case_analysis constraint on the ff_end2 register’s clock pin CP. The exception cannot be applied, and an EXC_0004 rule violation will be issued.

Feedback

**What Next**

The Violation Details section in the GUI shows the reason the -to objects are invalid.

For the first example scenario, the Violation Details section provides the following information: Figure 3: First Scenario Violation Details

The Reason, non-endpoint, is included in the table, which shows that the specified -to point is not a valid endpoint, and the set_false_path constraint was ignored.

For the second scenario, the “Violation Details” section provides the following information: Figure 4: Second Scenario Violation Details

The Reason, check disabled, is highlighted in the table, which means that the timing check at the specified -to point is disabled, and the set_false_path constraint was ignored.

To understand why the timing check is disabled at the -to point, use report_disable_timing -all_arcs on the -to cell, which is the endpoint specified in the constraint. This command may be useful for the case disabled, check disabled, sequential disabled, and node disabled Reason provided in the “Violation Details” section. For the example scenario, the command and its report are shown here:

rtl_shell> report_disable_timing -all_arcs [get_cells ff_end2]

**********
Report : disable_timing
Design : …
**********

Attributes
c - case-analysis
C - Conditional arc
d - default conditional arc
f - false net-arc
l - loop breaking
L - db inherited loopbreaking
m - mode
p - propagated constant
u - user-defined
U - User-defined library arcs

Cell or Port From To Sense Flag Reason
---------- ff_end2 CP QN rising_edge c CP = 0
ff_end2 CP Q rising_edge c CP = 0
ff_end2 CP CP clock_pulse_width_high c CP
ff_end2 CP CP clock_pulse_width_low c CP
ff_end2 CP D setup_clk_rise c CP = 0
ff_end2 CP D hold_clk_rise c CP = 0

The report shows that all the timing checks on the ff_end2 register were disabled due to a case analysis condition (flag column) that sets the CP pin on ff_end2 to 0 (reason column).

**FIX SUGGESTION**

It is best to revise the exceptions to specify only valid endpoints.

For the first example scenario, the correct way to specify the false path is to specify the data pin of the register as the endpoint:

set_false_path -from [get_pins ff1/CP] -to [get_pins ff2/D]

Feedback

For the second example scenario, you should verify the set_case_analysis constraint. If it is correct, the false path specification definition will need to be revisited. If it is incorrect, the case analysis command can be removed from the constraints.

**See Also**

set_false_path set_multicycle_path report_disable_timing set_case_analysis

---

## EXC-0005

(Error) An exception will create startpoints or endpoints and break other timing paths through those pins.

**Description**

A maximum or minimum delay exception forces the path startpoints or endpoints, because the -from or -to values are not typical timing path. This will break other paths from passing through these points. The ‘Violation Details’ located in the GUI shows the -from or -to pins that have been forced to be startpoints or endpoints. This exception will disable all other paths that would have passed through these points.

**ONE POSSIBLE SCENARIO**

Consider a scenario where a maximum delay exception is set on a timing path as follows:

set_max_delay 2 -from [get_pins ff_start/Q] -to [get_pins ff_end1/D]

The constraint specifies the ‘Q’ pin of a register which is not a valid timing startpoint. Any other paths that pass through this pin will be disabled. The path from ff_start to ff_end2 will be disabled because a new timing startpoint is created at ff_start/Q. In addition, the path from ff_start/CP to ff_end1 is blocked by the startpoint ff_start/Q. This exception will cause a EXC_0005 violation to be issued. Figure 1: Invalid Timing Startpoint

**FIX SUGGESTION**

Revise the exceptions to specify only valid startpoints and endpoints.

For the preceding example, the command specifies the maximum delay for the path by using the clock pin of the register, ff_start/CP, as the startpoint:

set_max_delay 2 -from [get_pins ff_start/CP] -to [get_pins ff_end1/D]

**See Also**

set_max_delay set_min_delay

---

## EXC-0007

(Warning) An exception has incomplete values.

Feedback

### Description

Both rise and fall values should be specified for maximum delay and minimum delay exceptions.

This check is performed only with the -rise and -fall options. There is no check if the specification is incomplete or if the -rise_from, -fall_from, -rise_to, or -fall_to options are used in the maximum or minimum delay exception.

The paths with the missing transition are checked if they meet the library specified setup or hold time. This may be an undesired analysis for unspecified transition.

### ONE POSSIBLE SCENARIO

Consider a scenario where a set_max_delay exception is specified with the -rise flag, but no constraint is included for the -fall flag:

set_max_delay -rise 8 -from [get_pins ff1/CP] -to [get_pins ff2/D]

The maximum delay exception forces a startpoint and endpoint for the rising path delays. Since no maximum delay constraint is set for the falling path; the path will be checked against the setup constraint for the capturing clock of register, ff2. In cases where a maximum or minimum delay constraint is specified for one transition and not the other, an EXC_0007 rule violation will be issued.

### FIX SUGGESTION

Provide the minimum and maximum delays, and rise and fall values for an exception.

For the preceding example, there are two ways the violation can be removed:

If different fall and rise constraints need to be set, then add another set_max_delay constraint with the -fall flag:

set_max_delay -rise 8 -from [get_pins ff1/CP] -to [get_pins ff2/D] set_max_delay -fall 9 -from [get_pins ff1/CP] -to [get_pins ff2/D]

If the fall constraint is the same as the rise constraint, then the -rise flag can be removed from the existing constraint:

set_max_delay 8 -from [get_pins ff1/CP] -to [get_pins ff2/D]

Or a -fall constraint can be added with the same delay as the rise constraint:

set_max_delay -rise 8 -from [get_pins ff1/CP] -to [get_pins ff2/D] set_max_delay -fall 8 -from [get_pins ff1/CP] -to [get_pins ff2/D]

### See Also

set_min_delay set_max_delay

Feedback

## EXC-0008

(Error) An exception is forcing startpoints or endpoints by breaking the clock network.

### Description

A maximum or minimum delay exception is forcing startpoints or endpoints. The -from or -to options specified are not timing startpoints or endpoints causing a break at the clock network. The ‘Violation Details’ section located in the GUI lists the pins that have been affected. This exception will break the clock network. An incomplete clock network may result in inaccurate timing analysis on the design.

### ONE POSSIBLE SCENARIO

Consider a scenario where a maximum delay constraint is set on an invalid startpoint:

set_max_delay 8 -from [get_pins inv/Z] -to [get_pins ff1/CP]

The inverter, inv/Z pin is not a valid timing startpoint; the set_max_delay command will create a startpoint at the specified pin and prevent the clock from propagating any further. This exception will cause an EXC_0008 rule violation to be issued. Figure 1: Invalid Timing Startpoint Preventing Clock Propagation

### FIX SUGGESTION

Revise the exceptions to specify only valid startpoints and endpoints.

For the preceding example, specify the maximum delay for the path using the clock source as the startpoint. Also, check your clock network to see if the maximum or minimum delay constraint can be removed. The example command is shown:

set_max_delay 8 –from [get_ports CLK] –to [get_pins ff1/CP]

### See Also

set_max_delay set_min_delay

## EXC-0009

(Warning) Inconsistent exception; min_delay must be less than max_delay.

### Description

Values for set_max_delay and set_min_delay must be consistent. The min rise delay must be less than the max rise delay. Similarly, the min fall delay must be less than the max fall delay.

Setup analysis on a design uses the worst-case operating condition, which produces the largest delay, to ensure that the setup time for the slowest path to each endpoint in the design is met. Similarly, hold analysis uses the best-case operating condition, which

Feedback

produces smallest delay, to ensure that the hold time for the fastest path to each endpoint in the design is met. It is expected that the max delay will always be larger than the min delay. If the min delay value is not smaller than the max delay, the min (hold) and max (setup) analysis will be performed on possibly incorrect values, resulting in optimistic timing.

**ONE POSSIBLE SCENARIO**

Consider a scenario where a max delay and min delay constraint are set on the same startpoint and endpoint, and the min delay is larger than the max delay:

set_max_delay 2 -from [get_pins ff1/CP] -to [get_pins ff2/D] set_min_delay 6 -from [get_pins ff1/CP] -to [get_pins ff2/D]

The min delay is set to 6 library timing units, and the max delay is set to 2 library timing units. Because the min delay is larger than the max delay, this will cause an EXC_0009 rule violation to be issued.

**What Next**

The ‘Violation Details’ section will give you the locations of the related set_max_delay and set_min_delay constraints with the problem.

**FIX SUGGESTION**

It is best to revise the exceptions with appropriate values for correct timing analysis. For the example scenario, the violation can be resolved by updating the constraints so that the max delay value is larger than the min delay:

set_max_delay 3 -from [get_pins ff1/CP] -to [get_pins ff2/D] set_min_delay 2 -from [get_pins ff1/CP] -to [get_pins ff2/D]

Make sure the values used make sense for your design.

**See Also**

set_max_delay set_min_delay

---

## EXC-0010

(Info) An exception has no corresponding min_delay for a max_delay exception.

A path constrained with a set_max_delay is also expected to have a corresponding set_min_delay exception. If the path specified in the from/through/to region of the set_max_delay is constrained through other means, then there is no risk associated with this violation. However, if the path is not constrained with a library check or a set_min_delay then the minimum timing on this path will not be checked.

Feedback

The purpose of this violation is to raise the awareness that you could have potentially missed specifying a constraint.

**ONE POSSIBLE SCENARIO**

Consider a scenario where a user sets a max delay constrain on a combinational path of the design.

set_max_delay 8 -from [get_ports in] -to [get_ports out]

Figure 1: Unconstrained Minimum Delay

Because this design has a purely combinational path and you did not specify input and output delays, its minimum delay is unconstrained. As a consequence, that minimum delay is never checked during timing analysis and could results in unexpected timing analysis.

Constraint consistency flags an EXC_0010 rule violation to make sure that the missing set_min_delay does not cause some paths to go unchecked by analysis.

**FIX SUGGESTION**

Specifying a set_min_delay constraint for the from/through/to corresponding to the set_max_delay constraint would eliminate this rule violation.

**See Also**

set_max_delay set_min_delay

---

## EXC-0011

(Info) An exception has no corresponding max_delay for a min_delay exception.

**Description**

A path constrained by set_min_delay is also expected to have a corresponding set_max_delay exception.

If the path specified in the from/through/to region of the set_min_delay is constrained through other means, then there is no risk associated with the EXC_0011 rule violation. However, if the path is not constrained with a library check or a set_max_delay then the maximum timing on this path will not be checked.

The purpose of this violation is to raise awareness of a potentially missing constraint in the design.

**ONE POSSIBLE SCENARIO**

Consider a scenario where a user sets a min delay constrain on a combinational path of the design.

set_min_delay 2 -from [get_ports in] -to [get_ports out]

Feedback

Because this design has a purely combinational path and you have not specified input and output delays, its maximum delay is unconstrained. As a consequence, the maximum delay is never checked during timing analysis. Constraint consistency flags an EXC_0011 rule violation to make sure that the missing set_max_delay does not cause some paths to go unchecked by analysis.

**What Next**

The "Violation Details" section also reports what constraints are currently applied on that path and what is missing.

**FIX SUGGESTION**

Specifying a set_max_delay constraint for the from/through/to corresponding to the set_min_delay constraint would eliminate this rule violation.

**See Also**

set_max_delay set_min_delay

---

## EXC-0012

(Info) An exception has incomplete values.

*Status*: Disabled by default

**Description**

An incomplete false path exception was specified. In most cases all combinations of rise/fall and setup/hold should be false if any of them are false. The violation detail section highlights which value is missing. If the current exception truly captures the intent of the designer and that not all the rise/fall and setup/hold are false for the specified path, this violation can be safely ignored. However, it is not a common practice that the user would want the path to be false only for a subset of those.

**FIX SUGGESTION**

It is best to re-express the false path exception for all the *rise/fall* and *setup/hold* combinations.

The current set of exceptions can be reported with the *report_exceptions* command:

rtl_shell> report_exceptions -from [get_pins {ff1/CP}] -to [get_pins {ff2/D}]

****************************************** Report : exceptions Design : EXC_0012 Scenario: default Version: … Date : … ******************************************

# ./src/scripts/constraints.tcl, line 3 set_false_path -setup -rise -from [get_pins {ff1/CP}] -to [get_pins {ff2/D}]

Feedback

**See Also**

set_false_path report_exceptions

---

## EXC-0013

(Info) An exception has incomplete values.

*Status*: Disabled by default

**Description**

An incomplete multicycle path exception was specified. In most cases all combinations of rise/fall and setup/hold should be specified if any of them are multicycle. The violation detail section will highlight which combination value is missing.

If the current exception truly captures the intent of the designer and that not all combinations of the rise/fall and setup/hold have a multicycle condition, this violation can be safely ignored. However, it is not a common practice that the user would want the path to have a multicycle value only for a subset of those.

**FIX SUGGESTION**

It is best to re-express the multicycle path exception for all the rise/fall and setup/hold combinations.

The currently set exceptions can be reported using the report_exceptions command:

rtl_shell> report_exceptions -from [get_pins {ff1/CP}] -to [get_pins {ff2/D}]

****************************************** Report : exceptions Design : EXC_0013 Scenario: default Version: ... Date : ... ******************************************

########################################################################## ## Exceptions that are dominant for at least one path: # ./src/scripts/constraints.tcl, line 3 set_multicycle_path 2 -setup -rise -from [get_pins {ff1/CP}] -to [get_pins {ff2/D}]

**See Also**

set_multicycle_path report_exceptions

---

# EXD

---

## EXD-0008

(Warning) The input_or_output delay at pin object_name is forcing a start_or_end point that blocks paths through this pin.

Feedback

**Description**

Use the set_input_delay or set_output_delay command to set the input or output delay on the reported pin location. This forces the reported pin to be either a timing startpoint or a timing endpoint. All paths through this pin are now blocked.

In the event no valid paths are going through this pin, this warning will not be issued.

Paths which should be crossing this pin will now be segmented. The path delays are now calculated based on the set_input_delay or set_output_delay command specifications. This calculation might result in incorrect timing analysis of the path.

**ONE POSSIBLE SCENARIO**

Consider a scenario shown in the following figure. where a clock is created on port CLK, while input delay is specified with respect to clock CLK on pin U1/Z:

create_clock -period 10 [get_ports CLK] set_input_delay -clock CLK 1 [get_ports A] set_input_delay -clock CLK 1 [get_ports B] set_input_delay -clock CLK 3 [get_pins U1/Z]

The set_input_delay command breaks any timing paths going through pin U1/Z as shown in Figure 1. The pin, U1/Z , becomes a startpoint for the rest of the design. Any path originating at pin U1/Z will now use “3” as the input delay rather than the actual propagated delay between ports “A” or “B” and pin U1/Z.

**What Next**

The “Violation Details” section located in the GUI reports the pin that is forced to become a path startpoint due to the input delay specification.

**FIX SUGGESTION**

Remove the set_input_delay command specification and use the propagated delay instead.

# FAE

## FAE-1001

(error) set_register_replication is not enabled during FM eco mode.

**Description**

The command is not enabled during formality eco mode, which is enabled using set_fm_eco_mode

Feedback

**What Next**

Command is not enabled in this mode so either exit the FM eco mode or drop the command.

---

## FAE-1002

(error) Error encountered during execution of region file '%s' in FM eco mode.

**Description**

Error encountered while executing region file which is being done during set_top_module in fm eco mode.

**What Next**

Verify the contents of region file.

---

## FAE-1003

(warning) It is already in FM eco mode.

**Description**

Tool is alredy set in formality eco mode. Check for correct region file.

**What Next**

Verify the region file path.

---

## FAE-1004

(warning) This command does not seem to be the first command. Execute elaborate, set_top_module, compile on design.

**Description**

set_fm_eco_mode keeps the tool in fm eco mode which does special actions in these commands so ideally set_fm_eco_mode should be the first command in the session. It seems that it is not the first command executed in current session so make sure to execute the impacted commands.

**What Next**

Either restart the session and execute the command again or start from elaborate again.

Feedback

## FAE-1005

(error) Error encountered during %s in FM eco mode.

**Description**

Error encountered during particular operation.

**What Next**

check the configuration.

## FAE-1006

(error) Verification checkpoints can't be enabled during FM eco mode.

**Description**

Verification checkpoint can't be enabled during formality eco mode, which is enabled using set_fm_eco_mode.

**What Next**

This operation is not enabled so either exit the FM eco mode or drop the command.

## FAE-1007

(error) Region file %s is not in correct format.

**Description**

Region file specified is not in correct encrypted formated.

**What Next**

Provide correct Region file.

## FAE-1008

(error) Required option '%s' for Targeted Synthesis is missing.

**Description**

Required option for Targeted Synthesis is missing.

**What Next**

Provide the required option to run Targeted Synthesis.

Feedback

## FAE-1009

(error) Invalid format of LLIB FRD file '%s'.

**Description**

The FRD file specified is not in the correct format.

**What Next**

Provide the FRD file in the correct format.

## FAE-1010

(error) Error encountered during loading ONET '%s' in TS flow.

**Description**

Encountered an error when loading the ONET (original netlist from full synthesis) in Targeted Synthesis flow.

**What Next**

Review the setup and the original netlist (ONET) provided as input to Targeted Synthesis flow.

## FAE-1011

(error) Error encountered during executing step '%s' in TS flow.

**Description**

Encountered an error when executing steps in Targeted Synthesis flow. Please review the TS setup before proceeding.

**What Next**

Review the setup and the options provided as input to Targeted Synthesis flow.

## FAE-1012

(error) FRD file '%s' does not include hybrid data file.

**Description**

The FRD file does not include the hybrid data file required for Targetes Synthesis flow.

Feedback

**What Next**

Provide a valid FRD file with region and hybrid data required for Targeted Synthesis.

---

## FAE-1013

(error) FRD file '%s' does not include region data file.

**Description**

The FRD file does not include the region data file required for Targetes Synthesis flow.

**What Next**

Provide a valid FRD file with region and hybrid data required for Targeted Synthesis.

---

## FAE-1015

(info) Extracting '%s' from '%s' .

**Description**

Exracting the reguired data for the flow.

**What Next**

This is an information only message. No action is needed.

---

## FAE-1020

(error) duplicate_registers is not enabled during FM eco mode.

**Description**

The command is not enabled during formality eco mode, which is enabled using set_fm_eco_mode

**What Next**

Command is not enabled in this mode so either exit the FM eco mode or drop the command.

---

# FILE

---

## FILE-001

(error) Unable to open file '%s' for reading; %s

Feedback

**Description**

The file cannot be read for the reason specified.

**What Next**

Examine the reason given. The file may not exist or may not be readable due to file permissions.

---

## FILE-002

(error) File '%s' cannot be found using search_path of: '%s'.

**Description**

The file cannot be found on the specified search_path.

**What Next**

Determine whether the file name or the search_path is incorrect, or whether the file does not exist at all.

---

## FILE-003

(error) Unable to open file '%s' for writing; %s.

**Description**

The file cannot be opened for writing for the reason specified.

**What Next**

Examine the reason given. The parent directory may not exist or write permission may be denied.

---

## FILE-004

(error) Unable to create directory '%s'; %s.

**Description**

The directory cannot be created for the reason specified.

**What Next**

Examine the reason given. The parent directory may not exist or write permission may be denied.

Feedback

## FILE-005

(error) Unable to access directory '%s'; %s

**What Next**

Check the directory path name.

## FILE-006

(error) Unable to remove directory '%s'; %s

**Description**

The tool was unable to remove the specified directory.

**What Next**

The directory could not be removed.

## FILE-007

(information) Loading %s file '%s'

**Description**

This informational message shows the full path name and type of a file being read into the application.

**What Next**

Use the full path name to help understand any problems encountered while reading the file.

## FILE-008

(error) Directory '%s' already exists

**Description**

The directory with the specified name already exists.

**What Next**

Try another name for the new directory.

Feedback

## FILE-009

(error) File '%s' already exists

**Description**

A file with the specified name already exists.

**What Next**

Try another name for the new file.

## FILE-010

(error) Unable to lock file '%s' for write @%s; %s.

**Description**

The file cannot be write-locked for the reason specified.

**What Next**

Examine the reason given. The file may be locked by another process or lock services may not be available on this server.

The reason “No locks available.” indicates a machine error. The ‘lockd’ process (a part of the NFS lock manager, which supports record locking operations on NFS files) is either not running on the server or has malfunctioned. ‘lockd’ may need to be started, or killed and then restarted. In some cases, a server reboot may be needed. These actions typically require the end-user to make a request to their local IT department.

## FILE-011

(error) Problem writing to file '%s'; %s.

**Description**

There was an error writing data to the specified file.

**What Next**

Examine the reason given. The disk may have ran out of space, or there was a system or network event that interrupted the write operation.

## FILE-012

(error) Directory '%s' already exists, '%s'.

Feedback

**Description**

The directory with the specified name already exists.

**What Next**

Try another name for the new directory.

---

## FILE-013

(error) Cannot open file '%s' for read.

**Description**

Unable to open file for read.

**What Next**

Check that the file exists. Check file/directory for read permissions.

---

## FILE-014

(error) Cannot open file '%s' for write.

**Description**

Unable to open output file for write.

**What Next**

Check file/directory for write permissions.

---

## FILE-015

(error) Basenames of original file and new file do not match. Original file: %s New file: %s

**Description**

The given filenames do not have matching basenames.

**What Next**

This check is done to avoid user error. If this is intended, please run update_cross_probing_files command with -force option to bypass this check.

---

## FILE-016

(error) File %s not found in the set of cross-probing files.

Feedback

**Description**

The file being requested for update does not appear to be in the set of Cross-Probing files.

**What Next**

Run report_cross_probing_files to retrieve the list of files known by Cross-Probing manager.

---

## FILE-017

(error) Unable to retreive the disk space for %s; %s.

**Description**

The directory cannot be created for the reason specified.

**What Next**

Examine the reason given. The directory may not exist or permission may be denied.

---

## FILE-018

(error) Unable to get working directory name; %s

**Description**

The current working directroy name could not be returned. The most common reason is the directory you are in no longer exists.

**What Next**

Examine the reason given. If your working directory no longer exists, change into a valid directory using absolute path and try again.

---

## FILE-019

(error) Decompressing failed; %s.

**Description**

Unable to decompress the file for the specified reason.

**What Next**

Check the compressed file.

Feedback

### FILE-020

(error) Command '%s' not executed due to '%s' error.

**Description**

The command run by you is not successfully executed due to the error message printed.

**What Next**

Please give a valid command. for example: if there is a directory "dir1" present and if someone creates directory using command "mkdir dir1" then error reported will be "mkdir: cannot create directory ‘dir1’: File exists"

### FILE-021

(info) File %s already exists. It will be overwritten.

**Description**

The command run by you is writing a file that already exists, so it will be overwritten.

## FLIB

### FLIB-1

(information) %s

**Description**

This is a general informational message.

**What Next**

No action required.

### FLIB-2

(warning) The fusion_lib library '%s' registry file error at line %d. %s

**Description**

The registry.dat file inside a fusion_lib library (directory) is created and updated by Library Compiler. Note that in the early phase of testing, there may be frequent incompatibility between versions. The following descriptions should help in diagnosing issues.

*Bad symbolink link.* Most likely the symbolic link points to a file no longer exists.

Feedback

*First line not recognizable.* The first line is of a fixed "key" format that users must not modify.

*File type not supported.* The fusion_lib only supports a limited number of file.

*Bad data format.* Each line consists of 2 or 3 tokens (fie_type file_name checksum). This line is invalid.

*File name already registered.* This is most likely a bug as two files cannot have the same name in a given fusion_lib directory.

*File missing or not readable.* The file has been removed or is not readable.

*Bad checksum in registry.* The registry checksum in the first line no longer agrees with the checksum generated from the rest of the file. This means a user may have modified the registry.

**What Next**

If the problem is a bad symbolic link because of a moved file, update the symbolic link. In general, the fusion_lib library needs to be re-checked and saved in order to update the registry. R&D.

---

## FLIB-3

(error) The fusion_lib library '%s' not qualified.

**Description**

This error message occurs after the fusion_lib checks failed either by check_fusion_lib command or internally called in create_fusion_lib or save_fusion_lib command (equivalent to check_library -logic_vs_physical). The default check is -logic_vs_physical and multiple library groups. When it is not qualified by the default check, save_fusion_lib cannot save it as qualified fusion_lib on disk. When the specified corner libraries are in multiple groups, a fusion library cannot be created. It is recommended using compile_fusion_lib to group and create. When there is no frame file, it will print out: No physical library found.

When the fusion library is not created, the check results are saved in autofix0.db3 at the current directory. When this message "Failed to qualify the fusion library due to multiple library groups" is shown it indicates the dbs are in multiple groups based on cell set.

The fusion_lib passing criteria depend on the policy specified by set_early_data_check_policy command:

1) error policy, default mode No missing physical cells reported in LIBCHK-211 and no missing logic or physical signal pins reported in LIBCHK-212.

2) tolerate policy Library data as is, and no severe errors below reported No missing logic or physical signal pins reported in LIBCHK-212.

Feedback

3) repair policy: Use the same criteria as tolerate policy, and apply check_library -auto_fix

In all policies, mismatched voltage_name/rail_name reported in LIBCHK-322 and missing pg_pins in logic libraries, i.e. non-pg_pin based dbs, will forbid fusion lib to be created.

**What Next**

Check the reported issues in LIBCHK-xxx tables for detail and correct the library accordingly or apply auto fix. In the case of multiple library groups, most likely the create_fusion_lib command failed as the input set of libraries may have multiple grouping scenario, It is recommended to run the compile_fusion_lib command when DB libraries need to be auto grouped. When mismatched voltage_name/rail_name is reported, please use rename_rail command to fix mismatched voltage_name. When mismatched pin/pg_pin direction is reported, please use set_attribute to fix. Please use add_pg_pin_to_db in 2013.12-SP5 lc_shell or common_shell (dc_shell) to convert the non-pg_pin based dbs. Use the following command to report the check results:

report_check_library_records -logic_vs_physical autofix0.db3

---

## FLIB-4

(error) Failed to save fusion library '%s': %s.

**Description**

This message indicates that an operation of saving the specified fusion library failed. The reason is given with the message.

**What Next**

This is just a summary message. Usually there will be other error message(s) before this one to indicate the actual problem. Please refer to those error messages for how to correct the issue.

---

## FLIB-5

(error) Failed to create fusion library '%s': %s.

**Description**

This message indicates that an operation of creating a fusion library failed. The reason is given with the message.

**What Next**

Correct the problem and try again.

Feedback

## FLIB-6

(error) Failed to %s ECO file '%s': %s.

**Description**

This message indicates that an operation of specified ECO file failed. The reason is given with the message.

**What Next**

Correct the problem and try again.

## FLIB-7

(error) Failed to save frame '%s' for fusion library '%s'.

**Description**

This message indicates that saving specified frame of fusion library failed. The reason is given with the message.

**What Next**

Correct the problem and try again.

## FLIB-8

(error) Failed to open fusion library '%s': %s.

**Description**

This message indicates that an operation of open the specified fusion library failed. The reason is given with the message.

**What Next**

Correct the problem and try again.

## FLIB-9

(information) ECO file '%s' was changed: %s.

**Description**

This message indicates that the specified ECO file had been changed outside Library Compiler. The actual change and tool's re-action are given with the message.

Feedback

**What Next**

Verify whether the change is expected.

---

## FLIB-9w

(warning) ECO file '%s' was changed: %s.

**Description**

This message indicates that the specified ECO file had been changed outside Library Compiler. The actual change and tool's re-action are given with the message.

**What Next**

Verify whether the change is expected.

---

## FLIB-10

(error) Unable to create a temporary work directory

**Description**

Library Compiler is unable to create a temporary work directory to store data. If the environment variable TMPDIR is present, please make sure it points to a writable disk location which is not full. If TMPDIR is missing, Library Compiler uses /usr/tmp, please make sure /usr/tmp is writable and not full.

**What Next**

Fix the problem and run again.

---

## FLIB-11

(information) The fusion_lib %s needs a refresh. %d file(s) have modified timestamps.

**Description**

The fusion_lib has been copied or moved inappropriately. As a result, timestamps on some files are altered. This triggers additional checks and performance overhead.

**What Next**

When copying using "cp", be sure to specify "-a" (for archive), rather than "-R":

cp -a fusion_lib_dir destination

When copying using "tar", preserve high-resolution time stamps using posix format.

Feedback

tar --posix -cf output.tar fusion_lib_dir

Alternatively, you may wish to refresh the fusion_lib using open_fusion_lib and save_fusion_lib commands in lc_shell.

---

## FLIB-12

(error) Cannot open fusion library '%s' as it had been changed since last save. Please save it again using Library Compiler first.

**Description**

This message indicates that the specified fusion library had been changed since last save and cannot be opened. Here, the change means any content or time stamp change of any relavent files in the fusion library.

**What Next**

Save the fusion library using Library Compiler and try again.

---

## FLIB-13

(error) Frame file using "reflib.ndm" as frame file name is NOT supported.

**Description**

The "reflib.ndm" is a reserved file name of directory-based ndm library. It usually resides under the ndm library directory. The frame file should be defined using the ndm library directory name.

If it's a file-based ndm file which has schema version prior to 1.200, please avoid using "reflib.ndm" as the file name.

**What Next**

For directory-based frame libray, please use the parent directory of reflib.ndm as the frame file name. For file-based frame library, please use a different file name as the frame file name.

---

## FLIB-14

(warning) Fusion library '%s' created or saved with inconsistency issues waived.

**Description**

This message indicates that a fusion library is created or saved with missing or mismatched issues waived between logic and physical libraries. These issues are reported

Feedback

in the library checker in LIBCHK tables above this message line. Please refer to FLIB-3 man page for qualification criteria under each check policy.

**What Next**

Investigate the reported missing or mismatched issues. If those cannot be waived for your design, please fix the library first. You may fix the issues by set_early_data_check_policy -config {repair} or explicitly use set_attribute or other utilities to fix.

---

## FLIB-15

(Error) %s

**Description**

This is a general Error message.

**What Next**

Please follow the message.

---

## FLIB-16

(warning) Cannot record command %s: %s.

**Description**

The specified command cannot be recorded. This means the effect of the command only persist until the fusion lib is closed.

**What Next**

Do not use this command as persistent eco command. If need to be persistent, require an enhancement first.

---

## FLIB-17

(warning) Mapping of lef pattern '%s' will %s: %s.

**Description**

This message occurred when lef pattern to post script mapping is duplicately specified. If the new post script is different from the old post script, the new post script will override the old one.

**What Next**

Check the content of physical_lib.lef_to_frame_postprocess_scripts and make sure it does not have duplicate lef pattern specified.

Feedback

## FLIB-18

(Warning) %s

**Description**

This is a general Warning message.

**What Next**

Please follow the message.

## FLIB-19

(error) Cannot create virtual frame as technology file not specified.

**Description**

This message indicates that the tool need to create virtual frame for a fusion lib, but there is no technology file specified.

**What Next**

Specify the -technology option or set application option *fusion_lib.use_default_technology_file* to true and try again.

## FLIB-19w

(warning) Use default technology file for creating virtual frame.

**Description**

This message indicates that the tool need to create virtual frame for a fusion lib, but there is no technology file specified. So the tool will use the default technology file for creating virtual frame.

**What Next**

If using the default technology file is not desired, please specify the -technology option and try again.

## FLIB-20

(error) Cannot add frame '%s', not a lib cell library.

**Description**

This message indicates that the specified frame is not a lib cell library.

Feedback

**What Next**

Specify a valid frame and try again.

---

## FLIB-21

(warning) The '%s' option will be ignored. Cannot be specified with '%s' option.

**Description**

The reported command option will be ignored as it cannot be specified with another option.

**What Next**

Look at the manpage for this command for more information on command options.

---

## FLIB-22

(warning) The '%s' option will be ignored.

**Description**

The reported command option will be ignored as it has no use in this run of the command.

**What Next**

Look at the manpage for this command for more information on command options.

---

## FLIB-23

(error) Must specify one or more of these options: %s.

**Description**

This command requires that one or more of the options in the list are specified.

**What Next**

Refer to the manual page for this command for detailed information on valid options.

---

## FLIB-24

(error) Cannot launch a job for CDPL run.

**Description**

This message indicates that the attemption to launch a job for CDPL run of the command was failed.

Feedback

**What Next**

Either make sure CDPL is configured correctly on local machine, or run the command without CDPL by removing the *-protocol* and *-parallel_jobs* options.

---

## FLIB-25

(error) Failed to add '%s' to the fusion lib: %s.

**Description**

This message indicates that adding a file to the fusion lib failed. The reason is given with the message.

---

## FLIB-25w

(warning) Not able to add '%s' to the fusion lib: %s.

**Description**

This message indicates that tool is not able to add a file to the fusion lib. The reason is given with the message.

**What Next**

Check whether there is typo in the db name. If it's a typo, then fix it and try again.

---

## FLIB-26

(warning) Skip duplicate %s '%s'.

**Description**

This message indicates that a duplicate file was specified. It will be ignored.

**What Next**

Check whether there is typo in the file name. If there is a typo, then fix it and try again.

---

## FLIB-27

(error) Please specify lib path for report_fusion_lib -summary

**Description**

Library Compiler is unable to report summary info without path, you should add libpath for this command or run open_fusion_lib at first

Feedback

**What Next**

Fix the problem and run again.

---

## FLIB-28

(error) The command %s needs an opened fusion lib

**Description**

Run open_fusion_lib command first before executing this command

**What Next**

Fix the problem and run again.

---

## FLIB-29

(error) The specified lib path is not consistent with the opened fusion lib path

**Description**

The specified lib path for report_fusion_lib is not consistent with the opened fusion lib path

**What Next**

Fix the problem and run again.

---

## FLIB-30

(warning) Data type '%s' is not supported for class '%s'; will use '%s'.

**Description**

This message indicates that the specified data type is not supported to be a user defined attribute of the specified class. Another data type will be used actually.

**What Next**

To avoid the warning message, use the allowed data type to replace the original data type.

---

## FLIB-31

(error) Invalid library name, the speicified library name %s doesn't exist

**Description**

The speicified library name doesn't exist

Feedback

### What Next

Fix the problem and run again.

---

## FLIB-32

(warning) Option '%s' is not supported for class '%s'; will be ignored.

### Description

This message indicates that the specified option is not supported for a user defined attribute of the specified class. The option will be ignored.

### What Next

To avoid the warning message, do not use the option when define the user attribute for the class.

---

## FLIB-33

(error) Failed to make the command %s persistent: %s.

### Description

This message indicates that the attempt to make the specified command persistent failed. This means the effect of the command only persist until the fusion lib is closed.

### What Next

Check the reported message for the reason of the fail.

---

## FLIB-34

(warning) Failed to make a Tcl recording of the command %s: %s.

### Description

This message indicates that the attempt to make a Tcl recording of the specified command failed. The consequence is: it cannot know what kind of operation had been performed to the fusion lib, and cannot restore the fusion lib in case a data corruption happens.

---

## FLIB-35

(error) Different types of physical files found in the group %s.

Feedback

**Description**

This message indicates that different types of physical files was found in the reported group. Although frames, LEFs and Milkyway libraries are allowed to be specified with the compile_fusion_lib command at the same time, they must be in different groups.

**What Next**

Please check the source of the physical files and make sure different types of physical files will be grouped into different groups.

---

## FLIB-36

(warning) %s as the -continue_on_error option is specified.

**Description**

This message indicates that an error condition is ignored as the -continue_on_error option is specified.

---

## FLIB-37

(error) File '%s' cannot be found using search_path of: '%s'.

**Description**

The file cannot be found on the specified search_path.

**What Next**

Determine whether the file name or the search_path is incorrect, or whether the file does not exist at all.

---

## FLIB-38

(warning) %s lib-cells of '%s' will be ignored as they are duplicate with the lib-cells in previous files.

**Description**

This message indicates that some or all lib-cells of the specified physical source file will be ignored as they are duplicate with the lib-cells in previous physical source files.

Here is the order with which the physical source files will be processed. For different kinds of physical source files, the tool processes them with following order: frame NDMs, LEFs, MW libraries. Among one kind of physical source files, the one listed earlier will be processed first.

Feedback

**What Next**

Please make sure there is no duplication among all physical source files, or check the duplication and make sure the ignore of those duplicates are expected.

---

## FLIB-39

(warning) '%s' will be ignored: %s.

**Description**

The message indicates that given source file for *compile_fusion_lib* will be ignored. The possible reasons which will be given in the message include:

1. There is no lib cell in the file

**What Next**

Check whether the ignorance of the given file is expected.

---

## FLIB-40

(error) '%s' is invalid: %s.

**Description**

The message indicates that given source file for *compile_fusion_lib* is invalid. The reason is given in the message.

**What Next**

Provide valid source files for *compile_fusion_lib*.

---

## FLIB-41

(warning) Remove eco file '%s': %s.

**Description**

This message indicates that the tool will remove the specified ECO file. The reason is given with the message.

Note that the fusion lib needs to be re-checked and saved by using save_fusion_lib command or "close_fusion_lib -save" command so that this message won't be reported again in the next open_fusion_lib.

If the remove of the eco file is not expected, please do not save the fusion lib. Just quit fusion lib session by "close_fusion_lib -force". The fusion lib will kept untouched. However, in the next open_fusion_lib, it will report this message again.

Feedback

## FLIB-42

(error) File '%s' was changed outside of Library Compiler.

**Description**

This message indicates that a specific file under the fusion library directory was changed outside of Library Compiler. This is an severe error since all fusion library modification should through Library Compiler.

**What Next**

Re-create the fusion lib using original data.

## FLIB-43

(error) Multiple designs are detected in etm flow.

**Description**

ETM flow will build a library from Extracted Timing Models (ETMs), it will merge multiple PVTs and modes for one design into a single library cell. For each session, will do the merge for only one design. You get this error message because multiple designs are detected in current session.

**What Next**

Only read DB files that contain the specific design you want to do the merge and create lib. You can start again by reading only related DB files.

## FLIB-44

(error) More than one instance of cell %s, mode %s, PVT '%s'

**Description**

For the ETM flow, each logical design with the same name is compared using the mode label and the PVT information from *add_db*. There can be only one unique combination of logical design name, mode, and PVT. You get this message if there is a duplicate.

**What Next**

Make sure you have applied the correct mode label to each logical library read into the ETM flow. Once you know which library has the wrong label, remove it, then re-read it with the correct label.

Feedback

## FLIB-45

(error) Library %s is not compatible with the etm flow:%s

**Description**

The named library has a problem construct which is not allowed in the etm flow, or is missing a required construct or attribute.

**What Next**

Either change flows, or modify the library source.

## FLIB-46

(error) Dbs added before are%s etm moded, while new added dbs are%s etm moded.

**Description**

New added dbs are not aligned with the dbs added before in terms of whether the dbs are paired etm modes or not.

**What Next**

Recreate the flib with all dbs added with or without etm modes.

## FLIB-47

(error) Cannot find a physical %s for '%s'.

**Description**

The specified object only exists in timing view. There is no physical view counterpart for it.

**What Next**

Make sure every timing view object has a physical view counterpart.

## FLIB-48

(error) Attribute value '%s' is not valid: %s.

**Description**

The specified attribute value is not valid. The reason is given with the message.

**What Next**

Specify a correct attribute value for setting.

Feedback

### FLIB-101

(information) Processing of %s NOT supported. Skipped.

**Description**

This information indicates a specific command or app option does NOT affect the Fusion Library creation, thus is NOT supported in LM script migration for LC.

**What Next**

No action required.

## FLT

### FLT-002

(information) Errors preprocessing compiled filter.

**Description**

This is a summary message generated after a filter expression has successfully parsed, but unsuccessfully processed because of an unknown identifier, type mismatch in a relation, or invalid operator in a relation.

**What Next**

Look at previous error messages to determine the problem with the filter expression. Correct the problems, and retry the operation.

### FLT-003

(error) while parsing filter expression: %s\n \tat '%s'

**Description**

A filter expression could not be successfully parsed, typically because of a syntax error. The point in the expression that caused the failure is shown along with the remainder of the expression.

**What Next**

Look at the man pages for filter expression syntax, and verify that your expression conforms to the syntax. Ensure that supported relation and logical operators are in use, that the expression is constructed of a series of relations separated by logical operations, etc.

Feedback

## FLT-005

(error) Unknown attribute '%s'.

### Description

Filters are evaluated within a context. Given the current context, an attribute which you entered is unknown.

A relation in a filter expression is very simple. For example, "area <= 2.4". In this case, the attribute is "area". If you were applying the filter to a pin collection, since "area" is not a valid pin attribute, this error would occur.

### What Next

Look at the man pages for the given command, and ascertain the valid values for attributes.

## FLT-006

(error) Type mismatch between '%s' and '%s'.

### Description

A relation in a your filter expression has an identifier and value with inconsistent types. The following simple rules apply:

```
Identifier              Type mismatch
        generates:              when value is:
        ------------------------------------------------------
        string                  n/a - never an error
        numeric literal         string, true, false
        boolean                 numeric literal, string
```

Note some important distinctions: the boolean words TRUE and FALSE are interpreted as strings in a string relation, or as boolean in a boolean relation. Similarly, the numeric literal 2.4E-9 is interpreted as a string in a string relation, or as a number in a numeric relation.

### What Next

Re-enter the filter with valid identifier/value relations.

## FLT-007

(error) Invalid operator '%s' for '%s' and '%s'.

Feedback

**Description**

A relation in a your filter expression has an identifier and value with consistent types but an invalid operator. The following simple rules apply:

Type: Invalid operators: ---------- string None. All operators ok. numeric literal =~, !~ boolean =~, !~, <, >, >=, <=

**What Next**

Re-enter the filter with a valid operator for the failed relation.

# FLW

## FLW-1101

(info) Design checksums: %s

**Description**

Starting print of Design checksum tokens for current step. This message appears in pair with FLW-1101.

**What Next**

Information. No action needed.

## FLW-1230

(error) Block %s does not have design view or is not in the current design library. Unable to generate a hierarchical checkpoint.

**Description**

In a hierarchical design all the blocks must have design view and should be in the current design library. The checkpoint will have only the current design, sub-designs/blocks will not be written to the checkpoint.

## FLW-1241

(error) Compile failed due to presence of %s cells in design '%s'.

**Description**

This message is reported when there are unresolved or unmapped cells. As such, Tool cannot proceed further with rest of the synthesis.

Feedback

**What Next**

- Use *analyze_mv_feasibility -format html* to know the detailed reasons of unmapped cells.
- Check strategy present using *report_mv_path*.
- Check if available library cells can fulfill operation conditions by running *report_mv_lib_cells*. See if library cells have correct *valid_purposes* in the report. Library cells with *none* purpose cannot be used by the Tool. Use *set_lib_cell_purpose* to make library cells available to be used by the Tool.

**See Also**

- analyze_mv_feasibility
- report_mv_lib_cells
- report_mv_path
- set_lib_cell_purpose

---

## FLW-1762

(error) Flow status set to Error by step: %s

**Description**

This indicates that synthesis flow encountered error during the given step and exited early.

**What Next**

Look for errors/warning prior to this error and fix them.

---

## FLW-2543

(warning) Flow step returns Error: %s

**Description**

This indicates that synthesis flow encountered error during the given step.

**What Next**

Look for errors/warning prior to this error and fix them.

Feedback

## FLW-3574

(warning) Option %s for command 'compile_fusion' is deprecated and should be replaced by the %s app option setting.

### Description

The command options *-no_autoungroup*, *-no_boundary_optimization*, and *-no_seq_output_inversion* are deprecated but will still be honored by compile_fusion.

### What Next

Users should instead use the associated app options to control the optimization behavior.

## FLW-5560

(error) The preceding synthesis command aborted unexpectedly. Please revert to a previously saved version of the database.

### Description

The previous optimization aborted with an unexpected error. Revert to last sane database to ensure correct results.

### What Next

Users should rewind to last saved database version and review the flow script.

## FLW-5561

(warning) Previously executed stage '%s' may be unexpected for the current flow stage '%s'

### Description

The optimization history is not the expected flow for the action you requested. Certain optimization stages must follow other specific stages, additionally certain actions may require specific design-states to be met. This error message can be made to stop the flow if changed to 'severe'.

### What Next

Users should review the compile flow script.

### See Also

- set_msg

Feedback

## FLW-5562

(info) For standard results, execute '%s', right after one of the following: %s.

**Description**

The optimization path is not the expected for achieving the best results.

**What Next**

Users should review the compile flow script.

## FLW-5563

(info) For standard results, run '%s' after having completed all of these stages: %s.

**Description**

The synthesis path is not the expected, for achieving best results.

**What Next**

Users should review the compile flow script.

## FLW-5564

(info) Verify the design '%s' before running '%s'.

**Description**

The synthesis state is not the expected, for achieving best results.

**What Next**

Users should review the compile flow script.

## FLW-8000

(info) Starting %s

**Description**

Starting message of a command, compile stage or step. This message appears in pair with FLW-8001.

**What Next**

Information. No action needed.

Feedback

## FLW-8001

(info) Ending %s

**Description**

Ending message of a command, compile stage or step. This message appears in pair with FLW-8000.

**What Next**

Information. No action needed.

## FLW-8003

(info) Running %s

**Description**

Information of the currently running step.

**What Next**

Information. No action needed.

## FLW-8004

(warn) Incomplete %s

**Description**

The incomplete flow has been started (as in FLW-8000), but no corresponding FLW-8001 is seen.

**What Next**

Users should review the flow.

## FLW-8100

(info) Time: %s / Session: %s / Command: %s / CPU: %s / Memory: %.0f MB

**Description**

Information of timestamp, elapsed time in the tool session, elapsed time and CPU time in the command and peak memory usage.

Feedback

**What Next**

Information. No action needed.

---

## FLW-9000

(info) %s

**Description**

This shows runtime summary for major engines at major breakpoints of core commands. Currently, it is printed at compile_fusion (final_place, final_opto), create_placement, place_opt, clock_opt, route_opt and hyper_route_opt.

**What Next**

Information. No action needed.

---

# FMT

---

## FMT-001

(information) FMT-DESIGN: 'write_correlation_files' completed with result = '%d'.

**Description**

This information message indicates whether 'write_correlation_files' completed successfully or not.

**What Next**

Please correct the error if the result is non-zero.

**See Also**

- set_consistency_settings_options

---

## FMT-002

(information) FMT-DESIGN: Found missing db-file '%s', successfully resolve to '%s'...\n

**Description**

This information message indicates the .db files stored inside nlib have been successfully resolved.

Feedback

Missing db-files could happen if we moved those .db files to other places after we built nlib.

In this case, we have to redirect the new .db files paths through 'set_consistency_settings_options -search_path {...}'.

**What Next**

No action needed.

**See Also**

- set_consistency_settings_options

---

## FMT-003

(Error) The original db-file '%s' has been moved or deleted! Please double-check.\n

**Description**

This information message indicates the .db files stored inside nlib have been deleted or moved to some other place.

Missing db-files could happen if we moved those .db files to other places after we built nlib.

In this case, we have to redirect the new .db files paths through 'set_consistency_settings_options -search_path {...}'.

**What Next**

Specify those .db files new paths with 'set_consistency_settings_options' and then retry.

**See Also**

- set_consistency_settings_options

---

## FMT-004

(Warning) Failed to find PVT [%f %f %f] in characterization points of lib '%s'.\n

**Description**

This information message indicates tool can't find the corresponding .db file which exactly matches with PVT criteria.

In this case, tool will choose the .db file which has closet PVT against the user's PVT.

Feedback

**What Next**

No action needed.

**See Also**

- set_consistency_settings_options

---

## FMT-005

(Error) Can not find flow-name '%s'! Ignore it...\n

**Description**

User could force write_correlation_files to skip specific flows if needed.

For example, option '-filter_flows XXX' could be used if user does not need to generate parasitics-files.

**What Next**

Please correct the error.

---

## FMT-006

(Warning) Found arc-problem with lib-cell '%s'\n.

**Description**

The arcs of this lib-cell could be empty of missed when we build this nlib.

Timing Qor could be impacted if arcs of lib-cell is incompleted.

**What Next**

Check whether we need to fix this issue or not.

---

## FMT-007

(Error) Scenario '%s' uses different SPEF<%s/%s> for early/late analysis.\n

**Description**

Due to incorrect extraction-setting(tlu+ or extraction-options), one scenario could use different SPEF-files

for early/late analysis and this could lead to bad Qor results.

Feedback

**What Next**

Correct the bad extraction-settings which could lead to different SPEF for early/late analysis.

---

## FMT-008

(Warning) High-fanout Net '%s'(threshold=%d) might lead to bad-Qor due to using ideal pin-caps for delay-calculation!\n

**Description**

Using ideal pin-cap for high-fanout net could lead to correlation issue.

**What Next**

Correct high-fanout settings and rerun the command again.

---

# FRAM

---

## FRAM-001

(warning) Ignore non-existent via definition '%s'.

**Description**

The specified via definition cannot be found in the design or the technology.

**What Next**

Check the specified object exists in either the design or the technology.

---

## FRAM-002

(info) Ignore default via definition '%s' specified in non-default via option.

**Description**

There is no need to specify default vias for this command. Default vias are automatically included.

**What Next**

Optionally remove the default via definition names from command option.

Feedback

## FRAM-003

(warning) Ignore design or library cell '%s' that has no physical data in design view.

**Description**

The specified design or library cell does not have physical data in the design view for the command to extract.

**What Next**

No action is needed if no physical data is expected. Otherwise, please check the LEF file and read_lef result.

## FRAM-003e

(error) Design or library cell '%s' that has no physical data in design view, cannot create frame view.

**Description**

The specified design or library cell does not have physical data in the design view for the command to extract.

**What Next**

No action is needed if no physical data is expected. Otherwise, please check the LEF file and read_lef result.

## FRAM-004

(warning) Ignore design or library cell '%s' that is not opened.

**Description**

The specified design or library cell is not opened.

**What Next**

Open the design and try again.

## FRAM-005

(warning) Non-manhattan '%s' is not supported. It is replaced by bbox.

**Description**

Non-manhattan shape is replaced by its bbox.

Feedback

## FRAM-006

(error) Current design is not a design view.

**Description**

This command requires that the current design is a design view. Either there is no current design available or the current design is not a design view.

**What Next**

Load the design view of this design.

## FRAM-007

(warning) Sub-row count is incorrect for block '%s'. Ignore extracting frame properties.

**Description**

The specified block has incorrect boundary or site information for extracting frame properties.

**What Next**

Check the specified block has correct data in either the design or the technology.

## FRAM-008

(warning) Skip parsing 'isCutArray' attribute for layer '%s' in block '%s'. Please check layer setting in technology file.

**Description**

Skip parsing attribute for shapes in via cut layers.

**What Next**

Check the specified block has reasonable number of shape in via cut layers, and the number of routing layer and via cut layer is also correct.

## FRAM-009

(warning) Parse the source-drain file failure.

**Description**

Fail to parse source-drain annotation.

Feedback

**What Next**

Check the source-drain file has correct format.

---

## FRAM-010

(error) Block '%s' is not in frame view. %s

**Description**

Specified block is reguired in frame view; if not, skip the process.

**What Next**

Check the specified block is in frame view.

---

## FRAM-011

(warning) For block '%s', there are pins smaller than the pin must connect area threshold on layer '%s', and those pins are not blocked by zero-spacing routing blockages.

**Description**

If the pin area of a port is not sufficient to the pin_must_connect_area_thresholds, then the zero-spacing routing blockages are not created for those pin shapes.

**What Next**

Check the setting of pin_must_connect_area_thresholds.

---

## FRAM-012

(warning) '%s' is not a valid layer name in tech file. Ignore the setting for option '%s'.

**Description**

The input layer name of the frame option does not exist in tech file, ignore the option setting.

**What Next**

Check the input layer name.

---

## FRAM-012e

(error) '%s' is not a valid layer name in tech file. Invalid setting for option '%s'.

Feedback

**Description**

The input layer name of the frame option does not exist in tech file.

**What Next**

Check the input layer name.

---

## FRAM-013

(warning) '%s' is not a via or polyCont layer for option '%s'. Ignore the setting on the layer.

**Description**

The input layer name is not a via or polyCont layer, ignore the setting on the layer.

**What Next**

Check the input layer name.

---

## FRAM-013e

(error) '%s' is not a via or polyCont layer for option '%s', which is invalid setting.

**Description**

The input layer name is not a via or polyCont layer, which is invalid setting.

**What Next**

Check the input layer name.

---

## FRAM-014

(warning) List of pins without %s.

**Description**

List the pins which miss via region or access edge in the library. Missing via region or access edge might impact routing QoR.

**What Next**

Check these pins and see if regenerate frame is needed.

---

## FRAM-015

(warning) List of cells without placement layers and constraints .

Feedback

**Description**

List the cells which miss geometry on tech file layers or constraints in the library. Missing these might impact placement QoR.

**What Next**

Please check the design and fix these issues.

---

## FRAM-016

(warning) '%s' is not a metal or poly layer for option '%s'. Ignore the setting on the layer.

**Description**

The input layer name is not a metal or poly layer, ignore the setting on the layer.

**What Next**

Check the input layer name.

---

## FRAM-016e

(error) '%s' is not a metal or poly layer for option '%s', which is invalid setting.

**Description**

The input layer name is not a metal or poly layer, which is invalid setting.

**What Next**

Check the input layer name.

---

## FRAM-017

(information) Use current library as the technology library.

**Description**

No technology library is set by -technology_lib option. Use current library as the technology library.

**What Next**

Please specify the -technology_lib value if the current library does not contain the correct technology information.

Feedback

## FRAM-018

(warning) List of pins with off-track or missing via regions.

**Description**

List the pins with off-track or missing via regions. Missing these might impact routing QoR.

**What Next**

Please check the via-region and wire track settings. Fix the via region not on track issues.

## FRAM-019

(warning) Cannot find site definition %s. Use default site definition instead.

**Description**

The site definition specified by option -wire_tracks_site cannot be found. Use default site definition instead.

**What Next**

Please check the value set by -wire_tracks_site.

## FRAM-020

(warning) The routing direction of layer %s is not defined for wire tracks calculation.

**Description**

The routing direction of the specific layer should be defined as vertical or horizontal. The wire tracks need to be calculated by horizontal or vertical routing direction.

**What Next**

Please check your technology file or site definition setting.

## FRAM-021

(warning) The option value of '%s' cannot be less than 0. Ignore the option setting on layer '%s'.

**Description**

The option value cannot be less than 0, ignore the setting on the specified layer.

Feedback

**What Next**

Check the input option value.

---

## FRAM-021e

(error) The option value of '%s' cannot be less than 0. Invalid option setting on layer '%s'.

**Description**

The option value cannot be less than 0, which is invalid setting on the specified layer.

**What Next**

Check the input option value.

---

## FRAM-022

(error) The maskName, layerNumber, or maskOrder of the layers in technology library %s don't match with the layers of reporting library %s.

**Description**

The maskName, layerNumber, or maskOrder of the layers of the technology library are different from those of the reporting library. Cannot use the information of the technology library to calculate the wire track position.

**What Next**

Please check your -technology_lib input and specify the correct technology library.

---

## FRAM-023

(error) Option -block is not specified in write_frame_options with -format icc2_shell.

**Description**

When using -format icc2_shell in write_frame_options, option -block is also required; if not, skip the process.

**What Next**

Specify option -block in write_frame_options command.

---

## FRAM-024

(warning) The %s frame view contains more than 100000 non-pin shapes (%d). This can increase routing runtime; you might want to regenerate the frame view.

Feedback

### Description

The frame view of the specified block contains more than 100000 non-pin shapes. The large number of obstructions can increase routing runtime.

### What Next

If this is not expected, regenerate the frame view. To reduce the number of obstructions in the frame view, try using the *remove_non_pin_shapes*, *merge_metal_blockage*, *block_all*, or *block_core_margin* options.

In lm_shell, these options are application options with the *lib.physical_model* prefix. In icc2_shell, these options are options of the *create_frame* command.

### See Also

- create_frame
- lib.physical_model.block_all
- lib.physical_model.block_core_margin
- lib.physical_model.merge_metal_blockage
- lib.physical_model.remove_non_pin_shapes

---

## FRAM-025

(warning) List of irregular cut shapes or routing blockages.

### Description

Frame has irregular cut shapes or routing blockages. A cut layer shape or regular routing blockage which does not meet any of the following conditions will be considered as iregular:

o is not rectangular o the width and height attributes does not match any of the following values o the minimum width of the layer o the default width of the layer o the values in cutHeightTbl and cutWidthTbl of the layer o the values of cut size of any simple via definitions of the layer

These shapes or blockages may cause long runtime issue during routing.

### What Next

Please set app option 'file.lef.non_real_cut_obs_mode' as true for lef files, or use layer mapping file to map zero spacing routing blockage on cut layers for gds files. Then, regenerate the frame.

Feedback

## FRAM-026

(information) Library %s contains frames which are converted from ICC FRAM.

**Description**

This library contains frames which are converted from the FRAM views of IC Compiler or Milkyway. For getting all the converted frames, please use the following command: get_lib_cells */*/frame -filter "is_from_icc_frame==true"

## FRAM-027

(warning) The minimum height of all lib_cells in library %s is not equal to the height of the default site '%s' of this library.

**Description**

Please check if it is expected that the minimum height of all lib_cells is not equal to the height of the default site definition of this library. This difference may cause issue in placement.

**What Next**

Correct the site definitions of the library if necessary.

## FRAM-028

(warning) The %s of lib_cell %s is not the multiplier of its site '%s' %s.

**Description**

The height/width of the lib_cell is not the multiplier of its site definition height/width. This will cause issue in placement.

**What Next**

Please associate the lib_cell to the correct site definition.

## FRAM-029

(warning) The value '%s' of '%s' option is improper for cell %s.

**Description**

The frame option value for the cell is set improperly. This will cause issue in routing.

Feedback

Improper frame options setting includes:

For the following design types: diode, end_cap, fill, filler, lib_cell, physical_only, well_tap • The value of 'trim_metal_blockage_around_pin' option should not be set as 'all'. • The value of 'block_all' option should not be set as 'true'.

**What Next**

Please correct the frame option value and regenerate the frame.

---

## FRAM-030

(warning) The terminal %s of cell %s is overlapped with other terminals of different port.

**Description**

The terminal is overlapped with other terminals of different port. This will cause router fail to connect this pin without DRC violation.

**What Next**

Please check the design, correct the terminal short issue and regenerate the fame view.

---

## FRAM-031

(warning) The port %s of cell %s has no physical pin shape. Router will not be able to connect this port.

**Description**

Physical shape is missing for the specific port. Router will not be able to connect this port.

Only if the port is a device bias port or the pin direction is internal, then it does not required to have physical pin shapes. However, if the library does not contain logic db information, then tool cannot distinguish device bias ports. So this message will still show up for such kind of ports.

**What Next**

Please fix the missing pin shape issue for the port and regenerate the frame.

---

## FRAM-032

(warning) The port %s of cell %s has inconsistent value of routing attribute '%s' between timing view '%s' and frame view '%s'.

Feedback

**Description**

The specific routing attribute is defined in timing view but it is not defined in frame view. This attribute is needed in frame view for routing purpose.

Checked attributes: is_secondary_pg, is_diode

The routing attribute of ports should be set to frame view correctly if you want to set the routing attribute manually for existing reference library using edit flow. Here is the Tcl example: set_attribute [get_lib_pins -of_objects [get_lib_cells */diode_cell/frame] -filter "name == diode_pin"] is_diode true

**What Next**

Please regenerate frame view or update the attributes in frame manually.

---

## FRAM-033

(warning) There are terminals blocked by blockages or metal shapes.

**Description**

There are terminals blocked by blockages or metal shapes. Router may not able to connect these blocked terminals without DRC violation.

**What Next**

Please check and remove the unnecessary blockage or shapes over the terminals.

---

## FRAM-034

(warning) There is no technology information for this library. The reporting will not be completed.

**Description**

There is no technology information for this library. The tool cannot do complete checking and reporting without the technology information.

**What Next**

Please set the technoloy information to the library either by binding a technology file or a technology library.

---

## FRAM-035

(warning) Invalid number specified for '%s', ignore the setting on layer '%s'.

Feedback

**Description**

The option value has invalid number, please see man page for the usage of the option.

**What Next**

Please update the input value.

---

## FRAM-035e

(error) Invalid number specified for '%s' on layer '%s'.

**Description**

The option value has invalid number, please see man page for the usage of the option.

**What Next**

Please update the input value.

---

## FRAM-036

(warning) The setting of option keepout_spacing_for_non_pin_shapes for layer %s is ignored because block_all or block_core_margin is applied on this layer.

**Description**

The tool does not support option keepout_spacing_for_non_pin_shapes when block_all or option block_core_margin is also applied on the same layer.

**What Next**

Please set option block_all as false when using option keepout_spacing_for_non_pin_shapes.

---

## FRAM-037

(warning) '%s' is not a library cell, there is no frame property.

**Description**

The implant width property, diffusion width height property, and source/drain annotation are only extracted in library cells. The report_frame_properties is also used for library cells.

**What Next**

Please assign library cells by option -library (required) and -block (optional) in report_frame_properties.

Feedback

## FRAM-038

(warning) The option value of '%s' cannot be less than 0. Ignore the option setting.

**Description**

The option value cannot be less than 0, ignore the option setting.

**What Next**

Check the input option value.

## FRAM-038e

(error) The option value of '%s' cannot be less than 0.

**Description**

The option value cannot be less than 0.

**What Next**

Check the input option value.

## FRAM-039

(warning) The internal port %s of cell %s has improper setting(s) of direction as internal and is_physical as false. Router will skip the terminals of this internal port.

**Description**

In general, internal ports from the logic library should not have physical terminals. If a port has a *direction* attribute of *internal* and an *is_physical* attribute of *false*, the router does not treat it as real metal and ignores its terminals.

**What Next**

Internal ports should not be formed as physical terminals in the design and frame views. You can modify the physical data (GDSII, OASIS, or LEF) directly or by using the *remove_terminal* command to change the terminals in the design view to shapes, as shown in the following script example:

```
create_workspace ...
read_gds ...

# Change terminals to shapes in the design view for internal ports
foreach_in_collection int_port [get_lib_pins -quiet -all -filter
 "direction==internal"] {
  set _block [get_object_name [get_attribute -quiet $int_port design]]
  set port_name [get_attribute -quiet $int_port name]
```

Feedback

```
  set block [get_lib_cells -quiet */${_block}/design]
  if {$block!=""} {
    current_block $block
    set int_term [get_terminal -quiet -filter "port.name==$port_name"]
    set _int_term [get_object_name $int_term]
    if {$int_term!=""} {
      remove_terminal $int_term
      echo -n "${_block}.design: Changed terminal objects \{$_int_term\}
 "
      echo "to shape objects for internal port
    }
  }
}

check_workspace ...
commit_workspace ...
```

## FRAM-040

(warning) Value '%s' for option '%s' is not valid. Specify one of: %s. Ignore the option setting on layer '%s'.

### Description

The option value is invalid, ignore the option setting on the specified layer.

### What Next

Check the input option value.

## FRAM-040e

(error) Value '%s' for option '%s' is not valid. Specify one of: %s. Invalid option setting on layer '%s'.

### Description

The option value is invalid on the specified layer.

### What Next

Check the input option value.

## FRAM-041

(warning) Value '%s' for option '%s' is not valid. Specify one of: %s. Tool will use default value '%s' for the option.

Feedback

**Description**

The option value is invalid, ignore the option setting and use the default value for the option.

**What Next**

Check the input option value.

---

## FRAM-041e

(error) Value '%s' for option '%s' is not valid. Specify one of: %s. The default value is '%s'.

**Description**

The option value is invalid.

**What Next**

Check the input option value.

---

## FRAM-042

(warning) Value '%s' for option '%s' is not a metal or poly layer. Tool will use default value '%s' for the option.

**Description**

The option value is not a metal or poly layer, ignore the option setting and use the default value for the option.

**What Next**

Check the input option value.

---

## FRAM-042e

(error) Value '%s' for option '%s' is not a metal or poly layer.

**Description**

The option value is not a metal or poly layer, which is invalid.

**What Next**

Check the input option value.

Feedback

## FRAM-043

(warning) '%s' has DRC violations.

**Description**

The frame has DRC violations.

**What Next**

Please update input data.

## FRAM-044

(error) Cannot open block %s.frame for write.

**Description**

check_frame_drc needs to have write permission for the frame block.

**What Next**

Please open the frame block with write permission.

## FRAM-045

(information) Block %s.frame, the feature of blocking core area is set on specific layers. New zero-spacing routing blockages and RC metal are added on layers:%s.

**Description**

The blocking core area feature is honored on specific layers. The tool creates zero-spacing routing blockages and RC metal in the frame view for the entire block except pins.

It is controlled by app_options lib.physical_model.block_all and lib.physical_model.block_core_margin in lm_shell, or controlled by create_frame options -block_all and -block_core_margin in icc2_shell.

**What Next**

Check if added zero-spacing routing blockage and RC metal on specific layers in frame view meets the original design intent. If not, you can modify app_options lib.physical_model.block_all and lib.physical_model.block_core_margin in lm_shell or modify create_frame options -block_all and -block_core_margin in icc2_shell to adjust/remove added zero-spacing routing blockage and RC metal.

Feedback

## FRAM-046

(information) Block %s.frame, the feature of blockage trimming by pin is set to 'touch' on specific layers. All metal shapes and routing blockages touch pins in design view are trimmed on layers:%s.

### Description

The blockage trimming by pin feature is set to 'touch' on specific layers. All metal shapes and routing blockages touch pins are trimmed and coverted to a zero-spacing routing blockage.

Trimming method depends on the option value.

It is controlled by app_options lib.physical_model.trim_metal_blockage_around_pin and lib.physical_model.preserve_metal_blockage in lm_shell, or controlled by create_frame options -trim_metal_blockage_around_pin and -preserve_metal_blockage in icc2_shell.

### What Next

Check if blockage trimming by pin on specific layers in frame view meets the original design intent. If not, you can modify app_options lib.physical_model.trim_metal_blockage_around_pin and lib.physical_model.preserve_metal_blockage in lm_shell, or modify create_frame options -trim_metal_blockage_around_pin and -preserve_metal_blockage in icc2_shell to adjust blockage trimming by pin.

## FRAM-047

(information) Block %s.frame, the feature of blockage trimming by pin is set to 'all' on specific layers. All metal shapes and routing blockages around pins in design view are trimmed on layers:%s.

### Description

The blockage trimming by pin feature is set to 'all' on specific layers. All metal shapes and routing blockages around pins are trimmed and coverted to a zero-spacing routing blockage.

Trimming method depends on the option value.

It is controlled by app_options lib.physical_model.trim_metal_blockage_around_pin and lib.physical_model.preserve_metal_blockage in lm_shell, or controlled by create_frame options -trim_metal_blockage_around_pin and -preserve_metal_blockage in icc2_shell.

Feedback

### What Next

Check if blockage trimming by pin on specific layers in frame view meets the original design intent. If not, you can modify app_options lib.physical_model.trim_metal_blockage_around_pin and lib.physical_model.preserve_metal_blockage in lm_shell, or modify create_frame options -trim_metal_blockage_around_pin and -preserve_metal_blockage in icc2_shell to adjust blockage trimming by pin.

---

## FRAM-048

(warning) Starting with the L-2016.03 release, the default value of app_options lib.physical_model.block_all will be changed from 'false' to 'auto'.

### Description

If you do not set app_options lib.physical_model.block_all explicitly and then the block_all behavior will be changed and it depends on design_type. For analog, black-box, and module design types, the tool creates zero-spacing routing blockages on all layers (true). For macro cells, the tool creates zero-spacing routing blockages only on used layers (used_layers). For all other design types, the tool does not create zero-spacing routing blockages(false).

### What Next

Set app_options lib.physical_model.block_all explicitly and then the tool behavior keeps the same.

---

## FRAM-049

(warning) Starting with the L-2016.03 release, the default value of -block_all option of create_frame command will be changed from 'false' to 'auto'.

### Description

If you do not set -block_all option of create_frame command explicitly and then the block_all behavior will be changed and it depends on design_type. For analog, black-box, and module design types, the tool creates zero-spacing routing blockages on all layers (true). For macro cells, the tool creates zero-spacing routing blockages only on used layers (used_layers). For all other design types, the tool does not create zero-spacing routing blockages(false).

### What Next

Set -block_all option of create_frame command explicitly and then the tool behavior keeps the same.

Feedback

## FRAM-050

(warning) There are color rules defined in the technology file, suggest to enable option 'color_based_dpt_flow' for frame creation.

### Description

The following color rules are defined in the technology file.

sameColorSpanTblXMinSpacing sameColorSpanTblYMinSpacing diffColorSpanTblXMinSpacing diffColorSpanTblYMinSpacing mask1SpanTblXMinSpacing mask1SpanTblYMinSpacing mask2SpanTblXMinSpacing mask2SpanTblYMinSpacing

If option color_based_dpt_flow is not enabled, the metal shapes beyond DRC distance will be generated as colorless shapes. By colorless, it means the attribute mask_constraint of the shape is undefined. (The metal shapes are added by option block_all or block_core_margin in frame creation.)

### What Next

In icc2_shell, enable -color_based_dpt_flow option of create_frame command.

In lm_shell, enable app_options lib.physical_model.color_based_dpt_flow.

## FRAM-050e

(error) There are color rules defined in the technology file, please enable option 'color_based_dpt_flow' for frame creation.

### Description

The following color rules are defined in the technology file.

sameColorSpanTblXMinSpacing sameColorSpanTblYMinSpacing diffColorSpanTblXMinSpacing diffColorSpanTblYMinSpacing mask1SpanTblXMinSpacing mask1SpanTblYMinSpacing mask2SpanTblXMinSpacing mask2SpanTblYMinSpacing

If option color_based_dpt_flow is not enabled, the metal shapes beyond DRC distance will be generated as colorless shapes. By colorless, it means the attribute mask_constraint of the shape is undefined. (The metal shapes are added by option block_all or block_core_margin in frame creation.)

### What Next

In icc2_shell, enable -color_based_dpt_flow option of create_frame command.

In lm_shell, enable app_options lib.physical_model.color_based_dpt_flow.

Feedback

## FRAM-051

(warning) Macro frame %s has colorless shapes on layers:%s. Those layers are with 'doublePattern*' rules in the technology file.

### Description

There are 'doublePattern*' rules defined for certain layers in the technology file. Shapes on those layers in 'module' or 'macro' design should be pre-colored already.

### What Next

Please pre-color shapes in 'module' or 'macro' design before frame creation.

## FRAM-052

(warning) Starting with the M-2016.12 release, the default value of frame option color_based_dpt_flow will be changed from 'false' to 'auto'.

### Description

If you do not set frame option color_based_dpt_flow explicitly and then the metal shapes beyond the DRC distance will be changed from colorless shapes to colored shapes when the following color rules are defined in the technology file. By colorless, it means the attribute mask_constraint of the shape is undefined. (The metal shapes are added by option block_all or block_core_margin in frame creation.)

sameColorSpanTblXMinSpacing sameColorSpanTblYMinSpacing diffColorSpanTblXMinSpacing diffColorSpanTblYMinSpacing mask1SpanTblXMinSpacing mask1SpanTblYMinSpacing mask2SpanTblXMinSpacing mask2SpanTblYMinSpacing

In icc2_shell, whether to create colored metal shapes beyond the DRC distance is controlled by -color_based_dpt_flow of create_frame command.

In lm_shell, whether to create colored metal shapes beyond the DRC distance is controlled by app_options lib.physical_model.color_based_dpt_flow.

### What Next

In icc2_shell, set -color_based_dpt_flow option of create_frame command explicitly and then the tool behavior keeps the same.

In lm_shell, set app_options lib.physical_model.color_based_dpt_flow explicitly and then the tool behavior keeps the same.

Feedback

## FRAM-053

(warning) Layer '%s' with mask name '%s' is not supported by option '%s'. Ignore the setting on the layer.

### Description

The mask name of the input layer is not supported by this frame option.

### What Next

Specify the layer with supported mask name for this frame option.

## FRAM-053e

(error) Layer '%s' with mask name '%s' is not supported by option '%s'.

### Description

The mask name of the input layer is not supported by this frame option.

### What Next

Specify the layer with supported mask name for this frame option.

## FRAM-054

(warning) Technology used to create frame-view and current technology have inconsistency: %s.

### Description

The reported part of the technology are inconsistent between the reference library and the design library.

### What Next

Update the library with consistent technology and try again.

### See Also

- create_lib
- open_lib
- read_tech_file

Feedback

## FRAM-055

(warning) The terminal '%s' of cell '%s' is on layer '%s' lower than the minimum pin layer.

**Description**

The terminal is on a layer lower than the minumum pin layer which is defined by the option -min_pin_layer.

**What Next**

Please update your design and regenerate frame.

## FRAM-056

(warning) The track color information of layer %s is not defined for color wire tracks calculation.

**Description**

The track color information of the specific layer should be defined before with create_track_pattern -mask_pattern. The color wire tracks need to be calculated using this information.

**What Next**

Please use create_track_pattern -mask_pattern to set the color of the wire tracks. See the man page of create_track_pattern for more information.

## FRAM-057

(information) '%s' has '%d' different geometries.

**Description**

There are different geometries between the two frames.

**What Next**

Please check input data.

## FRAM-058

(warning) List of terminals which are on improper color track. Terminal width smaller or equal to default width should be on the same color track, and terminal width larger than default width (fat pin) should be on inverse color track.

Feedback

**Description**

List the terminals which are on improper color track. These wrong setting might impact routing QoR.

**What Next**

Please check the color of the terminals and wire track settings. Fix the color inconsistent issue.

---

## FRAM-059

(warning) There are more than 10000 '%s'.

**Description**

There are more than 10000 different geometries between the two frames. Stop to check this type of error.

**What Next**

Please check input data.

---

## FRAM-060

(information) The error data is saved in '%s'.

**Description**

There are different geometries between the two frames. The error data will be saved in the path.

**What Next**

Please check the error data.

---

## FRAM-061

(error) The input layer of option %s is invalid.

**Description**

The input layer is not a valid layer. Please check your input.

**What Next**

Please check your input and specify the correct layer name.

Feedback

## FRAM-062

(warning) The siteDef %s is %f, thus the input track offset {%s} will be ignored when calculating the track position on layer '%s'.

**Description**

The specified track pattern offset value is bigger than the siteDef width/height. This value will be ignored when calculating the track position.

**What Next**

Please check your setting for the track pattern offset.

## FRAM-063

(warning) In block %s.frame, the port '%s' has many terminals on metal layer (Number of terminals: %d), which may increase the runtime of frame creation. Please check whether all the terminals are expected.

**Description**

The port has many terminals, which may increase the runtime of frame creation.

**What Next**

Please check the terminals for the port. Keep only the necessary terminals may improve the runtime of frame creation.

## FRAM-064

(warning) The port '%s' of cell '%s' has attribute 'pattern_must_join' set as true. However, it has multi-layer terminals on layers '%s', which conflicts the setting.

**Description**

Pattern_must_join is to perform must_join connection for lib pin in strict mode to use one single metal piece to join all the terminals. The reported port has pattern_must_join atrribute set as true but also has multi-layer terminals, which is a conlict setting.

**What Next**

Please update the pattern_must_join attribute or correct your design.

## FRAM-065

(warning) Regular routing blockage %s on metal layer %s touches terminal %s. For blockage purpose, please use zero-spacing routing blockage instead.

Feedback

### Description

Since router need to keep correct spacing with regular routing blockage, then regular routing blockage touches terminal might make router unable to connect the terminal without DRC violation. For blockage purpose, please use zero-spacing routing blockage instead or use option 'convert_metal_blocakge_to_zero_spacing' to convert those blockages into zero-spacing routing blockage during frame creation.

### What Next

Modify your design for set option 'convert_metal_blocakge_to_zero_spacing' and regenerate frame. See convert_metal_blocakge_to_zero_spacing(2).

---

## FRAM-066

(warning) Block %s.frame has a very large regular routing blockage %s on the %s metal layer; this might prevent the router from connecting to nearby terminals.

### Description

The tool issues this warning message when a block contains a very large regular routing blockage. For regular routing blockages, the router must honor the fat spacing rules. This can prevent the router from connecting to terminals near very large regular routing blockages without DRC violations.

### What Next

To improve the routability near a very large routing blockage, use a zero-spacing routing blockage, which does not need to honor the fat spacing rules, instead of a regular routing blockage.

To create a zero-spacing routing blockage, use the *-zero_spacing* option with the *create_routing_blockage* command. To convert regular routing blockages to zero-spacing routing blockages during frame view generation,

- In lm_shell, set the *lib.physical_model.convert_metal_blockage_to_zero_spacing* application option before running the *commit_workspace* command.
- In icc2_shell, use the *-convert_metal_blockage_to_zero_spacing* option with the *create_frame* command.

### See Also

- create_frame
- create_routing_blockage
- lib.physical_model.convert_metal_blockage_to_zero_spacing

Feedback

## FRAM-067

(warning) Invalid mode specified for '%s', mode should be one of %s. Ignore the setting on layer '%s'.

**Description**

The option value has invalid mode string, please see the message and man page for the usage of the option.

**What Next**

Please update the input value.

## FRAM-067e

(error) Invalid mode specified for '%s', mode should be one of %s. Invalid setting on layer '%s'.

**Description**

The option value has invalid mode string, please see the message and man page for the usage of the option.

**What Next**

Please update the input value.

## FRAM-068

(error) Input format for option '%s' is wrong; The correct format is '%s'.

**Description**

The input format for the option is wrong. Please see the message or man page for the correct input format.

**What Next**

Provide the correct intput format to the option.

## FRAM-069

(warning) There is no via region created in block '%s.frame'. Please check your design and settings.

Feedback

**Description**

There is no via region created for design, which may cause routing QoR issue.

**What Next**

Please check your design and settings and regenerate frame.

---

## FRAM-069e

(error) There is no via region created for port '%s' in block '%s.frame'. Please check your design and settings.

**Description**

There is no via region created for the port in the design, which may cause routing QoR issue.

**What Next**

Please check your design and settings and regenerate frame.

---

## FRAM-070

(warning)There are improper regular routing blockages which touch terminals or are extra-large. For blockage purpose, please use zero-spacing routing blockage instead.

**Description**

Since router needs to keep correct spacing with regular routing blockage, then regular routing blockage touches terminal or being extra-large might make router unable to connect the terminal without DRC violation. For blockage purpose, please use zero-spacing routing blockage instead or use option 'convert_metal_blocakge_to_zero_spacing' to convert those blockages into zero-spacing routing blockage during frame creation.

**What Next**

Modify your design for set option 'convert_metal_blocakge_to_zero_spacing' and regenerate frame. See convert_metal_blocakge_to_zero_spacing(2).

---

## FRAM-071

(warning) Error in %s.frame, the zero-spacing routing blockage around Z-shape is too small on layer '%s'. Z-shape object_class: '%s'; %s.

Feedback

**Description**

The zero-spacing routing blockage around Z-shape is too small for the keepout rule, which may cause routing QoR issue.

**What Next**

Please check your design and settings and regenerate frame.

---

## FRAM-072

(warning) In %s.frame, the zero-spacing routing blockage around Z-shape is too large on layer '%s'. Z-shape object_class: '%s'; %s.

**Description**

The zero-spacing routing blockage around Z-shape is too large for the keepout rule, which may cause routing QoR issue.

**What Next**

Please check your design and settings and regenerate frame.

---

## FRAM-073

(warning) Error in %s.frame, the zero-spacing routing blockage under Z-shape is too small on lower via layer '%s'. Z-shape object_class: '%s'; %s.

**Description**

The zero-spacing routing blockage under Z-shape is too small on the lower via layer, which may cause routing QoR issue.

**What Next**

Please check your design and settings and regenerate frame.

---

## FRAM-074

(warning) In %s.frame, the zero-spacing routing blockage under Z-shape is too large on lower via layer '%s'. Z-shape object_class: '%s'; %s.

**Description**

The zero-spacing routing blockage under Z-shape is too large on the lower via layer, which may cause routing QoR issue.

Feedback

**What Next**

Please check your design and settings and regenerate frame.

---

## FRAM-075

(warning) All terminals of the port '%s' of cell '%s' are on a single metal layer '%s'. However, the 'pattern_must_join' attribute is not set as true on this port, which makes conflicts.

**Description**

If all the terminals of a lib pin are on the same metal layer, then the 'pattern_must_join' attribute should be set as true on this lib pin.

**What Next**

Please update the 'pattern_must_join' attribute or correct your design.

---

## FRAM-076

(warning) Option value '%s' is set more than once in option '%s'. Keep the first setting value for frame creation, and skip the later setting value.

**Description**

If option value is specified with the same "key" for multiple times, the later setting value won't replace the first one.

For example:

set_app_options -name lib.physical_model.block_core_margin -value {{M1 0.5} {M2 0.3} {M1 0.3}} -as_user_default

The margin for M1 is the first value 0.5 for frame creation.

**What Next**

Please correct the duplicated settings in the option value list.

---

## FRAM-076e

(error) Option value '%s' is set more than once in option '%s'. Please correct the setting.

**Description**

The option value is specified with the same "key" for multiple times, please correct the setting.

Feedback

**What Next**

Please correct the duplicated settings in the option value list.

---

## FRAM-077

(warning) Via region %s of port %s is not associated to any via definition.

**Description**

There exists via region with no associated via definition, which might cause tool crash in further P&R stage. The missing via definition might be caused by removing via definition from library or tech after frame view creation.

**What Next**

Please regenerate the frame view.

---

## FRAM-078

(warning) %s %s in design %s is off track. Skip the end of preferred grid status reporting.

**Description**

The shape or regular blockage is either on the wrong color track or the center line of it is not aligned with the track. Zroute cannot do proper metal extension to such shapes/ blockages. Skip the end of preferred grid status reporting.

For off-track terminals, please see off-track condition using the report of report_lib -wire_track_colors.

**What Next**

Please check your design and track pattern setting.

---

## FRAM-079

(warning) %s %s in design %s has line ends in between low end grid and high end grid.

**Description**

The space in between the low end grid and high end grid is the potential space for cut metal insertion. However, the line end of the shape or regular blockage ends in between the low end grid and high end grid. This will cause issue when Zroute tries to do metal extension to meet the cut metal rule. The metal extension will either fail or waste routing resource.

Feedback

### What Next

Please check your design and fix the issue.

---

## FRAM-080

(warning) End of preferred grid setting on layer %s is invalid. Skip the checking for the shapes on this layer.

### Description

The end of preferred grid setting on the layer makes the low grids and high grids unable to make reasonable pairs. So the input end of preferred grid is invalid and the checking is skipped for the shapes and obstacles on this layer.

### What Next

Please check the track pattern, especially the low_offsets, high_offsets, low_steps, and high_steps values. See report_track_patterns(2)

---

## FRAM-081

(warning) Site '%s' has no end of preferred grid pattern on it. Skip the checking for the designs associated to this site.

### Description

There is no end of preferred grid pattern defined on the site. So the tool cannot do the checking for designs associated to this site. Please use create_track_pattern -site SITE \ -end_grid_high_steps {...} \ -end_grid_low_steps {...} \ -end_grid_high_offset {...} \ -end_grid_low_offset {...} to create the end of preferred grid pattern.

### What Next

Please create the correct track pattern with end of preferred grid information to the site. See create_track_pattern(2)

---

## FRAM-082

(error) The inconsistency between current technology and the technology used to create frame view is not acceptable for deriving via regions.

### Description

For deriving via regions, the only acceptable inconsistency between current technology and the technology used to create frame is contact code difference or via region creation

Feedback

related design rules. For other inconsistency, the whole frame should be regenerated instead of only update the via regions.

**What Next**

Update the library with consistent technology or regenerate the frame.

**See Also**

- read_tech_file
- check_workspace

---

## FRAM-083

(information) Block %s.frame, the feature of blocking via layers is set for the blocked metal layers. New zero-spacing routing blockages are added on layers:%s.

**Description**

The blocking via layers feature is honored for the blocked metal layers. The tool creates zero-spacing routing blockages on upper, lower or both via layers in the frame view for the entire block.

It is controlled by app_options lib.physical_model.create_zero_spacing_via_blockages_for_blocked_layers in lm_shell, or controlled by create_frame options -create_zero_spacing_via_blockages_for_blocked_layers in icc2_shell.

The blocked area on via layers are controlled by app_options lib.physical_model.block_all and lib.physical_model.block_core_margin in lm_shell, or controlled by create_frame options -block_all and -block_core_margin in icc2_shell.

**What Next**

Check if added zero-spacing routing blockage on specific via layers in frame view meets the original design intent. If not, you can modify app_options lib.physical_model.create_zero_spacing_via_blockages_for_blocked_layers in lm_shell or modify create_frame options -create_zero_spacing_via_blockages_for_blocked_layers in icc2_shell to adjust/remove added zero-spacing routing blockage.

---

## FRAM-084

(information) The feature of 'creat_zero_spacing_via_blockages_for_blocked_layers' and 'enable_via_regions_for_all_design_types' are conflict. Do not cut extra pin channel on upper via layer by pin direction.

Feedback

**Description**

When the feature of 'enable_via_regions_for_all_design_types' is specified, the tool checks pin direction and cuts extra channel on upper via layer if necessary. But when 'create_zero_spacing_via_blockages_for_blocked_layers' is specified, it intends to connect pins on the same layer without dropping vias. The two features are conflict, so the tool skip cutting extra pin channel on upper via layer by pin direction.

**What Next**

Check if added zero-spacing routing blockage on via layers in frame view meets the original design intent. If not, you can modify option 'create_zero_spacing_via_blockages_for_blocked_layers'. Or check if via regions are required for the design. If not, you can modify option 'enable_via_regions_for_all_design_types'.

---

## FRAM-085

(warning) There is no frame view in library '%s', skip derive_via_regions for this library.

**Description**

Via regions are only created on frame view. Thus, if there is no frame view in the library, then the derive via region process will be skipped.

**What Next**

Check your input library.

---

## FRAM-086

(warning) The boundary of block '%s' is not rectangular, skip 'layer_bbox' mode for layer '%s' in option pin_channel_boundary_modes.

**Description**

The tool does not support 'layer_bbox' mode for non-rectangular boundary in option pin_channel_boundary_modes. The option setting for the layer is skipped.

**What Next**

Check your input library.

---

## FRAM-086e

(error) The boundary of block '%s' is not rectangular, cannot set 'layer_bbox' mode for layer '%s' in option pin_channel_boundary_modes.

Feedback

**Description**

The tool does not support 'layer_bbox' mode for non-rectangular boundary in option pin_channel_boundary_modes.

**What Next**

Check your input library.

---

## FRAM-087

(warning) Library '%s' is an aggregate library. Please change current_lib to each sub-library and do the techfile replacement and derive_via_regions to each sub-library.

**Description**

In an aggregate library, each sub-library has its own associated technology. Thus, the techfile replacement and derive_via_regions operations should be done on each sub-library separately.

**What Next**

Please change current_lib to each sub-library and do the techfile replacement and derive_via_regions to each sub-library.

---

## FRAM-088

(information) Using technology '%s' to derive design library via region in the frame-views of library '%s'.

**Description**

Use the technology from the design library to derive via regions on the frame views in the reference library. These new derived via regions are only at the design library, and are not saved to the reference library. Zroute will honor these design-library via regions during routing.

---

## FRAM-089

(warning) Invalid pair value '%s' is specified for option '%s'. Ignore the setting on the layer.

**Description**

The input pair value in list has invalid format. The number or type may be incorrect. Ignore the setting on the spcified layer.

Feedback

**What Next**

Please check the command man page and correct the usage.

---

## FRAM-089e

(error) Invalid pair value '%s' is specified for option '%s' on the layer.

**Description**

The input pair value in list has invalid format. The number or type may be incorrect.

**What Next**

Please check the command man page and correct the usage.

---

## FRAM-090

(warning) In block '%s', layer '%s' has multiple islands for the boundary or more than 5000 shapes. Skip 'layer_boundary' mode on this layer in option pin_channel_boundary_modes and use defautl mode 'block_boundary' instead.

**Description**

For 'layer_boundary' mode in option pin_channel_boundary_modes, the tool does not support multiple islands or more than 5000 shapes. The tool will skip the option setting on this layer and use default mode 'block_boundary' instead.

**What Next**

Check your input library.

---

## FRAM-091

(warning) Input design %s does not belong to lib %s. Ingore this design.

**Description**

The input design of -cells option need to belong to the input library. Designs from other libraries are ignored.

**What Next**

Please correct your input.

---

## FRAM-092

(warning) create_frame command option %s will be obsolete in future release %s.

Feedback

**Description**

The command option will be obsolete in future release, please check the usage of this option.

**What Next**

Please check the usage of the command option.

---

## FRAM-093

(warning) The app_option lib.physical_model.* is set. For frame creation, please use "create_frame -honor_physical_model_app_options true". The app_options lib.physical_model.* will be obsolete in future release %s. Please use create_frame command options instead.

**Description**

The app_options lib.physical_model.* will be obsolete in future release, please use create_frame command options instead.

**What Next**

Please replace the usage of app_options lib.physical_model.* by create_frame command options.

---

## FRAM-094

(warning) When create_frame option %s is specified, the setting of option %s is ignored and using default value instead.

**Description**

The command does not support certain option combination. Please check the usage of command options.

**What Next**

Please verify the usage of command options.

---

## FRAM-094e

(error) When create_frame option %s is specified, cannot support option %s at the same time.

Feedback

**Description**

The command does not support certain option combination. Please check the usage of command options.

**What Next**

Please verify the usage of command options.

---

## FRAM-095

(warning) Block '%s' has non-rectlinear cell boundary. The '%s' mode in option via_region_trim_by_cell_boundary for layer '%s' will be ignored and use default mode 'all' instead.

**Description**

For 'horizontal' and 'vertical' mode in option via_region_trim_by_cell_boundary, the tool only support it when cell boundary is rectlinear. For non-rectlinear cell boundary, tool will use the default mode 'all' instead.

**What Next**

Please set 'all' or 'none' mode in option via_region_trim_by_cell_boundary for non-rectlinear cell boundary.

---

## FRAM-095e

(error) Block '%s' has non-rectlinear cell boundary. The tool cannot support the '%s' mode in option via_region_trim_by_cell_boundary for layer '%s'.

**Description**

For 'horizontal' and 'vertical' mode in option via_region_trim_by_cell_boundary, the tool only support it when cell boundary is rectlinear. For non-rectlinear cell boundary, tool only support the default mode 'all'.

**What Next**

Please set 'all' or 'none' mode in option via_region_trim_by_cell_boundary for non-rectlinear cell boundary.

---

## FRAM-096

(warning) For, block '%s, pin shape of '%s' is not a rectangle. The '%s' mode in option via_region_trim_by_cell_boundary will be ignored and use default mode 'all' instead.

Feedback

**Description**

For 'horizontal' and 'vertical' mode in option via_region_trim_by_cell_boundary, the tool only support it for rectangular pin. For non-rectangular pin, tool will use the default mode 'all' instead.

**What Next**

Please set 'all' or 'none' mode in option via_region_trim_by_cell_boundary for the layer which non-rectangular pin is on it.

---

## FRAM-096e

(error) For, block '%s, pin shape of '%s' is not a rectangle. The tool cannot support the '%s' mode in option via_region_trim_by_cell_boundary.

**Description**

For 'horizontal' and 'vertical' mode in option via_region_trim_by_cell_boundary, the tool only support it for rectangular pin. For non-rectangular pin, tool only support the default mode 'all'.

**What Next**

Please set 'all' or 'none' mode in option via_region_trim_by_cell_boundary for the layer which non-rectangular pin is on it.

---

## FRAM-097

(information) Skip creating frame view for block '%s', because the design_type is '%s'. %s

**Description**

Frame view is not necessary for some design_type, skip creating frame view for these designs.

**What Next**

Please check the design_type for the frame creation process.

---

## FRAM-098

(info) Block '%s' has null design bbox, skip the frame creation process.

**Description**

The design bbox is null, the block may have no physical data. Skip the frame creation process.

Feedback

**What Next**

Please check the block for the frame creation process.

---

## FRAM-099

(info) Block '%s' contains blkInst with null bbox. The refBlk is '%s' and it may have no physical data.

**Description**

The design bbox of the refernce block is null, it may have no physical data. Skip reading it in the frame creation process.

**What Next**

Please check the reference block for the frame creation process.

---

## FRAM-100

(warning) The technology information used to create frame-view is not available in library '%s'. Need to derive design library via region for this library.

**Description**

Technology information used to create frame-view is stored in the library in version later than K-2015.06-SP1. Reference library created using older version does not have the information, thus it is impossible to compare between the technology used to create frame-view and current technology. It is better to derive design library via region for this library.

**What Next**

Check if derive design library via regions are generated in later steps.

---

## FRAM-101

(warning) Frame '%s' has colorless shapes or colorless regular routing blockages on layers: '%s'. The number of mask setting is larger than one on those layers.

**Description**

The number of mask setting is larger than one for certain layers. Shapes and regular routing blockages on those layers should be pre-colored already.

**What Next**

Please pre-color shapes and regular routing blockages before frame creation.

Feedback

## FRAM-102

(warning) Drop colorless shapes and colorless regular routing blockages on DPT layers: '%s', when app option 'lib.setting.handle_noncolored_shapes' is in 'strict' mode.

**Description**

When 'lib.setting.handle_noncolored_shapes' is in 'strict' mode, drop colorless shapes and colorless regular routing blockages on dpt layers in frame creation.

**What Next**

Please pre-color shapes and regular routing blockages before frame creation.

## FRAM-103

(warning) Incorrect setting for port '%s' in option '%s', the setting is ignored for the port.

**Description**

The option value format is incorrect, and the setting is ignored for the port. The tool will use default option value for the port in frame creation.

**What Next**

Please look up the option man page and correct the option setting.

## FRAM-103e

(error) Incorrect setting for port '%s' in option '%s'.

**Description**

The option value format is incorrect for the port, user needs to correct the setting for frame creation.

**What Next**

Please look up the option man page and correct the option setting.

## FRAM-104

(warning) Fail to merge special router extension or patching net shape to pin '%s'.

Feedback

**Description**

Frame option "merge_extension_to_pin" or "merge_patch_to_pin" is turned on, but the tool cannot merge the special router extension or patching net shape to the pin. It may have color/net conflict or fail in deleting/creating shape/pin.

**What Next**

Please check the design or disable option "merge_extension_to_pin" or "merge_patch_to_pin".

---

## FRAM-105

(warning) Drop non-pin curved shape %s in frame view.

**Description**

The non-pin curved shapes in design view are dropped in frame view. Frame view only carries curved pin shapes from design view.

---

## FRAM-106

(error) Design '%s' has colorless shapes or colorless regular routing blockages on DPT layers: '%s'. Exit frame creation in 'strict' mode.

**Description**

When 'lib.setting.check_mask_constraints' is in 'strict' mode, exit frame creation when colorless shapes or colorless regular routing blockages are on DPT layers.

**What Next**

Please pre-color shapes and regular routing blockages on those layers before frame creation.

---

## FRAM-107

(warning) Frame '%s' has color shapes or color regular routing blockages on layers: '%s'. The number of mask setting is 0 or 1 on those layers.

**Description**

The number of mask setting is 0 or 1 for certain layers. Do not pre-color shapes and regular routing blockages on those layers.

Feedback

**What Next**

Please remove color constraints on shapes and regular routing blockages before frame creation.

---

## FRAM-108

(error) Design '%s' has color shapes or color regular routing blockages on non-DPT layers: '%s'. Exit frame creation in 'strict' mode.

**Description**

When 'lib.setting.check_mask_constraints' is in 'strict' mode, exit frame creation when color shapes or color regular routing blockages are on non-DPT layers.

**What Next**

Please remove color constraints on shapes and regular routing blockages before frame creation.

---

## FRAM-109

(information) Allow design library via region to extend by routing direction for port '%s' in block '%s'.

**Description**

Allow design library via region to extend by routing direction for the pin shape.

**What Next**

None

---

## FRAM-110

(warning) There are more than %d non-pin shapes on layer %s, drop those shapes in frame view for P&R runtime consideration.

**Description**

Too many shapes or routing blockages might cause long runtime issue in further P&R stage, the tool drops those non-pin shapes by default.

**What Next**

Please specify block_all or block_core_margin for adjacent metal layers, which can reduce the via layer shapes or routing blockages.

Feedback

If there are long PG terminals above the via layer, then the shapes and routing blockages under the terminals are preserved for DRC checking. It is fine to drop them if there is no DRC concern, but if you still need to keep those details on via layers, please change the internal threshold value.

---

## FRAM-111

(warning) There are more than %d RC metal shapes on layer %s, skip generating RC metal shapes in frame view for P&R runtime consideration.

**Description**

Too many RC metal shapes might cause long runtime issue in further P&R stage, the tool skips generating the RC metal shapes by default.

**What Next**

Please check if there are long PG terminals across the design, which will trim a large piece of RC metal as many fragmented shapes.

If you still need to output the RC metal shapes, please change the internal threshold value.

---

## FRAM-112

(warning) Option %s does not support backside layer, ignore layer %s of this option.

**Description**

Some frame view features do not support with backside layer.

**What Next**

Please remove backside layer from the option.

---

# GCR

---

## GCR-001

(error) Could not execute command '%s' because router is not initialized.

**Description**

The Custom Router is not initialized so the command cannot be executed.

Feedback

### What Next

This message doesn't apply to IC Compiler II, as the Custom Router will automatically initialize whenever the user executes a command which requires it.

### See Also

- route_custom

---

## GCR-002

(error) '%s' execution failed with message: '%s'

### Description

An internal Custom Router command has failed. The reason will be displayed.

### See Also

- route_custom

---

## GCR-003

(error) The router private command '%s' requires at least one of the %s features, which are not currently checked out.

### Description

One or more required licenses was not available.

### What Next

Ensure that the proper licenses have been installed. Check that other users are not using all the licenses.

### See Also

- route_custom

---

## GCR-004

(error) The router could not be initialized because there are no layers with material %s in the database.

Feedback

### Description

When reading the technology data, there were no layers with the specified material defined.

### What Next

Make sure the routes are on metal layers. Make sure the layers are defined in the technology data. For instructions, see the *Library Manager User Guide*.

### See Also

- route_custom

---

## GCR-006

(error) The license feature %s could not be checked out because %s.

### Description

Either the feature is not installed, or the license keys for the feature are in use by other users.

### What Next

The most common situation is that all licenses are taken by other users. Make sure the feature you want to use is installed correctly, or wait until other users have finished using the feature.

### See Also

- route_custom

---

## GCR-007

(info) The command was interrupted.

### Description

The user pressed Ctrl-C during a command.

### What Next

Rerun the command if desired.

### See Also

- route_custom

Feedback

## GCR-009

(info) The router was initialized with stop level %d.

### Description

This informational message indicates the stop level used to initialize the custom router.

### What Next

This is an informational message. Verify that the stop level is proper for your tasks and continue routing.

### See Also

- route_custom

## GCR-010

(error) The router could not be initialized.

### Description

There is a problem with the data and the Custom Router could not initialize itself.

### See Also

- route_custom

## GCR-011

(error) The router could not be initialized because an Open Access exception occurred: %s.

### Description

There is a problem with the database and the Custom Router tool could not initialize itself.

### What Next

Verify that your database has the correct permissions. Verify that network drives are connected and accessible on the system you are using.

### See Also

- route_custom

Feedback

## GCR-012

(error) The specified path style %s is not valid because %s.

### Description

The string entered does not match the valid path style values. Allowed style values are *truncate*, *extend*, *round*, and *variable*.

### What Next

Check your script for the correct spelling.

### See Also

- route_custom

## GCR-013

(error) The specified path segment end style %s is not valid because %s.

### Description

The string entered does not match the valid path styles. Allowed styles are *truncate*, *extend*, *variable*, *chamfer*, and *custom*.

### What Next

Check your script for the correct spelling.

### See Also

- route_custom

## GCR-014

(error) The router could not create a path segment because %s.

### Description

An invalid set of endpoints, an incorrect width, or an incorrect extension value is preventing the router from creating path segments.

### See Also

- route_custom

Feedback

## GCR-015

(error) The router could not create a path because %s.

### Description

An invalid point list (with duplicate points or collinear points), an incorrect width, or an incorrect extension value is preventing the router from creating path segments.

### See Also

- route_custom

## GCR-016

(error) The router could not create a %s via because %s.

### Description

The router tried to use an incorrect via definition name, the router could not access the via definition when trying to create the via, or the coordinates were illegal.

### See Also

- route_custom

## GCR-018

(error) The router could not be initialized because a general exception occurred.

### Description

A general exception error is preventing the router from initializing.

### See Also

- route_custom

## GCR-019

(warning) %s %s has no PR boundary defined.

### Description

The design does not have a boundary.

Feedback

**What Next**

Create a boundary for the design.

**See Also**

- route_custom

---

## GCR-020

(warning) %s rule for layer %s has a wire width of 0, which is not allowed.

**Description**

In the technology library, the min_width setting for the layer is set to 0.

**What Next**

In the technology library, set the min_width to a value > 0 for the layer.

**See Also**

- route_custom

---

## GCR-021

(warning) Layer %s has no minimum width constraint.

**Description**

The minimum width constraint is not defined for the layer in the technology data.

**What Next**

In the technology data, set the minimum width for the layer to a value greater than 0.

**See Also**

- route_custom

---

## GCR-022

(warning) Layer %s has a maximum width of %d which is less than the minimum width of %d.

**Description**

The maximum width for a layer must be greater than its minimum width.

Feedback

**What Next**

In the technology data, set the maximum width to a value greater than the minimum width value.

**See Also**

- route_custom

---

## GCR-023

(warning) Layer %s has no minimum spacing constraint.

**Description**

The minimum spacing constraint is not defined for the layer in the technology data.

**What Next**

In the technology data, define the minimum spacing constraint and set it to a value greater than 0 for the layer.

**See Also**

- route_custom

---

## GCR-024

(warning) %s rule for layer %s has a minimum spacing with value 0, which is not allowed.

**Description**

The minimum spacing setting for the layer is set to 0 in the technology data.

**What Next**

In the technology data, set the minimum spacing to a value greater than 0 for the layer.

**See Also**

- route_custom

---

## GCR-025

(warning) Layer %s has a minimum spacing of value 0 at 2D spacing table index (%d, %d), which is not allowed.

Feedback

### Description

The system is expecting a min_spacing value greater than 0 at the location in the 2D spacing table.

### What Next

In the technology library, set the min_spacing value greater than 0 at the location in the 2D table.

### See Also

- route_custom

---

## GCR-026

(warning) Layer %s has a minimum spacing of value 0 at 1D spacing table index %d, which is not allowed.

### Description

The system is expecting a min_spacing value greater than 0 at the location in the 1D table.

### What Next

In the technology library, set the min_spacing value greater than 0 at the location in the 1D table.

### See Also

- route_custom

---

## GCR-027

(warning) Layer %s has an invalid %s routing pitch %f. The pitch cannot be smaller than %f, which is minimum width plus minimum spacing.

### Description

An invalid pitch value is specified in the technology library.

### What Next

In the technology library, set the routing pitch to a value that is min_width plus min_spacing.

Feedback

**See Also**

- route_custom

---

## GCR-028

(warning) Layer %s has a manufacturing grid of 0, which is not allowed.

**Description**

The manufacturing grid in the technology library for this layer is 0.

**What Next**

In the controls section of the technology file, make sure the manufacturing grid resolution is greater than 0.

**See Also**

- route_custom

---

## GCR-029

(warning) Layer %s has 0 size minimum number cut table, which is not allowed.

**Description**

The via reliability constraint for the specified layer has no cut value.

**What Next**

Review the constraints and ensure the minimum number of cuts table for the specified layer is not empty.

**See Also**

- route_custom

---

## GCR-030

(warning) Layer %s has 0 value in minimum number cut table at index %d, which is not allowed.

**Description**

The via reliability constraint for the specified layer has a cut value of 0.

Feedback

### What Next

Review the constraints and ensure the minimum number of cuts table for the specified layer has values greater than 0.

### See Also

- route_custom

---

## GCR-031

(warning) Instance %s does not have a valid placement status.

### Description

The placement status of the instance is unplaced.

### What Next

Place the instance in the design, or change the placement status property in the property editor.

### See Also

- route_custom

---

## GCR-032

(warning) Pin %s does not have a valid placement status.

### Description

The pin belongs to an instance that has a status of unplaced.

### What Next

Place the pin and/or instance in the design, or change the placement status property in the property editor.

### See Also

- route_custom

---

## GCR-033

(warning) Net %s has no pin.

Feedback

### Description

The net is missing pins, so routes cannot be created. Routes requires pins to form connections. Pins can be top level pins or ports from the top level into lower level instances.

### What Next

Add pins on the specified net or create ports to your lower level instances.

### See Also

- route_custom

---

## GCR-034

(warning) %s %s has shielding gap %f smaller than minimum spacing %f.

### Description

The distance between the signal and shield is less than the default minimum spacing value.

### What Next

Check the shielding constraints and increase the gap value.

### See Also

- route_custom

---

## GCR-035

(warning) %s %s has shielding width %f smaller than minimum width %f.

### Description

The shield width is narrower than the default minimum width value.

### What Next

Check the shield constraints and ensure that the width value is at least equal to the minimum width value.

### See Also

- route_custom

Feedback

## GCR-036

(warning) %s has net match length constraint with max value %f less than minimum value %f.

### Description

The maximum value is less than the minimum value for the length constraint.

### What Next

Correct the minimum and maximum values for the length constraint.

### See Also

- route_custom

## GCR-037

(warning) The via stack limit cannot be less than 1.

### Description

The via stack limit constraint is set to a value less than 1 in the technology library.

### What Next

Check the technology library and set the via stack limit to a value of 1 or higher.

### See Also

- route_custom

## GCR-038

(error) The design has no via definitions that are suitable for routing.

### Description

The vias used in the design are not defined in the technology library.

### What Next

Review your technology library. Check that the vias are defined as desired, and create new via definitions if necessary.

### See Also

- route_custom

Feedback

## GCR-039

(error) %s has no valid routing layers.

**Description**

No valid routing layers are selected.

**What Next**

Check the valid routing layer constraint. Check the technology library for the layer definitions. Check that metal layers are selected for use.

**See Also**

- route_custom

## GCR-040

(warning) %s has no vertical routing layers.

**Description**

None of the eligible routing layers has vertical direction.

**What Next**

Check the layer directions in the layers constraint and on the router setup dialog. Ensure that at least one layer has vertical direction.

**See Also**

- route_custom

## GCR-041

(warning) %s has no horizontal routing layers.

**Description**

None of the eligible routing layers has horizontal direction.

**What Next**

Check the layer directions in the layers constraint and on the router setup dialog. Ensure that at least one layer has horizontal direction.

Feedback

**See Also**

- route_custom

---

## GCR-043

(warning) %s use via %s with cut %s number %d, which is not allowed.

**Description**

Either the number of row or the number of columns in the via is 0, which is not allowed.

**What Next**

Ensure your array vias have at least one cut per row and column.

**See Also**

- route_custom

---

## GCR-044

(warning) %s use via with minimum cut number %d at layer %s, which is not allowed.

**Description**

In the via reliability constraint, the number of cuts value for the specified layer is less than 1.

**What Next**

Check the constraints and set the number of cust value to 1 or greater in the via reliability constraint.

**See Also**

- route_custom

---

## GCR-045

(warning) Instance %s has no master.

**Description**

The instance does not have a master. The master could be unbound or missing from your library.

Feedback

**What Next**

Create a master cell for the instance. Check your library references and ensure that all the data is accessible.

**See Also**

- route_custom

---

## GCR-046

(warning) Instance %s is placed outside of design boundary.

**Description**

The instance has placed status but is incorrectly placed outside the design boundary.

**What Next**

Move the instance within the boundary, or enlarge the design boundary to enclose the instance.

**See Also**

- route_custom

---

## GCR-047

(warning) Pin %s is placed outside of design boundary.

**Description**

The pin is incorrectly placed.

**What Next**

Move the pin within the boundary, check the placement of its instance, or enlarge the design boundary to enclose the instance and all of its pins.

**See Also**

- route_custom

---

## GCR-048

(warning) Via constraint for %s contains via %s which has no valid via def. This will not be used by router.

Feedback

### Description

A via you selected in the Vias constraint has a missing or invalid via definition. The router is unable to use this via.

### What Next

In the Constraint Editor, check the via selections in the Vias constraint. Make sure that selected vias have valid via definitions or remove them from the Vias Constraint.

### See Also

- route_custom

---

## GCR-049

(warning) %s shorts with %s on layer %s at %s.

### Description

Two pins short circuited on the specified layer. Routing results may be affected.

### What Next

Change the placement of one instance so the pins no longer cause short circuitry.

### See Also

- route_custom

---

## GCR-050

(warning) Design %s has no net.

### Description

The design does not have any nets. You can only create routes in designs that have nets.

### What Next

Create nets in the design. If you are using a routing area, please ensure that it is defined correctly.

### See Also

- route_custom

Feedback

## GCR-053

(warning) Net %s shorts with net %s on layer %s at %s.

**Description**

Two shapes short circuited on the specified layer.

**What Next**

Check the overlapping shapes and move one of them top fix the short circuitry.

**See Also**

- route_custom

## GCR-054

(warning) Layer %s doesn't have a standard via available to move up.

**Description**

No standard via definition was found for the specified layer.

**What Next**

Select a standard via for the layer in the vias constraint.

**See Also**

- route_custom

## GCR-057

(warning) Layer %s has %s routing pitch %f bigger than %f, which is 10x(width+spacing).

**Description**

The given route pitch is very large compared to the width and spacing rules.

**What Next**

Check the pitch value in the grids constraint.

**See Also**

- route_custom

Feedback

## GCR-058

(error) Cannot define Group %s. Net is either NULL or pins have different net names.

### Description

The attempt to define a group from a topology_edge failed because a pin has a NULL net or a different net from the other pin.

### What Next

Check the definition of your toplogy routing to be sure it defines the correct end points.

### See Also

- route_custom

## GCR-059

(warning) Net %s has shielding gap %f but the clearance for width %f is %f in layer %s spacing table. Loss of shielding can occur.

### Description

The shielding gap conflicts with the layer width and layer spacing table.

### What Next

Check the gap parameters in the net shielding constraints.

### See Also

- route_custom

## GCR-061

(error) Shielding net %s but it does not contain any valid shield rules.

### Description

A net was specified for shielding in the command net list, but the net does not have a valid shield constraint associated with it.

### What Next

Check that you have selected the correct nets for routing. Check the shielding constraints to ensure that the net you wish to shield has a shield constraint defined.

Feedback

**See Also**

- route_custom

---

## GCR-062

(info) Custom Router Antenna Report

**Description**

This is the header of the antenna report.

**What Next**

This is an informational message and no action is required.

**See Also**

- route_custom

---

## GCR-063

(info) %d Net%s %sSuccessfully Modified to Correct Antenna Violations

**Description**

This message indicates how nets were successfully modified and/or could not be successfully modified to correct process antenna violations.

**What Next**

This is an informational message and no action is required. For nets which were not successfully modified, the net may have to be manually edited to correct the process antenna effect.

**See Also**

- route_custom

---

## GCR-064

(error) Unable to write detailed antenna report because file %s could not be opened.

**Description**

The attempt to write the report fail failed. The disk might be full, the directory might not have write permission, or the file might already exist and cannot be overwritten.

Feedback

### What Next

Check your disk space and the write permissions for the directory and report file.

### See Also

- route_custom

---

## GCR-066

(warning) Shielding net %s has no pin. No shield vias can be added.

### Description

The net selected to shield your connections does not have any valid place to insert the shield vias.

### See Also

- route_custom

---

## GCR-067

(warning) Net %s has %f shield gap, but coaxial shielding width %f requires %f clearance in layer %s spacing table. Please change shield constraint to split-coaxial.

### Description

A coaxial shield requires more space for the specified width.

### What Next

Check the coaxial shielding style constraint definition. Select the split-coaxial shielding style if possible or increase the shield gap value.

### See Also

- route_custom

---

## GCR-068

(warning) Width %.10g is not an even multiple of the layer manufacturing grid. Router will modify width to %.10g.

### Description

The width must be an even multiple of the manufacturing grid to aviod errors in precision.

Feedback

### What Next

The router automatically adjusts the width to the value shown.

### See Also

- route_custom

---

## GCR-069

(warning) Cannot complete Interactive Router solution at the specified point. Topology constraints on the net\n prevent the creation of the topology that would result from completing a route\n at the specified point with the specified start.\n If applicable, ensure that the route start point is within the Steiner geometry.

### Description

The Interactive Router cannot end at the specified point because of the given topology specifications.

### What Next

If attempting to begin or end the router at a topoogy node, ensure that the specified point is within the geometry of the node.

### See Also

- route_custom

---

## GCR-070

(warning) Cannot start Interactive Router from the specified point. Topology constraints on the net\n prevent the creation of the topology that would result from completing a route\n at the specified point.

### Description

The Interactive Router cannot begin at the specified point because of the given topology specifications.

### What Next

If attempting to being or end the router at a topology node, ensure that the specified point is withing the geometry of the node.

### See Also

- route_custom

Feedback

## GCR-071

(warning) Cannot start Interactive Router from the specified point. No object was selected or the selected object is not a routable object.

### Description

The Interactive Router did not detect a valid object at the selected point. Either there was no object or the object is not valid for routing.

### What Next

Click at an appropriate point, such as a pin, topology node, or wire and try again.

### See Also

- route_custom

## GCR-072

(warning) Cannot complete Interactive Router operation. Cannot find a legal snapping point to nearest routable target.

### Description

The Interactive Router did not detect a valid target point for routing.

### What Next

Click close to an appropriate point, such as a pin, topology node, or wire and try again.

### See Also

- route_custom

## GCR-073

(warning) Cannot complete Interactive Router operation. Cannot find a nearby routable target.

### Description

The Interactive Router did not detect a valid target point for routing.

### What Next

Click close to an appropriate point, such as a pin, topology node, or wire and try again.

Feedback

**See Also**

- route_custom

---

## GCR-074

(warning) Add Point event ignored. Specified point has previously been added.

**Description**

You clicked twice in the same area.

**What Next**

Do not click more than once in a small area. Zoom in closer for better precision.

**See Also**

- route_custom

---

## GCR-075

(warning) Cannot complete Interactive Router operation at specified point. The selected object is not a routable target.

**Description**

You have selected a non-routable object for interactive routing.

**What Next**

Click at an appropriate point, such as a pin, topology node, or wire and try again.

**See Also**

- route_custom

---

## GCR-076

(warning) Cannot complete Interactive Router operation. Cannot find a legal snapping point on specified routable target.

**Description**

The target you have selected does not have a legal snapping point. The target might be off grid, or might be blocked by other nearby objects.

Feedback

### What Next

Make sure the target is on grid. Check for nearby obstacles before adding routes.

### See Also

- route_custom

---

## GCR-077

(warning) Cannot provide an Interactive Router solution from the starting point. The starting pin/wire appears\n to be covered by a routing obstruction. Suggest attempting the route from a different point.

### Description

One or more obstacles might be blocking the starting pin or wire.

### What Next

Start the router from a different pin or wire.

### See Also

- route_custom

---

## GCR-078

(warning) Cannot provide an Interactive Router solution to the targeted pin/wire which appears\n to be covered by a routing obstruction. Suggest attempting the route to a different point.

### Description

One or more obstacles might be blocking the starting pin or wire.

### What Next

Try to end your route at a different pin or wire.

### See Also

- route_custom

Feedback

## GCR-079

(warning) Cannot provide an Interactive Router solution to the targeted pin/wire. Verify obstructions near the target for constraints.\n Alternatively, verify track patterns at the targeted object.

### Description

One or more obstacles might be blocking access to the target pin or wire. Track patterns may not permit routing to the target.

### What Next

Check nearby objects and review your track pattern definitions for conflicts.

### See Also

- route_custom

## GCR-080

(warning) Cannot provide an Interactive Router solution to the targeted pin/wire within resource limits. Routing failure occurs somewhere between\n the routing source and target. Verify for obstructions. Alternatively, verify that source and\n target are not separated by unroutable layers. Alternatively ensure that\n P2P guidance is orthogonal.

### Description

One or more obstacles might be preventing the router from creating a route between the specified source and target objects.

### What Next

Check for obstacles along the desired route. You might have to create a different route. Check that the source and target shapes are not separated by unroutable layers. Make sure the routes are in orthogonal directions only.

### See Also

- route_custom

## GCR-081

(warning) Cannot provide an Interactive Router solution from the designated pin/wire. Verify obstructions near the source for constraints.\n Alternatively, verify track patterns at the starting object.

Feedback

### Description

One or more obstacles might be preventing the router from creating a route between the specified source and target objects.

### What Next

Check for obstacles along the desired route. You might have to create a different route. Check that the source and target shapes are not separated by unroutable layers. Make sure the routes are in orthogonal directions only.

### See Also

- route_custom

---

## GCR-082

(warning) Cannot complete Interactive Router operation. Verify: valid routing layers, Non Default Rules, track patterns, pin accessibility.\n Alternatively, try setting the Ignore Samenet DRCs option.

### Description

Invalid routing layers, constraints violations, track patterns and obstacles might be preventing the routier from creating a route between the specified source and target objects.

### What Next

Check for obstacles along the desired route. Check for constraint violations and fix them. Check the track pattern definition for conflicts.

### See Also

- route_custom

---

## GCR-083

(warning) Cannot provide an Interactive Router solution from the starting point. The starting pin/wire location appears\n to conflict with local track pattern constraints.

### Description

A conflict exists between the starting point and the track pattern constraints.

### What Next

Review your track pattern definitions.

Feedback

### See Also

- route_custom

---

## GCR-085

(warning) Interactive solution could only be found by ignoring routing blockages.\n This may result in DRCs. The routing blockages preventing a clean route appear to be located \n near the targeted pin/wire.

### Description

Blockages near the target pin or wire are causing the router to ignore the routing blockages to route successfully.

### What Next

Review your DRC report carefully. The Interactive Router might be creating DRCs when it ignores a routing blockage.

### See Also

- route_custom

---

## GCR-086

(warning) Interactive solution could only be found by ignoring routing blockages.\n This may result in DRCs. The routing blockages preventing the generation of a clean route appear\n to be located near the source pin/wire.

### Description

Blockages near the origin pin or wire are causing the router to ignore the routing blockages to route successfully.

### What Next

Review your DRC report carefully. The Interactive Router might be creating DRCs when it ignores a routing blockage.

### See Also

- route_custom

Feedback

## GCR-087

(warning) Interactive solution could only be found by ignoring routing blockages or constraints like via redundancy. This may result in DRCs.

### Description

Blockages or constraints near the origin or target pin or wire are causing the router to fail. The router has chosen to ignore them to complete the route.

### What Next

Review your DRC report carefully. The Interactive Router might be creating DRCs when it ignores a routing blockage.

### See Also

- route_custom

## GCR-088

(warning) Point (%d,%d) is outside of the defined routing area. Point ignored.

### Description

The router is ignoring the point outside the defined routing area.

### What Next

Click inside the routing area and continue your routing.

### See Also

- route_custom

## GCR-089

(error) No contiguous path found between clicked points.

### Description

No contiguous path found between clicked points.

### What Next

Check the area being routed and click in an open space.

Feedback

**See Also**

- route_custom

---

## GCR-090

(error) Clicked shapes have no net definition.

**Description**

You clicked shapes that do not have assigned nets. The router cannot use these shapes as a starting or ending point.

**What Next**

Use the Property Editor to verify the shapes of the objects you want to use during routing.

**See Also**

- route_custom

---

## GCR-091

(error) P2P shielder doesn't currently support shielding on bit nets.

**Description**

The point-to-point shielding tool cannot shield individual bit nets of a bus or bundle.

**What Next**

Apply the shield to the entire bus or bundle.

**See Also**

- route_custom

---

## GCR-092

(warning) Cannot complete Interactive Router operation. There appears to be no possible routing path between the source and target.\n Verify: valid routing layers, Non Default Rules, track patterns, pin accessibility.

**Description**

One or more obstacles, unroutable layers between source and target shapes, or spacing violations are preventing the router from creating routes.

Feedback

### What Next

Check for obstacles, unroutable layers, and spacing constraints.

### See Also

- route_custom

---

## GCR-094

(warning) Width %.10g on original Interactive Router tandem route start layer has been increased to %.10g\n to match the maximum tandem routing width required.

### Description

The router had to adjust the width on a tandem route to match the required maximum width.

### What Next

Be aware of the change in width.

### See Also

- route_custom

---

## GCR-095

(warning) Cannot complete Interactive Router multi-wire routing. There appears to be no possible routing path between the sources and targets.\n Verify: valid routing layers, Non Default Rules, pin accessibility.\n Alternatively ensure that P2P guide points select accessible tapoff locations and adequate open corridor areas.\n Routed wire connections: %u, unrouted wire connections: %u.

### Description

One or more obstacles, unroutable layers between source and target shapes, or spacing violations are preventing the router from creating routes.

### What Next

Check for obstacles, unroutable layers, and spacing constraints. Also, for point-to-point routing, select accessible tapoff locations and corridors that have sufficient space.

### See Also

- route_custom

Feedback

## GCR-096

(warning) Tandem solution failed. However, a base route was found. To allow partial tandem solution,\n set the Allow Tandem Breaks option to true (preference: rtTandemAllowBreaks)

### Description

Your settings required that tandem shield be inserted without any breaks. There was at least one section where the tandem shield was blocked by routing, blockages or other obstacles on the layer above or below.

### What Next

The router will be able to create partial tandem shields if you set the preference rtTandemAllowBreaks to true. If partial tandem shields are okay, please set the preference as indicated.

### See Also

- route_custom

## GCR-097

(error) Shielding net %s but no shield net specified.

### Description

The net was specified for shielding in the command net list, but the net to use to create the shielding was not specified.

### What Next

Check that the net to be used for shielding is specified on the Add Shield toolbar, P2P Shield toolbar, and the Auto Route dialog box.

### See Also

- route_custom

## GCR-098

(warning) Tandem vias at corners were not inserted in order to respect stacking via constraints.

Feedback

### Description

The viaStackLimit constraint does not permit another via to be inserted at some location, so some shield vias were omitted.

### What Next

Check the viaStackLimit constraint. Reroute the connection being shielded to an area with more room for vias to be inserted.

### See Also

- route_custom

---

## GCR-099

(warning) There is no PR boundary or routing area defined in the design. Router will use boundary (%f %f) (%f %f) for routing.

### Description

The design does not have a boundary or a routing area defined, so the router computes the boundary box of all the shapes and pins.

### What Next

If this is not desired, be sure to define a PR boundary or routing area before starting the router.

### See Also

- route_custom

---

## GCR-100

(warning) Cannot switch to tandem routing mode in the middle of a routing operation. Routing operation has been aborted.

### Description

You attempted to switch to the Tandem routing mode in the middle of the routing operation. The router could not continue.

### What Next

Turn on the Tandem mode before starting a route operation.

Feedback

**See Also**

- route_custom

---

## GCR-111

(warning) Skip bit net %s for creating bus virtual pin, because only one steiner is found on it.

**Description**

The router found only one steiner on a bus net bit, so it could not create a bus virtual pin.

**What Next**

Check the bus net bit steiner definitions.

**See Also**

- route_custom

---

## GCR-112

(warning) Placement found %d images placed outside PR boundary (%g,%g)(%g,%g) These images will be ignored.

**Description**

This is an informational message and no action is required.

**What Next**

Check instance placement or referenced libraries

**See Also**

- route_custom

---

## GCR-113

(warning) Projection for '%s' failed because pin and/or routing path is blocked by obstruction or is outside corridors.

**Description**

Projection failed because pin and/or routing path is blocked by obstruction or is outside corridors

Feedback

**What Next**

Ensure blockages, corridors and/or pin alignments on pins and expected connecting paths, so remove ignore_constraints_for_straight_connection option and rerun route_custom.

**See Also**

- route_custom

---

## GCR-114

(warning) Bitnets of bus %s have different corridors assigned on layer %s, all corridors on layer %s are merged into one big corridor.

**Description**

The router combined one or more corridors into a single larger corridor on the specified layer.

**What Next**

Check your corridor definitions and define a single corridor per layer.

**See Also**

- route_custom

---

## GCR-115

(warning) Detected width, spacing, rule, or constraint errors. See details in file '%s'.

**Description**

One or more errors were detected while processing the constraints.

**What Next**

Review the errors in the specified output file and make corrections if possible.

**See Also**

- route_custom

---

## GCR-116

(warning) Shield constraint on '%s' is ignored. There is no shield net '%s' in database.

Feedback

**Description**

The net requested to be used for shielding was not found in the database.

**What Next**

Check that the shield net name is spelled correctly and review your design data.

**See Also**

- route_custom

---

## GCR-117

(warning) The Shielder could not create the shielding via at (%f %f) (%f %f) via due to DRC Error.

**Description**

When adding shielding, the router encountered a DRC error and did not create a via at the noted coordinate.

**What Next**

Visually check the area specified for other objects that might interfere with the shielding tool.

**See Also**

- route_custom

---

## GCR-118

(warning) Cannot shield net %s. It is not fully routed.

**Description**

Shielding is only supported on fully routed nets.

**What Next**

Make sure the net is fully routed before adding shielding.

**See Also**

- route_custom

Feedback

## GCR-119

(warning) Supply net %s has no top level shape and would be ignored for routing.

### Description

The router cannot use a supply net without one or more top-level shapes that can be used as connection points.

### What Next

Check the design data and the supply net definitions. Ensure that you have top-level shapes that can be used as connection points.

### See Also

- route_custom

## GCR-120

(info) Net %s configured %d multiport pins.

### Description

This informational message indicates that the net contained some pins configured as multiport.

### What Next

This is an informational message and no action is required.

### See Also

- route_custom

## GCR-121

(warning) Shielding net %s has no pin in specified edit window. No shield vias can be added.

### Description

The viewed area does not contain any pins from the supply net, so shield vias do not have connection points.

Feedback

### What Next

Try zooming out so that the viewing area includes the supply shapes and shield vias have connection points.

### See Also

- route_custom

---

## GCR-122

(warning) Net %s unable to process oaRoute with end type of %s.

### Description

The router does not support a topology route that ends at a via.

### What Next

Add a topology node or use an inst term for your route end-point.

### See Also

- route_custom

---

## GCR-123

(warning) Detected %d feedthrough connections and %d nearby pin-pin connections.

### Description

This is an informational message indicating how many feedthrough and nearby pin-pin connections were detected when the design was loaded.

### What Next

This is an informational message and no action is required. If the numbers seem unusual, review the design data.

### See Also

- route_custom

---

## GCR-124

(warning) Bus/Diffpair trunk %s has 3 or more terminal groups. P2P can only handle 2 terminal groups. All terminal groups are routed.

Feedback

### Description

During point-to-point interactive routing, the router can support two terminal groups. However, the design contains at least three terminal groups.

### What Next

Review the bus or diffpair and ensure there are no more than two terminal groups defined, then try to route again.

### See Also

- route_custom

---

## GCR-125

(warning) Bus/Diffpair trunk %s has predefined corridor. P2P corridor is suppressed.

### Description

The bus or diffpair already had a route-guide corridor defined. We cannot use the mouse-click corridor in this case.

### What Next

Check that the bus or diffpair route-guide corridor is defined as expected. Your mouse-click point-to-point corridor will not be used.

### See Also

- route_custom

---

## GCR-126

(info) Net:%s pin:[%s-%s] Defined as trunk found within image.

### Description

This informational message indicates that the router defined the given net and pin as a trunk.

### What Next

This is an informational message and no action is required.

### See Also

- route_custom

Feedback

## GCR-127

(info) %s:'%s' succeeded route : applied (%d) applied wire and (%d) routed wires

**Description**

This informational message indicates that the trunk route is fully routed.

**What Next**

This is an informational message and no action is required.

**See Also**

- route_custom

## GCR-128

(warning) %s:'%s' applied (%d) applied wire and (%d) routed wires

**Description**

This informational message indicates that the trunk router doesn't fully routed.

**What Next**

This is an informational message and no action is required.

**See Also**

- route_custom

## GCR-129

(info) legalizing bus '%s' trunk %s is succeeded.

**Description**

This informational message indicates that the trunk legalization is succeeded.

**What Next**

This is an informational message and no action is required.

**See Also**

- route_custom

Feedback

## GCR-130

(warning) legalizing bus '%s' trunk %s is failed and try to improve trunk.

**Description**

This informational message indicates that the trunk router doesn't fully routed.

**What Next**

This is an informational message and no action is required.

**See Also**

- route_custom

## GCR-131

(warning) Some '%s' bus bits may not be shielded due to missing constraint.
You may want to check the create_bus_routing_style -shield_placement value or custom.route.bus_intra_shield_placement app option.

**Description**

The results will not have shielding shapes.

**What Next**

Need to set shield_placement <style> in create_bus_routing_style

**See Also**

- route_custom

## GCR-132

(info) '%s' trunk will be shielded outside only because of the bus doesn't have any intra-shielding constraint.

**Description**

The results will not have interleave shielding shapes.

**What Next**

Need to set shield_placement <style> in create_bus_routing_style

Feedback

**See Also**

- route_custom

---

## GCR-133

(info) Initializing Custom Router.

**Description**

Custom Router is initializing.

**What Next**

The initialization will complete soon. No action is required.

**See Also**

- route_custom

---

## GCR-134

(info) Custom Router started.

**Description**

Custom Router has begun routing.

**What Next**

The results will display at completion. No action is required.

**See Also**

- route_custom

---

## GCR-135

(info) Custom Router finished.

**Description**

Custom Router has finished routing.

**What Next**

You can continue with other commands.

Feedback

**See Also**

- route_custom

---

## GCR-136

(warning) At least %u instances had invalid/unbound masters. Routing results may be incorrect.

**Description**

Some instances were in valid or unbound and could not be loaded by the router.

**What Next**

Review your design and library data to look for unbound instances. Files might have been moved or might not be readable. Correct if possible.

**See Also**

- route_custom

---

## GCR-137

(warning) %u instances could not be processed because of the placement status. Routing results may be incorrect.

**Description**

Some instances were unplaced or had an unknown placement status and could not be processed by the router.

**What Next**

Review your design and library data to look for instances with unplaced or unknown status. Correct if possible.

**See Also**

- route_custom

---

## GCR-138

(warning) Could not process via %s because its viaDef %s is not bound.

**Description**

Some via definitions were invalid or unbound and could not be loaded by the router.

Feedback

### What Next

Review your technology and library data to look for unbound vias masters. Files might have been moved or might not be readable. Correct if possible.

### See Also

- route_custom

---

## GCR-139

(error) TrackPattern %s with color %d exceeds maximum color of %d.

### Description

The router only supports up to three colors (upper and lower metal layer plus cut layer).

### See Also

- route_custom

---

## GCR-140

(error) Cannot use one-loop lengthening without Wire Matching or Length constraint.

### Description

Single-loop matching requires matching constraint (wireMatching or Length) assigned to the initial routed nets

### What Next

Set the Wire Matching or Length constraint to use the single-loop preference.

### See Also

- route_custom

---

## GCR-141

(error) Lengthen bounding box must be specified when using One-loop lengthening.

### Description

You must specify lengthening area (custom.route.match_box) to run single-loop matching.

Feedback

### What Next

Define the wire-lengthening area using the custom.route.match_box app option.

### See Also

- route_custom

---

## GCR-142

(warning) Net %s: Wire Matching or Length constraint must be specified to use One-loop lengthening.

### Description

Single-loop matching requires a matching constraint (wireMatching or Length) assigned to the net

### What Next

Set the Wire Matching or Length constraint for the net to use the single-loop preference.

### See Also

- route_custom

---

## GCR-143

(warning) Net %s: Routing is incomplete. One-loop lengthening does not support partially routed nets.

### Description

Because the single-loop matching is a post-route operation, all nets must be fully routed before using One-loop lengthening.

### What Next

Make sure the nets are fully routed before attempting One-loop lengthening.

### See Also

- route_custom

Feedback

## GCR-144

(warning) Net %s: Shapes on net must be located within One-loop lengthening bounding box.

### Description

The defined lengthening area (rtMatchBox) does not contain any shapes from the original routing, so there is no place to insert single detour wire for matching.

### What Next

Redefine the lengthening area by updating the rtMatchBox preference.

### See Also

- route_custom

## GCR-145

(warning) Net %s: One-loop lengthening does not have enough room to satisfy constraint within the specified bounding box.

### Description

Matching algorithm has exhaustively searched all possible single detour solutions based on the criteria you provided (loop min/max spacing, offset layer, on grid) within the defined lengthening area and cannot find a satisfactory solution.

### What Next

Enlarge the wire-lengthening area if possible. Check the other constraint values to ensure they are compatible with single loop matching.

### See Also

- route_custom

## GCR-146

(warning) Net:%s route_fromto_only preference is <on>. Skipping non-topology_edge [%s-%s][%s-%s].

### Description

You have requested to route only connections defined as routes. The specified non-route connection will be skipped.

Feedback

### What Next

If the connection must be routed, define it as a topology_edge using the topology commands.

### See Also

- route_custom

---

## GCR-147

(warning) Net %s will be protected and will not be routable by the batch router due to the presence of diagonal segment shapes. It will however be possible to modify the net using the Interactive Router.

### Description

Diagonal paths or path segments have prevented the Auto Router from creating routes.

### What Next

Use the Interactive Router to update your routes, or remove the diagonal path segments and try again.

### See Also

- route_custom

---

## GCR-148

(warning) Rules for layer '%s' are ignored because it has an undefined mask number.

### Description

When processing the rules and constraints, the router found an invalid mask number. The router could not process the rules for this layer.

### What Next

Check your technology library and verify the correct layers have been applied to all of your rules and constraints.

### See Also

- route_custom

Feedback

## GCR-149

(error) The router could not be initialized because there are more than 2 poly layers (%s) defined in the tech file.

### Description

The router current supports no more than two poly layers.

### What Next

Check your technology library and ensure only two poly layers are defined.

### See Also

- route_custom

## GCR-150

(error) The router could not be initialized because the custom router does not support routing with two poly layers (two poly layers "%s" are selected for %s).

### Description

The router can only route using one poly layer when there are two poly layers defined.

### What Next

Check your technology library and ensure only two poly layers are defined. Use the Object/Layer Panel to deselect one of the two poly layers.

### See Also

- route_custom

## GCR-151

(warning) Can't build the design channel map required to do device row cloning for the Interactive Router.\n The design appears to be too large. The current limit for the number of shapes to be processed is: %d.\n To increase this limit, set the rtMaxShapesForCloning preference to a higher value.

### Description

The router has encountered more shapes than the maximum number that your preference value permits.

Feedback

### What Next

Set the rtMaxShapesForCloning preference if possible, or define a smaller window to reduce the shape count.

### See Also

- route_custom

---

## GCR-152

(warning) Not all possible Interactive Router clone candidates were actually cloned. The limit of %d clones per Interactive Router operation has been reached.\n To increase this limit, set the rtMaxCloning preference to a higher value.

### Description

The router has reached the maximum number of clones based on the rtMaxCloning preference value.

### What Next

Increase the rtMaxCloning value if possible.

### See Also

- route_custom

---

## GCR-154

(warning) Only one poly layer could be selected.

### Description

You selected more than one poly layer for routing. The router only supports one poly layer.

### What Next

Check your selected layers and ensure only one poly layer is selected, then rerun your command.

### See Also

- route_custom

Feedback

## GCR-155

(error) Command '%s' cannot be run because the required license '%s' is not available.

**Description**

The command requires the specified license feature to be available.

**What Next**

Check your license installation to ensure the specified key is installed. Check that other users have not consumed all copies of the license. Contact Synopsys for assistance if you do not have the correct licenses available.

**See Also**

- route_custom

## GCR-156

(warning) Backup layers must be continuous for net %s.

**Description**

Backup layers must be continuous.

**What Next**

Check the backup layers for the specified net and ensure they are continuous.

**See Also**

- route_custom

## GCR-157

(warning) Backup layers must be subset of valid routing layers for net %s.

**Description**

Backups layers must be a subset of the valid routing layers. Layers not in the valid routing layers set for the net cannot be used as backup layers.

**What Next**

Check the valid routing layers for the specified net. Ensure that only layers listed in this set are used for backup.

Feedback

**See Also**

- route_custom

---

## GCR-158

(warning) The net %s must have at least 4 routing layers to setup backup layers.

**Description**

To specify backup layers, a net's valid routing layer set must include at least four routing layers.

**What Next**

Check the valid routing layers for the net and ensure that at least four layers are specified. Alternatively, do not use the backup feature.

**See Also**

- route_custom

---

## GCR-159

(warning) Track Regions on layer '%s' overlap and create infeasible track region setup, this region '%s' will be ignored.

**Description**

The tracks in track regions overlap on same layer and create an impossible setup on the specified layer. The router will ignore the region as specified.

**What Next**

Check the track regions at the global and local level and make sure they do not conflict or overlap unexpectedly for same layer.

**See Also**

- route_custom

---

## GCR-160

(warning) There are more than 64,000 shapes in a small area (%f, %f, %f, %f). The router will ignore some of the shapes.

Feedback

### Description

The router detected a very large number of shapes in a small area as shown. The router will ignore some of the shapes.

### What Next

Check that all pins and instances are properly placed. Check for overlapping placements. Check blockages in instance masters.

### See Also

- route_custom

---

## GCR-161

(error) The router could not be initialized because the routing area is outside the PR boundary (%f, %f, %f, %f).

### Description

The routing area you have defined is outside of the PR boundary. The router cannot proceed.

### What Next

Check the routing area definition and create an area which is within the PR boundary.

### See Also

- route_custom

---

## GCR-162

(error) The router could not be initialized because the routing area is null.

### Description

The routing area you have defined is effectively null (either the x or y dimension is zero or too small to be useful).

### What Next

Check the routing area definition and create an area which is larger than the track width and spacing rules.

### See Also

- route_custom

Feedback

## GCR-163

(error) The router could not be initialized because there are more than %d metal and cut layers defined.

### Description

The router only support a limited number of cut and metal layers. Your data contains more layers than the router can support.

### What Next

Check the technology library and ensure that the total number of cut and metal layers is within the noted limit.

### See Also

- route_custom

## GCR-164

(error) The router could not be initialized because there are no routing layers defined in the tech data.

### Description

The router did not find any metal or cut layers in the technology library.

### What Next

Check your technology library and ensure that the metal and cut layers are properly defined.

### See Also

- route_custom

## GCR-165

(error) The router could not be initialized because there are no valid via definitions in the tech data.

### Description

The router did not find any valid via definitions in the technology library.

Feedback

### What Next

Check your technology library and ensure that the via definitions are properly defined.

### See Also

- route_custom

---

## GCR-166

(warning) There is no user-function 'CustomViaParameterizeFunc' available for custom vias, custom vias will not be modified by router.

### Description

There is no user-function 'CustomViaParameterizeFunc' available for custom vias.

### What Next

Remove custom vias from the valid routing via definition, or provide the required function to use custom vias.

### See Also

- route_custom

---

## GCR-167

(warning) Found preroute shapes with MPT color or tech layer with default color. Provide active color tracks in layer %s to enable color routing methodology.

### Description

The preroute shapes contains shapes with multiple-patterning color or layer with default color.

### What Next

Please provide active color tracks for the specified layer to enable color routing.

### See Also

- route_custom

---

## GCR-168

(error) RPDL error: '%s'.

Feedback

### Description

While attempting to parse the RPDL extraction data, a syntax error occurred. The message inidicates the exact error.

### What Next

Examine the RPDL file and fix the indicated syntax error.

### See Also

- route_custom

---

## GCR-169

(info) Extracted RPDL: "%s".

### Description

The output indicates the extraction channels that were processed.

### What Next

This is an informational mesage. No action is needed.

### See Also

- route_custom

---

## GCR-170

(error) The routing area defined in %s contains an illegal value %s.

### Description

While processing the routing area in the custom.route.routing_area app option, an illegal string value was encountered.

### What Next

The routing area must consist of a Tcl list containing one or more rectangles or polygons. A rectangle can be specified by providing exactly two points: the lower right and upper left vertices. A polygon must contain a list of four or more points. Any non-numeric characters or incomplete entries in the option value will result in an error.

### See Also

- route_custom

Feedback

## GCR-171

(warning) Rules for layer '%s' are ignored because it is an equivalent layer.

### Description

If you have equivalent layers defined, the rules can only be processed for one of the layers.

### What Next

Check your technology library and ensure rules are not set for more than one layer of an equivalent set.

### See Also

- route_custom

## GCR-172

(warning) Pin centered at (%.10g,%.10g) on layer %s is off-track or off-colored.

### Description

Full colored routing for multipatterning layer requires pin to be on-track with mask color matching track color.

### What Next

Move pin to be on track. Change pin color to match track color.

### See Also

- route_custom

## GCR-173

(info) Converge mode [bus] has been activated and some potential DRC/shorts/runtime may be impacted, please turn on the %s and refer for detailed log.

### Description

Converge mode is kicked in during bus routing.

### What Next

Some potential DRC/shorts/runtime may be impacted.

Feedback

**See Also**

- route_custom

---

## GCR-174

(info) Converge mode [signal] has been activated and some potential DRC/shorts/runtime may be impacted, please turn on the %s and refer for detailed log.

**Description**

Converge mode is kicked in during signal routing.

**What Next**

Some potential DRC/shorts/runtime may be impacted.

**See Also**

- route_custom

---

## GCR-175

(warning) River style routing is not supported. The river constraint will be ignored for net %s.

**Description**

Custom router does not support river style routing. The router detected a river constraint in your data.

**What Next**

Remove the river constraint and try your routing again.

**See Also**

- route_custom

---

## GCR-176

(warning) %s style shielding is not supported. This shielding type will be ignored for net %s.

**Description**

Custom router does not support the specified shielding style. The constraint will be ignored.

Feedback

### What Next

Change your constraint to use parallel shielding or delete the constraint.

### See Also

- route_custom

---

## GCR-177

(warning) Min and max resistance constraints are not supported. The custom router will not attempt to follow the resistance constraint for net %s.

### Description

The custom router does not support min or max resistance constraints. The constraint will be ignored.

### What Next

You may continue routing and the router will ignore the constraint. You can delete the constraint to eliminate the warning message.

### See Also

- route_custom

---

## GCR-178

(warning) The router detected a wire resistance matching constraint, which is not supported. The constraint will be ignored for net %s.

### Description

The custom router does not support wire matching for resistance. The constraint will be ignored.

### What Next

You may continue routing and the router will ignore the constraint. You can delete the constraint to eliminate the warning message.

### See Also

- route_custom

Feedback

## GCR-179

(warning) The router detected a wire length matching-per-layer constraint where there are also more than two layers defined in the NDR layer constraint for net %s. The routing quality may not be acceptable.

### Description

The custom router supports wire length matching on a per layer basis, but it may not be able to produce good results when three or more layers are used to route the given net.

### What Next

For best result quality, create a non-default-rule specifying just two layers for any net which has this type of matching per layer constraint.

### See Also

- route_custom

## GCR-180

(warning) Topology constraints may only be used for bus routing. Other topology definitions have been ignored for net %s.

### Description

The custom router detected topology constraints and data for a net which is not part of a bus. These definitions will be ignored.

### What Next

Limit your topology node and edge definitions to nets which are included in a bus.

### See Also

- route_custom

## GCR-181

(info) Custom router requires all routing layers to have a discrete direction. The router has set layer %s to use direction %s.

### Description

Custom router detected that the layer direction was unknown. The router has assigned it to use the specified direction.

Feedback

### What Next

Check your technology definitions and ensure that all routing layers specify a direction of either verticial or horizontal.

### See Also

- route_custom

---

## GCR-182

(warning) Wire striping routing is not supported. The wire striping constraint will be ignored for net %s.

### Description

Custom router does not support wire striping routing. The router detected a wire striping constraint in your data.

### What Next

Remove the wire striping constraint and try your routing again.

### See Also

- route_custom

---

## GCR-183

(warning) Could not extract the routing pattern for net %s.

### Description

The routing pattern extraction was not able to successfully describe the net with the currently supported RPD syntax. The net was either dirty (extra jogs for instance) or simply not compatible with RPD syntax.

### What Next

Clean up the net if possible.

### See Also

- route_custom

Feedback

## GCR-184

(warning) Group shielding is only supported for the Add Shield flow. Shielding constraint will be ignored for net %s.

### Description

Group Shielding is only supported for the Add Shield flow. Auto-route flow is not supported at this time.

### What Next

Try the Add Shield flow in order to shield the group of nets.

### See Also

- route_custom

## GCR-185

(warning) Group shielding is only supported for net(s) belonging to a net group. Shielding constraint will be ignored for net %s.

### Description

Group Shielding is only supported for a group of nets.

### What Next

Create a net group and run the Add Shield command again.

### See Also

- route_custom

## GCR-186

(warning) Group shielding is only supported for a group of net(s) which are connected to macros. Shielding constraint will be ignored for net %s.

### Description

Group shielding is only supported for a group of net(s) which are connected to macros.

### What Next

Limit the the members of the net group to nets which are connected to macros.

Feedback

**See Also**

- route_custom

---

## GCR-187

(warning) Enclose Pin and Enclose Via options on net %s will be ignored for group shielding flow.

**Description**

Enclose Pin and Enclose Via options will be ignored for group shielding flow.

**See Also**

- route_custom

---

## GCR-188

(info) Custom Router violation summary.

**Description**

Custom Router violation summary

**What Next**

Check violation counts

**See Also**

- route_custom

---

## GCR-189

(info) Failed nets summary.

**Description**

Failed nets summary

**What Next**

Check failed net name

**See Also**

- route_custom

Feedback

## GCR-190

(warning) Bus '%s' has insufficient corridors at %s, hence bus route may leave some bus nets open. Please extend routing corridors to cover all pin area.

### Description

Bus trunk route has a unrouted trunk and may have some routing blockages or insufficient corridors.

### What Next

Check the bus net bit steiner definitions or corridor boundaries.

### See Also

- route_custom

## GCR-191

(warning) Bus '%s' can't be routed by projection because of %d nets over total %d nets can't be projected. Please checks corridors and/or blockages on pin and expected paths

### Description

Bus trunk route has a unrouted trunk and may have some routing blockages or insufficient corridors.

### What Next

Check the bus net bit steiner definitions or corridor boundaries.

### See Also

- route_custom

## GCR-192

(warning) Bus '%s' has insufficient corridors at %s %s, hence bus route may leave some bus nets open.

### Description

Bus trunk route has a unrouted trunk and may have some routing blockages or insufficient corridors.

Feedback

### What Next

Check the bus net bit steiner definitions or corridor boundaries.

### See Also

- route_custom

---

## GCR-233

(warning) '%s' is not a valid group for Lasso matching per layer. Inconsistent net's layer distribution.

### Description

When doing length matching per layer with Lasso matching style, it requires every net in the matching group to have the same initial routing layer distribution.

### What Next

Set min/max layer constraints on the nets of the matching group.

### See Also

- route_custom

---

## GCR-234

(warning) %s width of %f is incompatible with the allowedWidthRange constraint for layer %s. legal widths near this value are (%f %f)

### Description

used widths are incompatible with allowedWidthRange process constraint.

### What Next

user is expected to select one from suggested legal widths.

### See Also

- route_custom

---

## GCR-238

(warning) Length matching mismatches during routing for group %s. Matching percentage = %s.

Feedback

### Description

Length matching fails to find a feasible free space to lengthen wires in the group during MSCTS flow.

### What Next

User is expected to enlarge tolerance or adjust layout to leave more free space for length matching.

### See Also

- route_custom

---

## GCR-239

(error) The grouping rate %d%% is too low ( < %d%%) and not suitable for group planning flow. Please try other grouping size settings.

### Description

When enabling planning feature of custom router, it will try to group nets connected to boundary pins based on their pin locations in order to route similar topology. If the group rate is too low, it will stop custom router to avoid expected long runtime.

### What Next

User can turn off planning flow or remove repeaters in the design to improve planning results.

### See Also

- route_custom

---

## GCR-240

(error) Net %s is connected to standard cell pins and not suitable for group planning flow.

### Description

When enabling planning feature of custom router, it will try to group nets connected to boundary pins based on thier pin locations in order to route similar topology. Nets connected to standard cell pins are not allowed as they are almost impossible to group other pins togheter.

Feedback

### What Next

User can turn off planning flow or remove all nets connected to standard cell pins from the net collection to be routed.

### See Also

- route_custom

---

## GCR-241

(error) Option -planning should be specified when app option custom.route.plan_grouping_size is set.

### Description

When specify grouping size to one of the values {auto, small, medium, large} with app option custom.route.plan_grouping_size, user needs to enabling planning feature of custom router.

### What Next

Set app option custom.route.plan_grouping_size.

### See Also

- route_custom

---

## GCR-242

(error) App option custom.route.plan_grouping_size is not compatible with app option custom.route.transparent_hier.

### Description

Planning grouping size feature is not compatible with transparent hierarchy feature.

### What Next

Either turn off plan_grouping_size or turn off transparent_hier app options.

### See Also

- route_custom

Feedback

## GCR-243

(warning) Planning feature is not compatible with app option custom.route.transparent_hier. Will ignore -planning and continue routing with transparent hierarchy feature.

### Description

Planning feature is not compatible with transparent hierarchy feature. Router will ignore planning feature and continue with transparent hierarchy feature.

### What Next

User can turn off planning flow or remove all nets connected to standard cell pins from the net collection to be routed

### See Also

- route_custom

## GCR-245

(error) There are too many (%d) fragmented shapes on layer %s slicing the channel %s into discontinuous routing resources.

### Description

Custom route planner found many small shapes or vias in the channel such that makes it very difficult to plan bus-like patterns in auto grouping mode. Planner has to stop routing otherwise it will take extreme runtime to finish planning.

### What Next

User has to check the design to see if there are enough routing resources available in channel %s. For example, there may exist previous routing results in the design which should be cleared first before invoking planner.

### See Also

- route_custom

## GCR-246

(error) Violated rate %.2g%% exceeds max allowed violated rate %d%%.

Feedback

### Description

In auto grouping mode, custom route planner plans a lot of violated nets such that not suitable to continue routing because routing quality and runtime are expected to be poor. A violated net in planner is defined as either open (unplanned) or has DRC violations related to the net.

### What Next

User has to check the design if there are enough routing resources available. Or there are some difficult to route nets like power/ground or clock nets accidentally included in the routing nets.

### See Also

- route_custom

---

## GCR-247

(error) Open rate %.2g%% exceeds max allowed open rate %d%%.

### Description

In auto grouping mode, custom route planner generates many failed to plan nets such that not suitable to continue routing because routing quality and runtime are expected to be poor.

### What Next

User has to check if the pins of the routing nets are accessible or there exists many obstacles in the routing region such that router cannot route the nets without many bends.

### See Also

- route_custom

---

## GCR-248

(warning) Ignore net '%s' which connects to a standard cell.

### Description

Custom route planner is not suitable to route standard cells. It will not plan nets connected to standard cells and continue to plan the other nets.

Feedback

### What Next

User has to check if the nets are reasonable to connect to standard cells. Custom route planner works better if there are no standard cells in routing region because it may fragment routing resources and result in poor auto grouping results.

### See Also

- route_custom

---

## GCR-249

(error) There are no nets for planning.

### Description

Custom route planner does not find any nets to be planning.

### What Next

User has to check if the specified nets set is empty. Note custom route planner will ignore any nets connected to standard cells.

### See Also

- route_custom

---

## GCR-250

(error) Follow rate %.2g%% below custom.route.plan_min_follow_rate %.2g%%.

### Description

Autoroute follow rate (w.r.t. custom route planner) is less than the min value. Router will stop to avoid expected long runtime and scenic routes.

### What Next

User has to check if the design style is suitable for custom router planner such that it can generate high quality routing results which autoroute can follow most of planned results.

### See Also

- route_custom

Feedback

## GCR-251

(warning) Net %s is set to ignored %s.

### Description

Custom route planner checker detects errors on the net. The errors may cause serious routing problems and thus to be ignored in following planning and routing processes.

### What Next

Depends on the root cause, user has to fix the design errors reported eariler in the planner checking process.

### See Also

- route_custom

## GCR-252

(error) Found %d overlapped pins so %d nets are ignored. The first one is at %s of net %s.

### Description

Custom route planner checker found several overlapped pins of different nets in the design. The corresponding nets are set to be ignored in following planning and routing processes.

### What Next

User has to fix the overlapped pins reported by the planner checker.

### See Also

- route_custom

## GCR-253

(warning) Found %d off track pins. The first one is at %s of net %s.

### Description

Custom route planner checker found off track pins in the design. They can be handled in following planning and routing processes, but resulting routing shapes may be reported as DRC errors.

Feedback

### What Next

User has to make sure those off track pins are reasonable. If not, user may adjust track settings or move pin locations to on track locations.

### See Also

- route_custom

---

## GCR-254

(warning) Found %d pins have different pin width to track width. The first one is at %s of net %s.

### Description

Custom route planner checker found there are some pins have inconsistent pin width and track width settings. They can be handled in following planning and routing processes, but resulting routing shapes may be reported as DRC errors.

### What Next

User has to make sure those inconsistent pin width and track width pins are reasonable. If not, user may adjust track width or pin width settings.

### See Also

- route_custom

---

## GCR-255

(error) Found %d pins have short tracks so %d nets are ignored. The first one is at %s of net %s.

### Description

Custom route planner checker found there are short tracks on some pins. Short track on a pin means that the track length from the pin is too short to make router reserve enough routing resources before bending to another routing layer. Thus the net is set to be ignored in following planning and routing processes.

### What Next

User has to make sure the short track is reasonable. If not, user may adjust track settings to make longer track from the pin.

Feedback

**See Also**

- route_custom

---

## GCR-256

(warning) Found %d discontinuous tracks on layer %s. The first one is at %s.

**Description**

Custom route planner checker found there are some discontinuous tracks in the design. They can be handled in following planning and routing processes, but may result in undesired routing patterns.

**What Next**

User has to check if the discontinuous tracks are reasonable. If not, user may adjust track settings to get desired routign patterns.

**See Also**

- route_custom

---

## GCR-257

(warning) Found %d dangling shapes in the design. The first one is at %s of net %s.

**Description**

Custom route planner checker found there are some dangling shapes in the design. They can be handled in following planning and routing processes, but may affect grouping and routing quality.

**What Next**

User has to check if the dangling shapes are reasonable. If not, user may remove the dangling shapes in the design.

**See Also**

- route_custom

---

## GCR-258

(warning) Router is stopped because the runtime exceeds the user-defined maximum runtime %d seconds.

Feedback

### Description

Router stops because the runtime exceeds user-defined maximum allowed runtime by custom.route.max_routing_time app option.

### What Next

User can enlarge or reset custom.route.max_routing_time app option to change the early stop behavior.

### See Also

- route_custom

---

## GCR-259

(warning) Ignore net '%s' which connects to a macro cell.

### Description

Currently custom route planner cannot plan macro cells well in auto grouping mode. The nets connected to macro cells will be ignored and continue to plan the other nets.

### What Next

If user still wants to route the nets connected to macro cells, user has to flatten the macro pins into top-level IO pins.

### See Also

- route_custom

---

## GCR-260

(error) User defined group '%s' is invalid since %s.

### Description

In auto grouping mode of custom route planner, user defined groups have to meet the following conditions: 1. At least two nets in a group. 2. Pin counts of all nets in a group are equal. 3. Pin layers and directions of all nets in a group are the same. 4. One net cannot presence in more than one group.

### What Next

User has to modify group to meet the user defined group conditions.

Feedback

**See Also**

- route_custom

---

## GCR-261

(info) Routing layers have no routing pitch defined in technology file. The default pitch, which is the minimum width plus minimum spacing for each layer, will be used.

**Description**

There is no pitch value for any layer in the technology library.

**What Next**

In the technology library, set the routing pitch to a value that is min_width plus min_spacing.

**See Also**

- route_custom

---

## GCR-262

(warning) Found %d pins have different pin width to NDR width. The first one is at %s of net %s.

**Description**

Custom route planner checker found there are pins width mismatching thier NDR width.

**What Next**

This is just a warning. User can check if the NDR settings are reasonable.

**See Also**

- route_custom

---

## GCR-268

(warning) Found %d pins have different pin width to default width and NDR is not set. The first one is at %s of net %s.

**Description**

Custom route planner checker found there are pins width mismatching thier default width and NDR is not set.

Feedback

### What Next

This is just a warning. User can set NDR width to match the pin width.

### See Also

- route_custom

---

## GCR-269

(warning) Found %d no track pins. The first one is at %s of net %s.

### Description

Custom route planner checker found there are pins do not overlap with any track. They can be handled in following planning and routing processes, but resulting routing shapes may be reported as DRC errors.

### What Next

User has to make sure those no track pins are reasonable. If not, user may adjust track settings or move pin locations to on track locations.

### See Also

- route_custom

---

## GCR-270

(error) Found %d illegal width pins so %d nets are ignored. The first one is at %s of net %s.

### Description

Custom route planner checker found there are pins do not follow the Discrete Metal Widths Rule.

### What Next

User can open the error browser to check the legal widths of each pin and modify them.

### See Also

- route_custom

Feedback

## GCR-271

(error) Found %d pins violated min max layer rule so %d nets are ignored. The first one is at %s of net %s.

### Description

Custom route planner checker found there are pins do not in the valid layers.

### What Next

User can open the error browser to check the valid layer range.

### See Also

- route_custom

## GCR-272

(error) Selected user defined group '%s' does not appear in selected net.

### Description

In auto grouping mode of custom route planner, user difined group have to been selected to be routed. If nets in certain user-defined group is selected to be routed, all nets in this group would be routed with AG.

### What Next

User has to set these user-defined groups to be routed.

### See Also

- route_custom

## GCR-277

(error) Fail to parse topology. There is topology across the boundary of design or routing area.

### Description

The topology across the boundary is not the expected use model. Router may have trouble to follow such pattern.

Feedback

**What Next**

User adjust boundary or re-draw topology to make all of the topologies are inside the boundary of routing area or design.

**See Also**

- route_custom

---

## GCR-278

(info) '%s' ViaDef '%s' layer '%s' is not defined, so, this via definition will be ignored during routing. Check if it's routing layer function is defined correctly in technology.

**Description**

Mising Via Definition in GCR database.

**What Next**

Fix routing layer function defined in technology data attached to design

**See Also**

- route_custom

---

## GCR-280

(warning) Shorts are removed for %d nets, of which %d nets were fully routed.

**Description**

Shorts are removed in the end of Custom Route.

**What Next**

User can disable the short removal if user wanys to keep the short.

**See Also**

- route_custom

---

## GCR-281

(warning) Net %s is incomplete because of short removal.

Feedback

**Description**

Net is fully routed, but it is incomplete in the end due to the short removal.

**What Next**

User can disable the short removal if user wanys to keep the short.

**See Also**

- route_custom

---

## GCR-282

(info) [%s] Elapsed time: real = %s, cpu = %s, [sys = %s, usr = %s]

**Description**

Time stamp for the custom route stage.

**See Also**

- route_custom

---

## GCR-283

(info) [%s] Memory: total = %s, peak = %s.

**Description**

Memory information for the custom route stage.

**See Also**

- route_custom

---

## GCR-284

(info) Number of nets asked to route = %d.

**Description**

Number of nets asked to route

**See Also**

- route_custom

Feedback

## GCR-285

(info) Number of nets attempted to route = %d.

**Description**

Number of nets attempted to route

**See Also**

- route_custom

## GCR-286

(info) Number of nets rejected to route = %d.

**Description**

Number of nets rejected to route

**See Also**

- route_custom

## GCR-287

(error) Found %d signal pins on PG track so %d nets are ignored. The first one is at %s of net %s.

**Description**

Custom route planner checker found signal pins on PG track in the design.

**What Next**

User can open Error Browser to identify where are the error pins.

**See Also**

- route_custom

## GCR-288

(warning) Supply net %s is ignored for routing.

**Description**

Supply net is ignored for routing.

Feedback

**See Also**

- route_custom

---

## GCR-289

(warning) Failed to create trunk for net %s because no tracks are available on %s within search box.

**Description**

No tracks are available inside the search box.

**What Next**

Define tracks if there is no track in the design, or enlarge the search box to cover some tracks.

**See Also**

- route_custom

---

## GCR-292

(warning) User defined group '%s' is skipped since valid net count < 2.

**Description**

Some nets in the user define are not valid and the number of valid nets is less than 2. For example routed nets, PG nets and nets with less than 2 pins are not valid nets.

**What Next**

The group will not be routed, and user has to modify group to meet the user defined group conditions.

**See Also**

- route_custom

---

## GCR-315

(error) Failed to find user defined group '%s'. Please specify correct group name.

Feedback

### Description

The app option custom.route.plan_user_defined_groups takes a list of bundle names. This error indicates GCR failed to find the group with specified bundle names. The error would be also raised if user specifies object collection (like bundle itself) in the app option.

### What Next

Check if the bundle does exist or the bundle name is incorrect.

### See Also

- route_custom

---

## GCR-323

(warning) The pin or port '%s' to SuperNet '%s' is missing, it may be ignored during routing.

### Description

The pin object for the supernet is not found in GCR database. It is required to route supernet using GCR.

### What Next

check supernet creation step to see if specfied port or terminal is correctly specified. Run GCR in debug mode to see any error messages about it.

### See Also

- route_custom

---

## GCR-324

(warning) The '%s' pin for %s '%s' is missing, this %s will be ignored during %s routing.

### Description

Specified terminals are not found in GCR database, it may be either incorrectly specified during constraint creation or incorrectly parsed during data import into GCR.

### What Next

Check if the terminal exists in database. Run GCR in debug mode to check their dump into intent file and look for error messages in logfile for more information.

Feedback

**See Also**

- route_custom

---

## GCR-325

(warning) Skip net %s which is already fully connected.

**Description**

Custom route planner found a net which is already fully connected, so it would not try to plan and route it again.

**What Next**

User need to check if forgeting to remove preroutes or selecting wrong nets to route.

**See Also**

- route_custom

---

## GCR-330

(warning) Failed to route bus %s since %s.

**Description**

Routing fail in LH global routing

**What Next**

Adjust the design by hints in the warnning.

**See Also**

- route_custom

---

## GCR-350

(warning) A number of nets will not be routed because they have only one pin within the design prBoundary or within the specified routing area.\n Review %s for the complete net list.

**Description**

In order for GCR to route a signal net, two or more net pins must be present in the design or within the routing area if applicable.

Feedback

**What Next**

From the list of nets with only one pin in the design, review if these nets are correctly specified

**See Also**

- route_custom

---

## GCR-427

(warning) CTS net %s will not be matched because the original routing length is greater than net length limit.

**What Next**

User is expected to enlarge for net length matching target.

---

## GCR-428

(warning) NDR of CTS net %s is not properly set. Routing results may be incorrect.

**What Next**

User is expected to provide NDR rule for the CTS nets.

---

## GCR-429

(warning) Found unexpected routing layers of CTS net %s. Routing results may be incorrect.

**What Next**

User is expected to provide one NDR layer setting for the CTS nets.

---

## GCR-430

(error) This app option '%s' is only supported in 3DIC mode.

**Description**

This error indicates this option failed to work under this design session.

**What Next**

Check if the design session is 3DIC mode.

Feedback

**See Also**

- route_custom

---

## GCR-431

(error) Net '%s' cannot complete routing in balance mode feature. %s

**Description**

This error indicates the root cause failed to route this net in balance mode feature.

**What Next**

Check if the reason could be fixed.

**See Also**

- route_custom

---

## GCR-432

(warning) Net '%s' has unexpected result in balance mode feature. %s

**Description**

The warning indicates the result may be unexpected due to some reason.

**What Next**

Check if the reason could be fixed.

**See Also**

- route_custom

---

## GCR-433

(warning) Cannot connect pin/terminal %s for net %s. The pin/terminal is blocked by %s.

**What Next**

User is expected to modify placement results.

**See Also**

- route_custom

Feedback

# GDS

## GDS-001

(error) Library %s is not Found.

**Description**

The Given library is not Found.

## GDS-002

(error) Design %s with view type %s is not Found in the Library.

**Description**

The Given Design name with the specified viewtype is not Found in the Specified Library.

## GDS-003

(error) Technology Definition for Layer Number %d and Purpose %d not defined.

**Description**

The Given Design doesn't have a LAYER Definition for the Layer in question in the Associated Technology Library.

## GDS-004

(information) Text %s (%d, %d)is creating a short with already detected Pin %s.The current text will be ignored.

**Description**

The Text is having a coordinate which overlaps with a geometry with has been already detected as another pin. So this is creating a short and will be ignored.

## GDS-005

(error) Current Design %s is not Found.

**Description**

The current Design is not Found.

Feedback

## GDS-006

(error) Current Design is not of LAYOUT type.

**Description**

The current Design loaded is not of Layout type.

## GDS-007

(information) No Design has been Added or Modified in the current read_gds. So no tracing will happen.

**Description**

The current read_gds has not modified any Designs, so pin tracing won't happen as part of read_gds.

## GDS-008

(warn) Multiple Boundary definition found for Module %s, keeping the last definition.

**Description**

The current GDSII file has multiple boundary definition for a module.

## GDS-009

(error) Reading Directory is not supported. File %s is a directory.

**Description**

The input file is a directory.

## GDS-010

(warn) Layer mapping rule for "%d:%d" is defined multiple times. Only the last instance will be effective.

**Description**

Multiple Layer Mapping rule is not supported in GDSII read/write utility.

## GDS-011

(error) GDSII Parsing Error: %s

Feedback

**Description**

GDSII file has problem.

---

## GDS-012

(warn) Technology Definition for Layer Number %d and Purpose %d not defined.

**Description**

The Given Design doesn't have a LAYER Definition for the Layer in question in the Associated Technology Library.

---

## GDS-013

(warn) Layer Mapping for GDSII LAYER %d and DATATYPE %d not specified.

**Description**

For the GDSII LAYER and DATATYPE, layer mapping is not specified in the layer map file.

---

## GDS-014

(error) Library name has spaces.

**Description**

A library name had spaces.

**What Next**

Check that the given library name is valid simple path.

---

## GDS-015

(error) Library name specified %s cannot be same as workspace library.

**Description**

The Given library name is not allowed.

---

## GDS-016

(error) Wrong Block Map file Syntax.

Feedback

**Description**

The syntax of Block Map file is not allowed.

---

## GDS-017

(error) Wrong layer map file syntax at line %lu.

**Description**

The syntax of Layer Map file at the specified line number is not allowed.

**What Next**

Please check the man-page of the read_gds(2), write_gds(2) for the correct syntax.

---

## GDS-018

(warning) FIXEDMASK defined, geometry without MASK will be set as same_mask type.

**Description**

The module has the fixed mask attribute set but the geometry doesn't have a mask attribute defined. The geometry will be set as same_mask type.

---

## GDS-019

(error) The mask number exceeds the number of masks defined for the tech layer %d.

**Description**

The Mask number should not exceed the number of masks defined in the Tech layer.

---

## GDS-020

(error) Library '%s' is not a physical lib_cell library.

**Description**

The library specified for read GDSII must be a physical lib_cell library.

**What Next**

Choose a physical lib_cell library to read GDSII in.

Feedback

## GDS-021

(information) Read GDSII into %s library '%s'.

**Description**

This message reports the name of the library to accommodate GDSII data. It also indicates whether the library is an existing library or a new library.

**What Next**

No action required.

## GDS-022

(warn) The Application option "file.gds.port_type_map" is not defined before "read_gds" command.

**Description**

The Application option "file.gds.port_type_map" is not defined before "read_gds" command. GDSII generated libraries will not generate any PG pins. This may create problem in final library generation. Please check your setup.

## GDS-023

(warn) The Module '%s' does not have any port matching the PG names provided by "file.gds.port_type_map" application option

**Description**

The Module does not have any port matching the PG names provided by "file.gds.port_type_map" application option. GDSII generated Module will not have any PG pins. This may create problem in final library generation. Please check your setup.

## GDS-024

(warn) Text '%s' is not on routing layer. Tracing for the given text may result in identification of wrong pins.

**Description**

Text '%s' is not on routing layer. Tracing for this text has been enabled on all routing layers by setting the application option "file.gds.trace_unmapped_text".Also, routing layer to text layer mapping is not specified using "file.gds.text_layer_map" application option. Connectivity tracing for the text will be done on all layers. This may lead to erroneous

Feedback

result. Please specify routing layer to text layer mapping using "file.gds.text_layer_map". Please check your setup.

**What Next**

```
file.gds.port_type_map(3),
file.gds.trace_unmapped_text(3),
```

## GDS-025

(error) The %d is outside the allowed valid range for %s in layer map. Valid range is between %d and %d.

**Description**

Value specified outside valid range.

## GDS-026

(error) The special character %s is not allowed in %s field of layer map.

**Description**

The special character is not allowed in layer map field.

## GDS-027

(error) The input GDSII file name and the output ascii file name can't be same.

**Description**

GDSII file name and Ascii file name are same.

## GDS-028

(warning) The string '%s' is longer than %lu characters. Please use '-long_names' for proper output.

**Description**

The length of the string name to be written in the GDSII file is longer than the supported length. The name will be truncated.

**What Next**

The command "write_gds" needs to be run with "-long_names" option.

Feedback

## GDS-029

(Information) %s '%s' already exists (will %s).

**Description**

This message indicates that specified object already exists when reading a GDSII file into an existing library. The library manager shell, lm_shell, will take action according to the merge settings.

**What Next**

No action required.

## GDS-030

(information) Change %s name '%s' from GDS file to '%s' to resolve a conflict.

**Description**

When the object to be created by read GDS conflict with an existing object, there is an attempt to change the name of the object to be created to a unique name to resolve the conflict.

**What Next**

No action required.

## GDS-031

(warning) Merge action in '%s' will be ignored when read GDS into a new library.

**Description**

When read GDS into a new library, merge settings will be ignored.

**What Next**

Do not specify merge related options when read GDS into a new library.

## GDS-032

(warning) The width of the boundary %d is more than width %d of site width.

**Description**

The width of the boundary is more than width of the site def.

Feedback

**What Next**

May need to check the boundary or the site def of the lib-cell.

---

## GDS-033

(warning) The height of the boundary %d is more than height %d of site def.

**Description**

The height of the boundary is more than height of the site def.

**What Next**

May need to check the boundary or the site def of the lib-cell.

---

## GDS-034

(warning) The user specified view '%s' for lib-cell '%s' is not available.

**Description**

The user specified view for lib-cells is not available for this lib-cell design. The definition for the current lib-cell design will not be written to the GDSII file.

**What Next**

The user may need to change the view to be written to GDSII file for the lib-cell design using option "-lib_cell_view".

---

## GDS-035

(Error) The design '%s' is found to have mismatches. Use allow_design_mismatch option to write GDSII file.

**Description**

The design is found to have mismatches due to which allow_design_mismatch option needs to be used to write GDSII file. Please use report_design_mismatch user command initially to review all the mismatches. If the design mismatches are found to be fine, use the option allow_design_mismatch to write the GDSII file.

**What Next**

Use the command report_design_mismatch to review all the design mismatches. The user may then use allow_design_mismatch option to write the gds file.

Feedback

## GDS-036

(error) The mask number(%s) for the tech layer %d doesn't belong to a VIA Cut Layer.

**Description**

The Mask number should be associated with a VIA cut layer. The input layer number is not a VIA cut layer.

**What Next**

Please check the GDSII layer map.

## GDS-037

(warning) Unable to obtain mask value for surround on lower metal layer of via '%s'.

**Description**

This warning message is issued since the surround on lower metal layer of the via cannot touch any pin in the design. This could be a via that only connects two layers but is not on a pin.

**What Next**

Please check the via and the design data if needed.

## GDS-038

(warn) Zero width spur in the input point list {%s}.

**Description**

The Input point list has zero width spur. The polygon will not be added.

**What Next**

Please check the point list been passed.

## GDS-039

(error) Invalid layer-name '%s' specified at line %lu in the input layer-map file.

**Description**

A layer by the specified name does not exist. Please check the layers present in the associated technology file.

Feedback

## GDS-040

(error) Invalid value '%s' specified for '%s', at line %lu in the input layer-map file.

**Description**

The value specified for the flag is not a valid value. Please check the man-page of read_gds(2), write_gds(2) for the allowed values. The stream file will still be written out but ignoring the line with invalid value in the layer mapping file. This error will be reported in several scenarios. For example, the input value cannot be recognized due to type error or the use_type in newer version of the tool is not supported in the older version.

**What Next**

Please modify the layer mapping file for the invalid value or use the latest version of the tool.

## GDS-041

(warning) Ignoring the invalid layer map file syntax at line %lu.

**Description**

The syntax of Layer Map file at the specified line number is not allowed.

**What Next**

Please check the man-page of the read_gds(2), write_gds(2) for the correct syntax.

## GDS-042

(error) Line %lu: multi-pattern map entry with wild-card cannot co-exist with multi-pattern map entry with specified layer for same mask type.

**Description**

This message occurs when there is invalid line in the layer mapping file. Multi-pattern map entry with wildcard cannot co-exist with Multi-pattern map entry with specified layer for same mask type. For example, M1 * *:108 M1 12 12:108

**What Next**

Remove Multi-pattern map with wildcard or Multi-pattern map with specified layer.

## GDS-044

(error) The ICC2 layer number '%d', specified at line %lu is not a valid value for %s layer map format.

Feedback

**Description**

The ICC2 layer number specified should follow the limits below: icc2 : 0 - 32767 icc default : 0 - 255 icc extended : 0 - 4095

**What Next**

Use layer-numbers within the specified limits.

---

## GDS-045

(warning) Technology layer for layer number %d, does not exist. Ignoring the layer-map entry at line %lu.

**Description**

The technology file associated with the current library does not have a layer defined with the given layer-number.

---

## GDS-046

(warning) The magnification %f is different than the default value of 1.0. This will be ignored.

**Description**

The magnification specified is different from 1.0 and hence is ignored. Only magnification of 1.0 is supported now.

---

## GDS-047

(error) Invalid purpose-name '%s' specified at line %lu in the input layer-map file.

**Description**

A tech-layer purpose by the specified name does not exist. Please check the tech-layer purposes present in the associated technology file.

---

## GDS-048

(error) The simple via def '%s' does not have the lower overhang metal layer set.

**Description**

A simple via def has both the lower and upper metal overhang defined. A simple via_def cannot have either of them not defined .

Feedback

**What Next**

Use 'report_via_defs' to review the set of via_defs.

---

## GDS-049

(error) The simple via def '%s' does not have the upper overhang metal layer set.

**Description**

A simple via def has both the lower and upper metal overhang defined. A simple via_def cannot have either of them not defined .

**What Next**

Use 'report_via_defs' to review the set of via_defs.

---

## GDS-050

(information) Translating structure '%s' as cell '%s/%s'.

**Description**

The GDS structure being read is translated to the given cell name.

---

## GDS-051

(error) Input file '%s' is neither a GDSII nor an OASIS file.

**Description**

The input file can be either a GDSII file or an OASIS file.

---

## GDS-052

(warning) Ignoring the layer map entry at line %lu as it doesn't have usage_type as "area_fill".

SH DESCRIPTION Layer map entries without usage_type as "area_fill" is invalid in this command.

**What Next**

Specify usage_type as "area_fill" for every layer_map entry for passive_fill data.

Feedback

## GDS-053

(error) Input file '%s' is not a flat structured GDSII file.

**Description**

Only flat structured GDSII file can be given as input file when merge_action is replace_layer and append.

## GDS-054

(error) Input file '%s' is not a appropriate GDSII file.

**Description**

Illegal usage of the command.

## GDS-055

(error) Reading of input GDSII file failed.

**Description**

When reading of GDSII file fails, design will be restored to it's previous state.

## GDS-056

(information) no mapping information for system purpose %s in layer mapping file. Mapped to default data type.

**Description**

When there is explicit mapping of system purpose (-1, -2, -3 and -4) in layer mapping file, the tool will use that mapping; for example, with the mapping "all 32:-1 32:55", the tool maps database layer "32:-1" to GDS layer 32:55; if no mapping for these system purposes, the tool will follow the mapping of the purpose 0 of the same layer. For example, if the mapping has "all 32:0 32:70", system purpose "-1" will be mapped to GDS layer 32:70. Please note that GDS datatype can't be any data less than 0, the usage like "all 32:-1 132:-1" is invalid for GDS language, and the datatype will be taken as 0 like "A 32:-1 132 0".

**What Next**

If the implicit mapping as described above does not meet the requirement, you can explicitly map the system purpose (-1, -2, -3 and -4) in layer mapping file.

Feedback

## GDS-057

(warning) Found child design '%s' for option "-bbox_list".

**Description**

Option "-bbox_list" is only for flatten design, otherwise, it might cause GDS file inconsistency.

**What Next**

To flatten the design and try the option "-bbox_list" again.

## GDS-058

(error) Block %s is not Found in the Library.

**Description**

The given VIB block name is not Found in the current design of the Specified Library.

## GDS-059

(information) Outputting hierarchy '%s' from '%s' view.

**Description**

write_gds is generating the structure for the hierarchy.

## GDS-060

(error) command failed at outputting structure '%s'.

**Description**

write_gds failed at generating the content for the structure.

**What Next**

Please check what's the exact data caused the failure.

## GDS-061

(warning) The user specified view '%s' for cell '%s' is not available.

Feedback

**Description**

The user specified view for cells is not available for this cell. If all specified views cannot be found for the cell, design view will be output, if any. The definition for the cell will not be written to the GDSII file, if all specified views and design view cannot be found.

**What Next**

The user may need to change the view to be written to GDSII file for the cell using option "-switch_view_list".

---

## GDS-062

(information) Cell instance '%s' referred to reference design '%s' is retained and will refer to designs in reference library.

**Description**

It means the block to which the cell instance referred is retained in block mapping file, and the STRUCTURE record in GDSII file is skipped, the cell instance will be bound to corresponding reference design in reference library.

**What Next**

No action required.

---

## GDS-063

(error) Failed to write out Path '%s'.

**Description**

The path is irregular, please check its coordinates.

---

## GDS-064

(error) Input file '%s' is OASIS format file instead of GDSII.

**Description**

Command 'read_gds' only support GDSII format file.

**What Next**

Please use 'read_oasis' command to read in this file.

Feedback

## GDS-065

(error) Invalid value for option -mask_shift_layers.

**Description**

There can not be duplicated layers in the specified value.

## GDS-066

(error) Mask shifted cell '%s' does not have its corresponding mapping in the map file.

**Description**

There is not a corresponding mapping for the mask shifted cell in the map file.

**What Next**

Please add the corresponding mapping to the map file and then try again.

## GDS-067

(error) Mask shift property mismatch on cell '%s'.

**Description**

There is mask shift property mismatch between different map files.

**What Next**

Please correct the mask shift property and then try again.

## GDS-068

(error) Fail to remove the file '%s'.

**Description**

The gds file is removed if failure occurs.

**What Next**

Please ignore the file.

## GDS-069

(warning) Found uncolored multi-patterning shapes in the database which won't be output.

Feedback

**Description**

This warning message occurs when there are some uncolored multi-patterning shapes in the database, the check is controlled by the app option file.gds.ignore_uncolored_shapes or the syntax forbidNoMaskConstraint in technology file. write_gds will continue to write out all other shapes except the uncolored ones.

**What Next**

Please use command check_shapes -uncolored to find out all these shapes and color them, or turn off the app option mentioned above to output uncolored shapes. Note, if the syntax forbidNoMaskConstraint is 1, the value of the app option will be ignored.

---

## GDS-070

(error) Unsupported option '%s' to new engine of write_gds.

**Description**

It means the command option or app option isn't supported at current stage.

**What Next**

To achieve the option feature, turn off the file.gds.enable_new_writer to output GDSII layout file.

---

## GDS-071

(warning) The deprecated option '%s' is ignored when the new_writer app option is enabled.

**Description**

It means that the command option or app option is deprecated and ignored when turning on the app option "file.gds.enable_new_writer".

**What Next**

Remove the option setting

---

## GDS-072

(Information) Empty block %s is excluded in the GDSII file.

**Description**

This message indicates that the empty structure defintion(s) and instance(s) are excluded in the GDSII file for the empty block specified.

Feedback

**What Next**

No action required.

---

## GDS-073

(warning) GDSII Parsing Warning: %s

**Description**

GDSII file has problem. Ignore the problem geometory and continue parsing.

---

## GDS-074

(warning) GDSII Parsing Warning: %s

**Description**

GDSII file has problem. Ignore the problem geometory and continue parsing.

---

## GDS-075

(information) Output mask-shifted cell name according to rename files.

**Description**

This message occurs when -rename_cell is specified to write_gds and the rename files are only for mask-shifted cells.

**What Next**

No action required.

---

## GDS-076

(error) %s "%s" has an angle of %.3f degrees which is not supported currently.

**Description**

This message occurs when sref or aref specified an unsupported ANGLE value. Supported values are: 0/90/180/270 degrees.

**What Next**

Check the GDS file to find the "abnormal" object, change the ANGLE value of it, or delete the object.

Feedback

## GDS-077

(warning) Ignoring %s, because there is another design with the same name %s.

**Description**

This warning message occurs when app option "file.gds.output_first_same_name_cell" is on, write_gds only writes the first encountered cell.

**What Next**

Please turn off the app option "file.gds.output_first_same_name_cell" if you expect that write_gds renames the cells with the same name and writes them as different structures in the output GDSII file.

## GDS-078

(warning) Found uncolored shapes on multi-patterning layer in the database which won't be output.

**Description**

This warning message occurs when there are some uncolored shapes on multi-patterning layer in the database, the check is controlled by the app option file.gds.check_mask_constraints, value "loose". write_gds will continue to write out all other shapes except the uncolored ones.

**What Next**

Please use command check_shapes to find out all these shapes and color them, or turn off the app option mentioned above to output uncolored shapes.

## GDS-079

(warning) Found colored shapes on uni-patterning layers in the database.

**Description**

This warning message occurs when there are some colored shapes on uni-patterning layers in the database, the check is controlled by the app option file.gds.check_mask_constraints, value "loose". write_gds will continue to write out all shapes.

**What Next**

Please use command check_shapes to find out all these shapes.

Feedback

## GDS-080

(error) Found mismatch on shapes and layers for color check.

**Description**

This error message occurs when there is mismatch on shapes and layers for color check in the database, the check is controlled by the app option file.gds.check_mask_constraints, value "strict".

**What Next**

Please use command check_shapes to find out all these shapes and fix them.

## GDS-081

(error) The layer '%s' provided by "file.gds.text_layer_map" application option was not found.

**Description**

A layer by the specified name in "file.gds.text_layer_map" application option does not exist. Please check the layers present in the associated technology file.

**What Next**

```
file.gds.text_layer_map(3)
```

## GDS-082

(warning) Ignoring invalid value '%s' provided for app option file.gds.shape_use_for_pg_mesh.

SH DESCRIPTION Value provided for app option file.gds.shape_use_for_pg_mesh is invalid. Valid values are case-insensitive "stripe", "ring", "lib_cell_pin_connect" or "macro_pin_connect".

**What Next**

Specify correct values for app option.

## GDS-083

(error) The design %s is not of the 3dic_top type.

Feedback

**Description**

The design is not of the 3dic_top type

**What Next**

Make sure the design is 3dic_top type and run the command again

---

## GDS-084

(warning) object type is converted to all, pincode to none, use type to all in double pattern map rule.

**Description**

The layer mapping file in IC Compiler format supports double pattern map rule for supported masks. In the output rule, the object type is converted to all, pincode to none, use type to all no mattern what object type, pincode, use type are in original rule.

---

## GDS-085

(warning) Layer mapping file in IC Compiler II format does not contain the equivalent logical information for %s. This rule is skipped.

**Description**

The layer mapping file for MWNetType in IC Compiler II format does not contain the equivalent logical information as the original layer mapping file.

The IC Compiler II format file will skip the following IC Compiler special format MWNetType. * U for up conduction layer (upper layer in a via) * D for down conduction layer (lower layer in a via) * X for power or ground wires or contacts with a "signal" or "tie-off" routing type

The only supported use types are as follows. power, ground, clock, signal, and the sub types for them four; area_fill, boundary, hard_placement_blockage, soft_placement_blockage, routing_blockage, track, bridge_shape, keepout_region, instance_specific_mask, same_net_feedthrough_tsv, hi_voltage, lo_voltage, route_guide,single_row_via_ladder_pattern_must_join_allowed, routing_blockage,is_allow_metal_fill_only, routing_blockage,reserve_for_top_level_routing

---

## GDS-086

(warning) Layer mapping file in IC Compiler format does not contain the equivalent logical information for %s. This rule is skipped.

Feedback

**Description**

The layer mapping file in IC Compiler format does only support mask rules containing the following: same_mask, mask_one, mask_two and mask_three. This rule with other mask is skipped.

---

## GDS-087

(warning) Layer mapping file in IC Compiler format does not contain the equivalent use type for %s. This rule is skipped.

**Description**

The layer mapping file in IC Compiler format supports the following use types: use_all, power, clock, signal, ground, boundary, routing_blockage, hard_placement_blockage, soft_placement_blockage, hard_macro_placement_blockage, pin_blockage, pin_guide, route_guide, icc1_mw_net_type_u, icc1_mw_net_type_d, icc1_mw_net_type_x. Certain use types, for example, "power,net" are turned to a power net rule, "power, pin" to a power pin rule and "power,port" to a power port rule in IC Compiler II format.

Other examples of such use types are: power,net power,port power,pin ground,net ground,port ground,pin clock,net clock,port clock,pin signal,net signal,port signal,pin

---

## GDS-088

(warning) Layer mapping file in IC Compiler format does not contain an equivalent rule for %s.

**Description**

The layer mapping file in IC Compiler format does not support the following IC Compiler II layer mapping file syntax: [read_always { true | false [-mapped_only | -ignore_missing_layers ] } ] [blockage_as_zero_spacing {true|false}] [cell_prop_attribute attribute_number] [net_prop_attribute attribute_number] [pin_prop_attribute attribute_number] [via_prop_attribute attribute_number]

---

## GDS-110

(warning) Unplaced cell '%s' is written to the GDSII file.

**Description**

This warning is displayed upto 20 times when unplaced cells are present in the design. By default unplaced cells will be outputted to GDSII file.

Feedback

**What Next**

Please use option -skip_unplaced_cells if you do not need unplaced cells.

## GDS-111

(warning) Ignoring line %d in file '%s' for the reason: '%s'.

**Description**

This warning message occurs when there is something wrong in the marker layer map file, which was instroduced to duplicate routing data in database to specified GDSII/OASIS marker layer.

There are 4 types (layer, region, net and cell) in the marker layer map file. Each line indicates a mapping, the mapping starts with marker and ends with ";", any text after a semicolon "#" in each line is a comment.

For all the types, -FBlayer, -cad_layer and -data_type are required options. When the type value is not cell, the -layer option is also required option. When the type value is region, the -bbox is also required option. When the type value is net, the -net, -from and -to are also required options. When the type value is cell, the -cell is also required option.

The following statements are the example for all kinds of type. marker -type layer -layer METAL -FBlayer FB9 -cad_layer 135 -data_type 1999 ; #layer based marker -type region -layer METAL -FBlayer FB9 -bbox {{1294 479} {1415 507}} -cad_layer 222 -data_type 1999 ; #region based marker -type net -net VDD -layer METAL3 -from U458/ZN -to H1/H3/B1 -FBlayer FB6 -cad_layer 333 -data_type 1999 ; #net based marker -type cell -cell inst2 -FBlayer FB7 -cad_layer 444 -data_type 1999 ; #cell based

**What Next**

Please check and correct the marker layer map file according to the message about the warning reason.

# GRD

## GRD-0001

(Fatal) GPGPU is NOT supported on this platform or build.

**Description**

GPGPU (grdgenxo -gpu) is supported only on linux platform and dedicate released builds.

Feedback

**What Next**

If you are running grdgenxo on linux platform and still get this error, contact the Synopsys Support Center.

---

## GRD-0002

(Fatal) Fail to run grdgenxo on GPGPU.

**Description**

The grdgenxo tool was run with the *-gpu* option but failed to initialize gpu devices.

**What Next**

Contact the Synopsys Support Center.

---

## GRD-0003

(Fatal) grdgenxo -inc_clean is NOT to be used with any other command line options.

**Description**

The grdgenxo tool runs with the *-inc_clean* option to clean up the incremental grdgenxo work directory. Do not use the *-inc_clean* option with any other command line options.

**What Next**

Remove all other options besides the *-inc_clean* option and run the *grdgenxo -inc_clean* command again.

---

## GRD-0004

(Warning) grdgenxo -inc_clean should only be run once, before any job is submitted.

**Description**

Running the *grdgenxo -inc_clean* command cleans up the incremental grdgenxo work directory. This command should only be run one time before submitting other grdgenxo jobs.

**What Next**

If the *grdgenxo -inc_clean* command is being run before any other grdgenxo job is submitted, no action is needed.

If there are submitted grdgenxo jobs in the current working directory before the *grdgenxo -inc_clean* run, kill the jobs, rerun the *grdgenxo -inc_clean* command, and then submit the other grdgenxo jobs.

Feedback

## GRD-0005

(Fatal) Bad usage of -zirva option. Expected usage: grdgenxo -zirva [<config_file>] <itf_file>.

**Description**

Run "grdgenxo -zirva" with bad usage. Expected usage: grdgenxo -zirva [<config_file>] <itf_file>.

**What Next**

Correct grdgenxo command and run again.

## GRD-0006

(Fatal) INTERNAL Error: failed internal consistency check in 3D model generation.

**Description**

An internal consistency check has failed in 3D model generation.

**What Next**

Contact the Synopsys Support Center.

## GRD-0007

(Fatal) %s option requires filename.

**Description**

A filename is not supplied for a grdgenxo option.

**What Next**

Add a filename corresponding to the grdgenxo option mentioned in the error message and run again.

## GRD-0008

(Warning) The tech file to create TLU+ is created by metal composer. It should not be used for sign-off extraction purpose. Please contact foundry to get the official version.

**Description**

The ITF to create TLU+ is created by metal composer. It should not be used for sign-off extraction purpose, because the ITF may be not fully qualified by foundry yet.

Feedback

**What Next**

Please contact foundry to get the official version.

---

## GRD-0009

(Fatal) ITF file is NOT provided.

**Description**

An input ITF file was not provided to the grdgenxo tool.

**What Next**

Specify an ITF file by using the *-input* option with the *grdgenxo* command.

---

## GRD-0010

(Fatal) Do NOT use this option -goldxModel. It is used for experimenting GoldX model.

**Description**

This option "-goldxModel" for grdgenxo is only for SNPS testing internal models.

**What Next**

Remove "-goldxModel" in grdgenxo command.

---

## GRD-0011

(Fatal) Incorrect command line option: %s. Try "grdgenxo -usage" for help.

**Description**

An unknown option was used with the *grdgenxo* command.

Use the *grdgenxo -help* or *grdgenxo -usage* command to list the supported options for the grdgenxo tool.

**What Next**

Correct the syntax and resubmit the *grdgenxo* command.

---

## GRD-0012

(Fatal) Multiple ITF files supplied.

Feedback

**Description**

More than one ITF files are supplied for the grdgenxo run.

**What Next**

Remove redundant ITF files in the grdgenxo command line. Supply only one ITF file.

---

## GRD-0013

(Fatal) Incorrect command line option for launching %s.

**Description**

The grdgenxo tool calls external tools (for example, gds2cap, quickcap, rapid3d) to generate device models. This message indicates that arguments for those tools are not set correctly in the command line.

**What Next**

Refer to detailed messages in the log file and correct arguments for those external tools.

---

## GRD-0014

(Information) Done with all the structures in the configuration file!

**Description**

You are running "grdgenxo -config <config_file>". All the structures in config_file have been processed.

**What Next**

Information only. No action is needed.

---

## GRD-0015

(Fatal) Technology statements used in incremental run must be the same (%s vs. %s).

**Description**

The grdgenxo tool supports incremental runs only for ITF modifications that have localized capacitance effects. This error message indicates that some modifications in the ITF file are not supported for incremental grdgenxo runs.

Feedback

**What Next**

See the StarRC User Guide and Command Reference for more information on what kinds of modifications are supported for incremental runs. Modify the ITF file to comply with the incremental run rules.

Alternatively, run the grdgenxo tool without the incremental option.

---

## GRD-0016

(Fatal) Failed to open ITF file: %s.

**Description**

The grdgenxo tool failed to open the input ITF file.

**What Next**

Check the availability and access rights of your input ITF file. Address any issues, then resubmit the grdgenxo run. If it still fails, contact the Synopsys Support Center.

---

## GRD-0017

(Fatal) Failed to write to file: %s.

**Description**

The grdgenxo tool fails to write to a file.

**What Next**

Check the disk to make sure the disk is writable and not full. Address any issues, then resubmit the grdgenxo run. If it still fails, contact the Synopsys Support Center.

---

## GRD-0018

(Fatal) INTERNAL Error: failed to read from file: %s.

**Description**

The grdgenxo tool fails to read an internal file.

**What Next**

Resubmit the grdgenxo run. If it still fails, contact the Synopsys Support Center.

Feedback

## GRD-0019

(Fatal) Specify the old ITF file previously used to produce %s.nxtgrd database by using -old_itf <old_itf_filename> option.

**Description**

You are running the grdgenxo tool with the *-inc* option.

In incremental grdgenxo runs, the grdgenxo tool first tries to find the old ITF file from the previous grdgenxo run directory. If the grdgenxo tool cannot find it, the tool checks whether the *-old_itf* option was used to specify an ITF file.

This error message indicates that the grdgenxo tool cannot find the old ITF file, either from the previous grdgenxo run directory or from the *-old_itf* option.

**What Next**

Specify the old ITF file previously used to produce the nxtgrd database mentioned in the error message by using the *-old_itf <old_itf_filename>* option.

Alternatively, run the grdgenxo tool without the incremental option.

## GRD-0020

(Fatal) A different ITF file has been specified for this processor.

**Description**

You are running the grdgenxo tool with the *-inc* option.

In incremental grdgenxo runs, the grdgenxo tool checks that all grdgenxo processors are using the same ITF file. In this case, different ITF files have been specified for different processors.

**What Next**

Check the grdgenxo submission commands to ensure that all grdgenxo commands use the same ITF file.

## GRD-0021

(Fatal) Current working directory was created with an old version of grdgenxo. Delete it and run grdgenxo from scratch, or use an older version of grdgenxo (earlier than version A-2007.12) for the current working directory.

Feedback

### Description

The grdgenxo tool detects that the current working directory was created with a version earlier than version A-2007.12. The current grdgenxo version is not compatible with the working directory.

### What Next

Remove the current working directory and resubmit the grdgenxo run.

---

## GRD-0022

(Fatal) The input ITF file (%s) is different from the file used in the previous run (%s). Please remove the existing working directory and re-run grdgenxo from scratch.

### Description

You are running the grdgenxo tool with an existing grdgenxo working directory. However, the ITF file supplied for the current grdgenxo run is different from the one detected in the existing working directory. The grdgenxo tool supports continuous running with the existing working directory only when the new ITF is the same as the one used in the previous run.

### What Next

Remove the existing working directory and rerun the grdgenxo tool.

---

## GRD-0023

(Warning) The tech file is created by metal composer. It should not be used for sign-off extraction purpose. Please contact foundry to get the official version.

### Description

The ITF file used to generate this nxtgrd file was created by the metal composer function. The ITF and nxtgrd files should not be used for signoff extraction because they might not be fully qualified by the foundry.

### What Next

Contact the foundry to obtain official nxtgrd or ITF files.

---

## GRD-0024

(Fatal) Front-end process stacks till M1 are not equal for the two technologies. Please align all conductors, vias and dielectrics below M1.

Feedback

**Description**

You are running the grdgenxo tool with the *-reuse_device_models* option. The grdgenxo tool has found an issue during equivalence checking between the current ITF file and the ITF file under the reuse_device_models directory for the layers below M1.

**What Next**

Align the layer stacks below M1 between the current ITF and the ITF file under the reuse_device_models directory, then resubmit the grdgenxo run with the *-reuse_device_models* option.

Alternatively, run the grdgenxo tool without the *-reuse_device_models* option.

---

## GRD-0025

(Fatal) Invalid SCT file in reference technology.

**Description**

You are running the grdgenxo tool with the *-reuse_device_models* option. The grdgenxo tool has found an issue with the device models data file under the reuse_device_models directory.

**What Next**

Run the grdgenxo tool without the *-reuse_device_models* option to regenerate the device models.

---

## GRD-0026

(Fatal) Can NOT initialize SCT Table settings.

**Description**

The grdgenxo tool cannot generate the SCT table.

Possible reasons are as follows:

- You are running the grdgenxo tool with the existing working directory. The grdgenxo tool tries to reuse the SCT table in the existing working directory and finds that the SCT table was not correctly generated in the previous grdgenxo run.

- The path to another tool (such as the gds2cap, quickcap, or rapid3d tools) is not set correctly.

- Arguments for another tool are not set correctly.

Feedback

**What Next**

Check the log file for more information about the specific error condition. Address any issues and resubmit the grdgenxo run.

To address problems with SCT tables, remove all of the sct directories under the existing working directory and resubmit the grdgenxo run.

---

## GRD-0027

(Fatal) Can NOT generate SCT file. Check %s.failure for a list of the failing simulations and check the log files in the specified directories. Remove them before launching grdgenxo again.

**Description**

The grdgenxo tool cannot generate the SCT file.

**What Next**

Check the *.failure file mentioned in the error message for a list of the failing simulations and check the log files in the directories for those failing simulations. Remove the directories for the failing simulations and run the grdgenxo tool again.

---

## GRD-0028

(Fatal) Layer %s has a positive bottom thickness variation specified that is greater than the inter-layer conductor separation.

**Description**

Bottom thickness variation of conducting layer %s is positive and such large that it is greater than the inter-layer distance between this layer and conducting layer below. This makes the bottom of conducting layer %s extend into conducting layer below when bottom thickness variation is applied. StarRC does not support this kind of scenario.

**What Next**

Check BOTTOM_THICKNESS_VS_SI_WIDTH or ILD_VS_WIDTH_AND_SPACING table (depending on which table is used in current nxtgrd) to correct the violated numbers. Rerun grdgenxo and StarRC.

---

## GRD-0029

(Fatal) To use reluctance capability, both of the following must be specified: RELUCTANCE_FREQUENCIES, and RELUCTANCE_SUBSTRATE_RHO.

Feedback

**Description**

StarRC uses RELUCTANCE_SUBSTRATE_RHO to model non-ideal substrate effects by computing the skin depth of substrate. RELUCTANCE_SUBSTRATE_RHO must be specified with RELUCTANCE_FREQUENCIES. In this case, only RELUCTANCE_SUBSTRATE_RHO is specified.

**What Next**

Specify RELUCTANCE_FREQUENCIES in ITF and re-run grdgenxo.

---

## GRD-0030

(Information) Building %s.

**Description**

This information message indicates that 3D model generation by grdgenxo has correctly started.

**What Next**

Information only; no action is needed.

---

## GRD-0031

(Fatal) INTERNAL Error: failed internal consistency check in 3D model generation.

**Description**

An internal consistency check has failed in 3D model generation.

**What Next**

Contact the Synopsys Support Center.

---

## GRD-0032

(Fatal) Fatal error in SCT flow (mkdir %s): %s.

**Description**

StarRC cannot create a directory in the SCT flow.

**What Next**

Check the file system to see whether the disk is full or the connection is unstable. Address the issues and restart the grdgenxo run.

Feedback

## GRD-0033

(Warning) Nodes %s found in %s/%s were unexpected.

**Description**

Unexpected nodes have been detected during one or more simulations. The error message lists the unexpected nodes. The presence of these nodes might be detrimental for capacitance accuracy.

**What Next**

Review the list of nodes and the technology files and remove the problem nodes. Remove the folders containing the log file indicated in the message and relaunch the grdgenxo run.

## GRD-0034

(Severe) Failed to launch %s for %s, please check %s. End of %s.log: %s.

**Description**

The Field Solver simulation failed due to some unexpected reason. The grdgenxo tool cannot build three-dimensional model tables.

**What Next**

Review the information contained in the log file specified in the error message and perform the necessary fixes and cleanup. Rerun the grdgenxo tool to generate the models.

## GRD-0035

(Warning) Failed to launch field solver for %s, SCT table might be incomplete, please check %s.

**Description**

The Field Solver failed to launch for some simulations. Therefore the SCT table containing three-dimensional models might be incomplete and accuracy issues might result.

**What Next**

Check the specified log file and review why the Field Solver failed. Remove the folders of the failed simulations and rerun the grdgenxo tool.

## GRD-0036

(Fatal) Fatal error in SCT flow (while locking %s): %s.

Feedback

**Description**

StarRC cannot create or lock a file required for three-dimensional model generation.

**What Next**

Check the file system to see whether the disk is full or the connection is unstable. Address the issues and restart the grdgenxo run.

---

## GRD-0037

(Information) SCT file already generated and reused.

**Description**

This message indicates that the SCT file containing 3D models generated by grdgenxo has already been generated by a previous run of the tool. Consequently, grdgenxo skips 3D model generation and uses the previously-generated file.

**What Next**

If you would like to use the previously-generated SCT file, no action is required on your part. Otherwise, remove the SCT folder in grdgenxo working directory and rerun.

---

## GRD-0038

(Warning) SCT file failed to be generated in a previous run. See %s for additional info.

**Description**

The warning indicates that 3D model generation has failed in previous run of grdgenxo. The nxtgrd file will not be generated. The file indicated in the message contains additional instructions to follow to solve the problem.

**What Next**

Follow the instructions indicated in the file that is specified and rerun grdgenxo.

---

## GRD-0039

(Fatal) Could NOT initialize 3D model file generation.

**Description**

The grdgenxo tool cannot initialize the model which will contain 3D models. Therefore model generation cannot start.

Feedback

**What Next**

Check the file system to see whether the disk is full or the connection is unstable. Resolve any issues and resubmit the grdgenxo run.

---

## GRD-0040

(Fatal) Could NOT read field solver technology file (after %d attempts).

**Description**

The grdgenxo tool cannot read the field solver technology file that has been created by the tool that generates 3D models.

**What Next**

Check the file system to see whether the disk is full or the connection is unstable. Resolve any issues and resubmit the grdgenxo run.

---

## GRD-0041

(Fatal) Could NOT write field solver technology file.

**Description**

The grdgenxo tool cannot write the field solver technology file needed to generate 3D models.

**What Next**

Check the file system to see whether the disk is full or the connection is unstable. Resolve any issues and resubmit the grdgenxo run.

---

## GRD-0042

(Fatal) Cannot select Field-Poly layer for gate layer %s (missing or duplicated DEVICE_TYPE).

**Description**

The ITF file contains an invalid combination of field poly and gate layers. As a result, the grdgenxo tool cannot generate 3D models. Field poly and gate conductor layers should be defined together with unique DEVICE_TYPE statements to allow the grdgenxo tool to select a unique pair of gate and field poly layers for each device type.

Feedback

**What Next**

Review the syntax in the ITF file for gate and field poly conductors. Make sure that each gate has its corresponding field poly defined together with a unique DEVICE_TYPE statement.

---

## GRD-0043

(Warning) There were unexpected nodes found in the reports of the simulations launched for the 3D models generation. It may cause accuracy issues in 3D modeling. Please see %s.

**Description**

This message informs that unexpected nodes had been detected in one or more 3D model simulations. Unexpected nodes generally indicate a technology modeling issue such as parts of conductors that are not supposed to be created by the field solver. They can cause inaccuracies in capacitance calculation and should be removed.

**What Next**

Review the technology file (ITF or QTF files) and fix any modeling issue.

---

## GRD-0044

(Severe) Failed to check %s's version.

**Description**

The grdgenxo tool cannot determine the version of the field solver. You might be using an incompatible or old field solver.

**What Next**

Make sure that you have access and the correct environment variables configured to run the indicated version of the field solver. Clean up the model folder and rerun the grdgenxo tool.

---

## GRD-0045

(Severe) Failed to find %s's version.

**Description**

The grdgenxo tool cannot detect the version of the field solver. You might be using an incompatible or old field solver.

Feedback

**What Next**

Make sure that you have access and the correct environment variables configured to run the indicated version of the field solver. Clean up the model folder and rerun the grdgenxo tool.

---

## GRD-0046

(Severe) Failed to execute %s.

**Description**

grdgenxo failed to spawn the field solver simulation required to run 3D model generation.

**What Next**

Make sure that you have access and the correct environment variables configured to run the field solver, cleanup the model folder and rerun grdgenxo.

---

## GRD-0047

(Severe) Version of %s must be >= %s.%s.

**Description**

The grdgenxo tool has detected an incompatibility issue with the field solver version. To support the 3D model functionality, the field solver version should be higher that the one indicated in the message.

**What Next**

Make sure that you have access and the correct environment variables configured to run the indicated version of the field solver. Clean up the model folder and rerun the grdgenxo tool.

---

## GRD-0048

(Severe) Error: NOT supported differences in conductor %s between ITF and nxtgrd files.

**Description**

The grdgenxo tool detects a difference between the new ITF file and the old nxtgrd file that prevents the use of the grdgenxo update flow.

In the grdgenxo update flow, conductor layers must be consistent between the old nxtgrd file and the new ITF file.

Feedback

**What Next**

Check the differences between the new ITF and old nxtgrd files. If the differences are expected, you must regenerate the nxtgrd file without using the update flow.

---

## GRD-0049

(Warning) The tech file is created by metal composer. It should not be used for sign-off extraction purpose. Please contact foundry to get the official version.

**Description**

The ITF file used to generate this nxtgrd file was created by the metal composer function. These ITF and nxtgrd files should not be used for signoff extraction because they might not be fully qualified by the foundry.

**What Next**

Contact the foundry to obtain official nxtgrd or ITF files.

---

## GRD-0050

(Fatal) The %s statement cannot be encrypted in the tech file.

**Description**

The statement cannot be encrypted in the tech file.

**What Next**

Do not put the statement between #beginHide and #endHide

---

## GRD-0051

(Warning) The ITF includes both planar devices and FinFet devices. Is this intentional?

**Description**

The ITF includes both planar devices and FinFet devices. Is this intentional?

**What Next**

Check ITF correctness: both planar and FinFet device SCT models will be generated.

---

## GRD-0052

(Severe) Cannot open in read mode the QTF File %s.

Feedback

**Description**

The grdgenxo tool fails because it cannot open the specified QTF file for reading. The file might not exist or read access rights might not be set properly.

**What Next**

Check the path of the QTF file, verify that the file exists on the file system, and verify that you have the rights to read it. Correct the problems and resubmit the grdgenxo run.

---

## GRD-0053

(Severe) The QTF File is different from the previous run. Remove %s and re-run grdgenxo.

**Description**

The QTF FIle has been modified from the previous grdgenxo run. Therefore grdgenxo cannot proceed further.

**What Next**

Remove the QTF File in the nxtgrd folder and re-run grdgenxo.

---

## GRD-0054

(Severe) -gds2cap_arguments requires arguments.

**Description**

grdgenxo stops because it cannot find arguments specified after the -gds2cap_arguments statement in the command line.

**What Next**

Specify some arguments after -gds2cap_arguments or remove it from the command line.

---

## GRD-0055

(Information) gds2cap options have been set to "%s -qtfExtract %s".

**Description**

The listed options have been set for the gds2cap tool.

**What Next**

Information only; no action is required.

Feedback

## GRD-0056

(Fatal) Cannot open layer mapping file %s for metal composer

**Description**

grdgenxo cannot read the mapping file specified with option "-layer_mapping_file".

**What Next**

Provide correct mapping file.

## GRD-0057

(Warning) -quickcap_arguments: -g may cause simulation time to explode.

**Description**

The *-g* option has been specified for use with the QuickCap tool. This high-accuracy option might cause simulation to run more slowly.

**What Next**

If the use of the *-g* option is intentional, no action is needed.

## GRD-0058

(Information) QuickCap options have been set to "%s".

**Description**

The listed options have been set for the QuickCap tool.

**What Next**

Information only; no action is required.

## GRD-0059

(Severe) Could NOT encrypt rapid3d technology file.

**Description**

grdgenxo cannot encrypt the field solver technology file.

**What Next**

Check the file system to see whether disk is full or connection is stable. Make possible clean up and rerun the job.

Feedback

## GRD-0060

(Fatal) Usage is: itf2Linear <itf_file>.

**Description**

Usage not correct.

**What Next**

Follow the usage of: itf2Linear <itf_file>.

## GRD-0061

(Information) Generating Linear Cap Model.

**Description**

Indicating the job is running.

**What Next**

No further action is required.

## GRD-0062

(Information) Done.

**Description**

Indicate the model running is completed.

**What Next**

No further action is required.

## GRD-0063

(Fatal) INTERNAL Error: Array bound violation in SIDE_TANGENT, program terminated.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

Feedback

## GRD-0064

(Fatal) INTERNAL Error: File generation appears to be corrupt, please contact StarRC support.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## GRD-0065

(Fatal) File generation appears to be corrupt, please erase the file and re-run.

**Description**

Data file contains invalid number.

**What Next**

Remove this file and rerun the job. If issue still exist, Contact the Synopsys Support Center.

## GRD-0066

(Fatal) The GRD file version is earlier than 60. Please use an earlier version of grdgenxo or run grdgenxo again to get new binary file.

**Description**

grdgenxo version is different from the one used to generate the nxtgrd.

**What Next**

Use an earlier version of grdgenxo or run grdgenxo again to get new binary file.

## GRD-0067

(Fatal) INTERNAL Error: RELUCTANCE is expected.

**Description**

An internal consistency check has failed.

Feedback

**What Next**

Contact the Synopsys Support Center.

## GRD-0068

(Fatal) INTERNAL Error: END_RELUCTANCE is expected.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## GRD-0069

(Severe) Usage: grdgenxo -bin2ascii -i <bin.grd> -o <asc.grd>.

**Description**

An error has occurred with the usage of the *grdgenxo* command. The error might be caused by improper command syntax or by an unsuitable input file.

The correct syntax is as follows:

```
% grdgenxo -bin2ascii -i <input_grd_file> -o <ascii_grd_file>
```

where *-i* is the short form of the *-input* option and *-o* is the short form of the *-output* option of the *grdgenxo* command.

**What Next**

Correct the syntax of the *grdgenxo* command and resubmit the command.

## GRD-0070

(Severe) Error opening input file.

**Description**

Input file is not readable or is not a binary file.

**What Next**

Check the input file.

Feedback

## GRD-0071

(Severe) %s is NOT a binary grd file, exiting...

**Description**

Input file is not a binary file.

**What Next**

Check the input file.

## GRD-0072

(Severe) Usage: grdgenxo -prime_rail <nxtgrd>.

**Description**

An error has occurred with the usage of the *grdgenxo* command. The error might be caused by improper command syntax or by an unsuitable input file.

The correct syntax is as follows:

```
% grdgenxo -prime_rail <nxtgrd_file>
```

**What Next**

Correct the syntax of the *grdgenxo* command and resubmit the command.

## GRD-0073

(Severe) Usage: grdgenxo -add_sf <factor> -i <input_grd_file> -o <output_grd_file>.

**Description**

An error has occurred with the usage of the *grdgenxo* command. The error might be caused by improper command syntax or by an unsuitable input file.

The correct syntax is as follows:

```
% grdgenxo -add_sf <factor> -i <input_grd_file> -o <ascii_grd_file>
```

where *-i* is the short form of the *-input* option and *-o* is the short form of the *-output* option of the *grdgenxo* command. The argument of the *-add_sf* option must be a positive value representing the half-node scaling factor.

**What Next**

Correct the syntax of the *grdgenxo* command and resubmit the command.

Feedback

## GRD-0074

(Severe) ERROR: HALF_NODE_SCALE_FACTOR must be greater than 0.

**Description**

HALF_NODE_SCALE_FACTOR must be set to a number greater than 0.

**What Next**

Revise the HALF_NODE_SCALE_FACTOR value.

## GRD-0075

(Warning) Warning: The value of HALF_NODE_SCALE_FACTOR is 1.0 and will be ignored.

**Description**

StarRC ignores the HALF_NODE_SCALE_FACTOR because it is set to 1.0, which means no scaling should be applied.

**What Next**

No further action is required if the value of 1.0 is expected.

## GRD-0076

(Severe) Error: Orientation of ETCH_VS_WIDTH_AND_SPACING table must be the same or merely swapped between ITF and NXTGRD files.

**Description**

The grdgenxo tool detects an unsupported difference in the *ETCH_VS_WIDTH_AND_SPACING* table between the new ITF file and the old nxtgrd file.

In the grdgenxo update flow, the *ETCH_VS_WIDTH_AND_SPACING* table in the new ITF file and the old nxtgrd file must be the same; the only difference allowed is that the direction can be swapped.

**What Next**

Check the *ETCH_VS_WIDTH_AND_SPACING* tables in the new ITF and old nxtgrd files. If the difference is expected, you must regenerate the nxtgrd file without using the update flow.

Feedback

## GRD-0077

(Severe) Error: NOT supported differences in %s between ITF and NXTGRD files.

**Description**

The grdgenxo tool detects a difference between the new ITF file and the old nxtgrd file that prevents the use of the grdgenxo update flow.

**What Next**

Check the differences between the new ITF and old nxtgrd files. If the differences are expected, you must regenerate the nxtgrd file without using the update flow.

## GRD-0078

(Severe) Error: NOT supported differences in resistance between ITF and NXTGRD files, due to presence of reluctance.

**Description**

The grdgenxo tool detects a difference between the new ITF file and the old nxtgrd file that prevents the use of the grdgenxo update flow.

**What Next**

Check the differences between the new ITF and old nxtgrd files. If the differences are expected, you must regenerate the nxtgrd file without using the update flow.

## GRD-0079

(Severe) Error opening SCT file for reading.

**Description**

The input sct file cannot be opened for reading.

**What Next**

Check the existence of sct file and rerun the job.

## GRD-0080

(Severe) Error: Can NOT update encrypted nxtgrd file.

**Description**

The grdgenxo update flow does not support encrypted nxtgrd files.

Feedback

**What Next**

Decrypt the nxtgrd before running the update flow. Alternatively, regenerate the nxtgrd completely, without using the update flow.

---

## GRD-0081

(Severe) Error: Can NOT update nxtgrd file older than version 63.

**Description**

The version of the nxtgrd file is too old for the grdgenxo update flow.

**What Next**

Regenerate the nxtgrd completely, without using the update flow.

---

## GRD-0082

(Severe) Error in opening ITF file for updating.

**Description**

The input ITF file cannot be opened for reading.

**What Next**

Check the existence of the ITF file and rerun the job.

---

## GRD-0083

(Severe) Usage: grdgenxo -itf2itf -i <itf> -o <itf>.

**Description**

Use grdgenxo in a incorrect way. An error has occurred with the usage of the *grdgenxo* command. The correct syntax is as follows:

```
% grdgenxo -itf2itf -i <input_itf> -o <output_itf>
```

where *-i* is the short form of the *-input* option and *-o* is the short form of the *-output* option of the *grdgenxo* command.

**What Next**

Correct the syntax of the *grdgenxo* command and resubmit the command.

Feedback

## GRD-0084

(Severe) Error: Unable to write to %s.

**Description**

Cannot open output file to write.

**What Next**

Check the disk space and connection.

## GRD-0085

(Severe) Usage: grdgenxo -sct2sct_decrypt -i <sct> -o <sct>.

**Description**

Use grdgenxo in a incorrect way.

**What Next**

Follow the usage.

## GRD-0086

(Severe) Could NOT access %s.

**Description**

Cannot open output file to write.

**What Next**

Check the disk space and connection.

## GRD-0087

(Severe) %s is NOT a valid sct file.

**Description**

Input sct file is not valid.

**What Next**

Check the input file and rerun the job.

Feedback

## GRD-0088

(Severe) Usage: grdgenxo -nxtgrd2itf -i <bin.grd> -o <itf>.

**Description**

An error has occurred with the usage of the *grdgenxo* command. The correct syntax is as follows:

```
% grdgenxo -nxtgrd2itf -i <input_grd_file> -o <itf_file>
```

where *-i* is the short form of the *-input* option and *-o* is the short form of the *-output* option of the *grdgenxo* command.

**What Next**

Correct the syntax of the *grdgenxo* command and resubmit the command.

## GRD-0089

(Severe) %s is encrypted, exiting...

**Description**

The *-nxtgrd2itf* option of the *grdgenxo* command is not supported for nxtgrd files generated from encrypted ITF files.

**What Next**

Use a different nxtgrd file, or do not use the *-nxtgrd2itf* option.

## GRD-0090

(Severe) Error opening input nxtgrd file.

**Description**

Input file is not readable.

**What Next**

Check the input file.

## GRD-0091

(Fatal) Internal error in Boundary.C.

Feedback

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## GRD-0092

(Fatal) RPV_VS_SI_COVERAGE and RPV_VS_COVERAGE can not be applied to same layer.

**Description**

RPV_VS_SI_COVERAGE and RPV_VS_COVERAGE can not be applied to same layer.

**What Next**

Correct ITF and rerun StarRC.

---

## GRD-0093

(Warning) RPV_VS_SI_COVERAGE does not support trench contact virtual via.

**Description**

RPV_VS_SI_COVERAGE on trench contact virtual via layer is not allowed.

**What Next**

Correct ITF and rerun StarRC.

---

## GRD-0094

(Warning) RPV_VS_COVERAGE does not support trench contact virtual via.

**Description**

RPV_VS_COVERAGE on trench contact virtual via layer is not allowed.

**What Next**

Correct ITF and rerun StarRC.

---

## GRD-0095

(Fatal) Can NOT open configuration file %s.

Feedback

**Description**

Input configuration file is not readable.

**What Next**

Check the input file.

---

## GRD-0096

(Fatal) Invalid configuration input for model %s.

**Description**

Input configuration file is not correct for particular model.

**What Next**

Revise the configuration file and rerun the job.

---

## GRD-0097

(Fatal) Raphael run is NOT supported with GPGPU.

**Description**

Raphael run is NOT supported with GPGPU.

**What Next**

Unset Raphael run setting and rerun the job.

---

## GRD-0098

(Fatal) %s appears to be corrupt. Please erase that file and re-run.

**Description**

Data file contains invalid number.

**What Next**

Remove this file and rerun the job.

---

## GRD-0099

(Fatal) Can NOT create/open file handle %s.

Feedback

**Description**

grdgenxo cannot access the output file.

**What Next**

Check filesystem for space/connection.

---

## GRD-0100

(Fatal) Can NOT open file from file handle. Please check file system for space/connection.

**Description**

StarRC is not able to open a file to write.

**What Next**

Check the file system to see whether disk is full or connection is stable. Make possible clean up and rerun the job.

---

## GRD-0101

(Severe) INTERNAL Error: Wrong ascii header in the TLUPlus file.

**Description**

Ascii header in TLUPlus file is wrong.

**What Next**

Contact the Synopsys Support Center.

---

## GRD-0102

(Severe) INTERNAL Error: Cannot open file host_id.

**Description**

Internal host_id file cannot be opened.

**What Next**

Contact the Synopsys Support Center.

---

## GRD-0103

(Severe) INTERNAL Error: TLUPlus decode can NOT be run at this host.

Feedback

**Description**

This internal function can be run on specified hosts.

**What Next**

Contact the Synopsys Support Center.

---

## GRD-0104

(Severe) INTERNAL Error: Can NOT open file %s.

**Description**

The internal decode binary cannot open the input file.

**What Next**

Contact the Synopsys Support Center.

---

## GRD-0105

(Warning) INTERNAL Warning: For Synopsys internal use only. Do NOT distribute without R&D's permission.

**Description**

This binary decode can be used only within Synopsys

**What Next**

Contact the Synopsys Support Center.

---

## GRD-0106

(Severe) INTERNAL Error: Fail to read CapTable from attached file.

**Description**

The internal build decode cannot read CapTable from the specified file.

**What Next**

Contact the Synopsys Support Center.

---

## GRD-0107

(Severe) INTERNAL Error: Fail to read ResModel from attached file.

Feedback

**Description**

The internal binary decode fails to read ResModel from the specified file.

**What Next**

Contact the Synopsys Support Center.

---

## GRD-0108

(Severe) INTERNAL Error: Fail to read CapModel from attached file.

**Description**

The internal binary decode fails to read CapModel from the specified file.

**What Next**

Contact the Synopsys Support Center.

---

## GRD-0109

(Severe) INTERNAL Error: Fail to read additional CapInfo from attached file.

**Description**

The internal binary decode fails to read additional CapInfo from the specified file.

**What Next**

Contact the Synopsys Support Center.

---

## GRD-0110

(Severe) INTERNAL Error: Fail to read via info from attached file.

**Description**

The internal binary decode fails to read via info from the specified file.

**What Next**

Contact the Synopsys Support Center.

---

## GRD-0111

(Severe) INTERNAL Error: Format error in CapModel attached file %s.

Feedback

**Description**

The internal binary decode detects format error in CapModel in the specified file.

**What Next**

Contact the Synopsys Support Center.

---

## GRD-0112

(Severe) INTERNAL INFO: Usage: decode <file_name>.

**Description**

This is an internal binary related usage.

**What Next**

Contact the Synopsys Support Center.

---

## GRD-0113

(Fatal) INTERNAL Error: Conductors shorting in layer %s.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## GRD-0114

(Severe) INTERNAL Error: Read size %d exceeds the limit %d.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## GRD-0115

(Severe) INTERNAL Error: Check fails.

Feedback

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## GRD-0116

(Severe) Can NOT open file for GDS output.

**Description**

Cannot open output file for writing.

**What Next**

Check the file system and connection.

---

## GRD-0117

(Severe) Can NOT open %s.

**Description**

Cannot open input file for reading.

**What Next**

Check the input file existence.

---

## GRD-0118

(Severe) INTERNAL Error: Unrecognized record ID.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## GRD-0119

(Severe) Can NOT open GDS file to write.

Feedback

**Description**

Cannot open output file for writing.

**What Next**

Check the file system and connection.

---

## GRD-0120

(Fatal) INTERNAL Error: Ending besim index larger than size.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## GRD-0121

(Fatal) Can NOT open lock file handle. Either disk is full or a lock file from another run persists. Please remove *.llock from grd repository, or create some disk space.

**Description**

Can NOT open lock file handle.

**What Next**

Either disk is full or a lock file from another run persists. Please remove *.llock from grd repository, or create some disk space.

---

## GRD-0122

(Fatal) Lock file failure! Please remove *.llock from grd repository.

**Description**

Unable to lock the output file for writing.

**What Next**

Remove *.llock from grd repository.

---

## GRD-0123

(Fatal) Can NOT fdopen lock file.

Feedback

**Description**

Cannot open file for writing.

**What Next**

Check the file system and connection.

---

## GRD-0124

(Fatal) Grdgenxo can NOT proceed without STAR-RC2-TCAD license. Please make sure the license is up and running.

**Description**

STAR-RC2-TCAD license is not avalible.

**What Next**

Check the license setting and rerun the job.

---

## GRD-0125

(Warning) Database file %s is not ready for output. Please start another GRD process if there is no followup process.

**Description**

Data file is not generated, will be automated be running if there is followup process.

**What Next**

If there is no grdgenxo process is running, should start a new one. Otherwise, no actions is required.

---

## GRD-0126

(Fatal) There seems to be a bad GRD process. Please restart grdgenxo run.

**Description**

One grdgenxo process seems corrupted.

**What Next**

Start a new grdgenxo running process.

Feedback

## GRD-0127

(Fatal) File %s seems to be corrupted. Please remove the file and restart the grdgenxo run.

**Description**

Data file is corrupted due to some reasons.

**What Next**

Remove this specific file and rerun the job.

## GRD-0128

(Fatal) INTERNAL Error: Invalid table type.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## GRD-0129

(Fatal) Internal Error: dereferring to a unavailable via layer %s.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## GRD-0130

(Fatal) Can NOT open nxtgrd file %s for appending info

**Description**

grdgenxo fails to open nxtgrd file mentioned in error message above for appending info.

**What Next**

Check disk is accessible and has space for writing. If disk is in good state, contact the Synopsys Support Center.

Feedback

## GRD-0132

(Fatal) Can NOT create lock file.

**Description**

grdgenxo fails to create an internal file.

**What Next**

Check disk is accessible and has space for writing. If disk is in good state, contact the Synopsys Support Center.

## GRD-0133

(Fatal) SCT file failed to be generated. Exiting...

**Description**

grdgenxo finds errors in SCT file generation. There are possibly some information in log file explaining the reason why SCT file fails to be generated.

**What Next**

Check the log to find the detailed reason of this failure and guidance to fix the issue. If there is no any clues on the failure in the log, contact the Synopsys Support Center.

## GRD-0134

(Fatal) Can NOT open nxtgrd file for writing.

**Description**

grdgenxo fails to open nxtgrd file for writing.

**What Next**

Check disk is accessible and has space for writing.

## GRD-0135

(Information) Waiting for SCT file. If the current grdgenxo is the only one running, run several other grdgenxo processors in order to complete SCT file generation.

Feedback

**Description**

grdgenxo is waiting for SCT file to complete nxtgrd generation. If there is only one grdgenxo processor, you need run several other grdgenxo processors to accelerate SCT generation.

**What Next**

If there is only one grdgenxo processor, you need run several other grdgenxo processors to accelerate SCT generation. Otherwise, nothing is needed to be done.

---

## GRD-0136

(Fatal) Numerical instability in bottom thickness variation calculation for layer %s.

**Description**

In that layer, numerical number as NaN is calculated during generation.

**What Next**

Verify the thickness related tabel in ITF file for certain layer and rerun the job. If error still exists, contact the Synopsys Support Center.

---

## GRD-0139

(Fatal) %s is different from the previously defined ITF file. This is NOT supported by incremental grdgenxo. Do a complete run from scratch.

**Description**

grdgenxo supports incremental run with ITF modifications which have localized capacitance impact. The difference mentioned in error message above has substantial impact on nxtgrd models and is not supported by incremental grdgenxo.

**What Next**

Do a complete run from scratch

---

## GRD-0140

(Warning) The copy of old ITF file already under the directory of %s will be used, rather than the one specified using -old_itf %s.

**Description**

You are running grdgenxo with option "-inc". In incremental grdgenxo run mode, grdgenxo first tries to find old ITF from previous grdgenxo run directory. If grdgenxo cannot find it,

Feedback

grdgenxo turns to option "-old_itf" to get old ITF. In this case, grdgenxo finds old ITF from both the directory mentioned in message above and option "-old_itf". The copy of old ITF file under the specified directory will be used.

**What Next**

If the directory mentioned in message above is where you want to run incremental grdgenxo, nothing is needed to be done.

---

## GRD-0141

(Fatal) Usage is: inductancexo itffile.

**Description**

Use the binary in a wrong way.

**What Next**

Use the binary with above rules.

---

## GRD-0142

(Severe) INTERNAL Error: Peel exponent less than 1.0 may create unstable gridding (infinitely many sub filements).

**Description**

Peel exponent is an internal variable for controling filement generation when grdgenxo discretizing wires for inductance modeling. If you see this error in log, it means some unknown internal error occurs.

**What Next**

Contact the Synopsys Support Center.

---

## GRD-0143

(Severe) INTERNAL Error: Overlapping wires NOT allowed.

**Description**

The grdgenxo tool finds overlapped wires when preparing inductance models. Overlapped wires are not allowed and might be caused by internal errors.

**What Next**

Contact the Synopsys Support Center.

Feedback

## GRD-0145

(Warning) INTERNAL Warning: Conductors %s and %s at different potentials are (almost) touching.

### Description

The two conductors listed in the warning message are almost touching each other while they are modeled as different nets (having different potentials for capacitance extraction). The generated nxtgrd capacitance models associated with these conductors might be inaccurate.

### What Next

Contact the Synopsys Support Center.

## GRD-0146

(Fatal) INTERNAL Error: Bottom dielectric internal error for layer %s

### Description

For the conductor layer mentioned in the error message, the bottom dielectric layer associated with the conductor layer has been defined with ITF syntax *BOTTOM_DIELECTRIC_THICKNESS* abd *BOTTOM_DIELECTRIC_ER*. However, the grdgenxo tool cannot find the bottom dielectric layer because of an internal error.

### What Next

Contact the Synopsys Support Center.

## GRD-0147

(Fatal) Floating metal fill emulation does not work with sensitivity extraction!

### Description

For TLU flow, grdgenxo need emulate metal fill when generating cap/res models. The extractor (for example, extractor in ICC) using TLU need extract caps with metal fills considered although design does not include them yet at the extraction time. However, floating metal fill emulation in grdgenxo cannot work with sensitivity extraction which is activated by ITF syntax "VARIATION_PARAMETERS".

### What Next

Remove VARIATION_PARAMETERS in ITF and re-run grdgenxo.

Feedback

## GRD-0148

(Fatal) INTERNAL Error: Error in handling reluctance and floating metal fill for file %s.

**Description**

The grdgenxo tool detects floating metal fill in the file listed in the error message, which is not expected when grdgenxo is generating reluctance models.

**What Next**

Contact the Synopsys Support Center.

## GRD-0149

(Fatal) Incompatible/corrupt database file %s! Clean the offending files and re-run grdgenxo.

**Description**

The database file named in the error message is incompatible or corrupt.

**What Next**

Replace the specified file and resubmit the grdgenxo run. If the issue still occurs, contact the Synopsys Support Center.

## GRD-0150

(Fatal) Incompatibility in sensitivity database.

**Description**

The grdgenxo tool detects an incompatible sensitivity database. The database might be corrupted or incompatible with the current grdgenxo version.

**What Next**

Run the grdgenxo tool in an empty directory. If the issue still occurs, contact the Synopsys Support Center.

## GRD-0151

(Fatal) %s contains unrecognized field %s in GRD database

Feedback

**Description**

grdgenxo finds unrecognized fields in database file mentioned above. The database file might be corrupted or you are running grdgenxo with an incompatible old database.

**What Next**

Run grdgenxo in an empty directory. If the issue still occurs, contact the Synopsys Support Center.

---

## GRD-0152

(Fatal) INTERNAL Error: Unknown generic structure.

**Description**

grdgenxo fails to generate generic structure for simulation.

**What Next**

Contact the Synopsys Support Center.

---

## GRD-0153

(Fatal) INTERNAL Error: Failed to initialize the same_effective vectors.

**Description**

grdgenxo fails to initialize internal variables.

**What Next**

Contact the Synopsys Support Center.

---

## GRD-0154

(Fatal) INTERNAL Error: Failed to initialize vector: is_sens_computed.

**Description**

grdgenxo fails to initialize an internal variable.

**What Next**

Contact the Synopsys Support Center.

---

## GRD-0156

(Warning) "rapahel rc2" convergence is poor. Use these results with caution.

Feedback

**Description**

grdgenxo is calling raphael rc2 for simulation. "raphael rc2" reports poor convergence for simulated patterns.

**What Next**

Use these results with caution. Or Contact the Synopsys Support Center for further support.

---

## GRD-0157

(Fatal) INTERNAL Error: Raphael translation can NOT handle non-planar conductors.

**Description**

grdgenxo is calling raphael for simulation. grdgenxo forbids non-planar conductors when preparing inputs for raphael.

**What Next**

Contact the Synopsys Support Center.

---

## GRD-0158

(Severe) INTERNAL Error: Can NOT open file tmp.4in.

**Description**

grdgenxo failed to open an internal file "tmp.4in".

**What Next**

Check the disk is in good state and not full. Re-run grdgenxo. If the issue still occurs, contact the Synopsys Support Center.

---

## GRD-0159

(Information) User specified spacing dimension of %d is provided for layer %s.

**Description**

Informational message indicating the spacing for the layer specified by the user.

**What Next**

This message is for information only. Continue with your run.

Feedback

## GRD-0160

(Fatal) wrong syntax in mapping file: %s

**Description**

Wrong syntax in metal composer mapping file.

**What Next**

Check syntax for metal composer mapping file.

## GRD-0161

(Fatal) Duplicate layer name in metal composer : %s

**Description**

Layer name cannot be the same in one ITF file.

**What Next**

Check layer names in metal composer mapping file.

## GRD-0162

(Fatal) Fixed top layer %s is not a conducting layer.

**Description**

Fixed top layer must be a conducting layer.

**What Next**

Use conducting layer as fixed top layer in metal composer mapping file.

## GRD-0163

(Fatal) Cannot find layer %s in ITF file.

**Description**

Cannot find layer %s in ITF file.

**What Next**

Layers specified in Metal Composer mapping file must exists in ITF file. Check syntax of Metal Composer mapping file.

Feedback

## GRD-0164

(Warning) %s layer %s is very close to device layers, SCT tables need to be regenerated to ensure accuracy.

**Description**

Fixed top conducting layer, adjustable conducting layers and bottom fixed layer is not suggested to be too close to device layers. You need to regenerate all the SCT tables, instead of reuse of them.

**What Next**

Modify metal composer mapping file accordingly, or regenerated all the SCT tables for nxtgrd creatation.

## GRD-0165

(Fatal) Covertical layer %s is found in ITF file.

**Description**

Metal Composer does NOT support ITFs with covertical conducting layers.

**What Next**

Use ITFs with no covertical conducting layers.

## GRD-0166

(Fatal) %s

**Description**

An internal consistency check in Metal Composer has failed.

**What Next**

Contact the Synopsys Support Center.

## GRD-0167

(Fatal) Fixed bottom layer %s is not a conducting layer.

**Description**

Fixed bottom layer must be a conducting layer.

Feedback

**What Next**

Use conducting layer as fixed bottom layer in metal composer mapping file.

---

## GRD-0168

(Warning) There is no VIA between nearby conductor layer %s

**Description**

There is not VIA between nearby condutor layer in output ITF file.

**What Next**

Confirm mapping information in layer composer mapping file.

---

## GRD-0169

(Warning) Multiple VIA defination between layer %s

**Description**

There is more than 1 VIA layer defined between 2 conducting layers in the output ITF file.

**What Next**

Confirm mapping information in layer composer mapping file.

---

## GRD-0170

(Warning) VIA %s connecting layer %s and %s is not the VIA in original ITF.

**Description**

The 2 conductor layer come from the same input ITF and there does exist a VIA layer between those 2 layers in input ITF. The new specified VIA in output ITF is not that existing via.

**What Next**

Confirm mapping information in layer composer mapping file.

---

## GRD-0171

(Fatal) VIA %s does not connect with adjustable conducting layers or fixed top/bottom layers.

Feedback

**Description**

VIA layer specifed in layer composer mapping file should connect with adjustable conducting layers or fixed top/bottom layers.

**What Next**

Confirm VIA mapping information in layer composer mapping file.

---

## GRD-0172

(Fatal) MEASURED_FROM layer %s of layer %s is adjusted in mapping file.

**Description**

MEASURED_FROM layer is adjusted.

**What Next**

Confirm mapping information in layer composer mapping file.

---

## GRD-0173

(Fatal) wrong syntax in mapping file: %s, at least one conductor name is needed after layer_group

**Description**

Wrong syntax in metal composer mapping file.

**What Next**

Check syntax for metal composer mapping file.

---

## GRD-0174

(Fatal) %s layer %s is not one of BEOL layers.

**Description**

Either fixed_top_layer_name or fixed_bottom_layer_name defined in mapping file for layer composer should be one of BEOL layers.

**What Next**

Set either fixed_top_layer_name or fixed_bottom_layer_name to one of BEOL layers' name; or leave it empty, this will not keep the BEOL layers and will refill in all the BEOL layers based on mapping file users set.

Feedback

## GRD-0175

(Fatal) The ITF file %s contains encrypted sections cross one or more CONDUCTOR/ DIELECTRIC/VIA layers, stop layer composer.

**Description**

ITF partial encryption with #beginHide/#endHide contains one or more blocks of CONDUCTOR, DIELECTRIC or VIA. Layer composer is stopped to avoid decryption.

**What Next**

User could use #beginHide/#endHide inside each block defintions of CONDUCTOR, DIELECTRIC or VIA, then layer composer could be used without this error.

## GRD-0176

(Information) For this process, the accuracy goal of SCT models generation is set internally and overrides the -quickcap_arguments command.

**Description**

The accuracy goal of SCT models generation is set internally and overrides the -quickcap_arguments command.

**What Next**

Information only. Indicates the accuracy goal of SCT models generation is set internally.

## GRD-0180

(Fatal) TRENCH_CONTACT_EXTENSION can not be a non-positive value on layer %s.

**Description**

TRENCH_CONTACT_EXTENSION must be a positive value.

**What Next**

Specify a positive value to TRENCH_CONTACT_EXTENSION on the layer.

## GRD-0181

(Fatal) TRENCH_CONTACT_EXTENSION can not be specified more than once on layer %s.

Feedback

**Description**

TRENCH_CONTACT_EXTENSION can only be specified on a conductor layer once.

**What Next**

Modify the ITF file.

---

## GRD-0182

(Fatal) TRENCH_CONTACT_EXTENSION can not be specified on layer %s whose layer_type is not TRENCH_CONTACT.

**Description**

TRENCH_CONTACT_EXTENSION can only be specified on layer whose layer_type is TRENCH_CONTACT.

**What Next**

Modify the ITF file to only specify TRENCH_CONTACT_EXTENSION on conductor layers whose layer_type are TRENCH_CONTACT.

---

## GRD-0185

(Fatal) The FROM or TO layer %s of the via layer %s is not a valid conducting layer.

**Description**

The FROM or TO layer of a via layer must be a valid conducting layer.

**What Next**

Modify the ITF file.

---

## GRD-0186

(Fatal) The top height of two bottom conductors %s and %s of via %s are not same.

**Description**

The bottom layers of a via layer must have same top height.

**What Next**

Modify the ITF file.

Feedback

## GRD-0187

(Fatal) The bottom height of two top conductors %s and %s of via %s are not same.

**Description**

The top layers of a via layer must have same bottom height.

**What Next**

Modify the ITF file.

## GRD-0188

(Fatal) The fake via layer %s connects mutilple conducting layer at the bottom with different layer_type.

**Description**

When a via layer connects multiple conducting layers at the bottom, and one of them is TRENCH_CONTACT or ROUTING_VIA, all layers at the bottom must have same layer_type.

**What Next**

Update the ITF file.

## GRD-0189

(Fatal) The fake via layer %s connects mutilple conducting layer at the top with different layer_type.

**Description**

When a via layer connects multiple conducting layers at the top, and one of them is TRENCH_CONTACT or ROUTING_VIA, all layers at the top must have same layer_type.

**What Next**

Update the ITF file.

## GRD-0190

(Fatal) File %s is broken.

**Description**

File described in the message is broken.

Feedback

**What Next**

Check disk is accessible and has space for writing.

---

## GRD-0231

(Warning) INTERNAL Warning: The substrate charge distribution did NOT complete successfully.

**Description**

The substrate charge distribution did NOT complete successfully, and an internal model is used.

**What Next**

Contact the Synopsys Support Center.

---

## GRD-0232

(Fatal) INTERNAL Error: Can NOT create/open charge file handle %s.

**Description**

Cannot create/open internal charge distribution file.

**What Next**

Check filesystem for space/connection.

---

## GRD-0236

(Severe) INTERNAL Error: Wrong ascii header in the TLUPlus file.

**Description**

Ascii header in TLUPlus file is wrong.

**What Next**

Contact the Synopsys Support Center.

---

## GRD-0237

(Severe) INTERNAL Error: decode can NOT be run on this host.

**Description**

The internal build decode can be run on some specified hosts.

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## GRD-0238

(Severe) INTERNAL Error: Fail to read conductor infor from attached file.

**Description**

The internal binary decode fails to read conductor information from the specified file.

**What Next**

Contact the Synopsys Support Center.

---

## GRD-0242

(Warning) Cap unit is fixed to be 1fF and Res unit is fixed to be 1ohm. Ignore the unit setting in format file.

**Description**

This is a warning message indicating the unit of capacitance and resistance from the format file. Discard the unit setting in format file.

**What Next**

Verify that values of Resistance and Capacitance are in this unit.

---

## GRD-0243

(Warning) Cap unit is fixed to be 1fF. Ignore the CAP_UNIT setting in format file.

**Description**

This is a warning message indicating the unit of capacitance and resistance from the format file. Discard the unit setting in format file.

**What Next**

Verify that values of Resistance and Capacitance are in this unit.

---

## GRD-0244

(Warning) Res unit is fixed to be 1ohm. Ignore the RES_UNIT setting in format file.

Feedback

**Description**

This is a warning message indicating the unit of capacitance and resistance from the format file. Discard the unit setting in format file.

**What Next**

Verify that values of Resistance and Capacitance are in this unit.

---

## GRD-0245

(Fatal) Unknown operating condition: %s.

**Description**

Operating conditions other than 'MAX', 'NOM', or 'MIN' are not correct.

**What Next**

Change the operating conditions and run the case.

---

## GRD-0246

(Fatal) Format file %s from previous run have incorrent layer %s. Please remove the existing running directory and re-run grdgenxo from the scratch.

**Description**

Format file from the previous run have incorrect layers.

**What Next**

Correct the format file and re-run.

---

## GRD-0247

(Fatal) Unable to find layer %s.

**Description**

Unable to find the layer in format file.

**What Next**

Correct the format file and re-run.

---

## GRD-0248

(Fatal) Error in format file.

Feedback

**Description**

Current layer is not present. Correct the format file and re-run.

**What Next**

Correct the format file and re-run.

---

## GRD-0249

(Fatal) Width/Spacing table size can NOT exceed 16. Please check the format file.

**Description**

Width/Spacing table size can not be more than 16.

**What Next**

Correct the format file and re-run.

---

## GRD-0250

(Information) User specified width dimension of %d is provided for layer %s.

**Description**

Width for the layer is defined by the user. This is for information only. Ensure that this is what you intend.

**What Next**

No further work is required.

---

## GRD-0251

(Fatal) Number of WIDTH values in format file is different from NUMWIDTH value.

**Description**

Number of WIDTH values in format file is different from NUMWIDTH value.

**What Next**

Correct the format file and re-run.

---

## GRD-0252

(Fatal) Suspicious WIDTH value of %f is specified in the format file.

Feedback

**Description**

Number of width specified is either more than 100 or negative. Suspicious value.

**What Next**

Correct the width and re-run.

---

## GRD-0253

(Information) User specified width sampling points are provided for layer %s.

**Description**

Information message indicating number of width points specified for the layer.

**What Next**

This message is for information only. No work is required.

---

## GRD-0254

(Fatal) Number of SPACING values in format file is different from NUMSPACING value.

**Description**

Total number of SPACING values in format file is different from NUMSPACING value.

**What Next**

Correct the format file and re-run.

---

## GRD-0255

(Fatal) Suspicious SPACING value of %f is specified in the format file.

**Description**

Total number of spacing entries is either more than 100 or negative. Suspicious value.

**What Next**

Correct the format file and re-run.

---

## GRD-0256

(Information) User specified spacing sampling points are provided for layer %s.

Feedback

**Description**

Informational message indicating user specified spacing points.

**What Next**

This is for information only. No work is needed.

---

## GRD-0257

(Fatal) Unknown format keyword %s.

**Description**

Invalid keyword in the format file.

**What Next**

Correct the keyword and re-run. Consult the StarRC manual for valid keywords.

---

## GRD-0258

(Fatal) %s must be an integer.

**Description**

Number of spacing or width must be an integer.

**What Next**

Use integer and re-run.

---

## GRD-0259

(Information) CAP_UNIT = %fF; RES_UNIT = %fohm; opCond = %s.

**Description**

This is an informational message indicating capacitance unit, resistance unit, and operating condition.

**What Next**

This is for information only. No work needs to be done.

---

## GRD-0260

(Information) Checking the database: first pass...

Feedback

**Description**

Grdgenxo is checking the database.

**What Next**

Informational message only. No work needs to be done.

---

## GRD-0261

(Information) Checking the database: second pass...

**Description**

Grdgenxo is checking the database.

**What Next**

Informational message only. No work needs to be done.

---

## GRD-0263

(Information) Generating resistance tables...

**Description**

Informational message indicating generation of resistance tables in progress.

**What Next**

This is for information only. No work is required.

---

## GRD-0264

(Information) Generating additional capacitance information...

**Description**

Informational message indicating generation of additional capacitance information.

**What Next**

This is for information only. No work is needed.

---

## GRD-0265

(Fatal) FILL_TYPE %s for layer %s is NOT supported.

Feedback

**Description**

Fill type for the layer is neither floating nor grounded.

**What Next**

Only grounded and floating fills are supported. Correct the fill type and re-run.

---

## GRD-0266

(Information) Generating via information...

**Description**

Informational message indicating generation of via information.

**What Next**

Only for information. No work is needed.

---

## GRD-0267

(Information) Generating conductor layer information...

**Description**

Informational message indicating generation of conductor layer information.

**What Next**

Only for information. No work is needed.

---

## GRD-0268

(Information) Generating global variables...

**Description**

Informational message. Generating global variables such as version, capunit, resunit, etc.

**What Next**

For information only. No work is needed.

---

## GRD-0269

(Information) Layer = %s. BelowPlane = %s.

Feedback

**Description**

Informational message. Name of the layer and the plane below it.

**What Next**

Information only. Check if this is what you intend.

---

## GRD-0270

(Information) AbovePlane =

**Description**

Informational only - There is no plane above the layer.

**What Next**

Information only. Check if this is what you intend.

---

## GRD-0271

(Information) AbovePlane = %s.

**Description**

Name of the plane above layer.

**What Next**

Information only. Check if this is what you intend.

---

## GRD-0272

(Fatal) Final verification of the TLU file failed.

**Description**

The final step of the TLU file correctness could not be checked in the end.

**What Next**

Please check the TLU file correctness. If the TLU file is corrupted, remove the TLU file and relaunch grdgenxo one last time.

---

## GRD-0273

(Information) Another process is working on %s.

Feedback

**Description**

Another process is working on generating data for this file.

**What Next**

Information only. Indicates another process is working on a file.

---

## GRD-0274

(Information) Another process is working on this database. Leaving TLUPlus model output task to that process.

**Description**

Another process is working on generating data for this file.

**What Next**

Information only. Indicates another process is working on a file.

---

## GRD-0275

(Information) %s was successfully generated by another process.

**Description**

File was generated by another process.

**What Next**

This is for information only. No work is needed.

---

## GRD-0276

(Warning) %s seems to be corrupted. Regenerating it...

**Description**

File is corrupted. Regeneration is in progress.

**What Next**

This is warning only. No further work is required.

---

## GRD-0277

(Fatal) License lock file failure. Please remove *.llock from grd repository.

Feedback

**Description**

Grdgenxo encounterd license lock file failure.

**What Next**

Remove *.llock from grd repository and re-run. Contact Synopsys Support if the problem persists.

---

## GRD-0278

(Fatal) Lock file failure.

**Description**

Grdgenxo encountered a lock file failure.

**What Next**

Contact Synopsys Support if the problem persists.

---

## GRD-0279

(Information) Printing TLUPlus model file %s ...

**Description**

Data is being written into TLUPlus file.

**What Next**

Information only. No further acttion is required.

---

## GRD-0281

(Fatal) Fail to read CapTable from file %s.

**Description**

Failed to read capacitance table from the file under work directory.

**What Next**

Ensure that the capacitance tables are correct in the file.

---

## GRD-0282

(Fatal) Format error in file %s.

Feedback

**Description**

Format error in the file under work directory.

**What Next**

Correct the format error. If problem persists, contact the Synopsys Support Center.

---

## GRD-0283

(Fatal) %s appears to be corrupted. Remove that file and re-run grdgenxo.

**Description**

This file under work directory is corrupted. Remove that file and re-run grdgenxo.

**What Next**

Remove the corrupted file and re-run.

---

## GRD-0285

(Fatal) Number of conducting layers is limited to 100.

**Description**

The number of conducting layers in the ITF file is limited to 100.

**What Next**

If possible, modify the ITF file to contain fewer than 100 conducting layers. For assistance, contact the Synopsys Support Center.

---

## GRD-0286

(Fatal) The number of ER_VS_SI_SPACING entries %d exceeds the maximum number %d allowed for TLUPlus flow.

**Description**

The total number of spacing values in the *ER_VS_SI_SPACING* table is too large for the TLUPlus flow.

**What Next**

If possible, modify the *ER_VS_SI_SPACING* table in the ITF file to reduce the number of spacing values. For assistance, contact the Synopsys Support Center.

Feedback

## GRD-0287

(Fatal) Input format file has a %s value of %f different from previous runs. Please remove the existing running directory and re-run grdgenxo from scratch.

**Description**

CAP_UNIT in Input format file is different from previous runs.

**What Next**

Correct the unit in the format file or remove the existing running directory and re-run grdgenxo from scratch.

## GRD-0288

(Fatal) Input ITF file is different from %s used in previous runs. Please remove the existing running directory and re-run grdgenxo from scratch.

**Description**

Input ITF file is different from the one used in previous runs.

**What Next**

Remove the existing running directory and re-run grdgenxo from scratch.

## GRD-0289

(Fatal) Input format file is different from %s used in previous runs. Please remove the existing running directory and re-run grdgenxo from scratch.

**Description**

Input format file is different from the previous runs.

**What Next**

Remove the existing running directory and re-run grdgenxo from scratch.

## GRD-0290

(Fatal) %s used in previous runs is different from default format. Please remove the existing running directory and re-run grdgenxo from scratch.

**Description**

Format file used in previous runs is different from default format.

Feedback

**What Next**

Correct the format file then remove the existing running directory and re-run grdgenxo from scratch.

---

## GRD-0291

(Fatal) ITF file is too large. The input ITF file has more than 50000 lines.

**Description**

Input ITF files of more than 50000 lines are not supported.

**What Next**

If possible, modify the ITF file to contain fewer than 50000 lines. For assistance, contact the Synopsys Support Center.

---

## GRD-0292

(Fatal) Error in file %s.

**Description**

An error occurred in reading the ITF file. The file might be corrupt.

**What Next**

Correct the ITF file and resubmit the run.

---

## GRD-0293

(Fatal) Unknown direction: %s.

**Description**

Unknown keyword used for direction. Possible keywords can be BOTTOM, TOP, or LATERAL.

**What Next**

Correct the direction and re-run.

---

## GRD-0294

(Fatal) Previous running directory %s is out of date. Please remove the directory and re-run grdgenxo from scratch.

Feedback

**Description**

Previous running directory is out of date.

**What Next**

Remove the directory and re-run grdgenxo from scratch.

---

## GRD-0295

(Fatal) grdgenxo -itf2TLUPlus does NOT parse mapping file anymore. Mapping file is parsed in Astro.

**Description**

This feature is not supported.

**What Next**

Use the *grdgenxo -help* command to list the supported options for the grdgenxo tool.

---

## GRD-0296

(Fatal) ITF file is NOT provided.

**Description**

An input ITF file name must be provided in the grdgenxo command line, as follows:

```
% grdgenxo -i <input_itf> ...
```

where *-i* is the short form of the *-input* option.

Additional options are required to complete the grdgenxo command. Use the *grdgenxo -help* command to list the supported options for the grdgenxo tool.

**What Next**

Resubmit the *grdgenxo* command with the correct options.

---

## GRD-0297

(Warning) TLUPlus file name is not provided. Default file name %s will be used.

**Description**

An output file name is not provided in the command line, therefore the grdgenxo tool uses a default file name for the output TLUPlus file.

Feedback

The correct syntax for the TLUPlus flow is as follows:

```
% grdgenxo -itf2TLUPlus -i <input_itf> -o <tluplus_file>
```

where *-i* is the short form of the *-input* option and *-o* is the short form of the *-output* option of the *grdgenxo* command.

**What Next**

If the default file name is acceptable, no further action is needed. Correct the command line if you would like to specify the output file name.

---

## GRD-0298

(Information) Generating TLUPlus Cap Model...

**Description**

This message is for information only.

**What Next**

No further action is needed.

---

## GRD-0299

(Fatal) %s is reserved for grdgenxo running directory. Different TLUPlus file name must be provided.

**Description**

The name of TLUPlus output file is same as NXTGRD database directory.

**What Next**

Change the name of output TLUPLus file and re-run.

---

## GRD-0300

(Fatal) The ctf file %s doesn't contain the TLUPlus part. Please generate an nxtgrd file from your ctf file using -ctf2grd and run -res_update on the new GRD file. Commands to be launched are the following: grdgenxo -ctf2grd -i %s -o %s.GRD; grdgenxo -res_update %s -i %s.GRD -o %s

**Description**

An error has occurred with the usage of the *grdgenxo* command because the ctf file doesn't contain a TLUPlus part.

Feedback

**What Next**

Please generate an nxtgrd file from your ctf file using -ctf2grd and then run -res_update on the new GRD file.

The commands to launch are as follows:

```
% grdgenxo -ctf2grd -i <input_ctf_file> -o <output_grd_file>
% grdgenxo -res_update <updated_ITF> -i <output_grd_file> -o
 <updated_grd_file>
```

---

## GRD-0306

(Severe) INTERNAL Error: unmatched vector size.

**Description**

This is an internal check and unmatched vector size is encountered.

**What Next**

Contact the Synopsys Support Center.

---

## GRD-0307

(Severe) INTERNAL Error: Model name WOCN and WCN NOT found.

**Description**

This is an internal unit test related message.

**What Next**

Contact the Synopsys Support Center.

---

## GRD-0311

(Severe) INTERNAL Error: syntax error in file: %s : %d.

**Description**

Syntax error is found in -zirva configuration file.

**What Next**

Contact the Synopsys Support Center.

Feedback

## GRD-0312

(Severe) INTERNAL Error: Can NOT find center conductor named: %s in -zirva configuration file: %s.

**Description**

Center conductor named: %s cannot be found in -zirva configuration file.

**What Next**

Contact the Synopsys Support Center.

## GRD-0313

(Severe) Can NOT open -zirva configuration file %s.

**Description**

The tool cannot open the -zirva configuration file.

**What Next**

Check whether the -zirva configuration file is available.

## GRD-0314

(Warning) INTERNAL Warning: Bad option %s in file %s: %d.

**Description**

An unsupported option is found in the -zirva configuration file.

**What Next**

Check the file syntax and make corrections as needed. If the problem persists, contact the Synopsys Support Center.

## GRD-0315

(Severe) INTERNAL Error: Failed to generate Field Solver description file %s.

**Description**

The field solver input file cannot be opened.

**What Next**

Contact the Synopsys Support Center.

Feedback

## GRD-0316

(Warning) Can NOT initialize error file

**Description**

The grdgenxo tool cannot set up the error message system. This issue might be caused by a problem with the intallation package.

**What Next**

Contact the Synopsys Support Center.

## GRD-0319

(Fatal) INTERNAL Error: Fail to read nxtgrd file due to corrupted via data.

**Description**

StarRC failed to read the nxtgrd file, because an internal error happened while reading via descriptions.

**What Next**

Make sure that the nxtgrd file is not corrupted. If possible, regenerate the nxtgrd file using the latest grdgenxo version. If the error persists, contact the Synopsys Support Center.

## GRD-0323

(Severe) The nxtgrd file version number %d is not compatible with this release.

**Description**

The StarRC version used for this run does not support the specified nxtgrd file, which was generated using a much older StarRC version.

**What Next**

Regenerate the nxtgrd file using the newer StarRC version.

## GRD-0324

(Fatal) Can not open file to write internal Field Solver input file.

**Description**

StarRC can not open an output file to use when writing the Field Solver input file.

Feedback

**What Next**

Check whether the disk is full and resubmit the StarRC run. If the error still occurs, contact the Synopsys Support Center.

---

## GRD-0325

(Fatal) INTERNAL Error: effective thickness for subdielectric layer %s is less than 0.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## GRD-0326

(Warning) More than one raised diffusion layers have different RAISED_DIFFUSION_GATE_SIDE_CONFORMAL_ER defined or have the same RAISED_DIFFUSION_GATE_SIDE_CONFORMAL_ER but with different RAISED_DIFFUSION_THICKNESS defined in ITF. StarRC has difficulty to pair the raised diffusion layers with gate.

**Description**

StarRC detected more than one raised diffusion layers, and they have different RAISED_DIFFUSION_GATE_SIDE_CONFORMAL_ER defined or have the same RAISED_DIFFUSION_GATE_SIDE_CONFORMAL_ER but with different RAISED_DIFFUSION_THICKNESS defined.

**What Next**

Check whether this setting in ITF is intended. If this setting is intended, you don't need to take any action.

---

## GRD-0327

(Fatal) INTERNAL Error: error found while merging dielectric boxes.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

Feedback

## GRD-0328

(Fatal) INTERNAL Error: error found while processing dielectric boxes.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## GRD-0329

(Fatal) Conducting layer %s can not be placed on non-planar dielectric layer %s.

**Description**

A conducting layer cannot be placed on a nonplanar dielectric layer.

**What Next**

In the ITF file, check the properties of the reported conducting layer and dielectric layer.

## GRD-0330

(Fatal) Could not find MEASURED_FROM %s for %s.

**Description**

The MEASURED_FROM layer of the reported dielectric layer can not be found.

**What Next**

Check ITF to make sure the MEASURED_FROM layer of the reported dielectric layer is correctly defined.

## GRD-0331

(Fatal) Can not find FROM_LAYER %s for via %s.

**Description**

The layer name of FROM_LAYER for reported via layer can not be found.

**What Next**

Check ITF to make sure the layer in FROM_LAYER of reported via layer is correctly defined.

Feedback

## GRD-0332

(Fatal) Can not find TO_LAYER %s for via %s.

**Description**

The layer name of TO_LAYER for reported via layer can not be found.

**What Next**

Check ITF to make sure the layer in TO_LAYER of reported via layer is correctly defined.

## GRD-0333

(Fatal) Can not find dielectric %s when converting ITF to Field Solver format.

**Description**

When converting ITF information to field solver files, the StarRC tool detects that a specific dielectric layer is not found in the ITF file.

**What Next**

Check the ITF file to make sure that the named dielectric layer is specified correctly and resubmit the StarRC run. If the error still occurs, contact the Synopsys Support Center.

## GRD-0334

(Warning) INTERNAL Warning: no physical dielectric stack found

**Description**

When converting ITF information to field solver files, the StarRC tool detects that no dielectric layers are present.

**What Next**

Information only; if this structure is expected, no action is needed.

## GRD-0335

(Fatal) Fail to create internal file.

**Description**

StarRC can not create some internal file.

Feedback

**What Next**

Check whether the disk is full and rerun StarRC. If the error still ocurrs, contact Synopsys for more help.

---

## GRD-0336

(Fatal) Raised diffusion layer (%s) has edge offset while does not have thickness raise.

**Description**

The reported diffusion layer has raised diffusion etch but doesn't have raised diffusion thickness.

**What Next**

Check ITF to make sure when setting RAISED_DIFFUSION_ETCH, RAISED_DIFFUSION_THICKNESS is also set.

---

## GRD-0337

(Fatal) Failed to parse the nxtgrd file.

**Description**

An internal error has been encountered while loading the nxtgrd file.

**What Next**

Make sure that the nxtgrd file is not corrupted. If possible, regenerate the nxtgrd file using the latest grdgenxo version. If the error persists, contact the Synopsys Support Center.

---

## GRD-0338

(Information) (%d) ITF: %s.

**Description**

Information only.

**What Next**

No action is needed.

---

## GRD-0339

(Warning) (%d) ITF: WARNING: %s.

Feedback

**Description**

Information only.

**What Next**

Refer to the StarRC guide for a more detailed description. If this is acceptable, no action is needed.

---

## GRD-0340

(Information) No equal sign is found after MEASURED_FROM.

**Description**

StarRC did not find an equal sign after MEASURED_FROM.

**What Next**

The equal sign after MEASURED_FROM is optional. Information only.

---

## GRD-0341

(Fatal) The FROM layer %s of the TSV layer %s is not a valid conducting layer.

**Description**

Inside a TSV definition, the layer name specified for the FROM field does not match any CONDUCTOR definition.

**What Next**

Check for possible layer name misspelling or ITF completeness.

---

## GRD-0342

(Fatal) The TO layer %s of the TSV layer %s is not a valid conducting layer.

**Description**

Inside a TSV definition, the layer name specified for the TO field does not match any CONDUCTOR definition.

**What Next**

Check for possible layer name misspelling or ITF completeness.

Feedback

## GRD-0343

(Fatal) FROM layer %s and TO layer %s of TSV layer %s are located on the same side of the substrate.

**Description**

A through-silicon via (TSV) layer connects two conducting layers, which must be located on opposite sides of the substrate. In this run, two layers located on the same side of the substrate have incorrectly been provided.

**What Next**

Check for possible layer name misspelling.

If you want to define a standard via layer, use the *VIA* command instead of the *TSV* command.

## GRD-0344

(Severe) INTERNAL Error: %s.

**Description**

An internal consistency check has failed while reading the intermediate file sct/sct.

**What Next**

If you are reusing an existent sct file, try to regenerate it. Otherwise, contact the Synopsys Support Center.

## GRD-0345

(Fatal) Values in COV_X should be listed in ascending order.

**Description**

Inside a RPV_VS_COVERAGE table, values for COV_X are not in ascending order, which is not allowed.

**What Next**

Fix the ITF file.

## GRD-0346

(Fatal) Values in COV_Y should be listed in ascending order.

Feedback

**Description**

Inside a RPV_VS_COVERAGE table, values for COV_Y are not in ascending order, which is not allowed.

**What Next**

Fix the ITF file.

---

## GRD-0349

(Warning) Inconsistent ER_VS_SI_SPACING table value trend: Values in the table should be consistently either larger or smaller than nominal ER value for layer %s.

**Description**

Relative permittivity (ER) values specified in the *ER_VS_SI_SPACING* table do not follow expected behavior.

**What Next**

Check the ER values specified in the *ER_VS_SI_SPACING* command. If the values are correct, you can ignore this message.

---

## GRD-0350

(Information) Via %s has its bottom layer not defined or without the property LAYER_TYPE: DIFFUSION. TLI flow will be skipped for this VIA.

**Description**

The TLI flow only applies to via layers that connect difussions to metal layers, and so this via layer is ignored.

---

## GRD-0351

(Warning) Via layer %s will be converted to a TLI conductor but it has ETCH_VS_WIDTH_AND_LENGTH tables defined, which will be ignored for capacitance extraction.

**Description**

The TLI flow does not support ETCH_VS_WIDTH_AND_LENGTH tables, so these tables are ignored for via layers that are translated to TLI conductors.

Feedback

**What Next**

Either disable the TLI flow if ETCH_VS_WIDTH_AND_LENGTH tables are required, or ignore this warning message.

---

## GRD-0354

(Fatal) %s.

**Description**

An unknown option was used with the *grdgenxo* command.

Use the *grdgenxo -help* command to list the supported options for the grdgenxo tool.

**What Next**

Correct the syntax and resubmit the *grdgenxo* command.

---

## GRD-0355

(Fatal) Input format file has a %s value of %s different from previous runs. Please remove the existing running directory and re-run grdgenxo from scratch.

**Description**

Operating condition in format file is different from previous runs.

**What Next**

Correct the operating condition or remove the existing running directory and re-run grdgenxo from scratch.

---

## GRD-0357

(SEVERE) INTERNAL Error: %s appears corrupt. Erase that file and re-run the characterization.

**Description**

The internal file mentioned in the error message might be corrupted.

**What Next**

Erase the file mentioned in the error message and resubmit the grdgenxo command. If the issue still occurs, contact the Synopsys Support Center.

Feedback

## GRD-0358

(Severe) INTERNAL_Error: Failed to write nxtgrd file. Exiting...

**Description**

The grdgenxo tool failed to write the nxtgrd file.

**What Next**

Check that the disk is not full and that the directory is write-enabled. Resubmit the grdgenxo run. If the issue still occurs, contact the Synopsys Support Center.

## GRD-0359

(Fatal) The %s field for TSV layer %s must be strictly positive.

**Description**

Inside a TSV definition, a null or negative value is provided for a field that requires a value greater than zero.

**What Next**

This field must be fixed in the ITF file.

## GRD-0360

(Severe) %s.

**Description**

An error has occurred with the usage of the *grdgenxo* command. The error might be caused by improper command syntax or by an unsuitable input file.

Note that the grdgenxo update flow does not support encrypted nxtgrd files.

The correct syntax for the grdgenxo update flow is as follows:

```
% grdgenxo -res_update <updated_ITF> -i <input_grd_file> -o
 <updated_grd_file>
```

where *-i* is the short form of the *-input* option and *-o* is the short form of the *-output* option of the *grdgenxo* command.

**What Next**

Correct the syntax of the *grdgenxo* command, ensure that the nxtgrd file is not encrypted, and resubmit the command.

Feedback

## GRD-0361

(Fatal) (%d) ITF: ERROR: %s.

**Description**

The ITF file contains a syntax error which Grdgenxo cannot parse.

**What Next**

Refer to the StarRC guide for a description of the ITF syntax and correct the ITF file.

## GRD-0362

(Severe) (%d) ITF: ERROR: %s.

**Description**

The ITF file contains a syntax error which Grdgenxo cannot parse.

**What Next**

Refer to the StarRC guide for a description of the ITF syntax and correct the ITF file.

## GRD-0364

(Fatal) A temperature coefficient is defined for TSV layer %s but its nominal temperature is not specified.

**Description**

A layer-specific temperature coefficient is defined for this TSV layer, using CRT1 or CRT2. However, the corresponding nominal temperature is not specified, neither locally nor globally.

**What Next**

Nominal temperature can be defined locally using the T0 field inside the TSV definition, or globally using GLOBAL_TEMPERATURE.

## GRD-0365

(Fatal) Illegal Electro-Migration keywords were found inside the external Electro-Migration information file.

Feedback

### Description

You can provide most electromigration information either in the ITF file or in external files. However, the following statements (keywords) can only appear in the ITF file:

```
EM_INFORMATION
INCLUDE_EM_INFORMATION_FILE = <em_file>
END_OF_EM_INFORMATION
```

### What Next

Remove any occurrences of the *EM_INFORMATION*, *END_OF_EM_INFORMATION*, or *INCLUDE_EM_INFORMATION_FILE* keywords in the electromigration information file. Ensure that the syntax of the electromigration information file is correct.

---

## GRD-0366

(Fatal) "END_OF_EM_INFORMATION" statement expected at the end of EM information block.

### Description

The end of the electromigration information block must be marked with the *END_OF_EM_INFORMATION* statement.

### What Next

Correct the syntax of the electromigration block and resubmit the run.

---

## GRD-0367

(Fatal) An equal sign is expected after "INCLUDE_EM_INFORMATION_FILE".

### Description

The ITF file provided does not contain an equal sign after INCLUDE_EM_INFORMATION_FILE.

### What Next

Correct the syntax of INCLUDE_EM_INFORMATION_FILE in the ITF file. The expected syntax is "INCLUDE_EM_INFORMATION_FILE = file name".

---

## GRD-0368

(Fatal) A file name for the external Electro-Migration information file is expected.

Feedback

**Description**

The ITF file provided does not contain a file name after "INCLUDE_EM_INFORMATION_FILE = ".

**What Next**

Correct the syntax of INCLUDE_EM_INFORMATION_FILE in the ITF file. The expected syntax is "INCLUDE_EM_INFORMATION_FILE = file name".

---

## GRD-0370

(Fatal) Cannot use SCT script file %s.

**Description**

Cannot install the SCT script file specified in the message for SCT model generation.

**What Next**

Check that the SCT script file is correctly specified with the *-sct_script_path* option. Check that the disk is not full.

If the issue still exists, contact the Synopsys Support Center.

---

## GRD-0371

(Fatal) grdgenxo cannot proceed because of license failure. Please make sure either STAR-RC2-TCAD or Galaxy-Common license is up and running.

**Description**

This run requires either STAR-RC2-TCAD or Galaxy-Common license.

**What Next**

Ensure that you have the required license.

---

## GRD-0372

(Warning) SCT file not present in the folder containing the models to reuse: regenerating...

**Description**

The *grdgenxo -reuse_device_models* command is used, but the grdgenxo tool cannot find the device models in the reuse_device_models directory. Thereore, the tool ignores the *-reuse_device_models* option and regenerates the device models.

Feedback

**What Next**

To continue the run, ignore this warning message.

If you want to change the directory for the *-reuse_device_models* option, kill the grdgenxo jobs and restart the run with the correct directory.

---

## GRD-0373

(Information) Generating %s.

**Description**

The message indicates that 3D model generation has correctly started.

**What Next**

No action is needed from your part.

---

## GRD-0374

(Warning) The FROM layer %s of TSV layer %s is above the TO layer %s.

**Description**

StarRC expects the TSV layer to be defined from low to high. If it not the case, it swaps the FROM and TO layers internally.

**What Next**

The warning can be suppressed by swapping the TO layer with the FROM layer in the ITF file.

---

## GRD-0375

(Fatal) The option -profiler is obsolete.

**Description**

The *-profiler* option of the grdgenxo tool is obsolete.

**What Next**

Do not use the *-profiler* option with the *grdgenxo* command.

Feedback

## GRD-0376

(Warning) %s table(s) specified for layer %s together with more than %d %s. This might cause the simulation time to increase. Consider reducing the number of %s values.

### Description

When the ITF file contains one or more tables for a given layer, the grdgenxo tool uses the defined gate-to-contact spacings or conductor widths as a reference for building the 3D models. Simulation time increases as the number of different gate-to-contact spacing or conductor width values increases.

This warning message indicates that the number of different gate-to-contact spacing or conductor width values for a single layer is large.

### What Next

Review the list of gate-to-contact spacing or conductor width values in the mentioned tables. If possible, reduce the number of different gate-to-contact spacing or conductor width values.

## GRD-0377

(Warning) POLYNOMIAL_BASED_THICKNESS_VARIATION table consistency check: relative thickness variation changes more than 5%% (from %.2f%% to %.2f%%) between (density %.4f, width %.4f) and (density %.4f, width %.4f) for layer %s

### Description

The grdgenxo tool checks the *POLYNOMIAL_BASED_THICKNESS_VARIATION* table for consistency. The tool generates internal width and density sampling points in increasing order, then calculates thickness variations for each sampling point. For each combination of width and density, the grdgenxo tool checks that the relative thickness variation changes less than 5 per cent when width or density is increased to the next sampling point.

This warning message indicates that the grdgenxo tool detects violations between two neighboring sampling points.

### What Next

Check the information in the message carefully to ensure that the abrupt thickness variation change (>5%) is acceptable. If it is acceptable, you can ignore this warning message.

If it is not acceptable, revise the *POLYNOMIAL_BASED_THICKNESS_VARIATION* table and run the grdgenxo tool again.

Feedback

## GRD-0378

(Fatal) The maximum number of conductors that StarRC supports is limited to %i.

**Description**

There is more than %i CONDUCTOR declaration in the ITF file. StarRC does not support it.

**What Next**

Reduce the number of conductors in the ITF file or Contact the Synopsys Support Center.

## GRD-0379

(Fatal) INTERNAL Error: gap between subdielectric %s and %s for dielectric layer %s.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## GRD-0380

(Information) %s file already generated and reused.

**Description**

This message indicates that the TVM file containing 3D models generated by grdgenxo has already been generated by a previous run of the tool. Consequently, grdgenxo skips 3D model generation and uses the previously-generated file.

**What Next**

If you would like to use the previously-generated TVM file, no action is required on your part. Otherwise, remove the tvm folder in grdgenxo working directory and rerun.

## GRD-0381

(Warning) %s file failed to be generated in a previous run. See %s for additional info.

Feedback

**Description**

The warning indicates that 3D model generation has failed in previous run of grdgenxo. The nxtgrd file will not be generated. The file indicated in the message contains additional instructions to follow to solve the problem.

**What Next**

Follow the instructions indicated in the file that is specified and rerun grdgenxo.

---

## GRD-0382

(Fatal) Can NOT initialize %s Table settings.

**Description**

The grdgenxo tool cannot generate the 3D cap table decribed in the message.

Possible reasons are as follows:

- You are running the grdgenxo tool with the existing working directory. The grdgenxo tool tries to reuse the previous table in the existing working directory and finds that the table was not correctly generated in the previous grdgenxo run.

- The path to another tool (such as the gds2cap, quickcap, or rapid3d tools) is not set correctly.

- Arguments for another tool are not set correctly.

**What Next**

Check the log file for more information about the specific error condition. Address any issues and resubmit the grdgenxo run.

To address problems with SCT/DCM/TVM tables, remove all of the sct/dcm/tvm directories under the existing working directory and resubmit the grdgenxo run.

---

## GRD-0383

(Fatal) Can NOT generate %s file. Check %s.failure for a list of the failing simulations and check the log files in the specified directories. Remove them before launching grdgenxo again.

**Description**

The grdgenxo tool cannot generate the model file described in the message.

Feedback

**What Next**

Check the *.failure file mentioned in the error message for a list of the failing simulations and check the log files in the directories for those failing simulations. Remove the directories for the failing simulations and run the grdgenxo tool again.

---

## GRD-0384

(Fatal) The RVTV layer %s and the DIFFUSION layer %s are connected by the via layer %s and have no common DEVICE_TYPE.

**Description**

Grdgenxo creates models for RVTV layers based on the DEVICE_TYPE. If no DEVICE_TYPE is given, sct models are missing.

**What Next**

Please set at least one common DEVICE_TYPE for the RVTV layer and the DIFFUSION layers it is connected to through vias.

---

## GRD-0385

(Warning) The DEVICE_TYPE %s is %sin the RVTV layer %s%s

**Description**

Grdgenxo creates models for RVTV layers based on the DEVICE_TYPE. If no DEVICE_TYPE is given, grdgenxo deduces the DEVICE_TYPE from via connections.

**What Next**

Please remove the DEVICE_TYPE on the RVTV layer or set at least one common DEVICE_TYPE for the RVTV layer and the layers it is connected to through vias.

---

## GRD-0438

(Information) (%d) ITF: %s.

**Description**

Information only.

**What Next**

No action is needed.

Feedback

## GRD-0439

(Warning) (%d) ITF: WARNING: %s.

**Description**

Information only.

**What Next**

Refer to the StarRC guide for a more detailed description of ITF rules. If this is acceptable, no action is needed.

## GRD-0461

(Fatal) (%d) ITF: ERROR: %s.

**Description**

The ITF file contains a consistency error which grdgenxo cannot overcome.

**What Next**

Refer to the StarRC guide for a description of the ITF rules and correct the ITF file.

## GRD-0462

(Severe) (%d) ITF: ERROR: %s.

**Description**

The ITF file contains a consistency error which grdgenxo cannot overcome.

**What Next**

Refer to the StarRC guide for a description of the ITF rules and correct the ITF file.

## GRD-0463

(Warning) LINKED_TO statement between layers %s and %s cannot be satisfied because the layers are not capacitively equivalent

**Description**

LINKED_TO statement disables automatic linking and indicates two layers are candidate for explicit linking. However, grdgenxo will check if the candidate layers are effectively capacitively equivalent. These checks include the comparison of the conductors layer

Feedback

properties as well as the associated dielectrics properties. If one of these conditions is not satisfied, the LINKED_TO statement is discarded.

This warning message indicates that the grdgenxo tool cannot satisfy the LINKED_TO statement specified in the ITF.

**What Next**

Check if the conductor and associated dielectric properties are equivalent. Either fix the ITF by aligning them or remove the LINKED_TO statement to avoid the warning.

---

## GRD-0470

(Fatal) RC scaling file error : \n%s

**Description**

There is an error in the RC scaling file.

**What Next**

Correct the table.

---

## GRD-0471

(Fatal) SCT generation for this process requires a gds2cap supporting option %s.

**Description**

For generating SCT for the given process, grdgenxo needs to use a specific option for gds2cap. The version provided with gds2cap_path does not seem to provide this option.

**What Next**

Use a more recent version of gds2cap. Contact the support team to know which version contains this optiion.

---

## GRD-1002

(Warning) Option -old_sct_cfo will be removed in I-2013.12 release.

**Description**

The grdgenxo -old_sct_cfo command calls rapid3d to generate SCT models directly to characterize gate-diffusion caps (cfo). This option is obsolete starting with version I-2013.12.

Feedback

**What Next**

Do not use the -old_sct_cfo option in I-2013.12 and later versions.

---

## GRD-1003

(Fatal) %s file is complete but could not be renamed as %s. Move %s to %s and relaunch grdgenxo.

**Description**

This message indicates that the 3D model file generation has been successfully completed but the grdgenxo tool cannot rename it to complete the flow.

**What Next**

Check the file system to see whether the disk is full and if the network connection is stable. Make corrections as needed, rename the file as indicated in the message, and relaunch the grdgenxo tool.

---

## GRD-1004

(Warning) No previous nxtgrd run directory is found, -inc switch is ignored.

**Description**

The *grdgenxo -inc* command specifies that an incremental nxtgrd file update should be performed. However, the grdgenxo tool cannot find the previous nxtgrd run directory. Thereore, the tool ignores the *-inc* option and runs from the beginning.

**What Next**

To continue the run, ignore this warning message.

If you are running the grdgenxo tool from the wrong directory, kill the grdgenxo jobs and restart from the correct nxtgrd run directory.

---

## GRD-1005

(Fatal) In the process information keyword group of the ITF file, the process keyword %s is missing

**Description**

The process information keyword group is optional, but if one of the *PROCESS_FOUNDRY*, *PROCESS_NODE*, *PROCESS_VERSION*, or *PROCESS_CORNER* keywords is specified, then all four must be specified. Note that the *PROCESS_TYPE* keyword can exist independently from the other four.

Feedback

### What Next

Either add the missing keywords, or remove completely the process information keyword group.

For more information, see the reference pages for the *PROCESS_FOUNDRY*, *PROCESS_NODE*, *PROCESS_VERSION*, *PROCESS_CORNER*, and *PROCESS_TYPE* keywords in the *StarRC User Guide and Command Reference*.

---

## GRD-1006

(Fatal) Illegal value for the %s statement in the ITF file at line %i.

### Description

The input value of the process description keyword does not follow the requirements.

The argument of the *PROCESS_FOUNDRY*, *PROCESS_TYPE*, and *PROCESS_CORNER* keywords must be a one-word string without spaces and without special characters @, #, and $.

The argument of the *PROCESS_NODE* and *PROCESS_VERSION* keywords must be a nonnegative floating-point number.

### What Next

Correct the syntax of the process description statement.

### Examples

```
TECHNOLOGY = abc
PROCESS_FOUNDRY = fab_1802C
PROCESS_NODE = 130
PROCESS_TYPE = SOI
PROCESS_VERSION = 2.0
PROCESS_CORNER = TYPICAL
```

---

## GRD-1007

(Information) gds2cap options have been set to "%s %s".

### Description

Options for the gds2cap tool have been set to those specified in the message.

### What Next

Information only; no action is needed.

Feedback

## GRD-1008

(Fatal) Found failing simulations while generating 3D models. Could not write the log file. Please relaunch grdgenxo.

**Description**

The grdgenxo tool detected some failing simulations while generating 3D models. Consequently the tool could not write the log files correctly.

**What Next**

Review the cause of the failures in the log files specified in the SCT file, make the required corrections, and relaunch the grdgenxo run.

## GRD-1009

(Fatal) SCT_VERSION of the SCT file provided with -sct is older than the SCT_VERSION of the nxtgrd file.

**Description**

The grdgenxo tool cannot replace a new SCT file with an old SCT file.

**What Next**

Generate a new SCT file with a newwer grdgenxo version

## GRD-1011

(Fatal) The -bin2ascii option is not supported on a nxtgrd file with a bfs-encrypted header.

**Description**

You are running the grdgenxo -bin2ascii command. However, the nxtgrd file to be processed was generated by the grdgenxo -bfsEncrypt command. The grdgenxo tool cannot use the -bin2ascii option for a bfs-encryted nxtgrd file.

**What Next**

Contact the Synopsys Support Center.

## GRD-1012

(Fatal) The -res_update option is not supported on a nxtgrd file with a bfs-encrypted header.

Feedback

**Description**

You are running the grdgenxo -res_update command. However, the nxtgrd file to be processed was generated by the grdgenxo -bfsEncrypt command. The grdgenxo tool cannot use the -res_update option for a bfs-encryted nxtgrd file.

**What Next**

Run grdgenxo without the -res_update option.

---

## GRD-1014

(Fatal) VARIATION_PARAMETERS or SENSITIVITY is not supported starting from J-2014.12 release.

**Description**

VARIATION_PARAMETERS or SENSITIVITY are obsolete starting from J-2014.12 release.

**What Next**

Remove VARIATION_PARAMETERS or SENSITIVITY keywords from the ITF file.

---

## GRD-1015

(Fatal) INTERNAL Error: Can not find metal scheme %s.

**Description**

Can not find metal scheme.

**What Next**

Contact the Synopsys Support Center.

---

## GRD-1016

(Fatal) -print_process_stack is not allowed for encrypted ITF files.

**Description**

You are running grdgenxo with option -print_process_stack. The option cannot work for encrypted ITF file.

**What Next**

Run grdgenxo -print_process_stack with un-encrypted ITF.

Feedback

## GRD-1017

(Fatal) -viewer is not allowed for encrypted ITF files.

### Description

You are running grdgenxo with option -viewer. The option cannot work for encrypted ITF file.

### What Next

Run grdgenxo -viewer with un-encrypted ITF.

## GRD-1018

(Warning) ITF Process Header is needed for advanced process node. Please add process header information to ITF and rerun grdgenxo.

### Description

The ITF has met StarRC advanced modeling requirement and it needs following PROCESS header information in ITF.

*PROCESS_NODE*, *PROCESS_VERSION*, *PROCESS_FOUNDRY*, *PROCESS_TYPE*, and *PROCESS_CORNER*

### What Next

Add process header description statement into ITF.

### Examples

```
TECHNOLOGY = abc
PROCESS_FOUNDRY = fab_1802C
PROCESS_NODE = 130
PROCESS_TYPE = SOI
PROCESS_VERSION = 2.0
PROCESS_CORNER = TYPICAL
```

## GRD-1019

(Fatal) ITF Process Header is needed for advanced process node. Please add process header information to ITF and rerun grdgenxo.

### Description

The ITF has met StarRC advanced modeling requirement and it needs following PROCESS header information in ITF.

Feedback

*PROCESS_NODE*, *PROCESS_VERSION*, *PROCESS_FOUNDRY*, *PROCESS_TYPE*, and *PROCESS_CORNER*

**What Next**

Add process header description statement into ITF.

**Examples**

```
TECHNOLOGY = abc
PROCESS_FOUNDRY = fab_1802C
PROCESS_NODE = 130
PROCESS_TYPE = SOI
PROCESS_VERSION = 2.0
PROCESS_CORNER = TYPICAL
```

---

## GRD-1020

(Fatal) -print_conductors, -print_dielectrics and -print_vias is not allowed for encrypted ITF files.

**Description**

You are running grdgenxo with option -print_conductors or -print_dielectrics or -print_vias. The option cannot work for encrypted ITF file.

**What Next**

Run grdgenxo -print_conductors or -print_dielectrics or -print_vias with un-encrypted ITF.

---

## GRD-1021

(Warning) ITF Process Header is needed for advanced process node. Please add process header information to ITF and rerun grdgenxo. grdgenxo errors out with GRD-1019 outside Synopsys network.

**Description**

The ITF has met StarRC advanced modeling requirement and it needs following PROCESS header information in ITF.

*PROCESS_NODE*, *PROCESS_VERSION*, *PROCESS_FOUNDRY*, *PROCESS_TYPE*, and *PROCESS_CORNER*

**What Next**

Add process header description statement into ITF.

Feedback

### Examples

```
TECHNOLOGY = abc
PROCESS_FOUNDRY = fab_1802C
PROCESS_NODE = 130
PROCESS_TYPE = SOI
PROCESS_VERSION = 2.0
PROCESS_CORNER = TYPICAL
```

## GRD-1022

(Information) \nFollowing PROCESS HEADER was detected in ITF:%s.

### Description

Information of PROCESS HEADER in the ITF file.

### What Next

Information only. No action is needed.

## GRD-1023

(Fatal) The NXTGRD file is corrupted, please regenerate NXTGRD file.

### Description

The NXTGRD file is corrupted, maybe due to unstable machine or outdated version, please regenerate NXTGRD file and check again.

### What Next

Please rerun grdgenxo to generate NXTGRD again.

## GRD-1288

(Information) Detect the %s inconsistency while checking the ITF, %s.

### Description

Information message indicating the ITF inconsistency which could be ignored or not.

### What Next

This message is for information only. No work is required.

## GRD-3000

(Warning) No Field-Poly layer was found. 3D device models will not be generated.

Feedback

**Description**

Simulation has stopped because no Field-Poly layer is defined in the ITF File. Consequently, grdgenxo cannot build MOS devices and generate 3D device models.

**What Next**

Review the layer definition in the ITF File, add a conductor layer having LAYER_TYPE=FIELD_POLY defined and relaunch grdgenxo.

---

## GRD-3001

(Information) %s.

**Description**

Information only, indicating update of the TLUPlus file.

**What Next**

Information only; no action is needed.

---

## GRD-3002

(Fatal) %s.

**Description**

The error description depicts the problem with the run.

**What Next**

Resubmit the run after correcting the error.

---

## GRD-3003

(Severe) Usage: grdgenxo -itfkey_nxtgrd -i <grd_file> -o <key_file>.

**Description**

The *grdgenxo* command was used incorrectly. The correct usage is as follows:

```
% grdgenxo -itfkey_nxtgrd -i <grd_file> -o <key_file>
```

where *-i* is the shorthand form of the *-input* option and *-o* is the shorthand form of the *-output* option.

**What Next**

Modify the command and submit it again.

Feedback

## GRD-3004

(Severe) Usage: grdgenxo -itfkey_tluplus -i <tluplus_file> -o <key_file>.

**Description**

The grdgenxo command was used incorrectly. The correct usage for this application is as follows:

```
% grdgenxo -itfkey_tluplus -i <tluplus_file> -o <key_file>
```

where *-i* is the shorthand form of the *-input* option and *-o* is the shorthand form of the *-output* option.

**What Next**

Modify the command and submit it again.

## GRD-3005

(Severe) This feature is not supported for TluPlus version < 16

**Description**

This feature is supported only for TLUPlus versions 16 and later.

**What Next**

Generate the TLUPlus file using a newer version of the grdgenxo tool if you would like to use this feature.

## GRD-3006

(Severe) Syntax UPPER_LAYER_ONLY must be added with RPV_VS_SI_COVERAGE.

**Description**

Syntax UPPER_LAYER_ONLY must be added with RPV_VS_SI_COVERAGE.

**What Next**

Check if the table defined in ITF is correct. Add needed syntax and rerun StarRC.

## GRD-3007

(Severe) Usage: grdgenxo -checkgrd -i <binary_grd_file> -o <out_file>.

Feedback

**Description**

An error has occurred with the usage of the *grdgenxo* command. The error might be caused by improper command syntax or by an unsuitable input file.

The correct syntax is as follows:

```
% grdgenxo -checkgrd -i <binary_grd_file> -o <out_file>
```

where *-i* is the short form of the *-input* option and *-o* is the short form of the *-output* option of the *grdgenxo* command.

**What Next**

Correct the syntax of the *grdgenxo* command and resubmit the command.

---

## GRD-3008

(Fatal) The -checkgrd option is not supported on a nxtgrd file with a bfs-encrypted header.

**Description**

You are running the grdgenxo -checkgrd command. However, the nxtgrd file to be processed was generated by the grdgenxo -bfsEncrypt command. The grdgenxo tool cannot use the -checkgrd option for a bfs-encryted nxtgrd file.

**What Next**

Contact the Synopsys Support Center.

---

## GRD-3009

(Fatal) The via layer %s with SIDE_TANGENT is not a contact layer.

**Description**

When a via layer has SIDE_TANGENT, it should be a contact layer. A contact layer is a via layer connecting diffusion/substrate layer and metal layer which is above gate layer.

**What Next**

If the via layer is not a contact layer, remove SIDE_TANGENT in via description.

---

## GRD-3010

(Fatal) Three dimensional LATERAL_CAP_SCALING_VS_SI_SPACING table is not supported from release 2016.06 onwards.

Feedback

**Description**

Three dimensional LATERAL_CAP_SCALING_VS_SI_SPACING, which contains LENGTHS keyword along with SI_SPACINGS and VALUES, is not supported anymore. Two dimensional LATERAL_CAP_SCALING_VS_SI_SPACING table, without LENGTH keyword, is still supported.

**What Next**

Use either DIELECTRIC_FILL_EMULATION_VS_SI_SPACING or two dimensional LATERAL_CAP_SCALING_VS_SI_SPACING instead.

---

## GRD-3011

(Warning) DPT_MAX_SHIFT not supported anymore. All DPT_MAX_SHIFT values will be set to 0.

**Description**

DPT_MAX_SHIFT defined in ITF file will be set to 0. Consequently, any DPT_MAX_SHIFT will be ignored during grdgenxo run.

**What Next**

Remove any DPT_MAX_SHIFT definitions in the ITF File. It will not cause any impact in the final nxtgrd file.

---

## GRD-3012

(Fatal) The via layer %s with LAYER_TYPE=TALL_CONTACT is not a contact layer.

**Description**

When a via layer has LAYER_TYPE=TALL_CONTACT, it should be a contact layer. A contact layer is a via layer connecting diffusion/substrate layer and metal layer which is above gate layer.

**What Next**

If the via layer is not a contact layer, remove LAYER_TYPE=TALL_CONTACT in via description.

---

## GRD-3013

(Warning) The thickness of via layer %s with LAYER_TYPE=TALL_CONTACT is not very large.

Feedback

**Description**

When thickness of a via layer is not very large, the tall contact model is not needed. If LAYER_TYPE=TALL_CONTACT is added to via section, the high accuracy contact model in grdgenxo will be invoked, which will make generation time of NXTGRD file longer.

**What Next**

If the via layer is not a tall contact layer, remove LAYER_TYPE=TALL_CONTACT in via description.

---

## GRD-3014

(Fatal) The conductor layer %s with GATE_PITCH/GATE_WIDTH/ GATE_TO_CONTACT_SPACING is not a gate layer.

**Description**

When a conductor layer has GATE_PITCH/GATE_WIDTH/ GATE_TO_CONTACT_SPACING syntaxes, it should be a gate layer.

**What Next**

If the conductor layer is a gate layer, add LAYER_TYPE=GATE in conductor description. If the conductor layer is not a gate layer, remove GATE_PITCH/GATE_WIDTH/ GATE_TO_CONTACT_SPACING in conductor description.

---

## GRD-3015

(Severe) Usage: grdgenxo -f2f_itf -die1_itf <file> -die1_block_name <name> -die1_num_layers <num> -die2_itf <file> -die2_block_name <name> -die2_num_layers <num> -out_itf <file>

**Description**

An error has occurred with the usage of the *grdgenxo* command with the *-f2f_itf* option. The error might be caused by improper command syntax or by an unsuitable input file.

For the *-f2f_itf* mode of *grdgenxo* all the command-line options listed above are required. Die1 is the bottom die and die2 is the top die that gets flipped. The die input files can be either an ITF or NXTGRD.

*-dieN_num_layers* specifies the number of layers selected from each of the dies.

**What Next**

Correct the syntax of the *grdgenxo* command and resubmit the command.

Feedback

## GRD-3016

(Fatal) Incorrect value of -high_er option (should be > 10)\n Usage: grdgenxo -high_er <val> <itf_file>

### Description

An error has occurred with the usage of the *grdgenxo* command with the *-high_er* option. <val> must be a number > 10 specifying the dielectric permittivity above which the field solver accuracy is increased.

### What Next

Correct the syntax of the *grdgenxo* command and resubmit the command.

## GRD-3017

(Warning) Cannot delete intermediate file %s: %s.

### Description

An internal intermediate file could not be deleted.

### What Next

If no error happened latter and the NXTGRD was succesfully generated, you can ignore this warning message. Otherwise, the error reason can point to the failure root cause.

## GRD-3018

(Fatal) RCAPI_VERSION or RCAPI_PARAM_FILE is missing in the ITF header.

### Description

RCAPI_VERSION and RCAPI_PARAM_FILE must be specified together with RCAPI_LIBRARY_NAME in the ITF header.

### What Next

specify RCAPI_VERSION and RCAPI_PARAM_FILE together with RCAPI_LIBRARY_NAME in the ITF header.

## GRD-3019

(Information) Running incremental 3DIC NXTGRD flow with base NXTGRD "%s", 3DIC ITF "%s", and # of affected layers: %d.

Feedback

**Description**

Information only, indicating command line options of incremental 3DIC NXTGRD flow.

**What Next**

Information only; no action is needed.

---

## GRD-3020

(Fatal) Invalid number of affected layers(%s) specified on command line.

**Description**

Number of affected layers must be set to a number greater than or equal to zero.

**What Next**

Correct the syntax of the *grdgenxo* command and resubmit the command.

---

## GRD-3021

(Fatal) Missing number of affected layers on command line.

**Description**

Number of affected layers must be set to a number greater than or equal to zero.

**What Next**

Correct the syntax of the *grdgenxo* command and resubmit the command.

---

## GRD-3022

(Information) Using default number of affected layers(%d) for incremental 3DIC NXTGRD flow.

**Description**

Information only, indicating default value of number of affected layers is used.

**What Next**

Information only; no action is needed.

---

## GRD-3023

(Fatal) Can NOT open %s for reading.

Feedback

**Description**

Cannot open input file for reading.

**What Next**

Check the input file existence.

---

## GRD-3024

(Fatal) Can NOT open output file %s to write TLU input data.

**Description**

*grdgenxo* can not open an output file to use when writing the TLU input file.

**What Next**

Check the file existence.

---

## GRD-3025

(Information) Regenerate layer (%s) cap. model from for incremental 3DIC NXTGRD flow.

**Description**

Information only, show all conductor layers that will be regenerated.

**What Next**

Information only; no action is needed.

---

## GRD-3026

(Information) Checking SMC compatibility versus ITF file "%s".

**Description**

Information only, show action and reference ITF file.

**What Next**

Information only; no action is needed.

---

## GRD-3027

(Information) %s: compatible.

Feedback

**Description**

Information only, inform that ITF corner file is compatible.

**What Next**

Information only; no action is needed.

---

## GRD-3028

(Information) %s: not compatible.

**Description**

Information only, inform that ITF corner file is not compatible.

**What Next**

Information only; check the file smc_compatibility.summary for information on the incompatibilities found.

---

## GRD-3029

(Warning) SMC ITF compatibility checker detected issues with some of the input ITF files. \nCheck smc_compatibility.summary for details.

**Description**

Warning only, inform an SMC incompatible ITF was found.

**What Next**

Warning only; check the file smc_compatibility.summary for information on the incompatibilites that were found.

---

## GRD-3030

(Severe) %s is a binary file, exiting...

**Description**

Input is a binary file when ascii was expected.

**What Next**

Check input file.

Feedback

## GRD-3031

(Fatal) Found a discrepancy in the flipped RVTV technology. The %s of the %s layer is %f and %f in the flipped technology.

**Description**

Flipped technology doesn't match the initially constructed one.

**What Next**

Change the ITF or contact R&D

## GRD-3040

(Fatal) table name is missing in table %s at line %s.

**Description**

For mutliple table definition, table name must be set for each table.

**What Next**

Correct the syntax of the *grdgenxo* command and resubmit the command.

## GRD-3041

(Fatal) Duplidate table name %s in table %s on layer %s.

**Description**

Table name must be unique in the table.

**What Next**

Correct the syntax of the *grdgenxo* command and resubmit the command.

## GRD-3042

(Fatal) different number of tables in %s and %s on layer %s.

**Description**

Table number must be same between those two tables.

**What Next**

Correct the syntax of the *grdgenxo* command and resubmit the command.

Feedback

## GRD-3043

(Fatal) different table name (%s) in %s and %s on layer %s.

**Description**

Table names must be same between those two tables.

**What Next**

Correct the syntax of the *grdgenxo* command and resubmit the command.

## GRD-3044

(Fatal) Multiple table definition for RPSQ_VS_SI_WIDTH table is not allowed in RES_UPDATE_FILE.

**Description**

Multiple table definition for RPSQ_VS_SI_WIDTH table is not allowed in RES_UPDATE_FILE.

**What Next**

Use grdgenxo -res_update instead.

## GRD-3045

(Fatal) Table name %s is used in multiple %s tables.

**Description**

Table name must be unique in the table among all layers.

**What Next**

Correct the syntax of the *grdgenxo* command and resubmit the command.

## GRD-3046

(Fatal) %d of %d %s models are not ready after waiting for %d seconds, aborting run.

**Description**

grdgenxo needs results of some models to be available in order to continue with simulation of remaining models. The run aborts if results for prerequisite models are not found to be ready after waiting for an amount of time indicated in the error message.

Feedback

This will happen if machines generating prerequisite results are overloaded or generation of results is taking longer than expected for other reasons.

**What Next**

Launch grdgenxo one more time, keeping the files and directories from the aborting run intact.

---

## GRD-3047

(Fatal) Usage: grdgenxo -dp_config dp_config_file itf ...

**Description**

When using the -dp_config option to grdgenxo, the command-line syntax must be

```
grdgenxo -dp_config dp_config_file itf <... other options to
 grdgenxo ...>

That is, the first three arguments to grdgenxo must be the -dp_config
 keyword,
followed by the path to the DP config file, followed by the path to the
 ITF.
Any other command-line options should be specified after the ITF.
```

**What Next**

Change the order of your command-line options and run grdgenxo again.

---

## GRD-3048

(Fatal) %s is not an ITF file or it does not specify a TECHNOLOGY name

**Description**

The specified file is not an ITF file or it is an ITF file but does not specify a TECHNOLOGY name.

**What Next**

Run grdgenxo again specifying a valid ITF file in place of the file mentioned in the error message.

---

## GRD-3049

(Fatal) Another grdgenxo master process is executing in the current directory. Only one grdgenxo master process can be running in a directory at any given time.

Feedback

**Description**

grdgenxo prohibits running more than one process with the -dp_config option from a single directory at any given time to prevent data corruption.

**What Next**

Use a separate directory for each grdgenxo run.

---

## GRD-3050

(Fatal) Only one grdgenxo master process can be running in a directory at any given time. Process %d owned by user %s on host %s is currently running in this directory since %s.

**Description**

grdgenxo prohibits running more than one process with the -dp_config option from a single directory at any given time to prevent data corruption.

**What Next**

Use a separate directory for each grdgenxo run.

---

## GRD-3051

(Fatal) A grdgenxo DP run owned by %s with process ID %d on host %s is running in this directory since %s. Running more than one grdgenxo runs from the same directory is prohibited to prevent data corruption. If you believe this is an error, remove the .grdrun.info file from this directory and try again.

**Description**

grdgenxo prohibits running more than one grdgenxo run from the same directory to prevent corruption of on-disk data. This message will be seen when the user attempts to launch grdgenxo from a directory when another grdgenxo run with the -dp_config option is running in the same directory.

Multiple grdgenxo processes can be started in the same directory as long as there is no run with the -dp_config option launched from the same directory.

**What Next**

Use a separate directory for each grdgenxo run. If you believe this message is mistaken, remove the .grdrun.info file from the run directory and relaunch jobs. Do this only when you are absolutely sure than no other run is in progress.

Feedback

## GRD-3052

(Fatal) Invalid command-line option '%s' to grdgenxo. Command-line options to grdgenxo cannot contain the string '%s' as it is a reserved keyword.

**Description**

A command-line option specified to grdgenxo is invalid because it contained reserved keywords.

**What Next**

Remove the offending portion of the command-line option and try again. Reserved keywords have been chosen carefully to avoid conflicts in the most common usages of the the tool. Contact Synopsys technical support if removal of offending string is problematic.

## GRD-3053

(Fatal) Could NOT write command file for -sct_pattern_check.

**Description**

The grdgenxo tool cannot write the command file needed to check sct pattern.

**What Next**

Check the file system to see whether the disk is full or the connection is unstable. Resolve any issues and resubmit the grdgenxo run.

## GRD-3054

(Fatal) Via layer %s is associated with sidewall conformal dielectric %s and its FROM layer %s does not have an accepted layer type (gate, field_poly or pode_gate).

**Description**

A sidewall conformal dielectric (SW_T) can only be be defined for conductors or for vias above gate, field-poly, and pode and connecting to standard routing conductor layer.

**What Next**

Check ASSOCIATED_CONDUCTOR in the sidewall conformal dielectric and replace by a valid conductor or via layer. Check FROM definition in the VIA layer definition and ensure the specified conductor layer(s) have LAYER_TYPE defined as either GATE, FIELD_POLY or PODE_GATE. Contact Synopsys technical support.

Feedback

## GRD-3055

(Fatal) Via layer %s is associated with sidewall conformal dielectric %s and its TO layer %s has a special layer type (only regular conductors are valid).

### Description

A sidewall conformal dielectric (SW_T) can only be be defined for conductors or for vias above gate, field-poly, and pode and connecting to standard routing conductor layer.

### What Next

Check ASSOCIATED_CONDUCTOR in the sidewall conformal dielectric and replace by a valid conductor or via layer. Check TO definition in the VIA layer definition and ensure the specified conductor layer(s) does not have any LAYER_TYPE defined. Contact Synopsys technical support.

## GRD-3056

(Fatal) Sidewall conformal dielectric %s cannot be associated with non-rectangular via layer %s (only regular gate vias are valid).

### Description

A sidewall conformal dielectric (SW_T) can only be be defined for conductors or for rectangular gate vias. T-shape gate vias cannot have a sidewall conformal.

### What Next

Check ASSOCIATED_CONDUCTOR in the sidewall conformal dielectric and replace by a valid conductor or via layer. Remove TOP_EXTENSION_HEIGHT and TOP_EXTENSION_OFFSET in the via layer for it to be rectangular. Contact Synopsys technical support.

## GRD-3057

(Fatal) Conformal dielectric %s associated with via layer %s is not a sidewall conformal.

### Description

Only sidewall conformal dielectrics (SW_T) can be defined for gate vias. Other types of conformal dielectrics are not supported.

### What Next

Check in conformal dielectric definition SW_T>0 and TW_T and BW_T are either not defined or equal to 0. Contact Synopsys technical support.

Feedback

## GRD-3058

(Fatal) Conformal dielectric %s cannot be associated with via layer %s which has already the sidewall conformal %s (only 1 sidewall conformal allowed).

**Description**

Only 1 sidewall conformal dielectrics (SW_T) can be defined for gate vias.

**What Next**

Remove one or the other conformal associated with the gate via layer. Contact Synopsys technical support.

## GRD-3059

(Warning) The thickness of via layer %s between RVTV layer %s and conductor layer %s is greater than 0.001um

**Description**

Vias connecting RVTV conductor to frontside and backside layers are expected to be thin.

**What Next**

Check ITF to make sure the RVTV bottom or top layer heights are less than 0.001um in distance to conductor layers attached to it by via statements.

## GRD-3060

(Fatal) One or several 3D device modeling simulations failed. The 3D device modeling file was not written. Please see the %s file inside the grdgenxo working directory for a list of failed simulations.

**Description**

An error was encountered while running 3D device modeling simulations. This error can be due to disk space, gds2cap and quickcap setup issues, qtf layer aliases issue...

**What Next**

Please see the %s file inside the grdgenxo working directory for a list of failed simulations. Check the logs inside that directory to address the identified issue.

## GRD-3061

(Warning) Unknown Intel process type specified with tags %s: using the default process type for the specified technology.

Feedback

**Description**

The tool cannot recognize a known process type with tags %s. It will use the default process type settings for the technology specified.

**What Next**

Check if process type %s for is intentional.

---

## GRD-3062

(Fatal) Via layer %s connected to substrate is not allowed in RVTV technology

**Description**

In RVTV technologies, vias from/to substrate are not allowed.

**What Next**

Remove vias connected to substrate from ITF or fix FROM/TO statements. Fake welltaps or bulk ties might be defined in the runset and specified in the remove_layers section of StarRC mapping file.

---

## GRD-3500

(Fatal) Fatal error in Device Modeling flow. %s.

**Description**

Device Modeling flow is facing an error and cannot continue. Please fix the error based on the displayed error message.

**What Next**

Adjust the tool inputs based on the displayed error message. If you are running grdgenxo on linux platform and after fixing you still get this error, contact the Synopsys Support Center.

---

## GRD-3501

(Warning) %s.

**Description**

This a warning message from device model generation. Please revise the message and take action if unexpected.

Feedback

**What Next**

Adjust the tool inputs based on the displayed warning message. If the message is expected, ignore it.

---

## GRD-4000

(Fatal) The XMC file %s given by the -xmc option used in incremental run must be the same as the previously generated XMC file %s.

**Description**

The grdgenxo tool supports incremental runs for XMF flow using the -xmc option only if the given XMC file is the same as the one in the xmf directory.

**What Next**

Remove the xmf directory or the xmc file in that directory and re-run grdgenxo.

---

## GRD-4001

(Severe) INTERNAL Error: %s.

**Description**

An internal error appended during XMF generation

**What Next**

Please try to re-launch grdgenxo once again. Otherwise, contact the Synopsys Support Center.

---

## GRD-4002

(Severe) Usage: grdgenxo -nxtgrd_to_itf_wmin_smin_main -i <nxtgrd> -o <wmin&smin>.

**Description**

An error has occurred with the usage of the *grdgenxo* command. The error might be caused by improper command syntax or by an unsuitable input file.

The correct syntax is as follows:

```
% grdgenxo -nxtgrd_to_itf_wmin_smin_main -i <nxtgrd> -o <wmin&smin>
```

where *-i* is the short form of the *-input* option and *-o* is the short form of the *-output* option of the *grdgenxo* command.

Feedback

**What Next**

Correct the syntax of the *grdgenxo* command and resubmit the command.

---

## GRD-4003

(Fatal) The -dump_wmin_smin_from_nxtgrd option is not supported on a nxtgrd file with a bfs-encrypted header.

**Description**

You are running the grdgenxo -dump_wmin_smin_from_nxtgrd command. However, the nxtgrd file to be processed was generated by the grdgenxo -bfsEncrypt command. The grdgenxo tool cannot use the -dump_wmin_smin_from_nxtgrd option for a bfs-encryted nxtgrd file.

**What Next**

Contact the Synopsys Support Center.

---

## GRD-4004

(Fatal) The setting of the -xmf_linking option (%d), is different from the setting of the xmf linking of the XMC file (%d). Set the value of the -xmf_linking option to %d and re-run grdgenxo.

**Description**

Grdgenxo doesn't support different xmf linking settings. Set the value of the -xmf_linking option to the value in the XMC file.

**What Next**

Set the value of the -xmf_linking option to the value in the XMC file and re-run grdgenxo.

---

## GRD-4005

(Warning) When using -f2f_itf to merge one unencrypted die ITF with another encrypted die ITF, the OUT_ITF would still be encrypted.

**Description**

The warning indicates that users try to merge one encrypted ITF with another unecrypted one. Show the warning messages to notice users.

**What Next**

Still output the out_itf which is fully encrypted.

Feedback

## GRD-4006

(Fatal) SCALE_TO_SI_COVERAGE is specified when SAVC2 is not specified on line %d.

**Description**

In the VIA_COVERAGE table, SCALE_TO_SI_COVERAGE is only allowed when SAVC2 is specified.

**What Next**

Fix the VIA_COVERAGE table in the ITF file

## GRD-4007

(Fatal) The XMF flow is not supported with %s.

**Description**

The feature is not supported.

**What Next**

Contact the Synopsys Support Center.

## GRD-4010

(Fatal) An RPV table other than RPV_VS_AREA is spcified on the via layer %s which connects two conducting layers with LAYER_TYPE=STACKED_CONDUCTOR

**Description**

Only RPV, AREA and RPV_VS_AREA are supported on via layers which connect two conducting layers with LAYER_TYPE=STACKED_CONDUCTOR

**What Next**

Use supported syntax on the via layer indicated in this message.

## GRD-4011

(Fatal) The conducting layer %s with LAYER_TYPE=STACKED_CONDUCTOR does not connect with another same type conducting layer through a via.

**Description**

Conducting layers with LAYER_TYPE=STACKED_CONDUCTOR should appear in pairs and are connected with each other by a via layer.

Feedback

**What Next**

Update the ITF file with proper layer type settings.

---

## GRD-4012

(Fatal) LAYER_TYPE=STACKED_CONDUCTOR cannot be specified on %s which is not above all device layers.

**Description**

LAYER_TYPE=STACKED_CONDUCTOR can only be specified on Back-End-Of-Line conducting layers.

**What Next**

remove LAYER_TYPE=STACKED_CONDUCTOR from the indicated conducting layer block.

---

# GRF

---

## GRF-001

(Error) There are no arcs from pin '%s' to pin '%s' on cell '%s'.

**Description**

This error occurs when any arc does not exist between the two pins specified by the -from option and the -to option of set_disable_timing or remove_disable_timing commands.

**What Next**

Make sure the pins really exist on the cell or the library cell.

---

## GRF-002

(warning) Some timing arcs have been disabled for breaking timing loops\n \tor because of constant propagation.

**Description**

This message is displayed when timing arcs are disabled by tool to break combinational feedback loops or when propagating constant value due to case analysis. It is not displayed for arcs that are manually disabled with the *set_disable_timing* command.

Feedback

### What Next

If you want to manually break a timing loop, examine the design to see why there is combinational feedback, then choose a different point at which to break the loop. To do this, use the *set_disable_timing* command instead of letting the tool automatically break the loop.

---

## GRF-003

(warning) The master of the generated clock '%s' is not connected to any clock source.

### Description

There is no master clock at the -source pin of the generated clock. -source pin of a generated clock must be either the master clock source, or must be connected to a master clock source. If it is neither, the generated clock will have no period defined and be ignored for timing.

### What Next

Make sure that the master source from which the clock is generated is a clock source or is connected to a clock source.

---

## GRF-004

(warning) The master clock for generated clock '%s' is ambiguous.\n \tThere are multiple clocks on -source pin '%s'

### Description

The generated clock -source pin has more than one potential master clock on it. This tool will pick the potential master with the fewest levels of generated clocks from a primary master. If there is a tie, then the alphabetically first master clock will be chosen. Different tools in the flow may not pick the same master clock.

### What Next

Use the pin attribute "clocks" to see which master clock networks the -source pin is part of. Use the -master option to create_generated_clock to clearly define one master clock per generated clock.

---

## GRF-005

(warning) The generated clock '%s' has not been expanded,\n \tPlease create or activate its master clock.

Feedback

### Description

A generated clock will not expand if the master clock from which it is generated has not been created or activated. Also if the master clock was given with a -master_clock but does not reach the pin given with -source, this message will be given.

### What Next

Please create or activate the master of the generated clock or change the -source pin given. Use the *report_clock* command to see if the master clock is created or inactive.

### See Also

- create_clock
- report_clock

---

## GRF-006

(warning) The generated clock '%s' has non-unate master clock sense at the -source pin. The positive sense will be used.

### Description

A generated clock waveform is determined from the master clock sense at the -source pin. If the master clock is inverted at the -source pin, then the waveform for the generated clock is computed from the inverted master clock. For the given generated clock, both the inverting and non-inverting master clock waveforms reach the -source pin. The tool will use the non-inverting form of the master clock.

### What Next

Please change the -source pin to a location where the sense of the master clock is not ambiguous, or make other changes so one one sense of the master clock reaches the -source pin. The options to effect sense include using case analysis, set_disable_timing, or set_clock_sense.

### See Also

- create_generated_clock
- set_clock_sense
- report_clock

Feedback

## GRF-007

(warning) The generated clock '%s' with -duty_cycle is ignoring multiple edges of master clock '%s'.

### Description

The master clock of this generated clock has multiple rise edges in a single period. This generated clock uses the -duty_cycle option therefor the generated clock ignores all but the first rise and fall edges of the master clock.

### What Next

Verify the the correct master clock is used for this generated clock and that the generated clock waveform correctly models your circuit. If not change the clock definitions appropriately.

### See Also

- create_generated_clock
- report_clock

## GRF-008

(error) The requested clock sense '%s' at pin '%s' for clock '%s' does not exist.\n \tPropagating the '%s' sense of the clock through this pin.

### Description

There is a 'set_clock_sense' constraint at the pin given and the requested clock sense does not exist at the pin for the given clock. This may be due to a conflicting 'set_clock_sense' constraint on a previous pin or it may be because the only clock paths to that pin are not of the sense requested.

### What Next

Correct the 'set_clock_sense' constraints.

## GRF-009

(Warning) set_disable_timing ignores net timing arcs. Not disabling arc from '%s' to '%s'.

### Description

This warning occurs when a net timing arcs is given to set_disable_timing. To be consistent with other tools flow net arcs are not disabled by the command set_disable_timing.

Feedback

### What Next

Change your constraints to use set_disable_timing on the one of the pins of the net arc to get the desired result.

---

## GRF-010

(Warning) set_latch_loop_breaker does not have any effect on the '%s' pin because it is not a latch D pin.

### Description

This warning occurs when the pin specified in a *set_latch_loop_breaker* command is not the D pin of a latch. This inferrence occurs only during timing updates and not when the constraint is read. This constraint does not affect any latch loops.

### What Next

Change your constraints to use the correct pin in the *set_latch_loop_breaker* command to break latch loops as desired.

---

## GRF-011

(Error) Generated clock '%s' with source pin '%s' '%s' is not satisfiable; zero source latency will be used.

### Description

This is an error message whenever the clock network traversal can not find a path which satisfies the sense relationship defined by create_generated_clock command.

### What Next

Check for the generated clock definition to see if the generated clock is correctly defined. One example is that a divided_by 2 generated clock is driven by a inverter only. In this case, generated clock should be redefined with -invert with -divided_by 1. Another example is a divided_by 2 generated clock with preinverting. If master clock source pin is used as generated clock source pin, the warning message will be issued. In this case, generated clock source pin should be redefined to clock pin of divider.

---

## GRF-012

(Error) Cell '%s' is not bound and cannot be analyzed for timing.

Feedback

### Description

The reference to the cell was not resolved so the cell cannot be added to the timing graph for timing analysis.

### What Next

Resolve the reference first before running timing analysis.

---

## GRF-013

(Warning) Timing Arc from '%s' to '%s' has been disabled to break loops in network of clock '%s'

### Description

This message is displayed when timing arcs are disabled by tool to break loops in the source latency network of generated clocks which hop across sequential cells.

### What Next

If you want to manually break a timing loop, examine the design to see why there is a loop, then choose a different point at which to break the loop. To do this, use the *set_disable_timing* or *set_sense -stop* command instead of letting the tool automatically break the loop.

### See Also

- set_sense
- set_disable_timing

---

## GRF-014

(Error) Mode: %s, Generated clock '%s' has no path from master clock '%s'; zero source latency will be used.

### Description

This error occurs whenever the clock network traversal can not find a path from the master clock to the generated clock. The master clock did exists at the -source pin of the generated clock, but there is no path to the generated clock.

### What Next

Check the path between the master to the generated clock. Is there case values disabling the path? Is there unresolved library cell references blocking the path? Either fix the path or remove the generated clock.

Feedback

### GRF-015

(Warning) set_max_time_borrow 0 will treat '%s' pin latch loop breaker pin.

**Description**

This warning occurs when the constraint *set_max_time_borrow 0* is specified. The pin in constraint is treated as latch loop breaker pin. This inferrence occurs only during timing updates and not when the constraint is read. This constraint not affect any latch loops.

**What Next**

Change your constraints to use *set_latch_loop_breaker* constraint or timing_zero_max_time_borrow_latch_is_loop_breaker as false to break latch loops as desired.

## GRP

### GRP-1201

(warning) Cell %s is restricted from being ungrouped.

**Description**

Cell cannot be ungrouped due to no ungroup restrictions set on it. The no ungroup restriction is set either by set_ungroup, set_dont_touch, set_boundary_optimization, set_verification_priority or other constraints.

## GUI

### GUI-001

(Error) %s%s '%s' not found.

**Description**

The specified object does not exist.

### GUI-002

(Error) %s%s name is empty.

Feedback

**Description**

The name for the relevant object type [toolbar|menu|window type|hotkey,..] is empty.

---

## GUI-003

(Error) %sValid %s name cannot consist only of ampersands.

**Description**

Valid name for specified object type cannot consist only of ampersands. The object type can be a menu, toolbar, window type, hotkey, etc.

---

## GUI-004

(Error) %sTrailing '->' not allowed in menu name '%s'.

**Description**

Trailing '->' not allow in specified menu name.

---

## GUI-005

(Warning) %sHotkey '%s' is reserved. Please use another hotkey.

**Description**

Specified hotkey is reserved by application. Please use another hotkey.

---

## GUI-006

(Warning) %sHotkey '%s' is not supported.

**Description**

Specified hotkey is not supported. Please check spelling or use another hotkey. Shift modifier only work with letter or function hotkey. Thus, "Ctrl+Shift+4" is not supported.

---

## GUI-007

(Error) %sHotkey '%s' is invalid. Please check spelling.

**Description**

Specified hotkey is invalid. Please check spelling.

Feedback

## GUI-008

(Warning) %sMenu item '%s' already exist for %s '%s'.

**Description**

Specified menu item already exist for specified window type.

## GUI-009

(Warning) %sHotkey '%s' will not be set for menu '%s'. Hotkey is in use.

**Description**

Specified hotkey will not be set for specified menu. It is already in use by another menu item or tcl command.

## GUI-010

(Error) %sAnchor offset must be nonzero.

**Description**

Anchor offset value for specified command must be either a positive or negative integer.

## GUI-012

(Warning) %s: Specified dock side '%s' is invalid. Top docking side will be used.

**Description**

Specified dock side is invalid. Valid values include left|top|right|bottom.

## GUI-014

(Error) %s: menu item '%s' not found for window type '%s'.

**Description**

Specified menu item not found for specified window type or default window type if not specified.

## GUI-015

(Warning) %s: hotkey '%s' is already set for tcl command '%s'.

Feedback

**Description**

Specified hotkey is already set for an existing tcl command.

---

## GUI-016

(Warning) %s: hotkey not set, hotkey '%s' is already set for menu item '%s'.

**Description**

Specified hotkey has already been used by a menu item. Use -replace option to force hotkey to be set.

---

## GUI-017

(Warning) %s: hotkey '%s' is reserved by system function '%s'.

**Description**

Specified hotkey has already been used by an internal system function.

---

## GUI-018

(Error) None of the supported browsers %s are in the path

**Description**

This error message occurs when the application can not find a browser from the list of supported browsers in your environment.

**What Next**

Please correct the PATH to get one of the supported browsers. Please contact your system administrator to get the location of the browsers.

**See Also**

- gui_online_browser

---

## GUI-019

(Error) The default browser %s of your choice is not supported

**Description**

The current browser assigned to the variable gui_online_browser is not supported by your online help system.

Feedback

### What Next

Please assign a browser from the the following list %s to the variable gui_online_browser and try to bring up the helpsystem.

### See Also

- gui_online_browser

---

## GUI-020

(Info) No %s installation specified with variable %s

### Description

The application extension to ICC specified depends on the variable to point to the installation of that tool. The environment variable specified was not set so this extension to ICC will not be loaded.

### What Next

If you would like to load the ICC extension for that tool, then please setup the variable to point to the appropriate tool installation before running the ICC GUI.

---

## GUI-021

(Warning) Extension %s not loaded because extension loading is disabled

### Description

The loading of application extensions has been disabled. Therefore the specified extension will not be loaded.

### What Next

Re-enable extension loading if you want to have extensions loaded.

---

## GUI-022

(Error) %s installation directory specified in variable %s is not a directory. Extension can not be loaded from %s

### Description

The installation directory for the extension was not a directory, so the extension could not be loaded.

Feedback

### What Next

Correct the setting of the variable to specify the path to the installation of the tool and re-start the ICC GUI to get the extension loaded.

---

## GUI-023

(Warning) The %s installation specified does not contain an ICC extension setup file %s

### Description

The installation directory for the extension did not contain the expected ICC extension, so the extension could not be loaded.

### What Next

Correct the setting of the variable to specify the path to the installation of the tool and re-run ICC to get the extension loaded. If this version of the application does not support the ICC extension you may need to update to a newer version of this application.

---

## GUI-024

(Info) Loaded %s extension from %s

### Description

This message indicates that the ICC extension for the specified application was automatically loaded. The loading of the extension is controlled by the specification of the application installation for this application via a variable.

### What Next

There are no additional steps required to use this extension.

---

## GUI-025

(Error) Error sourcing %s extension setup file %s. %s : error_info is: %s

### Description

There was an error detected when loading the ICC extension for the specified application. Please contact the support staff for the application to get assistance in correcting the problem.

Feedback

**What Next**

Please ensure that the application installation specified is correct. If there is an error in the extension then please contact the support staff for that application to get a fix for their ICC extension.

---

## GUI-026

(Info) Visibility is turned %s for cells and cell contents because the task is set to %s

**Description**

The visibility options for cells and cell contents are turned OFF automatically when the task is set to Design Planning or turned ON automatically when the task is set to Block Implementation or All.

**What Next**

You can set the preference setVisibilityByTask to turn on or turn off this behavior. For example, to turn off this behavior, use the following command:

gui_set_pref_value -category layout -key setVisibilityByTask -value false

By default, this preference is set to true.

---

## GUI-027

(Error) All menu items may not be deleted from a window type menu

**Description**

All menu items may be deleted from a menu using the menu root name but not from a window type menu.

**What Next**

You can use the -root option to specify a menu root.

---

## GUI-028

(Error) A menu root named %s was not found.

**Description**

A menu root with the given name could not be found.

Feedback

## GUI-029

(Error) A menu root named %s was not found.

**Description**

A menu root with the given name could not be found.

## GUI-030

(Error) Specified view, %s, does not exist

**Description**

A view with the given name does not exist.

## GUI-031

(Error) Specified command, %s, does not exist

**Description**

The given name for the command option is invalid.

## GUI-032

(Error) Specified task, %s, does not exist

**Description**

The given name for the task option is invalid.

## GUI-033

(Error) Task item root: %s not found.

**What Next**

Check and correct the option value of the -item_root option.

## GUI-034

(Warning) Layers %s have no preferred routing direction.

Feedback

**Description**

Congestion data from those layers which have no preferred routing direction will not be displayed in the map.

**What Next**

If you would like to see the congestion data from those layers, then please setup the preferred routing direction on those layers.

---

## GUI-035

(Warning) Collection size limit reached

**Description**

The number of objects in the collection created by the gui command has exceeded the size limit.

**What Next**

Check the command is correct and if necessary increase the size limit.

---

## GUI-036

(Warning) not all objects could be represented as a collection

**Description**

Some of the selected or highlighted objects could not be returned as they are not representable in a collection.

**What Next**

Check the objects are correct and if necessary request gui collection support for the missing objects.

---

## GUI-037

(Error) Given color index is greater than max index: %d

**Description**

The given color index argument is too large.

**What Next**

Please ensure the the given index is no greater than the maximum color index allowed for the current or given Palette.

Feedback

## GUI-038

(Error) Given color name (%s) is not legal.

**Description**

The given color name is not a legal color name.

**What Next**

Please ensure the the given color name is a legal name for the current or given Palette.

## GUI-039

(Error) %s%s value '%s' is invalid.

**Description**

Specified filter value is invalid.

## GUI-040

(Warning) Anchor toolbar '%s' not found.

**Description**

Specified name for an anchor toolbar is invalid.

## GUI-041

(Warning) Anchor toolbar '%s' not found.

**Description**

Specified name for an anchor toolbar is invalid.

## GUI-042

(Error) Duplicate toolbar item: %s.

**What Next**

Check and correct the menu item name.

## GUI-043

(Error) The file %s could not be opened for write.

Feedback

**Description**

There was an error while opening the given file for writing.

**What Next**

Please check permissions for the file path.

---

## GUI-044

(Error) Specified task, %s, does not exist

**Description**

The given name for the task option is invalid.

---

## GUI-045

(Error) Specified file does not exist: %s.

**Description**

The given file could not be found or read.

---

## GUI-046

(Error) No tip files found in the given directory %s.

**Description**

The given directory could not be found or read, or there were no files found with ".html" file extension in the given directory.

---

## GUI-047

(Error) The given file %s already exists.

**Description**

The given output file already exists. Please move or delete the existing file.

---

## GUI-048

(Error) There was an error wrhile writing to %s.

Feedback

**Description**

There was an error while writing to the given output file and not all of the document could be written.

---

## GUI-049

(Error) Preset '%s' not found for category '%s'.

**Description**

Specified preset not found for specified category.

---

## GUI-050

(Error) Performance monitor could not be found.

**Description**

Performance monitor program executable files could not be found in the Synopsys tool installation.

---

## GUI-051

(Warning) 3D View requires X Server to have %s extension.

**Description**

3D View uses an OpenGL renderer which requires your X Server to have the GLX and RENDER extensions. Typically this is caused by using a vncserver which is old or not configured to have the GLX or RENDER extensions.

**What Next**

You can open a shell window in your X Server and run the command xdpyinfo. The list of extensions is listed near the top. You will need to see the following: GLX RENDER

---

## GUI-052

(Warning) 3D View requires a valid DISPLAY environment variable.

**Description**

3D View uses an OpenGL renderer which requires a connection to a valid X Server via the DISPLAY environment variable.

Feedback

### What Next

You can open a shell window in your X Server and run a simple X application like xclock. You can open a shell window in your X Server and run the command xdpyinfo. The list of extensions is listed near the top. You will need to see the following: GLX RENDER

---

## GUI-053

(Warning) 3D View requires a network connection to a X Server.

### Description

3D View uses an OpenGL renderer which requires a connection to a valid X Server via the DISPLAY environment variable.

### What Next

You can open a shell window in your X Server and run a simple X application like xclock. You can open a shell window in your X Server and run the command xdpyinfo. The list of extensions is listed near the top. You will need to see the following: GLX RENDER

---

## GUI-054

(Warning) 3D View Could not find a compatible X Visual in the X Server. The X Server needs to support 32 bit planes.

### Description

3D View requires the X Server to support 32 bit planes. OpenGL requests 24 bit planes PLUS 8 bits of Alpha. If the output of xdpyinfo for the X Server in question does not have at least one entry contianing "32 planes", then the X Server is not supported. That entry must also state "TrueColor

### What Next

If the output of xdpyinfo doesn't contain all these lines, then it is incompatible with the OpenGL based 3D view. GLX RENDER 32 planes

Here is an example of an X Visual output by xdpyinfo that is compatible with the OpenGL based 3D view:

default visual id: 0x21 visual: visual id: 0x21 class: TrueColor depth: 24 planes available colormap entries: 256 per subfield red, green, blue masks: 0xff0000, 0xff00, 0xff significant bits in color specification: 8 bits

Feedback

The output must contain at least one other entry that can support 32 planes:

visual: visual id: 0x47 class: TrueColor depth: 32 planes available colormap entries: 256 per subfield red, green, blue masks: 0xff0000, 0xff00, 0xff significant bits in color specification: 8 bits

---

## GUI-055

(Info) 3D View will not run with the -batch option (offscreen mode).

**Description**

3D View uses an OpenGL renderer which currently does not work in offscreen mode (-batch option).

**What Next**

Remove the -batch option when running fc_shell or icc2_shell

---

## GUI-056

(Warning) 3D View is missing OpenGL support library: libxcb-glx.so. Check your platform for QSC compliance.

**Description**

3D View uses an OpenGL renderer which is only supported on platforms compatible with CentOS 7 or greater and Suse 12 or greater.

**What Next**

Verify that you are running fc_shell or icc2_shell on a QSC-Q platform.

---

## GUI-057

(Warning) 3D View requires the Synopsys version of libGL.so.

**Description**

3D View uses an OpenGL renderer which is built by Synopsys specifically for this product. This error indicates the application is loading a version of libGL.so that wasn't supplied by Synopsys.

**What Next**

Make sure you are using the version of libGL.so supplied by Synopsys in the product installation. Make sure you don't have a LD_LIBRARY_PATH setting that finds your OS-installed libGL.so

Feedback

## GUI-058

(Warning) Given color has been matched to index: %d.

**Description**

The given color has been matched to index in a current palette.

**What Next**

Please ensure the the given color name is a legal name for the current or given Palette.

## GUI-059

(Warning) View "%s" is being disabled. See previous "3D View" warnings.

**Description**

3D Views are disabled when the runtime environemnt won't support a particular X Server, or an installation environemnt for OpenGL libraries isn't set up properly according to QSC-S

**What Next**

Read the "3D View" warnings prior to the message and take the suggested actions.

## GUI-060

(Warning) 3D View requires server glx version 1.4.

**Description**

3D View uses VKT9 which requires a glx version of 1.4.

**What Next**

You can open a shell window in your X Server and run glxinfo (mesa utilities) to determine current version.

## GUI-061

(Warning) The total number of visible errors exceed the set maximum for displayed errors.

**Description**

The maximum errors that are displayed in the layout are set using a preference key and can be changed using the same.

Feedback

**What Next**

You can use the command "gui_set_pref_value -category {Globals} -key {gui_error_browser_show_all_max} -value {val}" where val is the number of errors to be displayed in the layout and reload the database. However, increasing this value above the recommended limit (200000) might result in the application displaying slow downs depending on the available computational resources.

---

## GUI-062

(Warning) The application can display slow downs since the gui_error_browser_show_all_max value is greater than the recommended value (%d).

**Description**

Since a large number of errors are now being loaded into the error browser, the application can display a slow down depending upon the computational resources available.

**What Next**

Reset the value to the default value.

---

## GUI-063

(Warning) Since the desired value of the preference is out of bounds (minimum %s, maximum %s), it is being reset to the nearest bound.

**Description**

If the desired value is less than the minimum value of the preference, then the value is set equal to the minimum value. On the contrary, if it is greater than the maximum value of the preference, then the value is set equal to the maximum value.

---

## GUI-064

(Warning) Initialization of Tk can prevent use of other application features (See man page for details)

**Description**

If Tk is initialized then the GUI can no longer support close and reopen of display as Tk does not allow this feature.

That means user cannot call 'gui_stop -close' (to close gui) and then 'gui_start' (to restart gui) for example when switching from offscreen display to normal X11 display.

Feedback

It also means that save_checkpoint command cannot be used, as that command can only be run when GUI is closed and restore of the closed checkpoint, with the saved Tk state would fail.

## GUI-998

(Warning) %s

**Description**

A warning was issued for a non-fatal error when running the command.

**What Next**

If the error indicates a problem then fix it and re-run the command. If the error is benign then ignore the error.

## GUI-999

(Error) %s

**Description**

An error was found when running the command as desribed in the error message.

**What Next**

Fix the error and re-run the command.

# HBO

## HBO-3001

(Information) Reports boundary optimization settings on modules, instances and pins only for object specific controls and not during global control (compile.flow.boundary_optimization).

**Description**

This command reports the boundary_optimization restriction on objects in object_list. If the object_list is not specified, the command reports boundary optimization restrictions on the objects in the current design. The command reports the information on the objects which are restricted by set_boundary_optimization command and not during global control (compile.flow.boundary_optimization)

Feedback

# HCVR

## HCVR-001

(error) Could not open file %s for module %s.

**Description**

This error may occur during *expand_outline*. The Verilog file containing the module definition could not be opened. The *expand_outline* cannot expand that module without the original Verilog.

**What Next**

Locate the missing Verilog file and return it to its original location.

## HCVR-002

(error) File %s has been modified since it was read.

**Description**

This error may occur during *expand_outline*. If a source Verilog or DEF file has changed since the outline design was read with *read_verilog_outline* or *read_def*. The location of the modules within the file may have changed, so they cannot be found by *expand_outline*.

**What Next**

Restore the file to its original time stamp, or if you're sure the contents have not changed, try *expand_outline* with the -force option.

## HCVR-003

(error) File %s has changed size since it was read.

**Description**

This error may occur during *expand_outline*. The Verilog or DEF file containing the module definition has changed size since the outline design was created with *read_verilog_outline* or *read_def*. The location of the modules within the file may have changed, so they cannot be found by *expand_outline*.

**What Next**

Restore the file to its original size and time stamp, or if you're sure the contents have not changed, try *expand_outline* with the -force option.

Feedback

## HCVR-004

(warning) Files have changed since design was created. Attempting to re-read. Errors are likely.

### Description

This error may occur during *expand_outline*. The Verilog or DEF file containing the module definition has changed since the outline design was created with *read_verilog_outline* or *read_def*. By re-reading with the -force option, the *expand_outline* command is ignoring these changes. It is likely that the modules have moved within the file, so the re-read of those modules will fail.

### What Next

Restore the file to its original size and time stamp, or re-create the outline design from scratch with *read_verilog_outline* and *read_def*.

## HCVR-005

(error) Can't find module %s, file %s ended before line %u.

### Description

This error may occur during *expand_outline*. End of file was encountered while copying the module from the file. If *expand_outline* was run with the -force option, the fact that the file was truncated may have been ignored.

### What Next

Restore the file to its original size and time stamp, or re-create the outline design from scratch with *read_verilog_outline*.

## HCVR-006

(error) %s is an uncommitted partition.

### Description

This error may occur during *expand_outline*. The outline design contains uncommitted partitions. Expanding this design might create a very large design that might run out of memory.

### What Next

Commit the uncommitted partitions with the *commit_cell* command. Or ignore the partitions and use the -force option of *expand_outline*.

Feedback

## HCVR-007

(error) %s is not an outline design.

### Description

This error may occur during *expand_outline*. The current design is not an outline design created by *read_verilog_outline* or from an outline design with the *commit_cell* command. So it cannot be expanded.

### What Next

Change current_design to an outline design before expanding. Or if the design is already expanded, go to the next step of your design flow.

## HCVR-008

(warning) No leaf lib cells specified from reference library.

### Description

During set up for the *read_verilog_outline* command the reference libraries are scanned to identify macro cells, leaf cells and buffer cells. After this scan is complete, no leaf were identified. Without valid leaf cells, the capacity of *read_verilog_outline* will be very limited.

This error may occur during *expand_outline*. The current design is not an outline design created by *read_verilog_outline* or from an outline design with the *commit_cell* command. So it cannot be expanded.

### What Next

Check to see if the current library's reference library has been set with the *set_ref_libs* command. This setting is used to identify the libraries to scan.

If additional cell references are to be treated as leaf cells or buffers, the *-buffer_cells* and *-leaf_cells* options to *read_verilog_outline* may be used to modify the values found in the libraries. The *[-macro_cells* variable may be used to cause specified leaf or buffer cells to appear in the design along side the macros.

### See Also

- read_verilog_outline
- set_ref_libs

## HCVR-009

(error) %s does not contain outline modules.

Feedback

**Description**

This error may occur during *expand_outline*. The current design is empty or not an outline design created by *read_verilog_outline* or from an outline design with the *commit_cell* command. So it cannot be expanded.

**What Next**

Change current_design to an outline design before expanding. Or if the design is already expanded, go to the next step of your design flow.

---

## HCVR-010

(error) Expanded design %s already exists.

**Description**

This error may occur during *expand_outline*. A modified version of the expanded design already exists in memory. The *expand_outline command will not overwrite the existing design.*

**What Next**

Close the existing design with the *close_block* -force command. Don't forget to specify the .design suffix when closing the design. Then re-expand the outline design with *expand_outline*.

---

## HCVR-011

(error) Could not open DEF file %s.

**Description**

This error may occur during *expand_outline*. The DEF file used originally during read_def could not be opened. The *expand_outline* cannot expand DEF information without the original DEF.

**What Next**

Locate the missing DEF file and return it to its original location.

---

## HCVR-012

(warning) -%s_modules module '%s' not found in Verilog.

Feedback

### Description

This module was named in the -dense_modules, -port_modules, or -sparse_modules option to the *read_verilog_outline* command, but the named module was not found in the Verilog source. Because the module was not found, the -dense_modules, -port_modules, or -sparse_modules directive could not be applied to that module while reading the Verilog to create the outline view of the block.

The number of 'Module not found in Verilog' warnings is limited by the *plan.outline.message_limit* app option. Set this value to -1 to see all module warnings.

### What Next

Check to see if the module names passed to the -dense_modules, -port_modules, or -sparse_modules option were spelled correctly.

Verify that the named modules are in the Verilog source file. Verify that the named modules have not been commented out.

### See Also

- read_verilog_outline
- set_app_options

---

## HCVR-013

(warning) %d named -%s_modules not found in Verilog.

### Description

This module was named in the -dense_modules, -port_modules, or -sparse_modules option to the *read_verilog_outline* command, but the named module was not found in the Verilog source. Because the module was not found, the -dense_modules, -port_modules, or -sparse_modules directive could not be applied to that module while reading the Verilog to create the outline view of the block.

The number of 'Module not found in Verilog' warnings is limited by the *plan.outline.message_limit* app option. Set this value to -1 to see all module warnings.

Since the number of warnings was limited, this message reports the total number of modules not found in Verilog for this option.

### What Next

To see more of the modules named, but not found in Verilog, set the *plan.outline.message_limit* to a larger value:

```
prompt> set_app_options -global { plan.outline.message_limit 40 }
```

Feedback

To see all of the modules named, but not found in Verilog, set the *plan.outline.message_limit* to -1:

```
prompt> set_app_options -global { plan.outline.message_limit -1 }
```

Check to see if the module names passed to the -dense_modules, -port_modules, or -sparse_modules option were spelled correctly.

Verify that the named modules are in the Verilog source file. Verify that the named modules have not been commented out.

### See Also

- read_verilog_outline
- set_app_options

---

## HCVR-014

(error) Failed to create temporary file '%s'.

### Description

The tool could not create the temporary file used to expand the outline Verilog modules. Without a temporary file to hold the Verilog modules to be expanded, the *expand_outline* command fails.

This is most likely caused by a file permission problem, but there are a number of other reasons why the tool would not be able to create the temporary file: the path name might be invalid; there might be a file with the same name that could not be overwritten; you might not have permission to write to that directory; you might have exceeded your disk quota on that file system; the file system might be read-only; or the file system might be full.

### What Next

Verify that you can create that file in that directory and that text can be written to that file. Verify that there is sufficient free space on the disk.

To see the size of a file system, use the UNIX *df* command:

```
unix% df /tmp/.
```

Change the *shell.common.tmp_dir_path* application option setting to use a different path or file system to store store the temporary file. To change the temporary directory to /SCRATCH, use the *set_app_options* command:

```
prompt> set_app_options -list { shell.common.tmp_dir_path /SCRATCH }
```

Feedback

### See Also

- expand_outline
- set_app_options
- shell.common.tmp_dir_path

---

## HCVR-015

(error) Failed to save temporary file '%s'.

### Description

Failed to save the data in the temporary file used to expand the outline Verilog modules. Without a temporary file to hold the Verilog modules to be expanded, the *expand_outline* command fails.

Most likely the file system was full, but there are a number of other reasons why the data could not be saved to the temporary file.

The path name may be invalid. There may be a file with the same name that could not be over-written. The user may not have permission to write to that directory. The user may have exceeded their space quota on that file system. The file system may be read-only. Or the file system may not have enough available space to hold the contents of the temporary Verilog file.

### What Next

Verify that the user is able to create that file in that directory. Verify that text can be written to that file. Verify that there is sufficient free space on the disk.

To see the size of a file system, use the unix *df* command:

```
unix% df /tmp/.
```

Change the *shell.common.tmp_dir_path* app option setting to use a different path or file system to store store the temporary file. To change the temporary directory to /SCRATCH, use the *set_app_options* command:

```
prompt> set_app_options -list { shell.common.tmp_dir_path /SCRATCH }
```

### See Also

- expand_outline
- set_app_options
- shell.common.tmp_dir_path

Feedback

# HDL

## HDL-120

(Info) Operator associated with resources '%s' in design '%s' breaks the datapath extraction because %s.

**Description**

The datapath extraction flow identifies the node that breaks the datapath extraction to be extracted due to leakage.

Datapath leakage is when an internal operand is not wide enough to store the result of an operation; but the full result is required later. Operands with leakage must be binary and must be at the boundary of a DP block. Therefore, operands with leakage will break a potentially larger DP block into smaller DP blocks; resulting in suboptimal QoR.

Example of leakage due to truncation: module test( a, b, c, d, e, f, o1, o2, o3 ); input [10:0] a, b; input [5:0] c, d, e, f; output [8:0] o1; output [9:0] o2; output [7:0] o3; assign o1 = a + b + c; (add_9, add_9_2) assign o2 = f + o1; (add_10) assign o3 = e + o1; (add_11) endmodule

There is leakage due to truncation on the fanin of add_10.

If the leakage is caused by the driver/load sign mismatch, you may fix the RTL design to avoid mix signage on the driver and load. Here are two examples of leakage caused by driver/load sign mismatch: Example 1: module leakage_a (a, b, c, z);

input signed [7:0] a, b; input [7:0] c; output [9:0] z; wire signed [8:0] t;

assign t = a + b; // leakage: signed result used in wider unsigned expression assign z = t + c;

endmodule

Example 2: module leakage_b (a, b, c, z);

input [7:0] a, b; input signed [7:0] c; output signed [10:0] z; wire signed [8:0] t;

assign t = a + b; // leakage: unsigned result used in wider signed expression assign z = t + c;

endmodule

**What Next**

Refer to the 'Coding Guidelines for Datapath Synthesis' for information about how to improve RTL coding for datapath extraction. The coding guidelines can be found in SolvNet (https://solvnet.synopsys.com/retrieve/015771.html).

Feedback

## HDL-121

(Info) There is sequential cell between operator associated with '%s' and '%s' in design '%s'.

### Description

The sequential cells between DesignWare operators will result in these operators not being extracted into the same datapath block.

For datapath pipelining, you can place the pipeline registers at the input or output of the RTL datapath code and use retiming to move them to optimal locations.

Example: module test (clk, en, a, b, c, z); input clk, en; input [7:0] a, b; input [15:0] c; output [15:0] z; reg [15:0] prod;

always @ (posedge clk) begin if (en) begin prod <= a * b; end end assign z = prod + c; endmodule

There is a register between the multiplier and the adder. The following message will be issued:

Information: There is sequential cell in between operator associated with 'mult_11' and 'add_15' in design 'test'. (HDL-121)

The extracted DP block cannot contain sequential cell (for example: registers). The register between operators will break a potential larger DP block into smaller DP blocks; resulting in suboptimal QoR.

### What Next

You can modify your RTL design to place the pipeline register at the input or output of the RTL datapath code.

## HDL-122

(Info) Internal saturation on output signal '%s' of '%s' in design '%s' breaks the datapath extraction.

### Description

The comparator or selector associated with the saturation operation breaks the datapath extraction. There are other DesignWare operator followed by this saturation operation. It is suggested to perform saturation at the end of the datapath block. e.g. In the following design, the internal saturation breaks the datapath extraction: assign t = a*b+c; assign t2 = (t > 9'b111111111)? 9'b111111111 : t[8:0]; assign o = t2 + d; Message: Information: Internal saturation associates with 'add_11' in design 'test' breaks the datapath extraction. (HDL-122)

Feedback

It is suggested to move the saturation to the end: e.g. assign t = a*b+c+d; assign o = (t > 9'b111111111)? 9'b111111111 : t[9:0];

Another example of internal saturation: assign t = a*b+c; assign t2 = t[10]? 10'h3FF : t[9:0]; assign o = t2 + d; Message: Information: Internal saturation associates with 'add_11' in design 'test' breaks the datapath extraction. (HDL-122)

**What Next**

Try to move the saturation to the end of the datapath block. Refer to the 'Coding Guidelines for Datapath Synthesis' for information about how to improve RTL coding for datapath extraction. The coding guidelines can be found in SolvNet (https://solvnet.synopsys.com/retrieve/015771.html).

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## HDL-125

(Info) Cell '%s' in design '%s' cannot be extracted because it instantiates '%s', which could be inferred with operator '%s' instead.

**Description**

The cell cannot be extracted because it is an instantiated DW component. Only inferred DW operators can be extracted into datapath blocks. You can improve the HDL coding to infer the operator.

Example: module test(a,b,c,z); parameter wl = 5; input [wl-1:0] a,b,c; output [2*wl-1:0] z;

wire [wl-1:0] sum; assign sum = (b+a); DW02_mult #(wl, wl)U_mult(sum, c, 1'b0, z); endmodule

The instantiated DW02_mult can be replaced by an inferred multiplier.

---

## HDL-126

(Info) Missing possible extraction across cell '%s' in design '%s' and cell '%s' in design '%s'.

**Description**

The cells cannot be extracted into the same datapath block because they are in different design hierarchies.

Example: module top(a,b,c,d,e,o); input [8:0] a,b,c, d, e; output [32:0] o;

wire [19:0] t1; wire [17:0] t2;

assign t2 = a*b; bot UU1 (t2, c, d, t1); assign o = t1 * e;

endmodule

Feedback

module bot(a, b, c, o); input [17:0] a; input [8:0] b, c; output [19:0] o; assign o = a + b + c; endmodule

If auto-ungroup is disabled, the hierarchical boundary of design 'bot' will break the datapath extraction. If 'bot' is ungrouped, the operators in 'top' and 'bot' will be extracted into one datapath block.

Currently, small design hierarchy will be automatically ungrouped. If you see this message, auto-ungroup is either disabled or not happend on this deisgn. You may try to remove the hierarchical boundaries that block the extraction to see if there is a better chance of datapath extracion.

If one of the design mentioned in the message is inside the other design mentioned in the message, you can ungroup the sub-design to see if extraction could be improved. If the 2 designs mentioned in this message are in the same level of hierarchy, you need to ungroup both designs to see if extraction could be improved.

Note: We do not check leakage for possible extraction across hierarchy boundary. It's possible that even when the design is ungrouped, the operators can not be extracted into one datapath block due to leakage.

**What Next**

Try to remove the hierarchical boundaries that block the extraction, see if extraction could be improved.

---

## HDL-128

(Info) Constraints set on instance '%s' may be ignored.

**Description**

The timing constraint implied size_only restriction is set on unmapped DesignWare operator. The constraints may be ignored by the datapath optimization optimization. The constraint on the unmapped DesignWare operator may be lost.

**What Next**

Please try not to set timing constraint on unmapped DesignWare operator.

---

## HDL-129

(Info) Found gtech adder chain '%s' in design '%s'.

Feedback

**Description**

There is instantiated gtech adder chain of 'GTECH_ADD_AB' or 'GTECH_ADD_ABC' in the specified design. To improve QoR of the design, the gtech adder chain should be replaced by inferred DesignWare adder operator.

**What Next**

Try to use inferred DesignWare adder operator.

---

## HDL-132

(Info) The output of subtractor associated with resources '%s' is treated as signed signal.

**Description**

The output of a subtractor is treated as signed when sign extension happens on the unsigned subtractor operator.

Example of leakage due to sign mismatch, where the sign mismatch is caused by sign extension on unsiged subtractor: module test(a, b, c, z); input [7:0] a, b, c; output [17:0] z; wire [15:0] t; assign t = 0 - a; assign z = t + b; endmodule

To fix the leakage due to sign mismatch that is caused by sign extension on the output of unsigned subtractor, you can modify the design not to have extension on the subtractor output or to used signed operator for the subtractor and the driven operator.

**What Next**

Refer to the 'Coding Guidelines for Datapath Synthesis' for information about how to improve RTL coding for datapath extraction. The coding guidelines can be found in SolvNet (https://solvnet.synopsys.com/retrieve/015771.html).

---

## HDL-133

(Info) Gtech adder cell '%s' could potentially break the datapath extraction in design '%s'.

**Description**

There is instantiated gtech adder cell 'GTECH_ADD_AB' or 'GTECH_ADD_ABC' in the specified design. The gtech adder cells are driving and driven by DesignWare operator. This gtech adder cell can not be extracted into datapath block. It may help datapath extraction if the gtech adder cells are changed to inferred DesignWare adder operator,

**What Next**

Try to use inferred DesignWare adder operator.

Feedback

## HDL-168

(warning) The datapath block '%s' in design '%s' contains high fanout operator. Formality runtime might be long.

**Description**

The specified DP OP block contains operator that drives large number of fanouts. If the operator is extracted into datapath block. The Formality runtime might be long.

**What Next**

You could use set_verification_priority with '-low' option to prevent long formality runtime. \n

**Examples**

The following example set verification_priority for Formality runtime on design 'test':

prompt> *set_verification_priority -low test*

# HFS

## HFS-001

(error) -from is not specified or no valid driver pin is specified.

**Description**

The error is shown when -from is not specified or when no valid buffer tree driver is specified.

**What Next**

Specify the right pin names with -from.

## HFS-002

(warning) %s %s has been specified as %s. ignoring...

**Description**

The pin or port is not specified as expectation.

**What Next**

Please correct the specification.

Feedback

### HFS-003

(Error) Command option '%s' is not valid when app option '%s' is set to '%s'.

**Description**

The given command option is not compatible with the value of the app option. As a consequence, the command option is not supported.

**What Next**

Revisit the setting of the app option.

### HFS-004

(Error) Command option '%s' is not valid in combination whith option '%s'.

**Description**

The two command options cannot be used together.

**What Next**

Revisit the command options.

## HOPT

### HOPT-001

(Information) %s (Reference: %s) is treated as %s

**Description**

This message indicates whether sub-blocks/top-level design are modifiable or non-modifiable, along with reason, during Toplevel implementation flow.

### HOPT-003

(Information) %d sub-block instances are treated as non-modifiable.

**Description**

This message indicates the number of sub-blocks which are non-modifiable, during Toplevel implementation flow. For sub-blocks which are modifiable, tool issues HOPT-001 message for each block.

Feedback

## HOPT-005

(Warning) Setting planning as false for block '%s' since at level(s): %s it is already set as false.

### Description

This message indicates that for the concerned block there are multiple instantiations at different levels and at some level it is set to false for planning.

## HOPT-007

(Error) Toplevel sanity check has failed. Run check_hier_design -stage pre_placement command for detailed errors.

### Description

This message indicates that the hier checks have failed at top level. The issues may cause the top level flow to be incorrect or fail at some point. It is recommended to correct the reported errors before proceeding with the top level flow. Run check_hier_design -stage pre_placement command to see detailed errors.

### What Next

Run check_hier_design -stage pre_placement command to see detailed errors. If app-option top_level.continue_flow_on_check_hier_design_errors is set to true, command will continue even if top level design checks fail and HOPT-008 error message will be given.

### See Also

- check_hier_design
- top_level.continue_flow_on_check_hier_design_errors

## HOPT-008

(Error) Toplevel sanity check has failed but the command will continue. Run check_hier_design -stage pre_placement command for details.

### Description

This message is issued because the hier checks have failed at top level and the app-option top_level.continue_flow_on_check_hier_design_errors is set to true.
The issues may cause the top level flow to be incorrect or fail at some point. It is recommended to correct the reported errors before proceeding with the top level flow. Run check_hier_design -stage pre_placement command to see detailed errors.

Feedback

### What Next

Run check_hier_design -stage pre_placement command to see detailed errors.

### See Also

- check_hier_design
- top_level.continue_flow_on_check_hier_design_errors

---

## HOPT-010

(Warning) App-option "top_level.optimize_subblocks" is set to "none". All sub-blocks will be treated as non-modifiable

### Description

This message indicates that sub-block optimization has been disabled because of the setting of app-option "top_level.optimize_subblocks". Please set it to design in order to enable optimization inside sub-blocks.

---

## HOPT-011

(Error) App-option "top_level.optimize_subblocks" is set to "none". All sub-blocks will be treated as non-modifiable

### Description

This message indicates that sub-block optimization has been disabled because of the setting of app-option "top_level.optimize_subblocks". Flow will continue with disabling sub-blocks for optimization. Please set it to design in order to enable optimization inside sub-blocks.

---

## HOPT-012

(Warning) Found %d sub-block %s linked to design view and with optimization enabled; %s will be skipped

### Description

This message indicates that sub-blocks with optimization enabled but linked to design views, are being skipped for TIO specific operations.

Feedback

## HOPT-013

Warning: Found MIB block '%s' (%ld instances) with optimization enabled. It will be skipped

**Description**

This message indicates that MIB sub-blocks with optimization enabled, are being skipped for TIO specific operations.

## HOPT-014

(Error) There are no abstract instances with optimization enabled. Command will not proceed

**Description**

This message indicates that there are no sub-blocks with optimization enabled and linked to abstract views.

## HOPT-015

(Warning) There are no abstract instances with optimization enabled.

**Description**

This message indicates that there are no sub-blocks linked to abstract views that can be optimized. It may happen due to insufficient UNIX permissions also.

**What Next**

Check if required abstracts are enabled for optimization. Also check UNIX permissions of the design directories.

## HOPT-016

(Error) There are no sub-blocks with optimization enabled. Command will not proceed

**Description**

This message indicates that there are no sub-blocks with optimization enabled. This command will perform no operations.

Feedback

## HOPT-020

(Warning) Sub-blocks are found to be editable when not expected. Change set_editability to false for all sub-blocks before running this command.

### Description

This message is issued when editability setting for sub-block(s) is found to be true but current command expect all sub-blocks to be read-only. Change set_editability to false for all sub-blocks before running this command.

### What Next

Use report_editability command to check editability settings for all sub-blocks. Use set_editability command to update the editability of sub-blocks. For example:

prompt> *set_editability -blocks [get_blocks -hierarchical] -value false*

### See Also

- set_editability
- get_editability
- report_editability

## HOPT-021

(Warning) The default values of 'set_hierarchy_options' for %s '%s' in the context block '%s' are converted to 'set_editability' value of '%s'.

### Description

This warning message indicates that needed set_editability settings are missing for the listed blocks. Tool needs editability settings to understand the blocks that are expected to be modified by various applications and those that should be treated as non-editable. When set_editability settings are missing, link_block functionality triggers auto-conversion to infer user intent. The auto-conversion looks at user applied set_hierarchy_options settings. If user has not applied set_hierarchy_options tool will take default values of 'set_hierarchy_options' to maintain backward functional compatibility. The auto-conversion is tool's best effort to capture user intent. Warning message indicates user to review the tool derived settings and make needed changes using set_editability command. This auto-conversion of 'set_hierarchy_options' values into 'set_editability' is a temporary feature to enable easy transition. The auto-conversion feature will be deprecated in a future release.

Feedback

### What Next

Review the messages to make sure editability on listed blocks is as expected. Use 'report_editability' command to review editability settings and 'set_editability' command to make any changes to editability settings.

### See Also

- set_editability
- get_editability
- report_editability
- HOPT-025

---

## HOPT-022

(Warning) The 'set_hierarchy_options %s' usage is being deprecated; Use 'set_editability' command to control list of editable blocks.

### Description

This warning message comes for script backward compatibility, when user is specifying 'set_hierarchy_options' with argument '-enable_planning'.

### What Next

Review the messages to make sure editability on listed blocks is as expected. Use ‘report_editability’ command to review editability settings and 'set_editability' command to make any changes to editability settings.

### See Also

- set_editability
- get_editability
- report_editability

---

## HOPT-023

(Warning) Both 'set_editability' and 'set_hierarchy_options' settings present in the database for block '%s'. Ignoring the 'set_hierarchy_options' settings.

Feedback

### Description

This warning message comes when the database has both 'set_editability' setting and 'set_hierarchy_options' setting. Tool will take 'set_editability' setting; ignoring 'set_hierarchy_option' setting.

### What Next

Review the messages to make sure editability on listed blocks is as expected. Use ‘report_editability’ command to review editability settings and 'set_editability' command to make any changes to editability settings.

### See Also

- set_editability
- get_editability
- report_editability

---

## HOPT-024

(Warning) The 'report_hierarchy_options' usage is being deprecated for reporting hierarchical edit controls for the blocks. Use 'report_editability' command instead.

### Description

This warning message comes for script backward compatibility, when user is specifying 'report_hierarchy_options'.

### What Next

Use 'report_editability' command to report the hierarchical edit control for the blocks.

### See Also

- set_editability
- get_editability
- report_editability

---

## HOPT-025

(Warning) User specified 'set_hierarchy_options' settings for %s '%s' in the context block '%s' are converted to 'set_editability' value of '%s'.

Feedback

### Description

This warning message indicates that needed set_editability settings are missing for the listed blocks. Tool needs editability settings to understand the blocks that are expected to be modified by various applications and those that should be treated as non-editable. When set_editability settings are missing, link_block functionality triggers auto-conversion to infer user intent. The auto-conversion looks at user applied set_hierarchy_options settings. The auto-conversion is tool's best effort to capture user intent. Warning message indicates user to review the tool derived settings and make needed changes using set_editability command. This auto-conversion of 'set_hierarchy_options' values into 'set_editability' is a temporary feature to enable easy transition. The auto-conversion feature will be deprecated in a future release.

### What Next

Review the messages to make sure editability on listed blocks is as expected. Use 'report_editability' command to review editability settings and 'set_editability' command to make any changes to editability settings.

### See Also

- set_editability
- get_editability
- report_editability
- HOPT-021

---

## HOPT-030

(Error) Initialization for THO flow is already performed.

### Description

This message indicates that initialization as required by the THO flow is already performed through init_hier_optimization command. Subsequent calls to init_hier_optimization without commit is not allowed.

---

## HOPT-031

(Error) Initialization for THO flow is not performed. Nothing to commit.

### Description

This message indicates that commit_hier_optimization is called without prior init_hier_optimization. It is expected to use init_hier_optimization to initialize the necessary

Feedback

settings, perform necessary operations and then commit back the changes using commit_hier_optimization.

# IND

## IND-001

(error) ICV version does not support the ndm version of current ICC2 tool. ICV supported version is ('%d'.'%d') , while ICC2 supported version is('%d'.'%d'). Exiting...

**Description**

The specified operation failed because ICV version doesn't support the ICC2 ndm schema.

**What Next**

Please update to the compatible ICV version.

## IND-002

(error) IC Validator executable cannot be found.

**Description**

IC Validator executable cannot be found.

**What Next**

Please check that environment variable ICV_HOME_DIR is set and path to 'icv' executable is set.

## IND-003

(error) ICV run failed (%s).

**Description**

ICV run failed. Some of the possible reasons could be:

User fixable: - No ICV license. - User specified runset error. - Layer map file parse error. - Missign DESIGN/FRAME view error.

Internal errors: - Internal template runset error. - Command line error. - NDM error. - Fatal error.

Feedback

**What Next**

- No ICV license :=> Check that you have the valid ICV license. - User specified runset error :=> This could be a parse error in user specified foundry runset, or a include file missing. Please fix it. - Layer map file parse error :=> The layer map file must support valid GDS-stream IN/OUT syntax. - Missign DESIGN/FRAME view error :=> Make sure that are reference libraries are linked and all required views for child cells exist.

Internal Errors :=> Report the problem to Synopsys.

---

## IND-004

(error) Failed in saving design: %s.

**Description**

Design could not be saved to disk. This could be due to limited disk space or due to no write permissions on the currently opened design.

**What Next**

Check that sufficient disk space is available during the run.

---

## IND-005

(error) Current design/lib is not defined.

**Description**

No library/block is defined.

**What Next**

Open a valid block before calling this command.

---

## IND-006

(error) Design sanity error (%s).

**Description**

The error denotes that design is missing basic data items/attributes which are required for the command to work correctly. The data items could be related components like block, layer, tech file, net, etc.

**What Next**

Please define the required data items/attributes for appropriate component.

Feedback

## IND-007

(error) Fill does not exist in the design.

**Description**

This error denotes that you are doing post proessing on Fill which must exist in the design to begin with. The application is not able to locate fill cell inthe design.

**What Next**

Perhaps there is fill in the design to begin with or the fill cell name is not being recognized by the application. The top fill cell name should be "design.FILL". Also look for presence of file "design.FILL.design.ndm" in the block directory.

## IND-008

(error) ICV distributed processsing error (%s).

**Description**

An error occured while launching jobs through specified distibuted processing method.

**What Next**

Please check that specified resources are available.

## IND-009

(error) Fill does not exist.

**Description**

This error denotes that a particular internal stage in the run is expecting FILL, but FILL could not have been generated by earlier stage(s).

**What Next**

If this error is coming after ICV invocation, it is possible that ICV run failed. Please check error id IND-003 for possible causes of ICV error or report the issue to Synopsys.

## IND-010

(error) Invalid input error (%s).

Feedback

**Description**

An invalid option combination, or invalid value specified for command line options or application option values used by the application.

**What Next**

Please check man page for information on correct option value format and supported option combinations.

---

## IND-011

(error) Layer: %s, specified in option/app_option: %s is not a valid tech file layer name or a valid metal/via alias name.

**Description**

The layer specified must be a valid tech file layer. In some cases, layer alias names like m1, m2, m3 for mask layers metal1, metal2, metal3 can also be used in place of layer names.

**What Next**

Make sure that either the tech file layer name or alias name is used.

---

## IND-012

(error) No matching net found for user specified net/net pattern: "%s", for option: "%s".

**Description**

No matching net found for net name/ net regular expression pattern.

**What Next**

Provide a valid net net name/ net name pattern.

---

## IND-013

(error) No matching cell found for user specified cell name/ cell name pattern: "%s", for option: "%s".

**Description**

No matching cell found for cell name/ cell name pattern.

**What Next**

Provide a valid cell name/ cell name pattern.

Feedback

## IND-014

(error) No matching rules found for specified pattern: "%s" for option: "%s".

**Description**

No matching rules found for specified pattern.

**What Next**

check

## IND-015

(error) Failed in parsing layer map file: %s.

**Description**

Failed in parsing layer map file provided by user. The layer map file must be in GDS/oasis stream in format.

**What Next**

Please check guidelines on writing the layer map file.

## IND-016

(error) Incremental flow has not been initialized.

**Description**

The incremental DRC/FILL flow has not been initialized. Make sure that you have run fullchip signoff_check_drc/signoff_create_metal_fill atleast once before using the incremental (auto_eco) mode. If you have run signoff_check_drc without FILL, then it is necessary to re-run it with FILL, s.t initialization can be done for FILL data as well.

**What Next**

In some cases, incremental flow initialization data could be missing if you have re-created ndm block from LEF/GDS, or if the incremental flow was completely disabled during last signoff_check_drc/signoff_create_metal_fill call. To re-initialize the incremental flow, run signoff_check_drc/signoff_create_metal_fill by setting following environment variable: setenv "signoff_indesign_autoeco_flow_initialize" 1; signoff_check_drc; signoff_create_metal_fill; unsetenv "signoff_indesign_autoeco_flow_initialize"; Note that above operation will only do the initialization and exit. It will not run any DRC/FILL.

Feedback

## IND-017

(error) Fill runset not found.

**Description**

The error denotes that either the foundry fill runset has not been specified, or the runset file path is not valid.

**What Next**

The runset file should be set using following app options: signoff.create_metal_fill.runset

## IND-018

(error) DRC runset not found.

**Description**

The error denotes that either the foundry DRC runset has not been specified, or the runset file path is not valid.

**What Next**

The runset file should be set using following app options: signoff.check_drc.runset

## IND-019

(error) Missing required attribute: %s for layer(s): %s.

**Description**

This error denotes that a required layer info/attribute is missing.

**What Next**

Please initialize the missing required layer info/attribute.

## IND-020

(error) Missing required tech file attribute (%s).

**Description**

This error denotes that a required info/attribute for tech file data is missing.

**What Next**

Make sure that tech file has all the required components.

Feedback

## IND-021

(error) NDM object write error (%s).

**Description**

There was an internal error in creating/modifying a ndm object. Typically this problem should not happen.

**What Next**

Make sure that the design was in consistent state before the command was run. If the problem still persists, please report the problem to Synopsys.

## IND-022

(error) The user specified %s file: %s could not be read.

**Description**

The file path given by user could not be read.

**What Next**

Make sure that the file path is correct and the file has read permissions.

## IND-023

(error) Unable to write to user specified file: %s.

**Description**

Unable to write to user specified file.

**What Next**

Make sure that output file path is correct and you have write permissions and enough disk space for the file to be written.

## IND-024

(error) Internal file read/write error. File: %s.

**Description**

Failed in creating/reading an internal temporary file.

Feedback

**What Next**

This is an internal application error. Please report the problem to Synopsys.

---

## IND-025

(error) Fill could not be generated.

**Description**

No fill got generated.

**What Next**

Please check previous errors to trace the cause of fill not getting generated.

---

## IND-026

(error) DRC error data could not be generated.

**Description**

Failed in generating DRC error data. This could be either an application error, or the ICV-DRC run failed.

**What Next**

Please check if ICV-DRC run completed successfully. If not please check the ICV run log for reason of failure. If the ICV run was successful, please report the problem to Synopsys.

---

## IND-027

(error) NDM object read error (%s).

**Description**

Failed in reading data from a ndm object. This could possibly be due to ndm objects not being initialzed properly.

**What Next**

Please check if the data items are accessible using tcl APIs and if similar error is reported. You can also check if other applications are reporting similar errors. If above does not help, please report the problem to Synopsys.

---

## IND-028

(error) Incremental FILL/DRC flow initialization error.

Feedback

**Description**

There was error in initializing data required for incremental flow. Subsequent auto_eco runs may not work correctly. The current run, as well as any subsequent non auto_eco run will not be affected by this.

**What Next**

Please check the previous errors causing the initializing to fail.

---

## IND-029

(error) Internal application error : %s.

**Description**

This denotes an internal application error.

**What Next**

Please report the problem to Synopsys.

---

## IND-030

(error) %s Command Failed.

**Description**

The command failed. For more details, please review the corresponding log files and messages above.

**What Next**

Please review the corresponding log files and messages above.

---

## IND-101

(warning) The option/app_option: '%s' will be deprecated in future.

**Description**

This warning denotes that the command line option/ app_option will be deprecated in future.

**What Next**

It is likely that the option has been changed to new option. Please check command man page.

Feedback

## IND-102

(warning) The option/app_option: '%s' will be ignored (%s).

**Description**

This warning denotes that the command line option/ app_option will be ignored.

**What Next**

The option can be ignored if enabling of related option/app_option results in similar flow. If this is not intended, then check if the value of other options/app_options is appropriate as per your requirement. A user setting can also be ignored if the design does not have objects/object attributes that will either be affected, or will trigger a flow change, by the user setting.

## IND-103

(warning) The value of app_option: %s will be ignored because option(s): %s are not specified.

**Description**

This warning denotes that the app_option setting will be ignored because the setting makes sense only when a particular command line option is specified.

**What Next**

If this is not what you expected, please check man page for correct usage.

## IND-104

(warning) Unhandled/unidentified object(s) in design (%s). These will be ignored.

**Description**

Unhandled/unidentified object(s) in design which will be ignored.

**What Next**

If the application result is not affected by unhandled/unrecognized objects, then please ignore this warning. If the application result is affected by unhandled/unrecognized objects, then please report the issue to Synopsys.

## IND-105

(warning) An error occured in generating: %s.

Feedback

**Description**

An application error occured in generating a secondary file. This is mostly a summary/ reporting file. This does not affect the primary output.

**What Next**

Although primary output is unaffected, it is good to report the issue to Synopsys, so that this issue can be rectified in future.

---

## IND-106

(warning) The option/app_option: '%s' will be ignored (%s).

**Description**

The opion/app_option will be ignored since it is either not relevant to current flow, or it is not supported yet.

**What Next**

Enhancement may be needed to support the functionality.

---

## IND-107

(warning) No matching cells for cell name/pattern: '%s', specified for option/app_option: '%s'.

**Description**

There were no matching cells found for cell name/pattern specified by user.

**What Next**

Please check if the cell name/pattern is correct and the design has the relevant cells.

---

## IND-108

(warning) Skipping port '%s' in block '%s'. The antenna properties cannot be obtained.

**Description**

There were no properties attached to the port in the given block.

**What Next**

Please check if the port is not an internal port, non device bias type and non device layer.

Feedback

# INFSB

## INFSB-001

(Information) Infeasible path found from %s to %s in scenario %s.

**Description**

An Infeasible path is detected between the mentioned start and endpoint. Detected timing paths from specified startpoint to the endpoint will marked with path margins.

# INFT

## INFT-001

(warning) Technology for the current library will be overridden by the Technology created by the read_package_tech command.

**Description**

Technology already exists for the current library. It will be overridden by the Technology created by the read_package_tech command.

**What Next**

Ensure that the technology for the current library is as expected.

## INFT-002

(error) InFO-Tech Parser Error: '%s'.

**Description**

This error message indicates that there is some unexpected error in the InFO-Tech file.

**What Next**

Ensure that the file contents are correct.

## INFT-003

(warning) InFO-Tech Parser Warning: '%s'.

Feedback

**Description**

This warning message indicates that there is some unexpected but recoverable issue in the InFO-Tech file.

**What Next**

Ensure that the file contents are correct.

---

## INFT-004

(error) InFO-Tech Reader layer derivation error: '%s'.

**Description**

This error message indicates that there is some unexpected error while deriving InFO-Tech layer stack.

**What Next**

Ensure that the file contents are correct.

---

## INFT-005

(warning) TSV cut_layer name specified in -tsv option is not found in the layer stack.

**Description**

TSV cut_layer name specified in -tsv option is not found in name attribute of <LAYER> tag in the file. Please note that the name is case-sensitive.

**What Next**

Ensure that the cut layer name specified in read_info_tech -tsv option is correct.

---

## INFT-006

(error) Expected one or two file_names as input for the read_info_tech command.

**Description**

The read_package_tech command expects a single InFO-Tech file path or two file paths in the format "frontside_file_path backside_file_path".

**What Next**

Ensure that the read_package_tech command is according to the expected syntax.

Feedback

## INFT-007

(error) Cannot merge the given frontside and backside InFO-Tech files - '%s'.

**Description**

The read_package_tech command expects a common layer in frontside and backside InFO-Tech files for correct merge.

**What Next**

Check the InFO-Tech file and ensure that it is correct.

## INFT-008

(information) InFO-Tech Parser Information: '%s'.

**Description**

Information from the InFO-Tech parser.

**What Next**

Verify whether the information suits user needs.

## INFT-009

(error) %s named '%s' already exists.

**Description**

The bump cell reference library name provided in the -ref_lib_name option conflicts with existing library name.

**What Next**

Please provide a unique name for the library or use the -overwrite_bump_ref_lib option to overwrite the existing library.

## INFT-010

(warning) GDS Layer Number %u cannot be honored as layer_number for the layer %s.

**Description**

GDS Layer Number of the given layer number cannot be used as its layer_number because multiple layers have the same GDS Layer Number in the input Custom Compiler

Feedback

Layer Mapping file, specified by -input_cc_layer_mapping command option. The tool will auto-assign layer_number attribute to this layer.

**What Next**

Ensure that the input Custom Compiler layer mapping file is correct.

# ITF

## ITF-001

(error) No technology found for the current library.

**Description**

No Technology specified for the current library.

**What Next**

Ensure the existence of technology associated with current library.

## ITF-002

(warning) The following layers are not present in layer map file and will be dropped:%s

**Description**

The layer map file does not contain the indicated layers. They will not be considered for output stackup XML file.

**What Next**

Ensure the existence of layer in layer map file if it is not meant to be ignored in stackup XML output.

## ITF-003

(warning) The following layers are not present in current Technology:%s

**Description**

The current technology does not contain the indicated layers. The NDM tech layer name mentioned in the layer map file will be used in output XML file.

Feedback

**What Next**

Ensure the existence of layer in current technology. Check if the NDM layer name is not misspelled in the layer map file.

---

## ITF-004

(warning) ITF Layer name '%s' provided in Layer map file does not exist in ITF.

**Description**

No layer is present in the ITF file corresponding to the indicated name in layer map file.

**What Next**

Ensure that the layer map file contains the correct ITF layer name.

---

## ITF-005

(error) ITF File does not contain any dielectric layer. Layer elevation calculation is not possible.

**Description**

No dielectric layer is present in the layer map file. The elevation calculation assigns elevation 0 to bottommost dielectric layer.

**What Next**

Ensure that the input ITF file contains atleast one dielectric layer.

---

## ITF-006

(warning) The %s layer '%s' has no electrical properties defined in ITF.

**Description**

The input file does not contain the resistivity or permittivity information for the indicated layer.

**What Next**

Ensure that the layer indicated is assigned resitivity or permittivity value in the input ITF file.

---

## ITF-007

(warning) Using Conductivity of %s '%s' of '%s' as its conductivity.

Feedback

**Description**

The input file does not contain the resistivity information for the indicated via layer. In such a case, the conductivity of its start layer or stop layer is used as the conductivity of the via.

**What Next**

Ensure that the via layer indicated is assigned resitivity value in the input ITF file.

---

## ITF-008

(warning) Conductivity of '%s' were not available from its Start or Stop layers.

**Description**

The input ITF contains neither the resistivity information for the indicated via layer nor its start or stop layers.

**What Next**

Ensure that the via layer indicated is assigned resitivity value in the input ITF file.

---

## ITF-009

(warning) No color found for layer '%s' in Current Technology. Assigning the default color.

**Description**

Current Technology does not contain the layer color information for the indicated layer.

**What Next**

Ensure that the current technology has layer color defined for it.

---

## ITF-010

(warning) ITF layer map parser warning: %s

**Description**

The layer map file does not follow the expected syntax.

**What Next**

Ensure the layer map has correct syntax as described in read_itf -layer_map man page.s

Feedback

# LBDB

## LBDB-1

(error) The '%s' function requires %d arguments.

### Description

This message indicates that the number of arguments passed to the specified function or the complex attribute is wrong. The Library Compiler expects a specific number of arguments for various functions and complex attributes. For example, the define_cell_area and the capacitive_load_unit require two arguments. A common cause of this error is the argument's syntax definition. For example, the pin_equal complex attribute requires one quoted string as an argument instead of multiple strings

```
pin_equal("Q", "XQ");   wrong
        pin_equal("Q" "XQ");    wrong
        pin_equal("Q XQ");      correct
```

The following example shows an instance where this message occurs:

```
/* Try the wrong number of arguments to these functions. */
  define_cell_area(my_area,pad_slots,extra_junk);
  capacitive_load_unit(1, pf, pf);
```

The following is an example message:

```
Error: Line 23, The 'define_cell_area' function requires 2 arguments.
 (LBDB-1)
```

### What Next

Change the technology library to correct the number of arguments. Refer to the "Library Compiler Reference Manual" for the syntax description.

## LBDB-2

(error) The value of argument %d of the '%s' function\n \tis of the wrong type. A value of '%s' type is required.

### Description

This message indicates that the wrong value type has been set to the specified function argument. A common cause of this error is the alteration of argument order. For example, the capacitive_load_unit attribute expects a floating-point value then a unit of string type.

```
capacitive_load_unit( pf, 1);    wrong
  capacitive_load_unit( 1, pf);    correct
```

Feedback

The following example shows an instance where this message occurs:

```
capacitive_load_unit( pf, 1);
```

In this case, the arguments of the capacitive_load_unit need to be switched.

The following is an example message:

```
Error: Line 17, The value of argument 1 of the 'capacitive_load_unit'
 function
        is of the wrong type. A value of 'floating-point' type is
 required. (LBDB-2)
```

**What Next**

Change the technology library to correct the arguments. Refer to the "Library Compiler Reference Manual" for the syntax description.

---

## LBDB-3

(error) The '%s' value is invalid for the '%s' attribute.\n \tIts valid values are %s.

**Description**

This message indicates that the specified string value is not included in the list of accepted values. This error is often caused by a typo in the valid enumerated string value.

The following example shows an instance where this message occurs:

```
default_wire_load_mode : enclose;
```

In this case, the value should be 'enclosed'.

The following is an example message:

```
Error: Line 15, The 'enclose' value is invalid for the
 'default_wire_load_mode'
attribute.
        Its valid values are top, segmented and enclosed. (LBDB-3)
```

**What Next**

Change the technology library to correct the invalid value.

---

## LBDB-4

(error) The %s '%s' cannot be specified during\n \tthe update_lib command.

Feedback

### Description

Only library-level group statements can be added to an existing library using the update_lib command.

This message indicates that an invalid data (function, attribute, or group) has been found in the library to be added.

The following example shows an instance where this message occurs: Existing library

```
default_wire_load_mode      : top;
```

You cannot specify the default_wire_load_mode attribute to be added to an existing library.

The following is an example message:

```
Error: Line 8, The group 'default_wire_load_mode' cannot be specified
 during
        the update_lib command. (LBDB-4)
```

### What Next

Check the "Library Compiler User Guide" for valid data that can be added to existing libraries. Change the library to be updated appropriately.

---

## LBDB-5

(error) The library already has a '%s' attribute. It cannot be\n \toverwritten during the update_lib command.

### Description

Only library-level group statements can be added to an existing library using the update_lib command.

This message indicates that an attribute has been found in the library to be added.

### What Next

Remove the existing attribute from the technology file.

---

## LBDB-6

(error) The library already has a '%s' group; it cannot be\n \toverwritten during the update_lib command because it is permanent.

### Description

This message indicates that an existing group is being overwritten in the library; the existing group is permanent. You cannot overwrite existing groups because their

Feedback

definitions might already have affected subsequent group declarations in the library. You can, however, add new ones.

The following example shows an instance where this message occurs:

```
wire_load("05x05") {
        resistance : 0 ;
        capacitance : 1 ;
        area : 0 ;
        slope : 0.186 ;
        fanout_length(1,0.39) ;
    }
```

The following is an example message:

```
Error: Line 1, The library already has a '05x05' group; it cannot be
       overwritten during the update_lib command because it is permanent.
 (LBDB-6)
```

**What Next**

Change the library file to either remove the group or modify its name.

---

## LBDB-7

(information) A total of %d repeated messages are omitted.

**Description**

Repeated messages on buses, bundles, pin "collections" have been removed, only those corresponding to the "first" pins are shown. For the following examples:

bus(a) { bus_type : bit7to0; ... }

bundle (b) { member (x, y, z) ... }

pin (c_4 c_3 c_2 c_1) { /* pin "collections" */ ... }

messages will be shown for pins a[7], x, and c_4. This is shown

Line 330, Cell 'fd1', pin 'a[7]', The attribute 'max_transition' is not specified. (LBDB-605)

but this is omitted:

Line 330, Cell 'fd1', pin 'a[6]', The attribute 'max_transition' is not specified. (LBDB-605)

Note that if you use command suppress_message to suppress a certain warning message, that message is not generated at all, and so is not counted here. Lastly, some repeated messages may still show up, but non-repeating messages are never blocked.

Feedback

### What Next

To show all repeated messages, set the following variable before read_lib:

set lc_show_repeated_messages TRUE read_lib ...

---

## LBDB-8

(information) The %s '%s' group has been successfully %s\n \tin the library.

### Description

The *update_lib* command updates library groups in a given library. This message is for information purposes only and concerns cell and operating_conditions groups. However, if a wire-load model group is updated in a library, designs using that wire-load model must be reassigned a wire-load model.

The following is an example message:

```
Information: Line 1, The operating_conditions 'BCMIL' group has been
 successfully added
             in the library. (LBDB-8)
Information: Line 8, The cell 'lbdb8' group has been successfully added
             in the library. (LBDB-8)
```

### What Next

Use *set_wire_load_model* to set the wire-load model on designs using the modified wire-load models.

---

## LBDB-9

(error) The '%s' function requires %s arguments.

### Description

This message indicates that the number of arguments passed to the specified function or the complex attribute is wrong. The Library Compiler expects a specific number of arguments for various functions and complex attributes. For example, the define_cell_area and the capacitive_load_unit require two arguments. A common cause of this error is the argument's syntax definition. For example, the pin_equal complex attribute requires one quoted string as an argument instead of multiple strings

```
pin_equal("Q", "XQ");   wrong
        pin_equal("Q" "XQ");    wrong
        pin_equal("Q XQ");      correct
```

Feedback

The following example shows an instance where this message occurs:

```
/* Try the wrong number of arguments to these functions. */
  define_cell_area(my_area,pad_slots,extra_junk);
  capacitive_load_unit(1, pf, pf);
```

The following is an example message:

```
Error: Line 23, The 'define_cell_area' function requires 2 arguments.
 (LBDB-9)
```

**What Next**

Change the technology library to correct the number of arguments. Refer to the "Library Compiler Reference Manual" for the syntax description.

---

## LBDB-10

(error) Invalid character '%s' is detected in %s name.

**Description**

This message indicates that you specified an invalid name for the object. For cell, the name cannot contain '/' as it is a reserved character for hierarchy.

**What Next**

Change the library source file by correcting the name.

---

## LBDB-11

(warning) The same '%s' attribute is defined twice and is ignored the second time.

**Description**

This message indicates that an attribute has been registered twice. The Library Compiler ignores the second value.

**What Next**

Make sure that the attribute is defined only once.

---

## LBDB-12

(warning) The '%s' attribute is the wrong type for the '%s' object.

Feedback

### Description

This message indicates that an attribute's value had a wrong type. The Library Compiler ignores the value.

### What Next

Make sure that the attribute's value is of the correct type.

---

## LBDB-13

(warning) The '%s' enum has been defined twice and is being ignored.

### Description

This message indicates that an enumeration value registered with an attribute has been defined twice.

### What Next

Make sure that the enumeration value is defined only once.

---

## LBDB-14

(error) The library already has a '%s' group; the library\n \tcannot be overwritten unless the -overwrite option is specified.

### Description

This message indicates that a group has already been added by the *update_lib* command, and you are trying to overwrite it using the *update_lib* command again without specifying the -overwrite option.

The following is an example message:

```
Error: Line 2, The library already has a 'lbdb14' group; the library
        cannot be overwritten unless the -overwrite option is specified.
 (LBDB-14)
```

### What Next

Either specify the -overwrite option to the update_lib command if you want to overwrite the same group or change the name of the group.

---

## LBDB-15

(warning) an illegal value of '%s' %f is found, it cannot be %s than \ '%s' value '%s %f'.

Feedback

### Description

```
This message indicates that the values of capacitance models/attributes
 is not
correct. Note that when you encounter error LBDB-706 as well, then this
 screener
 is one cause for LBDB-706.

the capacitance values should meet the following rule:

Cmiller_rise = miller_cap_rise value
Cmiller_fall = miller_cap_fall value

pin_capacitance = rise_capacitance or fall_capacitance or if missing, use
 pin capacitance

1. Cmiller_rise/fall <= pin_capacitance;
```

The following example shows an instance where this message occurs:

```
pin(I)  {
    direction : input;
    rise_capacitance : 0.00263567;
    ...
    ccsn_first_stage () {
      miller_cap_rise : 0.0477054;
      ...
    }
    ...
  }
```

The following is an example message:

```
Warning: Line 364, Cell 'BUFFD0', pin 'I', an illegal value of
 'miller_cap_rise' 0.047705 is found, it cannot be more than
'capacitance' value '  0.002636'. (LBDB-15)
```

### What Next

Change the value of the attributes to meet the screener rules.

---

## LBDB-16

(warning) Found a duplicate %s attribute. Using the latest value.

### Description

This message indicates an attribute has been defined twice. The first definition is ignored.

Feedback

The following example shows an instance where this message occurs:

```
pin(Q) {
      direction : output;
      function : "1";
      function : "1";
    }
```

The following is an example message:

```
Warning: Line 59, Found a duplicate function attribute. Using the latest
 value. (LBDB-16)
```

### What Next

Remove the duplicate definition of the attribute.

---

## LBDB-17

(error) The library already has a type named '%s'. Type \n \tgroups can never be overwritten.

### Description

This message indicates that a type group already exists in the original library, and you are trying to overwrite it using the *update_lib* command again.

The following example shows an instance where this message occurs:

```
type(bus2) {
    base_type : array;
    data_type : bit;
    bit_width : 2;
    bit_from : 0;
    bit_to : 1;
    downto : false;
  }
```

The type named bus2 is defined in the library to be updated and the file used to update.

The following is an example message:

```
Error: Line 1, The Library already has a type named 'bus2'. Type
       groups can never be overwritten. (LBDB-17)
```

### What Next

Either remove the type group from the added library or change the name of the group in the file used for update.

Feedback

## LBDB-18

(warning) In the '%s' library, the environment attribute \n \t%s's value is out of range.

### Description

This message indicates that the nominal source voltage value defined in the library using nom_voltage attribute is less than 0. The library compiler ignores the value and takes the default nominal voltage. For a CMOS library, the default nominal voltage value is 5 volts.

The following example shows an instance where this message occurs:

```
nom_voltage : -5;
```

The following is an example message:

```
Warning: Line 29, In the library 'lbdb18', the environment attribute
        nom_voltage's value is out of range. (LBDB-18)
```

### What Next

Change the value of the nom_voltage attribute to be greater than zero.

## LBDB-19

(warning) Can't find a pin named '%s' in the '%s' cell.

### Description

This message indicates that one of the pin names in the list of the *pin_equal* or *pin_opposite* attribute or *gate_leakage* group is not defined in the cell. This often happens when there is a typo in the pin name.

The following example shows an instance where this message occurs:

```
cell(lbdb19) {
    area : 13;
    pin_opposite("Q","XQ");
    pin(Q, YQ) {
        direction : output;
        function : "1";
    }
  }

cell(lbdb19_1) {
    area : 13;
    pin(Q, YQ) {
        direction : input;
        function : "1";
    }
    leakage_current() {
```

Feedback

```
        gate_leakage(XQ) {
          input_high_value : 7.1;
          input_low_value : -8.7;
        }
    }
  }
```

The following is an example message:

```
Warning: Line 28, Can't find a pin named 'XQ' in the 'lbdb19' cell.
 (LBDB-19)
Warning: Line 218, Can't find a pin named 'XQ' in the 'lbdb19_1' cell.
 (LBDB-19)
```

### What Next

Check to see if there is a typo in the pin name. Otherwise, declare the pin in the technology library.

---

## LBDB-20

(warning) Cannot process 'pin_opposite' for the '%s' cell.

### Description

This message indicates that the pins used in the list of the *pin_opposite* attribute are not found in the cell.

The following example shows an instance where this message occurs:

```
cell(lbdb20) {
    area : 13;
    pin_opposite("Q","XQ");
    pin(Q, YQ) {
        direction : output;
        function : "1";
    }
    pin(YQ) {
        direction : output;
        function : "0";
    }
  }
```

In this case, the pins defined in the cell are 'Q' and 'YQ'. However, the pins listed in the pin_opposite attribute are 'Q' and 'XQ'.

The following is an example message:

```
Warning: Line 28, Can't process 'pin_opposite' for the 'lbdb20' cell.
 (LBDB-20)
```

Feedback

### What Next

Fix the pin names in the list of the pin_opposite attribute.

---

## LBDB-23

(error) There is a missing timing arc between pins '%s' and\n \t'%s' in the '%s' cell.

### Description

This message indicates there is a missing timing arc from an input or inout pin to an output pin.

For a combinational cell, the Library Compiler checks that

```
* An output port with a function statement has timing arcs to
          all functionally related inputs
        * An output port with a three_state attribute has timing arcs
          to all three_state related inputs
        * If all timing arcs are conditional, a default timing arc
          without any condition is required.
```

The following example shows an instance where this message occurs:

```
cell(lbdb23) {
    area : 1;
    pin(A) {
      direction : input;
      capacitance : 1;
    }
    pin(B) {
      direction : input;
      capacitance : 1;
    }
    pin(Z) {
      direction : output;
      function : "(A B)'";
      timing() {
        intrinsic_rise : 1.0;
        intrinsic_fall : 1.0;
        rise_resistance : 0.1;
        fall_resistance : 0.1;
        related_pin : "A";
      }
    }
  }
```

In this case, a timing arc is missing between the pin 'Z' and 'B'. To fix the problem, add the following timing group:

```
timing() {
        intrinsic_rise : 1.0;
```

Feedback

```
        intrinsic_fall : 1.0;
        rise_resistance : 0.1;
        fall_resistance : 0.1;
        related_pin : "B";
      }

The following is an example message:
Error: Line 12, There is a missing timing arc between pins 'A' and
        'Z' in the 'lbdb23' cell. (LBDB-23)
```

**What Next**

Add the missing timing group between the two pins.

---

## LBDB-23w

(warning) There is a missing timing arc between pins '%s' and\n \t'%s' in the '%s' cell.

**Description**

This message indicates there is a missing timing arc from an input or inout pin to an output pin.

For a combinational cell, the Library Compiler checks that

```
* An output port with a function statement has timing arcs to
          all functionally related inputs
        * An output port with a three_state attribute has timing arcs
          to all three_state related inputs
        * If all timing arcs are conditional, a default timing arc
          without any condition is required.
```

The following example shows an instance where this message occurs:

```
cell(lbdb23) {
    area : 1;
    pin(A) {
      direction : input;
      capacitance : 1;
    }
    pin(B) {
      direction : input;
      capacitance : 1;
    }
    pin(Z) {
      direction : output;
      function : "(A B)'";
      timing() {
        intrinsic_rise : 1.0;
        intrinsic_fall : 1.0;
        rise_resistance : 0.1;
        fall_resistance : 0.1;
```

Feedback

```
        related_pin : "A";
      }
    }
  }
```

In this case, a timing arc is missing between the pin 'Z' and 'B'. To fix the problem, add the following timing group:

```
timing() {
        intrinsic_rise : 1.0;
        intrinsic_fall : 1.0;
        rise_resistance : 0.1;
        fall_resistance : 0.1;
        related_pin : "B";
      }

The following is an example message:
Warning: Line 12, There is a missing timing arc between pins 'A' and
        'Z' in the 'lbdb23' cell. (LBDB-23w)
```

### What Next

Add the missing timing group between the two pins.

---

## LBDB-24

(warning) The '%s' symbol is used but is not defined.

### Description

This message indicates that in a symbol library, you assigned an undefined symbol name to a special symbol.

The following example shows an instance where this message occurs:

```
logic_1_symbol : "mylogic_1";
```

In this example, the symbol 'my_logic_1' is not defined. To fix the problem, define the symbol:

```
symbol(mylogic_1) {
    line(0,0,0,1.5);
    line(0,1.5,.5,2);
    line(0,1.5,-.5,2);
    pin(a,0,0,RIGHT);
  }
```

The following is an example message:

```
Warning: Line 38, The 'mylogic_1' symbol is used but is not defined.
 (LBDB-24)
```

Feedback

### What Next

Either define the symbol assigned to the special symbol, or correct the name if it is wrong.

---

## LBDB-27

(error) An invalid attribute '%s' is found.

### Description

This error message occurs when a bus *function*, *three_state*, or *state_function* attribute value is not valid.

This message is also used to notify you that it is not allowed to specify related_power_power or related_group_pin when a pin is short-pin and doesn't drive any timing arc in the cell.

This message is also used to notify you when the *level_shifter_enable_pin* attribute is not specified for the input pin of a level shifter cell, or when the *isolation_cell_enable_pin* attribute is not specified for the input pin of an isolation cell.

This message is also used to show that the attribute "antenna_diode_related_power_pins" and "antenna_diode_related_ground_pins" specify one or more pg pins that don't exist.

This message is also used to show that the *max_input_delta_overdrive_high* or *max_input_delta_underdrive_high* attribute is not specified for the input or inout pin, or is not specified for power switch, level shifter, macro(memory) or pad cell.

### What Next

Correct the value of the attribute and rerun the command.

---

## LBDB-27w

(warning) An invalid attribute '%s' is found.

### Description

This warning message occurs when a bus *function*, *three_state*, or *state_function* attribute value is not valid.

This message is also used to notify you that it is not allowed to specify related_power_power or related_group_pin when a pin is short-pin and doesn't drive any timing arc in the cell.

This message is also used to notify you when the *level_shifter_enable_pin* attribute is not specified for the input pin of a level shifter cell, or when the *isolation_cell_enable_pin* attribute is not specified for the input pin of an isolation cell.

Feedback

This message is also used to show that the attribute "antenna_diode_related_power_pins" and "antenna_diode_related_ground_pins" specify one or more pg pins that don't exist.

This message is also used to show that the *max_input_delta_overdrive_high* or *max_input_delta_underdrive_high* attribute is not specified for the input or inout pin, or is not specified for power switch, level shifter, macro(memory) or pad cell.

**What Next**

Correct the value of the attribute and rerun the command.

---

## LBDB-28

(error) The '%s' attribute is supplied with %d arguments.\n \tOnly %d arguments are expected.

**Description**

This message indicates that a command or an attribute got the wrong number of arguments.

The following example shows an instance where this message occurs:

```
define("lbdb28", "library", "string", "integer");
```

The following is an example message:

```
Error: Line 61, The 'define' attribute is supplied with 4 arguments.
        Only 3 arguments are expected. (LBDB-28)
```

**What Next**

Fix the arguments of the attribute.

---

## LBDB-29

(information) The '%s' attribute for %s group is official liberty syntax/already defined, this user define syntax is ignored.

**Description**

This message indicates that the user's defined attribute has been already defined or it is reserved by Library Compiler. The Library Compiler ignores the second definition. If the attribute has already supported by Library Compiler, the information can be ignored.

The following example shows an instance where this message occurs:

```
define("lbdb29", "pin", "string");
  define("lbdb29", "pin", "integer");
  define("area", "cell", "float");
```

Feedback

The following is an example message:

```
Information: Line 62, The 'lbdb29' attribute for pin group is official
 liberty syntax/already defined, this user define syntax is ignored.
 (LBDB-29)
Information:Line 100, The 'area' attribute for cell group is official
 liberty syntax/already defined, the user define syntax is ignored
 (LBDB-29)
```

### What Next

Users can ignore this information safely. or remove the redudant definition to eliminate the message.

---

## LBDB-30

(warning) There is a sequential timing arc with the\n \t%s non-clock pin for a related_pin attribute.

### Description

This message warns you if you specified any setup, hold, skew, removal, or edge-triggered timing arcs relative to a nonclock signal.

The following example shows an instance where this message occurs:

```
cell(lbdb30) {
    area : 9;
    pin(D) {
      direction : input;
      capacitance : 1;
      min_pulse_width_high : 1.0;
      min_pulse_width_low : 1.0;
      timing () {
        timing_type : rising_edge;
        intrinsic_rise : 1.0;
        intrinsic_fall : 1.0;
        rise_resistance : 0.1;
        fall_resistance : 0.1;
        related_pin : "CD";
      }
    }
    pin(CP) {
      direction : input;
      capacitance : 1;
    }
    pin(CD) {
      direction : input;
      capacitance : 1;
    }
    ff("IQ","IQN") {
      next_state : "D";
```

Feedback

```
      clocked_on : "CP";
      clear   : "CD'";
    }
  }
```

The following is an example message:

```
Warning: Line 1828, There is a sequential timing arc with the
        'CP' non-clock pin for a related_pin attribute. (LBDB-30)
```

### What Next

Make sure the related_pin value is a clock pin, or modify the timing group.

---

## LBDB-31

(information) The %s group has been successfully %s\n \tin the library.

### Description

This message informs you that a group has been successfully added or overwritten in a library using the *update_lib* command.

The following is an example message:

```
Information: Line 1, The type group has been successfully added
        in the library. (LBDB-31)
```

### What Next

No action is required.

---

## LBDB-32

(warning) The '%s' group has been defined multiple times in\n \tthe '%s' library. Using the last definition encountered.

### Description

The library contains more than one definition of a group. The Library Compiler issues this error message, ignores the previous definitions, and takes into consideration the last definition encountered.

The following example shows an instance where this message occurs:

```
wire_load_selection(lbdb32) {
    wire_load_from_area(27,100,"10x10");
    wire_load_from_area(10,25,"05x05");
  }
  wire_load_selection(lbdb32) {
    wire_load_from_area(27,10,"10x10");
```

Feedback

```
    wire_load_from_area(27,100,"10x10");
    wire_load_from_area(0,28,"05x05");
    wire_load_from_area(2,10,"15x10");
  }
```

The following is an example message:

```
Warning: Line 50, The 'lbdb32' group has been defined multiple times in
         the 'lib' library. Using the last definition encountered.
 (LBDB-32)
```

**What Next**

Change the group name if it is wrong, or delete the second definition.

---

## LBDB-34

(error) There is a syntax error in the related_bus_pins attribute's value.

**Description**

This message indicates there is a syntax error in the value of the related_bus_pins attribute. This might be caused by a typo.

The following example shows an instance where this message occurs:

```
related_bus_pins : {ADDR} ;
```

In this case, the 'ADDR' name is between parentheses. To fix the error, place the name between quotes.

The following is an example message:

```
Error: Line 164, There is a syntax error in the related_bus_pins
 attribute's value. (LBDB-34)
```

**What Next**

Check the "Library Compiler User Guide" for the correct syntax of the value, and fix the technology library source file.

---

## LBDB-35

(warning) Missing a timing arc of timing_type 'three_state_disable'\n \tbetween '%s' and '%s' pins in the '%s' cell.

**Description**

To describe the transition from 0->Z or 1->Z, use the timing arc with timing_type of *three_state_disable* for the pin with a three_state attribute.

Feedback

The following example shows an instance where this message occurs:

```
cell(BTS4)  {
        area : 3.0;
    pin(Z)  {
      direction : output;
      function : "A";
      three_state : "E";
      timing()  {
        intrinsic_rise : 1.0;
        intrinsic_fall : 1.0;
        rise_resistance : 0.1;
        fall_resistance : 0.1;
        related_pin : "A";
      }
      timing()  {
        intrinsic_rise : 1.0;
        intrinsic_fall : 1.0;
        rise_resistance : 0.1;
        fall_resistance : 0.1;
        related_pin : "E";
      }
    }
    pin(A)  {
      direction : input;
        capacitance : 1.0;
    }
    pin(E)  {
      direction : input;
        capacitance : 1.0;
    }
  }
```

The following is an example message:

```
Warning: Line 110, Missing a timing arc of timing_type
 'three_state_disable'
        between 'E' and 'Z' pins in the 'lbfb35' cell. (LBDB-35)
```

### What Next

Add the missing timing group between the two pins.

---

## LBDB-36

(error) The '%s' specifies wrong incorrect names.

### Description

This error message occurs when incorrect variable names are specified for a ff, ff_bank, latch, latch_bank, or a statetable group. This message also occurs when the number of variables is either less or more than the required number.

Feedback

Examples of incorrect variable names include the following:

- `The group name`

- `An empty string`

The following example shows an instance where this message occurs: The following example causes an error because it uses the variable group name and an empty string:

```
statetable("A", "statetable") {
ff("IQ", "") {

Error: Line 213, The 'statetable' specifies wrong variable names.
 (LBDB-36)
```

### What Next

Check the *Library Compiler User Guide* for the correct syntax, and fix the variable names of the group in the technology library.

---

## LBDB-37

(warning) The '%s' layer is defined multiple times:\n \tDeleting the old definition.

### Description

The symbol library contains more than one definition of a layer. The Library Compiler issues this warning message, deletes the previous definition, and takes into consideration the last definition encountered.

The following example shows an instance where this message occurs:

```
library("ds.sdb") {

  layer(lbdb37_layer) {
    set_font ("1_25.font");
    visible : TRUE ;
    line_width : 1 ;
    red : 65000 ;
    green : 33000 ;
    blue : 0 ;
  }

  layer(lbdb37_layer) {   /* A duplicate layer. */
    set_font ("1_25.font");
    visible : TRUE ;
    line_width : 1 ;
    red : 65000 ;
    green : 33000 ;
    blue : 0 ;
```

Feedback

```
  }
 }
```

The following is an example message:

```
Warning: Line 32, The 'lbdb37_layer' layer is defined multiple times:
        Deleting the old definition. (LBDB-37)
```

### What Next

In the symbol library file, change the layer name if it is wrong, or delete the second definition.

---

## LBDB-39

(error) The '%s' symbol is defined multiple times.

### Description

The symbol library contains more than one definition of a symbol. The Library Compiler issues this error message and deletes all the definitions.

The following example shows an instance where this message occurs:

```
symbol(lbdb39_dot) {
    line( -.25,-.25,.25,-.25);
    line(.25,.25,.25,-.25);
    line(.25,.25,-.25,.25);
    line( -.25,.25,-.25,-.25);
    line( -.25,-.25,.25,.25);
    line(.25,-.25,-.25,.25);
  }

  symbol(lbdb39_dot) {
    line( -.25,-.25,.25,-.25);
    line(.25,.25,.25,-.25);
    line(.25,.25,-.25,.25);
    line( -.25,.25,-.25,-.25);
    line( -.25,-.25,.25,.25);
    line(.25,-.25,-.25,.25);
  }
```

The following is an example message:

```
Error: Line 58, The 'lbdb39_dot' symbol is defined multiple times.
 (LBDB-39)
```

### What Next

Change the symbol name if it is wrong, or delete the second definition.

Feedback

## LBDB-40

(error) In the '%s' symbol, the '%s' and '%s' pins both\n \thave the '%s' direction and the same Y location.

### Description

The symbol library contains more than one pin with the same direction and the same Y location.

The following example shows an instance where this message occurs:

```
symbol("lbdb40") {
    set_minimum_boundary(0 , 0 , 8000, 12000);
    pin("P0", 1000, 10000, LEFT);
    pin("P1", 8000, 10000, LEFT);
    pin("P2", 8000, 10000, LEFT);
    line(0, 0, 0, 12000);
    line(0, 12000, 8000, 12000);
    line(8000, 12000, 8000, 0);
    line(8000, 0, 0, 0);
  }
```

The following is an example message:

```
Error: Line 15, In the 'lbdb40' symbol, the 'P2' and 'P0' pins both
        have the 'LEFT' direction and the same Y location. (LBDB-40)
```

### What Next

Change the direction or the Y location of any of the pins.

## LBDB-41

(error) In the '%s' symbol, the '%s' and '%s' pins both\n \thave the '%s' direction and the same X location.

### Description

The symbol library contains more than one pin with the same direction and the same X location.

The following example shows an instance where this message occurs:

```
symbol("lbdb41") {
    set_minimum_boundary(0 , 0 , 8000, 12000);
    pin("P0", 10000, 1000, DOWN);
    pin("P1", 10000, 8000, DOWN);
    pin("P2", 10000, 8000, DOWN);
    line(0, 0, 0, 12000);
    line(0, 12000, 8000, 12000);
```

Feedback

```
    line(8000, 12000, 8000, 0);
    line(8000, 0, 0, 0);
  }
```

The following is an example message:

```
Error: Line 15, In the 'lbdb41' symbol, the 'P2' and 'P0' pins both
        have the 'DOWN' direction and the same X location. (LBDB-41)
```

### What Next

Change the direction or the X location of any of the pins.

---

## LBDB-42

(error) In the '%s' symbol, the '%s' pin '%s' and\n \tthe '%s' pin '%s' are uncorrectly positioned.

### Description

This error indicates that one of the problems is encountered:

```
- Both pin P1 and P2 have same direction and same X and Y locations.
        - The pin P1 has a LEFT direction, the pin P2 has RIGHT
          direction, and P1 and P2 have the same Y location. However
          the X location of P2 is less or equal than to the X location
          of P1.
        - The pin P1 has a UP direction, the pin P2 has DOWN
          direction, and P1 and P2 have the same X location. However
          the Y location of P2 is greater or equal than to the Y location
          of P1.
```

The following example shows an instance where this message occurs:

```
symbol("lbdb41") {
    set_minimum_boundary(0 , 0 , 8000, 12000);
    pin("P0", 10000, 1000, DOWN);
    pin("P1", 10000, 8000, DOWN);
    pin("P2", 10000, 8000, DOWN);
    line(0, 0, 0, 12000);
    line(0, 12000, 8000, 12000);
    line(8000, 12000, 8000, 0);
    line(8000, 0, 0, 0);
  }
```

The following is an example message:

```
Error: Line 41, In the symbol 'lbdb42', the 'DOWN' pin 'P2' and
        the 'DOWN' pin 'P1' are uncorrectly positioned. (LBDB-42)
```

Feedback

### What Next

Change the direction or the locations of any of the pins.

---

## LBDB-43

(error) The '%s' pin is defined multiple times in the '%s' symbol.

### Description

The symbol library contains more than one definition of a pin in the specified symbol. The Library Compiler issues this error message and deletes all the definitions.

The following example shows an instance where this message occurs:

```
symbol("lbdb43") {
    set_minimum_boundary(0 , 0 , 8000, 12000);
    pin("P0", 10000, 1000, DOWN);
    pin("P0", 10000, 1000, DOWN);
    line(0, 0, 0, 12000);
    line(0, 12000, 8000, 12000);
    line(8000, 12000, 8000, 0);
    line(8000, 0, 0, 0);
  }
```

The following is an example message:

```
Error: Line 44, The 'P0' pin is defined multiple times in the 'lbdb43'
 symbol. (LBDB-43)
```

### What Next

Change the pin name if it is wrong, or delete the second definition.

---

## LBDB-46

(error) Found an invalid rotation.

### Description

This message indicates that the specified string value for the pin's rotation is not included in the list of accepted values. This error is often caused by a typo in the valid enumerated string value.

### What Next

Change the symbol library to correct the invalid rotation value. The valid values of the rotation are ANY_ROTATION, LEFT, RIGHT, UP, or DOWN.

Feedback

## LBDB-47

(error) The '%s' cell's pin '%s' has a timing arc that has\n \t%d matched timing arcs on the scaled_cell(%s,%s).

### Description

Each timing arc in the regular cell should have a matched timing arc on the same pin in the scaled_cell group. This message indicates that more than one such matching timing arc has been found in the library source.

The following example shows an instance where this message occurs:

```
library(lbdb47) {
    operating_conditions(WCCOM) {
        process : 1.5 ;
        temperature : 70 ;
        voltage : 4.75 ;
        tree_type : "worst_case_tree" ;
    }

    cell(AND) {
      area : 1;
      pin(A B) {
        direction : input;
        capacitance : 1;
      }
      pin(Z) {
        direction : output;
        function : "A B";
        timing() {
          intrinsic_rise : 0.1;
          intrinsic_fall : 0.1;
          rise_resistance : 0.1;
          fall_resistance : 0.1;
          slope_rise : 0.0;
          slope_fall : 0.0;
          related_pin : "A B";
        }
      }
    }

    scaled_cell(AND,WCCOM) {
      area : 1;
      pin(A B) {
        direction : input;
        capacitance : 1;
      }
      pin(Z) {
        direction : output;
        function : "A B";
        timing() {
```

Feedback

```
            intrinsic_rise : 0.1;
            intrinsic_fall : 0.1;
            rise_resistance : 0.1;
            fall_resistance : 0.1;
            slope_rise : 0.0;
            slope_fall : 0.0;
            related_pin : "A";
          }
          timing() {
            intrinsic_rise : 0.3;
            intrinsic_fall : 0.3;
            rise_resistance : 0.3;
            fall_resistance : 0.3;
            slope_rise : 0.0;
            slope_fall : 0.0;
            related_pin : "A ";
          }
        }
      }
    }
```

In this case, The timing arc between 'Z' and 'A' is defined once in the 'AND' cell and twice in the 'AND' scaled_cell. To fix the problem, remove one the timing arcs.

The following is an example message:

```
Error: Line 42, The 'AND' cell's pin 'Z' has a timing arc that has
       2 matched timing arcs on the scaled_cell(AND,WCCOM). (LBDB-47)
```

**What Next**

Find the duplicate timing group you do not need, and delete it.

---

## LBDB-48

(error) The '%s' pin has a %s group whose related_pin\n \tis the port itself.

**Description**

It is not possible to have a timing arc or a power table whose starting point and ending point are the same.

The following example shows an instance where this message occurs:

```
cell(lbdb48) {
      area : 1.0;
      pin ( O ) {
        direction : output;
        function : "1";
        timing() {
          timing_sense : non_unate;
          related_pin : "O";
```

Feedback

```
            rise_resistance : 0.1;
            fall_resistance : 0.1;
            intrinsic_rise : 5.0;
            intrinsic_fall : 0.0;
        }
      }
    }
```

The following is an example message:

```
Error: Line 43, The 'O' pin has a timing group whose related_pin
       is the port itself. (LBDB-48)
```

### What Next

Check the timing arc or the internal power group, and make the appropriate correction.

---

## LBDB-49

(error) No '%s' attribute has been specified for the\n \tlibrary. This attribute is needed in %s libraries.

### Description

This message indicates that you did not specify any of the following attributes in the physical library.

```
* distance_unit
        * capacitance_unit
        * resistance_unit
```

The following example shows an instance where this message occurs: Add any of the following examples of attributes to the library:

```
distance_unit : "1um";
```

The following is an example message:

```
Warning: No 'distance_unit' attribute has been specified for the
         library. This attribute is needed in technology libraries.
 (LBDB-49)
```

### What Next

Add the missing attribute to the library source file.

---

## LBDB-50

(error) In the '%s' cell, the '%s' input pin has a 'function' attribute.

Feedback

### Description

This message indicates that a pin declared as an input, implicitly by omitting the direction or explicitly where the direction is set to input, has a function attribute. Only output and inout ports can have a function attribute.

The following example shows an instance where this message occurs:

```
pin(Q) {
        function : "1";
    }
```

In this case, the direction of 'Q' is set implicitly to 'input' and 'Q' has a function statement defined. To fix the problem, either set the direction explicitly to 'output' or 'inout', or delete the function attribute.

The following is an example message:

```
Error: Line 55, In the 'lbdb50' cell, the 'Q' input pin has a 'function'
 attribute. (LBDB-50)
```

### What Next

Either change the direction of the port or remove the function attribute.

---

## LBDB-51

(warning) No '%s' attribute has been specified for the\n \t%s library. It is set to default value '%s'.

### Description

This message indicates that you did not specify any of the following attributes in the physical library. And a default value has been assigned.

```
* time_unit
```

The following example shows an instance where this message occurs: Add any of the following examples of attributes to the library:

```
distance_unit : 1um ;
```

The following is an example message:

```
Warning: No 'time_unit' attribute has been specified for the
         library. It is set to default value '1ns'. (LBDB-51)
```

### What Next

Make sure the default value is the desired value.

Feedback

## LBDB-53

(error) The '%s' attribute, which expects values of %s type,\n \tis being supplied a value of %s type.

### Description

This message indicate that the attribute has been supplied with the wrong value. Library Compiler ignores the value.

The following example shows an instance where this message occurs:

```
dont_touch : 1;
```

In this case, the 'dont_touch' attribute expects a boolean value such as true or false.

The following is an example message:

```
Error: Line 53, The 'dont_touch' attribute, which expects values of
 boolean  type,
       is being supplied a value of integer type. (LBDB-53)
```

### What Next

Check the "Library Compiler User Guide" for the correct type of attribute, and change the value accordingly.

## LBDB-54

(error) The define attribute has an invalid '%s' type.\n \tThe valid types are 'string', 'integer', 'float', and 'boolean'.

### Description

This message indicates that an invalid type is specified for the define attribute. Library Compiler ignores the invalid type.

The following example shows an instance where this message occurs:

```
library(lbdb54) {
  define( "glorp","library","my_type" );
}
```

The following is an example message:

```
Error: Line 2, The define attribute has an invalid 'my_type' type.
       The valid types are 'string', 'integer', 'float', and 'boolean'.
 (LBDB-54)
```

Feedback

### What Next

Change the type to string, integer, float, or boolean in the library.

---

## LBDB-55

(error) The '%s' technology license is not installed.

### Description

This message indicates that the specified technology license is missing during a read_lib command. Given the technology, the missing license feature is matched as follows:

```
* CMOS technology needs a Design-Compiler feature.
        * FPGA technology needs an FPGA-Compiler feature.
```

If your site is without a valid technology license, the library is read in, and all functional information is removed. All cells are black boxes, and optimization with this library is disabled.

The following is an example message:

```
Error: Line 2, The 'cmos' technology license is not installed. (LBDB-55)
```

### What Next

Make sure that you have a license and that the license is properly installed before next trying to read the specified technology library.

---

## LBDB-57

(error) The 'generic' technology can only be read by Synopsys.

### Description

The generic technology library supplies simple combinational and sequential cells that are useful for technology-independent component instantiation. This message indicates that you specified 'generic' as the technology type in the technology library source file. This technology type is reserved for Synopsys generic technology library. Only Synopsys is allowed to compile a generic technology library.

### What Next

Do not use 'generic' as the technology type in your technology library source. Choose the correct technology type from current supporting types, for example, CMOS or FPGA.

Feedback

## LBDB-58

(warning) The '%s' pin is a multicell_pad_pin, but\n \tit has no connection_class information.

### Description

This message indicates that you have a pin in a multicell pad with the multicell_pad_pin attribute defined but the connection_class attribute is missing.

If your library uses multicell pads, Design Compiler needs to know which pins on the various cells to connect to implement the pad properly. Two attributes give this information:

```
* The multicell_pad_pin attribute identifies the pins to
          connect to create a working multicell pad. Use this
          attribute to flag all the pins to connect on a pad or an
          auxiliary pad cell.
        * The connection_class attribute indicates the pins
          to be connected to pins on other cells.
```

The following example shows an instance where this message occurs:

```
cell (lbdb58) {
      pad_cell : true;
      area : 0.0;
      pin (A) {
        direction : input;
        capacitance : 1.0;
        fanout_load : 1.0;
      }
      pin (GZ) {
        direction : input;
        capacitance : 1.0;
        fanout_load : 1.0;
      }
      pin(Y) {
        direction : output;
        capacitance : 1.0;
        driver_type : open_drain;
        is_pad : true;
        multicell_pad_pin : true;
        slew_control : low;
        drive_current : 1.0;
        output_voltage  : STD_CMOS;     /* defined in the library */
        function : "A";
        three_state : "GZ";
        timing () {
          intrinsic_rise   : 0.0;
          rise_resistance  : 0.0;
          intrinsic_fall   : 1.0;
          fall_resistance  : 0.1;
          related_pin      : "A";
```

Feedback

```
        }
        timing () {
          intrinsic_rise   : 0.0;
          rise_resistance  : 0.0;
          intrinsic_fall   : 1.0;
          fall_resistance  : 0.1;
          related_pin      : "GZ";
        }
        max_transition : 0.1;
      }
    }
```

In this case, the 'Y' pin has the multicell_pad_pin attribute defined. To fix the problem, add the connection_class attribute.

The following is an example message:

```
Warning: Line 41, The 'Y' pin is a multicell_pad_pin, but
         has no connection_class information. (LBDB-58)
```

### What Next

Add the connection_class attribute to the specified pin.

---

## LBDB-59

(error) The '%s' found in the %s\n \tattribute is invalid.

### Description

This message indicates that the specified pin, bus, or bundle name defined in a related_inputs or related_outputs attribute does not exist in the library.

The following example shows an instance where this message occurs:

```
/* Specify non-existence output pin */
  internal_power(lbdb59) {
    values("0.1, 1.2, 0.3, 0.4");
    related_outputs : "X";
  }
```

The following is an example message:

```
Error: Line 46, The 'X' found in the related_inputs or related_outputs
        attribute is invalid. (LBDB-59)
```

### What Next

Add the port definition if it is missing in the library, or fix the name if it is a typographical error.

Feedback

## LBDB-60

(warning) The wire_load or wire_load_table '%s' group has no\n \t'%s'. Using the default value of (%d, %3.1f).

### Description

This message indicates that the specified attribute is not specified in the wire_load or wire_load_table group. The default value of the attribute is used. The missing attribute can be

```
* A fanout_length
        * A fanout_capacitance
        * A fanout_resistance
        * A fanout_area
```

The following example shows an instance where this message occurs:

```
wire_load_table("lbdb60") {
      fanout_length(1, 0.2) ;
      fanout_resistance(1, 0.17) ;
      fanout_area(1, 0.2) ;
    }
```

In this case, the fanout_capacitance is missing in the wire_load_table. If you do not want the default value, add the following attribute:

```
fanout_capacitance(1, 0.15);
```

The following is an example message:

```
Warning: Line 10, The wire_load or wire_load_table 'lbdb60' group has no
        'fanout_capacitance'. Using the default value of (1, 0.0).
 (LBDB-60)
```

### What Next

If you do not want to apply the default to the attribute indicated in the warning message, specify the attribute in the indicated group.

## LBDB-61

(warning) Template '%s' is defined in old library syntax.

### Description

This message indicates that a special symbol in the symbol library file is written using an old library syntax.

Feedback

### What Next

Check the "Library Compiler User Guide" for the correct syntax of symbols.

---

## LBDB-62

(error) The '%s' symbol is a duplicate '%s' template.

### Description

This message indicates that the specified symbol is a duplicate symbol in the symbol library file. Library Compiler ignores the duplicate symbol.

### What Next

Remove the duplicate symbol from the symbol library, or change the name if it is a typo.

---

## LBDB-66

(error) The '%s' attribute cannot be supplied a \n \tnonpositive value (%f).

### Description

This message indicates that the specified attribute cannot have a nonpositive value.

The following example shows an instance where this message occurs:

```
capacitive_load_unit( -1,pf );
```

The following is an example message:

```
Error: Line 18, The 'capacitive_load_unit' attribute cannot be supplied a
       nonpositive value (-1.000000). (LBDB-66)
```

### What Next

Change the value of the attribute to positive in the technology library file.

---

## LBDB-69

(error) Missing a %s name.

### Description

This message indicates that a group name is missing in the technology or symbol library file. The Library Compiler ignores the library.

Feedback

The following example shows an instance where this message occurs:

```
cell() {
    area : 9;
  }
```

The following is an example message:

```
Error: Line 63, Missing a cell name. (LBDB-69)
```

### What Next

Add the group name to the technology or symbol library file.

---

## LBDB-70

(Information) The '%s' group under %s group is official liberty syntax/already defined, this user define syntax is ignored.

### Description

The technology library contains more than one definition of a define_group attribute with the same name. or the group has already supported by Library Compiler. The Library Compiler issues this information message and deletes all the definitions.

The information can be ignored if the group has already supported by Library Compiler.

The following example shows an instance where this message occurs:

```
define_group(lbdb70, cell);
    define_group(lbdb70, cell);
```

The following is an example message:

```
Information: Line 58, The 'lbdb70' group under cell group is official
 liberty syntax / already defined, this user define syntax is ignored.
 (LBDB-70)
```

### What Next

Users can ignore this information safely. or remove the redudant definition to eliminate the message.

---

## LBDB-72

(error) Undefined module_pin '%s' referenced in pin_association\n \t'%s' in %s '%s'.

### Description

The name given to a pin_association group within a binding group or a state group names a pin that must be defined by a pin group within the module. This error arises when the

Feedback

name given to the pin_association group does not have a corresponding pin group on the module. For example, in the following synthetic library, the pin_association group named 'MY_B' attempts to bind the operator pin 'B' to the module pin 'MY_B'. However, the pin 'MY_B' is not defined for the module, which causes an error to occur.

```
library (example.sldb) {
   module (my_module) {
      design_library : "MYLIB";
      parameter(width) {
         hdl_parameter : TRUE;
      }
      pin (MY_A) {
         direction : input;
         bit_width : "width";
      }
      pin (MY_Z) {
         direction : output;
         bit_width : "width";
      }
      binding (b1) {
         bound_operator : "ADD_UNS_OP";
         pin_association(MY_A) { oper_pin : A ; }

         /* The following line creates an error: */
         pin_association(MY_B) { oper_pin : B ; }

         pin_association(MY_Z) { oper_pin : Z ; }
      }
   }
}
```

In this case you can fix the error by adding a pin group to the module:

```
pin (MY_B) {
         direction : input;
         bit_width : "width";
      }
```

**What Next**

Remove the pin_association group that refers to the nonexistent module pin. Alternately, you can add a module pin group with the name of the referenced pin in the pin_association group.

---

## LBDB-73

(error) The '%s' cell has more than one sequential \n \tfunction (seq, latch, or ff) declaration.

Feedback

### Description

This message indicates there is more than one declaration of a sequential function group.

### What Next

Leave only one sequential function group declaration; remove the rest of the declarations.

---

## LBDB-74

(error) The %s value, '%s', is either not defined or it is defined after this line.

### Description

This error message occurs when the *default_operating_conditions* or *default_wire_load* attribute refers to an undefined the operating_conditions or wire_load group name in the library.

This message can also report the *related_power_rail* attribute references and the undefined power_rail values in the power_supply group.

The following example shows the 05x05 wire_load group defined after the *default_wire_load* attribute and the resulting error message:

```
default_wire_load : 05x05;
  wire_load("05x05") {
    resistance : 0 ;
    capacitance : 1 ;
    area : 0 ;
    slope : 0.186 ;
    fanout_length(1,0.39) ;
  }

Error: Line 57, The default_wire_load value, '05x05',
has not been defined or it is defined after this line. (LBDB-74)
```

### What Next

Check your library to determine if you have defined the group name. If the group name exists in the library, make sure it is defined before it is used.

---

## LBDB-75

(warning) There is an extra timing arc between '%s' and \n \t'%s' pins in the '%s' cell.

### Description

Timing arcs are allowed for pins that are related to each other. To be related to each other, the input pin has to be in the *function* or *three_state* attribute of the output/input pin. This message is issued if the timing arc identified between pins does not fall into the

Feedback

categories described previously. Certain output transitions cannot result from a single input pin change when *function*, *three_state*, and *x_function* share input pin, because the x_function takes highest priority. This message is issued when such situation happens.

The following example shows an instance where this message occurs:

```
pin(Z) {
      direction : output;
      function : "B";
      timing() {
        intrinsic_rise : 1.0;
        intrinsic_fall : 1.0;
        rise_resistance : 0.1;
        fall_resistance : 0.1;
        related_pin : "A";
      }
   }
```

In this case, you defined a timing arc between the 'A' and 'Z pins, even though the 'A' pin is not functionally related to the 'Z' pin. Either remove the timing arc or change the function statement of 'Z' to include 'A'.

This is another example involving x_function:

```
pin(out) {
      direction : output ;
      function : "a0&s0 | a1&s1 | a2&s2" ;
      max_capacitance : 0.09 ;
      min_capacitance : 0.01 ;
      output_voltage : default ;
      x_function : "!(s0&!s1&!s2|!s0&s1&!s2|!s0&!s1&s2)" ;

      timing() {
        related_pin : "s0" ;
        sdf_cond : "a0===1'b0 && a1===1'b0 && a2===1'b1" ;
        timing_sense : negative_unate ;
        timing_type : combinational ;
        when : "!a0&!a1&a2" ;
      }
```

In this case, 's0', 's1' and 's2' are used in both function and x_function, no isolated change on 's0' can cause 01 or 10 transition on 'out', a timing arc from 's0' to 'out' requires simontaneous change of 's1' or 's2', LBDB-75 will warn on a timing arc from 's0' to 'out', and similarly, warn on timing arc from 's1' or 's2' as well.

The following is an example message:

```
Warning: Line 73, There is an extra timing arc between 'A' and
        'Z' pins in the 'lbd75' cell. (LBDB-75)
```

Feedback

### What Next

Check your library to see whether you have generated the timing group by mistake, whether the related_pin field is wrong or the function attribute value is not recognized.

---

## LBDB-76

(error) The '%s' %s cannot be specified here.

### Description

This message indicates that you specified an attribute outside its context. When processing the current context, Library Compiler flags invalid attributes.

It can also be used to indicate that you have specified an attribute/groups which conflicts with the other attributes/groups of the object. For example, if you specify 'io_type' attribute on a cell in the FPGA library which is based on fpga_isd information, the LBDB-76 will be issues and the 'io_type" attribute should be removed.

Another example is when ocv_centile_cell_rise|fall, ocv_centile_rise|fall_transition defined under a timing group, "sigma_type" attribute can NOT be defined for ocv_sigma_cell_rise|fall and ocv_sigma_rise|fall_transition under the same timing group.

The following example shows an instance where this message occurs:

```
pin(D) {
      direction : input;
      capacitance : 1.0;
      current_unit : "1mA";
    }
```

In this case, the 'current_unit' attribute is defined at the pin level. To fix the problem, move the attribute to the library context.

The following is an example message:

```
Error: Line 69, The 'current_unit' attribute cannot be specified here.
 (LBDB-76)
```

### What Next

Change the technology library source file to either delete the specified attribute or move the attribute to its correct context.

---

## LBDB-79

(error) The 'ripped_pin' name is not defined.

Feedback

### Description

This message indicates that you specified an undefined pin name for the ripped_pin attribute in a symbol library file.

The following example shows an instance where this message occurs:

```
symbol("lbdb79") {
         ripped_pin : "bus_pin" ;
         ripped_bits_property : "EDIF_property" ;
         line(0 , 50, 50, 0);
         line(50, 0, 0, -50);
         line(-50, 0 , 100, 0);
         pin("wire_pin", 100 , 0 , ANY_ROTATION);
  }
```

In this case, the 'bus_pin' pin is not defined.

The following is an example message:

```
Error: Line 69, The 'ripped_pin' name is not defined. (LBDB-79)
```

### What Next

Change the symbol library to either define the pin name before you use it in the ripped_pin attribute or fix the pin name if it is a typo.

---

## LBDB-80

(error) The '%s' port name is either undefined or \n \tits group definition is invalid.

### Description

This message indicates that the specified port name is either not defined in the library source file or Library Compiler did not read the port group definition successfully. This might often be caused by a problem in the pin bus or a bundle group definition.

The following example shows an instance where this message occurs:

```
cell (lbdb80) {
    area : 3.0;
    bus (D) {
      bus_type : BUS4;
      direction : input;
      capacitance : 1.0;
    }
    pin (CK) {
      direction : input;
      capacitance : 1.0;
    }
    ff_bank (IQ,IQN,4) {
      next_state : "D" ;
```

Feedback

```
      clocked_on : "CK";
    }
    pin (Q) {
      direction : output ;
      function : "D[0] CK" ;
    }
  }
```

In this case, the 'BUS4' type is not defined in the library. Thus, the 'D' bus is not recognized, and the next_state value is considered to be not defined. Fix the problem by adding the 'BUS4' type group definition.

The following is an example message:

```
Error: Line 186, The 'D' port name is either undefined or
        its group definition is invalid. (LBDB-80)
```

**What Next**

Change the library source file to define the port.

---

## LBDB-81

(error) The base type in the type group is invalid.

**Description**

This message indicates that the specified base_type name in the type group is invalid. This might be caused by a typo in the name. Library Compiler only supports the *array* base_type.

The following example shows an instance where this message occurs:

```
type(bus2) {
    base_type : afrray;
    data_type : bit;
    bit_width : 2;
    bit_from : 0;
    bit_to : 1;
    downto : false;
  }
```

In this case, there is a typo in the base_type value. change afrray to *array*.

The following is an example message:

```
Error: Line 30, The base type in the type group is invalid. (LBDB-81)
```

**What Next**

Check the "Library Compiler User Guide" and fix the name of the base_type.

Feedback

## LBDB-82

(error) The data type in the type group is invalid.

### Description

This message indicates that the specified data_type name in the type group is invalid. This might be caused by a typo in the name. Library Compiler only supports the *bit* data_type.

The following example shows an instance where this message occurs:

```
type(bus2) {
    base_type : array;
    data_type : int;    /* wrong */
    bit_width : 2;
    bit_from : 0;
    bit_to : 1;
    downto : false;
  }
```

In this case, the data_type value is invalid. change int to *bit*.

The following is an example message:

```
Error: Line 30, The data type in the type group is invalid. (LBDB-82)
```

### What Next

Check the "Library Compiler User Guide" and fix the name of the data_type.

## LBDB-83

(warning) The width value in the type group is invalid.\n \tIt is corrected.

### Description

This message indicates that the specified *bit_width* value in the type group is invalid. This might be caused by a typo in the value. The Library Compiler ignores the invalid value and computes the real value from the bit_from and bit_to fields.

The following example shows an instance where this message occurs:

```
type(bus2) {
    base_type : array;
    data_type : bit;
    bit_width : 3;
    bit_from : 0;
    bit_to : 1;
    downto : false;
  }
```

Feedback

In this case, the bit_width value is 3. To fix the problem, change it to 2 (bit_to - bit_from + 1).

The following is an example message:

```
Error: Line 30, The width value in the type group is invalid.
        It is corrected. (LBDB-82)
```

**What Next**

Change the library file, and fix the value of the bit_width.

---

## LBDB-86

(warning) The downto value in the type group is invalid.\n \tIt is corrected.

**Description**

This message indicates that the specified *downto* value in the type group is invalid. This might be caused by a typo in the value. The Library Compiler ignores the invalid value and computes the real value from the bit_from and bit_to fields.

The following example shows an instance where this message occurs:

```
type(bus2) {
    base_type : array;
    data_type : bit;
    bit_width : 2;
    bit_from : 0;
    bit_to : 1;
    downto : ffalse;
  }
```

In this case, the downto value has a typo, ffalse. To fix the problem, change ffalse to false. The Library Compiler corrects the value by setting to false bit_from value is less than the bit_to value. Otherwise, it is set to true.

The following is an example message:

```
Warning: Line 35, The downto value in the type group is invalid.
        It is corrected. (LBDB-86)
```

**What Next**

Change the library file, and fix the value of the downto field.

---

## LBDB-87

(error) The type group is missing its 'data_type' field.

Feedback

### Description

This message indicates that the data_type field is missing in the type group.

The following example shows an instance where this message occurs:

```
type(bus2) {
    base_type : array;
    bit_width : 2;
    bit_from : 0;
    bit_to : 1;
    downto : false;
  }
```

To fix the problem, add the data_type statement to the type group.

```
data_type : bit;
```

The following is an example message:

```
Error: Line 28, The type group is missing its 'data_type' field.
 (LBDB-87)
```

### What Next

Change the library file, and add the data_type field to the type group.

---

## LBDB-88

(error) The type group is missing its 'base_type' field.

### Description

This message indicates that the base_type field is missing in the type group.

The following example shows an instance where this message occurs:

```
type(bus2) {
    data_type : bit;
    bit_width : 2;
    bit_from : 0;
    bit_to : 1;
    downto : false;
  }
```

To fix the problem, add the base_type statement to the type group.

```
base_type : array;
```

The following is an example message:

```
Error: Line 28, The type group is missing its 'base_type' field.
 (LBDB-88)
```

Feedback

### What Next

Change the library file, and add the base_type field to the type group.

---

## LBDB-89

(error) The '%s' library attribute must be defined\n \tto specify a %s value.

### Description

This message indicates that an attribute needed by another group is missing in the library. Examples of attributes are

```
* The input_voltage value needs the voltage_unit attribute
          defined.
        * The slew_control value needs the time_unit attribute
          defined.
        * The drive_current value needs the current_unit attribute
          defined.
```

The following example shows an instance where this message occurs:

```
input_voltage(CMOS) {
    vil          : 0.8 ;
    vih          : 2.0 ;
    vimin        : -0.3 ;
    vimax        : VDD + 0.3 ;
   }
    cell(lbdb89) {
      area : 0.0;
      dont_touch : false;
      dont_use   : false;
      pad_cell : true;
      pin(PAD ) {
        is_pad : true;
        input_voltage : CMOS;
        direction   : input;
        capacitance : 3.0;
        fanout_load : 0.0;
      }
      pin(Y ) {
        direction   : output;
        function    : "PAD";
        max_fanout  : 16.0;
        timing() {
          intrinsic_fall  : 4.0;
          intrinsic_rise  : 4.0;
          fall_resistance : 0.0;
          rise_resistance : 0.0;
          related_pin  :"PAD ";
        }
```

Feedback

```
      }
    }
```

In this case, add the voltage_unit to the library as follows:

```
voltage_unit : "1V";
```

The following is an example message:

```
Error: Line 14, The 'voltage_unit' library attribute must be defined
       to specify a input_voltage value. (LBDB-89)
```

**What Next**

Add the missing attribute to the library.

---

## LBDB-90

(error) The type group is defined multiple times.

**Description**

This message indicates that the type group is defined multiple times in the technology library. This might be caused by a typo.

The following example shows an instance where this message occurs:

```
type(bus2) {
    base_type : array;
    data_type : bit;
    bit_width : 2;
    bit_from : 0;
    bit_to : 1;
    downto : false;
  }
  type(bus2) {
    base_type : array;
    data_type : bit;
    bit_width : 2;
    bit_from : 1;
    bit_to : 0;
    downto : true;
  }
```

In this case, rename the second type group to bus2_down.

The following is an example message:

```
Error: Line 36, The type group is defined multiple times. (LBDB-90)
```

Feedback

### What Next

Change the library file by either removing the multiple definitions but one or renaming the type groups.

---

## LBDB-91

(warning) The bus naming style format is invalid.\n \tUsing the default format.

### Description

This message indicates that you specified an invalid bus_naming_style format. The value must include %s and %d. Library Compiler ignores the invalid value and uses the default style "%s[%d]".

The following example shows an instance where this message occurs:

```
bus_naming_style : "%s$%s";
```

The following is an example message:

```
Warning: Line 28, The bus naming style format is invalid.
        Using the default format. (LBDB-91)
```

### What Next

Check the "Library Compiler User Guide" for the correct format of the bus_naming_style and fix the value.

---

## LBDB-92

(error) The bus subscript is out of bounds.

### Description

This message indicates that you specified a bus element whose index value falls outside the declared index range of the bus type.

The accessed bus element with an index outside the bus bounds is specified in function, three_state, ff, or latch statements.

The following example shows an instance where this message occurs:

```
type(bus2) {
    base_type : array;
    data_type : bit;
    bit_width : 2;
    bit_from : 0;
    bit_to : 1;
    downto : false;
```

Feedback

```
}
cell (lbdb92) {
  area : 31.0;
  bus (D) {
    bus_type : bus2;
    direction : input ;
    capacitance : 1.0 ;
  }
  pin (CK) {
    direction : input ;
    capacitance : 1.0 ;
  }
  ff_bank (IQ,IQN,2) {
    next_state : "D[3]" ;
    clocked_on : "CK" ;
  }
  bus (SO) {
    bus_type : bus2;
    direction : output ;
    function : "IQ" ;
  }
}
```

In this case, the 'D[3]' value assigned to next_state falls outside the bus2 range.

The following is an example message:

```
Error: Line 186, The bus subscript is out of bounds. (LBDB-92)
```

### What Next

Either change the bounds of the bus definition so that all subelements that are accessed fall within the range, or correct any bus indexing errors in the source library.

---

## LBDB-93

(error) The bus type is invalid.

### Description

This message indicates that you specified a bus name whose bus_type is invalid. This might be caused by an invalid base_type in the type group or a typo in the name. Library Compiler only supports the *array* base_type for bus groups.

The following example shows an instance where this message occurs:

```
type(bus2) {
    base_type : list;
    data_type : bit;
    bit_width : 2;
    bit_from : 0;
    bit_to : 1;
```

Feedback

```
    downto : false;
  }
  cell (lbdb93) {
    area : 31.0;
    bus (D) {
      bus_type : bus2;
      direction : input ;
      capacitance : 1.0 ;
    }
    pin (CK) {
      direction : input ;
      capacitance : 1.0 ;
    }
  }
```

In this case, the 'list' base_type of the 'bus2' type is invalid. Fix the problem by changing the 'list' to 'array'.

The following is an example message:

```
Error: Line 177, The bus type is invalid. (LBDB-93)
```

### What Next

Check the "Library Compiler User Guide" and fix the name of the base_type in the type group, or change the bus type name.

---

## LBDB-94

(warning) The ':' character is used in this bus naming style;\n \tthis makes it impossible to specify ranges of buses using ':'.

### Description

This message indicates that you specified a ':' in the string of the bus_naming_style. Library Compiler gets confused because the ':' is used to specify ranges of buses.

The following example shows an instance where this message occurs:

```
bus_naming_style : "%s:%d";
```

The following is an example message:

```
Warning: Line 28, The ':' character is used in this bus naming style;
         this makes it impossible to specify ranges of buses using ':'.
 (LBDB-94)
```

### What Next

Change the value of the bus_naming_style attribute in the technology library.

Feedback

## LBDB-95

(error) Buses have incompatible widths.

### Description

This message indicates that you specified a function or a three_state attribute of a group of buses with a different width.

The following example shows an instance where this message occurs:

```
cell (lbdb95) {
    area : 3.0;
    bus (D) {
      bus_type : bus2;
      direction : input;
      capacitance : 1.0;
    }
    pin (CK) {
      direction : input;
      capacitance : 1.0;
    }
    ff_bank (IQ,IQN,4) {
      next_state : "D";
      clocked_on : "CK";
    }
    pin (SO) {
      direction : output;
      function : "D[0] CK";
    }
  }
```

In this case, the number of bits in the ff_bank, which is 4, is different from the width of the 'D' bus, which is 2. Make sure that both values are the same.

The following is an example message:

```
Error: Line 186, Buses have incompatible widths. (LBDB-95)
```

### What Next

Change the library source file to make sure that all buses have the same width.

## LBDB-96

(error) The bus type name is missing.

### Description

This message indicates that you specified a bus with an invalid a bus_type.

Feedback

### What Next

Check the library source file, and correct the bus_type to the bus in the library.

---

## LBDB-97

(error) The bus type of a subelement is missing.

### Description

This message indicates that you specified a subelement whose parent is missing a bus type.

### What Next

Change the library, and fix the bus type.

---

## LBDB-98

(error) The type group name is not defined.

### Description

This message indicates that you specified a bus group with an undefined bus_type name in the technology library. This might be caused by a typo.

The following example shows an instance where this message occurs:

```
cell (lbdb98) {
    area : 3.0;
    bus (D) {
      bus_type : bus2;
      direction : input;
      capacitance : 1.0;
    }
   }
```

In this case, the 'bus2' is not defined in the library. To fix the problem, add the following type group:

```
type(bus2) {
    base_type : array;
    data_type : bit;
    bit_width : 2;
    bit_from : 0;
    bit_to : 1;
    downto : false;
  }
```

Feedback

The following is an example message:

```
Error: Line 177, The type group name is not defined. (LBDB-98)
```

**What Next**

Change the library file by adding the named type group.

---

## LBDB-99

(error) The pin direction is inconsistent with the bus or\n \tthe bundle parent direction.

**Description**

This message indicates that you specified a member of a bus or a bundle whose direction is different from its parent's direction.

The following example shows an instance where this message occurs:

```
bundle(nets) {
      members(n0, n1, n2, n3);
      direction : inout;
      pin (n0) {
        direction : output;
      }
    }
```

In this case, the 'nets' direction is 'inout', but the 'n0' direction is 'output'.

The following is an example message:

```
Error: Line 199, The pin direction is inconsistent with the bus or
        the bundle parent direction. (LBDB-99)
```

**What Next**

Change the library to fix the direction of the element, the bus, or the bundle.

---

## LBDB-101

(error) The piece number in piecewise linear mode is negative.

**Description**

This message indicates that you specified a negative piece number in either delay intercept attributes or pin resistance attributes in CMOS piecewise linear delay model library.

Feedback

The following example shows an instance where this message occurs:

```
piece_define("0 8 15");
        ...
  rise_delay_intercept(-11, "2.2");
```

In this case, the -11 piece number is invalid. To fix the problem, make the piece number 0, 1, or 2 as defined by the piece_define attribute.

The following is an example message:

```
Error: Line 172, The piece number in piecewise linear mode is negative.
 (LBDB-101)
```

### What Next

Modify the piece number to match the number of ranges in the piece_define attribute.

---

## LBDB-102

(warning) The '%s' piece is multiply\n \tdefined. Using the first one encountered.

### Description

This message indicates that you specified either delay intercept attributes or pin resistance attributes for the same piece number multiple times in a CMOS piecewise linear delay model library. Library Compiler ignores the later definitions.

The following example shows an instance where this message occurs:

```
rise_delay_intercept(1, "2.2");
        rise_delay_intercept(1, "2.2");
```

The following is an example message:

```
Warning: Line 173, The 'rise_delay_intercept(1)' piece is multiply
        defined. Using the first one encountered. (LBDB-102)
```

### What Next

Modify the piece number if it is a typo, or remove the second definition.

---

## LBDB-103

(warning) The piece number is greater than the number\n \tdefined with piece_define.

### Description

This message indicates that in a CMOS piecewise-linear delay model library you specified a piece number in either delay intercept attributes or pin resistance attributes greater than the number defined in the piece_define attribute.

Feedback

The following example shows an instance where this message occurs:

```
piece_define("0 8 15");
        ...
  rise_delay_intercept(3, "2.2");
```

In this case, the 3 piece number is invalid. To fix the problem, make the piece number 0, 1, or 2 as defined by the piece_define attribute.

The following is an example message:

```
Warning: Line 174, The piece number is greater than the number
        defined with piece_define. (LBDB-103)
```

**What Next**

Modify the piece number to match the number of ranges in the piece_define attribute.

---

## LBDB-104

(warning) The piecewise linear model is multiply defined.\n Using the first one encountered.

**Description**

This message indicates that you specified the *piece_define* attribute multiple times in a CMOS piecewise linear delay model library. Library Compiler ignores the later definitions.

The following example shows an instance where this message occurs:

```
piece_define("0 8 15");
  piece_define("0 8 15 25");
```

The following is an example message:

```
Warning: Line 13, The piecewise linear model is multiply defined.
        Using the first one encountered. (LBDB-104)
```

**What Next**

Remove the second definition of the attribute.

---

## LBDB-105

(warning) The timing arc has a negative %s\n \tspecified. Using the default value.

**Description**

This message indicates that you specified a negative drive resistance value of pin resistance attributes in a CMOS piecewise linear delay model library. Library Compiler

Feedback

assigns the *default_rise_pin_resistance* or the *default_fall_pin_resistance* value to the timing arc or the value 0.0 if no default value exists.

The following example shows an instance where this message occurs:

```
rise_pin_resistance(0, "-11.1");
```

The following is an example message:

```
Warning: Line 183, The timing arc has a negative rise_pin_resistance(0)
        specified. Using the default value. (LBDB-105)
```

**What Next**

Make the drive resistance value positive in the pin resistance attribute.

---

## LBDB-107

(warning) The %s '%s' is defined multiple times\n \tin the library. Using the last one encountered.

**Description**

This message indicates that the same name has been used for more than one object (for example, cells in a library or pins in a cell). Each object must have a unique name within its scope. In the case of a name conflict, Library Compiler ignores all except the last name encountered during the compilation. The compiled database contains the last object only.

The following example shows an instance where this message occurs:

```
statetable ( "D GN", "Q QB") {
        table : "L/H L : - - : L/H H/L,\
            -   H : - - : N   N";
      }

      statetable ( "D GN", "Q QB") {
        table : "L/H L : - - : L/H H/L,\
            -   H : - - : N   N";
      }
```

The following is an example message:

```
Warning: Line 987, The group 'statetable' is defined multiple times
        in the library. Using the last one encountered. (LBDB-107)
```

**What Next**

Update the library to give each object a unique name within its scope. Quite often, a typo is responsible for the name conflict.

Feedback

## LBDB-110

(warning) In the piecewise linear model, the first piece must\n \thave 0 length.

### Description

This message indicates that you specified the first piece range, in the piece_define attribute, not starting from zero. Library Compiler ignores the value, and resets it to zero.

The following example shows an instance where this message occurs:

```
piece_define("1 8 15");
```

The following is an example message:

```
Warning: Line 12, In the piecewise linear model, the first piece must
        have 0 length. (LBDB-110)
```

### What Next

Change the first piece range to zero.

## LBDB-111

(warning) In the piecewise linear model, a piece length\n \tsmaller than that of the previous piece has been found.

### Description

This message indicates that in the piece_define attribute you specified a piece length smaller than the previous one. The Library Compiler ignores the value and resets it to the previous value.

The following example shows an instance where this message occurs:

```
piece_define("0 18 15");
```

The following is an example message:

```
Warning: Line 12, In the piecewise linear model, a piece length
        smaller than that of the previous piece has been found.
 (LBDB-111)
```

### What Next

Change the piece ranges to be in ascending order.

Feedback

## LBDB-112

(error) The timing arc has only one segment of\n \t'%s'. It must have at least two if \n\ \tpiece_define has more than one piece.

### Description

This message indicates that in a CMOS piecewise linear delay model library you specified only one segment in either delay intercept attributes or pin resistance attributes. However, the *piece_define* defines more than one piece.

The following example shows an instance where this message occurs:

```
piece_define("0 8 15");
        ...
        rise_delay_intercept(0, "2.2");
```

In this case, only one rise_delay_intercept attribute is defined. To fix the problem, add two more rise_delay_intercept for piece 1 and 2.

```
rise_delay_intercept(1, "3.3");
        rise_delay_intercept(2, "4.4");
```

The following is an example message:

```
Error: Line 169, The timing arc has only one segment of
       'rise_delay_intercept'. It must have at least two if
        piece_define has more than one piece. (LBDB-112)
```

### What Next

Modify the piece number to match the number of ranges in the piece_define attribute.

## LBDB-117

(error) A list does not belong here.

### Description

This message indicates that you specified a list to an attribute whose type is not a list.

The following example shows an instance where this message occurs:

```
library(lbdb117) {
  cell(c) {
    area : 1.0;
    pin( a ) {
      direction : input;
      capacitance : 1.0;
    }
    pin( b ) {
```

Feedback

```
      direction : output;
      timing() {
        intrinsic_rise : 0.48;
        intrinsic_fall : 0.77;
        rise_resistance : 0.1443;
        fall_resistance : {0.0523,0.0523};
        slope_rise : 0.0;
        slope_fall : 0.0;
        related_pin : "a";
      }
    }
  }
}
```

In this case, the 'fall_resistance' attribute has a value of type list. Fix the problem by assigning a single value to the attribute.

The following is an example message:

```
Error: Line 14, A list does not belong here. (LBDB-117)
```

**What Next**

Correctthe value of the specified attribute.

---

## LBDB-119

(error) The '%s' pin on the %s could not be found on\n \tthe %s. There must be a one-to-one match.

**Description**

Each pin in the regular cell needs a matched pin in the scaled_cell group. This message indicates that a pin could not be matched in the library source.

The following example shows an instance where this message occurs:

```
library(lbdb119) {
    operating_conditions(WCCOM) {
        process : 1.5 ;
        temperature : 70 ;
        voltage : 4.75 ;
        tree_type : "worst_case_tree" ;
    }
    cell(AND) {
      area : 1;
      pin(A B) {
        direction : input;
        capacitance : 1;
      }
      pin(Z) {
        direction : output;
```

Feedback

```
        function : "A B";
        timing() {
          intrinsic_rise : 0.1;
          intrinsic_fall : 0.1;
          rise_resistance : 0.1;
          fall_resistance : 0.1;
          slope_rise : 0.0;
          slope_fall : 0.0;
          related_pin : "A B";
        }
      }
    }

    scaled_cell(AND,WCCOM) {
      area : 1;
      pin(A B C) {
        direction : input;
        capacitance : 1;
      }
      pin(Z) {
        direction : output;
        function : "A B";
        timing() {
          intrinsic_rise : 0.1;
          intrinsic_fall : 0.1;
          rise_resistance : 0.1;
          fall_resistance : 0.1;
          slope_rise : 0.0;
          slope_fall : 0.0;
          related_pin : "A B";
        }
      }
    }
  }
```

In this case, the 'C' pin exists in the scaled_cell but not in the parent cell.

The following is an example message:

```
Error: Line 100, The 'C' pin on the scaled_cell could not be found on
       the parent cell. There must be a one-to-one match. (LBDB-119)
```

**What Next**

Add the missing pin in the regular cell, or remove the extra pin in the scaled_cell.

---

## LBDB-120

(warning) Layers are ignored inside symbols.

Feedback

### Description

For a symbol library, Library Compiler ignores layer information within a symbol.

The following example shows an instance where this message occurs:

```
symbol(lbdb-120) {
    line( - flag_width / 2, - flag_height / 2, \
       - flag_width / 2, flag_height / 2) ;
    line( - flag_width / 2, flag_height / 2, \
         flag_width / 2, flag_height / 2) ;
    line( flag_width / 2, flag_height / 2, \
         flag_width / 2, - flag_height / 2) ;
    line( flag_width / 2, - flag_height / 2, \
         - flag_width / 2, - flag_height / 2) ;
    text( "Area", - string_length / 2 - char_width / 3, \
         - 0.75 / 2, "constraint_layer") ;
  }
```

In this case, the constraint_layer is used in the symbol.

The following is an example message:

```
Warning: Line 3453, Layers are ignored inside symbols. (LBDB-120)
```

### What Next

Remove the layer information from the symbol.

---

## LBDB-121

(error) The '%s' must be defined in a macro cell.

### Description

This message is reported when the specified group is defined in a cell not of type macro.

The group pg_setting_definition must be defined in macro cell.

The following is an example message:

```
Error: Line 546, There 'pg_setting_definition" must be defined in a macro
 cell.(LBDB-121)
```

### What Next

Change the cell type to macro.

---

## LBDB-122

(error) The %s '%s' is missing required values.

Feedback

### Description

This error indicates that the specified group is missing required attribute/group: 1) For a voltage_state_range_list group, at least one voltage_state_range is needed; 2) For a pg_setting_definition group, at least one pg_setting_value is needed; 3) For a pg_setting_definition group, default_pg_setting is needed; 4) For a pg_setting_value group, either pg_pin_condition or pg_setting_condition is needed; 5) For a pg_setting_transition group, either start_setting or end_setting is needed; 6) For a site group, width and height is needed; 7) For a routing_blockage group, either rectangle or polygon is needed; 8) For a routing_blockage group, layer_name is needed; 9) For a base_layer_density_view group, saved_layers is needed. 10) For a derived_layer group, one of and, or, xor, not and resize attribute is needed.

The following example shows an instance where this message occurs:

```
cell(CGNP) {
  area : 1;
  pg_setting_definition(read) {
  }
  ...
}
```

The following is an example message:

```
Error: Line 206, The pg_setting_definition 'read' is missing required
 values. (LBDB-122)
```

### What Next

Define the missing group/attribute.

---

## LBDB-123

(error) The state '%s' could not be found for the %s '%s'.

### Description

This message indicates that the specified state for pg_pin or pg_setting could not be found in the library.

The following example shows an instance where this message occurs:

```
/* Specify non-existence pg pin state */
  pg_setting_value(lbdb59) {
    pg_pin_active_state(VDD, none);
  }
```

The following is an example message:

```
Error: Line 46, The state 'none' could not be found for the pg pin 'VDD'.
 (LBDB-123)
```

Feedback

### What Next

Add the required state for the voltage name of the pg pin if it is missing in the library, or the pg setting in the cell, or fix the name if it is a typographical error.

---

## LBDB-124

(error) The pg setting instance pg_setting(%s, %s) is invalid.

### Description

The pg setting instance delaration must be referring to a pg_setting_definition and one of its pg_setting_value.

The following example shows an instance where this message occurs: The following example shows an incorrect pg setting instance:

```
cell(CGNP) {
  area : 1;
  pg_setting_definition(rw) {
    pg_setting_value(read) {
      ...
    }
  }
  mode_definition(rw) {
    mode_value(read) {
      ...
    }
    mode_value(write) {
        pg_setting(rw, write);
    }
  }
  ...
}
```

The following is an example message:

```
Error: Line 206, The pg setting instance pg_setting(rw, write) is
 invalid. (LBDB-124)
```

### What Next

Check for consistency between mode group and pg setting definitions, and pg setting instance declarations.

---

## LBDB-125

(warning) The %s '%s' in group %s is ignored.

Feedback

### Description

This message indicates that you specified a redundant attribute or group in the specified group.

The following example shows an instance where this message occurs: The following example shows a redundant pg_pin_active_state for VBP in this pg_setting_value group.

```
cell (CC) {
    pg_setting_definition(PD) {
        pg_setting_value(PV) {
            pg_pin_condition : "VDD * !VSS" ;
            pg_pin_active_state(VDD, on);
            pg_pin_active_state(VBP, pon);
        }
    }
}
```

The following is an example message:

```
Warning: Line 639, The pg_pin_active_state VBP in group pg_setting_value
 is ignored. (LBDB-125)
```

### What Next

Remove the redundant attribute from the parent group and rerun the command.

---

## LBDB-126

(error) The '%s' construct is not valid\n \tin '%s' libraries.

### Description

This message indicates that the specified construct is not valid in the specified library. In Library Compiler, certain constructs are valid only in certain types of libraries. For example, pin resistances are not valid when specified on timing arcs in nonlinear (table_lookup) libraries.

The following example shows an instance where this message occurs:

```
library(lbdb126) {
  delay_model : "generic_cmos";
  piece_define("0 8 15");

  cell(INVERTER) {
    area : 5.0;
    cell_power : 1.0;
    pin(A) {
      direction : input;
      capacitance : 1.0;
    }
```

Feedback

```
  }
}
```

In this case, the 'cell_power' construct is not valid in piecewise_cmos libraries.

The following is an example message:

```
Error: Line 155, The 'cell_power' construct is not valid
        in 'piecewise_cmos' libraries. (LBDB-126)
```

### What Next

Remove the invalid construct from the library source file

---

## LBDB-127

(error) The %s contains invalid variable '%s'.

### Description

This message indicates that the Boolean condition contains invalid operand. For pg_pin_condition, only pg pin and signal pin are allowed; For pg_setting_condition, only other pg_setting_definition and signal pin are allowed;

The following example shows an instance where this message occurs: The following example shows an incorrect pg_pin_condition in this pg_setting_value group, while X is neither a pg pin nor pin of the cell.

```
cell (CC) {
    pg_setting_definition(PD) {
        pg_setting_value(PV) {
            pg_pin_condition : "VDD * !VSS * X" ;
            pg_pin_active_state(VDD, on);
        }
    }
}
```

The following is an example message:

```
Warning: Line 639, The pg_pin_condition contains incorrect variable 'X'.
 (LBDB-127)
```

### What Next

Check the Boolean condition and remove invalid variable.

---

## LBDB-128

(error) The '%s' and '%s' attributes are both defined for this\n \t%s group.

Feedback

### Description

This error indicates some attributes are specified together in one group.

derived_layers groups require specification of either 1. and 2. or 3. not 4. xor 5. resize However, it is an error to specify more than one of them.

pg_setting_value groups require specification of either 1. pg_pin_condition and pg_pin_active_state 2. pg_setting_condition and pg_setting_active_state

However, it is an error to mix both the styles.

The following example shows an instance where this message occurs:

```
cell (CC) {
    pg_setting_definition(PD) {
        pg_setting_value(PV) {
            pg_pin_condition : "VDD * !VSS" ;
            pg_setting_condition : "PD1 * PD2";
        }
    }
}
```

The following is an example message:

```
Error: Line 104, The 'pg_pin_condition' and 'pg_setting_condition'
 attributes are both defined for this
       pg_setting_value group. (LBDB-128)
```

### What Next

Check the library source file to make sure that the previous requirement is met.

---

## LBDB-129

(error) The pg_setting_value '%s' contains circular pg_setting_active_state.

### Description

The pg_setting_active_state in the pg_setting_value has circular reference. A circular reference occurs when an pg_setting_value is related to itself through other pg_setting_value. Under this situation, the pg_setting_condition can never be resolved. Therefore, Library Compiler flags them with errors.

The following example shows an instance where this message occurs:

```
cell(lbdb140) {
        pg_setting_definition(PD) {
            ...
        }
        pg_setting_definition(PD_1) {
            ...
```

Feedback

```
        }
        pg_setting_definition(PD_2) {
            ...
        }
        pg_setting_definition(PD_3) {
            pg_setting_value(PV8) {
                pg_setting_condition : "PD_4 + PD_1" ;
                pg_setting_active_state(PD_4, PV1);
                pg_setting_active_state(PD_1, PV1);
            }
        }
        pg_setting_definition(PD_4) {
            pg_setting_value(PV1) {
                pg_setting_condition : "PD * !PD_3" ;
                pg_setting_active_state(PD, PV1);
                pg_setting_active_state(PD_3, PV8);
            }
        }

    }
```

The following is an example message:

```
Warning: Line 263, The pg_setting_value 'PV8' contains circular
 pg_setting_active_state. (LBDB-129)
```

### What Next

Check the library source file, and verify the questionable circular pg_setting_value.

---

## LBDB-130

(error) The pg_setting_condition of '%s' is always %s.

### Description

This error message indicates that the pg_setting_condition is always true or false, which means the pg_setting_value is always active or never active.

The following example shows an instance where this message occurs:

```
cell(lbdb140) {
        pg_setting_definition(PD) {
            pg_setting_value(PV1) {
                pg_pin_condition : "!VDD + VSS" ;
            }
        }
        pg_setting_definition(PD_2) {
            pg_setting_value(PV1) {
                pg_pin_condition : "VDD * !VSS" ;
            }
        }
```

Feedback

```
        pg_setting_definition(PD_3) {
            pg_setting_value(PV8) {
                pg_setting_condition : "PD_2 * PD" ;
                pg_setting_active_state(PD_2, PV1);
                pg_setting_active_state(PD, PV1);
            }
        }

    }
```

The following is an example message:

```
Warning: Line 263, The pg_setting_condition of 'PV8' is always false.
 (LBDB-130)
```

### What Next

Check the Boolean condition, and verify the problematic expression.

---

## LBDB-131

(error) The required state of %s with name '%s' has not been defined in the %s, or it is invalid.

### Description

This message indicates that the pg_pin_active_state or pg_setting_active_state of the specified object is not defined in the group, or it's an invalid state.

The following is an example message:

```
Error: Line 57, The required state of pg pin with name 'VDD" has not been
 defined in the pg_setting_value, or it is invalid.(LBDB-131)
```

### What Next

Add the correct state for this object.

---

## LBDB-132

(error) In the '%s' cell, the %s pin '%s' cannot have\n \ta '%s' attribute.

### Description

This message indicates that you specified a *clock* or a *prefer_tied* attribute on a inout pin.

The following example shows an instance where this message occurs:

```
cell(lbdb132) {
    area : 1;
    pin(A B C) {
```

Feedback

```
      direction : input;
      capacitance : 1;
    }
    pin(Z) {
      direction : inout;
      function : "A B";
      three_state : "C";
      clock : true;
      timing() {
        intrinsic_rise : 0.1;
        intrinsic_fall : 0.1;
        rise_resistance : 0.1;
        fall_resistance : 0.1;
        related_pin : "A B C";
      }
      timing() {
        timing_type : three_state_disable;
        intrinsic_rise : 0.1;
        intrinsic_fall : 0.1;
        rise_resistance : 0.1;
        fall_resistance : 0.1;
        related_pin : "C";
      }
    }
  }
```

In this case, the 'Z' pin whose direction is inout has a clok attribute set.

The following is an example message:

```
Error: Line 219, In the 'lbdb132' cell, the inout pin 'Z' cannot have
        a 'clock' attribute. (LBDB-132)
```

**What Next**

Remove the specified attribute, or change the direction of the pin to input.

---

## LBDB-135

(error) The attribute '%s' is required for at least one output pin as the related PG pin has attribute '%s'.

**Description**

For a macro cell, if output pins' PG pin has attribute 'permit_power_down', it is a requirement that at least one output pin associated to the PG pin has attribute 'alive_during_partial_power_down'.

The following example shows an instance where this message occurs:

```
cell(els) {
      is_macro_cell : true;
```

Feedback

```
    pg_pin (VDD) {
      pg_type : primary_power;
      voltage_name : VDD;
    }
    pg_pin (VDDL) {
      pg_type : primary_power;
      permit_power_down : true;
      voltage_name : VDDL;
    }
    pg_pin (VSS) {
      pg_type : primary_ground;
      voltage_name : VSS;
    }

    pin(EN) {
      direction : input;
      related_power_pin : VDDL;
      related_ground_pin : VSS;
    }
    pin(Y1) {
      direction : output;
      related_power_pin : VDDL;
      related_ground_pin : VSS;
    }
    pin(Y2) {
      direction : output;
      related_power_pin : VDDL;
      related_ground_pin : VSS;
    }
  }
```

In this case, all the output pins' related PG pins have the attribute 'permit_power_down', but there is no attribute 'alive_during_partial_power_down' in output pins ('Y1' and 'Y2').

The following is an example message:

```
Error: Line 326, Cell 'els', The attribute
 'alive_during_partial_power_down' is required for at least one output
 pin as the related PG pin has attribute 'permit_power_down' (LBDB-135)
```

**What Next**

Add required attribute to the output signal pin.

---

## LBDB-136

(warning) The '%s' attribute on the '%s' pin in\n \tthe '%s' cell is not valid on %s pins.\n \tThe attribute is ignored.

Feedback

### Description

This message indicates that the attribute on the specified pin is invalid. The attribute is ignored.

The following example shows an instance where this message occurs:

```
cell(lbdb136) {
      area : 9;
      pin(D) {
        direction : input;
        capacitance : 1;
      }
      pin(CP) {
        direction : input;
        capacitance : 1;
      }
      pin(CD) {
        direction : input;
        capacitance : 2;
      }
      ff("IQ","IQN") {
        next_state : "D";
        clocked_on : "CP";
        clear   : "CD'";
      }
      pin(Q) {
        direction : output;
        function : "IQ";
        min_period : 0;
      }
```

In this case, the min_period attribute is defined for an output pin.

The following is an example message:

```
Warning: Line 1848, The  'min_period'  attribute on the 'Q' pin in
        the 'lbdb136' cell is not valid on output pins.
        The attribute is ignored. (LBDB-136)
```

### What Next

Refer to the "Library Compiler User Guide" to determine the reason why the attribute is not valid. Make the correction.

---

## LBDB-137

(error) The attribute '%s' is required as its related PG pin has attribute '%s'.

Feedback

### Description

This message indicates that the required attribute on the specified pin is missing, since the related PG pin of the pin has specified attribute.

For an isolation cell, or macro cell with isolation enable pin, if the PG pin has attribute 'permit_power_down', it is a requirement that the output pin and isolation cell enable pin associated to the PG pin has attribute 'alive_during_partial_power_down'.

The following example shows an instance where this message occurs:

```
cell(els) {
      pg_pin (VDD) {
        pg_type : primary_power;
        voltage_name : VDD;
      }
      pg_pin (VDDL) {
        pg_type : primary_power;
        permit_power_down : true;
        voltage_name : VDDL;
      }
      pg_pin (VSS) {
        pg_type : primary_ground;
        voltage_name : VSS;
      }

      pin(EN) {
        isolation_cell_enable_pin : true;
        direction : input;
        related_power_pin : VDDL;
        related_ground_pin : VSS;
      }
      pin(Y) {
        direction : output;
        related_power_pin : VDDL;
        related_ground_pin : VSS;
      }
    }
```

In this case, the alive_during_partial_power_down is missing in pin EN and Y.

The following is an example message:

```
Error: Line 326, Cell 'els', pin 'EN', The attribute
 'alive_during_partial_power_down' is required as its related PG pin has
 attribute 'permit_power_down'. (LBDB-137)
Error: Line 346, Cell 'els', pin 'Y', The attribute
 'alive_during_partial_power_down' is required as its related PG pin has
 attribute 'permit_power_down'. (LBDB-137)
```

### What Next

Add required attribute to the signal pin.

Feedback

## LBDB-138

(warning) The timing arc is missing the piecewise data value\n \tfor '%s'. The value is interpolated if possible.\n \tOtherwise, the default value is used if it exists.

### Description

This message indicates that you did not specify the definition of either delay intercept attributes or pin resistance attributes of piece number in a CMOS piecewise linear delay model library.

The following example shows an instance where this message occurs:

```
library(test) {
  delay_model : "generic_cmos";
  piece_define("0 8 15");

  cell(lbdb138) {
    area : 5.0;
    pin(A) {
      direction : input;
      capacitance : 1.0;
    }
    pin(Z) {
      direction : output;
      function : "A'";

      timing() {
        intrinsic_rise : 5.0;
        intrinsic_fall : 2.0;
        slope_rise : 1.0;
        slope_fall : 2.0;

        rise_delay_intercept(0, "1.1");
        rise_delay_intercept(1, "2.2");
        rise_delay_intercept(2, "3.3");

        fall_delay_intercept(0, "1.1");
        fall_delay_intercept(1, "2.2");
        fall_delay_intercept(2, "3.3");

        rise_pin_resistance(0, "1.1");
        rise_pin_resistance(1, "2.2");

        fall_pin_resistance(0, "4.4");
        fall_pin_resistance(1, "4.4");

        related_pin : "A";
      }
    }
  }
}
```

Feedback

In this case, the value of the rise_pin_resistance attribute is missing for index 2.

The following is an example message:

```
Warning: Line 165, The timing arc is missing the piecewise data value
        for 'rise_pin_resistance(2)'. The value is interpolated if
 possible
        Otherwise, the default value is used if it exists. (LBDB-138)
```

**What Next**

Add the missing attribute to the specified piece number.

---

## LBDB-139

(error) Invalid delay model for the given technology.

**Description**

This message indicates that you specified an invalid value in the delay_model attribute.

The following example shows an instance where this message occurs:

```
delay_model : "lbdb139";
```

The following is an example message:

```
Error: Line 3, Invalid delay model for the given technology. (LBDB-139)
```

**What Next**

Refer to the "Library Compiler User Guide" for supported delay models. Change the value of the delay_model attribute to a valid one.

---

## LBDB-140

(warning) The '%s' cell contains circular timing arcs.\n \tThe '%s' pin is in one of the cycles.

**Description**

The pin in the library cell has circular timing arcs. A circular timing arc occurs when an inout or output pin is related to itself through other inout or output pins. A change in one pin belonging to a circular timing arc continually loops around.

Circular timing arcs do not make sense and are probably mistakes. Therefore, the Library Compiler flags them with warnings.

The following example shows an instance where this message occurs:

```
cell(lbdb140) {
      area : 9;
```

Feedback

```
    pin(D) {
      direction : input;
      capacitance : 1;
    }
    pin(CP) {
      direction : input;
      capacitance : 1;
    }

    ff("IQ","IQN") {
      next_state : "D";
      clocked_on : "CP";
    }
    pin(Q) {
      direction : output;
      function : "IQ";
      timing() {
        timing_type : rising_edge;
        intrinsic_rise : 1.0;
        intrinsic_fall : 1.0;
        rise_resistance : 0.1;
        fall_resistance : 0.1;
        related_pin : "CP";
      }
    }
    pin(Q2) {
      direction : output;
      function : "IQ";
      timing() {
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "Q3";
      }
    }
    pin(Q3) {
      direction : output;
      function : "IQ";
      timing() {
        intrinsic_rise : 1.0;
        intrinsic_fall : 1.0;
        rise_resistance : 0.1;
        fall_resistance : 0.1;
        related_pin : "Q2";
      }
    }
  }
```

The following is an example message:

```
Warning: Line 263, The 'lbdb140' cell contains circular timing arcs.
        The 'Q2' pin is in one of the cycles. (LBDB-140)
```

Feedback

### What Next

Check the library source file, and verify the questionable circular timing arcs.

---

## LBDB-141

(warning) The timing arc is missing the piecewise data value for\n \t'%s'. The default value is used if it exists.

### Description

This message indicates that you did not specify the definition of either delay intercept attributes or pin resistance attributes of the piece number defined in the piece_define in a CMOS piecewise linear delay model library.

The following example shows an instance where this message occurs:

```
library(test) {
  delay_model : "generic_cmos";
  piece_define("0");

  cell(lbdb138) {
    area : 5.0;
    pin(A) {
      direction : input;
      capacitance : 1.0;
    }
    pin(Z) {
      direction : output;
      function : "A'";

      timing() {
        intrinsic_rise : 5.0;
        intrinsic_fall : 2.0;
        slope_rise : 1.0;
        slope_fall : 2.0;

        rise_delay_intercept(0, "1.1");

        rise_pin_resistance(0, "1.1");
        fall_pin_resistance(0, "4.4");

        related_pin : "A";
      }
    }
  }
```

In this case, the value of the fall_delay_intercept attribute is missing for index 2.

Feedback

The following is an example message:

```
Warning: Line 170, The timing arc is missing the piece wise data value
 for
        'fall_delay_intercept(0)'. The default value is used if it
 exists. (LBDB-141)
```

### What Next

Add the missing attribute to the specified piece number.

---

## LBDB-142

(error) '%s' is an invalid value for the '%s'\n \tenumerated type attribute.

### Description

This message indicated that you specified a value that is not part of the enumerated literals for the specified attribute.

The following example shows an instance where this message occurs:

```
voltage_unit : "1";
```

The following is an example message:

```
Error: Line 19, '1' is an invalid value for the 'voltage_unit'
        enumerated type attribute. (LBDB-142)
```

### What Next

Refer to the "Library Compiler User Guide", and fix the invalid value.

---

## LBDB-143

(error) An invalid string is provided. The invalid string is\n \teither a blank string or a string that begins with a digit\n \tthat is unquoted.

### Description

A string begining with a digit must be enclosed in double quotes (""). In addition, blank strings are not valid in Library Compiler.

The following example shows an instance where this message occurs:

```
type(111) {
    base_type : array;
    data_type : bit;
    bit_width : 2;
    bit_from : 1;
    bit_to : 0;
```

Feedback

```
    downto : true;
  }
```

In this case, the 111 type name starts with a digit. To fix the problem, add quotes.

The following is an example message:

```
Error: Line 29, An invalid string is provided. The invalid string is
        either a blank string or a string that begins with a digit
        that is unquoted. (LBDB-143)
```

**What Next**

Check the library and correct the string. If the string begins with a digit, enclose the string in quotes. Remove any blank strings. For details on Library Compiler string rules, refer to the "Library Compiler Reference Manual".

---

## LBDB-144

(error) The '%s' pin name does not match the '%s' bus name.

**Description**

This message indicates that you specified a bus element name different from the bus name given in the bus_naming_style. Library Compiler cannot extract the element name properly.

**What Next**

Make sure that the bus_naming_style string matches the specified bus name.

---

## LBDB-145

(warning) The 'direction' attribute is missing in the %s '%s'.

**Description**

This message indicates that the direction attribute is missing in the bus or bundle group.

The following example shows an instance where this message occurs:

```
bundle(Q) {
      members(XQ, Q1);
/*      direction : output; */
      function : "1";
    }
```

The following is an example message:

```
Warning: Line 53, The 'direction' attribute is missing in the 'Q' bundle.
 (LBDB-145)
```

Feedback

### What Next

Add the direction attribute to the bus or bundle port in the technology library.

---

## LBDB-146

(error) The value for the update attribute must be "true" or "false".

### Description

This message indicates that, in a symbol library, you specified an invalid value to the *update* attribute in either an *annotate_symbol* or an *annotate* group. This might be caused by a typo because the Library Compiler expects a quoted string.

The following example shows an instance where this message occurs:

```
annotate_symbol() {
      value( "ANN_PAGE_NUM", "ANN_NUM_PAGES") ;
      format : "sheet:  %s of %s" ;
      x : RIGHT_X ;
      y : LOW_Y ;
      layer_name : "template_text_layer" ;
      update : true;
    }
```

The following is an example message:

```
Error: Line 95, The value for the update attribute must be "true" or
 "false". (LBDB-146)
```

### What Next

Check your symbol library file, and correct the value of the update attribute.

---

## LBDB-147

(error) A value must be specified for the annotate or\n \tthe annotate_symbol group.

### Description

This message indicates that, in a symbol library, you did not specify a value to either the *annotate* or the *annotate_symbol* group.

The following example shows an instance where this message occurs:

```
annotate_symbol() {
      format : "sheet:  %s of %s" ;
      x : RIGHT_X ;
      y : LOW_Y ;
      layer_name : "template_text_layer" ;
    }
```

Feedback

In this case, the value attribute is missing. To fix the problem, add the attribute,

```
value( "ANN_PAGE_NUM", "ANN_NUM_PAGES") ;
```

The following is an example message:

```
Error: Line 89, A value must be specified for the annotate or
        the annotate_symbol group. (LBDB-147)
```

**What Next**

Check your symbol library file, and add the value attribute.

---

## LBDB-148

(error) The '%s' pin is not found in the annotate_symbol group.

**Description**

This message indicates that, in a symbol library, you specified an invalid pin_name value for the *pin_name* attribute in an *annotate_symbol*. This might be caused by a typo.

The following example shows an instance where this message occurs:

```
symbol("lbdb148") {
    set_minimum_boundary(0 , 0 , 8000, 12000);
    pin("P0", 10000, 1000, DOWN);

    line(0, 0, 0, 12000);
    line(0, 12000, 8000, 12000);
    line(8000, 12000, 8000, 0);
    line(8000, 0, 0, 0);
    annotate_symbol() {
      value( "ANN_PAGE_NUM", "ANN_NUM_PAGES") ;
      pin_name : "P2";
      format : "sheet:  %s of %s" ;
      x : RIGHT_X ;
      y : LOW_Y ;
      layer_name : "template_text_layer" ;
    }

  }
```

In this case, the 'P2' name does not exist in the 'lbdb148' symbol. Change the name to P0.

The following is an example message:

```
Error: Line 89, The 'P2' pin is not found in the annotate_symbol group.
 (LBDB-148)
```

Feedback

### What Next

Check your symbol library file, and correct the value of the pin_name attribute.

---

## LBDB-149

(error) The X value is invalid in the annotate or\n \tthe annotate_symbol group.

### Description

This message indicates that, in a symbol library, you specified an invalid value to the *X* attribute in either an *annotate* or *annotate_symbol* group. This might be caused by a typo.

The following example shows an instance where this message occurs:

```
symbol("lbdb148") {
    set_minimum_boundary(0 , 0 , 8000, 12000);
    pin("P0", 10000, 1000, DOWN);

    line(0, 0, 0, 12000);
    line(0, 12000, 8000, 12000);
    line(8000, 12000, 8000, 0);
    line(8000, 0, 0, 0);
    annotate_symbol() {
      value( "ANN_PAGE_NUM", "ANN_NUM_PAGES") ;
      format : "sheet:  %s of %s" ;
      x : "";
      y : LOW_Y ;
      layer_name : "template_text_layer" ;
    }

  }
```

The following is an example message:

```
Error: Line 89, The X value is invalid in the annotate or
        the annotate_symbol group. (LBDB-149)
```

### What Next

Check your symbol library file, and correct the value of the X attribute.

---

## LBDB-150

(error) The format specification cannot have more than %d %%s specifications.

### Description

This message indicates that, in a symbol library, the format attribute has more than 10 string specifications. Library Compiler accepts only 10.

Feedback

The following example shows an instance where this message occurs:

```
annotate_symbol() {
      value( "A1","A2","A3","A4","A5","A6","A7","A8","A9","A10" ) ;
      format : "sheet:  %s of %s %s of %s %s of %s %s of %s %s of %s %s
 of %s " ;
      x : RIGHT_X ;
      y : LOW_Y ;
      layer_name : "template_text_layer" ;
    }
```

The following is an example message:

```
Error: Line 101, The format specification cannot have more than 10 %s
 specifications. (LBDB-150)
```

### What Next

Check your symbol library file, and reduce the number of specification in the format attribute.

---

## LBDB-151

(error) The format specification has %d %%s specification(s),\n \tbut only %d value(s) is/are specified.

### Description

This message indicates that, in a symbol library, the format specification does not match the number of values provided to either the *annotate* or the *annotate_symbol* group. This might be caused by a typo.

The following example shows an instance where this message occurs:

```
annotate_symbol() {
      value( "ANN_PAGE_NUM") ;
      format : "sheet:  %s of %s" ;
      x : RIGHT_X ;
      y : LOW_Y ;
      layer_name : "template_text_layer" ;
    }
```

In this case, the value attribute has only the string value, but the format refers to two %s. To fix the problem, either add the second string value in the value attribute or remove the second %s in the format attribute.

The following is an example message:

```
Error: Line 89, The format specification has 2 %s specification(s),
        but only 1 value(s) is/are specified. (LBDB-151)
```

Feedback

### What Next

Check your symbol library file, and correct the value of either the value attribute or the format attribute.

---

## LBDB-152

(error) The '%s' object type is invalid in the annotate\n \tor the annotate_symbol group.

### Description

This message indicates that, in a symbol library, you specified an invalid object type for the *object_type* attribute in either an *annotate* or *annotate_symbolgroup. This might be caused by a typo. Library Compiler accepts the following object types:*

```
* pin
        * design
        * port
        * cell
        * net
```

The following example shows an instance where this message occurs:

```
symbol("lbdb152") {
    set_minimum_boundary(0 , 0 , 8000, 12000);
    pin("P0", 10000, 1000, DOWN);

    line(0, 0, 0, 12000);
    line(0, 12000, 8000, 12000);
    line(8000, 12000, 8000, 0);
    line(8000, 0, 0, 0);
    annotate_symbol() {
      value( "ANN_PAGE_NUM", "ANN_NUM_PAGES") ;
      format : "sheet:  %s of %s" ;
      x : "RIGHT_X";
      y : LOW_Y ;
      layer_name : "template_text_layer" ;
      object_type : "library";
    }

  }
```

The following is an example message:

```
Error: Line 89, The 'library' object type is invalid in the annotate
        or the annotate_symbol group. (LBDB-152)
```

### What Next

Check your symbol library file, and correct the value of the object_type attribute.

Feedback

## LBDB-153

(error) A syntax error is found before the library or phys_library group.\n \tThe compilation is terminated.

### Description

This message follows lexical and syntax error messages issued when reading a library.

The following example shows an instance where this message occurs:

```
related_bus_pins : { ADDR";
```

In this case, a lexical error is encountered in the related_bus_pins value. Substitute the '{' for a "".

The following is an example message:

```
Error: A syntax error is found before the library or phys_library group.
        The compilation is terminated. (LBDB-153)
```

### What Next

Fix all lexical and syntax errors.

## LBDB-155

(warning) The %s_cell for the '%s' cell with '%s'\n \toperating_conditions is defined multiple times in the library.\n \tUsing the last one encountered.

### Description

This message indicates that a scaled_cell with the same operating_conditions is defined multiple times in the library. This might be caused by a typo in the name of the operating_conditions. Library Compiler uses the last cell encountered.

The following example shows an instance where this message occurs:

```
library(lbdb155) {
    operating_conditions(WCCOM) {
        process : 1.5 ;
        temperature : 70 ;
        voltage : 4.75 ;
        tree_type : "worst_case_tree" ;
    }
    cell(IVV) {
      area : 1;
      pin(A) {
        direction : input;
        capacitance : 1;
      }
```

Feedback

```
    pin(Z) {
      direction : output;
      function : "A'";
      timing() {
        intrinsic_rise : 0.1;
        intrinsic_fall : 0.1;
        rise_resistance : 0.1;
        fall_resistance : 0.1;
        slope_rise : 0.0;
        slope_fall : 0.0;
        related_pin : "A";
      }
    }
  }

  scaled_cell(IVV,WCCOM) {
    area : 1;
    pin(A) {
      direction : input;
      capacitance : 1;
    }
    pin(Z) {
      direction : output;
      function : "A'";
      timing() {
        intrinsic_rise : 0.3;
        intrinsic_fall : 0.3;
        rise_resistance : 0.3;
        fall_resistance : 0.3;
        slope_rise : 0.0;
        slope_fall : 0.0;
        related_pin : "A";
      }
    }
  scaled_cell(IVV,WCCOM) {
    area : 1;
    pin(A) {
      direction : input;
      capacitance : 1;
    }
    pin(Z) {
      direction : output;
      function : "A'";
      timing() {
        intrinsic_rise : 0.3;
        intrinsic_fall : 0.3;
        rise_resistance : 0.3;
        fall_resistance : 0.3;
        slope_rise : 0.0;
        slope_fall : 0.0;
        related_pin : "A";
      }
    }
```

Feedback

```
    }
  }
```

The following is an example message:

```
Warning: Line 327, The scaled_cell for the 'AND' cell with 'WCCOM'
         operating_conditions is defined multiple times in the library.
         Using the last one encountered. (LBDB-155)
```

**What Next**

Delete all the redundant scaled_cell groups, or fix the library if it is a typo.

---

## LBDB-156

(error) The LSI rounding digit and cutoff attributes\n \thave incompatible values.

**Description**

This message indicates that you specified invalid values for either the *lsi_rounding_digit* or the *lsi_rounding_cutoff* attribute. Library Compiler issues this error when the lsi_rounding_digit value is greater than the lsi_rounding_cutoff value multiplied by 10.01 or the lsi_rounding_cutoff value is greater than the lsi_rounding_digit value.

The following example shows an instance where this message occurs:

```
lsi_rounding_digit : 0.1;
   lsi_rounding_cutoff : 0.003;
```

In this case, the lsi_rounding_digit 0.1 is greater than (0.003 * 10.01). To fix the problem, assign the value 0.01 to lsi_rounding_digit.

```
lsi_rounding_digit : 0.01;
```

The following is an example message:

```
Error: Line 10, The LSI rounding digit and cutoff attributes
        have incompatible values. (LBDB-156)
```

**What Next**

Check the library, and fix either the value of the lsi_rounding_digit or the lsi_rounding_cutoff attribute.

---

## LBDB-157

(error) The '%s' cell name defined in the scaled_cell is not found.

Feedback

### Description

This message indicates that the technology library has a scaled_cell defined without a parent cell defined. Library Compiler uses the scaled_cell group to supply an alternate set of values for an existing cell. The choice is based on the set of operating conditions used.

The following example shows an instance where this message occurs:

```
library(lbdb157) {
    operating_conditions(WCCOM) {
        process : 1.5 ;
        temperature : 70 ;
        voltage : 4.75 ;
        tree_type : "worst_case_tree" ;
    }

cell(IV) {
  area : 1;
  pin(A) {
    direction : input;
    capacitance : 1;
  }
  pin(Z) {
    direction : output;
    function : "A'";
    timing() {
      intrinsic_rise : 0.1;
      intrinsic_fall : 0.1;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      slope_rise : 0.0;
      slope_fall : 0.0;
      related_pin : "A";
    }
  }
}
scaled_cell(IVV,WCCOM) {    /* Typo in the cell name. */
  area : 1;
  pin(A) {
    direction : input;
    capacitance : 1;
  }
  pin(Z) {
    direction : output;
    function : "A'";
    timing() {
      intrinsic_rise : 0.3;
      intrinsic_fall : 0.3;
      rise_resistance : 0.3;
      fall_resistance : 0.3;
      slope_rise : 0.0;
      slope_fall : 0.0;
      related_pin : "A";
```

Feedback

```
    }
  }
}
}
```

The following is an example message:

```
Error: Line 101, The 'IVV' cell name defined in the scaled_cell is not
 found. (LBDB-157)
```

**What Next**

Add the parent cell for the scaled_cell, or fix the scaled_cell name if it is a typo.

---

## LBDB-158

(error) The '%s' operating conditions is not found.

**Description**

This message indicates that the technology library has a scaled_cell with an undefined operating conditions.

The following example shows an instance where this message occurs:

```
library(lbdb158) {
    operating_conditions(WCCOM) {
        process : 1.5 ;
        temperature : 70 ;
        voltage : 4.75 ;
        tree_type : "worst_case_tree" ;
    }

cell(IV) {
  area : 1;
  pin(A) {
    direction : input;
    capacitance : 1;
  }
  pin(Z) {
    direction : output;
    function : "A'";
    timing() {
      intrinsic_rise : 0.1;
      intrinsic_fall : 0.1;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      slope_rise : 0.0;
      slope_fall : 0.0;
      related_pin : "A";
    }
  }
```

Feedback

```
}
scaled_cell(IV,WCCOM1) {    /* Typo in the operating_conditions name. */
  area : 1;
  pin(A) {
    direction : input;
    capacitance : 1;
  }
  pin(Z) {
    direction : output;
    function : "A'";
    timing() {
      intrinsic_rise : 0.3;
      intrinsic_fall : 0.3;
      rise_resistance : 0.3;
      fall_resistance : 0.3;
      slope_rise : 0.0;
      slope_fall : 0.0;
      related_pin : "A";
    }
  }
}
}
```

The following is an example message:

```
Error: Line 305, The 'WCCOM1' operating conditions is not found.
 (LBDB-158)
```

### What Next

Add the operating_conditions group if it is missing, or fix the operating_conditions name if it has a typo.

---

## LBDB-159

(error) Incomplete EDIF properties are specified in the '%s' symbol.\n \tSpecify 'edif_cell_name', 'edif_view_name', and 'edif_name_property'.

### Description

This message indicates that, in a symbol library, you specified incomplete attributes of the view identifier properties of the symbol for a cell. These attributes are *edif_cell_name*, *edif_view_name*, and *edif_name_property*. If Library Compiler encounters a symbol with any of these attributes, it must have all three of them declared.

The following example shows an instance where this message occurs:

```
symbol("lbdb159") {
    set_minimum_boundary(0 , 0 , 8000, 12000);
    pin("P0", 10000, 1000, DOWN);
```

Feedback

```
    line(0, 0, 0, 12000);
    line(0, 12000, 8000, 12000);
    line(8000, 12000, 8000, 0);
    line(8000, 0, 0, 0);
    edif_cell_name : "edifcell";
    edif_name_property : " edifproperty";
  }
```

In this case, the edif_view_name attribute is missing. To fix the problem, add

```
edif_view_name : "edifview";
```

The following is an example message:

```
Error: Incomplete EDIF properties are specified in the 'lbdb159' symbol.
       Specify 'edif_cell_name', 'edif_view_name', and
 'edif_name_property'. (LBDB-159)
```

### What Next

Change the symbol definition for the cell in the source text file of the symbol library. Either delete the view identifier property attributes that are there so that the symbol has none of those attributes or add appropriate view identifier property attributes, so that the symbol has all three of those attributes.

---

## LBDB-160

(error) It is not acceptable to set the technology to\n \t'%s' after it has been set to '%s'. The two\n \ttechnologies are incompatible.

### Description

This message indicates that you specified incompatible values for the technology attribute and the delay_model attribute.

The following example shows an instance where this message occurs:

```
library(lbdb160) {
    /* wrong technology and delay model combination */
    technology("cmos");
    delay_model : generic_ecl;
}
```

The following is an example message:

```
Error: Line 4, It is not acceptable to set the technology to
        'generic_ecl' after it has been set to 'cmos'. The two
        technologies are incompatible. (LBDB-160)
```

Feedback

### What Next

Check The Library Compiler User Guide Manual for compatible combinations, and fix either the value of the technology or the delay_model attribute.

---

## LBDB-161

(error) The '%s' bus needs to have its bus_type specified first.\n \tThe %s '%s' was found before the bus_type.

### Description

This message indicates that you specified a bus without its bus_type attribute, or Library Compiler rejected the bus_type value and considered it as not defined.

The following example shows an instance where this message occurs:

```
bus (D) {
/*    bus_type : bus2; */       /* remove the comment to fix the problem
 */
      direction : input;
      capacitance : 1.0;
    }
```

The following is an example message:

```
Error: Line 179, The 'D' bus needs to have its bus_type specified first.
       The attribute 'direction' was found before the bus_type.
 (LBDB-161)
```

### What Next

Add the bus_type attribute to the library, or fix the value of the attribute.

---

## LBDB-162

(error) An invalid area range is found in the wire_load_from_area attribute.

### Description

This message indicates that you specified an invalid area range. The problem is caused by any of these reasons:

```
* The min_area value is greater than the max_area value.
        * The min_area value is less than zero.
        * The max_area value is less or equal to zero.
```

The following example shows an instance where this message occurs:

```
wire_load("05x05") {
    resistance : 0 ;
```

Feedback

```
    capacitance : 1 ;
    area : 0 ;
    slope : 0.186 ;
    fanout_length(1,0.39) ;
  }
  wire_load_selection(test) {
    wire_load_from_area(27,10,"10x10");          /* problem */
    wire_load_from_area(27,100,"10x10");
    wire_load_from_area(0,28,"05x05");
    wire_load_from_area(2,10,"15x10");
  }
```

In this case, the minimum area value 27 is greater than the maximum area value 10.

The following is an example message:

```
Error: Line 51, An invalid area range is found in the wire_load_from_area
 attribute. (LBDB-162)
```

### What Next

Check the area values, and fix the problem.

---

## LBDB-163

(warning) The '%s' attribute value is\n \t %s (%3.1f). Using %3.1f instead.

### Description

This message indicates that you specified an attribute value that is out of the accepted range. The value is either less than the minimum value or greater than the maximum value.

The following example shows an instance where this message occurs:

```
k_volt_internal_power : -111.0;
```

In this case, the k_volt_internal attribute's value -111.0 is less than the minimum accepted value -100.0.

The following is an example message:

```
Warning: Line 9, The 'k_volt_internal_power attribute' attribute value is
         less than the minimum allowed  (-100.0). Using 0.0 instead.
 (LBDB-163)
```

### What Next

Change the attribute value to satisfy the value range.

Feedback

## LBDB-164

(error) The 'wire_load_from_area' range overlaps\n \tthe range in line %d.

### Description

This message indicates that you specified an overlapping area range in two *wire_load_from_area* attributes.

The following example shows an instance where this message occurs:

```
wire_load("05x05") {
    resistance : 0 ;
    capacitance : 1 ;
    area : 0 ;
    slope : 0.186 ;
    fanout_length(1,0.39) ;
  }

  wire_load_selection(test) {
    wire_load_from_area(0,20,"05x05");
    wire_load_from_area(10,25,"05x05");
  }
```

In this case, the area range [0,20] overlaps the area range [10,25].

The following is an example message:

```
Error: Line 12, The 'wire_load_from_area' range overlaps
        the range in line 13. (LBDB-164)
```

### What Next

Change the area range in the second *wire_load_from_area* attribute.

## LBDB-165

(warning) A range gap is found in the 'wire_load_selection'.\n \tThe 'min_area' is extended from %f to %f.

### Description

The wire_load_selection specified in this library has a gap. It means that two adjacent selector items are not overlapped. Library Compiler uses its built-in algorithm to fill the gap. It extends the lower bound of the selector that covers the bigger area toward the upper bound of the other selector item.

The following example shows an instance where this message occurs:

```
wire_load("05x05") {
    resistance : 0 ;
```

Feedback

```
    capacitance : 1 ;
    area : 0 ;
    slope : 0.186 ;
    fanout_length(1,0.39) ;
  }

  wire_load_selection(test) {
    wire_load_from_area(0,20,"05x05");
    wire_load_from_area(25,50,"05x05");
  }
```

In this case, there is a gap in area range from 20 to 25. Library Compiler corrects the problem as if you had the declaration

```
wire_load_from_area(20,50,"05x05");
```

The following is an example message:

```
Warning: Line 13, A range gap is found in the 'wire_load_selection'.
        The 'min_area' is extended from 25.000000 to 20.000000.
 (LBDB-165)
```

**What Next**

Ignore the warning if the Library Compiler's algorithm satisfies your modeling requirement. Otherwise, fix the library source code to match your wire_load_selection model.

---

## LBDB-166

(warning) A range does not start with 0.0 in the 'wire_load_selection'.\n \tThe 'min_area' is extended from %f to 0.0.

**Description**

The min_area value in the specified wire_load_selection attribute in this library does not start with 0.0. Library Compiler extends the lower bound of the selector to 0.0.

The following example shows an instance where this message occurs:

```
wire_load("05x05") {
    resistance : 0 ;
    capacitance : 1 ;
    area : 0 ;
    slope : 0.186 ;
    fanout_length(1,0.39) ;
  }

  wire_load_selection(test) {
    wire_load_from_area(10,25,"05x05");
  }
```

Feedback

The following is an example message:

```
Warning: Line 48, A range does not start with 0.0 in the
 'wire_load_selection'.
        The 'min_area' is extended from 10.000000 to 0.0. (LBDB-166)
```

### What Next

Fix the library source code to set the min_area to 0.0.

---

## LBDB-167

(error) The '%s' scaling factors group is not found.

### Description

The message indicates that you specified an invalid name for the scaling_factors group. This might be caused either by a typo in the name or by the group not being defined.

The following example shows an instance where this message occurs:

```
scaling_factors("IO_PAD_SCALING") {
        k_volt_intrinsic_rise : 0.846 ;
    }
    cell (lbdb167) {
       area : 0 ;
       scaling_factors : IO_PAD_SCALE ;
    }

In this case, there is a typo in the name.

The following is an example message:
Error: Line 20, The 'IO_PAD_SCALE' scaling factors group is not found.
 (LBDB-167)
```

### What Next

Add the scaling_factors group to the library, or correct the value of the scaling_factors if it is a typo.

---

## LBDB-168

(error) The %s timing constraint has only one value.\n \tOnly one 'intrinsic_rise' or 'intrinsic_fall' can be specified.

### Description

You specified invalid values for intrinsic_rise and intrinsic_fall in a timing group that describes a *recovery* or *removal* timing constraint. Library Compiler complains if each of these two constraints has

Feedback

```
* Both intrinsic_rise and intrinsic_fall values are assigned to zero.
        * Both intrinsic_rise and intrinsic_fall values are not
 specified.
        * Both intrinsic_rise and intrinsic_fall values are not zero.
```

The following example shows an instance where this message occurs:

```
pin(CD) {
      direction : input;
      capacitance : 2;
      timing() {
        timing_type : recovery_rising;
        /* both values are 0.0 */
        intrinsic_rise : 0.0;
        intrinsic_fall : 0.0;
        related_pin : "CP";
      }
      timing() {
        timing_type : removal_rising;
        /* both values not specified */
        related_pin : "CP";
      }
    }
```

The following is an example message:

```
Error: Line 56, The Recovery timing constraint has only one value.
        Only one 'intrinsic_rise' or 'intrinsic_fall' can be specified.
 (LBDB-168)
```

### What Next

Inspect the datasheet of the cell in question. If the asynchronous control signal is high-active, define the recovery and the removal constraints as the intrinsic_fall. If the asynchronous control signal is low-active, define the recovery and the removal constraints as the intrinsic_rise. Remove the other field (intrinsic_rise or intrinsic_fall) from the library source file.

---

## LBDB-169

(error) An invalid 'three_state_disable' timing type appears\n \ton the timing arc. Its parent output pin has no 'three_state' attribute.

### Description

This message indicates that you specified a *three_state_disable* timing_arc on a port that has no *three_state* attribute.

Feedback

The following example shows an instance where this message occurs:

```
cell(AND) {
    area : 1;
    pin(A B) {
      direction : input;
      capacitance : 1;
    }
    pin(Z) {
      direction : output;
      function : "A B";
      timing() {
        intrinsic_rise : 0.1;
        intrinsic_fall : 0.1;
        rise_resistance : 0.1;
        fall_resistance : 0.1;
        related_pin : "A B" ;
      }
      timing() {
        timing_type : three_state_disable;
        intrinsic_rise : 0.1;
        intrinsic_fall : 0.1;
        rise_resistance : 0.1;
        fall_resistance : 0.1;
        related_pin : "A";
      }
    }
  }
```

The following is an example message:

```
Error: Line 44, An invalid 'three_state_disable' timing type appears
         on the timing arc. Its parent output pin has no 'three_state'
 attribute. (LBDB-169)
```

**What Next**

Either delete the timing arc or correct the port group definition to add the three_state attribute.

---

## LBDB-170

(warning) The '%s' attribute's value is\n \t%s (%3.1f). Removing it.

**Description**

This message indicates that you specified an attribute value that is out of the accepted range. The value is either less than the minimum value or greater than the maximum value. Library Compiler ignores the value because it does not find a default value to replace it.

Feedback

The following example shows an instance where this message occurs:

```
pin(A) {
      direction : input;
      capacitance : 1;
      min_fanout : -1;
    }
```

In this case, the min_fanout attribute's value -1 is less than the minimum accepted value 0.0.

The following is an example message:

```
Warning: Line 283, The 'min_fanout' attribute value is
         less than the minimum allowed (0.0). Removing it. (LBDB-170)
```

### What Next

Change the attribute value to satisfy the value range.

---

## LBDB-171

(error) The '%s' wire load model is not defined, or \n \tit is defined after this line.

### Description

This message indicates that you specified a "wire_load_from_area" that refers to a "wire_load" group that is not defined or has been defined in the library after the current line.

The following example shows an instance where this message occurs:

```
wire_load("05x05") {
    resistance : 0 ;
    capacitance : 1 ;
    area : 0 ;
    slope : 0.186 ;
    fanout_length(1,0.39) ;
  }

  wire_load_selection(test) {
    wire_load_from_area(27,100,"10x10");
    wire_load_from_area(10,25,"05x05");
  }
```

The following is an example message:

```
Error: Line 47, The '10x10' wire load model is not defined, or
        it is defined after this line. (LBDB-171)
```

Feedback

### What Next

Check your library to see if you have defined the group. If you have done that, also check its position within the library to make sure it is defined before it is used.

---

## LBDB-172

(warning) The '%s' attribute is not specified. Using %4.2f.

### Description

This message indicates that an attribute is missing, and Library Compiler is reporting the default floating point or integer value it is setting. This message affects attributes with default values.

The following example shows an instance where this message occurs:

```
pin(D) {
      direction : input;
    }
```

The following is an example message:

```
Warning: Line 52, The 'capacitance' attribute is not specified. Using
 0.00. (LBDB-172)
```

### What Next

Either add the missing attribute to the technology library or ignore the message.

---

## LBDB-173

(error) The '%s' group requires one or more names.

### Description

This message indicates you specified no name for a group that requires at least one name.

The following example shows an instance where this message occurs:

```
pin() {
      direction : input;
      capacitance : 1.0;
   }
```

In this case, a pin name is required.

The following is an example message:

```
Error: Line 286, The 'pin' group requires one or more names. (LBDB-173)
```

Feedback

### What Next

Change your library to add the group name.

---

## LBDB-174

(error) The '%s' group requires %s names.

### Description

This error message occurs when a group is specified without its required number of names. This might be caused by a typographical error.

The following example shows an instance where this message occurs: In the following example, the ff group requires 2 names. To fix the problem, either remove the last element 2, or if it is a typographical error, change ff to ff_bank.

```
ff(IQ,IQN, 2)
```

### What Next

See the *Library Compiler User Guide* for the correct syntax of the specified group, and match the number of group names.

---

## LBDB-175

(warning) The bus_naming_style contains characters that\n \tare also used as delimiters in function, three_state, and related_pin\n \tattributes. Be sure to put spaces around the characters in the [%s] set\n \twhen supplying values for attributes of the types mentioned. \n \tAlso, surround subscripted pin names with double quotes in pin groups.

### Description

This message indicates that you specified delimiter characters in the *bus_naming_style* attribute such as "() +*|&!^'".

The following example shows an instance where this message occurs:

```
bus_naming_style : %s&%d;
```

The following is an example message:

```
Warning: Line 28, The bus_naming_style contains characters that
        are also used as delimiters in function, three_state, and
 related_pin
        attributes. Be sure to put spaces around the characters in the
 [&] set
        when supplying values for attributes of the types mentioned.
        Also, surround subscripted pin names with double quotes in pin
 groups. (LBDB-175)
```

Feedback

**What Next**

Change the delimiter character, or follow the rules specified in the message.

---

## LBDB-176

(error) Invalid bus syntax is detected in '%s'.

**Description**

This message indicates that you specified an invalid bus element. Library Compiler fails to extract the bus element information using the bus_naming_style attribute value.

**What Next**

Change the library by either fixing the bus_naming_style value or fixing the bus element representation.

---

## LBDB-177

(warning) The attribute '%s' is not specified Using '%s'.

**Description**

This message indicates that an attribute is missing, and Library Compiler is reporting the default string value it is setting. This message affects attributes with default values.

**What Next**

Either add the missing attribute to the technology library or ignore the message.

---

## LBDB-178

(error) Duplicated %s group. A group with similar '%s' attribute value is found at line %u

**Description**

This message indicates that this group conflicts with an earlier group because the attribute indicated has similar value in both of them.

The following example shows an instance where this message occurs:

```
receiver_capacitance() {
        receiver_capacitance_rise {
          segment : 1;
          ...
        }
        receiver_capacitance_rise {
          segment : 1;
```

Feedback

```
        ...
      }
      receiver_capacitance_rise {
        segment : 3;
        ...
      }
      ...
    }
```

In this case, the second receiver_capacitance_rise group is in error because its 'segment' attribute conflicts with the first group.

The following is an example message:

```
Error: Line 20014, Duplicated 'receiver_capacitance_rise' group. A group
 with
similar 'segment' attribute value is found at line 19876
```

### What Next

Change the library file by modify the indicated attribute.

---

## LBDB-179

(error) The '%s' group requires the '%s' attribute.\n \tEither the attribute is missing or the attribute has an invalid value.

### Description

This message indicates you specified a group without one of its required attributes. Library Compiler rejects the attribute definition if the attribute exists and has an invalid value.

The following example shows an instance where this message occurs:

```
memory() {
        type : random;
        address_width : 10;
        word_width : 8;
      }
```

In this case, the 'type' has an invalid value. To fix the problem, assign 'rom' or 'ram' to the type.

The following is an example message:

```
Error: Line 27, The 'memory' group requires the 'type' attribute.
        Either the attribute is missing or the attribute has an invalid
 value. (LBDB-179)
```

### What Next

Change the library file by adding the missing attribute to the group.

Feedback

## LBDB-180

(error) The '%s' attribute requires both\n \t'clear' and 'preset' attributes.

### Description

This message indicates that you specified the 'clear_preset_var1' or the 'clear_preset_var2' attribute or both in a sequential model group without specifying both the 'clear' and the 'preset' attributes.

The following example shows an instance where this message occurs:

```
ff("IQ","IQN") {
        next_state : "D";
        clocked_on : "CP";
        clear : "CD'";
        clear_preset_var1: "L";
      }
```

In this case, the 'preset' attribute is missing in the ff group.

The following is an example message:

```
Error: Line 98, The 'clear_preset_var1' attribute requires both
        'clear' and 'preset' attributes. (LBDB-180)
```

### What Next

Add the missing attribute to the group.

## LBDB-181

(error) The '%s' group, with both 'clear' and 'preset' attributes,\n \trequires 'clear_preset_var1' and/or 'clear_preset_var2' attributes.

### Description

This message indicates that you specified both the 'clear' and the 'preset' attribute in a sequential model group without specifying both the 'clear_preset_var1' or the 'clear_preset_var2' attributes.

The following example shows an instance where this message occurs:

```
ff("IQ","IQN") {
        next_state : "D";
        clocked_on : "CP";
        preset : "SD'";
        clear : "CD";
```

In this case, the 'clear_preset_var1' and or 'clear_preset_var2' attributes are missing.

Feedback

The following is an example message:

```
Error: Line 93, The 'ff' group, with both 'clear' and 'preset'
 attributes,
         requires 'clear_preset_var1' and/or 'clear_preset_var2'
 attributes. (LBDB-181)
```

### What Next

Add either 'clear_preset_var1' or 'clear_preset_var2' to the sequential group.

---

## LBDB-182

(error) Invalid %s name '%s' is detected. This name must be\n \tunique among all pin names, bus names, bundle names, and rail connection names.

### Description

This message indicates a duplicate pin, bus, bundle, a rail_connection, or a voltage_state_range name in the library.

The following example shows an instance where this message occurs:

```
pin(D) {
      direction : input;
      capacitance : 1.0;
    }
    pin(D) {
      direction : input;
      capacitance : 1.0;
    }
```

The following is an example message:

```
Error: Line 56, Invalid pin name 'D' is detected. This name must be
        unique among all pin names, bus names, and bundle names.
 (LBDB-182)
```

### What Next

Change the name of the pin, bus, bundle, rail_connection, voltage_state_range in the technology library.

---

## LBDB-183

(error) Missing '%s' group with '%s' attribute of '%s' value.

Feedback

### Description

This message indicates that a group with a specific attribute value is missing. This problem is associated with certain type of information that requires data to be given in a series of groups. These groups are "tagged" with a specific attribute with a consecutive value.

The following example shows an instance where this message occurs:

```
receiver_capacitance() {
        receiver_capacitance_rise {
          segment : 1;
          ...
        }
        receiver_capacitance_rise {
          segment : 3;
          ...
        }
        receiver_capacitance_rise {
          segment : 4;
          ...
        }
        ...
      }
```

In the example above, receiver_capacitance_rise with segment = 2 is missing.

The following is an example message:

```
Error: Line 20073, Missing 'receiver_capacitance_rise' group with
 'segment' attribute of value '2'.
```

### What Next

Fix the library by adding the missing group.

---

## LBDB-186

(error) Invalid '%s' pin name is detected in the '%s' bundle.

### Description

This message indicates that you specified an invalid pin name in a bundle group.

### What Next

Change the library source file by correcting the pin name.

Feedback

## LBDB-187

(error) The '%s' bundle needs to have its 'members' specified first.\n \tThe '%s' %s is found before a 'members'.

### Description

This message indicates that you specified the members attribute in a bundle group after another attribute. The *members* attribute has to be first in the bundle group.

The following example shows an instance where this message occurs:

```
bundle (DD) {
        direction : input;
        members(D1, D2);
        capacitance : 1;
      }
```

The following is an example message:

```
Error: Line 94, The 'DD' bundle needs to have its 'members' specified
 first.
       The 'direction' attribute is found before a 'members'. (LBDB-187)
```

### What Next

Change the library source file by specifying the members attribute first in the bundle group.

## LBDB-188

(error) The 'members' attribute is defined multiple times.

### Description

This message indicates that you specified the *members* attribute in a bundle group more than once. Library Compiler expects only one *members* attribute in the bundle group.

The following example shows an instance where this message occurs:

```
bundle (DD) {
        direction : input;
        members(D1, D2);
        members(D3, D4);
        capacitance : 1;
      }
```

The following is an example message:

```
Error: Line 95, The 'members' is defined multiple times. (LBDB-188)
```

Feedback

### What Next

Change the library source file by specifying only one members attribute in a bundle group.

---

## LBDB-189

(warning) Level shifters are required for this library.\n \tA level shifter is a buffer or inverter with differing\n \tconnection_class values specified between input and output\n \tpins. Design Compiler cannot produce valid designs using this\n \tlibrary if a level shifter component is not provided.

### Description

This warning message occurs when your library contains connection classes that cannot be used without level shifters. Design Compiler uses these components to shift a net from one connection class to another and does so to remove connection class violations from your design. Multi-input or multi-output cells with differing connection classes on inputs and outputs are not considered level shifters. Following this warning message, a list appears of those connection classes that do not have valid shifters between them.

This warning alerts you to situations where Design Compiler might not be able to validate your design for connection class. For example, suppose that Library Compiler warns that you are missing a level shifter with an input connection class of "a" and an output connection class of "b". If there is a pin or port in your design with a connection class of "b" connected to a driver with a connection class of "a", Design Compiler might be unable to validate this connection without a level shifting component that converts voltage levels from an input connection class of "a" to an output connection class of "b".

There are cases where this message can be too restrictive. For example, consider the case where you, the designer, know that all inputs to your design will only have connection class "a" and that all outputs will only have connection class "c". Meanwhile, for the purposes of our example, assume that all "internal" logic in the design will have connection class "b". If you defined a library with two level shifters (one with "a" on the input and "b" on the output, and the other with "b" on the input and "c" on the output), Design Compiler can validate the design because it can convert voltage levels from the inputs at level "a" to internal logic at level "b", and from internal logic at level "b" to the outputs at level "c". In this case, the warnings that you receive about missing level shifters would be too restrictive. However, you have an output that is at level "a", and is driven by internal logic at level "b", Design Compiler is not be able to validate the design unless it can replace the component driving the output port with a component that has level "a" on its output pin.

Be aware of the current limitations in the connection class validation and be advised that you can construct a library with connection rule restrictions that are too difficult for Design Compiler to solve. Future versions of the software might remove these restrictions, but Library Compiler currently informs you of the limitations on validating a design for connection class violations.

Feedback

The following example shows an instance where this message occurs:

```
cell(lbdb189) {
      /* Should only be connected to IO Cells only */
      area : 0.0;
      dont_touch : false;
      dont_use   : false;
      pad_cell : true;
      pin(PAD ) {
        is_pad : true;
        input_voltage : CMOS;
        direction   : input;
        capacitance : 1.0;
        fanout_load : 0.0;
      }
      pin(Y ) {
        connection_class : "iopcl";
        direction   : output;
        function    : "PAD";
        max_fanout  : 1.0;
        timing() {
          intrinsic_fall  : 1.0;
          intrinsic_rise  : 1.0;
          fall_resistance : 0.0;
          rise_resistance : 0.0;
          related_pin  :"PAD ";
        }
      }
    }
```

The following is an example message:

```
Warning: Line 1, Level shifters are required for this library.
        A level shifter is a buffer or inverter with differing
        connection_class values specified between input and output
        pins. Design Compiler cannot produce valid designs using this
        library if a level shifter component is not provided. (LBDB-189)
```

**What Next**

If this message appeared because you omitted some level shifting components from your library, add the components.

---

## LBDB-190

(warning) %s level shifter with input connection class '%s' \n and output connection class '%s' is needed.

Feedback

### Description

This message indicates that a (noninverting or inverting) level shifter is needed between the two connection classes. See the man page for error LBDB-189 for more information on how this message can occur.

The following example shows an instance where this message occurs:

```
cell(IOPCLBUF) {
      /* Should only be connected to IO Cells only */
      area : 0.000000;
      dont_touch : false;
      dont_use   : false;
      pad_cell : true;
      pin(PAD ) {
        is_pad : true;
        input_voltage : CMOS;
        direction   : input;
        capacitance : 35.000000;
        fanout_load : 0.000000;
      }
      pin(Y ) {
        connection_class : "iopcl";
        direction   : output;
        function    : "PAD";
        max_fanout  : 16.000000;
        timing() {
          intrinsic_fall  : 3.925000;
          intrinsic_rise  : 3.925000;
          fall_resistance : 0.000000; /* delay included in intrinsic */
          rise_resistance : 0.000000; /* delay included in intrinsic */
          related_pin  :"PAD ";
        }
      }
    }
```

The following is an example message:

```
Warning: Line 1, Noninverting level shifter with input connection class
 'iopcl'
and output connection class 'default' is needed. (LBDB-190)
```

### What Next

If you obtained this message because you omitted a level shifting component from your library, add the level shifting component.

---

## LBDB-191

(error) The inout '%s' pin, bus, or bundle has\n \tno 'three_state' function.

Feedback

### Description

The pin whose direction is specified as 'inout' has only a 'function' statement. The 'three_state' statement is also required by an "inout" pin.

The following example shows an instance where this message occurs:

```
pin(Z) {
      direction : inout;
      function : "A B";
    }
```

The following is an example message:

```
Error: Line 356, The inout 'Z' pin, bus, or bundle has
        no 'three_state' function. (LBDB-191)
```

### What Next

If the pin is actually an output pin, change the direction attribute to 'output'. Otherwise, add the 'three_state' statement to the pin.

---

## LBDB-192

(error) Illegal interdependence data specified in pin "%s".

### Description

timing arc with "interdependence_id" attribute is treated as interdependence data in pin. they are used for setup/hold pessimism reduction. There are some rules to specified the id for interdependence data (These rules applies to timing arcs with same condition):

1. Interdependece data can not be the first timing arc in this pin. This it to prevent potential backward compatibility issue.
2. Interdependece data should be in pair. That is, if there is a setup_rising interdependence data with id = 1 in the pin, a hold_rising interdependence data with same id = 1 is needed. Same case for timing type = setup_falling / hold_falling.
3. For one timing_type arcs, the interdependence id should be unique. E.g. If there are two setup_rising interdependence data with same id = 1, this error is issued.
4. The interdependence id starts from 1, and if multiple interdependence data defined for the pin, the id should be consecutive. E.g. id = 1,2,3 is okay, but 1, 2, 4 is not.

The following example shows an instance where this message occurs:

```
pin(D) {
        direction : input;
        timing() {
            timing_type : setup_rising;
```

Feedback

```
            fall_constraint("template") {
            ...
            }
            rise_constraint("template") {
            ...
            }
            related_in: "CLK";
        }
        timing() {
            timing_type : setup_rising;
            interdependence_id : 1;
            fall_constraint("template") {
            ...
            }
            rise_constraint("template") {
            ...
            }
            related_in: "CLK";
        }
        timing() {
            timing_type : hold_rising;
            fall_constraint("template") {
            ...
            }
            rise_constraint("template") {
            ...
            }
            related_in: "CLK";
        }
        timing() {
            timing_type : hold_rising;
            interdependence_id : 2;
            fall_constraint("template") {
            ...
            }
            rise_constraint("template") {
            ...
            }
            related_in: "CLK";
        }
    }
```

The following is an example message:

```
Error: Line 356, Illegal interdependence data specified in pin
 "D".(LBDB-192)
```

### What Next

Check the id value for interdependence data in the pin group, make sure all above requirments are satisfied.

Feedback

## LBDB-193

(error) Interdependence data is defined for wrong timing type arc in pin "%s".

### Description

Timing group with "interdependence_id" attribute is treated as interdependence data in pin. they are used to for setup/hold pessimism reduction. For now, It's only supported for these timing types: setup_rising, hold_rising, setup_falling, hold_falling.

The following example shows an instance where this message occurs:

```
pin(D) {
        direction : input;
        timing() {
            timing_type : setup_rising;
            fall_constraint("template") {
            ...
            }
            rise_constraint("template") {
            ...
            }
            related_in: "CLK";
        }
        timing() {
            timing_type : skew_rising; // not support type
            interdependence_id : 1;
            fall_constraint("template") {
            ...
            }
            rise_constraint("template") {
            ...
            }
            related_in: "CLK";
        }
        timing() {
            timing_type : hold_rising;
            fall_constraint("template") {
            ...
            }
            rise_constraint("template") {
            ...
            }
            related_in: "CLK";
        }
        timing() {
            timing_type : hold_rising;
            interdependence_id : 2;
            fall_constraint("template") {
            ...
            }
            rise_constraint("template") {
```

Feedback

```
            ...
            }
            related_in: "CLK";
        }
    }
```

The following is an example message:

```
Error: Line 356, Interdependence data is defined for wrong timing type
 arc in pin "D".(LBDB-193)
```

**What Next**

Check the interdependence data in the pin group, make sure they are defined for constraint arcs with right timing type.

---

## LBDB-200

(error) The cell has both the 'pad_cell' and the 'auxiliary_pad_cell' attributes.

**Description**

This message indicates that you specified both the *pad_cell* and the *auxiliary_pad_cell* attributes in the same cell.

The following example shows an instance where this message occurs:

```
cell(lbdb200){
      area : 0.000000;
      pad_cell : true;
      auxiliary_pad_cell : true;
    }
```

The following is an example message:

```
Error: Line 3577, The cell has both the 'pad_cell' and the
 'auxiliary_pad_cell' attributes. (LBDB-200)
```

**What Next**

Remove either the pad_cell attribute or the auxiliary_pad_cell attribute.

---

## LBDB-201

(error) A nonpad '%s' cell has a '%s' attribute.

**Description**

This message indicates that you specified a pad_type attribute in a nonpad cell.

Feedback

The following example shows an instance where this message occurs:

```
cell(lbdb201){
      area : 0.0;
      pad_type : clock;
      pin(Z)  {
        direction : output
        function : "A"
      }
      pin(A) {
        direction : input;
        capacitance : 1.0;
    }
  }
```

The following is an example message:

```
Error: Line 3577, A nonpad 'lbdb201' cell has a 'pad_type' attribute.
 (LBDB-201)
```

### What Next

Check the "Library Compiler User Guide" for information on pad cells. Either remove the invalid attribute or add the pad_type, the pad_cell, or the auxiliary_pad_cell attribute.

---

## LBDB-202

(error) A nonpad '%s' cell has a '%s' pin/pg_pin with a '%s' attribute.

### Description

This error message occurs when a pad pin/pg_pin attribute is specified in a nonpad cell.

The following example shows an instance where this message occurs: In the following example, there is an *is_pad* attribute at the pin level, but there is no pad attribute at the cell level, so the *read_lib* command issues the error message:

```
cell(lbdb202){
      area : 0.0;
      pin(Z)  {
        is_pad : true;
        direction : output
        drive_current : 2.0;
        function : "A"
      }
      pin(A) {
        direction : input;
        capacitance : 1.0;
    }
  }
```

Feedback

The following is an example message: The following is an example of the error message:

```
Error: Line 3581, A nonpad 'lbdb202' cell has a 'Z' pin/pg_pin
       with a 'is_pad' attribute. (LBDB-202)
```

### What Next

Either remove the invalid attribute or add the *pad_cell* attribute. See the Library Compiler documentation for information on pad cells.

### See Also

- read_lib

---

## LBDB-203

(error) The nonpad '%s' pin has a '%s' attribute.

### Description

This message indicates that you specified a pad cell with pad_type and pad_cell attributes defined and a nonpad pin with a hysteresis attribute.

The following example shows an instance where this message occurs:

```
cell(lbdb203){
    area : 0.0;
    pad_type : clock;
    pad_cell : true;
    pin(Z)  {
      direction : output;
      hysteresis : true;
      function : "A";
    }
    pin(A) {
      direction : input;
      capacitance : 1.0;
    }
  }
```

The following is an example message:

```
Error: Line 3584, The nonpad 'Z' pin has a 'hysteresis' attribute.
 (LBDB-203)
```

### What Next

Refer to the "Library Compiler User Guide" for information on pad cells. Either remove the pad cell attributes or add the pad pin attributes.

Feedback

## LBDB-204

(error) The '%s' %s pin cannot have a '%s' attribute.

### Description

This message indicates that you specified a pad pin attribute on a pin with an invalid direction.

The following example shows an instance where this message occurs:

```
cell(lbdb204){
    area : 0.0;
    pad_cell : true;
    pin(Z)  {
      direction : output;
      is_pad : true;
      hysteresis : true;
      function : "A";
    }
    pin(A) {
      direction : input;
      capacitance : 1.0;
    }
  }
```

The following is an example message:

```
Error: Line 3584, The 'Z' output pin cannot have a 'hysteresis'
 attribute. (LBDB-204)
```

### What Next

Refer to the "Library Compiler User Guide" for information on pad cells. Either change the pin's direction or move the attribute to the correct pin.

## LBDB-205

(warning) The '%s' pad pin is missing a '%s' attribute.

### Description

This message indicates that you specified an output or inout pad pin without a *drive_current* attribute.

The following example shows an instance where this message occurs:

```
cell(lbdb205){
    area : 0.0;
    pad_cell : true;
    pin(Z)  {
```

Feedback

```
      direction : output;
      is_pad : true;
      hysteresis : true;
      function : "A";
    }
    pin(A) {
      direction : input;
      capacitance : 1.0;
    }
  }
```

The following is an example message:

```
Warning: Line 3596, The 'Z' pad pin is missing a 'drive_current'
 attribute. (LBDB-205)
```

### What Next

Refer to the "Library Compiler User Guide" for information on pad cells and pins. Add the missing attribute to the pin.

---

## LBDB-206

(error) The '%s' pad cell has no pad pins.

### Description

This message indicates that you specified a pad cell with pad_type and pad_cell attributes defined but you did not specify pad pin attributes associated with any pin in the cell.

The following example shows an instance where this message occurs:

```
cell(lbdb206){
    area : 0.0;
    pad_type : clock;
    pad_cell : true;
    pin(Z)  {
      direction : output
      function : "A"
    }
    pin(A) {
      direction : input;
      capacitance : 1.0;
    }
  }
```

The following is an example message:

```
Error: Line 3575, The 'lbdb206' pad cell has no pad pins. (LBDB-206)
```

Feedback

### What Next

Refer to the "Library Compiler User Guide" for information on pad cells. Remove either the pad cell attributes or add the pad pin attributes.

---

## LBDB-207

(warning) The '%s' pad cell has more than one pad pin. \n

### Description

This message indicates that you specified more than one pad pin in a pad cell.

The following example shows an instance where this message occurs:

```
cell(lbdb207) {
    area : 0.0;
    pad_cell : true;
    bond_pads : 1;
    driver_sites : 1;
    pin(PAD ) {
      direction : input;
      is_pad : true;
      input_voltage : CMOS;
      capacitance : 1.0;
    }
    pin(PAD1 ) {
      direction : input;
      is_pad : true;
      input_voltage : CMOS;
      capacitance : 1.0;
    }
    pin(Y ) {
      direction : output;
      function : "PAD PAD1";
      timing() {
        intrinsic_fall : 1.0;
        intrinsic_rise : 1.0;
        fall_resistance : 0.50;
        rise_resistance : 0.50;
        related_pin :"PAD PAD1";
      }
    }
  }
```

In this case, there are two pad pins: 'PAD' and 'PAD1'.

The following is an example message:

```
Warning: Line 33, The 'lbdb207' pad cell has more than one pad pin.
 (LBDB-207)
```

Feedback

### What Next

Refer to the "Library Compiler User Guide" for information on pad cells.

---

## LBDB-208

(error) The %s '%s' pad cell cannot be a 'clock' pad.

### Description

This message indicates you specified an output pad pin with the is_pad attribute on a clock pad. Library Compiler accepts clock pads only on input pins.

The following example shows an instance where this message occurs:

```
cell(lbdb208) {
    area : 0.0;
    pad_cell : true;
    pad_type : clock;
    pin(PAD ) {
      direction : input;
      capacitance : 1.0;
    }
    pin(Y ) {
      direction : output;
      is_pad : true;
      function : "PAD";
      output_voltage : CMOS_OUT;
      timing() {
        intrinsic_fall : 1.0;
        intrinsic_rise : 1.0;
        fall_resistance : 0.0;
        rise_resistance : 0.0;
        related_pin :"PAD";
      }
    }
  }
```

The following is an example message:

```
Error: Line 65, The output 'lbdb208' pad cell cannot be a 'clock' pad.
 (LBDB-208)
```

### What Next

Refer to the "Library Compiler User Guide" for information on pad cells. Check the library source file, and specify the is_pad attribute to an input pin.

Feedback

## LBDB-209

(error) A '%s' attribute cannot be specified on a\n \tpin unless a pull_up or pull_down driver_type is specified.

### Description

This message indicates that you specified a *pulling_resistance* or a *pulling_current* attribute on a pin without specifying the pullup or pulldown driver type.

The following example shows an instance where this message occurs:

```
pin(A) {
      direction : output;
      pulling_resistance : 100;
      ...
    }
```

In this case, the driver_type is missing. To fix the problem, add the statement,

```
driver_type : pull_up;
```

The following is an example message:

```
Error: Line 355, A 'pulling_resistance' attribute cannot be specified on
 a
       pin unless a pull_up or pull_down driver_type is specified.
 (LBDB-209)
```

### What Next

Change the library file by either adding the driver_type attribute to the specified pin or removing the pulling_resistance or pulling_current attribute.

## LBDB-210

(error) The pulling_current value cannot be 0.0.

### Description

This message indicates that you specified a zero value for the *pulling_current* attribute.

The following example shows an instance where this message occurs:

```
pulling_current : 0.0;
```

The following is an example message:

```
Error: Line 357, The pulling_current value cannot be 0.0. (LBDB-210)
```

Feedback

### What Next

Change the value of the attribute to a nonzero value.

---

## LBDB-211

(error) The '%s' driver_type cannot be specified on a\n \tpin that already has a %s driver_type specified.

### Description

This message indicates that you specified an incompatible set of driver types on a pin. Library Compiler fails if the driver_type value includes the following combinations:

```
* pull_up and pull_down
        * pull_up and bus_hold
        * pull_down and bus_hold
        * open_drain and open_source
        * open_drain and bus_hold
        * open_source and bus_hold
```

The following example shows an instance where this message occurs:

```
cell(lbdb211) {
    area : 2;
    pin(A) {
      direction : input;
      capacitance : 1.0;
    }
    pin(Z2) {
      direction : output;
      function : "A";
      driver_type : "pull_up pull_down";
      timing() {
        intrinsic_rise : 0.1;
        intrinsic_fall : 0.1;
        related_pin : "A";
      }
    }
  }
```

The following is an example message:

```
Error: Line 84, The 'pull_down' driver_type cannot be specified on a
       pin that already has a 'pull_up' driver_type specified. (LBDB-211)
```

### What Next

Refer to the "Library Compiler User Guide" for driver_type information. Remove the incompatible values of the driver_type attribute.

Feedback

## LBDB-212

(error) The 'open_%s' driver_type cannot be specified on\n \tan input pin.

### Description

This message indicates that you specified an invalid *open_source* or *open_drain* driver_type value on an input_pin.

The following example shows an instance where this message occurs:

```
cell(lbdb212) {
    area : 2;
    pin(A) {
      direction : input;
      capacitance : 1.0;
      driver_type : "open_drain";
    }
    pin(Z2) {
      direction : output;
      function : "A";
      timing() {
        intrinsic_rise : 0.1;
        intrinsic_fall : 0.1;
        related_pin : "A";
      }
    }
  }
```

The following is an example message:

```
Error: Line 78, The 'open_drain' driver_type cannot be specified on
       an input pin. (LBDB-212)
```

### What Next

Refer to the "Library Compiler User Guide" for driver_type information. Remove the invalid value of the driver_type attribute.

## LBDB-213

(error) The '%s' area attribute cannot be specified\n \ton a pin of a cell that is not a pad cell.

### Description

This message indicates that you specified a user-defined cell area value for pad cells and applied it to a nonpad cell.

Feedback

The following example shows an instance where this message occurs:

```
define_cell_area(my_area,pad_slots);

    cell(NON_PAD_CELL) {
      area : 1;
      my_area : 10;
    }
```

The following is an example message:

```
Error: Line 29, The 'my_area' area attribute cannot be specified
       on a pin of a cell that is not a pad cell. (LBDB-213)
```

**What Next**

Change the library file, and either make the cell a pad cell or remove the pad information from the cell.

---

## LBDB-214

(error) The '%s' area attribute of '%s' type\n \tcannot be specified on a pin of a cell that is not a pad cell\n \tor an auxiliary pad cell.

**Description**

This message indicates that you specified a user-defined cell area value for pad cells or auxiliary pad cells and applied it to a nonpad cell.

The following example shows an instance where this message occurs:

```
define_cell_area(your_area,pad_driver_sites);

    cell(NON_PAD_CELL) {
      area : 1;
      your_area : 10;
    }
```

The following is an example message:

```
Error: Line 29, The 'your_area' area attribute of 'pad_driver_sites' type
       cannot be specified on a pin of a cell that is not a pad cell
       or an auxiliary pad cell. (LBDB-214)
```

**What Next**

Change the library file and either make the cell a pad cell or remove the pad information from the cell.

Feedback

## LBDB-215

(error) The '%s' attribute cannot be specified\n \ton the '%s' input pin.

### Description

This message indicates that you specified an invalid attribute on an input pin. Library Compiler fails if the following attributes are associated with an input pin.

```
* A driver_type attribute whose value is bus_hold
        * A slew_control attribute
        * An edge_rate_rise attribute
        * An edge_rate_fall attribute
        * An edge_rate_rise_load attribute
        * An edge_rate_fall_load attribute
        * A reference_resistance attribute
```

The following example shows an instance where this message occurs:

```
pin(B) {
            direction : input;
            capacitance : 1.0;
            fanout_load : 1.0;

            edge_rate_breakpoint_r0 : 0.000;
            edge_rate_breakpoint_f0 : 0.000;
            edge_rate_breakpoint_r1 : 0.010;
            edge_rate_breakpoint_f1 : 0.010;

            edge_rate_rise : 1.0;
            edge_rate_load_rise : 1.0;
            edge_rate_fall     : 1.0;
            edge_rate_load_fall : 1.0;
         }
```

The following is an example message:

```
Error: Line 110, The 'edge_rate_rise' attribute cannot be specified
       on the 'B' input pin. (LBDB-215)
```

### What Next

Check the library source file, and delete the invalid attribute from the input pin.

## LBDB-216

(error) The '%s' scaled_cell's '%s' pin has a different\n \tnumber of timing arcs than the corresponding pin on the '%s' cell.

Feedback

### Description

This message indicates that a timing group is missing in one of the output pins of either the specified cell or the scaled_cell.

The following example shows an instance where this message occurs:

```
library(lbdb216) {
    operating_conditions(WCCOM) {
        process : 1.5 ;
        temperature : 70 ;
        voltage : 4.75 ;
        tree_type : "worst_case_tree" ;
    }
    cell(IVV) {
      area : 1;
      pin(A) {
        direction : input;
        capacitance : 1;
      }
      pin(Z) {
        direction : output;
        function : "A'";
        timing() {
          intrinsic_rise : 0.1;
          intrinsic_fall : 0.1;
          rise_resistance : 0.1;
          fall_resistance : 0.1;
          slope_rise : 0.0;
          slope_fall : 0.0;
          related_pin : "A";
        }
      }
    }

    scaled_cell(IVV,WCCOM) {
      area : 1;
      pin(A) {
        direction : input;
        capacitance : 1;
      }
      pin(Z) {
        direction : output;
        function : "A'";
      }
    }
  }
```

In this case, add the following timing group to the 'Z' pin in the scaled_cell to fix the problem.

```
timing() {
          intrinsic_rise : 0.3;
```

Feedback

```
          intrinsic_fall : 0.3;
          rise_resistance : 0.3;
          fall_resistance : 0.3;
          slope_rise : 0.0;
          slope_fall : 0.0;
          related_pin : "A";
        }
```

The following is an example message:

```
Error: Line 40, The 'WCCOM' scaled_cell's 'Z' pin has a different
       number of timing arcs than the corresponding pin on the 'IVV'
 cell. (LBDB-216)
```

### What Next

Add the missing timing group to the specified output pin in either the cell or the scaled_cell.

---

## LBDB-217

(error) The '%s' cell's '%s' pin has a timing arc without\n \ta counterpart on the scaled_cell(%s,%s).

### Description

Each timing arc in the regular cell needs a matched timing arc on the same pin in the scaled_cell group. This message indicates that a matching timing arc is not found in the library source file.

The following example shows an instance where this message occurs:

```
library(lbdb217) {
    operating_conditions(WCCOM) {
        process : 1.5 ;
        temperature : 70 ;
        voltage : 4.75 ;
        tree_type : "worst_case_tree" ;
    }
    cell(AND) {
      area : 1;
      pin(A B) {
        direction : input;
        capacitance : 1;
      }
      pin(Z) {
        direction : output;
        function : "A B";
        timing() {
          intrinsic_rise : 0.1;
          intrinsic_fall : 0.1;
```

Feedback

```
            rise_resistance : 0.1;
            fall_resistance : 0.1;
            slope_rise : 0.0;
            slope_fall : 0.0;
            related_pin : "A B";
          }
        }
      }

      scaled_cell(AND,WCCOM) {
        area : 1;
        pin(A B) {
          direction : input;
          capacitance : 1;
        }
        pin(Z) {
          direction : output;
          function : "A B";
          timing() {
            intrinsic_rise : 0.1;
            intrinsic_fall : 0.1;
            rise_resistance : 0.1;
            fall_resistance : 0.1;
            slope_rise : 0.0;
            slope_fall : 0.0;
            related_pin : "A";
          }
          timing() {
            intrinsic_rise : 0.3;
            intrinsic_fall : 0.3;
            rise_resistance : 0.3;
            fall_resistance : 0.3;
            slope_rise : 0.0;
            slope_fall : 0.0;
            related_pin : "A ";
          }
        }
      }
    }
```

In this case, The timing arc between 'Z' and 'A' is defined once in the 'AND' cell and twice in the 'AND' scaled_cell. To fix the problem, remove one of the timing arcs.

The following is an example message:

```
Error: Line 42, The 'AND' cell's 'Z' pin has a timing arc without
       a counterpart on the scaled_cell(AND,WCCOM). (LBDB-217)
```

**What Next**

Find the missing timing group in the scaled_cell and add it, or remove one of the extra timing groups in the cell.

Feedback

## LBDB-218

(warning) The 'direction' of the '%s' scaled_cell pin \n \tdoes not match that of the same pin name on the '%s' cell.\n \tResetting the scaled_cell pin's direction to match both directions.

### Description

This message indicates that a direction attribute of a scaled_cell pin is different from the same pin on the primary cell. Library Compiler matches the direction of the scaled_cell pin to the direction of the primary pin.

The following example shows an instance where this message occurs:

```
cell(AND) {
    area : 1;
    pin(A B C) {
      direction : input;
      capacitance : 1;
    }
    pin(Z) {
      direction : inout;
      function : "A B";
      three_state : "C";
      timing() {
        intrinsic_rise : 0.1;
        intrinsic_fall : 0.1;
        rise_resistance : 0.1;
        fall_resistance : 0.1;
        related_pin : "A B C";
      }
      timing() {
        timing_type : three_state_disable;
        intrinsic_rise : 0.1;
        intrinsic_fall : 0.1;
        rise_resistance : 0.1;
        fall_resistance : 0.1;
        related_pin : "C";
      }
    }
    }
    scaled_cell(AND,WCCOM) {
      area : 1;
      pin(A B C) {
        direction : input;
        capacitance : 1;
      }
      pin(Z) {
        direction : output;
        function : "A B";
        timing() {
          intrinsic_rise : 0.1;
          intrinsic_fall : 0.1;
```

Feedback

```
            rise_resistance : 0.1;
            fall_resistance : 0.1;
            slope_rise : 0.0;
            slope_fall : 0.0;
            related_pin : "A B";
          }
        }
    }
```

The following is an example message:

```
Warning: Line 313, The 'direction' of the 'Z' scaled_cell pin
         does not match that of the same pin name on the 'AND' cell.
         Resetting the scaled_cell pin's direction to match both
 directions. (LBDB-218)
```

### What Next

Check the library source file, and make sure the directions are the same.

---

## LBDB-219

(error) The (%s,%s) scaled_cell has duplicate\n \ttiming arcs for the '%s' pin. It is unclear\n \twhich arc corresponds to the arcs in the '%s' cell.

### Description

This message indicates that a scaled_cell has duplicate timing arcs between two pins. Library Compiler fails to recognize the equivalent timing arc in the primary cell.

### What Next

Check the library source file, and remove the second timing arc.

---

## LBDB-220

(error) The '%s' %s_voltage group has no '%s' attribute\n \tspecified. This attribute is essential in defining valid voltages.

### Description

This message indicates that an input_voltage or an output_voltage group has one of its attributes missing. For an input_voltage group, Library Compiler fails if any of the vil, vih, vimin, or vimax attributes is missing. For an output_voltage group, Library Compiler fails if any of the vol, voh, vomin, or vomax attributes is missing.

The following example shows an instance where this message occurs:

```
output_voltage(CMOS_OUT) {
    vil : 1.5;
```

Feedback

```
    vih : 3.5;
    vimin : -0.3;
    vimax : VDD + 0.3;
  }
```

In this case, either there is a typo where the output_voltage is specified instead of input_voltage, or the vol, voh, vomin, and vomax are not specified for the output_voltage.

The following is an example message:

```
Error: Line 31, The 'CMOS_OUT' output_voltage group has no 'vol'
 attribute
       specified. This attribute is essential in defining valid voltages.
 (LBDB-220)
```

### What Next

Check the library source file, and add the missing attribute to the voltage group.

---

## LBDB-221

(warning) The '%s' %s_voltage group has a %s value,\n \twhich is %s %s.

### Description

This message indicates that an *input_voltage* or an *output_voltage* has inconsistent values for its attributes. Library Compiler issues warnings in these cases:

```
* vil value is less than vimin value.
        * vih value is greater than vimax value.
        * vol value is less than vomin value.
        * voh value is greater than vomax value.
```

The following example shows an instance where this message occurs:

```
input_voltage(CMOS) {
    vil : -1.5;
    vih : 3.5;
    vimin : -0.3;
    vimax : VDD + 0.3;
  }
```

In this case, the vil value is less than the vimin value.

The following is an example message:

```
Warning: Line 25, The 'CMOS' input_voltage group has a vil value,
        which is less than vimin. (LBDB-221)
```

Feedback

### What Next

Change the specified attribute value to be consistent with the minimum or the maximum values.

---

## LBDB-222

(error) The fpga_family attribute is required when\n \tthe fpga_cell_type attribute is specified.

### Description

This message indicates that in an fpga cell you specified an *fpga_cell_type* attribute without the *fpga_family* attribute.

The following example shows an instance where this message occurs:

```
cell( lbdb222 ) {
        area : 1;
        fpga_cell_type : CLB;

        pin( K ) {
            direction : input;
            capacitance : 0.0;
        }
        ...
    }
```

In this case, to fix the problem, add the statement to the cell.

```
fpga_family : "x4000";
```

The following is an example message:

```
Error: Line 28, The fpga_family attribute is required when
        the fpga_cell_type attribute is specified. (LBDB-222)
```

### What Next

Add the missing fpga_family attribute to the cell.

---

## LBDB-223

(error) The fpga_cell_type attribute is required when\n \tthe fpga_family attribute is specified.

### Description

This message indicates that in an fpga cell you specified an *fpga_family* attribute without the *fpga_cell_type* attribute.

Feedback

The following example shows an instance where this message occurs:

```
cell(lbdb223) {
        area : 1;
        fpga_family : "x4000";

        pin( K ) {
            direction : input;
            capacitance : 0.0;
        }
        ...
    }
```

In this case, to fix the problem, add the statement to the cell.

```
fpga_cell_type : CLB;
```

The following is an example message:

```
Error: Line 28, The fpga_cell_type attribute is required when
        the fpga_family attribute is specified. (LBDB-222)
```

### What Next

Add the missing fpga_cell_type attribute to the cell.

---

## LBDB-224

(error) The fpga_timing_type is required on all arcs \n \twhen fpga_cell_type and fpga_family are specified.

### Description

This message indicates that you did not specify *fpga_timing_type* groups in a cell with the *fpga_family* and the *fpga_cell_type* attributes.

The following example shows an instance where this message occurs:

```
cell(INV) {
      fpga_family : "x4000";
      fpga_cell_type : CLB;
      area : 1;
      pin(X) {
        function : "A'";
        direction : output;
        timing() {
          intrinsic_rise  : 0.1;
          intrinsic_fall  : 0.1;
          rise_resistance : 1.0;
          fall_resistance : 1.0;
          related_pin : "A";
        }
```

Feedback

```
      }
      pin(A) {
        direction : input;
        capacitance :  0.1;
        fanout_load :  0.1;
      }
    }
```

To fix the problem, add the attribute to the timing group.

```
timing() {
          fpga_timing_type : ICK;
          intrinsic_rise  : 0.1;
          intrinsic_fall  : 0.1;
          rise_resistance : 1.0;
          fall_resistance : 1.0;
          related_pin : "A";
        }
```

The following is an example message:

```
Error: Line 208, The fpga_timing_type is required on all arcs
        when fpga_cell_type and fpga_family are specified. (LBDB-224)
```

### What Next

Add the missing fpga_timing_type groups to all the pins of the cell.

---

## LBDB-225

(error) The fpga_timing_type is invalid when the fpga_cell_type\n \tand the fpga_family attributes are not defined on a cell.

### Description

This message indicates that you specified fpga_timing_type in a timing group of a pin without defining the fpga_family and the fpga_cell_type attributes at the cell level.

The following example shows an instance where this message occurs:

```
cell(lbdb225) {
      area : 1;
      pin(X) {
        function : "A'";
        direction : output;
        timing() {
          fpga_timing_type : ICK;
          intrinsic_rise  : 0.1;
          intrinsic_fall  : 0.1;
          rise_resistance : 1.0;
          fall_resistance : 1.0;
          related_pin : "A";
```

Feedback

```
        }
      }
      pin(A) {
        direction : input;
        capacitance :  0.1;
        fanout_load :  0.1;
      }
    }
```

To fix the problem, add the sample statements at the cell level.

```
fpga_family : "x4000";
      fpga_cell_type : CLB;
```

The following is an example message:

```
Error: Line 30, The fpga_timing_type is invalid when the fpga_cell_type
        and the fpga_family attributes are not defined on a cell.
 (LBDB-225)
```

### What Next

Add the missing attribute to the library file.

---

## LBDB-226

(error) The FPGA CLB cell type from the x2000 family requires\n \tA, B, C, D, K, X, and Y pins.

### Description

This message indicates that you specified an incomplete CLB FPGA cell with missing pins from the x2000 family.

The following example shows an instance where this message occurs:

```
cell(lbdb226) {
      fpga_family : "x2000";
      fpga_cell_type : CLB;
      area : 1;
      pin(X) {
        function : "A'";
        direction : output;
        timing() {
          fpga_timing_type : ICK;
          intrinsic_rise  : 0.1;
          intrinsic_fall  : 0.1;
          rise_resistance : 1.0;
          fall_resistance : 1.0;
          related_pin : "A";
        }
      }
```

Feedback

```
      pin(A) {
        direction : input;
        capacitance :  0.1;
        fanout_load :  0.1;
      }
    }
```

The following is an example message:

```
Error: Line 22, The FPGA CLB cell type from the x2000 family requires
        A, B, C, D, K, X, and Y pins. (LBDB-226)
```

### What Next

Add the missing pins to the cell.

---

## LBDB-227

(error) The FPGA IOB cell type from the x2000 family requires\n \tO, T, K, I, and PAD pins.

### Description

This message indicates that you specified an incomplete IOB FPGA cell with missing pins from the x2000 family.

The following example shows an instance where this message occurs:

```
cell(lbdb227) {
      fpga_family : "x2000";
      fpga_cell_type : IOB;
      area : 1;
      pin(X) {
        function : "A'";
        direction : output;
        timing() {
          fpga_timing_type : ICK;
          intrinsic_rise  : 0.1;
          intrinsic_fall  : 0.1;
          rise_resistance : 1.0;
          fall_resistance : 1.0;
          related_pin : "A";
        }
      }
      pin(A) {
        direction : input;
        capacitance :  0.1;
        fanout_load :  0.1;
      }
    }
```

Feedback

The following is an example message:

```
Error: Line 22, The FPGA IOB cell type from the x2000 family requires
        O, T, K, I, and PAD pins. (LBDB-227)
```

### What Next

Add the missing pins to the cell.

---

## LBDB-228

(error) The FPGA CLB cell type from the x3000 family requires \n \tA, B, C, D, E, K, EC, DI, RD, X, Y, and GSR pins.

### Description

This message indicates that you specified an incomplete CLB FPGA cell with missing pins from the x3000 family.

The following example shows an instance where this message occurs:

```
cell(lbdb228) {
      fpga_family : "x3000";
      fpga_cell_type : CLB;
      area : 1;
      pin(X) {
        function : "A'";
        direction : output;
        timing() {
          fpga_timing_type : ICK;
          intrinsic_rise  : 0.1;
          intrinsic_fall  : 0.1;
          rise_resistance : 1.0;
          fall_resistance : 1.0;
          related_pin : "A";
        }
      }
      pin(A) {
        direction : input;
        capacitance :  0.1;
        fanout_load :  0.1;
      }
    }
```

The following is an example message:

```
Error: Line 22, The FPGA CLB cell type from the x3000 family requires
        A, B, C, D, E, K, EC, DI, RD, X, Y, and GSR pins. (LBDB-228)
```

### What Next

Add the missing pins to the cell.

Feedback

## LBDB-229

(error) The FPGA IOB cell type from the x3000 family requires \n \tO, T, IK, OK, I, Q, PAD, and GSR pins.

### Description

This message indicates that you specified an incomplete IOB FPGA cell with missing pins from the x3000 family.

The following example shows an instance where this message occurs:

```
cell(lbdb229) {
      fpga_family : "x3000";
      fpga_cell_type : IOB;
      area : 1;
      pin(X) {
        function : "A'";
        direction : output;
        timing() {
          fpga_timing_type : ICK;
          intrinsic_rise  : 0.1;
          intrinsic_fall  : 0.1;
          rise_resistance : 1.0;
          fall_resistance : 1.0;
          related_pin : "A";
        }
      }
      pin(A) {
        direction : input;
        capacitance :  0.1;
        fanout_load :  0.1;
      }
    }
```

The following is an example message:

```
Error: Line 22, The FPGA IOB cell type from the x3000 family requires
        O, T, IK, OK, I, Q, PAD, and GSR pins. (LBDB-229)
```

### What Next

Add the missing pins to the cell.

## LBDB-230

(error) The FPGA CLB cell type from the x4000 family requires\n \tF1-F4, G1-G4, C1-C4, K, X, Y, XQ, YQ, GSR, CIN, and COUT pins.

Feedback

### Description

This message indicates that you specified an incomplete CLB FPGA cell with missing pins from the x4000 family.

The following example shows an instance where this message occurs:

```
cell(lbdb230) {
      fpga_family : "x4000";
      fpga_cell_type : CLB;
      area : 1;
      pin(X) {
        function : "A'";
        direction : output;
        timing() {
          fpga_timing_type : ICK;
          intrinsic_rise  : 0.1;
          intrinsic_fall  : 0.1;
          rise_resistance : 1.0;
          fall_resistance : 1.0;
          related_pin : "A";
        }
      }
      pin(A) {
        direction : input;
        capacitance :  0.1;
        fanout_load :  0.1;
      }
    }
```

The following is an example message:

```
Error: Line 22, The FPGA CLB cell type from the x4000 family requires
        F1-F4, G1-G4, C1-C4, K, X, Y, XQ, YQ, GSR, CIN, and COUT pins.
 (LBDB-230)
```

### What Next

Add the missing pins to the cell.

---

## LBDB-231

(error) The FPGA IOB cell type from the x4000 family requires\n \tO, T, IK, OK, I1, I2, PAD, and GSR pins.

### Description

This message indicates that you specified an incomplete IOB FPGA cell with missing pins from the x4000 family.

Feedback

The following example shows an instance where this message occurs:

```
cell(lbdb231) {
      fpga_family : "x4000";
      fpga_cell_type : IOB;
      area : 1;
      pin(X) {
        function : "A'";
        direction : output;
        timing() {
          fpga_timing_type : ICK;
          intrinsic_rise  : 0.1;
          intrinsic_fall  : 0.1;
          rise_resistance : 1.0;
          fall_resistance : 1.0;
          related_pin : "A";
        }
      }
      pin(A) {
        direction : input;
        capacitance :  0.1;
        fanout_load :  0.1;
      }
    }
```

The following is an example message:

```
Error: Line 22, The FPGA IOB cell type from the x4000 family requires
        O, T, IK, OK, I1, I2, PAD, and GSR pins. (LBDB-231)
```

**What Next**

Add the missing pins to the cell.

---

## LBDB-232

(error) The attribute '%s' cannot be %s while the pin has %s.

**Description**

The first specified attribute in the pin cannot be the value with the second specified attribute on the same pin has been defined.

For example, the *is_unconnected* attribute cannot be specified on a pin with *related_power_pin* or *related_ground_pin* attribute, or the pin is a feedthrough pin, i.e. in a *short* path.

The following example shows an instance where this message occurs:

```
cell(Block) {
    area : 2;
    pin(A) {
```

Feedback

```
      direction : input;
      capacitance : 1.0;
      related_power_pin : VDD ;
      related_ground_pin : VSS ;
      is_unconnected : true ;
    }
    ...

  }
```

The following is an example message:

```
Error: Line 84, The  attribute 'is_unconnected' cannot be true while
 the pin has related_power_pin/related_groud_pin or is feedthrough.
 (LBDB-232)
```

### What Next

Check the specification of the cell, remove the attribute that incorrecly defined.

---

## LBDB-233

(warning) The attribute '%s' should be defined while\n \tthe pin has no power supply and is not a feedthrough pin.

### Description

The specified attribute in the pin should be defined when the pin has no related_power_pin/related_ground_pin and is not a feedthrough pin if it should be unconnected.

Library Compiler will derive the attribute automatically during read_lib.

The following example shows an instance where this message occurs:

```
cell(Block) {
    area : 2;

    pin(Z) {
      direction : output;
      function : "A";
    }
  }
```

The following is an example message:

```
Warning: Line 84, The attribute 'is_unconnected' should be defined while
     the pin has no power supply and is not a feedthrough pin. (LBDB-233)
```

### What Next

Adding related power/ground pin if the pin should not be unconnected.

Feedback

## LBDB-235

(error) This %s voltage or power supply group is not defined.

### Description

This message indicates the specified voltage or the power supply associated with the signal level is not defined in the library source file. The voltage can either be input or output.

The following example shows an instance where this message occurs:

```
pin(PAD ) {
        is_pad : true;
        input_voltage : lbdb235;
        direction   : input;
        capacitance : 35.000000;
        fanout_load : 0.000000;
      }
      pin(PAD1 ) {
        is_pad : true;
        input_signal_level : VDD1;
        direction   : input;
        capacitance : 35.000000;
        fanout_load : 0.000000;
      }
```

In this case, the 'lbdb235' input_voltage is not defined. To fix the problem, add the following input_voltage:

```
input_voltage(lbdb235) {
    vil         : 0.8 ;
    vih         : 2.0 ;
    vimin       : -0.3 ;
    vimax       : VDD + 0.3 ;
   }
```

In addition, the 'VDD1' input_signal_level is not defined. To fix the problem, add the following power_supply group:

```
power_supply() {
    default_power_rail : VDD0;
    power_rail (VDD1, 5.0);
   }
```

The following is an example message:

```
Error: Line 23, This input voltage or power supply group is not defined.
 (LBDB-235)
Error: Line 30, This input voltage or power supply group is not defined.
 (LBDB-235)
```

Feedback

### What Next

Add the missing voltage or the power supply to the library file.

---

## LBDB-236

(error) An %s pin cannot specify an %s attribute.

### Description

This message indicates that there is a mismatch between the direction of a pin and the voltage attribute. Library Compiler fails if you specify an input_voltage to an output pin and an output_voltage to an input pin.

The following example shows an instance where this message occurs:

```
cell(lbdb236) {
    area : 0.0;
    pad_cell : true;
    pad_type : clock;
    pin(PAD ) {
      direction : input;
      is_pad : true;
      input_voltage : CMOS;
      capacitance : 1.0;
    }
    pin(Y ) {
      direction : output;
      is_pad : true;
      function : "PAD";
      input_voltage : CMOS;
      timing() {
        intrinsic_fall : 1.0;
        intrinsic_rise : 1.0;
        fall_resistance : 0.1;
        rise_resistance : 0.1;
        related_pin :"PAD";
      }
    }
  }
```

In this case, the 'Y' pin has an input_voltage specified. To fix the problem, change the voltage to an output_voltage attribute.

The following is an example message:

```
Error: Line 71, An output pin cannot specify an input_voltage attribute.
 (LBDB-236)
```

Feedback

### What Next

Check the library source file, and fix the invalid attribute, either the direction of the pin or the voltage attribute.

---

## LBDB-236w

(warning) An %s pin cannot specify an %s attribute.

### Description

This message indicates that there is a mismatch between the direction of a pin and the voltage attribute. Library Compiler fails if you specify an input_voltage to an output pin and an output_voltage to an input pin.

The following example shows an instance where this message occurs:

```
cell(lbdb236) {
    area : 0.0;
    pad_cell : true;
    pad_type : clock;
    pin(PAD ) {
      direction : input;
      is_pad : true;
      input_voltage : CMOS;
      capacitance : 1.0;
    }
    pin(Y ) {
      direction : output;
      is_pad : true;
      function : "PAD";
      input_voltage : CMOS;
      timing() {
        intrinsic_fall : 1.0;
        intrinsic_rise : 1.0;
        fall_resistance : 0.1;
        rise_resistance : 0.1;
        related_pin :"PAD";
      }
    }
  }
```

In this case, the 'Y' pin has an input_voltage specified. To fix the problem, change the voltage to an output_voltage attribute.

The following is an example message:

```
Warning: Line 71, An output pin cannot specify an input_voltage
 attribute. (LBDB-236w)
```

Feedback

### What Next

Check the library source file, and fix the invalid attribute, either the direction of the pin or the voltage attribute.

---

## LBDB-238

(warning) No '%s' attribute has been specified for the\n \tlibrary. This attribute is needed in %s libraries.

### Description

This message indicates that you did not specify any of the following attributes in the technology library.

```
* time_unit
        * capacitance_load_unit
        * pulling_resistance_unit
        * voltage_unit
        * current_unit
        * leakage_power_unit
```

The following example shows an instance where this message occurs: Add any of the following examples of attributes to the library:

```
capacitive_load_unit(0.042000,pf) ;
    voltage_unit  : "1V";
    current_unit  : "1mA";
    pulling_resistance_unit : "1kohm";
    time_unit : "1ns";
    leakage_power_unit : "1nW";
```

The following is an example message:

```
Warning: No 'capacitive_load_unit' attribute has been specified for the
         library. This attribute is needed in technology libraries.
 (LBDB-238)
```

### What Next

Add the missing attribute to the library source file.

---

## LBDB-239

(error) The '%s' driver_type cannot coexist\n \twith the '%s' attribute on the '%s' pin.

Feedback

### Description

The *driver_type* specified in the pin cannot coexist with the attribute on the same pin. For example, a pin with a *bus_hold driver_type* is tied to a DC source. A function attribute or *three_state* attribute cannot be specified on a *bus_hold* pin.

The following example shows an instance where this message occurs:

```
cell(lbdb239) {
    area : 2;
    pin(A) {
      direction : input;
      capacitance : 1.0;
    }
    pin(Z2) {
      direction : output;
      function : "A";
      driver_type : "bus_hold";
      timing() {
        intrinsic_rise : 0.1;
        intrinsic_fall : 0.1;
        related_pin : "A";
      }
    }
  }
```

The following is an example message:

```
Error: Line 84, The  'bus_hold' driver_type cannot coexist
        with the 'function' attribute on the 'Z2' pin. (LBDB-239)
```

### What Next

Check the specification of the cell, and make the appropriate change to the *driver_type* attribute or other attributes in the faulty pin group.

---

## LBDB-240

(error) The '%s' cell area attribute has already been defined.

### Description

This message indicates that the user-defined cell area is specified twice.

The following example shows an instance where this message occurs:

```
define_cell_area(my_area,pad_slots);
    define_cell_area(my_area,pad_slots);
```

Feedback

The following is an example message:

```
Error: Line 24, The 'my_area' cell area attribute has already been
 defined. (LBDB-240)
```

### What Next

Remove the second definition of the cell area attribute.

---

## LBDB-241

(warning) Multiple cell area definitions (%s, %s) map\n \tonto the same '%s' area. Using the last one encountered.

### Description

This message indicates that you specified multiple defined_cell_area attributes with the same area kind. Library Compiler uses the last one encountered.

The following example shows an instance where this message occurs:

```
define_cell_area(my_area,pad_slots);
    define_cell_area(your_area,pad_slots);
```

The following is an example message:

```
Warning: Line 24, Multiple cell area definitions (your_area, my_area) map
        onto the same 'pad_slots' area. Using the last one encountered.
 (LBDB-241)
```

### What Next

Delete the redundant define_cell_area definitions.

---

## LBDB-242

(warning) This '%s' timing arc has the same timing_type and\n \trelated_pin attributes as the timing arc on the line %d.

### Description

This message indicates that on the same pin you defined two timing groups with the same timing_type and related_pin attributes. Design Compiler considers the timing arcs as two separate arcs.

The following example shows an instance where this message occurs:

```
pin(QN) {
    direction : output;
    function : "IQN";
    timing() {
```

Feedback

```
      timing_type : rising_edge;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "CP";
    }
    timing() {
      timing_type : rising_edge;
      intrinsic_rise : 2.0;
      intrinsic_fall : 2.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "CP P";
    }
    timing() {
      timing_type : setup_rising;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "CP";
    }
  }
```

In this case, the 'QN' pin has the 'CP' related_pin and the 'rising_edge' timing_type multiply defined.

The following is an example message:

```
Warning: Line 591, This 'QN' timing arc has the same timing_type and
         related_pin attributes as the timing arc on the line 583.
 (LBDB-242)
```

### What Next

If it is a typo, change the library file, and delete the redundant timing groups.

---

## LBDB-243

(warning) The '%s' combinational cell has a '%s' pin with\n \ta sequential timing arc containing the '%s' timing_type.

### Description

This message indicates that you specified one of the cell's timing arcs as sequential, while the function of this cell is combinational. Library Compiler warns you if any of the following timing types is defined on a combinational cell:

```
* rising_edge
        * falling_edge
```

Feedback

```
        * preset
        * clear
        * setup_rising
        * setup_falling
        * hold_rising
        * hold_falling
        * recovery_rising
        * recovery_falling
        * skew_rising
        * skew_falling
        * removal_rising
        * removal_falling
```

The following example shows an instance where this message occurs:

```
cell(lbdb243) {
    area : 1;
    pin(A B) {
      direction : input;
      capacitance : 1;
    }
    pin(Z) {
      direction : output;
      function : "A B";
      timing() {
        timing_type : rising_edge;
        intrinsic_rise : 0.1;
        intrinsic_fall : 0.1;
        rise_resistance : 0.1;
        fall_resistance : 0.1;
        related_pin : "A B";
      }
    }
  }
```

In this case, the 'Z' pin has an invalid 'rising_edge' timing_type.

The following is an example message:

```
Warning: Line 84, The 'lbdb243' combinational cell has a 'Z' pin with
         a sequential timing arc containing the 'rising_edge'
 timing_type. (LBDB-243)
```

**What Next**

Change the library source file, and fix the timing_type of the specified pin.

---

## LBDB-246

(warning) The default_wire_load_selection is not defined.\n \tBy default, the '%s' wire_load_selection group is used.

Feedback

### Description

The *default_wire_load_selection* attribute is undefined in this library. The *default_wire_load_selection* attribute is required when multiple wire_load_selection groups are specified in a library. If there is only one wire_load_selection group defined in a library, Library Compiler automatically sets the *default_wire_load_selection* attribute to the only available wire_load_selection group.

The following example shows an instance where this message occurs:

```
wire_load("05x05") {
    resistance : 0 ;
    capacitance : 1 ;
    area : 0 ;
    slope : 0.186 ;
    fanout_length(1,0.39) ;
  }
  wire_load_selection(test) {
    wire_load_from_area(27,10,"10x10");
    wire_load_from_area(27,100,"10x10");
    wire_load_from_area(0,28,"05x05");
    wire_load_from_area(2,10,"15x10");
  }
```

The following is an example message:

```
Warning: Line 50, The default_wire_load_selection is not defined.
        By default, the 'test' wire_load_selection group is used.
 (LBDB-246)
```

### What Next

Set the *default_wire_load_selection* attribute to avoid this warning message, or ignore this message.

---

## LBDB-247

(error) The default_wire_load_selection is not defined.\n \tThe default_wire_load_selection attribute is required when more than one\n \twire_load_selection group is specified.

### Description

The *default_wire_load_selection* attribute is undefined in this library. Library Compiler requires this attribute because there is more than one wire_load_selection group specified.

The following example shows an instance where this message occurs:

```
wire_load("05x05") {
    resistance : 0 ;
    capacitance : 1 ;
    area : 0 ;
```

Feedback

```
    slope : 0.186 ;
    fanout_length(1,0.39) ;
  }

  wire_load_selection(test) {
    wire_load_from_area(0,25,"05x05");
  }
  wire_load_selection(test1) {
    wire_load_from_area(0,27,"05x05");
  }
```

To fix the problem, add this statement to the library:

```
default_wire_load_selection : test;
```

The following is an example message:

```
Error: Line 0, The default_wire_load_selection is not defined.
        The default_wire_load_selection attribute is required when more
 than one
        wire_load_selection group is specified. (LBDB-247)
```

### What Next

Set the *default_wire_load_selection* attribute to be one of the wire_load_selection groups.

---

## LBDB-250

(warning) The '%s' pin already has a 'pulling_resistance' value.\n \tThe 'pulling_current' causes the 'pulling_resistance' value to be overwritten.

### Description

This message indicates you specified both the *pulling_resistance* and the *pulling_current* attributes on the same pin. Library Compiler uses the pulling_current value and the nominal voltage to overwrite the pulling_resistance value.

The following example shows an instance where this message occurs:

```
pin(Y) {
        direction : output;
        capacitance : 1.0;
        driver_type : pull_up;
        is_pad : true;
        slew_control : low;
        drive_current : 1.0;
        output_voltage  : CMOS_OUT;
        function : "A";
        three_state : "GZ";
        pulling_resistance : 1000;
        pulling_current : 10;
     }
```

Feedback

The following is an example message:

```
Warning: Line 93, The 'Y' pin already has a 'pulling_resistance' value.
        The 'pulling_current' causes the 'pulling_resistance' value to be
 overwritten. (LBDB-250)
```

### What Next

If you do not want the pulling_resistance value overwritten, delete the pulling_current attribute.

---

## LBDB-251

(warning) The '%s' pin has a 'hysteresis' attribute but\n \tno 'input_voltage' attribute. Both attributes are needed.

### Description

This message indicates that you specified the hysteresis attribute, but you did not specify the input_voltage attribute in a pad cell.

The following example shows an instance where this message occurs:

```
pin(PAD ) {
      direction : input;
      is_pad : true;
      hysteresis : true;
      capacitance : 1.0;
    }
```

To fix the problem, add this statement to the 'PAD' pin:

```
input_voltage : CMOS; /* CMOS is defined in the library */
```

The following is an example message:

```
Warning: Line 53, The 'PAD' pin has a 'hysteresis' attribute but
        no 'input_voltage' attribute. Both attributes are required.
 (LBDB-251)
```

### What Next

Add the input_voltage attribute to the pin in the pad cell.

---

## LBDB-252

(error) The '%s' %s group already exists and\n \tcannot be overwritten.

Feedback

### Description

This message indicates that you multiply defined either an input_voltage or an output_voltage group. Library Compiler does not allow the overwriting of existing voltage groups.

The following example shows an instance where this message occurs:

```
input_voltage(CMOS) {
    vil : 1.5;
    vih : 3.5;
    vimin : -0.3;
    vimax : VDD + 0.3;
  }

  input_voltage(CMOS) {
    vil : 2.5;
    vih : 4.5;
    vimin : -0.3;
    vimax : VDD + 0.3;
  }
```

The following is an example message:

```
Error: Line 31, The 'CMOS' input_voltage group already exists and
       cannot be overwritten. (LBDB-252)
```

### What Next

Delete the second specification of the voltage group.

---

## LBDB-253

(error) The '%s' pin does not have all the %s slew rate attributes\n \tdefined. The '%s' attribute is missing.\n \tAll %s slew-rate attributes must be specified together as a group.

### Description

This message indicates that not all slew rate attributes are specified in a pin. Library Compiler expects the following rise slew rate attributes defined together:

```
* rise_current_slope_before_threshold
        * rise_time_before_threshold
        * rise_current_slope_after_threshold
        * rise_time_after_threshold
```

and the following fall slew rate defined together:

```
* fall_current_slope_before_threshold
        * fall_time_before_threshold
        * fall_current_slope_after_threshold
        * fall_time_after_threshold
```

Feedback

The following example shows an instance where this message occurs:

```
cell(lbdb253) {
    area : 0.0;
    pad_cell : true;
    bond_pads : 1;
    driver_sites : 1;
    pin(PAD) {
      is_pad : true;
      direction : output;
      output_voltage : CMOS_OUT;
      function : "Y ";
      drive_current : 1.0;
      slew_control : high;
      rise_time_before_threshold : 1.0;
      rise_current_slope_after_threshold : -0.1;
      rise_time_after_threshold : 1.0;
      fall_current_slope_before_threshold : -0.1;
      fall_time_before_threshold : 1.0;
      fall_current_slope_after_threshold : 0.1;
      fall_time_after_threshold : 1.0;
      timing() {
        intrinsic_fall : 1.0;
        intrinsic_rise : 1.0;
        fall_resistance : 0.1;
        rise_resistance : 0.1;
        related_pin :"Y";
      }
    }
    pin(Y ) {
      direction : input;
      capacitance : 1.0;
    }
  }
```

In this case, the 'rise_current_slope_before_threshold' attribute is not specified for the 'PAD' pin.

The following is an example message:

```
Error: Line 88, The 'PAD' pin does not have all the rise slew rate
 attributes
        defined. The 'rise_current_slope_before_threshold' attribute is
 missing.
        All rise slew-rate attributes must be specified together as a
 group. (LBDB-253)
```

**What Next**

Check the library source file, and add the missing slew rate attribute.

Feedback

## LBDB-254

(error) The '%s' edge rate related attribute \n \tcannot be specified on the '%s' output pin.

### Description

This message indicates that you specified an edge rate attribute on an output pin.

The following example shows an instance where this message occurs:

```
pin(Z) {
            direction : output;
            function : "(A & B)";
            max_fanout : 25;

            edge_rate_breakpoint_r0 : 0.0;
            edge_rate_breakpoint_f0 : 0.0;
            edge_rate_breakpoint_r1 : 0.1;
            edge_rate_breakpoint_f1 : 0.1;

            edge_rate_rise : 1.0;
            edge_rate_load_rise : 1.0;
            edge_rate_fall : 0.1;
            edge_rate_load_fall : 1.0;
        }
```

The following is an example message:

```
Error: Line 120, The 'edge_rate_breakpoint_r0' edge rate related
 attribute
        cannot be specified on the 'Z' output pin. (LBDB-254)
```

### What Next

Delete the attribute from the output pin group.

## LBDB-255

(error) The 'vhdl_name' attribute of '%s' is invalid VHDL or\n \tconflicts with another 'vhdl_name' attribute.

### Description

This message indicates that the cell or port *vhdl_name* attribute is either invalid VHDL (reserved VHDL name) or another cell or port has already been set to the same name.

The following example shows an instance where this message occurs:

```
cell(lbdb255) {
    area : 2;
    pin(A) {
```

Feedback

```
      vhdl_name : "pinvhdl";
      direction : input;
      capacitance : 1;
    }
    pin(B) {
      vhdl_name : "pinvhdl" ;
      direction : input;
      capacitance : 1;
    }
    pin(Z) {
      direction : output;
      function : "A^B";
      timing() {
        timing_sense : positive_unate;
        intrinsic_rise : 0.1;
        intrinsic_fall : 0.1;
        rise_resistance : 0.1;
        fall_resistance : 0.1;
        related_pin : "A";
      }
      timing() {
        timing_sense : non_unate;
        intrinsic_rise : 1.0;
        intrinsic_fall : 1.0;
        rise_resistance : 0.1;
        fall_resistance : 0.1;
        related_pin : "A B";
      }
    }
  }
```

In this case, both the 'A' and 'B' pin have the same value in the vhdl_name attribute.

The following is an example message:

```
Error: Line 93, The 'vhdl_name' attribute of 'pinvhdl' is invalid VHDL or
        conflicts with another 'vhdl_name' attribute. (LBDB-255)
```

**What Next**

If the attribute is invalid VHDL, change the library by fixing the value of the vhdl_name attribute. If the value of the attribute is redundant, either delete the second attribute or change its value.

---

## LBDB-256

(warning) The 'vhdl_name' attribute of '%s' is invalid VHDL or\n \tconflicts with another 'vhdl_name' attribute. Renamed to '%s'.

Feedback

### Description

This message indicates that the cell or port 'vhdl_name' attribute is either invalid VHDL or another cell or port has already been set to the same name.

Because this is an update_lib cell, the name is automatically renamed to a valid one.

The following example shows an instance where this message occurs:

```
cell (lbdb256) {
  area : 1;
  pin(IN) {
    vhdl_name : "port_X";
    direction : input;
    capacitance : 1;
  }
  pin(Z) {
    vhdl_name : "port_X";       /* rename after warning */
    direction : output;
    function : "IN'";
    timing () {
      intrinsic_rise : 0.1;
      intrinsic_fall : 0.1;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "IN";
    }
  }
}
```

The following is an example message:

```
Warning: Line 9, The 'vhdl_name' attribute of 'port_X' is invalid VHDL or
        conflicts with another 'vhdl_name' attribute. Renamed to
 'port_Xb'. (LBDB-256)
```

### What Next

Change the file to update by changing the value of the vhdl_name attribute if you are not satisfied with the renamed value.

---

## LBDB-257

(warning) The '%s' wire_load_selection group has been specified\n \twhile the default_wire_load_mode is %s 'top'. This causes \n \tthe 'wire_load_selection' group to work only for the top-level design.

### Description

The 'default_wire_load_mode' is defined as, or defaults to, 'top', which causes the 'wire_load_selection' group to only work for the top-level design.

Feedback

The following example shows an instance where this message occurs:

```
default_wire_load_mode : 'top';
  wire_load_selection(test) {
    wire_load_from_area(27,100,"10x10");
    wire_load_from_area(10,25,"05x05");
  }
```

The following is an example message:

```
Warning: The 'test' wire_load_selection group has been specified
        while the default_wire_load_mode is default to 'top'. This causes
        the 'wire_load_selection' group to work only for the top-level
 design. (LBDB-257)
```

**What Next**

Set the default_wire_load_mode to 'segmented' or 'enclosed' modes to work with wire_load_selection group.

---

## LBDB-258

(error) The '%s' driver_type cannot be specified\n \ton a inout pin without a three_state attribute.

**Description**

This message indicates that you specified a pull_up or a pull_down driver_type on an inout pin lacking a three_state attribute.

The following example shows an instance where this message occurs:

```
pin(A) {
      direction : inout;
      driver_type : pull_up;
      capacitance : 1;
      pulling_resistance : 100;
    }
```

The following is an example message:

```
Error: Line 354, The 'pull_up' driver_type cannot be specified
        on a inout pin without a three_state attribute. (LBDB-258)
```

**What Next**

Either delete the driver_type or add the three_state attribute.

---

## LBDB-259

(error) The '%s' refers to a nonexistent or empty '%s'.

Feedback

### Description

This message indicates that you specified a nonexistent or empty template group on a propagation delay in a nonlinear delay library.

The following example shows an instance where this message occurs:

```
fall_propagation(lbdb259_template) {
                values ("1, 2, 3, 4", "5, 6, 7, 8", \
                        "9, 10, 11, 12", "13, 14, 15, 16");
        }
```

To fix the problem, add the following group to the library file,

```
lu_table_template(lbdb259_template) {
        variable_1 : input_net_transition;
        variable_2 : output_net_length;
        index_1 ("0.1, 1.2, 2.3, 3.4");
        index_2 ("0.1, 1.2, 2.3, 3.4");
  }
```

The following is an example message:

```
Error: Line 111, The 'lbdb259_template' refers to a nonexistent or empty
 'lu_table_template'. (LBDB-259)
```

### What Next

Check the library source file, and either add the lu_table_template group if it is missing or correct the template name if it is a typo.

---

## LBDB-259w

(warning) The '%s' refers to a nonexistent or empty '%s'.

### Description

This message indicates that you specified a nonexistent or empty template group on a propagation delay in a nonlinear delay library.

The following example shows an instance where this message occurs:

```
fall_propagation(lbdb259_template) {
                values ("1, 2, 3, 4", "5, 6, 7, 8", \
                        "9, 10, 11, 12", "13, 14, 15, 16");
        }
```

To fix the problem, add the following group to the library file,

```
lu_table_template(lbdb259_template) {
        variable_1 : input_net_transition;
        variable_2 : output_net_length;
```

Feedback

```
        index_1 ("0.1, 1.2, 2.3, 3.4");
        index_2 ("0.1, 1.2, 2.3, 3.4");
  }
```

The following is an example message:

```
Warning: Line 111, The 'lbdb259_template' refers to a nonexistent or
 empty 'lu_table_template'. (LBDB-259w)
```

**What Next**

Check the library source file, and either add the lu_table_template group if it is missing or correct the template name if it is a typo.

---

## LBDB-260

(error) The '%s' in '%s' and '%s' \n \tin '%s' of the '%s' lu_table_template represent \n \tthe same unallowable meaning.

**Description**

This message indicates that the two variables in the lu_table_template represent the same unallowable meaning. Library Compiler fails if both the values of variable_1 and variable_2 are from the following set:

```
* total_output_net_capacitance
        * output_net_length
        * output_net_wire_cap
        * output_net_pin_cap
```

The following example shows an instance where this message occurs:

```
lu_table_template(lbdb260) {
        variable_1 : output_net_length;
        variable_2 : total_output_net_capacitance;
        index_2 ("0.1, 1.2, 2.3, 3.4");
  }
```

The following is an example message:

```
Error: Line 57, The 'output_net_length' in variable_1 and
 'total_output_net_capacitance'
        in variable_2 of the 'lbdb260' lu_table_template represent
        the same unallowable meaning. (LBDB-260)
```

**What Next**

Check the lu_table_template, and make the correction to either variable_1 or variable_2.

Feedback

## LBDB-261

(error) The %s has already been specified for '%s'%s. A duplicate is not allowed.

### Description

This message indicates that attribute or group information is specified multiple times. Library Compiler does not allow duplicate information.

### What Next

Check the library source file, and, if there is a duplicate, change either name of duplicated group or delete one group/attribute.

### Examples

The following example shows an instance where this message occurs:

```
lu_table_template(lbdb261_template) {
        variable_1 : constrained_pin_transition;
        variable_2 : related_pin_transition;
        index_1 ("0.1, 1.2, 2.3, 3.4");
        index_2 ("0.1, 1.2, 2.3, 3.4");
        index_2 ("0.1, 1.2, 2.3, 3.4");
  }
```

### Examples

```
Error: Line 43, The 'index_2' has already been specified for
 'lbdb261_template' at line 30. A duplicate is not allowed.(LBDB-261)
```

## LBDB-261w

(warning) The %s has already been specified for '%s'%s. Using the last definition encountered.

### Description

This message indicates that attribute or group information is specified multiple times. Library Compiler does not allow duplicate information and use last definition.

### What Next

Check the library source file, and, if there is a duplicate, change either name of group/attribute or delete one group/attribute.

### EXMAPLES

The following example shows two function definition are same

Feedback

```
pin(Q) {
             direction : output;
             function : "1";
             function : "1";
        }
```

### Examples

```
Warning: Line 43, The  attribute 'function'  has already been specified
 for group 'pin' at line 58. Using the last definition encountered.
 (LBDB-261w)
```

---

## LBDB-262

(error) The '%s' attribute has an invalid sequence of\n \tdata '%f , %f'. The values must be in\n \tmonotonically increasing order.

### Description

This message indicates that the set of data is not specified in monotonically increasing order.

The following example shows an instance where this message occurs:

```
lu_table_template(one_dimension) {
        variable_1 : input_net_transition;
        variable_2 : output_net_length;
        index_1 ("0.1");
        /* index does not in monotically increasing order */
        index_2 ("0.1, 4.2, 2.3, 3.4");
  }
```

The following is an example message:

```
Error: Line 22, The 'index_2' attribute has an invalid sequence of
        data '4.200000 , 2.300000'. The values must be in
        monotonically increasing order. (LBDB-262)
```

### What Next

Check your library and correct the order of the values. If the message shows identical values and the library file reveals tiny differences, you hit the resolution limit of single-precision floating-point numbers. Consider increasing the spacing between data points.

Feedback

If the error happens inside an output_current_rise or output_current_fall group, please refer to the latest version of "CCS Timing Library Characterization Guidelines" and try the following:

1. Reduce or set tstart = 0 (see Figure 4 in the Guidelines), and re-characterize the library.
2. If that does not work, further reduce the reference_time value. Subtract the same amount from all entries in index_3 (time) in the same vector group. Apply the same modifications to all vector groups having the same input_net_transition value (index_1 or index_2), within the same output_current_rise or output_current_fall group. Make sure no adjusted numbers become negative.

---

## LBDB-262w

(warning) The '%s' attribute has an invalid sequence of\n \tdata '%f , %f'. The values must be in\n \tmonotonically increasing order.

### Description

This message indicates that the set of data is not specified in monotonically increasing order.

The following example shows an instance where this message occurs:

```
lu_table_template(one_dimension) {
        variable_1 : input_net_transition;
        variable_2 : output_net_length;
        index_1 ("0.1");
        /* index does not in monotically increasing order */
        index_2 ("0.1, 4.2, 2.3, 3.4");
  }
```

The following is an example message:

```
Warning: Line 22, The 'index_2' attribute has an invalid sequence of
        data '4.200000 , 2.300000'. The values must be in
        monotonically increasing order. (LBDB-262w)
```

### What Next

Check your library and correct the order of the values. If the message shows identical values and the library file reveals tiny differences, you hit the resolution limit of single-precision floating-point numbers. Consider increasing the spacing between data points.

Feedback

If the warning happens inside an output_current_rise or output_current_fall group, please refer to the latest version of "CCS Timing Library Characterization Guidelines" and try the following:

1. Reduce or set tstart = 0 (see Figure 4 in the Guidelines), and re-characterize the library.
2. If that does not work, further reduce the reference_time value. Subtract the same amount from all entries in index_3 (time) in the same vector group. Apply the same modifications to all vector groups having the same input_net_transition value (index_1 or index_2), within the same output_current_rise or output_current_fall group. Make sure no adjusted numbers become negative.

---

## LBDB-263

(error) The '%s' attribute has a value '%f',\n which is less than '%f', the minimum required value of this attribute.

### Description

The value specified is less than the required minimum value for this attribute.

The following example shows an instance where this message occurs:

```
lu_table_template(invalid_template) {
        variable_1 : constrained_pin_transition;
        variable_2 : output_net_length;
        /* index value can not be less than zero */
        index_1 ("-0.1, 1.2, 2.3, 3.4");
        index_2 ("0.1, 1.2, 2.3, 3.4");
  }
```

The following is an example message:

```
Error: Line 35, The 'index_1' attribute has a value '-0.100000',
        which is less than '0.000000', the minimum required value
        of this attribute. (LBDB-263)
```

### What Next

Check the library source file and correct the problem. Find the minimum value for this attribute in the error message or in the Library Compiler reference manual.

---

## LBDB-264

(error) The '%s' is invalid in this look-up table.

Feedback

### Description

This message indicates that the specified index is not allowed in the current table because the current table is one dimensional.

The following example shows an instance where this message occurs:

```
lu_table_template(lbdb264) {
        variable_1 : constrained_pin_transition;
        index_2 ("0.1, 1.2, 2.3, 3.4");
  }
```

To fix the problem, change index_2 to index_1.

The following is an example message:

```
Error: Line 46, The 'index_2' is invalid in this delay look-up table.
 (LBDB-264)
```

### What Next

Check the library source file to determine whether you have the *variable_2* attribute specified before *index_2* or whether you are specifying *index_2* in a one-dimensional table or template.

---

## LBDB-265

(error) You cannot mix a cell delay table with\n \ta propagation delay table in the same timing group.

### Description

You can define cell delay table or propagation delay table within a timing group, but you cannot mix them together. For example, *cell_rise* and *propagation_fall* cannot be defined in a timing group. The same applies to *cell_fall* and *propagation_fall*.

The following example shows an instance where this message occurs:

```
cell (INVB) {
    area : 0.1;
    pin(Q) {
      direction : output;
      function :  "!A";
      timing() {
        related_pin : "A";
        rise_transition(fivebyfive) {
          index_1("0.000,0.200,0.400,0.600,0.800");
          index_2("0.100,0.200,0.300,0.400,0.500");
          values("0.100000, 0.100000, 0.100000, 0.100000, 0.100000", \
            "0.100000, 0.100000, 0.100000, 0.100000, 0.100000", \
            "0.100000, 0.100000, 0.100000, 0.100000, 0.100000", \
```

Feedback

```
            "0.100000, 0.100000, 0.100000, 0.100000, 0.100000", \
            "0.100000, 0.100000, 0.100000, 0.100000, 0.100000");
        }
        cell_rise(fivebyfive_prime) {
          index_1("0.000,0.200,0.400,0.600,0.800");
          values("-0.100000, -0.200000, 0.300000, 0.400000, 0.500000");
        }
        rise_propagation(fivebyfive_prime) {
          index_1("0.000,0.200,0.400,0.600,0.800");
          values("-0.100000, -0.200000, 0.300000, 0.400000, 0.500000");
        }
        fall_transition(fivebyfive) {
          index_1("0.000,0.200,0.400,0.600,0.800");
          index_2("0.100,0.200,0.300,0.400,0.500");
          values("0.100000, 0.100000, 0.100000, 0.100000, 0.100000", \
            "0.100000, 0.100000, 0.100000, 0.100000, 0.100000", \
            "0.100000, 0.100000, 0.100000, 0.100000, 0.100000", \
            "0.100000, 0.100000, 0.100000, 0.100000, 0.100000", \
            "0.100000, 0.100000, 0.100000, 0.100000, 0.100000");
        }
        cell_fall(fivebyfive) {
          index_1("0.000,0.200,0.400,0.600,0.800");
          index_2("0.100,0.200,0.300,0.400,0.500");
          values("0.100000, 0.100000, 0.100000, 0.100000, 0.100000", \
            "0.100000, 0.100000, 0.100000, 0.100000, 0.100000", \
            "0.100000, 0.100000, 0.100000, 0.100000, 0.100000", \
            "0.100000, 0.100000, 0.100000, 0.100000, 0.100000", \
            "0.100000, 0.100000, 0.100000, 0.100000, 0.100000");
        }
      }
      max_transition : 1.0;
    }
    pin(A) {
      direction : input;
      capacitance : 1.0;
    }
  }
```

The following is an example message:

```
Error: Line 1110, You cannot mix a cell delay table with
        a propagation delay table in the same timing group. (LBDB-265)
```

**What Next**

Make your selection, and use either a cell delay table or a propagation delay table.

---

## LBDB-266

(error) The '%s' attribute is needed in the specification.\n No default can be applied to this attribute.

Feedback

### Description

This error message occurs when the identified attribute is missing in the specification.

### What Next

Check your library, add the missing attribute, and run the command again.

For example, the *index_1* attribute is missing from the following specification:

```
lu_table_template(lbdb266_template) {
        variable_1 : constrained_pin_transition;
        index_2 ("0.1, 1.2, 2.3, 3.4");
  }
```

In the following example, since the *va_rise_constraint* is referring to a template that contains the *related_out_total_output_net_capacitance* related output loading variable, the *related_output_pin* attribute is required under the timing group.

```
lu_table_template(va_sup_hld_1) {
   variable_1 : related_out_total_output_net_capacitance;
   index_1("2.0, 3.0");
   variable_2 : constrained_pin_transition;
   index_2("3.0, 4.0");
 }
 ...
    timing () {
       timing_based_variation() {
         va_parameters(var1, var2);
         nominal_va_values(10.0, 20.0);
         va_rise_constraint(va_sup_hld_1) {
                        va_values(10.0, 19.0);
               ...
         }
   ...}
  ...}
 ...
```

---

## LBDB-266w

(warning) The '%s' attribute is needed in the specification.\n No default can be applied to this attribute.

### Description

This warning message occurs when the identified attribute is missing in the specification.

### What Next

Check your library, add the missing attribute, and run the command again.

Feedback

For example, the *index_1* attribute is missing from the following specification:

```
lu_table_template(lbdb266_template) {
        variable_1 : constrained_pin_transition;
        index_2 ("0.1, 1.2, 2.3, 3.4");
  }
```

In the following example, since the *va_rise_constraint* is referring to a template that contains the *related_out_total_output_net_capacitance* related output loading variable, the *related_output_pin* attribute is required under the timing group.

```
lu_table_template(va_sup_hld_1) {
   variable_1 : related_out_total_output_net_capacitance;
   index_1("2.0, 3.0");
   variable_2 : constrained_pin_transition;
   index_2("3.0, 4.0");
 }
 ...
    timing () {
       timing_based_variation() {
         va_parameters(var1, var2);
         nominal_va_values(10.0, 20.0);
         va_rise_constraint(va_sup_hld_1) {
                        va_values(10.0, 19.0);
               ...
         }
   ...}
  ...}
 ...
```

---

## LBDB-267

(error) The '%s' is missing for this timing arc.

### Description

All four delay look-up tables

```
* rise_propagation/cell_rise
        * fall_propagation/cell_fall
        * rise_transition
        * fall_transition
```

are required in the timing arc if the *delay_model* is *table_lookup*. If the *timing_type* is 'clear', define both fall tables. Defining the rise tables is optional. If the *timing_type* is 'preset', define both rise tables. Defining the fall tables is optional.

The following example shows an instance where this message occurs:

```
pin ( Y )    {
      direction : output;
      function : " (A+B)' ";
```

Feedback

```
      timing ()      {
        related_pin : A ;
        rise_propagation(prop) {
          values("0.100000, 0.100000, 0.100000",  \
                 "0.100000, 0.100000, 0.100000",  \
                 "0.100000, 0.100000, 0.100000",  \
                 "0.100000, 0.100000, 0.100000",  \
                 "0.100000, 0.100000, 0.100000");
        }
        rise_transition(tran) {
          values("0.000000, 0.10000");
        }
        fall_transition(tran) {
          values("0.000000, 0.100000");
        }
      }
    }
```

To fix the problem, add the attribute to the timing group,

```
fall_propagation(prop) {
          values("0.100000, 0.100000, 0.100000",  \
                 "0.100000, 0.100000, 0.100000",  \
                 "0.100000, 0.100000, 0.100000",  \
                 "0.100000, 0.100000, 0.100000",  \
                 "0.100000, 0.100000, 0.100000");
        }
```

The following is an example message:

```
Error: Line 144, The 'fall_propagation' is missing for this timing arc.
 (LBDB-267)
```

### What Next

Check the library source file to see if you missed the look-up table or put it in the wrong place.

---

## LBDB-268

(error) The '%s' cannot be specified in a timing arc with\n \tthe '%s' timing_type.

### Description

This message indicates that the timing arc with the indicated timing_type is not compatible with a *table_lookup* delay model.

The following example shows an instance where this message occurs:

```
rise_constraint(basic_template) {
                values ("1, 2, 3, 4", "5, 6, 7, 8", \
```

Feedback

```
                        "9, 10, 11, 12", "13, 14, 15, 16");
        }
```

The following is an example message:

```
Error: The 'rise_constraint' cannot be specified in a timing arc with
        the 'combinational' timing_type. (LBDB-268)
```

---

## LBDB-269

(error) You have both '%s' and '%s'\n \tspecified in this timing group.

### Description

You cannot specify both cell delay and propagation delay information in a timing group as indicated in the error message. For example, *cell_rise* and *propagation_fall* cannot be defined in a timing group. The same applies to *cell_fall* and *propagation_fall*.

You can use *intrinsic_rise* and *intrinsic_fall* to define default delay values when the delay model is a table_lookup.

The following example shows an instance where this message occurs:

```
cell (INVB) {
    area : 0.1;
    pin(Q) {
      direction : output;
      function :  "!A";
      timing() {
        related_pin : "A";
        rise_transition(fivebyfive) {
          index_1("0.000,0.200,0.400,0.600,0.800");
          index_2("0.100,0.200,0.300,0.400,0.500");
          values("0.100000, 0.100000, 0.100000, 0.100000, 0.100000", \
                 "0.100000, 0.100000, 0.100000, 0.100000, 0.100000", \
                 "0.100000, 0.100000, 0.100000, 0.100000, 0.100000", \
                 "0.100000, 0.100000, 0.100000, 0.100000, 0.100000", \
                 "0.100000, 0.100000, 0.100000, 0.100000, 0.100000");
        }
        cell_rise(fivebyfive_prime) {
          index_1("0.000,0.200,0.400,0.600,0.800");
          values(\
                 "-0.100000, -0.200000, 0.300000, 0.400000, 0.500000");
        }
        rise_propagation(fivebyfive_prime) {
          index_1("0.000,0.200,0.400,0.600,0.800");
          values(\
                 "-0.100000, -0.200000, 0.300000, 0.400000, 0.500000");
        }
        fall_transition(fivebyfive) {
          index_1("0.000,0.200,0.400,0.600,0.800");
          index_2("0.100,0.200,0.300,0.400,0.500");
```

Feedback

```
          values("0.100000, 0.100000, 0.100000, 0.100000, 0.100000", \
                 "0.100000, 0.100000, 0.100000, 0.100000, 0.100000", \
                 "0.100000, 0.100000, 0.100000, 0.100000, 0.100000", \
                 "0.100000, 0.100000, 0.100000, 0.100000, 0.100000", \
                 "0.100000, 0.100000, 0.100000, 0.100000, 0.100000");
        }
        cell_fall(fivebyfive) {
          index_1("0.000,0.200,0.400,0.600,0.800");
          index_2("0.100,0.200,0.300,0.400,0.500");
          values("0.100000, 0.100000, 0.100000, 0.100000, 0.100000", \
                 "0.100000, 0.100000, 0.100000, 0.100000, 0.100000", \
                 "0.100000, 0.100000, 0.100000, 0.100000, 0.100000", \
                 "0.100000, 0.100000, 0.100000, 0.100000, 0.100000", \
                 "0.100000, 0.100000, 0.100000, 0.100000, 0.100000");
        }
      }
      max_transition : 1.0;
    }
    pin(A) {
      direction : input;
      capacitance : 1.0;
    }
  }
```

The following is an example message:

```
Error: Line 1110, You have both 'rise_propagation' and 'cell_rise'
        specified in this timing group. (LBDB-269)
```

### What Next

Make your selection to use either a cell delay table or a propagation delay table.

---

## LBDB-270

(error) The '%s' has a count of %d, which does not match\n \tthe size %d specified.

### Description

The specification of the index implied the size of each axis of the look-up table. The syntax of *values* complex attribute should be organized in groups of floating-point values equal to the size of index_2. The total number of groups should be equal to the size of index_1. For a one-dimensional table, the size of index_2 is *1*.

The following example shows an instance where this message occurs:

```
lu_table_template(basic_template) {
        variable_1 : input_net_transition;
        variable_2 : output_net_length;
        index_1 ("0.1, 1.2, 2.3, 3.4");
        index_2 ("0.1, 1.2, 2.3, 3.4");
  }
```

Feedback

```
rise_propagation(basic_template) {
      values ("1, 2, 3, 4", "5, 6, 7, 8", \
              "9, 10, 11", "13, 14, 15, 16");
}
```

The following is an example message:

```
Error: Line 160, The 'values' have a count of 3, which does not match
        the size 4 specified. (LBDB-270)
```

**What Next**

Check the library source file, and correct the problem by grouping the values together in the required format.

---

## LBDB-271

(error) The '%s' has zero elements.

**Description**

This message indicates that one of the lu_table_template index complex attributes contains no value.

The following example shows an instance where this message occurs:

```
lu_table_template(missing_index) {
        variable_1 : output_net_length;
        index_1("");
  }
```

The following is an example message:

```
Error: Line 64, The 'index_1' has zero elements. (LBDB-271)
```

**What Next**

Check the library source file, and either add the missing value to the index attribute or delete the attribute if it is not used.

---

## LBDB-272

(warning) The '%s' attribute has a '%f' value,\n \twhich is less than '%f' the minimum recommended value of this attribute.

**Description**

This message indicates that the value assigned for the specified attribute is lower than the recommended minimum value. However, Library Compiler will maintain the values as is.

Feedback

The following example shows an instance where this message occurs:

```
lu_table_template(basic_template) {
        variable_1 : input_net_transition;
        variable_2 : output_net_length;
        index_1 ("0.1, 1.2, 2.3, 3.4");
        index_2 ("0.1, 1.2, 2.3, 3.4");
  }
  ...
        fall_transition(basic_template) {
                values ("-1, 2, 3, 4", "5, 6, 7, 8", \
                        "9, 10, 11, 12", "13, 14, 15, 16");
        }
```

The following is an example message:

```
Warning: Line 123, The 'values' attribute has a '-1.000000' value,
        which is less than '0.000000' the minimum recommended value of
        this attribute. (LBDB-272)
```

### What Next

Check the library source file, and make sure the value is correct. Refer to the "Library Compiler Reference Manual" for the attribute value requirements.

---

## LBDB-273

(error) The fanout_length complex attribute allows only 2\n \tor 5 arguments, and you specified %d arguments.

### Description

This message indicates that the *fanout_length* complex attribute has an invalid number of arguments. The attribute can have either the two standard arguments,

```
* fanout
        * length
```

or five arguments created automatically from the back-annotation information,

```
* fanout
        * length
        * average capacitance
        * standard deviation
        * number of nets
```

The following example shows an instance where this message occurs:

```
wire_load("05x05") {
    resistance : 0 ;
    capacitance : 1 ;
    area : 0 ;
```

Feedback

```
    slope : 0.186 ;
    fanout_length(1,0.39, 0.1) ;
  }
```

The following is an example message:

```
Error: Line 43, The fanout_length complex attribute allows only 2
        or 5 arguments, and you specified 3 arguments. (LBDB-273)
```

### What Next

Refer to the "Library Compiler User Guide" manual for more information about the fanout_length attribute. Check the library source file, and correct the format of the attribute.

---

## LBDB-274

(warning) The '%s' attribute has a '%f' value,\n \twhich is less than '%f' the minimum required value of this attribute.\n \tThe value is changed to the minimum value.

### Description

This message indicates that the value assigned for the specified attribute is lower than the required minimum value. Therefore, Library Compiler uses the minimum value for the attribute.

The following example shows an instance where this message occurs:

```
lu_table_template(basic_template) {
        variable_1 : output_pin_transition;
        variable_2 : connect_delay;
        index_1 ("0, 1");
        index_2 ("0, 1");
  }
  ...
        fall_transition_degradation(basic_template) {
                values ("-1, 2", "5, 8");
        }
```

The following is an example message:

```
Warning: Line 123, The 'values' attribute has a '-1.000000' value,
        which is less than '0.000000' the minimum required value of
        this attribute. The value is changed to the minimum value.
 (LBDB-274)
```

### What Next

Check the library source file, and correct the value if the minimum value is not acceptable. Refer to the "Library Compiler Reference Manual" for the attribute value requirements.

Feedback

## LBDB-275

(warning) The '%s' attribute has size 1, which is not \n \tused for any purpose with its associated variable.

### Description

This message indicates that the size of one of the *lu_table_template* attributes has size 1. Library Compiler expects the size of each dimension to be larger than *1*.

The following example shows an instance where this message occurs:

```
lu_table_template(lbdb275) {
        variable_1 : input_net_transition;
        variable_2 : output_net_length;
        index_1 ("0.1");
        index_2 ("0.1, 0.2, 0.3, 0.4");
  }
```

The following is an example message:

```
Warning: Line 20, The 'index_1' attribute has size 1, which is not
        used for any purpose with its associated variable. (LBDB-275)
```

### What Next

Check the library source file, and delete this dimension if it is not needed.

## LBDB-276

(error) The '%s' refers to an invalid '%s'.

### Description

This message indicates that the specified name refers to an invalid *lu_table_template* group. First, Library Compiler issues all the errors that make the group invalid and issues this message when the template is used.

The following example shows an instance where this message occurs:

```
lu_table_template(lbdb276_template) {
        variable_1 : constrained_pin_transition;
        variable_2 : output_net_length;
        index_1 ("-0.1, 1.2, 2.3, 3.4");
        index_2 ("0.1, 1.2, 2.3, 3.4");
  }

          rise_transition(invalid_template) {
                values ("1, 2, 3, 4", "5, 6, 7, 8", \
                        "9, 10, 11, 12");
          }
```

Feedback

```
In this case, the index_1 value is less than zero, so the look-up
table becomes invalid. The rise_transition timing information refers
to an invalid table.
```

The following is an example message:

```
Error: Line 165, The 'lbdb276_template' refers to an invalid
 'lu_table_template'. (LBDB-276)
```

**What Next**

Check the library source file, and correct the errors in the look-up table template to which this table refers.

---

## LBDB-278

(error) The 'timing' group with '%s' timing_type \n \tneeds at least one look-up table.

**Description**

This message indicates that in a nonlinear delay library, you did not specify a look-up table (rise_constraint or fall_constraint) for timing checks in a timing group. To be complete, Library Compiler expects at least one look-up table for the timing check groups.

The following example shows an instance where this message occurs:

```
pin ( D )    {
      direction : input;
      capacitance : 1;
      timing ()      {
        related_pin : CK ;
        timing_type : setup_rising ;
      }
  }
```

To fix the problem, define the template group and the rise_constraint group to the timing group,

```
lu_table_template(constraint) {
    variable_1 : constrained_pin_transition;
    index_1("0, 1, 2, 3, 4");
    variable_2 : related_pin_transition;
    index_2("0, 1, 2, 3, 4");
  }
  pin ( D )    {
      direction : input;
      capacitance : 1;
      timing () {
        related_pin : CK ;
        timing_type : setup_rising ;
        rise_constraint(constraint) {
```

Feedback

```
          values("1.0000, 1.0000, 1.0000, 1.0000, 1.0000",  \
                 "1.0000, 1.0000, 1.0000, 1.0000, 1.0000",  \
                 "1.0000, 1.0000, 1.0000, 1.0000, 1.0000",  \
                 "1.0000, 1.0000, 1.0000, 1.0000, 1.0000",  \
                 "1.0000, 1.0000, 1.0000, 1.0000, 1.0000");
        }
      }
  }
```

The following is an example message:

```
Error: Line 63, The 'timing' group with 'setup_rising' timing_type
        needs at least one look-up table. (LBDB-278)
```

**What Next**

Check the library source file, and correct the timing group. Add either a rise_constraint or fall_constraint table to the group.

---

## LBDB-279

(error) The 'timing' group with '%s' timing_type \n \tneeds one look-up table.

**Description**

This message indicates that you specified more than one table for the specified timing group. This group needs only one table, either *constraint_rise* or *constraint_fall*.

The following example shows an instance where this message occurs:

```
pin(cd) {
      direction : input;
      capacitance : 0.065
      timing() {
        related_pin : "cp";
        timing_type : recovery_rising;
        rise_constraint(constraint_template) {
          values("1, 2", "3, 4");
        }
        fall_constraint(scalar) {
          values("1");
        }
      }
    }
```

In this case, the 'cd' pin has both the rise_constraint and fall_constraint attribute. To fix the problem, delete the fall_constraint table.

The following is an example message:

```
Error: Line 61, The 'timing' group with 'recovery_rising' timing_type
        needs one look-up table. (LBDB-279)
```

Feedback

### What Next

Check the library source file, and correct the timing group. Delete from the group either the *constraint_rise* table or the *constraint_fall* table.

---

## LBDB-280

(error) The '%s' look-up table cannot use\n \t'%s' as its template. The look-up table is not\n \tcompatible with the template.

### Description

This message indicates that the specified look-up table is not compatible with the specified template. Variable types described in a look-up table template must be compatible with the usage of that template. For example, a template references variable types used for constraint checking ('related_pin_transition' and 'constrained_pin_transition') cannot be referenced in a cell delay description ('cell_rise' or 'cell_fall').

The following example shows an instance where this message occurs:

```
lu_table_template(basic_template) {
    variable_1 : input_net_transition;
    index_1("0, 1, 2, 3, 4");
    variable_2 : output_net_length;
    index_2("0, 1, 2");
  }
  lu_table_template(constraint) {
    variable_1 : constrained_pin_transition;
    index_1("0, 1, 2, 3, 4");
    variable_2 : related_pin_transition;
    index_2("0, 1, 2, 3, 4");
  }

        rise_constraint(basic_template) {
                values ("1, 2, 3, 4", "5, 6, 7, 8", \
                        "9, 10, 11, 12", "13, 14, 15, 16");
        }
```

To fix the problem, change the template value of the rise_constraint group from basic_template to constraint.

The following is an example message:

```
Error: Line 126, The 'rise_constraint' look-up table cannot use
        'basic_template' as its template. The look-up table is not
        compatible with the template. (LBDB-280)
```

Feedback

### What Next

Refer to the "Library Compiler User Guide" for more information about look-up tables. Change the library source file by referencing a different template in the look-up table description.

---

## LBDB-281

(error) The '%s' template has variables which\n \tshould not be used together.

### Description

This message indicates that you specified two incompatible variables in the same template.

The following example shows an instance where this message occurs:

```
lu_table_template(invalid_template) {
        variable_1 : constrained_pin_transition;
        variable_2 : output_net_length;
        index_1 ("0.1, 0.2, 0.3, 0.4");
        index_2 ("0.1, 0.2, 0.3, 0.4");
  }
```

The following is an example message:

```
Error: Line 32, The 'invalid_template' template has variables which
        should not be used together. (LBDB-281)
```

### What Next

Refer to the "Library Compiler User Guide" for more information on defining CMOS nonlinear timing model templates. Check the library source file to see if you made a mistake when you entered the variables strings.

---

## LBDB-282

(error) The '%s' attribute has a '%f' value\n \tthat is larger than the maximum allowed value of '%f'.

### Description

This message indicates that for the specified attribute, you defined a value that is larger than the maximum allowed value. The error message includes the maximum value for this attribute.

The following example shows an instance where this message occurs:

```
default_min_porosity : 91.0;
```

Feedback

The following is an example message:

```
Error: Line 53, The 'default_min_porosity' attribute has a '91.000000'
 value
        that is larger than the maximum allowed value of '90.000000'.
 (LBDB-282)
```

**What Next**

Refer to the "Library Compiler User Guide" for the maximum value required for the attribute, and correct the value in the library source file.

---

## LBDB-283

(error) The '%s' layer name is invalid.

**Description**

This message indicates that you specified an invalid name for the current attribute or function. The name should match a name previously defined in the resource group.

If the value is in the related_layer attribute and the layer is not valid. It may be because it is not one of the neighboring contact layer.

The following example shows an instance where this message occurs:

```
same_net_min_spacing(dummy, metal1, 0.1, TRUE);
```

The following is an example message:

```
Error: Line 104, The 'dummy' layer name is invalid.  (LBDB-283)
```

**What Next**

Check your library to see if you have an error in either previous layer definition or the current attribute / function.

---

## LBDB-284

(error) Missing 'routing_layers' attribute for this library\n \twhich is needed for specifying 'routing_track'.

**Description**

This message indicates that you specified a *routing_track* group without specifying the *routing_layers* attribute. Library Compiler expects the *routing_layers* complex attribute to be defined before you can specify the *routing_track* group. The *routing_track* group has a name to refer to a specific routing layer in the *routing_layers* complex attribute.

Feedback

The following example shows an instance where this message occurs:

```
routing_track(metal2) {
        tracks : 2;
        total_track_area : 0.2;
   }
```

To fix the problem, add the attribute to the library,

```
routing_layers(metal1,metal2);
```

The following is an example message:

```
Error: Line 65, Missing 'routing_layers' attribute for this library,
        which is needed for specifying 'routing_track'. (LBDB-284)
```

**What Next**

Check the library source file to see if you are missing the *routing_layers* attribute, which should be placed in the library before the *routing_track* group can be identified.

---

## LBDB-285

(error) In this 'routing_track' group, 'tracks' is 0 and\n \t'total_track_area' is %f, which is inconsistent.

**Description**

This message indicates that for a *routing_track* group, you specified the *total_track_area* attribute without specifying the *tracks* attribute. If you do not have tracks in a routing layer, you cannot have *total_track_area* for that layer either.

The following example shows an instance where this message occurs:

```
routing_layers(metal1,metal2);

   routing_track(metal2) {
        /* missing tracks */
        total_track_area : 1.5;
   }
```

To fix the problem, add the tracks attribute to the routing_track group,

```
tracks : 1;
```

The following is an example message:

```
Error: Line 266, In this 'routing_track' group, 'tracks' is 0 and
        'total_track_area' is 1.500000, which is inconsistent. (LBDB-285)
```

Feedback

### What Next

Check your library to see if it is missing the *tracks* attribute in this group. Do not have the *total_track_area* other than '0', or this group is not needed.

---

## LBDB-286

(error) No 'routing_track' information in the '%s' library.

### Description

This message indicates that, in the library, you specified the *default_min_porosity* attribute without specifying the *routing_track* group in any of the cells.

The following is an example message:

```
Error: Line 1, No 'routing_track' information in the 'lbdb286' library.
 (LBDB-286)
```

### What Next

Remove the *default_min_porosity* attribute or add *routing_track* information into the library.

---

## LBDB-287

(warning) No routability information in the '%s' %s.

### Description

This message indicates that one of the two cases occurred:

```
* You have the routing_layers attribute in a library and
          do not have routing_track group in the cell.
        * You have the default_min_porosity attribute in a
          library and do not have the routing_track information
          in a cell.
```

The following example shows an instance where this message occurs:

```
default_min_porosity : 0.0;
    routing_layers(metal1,metal2);

cell(AN2) {
  area : 2;
  pin(A) {
    direction : input;
    capacitance : 1;
  }
  pin(B) {
    direction : input;
    capacitance : 1;
```

Feedback

```
  }
  pin(Z) {
    direction : output;
    function : "A B";
    timing() {
      intrinsic_rise : 0.1;
      intrinsic_fall : 0.1;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "A";
    }
    timing() {
      intrinsic_rise : 0.1;
      intrinsic_fall : 0.1;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "B";
    }
  }
}
```

In this case, The library has the *default_min_porosity* attribute, but the 'AN2' cell has no *routing_track* information.

The following is an example message:

```
Warning: Line 63, No routability information in the 'AN2' cell.
 (LBDB-287)
```

### What Next

Check the library source file to remove the *routing_layers* attribute if the warning fits the first case. For the second case, check the cell to make sure it does not have any *routing_track* information on all routing layers (all of them are completely obstructed in this cell) when you have the *default_min_porosity* attribute.

---

## LBDB-288

(error) The '%f' 'total_track_area' value is larger than\n \tthe '%f' cell area value.

### Description

This message indicates that you specified a larger value for the *total_track_area* than the *cell area* value. The *total_track_area* value of a particular routing layer cannot be larger than the *cell area* value.

The following example shows an instance where this message occurs:

```
cell(IVP) {
  area : 1;
   routing_track(metal2) {
```

Feedback

```
        tracks : 1;
        total_track_area : 1.5;
    }
  pin(A) {
    direction : input;
    capacitance : 1;
  }
  pin(Z) {
    direction : output;
    function : "A'";
    timing() {
      intrinsic_rise : 0.1;
      intrinsic_fall : 0.1;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      slope_rise : 0.0;
      slope_fall : 0.0;
      related_pin : "A";
    }
  }
}
```

The following is an example message:

```
Error: Line 242, The 1.500000 'total_track_area' value is larger than
        the '1.000000' cell area value. (LBDB-288)
```

**What Next**

Check to see if you have entered the wrong information for either of these two attributes.

---

## LBDB-289

(error) '%s' has been used more than one time in the\n \tspecification.

**Description**

This message indicates that you specified the same routing layer name more than once in the routing_layers attribute. Library Compiler expects unique names.

The following example shows an instance where this message occurs:

```
routing_layers(metal2,metal2);
```

The following is an example message:

```
Error: Line 55, 'metal2' has been used more than one time in the
        specification. (LBDB-289)
```

**What Next**

Check to see if you entered the wrong information for this attribute.

Feedback

## LBDB-290

(error) '%s' should have at least one entry in it.

### Description

This message indicates that you specified an attribute without any entries. This complex attribute must contain at least one entry.

The following example shows an instance where this message occurs:

```
routing_layers();
```

The following is an example message:

```
Error: Line 18, 'routing_layers' should have at least one entry in it.
 (LBDB-290)
```

### What Next

Check to see if you have entered the required information for this complex attribute.

## LBDB-291

(warning) No routability information for the '%s' layer.\n \tIt is assumed to be completely obstructed.

### Description

The *routing_track* group of the indicated layer is not in the cell. It is assumed to be completely obstructed.

The following example shows an instance where this message occurs:

```
library(small_cmos1) {
    default_min_porosity : 0.0;
    routing_layers(metal1,metal2);

    cell(AN2) {
       area : 2;
       routing_track(metal2) {
         tracks : 2;
         total_track_area : 0.2;
       }
       pin(A) {
         direction : input;
         capacitance : 1;
       }
       pin(B) {
         direction : input;
         capacitance : 1;
```

Feedback

```
        }
        pin(Z) {
          direction : output;
          function : "A B";
          timing() {
             intrinsic_rise : 0.1;
             intrinsic_fall : 0.1;
             rise_resistance : 0.1;
             fall_resistance : 0.1;
             related_pin : "A";
          }
          timing() {
             intrinsic_rise : 0.1;
             intrinsic_fall : 0.1;
             rise_resistance : 0.1;
             fall_resistance : 0.1;
             related_pin : "B";
          }
        }
    }
}
```

In this case, 'metal1' is not used in any routing_track group.

The following is an example message:

```
Warning: Line 240, No routability information for the 'metal1' layer.
        It is assumed to be completely obstructed. (LBDB-291)
```

**What Next**

If the previous statement is not true, enter the *routing_track* information for the indicated layer. Otherwise, it is all right.

---

## LBDB-292

(error) The '%s' should be the %d%s attribute in the library.

**Description**

This message indicates that you specified the technology and delay_model in unacceptable order. For the Library Compiler to work correctly, the defined attribute must appear in this library at the indicated place in the error message.

The following example shows an instance where this message occurs:

```
library(lib) {
  technology : "cmos";
  delay_model : "generic_cmos";
}
```

Feedback

In this case, the delay_model attribute comes after the technology attribute. To fix the problem, switch the order of definition of the two attributes.

The following is an example message:

```
Error: Line 5, The 'delay_model' should be the 1st attribute in the
 library. (LBDB-292)
```

**What Next**

Modify the library source file to move the defined attribute to the correct place, as indicated in the error message.

---

## LBDB-292w

(warning) The '%s' should be the %d%s attribute in the library.

**Description**

This message indicates that you specified the technology and delay_model in unacceptable order. For the Library Compiler to work correctly, the defined attribute must appear in this library at the indicated place in the warning message.

The following example shows an instance where this message occurs:

```
library(lib) {
  technology : "cmos";
  delay_model : "generic_cmos";
}
```

In this case, the delay_model attribute comes after the technology attribute. To fix the problem, switch the order of definition of the two attributes.

The following is an example message:

```
Warning: Line 5, The 'delay_model' should be the 1st attribute in the
 library. (LBDB-292w)
```

**What Next**

Modify the library source file to move the defined attribute to the correct place, as indicated in the warning message.

---

## LBDB-293

(error) In this 'routing_track' group, the 'tracks' value\n \tis %d and the 'total_track_area' value is '0.0'. This is\n \tinconsistent.

Feedback

### Description

This message indicates that you specified the *tracks* attribute without specifying the *total_track_area* attribute in a *routing_track* group. If you have tracks in a routing layer, you need to have the *total_track_area* for that layer too.

The following example shows an instance where this message occurs:

```
routing_track(metal2) {
        tracks : 2;
        /* missing total_track_area */
   }
```

The following is an example message:

```
Error: Line 205, In this 'routing_track' group, the 'tracks' value
        is 2 and the 'total_track_area' value is '0.0'. This is
        inconsistent. (LBDB-293)
```

### What Next

Check the library source file to see if it is missing the *total_track_area* attribute in this group, and add it. Otherwise, set the *tracks* value to 0 or remove the routing_track group.

---

## LBDB-294

(warning) The UP or DOWN X pin coordinate does not lay on a grid.

### Description

This warning message appears when the library symbol has a pin that is not on a grid and is off by 0.01.

The following example shows an instance where this message occurs:

```
library(lbdb294) {
 SCALE = 1.0 / 8.0;

  symbol("gndsym") {
    set_minimum_boundary(0 * SCALE, 0 * SCALE, 40 * SCALE, 40 * SCALE);
    line(5 * SCALE, 10 * SCALE, 35 * SCALE, 10 * SCALE);
    line(0 * SCALE, 20 * SCALE, 40 * SCALE, 20 * SCALE);
    line(15 * SCALE, 2 * SCALE, 25 * SCALE, 2 * SCALE);
    line(20 * SCALE, 20 * SCALE, 20 * SCALE, 40 * SCALE);
    pin("gnd", 20 * SCALE, 40 * SCALE, ANY_ROTATION);
  }
}
```

In this case, the "gnd" pin has the Y coordinate off the grid.

Feedback

The following is an example message:

```
Warning: Line 24, The UP or DOWN X pin coordinate does not lay on a grid.
 (LBDB-294)
```

### What Next

Check the symbol library file, and correct the X coordinate.

---

## LBDB-295

(warning) The LEFT or RIGHT Y pin coordinate does not lay on a grid.

### Description

This warning message appears when the library symbol has a pin that is not on a grid and is off by 0.01.

The following example shows an instance where this message occurs:

```
library(lbdb295) {
 SCALE = 1.0 / 8.0;

  symbol("gnd") {
    set_minimum_boundary(0 * SCALE, 0 * SCALE, 40 * SCALE, 40 * SCALE);
    line(5 * SCALE, 10 * SCALE, 35 * SCALE, 10 * SCALE);
    line(0 * SCALE, 20 * SCALE, 40 * SCALE, 20 * SCALE);
    line(15 * SCALE, 2 * SCALE, 25 * SCALE, 2 * SCALE);
    line(20 * SCALE, 20 * SCALE, 20 * SCALE, 40 * SCALE);
    pin("Z", 128 * SCALE, 4 * SCALE, ANY_ROTATION);
    pin("gnd", 20 * SCALE, 40 * SCALE, ANY_ROTATION);
  }
}
```

In this case, the "Z" pin has the Y coordinate off the grid.

The following is an example message:

```
Warning: Line 24, The LEFT or RIGHT Y pin coordinate does not lay on a
 grid. (LBDB-295)
```

### What Next

Check the symbol library file, and correct the Y coordinate.

---

## LBDB-296

(warning) The '%s' pin of the '%s' auxiliary pad cell\n \tshould not have a function attribute.

Feedback

### Description

This message indicates that you specified a function attribute on a auxiliary pad cell.

Cells with a 'auxiliary_pad_cell' attribute are used together with other cells with a 'pad_cell' attribute to build a logical pad. They can also provide the pull-up, pull-down, buffer, or inverter functionality. They should not have real 'logic'. Design Compiler will not work correctly if you decompose your pad cell this way.

The following example shows an instance where this message occurs:

```
cell(lbdb296) {
      auxiliary_pad_cell : true;
      area : 1;
      general_drivers : 0;
      input_drivers : 8;
      pin(A) {
        direction : input;
        capacitance : 1;
      }
      pin(Z) {
        direction : output;
        function : "A'";
        timing() {
          intrinsic_rise : 0.1;
          intrinsic_fall : 0.1;
          rise_resistance : 0.1;
          fall_resistance : 0.1;
          related_pin : "A";
        }
      }
    }
```

The following is an example message:

```
Warning: Line 64, The 'Z' pin of the 'lbdb296' auxiliary pad cell
        should not have a function attribute. (LBDB-296)
```

### What Next

Check the technology source library. Either

```
1. remove the 'function' attribute from the cell under the
      following conditions:
        * If the cell is a buffer, no action is required.
        * If the cell is an inverter, invert the functionality of your
          'pad_cell', which is to be connected to this cell to form a
          pad cell, accordingly.
or
   2. remove the 'auxiliary_pad_cell' attribute, and model this cell
      as a regular cell but with the right 'connection_class' on the
      pin so it can be connected to the 'pad_cell' to form a pad as
      intended.
```

Feedback

## LBDB-297

(error) The '%s' pin is reserved and should not be used as a pin name.

### Description

Library Compiler uses several keywords internally for a sequential cell. The keywords are

```
* next_state
        * clocked_on
        * clocked_on_also
        * data_in
        * enable
        * enable_also
        * preset
        * clear
        * force_00
        * force_01
        * force_10
        * force_11
        * the variable1 and variable2 used in ff/latch/ff_bank/latch_bank

The following example shows an instance where this message occurs:
  pin(preset) {
    direction : input;
    capacitance : 1;
  }
```

The following is an example message:

```
Error: Line 182, The 'preset' pin is reserved and should not be used as a
 pin name. (LBDB-297)
```

### What Next

Modify your library to change the name of the pin.

## LBDB-298

(error) The '%s' attribute has a value %d, which is\n \tless than the minimum required value of this attribute %d.

### Description

This message indicates that the value specified is less than the required minimum value for this attribute.

Feedback

The following example shows an instance where this message occurs:

```
wire_load_table("40x40") {
      fanout_area(-11, 0.1);
   }
```

The following is an example message:

```
Error: Line 84, The 'fanout_area' attribute has a value -11, which is
        less than the minimum required value of this attribute 1.
 (LBDB-298)
```

**What Next**

Check the library source file, and correct the value of the attribute. If the minimum value for this attribute is not displayed in the error message, refer to the *Library Compiler User Guide Manual* for minimum attribute values.

---

## LBDB-299

(error) The library is missing 'operating_conditions' groups and 'default_operating_conditions' attribute.

**Description**

This error message occurs when the *default_operating_conditions* is undefined and there is no operating_conditions groups defined.

**What Next**

If you do not want Library Compiler to create the default operating_conditions for you, please check your library to define the relative operating_condition group and set it as the default_operating_conditions with the library-level attribute "default_operating_conditions".

---

## LBDB-300

(error) The '%s' related_outputs for the internal_power group has\n \tbeen specified. A duplicate is not allowed.

**Description**

This message indicates that you include the same output pin in more than one internal_power group.

The following example shows an instance where this message occurs:

```
internal_power(test_input) {
    values("0.1, 1.2, 2.3, 3.4");
    related_outputs : "Z";
   }
```

Feedback

```
   internal_power(test_input) {
    values("0.1, 1.2, 2.3, 3.4");
    related_outputs : "Z";
   }
```

The following is an example message:

```
Error: Line 41, The 'Z' related_outputs for the internal_power group has
        been specified. A duplicate is not allowed. (LBDB-300)
```

### What Next

Check the library source file, and delete the duplicate information from one of the internal_power groups.

---

## LBDB-301

(information) No internal_power information for the '%s' cell.

### Description

This message indicates that you did not define an internal_power group for the specified cell. Library Compiler expects at least one internal_power group so the internal power of the cell can be calculated.

The following example shows an instance where this message occurs:

```
cell(lbdb301) {
  area : 2;
  pin(A) {
    direction : input;
    capacitance : 1;
  }
  pin(B) {
    direction : input;
    capacitance : 1;
  }
  pin(Z) {
    direction : output;
    function : "A B";
    timing() {
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      slope_rise : 0.0;
      slope_fall : 0.0;
      related_pin : "A B";
    }
  }
  cell_leakage_power : 1;
}
```

Feedback

In this case, the 'lbdb301' cell is missing the internal_power group. To fix the problem, add this statement:

```
internal_power(output_by_cap_and_trans) {
     values(" 5.000000 ,  15.000000 ,  0.300000 ", \
            " 1.000000 ,  5.000000 ,  0.000000 ", \
            " 0.000000 ,  0.000000 ,  0.000000 ");
     related_outputs : "Z";
     related_inputs : "A B";
  }
```

The following is an example message:

```
Information: Line 191, No internal_power information for the 'lbdb301'
 cell. (LBDB-301)
```

### What Next

If you want to include the internal power of this cell in your design, add the required information into the library.

---

## LBDB-302

(error) The '%s' specified in the '%s' is an\n \t%s pin.

### Description

This message indicates that you specified either an input pin in a *related_outputs* attribute or an output pin in a *related_inputs* attribute.

The following example shows an instance where this message occurs:

```
cell(lbdb302) {
  area : 2;
  pin(A) {
    direction : input;
    capacitance : 1;
  }
  pin(B) {
    direction : input;
    capacitance : 1;
  }
  pin(Z) {
    direction : output;
    function : "A B";
    timing() {
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      slope_rise : 0.0;
      slope_fall : 0.0;
```

Feedback

```
      related_pin : "A B";
    }
  }
  cell_leakage_power : 1;
  internal_power(output_by_cap_and_trans) {
     values(" 5.000000 ,  15.000000 ,  0.300000 ", \
            " 1.000000 ,  5.000000 ,  0.000000 ", \
            " 0.000000 ,  0.000000 ,  0.000000 ");
     /* bad related port */
     related_outputs : "A";
     related_inputs : "Z B";
  }
}
```

In this case, both the related_outputs and the related_inputs attributes have a wrong pin name. To fix the problem, change the name from 'A' to 'Z' in the related_outputs attribute and 'Z' to 'A' in the related_inputs attribute.

The following is an example message:

```
Error: Line 272, The 'A' specified in the 'related_outputs' is an
        input pin. (LBDB-302)
Error: Line 273, The 'Z' specified in the 'related_inputs' is an
        output pin. (LBDB-302)
```

### What Next

Check the library source file, and make the necessary correction.

---

## LBDB-303

(error) You cannot specify more than one input pin or one bit\n \tof a bus or bundle input pin in the 'related_input'.

### Description

This message indicates that you specified more than one input pin or one bit of an input pin, in the case of bus and bundle groups, in the *related_input* attribute. Library Compiler allows only one input pin or one bit of a bus or bundle in the attribute.

The following example shows an instance where this message occurs:

```
cell(lbdb303) {
  area : 2;
  pin(A) {
    direction : input;
    capacitance : 1;
  }
  pin(B) {
    direction : input;
    capacitance : 1;
```

Feedback

```
  }
  pin(Z) {
    direction : output;
    function : "A+B";
    timing() {
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      slope_rise : 0.0;
      slope_fall : 0.0;
      related_pin : "A B";
    }
  }
  cell_leakage_power : 1;
  internal_power(output_by_cap_and_trans) {
     values(" 5.000000 ,  15.000000 ,  0.300000 ", \
            " 1.000000 ,  5.000000 ,  0.000000 ", \
            " 0.000000 ,  0.000000 ,  0.000000 ");
     related_outputs : "Z";
     /* more than one related input */
     related_input : "A B";
  }
}
```

In this case, the related_input attribute has two 'A' and 'B' input pins.

The following is an example message:

```
Error: Line 315, You cannot specify more than one input pin or one bit
        of a bus or a bundle input pin in the 'related_input'. (LBDB-303)
```

### What Next

Check the "Library Compiler User Guide" for more information on power analysis, and correct the problem. Make sure that the related_input attribute has only one input pin or one bit of an bus or bundle input pin and the related_inputs attribute has more than one pin.

---

## LBDB-304

(error) You can specify '%s'\n but not both, because they are mutually exclusive.

### Description

This message indicates that you specified a *related_input* attribute with a *related_outputs* attribute in an internal_power group. Library Compiler allows either a *related_input* attribute or the pair *related_outputs*, *related_inputs* attributes, but not both, since the

Feedback

information provided is mutually exclusive. Library Compiler supports the related_input, related_inputs, and the related_outputs attributes in the table as follows:

```
* A one dimensional table uses only related_outputs.
        * A one dimensional table uses only related_input.
        * A one dimensional table uses both related_inputs and
          related_outputs.
        * A two dimensional table uses both related_inputs and
          related_outputs.
```

The following example shows an instance where this message occurs:

```
cell(lbdb304) {
  area : 2;
  pin(A) {
    direction : input;
    capacitance : 1;
  }
  pin(B) {
    direction : input;
    capacitance : 1;
  }
  pin(Z) {
    direction : output;
    function : "A+B";
    timing() {
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      slope_rise : 0.0;
      slope_fall : 0.0;
      related_pin : "A B";
    }
  }
  cell_leakage_power : 1;
  internal_power(output_by_cap_and_trans) {
     values(" 5.000000 ,  15.000000 ,  0.300000 ", \
            " 1.000000 ,  5.000000 ,  0.000000 ", \
            " 0.000000 ,  0.000000 ,  0.000000 ");
     related_outputs : "Z";
     related_input : "A B";
  }
}
```

In this case, there is a typo. To fix the problem, add the character s to the related_input attribute name.

The following is an example message:

```
Error: Line 309, You can specify 'related_outputs/related_inputs or
 related_input'
        but not both, because they are mutually exclusive. (LBDB-304)
```

Feedback

### What Next

Check the library and correct the related_inputs attribute if it is a typo or delete the unnecessary information from the library.

---

## LBDB-305

(error) In the context, the '%s' valid value of the\n \t'%s' template is '%s'.\n \tYou specified '%s'.

### Description

This message indicates that you specified an invalid value for the variable_1 in the power_lut_template given the specified information in the internal_power group. Library Compiler expects

```
* A related_input attribute in an internal_power group for a
          input_transition_time value in variable_1 of the corresponding
          power_lut_template
        * A related_outputs attribute in an internal_power group for a
          total_output_net_capacitance value in variable_1 of the
          corresponding power_lut_template
```

The following example shows an instance where this message occurs:

```
power_lut_template(lbdb305_input) {
    variable_1 : input_transition_time;
    index_1 ("1.0, 2.0");
  }

cell(lbdb305) {
  internal_power(lbdb305_input) {
    index_1("1.0, 2.0");
    values("1.0, 2.0");
    related_outputs : "Z";
  }
}
```

The following is an example message:

```
Error: Line 27, In the context, the 'variable_1' valid value of the
        'lbdb305_input' template is 'total_output_net_capacitance'.
        You specified 'input_transition_time'. (LBDB-305)
```

### What Next

Refer to the "Library Compiler Reference Manual Volume 1" for more information on power modeling and analysis. Correct the attribute's value or its reference in the library.

Feedback

## LBDB-306

(warning) The '%s' preferred input voltage does not exist in this library.

### Description

This message indicates that you specified a input_voltage name to the *preferred_input_pad_voltage* attribute that is not described in the library or is defined after this attribute.

The following example shows an instance where this message occurs:

```
preferred_input_pad_voltage : "lbdb306";
  input_voltage(lbdb306) {
    vil : 1.5;
    vih : 3.5;
    vimin : -0.3;
    vimax : VDD + 0.3;
  }
```

in this case, the 'lbdb306' input_voltage is defined after the preferred_input_pad_voltage attribute.

The following is an example message:

```
Warning: Line 25, The 'lbdb306' preferred input voltage does not exist in
 this library. (LBDB-306)
```

### What Next

Check the preferred input pad voltage to be sure it is correct. If the input_voltage exists, make sure that the attribute is defined before the preferred_input_pad_voltage attribute. Otherwise, add the preferred input pad voltage to the library description.

## LBDB-307

(warning) The '%s' preferred output voltage does not exist in this library.

### Description

This message indicates that you specified an output_voltage name to the *preferred_output_pad_voltage* attribute that is not described in the library or is defined after this attribute.

The following example shows an instance where this message occurs:

```
preferred_output_pad_voltage : "lbdb307" ;
  output_voltage(lbdb307) {
    vol : 1.5;
    voh : 3.5;
    vomin : -0.3;
```

Feedback

```
    vomax : VDD + 0.3;
  }
```

in this case, the 'lbdb307' output_voltage is defined after the preferred_output_pad_voltage attribute.

The following is an example message:

```
Warning: Line 24, The 'lbdb307' preferred output voltage does not exist
 in this library. (LBDB-307)
```

### What Next

Check the preferred output pad voltage to be sure it is correct. If the output_voltage exists, make sure that the attribute is defined before the preferred_output_pad_voltage attribute. Otherwise, add the preferred output pad voltge to the library description.

---

## LBDB-308

(warning) Incomplete set of pads to support the '%s'\n \tpreferred output pad slew rate control.

### Description

This message indicates that in the library you specified the *preferred_output_pad_slew_rate_control* attribute but you did not define a complete set of pads to support it. Library Compiler expects at least one of the following set of pad cells in the library with the given slew rate control value:

```
* One output pad
        * One bidirectional pad
        * One three state pad
```

The following example shows an instance where this message occurs:

```
library (lbdb308) {
  preferred_output_pad_slew_rate_control : high;
  output_voltage(GENERAL) {
    vol          : 0.33 ;
    voh          : 3.7 ;
    vomin        : -0.3 ;
    vomax        : VDD + 0.3 ;
  }
  cell(OUTBUF) {
    area : 0.000000;
    dont_touch : false;
    dont_use   : false;
    pad_cell : true;
    pin(D ) {
      direction   : input;
      capacitance : 1.0;
```

Feedback

```
    }
    pin(PAD ) {
      is_pad : true;
      output_voltage : GENERAL;
      drive_current : 4.0;
      slew_control : high;
      direction   : output;
      function    : "D";
      timing() {
        intrinsic_fall  : 1.0;
        intrinsic_rise  : 1.0;
        fall_resistance : 0.1;
        rise_resistance : 0.1;
        related_pin  :"D";
      }
    }
  }
  cell(BBHS) {
    area : 0.000000;
    dont_touch : false;
    dont_use   : false;
    pad_cell : true;
    pin(E D ) {
      direction   : input;
      capacitance : 1.0;
    }
    pin(Y ) {
      direction   : output;
      timing() {
        intrinsic_fall  : 1.0;
        intrinsic_rise  : 1.0;
        fall_resistance : 0.1;
        rise_resistance : 0.1;
        related_pin  :"PAD ";
      }
    }
    pin(PAD ) {
      is_pad : true;
      input_voltage : CMOS;
      output_voltage : GENERAL;
      drive_current : 4.0;
      slew_control : high;
      direction   : inout;
      function    : "D";
      three_state : "E'";
      timing() {
        intrinsic_fall  : 1.0;
        intrinsic_rise  : 1.0;
        fall_resistance : 0.1;
        rise_resistance : 0.1;
        related_pin  :"E D";
      }
    }
```

Feedback

```
  }
}
```

In this case, the 'lbdb308' library has an output pad cell and a bidirectional pad cell, but it is missing a three_state pad cell.

The following is an example message:

```
Warning: Line 2, Incomplete set of pads to support the 'high'
        preferred output pad slew rate control. (LBDB-308)
```

### What Next

Check the preferred output pad slew rate control to be sure it is correct, and add pads to the library containing the preferred output pad slew rate control value.

---

## LBDB-309

(warning) In the '%s' gate, cannot degenerate the \n \t'%s' output due to excessive function size.

### Description

This message appears when the Library Compiler has determined that degenerating the given output is potentially too difficult. This determination is based on an evaluation of the function complexity at this output port.

### What Next

Either remove the *fpga_complex_degenerate* attribute from the library cell in question, or set the *fpga_degenerate_output* attribute to FALSE on this particular output. In either case, these actions delete the warning; they do not cause the degeneration software to attempt to degenerate the gate.

---

## LBDB-310

(information) Degenerating %d-input gates from '%s' component\n \tof the '%s' output.

### Description

This message indicates that the Library Compiler is starting to create degenerate functions based on the component named.

The following example shows an instance where this message occurs:

```
cell(lbdb310) {
    area : 1.0;
    fpga_complex_degenerate : true ;
    pin(D0 D1 D2 D3 S00 S01 S10 S11) {
      direction   : input;
```

Feedback

```
      capacitance : 1.0;
    }
    pin(Y) {
      direction : output;
      function :
 "(D0!(S00&S01)+D1(S00&S01))!(S10+S11)+(D2!(S00&S01)+D3(S00&S01))(S10+S11
)";
      timing() {
        rise_resistance : 0.5;
        fall_resistance : 0.5;
        related_pin  :  "D0 D1 D2 D3 S00 S01 S10 S11" ;
      }
    }
  }
```

The following is an example message:

```
Information: Degenerating 2-input gates from the 'lbdb310' component
        of the 'Y' output. (LBDB-310)
```

### What Next

No action is required.

---

## LBDB-311

(information) Degenerated %d %d-input gates from the '%s' component.\n \tA total of %d components have been degenerated from the '%s' output.

### Description

This message indicates that the Library Compiler has finished creating n-input degenerate functions based on the component and output named. The message tells you how the library size is growing and how long the degeneration process is taking.

The following example shows an instance where this message occurs:

```
cell(lbdb311) {
    area : 1.0;
    fpga_complex_degenerate : true ;
    pin(D0 D1 D2 D3 S00 S01 S10 S11) {
      direction   : input;
      capacitance : 1.0;
    }
    pin(Y) {
      direction   : output;
      function :
 "(D0!(S00&S01)+D1(S00&S01))!(S10+S11)+(D2!(S00&S01)+D3(S00&S01))(S10+S11
)";
      timing() {
        rise_resistance : 0.5;
        fall_resistance : 0.5;
```

Feedback

```
        related_pin  :  "D0 D1 D2 D3 S00 S01 S10 S11" ;
      }
    }
  }
```

The following is an example message:

```
Information: Degenerated 2 1-input gates from the 'lbdb311' component.
        A total of 2 components have been degenerated from the 'Y'
 output. (LBDB-311)
```

### What Next

No action is required.

---

## LBDB-312

(warning) Unable to do fpga complex degeneration on the '%s' gate because it %s.

### Description

This message occurs when you have defined *fpga_complex_degenerate* on a component for which this feature is not currently supported. Library Compiler issues this message for the following cells:

```
* Sequential devices: latches and flip flops
        * I/O pads
        * Cells with dont_use attribute
        * Cells with three_state attribute
```

### What Next

Remove the *fpga_complex_degenerate* attribute for this component, or set the attribute to FALSE.

---

## LBDB-315

(error) The '%s' attribute is incorrectly specified in the context.

### Description

The attribute indicated in the error message is not valid, because it is not compatible with the template to which this internal_power group refers. For example, the *related_input* attribute cannot be specified in a two-dimensional table.

The following example shows an instance where this message occurs:

```
power_lut_template(output_by_cap_and_trans) {
  variable_1 : total_output_net_capacitance;
  variable_2 : input_transition_time;
```

Feedback

```
  index_1 ("0.0, 5.0, 20.0");
  index_2 ("0.1, 1.00, 5.00");
}

cell(lbdb315) {
  area : 2;
  pin(A) {
    direction : input;
    capacitance : 1;
  }
  pin(B) {
    direction : input;
    capacitance : 1;
  }
  pin(Z) {
    direction : output;
    function : "A+B";
    timing() {
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      slope_rise : 0.0;
      slope_fall : 0.0;
      related_pin : "A B";
    }
  }
  cell_leakage_power : 1;
  internal_power(output_by_cap_and_trans) {
     values(" 5.000000 ,  15.000000 ,  0.300000 ", \
            " 1.000000 ,  5.000000 ,  0.000000 ", \
            " 0.000000 ,  0.000000 ,  0.000000 ");
     related_outputs : "Z";
     related_input : "A";
  }
}
```

The following is an example message:

```
Error: Line 315, The 'related_input' attribute is incorrectly specified
 in the context. (LBDB-315)
```

**What Next**

Refer to the Library Compiler manuals and the Design Power Reference manual for power modeling and analysis. Modify the library to correct the problem.

---

## LBDB-316

(error) Arithmetic overflow or exception is encountered\n \ton the '%s' attribute with the '%s' value.

Feedback

### Description

This message indicates that you specified an attribute's value that causes an arithmetic overflow or an exception. Library Compiler cannot handle the value.

The following example shows an instance where this message occurs:

```
fall_propagation ( table4x6 ) {
   index_1 ("2.00e+02 2.50e+02 3.50e+02 4.00e+02 ") ;
   index_2 ("4.48e-02 8.96e-02 1.34e-01 1.79e-01 2.24e-01 2.69e-01 ");
   values ("7.45e+01 9.56e+01 1.15e+02 1.34e+02 1.52e+02 1.71e+02 ", \
           "9.38e+01 1.20e+02 1.45e+02 1.67e+02 1.89e+02 2.10e+02 ", \
           "1.01e+02 1.32e+02 1.58e+02 1.83e+02 2.07e+02 2.29e+02 ", \
           "1.04e+02 1.43e-93 1.43e-93 1.43e-93 2.75e+226 9.30e+254 " ) ;
}
```

The following is an example message:

```
Error: Line 85, Arithmetic overflow or exception is encountered
        on the 'values' attribute with the '2.75e+226' value. (LBDB-316)
```

### What Next

Check the library source file, and correct the problem.

---

## LBDB-317

(error) It is invalid to specify the '%s' attribute\n \ton a pin within the bus or bundle group.

### Description

This message indicates that you specified a function attribute on a pin within a bus or a bundle group.

The following example shows an instance where this message occurs:

```
bundle (qn) {
    members(qn2);
    direction : output;
    function : "iq";
    pin (qn2) {
      function : "q2";
    timing() {
       related_pin   : "q2";
       intrinsic_rise :  0.1;
       intrinsic_fall :  0.1;
       rise_resistance : 1.0;
       fall_resistance : 1.0;
    }
  }
```

Feedback

The following is an example message:

```
Error: Line 67, It is invalid to specify the 'function' attribute
        on a pin within the bus or bundle group. (LBDB-317)
```

### What Next

Check the library source file, and correct the problem.

---

## LBDB-318

(warning) the user specified timing sense '%s' is different \n \tfrom the function calculated timing_sense '%s'.

### Description

```
In timing group of a pin, timing_sense should match the function of the
 pin.
```

The following example shows an instance where this message occurs:

```
cell(INV) {
  pin (Y) {
    timing () {
      related_pin : "A";
      timing_sense : positive_unate;
    }
    function : "!A"
  }
  ...
}
```

If timing_sense is specified, the Library Compiler calculates the value for it from the pin's logic function. And issues this warning if specified timing_sense is different from the calculated one. The user sepcified timing_sense will be stored in .db.

For example, the value calculated for an AND gate is positive_unate, the value for a NAND gate is negative_unate, and the value for an XOR gate is non_unate. For the above example, the value for an INV is negative_unate.

If timing_sense is not specified, the Library Compiler calculates and derives the value for it from the pin's logic function. The derived timing_sense will be stored in .db.

A function is unate if a rising (or falling) change on a positive unate input variable causes the output function variable to rise (or fall) or not change. A rising (or falling) change on a negative unate input variable causes the output function variable to fall (or rise) or not change. For a nonunate variable, further state information is required to determine the effects of a particular state transition.

Feedback

The following is an example message:

```
Warning: Line 123, Cell 'INV', pin 'Y', the user specified timing sense
'positive_unate' is different from the function calculated timing_sense
'negative_unate'. (LBDB-318)
```

### What Next

Correct the value of the timing_sense.

---

## LBDB-319

(Information) Voltage '%s' is used but not declared. A legacy value of '%g' is assumed.

### Description

This message indicates that the Voltage 'VDD'(or 'VCC', 'VSS') is used in the expression, but not declared in voltage_map. Thus a legacy value of '5'(or '5', '0') is assumed.

The following example shows an instance where this message occurs:

```
library (lbdb319) {
    voltage_map ( IOVDD1, 1.32 );
    voltage_map ( COREVDD1, 0.875 );
    voltage_map ( IOCOREGND1, 0 );
    input_voltage(cmos11) {
        vil : 0.35 * VDD ;
        vih : 0.65 * VDD ;
        vimin : -0.3 ;
        vimax : 1.32 ;
    }
}
```

The following is an example message:

```
Information: Line 34, Voltage 'VDD' is used but not declared. A legacy
 value of '5' is assumed. (LBDB-319)
```

### What Next

No action is required.

---

## LBDB-353

(warning) A nonsequential timing arc is specified with\n \trespect to the '%s' clock pin.

### Description

The cell has the *interface_timing* attribute set to TRUE. The interface timing policy requires that all timing arcs defined with respect to a clock pin be sequential (noncombinational) in nature.

Feedback

The following example shows an instance where this message occurs:

```
cell (lbdb353) {
    area    :   0.0;
    interface_timing : TRUE;
    pin (I1)    {
      direction   :   input;
      capacitance :   1.0;
      fanout_load :   1.0;
      timing () {
        timing_type : setup_rising;
        intrinsic_rise : 1.0;
        intrinsic_fall :  1.0;
        related_pin : "CLK";
      }
      timing () {
        timing_type : hold_rising;
        intrinsic_rise : 1.0;
        intrinsic_fall : 1.0;
        related_pin : "CLK";
      }
    }
    pin (CLK)    {
      direction   :   input;
      capacitance :   1.0;
      fanout_load :   1.0;
      clock : true;
      timing () {
        timing_type : skew_rising;
        intrinsic_rise : 1.0;
        related_pin : "CLK1";
      }
    }

    pin (CLK1)    {
      direction   :   input;
      capacitance :   1.0;
      fanout_load :   1.0;
      timing () {
        timing_type : skew_falling;
        intrinsic_rise : 1;
        related_pin : "CLK";
      }
    }
    pin(Q){
      direction   :   output;
      timing () {
        intrinsic_rise   : 1.0;
        rise_resistance  : 0.1;
        intrinsic_fall   : 1.0;
        fall_resistance  : 0.1;
        related_pin      : "CLK";
      }
```

Feedback

```
    }
  }
```

In this case, the 'Q' pin is missing the timing_type attribute in its timing group.

The following is an example message:

```
Warning: Line 62, A nonsequential timing arc is specified with
        respect to the 'CLK' clock pin. (LBDB-353)
```

### What Next

Make sure that all arcs with a clock pin as a related_pin are sequential.

---

## LBDB-354

(warning) A sequential timing arc is specified with respect\n \tto the '%s' nonclock pin.

### Description

The cell has the *interface_timing* attribute set to TRUE. The interface timing policy requires that all sequential timing arcs (except for clear and preset) be defined with respect to a pin that is labeled as a clock.

The following example shows an instance where this message occurs:

```
cell (lbdb354) {
    area    :   0.0;
    interface_timing : TRUE;
    pin (I1)    {
      direction   :   input;
      capacitance :   1.0;
      fanout_load :   1.0;
      timing () {
        timing_type : setup_rising;
        intrinsic_rise : 1.0;
        intrinsic_fall :  1.0;
        related_pin : "CLK";
      }
      timing () {
        timing_type : hold_rising;
        intrinsic_rise : 1.0;
        intrinsic_fall : 1.0;
        related_pin : "CLK";
      }
    }
    pin (CLK)    {
      direction   :   input;
      capacitance :   1.0;
      fanout_load :   1.0;
      clock : true;
      timing () {
```

Feedback

```
        timing_type : skew_rising;
        intrinsic_rise : 1.0;
        related_pin : "CLK1";
      }
    }

    pin (CLK1)    {
      direction   :   input;
      capacitance :   1.0;
      fanout_load :   1.0;
      timing () {
        timing_type : skew_falling;
        intrinsic_rise : 1;
        related_pin : "CLK";
      }
    }
    pin(Q){
      direction   :   output;
      timing () {
        intrinsic_rise   : 1.0;
        rise_resistance  : 0.1;
        intrinsic_fall   : 1.0;
        fall_resistance  : 0.1;
        related_pin      : "CLK";
      }
    }
  }
```

In this case, the 'CLK' pin has a timing group with a timing_group value skew_rising related to the 'CLK1' pin.

The following is an example message:

```
Warning: Line 40, A sequential timing arc is specified with respect
        to the 'CLK1'  nonclock pin. (LBDB-354)
```

### What Next

Make sure that all sequential arcs are defined with respect to a clock pin.

---

## LBDB-355

(warning) A skew constraint is specified for the nonclock '%s' pin.

### Description

A skew constraint is specified for a pin that is not a clock. All clock pins must have the *clock* attribute set to TRUE.

Feedback

The following example shows an instance where this message occurs:

```
cell (lbdb355) {
    area    :   0.0;
    interface_timing : TRUE;
    pin (I1)    {
      direction   :   input;
      capacitance :   1.0;
      fanout_load :   1.0;
      timing () {
        timing_type : setup_rising;
        intrinsic_rise : 1.0;
        intrinsic_fall :  1.0;
        related_pin : "CLK";
      }
      timing () {
        timing_type : hold_rising;
        intrinsic_rise : 1.0;
        intrinsic_fall : 1.0;
        related_pin : "CLK";
      }
    }
    pin (CLK)    {
      direction   :   input;
      capacitance :   1.0;
      fanout_load :   1.0;
      clock : true;
      timing () {
        timing_type : skew_rising;
        intrinsic_rise : 1.0;
        related_pin : "CLK1";
      }
    }

    pin (CLK1)    {
      direction   :   input;
      capacitance :   1.0;
      fanout_load :   1.0;
      timing () {
        timing_type : skew_falling;
        intrinsic_rise : 1;
        related_pin : "CLK";
      }
    }
    pin(Q){
      direction   :   output;
      timing () {
        intrinsic_rise   : 1.0;
        rise_resistance  : 0.1;
        intrinsic_fall   : 1.0;
        fall_resistance  : 0.1;
        related_pin      : "CLK";
      }
```

Feedback

```
    }
  }
```

In this case, the 'CLK1' pin has skew constraint timing group on the nonclock 'CLK' pin. Because Library Compiler failed to recognize the 'CLK' pin.

The following is an example message:

```
Warning: Line 52, A skew constraint is specified for the nonclock 'CLK1'
 pin. (LBDB-355)
```

### What Next

Make sure that skew constraints are defined for clock pins only. Check the library source file, and either set the *clock* attribute to TRUE on the pin, or remove the skew constraint.

---

## LBDB-356

(warning) Multiple %s constraints are specified between\n \tthe '%s' pin and the '%s' clock pin.

### Description

Design Compiler takes the worst case constraint when multiple constraints are specified.

The following example shows an instance where this message occurs:

```
cell (lbdb356) {
    area    :   0.0;
    interface_timing : TRUE;
    pin (I1)    {
      direction   :   input;
      capacitance :   1.0;
      fanout_load :   1.0;
      timing () {
        timing_type : setup_rising;
        intrinsic_rise : 1.0;
        intrinsic_fall : 1.0;
        related_pin : "CLK";
      }
      timing () {
        timing_type : hold_rising;
        intrinsic_rise : 1.0;
        intrinsic_fall : 1.0;
        related_pin : "CLK";
      }
      timing () {
        timing_type : hold_falling;
        intrinsic_rise : 1.0;
        intrinsic_fall : 1.0;
        related_pin : "CLK";
      }
```

Feedback

```
    }
    pin (CLK)    {
      direction   :   input;
      capacitance :   1.0;
      fanout_load :   1.0;
      clock : true;
    }
    pin(Q){
      direction   :   output;
      timing () {
        timing_type      : rising_edge;
        intrinsic_rise   : 1.0;
        rise_resistance  : 0.1;
        intrinsic_fall   : 1.0;
        fall_resistance  : 0.1;
        related_pin      : "CLK";
      }
    }
  }
```

In this case, the 'I1' pin has two hold constraints.

The following is an example message:

```
Warning: Line 31, Multiple setup or hold constraints are specified
        between the 'I1' pin and the 'CLK' clock pin. (LBDB-356)
```

**What Next**

PrimeTime supports multiple setup/hold constraints.

---

## LBDB-357

(warning) The '%s' cell has the interface_timing attribute,\n \tbut has no clock pin.

**Description**

According to interface timing specification policy, a library cell with the interface timing attribute set to TRUE must have at least one pin labeled as a clock.

The following example shows an instance where this message occurs:

```
cell (lbdb357) {
    area    :   0.0;
    interface_timing : TRUE;
    pin (I1)    {
      direction   :   input;
      capacitance :   1.0;
      fanout_load :   1.0;
    }
    pin(Q){
      direction   :   output;
```

Feedback

```
      function : "I1'";
      timing () {
        intrinsic_rise   : 1.0;
        rise_resistance  : 0.1;
        intrinsic_fall   : 1.0;
        fall_resistance  : 0.1;
        related_pin      : "I1";
      }
    }
  }
```

The following is an example message:

```
Warning: Line 6, The 'combo' cell has the interface_timing attribute,
        but has no clock pin. (LBDB-357)
```

### What Next

If the cell is combinational, remove the *interface_timing* attribute set on the cell. If the cell is sequential in nature, add the *clock* attribute to all clock pins of the cell. If the cell does not have a clock pin (like a RAM), choose the control pin with respect to which setups and holds are measured, and label it as a clock.

---

## LBDB-358

(error) It is not legal to specify the attribute '%s' in\n \tthe '%s' group in this context.

### Description

The indicated attribute cannot be specified in the indicated group because of the context. This situation can arise if the group can have several types and the attribute can only appear when the group is of specific type.

### What Next

Check your library to see if you have specified the wrong information for the group or if the attribute should be deleted from the group.

---

## LBDB-366

(warning) The '%s' attribute in char_config group is required for NLPM library. \n No default can be applied to this attribute.

### Description

From 2017.06 release, user will see LBDB-366 Warning when the library contains internal_power group, but NO internal_power_calculation attribute defined in library-level char_config group.

Feedback

The internal_power_calculation attribute takes enumerated values:

```
exclude_switching_on_rise
  exclude_switching_on_rise_and_fall
  include_switching
```

This attribute is critical for tools like PTPX to determine the meaning of internal_power table values.

**What Next**

Please define internal_power_calculation attribute in library-level char_config group when library contains NLPM data.

---

## LBDB-370

(error) The library contains a '%s' group,\n \tbut has no contain '%s' group.

**Description**

This message indicates that you specified only one transition degradation table in the library. Libraries that contain transition degradation tables must have tables for both rise and fall transitions.

The following example shows an instance where this message occurs:

```
rise_transition_degradation(trans_deg) {
        values("0.0, 0.6", \
               "1.0, 1.6");
}
```

In this case, the 'fall_transition_degradation' group is missing. To fix the problem, add this group to the library,

```
fall_transition_degradation(trans_deg) {
        values("0.0, 0.8", \
               "1.0, 1.8");
}
```

The following is an example message:

```
Error: Line 46, The library contains a 'rise_transition_degradation'
 group,
        but has no 'fall_transition_degradation' group. (LBDB-370)
```

**What Next**

Add the missing table group to the library description.

Feedback

## LBDB-371

(error) The '%s' lut group is missing the mandatory '%s' attribute.

### Description

This message indicates that you specified a lut group without the input_pins attribute in a FPGA library.

The following example shows an instance where this message occurs:

```
cell(lut5) {
    area : 1.0;
    lut(lbdb371) {
      /*      input_pins : "a b c d e"; */
    }
    pin(a b c d e) {
      direction : input;
      capacitance : 0.0;
    }
    pin(o) {
      direction : output;
      function : "lbdb371";
      timing() {
        related_pin : "a b c d e";
      }
    }
  }
```

The following is an example message:

```
Error: Line 98, The 'lbdb371' lut group is missing the mandatory
 'input_pins' attribute. (LBDB-371)
```

### What Next

Check the library source file, and add the missing attribute.

## LBDB-372

(error) The '%s' lut group attribute has an invalid '%s' value.

### Description

This message indicates that you specified a lut group with an invalid value for the input_pins attribute in a FPGA library.

The following example shows an instance where this message occurs:

```
cell(lut5) {
    area : 1.0;
```

Feedback

```
    lut(R) {
    input_pins : "";
    }
    pin(a b c d e) {
      direction : input;
      capacitance : 0.0;
    }
    pin(o) {
      direction : output;
      function : "R";
      timing() {
        related_pin : "a b c d e";
      }
    }
  }
```

The following is an example message:

```
Error: Line 98, The 'input_pins' lut group attribute has an invalid ''
 value. (LBDB-372)
```

### What Next

Check the library source file, and correct the value of the input_pins attribute.

---

## LBDB-373

(error) The '%s' cell has more than one lut group defined.

### Description

This message indicates that you specified more than one lut group in the FPGA library.

The following example shows an instance where this message occurs:

```
cell(lbdb373) {
    area : 1.0;
    lut(R) {
      input_pins : "a b c d e";
    }
    lut(R1) {
      input_pins : "a b c d e";
    }
    pin(a b c d e) {
      direction : input;
      capacitance : 0.0;
    }
    pin(o) {
      direction : output;
      function : "R";
      timing() {
        related_pin : "a b c d e";
```

Feedback

```
      }
    }
  }
```

The following is an example message:

```
Error: Line 96, The 'lbdb373' cell has more than one lut group defined.
 (LBDB-373)
```

### What Next

Check the library source file, and delete the extra lut groups.

---

## LBDB-374

(error) The '%s' input port is not in the '%s' lut.

### Description

This message indicates that you did not specify an input port in lut group.

The following example shows an instance where this message occurs:

```
cell(lbdb374) {
    area : 1.0;
    lut(R) {
      input_pins : "a b c d e";
    }
    pin(a b c d e f) {
      direction : input;
      capacitance : 0.0;
    }
    pin(o) {
      direction : output;
      function : "R";
      timing() {
        related_pin : "a b c d e";
      }
    }
  }
```

The following is an example message:

```
Error: Line 101, The 'f' input port is not in the 'lbdb374' lut.
 (LBDB-374)
```

### What Next

Check the library source file, and either add the input port to the lut group or delete the port.

Feedback

## LBDB-375

(error) The '%s' port is incorrectly listed as an input in the\n \t'%s' lut.

### Description

This message indicates that you specified a noninput port in the lut group.

The following example shows an instance where this message occurs:

```
cell(lbdb375) {
    area : 1.0;
    lut(R) {
      input_pins : "a b c d e o";
    }
    pin(a b c d e) {
      direction : input;
      capacitance : 0.0;
    }
    pin(o) {
      direction : output;
      function : "R";
      timing() {
        related_pin : "a b c d e";
      }
    }
  }
```

The following is an example message:

```
Error: Line 98, The 'o' port is incorrectly listed as an input in the
        'lbdb375' lut. (LBDB-375)
```

### What Next

Check the library source file, and delete the noninput port from the lut group.

## LBDB-376

(error) The '%s' port is invalid in a cell containing a lut group.

### Description

This message indicates that you specified an invalid port in a cell containing a lut group. Ports in a cell with a lut are either input or output.

The following example shows an instance where this message occurs:

```
cell(lbdb376) {
    area : 1.0;
    lut(R) {
      input_pins : "a b c d e ";
```

Feedback

```
    }
    pin(a b c d e) {
      direction : input;
      capacitance : 0.0;
    }
    pin(f) {
      direction : inout;
      capacitance : 0.0;
    }
    pin(o) {
      direction : output;
      function : "R";
      timing() {
        related_pin : "a b c d e";
      }
    }
  }
```

The following is an example message:

```
Error: Line 105, The 'f' port is invalid in a cell containing a lut
 group. (LBDB-376)
```

### What Next

Check the library source file, and correct the direction of the port.

---

## LBDB-377

(error) The '%s' lut group is invalid in a multiple output cell.

### Description

This message indicates that you specified more than one output port in a cell containing a lut group.

The following example shows an instance where this message occurs:

```
cell(lbdb377) {
    area : 1.0;
    lut(R) {
      input_pins : "a b c d e ";
    }
    pin(a b c d e) {
      direction : input;
      capacitance : 0.0;
    }
    pin(f) {
      direction : output;
      function : "R";
      timing() {
        related_pin : "a b c d e";
```

Feedback

```
      }
    }
    pin(o) {
      direction : output;
      function : "R";
      timing() {
        related_pin : "a b c d e";
      }
    }
  }
```

In this case, both 'f' and 'o' are output ports.

The following is an example message:

```
Error: Line 98, The 'R' lut group is invalid in a multiple output cell.
 (LBDB-377)
```

### What Next

Check the library source file, and delete the extra output ports.

---

## LBDB-378

(error) The '%s' lut group is invalid in a cell with no outputs.

### Description

This message indicates that you did not specify any output port in a cell containing a lut group.

The following example shows an instance where this message occurs:

```
cell(lbdb378) {
    area : 1.0;
    lut(R) {
      input_pins : "a b c d e ";
    }
    pin(a b c d e) {
      direction : input;
      capacitance : 0.0;
    }
  }
```

In this case, the 'lbdb378' has no output port.

The following is an example message:

```
Error: Line 98, The 'R' lut group is invalid in a cell with no outputs.
 (LBDB-378)
```

Feedback

### What Next

Check the library source file, and either add the output port or delete the lut group.

---

## LBDB-379

(error) The '%s' output port on the '%s' cell has no\n \tfunction attribute, or the function attribute is invalid with\n \tthe lut group on this cell.

### Description

This message indicates that either you specified an invalid function attribute on the output port a cell containing a lut group, or you did not define the function attribute.

The following example shows an instance where this message occurs:

```
cell(lbdb379) {
    area : 1.0;
    lut(R) {
      input_pins : "a b c d e ";
    }
    pin(a b c d e) {
      direction : input;
      capacitance : 0.0;
    }
    pin(o) {
      direction : output;
      timing() {
        related_pin : "a b c d e";
      }
    }
  }
```

In this case, The function attribute is missing in the 'o' port. To fix the problem, add the statement

```
function : "R";
```

The following is an example message:

```
Error: Line 113, The 'o' output port on the 'lbdb379' cell has no
        function attribute, or the function attribute is invalid with
        the lut group on this cell. (LBDB-379)
```

### What Next

Check the library source file, and either add or correct the function attribute.

Feedback

## LBDB-380

(error) The '%s' name is invalid for a lut group in the '%s'\n \tcell because a port on the design has the same name.

### Description

This message indicates that there is a name conflict between the specified lut group name and a port name. The names of the lut groups must be unique and must not conflict with existing port names.

The following example shows an instance where this message occurs:

```
cell(lbdb380) {
    area : 1.0;
    lut(R) {
      input_pins : "a b c d e";
    }
    pin(a b c d e) {
      direction : input;
      capacitance : 0.0;
    }
    pin(R) {
      direction : output;
      function : "R";
      timing() {
        related_pin : "a b c d e";
      }
    }
  }
```

The following is an example message:

```
Error: Line 98, The 'R' name is invalid for a lut group in the 'lbdb380'
        cell because a port on the design has the same name. (LBDB-380)
```

### What Next

Change the name of the lut group so that it is unique.

## LBDB-381

(error) The '%s' lut cell has %d input port(s). A lut\n \tcell must have at least 1 but no more than 9 input ports.

### Description

This message indicates that the specified lut cell has either too many or too few input ports. lut cells must have at least 1 but not more than 9 input ports.

Feedback

The following example shows an instance where this message occurs:

```
cell(lbdb381) {
    area : 1.0;
    lut(R) {
      input_pins : "a b c d e f g h i j";
    }
    pin(a b c d e f g h i j) {
      direction : input;
      capacitance : 0.0;
    }
    pin(o) {
      direction : output;
      function : "R";
      timing() {
        related_pin : "a b c d e f g h i j";
      }
    }
  }
```

The following is an example message:

```
Error: Line 98, The 'lbdb381' lut cell has 10 input port(s). A lut
        cell must have at least 1 but no more than 9 input ports.
 (LBDB-381)
```

### What Next

Edit the lut cell description in your library so that the cell has at least 1 but not more than 9 input ports.

---

## LBDB-382

(error) The '%s' cell is an invalid lut cell. Only\n \tsimple combinational cells are valid.

### Description

This message indicates that the specified cell is not a simple combinational cell, which is the only valid type of lut cell.

The following example shows an instance where this message occurs:

```
cell(lbdb382) {
    area : 1.0;
    lut(R) {
      input_pins : "D CP";
    }
    pin(D CP ) {
      direction : input;
      capacitance : 0.0;
    }
    ff("IQ","IQN") {
```

Feedback

```
      next_state : "D";
      clocked_on : "CP";
    }
    pin(o) {
      direction : output;
      function : "R";
      timing() {
        related_pin : "D CP";
      }
    }
  }
```

The following is an example message:

```
Error: Line 96, The 'lbdb382' cell is an invalid lut cell. Only
        simple combinational cells are valid. (LBDB-382)
```

### What Next

Remove the lut construct from this cell description, or modify the cell so that it is a simple combinational cell.

---

## LBDB-383

(error) The '%s' attribute is invalid on\n \tthe '%s' lut cell.

### Description

This message indicates that the specified attribute is not valid on the specified lut cell.

The following example shows an instance where this message occurs:

```
cell(lbdb383) {
    area : 1.0;
    fpga_lut_output : true;
    lut(R) {
      input_pins : "A B C D";
    }
    pin(A B C D) {
        direction : input;
        capacitance : 0.0;
    }
    pin(Z) {
        direction : output;
        function : "R";
        timing() {
            intrinsic_rise : 5.0;
            intrinsic_fall : 5.0;
            related_pin : "A B C D";
        }
    }
}
```

Feedback

The following is an example message:

```
Error: Line 25, The 'fpga_lut_output' attribute is invalid on
        the 'lbdb383' lut cell.  (LBDB-383)
```

### What Next

Remove the offending attribute from the lut cell.

---

## LBDB-384

(error) The lut marker cell attribute\n \t'%s' is invalid on the '%s' cell.

### Description

This message indicates that a lut marker cell attribute has been found on the specified cell, identifying it as a lut marker cell. However, the cell is not a valid candidate for a lut marker cell because it is not a single output buffer cell with 0 area.

Lut marker cells delimit the boundaries of luts and must be single-output buffer cells with 0 area.

The following example shows an instance where this message occurs:

```
cell(lbdb384) {
    area : 0.0;
    fpga_lut_insert_before_sequential : true;
    pin(A) {
      direction : input;
      capacitance : 0.0;
    }
    pin(Z) {
      direction : output;
      function : "A";
      timing() {
        intrinsic_rise : 0.0;
        intrinsic_fall : 0.0;
        related_pin : "A";
      }
    }
  }
```

The following is an example message:

```
Error: Line 126, The lut marker cell attribute
        'fpga_lut_insert_before_sequential' is invalid on the 'lbdb384'
 cell. (LBDB-384)
```

### What Next

Either remove the lut marker cell attribute from the cell, or edit the cell description to make it a valid marker cell; that is, a single output buffer cell with 0 area.

Feedback

## LBDB-385

(error) The lut marker '%s' cell is functionally invalid.\n \tLut marker cells must be single output buffers.

### Description

This message indicates that the specified marker cell is not a single output buffer, which is the only valid lut marker cell type.

The following example shows an instance where this message occurs:

```
cell(lbdb385) {
    area : 0.0;
    fpga_lut_output : true;
    pin(A) {
      direction : input;
      capacitance : 0.0;
    }
    pin(Z1) {
      direction : output;
      function : "A";
      timing() {
        intrinsic_rise : 0.0;
        intrinsic_fall : 0.0;
        related_pin : "A";
      }
    }
    pin(Z2) {
      direction : output;
      function : "A";
      timing() {
        intrinsic_rise : 0.0;
        intrinsic_fall : 0.0;
        related_pin : "A";
      }
    }
  }
```

In this case, there are two output ports in the cell.

The following is an example message:

```
Error: Line 143, The lut marker 'lbdb385' cell is functionally invalid.
        Lut marker cells must be single output buffers. (LBDB-385)
```

### What Next

Remove the marker cell attribute from the cell, or change the cell's functionality so that it is a single output buffer.

Feedback

## LBDB-386

(warning) The lut marker '%s' cell has a nonzero\n \t'%s' attribute. Marker cells must have \n \tzero area and zero delay.

### Description

This message indicates that the specified area or delay value in the library is nonzero, which is invalid for lut marker cells.

The following example shows an instance where this message occurs:

```
cell(lbdb386) {
    area : 0.0;
    fpga_lut_output : true;
    pin(A) {
      direction : input;
      capacitance : 0.0;
    }
    pin(Z) {
      direction : output;
      function : "A";
      timing() {
        intrinsic_rise : 1.0;
        intrinsic_fall : 0.0;
        related_pin : "A";
      }
    }
  }
```

In this case, the 'intrinsic_rise' value is 1.0. Change the value to zero to fix the problem.

The following is an example message:

```
Warning: Line 169, The lut marker 'lbdb386' cell has a nonzero
        'intrinsic rise delay' attribute. Marker cells must have
        zero area and zero delay. (LBDB-386)
```

### What Next

Change the offending nonzero value (area or delay, as appropriate) to zero.

## LBDB-387

(warning) The '%s' library has more than one lut output marker cell.

### Description

This message indicates that the specified library has too many lut output marker cells. Each library can have only one lut output marker cell.

Feedback

The following example shows an instance where this message occurs:

```
library(lbdb387) {
  ...
  cell(marker1) {
    area : 0.0;
    fpga_lut_output : true;
    pin(A) {
      direction : input;
      capacitance : 0.0;
    }
    pin(Z) {
      direction : output;
      function : "A";
      timing() {
        intrinsic_rise : 0.0;
        intrinsic_fall : 0.0;
        related_pin : "A";
      }
    }
  }

  cell(marker2) {
    area : 0.0;
    fpga_lut_output : true;
    pin(A) {
      direction : input;
      capacitance : 0.0;
    }
    pin(Z) {
      direction : output;
      function : "A";
      timing() {
        intrinsic_rise : 0.0;
        intrinsic_fall : 0.0;
        related_pin : "A";
      }
    }
  }
}
```

The following is an example message:

```
Warning: Line 56, The 'lbdb387' library has more than one lut output
 marker cell. (LBDB-387)
```

**What Next**

Check the library source file, and remove the duplicate output marker cells.

Feedback

## LBDB-388

(error) The '%s' attribute cannot be specified in a\n \ttiming arc that is not a timing constraint.

### Description

This message indicates that an attribute has been incorrectly specified. The specified attribute is allowed only on timing arcs that are timing constraints (for example, setup and hold).

The following example shows an instance where this message occurs:

```
pin(Q) {
      direction : output;
      function : "IQ";
      timing() {
        timing_type : falling_edge;
        intrinsic_rise : 1.0;
        intrinsic_fall : 1.0;
        when_end : "D0";
        rise_resistance : 0.1;
        fall_resistance : 0.1;
        related_pin : "CK2";
      }
    }
```

In this case, the 'falling_edge' timing_type value is not a constraint.

The following is an example message:

```
Error: Line 94, The 'when_end' attribute cannot be specified in a
        timing arc that is not a timing constraint. (LBDB-388)
```

### What Next

Check the library source file, and make the appropriate correction. For example, change the timing type to be a timing constraint.

## LBDB-389

(error) This timing arc has either '%s' or '%s'\n \tattribute but not both.

### Description

This message indicates that in a timing group, you defined only one attribute from the following sets:

```
* when attribute and sdf_cond attribute
        * when_start attribute and sdf_cond_start attribute
        * when_end attribute and sdf_cond_end attribute
```

Feedback

To define a state dependent timing arc, Library Compiler expects both attributes defined.

The following example shows an instance where this message occurs:

```
pin(Z) {
        direction : output;
        function : "A^B";
        timing() {
          when : "B'";
          timing_sense : positive_unate;
          intrinsic_rise : 0.1;
          intrinsic_fall : 0.1;
          rise_resistance : 0.1;
          fall_resistance : 0.1;
          related_pin : "A";
        }
```

In this case, only the 'when' attribute is specified. To fix the problem, add the statement,

```
sdf_cond : "!B";
```

The following is an example message:

```
Error: Line 88, This timing arc has either 'when' or 'sdf_cond'
        attribute but not both. (LBDB-389)
```

### What Next

Complete your specification by adding appropriate attributes to the timing group.

---

## LBDB-390

(warning) The when and/or sdf_cond attributes in this\n \ttiming arc are ignored.

### Description

This message indicates that you specified, in addition to the *when_start*, *when_end*, *sdf_cond_start*, or *sdf_cond_end* attributes in the timing (constraint) group, the *when* and *sdf_cond* attributes. Library Compiler ignores the *when* and *sdf_cond* attributes.

The following example shows an instance where this message occurs:

```
timing() {
      timing_type : setup_rising;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      sdf_edges : start_edge;
      when : " CD * SD ";
      sdf_cond : " SIG_2 == 1'b1 ";
      when_end : " CD * SD * Q' ";
      sdf_cond_end : " SIG_0 == 1'b1";
```

Feedback

```
      related_pin : "CP";
    }
```

In this case, both the when and sdf_cond pair and the when_end and sdf_cond_end are specified.

The following is an example message:

```
Warning: Line 836, The when and/or sdf_cond attributes in this
        timing arc are ignored. (LBDB-390)
```

### What Next

Remove the redundant *when* and/or *sdf_cond* attributes from the timing group.

---

## LBDB-391

(warning) The '%s' attribute on pin '%s' is not valid.\n \tThe attribute is ignored.

### Description

This message indicates that the attribute on the specified pin is invalid. The attribute is ignored. This situation occurs when other information specified on the same pin is incompatible with the attribute. For example, the *min_period* attribute is ignored if the *minimum_period* group is also specified on the same pin.

The following example shows an instance where this message occurs:

```
pin (CLK) {
    direction : input ;
    capacitance : 0 ;
    min_pulse_width_low  : 3 ;
    min_pulse_width_high : 3 ;
    min_period : 5.0;
    minimum_period() {
      constraint : 1.0;
      when : "D PRE";
      sdf_cond : "cond_1 == 1'b1";
    }
  }
```

The following is an example message:

```
Warning: Line 183, The 'min_period' attribute on the 'CLK' pin is not
 valid.
        The attribute is ignored. (LBDB-391)
```

### What Next

Refer to the *Library Compiler Reference Manual* to determine the reason why the attribute is not valid. Change the library source file and make the correction.

Feedback

## LBDB-392

(error) The ECL Technology is obsolete with v3.4b. Compilation terminated abnormally.

### Description

This error occurs when the ECL Technology is used. This technology is supported up to v3.4a but not for subsequent releases.

The following example shows an instance where this message occurs:

```
library(lbdb392) {
  technology(ecl);
  delay_model : "generic_ecl" ;
}
```

The following is an example message:

```
Error: The ECL Technology is obsolete with v3.4b. Compilation terminated
 abnormally. (LBDB-392)
```

### What Next

Use the supported technologies.

## LBDB-393

(error) Too many values entered for the driver_type attribute.

### Description

This error occurs when more than two values entered for a driver_type attribute in defining a simple attribute.

The following example shows an instance where this message occurs:

```
pin(y) {
      direction : inout;
      function : 1;
      three_state : "a";
      driver_type : "open_source pull_down resistive";
      timing() {
        timing_type : three_state_disable;
        related_pin : "a";
      }
      timing() {
        related_pin : "a";
      }
    }
```

In the driver_type line, three driver_types are specified for this pin.

Feedback

The following is an example message:

```
Error: Line 458, Too many values entered for the driver_type attribute.
 (LBDB-393)
```

**What Next**

Only inout pin allows for two driver_type values. Other pins can have only one value. During passing, we first catch the case where more than two are entered.

---

## LBDB-394

(error) The driver type %s is incompatible with the %s pin %s.

**Description**

Valid driver-type (programmable or non-programmable) and pin combinations (Y means valid):

non-programmable input output inout programmable
==========
pull_up Y Y Y pull_up_function pull_down Y Y Y pull_down_function open_drain N Y Y open_drain_function open_source N Y Y open_source_function bus_hold N N Y bus_hold_function resistive N Y Y resistive_function resistive_0 N Y Y resistive_0_function resistive_1 N Y Y resistive_1_function open_drain_with_pull_up N N Y open_drain_with_pull_down N N Y open_source_with_pull_up N N Y open_source_with_pull_down N N Y
==========

The following example shows an instance where this message occurs:

```
pin(a) {
      direction : input;
      capacitance : "1";
      driver_type : "open_source pull_up";
    }
```

The following is the example message:

```
Error: Line 177, The driver type open_source_with_pull_up is incompatible
        with the input pin a. (LBDB-394)
```

The following example shows another instance where this message occurs followed by the example message:

```
pin(A3) {
  direction : input;
  pull_up_function : "!A1 * !A2 * !A3";
  pull_down_function : "A1 * A2 * !A3";
  bus_hold_function : "A1 * !A2 * !A3";
  open_drain_function : "!A1 * A2 * !A3";
```

Feedback

```
  open_source_function : "!A1 * !A2 * A3";
  resistive_function : "A1 * A2 * A3";
  resistive_0_function : "A1 * !A2 * A3";
  resistive_1_function : "!A1 * A2 * A3";
  ...
 }
```

In this case, only pull_up_function and pull_down_function can be specified under an input pin. To avoid the error, please remove the rest of 6 programmable driver types.

The following is the example message:

```
Error: Line 217, The driver type open_drain_function is incompatible with
 the input pin A3. (LBDB-394)
```

### What Next

When single pull_up/down applys to an inout pin, it is for the input behavior; other single types applying to an inout pin are for the output behavior. Always comply with the above rules.

---

## LBDB-395

(warning) The timing arc with timing_type '%s'\n can only be specified on a pin with 'input' or 'inout' direction.

### Description

The timing_type non_seq_setup_rising, non_seq_setup_falling, non_seq_hold_rising and non_seq_hold_falling can only be specified on a timing arc on the input or the inout pins.

The following example shows an instance where this message occurs:

```
pin(Z) {
          direction : output;
          timing() {
            timing_type : non_seq_setup_rising;
            ...
          }
        }
```

The following is an example message: Warning: Line 144, The timing arc with timing_type 'non_seq_setup_rising' can only be specified on a pin with 'input' or 'inout' direction. (LBDB-395)

### What Next

Check the timing arc and make the correction to either the direction of the pin or delete the timing arc or move it to the appropriate place.

Feedback

## LBDB-396

(warning) The 'related_output_pin' attribute is only needed\n when the delay table refers to a template which uses the\n output loading of the related_output_pin in one of its dimension.

### Description

This can happen when the 'related_output_pin' is not needed in this timing arc or the template name used by this table is wrong or the template itself should use the output loading of the related_output_pin in one of its dimension.

The following example shows an instance where this message occurs:

```
lu_table_template(prop) {
          variable_1 :  output_net_length;
          index_1 ("1, 5, 10");
        }

        cell(A) {

          pin(O) {
            timing() {
              related_output_pin : "Z";
              rise_propagation(prop) {
                ...
              }
            }
          }
        }
```

The following is an example message: Warning: Line 144, The 'related_output_pin' attribute is only needed when the delay table refers to a template which uses the output loading of the related_output_pin in one of its dimension. (LBDB-396)

The timing arc is a rise_propagation table which refers to the template 'prop'. In the template, the only variable it depends on is the loading of the output pin. There is not dependency on the loading of pin 'Z' and there is no need to put the 'related_output_pin' in the timing group.

### What Next

Check the table or the template to see if the related_output_pin attribute is actually needed here. If not, just delete the attribute.

## LBDB-397

(error) There can only be one pin in the 'related_output_pin'\n attribute and the pin should be single bit.

Feedback

### Description

The "related_output_pin" attribute is used for figuring out the output loading to index into the table. It is not possible to determine which output pin to use for this purpose if there is more than one pin specified in the attribute. Multiple bits specification also have the same problem.

The following example shows an instance where this message occurs:

```
timing() {
          related_output_pin : "Z1 Z2";
          ...
        }
```

The following is an example message: Error: Line 144, There can only be one pin in the 'related_output_pin' attribute and the pin should be single bit. (LBDB-397)

There are two pins, Z1 and Z2, in the related_output_pin attribute. It is not possible to figure out which pin should be used to figuring out the loading to index into the timing table.

### What Next

Check the 'related_output_pin' attribute and make the correction according to the real electrical characteristics.

---

## LBDB-398

(error) The pin '%s' specified in the 'related_output_pin'\n attribute is not an output or inout pin.

### Description

The pin specified in the 'related_output_pin' attribute should have the direction 'output' or 'inout'.

The following example shows an instance where this message occurs:

```
pin(A) {
          direction : input;
          ...
        }
        pin(B) {
          timing() {
            related_output_pin : "A";
            ...
          }
        }
```

The following is an example message: Error: Line 144, The pin 'A' specified in the 'related_output_pin' attribute is not an output or inout pin. (LBDB-398)

Feedback

Pin 'A' is an input pin and cannot be put into the 'related_output_pin' attribute.

**What Next**

Check the 'related_output_pin' attribute to see if you have put the wrong pin in it or you forgot to put the correct pin direction on the pin specified in the 'related_output_pin' attribute.

---

## LBDB-399

(error) The timing_type '%s' is not supported\n with the 'related_output_pin' attribute.

**Description**

The timing_type supported with the 'related_output_pin' attribute are: 'setup_rising', 'setup_falling', 'hold_rising', 'hold_falling', 'non_seq_setup_rising', 'non_seq_setup_falling', 'non_seq_hold_rising', 'non_seq_hold_falling', 'skew_rising', 'skew_falling', 'removal_rising', 'removal_falling', 'recovery_rising' and 'recovery_falling'.

The following example shows an instance where this message occurs:

```
timing() {
          timing_type : rising_edge;
          related_output_pin : "A";
          ...
        }
```

The following is an example message: Error: Line 144, The timing_type 'rising_edge' is not supported with the 'related_output_pin' attribute.. (LBDB-399)

**What Next**

You may either put the wrong timing_type for the timing group or delete the timing group if it is not supported or the 'related_output_pin' attribute is not needed.

---

## LBDB-400

(warning) The '%s' operating condition has been defined \n \tmultiple times in the '%s' library. \n \tDesign Compiler will use the first definition only.

**Description**

The library contains more than one definition of an operating condition. Library Compiler issues this error message, records all definitions, but Design Compiler will only use the first definition encountered.

The following example shows an instance where this message occurs:

```
operating_condition(P1V1) {
    ...
```

Feedback

```
  }
  operating_condition(P1V1) {
    ...
  }
```

The following is an example message:

```
Warning: Line 50, The 'P1V1' operating condition has been defined
        multiple times in the 'sample-library' library.
        Design compiler will use the first one only. (LBDB-400)
```

**What Next**

Change the operating condition name if it is wrong, or delete all extra definitions.

---

## LBDB-401

(error) The '%s' is missing for this timing check.

**Description**

Both rise and fall constraint values are required in the nochange timing check.

The following example shows an instance where this message occurs:

```
pin ( EN )    {
      direction : input;
      timing ()      {
        timing_type : nochange_high_high;
        related_pin : CP ;
        rise_constaint(cons) {
          values("0.100000, 0.100000, 0.100000",  \
                 "0.100000, 0.100000, 0.100000",  \
                 "0.100000, 0.100000, 0.100000",  \
                 "0.100000, 0.100000, 0.100000",  \
                 "0.100000, 0.100000, 0.100000");
        }
      }
    }
```

To fix the problem, add the attribute to the timing group,

```
fall_constraint(cons) {
          values("0.100000, 0.100000, 0.100000",  \
                 "0.100000, 0.100000, 0.100000",  \
                 "0.100000, 0.100000, 0.100000",  \
                 "0.100000, 0.100000, 0.100000",  \
                 "0.100000, 0.100000, 0.100000");
        }
```

Feedback

The following is an example message:

```
Error: Line 104, The 'fall_constraint' is missing for this timing check.
 (LBDB-401)
```

### What Next

Check the library source file to see if you missed the rise or fall constraint value. Otherwise, use setup or hold check instead.

---

## LBDB-402

(error) The '%s' is missing for this internal_power group.

### Description

Internal_power groups require specification of either (1) both the rise_power and fall_power attributes; or (2) the power attribute.

The following example shows an instance where this message occurs:

```
pin ( EN )    {
      direction : input;
      internal_power ()      {
        rise_power(power_1d_temp) {
          values("0.100000, 0.100000, 0.100000");
        }
      }
    }
```

To fix the problem, add the attribute to the internal_power group.

```
fall_power(power_1d_temp) {
          values("0.100000, 0.100000, 0.100000");
        }
```

The following is an example message:

```
Error: Line 104, The 'fall_power' is missing for this internal_power
 group. (LBDB-402)
```

### What Next

Check the library source file to see if you missed the rise_power, fall_power, or power attribute.

---

## LBDB-403

(error) The '%s' and '%s' attributes are both defined for this\n \tinternal_power group.

Feedback

### Description

Internal_power groups require specification of either 1 both the rise_power and fall_power attributes; or 2. the power attribute.

However, it is an error to mix both the specifications.

The following example shows an instance where this message occurs:

```
pin ( EN )    {
      direction : input;
      internal_power ()      {
        power(power_1d_temp) {
          values("0.100000, 0.100000, 0.100000");
        }
        rise_power(power_1d_temp) {
          values("0.100000, 0.100000, 0.100000");
        }
      }
    }
```

To fix the problem, remove the rise_power attribute from the internal_power group, or remove power attribute and add the following fall_power attribute from the internal_power group.

```
fall_power(power_1d_temp) {
          values("0.100000, 0.100000, 0.100000");
        }
```

The following is an example message:

```
XXX
Error: Line 104, The 'rise_power' and 'power' attributes are both defined
 for this
       internal_power group. (LBDB-403)
```

### What Next

Check the library source file to make sure that the previous requirement is met.

---

## LBDB-404

(error) The '%s' lookup table in the input-associated\n \tinternal_power group cannot use '%s' as its template.

### Description

The rise_power, fall_power, or power lookup tables in a input-associated internal_power group must use 1-dimensional template with input_transition_time as its variable.

Feedback

The following example shows an instance where this message occurs:

```
power_lu_template(ok_temp) {
    variable_1 : input_transition_time;
    index_1("0, 1, 2, 3");
  }
  power_lu_template(err_temp) {
    variable_1 : total_output_net_capacitance;
    index_1("0, 1, 2, 3");
  }

        power(err_temp) {
                values ("1, 2, 3, 4");
        }
```

To fix the problem, change the template value of the power attribute from err_temp to ok_temp.

The following is an example message:

```
Error: Line 126, The 'power' lookup table in the input-associated
       internal_power group cannot use 'err_temp' as its template.
 (LBDB-404)
```

**What Next**

Refer to the *Library Compiler User Guide* for more information about lookup tables. Change the library source file by referencing a different template in the lookup table description.

---

## LBDB-405

(error) The 1-dimensional '%s' lookup table in the\n \tinout-associated internal_power group cannot use '%s' as its template.

**Description**

The 1-dimensional rise_power, fall_power, or power lookup tables in an inout-associated internal_power group must use a template with input_transition_time or total_output_net_capacitance as its variable.

The following example shows an instance where this message occurs:

```
power_lu_template(ok_temp) {
    variable_1 : input_transition_time;
    index_1("0, 1, 2, 3");
  }
  power_lu_template(ok2_temp) {
    variable_1 : total_output_net_capacitance;
    index_1("0, 1, 2, 3");
  }
```

Feedback

```
  power_lu_template(err_temp) {
    variable_1 : output_net_length;
    index_1("0, 1, 2, 3");
  }

        power(err_temp) {
                values ("1, 2, 3, 4");
        }
```

To fix the problem, change the template value of the power attribute from err_template to ok_temp or ok2_temp.

The following is an example message:

```
Error: Line 126, The 1-dimensional 'power' lookup table in the
        inout-associated internal_power group cannot use
        'basic_template' as its template. (LBDB-405)
```

### What Next

Refer to the *Library Compiler User Guide* for more information about lookup tables. Change the library source file by referencing a different template in the lookup table description.

---

## LBDB-406

(error) The 1-dimensional '%s' lookup table in the\n \t%s-associated internal_power group cannot use '%s' as its template.

### Description

The 1-dimensional rise_power, fall_power, or power lookup tables in an output- associated internal_power group must use a template with total_output_net_capacitance as its variable.

The following example shows an instance where this message occurs:

```
power_lu_template(err_temp) {
    variable_1 : input_transition_time;
    index_1("0, 1, 2, 3");
  }
  power_lu_template(ok_temp) {
    variable_1 : total_output_net_capacitance;
    index_1("0, 1, 2, 3");
  }

        power(err_temp) {
                values ("1, 2, 3, 4");
        }
```

Feedback

To fix the problem, change the template value of the power attribute from err_template to ok_temp.

The following is an example message:

```
Error: Line 126, The 1-dimensional 'power' lookup table in the
        output-associated internal_power group cannot use 'err_temp' as
 its template. (LBDB-406)
```

**What Next**

Refer to the *Library Compiler User Guide* for more information about lookup tables. Change the library source file by referencing a different template in the lookup table description.

---

## LBDB-407

(error) The 2-dimensional '%s' lookup table in the\n \t%s-associated internal_power group cannot use '%s' as its template.

**Description**

The 2-dimensional rise_power, fall_power, or power lookup tables in an output-associated internal_power group must use a template with input_transition_time and total_output_net_capacitance as its variables.

The following example shows an instance where this message occurs:

```
power_lu_template(ok_temp) {
    variable_1 : input_transition_time;
    index_1("0, 1, 2, 3");
    variable_2 : total_output_net_capacitance;
    index_2("0, 1, 2, 3");
  }
  power_lu_template(err_temp) {
    variable_1 : total_output_net_capacitance;
    index_1("0, 1, 2, 3");
    variable_2 : constrained_pin_transition;
    index_2("0, 1, 2, 3");
  }

        power(err_temp) {
                values ("1, 2, 3, 4", "5, 6, 7, 8", \
                        "9, 10, 11, 12", "13, 14, 15, 16");
        }
```

To fix the problem, change the template value of the power attribute from err_temp to ok_temp.

Feedback

The following is an example message:

```
Error: Line 126, The 2-dimensional 'power' lookup table in the
        output-associated internal_power group cannot use 'err_temp' as
 its template. (LBDB-407)
```

### What Next

Refer to the *Library Compiler User Guide* for more information about lookup tables. Change the library source file by referencing a different template in the lookup table description.

---

## LBDB-408

(error) The 3-dimensional '%s' lookup table in the\n \t%s-associated internal_power group cannot use '%s' as its template.

### Description

The 3-dimensional rise_power, fall_power, or power lookup tables in an output-associated internal_power group must use a template with input_transition_time, total_output_net_capacitance and equal_or_opposite_output_net_capacitance as its variables.

The following example shows an instance where this message occurs:

```
power_lu_template(ok_temp) {
    variable_1 : input_transition_time;
    index_1("0, 1, 2");
    variable_2 : total_output_net_capacitance;
    index_2("0, 1, 2");
    variable_3 : equal_or_opposite_output_net_capacitance;
    index_3("0, 1, 2");
  }
  power_lu_template(err_temp) {
    variable_1 : total_output_net_capacitance;
    index_1("0, 1, 2");
    variable_2 : constrained_pin_transition;
    index_2("0, 1, 2");
    variable_3 : equal_or_opposite_output_net_capacitance;
    index_3("0, 1, 2");
  }

        power(err_temp) {
                values ("1,  2,  3",  "4,  5,  6",  "7, 8, 9", \
                      "10, 11, 12", "13, 14, 15", "16, 17, 18", \
                       "19, 20, 21", "22, 23, 24", "25, 26, 27");
        }
```

To fix the problem, change the template value of the power attribute from err_temp to ok_temp.

Feedback

The following is an example message:

```
Error: Line 126, The 3-dimensional 'power' lookup table in the
        output-associated internal_power group cannot use 'err_temp' as
 its template. (LBDB-408)
```

### What Next

Refer to the *Library Compiler User Guide* for more information about lookup tables. Change the library source file by referencing a different template in the lookup table description.

---

## LBDB-409

(error) The %d-dimensional template used in '%s' lookup table\n \tis incompatible with the template used in '%s' lookup table.

### Description

Both rise_power and fall_power lookup tables must have the same dimension. One exception is when the internal_power is an arc from a tri-state pin (that is related_pin is a tri-state pin), the dimension of rise_power and fall_power can be different.

The following example shows an instance where this message occurs:

```
power_lu_template(1d_temp) {
    variable_1 : total_output_net_capacitance;
    index_1("0, 1, 2, 3");

  power_lu_template(2d_temp) {
    variable_1 : input_transition_time;
    index_1("0, 1, 2, 3");
    variable_2 : total_output_net_capacitance;
    index_2("0, 1, 2, 3");
  }

        rise_power(1d_temp) {
                values ("1, 2, 3, 4");
        }
        fall_power(2d_temp) {
                values ("1, 2, 3, 4", "5, 6, 7, 8", \
                        "9, 10, 11, 12", "13, 14, 15, 16");
        }
```

To fix the problem, use 1-dimensional or 2-dimensional lookup tables on both rise_power and fall_power attributes.

Feedback

The following is an example message:

```
Error: Line 126, The %d-dimensional template used in '%s' lookup table
        is incompatible with the template used in '%s' lookup table.
 (LBDB-409)
```

### What Next

Refer to the *Library Compiler User Guide* for more information about lookup tables. Change the library source file by referencing a different template in the lookup table description.

---

## LBDB-410

(error) The '%s' attribute is missing from the internal_power\n \tgroup.

### Description

The related_pin or related_bus_pins attributes are required in an internal_power group with a 2-dimensional or 3-dimensional lookup table. The equal_or_opposite_output attribute is required in an internal_power group with a 3-dimensional look-up table.

The following example shows an instance where this message occurs:

```
internal_power() {
        power(power_2d) {
                values ("1, 2, 3, 4", "5, 6, 7, 8", \
                        "9, 10, 11, 12", "13, 14, 15, 16");
        }
    }
```

To fix the problem, add the following line into the internal_power group.

```
related_pin : "A";
```

The following is an example message:

```
Error: Line 126, The 'related_pin or related_bus_pins' attribute is
 missing
        from the internal_power group. (LBDB-410)
```

### What Next

Change the library source file by specifying related_pin, related_bus_pins, or equal_or_opposite_output attribute in the internal_power group as needed.

---

## LBDB-411

(error) The internal_power group associated with pin '%s'\n \tcannot coexist with the cell-associated internal_power group in line %d.

Feedback

### Description

It is not allowed to use cell-associated and pin-associated internal_power group for the same pin.

The following example shows an instance where this message occurs:

```
cell(AN2) {
        internal_power(power_1d_temp) {
            related_input : "A";
            values("0.100000, 0.100000, 0.100000");
        }
        pin (A) {
            direction : input;
            internal_power () {
                power(power_1d_temp) {
                    values("0.100000, 0.100000, 0.100000");
                }
            }
        }
        ...
    }
```

To fix the problem, remove internal_power group associated with cell group or pin group.

The following is an example message:

```
Error: Line 126, The internal_power group associated with pin 'A'
        cannot co-exist with the cell-associated internal_power group in
 line 120. (LBDB-411)
```

### What Next

Refer to the *Library Compiler User Guide* for more information about power modeling. Change the library source file by using either the cell-associated style or pin-associated style internal_power on the specified pin.

---

## LBDB-412

(error) There is a missing internal_power relation between\n \tpins '%s' and '%s' in the '%s' cell.

### Description

This message indicates there is a missing internal_power relation from an input or inout pin to an output pin.

For a combinational cell, the Library Compiler checks that

```
* An output port with a function statement has internal_power relation
          to all functionally related inputs
```

Feedback

```
        * An output port with a three_state attribute has timing arcs
          to all three_state related inputs
```

However, Library Compiler will not check if internal_power information is completely missing in this output.

The following example shows an instance where this message occurs:

```
cell(AN2) {
    pin(A, B) {
      direction : input;
      capacitance : 1;
    }
    pin(Z) {
      direction : output;
      function : "A B";
      internal_power () {
        power (power_2d_temp) {
            values ("1, 2, 3, 4", "5, 6, 7, 8", \
                    "9, 10, 11, 12", "13, 14, 15, 16");
        }
        related_pin : "A";
      }
      timing() {
        ...
      }
    }
  }
```

In this case, an internal_power group is missing between the pin 'Z' and 'B'. To fix the problem, add the following internal_power group:

```
internal_power () {
        power (power_2d_temp) {
            values ("1, 2, 3, 4", "5, 6, 7, 8", \
                    "9, 10, 11, 12", "13, 14, 15, 16");
        }
        related_pin : "B";
      }

The following is an example message:
Error: Line 12, There is a missing internal_power relation between
        pins 'B' and 'Z' in the 'AN2' cell. (LBDB-412)
```

**What Next**

Add the missing internal_power group between the two pins.

Feedback

## LBDB-413

(error) There is an extra internal_power group between '%s' and\n \t'%s' pins in the '%s' cell.

### Description

Internal_power groups are allowed for pins that are related to each other. To be related to each other, the input pin has to be in the *function* or *three_state* attribute of the output or input pin. This message is issued if the internal_power group identified between pins does not fall into the categories previously described.

The following example shows an instance where this message occurs:

```
pin(Z) {
      direction : output;
      function : "B";
      internal_power () {
        power (power_2d_temp) {
            values ("1, 2, 3, 4", "5, 6, 7, 8", \
                    "9, 10, 11, 12", "13, 14, 15, 16");
        }
        related_pin : "A B";
      }
      timing() {
        ...
      }
   }
```

In this case, you defined a internal_power group between the 'A' and 'Z pins, even though the 'A' pin is not functionally related to the 'Z' pin. Remove the internal_power group or change the function statement of 'Z' to include 'A'.

The following is an example message:

```
Warning: Line 73, There is an extra internal_power group between 'A' and
        'Z' pins in the 'AN2' cell. (LBDB-413)
```

### What Next

Check your library to see whether you have generated the internal_power group by mistake, whether the related_pin field is wrong, or the function attribute value is not recognized.

## LBDB-414

(error) In this internal_power group, the '%s' output pin in\n \tthe '%s' attribute is not functionally equivalent or opposite to the '%s' pin.

Feedback

### Description

If an output is specified in the related_pin, related_bus_pins, or equal_or_opposite_output attributes, this output must be functionally equivalent or opposite of the output pin which owns the internal_power.

The following example shows an instance where this message occurs:

```
cell(AN2) {
    pin(A, B) {
      direction : input;
      capacitance : 1;
    }
    pin(Z) {
      direction : output;
      function : "A B";
      internal_power () {
        power (power_3d_temp) {
                values ("1, 2, 3", "4, 5, 6", "7, 8, 9", \
                        "10, 11, 12", "13, 14, 15", "16, 17, 18"\
                        "19, 20, 21", "22, 23, 24", "25, 26, 27");
        }
        related_pin : "A B";
        equal_or_opposite_output : "Y";
      }
      timing() {
        ...
      }
    }
    pin(Y) {
      direction : output;
      function : "A + B";
      timing() {
        ...
      }
    }
  }
```

To fix the problem, remove equal_or_opposite_output attribute from the internal_power group and use 2-dimensional lookup table; or make sure 'Y' is functionally equivalent or opposite to 'Z'.

The following is an example message:

```
Error: Line 126, In this internal_power group, the 'Y' output pin in
        the 'equal_or_opposite_output' attribute is not functionally
 equivalent
        or opposite to the 'Z' pin. (LBDB-414)
```

Feedback

### What Next

Refer to the *Library Compiler User Guide* for more information about lookup tables. Change the library source file by referencing a different template in the lookup table description.

---

## LBDB-415

(error) The '%s' attribute cannot be specified in this\n \tinternal_power group.

### Description

This message indicates you specified an attribute outside its context.

```
(1) An internal_power group with 1-dimensional look-up tables cannot
 specify
    related_pin, related_bus_pins, or equal_or_opposite_output
 attributes.
(2) An internal_power group with 2-dimensional look-up tables cannot
 specify
    an equal_or_opposite_output attribute.
```

One exception is when the internal_power is an arc from a tri-state pin (that is related_pin is a tri-state pin), the checker is diabled.

The following example shows an instance where this message occurs:

```
internal_power() {
        power(power_1d_temp) {
                values ("1, 2, 3, 4");
        }
        related_pin : "A";
    }
```

In this case, the 'related_pin' attribute is not allowed in 1-dimensional lookup table. To fix the problem, remove the related_pin attribute from internal_power group.

The following is an example message:

```
Error: Line 69, The 'related_pin or related_bus_pins' attribute cannot be
        specified in this internal_power group. (LBDB-415)
```

### What Next

Change the technology library source file to delete the specified attribute or move the attribute to its correct context.

---

## LBDB-416

(error) The '%s' cell is missing cell_leakage_power attribute.

Feedback

### Description

This message indicates there is one or more state-dependent leakage_power attribute, but no cell_leakage_power attribute specified as default value. When at least one state-dependent leakage power is specified in a cell, the cell must have also cell_leakage_power attribute defined.

The following example shows an instance where this message occurs:

```
cell(AN2) {
      leakage_power () {
        when : "A";
        value : 1.0;
      }
      ...
  }
```

In this case, a cell_leakage_power attribute is missing. To fix the problem, add the following attribute to the cell group:

```
cell_leakage_power : 0.5;

The following is an example message:
Error: Line 12, The 'AN2' cell is missing cell_leakage_power attribute.
 (LBDB-416)
```

### What Next

Add the missing timing group between the two pins.

---

## LBDB-417

(error) Pin '%s' has a timing arc that appears on only one of\n \tthe scaled cell (%s,%s) and the cell '%s'.

### Description

Timing arcs with the same pin pair and type must match by number between the scaled cell and the regular cell.

The following example shows an instance where this message occurs:

```
library(alib) {
    operating_conditions(WCCOM) {
        process : 1.5 ;
        temperature : 70 ;
        voltage : 4.75 ;
        tree_type : "worst_case_tree" ;
    }
    cell(AND) {
      area : 1;
```

Feedback

```
    pin(A B) {
      direction : input;
      capacitance : 1;
    }
    pin(Z) {
      direction : output;
      function : "A B";
      timing() {
        intrinsic_rise : 0.1;
        intrinsic_fall : 0.1;
        rise_resistance : 0.1;
        fall_resistance : 0.1;
        slope_rise : 0.0;
        slope_fall : 0.0;
        related_pin : "A B";
      }
    }
  }

  scaled_cell(AND,WCCOM) {
    area : 1;
    pin(A) {
      direction : input;
      capacitance : 1;
    }
    pin(B) {
      direction : input;
      capacitance : 1;
      timing() {
        intrinsic_rise : 0.3;
        intrinsic_fall : 0.3;
        rise_resistance : 0.3;
        fall_resistance : 0.3;
        slope_rise : 0.0;
        slope_fall : 0.0;
        related_pin : "A";
      }
    }
    pin(Z) {
      direction : output;
      function : "A B";
      timing() {
        intrinsic_rise : 0.3;
        intrinsic_fall : 0.3;
        rise_resistance : 0.3;
        fall_resistance : 0.3;
        slope_rise : 0.0;
        slope_fall : 0.0;
        related_pin : "A B";
      }
    }
  }
}
```

Feedback

In this case, The timing arc between 'B' and 'A' is defined only on the 'AND' scaled cell and not on the 'AND' scaled_cell. To fix the problem, remove one of the timing arcs.

The following is an example message:

```
Error: Ling 30, Pin 'A' has a timing arc that appears on only one of
        the scaled cell (AND,WCCM) and the cell 'AND'. (LBDB-417)
```

**What Next**

Find the timing group in the scaled_cell and remove the extra timing groups or add more timing groups in the parent cell.

---

## LBDB-418

(warning) The cell_degradation constraint is missing\n \tin this timing group in cell '%s'.

**Description**

This message informs the user that Library Compiler finds the cell_degradation constraint being described in at least one, but not all, timing arcs of the cell. If the cell_degradation constraint is specified in a cell, all timing arcs of the following types should specify cell_degradation constraint: combinational rising_edge falling_edge clear preset three_state_enable However, the cell_degradation constraint should not be defined in timing groups of the following timing types: setup_rising setup_falling hold_rising hold_falling recovery_rising recovery_falling removal_rising removal_falling skew_rising skew_falling non_seq_setup_rising non_seq_setup_falling non_seq_hold_rising non_seq_hold_falling nochange_high_high nochange_high_low nochange_low_high nochange_low_low

The following example shows an instance where this message occurs:

```
cell (DFF) {
  pin (D) {
    direction : input;
    ...
    timing () {  /* no warning */
      timing_type : setup_rising;
      related_pin : CP;
      rise_constraint (constr) {
        ...
      }
      rise_constraint (constr) {
        ...
      }
    }
  }
  pin (CP) {
    direction : input;
    ...
```

Feedback

```
    }
    pin (Q) {
      direction : output;
      timing () {  /* cell_degradation is defined */
        related_pin : CP;
        timing_type : rising_edge;
        cell_degradation (celldeg) {
          ...
        }
        cell_rise (delay) {
          ...
        }
        cell_fall (delay) {
          ...
        }
        ...
      }
      ...
    }
    pin (QN) {
      direction : output;
      timing () {  /* cell_degradation is NOT defined  --- LIBG-206 --- */
        related_pin : CP;
        timing_type : rising_edge;
        cell_rise (delay) {
          ...
        }
        cell_fall (delay) {
          ...
        }
        ...
      }
      ...
    }
    ...

}
```

In this case, the warning message will be issued on CP-to-QN timing arc.

The following is an example message:

```
Warning: Line 202, The cell_degradation constraint is missing
         in this timing group in cell 'DFF'. (LBDB-418)
```

**What Next**

Do either one of the following modification to the cell description: (1) Remove all cell_degradation constraints specified in the cell. (2) Make sure all delay arcs of the above timing types specify cell_degradation constraints.

Feedback

## LBDB-419

(warning) Found the obsolete and unsupported 'state' group in the'%s' cell;\n \tplease use 'ff' group or 'latch' group or 'statetable' instead.

### Description

The *state* group is not supported after v3.0. Instead, the sequential function should now be described in *ff* or *latch* group or *statetable*.

The following is an example message:

```
Warning: Line 159, Found the obsolete and unsupported 'state' group in
 the 'lbdbl419' cell;
        please use 'ff' group or 'latch' group or 'statetable' instead.
 (LBDB-419)
```

### What Next

If you have access to the technology library source file change the *state* group to *ff* or *latch* group. Otherwise contact the vendor and inform them about the problem.

## LBDB-420

(error) The index number '%d' must be less than\n \tthe address width '%d'.

### Description

This message indicates an index in the column_address or row_address attribute is greater than the address_width value.

The following example shows an instance where this message occurs:

```
memory() { /* Indicate this is a memory cell */
        type : ram;
        address_width : 4;
        word_width : 2;
        column_address : "0 6 "; /* index 6 > addr_width 4 */
      }
```

The following is an example message:

```
Error: Line 127, The index number '6' must be less than
        the address width '4'.  (LBDB-420)
```

### What Next

Correct the index in the technology library.

Feedback

## LBDB-421

(error) The column or row address width '%d' must be less than\n \tor equal to the address width '%d'.

### Description

This message indicates the width of the column_address or row_address attribute is greater than the address_width value.

The following example shows an instance where this message occurs:

```
memory() { /* Indicate this is a memory cell */
        type : ram;
        address_width : 4;
        word_width : 2;
        column_address : "0:4 "; /* col width 5 > addr_width 4 */
      }
```

The following is an example message:

```
Error: Line 127, The column/row address width '5' must be less than or
        equal to the address width '4'.  (LBDB-421)
```

### What Next

Reduce the number of indices in the column_address or row_address or correct the address_width value in the technology library.

## LBDB-423

(error) Address pins are not all used by column_address and row_address attributes.

### Description

This message indicates not all indices in the column_address and row_address attributes are used.

The following example shows an instance where this message occurs:

```
memory() { /* Indicate this is a memory cell */
        type : ram;
        address_width : 4;
        word_width : 2;
        column_address : "0 1";
        row_address : "3";              /* missing the index 2 */
      }
```

Feedback

The following is an example message:

```
Error: Line 122, Address pins are not all used by column_address and
 row_address attributes. (LBDB-423)
```

**What Next**

Correct the index in the technology library.

## LBDB-424

(warning) The number of address pins that overlap between column and row is '%d'.

**Description**

This message indicates that there are overlapping indices in the column_address and row_address attributes.

The following example shows an instance where this message occurs:

```
memory() { /* Indicate this is a memory cell */
        type : ram;
        address_width : 4;
        word_width : 2;
        column_address : "0:2";
        row_address : "2:3";            /* the index 2 is overlapping */
      }
```

The following is an example message:

```
Warning: Line 122, The number of address pins that overlap between column
 and row is 1. (LBDB-424)
```

**What Next**

Correct the index overlapping in the technology library.

## LBDB-425

(warning) Removing duplicate indices in the attribute.

**Description**

This message indicates an index in the column_address or row_address attribute is duplicated.

The following example shows an instance where this message occurs:

```
memory() { /* Indicate this is a memory cell */
        type : ram;
        address_width : 4;
```

Feedback

```
        word_width : 2;
        column_address : "0 3 3";
      }
```

The following is an example message:

```
Warning: Line 126, Removing duplicate indices in the attribute.
 (LBDB-425)
```

**What Next**

Correct the index in the technology library.

---

## LBDB-426

(error) The '%s' is missing for this inout pin.

**Description**

Both input_signal_level and the output_signal_level values are required in an inout pin.

The following example shows an instance where this message occurs:

```
pin ( BIDIR )    {
      direction : inout;
      input_signal_level : VDD1;
      }
    }
```

To fix the problem, add the attribute to the pin group,

```
output_signal_level : VDD2;
```

The following is an example message:

```
Error: Line 104, The 'output_signal_level' is missing for this inout pin.
 (LBDB-426)
```

**What Next**

Check the library source file to see if you missed the input_signal_level or the output_signal_level attributes. Otherwise, change the direction of the pin to input or output.

---

## LBDB-426w

(warning) The '%s' is missing for this inout pin.

**Description**

Both input_signal_level and the output_signal_level values are required in an inout pin.

Feedback

The following example shows an instance where this message occurs:

```
pin ( BIDIR )    {
      direction : inout;
      input_signal_level : VDD1;
      }
    }
```

To fix the problem, add the attribute to the pin group,

```
output_signal_level : VDD2;
```

The following is an example message:

```
Warning: Line 104, The 'output_signal_level' is missing for this inout
 pin. (LBDB-426w)
```

### What Next

Check the library source file to see if you missed the input_signal_level or the output_signal_level attributes. Otherwise, change the direction of the pin to input or output.

---

## LBDB-427

(error) All power supplies defined in the power_supply group\n \tmust exist in this operating_conditions group.

### Description

This message indicates that one or more power_rail attributes are missing in an operating_conditions group even though they were defined in the power_supply group.

The following example shows an instance where this message occurs:

```
power_supply() {
        default_power_rail : VDD0;
        power_rail(VDD1, 5.0);
        power_rail(VDD2, 3.3);
    }


    operating_conditions(MPSS) {
        process : 1.5 ;
        temperature : 70 ;
        voltage : 4.75 ;
        tree_type : "worst_case_tree" ;
        power_rail(VDD2, 2.9);
    }
```

To fix the problem, add the attribute to the operating_conditions group,

```
power_rail(VDD1, 4.9);
```

Feedback

The following is an example message:

```
Error: Line 14, All power supplies defined in the power_supply group
        must exist in this operating_conditions group. (LBDB-427)
```

### What Next

Check the library source file to see if you missed a power_rail attribute and correct it.

---

## LBDB-428

(error) All the pins in the '%s' cell with multiple power supplies\n \tmust have signal level attributes.

### Description

Either the input_signal_level or the output_signal_level attribute is missing in a pin within a multiple power supply cell.

The following example shows an instance where this message occurs:

```
power_supply() {
        default_power_rail : VDD0;
        power_rail(VDD1, 5.0);
        power_rail(VDD2, 3.3);
    }

cell(lbdb428) {
  area : 2;
  pad_cell : true;
  rail_connection(PV1, VDD1);
  rail_connection(PV2, VDD2);
  pin(A) {
    direction : input;          /* missing input_signal_level attribute
 */
    capacitance : 1;
  }
  pin(Z) {
    direction : output;
    function : "A";
    output_signal_level : VDD2;
    timing() {
      intrinsic_rise : 0.48;
      intrinsic_fall : 0.77;
      rise_resistance : 0.1443;
      fall_resistance : 0.0523;
      slope_rise : 0.0;
      slope_fall : 0.0;
      related_pin : "A";
    }
  }
```

Feedback

```
}
}
```

To fix the problem, add the attribute to the 'A' pin group,

```
input_signal_level : VDD1;
```

The following is an example message:

```
Error: Line 96, All the pins in the 'lbdb428' cell with multiple power
 supplies
       must have signal level attributes. (LBDB-428)
```

**What Next**

Check the library source file to see if you missed the input_signal_level or the output_signal_level attributes.

---

## LBDB-429

(warning) The timing arc from '%s' to '%s' is dormant.

**Description**

A timing arc from an input pin to an output pin is dormant if the signal change on the input pin can never cause the change on the output without simultaneous changes on any other input pins.

The following example shows an instance where this message occurs:

```
cell(a_cell) {
    pin(y) {
      function : "!a b"
      three_state : "!a !b | a b"
      timing() {
        related_pin : a;
      }
    }
  }
```

In this example, if only pin 'a' is changed, pin 'y' will not have the signal transitions between '0' and '1'. The combinational timing arc between 'a' and 'y' is dormant.

The following is an example message:

```
Warning: Line 50000, The timing arc from 'a' to 'y' is dormant.
 (LBDB-429)
```

Feedback

### What Next

This is caused by the input sharing between 'function', 'three_state', and 'x_function'. The signal transition is not physically possible and generally the timing arc should not be specified.

---

## LBDB-430

(error) The '%s' rail_connection in a cell with multiple power\n \tsupplies is missing an internal_power table.

### Description

This error message indicates that an output pin, regardless of the output_signal_level value, is missing an internal_power table for designated rail_connection.

The following example shows an instance where this message occurs:

```
power_supply() {
        default_power_rail : VDD0;
        power_rail(VDD1, 5.0);
        power_rail(VDD2, 3.3);
    }

cell(lbdb428) {
  area : 2;
  pad_cell : true;
  rail_connection(PV1, VDD1);
  rail_connection(PV2, VDD2);
  pin(A) {
    direction : input;
    capacitance : 1;
    input_signal_level : VDD1;
    internal_power () {
      power_level : VDD1;
      power(power_out1_d) {
        values (" 0.693418, 0.691911, 0.689730, 0.691771, 0.695229,
 0.696524");
      }
    }
  }
  pin(Z) {
    direction : output;
    function : "A";
    output_signal_level : VDD2;
    internal_power () {
      power_level : VDD2;
      power(power_out1_d) {
        values (" 0.693418, 0.691911, 0.689730, 0.691771, 0.695229,
 0.696524");
      }
```

Feedback

```
    }
    timing() {
      intrinsic_rise : 0.48;
      intrinsic_fall : 0.77;
      rise_resistance : 0.1443;
      fall_resistance : 0.0523;
      slope_rise : 0.0;
      slope_fall : 0.0;
      related_pin : "A";
    }
  }
}
}
```

To fix the problem, add the internal_power to the 'Z' pin.

```
internal_power () {
      power_level : VDD1;
      power(power_out1_d) {
        values (" 0.693418, 0.691911, 0.689730, 0.691771, 0.695229,
 0.696524");
      }
    }
```

The following is an example message:

```
Error: Line 96,  The 'VDD1' rail_connection in a cell with multiple power
        supplies is missing an internal_power table. (LBDB-430)
```

**What Next**

Check the library source file to see if you missed the power_level attribute or there is a typo in the name of the power_supply.

---

## LBDB-431

(error) The value for attribute '%s' is empty.

**Description**

This error message indicates that an value field is null or empty.

The following example shows an instance where this message occurs:

```
cell(GND_G_A) {
  area              : 1 ;
  cell_footprint    : GND_G ;
  dont_touch        : true ;
  dont_use          : true ;
  cell_leakage_power : 0.00;

  pin(PAD) {
```

Feedback

```
    direction             : output ;
    function              : "0" ;
    capacitance           : 0.000000 ;

    max_capacitance       : 999.989990 ;
  }

  internal_power(POWER_IO_A) {
    values ("0.00000  ,0.00000  ",\
           "0.00000  ,0.00000  ");
    related_inputs  : "";
    related_outputs : "PAD";
  }
}
```

To fix the problem, fill up value for related_inputs or remove related_inputs.

```
internal_power () {
      values ("0.00000  ,0.00000  ",\
           "0.00000  ,0.00000  ");
        related_inputs  : "";
        related_outputs : "PAD";
      }
```

The following is an example message:

```
Error: Line 722,  The value for attribute 'related_outputs' is
        empty. (LBDB-431)
```

**What Next**

Check the library source file to see if you missed giving the value. If so, fill it up with appropriate value.

---

## LBDB-432

(error) The non-Zero %d scalar value is not allowed in internal_power\n \tgroup within a cell group.

**Description**

This error message indicates that a non zero scalar value was defined in an internal_power group within a cell group.

The following example shows an instance where this message occurs:

```
cell(LBDB-432) {
  area             : 1 ;

  internal_power(scalar) {
        values("2.0");
        related_input : "";
```

Feedback

```
  }

   ....
}
```

To fix the problem, change the values to zero or move the internal_power to a pin group.

```
internal_power (scalar) {
      values ("0.00000");
        related_input  : "";
      }
```

The following is an example message:

```
Error: Line 174, The non-Zero 2.0 scalar value is not allowed in
 internal_power
        group within a cell group. (LBDB-432)
```

### What Next

Check the library source file to see if it was a typo. If so, change the value to zero for no power consumption or move the internal power to a pin group.

---

## LBDB-433

(error) The '%s' integrated gated clock cell has a '%s' pin with\n \ta combinational timing arc containing the '%s' timing_type.

### Description

This message indicates that you specified one of the cell's timing arcs as combinational, which should be sequential. While the function of this cell is integrated clock gated cell, the timing arc on the clock gate enable pin from the clock gate clock pin should be sequential timing type setup, hold or nochnage.

The following example shows an instance where this message occurs:

```
cell(CGNP) {
  area : 1;
  clock_gating_integrated_cell : "none_posedge";
  dont_use : true;
  pin(EN) {
    direction : input;
    capacitance : 0.017997;
    clock_gate_enable_pin : true;
    timing() {
      intrinsic_rise : 0.48;
      intrinsic_fall : 0.77;
      rise_resistance : 0.1443;
      fall_resistance : 0.0523;
      slope_rise : 0.0;
```

Feedback

```
      slope_fall : 0.0;
      related_pin : "EN";
    }
    timing() {
      intrinsic_rise : 0.22;
      intrinsic_fall : 0.22;
      rise_resistance : 0.1443;
      fall_resistance : 0.0523;
      slope_rise : 0.0;
      slope_fall : 0.0;
      related_pin : "CP";
    }
  }
  pin(CP) {
    direction : input;
    capacitance : 0.031419;
    clock_gate_clock_pin : true;
    min_pulse_width_low  : 0.319;
  }
  pin(Z) {
    direction : output;
    function : "CP + EN'";
    max_capacitance : 0.500;
    clock_gate_out_pin : true;
    timing() {
      intrinsic_rise : 0.48;
      intrinsic_fall : 0.77;
      rise_resistance : 0.1443;
      fall_resistance : 0.0523;
      slope_rise : 0.0;
      slope_fall : 0.0;
      related_pin : "EN";
    }
    timing() {
      intrinsic_rise : 0.22;
      intrinsic_fall : 0.22;
      rise_resistance : 0.1443;
      fall_resistance : 0.0523;
      slope_rise : 0.0;
      slope_fall : 0.0;
      related_pin : "CP";
    }
    internal_power (){
      rise_power(li4X3){
        index_1("0.0150, 0.0400, 0.1050, 0.3550");
        index_2("0.050, 0.451, 1.501");
        values("0.141, 0.148, 0.256",\
        "0.162, 0.145, 0.234",\
        "0.192, 0.200, 0.284",\
        "0.199, 0.219, 0.297");
      }
      fall_power(li4X3){
        index_1("0.0150, 0.0400, 0.1050, 0.3550");
```

Feedback

```
        index_2("0.050, 0.451, 1.500");
        values("0.117, 0.144, 0.246",\
        "0.133, 0.151, 0.238",\
        "0.151, 0.186, 0.279",\
        "0.160, 0.190, 0.217");
      }
      related_pin : "CP EN" ;
    }
  }
}
```

The following is an example message:

```
Error: Line 206, The 'CGNP' integrated gated clock cell has a 'EN' pin
 with
        a combinational timing arc containing the 'combinational'
 timing_type. (LBDB-433)
```

### What Next

Change the library source file, and fix the timing_type of the specified pin.

---

## LBDB-434

(error) The '%s' integrated gated clock cell has a '%s' pin with\n \ta sequential timing arc containing the '%s' timing_type.

### Description

This message indicates that you specified one of the cell's timing arcs as sequential, which should be combinational. While the function of this cell is integrated clock gated cell, the timing arc on the clock gate output pin from the clock gate clock/enable/test pin should be combinational timing type.

The following example shows an instance where this message occurs:

```
cell(CGNP) {
  area : 1;
  clock_gating_integrated_cell : "none_posedge";
  dont_use : true;
  pin(EN) {
    direction : input;
    capacitance : 0.017997;
    clock_gate_enable_pin : true;
    timing() {
      timing_type : nochange_high_low;
      intrinsic_rise : 0.4;
      intrinsic_fall : 0.4;
      related_pin : "CP";
    }
    timing() {
```

Feedback

```
      timing_type : nochange_low_low;
      intrinsic_rise : 0.4;
      intrinsic_fall : 0.4;
      related_pin : "CP";
    }
  }
  pin(CP) {
    direction : input;
    capacitance : 0.031419;
    clock_gate_clock_pin : true;
    min_pulse_width_low  : 0.319;
  }
  pin(Z) {
    direction : output;
    function : "CP + EN'";
    max_capacitance : 0.500;
    clock_gate_out_pin : true;
    timing() {
      timing_type : nochange_high_low;
      intrinsic_rise : 0.4;
      intrinsic_fall : 0.4;
      related_pin : "CP";
    }
    timing() {
      intrinsic_rise : 0.48;
      intrinsic_fall : 0.77;
      rise_resistance : 0.1443;
      fall_resistance : 0.0523;
      slope_rise : 0.0;
      slope_fall : 0.0;
      related_pin : "EN";
    }
    timing() {
      intrinsic_rise : 0.22;
      intrinsic_fall : 0.22;
      rise_resistance : 0.1443;
      fall_resistance : 0.0523;
      slope_rise : 0.0;
      slope_fall : 0.0;
      related_pin : "CP";
    }
    internal_power (){
      rise_power(li4X3){
        index_1("0.0150, 0.0400, 0.1050, 0.3550");
        index_2("0.050, 0.451, 1.501");
        values("0.141, 0.148, 0.256",\
        "0.162, 0.145, 0.234",\
        "0.192, 0.200, 0.284",\
        "0.199, 0.219, 0.297");
      }
      fall_power(li4X3){
        index_1("0.0150, 0.0400, 0.1050, 0.3550");
        index_2("0.050, 0.451, 1.500");
```

Feedback

```
        values("0.117, 0.144, 0.246",\
        "0.133, 0.151, 0.238",\
        "0.151, 0.186, 0.279",\
        "0.160, 0.190, 0.217");
      }
      related_pin : "CP EN" ;
    }
  }
}
```

The following is an example message:

```
Error: Line 206, The 'CGNP' integrated gated clock cell has a 'Z' pin
 with
        a sequential timing arc containing the 'nochange_high_low'
 timing_type. (LBDB-434)
```

### What Next

Change the library source file, and fix the timing_type of the specified pin.

---

## LBDB-435

(error) The '%s' integrated gated clock cell has a '%s' pin without\n \tspecified timing arcs.

### Description

This message indicates that you specified one of the cell's pin without timing arcs, while the function of this cell is integrated clock gated cell.

The following example shows an instance where this message occurs:

```
cell(CGNP) {
  area : 1;
  clock_gating_integrated_cell : "none_posedge";
  dont_use : true;
  pin(EN) {
    direction : input;
    capacitance : 0.017997;
    clock_gate_enable_pin : true;
  }
  pin(CP) {
    direction : input;
    capacitance : 0.031419;
    clock_gate_clock_pin : true;
    min_pulse_width_low  : 0.319;
  }
  pin(Z) {
    direction : output;
    function : "CP + EN'";
    max_capacitance : 0.500;
    clock_gate_out_pin : true;
```

Feedback

```
      timing() {
        timing_type : nochange_high_low;
        intrinsic_rise : 0.4;
        intrinsic_fall : 0.4;
        related_pin : "CP";
      }
      timing() {
        timing_type : nochange_low_low;
        intrinsic_rise : 0.4;
        intrinsic_fall : 0.4;
        related_pin : "CP";
      }
      timing() {
        intrinsic_rise : 0.48;
        intrinsic_fall : 0.77;
        rise_resistance : 0.1443;
        fall_resistance : 0.0523;
        slope_rise : 0.0;
        slope_fall : 0.0;
        related_pin : "EN";
      }
      timing() {
        intrinsic_rise : 0.22;
        intrinsic_fall : 0.22;
        rise_resistance : 0.1443;
        fall_resistance : 0.0523;
        slope_rise : 0.0;
        slope_fall : 0.0;
        related_pin : "CP";
      }
      internal_power (){
        rise_power(li4X3){
          index_1("0.0150, 0.0400, 0.1050, 0.3550");
          index_2("0.050, 0.451, 1.501");
          values("0.141, 0.148, 0.256",\
          "0.162, 0.145, 0.234",\
          "0.192, 0.200, 0.284",\
          "0.199, 0.219, 0.297");
        }
        fall_power(li4X3){
          index_1("0.0150, 0.0400, 0.1050, 0.3550");
          index_2("0.050, 0.451, 1.500");
          values("0.117, 0.144, 0.246",\
          "0.133, 0.151, 0.238",\
          "0.151, 0.186, 0.279",\
          "0.160, 0.190, 0.217");
        }
        related_pin : "CP EN" ;
      }
    }
  }
```

Feedback

The following is an example message:

```
Error: Line 206, The 'CGNP' integrated gated clock cell has a 'Z' pin
 without
        specified timing arcs. (LBDB-435)
```

**What Next**

Change the library source file, and add the missing timing arc of the specified pin.

---

## LBDB-436

(error) Illegal timing_model_type value '%s'.

**Description**

Currently the timing_model_type attribute can only take the value "stamp".

The following example shows an instance where this message occurs:

```
cell(CGNP) {
  area : 1;
  timing_type_model : stml;
  ...
}
```

The following is an example message:

```
Error: Line 206, Illegal timing_model_type value 'stml'. (LBDB-436)
```

**What Next**

Change or delete the attribute.

---

## LBDB-437

(warning) The generated_clock(%s) group is defined \n \tmultiple times.

**Description**

This message indicates that you specified the same generated_clock group multiple times. Only the last one is retained.

The following example shows an instance where this message occurs:

```
cell(CGNP) {
  area : 1;

  generated_clock(clock_A) {
    ...
  }
```

Feedback

```
  generated_clock(clock_A) {
    ...
  }
  ...
}
```

The following is an example message:

```
Warning: Line 206, The generated_clock(clock_A) group is defined
        multiple times. (LBDB-437)
```

**What Next**

Change the group name, or delete the duplicated group.

---

## LBDB-438

(error) The master pin is not specified in the generated_clock\n \tgroup.

**Description**

The master_pin attribute must be specified for a generated clock.

The following example shows an instance where this message occurs:

```
cell(CGNP) {
  area : 1;
  generated_clock(clockA) {
    clock_pin : CK;
    divided_by : 2;
  }
  ...
}
```

The following is an example message:

```
Error: Line 206, The master pin is not specified in the generated_clock
        group. (LBDB-438)
```

**What Next**

Add the master_pin attribute.

---

## LBDB-439

(error) The clock pin is not specified in the generated_clock\n \tgroup.

**Description**

The clock_pin attribute must be specified for a generated clock.

Feedback

The following example shows an instance where this message occurs:

```
cell(CGNP) {
  area : 1;
  generated_clock(clockA) {
    master_pin : CK;
    divided_by : 2;
  }
  ...
}
```

The following is an example message:

```
Error: Line 206, The clock pin is not specified in the generated_clock
        group. (LBDB-439)
```

**What Next**

Add the clock_pin attribute.

---

## LBDB-440

(error) The generated_clock divisor is less than 1.

**Description**

The divisor in the generated_clock group must be an integer greater than or equal to 1.

The following example shows an instance where this message occurs:

```
cell(CGNP) {
  area : 1;
  generated_clock(clockA) {
    clock_pin : CLK1;
    master_pin : CLK;
    divided_by : 0;
  }
  ...
}
```

The following is an example message:

```
Error: Line 206, The generated_clock divisor is less than 1. (LBDB-440)
```

**What Next**

Change the divisor value.

---

## LBDB-441

(error) The generated_clock multiplier is less than 1.

Feedback

### Description

The multiplier in the generated_clock group must be an integer greater than or equal to 1.

The following example shows an instance where this message occurs:

```
cell(CGNP) {
  area : 1;
  generated_clock(clockA) {
    clock_pin : CLK1;
    master_pin : CLK;
    multiplied_by : 0;
  }
  ...
}
```

The following is an example message:

```
Error: Line 206, The generated_clock multiplier is less than 1.
 (LBDB-441)
```

### What Next

Change the multiplier value.

---

## LBDB-442

(error) The generated_clock edge is less than 0.

### Description

The edge1, edge2, and edge3 parameter in the edges complex attribute in the generated_clock group must be non-negative integers.

The following example shows an instance where this message occurs:

```
cell(CGNP) {
  area : 1;
  generated_clock(clockA) {
    clock_pin : CLK1;
    master_pin : CLK;
    edges(-1, 2, 3);
  }
  ...
}
```

The following is an example message:

```
Error: Line 206, The generated_clock edge is less than 0. (LBDB-442)
```

Feedback

### What Next

Change the violating edge value.

---

## LBDB-443

(warning) The mode_definition(%s) group is defined\n \tmultiple times.

### Description

This message indicates that you specified the same mode_definition group multiple times. Only the last one is retained.

The following example shows an instance where this message occurs:

```
cell(CGNP) {
  area : 1;

  mode_definition(rw) {
    ...
  }
  mode_definition(rw) {
    ...
  }
  ...
}
```

The following is an example message:

```
Warning: Line 206, The mode_definition(rw) group is defined
        multiple times. (LBDB-443)
```

### What Next

Change the group name, or delete the duplicated group.

---

## LBDB-444

(error) The mode definition '%s' has no values defined.

### Description

A mode_definition group must have at least one mode value defined.

The following example shows an instance where this message occurs:

```
cell(CGNP) {
  area : 1;
  mode_definition(read) {
  }
```

Feedback

```
  ...
}
```

The following is an example message:

```
Error: Line 206, The mode definition 'read' has no values defined.
 (LBDB-444)
```

**What Next**

Define the mode values.

---

## LBDB-445

(warning) The mode_value(%s) group is defined\n \tmultiple times.

**Description**

This message indicates that you specified the same mode_value group multiple times. Only the last one is retained.

The following example shows an instance where this message occurs:

```
cell(CGNP) {
  area : 1;

  mode_definition(rw) {
    mode_value(read) {
      when : R;
      sdf_cond : "R == 1";
    }
    mode_value(read) {
      when : !R;
      sdf_cond : "R == 0";
    }
  }
}
```

The following is an example message:

```
Warning: Line 206, The mode_value(read) group is defined
        multiple times. (LBDB-445)
```

**What Next**

Change the group name, or delete the duplicated group.

---

## LBDB-446

(error) The sdf_cond attribute is not specified for the\n \tmode condition.

Feedback

### Description

This message indicates that you specified the when condition for the mode value, but did not specify the sdf_cond attribute.

The following example shows an instance where this message occurs:

```
cell(CGNP) {
  area : 1;
  mode_difinition(rw) {
    mode_value(read) {
      when : R;
    }
  }
  ...
}
```

The following is an example message:

```
Error: Line 206, The sdf_cond attribute is not specified for the
        mode condition. (LBDB-446)
```

### What Next

Add the missing sdf_cond attribute.

---

## LBDB-447

(information) The %s is defined multiple times for %s. All of them are kept.

### Description

This message indicates that you specified multiple attribute or group information. All of the attributes or group information are kept in db.

### What Next

Please make sure that it is your real intention to have multiple attribute or group information.

### EXMAPLES

The following example shows an instance where this message occurs:

```
cell(CGNP) {
  area : 1;

  pin (y) {
    timing() {
      mode(rw, read);
      mode(rw, write);
    }
```

Feedback

```
  }
}
```

### Examples

```
Information: Line 206, The mode is defined multiple times for timing. All
 of them are kept (LBDB-447)
```

---

## LBDB-448

(error) The Boolean condition overlaps with the condition\n \tspecified at line %d.

### Description

The two Boolean conditions must be mutually exclusive.

The following example shows an instance where this message occurs:

```
cell(CGNP) {
  area : 1;
  mode_definition(rw) {
    mode_value(read) {
      when : R;
      sdf_cond : "R == 1";
    }
    mode_value(write) {
      when : R;
      sdf_cond : "R == 1";
    }
  }
  ...
}
```

The following is an example message:

```
Error: Line 206, The Boolean condition overlaps with the condition
        specified at line 202. (LBDB-448)
```

### What Next

Check the conditions and make them mutually exclusive.

---

## LBDB-449

(error) The mode instance mode(%s, %s) is invalid.

### Description

The mohe instance delaration must be referring to a mode_definition and one of its mode_value

Feedback

The following example shows an instance where this message occurs:

```
cell(CGNP) {
  area : 1;
  mode_definition(rw) {
    mode_value(read) {
      ...
    }
    mode_value(werite) {
      ...
    }
  }
  pin (y) {
    timing () {
      mode(rw, latching);
      ...
    }
  }
  ...
}
```

The following is an example message:

```
Error: Line 206, The mode instance mode(rw, latching) is invalid.
 (LBDB-449)
```

### What Next

Check for consistency between mode group and mode value definitions, and mode instance declarations.

---

## LBDB-450

(error) Mismatched quotes in the sdf_cond string.

### Description

In the sdf_cond string, a double quoting is mismatched.

The following example shows an instance where this message occurs:

```
cell(CGNP) {
  area : 1;
  mode_definition(rw) {
    mode_value(read) {
      when : R;
      sdf_cond : "R == 1 && C ==
    }
  }
  ...
}
```

Feedback

The following is an example message:

```
Error: Line 206, Mismatched quotes in the sdf_cond string. (LBDB-450)
```

**What Next**

Add the missing quote or delete the extra quote.

---

## LBDB-451

(error) Undefined variable '%s'.

**Description**

The variable in the sdf_cond string must be a port or bus name.

The following example shows an instance where this message occurs:

```
cell(CGNP) {
  area : 1;
  mode_definition(rw) {
    mode_value(read) {
      when : R;
      sdf_cond : "read == 1";
    }
  }
  pin (R) {
    ...
  }
  ...
}
```

The following is an example message:

```
Error: Line 206, Undefined variable 'read'. (LBDB-451)
```

**What Next**

Eliminate undefined varibable.

---

## LBDB-452

(error) Fewer than two ports are specified in the\n \tshort list.

**Description**

A short complex attribute must have more than one parameter.

The following example shows an instance where this message occurs:

```
cell (c) {
  ...
```

Feedback

```
  short(A);
  pin (A) {...}
  pin (Y) {...}
}
```

The following is an example message:

```
Error: Line 206, Fewer than two ports are specified in the
        short list. (LBDB-452)
```

**What Next**

Add the missing port names.

---

## LBDB-453

(error) The parameter '%s' in the short list is not a\n \tport or bus name.

**Description**

The short complex attribute must take port or bus name as parameters.

The following example shows an instance where this message occurs:

```
cell(CGNP) {
  area : 1;
  short(R, T);

  pin(R) {...}
  ...
}
```

The following is an example message:

```
Error: Line 206, The parameter 'T' in the short list is not a
        port or bus name. (LBDB-453)
```

**What Next**

Eliminate undefined varibable.

---

## LBDB-454

(error) Unequal bus widths in the short list. \tspecified timing arcs.

**Description**

All members of a short list must be of the same width. That is, they must all be simple ports, or all be buses of the same width.

Feedback

The following example shows an instance where this message occurs:

```
cell(CGNP) {
  area : 1;
  short(A, B);
  bus(A) {
    bus_type : bus4;
  }
  bus(B) {
    bus_tyep : bus8;
  }
  ...
}
```

The following is an example message:

```
Error: Line 206, Unequal bus widths in the short list. (LBDB-454)
```

**What Next**

Use the same width members.

---

## LBDB-455

(error) Multiple drive arcs are defined on port %s.

**Description**

A port can have at most one drive arc defined.

The following example shows an instance where this message occurs:

```
pin(Y) {
  timing () {
    drive_arc : true;
    ...
  }
  timing () {
    drive_arc : true;
    ...
  }
  ...
}
```

The following is an example message:

```
Error: Line 206, Multiple drive arcs are defined on port Y. (LBDB-455)
```

**What Next**

Eliminate the extra ones.

Feedback

## LBDB-456

(error) The drive arc is not combinational.

### Description

Drive arc must be combinational.

The following example shows an instance where this message occurs:

```
pin(Z) {
    direction : output;
    timing() {
        timing_type : nochange_high_low;
        drive_arc : true;
        ...
    }
    ...
}
```

The following is an example message:

```
Error: Line 206, The drive arc is not combinational. (LBDB-456)
```

### What Next

Change the drive arc to combinational or delete it.

## LBDB-457

(error) Related pin is found on the drive arc.

### Description

Drive arc cannot have related pins.

The following example shows an instance where this message occurs:

```
pin(Z) {
    direction : output;
    timing() {
        timing_type : nochange_high_low;
        drive_arc : true;
        related_pin : A;
        ...
    }
    ...
}
```

The following is an example message:

```
Error: Line 206, Related pin is found on the drive arc. (LBDB-457)
```

Feedback

### What Next

Remove the related_pin attribute or unlabel the timing arc.

---

## LBDB-458

(error) Cell rise or fall table is missing on the\n \tdrive arc.

### Description

Drive arc requires cell rise and fall tables.

The following example shows an instance where this message occurs:

```
pin(Z) {
    direction : output;
    timing() {
        timing_type : nochange_high_low;
        drive_arc : true;
        cell_rise(table1) {
          ...
        }
    }
    ...
}
```

The following is an example message:

```
Error: Line 206, Cell rise or fall table is missing on the
        drive arc. (LBDB-458)
```

### What Next

Add the missing tables.

---

## LBDB-459

(error) The drive arc has tables that are not 1-D function\n \tof total_output_capacitance.

### Description

Drive arc requires delay tables to be 1-D function of total output capacitance.

The following example shows an instance where this message occurs:

```
pin(Z) {
    direction : output;
    timing() {
        timing_type : nochange_high_low;
        drive_arc : true;
        cell_rise(table2D) {
```

Feedback

```
          values("0.1, 0.2", "0.5, 0.8");
        }
    }
    ...
}
```

The following is an example message:

```
Error: Line 206, The drive rc has tables that are not 1-D function
        of total_output_capacitance. (LBDB-459)
```

**What Next**

Add the missing tables.

---

## LBDB-460

(error) Multiple tlatch groups are defined on\n \tpin '%s'.

**Description**

This message indicates that you specified multiple tlatch groups on a pin.

The following example shows an instance where this message occurs:

```
pin(D) {

  tlatch(EN) {
    ...
  }
  tlatch(E) {
    ...
  }
  ...
}
```

The following is an example message:

```
Error: Line 206, Muliple tlatch groups are defined on
        pin 'D'.  (LBDB-460)
```

**What Next**

Delete the duplicated group.

---

## LBDB-461

(error) The edge type is not specified for the tlatch.

Feedback

### Description

A tlatch group must have the edge_type attribute specified.

The following example shows an instance where this message occurs:

```
pin(D) {

  tlatch(EN) {
    tdisable : TRUE;
  }
  ...
}
```

The following is an example message:

```
Error: Line 206, The edge tyep is not specified for the tlatch.
 (LBDB-461)
```

### What Next

Add the missing attribute.

---

## LBDB-462

(error) Pin '%s' referred to in tlatch does not exist.

### Description

The tlatch group name must be the name of an existing enable pin.

The following example shows an instance where this message occurs:

```
cell (c) {
  pin(D) {
    tlatch(EN) {
      edge_type : rising;
    }
  }
  ...
  /* No such pin as EN */
}
```

The following is an example message:

```
Error: Line 206, Pin 'EN' referred to in tlatch does not exist.
 (LBDB-462)
```

### What Next

Add the missing attribute.

Feedback

## LBDB-463

(warning) The related pin '%s' for the non-sequential\n \tsetup/hold timing check is labeled a clock.

### Description

If the related pin for the non-sequential constraint arc is labeled a clock, then timing analysis will not propagate a clock signal through the cell.

The following example shows an instance where this message occurs:

```
cell (c) {
  pin(D) {
    timing() {
      timing_type : non_seq_setup_falling;
      related_pin : OSC;
      ...
    }
  }
  pin(OSC) {
    clock : true;
    ...
  }
}
```

The following is an example message:

```
Warning: Line 206, The related pin 'OSC' for the non-sequential
        setup/hold timing check is labeled a clock. (LBDB-463)
```

### What Next

Delete the "clock : true;" declaration in the pin group.

## LBDB-464

(error) The '%s' cell is a non-pad cell with x_function,\n \twhich is not supported by Library Compiler.

### Description

Library Compiler only supports x_function on pad cells. Library Compiler issues this error message if a non-pad cell defines x_function on one or more ports.

The following example shows an instance where this message occurs:

```
cell (lbdb464) {
    area    :   0.0;
    pin (D, CLK)    {
      direction   :   input;
```

Feedback

```
      capacitance :   1.0;
      fanout_load :   1.0;
    }
    pin(Q){
      direction   :   output;
      function : "IQ";
      x_function : CLK';
      timing () {
        timing_type : rising_edge;
        intrinsic_rise   : 1.0;
        rise_resistance  : 0.1;
        intrinsic_fall   : 1.0;
        fall_resistance  : 0.1;
        related_pin      : "CLK";
      }
    }
    ff (IQ, IQN) {
      next_state : D;
      clocked_on : CLK;
    }
  }
```

The following is an example message:

```
Error: Line 6, The 'lbdb464' cell is a non-pad cell with x_function,
 which is not supported by Library Compiler.(LBDB-464)
```

### What Next

Remove the x_function attribute defined for any output pin of the cell.

---

## LBDB-465

(error) The '%s' attribute is missing\n \tfrom the electromigration group.

### Description

The *related_pin* or *related_bus_pins* attribute is required in an electromigration group with a two-dimensional lookup table.

The following example shows an instance where this message occurs:

```
electromigration() {
        em_max_toggle_rate(em_2d) {
                values ("1, 2, 3, 4", "5, 6, 7, 8", \
                        "9, 10, 11, 12", "13, 14, 15, 16");
        }
    }
```

To fix the problem, add the following line into the electromigration group:

```
related_pin : "A";
```

Feedback

The following is an example message:

```
Error: Line 126, The 'related_pin or related_bus_pins' attribute is
 missing
        from the electromigration group. (LBDB-465)
```

### What Next

Change the library source file by specifying the *related_pin* or the *related_bus_pins* attribute in the electromigration group as needed.

---

## LBDB-466

(error) The '%s' attribute cannot be\n \tspecified in this electromigration group.

### Description

This message indicates that you specified an attribute outside its context. An electromigration group with one-dimensional lookup tables cannot specify *related_pin* or *related_bus_pins* attributes.

The following example shows an instance where this message occurs:

```
electromigration() {
        em_max_toggle_rate(em_1d_temp) {
                values ("1, 2, 3, 4");
        }
        related_pin : "A";
    }
```

In this case, the *related_pin* attribute is not allowed in the one-dimensional lookup table. To fix the problem, remove the *related_pin* attribute from the electromigration group.

The following is an example message:

```
Error: Line 69, The 'related_pin or related_bus_pins' attribute cannot be
        specified in this electromigration group. (LBDB-466)
```

### What Next

Change the technology library source file to delete the specified attribute or move the attribute to its correct context.

---

## LBDB-467

(error) The '%s' lookup table in the input-associated\n \telectromigration group cannot use '%s' as its template.

Feedback

### Description

The em_max_toggle_rate lookup table in an input-associated electromigration group must use a one-dimensional template with *input_transition_time* as its variable.

The following example shows an instance where this message occurs:

```
em_lut_template(ok_temp) {
    variable_1 : input_transition_time;
    index_1("0, 1, 2, 3");
  }
  em_lut_template(err_temp) {
    variable_1 : total_output_net_capacitance;
    index_1("0, 1, 2, 3");
  }

        em_max_toggle_rate(err_temp) {
                values ("1, 2, 3, 4");
        }
```

To fix the problem, change the template value of the *em_max_toggle_rate* attribute from *err_temp* to *ok_temp*.

The following is an example message:

```
Error: Line 126, The 'em_max_toggle_rate' lookup table in the
 input-associated
       electromigration group cannot use 'err_temp' as its template.
 (LBDB-467)
```

### What Next

For more information about lookup tables, see the *Library Compiler User Guide*. Change the library source file by referencing a different template in the lookup table description.

---

## LBDB-468

(error) The one-dimensional '%s' lookup table in the\n \tinout-associated electromigration group cannot use '%s' as its template.

### Description

The one-dimensional em_max_toggle_rate lookup table in an inout-associated electromigration group must use a template with *input_transition_time* or *total_output_net_capacitance* as its variable.

The following example shows an instance where this message occurs:

```
em_lut_template(ok_template) {
    variable_1 : input_transition_time;
    index_1("0, 1, 2, 3");
```

Feedback

```
  }
  em_lut_template(ok2_template) {
    variable_1 : total_output_net_capacitance;
    index_1("0, 1, 2, 3");
  }
  em_lut_template(err_template) {
    variable_1 : output_net_length;
    index_1("0, 1, 2, 3");
  }

        em_max_toggle_rate(err_template) {
                values ("1, 2, 3, 4");
        }
```

To fix the problem, change the template value of the *em_max_toggle_rate* attribute from err_template to ok_template or ok2_template.

The following is an example message:

```
Error: Line 126, The 1-dimensional 'em_max_toggle_rate' lookup table in
 the
        inout-associated electromigration group cannot use
        'err_template' as its template. (LBDB-468)
```

**What Next**

For more information about lookup tables, see the *Library Compiler User Guide*. Change the library source file by referencing a different template in the lookup table description.

---

## LBDB-469

(error) The one-dimensional '%s' lookup table in the\n \t%s-associated electromigration group cannot use '%s' as its template.

**Description**

The one-dimensional em_max_toggle_rate lookup table in an output-associated electromigration group must use a template with *total_output_net_capacitance* as its variable.

The following example shows an instance where this message occurs:

```
em_lut_template(err_template) {
    variable_1 : input_transition_time;
    index_1("0, 1, 2, 3");
  }
  em_lut_template(ok_template) {
    variable_1 : total_output_net_capacitance;
    index_1("0, 1, 2, 3");
  }
```

Feedback

```
        em_max_toggle_rate(err_template) {
                values ("1, 2, 3, 4");
        }
```

To fix the problem, change the template value of the *em_max_toggle_rate* attribute from *err_template* to *ok_template*.

The following is an example message:

```
Error: Line 126, The 1-dimensional 'em_max_toggle_rate' lookup table in
 the
        output-associated electromigration group cannot use
 'err_template' as its template. (LBDB-469)
```

**What Next**

For more information about lookup tables, see the *Library Compiler User Guide*. Change the library source file by referencing a different template in the lookup table description.

---

## LBDB-470

(error) The two-dimensional '%s' lookup table in the\n \t%s-associated electromigration group cannot use '%s' as its template.

**Description**

The two-dimensional em_max_toggle_rate lookup table in an output-associated electromigration group must use a template with *input_transition_time* and *total_output_net_capacitance* as its variables.

The following example shows an instance where this message occurs:

```
em_lut_template(ok_template) {
    variable_1 : input_transition_time;
    index_1("0, 1, 2, 3");
    variable_2 : total_output_net_capacitance;
    index_2("0, 1, 2, 3");
  }
  em_lut_template(err_template) {
    variable_1 : total_output_net_capacitance;
    index_1("0, 1, 2, 3");
    variable_2 : constrained_pin_transition;
    index_2("0, 1, 2, 3");
  }

        em_max_toggle_rate(err_template) {
                values ("1, 2, 3, 4", "5, 6, 7, 8", \
                        "9, 10, 11, 12", "13, 14, 15, 16");
        }
```

Feedback

To fix the problem, change the template value of the *em_max_toggle_rate* attribute from err_template to ok_template.

The following is an example message:

```
Error: Line 126, The 2-dimensional 'em_max_toggle_rate' lookup table in
 the
        output-associated electromigration group cannot use
 'err_template' as its template. (LBDB-470)
```

**What Next**

Refer to the *Library Compiler User Guide* for more information about lookup tables. Change the library source file by referencing a different template in the lookup table description.

---

## LBDB-471

(warning) The current_type cannot be the same as the one on line\n \t%d which is under the same electromigration() group. Using the last\n \tem_max_toggle_rate group encountered.

**Description**

Library Compiler considers em_max_toggle_rate groups with same current_type value attribute as duplicate group. If both em_max_toggle_rate groups don't specify the current_type attribue, they are also considered duplicate. Library Compiler ignores all duplicate em_max_toggle_rate groups except the last one encountered during the compilation. The compiled database contains the last one only.

The following example shows an instance where this message occurs:

```
pin(Q){
       ...
       electromigration() {
         em_max_toggle_rate(em_2d) {
                 current_type : average ;
                 values ("1, 2, 3, 4", "5, 6, 7, 8", \
                         "9, 10, 11, 12", "13, 14, 15, 16");
         }
         em_max_toggle_rate(em_2d) {
                 current_type : average ;
                 values ("21, 22, 23, 24", "25, 26, 27, 28", \
                         "29, 30, 31, 32", "33, 34, 35, 36");
         }
          related_pin : "A";
       }
  }
```

To fix the problem, remove one of the em_max_toggle_rate tables.

Feedback

```
pin(Q){
      ...
      electromigration() {
        em_max_toggle_rate(em_2d) {
                current_type : average ;
                values ("1, 2, 3, 4", "5, 6, 7, 8", \
                        "9, 10, 11, 12", "13, 14, 15, 16");
        }
        related_pin : "A";
      }
  }
```

The following is an example message:

```
Error: Line 11, The current_type cannot be the same as the one on line
      6 which is under the same electromigration() group.  Using the last
      em_max_toggle_rate group encountered. (LBDB-471)
```

### What Next

Change the library source file by removing any extra em_max_toggle_rate tables within the electromigration group.

---

## LBDB-472

(warning) The timing type of the timing arc from '%s' to\n \t'%s' is changed from '%s' to\n \t'%s'.

### Description

The functional relationships between the pin and the related pin(s) dictate that this timing arc is non-rising or non-falling. This is due to the input sharing between function, three_state, and x_function.

The following example shows an instance where this message occurs:

```
pin(io){
      direction : output;
      function : "!ap & !an";
      three_state : "ap & !an";
      x_function : "!ap & an";

      timing() {
        related_pin : "ap an";
        timing_type : three_state_disable;
        ...
      }
      ...
  }
```

To fix, change functions or change timing_types.

Feedback

```
pin(io){
      ...
      timing() {
        related_pin : "ap an";
        timing_type : three_state_disable_rise;
        ...
      }
      ...
  }
```

The following is an example message:

```
Warning: Line 9, The timing type of the timing arc from 'ap' to
        'io' is changed from 'three_state_disable' to
        'three_state_disable_rise'. (LBDB-472)
```

**What Next**

Change the library source file so that expected timing types are entered. In case of an error, change the functional description.

---

## LBDB-473

(error) The timing label '%s' is already used for \n \tanother timing arc in the same cell.

**Description**

You receive this error message when the given timing label is invalid because it was already used by another timing arc in the cell. Duplicate timing labels are not allowed within a cell.

**What Next**

Edit the timing label list so that there is not a label in the list that is used in another timing group in the cell.

---

## LBDB-474

(error) The timing group has incorrect number of labels.

**Description**

You receive this error message when the timing group at the specified line has either too many or too few specified labels.

**What Next**

Edit the timing label list in your library so that the timing group has the right number of labels specified.

Feedback

## LBDB-475

(error) The port '%s' is missing the attribute '%s'.

### Description

You receive this error message when the *rise_capacitance* attribute or the *fall_capacitance* attribute for a port is missing and the other attribute is specified. Both or neither of the attributes must be specified.

### What Next

Specify the missing attribute or remove the specified attribute.

## LBDB-476

(warning) The port '%s' does not have the attribute '%s' specified. The value %f will be assigned to the attribute.

### Description

You receive this error message when the *rise_capacitance* and *fall_capacitance* attributes are specified for a port, and the *capacitance* attribute is not. The maximum value between *rise_capacitance* and *fall_capacitance* will be assigned to the attrribute.

### What Next

If you accept the value assigned to the attribute referenced in the error message, no action is required on your part. If not, assign a value to the attribute.

## LBDB-477

(error) The table %s has invalid intermediate_values \n \tentry.

### Description

This message indicates that a intermediate_values table contain invalid values. The values have to be between zero and the corresponding values from the values table attribute.

### What Next

Edit the intermediate_values table so that it no longer contains invalid values.

## LBDB-478

(error) The intermediate_values cannot be used with \n \tcurrent setting of output delay threshold voltages.

Feedback

### Description

This message indicates that the output delay threshold voltages are not valid for the intermediate_values table. The intermediate_values table values are used to specify the values from first slew point to the output delay point. The assumption is that the output delay point lies between the two slew points. This error message is issued when Library Compiler detects that the output delay point does not lie between the two slew points and intermediate_values table was used.

### What Next

Stop using the intermediate_values table or edit the library-level attributes for threshold voltages (for delay and slew).

---

## LBDB-479

(error) when '%s' is specified '%s' has to be \n \t given as well.

### Description

This message indicates that a rise net delay table was given without a fall net delay table (or vice versa). Both tables are expected by the wire delay estimator.

### What Next

Edit the library source file to add the missing table or delete the incomplete table for the file to compile.

---

## LBDB-480

(error) Missing a timing arc of timing_type 'three_state_disable' between the '%s' and '%s' pins in the '%s' cell.

### Description

To describe the transition from 0->Z or 1->Z, use the timing arc with a timing_type of *three_state_disable* for the pin with a *three_state* attribute.

The following example shows an instance where this message occurs:

```
cell(BTS4)  {
        area : 3.0;
    pin(Z)  {
      direction : output;
      function : "A";
      three_state : "E";
      timing()  {
        intrinsic_rise : 1.0;
        intrinsic_fall : 1.0;
```

Feedback

```
        rise_resistance : 0.1;
        fall_resistance : 0.1;
        related_pin : "A";
      }
      timing()  {
        intrinsic_rise : 1.0;
        intrinsic_fall : 1.0;
        rise_resistance : 0.1;
        fall_resistance : 0.1;
        related_pin : "E";
      }
    }
    pin(A)  {
      direction : input;
        capacitance : 1.0;
    }
    pin(E)  {
      direction : input;
        capacitance : 1.0;
    }
  }
```

The following is an example message:

```
Error: Line 110, Missing a timing arc of timing_type
 'three_state_disable'
        between 'E' and 'Z' pins in the 'lbfb35' cell. (LBDB-480)
```

**What Next**

Add the missing timing group between the two pins. To change the message to a warning, set the environment variable *timing_check_all_errors* to 0.

---

## LBDB-481

(warning) Port %s has multiple default timing arcs.\n \tOnly one arc is used as the default arc.

**Description**

You receive this message because a port has multiple default timing arcs in the case of State Dependent Timing. This is probably due to two conditions: First, a timing arc with no WHEN statement has been given as a default arc. Second, one timing arc with a condition has been specified as a default timing arc by having its *default_timing* attribute set for the arc.

**What Next**

Examine the library source file, and remove all but one default timing arc.

Feedback

## LBDB-482

(error) Port %s has multiple timing \n \tarcs with default_timing attribute set.

### Description

You receive this message because a port has multiple timing arcs with the *default_timing* attribute set. A port with State Dependent Timing should have only one timing arc. A library with this error does not compile.

### What Next

Examine the library source file, and reset all but one timing arc of the *default_timing* attribute for the port.

## LBDB-483

(error) Cannot find include file '%s',%s.

### Description

You receive this message because one of the following: The *include()* attribute has been used to include a file but the file could not be located in the search path.

The file exists but it also includes another library file.

### What Next

Add the missing include file or do not use "nested" include file.

## LBDB-484

(error) No variables statement is specified in the '%s' poly_template.

### Description

You receive this message because you did not define a variables statement in the specified poly_template group. Library Compiler expects you to define one variables statement there.

The following example shows an instance where this message occurs: The following example shows a poly_template named *p1* that is missing a variables statement:

```
poly_template(p1) {
  variable_1_range("1,2");
  variable_2_range("3,4");
}
```

Feedback

To correct the problem in the example above, insert the following statement in the *p1* poly_template group:

```
variables("temperature, total_output_net_capacitance");
```

The following is an example message:

```
Error: Line 191, No variables statement is specified in the 'p1'
 poly_template. (LBDB-484)
```

**What Next**

Define a variables statement for the specified poly_template.

---

## LBDB-485

(error) Incompatible variable '%s' is used in the '%s' \n \tpoly_template.

**Description**

You receive this message because you defined an invalid variable in the variables statement in the poly_template group.

The following example shows an instance where this message occurs: In the following example, the *whatever* variable value is not valid in the *p1* poly_template.

```
poly_template(p1) {
  variables("whatever, total_output_net_capacitance");
  variable_1_range("1,2");
  variable_2_range("3,4");
}
```

To correct the problem in the example above, insert the following statement in the *p1* poly_template group:

```
variables("temperature, total_output_net_capacitance");
```

The following is an example message: Error: Line 191, Incompatible variable 'whatever' is used in the 'p1' poly_template. (LBDB-485)

**What Next**

Define a variable name for the specified poly_template.

---

## LBDB-486

(warning) Variable range outside variable \n \tdimension is defined in '%s' poly_template.

Feedback

### Description

You receive this message because you have defined a variable range outside the dimension of the poly_template.

The following example shows an instance where this message occurs: The following example shows a *variable_2_range* variable value that cannot be used in the *p1* poly_template.

```
poly_template(p1) {
  variables("total_output_net_capacitance");
  variable_1_range("1,2");
  variable_2_range("3,4");
}
```

Correct it by removing the *variable_2_range* statement.

The following is an example message: Warning: Line 191, Variable range outside variable dimension is defined in 'p1' poly_template. (LBDB-486)

### What Next

Remove the statement for the variable range that is not in use.

---

## LBDB-487

(error) No range information defined for variable_%d in\n\ \t'%s' poly_template.

### Description

You receive this message because you have not defined the range for one of the variables used in the poly_template.

The following example shows an instance where this message occurs: The following example shows a *p1* poly_template that is missing the *variable_1_range* statement.

```
poly_template(p1) {
  variables("temperature, total_output_net_capacitance");
  variable_2_range("3,4");
}
```

Correct it by adding the following statement in the poly_template group:

```
variable_1_range("1,2");
```

The following is an example message: Error: Line 191, No range information defined for variable_1 in 'p1' poly_template. (LBDB-487)

### What Next

Add the *variable_n_range* statement for the specified variable.

Feedback

## LBDB-488

(error) No variables statement specified in '%s' \n \tdomain of '%s' poly_template.

### Description

You receive this message because you have not defined a variables statement in this domain of the poly_template group.

The following example shows an instance where this message occurs: The following example shows a domain named *d1* in a poly_template named *p1* that is missing the variables statement:

```
poly_template(p1) {
  ......
  domain (d1) {
    variable_1_range("1,2");
    variable_2_range("3,4");
  }
}
```

Correct it by adding the following statement in the poly_template group:

```
variables("temperature, total_output_net_capacitance");
```

The following is an example message:

```
Error: Line 191, No variables statement specified in 'd1' domain of 'p1'
 poly_template. (LBDB-488)
```

### What Next

Define a variables statement for this domain in the poly_template.

## LBDB-489

(error) Incompatible variable '%s' used '%s' domain \n \tof '%s' poly_template.

### Description

You receive this message because you defined an invalid variable in the variables statement of this domain in the poly_template.

The following example shows an instance where this message occurs: The following example shows an invalid variable named *whatever* in the *p1* poly_template.

```
poly_template(p1) {
  ......
  domain(d1) {
    variables("whatever, total_output_net_capacitance");
    variable_1_range("1,2");
```

Feedback

```
    variable_2_range("3,4");
  }
}
```

Correct it by adding the following statement in the domain group:

```
variables("temperature, total_output_net_capacitance");
```

The following is an example message:

```
Error: Line 191, Incompatible variable 'whatever' used in 'd1' domain of
 'p1' poly_template. (LBDB-489)
```

**What Next**

Examine the name of the specified variable to determine the cause.

---

## LBDB-490

(warning) Variable range outside variable \n \tdimension is defined in '%s' domain of '%s' poly_template.

**Description**

You receive this message because you have defined a variable range outside the dimension of the specified domain of the poly_template.

The following example shows an instance where this message occurs: The following example shows a *variable_2_range* statement that is useless in the *d1* domain of the *p1* poly_template.

```
poly_template(p1) {
  ......
  domain(d1) {
    variables("total_output_net_capacitance");
    variable_1_range("1,2");
    variable_2_range("3,4");
  }
}
```

Correct it by removing the *variable_2_range* statement.

The following is an example message: Warning: Line 191, Variable range outside variable dimension is defined in 'd1' domain of 'p1' poly_template. (LBDB-490)

**What Next**

Remove the variable range statement not in use.

Feedback

## LBDB-491

(error) No range information defined for variable_%d in\n \t'%s' domain of '%s' poly_template.

### Description

You receive this message because you have not defined a range for the specified variable used in this domain of the poly_template.

The following example shows an instance where this message occurs: The following example shows the *d1* domain of the *p1* poly_template with the *variable_1_range* statement missing.

```
poly_template(p1) {
  ......
  domain(d1) {
    variables("temperature, total_output_net_capacitance");
    variable_2_range("3,4");
  }
}
```

Correct it by adding the following statement:

```
variable_1_range("1,2");
```

The following is an example message: Error: Line 191, No range information defined for variable_1 in 'd1' domain of 'p1' poly_template. (LBDB-491)

### What Next

Add a *variable_n_range* statement (substituting the specified variable number for *n*).

## LBDB-492

(error) Power rail mapping information missing in\n \t'%s' poly_template.

### Description

You receive this message because you have not defined the power rail mapping information for the *voltage2* variable in the poly_template.

The following example shows an instance where this message occurs: The following example shows the *p1* poly_template with the mapping statement missing.

```
poly_template(p1) {
  variables("temperature, voltage2, total_output_net_capacitance");
  ......
}
```

Feedback

Correct it by adding the following statement:

```
mapping("voltage2, VDD");
```

The following is an example message: Error: Line 191, Power rail mapping information missing in 'p1' poly_template. (LBDB-492)

**What Next**

Add the mapping statement in the poly_template.

---

## LBDB-493

(error) Power rail mapping information missing in\n \t'%s' domain of '%s' poly_template.

**Description**

You receive this message because you have not defined the power rail mapping information for the voltage2 variable in the specified domain of the specified poly_template.

The following example shows an instance where this message occurs: The following example shows the *d1* domain of the *p1* poly_template with the mapping statement missing.

```
poly_template(p1) {
  ......
  domain(d1) {
    variables("temperature, voltage2, total_output_net_capacitance");
    ......
  }
}
```

Correct it by adding the following statement:

```
mapping("voltage2, VDD");
```

The following is an example message: Error: Line 191, Power rail mapping information missing in 'd1' domain of 'p1' poly_template. (LBDB-493)

**What Next**

Add the mapping statement in this domain of the poly_template.

---

## LBDB-495

(error) Invalid number of dimensions in '%s' \n \t poly_template.

Feedback

### Description

You receive this message because the number of variables you have defined in the poly_template is invalid.

The following example shows an instance where this message occurs: The following example shows a variables statement that defines the wrong number of variables in the *p1* poly_template.

```
poly_template(p1) {
  variables("");
  ......
}
```

Correct it by changing the variables statement in the poly_template group to the following:

```
variables("temperature, total_output_net_capacitance");
```

The following is an example message: Error: Line 191, Invalid number of dimensions in 'p1' poly_template. (LBDB-495)

### What Next

Examine the variables statement to determine why the number of variables is invalid.

---

## LBDB-496

(error) Invalid number of dimensions in '%s' \n \t domain of '%s' poly_template.

### Description

You receive this message because the number of variables you have defined in the specified domain of the specified poly_template is invalid.

The following example shows an instance where this message occurs: The following example shows a variables statement that defines the wrong number of variables in the *d1* domain of the *p1* poly_template.

```
poly_template(p1) {
  ......
  domain (d1) {
    variables("");
    ......
  }
}
```

Correct it by changing the variables statement in this domain of the poly_template group to the following:

```
variables("temperature, total_output_net_capacitance");
```

Feedback

The following is an example message: Error: Line 191, Invalid number of dimensions in 'd1' domain of 'p1' poly_template. (LBDB-496)

**What Next**

Examine the variables statement to determine why the number of variables is invalid.

---

## LBDB-497

(error) Incomplete range of variable in '%s' \n \t poly_template.

**Description**

You receive this message because the variable statement of the specified poly_template has an incomplete range.

The following example shows an instance where this message occurs: The following example shows the *p1* poly_template with an incomplete *variable_1_range* statement.

```
poly_template(p1) {
  variables("temperature, total_output_net_capacitance");
  variable_1_range("1");
  ......
}
```

Correct it by changing the *variable_1_range* statement in the poly_template group to the following:

```
variable_1_range("1,2");
```

The following is an example message: Error: Line 191, Incomplete range of variable in 'p1' poly_template. (LBDB-497)

**What Next**

Examine all *variable_n_range* statements (substituting each existing variable number for *n*) of the specified poly_template to determine those that have an incomplete range.

---

## LBDB-498

(error) Incomplete range of variable in '%s' \n \t domain of '%s' poly_template.

**Description**

You receive this message because the variable statement in the specified domain of the specified poly_template has an incomplete range.

The following example shows an instance where this message occurs: The following example shows *d1* domain of the *p1* poly_template with an incomplete *variable_1_range* statement.

Feedback

```
poly_template(p1) {
  ......
  domain (d1) {
    variables("temperature, total_output_net_capacitance");
    variable_1_range("1");
    ......
  }
}
```

Correct it by changing the *variable_1_range* statement in the *d1* domain of the poly_template group to the following:

```
variable_1_range("1,2");
```

The following is an example message: Error: Line 191, Incomplete range of variable in 'd1' domain of 'p1' poly_template. (LBDB-498)

**What Next**

Examine all *variable_n_range* statements (substituting each existing variable number for *n*) in the specified domain of the specified poly_template to determine those that have an incomplete range.

---

## LBDB-499

(error) No orders information defined for the polynomial.

**Description**

You receive this message because you have not defined the *orders* statement for the polynomial.

The following example shows an instance where this message occurs: The following example shows a poly timing group with a missing *orders* statement.

```
rise_constraint(constraint) {
    coefs("1.2095, 0.2281, 0.0225, -0.0001, 0.0678, -0.0002, 0.0000,
 0.0000");
  }
```

Correct it by adding the following statement before the *coefs* statement:

```
orders("1,1,1");
```

The following is an example message: Error: Line 191, No orders information defined for the polynomial. (LBDB-499)

**What Next**

Add the *orders* statement before the *coefs* statement.

Feedback

## LBDB-500

(error) No orders information defined for '%s' \n \t domain of the polynomial.

### Description

You receive this message because you have not defined the *orders* statement for this domain of the polynomial.

The following example shows an instance where this message occurs: The following example shows the *d1* domain of a poly timing group with a missing *orders* statement.

```
rise_constraint(constraint) {
    ......
    domain (d1) {
      coefs("1.2095, 0.2281, 0.0225, -0.0001, 0.0678, -0.0002, 0.0000,
 0.0000");
    }
  }
```

Correct it by adding the following statement before the *coefs* statement:

```
orders("1,1,1");
```

The following is an example message: Error: Line 191, No orders information defined for 'd1' domain of the polynomial. (LBDB-500)

### What Next

Add the *orders* statement before the *coefs* statement.

## LBDB-501

(error) No coefs information defined for the polynomial.

### Description

You receive this message because you have not defined the *coefs* statement for the polynomial.

The following example shows an instance where this message occurs: The following example shows a poly timing group with a missing *coefs* statement.

```
rise_constraint(constraint) {
    orders("1,1,1");
  }
```

Correct it by adding the following statement after the *orders* statement:

```
coefs("1.2095, 0.2281, 0.0225, -0.0001, 0.0678, -0.0002, 0.0000,
 0.0000");
```

Feedback

The following is an example message: Error: Line 191, No coefs information defined for the polynomial. (LBDB-500)

**What Next**

Add the *coefs* statement after the *orders* statement.

---

## LBDB-502

(error) No coefs information defined for '%s' \n \t domain of the polynomial.

**Description**

You receive this message because you have not defined the *coefs* statement for the specified domain of the polynomial.

The following example shows an instance where this message occurs: The following example shows the *d1* domain of a poly timing group with a missing *coefs* statement.

```
rise_constraint(constraint) {
    ......
    domain (d1) {
      orders("1,1,1");
    }
  }
```

Correct it by adding the following after the *orders* statement:

```
coefs("1.2095, 0.2281, 0.0225, -0.0001, 0.0678, -0.0002, 0.0000,
 0.0000");
```

The following is an example message: Error: Line 191, No coefs information defined for 'd1' domain of the polynomial. (LBDB-502)

**What Next**

Add the *coefs* statement for the specified domain after the *orders* statement.

---

## LBDB-503

(Error) The low range setting is equal to or greater than the high range setting in %s.

**Description**

This warning message tells you that the values specified for the variable range setting statement are incorrect. Only temperature, voltages, and generic parameters can have the same low range and high range.

The following example shows an instance where this message occurs: The following example shows the p1 poly_template with an incomplete variable_1_range statement.

Feedback

```
poly_template(p1) {
  variables("temperature, total_output_net_capacitance");
  variable_1_range("100,1");
  ......
}
```

Correct it by changing the variable_1_range statement in the poly_template group to the following:

```
variable_1_range("1,100");
```

The following is an example message: Error: Line 191, The low range setting is equal to or greater than the high range setting in variable_1_range. (LBDB-503)

---

## LBDB-504

(warning) Range setting in domain at variable_%d_range conflicts with template setting. Use the template setting instead.

### Description

This warning tells you that the variable range setting statement has a range that conflicts with the range set previously in the template.

The following example shows an instance where this message occurs: The following example shows the p1 poly_template with an incorrect variable_1_range statement in the domain.

```
poly_template(p1) {
  variables("temperature, total_output_net_capacitance");
  variable_1_range("1,50");
  domain (d1) {
    variable_1_range("2,100");
  }
  ......
}
```

Correct it by changing the variable_1_range statement in the domain to the following:

```
variable_1_range("2,40");
```

The value 40 can be any number less than or equal to 50.

The following is an example message: Warning: Line 191, Range setting conflicts with template setting. (LBDB-504)

### What Next

Examine all variable_*n*_range statements. Substitute each existing variable number for *n* in the specified poly_template to determine the statements that conflict with each other.

Feedback

## LBDB-505

(error) Variable %s is not a voltage. Can only map voltages to a power rail.

### Description

This error message tells you that the first variable in mapping is not a voltage. Only voltages can be mapped to a power rail.

The following example shows an instance where this message occurs: The following example shows the p1 poly template with an incorrect mapping statement.

```
poly_template(p1) {
  variables("temperature, voltage,total_output_net_capacitance");
  variable_1_range("20,150");
  variable_2_range("1.2,3.2");
  variable_3_range("10,30");
  mapping(temperature,VDD1);
  ......
}
```

Fix the mapping statement by changing the variable_1_range statement in the domain to the following:

```
mapping(voltage,VDD1);
```

The following is an example message: Error: Line 191, Variable temperature is not a voltage (LBDB-505).

### What Next

Check the variables in the mapping and replace the first variable with a voltage.

## LBDB-506

(error) Variable %s is not defined in the %s.

### Description

This error message tells you that the first variable in mapping is not defined in the template.

The following example shows an instance where this message occurs: The following example shows the p1 poly_template with an incorrect mapping statement.

```
poly_template(p1) {
  variables("temperature, voltage,total_output_net_capacitance");
  variable_1_range("20,150");
  variable_2_range("1.2,3.2");
  variable_3_range("10,30");
  mapping(voltage1,VDD1);
```

Feedback

```
  ......
}
```

Correct it by changing the variable_1_range statement in the domain to the following:

```
mapping(voltage,VDD1);
```

The following is an example message: Error: Line 191, Variable voltage1 is not defined in the template (LBDB-506).

**What Next**

Define the variable in the template, or change the variable_1_range statement in the domain.

---

## LBDB-507

(error) Attempting to map to %s, which is defined in power_supply group. However, power_supply group is missing.

**Description**

This error message tells you that you have not defined a power_supply group before using its information.

The following example shows an instance where this message occurs: The following example shows the p1 poly_template with the mapping statement missing.

```
poly_template(p1) {
  variables("temperature, voltage2, total_output_net_capacitance");
  mapping("voltage2, VDD");
  ......
}
```

Correct it by adding the following statement before the template:

```
power_supply() {
    default_power_rail : VDD ;
  }
```

The following is an example message: Error: Line 191, Power_supply group missing but trying to map to %s, which is supposed to be defined in a power_supply group. (LBDB-507)

**What Next**

Add a power_supply group before the templates.

Feedback

## LBDB-508

(error) The '%s' port has hyperbolic low groups. Only input pin or bidirectional pin can have hyperbolic low groups.

### Description

Library Compiler issues this error message if one hyperbolic low group is defined within an output pin.

The following example shows an instance where this message occurs:

```
pin(Q){
      direction : output;
      ...
      hyperbolic_noise_low() {
        ...
      }
  }
```

To correct the problem, remove one of the hyperbolic low groups.

```
pin(Q){
      direction : output;
      ...
  }
```

The following is an example message:

```
Error: Line 6, The 'Q' port has hyperbolic groups.
        Only input pin or bidirectional pin can have hyperbolic low
 groups. (LBDB-508)
```

### What Next

Change the library source file by removing the hyperbolic low groups within the output pin group.

## LBDB-509

(error) The '%s' port has hyperbolic high groups.\n \tOnly input pin or bi-directional pin can have hyperbolic high groups.

### Description

Library Compiler issues this error message if one hyperbolic high group is defined within a output pin.

Feedback

The following example shows an instance where this message occurs:

```
pin(Q){
      direction : output;
      ...
      hyperbolic_noise_high() {
      }
  }
```

To fix the problem, remove one of the hyperbolic high groups.

```
pin(Q){
      direction : output;
      ...
  }
```

The following is an example message:

```
Error: Line 6, The 'Q' port has hyperbolic groups.
        Only input pin or bi-directional pin can have hyperbolic high
 groups. (LBDB-509)
```

**What Next**

Change the library source file by removing the hyperbolic high groups within the output pin group.

---

## LBDB-510

(error) The '%s' port has hyperbolic above_high groups.\n \tOnly input pin or bi-directional pin can have hyperbolic above_high groups.

**Description**

Library Compiler issues this error message if one hyperbolic above_high group is defined within a output pin.

The following example shows an instance where this message occurs:

```
pin(Q){
      direction : output;
      ...
      hyperbolic_noise_above_high() {
      }
  }
```

To fix the problem, remove one of the hyperbolic above_high groups.

```
pin(Q){
      direction : output;
      ...
  }
```

Feedback

The following is an example message:

```
Error: Line 6, The 'Q' port has hyperbolic groups.
        Only input pin or bi-directional pin can have hyperbolic
 above_high groups. (LBDB-510)
```

### What Next

Change the library source file by removing the hyperbolic above_high groups within the output pin group.

---

## LBDB-511

(error) The '%s' port has hyperbolic below_low groups.\n \tOnly input pin or bi-directional pin can have hyperbolic below_low groups.

### Description

Library Compiler issues this error message if one hyperbolic below_low group is defined within a output pin.

The following example shows an instance where this message occurs:

```
pin(Q){
      direction : output;
      ...
      hyperbolic_noise_below_low() {
      }
  }
```

To fix the problem, remove one of the hyperbolic below_low groups.

```
pin(Q){
      direction : output;
      ...
  }
```

The following is an example message:

```
Error: Line 6, The 'Q' port has hyperbolic groups.
        Only input pin or bi-directional pin can have hyperbolic
 below_low groups. (LBDB-511)
```

### What Next

Change the library source file by removing the hyperbolic below_low groups within the output pin group.

Feedback

## LBDB-512

(error) The '%s' port has noise hyperbolic groups.\n \tOnly one input/bi-directional port can have noise hyperbolic groups.

### Description

Library Compiler issues this error message if more than one noise hyperbolic is defined within a timing.

The following example shows an instance where this message occurs:

```
pin(Q){
      direction : output;
      ...
      noise_hyperbolic_low() {
        ...
      }
  }
```

To fix the problem, remove one of the noise hyperbolic groups.

```
pin(Q){
      direction : output;
      ...
  }
```

The following is an example message:

```
Error: Line 6, The 'Q' port has noise hyperbolic groups.
        Only input/bi-directional ports can have noise hyperbolic groups.
 (LBDB-512)
```

### What Next

Change the library source file by removing any extra noise hyperbolic groups within the pin group.

## LBDB-513

(error) The '%s' port has repeated noise hyperbolic groups. Only one such group is allowed.

### Description

Library Compiler issues this error message if more than one of the same noise hyperbolic group is defined within a timing.

Feedback

The following example shows an instance where this message occurs:

```
pin(Q){
      ...
      noise hyperbolic_low() {
        ...
      }
      noise hyperbolic_low() {
        ...
      }
  }
```

To correct the problem, remove extra noise hyperbolic groups.

```
pin(Q){
      ...
      noise hyperbolic_low() {
        ...
      }
  }
```

The following is an example message:

```
Error: Line 6, The 'Q' port has repeated noise hyperbolic groups.
        Only one such group is allowed. (LBDB-513)
```

### What Next

Change the library source file by removing any extra noise hyperbolic groups within the timing group.

---

## LBDB-514

(error) The constraint timing arc has noise immunity groups.\n \tOnly non-constraint timing arc can specify such info.

### Description

Library Compiler issues this error message if noise immunity info is defined within a constraint timing arc.

The following example shows an instance where this message occurs:

```
pin(Q){
      direction : output;
      ...
      timing() {
         timing_type: setup_rising;
         ...
         noise_immunity_low() {
           ...
         }
```

Feedback

```
        }
    }
```

To fix the problem, remove the noise immunity tables/polys.

```
pin(Q){
       direction : output;
       ...
       timing() {
          timing_type: setup_rising;
          ...
       }
    }
```

The following is an example message:

```
Error: Line 6, The timing arc has noise immunity groups.
         Only non-constraint timing arc can specify such info. (LBDB-514)
```

**What Next**

Remove the noise immunity tables/polys.

---

## LBDB-515

(error) The '%s' port has repeated noise immunity groups inside timing group. Only one such group is allowed.

**Description**

Library Compiler issues this error message if more than one noise immunity is defined within a timing group.

The following example shows an instance where this message occurs:

```
pin(Q){
       ...
       timing() {
          ...
          noise_immunity_low() {
            ...
          }
          noise_immunity_low() {
            ...
          }
       }
    }
```

To correct the problem, remove extra noise immunity groups.

```
pin(Q){
       ...
```

Feedback

```
      timing() {
      ...
        noise_immunity_low() {
        ...
        }
      }
  }
```

The following is an example message:

```
Error: Line 6, The 'Q' port has repeated noise immunity groups inside
 timing group.
        Only one such group is allowed. (LBDB-515)
```

**What Next**

Change the library source file by removing any extra noise immunity tables or polys within the timing group.

---

## LBDB-516

(error) The '%s' group uses\n \ttemplate '%s' which does not contain 'iv_output_voltage' as its variable.

**Description**

The iv characteristics group must use a 1-d iv characteristics template with *iv_output_voltage* as its variable.

The following example shows an instance where this message occurs:

```
iv_lut_template(ok_temp){
        variable_1 : iv_output_voltage;
        index_1("0.5,1.0,1.5,2.0,2.5");
    }
    iv_lut_template(err_temp){
        variable_1 : input_transition_time;
        index_1("0.5,1.0,1.5,2.0,2.5");
    }
    ...
         steady_state_current_high(err_temp) {
                values("1.385,2.554,3.722,4.891,6.059");;
        }
```

To fix the problem, change the template value of the *steady_state_current_high* attribute from *err_temp* to *ok_temp*.

The following is an example message:

```
Error: Line 126, The 'steady_state_current_high' group in the timing
       group cannot use 'err_temp' as its template. (LBDB-516)
```

Feedback

### What Next

Change the library source file by including 'iv_output_voltage' in the template.

---

## LBDB-517

(error) The '%s' lookup table in the\n \ttiming group cannot use '%s' as its template.

### Description

The noise characteristics group must use a 2-d noise characteristics template with *input_noise_width, total_output_net_capacitance* as its variables.

The following example shows an instance where this message occurs:

```
noise_lut_template(ok_temp){
        variable_1 : input_noise_width;
        variable_2 : total_output_net_capacitance;
        index_2("1.0,2.0,4.0,6.0,10.0");
        index_2("0.5,1.0,1.5,2.0,2.5");
    }
    noise_lut_template(err_temp){
        variable_1 : input_net_transition;
        variable_2 : total_output_net_capacitance;
        index_2("1.0,2.0,4.0,6.0,10.0");
        index_2("0.5,1.0,1.5,2.0,2.5");
    }
    ...
         noise_immunity_high(err_temp) {
                values("0.733,1.073,1.412,1.752,2.092",\
                           "0.873,1.214,1.554,1.894,2.234",\
                           "1.095,1.442,1.787,2.132,2.477",\
                           "1.298,1.648,1.996,2.343,2.691",\
                           "1.703,2.060,2.414,2.766,3.119");
        }
```

To fix the problem, change the template value of the *noise_immunity_high* attribute from *err_temp* to *ok_temp*.

The following is an example message:

```
Error: Line 126, The 'noise_immunity_high' lookup table in the timing
       group cannot use 'err_temp' as its template. (LBDB-517)
```

### What Next

Change the library source file by including 'input_noise_width' and 'total_output_net_capacitance' in the template.

Feedback

## LBDB-519

(error) The timing arc has steady_state_resistance_low/high/above_high/below_low attribute specified.\n \tOnly combo, enable, disable, rising_edge, falling_edge, clear and preset timing arc can specify such attributes.

### Description

Library Compiler issues this error message if any steady_state_resistance_low/high/above_high/below_low attribute is defined within a timing arc which is not combo, disable or enable.

The following example shows an instance where this message occurs:

```
pin(Q){
       ...
       timing() {
         timing_type : setup_rising;
         ...
         steady_state_resistance_high() {
            ...
         }
       }
  }
```

To fix the problem, remove the steady_state_resistance attributes.

```
pin(Q){
       ...
       timing() {
         timing_type : setup_rising;
         ...
       }
  }
```

The following is an example message:

```
Error: Line 6, The timing arc has
 steady_state_resistance_low/high/above_high/below_low attribute
 specified.
        Only combo, enable, disable, rising_edge, falling_edge, clear and
 preset timing arc can specify such attributes. (LBDB-519)
```

### What Next

Change the library source file by removing such attributes inside the timing() group.

## LBDB-520

(error) The timing arc has steady_state_current_high/low specified.\n \tOnly combo, rising_edge, falling_edge, preset and clear timing arcs can specify such info.

Feedback

### Description

Library Compiler issues this error message if any steady_state_current_high/low group is defined within a timing group which is not combo, disable or enable.

The following example shows an instance where this message occurs:

```
pin(Q){
      ...
      timing() {
        timing_type : setup_rising;
        ...
        steady_state_current_high(iv1x5) {
           ...
        }
      }
  }
```

To fix the problem, remove the steady_state_current_high group.

```
pin(Q){
      ...
      timing() {
        timing_type : setup_rising;
        ...
      }
  }
```

The following is an example message:

```
Error: Line 6, The timing arc has steady_state_current_high/low
 specified.
       Only combo, rising_edge, falling_edge, preset and clear timing
 arcs can specify such info. (LBDB-520)
```

### What Next

Change the library source file by removing such groups inside the timing() group.

---

## LBDB-521

(error) The timing arc has steady_state_current_tristate specified.\n \tOnly enable or disable timing arc can specify such info.

### Description

Library Compiler issues this error message if any steady_state_current_tristate group is defined within a timing group which is not disable or enable.

Feedback

The following example shows an instance where this message occurs:

```
pin(Q){
      ...
      timing() {
        timing_type : setup_rising;
        ...
        steady_state_current_tristate(iv1x5) {
           ...
        }
      }
  }
```

To fix the problem, remove the steady_state_current_tristate group.

```
pin(Q){
      ...
      timing() {
        timing_type : setup_rising;
        ...
      }
  }
```

The following is an example message:

```
Error: Line 6, The timing arc has steady_state_current_tristate
 specified.
        Only enable or disable timing arc can specify such info.
 (LBDB-521)
```

### What Next

Change the library source file by removing such groups inside the timing() group.

---

## LBDB-522

(error) The '%s' group in the\n \ttiming arc cannot use '%s' as its template.

### Description

The noise characteristics group must use a 3-d noise characteristics template with *input_noise_width, input_noise_height, total_output_net_capacitance* as its variables.

The following example shows an instance where this message occurs:

```
...
    propagation_lut_template(ok_temp){
        variable_1 : input_noise_width;
        variable_2 : input_noise_height;
        variable_3 : total_output_net_capacitance;
        index_1("1.0,2.0,4.0,6.0,10.0");
        index_2("0.5,1.0,1.5,2.0,2.5");
```

Feedback

```
        index_3("0.5,1.0,1.5,2.0,2.5");
    }
    noise_lut_template(err_temp){
        variable_1 : input_net_transition;
        variable_2 : input_noise_height;
        variable_3 : total_output_net_capacitance;
        index_1("1.0,2.0,4.0,6.0,10.0");
        index_2("0.5,1.0,1.5,2.0,2.5");
        index_3("0.5,1.0,1.5,2.0,2.5");
    }
    ...
         propagation_noise_width_high(err_temp) {
                values("0.733,1.073,1.412,1.752,2.092",\
                           "0.873,1.214,1.554,1.894,2.234",\
                           "1.095,1.442,1.787,2.132,2.477",\
                           "1.298,1.648,1.996,2.343,2.691",\
                           ...
                           "1.703,2.060,2.414,2.766,3.119");
        }
```

To fix the problem, change the template value of the *propagation_noise_width_high* attribute from *err_temp* to *ok_temp*.

The following is an example message:

```
Error: Line 126, The 'propagation_noise_width_high' group in the timing
       arc cannot use 'err_temp' as its template. (LBDB-522)
```

**What Next**

Change the library source file by including 'input_noise_width', 'input_noise_height' and 'total_output_net_capacitance' in the template.

---

## LBDB-523

(error) The related '%s' group is not specified for\n \tthe %s' group in the timing arc.

**Description**

The noise propagation width and height group must coexist for the same region(low, high, below_low, above_high).

The following example shows an instance where this message occurs:

```
...
        timing() {
                propagation_noise_width_high(ok_temp) {
                values("0.733,1.073,1.412,1.752,2.092",\
                           "0.873,1.214,1.554,1.894,2.234",\
                           "1.095,1.442,1.787,2.132,2.477",\
                           "1.298,1.648,1.996,2.343,2.691",\
                           ...
```

Feedback

```
                    "1.703,2.060,2.414,2.766,3.119");
                }
                propagation_noise_width_low(ok_temp) {
                values("0.733,1.073,1.412,1.752,2.092",\
                    "0.873,1.214,1.554,1.894,2.234",\
                    "1.095,1.442,1.787,2.132,2.477",\
                    "1.298,1.648,1.996,2.343,2.691",\
                    ...
                    "1.703,2.060,2.414,2.766,3.119");
                propagation_noise_height_low(ok_temp) {
                values("0.733,1.073,1.412,1.752,2.092",\
                    "0.873,1.214,1.554,1.894,2.234",\
                    "1.095,1.442,1.787,2.132,2.477",\
                    "1.298,1.648,1.996,2.343,2.691",\
                    ...
                    "1.703,2.060,2.414,2.766,3.119");
        }
        ...
```

To fix the problem, add the *propagation_noise_width_high* group in the timing group.

The following is an example message:

```
Error: Line 126, The related 'propagation_noise_height_high' group is not
 specified
       for the 'propagation_noise_width_high' group in the timing arc.
 (LBDB-523)
```

### What Next

Add the missiong propagation group in the timing group.

---

## LBDB-524

(error) Invalid variable '%s' is used in the '%s' \n \tpoly_template refered by %s group.

### Description

You receive this message because you defined an invalid variable in the variables statement in the poly_template group refered by leakage_power or pin_capacitance group. The valid variables can only be temperature voltage, and power_rails.

The following example shows an instance where this message occurs: In the following example, the *whatever* variable value is not valid in the *p1* poly_template.

```
poly_template(p1) {
  variables("temperature, total_output_net_capacitance");
  variable_1_range("1,2");
  variable_2_range("3,4");
}
```

Feedback

To correct the problem in the example above, insert the following statement in the *p1* poly_template group:

```
variables("temperature, voltage");
```

The following is an example message: Error: Line 191, Invalid variable 'total_output_net_capacitance' is used in the 'p1' poly_template refered by leakage_power group.(LBDB-524)

**What Next**

Define a variable name for the specified poly_template.

---

## LBDB-525

(error) Only polynomial format is supported for '%s' group.

**Description**

You receive this message because you defined an non-polynomial group, such as pin_capacitance() group, leakagage_power() group.

**What Next**

Please redefine these groups using polynomials.

---

## LBDB-526

(warning) The always on cell has no always on pin related to backup power or ground.

**Description**

If the cell is an always_on buffer/invertor cell, then backup power/ground is the preferred related pg pin.

This message is showed up when the cell has backup power/ground, but there is no singal pin related to backup pg pins.

The following example shows an instance where this message occurs:

```
cell(test) {
   always_on : TRUE;
   pg_pin(PWR) {
   voltage_name : VDD;
   pg_type : primary_power;
   }
   pg_pin(GND) {
   voltage_name : VSS;
   pg_type : primary_ground;
   }
```

Feedback

```
    pg_pin(GND1) {
    voltage_name : VSS1;
    pg_type : backup_ground;
    }
    pg_pin(VDDI) {
    voltage_name : VDDH;
    pg_type : backup_power;
    }
    ...
    pin(A) {
         direction : input;
         related_power_pin : PWR;
         related_ground_pin : GND;
       ...
    }
    pin(Z) {
         direction : output;
         function : A;
         related_power_pin : VDD;
         related_ground_pin : GND;
      ...
    }
    ...
}
```

Both input pin A and output pin Z should connect to backup PG.

The following is an example message:

```
Warning: Line 335, Cell "test", The always on cell has no always on pin
 related to backup power or ground. (LBDB-526)
```

### What Next

Check the library source file, and make the necessary correction.

---

## LBDB-527

(warning) is_clock_isolation_cell attribute is not supported. It's marked as dont_use, dont_touch.

### Description

Currently if an isolation cell is modeled using the is_clock_isolation_cell attribute, it can not be auto inferred by Galaxy design tools (e.g. Power Compiler). Thus the cell is being marked as dont_touch and dont_use during read_lib.

The following is an example message:

```
Warning: Line 1023, Cell 'A', is_clock_isolation_cell attribute is not
 supported. (LBDB-527)
```

Feedback

### What Next

Warning only. No action is required.

---

## LBDB-530

(error) The pin '%s' used in\n \t%s '%s' cannot specify the '%s' %s.

### Description

This error message occurs when the pin with "is_isolated" attribute is used in the expression of isolation_enable_condition.

The following example shows an instance where this message occurs: The following is an example of incorrect input that causes this error message:

```
cell (lbdb530) {
    ...
    pin (EN) {
      direction    :   input;
      capacitance :   1.0;
      fanout_load :   1.0;
      is_isolated :   true;
    }
    pin(Q0){
        direction   :   output;
        function : "I1";
        is_isolated :   true;
        isolation_enable_condition : "EN";

    }
    ...
  }
```

In this case, pin EN used in the attribute isolation_enable_condition of pin Q0 has been specified with is_isolated attribute. To fix the error, remove is_isolated attribute from pin EN.

The following is an example message:

```
Error: Line 179, Cell 'lbdb530', pin 'Q0', The pin 'EN' used in
        attribute 'isolation_enable_condition' cannot specify the
 'is_isolated' attribute. (LBDB-530)
```

### What Next

Check the library source file and remove is_isolated attribute from the pin, or don't use the pin in the expression of isolation_enable_condition.

Feedback

## LBDB-541

(error) The pin '%s' in the short list is not an 'inout' pin.

### Description

The port in "short" attribute must be direction "inout".

The following example shows an instance where this message occurs:

```
cell (c) {
  ...
  short(A, Y);
  pin (A) {
    direction : input ;
    ...
  }
  pin (Y) {
    direction : input ;
    ...
  }
}
```

The following is an example message:

```
Error: Line 145, Cell 'c', The pin 'A' in the short list is not an
 'inout' pin. (LBDB-541)
```

### What Next

Correct the port direction.

## LBDB-542

(error) The pin '%s' in the short list is missing pin capacitance attribute.

### Description

The port in "short" attribute must be specified with "capacitance" (or "rise_capacitance/ fall_capacitance") attribute.

The following example shows an instance where this message occurs:

```
cell (c) {
  ...
  short(A, Y);
  pin (A) {
    ...
  }
  pin (Y) {
    capacitance : 0.01 ;
    ...
```

Feedback

```
  }
}
```

The following is an example message:

```
Error: Line 145, Cell 'c', The pin 'A' in the short list is missing pin
 capacitance attribute. (LBDB-542)
```

**What Next**

Add the pin capacitance attribute on the port.

---

## LBDB-543

(error) Found an inconsistant pin capacitance for the shorted pin.

**Description**

The ports in same “short” attribute must be specified with identical "capacitance" (or “rise_capacitance/fall_capacitance”) attribute.

The following example shows an instance where this message occurs:

```
cell (c) {
  ...
  short(A, Y);
  pin (A) {
    capacitance : 0.01 ;
    ...
  }
  pin (Y) {
    capacitance : 0.02 ;
    ...
  }
}
```

The following is an example message:

```
Error: Line 145, Cell 'c', Pin 'Y', Found an inconsistant pin capacitance
 for the shorted pin. (LBDB-543)
```

**What Next**

Correct the pin capacitance attribute on the port.

---

## LBDB-544

(error) The "retention_equivalent_cell" attribute can not be specified in a retention cell.

Feedback

### Description

This message indicates that both attribute "retention_cell" and "retention_equivalent_cell" are specified on the cell. The "retention_equivalent_cell" is for a non-retention cell to point to its retention equivalent cell. So it can only be specified on a non retention cell.

The following example shows an instance where this message occurs:

```
cell (ret_cell) {

    retention_cell : test;
    retention_equivalent_cell : my_ret_cell;

    ...

  }
```

The following is an example message:

```
Error: Line 149, Cell 'ret_cell', The "retention_equivalent_cell"
 attribute can not be specified in a retention cell. (LBDB-544)
```

### What Next

Check the library source file, remove the retention_equivalent_cell attribute.

---

## LBDB-545

(error) The "retention_equivalent_pin" attribute can not be specified in cell without "retention_equivalent_cell" attribute.

### Description

This message indicates that the pin has attribute "retention_equivalent_pin", but no cell level attribute "retention_equivalent_cell" specified. "retention_equivalent_pin" can only be used together with "retention_equivalent_cell" to describe the pin name.

The following example shows an instance where this message occurs:

```
cell (cell) {

    /* retention_equivalent_cell : my_ret_cell; */

    pin(D) {
      retention_equivalent_pin : DD;
      ...
    }

    ...

  }
```

Feedback

The following is an example message:

```
Error: Line 149, Cell 'ret_cell', Pin 'D', The "retention_equivalent_pin"
 attribute can not be specified in cell without
 "retention_equivalent_cell" attribute. (LBDB-545)
```

**What Next**

Check the library source file, add the retention_equivalent_cell attribute.

---

## LBDB-546

(warning) The "retention_equivalent_cell" can not be specified in a non-black-box cell.

**Description**

This message indicated that cell is not a black-box cell, but is specified with "retention_equivalent_cell" attribute. The "retention_equivalent_cell" is desired for complex cell, so the cell should be black-box.

**What Next**

Check the library source file, remove the retention_equivalent_cell attribute.

---

## LBDB-547

(warning) The cell referenced by "retention_equivalent_cell" attribute must be a retention cell.

**Description**

This message indicates that the cell referenced by "retention_equivalent_cell" is not a retention cell, it is missing the "retention_cell" attribute, or the attribute is pointing to a wrong cell.

The following example shows an instance where this message occurs:

```
cell (cell) {

    retention_equivalent_cell : my_ret_cell;

    ...

  }

  cell (my_ret_cell) {

    /* retention_cell : test; */

    ...
```

Feedback

```
}
```

The following is an example message:

```
Warning: Line 149, Cell 'my_ret_cell', The cell referenced by
 "retention_equivalent_cell" attribute must be a retention cell.
 (LBDB-547)
```

### What Next

Check the library source file, add the retention_cell attribute, or correct the retention_equivalent_cell attribute.

---

## LBDB-548

(warning) The cell referenced by "retention_equivalent_cell" attribute can't have dont_use attribute.

### Description

This message indicates that the cell referenced by "retention_equivalent_cell" is a dont_use cell. This is wrong for optimization tools.

The following example shows an instance where this message occurs:

```
cell (cell) {

    retention_equivalent_cell : my_ret_cell;

    ...

  }

  cell (my_ret_cell) {

    dont_use: true;

    ...

  }
```

The following is an example message:

```
Error: Line 149, Cell 'my_ret_cell', The cell referenced by
 "retention_equivalent_cell" attribute can't have dont_use attribute.
 (LBDB-548)
```

### What Next

Check the library source file, remove the dont_use attribute, or correct the retention_equivalent_cell attribute.

Feedback

## LBDB-549

(warning) The cell referenced by "retention_equivalent_cell" attribute has inconsistent bias pg_pin setting with non-retention cell.

### Description

If non-retention cell has bias pg_pin, its retention equivalent cell should also have bias pg_pin; If non-retention cell has no bias pg_pin, its retention equivalent cell should also have no bias pg_pin; This message indicates that the non-retention cell and its retention equivalent cell have different bias pg_pin setting.

The following example shows an instance where this message occurs:

```
cell (cell) {

    retention_equivalent_cell : my_ret_cell;

    /*
    pg_pin(VBP) {
      pg_type: pwell;
    }
    */
    ...

  }

  cell (my_ret_cell) {

    pg_pin(VBP) {
      pg_type: pwell;
    }

    ...

  }
```

The following is an example message:

```
Error: Line 149, Cell 'my_ret_cell', The cell referenced by
 "retention_equivalent_cell" attribute has inconsistent bias pg_pin
 setting with non-retention cell. (LBDB-549)
```

### What Next

Check the library source file, correct the bias pg_pin setting.

## LBDB-550

(warning) %s should be consistent with %s.

Feedback

### Description

This message occurs when 2 or more related LVF table values should be consistent with each other.

```
The rules are issuing Warning when:
  Rule 1: mean_shift * skewness  < 0 (obsolete)
  Rule 2: skewness * (sigma_late - simga_early) < 0
  Rule 3: (std_dev - sigma_early) * (std_dev - sigma_late) > 0
```

For example, the corresponding values of skewness * (sigma_late - simga_early) should be negative.

The following example shows the inconsistent ocv_sigma_cell_fall early/late value and ocv_skewness_cell_fall table(ocv_std_dev_cell_fall value):

```
lu_table_template("del_1_8_7") {
    variable_1 : "input_net_transition" ;
    index_1("1, 2, 3, 4, 5, 6, 7, 8");
    variable_2 : "total_output_net_capacitance" ;
    index_2("1, 2, 3, 4, 5, 6, 7");
  }
 ...
 timing() {
   ...
    ocv_skewness_cell_fall(del_1_8_7) {
          index_1("0.004626, 0.009397, 0.01909, 0.03877, 0.07875, 0.16,
 0.3249, 0.66");
          index_2("0.000253, 0.0007021, 0.001949, 0.005408, 0.01501,
 0.04166, 0.1156");
          values("0.0003315, 0.0004154, 0.0006503, 0.001309, 0.003135,
 0.008191, 0.02237",\
                 "0.0003667, 0.0004518, 0.000687, 0.001345, 0.003174,
 0.008257, 0.02242",\
                 "0.0005104, 0.0005502, 0.0007653, 0.001423, 0.003261,
 0.00832, 0.0224",\
                 "0.0009744, 0.0009964, 0.001041, 0.001585, 0.003421,
 0.008503, 0.02259",\
                 "0.001875, 0.001913, 0.001975, 0.002119, 0.00375,
 0.008838, 0.02296",\
                 "0.003633, 0.003697, 0.003811, 0.004024, 0.004596,
 0.009506, 0.02362",\
                 "0.007095, 0.007179, 0.007403, 0.007758, 0.008299,
 0.01093, 0.02499",\
                 "0.01384, 0.0141, 0.01444, 0.01506, 0.01599, 0.01718,
 0.0278");
       }
   ...
 }

Warning: Line 1912, Cell 'STN_INV_1', pin 'X', related_pin 'A',
 when '', Rule 2, ocv_std_dev_cell_fall value [0.02237], the
 ocv_skewness_cell_fall value [0.02237] should be consistent with
```

Feedback

```
ocv_sigma_cell_fall early/late value [0.02411/-0.001501] (place: 6).
(LBDB-550)
```

**What Next**

Check the charaterization data which form the distribution and fix the values. These checks could be skipped if thresholds defined by global variable: - lc_lvf_skewness_threshold# if abs(skewness)<lc_lvf_skewness_threshold, then any skewness related checks will be skipped - lc_lvf_sigma_threshold#if abs(sigma_early-sigma_late)<lc_lvf_sigma_threshold, then any sigma related checks will be skipped - lc_lvf_normalized_skewness_threshold# if abs(skewness/std_dev)^3)<lc_lvf_normalized_skewness_threshold, then any skewness related checks will be skipped

See more on UIL-550 man page.

---

## LBDB-551

(warning) The cell referenced by "retention_equivalent_cell" attribute can not be a non-black-box cell.

**Description**

This message indicated that retention equivalent cell is not a black-box cell. The cell referenced by "retention_equivalent_cell" is desired for a complex retention cell, so the cell should be black-box.

**What Next**

Check the library source file, correct the retention_equivalent_cell attribute.

---

## LBDB-552

(warning) Can't find pin '%s' in the retention_equivalent_cell '%s'.

**Description**

For the non-retention cell with retention_equvialent_cell attribute, if any pin has no retention_equvialent_pin attribute, then the pin with same name should be exist in the referenced retention equivalent cell

For the retention equivalent pin referenced by retention_equvialent_pin attribute, the specified pin name should be existed in the retention equivalent cell

The following example shows an instance where this message occurs:

```
cell (cell) {

    retention_equivalent_cell : my_ret_cell;
```

Feedback

```
    pin(D) {
      ...
    }
    pin(E) {
      retention_equivalent_pin : EE;
      ...
    }

    ...

  }

  cell (my_ret_cell) {

    /*
    pin(D) {
      ...
    }
    pin(EE) {
      ...
    }
    */

    ...

  }
```

The following is an example message:

```
Warning: Line 139, Cell 'my_ret_cell', Pin 'D', Can't find pin 'D' in the
 retention_equivalent_cell 'my_ret_cell' (LBDB-552)
Warning: Line 149, Cell 'my_ret_cell', Pin 'E', Can't find pin 'EE'in the
 retention_equivalent_cell 'my_ret_cell' (LBDB-552)
```

### What Next

Check the library source file, correct the retention_equivalent_pin attribute.

---

## LBDB-553

(warning) The '%s' %s should not coexist with the '%s' %s.

### Description

The first specified attribute/group should not coexist with the second specified attribute/group on the same group. For example, the *char_when_rise* attribute should not be specified with *char_when* attribute on the same timing group.

Feedback

The following example shows an instance where this message occurs:

```
cell(lbdb553) {
      Pin (Y) {
         timing () {
          related_pin : "IN";
            when : "A";
            char_when : "A*B";
            char_when_rise : "A*B";
            ...
         }
         ...
      }
      ...
   }
```

The following is an example message:

```
Warning: Line 2181, Cell 'lbdb553', pin 'Y', The 'char_when_rise'
 attribute should not coexist with the 'char_when' attribute. (LBDB-553)
```

### What Next

Check the library source file, and remove the redundant one.

---

## LBDB-555

(Error) The %s attribute is not defined for the library when %s is defined.

### Description

This is for when a default attribute is required to be defined for the library but is not defined when certain attribute or group is defined.

The following is an example message:

```
Error: Line 10, The distance_unit attribute is not defined for the
 library when
default_ocv_derate_distance_group or ocv_derate_distance_group is
 defined. (LBDB-555)
```

### What Next

Check the library source file, define the default attribute for the library.

---

## LBDB-556

(Error) The '%s' ocv_derate group referenced by %s contains non-distance-based tables.

Feedback

### Description

This is for when a default_ocv_derate_distance_group/ocv_distance_group attribute referenced an ocv_derate group, at least one ocv_derate_factor in that ocv_derate group used ocv_table_template that has "variable_1 : path_depth;" defined.

The following is an example message:

```
Error: Line 110, The 'ocv' ocv_derate group referenced by
 ocv_derate_group contains non-distance-based tables. (LBDB-556)
```

### What Next

Check the library source file, use ocv_derate group that contain only distance-based tables.

---

## LBDB-557

(error) The '%s' is missing in '%s' group.

### Description

Required content is missing for the group.

The following example shows an instance where this message occurs:

```
cell ( test_cell )    {
      ocv_derate(ocv) {
        /* require ocv_derate_factors defined here */
      }
      ... ...
    }
```

The following is an example message:

```
Error: Line 144, The 'ocv_derate_factors' is missing in 'ocv_derate'
 group. (LBDB-557)
```

### What Next

Check the library source file to see if you missed the required content for the specified group.

---

## LBDB-558

(error) The '%s' %s group referenced by %s attribute is not found.

Feedback

### Description

This message indicates that the attribute cross-referenced by its value an undefined group with specific name.

The following example shows an instance where this message occurs:

```
library(lbdb158) {
cell(IV) {
  ocv_derate_group : a_ocv_derate_not_defined;
  area : 1;
  pin(A) {
    direction : input;
    capacitance : 1;
  }
  pin(Z) {
    direction : output;
    function : "A'";
    timing() {
      intrinsic_rise : 0.1;
      intrinsic_fall : 0.1;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      slope_rise : 0.0;
      slope_fall : 0.0;
      related_pin : "A";
    }
  }
}
}
```

The following is an example message:

```
Error: Line 35, The 'a_ocv_derate_not_defined' ocv_derate group
 referenced by ocv_derate_group attribute
is not found. (LBDB-558)
```

### What Next

Add the group if it is missing, or fix the attribute value if it has a typo.

---

## LBDB-559

(warning) The '%s' table used different indices from the '%s' table.

### Description

This message occurs when a look-up table used different indices from its master table. Using same indices between the tables is prefered.

Feedback

The following example shows the ocv_sigma_cell_rise table used different indices from cell_rise table:

```
lu_table_template(del_0_2_2) {
  variable_1 : input_net_transition;
  index_1("7.500000,325.000000");
  variable_2 : total_output_net_capacitance;
  index_2("1.100,10.000");
 }
 lu_table_template(del_1_2_2) {
  variable_1 : input_net_transition;
  index_1("17.500000,25.000000");
  variable_2 : total_output_net_capacitance;
  index_2("3.518,12.418");
 }
 ...
 timing() {
   ...
   ocv_sigma_cell_rise(del_1_2_2) {
     values( " 95.492787, 104.196860",\
             " 221.983170, 258.953160");
   }
   cell_rise(del_0_2_2) {
     values( " 95.492787, 104.196860",\
             " 221.983170, 258.953160");
   }
   ...
 }

Warning: Line 191, The 'ocv_sigma_cell_rise' table used different indices
 from the 'cell_rise' table. (LBDB-559)
```

### What Next

Charaterize the library using same indices for the look-up tables.

---

## LBDB-560

(error) The pin '%s' specified in the '%s' attribute is not a(n) %s pin.

### Description

This error message occurs when a pin specified by this attribute has direction not of required type.

The following example shows an instance where this message occurs: The following example shows the attribute which needs a internal pin.

```
mode_value ( A2I ) {
    mode_value_internal_pin : A1 ;
    when : "EN" ;
```

Feedback

```
    sdf_cond : "EN=1" ;
}
```

The following is an example message:

```
Error: Line 191, The pin 'A1' specified in the 'mode_value_internal_pin'
 attribute is not a(n) internal pin. (LBDB-560)
```

**What Next**

Specify a correct pin for this attribute.

---

## LBDB-561

(error) The event instance event(%s, %s) is invalid.

**Description**

The event instance delaration must be referring to a mode_definition and one of its event_defintion.

The following example shows an incorrect event instance:

```
cell(CGNP) {
  area : 1;
  mode_definition(rw) {
    mode_value(read) {
      ...
    }
    mode_value(write) {
      ...
    }
    event_definition (I2A) {
        start_mode : "read" ;
        end_mode : "write" ;
    }

  }
    dynamic_current () {
      event(rw, latching);
      ...
    }
  ...
}
```

The following is an example message:

```
Error: Line 206, The event instance event(rw, latching) is invalid.
 (LBDB-561)
```

Feedback

### What Next

Check for consistency between mode group and event definitions, and event instance declarations.

---

## LBDB-562

(error) '%s' only allows single pin Boolean-expression.

### Description

This error message occurs when Boolean expression contains multiple pins defined for the specified attribute.

The following example shows an incorrect Boolean expression.

```
cell(CGNP) {
  area : 1;
  mode_definition(rw) {
    mode_value(read) {
      ...
    }
    mode_value(write) {
      ...
    }
    event_trigger () {
        trigger_transition : "clock + EN";
    }
  }
  ...
}
```

The following is an example message:

```
Error Line 272, "trigger_transition" only allows single pin
 Boolean-expression. (LBDB-562)
```

### What Next

Check the Boolean expression.

---

## LBDB-563

(error) The '%s' must be defined within a mode_defintion group with mutually_exclusive_mode_values is true.

### Description

This error message occurs when specified attribute defined within a mode_defintion group which has no attribute mutually_exclusive_mode_values or its value is false.

Feedback

The following example shows an instance where this message occurs: The following example shows an incorrect mode_defintion group.

```
cell(CGNP) {
  area : 1;
  mode_definition(rw) {
    default_mode : "read" ;
    mode_value(read) {
      ...
    }
    mode_value(write) {
      ...
    }
  }
  ...
}
```

The following is an example message:

```
Error: Line 540, The 'default_mode' must be defined within a
 mode_defintion
group with mutually_exclusive_mode_values is true. (LBDB-563)
```

**What Next**

Add mutually_exclusive_mode_values : true or remove the attribute in the mode_defintion group.

---

## LBDB-564

(information) The event instance is defined multiple times for the same group.

**Description**

This message indicates that you specified multiple event instances for a power group. All the event instances will be retained.

The following example shows an instance where this message occurs:

```
cell(CGNP) {
  area : 1;

  dynamic_current () {
      event(rw, A2I);
      event(rw, I2S);
  }
}
```

The following is an example message:

```
Information: Line 206, The event instance is defined multiple times
        for the same group. (LBDB-564)
```

Feedback

### What Next

Please make sure that it is your real intention to have multiple events that apply to the power group.

---

## LBDB-566

(warning) the macro cell has the switch_cell_type: coarse_grain. It is marked as dont_touch and dont_use.

### Description

If macro cell has switch_cell_type, it should be fine_grain. This message means that this macro cell has the switch_cell_type: coarse_grain, it is marked as dont_touch and dont_use, it can not be auto inferred by galaxy design tools (e.g. ICC1/ICC2 router).

The following example shows an instance where this message occurs:

```
cell(A) {
    is_macro_cell : true;
    switch_cell_type : coarse_grain;
  }
```

The following is an example message:

```
Warning: Line 1023, Cell 'A', the macro cell has the switch_cell_type:
 coarse_grain. It is marked as dont_touch and dont_use. (LBDB-566)
```

### What Next

Correct the "switch_cell_type" to fine_grain for the macro cell.

---

## LBDB-567

(error) The swing of the slew/transition is more than 100% which is not possible.

### Description

This error message occurs when (slew_upper_threshold_pct_rise - slew_lower_threshold_pct_rise) / slew_derate_from_library > 100 or (slew_upper_threshold_pct_fall - slew_lower_threshold_pct_fall) / slew_derate_from_library > 100. If a slew after considering the slew_derate factor is more than 100%, such a waveform slew for any characterization waveform points to waveform going beyond the rail to rail swing which is not possible.

The following example shows an instance where this message occurs:

```
library( MyLib ) {
  slew_derate_from_library : 0.5;
  slew_lower_threshold_pct_rise : 20.0;
```

Feedback

```
  slew_upper_threshold_pct_rise : 80.0;
  ...
}
```

From the above information the slew/transition has a swing of more than 100% which is not possible.

The following is an example message:

```
Error: Line 140, The swing of the slew/transition is more than 100% which
 is not possible. (LBDB-567)
```

### What Next

correct the slew thresholds or slew derate such that its swing is less than rail-to-rail.

---

## LBDB-567w

(warning) The swing of the slew/transition is more than 100% which is not possible.

### Description

This warning message occurs when (slew_upper_threshold_pct_rise - slew_lower_threshold_pct_rise) / slew_derate_from_library > 100 or (slew_upper_threshold_pct_fall - slew_lower_threshold_pct_fall) / slew_derate_from_library > 100. If a slew after considering the slew_derate factor is more than 100%, such a waveform slew for any characterization waveform points to waveform going beyond the rail to rail swing which is not possible.

The following example shows an instance where this message occurs:

```
library( MyLib ) {
  slew_derate_from_library : 0.5;
  slew_lower_threshold_pct_rise : 20.0;
  slew_upper_threshold_pct_rise : 80.0;
  ...
}
```

From the above information the slew/transition has a swing of more than 100% which is not possible.

The following is an example message:

```
Warning: Line 140, The swing of the slew/transition is more than 100%
 which is not possible. (LBDB-567w)
```

### What Next

correct the slew thresholds or slew derate such that its swing is less than rail-to-rail.

Feedback

## LBDB-568

(warning) Cell '%s' has %d async signal pins, user must specify one async signal pin for level shifter cell or isolation cell with more than 3 pins. It's marked as dont_use, dont_touch.

### Description

This warning message occurs when the level shifter or isolation cell has more than 3 pins, but the number of async signal pin is not 1. The cell will be marked as dont_use, dont_touch except one condition for level shift with more than 3 pins. The condition is: isDifferentLS is true, and including 3 input pins, 1 output pin, 1 level shifter enable pin.

The following is an example message:

```
Warning: Line 191, Cell 'LS', Cell 'LS' has 0 async signal pins, user
 must
specify one async signal pin for level shifter cell or isolation cell
 with
more than 3 pins. It's marked as dont_use, dont_touch.
```

### What Next

This is only a warning message. No action is required.

You could correct the modeling in the previous example to specify the cell with exact 1 async signal pin.

## LBDB-569

(warning) The std cell has the switch_cell_type: fine_grain. It is marked as dont_touch and dont_use.

### Description

If std cell has switch_cell_type, it should be coarse_grain. This message means that this std cell has the switch_cell_type: fine_grain, it is marked as dont_touch and dont_use, it can not be auto inferred by galaxy design tools (e.g. ICC1/ICC2 router).

The following example shows an instance where this message occurs:

```
cell(A) {
    switch_cell_type : fine_grain;
  }
```

The following is an example message:

```
Warning: Line 1023, Cell 'A', The std cell has the switch_cell_type:
 fine_grain. It is marked as dont_touch and dont_use. (LBDB-569)
```

Feedback

### What Next

Correct the "switch_cell_type" to coarse_grain for the std cell.

---

## LBDB-580

(error) This retention cell is not a valid sequential device in normal mode.

### Description

```
When retention pin disable values are appied, the sequential elements do
 not function.
This may be due to user was intended to model a zero-pin retention cell
 but put a wrong
retention_pin attribute on the clock signal.
```

The following example shows an instance where this message occurs:

```
library(test) {
 ...
  cell (zpr) {
    ...
    pin(clk) {
      direction : input ;
      related_ground_pin : vss ;
      related_power_pin : vcc_in ;
      retention_pin(save_restore, 1); /* wrong attribute, should be
 removed */
      ...
    }

    latch(IQ1,IQN1) {
      enable : "!clk" ;
      data_in : "d" ;
      power_down_function : "!vcc+vss" ;
    }
    latch(IQ2,IQN2) {
      enable : "clk" ;
      data_in : "IQ1" ;
      power_down_function : "!vcc_in+vss" ;
    }

    retention_condition() {
      required_condition : "!clk";
      power_down_function : "!vcc+vss" ;
    }

    ...
  }
}
```

Feedback

The following is an example message:

```
Error: Line 3, Cell 'zpr', This retention cell is not a valid sequential
 device in normal mode.
```

### What Next

```
Remove the "retention_pin" attribute in this cell.
```

---

## LBDB-581

(warning) The '%s' pg_pin '%s' is using '%s' logic in power_down_function.

### Description

```
The pg_pin specified as power pg_type should have negative logic in
 power_down_function;
The pg_pin specified as ground pg_type should have positive logic in
 power_down_function.

This message indicate that the pg_pin logic in the power_down_function is
 wrong.
```

The following example shows an instance where this message occurs:

```
library(test) {
 ...
  cell (a) {
    ...
    pg_pin(vdd) {
      pg_type : primary_power;
      ...
    }
    pg_pin(vss) {
      pg_type : primary_ground;
      ...
    }

    pin(out) {
      power_down_function : "vdd+vss" ;
      ...
    }

    ...
  }
}
```

The following is an example message:

```
Error: Line 300, Cell 'a', pin 'out', The 'power' pg_pin 'vdd' is using
 'positive' logic in power_down_function.
```

Feedback

### What Next

Correct the pg_pin logic in power_down_function.

---

## LBDB-582

(error) The 'true' is_pass_gate attribute is conflicting with \n \tthe 'true' is_inverting attribute for the '%s' group.

### Description

This message indicates you specified 'true' is_pass_gate attribute and 'true' is_inverting attribute simltaneously for the same CCB (i.e., the same ccsn_first_stage, ccsn_last_stage, input_ccb or output_ccb group), which causes conflict.

The following example shows an instance where this message occurs.

```
library (test) {
   ...
   cell(inv) {
     ...
     pin(A) {
       direction : input;
       input_ccb(iccb) {
         is_inverting : true;
         is_pass_gate : true;
         ...
       }
       ...
    }
    ...
 }
```

The following is an example message:

```
Error: Line 74, Cell 'inv', pin 'A', The 'true' is_pass_gate attribute is
 conflicting with
the 'true' is_inverting attribute for the 'input_ccb' group. (LBDB-582)
```

### What Next

Change the library file by either remove the is_pass_gate attribute from the CCB or specify 'false' is_inverting attribute for the CCB.

---

## LBDB-586

(error) This timing group is missing NLDM '%s' data to work with 'propagating_ccb' attribute at line %u.

Feedback

### Description

In the referenced ccs noise data modeling (hereinafter "ccb modeling"), there is a synergy between NLDM delay data and input_ccb/output_ccb data referenced by the attribute propagating_ccb. However, a required data (cell_rise, cell_fall, rise_transition, fall_transition) is not found, as indicated by the message.

### Examples

Error: Line 1564, Cell 'INV_X8M_A9TL', pin 'Y', This timing group is missing NLDM 'cell_rise' data to work with 'propagating_ccb' attribute at line 2815. (LBDB-586)

### What Next

Add missing data.

---

## LBDB-587

(warning) attribute '%s'. Input slew (vector/index_1) not matching between '%s' at line %u of 'input_ccb' stage (line %u) and '%s' at line %u of 'output_ccb' stage (line %u). Mismatch %s.

### Description

In a two-stage propagation_ccb usage, both referenced ccb stages must have the same set of input slew values (vector/index_1 of respective output_voltage_rise/fall groups), matching in size and values.

Warning: Line 5085, Cell 'AND2_X2M_A9TL', pin 'Y', attribute 'propagating_ccb'.
Input slew (vector/index_1) not matching between 'output_voltage_rise' at line 3118 of 'input_ccb' stage (line 3017) and 'output_voltage_fall' at line 6424 of 'output_ccb' stage (line 6382). Mismatch in size. (LBDB-587)

### What Next

Correct characterization procedure.

---

## LBDB-588

(error) attribute '%s'. The referenced '%s' stage at line %u has more than one load value in its 'vector/index_2' of '%s' at line %u.

### Description

The issue is related to a two-stcge input_ccb/output_ccb configuration. The input_ccb stage has more than one index_2 value ("total_output_net_capacitance") among the vectors of its output_voltage_rise/fall group. However, this stage drives a fixed internal oad

Feedback

(i.e. the output ccb stage), no internal load variation is currently supported. Hence there can only be one index_2 value.

This problem may be caused by the use of the same input_ccb in a single-stage propagating_ccb of another timing group.

**Examples**

Error: Line 7492, Cell 'custom_cell_xwy', pin 'nx2', attribute 'propagating_ccb'. The referenced 'input_ccb' stage at line 488 has more than one load value in its 'vector/index_2' of 'output_voltage_rise' at line 561. (LBDB-588)

**What Next**

Correct the characterization procedure.

---

## LBDB-589

(warning) The voltage value of middle 3 points should be consistent with that of delay/slew measure points in %s group.

**Description**

In referenced CCB noise modeling, the voltage value of middle three points (stored in attribute "values") in each vector group of output_voltage_rise group should be slew_lower_threshold_pct_rise, output_threshold_pct_rise, slew_upper_threshold_pct_rise of Vdd, and for output_voltage_fall group, slew_upper_threshold_pct_fall, output_threshold_pct_fall, slew_lower_threshold_pct_fall of Vdd.

The following example shows an instance where this message occurs:

```
library("test_library") {
  ...
  output_threshold_pct_fall : 50 ;
  output_threshold_pct_rise : 50 ;
  slew_lower_threshold_pct_fall : 30 ;
  slew_lower_threshold_pct_rise : 30 ;
  slew_upper_threshold_pct_fall : 70 ;
  slew_upper_threshold_pct_rise : 70 ;
  voltage_map(VDD, 0.765);
  ...

  cell(BUFX0P5BV0LI35P) {
    ...
    pin(A) {
      ...
      related_power_pin : VDD ;

      input_ccb(INVX1BV0LI35P__nco_0_FR_RF:a:z) {
```

Feedback

```
        ...
        output_voltage_fall() {

          vector(ccsn_wf1x1x5) {
            index_1("0.001000");
            index_2("0.000500");
            index_3("0.00430831, 0.00566391, 0.00721167, 0.00889193,
 0.0114251");
            values("0.688500, 0.535500, 0.482500, 0.229500, 0.076500");
          }
          ...
        }
        ...
      }
      ...
    }
    ...
  }
  ...
}
```

The following is an example message:

```
Warning: Line 1193, Cell 'BUFX0P5BV0LI35P', pin 'A', The voltage value of
 middle
 3 points should be consistent with that of delay/slew measure points in
output_voltage_fall group. (LBDB-589)
```

**What Next**

Fix the mistake. In the example above, the middle value should be: 0.382500.

---

## LBDB-590

(warning) There is only %d index value(s) in '%s' for this %s table.

**Description**

To be able to calculate value that not on grid, there's minimal index values requirement on specific tables.

For pg_current in dynamic_current, the indices except time (the last index) in all vectors will be grouped and checked. If it has only one index, which usually is input_net_transition, at least three index values are needed. If it has multiple indices, then for each index, at least two index values are needed.

The following example has only one input_net_transition value of 5.1 in the pg_current table, and two total_output_net_capacitance values of 0.1 and 0.3.

```
pg_current(V2) {
        vector(test_2) {
```

Feedback

```
          reference_time : 93.2;
          index_1("5.1");
          index_2("0.1");
          index_3("8.2 8.3 9.0");
          values("3.78 92.4 100.1");
         }
        vector(test_2) {
          reference_time : 93.2;
          index_1("5.1");
          index_2("0.3");
          index_3("8.2 9.4 9.8");
          values("1.78 12.4 110.1");
         }
```

```
The following is an example message:
Warning: Line 518, Cell 'OR2', There is only 1 index value(s) in
 'input_net_transition' for this pg_current table. (LBDB-590)
```

**What Next**

Add more index values.

---

## LBDB-591

(warning) The pg_current values are irregular, peak point is at beginning or end.

**Description**

This message is issued since the current waveform("values(...)") in the pg_current vector is abnormal, it has the peak value at beginning or end. Note: peak is the largest absolute value. This constraint will only be applied while output is switching, and if the peak value is under 1uA, which is closing to leakage current, the constraint is not needed.

The following example has the peak value at end.

```
vector(test_1) {
     reference_time : 23.7;
     index_1("0.8");
     index_2("0.7");
     index_3("10.4");
     index_4("8.2 8.5 9.1 9.4 9.8");
     values("0.7 34.6 3.78 92.4 100.1");
    }
```

The following is an example message:

```
Warning: Line 520, Cell 'OR2', The pg_current values are irregular, peak
 point is at beginning or end. (LBDB-591)
```

**What Next**

Check the characterization data.

Feedback

## LBDB-592

(warning) Too few value points in this vector, at least '%d' values needed.

### Description

This message is issued since the current waveform("values(...)") in the pg_current vector has too few points. At least three points are needed. The only exception is that all current values are under the threshold (1uA), which is closing to leakage current, then the requirement is not needed.

The following example has the peak value at end.

```
vector(test_1) {
     reference_time : 23.7;
     index_1("0.8");
     index_2("0.7");
     index_3("10.4");
     index_4("8.2 8.5");
     values("0.7 34.6");
    }
```

The following is an example message:

```
Warning: Line 520, Cell 'OR2', Too few value points in this vector, at
 least '3' values needed. (LBDB-592)
```

### What Next

Check the characterization data.

## LBDB-593

(error) value list of '%s' attribute not full range, or not monotonically %s.

### Description

The value list of this attribute is either not full range or monotonic. The values are required to:

start at 0.0, and increase monotonically to 100.0, or

start at 100.0, and decrease monotonically to 0.0

The following example has two issues: the 3rd value is less than the 2nd, and the last one is not 100.0;

```
receiver_trip_threshold_pct_rise(0.0, 40.0, 30.0, 60.0, 75.0,
 85.0, 99.0);
```

Feedback

The following is an example message:

```
Error: Line 74, value list of 'receiver_trip_threshold_pct_rise'
 attribute not full range, or not monotonically increasing.
```

### What Next

Fix the problem.

---

## LBDB-594

(error) Unnamed CCB stage.

### Description

All input_ccb & output_ccb groups must be given names. Named ccsn stages are placed only at pin group level. They are referenced, if needed, in acitve_input_ccb, active_output_ccb, & propagating_ccb attributes at timing group level.

The following "input_ccb" example does not have a name.

```
library (test) {
   ...
   cell(inv) {
      ...
     pin(A) {
       direction : input;
       input_ccb() {
         ...
       }
       ...
    }
    ...
 }
```

The following is an example message:

```
Error: Line 74, Cell 'inv', pin 'A', Unnamed CCB stage. (LBDB-594)
```

### What Next

Add name to the input_ccb or output_ccb group.

---

## LBDB-595

(error) CCB stage name '%s' already used by the %s group at line %u.

Feedback

### Description

The name shown in the message is already used by another input_ccb or output_ccb group in the same pin group. Please replace it with a unique name, and update affected active_input_ccb, active_output_ccb, and propagating_ccb attributes.

The following example shows an instance where this message occurs. The CCB stage name *b* is used twice under pin b:

```
pin(b) {
     direction : inout;
     input_ccb(b) {
       ...
     }
     output_ccb(b) {
       ...
     }
     ...
   }
```

The following is an example message:

```
Error: Line 1813, Cell 'dup_names', pin 'I', CCB stage name 'b' already
used by the input_ccb group at line 1476. (LBDB-595)
```

### What Next

Modify to make sure names are unique. For example:

```
pin(b) {
     direction : inout;
     input_ccb(b_in) {
       ...
     }
     output_ccb(b_out) {
       ...
     }
     ...
   }
```

---

## LBDB-596

(error) The value of min_capacitance is greater than that of max_capacitance.

### Description

This information occurs when the value of min_capacitance is greater than that of max_capacitance.

Feedback

The following example shows an instance where this message occurs:

```
pin (Z) {
  direction : "output";
  function : "A";
  output_signal_level : "VDD";
  max_capacitance : 0.057841;
  min_capacitance : 0.557841;
 }
```

The following is an example message:

```
Error: Line 258, Cell 'BUFX1', pin 'Z', The value of min_capacitance is
 greater than that of max_capacitance. (LBDB-596)
```

**What Next**

Make sure the value of min_capacitance is not greater than that of max_capacitance.

---

## LBDB-597

(error) CCB group '%s' has been utilized by attribute '%s' at line %u.

**Description**

For a given timing group, a particular CCB stage group may only be referenced once among attributes propagating_ccb, active_input_ccb, and active_output_ccb. It is an error to be referenced more than once.

The following example shows an instance where this message occurs. The input_ccb group *ccb_a* is referenced twice in the same timing group.

```
library(test) {
    cell test(inv) {
      pin(a) {
        direction : in;
          input_ccb(ccb_a) {
          ...
          }
        ...
      }
      pin(out) {
        direction : out;
        output_ccb(ccb_out) {
          ...
          }
        timing () {
          related_pin : in;
          active_input_ccb("ccb_a");
          propagating_ccb("ccb_a", "ccb_out");
        }
      }
```

Feedback

```
    }
  }
```

The following is an example message:

```
Error: Line 3209, Cell 'inv', pin 'ZN', CCB group 'ccb_a' has been
 utilized by attribute 'propagating_ccb' at line 3210. (LBDB-597)
```

**What Next**

Remove the incorrect usage.

---

## LBDB-598

(error) CCB group '%s' referenced by attribute '%s' not found in pin '%s' at line %u.

**Description**

A CCB stage referenced in attributes propagating_ccb, active_input_ccb, and active_output_ccb must be located at the input pin (related_pin) or the output pin, depending on the context.

The following example shows an instance where this message occurs:

```
library(test) {
    cell test(and2) {
      pin(a) {
        direction : in;
        input_ccb(ccb_a) {
        ...
        }
      ...
      }
      pin(b) {
        direction : in;
        input_ccb(ccb_b) {
        ...
        }
      ...
      }
      pin(o1) {
        direction : out;
        output_ccb(ccb_o1) {
          ...
          }
        ...
      }
      pin(o2) {
        direction : out;
        output_ccb(ccb_o2) {
          ...
          }
```

Feedback

```
        timing () {
          related_pin : a;
          propagating_ccb("ccb_a", "ccb_o1");
        }
      }
    }
  }
```

The following is an example message:

```
Error: Line 759, Cell 'and2', pin 'o2', CCB group 'ccb_o1' referenced by
 attribute 'propagating_ccb' not found in pin 'o2' at line 70. (LBDB-598)
```

### What Next

Fix the mistake. In the example above, attribute propagating_ccb should be specified as:

```
propagation("ccb_a", "ccb_o2");
```

---

## LBDB-599

(error) In attribute '%s', ccb '%s' referenced is of the wrong type.

### Description

The ccb referenced is not of the correct type for one of the following reasons:

1. The 1st parameter of propagating_ccb, or all parameters in active_input_ccb must reference a input_ccb of the related pin, not a output_ccb.
2. The 2nd parameter of propagating_ccb, if present, must reference a output_ccb in the current (output/inout) pin, not a input_ccb.
3. Finally, note that only one ccb can drive the output. If the attribute mentioned is 'active_output_ccb', remove the offending output_ccb because:

1. Another output_ccb has already been given in the same attribute. In fact, the only one valid possibility of more than 1 output_ccb's being referenced is the two output_ccb's in which one has the stage_type:pull_up and the other has the stage_type:pull_down.
2. The 'propagating_ccb' attribute is present.

The following example shows an instance where this message occurs:

```
library(test) {
    cell test(complex) {
      pin(a) {
        direction : inout;
        input_ccb(ccb_a_in) {
        ...
        }
        output_ccb(ccb_a_out) {
```

Feedback

```
        ...
        }
      ...
      }
      pin(out) {
        direction : out;
        output_ccb(ccb_o2) {
          ...
          }
        timing () {
          related_pin : a;
          active_input_ccb("ccb_a_out");
        }
      }
    }
  }
```

The following is an example message:

```
Error: Line 3175, Cell 'complex', pin 'out', In attribute
 'active_input_ccb', CCS noise stage 'ccb_a_out' referenced is of the
 wrong type. (LBDB-599)
```

### What Next

Fix the mistake.

---

## LBDB-600

(error) The '%s' direction cannot be specified on a\n \tpin.

### Description

This message indicates that you specified an incompatible direction value on a pin. Library Compiler fails if the direction value includes the following combinations:

```
* tristate
```

The following example shows an instance where this message occurs:

```
cell(lbdb600) {
    area : 2;
    pin(A) {
      direction : tristate;
      capacitance : 1.0;
    }
    ...
  }
```

Feedback

The following is an example message:

```
Error: Line 84, The 'tristate' direction cannot be specified on a
       pin. (LBDB-600)
```

### What Next

Refer to the "Library Compiler User Guide" for direction information. Remove the incompatible values of the direction attribute.

---

## LBDB-601

(error) related_pin is illegal in the timing group. \ Timing arc info is missing.

### Description

This message indicates that you specified an illegal related_pin in a timing group.

The following example shows an instance where this message occurs:

```
timing() {
                /* noise immunity */
                noise_immunity_high(noise5x5){
                    index_2("0.362,0.725,1.087,1.449,1.812");
                    values("0.733,1.073,1.412,1.752,2.092",\
                           "0.873,1.214,1.554,1.894,2.234",\
                           "1.095,1.442,1.787,2.132,2.477",\
                           "1.298,1.648,1.996,2.343,2.691",\
                           "1.703,2.060,2.414,2.766,3.119");
                }
                related_pin: "a";
            }
```

The following is an example message:

```
Error: Line 120, related_pin is illegal in the timing group. \
 Timing arc info is missing.  (LBDB-601)
```

### What Next

Refer to the "Library Compiler User Guide" for related_pin information.

---

## LBDB-602

(warning) The units of time, capacitance, voltage and current are not consistent.

### Description

This message indicates that you specified an inconsistent units in the library.

Feedback

The following example shows an instance where this message occurs:

```
library (mi333) {
    ...
   time_unit                    : "1ns";
   voltage_unit                 : "1mV";
   current_unit                 : "1mA";
   pulling_resistance_unit      : "1kohm";
   capacitive_load_unit  (1, pf);
    ...
```

The following is an example message:

```
Warning: Line 120, The units of time, capacitance, voltage and current
 are not consistent. (LBDB-602)
```

**What Next**

Refer to the "Library Compiler User Guide" for unit information. Modify the realted unit values to make them consistent.

---

## LBDB-603

(error) The '%s' attribute cannot be supplied a \n \tnonpositive value (%d).

**Description**

This message indicates that the specified attribute cannot have a nonpositive value.

The following example shows an instance where this message occurs:

```
orders ( 0, 2, 3 );
```

The following is an example message:

```
Error: Line 18, The 'orders' attribute cannot be supplied a
       nonpositive value (0) (LBDB-603)
```

**What Next**

Change the value of the attribute to positive in the technology library file.

---

## LBDB-604

(error) '%s' attribute should be less than '%s' attribute.

**Description**

This message indicates that the specified attributes are not consistent.

Feedback

The following example shows an instance where this message occurs:

```
slew_lower_threshold_pct_rise: 80.0
   slew_upper_threshold_pct_rise: 20.0
```

The following is an example message:

```
Error: Attribute 'slew_lower_threshold_pct_rise' should be less
       than 'slew_upper_threshold_pct_rise'. (LBDB-604)
```

**What Next**

Change the value of the attributes to be consistent in the technology library file.

---

## LBDB-605

(warning) The attribute '%s' is not specified.

**Description**

This message indicates that an attribute is missing.

The following example shows an instance where this message occurs:

```
slew_lower_threshold_pct_rise: 20.0
```

The following is an example message:

```
Warning: Line 18, The attribute 'slew_lower_threshold_pct_rise' is not
 specified. (LBDB-605)
```

**What Next**

Either add the missing attribute to the technology library or ignore the message.

---

## LBDB-606

(error) The invalid '%s' value is encountered\n \ton the '%s' attribute.

**Description**

This message indicates that you specified an invalid attribute's value

The following example shows an instance where this message occurs:

```
fall_propagation ( table4x6 ) {
   index_1 ("2.00e+02 2.50e+02 3.50e+02 4.00e+02 ") ;
   index_2 ("4.48e-02 8.96e-02 1.34e-01 1.79e-01 2.24e-01 2.69e-01 ");
   values ("Look.here 9.56e+01 1.15e+02 1.34e+02 1.52e+02 1.71e+02 ", \
           "9.38e+01 1.20e+02 1.45e+02 1.67e+02 1.89e+02 2.10e+02 ", \
           "1.01e+02 1.32e+02 1.58e+02 1.83e+02 2.07e+02 2.29e+02 ", \
```

Feedback

```
            "1.04e+02 1.43e-93 1.43e-93 1.43e-93 2.75e+226 9.30e+254 " ) ;
}
```

The following is an example message:

```
Error: Line 85, The invalid 'Look.here' value is encountered
        on the 'values' attribute. (LBDB-606)
```

**What Next**

Check the library source file, and correct the problem.

---

## LBDB-607

(warning) The pin '%s' does not have a internal_power group.

**Description**

For a signal pin, it should have one internal_power table.

The following example shows an instance where this message occurs:

```
cell ( cell1 ) {
    pg_pin (VDD) {
      pg_type : "primary_power";
      related_bias_pin : "VDDB";
      voltage_name : "VDD";
    }
    pg_pin (VDDB) {
      pg_type : "nwell";
      physical_connection : "device_layer";
      voltage_name : "VDDB";
    }
    pg_pin (VSS) {
      pg_type : "primary_ground";
      related_bias_pin : "VSSB";
      voltage_name : "VSS";
    }
    pg_pin (VSSB) {
      pg_type : "pwell";
      physical_connection : "device_layer";
      voltage_name : "VSSB";
    }
     pin(A) {
        ...
     }
     pin(B) {
        ...
     }
     pin ( CP ) {
        direction : input ;
        related_bias_pin : "VDDB VSSB";
```

Feedback

```
        related_ground_pin : "VSS";
        related_power_pin : "VDD";

         ...
      }
      ...
```

To fix the problem, add internal_power group under pin CP.

```
internal_power () {
        related_pg_pin : "VDD";
        when : "(!A&B)";
        power ("power_1") {
          index_1("0.002, 0.0263908, 0.0995546, 0.2215, 0.392218,
 0.611718, 0.88");
          values("0.0002916, 0.0005488, 0.0015848, 0.0024696, 0.0029070,
 0.0031031, 0.0032115");
        }

}
```

The following is an example message:

```
Warning: Line 126, The pin 'CP' does not have a internal_power group.
 (LBDB-607)
```

### What Next

Add the internal_power table.

---

## LBDB-608

(Warning) Library cell '%s' has a valid function-id,\n \tbut it has also been annotated with the user_function_class\n \tattribute. Resolving this conflict by ignoring the\n \tuser_function_class attribute for this library cell.

### Description

The specified cell has been annotated with the *user_function_classr* attribute even though it has a valid function-id. Library Compiler can generate a function-id for most cells in a technology library. However, there may be a few complex sequential or combinational cells that Library Compiler cannot successfully generate a function-id for. In addition, some black-box cells in the library do not have any function information. The *user_function_class* attribute is intended for such cells that otherwise would not have a function-id. The *user_function_class* attribute is not intended for library cells that already have a valid function-id.

Feedback

### What Next

This warning message indicates that the *user_function_class* attribute for this library cell will be ignored. No further action is required. To turn off this warning message in the future, make sure that a *user_function_class attribute* is not set on this library cell.

---

## LBDB-609

(Warning) Library cells '%s' and '%s' have\n \tdifferent pin information, but they have been assigned the\n \tsame user_function_class. Resolving this conflict by ignoring\n \tthe user_function_class attribute on the latter library cell.

### Description

Two library cells that are assigned the same *user_function_class* attribute should have the exact same pin information. That is, they must have the same number of pins and the same pin names. This requirement is necessary so that Design Compiler can establish pin-to-pin correspondence between two cells in the same *user_function_class* when it replaces a reference to one library cell by a reference to the other library cell.

This warning message indicates that an inconsistency was found between the pin information of two library cells in the same *user_function_class*. To resolve this conflict, the *user_function_class* attribute for the second library cell will be ignored.

### What Next

This warning message indicates that the *user_function_class* attribute for the second library cell will be ignored. No further action is required. To turn off this warning message in the future, make sure that all library cells in a given *user_function_class* attribute have the same pin information.

---

## LBDB-610

(warning) The units of %s are not consistent.

### Description

This message indicates that you specified an inconsistent units in the library.

The following example shows an instance where this message occurs:

```
library (mi333) {
    ...
   time_unit                    : "1ns";
   voltage_unit                 : "1V";
   current_unit                 : "1mA";
   resistance_unit              : "1kohm";
   capacitive_load_unit  (1, pf);
    ...
```

Feedback

The following is an example message:

```
Warning: Line 120, The units of time, capacitance, resistance are not
 consistent. (LBDB-610)
```

### What Next

Refer to the "Library Compiler User Guide" for unit information. Modify the realted unit values to make them consistent.

---

## LBDB-611

(warning) The '%s' %s group is not used by any %s pin in the library.

### Description

This message indicates that the library pins does not use the defined input/output voltage groups.

The following example shows an instance where this message occurs:

```
input_voltage(CMOS) {
    vil : 1.5;
    vih : 3.5;
    vimin : -0.3;
    vimax : VDD + 0.3;
  }

  input_voltage(CMOS) {
    vil : 2.5;
    vih : 4.5;
    vimin : -0.3;
    vimax : VDD + 0.3;
  }
```

The following is an example message:

```
Error: Line 31, The 'CMOS' input_voltage  group is not used by any input
 pin in the library. (LBDB-611)
```

### What Next

For input_voltage group, add "input_voltage" attrribute to input library pins. For output_voltage group, add "output_voltage" attrribute to output library pins.

---

## LBDB-612

(information) The '%s' attribute value is\n \t %s (%3.1f).

Feedback

### Description

This message indicates that you specified an attribute value that maybe out of the accepted range. The value is either less than the minimum value or greater than the maximum value. Unlike LBDB-163, it does not reset the value.

The following example shows an instance where this message occurs:

```
height_coefficient : -0.01;
```

In this case, the height_coefficient's value -0.01 is less than the minimum accepted value 0.0. However, it is not reset.

The following is an example message:

```
Warning: Line 109, The 'height_coefficient' attribute value is
         less than 0.0. (LBDB-612)
```

---

## LBDB-612w

(warning) The '%s' attribute value (%g) is\n \t %s (%g).

### Description

This message indicates that you specified an attribute value that maybe out of the accepted range. The value is either less than the minimum value or greater than the maximum value. Unlike LBDB-163, it does not reset the value.

The following example shows an instance where this message occurs:

```
voltage_map(VDDL, 1.00) ;
  ...
  cell (A) {
    ...
    pin(I)  {
      ...
      input_voltage_range (0.4, 3.3);
      related_power_pin : VDDL;
      max_input_delta_overdrive_high : 0.05;
      max_input_delta_underdrive_high : 0.8;
    }
  }
```

In this case, the max_input_delta_underdrive_high's value (0.8) is greater than the input_voltage_range allowed (0.6, which is the difference of voltage value between related_power_pin and the lower bound of input_voltage_range). However, it is not reset.

The following is an example message:

```
Warning: Line 79, The 'max_input_delta_underdrive_high' attribute value
 (0.8) is
         greater than the input_voltage_range allowed (0.6). (LBDB-612w)
```

Feedback

### What Next

Change the attribute value to satisfy the value range.

---

## LBDB-613

(error) The '%s' group requires the '%s' %s.\n \tEither the attribute(or group) is missing or the attribute(or group) has an invalid value.

### Description

This message indicates a group is missing, or you specified a group without one of its required attributes(or groups, or vectors). Library Compiler rejects the attribute(group) definition if the attribute(group) is missing, or exists but has an invalid value.

The following example shows an instance where this message occurs:

```
memory() {
        type : random;
        address_width : 10;
        word_width : 8;
      }
```

In this case, the 'type' has an invalid value. To fix the problem, assign 'rom' or 'ram' to the type.

The following is an example message:

```
Error: Line 27, The 'memory' group requires the 'type' attribute.
 Either the attribute(or group) is missing or the attribute(or group)has
 an invalid value. (LBDB-613)
```

Here is another example shows where this message occurs:

```
ccsn_last_stage() {
          output_voltage_fall () {
          }
          propagated_noise_low () {
          }
          propagated_noise_high () {
          }
 }
```

In this case, 'output_voltage_rise' group is missing. In addition, 'vector' data tables are also missing for output_voltage_fall, propagated_noise_low, propagated_noise_high. These need to be added to fix the problem.

The following is an example message.

```
Error: Line 54, The 'ccsn_last_stage' group requires the
 'output_voltage_rise' group.
```

Feedback

```
 Either the attribute(or group) is missing or the attribute(or group)has
 an invalid value. (LBDB-613)
Error: Line 54, The 'ccsn_last_stage' group requires the
 'output_voltage_fall' group.
 Either the attribute(or group) is missing or the attribute(or group)has
 an invalid value. (LBDB-613)
Error: Line 54, The 'ccsn_last_stage' group requires the
 'propagated_noise_low' group.
 Either the attribute(or group) is missing or the attribute(or group)has
 an invalid value. (LBDB-613)
Error: Line 54, The 'ccsn_last_stage' group requires the
 'propagated_noise_high' group.
 Either the attribute(or group) is missing or the attribute(or group)has
 an invalid value. (LBDB-613)
```

### What Next

Change the library file by adding the missing attribute(or group, or vector) to the group.

---

## LBDB-614

(error) The '%s' has a count of %d, which does not match\n \tthe number of "%s" specified.

### Description

The specification of the variables implied the number of indices of a template (or template domain).

The following example shows an instance where this message occurs:

```
lu_table_template(basic_template) {
        variable_1 : input_net_transition;
        index_1 ("0.1, 1.2, 2.3, 3.4");
        index_2 ("0.1, 1.2, 2.3, 3.4");
  }
```

The following is an example message:

```
Error: Line 160, The 'index_*' have a count of 2, which does not match
        the number of 'varible_*' specified. (LBDB-614)
```

### What Next

Check the library source file, and make the number of variables and indices to\n \tthe same.

---

## LBDB-615

(warning) Connect pin '%s' to the default_power_supply '%s'\n \tdefined in the power_supply() group in library.

Feedback

### Description

Either the input_signal_level or the output_signal_level attribute is missing in a pin within a multiple power supply cell. By default, LC connect this pin to the default_power_supply defined in the power_supply() group in library.

The following example shows an instance where this message occurs:

```
power_supply() {
        default_power_rail : VDD0;
        power_rail(VDD1, 5.0);
        power_rail(VDD2, 3.3);
    }

cell(lbdb428) {
  area : 2;
  pad_cell : true;
  rail_connection(PV1, VDD1);
  rail_connection(PV2, VDD2);
  pin(A) {
    direction : input;          /* missing input_signal_level attribute,
 will use default VDD0 */
    capacitance : 1;
  }
  pin(Z) {
    direction : output;
    function : "A";
    output_signal_level : VDD2;
    timing() {
      intrinsic_rise : 0.48;
      intrinsic_fall : 0.77;
      rise_resistance : 0.1443;
      fall_resistance : 0.0523;
      slope_rise : 0.0;
      slope_fall : 0.0;
      related_pin : "A";
    }
  }
}
}
```

The following is an example message:

```
Warning: Line 96, Connect pin 'A' to the default_power_supply 'VDD0'
       defined in the power_supply() group in library. (LBDB-615)
```

### What Next

Check the library source file to see if you missed the input_signal_level or the output_signal_level attributes.

Feedback

## LBDB-616

(error) Found the timing arc '%s' has lookup table or poly template \n \twhich is more than %s dimensional in a '%s' timing group in the library

### Description

The related_pin and constraint_pin are the same in the specified timing group; therefore, there cannot be a two-dimension or more lookup table/poly template based on related_pin_transition and constraint_pin_transition because they are the same pin and one pin cannot have two different values at the same time.

Note that the min_pulse_width timing group can be one- or two-dimensional, because the arc can be indexed by the output load.

The following example shows an instance where this message occurs:

```
lu_table_template  (table_2)   {
   variable_1  :  constrained_pin_transition  ;
   variable_2  :  related_pin_transition  ;
   index_1 ("0.00, 0.25, 0.50") ;
   index_2 ("0.00, 0.50, 2.00") ;
}

 ...

    timing() {
      related_pin : CP ;
      timing_type : min_pulse_width ;
      rise_constraint(table_2) {
          index_1 ("0.00, 1.00, 2.50") ;
          index_2 ("0.00, 1.00, 2.50") ;
          values (
                  "4.0, 5.0, 6.0"
                  "4.2, 5.2, 6.2"
                  "4.4, 5.4, 6.4"
          );
      }
      fall_constraint(table_2) {
          index_1 ("0.00, 1.00, 2.50") ;
          index_2 ("0.00, 1.00, 2.50") ;
          values (
                  "4.0, 5.0, 6.0" \
                  "4.2, 5.2, 6.2" \
                  "4.4, 5.4, 6.4" \
          );
      }
```

Feedback

The following is an example message:

```
Error: Line 160, Found the timing arc 'rise_constraint' has lookup table
 or poly template which is more than one dimensional in a min_pulse_width
 timing group in the library (LBDB-616)
```

### What Next

Modify the timing arc based on a one-dimensional lookup table or poly template.

---

## LBDB-617

(warning) The related pin in a '%s' timing group is specified as '%s', not \n \t the pin '%s' itself. It is reset to pin '%s'.

### Description

In the timing group reported, since the related_pin and constraint_pin are the same, the related_pin attribute is optional and has be to set to the pin itself.

The following example shows an instance where this message occurs:

```
pin(CP) {
   ...
    timing() {
      related_pin : D ;
      timing_type : min_pulse_width ;
      rise_constraint(table_2) {
          index_1 ("0.00, 1.00, 2.50") ;
          index_2 ("0.00, 1.00, 2.50") ;
          values (\
                  "4.0, 5.0, 6.0"\
                  "4.2, 5.2, 6.2"\
                  "4.4, 5.4, 6.4"\
          );
      }
      fall_constraint(table_2) {
          index_1 ("0.00, 1.00, 2.50") ;
          index_2 ("0.00, 1.00, 2.50") ;
          values (\
                  "4.0, 5.0, 6.0"\
                  "4.2, 5.2, 6.2"\
                  "4.4, 5.4, 6.4"\
          );
      }
  }

need to delete related_pin attribute or set it as
related_pin : CP ;
```

Feedback

The following is an example message:

```
Warning: Line 160, The related pin in a min_pulse_width timing group
 is specified as 'D', not the pin 'CP' itself. It is reset to pin 'CP'.
 (LBDB-617)
```

### What Next

Either delete the related_pin attribute or set the attribute to the pin's name.

---

## LBDB-618

(error) The min/max_clock_tree_path timing arc contains illegal rise/fall_propagation group.

### Description

In a min/max_clock_tree_path timing arc, only cell_rise group or cell_fall group or both are allowed

The following example shows an instance where this message occurs:

```
pin(CP) {
   ...
    timing() {
      related_pin : CLK ;
      timing_type : min_clock_tree_path ;
      rise_propagation(table_2) {
          index_1 ("0.00, 1.00, 2.50") ;
          values (\
                  "4.0, 5.0, 6.0"\
          );
      }
      fall_propagation(table_2) {
          index_1 ("0.00, 1.00, 2.50") ;
          index_2 ("0.00, 1.00, 2.50") ;
          values (\
                  "4.0, 5.0, 6.0"\
          );
      }
  }

need to delete both rise_propagation and fall_propagation lookup table
 and replace with
cell_rise and cell_fall lookup tables
```

The following is an example message:

```
Warning: Line 160, the min/max_clock_tree_path timing arc containts
 illegal rise/fall_progagation group (LBDB-618)
```

Feedback

### What Next

Delete the illegal rise/fall_propagation group

---

## LBDB-619

(error) The '%s' attribute cannot be specified on the %s %s.

### Description

This error message occurs when you specify an invalid attribute for an object. As shown in the following example, the *output_signal_level_high* attribute is invalid for a pin object, causing the error message:

```
cell(lbdb600) {
    area : 2;
    pin(A) {
      direction : input;
      capacitance : 1.0;
      output_signal_level_high : 1.2;
      ...
    }
    ...
  }

Error: Line 84, The 'output_signal_level_high' attribute cannot be
 specified on the
       pin 'A'. (LBDB-619)
```

### What Next

Remove the attribute from the pin and rerun the command.

---

## LBDB-619w

(warning) The '%s' attribute cannot be specified on the %s %s.

### Description

This warning message occurs when you specify an invalid attribute for an object. As shown in the following example, the *output_signal_level_high* attribute is invalid for a pin object, causing the warning message:

```
cell(lbdb600) {
    area : 2;
    pin(A) {
      direction : input;
      capacitance : 1.0;
      output_signal_level_high : 1.2;
      ...
```

Feedback

```
    }
    ...
  }

Warning: Line 84, The 'output_signal_level_high' attribute cannot be
 specified on the
       pin 'A'. (LBDB-619w)
```

### What Next

Remove the attribute from the pin and rerun the command.

---

## LBDB-620

(warning) The lookup table domain '%s' is defined multiple times\n \twithin '%s'. Using the last one encountered.

### Description

This message indicates that the same lookup table domain has been defined for multiple times within one timing/power arc. In the case of a domain conflict, Library Compiler ignores all except the last domain encountered during the compilation. The compiled database contains the last domain only.

The following example shows an instance where this message occurs:

```
cell_rise(li7X7){
     domain(D1) { ... }
     domain(D1) { ... }
     domain(D2) { ... }
   }
```

The following is an example message:

```
Warning: Line 987, The lookup table domain 'D1' is defined multiple times
         within 'cell_rise'. Using the last one encountered. (LBDB-620)
```

### What Next

Make sure that only define one lookup table for each domain for each timing/power arc.

---

## LBDB-621

(error) Invalid calc_mode '%s' is detected. This name must be\n \tunique among domains within one lu_table_template.

### Description

This message indicates a duplicate calc_mode defined among domains within one lu_table_template.

Feedback

The following example shows an instance where this message occurs:

```
lu_table_template(li7X7){
    domain(D1) {
      calc_mode : "CM1";
      . . .
    }
    domain(D2) {
      calc_mode : "CM1";
      . . .
    }
  }
```

The following is an example message:

```
Error: Line 56, Invalid calc_mode 'CM1' is detected. This name must be
        unique among domains within one lu_table_template. (LBDB-621)
```

**What Next**

Change the name of the pin, bus, bundle, rail_connection in the technology library.

---

## LBDB-622

(error) The '%s' attribute is missing for the %s %s.

**Description**

This error message occurs because either the *max_input_noise_width* or the *min_input_noise_width* attribute is missing when both attributes are required.

This message also occurs when *input_signal_level* and *output_signal_level* are not defined in the input/output pin of a level shifter or isolation cell.

**What Next**

Check the library source file, and add the missing attribute.

---

## LBDB-623

(error) The '%s' attribute is larger than the %s '%s' %sfor the %s %s.

**Description**

This error message occurs when the *min_input_noise_width* attributed value is larger than the *max_input_noise_width* attribute value. Library Compiler requires that the *min_input_noise_width* value be no larger than the *max_input_noise_width* value.

The following example shows *min_input_noise_width* set at a higher value than *max_input_noise_width* and the resulting error message.

Feedback

```
pin(S) {
  min_input_noise_width : 2.0;
  max_input_noise_width : 1.0;
  ...
  }

Error: Line 18, The 'min_input_noise_width' attribute is larger than
the 'min_input_noise_width' attribute for the pin '%s'.  (LBDB-623)
```

This error message also occurs when any of the following are true:

- The value of **output_signal_level_high** is greater than **output_signal_level_low**.
- The value of **output_signal_level_high** is greater than **output_signal_level**.
- The value of **input_signal_level_high** is greater than **input_signal_level_low**.
- The value of **input_signal_level_high** is greater than **input_signal_level**.

### What Next

Change all related values to conform to the requirement and rerun the command.

---

## LBDB-624

(error) The output/inout pin '%s' has illegal 'tied-off' timing arcs.

### Description

If a output pin is not tied to "high"(the function attribute is set to "1") or "low"(the function attribute is set to "0"), its time arcs should not set the 'tied-off' attribute to true.

The following example shows an instance where this message occurs:

```
pin(S) {
   function : "A B";
   ...
    timing() {
      tied_off : true;
      steady_state_current_high(table_1) {
          index_1 ("0.00, 1.00, 2.50") ;
          values (\
                  "4.0, 5.0, 6.0"\
                  ...
          );
      }
      ...
  }
```

Feedback

```
need to delete the 'tied-off' attribute.
```

The following is an example message:

```
Warning: Line 160,  The output pin 'S' has illegal 'tied-off' timing
 arcs. (LBDB-624)
```

### What Next

Delete the 'tied-off' attribute.

---

## LBDB-625

(error) The output/inout pin '%s' has a 'tied-off' timing arc containing illegal '%s' %s.

### Description

If a output pin is tied to "high"(the function attribute is set to "1"), only steady_state_current_high group is allowed in its timing arc whose 'tied_off' attribute is set to true.

The following example shows an instance where this message occurs:

```
pin(S) {
   function : "1";
   ...
    timing() {
      tied_off : true;
      steady_state_current_high(table_1) {
          index_1 ("0.00, 1.00, 2.50") ;
          values (\
                  "4.0, 5.0, 6.0"\
                  ...
          );
      }
      ...
  }

need to delete the 'steady_state_current_high' table.
```

The following is an example message:

```
Warning: Line 160,  The output pin 'S' has a 'tied-off' timing arc
 containing illegal 'steady_state_current_high' group.  (LBDB-625)
```

### What Next

Delete the illegal noise_immunity/noise_propagation groups.

Feedback

## LBDB-626

(error) The 'tied-off' timing arc contains illegal noise_immunity/noise_propagation group.

### Description

In a timing arc whose tied_off attribute is set to true, for noise information, only steady_state_current_low group or steady_state_current_high group is allowed

The following example shows an instance where this message occurs:

```
pin(CP) {
   ...
    timing() {
      tied_off : true;
      noise_immunity_high(table_1) {
          index_1 ("0.00, 1.00, 2.50") ;
          index_2 ("0.00, 1.00, 2.50") ;
          values (\
                  "4.0, 5.0, 6.0"\
                  ...
          );
      }
      ...
  }

need to delete the 'noise_immunity_high' table.
```

The following is an example message:

```
Warning: Line 160, the 'tied-off' timing arc containts illegal
 noise_immunity/noise_propagation group (LBDB-626)
```

### What Next

Delete the illegal noise_immunity/noise_propagation groups.

## LBDB-627

(error) The '%s' group of the '%s' timing arc\n \tis not referencing the compatible template.

### Description

In the specified min_pulse_width timing arc, the only valid templates are as follows:

1. "constrained_pin_transition" (one-dimensional)
2. "related_pin_transition" (one-dimensional)
3. "constrained_pin_transition" + "related_out_total_output_net_capacitance"
4. "related_pin_transition" + "related_out_total_output_net_capacitance"

Feedback

The following example shows an instance where this message occurs:

```
lu_table_template  (table_1)   {
   variable_1  :  related_pin_transition  ;
   index_1 ("0.10, 0.25, 0.50") ;
}
lu_table_template  (table_2)   {
   variable_1  :  related_out_total_output_net_capacitance;
   index_1 ("0.10, 0.25, 0.50") ;
}

 ...

    timing() {
      related_pin : CP ;
      timing_type : min_pulse_width ;
      rise_constraint(table_2) {
          index_1 ("0.50, 1.00, 2.50") ;
          values (\
                  "4.0, 5.0, 6.0"\
          );
      }
      fall_constraint(table_1) {
          index_1 ("0.50, 1.00, 2.50") ;
          values (\
                  "4.0, 5.0, 6.0"\
          );
      }
```

change the template table_2 to

```
lu_table_template  (table_2)   {
   variable_1  :  constrained_pin_transition  ;
   index_1 ("0.10, 0.25, 0.50") ;
}
```

The following is an example message:

```
Error: Line 160, The 'rise_constraint' group of the 'min_pulse_width'
 timing arc\n
\tis not referencing the compatible template.  (LBDB-627)
```

**What Next**

Change the variable in the template accordingly.

---

## LBDB-628

(error) The value of timing_sense attribute is invalid for \n \tthe timing arc whose timing_sense should be '%s' instead.

Feedback

### Description

The timing arc with feed_through_type attribute must have 'positive_unate' timing_sense.

The following example shows an instance where this message occurs:

```
...
        pin(Z) {
          direction : inout;
          timing() {
            feed_through_type : short;
            timing_sense : negative_unate;
            ...
          }
        }
        ...
```

In this case, the timing sense should be "positive_unate".

The following is an example message:

```
Error: Line 256, The value of timing_sense attribute is invalid for the
        timing arc whose timing_sense should be 'positive_unate' instead.
 (LBDB-628)
```

### What Next

Check the timing arc and make the correction accordingly.

---

## LBDB-629

(error) The parent '%s' of the user-defined attribute is not defined.

### Description

For define(name, parent, type), parent must be the name of a user-defined group or a predefined library group.

The following example shows an instance where this message occurs:

```
library(test) {
        define(a, test, string);
        ...
    }
```

Since "test" is not defined, we can define a new user-defined group call "test" by adding the following statement: define_group(test, library);

The following is an example message:

```
Error: Line 256, The parent 'test' of the user-defined attribute is not
 defined. (LBDB-629)
```

Feedback

### What Next

Either change the parent name or define a new user-defined group with the parent name.

---

## LBDB-630

(error) The 'user_parameters' group is not specified.

### Description

This message indicates that the 'user_parameters' group is not specified. Library Compiler errors out because some generic parameters(parameter1..5) are used in either poly_template or power_poly_template, and the 'user_parameters' must be defined to contain the definition of those generic parameters.

The following example shows an instance where this message occurs:

```
user_parameters() {
      parameter1 : 0.80;
    }
```

The following is an example message:

```
Error: Line 52, The 'user_parameters' group is not specified. (LBDB-630)
```

### What Next

Add the missing attribute to the 'user_parameters' group. the message.

---

## LBDB-631

(error) The '%s' attribute in the 'user_parameters' group is not specified.

### Description

This message indicates that an attribute is missing, and Library Compiler errors out because this attribute is used in either poly_template or power_poly_template.

The following example shows an instance where this message occurs:

```
user_parameters() {
      parameter1 : 0.80;
    }
```

The following is an example message:

```
Error: Line 52, The 'parameter2' attribute in  the 'user_parameters'
 group is not specified. (LBDB-631)
```

Feedback

### What Next

Add the missing attribute to the 'user_parameters' group. the message.

---

## LBDB-632

(error) The power rail name '%s' is invalid.

### Description

This message indicates that you specified an invalid name for the current power rail. The name should not conflict with the generic parameters(parameter1..paramter5).

The following example shows an instance where this message occurs:

```
power_supply() {
          default_power_rail : VDD0;
          power_rail(parameter1, 4.95);
 }
```

The following is an example message:

```
Error: Line 104, The power rail name 'parameter1' layer name is invalid.
 (LBDB-632)
```

### What Next

User a different name that is not conflicting with parameter1..paramter5.

---

## LBDB-633

(error) The %s value, '%s', has not been\n \tdefined in the %s.

### Description

This message indicates that the referred value is undefined in the library.

The following example shows an instance where this message occurs:

```
part(a) {
    valid_speed_grades ("A","B","C");
    valid_step_levels ("step0","step1","step3");
    default_step_level : "step0";
    speed_grade("D") {
        step_level("step0");
    }
  }
```

In this case, the "D" valid_speed_grade is not defined.

Feedback

The following is an example message:

```
Error: Line 57, The valid_speed_grade value, 'D',
       has not been defined in the part. (LBDB-74)
```

**What Next**

Add the referred value in the library.

---

## LBDB-634

(error) The "variables" definition is conflicting with the main template.

**Description**

This message indicates that you specified an invalid 'variables' for the current poly template domain. It's required that a poly template define the same 'variables' as that of its main poly template.

The following example shows an instance where this message occurs:

```
poy_template(T3) {
       variables (input_net_transition, voltage, temperature);
       . . .
    domain(D1) {
       variables (input_net_transition, voltage);
       . . .
    }
 }
```

The following is an example message:

```
Error: Line 124, The "variables" definition is conflicting with its main
 template. (LBDB-634)
```

**What Next**

User the same 'variables' definition as that of its main poly template.

---

## LBDB-635

(error) The 'tied-off' timing arc contains illegal related_pin attribute.

**Description**

In a timing arc whose tied_off attribute is set to true, related_pin attribute is not allowed.

The following example shows an instance where this message occurs:

```
pin(CP) {
   ...
```

Feedback

```
    timing() {
      tied_off : true;
      ...
      related_pin : A;
      }
      ...
  }

need to delete the 'related_pin' attribute.
```

The following is an example message:

```
Warning: Line 160, the 'tied-off' timing arc containts illegal
 related_pin attribute. (LBDB-635)
```

**What Next**

Delete the related_pin attribute.

---

## LBDB-636

(error) The '%s' attribute value is invalid for the timing arc.

**Description**

This message indicates for a timing arc that the attribute value is invalid. For example, clock_gate_check attribute can only be defined to "true" for the setup/hold/nochange timing arcs.

**What Next**

Remove the attribute from the timing arc.

---

## LBDB-637

(error) The %s-associated '%s' lookup table\n \tcannot use '%s' as its template.

**Description**

This message can be used as generic error message template. For frequency-base max_cap, it means that: 1. the max_cap lookup table in an input-associated max_cap group must use a one-dimensional template with *frequency* as its variable. 2. the max_cap lookup table in an ouput-associated max_cap group must use a one-dimensional template with *frequency* as its variable, or a two-dimensional template with *frequency* and *total_output_net_capacitance* as its variables with *frequency* being the first variable.

The following example shows an instance where this message occurs:

```
maxcap_lut_template(ok_temp) {
    variable_1 : frequency;
```

Feedback

```
    index_1("0, 1, 2, 3");
  }
  maxcap_lut_template(err_temp) {
    variable_1 : input_transition_time;
    index_1("0, 1, 2, 3");
  }

        max_cap(err_temp) {
                values ("1, 2, 3, 4");
        }
```

To fix the problem, change the template value of the *max_cap* attribute from *err_temp* to *ok_temp*.

The following is an example message:

```
Error: Line 126, The input-associated 'max_cap' lookup table
       cannot use 'err_temp' as its template. (LBDB-637)
```

**What Next**

For more information about lookup tables, see the *Library Compiler User Guide*. Change the library source file by referencing a different template in the lookup table description.

---

## LBDB-638

(error) It is invalid to specify the '%s' %s \n \ton an %s pin.

**Description**

This message indicates that you specified a invalid group on a pin.

The following example shows an instance where this message occurs:

```
pin (p) {
    direction : input;
    ...
    max_cap (test) {
      ...
    }
    ...
  }
```

The following is an example message:

```
Error: Line 67, It is invalid to specify the 'max_cap' group
        on an input pin. (LBDB-638)
```

**What Next**

Check the library source file, and correct the problem.

Feedback

## LBDB-639

(warning) The '%s' %s in this timing arc are ignored.

### Description

This message indicates that you specified a redundant attribute or group for the sequential half-nate timing arcs.

In the following example, both the cell_fall group and the fall_transition group are redundant for the sequential half-unate timing arc.

```
timing() {
  timing_type : rising_edge;
  timing_sense : positive_unate;
  cell_rise( f_ocap ){
    index_1 ( "0.0000, 1.0000, 512.0000, 7999.9995, 15999.9990,
 31999.9980, 63999.9961");
    values ( "25.0000, 35.0000, 5145.0000, 80024.9922, 160024.9844,
 320024.9688, 640024.9375");
  }
  rise_transition( f_ocap ){
    index_1 ( "0.0000, 1.0000, 512.0000, 7999.9995, 15999.9990,
 31999.9980, 63999.9961");
    values ( "15.0000, 25.0000, 5135.0000, 80014.9922, 160014.9844,
 320014.9688, 640014.9375");
  }
  cell_fall( f_ocap ){
    index_1 ( "0.0000, 1.0000, 512.0000, 7999.9995, 15999.9990,
 31999.9980, 63999.9961");
    values ( "25.0000, 35.0000, 5145.0000, 80024.9922, 160024.9844,
 320024.9688, 640024.9375");
  }
  fall_transition( f_ocap ){
    index_1 ( "0.0000, 1.0000, 512.0000, 7999.9995, 15999.9990,
 31999.9980, 63999.9961");
    values ( "15.0000, 25.0000, 5135.0000, 80014.9922, 160014.9844,
 320014.9688, 640014.9375");
  }
}

Warning: Line 639, The cell_fall group in this
   timing arc are ignored. (LBDB-390)
Warning: Line 639, The fall_transition group in this
   timing arc are ignored. (LBDB-390)
```

### What Next

Remove the redundant groups from the timing arc and rerun the command.

Feedback

## LBDB-640

(error) The '%s' attribute %s \n \tin this %s group.

### Description

This message indicates you need to specify or remove an attribute from the group according to the template it references as follows: 1. 1-dimentional template does not allow the 'related_pin' attribute. 2. 2-dimentional template requires the 'related_pin' attribute.

The following example shows an instance where this message occurs:

```
max_cap(maxcap_1d_template) {
                values ("1, 2, 3, 4");
                related_pin : "A";
        }
```

In this case, the 'related_pin' attribute is not allowed in 1-dimensional lookup table. To fix the problem, remove the related_pin attribute from the max_cap group.

The following is an example message:

```
Error: Line 69, The 'related_pin' attribute cannot be specified
        in this max_cap group. (LBDB-640)
```

### What Next

Change the technology library source file to add or delete the specified attribute.

## LBDB-641

(error) The '%s' %s has some %s groups \n \tmissing the %s attribute.

### Description

This message indicates there is one or more leakage_power(internal) groups without the power_level attribute. When at least one leakage_power(internal_power) is specified with the power_level attribute in a cell(pin), all the leakage_power(internal_power) in the cell must also have the power_level attribute defined.

It can also be used to notify a user that teh related_pg_pin attribute is missing in the leakage_power/internal_power groups.

The following example shows an instance where this message occurs:

```
cell(AN2) {
      leakage_power () {
        power_level : VDD1;
        value : 1.0;
      }
```

Feedback

```
      leakage_power () {
        /* power_level : VDD2; */
        value : 2.0;
      }
      ...
  }
```

In this case, the power_level attribute is missing for the 2nd leakage_power. To fix the problem, add the attribute to the leakage_power group:

```
cell(AN2) {
      leakage_power () {
        power_level : VDD1;
        value : 1.0;
      }
      leakage_power () {
        power_level : VDD2;
        value : 2.0;
      }
      ...
  }
```

The following is an example message:

```
Error: Line 12, The 'AN2' cell has some leakage_power groups
        missing the power_level attribute. (LBDB-641)
```

### What Next

Add the missing attribute for the leakage_power group.

---

## LBDB-642

(warning) The group '%s' is defined multiple times\n \tin group '%s'. Using the last one encountered.

### Description

Some groups require that no more than one same group be defined in their scope.

Some groups require when statement, and only one of them can be defined without when statement in their scope.

The following example shows an instance where this message occurs:

```
pin_capacitance ()  {
     rise_capacitance_range() { ... }
     rise_capacitance_range() { ... }
   }
```

Feedback

The following is the example message:

```
Warning: Line 987, The group 'rise_capacitance_range' is defined multiple
 times
        in group 'pin_capacitance'. Using the last one encountered.
 (LBDB-642)
```

The following example shows another instance where this message occurs:

```
cell(test) {
   ...
   leakage_current ()  {
     when : A;
     pg_current(VDD) {
       value : 5.3;
     }
   }
   leakage_current ()  {
     when : A1;
     pg_current(VDD) {
       value : 15.3;
     }
   }
   ...
   leakage_current ()  {
     pg_current(VDD) {
       value : 0.3;
     }
   }
   leakage_current ()  {
     pg_current(VDD) {
       value : 8.3;
     }
   }
   ...
Only one of leakage_current groups with no 'when' statement can be
defined under a cell.  If there are more than one group with no
'when' statement defined, then they are duplicated groups.
```

The following is the example message:

```
Warning: Line 138, The group 'leakage_current' is defined multiple times
        in group 'cell'. Using the last one encountered. (LBDB-642)
```

**What Next**

Remove the extra objects.

---

## LBDB-643

(error) '%s' is used more than once inside cell '%s'.

Feedback

### Description

This error message occurs when unique *power_gating_pin* attribute value is specified more than once within a cell.

The following example shows power_pin_1 specified twice within a cell and the resulting error message:

```
cell (BUF) {
    pin(A) {
      power_gating_pin (power_pin_1, "0") ;
      direction : input;
    }
    pin(B) {
      power_gating_pin (power_pin_1, "0") ;
      direction : output;
    }
  }

Error: Line 56, 'power_pin_1' is used more than once inside cell 'BUF'.
(LBDB-643)
```

### What Next

Make sure that each unique *power_gating_pin* value is specified only once.

---

## LBDB-644

(warning) The cell_leakage_power attribute of the '%s' cell is redundant\n \tand not used in the leakage_power modeling.

### Description

This message indicates one of the following two cases are true: 1. there is power_level specific leakage_power groups in the cell 2. the cell has no power_level specific leakage_power groups, and it has default leakage_power group(leakage_power group without 'when' statement).

The following example shows an instance where this message occurs:

```
cell(AN2) {
      cell_leakage_power : 0.5;
      leakage_power () {
        power_level : VDD1;
        when : "A";
        value : 1.0;
      }
      ...
  }
```

Feedback

In this case, the cell has power_level specific leakage_power groups, thus the cell_leakage_power attribute is redundant. To fix the problem, remove the attribute from the cell group.

The following example shows another instance where this message occurs:

```
cell(AN2) {
      cell_leakage_power : 0.5;
      leakage_power () {
        when : "A";
        power(LKP_T) {
           orders ("1, 1");
           coefs  ("1, 1, 1, 1");
           domain (D1) {
                 orders ("1, 1");
                 coefs  ("1, 1, 1, 1");
           }
        }
      }
      ...
  }
```

In this case, a cell_leakage_power attribute is redundant. To fix the problem, remove the attribute from the cell group:

Warning: Line 12, The cell_leakage_power attribute of the 'AN2' cell is redundant and not used in the leakage_power modeling. (LBDB-644)

**What Next**

Remove the 'cell_leakage_power' from the cell.

---

## LBDB-645

(error) Cannot find the '%s' %s in the %s.

**Description**

This error message occurs when a power_rail name attached to a *related_power_rail* or *rail_value* attribute does not exist in the power_supply.

This error message also occurs when the *default_power_rail* attribute is not defined in the power_supply group.

The following code does not contain VDD4 in the power_supply, which results in the error message.

```
library(libg5) {
    ...
    power_supply(pw) {
      default_power_rail : VDD;
```

Feedback

```
        power_rail (VDD1, 2.0);
        power_rail (VDD2, 2.5);
        power_rail (VDD3, 3.0);
        power_rail (VSS1, 3.0);
        power_rail (VSS2, 3.0);

        mapping(VSS1) {
                related_power_rail : "VDD1 VDD2";
        }
        mapping(VSS2) {
                related_power_rail : "  VDD4 VDD3       ";
        }
      }

 }

Error: Line 403, Cannot find the 'VDD4' power_rail in the power_supply.
(LBDB-645)
```

### What Next

Check your library for an incorrect power_rail name, correct the name, and run the command again.

---

## LBDB-646

(error) No %s information is defined before the %s '%s'.

### Description

This error message occurs when you have not defined the power_supply statement for the power_level.

This message also occurs when the power_supply group is missing for a cell rail_connections because some pins of the cell do not define the *input_signal_level* (for an input pin) or the *output_signal_level* (for an output pin).

The following example shows a library with a missing power_supply statement before the cell containing the power_level attribute and the resulting error message.

```
library(lib) {
  ...
  cell (A)
    ...
    leakage_power() {
        power_level : "VDD";
        ...
    }
    ...
 }
```

Feedback

```
Error: Line 191, No power_supply information is defined before the
power_level 'VDD'. (LBDB-646)
```

### What Next

Add the power_supply statement before the cell statement that contains the power_level, as shown in the following example:

```
power_supply() {
        default_power_rail : VDD;
        power_rail (VDD, 2.0);
        ...
  }
```

---

## LBDB-647

(warning) Found the obsolete and unsupported '%s' %s in the '%s' cell; use '%s' instead.

### Description

The *rail_connection* attribute is not supported after 2004.12 Beta version. Instead, the information should now be described in the power_pin group.

The following is an example of the error message:

```
Warning: Line 159, Found the obsolete and unsupported 'rail_connection'
attribute in the 'sample' cell; use 'power_pin' group instead. (LBDB-647)
```

### What Next

If you have access to the technology library source file, change the *rail_connection* attribute to power_pin group. Otherwise, contact the vendor and inform them of the problem.

---

## LBDB-648

(error) The power_pin definitions of cell '%s' do not follow the mapping rules specified in the power_supply group.

### Description

This error message occurs when the definitions of the power_pins do not follow the mapping rules in the power_supply group. The following is an example of this error message:

```
Warning: Line 159, The power_pin definitions of cell '%s' do not follow
the mapping rules in the power_supply group. (LBDB-648)
```

Feedback

### What Next

Either change the mapping rules in the power_supply group or redefine the power_pins and run the command again.

---

## LBDB-649

(error) The '%s' domain's power rail mapping information conflicts with its main template.

### Description

This error message occurs when the power rail mapping information is incorrectly defined in the domain of the poly_template. The information in poly_template domain must be exactly the same as the poly_template.

The following example shows the *d1* domain of the *p1* poly_template with the incorrect mapping statement and the resulting error message:

```
poly_template(p1) {
  ...
  mapping(voltage1, VDD1);
  mapping(voltage, VDD0);
  ...
  domain(d1) {
    ...
    mapping(voltage1, VDD2);
    mapping(voltage, VDD0);
    ...
  }
}

Error: Line 191, The 'd1' domain's power rail mapping information
conflicts with its main template. (LBDB-649)
```

### What Next

Modify the mapping statements in this domain of the poly_template to match those of the poly_template.

The following example shows the correct *d1* domain mapping statements:

```
mapping(voltage1, VDD1);
    mapping(voltage, VDD0);
```

---

## LBDB-650

(error) The '%s' cell is not a valid %s.

Feedback

### Description

This error message occurs when the level shifter, isolation cell or clock isolation cell is incorrectly modeled. A level shifter or an isolation cell must satisfy one of the following sets of conditions:

- It has 3 pins, including 1 input pin, 1 level_shifter_enable_pin/ isolation_cell_enable_pin, 1 output pin.
- It has 2 pins, including 1 input pin and 1 output pin.

A clock isolation cell must satisfy:

- It has 3 pins, including 1 clock_isolation_cell_clock_pin, 1 isolation_cell_enable_pin, 1 output pin.

  The following example shows the level shifter with the incorrect modeling and the resulting error message:

```
cell(LS) {
 is_level_shifter : true;
 ...
 pin(a) {
        ...
 }
 pin(b) {
        ...
 }
 pin(c) {
        ...
 }
 pin(d) {
        ...
 }
}
Error: Line 191, The 'LS' cell is not a valid level shifter.
 (LBDB-650)
```

### What Next

Correct the modeling information of the level shifter or isolation cell to satisfy the requirements.

You could correct the modeling in the previous example by removing one of the pins specified in the cell.

---

## LBDB-651

(error) The %s%s '%s' contains the \n \tconflicting '%s'/'%s' values.

Feedback

### Description

This error message occurs when the *input_signal_level* and *output_signal_level* values are modeled incorrectly for the level shifter or isolation cell.

For a level shifter, *input_signal_level* in input pin can't be the same as *output_signal_level* in output pin. Meaning that the rail names can't be the same in a level shifter.

For an isolation cell, the value of *input_signal_level* in input pin can't be different from the value of *output_signal_level* in output pin. Meaning that the rail values can't be difference in an isolation cell.

The following example shows the level-shifter with the incorrect modeling and the resulting error message:

```
cell(LS) {
 is_level_shifter : true;
 ...
 pin(in) {
        ...
        input_signal_level : VDD2
 }
 pin(out) {
        ...
        output_signal_level : VDD2
 }
}

Error: Line 191, The pins of level shifter 'LS' contain the
conflicting 'input_signal_level'/'output_signal_level' values. (LBDB-651)
```

The following example shows the isolation cell with the incorrect modeling and the resulting error message:

```
library(libdb651) {
  ...
  /* operation conditions */
  operating_conditions(5v_1v) {
    ...
    power_rail (VDDH, 5);    /* high power */
    power_rail (VDDL, 1);    /* low power */
    ...
  }
  ...
  default_operating_conditions : 5v_1v;

  ...
cell(ISO) {
 is_isolation_cell : true;
 ...
 pin(in) {
        ...
```

Feedback

```
        input_signal_level : VDDH
 }
 pin(out) {
        ...
        output_signal_level : VDDL
 }
}

Error: Line 191, The pins of isolation cell 'ISO' contain the
conflicting 'input_signal_level'/'output_signal_level' values. (LBDB-651)
```

### What Next

For the first example, modify the modeling information of the level shifter to meet the requirements.

To correct the modeling of the example above, make the *input_signal_level* string "VDD1" of the input pin different from the *output_signal_level* string "VDD2" of the output pin as follows:

```
cell(LS) {
 is_level_shifter : true;
 ...
 pin(in) {
        ...
        input_signal_level : VDD1
 }
 pin(out) {
        ...
        output_signal_level : VDD2
 }
}


For the second example, modify the modeling information of the isolation
 cell
to meet the requirements.

To correct the modeling of the example above, make the
input_signal_level value "5" of the input pin
same as the output_signal_level value "1" of the
output pin.  You can assiged both to the same pin like
"VDDH", so that both will refer to same value "5" in
operating_conditions as follows.

library(libdb651) {
  ...
  /* operation conditions */
  operating_conditions(5v_1v) {
    ...
    power_rail (VDDH, 5);    /* high power */
    power_rail (VDDL, 1);    /* low power */
```

Feedback

```
    ...
  }
  ...
  default_operating_conditions : 5v_1v;

  ...
cell(ISO) {
 is_isolation_cell : true;
 ...
 pin(in) {
        ...
        input_signal_level : VDDH
 }
 pin(out) {
        ...
        output_signal_level : VDDH
 }
}
```

## LBDB-652

(error) The %s '%s' has incompatible '%s' and '%s'.

### Description

This error message occurs when the user specified attributes are incompatible. For example, specifying both *pulse_clock* and *generated_clock* inside a cell.

```
Error: Line 159, The cell 'A' has incompatible 'pulse_clock'
and 'generated_clock'.  (LBDB-652)
```

### What Next

Either remove the *pulse_clock*attribute or the *generated_clock* group for the cell.

## LBDB-653

(error) The index of '%s' can define only one value.

### Description

This error message occurs because the specified index cannot contain more than one value. This restriction applies to the dimension input_net_transition, and total_output_net_capacitance in the vector group.

The following example shows an index with more than one value and the resulting error message.

```
output_current_template(CCS_T) {
    variable_1 : input_net_transition;
```

Feedback

```
    variable_2 : total_output_net_capacitance;
    variable_3 : time;
  }
  vector(CCS_T) {
        reference_time : 0.5;
        index_1 ("0.12");
        index_2 ("0.1, 1.2");
        index_3 ("0.01, 0.02, 0.03");
        values (" ... ");
  }

Error: Line 46, The index of 'total_output_net_capacitance' can
only define one value. (LBDB-653)
```

To correct the error, change index_2 to ("0.1");

**What Next**

Check the library source file and remove the extra values in the index definition.

---

## LBDB-654

(error) The values of '%s' and '%s' are not consistent in %s '%s'.

**Description**

This error message occurs when the values of the *reference_time* and *input_net_transition* attributes are inconsistent inside the output_current_rise or output_current_fall group. Inside each output_current_rise or output_current_fall group, only the same pair of *reference_time* and *inout_net_transition* values is allowed.

Note: In 2007.03 release, the message is enhanced to also report inconsistent input_voltage_range and output_voltage_range values in a level shifter. Assuming that a level shifter defines the following information: level_shifter_type : <lst_val>; input_voltage_range(<<ivrl_val>, <ivrh_val>); output_voltage_range(<ovrl_val>, <ovrh_val>); then it must satisfy the following requirements: - if <lst_val> = HL, then <ivrh_val> > <ovrl_val>. - if <lst_val> = LH, then <ivrl_val> < <ovrh_val>. Otherwise, LC will issue LBDB-654 error.

The following is an example of inconsistent *reference_time* and *input_net_transition* values and the resulting error message:

```
output_current_template(CCT) {
  variable_1 : input_net_transition;
  variable_2 : total_output_net_capacitance;
  variable_3 : time;
 }
 . . .
 output_current_rise() {
   vector(CCT) {
```

Feedback

```
      reference_time : 0.11; <=== reference_time = 0.11 with
 input_net_transition = 0.1
      index_1 ("0.1");
      index_2 ("1");
      index_3 ("1, 2, 3");
      values ("1, 2, 3");
   }
    vector(CCT) {
      reference_time : 0.12; <=== reference_time = 0.12 with
 input_net_transition = 0.1
      index_1 ("0.1");
      index_2 ("2");
      index_3 ("1, 2, 3");
      values ("1, 2, 3");
   }
 }

Error: Line 98, The values of 'reference_time' are not consistent in
group 'output_current_rise'. (LBDB-654)
```

### What Next

Check the library source file, and correct the values of the *reference_time* attribute.

---

## LBDB-655

(error) The vectors are not dense in group '%s'.

### Description

This error message occurs because Library Compiler requires a "complete grid" of vector groups inside each output_current_rise or output_current_fall group. The vector groups must form a dense MxN grid, with M unique input_net_transition values, and N unique total_output_net_capacitance values. The vector groups should have exact MxN number and any duplicate input_net_transition and total_output_net_capacitance pair will cause this error.

A similar dense grid requirement is also imposed on the output_voltage_rise and output_voltage_fall groups of input_ccb and output_ccb. (This requirement does not apply to those of ccsn_first_stage and ccsn_last_stage).

The following example shows an output_current_rise group without a dense vector and resulting error message.

```
output_current_template(CCT) {
  variable_1 : input_net_transition;
  variable_2 : total_output_net_capacitance;
  variable_3 : time;
 }
 . . .
 output_current_rise() {
```

Feedback

```
   vector(CCT) {
      reference_time : 0.11;
      index_1 ("0.1");
      index_2 ("1");
      index_3 ("1, 2, 3");
      values ("1, 2, 3");
   }
    vector(CCT) {
      reference_time : 0.11;
      index_1 ("0.2");
      index_2 ("2");
      index_3 ("1, 2, 3");
      values ("1, 2, 3");
   }
   /* need vectors for (0.1, 2) and (0.2, 1) */
 }

Error: Line 198, The vectors are not dense in group
 'output_current_rise'.
(LBDB-655)
```

### What Next

Check the library source file and add a vector for each input_net_transition and total_output_net_capacitance pair.

---

## LBDB-656

(error) The lu_table_template '%s' referred by '%s' in line %d is invalid.

### Description

This error message occurs because the variable of the referred lu_table_template can only include the *input_net_transition* variable for the pin based receiver model. For the timing arc based receiver model, the variable of the referred lu_table_template can include *input_net_transiton* and *total_output_net_capacitance*.

The following example shows an invalid lu_table_template reference and the resulting error message:

```
lu_table_template(T1) {
  variable_1 : input_net_transition;
  variable_2 : total_output_net_capacitance;
  . . .
 }
 . . .
 pin(A) {

 receiver_capacitance() {
   receiver_capacitance1_rise(T1) {
      index_1 ("0.1, 0.1");
```

Feedback

```
      index_2 ("1, 2");
      values ("1, 2", "2.5, 3");
   }
   . . .
 }
```

```
Error: Line 298, The lu_table_template 'T1' referred by
 'receiver_capacitance1_rise' in line 1239 is invalid. (LBDB-656)
```

### What Next

Check the library source file, and correct the variable(s) of the corresponding lu_table_template.

---

## LBDB-657

(error) Conflicting receiver model found in pin '%s' (pin-based) and in timing arc '%s-%s' (arc-based).

### Description

This error message occurs because when an input (or inout) pin A defines a receiver_capacitance() group, then no timing arc whose from pin is A, can define receiver_capacitance1_rise/receiver_capacitance1_fall/ receiver_capacitance2_rise/ receiver_capacitance2_fall.

The following is an example of an incorrect definition and the resulting error message:

```
pin(A) {
   direction : input;
   receiver_capacitance() { . . .}
   . . .
 }
 pin(Y) {
   direction : output;
   timing() {
     receiver_capacitance1_rise(T1) { . . .}
     receiver_capacitance1_fall(T1) { . . .}
     receiver_capacitance2_rise(T1) { . . .}
     receiver_capacitance2_fall(T1) { . . .}
     . . .
   }
   . . .
 }
```

Error: Line 398, Conflicting receiver model found in pin 'A' (pin-based) and in timing arc 'A-Y' (arc-based). (LBDB-657)

Feedback

### What Next

Check the library source file and remove the duplicated receiver model. Usually it is best practice to use the timing arc based receiver model.

---

## LBDB-658

(error) Both variable and range of poly_template '%s' should be defined before domain '%s' is defined.

### Description

You receive this message a domain can't be defined because either the variable or range statement of the main poly_template is missing.

The following example shows an instance where this message occurs: The following example shows the *p1* poly_template is missing both variable and range information.

```
poly_template(p1) {
 poly_template(PT)  {
  domain(D1)  {
    variables("temperature, total_output_net_capacitance");
    variable_1_range(0, 40);
    variable_2_range(1, 2);
  }
  ......
}
```

Correct it by add *variables, variable_1_range and variable_2_range statement in the poly_template group.*

The following is an example message: Error: Line 191, Both variable and range of poly_template 'PT' should be defined before domain 'D1' is defined. (LBDB-658)

### What Next

Make sure that *variables* and all *variable_n_range* statements exist before defining domains.

---

## LBDB-659

(error) The %s %s cannot coexist\n \twith the '%s' %s on the '%s' pin.

### Description

The first specified attribute/group in the pin cannot coexist with the second specified attribute/group on the same pin. For example, the *is_three_state* attribute cannot be specified on a pin with *three_state* attribute.

Feedback

The following example shows an instance where this message occurs:

```
cell(lbdb239) {
    area : 2;
    pin(A) {
      direction : input;
      capacitance : 1.0;
    }
    ...

    pin(Z) {
      direction : output;
      function : "A";
      three_state : "B"

      is_three_state : false; /* error */
      ...
    }
  }
```

The following is an example message:

```
Error: Line 84, The  "false" 'is_three_state' attribute cannot coexist
        with the 'three_state' attribute on the 'Z' pin. (LBDB-659)
```

**What Next**

Check the specification of the cell, and make the appropriate change to the *driver_type* attribute or other attributes in the faulty pin group.

---

## LBDB-660

(error) It is invalid to specify the '%s' %s on the '%s' %s%s.

**Description**

This message indicates that you have specified an invalid attribute or group for a cell.

The following example shows an instance where this message occurs:

```
cell (p) {
    clocked_cell : rising_edge;
    pin(clk) {
      clock : true;
    }

    pin(out) {
      function : "clk";
    }
  }
```

To correct the error, add at least 1 combinational timing arc between the *clk* and *out* pins.

Feedback

The following is an example message:

```
Error: Line 67, It is invalid to specify the 'clocked_cell' attribute on
 the 'p' cell. (LBDB-660)
```

### What Next

Check the library source file, and ensure that the following conditions are met:

- The *is_isolated attribute* is specified only on the macro-cell pins.
- When the *clocked_cell* attribute is specified, the corresponding cell has
  - only 1 clock-pin, with the *clock* attribute set to *true*.
  - at least 1 timing arc between the clock pin and an output pin.

---

## LBDB-661

(error) The 'generic' integrated clock gating cell should not define the 'statetable' group without 'ff/latch' group.

### Description

This message indicates that you specified the 'generic' integrated gating clock cell with 'statetable' definition without 'ff/latch' group definition. You should define 'statetable' group with 'ff' or 'latch' group.

The following example shows an instance where this message occurs:

```
cell(CGNP) {
  area : 1;
  clock_gating_integrated_cell : "generic";

  statetable(" CP EN ", "IQ ") {
     table : " L  L  : - : L ,\
               L  H  : - : H ,\
               H  -  : - : N ";
  }
  ...
}

The modified cell description should be :

cell(CGNP) {
  area : 1;
  clock_gating_integrated_cell : "generic";

  statetable(" CP EN ", "IQ ") {
     table : " L  L  : - : L ,\
               L  H  : - : H ,\
               H  -  : - : N ";
```

Feedback

```
  }
  latch("IQ","IQN") {
    enable : "CP'";
    data_in : "EN";
  }
  ...
}
```

The following is an example message:

```
Error: Line 206, The 'generic' integrated clock gating cell should not
 define the 'statetable' group without 'ff/latch' group.(LBDB-661)
```

**What Next**

Change the library source file, add 'ff' group or 'latch' group definition.

---

## LBDB-662

(warning) The %s is not defined. operating_conditions "nom_pvt" is created \n \tand set as the default_operating_conditions.

**Description**

This warning message occurs when the *default_operating_conditions* is undefined and there is eith no operating_conditions groups defined or multiple operating_conditions defined.

**What Next**

If you do not want Library Compiler to create the default operating_conditions for you, please check your library to define the relative operating_condition group and set it as the default_operating_conditions with the library-level attribute "default_operating_conditions".

---

## LBDB-663

(warning) The %s is not defined. operating_conditions '%s' is set\n \tas the default_operating_conditions.

**Description**

This warning message occurs when the *default_operating_conditions* is undefined and there is extactly 1 operating_conditions group defined in the library.

**What Next**

If you do not want Library Compiler to automatically set the default operating_conditions for you, please check your library to define the relative operating_condition group

Feedback

and set it as the default_operating_conditions with the library-level attribute "default_operating_conditions".

## LBDB-664

(error) The '%s' attribute of the '%s' has a value '%g',\n \twhich should be %s for the %s.

### Description

The value specified is equal to a threshold value 0.0 for the attribute.

The following example shows an instance where this message occurs:

```
output_current_rise() {
        vector(current_template_8x7) {
           reference_time : 0.0140609;
           index_1("0");
           index_2("0.019368");
           index_3("0.1462, 0.175, 0.19, 0.205, 0.235, 0.25, 0.265,
 0.275, 0.29, 0.305, 0.32, 0.35, 0.406591");
           values("0.0, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0, 1.1,
 1.2, 1.3");
        }
      }
```

The following is an example message:

```
Error: Line 1897, Cell 'A', pin 'X', The 'values' attribute of the
 'vector' has a value '0',
       which should be non-zero for the output_current_rise vector.
 (LBDB-664)
```

### What Next

Check the library source file and correct the problem.

## LBDB-665

(error) The %s value of the '%s' attribute of the '%s' is the peak value.

### Description

The frist value specified in the specified attribute is a peak value, i.e., the maximun value for the values attribute of the vector of an output_current_rise group, or the minimum value for the values attribute of the vector of an output_current_fall group.

The following example shows an instance where this message occurs:

```
output_current_rise(current_template_8x7) {
        vector(current_template_8x7) {
           reference_time : 0.0140609;
```

Feedback

```
          index_1("0");
          index_2("0.019368");
          index_3("0.1462, 0.175, 0.19, 0.205, 0.235, 0.25, 0.265,
 0.275, 0.29, 0.305, 0.32, 0.35, 0.406591");
          values("1.8, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0, 1.1,
 1.2, 1.3");
       }
     }
```

The following is an example message:

```
Error: Line 35, The first value of the 'values' attribute of the 'vector'
 is the peak value. (LBDB-665)
```

### What Next

Check the library source file and correct the problem.

---

## LBDB-666

(warning) The '%s' has the same %s %g as the '%s' at line %d, \n \tbut the %s does not increase(from %g to %g) \n \twith increasing %s (from %g to %g).

### Description

For a timing arc, within each output_current_rise/fall group. for a given total_output_net_capacitance(load), teh reference_times should increase with the increasing input_transition_time(slew). This check is only useful for the signoff timing analysis with the 2004.12 and the relative sevice pack releases of PrimeTime.

The following example shows an instance where this message occurs:

```
output_current_rise(current_template) {
               ...
              vector(current_template) {
                 reference_time : 0.0140609;
                 index_1("0.1"); /* load */
                 index_2("0.019368"); /* slew */
                 index_3("0.1462, 0.175, 0.19, 0.205, 0.235, 0.25,
 0.265, 0.275, 0.29, 0.305, 0.32, 0.35, 0.406591");
                 /* LBDB-664 */
                 values("0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9,
 1.0, 1.1, 1.2, 1.3");
              }
              vector(current_template) {
                 reference_time : 0.0319875;
                 index_1("0.1"); /* load */
                 index_2("0.049573"); /* slew */
                 index_3("0.160795, 0.185, 0.2, 0.23, 0.24, 0.255, 0.27,
 0.285, 0.295, 0.31, 0.325, 0.34, 0.355, 0.37, 0.4
175");
```

Feedback

```
                values("0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9,
 1.0, 1.1, 1.2, 1.3");
              }
               ...
        }
```

The following is an example message:

```
Warning : Line 357, The 'vector' has the same
 total_output_net_capacitance 0.000000 as the 'vector' at line 364,
        but the reference_time does not increase(from 0.031987 to
 0.026168)
        with increasing input_net_transition (from 0.049573 to 0.094881).
 (LBDB-666)
```

### What Next

Check the library source file and correct the problem.

---

## LBDB-667

(warning) This '%s' group has a value (%g) in 'vector/index_1' that does not match any values of 'index_1' in '%s' at line %u. Parent '%s' is at line %u.

### Description

In the referenced ccs noise data modeling, output_voltage_rise and output_voltage_fall are characterized in ways to be closer to timing characterization. Their "input slew" (vector/index_1) and "output load" (vector/index_2) values form a 2D grid, similar to various timing lookup tables.

To best match timing behavior, "input slew" values of all CCB's should match points given in index_1 of respective driver waveform. See also LBDB-979 for additional checks on CCB's involved in propagating noise.

The following example shows an instance where this message occurs:

```
library(test) {
  ...
  normalized_driver_waveform(ndw1) {
    driver_waveform_name : "driver_waveform_default_fall" ;
    index_1("0.003209, 0.0133113, 0.033516, 0.0726625, 0.152218");
    ...
  }
  ...
  cell (flop) {
    ...
    pin(A) {
      ...
      driver_waveform_fall : driver_waveform_default_fall ;
      ...
```

Feedback

```
        input_ccb(ccb1) {
          ...
          output_voltage_fall() {
            vector(vec1) {
              index_1("0.013209,...");
              ...
          }
          ...
        }
        ...
      }
      ...
    }
}
```

### Examples

Warning: Line 28818, Cell 'ICG', pin 'SE', This 'output_voltage_fall' group has a value (0.00863453) in 'vector/index_1' that does not match any values of 'index_1' in 'normalized_driver_waveform' at line 261. Parent 'input_ccb' is at line 28776. (LBDB-667)

### What Next

Correct the characterization procedure.

---

## LBDB-668

(warning) related_pin '(%s)', when '(%s)', slew/load '(%g/%g)', The final signal voltage of the %s is %g, which does not reach within the 5 percent of the rail voltage %g.

### Description

For each output_current_rise/fall vector, the final signal voltage must be within the 5% of the final rail voltage. This can be described with the following formulas:

If output_signal_level_low and output_signal_level_high are specified, volt_low = output_signal_level_low; volt_high = output_signal_level_high; Else volt_low = VSS; volt_high = VDD;

1. For the output_current_rise vector, Vfinal = volt_low + (0.5/Cout)*(I2+I1)*(T2-T1)+... +(0.5/Cout)*(In+In-1)*(Tn-Tn-1) and fabs(Vfinal - volt_high) must NOT be greater than 0.02 (volt_high - volt_low) 2. For the output_current_fall vector, Vfinal = volt_high + (0.5/Cout)*(I2+I1)*(T2-T1)+...+(0.5/Cout)*(In+In-1)*(Tn-Tn-1) and fabs(Vfinal - volt_low) must NOT be greater than 0.02 (volt_high - volt_low)

The following is an example message:

```
Error : Line 373, cell 'TEST', pin 'Q', related_pin '(CP)', when '(!D)',
 slew/load '(0.0088/0.0012)', The final signal voltage of the vector is
 1.200000, which does not reach within the 5 percent of the rail voltage
 1.000000. (LBDB-668)
```

Feedback

### What Next

Check the library source file and correct the problem.

---

## LBDB-669

(error) The final signal voltage of the %s is %g and it does not reach beyond the 2nd \n \tslew threshold voltage, which is %g for the %s.

### Description

For each output_current_rise/fall vector, the final signal voltage must reach the 2nd slew threshold(i.e., the maximum of the slew thresholds for the output_current_rise vectors, and the minimum of the slew thresholds for the output_current_fall vectors) This can be described with the following formulas:

If output_signal_level_low and output_signal_level_high are specified, volt_low = output_signal_level_low; volt_high = output_signal_level_high; Else volt_low = VSS; volt_high = VDD;

1. For the output_current_rise vector, Vfinal = volt_low + (0.5/Cout)*(I2+I1)*(T2-T1)+... +(0.5/Cout)*(In+In-1)*(Tn-Tn-1) Verr = volt_low + MAX(slew_lower_threshold_pct_rise, slew_upper_threshold_pct_rise, output_threshold_pct_rise) * (volt_high-volt_low) * 0.01

Vfinal must NOT be less than Verr. 2. For the output_current_fall vector, Vfinal = volt_high + (0.5/Cout)*(I2+I1)*(T2-T1)+...+(0.5/Cout)*(In+In-1)*(Tn-Tn-1) Verr = volt_low + MIN(slew_lower_threshold_pct_fall, slew_upper_threshold_pct_fall, output_threshold_pct_fall) * (volt_high-volt_low) * 0.01

Vfinal must NOT be greater than Verr.

The following is an example message:

```
Error : Line 373, The final signal voltage of the vector is 0.800000 and
 it does not reach beyond the 2nd
        slew threshold voltage, which is 1.000000 for the
 output_current_fall vector. (LBDB-669)
```

### What Next

Check the library source file and correct the problem.

---

## LBDB-670

(information) The '%s' has '%s' %s, \n \twhich has less than %d significant digits.

Feedback

### Description

The values in the current/receiver_capacitance tables should have at least 4 significant digits. The value of intrinsic_resistance and intrinsic_capacitance should have at least 2 significant digits if there is no off channel resistance (i.e. greater than 1Mohm) in the library.

The following example shows an instance where this message occurs:

```
output_current_rise(current_template) {
        ...
        vector(current_template) {
           reference_time : 0.0961844;
           index_1("0");
           index_2("0.170393");
           index_3("0.254379, 0.295, 0.31, 0.325, 0.34, 0.355, 0.385,
 0.4, 0.415, 0.43, 0.445, 0.475, 0.543056");
           values("0.1234, 0.2345, 0.3456, 0.4567, 0.5678, 0.6789,
 0.7890, 0.7891, 0.0001, 0.7892, 0.7893, 0.7894, 0.7895");
        }
        ...
    }
```

The following is an example message:

```
Information : Line 373, The 'vector' has 'values' 0.4,
        which has less than 4 significant digits. (LBDB-670)
```

### What Next

Check the library source file and correct the problem.

---

## LBDB-671

(warning) The '%s' has %s adjacent '%s' (%.8g, %.8g).

### Description

The adjacent 'values' in the vectors fo the outpur_current_rise/fall tables are identical.

The following example shows an instance where this message occurs:

```
output_current_rise(current_template) {
        ...
        vector(current_template) {
           reference_time : 0.0961844;
           index_1("0");
           index_2("0.170393");
           index_3("0.254379, 0.295, 0.31, 0.325, 0.34, 0.355, 0.385,
 0.4, 0.415, 0.43, 0.445, 0.475, 0.543056");
           values("0.1234, 0.2345, 0.2345, 0.4567, 0.5678, 0.6789,
 0.7890, 0.7891, 0.0001, 0.7892, 0.7893, 0.7894, 0.7895");
```

Feedback

```
        }
        ...
    }
```

The following is an example message:

```
Error : Line 373, The 'vector' has identical adjacent 'values' (0.2345,
 0.2345). (LBDB-671)
```

**What Next**

Check the library source file and correct the problem.

---

## LBDB-672

(information) There are more than 1 operating_conditions defined in the library.

**Description**

The message is to notify users that there are more than 1 operating_conditions defined in the library. For each library, only 1 operating_condtions group is necessary since it defined the pvt used for characterizing the library, and this operating_condtion is also the "default_operating_conditions" of the library.

---

## LBDB-673

(warning) The timing arc does not have full receiver modeling information.

**Description**

This message indicates that the timing arc does not have all the following receiver modeling information: receiver_capacitance1_rise, receiver_capacitance1_fall, receiver_capacitance2_rise, receiver_capacitance2_fall

Although it is usually valid to specify only half of the receiver modeling information for the half-nate timing arcs. However, in some corner cases, we may need the full receiver modeling information.

In the following example,the receiver_capacitance1_fall group and the receiver_capacitance2_fall group are not specified for the half-unate timing arc.

```
timing() {
  timing_type : rising_edge;
  timing_sense : positive_unate;
  ...
  receiver_capacitance1_rise(basic_template_8x7) {
              index_1 ("0, 0.00081298, 0.0188997, 0.0460298, 0.0912466,
 0.18168, 0.452981, 0.905149");
              index_2 ("0.019368, 0.049573, 0.0948806, 0.170393,
 0.321418, 0.774494, 1.52962");
```

Feedback

```
                values ("0.00165664, 0.00165664, 0.00165936, 0.00166126,
 0.00166211, 0.00166425, 0.00167038", \
                        "0.00165664, 0.00165664, 0.00165936, 0.00166126,
 0.00166211, 0.00166425, 0.00167038", \
                        "0.00165664, 0.00165664, 0.00165936, 0.00166126,
 0.00166211, 0.00166425, 0.00167038", \
                        "0.00165664, 0.00165664, 0.00165936, 0.00166126,
 0.00166211, 0.00166425, 0.00167038", \
                        "0.00165664, 0.00165664, 0.00165936, 0.00166126,
 0.00166211, 0.00166425, 0.00167038", \
                        "0.00165664, 0.00165664, 0.00165936, 0.00166126,
 0.00166211, 0.00166425, 0.00167038", \
                        "0.00165664, 0.00165664, 0.00165936, 0.00166126,
 0.00166211, 0.00166425, 0.00167038");
  }
  receiver_capacitance2_rise(basic_template_8x7) {
                index_1 ("0, 0.00081298, 0.0188997, 0.0460298, 0.0912466,
 0.18168, 0.452981, 0.905149");
                index_2 ("0.019368, 0.049573, 0.0948806, 0.170393,
 0.321418, 0.774494, 1.52962");
                values ("0.00165664, 0.00165664, 0.00165936, 0.00166126,
 0.00166211, 0.00166425, 0.00167038", \
                        "0.00165664, 0.00165664, 0.00165936, 0.00166126,
 0.00166211, 0.00166425, 0.00167038", \
                        "0.00165664, 0.00165664, 0.00165936, 0.00166126,
 0.00166211, 0.00166425, 0.00167038", \
                        "0.00165664, 0.00165664, 0.00165936, 0.00166126,
 0.00166211, 0.00166425, 0.00167038", \
                        "0.00165664, 0.00165664, 0.00165936, 0.00166126,
 0.00166211, 0.00166425, 0.00167038", \
                        "0.00165664, 0.00165664, 0.00165936, 0.00166126,
 0.00166211, 0.00166425, 0.00167038", \
                        "0.00165664, 0.00165664, 0.00165936, 0.00166126,
 0.00166211, 0.00166425, 0.00167038");
  }
  ...
}

Warning: Line 673, The timing arc does not have full receiver modeling
 information. (LBDB-390)
```

### What Next

If possoble, add the missing receiver modeling information in the library.

Feedback

## LBDB-674

(information) The 1st pair of (%s, %s) index of timing arc\n \t'%s' is (%g, %g) which is larger than recommended (%g, %g).

### Description

This information occurs when the 1st index values of the timing arc are larger than the recommended values.

The following example shows an instance where this message occurs:

```
cell_rise(template) {
      index_1("1.0 ,2.0,4.0,6.0,10.0"); /* total_output_net_capacitance
 */
      index_2("0.362,0.725,1.087,1.449,1.812"); /* input_transition_time
 */
      ...
  }
```

The following is an example message:

```
Information: Line 20068, The 1st pair of
 (total_output_net_capacitance,input_transition_time) index of timing arc
       'cell_rise' is (1.0, 0.362) which is larger than recommended (0,
 0).  (LBDB-674)
```

### What Next

Make sure the 1st index values are not larger than the recommended.

## LBDB-675

(information) The 1st %s index of timing arc\n \t'%s' is %g, which is larger than recommended %g.

### Description

This information occurs when the 1st index value of the timing arc is larger than the recommended value.

The following example shows an instance where this message occurs:

```
cell_rise(template) {
      index_1("1.0 ,2.0,4.0,6.0,10.0"); /* total_output_net_capacitance
 */
      ...
  }
```

Feedback

The following is an example message:

```
Information: Line 20068, The 1st total_output_net_capacitance index of
 timing arc
        'cell_rise' is 1.0, which is larger than recommended 0.
 (LBDB-675)
```

### What Next

Make sure the 1st index value is not larger than the recommended.

---

## LBDB-676

(information) The significant digits of values in timing arc '%s' is %d,\n \twhich is less than recommended %d.

### Description

This information occurs when the significant digits of values in the specified table is less than the recommended number.

The following example shows an instance where this message occurs:

```
cell_rise(template) {
      ...
      values("0.459,0.651,0.843,1.034,1.225",\
      ...);
  }
```

The following is an example message:

```
Information: Line 454, The significant digits of values in timing arc
 'cell_rise' is 4,
        which is less than recommended 5. (LBDB-676)
```

### What Next

Make sure the significant digits is not less than the recommended.

---

## LBDB-677

(warning) The table size of timing arc '%s' is %dx%d,\n \twhich is less than recommended %dx%d.

### Description

This information occurs when the table size is less than the recommended number.

Feedback

The following example shows an instance where this message occurs:

```
cell_rise(template) {
      index_1("1.0 ,2.0,4.0,6.0,10.0"); /* total_output_net_capacitance
 */
      index_2("0.362,0.725,1.087,1.449,1.812"); /* input_transition_time
 */
      ...
  }
```

The following is an example message:

```
Warning: Line 446, The table size of timing arc 'cell_rise' is 5x5,
        which is less than recommended 7x7. (LBDB-677)
```

**What Next**

Make sure the table size is not less than the recommended.

---

## LBDB-678

(warning) The table size of timing arc '%s' is %d,\n \twhich is less than recommended %d.

**Description**

This information occurs when the table size is less than the recommended number.

The following example shows an instance where this message occurs:

```
cell_rise(template) {
      index_1("1.0 ,2.0,4.0,6.0,10.0"); /* total_output_net_capacitance
 */
      ...
  }
```

The following is an example message:

```
Warning: Line 446, The table size of timing arc 'cell_rise' is 5,
        which is less than recommended 7. (LBDB-678)
```

**What Next**

Make sure the table size is not less than the recommended.

---

## LBDB-679

(error) Can not find the %s of the %spin%s.

Feedback

### Description

This information occurs when pin's max_capacitance/min_capacitance or the related_pin's max_transition information can not be found.

The following example shows an instance where this message occurs:

```
cell ( INV ) {
    area : 1 ;
    pin ( A ) {
      direction : input ;
      capacitance : 1 ;
      fanout_load : 1 ;
      ...
    }
    pin ( Z ) {
      related_pin :  A ;
      direction : output ;
      capacitance : 1 ;
      fanout_load : 1 ;
      ...
    }
  }
```

The following is an example message:

```
Error: Line 446, Can not find the max_transition of the input pin 'A'.
 (LBDB-679)
Error: Line 446, Can not find the min_capacitance of the  pin 'Z'.
 (LBDB-679)
Error: Line 446, Can not find the max_capacitance of the  pin 'Z'.
 (LBDB-679)
```

### What Next

Make sure the max_transition/max_capacitance/min_capacitance information are defined in the source .lib file.

---

## LBDB-680

(warning) The max value of %s index of group '%s' is %g,\n \twhich is less than max %s of pin '%s', %g.

### Description

This information occurs when the max value of the specified index is less than the max constraint for the pin.

Feedback

In 10nm library, the index of input_net_transition of CCS receiver model will be checked using these ways in order of priority:

1) If driver waveform is defined, choose driver waveform matching the max_transition of related input pin, masure the transition time used in characterization of C1/C2 respectively from driver waveform, then use the restored transition time as constraint.

2) If driver waveform is NOT defined, use 80% of the max_transition value of related input pin as constraint.

The following example shows an instance where this message occurs:

```
pin(y) {
     direction   : input;
     min_capacitance : 0.361;
     max_capacitance : 1.813;
     max_transition : 0.5;
     cell_rise(template) {
       index_2("0.362,0.725,1.087,1.449,1.812"); /* total_out_net_cap  */
       ...
     }
     ...
   }
```

The following is an example message:

```
Warning: Line 762, The max value of total_out_net_cap index of timing arc
 'cell_rise' is 1.812,
        which is less than max total_out_net_cap of pin 'y', 1.813.
 (LBDB-680)

Warning: Line 1117, Cell 'dfnrb4', The max value of 1st index of group
 'receiver_capacitance1_fall' is 1755.58,
        which is less than max transition restored from driver_waveform
 of pin 'CP', 2369.48. (LBDB-680)

Warning: Line 78450, Cell 'r0hd_eln20_cbgclnbt06p00', The max value of
 1st index of group 'receiver_capacitance1_rise' is 0.4035,
        which is less than max transition with a 20.0 percent tolerance
 of pin 'root_clk_off', 1.2. (LBDB-680)
```

**What Next**

Make sure the max index value is not less than the constraint.

---

## LBDB-681

(warning) The min value of %s index of timing arc '%s' is %g,\n \twhich is bigger than min %s of pin '%s', %g.

Feedback

### Description

This information occurs when the min value of the specified index is larger than the min constraint for the pin.

The following example shows an instance where this message occurs:

```
pin(y) {
     direction   : input;
     min_capacitance : 0.361;
     max_capacitance : 1.813;
     max_transition : 0.5;
     cell_rise(template) {
       index_2("0.362,0.725,1.087,1.449,1.812"); /* total_out_net_cap  */
       ...
     }
     ...
   }
```

The following is an example message:

```
Warning: Line 762, The min value of total_out_net_cap index of timing arc
 'cell_rise' is 0.362,
        which is larger than min total_out_net_cap of pin 'y', 0.361.
 (LBDB-681)
```

### What Next

Make sure the min index value is not larger than the constraint.

---

## LBDB-682

(warning) The values in timing arc '%s' are nonmonotonic %g, %g,\n \twhen %s = %g.

### Description

This information appears when the delay values in the specified table do not increase monotonically with the increasing capacitance.

The following example shows an instance where this message occurs:

```
cell_rise(template) {
      index_1("1.0 ,2.0,4.0,6.0,10.0"); /* input_transition_time */
      index_2("0.362,0.725,1.087,1.449,1.812"); /*
 total_output_net_capacitance */
      values ("1.405, 1.116...")
  }
```

Feedback

The following is an example message:

```
Warning: Line 762, The values in timing arc 'cell_rise' are non monotonic
 1.405, 1.116,
        when input_net_transition = 1.    (LBDB-682)
```

**What Next**

Make sure that the delay values increase monotonically with the increasing capacitance.

---

## LBDB-683

(warning) The table is a load independent table.

**Description**

This information occurs when the template of one table do not have the total_output_net_capacitance index where it is needed.

The following example shows an instance where this message occurs:

```
lu_table_template(transition1x5){
    variable_1 : input_net_transition;
    index_1("0.5,1.0,1.5,2.0,2.5");
 }
 ...
 cell (AND) {
   pin(y) {
      ...
      cell_rise(transition1x5) {
        index_1("0.362,0.725,1.087,1.449,1.812"); /* input_net_transition
 */
        ...
      }
      ...
   }
   ...
 }
```

The following is an example message:

```
Warning: Line 762, The table is a load independent table. (LBDB-683)
```

**What Next**

Make sure the table use the correct template which is with total_output_net_capacitance index.

---

## LBDB-684

(warning) The table is a scalar table.

Feedback

### Description

This information occurs when the template of one table do not have any index where it is needed.

The following example shows an instance where this message occurs:

```
lu_table_template(scalar_template){
 }
 ...
 cell (AND) {
   pin(y) {
     ...
     cell_rise(scalar_template) {
        values ("...");
     }
     ...
   }
   ...
 }
```

The following is an example message:

```
Warning: Line 762, The table is a scalar table. (LBDB-684)
```

### What Next

Make sure the table use the correct template which is with index.

---

## LBDB-685

(warning) The voltage range of %s is %g->%g,\n \twhich is less than recommended %g->%g.

### Description

This information occurs when the range of voltage index for IV curves is less than the recommended.

The following example shows an instance where this message occurs:

```
...
   iv_lut_template ("LUT_TEMPLATE_13570_t") {
        variable_1 : "iv_output_voltage";
        index_1("-1.500000, -0.611850, -0.019800, 0.009900, 0.246750,
 0.868350, 1.016400, 1.578900, 2.526300, 3.000000");
   }
     ...
    timing () {
      steady_state_current_low ("LUT_TEMPLATE_13570_t") {
                    values("-1.500000, -0.611850, -0.019800, 0.009900,
 0.246750, 0.868350, 1.016400, 1.578900, 2.526300, 3.000000");
```

Feedback

```
      }
    }
```

The following is an example message:

```
Warning: Line 780, The voltage range of steady_state_current_low is
 -1.5->3,
        which is less than recommended -5->10. (LBDB-685)
```

### What Next

Make sure the voltage range is not less than the recommended.

---

## LBDB-686

(warning) There is potential extrapolation problem\n \t%s[%g, %g, %g]=%g (> %g).

### Description

This information occurs when the noise value extrapolated with the boundary values is larger than the recommended zero value.

The following example shows an instance where this message occurs:

```
propagated_noise_width_below_low(my_noise_propagation){
      index_1("0.00250, 0.00500, 0.01000, 0.01500, 0.02500, 0.03750,
 0.05000"); /*total_output_net_capacitance*/
      index_2("0.400,1.000,1.500,2.000"); /*input_noise_width*/
      index_3("0.36000,0.54000,0.72000,0.90000");
      values("0.025, 0.024, 0.010, 0.013",\
             "1.071, 0.342, 0.121, 0.222",\
             "0.489, 0.565, 0.789, 0.750",\
             "0.282, 0.021, 0.279, 0.118",\
             "0.034, 0.042, 0.009, 0.015",\
             "1.068, 0.346, 0.121,...);
  }
```

The following is an example message:

```
Warning: Line 215, There is potential extrapolation problem
        Width[0.05, 1, 0]=0.376 (> 0.1). (LBDB-686)
```

### What Next

Make sure the boundary values in the table correct.

---

## LBDB-687

(warning) There is potential extrapolation problem\n \t%s[%g, %g]=%g (> %g).

Feedback

### Description

This information occurs when the noise value extrapolated with the boundary values is larger than the recommended zero value.

The following example shows an instance where this message occurs:

```
propagated_noise_width_below_low(my_noise_propagation){
      index_1("0.400,1.000,1.500,2.000"); /*input_noise_width*/
      index_2("0.36000,0.54000,0.72000,0.90000");
      values("0.025, 0.024, 0.010, 0.013",\
             "1.071, 0.342, 0.121, 0.222",\
             "0.489, 0.565, 0.789, 0.750",\
             "0.282, 0.021, 0.279, 0.118");
  }
```

The following is an example message:

```
Warning: Line 215, There is potential extrapolation problem
        Width[1, 0]=2.529 (> 0.1). (LBDB-687)
```

### What Next

Make sure the boundary values in the table correct.

---

## LBDB-688

(warning) The curve is curling upwards at the end.\n \twhen total_output_net_capacitance=%g.

### Description

This information occurs when the immunity curve curls upwards at the end of the curve.

The following example shows an instance where this message occurs:

```
noise_immunity_high (my_noise_reject) {
    values ("1.3, 0.8, 0.7, 0.6, 0.55", \
            "1.5, 0.9, 0.8, 0.65, 0.6", \
            "1.5, 0.9, 0.8, 0.65, 0.6", \
            "1.5, 0.9, 0.8, 0.65, 0.6", \
            "-1.5, 1.9, 0.8, 0.65, 0.6") ;
  }
```

The following is an example message:

```
Warning: Line 178, The curve is curling upwards at the end.
        when total_output_net_capacitance=0.1. (LBDB-688)
```

Feedback

### What Next

Make sure each last 2 values with the same total_output_net_capacitance index are almost identical.

---

## LBDB-689

(warning) The curve is not leveling off at the end.\n \twhen total_output_net_capacitance= %g.

### Description

This information occurs when the immunity curve is not leveling off at the end.

The following example shows an instance where this message occurs:

```
noise_immunity_high (my_noise_reject) {
    values ("1.3, 0.8, 0.7, 0.6, 0.55", \
            "1.5, 0.9, 0.8, 0.65, 0.6", \
            "1.5, 0.9, 0.8, 0.65, 0.6", \
            "1.5, 0.9, 0.8, 0.65, 0.6", \
            "-1.5, 1.9, 0.8, 0.65, 0.6") ;
  }
```

The following is an example message:

```
Warning: Line 178, The curve is not leveling off at the end.
        when total_output_net_capacitance=0. (LBDB-689)
```

### What Next

Make sure each last 2 values with the same total_output_net_capacitance index are almost identical.

---

## LBDB-690

(error) The current polarity is reversed.

### Description

This information occurs when the current polarity is reversed.

The following example shows an instance where this message occurs:

```
steady_state_current_low(my_current_low) {
    values("0.1, 0.05, 0, -0.1, -0.25, -1, -1.8");
  }
```

The following is an example message:

```
Error: Line 198, The current polarity is reversed. (LBDB-690)
```

Feedback

### What Next

Make sure the skew of the current at middle point is correct.

---

## LBDB-691

(error) There is negative or zero values in the table.

### Description

This information occurs when the value in the table is negative or zero where it should be positive.

The following example shows an instance where this message occurs:

```
propagated_noise_height_below_low(my_noise_propagation){
  ...
          values ("-1.4 ..");
```

The following is an example message:

```
Error: Line 198, There is negative values in the table. (LBDB-691)
```

### What Next

Make sure the values in the table are positive.

---

## LBDB-692

(warning) The max height of the noise table %s is %g,\n \twhich is less than recommended %g.

### Description

This information occurs when the max height of the noise table is less than the recommended.

The following example shows an instance where this message occurs:

```
propagated_noise_height_below_low(my_noise_propagation){
                index_1("0.00250, 0.00500, 0.01000, 0.01500, 0.02500,
 0.03750, 0.05000");
                index_2("0.400,1.000,1.500,2.000");

 index_3("0.36000,0.54000,0.72000,0.90000"); /*input_noise_height*/
```

The following is an example message:

```
Warning: Line 248, The max height of the noise table
 propagated_noise_height_below_low is 0.9,
        which is less than recommended 5. (LBDB-692)
```

Feedback

### What Next

Make sure the max height is not less than the recommended.

---

## LBDB-693

(warning) The max width of the noise table %s is %g,\n \twhich is less than recommended %g.

### Description

This information occurs when the max width of the noise table is less than the recommended.

The following example shows an instance where this message occurs:

```
propagated_noise_height_below_low(my_noise_propagation){
                index_1("0.00250, 0.00500, 0.01000, 0.01500, 0.02500,
 0.03750, 0.05000");
                index_2("0.400,1.000,1.500,2.000"); /*input_noise_width*/
                index_3("0.36000,0.54000,0.72000,0.90000");
```

The following is an example message:

```
Warning: Line 281, The max width of the noise table
 propagated_noise_width_low is 2,
        which is less than recommended 4. (LBDB-693)
```

### What Next

Make sure the max width is not less than the recommended.

---

## LBDB-694

(warning) The width range of %s is %g->%g,\n \twhich is less than recommended %g->%g.

### Description

This information occurs when the width index does not have enough range.

The following example shows an instance where this message occurs:

```
noise_immunity_above_high (my_noise_reject_outside_rail) {
    index_1("0, 0.1, 2"); /*input_noise_width*/
    values ("1, 0.8, 0.5", \
            "1, 0.8, 0.5", \
            "1, 0.8, 0.5");
  }
```

Feedback

The following is an example message:

```
Warning: Line 192, The width range of noise_immunity_above_high is 0->2,
        which is less than recommended 2->4. (LBDB-694)
```

### What Next

Make sure the width index have the enough range.

---

## LBDB-695

(error) The pin '%s' misses DC noise margin.

### Description

This information occurs when one pin does not DC noise margin information.

The following example shows an instance where this message occurs:

```
pin(a) {
            direction   : input;
            capacitance : 1.000;
        }
```

The following is an example message:

```
Error: Line 794, The pin 'a' misses DC noise margin. (LBDB-695)
```

### What Next

Make sure there is DC noise margin information defined in one pin if no other noise information.

---

## LBDB-696

(warning) The timing arc misses noise %s information.

### Description

This information occurs when the timing arc does not have specified noise information.

The following is an example message:

```
Warning: Line 137, The timing arc misses noise
 propagated_noise_width_high information. (LBDB-696)
```

### What Next

Make sure there is complete noise information defined in the timing arc.

Feedback

## LBDB-697

(warning) The timing arc %s->%s misses noise %s_%s information.

### Description

This information occurs when the timing arc does not have specified noise information.

The following is an example message:

```
Warning: Line 137, The timing arc DA[0]->QB[0] misses noise
 propagated_noise_width_above_high information. (LBDB-697)
```

### What Next

Make sure there is complete noise information defined in the timing arc.

## LBDB-698

(warning) The values in noise table '%s' are non monotonous\n \t%g, %g, when %s = %g, %s = %g.

### Description

This information occurs when the delay values in specified table do not decrease monotonously with increasing capacitance.

The following example shows an instance where this message occurs:

```
propagated_noise_width_below_low(my_noise_propagation){
                index_1("0.00250, 0.00500, 0.01000, 0.01500, 0.02500,
 0.03750, 0.05000");
                index_2("0.400,1.000,1.500,2.000");
                index_3("0.36000,0.54000,0.72000,0.90000");
                values("0.000, 0.024, 0.010, 0.013",\
                       "1.071, 0.342, 0.121, 0.222",\
                       "0.489, 0.565, 0.789, 0.750",\
                       "0.282, 0.021, 0.279, 0.118",\
                       "1.000, 0.042, 0.009, 0.015",\
                       "1.068, 0.346, 0.121, 0.248",\
                       "0.497, 0.584, 0.786, 0.777",\
                       "0.271, 0.029, 0.281, 0.120",\
                       "0.055, 0.074, 0.008, 0.017",\
                       "1.076, 0.356, 0.121, 0.306",\
                       "0.481, 0.612, 0.783, 0.778",\
                       "0.257, 0.045, 0.288, 0.103",\
                       "0.072, 0.102, 0.006, 0.018",\
                       "1.098, 0.367, 0.121, 0.359",\
                       "0.506, 0.646, 0.779, 0.770",\
                       "0.248, 0.057, 0.295, 0.105",\
                       "0.102, 0.161, 0.002, 0.018",\
```

Feedback

```
                        "1.162, 0.392, 0.118, 0.437",\
                        "0.571, 0.707, 0.775, 0.763",\
                        "0.238, 0.076, 0.311, 0.107",\
                        "0.142, 0.202, 0.002, 0.059",\
                        "0.389, 0.426, 0.102, 0.541",\
                        "0.673, 0.775, 0.772, 0.768",\
                        "0.235, 0.094, 0.329, 0.109",\
                        "0.180, 0.246, 0.005, 0.071",\
                        "0.434, 0.463, 0.090, 0.644",\
                        "0.705, 0.835, 0.758, 0.783",\
                        "0.235, 0.108, 0.345, 0.111");
            }
```

The following is an example message:

```
Warning: Line 248, The values in noise table
 'propagated_noise_height_below_low' are non monotonous
        0.025, 1.000, when input_noise_width = 0.4, input_noise_height =
 0.36. (LBDB-698)
```

### What Next

Make sure the delay values decrease monotonously with increasing capacitance.

---

## LBDB-699

(error) The 'generic' integrated clock gating cell '%s' should only have at most 1 inout/output pin with \n \tattribute 'clock_gate_out_pin' set to true.

### Description

This message indicates that you specified the 'generic' integrated gating clock cell can only have 1 inout/output pin with 'clock_gate_out_pin : true'.

The following example shows an instance where this message occurs:

```
cell(CGNP) {
  area : 1;
  clock_gating_integrated_cell : "generic";
  ...
  pin(O1) {
                direction : output;
                clock_gate_out_pin : true;
                ...
  }
  pin(O2) {
                direction : output;
                clock_gate_out_pin : true;
                ...
  }
  ...
}
```

Feedback

```
The modified cell description can be :

cell(CGNP) {
  area : 1;
  clock_gating_integrated_cell : "generic";
  ...
  pin(O1) {
                direction : output;
                clock_gate_out_pin : true;
                ...
  }
  ...
}
```

The following is an example message:

```
Error: Line 206, The 'generic' integrated clock gating cell 'CGNP' should
 only have at most 1 inout/output pin with \n
        attribute 'clock_gate_out_pin' set to true. (LBDB-699)
```

### What Next

Change the library source file.

---

## LBDB-700

(error) The pin '%s' of the 'generic' integrated clock gating cell '%s' can not define\n \t attribute 'state_function'. It should use attribute 'function" instead.

### Description

This message indicates that you specified attribute 'state_function' in the 'generic' integrated gating clock cell. It should be changed as 'function'.

The following example shows an instance where this message occurs:

```
cell(CGNP) {
  area : 1;
  clock_gating_integrated_cell : "generic";
  ...
  pin(O1) {
                direction : output;
                state_function : "!(IQ + SE) * CP_IN"
                ...
  }
  ...
}

The modified cell description can be :

cell(CGNP) {
```

Feedback

```
  area : 1;
  clock_gating_integrated_cell : "generic";
  ...
  pin(O1) {
                direction : output;
                function : "!(IQ + SE) * CP_IN"
                ...
  }
  ...
}
```

The following is an example message:

```
Error: Line 206, The pin 'O1' of the 'generic' integrated clock gating
 cell 'CGNP' can not define attribute 'state_function'. It should use
 attribute 'function" instead. (LBDB-700)
```

**What Next**

Change the library source file by replacing "state_function" to "function".

---

## LBDB-701

(error) The %scell '%s' can not define pin functions of both '%s' and '%s'.

**Description**

Assume the integrated clock gating cell is modeled as :

... clock_gating_integrated_cell : "generic"; latch("IQ", "IQN") { data_in : "EN1 * EN2" ; enable : "CLK"' ; } ...

This message indicates that you specified pin functions for either one of the following 2 cases, which is invalid: 1. the same pin function with both IQ and IQN. for example, pin(Y1) { function : "IQ * (A + IQN)"; ... }

1. the different pin functions involved with IQ and IQN. for example, pin(Y1) { function : "IQ * A"; ... } pin(Y2) { function : "IQN * A"; ... }

The following is an example message:

```
Error: Line 206, The integrated clock gating cell '%s' can not define pin
 functions of both '%s' and '%s'. (LBDB-701)
```

**What Next**

Change the library source file by replacing "IQN" to "IQ"'.

Feedback

## LBDB-702

(error) The cell should not define the 'statetable' group. \n \tIt should use 'ff' group or 'latch' group instead.

### Description

This message indicates that you specified statetable and the pin function of "...IQ..." or "...IQN..."(excluding "IQ" and IQN", where IQ and IQN are the outptut of sequential element. You should define 'ff/latch' group instead.

The following example shows an instance where this message occurs:

```
cell(A) {
  area : 1;
  ...
  statetable(" CP EN ", "IQ ") {
     table : " L  L  : - : L ,\
               L  H  : - : H ,\
               H  -  : - : N ";
  }

  pin(Y) {
        function : "IQ * CP";
        ...
  }
  ...
}

The modified cell description should be :

cell(A) {
  area : 1;

  latch("IQ","IQN") {
    enable : "CP'";
    data_in : "EN";
  }
  ...
  pin(Y) {
        function : "IQ * CP";
        ...
  }
}
```

The following is an example message:

```
Error: Line 206, The cell should not define the 'statetable' group.
        It should use 'ff' group or 'latch' group instead. (LBDB-702)
```

Feedback

**What Next**

Change the library source file, and replace the statetable group with the ff/latch group of the specified cell.

---

## LBDB-703

(error) An invalid %s '%s' is found in the '%s' group.

**Description**

The message is to notify users that either the attribute/group should not be defined in the relative parent group, or the value of the attribute/group need to be corrected.

**What Next**

Remove the attribute/group or correct the value of the attribute/group and rerun the command.

---

## LBDB-704

(warning) The standard cells '%s' and '%s' does not have the same pg_pin configuration.

**Description**

The message is to notify users that all the standard cells(cells having 1 primary_power pg_pin and 1 primary_ground pg_pin) should have exactly the same pg_pin configuration: - all the primrary_power pg_pins of these cells have the same "voltage_name" - all the primrary_ground pg_pins of these cells have the same "voltage_name"

**What Next**

Correct the "voltage_name" attribute values of the pg_pins of the standard cells and rerun the command.

---

## LBDB-705

(warning) The '%s' %s group has been defined multiple times in\n \tthe '%s' %s group. Using the last definition encountered.

**Description**

The library contains more than one definition of a group. The Library Compiler issues this error message, ignores the previous definitions, and takes into consideration the last definition encountered.

Feedback

The following example shows an instance where this message occurs:

```
cell(sample) {
        ...
        pg_pin(A) {
          ...
        }
        pg_pin(A) {
          ...
        }
        ...
    }
```

The following is an example message:

```
Warning: Line 50, The 'A' pg_pin group has been defined multiple times in
        the 'sample' cell group. Using the last definition encountered.
 (LBDB-705)
```

**What Next**

Change the group name if it is wrong, or delete the second definition.

---

## LBDB-706

(error) %s analysis failed during CCS or CCB Noise compilation

**Description**

This error message reports that the ccsn_first_stage, ccsn_last_stage, input_ccb, or output_ccb at the line indicated cannot be compiled successfully. The analysis of an output_voltage_rise/fall data set fails.

If there is an LBDB-953 warning also associated with this stage, please consult the man page on LBDB-953 on how to debug the dc_current table.

If there is an LBDB-15 (or LBDB-954w, LBDB-955 specifically for N10) warning also associated with this stage, please consult the corresponding man page on how to correct the miller capacitance values.

If the cell is a level shifter, or has multiple pg_pins (primary_power, backup_power, or internal_power) of different voltages, LDBD-706 is likely to occur on those low-to-high level-shifting ccs noise stages. Please look for LBDB-939 warning to get more details.

Otherwise, make sure all capacitance values (miller_cap_rise/fall, and total_output_net_capacitance) are reasonable. Look for an LBDB-955 warning on miller_cap_rise/fall. Check to make sure index_2 (total_output_net_capacitance) values of all vector groups within the output_voltage_rise/fall group are smaller than the largest output load of delay and slew lookup tables.

Feedback

Finally, examine index_1 (input_net_transition) values of all vector groups within the output_voltage_rise/fall group. They should not be smaller than the second smallest index_1 (input_net_transition) values of delay and slew lookup tables.

**What Next**

Correct the CCS or CCB Noise data accordingly.

---

## LBDB-706w

(warning) %s analysis failed during CCS or CCB Noise compilation

**Description**

This warning message reports that the ccsn_first_stage, ccsn_last_stage, input_ccb, or output_ccb at the line indicated cannot be compiled successfully. The analysis of an output_voltage_rise/fall data set fails.

If there is an LBDB-953 warning also associated with this stage, please consult the man page on LBDB-953 on how to debug the dc_current table.

If there is an LBDB-15 (or LBDB-954w, LBDB-955 specifically for N10) warning also associated with this stage, please consult the corresponding man page on how to correct the miller capacitance values.

If the cell is a level shifter, or has multiple pg_pins (primary_power, backup_power, or internal_power) of different voltages, LDBD-706w is likely to occur on those low-to-high level-shifting ccs noise stages. Please look for LBDB-939 warning to get more details.

Otherwise, make sure all capacitance values (miller_cap_rise/fall, and total_output_net_capacitance) are reasonable. Look for an LBDB-955 warning on miller_cap_rise/fall. Check to make sure index_2 (total_output_net_capacitance) values of all vector groups within the output_voltage_rise/fall group are smaller than the largest output load of delay and slew lookup tables.

Finally, examine index_1 (input_net_transition) values of all vector groups within the output_voltage_rise/fall group. They should not be smaller than the second smallest index_1 (input_net_transition) values of delay and slew lookup tables.

**What Next**

Correct the CCS or CCB Noise data accordingly.

---

## LBDB-707

(information) Compiling CCS Noise data --- %%d...

Feedback

### Description

This message is for information purposes only and it is to show the process of compiling CCS noise data.

The following is an example message:

```
Information: Compiling CCS Noise data --- 10%... (LBDB-707)
Information: Compiling CCS Noise data --- 20%... (LBDB-707)
 ...
Information: Compiling CCS Noise data --- 100%... (LBDB-707)
```

---

## LBDB-708

(error) The timing arc has '%s' timing_sense, which is not consistent with \n \tthe "is_inverting" attribute value(s) of its %s group(s).

### Description

```
In the following descriptions, "input ccb" stands for input_ccb in new
 noise
model or ccsn_first_stage in old noise model, and "output ccb" stands for
output_ccb in new noise model or ccsn_last_stage in old noise model.

This message is to indicate that the "is_inverting" attribute value of
 the
"input ccb"/"output ccb" group(s) defined in the timing arc is not
 consistent
with the timing_sense of the arc.

The correct configuration should be:
1. If the timing sense is "positive_unate", then it can only contain
 either 1
"input ccb" with "FALSE" "is_inverting" attribute,
or 1 "input ccb" with "TRUE" "is_inverting" attribute and 1 "output ccb"
 with
"TRUE" "is_inverting" attribute,
or 1 "input ccb" with "FALSE" "is_inverting" attribute and 1 "output ccb"
 with
"FALSE" "is_inverting" attribute.
2. If the timing sense is "negative_unate", then it can only contain
 either 1
"input ccb" with "TRUE" "is_inverting" attribute,
or 1 "input ccb" with "TRUE" "is_inverting" attribute and 1 "output ccb"
 with
"FALSE" "is_inverting" attribute,
or 1 "input ccb" with "FALSE" "is_inverting" attribute and 1 "output ccb"
 with
"TRUE" "is_inverting" attribute.
```

Feedback

The following is an example message:

```
Error: Line 50, The timing arc has 'negative_unate' timing_sense, which
 is not
consistent with the "is_inverting" attribute values of its
 input_ccb/output_ccb
group(s). (LBDB-708)
```

## LBDB-709

(error) '%s' cannot be used in the timing arc.

### Description

This particular timing arc cannot have the aforementioned group or attribute because it is not a delay arc, or its timing sense is not positive or negative unate. (Non-unate timing sense typically occurs in sequential arcs.)

The following is an example message:

Error: Line 2077, Cell 'LL_SSYNCCFD1QSX010', pin 'Q', 'ccsn_first_stage' cannot be used in the timing arc.

### What Next

correct the timing type or sense. Or remove the group or attribute.

## LBDB-710

(error) The table size of '%s' is %dx%d,\n \twhich is less than the required %dx%d.

### Description

This information occurs when the table size is less than the required number.

The following example shows an instance where this message occurs:

```
dc_current(template) {
      index_1("1.0") ; /* input_voltage */
      index_2("0.362"); /* output_voltage */
      ...
  }
```

The following is an example message:

```
Warning: Line 446, The table size of 'dc_current' is 5x5,
        which is less than the required 6x6. (LBDB-710)
```

### What Next

Make sure the table size is not less than the recommended.

Feedback

## LBDB-711

(error) a %s pin cannot specify the '%s' %s.

### Description

This error message occurs when there is a mismatch between the direction of a pin and the CCS Noise data. Library Compiler fails if the following rules are not satisfied:

```
1. An input pin can only specify at least 1 ccsn_first_stage group.

2. An inout pin can specify both ccsn_first_stage group and
 ccsn_last_stage group.

3. An output pin can only specify ccsn_last_stage group.
```

The following example shows an instance where this message occurs: The following is an example of incorrect input that causes this error message:

```
cell(lbdb711) {
    ...
    pin(Y ) {
      direction : output;
      ...
      ccsn_first_stage() {
        ...
      }
    }
  }
```

In this case, the Y pin has the ccsn_first_stage group specified. To fix the error, change ccsn_first_stage to ccsn_last_stage.

### What Next

Check the library source file and fix either the invalid CCS Noise data, or the direction of the pin.

## LBDB-712

(warning) an %s pin should either specify at least one non-static\n \t'%s' in the pin group or specify noise data for all of its timing arcs.

### Description

This warning message occurs because: 1. an inout or input pin should have: 1.1 at least one non-static ccsn_first_stage group in the pin group or define non-static ccs noise first stage data for all of its timing groups for CCSN modeling. 1.2 at least one non-static input_ccb group in the pin group for referenced CCSN modeling. 2. an inout or output pin should have: 2.1 at least one non-static ccsn_last_stage group in the pin group or define

Feedback

non-static ccs noise last stage data for all of its timing groups for CCSN modeling. 2.2 at least one non-static output_ccb group in the pin group for referenced CCSN modeling.

A static CCS noise model captures the behaviors of a static CCB (channel connected block). A static CCB refers to a CCB that does not have any input terminal or the voltage level at the input terminal does not affect the output current of the CCB. For example, a static CCS noise model can be characterized for a CMOS NAND gate when both inputs are tied to ground. Such static CCS noise models can be used in noise analysis with case analysis, but they cannot be used for propagating crosstalk delay or noise waveforms through the cell under study. See the *Make CCS Noise User Guide* for detailed information.

The following example shows an instance where this message occurs: In the following example, the Y pin does not have the ccsn_last_stage group specified, so the tool issues this warning message. To fix the problem, add the ccsn_last_stage group into the pin group.

```
cell(good) {
    ...
    pin(Y ) {
      direction : output;
      ...
      ccsn_last_stage() {
        ...
      }
      }
    }
  }
  cell(lbdb712) {
    ...
    pin(Y ) {
      /* no ccs noise information inside the pin group */
      direction : output;
      ...
      /* not all timing arcs inside the pin group has ccs noise data */
    }
  }
```

**What Next**

This is only a warning message. No action is required.

Feedback

However, you can do any of the following:

- Check the library source file and fix either the invalid CCS Noise data, or the direction of the pin.
- Safely ignore this warning if the following three categories of cells are identified in a library:
  - inverters: A cell always has 1 input pin and 1 output pin with the function as "!input" or "input'"
  - NAND: A cell always has 2 input or more and 1 output pin with the function as "!(input1 * input2 * ... input(n))" or "(input1 * input2 * ... input(n))'"
  - NOR: A cell always has 2 input or more and 1 output pin with the function as "!(input1 + input2 + ... input(n))" or "(input1 + input2 + ... input(n))'"

---

## LBDB-713

(error) The '%s' group with %s '%s' attribute cannot specify following CCS or CCB Noise data: %s.

### Description

This message indicates that if a ccsn_first_stage, ccsn_last_stage, input_ccb, or output_ccb group has 'is_needed' attribute set to FALSE, then it is not valid to specify the following noise data:

stage_type, is_inverting, miller_cap_rise, miller_cap_fall, dc_current, output_voltage_rise, output_voltage_fall, propagated_noise_low, propagated_noise_high.

The following example shows an instance where this message occurs:

```
cell(bad) {
    ...
    pin(Y ) {
      direction : output;
        ...
        ccsn_last_stage() {
                is_needed : FALSE;
                stage_type : both
                ...
        }
      }
    }
  }
```

Feedback

The following is an example message:

```
Error: Line 71, The 'ccsn_last_stage' group with FALSE 'is_needed'
 attribute
cannot specify following CCS or CCB Noise data: 'stage_type'. (LBDB-713)
```

**What Next**

Check the library source file, and remove the redundant CCS/CCB Noise data.

---

## LBDB-714

(error) The '%s' group cannot specify the '%s' attribute with value '%s'.

**Description**

This message can be used to indicate that the output pin of a tie-off cell(pin function= "1" | "0", or driver_type = "pull_up" | "pull_down") specify incorrect value for the stage_type attribute of the ccsn_first_stage/ccsn_last_stage groups defined on the pin, which must satisfy the following rules: 1. If the driver pin with driver_type = pull_up or function = "1", then its ccsn_first_stage/ccsn_last_stage group must have the stage_type attribute with value "PULL_UP". 2. If the driver pin with driver_type = pull_down or function = "0", then its ccsn_first_stage/ccsn_last_stage group must have the stage_type attribute with value "PULL_DOWN".

The following example shows an instance where this message occurs:

```
cell(bad) {
    ...
    pin(Y ) {
        direction : output;
        function : 1;
        driver_type : "pull_up";
        ...
        ccsn_last_stage() {
                is_needed : FALSE;
                stage_type : "pull_down";
                ...
        }
      }
    }
  }
```

In this case, the 'Y' pin specify 'pull_down' stage_type attribute in its 'ccsn_last_stage' group. To fix the problem, change the 'pull_down' to 'pull_up'.

The following is an example message:

```
Error: Line 71, The 'ccsn_last_stage' group cannot specify the
 'stage_type' attribute with value 'PULL_DOWN'. (LBDB-714)
```

Feedback

**What Next**

Check the library source file, and correct the incorrect 'stage_type' value.

---

## LBDB-716

(error) The %s %g of '%s' attribute is %s %s = %g.

**Description**

This message indicates that the first/last/some value of the specific attribute is greater than the upper bound value, or lower than the lower bound value.

**What Next**

Check your library and correct the value.

---

## LBDB-717

(error) The value %g of '%s' attribute is %s %g.

**Description**

This message indicates that the value of the specific attribute is greater than the upper bound value, or less than the lower bound value.

**What Next**

Check your library and correct the value.

---

## LBDB-718

(warning) The %s %g of '%s' attribute is %s %s = %g.

**Description**

This message indicates that the first/last/some value of the specific attribute is greater than the upper bound value, or lower than the lower bound value. Note that this is just a warning, not an error.

**What Next**

Check your library and correct the value.

---

## LBDB-719

(error) The cell must have at least 1 'primary_ground' pg_pin and at least 1 'primary_power' pg_pin. It's being marked as dont_use, dont_touch.

Feedback

### Description

This error message occurs when a primary_ground pg_pin or a primary_power pg_pin is missing. Cell in the following list could occurs this message, if it doesn't have at least 1 primary_ground pg_pin and at least 1 primary_power pg_pin. (1) The cell is a macro cell. (2) The cell has function info. (except for tie-off cell) (3) The cell has timing/noise/power info.

### What Next

Add the missing primary_ground pg_pin or primary_power pg_pin and run the command again.

---

## LBDB-720

(error) The '%s' %s cannot be specified in the library\n \tbased on pg_pin.

### Description

This message indicates that you specified the syntax(power_supply, rail_connection, input/output_signal_level, power_level, etc.) which is incompatible with the syntax based on pg_pin such as voltage_map, pg_pin, related_power/ground_pin, related_pg_pin).

The following example shows an instance where this message occurs:

```
cell(lbdb720) {
    area : 2;
    pin(A) {
      direction : input;
      capacitance : 1.0;

      input_signal_level : VDD;

      related_power_pin : VDD;
      related_ground_pin : VSS;

    }
    ...
  }
```

The following is an example message:

```
Error: Line 84, The 'input_signal_level' attribute cannot be specified in
 the library
        based on pg_pin. (LBDB-720)
```

Feedback

### What Next

Refer to the "Library Compiler User Guide" for related information and emove the incompatible values,

---

## LBDB-720w

(warning) The '%s' %s cannot be specified in the library\n \tbased on pg_pin.

### Description

This message indicates that you specified the syntax(power_supply, rail_connection, input/output_signal_level, power_level, etc.) which is incompatible with the syntax based on pg_pin such as voltage_map, pg_pin, related_power/ground_pin, related_pg_pin).

The following example shows an instance where this message occurs:

```
cell(lbdb720) {
    area : 2;
    pin(A) {
      direction : input;
      capacitance : 1.0;

      input_signal_level : VDD;

      related_power_pin : VDD;
      related_ground_pin : VSS;

    }
    ...
  }
```

The following is an example message:

```
Warning: Line 84, The 'input_signal_level' attribute cannot be specified
 in the library
        based on pg_pin. (LBDB-720w)
```

### What Next

Refer to the "Library Compiler User Guide" for related information and emove the incompatible values,

---

## LBDB-721

(warning) The value %f of 'nom_voltage' is not the same as the value\n \t%f of 'voltage' of the default operating_conditions '%s'.

Feedback

### Description

This message indicates that the nom_voltage valye is not equal to the value of "voltage" attribute of the default operating_conditions group of the library.

The following example shows an instance where this message occurs:

```
nom_voltage     : 1.09;

  operating_conditions(sample) {
    process     : 1;
    temperature : 85;
    voltage     : 1.08;
    tree_type   : balanced_tree
  }
```

The following is an example message:

```
Warning: Line 52, The value 1.09 of 'nom_voltage' is not the same as the
 value
        1.08 of 'voltage' of the default operating_conditions '%s'.
 (LBDB-721)
```

### What Next

Correct the nominal_voltage value to make it the same as the value of the voltage attribute of the default operating_conditions group.

---

## LBDB-722

(error) The voltage value %f of the 1st voltage_map '%s' is not the same as\n \tthe value of 'voltage' of the default operating_conditions '%s'.

### Description

This message indicates the 1st specified voltage_map attribute specifiy the different voltage value from that of the 'voltage' of the default operating_consitions group if there are only two voltage_map specified in library. They should be equal.

The following example shows an instance where this message occurs:

```
voltage_map(VDD, 1.2);
      voltage_map(VSS, 0);
      operating_conditions(sample) {
        process     : 1;
        temperature : 85;
        voltage     : 1.08;
        tree_type   : balanced_tree
      }
      default_operating_conditions : sample;
```

Feedback

In this case, there are two voltage_map specified, and the voltage value of the 1st voltage_map 'VDD1' voltage_map is 1.2, while the voltage value of default operating_conditions 'sample' is 1.08. To Fix the problem, update the voltage_map 'VDD' as follows:

```
voltage_map(VDD, 1.08);
      voltage_map(VSS, 0);
      operating_conditions(sample) {
        process     : 1;
        temperature : 85;
        voltage     : 1.08;
        tree_type   : balanced_tree
      }
      default_operating_conditions : sample;
```

The following is an example message:

```
Error: Line 23, The voltage value 1.2 of the 1st voltage_map 'VDD' is not
 the same as\n
\tthe value of 'voltage' of the default operating_conditions 'sample'.
 (LBDB-722)
```

**What Next**

Correct the voltage value of the 1st voltage_map.

---

## LBDB-723

(error) The last variable '%s' is '%s',\n \twhich is wrong. The value must be "time".

**Description**

This message indicates that the value of the last variable can only be "time" in pg_current_template. pg_current_template can have one variable up to 4 variables, and input_net_transition, time, and total_output_net_capacitance are 3 avaiable values can be assigned. No matter what size of template is, the last variable must be "time".

The following example shows an instance where this message occurs:

```
pg_current_template(basic_template) {
    variable_1 : input_net_transition;
    variable_2 : time;
    variable_3 : total_output_net_capacitance;
    variable_4 : total_output_net_capacitance;
  }
```

To fix the problem, exchange the values between variable_2 and variable_4.

Feedback

The following is an example message:

```
Error: Line 388, The last variable variable_4 is
 total_output_net_capacitance,
        which is wrong.  The value must be "time". (LBDB-723)
```

### What Next

If the last variable is not "time", change the variable, which is assigned to "time", to the last.

---

## LBDB-724

(error) There is no voltage_map defined for ground voltage value 0.

### Description

This message indicates the library does not define the voltage_map for the ground voltage value 0. In the libraries based on the new power modeling syntax ( voltage_map, pg_pin, ..., etc.), this condition myst be satisfied.

The following example shows an instance where this message occurs:

```
voltage_map(VDD, 1.08);
      voltage_map(VDD1, 1.1);

      operating_conditions(sample) {
        process     : 1;
        temperature : 85;
        voltage     : 1.08;
        tree_type   : balanced_tree
      }
      default_operating_conditions : sample;
```

In this case, there is no voltage_map defined for ground voltage value 0. To fix the problem, add the following voltage_map attribute at the library level:

```
voltage_map(VDD, 1.08);
      voltage_map(VDD1, 1.1);
      voltage_map(VSS, 0);

      operating_conditions(sample) {
        process     : 1;
        temperature : 85;
        voltage     : 1.08;
        tree_type   : balanced_tree
      }
      default_operating_conditions : sample;
```

Feedback

The following is an example message:

```
Error: Line 23, There is no voltage_map defined for ground voltage value
 0. (LBDB-724)
```

**What Next**

Add the voltage_map for the ground voltage value 0.

---

## LBDB-725

(warning) Connect pin '%s' to the default %s pg_pin '%s'.

**Description**

Either the related_power_pin or the related_ground_pin attribute is missing in a pin. By default, LC connect this pin to the default related_power_pin/related_ground_pin.

The following example shows an instance where this message occurs:

```
voltage_map(VDD0, 1.8);
   voltage_map(VDD1, 1.9);
   voltage_map(VDD2, 23);
   voltage_map(VSS, 0);

cell(lbdb428) {
  area : 2;
  pad_cell : true;
  pg_pin(PV1) {
    voltage_name : VDD1;
    pg_type : primary_power;
  }
  pg_pin(PV2) {
    voltage_name : VDD2;
    pg_type : primary_power;
  }
  pg_pin(VSS) {
    voltage_name : VSS;
    pg_type : primary_ground;
  }
  pin(A) {
    direction : input;
    capacitance : 1;
    /* missing related_power_pin attribute, will use default PV1 */
    related_ground_pin : VSS;
  }
  pin(Z) {
    direction : output;
    function : "A";
    related_power_pin : PV1;
    related_ground_pin : VSS;
    timing() {
```

Feedback

```
      intrinsic_rise : 0.48;
      intrinsic_fall : 0.77;
      rise_resistance : 0.1443;
      fall_resistance : 0.0523;
      slope_rise : 0.0;
      slope_fall : 0.0;
      related_pin : "A";
    }
  }
}
```

The following is an example message:

```
Warning: Line 96, Connect pin 'A' to the default power pg_pin 'PV1'.
 (LBDB-725)
```

### What Next

Check the library source file to see if you missed the related_power_pin or the related_ground_pin attributes.

---

## LBDB-726

(error) All the pins in the '%s' cell with more than 2 %s pg_pins\n \tmust have '%s' attribute.

### Description

The related_power_pin (or the related_ground_pin) attribute is missing in a pin within a cell with more than 2 power pg_pins (or ground pg_pins) respectively.

The following example shows an instance where this message occurs:

```
voltage_map(VDD0, 1.8);
   voltage_map(VDD1, 1.9);
   voltage_map(VDD2, 23);
   voltage_map(VSS, 0);

cell(lbdb428) {
  area : 2;
  pad_cell : true;
  pg_pin(PV1) {
    voltage_name : VDD1;
    pg_type : primary_power;
  }
  pg_pin(PV2) {
    voltage_name : VDD2;
    pg_type : primary_power;
  }
  pg_pin(VSS) {
    voltage_name : VSS;
    pg_type : primary_ground;
  }
```

Feedback

```
  pin(A) {
    direction : input;
    capacitance : 1;
    /* missing related_power_pin attribute, will use default PV1 */
    related_ground_pin : VSS;
  }
  pin(Z) {
    direction : output;
    function : "A";
    related_power_pin : PV1;
    related_ground_pin : VSS;
    timing() {
      intrinsic_rise : 0.48;
      intrinsic_fall : 0.77;
      rise_resistance : 0.1443;
      fall_resistance : 0.0523;
      slope_rise : 0.0;
      slope_fall : 0.0;
      related_pin : "A";
    }
  }
}
```

To fix the problem, add the attribute to the 'A' pin group,

```
related_power_pin : VDD1;
```

The following is an example message:

```
Error: Line 96, All the pins in the 'lbdb726' cell with more than 2 power
 pg_pins
        must have 'related_power_pin' attribute. (LBDB-726)
```

### What Next

Check the library source file to see if you missed the related_power_pin or the related_ground_pin attributes.

---

## LBDB-727

(error) The %s%s '%s' contains the conflicting %s \n \t'%s' for pin '%s' and %s '%s' for pin '%s'.

### Description

This error message occurs when the *input_signal_level*, *output_signal_level*, or *related_power_pin* values is modeled incorrectly for the level shifter or isolation cell. See detail checking as described below.

Liberty supports two kinds of PG syntax: old and new.

Feedback

Rule for a level shifter in the old syntax:

The rail name of *input_signal_level* in input pin cannot be the same as rail name of *output_signal_level* in output pin.

Rule for a level shifter in the new syntax:

If *input_signal_level* and *related_power_pin* both present in input pin, then Library Compiler will choose the value of *input_signal_level* for comparison. Either *input_signal_level* or *related_power_pin* in input pin cannot refer to the same voltage name as what *related_power_pin* refers to in output pin.

Rule for an isolation cell in the old syntax:

The rail value of *input_signal_level* in input pin shall be the same as the rail value of *output_signal_level* in output pin.

Rule for an isolation cell in the new syntax:

The value of either *input_signal_level* or *related_power_pin* in input pin shall refer to the same voltage value as what *related_power_pin* refers to in output pin.

The following example shows a level shifter with incorrect modeling and the resulting error message in the new PG syntax:

```
library (test) {
 ...
 voltage_map( VSS, 0.0);
 voltage_map( VDDH, 0.9);
 voltage_map( VDDL, 0.7);
 ...
 cell(LS) {
 ...
 pg_pin(GND) {
 voltage_name : VSS;
 pg_type : primary_ground;
 }
 pg_pin(VDD2) {
 voltage_name : VDDH;
 pg_type : primary_power;
 }
 pg_pin(VDD1) {
 voltage_name : VDDL;
 pg_type : primary_power;
 }

 is_level_shifter : true;
 ...
 pin(in) {
        ...
        input_signal_level : VDDL;
        related_power_pin : VDD1;  /* discard */
```

Feedback

```
        related_ground_pin : GND;
 }
 pin(out) {
        ...
        related_power_pin: VDD1;
        related_ground_pin : GND;
 }
}
```

In this case, input_signal_level "VDDL", which is the same voltage name as what related_power_pin "VDD1" is referring to.

```
Error: Line 191, The level shifter 'LS' contain the conflicting
 input_signal_level
        'VDDL' for pin 'in' and related_power_pin 'VDD1' for pin 'out'.
 (LBDB-727)
```

The following example shows the isolation cell with the incorrect modeling and the resulting error message in new pg syntax:

```
library (libdb727) {
 ...
 voltage_map( VDD, 0.7);
 voltage_map(VDDH, 0.80); /* high power */
 voltage_map(VDDL, 0.4); /* low power */
 voltage_map(VSS, 0.0);   /* primary ground */
 ...
 cell(ISO) {
 ...
 pg_pin(VDD1) {
 voltage_name : VDDH;
 pg_type : primary_power;
 }
 pg_pin(VDD2) {
 voltage_name : VDDL;
 pg_type : primary_power;
 }

 is_isolation_cell : true;
 ...
 pin(in) {
        ...
        related_power_pin : VDD1;
        related_ground_pin : GND;
 }
 pin(out) {
        ...
        related_power_pin: VDD2;
        related_ground_pin : GND;
 }
}
```

Feedback

In this case, related_power_pin 'VDD1' is referred to voltage value 0.8 and related_power_pin 'VDD2' is referred to voltage value 0.4. So, the valtage value 0.8 in input pin is different from valtage value 0.4 in output pin, which is wrong.

```
Error: Line 191, The isolation cell 'ISO' contain the conflicting
 related_power_pin
        'VDD1' for pin 'in' and related_power_pin 'VDD2' for pin 'out'.
 (LBDB-727)
```

The following example shows the level shifter with the incorrect modeling and the resulting error message in the old PG syntax:

```
cell(LS) {
 is_level_shifter : true;
 ...
 pin(in) {
        ...
        input_signal_level : VDD2

 }
 pin(out) {
        ...
        output_signal_level : VDD2

 }
}

Error: Line 191, The level shifter 'LS' contain the conflicting
 input_signal_level 'VDD2' for pin 'in' and output_signal_level 'VDD2'
 for pin 'out'. (LBDB-727)
```

The following example shows the isolation cell with the incorrect modeling and the resulting error message in old pg syntax:

```
library(libdb727) {
  ...
  /* operation conditions */
  operating_conditions(5v_1v) {
    ...
    power_rail (VDDH, 5);    /* high power */
    power_rail (VDDL, 1);    /* low power */
    ...
  }
  ...
  default_operating_conditions : 5v_1v;

  ...
cell(ISO) {
 is_isolation_cell : true;
 ...
 pin(in) {
        ...
```

Feedback

```
        input_signal_level : VDDH
 }
 pin(out) {
        ...
        output_signal_level : VDDL
 }
}

Error: Line 191, The isolation cell 'ISO' contain the conflicting
 input_signal_level 'VDDH' for pin 'in' and output_signal_level 'VDDL'
 for pin 'out'. (LBDB-727)
```

### What Next

For the first example, modify the modeling information of the level shifter to meet the requirements.

To correct the modeling of the example above, make the *input_signal_level* value of the input pin different from the *related_power_pin* value of the output pin as follows:

```
cell(LS) {
 is_level_shifter : true;
 ...
 pin(in) {
        ...
        input_signal_level : VDDL;
        related_power_pin: VDD1;
        related_ground_pin : GND;
 }
 pin(out) {
        ...
        related_power_pin: VDD2;
        related_ground_pin : GND;
 }
}


For the second example, modify the modeling information of the isolation
 cell to meet
the requirements.

To correct the modeling of the example above, make the
related_power_pin value of the input pin
same as the related_power_pin
value of the output pin as follows:

library (libdb727) {
 ...
 voltage_map( VDD, 0.7);
 voltage_map(VDDH, 0.80); /* high power */
 voltage_map(VDDL, 0.4); /* low power */
 voltage_map(VSS, 0.0);   /* primary ground */
 ...
```

Feedback

```
 cell(ISO) {
 ...
 pg_pin(VDD1) {
 voltage_name : VDDH;
 pg_type : primary_power;
 }
 pg_pin(VDD2) {
 voltage_name : VDDL;
 pg_type : primary_power;
 }

 is_isolation_cell : true;
 ...
 pin(in) {
        ...
        related_power_pin : VDD1;
        related_ground_pin : GND;
 }
 pin(out) {
        ...
        related_power_pin: VDD1;
        related_ground_pin : GND;
 }
}
```

For the third example, modify the modeling information of the level shifter to meet the requirements.

To correct the modeling of the example above, make the *input_signal_level* string of the input pin different from the *output_signal_level* string "VDD2" of the output pin as follows:

```
cell(LS) {
 is_level_shifter : true;
 ...
 pin(in) {
        ...
        input_signal_level : VDD1
 }
 pin(out) {
        ...
        output_signal_level : VDD2
 }
}
```

```
For the fourth example, modify the modeling information of the isolation
 cell
to meet the requirements.

input_signal_level 'VDDH' is referred to voltage value
'5' and output_signal_level 'VDDL' is referred to
voltage value '1'.  The voltage value is '5' in input
pin and the voltage value is '1' in output pin, which
```

Feedback

```
is wrong.

To correct it, you can assiged both to the same voltage
like "VDDH", so that both will refer to same value "5"
in operating_conditions as follows.


library(libdb727) {
  ...
  /* operation conditions */
  operating_conditions(5v_1v) {
    ...
    power_rail (VDDH, 5);    /* high power */
    power_rail (VDDL, 1);    /* low power */
    ...
  }
  ...
  default_operating_conditions : 5v_1v;

  ...
cell(ISO) {
 is_isolation_cell : true;
 ...
 pin(in) {
        ...
        input_signal_level : VDDH
 }
 pin(out) {
        ...
        output_signal_level : VDDH

 }
}
```

## LBDB-728

(error) You can not specify the same pin '%s' twice\n \tin '%s'.

### Description

This message indicates that you specified the same pin more than one in either *related_inputs* or *related_outputs*. Each pin can be only specifed once.

The following example shows an instance where this message occurs:

```
cell(lbdb728) {
  area : 2;
  pin(A) {
    direction : input;
    capacitance : 1;
  }
  pin(B) {
```

Feedback

```
    direction : input;
    capacitance : 1;
  }

  pin(Z) {
    direction : output;
    function : "A B";
    timing() {
     ...
    }
  }
  dynamic_current() {
    when : "A";
    related_inputs : "B B";
    related_outputs : "Z Z"
    typical_capacitances(0.3 0.4);
    switching_group() {
     ...
    }
  }
  ...
}
```

In this case, both related_outputs and related_inputs, have specified the same pin name twice. To fix the problems, change the name from 'Z Z' to 'Z' in the related_outputs, change the name from 'B B' to 'B' in the related_inputs.

The following is an example message:

```
Error: Line 272, You can not specify the same pin 'Z' twice
        in 'related_outputs'. (LBDB-728)
Error: Line 572, You can not specify the same pin 'B' twice
        in 'related_inputs'. (LBDB-728)
```

### What Next

Check the library source file, and make the necessary correction.

---

## LBDB-729

(warning) You can't specify the same pin '%s' in different\n \trelated_output attributes which are under the same intrinsic_parasitic group.

### Description

If there are more than one *intrinsic_resistance* group under a *intrinsic_parasitic* group, then no *related_output* can have the same pin. However, the rule does not apply to closed channel, where the value specified in *intrinsic_resistance* group is greater than 1M Ohm.

Feedback

The following example shows an instance where this message occurs:

```
library(my) {
  /* unit attributes */
  time_unit : "1ns";
  capacitive_load_unit (1.0,pf);
  ...
 cell(lbdb729) {
  area : 2;
  pin(A) {
    direction : input;
    capacitance : 1;
  }
  pin(B) {
    direction : input;
    capacitance : 1;
  }
  pin(Z) {
    direction : output;
    ...
  }
  pin(ZN) {
    direction : output;
    ...
  }
  intrinsic_parasitic() {
    /* default state */
    intrinsic_resistance(<pg_name>) {
       related_output : "ZN";
       value : 9.0;
    }
    intrinsic_resistance(<pg_name>) {
       related_output : "ZN";
       value : 920.0;
    }
    intrinsic_resistance(<pg_name>) {
       related_output : "ZN";
       value : 1001.0;
    }
    intrinsic_capacitance(<pg_name>) {
       value : 8.2;
    }
  }
  ...
 }
}
```

Based on timing and capacitance unit, we know the resistance unit is kilo. In this case, all intrinsic_resistance groups under a intrinsic_parasitic have same related_output 'ZN'. To fix the problem, change the name 'ZN' to 'Z' in either the group which has value '9.0' or the group which has value '920.0' because both are opened channel. No need to change the

Feedback

name in the group which has value '1001.0' because it is greater than 1M ohm, which is closed channel.

The following is an example message:

```
Warning Line 272, You can't specify the same pin 'ZN' in different
        related_output attributes which are under the same
 intrinsic_parasitic group. (LBDB-729)
```

### What Next

Check the library source file, and make the necessary correction.

---

## LBDB-730

(error) The pin '%s' specified in the '%s' is neither\n \tan %s pin nor an inout pin.

### Description

This message indicates that direction of the pin you specified in a *related_inputs*, *related_outputs*, *related_output*, *index_output* or *gate_leakage* is wrong.

The following example shows an instance where this message occurs:

```
cell(lbdb730) {
  area : 2;
  pin(A) {
    direction : input;
    capacitance : 1;
  }
  pin(B) {
    direction : input;
    capacitance : 1;
  }
  pin(Z) {
    direction : output;
    function : "A B";
    timing() {
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      slope_rise : 0.0;
      slope_fall : 0.0;
      related_pin : "A B";
    }
  }
  leakage_current() {
     when : "A & !B & Z";
     pg_current(V1) {
      ...
     }
```

Feedback

```
      ...
      gate_leakage(Z) {  /* must be input or inout pin */
        input_high_value : 2.1;
        input_low_value : -1.7;
      }
    }
    dynamic_current() {
      when : "A";
      related_inputs : "Z";
      related_outputs : "B"
      typical_capacitances(0.3);
      switching_group() {
        input_switching_condition(fall);
        output_switching_condition(rise);
        pg_current(<pg_name>) {
          vector(<lu_template_name>) {
            reference_time : 93.2;
            index_output : "A";
            index_1("5.1");
            index_2("0.3");
            index_3("8.2 9.4 9.8");
            values("1.78 12.4 110.1");
          }
        }
      ...
      }
    }
    intrinsic_parasitic() {
      /* default state */
      intrinsic_resistance(<pg_name>) {
         related_output : "B";
         value : 9.0;
      }
      intrinsic_capacitance(<pg_name>) {
         value : 8.2;
      }

    }
    ...
}
```

In this case, all related_output, related_outputs, related_inputs, index_output and gate_leakage have a wrong pin name. To fix the problems, change the name from 'Z' to either 'A' or 'B' in the related_inputs, change the name from 'B' to 'Z' in the related_outputs, change the name from 'A' to 'Z' in the index_output, change the name from 'B' to 'Z' in the related_output and change the name from 'Z' to either 'A' or 'B' in the gate_leakage.

The following is an example message:

```
Error: Line 272, The pin 'Z' specified in the 'related_inputs' is neither
        an input pin nor an inout pin. (LBDB-730)
```

Feedback

```
Error: Line 471, The pin 'B' specified in the 'related_outputs' is
 neither
        an output pin nor an inout pin. (LBDB-730)
Error: Line 526, The pin 'A' specified in the 'index_output' is neither
        an output pin nor an inout pin. (LBDB-730)
Error: Line 890, The pin 'B' specified in the 'related_output' is neither
        an output pin nor an inout pin. (LBDB-730)
Error: Line 749, The pin 'Z' specified in the 'gate_leakage' is neither
        an input pin nor an inout pin. (LBDB-730)
```

### What Next

Check the library source file, and make the necessary correction.

---

## LBDB-731

(error) The pin '%s' specified in the '%s' can not\n \tbe found in '%s'.

### Description

Whatever the pin you specified in *index_output* must be matched to one of pins that you specified in *related_outputs*.

This message indicates that the pin you specified in a *index_output* is wrong because it can not be found in *related_outputs* which is defined under the same dynamic_current group as where *index_output* is defined.

The following example shows an instance where this message occurs:

```
cell(lbdb731) {
  area : 2;
  pin(A) {
    direction : input;
    capacitance : 1;
  }
  pin(B) {
    direction : input;
    capacitance : 1;
  }
  pin(Z) {
    direction : output;
    ...
  }
  pin(ZN) {
    direction : output;
    ...
  }
  dynamic_current() {
    when : "A";
    related_inputs : "B";
    related_outputs : "Z"
    typical_capacitances(0.3);
```

Feedback

```
    switching_group() {
      input_switching_condition(fall);
      output_switching_condition(rise);
      pg_current(<pg_name>) {
        vector(<lu_template_name>) {
          reference_time : 93.2;
          index_output : "ZN";
          index_1("5.1");
          index_2("0.3");
          index_3("8.2 9.4 9.8");
          values("1.78 12.4 110.1");
        }
      }
     ...
    }
  }
  ...
}
```

In this case, index_output have a wrong pin 'ZN' because it does not match the pin 'Z', which is specified in related_outputs. To fix the problem, change the name from 'ZN' to 'Z' in index_output.

The following is an example message:

```
Error: Line 272, The pin 'ZN' specified in the 'index_output' can not
        be found in 'related_outputs'. (LBDB-731)
```

**What Next**

Check the library source file, and make the necessary correction.

---

## LBDB-732

(error) You cannot specify more than one output pin or one bit\n \tof a bus or bundle output pin in the '%s'.

**Description**

Only single output pin is allowed to be specified in a *index_output* and a *related_output*.

This message indicates that you specified multiple pins either in a *index_output* or in a *related_output*.

The following example shows an instance where this message occurs:

```
cell(lbdb732) {
  area : 2;
  pin(A) {
    direction : input;
    capacitance : 1;
  }
```

Feedback

```
    pin(B) {
      direction : input;
      capacitance : 1;
    }
    pin(Z) {
      direction : output;
      ...
    }
    pin(ZN) {
      direction : output;
      ...
    }

    dynamic_current() {
      when : "A";
      related_inputs : "B";
      related_outputs : "Z ZN"
      typical_capacitances(0.3 0.4);
      switching_group() {
        input_switching_condition(fall);
        output_switching_condition(rise fall);
        pg_current(<pg_name>) {
          vector(<lu_template_name>) {
            reference_time : 93.2;
            index_output : "Z ZN";
            index_1("5.1");
            index_2("0.3");
            index_3("8.2 9.4 9.8");
            values("1.78 12.4 110.1");
          }
        }
       ...
      }
    }
    ...
    intrinsic_parasitic() {
      /* default state */
      intrinsic_resistance(<pg_name>) {
         related_output : "Z ZN";
         value : 9.0;
      }
      intrinsic_capacitance(<pg_name>) {
         value : 8.2;
      }

    }

}
```

In this case, index_output and related_output have specified "Z ZN" which is wrong. To fix the problem, remove either 'Z' or 'ZN' from "Z ZN".

Feedback

The following is an example message:

```
Error: Line 272, You cannot specify more than one output pin or one bit
        of a bus or bundle output pin in the 'index_output'. (LBDB-732)
Error: Line 262, You cannot specify more than one output pin or one bit
        of a bus or bundle output pin in the 'related_output'. (LBDB-732)
```

**What Next**

Check the library source file, and make the necessary correction.

---

## LBDB-733

(error) An unbuffered %s pin '%s' should specify at least 1 'ccsn_first_stage' group.

**Description**

This message indicates that the following rule is not satisfied: If is_unbuffered is set to true on an inout/output pin, there should be at least 1 ccsn_first_stage group in the pin group.

The following example shows an instance where this message occurs:

```
cell(LAQM1RA) {
    ...
    pin (Q) {
      is_unbuffered : true;
      direction : "output";
      function : "IQ";
      ...
    }
  }
```

In this case, is_unbuffered of output pin 'Q' is true but the pin 'Q' has no 'ccsn_first_stage' group specified. To fix the problem, add the 'ccsn_first_stage' group into the pin 'Q'.

The following is an example message:

```
Error: Line 1620, Cell 'LAQM1RA', pin 'Q', An unbuffered output pin 'Q'
 should specify at least 1 'ccsn_first_stage' group. (LBDB-733)
```

**What Next**

Check the library source file, and add ccsn_first_stage or change the value of is_unbuffered.

---

## LBDB-734

(warning) an %s pin should specify at least 1 non-static '%s' groups.

Feedback

### Description

This warning message occurs when an input or inout pin does not have non-static ccsn_first_stage (or input_ccb for referenced CCSN modeling) group in the pin group and there are no related timing arcs referring this pin as the related_pin.

A static CCS noise model captures the behaviors of a static CCB (channel connected block). A static CCB refers to a CCB that does not have any input terminal or the voltage level at the input terminal does not affect the output current of the CCB. For example, a static CCS noise model can be characterized for a CMOS NAND gate when both inputs are tied to ground. Such static CCS noise models can be used in noise analysis with case analysis, but they cannot be used for propagating crosstalk delay or noise waveforms through the cell under study. See the *Make CCS Noise User Guide* for detailed information.

The following example shows an instance where this message occurs: The following is an example of incorrect input that causes this warning message:

```
cell(sample) {
    ...
    pin (A) {
        direction : input;
    }

    pin(Y ) {
      direction : output;
      ...
      ccsn_last_stage() {
        ...
      }

    }
  }
```

In this example, the A pin does not have the ccsn_first_stage group specified, and there are no associated timing arcs specifying ccs_first_stage groups. To fix the problem, add the non-static ccsn_first_stage group into pin A.

### What Next

Check the library source file and fix either the invalid CCS Noise data, or the direction of the pin.

---

## LBDB-735

(error) '%s' can't be specified in '%s' if power cell type is macro.

Feedback

### Description

Currently in Liberty syntax, we support two power cell types for CCS power model. One is 'stdcell' and another is 'macro'. For a 'macro' cell type, there is one restriction. You can not specify *related_outputs* in *dynamic_current* group and *output_switching_condition* in *switching_group* group if cell type in CCS power is 'macro'.

This message indicates that you either specified *related_outputs* or *output_switching_condition* when cell type is 'macro'.

The following example shows an instance where this message occurs:

```
cell(lbdb735) {
  area : 2;
  pin(A) {
    direction : input;
    capacitance : 1;
  }
    ...
  power_cell_type : macro;
  dynamic_current() {
    ...
    related_outputs : "Z ZN";                /* remove this */
    typical_capacitances(0.3 0.4);
    switching_group() {
      input_switching_condition(fall);
      output_switching_condition(rise fall); /* remove this */
      pg_current(<pg_name>) {
        vector(<lu_template_name>) {
        ...
        }
        ...
      }
     ...
    }
  }
  ...
}
```

In the case above, related_outputs and output_switching_condition are sepcified when cell type is 'macro', which is wrong. To fix the problem, we shall remove both attributes.

The following is an example message:

```
Error: Line 272, 'related_outputs' can't be specified in
 'dynamic_current' if power cell type is macro. (LBDB-735)
Error: Line 292, 'output_switching_condition' can't be specified in
 'switching_group' if power cell type is macro. (LBDB-735)
```

Feedback

### What Next

Check the library source file, and make the necessary correction. You can remove *output_switching_condition* or *related_outputs* attribute to avoid the error if you specify any of them.

---

## LBDB-736

(error) Size of '%s' must be the same as size of related_outputs.

### Description

Size of *output_switching_condition* specified in *switching_group* and size of *related_outputs* specified in *dynamic_current*, they must be identical, which is the same for *typical_capacitances*. Meaning that size of *typical_capacitances* specified in *dynamic_current* must be the same size as what *related_outputs* has.

This message indicates that either size is different between *related_outputs* and *typical_capacitances* or size is different between *related_outputs* and *output_switching_condition*.

The following example shows an instance where this message occurs:

```
cell(lbdb736) {
  area : 2;
  pin(A) {
    direction : input;
    capacitance : 1;
  }
  pin(B) {
    direction : input;
    capacitance : 1;
  }
  pin(Z) {
    direction : output;
    ...
  }
  pin(ZN) {
    direction : output;
    ...
  }

  dynamic_current() {
    when : "A";
    related_inputs : "B";
    related_outputs : "Z";
    typical_capacitances(0.3 0.4);
    switching_group() {
      input_switching_condition(fall);
      output_switching_condition(rise fall);
      pg_current(<pg_name>) {
```

Feedback

```
          vector(<lu_template_name>) {
            ...
        }
       ...
     }
   }
   ...
   }
}
```

In this case, size of related_outputs is 1 ("Z"), but size of typical_capacitances is 2 (0.3 and 0.4) and size of output_switching_condition is 2 (rise and fall) as well. To fix the problem, you can increase size of related_outputs to 2. You can change "Z" to "Z ZN".

The following is an example message:

```
Error: Line 272, Size of 'typical_capacitances' must be the same as size
 of related_outputs. (LBDB-736)
Error: Line 282, Size of 'output_switching_condition' must be the same as
 size of related_outputs. (LBDB-736)
```

**What Next**

Check the library source file, and make the necessary correction. Size of *output_switching_condition*, *related_outputs* and *typical_capacitances* must be identical.

---

## LBDB-737

(error) '%s' is required in '%s' if index_output is specified\n \tin one of pg_current groups.

**Description**

For expanded ccs power, if *index_output* is specified in one of *vector* groups, then all of the *vector* groups under the same *pg_current* group must define an *index_output*.

For compact ccs power, if *index_output* is specified in one of *compact_ccs_power* groups, then all of the *compact_ccs_power* groups under the same *pg_current* group must define an *index_output*.

In the case like this, *typical_capacitances* attribute is required in *dynamic_current* group.

This message indicates that there no *typical_capacitances* attribute can be found in *dynamic_current* when *index_output* is defined.

The following example shows an instance where this message occurs:

```
cell(lbdb737) {
  area : 2;
  pin(A) {
    direction : input;
    capacitance : 1;
```

Feedback

```
  }
  pin(B) {
    direction : input;
    capacitance : 1;
  }
  pin(Z) {
    direction : output;
    ...
  }
  pin(ZN) {
    direction : output;
    ...
  }

  dynamic_current() {
    when : "A";
    related_inputs : "B";
    related_outputs : "Z ZN"
    switching_group() {
      ...
      pg_current(<pg_name>) {
        vector(<lu_template_name>) {
          reference_time : 93.2;
          index_output : "Z";
          index_1("5.1");
          index_2("0.3");
          index_3("8.2 9.4 9.8");
          values("1.78 12.4 110.1");
        }
        ...
      }
     ...
    }
  }
  ...
}
```

In this case, index_output is specified, but no typical_capacitances attribute is specified. To fix the problem, add the line like this typical_capacitances(0.1); in the dynamic_current group.

The following is an example message:

```
Error: Line 272, 'typical_capacitances' is required in 'dynamic_current'
 if index_output is specified
       in one of pg_current groups. (LBDB-737)
```

**What Next**

Check the library source file, and make the necessary correction. Add *typical_capacitances* attribute to avoid the error.

Feedback

## LBDB-738

(error) In '%s' group, size of '%s' must be larger than one\n \tif index_output is defined in one of pg_current groups.

### Description

For expanded ccs power, if *index_output* is specified in one of *vector* groups, then all of the *vector* groups under the same *pg_current* group must define an *index_output* attribute.

For compact ccs power, if *index_output* is specified in one of *compact_ccs_power* groups, then all of the *compact_ccs_power* groups under the same *pg_current* group must define an *index_output* attribute.

In the case like this, the size of *related_outputs* must be larger than 1.

This message indicates that the size of *related_outputs* is less than 2, when *index_output* is defined.

The following example shows an instance where this message occurs:

```
cell(lbdb738) {
  area : 2;
  pin(A) {
    direction : input;
    capacitance : 1;
  }
  pin(B) {
    direction : input;
    capacitance : 1;
  }
  pin(Z) {
    direction : output;
    ...
  }
  pin(ZN) {
    direction : output;
    ...
  }

  dynamic_current() {
    related_outputs : "Z"
    ...
    switching_group() {
      ...
      pg_current(<pg_name>) {
        vector(<lu_template_name>) {
          reference_time : 93.2;
          index_output : "Z";
          index_1("5.1");
          index_2("0.3");
          index_3("8.2 9.4 9.8");
```

Feedback

```
          values("1.78 12.4 110.1");
        }
        ...
      }
    ...
```

In this case, size of related_outputs is 1 ("Z"), when index_output is defined, which is wrong. To fix problem, we shall increase size of related_outputs. We can change "Z" to "Z ZN".

The following is an example message:

```
Error: Line 272, In 'dynamic_current' group, size of 'related_outputs'
 must be larger than one
       if index_output is defined in one of pg_current groups. (LBDB-738)
```

**What Next**

Check the library source file, and make the necessary correction. Increase the size of *related_outputs* up to at least 2 to avoid the error.

---

## LBDB-739

(error) '%s' attribute is required for this '%s' group.

**Description**

For expanded ccs power, all *vector* groups under the same *pg_current* group must specify the same template. If there is no any *index_output* attribute specified in these *vector* groups and template used for these *vector* groups contains only one total_output_net_capacitance, then *related_outputs* attribute is required in *dynamic_current*.

For expanded ccs power, all *compact_ccs_power* groups under the same *pg_current* group must specify the same template. If there is no any *index_output* attribute specified in these *compact_ccs_power* groups and template used for these *compact_ccs_power* groups contains only one total_output_net_capacitance, then *related_outputs* attribute is required in *dynamic_current*.

This message indicates that you must specify *related_outputs* attribute in *dynamic_current* because the above rule is applied.

The following example shows an instance where this message occurs:

```
pg_current_template(test_2) {
    variable_1 : input_net_transition;
    variable_2 : total_output_net_capacitance;
    variable_3 : time;
    index_1("0.6 0.9");
    index_2("0.6 0.9");
```

Feedback

```
    index_3("0.3 4.7");
  }
  ...
  cell(lbdb739) {
    area : 2;
    pin(A) {
      direction : input;
      capacitance : 1;
    }
    pin(B) {
      direction : input;
      capacitance : 1;
    }
    pin(Z) {
      direction : output;
      ...
    }
    pin(ZN) {
      direction : output;
      ...
    }

    dynamic_current() {
      when : "A";
      related_inputs : "B";
      ...
      switching_group() {
        input_switching_condition(fall);
        output_switching_condition(rise fall);
        pg_current(<pg_name>) {
          vector(test_2) {
            reference_time : 93.2;
            index_1("5.1");
            index_2("0.3");
            index_3("8.2 9.4 9.8");
            values("1.78 12.4 110.1");
          }
          ...
        }
       ...
      }
    }
    ...
  }
```

In this case, there is no index_output specified in any vector under a pg_current group. Also, the template (test_2) that vector referred to contains only one total_output_net_capacitance (variable_2). In the case like this a related_outputs attribute is required to be defined in dynamic_current group. To fix the problem, we need to add a related_outputs attribute within a dynamic_current group.

Feedback

The following is an example message:

```
Error: Line 272, 'related_outputs' attribute is required for this
 'dynamic_current' group. (LBDB-739)
```

**What Next**

Check the library source file, and make the necessary correction.

---

## LBDB-740

(error) Size of '%s' in this '%s' group must be two.

**Description**

For expanded ccs power, all *vector* groups under the same *pg_current* group refer to the same *pg_current_template*.

For compact ccs power, a *compact_ccs_power* group refers to a *compact_lut_template*.

In the pg_current_template or compact_lut_template, if there are two total_output_net_capacitance, then size of *related_outputs* in *dynamic_current* must be two.

This message indicates that the size of *related_outputs* is not two.

The following example shows an instance where this message occurs:

```
pg_current_template(test_1) {
    variable_1 : input_net_transition;
    variable_2 : total_output_net_capacitance;
    variable_3 : total_output_net_capacitance;
    variable_4 : time;
    index_1("0.6 0.9");
    index_2("0.6 0.9");
    index_3("0.3 4.7");
    index_4("4.5 6.7 7.2");
 }
 ...
 cell(lbdb740) {
   area : 2;
   pin(A) {
     direction : input;
     capacitance : 1;
   }
   pin(B) {
     direction : input;
     capacitance : 1;
   }
   pin(Z) {
     direction : output;
     ...
```

Feedback

```
}
pin(ZN) {
  direction : output;
  ...
}

dynamic_current() {
  when : "A";
  related_inputs : "B";
  related_outputs : "Z"
  switching_group() {
    ...
    pg_current(<pg_name>) {
      vector(test_1) {
        reference_time : 93.2;
        index_1("5.1");
        index_2("0.3");
        index_3("0.3");
        index_4("8.2 9.4 9.8");
        values("1.78 12.4 110.1");
      }
      ...
    }
   ...
```

In this case, template 'test_1' contains two total_output_net_capacitance values, but size of related_outputs is 1 ("Z"), which is wrong. To fix the problem, you might want to increase size of related_outputs to two. You can change "Z" to "Z ZN".

The following is an example message:

```
Error: Line 262, Size of 'related_outputs' in this 'dynamic_current'
 group must be two. (LBDB-740)
```

### What Next

Check the library source file, and make the necessary correction.

---

## LBDB-741

(error) Two switching_group groups in line %d and %d are\n \toverlapping.

### Description

If either min_input_switching_count or max_input_switching_count specified in switching groups, and the switching count number they covered are overlapping, then these *switching_group* groups are considered overlapped.

If there is no min_input_switching_count or max_input_switching_count, then two *switching_group* groups are considered overlapped if both *input_switching_condition* and *output_switching_condition* are overlapped.

Feedback

If one of followings are true, then *output_switching_condition* is considered to be overlapping condition : + *output_switching_condition* is undefined on both *switching_group* groups. + *output_switching_condition* is defined on both *switching_group* groups and values of them are indentical.

If one of followings are true, then *input_switching_condition* is considered to be overlapping condition : + *input_switching_condition* is undefined on both *switching_group* groups. + *input_switching_condition* is defined on both *switching_group* groups and values of them are indentical. + values of them are difference, but one *input_switching_condition* is undefined, and another has value either "rise" or "fall".

This message indicates that two *switching_group* groups are overlapped because *input_switching_condition* and *output_switching_condition* both are in overlapping condition.

The following example shows an instance where this message occurs:

```
cell(lbdb741) {
  area : 2;
  pin(A) {
    direction : input;
    capacitance : 1;
  }
  pin(B) {
    direction : input;
    capacitance : 1;
  }
  pin(Z) {
    direction : output;
    ...
  }
  pin(ZN) {
    direction : output;
    ...
  }
  power_cell_type : stdcell;
  dynamic_current() {
    when : "A";
    related_inputs : "B";
    related_outputs : "Z ZN"
    typical_capacitances(0.3 0.4);
    switching_group() {
      input_switching_condition(fall);
      output_switching_condition(rise fall);
     ...
    }
    switching_group() {
      output_switching_condition(rise fall);
     ...
    }
```

Feedback

```
  }
  ...
```

In this case, both switching_group have same output_switching_condition, and one input_switching_condition has value "fall" and another is undefined. input_switching_condition is overlapping and which is same as output_switching_condition, so two switching_group groups are overlapped. To fix the problem, you might want to change the value to "fall fall" in one of output_switching_condition groups.

```
type(bus6) {
  base_type : array ;
  data_type : bit ;
  bit_width : 6 ;
  bit_from : 5 ;
  bit_to : 0 ;
  downto : true ;
}

cell(lbdb741) {
  bus(sel) {
    bus_type : bus6 ;
    direction : input ;
    capacitance : 2 ;
    related_power_pin : V1;
    related_ground_pin : G1;
  }

  bundle(C) {
    members(Cx, Cy, Cz);
    direction : input;
    capacitance : 2 ;
    related_power_pin : V1;
    related_ground_pin : G1;

  }
  area : 2;
  pg_pin(V1) {
     voltage_name : VDD1;
     pg_type : primary_power;
  }
  pg_pin(V2) {
     voltage_name : VDD2;
     pg_type : backup_power;
  }
  pg_pin(G1) {
     voltage_name : GND1;
     pg_type : primary_ground;
  }
  pg_pin(G2) {
     voltage_name : GND2;
     pg_type : backup_ground;
```

Feedback

```
  }
  power_cell_type : macro;
  dynamic_current() {
    related_inputs : "C sel";
    switching_group() {
      min_input_switching_count : 1;
      max_input_switching_count : 6;
      ...
    }
    switching_group() {
      min_input_switching_count : 5;
      max_input_switching_count : 9;
      ...
    }
    ...
  }
}

In this case, the min_input_switching_count and max_input_switching_count
 are
specified in "macro" cell type.  If that is the case, we need to check if
 the number
they covered are overlapping.  We don't worry about
 input_switching_condition or
output_switching_condition here because these attributes can't be
 sepcified
if either min_input_switching_count or max_input_switching_count is
 sepcified.

'C' is 6 bits bus and 'sel' is 3 bits bundle.  The total bits are 9.
For the first switching_group, the switching count is from 1 to 6, and
 for the
second switching_group, the switching count is from 5 to 9.  The number 5
 and 6
are covered in both groups, which means the two groups are overlapping.
 To fix the
problem, please change number 5 to 7 in the second switching_group or
 change number
6 to 4 in the first switching_group.
```

### SH EXAMPLE MESSAGE

```
Error: Line 100, Two switching_group groups in line 170 and 189 are
       overlapping. (LBDB-741)
```

### What Next

Check the library source file, and make the necessary correction.

Feedback

## LBDB-742

(error) Only one total_output_net_capacitance is allowed for the\n \ttemplate specified in this %s.

### Description

For expanded ccs power, if *index_output* is specified in one of *vector* groups, then all of the *vector* groups under the same *pg_current* group must define an *index_output* attribute.

For compact ccs power, if *index_output* is specified in one of *compact_ccs_power* groups, then all of the *compact_ccs_power* groups under the same *pg_current* group must define an *index_output* attribute.

In the case like this, the template , which these *vector*/*compact_ccs_power* groups referred to, must contains only one total_output_net_capacitance.

This message indicates that the template contains either more than one or less than one total_output_net_capacitance.

The following example shows an instance where this message occurs:

```
pg_current_template(test_1) {
    variable_1 : input_net_transition;
    variable_2 : total_output_net_capacitance;
    variable_3 : total_output_net_capacitance;
    variable_4 : time;
    index_1("0.6 0.9");
    index_2("0.6 0.9");
    index_3("0.3 4.7");
    index_4("4.5 6.7 7.2");
 }

 pg_current_template(test_2) {
    variable_1 : input_net_transition;
    variable_2 : total_output_net_capacitance;
    variable_3 : time;
    index_1("0.6 0.9");
    index_2("0.6 0.9");
    index_3("0.3 4.7");
 }
 ...
 cell(lbdb742) {
   area : 2;
   pin(A) {
     direction : input;
     capacitance : 1;
   }
   pin(B) {
     direction : input;
     capacitance : 1;
   }
```

Feedback

```
pin(Z) {
  direction : output;
  ...
}
pin(ZN) {
  direction : output;
  ...
}

dynamic_current() {
  when : "A";
  related_inputs : "B";
  related_outputs : "Z ZN";
  typical_capacitances(0.3 0.4);
  switching_group() {
    input_switching_condition(fall);
    output_switching_condition(rise fall);
    pg_current(<pg_name>) {
      vector(test_1) {
        reference_time : 93.2;
        index_output : "Z";
        index_1("5.1");
        index_2("0.3");
        index_3("0.3");
        index_4("8.2 9.4 9.8");
        values("1.78 12.4 110.1");
      }
    }
   ...
```

In this case, index_output is specified in vector, and the vector is referred to template 'test_1', which contains two total_output_net_capacitance values. That is wrong. To fix the problem, we can change referred template to 'test_2'.

The following is an example message:

```
Error: Line 272, Only one total_output_net_capacitance is allowed for the
       template specified in this vector. (LBDB-742)
```

**What Next**

Check the library source file, and make the necessary correction.

---

## LBDB-743

(error) All %s under a pg_current group must specify\n \tsame template.

**Description**

All *vector* or *compact_ccs_power* groups under the same *pg_current* group must specify the same template.

Feedback

This message indicates that you specified different templates in /fBvector or *compact_ccs_power* groups, which are all under same *pg_current*.

The following example shows an instance where this message occurs:

```
pg_current_template(test_1) {
    variable_1 : input_net_transition;
    variable_2 : total_output_net_capacitance;
    variable_3 : total_output_net_capacitance;
    variable_4 : time;
    index_1("0.6 0.9");
    index_2("0.6 0.9");
    index_3("0.3 4.7");
    index_4("4.5 6.7 7.2");
 }

 pg_current_template(test_2) {
    variable_1 : input_net_transition;
    variable_2 : total_output_net_capacitance;
    variable_3 : time;
    index_1("0.6 0.9");
    index_2("0.6 0.9");
    index_3("0.3 4.7");
 }
 ...
 cell(lbdb743) {
   area : 2;
   ...
   dynamic_current() {
      ...
     switching_group() {
       ...
       pg_current(<pg_name>) {
         vector(test_2) {
           reference_time : 93.2;
           index_1("5.1");
           index_2("0.3");
           index_3("8.2 9.4 9.8");
           values("1.78 12.4 110.1");
         }
         vector(test_1) {
           reference_time : 93.2;
           index_1("5.1");
           index_2("0.3");
           index_3("0.3");
           index_4("8.2 9.4 9.8");
           values("1.78 12.4 110.1");
         }
         ...
       }
      ...
```

Feedback

In this case, two vectors refer to different templates, one is 'test_1', and another is 'test_2'. This is wrong. To fix the problem, change 'test_1' to 'test_2' in second vector.

The following is an example message:

```
Error: Line 272, All vectors under a pg_current group must specify
       same template. (LBDB-743)
```

**What Next**

Check the library source file, and make the necessary correction.

---

## LBDB-744

(error) index_output is required for all %s in this\n \tpg_current group.

**Description**

For expanded ccs power, if *index_output* is specified in one of *vector* groups, then all of the *vector* groups under the same *pg_current* group must define an *index_output* attribute.

For compact ccs power, if *index_output* is specified in one of *compact_ccs_power* groups, then all of the *compact_ccs_power* groups under the same *pg_current* group must define an *index_output* attribute.

This message indicates that *index_output* attribute is not defined in all *vector*/*compact_ccs_power* groups under a *pg_current*.

Also to see, LBDB-737, LBDB-738, and LBDB-742.

The following example shows an instance where this message occurs:

```
cell(lbdb744) {
  area : 2;
  pin(A) {
    direction : input;
    capacitance : 1;
  }
  pin(B) {
    direction : input;
    capacitance : 1;
  }
  pin(Z) {
    direction : output;
    ...
  }
  pin(ZN) {
    direction : output;
    ...
  }

  dynamic_current() {
```

Feedback

```
    when : "A";
    related_inputs : "B";
    related_outputs : "Z ZN"
    typical_capacitances(0.3 0.4);
    switching_group() {
      input_switching_condition(fall);
      output_switching_condition(rise fall);
      pg_current(VDD) {
        vector(<lu_template_name>) {
          reference_time : 93.2;
          index_output : "ZN";
          index_1("5.1");
          index_2("0.3");
          index_3("8.2 9.4 9.8");
          values("1.78 12.4 110.1");
        }
         vector(<lu_template_name>) {
          reference_time : 93.2;
          index_output : "Z";
          index_1("5.1");
          index_2("0.3");
          index_3("8.2 9.4 9.8");
          values("1.78 12.4 110.1");
        }
        vector(<lu_template_name>) {
          reference_time : 93.2;
          index_1("5.1");
          index_2("0.3");
          index_3("8.2 9.4 9.8");
          values("1.78 12.4 110.1");
        }
      }
     ...
    }
  }
  ...
```

In this case, index_output is not specified in third vector, which is wrong. All vectors under pg_current(VDD) must specify index_output, because index_output has beed defined in one of vectors. To fix the problem, add index_output attribute to the third vector.

The following is an example message:

```
Error: Line 272, index_output is required for all vectors in this
       pg_current group. (LBDB-744)
```

**What Next**

Check the library source file, and make the necessary correction.

Feedback

## LBDB-745

(error) Under this dynamic_current, pin '%s' is specified in\n \t'%s' and '%s'.

### Description

/fBwhen, *related_outputs*, and *related_inputs* can't specify the same pin if they are under the same *dynamic_current* group.

This message indicates that you specified the same pin for two of following attributes : /fBwhen, *related_outputs*, and *related_inputs*

The following example shows an instance where this message occurs:

```
cell(lbdb745) {
  area : 2;
  pin(A) {
    direction : input;
    capacitance : 1;
  }
  pin(B) {
    direction : input;
    capacitance : 1;
  }
  pin(Z) {
    direction : output;
    ...
  }
  pin(ZN) {
    direction : output;
    ...
  }
  pin(ZN1) {
    direction : output;
    ...
  }
  dynamic_current() {
    when : "B + ZN'";
    related_inputs : "B";
    related_outputs : "Z ZN"
    typical_capacitances(0.3 0.4);
    switching_group() {
  ...
}
```

In this case, pin B is specified in both when and related_inputs, which is wrong. Same for pin ZN, which is specified in both when and related_outputs. To fix the problem, change "B" to "A" in related_inputs, and change "Z ZN" to "Z ZN1" in related_outputs.

Feedback

The following are example messages:

```
Error: Line 272, Under this dynamic_current, pin 'B' is specified in
 'related_inputs' and 'when'. (LBDB-745)

Error: Line 272, Under this dynamic_current, pin 'ZN' is specified in
 'related_outputs' and 'when'. (LBDB-745)
```

**What Next**

Check the library source file, and make the necessary correction.

---

## LBDB-746

(error) The %s of cell '%s'\n \tshould be specified before the %s '%s' is defined.

**Description**

You receive this message because the switch_cell_type is not defined before specifying the attributes related to switch cells.

The following example shows an instance where this message occurs: The following example shows the *switch_cell_type* is missing.

```
cell(sample) {
 pin(Y)  {
        switch_function : "sp";
        ...
 }
  ......
}
```

Correct it by add *fBswitch_cell_type attribute in the cell group.*

The following is an example message: Error: Line 191, The switch_cell_type attribute of cell 'sample' should be specified before the switch_function 'sp' is defined. (LBDB-746)

**What Next**

Make sure that *switch_cell_type* attribute exists before defining other attributes related switch cells.

---

## LBDB-747

(warning) %s on input pin and %s on output pin\n \thave same power rail value but different rail name.

**Description**

This message will only be applied for an isolation cell.

Feedback

In Liberty, we support two kinds (old and new) of pg syntax. If *input_signal_level* and *related_power_pin* both presented in new syntax, then *input_signal_level* will be chosen, and *related_power_pin* will be discarded.

If you are in new pg syntax, the message indicates that voltage name of either *input_signal_level* or *related_power_pin* in input pin is different from voltage name of *related_power_pin* in ouput pin, but both are referring to the same voltage value.

If you are in old pg syntax, the message indicates that rail name of *input_signal_level* in input pin is different from rail name of utput_signal_level *in ouput pin, but both rail names are referring to the same voltage value.*

The following example shows an instance where this message occurs:

```
/* This is an example of old pg syntax. */
library(libdb747) {
 ...
   power_supply() {
        default_power_rail : VDD;
        power_rail (VDDH, 3); /* high power */
        power_rail (VDDL, 3); /* low power */
        power_rail (VSS, 0.0);   /* primary ground */
  }
  ...
  /* operation conditions */
  nom_process   : 1;
  nom_temperature       : 25;
  nom_voltage   : 1.0;
  operating_conditions(3v_3v) {
    process     : 1;
    temperature : 25;
    voltage     : 1.0;
    tree_type   : balanced_tree
    power_rail (VDDH, 3);    /* high power */
    power_rail (VDDL, 3);    /* low power */
    power_rail (VSS, 0.0);   /* primary ground */
  }
  default_operating_conditions : 3v_3v;

 ...
cell (LVLHLEHX2M) {
  ...
  rail_connection (VDD, VDDL);
  area : 2.600000;
  is_isolation_cell : true;
  pin(A) {
    direction : input;
    input_signal_level : VDDH;
    capacitance : 0.1859;
    internal_power() {
    ...
  }
```

Feedback

```
 pin(Y) {
   direction : output;
   output_signal_level : VDDL;
   capacitance : 0.0;
   function : "(A & EN)";
   internal_power() {
     power_level : VDDL;
  ...
```

In this case, input_signal_level in input pin "A" is specified "VDDH" as power rail, which is different from "VDDL" specified in output_signal_level in output pin "Y". However, both "VDDH" and "VDDL" are referring to same voltage value "3" as defined in operating_conditions group.

The following is the example message:

```
Warning: Line 100, input_signal_level on input pin and
 output_signal_level on output pin
        have same power rail value but different rail name. (LBDB-747)
```

The following example shows another instance where this message occurs:

```
/* This is an example of new pg syntax */
library (libdb747) {
 ...
 voltage_map( VDD, 0.7);
 voltage_map(VDDH, 0.8); /* high power */
 voltage_map(VDDL, 0.8); /* low power */
 voltage_map(VSS, 0.0);   /* primary ground */
 ...
 cell(ISO) {
 ...
 pg_pin(VDD1) {
 voltage_name : VDDH;
 pg_type : primary_power;
 }
 pg_pin(VDD2) {
 voltage_name : VDDL;
 pg_type : primary_power;
 }

 is_isolation_cell : true;
 ...
 pin(in) {
        ...
        input_signal_level : VDDH;
        related_power_pin : VDD1;  /* discard */
        related_ground_pin : GND;
 }
 pin(out) {
        ...
        related_power_pin: VDD2;
        related_ground_pin : GND;
```

Feedback

```
 }
}

In this case, input_signal_level 'VDDH' is referred to voltage value 0.8
 and
related_power_pin 'VDD2' is referred to voltage value 0.8, too.  The
 voltage
value 0.8 in input pin is same as valtage value in output pin, but their
voltage names are different, one is VDDH, and the other is VDDL.

The following is the example message:
Warning: Line 100, input_signal_level on input pin and related_power_pin
 on output pin
        have same power rail value but different rail name. (LBDB-747)
```

### What Next

Check the library source file, and make the necessary correctiorn if it is needed.

---

## LBDB-748

(error) There are less than 4 vectors specified in group '%s'.

### Description

This error message occurs because Library Compiler requires at least 4 vectors inside each output_current_rise or output_current_fall group.

The following example shows an output_current_rise group without a dense vector and resulting error message.

```
output_current_template(CCT) {
  variable_1 : input_net_transition;
  variable_2 : total_output_net_capacitance;
  variable_3 : time;
 }
 . . .
 output_current_rise() {
   vector(CCT) {
      reference_time : 0.11;
      index_1 ("0.1");
      index_2 ("1");
      index_3 ("1, 2, 3");
      values ("1, 2, 3");
   }
   /* need at least additional 3 vectors to compile */
 }

Error: Line 198, There are less than 4 vectors specified in group
 'output_current_rise'.  (LBDB-748)
```

Feedback

### What Next

Check the library source file and add additional vectors.

---

## LBDB-749

(error) The size of %s should be at least %d.

### Description

The specification of the size of the index is less than the minim index size.

The following example shows an instance where this message occurs:

```
lu_table_template(basic_template) {
        variable_1 : input_net_transition;
        index_1 ("1");
  }

          receiver_capacitance1_rise(basic_template) {
                index_1 ("500");
                values ("1.3401");
          }
```

The following is an example message:

```
Error: Line 160, The size of index_1 should be at least 2. (LBDB-749)
```

### What Next

Check the library source file, and correct the problem by inceasing the size of the specified index.

---

## LBDB-750

(warning) Overwrite '%s' by default '%s' value %f defined\n \tin 'operating_conditions'.

### Description

There are two pg pin syntaxes in liberty: one is old syntax such as power_rail or rail_connection, another is new syntax such as voltage_map or pg_pin.

If new pg_pin syntax has been defined in .lib file , all PVT (process, voltage and temperature) defined in default_operating_conditions will be copied over to nom PVT.

For .lib/.db file without pg pin syntaxes, all PVT (process, voltage and temperature) defined in default_operating_conditions will be copied over to nom PVT.

The warning message indicates that whatever default nom PVT values you defined have been overwritten by values of default_operating_conditions.

Feedback

The following example shows an instance where this message occurs:

```
library(libdb750) {
 ...
  nom_process : 1.3
  nom_temperature : 20.0;
  nom_voltage : 4.0;

  operating_conditions ( TYPICAL ) {
    process : 1.0 ;
    temperature : 22.0 ;
    voltage : 3.0 ;
    tree_type : balanced_tree ;
  }
  default_operating_conditions :  TYPICAL;
   ...
  voltage_map(VDD1, 4.0); /* new pg pin syntax */
  voltage_map(VDD2, 4.5);
   ...
  cell (and2) {
   ...
  pg_pin(V1) { /* new pg pin syntax */
     voltage_name : VDD1;
     pg_type : primary_power;
  }
  pg_pin(V2) {
     voltage_name : VDD2;
     pg_type : backup_power;
  }
  ...
```

In this case, nom_process will be overwritten by "process" value 1.0, which is defined in default operating_conditions group. nom_temperature will be overwritten by "temperature" value 22.0, which is defined in default operating_conditions group. nom_voltage will be overwritten by "voltage" value 3.0, which is defined in default operating_conditions group.

The following is an example message:

```
Warning: Line 100, Overwrite 'nom_process' by default 'process' value 1.0
 defined
        in 'operating_conditions'. (LBDB-750)
Warning: Line 100, Overwrite 'nom_temperature' by default 'temperature'
 value 22.0 defined
        in 'operating_conditions'. (LBDB-750)
Warning: Line 100, Overwrite 'nom_voltage' by default 'voltage' value 3.0
 defined
        in 'operating_conditions'. (LBDB-750)
```

## LBDB-751

(error) The %s group with name '%s', has not been defined in the %s.

Feedback

### Description

This message indicates that the referred group name is undefined in the library.

The following is an example message:

```
Error: Line 57, The base_curves group with name 'AND2",
       has not been defined in the library.(LBDB-751)
```

### What Next

Add the referred group in the library. Or remove the reference

---

## LBDB-752

(error) The '%s' attribute is missing in compressed_lut_template '%s'.

### Description

This message indicates that the attribute is not defined in the compressed_lut_template group.

The following is an example message:

```
Error: Line 58, The 'base_curves_group' attribute is missing in
 compressed_lut_template 'LTT3'.(LBDB-752)
```

### What Next

Add the attribute in compressed_lut_template.

---

## LBDB-753

(error) The '%s' base curve parameter is missing in compressed_lut_template '%s' index_3.

### Description

For ccs timing base curves, these six elementary curve parameters are must needed: init_current, peak_current, peak_voltage, peak_time, left_id, right_id. Additional parameters are also allowed, the index_3 value list can contains more than six parameters.

The following example shows an instance where this message occurs:

```
compressed_lut_template(LTT3) {
        variable_1 : input_net_transition;
        variable_2 : total_output_net_capacitance;
        variable_3 : curve_parameters;
        index_1 ("0.1, 0.2");
        index_2 ("1.0, 1.2");
```

Feedback

```
        index_3 ("init_current, peak_current, peak_voltage, left_id,
 right_id");
    }
```

The following is an example message:

```
Error: Line 60, The 'peak_time' base curve parameter is missing in
 compressed_lut_template 'LTT3' index_3.(LBDB-753)
```

**What Next**

Add the missing elementary curve parameter in compressed_lut_template index_3.

---

## LBDB-754

(error) Redudant curve_x defined in base_curves '%s'.

**Description**

In a base_curves group, only one curve_x definition is allowed, otherwise it brings confusing to users.

The following example shows an instance where this message occurs:

```
base_curves(AND2_BC) {
        base_curve_type : ccs_timing_half_curve;
        curve_x ("0.1, 0.5, 0.9");
        curve_x ("0.2, 0.5, 0.8");
        curve_y (1, "0.8, 0.5, 0.2");
        curve_y (2, "0.85, 0.5, 0.15");
        ...
    }
```

The following is an example message:

```
Error: Line 62, Redudant curve_x defined in base_curves
 'AND2_BC'.(LBDB-754)
```

**What Next**

Remove redudant curve_x definition in base curves group, left one curve_x is enough.

---

## LBDB-755

(error) Illegal base curve data is specified in '%s'.

Feedback

### Description

The syntax of base_curves group:

```
base_curves("name") {
    base_curve_type : ccs_half_curve;
    curve_x ("float..., float");
    curve_y (curve_id, "float..., float";
    ...
}
```

Following are rules for base curve data: For curve data specified in curve_x: 1. At least 3 points are specified; 2. The value should between 0 and 1; 3. The values should be monotonic increasing;

For curve_data specified in curve_y: 1. number of points should be same as that in curve_x; 2. curve_y should be defined after curve_x; 3. valid boundary is [-inf, 1] for both compact CCS timing and compact CCS power.

The following example shows an instance where this message occurs:

```
base_curves(AND2_BC) {
        base_curve_type : ccs_timing_half_curve;
        curve_x ("0.1, 0.5, 1.1");
        curve_y (1, "0.8, 0.5, 0.2, 0.1");
        curve_y (2, "0.85, 0.5, 0.15");
        ...
    }
```

The following is an example message:

```
Error: Line 63, Illegal base curve data is specified in
 'curve_x'.(LBDB-755)
Error: Line 64, Illegal base curve data is specified in
 'curve_y'.(LBDB-755)
```

### What Next

Check base curve data according these rules, fix error.

---

## LBDB-756

(error) No curve_x definition found before curve_y in the base_curves '%s'.

### Description

In a base_curves group, Only and Must have one curve_x defined before curve_y.

The following example shows an instance where this message occurs:

```
base_curves(AND2_BC) {
        base_curve_type : ccs_timing_half_curve;
```

Feedback

```
        curve_y (1, "0.8, 0.5, 0.2");
        curve_y (2, "0.85, 0.5, 0.15");
        ...
    }
```

The following is an example message:

```
Error: Line 65, No curve_x definition found before curve_y in the
 base_curves 'AND2_BC'.(LBDB-756)
```

**What Next**

Add the curve_x definition before the curve_y, otherwise, the base curve data is incomplete.

---

## LBDB-757

(error) No curve_y definition found in the base_curves '%s'.

**Description**

In a base_curves group, base curves are demonstrated by curve_x and curve_y together. curve_y can't be absent.

The following example shows an instance where this message occurs:

```
base_curves(AND2_BC) {
        base_curve_type : ccs_timing_half_curve;
        curve_x ("0.1, 0.5, 0.9");
    }
```

The following is an example message:

```
Error: Line 63, No curve_y definition found in the base_curves
 'AND2_BC'.(LBDB-757)
```

**What Next**

Add the curve_y definition after the curve_x, in the base curves group.

---

## LBDB-758

(error) Duplicate or negative curve_id '%d' is specified.

**Description**

In a base_curves group, the curve_id in curve_y should satisfy following rules: 1. curve_id should be an interger that greater than 0; 2. curve_id should be unique in the base_curves group;

Feedback

The following example shows an instance where this message occurs:

```
base_curves(AND2_BC) {
        base_curve_type : ccs_timing_half_curve;
        curve_x ("0.1, 0.5, 1.1");
        curve_y (-1, "0.8, 0.5, 0.2, 0.1");
        curve_y (2, "0.85, 0.5, 0.15");
        curve_y (4, "0.8, 0.5, 0.2, 0.1");
        curve_y (4, "0.85, 0.5, 0.15");
        ...
    }
```

The following is an example message:

```
Error: Line 63, Duplicate or negative curve_id '-1' is
 specified.(LBDB-758)
Error: Line 60, Duplicate or negative curve_id '4' is
 specified.(LBDB-758)
```

**What Next**

Check rules above, correct the curve_id data in curve_y.

---

## LBDB-759

(error) Bad init_current value '%g' is specified in '%s'.

**Description**

Init_current is one of six parameters to model compressed ccs timing base curve. In compressed_ccs_rise group, the init_current values should >= 0; And in compressed_ccs_fall group, the init_current values should <= 0;

The following example shows an instance where this message occurs:

```
library(name) {
  ...
  base_curves (ctbct1){
    base_curve_type : ccs_timing_half_curve;
    curve_x("0.2, 0.5, 0.8");
    curve_y(1, "0.8, 0.5, 0.2");
    curve_y(2, "0.75, 0.5, 0.35");
    curve_y(3, "0.85, 0.5, 0.15");
    curve_y(4, "0.85, 0.5, 0.15");
  }
  compressed_lut_template(LTT3) {
    variable_1 : input_net_transition;
    variable_2 : total_output_net_capacitance;
    variable_3 : curve_parameters;
    index_1 ("0.1, 0.2");
    index_2 ("1.0, 2.0");
```

Feedback

```
    index_3 ("init_current, peak_current, peak_voltage, peak_time,
 left_id, right_id");
    base_curves_group: "ctbct1";
  }

  cell(cell_name) {
    pin(pin_name) {
        timing() {
            compressed_ccs_rise("LTT3") {
              values("0, 0.5, 0.6, 0.8, 1, 3", \
                     "-0.15, 0.55, 0.65, 0.85, 2, 4", \
                     "0.2, 0.6, 0.7, 0.9, 3, 2", \
                     "0.25, 0.65, 0.75, 0.95, 4, 1");
            } /* end of compressed_ccs_rise */
        }
    }
  }
  ...
}
```

The following is an example message:

```
Error: Line 1163, Bad init_current value '-0.15' is specified in
 'compressed_ccs_rise'.(LBDB-759)
```

**What Next**

According the rule above, correct the init_current value.

---

## LBDB-760

(error) Bad peak_current value '%g' is specified in '%s'.

**Description**

peak_current is one of six parameters to model compressed ccs timing base curve. In compressed_ccs_rise group, the peak_current values should > 0; And in compressed_ccs_fall group, the peak_current values should < 0;

The following example shows an instance where this message occurs:

```
library(name) {
  ...
  base_curves (ctbct1){
    base_curve_type : ccs_timing_half_curve;
    curve_x("0.2, 0.5, 0.8");
    curve_y(1, "0.8, 0.5, 0.2");
    curve_y(2, "0.75, 0.5, 0.35");
    curve_y(3, "0.85, 0.5, 0.15");
    curve_y(4, "0.85, 0.5, 0.15");
  }
  compressed_lut_template(LTT3) {
```

Feedback

```
    variable_1 : input_net_transition;
    variable_2 : total_output_net_capacitance;
    variable_3 : curve_parameters;
    index_1 ("0.1, 0.2");
    index_2 ("1.0, 2.0");
    index_3 ("init_current, peak_current, peak_voltage, peak_time,
 left_id, right_id");
    base_curves_group: "ctbct1";
  }

  cell(cell_name) {
    pin(pin_name) {
        timing() {
            compressed_ccs_rise("LTT3") {
              values("0.1, 0, 0.6, 0.8, 1, 3", \
                     "0.15, -0.55, 0.65, 0.85, 2, 4", \
                     "0.2, 0.6, 0.7, 0.9, 3, 2", \
                     "0.25, 0.65, 0.75, 0.95, 4, 1");
            } /* end of compressed_ccs_rise */
        }
    }
  }
  ...
}
```

The following are example messages:

```
Error: Line 1163, Bad peak_current value '0' is specified in
 'compressed_ccs_rise'.(LBDB-760)
Error: Line 1163, Bad peak_current value '-0.55' is specified in
 'compressed_ccs_rise'.(LBDB-760)
```

**What Next**

According the rule above, correct the peak_current value.

---

## LBDB-761

(error) Non-integer base curve id '%s' is specified in '%s'.

**Description**

Curve_id should be integers because it is defined with integer data type.

For compact ccs timing, curve_id is the data corresponding to left_id or right_id in compressed_ccs_rise/compressed_ccs_fall group.

For compact ccs power, curve_id is the data corresponding to bc_id* in compact_ccs_power group.

Feedback

The following example shows an instance where this message occurs:

```
library(name) {
  ...
  base_curves (ctbct1){
    base_curve_type : ccs_timing_half_curve;
    curve_x("0.2, 0.5, 0.8");
    curve_y(1, "0.8, 0.5, 0.2");
    curve_y(2, "0.75, 0.5, 0.35");
    curve_y(3, "0.85, 0.5, 0.15");
    curve_y(4, "0.85, 0.5, 0.15");
  }
  compressed_lut_template(LTT3) {
    variable_1 : input_net_transition;
    variable_2 : total_output_net_capacitance;
    variable_3 : curve_parameters;
    index_1 ("0.1, 0.2");
    index_2 ("1.0, 2.0");
    index_3 ("init_current, peak_current, peak_voltage, peak_time,
 left_id, right_id");
    base_curves_group: "ctbct1";
  }

  cell(cell_name) {
    pin(pin_name) {
        timing() {
            compressed_ccs_rise("LTT3") {
              values("0.1, 0, 0.6, 0.8, 1, 3", \
                     "0.15, -0.55, 0.65, 0.85, 2, 4", \
                     "0.2, 0.6, 0.7, 0.9, 3, 2", \
                     "0.25, 0.65, 0.75, 0.95, 4, 1");
            } /* end of compressed_ccs_rise */
        }
    }
  }
  ...
}
```

The following is an example message:

```
Error: Line 1163, Non-integer curve_id(lef_id/right_id) '3.5' is
 specified in 'compressed_ccs_rise'.(LBDB-761)
Error: Line 1163, Non-integer curve_id(lef_id/right_id) '2.0' is
 specified in 'compressed_ccs_rise'.(LBDB-761)
```

**What Next**

According the rule above, correct the left_id/right_id value in compressed_ccs_rise/ compressed_ccs_fall group, or the bc_id* value in compact_ccs_power group.

Feedback

## LBDB-762

(error) Undefined base curve id '%d' is referenced in '%s'.

### Description

Curve_id must be predefined in previous base_curve group.

For compact ccs timing, curve_id is the data corresponding to left_id or right_id in compressed_ccs_rise/compressed_ccs_fall group.

For compact ccs power, curve_id is the data corresponding to bc_id* in compact_ccs_power group.

The following example shows an instance where this message occurs:

```
library(name) {
  ...
  base_curves (ctbct1){
    base_curve_type : ccs_timing_half_curve;
    curve_x("0.2, 0.5, 0.8");
    curve_y(1, "0.8, 0.5, 0.2");
    curve_y(2, "0.75, 0.5, 0.35");
    curve_y(3, "0.85, 0.5, 0.15");
    curve_y(4, "0.85, 0.5, 0.15");
  }
  compressed_lut_template(LTT3) {
    variable_1 : input_net_transition;
    variable_2 : total_output_net_capacitance;
    variable_3 : curve_parameters;
    index_1 ("0.1, 0.2");
    index_2 ("1.0, 2.0");
    index_3 ("init_current, peak_current, peak_voltage, peak_time,
 left_id, right_id");
    base_curves_group: "ctbct1";
  }

  cell(cell_name) {
    pin(pin_name) {
        timing() {
            compressed_ccs_rise("LTT3") {
              values("0.1, 0, 0.6, 0.8, 1, 31", \
                     "0.15, -0.55, 0.65, 0.85, 7, 4", \
                     "0.2, 0.6, 0.7, 0.9, 3, 2", \
                     "0.25, 0.65, 0.75, 0.95, 4, 1");
            } /* end of compressed_ccs_rise */
        }
    }
  }
  ...
}
```

Feedback

The following is an example message:

```
Error: Line 1163, Undefined curve_id(lef_id/right_id) '31' is referenced
 in 'compressed_ccs_rise'.(LBDB-762)
Error: Line 1163, Undefined curve_id(lef_id/right_id) '7' is referenced
 in 'compressed_ccs_rise'.(LBDB-762)
```

**What Next**

According the rule above, correct the left_id/right_id value in compressed_ccs_rise/ compressed_ccs_fall group, or the bc_id* value in compact_ccs_power group.

---

## LBDB-763

(error) Incompatible %s data specified in %s.

**Description**

For ccs timing and power data, there are two models that can be used. One is expanded model, the other is compact model.

Expanded ccs timing model is specified by output_current_{rise|fall}. Compact ccs timing model is specified by compact_ccs_{rise|fall}. Expanded ccs power model is specified by vector under pg_current group. Compact ccs power model is specified by compact_ccs_power under pg_current group.

Currently, only one model is supported in one library. If both of them are used in the library, the error is issued.

Users can't specify expanded ccs timing/power model for some cells in the library, while compact ccs timing/power model for other cells.

The following example shows an instance where this message occurs:

```
library(name) {
  ...

  cell(cell_name1) {
    pin(pin_name) {
        timing() {
            output_current_rise() {
              vector("CT3") {
                ...
              }
              ...
            }
            output_current_fall() {
              vector("CT3") {
                ...
              }
              ...
```

Feedback

```
            }
          }
        }
      }
      cell(cell_name2) {
        pin(pin_name) {
            timing() {
              compact_ccs_rise("LTT3") {
                values("..."..., "...");
              }
              compact_ccs_fall("LTT3") {
                values("..."..., "...");
              }
          }
        }
      }
      ...
}
```

The following is an example message:

```
Error: Line 1, Un-Compatible CCS timing data specified in timing
 arc.(LBDB-763)
```

**What Next**

Keep only one ccs timing/power model in the library.

---

## LBDB-764

(error) Redundant base_curves group '%s' with '%s' type defined in library.

**Description**

Since the compressed ccs data (for the time being, it only support compressed ccs timing) is consumed by PT, to ensure the performance of PT, guarantee the ccs accuracy and ease the implementation of PT support and implementation, we only support one base curve group with specific base curve type.

The following example shows an instance where this message occurs:

```
library(name) {
  ...

  base_curves(bct1) {
    base_curve_type : ccs_timing_half_curve;
    curve_x ("0.1, 0.5, 0.9");
    curve_y (1, "0.2, 0.5, 0.8");
    ...
  }
  base_curves(bct2) {
    base_curve_type : ccs_timing_half_curve;
```

Feedback

```
    curve_x ("0.1, 0.5, 0.9");
    curve_y (1, "0.33, 0.55, 0.88");
    ...
  }
  ...
}
```

The following is an example message:

```
Error: Line 165, Redundant base_curves group 'bct2' with
 'ccs_timing_half_curve' type defined in library.(LBDB-764)
```

**What Next**

Remove the base_curves group or combine same base curve type groups into one base_curves group if possible. Leave only one base_curves group with the base curve type.

---

## LBDB-765

(error) Bad peak_voltage value '%g' is specified in '%s',\n \tLegal vale should be in range of [%g, %g].

**Description**

peak_voltage is one of six parameters to model compact ccs timing base curve. In [va_]compact_ccs_rise/fall group, the peak_voltage values should less than Vdd of this port.

Note that the ways to define VDD are different between pg pin library, power-rail library, non-pg pin library. details may need to refer to application notes for pg pin supporting.

The following example shows an instance where this message occurs:

```
library(name) {
  ...
  operating_conditions(typical) {
    process     : 1;
    temperature : 25;
    voltage     : 1.0;
    tree_type   : balanced_tree
  }
  default_operating_conditions : typical;

  base_curves (ctbct1){
    base_curve_type : ccs_timing_half_curve;
    curve_x("0.2, 0.5, 0.8");
    curve_y(1, "0.8, 0.5, 0.2");
    curve_y(2, "0.75, 0.5, 0.35");
    curve_y(3, "0.85, 0.5, 0.15");
    curve_y(4, "0.85, 0.5, 0.15");
```

Feedback

```
  }
  compressed_lut_template(LTT3) {
    variable_1 : input_net_transition;
    variable_2 : total_output_net_capacitance;
    variable_3 : curve_parameters;
    index_1 ("0.1, 0.2");
    index_2 ("1.0, 2.0");
    index_3 ("init_current, peak_current, peak_voltage, peak_time,
 left_id, right_id");
    base_curves_group: "ctbct1";
  }

  cell(cell_name) {
    pin(pin_name) {
        timing() {
            compact_ccs_rise("LTT3") {
              values("0.1, 0, 0.6, 0.8, 1, 3", \
                     "0.15, 1.55, 0.65, 0.85, 2, 4", \
                     "0.2, 0.6, 0.7, 0.9, 3, 2", \
                     "0.25, 0.65, 0.75, 0.95, 4, 1");
            } /* end of compressed_ccs_rise */
        }
    }
  }
  ...
}
```

The following is an example message:

```
Error: Line 1163, Bad peak_voltage value '1.55' is specified in
 'compact_ccs_rise',
        Legal value should be in range of [0, 1.0].(LBDB-765)
```

**What Next**

According the rule above, correct the peak_voltage value.

---

## LBDB-766

(warning) The '%s' attribute of %s '%s' overwrites the value specified \tin the '%s' attrbiute.

**Description**

This message indicates that the latest value overwrites the previous value defined for the same attribute.

The following is an example message:

```
Error: Line 100, The 'retention_pin' attribute of pin 'A' overwrites the
 value specified
        in the 'map_to_logic' attrbiute. (LBDB-766)
```

Feedback

### What Next

Make sure that it is what have been expected.

---

## LBDB-767

(error) No mandatory attribute '%s' defined in sensitization group '%s'.

### Description

sensitization group is to describe the complete state patterns (by vector attributes) for a specific list of pins (by pin_names attribute). vector and pin_names are mandatory attributes in the group.

The following example shows an instance where this message occurs:

```
library(name) {
  ...
  sensitization (my_sensitization){
    pin_names("A, B, Y");
  }

  ...
}
```

The following is an example message:

```
Error: Line 11, No mandatory attribute 'vector' defined in sensitization
 group 'my_sensitization'.(LBDB-767)
```

### What Next

According to message, add pin_names or vector to make the sensitization group complete.

---

## LBDB-768

(error) Duplicate pin name '%s' specified in '%s'.

### Description

This error is issued when duplicate pin name specified in sensitization group pin_names, or cell/timing group pin_name_map attribute. Pin names in the attribute are used to generate stimuli from sensitization group for timing arc.

The following example shows an instance where this message occurs:

```
library(name) {
  ...
  sensitization (my_sensitization){
```

Feedback

```
    pin_names("A, B, B, Y");
    vector (0, "0 0 0 0");
    vector (1, "0 0 0 1");
    ...
  }

  ...
}
```

The following is an example message:

```
Error: Line 12, Duplicate pin name 'B' specified in
 'pin_names'.(LBDB-768)
```

### What Next

Correct the duplicate pin name definition, if the number of pin is changed because of the fix, other sensitization related attributes may also need to be updated.

---

## LBDB-769

(error) %s id (%d) specified sensitization vector.

### Description

This error is issued when duplicate or negative vector id specified in vector attribute. The vector attribute is composed by: vector(vector_id, vector_string); The vector id should greater than or equal to zero, and unique in current sensitization group.

The following example shows an instance where this message occurs:

```
library(name) {
  ...
  sensitization (my_sensitization){
    pin_names("A, B, C, Y");
    vector (0, "0 0 0 0");
    vector (1, "0 0 0 1");
    vector (1, "0 0 1 1");
    vector (-1, "0 0 1 0");
    ...
  }

  ...
}
```

The following is an example message:

```
Error: Line 14, Duplicate id (1) specified in sensitization
 vector.(LBDB-769)
Error: Line 15, Negative id (-1) specified in sensitization
 vector.(LBDB-769)
```

Feedback

### What Next

Correct the vector id in the vector attribute, according to the rule above.

---

## LBDB-770

(error) No pin_names attribute defined before the vector.

### Description

In sensitization group, pin_names attribute should be declared before all vectors. The error is issued when processing vector but find no pin_names attribute defined before in the sensitization group.

The following example shows an instance where this message occurs:

```
library(name) {
  ...
  sensitization (my_sensitization){
    vector (0, "0 0 0 0");
    pin_names("A, B, C, Y");
    vector (1, "0 0 0 1");
    vector (2, "0 0 1 0");
    vector (3, "0 0 1 1");
    ...
  }

  ...
}
```

The following is an example message:

```
Error: Line 13, No pin_names attribute defined before the
 vector.(LBDB-770)
```

### What Next

Add pin_names attribute if it's missing, or reorder the pin_names attribute to the first declaration in the sensitization group.

---

## LBDB-771

(error) Number of elements in the vector string (%d) is different from number of pin (%d).

### Description

This error is issued when the number of elements in vector string is different the number of pins defined in pin_names attribute. The vector attribute is composed by: vector(vector_id, vector_string);

Feedback

The following example shows an instance where this message occurs:

```
library(name) {
  ...
  sensitization (my_sensitization){
    pin_names("A, B, C, Y");
    vector (0, "0 0 0 0");
    vector (1, "0 0 0 1");
    vector (2, "0 0 1 0");
    vector (3, "0 1 1");
    ...
  }

  ...
}
```

The following is an example message:

```
Error: Line 17, Number of elements in the vector string (3) is different
 from number of pins (4).(LBDB-771)
```

### What Next

Correct the vector string in the vector attribute, make it's elements number equal to number of pin in pin_names.

---

## LBDB-772

(error) Number of pins (%d) in the %s pin_name_map \tis different from that (%d) in sensitization master '%s'.

### Description

This error is issued when run into following situations: 1. Number of pin in cell->pin_name_map is different from that in cell->sensitization_master->pin_names. 2. Number of pin in timing->pin_name_map is different from that in timing->sensitization_master->pin_names. 3. When there is timing->pin_name_map defined, but there is only cell->sensitization_master attribute, Number of pin in timing->pin_name_map is different from that in cell->sensitization_master->pin_names.

When there is pin_name_map defined in cell or timing, pins in the attribute is used to generate stimuli for characterization, instead of pin_names in sensitization master. but the number of pins should be the same, otherwise vectors in the sensitization_master can not be mapped to these pins properly.

The following example shows an instance where this message occurs:

```
library(name) {
  ...
  sensitization (my_sensitization){
    pin_names("A, B, C, Y");
```

Feedback

```
    vector (0, "0 0 0 0");
    vector (1, "0 0 0 1");
    vector (2, "0 0 1 0");
    vector (3, "0 0 1 1");
    ...
  }

  cell (my_cell) {
    sensitization_master : my_sensitization;
    pin_name_map("A, B, Z");
    pin (A) {
        ...
    }
    ...
  }
  ...
}
```

The following is an example message:

```
Error: Line 1170, Number of pins (3) in the cell pin_name_map
        is different from that (4) in sensitization master
 'my_sensitization'.(LBDB-772)
```

### What Next

Correct the vector string in the vector attribute, make it's elements number equal to number of pin in pin_names.

---

## LBDB-773

(error) Illegal pin name '%s' specified in %s level sensitization attribute.

### Description

This error is issued when pin name specified in cell/timing level sensitizaiotn attribute is an illegal(undefined) pin in the cell: This screener rule checks following pin names: 1. pin names specified in cell->pin_name_map attribute. 2. when there is no cell->pin_name_map, pin_names in cell->sensitization_master. 3. pin names in timing->pin_name_map attribute. 4. when there is no timing->pin_name_map, pin_names in timing->sensitization_master.

The following example shows an instance where this message occurs:

```
library(name) {
  ...
  sensitization (my_sensitization){
    pin_names("A, B, C");
    vector (0, "0 0 0");
    vector (1, "0 0 1");
    vector (2, "0 1 0");
```

Feedback

```
    vector (3, "0 1 1");
    ...
  }

  cell (my_cell) {
    sensitization_master : my_sensitization;
    pin_name_map("A, B, Z");
    pin (A) {
        ...
    }
    pin (B) {
        ...
    }
    pin (Y) {
        ...
    }
  }
  ...
}
```

The following is an example message:

```
Error: Line 1170, Illegal pin name 'Z' specified in cell level
 sensitization attribute.(LBDB-773)
```

**What Next**

Check pin names in the sensitization attributes(pin_name_map, or pin_names in sensitization_master), make sure all pin names are real pin in the cell.

---

## LBDB-774

(error) Incomplete sensitization info in the timing arc.

**Description**

This error is issued when there is wave_rise/wave_fall attribute defined in the timing group, but none of following attributes found: 1. pin_name_map attribute in current timing group. 2. sensitization_master attribute in current timing group. 3. pin_name_map attribute in current cell. 2. sensitization_master attribute in current cell.

Since there is no way to get the pin names for the wave_rise/wave_fall, the sensitization information for the timing is incomplete.

The following example shows an instance where this message occurs:

```
library(name) {
  ...
  sensitization (my_sensitization){
    pin_names("A, B, C");
    vector (0, "0 0 0");
```

Feedback

```
    vector (1, "0 0 1");
    vector (2, "0 1 0");
    vector (3, "0 1 1");
    ...
  }

  cell (my_cell) {
    ...
    pin (A) {
        timing() {
            wave_rise (1, 3);
            wave_fall (2, 0);
            /* other timing arc data */
        }
    }
  }
  ...
}
```

The following is an example message:

```
Error: Line 1170, Incomplete sensitization info in the timing
 arc.(LBDB-774)
```

### What Next

Define pin_name_map or sensitization_master attribute either in the timing arc or in the cell, make sure the sensitization information in this timing group is complete.

---

## LBDB-775

(warning) No sensitization_master defined, sensitization vector is derived from vector id in the timing arc.

### Description

This warning is issued when there is wave_rise/wave_fall attribute defined in the timing group, But no sensitization_master attribute defined either in the timing group or current cell.

This is a simplified usage model of sensitization, that sensitization vector used to get from sensitization group by vector id, but is derived from vector id and number of pin in pin_name_map by binary scale directly in such cases.

Example:

```
timing() {
    pin_name_map("A, B, C, Z");
    wave_rise (5, 8, 15);
}
```

Here '5' implies vector string of "0 1 0 1", '8' implies vector string of "1 0 0 0", and so on.

The following example shows an instance where this message occurs:

```
library(name) {
  ...

  cell (my_cell) {
    ...
    pin (A) {
        timing() {
            pin_name_map("A, B, Y");
            wave_rise (1, 3);
            wave_fall (2, 0);
            /* other timing arc data */
        }
    }
  }
  ...
}
```

The following is an example message:

```
Warning: Line 1170, No sensitization_master defined, sensitization vector
 is derived from vector id in the timing arc.(LBDB-775)
```

### What Next

If this is what the users wanted, just ignore this warning. otherwise, specify valid sensitization_master attribute in the timing group or current cell.

---

## LBDB-776

(error) Vector id (%d) in '%s' is undefined in sensitization group '%s'.

### Description

When there is sensitization master defined for the timing arc, Vector id specified in wave_rise and wave_fall should be predefined in the sensitization group, otherwise, this error is issued.

The following example shows an instance where this message occurs:

```
library(name) {
  sensitization (my_sensitization){
    pin_names("A, B, C");
    vector (0, "1 0 0");
    vector (1, "0 0 1");
    vector (2, "1 1 0");
    vector (3, "0 1 1");
  }
  ...
```

Feedback

```
  cell (my_cell) {
    ...
    pin (A) {
        timing() {
            sensitization_master : my_sensitization;
            pin_name_map ("A, B, Y");
            wave_rise (1, 3);
            wave_fall (2, 4);
            /* other timing arc data */
        }
    }
  }
  ...
}
```

The following is an example message:

```
Error: Line 1181, Vector id (4) in 'wave_fall' is undefined in
 sensitization group 'my_sensitization'(LBDB-776)
```

### What Next

Modify vector id in the wave_rise/wave_fall, make sure every vector id is predefined in sensitization master group.

---

## LBDB-777

(error) Vector id (%d) for implied vector string in '%s' is out of range (0~%d).

### Description

When there is no sensitization master defined either in the timing group or current cell, Vector string is implied by vector id and number of pin in pin_name_map. So the vector id should greater than or equal to zero, and less than 2^(number of pin in pin_name_map).

For example:

```
timing() {
    pin_name_map("A, B, C, Z");
    wave_rise (5, 8, 17);
}
```

Here number of pin in pin_name_map is 4, the vector id should < (2^4 = 16), so the vector id = 17 is out of range, and is not allowed.

The following example shows an instance where this message occurs:

```
library(name) {
  ...
  cell (my_cell) {
    ...
```

Feedback

```
    pin (A) {
        timing() {
            sensitization_master : my_sensitization;
            pin_name_map ("A, B, Y");
            wave_rise (1, 7);
            wave_fall (2, 8);
            /* other timing arc data */
        }
    }
  }
  ...
}
```

The following is an example message:

```
Error: Line 1181, Vector id (8) for implied vector string in 'wave_fall'
 is out of range (0~7)(LBDB-777)
```

### What Next

Modify vector id in the wave_rise/wave_fall, make sure every vector id is in the legal range.

---

## LBDB-778

(error) index (%d) specified in '%s' is out of range (1~%d).

### Description

The wave_rise_sampling_index/wave_fall_sampling_index is to define from which transition, (instead of default last transition) to the transition of the output pin, the delay is measured.

If the number of elements in wave_rise/wave_fall is N, then The index value should between (1 ~ N-1).

The following example shows an instance where this message occurs:

```
library(name) {
  ...
  cell (my_cell) {
    ...
    pin (A) {
        timing() {
            sensitization_master : my_sensitization;
            pin_name_map ("A, B, Y");
            wave_rise (1, 4, 7);
            wave_fall (2, 5, 8);
            wave_rise_sampling_index : 3;
            /* other timing arc data */
        }
    }
```

Feedback

```
  }
  ...
}
```

The following is an example message:

```
Error: Line 1181, index (3) specified in 'wave_rise_sampling_index' is
 out of range (1~2)(LBDB-778)
```

**What Next**

Modify sampling index accordingly, make sure it's in the legal range.

---

## LBDB-779

(error) Number of elements (%d) specified in '%s' is out of range (1~%d).

**Description**

The wave_rise_timing_interval/wave_fall_timing_interval attributes are for special cases that customers want to control the time interval between transitions. which is used to characterize some special-purpose cells and pessimistic timing characterization.

If the number of elements in wave_rise/wave_fall is N, then Number of elements in these two attributes should between (1 ~ N-1);

The following example shows an instance where this message occurs:

```
library(name) {
  ...
  cell (my_cell) {
    ...
    pin (A) {
        timing() {
            sensitization_master : my_sensitization;
            pin_name_map ("A, B, Y");
            wave_rise (1, 4, 7);
            wave_fall (2, 5, 8);
            wave_rise_timing_interval(0.0, 0.5, 1.0);
            /* other timing arc data */
        }
    }
  }
  ...
}
```

The following is an example message:

```
Error: Line 1181, Number of elements (3) specified in
 'wave_rise_sampling_index' is out of range (1~2)(LBDB-779)
```

Feedback

### What Next

Modify the wave_rise_timing_interval/wave_fall_timing_interval attribute values, accordingly to the rule above.

---

## LBDB-780

(error) The '%s' %s is not %s '%s' %s.

### Description

This error message occurs when you specify an invalid 'std_cell_main_rail' pg_pin in a level shifter not satisfying one of the following constraints: - the pg_pin must be a 'primary_power' or 'primary_ground' pg_pin. - if the pg_pin is 'primary_power', the pg_pin must be specified as the "related_power_pin" of one of the signal pins of the level shifter. - if the pg_pin is 'primary_ground', the pg_pin must be specified as the "related_ground_pin" of one of the signal pins of the level shifter.

As shown in the following example, the 'VDD' pg_pin is invalid because it is not a 'primary power' pg_pin.

```
cell(ls) {
    is_level_shifter : true;

    pg_pin(VDD) {
        voltage_name : VDDL;
        pg_type : backup_power;

        std_cell_main_rail : true;
    }
    ...
  }

Error: Line 84, The 'VDD' pg_pin is not a 'primary_power' pg_pin.
 (LBDB-780)
```

### What Next

Modify the cell accordingly to satisfy the constraints.

---

## LBDB-781

(error) The timing arc with feed_through_type '%s'\n \t%s specify the CCS Timing Current waveform vectors.

### Description

The timing arc with feed_through_type attribute must satisfy the following requirements: 1. if the feed_through_type attribute is "short", then the timing arc must not have CCS Timing

Feedback

current waveform vectors. 2. if the feed_through_type attribute is "wire" or "gate", then the timing arc must have CCS Timing current waveform vectors.

The following example shows an instance where this message occurs:

```
pin(Z) {
          direction : output;
          timing() {
            /* no ccs information defined in the timing arc */
            feed_through_type : wire;
            ...
          }
        }
```

The following is an example message: Error: Line 144, The timing arc with feed_through_type 'short' cannot specify the CCS Timing Current waveform vectors. (LBDB-781)

**What Next**

Check the timing arc and make the correction accordingly.

---

## LBDB-782

(error) The pin '%s' should be an 'inout' pin because of the associated \n \ttiming arc with feed_through_type attribute.

**Description**

The timing arc with feed_through_type attribute can only be specified between inout pins.

The following example shows an instance where this message occurs:

```
pin (A) {
          direction : output;
          ...
        }
        pin(Z) {
          direction : output;
          timing() {
            feed_through_type : short;
            ...
            related_pin : "A";
          }
        }
```

The following is an example message: Error: Line 144, The pin 'A' should be an 'inout' pin because of the associated timing arc with feed_through_type attribute. (LBDB-782) Error: Line 148, The pin 'Z' should be an 'inout' pin because of the associated timing arc with feed_through_type attribute. (LBDB-782)

Feedback

### What Next

Check the timing arc and make the correction accordingly.

---

## LBDB-783

(error) In cell '%s', the %s timing arc from pin '%s' to '%s' with feed_through_type '%s'\n \tdoes not have the required timing arc from pin '%s' to '%s' of the same timing_type\n \t, timing_sense and feed_though_type.

### Description

The timing arc with feed_through_type attribute 'short' or 'wire' must have the relative backward timing arc with the same timing_type and feed_through_type for the same cell.

The following example shows an instance where this message occurs:

```
cell(T) {
        ...
        pin(Z) {
          direction : output;
          /* no timing arc Z->A with timing_type = combinational and
 feed_through_type= wire exists in the cell T */
          timing() {
            timing_type : combinational;
            feed_through_type : wire;
            ...
            related_pin "A";
          }
        }
        ...
     }
```

The following is an example message: Error: Line 144, In cell 'T', the combinational timing arc from pin 'A' to 'Z' with feed_through_type '%s' does not have the required timing arc from pin 'Z' to 'A' of the same timing_type, timing_sense and feed_though_type. (LBDB-783)

### What Next

Check the timing arcs and make the correction accordingly.

---

## LBDB-784

(error) The %s value of the '%s' attribute is %g,\n \twhich is %s the required value %g.

### Description

The minimum/maximum value of the attribute is less/greater than the required value.

Feedback

The following example shows an instance where this message occurs:

```
...
  current_unit : "1mA" ;
  ...
  dc_current (ccsn_dc_29x29) {
        index_1 ("...");
        index_2 ("...");
        /* current unit = 1mA, and the max value = 0.005 */
        values ("0.0005, ...., 0.00091);
  }
```

The following is an example message:

```
Error: Line 35, The maximum value of the 'values' attribute is 0.0005,
        which is less than required value 0.001. (LBDB-784)
```

**What Next**

Check the library source file and correct the problem.

---

## LBDB-785

(warning) The '%s' attribute is not defined in the library. \n \tUsing %f as the default value.

**Description**

You receive this message because the relative attribute is not defined in the library, so LC will create the attribute with the predefined default value.

The following is an example message: Warning: Line 85, The 'nom_voltage' attribute is not defined in the library. Using 5 as the default value. (LBDB-785)

**What Next**

Examine the library source file, and add the correct value for the attribute.

---

## LBDB-786

(error) The attribute valus is not a switch pin.

**Description**

The value of related_switch_pin must be referring to either a switch pin or an internal pin as defined below.

switch pin : switch_pin attribute is set to "true" within a pin. internal pin : related_pin attribute is referred to a switch pin.

Feedback

The following example shows an instance where this message occurs:

```
dc_current (ccsn_dc_29x29) {
    related_switch_pin : D;
    ...
 }
 pin(CTL) { /* this is a switch pin as switch_pin is set to "true" */
    capacitance : 0.5;
    direction : input;
    switch_pin : true;
    ...
 }
 pin(int_1) { /* this is an internal pin, which refers to a switch pin by
                 related_pin attribute */
    direction : internal;
    timing () {
      related_pin : CTL;
      ...
    }
    ...
 }
 pin(D) {
    capacitance : 0.5;
    direction : input;
    ...
  }
```

In this case, related_switch_pin is pointing to pin "D", which is not either an internal pin or a switch pin. The way to correct it is to specify related_switch_pin to either pin "CTL" or pin "int_1".

The following is an example message:

```
Error Line 272, The attribute valus is not a switch pin. (LBDB-786)
```

**What Next**

Check the library source file, and make the necessary correction.

---

## LBDB-787

(error) Cell(%s): The 'internal_node' in the '%s' port\n \tcan't be specified because no statetable is allowed in generic type\n \tof clock gating integrated cell.

**Description**

If attribute 'clock_gating_integrated_cell' is generic, then no statetable is allowed. 'internal_node' attribute must match one of the internal node names in statetable. Since there is no statetable is allowed, there is no 'internal_node' attribute can be specified in a port, either.

Feedback

The following example shows an instance where this message occurs:

```
cell(ldbd-787) {
    ...
  cell_leakage_power : 0.023529479 ;
  clock_gating_integrated_cell : "generic" ;
    ...

  pin(Q) {
    direction : internal;
    internal_node : "Q1";
    timing() {
      timing_type : rising_edge;
      intrinsic_rise : 1.34;
      intrinsic_fall : 1.54;
      rise_resistance : 0.0718;
      fall_resistance : 0.0347;
      related_pin : "CP";
    }
  }
}
```

In this case, Q1 in internal_node attribute shall not be specified because there is no statetable in the cell.

The following is an example message:

```
Error: Line 139, Cell(lbdb-787): The 'internal_node' in the 'Q' port
        can't be specified because no statetable is allowed in generic
 type
        of clock gating integrated cell. (LBDB-787)
```

**What Next**

Remove the internal_node attribute to fix the problem.

---

## LBDB-788

(warning) No default '%s" group defined in '%s'.

**Description**

This warning is issued when there is some default group is missing in library, but no default can be assumed to it.

For example, if there is no default driver waveform specified, This warning will be reported.

The following is an example message:

```
Warning Line 10, No default 'normalized_driver_waveform" group defined in
 'library'.(LBDB-788)
```

Feedback

### What Next

Check the library source file, add default group.

---

## LBDB-789

(warning) The '%s' attribute in the '%s' table has\n \tless than %d '%s" points,that is point between %g to %g.

### Description

The values of the corresponding attribute has no enough type of points.

For example, the normalized voltage points for driver waveform need at lease 1 start points(0~0.05) and end points(0.95~1.0) for accuracy.

The following example shows an instance where this message occurs:

```
normalized_driver_waveform (dw01) {
    index_1 ("1.0");    /* input net transition */
    index_2 ("0.1, 0.3, 0.5, 0.7, 0.9"); /* normalized voltage*/
    ...
 }
```

In this case, the index_2 of normalized_driver-waveform is normalized_voltage, It would be better to supply at leat one start points and end points for accuracy purpose.

The following is an example message:

```
Warning Line 27, The 'index_2' attribute in the
 'normalized_driver_waveform' table
       has less than 1 'start" points, that is between '0' to '0.05'.
 (LBDB-789)
Warning Line 27, The 'index_2' attribute in the
 'normalized_driver_waveform' table
       has less than 1 'end" points, that is between '0.95' to '1.0'.
 (LBDB-789)
```

### What Next

Check the library source file, add some example points for the attribute.

---

## LBDB-790

(error) '%s' only allows single %s pin name.

### Description

Only one single pg pin name can be specified in intrinsic_resistance, intrinsic_capacitance and pg_current or only one single pin name can be specified in gate_leakage.

Feedback

The following example shows an instance where this message occurs:

```
pg_pin(V1) {
     voltage_name : VDD1;
     pg_type : primary_power;
  }
  pg_pin(V2) {
     voltage_name : VDD2;
     pg_type : backup_power;
  }
  pg_pin(G1) {
     voltage_name : GND1;
     pg_type : primary_ground;
  }
  pg_pin(G2) {
     voltage_name : GND2;
     pg_type : backup_ground;
  }

 pin(A1) {
  direction : input;
  ...
   }
 pin(A2) {
  direction : input;
  ...
   }
 pin(A3) {
  direction : input;
  ...
   }
 pin(ZN1) {
  direction : output;
  ...
 }
 pin(ZN) {
  direction : output;
  ...
 }
 ...
  dynamic_current() {
    when : "A1";
    related_inputs : "A2 A3";
    related_outputs : "ZN ZN1";
    typical_capacitances(0.3 0.4);
    switching_group() {
      input_switching_condition(fall);
      output_switching_condition(rise fall);
      pg_current(V2 G1) {
        vector(test_2) {
          reference_time : 193.2;
          index_output : "ZN1";
          index_1("15.1");
```

Feedback

```
          ...
        }
      ...
    }
   ...
 }

leakage_current() {
   pg_current(V1) {
     value : 4.5;
   }
   pg_current(G1 G2) {
     value : -4.5;
   }
  ...
   gate_leakage(A2 A3) {
     input_high_value : 1.0;
     input_low_value : -10.0;
   }
   gate_leakage() {
     input_high_value : 3.0;
     input_low_value : -40.0;
   }


}
intrinsic_parasitic() {
  when : "A1 & A2 & ZN";
  intrinsic_resistance(G1 G1) {
     related_output : "ZN";
     value : 9.0;
  }
  intrinsic_resistance(G1) {
     related_output : "ZN1";
     value : 62.2;
  }
  intrinsic_capacitance(G2 G2) {
     value : 31.47;
  }
}
```

In this case, the pg_current within switching_group has two pg pins, V2 and G1, which is wrong. The pg_current within leakage_current has two pg pins, G1 and G2, which is wrong. The intrinsic_resistance has two pg names, G1 and G1, which is wrong. The intrinsic_capacitance has two pg names, G2 and G2, which is wrong. The first gate_leakage has two pin names A3 and A2, which is wrong, and the second gate_leakage has no name there, which is wrong, too. You might want to change all of them to one PG or pin name to avoid the error.

Feedback

The following is an example message:

```
Error Line 272, 'pg_current' only allows single pg pin name. (LBDB-790)
Error Line 222, 'gate_leakage' only allows single pin name. (LBDB-790)
```

**What Next**

Check the library source file, and make the necessary correction.

---

## LBDB-791

(error) pg_pin group is required for ccs power cell.

**Description**

If the cell is a ccs power cell, which contains dynamic_current, leakage_current or intrinsic_parasitic, then pg_pin group is required.

The following example shows an instance where this message occurs:

```
cell (AND3) {
  ...
  power_cell_type : stdcell;
  dynamic_current() {
    when : "A1";
    related_inputs : "A2 A3";
    related_outputs : "ZN ZN1";
    typical_capacitances(0.3 0.4);
    switching_group() {
    ...
    }
    ...
  } ...
}
```

In this case, cell, AND3, is a ccs power cell. The pg_pin must be specified within cell to fix the error as shown below :

```
cell (AND3) {
  pg_pin(V1) {
     voltage_name : VDD1;
     pg_type : primary_power;
  }
  pg_pin(V2) {
     voltage_name : VDD2;
     pg_type : backup_power;
  }
  pg_pin(G1) {
     voltage_name : GND1;
     pg_type : primary_ground;
  }
  pg_pin(G2) {
```

Feedback

```
    voltage_name : GND2;
    pg_type : backup_ground;
  }

  power_cell_type : stdcell;
  dynamic_current() {  ...
```

The following is an example message:

```
Error Line 272, pg_pin group is required for ccs power cell. (LBDB-791)
```

**What Next**

Check the library source file, and make the necessary correction.

---

## LBDB-792

(error) leakage current in simplified format must be >= 0.

**Description**

If there is no pg_current within leakage_current group, we defined that as simplified format. Please refer to users guide for details.

If leakage_current is in simplifed format, then the value of leakage current must be zero or positive floating number.

The same rule is applied to va_leakage_current.

The following example shows an instance where this message occurs:

```
cell (OR2) {
  pg_pin(V1) {
    voltage_name : VDD1;
    pg_type : primary_power;
  }
  pg_pin(G1) {
    voltage_name : GND1;
    pg_type : primary_ground;
  }
  leakage_current() {
    when : "A1 & !A2 & ZN";
    value : 3.1;
  }
  leakage_current() {
    when : "A1 & !A2 & !ZN";
    value : 0.0;
  }
  leakage_current() {
    /* default state */
    value : -8.1;
  }
```

Feedback

```
  ...
}
```

In this case, leakage current -8.1 is not allowed. The value must be >= 0 in simplified format.

The following is an example message:

```
Error Line 272, leakage current in simplified format must be >= 0.
 (LBDB-792)
```

**What Next**

Check the library source file, and make the necessary correction.

---

## LBDB-793

(error) Name specified in '%s' is not a valid %s pin.

**Description**

You got this message because an invalid PG or signal pin is used.

The pin name specified in intrinsic_capacitance, intrinsic_resistance, or pg_current group must be referred to a valid PG pin defined in cell, and a pg name is required for these groups.

The pin name specified in gate_leakage group must be referred to a valid signal pin defined in cell, and a pin name is required for gate_leakage.

The following example shows an instance where this message occurs:

```
cell (AND3) {
  ...
  pg_pin(V1) {
     voltage_name : VDD1;
     pg_type : primary_power;
  }
  pg_pin(V2) {
     voltage_name : VDD2;
     pg_type : backup_power;
  }
  pg_pin(G1) {
     voltage_name : GND1;
     pg_type : primary_ground;
  }
  pg_pin(G2) {
     voltage_name : GND2;
     pg_type : backup_ground;
  }
 ...
 dynamic_current() {
```

Feedback

```
  when : "A1";
  related_inputs : "A2 A3";
  related_outputs : "ZN ZN1";
  typical_capacitances(0.3 0.4);
  switching_group() {

    input_switching_condition(fall);
    output_switching_condition(rise fall);

    pg_current(V5) {     /* error */
      vector(test_2) {
      ...
    }
    pg_current() {       /* error */
    ...
  ...
}
 ...
 leakage_current() {
   when : "A1 & !A2 | ZN";
   pg_current(V5) {      /* error */
     value : 4.5;
   }
   pg_current(V2) {
     value : 4.5;
   }
   pg_current(G2) {
     value : -3.5;
   }
   pg_current() {       /* error */
     value : 4.5;
   }
}
...
intrinsic_parasitic() {
  when : "A1 & A2 & ZN";
  intrinsic_resistance() {    /* error */
     related_output : "ZN";
     value : 9.0;
  }
  intrinsic_resistance(V5) {   /* error */
     related_output : "ZN1";
     value : 62.2;
  }
  intrinsic_capacitance(G2) {
     value : 31.47;
  }
  intrinsic_capacitance() {    /* error */
     value : 31.47;
  }
  intrinsic_capacitance(V5) {  /* error */
     value : 31.47;
  }
```

Feedback

```
    ...
  }
}
```

Please check the lines marked as 'error'. For these lines, they are either refers to an empty pg name, or refers to a invalid pg pin name, V5. Both are wrong. To fix the problems, please change V5 to a valid pg pin like V1, and also give a valid pg name for the empty ones.

The following is an example message:

```
Error Line 272, Name specified in 'pg_current' is not a valid PG pin.
 (LBDB-793)
```

### What Next

Check the library source file, and make the necessary correction.

---

## LBDB-794

(warning) The value of pg_current should be '%s'.

### Description

This is rule for leakage_current groups. If the PG pin is a power or internal ground pin, then the value of leakage current should be either zero or positive floating number. If the PG pin is a ground or internal power pin, then the value of leakage current should be either zero or negative floating number.

The following example shows an instance where this message occurs:

```
cell (AND3_1) {
  pg_pin(V1) {
     voltage_name : VDD1;
     pg_type : primary_power;
  }
  pg_pin(V2) {
     voltage_name : VDD2;
     pg_type : backup_power;
  }
  pg_pin(V3) {
     voltage_name : VDD3;
     pg_type : internal_ground;
  }
  pg_pin(G1) {
     voltage_name : GND1;
     pg_type : primary_ground;
  }
  pg_pin(G2) {
     voltage_name : GND2;
     pg_type : backup_ground;
```

Feedback

```
   }
   pg_pin(G3) {
      voltage_name : GND3;
      pg_type : internal_power;
   }
   leakage_current() {
      when : "A1 & !A2 & ZN";
      pg_current(V1) {
        value : -4.5;
      }
      pg_current(V2) {
        value : -4.5;
      }
      pg_current(G2) {
        value : 4.5;
      }
      pg_current(V3) {
        value : -4.5;
      }
      pg_current(G1) {
        value : 4.5;
      }
      pg_current(G3) {
        value : 4.5;
      }
   }
   ...
}
```

In this case, V1, V2 and V3 should be positive or zero, and G1, G2 and G3 should be negative or zero based on the pg_type.

The following is an example message:

```
Warning Line 272, The value of pg_current should be '>=0'. (LBDB-794)
```

**What Next**

Check the library source file, and make the necessary correction.

---

## LBDB-795

(error) Only one PG pin can be omitted in %s group.

**Description**

All the PG pins except one shall be specified in leakage_current. This rule is also applied to va_leakage_current group.

Feedback

The following example shows an instance where this message occurs:

```
cell (AND3) {

  pg_pin(V1) {
     voltage_name : VDD1;
     pg_type : primary_power;
  }
  pg_pin(V2) {
     voltage_name : VDD2;
     pg_type : backup_power;
  }
  pg_pin(G1) {
     voltage_name : GND1;
     pg_type : primary_ground;
  }
  pg_pin(G2) {
     voltage_name : GND2;
     pg_type : backup_ground;
  }
  leakage_current() {
     when : "A1 & A2 & !ZN";
     pg_current(V1) {
       value : 4.5;
     }
     pg_current(V2) {
       value : 4.5;
     }
  }
  ...
}
```

There are 4 pg pins in the cell AND3, and based on rule only one pg pin can be omitted in leakage_current group. Meaning you have to specify at least three pg pins for this case. However, there are only two pg pins (V1 and V2) specified in leakage_current. To fix the problem, please specify at least one of ground pins for leakage_current as shown below.

```
leakage_current() {
     when : "!A1 & A2 & ZN";
     pg_current(V1) {
       value : 4.5;
     }
     pg_current(V2) {
       value : 4.5;
     }
     pg_current(G2) {
       value : -4.5;
     }
  }
```

Feedback

The following is an example message:

```
Error Line 272, Only one PG pin can be omitted in leakage_current group.
 (LBDB-795)
```

### What Next

Check the library source file, and make the necessary correction.

---

## LBDB-796

(warning) The leakage current of all PG pins and gate\n \tleakage current of all input/inout pins must be summed up to zero.

### Description

This rule applies to leakage_current and va_leakage_current groups if leakage current of all PG pins are specified.

If no gate_leakage are specified, the leakage current of all PG pins should add up to zero. If gate_leakage of all input/inout pins are specified, the leakage current of all PG pins and gate leakage current of all input/inout pins should add up to zero.

Total leakage current must add up to zero. A small error of +/- 1e-10 A (0.1 nA) is allowed. Larger errors must still be within a tolerance of 1.0e-6 * (total of absolute values of leakage current). The tolerance is calculated as follows:

```
| total of leakage current |
       ------------------------------  <= 1.0e-06
        total of | leakage current |
```

The following example shows an instance where this message occurs: In the example below, the total of leakage current is -0.000028. Based on the rule above, it will be like | -0.000028 | / 27.000028 = 1.03 * 1.0e-6 The result is greater than 1.0e-6. To fix the problem, we can change the G3 current to -4.500020.

```
cell (AND3_1) {
  pg_pin(V1) {
     voltage_name : VDD1;
     pg_type : primary_power;
  }
  pg_pin(V2) {
     voltage_name : VDD2;
     pg_type : backup_power;
  }
  pg_pin(V3) {
     voltage_name : VDD3;
     pg_type : internal_ground;
  }
  pg_pin(G1) {
     voltage_name : GND1;
```

Feedback

```
      pg_type : primary_ground;
   }
   pg_pin(G2) {
      voltage_name : GND2;
      pg_type : backup_ground;
   }
   pg_pin(G3) {
      voltage_name : GND3;
      pg_type : internal_power;
   }
   leakage_current() {
      when : "A1 & A2 & ZN";
      pg_current(V1) {
        value : 4.5;
      }
      pg_current(V2) {
        value : 4.5;
      }
      pg_current(G2) {
        value : -4.5;
      }
      pg_current(V3) {
        value : 4.5;
      }
      pg_current(G1) {
        value : -4.5;
      }
      pg_current(G3) {
        value : -4.500028;
      }
 }
```

The following is an example message:

```
Warning: Line 246, The leakage current of all PG pins and gate leakage
         current of all input/inout pins must be summed up to zero.
 (LBDB-796)
```

### What Next

Check the library source file and make the necessary correction.

---

## LBDB-797

(warning) The leakage_current is in wrong format.

### Description

Please also refer to LBDB-798.

The leakage_cuurent only allow either regular or simplified format. The regular format shall contain pg_current group, and the simplified format shall contain value attribute.

Feedback

The following example shows an instance where this message occurs:

```
leakage_current() {
     when : "A1 & !A2 & !ZN";
   }
```

In this case, leakage_current is neither in regular nor in simplified format. To avoid the warning, please change it to either of following formats:

```
simplified format :
   leakage_current() {
     when : "A1 & !A2 & !ZN";
     value : 0.0;
   }

  regular format :
   leakage_current() {
     when : "A1 & !A2 & !ZN";
     pg_current(V1) {
       value : 0.0;
     }
   }
```

The following is an example message:

```
Warning Line 272, The leakage_current is in wrong format. (LBDB-797)
```

**What Next**

Check the library source file, and make the necessary correction.

---

## LBDB-798

(error) The leakage current groups have mix format.

**Description**

Please also refer to LBDB-797.

The leakage_cuurent only allow either regular or simplified format. The regular format shall contain pg_current group, and the simplified format shall contain value attribute. And, two formats can not be mixed.

The following example shows an instance where this message occurs:

```
cell (test) {
  ...
  leakage_current() {
     pg_current(V1) {
       value : 4.5;
     }
     pg_current(G1) {
```

Feedback

```
        value : -4.5;
      }
      pg_current(G2) {
        value : -14.5;
      }
      pg_current(V2) {
        value : 14.5;
      }
  }
  leakage_current() {
     when : "!A1 & !A2 & ZN";
     value : 0.0;
  }
  ...
}
```

In this case, first leakage_current is in regular format and second leakage_current is in simplified format, which is wrong. Two formats can not be mixed under the same cell.

The following is an example message:

```
Error Line 272, The leakage current groups have mix format. (LBDB-798)
```

**What Next**

Check the library source file, and make the necessary correction.

---

## LBDB-799

(error) The multiple PG pins can not be simplified format\n \tin %s.

**Description**

The simplified format in leakage_current means no pg_current group specified under this leakage_current. If a cell have more than one power or ground pin, then we can not use simplified format for leakage_current.

The rule is also applied to va_leakage_current.

The following example shows an instance where this message occurs:

```
cell (test) {
  pg_pin(V1) {
     voltage_name : VDD1;
     pg_type : primary_power;
  }
  pg_pin(G1) {
     voltage_name : GND1;
     pg_type : primary_ground;
  }
  pg_pin(V2) {
     voltage_name : VDD2;
```

Feedback

```
      pg_type : backup_ground;
   }
   leakage_current() {
      when : "A1 & !A2 | ZN";
      value : 3.1;
   }
  ...
}
```

In this case, the cell 'test' has more than one power pin, and leakage_current is in simplified format, which is wrong. To fix the problem, change the simplified format to regular format as shown below.

```
leakage_current() {
     when : "A1 & !A2 | ZN";
     pg_current(V1) {
       value : 1.1;
     }
     pg_current(G1) {
       value : -3.1;
     }
     pg_current(V2) {
       value : 2.0;
     }
   }
```

The following is an example message:

```
Error Line 272, The multiple PG pins can not be simplified format
        in leakage_current group. (LBDB-799)
```

### What Next

Check the library source file, and make the necessary correction.

---

## LBDB-800

(error) In group '%s', the number of differnt\n \t'%s' values in the vectors is %d, which should \n \tbe at least %d.

### Description

This error message occurs because Library Compiler requires the vectors should contain at least 2 different slew values and 2 different load values.

The following example shows an output_current_rise group with vectors containing only 1 slew value, which is 1.

```
output_current_template(CCT) {
  variable_1 : input_net_transition;
  variable_2 : total_output_net_capacitance;
```

Feedback

```
 variable_3 : time;
}
. . .
output_current_rise() {
  vector(CCT) {
     reference_time : 0.11;
     index_1 ("0.1");
     index_2 ("1");
     index_3 ("1, 2, 3");
     values ("1, 2, 3");
  }
   vector(CCT) {
     reference_time : 0.11;
     index_1 ("0.2");
     index_2 ("1");
     index_3 ("1, 2, 3");
     values ("1, 2, 3");
  }
}
```

```
Error: Line 198, In group 'output_current_rise', the number of differnt
        'total_output_net_capacitance' values in the vectors is 1, which
 should
       be at least %d. (LBDB-800)
```

**What Next**

Check the library source file and add a vector for each input_net_transition and total_output_net_capacitance pair.

---

## LBDB-801

(error) %s and %s attributes must be specified in pair.

**Description**

The max_input_switching_count and min_input_switching_count attributes must be defined in pair within a switching_group. Meaning that if max_input_switching_count is defined, then min_input_switching_count must be defined under the same switching_group and vice versa.

The following example shows an instance where this message occurs:

```
power_cell_type : macro;
  dynamic_current() {
    ...
    switching_group() {
      min_input_switching_count : 1;
      ...
    }
  ...}
```

Feedback

In this case, min_input_switching_count is defined, but there is no max_input_switching_count, which is wrong. To fix the problem, you can either remove min_input_switching_count from the switching_group or specify max_input_switching_count within switching_group.

The following is an example message:

```
Error Line 272, min_input_switching_count and max_input_switching_count
 attributes must be specified in pair. (LBDB-801)
```

**What Next**

Check the library source file, and make the necessary correction.

---

## LBDB-802

(error) %s can not be defined within a %s\n \tgroup where %s is defined.

**Description**

You can only specify either max_input_switching_count and min_input_switching_count or input_switching_condition within a switching_group group. Meaning max_input_switching_count/ min_input_switching_count and input_switching_condition shall not present under the same switching_group group.

ALSO SEE LBDB-801

The following example shows an instance where this message occurs:

```
switching_group() {
      min_input_switching_count : 1;
      max_input_switching_count : 9;
      input_switching_condition(rise);
      ...
   }
```

In this case, min_input_switching_count and max_input_switching_count are defined, and also input_switching_condition is defined under the same switching_group where min_input_switching_count and max_input_switching_count are defined. This is wrong. To fix the problem, please remove either input_switching_condition or min_input_switching_count and max_input_switching_count.

The following is an example message:

```
Error Line 272, input_switching_condition can not be defined within a
 switching_group
        group where min_input_switching_count or
 max_input_switching_count is defined. (LBDB-802)
```

Feedback

### What Next

Check the library source file, and make the necessary correction.

---

## LBDB-803

(error) %s and %s are required for all %s\n \tgroups within this %s group.

### Description

If min_input_switching_count and max_input_switching_count are specified in one of switching_group groups within a dynamic_current group, then they must be defined in the rest of switching_group groups.

This message indicates that min_input_switching_count and max_input_switching_count are not defined in all switching_group groups within a dynamic_current.

The following example shows an instance where this message occurs:

```
cell(lbdb803) {
    ...
  dynamic_current() {
     ...
   switching_group() {
      pg_current(V1) {
        vector(test_3) {
          reference_time : 193.2;
          index_1("1.1");
          index_2("18.2 18.3 19.0");
          values("13.78 192.4 1100.1");
         }
        vector(test_3) {
          reference_time : 193.2;
          index_1("1.8");
          index_2("18.2 19.4 19.8");
          values("11.78 112.4 1110.1");
         }
        vector(test_3) {
          reference_time : 193.2;
          index_1("2.1");
          index_2("18.2 19.4 19.8");
          values("12.78 122.4 1120.1");
         }
         ...
       }
    }
   switching_group() {
      min_input_switching_count : 1;
      max_input_switching_count : 9;
      pg_current(V2) {
        vector(test_3) {
```

Feedback

```
          ...
          }
        vector(test_3) {
          ...
          }
          ...
       }
    }
  }
  ...
```

In this case, min_input_switching_count and max_input_switching_count are defined in second switching_group, but not in first switching_group. This is wrong. min_input_switching_count and max_input_switching_count are required for all switching_group groups if they are defined in one of switching_group under the same dynamic_current. To fix the problem, please define min_input_switching_count and max_input_switching_count in first switching_group.

The following is an example message:

```
Errorr Line 272, min_input_switching_count and max_input_switching_count
 are required for all switching_group
        groups within this dynamic_current group. (LBDB-803)
```

### What Next

Check the library source file, and make the necessary correction.

---

## LBDB-804

(error) The switching count shall cover all bits in related_inputs.

### Description

The max_input_switching_count and min_input_switching_count attributes defined in a dynamic_current shall cover all bits specified in related_inputs.

The following example shows an instance where this message occurs:

```
type(bus6) {
  base_type : array ;
  data_type : bit ;
  bit_width : 6 ;
  bit_from : 5 ;
  bit_to : 0 ;
  downto : true ;
}
cell (libdb804) {
  bus(sel) {
    bus_type : bus6 ;
    direction : input ;
```

Feedback

```
    ...
  }
  bundle(C) {
    members(Cx, Cy, Cz);
    direction : input;
    ...
  }
 pin(A1) {
  direction : input;
 ...
   }
 pin(A2) {
  direction : input;
  ...
   }
 pin(A3) {
  direction : input;
   ...
   }
  ...
  power_cell_type : macro;
  dynamic_current() {
    when : "A1";
    related_inputs : "Cx A3 Cy sel[0:3] A2";
    switching_group() {
      min_input_switching_count : 1;
      max_input_switching_count : 3;
      ...
    }
    switching_group() {
      min_input_switching_count : 5;
      max_input_switching_count : 8;
      ...
    }
  }
  ...
```

In this case, total number of bits in related_inputs is 8, so the switching count we specified in min_input_switching_count and max_input_switching_count shall cover number from 1 up to 8. The first switching group covers number from 1 to 3 and second switching group covers number from 5 to 8. The number 4 is not covered by any switching count within the dynamic_current. This is wrong. To fix the problem, change the third switching_group as follows :

```
switching_group() {
      min_input_switching_count : 4;
      max_input_switching_count : 4;
      ...
    }
```

Feedback

The following is an example message:

```
Error Line 272, The switching count shall cover all bits in
 related_inputs. (LBDB-804)
```

**What Next**

Check the library source file, and make the necessary correction.

---

## LBDB-805

(error) The value of %s is invalid.

**Description**

The message indicates that you are breaking one of following rules :

1> The value of min_input_switching_count must be greater than 0. 2> The value of max_input_switching_count must be greater than or equal to min_input_switching_count. 3> The value of max_input_switching_count must be less than or equal to total number of bits in related_inputs

The following example shows an instance where this message occurs:

```
type(bus6) {
  base_type : array ;
  data_type : bit ;
  bit_width : 6 ;
  bit_from : 5 ;
  bit_to : 0 ;
  downto : true ;
}
cell (libdb804) {
  bus(sel) {
    bus_type : bus6 ;
    direction : input ;
    ...
  }
  bundle(C) {
    members(Cx, Cy, Cz);
    direction : input;
    ...
  }
 pin(A1) {
  direction : input;
 ...
   }
 pin(A2) {
  direction : input;
  ...
   }
 pin(A3) {
```

Feedback

```
direction : input;
 ...
 }
...
power_cell_type : macro;
dynamic_current() {
  when : "A1";
  related_inputs : "Cx A3 Cy sel[0:3] A2";
  switching_group() {
    min_input_switching_count : 0;
    max_input_switching_count : 9;
    ...
  }
  switching_group() {
    min_input_switching_count : 8;
    max_input_switching_count : 6;
    ...
  }
}
...
```

In this case, total number of bits in related_inputs is 8, so the switching count we specified in min_input_switching_count and max_input_switching_count shall cover only from 1 up to 8. 0 in min_input_switching_count is not a valid number, and 9 in max_input_switching_count is not a valid number, either. In the second switching_group, min_input_switching_count is greater than max_input_switching_count, which is wrong.

The following is an example message:

```
Error Line 272, The value of min_input_switching_count is invalid.
 (LBDB-805)
Error Line 272, The value of max_input_switching_count is invalid.
 (LBDB-805)
```

**What Next**

Check the library source file, and make the necessary correction.

---

## LBDB-806

(error) table_lookup is required model in ccs.

**Description**

The table_lookup is required delay_model for following ccs syntax forms :

ccs timing (driver and receiver), compact ccs timing, ccs noise, and ccs power

The message indicates that either there is no delay_model in your library or the delay_model you defined is not "table_lookup".

Feedback

The following example shows an instance where this message occurs:

```
library(libdb806) {
  ...
  cell(test_1) {
    pin(Z) {
      direction : output;
      timing() {
        compact_ccs_rise(template) {
          ....
        }
        compact_ccs_fall(template) {
          ....
        }
      }
    }
    ...
  }
  ...
}
```

In this case, ccs compact is defined but there is no delay_model specified, which is wrong. This is the fix to avoid the error message.

```
library(libdb806) {
  delay_model : table_lookup;
  ...
  cell(test_1) {
    ...
}
```

The following is an example message:

```
Error Line 272, table_lookup is required model in ccs. (LBDB-806)
```

**What Next**

Check the library source file, and make the necessary correction.

---

## LBDB-807

(error) The power_down_function is not allowed in test_cell.

**Description**

The power_down_function attribute can not be specified under test_cell group.

The following example shows an instance where this message occurs:

```
cell(libdb807) {
  ...
    test_cell() {
```

Feedback

```
      pin(QN) {
        direction : output;
        power_down_function : "!VDD + VSS";
        function : "IQN";
        signal_type : test_scan_out_inverted;
      }
    }
  ...
}
```

In this case, power_down_function attribute is defined under a test_cell, which is wrong. You can avoid the message by comment out power_down_function attribute.

The following is an example message:

```
Error Line 272, The power_down_function is not allowed in test_cell.
 (LBDB-807)
```

**What Next**

Check the library source file, and make the necessary correction.

---

## LBDB-808

(error) The %s can't be specified if %s is not defined.

**Description**

The nominal tables must be specified if variation-aware tables are there. Meaning the following variation-aware tables can be specified only if the corresponding nominal tables are defined.

va tables : va_compact_ccs_rise , va_compact_ccs_fall va_receiver_capacitance1_rise , va_receiver_capacitance2_rise va_receiver_capacitance1_fall , va_receiver_capacitance2_fall va_rise_constraint, va_fall_constraint and va_leakage_current

corresponding nominal tables : compact_ccs_rise , compact_ccs_fall receiver_capacitance1_rise , receiver_capacitance2_rise receiver_capacitance1_fall , receiver_capacitance2_fall rise_constraint, fall_constraint and leakage_current

For the leakage_current, we need to consider when statement. Meaning that va_leakage_current can be defined only if leakage_current with the same state condition as what va_leakage_current has and it is defined within the same cell as where va_leakage_current is deinfed.

The same rule is applied to gate_leakage. The gate_leakage in va_leakage_current can't be specified if gate_leakage with same pin name is not defined in leakage_current, and both va_leakage_current and leakage_current are under the same state condition.

Feedback

The following example shows an instance where this message occurs:

```
pin(QN) {
   ...
   pin_based_variation() {
      va_parameters(var1, var2);
      nominal_va_values(10.0, 20.0);
      va_receiver_capacitance1_rise ( pinTB2INVXC_rise_1 ) {
        va_values(10.0, 21.0);
        values ( \
          "1.100, 1.100, 1.100");
      }
      ...
    }
  }
```

In this case, there is no receiver_capacitance specified within QN pin, so it is wrong to define va_receiver_capacitance1_rise table.

The following is the example message:

```
Error: Line 341, The va receiver cap can't be specified if nominal
 receiver cap is not defined. (LBDB-808)
```

The following example shows another instance where this message occurs:

```
leakage_current() {
     when : "B1";
     ...
   }
   leakage_current() {
     when : "C1";
     ...
   }
   leakage_current() {
     when : "D1";
     ...
   }
   leakage_current() { /* default */
     ...
   }
   cell_based_variation() {
      va_parameters(var1, var2);
      nominal_va_values(10.0, 20.0);
      va_leakage_current() {
        when : "A1";
        va_values(10.0, 21.0);
        ...
      }
      ...
    }
  }
```

Feedback

In this case, there is no "when statement" A1 in leakage_current, but "when : A1" is defined in va_leakage_current, which is wrong.

The following is an example message:

```
Error: Line 246, The va_leakage_current can't be specified if
 leakage_current with the same state condition is not defined.
 (LBDB-808)
```

The following example shows another instance where this message occurs:

```
leakage_current() {
     when : "A2";
     pg_current(V1) {
       value : 4.5;
     }
     pg_current(G1) {
       value : -4.5;
     }
     gate_leakage(A2) {
       input_high_value : 7.1;
     }
   }
   cell_based_variation() {
      va_parameters(var1);
      nominal_va_values(10.0);
      va_leakage_current() {
        when : "A2";
        va_values(11.0);
        pg_current(V1) {
          value : 4.5;
        }
        pg_current(G1) {
          value : -4.5;
        }
        gate_leakage(A1) {
         input_high_value : 9.3;
         input_low_value : -8.7;
        }
      }
      ...
    }
  }
```

The gate_leakage "A1" is defined under va_leakage_current, but it is not defined under leakage_current, where va_leakage_current and leakage_current have same state condition (when : "A2"). This violate the rule.

The following is the example message:

```
Error: Line 161, The gate_leakage in va_leakage_current can't be
 specified if gate_leakage in leakage_current is not defined.  (LBDB-808)
```

Feedback

### What Next

Check the library source file, and make the necessary correction.

---

## LBDB-809

(error) The left_id and right_id must be either defined in\n \tpairs or non-defined.

### Description

The left_id and right_id are optoinal attributes for variation-aware compact table. If they are defined, then they must be defined in pairs.

The following example shows an instance where this message occurs:

```
library(libdb809) {
 ...
 compact_lut_template(va_test){
  variable_1 : input_net_transition ;
  variable_2 : total_output_net_capacitance ;
  variable_3 : curve_parameters;
  index_1 ( "5.236700e-02, 6.042500e-02, 8.710600e-02" ) ;
  index_2 ( "1.000000e-03, 1.511000e-03" ) ;
  index_3 ("init_current, peak_current, peak_voltage, peak_time,
 left_id");
  base_curves_group : "tt";
 }
 ...
   timing () {
     ...
      timing_based_variation() {
        va_parameters(var1, var2);
        nominal_va_values(10.0, 20.0);
        va_compact_ccs_rise(va_test) {
         va_values(10.0, 21.0);
         values ( \
            "0.01, 3.45, 3.20, 1.3, 1", "1.01, 2.45, 4.20, 2.3, 1",
 "2.01, 3.35, 5.20, 3.3, 1","3.01, 4.45, 6.20, 4.3, 1","4.01, 5.45, 7.20,
 5.3, 1", "6.01, 7.45, 8.20, 6.3, 1");
       }
     }
   ...
}
```

In this case, va_compact_ccs_rise is referring to a template, va_test, which has only left_id without right_id. This is wrong. To avoid the error message, you shall either remove left_id from the template or define right_id in the template.

The following is an example message:

```
Error: Line 813, The left_id and right_id must be either defined in
        pairs or non-defined. (LBDB-809)
```

Feedback

### What Next

Check the library source file, and make the necessary correction.

---

## LBDB-810

(error) The %s can't be empty.

### Description

There must be at least one variable specified in va_parameters attribute.

The following example shows an instance where this message occurs:

```
pin_based_variation() {
        va_parameters();
        ...
    }
```

In this case, there is no value inside va_parameters, which is wrong. You can specified a list of variables in va_parameters(), and the list can't be empty.

The following is an example message:

```
Error: Line 1029, The va_parameters can't be empty. (LBDB-810)
```

### What Next

Check the library source file, and make the necessary correction.

---

## LBDB-811

(error) The %s must be defined before %s is defined.

### Description

The va_parameters can be defined within library(), timing_based_variation(), pin_based_variation() or cell_based_variation() group.

The va_parameters is referred by nominal_va_values and va_values, and must be defined before nominal_va_values and va_values are defined.

The following example shows an instance where this message occurs:

```
library(libdb811) {
  ...
    pin_based_variation() {
        nominal_va_values(10.0, 20.0);
        ...
        va_receiver_capacitance1_rise ( pinTB2INVXC_rise_1 ) {
         va_values(10.0, 21.0);
```

Feedback

```
          values ( \
             "1.100, 1.100, 1.100");
          }
      }
    ...
    va_parameters(var1, var2);
}
```

In this case, there is no va_parameters attribute defined inside either library() or pin_based_variation() before nominal_va_values and va_values. To fix the problem, please move va_parameters(var1, var2) to the line above pin_based_variation().

```
library(libdb811) {
  va_parameters(var1, var2);
  ...
    pin_based_variation() {
        nominal_va_values(10.0, 20.0);
        ...
        va_receiver_capacitance1_rise ( pinTB2INVXC_rise_1 ) {
         va_values(10.0, 21.0);
         values ( \
            "1.100, 1.100, 1.100");
        }
    }
  ...
}
```

Or, you can move the va_parameters(var1, var2) to the line above nominal_va_values(10.0, 20.0).

```
library(libdb811) {
  ...
    pin_based_variation() {
        va_parameters(var1, var2);
        nominal_va_values(10.0, 20.0);
        ...
        va_receiver_capacitance1_rise ( pinTB2INVXC_rise_1 ) {
         va_values(10.0, 21.0);
         values ( \
            "1.100, 1.100, 1.100");
        }
    }
  ...
}
```

The following is an example message:

```
Error: Line 626, The va_parameters must be defined before
 nominal_va_values is defined. (LBDB-811)
Error: Line 638, The va_parameters must be defined before va_values is
 defined. (LBDB-811)
```

Feedback

### What Next

Check the library source file, and make the necessary correction.

---

## LBDB-812

(error) The size of %s and %s must be identical.

### Description

When the above-mentioned attributes are specified, they must be of the same size.

This rule applies to:

1. receiver_capacitance_fall_threshold_pct and receiver_capacitance_rise_threshold_pct.
2. va_parameters, nominal_va_values, and va_values.

The following example shows an instance where this message occurs:

```
library(libdb821) {
  ..
  receiver_capacitance_rise_threshold_pct("0 50 60 70 80 90 100 ");
  receiver_capacitance_fall_threshold_pct("100 50 40 30 20 10 ");
  ...
}
```

The two attribtues are of different sizes (7 versus 6).

### Examples

The following is an example message:

```
Error: Line 18, The size of receiver_capacitance_fall_threshold_pct and
 receiver_capacitance_rise_threshold_pct must be identical. (LBDB-812)
```

### What Next

Check the library source file, and make the necessary correction.

---

## LBDB-813

(error) The values in %s must be unique.

### Description

This error message occurs when the specified variables in va_parameters are not unique or when the variable name in ff/latch/ff_bank/latch_bank is not unique.

Feedback

The following example shows an instance where this message occurs: The following example is incorrect, because there are 2 var2 entries in va_parameters:

```
va_parameters(var1, var2, var2, var4); i
```

This causes the following error message:

```
Error: Line 363, The values in va_parameters must be unique.
(LBDB-813)
```

In this example, the second and the third ff have the same variable name of IQ1, which is incorrect. All variable names in an ff group within a cell must be unique.

```
cell (LBDB-813) {
   ...
   ff ("IQ","IQN") {
          ...      }
   ff ("IQ1","IQN1") {
          ...      }
   ff ("IQ1","IQN2") {
          ...      }
   ...
}
```

This results in the following error message:

```
Error: Line 363, The values in ff/latch/ff_bank/latch_bank must
be unique.  (LBDB-813)
```

**What Next**

Check the library source file, and make the necessary correction.

---

## LBDB-814

(error) The values in %s must be the same as defined values.

**Description**

The following predefined parameters can be specified in *va_parameters*:

- The parameters defined in default_operating_conditions, such as process, temperature, and voltage
- The voltage names defined in voltage_map

If these predefined parameters are specified in *va_parameters*, then the values defined in *nominal_va_values* must match the values of the predefined parameters.

Feedback

If both *default_operating_conditions* and *voltage_map* are defined, and *voltage* is defined in *va_parameters*, then Library Compiler considers *voltage* as a user-defined parameter, an exception to the rule.

**What Next**

Check the library source file, and make sure the values are correct.

In the following example, the predefined parameter *voltage* is specified in *va_parameters*, and the corresponding value in *nominal_va_values* is defined as 15.1. This value is incorrect because it will be the same value of 9.0 as defined in *default_operating_conditions*, WCCOM. For the *process* predefined parameter, 35.0 in *nominal_va_values* is incorrect, because it is not the same value of 1.5 as defined in *default_operating_conditions*.

```
library(libdb814) {
    operating_conditions(WCCOM) {
        process : 1.5 ;
        temperature : 45 ;
        voltage : 9.0 ;
        tree_type : "worst_case_tree" ;
    }
    default_operating_conditions : WCCOM;
    ...
      timing_based_variation() {
        va_parameters(voltage, var2, process, temperature);
        nominal_va_values(15.1, 25.0, 35.0, 45.0);
        ...
      }
    ...
}
```

In the example below, both *default_operating_conditions* and *voltage_map* are defined. The *voltage* is user-defined parameter, (not a predefined parameter), which means you can specify any voltage value in *nominal_va_values*. However, VDD is a predefined parameter in *voltage_map*. If you specified it in *va_parameters*, then the corresponding value of 10.0 in *nominal_va_values* will be the same as the value of 9.0 defined in *voltage_map*. The values are not the same, so the error message is generated.

```
library(libdb814) {
    ...
    voltage_map( VDD, 9.0);
    voltage_map(VDDH, 4.91);
    voltage_map(VDDL, 4.80);
    voltage_map(VSS, 0.0);

    operating_conditions(WCCOM) {
        process : 1.5 ;
        temperature : 45 ;
        voltage : 9.0 ;
        tree_type : "worst_case_tree" ;
```

Feedback

```
    }
    default_operating_conditions : WCCOM;
    ...
      timing_based_variation() {
        va_parameters(voltage, var2, VDD, temperature);
        nominal_va_values(15.1, 25.0, 10.0, 45.0);
        ...
      }
    ...
}
```

## LBDB-815

(error) The number of %s in %s must be exactly twice the number of the values in va_parameters.

### Description

This error message occurs when the number of variation tables does not equal twice the number of the values in *va_parameters*.

The following are the possible variation tables that you can specify under *pin_based_variation*:

```
va_receiver_capacitance1_rise
va_receiver_capacitance2_rise
va_receiver_capacitance1_fall
va_receiver_capacitance2_fall
```

in -25i

The following are the possible variation tables that you can specify under *timing_based_variation*:

```
va_receiver_capacitance1_rise
va_receiver_capacitance2_rise
va_receiver_capacitance1_fall
va_receiver_capacitance2_fall
va_rise_constraint
va_fall_constraint
va_compact_ccs_rise
va_compact_ccs_fall
```

The total number of these tables must be exactly twice number of values in *va_parameters*.

The following example shows an instance where this message occurs: In the following example, there are 5 *va_receiver_capacitance1_rise groups* under *timing_based_variation*. Since the size of va_parameters is 2, var1 and var2, the total number of *va_receiver_capacitance1_rise* groups will be 4 instead of 5.

Feedback

```
timing_based_variation() {
  va_parameters(var1, var2);
  nominal_va_values(15.0, 25.0);
  va_receiver_capacitance1_rise ( va_TB2INVXC_rise_1 ) {
    va_values(16.0, 25.0);
    values ( \
      "1.100, 1.100", "1.100, 1.100", "1.100, 1.100");
  }
  va_receiver_capacitance1_rise ( va_TB2INVXC_rise_1 ) {
    va_values(14.0, 25.0);
    values ( \
      "1.200, 1.200", "1.200, 1.200", "1.200, 1.200");
  }
  va_receiver_capacitance1_rise ( va_TB2INVXC_rise_1 ) {
    va_values(15.0, 26.0);
    values ( \
      "1.300, 1.300", "1.300, 1.300", "1.300, 1.300");
  }
  va_receiver_capacitance1_rise ( va_TB2INVXC_rise_1 ) {
    va_values(15.0, 24.0);
    values ( \
      "1.400, 1.400", "1.400, 1.400", "1.400, 1.400");
  }
  va_receiver_capacitance1_rise ( va_TB2INVXC_rise_1 ) {
    va_values(15.0, 23.0);
    values ( \
      "1.400, 1.400", "1.400, 1.400", "1.400, 1.400");
  }
 ...
```

**What Next**

Check the library source file, make the necessary corrections, and run the command again.

---

## LBDB-816

(error) %s in this group and %s defined in %s\n \tshall have identical values on all but one.

**Description**

The values you specified in va_values and nominal_va_values must be identical except for one value.

The following example shows an instance where this message occurs:

```
pin_based_variation() {
  va_parameters(var1, var2, var3, var4);
  nominal_va_values(15.0, 25.0, 35.0, 45.0);
  va_receiver_capacitance1_rise ( pinTB2INVXC_rise_1 ) {
    va_values(15.0, 25.0, 35.0, 45.0);
    values ( \
```

Feedback

```
      "1.100, 1.100, 1.100");
  }

  va_receiver_capacitance1_rise ( pinTB2INVXC_rise_1 ) {
    va_values(14.0, 25.0, 35.0, 45.0);
    values ( \
      "1.200, 1.200, 1.200");
  }

  va_receiver_capacitance1_rise ( pinTB2INVXC_rise_1 ) {
    va_values(15.0, 26.0, 36.0, 45.0);
    values ( \
      "1.300, 1.300, 1.300");
  }
   ...
```

In the first va_receiver_capacitance1_rise, all values in nominal_va_values and va_values are the same, which is wrong. There must be one value difference.

In the second va_receiver_capacitance1_rise, var1 has different value between va_values and nominal_va_values, which is okay.

In the third va_receiver_capacitance1_rise, var2 and var3 have different values between va_values and nominal_va_values, which is wrong. Only one value is allowed to be different.

The following is an example message:

```
Error: Line 381, va_values in this group and nominal_va_values defined in
 pin_based_variation
        shall have identical values on all but one.      (LBDB-816)
```

**What Next**

Check the library source file, and make the necessary correction.

---

## LBDB-817

(error) For %s groups in %s, one characterization point\n \tis required and up to two points are allowed for each parameter.

**Description**

At least one characterization point is required for each parameter, and only up to two points are allowed. The value of characterization point must be different from the nominal value. If there are two points, then one must be greater than nominal value, and the other must be less than nominal value.

This rule also applies for all va_leakage_current groups under the same state condition.

Feedback

The following example shows an instance where this message occurs:

```
timing_based_variation() {
  va_parameters(var1, var2);
  nominal_va_values(15.0, 25.0);
  va_compact_ccs_rise(va_lut) {
     va_values(16.0, 25.0);
     values ( ... );
  }
  va_compact_ccs_rise(va_lut) {
     va_values(19.0, 25.0);
     values ( ... );
  }
  va_compact_ccs_rise(va_lut) {
     va_values(15.0, 26.0);
     values ( ... );
  }
  va_compact_ccs_rise(va_lut) {
     va_values(15.0, 24.0);
     values ( ... );
  }
  ...
}
```

For the parameter var1, there are two variations (19.0 and 16.0), and both of them are greater than nominal value 15.0, which is wrong.

The following example shows another instance where this message occurs:

```
timing_based_variation() {
        va_parameters(var1, var2);
        nominal_va_values(15.0, 25.0);
            va_rise_constraint(va_sup_hld) {
                va_values(16.0, 25.0);
                index_1(" 0.006, 0.04, 0.1, 0.2, 1");
                index_2(" 0.006, 0.04, 0.1, 0.2, 1");
                values( ...);
                }
            va_rise_constraint(va_sup_hld) {
                va_values(14.0, 25.0);
                index_1(" 0.006, 0.04, 0.1, 0.2, 1");
                index_2(" 0.006, 0.04, 0.1, 0.2, 1");
                values( ...);
                }
            va_rise_constraint(va_sup_hld) {
                va_values(16.0, 25.0);
                index_1(" 0.006, 0.04, 0.1, 0.2, 1");
                index_2(" 0.006, 0.04, 0.1, 0.2, 1");
                values( ...);
                }
           va_rise_constraint(va_sup_hld) {
                va_values(15.0, 24.0);
                index_1(" 0.006, 0.04, 0.1, 0.2, 1");
```

Feedback

```
                index_2(" 0.006, 0.04, 0.1, 0.2, 1");
                values( ...);
                }
          ...
```

For parameter var1, there are three variation values (16.0, 14.0 and 16.0 again). This is wrong because only up to 2 points are allowed. For parameter var2, there is only one variation value (24.0), which is okay.

The following example shows another instance where this message occurs:

```
cell_based_variation() {
        va_parameters(var1, var2);
        nominal_va_values(15.0, 25.0);
            va_leakage_current() {
                when : "A1";
                va_values(17.0, 25.0);
                 ...
                }
            va_leakage_current() {
                when : "A1";
                va_values(13.0, 25.0);
                 ...
                }
            va_leakage_current() {
                when : "!A1";
                va_values(17.0, 25.0);
                 ...
                }
            va_leakage_current() {
                when : "!A1";
                va_values(15.0, 29.0);
                 ...
                }
            va_leakage_current() {  /* default state */
                va_values(16.0, 25.0);
                 ...
                }
            va_leakage_current() {  /* default state */
                va_values(15.0, 26.0);
                 ...
                }
       } /* end of cell_based_variation */
         ...
```

Under the when statement "A1", there is no characterization point for var2, which is wrong, and var1 have two points (17.0 and 13.0), which are okay. For the cases with state condition "!A1" and without any state condition (default state), both are okay.

The following is an example message:

```
Error: Line 552, For va_rise_constraint groups in timing_based_variation,
 one characterization point
```

Feedback

```
        is required and up to two points are allowed for each parameter.
(LBDB-817)
```

### What Next

Check the library source file, and make the necessary correction.

---

## LBDB-818

(error) At least, one of the variation-aware groups is absent.

### Description

This message indicates that at least you are missing one of the variation-aware tables. Here is the rule : if both of nominal tables are defined, then either no variation-aware table is required or all variation-aware tablea are required. Meaning that if all nominal tables are defined, then you can't not just define partial of correspoding variation-aware tables.

The following example shows an instance where this message occurs:

```
timing() {
      ...
        fall_constraint(sup_hld) {
           index_1(" 0.006    ,  0.04     ,  0.1      ,  0.2      ,  1
     ");
           index_2(" 0.006    ,  0.04     ,  0.1      ,  0.2      ,  1
     ");
           values( ...
        }
        rise_constraint(sup_hld) {
           index_1(" 0.006    ,  0.04     ,  0.1      ,  0.2      ,  1
     ");
           index_2(" 0.006    ,  0.04     ,  0.1      ,  0.2      ,  1
     ");
           values( ...
        }
        timing_based_variation() {
        ...
           va_rise_constraint(va_sup_hld) {
             ...
             values( ...
        ...}
     ...}
,,,}
```

In this case, both nominal tables (fall_constraint and rise_constraint) are defined and only one of variation-aware tables (va_rise_constraint) is defined, which is wrong. To fix the problem, add va_fall_constraint:

```
timing() {
      ...
```

Feedback

```
      fall_constraint(sup_hld) {
         index_1(" 0.006    ,  0.04     ,  0.1      ,  0.2      ,  1
");
         index_2(" 0.006    ,  0.04     ,  0.1      ,  0.2      ,  1
");
         values( ...
      }
      rise_constraint(sup_hld) {
         index_1(" 0.006    ,  0.04     ,  0.1      ,  0.2      ,  1
");
         index_2(" 0.006    ,  0.04     ,  0.1      ,  0.2      ,  1
");
         values( ...
      }
      timing_based_variation() {
      ...
         va_rise_constraint(va_sup_hld) {
           ...
           values( ...
      ...}
         va_fall_constraint(va_sup_hld) {
           ...
           values( ...
      ...}

   ...}
...}
```

or remove va_rise_constraint from the timing arc :

```
timing() {
     ...
       fall_constraint(sup_hld) {
          index_1(" 0.006    ,  0.04     ,  0.1      ,  0.2      ,  1
");
          index_2(" 0.006    ,  0.04     ,  0.1      ,  0.2      ,  1
");
          values( ...
       }
       rise_constraint(sup_hld) {
          index_1(" 0.006    ,  0.04     ,  0.1      ,  0.2      ,  1
");
          index_2(" 0.006    ,  0.04     ,  0.1      ,  0.2      ,  1
");
          values( ...
       }
       timing_based_variation() {
       ...
   ...}
...}
```

Feedback

The following is an example message:

```
Error: Line 552 At least, one of the variation-aware groups is absent.
 (LBDB-818)
```

### What Next

Check the library source file, and make the necessary correction.

---

## LBDB-819

(warning) The %s %s is not present, so\n \taccuracy can be deteriorated.

### Description

You will get the warning due to the following reason : - If CCS timing driver model is defined, then CCS timing receiver model is expected to be there, too.

The following example shows an instance where this message occurs:

```
...
    pin(I) {
      direction : input ;
      capacitance : 0.0014397 ;
      rise_capacitance : 0.0013000 ;
      fall_capacitance : 0.0012000 ;
    }
    pin(Z) {
      direction : output ;
      max_capacitance : 0.17890 ;
      function : "I" ;

      timing() {
        related_pin : "I" ;
        timing_sense : positive_unate ;
        cell_rise(delay_template_7x7) {
          ...
        }
        rise_transition(delay_template_7x7) {
          ...
        }
        cell_fall(delay_template_7x7) {
                  ...
        }
        fall_transition(delay_template_7x7) {
                   ...
        }
        output_current_fall() {
           ...
        }
        output_current_rise() {
           ...
```

Feedback

```
        }
        ...
      }
    ...
```

In this case, ccs receiver data for pin 'I' is expected because ccs driver model is defined.

The following is an example message:

```
Warning: Line 3186, The CCS receiver data is not present, so
        accuracy can be deteriorated. (LBDB-819)
```

**What Next**

Check the library source file, and make the necessary correction.

---

## LBDB-819e

(error) The %s %s is not present, so\n \taccuracy can be deteriorated.

**Description**

If CCS timing driver model is defined, then CCS timing receiver model is expected to be there, too. If both pin-based and arc-based receiver model is missing for the pin in a CCS library, you will get this error.

The following example shows an instance where this message occurs:

```
...
    pin(I) {
      direction : input ;
      capacitance : 0.0014397 ;
      rise_capacitance : 0.0013000 ;
      fall_capacitance : 0.0012000 ;
    }
    pin(Z) {
      direction : output ;
      max_capacitance : 0.17890 ;
      function : "I" ;

      timing() {
        related_pin : "I" ;
        timing_sense : positive_unate ;
        cell_rise(delay_template_7x7) {
          ...
        }
        rise_transition(delay_template_7x7) {
          ...
        }
        cell_fall(delay_template_7x7) {
                  ...
        }
```

Feedback

```
        fall_transition(delay_template_7x7) {
                  ...
        }
        output_current_fall() {
           ...
        }
        output_current_rise() {
           ...
        }
        ...
      }
    ...
```

In this case, ccs receiver data for pin 'I' is expected because ccs driver model is defined.

The following is an example message:

```
Warning: Line 3186, The CCS receiver data is not present, so
        accuracy can be deteriorated. (LBDB-819e)
```

**What Next**

Check the library source file, and add the missing receiver model.

---

## LBDB-820

(error) The %s is preferred pg type.

**Description**

If the cell is an always_on cell, then for all signal pins, the backup_power is preferred pg_type that related_power_pin shall refer to, and backup_ground is preferred pg_type that related_ground_pin shall refer to.

The following example shows an instance where this message occurs:

```
cell(test) {
   always_on : TRUE;
   pg_pin(PWR) {
   voltage_name : VDD;
   pg_type : primary_power;
   }
   pg_pin(GND) {
   voltage_name : VSS;
   pg_type : primary_ground;
   }
   pg_pin(GND1) {
   voltage_name : VSS1;
   pg_type : backup_ground;
   }
   pg_pin(VDDI) {
   voltage_name : VDDH;
```

Feedback

```
    pg_type : backup_power;
    }
    ...
    pin(A1) {
         always_on : TRUE;
         direction : input;
         related_power_pin : PWR;
         related_ground_pin : GND1;
       ...
    }
    pin(A2) {
         direction : input;
         related_power_pin : VDDI;
         related_ground_pin : GND;
      ...
    }
    ...
}
```

For the always_on pin, A1 , the related_power_pin is pointing to PWR, which is not backup_power. For the non always_on pin, A2, the related_ground_pin is pointing to GND, which is not backup_ground either. Both of them shall connect to backup PG.

The following is an example message:

```
Error: Line 335, The backup_power is preferred pg type. (LBDB-820)
Error: Line 385, The backup_ground is preferred pg type. (LBDB-820)
```

**What Next**

Check the library source file, and make the necessary correction.

---

## LBDB-821

(error) The always_on pin is required for always_on cells.

**Description**

An always_on cell requires at least one or more always_on pin under it.

The following example shows an instance where this message occurs:

```
cell(test) {
    always_on : TRUE;
    ...
    pin(sp) {
         direction : input;
         related_power_pin : PWR;
         related_ground_pin : GND;
       ...
    }
    pin(a) {
```

Feedback

```
        ...
    }
    ...
}
```

Since 'test' cell is an always_on cell, the always_on attribute is required to be specified at least in one of pins under 'test' cell.

The following is an example message:

```
Error: Line 1548, The always_on pin is required for always_on cells.
 (LBDB-821)
```

**What Next**

Check the library source file, and make the necessary correction.

---

## LBDB-822

(error) A pin can only has either non-programmable or\n \tprogrammable driver types.

**Description**

Under a pin, there can only be either non-programmable driver type or programmable driver types, but not both.

The following example shows an instance where this message occurs:

```
pin(ZN) {
  direction : inout;
  pull_up_function : "!A1 * !A2 * !A3";
  pull_down_function : "A1 * A2 * !A3";
  bus_hold_function : "A1 * !A2 * !A3";
  driver_type : pull_up;
  ...
 }
```

In pin ZN, there are three programmable driver types, pull_up_function, bus_hold_function, and pull_down_function. Also, the non-programmable driver type, "driver_type : pull_up", is defined under the same pin, which is wrong. Please remove either programmable driver types or non-programmable driver type from the pin.

The following is an example message:

```
Error: Line 104, A pin can only has either non-programmable or
        programmable driver types. (LBDB-822)
```

**What Next**

Check the library source file, and make the necessary correction.

Feedback

## LBDB-823

(information) The %s %s is not present, so\n \taccuracy can be deteriorated.

### Description

You will get the message due to one of following reasons : - ccs model is expected to be defined to improve the data accuracy.

The following example shows an instance where this message occurs:

```
...
      timing() {
        related_pin : "I" ;
        timing_sense : positive_unate ;
        cell_rise(delay_template_7x7) {
          ...
        }
        rise_transition(delay_template_7x7) {
          ...
        }
        cell_fall(delay_template_7x7) {
                  ...
        }
        fall_transition(delay_template_7x7) {
                   ...
        }
        ...
      }
```

In this case, there is no ccs timing model but NLDM, and we expected ccs timing models for better accuracy.

The following is an example message:

```
Information: Line 582, The CCS data output_current_rise is not present,
 so
        accuracy can be deteriorated. (LBDB-823)
Information: Line 582, The CCS data output_current_fall is not present,
 so
        accuracy can be deteriorated. (LBDB-823)
```

### What Next

Check the library source file, and make the necessary correction.

## LBDB-824

(error) This static '%s' group cannot be specified in a '%s' group at line %u.

Feedback

### Description

This message indicates that a static ccsn_*_stage or input/output_ccb group has been specified in a timing arc or input/internal pin. This is not allowed.

A static CCS/CCB noise model captures the behaviors of a static channel connected block. A static channel connected block does not have any input terminal or the voltage level at the input terminal does not affect the output current of the block. It cannot be used for propagating crosstalk delay or noise waveforms through the cell under study. Consequently, the only use for a static noise model is for a tie-off pin that is permanently 0 or 1. Please refer to "CCS Noise Library Characterization Guide" for detailed information.

The following is an example message:

```
Error: Line 27, The 'timing' group cannot specify a static
 'ccsn_first_stage' group. (LBDB-824)
```

### What Next

Replace the aforementioned data with a non-static ccsn_*_stage or *_ccb group.

---

## LBDB-826

(error) The %s entry cannot be found for %s(%s) of cell(%s).

### Description

This message indicates that you do not specify the relative entry in the map file required by add_pg_pin_to_db command.

The following example shows an instance where this message occurs:

The following is an example message:

```
Error: The pg_to_voltage_map entry cannot be found for pg_pin(VDD) of
 cell(sample). (LBDB-826)
```

---

## LBDB-826w

(warning) The %s entry cannot be found for %s(%s) of cell(%s).

### Description

This warning message occurs when there is no related entry for *pg_pin*, *rail_connection*, or other attributes specified in the map file required by the *add_pg_pin_to_db* command.

Feedback

The following example shows an instance where this message occurs:

The following is an example message:

```
Warning: The pg_to_voltage_map entry cannot be found for pg_pin(VDD) of
 cell(sample). (LBDB-826w)
```

**What Next**

This is only a warning message. No action is required.

However, you can eliminate this warning message by adding the required entry to the map file.

---

## LBDB-827

(error) The input library '%s' has not been read in.

**Description**

This message indicates that the input library has not been read in.

The following example shows an instance where this message occurs:

The following is an example message:

```
Error: The input library 'a.db' has not been read in. (LBDB-827)
```

---

## LBDB-828

(error) The input library '%s' is pg_pin-based library.

**Description**

This message indicates that the input library is pg_pin-based.

The following example shows an instance where this message occurs:

The following is an example message:

```
Error: The input library 'a.db' is pg_pin-based library. (LBDB-828)
```

**What Next**

The input library is already pg_pin based, and no need to be converted by the utility. However, if you want to add or update the library with pg or power management attributes, you should specify a tcl file with update_lib_model and associated commands.

Feedback

## LBDB-828w

(warning) The input library '%s' is pg_pin-based library.

**Description**

This message indicates that the input library is pg_pin-based.

The following example shows an instance where this message occurs:

The following is an example message:

```
Warning: The input library 'a.db' is pg_pin-based library. (LBDB-828w)
```

**What Next**

The input library is already pg_pin based, and only incremental update will be performed.

## LBDB-829

(error) The input library '%s' does not have default operating_conditions.

**Description**

This message indicates that the input library does not have default operating_conditions.

The following example shows an instance where this message occurs:

The following is an example message:

```
Error:  The input library 'A.db' does not have default
 operating_conditions. (LBDB-829)
```

## LBDB-830

(error) The input library '%s' does not have 'voltage' attribute in the default operating_conditions '%s'.

**Description**

This message indicates that the input library does not have default operating_conditions.

The following example shows an instance where this message occurs:

The following is an example message:

```
Error:  The input library 'A.db' does not have 'voltage' attribute  in
 trhe default operating_conditions 'nom_pvt'. (LBDB-830)
```

Feedback

## LBDB-831

(error) The converted voltage_map table is empty.

**Description**

This message indicates that converted voltage_map table is empty.

The following example shows an instance where this message occurs:

The following is an example message:

```
Error:  The converted voltage_map table is empty. (LBDB-831)
```

## LBDB-832

(error) The number of pg_pin's generated for cell(%s) is %d, which is less than 2.

**Description**

This message indicates that the number of generated pg_pin's for the cell is less than the required minimum value (2).

The following example shows an instance where this message occurs:

The following is an example message:

```
Error:  The number of pg_pin's generated for cell(A) is 1, which is less
 than 2. (LBDB-832)
```

## LBDB-833

(error) The related_pg_pin for %s %s power_level(%s) of cell(%s) cannot be found.

**Description**

This message indicates that the related_pg_pin entry cannot be found for internal_power/ leakage_power group.

The following example shows an instance where this message occurs:

The following is an example message:

```
Error:  The related_pg_pin for cell-level internal_power power_level(VDD)
 of cell(A). (LBDB-833)
```

## LBDB-834

(error) The related_pg_pin for %s %s of cell(%s) cannot be found.

Feedback

### Description

This message indicates that the related_pg_pin entry cannot be found for internal_power/ leakage_power group.

The following example shows an instance where this message occurs:

The following is an example message:

```
Error:  The related_pg_pin for cell-level internal_power of cell(A).
 (LBDB-834)
```

## LBDB-835

(error) For cell(%s), the %s(%s) which is the %s of pin(%s) cannot be found.

### Description

This message indicates that you do not specify the power/ground pg_pin of a cell in the map file required by add_pg_pin_to_db command.

The following example shows an instance where this message occurs:

The following is an example message:

```
Error: For cell(sample), the pg_pin(VDD) which is the related_power_pin
 of pin(A) cannot be found. (LBDB-835)
```

## LBDB-836

(warning) Unable to convert to %s from pin(%s) of cell(%s).

### Description

This message indicates that add_pg_pin_to_db or update_lib_model command cannot 1) convert to pg_pin from the signal pin. In the following cases, it cannot be converted to pg_pin: a) pin(VDD) with timing, internal_power, ccsn_first_stage/ccsn_last_stage or receiver_capacitance group, where VDD is a PG pin in FRAM view. b) pin(VDD) referenced in function/statetable expressions in the cell. c) pin(VDD) referenced in the when or related_pin of the other pin's internal_power groups or when in other pin's ccsn_first_stage, ccsn_last_stage or receiver_capacitance groups or leakage_power, dynamic_current, leakage_current or intrinsic_parasitic groups in the cell. 2) generate related_power_pin/related_ground_pin for the specfied signal pin.

The following example shows an instance where this message occurs:

The following is an example message:

```
Warning: Unable to convert to pg_pin from pin(VDD) of cell(r1hd).
 (LBDB-836)
```

Feedback

### What Next

You can correct the pin group as described above. For example, if pin(VDD) has ccsn_first_stage group and you deem it is a pg_pin, you should remove the group under the pin to make the conversion. However, if it is an analog pin, you should not remove such a group and no conversion to pg_pin will be performed.

---

## LBDB-837

(error) The %s for pin(%s) of cell(%s) cannot be found.

### Description

This message indicates that the related_power_pin/related_ground_pin cannot be found for the specified pin.

The following example shows an instance where this message occurs:

The following is an example message:

```
Error:  The related_power_pin for pin(A) of cell(sample) cannot be found.
 (LBDB-837)
```

---

## LBDB-838

(error) Failed to find the file name for %s option.

### Description

In add_pg_pin_to_db command, some options have dependency and requirement. (1) An output db file name must be specified in -output option and must be different from the input db file. (2) If -mw_library_name is not specified, -pg_map_file must be specified, and vice versa. (3) If the Milkyway library does not cover all cells in db, -pg_map_file must be specified. (4) the map file name must be valid and exists in disk.

The following example shows an instance where this message occurs:

```
add_pg_pin_to_db input.db -output pg_pin.db
```

In this case, neither -pg_map_file nor -mw_library_name is specified. You will get the second error message in the following example.

The following is an example message:

```
Error: Failed to find the file name for -output option. (LBDB-838)
Error: Failed to find the file name for either -pg_map_file or
 -mw_library_name. (LBDB-838)
Error: Failed to find the file name for -pg_map_file option. (LBDB-838)
```

Feedback

### What Next

Check the options to this command, and specify all the necessary ones.

---

## LBDB-839

(warning) Partial Milkyway library for the input library file.

### Description

In add_pg_pin_to_db command, when you specify -mw_library_name and the Milkyway library does not cover all cells in the input db, this warning will occur. This includes missing cells, as well as missing and mismatched pins. This is shown after check_library command is invoked under the hood, and you will also see Logic vs. physical library check summary: Number of cells missing in logic library: <number of missing cells>

The following example shows an instance where this message occurs:

```
add_pg_pin_to_db input.db -mw_library_name mwlib -pg_map_file pg.map
 -output pg_pin.db

In this case, Milkyway library mwlib does not cover all cells in
 input.db.
You will get the second error message in the following example.
```

The following is an example message:

```
Logic vs. physical library check summary:
Number of cells missing in logic library:       23
Logic library is INCONSISTENT with physical library.
Warning: Partial Milkyway library for the input library file. (LBDB-839)
```

### What Next

First you need to check if the Milkyway library you specified is the correct one. Otherwise, you should specify mapping in the map file for those cells that do not exist or exist but have missing or mismatched pins in the Milkyway library. Otherwise, later checking in the flow will find such cells are not specified in the map file and pg_pin based db will not be generated. If you want to get a list of the missing cells and/or missing and mismatched pins, run check_library command for logic vs. physical library cross checking with default options. You do not need to use set_check_library_options command to set options. You only need to specify logic and Milkyway library names in check_library command.

---

## LBDB-840

(warning) No %s found in map file.

Feedback

### Description

In the map file, if any of the following occurs: (1) No BEGIN <section_name> or END <section_name> (2) incorrect <section_name> (3) incorrect key words in title line this message will be printed out. Valid section names include: PG_PIN_MAP PG_TO_VOLTAGE_MAP VOLTAGE_MAP POWER_DOWN_FUNCTION_MAP

Key words in title line include: (1) PG_TO_VOLTAGE_MAP section cell, pg_pin, voltage_name, pg_type and direction where key word direction is optional. (2) PG_PIN_MAP section cell, pin, rail_connection and pg_pin (3) VOLTAGE_MAP section voltage_name and voltage_value (4) POWER_DOWN_FUNCTION_MAP cell, pin and power_down_function

The key words in each title line should be in the above order and delimited by Tab or space(s). Invalid record will be ignored. If BEGIN or END section line or title line is missing, the whole section will be ignored. In the map file, the following lines are legal but will be ignored: (1) a comment line starting with # is a comment line and (2) a blank line

The following example shows an instance where this message occurs:

```
BEGIN PG_TO_VOLTAGE_MAP
cell            pg_pin          voltage_name    pg_type
ADDFHX1         VDD             VDD             power
ADDFHX1         VSS             VSS             primary_ground
END PG_TO_VOLTAGE_MAP
```

The following is an example message:

```
Warning: Line 14, No END section found in map file. (LBDB-840)
Warning: Line 20, No BEGIN section found in map file. (LBDB-840)
Warning: Line 44, No title line found in map file. (LBDB-840)
```

### What Next

You should correct the syntax errors in the map file.

---

## LBDB-841

(error) Failed to read input library '%s'.

### Description

This message occurs when one or more of the following exist: 1) the input library file does not exist 2) the file cannot open for read 3) the library is not a valid technology library 4) the library file cannot compile. 5) .lib file loaded by read_db

The following example shows an instance where this message occurs:

Feedback

The following is an example message:

```
Error: Failed to read input library 'rail.db'. (LBDB-841)
```

### What Next

Check for the input library file name and location to make sure it exists with read permission. If it does, check if the input library is a valid technology library. If the library cannot compile, please check the Error messages during compilation, and correct them.

---

## LBDB-842

(warning) Number of rail_connections %d for cell '%s' is not equivalent to number of power pins %d in %s.

### Description

In cell group in db, rail_connection is used to specify power rails, not ground or internal pins. Therefore, if a rail_connection is for ground, such as VSS in the following example, the rail_connection is not valid.

The following example shows an instance where this message occurs:

```
cell (test) {
        rail_connection(VDD, VDD);
        rail_connection(VDD_AUX, VDD1);
        rail_connection(VSS, VSS);
        ...
    }

    In Milkyway library, cell test has VDD and VSS, but no VDD1.
```

The following is an example message:

```
Warning: Number of rail_connections 3 for cell 'test' is not equivalent
 to number of power pins 1 in Milkyway. (LBDB-842)
```

### What Next

If you have more rail_connections in db than power pins in Milkyway or map file, check if the extra rail is proper or not. If not, you should remove it from db. Otherwise, there is mismatch between db and Milkyway (or map file if the cell is from map file) and the command will stop conversion for a new pg_pin db.

---

## LBDB-843

(error) %s cell '%s' does not have %s pg_pins.

Feedback

### Description

Each cell with NLDM, CCS-T and CCS-N has at least 2 pg_pins (1 power and 1 ground) while a load or pull_up (tieoff high) or pull_down (tieoff low) cell has at least one pg_pin as documented in LC manual. Further, a pull_up cell should have a power pg_pin while a pull_down cell should have a ground pg_pin.

The following example shows an instance where this message occurs:

```
BEGIN PG_TO_VOLTAGE_MAP
cell            pg_pin          voltage_name    pg_type
tieoff_pullup   VDD             VDD             primary_power
tieoff_pulldown VSS             VSS             primary_power
END PG_TO_VOLTAGE_MAP
where tieoff_pulldown cell does not have a ground pg_pin.
```

The following is an example message:

```
Error: Tieoff cell 'tieoff_pulldown' does not have ground pg_pins.
 (LBDB-843)
Error: cell 'ADDFHX4' pin 'A' does not have related_power pg_pins.
 (LBDB-843)
```

### What Next

Check and specify the correct pg_pin in the map file: 1) for a cell with NLDM, CCS-T and CCS-N, check if there are at least 1 power and 1 ground pg_pin's, and 1 power and 1 ground pg_pin's for each signal pin. 2) for a load cell, check if there are at least 1 pg_pin 3) for a pull_up cell, check if there is 1 power pg_pin 4) for a pull_down cell, check if there is 1 ground pg_pin

---

## LBDB-843w

(warning) %s cell '%s' does not have %s pg_pins.

### Description

Each cell with NLDM, CCS-T and CCS-N has at least 2 pg_pins (1 power and 1 ground) while a load or pull_up (tieoff high) or pull_down (tieoff low) cell has at least one pg_pin as documented in LC manual. Further, a pull_up cell should have a power pg_pin while a pull_down cell should have a ground pg_pin.

The following example shows an instance where this message occurs:

```
BEGIN PG_TO_VOLTAGE_MAP
cell            pg_pin          voltage_name    pg_type
tieoff_pullup   VDD             VDD             primary_power
tieoff_pulldown VSS             VSS             primary_power
END PG_TO_VOLTAGE_MAP
where tieoff_pulldown cell does not have a ground pg_pin.
```

Feedback

The following is an example message:

```
Warning: Tieoff cell 'tieoff_pulldown' does not have ground pg_pins.
 (LBDB-843w)
```

### What Next

Check and specify the correct pg_pin in the map file: 1) for a cell with NLDM, CCS-T and CCS-N, check if there are at least 1 power and 1 ground pg_pin's, and 1 power and 1 ground pg_pin's for each signal pin. 2) for a load cell, check if there are at least 1 pg_pin 3) for a pull_up cell, check if there is 1 power pg_pin 4) for a pull_down cell, check if there is 1 ground pg_pin

---

## LBDB-844

(warning) %s '%s' does not exist in input db. Remove it from map.

### Description

When a cell and/or pin specified in the map file does not exist in the input db, it will be ignored and removed from the map. The cell may come from the map file or Milkyway. If it comes from Milkyway, no line number is printed out.

The following example shows an instance where this message occurs:

The following is an example message:

```
Warning: Line 24, Cell 'test' does not exist in input db. Remove it from
 map. (LBDB-844)
```

### What Next

Check if the cell and/or pin is what you want. If so, you need to check the input db for the missing cell. Otherwise, ignore it or remove it from the map file.

---

## LBDB-845

(error) %s '%s' is missing in %s.

### Description

This error message occurs when a .db cell and/or pin is missing in the map file or its Milkyway library.

The following is an example message:

```
Error: Cell 'test' is missing in map file. (LBDB-845)
Error:  'Map data or FRAM' is missing in input. (LBDB-845)
```

Feedback

### What Next

Add the mapping in the map file or specify the correct Milkyway library for the missing cells to generate a complete pg_pin based .db file. You can use the * (asterisk) wildcard character for all cells that are not explicitly specified in the map file. For dirty flow and on-the-fly PG library updates, input either Milkyway library or map data for non-rail_based .dbs. In the case of incremental updates to a PG library, specify a map file or a Tcl file for maps.

---

## LBDB-845w

(warning) %s '%s' is missing in %s.

### Description

If a power management attribute is missing in the cell, this message will occur. If a db cell and/or pin is missing in the map file or its Milkyway library, this message will be printed out. You need to add its mapping in the map file to generate a complete pg_pin based db file.

The following example shows an instance where this message occurs:

The following is an example message:

```
Warning: Cell 'LS16FROM16TO20' 'level_shifter_type' is missing in
 library'. (LBDB-845w)
```

### What Next

You need to add the missing attribute for power management cells; otherwise, the cells will not be used for UPF. You need to add the mapping in the map file or specify the correct Milkyway library for the missing cells to generate a complete pg_pin based db file. You can use wild card "*" for all the cells that are not explicitly specified in the map file.

---

## LBDB-846

(warning) Cell '%s' has %d %s pins in Milkyway library.

### Description

If a cell has multiple power or ground pins in Milkyway and db, you should specify related_power_pin and related_ground_pin for its signal pins.

The following example shows an instance where this message occurs:

```
BEGIN PG_PIN_MAP
cell    pin rail_connection pg_pin
test    LSI VDD             VDD
test    LSO VDD_AUX         VDD_AUX
```

Feedback

```
test    LSI -               VSS
test    LSO -               VSS
END PG_PIN_MAP
where cell test has 2 power pins: VDD and VDD_AUX.
```

The following is an example message:

```
Warning: Cell 'test' has 2 power pins in Milkyway library. (LBDB-846)
```

**What Next**

Check PG_PIN_MAP section in the map file to see if related_power_pin and related_ground_pin's are specified for these cells that have multiple P/G pins. This is only a warning message. In the flow that follows this message, there are other checkings that will catch error if you did not specify related_power_pin and related_ground_pin for these cells.

---

## LBDB-847

(error) Too many values (%d) specified in the table.

**Description**

This message indicates that there are too many values specified in the table. (number = index1_size * index2_size [* index3_size] [*index4_size]). The number exceed the limit (32767) that Library Compiler current handles.

The following is an example message:

```
Error: Line 307, Too many values (45000) specified in the table.
 (LBDB-847)
```

**What Next**

Split the table, to make each table's size less than the limit.

---

## LBDB-848

(error) There is no '%s' %s defined for '%s' %s.

**Description**

This message indicates the library does not define the voltage_map for the voltage_name specified in a pg_pin, or input_signal_level/output_signal_level defined in a pin, or the active state of pg pin/pg setting in pg_setting_value group.

The following example shows an instance where this message occurs:

```
voltage_map(VDD, 1.08);
      voltage_map(VDD1, 1.1);
```

Feedback

```
      voltage_map(VSS, 0.0);

        ...
      cell(sample) {
        pg_pin(VDD) {
                voltage_name : VDD2;
                ...
        }
        ...
      }
      ...
```

In this case, there is no voltage_map defined for voltage_name VDD2. To fix the problem, add the voltage_map attribute for VDD2 at the library level:

```
voltage_map(VDD, 1.08);
      voltage_map(VDD1, 1.1);
      voltage_map(VDD2, 1.2);
      voltage_map(VSS, 0);

        ...
      cell(sample) {
        pg_pin(VDD) {
                voltage_name : VDD2;
                ...
        }
        ...
      }
      ...
```

The following is an example message:

```
Error: Line 23, There is no 'VDD2' voltage_map defined for 'VDD2'
 voltage_name. (LBDB-848)
```

### What Next

Add the missing voltage_map.

---

## LBDB-848w

(warning) There is no '%s' %s defined for '%s' %s.

### Description

This message indicates the library does not define the voltage_map for the voltage_name specified in a pg_pin, or input_signal_level/output_signal_level defined in a pin, or the active state of pg pin/pg setting in pg_setting_value group.

Feedback

The following example shows an instance where this message occurs:

```
voltage_map(VDD, 1.08);
      voltage_map(VDD1, 1.1);
      voltage_map(VSS, 0.0);

        ...
      cell(sample) {
        pg_pin(VDD) {
                voltage_name : VDD2;
                ...
        }
        ...
      }
      ...
```

In this case, there is no voltage_map defined for voltage_name VDD2. To fix the problem, add the voltage_map attribute for VDD2 at the library level:

```
voltage_map(VDD, 1.08);
      voltage_map(VDD1, 1.1);
      voltage_map(VDD2, 1.2);
      voltage_map(VSS, 0);

        ...
      cell(sample) {
        pg_pin(VDD) {
                voltage_name : VDD2;
                ...
        }
        ...
      }
      ...
```

The following is an example message:

```
Warning: Line 23, There is no 'VDD2' voltage_map defined for 'VDD2'
 voltage_name. (LBDB-848w)
```

### What Next

Add the missing voltage_map.

---

## LBDB-849

(warning) The '%s' group is overwritten by the '%s' group on line %d.

### Description

The same group has been defined multiple times and only the last one will be recoded in the library db.

Feedback

The following example shows an instance where this message occurs:

The following is an example message:

```
Warning: The 'dc_current' group is overwritten by the 'dc_current' group
 on line 120. (LBDB-849)
```

**What Next**

Check the dc_current groups for wrong information and fix.

---

## LBDB-850

(warning) Missing %s value under section %s in map file.

**Description**

If data entered in map file is incomplete, this message will be printed out. In PG_TO_VOLTAGE_MAP section, there are four fields: cell, pg_pin, pg_type and voltage_name. If one or more fields are left blank, you will see missing data in this section. The same is true for other sections. In VOLTAGE_MAP, for multi-rail library you should specify voltage_map entry with voltage_value = 0. Invalid record will be ignored. In POWER_DOWN_FUNCTION_MAP, if power_down_function values are missing for some cell/pins, the default values !VDD1+!VDD2+...+VSS1+VSS2 will be used where VDDi and VSSi are pg_pins of the cell.

The following example shows an instance where this message occurs:

```
BEGIN PG_TO_VOLTAGE_MAP
cell            pg_pin          voltage_name    pg_type
ADDFHX1         VDD             VDD
ADDFHX1         VSS             VSS             primary_ground
END PG_TO_VOLTAGE_MAP
```

The following is an example message:

```
Warning: Line 14, Missing pg_type value under section PG_TO_VOLTAGE_MAP
 in map file. (LBDB-850)
Warning: Line 20, Missing voltage value under section VOLTAGE_MAP in map
 file. (LBDB-850)
```

**What Next**

You should enter values for each field in the map file. Do not leave blank in any field except optional direction field in PG_TO_VOLTAGE_MAP section.

---

## LBDB-851

(error) %s section %s in map file.

Feedback

### Description

If the whole map section is missing (neither in map file nor derivable) or no pg_pins are available, this message will be printed out. For single-rail library, only PG_TO_VOLTAGE_MAP is required if no Milkyway library. If there is complete Milkyway library to cover all the cells in the input db, no map file is required. In this case, VOLTAGE_MAP will be derived from the input db that includes one voltage_map for nominal voltage and the other one for ground (voltage value = 0). For multi-rail library, PG_PIN_MAP, PG_TO_VOLTAGE_MAP, VOLTAGE_MAP and POWER_DOWN_FUNCTION_MAP sections are all required. If there are rail_connection's, PG_PIN_MAP is required in any case. Please note that invalid entries in the map section are ignored and removed. So even though these entries exist in the map section, the final map table may not include such entries. If pg_pins defined in FRAM have the same name pins in db as signal pins and these pins are unable to be converted to pg_pins due to complexity of these pins such as having timing, noise or power groups, most likely analog pins, they will stay as signal pins in db and no pg_pins of the same names are added.

The following example shows an instance where this message occurs:

```
BEGIN PG_TO_VOLTAGE_MAP
cell    pg_pin  voltage_name    pg_type
*       VDD     VDD             primary_power
*       VSS     VSS             primary_ground
*       VSS1    VSS1            backup_ground
END PG_TO_VOLTAGE_MAP
BEGIN POWER_DOWN_FUNCTION_MAP
cell    pin     power_down_function
test    test    VSS
*       *       !VDD+VSS+VSS1
END POWER_DOWN_FUNCTION_MAP
```

The following is an example message:

```
Error: Missing section PG_PIN_MAP in map file. (LBDB-851)
Error: Missing section VOLTAGE_MAP in map file. (LBDB-851)
```

### What Next

Enter the missing section. For multi-rail library, specify all the four map sections with complete data. You can use "*" in cell and/or pin fields for all the others that are not listed in the map section. For single-rail library, specify at least PG_TO_VOLTAGE_MAP section with complete pg_pin, voltage_name and pg_type for each cell. Otherwise, check if those complex pins are analog pins or pg_pins. If pg_pins, remove the defined groups or attributes that make the pins complex so that they can be converted to pg_pins.

Feedback

## LBDB-852

(error) %s found in %s is missing in %s.

### Description

This message occurs during cross checking of map sections. (1) PG_PIN_MAP and PG_TO_VOLTAGE_MAP It is two-way checking. It is to check consistency of pg_pin associated with a cell between the two map sections. If a pg_pin for a cell exists in one map section but either the pg_pin or cell is missing in the other section, you will see this message. (2) PG_PIN_MAP and POWER_DOWN_FUNCTION_MAP It is to check signal pin consistency between the two sections. If a pin for a cell exists in POWER_DOWN_FUNCTION_MAP but is missing in PG_PIN_MAP, you will see this message. This checking is for multi-rail library. (3) PG_TO_VOLTAGE_MAP and VOLTAGE_MAP It is to check voltage_name consistency between the two sections. If the voltage_name for the nominal voltage is specified in VOLTAGE_MAP but is missing in PG_TO_VOLTAGE_MAP, you will see a message saying so. Please also refer to LBDB-854. In VOLTAGE_MAP, you can have more entries than necessary.

The following example shows an instance where this message occurs:

```
BEGIN PG_PIN_MAP
cell    pin rail_connection pg_pin
ADDFHX1 A       -   VSS
ADDFHX1 A       -   VDD
ADDFHX1 A       -   VDD1
END PG_PIN_MAP
BEGIN PG_TO_VOLTAGE_MAP
cell    pg_pin  voltage_name    pg_type
ADDFHX1 VDD     VDD primary_power
ADDFHX1 VSS     VSS primary_ground
END PG_TO_VOLTAGE_MAP
```

The following is an example message:

```
Error: Line 14, Cell 'test' pg_pin 'VDD1' found in PG_PIN_MAP is missing
 in PG_TO_VOLTAGE_MAP. (LBDB-852)
```

### What Next

Check if the missing data (cell, pg_pin or pin) is what is required in the map section or extra data in the section where it exists. In the following example, if it is a multi-rail library and pg_pin VDD1 is what you should need in the library, you should list its entry in PG_TO_VOLTAGE_MAP, in the form of ADDFHX1 VDD1 VDD1 backup_power Otherwise, if it is a single-rail library, you should remove the entry for VDD1 from PG_PIN_MAP.

Feedback

## LBDB-853

(warning) Voltage value %g for voltage_name '%s' is replaced by %g from input library.

### Description

In add_pg_pin_to_db, it takes the data in the input db as higher priority. Therefore, if you defined a voltage_map in the map file while there is a different voltage value in the db for this voltage_name, the user defined value will be replaced by the value from the db, e.g. power_rail. The voltage value from the input db is taken in the following order: 1. If the library db has defined default_operating_conditions, then the specified voltage value in operating_condition will be used as the nominal voltage. 2. Otherwise, nom_voltage value defined at the library level will be used as the nominal voltage.

The following example shows an instance where this message occurs:

```
BEGIN VOLTAGE_MAP
voltage_name    voltage_value
VDD             1.1
VSS             0.0
END VOLTAGE_MAP
in default_operating_conditions,
   power_rail (VDD, 1.2)
```

The following is an example message:

```
Warning: Voltage value 1.1 for voltage_name 'VDD' is replaced by 1.2 from
 input library. (LBDB-853)
```

## LBDB-854

(warning) voltage_name '%s' for nominal voltage not referenced in %s.

### Description

This warning message occurs when you specify a voltage_map with the nominal voltage (or in the case of a single-rail or 1P1G library, a default voltage_map for the nominal voltage is automatically derived from the input db), but this voltage name is not referenced in the PG_TO_VOLTAGE_MAP table for add_pg_pin_to_db/lib or pg_pin group in .lib.

The following example shows an instance where this message occurs: The following is an example:

```
BEGIN PG_TO_VOLTAGE_MAP
cell    pg_pin  voltage_name   pg_type
test    VDD1    VDD1           primary_power
test    VSS     VSS            primary_ground
END PG_TO_VOLTAGE_MAP

BEGIN VOLTAGE_MAP
```

Feedback

```
voltage_name    voltage_value
VDD             1.2
VDD1            1.08
VSS             0.0
END VOLTAGE_MAP
where 1.2 is nominal voltage.
```

The following is an example message:

```
Warning: voltage_map 'VCC' for nominal voltage not referenced in
 PG_TO_VOLTAGE_MAP. (LBDB-854)
Warning: voltage_name 'VCC' for nominal voltage not referenced in pg_pin
 group.  (LBDB-854)
```

**What Next**

Check if this nominal voltage and voltage name are correct. If so, add its mapping entry from pg_pin to voltage_map with the nominal voltage in PG_TO_VOLTAGE_MAP or reference the voltage_name in a pg_pin group in .lib. If you have not specified the correct voltage name for the nominal voltage in voltage_map, correct the voltage name. If you have not specified the voltage_map in the map file for the nominal voltage and it is derived by the utility, specify the voltage map for the nominal voltage using the correct voltage name from PG_TO_VOLTAGE_MAP.

---

## LBDB-855

(warning) Invalid %s found under section %s in map file.

**Description**

In the map file, if any of the following is incorrect, (1) pg_type (2) pg_pin direction (3) voltage value (4) wild card or - in wrong field (5) incorrect section name or keywords in title line (6) pin direction in POWER_DON_FUNCTION_MAP (7) duplicate data specified on the same cell/pin (8) incorrect rail_connection this message will occur.

Valid pg_type values include: primary_power primary_ground backup_power backup_ground internal_power internal_ground Valid pg_pin direction values include input output inout internal nwell pwell deepnwell deeppwell

Valid voltage values are non-negative floating point numbers. Valid pin directions in POWER_DOWN_FUNCTION_MAP are output or inout. Valid power_down_functions are double quoted strings or unquoted strings if no space within the strings and the strings must contain only the power and ground pin names. rail_connection value should be derived from input .lib/db for the cell. If no rail_connection, please enter -.

Wild cards "*" and "-" can only be entered in some specific fields: (1) PG_PIN_MAP section "*" can be entered in cell and pin fields, and "-" can be entered in pin and rail_connection fields. (2) POWER_DOWN_FUNCTION_MAP section "*" can be entered in cell and pin fields. (3) PG_TO_VOLTAGE_MAP section "*" can be entered in cell

Feedback

field only, and "-" can be entered in optional direction field. (4) VOLTAGE_MAP section No "*" or "-" can be entered in any field in this section. Caret "^" cannot be entered in the first entry of each map section. In checking title lines, the key words and their order should be exactly the same as mentioned in the spec. If an invalid value is entered or the value is entered in the wrong field in the map file, the whole record will be ignored for all fields except optional field in which only optional value is ignored. For example, if the pg_pin direction value is invalid, only the invalid value in this field is ignored. However, if the value is invalid in other fields, the whole line of record wil be ignored. (5) POWER_MANAGEMENT_ATTRIBUTE_MAP section If you specify duplicate entries on the same cell/pin/attribute, the last one will be ignored. For example, if you specify retention_2 sv retention_pin (save, "1") retention_2 sv retention_pin (restore, "0") you will receive this message and the second entry will be ignored. If you want to specify pin SAVE as both save and restore, specify it in this way, retention_2 sv retention_pin (save_restore, "0")

The following example shows an instance where this message occurs:

```
BEGIN PG_TO_VOLTAGE_MAP
cell            pg_pin          voltage_name    pg_type
ADDFHX1         VDD             VDD             power
ADDFHX1         VSS             VSS             primary_ground
GEND PG_TO_VOLTAGE_MAP
```

The following is an example message:

```
Warning: Line 14, Invalid pg_type found under section PG_TO_VOLTAGE_MAP
 in map file. (LBDB-855)
Warning: Line 20, Invalid wild card found under section PG_PIN_MAP in map
 file. (LBDB-855)
```

### What Next

You should correct the invalid values in the map file. For example, in pg_type field, if you enter power, you will receive this message. The correct pg_type should be primary_power.

---

## LBDB-856

(error) For pin(%s) of cell(%s), the %s entry for %s of pin(%s) cannot be found.

### Description

This message indicates that you do not specify the relative_power_pin/related_ground_pin of the signal pin in the PG_PIN_MAP section of the map file used by add_pg_pin_to_db command.

Feedback

The following example shows an instance where this message occurs:

The following is an example message:

```
Error: For pin(A) of cell(sample), the pg_pin_map entry for
 related_power_pin of pin(A) cannot be found. (LBDB-856)
```

## LBDB-857

(error) Missing %s under section %s.

### Description

You will receive this Error if the following occurs: (1) VOLTAGE_MAP for multi-rail library if you do not specify voltage_map entry with voltage_value = 0 (2) POWER_MANAGEMENT_ATTRIBUTE_MAP if you do not specify a complete set of power management attributes in the map file, or the input .lib has partial power management attributes but you do not specify the ones that are missing in the library

The following example shows an instance where this message occurs:

```
BEGIN VOLTAGE_MAP
voltage_name    voltage_value
VDDlow      1.6
VDDhigh     2.0
VSS             0.1
END VOLTAGE_MAP
```

The following is an example message:

```
Error: Missing voltage value 0 under section VOLTAGE_MAP. (LBDB-857)
Error: Missing power_pin_class for cell 'retention' pin 'sv' under
 section POWER_MANAGEMENT_ATTRIBUTE_MAP. (LBDB-857)
```

### What Next

You should enter values for each field in the map file. Do not leave blank in any field except optional direction field in PG_TO_VOLTAGE_MAP section. In VOLTAGE_MAP, for multi-rail library you should specify voltage_map entry with voltage_value = 0. In POWER_MANAGEMENT_ATTRIBUTE_MAP, specify the missing power attributes. In converting power_gating_pin, it uses power_pin_class and map_to_logic attributes in DB, and change power_pin_class = 6, 7, 8 to pin class "save", "restore" and "save_restore", respectively, and take the value of map_to_logic as <disable_value>. If the input .lib does not have power_pin_class = 6, 7 or 8, you should specify retention_pin_class as in the example RET_1 sv retention_pin (save, "0")

## LBDB-858

(error) The timing arc %s->%s with when condition "%s" doesn't have ccs noise model.

Feedback

### Description

For every timing arc, there must be one of the followings: (1) arc-based ccs noise models with the same "when" condition as the "when" of the timing arc, (2) pin-based ccs noise models with the same "when" condition as the "when" of the timing arc, or (3) pin-based ccs noise models with default (i.e., no "when") "when" condition.

For example,

```
pin(CO) {
  timing() {
    when : "!B&A";
    ...
    ccsn_first_stage() {
      /* no when condition */
      ...
    } /* qualified model 1 */
  }

  ccsn_last_stage() {
    when : "!B&A";
    ...
  } /* qualified model 2 */

  ccsn_last_stage() {
    /* no when condition */
    ...
  } /* qualified model 3 */

  ...
}
```

when condition "default" means that this timing arc has no "when" attribute.

The following is an example message:

```
Error: The timing arc CI->CO with when condition "!B&A" doesn't have ccs
 noise model. (LBDB-858)
```

### What Next

Add ccs noise model with the qualified "when" condition.

---

## LBDB-859

(error) There is no cross point in this %s group with input_net_transition %s.

### Description

For expanded ccs power, if *pg_current* is represented as a sparse cross table, if there is no *input_net_transition*, then there must be one and only one cross point in one of the

Feedback

vectors. if *input_net_transition* is specified, then one and only one cross point is required for each *input_net_transition*.

For compact ccs power, if *pg_current* is represented as a sparse cross table, then the typical capacitance of the pin specified in "index_output" must be one of the values in index with variable "total_output_net_capacitance".

The following example shows an instance where this message occurs:

```
dynamic_current() {
    ...
    related_outputs : "Q QN QN1 QN2";
    typical_capacitances(10.0, 12.0, 14.0, 16.0);
    switching_group() {
      ...
      pg_current(VDD) {
        vector(CCS_power_1) {
          index_output : "QN1";
          index_1 ("0.01");
          index_2 ("14.0"); /* cross point */
          ...
        }
        vector(CCS_power_1) {
          index_output : "QN2";
          index_1 ("0.02");
          index_2 ("11.0"); /* non cross point */
          ...
        }
      }
      ...
    }
  }
```

In this case, pg_current has only two vectors. There is no cross point with input_net_transition 0.02. So error message LBDB-859 will be issued.

The following is an example message:

```
Error: There is no cross point in this pg_current group with
 input_net_transition 0.02. (LBDB-859)
```

**What Next**

Add cross point for each input_net_transition.

---

## LBDB-860

(warning) This cross point vector is inconsistent with a previous one with the same input_net_transition on Line %d.

Feedback

### Description

If two cross point *vector* groups have the same *index* value of *input_net_transition*, then waveform of the I-t curve and *reference_time* must be the same for both vectors.

Two I-t curve waveforms are "equal" means,

Step1. Use the 1st waveform as a reference. For each time point in the reference waveform, get the current from the 2nd waveform. (may involve linear interpolation).

Step2. Take the sum of absolute difference on currents for each time point at the reference waveform. Assume total num of points in the reference waveform is N.

Step3. Calculate relative waveform difference as Sum_of_difference_on_current / ( peak_from_reference * N ), which should <= 2%.

Step4. Calculate relative peak difference as ABS (peak_from_2nd - peak_from_reference ) / peak_from_reference, which should <= 2%.

The following example shows an instance where this message occurs:

```
dynamic_current() {
    ...
    related_outputs : "Q QN";
    typical_capacitances(10.0, 12.0);
    switching_group() {
      ...
      pg_current(VDD) {
        vector(CCS_power_1) {
          index_output : "Q";
          reference_time : 0.01;
          index_1 ("0.02"); /* input_net_transition */
          index_2 ("10.0"); /* total_output_net_capacitance */
          index_3 ("0.000, 0.0873, 0.135, 0.764"); /* time */
          values ("0.002, 0.009, 0.134, 0.546");
        }
        vector(CCS_power_1) {
          index_output : "QN";
          reference_time : 0.04;
          index_1 ("0.02"); /* input_net_transition */
          index_2 ("12.0"); /* total_output_net_capacitance */
          index_3 ("0.000, 0.0873, 0.135, 0.764"); /* time */
          values ("0.002, 0.009, 0.134, 0.546");
        }
      }
      ...
    }
  }
```

In this case, pg_current is a sparse cross table, two vectors are represented as cross points, and both have the same input_net_transition 0.02. In such a case, I-t curve

Feedback

waveform and reference_time (0.01 and 0.04, not identical. Issue a warning.) must be the same.

The following is an example message:

```
Warning: This cross point vector is inconsistent with a previous one with
 the same input_net_transition on line 57191. (LBDB-860)
```

**What Next**

Correct one of the cross points to make them consistent.

---

## LBDB-861

(error) This vector is repeated with a previous one under the same pg_current group.

**Description**

Under the same *pg_current* group, no two vectors can have the same *index_output* and values of all *index* attributes except for the last *index* (time). Otherwise, this vector will be regarded as repeated.

The following example shows an instance where this message occurs:

```
pg_current(VDD) {
        vector(CCS_power_1) {
          index_output : "QN1";
          reference_time : 0.01;
          index_1 ("0.01"); /* input_net_transition */
          index_2 ("14.0"); /* total_output_net_capacitance */
          index_3 ("0.001, 0.003, 0.006"); /* time */
          ...
        }
        vector(CCS_power_1) {
          index_output : "QN1";
          reference_time : 0.01;
          index_1 ("0.01"); /* input_net_transition */
          index_2 ("14.0"); /* total_output_net_capacitance */
          index_3 ("0.004, 0.007, 0.009"); /* time */
          ...
        }
      }
```

In this case, index_output (QN1), input_net_transition (0.01), and total_output_net_capacitance(14.0) are all the same between two vectors. So error message LBDB-861 will be issued.

The following is an example message:

```
Error: This vector is repeated with a previous one under the same
 pg_current group. (LBDB-861)
```

Feedback

### What Next

Correct one of the vectors to make them different.

---

## LBDB-862

(error) The dense table under the pg_current group is incomplete with input_net_transition %s.

### Description

If a <template> with two *total_output_net_capacitance* variables is applied to all vectors under a *pg_current* group, then all possible combination of capacitances between two output pins must be specified if all of them have the same *input_net_transition*.

The following example shows an instance where this message occurs:

```
pg_current(VDD) {
        vector(CCS_power_dense) {
          reference_time : 0.01;
          index_1 ("0.01"); /* input_net_transition */
          index_2 ("1.0"); /* total_output_net_capacitance */
          index_3 ("2.0"); /* total_output_net_capacitance */
          index_4 ("0.000, 0.0873, 0.135, 0.764"); /* time */
          values ("0.002, 0.009, 0.134, 0.546");
        }
        vector(CCS_power_dense) {
          reference_time : 0.01;
          index_1 ("0.01"); /* input_net_transition */
          index_2 ("1.5"); /* total_output_net_capacitance */
          index_3 ("2.5"); /* total_output_net_capacitance */
          index_4 ("0.100, 0.0873, 0.135, 0.764"); /* time */
          values ("0.113, 0.110, 0.243, 0.657");
        }
        vector(CCS_power_dense) {
          reference_time : 0.01;
          index_1 ("0.01"); /* input_net_transition */
          index_2 ("1.5"); /* total_output_net_capacitance */
          index_3 ("2.0"); /* total_output_net_capacitance */
          index_4 ("0.000, 0.0873, 0.135, 0.764"); /* time */
          values ("0.224, 0.221, 0.358, 0.769");
        }
      }
```

In this case, all possible combinations of (c1, c2) with input_net_transition 0.01 are (1.0, 2.0), (1.5, 2.5), (1.0, 2.5) and (1.5, 2.0). But (1.0, 2.5) is missing. So error message LBDB-862 will be issued.

Feedback

The following is an example message:

```
Error: The dense table under the pg_current group is incomplete with
 input_net_transition 0.01. (LBDB-862)
```

**What Next**

Add vectors to make the dense table complete.

---

## LBDB-863

(error) Size or values of total_output_net_capacitance are not identical for output pin %s.

**Description**

The size and values of *total_output_net_capacitance* for the same output pin or *index_output* (cross type) shall be identical for all vectors which have different *input_net_transition* index values.

In other words, every fixed (without tolerance) input_net_transition needs the same set of total_output_net_capacitance.

In the message, "output pin" means index_output (cross type) or all pins in "related_outputs" (dense type and diagonal type).

The following example shows an instance where this message occurs:

```
dynamic_current() {
    ...
    related_outputs : "Q QN";
    typical_capacitances(10.0, 12.0);
    switching_group() {
      ...
      pg_current(VDD) {
        vector(CCS_power_1) {
          index_output : "Q";
          reference_time : 0.01;
          index_1 ("0.01"); /* input_net_transition */
          index_2 ("5.0"); /* total_output_net_capacitance */
          ...
        }
        vector(CCS_power_1) {
          index_output : "Q";
          reference_time : 0.01;
          index_1 ("0.01"); /* input_net_transition */
          index_2 ("10.0"); /* total_output_net_capacitance */
          ...
        }
        vector(CCS_power_1) {
          index_output : "Q";
          reference_time : 0.01;
          index_1 ("0.01"); /* input_net_transition */
```

Feedback

```
    index_2 ("15.0"); /* total_output_net_capacitance */
    ...
  }
  vector(CCS_power_1) {
    index_output : "QN";
    reference_time : 0.01;
    index_1 ("0.01"); /* input_net_transition */
    index_2 ("6.0"); /* total_output_net_capacitance */
    ...
  }
  vector(CCS_power_1) {
    index_output : "QN";
    reference_time : 0.01;
    index_1 ("0.01"); /* input_net_transition */
    index_2 ("12.0"); /* total_output_net_capacitance */
    ...
  }
  vector(CCS_power_1) {
    index_output : "QN";
    reference_time : 0.01;
    index_1 ("0.01"); /* input_net_transition */
    index_2 ("18.0"); /* total_output_net_capacitance */
    ...
  }
  vector(CCS_power_1) {
    index_output : "Q";
    reference_time : 0.01;
    index_1 ("0.02"); /* input_net_transition */
    index_2 ("5.0"); /* total_output_net_capacitance */
    ...
  }
  vector(CCS_power_1) {
    index_output : "Q";
    reference_time : 0.01;
    index_1 ("0.02"); /* input_net_transition */
    index_2 ("11.0"); /* total_output_net_capacitance */
    ...
  }
  vector(CCS_power_1) {
    index_output : "Q";
    reference_time : 0.01;
    index_1 ("0.02"); /* input_net_transition */
    index_2 ("15.0"); /* total_output_net_capacitance */
    ...
  }
  vector(CCS_power_1) {
    index_output : "QN";
    reference_time : 0.01;
    index_1 ("0.02"); /* input_net_transition */
    index_2 ("6.0"); /* total_output_net_capacitance */
    ...
  }
  vector(CCS_power_1) {
```

Feedback

```
          index_output : "QN";
          reference_time : 0.01;
          index_1 ("0.02"); /* input_net_transition */
          index_2 ("12.0"); /* total_output_net_capacitance */
          ...
        }
        vector(CCS_power_1) {
          index_output : "QN";
          reference_time : 0.01;
          index_1 ("0.02"); /* input_net_transition */
          index_2 ("18.0"); /* total_output_net_capacitance */
          ...
        }
        vector(CCS_power_1) {
          index_output : "QN";
          reference_time : 0.01;
          index_1 ("0.02"); /* input_net_transition */
          index_2 ("24.0"); /* total_output_net_capacitance */
          ...
        }
      }
      ...
    }
  }
```

When input_net_transition is 0.01 and output pin is Q, total_output_net_capacitance size=3, values = (5.0, 10.0, 15.0). /*group1*/ When input_net_transition is 0.01 and output pin is QN, total_output_net_capacitance size=3, values = (6.0, 12.0, 18.0). /*group2*/ When input_net_transition is 0.02 and output pin is Q, total_output_net_capacitance size=3, values = (5.0, 11.0, 15.0). /*group3*/ When input_net_transition is 0.02 and output pin is QN, total_output_net_capacitance size=4, values = (6.0, 12.0, 18.0, 24.0). /*group4*/ group1 and group3 have different values. group2 and group4 have different sizes. So error message LBDB-863 will be issued.

The following is an example message:

```
Error: Size or values of total_output_net_capacitance are not identical
 for output pin Q. (LBDB-863)
Error: Size or values of total_output_net_capacitance are not identical
 for output pin QN. (LBDB-863)
```

**What Next**

Change the size or values of total_output_net_capacitance with a input_net_transition value to make the size and values consistent.

---

## LBDB-864

(error) The design '%s' has separate intrinsic_parasitic groups.

Feedback

### Description

This error occurs when there are two intrinsic_parasitic groups in the same cell, where the only difference is that one group has only intrinsic_resistance, and the other group has only intrinsic_capacitance.

However, it is allowed that only intrinsic_resistance or intrinsic_capacitance is defined for the intrinsic_parasitic group.

The following example shows an instance where this message occurs:

```
cell (AND3) {
  ...
  intrinsic_parasitic() {
    /* default state */
    intrinsic_resistance(G1) {
       related_output : "ZN";
       value : 9.0;
    }
  }

  intrinsic_parasitic() {
    /* default state */
    intrinsic_capacitance(G2) {
       value : 8.2;
    }
  }
}
```

In this case, cell 'AND3' has both an intrinsic_parasitic group with only intrinsic_resistance and an intrinsic_parasitic group with only intrinsic_capacitance. So error message LBDB-864 will be issued.

The following is an example message:

```
Error: The design 'AND3' has separate intrinsic_parasitic groups.
 (LBDB-864)
```

### What Next

Combine the separate intrinsic_parasitic groups into one group.

---

## LBDB-868

(warning) The estimated time when output signal reaches\n \tVDD level is out of range [%g, %g].

Feedback

### Description

The time when output signal reaches VDD level is estimated as T = reference time + NLDM delay Reference time means the time when input signal reaches VDD level. T has to be within the range of the current waveform (min and max of time in current waveform).

The way to evaluate NLDM delay is to lookup the cell_rise/cell_fall table in the corresponding timing arc with the given input_net_transition and total_output_net_capacitance in the vector.

If the given input_net_transition and total_output_net_capacitance can not be found in the cell_rise/cell_fall table, use the nearest index to get the delay value.

The following example shows an instance where this message occurs:

```
lu_table_template (table_dodelaycell_rise_0k0aa01m1) {
  variable_1 : total_output_net_capacitance
  variable_2 : input_net_transition
  index_1 (" 0.006404, 0.055194, 0.147118 ");
  index_2 (" 0.060000, 0.400000, 0.800000 ");
}
pg_current_template(t1) {
  variable_1 : input_net_transition;
  variable_2 : total_output_net_capacitance;
  variable_3 : time;
  index_1("0.6 0.9");
  index_2("0.6 0.9");
  index_3("0.3 4.7");
}

cell (AND3) {
  dynamic_current() {
    ...
    related_inputs : "A2 A3";
    related_outputs : "ZN ZN1";
    typical_capacitances(0.3 0.4);
    switching_group() {
      ...
      pg_current(V2) {
        vector(t1) {
          reference_time : 15.6;
          index_output : "ZN1";
          index_1("0.5");
          index_2("0.043");
          index_3("18.2 18.3 19.0");
          values("13.78 192.4 1100.1");
        }
      }
    }
  }

  pin(ZN1) {
```

Feedback

```
      ...
      timing() {
        related_pin : "A1 A2"
        cell_rise( scalar ) {
          values("0.28"); }
        cell_fall( scalar ) {
          values("0.0"); }
        ...
      }
      timing() {
        related_pin : "A3"
        cell_rise ("table_dodelaycell_rise_0k0aa01m1") {
        values (\
          "0.1729, 0.2200, 0.2397 ",\
          "0.3494, 0.4012, 0.4273 ",\
          "0.6499, 0.7018, 0.7281 "\
        );
      }
      cell_fall( scalar ) {
        values("0.0"); }
      }
    }
  }
```

In this case, the information on the given current waveform is, (1) the time range is [18.2, 19.0]; (2) reference time is 15.6; (3) input pins are "A2 A3", output pin is ZN1; (4) input_net_transition is 0.5; (5) total_output_net_capacitance is 0.043.

In the section of pin(ZN1), (1) the first timing arc is specified with input pin "A1 A2", overlaped with "A2 A3" in the current waveform. So the delay is needed. delay1 = MAX(0.28, 0.0) = 0.28.

(2) the second timing arc is specified with input pin "A3", overlaped with "A2 A3" in the current waveform. So the delay is needed.

Look up cell_rise table with (input_net_transition=0.5, total_output_net_capacitance=0.043). The nearest index is values[1][1] = 0.4012. delay2 = MAX(0.4012, 0.0) = 0.4012.

max_delay = MAX(delay1, delay2) = 0.4012.

reference_time + max_delay = 15.6+0.4012 = 16.0012. It isn't within the range [18.2, 19.0]. So warning message LBDB-868 will be issued.

The following is an example message:

```
Warning: The estimated time when output signal reaches VDD level is out
 of range [18.2, 19.0]. (LBDB-868)
```

**What Next**

Extend current waveform, or change reference time and delay.

Feedback

## LBDB-869

(information) Failure rate of the library cells will be overwritten.

### Description

Failure rate is a Design-For-Yield concept, it is a integer number, value from 0 ~ 1e9, means the number of failures per billion.

In Liberty syntax, there are two models can describe the cell yield, under the same cell level group, functional_yield_metric().

```
cell (my_cell) {
  ...
  functional_yield_metric() {
    /* 1st method */
    average_number_of_faults(<template>) {
      values("float...float");
    }
    /* 2nd method */
    critical_area_table(<template>) {
      defect_type:enum(short, open, short_and_open);
      related_layer:<layer_name>;
      index_1("float...float"); /*particle diameter array*/
      values("float...float"); /*critical area values */
    }
  }
  ...
}
```

Using the 1st method can defined the average failure rate and derive a cell failure rate directly by Library Compiler.

The 2nd method, besides the critical area table in .lib, user need particle distribution function file, using command calculate_caa_based_yield2db to calculate out the failure rate.

This message is reported when the command calculate_caa_based_yield2db is called to calculate the CAA based yield (failure rate), while the failure rate attribute is already exist in the library(db file), regardless the failure rate is calculated by 1st or 2nd method before.

The following is an example message:

```
Information: Failure rate of the library cells will be overwritten.
 (LBDB-869)
```

### What Next

This is just an information to update the status, you can just ignore it.

Feedback

## LBDB-870

(information) Redundant cross point vector with input_net_transition %s is removed.

### Description

If there are more than one cross point vectors with the same input_net_transition under a pg_current group, read_lib will only keep the first one. Others will be removed.

The following example shows an instance where this message occurs:

```
dynamic_current() {
    ...
    related_outputs : "Q QN QN1 QN2";
    typical_capacitances(10.0, 12.0, 14.0, 16.0);
    switching_group() {
      ...
      pg_current(VDD) {
        vector(CCS_power_1) {
          index_output : "QN1";
          index_1 ("0.01");
          index_2 ("14.0"); /* cross point */
          ...
        }
        vector(CCS_power_1) {
          index_output : "QN2";
          index_1 ("0.01");
          index_2 ("16.0"); /* cross point */
          ...
        }
      }
      ...
    }
  }
```

In this case, there are two cross points with input_net_transition 0.01. So the second one will be removed.

The following is an example message:

```
Info: Redundant cross point vectors with input_net_transition 0.01 are
 removed. (LBDB-870)
```

### What Next

It's just an information. Change in .lib is not required. However, in order to remove this message, keep only one cross point vector with the specified input_net_transition.

Feedback

## LBDB-871

(error) The difference between the first point \n \t current value(%f) and the peak current value (%f) is less than 0.001%%.

### Description

This error is issued when init current value is numerically identical to the peak current in CCS Timing or Compact CCS Timing data. The tolerance is set to 0.001%, that is: if fabs((I(first_point) - I(peak_point))/I(peak_point)) < 0.001%,

This could potentially cause larger errors in the calculation if data is truncated during reading, when these two values are too close.

The following example shows an instance where this message occurs:

```
output_current_rise() {
        vector(ccs) {
        reference_time : 1.493117e+00;
        index_1(2.986235e+00);
        index_2(2.500000e+01);
        index_3("8.216709e+00, 8.391117e+00, 1.775112e+01, 3.249112e+01,
 4.366012e+01, 6.033512e+01, 7.932612e+01, 9.818012e+01, 1.184101e+02,
 1.420501e+02, 1.728901e+02, 1.851275e+02");
        values( "4.299772e-01, 4.299781e-01, 4.117592e-01, 3.890655e-01,
 3.539464e-01, 2.690953e-01, 1.685153e-01, 9.581405e-02, 4.934546e-02,
 2.188565e-02, 7.340714e-03, 5.989923e-03");
        }
      }
```

The following is an example message:

```
Error: Line 3383, The first point current value(0.429977) is almost
        the same as peak point current value(0.429978). (LBDB-871)
```

### What Next

Check the library source file and correct the data, most likely, you may need to re-characterize the library cells for CCS timing data.

## LBDB-872

(warning) Curve parameters in %s are not exact for compact CCS power.

Feedback

### Description

This warning message occurs when the curve parameters are not exact for compact CCS power. If a compact_lut_template is for compact CCS power, the value of the curve_parameters index (the last index) must be the following:

```
init_time, init_current, bc_id1, point_time1, point_current1, bc_id2,
[point_time2, point_current2, bc_id3, ...], end_time, end_current
```

This is a pattern instead of a specified series because the compact CCS power table varies in size.

The following example shows an instance where this message occurs: The following example results in this warning message:

```
compact_lut_template(t1) {
    variable_1 : input_net_transition;
    variable_2 : total_output_net_capacitance;
    variable_3 : curve_parameters;
    index_1 : ("...");
    index_2 : ("...");
    index_3 : ("init_time, init_current, bc_id1");
  }
```

The values in index_3 are not exact for compact CCS power.

### What Next

This is only a warning message. No action is required.

However, to avoid this warning message you can change the values in the curve_parameters index and run the command again.

### See Also

- LBDB-873
- LBDB-876
- LBDB-877

---

## LBDB-873

(error) Invalid size of data for I-t curve #%d in "values".

### Description

In compact ccs power, an I-t curve is always described with the pattern init_time, init_current, bc_id1, point_time1, point_current1, bc_id2, [point_time2, point_current2, bc_id3, ...], end_time, end_current

Feedback

So the size of data within a pair of quotation (e.g., data for an I-t curve) should be able to reprented as 8+3i (i>=0).

"I-t curve #?" means the order of this curve in "values". For example, "I-t curve #2" means the 2nd curve in "values".

The following example shows an instance where this message occurs:

```
compact_ccs_power(t1) {
    ...
    values("0.0, 1, 0.32, 1.08, 2, 0.87, -0.33, 3, 1.03, 0.0, 4, 1.50,
 0.67", /* size = 13 */\
           "0.0, 0.0, 1, 0.28, 0.93, 2, 0.45, 0.28", /* size = 8 */\
           "0.0, 0.0, 1, 0.36, 1.05, 2, 0.52, -0.49, 3, 0.88, 0.22",/*
 size = 11 */\
           "0.0, 0.0, 1, 0.33, 1.31, 2, 0.61, -0.83, 3, 0.96, 0.66" /*
 size = 11 */);
  }
```

In this example, size of the first line in "values" is 13, which is invalid. Sizes of other lines are valid.

The following is an example message:

```
Error: Line 388, Invalid size of data for I-t curve #1 in "values".
 (LBDB-873)
```

**What Next**

Change size and data in "values".

---

## LBDB-874

(error) Time values in I-t curve #%d in "values" should be monotonically increasing.

**Description**

In compact ccs power, an I-t curve is always described with the pattern init_time, init_current, bc_id1, point_time1, point_current1, bc_id2, [point_time2, point_current2, bc_id3, ...], end_time, end_current

Time values with corresponding parameter "init_time", "point_time*" and "end_time" in an I-t curve should be monotonically increasing.

"I-t curve #?" means the order of this curve in "values". For example, "I-t curve #2" means the 2nd curve in "values".

The following example shows an instance where this message occurs:

```
compact_ccs_power(t1) {
    ...
```

Feedback

```
    values("0.0, 0.0, 1, 1.32, 1.08, 2, 0.87, -0.33, 3, 1.03, 0.0, 4,
 1.50, 0.67", ...
  }
```

In this example, time-related data are {0.0, 1.32, 0.87, 1.03, 1.50}, which are not monotonically increasing.

The following is an example message:

```
Error: Line 388, Time values for I-t curve #1 in "values" should be
 monotonically increasing. (LBDB-874)
```

### What Next

Change values of time-related data in "values".

---

## LBDB-875

(warning) %s value %f in I-t curve #%d is too small and may be reset 0.

### Description

In a I-t curve, all time values with corresponding parameter "init_time", "point_time*" and "end_time" can't be too diverse. Otherwise, small values will be reset 0. Similarly, all current values with corresponding parameter "init_current", "point_current*" and "end_current" can'be too diverse, either. "I-t curve #?" means the order of this curve in "values". For example, "I-t curve #2" means the 2nd curve in "values".

The following example shows an instance where this message occurs:

```
compact_ccs_power(t1) {
    ...
    values("0.0, 0.0, 1, 1.32, 1.08, 2, 0.87, -0.33, 3, 1.03, 0.0, 4,
 1e+100, 0.67", ...
  }
```

In this example, "end_time" is 1e+100, which will make other non-zero time values 1.32, 0.87 and 1.03 reset 0.

The following is an example message:

```
Warning: Line 388, Time value 1.32 in I-t curve 1 is too small and may be
 reset 0. (LBDB-875)
```

### What Next

Change time or current values.

Feedback

## LBDB-876

(error) Variables are specified behind indexes in compact_lut_template.

### Description

compact_lut_template requires variables are specified before indexes. Other templates don't have this requirement.

The following example shows an instance where this message occurs:

```
compact_lut_template (t1) {
    index_1 ("0, 1");
    index_2 ("0, 1");
    index_3 ("init_time, init_current, bc_id1, point_time1,
 point_current1,
              bc_id2, [point_time2, point_current2, bc_id3, ...],
              end_time, end_current");
    variable_1 : input_net_transition;
    variable_2 : total_output_net_capacitance;
    variable_3 : curve_parameters;
  }
```

The following is an example message:

```
Error: Line 388, Variables are specified behind indexes in
 compact_lut_template. (LBDB-876)
```

### What Next

Put all variables in front of indexes.

## LBDB-877

(error) Variables in compact_lut_template are invalid.

### Description

Variables in compact_lut_template must satisfy, (1) The last variable is "curve_parameters"; (2) The variables before the last one are "input_net_transition" and "total_output_net_capacitance".

If this template is applied in compact ccs timing or va compact ccs timing, (3) There should be exactly one "input_net_transition" and one "total_output_net_capacitance".

If this template is applied in compact ccs power, (3) There should be at most one "input_net_transition" and at most two "total_output_net_capacitance".

Feedback

The following example shows an instance where this message occurs:

```
compact_lut_template(t1) {
    variable_1 : input_net_transition;
    variable_2 : curve_parameters;
    variable_3 : total_output_net_capacitance;
  }
```

To fix the problem, exchange the values between variable_2 and variable_3.

The following is an example message:

```
Error: Line 388, Variables in compact_lut_template are invalid.
 (LBDB-877)
```

**What Next**

Make necessary change in variable and index.

---

## LBDB-878

(error) One point current value(%f) is almost\n \tthe same as the adjacent point current value(%f).

**Description**

This error is issued when current value in one point is numerically identical to that in the adjacent point in Compact CCS Power data. The tolerance is set to 0.001%, that is: if fabs((I(first_point) - I(second_point))/I(second_point)) < 0.001%,

This could potentially cause larger errors in the calculation if data is truncated during reading, when these two values are too close.

The following example shows an instance where this message occurs:

```
compact_ccs_power(t1) {
    ...
    values("0.0, 0.0, 1, 0.28, 0.93, 2, 0.45, 0.929999", ...
  }
```

The following is an example message:

```
Error: Line 388, One point current value(0.93) is almost the same as the
 adjacent point current value(0.929999). (LBDB-878)
```

**What Next**

Check the library source file and correct the data, most likely, you may need to re-characterize the library cells for CCS power data.

Feedback

## LBDB-879

(error) Pin '%s' can't be a pll %s pin.

### Description

This message is issued because pin direction doesn't match the required direction of pll reference/feedback/output pin. If a pin is set "is_pll_reference_pin" or "is_pll_feedback_pin" true, it should be an input pin. If a pin is set "is_pll_output_pin" true, it should be an output pin.

The following example shows an instance where this message occurs:

```
cell (AND2) {
  is_pll_cell : true;
  pin (A1) {
    direction : input;
    is_pll_output_pin : true;
  }
  ...
}
```

The following is an example message:

```
Error: Pin 'A1' can't be a pll output pin. (LBDB-879)
```

### What Next

Specify the attribute "is_pll_reference_pin", "is_pll_feedback_pin" or "is_pll_output_pin" under another pin with consistent direction.

## LBDB-880

(error) More than one exclusive pll pin attributes are specified in pin '%s'.

### Description

In a cell, pll reference pin, pll feedback pin and pll output pin must be different pins. Therefore in a pin group, at most one of the three attributes "is_pll_reference_pin", "is_pll_feedback_pin" and "is_pll_output_pin" can be set true.

The following example shows an instance where this message occurs:

```
cell (AND2) {
  is_pll_cell : true;
  pin (ZN1) {
    direction : input;
    is_pll_reference_pin : true;
    is_pll_feedback_pin : true;
  }
```

Feedback

```
  ...
}
```

The following is an example message:

```
Error: More than one exclusive pll pin attributes are specified in pin
 'ZN1'. (LBDB-880)
```

**What Next**

Move all pll pin specifications except one to other pins.

---

## LBDB-881

(error) Pll cell '%s' has wrong pll %s pin defined.

**Description**

If a cell is specified as a pll cell, then the cell group should contain one and only one pll reference pin, one and only one pll feedback pin, and one or more pll output pins.

The following example shows an instance where this message occurs:

```
cell (AND2) {
  is_pll_cell : true;
  pin (ZN1) {
    direction : input;
    is_pll_reference_pin : true;
  }
}
```

The following is an example message:

```
Error: Pll cell 'AND2' has wrong pll feedback pin defined. (LBDB-881)
Error: Pll cell 'AND2' has wrong pll output pin defined. (LBDB-881)
```

**What Next**

Correct pins under the pll cell.

---

## LBDB-882

(warning) Pll pin tags in non-pll cell '%s' will be ignored.

**Description**

If a cell is not set "is_pll_cell" true, then pin level attributes "is_pll_reference_pin", "is_pll_feedback_pin" and "is_pll_output_pin" will be ignored.

Feedback

The following example shows an instance where this message occurs:

```
cell (AND2) {
  pin (ZN1) {
    direction : input;
    is_pll_reference_pin : true;
  }
}
```

The following is an example message:

```
Warning: Pll pin tags in non-pll cell 'AND2' will be ignored. (LBDB-882)
```

**What Next**

Remove pll pin tags from the non-pll cell.

---

## LBDB-883

(error) More than one physical only cell types are specified in design '%s'.

**Description**

In a pin group, at most one of the three attributes "is_decap_cell", "is_filler_cell" and "is_tap_cell" can be set true.

The following example shows an instance where this message occurs:

```
cell (S1CAP1) {
  is_filler_cell : true;
  is_decap_cell : true;
  ...
}
```

The following is an example message:

```
Error: More than one physical only cell types are specified in design
 'S1CAP1'. (LBDB-883)
```

**What Next**

Remove all physical only cell type specifications except one.

---

## LBDB-884

(error) Physical only cell '%s' can't contain I/O pins.

**Description**

If one of the three attributes "is_decap_cell", "is_filler_cell" and "is_tap_cell" is set true, this cell is a physical only cell. It should contain pg-pin only and has no signal pins in it.

Feedback

The following example shows an instance where this message occurs:

```
cell (S1CAP1) {
  is_decap_cell : true;
  pin(A1) {
    ...
  }
  ...
}
```

The following is an example message:

```
Error: Physical only cell 'S1CAP1' can't contain I/O pins. (LBDB-884)
```

**What Next**

Remove I/O pins in this cell.

---

## LBDB-885

(error) There is no bias pin named '%s' related to this\n \tsignal pin's related power or ground pin.

**Description**

This error message is reported when the association of power/ground pin and bias pin in signal is incorrect.

A signal pin's related bias pin must one of the related bias pins of this signal pin's related power/ground pin.

The following is an example message:

```
Error: Line 546, There is no bias pin named vpw related to this signal
 pin's
related power/ground pin.(LBDB-885)
```

**What Next**

Check if this related bias pin is related to this signal pin's power pin or ground pin.

---

## LBDB-885w

(warning) There is no bias pin named '%s' related to this\n \tsignal pin's related power or ground pin.

**Description**

This warning message is reported when the association of power/ground pin and bias pin in signal is incorrect.

Feedback

A signal pin's related bias pin must one of the related bias pins of this signal pin's related power/ground pin.

The following is an example message:

```
Waring: Line 546, There is no bias pin named vpw related to this signal
 pin's
related power/ground pin.(LBDB-885w)
```

**What Next**

Check if this related bias pin is related to this signal pin's power pin or ground pin.

---

## LBDB-886

(error) Bias pg_pin '%s' can not be properly associated to\n \tthe pg_pin.

**Description**

This error message is reported when bias pin %s can not be properly associated to a pg_pin.

Power type pg_ppin should be associated with a nwell type bias pg pin; ground type pg_pin should be associated with a pwell type bias pg pin.

The following example shows an instance where this message occurs:

```
pg_pin(PWR) {
    voltage_name : VDD;
    pg_type : primary_power;
    related_bias_pin : "vpw"
  }
  pg_pin(vpw) {
    voltage_name : VDDL;
    pg_type : pwell;
    bias_connection : routing_pin;
  }
```

The following is an example message:

```
Error: Line 546, Bias pg_pin vpw can not be properly associated to the
pg_pin.(LBDB-886)
```

**What Next**

Check the pg_type of the pg_pin to be assoicated and the pg_type of the bias pg_pin.

---

## LBDB-887

(warning) Only one (P or N)well bias pin exists.

Feedback

### Description

This message is indicates that only one bias pin with pg_type P or N well is exists. Generally, pwell and nwell bias pins should be exsit as a pair.

The following is an example message:

```
Warning: Line 546, Only one (P or N)well bias pin vpw exists.(LBDB-887)
```

### What Next

Delete the odd bias pin or add a new bias pin to be couple with the odd one.

---

## LBDB-888

(error) physical_connection is associated with an invalid pg_type.

### Description

This error message occurs when the physical_connection is associated with an invalid pg_type. The pg_type of a bias PG pin must be one of the predefined bias types: pwell, nwell, deeppwell, or deepnwell.

The following is an example message:

```
Error: Line 546,  physical_connection is associated with an invalid
 pg_type.
(LBDB-888)
```

### What Next

Make sure the pg_type of the bias PG pin is one of the valid types listed above.

---

## LBDB-889

(error) The pg_type of bias pg pin '%s' to be related is invalid.

### Description

This error message is reported when a bias pg pin is related to a pg_pin or a signal pin, its type_type is not one of the predefined bias types: pwell|nwell|deeppwell|deepnwell.

The following is an example message:

```
Error: Line 546, The pg_type of bias pg pin vpw to be related
is invalid.(LBDB-889)
```

### What Next

Check the pg_type of the bias pg pin.

Feedback

## LBDB-890

(warning) Isolation cell '%s' contains more than two input pins.

### Description

The following example shows an instance where this message occurs:

The following is an example message:

```
Warning: Line 202, Isolation cell 'acell' contains more than two input
 pins. (LBDB-890)
```

### What Next

Delete unwanted input pins to keep the input pin number is no less than 2.

## LBDB-891

(error) Retention cell '%s' has no retention pin.

### Description

This error message occurs because a retention cell (identified by the *retention_cell* or *power_gating_cell* attribute), must have at least one control pin (identified by the retention_pin or power_gating_pin function).

The following example shows the correct control pin in the retention cell:

```
cell(retention_dff) {
  ...
  retention_cell : "my_retention_dff" ;
  clock_gating_integrated_cell : "generic";

  pin(SAVE) {
    ...
    direction : input;
  retention_pin (save, "0") ;
  }

  pin(RESTORE) {
    ...
    direction : input;
  retention_pin (restore, "0") ;
  }
}
```

The following is an example message:

```
Error: Retention cell 'retention_dff' has no retention pin. (LBDB-891)
```

Feedback

### What Next

Add retention pins to the given cell and run the command again.

---

## LBDB-892

(error) Pin name '%s' is duplicated with pg pin name.

### Description

Names of Signal pins should be different with pg pin names and vice-versa.

The following is an example message:

```
Error: Pin name 'VSS' is duplicated with pg pin name.(LBDB-892)
```

### What Next

Change the duplicated signal pin name or pg pin name.

---

## LBDB-894

(warning) No scaling is done in '%s' because nominal process can not be found.

### Description

This message is issued because scaling feature is turned on and there is no nominal process defined in the library.

Nominal process is selected in this order of precedence:

(1) if "default_operating_conditions" is specified, nominal process is taken from attribute "process" in the default "operating_conditions" group; (2) nominal process is taken from library-level attribute "nom_process".

The following example shows an instance where this message occurs:

The following is an example message:

```
Warning: No scaling is done because nominal process can not be found.
 (LBDB-894)
```

### What Next

Define nominal process in library to continue scaling.

---

## LBDB-895

(warning) "values" in "%s" group won't be scaled because "%s" is not found.

Feedback

### Description

This message is issued because scaling feature is turned on and there is no corresponding k-factor defined for the specified attribute.

Here is the mapping table of supported k-factors and db attributes.

Nominal process is selected in this order of precedence:

k-factor name db_attribute_name k_process_cell_rise timing -> cell_rise -> values k_process_cell_fall timing -> cell_fall -> values k_process_rise_propagation timing -> rise_propagation -> values k_process_fall_propagation timing -> fall_propagation -> values k_process_rise_transition timing -> rise_transition -> values k_process_fall_transition timing -> fall_transition -> values k_process_setup_rise timing -> rise_constraint ->values (when "timing_type" is "setup_rising" or "setup_falling") k_process_setup_fall timing -> fall_constraint ->values (when "timing_type" is "setup_rising" or "setup_falling") k_process_hold_rise timing -> rise_constraint ->values (when "timing_type" is "hold_rising" or "hold_falling") k_process_hold_fall timing -> fall_constraint ->values (when "timing_type" is "hold_rising" or "hold_falling") k_process_recovery_rise timing -> rise_constraint ->values (when "timing_type" is "recovery_rising" or "recovery_falling") k_process_recovery_fall timing -> fall_constraint ->values (when "timing_type" is "recovery_rising" or "recovery_falling") k_process_removal_rise timing -> rise_constraint ->values (when "timing_type" is "removal_rising" or "removal_falling") k_process_removal_fall timing -> fall_constraint ->values (when "timing_type" is "removal_rising" or "removal_falling")

The following example shows an instance where this message occurs:

The following is an example message:

```
Warning: "values" in "cell_rise" group won't be scaled because
 "k_process_cell_rise" is not found. (LBDB-895)
```

### What Next

Add corresponding k-factor.

---

## LBDB-896

(error) Invalid %s found under section %s in map file.

### Description

This error message occurs if any the following are incorrect in the map file:

- `voltage_name`
- `voltage value`
- `power management attributes`

Feedback

A valid voltage_name must be a valid string that starts with either an uppercase character A-Z or a lowercase character a-z.

The voltage_name in the PG_TO_VOLTAGE_MAP section must be the name used in VOLTAGE_MAP. The name could be a rail name, and can be the same as the pg_pin name. You cannot use an invalid value such as "-" or "*" in the voltage_name field in PG_TO_VOLTAGE_MAP or VOLTAGE_MAP section.

Valid voltage values are non-negative floating point numbers, for example, 0.9, that are valid voltage values in the input library.

For valid power management attributes see the *Library Compiler Modeling Timing, Signal Integrity, and Power in Technology Libraries User Guide*. The following are some examples of power management attributes:

- valid switch_cell_type is coarse_grain and fine_grain
- valid level_shifter_type is HL, LH and HL_LH
- always_on pins are related to backup power
- a switch cell has VVDD+VDD or VVSS+VSS and VVDD has pg_function where VVDD is virtual VDD;
- a valid retention_cell must be a sequential cell with retention_cell and retention_pin attributes

The following example shows an instance where this message occurs: The following example shows a "-" in the voltage name field, resulting in the error message:

```
BEGIN PG_TO_VOLTAGE_MAP
cell            pg_pin          voltage_name    pg_type
ADDFHX1         VDD             -               power
ADDFHX1         VSS             VSS             primary_ground
END PG_TO_VOLTAGE_MAP
```

**What Next**

Correct the invalid values or voltage_name in the map file. For example, if there is a "-" in the voltage_name field, this message occurs. You must input a valid voltage_name; for example, VDD.

---

## LBDB-897

(error) The input library '%s' has pg_pin's defined as signal pins.

Feedback

### Description

When pg_pin's are as signal pins in the library, this message occurs. add_pg_pin_to_db abd add_pg_pin_to_lib have limitation. In the above case, the utility will stop generation of map template and pg_pin based .lib/db.

The following example shows an instance where this message occurs:

The following is an example message:

```
Error: The input library 'test.lib' has pg_pin's defined as signal pins.
 (LBDB-897)
```

### What Next

Check the contents of the library to see if there are pg_pin's or rail names or in/output_signal_level as signal pins. If so, remove these pin groups in your .lib and try to rerun the utility add_pg_pin_to_lib.

---

## LBDB-898

(warning) both 'stage_type : pull_up' and 'stage_type : pull_down' '%s' groups are specified with an identical or overlapping when condition.

### Description

This warning message occurs when both "stage_type : pull-up" and "stage_type : pull_down" ccs_first_stage/ccs_last_stage groups are specified with an identical or overlapping when condition.

If a first (last) CCB can physically make both rising and falling output transition under the same side-pin condition, then the stage_type of the CCB should be marked as both. The CCSN data should not be split into two stage_type pull_up/pull_down. Otherwise, the characterized result will not be correct.

The following example shows incorrect usage of the *stage_type* attribute, resulting in the warning message:

```
cell(bad) {
    ...
        Pin (A) {
        ...
        ccsn_first_stage() {
        stage_type : PULL_UP;
        when : "B";
                }

        ccsn_first_stage() {
        stage_type : PULL_DOWN;
        when : "B";
```

Feedback

```
                    }
            }

    }
```

In this example, pin 'A' has both "stage_type : pull-up" and "stage_type : pull_down" ccs_first_stage groups specified with an identical or overlapping when condition.

**What Next**

Check the library source file and correct the incorrect *stage_type* attribute.

---

## LBDB-899

(warning) there is no CCS Noise information.

**Description**

This warning message occurs when checking cells for CCS Noise information. The tool determined that there are some cells that have CCS Noise data, but no CCS Noise information is defined for the specified cell.

**What Next**

This is only a warning message. No action is required.

If CCS Noise information is not needed for the cell, you can ignore this message. Otherwise, you can add CCS Noise data for the cell and run the command again.

---

## LBDB-900

(error) The %f value of the power distribution table is not between 0 and 1.

**Description**

This error message occurs when the values of the power distribution tables are not between 0 and 1.

**What Next**

Modify the values of the power distribution tables to meet the requirements. Make sure the values are between 0 and 1.

**See Also**

- LBDB-901
- LBDB-902
- LBDB-903

Feedback


- LBDB-904
- LBDB-905
- LBDB-906
- LBDB-907


## LBDB-901

(error) Missing power distribution tables in %s.

**Description**

This error message occurs when a cell contains one leakage power group with power distribution tables, but not all leakage power groups have power distribution tables, or when a pin contains one internal power group with power distribution tables, but not all internal power groups have power distribution tables.

**What Next**

Make sure that all leakage power groups have power distribution tables in a cell, or that all internal power groups have power distribution tables in a pin. Add the missing distribution tables in the .lib file before reading it in.

## LBDB-902

(error) The sum of the values in all the distribution tables\n \tattached to %s table is not equal to 1.

**Description**

This error message occurs when the sum of all values in the distribution tables are not equal to 1. Each value in the distribution table must be between 0 and 1, and the sum of them should be 1.

**What Next**

Check all the values in all distribution tables attached to the specified leakage power or internal power to make sure that each value is between 0 and 1, and the sum of them is 1. Update the values as needed to meet the requirements.

## LBDB-903

(error) In an internal power group, the number of values for power and power\n \tdistribution tables are not the same.

Feedback

### Description

This error message occurs when the number of values in the paired power and power distribution tables are not the same in the internal power group. For example, there are 5 values in the rise power table, but there are 6 values in the paired rise power distribution table.

### What Next

Check the internal power group for the number of values between the power group and paired power distribution table, rise power group and paired rise power distribution table, or fall power group and paired fall power distribution table. Make sure the number is the same.

---

## LBDB-904

(error) The %s distribution tables are not paired with the %s tables in\n \tthe internal power group.

### Description

This error message occurs when power, rise, or fall power distribution tables are not paired with the related power group. For example, when the rise power distribution table is found in the internal power group but no rise power table is found, the tool issues this error message.

### What Next

Check the power distribution tables in the internal power group. When the power, rise power, or fall power distribution tables are found, there must be paired power, rise power, fall power groups. If this is not the case, remove the unpaired power distribution tables or add paired power, rise power, fall power groups before reading in the .lib file.

---

## LBDB-905

(error) Indices for the distribution tables are not matched with the\n \tpaired rise power, fall power, or power groups.

### Description

This error message occurs when the indices for the distribution tables are not matched with the paired rise power, fall power, or power groups.

### What Next

In the internal power group, check the power template and indices for distribution tables and its paired rise power, fall power, or power groups. If the power template is used, they should be the same for distribution tables and its paired rise power, fall_power, or power

Feedback

groups. Index_1 and index_2 for distribution tables and its paired rise power, fall power, or power groups should also be the same. If necessary, update the power template and indices to meet the requirements.

## LBDB-906

(error) Missing related_ground_pin in the power distribution table.

**Description**

This error message occurs when no related_ground_pin is specified in the power distribution table.

**What Next**

Check the power distribution table to determine if the related_ground_pin attribute is specified. If necessary, add the missing attribute before reading in the .lib file.

## LBDB-907

(error) Invalid ground_pin %s with incorrect pg_type.

**Description**

This error message occurs when the ground_pin is not a valid PG pin defined for the cell with pg_type of primary_ground, backup_ground, or internal_ground.

**What Next**

Check the pg_type of the related_ground_pin in the power distribution table to see if it is primary_ground, backup_ground, or internal_ground. If the related_ground_pin is not specified as one of the 3 types, it is not a valid ground pin in the power distribution table. Re-specify a valid ground pin for power distribution.

## LBDB-908

(error) The logic (%s) represented by %s 'when' attribute on line %u is incompatible with or is a subset of the logic (%s) represented by %s '%s' attribute on line %u.

**Description**

This error message indicates that the "when" attribute in the noise data group is a subset of the "when" attribute in the current timing group or incompatible with the function attribute of the pin. Note that a noise group is a ccsn_first_stage or ccsn_last_stage for the old format, or an input_ccb or output_ccb referenced in active_input_ccb, active_output_ccb, or propagating_ccb for the new format.

Feedback

The "when" condition in the noise group can only be equal or superset of the "when" condition in the current timing group. Otherwise, the characterized result is not correct.

The following example results in this error. The "when" condition "A" in the ccsn_first_stage is a subset of the "when" condition "A+B" of the timing group. When (A=0 & B=1), the timing group is active, but the ccsn_first_stage group is disabled. This behavior contradicts the fact the ccsn_first_stage is supposed to propagate noise.

```
cell(bad) {
      pin (Out) {
         function : "(A+B) C";
         timing () {
     related_pin : C;
            when : "A+B";
            ccsn_first_stage() {
               when : "A";
               ...
            }
            ...
         }
         ...
      }
      ...
   }
```

The following is the example message:

Error: Cell 'bad', pin 'A', the logic represented by the 'when' attribute (A) in 'ccsn_first_stage' group is a subset of the logic represented by the 'when' attribute (A+B) in the timing group.

Note: OUTPUT PINS NOT PERMITTED You may not use an output pin in the "when" attribute. You need to replace the pin with the expression stored in its "function" attribute.

**What Next**

Check the library source file, and correct the incorrect "when" attribute.

---

## LBDB-909

(warning) The cell should have at least 1 primary_ground pg_pin and at least 1 primary_power pg_pin. It's being marked as dont_use, dont_touch.

**Description**

This warning message occurs because in the new libraries based on pg_pin, all of the cells must have at least 1 primary_ground pg_pin and at least 1 primary_power pg_pin.

Feedback

### What Next

This is only a warning message. No action is required if users do not want to modify the cell pg pin modeling.

However, if users want to avoid this warning message, add the missing primary_ground pg_pin or primary_power pg_pin and run the command again.

---

## LBDB-910

(warning) The cell skips partial PG requirement checking. It's being marked as dont_use, dont_touch.

### Description

This warning message advises you that cells in the specified category skip PG pin requirement checking. The cells could have partial PG pins, meaning the cells are without a primary_power pg_pin or a primary_ground pg_pin. Currently, there is only one type ETM cell in the category.

### What Next

This is only a warning message. No action is required.

However, to avoid this warning message, add the missing primary_ground pg_pin or primary_power pg_pin and run the command again.

---

## LBDB-911

(error) The attribute '%s' violates the variation-aware consistence rule.

### Description

This warning message advises you that the variation data in the library violates the consistence rule. If any variation attribute contains N/2N data entries, all variation attributes in the same library must also contain N/2N entries at the same N/2N process points (va_values). The number of data entries must be either N or 2N for any variation attribute. This checking rule includes the following models:

- `VA Timing Constraint Model`
- `VA Compact CCS Timing Driver Model`
- `VA CCS Timing Receiver Model (arc-based and pin-based)`

Feedback

### What Next

Add the missing variation attribute or correct the process point (va_values). This message appears only once on the first violation entry. Check the other entries in the variation group for violations.

---

## LBDB-912

(error) The variation needs to be consistent across corresponding arcs.

### Description

This error message occurs when a variation inconsistency is found among the timing arcs. If the timing_based_variation group exists in a timing arc, all its corresponding timing arcs must also contain the timing_based_variation group. Otherwise, none of the arcs can contain the timing_based_variation group.

### What Next

Add the missing timing_based_variation in the corresponding timing arcs and run the command again.

---

## LBDB-913

(warning) is missing '%s' pg_pin, so it will become a black-box cell for multivoltage functional optimization flow. It is being marked dont_touch, dont_use.

### Description

This warning message occurs when a cell has 2 or more power or ground pins, but there is no backup power or backup ground pin.

The following conditions apply to retention and isolation cells:

- If cell has 2 or more power pins, there must be a backup power pin.
- If cell has 2 or more ground pins, there must be a backup ground pin.

If the conditions, are not met, the tool issues this warning message. The cell is black box for the multivoltage flow and is being marked as a dont_touch, dont_use (d,u). The black box cannot be recognized during functional optimization by the tools (Power Compiler and IC Compiler).

### What Next

This is only a warning message. No action is required.

However, if the result is not what you intended, add or correct the backup_ground pg_pin or backup_power pg_pin and run the command again.

Feedback

## LBDB-913e

(error) is missing '%s' pg_pin, so it will become a black box cell for multivoltage functional optimization flow. This cell should be marked as dont_touch, dont_use.

### Description

This error message occurs when a cell has 2 or more power or ground pins, but there is no backup power or backup ground pin.

The following conditions apply to retention and isolation cells:

- If the cell has 2 or more power pins, there must be a backup power pin.
- If the cell has 2 or more ground pins, there must be a backup ground pin.

If the above conditions are not met, the tool issues this error message. The cell is black box for the multivoltage flow and should be marked as dont_touch, dont_use (d,u). The black box cannot be recognized during functional optimization by the tools (Power Compiler and IC Compiler).

### What Next

If the result is not what you intended, add or correct the backup_ground pg_pin or backup_power pg_pin and run the command again. Alternatively, you can mark this cell as dont_touch, dont_use.

## LBDB-914

(warning) is missing related_power_pin to a '%s' pg_pin, so it will become a black box cell for multivoltage functional optimization flow. It is being marked as dont_touch, dont_use.

### Description

This warning message occurs when a cell has no related_power_pin, or when the related_power_pin relates to an incorrect pg_pin.

For retention cells, if save or save_restore pins exist, they should be related to the backup power or ground, depending on whether backup power or ground is available. All other signal pins should be related to the primary power or ground. For isolation cells, if a backup power pin exists, the output pin must be related to backup power through a related_power_pin.

The cell is black box for the multivoltage flow and is marked as dont_touch, dont_use (d,u). it cannot be recognized during functional optimization by the tools (Power Compiler and IC Compiler).

Feedback

The following example shows an instance where this message occurs:

```
cell(Isolation_Cell2){
  area : 1.0;
is_isolation_cell : true;
dont_touch : true;
dont_use : true;

pg_pin(VDDB) {
  voltage_name : VDDB;
  pg_type : backup_power;
}

pg_pin(VDD) {
  voltage_name : VDD;
  pg_type : primary_power;
}

pg_pin(VSS) {
  voltage_name : VSS;
  pg_type : primary_ground;
}

  leakage_power() {
    when : "!A";
    value : 1.5;
    related_pg_pin : VDD;
  }
  leakage_power() {
    value : 0.5;
    related_pg_pin : VDD;
  }

pin(A) {
  direction : input;
  related_power_pin : VDD;
  related_ground_pin : VSS;
  isolation_cell_data_pin : true;
  capacitance : 1.0;
  internal_power() {
    rise_power (scalar) { values ("0.0");}
    fall_power (scalar) { values ("0.0");}
  }
}

pin(EN) {
  direction : input;
  related_power_pin : VDD;
  related_ground_pin : VSS;
  isolation_cell_enable_pin : true;
  capacitance : 1.0;
  internal_power() {
    rise_power (scalar) { values ("0.0");}
```

Feedback

```
    fall_power (scalar) { values ("0.0");}
  }
}
pin(Y) {
  direction : output;
  /*VDD is a primary power pin. For isolation output pin,
    it should related to backup power, issue LBDB-914 warning message */
  related_power_pin : VDD;
  related_ground_pin : VSS;
  function : "A * EN";
  power_down_function : "!VDD + VSS";
  timing() {
    related_pin : "A EN";
        cell_rise(scalar) { values ("0.1");}
        rise_transition (scalar) { values ("0.1");}
        cell_fall(scalar) { values ("0.1");}
        fall_transition (scalar) { values ("0.1");}
  }
  internal_power() {
    related_pin : A;
    related_pg_pin : VDD;
        rise_power (scalar) { values ("0.0");}
        fall_power (scalar) { values ("0.0");}
  }

    }/* end pin group*/
  }
```

The following is an example message:

```
Warning: Line 192, Cell 'Isolation_Cell2', pin 'Y', Cell
 'Isolation_Cell2', pin 'Y', is missing related_power_pin to a
 'backup_power' pg_pin, so it will become a black box cell for
 multivoltage functional optimization flow. It is being marked as
 dont_touch, dont_use.  (LBDB-914)
```

### What Next

Check the library source file to add or correct related_power_pin attributes.

---

## LBDB-914e

(error) is missing related_power_pin to a '%s' pg_pin, so it will become a black box cell for multivoltage functional optimization flow. This cell should be marked as dont_touch, dont_use.

### Description

This error message occurs when a cell has no related_power_pin, or when the related_power_pin relates to an incorrect pg_pin.

Feedback

For retention cells, if save or save_restore pins exist, they should be related to the backup power or ground, depending on whether backup power or ground is available. All other signal pins should be related to the primary power or ground. For isolation cells, if a backup power pin exists, the output pin must be related to backup power through a related_power_pin.

The cell is black box for the multivoltage flow and should be marked as dont_touch, dont_use (d,u). The cell cannot be recognized during functional optimization by the tools (Power Compiler and IC Compiler).

**What Next**

If the result is not what you intended, add or correct the related_power_pin and run the command again. Alternatively, you can mark this cell as dont_touch, dont_use.

---

## LBDB-915

(warning) is missing related_ground_pin to a '%s' pg_pin, so it will become a black box cell for multivoltage functional optimization flow. It is being marked as dont_touch, dont_use.

**Description**

This warning message occurs when a cell has no related_ground_pin, or when the related_ground_pin is related to an incorrect pg_pin.

For retention cells, if save and restore pins exist, they should be related to the backup power or ground, depending on whether backup power or ground is available. All other signal pins should be related to the primary power or ground.

The cell is black box for the multivoltage flow and is marked as dont_touch, dont_use (d,u). It cannot be recognized during functional optimization by the tools (Power Compiler and IC Compiler).

The following example shows an instance where this message occurs:

```
cell(retention_dff) {
  area : 1.0;
  retention_cell : "my_retention_dff" ;

  pg_pin(VDDB) {
    voltage_name : VDDB;
    pg_type : backup_power;
  }
  pg_pin(VDD) {
    voltage_name : VDD;
    pg_type : primary_power;
  }
  pg_pin(VSS) {
    voltage_name : VSS;
    pg_type : primary_ground;
```

Feedback

```
  }
  pg_pin(VSS1) {
    voltage_name : VSS1;
    pg_type : backup_ground;
  }

  pin(D) {
    direction : input;
    capacitance : 1;
    nextstate_type : data ;
    related_power_pin : VDD;
    related_ground_pin : VSS;
      internal_power() {
        rise_power(scalar) { values ("1.0");}
        fall_power(scalar) { values ("1.0");}
      }
  }

  pin(CP) {
    direction : input;
    capacitance : 1;
    related_power_pin : VDD;
    related_ground_pin : VSS;
      internal_power() {
        rise_power(scalar) { values ("1.0");}
        fall_power(scalar) { values ("1.0");}
      }
     }

  pin(SAVE) {
    direction : input;
    capacitance : 1.0;
    nextstate_type : data ;
    related_power_pin : VDD;
    /* backup power exists, SAVE/SAVE_RESTORE retention pin must use
 backup power,
     issues LBDB-915 warning message*/
    related_ground_pin : VSS;
    retention_pin (save, "0") ;
      internal_power() {
        rise_power(scalar) { values ("1.0");}
        fall_power(scalar) { values ("1.0");}
      }
  }

  pin(RESTORE) {
    direction : input;
    capacitance : 1.0;
    nextstate_type : data ;
    related_power_pin : VDDB;
    /*RESTORE retention pin can use primary or backup ground as RGP, VDD
 is a primary power,
     issues LBDB-915 warning message */
```

Feedback

```
    related_ground_pin : VDD;
    retention_pin (restore, "0") ;
      internal_power() {
        rise_power(scalar) { values ("1.0");}
        fall_power(scalar) { values ("1.0");}
      }
  }

  ff("IQ", "IQN") {
    next_state : "D & (!SAVE & !RESTORE)" ;
    clocked_on : "CP" ;
  }

  pin(Q) {
    direction : output;
    function : "IQ";
    power_down_function : "!VDD + !VDDB + VSS";
    max_capacitance : 5.0 ;
    related_power_pin : VDD;
    related_ground_pin : VSS;
      timing() {
        related_pin : CP;
        timing_type : rising_edge;
        cell_rise(scalar) { values ("0.1");}
        rise_transition (scalar) { values ("0.1");}
        cell_fall(scalar) { values ("0.1");}
        fall_transition (scalar) { values ("0.1");}
      }
      internal_power() {
        related_pin : "D SAVE RESTORE CP";
        rise_power(scalar) { values ("1.0");}
        fall_power(scalar) { values ("1.0");}
      }
  }
  cell_leakage_power : 0.5 ;
  leakage_power() {
    when  : "!SAVE & !RESTORE" ;
    value : 0.7 ;
  }
  leakage_power() {
    when  : "SAVE" ;
    value : 0.1 ;
  }
 }
```

The following is an example message:

```
Warning: Line 103, Cell 'retention_dff', pin 'SAVE', is missing
 related_ground_pin to a 'backup_ground' pg_pin, so it will become a
 black box cell for multivoltage functional optimization flow. It is
 being marked as dont_touch, dont_use. (LBDB-915)
Warning: Line 116, Cell 'retention_dff', pin 'RESTORE', is missing
 related_ground_pin to a 'primary_ground' pg_pin, so it will become a
```

Feedback

```
black box cell for multivoltage functional optimization flow. It is
being marked as dont_touch, dont_use. (LBDB-915)
```

**What Next**

Check the library source file to add or correct related_ground_pin attributes.

---

## LBDB-915e

(error) is missing related_ground_pin to a '%s' pg_pin, so it will become a black box cell for multivoltage functional optimization flow. This cell should be marked as dont_touch, dont_use.

**Description**

This error message occurs when a cell has no related_ground_pin, or when the related_ground_pin is related to an incorrect pg_pin.

For retention cells, if save and restore pins exist, they should be related to the backup power or ground, depending on whether backup power or ground is available. All other signal pins should be related to the primary power or ground.

The cell is black box for the multivoltage flow and should be marked as dont_touch, dont_use (d,u). It cannot be recognized during functional optimization by the tools (Power Compiler and IC Compiler).

**What Next**

If the result is not what you intended, add or correct the related_ground_pin and run the command again. Alternatively, mark this cell as dont_touch, dont_use.

---

## LBDB-916

(warning) pin '%s' and pin '%s' have different related_power_pin settings, so it will become a black box cell for multivoltage functional optimization flow. It is being marked as dont_touch/dont_use.

**Description**

This warning message occurs when the 2 specified pins do not relate to the same related_power_pin.

For a level-shifter cell, the related power pin of enable pin should be either same as the related power pin of output pin or data pin, or a backup power pin which is not of std_cell_main_rail.

Feedback

The cell is black box for the multivoltage flow and is marked as dont_touch, dont_use (d,u). It cannot be recognized during functional optimization by the tools (Power Compiler and IC Compiler).

The following example shows an instance where this message occurs:

```
cell (LVLHLEHX2){
  cell_footprint : lvlhleh

  pg_pin(VDDI) {
 voltage_name : "VDDH";
 pg_type : primary_power;
  }
  pg_pin(VDDO) {
 voltage_name : "VDDL";
 pg_type : primary_power;
  }
  pg_pin(VSS) {
 voltage_name : "VSS";
 pg_type : primary_ground;
  }

  is_level_shifter : true ;

  area : 5.883500;
  pin(A) {
    direction : input;
    input_signal_level : VDDH;
    related_power_pin : VDDI;
    related_power_pin : VDDO; /* warning : LBDB-16 */
    related_ground_pin : VSS;
    capacitance : 0.001462;
    ...
  }
  pin(EN) {
    direction : input;
    input_signal_level : VDDH;
    related_power_pin : VDDI;
    related_ground_pin : VSS;
    level_shifter_enable_pin : true ;
    capacitance : 0.001699;
    internal_power();
    ...
  }
  pin(Y) {
    direction : output;
    output_signal_level : VDDL;
    related_power_pin : VDDO;
    related_ground_pin : VSS;
    capacitance : 0.0;
    function : "(A EN)";
    ...
  }
```

Feedback

```
}
```

The following is an example message:

```
Warning: Line 308, Cell 'LVLHLEHX2', pin 'Y' and pin 'EN' have different
 related_power_pin settings, so it will become a black box cell for
 multivoltage functional optimization flow. It is being marked as
 dont_touch/dont_use. (LBDB-916)
```

**What Next**

Check the library source file to correct related_power_pin attributes.

---

## LBDB-916e

(error) pin '%s' and pin '%s' have different related_power_pin settings, so it will become a black box cell for multivoltage functional optimization flow. This cell should be marked as dont_touch, dont_use.

**Description**

This error message occurs when the 2 specified pins do not relate to the same related_power_pin.

For a level-shifter cell, the enable pin and the output pin must be related to the same power pin.

The cell is black box for the multivoltage flow and should be marked as dont_touch, dont_use (d,u). It cannot be recognized during functional optimization by the tools (Power Compiler and IC Compiler).

**What Next**

If the result is not what you intended, make sure the specified pins have the same related_power_pin and run the command again. Alternatively, you can mark this cell as dont_touch, dont_use.

---

## LBDB-917

(warning) no pg_pin or signal pin in this pad cell.

**Description**

This message indicates this pad cell hasn't a signal pin or pg_pin.

The following example shows an instance where this message occurs:

```
cell(AN2) {
      pad_cell : true;
```

Feedback

```
    ...
}
```

In this case, the IO cell has no signal pin or pg_pin. To fix the problem, add one signal pin or pg_pin with attribute "is_pad : true".

Warning: Line 12, Cell 'AN2', no pg_pin or signal pin in this pad cell. (LBDB-917)

**What Next**

Add one signal pin or pg_pin with attribute "is_pad : true".

---

## LBDB-918

(error) The cell '%s' has a retention pin, but it is not a retention cell.

**Description**

This error message occurs because retention pin and retention cell attributes are considered in pairs. A retention cell is defined with the *retention_cell* or *power_gating_cell* attribute. The cell must have at least one retention control pin, which is defined with the *retention_pin* or *power_gating_pin* attribute.

This means that if one of the pins within a cell is a retention control pin, then the cell might be a retention cell. The retention pin and cell attributes are considered in pairs.

The following example shows an instance where this message occurs: In the following example, there is a *retention_pin* attribute, but there is no *retention_cell* attribute. Remove the comment from the *retention_cell* attribute to correct the problem.

```
cell(retention_dff) {
  ...
  /* retention_cell : "my_retention_dff"; */
  clock_gating_integrated_cell : "generic";

  pin(SAVE) {
    ...
    direction : input;
    retention_pin (save, "0");
  }

  pin(RESTORE) {
    ...
    direction : input;
    retention_pin (restore, "0");
  }
}
```

Feedback

The following is an example message: The above example results in the following error message:

```
Error : The cell 'retention_dff' has a retention pin, but it is not a
retention cell. (LBDB-918)
```

### What Next

Resolve the issue by either removing the *retention_pin* attribute or adding a *retention_cell* attribute.

### See Also

- LBDB-891

---

## LBDB-919

(warning) Retention cell '%s' is a zero pin design.

### Description

This warning message occurs because a retention cell (identified by the *retention_cell* or *power_gating_cell* attribute), might not have a retention pin.

The following example shows an instance where this message occurs: The following design does not have a retention pin, so it is a zero pin retention cell:

```
cell(retention_dff) {
  ...
  retention_cell : "my_retention_dff" ;

  pin(SAVE) {
    ...
    direction : input;
  }

  pin(RESTORE) {
    ...
    direction : input;
  }
}
```

The following is an example message:

```
Warning: Retention cell 'retention_dff' is a zero pin design (LBDB-919)
```

### What Next

Check that the design is a real zero pin retention cell.

Feedback

### See Also

- LBDB-891
- LBDB-918

---

## LBDB-920

(warning) The is_inverting attribute of ccsn_first_stage and ccsn_last_stage are not matched with the arc's timing_sense type.

### Description

This warning message occurs when the timing arc's CCS noise data does not match with the timing_sense of this arc. The tool expects the following matching between arc-based noise data and timing_sense:

- When both *ccsn_first_stage* and *ccsn_last_stage* have is_inverting = TRUE, the expected timing_sense is pos_unate.
- When both *ccsn_first_stage* and *ccsn_last_stage* have is_inverting = FALSE, the expected timing_sense is pos_unate.
- When either *ccsn_first_stage* or *ccsn_last_stage* has is_inverting = TRUE, and the other has is_inverting = FALSE, the expected timing_sense is neg_unate.

  The following example shows an instance where this message occurs: The incorrect input below causes this warning message:

```
pin(CO) {
  timing() {
    timing_sense : neg_unate ;
    ccsn_first_stage() {
      is_inverting = true;
      ...
    } /* qualified model 1 */
    ccsn_first_stage() {
      is_inverting = true;
      ...
    } /* qualified model 1 */
  }
  ...
}
```

In this example, the CO pin has a two stage arc, and both the first and last stages are inverting, so a positive arc is expected. To fix the problem, change timing_sense to positive.

Feedback

The following is an example message: The following is an example of the warning message:

```
Warning: The is_inverting attribute of ccsn_first_stage and
ccsn_last_stage are not matched with the arc's timing_sense type.
```

**What Next**

This is only a warning message. No action is required.

However, if the result is not what you intended, correct the noise modeling. If the noise modeling comes from the noise characterization tool, check with the noise characterization tool to determine why this type of data was generated.

---

## LBDB-921

(warning) %s is missing in timing arc in pin %s.

**Description**

This warning message occurs when a timing arc has *receiver_capacitance_rise* or *receiver_capacitance_fall*, but not both.

The following is an example message: The following is an example of the warning message:

```
Warning: Line 206, receiver_capacitance_fall is missing in timing arc
in pin A.(LBDB-921)
```

**What Next**

This is only a warning message. No action is required.

---

## LBDB-922

(warning) In '%s' group, ccsn_first_stage at line %u and\n \tccsn_last_stage at line %u have different values for related_ccb_node.

**Description**

related_ccb_node of the ccsn_first_stage and ccsn_last_stage in the same timing group should be identical.

The following example shows an instance where this message occurs:

```
library(test) {
  cell (flop) {
    ...
    pin (CP) {
      ...
```

Feedback

```
    }
    pin (Q) {
      ...
      timing () {
        ...
        related_pin :CP;
        ccsn_first_stage() {
          related_ccb_node : "net3:3";
          ...
        }
        ccsn_last_stage() {
          related_ccb_node : "net7:5";
          ...
        }
      }
    }
    ...
  }
}
```

The following is an example message:

```
Warning: Line 32, In 'timing' group, ccsn_first_stage at line 35 and
        ccsn_last_stage at line 39 have different values for
 related_ccb_node. (LBDB-922)
```

### What Next

Use same values for the related_ccb_node of the ccsn_first_stage and ccsn_last_stage in the same timing group.

---

## LBDB-923

(error) %s but not %s is needed in this timing arc %s.

### Description

This error message occurs when a mismatched *receiver_capacitance_rise* exists in a timing arc. For example, when timing_type = combinational_rise and timing_sense = positive_unate, then *receiver_capacitance_rise* is needed. The error occurs when the timing arc has only *receiver_capacitance_fall*.

The following is an example message: The following is an example of the error message:

```
Error: Line 206, receiver_capacitance_rise but not
receiver_capacitance_fall is needed in timing arc.(LBDB-921)
```

Feedback

### What Next

Correct the timing modeling. If the timing modeling comes from the timing characterization tool, check with the timing characterization tool to determine why this type of data was generated.

---

## LBDB-924

(warning) There are more than %d %s in %s. \n

### Description

This message indicates that the object number is not as expected.

Only one pg_pin can be specified in pg_function Boolean expression of a non-macro switch cell, otherwise the cell will be marked dont use.

The following example shows an instance where this message occurs:

```
cell(lbdb136) {
      area : 9;
      pg_pin(VDD) {
        pg_type : primary_power;
        voltage_name :VDD;
      }
      pg_pin(VSS) {
        pg_type : primary_ground;
        voltage_name :VSS;
      }
      pg_pin(VVDD) {
        pg_type : internal_power;
        voltage_name :VVDD;
        switch_function : "sleep";
        pg_function : "!VDD+VSS";
      }
    }
```

In this case, the pg_function is composed by two pg_pin, which suppose to be 1.

The following is an example message:

```
Warning: Line 1848, There are more than 1 pg_pin in pg_function.
 (LBDB-924)
```

### What Next

Refer to the "Library Compiler User Guide" to determine the object number. Make the correction.

Feedback

## LBDB-925

(warning) The %s of %s%s%s should be %s. \n

### Description

This message indicates that the value or attribute is not as expected.

For non-macro switch cell, the pg_pin specified in pg_function of a power pg_pin must be primary/backup power, the pg_pin specified in pg_function of a ground pg_pin must be primary/backup ground.

For macro switch cell, the pg_pin specified in pg_function of a power pg_pin must be primary/internal/backup power, the pg_pin specified in pg_function of a ground pg_pin must be primary/internal/backup ground.

The following example shows an instance where this message occurs:

```
cell(lbdb136) {
      area : 9;
      pg_pin(VDD) {
        pg_type : primary_power;
        voltage_name :VDD;
      }
      pg_pin(VSS) {
        pg_type : primary_ground;
        voltage_name :VSS;
      }
      pg_pin(VVDD) {
        pg_type : internal_power;
        voltage_name :VVDD;
        switch_function : "sleep";
        pg_function : "!VDD+VSS";
      }
    }
```

In this non-macro case, the pg_function is specified on pg_pin VVDD, whose pg_type is internal_power. So the pg_function should be composed by pg_pin whose pg_type is primary_power or backup_power.

The following is an example message:

```
Warning: Line 848, The pg_type of pin VDD in pg_function should be
 primary_power or backup_power. (LBDB-925)
```

### What Next

Refer to the "Library Compiler User Guide" to determine the value or attribute. Make the correction.

Feedback

## LBDB-926

(warning) The %s '%s' is missing the attribute '%s'.

### Description

You receive this message when the required attribute is missing.

The following example shows an instance where this message occurs:

```
cell(lbdb926) {
      area : 9;
      pg_pin(VDD) {
        pg_type : primary_power;
        voltage_name :VDD;
      }
      pg_pin(VSS) {
        pg_type : primary_ground;
        voltage_name :VSS;
      }
      pg_pin(VVDD) {
        pg_type : internal_power;
        voltage_name :VVDD;
      }
    }
```

In this case, there is no pg_function when pg_type is internal_power

The following is an example message:

```
Warning: Line 848, The pg_pin 'VVDD' is missing the attribute
 'pg_function'. (LBDB-926)
```

This is another example:

```
cell(lbdb926) {
      area : 9;
      pg_pin(VDD) {
        pg_type : primary_power;
        voltage_name :VDD;
      }
      pg_pin(VSS) {
        pg_type : primary_ground;
        voltage_name :VSS;
      }
      pg_pin(VVDD) {
        pg_type : internal_power;
        voltage_name :VVDD;
        switch_function : "sleep";
      }
    }
```

Feedback

In this case, there is only switch_function in pg_pin VDD, which should coexist with pg_function.

The following is an example message:

```
Warning: Line 848, The pg_pin 'VVDD' is missing the attribute
 'pg_function'. (LBDB-926)
```

**What Next**

Specifiy the missing attribute.

---

## LBDB-927

(warning) It is invalid to specify the '%s' %s \n \ton the '%s' %s%s.

**Description**

This message indicates that you specified an invalid attribute/group on a cell/pin.

The following example shows an instance where this message occurs:

```
cell (p) {
    clocked_cell : rising_edge;
    pin(clk) {
        clock : true;
    }

    pin(out) {
        function : "clk";
    }
  }
```

To solve this issue, we need to add "combinational" timing arcs between "clk" pin and "out" pin.

The following is an example message:

```
Warning: Line 67, It is invalid to specify the 'clocked_cell' attribute
        on the 'p' cell. (LBDB-927)
```

**What Next**

Check the library source file, and correct the problem. For example, in order to specify the "clocked_cell" attribute, the cell needs to satisfy the following 2 conditions: - has only 1 clock pin whose "clock" attribute is set to "true". - has at least 1 timing arc between the clock pin and an output pin.

Feedback

## LBDB-928

(warning) The pin '%s' specified in the '%s' is neither\n \tan %s pin nor an inout pin.

### Description

This message indicates that direction of the pin you specified in a *related_inputs*, *related_outputs*, *related_output*, *index_output* or *gate_leakage* is wrong.

The following example shows an instance where this message occurs:

```
cell(lbdb730) {
  area : 2;
  pin(A) {
    direction : input;
    capacitance : 1;
  }
  pin(B) {
    direction : input;
    capacitance : 1;
  }
  pin(Z) {
    direction : output;
    function : "A B";
    timing() {
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      slope_rise : 0.0;
      slope_fall : 0.0;
      related_pin : "A B";
    }
  }
  leakage_current() {
     when : "A & !B & Z";
     pg_current(V1) {
      ...
     }
     ...
     gate_leakage(Z) {  /* must be input or inout pin */
       input_high_value : 2.1;
       input_low_value : -1.7;
     }
  }
  dynamic_current() {
    when : "A";
    related_inputs : "Z";
    related_outputs : "B"
    typical_capacitances(0.3);
    switching_group() {
      input_switching_condition(fall);
```

Feedback

```
        output_switching_condition(rise);
        pg_current(<pg_name>) {
          vector(<lu_template_name>) {
            reference_time : 93.2;
            index_output : "A";
            index_1("5.1");
            index_2("0.3");
            index_3("8.2 9.4 9.8");
            values("1.78 12.4 110.1");
          }
        }
       ...
      }
    }
    intrinsic_parasitic() {
      /* default state */
      intrinsic_resistance(<pg_name>) {
         related_output : "B";
         value : 9.0;
      }
      intrinsic_capacitance(<pg_name>) {
         value : 8.2;
      }

    }
    ...
}
```

In this case, all related_output, related_outputs, related_inputs, index_output and gate_leakage have a wrong pin name. To fix the problems, change the name from 'Z' to either 'A' or 'B' in the related_inputs, change the name from 'B' to 'Z' in the related_outputs, change the name from 'A' to 'Z' in the index_output, change the name from 'B' to 'Z' in the related_output and change the name from 'Z' to either 'A' or 'B' in the gate_leakage.

The following is an example message:

```
Warning: Line 272, The pin 'Z' specified in the 'related_inputs' is
 neither
        an input pin nor an inout pin. (LBDB-928)
Warning: Line 471, The pin 'B' specified in the 'related_outputs' is
 neither
        an output pin nor an inout pin. (LBDB-928)
Warning: Line 526, The pin 'A' specified in the 'index_output' is neither
        an output pin nor an inout pin. (LBDB-928)
Warning: Line 890, The pin 'B' specified in the 'related_output' is
 neither
        an output pin nor an inout pin. (LBDB-928)
Warning: Line 749, The pin 'Z' specified in the 'gate_leakage' is neither
        an input pin nor an inout pin. (LBDB-928)
```

Feedback

### What Next

Check the library source file, and make the necessary correction.

---

## LBDB-929

(error) The %s cell is invalid for missing %s%s. It's being marked as dont_use.\n

### Description

This message indicates that there are some attribute/group missing for the cell.

The following example shows an instance where this message occurs:

```
cell(lbdb136) {
      switch_cell_type : coarse_grain;
      area : 9;
      pg_pin(VDD) {
        pg_type : primary_power;
        voltage_name :VDD;
      }
      pg_pin(VSS) {
        pg_type : primary_ground;
        voltage_name :VSS;
      }
      pg_pin(VVDD) {
        pg_type : internal_power;
        voltage_name :VVDD;
        pg_function : "VDD";
      }
    }
```

In this case, the switch cell has no internal_power|ground pg_pin with both "switch_function" and "pg_function".

The following is an example message:

```
Error: Line 820, The switch cell is invalid for missing
 internal_power|ground
pg_pin with both switch_function and pg_function. It's being marked as
 dont_use. (LBDB-929)
```

### What Next

Refer to the "Library Compiler User Guide" to sepcify the attribute/group. Make the correction.

---

## LBDB-930

(warning) The %s cell is invalid for missing %s%s. \n

Feedback

### Description

This message indicates that there are some attribute/group missing for the cell.

The following example shows an instance where this message occurs:

```
cell(lbdb136) {
      switch_cell_type : coarse_grain;
      area : 9;
      pg_pin(VDD) {
        pg_type : primary_power;
        voltage_name :VDD;
      }
      pg_pin(VSS) {
        pg_type : primary_ground;
        voltage_name :VSS;
      }
      pg_pin(VVDD) {
        pg_type : internal_power;
        voltage_name :VVDD;
        pg_function : "VDD";
      }
    }
```

In this case, the switch cell has no internal_power|ground pg_pin with both "switch_function" and "pg_function".

The following is an example message:

```
Warning: Line 820, The switch cell is invalid for missing
 internal_power|ground
pg_pin with both switch_function and pg_function. (LBDB-930)
```

### What Next

Refer to the "Library Compiler User Guide" to sepcify the attribute/group in source lib file. Make the correction and regenerate a new db file.

---

## LBDB-931

(error) The %s number is less than %d in %s.

### Description

This message indicates that the object number is not as expected.

The following example shows an instance where this message occurs:

```
cell(lbdb136) {
      area : 9;
      is_level_shifter : TRUE;
      pin_opposite("A", "BB");
```

Feedback

```
      contention_condition : "A";
      pin(A) {
        direction : input;
        level_shifter_data_pin : TRUE;
      }
      pin(BB) {
        direction : input;
        level_shifter_data_pin : TRUE;
      }
      pin(Y) {
        direction : output;
        function : "A+BB"
      }
    }
```

In this case, the contention_condition is composed by one pin, which suppose to be at least 2.

The following is an example message:

```
Error: Line 1848, The pin number is less than 2 in contention_condition
 for a differential level shifter. (LBDB-931)
```

**What Next**

Refer to the "Library Compiler User Guide" to determine the object number. Make the correction.

---

## LBDB-932

(warning) The %s%s%s%s is missing%s.

**Description**

This message indicates that there is object/group missing.

The following example shows an instance where this message occurs:

```
cell(lbdb136) {
      area : 9;
      is_level_shifter : TRUE;
      pin_opposite("A", "BB");
      contention_condition : "!(A^BB)";
      pin(A) {
        direction : input;
        level_shifter_data_pin : TRUE;
      }
      pin(BB) {
        direction : input;
        level_shifter_data_pin : TRUE;
      }
      pin(Y) {
```

Feedback

```
        direction : output;
        function : "A+BB"
      }
    }
```

In this case, there should be the timing constraints group between A and BB.

The following is an example message:

```
Warning: Line 1848, The nonsequential timing constraints group between
 pin A and pin BB is missing for a differential level shifter. (LBDB-932)
```

### What Next

Refer to the "Library Compiler User Guide" to determine if the object/group is required. Make the correction.

---

## LBDB-933

(error) The value of '%s' is %g, and is not in the range of %s which is between %g and %g.

### Description

This message indicates that the attribute value you have specified is not in the required range. For example: For input_signal_level, Library Compiler will check if it is in pin level "input_voltage_range". If there is no pin level "input_voltage_range", check the range of cell level "input_voltage_range"; . For output_signal_level, Library Compiler will check if it is in pin level "output_voltage_range". If there is no pin level "output_voltage_range", check the range of cell level "output_voltage_range"; .

The following example shows an instance where this message occurs:

```
cell (ls) {
    input_voltage_range(0.5, 1.2);
    pin(A) {
      input_signal_level : 1.3;
    }

    pin(Y) {
      function : "A";
    }
  }
```

To correct the error, change the input_signal_level to 1.2, which is in the range of (0.5 ~ 1.2)

Feedback

The following is an example message:

```
Error: Line 77, Cell 'ls', pin 'Y', The value of 'input_signal_level' is
 1.3, and is not in the range of input_voltage_range which is between 0.5
 and 1.2. (LBDB-933)
```

### What Next

Check the library source file, and ensure the attribute value is consistent with corresponding range.

---

## LBDB-934

(warning) Overwrite '%s' defined in 'operating_condition' by '%s' value %f.

### Description

In existing .db files, nom PVT can be copied over to all PVT (process, voltage and temperature) defined in default_operating_conditions.

The warning message indicates that whatever PVT defined in default_operating_conditions have been overwritten by values of nom PVT.

The following example shows an instance where this message occurs:

```
library(libdb934) {
 ...
  nom_process : 1.3
  nom_temperature : 20.0;
  nom_voltage : 4.0;

  operating_conditions ( TYPICAL ) {
    process : 1.0 ;
    temperature : 22.0 ;
    voltage : 3.0 ;
    tree_type : balanced_tree ;
  }
  default_operating_conditions :  TYPICAL;
   ...
  voltage_map(VDD1, 4.0); /* new pg pin syntax */
  voltage_map(VDD2, 4.5);
   ...
  cell (and2) {
   ...
  pg_pin(V1) { /* new pg pin syntax */
     voltage_name : VDD1;
     pg_type : primary_power;
  }
  pg_pin(V2) {
     voltage_name : VDD2;
     pg_type : backup_power;
```

Feedback

```
  }
  ...
```

In this case, process defined in default operating_conditions group will be overwritten by "nom_process" value 1.3. Temperature defined in default operating_conditions group will be overwritten by "nom_temperature" value 20.0. Voltage defined in default operating_conditions group will be overwritten by "nom_voltage" value 4.0.

The following are example messages:

```
Warning: Line 100, Overwrite 'process' defined in 'operating_conditions'
 by default 'nom_process' value 1.3. (LBDB-934)
Warning: Line 100, Overwrite 'voltage' defined in 'operating_conditions'
 by default 'nom_voltage' value 20.0. (LBDB-934)
Warning: Line 100, Overwrite 'temperature' defined in
 'operating_conditions' by default 'nom_temperature' value 4.0.
 (LBDB-934)
```

---

## LBDB-935

(error) Attribute %s is not allowed in this ccs noise stage.

### Description

input_signal_level and output_signal_level are used to provide additional information on the operation of a CCS noise stage, but not to override voltages that can be determined from the pin, or related pin.

The following usage is disallowed:

```
input_signal_level in a pin-based ccsn_first_stage group
output_signal_level in a pin-based ccsn_last_stage group
input_signal_level or output_signal_level in an arc-based
 ccsn_first_stage
or ccsn_last_stage group

The following example shows an instance where this message occurs:
library(lbdb935) {
  ...
  voltage_map(COREVDD1,1.080000);
  voltage_map(COREGND1,0.000000);
  voltage_map(COREVDD2,0.900000);
  voltage_map(COREVDD3,0.990000);
  ...
  cell (INV_PIN_BASED) {
    ...
    pg_pin(VDD) {
      voltage_name : VDD1;
      ...
    }
    pg_pin(VDDL) {
```

Feedback

```
      voltage_name : VDD2;
      ...
    }
    pg_pin(VSS) {
      voltage_name : GND1;
      ...
    }
    pin (I) {
      direction : input;
      related_power_pin : VDDL;
      ccsn_first_stage {
        input_signal_level : COREVDD1;
        ...
      }
      ...
    }
    pin (Z) {
      direction : output;
      related_power_pin : VDD2;
      ccsn_last_stage {
        output_signal_level : COREVDD1;
        ...
      }
      ...
    }

  ...
```

The input signal level of the ccsn_first_stage group is determined by related_power_pin of pin I. The output signal level of the ccsn_last_stage group is determined by the related_pin of pin Z.

The following are example messages:

```
Error: Line 4303, Cell 'INV_PIN_BASED', pin 'I', Attribute
 input_signal_level not allowed in this ccs noise stage (LBDB-935)

Error: Line 4455, Cell 'INV_PIN_BASED', pin 'Z', Attribute
 output_signal_level not allowed in this ccs noise stage (LBDB-935)
```

---

## LBDB-936

(error) %s is an invalid signal value.

### Description

The signal level referenced by input_signal_level or output_signal_level of a CCS or CCB noise stage must already be defined in a pg_pin. Also the signal level must has a positive and non-zero voltage.

Feedback

The following example shows an instance where this message occurs:

```
library(lbdb936) {
  ...
  voltage_map(COREVDD1,1.080000);
  voltage_map(COREGND1,0.000000);
  voltage_map(COREVDD2,0.900000);
  voltage_map(COREVDD3,0.990000);
  ...
  cell (INV_BAD_SIGNAL_LEVEL) {
    ...
    pg_pin (VDD) {
      voltage_name : COREVDD1;
      ...
    }
    pg_pin (VDDL) {
      voltage_name : COREVDD2;
      ...
    }
    pg_pin (VSS) {
      voltage_name : COREGND1;
      ...
    }
    pin (I) {
      direction : input;
      related_ground_pin : VSS;
      related_power_pin : VDDL;
      ...
      ccsn_first_stage () {
        ...
        output_signal_level : COREVDD3;
...
    }
    ...
  }
  ...
```

The output_signal_level COREVDD3 is not declared in any pg_pins of cell INV_BAD_SIGNAL_LEVEL

The following is an example message:

```
Error: Line 187, Cell 'INV_BAD_SIGNAL_LEVEL', pin 'I', COREVDD3 is an
 invalid signal level. (LBDB-936)
```

**What Next**

Correct the problem.

---

## LBDB-937

(error) Arc-based level-shifting CCS noise stages have inconsistent voltages.

Feedback

### Description

This error message concerns arc-based, two-stage noise models ( ccsn_first_stage/ ccsn_last_stage pair inside a timing group) that are level shifting. The voltage value given in the output_signal_level attribute of the first stage does not agree with that from the input_signal_level attribute of the last stage.

The following example shows an instance where this message occurs:

```
library(lbdb937) {
  ...
  voltage_map(VDD1,1.080000);
  voltage_map(GND1,0.000000);
  voltage_map(VDD2,0.900000);
  ...
  cell (INV_ARC_LS) {
    ...
    pg_pin (VDD) {
      voltage_name : VDD1;
      ...
    }
    pg_pin (VDDL) {
      voltage_name : VDD2;
      ...
    }
    pg_pin (VSS) {
      voltage_name : GND1;
      ...
    }
    pin (I) {
      direction : input";
      related_power_pin : VDDL;
      ...
   }
   pin (Z) {
     direction : output;
     related_power_pin : VDD;
     timing () {
       related_pin : I;
       ccsn_first_stage {
         output_signal_level: VDD2;
       }
       ccsn_last_stage {
         input_signal_level: VDD1;
       }
     ...
     }
  ...
```

The first stage has an output level of 0.9V, but the last stage has an input level of 1.08V. Correct the problem.

Feedback

The following is an example message:

```
Error: Line 1255, Cell 'INV_ARC_LS', pin 'Z', arc-based level-shifting
 CCS noise stages have inconsistent voltages" (LBDB-937)
```

## LBDB-938

(error) The related_pins in this timing group have incompatible input signal levels.

### Description

A timing group may be shared by multiple input pins by putting the names of the input pins in the related_pin attribute. For CCS noise model sharing to be valid, the inputs must have the same signal levels.

The following example shows an instance where this message occurs:

```
library(lbdb938) {
  ...
  voltage_map(VDD1,1.080000);
  voltage_map(GND1,0.000000);
  voltage_map(VDD2,0.900000);
  ...
  cell (BAD_RELATED_PINS) {
    ...
    pg_pin (VDD) {
      voltage_name : VDD1;
      ...
    }
    pg_pin (VDDL) {
      voltage_name : VDD2;
      ...
    }
    pg_pin (VSS) {
      voltage_name : GND1;
      ...
    }
    pin (E) {
      direction : "input";
      level_shifter_enable_pin : true;
      related_ground_pin : "VSS";
      related_power_pin : "VDD";
      ...
    }
    pin (I) {
      direction : "input";
      level_shifter_data_pin : true;
      related_ground_pin : "VSS";
      related_power_pin : "VDDL";
      ...
    }
    pin (Z) {
```

Feedback

```
      direction : output;
      related_power_pin : VDDL;
      timing () {
        function : "I' E'";
        related_pin : "I E";  /* two pins */
        ccsn_first_stage () { /* this is a shared arc-based noise model
 */
        ...
      }
     ...
    }
  ...
```

The ccsn_first_stage is shared by input pins E and I. However, pins E and I have different signal levels, which is illegal.

The following is an example message:

```
Error: Line 198, Cell 'BAD_RELATED_PINS', pin 'Z', The related_pins in
 this timing group have incompatible input signal levels" (LBDB-938)
```

---

## LBDB-939

(warning) This CCS noise stage group is missing %s attribute

### Description

A level shifter cell is likely to have a CCS noise stage group that is level shifting. This particular group is assumed to be single voltage. If it is actually a low-to-high level shifting stage, you will also most likely encounter errors LBDB-706 and LBDB-953 as well. See section EXAMPLE MESSAGE at the end.

The following example shows a liberty modeling snippet where this message occurs:

```
voltage_map("VDD1",1.080000);
  voltage_map("VSS", 0.000000);
  voltage_map("VDD2",0.900000);

  cell (low_to_high_LS) {
    is_level_shifter : true;
    level_shifter_type : "LH";
    input_voltage_range(0.900000,1.320000);
    output_voltage_range(0.900000,1.320000);
    ...
    pg_pin (VDD1) {
      pg_type : "primary_power";
      voltage_name : "VDD1";
      std_cell_main_rail : true;
    }
    pg_pin (VDD2) {
      pg_type : "primary_power";
      voltage_name : "VDD2";
```

Feedback

```
    }
    pg_pin (VSS) {
      pg_type : "primary_ground";
      voltage_name : "VSS";
    }
    pin (I) {
      direction : "input";
      level_shifter_data_pin : true;
      related_ground_pin : "VSS";
      related_power_pin : "VDD2";
      ccsn_first_stage () {
        /* output_signal_level : "VDD1"; */
        ...
      }
      ...
    }
    pin (Z) {
      direction : "output";
      function : "I'";
      related_ground_pin : "VSS";
      related_power_pin : "VDD1";
      ...
      ccsn_last_stage () {
        /* input_signal_level : "VDD2"; */
        ...
      }
      ...
  }
```

The ccsn_first_stage group is missing the output_signal_level attribute. The ccsn_last_stage group is missing the input_signal_level attribute

The following is an example message:

```
Warning: Line 180, Cell 'low_to_high_LS', pin 'I', This CCS noise stage
 group is missing output_signal_level (LBDB-939)
Error: Line 180, Cell 'low_to_high_LS', pin 'I', output_voltage_fall
 analysis failed during CCS Noise compilation (LBDB-706)
Warning: Line 332, Cell 'low_to_high_LS', pin 'Z', This CCS noise stage
 group is missing input_signal_level (LBDB-939)
```

**What Next**

Add the missing input_signal_level or output_signal_level with a correct voltage name.

---

## LBDB-940

(error) The level shifter cell contains invalid related power/ground pin configuration for input pin '%s' and output pin '%s'.

Feedback

### Description

The tool issues this error message if the related power and ground pin configuration for the specified input and ouput pins is modeled incorrectly for a level-shifter cell.

Liberty supports the following types of power and ground pin configurations for the input and output pins of a level shifter:

- Power level shifters

  The related power pins are different and the related ground pins are the same.

- Ground level shifters

  The related ground pins are the same and the related power pins are different.

- Power and ground level shifters

  Both the related power pins and the related ground pins are different.

This error occurs when both the related power pins and the related ground pins are the same for the input and output of a level-shifter cell.

The following example shows an instance where this message occurs: The following example shows a level shifter with incorrect modeling and the resulting error message in the PG syntax:

```
library (test) {
  ...
  voltage_map( VSS, 0.0);
  voltage_map( VDDH, 0.9);
  voltage_map( VDDL, 0.7);
  ...
  cell(LS) {
  ...
    pg_pin(GND) {
      voltage_name : VSS;
      pg_type : primary_ground;
    }
    pg_pin(VDD2) {
      voltage_name : VDDH;
      pg_type : primary_power;
    }
    pg_pin(VDD1) {
      voltage_name : VDDL;
      pg_type : primary_power;
    }

    is_level_shifter : true;
    ...
    pin(in) {
      ...
      related_power_pin : VDD1;
```

Feedback

```
      related_ground_pin : GND;
    }
    pin(out) {
      ...
      related_power_pin: VDD1;
      related_ground_pin : GND;
    }
  }
  ...
}

Error: Line 191, Cell 'LS', The level shifter cell contains invalid
related power/ground pin configuration for input pin 'in' and output
pin 'out'. (LBDB-940)
```

### What Next

Modify the modeling information of the level shifter to meet the requirements.

To correct the modeling of the previous example, make the *related_power_pin* value of the input pin different from the *related_power_pin* value of the output pin as follows:

```
cell(LS) {
  is_level_shifter : true;
  ...
  pin(in) {
    ...
    related_power_pin: VDD1;
    related_ground_pin : GND;
  }
  pin(out) {
    ...
    related_power_pin: VDD2;
    related_ground_pin : GND;
  }
}
```

---

## LBDB-941

(error) Attribute "scan_start_pin" can only be specified under an output/inout bus/bundle.

### Description

This message indicates that the direction of the bus/bundle with attribute "scan_start_pin" is not output/inout.

The following example shows an instance where this message occurs:

```
cell (mb_cell) {

    bundle(Q) {
      members(Q0, Q1)
```

Feedback

```
     direction       : internal;
      scan_start_pin : Q0;
      ...
    }

    ...

  }
```

The following is an example message:

```
Error: Line 253, Cell 'SDFF2_SO', pin 'Q', Attribute "scan_start_pin" can
 only be specified under an output/inout bus/bundle. (LBDB-941)
```

**What Next**

Check the library source file, and make sure the attribute "scan_start_pin" is specified under the correct bus/bundle.

---

## LBDB-942

(error) Can't find single-bit scan input/output pin in the multi-bit scan cell.

**Description**

This message indicates that the multi-bit scan cell with attribute "scan_start_pin" must have both single-bit scan input pin and scan output pin defined. The single-bit scan input/output pin should be specified with signal_type "test_scan_in"/"test_scan_out" in test_cell.

The following example shows an instance where this message occurs:

```
cell (mb_cell) {

    pin(SI) {
        direction          : input;
        ...
    }
    pin(SO) {
        direction          : output;
        ...
    }

    bundle(Q) {
        members(Q0, Q1)
        direction       : output;
        scan_start_pin : Q0;
        ...
    }

    test_cell() {
        pin(SI) {
            direction          : input;
```

Feedback

```
            signal_type         : "test_scan_in";
            ...
        }
        pin(SO) {
            direction           : output;
            /* signal_type      : "test_scan_out"; */
            ...
        }
    }
    ...

}
```

The following is an example message:

```
Error: Line 149, Cell 'SDFF2_SO', Can't find single-bit scan input/output
 pin in the multi-bit scan cell. (LBDB-942)
```

### What Next

Check the library source file, and make sure the correct signal_type attribute is specified in test_cell.

---

## LBDB-943

(error) Can't find attribute "single_bit_degenerate" in the multi-bit scan cell.

### Description

This message indicates that the attribute "single_bit_degenerate" is missing for the multi-bit scan cell with attribute "scan_start_pin" on bus/bundle. To mapping a complex multi-bit scan cell with serial scan chain and dedicate scan output, a single-bit degenerate cell is needed for mapping the multi-bit cell to N-slice single-bit cell.

The following example shows an instance where this message occurs:

```
cell (mb_cell) {

    /* single_bit_degenerate : sb_cell; */

    bundle(Q) {
      members(Q0, Q1)
     direction       : output;
      scan_start_pin : Q0;
      ...
    }

    ...

  }
```

Feedback

The following is an example message:

```
Error: Line 149, Cell 'SDFF2_SO', Can't find attribute
 "single_bit_degenerate" in the multi-bit scan cell. (LBDB-943)
```

**What Next**

Check the library source file, and make sure the attribute "single_bit_degenerate" is specified under the cell

---

## LBDB-944

(error) The value of "scan_start_pin" can only be the first or last bit pin of the bus/bundle.

**Description**

This message indicates that the value of attribute "scan_start_pin" can only be the first/last bit pin of the bus/bundle.

The following example shows an instance where this message occurs:

```
cell (mb_cell) {

    bundle(Q) {
      members(Q0, Q1, Q2, Q3)
     direction       : output;
      scan_chain_path : Q2;
      ...
    }

    ...

  }
```

The following is an example message:

```
Error: Line 199, Cell 'mb_4_scanin_gated_scanout1', pin 'Q', The value
 of "scan_start_pin" can only be the first or last bit of the parent
 bus/bundle. (LBDB-944)
```

**What Next**

Check the library source file, and make sure the value of attribute "scan_start_pin" is the first/last bit pin of the bus/bundle.

---

## LBDB-945

(warning) Since the function of the cell is understood the "single_bit_degenerate" attribute will be ignored.

Feedback

### Description

This message indicates that the "single_bit_degenerate" attribute is not needed for this multi-bit cell function specification, and will be ignored in function extraction. User should remove it from the Liberty file.

The function of this multi-bit scan cell can be recognized, and the corresponding single-bit cell can be degenerated from the multi-bit cell itself.

This attribute is only required when the multi-bit cell function is too complex, that corresponding single-bit cell can't be denegerated from the multi-bit cell.

### What Next

Check the library source file, and remove attribute "single_bit_degenerate".

---

## LBDB-946

(error) The voltage value %f of the insulated PG pin voltage_map '%s' is not the same as the value %f of voltage of the tied to power/ground PG pin voltage_map '%s'.

### Description

This message indicates the insulated well PG pin voltage_map attribute specify the different voltage value from that of the voltage of the tied to power/ground PG pin. They should be equal.

The following example shows an instance where this message occurs:

```
library(insulatedPgpin) {
  ...
   voltage_map (vdd, 1.0);
   voltage_map (gnd, 0.0);
   voltage_map (vddo, 1.0);
   voltage_map (gndo, 0.0);
   voltage_map (vdds, 0.0);
   voltage_map (gnds, 1.0);
   voltage_map (vdds', 1.0);
   voltage_map (gnds', 1.0);
cell (insulated_well) {
   area : 0.3264;
   pg_pin(gndo){
    pg_type : backup_ground;
    voltage_name : gndo;
    related_bias_pin : gnds';
   }

   pg_pin(vddo){
    pg_type : backup_power;
    voltage_name : vddo;
    related_bias_pin : vdds';
```

Feedback

```
    }
    pg_pin(gnds') {
     pg_type : pwell;
     voltage_name : gnds';
     is_insulated : TRUE;
     tied_to : gndo;
     direction : internal;
     }
     pg_pin(vdds') {
     pg_type : nwell;
     voltage_name : vdds';
     is_insulated : TRUE;
     tied_to : vddo;
     direction : internal;
     }
     pg_pin(gnds) {
     pg_type : pwell;
     voltage_name : gnds;
     direction : internal;
     physical_connection : device_layer;
     }
     pg_pin(vdds) {
     pg_type : nwell;
     voltage_name : vdds;
     direction : internal;
     physical_connection : device_layer;
     }
     }
} /* end library */
```

The following is an example message:

```
Error: Line 58, Cell 'insulated_well', pin 'gnds'', The voltage value
 1.000000 of the insulated PG pin voltage_map 'gnds'' is not the same
 as the value 0.000000 of voltage of the tied to power/ground PG pin
 voltage_map 'gndo'. (LBDB-946)
```

### What Next

Correct the voltage value of the insulated well PG pin's voltage_map.

---

## LBDB-947

(warning) is an insulated well but not PG pin is tied to. The cell is being marked dont_touch/dont_use.

### Description

This warning message occurs when the insulated substrate well PG pin do not tie off to other PG pin.

For a insulated substrate well PG pin, it must be tied off to a power/ground PG pin.

Feedback

The cell is black box for the multivoltage flow and is marked as dont_touch, dont_use (d,u). It cannot be recognized during functional optimization by the tools (Power Compiler and IC Compiler).

The following example shows an instance where this message occurs:

```
library(insulatedPgpin) {
 ...

   voltage_map (vdd, 1.0);
   voltage_map (gnd, 0.0);
   voltage_map (vddo, 1.0);
   voltage_map (gndo, 0.0);
   voltage_map (vdds, 0.0);
   voltage_map (gnds, 1.0);
   voltage_map (vdds', 1.0);
   voltage_map (gnds', 0.0);
cell (insulated_well) {
   area : 0.3264;

    pg_pin(vdds') {
    pg_type : nwell;
    voltage_name : vdds';
    is_insulated : TRUE;
    direction : internal;
/*    physical_connection : device_layer;*/
    }
    }
} /* end library */
```

The following is an example message:

```
Warning: Line 65, Cell 'insulated_well', pin 'vdds'' is an insulated well
 but not PG pin is tied to. The cell is being marked dont_touch/dont_use.
 (LBDB-947)
```

**What Next**

This is only a warning message. No action is required.

However, if the result is not what you intended, make sure the specified PG pins have the tied power/ground PG pin and run the command again.

---

## LBDB-948

(error) Insulated substrate well PG pin is associated with an invalid pg_type.

Feedback

### Description

This error message occurs when the insulated substrate well PG pin is associated with an invalid pg_type. The pg_type of a bias PG pin must be one of the predefined bias types: pwell, nwell, deeppwell, or deepnwell.

The following example shows an instance where this message occurs:

```
library(insulatedPgpin) {

   voltage_map (vdd, 1.0);
   voltage_map (gnd, 0.0);
   voltage_map (vddo, 1.0);
   voltage_map (gndo, 0.0);
   voltage_map (vdds, 0.0);
   voltage_map (gnds, 1.0);
   voltage_map (vdds', 1.0);
   voltage_map (gnds', 0.0);
cell (insulated_well) {
   area : 0.3264;


   pg_pin(gnds') {
    pg_type : internal_power;
    voltage_name : gnds';
    is_insulated : TRUE;
    tied_to : gndo;
    direction : internal;
    /*physical_connection : device_layer;*/
    }
  ...
  }
} /* end library */
```

The following is an example message:

```
Error: Line 54, Cell 'insulated_well', pin 'gnds'', Insulated substrate
 well PG pin is associated with an invalid pg_type. (LBDB-948)
```

### What Next

Make sure the pg_type of the bias PG pin is one of the valid types listed above.

---

## LBDB-949

(error) Exactly 5 measurement points are required for this vector group inside %s.

### Description

When AWP mode is enabled on CCS noise modeling, each vector group inside a output_voltage_rise or output_voltage_fall group must have exactly 5 time-voltage point

Feedback

in waveform measurement. Note that time and voltage values are stored in "index_3" and "values" respectively.

The following example shows an instance where this message occurs:

```
ccsn_first_stage () {
    is_needed : "true";
    stage_type : "both";
    is_inverting : "true";
    miller_cap_fall : "0.00266914";
    miller_cap_rise : "0.00252059";
    output_voltage_rise() {
      vector("ccsn_ov") {
        index_1("0.01");
 index_2("0.0407");
        index_3("0.0137562, 0.0212768, 0.0403229, 0.0554306");
        values("0.081, 0.243, 0.567, 0.729");
      }
      ...
    }
    ...
  }
```

The following is an example message:

```
Error: Line 19484, Cell "test_ccsn", pin "z", Exactly 5 measurement
 points are required for this vector group in output_voltage_rise
```

**What Next**

Correct the problem in the characterization setup.

---

## LBDB-950

(error) The 5 voltage points in this vector group is not consistent with those of the first group.

**Description**

When AWP mode is enabled on CCS Noise modeling, all measurements within an output_voltage_rise or output_voltage_fall group must have the same voltage values. Note that voltage values are store in attribute "values" of a "vector" group.

The following example shows an instance where this message occurs:

```
voltage_map("vddfx", 0.8100);
  ...
  cell (test) {
    ...
    pin (z) {
      direction : output;
      related_power_pin : vddfx;
```

Feedback

```
      ...
      ccsn_last_stage () {
        is_needed : "true";
        stage_type : "both";
        is_inverting : "true";
        miller_cap_fall : "0.00266914";
        miller_cap_rise : "0.00252059";
        output_voltage_rise() {
          vector("ccsn_ov") {
            index_1("0.01");
          index_2("0.0407");
            index_3("0.0137562, 0.0212768, 0.0299644, 0.0403229,
0.0554306");
            values("0.081, 0.243, 0.405, 0.567, 0.729");
          }
  vector("ccsn_ov") {
            index_1("0.01");
            index_2("0.0263933");
            index_3("0.0150596, 0.0229561, 0.0316666, 0.0417733,
0.0562343");
            values("0.091, 0.253, 0.410, 0.562, 0.719");
          }
        ...
      }
    ...
```

The *"values"* attributes have different numbers.

The following is an example message:

```
Error: Line 19484, Cell "test_ccsn", pin "z", the 5 voltage points given
 in the "values" attribute should be of specific values.
```

### What Next

Correct the problem in the characterization setup.

---

## LBDB-951

(warning) The 5 voltage points should be of specific values in %s group.

### Description

When AWP mode is enabled on CCS noise modeling, the five voltage points (stored in attribute "values") in each vector group of output_voltage_rise group should be 10%, 30%, 50%, 70%, 90% of Vdd, and for output_voltage_fall group, 90%, 70%, 50%, 30%, 10% of vdd.

The following example shows an instance where this message occurs:

```
voltage_map("vddfx", 0.8100);
  ...
```

Feedback

```
 cell (test) {
   ...
   pin (z) {
     direction : output;
     related_power_pin : vddfx;
     ...
     ccsn_last_stage () {
       is_needed : "true";
       stage_type : "both";
       is_inverting : "true";
       miller_cap_fall : "0.00266914";
       miller_cap_rise : "0.00252059";
       output_voltage_rise() {
         vector("ccsn_ov") {
           index_1("0.01");
         index_2("0.0407");
           index_3("0.0137562, 0.0212768, 0.0299644, 0.0403229,
0.0554306");
           values("0.089, 0.243, 0.405, 0.567, 0.729");
         }
         ...
       }
output_voltage_fall() {
  vector("ccsn_ov") {
           index_1("0.01");
           index_2("0.0263933");
           index_3("0.0150596, 0.0229561, 0.0316666, 0.0417733,
0.0562343");
           values("0.729, 0.567, 0.405, 0.243, 0.089");
         }
       ...
     }
   ...
```

The following is an example message:

```
Error: Line 19484, Cell "test_ccsn", pin "z", the 5 voltage points given
 in the "values" attribute should be of specific values.
```

**What Next**

Correct the problem in the characterization setup.

---

## LBDB-952

(warning) Incomplete when condition coverage for CCS receiver capacitance model.

**Description**

This warning message occurs when pin-based/arc-based receiver capacitance has incomplete "when" condition coverage

Feedback

The following example shows an instance where this message occurs:

```
...
    pin(I) {
      direction : input ;
      receiver_capacitance () {
        when : "A*!B";
        ...
      }
      receiver_capacitance () {
        when : "!A*B";
        ...
      }
      /*
      receiver_capacitance () {
        when : "!A*!B";
        ...
      }
      */
    }
    pin(Z) {
      direction : output ;
      function : "A*B*I" ;
      timing() {
        related_pin : "I" ;
        when : "A*B";
        receiver_capacitance_*() {...}
        ...
      }
    }
    ...

In the example above, input pin"I" does not have a receiver capacitance
 model
for the when condition "!A*!B".

The following is an example message:
The following is an example of the warning message:

Warning: Line 206, cell "test", pin "I", Incomplete when condition
 coverage for CCS receiver capacitance model. (LBDB-952)
```

### What Next

Complete the receiver model. Either introduce a receiver capacitance model with the missing "when" condition, or add a default model (without the "when" attribute). If the timing modeling comes from the timing characterization tool, check with the timing characterization tool to determine why this type of data was generated.

Feedback

## LBDB-953

(warning) CCS/CCB noise output DC swing (%g, %g) is not within %.1f%% of rail voltages (%g, %g)

### Description

For a normal CMOS circuit, the DC output swing is rail to rail. This CCS/CCB noise tage has abnormal output swing according to its dc_current table.

If either the output low has a significant offset to ground, or the output high has a significant droop with respect to Vdd, or both, the output is deemed to have inadequate swing. The problem may be caused by cell characterization issues using improper supply voltage, or it may be due to serious DC leakage in the design. An LBDB-706 error may also be reported.

If the output high is larger than Vdd, the problem is caused by improper voltage used in cell characterization.

Note that if the stage_type attribute is "pull_up" or "pull_down", the output starting voltage is not analyzed, and assumed to be 0 or Vdd respectively.

The following is an example message:

```
Error: Line 1934, Cell "test", pin "a", DC output swing (0.023, 0.849) is
not within 5.0% of rail voltages (0.0, 0.9);
```

### What Next

SolvNet article 2069871 has explanation about LBDB-706. Apply the same demonstration of example 4 using the dc_current group where one of the LBDB-706/LBDB-953 occurs to understand why the CCS/CCB noise output DC swing is not within 5% of rail voltages.

## LBDB-954

(error) an invalid zero or negative capacitance value of '%s' is found.

### Description

This message indicates zero or negative value is found of the capacitance attributes.

This checker is triggered when lc_enable_10nm_mode set true, and capacitance attributes are specified under connected input pin, otherwise it's skipped and another warning message LBDB-163 reported.

Consider of single precision for values stored in .db, the checker uses a tolerance of 1e-6, if value < 1e-6, LC treats it less than or equal to 0.

Feedback

LC checks these capacitance values:

Pin level attributes: capacitance, rise_capacitance, fall_capacitance:

There is a special case: when there is no pin capacitance attribute defined in .lib, and library level input/inout/outpu default capacitances equal to zero, e.g. library (my_library) { default_input_pin_cap : 0 ; default_output_pin_cap : 0 ; default_inout_pin_cap : 0 ; .... cell (my_cell) { ... pin (ABC) { direction : input; /* no capacitance attribute defined */ ... } /* end pin 'ABC' */ } /* end cell */ } In this case, input pin 'ABC' capacitance will inherit from library level attribute "default_input_pin_cap" which happens to be zero with a LBDB-172 warning, e.g. Warning: Line 100, Cell 'my_cell', pin 'ABC', The 'capacitance' attribute is not specified. Using 0.00. (LBDB-172)

Then LBDB-954 error will be reported for pin 'ABC' capacitance. Users can define pin capacitance attribute to override the value in library "default_input_pin_cap".

The following example shows an instance where this message occurs:

```
pin (I) {
…
direction : input;
capacitance : 0.0; /*wrong*/
rise_capacitance: 0.0000001; /*<1e-6, wrong*/
fall_capacitance: 0.001; /*correct*/
}
```

The following is an example message:

```
Error: Line 385, Cell 'HP_SHDFFNX1', pin 'I', an invalid zero or negative
 capacitance value of 'capacitance' is found. (LBDB-954)
Error: Line 386, Cell 'HP_SHDFFNX1', pin 'I', an invalid zero or negative
 capacitance value of 'rise_capacitance' is found. (LBDB-954)
```

**What Next**

Change the value of the attributes to positive, make sure value >= 1e-6.

---

## LBDB-954w

(warning) an invalid zero or negative capacitance value of '%s' is found.

**Description**

This message indicates zero or negative value is found in the capacitance model.

Consider of single precision for values stored in .db, the checker uses a tolerance of 1e-6, if value < 1e-6, LC treats it less than or equal to 0 .

Feedback

LC checks these capacitance values:

1. the first value of N-cap model under pin or timing level.
2. receiver_capacitance1_rise|fall value under pin or timing level.
3. miller_cap_rise/miller_cap_fall under first/last_ccsn_stage or input/output_ccb group, this checker is controlled by lc_enable_10nm_mode.

The following example shows an instance where this message occurs:

```
pin(I)  {
   ...
receiver_capacitance() {
 receiver_capacitance1_fall(LTT1) {
 index_1 ("0.05,0.075,0.100,0.225,0.400");
 values (\
 "0.0,1.960000e-03,1.992000e-03,2.031000e-03,2.185000e-03"\
 );
}
pin(Z)  {
    direction : output;
    function : "I";
    timing() {
      related_pin : "I";
 ...
 receiver_capacitance1_fall (LTT2) {
 index_1 ("0.05,0.075,0.100,0.225,0.400");
 index_2 ("0.010,0.025,0.040,0.075");
 values (\
 "1.947000e-03,1.959000e-03,1.988000e-03,2.016000e-03",\
 "1.960000e-07,1.969000e-03,1.994000e-03,2.020000e-03",\
 "1.992000e-03,1.997000e-03,2.012000e-03,2.030000e-03",\
 "2.031000e-03,2.032000e-03,2.035000e-03,2.043000e-03",\
 "2.185000e-03,2.184000e-03,2.179000e-03,2.171000e-03"\
 );
 }
```

The following is an example message:

```
Warning: Line 374, Cell 'BUFFD0', pin 'I', an invalid zero or negative
 capacitance value of 'receiver_capacitance1_fall' is found. (LBDB-954w)
Warning: Line 415, Cell 'BUFFD0', pin 'Z', an invalid zero or negative
 capacitance value of 'receiver_capacitance1_fall' is found. (LBDB-954w)
```

**What Next**

Change the value of the attributes to positive, make sure value >= 1e-6.

---

## LBDB-955

(warning) an illegal value of '%s' %f is found, it cannot be %s than \ '%s' value '%s %f'.

Feedback

### Description

```
This message indicates that the values of all capacitance
 models/attributes is not correct.

the capacitance values should meet the following 4 rules:

C1 = receiver_capacitance1_[rise|fall] values
C2 = receiver_capacitance2_[rise|fall] values
Cmiller_rise = miller_cap_rise value
Cmiller_fall = miller_cap_fall value

pin_capacitance = rise_capacitance or fall_capacitance or if missing, use
 pin capacitance

1. C1 < (2 * pin_capacitance);
2. C2 <= (5 * pin_capacitance)
3. Cmiller_rise/fall >= 10% * pin_capacitance;
4. Cmiller_rise/fall < pin_capacitance;
```

The following example shows an instance where this message occurs:

```
pin(I)  {
    direction : input;
    capacitance : 0.002;
    receiver_capacitance() {
 receiver_capacitance1_rise(LTT1) {
 index_1 ("0.05,0.075,0.100,0.225,0.400");
 values (\
 "0.002,0.005,1.992000e-03,2.031000e-03,2.185000e-03"\
 );
}
```

The following is an example message:

```
Warning: Line 364, Cell 'BUFFD0', pin 'I', an illegal value of 'values'
 0.005000 is found, it cannot be more than '2 * capacitan' value '2 *
 0.002000'. (LBDB-955)
```

### What Next

Change the value of the attributes to meet the screener rules.

---

## LBDB-956

(warning) 'miller_cap_rise' cannot more than 5 times value of 'miller_cap_fall'.

### Description

This message indicates that the difference of miller_cap_rise and miller_cap_fall is wrong.Cmiller_rise/Cmiller_fall must be less than 5.

Feedback

The following example shows an instance where this message occurs:

```
pin(I) {
      direction : input;
      max_transition : 2100.0;
      capacitance : 0.002000;
      fanout_load : 1;

      /* test pin-level */
        ccsn_first_stage ( ) {
          is_needed     : true;
          is_inverting    : true;
          stage_type      : both;
          miller_cap_rise : 0.0021;
          miller_cap_fall : 0.0002;
    ...
  }
```

The following is an example message: Warning: Line 106, Cell 'inv0d0', pin 'I', 'miller_cap_rise' cannot more than 5 times value of 'miller_cap_fall'. (LBDB-956)

### What Next

Change the value of the attributes to non-zero, make sure Cmiller_rise/Cmiller_fall < 5.

---

## LBDB-957

(warning) No driver waveform defined in the %s.

### Description

In cell characterization, the shape of the Waveform driving the characterized circuit can have a significant impact on the final results.

This message is reported when there is no "normalized_driver_waveform" group defined in the library. Or there is normalized_driver_waveform group in library, but a library cell does not have driver waveform defined By any of following cell level or pin level attributes: driver_waveform, driver_waveform_rise and driver_waveform_fall.

The checkers is for 10nm library.

The following is an example message:

```
Missing normalized_driver_waveform in library:
Warning: Line 6, No driver waveform defined in the library. (LBDB-957)

Missing driver_waveform, driver_waveform_rise, driver_waver_fall in cell:
Warning: Line 594, Cell 'TCAINVXC', No driver waveform defined in the
 cell. (LBDB-957)
```

Feedback

### What Next

Define normalized_driver_waveform group in library or define driver_waveform, driver_waveform_rise, driver_waver_fall in cell or pin level.

---

## LBDB-958

(warning) dc_current values not correct for tied output

### Description

For an output pin that is tied to low or high, the dc_current table of its ccsn_last_stage must be of a special form. In a strict sense, this ccsn stage does not have an input (because the function is 0 or 1). To conform the current data to the input/output format, we repeat the same values for all input voltages, as shown below:

```
dc_current {
   ...
   values( \\
      "0.0672194, 0.0647206, 0.0628569, 0.0621341 ..." \\
      "0.0672194, 0.0647206, 0.0628569, 0.0621341 ..." \\
      "0.0672194, 0.0647206, 0.0628569, 0.0621341 ..." \\
      ...)
   ...
}
```

The following is an example message:

```
Warning: Line 214, Cell 'hptiehi', pin 'Z', dc_current values not correct
 for tied output. (LBDB-958)
```

### What Next

Review the "values" attribute of the affected "dc_current" group, root cause the differences in row values, and re-characterize with correct settings.

---

## LBDB-959

(warning) suspicious dc_current value (%g) detected at row %d, entry %d

### Description

The "values" attribute in dc_current group contains complex, two-dimensional numerical data. A potentially bad value has been identified at the shown location. An example is given below:

```
dc_current {
   ...
   values( \\
      "0.270561, 0.111316, ... 0.0110591, -1.465e-05, ...",\\ (1st row)
```

Feedback

```
      ...
      "0.348481, 0.137316, ... -0.0899564, 1.45095e+31, ...", \\ (25th
 row)
      ...)
   ...
}
```

The 23rd entry of the 25th row is abnormally large (1.45095e+31). A subsequent LBDB-706 error is possible.

The following is an example message: Warning: Line 120, Cell 'inv', pin 'Z', suspicious dc_current value (1.45095e+31) detected at row 25, entry 23 (LBDB-959)

**What Next**

Review the "values" attribute of the affected "dc_current" group. Re-generate the library with corrected data.

---

## LBDB-960

(error) "%s" is not a valid characterization model for %s%s.

**Description**

This error is issued when <char_model> is not one of the valid enumerated values for all attributes, a specific attribute, or a specific value of a atribute.

The following is an example message:

```
Error: Line 10, "invalid_char_model" is not a valid characterization
 model for all attributes. (LBDB-960)
```

**What Next**

Check the library source file, correct the <char_model> definition.

---

## LBDB-961

(warning) Attribute driver_waveform, driver_waveform_rise, driver_waveform_fall defined outside of char_config group.

**Description**

This error is issued when char_config is defined but there is driver_waveform, driver_waveform_rise, driver_waveform_fall defined outside of char_config group.

If char_config is defined, all driver_waveform attributes should be defined inside char_config group.

Feedback

The following is an example message:

```
Warning: Line 10, Attribute driver_waveform, driver_waveform_rise,
 driver_waveform_fall defined outside of char_config group. (LBDB-961)
```

**What Next**

Check the library source file, remove driver_waveform attributes which defined outside char_config group.

---

## LBDB-962

(error) "%s" is not a valid selection method.

**Description**

For these two attributes:

default_value_selection_method(<char_model>, <method>);
merge_selection(<char_model>, <method>);

This error is issued when <method> is not one of these enumerated values: any min max average min_mid_table max_mid_table follow_delay

The following is an example message:

```
Error: Line 10, "invalid_method" is not a valid selection method.
 (LBDB-962)
```

**What Next**

Check the library source file, correct the <method> definition.

---

## LBDB-963

(warning) Missing driver_waveform_name in normalized_driver_waveform group.

**Description**

When char_config() group is defined, the default driver_waveform should be defined by using "all" as <char_model> in driver_waveform definition. In this case, all normalized_driver_waveform group should have a driver_waveform_name.

The following is an example message:

```
Warning: Line 10, Missing driver_waveform_name in
 normalized_driver_waveform group. (LBDB-963)
```

Feedback

### What Next

Check the library source file, either add driver_waveform_name to all normalized_driver_waveform group, or delete the normalized_driver_waveform which do not have driver_waveform_name.

---

## LBDB-964

(warning) default char_config attribute %s is not defined for the library.

### Description

For characterization configuration, each attribute may have a default value in the library level. If it is intended not to use a default value for the whole library, this warning can be ignored.

The following is an example message:

```
Error: Line 10, default char_config attribute dirver_waveform for nldm
 model is not defined for the library. (LBDB-964)
```

### What Next

Check the library source file, define the default attribute for the library.

---

## LBDB-965

(warning) default char_config attribute %s is not defined for the library when "%s" is defined.

### Description

This is for when a default char_config attribute is required defined for the library but is not defined. Sometimes a default attribute is required to be defined if char_config group is used. Sometimes a default attribute is required to be defined if another default attribute is defined.

The following is an example message:

```
Warning: Line 10, default "ccs_timing_segment_voltage_tolerance_rel"
 attribute is not defined for the library when char_config is defined.
 (LBDB-965)
```

### What Next

Check the library source file, define the default attribute for the library.

Feedback

## LBDB-966

(information) The '%s' pg_pin is not '%s' of any signal \n \t pin on the cell.

### Description

This information is issued when a pg_pin group satisfies the following conditions: - the pg_pin must be a 'primary_power' or 'primary_ground' pg_pin. - the pg_pin must be a 'std_cell_main_rail' pg_pin. - if the pg_pin is 'primary_power', the pg_pin is not the related power pin of any signal pin. - if the pg_pin is 'primary_ground', the pg_pin is not the related ground pin of any signal pin. The 'voltage_name' is optional for this pg_pin.

The following example shows an instance where this message occurs:

```
cell(ls) {
    is_level_shifter : true;

    pg_pin(VDD) {
        voltage_name : VDDL;
        pg_type : primary_power;
        std_cell_main_rail : true;
    }
    ...
  }
```

The following is an example message:

```
Information: Line 84, The 'VDD' pg_pin is not a 'related_power_pin'
 pg_pin of any signal pin on the cell. (LBDB-966)
```

### What Next

If any screener rule fails on 'voltage_name', delete the 'voltage_name' under the pg_pin.

## LBDB-967

(error) The delay trip point "%s" is not in the range between slew lower trip point "%s" and slew upper trip point "%s".

### Description

This message indicates that the attribute value you have specified is not in the required range. For the 4 delay trip point attributes below, their value should be in the range of slew lower and upper trip point. input_threshold_pct_rise input_threshold_pct_fall output_threshold_pct_rise output_threshold_pct_fall

Below are the attributes for slew upper/lower trip point. slew_lower_threshold_pct_rise slew_upper_threshold_pct_rise slew_lower_threshold_pct_fall slew_upper_threshold_pct_fall

Feedback

For example: slew_lower_threshold_pct_rise < input_threshold_pct_rise < slew_upper_threshold_pct_rise slew_lower_threshold_pct_fall < input_threshold_pct_fall < slew_upper_threshold_pct_fall slew_lower_threshold_pct_rise < output_threshold_pct_rise < slew_upper_threshold_pct_rise slew_lower_threshold_pct_fall < output_threshold_pct_fall < slew_upper_threshold_pct_fall

The following example shows an instance where this message occurs:

```
library (test) {
    slew_lower_threshold_pct_rise : 30.0;
    slew_upper_threshold_pct_rise : 70.0;
    slew_lower_threshold_pct_fall : 30.0;
    slew_upper_threshold_pct_fall : 70.0;
    input_threshold_pct_rise : 50.0;
    input_threshold_pct_fall : 50.0;
    output_threshold_pct_rise : 50.0;
    output_threshold_pct_fall : 80.0;

  }
```

To correct the error, change the output_threshold_pct_fall to 50.0, which is in the range of (30.0 ~ 70.0)

The following is an example message:

```
Error: Line 30, The delay trip point "output_threshold_pct_fall" is not
 in the range between
  slew lower trip point "slew_lower_threshold_pct_fall" and slew upper
 trip point
  "slew_upper_threshold_pct_fall". (LBDB-967)
```

**What Next**

Check the library source file, and ensure the attribute value is consistent with corresponding range.

---

## LBDB-968

(Error) This %s group does not have a driver waveform associated with input %s.

**Description**

This library uses input_ccb and output_ccb for noise/timing applications, instead of ccsn_first_stage and ccsn_last_stage. An associated change is that all output_voltage_rise &_voltage_fall data (in input_ccb & output_ccb) are obtained with a driver waveform that is consistent with timing characterization.

The driver waveform must be specified at the pin or cell level using driver_waveform, driver_waveform_rise, or driver_waveform_fall attribute. If these attributes are missing, a

Feedback

default normalized_driver_waveform group (without the "driver_waveform_name" attribute) needs to be present at the library level.

This message indicates the above search fails to find a waveform.

**Examples**

Line 786, Cell 'inv0d0', pin 'ZN', This output_ccb group does not have a driver waveform associated with input fall. (LBDB-968)

**What Next**

Check the library source file, add an appropriate cell-level driver_waveform attribute or a default normalized_driver_waveform group at the library level.

---

## LBDB-969

(error) The '%s' group should not be used in a non-retention cell.

**Description**

This error message occurs because user specifies following groups in a non-retention cell. retention_condition clock_condition clear_condition preset_condition Retention cell must have cell level attribute "retention_cell", otherwise it is a non-retention cell. These groups can only work in a retention cell.

The following example shows an instance where this message occurs: In the following example, there is a *retention_condition* group, but there is no *retention_cell* attribute. Remove the comment from the *retention_cell* attribute to correct the problem.

```
cell(retention_dff) {
  ...
  /* retention_cell : "my_retention_dff"; */

  retention_condition() {
    required_condition : "!sleep";
    power_down_function : "!vcc+vss" ;
  }

}
```

The following is an example message: The above example results in the following error message:

```
Error: Line 192, Cell 'retention_latch_with_async_preset_clear', The
 'retention_condition'
  group should not be used in a non-retention cell. (LBDB-969)
```

**What Next**

Resolve the issue by either removing these groups or adding a *retention_cell* attribute.

Feedback

## LBDB-970

(error) Invalid parent group '%s' of the user defined %s '%s'.

### Description

This message indicates that the group where the user defined attribute belongs to is not valid.

The following example shows an instance where this message occurs:

```
library(test) {
  define(user_defined_attribute, user_defined_group, string);
  define_group(another_user_defined_group, user_defined_group);
  ...
}
```

The following is an example message:

```
Error: Line 2, Invalid parent group 'user_defined_group' of the user
 defined attribute 'user_defined_attribute'.  (LBDB-970)
Error: Line 3, Invalid parent group 'user_defined_group' of the user
 defined group 'another_user_defined_group'.  (LBDB-970)
```

### What Next

Check the parent group, define it before this line or fix the group name it is a typo.

## LBDB-971

(error) User defined group '%s' cannot use the same name as its parent group.

### Description

This message indicates that the user defined group is using same name as its parent group in define_group function.

The following example shows an instance where this message occurs:

```
library(test) {
  define_group(user_defined_group, library);
  define_group(user_defined_group, user_defined_group);
  ...
}
```

The following is an example message:

```
Error: Line 2, User defined group 'user_defined_attribute' cannot use the
 same name as its parent group.  (LBDB-971)
```

Feedback

### What Next

Check the define_group, using different name between the user defined group and its parent group.

---

## LBDB-972

(error) The scan pin '%s' should not be used in the non-scan mode function.

### Description

The test_cell is describing the non-scan mode of the scan cell, scan pin should not be used in the test_cell function. Only non-scan pin is allowed. Currently, scan input pin and scan enable pin is not allowed to be used in the ff group under test_cell.

The following example shows an instance where this message occurs:

```
cell(test) {
  ...
    test_cell() {
      ...
      pin(SD) {
        direction : input ;
        signal_type : test_scan_in ;
      }
      pin(SE) {
        direction : input ;
        signal_type : test_scan_enable ;
      }
      ...
      ff(IQ, IQN) {
        clocked_on : "CK" ;
        next_state : "((RT&((SE&SD)|(!SE&D))&SL)|(IQ&!RT&SL))" ;
      }
      ...
    }
  ...
}
```

In this case, the scan input pin "SD" and scan enable pin "SE" are used in the ff group "next_state" attribute inside test_cell. This is not allowed, user should correct the next_state attribute, remove the scan pin inside.

The following is an example message:

```
Error: Line 2759, Cell 'scan_cell', test_cell, The scan pin 'SE' should
 not be used in the non-scan mode function. (LBDB-972)
Error: Line 2759, Cell 'scan_cell', test_cell, The scan pin 'SD' should
 not be used in the non-scan mode function. (LBDB-972)
```

Feedback

### What Next

Check the test_cell function and scan pin, remove the scan pin in function description.

---

## LBDB-973

(error) Attribute '%s' should be '%s',\n \t\twhich is conflict with user sepcified info.

### Description

For multibit scan cell with dedicated scan output pin, user should specify attribute "scan_start_pin" to indicate the scan chain direction, and attribute "scan_pin_inverted" to indicate if the scan signal shifting between sequential elements is inverted. In the mean time, LC will also auto-derive these two attributes. For each MB scan cell with auto-derived scan_start_pin/scan_pin_inverted, if user also specifies these attributes explicitly, a consistency check will be performed. This message reports the conflict if the user-given values are different from the derived values.

The following example shows an instance where this message occurs:

```
cell(test) {

  ...
  statetable (" D  CP  SE  SI " ,         "IQ IQN" )
    { table : " -  ~R     -    -  : - - :  N  N ,    \
                H/L  R     L    -  : - - : H/L H/L,   \
                 -   R    H   H/L  : - - : H/L L/H";
    }

  bundle(Q){
    scan_start_pin : Q1; /* This is wrong, should be Q0 */
    scan_pin_inverted : true;

    pin(Q0) {
      internal_node : "IQ" ;
      input_map : " D0  CP  SE  SI";
      ...
    }
    pin(Q1) {
      internal_node : "IQ" ;
      input_map : " D1  CP  SE  Q0";}
      ...
    }
    ...
  }
  pin(SO) {
    state_function : "SE * Q1" ;
    ...
  }
  ...
}
```

Feedback

The attribute "scan_start_pin" should be "Q0"

The following is an example message:

```
Error: Line 437, Cell 'test', pin 'Q', Attribute 'scan_start_pin' should
 be 'Q0',
        which is conflict with user sepcified info.  (LBDB-973)
```

### What Next

Check library file, correct attribute "scan_start_pin" and "scan_pin_inverted"

---

## LBDB-974

(warning) The logic represented by the '%s' attribute (%s)\n \tis not equal or a subset of the logic represented by the '%s' attribute (%s)

### Description

This error message indicates that the 'char_when', 'char_when_rise' or 'char_when_fall' attribute should be equal or a subset of the "when" attribute.

The following example shows an instance where this message occurs: The following example results in this warning. The "char_when" condition "A" in the receiver_capacitance is a superset of the "when" condition "A*B".

```
cell(bad) {
      Pin (A) {
         receiver_capacitance () {
            when : "A*B";
            char_when : "A";
            ...
         }
         ...
      }
      ...
   }
```

The following is an example message:

```
Warning: Line 84, The logic represented by the 'char_when' attribute (A)
        is not equal or a subset of the logic represented by the 'when'
 attribute (A*B). (LBDB-974)
```

### What Next

Check the library source file, and correct the incorrect attribute.

Feedback

## LBDB-975

(information) The "scan_start_pin" attribute is not needed for multi-bit function extraction and is ignored.

### Description

This message indicates that the multi-bit scan cell can derive the mulbit-bit mapping info from cell function, attribute "scan_star_pin" is useless and should be removed from the cell. Attribute 'scan_start_pin' is only needed in multi-bit scan cell with single-bit scan output pin.

The following example shows an instance where this message occurs:

```
cell (test) {
  ...
  statetable (" D  CP SE SI ",         "QN   QT") {
     table : "  - ~R   -  -  : -  - :  N    N, \
               H/L R   L  -  : -  - : L/H  L/H, \
                 -  R   H H/L : -  - : L/H  H/L";
  }
  bundle(QN) {
    members (QN1, QN2);
    direction : output;
    scan_start_pin : QN1;
    ...
    pin(QN1) {
      internal_node : "QN";
      input_map : "D1 CP SE SI";
      max_capacitance : 0.0724592;
    }
    pin(QN2) {
      internal_node : "QT";
      input_map : "D2 CP SE QN1";
      max_capacitance : 0.0724218;
    }
  }

  /* no single-bit scan output pin */

  ...
}
```

The following is an example message:

```
Information: Line 303, Cell 'test', The "scan_start_pin" attribute is not
 needed for multi-bit function extraction and is ignored. (LBDB-975)
```

### What Next

Check the library source file, and remove attribute "scan_start_pin".

Feedback

## LBDB-976

(Warning) The multi-bit retention cell should be modeled with\n \treference_input attribute for multi-bit pin mapping.

### Description

The mutli-bit retention scan cell must be modeled with 1-pair of sequential groups and reference_input, then the 1-pair of sequential can be used for the slice modeling for function processing. Warning if it is wrongly modeled with N-pairs sequential groups.

The multi-bit cell function cannotbe recognized by Design Compiler during banking/de-banking flow.

The following example shows an instance where this message occurs:

```
cell(test) {
  ...
    latch(iq1_1,iqn1_1) {
      enable : "(CK * RET)'" ;
      data_in : "D[0] * !SE + SI * SE" ;
      power_down_function : "!VDD+VSS" ;
    }
    latch(iq1_2,iqn1_2) {
      enable : "(CK * RET)" ;
      data_in : "iq1_1" ;
      power_down_function : "!VDDB+VSS" ;
    }
    latch(iq2_1,iqn2_1) {
      enable : "(CK * RET)'" ;
      data_in : "D[1] * !SE + iq1_2 * SE" ;
      power_down_function : "!VDD+VSS" ;
    }
    latch(iq2_2,iqn2_2) {
      enable : "(CK * RET)" ;
      data_in : "iq2_1" ;
      power_down_function : "!VDDB+VSS" ;
    }
    bundle(Q) {
   members(Q[0], Q[1]);
   direction : output ;
   pin(Q[0]) {
    function : iq1_2;
        ...
   }
   pin(Q[1]) {
    function : iq2_2;
        ...
   }
  }
  ...
}
```

Feedback

In this case, the cell is using 2 pairs of latch groups (4 latches), which can't be analyaze in function processing. User should correct the cell modeling with 1-pair of latch groups and reference_input as below.

```
cell(test) {
  ...
    latch("pa pb", iq1, iqn1) {
      enable : "(CK * RET)'" ;
      data_in : "pa * !SE + pb * SE" ;
    power_down_function : "!VDD+VSS" ;
    }
    latch("pc", iq2, iqn2) {
      enable : "(CK * RET)" ;
      data_in : "pc" ;
    power_down_function : "!VDDB+VSS" ;
    }
   bundle(Q) {
   members(Q[0], Q[1]);
   direction : output ;
   pin(Q[0]) {
    function : iq2;
        reference_input : "QT[0]";
   }
   pin(Q[1]) {
    function : iq2;
        reference_input : "QT[1]";
   }
  }
    bundle(QT) {
   members(QT[0], QT[1]);
      direction : internal;
      pin(QT[0]) {
        reference_input : "D[0] SI";
        function : iq1;
      }
      pin(QT[1]) {
        reference_input : "D[1] Q[0]";
        function : iq1;
      }
    }
  ...
}
```

The following is an example message:

```
Warning: Line 32, Cell 'MB2_scan_retention_1control_master_slave_design',
 The multi-bit retention cell should be modeled with
 reference_input attribute for multi-bit pin mapping. (LBDB-976)
```

### What Next

Check the cell function, correct the cell modeling with reference_input.

Feedback

## LBDB-977

(warning) In attribute '%s', CCB groups '%s' and '%s' have different values for related_ccb_node.

### Description

related_ccb_node of the CCB noise stages specified in propagating_ccb should be identical.

The following example shows an instance where this message occurs:

```
library(test) {
  cell (flop) {
    ...
    pin (CP) {
      ...
      input_ccb("CCB_CP2") {
        related_ccb_node : "net3:3";
      }
      ...
    }
    pin (Q) {
      ...
      output_ccb("CCB_Q1") {
        related_ccb_node : "net7:5";
      }
      timing () {
        related_pin : CP;
        active_input_ccb ("CCB_CP1");
        propagating_ccb("CCB_CP2", "CCB_Q1");
      }
      ...
    }
    ...
  }
}
```

The following is an example message:

```
Warning: Line 32, In attribute 'propagating_ccb', CCB groups 'CCB_CP2'
 and
 'CCB_Q1' have different values for related_ccb_node.
```

### What Next

Use same values for the related_ccb_node of the CCB noise stages specified in propagating_ccb.

Feedback

## LBDB-978

(error) Old CCSN stage noise model cannot co-exist with new CCB noise model.

### Description

If the presence of input_ccb or output_ccb is detected, the library is assumed to use the new CCB format. Any ccsn_first_stage or ccsn_last_stage will trigger an error.

The following is an instance where this error occurs:

```
library (test) {
  ...
  cell (flop) {
    pin (D) {
      input_ccb("CCB_D1") {
        is_needed : true;
        related_ccb_node : "net1:5";
      }
      ccsn_first_stage() {
        is_needed : true;
        related_ccb_node : "net1:5";
      }
    }
  }
  ...
}
```

The following is an example message:

```
Error: Line 9, Cell 'flop', pin 'D', Old CCSN stage noise model cannot
 co-exist
with new CCB noise model. (LBDB-978)
```

### What Next

Use either old CCSN stage noise model or new CCB noise model.

## LBDB-979

(warning) attribute '%s'. '%s' stage at\ line %u has a value (%g) in '%s/vector/%s' (line %u) that does\ not match any values of '%s' in '%s' at line %u.

### Description

In the referenced ccs noise data modeling, output_voltage_rise and output_voltage_fall are characterized in ways to be closer to timing behavior. Their "input slew" (vector/index_1) and "output load" (vector/index_2) values form a 2D grid, similar to various timing lookup tables.

Feedback

To best match timing behavior, the ccb's referenced by the 'propagating_ccb' attribute need to meet the following guideline on input slew and output load values:

The starting/ending values must match those of NLDM delay data (cell_rise/cell_fall). See LBDB-980

Intermediate values may be sparse, but they have to match some entries in the delay data.. This message, LBDB-979, addresses the mimsatch.

Note that there is a different requirement one the output load of input_ccb in an input_ccb/output_ccb two-stage configuration. See LBDB-588.

The following example shows an instance where this message occurs:

```
library(test) {
  ...
  normalized_driver_waveform(ndw1) {
    driver_waveform_name : "driver_waveform_default_fall" ;
    index_1("0.003209, 0.0133113, 0.033516, 0.0726625, 0.152218,
 0.31133");
    ...
  }
  ...
  cell (flop) {
    ...
    pin(A) {
      ...
      driver_waveform_fall : driver_waveform_default_fall ;
      ...
      input_ccb(ccb1) {
        ...
        output_voltage_fall() {
          vector(vec1) {
            index_1("0.013209");
            ...
          }
          ...
        }
        ...
      }
    ...
    }
    pin (Y) {
      ...
      timing() {
        related_pin : "A" ;
        ...
        propagating_ccb(ccb1, ccb2);
      }
      ...
    }
    ...
```

Feedback

```
  }
}
```

### Examples

Warning: Line 10287, Cell 'xyz', pin 'nx2', attribute 'propagating_ccb'. 'input_ccb' stage at line 2321 has a value (0.005) in 'output_voltage_rise/vector/index_1' (line 2380) that does not match any values of 'index_1' in 'cell_fall' at line 9728. (LBDB-979)

### What Next

Correct the characterization procedure.

---

## LBDB-980

(warning) attribute '%s'. '%s' stage at line %u has a %s value (%g) in '%s/vector/%s' (line %u) that does not match the %s value of '%s' in '%s' at line %u.

### Description

In the referenced ccs noise data modeling, output_voltage_rise and output_voltage_fall are characterized in ways to be closer to timing behavior. Their "input slew" (vector/index_1) and "output load" (vector/index_2) values form a 2D grid, similar to various timing lookup tables.

To best match timing behavior, the ccb's referenced by the 'propagating_ccb' attribute need to meet the following guideline on input slew and output load values:

The starting/ending values must match those of NLDM delay data (cell_rise/cell_fall). This message, LBDB-980, addresses the mismatch.

Intermediate values may be sparse, but they have to match some entries in the delay data. See LBDB-979.

Note that there is a different requirement one the output load of input_ccb in an input_ccb/output_ccb two-stage configuration. See LBDB-588.

The following example shows an instance where this message occurs:

```
library(test) {
  ...
  normalized_driver_waveform(ndw1) {
    driver_waveform_name : "driver_waveform_default_fall" ;
    index_1("0.003209, 0.0133113, 0.033516, 0.0726625, 0.152218");
    ...
  }
  ..
  cell (flop) {
    ...
    pin(A) {
      ...
```

Feedback

```
      driver_waveform_fall : driver_waveform_default_fall ;
      ...
      input_ccb(ccb1) {
        ...
        output_voltage_fall() {
          vector(vec1) {
            index_1("0.002209");
            ...
          }
          ...
        }
        ...
    }
    ...
    pin (Y) {
      ...
      timing() {
        related_pin : "A" ;
        ...
        propagating_ccb(ccb1, ccb2);
      }
      ...
    }
    ...
  }
}
```

### Examples

Warning: Line 7492, Cell 'xyz', pin 'nx2', attribute 'propagating_ccb'. 'input_ccb' stage at line 488 has a min value (0.005) in 'output_voltage_rise/vector/index_1' (line 561) that does not match the first value of 'index_1' in 'cell_fall' at line 6933. (LBDB-980)

### What Next

Correct the characterization procedure.

---

## LBDB-981

(warning) For the referenced ccs noise data modeling, each delay timing arc needs to have valid CCB model.

### Description

```
For the referenced ccs noise data modeling, each delay timing arc needs
 to have:
1.  at least one active input_ccb which is being referenced by
 active_input_ccb
or propagating_ccb
2.  one or two active output_ccb which is being referenced by
 active_output_ccb
```

Feedback

```
or propagating_ccb, in case of two output_ccb, one is of
 stage_type:pull_up, the
other is of stage_type:pull_down
If any of above is not satisfied, the warning is issued.
```

The following example shows an instance where this message occurs:

```
library(test981) {
  ...
  cell(BUF) {
    pin(Z) {
      timing() {
      }
      output_ccb(BUFX) {
      ...
      }
    }
  }
}
```

The following is an example message:

```
Warning: Line 223, Cell 'BUF', pin 'Z', For the referenced ccs noise data
modeling, each delay timing arc needs to have valid CCB model. (LBDB-981)
```

### What Next

Resolve the issue by providing appropriate CCB combinations for each timing.

---

## LBDB-982

(warning) is a '%s' class retention pin but is missing all recommended attributes.

### Description

This message occurs in either of following situations: 1. Missing following 'save' related retention attributes in a save class retention pin. save_action save_condition

1. Missing following 'restore' related retention attributes in a restore class retention pin. restore_action restore_condition restore_edge_type
2. Missing following 'save' or 'restore' related retention attributes in a save_restore class retention pin. save_action save_condition restore_action restore_condition restore_edge_type

The following example shows an instance where this message occurs: In the following example, this pin is a *save* class retention pin, but retention attributes *save_action* and *save_condition* are missing. To correct the problem, add both *save_action* and *save_condition* in this retention pin.

Feedback

```
cell(retention_dff) {
  ...
  retention_cell : "my_retention_dff";

  pin(SAVE) {
    retention_pin (save, 1);
    /* save_action : L; */
    /* save_condition : CK; */
  }

}
```

The following is an example message: The above example results in the following error message:

```
Warning: Line 192, Cell 'retention_dff', Pin 'SAVE', is a 'save' class
 retention pin but is missing all recommended attributes.(LBDB-982)
```

**What Next**

Resolve the issue by add the missing retention attributes in the retention pin.

---

## LBDB-983

(warning) Found inconsistency between attribute '%s' and retention pin disable value.

**Description**

Retention pin disable value and save_action/restore_action must be consistent as below: 1. For a save class retention pin When save_action is "L" or "R", disable value should be "1" When save_action is "H" or "F", disable value should be "0" 2. For a restore class retention pin When restore_action is "L", disable value should be "1" When restore_action is "H", disable value should be "0" When restore_action is "R", and restore_edge_type is "leading", disable value should be "0" When restore_action is "F", and restore_edge_type is "leading", disable value should be "1" When restore_action is "R", and restore_edge_type is "trailing" or missing, disable value should be "1" When restore_action is "F", and restore_edge_type is "trailing" or missing, disable value should be "0"

The following example shows an instance where this message occurs: In the following example, this pin is a *save* class retention pin, and *save_action* is 'L', but the disable value is not '1'. To correct the problem, change disable value to '1'.

```
cell(retention_dff) {
  ...
  retention_cell : "my_retention_dff";

  pin(SAVE) {
    retention_pin (save, 0);
    save_action : L;
```

Feedback

```
    save_condition : CK;
  }

}
```

The following is an example message: The above example results in the following error message:

```
Warning: Line 192, Cell 'retention_dff', Pin 'SAVE', Found inconsistency
 between attribute 'save_action' and retention pin disable
 value.(LBDB-983)
```

### What Next

Resolve the issue by correct the save_action/restore_action attribute or retention pin disable value.

---

## LBDB-984

(warning) Missing attribute required_condition under retention_condition group of the zero-pin retention cell, this cell will be black-box for multi-voltage functional flow. It is marked as dont_touch and dont_use.

### Description

```
For zero-pin retention cell, user must specify group retention_condition
and sub-attribute required_condition, otherwise the cell will be marked
as black-box.
This attribute is important to client tools in power opimization flow.
```

The following example shows an instance where this message occurs:

```
library(test) {
 ...
  cell (zpr) {
    ...
    latch(IQ1,IQN1) {
      enable : "!clk" ;
      data_in : "d" ;
      power_down_function : "!vcc+vss" ;
    }
    latch(IQ2,IQN2) {
      enable : "clk" ;
      data_in : "IQ1" ;
      power_down_function : "!vcc_in+vss" ;
    }

    retention_condition() {
      /* required_condition : "!clk"; */
      power_down_function : "!vcc+vss" ;
    }
```

Feedback

```
    ...
  }
}
```

The following is an example message:

```
Warning: Line 3, Cell 'zpr', Missing attribute required_condition under
 retention_condition group
of the zero-pin retention cell, this cell will be black-box for
 multi-voltage
functional flow. It is marked as dont_touch and dont_use. (LBDB-984)
```

### What Next

```
Add the missing retention_condition and sub-attribute required_condition
```

---

## LBDB-985

(error) The pin '%s' specified in required_condition attribute should be an input signal pin powered by backup power or backup ground.

### Description

```
For zero-pin retention cell, the pins specified in required_condition
 attribute
must be input signal pins powered by backup_power or backup_ground.

For other types of retention cell (like 1-pin or 2-pin), the pins
 specified in
required_condition attribute must be retention pins, or input signal pins
 powered by
backup_power or backup_ground.
```

The following example shows an instance where this message occurs:

```
library(test) {
 ...
  cell (zpr) {
    ...
    pin(clk) {
      direction : input ;
      related_ground_pin : VSS ;
      related_power_pin : VDD_BAK ;
      ...
    }

    pin(d) {
      direction : input ;
      related_ground_pin : VSS ;
      related_power_pin : VDD ;
      ...
```

Feedback

```
    }

    latch(IQ1,IQN1) {
      enable : "!clk" ;
      data_in : "d" ;
      power_down_function : "!VDD+VSS" ;
    }
    latch(IQ2,IQN2) {
      enable : "clk" ;
      data_in : "IQ1" ;
      power_down_function : "!VDD_BAK+VSS" ;
    }

    retention_condition() {
      required_condition : "!clk + d"; /* d is incorrect */
      power_down_function : "!VDD+VSS" ;
    }

    ...
  }
}
```

The following is an example message:

```
Error: Line 3, Cell 'zpr', The pin 'd' specified in required_condition
 attribute should be
 an input signal pin powered by backup power or backup ground. (LBDB-985)
```

### What Next

```
Add correct pins specified in required_condition under
 retention_condition group of the cell.
```

---

## LBDB-986

(warning) The '%s' pin '%s' specified in required_condition attribute is not using the '%s' value.

### Description

```
For zero-pin retention cell, the clock pin logic in required_condition
 must be
the disable value.
For zero-pin retention cell, the async preset/clear pin logic in
 required_condition
must be the inactive value.

For 2-pin retention cell, the "save" signal logic in required_condition
 must using
the disable value.
```

Feedback

```
For 1-pin retention cell, the "save_restore" signal logic in
 required_condition must using
the enable value.
```

The following example shows an instance where this message occurs:

```
library(test) {
 ...
  cell (zpr) {
    ...
    pin(clk) {
      direction : input ;
      related_ground_pin : vss ;
      related_power_pin : vcc_in ;
      ...
    }

    latch(IQ1,IQN1) {
      enable : "!clk" ;
      data_in : "d" ;
      power_down_function : "!vcc+vss" ;
    }
    latch(IQ2,IQN2) {
      enable : "clk" ;
      data_in : "IQ1" ;
      power_down_function : "!vcc_in+vss" ;
    }

    retention_condition() {
      required_condition : "clk"; /* "clk" is incorrect, should be "clk"
 */
      power_down_function : "!vcc+vss" ;
    }

    ...
  }
}
```

The following is an example message:

```
Warning: Line 3, Cell 'zpr', The 'clock' pin 'clk' specified in
 required_condition attribute is not using the 'disable' value.
 (LBDB-986)
```

### What Next

```
Add correct the pin logic specfied in required_condition of
 retention_condition.
```

---

## LBDB-987

(warning) Can't find '%s' in zero-pin retention cell.

Feedback

### Description

```
Zero-pin retention cell with master-slave latches must have an alive
 latch
to save data in the retention mode.

The alive latch is the latch contains backup power or backup ground
in its power_down_function.

For zero-pin retention cell modeled with ff grorup, the ff group also
 needs
power_down_function attribute with backup power or backup ground.
```

The following example shows an instance where this message occurs:

```
library(test) {
 ...
  cell (zpr) {
    ...
    pin(clk) {
      direction : input ;
      related_ground_pin : vss ;
      related_power_pin : vcc_in ;
      ...
    }

    latch(IQ1,IQN1) {
      enable : "!clk" ;
      data_in : "d" ;
      power_down_function : "!vcc+vss" ;
    }
    latch(IQ2,IQN2) {
      enable : "clk" ;
      data_in : "IQ1" ;
      power_down_function : "!vcc+vss" ;  /* should be "!vcc_in + vss" */
    }

    retention_condition() {
      required_condition : "!clk";
      power_down_function : "!vcc+vss" ;
    }

    ...
  }
}
```

The following is an example message:

```
Warning: Line 3, Cell 'zpr', Can't find 'alive latch' in zero-pin
 retention cell.
```

Feedback

### What Next

Add or correct the power_down_function in the ff/latch group.

---

## LBDB-988

(error) Can't specify "retention_pin" attribute in zero-pin retention cell.

### Description

Zero-pin retention cell is a retention cell with no retention pin.
User cannot specify "retention_pin" attribute on clock pin.

The following example shows an instance where this message occurs:

```
library(test) {
 ...
  cell (zpr) {
    ...
    pin(clk) {
      direction : input ;
      related_ground_pin : vss ;
      related_power_pin : vcc_in ;
      retention_pin(save_restore, 1); /* wrong attribute, should be
 removed */
      ...
    }

    latch(IQ1,IQN1) {
      enable : "!clk" ;
      data_in : "d" ;
      power_down_function : "!vcc+vss" ;
    }
    latch(IQ2,IQN2) {
      enable : "clk" ;
      data_in : "IQ1" ;
      power_down_function : "!vcc_in+vss" ;
    }

    retention_condition() {
      required_condition : "!clk";
      power_down_function : "!vcc+vss" ;
    }

    ...
  }
}
```

The following is an example message:

```
Error: Line 3, Cell 'zpr', Can't specify "retention_pin" attribute in
 zero-pin retention cell.
```

Feedback

### What Next

```
Remove the "retention_pin" attribute in this cell.
```

---

## LBDB-989

(error) the cell has bad antenna diode type.

### Description

This message means that value of attribute "antenna_diode_type" doesn't meet the following requirement.

(1) The cell is not an antenna cell. Antenna cell signal pin should be direction input or inout.

(2) If diode_cell_type is "power-ground", the signal pin must specify "antenna_diode_related_power_pins" and "antenna_diode_related_ground_pins".

If diode_cell_type is "power", the signal pin must specify "antenna_diode_related_power_pins".

If diode_cell_type is "ground", the signal pin must specify "antenna_diode_related_ground_pins".

The following is an example message:

```
Error: Line 1023, Cell 'ANTENNA', pin 'a', the cell has bad diode cell
 type. (LBDB-989)
```

### What Next

Correct either the signal pin or "antenna_diode_type" attribute.

---

## LBDB-989w

(warning) the antenna cell has more than one signal pin. It is marked as dont_touch and dont_use.

### Description

Antenna cell should have only one signal pin. This message means that this antenna cell has more than one signal pin, it is marked as dont_touch and dont_use, it can not be auto inferred by galaxy design tools (e.g. ICC1/ICC2 router).

The following is an example message:

```
Warning: Line 1023, Cell 'ANTENNA', the antenna cell has more than one
 signal pin. It is marked as dont_touch and dont_use. (LBDB-989w)
```

Feedback

### What Next

Correct either the signal pin or "antenna_diode_type" attribute.

---

## LBDB-990

(error) the pin has bad antenna diode type.

### Description

This message means that pin-level attribute "antenna_diode_type" doesn't meet the following requirement.

If antenna_diode_type is "power-ground", the signal pin must specify "antenna_diode_related_power_pins" and "antenna_diode_related_ground_pins".
If antenna_diode_type is "power", the signal pin must specify "antenna_diode_related_power_pins". If antenna_diode_type is "ground", the signal pin must specify "antenna_diode_related_ground_pins".

The following is an example message:

```
Error: Line 1023, Cell 'HM_MACRO', PIN 'Z', the pin has bad antenna diode
 type. (LBDB-990)
```

### What Next

Correct attribute "antenna_diode_type".

---

## LBDB-991

(error) attribute antenna_diode_type is defined in both cell and pin.

### Description

This message means the cell specifies attribute "antenna_diode_type", and one or more of its signal pins also specifies attribute "antenna_diode_type".

Cell-level attribute "antenna_diode_type" is used in a dedicated antenna diode cell, which basically has only one signal pin for static diffusion.

If a cell has a build-in diode port for static diffusion protection, this port should specify pin-level attribute "antenna_diode_type".

So cell-level attribute "antenna_diode_type" and pin-level attribute "antenna_diode_type" can't coexist in one cell.

The following is an example message:

```
Error: Line 1023, Cell 'ANTENNA', attribute "antenna_diode_type" is
 defined in both cell and pin. (LBDB-991)
```

Feedback

### What Next

Keep at most one "antenna_diode_type" attribute in this cell.

---

## LBDB-992

(error) The '%s' related retention attributes can't be specified in a %s retention pin.

### Description

This error message occurs in either of following situations: 1. user specifies following 'save/restore' related retention attributes in a non-retention pin. save_action save_condition restore_action restore_condition restore_edge_type 2. user specifies following 'save' related retention attributes in a restore class retention pin. save_action save_condition 3. user specifies following 'restore' related retention attributes in a save class retention pin. restore_action restore_condition restore_edge_type

Retention pin must have pin level attribute "retention_pin", and the retention pin class spcified in it must be consistent with retention attributes.

In the following example, this pin has *save_action* and *save_condition* attributes, but the retention pin class in *retention_pin* attribute is *restore* To correct the problem, either change the 'restore' to 'save' in the retention_pin attribute, or use 'restore' retention attribute instead of 'save' in this pin.

```
cell(retention_dff) {
  ...
  retention_cell : "my_retention_dff";

  pin(SAVE) {
    retention_pin (restore, 1);
    save_action : L;
    save_condition : CK;
  }

}
```

The above example results in the following error message:

```
Error: Line 192, Cell 'retention_dff', Pin 'SAVE', The 'save' related
 retention attributes
  can't be specified in a restore class retention pin. (LBDB-992)
```

### What Next

Resolve the issue by add/correct the *retention_pin* attribute, or use correct save/restore retetion attributes.

Feedback

## LBDB-993

(error) The pin '%s' specified in required_condition attribute should be a retention pin, or an input signal pin powered by backup power or backup ground.

### Description

```
For zero-pin retention cell, the pins specified in required_condition
 attribute
must be input signal pins powered by backup_power or backup_ground.

For other types of retention cell (like 1-pin or 2-pin), the pins
 specified in
required_condition attribute must be retention pins, or input signal pins
 powered by
backup_power or backup_ground.
```

The following example shows an instance where this message occurs:

```
library(test) {
 ...
  cell (nonzpr) {
    ...
    pin(save) {
      direction : input ;
      retention_pin(save, "1");
      related_ground_pin : VSS ;
      related_power_pin : VDD_BAK ;
      ...
    }

    pin(d) {
      direction : input ;
      related_ground_pin : VSS ;
      related_power_pin : VDD ;
      ...
    }

    latch(IQ1,IQN1) {
      enable : "!clk" ;
      data_in : "d" ;
      power_down_function : "!VDD+VSS" ;
    }
    latch(IQ2,IQN2) {
      enable : "clk" ;
      data_in : "IQ1" ;
      power_down_function : "!VDD_BAK+VSS" ;
    }

    retention_condition() {
      required_condition : "!save + d"; /* d is incorrect */
      power_down_function : "!VDD+VSS" ;
```

Feedback

```
    }

    ...
  }
}
```

The following is an example message:

```
Error: Line 3, Cell 'nonzpr', The pin 'd' specified in required_condition
 attribute should be
a retention pin, or an input signal pin powered by backup power or backup
 ground. (LBDB-993)
```

### What Next

```
Add correct pins specified in required_condition under
 retention_condition group of the cell.
```

---

## LBDB-994

(error) In attribute '%s', The ccsn noise stage referenced is insufficient for this timing.

### Description

The ccsn noise stage referenced(thereafter, "ccb") in active_output_ccb must be pull-up/down compatible with the timing. The valid ccb(s) with proper stage_type could be any one of the following:

1. when output of the timing could be rise and fall. (e.g. timing_type:"combinational")

1. one ccb with stage_type:both

2. two ccb's, one with stage_type:pull_up, the other with stage_type:pull_down

    a. when output of the timing always rise. (e.g. timing_type:"combinational_rise")

3. one ccb with stage_type:both

4. one ccb with stage_type:pull_up

    a. when output of the timing always fall. (e.g. timing_type:"combinational_fall")

5. one ccb with stage_type:both

6. one ccb with stage_type:pull_down

The following example shows an instance where this message occurs:

```
library(test) {
    cell(INV) {
      pin(X) {
        direction : "output" ;
        ...
```

Feedback

```
        output_ccb(o_rise) {
          stage_type : pull_up ;
          ...
        }
        timing() {
          timing_type : "combinational" ;
          ...
          active_output_ccb(o_rise);
        }
      }
    }
  }
```

The following is an example message:

```
Error: Line 3401, Cell 'INV', pin 'X', In attribute 'active_output_ccb',
 The
ccsn noise stage referenced is insufficient for this timing. (LBDB-994)
```

### What Next

Provide the valid ccb(s) accordingly.

---

## LBDB-995

(error) it's not allowed to specify both intrinsic_capacitance and total_capacitance in macro cell intrinsic_parasitic group.

### Description

If cell is macro cell, user can specify only intrinsic_capacitance or total_capacitance, but not both.

The following example shows an instance where this message occurs:

```
cell (my_macro_cell) {
  ...
  power_cell_type : macro;
  intrinsic_parasitic() {
    when : "A1";
    total_capacitance(V2) {
       value : 9.0;
    }
    intrinsic_capacitance(G1) {
       value : 62.2;
    }
  }
}
```

Feedback

In this case, cell 'my_macro_cell' is a macro cell, both total_capacitance and intrinsic_capacitance are specified in intrinsic_parasitic. So error message LBDB-995 will be issued.

The following is an example message:

```
Error: Line 115, Cell 'my_macro_cell', it's not allowed to specify
 both intrinsic_capacitance and total_capacitance in macro cell
 intrinsic_parasitic group. (LBDB-995)
```

**What Next**

Remove intrinsic_capacitance or total_capacitance group.

---

## LBDB-996

(error) The timing arc's fpga_arc_condition attribute contains \n \tinvalid format '%s.%s'.

**Description**

This message indicates that you specified invalid <pin_name>.<pin_status> or <fpga_cond_name>.<fpga_cond_value_name> in the fpga_arc_condition attribute of the timing arc.

The following example shows an instance where this message occurs:

```
cell(test) {

  ...
  pin(A) {
        ...
  }
  pin(B) {
        ...
        timing() {
                ...
                fpga_arc_condition: "wrong_pin.CONNECTED";
                ...
        }
        ...
  }
  ...
}
```

We can correct it by replace "wrong_pin" with "A".

The following is an example message:

```
Error: Line 206, The timing arc's fpga_arc_condition attribute contains
        invalid format 'wrong_pin.CONNECTED'. (LBDB-996)
```

Feedback

### What Next

Modify <A>.<B> to make it as a valid <pin_name>.<pin_status> or <fpga_cond_name>.<fpga_cond_value_name>.

---

## LBDB-997

(warning) The fpga_condition(%s) group is defined\n \tmultiple times.

### Description

This message indicates that you specified the same fpga_condition group multiple times. Only the last one is retained.

The following example shows an instance where this message occurs:

```
cell(CGNP) {
  area : 1;

  fpga_condition(cond1) {
    ...
  }
  fpga_condition(cond1) {
    ...
  }
  ...
}
```

The following is an example message:

```
Warning: Line 206, The fpga_conditino(cond1) group is defined
        multiple times. (LBDB-997)
```

### What Next

Change the group name, or delete the duplicated group.

---

## LBDB-998

(error) The fpga condition '%s' has no values defined.

### Description

A fpga_condition group must have at least one fpga condition value defined.

The following example shows an instance where this message occurs:

```
cell(CGNP) {
  area : 1;
  fpga_condition(cond1) {
  }
```

Feedback

```
  ...
}
```

The following is an example message:

```
Error: Line 206, The fpga condition 'cond1' has no values defined.
 (LBDB-998)
```

### What Next

Define the fpga condition values.

---

## LBDB-999

(warning) The fpga_condition_value(%s) group is defined\n \tmultiple times.

### Description

This message indicates that you specified the same fpga_condition_value group multiple times. Only the last one is retained.

The following example shows an instance where this message occurs:

```
cell(CGNP) {
  area : 1;

  fpga_condition(cond1) {
    fpga_condition_value(normal) {
      fpga_arc_condition : "operation_mode=normal";
    }
    fpga_condition_value(normal) {
      fpga_arc_condition : "operation_mode=arithmetic";
    }
  }
}
```

The following is an example message:

```
Warning: Line 206, The fpga_condition_value(normal) group is defined
        multiple times. (LBDB-999)
```

### What Next

Change the group name, or delete the duplicated group.

---

## LBDB-1000

(error) The fpga_arc_condition attribute is not specified\n \tfor the fpga_condition_value.

Feedback

### Description

This message indicates that you have not specified fpga_arc_condition attribute within fpga_condition_value.

The following example shows an instance where this message occurs:

```
cell(CGNP) {
  area : 1;
  fpga_condition(cond1) {
    fpga_condition_value(normal) {
      fpga_arc_condition : "operation_mode=normal";
    }
  }
  ...
}
```

The following is an example message:

```
Error: Line 206, The fpga_arc_condition attribute is not specified for
 the
        fpga_condition_value. (LBDB-1000)
```

### What Next

Add the missing fpga_arc_condition attribute.

---

## LBDB-1001

(error) Found one or more cells defined before '%s' group.

### Description

This message indicates the named group in phys_library is not defined before all cell groups.

The following example shows an instance where this message occurs:

```
phy_library(test) {
        ...
        macro (cell1) {
          ...
        }
        resource ( std_cell ) {
          ...
        }
    }
```

The following is an example message:

```
Error: Line 59, Found one or more cells defined before 'resource' group.
 (LBDB-1001)
```

Feedback

### What Next

Move this group inside the physical library so that it is defined in front of all cell groups.

---

## LBDB-1002

(error) The resource name is not supported.

### Description

Library Compiler accepts resource with the following names: - std_cell - array

This message indicates the resource group is using a name other than mentioned above.

The following example shows an instance where this message occurs:

```
phys_library(test) {
        ...
        resource ( wrong_name ) {               /* ERROR */
          ...
        }
    }
```

The following is an example message:

```
Error: Line 5, The resource name is not supported. (LBDB-1002)
```

### What Next

Change the resource group name to either "std_cell" or "array".

---

## LBDB-1003

(error) The resource group does not contain any layer definition.

### Description

Library Compiler requires that one or more layer definition be defined in the resource group. A layer definition can be one of the following functions / groups: - device_layer - contact_layer - overlap_layer - routing_layer

This message indicates there is none of the above defined in the resource group.

The following example shows an instance where this message occurs:

```
phys_library(test) {
        resource ( std_cell ) {          /* ERROR */
        }
        ...
    }
```

Feedback

The following is an example message:

```
Error: Line 3, The resource group does not contain any layer definition.
 (LBDB-1003)
```

**What Next**

Add layer information into resource group.

---

## LBDB-1004

(error) The resource group does not contain any site / tile definition.

**Description**

Library Compiler requires that one or more site / tile groups be defined in the resource group.

This message indicates there is neither any site nor any tile groups defined in the resource group.

The following example shows an instance where this message occurs:

```
phys_library(test) {
        resource ( std_cell ) {          /* ERROR */
        }
        ...
    }
```

The following is an example message:

```
Error: Line 3, The resource group does not contain any site / tile
 definition. (LBDB-1004)
```

**What Next**

Add site information into resource group.

---

## LBDB-1005

(warning) Found a duplicate %s attribute. Using the first value.

**Description**

This message indicates an attribute has been defined twice. The second definition is ignored.

The following example shows an instance where this message occurs:

```
phys_library(test) {
        ...
```

Feedback

```
        macro(and2) {
          ...
        }
        macro(and2a) {
          ...
        }
        macro(and2a1) {
          ...
          eq_cell : and2a;
          eq_cell : and2;
          ...
        }
    }
```

The following is an example message:

```
Warning: Line 34, Found a duplicate eq_cell attribute. Using the first
 value. (LBDB-1005)
```

**What Next**

Remove the duplicate definition of the attribute.

---

## LBDB-1006

(error) Cannot accept the '%s' pin as the logically equivalent pin.

**Description**

This message indicates the eq_pin attribute value is an unacceptable pin name. An unacceptable eq_pin is defined as either - a name equivalent to the current pin name; or - a pin that is not already defined.

The following example shows an instance where this message occurs:

```
phys_library(test) {
        ...
        macro(and2) {
          ...
          pin(a) {
            eq_pin : a1;        /* error */
            ...
          }
          pin(a1) {
          }
        }
    }
```

The following is an example message:

```
Error: Line 34, Cannot accept the 'a1' pin as the logically equivalent
 pin. (LBDB-1006)
```

Feedback

### What Next

Check the pin name and change it. Also make sure the pin specified in the eq_pin attribute is pre-defined.

---

## LBDB-1007

(error) The %s name '%s' is an undefined layer name.

### Description

This message indicates the obs / geometry group name is not a legal layer name.

The following example shows an instance where this message occurs:

```
phys_library(test) {
        ...
        macro(and2) {
          ...
          pin(a) {
            ...
            obs(new_layer) {    /* error */
              ...
            }
          }
        }
    }
```

The following is an example message:

```
Error: Line 34, The obs name 'new_layer' is an undefined layer name.
 (LBDB-1007)
```

### What Next

Use a pre-defined layer name as the name for the obs / geometry group.

---

## LBDB-1008

(error) The %s function has inconsistent number of arguments.

### Description

This message indicates the path / polygon / rectangle function has wrong number of arguments. In Library Compiler, - a path function requires odd number of arguments no less than 3; - a polygon function requires even number of arguments no less than 8; - a rectangle function requires 4 arguments.

Feedback

The following example shows an instance where this message occurs:

```
phys_library(test) {
        ...
        macro(and2) {
          ...
          pin(a) {
            ...
            obs(new_layer) {
              rectangle(0.1, 0.3);  /* error */
            }
          }
        }
    }
```

The following is an example message:

```
Error: Line 34, The rectangle function has inconsistent number of
 arguments. (LBDB-1008)
```

### What Next

Make sure the right argument is used for each function.

---

## LBDB-1009

(error) Non-positive number found in the size function.

### Description

This message indicates either the width or the height of the size function is a non-positive number.

The following example shows an instance where this message occurs:

```
phys_library(test) {
        ...
        macro(and2) {
          ...
          size (-1.5, 2.0);      /* error */
        }
    }
```

The following is an example message:

```
Error: Line 34, Non-positive number found in the size function.
 (LBDB-1009)
```

### What Next

Make sure both width or height of the size function are positive numbers.

Feedback

## LBDB-1010

(error) The site '%s' is not defined in the resource group.

### Description

This message indicates name of the site attribute is not a pre-defined site.

The following example shows an instance where this message occurs:

```
phys_library(test) {
        ...
        macro(and2) {
          ...
          site : new_site;      /* error */
        }
    }
```

The following is an example message:

```
Error: Line 34, The site 'new_site' is not defined in the resource group.
 (LBDB-1010)
```

### What Next

Make sure the site attribute uses a pre-defined site.

## LBDB-1011

(error) Cannot accept the '%s' macro as the equivalent cell.

### Description

This message indicates the eq_cell attribute value is an unacceptable macro name. An unacceptable macro name is defined as either - a name equivalent to the current macro name; or - a macro that is not already defined.

The following example shows an instance where this message occurs:

```
phys_library(test) {
        ...
        macro(and2a1) {
          ...
          eq_cell : and2;               /* error */
          ...
        }
        macro(and2) {
          ...
        }
    }
```

Feedback

The following is an example message:

```
Error: Line 34, Cannot accept the 'and2' macro as the electrically
 equivalent cell. (LBDB-1011)
```

### What Next

Check the macro name and change it. Also make sure the macro specified in the eq_cell attribute is pre-defined.

---

## LBDB-1012

(error) The layer name, '%s', in the '%s' group is not unique.

### Description

Each via group can specify up to 3 geometry groups with the 'layer1', 'contact' and 'layer2' groups. The name of each group gives the name of the layer on which all geometry shapes are defined. The 3 geometry groups must be defined on different layers. This error message indicates that 2 or more geometry groups are specified on the same layer.

The following example shows an instance where this message occurs:

```
phys_library(test) {
        ...
        resource(std_cell) {
          ...
          via (test) {
            ...
            layer1(met1) {
                ...
            }
            contact(con1) {
                ...
            }
            layer2(met1) {      /* error */
                ...
            }
          ...
          }
        }
    }
```

The following is an example message:

```
Error: Line 34, The layer name, 'met1', in the 'layer2' group is not
 unique. (LBDB-1012)
```

Feedback

### What Next

Check the layer names on the 'layer1', 'contact' and 'layer2' groups and make sure they call for different layer names.

---

## LBDB-1013

(error) The %s function has inconsistent number of arguments.

### Description

This message indicates the via / via_iterate function has wrong number of arguments. In Library Compiler, - a via function requires 3 arguments: a name followed by x / y coordinates; - a via_iterate function requires 7 arguments: 2 integers (for x-iteration and y-iteration), 2 real numbers (for x-spacing and y-spacing), a name and another 2 real numbers for the first x / y coordinates.

The following example shows an instance where this message occurs:

```
phys_library(test) {
        ...
        macro(and2) {
          ...
          pin(a) {
            ...
            obs(new_layer) {
              via(0.1, 0.3);  /* error */
            }
          }
        }
    }
```

The following is an example message:

```
Error: Line 34, The via function has inconsistent number of arguments.
 (LBDB-1013)
```

### What Next

Make sure the right argument is used for each function.

---

## LBDB-1014

(error) The via group does not have a 'contact' group.

### Description

Each via group is required to have a 'contact' group defined for the geometry in the contact layer. This error message indicates that the 'contact' group is missing from the current via group.

Feedback

The following example shows an instance where this message occurs:

```
phys_library(test) {
        ...
        resource(std_cell) {
          ...
          via (test) {  /* error */
            layer1(met1) {
                ...
            }
            layer2(met2) {
                ...
            }
          ...
          }
        }
    }
```

The following is an example message:

```
Error: Line 34, The via group does not have a 'contact' group.
 (LBDB-1014)
```

**What Next**

Make sure the 'contact' group is defined in current 'via' group.

---

## LBDB-1015

(error) One or more rectangles are required in this group.

**Description**

The 'layer1', 'contact' and 'layer2' sub-groups inside a 'via' group define the geometry shapes for the via. This error message indicates that the current sub-group has no rectangle specified.

The following example shows an instance where this message occurs:

```
phys_library(test) {
        ...
        resource(std_cell) {
          ...
          via (test) {
            layer1(con1) { }    /* error */
            ...
          }
        }
    }
```

Feedback

The following is an example message:

```
Error: Line 34, One or more rectangles are required in this group.
 (LBDB-1015)
```

### What Next

Make sure there are at least 1 rectangle defined in the current sub-group. Otherwise, remove the sub-group altogether.

---

## LBDB-1016

(error) The routing_grid function has inconsistent number of arguments.

### Description

This message indicates the routing_grid function has wrong number of arguments. In Library Compiler, - a routing_grid function requires 6 arguments: x-starting-coordinate, number-of-columns, x-space, y-start-coordinate, number-of-rows, y-space.

The following example shows an instance where this message occurs:

```
phys_library(test) {
      ...
      array(ABC) {
        routing_grid(100, 200);
        ...
      }
      ...
    }
```

The following is an example message:

```
Error: Line 34, The routing_grid function has inconsistent number of
 arguments. (LBDB-1016)
```

### What Next

Make sure the right arguments are used for each function.

---

## LBDB-1017

(error) The floorplan group contains mismatching site_array group.

### Description

This message indicates that either: - the default floorplan (with no given name) is missing site_array with "regular" placement_rule; or - the non-default floorplan contains site_array with "regular" placement_rule.

Feedback

The default floorplan should contain one or more "regular" site_array groups. The non-default floorplan should not contain any "regular" site_array group.

The following example shows an instance where this message occurs:

```
phys_library(test) {
      ...
      array(ABC) {
        floorplan() {
          site_array(new) {
            placement_rule : can_place;
            ...
          }
        }
        ...
      }
      ...
    }
```

The following is an example message:

```
Error: Line 34,  The floorplan group contains mismatching site_array
 group.  (LBDB-1017)
```

**What Next**

Check all site_array groups under the current floorplan and make sure the above rules are followed.

---

## LBDB-1018

(warning) The floorplan group does not contain can_place/cannot_occupy\n \t site_array groups.

**Description**

This message indicates that the current floorplan group only contain site_array groups with "regular" placement_rule.

The following example shows an instance where this message occurs:

```
phys_library(test) {
      ...
      array(ABC) {
        floorplan() {
          site_array(new) {
            placement_rule : regular;
            ...
          }
        }
        ...
      }
```

Feedback

```
      ...
    }
```

The following is an example message:

```
Warning: Line 34,  The floorplan group does not contain
 can_place/cannot_occupy
                site_array groups.  (LBDB-1018)
```

**What Next**

Make sure this is the correct physical library technology data.

---

## LBDB-1019

(error) The array group does not contain base floorplan.

**Description**

This message indicates that the current array group contain no base floorplan data. An array group must have a base floorplan group - i.e. a floorplan group defined with no given name.

The following example shows an instance where this message occurs:

```
phys_library(test) {
      ...
      array(ABC) {
        floorplan(NEW) {
            ...
        }
      }
        ...
      }
      ...
    }
```

The following is an example message:

```
Error: Line 34,  The floorplan group does not contain base floorplan.
 (LBDB-1018)
```

**What Next**

Add base floorplan data into this array group.

---

## LBDB-1020

(error) The '%s' function has inconsistent number of arguments.

Feedback

### Description

This message indicates the iterate function in the current site_array has wrong number of arguments. In Library Compiler, - a iterate function requires 4 numerical arguments.

The following example shows an instance where this message occurs:

```
phys_library(test) {
      ...
      array(ABC) {
        site_array(test) {
          iterate (100, 200);
          ..
        }
        ...
      }
      ...
    }
```

The following is an example message:

```
Error: Line 34, The iterate function has inconsistent number of
 arguments. (LBDB-1020)
```

### What Next

Make sure the right arguments are used for each function.

---

## LBDB-1021

(error) The resource group does not contain any via definition.

### Description

Library Compiler requires that one or more via groups be defined in the resource group.

This message indicates there is no via groups defined in the resource group.

The following example shows an instance where this message occurs:

```
phys_library(test) {
        resource ( std_cell ) {          /* ERROR */
        }
        ...
    }
```

The following is an example message:

```
Error: Line 3, The via group does not contain any site definition.
 (LBDB-1021)
```

Feedback

### What Next

Add via information into resource group.

---

## LBDB-1022

(error) The resource group must be defined before the\n \ttopological_design_rules group.

### Description

Library Compiler requires all physical library plib file to specify the resource group before specifying the topological_design_rules group.

This message indicates there is no resource group defined before the topological_design_rules group.

The following example shows an instance where this message occurs:

```
phys_library(test) {
        topological_design_rules ( ) {          /* ERROR */
        }
        ...
    }
```

The following is an example message:

```
Error: Line 2, The resource group must be defined before the
               topological_design_rules group.  (LBDB-1022)
```

### What Next

Add or move the resource group to the front of the topological_design_rules group.

---

## LBDB-1023

(warning) No '%s' group defined in the topological_rules group.

### Description

There is no specified group information in the current library.

The following example shows an instance where this message occurs:

```
phys_library(test) {
      resource(std_cell) {
        ...
      }
      topological_rules() {
      }
      ...
    }
```

Feedback

The following is an example message:

```
Warning: Line 34,  No via_rule group defined in the topological_rules
 group. (LBDB-1023)
```

**What Next**

Make sure the library is correct.

---

## LBDB-1024

(error) The %s function has inconsistent number of arguments.

**Description**

This message indicates the min_generate_via_size function has wrong number of arguments. In Library Compiler, it requires 2 floating numbers in the min_generated_via_size function.

The following example shows an instance where this message occurs:

```
phys_library(test) {
        min_generated_via_size (0.1, 0.2, 0.1);
        ...
    }
```

The following is an example message:

```
Error: Line 2, The min_generated_via_size function has inconsistent
 number of arguments. (LBDB-1024)
```

**What Next**

Make sure the right argument is used for the function.

---

## LBDB-1025

(error) The '%s' function cannot be supplied a \n \tnonpositive value.

**Description**

This message indicates that the specified function cannot have a nonpositive value.

The following example shows an instance where this message occurs:

```
min_generate_via_size(-1, 0.3);
```

The following is an example message:

```
Error: Line 18, The 'min_generate_via_size' function cannot be supplied a
       nonpositive value. (LBDB-1025)
```

Feedback

### What Next

Change all value of the function to positive in the physical library file.

---

## LBDB-1026

(error) The '%s' function has inconsistent arguments.

### Description

This message indicates the same_net_min_spacing function has wrong arguments. In Library Compiler, it requires 2 layer names followed by a floating numbers and a boolean value (TRUE or FALSE) in the same_net_min_spacing function.

The following example shows an instance where this message occurs:

```
phys_library(test) {
        same_net_min_spacing (metal1, metal2, 0.1);
        ...
    }
```

The following is an example message:

```
Error: Line 2, The 'same_net_min_spacing" function has inconsistent
 arguments. (LBDB-1026)
```

### What Next

Make sure the right argument is used for the function.

---

## LBDB-1027

(error) The '%s' function has inconsistent arguments.

### Description

This message indicates the contact_spacing function has wrong arguments. In Library Compiler, it requires 2 floating numbers in the contact_spacing function.

The following example shows an instance where this message occurs:

```
phys_library(test) {
        ...
        contact_spacing (0.1, 0.2);
        ...
    }
```

The following is an example message:

```
Error: Line 158, The 'contact_spacing" function has inconsistent
 arguments. (LBDB-1027)
```

Feedback

### What Next

Make sure the right argument is used for the function.

---

## LBDB-1028

(error) The '%s' layer name is of invalid layer type. A %s name is expected.

### Description

This message indicates that you specified an layer name with wrong type for the current attribute or function. The name should be a routing_layer or contact_layer name as indicated in the error message.

The following example shows an instance where this message occurs:

```
contact_min_spacing(metal1, contact1, 0.1);
```

The following is an example message:

```
Error: Line 104, The 'metal1' layer name is of invalid layer type.  A
 contact_layer name is expected.  (LBDB-1028)
```

### What Next

Check your library to see if you have an error in either previous layer definition or the current attribute / function.

---

## LBDB-1029

(error) There is no rectangle defined in the contact_formula group.

### Description

There is no rectangle defined in the current contact_formula group. Library Compiler requires at least 1 rectangle being specified in each contact_formula group.

The following example shows an instance where this message occurs:

```
via_rule_generate(GEN1) {
      contact_formula(con1) {   /* error */
        routing_direction : horizontal;
        resistance : 0.2;
      }
    }
```

The following is an example message:

```
Warning: Line 34,  There is no rectangle defined in the  contact_formula
 group. (LBDB-1029)
```

Feedback

### What Next

Define the rectangle shape for the contact_formula group.

---

## LBDB-1030

(error) The '%s' attribute value is larger than the '%s' attribute value.

### Description

This message indicates that the 'min_wire_width' value is larger than 'max_wire_width' value. Library Compiler requires that the 'min_wire_width' is no larger than 'max_wire_width' value.

The following example shows an instance where this message occurs:

```
routing_layer_rule(met1) {
      min_wire_width : 2.0;
      max_wire_width : 1.0;
      ...
    }
```

The following is an example message:

```
Error: Line 18, The 'min_wire_width' attribute value is larger than the
 'max_wire_width' attribute value.  (LBDB-1030)
```

### What Next

Change all related values follow the requirement.

---

## LBDB-1031

(error) The wire_rule group is empty.

### Description

This message indicates the wire_rule group does not have any layer_rule / via group or same_net_min_spacing function defined. Library Compiler requires that the wire_rule group not be empty.

The following example shows an instance where this message occurs:

```
wire_rule(rule1) {
    }
```

The following is an example message:

```
Error: Line 34, The wire_rule group is empty. (LBDB-1031)
```

Feedback

### What Next

Make sure the the wire_rule group is not empty.

---

## LBDB-1032

(error) Cannot accept the '%s' pin as the must_join pin.

### Description

This message indicates the must_join pin attribute value is an unacceptable pin name. An unacceptable must_join is defined as either - a name equivalent to the current pin name; or - a pin that is not already defined.

The following example shows an instance where this message occurs:

```
phys_library(test) {
        ...
        macro(and2) {
          ...
          pin(a) {
            must_join : a1;     /* error */
            ...
          }
          pin(a1) {
          }
        }
    }
```

The following is an example message:

```
Error: Line 34, Cannot accept the 'a1' pin as the must_join pin.
 (LBDB-1032)
```

### What Next

Check the pin name and change it. Also make sure the pin specified in the must_join attribute is pre-defined.

---

## LBDB-1033

(error) The '%s' function has inconsistent arguments.

### Description

This message indicates the ranged_spacing function has wrong arguments. In Library Compiler, it requires 3 floating numbers in the ranged_spacing function.

Feedback

The following example shows an instance where this message occurs:

```
phys_library(test) {
        ranged_spacing (0.1, 0.2);
        ...
    }
```

The following is an example message:

```
Error: Line 2, The 'ranged_spacing" function has inconsistent arguments.
 (LBDB-1033)
```

### What Next

Make sure the right argument is used for the function.

---

## LBDB-1034

(warning) The '%s' %s group already exists and\n \tcannot be overwritten. The first definition will be used.

### Description

This message indicates that you multiply defined a site group. Library Compiler does not allow the overwriting of existing site groups.

The following example shows an instance where this message occurs:

```
site(IOsite) {
         ...
        }
        site(IOsite) {
         ...
        }
```

The following is an example message:

```
Error: Line 31, The 'IOsite' site group already exists and
       cannot be overwritten. The first definition will be used.
 (LBDB-1034)
```

### What Next

Delete the second specification of the site group.

---

## LBDB-1035

(error) The '%s' array must have '%d' entries.

Feedback

### Description

You receive this message because the specified array has a different size from the one expected.

Where *N* is the number of routing layers, the following criteria are set:

```
plate_cap array must have ((N * (N-1)) / 2) entries
overlap_wire_ratio must have (N * (N-1)) entries
adjacent_wire_ratio must have N entries
wire_ratio_x must have N entries
wire_ratio_y must have N entries
```

The following example shows an instance where this message occurs:

```
plate_cap( "0.05, 0.06, 0.03" );
```

The following is an example message: Error: Line 127, The 'plate_cap' array must have '6' entries. (LBDB-1035)

### What Next

Correct the values in the specified array.

---

## LBDB-1036

(error) The sum of values in the '%s' array attribute does not meet the requirement.

### Description

You receive this message because the sum of ratios does not match the specified criteria. The following criteria are set:

*overlap_wire_ratio*: the sum of value[i(N-1)] to value[i(N-1)+(N-2)] must not exceed 100.0 (100 percent); where *N* is the number of routing layers and *i* is the routing layer id in the range of i=0..N-1.

*adjacent_wire_ratio*: n/a (not applicable);

*wire_ratio_x*: the sum of all values must equal 100.0 (100 percent);

*wire_ratio_y*: the sum of all values must equal 100.0 (100 percent).

The following example shows an instance where this message occurs:

```
wire_ratio_x( "40.1, 20.0, 20.5" );
```

The following is an example message: Error: Line 127, The sum of values in the 'wire_ratio_x' array attribute does not meet requirement. (LBDB-1036)

Feedback

### What Next

Correct the values in the specified array.

---

## LBDB-1037

(error) Cannot accept the '%s' as the default_routing_wire_model value.

### Description

You receive this message because the default_routing_wire_model attribute value is an unacceptable routing_wire_model name, that is, one that is a routing_wire_model that has not already been defined.

The following example shows an instance where this message occurs:

```
phys_library(test) {
        ...
        resource(...) {
          ...
          default_routing_wire_model : model_1;          /* error */
          ...
          routing_wire_model(model_1) {
          ...
          }
        }
    }
```

The following is an example message: Error: Line 34, Cannot accept the 'model_1' as the default_routing_wire_model value. (LBDB-1037)

### What Next

Change the routing_wire_model name. Ensure that the routing_wire_model specified in the default_routing_wire_model attribute is predefined.

---

## LBDB-1038

(error) The plate_cap attribute is not predefined.

### Description

You receive this message because the *plate_cap* attribute is not predefined. It has not been defined before the routing_wire_model() group is defined.

The following example shows an instance where this message occurs:

```
phys_library(test) {
    ...
    resource(...) {
```

Feedback

```
        ...
        routing_wire_model(model_1) { /* error: LBDB-1038 */
        ...
        }
        plate_cap ("0.05, 0.04, 0.06");
    }
}
```

The following is an example message: Error: Line 34, The plate_cap attribute is not predefined. (LBDB-1038)

**What Next**

Ensure that *plate_cap* is defined before the routing_wire_model() group is defined.

---

## LBDB-1039

(error) The '%s' attribute has an out-of-range '%f' ratio.

**Description**

You receive this message because a percentage ratio in an array attribute is larger than 100.0 (100%) or less than 0 (0%); it has not been accepted.

The following example shows an instance where this message occurs:

```
phys_library(test) {
        ...
        resource(...) {
          ...
          routing_wire_model(model_1) {
            ...
            wire_ratio_x ("21.3, 305.4, 19.4"); /* error */
            ...
          }
        }
    }
```

The following is an example message: Error: Line 34, The 'wire_ratio_x' attribute has an out-of-range '305.4' ratio. (LBDB-1039)

**What Next**

Ensure that all values are between 0.0 (0%) and 100.0 (100%).

---

## LBDB-1040

(error) The %s function has an inconsistent number of arguments.

Feedback

### Description

You receive this message because the *obs_clip_box* or *keepout_clip_box* function has the wrong number of arguments. In Library Compiler, the specified command requires four floating numbers in the *obs_clip_box* or *keepout_clip_box* function.

The following example shows an instance where this message occurs:

```
keepout_clip_box (0.1, 0.2, 0.1);
        ...
```

The following is an example message: Error: Line 25 The keepout_clip_box function has an inconsistent number of arguments. (LBDB-1040)

### What Next

Ensure that the right argument is used for the specified function.

---

## LBDB-1041

(error) The %s function has inconsistent arguments.

### Description

You receive this message because the *gds2_layer_map* function has the wrong number or type of arguments. In Library Compiler, the specified function requires an integer, a string, and an integer in the *gds2_layer_map* function.

The following example shows an instance where this message occurs:

```
gds2_layer_map (0.1, METAL1, 1);
```

The following is an example message: Error: Line 30, The gds2_layer_map function has inconsistent arguments. (LBDB-1041)

### What Next

Ensure that the right argument is used for the function.

---

## LBDB-1042

(warning) The dist_conversion_factor value '%d' is not allowed. Using default value '%d'.

### Description

You receive this message because you specified a dist_conversion_factor value that either is not a multiple of 100 or may be a nonpositive number.

Feedback

The following example shows an instance where this message occurs:

```
dist_conversion_factor : 123;
```

The following is an example message: Warning: Line 85, The dist_conversion_factor value '123' is not allowed. Using default value '1000'. (LBDB-1042)

**What Next**

Examine the library source file, and correct the dist_conversion_factor value.

---

## LBDB-1043

(error) The gds2_conversion_factor value '%d' is not allowed.

**Description**

You receive this message because you specified a *gds2_conversion_factor* value that is not a multiple of 100s or may be a nonpositive number.

The following example shows an instance where this message occurs:
gds2_conversion_factor : -100;

The following is an example message: Warning: Line 85, The gds2_conversion_factor value '-100' is not allowed. (LBDB-1043)

**What Next**

Examine the library source file, and correct the *gds2_conversion_factor* value.

---

## LBDB-1044

(error) The process resource group must be defined after the \n \tresource group and the topological design rules group.

**Description**

This message indicates that there is no resource group and/or topological design rules group defined before the process resource group. Library Compiler requires all physical library plib files to specify the resource group and the topological_design_rules group before specifying the process_resource group.

The following example shows an instance where this message occurs:

```
phys_library(test) {
        process_resource (...) {                /* ERROR */
        }
        ...
    }
```

Feedback

The following is an example message:

```
Error: Line 2, The process_resource group must be defined after the
               resource group and the topological_design_rules group.
 (LBDB-1044)
```

### What Next

Add or move the resource group and the topological design rules group to the front of the process resource group.

---

## LBDB-1045

(error) The %s name '%s' is an undefined %s name.

### Description

This error message means that the specified process routing layer, process via, or process wire rule name was not predefined.

The following example shows an instance where this message occurs:

```
phys_library(test) {
        resource(...) {
          routing_layer (met1) {
          }
        }
        process_resource(...) {
          process_routing_layer(METAL1) { /* ERROR: METAL 1 is not
 defined.  Use met1 instead */
          }
        ...
        macro(and2) {
          ...
          pin(a) {
            ...
            obs(new_layer) {    /* error */
              ...
            }
          }
        }
    }
```

The following is an example message:

```
Error: Line 34, The process_routing_layer name 'METAL1' is an undefined
 routing_layer name. (LBDB-1045)
```

### What Next

Use the *read_lib* command and specify a predefined routing layer, via, or process wire rule.

Feedback

## LBDB-1046

(warning) The boundary_layer attribute is not defined in\n \t gds2_extraction_rules.

### Description

This warning message tells you that you used the *read_lib* command and specified the *boundary_layer* attribute, but the *boundary_layer* attribute is not defined in gds2_extraction_rules. In Library Compiler, the *boundary_layer* attribute specifies a layer ID for GDSII extraction to look for the designated macro boundary for the macro size in the PLIB file. When this attribute is missing, the macro size in the PLIB file is determined by the bounding box of all geometries.

The following is an example message: Warning: The boundary_layer attribute is not defined in gds2_extraction_rules. (LBDB-1046)

### What Next

Make sure you did not intend to specify the boundary layer. If you did intend to specify the boundary layer, define the *boundary_layer* attribute and reexecute the command.

### See Also

- read_lib

## LBDB-1047

(error) The '%s' function has inconsistent arguments.

### Description

This message indicates the lateral_oxide function has wrong arguments. In Library Compiler, it requires 2 floating numbers in the lateral_oxide function.

The following example shows an instance where this message occurs:

```
lateral_oxide (0.1, 0.2, 0.3);
```

The following is an example message:

```
Error: Line 2, The 'lateral_oxide" function has inconsistent arguments.
 (LBDB-1047)
```

### What Next

Make sure the right argument is used for the function.

Feedback

## LBDB-1048

(error) The '%s' permittivity value is less than 1.00.

### Description

The permittivity value must be at least 1.00. It also includes the second value in lateral_oxide() complex attribute.

The following example shows an instance where this message occurs:

```
lateral_oxide ( 3.9, 0.99 );
```

The following is an example message:

```
Error: Line 46, The 'lateral_oxide' permittivity value is less than 1.00.
  (LBDB-1048)
```

### What Next

Check your library and use correct permittivity value.

## LBDB-1049

(error) The '%s' thickness value is less than zero.

### Description

The thickness value must be positive. It also applies to the first value in lateral_oxide() complex attribute.

The following example shows an instance where this message occurs:

```
lateral_oxide ( -3.9, 3.99 );
```

The following is an example message:

```
Error: Line 46, The 'lateral_oxide' thickness value is less than zero.
 (LBDB-1049)
```

### What Next

Check your library and use correct thickness value.

## LBDB-1050

(error) There is no rectangle defined in the via_contact_layer group.

Feedback

### Description

There is no rectangle defined in the current via_contact_layer group. Library Compiler requires at least 1 rectangle being specified in each via_contact_layer group.

The following example shows an instance where this message occurs:

```
default_via_generate(GEN1) {
      via_contact_layer(con1) {          /* error */
        resistance : 0.2;
      }
    }
```

The following is an example message:

```
Warning: Line 34,  There is no rectangle defined in the
 via_contact_layer group. (LBDB-1050)
```

### What Next

Define the rectangle shape for the via_contact_layer group.

---

## LBDB-1051

(error) The %s attribute is obsolete.

### Description

This message indicates the named attribute is obsolete and need to be changed accordingly.

The following example shows an instance where this message occurs:

```
lateral_oxide_thickness : 0.5 ;
```

The following is an example message:

```
Error: Line 34, The lateral_oxide_thickness attribute is obsolete.
 (LBDB-1051)
```

### What Next

Refer to the Library Compiler User Guide for the newest attribute names.

---

## LBDB-1052

(error) The 'layer_antenna_factor' function has inconsistent arguments.

Feedback

### Description

This message indicates the layer_antenna_factor function has wrong arguments. In Library Compiler, it requires one layer name followed by a floating number in the layer_antenna_factor function.

The following example shows an instance where this message occurs:

```
phys_library(test) {
        ...
        layer_antenna_factor (metal1, metal2, 0.1);
        ...
    }
```

The following is an example message:

```
Error: Line 25, The 'layer_antenna_factor' function has inconsistent
 arguments. (LBDB-1052)
```

### What Next

Make sure the right argument is used for the function.

---

## LBDB-1053

(error) The '%s' group contains conflicting layer specification styles,\n \t'%s' and '%s'.

### Description

This message indicates the gds2 layer id definition styles are found. For example, you should either use obs_layer_map or obs_layer_id to specify geometry layer id and/or data type, but never use both functions in the same plib/pplib file.

The following example shows an instance where this message occurs:

```
phys_library(test) {
        ...
        gds2_extraction_rules() {
          obs_layer_map ( "14, 16, 18, 20", "07, 08, 09, 10");
          obs_layer_id (22, 0, 11, 0);
        ...
    }
```

The following is an example message:

```
Error: Line 25, The 'gds2_extraction_rules' group contains conflicting
 layer specification styles,
        'obs_layer_map' and 'obs_layer_id'. (LBDB-1053)
```

### What Next

Make sure only one style is used in layer id definition.

Feedback

## LBDB-1054

(warning) The port '%s' does not have the attribute '%s' specified. The value (%f, %f) will be assigned to the attribute.

### Description

You receive this error message when the *rise_capacitance* and *fall_capacitance* attributes are specified for a port, but the *rise_capacitance_range* and *fall_capacitance_range* attribute are not. (*rise_capacitance*, *rise_capacitance*) will be assigned to attribute *rise_capacitance_range\ and (fall_capacitance, fall_capacitance) will be assigned to attribute fall_capacitance_range respectively.*

### What Next

If you accept the value assigned to the attribute referenced in the error message, no action is required on your part. If not, assign a value to the attribute.

## LBDB-1055

(error) The '%s' function has inconsistent arguments.

### Description

This message indicates the contact_min_spacing function, min_overhang function or diff_net_min_spacing function has wrong arguments. In Library Compiler, it requires 2 layer names followed by a floating numbers in the contact_min_spacing function, min_overhang function or diff_net_min_spacing function.

The following example shows an instance where this message occurs:

```
phys_library(test) {
        ...
        topological_design_rules() {
          contact_min_spacing (contact1, 0.5);
          min_overhang (metal1, contact1, 0.3);
          diff_net_min_spacing (metal1, contact1, 0.6);
          ...
        }
        ...
    }
```

The following is an example message:

```
Error: Line 245, The 'contact_min_spacing" function has inconsistent
 arguments. (LBDB-1027)
```

### What Next

Make sure the right argument is used for the function.

Feedback

## LBDB-1056

(warning) The '%s' value '%d' is not allowed. Using default value '%d'.

### Description

You receive this message because you specified a conversion factor value that either is not a multiple of 100 or may be a nonpositive number. A conversion factor can be - capacitance_conversion_factor, or - resistance_conversion_factor, or - current_conversion_factor, or - inductance_conversion_factor.

The following example shows an instance where this message occurs:

```
capacitance_conversion_factor : 123;
```

The following is an example message: Warning: Line 85, The 'capacitance_conversion_factor' value '123' is not allowed. Using default value '1000'. (LBDB-1056)

### What Next

Examine the library source file, and correct the conversion factor value.

## LBDB-1057

(error) Found milkyway_layer_map group defined before '%s' group.

### Description

This message indicates the named group in phys_library is not defined before milkyway_layer_map group.

The following example shows an instance where this message occurs:

```
phy_library(test) {
        ...
        milkyway_layer_group () {
         ...
        }
        resource ( std_cell ) {
          ...
        }
    }
```

The following is an example message:

```
Error: Line 59, Found milkyway_layer_group defined before 'resource'
 group. (LBDB-1001)
```

Feedback

### What Next

Move this group inside the physical library so that it is defined in front of milkyway_layer_map group.

---

## LBDB-1058

(error) The '%s' color name is invalid.

### Description

This message indicates that you specified an invalid color name for the current attribute or function. The color name should match a color name previously defined in the color groups.

The following example shows an instance where this message occurs:

```
color : foo ;
```

The following is an example message:

```
Error: Line 104, The 'foo' color name is invalid.  (LBDB-1058)
```

### What Next

Check your library to see if you have an error in either previous color definition or the current attribute / function.

---

## LBDB-1059

(error) The '%s' '%s' name is invalid.

### Description

This message indicates that you specified an invalid line_style/stipple name for the current attribute or function. The color name should match a line_style/stipple name previously defined in the line_style/stipple groups.

The following example shows an instance where this message occurs:

```
stipple : foo ;
```

The following is an example message:

```
Error: Line 104, The 'foo' 'stipple' name is invalid.  (LBDB-1059)
```

### What Next

Check your library to see if you have an error in either previous line_style/stipple definition or the current attribute / function.

Feedback

## LBDB-1060

(error) The tile '%s' is not defined in the resource group.

### Description

This message indicates name of the tile attribute is not a pre-defined tile.

The following example shows an instance where this message occurs:

```
phys_library(test) {
        ...
        macro(and2) {
          ...
          in_tile : new_tile;    /* error */
        }
    }
```

The following is an example message:

```
Error: Line 34, The tile 'new_tile' is not defined in the resource group.
 (LBDB-1060)
```

### What Next

Make sure the tile attribute uses a pre-defined tile.

## LBDB-1061

(error) The %s shape has zero or negative area.

### Description

This message indicates the path / polygon / rectangle function has zero or negative size. That is not a real geometrical shape.

The following example shows an instance where this message occurs:

```
phys_library(test) {
        ...
        macro(and2) {
          ...
          pin(a) {
            ...
            obs(new_layer) {
              rectangle(0.1, 0.3, 0.1, 0.3);  /* error */
            }
          }
        }
    }
```

Feedback

The following is an example message:

```
Error: Line 34, The rectangle shape has zero area. (LBDB-1061)
```

**What Next**

Make sure it is a real geometry.

---

## LBDB-1062

(error) The '%s' via name is invalid.

**Description**

This message indicates that you specified an invalid name for the current attribute or function. The via name should match a name previously defined in the resource group.

The following example shows an instance where this message occurs:

```
reference_cut_table( my_template ) {
        values ( "via12, via12a, via12b");
   }
```

The following is an example message:

```
Error: Line 104, The 'via12a' via name is invalid.  (LBDB-1062)
```

**What Next**

Check your library to see if you have an error in either previous layer definition or the current attribute / function.

---

## LBDB-1063

(warning) The argument value %d of the '%s' function\n \tis out of range. It is expected to be between 0 and 1.

**Description**

This message indicates that the out-of-range value has been set to the specified function.

```
pattern( 0, 1, 0, 1 );                /* correct */
```

The following example shows an instance where this message occurs:

```
pattern( 0, 20, 0, 1 );               /* incorrect */
```

The following is an example message:

```
Warning: Line 17, The argument value 20 of the 'pattern' function
        is out of range.  It is expected to be 0 or 1.  (LBDB-1063)
```

Feedback

### What Next

Change the physical library to correct the arguments.

---

## LBDB-1064

(error) Invalid attribute '%s' value '%d' is detected. The value must be\n \tunique among all '%s' objects.

### Description

This message indicates a non-unique id is found. - All via are required to have a unique via_id value. - All mask layers (device_layer, poly_layer, routing_layer, cont_layer) must have a unique mw_map id specified in its corresponding stream_layer group.

The following example shows an instance where this message occurs:

```
via (via12) {
    via_id : 12 ;
    ...
  }
  via (via23) {
    via_id : 12 ;
    ...
  }
```

The following is an example message:

```
Error: Line 56, Invalid attriute 'via_id' value '12' is detected.  The
 value must be
        unique among all 'via' objects. (LBDB-1064)
```

### What Next

Make sure the specified id value is unique among all named objects.

---

## LBDB-1065

(warning) No '%s' attribute has been specified for the\n \t%s '%s'. It is set to default value '%d'.

### Description

This message indicates that the library, perhaps an older PLIB file, does not contain the newly required object id's and default values are assigned to it. - All via are required to have a unique via_id value, from 1 to 255. - All mask layers (device_layer, poly_layer, routing_layer, cont_layer) must have a unique mw_map id specified in its corresponding stream_layer group. Datatype in mw_map is always default to 0.

Feedback

The following example shows an instance where this message occurs:

```
via (via12) {
    ...
  }
```

The following is an example message:

```
Warning: Line 56, No 'via_id' attribute has been specified for the
        via 'via12'.  It is set to default value '12'.  (LBDB-1065)
```

**What Next**

Enhance PLIB file to - specify a unique via_id for each via specification; - specify a corresponding stream_layer (with mw_map attribute) for each mask layer. These stream_layer should have unique mw_map layer id among them.

---

## LBDB-1066

(warning) Cannot locate corresponding layer id and layer datatype\n \tfor display_layer '%s'. It is set to layer id %d and layer datatype %d.

**Description**

This message indicates that the library, perhaps an older PLIB file, does not contain the newly required matching stream_layer object and mw_map(id, datatype) attribute values for all display_layer object. A unique mw_map(id, datatype) is automatically assigned to it.

The following example shows an instance where this message occurs:

```
display_layer (m1_pg) {
    ...
  }
```

The following is an example message:

```
Warning: Line 56,  Cannot locate corresponding layer id and layer
 datatype
for display_layer 'm1_pg'. It is set to layer id 23 and layer datatype
 '3'. (LBDB-1066)
```

**What Next**

Enhance PLIB file to - specify a corresponding stream_layer (with mw_map attribute) for each display layer. These stream_layer should have unique mw_map(id, datatype) value pair.

---

## LBDB-1067

(error) The '%s' attribute value(s) is out of range.\n

Feedback

### Description

This message indicates that the gds_map or mw_map attribute has out-of-range values.

The acceptable value ranges are:

id <= 255 datatype <= 255 where mw_map ( id , datatype ); gds_map ( id, datatype );

The following example shows an instance where this message occurs:

```
mw_map ( 23, 256 );
```

In this case, the datatype value 256 is out of range.

The following is an example message:

```
Error: Line 912, The 'mw_map' attribute' attribute value is out of range.
 (LBDB-1067)
```

### What Next

Change the attribute value to satisfy the value range.

---

## LBDB-1068

(warning) Floating-point value %s in '%s' attribute is too precise and\n \twill be truncated to %s.

### Description

This warning message occurs when the specified floating-point number cannot be completely stored in the database and must be rounded off.

When storing floating-point numbers in the database, the number is scaled by multiplying the floating-point number by dist_conversion_factor. Digits to the right of the decimal point are dropped. When nonzero digits are dropped, this message appears.

The following example shows an instance where this message occurs:

```
min_width : 0.1155 ;
```

The following is an example message:

```
Warning: Floating-point value 0.11550 in 'min_width' attribute is too
 precise and
         will be truncated to 0.115. (LBDB-1068)
```

### What Next

This is a warning message only. No action is required.

Feedback

However, if the result is not what you intended, you can increase the dist_conversion_factor or reduce the number of digits in the floating-point number, and then run the command again.

## LBDB-1069

(error) The %s object '%s' has a width %d and height %d\n \tthat is inconsistent with the pattern size %d.

### Description

This error message occurs when the pattern length of the specified section does not match the size specified by the pattern width and height. The number of elements in the pattern list must equal the product of the pattern width and height.

The following example shows an instance where this message occurs:

```
stipple ( S1 ) {
        width : 3 ;
        height : 2 ;
        pattern ( 1, 1, 0, 0 );
     }
```

The following is an example message:

```
Error: Line 67, Stipple object 'S1'  has a width 3 and height 2
        that is inconsistent with the pattern size 4. (LBDB-1069)
```

### What Next

Change the width and height attribute values or add or remove Boolean values to the pattern list. After making your changes, run the command again.

## LBDB-1070

(warning) Layer '%s' has a pitch %.4f that is less than the minimum spacing\n \tand width sum %.4f.

### Description

This warning message occurs when the specified Layer section contains an invalid pitch, minimum spacing or minimum width. The pitch must be greater than or equal to the sum of the layer's minimum spacing and minimum width.

The following example shows an instance where this message occurs:

```
layer ( M1 ) {
        min_width : 0.10 ;
        spacing : 0.12 ;
```

Feedback

```
        pitch : 0.15 ;
    }
```

The following is an example message:

```
Warning: Line 24, Layer 'M1' has a pitch 0.1500 that is less than the
 minimum spacing
and width sum 0.2200.  (LBDB-1069)
```

**What Next**

This is a warning message only. No action is required.

However, if the result is not what you intended, change the pitch, minimum spacing, or minimum width to meet the requirement that the pitch be greater than or equal to the sum of the layer's minimum spacing and minimum width.

---

## LBDB-1071

(warning) Routing_layer '%s' has invalid spacing_table value %.4f.

**Description**

This warning message occurs when the specified routing_layer contains an invalid spacing_table value. The spacing_table values must be equal to or greater than the routing_layer spacing value.

The following example shows an instance where this message occurs:

```
layer ( M1 ) {
        spacing : 0.12 ;
        spacing_table (1d_table) {
          values (" 0.1 , 0.15, 0.24" );
        }
    }
```

The following is an example message:

```
Warning: Line 235, Routing_layer 'M1' has invalid spacing_table value
 0.100. (LBDB-1071)
```

**What Next**

This is only a warning message. No action is required.

However, it is best practice to change the spacing_table value of the specified routing_layer so the values are equal to or greater than the layer spacing value. After making your changes, run the command again.

Feedback

## LBDB-1072

(error) Routing_layer '%s' has invalid spacing_table value %.4f.

### Description

This warning message occurs when the specified routing_layer contains an invalid spacing_table value. The spacing_table values must be equal to or greater than the routing_layer spacing value. And values[0] = spacing.

The following example shows an instance where this message occurs:

```
layer ( M1 ) {
        spacing : 0.12 ;
        spacing_table (1d_table) {
          values (" 0.1 , 0.15, 0.24" );
        }
     }
```

The following is an example message:

```
Error: Line 235, Routing_layer 'M1' has invalid spacing_table value
 0.100. (LBDB-1072)
```

### What Next

Change the spacing_table value so the value in the list are equal to or greater than the layer spacing value. After making your changes, run the command again.

## LBDB-1073

(warning) Layer '%s' has a pitch %.4f that does not match the recommended\n \twire-to-via pitch %.4f.

### Description

This warning message occurs when the pitch of the specified layer does not match the recommended pitch. The wire-to-via pitch is the recommended pitch and provides the best wire track resources for routing.

The wire-to-wire pitch is more aggressive, the via-to-via is more conservative and neither are recommended. The metal layer pitch must match the wire-to-via pitch in the routing direction.

The horizontal wire-to-via pitch is calculated as follows:

```
(min spacing) + (min width)/2 + (via metal width)/2
```

```
The vertical wire-to-via pitch is calculated as follows:
```

Feedback

```
    (min spacing) + (min width)/2 + (via metal height)/2
```

The via metal width and height are based on the larger of the lower and upper default contact codes.

The following example shows an instance where this message occurs:

```
layer ( metal3 ) {
        spacing : 1.9;
        min_width : 1.8;
        pitch : 4.5;
        ...
     }

     via (via23) {
        ...
        via_layer(metal3) {
          rectangle ( -1.1, -1.1, 1.1, 1.1 );
        }
     }
     via (via34) {
        ...
        via_layer(metal3) {
          rectangle  ( -1.1, -1.1, 1.1, 1.1 );
        }
     }
```

The following is an example message:

```
Warning : Line 36, Layer 'metal3' has a pitch 4.500 that does not match
 the recommended
        wire-to-via pitch 3.900. (LBDB-1073)
```

### What Next

This is only a warning message. No action is required.

You can leave the pitch of the specified layer at the current setting, or you can change the pitch of the layer to the recommended pitch.

---

## LBDB-1074

(warning) Layer '%s' has a pitch %.4f that is not aligned with\n \tbottom layer pitch %.4f.

### Description

This warning message occurs when the pitch of the specified layer is not double or triple the pitch of the lower metal layer with the same routing direction. Doubling or tripling the pitch value, maximizes routing resources and minimizes overlap with other metal layers.

Feedback

The following example shows an instance where this message occurs:

```
layer ( metal1 ) {
        pitch : 1.9;
        ...
     }
        ...
     layer ( metal3 ) {
        pitch : 3.9;
        ...
     }
```

The following is an example message:

```
Warning : Line 36, Layer 'metal3' has a pitch 3.900 that is not aligned
 with
        bottom layer pitch 1.900. (LBDB-1074)
```

**What Next**

This is only a warning message. No action is required.

You can either leave the pitch of the specified layer unchanged, or change the pitch to the doubled or tripled pitch and run the command again.

---

## LBDB-1075

(error) The attribute '%s' has %d entries, which exceeds the maximal\n \tallowed 8 entries.

**Description**

This error message occurs when the specified index array in routing_layer/cont_layer group that is too large. The index array size must be beween 1 and 8 inclusive.

The following example shows an instance where this message occurs:

```
layer ( metal1 ) {
        spacing_table ( 1d_stable ) {
          index_1 (0.0, 1.3, 2.4, 4.5, 6.6, 7.1, 9.3, 12.2, 25.0, 33.9);
        }
        ...
     }
```

The following is an example message:

```
Warning : Line 36, The attribute 'index_1' has 10 entries, which exceeds
 the maximal
        allowed 8 entries. (LBDB-1075)
```

**What Next**

Change the index array so that it does not exceed 8 entries. Then run the command again.

Feedback

## LBDB-1076

(warning) The attribute '%s' has invalid threshold values %.4f.

### Description

This warning message occurs when the specified table index contains an invalid threshold values. The threshold list must meet the following requirements:

- The first value must be between 0 and the default_routing_width inclusively.
- The second value must be greater than the layer min_width.

The following example shows an instance where this message occurs:

```
layer ( metal1 ) {
        min_width : 1.5 ;
        spacing_table ( 1d_stable ) {
          index_1 (0.0, 1.3, 2.4, 4.5, 6.6, 7.1, 9.3, 12.2, 25.0, 33.9);
          ...
        }
        ...
     }
```

The following is an example message:

```
Warning : Line 36, The attribute 'index_1' has invalid index value 1.3.
 (LBDB-1076)
```

### What Next

This is only a warning message. No action is required.

However, it is best practice to change the index values to meet the above requirements. After making your changes, run the command again.

## LBDB-1077

(error) The attribute '%s' has invalid threshold values %.4f.

### Description

This warning message occurs when the specified table index contains an invalid threshold values. The threshold list must meet the following requirements:

- The first value must be between 0 and the min_width inclusively.
- The second value must be greater than the layer min_width.

Feedback

The following example shows an instance where this message occurs:

```
layer ( metal1 ) {
        min_width : 1.5 ;
        spacing_table ( 1d_stable ) {
          index_1 (0.0, 1.3, 2.4, 4.5, 6.6, 7.1, 9.3, 12.2, 25.0, 33.9);
          ...
        }
        ...
     }
```

The following is an example message:

```
Error : Line 36, The attribute 'index_1' has invalid index value 1.3.
 (LBDB-1077)
```

### What Next

Change the index values to meet the above requirements. After making your changes, run the command again.

---

## LBDB-1078

(error) Layer '%s' has a default_routing_width value %.4f that is less than its\n \tmin_width value %.4f.

### Description

This error message occurs when the default width or minimum width of the specified Layer section is invalid. The default width (default_routing_width) of each layer must be greater than or equal to its minimum width (min_width).

The following example shows an instance where this message occurs:

```
layer ( metal1 ) {
        min_width : 1.5 ;
        default_routing_width : 1.3;
          ...
        }
        ...
     }
```

The following is an example message:

```
Error : Line 36, Layer 'metal1' has a default_routing_width value 1.3000
 that is less than its
        min_width value 1.5000. (LBDB-1078)
```

Feedback

### What Next

Change the default_routing_width or min_width of the specified Layer section so that the default_routing_width is greater than or equal to the min_width. After making your changes, run the command again.

---

## LBDB-1079

(warning) The attribute '%s' has rounded threshold value %.4f.

### Description

This warning message occurs when the threshold value from the index array contains rounded values. It is best practice to avoid ending threshold values with a 0 digit in the most precise position. This practice minimizes the confusion between tools that interpret the thresholds as exclusive, rather than inclusive, lower bounds.

The following example shows an instance where this message occurs:

```
layer ( metal1 ) {
        spacing_table ( 1d_spacing ) {
          index_1 ( "0.0, 1.500, 3.015" );
          ...
        }
        ...
     }
```

The following is an example message:

```
Warning : Line 36, The attribute 'index_1' has rounded threshold value
 1.5000. (LBDB-1079)
```

### What Next

This is only a warning message. No action is required.

However, you can either increase the rounded threshold value by one grid resolution value and run the command again, or you can leave the threshold value at the current setting.

---

## LBDB-1080

(warning) Attribute '%s' has insignificant differences between\n \tthreshold values %.4f and %.4f.

### Description

This warning message occurs when the index threshold contains insignificant differences between the threshold values. It is best practice to have thresholds differ by more than half of the min_width and more than ten percent of the greater threshold. If they do not differ by

Feedback

these amounts, then the table values can contain redundant data, be unnecessarily large, and increase router run-time dramatically.

The following example shows an instance where this message occurs:

```
layer ( metal1 ) {
        min_width : 1.5;
        spacing_table ( 1d_spacing ) {
          index_1 ( "0.0, 1.905, 1.915" );
          ...
        }
        ...
     }
```

The following is an example message:

```
Warning : Line 36, The attribute 'index_1' has insignificant differences
 between
        threshold values 1.9050 and 1.9150. (LBDB-1080)
```

**What Next**

This is only a warning message. No action is required.

However, you can check the table values for redundant data. If you find redundant data, then reduce the index dimension and the number of threshold values. After making your changes, rerun the command.

---

## LBDB-1081

(warning) Routing layer '%s' does not contain positive min_enclosed_area value.

**Description**

The value of the *minEnclosedArea* attribute for default metal layers should be greater than 0. Other values can affect the quality of the application results and can cause excessive design rule constraint violations during physical verification.

The following example shows an instance where this message occurs:

```
layer ( metal1 ) {
          ...
     }
```

The following is an example message:

```
Warning : Line 36, Routing layer 'metal1' does not contain positive
 min_enclosed_area value. (LBDB-1081)
```

**What Next**

This is only a warning message. No action is required.

Feedback

However, it is best practice to check the min_enclosed_area or min_enclosed_area_table attribute value of the specified layer, and then change the value to a value greater than 0, if necessary. After making your changes, run the command again.

## LBDB-1082

(error) The u_shaped_wire_spacing attribute value %.4f in routing layer '%s'\n \tis less than spacing value %.4f.

### Description

This message indicates that the u_shaped_wire_spacing value is less than the spacing value, which is not allowed. It is required that u_shaped_wire_spacing attribute value be no less than the spacing attribute value of the same routing layer.

The following example shows an instance where this message occurs:

```
layer ( metal1 ) {
          spacing : 1.4;
          u_shaped_wire_spacing : 1.0;
          ...
     }
```

The following is an example message:

```
Error : Line 45, The u_shaped_wire_spacing attribute value 1.0000 in
 routing layer 'metal1'
        is less than the spacing attribute value 1.400. (LBDB-1082)
```

### What Next

Change u_shaped_wire_spacing value so that it is equal or more than the spacing value. Then run the command again.

## LBDB-1083

(error) The same_net_min_spacing attribute value %.4f in routing layer '%s'\n \tis more than spacing value %.4f.

### Description

This message indicates that the same_net_min_spacing value is more than the spacing value, which is not allowed. It is required that same_net_min_spacing be no more than the spacing attribute value of the same routing layer.

The following example shows an instance where this message occurs:

```
layer ( metal1 ) {
          spacing : 1.4;
          same_net_min_spacing : 1.7;
```

Feedback

```
        ...
    }
```

The following is an example message:

```
Error : Line 45, The same_net_min_spacing attribute value 1.7000 in
 routing layer 'metal1'
       is more than the spacing attribute value 1.400. (LBDB-1083)
```

**What Next**

Change same_net_min_spacing value so that it is equal or less than the spacing value. Then run the command again.

---

## LBDB-1084

(warning) Tile '%s' has size %.4f that is not a multiple of\n \t%s routing layer pitch %.4f.

**Description**

This warning message occurs when: - in horizontal floorplan, tile width is not consistent with first vertical routing layer pitch; or - in vertical floorplan, tile height is not consistent with first horizontal routing layer pitch.

It is recommended that the tile dimension lines up with pitch to maximize utilization.

The following example shows an instance where this message occurs:

```
layer ( metal2 ) {
          routing_direction : vertical ;
          pitch : 1.4;
          ...
    }
       ...
    tile (unit) {
          size ( 1.6, 5.5 );
          ..
    }
```

The following is an example message:

```
Warning : Line 45, Tile 'unit' has size 1.6000 that is not a multiple of
        metal2 routing layer pitch 1.4000. (LBDB-1084)
```

**What Next**

This is a warning message only. No action is required.

However, it is best practice to either change the size value in question to the suggested pitch, or change the specified routing layer pitch to match tile dimension. After making your changes, run the command again.

Feedback

## LBDB-1085

(error) Cannot find via defined for contact layer '%s'.

### Description

This error message occurs when there is no via defined with a specific contact layer between 2 metal layers.

At least one via must be defined between 2 metal layers in a library.

The following is an example message:

```
Error : Cannot find via defined for contact layer 'via12'. (LBDB-1085)
```

### What Next

Define at least one via for the contact layer. Run the command again.

## LBDB-1086

(warning) The %s attribute value in via '%s' specifies\n \tnon-positive metal enclosure.

### Description

This warning message occurs when the specified via section contains non-positive metal enclosure dimensions or non-positive enclosure metal rectangle. If the non-positive metal enclosure values are set intentionally, this warning can be ignored. Otherwise, the values may cause the router to create enclosures that violate design rule constraints, such as completely enclosed spaces or donuts.

The following example shows an instance where this message occurs:

```
via ( via12 ) {
        via_layer( metal1 ) {
          enclosure ( 0, 0 );
        }
     }
```

The following is an example message:

```
Warning : Line 45, The enclosure attribute value in via 'via12' specifies
        non-positive metal enclosure. (LBDB-1086)
```

### What Next

This is a warning message only. No action is required.

However, if the zero metal enclosure values are not intentional, you can change either the rectangle coordinates or change enclosure value in via group to avoid potential design rule violations.

Feedback

## LBDB-1088

(warning) There are more than 255 layers in this library.

### Description

This warning message occurs when more than 255 layers are found in the current library. It does not include display layers. Currently Milkyway database can accommodate up to 255 layers. Upper layers exceeding the limit will be ignore in the database.

The following is an example message:

```
Warning : There are more than 255 layers in this library. (LBDB-1088)
```

### What Next

This is a warning message only. No action is required.

However, it is best practice to make sure that the total number of layers do not exceed the limit. After making your changes, run the command again.

## LBDB-1089

(warning) '%s' and '%s' layers in %s attribute are not\n \tadjacent layers.

### Description

This warning message occurs when the 2 layers specified in: - same_net_min_spacing - diff_net_min_spacing - contact_min_spacing - corner_min_spacing - min_overhang - min_enclosure - end_of_line_enclosure complex attributes are not adjacent layers.

It is recommended that the 2 layers must be adjacent routing or contact layers.

The following example shows an instance where this message occurs:

```
diff_net_min_spacing ( met1, met4, 1.4 );
```

The following is an example message:

```
Warning : Line 45, 'met1' and 'met3' layers in diff_net_min_spacing
 attribute are not
        adjacent layers. (LBDB-1089)
```

### What Next

This is a warning message only. No action is required.

However, it is best practice to follow the above guideline. After making your changes, run the command again.

Feedback

## LBDB-1090

(warning) Cannot find matching stream_layer for layer '%s'.

### Description

This warning message occurs when the layer is specified in resource layer section, but no stream_layer with identical name can be found in milkyway_layer_map group.

It is recommended that a stream_layer is defined for each layer found in resource group.

The following is an example message:

```
Warning : Cannot find matching stream_layer for layer 'metal1'.
 (LBDB-1090)
```

### What Next

This is a warning message only. No action is required.

However, it is best practice to add stream_layer definition for the specified layer. After making your changes, run the command again.

## LBDB-1091

(warning) Reset density value from %4.1f to %4.1f.

### Description

Density value defined in density_property in PLIB file should be between 0.0 and 100.0. Else, it will be reset to 0.0 or 100.0.

The following is an example message:

```
Warning: Line 12, Reset density value from 101.6 to 100.0. (LBDB-1091)
```

### What Next

Revise density value if the reset value is not expected.

## LBDB-1092

(warning) Layer '%s' attribute '%s' has a value of %g.

### Description

This warning message occurs when the value of the specified layer attribute may not be optimal.

Feedback

The values can affect the quality of the application results and can cause excessive design rule constraint violations during physical verification.

For a set of rule, completeness of rule parameters should be maintained:

For enclosed via min edge length rule, if enclosed_via_min_edge_length is specified, min_edge_length should be specified, and enclosed_via_min_edge_length > min_edge_length > 0

For U-shape rule, if you specify u_shaped_min_length or u_shaped_wire_spacing, you should also specify u_shaped_min_depth and min_spacing. And u_shaped_wire_spacing > min_spacing.

For multiple min area rule, "min_area" attribute is needed if "min_polygon_area_rule() -> polygon_min_area" is specified and min_polygon_area_rule()->polygon_min_area > min_area. If both "min_polygon_area_rule() -> polygon_min_area" and "min_polygon_area_rule() -> max_edge_length" are specified, "min_length" should be specified. And min_polygon_area_rule() -> max_edge_length >= min_length

For non-preferred direction routing rule, if you specify non_preferred_dir_width, it will satisfy non_preferred_dir_width > default_routing_width. If you specify both non_preferred_dir_width and non_preferred_dir_max_length, it will satisfy non_preferred_dir_max_length > non_preferred_dir_width-default_routing_width.

For min_spacing rule in routing_layer, if you specify min_spacing(x_min_spacing, y_min_spacing), it will satisfy x_min_spacing >= spacing and y_min_spacing >= spacing.

For jog wire corner to corner spacing rule, if you specify aligned_jog_width and unaligned_jog_diag_width, it will satisfy aligned_jog_width >= min_width and unaligned_jog_diag_width >= min_width.

For [x|y]_min_spacing_table, [x|y]_max_spacing_table, the first value, i.e., value[0], will satisfy value[0] >= spacing.

For [x|y]_max_spacing_check_range_table, the first value, i.e., value[0], will satisfy value[0] >= spacing*2 + min_width.

The following is an example message:

```
WARNING : Layer 'M1' attribute 'polygon_min_area' has a value of 0.03.
 (LBDB-1092)
```

**What Next**

This is only a warning message. No action is required.

However, it is best practice to check the attribute value of the specified layer, and then change the value to a value greater than 0, if necessary. After making your changes, run the command again.

Feedback

## LBDB-1093

(warning) Layer '%s' attribute '%s' is missing.

### Description

This warning message occurs when the specified section is missing an important attribute. If the attribute is not specified in PLIB file, the default value is 0 for number attributes, or an empty string for string attributes.

For multiple min area rule, "min_area" attribute is needed if "min_polygon_area_rule() -> polygon_min_area" is specified and min_polygon_area_rule()->polygon_min_area > min_area. If both "min_polygon_area_rule() -> polygon_min_area" and "min_polygon_area_rule() -> max_edge_length" are specified, "min_length" should be specified.

For U-shape rule, if you specify u_shaped_min_length or u_shaped_wire_spacing, you should also specify u_shaped_min_depth and min_spacing.

For enclosed via min edge length rule, if enclosed_via_min_edge_length is specified, min_edge_length should be specified, and enclosed_via_min_edge_length > min_edge_length > 0 If both enclosed_via_min_edge_length and min_edge_length are specified, enclosed_via_adjacent_edge_length should be specified.

For 2-line end rule, if end_to_end_min_spacing is specified, end_of_line_metal_max_width should be specified.

For end of line rule, if end_of_line_side_keepout_length is specified, end_of_line_metal_max_width should be specified.

For line end depth rule, if end_of_line_min_length is specified, end_of_line_metal_max_width should be specified.

The following is an example message:

```
WARNING : Layer 'M1' attribute 'min_area' is missing. (LBDB-1093)
```

### What Next

This is only a warning message. No action is required.

However, if you do not want to use the default value for the attribute, add the attribute with the desired value to the specified section in PLIB file and rerun the command.

## LBDB-1094

(error) The table '%s' can supports 'scalar' template only.

Feedback

### Description

This error message is reported for some table groups, that currently only a scalar value is supported for them by the tool. But found they are using a non-scalar template name.

Following two groups are under the restriction for now:

```
pgate_antenna_ratio(template_name) {
}
ngate_antenna_ratio(template_name) {
}
```

The following is an example message:

```
Error: Line 531, The table 'pgate_antenna_ratio' can supports 'scalar'
 template only. (LBDB-1094)
```

### What Next

Modify the plib file accordingly, change the template name to 'scalar' and specify only one value for the table.

Example:

```
antenna_rule("M1_ANTENNA_RULE")
    ...
    pgate_antenna_ratio("antenna_tempalte1") {
        index_1 ("0.0 1.0 2.0 3.0")
        values ("200 200 400 600");
    }
}
```

Modify the group to:

```
antenna_rule("M1_ANTENNA_RULE")
    ...
    pgate_antenna_ratio("scalar") {
        values ("200");
    }
}
```

---

## LBDB-1095

(error) The '%s' should coexist with '%s' in '%s' group.

### Description

This error message is reported when there is restriction that two tables/functions/attributes should coexist the group, but only one found.

Feedback

Example:

In antenna_rule group, the pgate_antenna_ratio table should coexist with ngate_antenna_ratio. If any of the other is not specified when one is present, the error is reported.

```
antenna_rule (M1_mode1_d4) {
    antenna_ratio ("antenna_tempalte") {
        index_1("0, 0.05 0.06 0.6");
        values("100 100 300 500");
    }
    pgate_antenna_ratio("scalar") {
        values("100");
    }
    layer_antenna_factor("M1", 1.0);
}
```

The following is an example message:

```
Error: Line 546, The 'pgate_antenna_ratio' should coexist with
 'ngate_antenna_ratio' in 'antenna_rule' group. (LBDB-1095)
```

**What Next**

Modify the plib file accordingly, specify both tables/attributes/functions in the group. or remove the existing one.

---

## LBDB-1096

(error) There is unsupported antenna data in the group.

**Description**

This error message is reported when there is unsupported antenna data found in the antenna_rule group. Mainly, the error is issued for following situations:

1. Usage of pgate_antenna_ratio and ngate_antenna_ratio
   pgate_antenna_ratio and ngate_antenna_ratio can't be supported when:
   antenna_accumulation_calculation_method : accumulative_ratio; with
   routing_layer_calculation_method : top_area|side_wall_area;

Note that when reading a plib file to Milkyway library, antenna_accumulation_calculation_method is default to accumulative_ratio and routing_layer_calculation_method is default to top_area;

So, if you want to define pgate_antenna_ratio and ngate_antenna_ratio in the antenna_rule group, you can't leave these two attributes undefined, otherwise, this error will be issued too.

Feedback

1. Usage of antenna area rule The antenna_rule group is defined as antenna area rule, when antenna_ratio_calculation_method == total_routing_area. And there is restrcitions to antenna_accumulation_calculation_method and geometry_calculation_method, according to current tool support. Following combinations are supported now, this error will be issued for other combinations. mode 1: calculate metal area ignore all lower layer segments. antenna_accumulation_cm = single_layer, geometry_cm = connected_only; mode 2: calculate metal area includes lower layer segments to input pins. antenna_accumulation_cm = accumulative_area, geometry_cm = connected_only; mode 3: calculate metal area includes all lower layer segments. antenna_accumulation_cm = accumulative_area, geometry_cm = all_geometries;

Besides above three calculation methods, the antenna_ratio should using scalar template and have only one value defined as max allowable antenna area, attribute max_diode_insertion_distance is used to define max allowable distance from inserted diode to gate.

other attributes and groups defined in antenna area rule are totally ignored during read_plib and not stored into database.

The following is an example message:

```
Error: Line 546, There is unsupported antenna data in the
 group.(LBDB-1096)
```

**What Next**

Modify the plib file accordingly to rule above.

---

## LBDB-1097

(error) Layer '%s' attribute '%s' and '%s' are different.

**Description**

x_min_spacing_table, y_min_spacing_table, x_max_spacing_table, y_max_spacing_table, x_max_spacing_check_range_table, y_max_spacing_check_range_table currently require the content in index_1 and index_2 are the same.

If both exist, the dimension and content in index_1[] in fat_wire_via_keepout_enclosure_table and in fat_wire_via_keepout_edge_table should be the same.

If both exist, the dimension and content in index_1[] in fat_wire_via_keepout_enclosure_table and in fat_wire_via_spacing_threshold_table should be the same.

Feedback

The following is an example message:

```
Error : Layer 'M1' attribute 'x_min_spacing/index_1' and
 'x_min_spacing/index_2' are different. (LBDB-1097)
```

**What Next**

Unify the different elements.

---

## LBDB-1098

(error) Layer '%s' has wrong '%s'.

**Description**

Values in x_legal_width and y_legal_width should satisfy, (1) the width values are in ascending order; (2) the first value >= min_width; (3) the last value <= max_width if max_width is specified; (4) default_routing_width is among the values.

The following is an example message:

```
Error : Layer 'M1' has wrong 'x_legal_width'. (LBDB-1098)
```

**What Next**

Re-specify x_legal_width or y_legal_width.

---

## LBDB-1099

(error) Layer '%s' '%s' and '%s' can't co-exist.

**Description**

fat_wire_via_keepout_edge_table and fat_wire_via_spacing_threshold_table can't co-exist in a cont_layer.

The following is an example message:

```
Error : Layer 'M1' 'fat_wire_via_keepout_edge_table' and
 'fat_wire_via_spacing_threshold_table' can't co-exist. (LBDB-1099)
```

**What Next**

Remove at least one of the two given attributes/functions/groups.

---

## LBDB-1100

(information) Dummy cell '%s' defined in the library.

Feedback

### Description

This info message is reported when there is dummy cell defined in an incremental plib file (that is, the library attribute is_incremental_library : true;)

A dummy cell is identified by following features: 1. The cell name has wildcard '*' used, and the '*' is the last character in the cell name, for example, cell name is "*", or "AND*". 2. There is no cell geometries (macro obs, pin port) defined in the cell.

A dummy cell is expected only in an incremental plib file, and it's used to update cell/pin attributes for FRAM view cells when read plib data to Milkyway database.

### What Next

Users do not need to do anything.

---

## LBDB-1101

(error) There is geometry data defined in dummy cell '%s'.

### Description

This warning message is reported when there is geometry data defined in the dummy cell.

A dummy cell is identified by following features: 1. The cell name has wildcard '*' used, and the '*' is the last character in the cell name, for example, "*" or "AND*". 2. There is no cell geometries (macro obs, pin port) defined in the cell.

A dummy cell is expected only in an incremental plib file, and it's used to update cell/pin attributes for a set of FRAM view cells (pattern matched cells) when read plib data to Milkyway database.

The following is an example message:

```
Error: Line 546, There is geometry data defined in dummy cell
 '*'.(LBDB-1101)
```

### What Next

Remove all geometry data from the cell (macro obs, pin port) if users want to use dummy cell to update cell/pin attributes for a set of cells. Otherwise, rename the cell name to avoid it match with dummy cell naming rule.

---

## LBDB-1111

(error) Incorrect number of arguments specified in '%s' attribute.

Feedback

### Description

This message indicates the number of arguments for an attribute does NOT meet one of the requirements: minimum/maximum number, even/odd number, multiples of certain number, etc.

The following example shows an instance where this message occurs:

```
library(test) {
        ...
        cell(testcell) {
          example(1, 2);            /* line 20 */
        }
    }
```

The requirement is the number of argument of 'example' attribute should be odd and bigger than 2.

The following is an example message:

```
Error: Line 20, Cell 'cell1', Incorrect number of arguments specified in
 'example' attribute. (LBDB-1111)
```

### What Next

Make sure the correct argument is used for the attribute.

---

## LBDB-1112

(error) Different number of arguments found in '%s' attributes declared under this %s group.

### Description

This message indicates this multiply declared attribute under the same parent group have different number of Arguments. The numbers of arguments should be identical among them.

The following example shows an instance where this message occurs:

```
library(test) {
        ...
        cell(testcell) {            /* line 19 */
          example(1, 2, 3);
          example(1, 2, 3, 4, 5);
        }
    }
```

The following is an example message:

```
Error: Line 19, Cell 'cell1', Different number of arguments found in
 'example' attributes declared under this 'cell' group. (LBDB-1112)
```

Feedback

### What Next

Make sure the correct argument is used for the attributes.

---

## LBDB-1150

(error) Invalid value in end_of_line_via_wire_rule.

### Description

Name the five values in end_of_line_via_wire_rule as layer1, layer2, end_of_line_via_wire_min_width, end_of_line_via_wire_min_spacing, end_of_line_via_ortho_wire_max_threshold, respectively.

Suppose layer1 is routing_layer, and layer2 is cont_layer. default_via is the default via on layer2. Taking min_width and min_spacing specification from layer1, and compute cut_width by the geometry of default_via.

If there is another design rule for this layer pair

min_enclosure (string, string, float ); /* layer1, layer2, min_enclosure */

Take min_enclosure from this specification. Otherwise let min_enclosure=0.

Values should satisfy, (1) end_of_line_via_wire_min_width > min_width (2) end_of_line_via_wire_min_width >= cut_width + 2* min_enclosure (3) end_of_line_via_wire_min_spacing >= min_spacing + min_enclosure (4) end_of_line_via_ortho_wire_max_threshold >= minWidth

The following is an example message:

```
Error : Invalid value in end_of_line_via_wire_rule. (LBDB-1150)
```

### What Next

Change the value to satisfy all the 4 restrictions.

---

## LBDB-1151

(error) "%s" has invalid value %f.

### Description

same_segment_center_min_spacing, same_net_center_min_spacing, diff_segment_center_min_spacing, diff_net_center_min_spacing in cont_layer() group or in topological_design_rules() group should satifsy, [same|diff]_[segment|net]_center_min_spacing >= min_cut_spacing + cut_width

The value of min_cut_spacing and cut_width is explained as follows.

Feedback

As for [same|diff]_[segment|net]_center_min_spacing in cont_layer() group, let "default_via" to be the default via of this cont_layer. Then min_cut_spacing is taken from contact_spacing specification. Cut_width is taken from the geometry of default_via.

As for [same|diff]_[segment|net]_center_min_spacing in topological_design_rules() group, we only check if both layer1 and layer2 are cont_layers. Let "default_via_1" to be the default via on layer1, "default_via_2" to be the default via on layer2. min_cut_spacing is taken from the function in topological_design_rule():

contact_min_spacing (layer1, layer2, min_cut_spacing)

Cut_width_1 and cut_width_2 are taken from the geometry of default_via_1 and default_via_2, respectively. Let cut_width = (cut_width_1+cut_width_2)/2.

The following is an example message:

```
Error : "diff_segment_center_min_spacing" has invalid value 0.12.
 (LBDB-1151)
```

**What Next**

Change the value to satisfy the restriction.

---

## LBDB-1152

(error) The '%s' attribute has an invalid sequence of\n \tdata '%g , %g'. The values must be in decreasing order.

**Description**

This message indicates that the set of data is not specified in decreasing order. Content in "values" in min_metal_spacing_table must be in descending order.

The following is an example message:

```
Error: Line 22, The 'values' attribute has an invalid sequence of
        data '1.200000 , 2.300000'. The values must be in decreasing
 order. (LBDB-1152)
```

**What Next**

Check your library and correct the order of the values.

---

## LBDB-1153

(error) '%s' has 0 or more than %d arguments.

Feedback

### Description

[x|y]_legal_width requires that the argument number is greater than 0 and less than or equal to 16.

The following is an example message:

```
Error: Line 22, 'x_legal_width' has 0 or more than 16 arguments.
 (LBDB-1153)
```

### What Next

Add or reduce arguments into the specified function.

---

## LBDB-1154

(error) There is no noise information in the library.

### Description

This information occurs when checking library for noise information, but there is no noise information in the library at all.

### What Next

If noise information is not needed, cancel the noise checking; If noise information is needed, add I-V characteristics, noise immunity data and noise propagation data.

---

## LBDB-1155

(error) Invalid %s found under section %s in map file.

### Description

In the map file, if any of the following is incorrect, (1) voltage_name (2) voltage value (3) power management attributes this message will occur.

Valid voltage_name should be a valid string that starts with a character a-z or A-Z. The voltage_name in PG_TO_VOLTAGE_MAP section should be one in VOLTAGE_MAP. It could be a rail name, and could be the same as pg_pin name. But never input an invalid value such as "-" or "*" in voltage_name field in PG_TO_VOLTAGE_MAP or VOLTAGE_MAP section. Valid voltage values are non-negative floating point numbers, for example, 0.9, that are valid voltage values in the input library. For valid power management attributes, please refer to related documents or user guide. The following are only a few examples: valid switch_cell_type is coarse_grain and fine_grain; valid level_shifter_type is HL, LH and HL_LH; always_on pins are related to backup power; a switch cell has VVDD+VDD or VVSS+VSS, and. VVDD has pg_function where VVDD is virtual VDD.

Feedback

The following example shows an instance where this message occurs:

```
BEGIN PG_TO_VOLTAGE_MAP
cell            pg_pin          voltage_name    pg_type
ADDFHX1         VDD             -       power
ADDFHX1         VSS             VSS             primary_ground
GEND PG_TO_VOLTAGE_MAP
```

The following is an example message:

```
Error: Line 14, Invalid voltage_name found under section
 PG_TO_VOLTAGE_MAP in map file. (LBDB-1155)
```

**What Next**

You should correct the invalid values and/or voltage_name in the map file. For example, in voltage_name field, if you enter "-", you will receive this message. You must input a valid voltage_name, e.g. VDD.

---

## LBDB-1156

(warning) private variable '%s' is enabled. Do not use the compiled DB in design flow.

**Description**

This message indicates that a pribate variable is enabled when compiling the DB. The compiled DB file may cause design flow fail.

**What Next**

Disable the private variable and recompile the DB file.

---

## LBDB-1157

(warning) The direction of pg_pin '%s' can only be input or inout when pg_type is '%s'!

**Description**

You receive this message when pg_type is primary_power or primary_ground but the pg_pin direction is output.

The following example shows an instance where this message occurs:

```
cell(lbdb1157) {
      area : 9;
      pg_pin(VDD) {
        direction : output ;
        pg_type : primary_power;
        voltage_name :VDD;
      }
      pg_pin(VSS) {
```

Feedback

```
        pg_type : primary_ground;
        voltage_name :VSS;
      }
    }
```

In this case, direction of pg_pin VDD is output but its pg_type is primary_power.

The following is an example message:

```
Warning: Line 848, The direction of pg_pin 'VDD' can only be input or
 inout when pg_type is 'primary_power'! (LBDB-1157)
```

**What Next**

Correct the direction or pg_type

---

## LBDB-1158

(error) Multiple '%s' groups found when same group with type 'both' is defined at line %d!

**Description**

You receive this message when more than one specified group are specified and the 'type' attribute in one of the group is 'both'.

The following example shows an instance where this message occurs:

```
cell(lbdb1158) {
    tap_cell_properties() {
        type : both;
        coverage_distance (23, 10) ;
        min_row_edge_distance : 12;
        coverage_pattern_left : left_full ;
        coverage_pattern_right : right_full;
    }
    tap_cell_properties() {
        type : nwell ;
        coverage_distance (10, 23) ;
        min_row_edge_distance : 16;
        coverage_pattern_left : left_full ;
        coverage_pattern_right : right_full;
  }
}
```

In this case, two tap_cell_properties are specified

The following is an example message:

```
Error : Line 1000, Multiple 'tap_cell_properties' groups found when same
 group with type 'both' is defined at line 1001! (LBDB-1158)
```

Feedback

### What Next

Remove one of the group.

---

## LBDB-1159

(error) The value of '%s' cannot be less than %d!

### Description

You receive this message when the value in the attribute is less than minimum allowed number.

The following example shows an instance where this message occurs:

```
cell(lbdb1158) {
    tap_cell_properties() {
        type : both;
        coverage_distance (-1, 10) ;
        min_row_edge_distance : 12;
        coverage_pattern_left : left_full ;
        coverage_pattern_right : right_full;
    }
}
```

In this case, value in coverage_distance is less than 0

the following is an example message:

```
Error : Line 1000,  The value of 'coverage_distance' cannot be less than
 0!
```

### What Next

Remove one of the group.

---

## LBDB-1161

(error) The attribute '%s' is required for %s.

### Description

This message indicates that the required attribute on the specified pin is missing.

For single rail no enable isolation cell, it is a requirement that the output pin has attribute 'alive_during_partial_power_down'.

The following example shows an instance where this message occurs:

```
cell(Isolation_Cell) {
  is_isolation_cell : true;
```

Feedback

```
  pg_pin(DVDDB) {
    voltage_name : DVDDB;
    pg_type : primary_power;
  }
  pg_pin(VSS) {
    voltage_name : VSS;
    pg_type : primary_ground;
  }
  pin(D) {
    direction : input;
    related_power_pin : DVDDB;
    related_ground_pin : VSS;
    isolation_cell_data_pin : true;
  }
  pin(Z) {
    direction : output;
    related_power_pin : DVDDB;
    related_ground_pin : VSS;
    function : "D";
    clamp_0_function : "!DVDDB";
    power_down_function : "VSS";
      timing() {
        related_pin : "D";
        …
      } /* end timing group */
  }/* end pin group*/
}/*end cell group*/
```

In this case, the alive_during_partial_power_down is missing in pin Z.

The following is an example message:

```
Error: Line 326, Cell 'Isolation_Cell', pin 'Z', The attribute
 'alive_during_partial_power_down' is required for single rail no enable
 isolation cell. (LBDB-137)
```

**What Next**

Add required attribute to the output pin.

---

## LBDB-1162

(error) The %s pin '%s' in %s cannot be used in %s!

**Description**

This message indicates that the pin name should not be in the function.

If pg_function is MUX21, the signal pin used in switch_function should be a different signal pin from pg_function.

Feedback

The following example shows an instance where this message occurs:

```
cell(LBDB-1162) {
switch_cell_type : coarse_grain;
   pg_pin (VDD) {
        direction : output;
        pg_function : "VDDB * A + !A * VDDA";
        pg_type : internal_power;
        switch_function : "!(A|B)";
        voltage_name : "VDD";
   }

}/*end cell group*/
```

In this case, A cannot be used in switch_function as A is used in pg_function

The following is an example message:

```
Error: Line 326, The signal pin 'A' in switch_function cannot be used in
 pg_function! (LBDB-1162)
```

**What Next**

remove the signal pin from switch_function.

---

## LBDB-1162w

(warning) The %s pin '%s' in %s cannot be used in %s!

**Description**

This message indicates that the pin name should not be in the function.

If pg_function is MUX21, the signal pin used in switch_function should be a different signal pin from pg_function.

The following example shows an instance where this message occurs:

```
cell(LBDB-1162w) {
switch_cell_type : coarse_grain;
   pg_pin (VDD) {
        direction : output;
        pg_function : "VDDB * A + !A * VDDA";
        pg_type : internal_power;
        switch_function : "!(A|B)";
        voltage_name : "VDD";
   }

}/*end cell group*/
```

In this case, A cannot be used in switch_function as A is used in pg_function

Feedback

The following is an example message:

```
Warning: Line 326, The signal pin 'A' in switch_function cannot be used
 in pg_function! (LBDB-1162w)
```

**What Next**

remove the signal pin from switch_function.

---

## LBDB-1163

(error) Invalid pg_function '%s' found for switch cell!

**Description**

This message indicates that pg_function format is incorrect.

For switch cell, pg_function can only be a single pg_pin, or two pg_pins and one or more signal pins in a MUX21 format (signal pins are the select pins).

The following example shows an instance where this message occurs:

```
cell(LBDB-1162) {
switch_cell_type : coarse_grain;
   pg_pin (VDD) {
        direction : output;
        pg_function : "VDDB * VDDA";
        pg_type : internal_power;
        switch_function : "!(A|B)";
        voltage_name : "VDD";
   }

}/*end cell group*/
```

In this case, there are two pg pins in pg_function and it is not in MUX21 format

The following is an example message:

```
Error: Line 326, Invalid pg_function 'VDDB * VDDA' found for switch cell!
 (LBDB-1163)
```

**What Next**

only write one pg pin in pg_function or write the function in MUX21 format.

---

## LBDB-1163w

(warning) Invalid pg_function '%s' found for switch cell!

Feedback

### Description

This message indicates that pg_function format is incorrect.

For switch cell, pg_function can only be a single pg_pin, or two pg_pins and one or more signal pins in a MUX21 format (signal pins are the select pins).

The following example shows an instance where this message occurs:

```
cell(LBDB-1163w) {
switch_cell_type : coarse_grain;
   pg_pin (VDD) {
        direction : output;
        pg_function : "VDDB * VDDA";
        pg_type : internal_power;
        switch_function : "!(A|B)";
        voltage_name : "VDD";
   }

}/*end cell group*/
```

In this case, there are two pg pins in pg_function and it is not in MUX21 format

The following is an example message:

```
Warning: Line 326, Invalid pg_function 'VDDB * VDDA' found for switch
 cell! (LBDB-1163w)
```

### What Next

only write one pg pin in pg_function or write the function in MUX21 format.

---

## LBDB-1164

(error) Multiple %s '%s' groups are defined at line %d!

### Description

You receive this message when more than one specified group are specified.

The following example shows an instance where this message occurs:

```
cell(lbdb1164) {
     site(site_def_1) {
         width: 2;
         height: 3;
         type: pad;
         symmetry (x);
         is_default: true;
     }
     site(site_def_2) {
         width: 2;
```

Feedback

```
            height: 3;
            type: pad;
            symmetry (y);
            is_default: true;
        }

}
```

In this case, two site are specified and they are both default.

The following is an example message:

```
Error : Line 1000, Multiple default 'site' groups are defined at line
 1001! (LBDB-1164)
```

**What Next**

Remove one of the group.

---

## LBDB-1165

(error) The size of %s [%d] should be an integral multiple of the size of %s [%d]!

**Description**

When the above-mentioned attributes are specified, the size should be an integral multiple of some attribute.

The following example shows an instance where this message occurs:

```
library(libdb1165) {
  ..
  site (unitTile) {
    track_pattern (M1) {
        type : uniform;
        direction: horizontal;
        mask_pattern (“mask_one”, “mask_two”, “mask_one”, “mask_two”);
        spacing : 0.2;
        offsets (“0.3”);
        widths(“0.1”);
        reserved_width_flags(“true”);
        grid_low_offsets(“0.1, 0.2, 0.3, 0.4”);
        grid_high_offsets(“0.1, 0.2, 0.3, 0.4”);
        grid_low_steps(“0.3”, “0.2, 0.4”, “0.2, 0.4”, “0.3, 0.4” );
        grid_high_steps(“0.3, 0.4”, “0.2, 0.4”, “0.2, 0.4”, “0.3, 0.4” );
    }
    ...
  }
```

grid_low_steps has size 7 but mask_pattern has size 4. 7 is not integral multiple of 4.

Feedback

### Examples

The following is an example message:

```
Error: Line 18,  The size of grid_low_steps [7] should be an integral
 multiple of the size of mask_pattern [4]! (LBDB-1165)
```

### What Next

Check the library source file, and make the necessary correction.

---

## LBDB-1166

(error) The size of '%s' in group '%s' should be %s!

### Description

When the above-mentioned attributes are specified, the size should be required size. For track_pattern, 1) if the type is uniform, 1.a) the size of 'offsets', and 'reserved_width_flags' if specified should be 1. 1.b) the size of 'widths' can be either 1 or same as size of 'mask_pattern'. 2) if the type is non-uniform, the size of 'mask_pattern', 'widths' and 'reserved_width_flags' if specified should be same as size of 'offsets'. 3) the size of 'grid_low_offsets' and 'grid_high_offsets' if specified should be same as size of 'mask_pattern'.

The size of resize under derived_layer should be either 2 or 3 or 5. first parameter is layer name and other parameters are float numbers. following are correct resize format. 1) resize(poly, 0.005); If one float value is provided, the left, bottom, right and top are each expanded by the distance 0.005. 2) resize(poly, 0.005, 0.00); If two float values are provided, the left and right edges are each expanded by the first distance 0.005 and the top and bottom edges are each expanded by the second distance 0.00 3) resize(poly, 0.005, 0.10, 0.002, 0.0); If four values are provided, the left, bottom, right, and top edges are expanded the respective distances.

The following example shows an instance where this message occurs:

```
library(libdb1166) {
  ..
  site (unitTile) {
    track_pattern (M1) {
        type : uniform;
        direction: horizontal;
        mask_pattern (“mask_one”, “mask_two”, “mask_one”, “mask_two”);
        spacing : 0.2;
        offsets (“0.3 0.2”);
        widths(“0.1”);
        reserved_width_flags(“true”);
        grid_low_offsets(“0.1, 0.2, 0.3, 0.4”);
        grid_high_offsets(“0.1, 0.2, 0.3, 0.4”);
        grid_low_steps(“0.3 0.1”, “0.2, 0.4”, “0.2, 0.4”, “0.3, 0.4” );
```

Feedback

```
        grid_high_steps("0.3, 0.4", "0.2, 0.4", "0.2, 0.4", "0.3, 0.4" );
    }
    ...
  }
```

offsets has size 2 but its size should be 1. of 4.

### Examples

The following is an example message:

```
Error: Line 18,  The size of 'offsets' in group 'track_pattern' should be
 1! (LBDB-1166)
```

### What Next

Check the library source file, and make the necessary correction.

---

## LBDB-1167

(error) Invalid single site '%s' defined in attribute '%s'!

### Description

The single site specified in attribute 'overlapping_site_mapping' should be exist in attribute 'site_list' in same 'row_pattern' group.

The following example shows an instance where this message occurs:

```
library(libdb1167) {
  ..
  row_pattern() {
     site_list ("unitH182, unitH210, unitH210");
     overlapping_site_alignment("unitH392", "unitH192, unitH210", R0);
     overlapping_site_alignment("unitH392MX", "unitH182, unitH210", MX);
     first_row_orientation: R0;
     last_site_row_index: 0;
  };
 ...
}
```

unitH192 defined in overlapping_site_mapping doesn't exist in 'site_list' of 4.

### Examples

The following is an example message:

```
Error: Line 18,  Invalid single site 'unitH192' defined in attribute
 'overlapping_site_alignment'! (LBDB-1167)
```

### What Next

Check the library source file, and make the necessary correction.

Feedback

## LBDB-1169

(warning) The values in '%s' are not decreasing %g, %g,\n \twhen %s = %g, %s = %g, %g.

### Description

This information appears when the values in the specified table do not decrease monotonically with the increasing output voltage from voltage value of VSS through VDD.

The following example shows an instance where this message occurs:

```
dc_current(template) {
    index_1("-0.675, -0.3375, -0.135, -0.0675, 0, 0.03375, 0.0675,
 0.10125, 0.135, 0.16875, 0.2025, 0.23625, 0.27, 0.30375, 0.3375,
 0.37125, 0.405, 0.43875, 0.4725, 0.50625, 0.54, 0.57375, 0.6075,
 0.64125, 0.675, 0.7425, 0.81, 1.0125, 1.35"); /* input_voltage */
    index_2("-0.675, -0.3375, -0.135, -0.0675, 0, 0.03375, 0.0675,
 0.10125, 0.135, 0.16875, 0.2025, 0.23625, 0.27, 0.30375, 0.3375,
 0.37125, 0.405, 0.43875, 0.4725, 0.50625, 0.54, 0.57375, 0.6075,
 0.64125, 0.675, 0.7425, 0.81, 1.0125, 1.35"); /* output_voltage */
    values("...0.0166949, 0.016715,...")
  }
```

The following is an example message:

```
Warning: Line 762, Cell 'A', pin 'clk', The values in 'dc_current' are
 not decreasing 0.0166949, 0.016715,
        when input_voltage = 0.0675, output_voltage = 0.16875, 0.2025.
 (LBDB-1169)
```

### What Next

Make sure that the values decrease monotonically with the increasing output voltage from voltage value of VSS through VDD.

## LBDB-1170

(error) The '%s' attribute is not specified.

### Description

This message indicates that an attribute is missing. start_site_id should be specified for all cells in a cell group. For the cells in a cell group, you can either specify start_site_id for them all, or not specify it at all.

The following example shows an instance where this message occurs:

```
library ( prf ) {
    ...
    cell (A) {
```

Feedback

```
      cell_group: A;
      start_site_id (1, 2);
    }
    cell (A1) {
      cell_group: A;
    }
  }
```

The following is an example message:

```
Error: Line 52, Cell 'A1', The 'start_site_id' attribute is not
 specified. (LBDB-1170)
```

**What Next**

Add the missing attribute.

---

## LBDB-1171

(error) The '%s' attribute is specified without '%s' attribute specified.

**Description**

This message indicates that an attribute is specified without another attribute specified in a cell group.

The following example shows an instance where this message occurs:

```
library ( prf ) {
    ...
    cell (A) {
      delta_tap_distance (-0.18,0.27,0.09,0.09,0.09,0.09,-0.18,0.27);
    }
  }
```

The following is an example message:

```
Error: Line 52, Cell 'A', The 'delta_tap_distance' attribute is specified
 without 'tap_boundary_wall_cell_properties' attribute specified.
 (LBDB-1171)
```

**What Next**

Add the missing attribute.

---

## LBDB-1172

(error) The '%s' attribute is specified without '%s' attribute specified in the referred '%s ( %s )' group at line %u.

Feedback

### Description

This message indicates that an attribute is specified without another attribute specified. One case is that both 'delta_tap_distance' and 'tap_boundary_wall_cell_properties : <prop_name>' attributes are specified in a cell group, however in the referred 'tap_boundary_wall_cell_properties( <prop_name> )' group in the library, the 'tap_distance' attribute is not specified.

The following example shows an instance where this message occurs:

```
library ( prf ) {
    tap_boundary_wall_cell_properties( AAA ) {
    /* tap_distance (TTT, TTT, 0, 0, 0, 0, TTT, TTT); */
    }
    ...
    cell (A) {
      tap_boundary_wall_cell_properties: AAA;
      delta_tap_distance (-0.18,0.27,0.09,0.09,0.09,0.09,-0.18,0.27);
    }
  }
```

The following is an example message:

```
Error: Line 68, Cell 'A', The 'delta_tap_distance' attribute is
 specified without 'tap_distance' attribute specified in the referred
 'tap_boundary_wall_cell_properties ( AAA )' group at line 23.
 (LBDB-1172)
```

### What Next

Add the missing attribute.

---

## LBDB-1173

(error) Number of elements in %s (%u) is not same as that in %s (%u) at line %u.

### Description

This message indicates that the size of 'delta_tap_distance' is not same as the size of 'tap_distance' in the referred 'tap_boundary_wall_cell_properties()' group.

The following example shows an instance where this message occurs:

```
library ( prf ) {
    tap_boundary_wall_cell_properties( AAA ) {
      tap_distance (TTT, TTT, 0, 0, 0, 0, TTT, TTT);
    }
    ...
    cell (A) {
      tap_boundary_wall_cell_properties: AAA;
      delta_tap_distance (-0.18,0.27,0.09,0.09);
```

Feedback

```
    }
  }
```

The following is an example message:

```
Error: Line 52, Cell 'A', Number of elements in delta_tap_distance (4) is
 not same as that in tap_distance (8) at line 24. (LBDB-1173)
```

**What Next**

Add the missing attribute.

---

## LBDB-1174

(error) Number of elements in %s (%u) is neither 8 nor 4.

**Description**

This message indicates that the size of 'tap_distance' is neither 8 nor 4.

The following example shows an instance where this message occurs:

```
library ( prf ) {
    tap_boundary_wall_cell_properties( AAA ) {
      tap_distance (0, 0);
    }
    ...
  }
```

The following is an example message:

```
Error: Line 52, Number of elements in tap_distance (2) is neither 8 nor
 4. (LBDB-1174)
```

**What Next**

Provide the correct number of elements for the attribute.

---

## LBDB-1175

(warning) The vector has duplicate indices for input_net_transition and total_output_net_capacitance pair with other vector.

**Description**

This error message occurs because under the output_voltage_rise|fall, some vector's total_output_net_capacitance and input_net_transition are duplicate with other vector.

The following example shows an output_current_rise group without a dense vector and resulting error message.

Feedback

```
output_voltage_template(ccsnov) {
  variable_1 : input_net_transition;
  variable_2 : total_output_net_capacitance;
  variable_3 : time;
 }
 . . .
 output_voltage_rise() {
   vector(ccsnov) {
      index_1 ("0.1");
      index_2 ("1");
      index_3 ("1, 2, 3");
      values ("1, 2, 3");
   }
    vector(ccsnov) {
      reference_time : 0.11;
      index_1 ("0.1");
      index_2 ("1");
      index_3 ("3, 4, 6");
      values ("1, 2, 3");
   }
 }

Error: Line 198, The vector has duplicate indices for
 input_net_transition and total_output_net_capacitance with other vector.
 (LBDB-1175)
```

### What Next

Check the library source file and correct the vector for input_net_transition and total_output_net_capacitance pair.

---

## LBDB-1176

(error) The '%s' in '%s' is invalid.

### Description

This message indicates that one of the value in 'tap_distance' is invalid. Each value in 'tap_distance' can be a name of pre-defined 'tap_rule' group, or direct number.

The following example shows an instance where this message occurs:

```
library ( prf ) {
    tap_rule( TTT ) {
      default_tap_distance : 15;
      tap_distance_layer_rule (LUP_075U, 10);
      tap_distance_layer_rule (LUP_045U, 3);
    }
    tap_boundary_wall_cell_properties( AAA ) {
      tap_distance (TTT, TTT, 0, 0, 0, 0, TTT, XXX);
    }
```

Feedback

```
    ...
  }
```

The following is an example message:

```
Error: Line 23, The 'XXX' in 'tap_distance' is invalid. (LBDB-1176)
```

**What Next**

Provide a name of pre-defined 'tap_rule' group or direct number for each value in 'tap_distance'.

---

## LBDB-1177

(error) The '%s' in '%s' refers to a '%s' group which is defined at line %u after this line.

**Description**

This message indicates that one of the value in 'tap_distance' refers to a 'tap_rule' group after this line.

The following example shows an instance where this message occurs:

```
library ( prf ) {
    tap_boundary_wall_cell_properties( AAA ) {
      tap_distance (TTT, TTT, 0, 0, 0, 0, TTT, TTT);
    }
    tap_rule( TTT ) {
      default_tap_distance : 15;
      tap_distance_layer_rule (LUP_075U, 10);
      tap_distance_layer_rule (LUP_045U, 3);
    }
    ...
  }
```

The following is an example message:

```
Error: Line 23, The 'TTT' in 'tap_distance' refers to a 'tap_rule' group
 which is defined at line 25 after this line. (LBDB-1177)
```

**What Next**

Define tap_rule group before using it.

---

## LBDB-1178

(warning) The '%s' under '%s' group must have at least one '%s' which is defined in '%s' group.

Feedback

### Description

This message indicates that 'default_pg_setting' should be specified in one pg_setting under mode_value group for each mode_definition group.

The following example shows an instance where this message occurs:

```
cell(test) {
   pg_setting_definition(PD) {
      default_pg_setting : PV1;
      pg_setting_value(PV1) {
        ...
      }
      pg_setting_value(PV2) {
        ...
      }

   }
   mode_definition(rw) {
      mode_value(read) {
         pg_setting(PD, PV2);
      }
   }
}
```

The following is an example message:

```
Warning: Line 35, The 'pg_setting' under 'mode_value' group must have at
least one 'default_pg_setting' which is defined in
'pg_setting_definition' group. (LBDB-1178)
```

### What Next

Add the group if it is missing, or fix the attribute value if it has a typo.

---

## LBDB-1179

(error) The '%s' attribute for %s '%s' is specified without '%s' attribute for the same %s.

### Description

This message indicates that an attribute is specified without another attribute specified. One case is that there is no corresponding "extra_site_list" for the same name site-array when relative_shift" for the named site-array is specified

The following example shows an instance where this message occurs:

```
library ( prf ) {
    row_pattern (demo) {
       site_list (core);
       extra_site_list (“halo”, R0);
       relative_shift (“halo1”, “-0.025, 0”);
```

Feedback

```
        ......
    }
  }
```

The following is an example message:

```
Error: Line 68,The 'relative_shift' attribute for site-array 'halo1' is
 specified without 'extra_site_list' attribute for the same site-array.
```

**What Next**

Add the missing attribute.

---

## LBDB-1180

(error) An invalid zero or negative value (%f) of '%s' is found.

**Description**

This message indicates zero or negative value is found for the attribute.

library (my_library) { lifetime_profile_definition ( lifetime_profile_1 ) { stress_time : 10; stress_voltage : 1.1; stress_temperature : 125; signal_probability : 0.5; activity_factor : 0.5; stress_mode: HCI; /*stress_clock_period cannot be 0 and should be larger than 0*/ stress_clock_period: 0; stress_slew_index: 2; stress_load_index: 0; }

}

The following is an example message:

```
Error: Line 136, An invalid zero or negative value (0.000000) of
 'stress_clock_period' is found. (LBDB-1180)
```

**What Next**

Change the value of the attributes to positive, make sure value >= 1e-6.

---

## LBDB-1181

(warning) There is no %s specified on %s pin, may affect STA accuracy.

**Description**

This warning message occurs when a internal pin has no related_power_pin/ related_ground_pin attribute.

The following example shows an instance where this message occurs:

```
cell(test_LBDB-1181){
  area : 1.0;
```

Feedback

```
pg_pin(VDDB) {
  voltage_name : VDDB;
  pg_type : backup_power;
}

pg_pin(VDD) {
  voltage_name : VDD;
  pg_type : primary_power;
}

pg_pin(VSS) {
  voltage_name : VSS;
  pg_type : primary_ground;
}


pin(A) {
  direction : internal;
}
}
```

The following is an example message:

```
Warning: Line 192, Cell 'test_LBDB-1181', pin 'A', There is no
 related_power_pin specified on internal pin, may affect STA accuracy.
 (LBDB-1181)
Warning: Line 193, Cell 'test_LBDB-1181', pin 'A', There is no
 related_ground_pin specified on internal pin, may affect STA accuracy.
 (LBDB-1181)
```

### What Next

Check the library source file to add or correct related_power_pin/related_ground_pin attributes.

---

## LBDB-1210

(error) Incomplete when condition coverage ("%s") for the '%s' group%s.

### Description

This message occurs when internal_power, leakage_power or timing group has incomplete "when" condition coverage

The following example shows an instance where this message occurs:

```
...
    pin(Z) {
      direction : output ;
      timing() {
        related_pin : "I" ;
        when : "A*B";
```

Feedback

```
      ...
    }
    timing() {
      related_pin : "I" ;
      when : "A*!B";
      ...
    }
    timing() {
      related_pin : "I" ;
      when : "!A*B";
      ...
    }
    /*
    timing() {
      related_pin : "I" ;
      when : "!A*!B";
      ...
    }
    */
    ...
  }
  ...
```

```
In the example above, the arc from 'I' to 'Z' does not have a timing
 model
for the when condition "!A*!B".

The following is an example of the message:

Error: Line 206, cell 'test', pin 'Z', Incomplete when condition
 coverage("A + B") for the 'timing' group of related_pin 'I'. (LBDB-1210)
```

### What Next

Complete the model. Either introduce a new group with the missing "when" condition, or add a default “when” condition.

---

## LBDB-1211

(error) The value for 'parameter_value_delta' index size must be one.

### Description

Currently only support one dimension for 'parameter_value_delta'.

The following example shows an instance where this message occurs:

```
lu_table_template (lle_delta_template_3x3x1) {
    variable_1 : input_net_transition;
    variable_2 : total_output_net_capacitance;
    variable_3 : parameter_value_delta;
    index_1 ("0, 0.0625, 1");
```

Feedback

```
    index_2 ("0, 0.0625, 1");
    index_3 ("0,1");
  }
  ...
  lle_delta_cell_rise (lle_delta_template_3x3x2) {
    related_params : "vth";
    related_devices : "tr2";
    index_1 ("0.0018, 0.0743, 1.15149");
    index_2 ("9.35e-05, 0.0025, 0.039");
    index_3 ("0.050,0.06");
    values ("0.000100", "0.000200", "0.000300", \
            "0.001000", "0.002000", "0.003000", \
            "0.010000", "0.020000", "0.030000");
  }
```

The following is an example message:

```
Error: Line 144, The value for 'parameter_value_delta' index size must be
 one. (LBDB-1211)
```

### What Next

Check the library source file to see if you defined the 'parameter_value_delta' not one dimension for the specified group.

---

## LBDB-1212

(warning) The timing arc from '%s' to '%s' is missing possible condition: "%s".

### Description

This message indicate some possible conditions are missing for this timing arc, and there is no default arc.

The following example shows an instance where this message occurs:

```
pin(Z) {
    direction : output;
    function : "(S' * A+S * B)";
    timing () {
      related_pin : "A";
      when :     "B * !S";
      sdf_cond : "B * !S";
      cell_rise(scalar)       { values ( "0.1");}
      rise_transition(scalar) { values ( "0.1");}
      cell_fall(scalar)       { values ( "0.1");}
      fall_transition(scalar) { values ( "0.1");}
    }
```

Feedback

In this case, the condition "B' S'" is missing for this timing arc from 'A' to 'Z'. To fix the problem, add the following timing group:

```
timing () {
      related_pin : "A";
      when :     "!B * !S";
      sdf_cond : "!B * !S";
      cell_rise(scalar)       { values ( "0.1");}
      rise_transition(scalar) { values ( "0.1");}
      cell_fall(scalar)       { values ( "0.1");}
      fall_transition(scalar) { values ( "0.1");}
    }
```

```
The following is an example message:
Warning: Line 190, Cell 'MUX21', pin 'Z', The timing arc from 'A' to 'Z'
 is missing possible condition: "B' S'". (LBDB-1212)
```

### What Next

Add the timing arcs with missing conditions between the two pins.

---

## LBDB-1213

(error) The value '%s' for 'related_devices' of lle delay must be listed in 'lle_devices' definition.

### Description

The following example shows an instance where this message occurs:

```
cell ( cell_a ) {
      lle_devices: "tr2 tr1";
      lle_params: "vth vth1";
  ...
  }
  lle_delta_cell_rise (lle_delta_template_3x3x2) {
    related_params : "vth";
    related_devices : "tr3";
    index_1 ("0.0018, 0.0743, 1.15149");
    index_2 ("9.35e-05, 0.0025, 0.039");
    index_3 ("0.050,0.06");
    values ("0.000100", "0.000200", "0.000300", \
            "0.001000", "0.002000", "0.003000", \
            "0.010000", "0.020000", "0.030000");
  }
```

The following is an example message:

```
Error: Line 144, The value 'tr3' for 'related_devices' of lle delay must
 be listed in 'lle_devices' defined. (LBDB-1213)
```

Feedback

### What Next

Check the value for 'related_devices' to see if your definition is in 'lle_devices' allowable values.

---

## LBDB-1214

(error) The value '%s' for 'related_params' of lle delay must be listed in 'lle_params' definition.

### Description

The following example shows an instance where this message occurs:

```
cell ( cell_a ) {
      lle_devices: "tr2 tr1";
      lle_params: "vth vth1";
  ...
  }
  lle_delta_cell_rise (lle_delta_template_3x3x2) {
    related_params : "vth0";
    related_devices : "tr2";
    index_1 ("0.0018, 0.0743, 1.15149");
    index_2 ("9.35e-05, 0.0025, 0.039");
    index_3 ("0.050,0.06");
    values ("0.000100", "0.000200", "0.000300", \
            "0.001000", "0.002000", "0.003000", \
            "0.010000", "0.020000", "0.030000");
  }
```

The following is an example message:

```
Error: Line 144, The value 'vth0' for 'related_params' of lle delay must
 be listed in 'lle_params' defined. (LBDB-1214)
```

### What Next

Check the value for 'related_params' to see if your definition is in 'lle_params' allowable values.

---

## LBDB-1215

(error) The '%s' group is missing the table definition for %s %s,%s %s.

### Description

The following example shows an instance where this message occurs:

```
cell ( cell_a ) {
      lle_devices: "tr2 tr1";
      lle_params: "vth";
```

Feedback

```
  ...
  }
  lle_delta_cell_rise (lle_delta_template_3x3x2) {
    related_params : "vth";
    related_devices : "tr1";
    index_1 ("0.0018, 0.0743, 1.15149");
    index_2 ("9.35e-05, 0.0025, 0.039");
    index_3 ("0.050,0.06");
    values ("0.000100", "0.000200", "0.000300", \
            "0.001000", "0.002000", "0.003000", \
            "0.010000", "0.020000", "0.030000");
  }
```

The following is an example message:

```
Error: Line 27, Cell 'cell_a', pin 'z', The 'lle_delta_cell_rise' group
 is missing the table definition for related_devices tr2,related_params
 vth. (LBDB-1215)
```

### What Next

Check lle_delta group to see if you are missing the table definition for all lle_devices and lle_params specified in cell.

---

## LBDB-1216

(warning) is missing internal_power with when condition: "%s" from '%s' to '%s'.

### Description

This message indicate some possible conditions are missing for this timing arc, and there is no default arc.

The following example shows an instance where this message occurs:

```
pin(Z) {
    direction : output;
    function : "(S' * A+S * B)";
    internal_power () {
      related_pin : "A";
      when :     "B * !S";
    }
```

In this case, the internal_power condition "B' S'" is missing for this arc from 'A' to 'Z'. To fix the problem, add the following internal_power group:

```
internal_power () {
      related_pin : "A";
      when :     "!B * !S";
    }

The following is an example message:
```

Feedback

```
Warning: Line 190, Cell 'MUX21', pin 'Z', is missing internal_power with
 when condition: "B' S'" from 'A' to 'Z'. (LBDB-1216)
```

### What Next

Add the internal_power group with missing conditions between the two pins.

---

## LBDB-1217

(warning) is missing internal_power with when condition: "%s".

### Description

This message indicate some possible conditions are missing for this input, and there is no default arc.

The following example shows an instance where this message occurs:

```
pin(Z) {
    direction : output;
    function : "a&b&c";
    ...
    }
    pin(a){
    direction : input;
    internal_power(){
      when :   "!b&c";
    }
    internal_power(){
      when :   "!b&!c";
    }
   }
```

In this case, the internal_power condition "b&!c" is missing for this input pin 'a'. To fix the problem, add the following internal_power group:

```
internal_power () {
      when :     "b&!c";
    }

The following is an example message:
Warning: Line 190, Cell 'MUX21', pin 'a', is missing internal_power with
 when condition: "b&!c".(LBDB-1217)
```

### What Next

Add the internal_power group with missing conditions for input pin's dynamic power.

---

## LBDB-1218

(warning) The internal_power group with when condition "%s" from '%s' to '%s' is invalid.

Feedback

### Description

A arc from an input pin to an output pin for internal_power is dormant if the signal change on the input pin can never cause the change on the output without simultaneous changes on any other input pins.

The following example shows an instance where this message occurs:

```
cell(a_cell) {
    pin(Z) {
    direction : output;
    function : "A * B";
    internal_power () {
       related_pin : "A";
       when :     "!B";
       rise_power(scalar)       { values ( "0.1");}
       fall_power(scalar) { values ( "0.1");}
    }
   }
  }
```

In this example, if only pin 'A' is changed, pin 'Z' will not have the signal transitions between '0' and '1'. The combinational arc between 'A' and 'Z' is invalid.

The following is an example message:

```
Warning: Line 50000, The internal_power group with when condition "!B"
 from 'A' to 'Z' is invalid. (LBDB-1218)
```

### What Next

This is caused by the input sharing between 'function', 'three_state', and 'x_function'. The signal transition is not physically possible and generally the arc should not be specified.

---

## LBDB-1219

(warning) The internal_power group with when condition "%s" is invalid.

### Description

A arc from an input pin for internal_power(dynamic power) is dormant if the signal change on the input pin can never cause the change on the output without simultaneous changes on any other input pins.

The following example shows an instance where this message occurs:

```
cell(a_cell) {
    pin(Z) {
    direction : output;
    function : "a&b&c";
    ...
```

Feedback

```
    }
    pin(a){
    direction : input;
    internal_power(){
      when :   "b&c";
    }
  }
```

In this example, if input pin 'a' is changed, pin 'Z' will not have the signal transitions between '0' and '1'. So the input pin 'a' for internal_power(dynamic_power) is dormant.

The following is an example message:

```
Warning: Line 50000, The internal_power group with when condition "b&c"
 is invalid. (LBDB-1219)
```

### What Next

This is caused by the input sharing between 'function', 'three_state', and 'x_function'. The signal transition is not physically possible and generally the arc should not be specified.

---

## LBDB-1220

(warning) is missing leakage power possible condition: "%s".

### Description

This message indicate some possible conditions are missing for this cell, and there is no default arc.

The following example shows an instance where this message occurs:

```
cell(std) {
    leakage_power () {
      when :     "B";
    }
```

In this case, the leakage power condition "B'" is missing for this cell 'std'. To fix the problem, add the following leakage_power group:

```
leakage_power () {
      when :     "!B ";
    }

The following is an example message:
Warning: Line 190, Cell 'std', is missing leakage power possible
 condition: "B'". (LBDB-1220)
```

### What Next

Add the leakage_power group with missing conditions for the cell.

Feedback

## LBDB-1221

(error) non input signal pin is not allowed to be specified in bump cell.

### Description

This message indicate bump cell have not only input signal pins.

The following example shows an instance where this message occurs:

```
cell (simple_front_side_bump) {
    bump_cell: true;
    pin (PAD) {
      direction : output ;
      bump_type: front_side;
    }
  }
```

In this case, the pin "pad" direction is not input signal pin. To fix the problem, change direction to input:

```
cell (simple_front_side_bump) {
    bump_cell: true;
    pin (PAD) {
      direction : input ;
      bump_type: front_side;
    }
  }

The following is an example message:
Error: Line 190, Cell 'simple_front_side_bump', pin 'PAD', non input
 signal pin is not allowed to be specified in bump cell. (LBDB-1221)
```

### What Next

Change pin to input signal pin for the bump cell.

## LBDB-1222

(error) is missing bump_type on signal pin of bump cell.

### Description

This message indicate signal pins in bump cell has not bump_type attribute.

The following example shows an instance where this message occurs:

```
cell (simple_front_side_bump) {
    bump_cell: true;
    pin (PAD) {
      direction : input ;
```

Feedback

```
    }
  }
```

In this case, the pin "pad" has no bump_type attribute. To fix the problem, add the following bump_type attribute:

```
cell (simple_front_side_bump) {
    bump_cell: true;
    pin (PAD) {
      direction : input ;
      bump_type: front_side;
    }
  }
```

```
The following is an example message:
Error: Line 190, Cell 'simple_front_side_bump', pin 'PAD', is missing
 bump_type on signal pin of bump cell. (LBDB-1222)
```

**What Next**

Add bump_type attribute for the bump cell signal pin.

---

## LBDB-1223

(error) the "is_pad" attribute is not allowed to be specified in signal pin of bump cell.

**Description**

This message indicate bump cell contains is_pad pin attributes.

The following example shows an instance where this message occurs:

```
cell (simple_front_side_bump) {
    bump_cell: true;
    pin (PAD) {
      direction : input ;
      bump_type: front_side;
      is_pad   : true;
    }
  }
```

In this case, contains is_pad attributes. To fix the problem, remove the following is_pad attribute:

```
cell (simple_front_side_bump) {
    bump_cell: true;
    pin (PAD) {
      direction : input ;
      bump_type: front_side;
    }
  }
```

Feedback

```
The following is an example message:
Error: Line 190, Cell 'simple_front_side_bump', pin 'PAD', the "is_pad"
 attribute is not allowed to be specified in signal pin of bump cell.
 (LBDB-1223)
```

**What Next**

Remove pin's is_pad attribute for the bump cell.

---

## LBDB-1224

(error) pg_pin is not allowed to be specified in bump cell.

**Description**

This message indicate bump cell contains pg_pin.

The following example shows an instance where this message occurs:

```
cell (simple_front_side_bump) {
    bump_cell: true;
    pin (PAD) {
      direction : input ;
      bump_type: front_side;
    }
    pg_pin(VDD){
    ..
    }
  }
```

In this case, contains pg_pin. To fix the problem, remove the following pg_pin attribute:

```
cell (simple_front_side_bump) {
    bump_cell: true;
    pin (PAD) {
      direction : input ;
      bump_type: front_side;
    }
  }

The following is an example message:
Error: Line 190, Cell 'simple_front_side_bump', pg_pin is not allowed to
 be specified in bump cell. (LBDB-1224)
```

**What Next**

Remove pg_pin for the bump cell.

---

## LBDB-1225

(error) the "%s" attribute is not allowed to be specified in bump cell.

Feedback

### Description

This message indicate bump cell contains pad_cell or auxiliary_pad_cell attributes.

The following example shows an instance where this message occurs:

```
cell (simple_front_side_bump) {
    bump_cell: true;
    auxiliary_pad_cell : true;
    pin (PAD) {
      direction : input ;
      bump_type: front_side;
    }
  }
```

In this case, the pin "pad" contains auxiliary_pad_cell. To fix the problem, remove the following auxiliary_pad_cell attribute:

```
cell (simple_front_side_bump) {
    bump_cell: true;
    pin (PAD) {
      direction : input ;
      bump_type: front_side;
    }
  }

The following is an example message:
Error: Line 190, Cell 'simple_front_side_bump', the "auxiliary_pad_cell"
 attribute is not allowed to be specified in bump cell. (LBDB-1225)
```

### What Next

Remove auxiliary_pad_cell/pad_cell for the bump cell.

---

## LBDB-1230

(error) The value '%s' for 'related_devices' of lle leakage power must be listed in 'lle_devices' definition.

### Description

The following example shows an instance where this message occurs:

```
cell ( cell_a ) {
      lle_devices: "tr1 tr2";
      lle_params: "vth vth1";
   ...
   leakage_power() {
        when : "(!A) | (A)" ;
        value : 0.00644538 ;
  lle_delta_leakage_power( lle_template_sample ) {
   related_params : "vth" ;
```

Feedback

```
   related_devices : "tr3" ;
   index_1("-0.1,0.1") ;
   values("4.20784,4.20785");
  }
     }
   }
```

The following is an example message:

```
Error: Line 144, The value 'tr3' for 'related_devices' of lle leakage
 power must be listed in 'lle_devices' defined. (LBDB-1230)
```

### What Next

Check the value for 'related_devices' to see if your definition is in 'lle_devices' allowable values.

---

## LBDB-1231

(error) The value '%s' for 'related_params' of lle leakage power must be listed in 'lle_params' definition.

### Description

The following example shows an instance where this message occurs:

```
cell ( cell_a ) {
      lle_devices: "tr1 tr2";
      lle_params: "vth vth1";
   ...
   leakage_power() {
        when : "(!A) | (A)" ;
        value : 0.00644538 ;
  lle_delta_leakage_power( lle_template_sample ) {
   related_params : "vth2" ;
   related_devices : "tr2" ;
   index_1("-0.1,0.1") ;
   values("4.20784,4.20785");
   }
     }
   }
```

The following is an example message:

```
Error: Line 144, The value 'vth2' for 'related_params' of lle leakage
 power must be listed in 'lle_params' defined. (LBDB-1231)
```

### What Next

Check the value for 'related_params' to see if your definition is in 'lle_params' allowable values.

Feedback

## LBDB-1232

(error) The '%s' group is missing the table definition for %s %s,%s %s.

### Description

The following example shows an instance where this message occurs:

```
cell ( cell_a ) {
      lle_devices: "tr2 tr1";
      lle_params: "vth";
   ...
   leakage_power() {
        when : "(!A) | (A)" ;
        value : 0.00644538 ;
  lle_delta_leakage_power( lle_template_sample ) {
   related_params : "vth" ;
   related_devices : "tr2" ;
   index_1("-0.1,0.1") ;
   values("4.20784,4.20785");
   }
     }
   }
```

The following is an example message:

```
Error: Line 27, Cell 'cell_a', leakage_power, The
 'lle_delta_leakage_power' group is missing the table definition for
 related_devices tr1,related_params vth. (LBDB-1232)
```

### What Next

Check lle_delta group to see if you are missing the table definition for all lle_devices and lle_params specified in cell.

## LBDB-1233

(warning) The same timing arc with different sdf_cond has already been specified%s.

### Description

This message indicates the same timing arc with different sdf_cond is specified multiple times.

### What Next

Check the library source file if there is a duplicate, gave the warning

### EXMAPLES

The following example shows two timing arc definition are same with different sdf_cond.

Feedback

```
pin(Q) {
            timing(){
               timing_type: rising_edge;
               related_pin: "B";
               sdf_cond : "sdf1";
               when: "A";
             }

            timing(){
               timing_type: rising_edge;
               related_pin: "B";
               sdf_cond : "sdf2";
               when: "A";
             }
        }
```

### Examples

```
Warning: Line 456, Cell 'sass', pin 'CLKA', The same timing arc with
 different sdf_cond has already been specified at line 123.(LBDB-1233)
```

---

## LBDB-1234

(error) Invalid attribute '%s' value '%s' is detected. The value %s.

### Description

This message indicates that you specified an invalid value.

```
lib ( liberty_lde )    {
      lde_boundary_type_names ("w1", "w2");
   ... ...
      cell ( cell_a ) {
     lde_boundary_count : 0 ;
     lde_boundary_types ( "w1" ) ;
        pin(A) {
    timing () {
   lde_delta ( ) {
   ...
   }
    }
  }
      }
      ... ...
  }
```

The following is an example message:

```
Error: Line 12, Invalid attribute 'lde_boundary_count' value '0' is
 detected. The value must be positive integer. (LBDB-1234)
```

Feedback

### What Next

Check the library source file, and correct the attribute value in the library.

---

## LBDB-1235

(error) Invalid attribute '%s' is detected. The size must be same with lde_boundary_count : %d.

### Description

This message indicates that you specified an invalid attribute size.

```
lib ( liberty_lde )    {
   lde_boundary_type_names ("w1", "w2");
      cell ( cell_a ) {
     lde_boundary_count : 2 ;
     lde_boundary_types ( "w1" ) ;
        pin(A) {
    timing () {
   lde_delta ( ) {
   ...
   }
    }
  }
     }
     ... ...
    }
```

The following is an example message:

```
Error: Line 22, Cell 'cell_a', Invalid attribute 'lde_boundary_types' is
 detected. The size must be same with lde_boundary_count : 2. (LBDB-1235)
```

### What Next

Check the library source file, and correct the attribute size in the library.

---

## LBDB-1236

(error) The '%s' group is missing the table definition for lde_boundary_id:%d, lde_boundary_type:%s.

### Description

The following example shows an instance where this message occurs:

```
lib ( liberty_lde )    {
   lde_boundary_type_names ("w1", "w2");
      cell ( cell_a ) {
     lde_boundary_count : 1 ;
```

Feedback

```
     lde_boundary_types ( "w1" ) ;
        pin(z) {
    timing () {
   lde_delta ( ) {
    lde_boundary_id : 1 ;
    lde_boundary_type : "w2" ;
   ...
   }
    }
  }
      }
      ... ...
    }
```

The following is an example message:

```
Error: Line 29, Cell 'cell_a', pin 'z', The 'lde_delta' group is missing
 the table definition for lle_boundary_id:1, lle_boundary_type:w1.
 (LBDB-1236)
```

**What Next**

Check lde_delta group to see if you are missing the table definition for all lde_boundary_count and lde_boundary_type_names specified in cell.

---

## LBDB-1237

(error) The '%s' that required by liberty lde delta model is missing in %s '%s'.

**Description**

Defined lde_delta groups in .lib, but the dependent attribute is missing in the library or cell.

The following example shows an instance where this message occurs:

```
lib ( liberty_lde )    {
   /*lde_boundary_type_names miss*/
      cell ( cell_a ) {
     lde_boundary_count : 1 ;
     lde_boundary_types ( "w1" ) ;
        pin(A) {
    timing () {
   lde_delta ( ) {
     lde_boundary_id : 1 ;
     lde_boundary_type : "w1" ;
   ...
   }
    }
  }
      }
      ... ...
    }
```

Feedback

The following is an example message:

```
Error: Line 3, The 'lde_boundary_type_names' that required by liberty lde
 delta model is missing in library 'liberty_lde'. (LBDB-1237)
```

### What Next

Check the library source file to see if you missed the required content for the specified group.

---

## LBDB-1237w

(warning) The '%s' is defined in %s '%s', but no lde_delta use it.

### Description

Defined lde_boundary_type_names attribute in liberty_lde library, but there is no lde_delta group in all timing arcs.

The following example shows an instance where this message occurs:

```
lib ( liberty_lde )    {
   lde_boundary_type_names ( "w1",  "w2" )
      cell ( cell_a ) {
     lde_boundary_count : 1 ;
     lde_boundary_types ( "w1" ) ;
        pin(A) {
    timing () {
   /* No lde_delta in timing arc*/
   ...
   }
    }
  }
      }
      ... ...
    }
```

The following is an example message:

```
Warning: Line 3, The 'lde_boundary_type_names' is defined in library
 'libertyLdeMulti', but no lde delta use it. (LBDB-1237w)
```

### What Next

Check the library source file to see if you need the content that not used for the specified group.

---

## LBDB-1238

(error) The number of '%s' in group '%s' should be %s.

Feedback

### Description

When the above-mentioned attributes are specified, the number of the elements should be required size. For track_pattern, the number of <list_of_offsets> in row_grid_low_steps and row_grid_high_steps should be same as the size of grid_low_offsets.

The following example shows an instance where this message occurs:

```
library(libdb1166) {
  ..
  row_pattern (unit10_unit12) {
    track_pattern (M1) {
        type : uniform;
        direction: horizontal;
        mask_pattern (“mask_one”, “mask_two”, “mask_one”, “mask_two”);
        spacing : 0.2;
        offsets (“0.3 0.2”);
        widths(“0.1”);
        reserved_width_flags(“true”);
        grid_low_offsets(“0.1, 0.2, 0.3, 0.4”);
        grid_high_offsets(“0.1, 0.2, 0.3, 0.4”);
        row_grid_low_steps("unit10", “0.3 0.1”, “0.2, 0.4”, “0.2, 0.4”);
        row_grid_high_steps("unit12", “0.3, 0.4”, “0.2, 0.4”, “0.2, 0.4”,
 “0.3, 0.4” );
    }
    ...
  }
```

row_grid_low_steps has number of <list_of_offsets> as 3 but its size should be 4. of 4.

### Examples

The following is an example message:

```
Error: Line 18,  The number of '<list_of_offsets> in row_grid_low_steps'
 in group 'track_pattern' should be 4. (LBDB-1238)
```

### What Next

Check the library source file, and make the necessary correction.

# LBMG

## LBMG-001

(error) %s:%s doesn't exist in base library.

Feedback

### Description

This message indicates that some groups in model library do not exist in base library. These groups are required for merge process.

### What Next

Find the correct base library.

### Examples

```
Error: cell: cell_1 doesn't exist in base library. (LBMG-001)
```

---

## LBMG-002

(warning) Conflict found, use group/attribute %s %s in %s library.

### Description

This message indicates that conflict group/attribute is found between base and model library. But the conflict could be resolved by using the one in base/model library.

### What Next

No action is required.

### Examples

```
Warning: Conflict found, use group/attribute ocv_derate_group in base
 library. (LBMG-002)
```

---

## LBMG-003

(error) Conflict error found, group/attribute %s %s is diffenrent between base and model library.

### Description

This message indicates that conflict group/attribute is found between base and model library.

### What Next

Update the group/attribute in base/model library.

### Examples

```
Error: Conflict error found, group/attribute ocv_derate is diffenrent
 between base and model library. (LBMG-003)
```

Feedback

## LBMG-004

(warning) New %s found, update %s '%s' from partial lib into base library.

### Description

This message indicates that a new group/attribute is found in partial library. Copy the group/attribute from partial lib into base library.

### What Next

No action is required.

### Examples

```
Warning: Line 3, New group found, update group 'cell’ from partial lib
 into base library. (LBMG-004)
```

## LBMG-005

(warning) Conflict %s found, update %s '%s' from partial lib into base library.

### Description

This message indicates that conflict group/attribute is found between base and partial library. Update the group/attribute from partial lib into base library.

### What Next

No action is required.

### Examples

```
Warning: Line 6, Cell 'IV', pin 'A', Conflict attribute found, update
 attribute 'capacitance' from partial lib into base library. (LBMG-005)
```

## LBMG-006

(warning) Conflict %s found, keep %s '%s' in base library.

### Description

This message indicates that conflict group/attribute is found between base and partial library. Keep the group/attribute in base library.

### What Next

No action is required.

Feedback

### Examples

```
Warning: Line 6, Cell 'IV', pin 'A', Conflict attribute found, keep
 attribute 'direction' in base library. (LBMG-006)
```

## LBMG-007

(warning) Unable to merge attribute/group '%s' of %s.

### Description

This message indicates that the group/attribute can't be merged. This is current limitation.

### What Next

No action is required.

### Examples

```
Warning: Line 6, Cell 'IV', Unable to merge attribute/group
 'userDefinedAttribute' of cell. (LBMG-007)
```

## LBMG-008

(error) Unable to merge attribute/group '%s' of %s.

### Description

This message indicates that the group/attribute can't be merged. The matching group for partial lib can't be found in base library accordingly. LC is using below method to match a group for merge - "group_id" user defined attribute - "original_name" user defined attribute - group name - group key attributes User will get this message if can't find a match group.

### Examples

base.db

cell(AAA) { ... }

partial.lib cell(BBB) { original_name : AA }

### What Next

Fix the partial.lib for wrong specification. Check if the "group_id"/"original_name"/group name/group key attributes are specified correctly. In this case, original_name should be "AAA".

### Examples

```
Error: Line 6, Cell 'BBB', Unable to merge attribute/group 'cell' of
 library. (LBMG-008)
```

Feedback

# LBSYN

## LBSYN-1

(error) The parameter '%s' has been defined multiple times \n \tin module '%s'.

**Description**

You receive this message during reading of a library, if the *read_lib* command finds that the same parameter has been defined more than once in the synthetic module. Each parameter must have a unique definition.

**What Next**

Edit the .sl file to remove the duplicate parameter definition from the synthetic module. Then reexecute *read_lib*.

## LBSYN-2

(error) Pin association '%s' must specify exactly one of "oper_pin" or "value".

**Description**

You receive this message if the library being read contains a pin association with multiple specifications. The pin association must have either one oper_pin or a value.

**What Next**

Locate the specified pin association in the synthetic library file, and edit the file so that the pin association specifies either one oper_pin or a value. Then reexecute *read_lib*.

## LBSYN-3

(error) Binding '%s' must specify "bound_operator".

**Description**

You receive this message during reading of a library, if the *read_lib* command finds a binding that does not have a bound_operator specification in the binding group in the current processed module. Each binding must have a bound_operator specification.

**What Next**

Edit the file to add the missing bound_operator specification. Then reexecute *read_lib*.

Feedback

## LBSYN-4

(error) The module or component '%s' requires a design_library attribute.

**Description**

Each module and component in a synthetic library must have a design_library attribute that identifies the design library where its implementation source is stored. The library should correspond to a library identified in the design_design_lib command.

**What Next**

Supply a design_library attribute.

## LBSYN-5

(error) Operator pin '%s' must either be "input" or "output".

**Description**

Synthetic library operators can only represent combinational functions.As such, it does not make sense to have inout pins on operators. Functions with inout pins must be instantiated directly as synthetic library modules.

**What Next**

Do not use operators with inout pins. Use modules if inout pins are really required.

## LBSYN-6

(error) Operator '%s' data_class must either be "unsigned" or "signed".

**Description**

Synthetic library operators can only have a data_class value of "signed" or "unsigned". Other values are illegal.

## LBSYN-7

(error) Module pin '%s' must have a 'bit_width' attribute.

**Description**

'Bit_width' is a required attribute on all synthetic library module pins.

Feedback

## LBSYN-8

(error) The value "%s" in group %s(%s) is not legal.

**Description**

Only a string of 1's and 0's is allowed in the attribute. The only exception is that "unbound" can occur in constraint groups.

**What Next**

To fix the syntax error, use a legal value.

## LBSYN-9

(error) The value attribute is missing in group %s(%s).

**Description**

A value attribute is required in this group.

**What Next**

Please supply a value attribute.

## LBSYN-10

(error) Parameter '%s' cannot have both a constant attribute\n and a formula attribute.

**Description**

The constant attribute is used to assigned the parameter a typed constant value. The formula attribute is used to assigned the parameter a computed expression. Only one of these can be used to assign the parameter a value.

**What Next**

Eliminate either the constant or the formula attribute.

## LBSYN-11

(error) Parameter '%s' has a constant attribute\n without a type attribute.

**Description**

A parameter that sets its value using the constant attribute must have a type attribute to declare the constant's type.

Feedback

**What Next**

Add a type attribute.

---

## LBSYN-12

(warning) In parameter '%s', the type attribute is ignored.

**Description**

The formula attribute ignores the type declaration in the type attribute. The formula's type is determined from the equation.

**What Next**

Use the constant attribute to declare typed parameters.

---

## LBSYN-13

(error) The string %s can not be evaluated,\n \ttherefore it can't be used for the bit_width value on pin %s.

**Description**

This message indicates that the given string contained parameter(s) that were not previously defined in the current module's scope. Therefore, the string can not be used as the value for the bit_width attribute on the given pin.

**What Next**

Modify the .sl file, by creating the desired parameter group in the module that contains the given pin group. The parameter group should be defined prior to the definition of the pin group.

---

## LBSYN-15

(error) The binding group %s in the module %s does not\n \tcontain any pin_association groups.

**Description**

This message indicates that the given binding group in the stated module does not contain any pin_association groups. At least one pin_association group is required for a binding group. If this is a sequential binding group, at a minimum the clock pin must be connected to the clock input of the operator in each state.

Feedback

### What Next

Modify the .sl file, by creating the necessary pin_association group(s) for the binding. The pin_association group statement defines how the module pins are associated with synthetic operator pins and constant values. See the DesignWare Developer Guide for further information on this topic.

---

## LBSYN-16

(warning) The parameter %s does not have the\n \thdl_parameter attribute set true, nor does it have a formula.

### Description

This message indicates that the given parameter group does not have its hdl_parameter attribute set to "true", nor does it have a formula defined for it. A parameter must have either a formula or the hdl_parameter attribute set to "true" in order for this parameter to be referenced successfully.

### What Next

Modify the .sl file to include a formula and/or set the hdl_parameter attribute to "true". See the DesignWare Developer Guide for further information on this topic.

---

## LBSYN-17

(warning) the parameter %s formula may not evaluate\n \tcorrectly during a compile operation because of: %s.

### Description

This message indicates that the given formula potentially has an error in it which might prevent this parameter from being evaluated during run time. The reason given may provide helpful information for correcting the problem.

Certain static checks, such as syntax, are performed on formulas within parameters when a library is read in. During this process, a formula is evaluated with values that are available statically. Variables within the formula which are not known until a compile is executed can not be checked.

### What Next

Check the formula attribute within the given parameter for the problem indicated by the message. Edit the .sl file as necessary to correct the problems.

Feedback

## LBSYN-18

(error) Substitute Licenses are for internal use only.

### Description

The "Substitute License" feature is for internal Synopsys use only. This feature is not available to Synopsys customers.

### What Next

Remove the "substitute_license" group from your implementation description or acquire a "Synopsys" license.

## LBSYN-19

(error) The "substitute_license" group requires an "original_license" attribute.

### Description

The "substitute_license" group for synthetic module implementations requires at least an "original_license" and a "substitute_license" attribute. An example "substitute_license" group is:

```
substitute_license () {
        original_license : lic_a;
        substitute_license : lic_b;
        valid_modules : "DW01_mult DW01_add DW01_addsub";
    }
```

### What Next

Either add the "original_license" attribute or delete the "substitute_license" group.

## LBSYN-20

(error) The "substitute_license" group requires a "substitute_license" attribute.

### Description

The "substitute_license" group for synthetic module implementations requires at least an "original_license" and a "substitute_license" attribute. An example "substitute_license" group is:

```
substitute_license () {
        original_license : lic_a;
        substitute_license : lic_b;
        valid_modules : "DW01_mult DW01_add DW01_addsub";
    }
```

Feedback

### What Next

Either add the "substitute_license attribute or delete the "substitute_license" group.

---

## LBSYN-21

(warning) The "substitute_license" group contains no "valid_modules" attribute.

### Description

No "valid_modules" attribute has been specified for this "substitute_license" group. This will allow any modules licensed with the "original_license" to be instantiated in this implementation without consuming the license. An example "substitute_license" group is:

```
substitute_license () {
        original_license : lic_a;
        substitute_license : lic_b;
        valid_modules : "DW01_mult DW01_add DW01_addsub";
    }
```

### What Next

If improved protection is desired use the "valid_modules" attribute to restrict the modules to which this license substitution will be applied.

---

## LBSYN-22

(error) Original license must be DesignWare-Foundation.

### Description

The "Original License" feature must be DesignWare or DesignWare-Foundation license. We only support Foundation license substitution.

### What Next

Correct your Original license name. Or delete "susbstitute_license" group from your implementation declaration.

---

## LBSYN-23

(error) Only one generator group can exist in a library.

### Description

A synthetic library can have at most one generator group specifying information used during creation of library implementations.

Feedback

### What Next

Remove generator groups from the synthetic library. Separate implementations requiring different generators from a single synthetic library into multiple synthetic libraries. In each new synthetic library only one generator should exist with information used to generate all implementations in the new synthetic library.

---

## LBSYN-24

(error) The "generator" group requires the \n \t'%s' attribute.

### Description

The "generator" group for synthetic libraries requires a minimum set of attributes. This set consists of the following attributes: "executable_os_variable", "generator_interface", and "contact_upon_failure".

### What Next

Either add the required attributes to the "generator" group or delete the "generator" group.

---

## LBSYN-25

(error) The "generator" group requires a "generator_interface" attribute.

### Description

The "generator" group for synthetic libraries requires at least an "executable_os_variable" attribute, a "generator_interface" attribute, and a "contact_upon_failure" attribute. An example "generator" group is:

```
generator (mult_gen) {
        generator_interface : file;
        executable_os_variable : "/usr/local/bin/mgen";
        contact_upon_failure : "MGEN Customer Support Hotline";
    }
```

### What Next

Either add the "generator_interface" attribute or delete the "generator" group.

---

## LBSYN-26

(error) The "generator" group requires a "contact_upon_failure" attribute.

Feedback

### Description

The "generator" group for synthetic libraries requires at least an "executable_os_variable" attribute, a "generator_interface" attribute, and a "contact_upon_failure" attribute. An example "generator" group is:

```
generator (mult_gen) {
        generator_interface : file;
        executable_os_variable : "/usr/local/bin/mgen";
        contact_upon_failure : "MGEN Customer Support Hotline";
    }
```

### What Next

Either add the "contact_upon_failure" attribute or delete the "generator" group.

---

## LBSYN-27

(error) Only one "verilog" subgroup may exist in a "generator" group.

### Description

A "generator" group may have at most one "verilog" subgroup specifying information used during processing of generator produced Verilog netlists.

### What Next

Remove "verilog" groups from the generator group leaving at most one "verilog" group.

---

## LBSYN-28

(error) Only one "vhdl" subgroup can exist in a "generator" group.

### Description

A "generator" group can have at most one "vhdl" subgroup specifying information used during processing of generator produced VHDL netlists.

### What Next

Remove "vhdl" groups from the generator group leaving at most one "vhdl" group.

---

## LBSYN-29

(error) The attribute "output_format" requires either a "vhdl" or a "verilog" subgroup.

Feedback

### Description

When the "output_format" attribute specifies "soft_macro", a "vhdl" subgroup or a "verilog" subgroup is required in the "generator" group.

### What Next

Add a "verilog" or "vhdl" subgroup to the generator group. Change the value of the "output_format" to something other than "soft_macro". Removing the output_format attribute from a generator group will not solve the error because soft_macro is the default output format.

---

## LBSYN-30

(error) Specify either a "vhdl" or "verilog" group, but not both.

### Description

When the "output_format" attribute specifies "soft_macro", a "vhdl" subgroup or a "verilog" subgroup is required in the "generator" group.

### What Next

Add a "verilog" or "vhdl" subgroup to the generator group. Change the value of the "output_format" to something other than "soft_macro". Removing the output_format attribute from a generator group will not remove the error because soft_macro is the default output format.

---

## LBSYN-31

(information) Generator '%s' assumed to create DB output.

### Description

Neither a "vhdl" group nor a "verilog" group were found within the "generator" group. Thus, the netlist created by the generator is assumed to be a .db file.

### What Next

If the generator does not produce DB files, then please add either a "verilog" or "vhdl" subgroup to the generator group. Alternatively, add the "output_format" attribute with the value "hard_macro" to the generator group to specify a hard macro. The default output format expected from the generator is a soft macro.

---

## LBSYN-32

(error) The formula for parameter '%s' requires a string \n\ expression.

Feedback

**Description**

The expression used in the formula should be a double-quoted string.

**What Next**

Modify the formula attribute to the form 'formula : "expression"'.

---

## LBSYN-33

(error) Only one "sdf" subgroup may exist in a\n \t"generator" group.

**Description**

A "generator" group may have at most one "sdf" subgroup specifying information used during processing of generator produced netlists.

**What Next**

Remove "sdf" groups from the generator group leaving at most one "sdf" group.

---

## LBSYN-34

(warning) The module group '%s' has no output pins defined.\n

**Description**

Synthetic library module declarations must contain at least one output pin group. Additionally, a warning is generated if no input pin groups are declared within the synthetic module.

**What Next**

Add an output pin group to the synthetic module.

---

## LBSYN-35

(warning) The module group '%s' has no input pins defined.\n

**Description**

Synthetic library module declarations should generally contain at least one input pin group. This message can also indicate that a synthetic module has been declared with only one port which has the direction attribute value of "inout." Additionally, a warning is generated if no output pin groups are declared within the synthetic module.

**What Next**

Add an input pin group to the synthetic module.

Feedback

## LBSYN-36

(error) The pin '%s' in module group '%s' has an improper\n \tdirection specified.

**Description**

Pin groups within synthetic library modules must specify a direction. This direction attribute can specify one of three choices: "input", "output", or "inout". For example a valid direction attribute within a pin specification might appear as:

pin (A) { direction : input; bit_width : "width"; }

**What Next**

Change the value of the direction attribute to "input", "output", or "inout".

## LBSYN-37

(warning) The pin associations for the module %s do not\n \tbind to any output pins.

**Description**

One of the pin associations for a module should be bound to an output pin of the module. Note that only one pin association in all of the binding groups needs to be bound to an output pin. Furthermore, module ports of types output and inout are considered valid for pin associations.

**What Next**

Specify a pin association group for the module that names a module pin with the direction of output or inout.

## LBSYN-38

(warning) The pin associations for the module %s do not\n \tbind to any input pins.

**Description**

One of the pin associations for a module should be bound to an input pin of the module. Note that only one pin association in all of the binding groups needs to be bound to an input pin.

**What Next**

Specify a pin association group for the module that names a module pin with the direction of input or inout.

Feedback

## LBSYN-39

(warning) The sequential bindings allow states '%s' and '%s'\n \tto simultaneously use pin '%s' on module '%s'.

### Description

This warning indicates that non-clock pins bound to one or more nonexclusive states have been found in the synthetic library description.

### What Next

Add resources to the module and use these resources within the binding to exclude the possibility of pins being simultaneously used.

## LBSYN-40

(Error) The '%s' group named '%s' is multiply defined.

### Description

Two or more groups exist within the parent of the same type and the same name.

For example in the fragment of synthetic library code:

: : module (ram_wrapper) { resource(address_lines) {} resource(data_in_lines) {} resource(address_lines) {} resource(address_lines) {} : :

The resource group address_lines has been defined three times.

### What Next

Give each group of the same type a unique name or remove redundant groups.

## LBSYN-41

(Warning) The resource named '%s' is used but not defined\n \tin the synthetic module '%s'.

### Description

A use_resource group refers to a resource which has not been defined with a resource group.

In the following fragment of synthetic library code a single resource token, r_only, is defined. However, a resource token is used, undefined_resource_token, which has not been defined.

Feedback

: : module (ram_wrapper) { resource(r_only) {} /* This is the only resource defined. */ binding (b0) { : : state () { : : use_resource(undefined_resource_token) {} } } }

**What Next**

Add a new resource group to the module, or remove the use_resource group that refers to the nonexistent resource.

---

## LBSYN-42

(Warning) Operator '%s' has an excessive number (%d total)\n \tof bindings.

**Description**

The synthetic operator has a large number of either sequential or combinational bindings. A large number of bindings decreases performance, but does not otherwise affect functionality. A large set of bindings usually results from representing each combination of permutable pin with a binding.

Sequential bindings are considered separately from combinational bindings, and in both cases this warning is issued when more than 32 bindings exist.

**What Next**

Verify the set of permutable inputs as specifed through the 'permutable_inputs' attribute. Eliminate unnecessary synthetic bindings.

---

## LBSYN-43

(Error) Unable to find the operator pin '%s' named by an\n \tunbound_oper_pin group in binding '%s' in\n \tmodule '%s'.

**Description**

The unbound_oper_pin group found within a binding group (or a state group within a binding group for sequential bindings) names a pin on the operator that does not exist. Note that the pin names are case-sensitive, so this error can occur if an operator has a pin named "CLK" while the unbound_oper_pin group refers to a pin named "clk". The following synthetic library fragment suffers from this error:

: : operator(oper_0) { data_class : "signed"; pin(CLK) { direction : input; } pin(A) { direction : input; } pin(Z) { direction : output; } }

Feedback

module (module_0) { : : binding(binding_0) { bound_operator : "oper_0"; unbound_oper_pin(clk) { value : "1"; } /* Error: There is no pin named clk (lower case) in oper_0. */ : : } } : :

**What Next**

Assuming that the name of the pin in the operator group is correct, change the name of the pin in the unbound_oper_pin group. Another possibility is to remove the unbound_oper_pin group from the synthetic library if it is unnecessary.

---

## LBSYN-44

(Error) Multiple implementation groups named '%s'\n \tfound in synthetic module '%s'.

**Description**

One or more implementation groups were found to have the same name. Each implementation group used in a set of synthetic libraries must have a unique group name within the context of the implementation's associated module.

**What Next**

Rename the implementation groups in the synthetic library source file (.sl) with unique names.

---

## LBSYN-45

(Error) Input pin '%s' cannot have both the 'sreset' and the\n \t'areset' attribute in the synthetic module '%s'.

**Description**

A synthetic library module pin can only have either the sreset attribute or the areset attribute; not both.

In the following synthetic library source fragment, the pin 'a' is an error, while all other module pins properly specify the sreset and areset attributes.

```
:
          :
      module (module_with_sequential_bindings) {
         pin(a) {
            direction : input;
            bit_width : "width_a";
            sreset    : TRUE;
            areset    : TRUE;
         }
         pin(b) {
            direction : input;
```

Feedback

```
            bit_width : "width_b";
            sreset    : TRUE;
            areset    : FALSE;
         }
         pin(c) {
            direction : input;
            bit_width : "width_c";
         }
            :
            :

   }
      :
      :
```

### What Next

Remove either or both of the 'sreset' or 'areset' groups from the pin group. Another solution is to set one of the attributes' values to FALSE.

---

## LBSYN-46

(Error) The binding '%s' in module '%s'\n \tcontains a constraint specifying a nonexistent pin '%s'\n \tof the operator '%s'.

### Description

Each synthetic library binding group is bound to an operator, and each binding group might contain a constraint group which refers to pins on the synthetic library operator. This error occurs when a constraint refers to a pin on an operator pin that does not exist.

In the following synthetic library source fragment, the binding binding_0 is bound to the synthetic operator oper_0. Furthermore, the binding group binding_0 specifies a constraint group that does not refer to any pin found in the operator oper_0.

: : operator(oper_0) { data_class : "signed"; pin(a) { direction : input; } pin(z) { direction : output; } }

module (module_0) { : : binding(binding_0) { bound_operator : "oper_0"; constraint(BogusPin) { value : "01"; } /* Error: There is no pin named BogusPin in oper_0. */ : : } } : :

### What Next

Modify the constraint to specify a valid operator pin or remove the constraint group altogether.

Feedback

## LBSYN-47

(Warning) The number of permutable inputs (%d) and the\n \tnumber of bindings (%d) do not agree for operator '%s'\n \tin the module '%s'.

**Description**

Each possible ordering of of permutable input pins requires a unique binding. If the total number of bindings found does not equal or exceed the number of permutable inputs factorial (such as, n!), one or more bindings specifying various permutations have not been specified in the synthetic library.

Note that only bindings referring to the same operator within single modules are counted.

**What Next**

Add binding groups to the synthetic library. One binding group is necessary to representing one pin permutation, and all possible pin permutations require a representative binding group.

Alternately, remove the permutable_inputs attribute from the operator group.

## LBSYN-49

(Warning) The binding '%s' in module '%s' refers to an\n\ \toperator, '%s', which is not found in the synthetic library list.

**Description**

The binding group found within the module attempts bind to an operator which does not exist. This is only a warning as binding groups facilitate the creation of an implementation for the operator. A binding which specifies an operator that does not exist provides no useful information to the synthesis tool and is ignored.

**What Next**

Check the operator spelling within the binding group for correctness. Also check the synthetic library or libraries to ensure that the operator has been declared with an operator group.

If the binding refers to an operator which has been removed from the synthetic library, then perhaps the binding group should also be removed from the synthetic library.

## LBSYN-57

(Error) Multiple clock pins exist on module '%s'.

Feedback

**Description**

Only one pin group within a module may specify the 'clock_pin' attribute.

**What Next**

If the module only has combinational bindings, then remove all clock pin attributes from the module as they are unnecessary. If the module contains sequentail bindings or a mixture of sequential and combinational bindings, then remove all but one clock pin attributes from the module's pin groups.

---

## LBSYN-58

(Error) Unbound operator pin value must evaluate to '1'\n \tfor operator pin '%s' in binding '%s' in\n \tmodule '%s'.

**Description**

You receive this message because the *check_synlib* or *compile* command has found that the specified pin's unbound operator pin value does not evaluate to 1. It must have a value of 1.

**What Next**

Locate the specified binding and pin in the appropriate file, and edit the file so that the unbound operator pin value evaluates to 1. Then reexecute *check_synlib* or *compile*.

---

## LBSYN-59

(Warning) The width function is only supported in module group\n \tparameter formulas.

**Description**

Support for the width function only exists for parameters within module groups.

**What Next**

Remove width functions from formulae which are not found in module groups.

Please refer to the DesignWare Developer Guide.

---

## LBSYN-60

(Warning) Module pin '%s' in state '%s'\n \tof binding '%s' for module '%s' is not bound to any\n \toperator pin and the module pin does not specify a\n \tstall value.

Feedback

### Description

This message only occurs for modules containing sequential bindings. One or more states in these bindings does not have all module input pins either mapped to an operator pin or defined with a stall value.

### What Next

All control pins such as write enable must be fully specified at all times. Either add a stall value to the pin's definition in the module or add a pin_association in the state group.

For data pins it is optional to force a default value. If this is the intended usage this message can be ignored.

Please refer to the DesignWare Developer Guide.

---

## LBSYN-61

(Error) Module '%s' is of type '%s',\n \twhich is not a valid type.

### Description

The value of the module_type attribute for this module specifies an incorrect type. Incorrect types are those types which are reserved for internal use. This error message may report a module has a type other than that found in the synthetic library source file (.sl). This indicates that the type specified in the synthetic library source file was not a recognized type.

### What Next

Note that the module_type attribute is optional.

Please refer to the DesignWare Developer Guide.

---

## LBSYN-62

(Error) Operator '%s' is of type '%s',\n \twhich is not a valid type.

### Description

The value of the operator_type attribute for this operator specifies an incorrect type. Incorrect types are those types which are reserved for internal use. This error message may report an error for an operator which has a type other than that found in the synthetic library source file (.sl). This indicates that the type specified in the synthetic library source file was not a recognized type.

### What Next

Note that the operator_type attribute is optional.

Feedback

Please refer to the DesignWare Developer Guide.

## LBSYN-63

(error) Invalid formula '%s' for parameter '%s'.

**Description**

You receive this error message because the formula of the specified parameter contains the parameter itself. This will cause an infinite loop during evaluation of the formula.

**What Next**

Verify that you have specified the correct parameter name and that the formula is accurate.

See the *DesignWare Developer Guide* for more information.

**See Also**

- read_lib

# LCSH

## LCSH-1

(information) Running common shell Library Compiler.

**Description**

This message indicates that current lc_shell is invoked from common shell executable. Or read_lib command is compiling library using common shell based Library Compiler engine.

Since 2014.09, new library compiler is released on amd64 and suse64. for sparc64 platform, library compiler engine is still common shell based. so this information will always showup in 2014.09 lc_shell banner and read_lib command.

read_lib command in DC and ICC by default call new library compiler to compile .lib file into .db. if new library compiler is not defined with SYNOPSYS_LC_ROOT, it still call common shell library compiler to compile the library, with the information reported.

New library compiler currently does not support symbol and synthetic library's compilation natively, so users will also see this information when compile these two types of libraries.

Feedback

### What Next

Users can ignore this information for sparc64; ingore this information when compile symbol and synthetic library. For messages reported on amd64/suse64, users can choose to install lc, and set SYNOPSYS_LC_ROOT to lc installation directory, then read_lib won't report this message.

### Examples

```
Information: Running common shell Library Compiler. (LCSH-1)
```

---

## LCSH-2

(error) No common shell Library Compiler found.

### Description

This message is issued when LC try to call common shell Library Compiler to perform a task.

Common shell Library compiler is needed when: 1. Users explicitly set lc_enable_common_shell_lc true before read_lib. 2. When compile symbol/synthetic libraries.

### What Next

Users need to set SYNOPSYS_SYN_ROOT to a synthesis installation directory in which common shell executable resides, then re-start lc_shell again to run the command.

### Examples

```
Error: No common shell Library Compiler found. (LCSH-1)
```

---

## LCSH-4

(warning) No SYNOPSYS_SYN_ROOT found, some commands are not available.

### Description

This message indicates that some commands are not available since LC cannot find common shell Library Compiler installation by environment variable SYNOPSYS_SYN_ROOT.

The affected commands are: add_pg_pin_to_db add_pg_pin_to_lib lib2saif

### What Next

Please set SYNOPSYS_SYN_ROOT environment variable to synthesis installation directory where common shell executable resides, then re-start lc_shell again to run those commands.

Feedback

### Examples

```
Warning: No SYNOPSYS_SYN_ROOT found, some commands are not available.
 (LCSH-4)
```

## LCSH-5

(fatal) %s is not enabled.

### Description

The application tried to reserve the specified license, but it was not available.

### What Next

Verify that the key file is up to date and that sufficient licenses are available.

# LDB

## LDB-1

(error) Cannot delete views from the non-view-based library .db '%s'.

### Description

This error message occurs because views can only be deleted in a view-based library .db file.

### What Next

Convert the original .db file into a view-based .db file, and read in the file again.

## LDB-2

(error) Cannot delete view '%s'.

### Description

This error message occurs when the specified view cannot be deleted because it's not a valid view name.

### What Next

Check the view name and correct as needed.

Feedback

## LDB-3

(error) Failed to load the skeleton view to get all cell names.

**Description**

This error message occurs because when reading a view for unspecified cells, the tool reads the skeleton view node to get all cell names. If the skeleton view node cannot be read, this step fails and nothing is loaded.

**What Next**

Make sure the skeleton view node can be read successfully by ldbatt.

## LDB-4

(error) Failed to load view node '%s'.

**Description**

This error message occurs when there is incorrect data in the input view-based .db file.

The pattern of the view node name is as follows:

```
<path_name>/<view_based_db_file_name>/<lib_name>/<view_name>
```

**What Next**

Make sure the content in the specified view node is correct (by ldbatt).

## LDB-5

(warning) Cannot read cell '%s'.

**Description**

This error message occurs when the cell name doesn't exist in the library.

**What Next**

Correct cell name.

# LED

## LED-001

(error) %s

Feedback

**Description**

Layout editing internal error.

**What Next**

Please send the log to Synopsys.

---

## LED-002

(error) Placement status cannot be set on object type '%s'.

**Description**

Placement status can only be set on cells and ports.

---

## LED-003

(error) The object '%s' does not exist.

**Description**

The object with specified name does not exist.

---

## LED-004

(error) The layer '%s' cannot be found.

**Description**

The layer with the given name cannot be found in the technology database.

---

## LED-005

(error) The path cannot be found.

**Description**

The routing path cannot be found. The possible reason is that the area is too dense.

---

## LED-006

(error) No valid objects specified.

**Description**

No valid objects were specified for processing.

Feedback

## LED-007

(warning) Maximum processing time reached.

**Description**

DRC is terminated before completion, because the maximum processing time is reached.

## LED-008

(warning) Maximum shape number reached.

**Description**

DRC is not performed because the number of shapes is over the maximum shape number.

## LED-009

(error) No valid anchor specified.

**Description**

No valid anchor objects were specified for processing.

## LED-010

(error) Specified pins are on multiple soft macros or blocks

**Description**

Only pins/port on the same soft macro/block can be aligned at the same time

## LED-011

(error) No tracks found in specified region

**Description**

There are no tracks for a given width found in specified region

## LED-012

(error) Some nets have invalid number of pins

**Description**

The message is reported if net has 0 or 1 pin.

Feedback

## LED-013

(error) Specified track number is invalid

**Description**

There are no track with specified number for a given layer/width

## LED-014

(error) Cannot fit spacified nets on a track

**Description**

The nets bounding boxes overlap too much. Either reoder or drop some nets.

## LED-015

(error) Net bounding box is invalid

**Description**

There could be multiple reasons for failure. The net might have no pins or pins were filtered out. The pullbacks or gaps specified for trunks are too large.

## LED-016

(error) Topology plan contains invalid edges.

**Description**

Some edge might not have valid shape_layers attribute defined.

## LED-017

(error) No nets specified on edge or plan

**Description**

The plan or edge needs to be associated with either nets, bundles or supernets.

## LED-018

(error) Topology plan contains cycles

Feedback

**Description**

The plan edges should not form cycles in topology graph.

---

## LED-019

(warning) Change in placement status on RP blockage %s is disabled.

**Description**

Change in placement status on RP blockage is disabled. Change placement status of associated RP group.

---

## LED-020

(error) Cannot assign objects

**Description**

The message might be triggered in several situations: e.g. if option -keep is used and all target bumps are already assigned. e.g. if nets are skipped due to various reasons.

---

## LED-021

(warning) Some objects were skipped.

**Description**

The operation is not supported for some object types.

---

## LED-022

(warning) '%s' is logical only. Placement status is not available.

**Description**

Placement status cannot be set on a logical only cell/port.

---

## LED-023

(warning) One of the edited objects is outside of design boundary.

**Description**

One of the edited or created objects is outside of design boundary.

Feedback

## LED-024

(warning) The process was interrupted.

**Description**

The process was interrupted by user.

## LED-025

(warning) Some selected object cannot be edited.

**Description**

The selection set might contains fixed or other non editable object.

## LED-026

(warning) Swap action is not performed because %s.

**Description**

Swap cannot be performed on the given objects.

## LED-027

(warning) Cannot %s on the given objects. %s.

**Description**

This message informs you that the given operation cannot be performed. No modification is made as a result.

## LED-028

(warning) Site def associated with given cell does not match cell size.

**Description**

Snapping accuracy of the given object is not guaranteed.

**What Next**

Please run derive_cell_snap_data command to derive an additional cell snapping data.

Feedback

**See Also**

- derive_cell_snap_data

---

## LED-029

(warning) Ambiguity detected when following several objects mask.

**Description**

When using -follow* option several objects were discovered with contradicting mask colors. As the result in such ambiguous situation(s) object mask were not changed according to options.

---

## LED-030

(warning) Target rectangle is too small for %s

**Description**

Target rectangle is too small to fit the object. In case of path object, it might be due to large end caps.

**What Next**

Please specify larger rectangle.

---

## LED-031

(warning) Net '%s' has been unlocked/unfixed

**Description**

physical_status attribute of a net has been set to unrestricted.

---

## LED-032

(warning) Net '%s' has routing layer constraints: min_layer_name=%s max_layer_name= %s

**Description**

This warning indicates that routing is forced outside of max/min routing layers range defined on a net.

Feedback

## LED-033

(warning) cutTblSize does not match actual size of cutNameTbl (or cutNameTbl is missing) for layer '%s' in technology file

**Description**

This warning indicates that cutTblSize contradict to actual size of cutNameTbl. cutTblSize is overridden by actual size of cutNameTbl for a purpose of via generation in interactive routing tools.

**What Next**

Fix technology file to remove this warning.

## LED-034

(warning) Cannot set net estimation rule for the selected topology plan/edge.

**Description**

None of the input estimation rules was found.

**What Next**

Set valid net estimation rules that have already been defined for the selected topology plan/edge.

## LED-035

(warning) Only some of the input net estimation rules were set; cannot set all net estimation rules for the selected topology plan/edge.

**Description**

Only some but not all of the input estimation rules were found.

**What Next**

Make sure all the input net estimation rules are valid and defined for the selected topology plan/edge.

## LED-036

(warning) Topology edge '%s' is skipped.

Feedback

**Description**

The edge might not have valid shape_layers attribute defined or edge length is zero.

---

## LED-040

(warning) Specified leg extension do not reach destination point.

**Description**

Usually it means that sum of all horizontal or vertical extensions does not equal 1 (or 100%). For example, if -layer_legs argument specifies three layers: horizontal, vertical and again horizontal, then the following extensions are considered correct: {0.3 1 0.7}, {1.3 1 -0.3} while these will trigger a warning: {0.3 1 0.8}, {0.3 1 0.6}, {0.3 0.9 0.7}.

**What Next**

The message can be ignored if extensions do not reach destination on purpose, otherwise it needs to be fixed.

---

## LED-041

(warning) %u net(s) were skipped

**Description**

Some nets are skipped due to obstructions or other errors.

**What Next**

Please rerun the command with -verbose switch to get more details about failure.

---

## LED-042

(warning) The coordinate list is unsorted.

**Description**

The specified track numbers or coordinates are not in ascending/descending order.

---

## LED-043

(warning) Option %s is not applicable for annotation_shapes of type %s.

**Description**

Specified option is not applicable for annotation_shapes of given type and will be ignored.

Feedback

**What Next**

The message can be ignored.

---

## LED-044

(warning) The shielding %s of signal bit %s is skipped

**Description**

The shielding net is skipped due to presence of power/ground nets in the vicinity.

**What Next**

if skipping the shielding is not desired, it can be turned off using app option create_trunk.allow_pg_as_shield

---

## LED-050

(error) Net specified in -net option is not valid.

**Description**

Cannot find the net specified with -net option.

**What Next**

Check that net specified in the -net option is actually present.

---

## LED-051

(error) Pin/port for specified net '%s' has multiple terminals on interconnect layer(s).

**Description**

The only way for command to work in -net mode is that specified net should have exactly two pin/ports and each of them should have exactly one terminal (on interconnect layers). Terminals on via layers are ignored.

**What Next**

Check that net specified in the -net option is what was intended. Try to use -pins/-shapes options if net is specified correctly but resistance cannot be calculated b/c of ambiguity in determining start/stop points for resistance estimation.

---

## LED-052

(error) Incorrect number of pins/ports for specified net '%s', should be exactly two.

Feedback

**Description**

The only way for command to work in -net mode is that specified net has exactly two pin/ports and each of them should have exactly one terminal.

**What Next**

Check that net specified in the -net option is what was intended. Try to use -pins/-shapes options if net is specified correctly but resistance cannot be calculated b/c of ambiguity in determining start/stop points for resistance estimation.

---

## LED-053

(error) Invalid pin/port '%s' specified.

**Description**

Specified pin/port is invalid.

**What Next**

Make sure that pin/port in -pins option is specified correctly. Check alternative ways to measure resistance with either -net or -shapes option.

---

## LED-054

(error) Both pins/ports specified in -pins options should belong to the same net.

**Description**

For correct work of command in -pins option mode, exactly two pins/ports should be specified and both should belong to the same net.

**What Next**

Check that pins/ports specified in the -pins option is what was intended. Check alternative ways to measure resistance with either -net or -shapes option.

---

## LED-055

(error) Specified pin/port '%s' has multiple terminals on interconnect layer(s).

**Description**

The only way for command to work in -pins mode is that both specified pins/ports should have exactly one terminal (on interconnect layers). Terminals on via layers are ignored.

Feedback

**What Next**

Check that pin/port specified in the -pins option is what was intended. Check alternative ways to measure resistance with either -net or -shapes option.

---

## LED-056

(error) Incorrect number of pins/ports specified with -pins option.

**Description**

There should be exactly two pins/ports in -pins option mode.

**What Next**

Check that specified pins/ports in -pins option is what was intended. Check alternative ways to measure resistance with either -net or -shapes option.

---

## LED-057

(error) Shapes specified in -shapes option should belong to the same net.

**Description**

For proper work of option -shapes both shapes should belong to the same net.

**What Next**

Check that that shapes specified in -shapes options belong to the same net.

---

## LED-058

(error) Layer specified in '%s' option is incorrect or option is missing.

**Description**

Command working in -shapes option mode requires specification of both -start_layer and -stop_layer options. Both options require correct layer as argument.

**What Next**

Make sure that layer specified in -start_layer/-stop_layer is what was intended.

---

## LED-059

(error) Layer '%s' specified in '%s' option is incorrect or not found.

Feedback

**Description**

Options 'HorizontalLayer" and "VerticalLayer" require layers with specific routing direction.

**What Next**

Make sure that specified layer has proper routing direction.

---

## LED-060

(error) Shapes specified with -shapes option should have net associated.

**Description**

Command working in -shapes option mode requires all shapes to be associated with the net (same net for all of them).

**What Next**

Make sure that all shapes specified with -shapes option have net associated with them.

---

## LED-061

(error) Topology shape '%s' points are not valid.

**Description**

Either end nodes are not set or an 'origin' attribute of end nodes is not defined.

**What Next**

Make sure that specified topology edge has both end nodes assigned.

---

## LED-062

(error) Merge stream file %s does not exist

**Description**

Merge stream file does not exist by the value from signoff.physical.merge_stream_files

**What Next**

signoff.physical.merge_stream_files should either be reset or updated to include a correct stream file

---

## LED-063

(error) The shape_layers are not defined for topology edge '%s'.

Feedback

**Description**

The shape_layers are not defined for specified topology edge.

**What Next**

Set shape_layers for the specified topology edge.

---

## LED-064

(error) Cannot get net estimation rule layers for topology edge '%s'.

**Description**

The layersr defined by the net estimation rule are not set.

**What Next**

Define a valid layers for the topology edge related net estmation rule.

**See Also**

- report_net_estimation_rules
- set_net_estimation_rule

---

## LED-065

(error) Cannot find tap point.

**Description**

It was not possible to locate tap point (to PG mesh) from specified pin using physical connectivity tracing.

**What Next**

Make sure that specified pin is connected to PG mesh. Check what is connected to specified pin using command get_connected_routing <pin>.

---

## LED-066

(error) More than one tap point found.

**Description**

By using physical connectivity tracing starting from specified pin more than one tap point was found on PG mesh.

Feedback

**What Next**

It is still possible to measure resistance from the pin to specific tap point, but -shape option should be used instead. All shapes in the connectivity route from pin to tap should be explicitly specified with the -shapes option. To see all connected routing up to the tap points use command get_connected_routing <pin> -hierarchical -stop_on_shape_use {stripe ring}.

---

## LED-067

(error) No valid menu item %s.

**Description**

Specified menu item did not exist in current menus.

**What Next**

Please try to use the following command to create the menu item and then update the gui preference. gui_create_menu -tcl_cmd "testcmd" -menu "Test->test" gui_set_pref_value -category topologyToolbox -key userButtonsList -value "Test->test"

---

## LED-068

(error) Current design library scale factor is not compatible with the session scale factor.

**Description**

Impossible to create thermal component dedicated library because current design scale factor is not compatible with the session scale factor.

**What Next**

Set app option design.is_3dic_mode to true at the very start of the script to catch this incompatibility when it is first created or even suppress this incompatibility completely in some cases.

Set design.session_scale_factor explicitly to the preferred value to avoid using implicit default value when created new library. App option design.session_scale_factor should be set with no libraries currently opened, otherwise it does not have any effect.

For example, use those lines at the beginning of a script to set default session scale factor to 4000 (by default it is 10000):

set_app_options -name design.is_3dic_mode -value true

set_app_options -name design.session_scale_factor -value 4000

Feedback

## LED-101

(information) %d nets processed.

**Description**

This informational message shows how many nets were processed during automatic bundle generation.

## LED-102

(information) There are %d new bundles defined.

**Description**

This informational message shows how many bundles defined in the design.

## LED-103

(information) Some pin terminals are skipped

**Description**

This informational message indicates that terminal is skipped because parent pin is selected

## LED-104

(information) Some port terminals are skipped

**Description**

This informational message indicates that terminal is skipped because parent port is selected

## LED-105

(information) Some block terminals are skipped

**Description**

This informational message indicates that terminal is skipped because parent instance is selected

Feedback

## LED-106

(information) Some block pins are skipped

**Description**

This informational message indicates that pin is skipped because parent instance is selected

## LED-107

(information) Library cell object(s) are skipped

**Description**

This informational message indicates that object is skipped because parent instance is a library cell

## LED-108

(information) Some site rows are skipped

**Description**

This informational message indicates that site row is skipped because parent site array is selected

## LED-109

(information) Uneditable block object(s) are skipped

**Description**

This informational message indicates that object is skipped because parent instance is an uneditable block. Please see 'set_editability' command man page for more details.

## LED-110

(warning) %s

**Description**

Layout editing warning.

**What Next**

Take note of warning and take any necessary action to correct.

Feedback

## LED-111

(information) Incomplete extensions are split proportionally

**Description**

The message is reported if -extension option is omitted or number of extensions is less than number of layers specified by -layers_legs. Incomplete extensions are calculated automatically. For example, when -extension argumnet is missing, the calculated extensions for single layer it will be {1}, for two orthogonal segments it will be {1 1}, for three orthogonal segments it sill be {0.5 1 0.5}, etc.

## LED-112

(information) %u net(s) violate given freedom options

**Description**

The message is reported when trunk segment is shifted away from source/destination point. The tollerence distance is specified by -horizontal_freedom/-vertical_freedom.

## LED-113

(information) The layer '%s' has duplicated mask '%s'

**Description**

The given layer has a mask which is already assigned to other layer.

## LED-114

(error) No IC Validator Live DRC rules are selected

**Description**

No IC Validator Live DRC rules are selected.

**What Next**

Please select IC Validator Live DRC rules to check

## LED-115

(error) Application option '%s' is required in order to run %s

Feedback

**Description**

Application option is required for the specified functionality.

**What Next**

Please set a valid value to the specified Application option

---

## LED-116

(error) More than %d polygons detected. Cannot run IC Validator Live DRC. IC Validator In-Design is recommended.

**Description**

IC Validator Live DRC hard polygon limit was exceeded.

**What Next**

Please run IC Validator Live DRC on smaller region

---

## LED-117

(warning) More than %d polygons detected. IC Validator Live DRC performance can be affected. IC Validator In-Design is recommended.

**Description**

IC Validator Live DRC soft polygon limit was exceeded.

**What Next**

Please run IC Validator Live DRC on smaller region or run the IC Validator InDesign .

---

## LED-118

(information) Some nodes are skipped

**Description**

This informational message indicates that topology node is skipped because related topology edge is selected

---

## LED-119

(information) Some repeaters are skipped

Feedback

**Description**

This informational message indicates that topology repeater is skipped because related topology edge is selected

---

## LED-120

(information) No IC Validator Live DRC error environment file was generated

**Description**

This informational message indicates the error environment file was missing after an on-demand merge IC Validator Live DRC run.

**What Next**

Please check your ICV engine settings, or send the log to Synopsys

---

## LED-123

(information) Some bump regions are skipped

**Description**

This informational message indicates that bump region is skipped because related pseudo bumps or tsv is selected

---

## LED-200

(warning) %s

**Description**

Layout editing warning.

**What Next**

Take note of warning and take any necessary action to correct.

---

# LEFPARS

---

## LEFPARS-1

(error) %s

Feedback

**Description**

A syntax error occured in LEF parser.

**What Next**

Check reported line and token in the LEF file with LEFDEF Reference Manual.

# LEFR

## LEFR-001

(error) Cannot find lef file '%s'.

**Description**

The file, specified in the read_lef command, cannot be read.

**What Next**

Check the existence of the file in the search_path using the *which* command.

## LEFR-002

(error) read lef file '%s' failed.

**Description**

read lef file failed.

**What Next**

Check lef file syntax and content.

## LEFR-003

(error) Cannot create %s '%s'.

**Description**

There is an error creating the specified object in the database. For VIA/VIARULE GENERATE, it may due to incorrect definition in LEF. For other objects, it is due to internal error.

**What Next**

Check the LEF definition corresponding to the specified object.

Feedback

## LEFR-004

(error) Line %d, Cannot find %s '%s'.

**Description**

There is an error finding the specified object in the database.

**What Next**

Check the LEF definition corresponding to the specified object.

## LEFR-005

(error) Line %d, unknown or wrong %s '%s'.

**Description**

There is an syntax error of the reported type in the reading LEF file.

**What Next**

Check syntax related to the reported type in the lef file.

## LEFR-006

(warning) Line %d, Macro '%s' has duplicate definition of PIN '%s', the first one is used.

**Description**

The reported PIN has been defined more than once. The later definition will be ignored.

**What Next**

Check lef file content related to the reported port.

## LEFR-006e

(error) Line %d, Macro '%s' has duplicate definition of PIN '%s'.

**Description**

The reported PIN has been defined more than once. Library developer must clean the duplicated data.

**What Next**

Check lef file content related to the reported port.

Feedback

## LEFR-007

(error) memory allocation error.

**Description**

There is a memory allocation or reallocation error when reading def file.

**What Next**

Check available machine memory and determine if there is enough memory to hold the design.

## LEFR-008

(warning) Line %d, macro %s has site statement '%s' with site pattern.

**Description**

Site patterns are not supported. It will be ignored.

**What Next**

Remove the site pattern from the LEF file.

## LEFR-009

(error) Line %d, Macro '%s', %s '%s' cannot be created.

**Description**

The indicated macro object could not be created in the database. It could be either incorrect definition or system error.

**What Next**

Check the macro object definition in the LEF file.

## LEFR-010

(warning) Line %d, Via definition '%s' has shapes on invalid layer '%s'.

**Description**

The indicated via definition has shapes on an invalid layer. All via shapes must be on an interconnect, local_interconnect, via_cut, or local_via_cut layer.

Feedback

**What Next**

Check the layer_type attribute of the indicated layer. Correct either the layer_type by reloading the correct technology file, or correct the LEF input file.

---

## LEFR-010e

(error) Line %d, Via definition '%s' has shapes on invalid layer '%s'.

**Description**

The indicated via definition has shapes on an invalid layer. All via shapes must be on an interconnect, local_interconnect, via_cut, or local_via_cut layer.

**What Next**

Check the layer_type attribute of the indicated layer. Correct either the layer_type by reloading the correct technology file, or correct the LEF input file.

---

## LEFR-011

(warning) Line %d, Invalid bus bit characters '%s' are specified; using default characters '[]'.

**Description**

The bus bit characters specified in the LEF file are invalid. Bus bit character strings must consist of exactly two characters. The default bus bit characters '[]' will be used.

**What Next**

Verify the bus information. If not correct, fix the LEF file's bus bit characters and read it again.

---

## LEFR-012

(error) The library has no technology information.

**Description**

The library was created without a technology file. The library must have technology information to correctly read the LEF file.

**What Next**

Load a technology file into the library using the *read_tech_file* command.

Feedback

## LEFR-013

(warning) Line %d, LEF distance unit precision '%d' is not supported.

**Description**

The database supports distance precisions up to and including 10000. By reading a LEF file with higher precision, you may have round-off errors in your distance values.

**What Next**

Check your LEF file for distance values requiring precision greater than 10000. These values will be rounded.

## LEFR-014

(warning) Missing design view for timing cell '%s'.

**Description**

The LEF and .db files are mismatched. The indicated cell has a timing view (loaded from the .db file), but no design view (loaded from the LEF file).

**What Next**

Check that you have loaded the correct LEF and .db files.

## LEFR-015

(warning) Line %d, Failed to change layer '%s' direction from %s to %s.

**Description**

The layer's current routing direction is already set and it differs from the direction specified in the input LEF file. The current routing direction is not changed.

**What Next**

Check that you have loaded the correct LEF file. Check that the LEF file matches the technology file which was loaded into the library.

## LEFR-016

(information) Line %d, Converted site definition '%s' to '%s'.

Feedback

**Description**

The indicated site definition name was changed to the new name, as specified in the *-convert_sites* option. All macros associated with site definition will be associated with the newly named site definition.

**What Next**

Verify that you have specified the correct site definition names.

---

## LEFR-017

(warning) Line %d, Ignoring unsupported syntax '%s' on %s '%s'.

**Description**

The indicated syntax is not supported and will not be annotated on the object.

**What Next**

Check whether you can ignore this warning message or need to import the data using additional commands.

---

## LEFR-018

(error) Line %d, Site '%s' already exists having width %s and height %s, which conflicts with the LEF site width %s and height %s.

**Description**

The indicated site already exists in the library, and its dimensions do not match the LEF site definition. The LEF site definition cannot overwrite the existing site in the library. The existing library site is not modified.

**What Next**

Check whether you have read the correct LEF file. Check whether the LEF or existing library site definition is correct.

---

## LEFR-019

(information) Creating rectilinear boundary for macro '%s' using its overlap shapes.

**Description**

The macro contains overlap shapes. Its boundary will be set according to the overlap shapes defined in the macro.

Feedback

**What Next**

Check the boundary of the macro.

---

## LEFR-020

(warning) Macro '%s' has multiple disjoint overlap shapes; ignoring all but the first set of contiguous shapes.

**Description**

The macro contains multiple disjoint overlap shapes. Its boundary will be set according to the first set of contiguous overlap shapes defined in the macro. All other overlap shapes are ignored.

**What Next**

Check the boundary of the macro.

---

## LEFR-021

(error) Line %d, %s '%s', Layer '%s' does not exist in the library technology information.

**Description**

The indicated layer was not defined in the technology file which was loaded into the library. All layers must be specified in the loaded technology file.

All geometries defined on this missing layer will be ignored.

**What Next**

Manually add the missing layer into the technology file, load the technology file, and then read the LEF file.

---

## LEFR-022

(information) Read LEF into %s library '%s'.

**Description**

This message reports the name of the library to accomodate LEF data. It also indicates whether the library is an existing library or a new library.

**What Next**

No action required.

Feedback

## LEFR-023

(warning) Macro '%s', Pin '%s', The antenna area value %g is invalid; %s.

**Description**

The specified value is invalid for the indicated reason. Area values must be non-negative. Furthermore area values are invalid if the value is too large, causing an arithmetic overflow error, when converted into internal database units. This value will be ignored.

**What Next**

Correct the LEF file so that it contains a valid area value.

## LEFR-024

(warning) Line %d, Via definition '%s' is invalid; %s.

**Description**

The via definition is invalid for the indicated reason. All LEF via definitions must contain shapes on two adjacent routing layers. This via definition is ignored.

**What Next**

Check the LEF via definition. If the via definition appears to be correct, check the Technology File's layer definitions, and make sure they match up with the LEF file's layer definitions.

## LEFR-025

(Information) Line %d, %s '%s' already exists in %s (will %s).

**Description**

This message indicates that specified object already exists when reading a LEF file into an existing library. The library manager shell, lm_shell, will take action according to the merge settings.

**What Next**

No action required.

## LEFR-026

(information) Line %d, Change %s name '%s' from LEF file to '%s' to resolve a conflict.

Feedback

**Description**

When the object to be created by read LEF conflict with an existing object, there is an attempt to change the name of the object to be created to a unique name to resolve the conflict.

Here are some situations that this renaming will occur when a conflict found:

- read LEF into an existing library with "-merge add" option specified
- read LEF into a new library and *file.lef.auto_rename_conflict_sites* app option is true

For most object types, conflict means that an object with same name already exists. However, for site definition object type, the conflict means a same name site definition exists with a different size. If a same name site definition exists with a same size, it won't be considered as a conflict, and the symmetry information will be appended to the existing site definition.

**What Next**

No action required.

---

## LEFR-027

(error) Library name '%s' cannot be same as workspace name.

**Description**

The library name specified for read LEF cannot be the workspace name.

**What Next**

Choose a library name other than the workspace name to read LEF in.

---

## LEFR-028

(error) Library '%s' is not a physical lib_cell library.

**Description**

The library specified for read LEF must be a physical lib_cell library.

**What Next**

Choose a physical lib_cell library to read LEF in.

---

## LEFR-029

(warning) '%s' will be ignored when read LEF into a new library.

Feedback

**Description**

When read LEF into a new library, merge settings will be ignored.

**What Next**

Do not specify merge related options when read LEF into a new library.

---

## LEFR-030

(warning) Line %d, Cannot update or overwrite the '%s' via definition.

**Description**

The existing via definition has different type with the via definition to be created. The library manager shell, lm_shell, cannot overwrite or update it.

**What Next**

Try to remove the existing via definition before creating the new one.

---

## LEFR-031

(warning) Line %d, Invalid maximum mask number '%d' will be ignored for layer '%s'.

**Description**

The maximum mask number specified in the LEF file for the specified layer is not one of the valid values (2 or 3). The specified value is ignored and the tool behaves as though a maximum mask number was not specified for the layer.

**What Next**

Modify the maximum mask number in the LEF file and read it again.

---

## LEFR-032

(warning) Line %d, LAYER '%s' has geometry with MASK '%d' defined, which exceeds maximum MASK number '%d'.

**Description**

This message was issued because the geometry has defined a larger MASK on the LAYER which exceeds the maximum mask number defined in LAYER section. The invalid MASK will be ignored and the geometry will be treated as no_mask.

**What Next**

Correct MASK number in geometry in the LEF file and read it again.

Feedback

## LEFR-033

(warning) Line %d, Macro '%s', FIXEDMASK is defined but there has geometry without a mask.

**Description**

This message was issued because the FIXEDMASK statement exists for the macro or the library, but no mask is specified for some geometries.

**What Next**

Specify a mask number for each geometry for the macro if FIXEDMASK statement is on.

## LEFR-034

(error) Line %d, Macro '%s' does not specify site and no default site to use either.

**Description**

Macro doesn not specify site and can not find default site to use.

**What Next**

Either specify site in Macro or set isDefault in technology file tile section.

## LEFR-035

(warning) Line %d, Macro '%s' does not specify site, using default site '%s'.

**Description**

The macro doesn't specify site and we find default site to use.

**What Next**

Check whether the default site is proper to use.

## LEFR-036

(error) Line %d, '%s' is a version %g or later syntax. Your LEF file is defined with version %g.

**Description**

The given syntax is not supported with the version defined in the LEF file.

Feedback

### What Next

Use a higher version for the LEF file that support the given syntax.

---

## LEFR-037

(warning) Line %d, 'SYMMETRY' is not defined for macro '%s' , only 'N' rotation is allowed.

### Description

According to the LEF syntax, when there is no symmetry definition for a macro, only the N orientation is allowed.

### What Next

Check whether the default orientation is proper to use.

---

## LEFR-038

(warning) Removed unused site definition '%s'.

### Description

Indicates that a site definition is removed. This site definition had been renamed to avoid a conflict with an existing site definition. However, after read all data in the LEF file, there is no macro refers to the site definition. So the site defintion is removed.

### What Next

If the site definition is really needed, please change its name to avoid conflict with an existing site definition.

### See Also

- file.lef.auto_rename_conflict_sites
- LEFR-039

---

## LEFR-039

(warning) Unused conflict site definition will be removed when file.lef.auto_rename_conflict_sites is true.

### Description

When *file.lef.auto_rename_conflict_sites* is true and there is a conflict site definition, if the conflict site definition is not referred by any macros, it will be removed eventually. The message is reported to warn about this situation.

Feedback

### What Next

If the situation really happen, there will be *LEFR-038* messages reported. Please refer to the man page of that message for suggestions.

### See Also

- file.lef.auto_rename_conflict_sites
- LEFR-038

---

## LEFR-040

(warning) Layer '%s', The rule '%s' is already in the TF, LEF will skip this rule.

### Description

During read_lef, if TF exists during create_workspace, TF will be regarded as golden by default. If a rule already existed in the TF, LEF will skip it and proceed to next.

### What Next

Using option*"-tech_rules overwrite"* in read_lef can overwrite the rule in the TF.

---

## LEFR-041

(error) Cannnot read LEF file to workspace library.

### Description

The library specified for read LEF cannot be the workspace library.

### What Next

Choose a library name other than the workspace name to read LEF in.

---

## LEFR-042

(warning) Line %d, Skipped overlapping density rectangle '%s' in layer '%s' Macro '%s'.

### Description

This messeage reported when a rectangle overlaps with any of the previous ones on the same layer, the latter rectangle will be skipped.

### What Next

Check the density rectangles, resize the rectangle if needed, then reload the file.

Feedback

## LEFR-043

(warning) Line %d, Ignoring the '%s' site specification for the '%s' block, %s.

### Description

You can specify a nondefault site for a block only if its *design_type* attribute has one of the following settings: *lib_cell*, *diode*, *filler*, *well_tap*, *end_cap*, or *corner*.

The tool issues this warning if you specify a nondefault site for a block whose *design_type* attribute does not have one of these settings. The tool behavior in this case depends on whether a default site is defined.

- If the default site is not defined, the tool ignores the site specification for the block, and the block will not have a site specification.
- If the default site is defined, but differs from the specified site, the tool sets the default site on the block.

### What Next

Check the default site definition and verify that the tool behaves as expected.

## LEFR-044

(information) Line %d, Convert fixed via definition '%s' to simple via definition.

### Description

The specified custom via definition is converted to a simple via definition for it meets following conditions: - it contains no polygons - it has uniform mask pattern - it contains a full cut pattern

### What Next

No action required.

## LEFR-045

(warning) Line %d, The mask pattern for via definition '%s' is invalid.

### Description

The mask pattern for the specified via definition is invalid. Here are valid mask pattern specifications: 1. There is no mask value specified for all via cuts. In this case, the mask pattern will be alternating. 2. The mask values of all via cuts are same. In this case, the mask pattern will be uniform. 3. The mask values of via cuts cycling through one row from

Feedback

left to right, and cycling through the start cuts of each columns from bottom to up. In this case, the mask pattern will be alternating.

**What Next**

Correct the mask specification of the via definition and read the LEF again.

---

## LEFR-046

(warning) Line %d, Layer '%s' has no MASK defined, maximal supported value will be used.

**Description**

This message was issued because the layer has geometry with MASK value defined, but the layer has no maximal MASK value specified. This usually happens while the layer is defined by technology file(TF). A default maximal MASK (current is 3 for metal layer and 15 for via layer) value will be applied to the layer then.

**What Next**

If maximal MASK is not feasiable for this layer, please define a technology LEF file with correct MASK setting and update the library.

---

## LEFR-047

(warning) Macro '%s', blockages connected to port '%s' will not be converted as it will cause the port to be short with port '%s'.

**Description**

Some blockages, if converted to real metal shapes, will cause two ports to be short. So connectivity tracer will ignore these blockages.

**What Next**

This is a warning message. Please verify that the result is expected.

---

## LEFR-048

(warning) Line %d, Macro '%s' has no overlap shapes.

**Description**

The option -cell_boundary by_overlap_layer is specified. But there is no overlap shapes for the specified macro.

Feedback

**What Next**

This is a warning message and won't stop the lef reader. However, please verify that the result is expected.

---

## LEFR-049

(warning) Line %d, %s '%s' of macro '%s' will be ignored by LEF connectivity tracer.

**Description**

The geometry of the specified object has edge(s) that not parallel to x-axis or y-axis. This kind of objects will be ignored by the connectivity tracer although they can still be read in by read_lef.

**What Next**

This is a warning message. Please verify that the result is expected.

---

## LEFR-050

(warning) Line %d, Ignoring %s '%s'; %s.

**Description**

The object will be ignored for the indicated reason.

**What Next**

Check whether you can ignore this warning message.

---

## LEFR-051

(warning) Failed to set attribute %s on %s '%s'.

**Description**

The message indicated that set an attribute on specified object failed.

**What Next**

Check whether this warning message can be ignored.

---

## LEFR-052

(warning) Line %d, Ignoring %s on %s '%s'; %s.

Feedback

**Description**

The attribute on specified object will be ignored for the indicated reason.

**What Next**

Check whether you can ignore this warning message.

---

## LEFR-053

(Information) Line %d, %s '%s' of %s '%s' does not exist (will %s).

**Description**

This message indicates that specified object does not exist when reading a LEF file into an existing library. The library manager shell, lm_shell, will take action according to the merge settings.

**What Next**

No action required.

---

## LEFR-054

(Information) Updated %s %s.

**Description**

This message indicates database is updated by reading LEF.

Usually the updating happens when there are an existing object in database which having same name with a objects in LEF and user requires to updating some attributes of the existing database object by LEF.

Updating behavior of read_lef could be controlled by options, e.g. -merge_action attributes_only

This message will show which attribute of which object is updated and the value change in most situations, e.g. Information: Updated direction of pin 'pinA' in macro 'macroA' from 'out' to 'inout'. (LEFR-054)

This message will also show additional notification when the updating is caused by a "not set" statements in LEF which has a default value in LEF specification. Information: Updated site_name of macro 'macroA' from 'CORE_1' to 'unit'(default value when not specifying site name of macro in LEF). (LEFR-054)

For antenna attributes, tool will only show a simple info to notify user they may be changed when updating attributes of a existing macro pin by LEF: Information: Updated antenna properties of pin 'pinA' in macro 'macroA' , if there is any change. (LEFR-054)

Feedback

### What Next

No action required if all the updating is as user expected.

---

## LEFR-055

(Warning) Line %d, eeq_class attribute on ports of LEF pin %s will not be set as attribute value exceeds its range.

### Description

This message indicates that number of ports within a LEF pin exceeds count 256 which determine value for attribute eeq_class on ports.

Currently only 256 values (0-255) are supported for this attribute.

### What Next

Support for exceeding this range can be discussed as required.

---

## LEFR-056

(error) Line %d, No geometry for %s '%s' of macro '%s'.

### Description

The specified object does not have any geometry. For signal pin, the geometry must be defined.

### What Next

Correct the source LEF file and try again.

### See Also

- file.lef.allow_empty_pin
- LEFR-057

---

## LEFR-057

(warning) Line %d, No geometry for %s '%s' of macro '%s'.

### Description

The specified object does not have any geometry. For Power/Ground pin, the geometry can be omitted. This is just a warning.

Feedback

### What Next

Double check whether the object without any geometry is expected.

### See Also

- file.lef.allow_empty_pin
- LEFR-056

---

## LEFR-058

(warning) %s because of app option '%s' was set to '%s'.

### Description

This warning message indicates that some special behavior were enabled for the read_lef command.

### What Next

Double check whether the behavior enabled by the setting is desired.

---

## LEFR-059

(warning) Line %d, %s on %s '%s' already defined (will %s).

### Description

This message indicates that a duplicate attribute definition was found. The library manager shell, lm_shell, will take the action reported in this message.

### What Next

If the specified action is not expected, please correct the input LEF file and try again.

---

## LEFR-060

(error) Length unit of technology is not 'micron', cannot read lef file.

### Description

LEF file can only be read into a tech with length unit as 'micron'.

### What Next

Please use a tech with length unit as 'micron'.

Feedback

## LEFR-061

(error) Line %d, Invalid %s '%s'; %s.

### Description

The specified object is invalid for the indicated reason.

### What Next

Correct the input LEF file and try again.

## LEFR-062

(Information) Line %d, Set is_zero_spacing for %d/%d non-real cut OBS in macro %s.

### Description

This message indicates that some or all cut routing blockages in the specified macro were set as zero-spacing.

### What Next

No action required.

### See Also

- file.lef.non_real_cut_obs_mode

## LEFR-063

(warning) Ignoring PROPERTY %s on %s '%s'; not defined in PROPERTYDEFINITIONS.

### Description

For any PROPERTY statement, the definition of it must be previously specified in the PROPERTYDEFINITIONS section. In following example, it defines a property "myProp" on macro with integer type in PROPERTYDEFINITIONS section. Then a PROPERTY statement is specified in a MACRO section to really set the value to "myMacro" macro.

```
PROPERTYDEFINITIONS
  MACRO myProp INTEGER ;
END PROPERTYDEFINITIONS


MACRO myMacro
  ...
  PROPERTY myProp 100 ;
END myMacro
```

Feedback

If a property was not defined in the PROPERTYDEFINITIONS section, then any use of this property will be ignored with LEFR-063 warning message. In above example, if "myProp" was not defined in PROPERTYDEFINITIONS section, then following message will be reported for "PROPERTY myProp 100" statement:

```
Warning: Ignoring PROPERTY myProp on macro 'myMacro'; not defined in
PROPERTYDEFINITIONS. (LEFR-063)
```

**What Next**

Add property definition to PROPERTYDEFINITIONS section for the property.

---

## LEFR-064

(information) Changed the site definition for the %s library cell to %s from %s.

**Description**

The *read_lef* command issues this message when it changes the site definition for a library cell. In the LEF file, a library cell is defined with a MACRO statement, and its site definition is specified in the SITE attribute.

By default the *read_lef* command changes the site definition for a library cell in either of the following situations:

- The size of the default site definition matches the size of the library cell

  In this case, the *read_lef* command assigns the default site definition to the library cell.

- The size of the site definition in the LEF file does not match the size of the library cell

  In this case, the *read_lef* command sets the site definition for the library cell to the smallest site definition that matches the size of the library cell.

If the size of the default site definition does not match the size of the library cell, but its SITE attribute does, the command does not change the site definition for the library cell.

You can use the *-preserve_lef_cell_site* option to keep site definition from input LEF.

**Example Message**

```
Information: Changed the site definition for the AND library cell
to unit from CORE. (LEFR-064)
```

**What Next**

If necessary, you can explicitly change the site definition for a cell by using the *set_cell_site* or *set_attribute* command.

Feedback

### See Also

- read_lef
- set_attribute
- set_cell_site

---

## LEFR-065

(error) The height or width of MACRO %s is not a multiple of any site definition in the technology data.

### Description

The *read_lef* command issues this error message if it cannot find a site definition that matches the size of the specified MACRO in the technology data associated with the library workspace.

If a cell does not have a site definition, the placer cannot automatically put the cell on any row.

### Example Message

```
Error: The height or width of MACRO AN2X3 is not a multiple of any
site definition in the technology data. (LEFR-065)
```

### What Next

Check the size of the specified MACRO, or add a new site definition for this type of cell.

### See Also

- create_site_def
- read_lef

---

## LEFR-066

(warning) Line %d, OBS on layer %s has %s %g; the blockages will be %s in the frame view.

### Description

The *read_lef* command issues this warning message when an OBS definition has a nonzero SPACING or DESIGNRULEWIDTH attribute.

Feedback

A nonzero SPACING or DESIGNRULEWIDTH attribute causes the blockage generated in the frame view to be bloated or trimmed. This might not fully match the intent, and could cause DRC violations.

### Example Message

```
Warning: Line 8598, OBS on layer M0 has SPACING 0.5; the blockages will
be bloated in the frame view. (LEFR-066)
```

### What Next

Update the OBS definition in the LEF file to remove the SPACING or DESIGNRULEWIDTH attribute.

### See Also

- read_lef

---

## LEFR-066e

(error) Line %d, OBS on layer %s has %s %g; the blockages will be %s in the frame view.

### Description

When the *lib.workspace.library_developer_mode* application option is *true*, the *read_lef* command issues this error message if an OBS definition has a nonzero SPACING or DESIGNRULEWIDTH attribute.

A nonzero SPACING or DESIGNRULEWIDTH attribute causes the blockage generated in the frame view to be bloated or trimmed. This might not fully match the intent, and could introduce DRC violations.

The library developer should fix this issue before the cell is used in designs.

Note that the DESIGNRULEWIDTH attribute is not supported for OBS definitions for 20-nm and below process nodes.

### Example Message

```
Error: Line 8598, OBS on layer M0 has SPACING 0.5; the blockages will
be bloated in the frame view. (LEFR-066)
```

### What Next

Update the OBS definition in the LEF file to remove the SPACING or DESIGNRULEWIDTH attribute.

Feedback

### See Also

- read_lef
- lib.workspace.library_developer_mode

---

## LEFR-067

(warning) Line %d, OBS %s on layer %s does not have a "SPACING 0" attribute but overlaps or touches pin %s.

### Description

The *read_lef* command issues this warning message when an OBS statement defines a regular blockage that overlaps or touches a pin.

When a regular blockage overlaps or touches a pin, the router might not be able to connect the pin without DRC violations because the blockage reduces the available routing resources and might even block the routing. In addition, this type of blockage might increase the routing runtime.

### Example Message

```
Warning: Line 5590, OBS (1,1 2,2) on layer M0 does not have a
"SPACING 0" attribute but overlaps or touches pin AO222D0/A. (LEFR-067)
```

### What Next

Add the SPACING 0 attribute to the OBS LAYER or redesign the blockage.

### See Also

- read_lef

---

## LEFR-067e

(error) Line %d, OBS %s on layer %s does not have a "SPACING 0" attribute but overlaps or touches pin %s.

### Description

When the *lib.workspace.library_developer_mode* application option is *true*, the *read_lef* command issues this error message when an OBS statement defines a regular blockage that overlaps or touches a pin.

When a regular blockage overlaps or touches a pin, the router might not be able to connect the pin without DRC violations because the blockage reduces the available routing resources and might even block the routing. In addition, this type of blockage might increase the routing runtime.

Feedback

The library developer should fix this issue before the cell is used in designs.

### Example Message

```
Error: Line 5590, OBS (1,1 2,2) on layer M0 does not have a
"SPACING 0" attribute but overlaps or touches pin AO222D0/A. (LEFR-067)
```

### What Next

Add the SPACING 0 attribute to the OBS LAYER or redesign the blockage.

### See Also

- read_lef
- lib.workspace.library_developer_mode

---

## LEFR-068

(warning) Line %d, OBS %s on layer %s is less than the fat spacing, %s, of pin %s.

### Description

The *read_lef* command issues this warning message when an OBS statement defines a large blockage that is the minimum spacing or less away from the specified pin.

If a large blockage is not a zero-spacing blockage, it might cause different-net spacing violations during routing.

### Example Message

```
Warning: Line 5590, OBS (1,1 2000,2000) on layer M0 is less than the fat
spacing, 0.056, of pin AO222D0/A. (LEFR-068)
```

### What Next

If the blockage can be a zero-spacing blockage, add the "SPACING 0" attribute to the OBS LAYER; otherwise, redesign the blockage.

### See Also

- read_lef

---

## LEFR-068e

(error) Line %d, OBS %s on layer %s is less than the fat spacing, %s, of pin %s.

Feedback

### Description

When the *lib.workspace.library_developer_mode* application option is *true*, the *read_lef* command issues this error message when an OBS statement defines a large blockage that is the minimum spacing or less away from the specified pin.

If a large blockage is not a zero-spacing blockage, it might cause different-net spacing violations during routing.

The library developer should fix this issue before the cell is used in designs.

### Example Message

```
Error: Line 5590, OBS (1,1 2000,2000) on layer M0 is less than the fat
spacing, 0.056, of pin AO222D0/A. (LEFR-068)
```

### What Next

If the blockage can be a zero-spacing blockage, add the "SPACING 0" attribute to the OBS LAYER; otherwise, redesign the blockage.

### See Also

- read_lef
- lib.workspace.library_developer_mode

---

## LEFR-069

(warning) The size of the %s library cell does not match its assigned site %s.

### Description

The *read_lef* command issues this warning message when the height or width of the specified library cell is not an integer multiple of its assigned site. In the LEF file, the library cell is defined by a MACRO statement, and the site definition is specified in the SITE attribute.

When the library cell size does not match the site size, the placer cannot place the cell on the site row.

### Example Message

```
Warning: The size of the AND library cell does not match its assigned
site CORE. (LEFR-069)
```

### What Next

Change the site definition by using the *set_cell_site* or *set_attribute* command.

Feedback

### See Also

- read_lef
- set_attribute
- set_cell_site

## LEFR-069e

(error) The size of the %s library cell does not match its assigned site %s.

### Description

When the *lib.workspace.library_developer_mode* application option is *true*, the *read_lef* command issues this error message when the height or width of the specified library cell is not an integer multiple In the LEF file, the library cell is defined by a MACRO statement, and the site definition is specified in the SITE attribute.

When the size of a library cell does not match its site definition, the placer cannot place the cell on the site row.

### Example Message

```
Error: The size of the AND library cell does not match its assigned
site CORE. (LEFR-069e)
```

### What Next

Change the site definition by using the *set_cell_site* or *set_attribute* command.

### See Also

- read_lef
- set_attribute
- set_cell_site
- lib.workspace.library_developer_mode

## LEFR-070

(information) The error data is saved in %s; use the error browser to view the errors.

### Description

If the *read_lef* command identifies issues, it saves the information in an error data file. You can use the error browser in the GUI to load the error data and view the issues.

Feedback

### What Next

Check the error data and fix the issues.

### See Also

- read_lef

---

## LEFR-071

(error) The library workspace has %d cell layout errors.

### Description

When the *lib.workspace.library_developer_mode* application option is *true*, the *read_lef* command issues this error message when it finds errors in the physical layout of one or more cells. These errors might cause DRC violations in the generated frame views.

The library developer should fix these issues before using the LEF file to generate a reference library.

### What Next

Fix the issues in the MACRO definitions in the LEF file and then read the LEF file again.

### See Also

- read_lef
- lib.workspace.library_developer_mode

---

## LEFR-072

(warning) The library workspace has %d cell layout warnings.

### Description

The *read_lef* command issues this warning message when it finds issues in the physical layout of one or more cells. These issues might cause DRC violations in the generated frame views.

### What Next

Either fix the issues before running the *check_workspace* command or adjust the frame view generation application options to avoid the issues.

Feedback

**See Also**

- configure_frame_options
- read_lef

---

## LEFR-074

(warning) Cannot create %s '%s'.

**Description**

This warning is issued since the specified object has incorrect definition, thus it cannot be created in the database.

**What Next**

Check the LEF definition corresponding to the specified object.

---

## LEFR-075

(warning) Line %d, %s '%s', Layer '%s' does not exist in the library technology information, it will be ignored.

**Description**

The indicated layer was not defined in the technology file which was loaded into the library.

If this is a LAYER in LEF file, it must be specified in the loaded technology file before it is used. If this is a VIA in LEF file, the VIA that the layer is used will not be created.

**What Next**

Manually add the missing layer into the technology file, load the technology file, and then read the LEF file.

---

## LEFR-076

(warning) '%lu' cell(s) read in through previous read_lef still have error(s).

**Description**

Current read_lef has no error found, but some cells read in through previous read_lef have marked as having error. If any cell has error, create_frame will not proceed until you fix the error.

**What Next**

You can use *read_lef -merge_action overwrite* to overwrite the cell with correct definiton.

Feedback

## LEFR-077

(error) : Failed to create cell group '%s' for cell list (%s)!

### Description

This message indicates that the cell group is not created successfully from EEQ in input LEF file.

### What Next

Please check the error message printed by create_cell_group and correct them. For example, if the cell group already exists, please use command remove_cell_group to remove the cell group and redo read_lef.

### See Also

- read_lef

## LEFR-078

(information) : Cell group '%s' is created successfully for cell list (%s)!

### Description

This message indicates that the cell group is created successfully from EEQ in in input LEF file. Users can check the result by *report_cell_groups*.

### What Next

No action required

### See Also

- read_lef

## LEFR-079

(information) : Line %d, %s %s has same structure as existing '%s' in technology, it will be ignored.

### Description

This message indicates that the via_def generated from VIA/VIARULE GENERATE has same structure as existing same name via_def in the technology of the library, read_lef will drop the via_def, even the merge_action is *add*.

Feedback

Redundant via_def will bring duplicate via regions in frame, and consequently impacts the performance of router/placement.

**What Next**

No action required

**See Also**

- read_lef
- get_via_defs

---

## LEFR-080

(warning) : Line %d of '%s', layer number %d is not found in library technology, mapping of LEF layer ‘%s’ will be ignored.

**Description**

This message indicates that the layer number in the mapping is not found in current technology, thus the mapping will be ignored.

Mapped layer number is defined by the second value in the line, if it is not a number, 0 will be used.

**What Next**

Put the correct layer number in the mapping.

---

## LEFR-081

(warning) : Line %d of '%s', unrecognized data line '%s'.

**Description**

This message indicates that the data line in the mapping file is not in the format of "layer_name layer_number", thus it will be ignored.

An example mapping is: M1 31

If it is a comment line, add ';' at beginning: ; this is a comment

**What Next**

Remove the line or add ';' at beginning.

Feedback

## LEFR-082

(warning) : Line %d, invalid center-to-center spacing '%g' found in VIARULE '%s'.

**Description**

This message indicates that the SPACING of the specified VIARULE is equal or smaller than cut size, this VIARULE will not be converted to a via_def.

The SPACING value defined in VAIRULE is center-to-center spacing, it should be larger than the cut size.

For example: Below VIARULE has SPACING 0.8 x 0.7, while is smaller than the cut size of 3.0 x 3.0.

VIARULE INVALID_VIA GENERATE LAYER M1 ; ENCLOSURE 0.5 0.5 ; LAYER M2 ; ENCLOSURE 0.6 0.7 ; LAYER VIA1 ; RECT -1.5000 -1.5000 1.5000 1.5000 ; SPACING 0.8 BY 0.7 ; END INVALID_VIA

Warning: Line 12, invalid center-to-center spacing '0.8' found in VIARULE 'INVALID_VIA'. (LEFR-082)

**What Next**

Correct the value and read in the LEF file again.

## LEFR-083

(warning) Line %d, Macro '%s', geometries on layer %s mask ignored.

**Description**

This message was issued because the geometry has incorrect MASK specified: either it has no MASK specified and the LAYER is a multi-pattern layer; or it has MASK specified and the LAYER is a uni-pattern layer. The geometry will be dropped.

**What Next**

Correct the Mask of the geometry to conform to the layer and try again.

## LEFR-083e

(error) Line %d, Macro '%s', failed to read geometries on layer %s mask.

**Description**

This message was issued because the geometry has incorrect MASK specified: either it has no MASK specified and the LAYER is a multi-pattern layer; or it has MASK specified and the LAYER is a uni-pattern layer.

Feedback

**What Next**

Correct the Mask of the geometry to conform to the layer and try again.

---

## LEFR-084

(warning) Cannot create %s '%s'; %s.

**Description**

The specified object cannot be created in the design for the specified reason.

**What Next**

If the layer does not exist, define the layer in the technology or correct the layer name in the LEF input file.

If the geometry is invalid, correct the geometry definition in the LEF input file.

If the via cut pattern is invalid, correct the via cut pattern in the LEF input file.

---

## LEFR-084e

(error) Cannot create %s '%s'; %s.

**Description**

The specified object cannot be created in the design for the specified reason.

**What Next**

If the layer does not exist, define the layer in the technology or correct the layer name in the LEF input file.

If the geometry is invalid, correct the geometry definition in the LEF input file.

If the via cut pattern is invalid, correct the via cut pattern in the LEF input file.

---

## LEFR-085

(error) The ISDUMMYMETAL property value of macro '%s' is invalid: %s.

**Description**

This message indicates that an invalid ISDUMMYMETAL property value was specified for given macro. A valid ISDUMMYMETAL property value must begin with the keyword ISDUMMYMETAL, succeeded by a series of floating point numbers, and end with a ";". The number of floats must be multiple of 4, and at least 4. Every 4 floats represent

Feedback

the lower-left and the upper-right coordinates of a rectangle. The rectangles should not duplicate. There should be at most one ISDUMMYMETAL preoprty per macro.

**What Next**

Correct the ISDUMMYMETAL property value and try again.

---

## LEFR-085w

(warning) The ISDUMMYMETAL property value of macro '%s' is invalid: %s.

**Description**

This message indicates that an invalid ISDUMMYMETAL property value was specified for given macro. A valid ISDUMMYMETAL property value must begin with the keyword ISDUMMYMETAL, succeeded by a series of floating point numbers, and end with a ";". The number of floats must be multiple of 4, and at least 4. Every 4 floats represent the lower-left and the upper-right coordinates of a rectangle. The rectangles should not duplicate. There should be at most one ISDUMMYMETAL preoprty per macro.

**What Next**

This is a warning message and tool will take action accordingly. If this is not expected, please correct the ISDUMMYMETAL property value and try again.

---

## LEFR-086

(error) The NOCELLPATCH property value of macro '%s' is invalid: %s.

**Description**

This message indicates that an invalid NOCELLPATCH property value was specified for given macro. A valid NOCELLPATCH property value must begin with the keyword NOCELLPATCH, succeeded by cell patch type name, a layer name and a float offset number, and end with a ";". Only type "OBS" is supported. The float represent the shrinking value for x to the lower-left and the upper-right coordinates of the macro boundary. There should be at most one NOCELLPATCH OBS property per layer of a macro.

An example NOCELLPATCH is: PROPERTY NOCELLPATCH "NOCELLPATCH OBS M1 0.013 ;" ;

**What Next**

Correct the NOCELLPATCH property value and try again.

Feedback

## LEFR-086w

(warning) The NOCELLPATCH property value of macro '%s' is invalid: %s.

**Description**

This message indicates that an invalid NOCELLPATCH property value was specified for given macro. A valid NOCELLPATCH property value must begin with the keyword NOCELLPATCH, succeeded by cell patch type name, a layer name and a float offset number, and end with a ";". Only type "OBS" is supported. The float represent the shrinking value for x to the lower-left and the upper-right coordinates of the macro boundary. There should be at most one NOCELLPATCH OBS property per layer of a macro.

An example NOCELLPATCH is: PROPERTY NOCELLPATCH "NOCELLPATCH OBS M1 0.013 ;" ;

**What Next**

This is a warning message and tool will take action accordingly. If this is not expected, please correct the NOCELLPATCH property value and try again.

## LEFR-087

(error) The LEF58_LAYERMASKSHIFT property value of macro '%s' is invalid: %s.

**Description**

This message indicates that an invalid LEF58_LAYERMASKSHIFT property value was specified for given macro. A valid LEF58_LAYERMASKSHIFT property value must begin with the keyword LAYERMASKSHIFT, succeeded by layer names, and end with a ";". There should be at most one LAYERMASKSHIFT property per layer of a macro.

An example LEF58_LAYERMASKSHIFT is: PROPERTY LEF58_LAYERMASKSHIFT "LAYERMASKSHIFT M1 M2 ;" ;

**What Next**

Correct the LEF58_LAYERMASKSHIFT property value and try again.

## LEFR-087w

(warning) The LEF58_LAYERMASKSHIFT property value of macro '%s' is invalid: %s.

**Description**

This message indicates that an invalid LEF58_LAYERMASKSHIFT property value was specified for given macro. A valid LEF58_LAYERMASKSHIFT property value must begin

Feedback

with the keyword LAYERMASKSHIFT, succeeded by layer names, and end with a ";". There should be at most one LAYERMASKSHIFT property per layer of a macro.

An example LEF58_LAYERMASKSHIFT is: PROPERTY LEF58_LAYERMASKSHIFT "LAYERMASKSHIFT M1 M2 ;" ;

**What Next**

This is a warning message and tool will take action accordingly. If this is not expected, please correct the LEF58_LAYERMASKSHIFT property value and try again.

# LEFW

## LEFW-001

(error) The specified design '%s' has neither frame view nor design view.

**Description**

User can only specify a design which has either a frame or a design view.

**What Next**

Check that you have specified a correct and supported design.

## LEFW-002

(error) Cannot find layer of layer number %d.

**Description**

The layer of specified layer number cannot be found. One possible cause is that the technology for creating the library is different with current technology.

**What Next**

Fix the incorrect data and try again.

## LEFW-003

(error) Fail to write %s '%s' of %s '%s'.

**Description**

This error message indicates which object or attribute was failed for writing.

Feedback

**What Next**

Fix the incorrect data and try again.

---

## LEFW-004

(information) Block '%s' has %s view with multiple labels, will write out empty label only.

**Description**

The reported block has multiple user labels. This message indicates that only the empty user label will be written out.

**What Next**

No action required.

---

## LEFW-005

(warning) Block '%s' has no %s view with empty label, will ignore it.

**Description**

The reported block has no frame view with empty label. It will be ignored.

**What Next**

Please create frame view for the specifid block and try again. Or if non empty label need to be written out, please use -design option to specify the label to be written.

---

## LEFW-006

(warning) The frame view with non-empty label '%s' will be written out for block '%s'.

**Description**

A non-empty label block will be written out by the write_lef *command. The output LEF file will not contain label information.*

**What Next**

No action required.

---

## LEFW-007

(warning) Use %s %s as the WRONGDIRECTION width for layer '%s'.

Feedback

**Description**

Usually, if preferred routing direction is horizontal, xDefaultWidth will be written out as the WRONGDIRECTION width; if preferred routing direction is vertical, yDefaultWidth will be written out as the WRONGDIRECTION width. This warning message indicates that the preferred routing direction not exist, or is not horizontal or vertical, or corresponding xDefaultWidth/yDefaultWidth not exist. In that case, the tool will write out the specified width as the WRONGDIRECTION width.

**What Next**

Please verify whether the reported width is truly the WRONGDIRECTION width for the specified layer.

---

## LEFW-008

(warning) Terminals with eeq_class %d of block '%s' will be outputted in different PORT sections.

**Description**

Usually, terminals with same eeq_class will be outputted in one PORT section so that when read back the LEF, proper eeq_class can be derived. However, for the terminals of the reported block and eeq_class, they have different classes. So they will be outputted in different PORT section as every PORT section can have only one class. That means the eeq_class information of these terminals will be lost.

**What Next**

Please correct the eeq_class or class attribute for the reported terminals and try again.

---

## LEFW-009

(warning) Some syntax will not be outputted for lower version!

**Description**

User asks to output lower version than 5.8. The message is issued out.

**What Next**

Check whether the output is wanted.

---

## LEFW-010

(warning) Macro '%s', layer geometry %s %s mask%s.

Feedback

**Description**

This message was issued because the geometry has incorrect MASK specified: either it has no MASK specified and the LAYER is a multi-pattern layer; or it has MASK specified and the LAYER is a uni-pattern layer.

**What Next**

No action required.

---

## LEFW-010e

(error) Macro '%s', failed to write layer geometry %s %s mask%s.

**Description**

This message was issued because the geometry has incorrect MASK specified: either it has no MASK specified and the LAYER is a multi-pattern layer; or it has MASK specified and the LAYER is a uni-pattern layer.

**What Next**

Correct the mask of the geometry to conform to the layer and try again.

---

## LEFW-011

(warning) The LEF syntax does not support multiple spacings for non-default rule %s layer %s.

**Description**

The LEF non-default rule syntax supports only one spacing per layer. The specified rule has multiple spacings, along with weights and effort levels, on the specified layer. In these cases, the output LEF file will only include the first spacing of the layer.

**What Next**

To export the complete set of non-default rules, use the *report_routing_rules -output* command to write out the rules to a script.

---

# LGA

---

## LGA-4000

(information) Found %d gaps

Feedback

**Description**

This message shows the number of identified gaps during gap analysis

**What Next**

This is an informational message, no action is required.

# LGL

## LGL-001

(error) Placement attributes of instance %s do not match any sites.

**Description**

Leaglization could not find any sites in the floorplan with the same site definition, voltage area and exclusive move bound as this cell instance.

**What Next**

Find the site definition the reference cell. Check that the design contains site rows with this site definition. Next, check that the voltage area of the cell includes valid sites. If the voltage area is embedded inside another voltage area, make sure the stacking order allows it to be seen, *i.e.*, it is not completely covered by other voltage areas. Finally, if the cell belongs to an exclusive move bound, make sure it also includes valid sites in the proper voltage area.

## LGL-002

(warning) Spare cell instance %s is not placed.

**Description**

This cell is a spare cell but is marked as unassigned. Legalization will ignore this cell.

**What Next**

Spare cells should be placed by the *create_placement* or other commands which call *create_placement* internally. If this was done, and the cells are still unplaced, then contact the developers.

## LGL-003

(warning) Standard cell instance %s is not placed.

Feedback

**Description**

This cell is marked as unassigned. Legalization requires that all non-spare standard cells have a valid placement, otherwise this cell will be ignored during legalization.

**What Next**

Make sure that *create_placement* was run previously, either as a stand-alone command or from within another command, and succeeded.

---

## LGL-004

(error) Cell instance %s is not placed.

**Description**

This cell has not been placed in the floorplan.

**What Next**

Make sure the cell is placed in the floorplan.

---

## LGL-005

(warning) Cell instance %s has no reference cell and cannot be legalized.

The named cell has no reference cell. It will be assumed to be a blockage.

**What Next**

This problem is usually caused when a design is opened and the reference library is missing. Make sure your input is data correct.

---

## LGL-006

(warning) Reference %s has no design and cannot be legalized.

**Description**

The named reference cell has no design associated with it. Cell instances with this reference will be assumed to be blockage.

---

## LGL-007

(error) Site row %s overlaps %s with the same site definition.

Feedback

**Description**

Site rows are not allowed to overlap.

**What Next**

Check your floorplan to find overlapping rows. Modify your DEF file to eliminate the overlap.

---

## LGL-008

(error) Could not initialize physical design checker.

**Description**

The legality subsystem could not start the physical design checker.

**What Next**

Look for PDC-NNN messages for more information.

---

## LGL-009

(error) Site rows with site definition %s are misaligned.

**Description**

All the site rows with the same site definition must be aligned such that there is no vertical offset between sites.

**What Next**

Change the floorplan so that the rows align properly. Ensure that the distance between x-origins of the different rows is zero modulo the site width.

---

## LGL-010

(Error) Incompatible site rows overlap.

**Description**

Site rows with different site definitions cannot overlap unless the site definitions are compatible. A site definition is compatible with another site definition if its dimensions are an integral multiple of the dimensions of the other site definition. Rows with compatible site definitions can overlap if their sites are aligned.

Feedback

**What Next**

Change the floorplan so that the incompatible rows do not overlap. If the rows should be compatible, make sure they are aligned properly.

---

## LGL-011

(error) Design has no site rows.

**Description**

There are no site rows defined for the design. Site rows are required for placement.

---

## LGL-012

(error) Cell %s is not in a proper voltage area.

**Description**

This message is indicating that this cell instance is not inside its corresponding voltage area.

**What Next**

Move this cell instance into the correct voltage area boundary manually or through command legalize_placement.

---

## LGL-013

(error) This design is not MV ready, please check the UPF setting for your design.

**Description**

This message is indicating that some UPF problem has been detected, and the run cannot proceed.

**What Next**

Check the UPF setting and correct the problem.

---

## LGL-014

(error) Cell %s is not in a proper movebound.

**Description**

This message is indicating that this cell instance is not inside its corresponding movebound.

Feedback

**What Next**

Move this cell instance into the correct movebound boundary manually or through command legalize_placement.

---

## LGL-015

(error) cell instance %s is illegal for all sites.

**Description**

This message is indicating that the legalization has detected that the above cell instance is illegal for all sites, in other words, this cell instance cannot be placed legally within this design.

As below are some frequently encountered situations which may cause this issue:

(1) This cell belongs to multiple exclusive bounds, voltage areas at the same time, but these exclusive bounds and/or voltage areas do not have "common area".

(2) This cell may be specified with one site master, but there is no such type of site rows within this design.

(3) This cell may contain some complex pin shape, which may be illegal for all pnet checking.

**What Next**

Customer may need to double check the design constraint setup, make sure that each cell instance can have some "legal locations" to be placed into.

---

## LGL-016

(error) Cell %s violates one or more pnets.

**Description**

This message is indicating that this cell instance is violating one or more pnet straps in the design.

**What Next**

Run command legalize_placement.

---

## LGL-017

(error) Cell %s and cell %s are violating one or more standard cell spacing rules.

Feedback

**Description**

This message is indicating that this cell instance and one other neighbor cell are violating one or more standard cell spacing rules.

**What Next**

Run command legalize_placement.

---

## LGL-018

(error) Cell %s and cell %s are violating CTS cell spacing rules.

**Description**

This message is indicating that this cell instance and one other neighbor cell are violating CTS cell spacing rules.

**What Next**

Re-run the CTS commands.

---

## LGL-019

(error) Cell %s is unplaced and will not get processed by checker or legalizer

**Description**

This message is indicating that this cell instance is unplaced.

**What Next**

Assign one initial location for this cell instance.

---

## LGL-020

(error) Cell %s is overlapping with cell %s .

**Description**

This message is indicating that this cell instance and one other neighbor cell are overlapping each other.

**What Next**

Run command legalize_placement.

Feedback

## LGL-021

(error) macro cell %s is overlapping with macro cell %s .

**Description**

This message is indicating that this macro cell and one other neighbor cell are overlapping each other.

**What Next**

Move the macro cell to remove the overlapping.

## LGL-022

(error) cell %s is overlapping with blockage %s .

**Description**

This message is indicating that this cell instance is overlapping with a blockage.

**What Next**

Run command legalize_placement.

## LGL-023

(error) Cell %s is not in a proper row.

**Description**

This message is indicating that this cell instance is not inside a proper row.

**What Next**

Run command legalize_placement.

## LGL-024

(error) Cell %s has an illegal orientation.

**Description**

This message is indicating that this cell instance has an illegal orientation.

**What Next**

Run command legalize_placement.

Feedback

## LGL-025

(warning) Site master "%s" has both X-Symmetry and Y-Symmetry. Usually, this may not be correct, and may cause the legalization to fail.

### Description

This message is indicating that the legalization has detected that the above site master has both X-Symmetry and Y-Symmetry at the same time. For a normal design, usually, the site master may have only one "site symmetry", either "X-Symmetry" or "Y-Symmetry". A site master having both "X-Symmetry" and "Y-Symmetry" at the same time may cause the legalization to fail.

### What Next

Please double-check the design and library setting and make sure the setting is correct.

## LGL-026

(information) Using "Y-Symmetry" for site master "%s" in legalization because this is a horizontal design.

### Description

This message is indicating that the legalization has detected that the above site master has both X-Symmetry and Y-Symmetry at the same time. Usually, for a normal horizontal design, the site master may have "site symmetry" of "Y-Symmetry" only.

### What Next

Please double-check the design and library setting and make sure the setting is correct.

## LGL-027

(information) Using "X-Symmetry" for site master "%s" in the legalization because this is a vertical design.

### Description

This message is indicating that the legalization has detected that the above site master has both X-Symmetry and Y-Symmetry at the same time. Usually, for a normal vertical design, the site master may have "site symmetry" of "X-Symmetry" only. A site master having both "X-Symmetry" and "Y-Symmetry" at the same time may cause the legalization to fail.

### What Next

Please double-check the design and library setting and make sure the setting is correct.

Feedback

## LGL-028

(error) This design has both horizontal site rows and vertical site rows, which cannot be supported.

**Description**

This message is indicating that the legalization has detected that this design has both horizontal site rows and vertical site rows at the same time.

This type of designs cannot be supported by the legalization at this time. One design can have either horizontal site rows or vertical site rows at one time.

**What Next**

Please double-check the site row specification.

## LGL-029

(error) The legalization cannot place the block because it is over capacity (density is %5.1f%%).\n \tPlease enlarge the size of the legalization working area and/or reduce the number of the contained cells.

**Description**

This message is indicating that the legalization has detected that the standard cells area is over utilized (more than 100%) in this area, so the legalization cannot produce a legal placement for your block.

**What Next**

Increase the legalization working area (for example, adding more site rows, reducing placement blockage areas, etc), or reduce the number of movable standard cells within this area.

## LGL-030

(warning) Cell instance %s (lib_cell: %s) does not have initial location.\n \tAssign it to the low left corner of the core area.

**Description**

This message is indicating that the legalization has detected that the input cell instance does not have initial location. It is recommended that all cell instances should contine initial locations before running legalization.

Feedback

**What Next**

Check the design flow, make sure all cell instances have initial locations before running legalization.

---

## LGL-031

(warning) Site master "%s" has neither X-Symmetry nor Y-Symmetry. The "legal orientations" for the standard cells will be limited.

**Description**

This message is indicating that the legalization has detected that the above site master has neither X-Symmetry nor Y-Symmetry. For a normal design, usually, the site master may have either "X-Symmetry" or "Y-Symmetry".

If a site master does not have either of "X-Symmetry" and "Y-Symmetry", the standard cells will have limited legal orientations because the standard cells will not be allowed to flip within the site rows.

**What Next**

Please double-check the design and library setting and make sure the setting is correct.

---

## LGL-032

(information) The Variant-aware Legalization started.

**Description**

This message is indicating that the Variant-aware Legalization is running.

**What Next**

Check the Variant-aware Legalization result.

---

## LGL-033

(information) The Variant-aware Legalization succeeded.

**Description**

This message is indicating that the Variant-aware Legalization finished successfully.

**What Next**

Check the Variant-aware Legalization result.

Feedback

## LGL-034

(error) A legal placement could not be found.

**Description**

This message is indicating that the legalization cannot find a legal placement for this block. You probably have to enlarge the core area and re-try.

**What Next**

Check the design setting, and make sure all of the constraints are correct. If no problem has been found, then you may need to enlarge the core area.

## LGL-035

(error) Cell %s has a Gemini violation.

**Description**

This message is indicating that this cell instance is violating one Gemini constraint in the design.

**What Next**

Run command legalize_placement.

## LGL-036

(warning) Cell %s is out of the core area.

**Description**

This message is indicating that this cell instance is out of the core area.

**What Next**

Move this cell into the core area manually or run command legalize_placement.

## LGL-037

(information) Fixed cell %s has been swapped because it has a "gemini_swap_only" attribute.

**Description**

This message is indicating that this cell instance has been swapped to meet the Gemini constraint in the design. Usually, the legalization should not touch a "fixed cell". However,

Feedback

if a fixed cell contains "gemini_swap_only" attribute, then it is expected that this cell can be "swapped".

**What Next**

Run command legalize_placement.

---

## LGL-038

(information) %s been swapped because attribute "gemini_swap_only" has been set.

**Description**

This message is indicating that some fixed cell instances have been swapped to meet the Gemini constraint in the design.

**What Next**

Run command check_legality.

---

## LGL-039

(error) Cell %s violates CPODE L2 rule

**Description**

This message is indicating that this cell instance is violating CPODE L2 rule.

**What Next**

Run command legalize_placement.

---

## LGL-040

(error) Color violation detected between cell %s and cell %s .

**Description**

This message is indicating that color violation is found between two abutted cells.

**What Next**

Run command legalize_placement.

---

## LGL-041

(error) Cell %s violates OD L5 rule

Feedback

**Description**

This message is indicating that this cell instance is violating OD L5 rule.

**What Next**

Run command legalize_placement.

---

## LGL-042

(Warning) Advanced rule query engine not initialized. No advanced rule checking performed.

**Description**

This message is indicating that there was an error in initialization of the advanced rule query engine and thus no advanced rule checking will be performed by optimization. This may result in illegal placement.

**What Next**

Review the design and floorplan preparation.

---

## LGL-043

(error) Cell %s pin color alignment check.

**Description**

This message is indicating that this cell instance is violating pin color alignment in its current location.

**What Next**

Run command legalize_placement.

---

## LGL-044

(error) Design has vertical site rows.

SH DESCRIPTION There are vertical site rows defined for the design. Vertical site rows are not currently supported.

---

## LGL-045

(error) Site rows with site definition '%s' are not covered by base site rows (%s).

Feedback

**Description**

If site rows with different compatible site definitions overlap, then there must be a base site definition whose rows cover all other site rows. When there are multiple site definitions with overlapping rows, the base site definition is determined to be the one with the smallest dimensions whose rows occupy the largest area. All other site definitions must be have dimensions that are multiples of the base.

**What Next**

Modify the floorplan so that there is a base site definition whose rows completely cover the rows of compatible site definitions.

---

## LGL-046

(Warning) Library cell

**Description**

The warning message tells user that there are inconsistent setup between design and library. The site def used in current design has different width or height compared to same site def name used in library. User should have consistent setup between design and library, otherwise it may cause legalization error. If any placeable cell instance is from this library cell, it will result in LGL-047 error and legalization will stop.

**What Next**

Please make sure site def width and height are same between design and libraries.

---

## LGL-047

(Error) Placeable cell %s has inconsistent site def defined in design and library.

**Description**

This error tells user that the site def definition used in current design is not consistent with library. For example, site def defined in current design has width 1240, height 2480. However, the site def used in library cell has width 500, height 1000. When this cell is not fixed, which is placeable, but their definitions are inconsistent between design and library, legalize_placement cannot place it because it cannot place a cell with width 500 and height 1000 into current design's row with width 1240, height 2480.

**What Next**

Please refer previous warning message LGL-046 to see which library cell and the detail information

Feedback

## LGL-048

(Warning) Library cells %s and %s have different supply architectures.

**Description**

Based on their site definitions and the corresopnding site rows, the two library cells are intended to be placeable together in the same row. In the normal R0 orientation, however, one cell appears to have its ground pin at the bottom while the other has its power pin at the bottom creating a conflict.

**What Next**

The most common cause of this error is incompatible libraries which use the same site definition being loaded together. Either they need to use different site definitions, or they cannot both be present at the same time.

## LGL-049

(Warning) Cannot determine %s supply pin for library cell %s.

**Description**

When examining the geometries of the power and ground pins, it could not be determined which is at the top or bottom of the cell.

**What Next**

In most cases this warning can be ignored. This information is only used when the LEF/DEF information is insufficient to determine the legal rows for multi-row cells. It may indicate a problem in library preparation, or this cell may have a special purpose for which the pin geometries are unusual but correct.

## LGL-050

(Warning) Library cell %s is missing supply pins. Its power structure cannot be determined.

**Description**

When examining the geometries of the power and ground pins, one or both waas missing. Subsequent analysis will assume that the ground rail lies along the bottom of the cell when in the normal (R0) orientation.

**What Next**

This information is only used when the LEF/DEF information is insufficient to determine the legal rows for multi-row cells. It may indicate a problem in library preparation.

Feedback

## LGL-051

(warning) Could not find legal location for cell instance %s(%s).

**Description**

This message is indicates that a legal location could not be found for this cell during legalization. There are many possible causes for this. Below is a partial list.

(1) The legalizer may be trying to place the cell in a region that is over-constrained.

(2) This cell belongs to multiple exclusive bounds, voltage areas at the same time, but these exclusive bounds and/or voltage areas do not have "common area".

(3) This cell may be specified with one site definition, but there is no such type of site rows within this design.

(4) This cell may contain some complex pin shape, which may be illegal for all sites.

**What Next**

Check the design setup with respect to bounds or voltage areas. If the pass-rate of the reference of this cell is very low or zero (see command 'analyze_lib_cell_placement'), then check the design rule setup (disable individual rules and rerun legalization).

## LGL-052

(warning) Unable to swap lib cell from %s to %s (variant index %d) on cell %s due to color alignment check.

**Description**

This warning message indicated that there is one lib cell cannot be swapped to other lib cell to fix color alignment violation, and the corresponding lib cell and its variant index number failed due to color alignment check. User will only see this warning message when all variants defined in the variant cell group cannot fix current color alignment violation by swapping. User must go back to check all variants defined in the cell group and investigate why all variants cannot fix the violation.

When variant swapping failed, legalizer will start a normal legalization by moving cells to fix the rest violations. If users don't want to have additional legalization run, they can use *set_app_option -name place.legalize.enable_full_variant_check -value true* to make legalizer performs real check on all queries to make cells defined in invalid cell group will be moved automatically. However, by using that app option, it may have runtime impact depends on the query and variant status.

Feedback

### What Next

report_cell_groups set_app_option -name place.legalize.enable_full_variant_check -value true

---

## LGL-053

(warning) Unable to swap lib cell from %s to %s (variant index %d) on cell %s due to pnet check.

### Description

This warning message indicated that there is one lib cell cannot be swapped to other lib cell to fix color alignment violation. To fix color alignment by swapping lib cells, legalizer will check both color alignment and pnet, not only color alignment. If user receives this message, but there is no LGL-052 message at all, that means this lib cell has color matched, but it will cause pnet violation so legalizer will not be able to swap this lib cell. If user receives both LGL-052 and LGL-053 messages, that means this variant lib cell is not valid to swap by color alignment check and pnet check.

User will only see this warning message when all variants defined in the variant cell group cannot fix current color alignment violation by swapping. User must go back to check all variants defined in the cell group and investigate why all variants cannot fix the violation.

When variant swapping failed, legalizer will start a normal legalization by moving cells to fix the rest violations. If users don't want to have additional legalization run, they can use *set_app_option -name place.legalize.enable_full_variant_check -value true* to make legalizer performs real check on all queries to make cells defined in invalid cell group will be moved automatically. However, by using that app option, it may have runtime impact depends on the query and variant status.

### What Next

report_cell_groups set_app_option -name place.legalize.enable_full_variant_check -value true

---

## LGL-054

(information) %s moved %d cells (%.1f%%) with an average displacement of %.3lf and a maximum of %.3lf.

### Description

This message indicates how many cells where moved and how much they where moved during a certain phase of the legalizer. The average displacement is the average over all cells that moved.

Feedback

## LGL-055

(error) There %s %d cell%s which cannot be legalized.

### Description

This message summarizes the amount of cells for which no legal location could be found. Message LGL-051 indicates which cells are affected.

### What Next

Please check whether the design is over-utilized or constraints are not correctly applied.

## LGL-056

(warning) Could not find layer '%s' in the design technology library. Ignoring shapes on this layer for pin '%s/%s'.

The given library cell pin has shapes on a layer which is not present in the design's technology library.

### What Next

Make sure that all library cells are bound to the same tech library (reference) as the current design.

## LGL-057a

(warning) Unbound instance '%s' is ignored by the legalizer.

### Description

The instance shown is unbound, that is, it is not mapped to any library cell, or it is not mapped to a library cell with a physical view. As such, it cannot be legalized, so the legalizer is ignoring it. A summary of how many instances are unbound is covered by LGL-057. By default, 10 of these will be displayed when the legalizer is initialized.

### What Next

Make sure that all cells in the design are bound to a reference library cell.

### See Also

- LGL-057

Feedback

## LGL-057

(warning) %d cell%s not bound to any reference.

The given number of cells are unbound, i.e. they are not mapped to any library cell.

**What Next**

Make sure that all cells in the design are bound to a reference library cell.

## LGL-058

(error) The number of references must be greater than 0 for the command.

The command requires the number of references to be greater than 0.

**What Next**

Ensure that the number of reference value is greater than 0.

## LGL-059

(Error) Site '%s' has a height (%s) which is not a multiple of the base site '%s' height (%s).

**Description**

All sites need to have a height which is a integer multiple of the base site height. All other site configurations are not accepted by the legalizer. An exception are half height sites used in inbound flows. This rule also applies to non-overlapping site rows.

**What Next**

Please correct the site row definitions.

## LGL-060

(warning) Could not find legal location for cell instance %s(%s).

**Description**

This message is indicates that a legal location could not be found for this cell. There are many possible causes for this. Below is a partial list.

(1) The current command may be trying to place the cell in a region that is over-constrained.

(2) This cell belongs to multiple exclusive bounds, voltage areas at the same time, but these exclusive bounds and/or voltage areas do not have "common area".

Feedback

(3) This cell may be specified with one site definition, but there is no such type of site rows within this design.

(4) This cell may contain some complex pin shape, which may be illegal for all sites.

**What Next**

If the cause is reason (1) above, the command will usually adapt and this message is informational only. Otherwise, there may be a problem with the design. If 'legalize_placement' also fails, then further investigation is required. If legalization succeeds, then the message may be ignored.

---

## LGL-061

(warning) Cell '%s' has an orientation '%s' with 90 degree rotation. Most rule checks are disabled for this cell.

**Description**

The given cell is 90 degree rotated. The cell is not legalizable.

**What Next**

You need to change the orientation of the cell to R0, MX, MY, or R180.

---

## LGL-062

(warning) No site rows found for site '%s'.

**Description**

The given site definition has no site rows defined. The given site definition will be ignored and all cells referring to that site won't get legalized.

**What Next**

You need to define site rows for the given site definition.

---

## LGL-063

(error) There is no site row or array found in the design.

**Description**

No site definitions with rows were found in the design. The design cannot be legalized.

Feedback

**What Next**

You need to define site definitions with site rows for all sites referred to by the cells in the design.

---

## LGL-064

(error) Unable to find site rows to legalize cell '%s'.

**Description**

No site rows were found to legalize the given cell.

**What Next**

You need to define site definitions with site rows for all sites referred to by the cells in the design.

---

## LGL-065

(error) Unable to legalize cell '%s' because cell width '%s' is not multiple of site (%s) width '%s'.

**Description**

The width of the given cell is not a multiple of the width of the associated site defintion.

**What Next**

You need to assign a suitable site definition to the lib cell of the given cell.

---

## LGL-067

(error) Cell %s with reference %s is is misaligned.

**Description**

This message is indicating that this cell instance is not aligned to a site row.

**What Next**

Make sure that all cells are properly placed and aligned to site rows before running the command that generated the error message.

---

## LGL-068

(Error) TPO width error encountered near %s.

Feedback

**Description**

You received this error message because tool encounters unexpected condition during TPO width checking around indicated cell. If you encounter this error running filler insertion, the possible cause is design is not legality clean before.

**What Next**

1. If running filler insertion, please ensure check_legality is clean before filler insertion.
   2. Please use below option to specify cells whose TPO layer not exntend to left or right edge of cell: set_app_options -name place.legalize.cpode_exception_cells -value "BOUNDARY TAP"

---

## LGL-069

(warning) Library cell %s has missing vertical abutment indexes

**Description**

This library cell provides only partial vertical abutment indexes. Missing indexes will be ignored.

**What Next**

Make sure library cell contains vertical abutment indexes on both top and bottom edges

---

## LGL-070

(warning) Library cell %s has missing vertical abutment indexes; duplicate missing indexes from other side.

**Description**

This library cell provides only partial vertical abutment indexes. Missing indexes will be duplicated from the top/bottom side with indexes.

**What Next**

Make sure library cell contains vertical abutment indexes on both top and bottom edges

---

## LGL-071

(warning) %s%s%s It cannot be used to reduce displacement during variant aware legalization.

Feedback

**Description**

The reported library cells are part of a common cell group (variant definition), but these library cells are not compatible, i.e. one cannot replace the other. The legalizer won't use these library cells as variants for each other.

**What Next**

Make sure that the library preparation and cell group definiton only references library cells that can be used for sizing a cell. Use the 'report_cell_groups' command to get an overview of the defined sets of variants.

---

## LGL-072

(warning) %s '%s' has cells attached but does not overlap the core area.

**Description**

This message is printed for move bounds and voltage areas. The given bound or voltage area doesn't overlap the core area and hence the cells attached to it cannot be legalized.

**What Next**

Make sure that either the bound or voltage area doesn't have any cells attached or it overlaps the core area (standard cell placement area).

---

## LGL-073

(error) The advanced checker/legalizer has detected non-abutting or non-aligned site row %s at coordinate {%s}.

**Description**

A site row configuration has been found which is not supported by default.

**What Next**

Try using the app option place.legalize.legalize_non_uniform_va_exbound

---

## LGL-074

(warning) No filler cells present in the reference library of the design.

There are no cells in the design marked as filler cells. Neither does any cell present in the reference library has filler design type, nor has the user provided list of filler cells. This may lead to gaps between the standard cells after placement, and therefore DRC violations or discontinuity in PG rails connections.

Feedback

**What Next**

Make sure that there are appropriately defined filler cells in the reference library to avoid any design violations.

---

## LGL-075

(warning) Spacing rules are being applied to lib/CTS cell %s.

Spacing rules on a lib/CTS cell will result in rules being directed to each and every cell having the same lib cell. If applied to a larger number of library or CTS cells might cause a longer runtime in AL.

**What Next**

Reduce number of lib cells having spacing rules while running advanced legalizer.

---

## LGL-076

(information) Via ladder spacing app_option or command being used which is not required during AL.

App_options and command related to via_ladder_spacing are for classic legalizer and there is no need to use them while running AL. Advanced legalizer supports these setting in a more advanced manner.

---

## LGL-077

(warning) Smallest size of filler cells present in the library is %.2f, which is larger than the smallest site width.

1X size filler cell unavailable in the library to fill gaps. The lack of fillers sized = 1 typically cause extra constraints to legalization as the legalizer can't leave a gap 1X wide.

**What Next**

Check the filler cell definitions in the reference libraries.

---

## LGL-078

(error) Site symmetry is set incorrectly : Y symmentry = %d, X symmentry = %d.

Default site symmentry is "Y" after ICC2 floorplan creation. Since, symmetry determines legal orientations during legalization, incorrect symmetry will result in failure during filler cells insertion in Advanced Legalizer.

Feedback

**What Next**

Set the site symmetry correctly before running AL.

---

## LGL-079

(warning) Vertical keepout will be considered as place.legalize.advanced_ignore_vertical_keepout_margin is false. This may cause runtime and displacement issues during legalization.

**Description**

Vertical keepouts cause significant displacement degradations during legalization. It may lead to over constraining. So the legalizer aborts if many cells have vertical keepout margins.

**What Next**

Turn on the app option place.legalize.advanced_ignore_vertical_keepout_margin

---

## LGL-080

(warning) The bound %s of type %s has a high utilization of %.3g which could lead to high runtime.

**Description**

The VA or bounds with this high density could lead to longer runtime.

**What Next**

Stop the run.

---

## LGL-081

(warning) The command set_legalizer_preroute_keepout is not supported by the advanced legalizer.

**Description**

The preroute keepout command is not supported by the advanced legalizer and nplDRC.

**What Next**

Do not run this command.

Feedback

## LGL-082

(warning) Stream place is not required in Advanced Legalizer

**Description**

Stream place is no longer needed in Advanced legalizer. Stream place replaces single large displacement with a stream of small displacements but Advanced Legalizer is taking care of this already.

## LGL-083

(error) The cell %s is unplaced and outside the core area.

**Description**

When the cell is present outside the core area and is unplaced, it may lead to longer runtime.

## LGL-084

(warning) For color rules defined in tech file, the design has layer with colored routing tracks = %s, and/or colored stdcells' pins = %s.

Color rules for standard cells' pins in the frame and routing tracks on a layer at design level should not be mismatched, i.e., either both or neither must be colored.

## LGL-085

(Error) Filler prediction failed due to off-site-row cells.

**Description**

You received this error message because tool encounters unexpected condition during OD Edge Rule checking due to off-site-row cells. If you encounter this error during check_legality, the possible cause is design is not legalized before.

**What Next**

If running check_legality, please ensure legalize_placement is run before.

## LGL-086

(warning) Cell %s is unplaceable after %d trials.

Feedback

**Description**

This message is printed if a cell cannot be legalized after specified number of tries during legalization.

**What Next**

Check the library-cell and the floorplan to see why the cell difficult to legalize. For unlimited number of trials to legalize, set app-option place.legalize.limit_legality_checks to false.

---

## LGL-087

(warning) %lu cells with invalid orientation present in the design.

**Description**

If a cell has bad orientation, it will not be legalized.

**What Next**

Check the library-cell and the floorplan to see why the cell has invalid orientation. If such cells are present in the design, run create_placement before legalizer.

---

## LGL-088

(warning) Ignoring cell '%s' from IR-drop optimization%s.

**Description**

The legalizer cannot move the specified cell due to the given reason. IR-drop optimization will be limited around this cell.

**What Next**

This message might, for example, show up after clock tree synthesis where most of the sequential cells are marked as fixed. In this case, this user can ignore the warning but should be aware that the IR-drop optimization has limited room to move cells.

---

## LGL-089

(warning) cellmap ignored instance with missing physical info: %s

**Description**

cellmap is asked by the client to operate on the instance (e.g. add/remove) that doesn't have needed physical info. For instance, the instance may lack a reference (i.e. it is set as NULL) or the reference doesn't have proper definition of sites assigned.

Feedback

### What Next

If the instance reported should not have been ignored then the user is to verify that proper physical data present either in the reference library or the given block.

---

## LGL-090

(warning) Via ladder constraint %s of lib cell %s does not match its pin structure, so the via ladder check on the cell instances derived from this lib cell with the given via ladder constraint is bypassed.

### Description

You received this warning message because there exists a lib cell, denoted by $lc, in your design having a short pin such that advanced legalizer is not able to grow a via ladder on it based on the given via ladder constraint, because advanced legalizer does NOT support automatically pin extension. Advanced legalizer will automatically bypass the via ladder checks on those cell instances of the lib cell $lc, and still perform via ladder checks on other normal cells if the app option place.legalize.enable_via_ladder_checks was enabled.

### What Next

You can take one of the following actions to fix this warning. 1. Modify the mismatched via ladder constraints such that advanced legalizer can grow via ladders on pins without pin extension. For example, decrease the number of cut rows/cuts per row. 2. Modify the cell layout by manually extending the short pins

---

## LGL-091

(warning) Found duplicate spacing rule %s(%d) - %s(%d) %u - %u in library '%s'.

### Description

The tool picks up spacing rules first from the design library, and then from the reference libraries. This message indicates that duplicate definitions of spacing rules were found. The occurance of this situation has no impact and QoR.

---

## LGL-092

(warning) Ignoring already existing max. horizontal width label '%s' of library '%s'.

### Description

The tool picks up maximum horizontal with labels first from the design library, and then from the reference libraries. This message indicates that duplicate definitions of labels were found (see PRF flow).

Feedback

## LGL-093

(warning) Ignoring vertical abuttment rules from reference library '%s'.

**Description**

The tool picks up vertical abuttment rules first from the design library, and then from the reference libraries. This message indicates that abuttment rule definitions were found in the design library as well as in the given reference libary. The ones from the reference library are ignored.

## LGL-094

(warning) Routing layer '%s' has a non uniform routing pitch.%s

**Description**

Certain runtime improvement settings and optimizations require that lower routing layers have a constant track pitch.

## LGL-095

(warning) Lib-cell '%s' has %s '%s' which is not a multiple of the VT site %s '%s'. VT information will be ignored.

**Description**

The given library cell has a width or height which is not an integer multiple of the considered VT site map. The library cell will hence be ignored for any VT implant checks.

## LGL-096

(information) Lib-cell '%s' is vertical stacked length exception lib-cell.

**Description**

The given library cell is ignored for the vertical stacked length design rule check.

## LGL-097

(warning) The safety register '%s' is marked as fixed but requires spacing for tap cell insertion.

Feedback

**Description**

A safety register rule hass been defined for the given cell which requires tap cells next to it. Since the cell is fixed, the legalizer cannot move the cell to enforce enough space net to the cell.

---

## LGL-098

(warning) The design contains library cells, example %s, having very low legalization pass rates.

The design has cells with zero or low pass rates. This will cause a longer runtime during legalization because a cell with low pass rate would mean that as compared to the number of sites tested, it is placeable at a very small number of sites in any orientation.

**What Next**

Run analyze_lib_cell_placement to identify problem cells.

---

## LGL-099

(warning) On-by-default app options : place.legalize.enable_cell_spreading is set as false = %s ; and place.legalize.enable_cell_ordering is set as false = %s in the design.

**Description**

On-by-default app options for enable_cell_spreading and enable_cell_ordering, both are required to be always true in advanced legalizer.

**What Next**

Turn on these app options before running AL.

---

## LGL-100

(warning) Metal1/Metal2 PG rails in the design at coordinates {%s} have shape type = %s.

M1/M2 PG rails should have only one of the following shape types : lib_cell_pin_connect or follow_pin or stripe. If the rails has shape other than these, say "user_route", the legalizer considers it route specific to a specific pin and not the primary PG grid. Any other PG shapes other than the three mentioned above cause long runtime/hang while running advanced legalizer.

**What Next**

Check the design and set the PG rails properly.

Feedback

## LGL-101

(warning) The cell '%s' has incorrect site definition %s.

Lib-cells used in this design either do not have a site association, or have a site def for which there are no site rows nor site arrays in the design. These cells cannot be placed anywhere.

## LGL-102

(warning) For overlapping site rows, the dimensions of their site definitions, namely %s and %s, are not an integral multiple of the dimensions of base site def.

If overlapping site rows are present in the design, the height or width of their respective site defs should be integral multiple of the base/unit site def which is determined as the site def with the smallest dimensions.

## LGL-103

(warning) Site rows with site definition %s are misaligned.

**Description**

All the site rows with the same site definition must be aligned such that there is no vertical offset between sites.

**What Next**

Change the floorplan so that the rows align properly. Ensure that the distance between x-origins of the different rows is zero modulo the site width.

## LGL-104

(warning) Library cell %s doesn't have sufficient allowable orientations.

**Description**

Allowable orientations on lib cells and possible legal orientations on the site rows have no legal orientation in common. The cell won't find a legal placable orientation, hence causing longer runtime.

**What Next**

Set place.legalize.enable_allowable_orient false.

Feedback

## LGL-105

(warning) Cell map created for block %s is invalid.

### Description

Current design has one or more invalid cell maps, which might cause the command to assert. Cell map data is not being generated properly. This might occur due to variety of reasons realted to bad/dirty design inputs (such as, missing physical information).

### What Next

Check design before running.

## LGL-106

(Warning) For color rules defined in tech file, the design has layer %s with colored routing tracks = %s, but colored stdcells' pins = %s.

Color rules for standard cells' pins in the frame and routing tracks on a layer at design level should not be mismatched, i.e., either both or neither must be colored.

### What Next

Check the color contraints of the afrementioned layer in this design or revise your app option settings.

### See Also

- LGL-084
- place.legalize.enable_pin_color_alignment_check
- place.legalize.pin_color_alignment_layers

## LGL-107

(warning) Overlapping site rows having site defs %s and %s not completely used in the design floorplan.

Overlapping site rows must cover the entire site array.

## LGL-108

(warning) Spare cells present in the design at {%s} are pre-clumped.

Feedback

Clumping of spare cells together might cause high local density in the design, hence causing longer runtime.

## LGL-109

(warning) Direction for routing layer %s is not specified.

Routing layer must have a direction (horizontal or vertical) defined.

## LGL-110

(warning) Zero spacing blockage %s is present on PG net %s.

Large zero-spacing blockages not excluded from PG net will hang the legalizer engine.

**What Next**

Please set net type (eg., clock, reset, scan, signal, tie_high, tie_low) for zero spacong blockages.

## LGL-111

(warning) %s

**Description**

This message is to communicate to the user that for the specified cell (or reference cell) cellmap runs into lengthy search operation to find legal location.

**What Next**

The user needs to be aware that layout internals of the particular cell (or reference cell) may interacts with existing routing that limits available locations where the cell (or reference cell) can be legally placed. Also the user needs to be aware that lacking appropriate filler cells in the library may manifest itself in a similar way. In each case the cell (or reference cell) needs to be looked at from this perspective. Refining settings for cellmap checks may be needed.

**See Also**

- LGL-113

## LGL-112

(warning) %s

Feedback

### Description

This message is to communicate to the user that for the specified cell (or reference cell) cellmap runs into lengthy search operation to find legal location (iterating rows).

### What Next

The user needs to be aware that layout internals of the particular cell (or reference cell) may interacts with existing routing that limits available locations where the cell (or reference cell) can be legally placed. Also the user needs to be aware that lacking appropriate filler cells in the library may manifest itself in a similar way. In each case the cell (or reference cell) needs to be looked at from this perspective. Refining settings for cellmap checks may be needed.

### See Also

- LGL-113

---

## LGL-113

(warning) %s

### Description

This message serves to print out Legality Checks Profile that captures effort and statistics for the specified cell (or reference cell) when cellmap does search to find legal location.

### What Next

The user needs to be aware that layout internals of the particular cell (or reference cell) may interact with existing routing that limits available locations where the cell (or reference cell) can be legally placed. Also the user needs to be aware that lacking appropriate filler cells in the library may manifest itself in a similar way. In each case the cell (or reference cell) needs to be looked at from this perspective. Refining settings for cellmap checks may be needed.

### See Also

- LGL-111
- LGL-112

---

## LGL-114

(information) %s

Feedback

### Description

This message serves to print out Legality Checks Profile for each executed FLL (find legal location) query by cellmap along with additional debug information like number of checks, elapsed CPU and Wall time.

### What Next

The user can grep these messages from the log file and analyze specific queries in terms of how long each query took to run and how many checks were made in the process along with Legality Checks Profile statistics.

### See Also

- LGL-113

---

## LGL-115

(warning) App option place.legalize.enable_pin_color_alignment_check is set to false.

### Description

enable_pin_color_alignment_check must be set to true while running AL, which enables legalizer to automatically align cell pin/shapes to track on the given metal layer.

### What Next

Turn on the app option before running AL.

---

## LGL-116

(warning) Track offset on layer %s is set incorrectly. One example library cell with shape to track mis-alignment: '%s'.

### Description

Lib cells' pins are not center aligned with routing tracks on layer %s.

### What Next

Set the following app options before running AL. 1) place.legalize.enable_pin_color_alignment_check to true to enable legalizer to automatically align cell pin/shapes to track on the given metal layer. 2) If above is already true, set place.legalize.disable_advanced_legalizer_rules -value {pg_drc} to bypass the issue from mis-aligned tracks if want to go with dirty design.

Feedback

## LGL-117

(Information) Setting orientation for the cell %s to R0.

**Description**

This message indicates that this cell has Invalid orientation which might cause problem during legalization.

**What Next**

Check the library-cell and the floorplan to see why the cell has invalid orientation. If such cells are present in the design, run create_placement before legalizer.

## LGL-118

(information) Enabling the app option place.legalize.enforce_1x_min_filler

**Description**

This message indicates that the design does not contain filler cells. As a result, it might cause long runtime or hang at the legalization or check_legality in TPO, VT, OD rules.

**What Next**

Please doube check the design and the libraries for filler cells.

## LGL-119

(information) The bound %s of type %s has utilization > 100%. Hence leaving all the cells inside it unlegalized.

**Description**

This message indicates that this bound has very high utilization. Any bound with utilization this high cannot be legalized. It will take a lot of runtime and at the end leave cells unlegalized.

**What Next**

Increase the size of this bound or decrease the number of cells present in it.

## LGL-120

(information) Trying to legalize the cell %s in this gap by disabling allowable orient.

Feedback

**Description**

This message indicates that for this gap, this lib cell might not have sufficient allowable orientations. As a result of which, legalizer could not find a legal orientation for this gap.

**What Next**

Please double-check the design and library and make sure the there are asufficient allowable orientations for a given lib cell.

---

## LGL-121

(information) Bound has utilization > 120%. Recomputing total density by removing the placeable and total area of the bound.

**Description**

This messages indicates that the floorplan has utilization greater than 100%. There are bounds having density greater than 120% which might be causing high overall density. Therefore leaving those bounds unlegalized and removing their area from the total floorplan area.

**What Next**

Increase the total area of the bound or decrease its utilization.

---

## LGL-122

(information) Enabling the app option %s locally.

**Description**

This messages indicates that the app option place.legalize.enable_cell_spreading and place.legalize.enable_cell_ordering should not be turned off during advanced legalizer.

**What Next**

Turn on the app options before running legalize_placement for advanced legalizer.

---

## LGL-123

(error) Advanced Legalizer detects incorrect data input; legalization aborts

**Description**

This messages indicates that advanced legalizer fails to initialize engine because of incorrect data input.

Feedback

**What Next**

Check the input data and run check_legalizer_sanity to get warnings related to incorrect data.

---

## LGL-124

(error) Basic rule engine initialization fails

**Description**

This message indicates that advanced legalizer failed to initialize basic rule engine because of incorrect data.

**What Next**

Check for incorrect data before running advanced legalizer.

---

## LGL-125

(warning) Voltage area '%s' has an empty effective area but cells assigned.

**Description**

The given voltage area has no effective physical area but at least one cell is assigned to it. All cells assigned to such a voltage area cannot be legalized.

**What Next**

Correct the voltage area setup. Also check the stacking order which impacts the effective area of the voltage areas.

---

## LGL-126

(error) The legalization cannot place the block because it has no placeable area.\n \tPlease remove some blockages from the design to be able to place cells.

**Description**

This message is indicating that the legalization has detected that there is no placeable area in this floor plan, so the legalization cannot produce a legal placement for your block.

**What Next**

Remove some bloackages from the block.

Feedback

## LGL-127

(information) Lib-cell '%s' is a user-defined filler cell.

**Description**

The given library cell is considered a filler cell due to app-option place.legalize.filler_lib_cells.

## LGL-128

(information) Via Ladder constraint %s failed to find solution due to PG shapes on layer %s.

**Description**

Tool could not find legal location for this via ladder constraint. The root cause is the PG layer report in the message.

**What Next**

Review the PG structure and via ladder constraint. Fix at least one of them to solve the problem.

## LGL-129

(warning) Via Ladder check on libCell %s is ignored after failure threshold.

**Description**

Via ladder failure on libCell meet the threshold, so tool will ignore the libCell.

**What Next**

Review the PG structure and via ladder constraint.

## LGL-130

(information) Legalizer threading is enabled with %s effort

**Description**

This is an informational message emitted during legalizer initialization reminding you that you requested the legalizer use threading.

Feedback

**See Also**

- LGL-131
- place.legalize.enable_threaded_advanced_legalizer

---

## LGL-131

(information) Legalizer used up to %d thread%s

**Description**

This is an informational message emitted when the legalizer completes telling you the peak number of threads used for legalization.

**See Also**

- LGL-130
- place.legalize.enable_threaded_advanced_legalizer

---

## LGL-132

(error) Initialization of the command failed.

**Description**

This message indicates that command failed because site def grids could not get initialised as part of Basic Rule Engine initialisation.

**What Next**

Check for incorrect site definitions.

---

## LGL-133

(warning) This design has spacing rules applied on lib cells, causing over constraints. Spacing rule constraints have been relaxed.

**Description**

This message indicates that the design has spacing rules applied to lib cells. As a result, they are over constrained to be legalized. The spacing rule max constraints have been optimized according to thier utilizations.

**What Next**

Relax the applied spacing rules.

Feedback

## LGL-134

(warning) The library cell "%s" has implant layers missing in frame.

**Description**

This message indicates that the design has lib cells with no implant layer which can cause hang or long runtime.

**What Next**

Rectify the design by assigning the implant layers to the lib cell. Turn on the app option place.legalize.use_virtual_implant_layer.

## LGL-135

(warning) Using virtual implant layer for lib cell "%s" not having implant layer. Enabling the app option place.legalize.use_virtual_implant_layer locally.

**Description**

This message indicates that there were some lib cells that had no implant layer assigned to them. To avoid hang a virtual implant layer has been assigned to them.

**What Next**

Provide implant layer information for all such lib cells.

## LGL-136

(warning) Design has high local density %s which may cause higher runtime.

**Description**

This message indicates that the design has high local density.

**What Next**

You may need to check your placement constraints.

## LGL-137

(warning) Pre clumping of spare cells resulted in an increased density of %s.

**Description**

This message indicates that the design has high spare cell density.

Feedback

**What Next**

You may need to check your placement constraints.

---

## LGL-138

(warning) Missing fillers for %s layer.

No fillers found for the given layer. It can cause legalizer to hang.

**What Next**

Check the filler cells constraints on the cell lib.

---

## LGL-139

(warning) Duplicate color variant %s has been detected. This library cell will be ignored by the legalizer.

Multiple identical color variants make variant legalization useless because any variant swap attempted by Advanced Legalizer would still give the same result.

---

## LGL-140

(warning) Lib-cell '%s' has %s '%s' which is not a multiple of the OD site %s '%s'. OD information will be ignored.

**Description**

The given library cell has a width or height which is not an integer multiple of the considered OD site map. The library cell will hence be ignored for any OD implant checks.

---

## LGL-141

(warning) Via ladder constraint %s of lib cell %s is blocked by cell's internal objects, so the via ladder check on the cell instances derived from this lib cell with the given via ladder constraint is bypassed.

**Description**

You received this warning message because there exists a lib cell, denoted by $lc, other objects inside the lib cell generate DRC violation against the via ladder constraint and block via ladder insertion. Advanced legalizer will automatically bypass the via ladder checks on those cell instances of the lib cell $lc, and still perform via ladder checks on other normal cells if the app option place.legalize.enable_via_ladder_checks is enabled.

Feedback

**What Next**

You can take one of the following actions to fix this warning. 1. Modify the mismatched via ladder constraints such that advanced legalizer can grow via ladders on pins without causing DRC violation against other objects inside cell. For example, decrease the number of cut rows/cuts per row. 2. Modify the cell layout by manually removing objects block via ladder construction.

---

## LGL-142

(error) Cell '%s' has a height (%s) which is not a multiple of the associated site definition '%s' height (%s). Cell is skipped.

**Description**

The height of the given cell is not a multiple of the height of the associated site defintion.

**What Next**

You need to regenerate the corresponding lib cell properly.

---

## LGL-143

(error) No special cell has been specified.

**Description**

This message is indicating that there are no cells defined while setting special cell legalization constraints. Special cell legalization won't take place.

**What Next**

Correctly specify cells constraint.

---

## LGL-144

(error) Both X max displacement and Y max displacement can't be zero simultaneously.

**Description**

No correct direction set for moving cells while setting special cell legalization constraints. Special cell legalization won't take place.

**What Next**

Correctly specify at least one of, X or Y direction displacement for special cell movement.

Feedback

## LGL-145

(Warning) Cell '%s' of ref '%s' does not have a site association compatible with floorplan. Cell cannot be snapped to row.

**Description**

Site information in the listed lib_cell is either missing or is not compatible with the site_def(s) of the site rows in the current design. Instance cannot be snapped to row for placement.

**What Next**

Check lib_cell properties.

## LGL-146

(Warning) Cell '%s' has an invalid orientation '%s'. The cell will be ignored.

**Description**

The given cell has an orientation which is either not defined or has a 90 degree rotation. Allowed orientations are R0, MX, R180, and MY.

**What Next**

Use the 'orientation' attribute to define a valid orientation on the cell.

## LGL-147

(error) Options for report_special_cell_legalization : -cell = %s, -all = %s, -summary = %s.

**Description**

Not all options can be specified simultaneously for reporting special cell constraints. Specify only one. If none is provided, summary is printed.

**What Next**

Correctly specify report special cells constraint.

## LGL-148

(error) None of the options for remove_special_cell_legalization have been set.

Feedback

**Description**

Both the options (-cell and -all) can't be simultaneously set to false for removing special cell constraints. Specify at least one.

**What Next**

Correctly specify remove special cells constraint.

---

## LGL-149

(warning) Site master "%s" has no Y-Symmetry. The "legal orientations" for the standard cells will be limited.

**Description**

This message is indicating that the legalization has detected that the above site master has no Y-Symmetry. For a normal design, usually, the site master should have "Y-Symmetry".

If a site master does not have "Y-Symmetry", the standard cells will have limited legal orientations because the standard cells will not be allowed to flip within the site rows.

**What Next**

Please double-check the design and library setting and make sure the setting is correct.

---

## LGL-150

(information) %s

**Description**

This message prints information about found lib cells and filler cells in the reference libraries.

---

## LGL-151

(warning) Filler cell '%s' has no VT implant layer shapes.

**Description**

The given filler cell has no shapes on any implant layer. So this lib cell cannot be used for virtual filler insertion.

**What Next**

Make sure that the frame view has VT implant layer shapes.

Feedback

## LGL-155

(warning) Filler lib cell %s is set to dont_use and will be ignored during legalization.

**Description**

Fillers with dont_use attribute set to true will be ignored during legalization/check legality.

**What Next**

To use this filler lib cell, mark dont_use attribute to false.

## LGL-156

(warning) Design %s has no EM via ladders defined. Disabling via ladder related checks.

**Description**

No EM via ladder definitions have been found. Disabling the via ladder related rule checks for this design.

**What Next**

Enable Electromigration via ladder rule for the design.

## LGL-157

(warning) Layer %s has no proper routing tracks.

**Description**

Tech layer should have routing tracks. No pins can be routed in this layer.

**What Next**

Repair this layer by adding routing tracks to it.

## LGL-158

(warning) Layer '%s' not found.

**Description**

The layer whose debug parameters were passed by user is not available for this design.

**What Next**

Correct the layer name specification.

Feedback

## LGL-159

(warning) The available area of Voltage Area shapes%s '%s' is smaller than the cell area for cell '%s'. These shapes are ignored for legalization.

**Description**

This Voltage Area or Move Bound shape has area less than any cell that belong to this region. Hence this shape is ignored for any legalization.

**What Next**

Increase the area of this shape.

## LGL-160

(warning) Cannot enable fast_pg rule due to missing vertical M1 (mask metal2) layer.

**Description**

Since this design does not have vertical M1 layer, fast_pg rule cannot be enabled here.

**What Next**

Disable fast pg rule check

## LGL-161

(warning) Cross row VT check is not supported for inbound (N12) designs.

**Description**

The Advanced Legalizer does not support cross row VT checks for inbound designs. Cross row VT checks are not performed.

**What Next**

Disable cross row VT checks

## LGL-162

(warning) Reference %s pin %s does not have port_type attribute set.

**Description**

The pin does not have port type set. Hence it cannot be used in Advanced Legalizer.

Feedback

**What Next**

Set the port type for this pin in the library cell.

---

## LGL-163

(warning) Failed to find legal location for %s (%s) after %d trials.

**Description**

This cell has exceeded the maximum failed trials for any one cell to find legal location in Advanced Legalizer.

**What Next**

Increase the max trials for legalizer by using the app option place.legalize.max_legality_failures.

---

## LGL-164

(warning) After %d legalization trials, instance %s (%s) has the following legality failures:

**Description**

This message is printed for a cell after a given interval of legalization trials (default interval : 40000). This signifies that the cell has high failure rate.

**What Next**

There is a detailed tally of the number of failures for each rule after this message. Debug to find out reason for rule failures.

---

## LGL-165

(information) Printing the legality failure profile for this legalization.

**Description**

This legality failure profile is printed when the app option place.legalize.profile_legality_failures is set to true. It prints the fail count for each rule in this legalize_placement run.

**What Next**

Check why these rules are failing and accordingly make changes in the design.

Feedback

## LGL-166

(warning) Unable to set layer shape mask fixed for cell '%s'.

**Description**

Failed to set layer shape mask fixed for the block instance in NDM.

## LGL-167

(warning) Unable to set blockage mask fixed for cell '%s'.

**Description**

Failed to set blockage mask fixed for the block instance in NDM.

## LGL-168

(warning) Unable to save %s file %s.

**Description**

File could not be saved as number of report files already saved in this directory exceed 10000.

**What Next**

Cleanup the directory as the report exceeds limit.

## LGL-169

(warning) %s than 2 spacing rule labels are found from %s, from line %d %s, %s.

**Description**

For any given spacing rule, only 2 labels are required. If there are more than 2 labels, only 2 are useful and if there are less than 2 labels, that spacing rule cannot be applied. Hence, the rule is ignored.

**What Next**

Redefine the spacing rules providing only 2 labels for each rule.

## LGL-170

(warning) No rule engine found. Pin Access Optimization is not executed.

Feedback

**Description**

Basic engine is not initialized for this design. Cannot execute Pin Access Optimization without basic engine.

**What Next**

Rerun the design with place.legalize.enable_early_data_support to check the reason for basic engine failure and rectify the design accordingly.

---

## LGL-171

(warning) No base site definition found. Pin Access Optimization is not executed.

**Description**

Basic site def is not set for this design. Pin Access Optimization cannot be executed without basic site def.

**What Next**

Rerun the design with place.legalize.enable_early_data_support to check the reason for no basic site def.

---

## LGL-172

(warning) No grid found with non-zero width. Pin Access Optimization is not executed.

**Description**

This design does not have a grid with non-zero width. Pin Access Optimization cannot be executed without a non-zero width grid

---

## LGL-173

(warning) Pin access %s encountered some internal issue. Reverting changes done by it.

**Description**

Failed to commit or uncommit a cell or move during pin access optimization. Restoring the cell placement to original state.

---

## LGL-174

(information) Pin access %s used up to %d thread%s.

Feedback

**Description**

This is an informational message emitted when the pin access optimization completes telling you the peak number of threads used for it.

---

## LGL-175

(warning) No characterization data in memory. Will use default (method 0) for legalization library characterization.

**Description**

No characterization data found for legalization library characterization.

---

## LGL-176

(warning) There is no legality library to link to the existing legalizer.

**Description**

Legality library unavailable for the current legalizer.

**What Next**

Create a legality library for this legalizer.

---

## LGL-177

(warning) Existing legality information is compiled from different %s.

**Description**

The current legality library compiled does not match the design or rule setting.

**What Next**

Recompile the legality library for this design and rule setting.

---

## LGL-178

(warning) Resetting existing compiled legality information.

**Description**

The existing legality information is for different design or rule setting. Hence resetting it.

Feedback

## LGL-179

(warning) There %s %d NDM instance%s out of sync with ALMap's final state (existence %d; reference %d; origin %d; orient %d).

**Description**

There are instances that are not in sync with the final state after restoring states. There could be multiple reasons for sync failure like 1. Origin mismatch 2. Orient mismatch 3. Library cell mismatch 4. Cell deleted in Advanced Legalizer but not in NDM

## LGL-180

(warning) Library cell %s fails compilation/processing. Skipping.

**Description**

The library cell has failed processing or legality compilation because of any of the following reasons. 1. Library cell name not unique. A library cell with same name already exists with same name. 2. Library cell is not standard. 3. Legality file is unreadable.

## LGL-181

(warning) Design %s has Short Pins. Advanced Legalizer automatically adjusts DRC rule sets.

**Description**

This design has short pins. Advanced Legalizer will automatically adjust DRC rules to support them.

## LGL-182

(warning) Overwriting CTS spacing constraints (old x=%g y=%g) (new x=%g y=%g) for cell '%s (%s)'.

**Description**

Updating the CTS spacing contraints for this cell.

## LGL-183

(warning) Site '%s' of cell '%s' has no grid width or grid height.

Feedback

**Description**

This cell's site map is not initialized properly. Hence the cell cannot be placed anywhere.

**What Next**

Check the instance data carefully and run the design again.

---

## LGL-184

(warning) Cannot place cell %s into %s %s.

**Description**

Cannot find site row in Move Bound or Voltage Area matching the site definition for this cell.

---

## LGL-185

(warning) Layer %s %s is not a 2 dimentional table. TF information will be ignored.

**Description**

This table is not a 2 dimentional table. Hence removing the information as it cannot be used.

**What Next**

Create a 2 dimentional table.

---

## LGL-186

(warning) Cannot place cell %s because the design does not contain site rows with site definition %s.

**Description**

This cell cannot be placed as there are no site rows available for this site definition.

---

## LGL-187

(warning) Layer has non-constant pitch at the track location %f.

**Description**

This layer has non constant pitch at the mentioned track location. But the pitches are integral multiple of each other. Choosing the smaller one.

Feedback

## LGL-188

(warning) Cannot find library cell %s in the 2D rule query system.

**Description**

The 2D query system does not have a library cell. Will not be able to process the VT type for this design.

## LGL-189

(warning) Unable to process non-standard pin shapes of %s (%s).

**Description**

Advanced Legalizer can only process pin shapes of type NDM_OBJ_TYPE_POLYGON and NDM_OBJ_TYPE_RECT. Any other pin shapes cannot be processed.

**What Next**

Only use pins with shapes NDM_OBJ_TYPE_POLYGON and NDM_OBJ_TYPE_RECT for this design.

## LGL-190

(warning) Empty repelling bound for %s '%s' is ignored.

**Description**

Cannot build safety bounds as the repelling bound for heirarchy/group does not have any shapes.

## LGL-191

(warning) Library cell %s of site definition %s contains no element site definition.

**Description**

Cannot build lib cell info for VT or OD rules for this library cell as it does not have an element site definition.

## LGL-192

(warning) Library cell %s's %s %d is not an integral multiple of site %s %d.

Feedback

**Description**

This library cell's width/height is not an integral multiple of the site width/height. Cannot build library cell information for VT and OD rules for this library cell.

---

## LGL-193

(warning) Library cell %s of height %d is not an integral multiple of its element site definition %s of height %d.

**Description**

Cannot build library cell information for VT or OD rules as it's height is not an integral multiple of the element site definition's height.

---

## LGL-194

(warning) Cannot find%s pin %s from cell %s.

**Description**

Cannot process power grid blockage rules as the pin for this cell is missing.

---

## LGL-195

(warning) Unable to legalize cell %s.

**Description**

The cell could not be legalized.

---

## LGL-196

(warning) Library cell %s has no corresponding site definition.

**Description**

This library cell has no site definition. Hence it cannot be processed or used.

**What Next**

Define a site definition for this library cell and rerun the design.

---

## LGL-197

(warning) Multiple site id alignment app options are set. Please check the flow used in below message and review app options setup.

Feedback

**Description**

There is a problem with the app option set up as multiple site id allignment app options should be set. Site id should have a unique value set.

**What Next**

Set only one app option for site id allignment.

---

## LGL-198

(warning) Invalid Site Id Alignment Flow. Please check the app options setup.

**Description**

Site Id align should not have 1 set to it. This is invalid site id alignment flow.

---

## LGL-199

(warning) The max number of variants within all cell groups is %d, but the site row cycle is set to %d at This may be design setup issue and may cause legalization failure. Please review cell groups and check design setup.

**Description**

The current design has a setup failure and can cause legalization failure. The max number of variants within all cell groups is not eqaul to site row cycle.

---

## LGL-200

(warning) Variant Id/N-Variants Id alignment is enabled, but "place.legalize.enable_variant_aware" is disabled.

**Description**

The app option place.legalize.enable_variant_aware should be set if variant id allignment is enabled.

---

## LGL-201

(warning) Illegal region filter option.

**Description**

The pin access region filter to be optimized is illegal.

Feedback

## LGL-202

(warning) Invalid %s in golden DRC marker file. Ignoring the file.

**Description**

Invalid syntax/bbox in the DRC marker file. This file cannot be used for processing the DRC regions. Hence ignoring this file.

## LGL-203

(warning) Have tried %d filler combinations for the gap...

**Description**

This message is printed after every 1000 legalizer trials of the filler combinations for the gap.

## LGL-204

(warning) Site row not found at {%g %g}.

**Description**

No site row is present at the given coordinate. Hence cannot place the cell here.

## LGL-205

(warning) Unable to fill gap at {%g %g}.

**Description**

Could not fill these gaps because of the following reasons: 1. Number of trials exceed the threshold trials. 2. Could not find an instance to fill the gap 3. Could not build a new instance with similar cell design as the gap

## LGL-206

(warning) The app option %s is not set.

**Description**

The value of this app option returned is the default value.

**What Next**

Set the app option if required.

Feedback

## LGL-207

(warning) Via0 alignment rule will be ignored.

**Description**

The via0 allignment rule is ignored because of the following reasons: 1. The app option is not set. 2. Could not find the name of the via0 allignment grid specified by user.

**What Next**

Set the app option if not set already, and if the app option is set, pass the grid which exists in this design.

## LGL-208

(warning) Max %s inbound allignment rules/length checks will be ignored.

**Description**

The specified rules or checks have been ignored because of the app option not set.

## LGL-209

(warning) There are no inbound cells in the design. CPOR7 rules will be ignored.

**Description**

CPOR7 rules cannot be checked without Inbound cells. This design does not have inbound cells.

## LGL-210

(warning) %s tech layer was not found.

**Description**

Metal tech layer not found for this design. Fast PG engine and other PG related parameters will not be initialised and used in this flow.

## LGL-211

(warning) Cell %s %sOrientation %d cannot be recognized in Cell Library DRC Soft Rule.

**Description**

In cell library DRC soft rule, the cell's orientation could not be recognized.

Feedback

## LGL-212

(warning) %s rule info has already been constructed, skipped...

**Description**

This engine's rule info has already been constructed. Skipping the initialization again.

## LGL-213

(warning) Cannot find cell edge alignment grid named %s; cell edge alignment rule will be ignored.

**Description**

Cell edge alignmnet rules are being ignored. Cell edge alignment grid names not found.

## LGL-214

(warning) Issue found in calculating ML prediction accuracy. Skipping it.

**Description**

Could not calculate ML prediction accuracy.

## LGL-215

(warning) Skipping ML prediction accuracy as zero floor area.

**Description**

Cannot calculate ML prediction accuracy as the floor area is zero.

## LGL-216

(warning) Evaluation of command returned error code %d, command was %s.

**Description**

The command returned error with the given code.

## LGL-217

(warning) Skipping route-aware pin-access check due to large displacement for inst: %s.

Feedback

**Description**

The instance shows a large displacement. Hence skipping route aware pin access check for it.

---

## LGL-218

(warning) Unknown 2D filler rule %s, will be ignored.

**Description**

This filler rule is not known to Advanced Legalizer, hence will be skipped.

---

## LGL-219

(warning) M0PO max length check will be skipped as %s.

**Description**

The M0PO max length check is skipped because of one of the following reasons: 1. There is no layer with mask name metal0 in the design. 2. M0 layer has no max length info in the tech file.

---

## LGL-220

(error) 3D grid (%u, %u, %u) cannot connect to %s-layer grid due to %s %f".

**Description**

3D grid could not connect to upper/lower layer because of X/Y rule failure.

---

## LGL-221

(error) 3D grid (%u, %u, %u) cannot connect to %s-layer grid due to wrong index (%d, %d, %u).

**Description**

Because of wrong index passed, the #D grid cannot connect to upper/lower layer.

---

## LGL-222

(error) Layer %s not found %s.

**Description**

Could not find any layer in Advanced Legalizer to match the layer name given.

Feedback

**What Next**

Please check the layer name passed.

---

## LGL-223

(error) Library cell %s not found %s.

**Description**

Could not find a library cell with same name as this in the Advanced Legalizer.

---

## LGL-224

(error) Unable to obtain base site definition information.

**Description**

The base site definition information is missing. Legalizer will not function properly if base site definition is not set.

---

## LGL-225

(error) Unable to extract horizontal legal pattern for the library cell %s.

**Description**

Cannot build legal pattern for this Advanced Legalizer as the tool could not extract the horizontal legal pattern for this given library cell.

---

## LGL-226

(error) Base site def %s %s %g is not a multiple of grid %s %g.

**Description**

It might be an invalid legalizer design as base site def should always be an integral multiple of the grid width/height.

**What Next**

Rerun the design with place.legalize.enable_early_data_handling set to true to check if this design is valid or not. Can also use chck_legalizer_sanity command instead of legalize_placement.

Feedback

## LGL-227

(error) Cell %s has an invalid orientation %s.

**Description**

Cell does not have an orientation set or the orientation set is invalid for Advanced Legalizer. Cannot set keepout for this cell.

**What Next**

Set a valid orientation for this cell.

## LGL-228

(error) Cell Map initialization failed.

**Description**

Cell map could not be initialized properly because of basic engine failure.

## LGL-229

(error) compile_legality failed.

**Description**

This command failed as either the pg_drc rules are not set or legal map initialization failed.

## LGL-230

(error) Found a non-regular via color %d.

**Description**

The via color found is non regular.

## LGL-231

(error) Found zero pitch %d.

**Description**

The layer has zero pitch.

Feedback

## LGL-232

(error) Unable to snap cell %s to row.

**Description**

The cell could not be snapped to row. Cannot continue legalization.

## LGL-234

(error) Pin access %s detected incorrect data input. Aborting.

**Description**

Pin access optimization or cell spreader were aborted because the engines could not be initialized properly. Some incorrect data has been passed in the design.

**What Next**

Check for reason of engine initialization failure and find out the incorrect data which led to failure.

## LGL-235

(error) Could not find hybrid weight for cell '%s'.

**Description**

Could not get the hybrid weightage for this cell. Hence cannot perform hybrid spreading for the cell.

## LGL-236

(error) Lib-cell '%s' is unplaceable. %d success after %d trials; passing rate %g.

**Description**

This lib-cell is unplaceable. After performing a lot of trials, the success rate or pass rate is very low. Cells of this lib-cell will remain unlegalized during this legalization run.

**What Next**

Run check_legality for cells of this lib-cell to get the rules which are failing for this cell.

## LGL-237

(error) Tcl ""%s"" execution failed.

Feedback

**Description**

Could not execute the tcl.

---

## LGL-238

(error) Cell from cell map is NULL or invalid.

**Description**

Cell passed from cell map is either NULL or invalid. Could not obtain cell's block instance and hence invalid.

---

## LGL-239

(error) The via ladder %s has a locked/fixed shape %s. Ignore removing this via ladder.

**Description**

This via ladder will not be removed as it has locked or fixed shape.

---

## LGL-240

(error) NYPD legalizer is not initialized.

**Description**

Advanced Legalizer(NYPD legalizer) has not been initialized yet. Rerun the flow after initializing NYPD legalizer object.

**What Next**

First initialize the NYPD legalizer before calling this functionality.

---

## LGL-241

(error) Unable to uncommit cell %s.

**Description**

Could not uncommit cell for the current location. There could be different reasons for not being able to uncommit it depending on different rule engines. eg. another cell might be placed at the location where we trying to uncommit this cell.

---

## LGL-242

(error) Found an MX_site cell %s not placed on an MX site %s.

Feedback

**Description**

The cell cannot be placed at this site as the cell has MX orientation which does not match the site orientation. The cell will be legalized at another location which matches the orientation.

---

## LGL-243

(error) App option place.legalize.enable_machine_learning is not turned on.

**Description**

The app option place.legalize.enable_machine_learning needs to be set for the Advanced Legalizer to continue lib cell legality query.

**What Next**

Enable the app option place.legalize.enable_machine_learning and rerun the command.

---

## LGL-244

(error) Got unknown option

**Description**

Passed an option that does not match the requirements/criterion of spacing rule/label definition.

**What Next**

Re define the spacing rule/label with proper options.

---

## LGL-245

(error) Range %s

**Description**

While specifing the spacing rule, the range between labels should always be a non negative integer. Except for adjacent rows, that too only for horizontal spacing.

**What Next**

Re define the spacing rule, making sure the spacing range is a non negative integer.

---

## LGL-246

(error) Range %s

Feedback

**Description**

The given spacing range should be in range of the maximum spacing that can be used for Advanced Legalizer. If the range is specified in micron rather than site width, then it should be within maximum spacing * site width.

**What Next**

Redefine the spacing label considering the maximum constraints.

---

## LGL-247

(error) Range %s

**Description**

The spacing rule range specified in microns must always be a non-negative real number.

**What Next**

Redefine the spacing rule with proper range in micron.

---

## LGL-248

(error) Got unknown command

**Description**

Unknown data passed to the spacing rule command. Could be: 1. Read string not required for the option. 2. Unknown option passed along with the command. 3. Unknown command called. 4. Unknown side passed.

**What Next**

Rerun teh script with appropriate data/commands.

---

## LGL-249

(error) Cannot specify

**Description**

Out of vertical and horizontal, only one direction of spacing label can be specified in one command.

**What Next**

Redefine the spacing labels, such that there is different spacing label for horizontal and vertical spacing label.

Feedback

## LGL-250

(error) Neither -side nor -vertical_side is specified from line %d

**Description**

Cannot make the spacing labels and spacing rules till the sides are not specified.

**What Next**

Redefine the spacing label with sides defined.

## LGL-251

(error) The %s label

**Description**

Cannot define the spacing rules if the label is not defined.

**What Next**

Redefine the spacing labels.

## LGL-252

(error) Label

**Description**

Labels have different polarity, i.e both labels should either be horizontal or should be vertical. Mismatch is not allowed.

**What Next**

Redefine the rule with labels having same polarity.

## LGL-253

(error) -adjacent_row cannot be specified with -halo together, from line %d

**Description**

The two options -adjacent_row and -halo cannot be specified by the user together.

## LGL-254

(error) Cannot open vertical spacing rule configuration file name

Feedback

**Description**

Cannot load spacing rule information as the vertical spacing rule configuration files could not be opened.

---

## LGL-255

(error) LegalMap val %d at index %d,%d is not a valid cellId.

**Description**

The cell id value in legal map at the given index is not valid. Cannot check for overlap at this location.

---

## LGL-256

(error) Range is not specified for line %d

**Description**

Cannot define a spacing rule without the range. Hence this spacing rule will be ignored.

**What Next**

Redefine the spacing rule with maximum and minimum spacing range.

---

## LGL-257

(error) Unable to read boundary leakage side file.

**Description**

Boundary leakage details will not be available as the boundary leakage side file could not be read.

---

## LGL-258

(error) Unable to commit ignoreCell %s %s %s.

**Description**

Unable to commit cell because of: 1. Region failure. 2. Rule failure.

---

## LGL-259

(error) File '%s' is not of expected type.

Feedback

**Description**

File type does not match the expected type.

---

## LGL-260

(error) Empty _rtLayers.

**Description**

Cannot build the via coloring map and tracks as the rt layers vector is empty.

---

## LGL-261

(error) Unknown violation type. Violation rule = %s.

**Description**

This violation rule does not have any such violation type.

---

## LGL-262

(error) 2D checker/legalizer does not support a site row (%s) whose width is not an even multiple of the grid width (%s).

**Description**

Any site row which has site width odd multiple of the grid width cannot be used in Advanced Legalizer. Cannot build basic engine with such site rows. Hence an invalid design.

**What Next**

Make sure the site row's width is an even multiple of the grid width.

---

## LGL-263

(error) Unable to restore %s (ref %s) to original legal state %d (last state %d; engine %s) Cell siteMap %s Lib siteDef %s Cell origin %d %d orient %s libCell %s.

**Description**

Cannot restore the cell to the original legal state from the last state.

---

## LGL-264

(error) Unable to read equiv lib cell file %s.

Feedback

**Description**

The equivalent lib cell cannot be read.

---

## LGL-265

(error) Library cell %s's height is a not multiple of the site height.

**Description**

Cannot place a cell whose height is not an integral multiple of the base site height. There is no site row in this Advanced Legalizer where it can be placed.

**What Next**

Change the library cell's height to integra multiple of the base site height.

---

## LGL-266

(error) Unable to run DRC prediction.

**Description**

Cannot run drc predictions

---

## LGL-267

(error) Invalid DRC prediction model.

**Description**

The drc prediction model is not valid. Cannot continue drc predictions.

---

## LGL-268

(error) No DRC error file.

**Description**

Could not find any DRC file.

---

## LGL-269

(error) Illegal row height specified.

Feedback

**Description**

Row height specified here is illegal. Cannot generate 2D multi height gaps.

---

## LGL-270

(error) Failed to generate new mask vec.

**Description**

Could not generate new mask vector.

---

## LGL-271

(error) The app option place.legalize.max_horizontal_pin_density must be a set with pair of non-zero positive integers {%d %d}.

**Description**

The app option should get both the values for pin density, positive integral values. Cannot initialize horizontal pin density.

**What Next**

Re define the app option with proper values.

---

## LGL-272

(error) Duplicated maximum cell group constraints in PRF & json_glc_lib_cell; following PRF constraints & skipping the other!!

**Description**

These constraints are already present in the PRF. Hence skipping it.

---

## LGL-273

(error) No %s height filler library cells are specified to fill %s height gaps.

**Description**

Cannot insert filler full/half gaps as no full/half height filler library cells specified.

---

## LGL-274

(error) Unable to legalize %s due to max displacement constraint (%gum > %gum).

Feedback

**Description**

The cell is surpassing the maximum displacement constraints. Hence it cannot be legalized.

---

## LGL-275

(error) Unable to read cell library drc side file.

**Description**

Could not find data in the file passed for cell library dc side.

---

## LGL-276

(error) Failed to insert BBox{ {%d, %d} {%d, %d} }, keyX = %d, keyY = %d, lay = %lu.

**Description**

Could not insert bbox.

---

## LGL-277

(error) Failed to insert Obj(%s), keyX = %d, keyY = %d, lay = %lu.

**Description**

Failed to insert object.

---

## LGL-278

(error) Failed to insert Mask(%d) of BBox{ {%d, %d} {%d, %d} }, keyX = %d, keyY = %d, lay = %lu.

**Description**

Failed to insert mask.

---

## LGL-279

(error) The list of OD.S.17 forbidden spacing range {min, max} is empty. Ignore OD.S.17 checking.

**Description**

OD.S.17 spacing forbidden range is not found.

Feedback

## LGL-280

(error) Dynamic OD length constriants' techfields should be 0 or 5!!

**Description**

The value of dynamic OD length constraints should either by 0 or 5. Any other value is invalid.

## LGL-281

(error) Cannot find dummy cell for filler lib cell %s.

**Description**

No dummy cell has been created for this filler library cell.

## LGL-282

(error) No basic site map is found.

**Description**

Basic site map not found for this Advanced Legalizer. Cannot continue with the design.

## LGL-283

(error) The list of OD proximity values is empty. Ignore clock min jog check.

**Description**

Clock min jog check will be ignored as the OD proximity value list is empty.

## LGL-284

(error) Error in processing place.legalize.VT_groups app options. Layer %s is not a valid layer for VT group.

**Description**

The app option cannot be processed as the value passed by the user is not valid for the VT group.

## LGL-285

(error) VT group %s is not defined . %s.

Feedback

**Description**

Cannot continue this check as the VT group is not defined.

---

## LGL-286

(error) Layer %s is not a valid implant layer.

**Description**

The specified layer is not an implant layer.

---

## LGL-287

(error) PG blockage spec {%s} is not a valid specification (e.g. VR:name|LIBCELL:name layerName blockRate).

**Description**

The PG blockage specification is not valid. It should be of the format as shown.

---

## LGL-288

(error) Unable to read via ladder analysis file vialadderAnalysisFile.

---

## LGL-289

(error) Via ladder rule %s has unmatched pin region W x H = %d x %d!

**Description**

The pin region does not match the specification of the via ladder rule.

---

## LGL-290

(error) Cannot restore %s to its original library cell %s.

**Description**

Could not commit the instance with swapped library cell.

---

## LGL-291

(error) The PT-ECO link has already been initiated; must terminate it first.

Feedback

**Description**

The PT ECO has already been initialized. To re initialize it, the already existing PT ECO has to be terminated.

**What Next**

Terminate the already existing PT ECO.

---

## LGL-292

(error) Trial placement must be {instance_name reference_name} list pair.

**Description**

The trial placement passed by the user should be a pair of two objects: instance name and reference name. Cannot execute trial placement for the passed list.

**What Next**

Pass the list according to the specified format.

---

## LGL-293

(error) Unable to find 2D checker cell named %s.

**Description**

The PT ECO link cannot find any instance with same name in advanced legalizer.

---

## LGL-294

(error) Unable to %s trial placement %s file %s.

**Description**

PT ECO link could not read/write the input/output file as the file name is not correct.

---

## LGL-295

(error) Instance at (%f, %f) is not covered by valid routing tracks.

**Description**

No valid routing tracks are present at the given location. Hence the instance cannot be covered by the tracks.

Feedback

## LGL-296

(error) Instance %s is not covered by valid %s routing tracks.

**Description**

No valid routing tracks are present at the location where this instance is placed. Hence it cannot be covered by the tracks.

## LGL-297

(error) Wrong via %s.

**Description**

The via does not match the requirement.

## LGL-298

(error) No bot pattern.

**Description**

No bot pattern.

## LGL-299

(error) The PT-ECO link has not been initiated; %s.

**Description**

Cannot perform the operation as teh PT ECO link has not been initialized yet.

## LGL-300

(error) Invalid sameColorCutCenterMinSpacing.

**Description**

Invalid spacing constraint.

## LGL-301

(error) Cannot find sameColorCutCenterMinSpacigTbl in V1.

Feedback

**Description**

The spacing table is not found for this Advanced Legalizer.

---

## LGL-302

(error) Invalid AON cells spec %s.

**Description**

The AON cell spec passed by user from the app option "place.legalize.AON_cell_placement_spec" has invalid data. There should be 3 tokens for every entry in the list.

---

## LGL-303

(error) Unable to find %s %s from AON placement spec.

**Description**

The attribute passed in the AON placement spec is not present in the Advanced Legalizer.

**What Next**

Check the design for this attribute.

---

## LGL-304

(error) Invalid AON cell pin keepout spec %s.

**Description**

The AON cell pin keepout spec passed by user from the app option "place.legalize.AON_pg_keepout_spec" has invalid data. There should be 3 tokens for every entry in the list

---

## LGL-305

(error) Unable to find %s %s from AON cell pin keepout spec.

**Description**

The attribute passed in the AON placement spec is not present in the Advanced Legalizer.

**What Next**

Check the design for this attribute.

Feedback

## LGL-306

(error) Metal/via tech layer was not found.

**Description**

No metal or tech layer found. The LA rules engine cannot be initialized.

## LGL-307

(error) Movable region size should be -1 or positive values, but its current value: %d.

**Description**

The movable region size should either be -1 or some positive value. It is currently set to 0, which is not a valid movable region size.

## LGL-308

(error) For the id (%d), cross cell subnet is out of range.

**Description**

The value of cross cell subnet is out of range for this id.

## LGL-309

(error) Cannot find library cell info for cell %s.

**Description**

The LA info map does not contain any library cell info for the given cell.

## LGL-310

(error) Tech layer has no minimum width information.

**Description**

This tech layer does not have a minimum width information. Using default value for this tech layer.

## LGL-311

(error) Layer's minimum widths from the tech file are not correct.

Feedback

**Description**

The tech file does not have valid minimum width for the layers.

---

## LGL-312

(error) Boundary values are not set in getSpecificBox function.

**Description**

The boundary values should be set for specific box. Asserting.

---

## LGL-313

(error) 3D layer %u has ZERO grid size: %lu x %lu, rt_pin_access is not enabled!

**Description**

Rt pin access is not enabled as the 3D layer has zero grid size.

---

## LGL-314

(error) Pin layer has no upper grid connection at (%u, %u).

**Description**

The pin layer should have an upper grid connection at the given coordinates.

---

## LGL-315

(error) 3D grid connection failed (broken num: %d).

**Description**

The connection for 3D grid failed.

---

## LGL-316

(error) Unable to compile the library cell %s.

**Description**

The legality compiler could not compile the given library cell because: 1. It could not read the legality file. 2. Could not find site map for this library cell.

Feedback

## LGL-317

(error) Unable to write legality compile file %s.

**Description**

Could not write legality compile file because either the pg_drc data was missing or the file name was incorrect.

## LGL-318

(error) There is no legality compilation information.

**Description**

The legality compilation information is missing.

## LGL-319

(error) Unable to read legality library file %s.

**Description**

The legality compiler cannot read library file because either the file name is wrong.

## LGL-320

(error) PG data changed since library %s. Need to regenerate library compiler file.

**Description**

The data saved previously has changed. The library compiler file needs to regenerated.

## LGL-321

(error) Fail to open PURE %s file: %s.

**Description**

Could not open the file.

## LGL-322

(error) Syntax is wrong in %s layers in PURE!!!

Feedback

**Description**

Wrong syntax for the specified layer.

---

## LGL-323

(error) Order of %s %d is incorrect.

**Description**

Order of the id is not valid for the shape/pattern.

---

## LGL-324

(error) Too many features > 100.

**Description**

Features should not exceed 100.

---

## LGL-325

(error) Layer name: %s has no mapping layer number.

**Description**

The layer number passed from the file has no layer number. Cannot process this layer.

---

## LGL-326

(error) No %s in pattern, patID = %d.

**Description**

No layer/box in the specified pattern.

---

## LGL-327

(error) NO layers in PURE!!!

**Description**

No layers found from the user passed file.

Feedback

## LGL-328

(error) Orientation %s is not supported in PURE!

**Description**

This orientation is not supported in the flip graph. Only MX, MY and R180 are allowed.

## LGL-329

(error) Cannot initialize graph with no layers!!!!

**Description**

There are no layers present, hence cannot initialize the pattern graph.

## LGL-330

(error) Illegal pattern index %d to add violation box!!!!

**Description**

Cannot add violation box this pattern index.

## LGL-331

(error) No box id in pattern graph!!!

**Description**

The pattern graph has no box.

## LGL-332

(error) This pattern is not implemented yet...

**Description**

The specified pattern is not specified yet. Cannot continue.

## LGL-333

(error) The layer %d is not defined yet.

**Description**

The specified layer has not been defined yet.

Feedback

## LGL-334

(error) ShapeBBox is not covered by patBBox. Please modify patBBox %s to cover %s.

**Description**

The pattern bbox should cover the shape bbox. Modify the pattern bbox to cover it.

## LGL-335

(error) There is no site def defined for the block.

**Description**

This block does not have any site def defined. Advanced Legalizer cannot function for such block.

## LGL-336

(information) Adjusting %s to a multi-height library cell.

**Description**

The height of this library cell is more than equal the base site height. So marking this library cell as multi height.

## LGL-337

(warning) No cells selected for IR-drop optimization.

**Description**

There are no cells for IR drop optimization.

## LGL-338

(error) Cannot get value for max group length from library '%s'.

**Description**

The library has no value for max group length.

## LGL-339

(warning) CPODE Layer %s has no corresponding tech layer.

Feedback

**Description**

There is no tech layer for the CPODE layer obtained from the tech file.

---

## LGL-340

(warning) The max length of rule max_cpode_length is not specified by app option max_vertical_stacked_length.

**Description**

Cannot initialize CPODE rules as the app option place.legalize.max_vertical_stacked_length does not specify the max length of max_cpode_length.

---

## LGL-341

(warning) The max length of rule max_cpode_length is not positive from tech file, so cpode check will be ignored.

**Description**

The value of max cpode length specified from tech file is not positive. Cpode rule check will be ignored.

---

## LGL-342

(warning) Max parallel run spacing %g is less than the site width %g.

**Description**

Max parallel run spacing should not be less than the site width.

---

## LGL-343

(warning) AL should get a hybrid filler at {%g %g}, but got another cell instead (cell = %s, lib = %s, orient = %s, bbox = {{%g %g} {%g %g}}), indicating this cell might not be on site.

**Description**

At the given location, there should be a hybrid filler but there is another instance placed here. This should be resolved before filler insertion.

---

## LGL-344

(warning) User rule %s will be overidden.

Feedback

**Description**

This user rule has been overidden.

---

## LGL-345

(warning) Repeated layers during graph construction!

**Description**

The layer has already been included in the pattern graph.

---

## LGL-346

(warning) Anchor Point is invalid; it should be located at a shape's corner.

**Description**

The anchor point should always be present at the corner of the shape. This point is invalid.

---

## LGL-347

(warning) Misaligned gap found; offending cell is %s ref %s origin {%g %g}, gapX1IndexU %d gapX2IndexU %d gapYIndexU %d gapX1IndexL %d gapX2IndexL %d gapYIndexL %d.

**Description**

The found gap is misaligned.

---

## LGL-348

(warning) Could not find nypdLibCell for VL excluded libCell %s.

**Description**

No nypdLibCell found in Advanced Legalizer with same name as the given VL excluded library cell.

---

## LGL-349

(warning) %d out of %d fixed cells need TPT recoloring!

**Description**

Out of the total fixed cells, these many cells need TPT recoloring.

Feedback

## LGL-350

(warning) There are %d fixed cells w/ TPT coloring violations.

**Description**

These many fixed cells show TPT colouring violations.

## LGL-351

(warning) More than 2 values in the list of OD.S.17 forbidden spacing range {min, max} are found. Only the first 2 values will be used.

**Description**

Only two values should be passes along the OD.S.17 spacing range. The user has passed more than two values so only first two will be used.

## LGL-352

(warning) There are hybrid fillers in the filler library but the app option "%s" %s. Otherwise, hybrid fillers would be treated as normal fillers, resulting in jog violations.

**Description**

Hybrid filler are present in the filler library, but because of this app option, these hybrid fillers would be treated as normal fillers.

## LGL-353

(warning) There is no special filler in the filler library but the app option "place.legalize.enable_1cpp_filler_rule_check" is enabled. Thus, the boundary transition rule check is "automatically" disabled, so that only filler1 and filler1 abutment rule is considered.

**Description**

Only the above specified rules are enabled as there are no special fillers, and this app option is enabled.

## LGL-354

(warning) No instance in the design or No POD in library cells!!

Feedback

**Description**

No POD in the library cells.

---

## LGL-355

(warning) More than 2 proximity values in the list are found. Only the first 2 values will be used.

**Description**

Only two values should be present in the list. Only first two proximity values will be used.

---

## LGL-356

(warning) Select %d as major POD!

**Description**

Select major POD value as the one specified.

---

## LGL-357

(warning) Improper setting of minWidth uJog = %d (%g um) > lJog = %d (%g um). Should set to equal.

**Description**

Improper setting of minWidth Jog. Will be set to eqaul.

---

## LGL-358

(warning) Improper setting of min 1-fin diffusion length = %d (%g um) > lJog = %d (%g um). Should set to equal.

**Description**

The setting for min 1 fin diffusion length is not correct. Should be set to equal.

---

## LGL-359

(warning) No OD related tech field was set. AL will automatically omit OD checking.

**Description**

Advanced Legalizer will skip OD rules checking as the OD related fields are not set.

Feedback

## LGL-360

(warning) No LUP shape of name =

**Description**

There is no LUP shape on the layout with the name as this.

## LGL-361

(warning) Additional power management user lib type name

**Description**

This lib type has already been registered in the Advanced Legalizer. Hence this will be skipped.

## LGL-362

(warning) Additional power management lib type name

**Description**

This librry type is not present in the Advanced Legalizer. Hence the lib cell list will be ignored.

## LGL-363

(warning) A tap uncover violation at {{%g %g} {%g %g}} is suppressed due to %s.

**Description**

The violation has been suppressed becuase of the following reasons: 1. The app option place.legalize.suppress_bndry_cover_violation. 2. Drop edge with given bbox.

## LGL-364

(warning) The user tap rule table

**Description**

The table is empty or invalid. Hence the user tap rule will be ignored.

Feedback

## LGL-365

(warning) The user tap rule table one but an odd number of numbers, so the last number will be ignored.

**Description**

The number of values in the tap rule table should have even number ov values. If the number of values passed by user is odd,we will ignore the last value.

## LGL-366

(warning) Additional power management lib-cell name

**Description**

The library cell passed by the user through app option is not found in the cell library for this Advamced Legalizer.

## LGL-367

(warning) Tap cover distance in site %d is too much, should be less than %d.

**Description**

The tap cover distance is very high.

## LGL-368

(warning) No rule is specified. Please set rules via "set_tap_boundary_wall_cell_rules" before checking.

**Description**

There are no rules specified for checking the chip finishing. The rule set_tap_boundary_wall_cell_rules should be set before checking for rules.

## LGL-369

(warning) Tap distance is not a positive number. Please check the rule setting by "set_tap_boundary_wall_cell_rules".

**Description**

Tap distance should always be a positive number.

Feedback

## LGL-370

(warning) The tap coverage distance is not defined in either rule container or app option (place.legalize.tap_cover_distance <micron>). The tap coverage checking might fail.

**Description**

The tap coverge might fail as the tap coverage is not defined.

## LGL-371

(warning) Failed to uncommit cell '%s'.

**Description**

Advanced Legalizer could not uncommit this cell.

## LGL-372

(warning) Characterization data for library cell '%s' not found.

**Description**

No information found for this library cell in Advanced Legalizer.

## LGL-373

(warning) Unable to find characterized lib-pin shape %s. Cell: %s pin: %s.

**Description**

Could not find characterization data for library pin shape.

## LGL-374

(warning) Illegal access point at %s for shape %s.

**Description**

There is no access point for this shape at the given coordinates.

## LGL-375

(warning) No site definition found. Unable to extract local connectivity.

Feedback

**Description**

Could not extract local connectivity as there is no site definition for this pin access optimizer.

---

## LGL-376

(warning) Duplicate cell found in move set. Aborting move.

**Description**

Cannot normalize cells as a duplicate cell os found in the move set.

---

## LGL-377

(warning) Debug options are defined.

**Description**

Debug options are defined by debug app option.

---

## LGL-378

(warning) Failed to read value '%s' for option '%s'.

**Description**

Could not read values for the given option.

---

## LGL-379

(warning) Couldn't find option '%s'.

**Description**

The given debug option in the app option value is not defined.

---

## LGL-380

(error) User Rule JSON BBox has no complete L B T R values.

**Description**

Update the JSON bbox for complete information.

Feedback

## LGL-381

(error) Syntax is not correct for User Rule JSON: (name, value)=(%s, %s).

**Description**

No data passed for either name or value of the user rule JSON.

## LGL-382

(error) User Rule mode %s is not defined.

**Description**

Mode for this user rule is not defined.

## LGL-383

(error) User Rule feature %s is not defined.

**Description**

Feature for this user rule is not defined.

## LGL-384

(error) User Rule JSON BBox has no layers.

**Description**

No layers defined for the user rule JSON BBox.

## LGL-385

(error) Unknown syntax!!!

**Description**

Cannot extract user rule information from JSON as the syntax is not correct.

## LGL-386

(error) Syntax wrong for inputFileName: %s.

**Description**

Cannot read the data from file as the file name is not correct.

Feedback

## LGL-387

(error) ShapeOrient only support 1 orient.

**Description**

Syntax error in the shape orient. Only 1 orient is supported.

## LGL-388

(error) Syntax not correct at the line %s.

**Description**

The syntax at this line does not match the required syntax, hence closing the file without reading data.

## LGL-389

(warning) Want to get a non-initialized flipped graph.

**Description**

The user wants to get non initialized flipped graph.

## LGL-390

(warning) Gui is not running.

**Description**

Could not highlight the bbox as the gui is not running. Run the gui before executing this command.

## LGL-391

(warning) No cells to be legalized/checked in %s.

**Description**

This VA or MB does not have any movable cell to legalize or check.

## LGL-392

(warning) Removing the cell %s as it belongs to high local density area.

Feedback

**Description**

The design has high local density which may lead to high runtime. So removing the cells in the high local density area to ensure reduced runtime. The area occupied by these cells will be replaced by a blockage and the design remains unlegalized.

**What Next**

Decrease local density for the mentioned location and rerun AL.

---

## LGL-393

(warning) Library cell %s has a VT layer %s (%d) that does not exist in Tech File.

**Description**

The layer number for this library cell does not match in the tech file.

---

## LGL-394

(warning) Invalid value of vertical abutment. Insert spaces between values for valid input.

**Description**

The value of vertical abutment input by user is invalid. The vaue exceeds the maximum possible value. Insert space between abutment values to ensure correct input.

---

## LGL-396

(information) Removing unfixed physical only cell %s.

**Description**

Removing the unfixed physical only cells as the app option place.legalize.exclude_unfixed_physical_only_cells is enabled.

---

## LGL-397

(Warning) No site rows found in floorplan.

**Description**

This message is indicating that there was an error when analyzing site row configuration in current block.

**What Next**

Review the design and floorplan preparation.

Feedback

## LGL-398

(Error) Site row header %s has missing site definition.

**Description**

The specified site row header has missing site definition. Such designs with missing site definition are not valid for legalize_placement and check_legalizer_sanity.

**What Next**

Define a site definition for the site row header.

## LGL-399

(Warning) Option design.sort_em_via_ladders is enabled. The priority of EM via ladder constraints on cell pin is modified.

**Description**

This warning message is to remind user the priority of EM via ladder constraints on cell pin is modified. Tool would try to preserve more routing resource for EM via ladder insertion during legalize_placement. It is expected to see larger cell displacement or longer runtime.

**What Next**

If you observe any unexpected cell displacement or runtime increase, please turn off app_option design.sort_em_via_ladders so that the tool would honor the original EM via ladder constraint priority.

## LGL-400

(Warning) App option place.legalize.enable_early_data_support is set to true. Early data handling is enabled for Advanced Legalizer.

**Description**

The app option when set to true, detects and handles dirty input. It may lead to bailing out of legalizer at an early stage leading to unlegalized instances and could also involve making changes in the design to prevent any crash, hang or long runtime condition that may occur because of the dirty data input.

**What Next**

Check for early data warning and error messages in the log if any and make changes in the input accordingly.

Feedback

## LGL-401

(Error) No mask shift layer definition in Cell %s NDM %s.

There is a conflict in the setting of color-shifting flow between the app option value and the library cell attribute. Color-shifting flow would not be performed. Please reexamine your app option setting regarding color-shifting flow and the attribute of the library cell.

**What Next**

Revise the aforementioned app option settings or reassign the attributes for the library cells.

## LGL-402

(error) Found MX_site cell %s not placed on MX site %s origin {%g %g} orient %s.

**Description**

This message is indicating that MX_site cell not placed correctly on MX site.

**What Next**

Run command legalize_placement.

## LGL-403

(Warning) All non-default voltage areas are ignored due to app option 'place.legalize.ignore_voltage_areas'.

**Description**

This message is indicating that the app option 'place.legalize.ignore_voltage_areas' has been turned on and all non-default voltage areas are ignored during legalization and check_legality. Even if cells are not assigned to a specific voltage area, they can be placed within a non-default voltage area.

## LGL-404

(Warning) auto_stagger is enabled in router for via ladder, but it is not enabled in legalizer.

**Description**

route.auto_via_ladder.auto_stagger is enabled, but place.legalize.via_ladder_auto_stagger is not enabled.

Feedback

**What Next**

The suggestion is to set the two options consistently.

---

## LGL-405

(Warning)

**Description**

place.legalize.user_route_via_pillar_map is an obsolete app option that is no longer needed. This app option will be removed in the next release. Please remove this setting before the next release.

**What Next**

The suggestion is to remove the usage of app option place.legalize.user_route_via_pillar_map

---

## LGL-406

(Warning) Single height rectilinear lib-cell '%s' found. This lib-cell is considered as a rectangular cell.

**Description**

This message is indicating that there is a setup issue regarding the site definition of a rectilinear cell. Rectilinear cells need to be multi-height with respect to its sitedef.

**What Next**

Review the sitedef setup.

---

## LGL-407

(Warning) Not a recommended usage for color-shifting: app option place.legalize.%s and cell %s attribute %s. Use app option definition.

There is a conflict in the setting of color-shifting flow between the app option value and the library cell attribute. Color-shifting flow would adopt the app option setting regardless of the attributes from the library cell. Please reexamine your app option setting regarding color-shifting flow and the attribute of the library cell.

**What Next**

Revise the aforementioned app option settings or reassign the attributes for the library cells.

Feedback

## LGL-408

(Error) Inconsistent between app option %s and cell %s NDM %s.

There is a conflict in the setting of color-shifting flow between the app option value and the library cell attribute. Color-shifting flow would not be performed. Please reexamine your app option setting regarding color-shifting flow and the attribute of the library cell.

**What Next**

Revise the aforementioned app option settings or reassign the attributes for the library cells.

## LGL-409

(Warning) No mask shift layer definition in Cell %s NDM %s, use app option definition.

There is a conflict in the setting of color-shifting flow between the app option value and the library cell attribute. Color-shifting flow would not be performed. Please reexamine your app option setting regarding color-shifting flow and the attribute of the library cell.

**What Next**

Revise the aforementioned app option settings or reassign the attributes for the library cells.

## LGL-410

(warning) For cpode_spacing in AL, could not open the side file %s.

**Description**

Users need to define side file for cpode_spacing.

**What Next**

Please set the app_option place.legalize.cpode_spacing_side_file. You can set it to the directory name or use this format<dir>/*.txt.

## LGL-411

(warning) For cpode_spacing in AL, there's missing information for cpode soft placement constraints. Need to specify target cells, attacking types and two kinds of keepout distance \n

**Description**

Users need to set all four app_options related to cpode_spacing.

Feedback

**What Next**

Please check if all app_options are set correctly:
place.legalize.cpode_soft_placement_constraint_attacking_types
place.legalize.cpode_soft_placement_constraint_target_cells
place.legalize.cpode_soft_placement_constraint_target_cell_keepout_distance
place.legalize.cpode_soft_placement_constraint_attacking_cell_keepout_distance

---

## LGL-412

(warning) For cpode_spacing in AL, the list number of target cells, attacking types and keepout distance do not match.

**Description**

Users need to define same types of target cells, attacking types and corresponding keepout distance.

**What Next**

Please check if the type number in app_options are same:
place.legalize.cpode_soft_placement_constraint_attacking_types
place.legalize.cpode_soft_placement_constraint_target_cells
place.legalize.cpode_soft_placement_constraint_target_cell_keepout_distance
place.legalize.cpode_soft_placement_constraint_attacking_cell_keepout_distance

---

## LGL-413

(warning) For cpode_spacing in AL, unable to find user defined group name %s in target cells and attacking types.

**Description**

The group name in cpode_spacing app_options do not match.

**What Next**

Please check if group name in app_options are set correctly:
place.legalize.cpode_soft_placement_constraint_attacking_types
place.legalize.cpode_soft_placement_constraint_target_cells
place.legalize.cpode_soft_placement_constraint_target_cell_keepout_distance
place.legalize.cpode_soft_placement_constraint_attacking_cell_keepout_distance

---

## LGL-414

(information) Found redundancy group '%s'.

Feedback

**Description**

Prints the names of the safety register redundancy groups that are considered by legalization.

---

## LGL-415

(information) Found safety_error_code_group '%s'.

**Description**

Prints the names of the safety error-code groups that are considered by legalization.

---

## LGL-416

(information) Found failsafe_fsm_group '%s'.

**Description**

Prints the names of the safety fail-safe groups that are considered by legalization.

---

## LGL-417

(information) Vertical abutment prohibit pattern %d-%d accepted.

**Description**

Prints the pattern pairs of prohibited vertical abutments of std cells. The legalizer will consider these patterns as a hard placement rule and avoids vertical abutment of std cells with these given patterns.

---

## LGL-418

(information) Reducing rule set for cell '%s'.

**Description**

In early data mode when the strategy is set to limit_legalize for the zero-pass-rate legalization check, the legalizer will reduce the rule set to overlap-free and cell-on-site. This message indicates which cells are affected by this rule set reduction.

---

## LGL-419

(information) Pass-rate based cell priority: %s.

Feedback

**Description**

This is a debug message for lib-cell pass-rate based cell priority definition.

---

## LGL-420

(Error) CTS is trying to legalize %u unplaced cell%s.

This error message indicates a flow problem during CTS. CTS should never try to legalize cells without a placed location.

---

## LGL-421

(warning) App option '%s' is enabled.

**Description**

This messages indicates that the given app option is enabled which might have a negative impact on the legalization result.

**What Next**

Check whether this setting is required.

---

## LGL-422

(information) Short variant '%s' is wider than the M1 PG pitch. This lib-cell is allowed to be placed under M1 PG shapes.

When the app options place.legalize.avoid_wide_variants and place.legalize.limit_wide_variants are turned on, wide variants (in X direction) are only allowed to be placed under M1 PG shapes. The shorter varaints are not allowed to be placed under PG shapes. However, if the M1 PG pitch is smaller than the width of the shorter variant(s), then an exception applies and the shorter variants are allowed to straddle the PG shapes.

**What Next**

None

---

## LGL-460

(Warning) Lib cell %s does not have any primary PG pin.

Feedback

**Description**

This message indicates that this lib cell does not have PG pins with necessary attribute "pg_type == primary". It could lead to false track capacity violation on multi-height cells.

**What Next**

Please double-check the design and library and make sure all lib cells have necessary primary PG pins.

---

## LGL-461

(Error) Legalizer encounter internal memory error... %s : %s.

**Description**

This message indicates that during legalization, some internal memory error related to the indicated object(s)/condition(s).

**What Next**

Follow message suggestion to remedy the issue if any. Run check_database before saving/continuing design and report this to Synopsys.

---

## LGL-462

(Error) Legalizer exceed max allowed internal memory errors.

**Description**

This message indicates that during legalization,more than the number of allowed internal memory errors happened and therefore command exits

**What Next**

Look for earlier error messages to remedy the issue if possible. Run check_database before saving/continuing design and report this to Synopsys.

---

## LGL-463

(Warning) site '%s' polarity undetermined.

**Description**

You received this message because the named site polarity has not been determined.

**What Next**

Please file a bug report.

Feedback

## LGL-464

(warning) There are %n non-default %sLegalizer/Chip-Finishing app-options in use.

The design has non-default legalizer/chip-finishing app_options, which may lead to unexpected behavior.

**What Next**

Cross-check non-default place.legalize.* and chipfinishing.* app_options, and please check with CAE on hidden non-default settings being used.

## LGL-465

(warning) The specified region is too big; it is more than %.2f percent of core area.

The region provided by cellmap client is too big, which can lead to long-runtime issue.

**What Next**

Ask the client to review the usage of the region-based APIs from the stack trace below the message.

## LGL-466

(information) cellmap bailed out after %n trials.

We have a trial limit from cellmap runtime improvement for non-region-based cellmap query/commit.

**What Next**

Please use a region-based cellmap API if a far-away legal location is acceptable.

## LGL-467

(warning) CLO_ERR: LocalOTF for this thread does not exist!!!

The client did not setup CLO or ALMap to use in lock-free mode during multi-threading

**What Next**

Open a bug so that this can be addressed

## LGL-468

(Warning) via %s doesn't have lower metal layer.

Feedback

This message indicate that via doesn't have connected lower metal layer.

**What Next**

Please check via has associated upper and lower metal layer.

---

## LGL-470

(information) Adding AON placement spec: %s

An AON placement specification is added as a legalization constraint

**What Next**

None

---

## LGL-471

(Error) Cell %s has ndm-cm mismatch. %s

**Description**

This message indicates that there is inconsistency between NDM and CM in terms of the cell and describes its details.

**What Next**

None

---

## LGL-472

(Error) Cell %s has no ref block, but it was loaded into cellmap

**Description**

The cell was loaded into cellmap, but it has no reference block.

**What Next**

None

---

## LGL-473

(info) %s

**Description**

The message prints summary of NDM and cellmap consistency check

Feedback

**What Next**

None

---

## LGL-474

(warning) In vertical abutment rule specification, the portion of pattern exceeding the width of cell %s must be an even number. Pattern %s is ignored.

**Description**

While specifying the vertical abutment rule, if the pattern length is larger than the cell site width, the portion exceeding the cell width must be an even number.

**What Next**

Re define the vertical abutment rule.

---

## LGL-475

(Warning) AL-Cellmap cannot commit app_fixed cell %s (lib %s) at %s %s, which could lead to cell overlaps after the flow.

AL-Cellmap expects no overlaps between app_fixed cells in the incoming database; otherwise, cell overlaps after AL-Cellmap are expected. To fix the error, please ensure all app_fixed cells are legal before running the flow.

**What Next**

To detect all violating app_fixed cells, please enable the app_option "place.legalize.enable_check_legality_before_and_after_cellmap". Clean up all violations and rerun the flow to see the cell overlapping issue is resolved.

---

## LGL-476

(Warning) A large %d-row-height cell %s (Lib %s) is detected in CLO, which may cause long runtime to find a legal location.

When CLO legalizes a big cell, typically it leads to long runtime.

**What Next**

If there is a freezing issue, please add dont_use on the big Lib cell(s).

Feedback

## LGL-477

(Warning) app option place.legalize.v0vt_layer_abutment_edge_distance will be deprecated in the future release. Please use place.legalize.v0_layer_abutment_edge_distance, place.legalize.v1_layer_abutment_edge_distance, and place.legalize.vt_layer_abutment_edge_distance instead

### Description

place.legalize.user_route_via_pillar_map is an obsolete app option that is no longer needed. This app option will be removed in the next release. Please remove this setting before the next release.

### What Next

Please replace it with suggested app option instead

## LGL-478

(warning) lib-pin %s at (%s) has %s access points, but the Via ladder constraint %s needs %s access points, this Via ladder constraint is infeasible at current location.

### Description

You received this warning message since this lib pin $lp at current location is not able to grow a via ladder on it based on the given via ladder constraint, because advanced legalizer does NOT support automatically pin extension.

### What Next

You can take one of the following actions to fix this warning. 1. Modify the mismatched via ladder constraints such that advanced legalizer can grow via ladders on pins without pin extension. For example, decrease the number of cut rows/cuts per row. 2. Modify the cell layout by manually extending the short pins. 3. If you want to bypass this via ladder constraint, please enable place.legalize.enable_via_ladder_early_data_support. Tool will automatically bypass the via ladder checks on those cell instances of the lib cell $lc, and still perform via ladder checks on other normal cells (see LGL-090).

# LIB

## LIB-001

(error) Current library is not defined.

Feedback

**Description**

The current library is not defined. Some commands require that the current library is set.

**What Next**

You must open or create a library and set current library with the current_lib command

---

## LIB-002

(error) Empty reference library path.

**Description**

A reference library path was blank or an empty string.

**What Next**

Check that the given reference library path has a value that can be found on the search_path.

---

## LIB-003

(error) Reference library name %s in library path %s has the same name as path %s.

**Description**

The library name is the last part of each reference library path. Each reference library path in a reference library list must have a unique library name.

**What Next**

Check the list of reference library paths for libraries that end in the same name. Each library name must be unique. Libraries ending in the same name must be renamed. Once the library with the duplicate name has been renamed, the library may be added to the reference library path list.

---

## LIB-004

(error) Cannot find reference library path %s.

**Description**

The reference library given in the -before or -remove argument cannot be found in this library's reference library path list.

Feedback

**What Next**

Check the ref_libs in the library with the get_attribute command or the report_ref_libs command. Use either the library name (the last part of the path), the reference library path, or the full path name to that library for the set_ref_libs command's -before or -remove argument.

---

## LIB-005

(error) Library name cannot contain ":", "|", or white spaces.

**Description**

The ':' and '|' character is used to delimit library and design names. It cannot be used for library name.

---

## LIB-006

(error) Library '%s' already exists

**Description**

This error message occurs because a library of the specified name already exists.

**What Next**

Use another name for the new library or remove the library from the file system and rerun the command.

---

## LIB-007

(error) Cannot load technology file '%s'.

**Description**

The specified technology file cannot be loaded.

**What Next**

Make sure the path points to a valid technology file.

---

## LIB-008

(Error) %s failed. Library '%s' is modified.

**Description**

The specified operation failed because the library contains modified and un-saved data.

Feedback

**What Next**

Try again with the -force option.

---

## LIB-009

(error) Library directory '%s' already exists

**Description**

A library directory with the specified name already exists.

**What Next**

Use another name for the new library or remove the library from the file system and rerun the command.

---

## LIB-010

(error) Library name is blank or empty.

**Description**

A library name was blank or an empty string.

**What Next**

Check that the given library name is valid simple path.

---

## LIB-011

(error) Library '%s' is not found.

**Description**

The specified library is not found on the current search_path.

**What Next**

Check that the given library name has a valid value that can be found on the search_path.

---

## LIB-012

(error) %s failed. Library '%s' is in read mode.

**Description**

The specified operation failed because the library is open in read mode.

Feedback

**What Next**

open the library again in edit mode and try again.

---

## LIB-013

(error) %s failed because the library cell library '%s' cannot be modified.

**Description**

The specified operation failed because the library is a library cell library. Library cell libraries are read-only in icc2_shell. They can be modified only in the library manager shell, lm_shell.

**What Next**

Use the library manager shell to make changes to this library.

---

## LIB-014

(Information) Re-setting technology section of library '%s' with '%s'.

**Description**

The technology section of the specified library is being replaced.

---

## LIB-015

(Information) Adding design '%s' to library '%s'.

**Description**

The specified design.view is being added the the library.

---

## LIB-016

(Information) Removing design '%s' from library '%s'.

**Description**

The specified design.view is being removed the the library.

---

## LIB-017

(Information) Incrementing open_count of library '%s' to %d.

Feedback

**Description**

The library is already in memory. Each open_lib command on a library will increase a library's open_count by one.

---

## LIB-018

(Information) Decrementing open_count of library '%s' to %d.

**Description**

The library is still in memory. Each close_lib command will decrease a library's open_count by one. The library is removed from memory when its open_count drops to zero.

---

## LIB-019

(Error) rebuild_lib cannot be used on single-file format library.

**Description**

rebuild_lib can only be used on directory-format library to rebuild a library's design catalog based on the directory contents.

**What Next**

Make sure the library is in directory format and try again.

---

## LIB-020

(Error) Technology replacement failed. Technology is incompatible.

**Description**

The technology data must be compatible with the existing technology data in the library.

**What Next**

Make sure the replacement technology data is compatible.

---

## LIB-021

(Error) Cannot copy library to self.

**Description**

The source and destination library are the same. Copying a library to itself is not allowed.

Feedback

**What Next**

Try a different destination library name.

---

## LIB-022

(Error) Cannot move library to self.

**Description**

The source and destination library are the same. Moving a library to itself is not allowed.

**What Next**

Try a different destination library name.

---

## LIB-023

(Warning) Library '%s' is being opened in read mode because it is a library cell library.

**Description**

Library cell libraries are read-only in icc2_shell. They can be modified only in the library manager shell, lm_shell.

**What Next**

Use the library manager shell to make changes to this library.

---

## LIB-024

(Error) %s failed. Library '%s' is modified.

**Description**

The specified operation failed because the library contains modified and un-saved data.

**What Next**

Try again with the -force option.

---

## LIB-025

(Error) technology library '%s' is invalid.

**Description**

A technology library must be a valid library and it must have a technology section.

Feedback

**What Next**

Try again with a library with a valid technology section and that is accessible through the current search_path setting.

---

## LIB-026

(error) Reference library path '%s' is the same as the library.

**Description**

The reference library path cannot point to the library itself.

**What Next**

Make sure the reference library path is not the same as the library path and try again.

---

## LIB-027

(error) Reference library path '%s' is not a valid library

**Description**

The reference library path does not point to a valid library given the current search_path setting.

In case that physical source files were provided to automatically create reference libraries, a technology file must also be specified. If no technology file specified, then the tool will treat all files provided as full NDM reference libraries, thus report this message.

**What Next**

Check that the given reference library path has a value that can be found on the search_path. If auto reference libraries creation is desired, then please also provide a technology file.

---

## LIB-028

(Error) rebuild_lib cannot be used on library that hasn't been saved to disk.

**Description**

rebuild_lib can only be used on directory-format library to rebuild a library's design catalog based on the on-disk directory contents.

**What Next**

If you haven't saved the library to disk, you don't need rebuild_lib.

Feedback

## LIB-029

(Error) Library already has a technology section.

**Description**

read_tech_file cannot be used to replace the technology section of a library. It will fail if a library already has or associates with a technology section.

**What Next**

Make sure the library does not already contain a technology section.

## LIB-030

(Error) Library '%s' has no technology information.

**Description**

The library does not have a technology section. copy or move a designs into this library will fail.

**What Next**

Check that the library contains a technology section. Use read_tech_file to create a technology section.

## LIB-031

(Error) Library '%s' and '%s' contain incompatible technology information.

**Description**

The specified libraries contain incompatible technology sections. copy or move designs between these two libraries will fail.

**What Next**

Check that the two libraries contains compatible technology sections.

## LIB-032

(information) Overwriting library '%s'.

**Description**

The library is being overwritten. It will be removed from the ref_lib list of all currently opened libraries that reference it, and be replaced with a new library.

Feedback

**What Next**

Showing the library name to help understand when a library is being overwritten.

---

## LIB-033

(Information) Promoting open_mode of library '%s' from read to edit.

**Description**

The library is already open in read mode. Subsequent open_lib -edit will promote the library open_mode from read to edit.

---

## LIB-034

(error) Cannot write out technology file '%s'.

**Description**

The specified library cannot write out technology file.

**What Next**

Make sure the specified library was valid.

---

## LIB-035

(warning) Library '%s' is not being saved because it is in read mode.

**Description**

The specified library is not saved during save_lib -all because it is open in read mode.

**What Next**

open this library in edit mode and try again.

```
prompt> open_lib <lib_name> -edit
```

**See Also**

- open_block

---

## LIB-036

(error) Library '%s' is referenced by another library '%s'.

Feedback

### Description

A library cannot be closed or removed if it is on another library's reference library list. The higher-level library that references it must be closed first.

### What Next

Remove the library that references this library first, and then remove this library. Or force the library to be purged from memory with the *-purge* option.

```
prompt> close_lib -force ...
```

---

## LIB-037

(information) Replacing %s file '%s' with '%s'

### Description

This informational message shows the full path name and type of a file being read into the application, replacing an existing one.

### What Next

Use the full path name to help understand any problems encountered while reading the file.

---

## LIB-038

(error) Cannot open %s library '%s' as reference library.

### Description

Physical libraries can be generated from Library Manager frame flow or Library Compiler create_physical_lib. In order to use such libraries as reference libraries with physical data only, the libraries must be prepared using Library Manager “create_workspace -flow physical_only” or Library Compiler “create/compile_fusion_lib -frame” commands. Direct usage will no longer be supported starting in T-2022.03-SP3 release.

---

## LIB-039

(error) Cannot open logical library '%s' in '%s' flow.

### Description

You can save a library from the library manager shell (lm_shell) prior to checking the workspace using *save_lib*. If the source of the library was a DB or .lib file, this is considered a logical source library. These libraries can be reloaded into the library

Feedback

manager at a later time, but they cannot into a workspace in any of these flows: *edit*, *physical_only* or *organize_by_pvt*.

---

## LIB-041

(error) Cannot open %s library '%s' outside of a workspace.

**Description**

Only complete NDM files can be opened outside of a workspace, and they will be read-only. Other NDM file types (such as frame files) cannot be opened outside of the workspace.

**What Next**

If you are trying to change a frame NDM, create a workspace with the frame flow, and load the ndm file using *read_ndm*.

---

## LIB-042

(warning) Library '%s' cannot be added to the aggregate library.

**Description**

Design libraries or aggregate libraries cannot be added to an aggregate library. Only libraries which were written by the library manager, lm_shell, using *commit_workspace*, using a flow other than the *aggregate* flow, can be added to an aggregate.

---

## LIB-043

(information) Library '%s' has been added to the aggregate library.

**Description**

This is an informational message that you will see in the library manager shell, using the aggregate flow, indicating that a lib-cell library has been added to the aggregate library, usually after using *open_lib*.

---

## LIB-044

(information) Library '%s' has been removed from the aggregate library.

**Description**

This is an informational message that you will see in the library manager shell, using the aggregate flow, indicating that a lib-cell library has been removed from the aggregate library, usually after using *close_lib*.

Feedback

## LIB-045

(error) Can not save %s library '%s'.

**Description**

This message is reported when you try to save a workspace library or uncommitted logic library in Library Manager.

**What Next**

Can not save workspace library or uncommitted logic library.

## LIB-046

(error) Library '%s' cannot be used for the the edit flow

**Description**

When creating a workspace in the library manager shell (lm_shell) with the edit flow, only a library previously committed by lm_shell can be specified. Design libraries, aggregate libraries, or libraries saved from the lm_shell with *save_lib* cannot be used in this context.

## LIB-047

(warning) library '%s' is already in the aggregate library.

**Description**

This lib-cell library was already in the aggregate library. It cannot be added to the aggregate library multiple times.

## LIB-048

(Information) Technology file was not replaced. '%s' is identical to the current technology file.

**Description**

This informational message tells that technology file was not replaced. The replacement technology file contains identical content as the current technology file.

## LIB-049

(error) Block '%s' is open. Technology file cannot be replaced while blocks are opened in the library.

Feedback

**Description**

It is an error to replace the technology section in a library in which there are open blocks. As technology file replacement could clobber in-memory design data. All blocks must be closed during technology file replacement.

**What Next**

close all opened block and try again

---

## LIB-050

(error) Library '%s' is also bound to this tech section. Technology file cannot be replaced if the tech section is bound to more than one library.

**Description**

It is an error to replace the technology section in a library if the technology section are referenced by other libraries in memory, as technology file replacement could clobber in-memory library data. All other libraries that reference this technology section must be closed during technology file replacement.

**What Next**

close the other libraries and try again.

---

## LIB-051

(error) Version '%s' is not supported. Use report_versions to list available versions.

**Description**

The specified version is not supported. save_lib can only save to specific release versions. Please use report_versions to list the versions available.

**What Next**

Try again with a valid release version.

---

## LIB-052

(error) Cannot close a member library of an aggregate library.

**Description**

This message indicates that you tried to close a member library of an aggregate library but the closing of a member library was not allowed. When an aggregate library is opened, all member libraries of the aggregate library are also opened automatically and will be closed

Feedback

when the aggregate library is closed. If a member library is specifically closed using the close_lib or remove_lib command, the member library will be removed from the containing aggregate library. However, the removal of a member library from an aggregate library can only be done in the aggregate flow in the library manager shell.

**See Also**

- create_workspace

---

## LIB-053

(error) Specified length precision '%d' does not match that of already opened libraries '%d'.

**Description**

This message indicates that you specified the length precision for the new library in create_lib command using "-scale_factor" option. However, libraries with a different length precision are already open. This is not allowed. All libraries open at the same time must have the same length precision. Similarly, the length precision of any reference libraries used for create_lib, must be identical to the length precision for the new library.

**What Next**

Here are a few options, 1) Provide the correct length precision or remove the option, in which case length precision will be inherited from previously opened libraries. 2) If the libraries already open are not needed, please close them and rerun the command.

**See Also**

- create_lib

---

## LIB-054

(error) Specified length precision '%d' is not valid.

**Description**

The specified length precision is not a valid value.

**What Next**

The valid values for length precision are 100, 200, 1000, 2000, 4000, 10000 and 20000

**See Also**

- create_lib

Feedback

## LIB-055

(error) Length precision '%u' in the input file is not compatible with the length precision '%u' used in the current library.

### Description

The input file (DEF, LEF or GDS) cannot be read in as it will result in data being lost when converting length numbers from the file to the internal data in the library.

### What Next

Please choose a different input file which has a length precision that can be used to translate the numbers without loss of data.

### See Also

- create_lib
- read_def
- read_lef

## LIB-056

(error) Length precision '%d' specified for output file is not compatible with the length precision '%d' used in the current library.

### Description

The file (DEF, LEF or GDS) cannot be written out with the specified length precision, as it will result in data being lost when converting length numbers from internal data in the library for output.

### What Next

Please specify a compatible length precision, which can preserve all data.

### See Also

- create_lib
- write_def
- write_lef
- write_gds

Feedback

## LIB-057

(error) The length precision of the library being opened '%s' does not match that of already opened libraries '%d'.

### Description

This message indicates that the library being opened has a scale factor different from libraries previously opened. This can happen for a library being opened explicitly, using *open_lib*, or implicitly if it is being set as a reference library for a main library using either *create_lib* or *set_ref_libs* commands. This is not allowed. All libraries opened at the same time must have the same length precision. All reference libraries must have the same length precision as the main library.

### What Next

Here are a few options, 1) If the library is used as a reference library, remove it from the list of specified reference libraries. 2) Choose reference libraries with the correct scale factor. 3) Close previously opened libraries if they are no longer needed.

### See Also

- open_lib
- create_lib
- set_ref_libs

## LIB-058

(warning) Length precision '%d' specified for output file can result in loss of data because it is less than the precision used in the library '%d'.

### Description

The file (DEF, LEF or GDS) should not be written out with the specified length precision, as it will result in data being lost when converting length numbers from internal data in the library for output.

### What Next

Please specify a compatible length precision, which can preserve all data.

### See Also

- create_lib
- write_def

Feedback

- write_lef
- write_gds

---

## LIB-059

(Error) technology library '%s' does not match any library on the reference library list.

**Description**

The technology library must be specified either as a simple path that matches the tail portion of an reference library entry or as a relative or absolute path that exactly match an entry on the reference library list.

**What Next**

Add a matching entry of the technology library on the reference library list.

**Examples**

create_lib -use_technology_lib technology_lib.nlib -ref_libs { technology_lib.nlib }

create_lib -use_technology_lib ../nlibs/technology_lib.nlib -ref_libs { ../nlibs/technology_lib.nlib }

create_lib -use_technology_lib technology_lib.nlib -ref_libs { ../nlibs/technology_lib.nlib }

---

## LIB-060

(error) Version '%s' is not supported. Supported version range must be within %s and %s.

**Description**

The specified version is either not in the supported format, or outside the supported range. Schema version must be specified in <major>.<minor> format and between the last compatible schema version and the current schema version.

**What Next**

Try again with a valid schema version format within the allowable range.

---

## LIB-061

(information) Version '%s' is truncated to '%s'.

Feedback

### Description

The input version cannot be found, to truncate the input version string to the character '-' from the last one to the first. The truncation will stop when the rest string is legal or without '-' in the rest.

### What Next

To input legal version. User can get legal versions by report_version.

---

## LIB-062

(error) The length precision of the library being opened '%s' does not allow its data to be scaled correctly.

### Description

This message indicates that the library being opened has a scale factor that cannot be correctly scaled for the *design.session_scale_factor*. This can happen for a library being opened explicitly, using *open_lib*, or implicitly if it is being set as a reference library for a main library using either *create_lib* or *set_ref_libs* commands. This is not allowed. The *design.session_scale_factor* must be a multiple of the length precision of the library being opened.

### What Next

Here are a few options, 1) Set *design.session_scale_factor* to a different value which is a multiple of length precision of all libraries needed. Please note if *design.session_scale_factor* is changed, all open libraries must be closed and reopened. 2) If the library is used as a reference library, remove it from the list of specified reference libraries. 3) Choose reference libraries with the correct scale factor. 4) Close previously opened libraries if they are no longer needed.

### See Also

- open_lib
- create_lib
- set_ref_libs
- design.is_3dic_mode
- design.session_scale_factor

Feedback

## LIB-063

(info) The library must be saved, closed and reopened for any change in value of half_node_scale_factor attribute to take effect.

### Description

The library must be saved, closed and reopened for any change in value of *half_node_scale_factor* attribute to take effect. Further, the attibute *half_node_scale_factor* can only be set when no designs from the library are open.

### What Next

Please close all designs from this library before changing *half_node_scale_factor*. After changing the attribute, save, close and reopen the library to continue.

### See Also

- open_lib
- design.is_3dic_mode
- design.session_scale_factor

## LIB-064

(error) Specified scale factor '%d' is not compatible with the session scale factor '%d'.

### Description

This message indicates that the specified the scale factor is not compatible with the session scale factor. Scale factor is also known as length precision. The session scale factor is specified using the app option *design.session_scale_factor*. The session scale factor must be a mutliple of the scale factor for the library.

### What Next

Please specify a scale factor so that the session scale factor is a multiple of that.

### See Also

- create_lib
- design.session_scale_factor
- design.allow_multiple_scale_factors

Feedback

## LIB-065

(error) Required argument '%s' was not found.

### Description

This message indicates that the *-scale_factor* argument must be specified due to the app options being used. If either of *design.allow_multiple_scale_factors* or *design.is_3dic_mode* app options are set to true and *design.session_scale_factor* is also set, the *-scale_factor* must be specified when running *create_lib*.

### What Next

Please specify the -scale_factor argument

### See Also

- create_lib
- design.session_scale_factor
- design.allow_multiple_scale_factors
- design.is_3dic_mode

## LIB-066

(error) Specified scale factor '%d' does not match the session scale factor '%d'.

### Description

For converting libraries from one scale factor to another, the session scale factor is the library scale factor. A different scale factor is not allowed. The session scale factor is specified using the app option *design.session_scale_factor*.

### What Next

Please specify a scale factor so that it is same as the session scale factor or remove the argument from the command.

### See Also

- create_lib
- design.session_scale_factor
- design.allow_multiple_scale_factors

Feedback

## LIB-067

(warning) Length precision '%d' specified in the input file can result in loss of data because it is more than the precision used in the library '%d'.

### Description

The file (DEF) being read in has distance data in length precision higher than the design library. It can potentially result in data being lost when converting length numbers to internal data in the library during read.

### What Next

Please use a library with appropriate scale factor to avoid data loss.

### See Also

- create_lib
- read_def

## LIB-068

(error) Cannot open library '%s' because it has potential risk of precision loss.

### Description

Physical library created with potential risk of precision loss is not allowed to be opened and used. The physical lib could be created with smaller scale factor than its tech file's length precision.

### What Next

Regenerate the physical library and make sure its scale factor is equal or larger than its tech file's length precision.

## LIB-068w

(Warning) The library '%s' opened has potential risk of precision loss.

### Description

Physical library created with potential risk of precision loss is not allowed to be opened and used, unless lib.setting.allow_ref_lib_with_lossy_scale_factor has been turned on. The physical lib could be created with smaller scale factor than its tech file's length precision.

Feedback

**What Next**

Regenerate the physical library and make sure its scale factor is equal or larger than its tech file's length precision.

---

## LIB-069

(Warning) The physical library '%s' is being created with potential risk of precision loss.

**Description**

The physical lib is being created with smaller scale factor than its tech file's length precision, which could lead to precision loss.

**What Next**

Regenerate the physical library and make sure its scale factor is equal or larger than its tech file's length precision.

---

## LIB-070

(error) No such PVT configuration rule '%s'

**Description**

The rule you specified to *set_pvt_configuration -rule* does not exist. When this error occurs, for convenience, the application will show the current configuration.

**What Next**

Check for spelling errors and re-enter the command to change the current rule and apply changes to it.

---

## LIB-071

(information) %s PVT configuration rule '%s'

**Description**

This is an informational message indicating that the *set_pvt_configuration* command either created a new rule or deleted an existing rule because you removed all constraints from it.

---

## LIB-072

(error) No panes in lib '%s' matched your PVT configuration

Feedback

**Description**

You created a PVT configuration using *set_pvt_configuration*. Now, when opening a reference library with *open_lib*, the information from the configuration is used to determine which logical panes of data are active. Your configuration has deactivated all of the panes from this library, effectively rendering the library useless to timing, power, and optimization. The library fails to open.

**What Next**

It's possible that you loaded a library that simply does not have any desirable PVTs. If this message is unexpected, consult the configuration (using *set_pvt_configuration* with no arguments) and determine why all of the panes were filtered out. Then adjust the configuration rules accordingly to enable the desired PVTs.

---

## LIB-073

(warning) PVT configuration changes are not applied to open libs

**Description**

You made a change to the PVT configuration using *set_pvt_configuration* and there were one or more reference libraries already open. The PVT configuration is only applied at the time the reference library is opened with *open_lib*. Incremental changes to libraries already opened is not possible.

**What Next**

If you wanted to change the available PVTs for the libraries that are already open, you need to first close these libraries, then reopen them after the configuration has been updated.

---

## LIB-074

(error) Cannot create PVT configuration rule named '%s': %s

**Description**

The name you specified to *set_pvt_configuration -name* is invalid. The message will say why. Typically, the name you supplied is already in use as a rule name, or it is empty or all whitespace.

**What Next**

Check for spelling errors and re-enter the command to create the rule.

Feedback

## LIB-075

(warning) Library '%s' has no reference Libraries.

**Description**

The Libaries has no reference Libraries.

**What Next**

The reference library list can be set using set_ref_lib command.

## LIB-080

(warning) Cannot find .frame/LEF/Milkyway input, the auto reference library creation will be disabled.

**Description**

When link_library is specified, those db files should either 1) be covered by full NDM reference libraries, or 2) there should have .frame/LEF/Milkyway physical library view specified, so that auto reference library can be created. This message indicates that no physical library view can be found, so that the auto reference library creation will be disabled.

**What Next**

To create reference library automatically, please provide physical library views corresponding to the link_library.

## LIB-081

(warning) No db files from link_library to create reference libraries, will create a physical-only reference library.

**Description**

For automatic reference library creation, the logical data comes from the db files specified in link_library setting. If full NDM reference libraries also specified, then those db files already in the full NDM reference libraries will be excluded from auto reference library creation.

This message indicates that either no db files specified in link_library, or all db files specified in link_library are already contained in full NDM reference libraries. In that case, no db files will be used for auto reference library creation, thus a physical-only reference library will be created.

Feedback

**What Next**

If creating physical-only reference library is not desired, please provide necessary db files in link_library setting.

---

## LIB-082

(error) Failed to create reference library %s, please check log file %s for detailed error.

**Description**

This message indicates that the reported reference library was not created successfully from auto reference library creation.

**What Next**

Please refer to the error message printed prviously, or indicated log file for detailed error information.

---

## LIB-083

(warning) There are %d %s not used for creating cell libraries: %s.

**Description**

This message reports a list of files that provided but not used for cell libraray creation. This could be caused by not reading the files successfully. For db files, it could be caused by no corresponding physical data.

**What Next**

Please check messages printed previously, or the log files under the log directory to know the detailed information.

---

## LIB-084

(information) Auto created reference libraries are up-to-date, no need to rebuild.

**Description**

This message indicates that all automatically created reference libraries are up-to-date. There is no need to rebuild them.

**What Next**

This is an informational message, no action required.

Feedback

## LIB-085

(information) Need to rebuild auto created reference library: %s.

**Description**

This message indicates that automatically created reference libraries need to be rebuilt. The reason was given along with the message itself.

**What Next**

This is an informational message, no action required.

## LIB-086

(information) Rebuild of auto created reference libraries succeeded.

**Description**

This message indicates that rebuild of the automatically created reference libraries succeeded.

**What Next**

This is an informational message, no action required.

## LIB-087

(warning) Skip the rebuild of the auto created reference libraries: %s.

**Description**

This message indicates that the auto created reference libraries need to be rebuild. However, some pre-conditions cannot be met so the rebuild was skipped. Most likely, the source files for rebuild the auto created reference libraries cannot be found.

**What Next**

To rebuild the auto created reference libraries, please make sure that all source files for building them can be found using search_path.

## LIB-088

(error) Failed to rebuild the auto created reference libraries.

**Description**

This message indicates that the rebuild of the auto created reference libraries failed.

Feedback

**What Next**

Please refer to the previously reported error messages for detailed information.

---

## LIB-089

(error) %s '%s' is not valid: %s.

**Description**

This message indicates that the specified file is not valid. The reason is reported with the message itself.

---

## LIB-090

(error) %s '%s' must be specified: %s.

**Description**

This message indicates that required option must be specified. The reason was reported alone with the message itself.

---

## LIB-091

(error) Failed to convert Milkyway libraries to frame view.

**Description**

This message indicates that convert Milkyway libraries to frame view failed.

**What Next**

Please refer to previously reported error messages for detailed information.

---

## LIB-092

(error) Failed to create %d (total %d) reference libraries, please check log file '%s' for detailed error.

**Description**

This message indicates that auto reference libraries creation failed. There are some reference libraries not created successfully.

**What Next**

Check the reported log file for detailed errors.

Feedback

## LIB-093

(information) Successfully built %d reference libraries: %s.

**Description**

This message indicates that auto reference library creation was succeeded.

**What Next**

This is an informational message. No action required.

## LIB-094

(information) Successfully converted %d Milkyway libraries: %s.

**Description**

This message indicates that the Milkyway library to frame conversion was succeeded.

**What Next**

This is an informational message. No action required.

## LIB-095

(error) Failed to create frame library %s, please check log file '%s' for detailed error.

**Description**

This message indicates that the reported frame library was not created successfully from corresponding Milkyway library.

**What Next**

Please refer to the error message printed prviously for detailed error information.

## LIB-096

(error) Cannot launch %s to %s.

**Description**

This message indicates that launching a specific executable for a specific job failed.

**What Next**

If the executable is IC Compiler shell, please make sure it is a valid executable.

Feedback

## LIB-097

(information) db files specified in link_library already covered by full NDM reference libraries, the auto reference library creation will be disabled.

### Description

When link_library is specified, those db files should either 1) be covered by full NDM reference libraries, or 2) there should have .frame/LEF/Milkyway physical library view specified, so that auto reference library can be created. This message indicates that db files already covered by full NDM reference libraries, so that the auto reference library creation will be disabled. The full NDM reference libraries specified will be used directly.

### What Next

This is an informational message, no action is required.

## LIB-098

(information) Use existing reference libraries.

### Description

There should be an attempt to rebuild the auto reference libraries and the attemp was failed. This message indicates that existing reference libraries will be used.

## LIB-099

(warning) Override the previous label mapping, now db '%s' attaches to %s label '%s'.

### Description

This message occurred when users attached the same db name with 2 or more different labels by setting app option lib.configuration.process_label_mapping. For example: When users set the process label mapping with the below command: set_app_options -name lib.configuration.process_label_mapping -value {{label1 {tcbn90ghplt_ccs.db}} {label2 {tcbn90ghplt_ccs.db}}}

The LIB-099 warning message will be displayed as below: Warning: Override the previous label mapping, now db 'tcbn90ghplt_ccs.db' attaches to process label 'label1'.(LIB-099)

### What Next

Please correct the issue of the app option(lib.configuration.process_label_mapping) setting, and then set the app option with the corrected setting.

Feedback

## LIB-100

(warning) link_library does not cover all db files of the library '%s'.

### Description

This message is reported when lib.configuration.force_update_timing_views app option is set to true. It indicates that the db files specified in the link_library app var does not cover all db files of a source library. So, one or more db files in the source library will missing in the generated cell library.

### What Next

Please verify that the missing of db files is expected.

## LIB-101

(error) Prebuilt library '%s' cannot be in the output directory '%s'.

### Description

This message is issued when the *lib.configuration.force_update_timing_views* application option is set to *true* and the reported prebuilt library is in the output directory specified with *lib.configuration.local_output_dir* application option.

### What Next

Put the prebuilt library into another directory, or specify a different output directory with the *lib.configuration.local_output_dir* application option.

## LIB-102

(error) Physical source file '%s' cannot be associated with two assembly scripts: '%s' and '%s'.

### Description

This message indicates that the value specified for *lib.configuration.assembly_scripts* application option is not valid. The reported physical source file is associated with more than one assembly script. This is not allowed as the tool will not able to determine which assembly script should be used to assembly the data.

### What Next

Revise the app option value to make sure every physical source file only be associated with one assembly script.

Feedback

## LIB-103

(error) Application option '%s' has an invalid value.

**Description**

This message indicates that the specified application option has an invalid value.

**What Next**

Provided a correct value for the application option.

## LIB-104

(error) Cannot determine assembly script: physical source file '%s' associated with assembly script '%s', physical source file '%s' associated with assembly script '%s'.

**Description**

This message indicates that two physical source files of one lib have different assembly scripts. In this case, tool cannot determine which assembly script to use.

**What Next**

Revise the lib.configuration.assembly_scripts application option to have correct mapping between assembly scripts and physical source files.

## LIB-105

(error) Failed to group source data, please check log file %s for detailed error.

**Description**

This message indicates that the tool failed to group the source data, .db files from link_library and physical files from -ref_libs option, into groups that for reference libraries creation.

**What Next**

Please refer to the indicated log file for detailed error information.

## LIB-106

(warning) %s will not be persistent because '%s' is a library cell library, so the setting will be lost when the session ends.

Feedback

**Description**

The specified operation will apply in this session only because the library is a library cell library. Library cell libraries are read-only in icc2_shell. They can be modified only in the library manager shell, lm_shell. To have the setting persist without using the library manager shell, you can make the setting in a script that is called at each new session.

**What Next**

Use the library manager shell to make permanent changes to this library. To have the setting persist without using the library manager shell, you can make the setting in a script that is called at each new session.

---

## LIB-107

(warning) Found duplicated db file '%s' in the link_library. The duplicated one will be ignored.

**Description**

The reported db file is duplicated specified in the link_library. Only the first one will be used. All duplicates will be ignored.

**What Next**

Remove the duplicated db files in the link_library.

---

## LIB-108

(information) Can reuse library %s, skip %s.

**Description**

This message indicates that specified library is up-to-date. The specified action can be skipped.

**What Next**

This is an informational message, no action required.

---

## LIB-109

(information) Cannot reuse library %s: %s.

**Description**

This message indicates that specified library cannot be reused for current operation. The reason is given in the message.

Feedback

**What Next**

This is an informational message, no action required.

---

## LIB-110

(error) No opened physical library to save.

**Description**

There is no opened physical library in memory, so no library for this command to save.

**What Next**

Check if you need to create or open a library before save it to disk.

---

## LIB-111

(Error) Technology data in %s and currrent physical library's technology data is not compatible

**Description**

Technology data in the library which is reading and current physical library's technology data must be compatible, otherwise *read_ndm* will fail.

**What Next**

Try another library with compatible technology data.

---

## LIB-112

(warning) Current physical library's technology data replaced by the technology data from '%s'

**Description**

*read_ndm* will replace the current physical library's technology data with technology data from the library which is reading. The replacement occurs when the two technology data are compatible.

---

## LIB-113

(error) Cannot open or create multiple libraries, library '%s' has already been opened.

Feedback

**Description**

Only one physical library is allowed to be opened or created at the same time. You can only create or open another physical library after closing current library.

**What Next**

Close the current the library, and then create or open the new library.

---

## LIB-114

(error) Cannot open a none library cell library '%s'.

**Description**

Only library cell library is allowed to be opened with this command. You will see this error if you are trying to open a none library cell library with this command, such as design library.

**What Next**

Open a library cell library instead.

---

## LIB-115

(warning) Set app option '%s' has no effect when app option '%s' is %s.

**Description**

Set first app option has no effect when the second app option has specified value.

**What Next**

Make sure the second app option has a proper value, then set the first app option again.

---

## LIB-116

(warning) Ignore the application option '%s': %s.

**Description**

The reported application option will be ignored for the given reason.

**What Next**

Refer to the man page of the corresponding application option and make sure it has a proper value.

Feedback

## LIB-117

(warning) Library configuration will not be performed: %s.

**Description**

This warning message indicates that the library configuration will not be performed. The possible reason is that technology file is not specified.

This message will only be reported when link_library application variable is specified.

**What Next**

Make sure technology file is specified if need to invoke library configuration.

## LIB-118

(error) Reference library path '%s' will introduce cyclic reference libraries on the library.

**Description**

The reference library path should not introduce cyclic reference libraries as it will lead to opening a library infinite times. Cyclic reference libraries structure: library 'MID' is reference library to library 'TOP' and again library 'TOP' is reference library ot library 'MID'.

**What Next**

Make sure the reference library path doesn't create any cyclic reference libraries. To make sure no cyclic reference libraries are formed, add the reference libraries bottom-up (BOT->MID->TOP) approach.

## LIB-119

(error) Invalid value '%s' specified for application option '%s': %s.

**Description**

This message indicates that the specified application option has an invalid value.

**What Next**

Provided a correct value for the application option.

## LIB-120

(error) Invalid value '%s' specified for command option '%s': %s.

Feedback

**Description**

This message indicates that the specified command option has an invalid value.

**What Next**

Provided a correct value for the command option.

---

## LIB-121

(error) Current working directory is not writable.

**Description**

The command being executed need a writable current working directory.

**What Next**

Change current working directory to a writable directory and try again.

---

## LIB-122

(information) There are missing or invalid reference libraries. Will try to rebuild.

**Description**

This message indicates that some or all reference libraries of the library to be opened is missing or invalid. It will try to rebuild reference libraries.

**What Next**

This is an informational message, no action required.

---

## LIB-123

(warning) Local output directory '%s' containing cell libraries that are used by following processes: %s. Cannot remove it now. Please remove it manually when those processes exit.

**Description**

This warning message will reported when *lib.configuration.remove_local_output_dir* application option is true, but it cannot remove the local output directory because there are other processes are using the cell libraries in it.

**What Next**

Check the processes reported in the message. When they all exit, remove the local output directory manually.

Feedback

## LIB-124

(warning) Cannot use '%s' as local output directory. Library configuration skipped.

**Description**

This warning message indicates that the value of *lib.configuration.local_output_dir* application option is not valid, and the library configuration will be skipped. Usually, this means the directory is a volatile cache of another process.

**What Next**

Set *lib.configuration.local_output_dir* to another proper directory and try again.

## LIB-125

(error) Attribute '%s' cannot be imported: You defined it as %s; db defines it as '%s'.

**Description**

You receive this error message because while reading an attribute from db, there was a mismatch between the type (integer, string, etc.) that db defines for the attribute and the type that PrimeTime defines for the attribute. Usually, this is a user-defined attribute, defined with the *-import* option on *define_user_attribute*, and the types do not match.

**What Next**

Rerun PrimeTime and define the attribute correctly. You might need to use other Synopsys applications to determine the actual type of the attribute.

## LIB-126

(information) Cannot use '%s' as local output directory. Library configuration skipped.

**Description**

This message indicates that the tool cannot use the specified directory as the local output directory, and the library configuration will be skipped. Usually, this means the directory is already a volatile cache of another process.

**What Next**

This is an informational message and no action required.

## LIB-127

(Information) Find %s '%s' from %s.

Feedback

**Description**

This message indicates where a specific executable was found.

---

## LIB-128

(error) Cannot find %s '%s' from %s: %s.

**Description**

This message indicates that a specific executable cannot be found from the reported location.

**What Next**

Check whether the related environment variable is correct.

---

## LIB-129

(error) Library configuration skipped, since "%s" are different between master and current session.

**Description**

This message indicates that current session changed link_library or physical libraries while the shared volatile cache directory is using by the other ICC2 sessions.

**What Next**

Run the library configuration flow again while there is no other session using the volatile cache directory.

---

## LIB-130

(information) Library configuration is pending, the master session is running library configuration with the same configuration as the current session.

**Description**

This message indicates that the current session has the same link_library and physical libraries' setting as the master session which is running library configuration to build the reference libs in the shared volatile cache directory.

**What Next**

This is an informational message and no action required.

Feedback

## LIB-131

(warning) Library '%s' is not in the reference library list of current_lib '%s'.

**Description**

The desired library must be a reference library of current_lib.

**What Next**

Either use current_lib switch to the design library which has desired library as reference library, or add the desired library to the current lib's reference library list.

## LIB-132

(warning) None of panes in lib '%s' matched the specification.

**Description**

The *get/set_lib_timing*, *get/set_lib_pin_capacitance*, *get/set_lib_max_cap_table*, *get/set_lib_max_trans_table* commands issue this error message when they cannot find a pane in the desired library that matched the voltage, temperature, process number, and process label data.

**What Next**

Either use *report_lib* to find out the exactly values of voltage, temperature, process number, and process label data, or simply delete the option -voltage/-number/-label/-temperature if you don't know the exactly values.

## LIB-133

(warning) The objects do not have specified constraint table.

**Description**

The *get/set_lib_timing* commands issue this error message when they cannot find the desired constraint table.

**What Next**

Check the object with input options again. Use *get_lib_timing -table setup_hold */** to see if the desired table are there.

## LIB-134

(error) The option -shift_value must be followed by a non-zero value.

Feedback

**Description**

The *set_lib_timing* command issues this error message when giving zero to -shift_value.

**What Next**

Gives a non-zero value to -shift_value and run the command again.

---

## LIB-135

(warning) The command only supports setting scalar, 1-D, and 2-D table.

**Description**

The *set_lib_timing* command issues this warning message when the object is the table that dimension greater than two. The table value will keep the same.

**What Next**

Use *set_lib_timing* on the table which has dimension smaller or equal to two.

---

## LIB-136

(error) The objects do not exist in reference libraries.

**Description**

The *get/set_lib_pin_capacitance*, *get/set_lib_max_cap_table*, *get/set_lib_max_trans_table* commands issue this error message when they cannot find the desired object in reference libraries.

**What Next**

Make sure the desired objects are in the reference libraries.

---

## LIB-137

(warning) There is no %s table on this lib pin.

**Description**

The *get/set_lib_max_cap_table*, *get/set_lib_max_trans_table* commands issue this warning message when there is no max_cap/max_trans table on the specified library pin.

**What Next**

Make sure the specified library pin contains max_cap/max_trans table.

Feedback

## LIB-138

(warning) The desired table on the specified library pin is not a 1-d frequency based %s table.

### Description

The *get/set_lib_max_cap_table*, *get/set_lib_max_trans_table* commands issue this warning message when the desired table on the specified library pin is not a 1-d frequency based max_cap/max_trans table. The *get/set_lib_max_cap_table*, *get/set_lib_max_trans_table* commands are only supporting get/set 1-d frequency based max_cap/max_trans table.

### What Next

Make sure the desired table on the specified library pin is a 1-d frequency based max_cap/max_trans table.

## LIB-139

(error) When -index_1 option is specified, the -value option must be a single value.

### Description

The *set_lib_max_cap_table* command issues this error message when specified -index_1 option but the argument for -value is not a single value. Using -index_1 only when you want to modify a value in max_cap table at certain index (frequency). You must to specify a single value for -value.

### What Next

Give a single value for -value and run the command again.

## LIB-140

(warning) Invalid -index_1 value '%f', only accepts existing table index

### Description

The *set_lib_max_cap_table*, *set_lib_max_trans_table* commands issue this warning message when specified -index_1 option but none of that index_1 value is in the table index.

### What Next

Use *get_lib_max_cap_table*, *get_lib_max_trans_table* to see all the index_1 values in the table.

Feedback

## LIB-141

(error) -value list size is '%lu', which is different to table width '%d'

**Description**

The *set_lib_max_cap_table*, *set_lib_max_trans_table* commands issue this error message when the given list of -value size is not equal to table width.

**What Next**

Use *get_lib_max_cap_table*, *get_lib_max_trans_table* to check the table size.

## LIB-142

(information) There is no lib data change in the scope of current_lib: '%s'

**Description**

The *report_lib_changes* command issues this informational message when there is no lib data change related to the reference libraries of current library.

**What Next**

This is an informational message and no action required.

## LIB-143

(information) All lib data changes in the scope of current_lib are reverted back to original values.

**Description**

The *revert_lib_changes* command issues this informational message when it reverted all modified data within the scopt of current_lib back to original values successfully.

**What Next**

This is an informational message and no action required.

## LIB-144

(information) Previous set lib data command attachment is now deleted

**Description**

This informational message means there was an attachment recording the operations of *set_lib_timing*, *set_lib_pin_capacitance*, *set_lib_max_cap_table* and/or

Feedback

*set_lib_max_trans_table*; and now the attachment has been deleted because you reverted all lib changes and saved it.

**What Next**

This is an informational message and no action required.

---

## LIB-145

(information) Set lib data command(s) found, are saved into attachment.

**Description**

This informational message means there were operations of *set_lib_timing*, *set_lib_pin_capacitance*, *set_lib_max_cap_table* and/or *set_lib_max_trans_table* in the session. The operations were recorded and saved in an attachment. These operations will be reapply when you use *open_lib* to open the library again.

**What Next**

This is an informational message and no action required.

---

## LIB-146

(information) Reapplying previous lib data changes recorded in '%s'

**Description**

This informational message means there were operations of *set_lib_timing*, *set_lib_pin_capacitance*, *set_lib_max_cap_table* and/or *set_lib_max_trans_table* saved inside the design library, and these operations are re-applying.

**What Next**

This is an informational message and no action required.

---

## LIB-147

(warning) The attachment recorded lib data change(s) failed to compile, so the previous lib data changes are not reapplied.

**Description**

This warning message is issued when there is an attachment recorded operations of *set_lib_pin_capacitance*, *set_lib_max_cap_table*, *set_lib_timing* and/or *set_lib_max_trans_table*, but the attachment can not be compiled successfully during *open_lib*, so that the previous lib data changes are not able to reapply. The reason could

Feedback

be 1. Too old Library Compiler version (require at least 19.12 version). 2. Attachment format is invalid.

**What Next**

Make sure your $SYNOPSYS_LC_ROOT is pointing to latest version Library Compiler. Make sure the attachment (lib.setLibDataCmd) format is valid Liberty format.

---

## LIB-148

(warning) Library data change is applied, but not recorded.

**Description**

The commands *set_lib_pin_capacitance*, *set_lib_max_cap_table*, *set_lib_timing*, *set_lib_max_trans_table* issue this warning message when the settings are applied without recorded. *set_lib_pin_capacitance*, *set_lib_max_cap_table*, *set_lib_timing*, *set_lib_max_trans_table* operations will only be recorded in a design library.

**What Next**

Use *current_lib* to see if current library is a design library.

---

## LIB-149

(error) The input of option -index_1 is invalid. Please make sure the values in the list are monotonically increasing.

**Description**

The commands *set_lib_max_cap_table*, *set_lib_max_trans_table* issue this error message when option -index_1 is specified with a invalid list of values.

**What Next**

Make sure the values in the list are monotonically increasing.

---

## LIB-150

(error) The length of the input of option -index_1 and -value should be the same.

**Description**

The commands *set_lib_max_cap_table* and *set_lib_max_trans_table* issue this error message when give the option -index_1 and the option -value two different length input lists.

Feedback

**What Next**

Make sure the the input lists of the option -index_1 and -value have same length.

---

## LIB-151

(information) Successfully configured DB+Frame reference library: %s.

**Description**

This message indicates that DB+Frame reference library-configuration succeeded.

**What Next**

This is an informational message. No action required.

---

## LIB-152

(error) Cache-free library name must be a simple name, it cannot be absolute or relative path.

**Description**

The '~' and '/' character are used in absolute and relative path. It cannot be used in name of cache-free library.

**What Next**

Specify a simple name for the cache-free library.

---

## LIB-153

(warning) We do not recommend a mixture of library specific rules and non-library specific rules.

**Description**

A -library rule is added and there are non-library rules, or if a non -library rule is added and there are -library rules. For example: set_pvt_configuration -library "libA libB*" -temperatures 125 -voltages 1.05 set_pvt_configuration -add -temperatures -25 Such setting may lead all panes in a library being eliminated. For example: lib 1 name : rose pane0: voltage 1.1 pane1: voltage 1.8 pane2: voltage 2.0 lib 2 name : rose2 pane0: voltage 1.2 Pane1: voltage 2.0 lib 3 name : daisey pane0: voltage 1.3 pane1: voltage 1.5

For the below mix setting(not recommend), all the panes of rose2 should be filtered. set_pvt_configuration -lib rose -voltage 1.1 set_pvt_configuration -add -voltage 1.3 The setting is equal to the below setting: set_pvt_configuration -lib rose -voltage 1.1 set_pvt_configuration -add -lib * -voltage 1.3

Feedback

### What Next

Specify all rules with -library or without -library option For example: set_pvt_configuration -library "libA libB*" -temperatures 125 -voltages 1.05 set_pvt_configuration -add -library "moreLib" -temperatures -25 or set_pvt_configuration -temperatures 125 -voltages 1.05 set_pvt_configuration -add -temperatures -25

---

## LIB-154

(warning) The library option must be a file name (it cannot be a path): change the path "%s" to "%s".

### Description

The file name specified with the -library option must be file name only, i.e. it cannot contain any path. For example: For the following reference library: tcbn90ghplvtwc_ccs, 2 panes: Pane 0: process = "" (1), voltages: 0.9; temperatures: 125 Pane 1: process = "" (1), voltages: 1.1; temperatures: -40 Here is a correct setting with library option to set pvt for pane0: set_pvt_configuration -library tcbn90ghplvtwc_ccs -voltages 0.9

But if users set the library option as below(bad setting): set_pvt_configuration -library ./ref/tcbn90ghplvt_ccs.ndm -voltages 0.9 The tool will issue LIB-154 and try to fix the issue by changing the path name to file name. For example: Warning: The library option must be a file name (it cannot be a path): change the path "./ref/tcbn90ghplvt_ccs.ndm" to "tcbn90ghplvt_ccs". (LIB-154) Current PVT Configuration: rule_1: Process Labels : <All included> Process Numbers: <All included> Voltages : 0.9 Temperatures : <All included> Libraries : "tcbn90ghplvt_ccs"

### What Next

Specify a file name instead of path name for the library option.

---

## LIB-155

(error) Cannot merge reference library name '%s' from ref_lib list of library '%s' to ref_lib list of library '%s' because it may be overshadowed by reference library name '%s'.

### Description

This message indicates that merging the ref_lib from the ref_lib list of a sub-library to it's parent during uncommit_block could cause an issue, because a ref_lib of the parent would be used for a lib-cell instead of the sub-library. present in the sub-library would change positions relative to a ref_lib present in both libraries.

For example: lib1 has ref_libs {A C}, and sub_lib2 has ref_libs {B C}, and both B and C have the a lib-cell with the same name but different versions. Merging the ref_libs from

Feedback

sub_lib2 into lib1 would cause the ref_libs to be {A C B}, which may cause issues with ref_lib C overshadowing the element of ref_lib B

**What Next**

Try to manually merge the ref_libs in such a way that avoids one ref_lib overshadowing another for your project.

**See Also**

- uncommit_block

---

## LIB-156

(Error) parasitic tech library '%s' does not match any library on the reference library list.

**Description**

The parasitic tech library must be specified either as a simple path that matches the tail portion of an reference library entry or as a relative or absolute path that exactly match an entry on the reference library list.

**What Next**

Add a matching entry of the parasitic tech library on the reference library list.

**Examples**

create_lib -use_parasitic_tech_lib parasitic_tech_lib.nlib -ref_libs { parasitic_tech_lib.nlib }

create_lib -use_technology_lib ../nlibs/parasitic_tech_lib.nlib -ref_libs { ../nlibs/parasitic_tech_lib.nlib }

---

## LIB-157

(Error) parasitic tech library '%s' is invalid.

**Description**

A parasitic tech library must be a valid library and it must have a parasitic tech section.

**What Next**

Try again with a library with a valid parasitic tech section and that is accessible through the current search_path setting.

Feedback

## LIB-158

(error) %s library has dedicated parasitic tech library %s.

**Description**

Library with dedicated parasitic tech library can not create local parasitic. First remove dedicated parasitic tech library from reference library list.

**What Next**

Check "set_ref_libs" command to remove dedicated parasitic tech library from reference list

## LIB-159

(error) Library '%s' is referenced by another library as parasitic tech lib.

**Description**

A parasitic techs can not be replaced if library is, another library's parasitic tech lib.

**What Next**

Remove library from another library's reference list or close another library.

## LIB-160

(error) Cannot write library '%s' which has timing data.

**Description**

The write_physical_lib command can only write physical library. The given library has timing data so cannot be written.

**What Next**

Use Library Manager to write library with timing data.

## LIB-161

(warning) Open library in read only mode, only physical library can be opened in edit mode.

**Description**

This message indicates that the given library is not a physical only library, and the -read option was not specified to open it. Library Compiler does not support modify a library

Feedback

with logical information. So the library was opened in read only mode. You can query information from it. But please do not use write_physical_lib with it.

**What Next**

This is a warning message. No action needed.

---

## LIB-162

(error) Mask constraints check on library '%s' failed.

**Description**

This message indicates that mask constraints check on reported library failed. There is either uncolored shapes on multi-pattern layers, or colored shapes on uni-pattern layers. As the *lib.setting.check_mask_constraints* is set to *strict*, the current command will error out.

If the reported library is a library cell library, then the check is performed on it directly. If the reported library is a design library, then the check is performed on all the reference libraries of it.

**What Next**

Either rebuild the reported library, or set *lib.setting.check_mask_constraints* to *loose* or *none*.

---

## LIB-163

(error) Mixture use of library configuration and fusion library not supported.

**Description**

This message indicates that there is fusion library specified for library configuration. This is not allowed.

**What Next**

Either use library configuration for all reference libraries, or create fusion libraries for all reference libraries. The mixture use of them are not supported.

---

## LIB-164

(error) This library contains advanced technology not supported by this product. Please contact your Synopsys representative for more information.

Feedback

**Description**

This library contains advanced technology not supported by this product.

**What Next**

Please contact your Synopsys representative for more information.

---

## LIB-165

(error) Pane id %d is invalid for cell library '%s'.

**Description**

The pane id specified for the *-paneId* option is invalid. It is greater than the maximum pane id of the given cell library.

**What Next**

If the pane id was specified at the shell directly, please specify a valid pane id and try again. The maximum pane id can be got from the output of the *report_lib* command.

If *-paneId* option was not used for setting the lib data, then this error message must be reported during open a design library. It means the cell library had been changed since last time the lib data was set with the design library. There are fewer panes in the new cell library. And most likely, the change of the cell library, as the reference library of the design library, didn't through the *set_ref_libs* command. Please re-run the lib data setting with the design library to fix the error. After saving the design library again, it should have no this kind of error messages when open the design library next time.

---

# LIBCHK

---

## LIBCHK-001

(warning) Missing option %s when %s is specified.

**Description**

This message occurs when an option is specified but its associated option is missing. -optimization {power} requires -criteria {lvth_groups=lvt_name_list} to be associated with it. -single_mode requires set_lib_group -single_mode_lib to be associated with it. Since -all and -logic include -optimization {power}, when you specify -all or -logic, you also need to specify -criteria. -tolerance {type relative_tolerance absolute_tolerance} should be specified together. If only the type is specified without tolerance value, this message also occurs.

Feedback

If -analyze {table_trend} is specified without -criteria {trend=trend_char}, all trends will be checked and reported. If the associated option is not specified, the check as specified by the primary option will be ignored

**What Next**

Please specified the missing option.

**Examples**

```
Warning: Missing option -criteria {lvth_groups=lvt_name_list} when -all
 is specified. (LIBCHK-001)
```

---

## LIBCHK-002

(warning) Specified variable or command option '%s' ignored.

**Description**

This message occurs when an option or its value or variable is specified incorrectly in syntax or location in the following cases, and will be ignored.

1) for variable lc_check_lib_keep_line_number, when

```
set lc_check_lib_keep_line_number true
 is issued after read_lib or input library is a .db file.
```

2) for command options

```
a) -group_attribute or -criteria
  When the attribute or group specified in -group_attribute or -criteria
 is not found in the library.

b) -tolerance {...} -report_all
  When both options are specified, -tolerance will be ignored
```

c) -compare with other options such as -validate

```
When -compare and other options such as -validate are specified either in
 one command or separate, and there is only one library exists.
```

d) -criteria {VDD=(val1,val2,...)}, e.g. VDD=(0.7,0.8)

```
When specifying -analyze {pvt_interpolation} with interpolated voltages
 by -criteria, do not add a space between the values unless double quote
 is used. For example, both -criteria {VDD=(0.7,0.8)} and -criteria
 {"VDD=(0.7, 0.8)"} are correct.
```

e) create_scaling_lib_group -exclude_voltage_names railNames

```
When the rail names (voltage_name) specified do not exist in the library,
 for example, -exclude_voltage_names "VDDxx".
```

Feedback

f) -logical_library_name libNames -ctpm_file ctpmFileNames

```
When the check library option sensitivity_lib_vs_ctpm is false,
 -ctpm_file will be ignored.
```

**What Next**

If you set lc_check_lib_keep_line_number true, but still do not see line numbers reported, please make sure this variable is set before read_lib command and the library, if specified in -logic_library_name, is exactly the same as the File name with Path name when you see after issuing read_lib and list_lib or do not explicitly specify the library in -logic_library_name option in which case, the runtime library will be checked. If the group/attribute name specified in -group_attribute or -criteria option is not found in the library, it will be ignored. To compare libraries, more than one libraries should be specified. When you specify -report_all, do not specify -tolerance because report_all just use 0 tolerance internally. If you specify create_scaling_lib_group or set_lib_group command but no options are specified for multiple libraries, this lib group will be ignored.

In the case of unmatched voltage_names in above d) or e), please find an existing voltage_name from the library and update the option specification.

**Examples**

```
Warning: Specified variable or option 'lc_check_lib_keep_line_number'
 ignored. (LIBCHK-002)
Warning: Specified variable or option '-criteria {transitionxxx
 min_value=0.1 max_value=0.2} ignored. (LIBCHK-002)
```

---

## LIBCHK-003

(warning) No check done for logic library checking. Please specify valid options and libraries.

**Description**

This message occurs when any of the following occurs: 1) No logic library specified or available at runtime for logic related checks including default checks. 2) Less than 2 libraries specified or available at runtime for -compare option. 3) Invalid cell names specified in -cells option. 4) Invalid voltage_name in create_scaling_lib_group -exclude_voltage_names option. 5) -logic_vs_physical specified but -logic_library_name not specified or no logic library available at runtime.

**What Next**

Please specify the missing libraries or option based on above scenario.

**Examples**

```
Warning: No check done for logic library checking. Please specify valid
 options and libraries. (LIBCHK-003)
```

Feedback

## LIBCHK-004

(error) The specified logical libraries must be %s for check logic option %s.

### Description

This message occurs when specified logic libraries are incorrect types for the check option specified by set_check_library_options.

### What Next

Please check the types of specified libraries.

### Examples

```
Error: The specified logical libraries must be sensitivity libs for check
 logic option -sensitivity_lib_vs_ctpm. (LIBCHK-004)
```

## LIBCHK-005

(error) The number of %s and %s must be same for check logic option %s.

### Description

This message occurs when the number of items in two lists to be checked is different for the check option specified by set_check_library_options.

### What Next

Please check the number of specified libraries or files.

### Examples

```
Error: The number of specified sensitivity libraries and ctpm files must
 be same for check logic option -sensitivity_lib_vs_ctpm. (LIBCHK-005)
```

## LIBCHK-100

(warning) %s data found. Please refer to %s.

### Description

This message occurs when there are more serious issues that are found in the checking result listed in the table that follows. It redirects you to the message ID on next line that will give you detailed information.

### What Next

Please refer to the man page of the recommended message ID for details.

Feedback

### Examples

```
Warning: Mismatched data found. Please refer to LIBCHK-110. (LIBCHK-100)
```

## LIBCHK-101

(error) %s data found. Please refer to %s.

### Description

This message occurs when there are more serious issues are found in the checking result listed in the table that follows. It redirects you to the message ID on next line that will give you detailed information.

### What Next

Please refer to the man page of the recommended message ID for details.

### Examples

```
Error: Mismatches data found. Please refer to LIBCHK-212. (LIBCHK-101)
```

## LIBCHK-110

(information) List of cells with missing views

### Description

This message is for reporting cells with missing views in physical library. All the physical cells with CEL/FRAM view missing will be reported in this table. In this report table, there are three columns: Cell name, CEL and FRAM. In the table, the CEL and FRAM columns list the cell name:version number. An X marks the view that is missing.

### What Next

If FRAM view is missing, it is more critical that missing CEL view. You should use extract BPV command extract_blockage_pin_via to generate FRAM view.

### Examples

```
List of cells with missing views
------------------------------------------------------------------------
--------
Cell name                       CEL                     FRAM
------------------------------------------------------------------------
--------
dffs_1                          dffs_1:2                X
tiehi_1                         tiehi_1:2               X
------------------------------------------------------------------------
--------
```

Feedback

### Examples

```
Information: List of cells with missing views (LIBCHK-110)
```

---

## LIBCHK-111

(warning) List of cells with mismatched views

### Description

This message is for reporting cells with mismatched veiwewss in physical library. All the physical cells with CEL/FRAM view mismatched will be reported in this table. In this report table, there are five columns: Cell name, CEL Version, CEL Modified time, FRAM Version and FRAM Modified time. the CEL version and FRAM version columns list the version numbers, and the CEL and FRAM modified time columns list the time when the view was last modified. If a cell has an earlier FRAM view than its CEL view, it is marked as mismatched. The time checked is the internal view creation or modification time in the Milkyway database, not the UNIX time. No mismatch check is performed on the cell content.

### What Next

If a cell has an earlier FRAM view than its CEL view, please chec is its FRAM view is not updated. You may need to regenerate FRAM view by extracting BPV. If you read FRAM views from the LEF and then stream in CEL views, the views are shown as mismatched. You can ignore this report in this case.

### Examples

```
List of cells with mismatched views
------------------------------------------------------------------------
--------
                              CEL          CEL           FRAM         FRAM
Cell name                  Version   Modified time     Version   Modified
 time
------------------------------------------------------------------------
--------
FILL8                        2    Oct 24 19:17:9 2000    1    Oct 24
 19:14:33 2000
FILL4                        2    Oct 24 19:17:9 2000    1    Oct 24
 19:14:33 2000
ANTENNA                      2    Oct 24 19:17:23 2000   1    Oct 24
 19:14:43 2000
------------------------------------------------------------------------
--------
```

### Examples

```
Warning: List of cells with mismatched views (LIBCHK-111)
```

Feedback

## LIBCHK-112

(information) List of cells with same names

### Description

This message is for reporting cells with same names among the specified main or design library and all physical reference libraries linked to the specified library. All the physical cells with same name will be reported in this table. In this report table, there are two columns: Cell name and Library list in precedence order. If there are cells with the same name in multiple reference libraries, the tool uses the first cell in the reference control file and ignores the remaining cells. This option does not check naming among the logic libraries.

### What Next

This is information only. If there are cells with the same name in multiple reference libraries, the tool uses the first cell in the reference control file and ignores the remaining cells. This option does not check naming among the logic libraries.

### Examples

```
List of cells with same names
-------------------------------------------------------------
   Cell Name                 Library list in precedence order
-------------------------------------------------------------
   XOR3                      mainlib ref ref1 ref3 ref5
   DFF                       ref ref1 ref3
-------------------------------------------------------------
```

### Examples

```
Information: List of cells with same names (LIBCHK-112)
```

## LIBCHK-113

(information) List of signal EM data

### Description

This message is for reporting signal EM data for routing layers in physical library. All EM data will be reported in this table. In this report table, there are three columns: Layer name, Current model and Model type. If no signal EM data at all in the library, it will not print the table but print a message: The library is missing signal EM data.

### What Next

If the signal EM data is not complete, for example, it exists only in some layers but not all layers, you should check and reload the missing data.

Feedback

### Examples

```
List of signal EM data (LIBCHK-113)
-------------------------------------------------------------------
     Layer name      Current model       Model type
-------------------------------------------------------------------
     METAL1              peak                table
     METAL1              rms                 table
     METAL1              average             table
-------------------------------------------------------------------
```

### Examples

```
Information: List of signal EM data (LIBCHK-113)
```

---

## LIBCHK-114

(warning) The library is missing signal EM data.

### Description

This message occurs when there is no signal EM data for routing layers at all in the physical library.

### What Next

For flows using signal EM rules for EM analysis with different temperature corners, signal EM data is required. You should check and reload the data into the physical library.

### Examples

```
Warning: The library is missing signal EM data. (LIBCHK-114)
```

---

## LIBCHK-115

(information) List of antenna rules

### Description

This message is for reporting antenna rules in the layers in the specified main library. In this report table, there are six columns: Layer name, Mode, Diode mode, Default metel ratio, Default cut ratio and Max ratio.

### What Next

This is information only.

### Examples

```
List of antenna rules
-------------------------------------------------------------------------
```

Feedback

```
                                        Default     Default
  Layer Name    Mode   Diode mode   metal ratio  cut ratio   Max ratio
-------------------------------------------------------------------------
  M1             1         3           500.00      20.00       500.00
  M2             1         3           500.00      20.00       500.00
-------------------------------------------------------------------------
```

### Examples

```
Information: List of antenna rules (LIBCHK-115)
```

## LIBCHK-116

(warning) The library is missing antenna rules.

### Description

This message occurs when no antenna rules exist in the physical library.

### What Next

Please check and load the antenna rule file after FRAM view is created.

### Examples

```
Warning: The library is missing antenna rules. (LIBCHK-116)
```

## LIBCHK-117

(warning) List of cells missing antenna property

### Description

This message is for reporting missing antenna property for cells in physical library. All the cells that are missing antenna property will be reported in this table. In this table, it lists cell names and pin names missing the antenna property and the missing property type. If an input pin is missing the gate size, it is counted as missing the property. If an output pin is missing diode pro tection, it is counted as missing the property. If a macro is missing the hierarchical antenna property, then it is counted as missing the property. It is considered good practice to specify hierarchical antenna properties for macros.

### What Next

For antenna fix, please check and load the antenna properties (CLF) after FRAM view is created.

### Examples

```
List of cells missing antenna property
-------------------------------------------------------------------------
--------
```

Feedback

```
   Cell name             Cell type  Missing property     Pin name(s)
-------------------------------------------------------------------------
--------
   TLATNTSCAX1           StdCell    DiodeProt            ECK
   TLATNTSCAX4           StdCell    DiodeProt            ECK
   XOR2X4                StdCell    DiodeProt            Y
   XOR2XL                StdCell    DiodeProt            Y
-------------------------------------------------------------------------
--------
```

### Examples

```
Warning: List of cells missing antenna property (LIBCHK-117)
```

---

## LIBCHK-118

(information) List of cells with rectilinear boundaries

### Description

This message is for reporting cells with rectilinear boundaries in physical library. All cells with rectilinear boundaries will be reported in this table. In this report table, there are four columns: Cell name, Cell type, Number of points and Coordinate.

### What Next

This is information only. No further action is required.

### Examples

```
List of cells with rectilinear boundaries
-------------------------------------------------------------------------
  Cell Name   Cell type  Number of points                   Coordinate
-------------------------------------------------------------------------
  datapath    Macro            17    ( 78.750,   0.000) ( 78.750,
 490.760)
                                     ( 44.435, 490.760) ( 44.435,
 915.035)
                                     ( 27.440, 915.035)
 ( 27.440,1143.605)
                                     (  0.000,1143.605) (
  0.000,1313.200)
                                     (677.295,1313.200)
 (677.295,1143.605)
                                     (649.855,1143.605) (649.855,
 915.035)
                                     (632.860, 915.035) (632.860,
 490.760)
                                     (598.545, 490.760) (598.545,
 0.000)
                                     (  0.017,   0.000)
-------------------------------------------------------------------------
```

Feedback

### Examples

```
Information: List of cells with rectilinear boundaries (LIBCHK-118)
```

---

## LIBCHK-119

(information) List of physical only cells

### Description

This message is for reporting physical-only cells in the physical library. In Milkyway, physical-only cells are defined as FillerCell: Filler cell IOPadCell: Pad filler (IO Pad) cell CornerCell: Corner (pad) cell ChipCell: Chip cell FlipChipPadCell:Flip chip pad (bump) cell CoverCell: Cover cell TapCell: Tap cell PGPinOnlyCell: PG-pin-only cell (containing only PG ports) In this report table, there are three columns: Cell name, Cell type and Physical library.

### What Next

Please verify that the cell types of the physical only cells are marked correctly in the physical library.

### Examples

```
List of physical only cells
-------------------------------------------------------------------------
Cell name                  Cell type             Physical library
-------------------------------------------------------------------------
FILL8                      FillerCell            tsmc18_cg_LM
FILL64                     FillerCell            tsmc18_cg_LM
-------------------------------------------------------------------------
```

### Examples

```
Information: List of physical only cells (LIBCHK-119)
```

---

## LIBCHK-120

(information) List of main and reference libraries

### Description

This message is for reporting each libraries (main and reference physical libraries) and their full path locations, along with unit tile name and size. In this report table, there are four columns: Library name, Path, Unit tile and Tile size.

Feedback

### What Next

This is information only. No further action is required. Note: if tile patterns listed in List of tile patterns are not consistent with tiles in List of main and reference libraries, it is recommended that users redo library prep to use same tile. Please refer to example 2.

### Examples

```
Example 1:
Information:         List of main and reference libraries (LIBCHK-120)
------------------------------------------------------------------------
---------
Library name       Path                                        Unit tile
 Tile size
------------------------------------------------------------------------
---------
tsmc18_cg_LM       /remote/reg_designs/LIB_CENTER/180nm/san unit
 0.660x5.040
------------------------------------------------------------------------
---------

Example 2:
Information:         List of main and reference libraries(LIBCHK-120)
------------------------------------------------------------------------
---------
Library name       Path                                        Unit tile
 Tile size
------------------------------------------------------------------------
---------
cb18os120_unit010  /remote/dept5532b/styao/design              unit010
 0.560x5.600
------------------------------------------------------------------------
---------

                 List of tile patterns
------------------------------------------------------------------------
--------
 Tile pattern     Unit tile     Location     Orientation          Tile
 size
 1                   unit         (0,0)    R0|R0_MX
 0.560x0.560
 2                   unit         (-0.28,0)R0|R0_MX
 0.560x0.560
------------------------------------------------------------------------
--------
```

### Examples

```
Information: List of main and reference libraries (LIBCHK-120)
```

Feedback

## LIBCHK-121

(information) List of placement properties

### Description

This message is for reporting placement properties for cells in physical library. In this report table, there are seven columns: Cell name, PR boundary, Cell Height, Symmetry, Orientation, Tile pattern and Remarks. PR boundary is represented by its lower left and upper right coordinates. Cell height is relative to the unit tile height, such as 1xH for single height. Cell symmetry includes: rxy - RXYsymmetry xy - XYsymmetry y - Ysymmetry x - Xsymmetry r - Rsymmetry a - Asymmetry Cell orientation includes: R0_MX R0_MY R180 R90_MY R90 R270 R90_MX Number in Tile pattern denotes the serial number in the associated List of tile patterns table. Remarks notes that the PR boundary mismatches the tile pattern. For macros, this table reports cell boundary and height. The unit tile size is reported in the format of width x height.

### What Next

For standard cells, if mismatches are found between PR boundary and the tile pattern (Mismatch in Remarks column), please use the cmSetMultiHeightProperty command to set tile patterns for multi-height cells.

### Examples

```
List of tile patterns
--------------------------------------------------------------------------------
--------
 Tile pattern     Unit tile     Location     Orientation          Tile
 size
 1                   unit         (0,0)      R0|R0_MX
 0.660x5.040
--------------------------------------------------------------------------------
--------

         List of placement properties
 ------------------------------------------------------------------------------
--------------------
 Cell name      PR boundary       Height  Symmetry  Orientation
 Tile pattern  Remarks
 ------------------------------------------------------------------------------
--------------------
 TLATNTSCAX1    (0,0)(9.24,5.04)     1xH  xy        R0|R0_MX          1
 NOR2X1         (0,0)(1.98,5.04)     1xH  xy        R90_MY            1
                       Mismatch
--------------------------------------------------------------------------------
--------------------
```

### Examples

```
Information: List of placement properties (LIBCHK-121)
```

Feedback

## LIBCHK-122

(information) List of routing properties

### Description

This message is for reporting routing properties for routing layers in physical library. All routing properties for layers will be reported in this table. In this report table, there are six columns: Layer, Preferred direction, Track direction, Offset, Pitch and Remarks. Remarks column denotes if the routing property on this layer has issues. If no issues are found, you will see OK in this Remarks; otherwise, it will note what issue is found, for example, if the offset is 0.46*pitch, which is neither 0 nor half pitch, offset=0.46pitch will be noted.

### What Next

If remarks other than OK are noted, such as offset=0.46pitch, please check the issue and correct it. In the case of offset!=0.5*pitch, please reset the offset to be either 0 or half pitch. If blanks are left in Preferred direction, that denotes the preferred routing direction has not been set. Please define the preferred routing direction. If less than three routing layers are listed in this table, it is an error. Please check technology file and if some routing layers are missing, define them in the technology file and reload it.

### Examples

```
List of routing properties
-------------------------------------------------------------------------
              Preferred      Track
   Layer      direction    direction   Offset    Pitch   Remarks
-------------------------------------------------------------------------
   METAL1         H            H       0.280     0.560   OK
   METAL2         V            V       0.330     0.660   OK
   METAL3                      H       0.280     0.560   OK
   METAL4                      V       0.330     0.660   OK
   METAL5                      H       0.280     0.610   offset=0.46pitch
   METAL6                      V       0.380     0.950   offset=0.40pitch
-------------------------------------------------------------------------
```

### Examples

```
Information:  List of routing properties (LIBCHK-122)
```

## LIBCHK-123

(information) List of cell and pin properties

### Description

This message is for reporting cell and pin properties for cells in physical library. In this report table, there are six columns: Cell name, Cell type, Pin name, Pin direction, Pin type

Feedback

and Multi PG. If a cell has either multiple power pins or multiple ground pins, multiPG will be noted in the last column. At the end of the table, it also reports the number of cells with multiple P/G pins.

### What Next

If there is not any power or ground pins for a cell, it is an error. Please check the cell and may need to rerun extract BPV.

### Examples

```
List of cell and pin properties
------------------------------------------------------------------------
------------
   Cell name            Cell type     Pin name      Pin direction  Pin
 type  Multi PG
------------------------------------------------------------------------
------------
   XOR2XL               StdCell       Y             Output         signal
                                      VSS                          ground
                                      B             Input          signal
                                      A             Input          signal
                                      VDD                          power
   NAND4BBX1            StdCell       BN            Input          signal
                                      Y             Output         signal
                                      VSS                          ground
                                      C             Input          signal
                                      AN            Input          signal
                                      D             Input          signal
                                      VDD                          power
------------------------------------------------------------------------
------------
```

### Examples

```
Information:  List of cell and pin properties (LIBCHK-123)
```

---

## LIBCHK-124

(warning) List of pins without optimal routeability

### Description

This message is for reporting pins without optimal routeability in physical library. All pins without optimal routeability will be reported in this table. Optimal routeability means pin's on-track accessibility of defined wire tracks. It also reports the total number of pins without optimal on-track routability and lists pin names, directions, layers, and tracks for each cell of this issue. In the track column, an H denotes that the pin is not accessible on a horizontal track, a V denotes the pin is not accessible on a vertical track, and H&V denotes that the pin is not accessible on both horizontal and vertical tracks.

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### What Next

If a pin is reported as not accessible on track, the pin might be routed by off-track wire during detailed routing. If too many pins do not have any on-track routability, adjust the offset values of the wire track (0 or half pitch preferred) and run the the create_lib_track command to reduce the number of pins without optimal accessibility. If routability checking is not successful, the cause may be no unit tile.

### Examples

```
List of pins without optimal routeability
-------------------------------------------------------------
   Cell Name          Pin Name  Layer     Direction   Track
-------------------------------------------------------------
   TLATNSRX2          D         METAL1    Input       H&V
   SDFFSRHQX2         SE        METAL1    Input       H&V
   NOR4BBXL           C         METAL1    Input       H&V
   DFFRHQX1           D         METAL1    Input       H&V
   AND4X2             C         METAL1    Input       H&V
-------------------------------------------------------------
```

### Examples

```
Warning: List of pins without optimal routeability (LIBCHK-124)
```

---

## LIBCHK-125

(warning) List of cells with DRC violation

### Description

This message is for reporting cells with DRC violation in physical libraries. All cells with DRC violation will be reported in this table. It checks design rules such as wire minWidth, minEdgeLength, minEnclosedArea, minSpacing, cutWidth, cutSpacing, and minEnclosure. In this report table, there are three columns: Cell Name, Cell type and Error cell. Error cell contains detailed information for DRC violation of the FRAM view cell.

### What Next

Check the error cells for details of DRC violations. If a cell has DRC violation, it may not be used in the design without correction.

### Examples

```
List of cells with DRC violation
-------------------------------------------------------------
     Cell name                    Cell type      Error cell
-------------------------------------------------------------
     ONEPIN                       Standard       ONEPIN.err
-------------------------------------------------------------
```

Feedback

### Examples

```
Warning: List of cells with DRC violation (LIBCHK-125)
```

## LIBCHK-126

(warning) List of cells with metal density error

### Description

This message is for reporting cells with metal density error in physical library. All cells with metal density error will be reported in this table. It checks macro metal density data for the cells, i.e. it checks if a cell has at least one of the following errors: (1) Metal density data in CEL view and FRAM view are inconsistent (2) Metal density windows do not cover the entire area on the layer In this report table, there are three columns: Cell Name, Inconsistency and the layer on which the density windows do not cover the entire area. Y denotes Yes or Inconsistency exists and N denotes No consistency.

### What Next

If the density windows do not cover the entire area, you may need to add metal fill or smooth metal wire density. If metal density data in CEL view and FRAM view are inconsistent, please check FRAM vs. CEL view.

### Examples

```
List of cells with metal density error
------------------------------------------------------------------------
   Cell name          Inconsistency       Not cover entire area
------------------------------------------------------------------------
   ADDFX1MTR          Y                   M4
   OA21X1MTR          N                   M1
                                          M2
------------------------------------------------------------------------
```

### Examples

```
Warning: List of cells with metal density error (LIBCHK-126)
```

## LIBCHK-130

(information) Physical library checks FAILED.

### Description

This message is a concluding statement after physical checks are done. When any issue is found, for example, no physical libary to check, or no cells need to check, this message will occur. Otherwise, it reports Physical library checks PASSED.

Feedback

### Examples

```
Information: Physical library checks FAILED. (LIBCHK-130)
```

### What Next

You are advised to go back to the check report to see inconsistent results and resolve them as directed by the associated LIBCHK and FRAM man pages.

If you see the below message IDs in the check report, go to the related messages for details in

```
LIBCHK-112
LIBCHK-114
LIBCHK-116
LIBCHK-125
FRAM-010
FRAM-014
FRAM-015
FRAM-018
FRAM-019
FRAM-020
FRAM-022
FRAM-027
FRAM-028
FRAM-029
FRAM-030
FRAM-031
FRAM-033
FRAM-034
FRAM-039
FRAM-055
FRAM-056
FRAM-058
FRAM-062
FRAM-064
FRAM-070
FRAM-075
FRAM-078
FRAM-079
FRAM-080
FRAM-081
```

Any of the related LIBCHK FRAM messages will lead to the conclusion of some issues found during the checking.

### See Also

- LIBCHK-112
- LIBCHK-114

Feedback


- LIBCHK-116
- LIBCHK-125
- FRAM-010
- FRAM-014
- FRAM-015
- FRAM-018
- FRAM-019
- FRAM-020
- FRAM-022
- FRAM-027
- FRAM-028
- FRAM-029
- FRAM-030
- FRAM-031
- FRAM-033
- FRAM-034
- FRAM-039
- FRAM-055
- FRAM-056
- FRAM-058
- FRAM-062
- FRAM-064
- FRAM-070
- FRAM-075
- FRAM-078
- FRAM-079

Feedback

- FRAM-080
- FRAM-081

## LIBCHK-140

(warning) List of inconsistent ICC2 and %s attributes in %s class

### Description

This table lists ICC2 and MW/LEF/GDS attributes with inconsistent values in lib_cell/ mw_cel class. In this table, there are five columns: Library, Cell name, Attribute name, ICC2 value and MW/LEF/GDS value.

### Examples

```
Warning: List of inconsistent ICC2 and MW attributes in lib_cell class
 (LIBCHK-140)
-------------------------------------------------------------------------
----------------------
Library                    Cell name           Attribute name      ICC2
 value      MW value
-------------------------------------------------------------------------
----------------------
testlib                    AND2X1_LVT          allowable_orientation
                                                                    {R0
 R180 MX MY} {N FN FS}
testlib                    AND2X1_LVT          area
 2.03315   2.04111
-------------------------------------------------------------------------
----------------------
```

## LIBCHK-141

(warning) List of inconsistent ICC2 and %s attributes in %s class

### Description

This table lists ICC2 and MW/LEF/GDS attributes with inconsistent values in port/lib_pin class. In this table, there are five columns: Cell name, Object name, Attribute name, ICC2 value and MW/LEF/GDS value.

### Examples

```
Warning: List of inconsistent ICC2 and MW attributes in port class
 (LIBCHK-141)
-------------------------------------------------------------------------
----------------------
Cell name                    Object name         Attribute name     ICC2
 value      MW value
```

Feedback

```
------------------------------------------------------------------------
----------------------
XOR3X2_LVT                    A1                    direction           input
  inout
XOR3X2_LVT                    A1                    is_diode            false
  true
------------------------------------------------------------------------
----------------------
```

## LIBCHK-142

(warning) List of inconsistent ICC2 and %s attributes in %s class

### Description

This table lists ICC2 and MW attributes with inconsistent values in terminal class. In this table, there are five columns: Cell name, Object name, Attribute name, ICC2 value and MW value. users can find the MW value from import_icc_fram_lib.

### Examples

```
Warning: List of inconsistent ICC2 and MW attributes in terminal class
 (LIBCHK-142)
------------------------------------------------------------------------
----------------------
Cell name                     Object name           Attribute name      ICC2
 value      MW value
------------------------------------------------------------------------
----------------------
XOR3X2_LVT                    A1                    access_direction    all
XOR3X2_LVT                    A2                    is_fixed
 false   true
------------------------------------------------------------------------
----------------------
```

## LIBCHK-143

(warning) List of inconsistent ICC2 and %s attributes in %s class

### Description

This table lists ICC2 and MW attributes with inconsistent values in shape class. In this table, there are five columns: Cell name, Object name, Attribute name, ICC2 value and MW value. Users can find the MW value from import_icc_fram_lib.

### Examples

```
Warning: List of inconsistent ICC2 and MW attributes in shape class
 (LIBCHK-142)
------------------------------------------------------------------------
--------------------------
```

Feedback

```
Cell name                    Object name         Attribute name      ICC2
 value     MW value
-------------------------------------------------------------------------
---------------------------
HSBULT08_AOI21_MMY2_4        RECT_159_4          owner               B
          B_1
HSBULT08_AOI21_MMY2_4        RECT_159_5          owner               B_1
          B_2
HSBULT08_AOI21_MMY2_4        RECT_2030_6 / RECT_2030_0
                                                 mask_constraint
 mask_two       no_mask
HSBULT08_AOI21_MMY2_4        RECT_2030_7 / RECT_2030_1
                                                 mask_constraint
 mask_two       no_mask
-------------------------------------------------------------------------
----------------------
```

## LIBCHK-144

(warning) List of inconsistent ICC2 and %s attributes in %s class

### Description

This table lists ICC2 and MW attributes with inconsistent values in routing_blockage class. In this table, there are five columns: Cell name, Object name, Attribute name, ICC2 value and MW value. Users can find the MW value from import_icc_fram_lib.

### Examples

```
Warning: List of inconsistent ICC2 and MW attributes in routing_blockage
 class (LIBCHK-142)
-------------------------------------------------------------------------
---------------------------
Cell name                    Object name         Attribute name      ICC2
 value     MW value
-------------------------------------------------------------------------
---------------------------
HSBULT08_AOI21_MMY2_4        RB_1                mask_constraint
 mask_one       no_mask
HSBULT08_AOI21_MMY2_4        RB_7 / RB_2         mask_constraint
 mask_two       no_mask
HSBULT08_AOI21_MMY2_4        RB_8 / RB_3         mask_constraint
 mask_two       no_mask
-------------------------------------------------------------------------
----------------------
```

## LIBCHK-145

(warning) List of missing or mismatched placement rules!

Feedback

### Description

This table reports missing or mismatched placement rules in library level. The report table has the following columns:

```
placement_rules
<ndm lib 1>
<ndm lib 2>
```

### What Next

It is recommended that missing rules be added and mismatched models be corrected.

### Examples

The following is an example:

```
Warning: List of missing or mismatched placement rules. (LIBCHK-145)
lib1: realLib1
lib2: realLib2
-------------------------------------------------------------------------
-----------
placement_rules                        lib1                          lib2
-------------------------------------------------------------------------
-----------
max_horizontal_label_width             (W4, 0.6)
 missing
max_horizontal_label_width             (W2, 0.1)                     (W2,
 0.6)
-------------------------------------------------------------------------
-------
```

---

## LIBCHK-210

(information) List of cells missing in logic library

### Description

This message reports missing cells in logic library in logic vs. physical library checking. All the physical cells except physical only cells that are missing in logic library will be reported in this table. In this report table, there are three columns: Cell name, Cell type and Physical library. Cell types include lib_cell, macro, module, pad, corner, cover, flip_flop, analog, diode, filler, fill, etc.

### What Next

If some cells are found missing in specified logic library (.db), please check if you have not specified all the db files since logic cells may exist in different db files while the physical cells exist in one library.

Feedback

### Examples

```
List of cells missing in logic library (LIBCHK-210)
------------------------------------------------------------------------
Cell name                  Cell type             Physical library
------------------------------------------------------------------------
AND2                       lib_cell              phys_lib
NOR1                       lib_cell              phys_lib
------------------------------------------------------------------------

Number of cells missing in logic library:   3 out of 498
```

### Examples

```
Information: List of cells missing in logic library (LIBCHK-210)
```

---

## LIBCHK-211

(warning) List of cells missing in physical library

### Description

This message reports missing cells in physical library in logic vs. physical library checking. All the logic cells that are missing in physical library will be reported in this table. In this report table, there are three columns: Cell name, Cell type and Logic library. The cell types include pad_cell, filler_cell, tap_cell, decap_cell, pll_cell, level_shifter, isolation_cell, retention_cell, memory_cell, always_on, macro_cell and memory_cell, and std_cell. If no missing cells, no table will be printed out, but Number of cells missing in physical library: 0 (out of total) will be printed out.

### What Next

If some cells are found missing in the specified physical library, please check if you have complete physical libraries specified since physical cells may exist in different library. If the cells are used in the design, the physical cells have to exist in the physical library that you are using. If the cells are not expected to be part of the physical library, then the incoming netlist must not have any of the missing cells. For example, by setting the dont_use attribute during synthesis.

In fusion library, when you see

missing physical cells reported in LIBCHK-211

without LIBCHK-211 table, please run one of the following to get detailed report in the table format. 1) check_fusion_lib 2) set_app_options -name fusion_lib.check_fusion_lib_verbose -value true before create_fusion_lib 3) report_check_library_records autofix0.db3 -libchk_codes LIBCHK-211 where autofix0.db3 is located in fusion lib dir or current dir.

Feedback

### Examples

```
List of cells missing in physical library (LIBCHK-211)
------------------------------------------------------------------------
Cell name                  Cell type              Logic library
------------------------------------------------------------------------
gaard21                    std_cell            memory.lib
gaard22                    std_cell            memory.lib
------------------------------------------------------------------------

 Number of cells missing in physical library:   2 (out of 498)
```

### Examples

```
Warning: List of cells missing in physical library (LIBCHK-211)
```

---

## LIBCHK-212

(warning) List of signal and PG pins missing in %s library

### Description

This message reports missing signal pins and pg_pins in logical (or physical) cells in logic vs. physical library checking. All the logical (or physical) pins that are missing in physical (or logical) cells will be reported in this table. In this report table, there are four columns: Physical (or Logic) library, Cell name, Pin name and direction It also checks pg_pin's in logic library and PG pins in physical library, and reports them in the same table. Bias pins with physical_connection : device_layer defined in logic library, if missing in frame view, will not be reported in LIBCHK-212 as these are internal and not required in frame.

### What Next

If some signal pins excluding internal direction are found missing in logic cells, it is a real issue and you need to correct the inconsistency. If pg_pins are missing in the logic library, it indicates that the logic library is not pg_pin based. UPF flow requires pg_pin based library.

In fusion library, when you see

missing logical pins reported in LIBCHK-212

without LIBCHK-212 table, please run one of the following to get detailed report in the table format. 1) check_fusion_lib 2) set_app_options -name fusion_lib.check_fusion_lib_verbose -value true before create_fusion_lib 3) report_check_library_records autofix0.db3 -libchk_codes LIBCHK-212 where autofix0.db3 is located in fusion lib dir or current dir.

Feedback

### Examples

```
Warning: List of signal and PG pins missing in logic library (LIBCHK-212)
-------------------------------------------------------------------------
----
Physical library        Cell name       Pin name        direction
-------------------------------------------------------------------------
----
NLC13IOplLib_3v         pnl_go          A               in
-------------------------------------------------------------------------
----

Number of cells missing pins in logic library:   1
```

### Examples

```
Warning: List of signal and PG pins missing in logic library (LIBCHK-212)
```

---

## LIBCHK-212e

(error) List of signal and PG pins missing in %s library

### Description

This message reports missing signal pins and pg_pins in logical (or physical) cells in logic vs. physical library checking. All the logical (or physical) pins that are missing in physical (or logical) cells will be reported in this table. It also reports same name physical MUSTJOIN pins if defined in DB as such pins are not logical signal pins, which indicates the pins should be either removed from logic library or added in physical library as logical signal pins. In this report table, there are four columns: Physical(or Logic) library, Cell name, Pin name and direction It also checks pg_pin's in logic library and PG pins in physical library, and reports them in the same table.

### What Next

If some signal pins are found missing in logic cells, it is a real issue and you need to correct the inconsistency. If pg_pins are missing in the logic library, it indicates that the logic library is not pg_pin based. UPF flow requires pg_pin based library.

In fusion library, when you see

missing logical pins reported in LIBCHK-212e

without LIBCHK-212e table, please run one of the following to get detailed report in the table format. 1) check_fusion_lib 2) set_app_options -name fusion_lib.check_fusion_lib_verbose -value true before create_fusion_lib 3) report_check_library_records autofix0.db3 -libchk_codes LIBCHK-212e where autofix0.db3 is located in fusion lib dir or current dir.

Feedback

### Examples

```
Error: List of signal and PG pins missing in logic library (LIBCHK-212e)
-------------------------------------------------------------------------
----
Physical library        Cell name       Pin name        direction
-------------------------------------------------------------------------
----
NLC13IOplLib_3v         pnl_go          A               in
av16_lib                a16_fillprog_a2 A.1             in
-------------------------------------------------------------------------
----

Number of cells missing pins in logic library:   2
```

### Examples

```
Warning: List of signal and PG pins missing in logic library
 (LIBCHK-212e)
```

---

## LIBCHK-213

(error) List of (pg_)pins mismatched in logic and physical libraries

### Description

This message reports pins/pg_pins mismatched in either direction or (pg_)pin type between the logic and physical cells during logic vs. physical library checking. All the mismatched pins will be reported in this table. In this report table, at the header of this table, the logic and physical library full path file names are listed followed by the table contents with these columns: Cell name, Pin name, and Pin direction/types in logic and physical libraries. In Pin direction/type column, it lists values for both logic and physical libraries. Pin type can be port_type, pg_type, bias_type or bus_pin. In the pin type, bus_pin in .db denotes it is a bus or bundle, i.e. not bit blasted, and bus_port in .frame denotes it is a bus port.

For fusion lib, column pin_type_direction will show the pin direction or pg_type mismatch. If pg_types are mismatched these will be reported; otherwise mismatched directions will be reported. If only the PG pin direction is not defined in the logic library, the LIBCHK-213w warning message is reported instead of the LIBCHK-213 error.

The following list the pg_type mapping between logic and physical libraries:

```
-------------------------------------------------------------------------
--------------
        Logic(DB)                                   Physical
pg_type         sub type                       port_type   sub type
-------------------------------------------------------------------------
--------------
primary_power   std_cell_main_rail : true or
```

Feedback

```
                the only primary_power in cell  power       primary
primary_power   All others                      power       primary,
 is_secondary_pg
primary_ground  std_cell_main_rail : true or
                the only primary_ground in cell ground      primary
primary_ground  All others                      ground      primary,
 is_secondary_pg
backup_power                                    power       backup,
 is_secondary_pg
backup_ground                                   ground      backup,
 is_secondary_pg
internal_power                                  power       internal
internal_ground                                 ground      internal
nwell           physical_connection:routing_pin nwell       routing_pin,
 is_secondary_pg
deepnwell       physical_connection:routing_pin deep_nwell  routing_pin,
 is_secondary_pg
pwell           physical_connection:routing_pin pwell       routing_pin,
 is_secondary_pg
deeppwell       physical_connection:routing_pin deep_pwell  routing_pin,
 is_secondary_pg
nwell           physical_connection:device_layer nwell      device_layer
deepnwell       physical_connection:device_layer deep_nwell device_layer
pwell           physical_connection:device_layer pwell      device_layer
deeppwell       physical_connection:device_layer deep_pwell device_layer
------------------------------------------------------------------------
--------------
```

### What Next

If some pins are found mismatched in either pin direction or pin type between logic and physical cells, it is a real issue and you need to resolve the inconsistencies in the cells in either the logic or physical library or both. If the pin/port type in physical library is not defined, i.e. invalid type or inconsistent with DB, you may need to set_attribute on port_type or pg_type. bias_type is not settable by set_attribute. You can verify pin attribute value in .frame by get_attribute or report_attributes.

### Examples

```
List of pins mismatched in logic and physical libraries (LIBCHK-213)
------------------------------------------------------------------------
----
Cell name               Pin name        Pin direction/type
------------------------------------------------------------------------
----
ANTENNAHVT               I               signal(Logic#1)  diode(Physi#1)
------------------------------------------------------------------------
----

Number of cells with mismatched pins:   1
```

Feedback

### Examples

```
Error: List of (pg_)pins mismatched in logic and physical libraries
 (LIBCHK-213)
```

---

## LIBCHK-213w

(warning) List of (pg_)pins mismatched in logic and physical libraries

### Description

This message reports pins/pg_pins mismatched in either direction or pin type between the logic and physical cells during logic vs. physical library checking. All the mismatched pins will be reported in this table. In this report table, the logic and physical library names are listed followed by the table contents with these columns: Cell name, (pg_)pin name, and Pin direction/types in logic and physical libraries. In Pin direction/type columns, it lists values for both logic and physical libraries. Pin type can be port_type, pg_type, bias_type or bus_pin. In the pin type, bus_pin in .db denotes it is a bus or bundle, i.e. not bit blasted, and bus_port in .frame denotes it is a bus port.

The following list the pg_type mapping between logic and physical libraries:

```
------------------------------------------------------------------------
--------------
        Logic(DB)                                    Physical
pg_type         sub type                          port_type   sub type
------------------------------------------------------------------------
--------------
primary_power   std_cell_main_rail : true or
                the only primary_power in cell  power       primary
primary_power   All others                      power       primary,
 is_secondary_pg
primary_ground  std_cell_main_rail : true or
                the only primary_ground in cell ground      primary
primary_ground  All others                      ground      primary,
 is_secondary_pg
backup_power                                    power       backup,
 is_secondary_pg
backup_ground                                   ground      backup,
 is_secondary_pg
internal_power                                  power       internal
internal_ground                                 ground      internal
nwell           physical_connection:routing_pin nwell       routing_pin,
 is_secondary_pg
deepnwell       physical_connection:routing_pin deep_nwell  routing_pin,
 is_secondary_pg
pwell           physical_connection:routing_pin pwell       routing_pin,
 is_secondary_pg
deeppwell       physical_connection:routing_pin deep_pwell  routing_pin,
 is_secondary_pg
nwell           physical_connection:device_layer nwell      device_layer
```

Feedback

```
deepnwell       physical_connection:device_layer deep_nwell device_layer
pwell           physical_connection:device_layer pwell      device_layer
deeppwell       physical_connection:device_layer deep_pwell device_layer
------------------------------------------------------------------------
--------------
```

### What Next

If the directions of pg_pins are not defined in the logic library while defined in the physical library, this is a warning. In other cases, you need to resolve the inconsistencies in the cells in either the logic or physical library or both. If the pin/port type in physical library is not defined, i.e. invalid/unset type or inconsistent with DB, you may need to set_attribute on port_type, or pg_type. bias_type is not settable by set_attribute. You can verify pin attribute value in .frame by get_attribute or report_attributes.

### Examples

```
List of pins mismatched in logic and physical libraries (LIBCHK-213w)
------------------------------------------------------------------------
----
Cell name               (pg_)pin name        Pin direction/type
------------------------------------------------------------------------
----
ADDF_X1M_A9TR       VDD           unknown(Logic#1)  inout(Physi#1)
ADDF_X1M_A9TR       VSS           unknown(Logic#1) inout(Physi#1)
------------------------------------------------------------------------
----

Number of cells with mismatched pins:   1
```

### Examples

```
Warning: List of (pg_)pins mismatched in logic and physical libraries
 (LIBCHK-213w)
```

---

## LIBCHK-214

(information) List of bus naming styles

### Description

This message reports bus delimiters in logic and physical libraries. There are three fields in this report table: Library name, Library type and Bus naming style. Library type is either physical library or logic library. If there is no bus delimiter in the library, the Bus naming style field will have "Undefined" in the report. To report bus naming styles, please set -bus_delimiter to set_check_library_options command. it is often true that logic libraries do not specify the bus delimiter.

Feedback

### What Next

If inconsistent between logic and physical libraries, users may need to set it to a consistent one in the physical library by the app option design.bus_delimiters.

### Examples

```
List of bus naming styles (LIBCHK-214)
-------------------------------------------------------------------
Library name                 Library type        Bus naming style
-------------------------------------------------------------------
memory.lib                   Logic library       Undefined
tsmc18_xx                    Physical library    []
-------------------------------------------------------------------
```

### Examples

```
Information: List of bus naming styles (LIBCHK-214)
```

---

## LIBCHK-215

(warning) List of cells with cell_footprint attribute

### Description

This message is to check that the cell PR boundary or Cell boundary for Macro in the physical library is consistent among a class of cells with the same cell_footprint attribute specified in the logic library. This option reports cell names and their PR boundaries grouped by cell_footprint string. If PR boundaries for the same cell_footprint are different, the last field (Compatibility) marks it as Incompatible. To check cell_footprint, please set -cell_footprint to set_check_library_options command.

### What Next

If the cells are used in the design, please check their PR boundaries or cell boundaries for inconsistency. For example, if the PR boundaries of the cells with the same cell_footprint are different, you can not use them during optimization by swapping these cells.

### Examples

```
List of cells with cell_footprint attribute (LIBCHK-215)
-------------------------------------------------------------------------
---
Logic library name  cell_footprint Cell name  PR boundary
 Compatibility
-------------------------------------------------------------------------
---
tcbn90ghvtbc        TIEHHVT        TIEHHVT    {0 0} {0.64 2.52}
tcbn90ghvtbc        TIELHVT        TIELHVT    {0 0} {0.64 2.52}
tcbn90ghvtbc        an2d1          AN2HVTD0   {0 0} {1.6 2.52}
 Incompatible
```

Feedback

```
tcbn90ghvtbc        an2d1          AN2HVTD1    {0 0} {1.6 2.52}
 Incompatible
tcbn90ghvtbc        an2d1          AN2HVTD2    {0 0} {1.92 2.52}
 Incompatible
------------------------------------------------------------------------
---

Number of incompatible cell_footprints in the libraries: 1 out of 3
```

### Examples

```
Warning: List of cells with cell_footprint attribute (LIBCHK-215)
```

---

## LIBCHK-216

(warning) List of cells with inconsistent area

### Description

This message is to check the area attribute of cells in the logic library versus the actual area by cell PR boundary in the physical library. For comparison, the ratio of a stdCell PRBoundary in the rectangle (or Macro CellBoundary in the polygon) in the FRAM view to the cell area in the .db is used. If the ratio is the same for all the cells in the library, the cell areas are considered to be consistent. Otherwise, if the ratio for a cell deviates from the normal or average ratio for this library by a margin of 5%, it is counted as an inconsistent area. and will be reported in the table. The average area ratio is calculated by dividing the total cell area ratios by the number of cells. There is no area check for pad cells. A pad cell is a special cell at the chip boundary with the pad_cell attribute in a logic cell. Pad cells are not checked because they are not used as internal gates and always have an area of 0. To check cell area, please set -cell_area to set_check_library_options command.

### What Next

If the cell areas are inconsistent between logic and physical libraries, it will affect timing and placement accurary. The bigger the area deviation, the worse the accuracy and correlation.

### Examples

```
Average cell area ratio physical/logic in the library:  0.9978
Number of cells with inconsistent area: 9

     List of cells with inconsistent area (LIBCHK-216)
------------------------------------------------------------------------
Logic library name           Cell name             Area ratio phys/logic
------------------------------------------------------------------------
slow                         AFCSHCONX4            0.7966
slow                         AFCSHCONX2            0.8431
slow                         AFHCONX2              1.1167
slow                         AFHCINX4              0
```

Feedback

```
------------------------------------------------------------------------

Number of cells with inconsistent area: 4
Average cell area ratio physical/logic in the library:  0.9978
```

### Examples

```
Warning: List of cells with inconsistent area (LIBCHK-216)
```

---

## LIBCHK-217

(warning) List of cells mismatched in logic and physical libraries

### Description

This message is for reporting cells mismatched in antenna diode type between the logic and physical cells during logic vs. physical library checking. All the mismatched pins will be reported in this table. In this report table, at the header of this table, the logic and physical library names are listed followed by the table contents with two columns: Cell name, Antenna Diode type. In Antenna Diode type column, it lists values for both logic and physical libraries.

The mapping table between .db and FRAM view is as follows.

Type name in Logic Type name in Physical power_and_ground AntennaDiodeTypePowerAndGround power AntennaDiodeTypePower ground AntennaDiodeTypeGround

### What Next

If some cells are found mismatched in antenna diode type between logic and physical cells, command "update_mw_port_by_db -antenna_diode_type" may correct it.

### Examples

```
List of cells mismatched in logic and physical libraries
Logic library:    saed90nm_max
Physical library: ./design
------------------------------------------------------------------------
----
                          Antenna Diode Type
Cell name              Logic               Physical
------------------------------------------------------------------------
----
cell1                  power_and_ground    AntennaDiodeTypePower
cell2                  ground              missing
------------------------------------------------------------------------
----
```

Feedback

### Examples

```
Warning: List of cells mismatched in logic and physical libraries
 (LIBCHK-217)
```

---

## LIBCHK-220

(information) Logic library is inconsistent with physical library

### Description

This message is a concluding statement after logic vs. physical cross checking is done. When any inconsistency, for example, missing cells and mismatched pins, is found during cross checking, this message will occur. Otherwise, it reports Logic vs. physical library consistency checks PASSED.

### What Next

This is for your information. You need to go back to the check report to see inconsistent results and resolve them as directed by related man pages (LIBCHK).

### Examples

```
Information: Logic library is inconsistent with physical library
 (LIBCHK-220)
```

---

## LIBCHK-300

(warning) List of inconsistent library group data

### Description

This message is for reporting inconsistent data in logic libraries during logic vs. logic library checking. In the associated report table, it reports inconsistent values of the attributes at the library level. The attributes include. a) library version b) library type (pg_pin based or non-pg_pin based) c) units of timing, power,capacitance, etc. d) trip points e) base curve_x

For single mode etm checking, inconsistencies and/or incorrect attribute values in the following are reported in this table: leading to Error (LIBCHK-366): nom_voltage, nom_temp, nom_process time_unit, time_unit_name, time_scale current_unit capacitive_load_units slew_lower_threshold_pct_rise slew_lower_threshold_pct_fall slew_upper_threshold_pct_rise slew_upper_threshold_pct_fall input_threshold_pct_rise input_threshold_pct_fall output_threshold_pct_rise output_threshold_pct_fall slew_derate_from_library leading to Error: k_process_cell_rise k_process_cell_fall k_process_rise_transition k_process_fall_transition

Feedback

### What Next

If some inconsistencies are reported at the library level, please check the libraries and correct them. Some attributes should be consistent cross the libraries. If the libraries are used for timing scaing, trip points should be consistent, and if they are used for UPF or MV flow, the libraries should be pg_pin based. For single mode etm checking, user needs to correct the issues leading to Errors.

### Examples

```
List of inconsistent library group data
-------------------------------------------------------------------------
-----
Library name              SR60_W_SVT_UHD.db      SR60_S_SVT_UHD.db
-------------------------------------------------------------------------
-----
Voltage unit              1V                     1mV
base_curves/curve_x       "0.2, 0.5, 0.8"        "0.2, 0.5, 0.9"
-------------------------------------------------------------------------
-----
```

### Examples

```
Warning: List of inconsistent library group data (LIBCHK-300)
```

---

## LIBCHK-301

(warning) List of incompliant operating conditions

### Description

This message is to report incompliant operating conditions in the library group. In this report table, nom_voltage(V), nom_temperature(T), nom_process(P), process_label(PL) if defined, voltage_map's and default_operating_conditions of each library are reported.

1) Single rail libraries: at least one of process (P), voltage (V) and temperature (T) should be different.

2) Multi rail libraries: at least one of process (P), voltage (V), temperature (T) and voltage in voltage_map should be different. At least one of nominal PVTs are different, or if nominal PVTs are identical, voltage_map should be different among the lib group. The number of rails counts all the unique power voltage_name's including input_signal_level that are in use in the cells.

process_label, if defined, is also taken as one of the P,V,T and PL. Libraries with the same PVT and voltage_map are counted as duplicate corners.

If default_operating_condition is not specified, Library Compiler will add a default name nom_pvt by creating one from nominal PVTs.

Feedback

### What Next

For library scaling, the operating conditions in the same scaling library group should be different. For MCMM and MV/UPF, the nominal voltages should be different under different operating conditions. For -compare, the operating_consitions should be identical. For all others, this is information only. No further action is required. If there are duplicate corners, please remove duplications for scaling or MCMM. For voltage scaling, it is recommended that the voltages should be no more than 20% apart for good accuracy

### Examples

```
Warning: List of incompliant operating conditions (LIBCHK-301)
-------------------------------------------------------------------------
Library name        lib#1              lib#2              lib#3
-------------------------------------------------------------------------
PL/P/V/T            SH/0.825/125       SS/0.675/125       SH/0.825/125
voltage_map         (VBN,0)            (VBN,0)            (VBN,0)
voltage_map         (VBP,0.825)        (VBP,0.675)        (VBP,0.825)
voltage_map         (VDD,0.825)        (VDD,0.675)        (VDD,0.825)
voltage_map         (VSS,0)            (VSS,0)            (VSS,0)
default_opcon       FFGNP0P825V125C    SSGNP0P675V125C    FFG0P825V125C
-------------------------------------------------------------------------
```

### Examples

```
Warning: List of incompliant operating conditions (LIBCHK-301)
```

---

## LIBCHK-302

(information) Voltages for library scaling are more than 20%% apart.

### Description

This message is for voltage scaling, the voltages should be no more than 20% apart for good accuracy.

### What Next

This is information only. However, for better accuracy, it is recommended that the voltages should be no more than 20% apart for voltage scaling.

### Examples

```
Information: Voltages for library scaling are more than 20% apart.
 (LIBCHK-302)
```

---

## LIBCHK-303

(warning) Missing libraries in scaling groups

Feedback

### Description

This table reports missing corner libraries in the scaling library group. It indicates the missing libraries by operating conditions (temperature and voltages on rails)

### What Next

User needs to add the missing libraries into the group to make a full use of all the libraries in the group for better scaling accuracy.

### Examples

```
Warning: Missing libraries in scaling groups  (LIBCHK-303)
-------------------------------------------------
Group name          Temp          VDD1  VDD2
-------------------------------------------------
grp1                -40.0          1.10  1.21
-------------------------------------------------
```

---

## LIBCHK-303i

(information) Table of library formation

### Description

This table shows library formation in terms of temperature(T) and rail voltages(V) values in the libraries in the scaling library group. In the table, library index will be shown if one exists at the grid (V/T) or dash(-) if nonexistent. The library indexes are from the library order in the input library list starting from 1. For example, if there are 10 libraries, the indexes will be 1, 2, ..., 10.

### What Next

The PVT values on each library can be referenced for scaling or PVT interpolation. User needs to add the missing libraries into the group to make a full use of all the libraries in the group for better scaling accuracy.

### Examples

```
Information: Table of library formation (LIBCHK-303i)
-------------------------------------------------------------------------
--------------------
 Temp\\VDD  0.500  0.550  0.600  0.675  0.720  0.750  0.800  0.825  0.880
  0.900  1.000  1.050
-------------------------------------------------------------------------
--------------------
      -40 lib#15    -   lib#26 lib#5  lib#12    -      -   lib#18 lib#31
 lib#21    -   lib#32
        0 lib#33    -   lib#13 lib#20 lib#14    -      -   lib#4  lib#1
 lib#9     -   lib#24
```

Feedback

```
      25    -   lib#38    -      -      -   lib#27 lib#10    -      -
   -   lib#34    -
      85    -   lib#25    -      -      -   lib#23 lib#16    -      -
   -   lib#37    -
     125 lib#19    -   lib#7  lib#6  lib#8     -      -   lib#17 lib#30
 lib#36    -   lib#2
     150 lib#11    -   lib#35 lib#22 lib#3     -      -   lib#29 lib#28
   -      -      -
-------------------------------------------------------------------------
---------------------
In the above example, if an interpolated voltage on VDD is 0.525 at
 Temperature -40,
the quadratic value reported in LIBCHK-358 at Temperature -40 will be
 from 3 libraries:
15, 26, 5, while the linear value will be from 2 libraries: 15, 26.
```

## LIBCHK-304

(information) Valid scaling library groups

### Description

This table reports valid scaling library groups that meet the library formations. In the case of a complete library list are not specified, it will list the subset of the libraries meeting the formation. In the table, it also reports temperature (T) and voltage (V) ranges and adjacent voltage difference percentage in the group for scaling, or individual T & V values for exact match.

### What Next

User can pick any library group from the list in the case of multiple groups are found, but it is recommended to pick the one with maximum number of libraries and maximum scaling dimensions.

### Examples

```
Information: Valid scaling library groups (LIBCHK-304)
-------------------------------------------------------------------------
----------------
Group name Library list      T range  V range  Max delta V
-------------------------------------------------------------------------
----------------
grp1     v1t1.db v2t1.db v1t2.db v2t2.db  -40 to 125 0.9 to 1.2 33.3%
grp2       lib21 lib12 lib13     -40, 125 0.9, 1.2
-------------------------------------------------------------------------
----------------
```

## LIBCHK-305

(error) Library group '%s' not ready for %s.

Feedback

### Description

The library group is not ready for scaling because of at least one of the following reasons: 1) Library data consistency checks by check_library not passed 2) Originally defined scaling library group is missing some corners or some corner libraries are removed to generate reduced subsets that meet the formation. 3) When a library is specified in multiple library groups by create_scaling_lib_group command, or a user defined lib group has multiple groups in terms of cells. The libraries at this issue can be seen in Library grouping table LIBCHK-308.

The library group is not ready for voltage interpolation if no enough libraries form on one dimension, i.e. one variable rail (e.g. VDD) to interpolate on with other dimensions (e.g. VDD2) fixed. For quadratic interpolation, it needs at least 3 libraries with different voltage values on on rail while others are fixed.

### What Next

Also refer to comments in the output script by write_scaling_lib_group, and fix library data consistency and/or add the missing corner libraries if not for -exact_match_only.

For PVT interpolation, please specify enough libraries with >=3 voltage values on a rail/voltage_name while other dimensions are fixed. To exclude rails not used for scaling/pvt_interpolation or dependent rails (most likely a bias pin), please use create_scaling_lib_group -exclude_voltage_names {VBP}, for example.

### Examples

```
Error:Library group 'grp1' not ready for scaling.
```

---

## LIBCHK-306

(warning) link_library %s '%s'.

### Description

This message occurs when 1) link_library has extra libraries 2) link_library is missing a library from groups In PrimeTime, it is required that 1) one-and-only-one library in SLG to be in link_library 2) one library cannot be in more than one DSLGs. In ICC/DC, it is recommended but not required.

### What Next

Remove the extra libraries from link_library or add a missing library from the group.

### Examples

```
Warning: link_library has extra library: 'p1v1.db' (LIBCHK-306)
```

Feedback

## LIBCHK-307

(information) List of miscellaneous inconsistent library data

### Description

This message is reporting different values in the same group/attributes that are not reported in any other tables when comparing 2 dbs. In this report table, Path name consists of hierarchical group/object names. If the attribute is at the library level, it is the library_name. This is mainly for internal use.

### What Next

If difference is reported for the attribute, please open the db files to view the values to see if it is expected.

### Examples

```
List of miscellaneous inconsistent library data
-------------------------------------------------------------------------
-----------------
Path name                      Object              Attribute
 Value(lib#1)   Value(lib#2)
-------------------------------------------------------------------------
-----------------
mycell/Q3/                     bag_contents(CP)    active        2
      1
-------------------------------------------------------------------------
-----------------
```

### Examples

```
Information: List of miscellaneous inconsistent library data (LIBCHK-307)
```

## LIBCHK-308

(information) Library grouping table

### Description

This message is for reporting list of libraries with group name and group number they belong to. When a library is specified in multiple library groups by create_scaling_lib_group command, or a user defined lib group has multiple groups in terms of cells, the group number will include sub-group number, e.g. 2.1.

### What Next

For library scaling, the operating conditions in the same scaling library group should be different in nominal voltage, temperature and/or process. For MCMM and MV/UPF, the nominal voltages should be different under different operating conditions. For -compare,

Feedback

the operating_consitions should be identical. For all others, this is information only. No further action is required. For voltage scaling, it is recommended that the voltages should be no more than 20% apart for good accuracy

### Examples

```
When user defines
set search_path "dbs"
create_scaling_lib_group -exact -name SVT
"SVT_ts28nmhsdst_ss0p81v0c.db \
 SVT_ts28nmhsdst_ss0p81v125c.db  \
 SVT_ts28nmhsdst_tt0p9v25c.db  \
 SVT_ts28nmhsdst_tt0p9v85c.db"
create_scaling_lib_group -exact -name LVT
"SVT_ts28nmhsdst_ss0p81v0c.db \
 LVT_ts28nmhldst_ff0p99vn40c.db \
 LVT_ts28nmhldst_tt0p9v85c.db"

check_library report will be

Error: set_lib_group or create_scaling_lib_group -name "LVT" has multiple
 lib groups (LIBCHK-305)

Information: Library grouping table (LIBCHK-308)
-------------------------------------------------------------------------
-------
Group name Number Library name           Library file name
-------------------------------------------------------------------------
-------
SVT         1    ts28nmhsdst_ss0p81v0c    dbs/SVT_
 ts28nmhsdst_ss0p81v0c.db
SVT         1    ts28nmhsdst_ss0p81v125c
 dbs/SVT_ts28nmhsdst_ss0p81v125c.db
SVT         1    ts28nmhsdst_tt0p9v25c
 dbs/SVT_ts28nmhsdst_tt0p9v25c.db
SVT         1    ts28nmhsdst_tt0p9v85c
 dbs/STD_SVT_ts28nmhsdst_tt0p9v85c.db
LVT         2.1  ts28nmhsdst_ss0p81v0c
 dbs/SVT_ts28nmhsdst_ss0p81v0c.db
LVT         2.2  ts28nmhldst_ff0p99vn40c
 dbs/LVT_ts28nmhldst_ff0p99vn40c.db
LVT         2.2  ts28nmhldst_tt0p9v85c
 dbs/LVT_ts28nmhldst_tt0p9v85c.db
-------------------------------------------------------------------------
-------
```

## LIBCHK-309

(warning) Library# %s are duplicate in %s.

Feedback

### Description

This message is to report duplicate libraries in the library group. The library# can be found in LIBCHK-308 table. Duplicate libraries are those that have identical process (P), voltage (V) and temperature (T), voltage_map's and process_label (PL) if defined.

### What Next

For library scaling, the operating conditions in the same scaling library group should be different. For MCMM and MV/UPF, the nominal voltages should be different under different operating conditions. For -compare, the operating_consitions should be identical. For all others, this is information only. No further action is required. If there are duplicate corners, please remove duplications for scaling or MCMM. Please refer to LIBCHK-301 for a list of PVTs at each library.

### Examples

```
Warning: Library# 1,3 are duplicate in 1/0.81/110 (LIBCHK-309)
```

### See Also

- LIBCHK-301

---

## LIBCHK-310

(information) List of cells missing in logic libraries

### Description

This message is for reporting cells missing in one or more logic libraries during logic vs. logic library checking. All the logic cells that are missing will be reported in this table.

### What Next

If cells are found missing in one or more of the libraries, please check why these are existing in onle library but missing in the other. If such cells are used in design, these cells have to be in all the libraries. However, for library scaling, missing cells in link_library compared to other libraries in the same group will not cause failure(LIBCHK-362).

### Examples

```
List of cells missing in logic libraries
------------------------------------------------------------------------
-----
Cell name           SR60_W_SVT_UHD.db           SR60_S_SVT_UHD.db
------------------------------------------------------------------------
-----
nsdffnqs_f4_svt     missing                     existing
------------------------------------------------------------------------
-----
```

Feedback

### Examples

```
Warning: List of cells missing in logic libraries (LIBCHK-310)
```

---

## LIBCHK-311

(warning) List of cells with mismatched or missing attributes in logic libraries

### Description

This message is for reporting mismatched cell data in logic libraries during logic vs. logic library checking. Mismatched cells are those that have inconsistent attribute values at the cell level. For characterization values (e.g. cell_leakage_power), there are calculated absolute and relative errors along with respective values in each library. They are listed in the columns on the right-hand side. For check_library -upf -mcmm, cell_leakage_power value = 0 is reported in this table. For single mode etm checking, inconsistencies and/or incorrect attribute values in the following are reported in this table: leading to Error (LIBCHK-366): timing_model_type is_macro_cell not leading to Error: area dont_use/touch mode_definition

Please note that the following attributes, if defined as false in .lib, will be reported as missing as these do not exist in compiled .db: dont_use, dont_touch, preferred and interface_timing. The above mentioned attributes when specified as false in the .lib will not be saved in db.

### What Next

Mismatched cells will cause issues if used in design. The same name cells existing in different libraries should have the same values for attributes such as function, dont_use and dont_touch, and same attributes should exist in all the libraries. For single mode etm checking, user needs to correct the issues leading to Errors.

### Examples

```
Warning: List of cells with mismatched or missing attributes in logic
 libraries (LIBCHK-311)
------------------------------------------------------------------------
---------
                                        Attribute value
 Error
Cell name           Attribute           lib#1       lib#2       Absolute
 Relative
------------------------------------------------------------------------
---------
TIEOFF1             function            "A * B"     "A * B + c"
std_cell_inv        pin_order           "A_Y_Z"     "A_Y"
mycell1             cell_leakage_power  56.356129   18.580099   -37.77603
 -67.03%
------------------------------------------------------------------------
---------
```

Feedback

### Examples

```
Warning: List of cells with mismatched or missing attributes in logic
 libraries (LIBCHK-311)
```

## LIBCHK-312

(information) List of cell classification

### Description

This message is for reporting cell classification of logic libraries. In this report table, numbers of total cells, inverters, buffers, level shifters, isolation cells, retention cells, switch cells, and always on cells in each library For library scaling, number of single and multi-rail cells in pg_pin based libraries is also reported. are reported.

### What Next

For UPF flow, special cells such as level shifters should exist in the library. in general, the library should have inverters. For library scaling, single and multi-rail cells should be in separate libraries. For all other cases, this is information only. No further action is required.

### Examples

```
List of cell classification
-------------------------------------------------------------------------
-----
Library name        SR60_W_SVT_UHD.db         SR60_S_SVT_UHD.db
-------------------------------------------------------------------------
-----
Total number               3                         3
Inverter                   2                         1
Buffer                     0                         0
Level shifter              0                         0
Differential level shifter 0                        0
Isolation cell             0                         0
Clock Isolation cell       1                         1
Retention cell             0                         0
Switch cell                0                         0
Always on cell             0                         0
-------------------------------------------------------------------------
-----
```

### Examples

```
Information: List of cell classification (LIBCHK-312)
```

## LIBCHK-313

(warning) List of function groups missing high threshold_voltage_group cells

Feedback

### Description

This message will occur when high threshold_voltage_group for the same function group. In this report table, the following columns are listed: function, Library name and Example cell name.

### What Next

If some cells are found missing high threshold_voltage_group in logic cells, please check if the specified target_library is correct or missing some libraries in target_library.

### Examples

```
Number of function groups missing high threshold_voltage_group:   14 (out
 of 148)

      List of function groups missing high threshold_voltage_group cells
------------------------------------------------------------------------
----------------
function                                Library name
 Example cell name
------------------------------------------------------------------------
----------------
A1 A2                                   tcbn45lpbwplvtwc
 AN2D0BWPLVT
(A1 A2) A3                              tcbn45lpbwplvtwc
 AN3D0BWPLVT
((A1 A2)+B)+C                           tcbn45lpbwplvtwc
 AO211D0BWPLVT
(A1 A2)+B                               tcbn45lpbwplvtwc
 AO21D0BWPLVT
((A1 A2)+(B1 B2))+C                     tcbn45lpbwplvtwc
 AO221D0BWPLVT
------------------------------------------------------------------------
----------------
```

### Examples

```
Warning: List of function groups missing high threshold_voltage_group
 cells (LIBCHK-313)
```

---

## LIBCHK-314

(information) List of mapping multi bit to single bit cells

### Description

This message occurs when checking multi bit mapping to single bit cells. It shows the mapping between multi bit cells and single bit cells. In this report table, the following columns are listed for each multibit function as a mapping between multi bit cells and single bit cells with library index: Multi bit cell(lib#) and Single bit cell(lib#). Blackbox

Feedback

multibit(MB) cells with single_bit_degenerate attribute will be reported in a separate table of the same LIBCHK type.

### What Next

If mapping single bit cells are found missing as reported in LIBCHK-314w, please check if the specified target_library or lc_single_bit_degenerate_cell_library is correct or missing some libraries in the library list.

### Examples

```
List of mapping multi bit to single bit cells (LIBCHK-314)
-------------------------------------------------------------------------
-------
Multi bit cell(lib#)                    Single bit cell(lib#)
-------------------------------------------------------------------------
-------
mb_4_scanin_gated_scanout1(lib#2)       sb_gated_scanout1(lib#2)
-------------------------------------------------------------------------
-------
```

### Examples

```
Information: List of mapping multi bit to single bit cells (LIBCHK-314)
```

---

## LIBCHK-314w

(warning) List of missing/mapping multi bit to single bit cells

### Description

This message will occur when checking multi bit mapping to single bit cells. When single bit cells are not found in the libraries, it will be reported as missing single bit cells. In this report table, the following columns are listed for each multibit function as a mapping between multi bit cells and single bit cells with library index: Multi bit cell(lib#) and Single bit cell(lib#). Blackbox multibit (MB) cells with single_bit_degenerate attribute will be reported in a separate table of the same LIBCHK type.

### What Next

If mapping single bit cells are found missing, please check if the specified target_library or lc_single_bit_degenerate_cell_library is correct or missing some libraries in the library list.

### Examples

```
List of missing/mapping multi bit to single bit cells (LIBCHK-314w)
-------------------------------------------------------------------------
-------
Multi bit cell(lib#)                    Single bit cell(lib#)
-------------------------------------------------------------------------
-------
```

Feedback

```
STL_FSDPQOMB_D_1(lib#1)                  missing
mb_4_scanin_gated_scanout1(lib#2)        sb_gated_scanout1(lib#2)
-------------------------------------------------------------------------
-------

Number of multibit cells missing single bit cells: 1 (out of 2)
```

### Examples

```
Warning: List of missing/mapping multi bit to single bit cells
 (LIBCHK-314w)
```

## LIBCHK-315

(warning) List of data models and check policy in cells

### Description

This message occurs when there are inconsistent early data models and/or policies in the cells. Similar to report_early_data_check_records that is reporting for single DB the early library data on the lib cells, check_library will report these data recorded in each DBs excluding error codes in the LIBCHK-315 table.

### What Next

Early data models tolerated or repaired in the library compilation should be used for early stage in the design, and should not be used for sign-off.

### Examples

```
Warning: List of data models and check policy in cells (LIBCHK-315)
-------------------------------------------------------------------
                                          Policy
Cell name   Data model        lib#1       lib#2
-------------------------------------------------------------------
BUF1      nldm           tolerate repair
ISO1      power_aware    repair  error
ISO1      ccsn           tolerate repair
-------------------------------------------------------------------
```

### Examples

```
Warning: List of data models and check policy in cells (LIBCHK-315)
```

## LIBCHK-320

(warning) List of pins missing in logic libraries

Feedback

### Description

This message is for reporting missing signal pins in logic cells during logic vs. logic library checking. All the logic pins that are missing in a cell but existing in other cells will be reported in this table. In this report table, the following columns are listed: Cell name, Pin name and each logic library names.

### What Next

If some signal pins are found missing in logic cells, it is a real issue and therefore an Error message LIBCHK-101 will occur to direct you to correct the cells that are missing these pins in the logic library.

### Examples

```
List of pins missing in logic libraries
-------------------------------------------------------------------------
-----
Cell name       Pin name SR60_W_SVT_UHD.db         SR60_S_SVT_UHD.db
-------------------------------------------------------------------------
-----
tieofflx_svt    AA    missing                  existing
tieofflx_svt    ZZ    missing                  existing
-------------------------------------------------------------------------
-----
```

### Examples

```
Warning: List of pins missing in logic libraries (LIBCHK-320)
```

---

## LIBCHK-321

(warning) List of pins with mismatched or missing attributes in logic libraries

### Description

This message occurs when the tool finds mismatched pins in the logic cells during logic versus logic library checking. Mismatched pins are those that have inconsistent pin attributes such as pin direction, function, related_power_pin, and so forth. For characterization values (e.g. capacitancer), there are calculated absolute and relative errors along with respective values in each library. They are listed in the columns on the right-hand side. For -single_mode_etm checking, inconsistencies across the ETM models in pin direction, related_power_pin, related_ground_pin, tlatch, capacitance, max_fanout, max_capacitance, max_transition, min_capacitance, min_transition, function are reported here. And the issues in pin direction, related_power_pin, related_ground_pin, tlatch lead to Error LIBCHK-366. Pin level user defined attributes also are reported in this table if mismatched or missing. The report table has the following columns:

```
Cell name
Pin name
```

Feedback

```
Attribute/Group name
Attribute value of each of the logic libraries
Absolute and Relative errors
```

### What Next

However, mismatched pins will cause issues if the cells are used in the design. The same name pins should exist in the same name cells in different libraries with consistent values for attributes such as function, direction, related_power_pin, and related_ground_pin, and the same attributes should exist for the same pins in the same name cells. Characterization values such as capacitance can be different. For mcmm, mismatched pin_number's will cause MV-087 Error.

### Examples

The following is an example of a mismatched pins list:

```
Warning: List of pins with mismatched or missing attributes in logic
 libraries (LIBCHK-321)
-------------------------------------------------------------------------
---------------
                     Subgroup/              Attribute value
 Error
Cell name Pin name   Attribute           lib#1       lib#2
 Absolute  Relative
-------------------------------------------------------------------------
---------------
tieoff    lo         pin_number          0           1
std_inv   Y          related_ground_pin  missing     VSS
MX02D1    I1         fall_capacitance_upper
                                         0.003505    0.003509
 0.0000040 0.11%
MX02D1    S          fall_capacitance_upper
                                         0.005593    0.005586
 -0.000007 -0.13%
MX02D1    I0         fall_capacitance_lower
                                         0.003723    0.003727
 0.0000040 0.11%
MX02D1    I1         fall_capacitance_lower
                                         0.003505    0.003509
 0.0000040 0.11%
MX02D1    I0         fall_capacitance    0.003723    0.003727
 0.0000040 0.11%
MX02D1    I1         fall_capacitance    0.003505    0.003509
 0.0000040 0.11%
MX02D1    S          fall_capacitance    0.005593    0.005586
 -0.000007 -0.13%
MX02D1    I1         capacitance         0.003596    0.003598
 0.0000020 0.06%
CKGTPLS   CK         ecsm_capacitance    missing     existing
-------------------------------------------------------------------------
---------------
```

Feedback

## LIBCHK-322

(warning) List of pg_pins with mismatched or missing attributes in logic libraries

### Description

This information message occurs when the tool finds pg_ pins with mismatched or missing attributes in the logic cells during logic versus logic library checking. Mismatched pg_pins are those that have inconsistent pg_pin attributes.

The report table has the following columns:

```
Cell name
pg_pin name
Attribute/Group name
Attribute value of each of the logic libraries
```

### What Next

Mismatched pg_pins will cause issues if the cells are used in the design. The same name pg_pins should exist in the same name cells in different libraries with consistent values for attributes such as direction, pg_function and switch_function. The same attributes should exist for the same pg_pins in the same name cells.

### Examples

The following is an example of mismatched pg_pins list:

```
List of pg_pins with mismatched or missing attributes in logic libraries
-------------------------------------------------------------------------
-----
                            Attribute                  Attribute value
Cell name       pg_pin name  /Group                non_pg_lib      pg_lib
-------------------------------------------------------------------------
-----
LVLLHCD         VDD         std_cell_main_rail     missing         true
FOOTER          VSS         pg_function            missing         "TVSS"
-------------------------------------------------------------------------
-----
```

## LIBCHK-323

(warning) List of pg_pins missing in logic libraries

### Description

This message is for reporting missing pg_pins in logic cells during logic vs. logic library checking. All the logic pg_pins that are missing in logic cells will be reported in this table. In this report table, the following columns are listed: Cell name, pg_pin and each logic library names.

Feedback

### What Next

If pg_pins are missing in the logic library, it indicates that the logic library is not pg_pin based, and you need to use add_pg_pin_to_db utility to convert it into pg_pin based db.

### Examples

```
List of pg_pins missing in logic libraries
-------------------------------------------------------------------------
-----
Cell name       pg_pin          SR60_W_SVT_UHD.db
 SR60_S_SVT_UHD.db
-------------------------------------------------------------------------
-----
tieofflx_svt    __VSS    missing                  existing
tieofflx_svt    __VDD    missing                  existing
-------------------------------------------------------------------------
-----
```

### Examples

```
Warning: List of pg_pins missing in logic libraries (LIBCHK-323)
```

---

## LIBCHK-324

(warning) List of missing or mismatched models in pin groups

### Description

This table reports missing or mismatched models in pin groups including receiver_capacitance and its subgroups. The report table has the following columns:

```
Cell name
Pin name
when condition
Group name
Attribute values in each library
```

### What Next

It is recommended that missing models be added and mismatched models be corrected.

### Examples

The following is an example:

```
Warning: List of missing or mismatched models in pin groups (LIBCHK-324)
-------------------------------------------------------------------------
-----------
Cell name   Pin name    when        Group       Attribute   lib#1
 lib#2
```

Feedback

```
------------------------------------------------------------------------
-----------
AN2          A1           !A2          receiver_capacitance
                                                  sensitivity
 missing
------------------------------------------------------------------------
-------
```

## LIBCHK-330

(warning) List of timing arcs missing or existing in libraries

### Description

This message is for reporting timing arcs missing or unexpected existing in the logic libraries. For -single_mode_etm, if a mode is already defined in timing group, it also is reported. This is an Error for this check for ETM.

In the fields for the libraries with duplicate arcs, you will see "duplicate".

### What Next

For timing scaling, timing arc information should exist with consistent timing data in each library. For other cases, this is information only. No further action is required. For -single_mode_etm, the ETM cannot have mode definition in an arc. However, for library scaling, missing arcs in link_library compared to other libraries in the same group will not cause failure(LIBCHK-362).

### Examples

```
List of timing arcs missing or existing in libraries
------------------------------------------------------------------------
-----
Cell name     Arc             SR60_W_SVT_UHD.db         SR60_S_SVT_UHD.db
------------------------------------------------------------------------
-----
tbufl_f7_svt  gz_y            existing                  missing
------------------------------------------------------------------------
-----
```

### Examples

```
Warning: List of timing arcs missing in libraries (LIBCHK-330)
```

## LIBCHK-331

(warning) List of timing arcs mismatched in logic libraries

Feedback

### Description

This message is for reporting mismatched arcs including missing timing models in logic libraries. It also includes inconsistent output load indices, and voltage indexes for dc_current. When the library is missing either NLDM or CCST models when -validate timing or -scaling timing is specified or missing either NLDM or CCSN model when -validate noise is, this will be checked and reported. It includes 1) completely missing models from the whole library, reported in LIBCHK-332. 2) partially missing models (some arcs missing models), reported in LIBCK-331.

For half unate timing arcs with some timing_type and timing_sense combinations, only fall or rise groups are required. For example, in timing arcs with timing_type : clear, only fall groups are required and rise groups are optional. Likewise, in timing arcs with timing_type : preset, only rise groups are required and fall groups are optional. Those optional groups, if missing, will not be reported.

For -scaling noise, input_voltage and output_voltage indices should be in the same percentage of nominal voltages.

For -validate receiver_capacitance, it reports missing receiver_capacitance models on arcs. For the arcs missing rcv cap, the rcv cap on default arc, if defined, will be used, otherwise pin based rcv cap with default or matched when condition will be used. For better accuracy and correlation with SPICE, arc based rcv should be defined.

### What Next

For timing scaling and timing/delay calculations, consistent timing model structures should exist in the libraries for each timing arc, and for timing scaling, index_[1|2]_size should be consistent and not a 3D table (without index_3). For noise scaling, input_voltage and output_voltage indices should be in the same percentage of nominal voltages.

### Examples

```
List of timing arcs mismatched in logic libraries (LIBCHK-331)
-------------------------------------------------------------------------
---------------------------------

      W_SVT_UHD.db S_SVT_UHD.db
Cell name           pin/related_pin when  timing_type   Group/Attribute
         lib#1   lib#2
-------------------------------------------------------------------------
---------------------------------
sdffcqs_f2_hp_tw_dh sq/clk          !scan rising_edge   cell_rise
         missing       existing
sdffcqs_f2_hp_tw_dh q/clk           scan  rising_edge   compact_ccs_rise
         missing       existing
p210_3d             B/A                 combinational
 cell_rise/index_3_size    2             2
p210_3d             B/A                 combinational
 cell_rise/index_1_size    3             2
```

Feedback

```
DFFQ_X1             D/CK                   setup_rising
 receiver_capacitance      missing       existing
-------------------------------------------------------------------------
--------------------------------
```

### Examples

```
Warning: List of timing arcs mismatched in logic libraries (LIBCHK-331)
```

## LIBCHK-332

(warning) Library '%s' is missing CCS %s models.

### Description

This warning message occurs during logic versus logic library checking, when no driver or receiver model exists in the library.

### What Next

Receiver models and driver models (compressed or original format) must exist in the libraries for timing scaling. Please add the missing models. For all other cases, this is information only.

## LIBCHK-334

(warning) List of missing models or invalid values in timing groups

### Description

This table reports missing models or invalid values in subgroups in timing arcs. Invalid values can be all zeros in a NLDM table. This table has the following columns: Cell name, pin/related_pin, timing_type, when, Group/Attribute and values or missing in each library.

### What Next

It is recommended that missing models be added and mismatched models be corrected.

### Examples

```
Warning: List of missing models or invalid values in timing groups
 (LIBCHK-334)
-------------------------------------------------------------------------
--------
Cell name pin/related_pin timing_type   when  Group/Attribute  lib#1
 lib#2
-------------------------------------------------------------------------
--------
AIOI21    ZN/A1           combinational !A2   cell_rise
 missing
```

Feedback

```
CQAN2X    Z/B             combinational !A1   timing
 missing
AIOI21    ZN/A2           combinational !A1   cell_rise/values
 "0,0,0"
------------------------------------------------------------------------
--------
```

### Examples

```
Warning: List of missing models or invalid values in timing groups
 (LIBCHK-334)
```

## LIBCHK-340

(warning) List of inconsistent CCS noise models

### Description

This message is for reporting inconsistent CCS noise models in logic libraries. It reports: a) inconsistent input_voltage and output_voltage indices for CCS noise models. They should be in the same percentage of nominal voltages. b) inconsistent conditional (when) pin models (when order). c) inconsistent CCS noise models for all pins and arcs. This report table includes the following columns: Cell name, Pin name, Group name, Attrbute name and attribute value of each library.

### What Next

For CCS noise scaling, consistent noise models are required. For other cases, it is for information only.

### Examples

```
List of inconsistent CCS noise models
------------------------------------------------------------------------
-----
                                      Attribute           Attribute value
Cell name Pin name  Group name        /variable    SR60_W_SVT_UHD
 SR60_S_SVT_UHD
------------------------------------------------------------------------
-----
inv0d0    ZN        ccsn_first_stage  input_net_transition
                                                 "175.0"
 "980.0"
inv0d0    ZN        ccsn_last_stage   total_output_net_capacitance
                                                 "0.0"             "0.0"
------------------------------------------------------------------------
-----
```

### Examples

```
Warning: List of inconsistent CCS noise models (LIBCHK-340)
```

Feedback

## LIBCHK-341

(warning) List of inconsistent power models.

### Description

This warning message occurs when there are inconsistent power models in the logic library, or power value = 0 in the case of check_library -upf -mcmm. For characterization values (e.g. internal_power), there are calculated absolute and relative errors as well as error type along with respective values in each library. They are listed in the columns on the right-hand side. These values are only shown when library_features is defined with report_power_calculation for power and report_delay_calculation for delay/slew, or source .libs are read in by read_lib.

It reports the following inconsistencies:

- `Inconsistent dynamic_current groups for each cell`
- `Inconsistent intrinsic_parasitic tables for each cell`
- `Inconsistent leakage_current groups for each cell`
- `Inconsistent power_cell_type attribute for each cell`
- `Inconsistent internal_power tables for each pin or cell`
- `Inconsistent leakage_power tables for each cell`

The report table includes the following columns:

```
Cell name
Group name
Attribute name
Attribute value of all logic libraries
Absolute and Relative errors and type
The absolute and relative errors are only reported when there are <=2
 libraries, and the Type includes
NLPM_IP - internal_power values
NLPM_LP - leakage_power value
CCS_POWER - CCS power values
GRID   - index_1/2 mismatch/missing
```

### What Next

This is only a warning message. No action is required.

However, consistent power models are required for power scaling. For other cases, the report is for information only.

Feedback

### Examples

The following is an example of a list of inconsistent power models:

```
List of inconsistent power models
-------------------------------------------------------------------------
-----
                                                          Attribute value
Cell name  Group            Attribute       SR60_W_SVT_UHD
 SR60_S_SVT_UHD
-------------------------------------------------------------------------
-----
ap001_svt  leakage_power    value           0.000000         2.686740
an02d1     dynamic_current  related_inputs  "A1"             "A0"
-------------------------------------------------------------------------
-----

List of inconsistent power models. (LIBCHK-341)
index_1:    input_transition_time
-------------------------------------------------------------------------
-------------------------------------------------

       Attribute value               Error
Cell name Group           when                Group/Attribute
 (index_1,index_2) lib#1     lib#2     Absolute  Relative Type
-------------------------------------------------------------------------
-------------------------------------------------
ap001_svt leakage_power   A+B                 value
       0.000000  2.686740
usb30io_io8kv_ns
          wpu_n/internal_power
                          hold_h_n&sig_in&ds[0]'&ds[1]'
                                              rise_power/values
                                                                0.0208537
       0.003811  0.0010794 -0.002731 -71.68%  NLPM_IP
                                                                0.0417187
       0.0041157 0.0012722 -0.002843 -69.09%  NLPM_IP
                                                                0.0834375
       0.004725  0.0016579 -0.003067 -64.91%  NLPM_IP
                                                                0.166875
       0.0049465 0.0016862 -0.003260 -65.91%  NLPM_IP
                                                                0.33375
       0.0053895 0.0017428 -0.003646 -67.66%  NLPM_IP
                                                                0.6675
       0.0062755 0.001856  -0.004419 -70.42%  NLPM_IP
                                                                1.335
       0.0080475 0.0020824 -0.005965 -74.12%  NLPM_IP
-------------------------------------------------------------------------
-------------------------------------------------
```

Feedback

## LIBCHK-342

(warning) List of inconsistent data between CCS noise and NLDM models

### Description

This is validation between NLDM delay/slews and those derived from CCSN with NLDM as reference. This is a warning message about CCSN accuracy. It may render library cells with the reported accuracy issue unusable causing a loss of accuracy during STA AWP consumption for the reported timing arc(s). This message occurs when there are inconsistent delay and slew data between CCS noise and NLDM models in the logic library. The default relative and absolute tolerance values for delay are 3% and 3ps, respectively, and for slew, 5% and 5ps. Please refer to report_check_library_options commands for a list of default tolerances.

In this table, N/A, if reported, refers to the following cases: 1) CCSN model does not exist (i.e. not defined in .lib) on that timing arc. 2) CCSN model is defined in .lib but no valid CCSN data are extracted in .db on that timing arc. 3) referenced CCSN model is defined, i.e. input_ccb/output_ccb defined under pin, but active_output_ccb is missing in the arc for pin based CCSN.

When the tcl variable lc_check_lib_use_tolerance_rel_to_inslew is set true, a new passing criterion is added that is Absolute_ error / input_slew, or difference between CCSN and NLDM relative to input slew, is <= 2% of input slew for delay and 3% for output slew. If this is met it also passes.

The load index is directly taken from NLDM table in the library.

Transition values may show misalignment for ETM macro lib cells on output pins. This is expected because net parasitic effects are incorporated into CCSN and NLDM models in different ways. Even with these warnings, the extracted CCSN models can help top level timing accuracy using an ETM library containing CCST and CCSN. There is no option in check_library to skip transition checks.

### What Next

If large discrepancy is reported between CCSN and NLDM, please double check .lib file to make sure valid CCSN model is specified correctly. And if no CCSN model is defined in .lib please recharacterize .lib with the required model.

### Examples

The following is an example of a list of inconsistent CCSN vs. NLDM models:

```
List of inconsistent data between CCS noise and NLDM models (LIBCHK-342)
------------------------------------------------------------------------
-------------------------------------------------

               mylib                      Error
```

Feedback

```
Cell name     pin/related_pin  timing_type    when   Group
 index_1,index_2   (NLDM)       (CCSN)       Absolute   Relative
-------------------------------------------------------------------------
-------------------------------------------------
mycell1       Z/C              combinational  !A&B   cell_rise
 "0.002,0.0005"    0.0127228   0.00456662  -0.008156  -64.11%
mycell1       Z/C              combinational  !A&B   rise_transition

 "0.002,0.0005"    0.022014    0.00171184  -0.020302  -92.22%
mycell1       Z/C              combinational  !A&B   fall_transition

 "0.002,0.0106"   0.1157      0.066054    -0.049646  -42.91%
mycell2       Q/E              rising_edge           cell_rise *,*
        0.0655981   N/A         N/A        N/A
mycell2       Q/E              rising_edge           rise_transition *,*
        0.00602941  N/A         N/A        N/A
-------------------------------------------------------------------------
-------------------------------------------------

Below is an example report when user sets the Tcl variable to control the
 relative/absolute tolerance:
set lc_check_lib_use_tolerance_rel_to_inslew true

Warning: List of inconsistent data between CCS noise and NLDM models
 (LIBCHK-342)
index_1:  input_net_transition
index_2:  total_output_net_capacitance
-------------------------------------------------------------------------
-----------------------------------------------------------------

                mylib2                          Error
Cell name   pin/related_pin  timing_type    when   Group
 index_1,index_2  (NLDM)     (CCSN)     Absolute    Relative    Abs/inslew
-------------------------------------------------------------------------
-----------------------------------------------------------------
INV1  Z/I              combinational         rise_transition
 0.112,0.00077  0.0353892 0.251844   0.216454    611.64%     193.26%
-------------------------------------------------------------------------
-----------------------------------------------------------------
```

## LIBCHK-343

(warning) List of sensitive CCSN models

### Description

The CCSN models of some cells are very sensitive in delay and slew to their input driving waveform. Small distortion of the input driving waveform tail can lead to a significantly different output waveform. A library cell is considered sensitive if the difference of delay or slew of output responses of the library cell to an input waveform versus a tail-distorted version of the input waveform is larger than the given tolerances. The default relative and

Feedback

absolute tolerance values for delay are 2% and 5ps, respectively, and for slew, 3% and 7.5ps. In the table, N/A, if any, refers to that CCSN model does not exist. The load index is directly taken from NLDM table in the library.

### What Next

This is a warning message about CCSN sensitivity. To drive sensitive cells, it is recommended that a driver cell with better full swing (>95% VDD) be used. And library developer may choose more proper CCB cut to minimize the issue. Otherwise, it may render library cells with the reported accuracy issue unusable.

### Examples

The following is an example of the report:

```
List of sensitive CCSN models (LIBCHK-343)
-------------------------------------------------------------------------------
-------------------------------------------------

 tsmc_cln40g_sclmp_pt_fast_125c_0p99v      Error
Cell name     pin/related_pin  timing_type    when   Group
 index_1,index_2   (Normal)   (Clamped)      Absolute    Relative
-------------------------------------------------------------------------------
-------------------------------------------------
XOR3X05LMP_BZA_P21
              Z/C              combinational  !A&B   cell_rise
 "0.002,0.0005"    0.0127228   0.00456662  -0.008156   -64.11%
XOR3X05LMP_BZA_P21
              Z/C              combinational  !A&B   rise_transition

 "0.002,0.0005"    0.022014    0.00171184  -0.020302   -92.22%
XOR3X05LMP_BZA_P21
              Z/C              combinational  !A&B   fall_transition

 "0.002,0.0106444" 0.1157      0.066054    -0.049646   -42.91%
-------------------------------------------------------------------------------
-------------------------------------------------
```

## LIBCHK-344

(warning) Table of attribute value range analysis

### Description

This table lists scalar attributes and their values that are out of range as defined in set_check_library_option. In the table, the following fields are listed: Cell name, pin/subgroup, timing_type, when, Attribute, value and line_num where the second field can be pin or a subgroup name under the cell such as leakage_power. When an entry has no value in a field, it is blank. If the library is loaded from source .lib and the tcl var

Feedback

lc_check_lib_keep_line_number is set true, line_num field will show the line numbers where the attribute group starts.

This table also reports dc_resistance_high/low derived from dc_current tables in CCSN V1 and V2: 1) for 1-stage CCSN, check ccsn_first_stage/input_ccb 2) for multi- or 2-stage CCSN, check ccsn_last_stage/output_ccb only

**What Next**

Please check the attributes that have values out of range and correct them as needed. User should check the values and find out if characterization or setting have done correctly.

**Examples**

The following is an example table:

```
Warning: Table of attribute value range analysis (LIBCHK-344)
-------------------------------------------------------------------------------
---------------------------
Cell name           pin/subgroup    timing_type   when      Attribute
       value     line_num
-------------------------------------------------------------------------------
---------------------------
STN_AN6_1           leakage_power                 A1 * A2  value
      0.00502   1048
STN_AN6_1           A4
 rise_capacitance    0.000598  1739
STN_AN6_1           A5                                      capacitance
       0.000625  1882
STN_FDNQ_1          Q
 ccsn_last_stage/miller_cap_rise

       0.000246  16064
-------------------------------------------------------------------------------
---------------------------
```

## LIBCHK-345

(warning) Table of %s value ratios.

**Description**

This table lists the dynamic_current groups, which has output switching, that:
1. The ratio of peak pg_current(power) to peak pg_current(ground) are out of range as defined in set_check_library_options -analyze {pg_current} \ -criteria {min_power_to_ground_current_ratio=...} .

Feedback

1. The ratio of time constant of CCSP to time constant of NLDM/CCST are out of range as defined in set_check_library_options -analyze {ccsp_time_constant} \ -criteria { max_nldm_to_ccsp_time_const_ratio=... \ max_ccst_to_ccsp_time_const_ratio=...}

The report table includes the following columns:

```
Common columns:
  Cell name
  Group name
  When condition
  Attribute name
  index_1/index_2
  line_num1(optional)
  line_num2(optional)

For peak current check:
  Peak pg_current for Power
  Peak pg_current for Ground
  Ratio of Power/Ground
  Ratio of Ground/Power

For time constant check:
  Time of CCSP
  Time of NLDM/CCST
  Ratio of time constant
  Type : NLDM or CCST
```

If the library is loaded from source .lib and the tcl var lc_check_lib_keep_line_number is set true, line_num field will show the line numbers where the group starts.

**What Next**

User should check the reported group and verify if the value is correct.

**Examples**

The following is an example of a table of PG Current value ratios:

```
Warning: Table of PG Current value ratios (LIBCHK-345)
------------------------------------------------------------------------
------------------------------------------------------------
Cell name Group                    when        Group/Attribute line_num1
 line_num2 (index_1,index_2) Power    Ground    (P/G)      (G/P)
------------------------------------------------------------------------
------------------------------------------------------------

XOR4XD8BWP12T
          related_outputs:Z/related_inputs:A4/input:rise|output:fall
                                  A1&A2&A3   pg_current/peak_value
                                                          10978
 11061    0.0017,0.00205    0.786786  0.980786  0.8022    1.247

         0.0017,0.06806    0.614211  2.34515   0.2619    3.818
```

Feedback

The following is an example of a table of CCSP Time constant ratios:

```
Warning: Table of CCSP Time constant value ratios. (LIBCHK-345)
-----------------------------------------------------------------------
-----------------------------------------------------------------
Cell name Group                    when        Group/Attribute line_num1
 line_num2 (index_1,index_2)  CCSP     NLDM/CCST  Ratio    Type
-----------------------------------------------------------------------
-----------------------------------------------------------------
XOR4XD8BWP12T
          related_outputs:Z/related_inputs:A4/input:rise|output:rise
                           !A1&!A2&!A3         pg_current/time 10463
 40914     0.2109,0.00205    0.714328  0.0254    28.12     CCST

         0.2109,0.06806    0.714328  0.058     12.32     CCST
```

## LIBCHK-346

(warning) List of inconsistent data between CCS noise and CCS timing models

### Description

This is validation between the times at the trip points derived from CCST and those derived from CCSN with CCST as reference. This message occurs when there are inconsistent data between CCS noise and CCS timing models in the logic library. The default relative and absolute tolerance values for all trip points are 5% and 0.005ns, respectively.

### What Next

This is a warning message about CCSN vesus CCST accuracy. It may render library cells with the reported accuracy issue unusable.

### Examples

The following is an example of a list of inconsistent CCS noise and CCS timing models:

```
List of inconsistent data between CCS noise and CCS timing models
 (LIBCHK-346)
-----------------------------------------------------------------------
-------------------------------------------------

        ss_0p5225v_m25c                 Error
Cell name     pin/related_pin  timing_type    when   point
 index_1,index_2   (CCST)       (CCSN)      Absolute    Relative
-----------------------------------------------------------------------
-------------------------------------------------
INV_X8M_A9TL  Y/A              combinational         rise_0.15
 0.003209,0.448719 0.108767    0.107171    -0.001595   -1.47%
INV_X8M_A9TL  Y/A              combinational         rise_0.15
 0.003209,0.904573 0.214978    0.211207    -0.003771   -1.75%
```

Feedback

```
INV_X8M_A9TL  Y/A              combinational         rise_0.15
 0.0133113,0.448719

        0.113749    0.112348    -0.001400   -1.23%
-------------------------------------------------------------------------
-------------------------------------------------
```

## LIBCHK-347

(warning) List of inconsistent receiver_capacitance C1/C2

### Description

This is validation between C1/C2 converted from C1/Cn and original receiver_capacitance1/2_rise/fall. When the difference exceeds the default or specified capacitance tolerance, the values will be reported.

### What Next

This is a warning message about n-receiver_capacitance accuracy.

### Examples

The following is an example of the report.

```
#BEGIN_XCHECK_C1C2

Warning: List of inconsistent receiver_capacitance C1/C2 (LIBCHK-347)
-------------------------------------------------------------------------
--------------------------------

 values                Error
Cell name     pin/related_pin  timing_type   when    Attribute (C1/C2)
 (Cn converted) Absolute  Relative
-------------------------------------------------------------------------
--------------------------------
DBLVT16       X/A1             combinational A2      fall_c2   0.00310165
  0.00413994  0.0010382 33.48%
MUX2_12       X/S              combinational D0&!D1  rise_c2   0.00471183
  0.00585422  0.0011423 24.25%
MUX2_8        X/S              combinational D0&!D1  rise_c2   0.00332178
  0.00448415  0.0011623 34.99%
-------------------------------------------------------------------------
--------------------------------

#END_XCHECK_C1C2
```

## LIBCHK-348

(warning) List of invalid values in LLE models.

Feedback

### Description

This message occurs when one or more of the following are found for LLE tables. If "set_check_library_options -verbose" is used, additional LIBCHK-348-W* warnings appear with more details, otherwise, only a summary table appears.

Below (1) to (5) are "Value checks": (1) partial zero values (in the same table, some values are zero and some values are nonzero), verbose warning code LIBCHK-348-W1 (2) mix of postive and negative values (in the smae table, if some values are positive and some values are negative), verbose warning code LIBCHK-348-W2 (3) values not increasing or decreasing according to “vth”, u0” and the index_1 value, verbose warning code LIBCHK-348-W53 (4) sum of lle value and corresponding nldm value is not increasing or decreasing, according to “vth”, u0” and the index_1 value, verbose warning code LIBCHK-348-W67 or W68 (5) a value is more than a relative percentage threshold of its associated nominal timing value, and the nominal timing value is greater than 1e-6 (to avoid dividing with a very small value) verbose warning code LIBCHK-348-W69

Below (11) to (15) are "structure checks": (11) For a cell with one or more LLE table, a nominal NLDM delay/slew arc has no corresponding LLE table. The “Value checks #1 to #5” and further “Structure checks” will be skipped for this missing LLE table. verbose warning code LIBCHK-348-W100 (12) the list of index_1 (which denotes the delta value of “vth” or “u0”) has more than one value (if this happens, only the first value will be used for checking. The other values will be ignored.), or that value is zero. In other words, index_1 should have only one value of nonzero. verbose warning code LIBCHK-348-W31, if it has more than one value verbose warning code LIBCHK-348-W51, and “Value checks #3” will be skipped, if the value is zero (13) the “related_params” string must be either “vth” or “u0” verbose warning code LIBCHK-348-W5, and “Value checks #3, #4, #5” will be skipped (14) an LLE timing table with its index_2 indices not as a subset of (or not found in) its associated nominal (NLDM) timing table’s index_1 indices, here the assumption is that this pair of indices (LLE timing table’s index_2 and its nominal timing table’s index_1 are for the same parameter, which is either input_net_transition or total_output_net_capacitance). verbose warning code LIBCHK-348-W65, and “Value checks #4 and #5” will be skipped (15) an LLE timing table with its index_3 indices not as a subset of (or not found in) its associated nominal (NLDM) timing table’s index_2 indices, here the assumption is that this pair of indices (LLE timing table’s index_3 and its nominal timing table’s index_2 are for the same parameter, which is either input_net_transition or total_output_net_capacitance). verbose warning code LIBCHK-348-W65, and “Value checks #4 and #5” will be skipped

### What Next

You are advised to go back to the above check e.g. report LIBCHK-348 to see inconsistencies across the libraries and resolve them as directed by the associated LIBCHK man pages.

Feedback

### Examples

```
Warning: List of invalid values in LLE models (LIBCHK-348)
------------------------------------------------------------------------
---------
Cell name   pin/related_pin when        timing_type Group/Attribute
 lib#1    line_num
------------------------------------------------------------------------
---------
AND2X4   Y/A               combinational lle_delta_cell_rise/values
 "0.00123266,0.00134349,..." 328
AND2X4  Y/A               combinational lle_delta_rise_transition/values
 "0.000786874,0.00742903,..." 361
AND2X4  Y/A               combinational lle_delta_cell_fall/values
 "8.36731e-07,1.83673e-06,..." 416
AND2X4  Y/A               combinational lle_delta_fall_transition/values
 "6.71382e-09,2.00671e-06,..." 427
------------------------------------------------------------------------
---------
```

## LIBCHK-349

(information) Table of invalid values in aged models

### Description

This table reports two types of value range analysis results. Type deviated is for those aged NLDM table values out of certain range from nominal values. The certain range can be specified by -tolerance option of set_check_library_options. Type identical is for aged values very close to nominal values. The default identical range is 1e-6. This table has the following columns: Cell name, pin/related_pin, timing_type, when, Group, Attribute, index_1, index_2, nominal, aged, Absolute, Relative, Type

### What Next

It is recommended that aged values out of range from nominal should be corrected and the identical values should be noticed.

### Examples

```
Information: Table of invalid values in aged models (LIBCHK-349)
------------------------------------------------------------------------
------------------------------------------------------------------------
Cell name pin/related_pin timing_type   when  Group            Attribute
  index_1   index_2  nominal   aged      Absolute   Relative   Type
------------------------------------------------------------------------
------------------------------------------------------------------------
AIOI21    ZN/A1           combinational !A2   aged(profile_1)  cell_rise
  0.0026    0.0172   0.099635  0.109646  0.010011   10.05%     deviated
CQAN2X    Z/B             combinational !A1   aged(profile_2)  cell_fall
  0.2385    0.2791   0.008123  0.009123  0.001      12.31%     deviated
```

Feedback

```
CQAN2X    Z/C             combinational !A2   aged(profile_3)  cell_fall
  0.2385    0.2791   0.008085  0.008085  0          0.00%       identical
------------------------------------------------------------------------
------------------------------------------------------------------------
```

### Examples

```
Information: Table of invalid values in aged models (LIBCHK-349)
```

---

## LIBCHK-350

(warning) List of inconsistent data between different timing models

### Description

This message is for validating library data between CCS and NLDM models. It reports: a) inconsistent output load indices between CCS and NLDM models, and inconsistent load indices for CCS drive/receiver b) inconsistent NLDM delay and slew indices and values with a default or user specified tolerance. c) inconsistent delays between NLDM and CCS timing with a user specified tolerance for both compact CCS and expanded CCS d) inconsistent data between two CCS driver model current waveforms including compact CCS with expanded CCS data with a tolerance e) inconsistent between CCS power data and NLPM data (not in 2008.09 releases) The load index is directly taken from NLDM table in the library. In this report table, including following columns: Cell name, Group name, Attrbute name and attribute value of all liogic libraries.

### What Next

If the difference is significantly larger between the models, please rerun the simulation to correct the libraries. For base vs. sensitivity libraries, when inconsistent hold constraint values are reported, please check for missing ocv_sigma tables when hold_constraint_sigma_folding_factor is defined.

### Examples

```
Warning: List of inconsistent data between different models (LIBCHK-350)
Cell name:  dtmuxi3_d10m
index_1:    total_output_net_capacitance
index_2:    input_net_transition
------------------------------------------------------------------------
--------------------------------

 gold_timing_index_lib     Error
pin/related_pin timing_type   when  Group/Attribute (index_1,index_2)
 (NLDM)     (CCS)     Absolute Relative
------------------------------------------------------------------------
--------------------------------
y/s2            combinational       cell_rise/values
                                                        (1.83,0.5)
 18.016    18.2206   0.20456  1.1%
```

Feedback

```
                                                                    (1.83,4)
 19.5332    19.8679    0.33468   1.7%
                                                                    (1.83,12)
 23.9233    28.2093    4.286     18%
-------------------------------------------------------------------------------
---------------------------------
```

### Examples

```
Warning: List of inconsistent data between different models (LIBCHK-350)
```

---

## LIBCHK-351

(information) List of inconsistent constructs and values

### Description

This message is for reporting inconsistent liberty syntax constructs and/or values of the two logic libraries

### What Next

This is information only. No further action is required.

### Examples

```
Information: List of inconsistent constructs and values (LIBCHK-351)
```

---

## LIBCHK-352

(information) Table of timing variation analysis

### Description

This table lists LVF analysis results for delay, slew and constraint models if LVF/OCV models exist in the library and -analyze {lvf} is specified. Alternatively this table can report variation aware analysis results for driver, receiver and constraint models if VA models exist in the library and -analyze {nominal_vs_sigmal} is specified. For LVF analysis, it reports all grid points where the ratio of sigma to nominal values or ratio of early (or late) sigma to late to early sigma values exceeds the specified minimum ratio. For a set of va_values, there are normally three values: nominal, -Sigma and +Sigma. From the three values for each pair of indices, a linear model is used to find: a) monotonicity (trend) including - delays increase as slew and load increase. - delays increase/decrease as va_parameters increase. b) significance of each va_parameters

As trial and error procedure, the following statistics will be used in the study on the VA models. iThe linear model is used to roughly measure how significant each va_parameters affects on delay. 1) If slope ~= 0, va_parameter is insignificant (not sensitive) 2) If slope > 0, monotonously increasing 3) If slope < 0, monotonously decreasing 4) Alternating

Feedback

slopes may indicate issues 5) Trend: If Dnom > both D+/-sigma, or If Dnom < both D+/-sigma, not monotonous, where D is Delay. 6) coefficient of correlation to measure how closely the va_parameters are related to delays. Std error is used to measure how much va_parameter and delay points deviate from the linear model.

This report table includes the following columns for VA: pin/related_pin, timing_type, when, Group, va_values, (index_1,2), nomimal, -Sigma, +Sigma, Slope, Std error and Trend
In Trend column, / is monotonously increasing, \ is monotonously decreasing, ^ is non-monotonous up, V is non-monotonous down and - is flat. For LVF, it has nominal, early, late, sigma/nom, and sigma/sigma fields instead.

### What Next

If the difference is significantly larger between the models, please rerun the simulation to correct the libraries.

### Examples

```
Information: Table of LVF model analysis (LIBCHK-352)
index_1:  related_pin_transition
index_2:  constrained_pin_transition
-------------------------------------------------------------------------
-----------------------------------------------

 values                 Error
Cell name  pin/subgroup timing_type  when   Group     index_1,index_2
 nominal  early  late  sigma/nom sigma/sigma
-------------------------------------------------------------------------
-----------------------------------------------
SDFKRPQD1  Q/CP       rising_edge    C&!D    cell_fall   0.005,0.00019
 0.189137 0.099  0.135   0.7167    1.3654
HDBUL      D/CK       hold_rising  !SE&!SI fall_constraint
                  0.0135,0.3618  0.037298 0.0186 0.0186  0.4995    1.0000
-------------------------------------------------------------------------
-----------------------------------------------

Number of characterization values meeting the criteria in library 1:   2
 (out of 896)
```

### Examples

```
Information: Table of LVF model analysis (LIBCHK-352)
```

---

## LIBCHK-353

(information) Statistics per characterization model

### Description

This information message reports the statistical analysis results of characterizations, for example, between CCS and NLDM models or between the same types of models

Feedback

in the libraries, using the golden NLDM or first library as the base for CCST/CCSN vs. NLDM comparision. It occurs on -char_integrity or -compare value option. In counting the grid points, for example, if a 7x7 NLDM table grids mismatch 7X7 CCS, it should be counted as 49, not 98. And delay/slew values passed (example below) do not include failed grid points. Number of CCSN delay values passed 66054 This table only includes characterization models or derived types: delay, slew, constraint, receiver_cap, ccs driver, voltage range, ccsn (dc_current), internal_power, and leakage_power. The columns in the table include the following:

- Group type: Delay, Slew, Constraint, Receiver capacitance
- Mean of differences: Absolute, Relative
- Standard deviation: Absolute, Relative
- Max outlier: the maximum value of the deviations
- Number of grid points: Passed, Failed

**What Next**

If the mean is far from 0, the two models or libraries have significant differences. If the standard deviation is significantly large, the differences of these two are dispersed. In any case, small numbers of mean and deviation, close to 0, should be expected if the characterizations are generated under the same operating conditions.

**Examples**

The following is an example report of the statistical analysis results of characterization models:

```
Information: Statistics per characterization model (LIBCHK-353)
------------------------------------------------------------------------
----------
               Mean of differences    Std deviation
 Number of grids
Group type      Absolute Relative   Absolute Relative   Max outlier  Pass
    Fail
------------------------------------------------------------------------
----------
CCST delay      0.000463534 0.14%      0.000636014 0.13%   0.00412124
 300     0
CCST slew       0.000492754 0.53%      0.000772329 0.44%   0.00368556
 300     0
CCSN delay      0        0.00%         0        0.00%      0            0
       300
```

Feedback

```
CCSN slew        0        0.00%          0        0.00%       0             0
        300
Sensitivity delay 0       0.00%          0        0.00%       0             0
        300
Sensitivity slew 0        0.00%          0        0.00%       0             0
        300
Voltage range    0        0.00%          0        0.00%       0             8
        0
-------------------------------------------------------------------------------
----------
```

## LIBCHK-354

(information) Table of characterization table trend analysis.

### Description

This information message presents a table of single characterization table trend analysis results for such as delay, slew, driver, receiver and constraint models. Along with the detailed reports in the table, a summary is reported at the end of the report with Number of cells checked 10 ;# cells with the specified group Number of cells conformed 6 ;# cells conforming to the criteria Cell conforming rate 60% Number of trends checked 435 ;# total number of trends in the specified group Number of trends conformed 18 Trend conforming rate 4.13793% ;# number of trends conforming to the criteria

The trends include the following:

```
/       monotonously increasing
\\\       monotonously decreasing
^       non-monotonous up
V       non-monotonous down
-       flat
M       Multiple peaks
W       Multiple troughs
N       2 peaks with 2nd not turning low
```

### What Next

This is an information message only. No action is required.

However, if the trend indicates that the values do not increase as slew/load indices, please check that the characterizations have been done correctly.

### Examples

The following is an example table:

```
Table of characterization table trend analysis (LIBCHK-354)
index_1:    related_pin_transition
index_2:    output_net_capacitance
```

Feedback

```
-------------------------------------------------------------------------
---------------------------
Cell name   pin/subgroup  timing_type    when    Group
 index_1,index_2   values         Trend
-------------------------------------------------------------------------
---------------------------
sdffcqs     q/clk          min_pulse_width        fall_constraint
                                                          1.11,*
        "66.528,..."   V
sdffcqs     q/clrz         clear                  cell_fall   1.11,*
        "50.8109,..."  /
-------------------------------------------------------------------------
---------------------------


Below table is for sensitivity check
Cell name   pin/subgroup when        timing_type Group       param_name
 pert_value    index_1  index_2     values                Trend
-------------------------------------------------------------------------
-----------------------------------------------
cell1       A1/receiver_capacitance
                       A2&!B
 receiver_capacitance_rise

 delvtrandn_mos_lvt

 "-0.01,0,0.01"                     "1.0005,0,0.000853384" V

cell1       ZN/A1                  combinational
                                              rise_transition

 ids0multn_mos_lvt

 "-0.1,0,0.1" 0.0026    0.00172     "8.588e-05,0,2.19e-05" V


param_name is attribute under sensitivity model.
pert_value is perturbation_value under sensitivity model for same
 param_name. 0 is for nominal.
values is points extract from same param_name and different pert_value.
 0 is for nominal value.
For receiver capacitance, index_1 and index_2 are empty.
For NLDM value, the values are for each combination of slew/load.
The table shows all the sensitivity data which does not satisfy monotonic
 increase or decrease.
```

## LIBCHK-355

(warning) Table of characterization table bound analysis.

Feedback

### Description

This warning message presents a table of single characterization table value bound analysis results for delay, slew, driver, receiver, constraint and power tables. The bound include upper bound (max value) and lower bound (min value).

The violation (table values out of bound) will be triggered and reported when 1) table-value > upper bound or 2) table-value < lower bound

If no violate found for upper criteria or lower criteria, '-' will be reported, otherwise the peak value (largest violation for upper criteria or smallest violation for lower criteria, or the larger of the absolute values if both violated) will be reported in the table.

The index field indicate the grid point that peak violation located, and the '*' in index field indicate which index direction is scaned during the analysis.

### What Next

This warning message will show the value tables that have value out of the upper bound or lower bound. User should check the value and find out if characterization have done correctly.

### Examples

The following is an example table:

```
Table of characterization table bound analysis (LIBCHK-355)
-------------------------------------------------------------------------
----------------------------------------
Cell name          pin/related_pin  timing_type         when   Group
 index_1, index_2    Min value  Max value
-------------------------------------------------------------------------
----------------------------------------
T11ZNRBN1C1F4ZSE   IO/IN            3_state_enable_fall
 CFO&!CHDRV&!IE&NRST
                                                              cell_fall

 "1.5, 10.9009*"     -          3.81301

 "2.5, 10.9009*"     -          3.85745
-------------------------------------------------------------------------
----------------------------------------
```

## LIBCHK-356

(warning) Table of characterization table slope analysis.

Feedback

### Description

This warning message presents a table of single characterization table value slope analysis results for delay, slew, driver, receiver, constraint and power tables. The slope include the max slope and min slope, and defined as:

```
Slope=((V2-V1)/(S2-S1))*(Srange/Vrange)
V2, V1: the values of two adjacent points in the table;
S2, S1: the index values of two adjacent points in the table;
Vrange = V_largest - V_smallest
Srange = S_largest - S_smallest
```

When checking the slope, violation will be triggered and reported when: 1) curve-slope > max slope or 2) curve-slope < min slope

If no violate found for upper criteria or lower criteria, '-' will be reported, otherwise the peak value (largest violation for upper criteria or smallest violation for lower criteria, or the larger of the absolute values if both violated) will be reported in the table.

The index field indicate the grid point that peak violation located, and the '*' in index field indicate which index direction is scaned during the analysis.

### What Next

This warning message will show the value tables that have data slope out of the max slope or min slope. User should check the value and find out if characterization have done correctly.

### Examples

The following is an example table:

```
Table of characterization table slope analysis. (LIBCHK-356)
-------------------------------------------------------------------------------
----------------------------------------
Cell name          pin/related_pin  timing_type         when   Group
 index_1, index_2    Min slope  Max slope
-------------------------------------------------------------------------------
----------------------------------------
T11ZNRBN1C1F4ZSE   OUT/IE           combinational              cell_rise

 "0.05, 2.72*"       -          5.44444
-------------------------------------------------------------------------------
----------------------------------------
```

## LIBCHK-357

(information) Table of characterization table index analysis.

Feedback

### Description

This information message presents a table of single characterization table index analysis results for delay, slew, driver, receiver and constraint models. It lists all the index tables that meet index[i+1]/index[i] > x-factor specified.

### What Next

This is an information message only. However, if the index analysis indicates that the index values increase extremely, the accuracy may suffer.

### Examples

The following is an example table:

```
Table of characterization table index analysis (LIBCHK-357)
index_1:    related_pin_transition
index_2:    output_net_capacitance
-------------------------------------------------------------------------
--------------------------------
Cell name     pin/related_pin  timing_type    when   Group/attribute
 value                    Max ratio
-------------------------------------------------------------------------
--------------------------------
T11ZNRBN1C1F4ZSE
              OUT/IE           combinational  CFO&!CHDRV&!IN&IO&NRST)
                                                    cell_rise/index_2
 "0.03,2.72,...           90.6667
-------------------------------------------------------------------------
--------------------------------
```

---

## LIBCHK-358

(warning) Table of interpolation accuracy analysis

### Description

This interpolation accuracy analysis is performed by fitting the rows and columns of a 2-D NLDM timing model (delay/slew/constraint) with (bi)cubic and linear models, and comparing the values at mid index points with one index fixed. For example, in a 7x7 table, there would be 6 checks along each row and 6 checks along each column at mid index points. If the difference between the two models exceeds the specified tolerances, the mid grid (index values) and the interpolated values will be reported.

### What Next

If the interpolation difference is too large, the library needs to be enhanced by either increasing table size or using better spacing between the indexes.

For PVT interpolation using quadratic models, please specify enough libraries with >=3 voltage values on a rail/voltage_name while other dimensions are fixed. To exclude

Feedback

rails not used for scaling/pvt_interpolation or dependent rails (most likely a bias pin), please use create_scaling_lib_group -exclude_voltage_names {VBP}, for example. PVT interpolation works on NLDM delay, transition, constraint tables. When no quadratic interpolation is derived but linear one is in the case of only 2 voltage values on a rail/voltage_name, a dash (-) will be output in the related fields (Qquadratic, Absolute and Relative errors) in LIBCHK-358 and csv file.

**Examples**

The following is an example table:

```
Warning: Table of interpolation accuracy analysis (LIBCHK-358)
index_1:  related_net_transition
index_2:  total_output_net_capacitance
------------------------------------------------------------------------
---------------------------------------

 Interpolation                 Error
Cell name  pin/related_pin  timing_type  when   Group
 index_1,index_2 (Bicubic) (Linear)  Absolute Relative
------------------------------------------------------------------------
---------------------------------------
sdffcqs_f2 q/clk            rising_edge  !scan  cell_rise  1.11,3.71
  0.06724   0.07724   0.0100   14.87%
------------------------------------------------------------------------
---------------------------------------

For -analyze {pvt_interpolation},
Warning: Table of interpolation accuracy analysis (LIBCHK-358)
------------------------------------------------------------------------
------------------------------------------------------------------------
---------------
Cell name   pin/related_pin when        timing_type Group/Attribute
 index_1     index_2     Temperature VDD1        Quadratic   Linear
 Absolute    Relative
------------------------------------------------------------------------
------------------------------------------------------------------------
---------------
AN2D0       Z/A2                        combinational

 fall_transition/values

 0.0011      0.00025     -40         0.55        0.0348849   0.0467479
 0.011863    34.01%
------------------------------------------------------------------------
------------------------------------------------------------------------
---------------
```

## LIBCHK-359

(warning) Table of partial voltage ranges on timing arcs

Feedback

### Description

This message presents a table of partial voltage ranges/swings on timing arcs. It lists all the timing arcs with partial voltage ranges that exceed the voltage threshold/tolerance specified.

### What Next

If a driver cell whose output voltage never reaches full voltage swing and this cell drives a sensitive cell, there will be an issue.

### Examples

The following is an example table:

```
Warning: Table of partial voltage ranges on timing arcs (LIBCHK-359)
------------------------------------------------------------------------
------------------
Cell name   pin/related_pin timing_type    when   VDD   Voltage range
 Relative error Type
------------------------------------------------------------------------
------------------
XOR3X05LMP  z/a             combinational  !a     0.99  0.980
 -1.01%         CCSN
sdffcqs_f2  q/clk           rising_edge    !a     1.20  1.122
 -6.50%         CCST
------------------------------------------------------------------------
------------------
```

## LIBCHK-360

(information) Logic library inconsistencies found for MCMM.

### Description

This information message occurs when any inconsistency in MCMM is found during cross checking the libraries. MCMM requires the following:

```
a)      Same pin (pg_pin) names, directions, types, functions and
 voltage_name
b)      Same in/output_signal_level for rail-based library and same
 related_power_pin/related_ground_pin for pg_pin-based library
c)      Same derived pin_number attribute values
d)      Different PVT values for different characterizations between
 two libraries for different operation conditions, i.e. libraries with
 same-name cells have different nominal PVT values.
e)      Same cell attributes: dont_use, dont_touch and black box
f)      Same power_down_function for output/inout pins
g)      Power data (leakage_power etc.) exist for all opconds with
 non-zero value
h)      Same threshold_voltage_group attribute values for same cells
 across corner libraries
```

Feedback

### What Next

You are advised to go back to the check report to see inconsistent results and resolve them as directed by the associated LIBCHK man pages.

If you see "inconsistencies found" in the summary report, go to the related tables for details in

```
LIBCHK-300
LIBCHK-301
LIBCHK-310
LIBCHK-311
LIBCHK-312
LIBCHK-320
LIBCHK-321
LIBCHK-322
LIBCHK-323
LIBCHK-330
LIBCHK-331
LIBCHK-340
```

Any of the related LIBCHK messages will lead to the conclusion of inconsistencies found.

### See Also

- LIBCHK-300
- LIBCHK-301
- LIBCHK-310
- LIBCHK-311
- LIBCHK-312
- LIBCHK-320
- LIBCHK-321
- LIBCHK-322
- LIBCHK-323
- LIBCHK-330
- LIBCHK-331
- LIBCHK-340

---

## LIBCHK-361

(information) Logic library inconsistencies found for UPF.

Feedback

### Description

This message occurs after the checking of the libraries is completed. When any inconsistency in the UPF flow is found during the checking, this message appears. UPF requires the following:

```
a)      Same pin (pg_pin) names, directions, types, functions and
 voltage_name
b)      Same in/output_signal_level for rail-based library and same
 related_power_pin/related_ground_pin for pg_pin-based library
c)      Same derived pin_number attribute values
d)      Different PVT values for different characterizations
e)      NLDM or CCS model under different voltage conditions exists
f)      Level shifter, isolation, retention and switch cells exist with
 consistent and complete attributes
g)      voltage_map and pg_pin groups exist in multi P/G pg_pin based
 library with consistent attributes
h)      Same power_down_function for output/inout pins
i)      Power data (leakage_power etc.) exist for all opconds with
 non-zero value
j)      power_down_function must be specified on all output/inout pins
k)      always_on inverter and always_on buffer cells must be specified
l)      input/output_voltage_range specified on the same pin must have
 identical values among all PVT libraries
m)      PG/PM checking for on-the-fly UPF library update
n)      Same threshold_voltage_group attribute values for same cells
 across corner libraries
o)      max_input_delta_overdrive/underdrive_high specified on the same
 pin must have identical values among all PVT libraries
```

### What Next

You are advised to go back to the above check report to see inconsistent results and resolve them as directed by the associated LIBCHK man pages.

If you see "inconsistencies found" in the summary report, go back to the related tables for details in

```
LIBCHK-300
LIBCHK-301
LIBCHK-302
LIBCHK-310
LIBCHK-311
LIBCHK-312
LIBCHK-320
LIBCHK-321
LIBCHK-322
LIBCHK-323
LIBCHK-330
LIBCHK-331
LIBCHK-332
LIBCHK-340
```

Feedback

Any of the related LIBCHK messages will lead to the conclusion of inconsistencies found.

**See Also**

- LIBCHK-300
- LIBCHK-301
- LIBCHK-310
- LIBCHK-311
- LIBCHK-312
- LIBCHK-320
- LIBCHK-321
- LIBCHK-322
- LIBCHK-323
- LIBCHK-330
- LIBCHK-331
- LIBCHK-332
- LIBCHK-340

---

## LIBCHK-362

(information) Logic library inconsistencies found for %s scaling.

**Description**

This information message occurs after the cross checking of the libraries is completed. When the tool finds any inconsistencies, such as CCS timing scaling, CCS noise scaling, or power scaling during cross checking, this message appears.

**What Next**

This is an information only message. No action is required.

However, you are advised to go back to the above check report to see the inconsistent results and resolve them as directed by the associated LIBCHK man pages.

When you see "inconsistencies found for CCS timing scaling" in the summary report, go back to the related tables for details in

```
LIBCHK-300
LIBCHK-301
```

Feedback

```
LIBCHK-302
LIBCHK-310
LIBCHK-311
LIBCHK-320
LIBCHK-321
LIBCHK-330
LIBCHK-331
LIBCHK-332
LIBCHK-340
LIBCHK-350
```

Any of the related LIBCHK man pages will lead to this conclusion.

When you see "inconsistencies found for CCS noise scaling" in the summary report, go back to the related tables in

```
LIBCHK-300
LIBCHK-301
LIBCHK-302
LIBCHK-310
LIBCHK-311
LIBCHK-320
LIBCHK-321
LIBCHK-330
LIBCHK-331
LIBCHK-340
```

Any of the related man pages (LIBCHK) will lead to this conclusion.

When you see "inconsistencies found for power scaling" in the summary report, go back to the related tables in

```
LIBCHK-300
LIBCHK-301
LIBCHK-302
LIBCHK-310
LIBCHK-311
LIBCHK-320
LIBCHK-321
LIBCHK-330
LIBCHK-331
LIBCHK-341
```

Any of the related LIBCHK man pages will lead to this conclusion.

**See Also**

- LIBCHK-300
- LIBCHK-301
- LIBCHK-302

Feedback

- LIBCHK-310
- LIBCHK-311
- LIBCHK-320
- LIBCHK-321
- LIBCHK-330
- LIBCHK-331
- LIBCHK-332
- LIBCHK-340
- LIBCHK-350

---

## LIBCHK-363

(information) Logic library inconsistencies found for %s comparison.

### Description

This information message occurs after cross checking between the two libraries is completed. When any the tool finds inconsistencies in the Liberty syntax construct comparison or value comparison during cross checking, this message will appears. In construct comparison, the missing or existing data is reported. In value comparison, the characterization values are reported.

### What Next

This is an information only message. No action is required.

However, you are advised to go back to the above check report to see inconsistent results and resolve them as directed by the associated LIBCHK man pages.

When you see "inconsistencies found" for the construct or value comparison in the summary report, go back to the related tables for details in

```
LIBCHK-300
LIBCHK-301
LIBCHK-310
LIBCHK-311
LIBCHK-312
LIBCHK-320
LIBCHK-321
LIBCHK-330
LIBCHK-331
LIBCHK-340
LIBCHK-341
```

Feedback

Any of the related LIBCHK man pages will lead to this conclusion.

**See Also**

- LIBCHK-300
- LIBCHK-301
- LIBCHK-310
- LIBCHK-311
- LIBCHK-312
- LIBCHK-320
- LIBCHK-321
- LIBCHK-330
- LIBCHK-331
- LIBCHK-340
- LIBCHK-341

---

## LIBCHK-364

(information) Logic library inconsistencies found for %s validation.

**Description**

This information message occurs after validation checking between the two libraries or between two models (NLDM versus CCST/CCSN) is completed. When the tool finds any inconsistency in validation during validation checking, this message appears. If grid points are mismatched, the tool will not validate the delay, slew, and constraint values. In this case, you can ignore this message.

Missing related or checked attributes, e.g. receiver_capacitance in set_check_library_options -validate receiver_capacitance is also counted as failing.

**What Next**

Please review the check report to see the inconsistent results and resolve them as directed by the related LIBCHK man pages.

When you see "inconsistencies found" in the summary report, review the related tables for details in the following messages:

```
LIBCHK-300
LIBCHK-301
```

Feedback

```
LIBCHK-310
LIBCHK-311
LIBCHK-320
LIBCHK-321
LIBCHK-330
LIBCHK-331
LIBCHK-350
```

Any of the related LIBCHK man pages will lead to this conclusion.

### See Also

- LIBCHK-300
- LIBCHK-301
- LIBCHK-310
- LIBCHK-311
- LIBCHK-320
- LIBCHK-321
- LIBCHK-330
- LIBCHK-331
- LIBCHK-350

---

## LIBCHK-365

(information) Statistics of all characterization data per library

### Description

This information table reports the overall statistical analysis results of library checking with one library (group) in one row. This only includes characterization model types listed in LIBCHK-353: delay, slew, constraint, receiver_cap, ccs driver, voltage range, ccsn (dc_current), internal_power, and leakage_power. The columns in the table include the following:

- `Library name`
- `Abs/Rel mean error: Absolute/Relative mean error`
- `Abs std dev: Absolute/Relative standard deviation`
- `Cell pass rate: ratio of number of characterization model check passed cells to number of total cells checked`

Feedback

### What Next

If the mean error is far from 0, the models are not so accurate. If the standard deviation is significantly large, the models are dispersed, i.e. the deviation is significant. In any case, small numbers of mean error and deviation, close to 0, should be expected for high accuracy.

### Examples

The following is an example:

```
Information: Statistics of all characterization data per library
 (LIBCHK-365)
------------------------------------------------------------------------
-------------
Library name    Abs mean error Rel mean error  Abs std dev Rel std dev
 Cell pass rate
------------------------------------------------------------------------
-------------
library_typ     0.000344       1.04%           0.00176     16.86%
 79.00%
test_typ        0.000688       2.08%           0.00352     33.72%
 56.12%
------------------------------------------------------------------------
-------------
```

---

## LIBCHK-366

(error) Logic library inconsistencies found for single mode macros.

### Description

This message occurs after the checking of the single mode libraries is completed. When any incorrect data or inconsistency in the single mode libraries such as ETMs is found during the checking, this message appears. Single mode cells including ETM require the following:

```
a)      Each single model cell, e.g. ETM, has to have timing_model_type
 (for ETM, the value must be extracted) and is_macro_cell defined
b)      The single mode libraries for different modes have to have the
 same PVT
c)      Consistencies in specific library/cell/pin level attributes
 across single mode libraries at same PVTs
For details, please refer to related LC document or App Note.
The checks in check_library for ETM is consistent with merge_models
 command in PrimeTime.
Please note that for other single model cells than ETM,
 timing_model_type : extracted is not required.
```

Feedback

### What Next

You are advised to go back to the above check report to see inconsistent results and resolve them as directed by the associated LIBCHK man pages.

```
LIBCHK-300
LIBCHK-301
LIBCHK-310
LIBCHK-311
LIBCHK-320
LIBCHK-321
LIBCHK-322
LIBCHK-330
LIBCHK-331
```

Any of the related LIBCHK messages will lead to the conclusion of inconsistencies found.

---

## LIBCHK-367

(error) Logic library inconsistencies found for cell electromigration.

### Description

This message occurs after checking of cell electromigration/em_max_toggle_rate across corner libraries. When structural inconsistency in the EM groups is found during the checking, this message appears.

### What Next

You are advised to go back to the above check e.g. report LIBCHK-321 to see inconsistencies across the libraries and resolve them as directed by the associated LIBCHK man pages. If the electromigration table is not characterized for the expected PVT corners, in PrimePower, user can use set_em_scaling_factor command to set a scaling factor on the cell max toggle rate; however, this may still compromise the accuracy.

### Examples

```
Warning: List of pins with mismatched or missing attributes in logic
 libraries (LIBCHK-321)
------------------------------------------------------------------------
------------------
                      Subgroup/              Attribute value
 Error
Cell name Pin name    Attribute           lib#1       lib#2
 Absolute  Relative
------------------------------------------------------------------------
------------------
ad3       y           em_max_toggle_rate/current_type:2
                                          existing    missing
ad3       y           em_max_toggle_rate/current_type:2/current_type
                                          peak        missing
```

Feedback

```
ad3       y           electromigration/related_pins:a
                                          existing    missing
------------------------------------------------------------------------
------------------
```

## LIBCHK-368

(error) Base and model library inconsistencies found for sensitivity.

### Description

This message occurs when one or more of the following inconsistencies between base library and model file are found after checking for base library and model sidefile with sensitivity data: (1) inconsistent time/cap/power/energy units and slew_derates (2) mismatched/missing pin names Such inconsistencies will impact tools consuming such libraries.

### What Next

You are advised to go back to the above check e.g. report LIBCHK-321 to see inconsistencies across the libraries and resolve them as directed by the associated LIBCHK man pages.

### Examples

```
Error: Base and model library inconsistencies found for sensitivity
 (LIBCHK-368)
```

## LIBCHK-368w

(warning) Base and model library inconsistencies found for sensitivity.

### Description

This message occurs when one or more of the following inconsistencies between base library and model file are found after checking for base/main library and model sidei file with sensitivity data: (1) inconsistent time/cap/power/energy units and slew_derates (2) mismatched/missing pin names (3) missing cells in either base lib or side file (4) inconsistences for timing arcs: 1 NLDM table in main lib, 1 NLDM table in the side file and N sensitivity table in the side file (5) inconsistent table indices in NLDM and receiver_cap, respectively (6) NLDM nominal value in the side file has larger than 1ps/1% gap vs. main library NLDM (7) CCS pin-based receiver cap value in the side-file has larger than 1fF/1% gap vs. main library receiver cap. (8) missing sensitivity block for timing arc or receiver_capacitance (9) whole sensitivity table has values of 0's

Such inconsistencies will impact tools consuming such libraries.

Feedback

### What Next

You are advised to go back to the above check e.g. report LIBCHK-321 to see inconsistencies across the libraries and resolve them as directed by the associated LIBCHK man pages.

### Examples

```
Warning: Base and model library inconsistencies found for sensitivity
 (LIBCHK-368w)
```

---

## LIBCHK-369

(error) Missing aged values or invalid aged values found.

### Description

This message occurs when missing aged values or existing zero values. Missing values includes missing aged models, missing aged table values, or inconsistent indexes.

### What Next

Please correct the invalid values or check characterization setting.

### Examples

```
Error: Missing aged values or invalid aged values found. (LIBCHK-369)
```

---

## LIBCHK-369i

(information) Aged values exceeding certain range from nominal or identical to nominal found

### Description

This message occurs when aged table values exceeding a tolerant range from nominal values or very close to nominal values.

### What Next

This is an information message.

### Examples

```
Information: Aged values exceeding certain range from nominal or
 identical to nominal found (LIBCHK-369i)
```

---

## LIBCHK-369w

(warning) Non-monotonicity aged values with lifetime profiles found.

Feedback

### Description

This message occurs when aged table values violating certain monotonicity table trend.

### What Next

Non-monotonicity aged values should be correct.

### Examples

```
Warning: Non-monotonicity aged values with lifetime profiles found.
 (LIBCHK-369w)
```

---

## LIBCHK-370

(warning) Table of non-monotonicity with lifetime profiles

### Description

This message presents a table of monotonicity check results for aged table values. The variable is one of parameters in lifetime profiles.

The trends include the following:

```
/      monotonously increasing
\\\      monotonously decreasing
^      non-monotonous up
V      non-monotonous down
-      flat
M      Multiple peaks
W      Multiple troughs
N      2 peaks with 2nd not turning low
```

### What Next

Aged values violating monotonicity criteria should be corrected.

### Examples

The following is an example table:

```
Warning: Table of non-monotonicity with lifetime profiles (LIBCHK-370)
------------------------------------------------------------------------
------------------------------------------------------------------------
-----------------
Cell name   pin/related_pin  when      timing_type  Group/Attribute
 variable_name       variable_values    index_1  index_2   values
             Trend
------------------------------------------------------------------------
------------------------------------------------------------------------
-----------------
cell1       ZN/A1           A2&!B
 aged(profile_group_1)/cell_rise
```

Feedback

```
         stress_temperature  "135, 136, 137"     0.0026  0.00172
 "1.0005,0.0035,0.00853384"  V
cell1       ZN/A1           A2&!B
 aged(profile_group_2)/cell_fall

         stress_voltage      "0.8, 0.9, 1.05"    0.0029  0.00193
 "2.0005,2.0035,0.00853384"  ^
-------------------------------------------------------------------------
-------------------------------------------------------------------------
-------------------

The table shows all the aged data which does not satisfy monotonic
 increase or decrease.
```

## LIBCHK-374

(error) List of aged tables with zero values

### Description

This table reports aged NLDM table whose values contain zero values. This table has the following columns: Cell name, pin/related_pin, timing_type, when, Group, Attribute, lib#1

### What Next

It is recommended that zero aged values should be corrected.

### Examples

```
Error: List of aged tables with zero values (LIBCHK-374)
-------------------------------------------------------------------------
---------------------------
Cell name pin/related_pin timing_type   when  Group     Attribute
 lib#1
-------------------------------------------------------------------------
---------------------------
AIOI21    ZN/A1           combinational !A2
 aged(lifetime_profile_1)/cell_rise
                                                      values
 "-1.228e-05,-1.165e-05,..."
CQAN2X    Z/B             combinational !A1
 aged(lifetime_profile_2)/cell_fall
                                                      values
 "-1.328e-05,-1.665e-05,..."
-------------------------------------------------------------------------
---------------------------
```

### Examples

```
Error: List of aged tables with zero values (LIBCHK-374)
```

Feedback

## LIBCHK-375

(warning) List of invalid LLE leakage_power

### Description

This table reports invalid values of LLE leakage_power. This table has the following columns: Cell name, Group, when, related_pg_pin, Subgroup, Attribute, lib#1, Type

The Type column lists the invalid type. There are five invalid types for LLE leakage_power checks: partial zero, all zero, deviated, increasing, decreasing. (1) partial zero: in a lle_delta_leakage_power group, if one value is zero and the other is non-zero, the Type is partial zero. (2) all zero: in a lle_delta_leakage_power group, if all values are zero, the Type is all zero. (3) deviated: a value out of maximum absolute threshold and relative percentage threshold. The relative percentage is lle delta leakage power value to its associated nominal leakage power value. The absolute threshold max_lle_delta_leakage_value and relative threshold max_lle_delta_leakage_to_nominal_ratio can be specified by -criteria option of set_check_library_options. (4) increasing: the leakage_power value is monotonously increasing according to related_params. (5) decreasing: the leakage_power value is monotonously decreasing according to related_params.

The Type "increasing" and "decreasing" are for monotonicity checks. The correct monotonicity: (1) If related_params is vth, the leakage_power value should be decreasing according to related_params. (2) If related_params is u0, the leakage_power value should be increasing according to related_params. In lle_delta_leakage_power model, attribute "index_1" is delta related_params and attribute "values" is delta leakage_power. (1) Decreasing monotonicity is equal to: positive index_1 and negative value; negative index_1 and positive value. (2) Increasing monotonicity is equal to: positive index_1 and positive value; negative index_1 and negative value. For monotonicity checks, note that if the delta leakage power value is smaller than certain absolute and relative tolerance, the value will not be checked and reported. The default absolute tolerance is 1e-6 and relative tolerance is 0.01. The tolerance can be specified by user. e.g. set_check_library_options -tolerance {leakage_lle 0.02 1e-5}

### What Next

It is recommended that invalid LLE leakage power values should be corrected.

### Examples

```
Warning: List of invalid LLE leakage_power (LIBCHK-375)
-------------------------------------------------------------------------------
-------------------------------------
Cell name   Group          when        related_pg_pin Subgroup
 Attribute   index_1       lib#1       Type
-------------------------------------------------------------------------------
-------------------------------------
```

Feedback

```
AIOI21      leakage_power  !CK&!E      VDD
 lle_delta_leakage_power(vth,tr1)
                                                                    values
     -0.03         -0.0147613  deviated
AIOI21      leakage_power  !CK&!E      VDD
 lle_delta_leakage_power(vth,tr1)
                                                                    values
     -0.03         -0.0147613  increasing
AIOI21      leakage_power  !CK&!E      VDD
 lle_delta_leakage_power(vth,tr2)
                                                                    values
     "-0.03,0.03"  "0,-1.32462e-08"

                              partial zero
AIOI21      leakage_power  !CK&E       VDD
 lle_delta_leakage_power(vth,tr3)
                                                                    values
     "-0.03,0.03"  "0,0"       all zero
--------------------------------------------------------------------------------
--------------------------------------
```

### Examples

```
Warning: List of invalid LLE leakage_power (LIBCHK-375)
```

## LIBCHK-378

(warning) Invalid values found in LLE delta leakage power models.

### Description

This message occurs when one or more of the following invalid values found in lle delta leakage power models: (1) partial or all zero values (2) values out of maximum thresholds (3) values violating certain monotonicity according to related params vth or u0

### What Next

You are suggested to check the detailed report in table LIBCHK-375 which relates to the above three kinds of invalid values.

### Examples

```
Warning: Invalid values found in LLE delta leakage power models
 (LIBCHK-378)
```

## LIBCHK-400

(error) The %s of the file is %s.

Feedback

### Description

This message occurs when any property of the file is incorrect.

### What Next

Please check the properties of the file such as format.

### Examples

```
Error: The format of the file is wrong. (LIBCHK-400)
```

## LIBCHK-408

(information) Library file and CTPM file grouping table

### Description

This message is for reporting list of library and CTPM file pairs with group number.

### What Next

For checking consistency between sensitivity libraries and ctpm files, one library and one ctpm are grouped together according to the specified order. For example, if there are specified libraries sens1.db, sens2.db and ctpm files ctpm1, ctpm2, one group is sens1.db and ctpm1, the other group is sen2.db and ctpm2.

### Examples

```
set_check_library_options -sensitivity_lib_vs_ctpm
check_library -logic_library_name "sens1.db sens2.db" -ctpm_file "ctpm1,
 ctpm2"

Information: Library file and CTPM file grouping table (LIBCHK-408)
--------------------------------------------------------------------------
---------------------
Group#   Library name                  Library file name
 CTPM file name
--------------------------------------------------------------------------
---------------------
1        ssgnp_0p600v_125c_cworst_t    side_file_125c.db
 pipeclean_125.ctpm
2        ssgnp_0p600v_m25c_cworst_t    side_file.db
 pipeclean.ctpm
--------------------------------------------------------------------------
---------------------
```

## LIBCHK-410

(warning) Cell %s is not found in %s.

Feedback

### Description

This message occurs when the cell is not found in the specified file.

### What Next

Please check if the cell name is correct.

### Examples

```
Warning: Cell A0211 is not found in the ctpm file. (LIBCHK-410)
```

---

## LIBCHK-430

(warning) List of timing arcs missing in sensitivity libraries or ctpm files

### Description

This message is for reporting timing arcs missing in the sensitivity libraries and ctpm files. For -sensitivity_lib_vs_ctpm check, if timing arcs in ctpm are not found in db, report missing in sens_db, if timing arcs in db are not found in ctpm, report missing in ctpm.

### Examples

```
List of timing arcs missing in sensitivity libraries or ctpm files
--------------------------------------------------------------------------------
--------
Cell name     pin/related_pin   timing_sense      when         sens_db
 ctpm
--------------------------------------------------------------------------------
--------
A0211         C/Z               positive_unate    !A1&!A2&!C   existing
 missing
--------------------------------------------------------------------------------
--------
```

### Examples

```
Warning: List of timing arcs missing in sensitivity libraries or ctpm
 files(LIBCHK-430)
```

---

## LIBCHK-431

(error) Failed to collect timing arc data from %s.

### Description

This message occurs when no timing arc data is collected from specified file.

Feedback

### What Next

Please check if the file has timing arcs and the format is correct to be parsed.

### Examples

```
Error: Failed to collect timing arc data from pipeclean.ctpm.
 (LIBCHK-431)
```

## LIBCHK-431w

(warning) Failed to collect timing arc data from %s.

### Description

This message occurs when no timing arc data is collected from specified file.

### What Next

Please check if the file has timing arcs and the format is correct to be parsed.

### Examples

```
Warning: Failed to collect timing arc data from pipeclean.ctpm.
 (LIBCHK-431w)
```

# LIBG

## LIBG-1

(error) The Synopsys database is corrupted. The library is not created.

### Description

A file is read in but no library is found in the file. It is most likely that the file is empty.

The following is an example message:

```
Error: The Synopsys database is corrupted. The library is not created.
 (LIBG-1)
```

### What Next

Check you library file to make sure it is a valid library.

## LIBG-2

(error) An invalid '%s' function string in the '%s' cell.

Feedback

### Description

This error message occurs when the string given is not valid for representing a Boolean logic defined in Synopsys format.

The following example shows an instance where this message occurs: This example causes an error message because the closing parenthesis is missing in the function string:

```
cell(libg2) {
  area : 1 ;
  pin (I0) {
        direction : input ;
        capacitance : 0 ;
  }
  pin ("Y") {
        direction : output ;
        function : "(I0" ;
        timing() {
                related_pin     : "I0" ;
                intrinsic_rise  : 1.0 ;
                rise_resistance : 0.0 ;
                intrinsic_fall  : 1.0 ;
                fall_resistance : 0.0 ;
        }
  }
```

The resulting error message lists the cause of the error:

```
Error: Line 400, An invalid '(IO' function string in the 'libg2' cell.
 (LIBG-2)
```

### What Next

See the Library Compiler documentation to find out the valid format of the string.

Constant Boolean expressions do not allow multiple constants with operators; for example, the following expressions are not allowed:

```
function/when : "10" ;

function/when : "1 ^ 0" ;
```

Specify constant Boolean expressions as follows:

```
function/when : "0" ;

function/when : "1" ;
```

---

## LIBG-3

(error) A bad '%s' pin name in the '%s' cell.

Feedback

### Description

This error gets detected during a *read_lib* and an *update_lib* commands. The pin name does not either exist or a previous error in any of the attributes of the pin group is detected. In this case the pin group is not recognized and the pin name is labeled as bad.

The following example shows an instance where this message occurs:

```
cell (libg3) {
  area : 4
  pin(A) {
    direction : input
    capacitance : 1
    fanout_load : 1.0
  }
  pin(EN) {
    direction : input
    capacitance : 1
    fanout_load : 1.0
  }
  pin (Z) {
    direction : output
    function : "A"
    three_state : "E";       /* <-- This is wrong */
    max_fanout : 10
 }
```

The following is an example message:

```
Error: Line 426, A bad 'E' pin name in the 'libg3' cell. (LIBG-3)
```

### What Next

Check your library for a wrong pin name in any related attribute or for an incorrect pin group.

---

## LIBG-4

(error) In the '%s' cell, the '%s' noninput pin \n \tcannot be used in the function.

### Description

The string associated with a *function* and *three_state* attributes should include only be of *input* or *inout* directions.

The following example shows an instance where this message occurs:

```
cell (libg4) {
  area : 4
  pin(A) {
    direction : input
    capacitance : 1
```

Feedback

```
    fanout_load : 1.0
  }
  pin(EN) {
    direction : input
    capacitance : 1
    fanout_load : 1.0
  }
  pin (Z1) {
    direction : output
    function : "A"
    max_fanout : 10
 }
  pin (Z) {
    direction : output
    function : "Z1"           /* <-- This is wrong */
    three_state : "Z1";       /* <-- This is wrong */
    max_fanout : 10
 }
```

The following is an example message:

```
Error: Line 431, In the 'libg4' cell, the 'Z1' noninput pin
        cannot be used in the function. (LIBG-4)
```

### What Next

Check your library for a wrong direction of the pin in the function attribute. Another possibility is to check the use of the wrong pin name.

---

## LIBG-5

(error) Cannot find the '%s' pin in the '%s' cell.

### Description

This error is issued when a pin name attached to a *related_pin* or a *related_bus_pins* attribute does not exist in the cell.

The following example shows an instance where this message occurs:

```
cell(libg5) {
  area : 1 ;
  pin (I0) {
        direction : input ;
        capacitance : 0 ;
  }
  pin ("Y") {
        direction : output ;
        function : "I0" ;
        timing() {
                related_pin     : "IO" ; /* typo zero replaced by letter
 O */
```

Feedback

```
                intrinsic_rise  : 1.0 ;
                rise_resistance : 0.0 ;
                intrinsic_fall  : 1.0 ;
                fall_resistance : 0.0 ;
        }
  }
```

The following is an example message:

```
Error: Line 403, Cannot find the 'IO' pin in the 'libg5' cell. (LIBG-5)
```

**What Next**

Check your library for a wrong pin name.

---

## LIBG-6

(error) Missing a related_pin for the '%s' pin timing group\n \tin the '%s' cell.

**Description**

This error is issued when the *related_pin* or the *related_bus_pins* attribute is missing in a timing group.

The following example shows an instance where this message occurs:

```
cell(libg6) {
  area : 1 ;
  pin (I0) {
        direction : input ;
        capacitance : 0 ;
  }
  pin ("Y") {
        direction : output ;
        function : "I0" ;
        timing() {
           frelated_pin : "I0" ; /* typo f inserted before attribute */
           intrinsic_rise       : 1.0 ;
           rise_resistance : 0.0 ;
           intrinsic_fall  : 1.0 ;
           fall_resistance : 0.0 ;
        }
  }
```

The following is an example message:

```
Error: Line 399, Missing a related_pin for the 'Y' pin timing group
        in the 'libg6' cell. (LIBG-6)
```

**What Next**

Check your library for a typo in the attribute or add it if it is missing.

Feedback

## LIBG-8

(error) In the '%s' cell, the '%s' input drives more than one function.

### Description

This error is issued when attribute three_state on a port has common input ports with attribute inside ff, ff_bank, latch, latch_bank, or seq group.

The following example shows an instance where this message occurs:

```
cell(libg8) {
         area : 6.50;
         pin(D TE) {
           direction : input;
           capacitance : 3.0;
         }
        pin(CLK) {
                direction : input;
                capacitance : 1.5;
        }
        ff("IQ","IQN") {
                next_state : "D";
                clocked_on : "CLK";
                preset     : "TE";
                }
        pin(Q) {
                direction : output;
                function : "IQ";
                three_state : "D";   /* wrong, D is shared with
 next_state. */
        }
        pin(QN) {
                direction : output;
                function : "IQN";
        }
      }
```

The following is an example message:

```
Error: Line 411, In the 'libg8' cell, the 'D' input drives more than one
 function. (LIBG-8)
```

### What Next

Either delete one of the attributes or associate a unique string to each of them.

## LIBG-10

(warning) Failed to recognize the functionality of cell '%s'.

Feedback

### Description

The warning message is to notify users that Library Compiler fails to synthesize a netlist structure representing functionality of this cell. In order to other cells to recognize the cell, user can specifiy the "user_function_class" attribute either in the .lib file or in Design Compiler.

Note, although the function id of the cell is unknown, it may not be black-box. That is why we are saying that it may not be recognized by Design Compiler.

This message can be used to identify a cell for which pin-class conflict exists. Pin-class is defined as the set of pins which can be swapped without changing the functionality of the gate and is used for DC mapping. In the following simple example, we have PO = ~(PAD * PI) and Y=PAD, and it will cause pin-class conflict because: After parsing the 1st function, we have pin_class(PAD) = pin(PI) because PAD and PI can be exchangable without changing the functionality of the pin PO. However it will break the functionality of the 2nd function cause Y=PAD and Y=PI are totally different.

When pin-class conflict happens, the function id of the cell is set to unknown.

The following example shows an instance where this message occurs:

```
cell(libg28) {
          area : 1;
          ...
          pin(PAD) {
            direction : input;
            ...
          }
          pin(PI) {
            direction : input;
            ...
          }

          pin(PO) {
            direction : output;
            capacitance : 0;
            ...
            function: "!(PAD * PI)"
          }
          pin(Y) {
            direction : output;
            capacitance : 0;
            ...
            function: "PAD"
          }
        }
```

The following is an example message:

```
Warning: Line 34,   Failed to recognize the functionality of cell '%s'.
 (LIBG-10)
```

Feedback

### What Next

Check your library.

---

## LIBG-16

(warning) The '%s' Pin/bus on the '%s' cell has no 'function' attribute.\n \tThe cell becomes a black box.

### Description

This warning is issued when any output of a given cell does not have a valid function(missing or wrong information associated with the function of the pin). A function can be defined either by a *function* attribute, a *three_state* attribute, a *memory_read* group, or an *internal_node* attribute.

The following example shows an instance where this message occurs:

```
cell(libg16) {
  area : 1 ;
  pin (I0) {
        direction : input ;
        capacitance : 0 ;
  }
  pin ("Y") {
        direction : output ;
/*      function : "I0" ;  */
        timing() {
                related_pin     : "I0" ;
                intrinsic_rise  : 1.0 ;
                rise_resistance : 0.0 ;
                intrinsic_fall  : 1.0 ;
                fall_resistance : 0.0 ;
        }
  }
```

The following is an example message:

```
Warning: Line 39, The 'Y' Pin/bus on the 'libg16' cell has no 'function'
 attribute.
        The cell becomes a black box. (LIBG-16)
```

### What Next

Check your library for missing or wrong attributes.

---

## LIBG-17

(error) The 'function' of the '%s' pin/bus on the '%s' cell\n \tcan only be '%s' or '%s'.

Feedback

### Description

This error is issued when either the noninverted output (first variable) or the inverted output (second variable) of a sequential model, defined by a ff, a ff_bank, a latch, or a latch_bank group are not assigned to a primary output.

A function statement of each primary output must include either the first variable or the second variable.

The following example shows an instance where this message occurs:

```
cell(libg17) {
        area : 6.50;
        pin(CLK) {
                direction : input;
                capacitance : 1.5;
        }
        pin(SET) {
                direction : input;
                capacitance : 3.0;
        }
        ff("IQ","IQN") {
                next_state : "IQ'";
                clocked_on : "CLK";
                preset     : "SET'";
        }
        pin(Q) {
                direction : output;
                function : "IQ1";       /* wrong, should be IQ */
        }
        pin(QN) {
                direction : output;
                function : "IQN";
        }
      }
```

The following is an example message:

```
Error: Line 23, The 'function' of the 'Q'  in/bus on the 'libg17' cell
        can only be 'IQ' or 'IQN'. (LIBG-17)
```

### What Next

Make the appropriate correction in your library, as indicated in the error message. For more information on sequential models, refer to the *Library Compiler Reference* and *User Guide Manuals*.

---

## LIBG-18

(information) The '%s' equation on the '%s' cell is not recognized.

Feedback

### Description

This information is issued when the function is not recognized. The function can be a complex *state_function*, a corrupted string in a *three_state*, a *clocked_on*, or a *clocked_on_also* attribute.

The following example shows an instance where this message occurs:

```
cell(libg18) {
        area : 6.50;
        pin(CLK) {
                direction : input;
                capacitance : 1.5;
        }
        pin(SET) {
                direction : input;
                capacitance : 3.0;
        }
        ff("IQ","IQN") {
                next_state : "IQ'";
                clocked_on : "CLK";
                preset     : "SET'";
        }
        pin(Q) {
                direction : output;
                function : "IQ";
        }
        pin(QN) {
                direction : output;
                function : "IQN";
                three_state : "Q"; /* non-input pin not used in
 function*/
        }
      }
```

The following is an example message:

```
Information: Line 218, The 'three_state' equation on the 'libg18' cell is
 not recognized. (LIBG-18)
```

### What Next

Check your library to determine under which case the information is given. It it is due to the previous error LIBG-4, fix the library. Otherwise, ignore it.

---

## LIBG-19

(warning) Failed to recognize the functionality of cell '%s'.

Feedback

### Description

Library Compiler fails to synthesize a netlist structure representing functionality of this cell. If the function of a cell is too complex for LC to recognize, the cell becomes a black box. In order to other cells to recognize the cell, user can specifiy the "user_function_class" attribute either in the .lib file or in Design Compiler.

### What Next

To specify "user_function_class" attribute as described above.

---

## LIBG-20

(error) The '%s' equation on the '%s' cell evaluates\n \tto a '%s' constant.

### Description

This message is generated when Library Compiler encounters a three-state cell whose enable signal string is 1 or always evaluates to 1.

The following example shows an instance where this message occurs:

```
cell(libg20) {
        area : 6.50;
        pin(CLK) {
                direction : input;
                capacitance : 1.5;
        }
        pin(SET) {
                direction : input;
                capacitance : 3.0;
        }
        ff("IQ","IQN") {
                next_state : "IQ'";
                clocked_on : "CLK";
                preset     : "SET'";
        }
        pin(Q) {
                direction : output;
                function : "IQ";
        }
        pin(QN) {
                direction : output;
                function : "IQN";
                three_state : "1";
        }
      }
```

Feedback

The following is an example message:

```
Error: Line 218, The 'three_state' equation on the 'libg20' cell
 evaluates
        to a '1' constant. (LIBG-20)
```

### What Next

Verify the cell and check the library description of the offending cell for correctness.

---

## LIBG-21

(error) The derived equality relationship between '%s' and \n \t'%s' pins conflicts with the user-specified 'pin_opposite' for the '%s' cell.

### Description

This error is generated if a cell is read by the library compiler with a pin_opposite attribute that conflicts with the derived equality of the pins. The function statements associated with the pins are the same.

The following example shows an instance where this message occurs:

```
cell(libg21) {
        area : 6.50;
        pin(CLK) {
                direction : input;
                capacitance : 1.5;
        }
        pin(SET) {
                direction : input;
                capacitance : 3.0;
        }
        ff("IQ","IQN") {
                next_state : "IQ'";
                clocked_on : "CLK";
                preset     : "SET'";
        }
        pin(Q) {
                direction : output;
                function : "IQ";
        }
        pin(QN) {
                direction : output;
                function : "IQ";        /* typo, it should be IQN */
        }
          pin_opposite("Q","QN");         /* No, they're equal. */
      }
```

Feedback

The following is an example message:

```
Error: Line 215, The derived equality relationship between 'QN' and
        'Q' pins conflicts with the user-specified 'pin_opposite' for the
 'libg21' cell. (LIBG-21)
```

### What Next

Check your library for a typo in the function strings of the cell or remove the pin_opposite attribute.

---

## LIBG-22

(error) The derived opposite relationship between '%s' and \n \t'%s' pins conflicts with the user-specified 'pin_equal' for the '%s' cell.

### Description

This error is generated if a cell is read by the library compiler with a pin_equal attribute that conflicts with the derived opposite relationship of the pins . The function statements associated with the pins are opposite.

The following example shows an instance where this message occurs:

```
cell(libg22) {
        area : 6.50;
        pin(CLK) {
                direction : input;
                capacitance : 1.5;
        }
        pin(SET) {
                direction : input;
                capacitance : 3.0;
        }
        ff("IQ","IQN") {
                next_state : "IQ'";
                clocked_on : "CLK";
                preset     : "SET'";
        }
        pin(Q) {
                direction : output;
                function : "IQ";
        }
        pin(QN) {
                direction : output;
                function : "IQN";
        }
          pin_equal("Q QN");         /* No, they're opposite. */
      }
```

Feedback

The following is an example message:

```
Error: Line 215, The derived opposite relationship between 'QN' and
        'Q' pins conflicts with the user-specified 'pin_equal' for the
 'libg22' cell. (LIBG-22)
```

**What Next**

Check your library for a typo in the function strings of the cell or remove the pin_equal attribute.

---

## LIBG-24

(information) Unable to honor the 'prefer_tied' attribute onthe '%s' pin\n \tof the '%s' cell.

**Description**

Library Compiler honors as many prefer_tied attributes as possible while it is still able to implement D flip flop functionality. However, it cannot honor all of them. For example, if the library developer specified *prefer_tied : 0* on all the inputs, Library Compiler honors as many as possible and the rest are ignored. If they are ignored, this message is issued during read_lib.

The prefer_tied attribute also can be used to indicate which pin(s) are tied to fixed logic during three_state degeneration for three-state pad cell.

The following example shows an instance where this message occurs:

```
cell(libg24)  {
        area : 0.0;
        pad_cell : true;
        pin(Z)  {
            is_pad : true;
            direction : output;
            drive_current : 2.0;
            function : "A";
            three_state : "en tn";
            timing()  {
                timing_type : three_state_disable;
                intrinsic_rise : 0.1;
                intrinsic_fall : 0.1;
              related_pin : "tn";
            }
            timing()  {
                timing_type : three_state_disable;
                intrinsic_rise : 0.1;
                intrinsic_fall : 0.1;
              related_pin : "en";
            }
            timing()  {
                intrinsic_rise : 0.1;
```

Feedback

```
                intrinsic_fall : 0.1;
                related_pin : "tn";
            }
            timing()  {
                intrinsic_rise : 0.1;
                intrinsic_fall : 0.1;
                related_pin : "en ";
            }
            timing()  {
                intrinsic_rise : 0.1;
                intrinsic_fall : 0.1;
                related_pin : "A";
            }
        }
        pin(A)  {
            direction : input;
            capacitance : 0.0;
        }
        pin(en)  {
            direction : input;
            capacitance : 0.500;
        }
        pin(tn)  {
            prefer_tied : "0";  /* in this case the value should be 1 */
            direction : input;
            capacitance : 0.500;
        }
    }
```

The following is an example message:

```
Information: Line 306, Unable to honor the 'prefer_tied' attribute on the
  'tn' pin
       of the 'libg24' cell. (LIBG-24)
```

**What Next**

Check your library for the use of the prefer_tied attribute.

---

## LIBG-25

(warning) Duplicated test signals of '%s' type is found on\n \tthe '%s' library cell.

**Description**

The test_cell has more than one of the following valid signal_types: - test_scan_in - test_scan_in_inverted - test_scan_enable - test_scan_enable_inverted - test_scan_out - test_scan_out_inverted - test_scan_clock - test_scan_clock_a - test_scan_clock_b - test_clock

Feedback

This warning informs you that the port in the test_cell pointed to has more than one signal_type that are listed previously.

The following example shows an instance where this message occurs:

```
cell(libg25) {
            area : 12;
            pin(D) {
              direction : input;
              capacitance : 1;
            }
            pin(CK,CK2,IH) {
              direction : input;
              capacitance : 1;
            }
            pin(A,B) {
              direction : input
                capacitance : 2;
            }
            pin(SI) {
              direction : input;
              capacitance : 1;
              timing() {
                timing_type : setup_falling;
                intrinsic_rise : 2.8000;
                intrinsic_fall : 2.8000;
                related_pin : "A";
              }
              timing() {
                timing_type : hold_falling;
                intrinsic_rise : 1.2000;
                intrinsic_fall : 1.2000;
                related_pin : "A";
              }
            }
            pin(Q) {
              direction : output;
              timing() {
                timing_type : rising_edge;
                intrinsic_rise : 2.5500;
                intrinsic_fall : 2.4000;
                rise_resistance : 0.0700;
                fall_resistance : 0.0300;
                related_pin : "CK IH CK2";
              }
              timing() {
                timing_type : falling_edge;
                intrinsic_rise : 2.5500;
                intrinsic_fall : 2.4000;
                rise_resistance : 0.0700;
                fall_resistance : 0.0300;
                related_pin : "B";
              }
```

Feedback

```
    }
    pin(XQ) {
      direction : output;
    }
    test_cell(){
      pin(D CK){
        direction : input;
      }
      pin(IH CK2){              /* duplicate test_clock */
        direction : input;
        signal_type : "test_clock";
      }
      pin(SI){
        direction : input;
        signal_type : "test_scan_in";
      }
      pin(A){
        direction : input;
        signal_type : "test_scan_clock_a";
      }
      pin(B){
        direction : input;
        signal_type : "test_scan_clock_b";
      }
      state("IQ","IQN"){
        clocked_on : "CK CK2";
        next_state : "D";
      }
      pin(Q){
        direction : output;
        function : "IQ";
        signal_type : "test_scan_out";
      }
      pin(XQ){
        direction : output;
        function : "IQN";
        signal_type : "test_scan_out_inverted";
      }
    }
  }
```

In this case, there are two test_clock attributes for both 'IH' and 'CK2' pins.

The following is a message example:

```
Warning: Line 67, Duplicated test signals of test_clock type is found on
        the 'libg25' library cell. (LIBG-25)
```

**What Next**

Vendors/Library developers should check the test_cell pointed to and make sure that it is the behavior they meant to have. Otherwise,r delete the duplicate signal_type.

Feedback

If a designer finds this problem, he should contact the vendor or library developer and explain the problem. This problem can only be fixed by the vendor/library developer.

---

## LIBG-26

(error) The '%s' test cell pin on the '%s' cell\n \thas no 'function' attribute. The test cell is removed.

### Description

The specified pin has neither a test *signal_type* nor a *function* attribute. This incomplete functional description results in a black-box test cell, which is removed from the library.

The following example shows an instance where this message occurs:

```
cell(libg26) {
            area : 12;
            pin(D) {
              direction : input;
              capacitance : 1;
            }
            pin(CK IH) {
              direction : input;
              capacitance : 1;
            }
            pin(A,B) {
              direction : input
                capacitance : 2;
            }
            pin(SI) {
              direction : input;
              capacitance : 1;
              timing() {
                timing_type : setup_falling;
                intrinsic_rise : 2.8000;
                intrinsic_fall : 2.8000;
                related_pin : "A";
              }
              timing() {
                timing_type : hold_falling;
                intrinsic_rise : 1.2000;
                intrinsic_fall : 1.2000;
                related_pin : "A";
              }
            }
            pin(Q) {
              direction : output;
              timing() {
                timing_type : rising_edge;
                intrinsic_rise : 2.5500;
                intrinsic_fall : 2.4000;
```

Feedback

```
        rise_resistance : 0.0700;
        fall_resistance : 0.0300;
        related_pin : "CK IH";
      }
      timing() {
        timing_type : falling_edge;
        intrinsic_rise : 2.5500;
        intrinsic_fall : 2.4000;
        rise_resistance : 0.0700;
        fall_resistance : 0.0300;
        related_pin : "B";
      }
    }
    pin(XQ) {
      direction : output;
    }
    test_cell(){
      pin(D CK){
        direction : input;
      }
      pin(IH ){
        direction : input;
        signal_type : "test_clock";
      }
      pin(SI){
        direction : input;
        signal_type : "test_scan_in";
      }
      pin(A){
        direction : input;
        signal_type : "test_scan_clock_a";
      }
      pin(B){
        direction : input;
        signal_type : "test_scan_clock_b";
      }
      state("IQ","IQN"){
        clocked_on : "CK CK2";
        next_state : "D";
      }
      pin(Q){
        direction : output;
      }
      pin(XQ){
        direction : output;
        function : "IQN";
        signal_type : "test_scan_out_inverted";
      }
    }
  }
```

Feedback

In this case, the pin 'Q' needs to have a function : "IQ" or signal_type : "test_scan_out" statement

The following is a message example:

```
Error: Line 385, The 'Q' test cell pin on the 'libg26' cell
        has no 'function' attribute. The test cell is removed. (LIBG-26)
```

**What Next**

Add a proper *function* attribute or *signal_type* attribute to the faulty pin group.

---

## LIBG-27

(warning) The '%s' pin is eliminated from the '%s' pin \n \tfunction on the '%s' cell. The cell becomes a black box.

**Description**

This message is issued to warn users about the redundant ports in a function string. Redundant ports are ports which are eliminated from the boolean function due to normal boolean simplification.

The following example shows an instance where this message occurs:

```
cell(libg27) {
  area : 10;
  pin(D) {
    direction : input;
    capacitance : 1;
  }
  pin(CP) {
    direction : input;
    capacitance : 1;
  }

  statetable ( "D CP", "Q QB") {
    table : "L/H R  : - - : L/H H/L,\
             -   ~R : - - : N   N";
  }

  pin(Q) {
    direction : output;
    internal_node : "Q";
  }
  pin(QB) {
    direction : output;
    state_function : "D * 0";
  }
}
```

Feedback

The following is an example message:

```
Warning: Line 448, The 'D' pin is eliminated from the 'QB' pin
        function on the 'libg27' cell. The cell becomes a black box.
 (LIBG-27)
```

### What Next

Check the library for a possible error in the function string.

---

## LIBG-28

(warning) The '%s' three state pin has no 'function' or\n \tthe function is too complex to be recognized;\n \tthe '%s' cell becomes a black box.

### Description

This message identifies a cell for which one of its output pins with a three_state attribute has no valid function defined, or its function is too complex to be mapped onto. In this case the library compiler treats the cell as a black box.

The following example shows an instance where this message occurs:

```
cell(libg28) {
          area : 1;
          pin(A) {
            direction : input;
            capacitance : 0;
          }
          pin(TA) {
            direction : output;
            three_state : "A'"; /* missing a function attribute */
          }
        }
```

The following is an example message:

```
Warning: Line 34, The 'TA' three state pin has no 'function' or
        the function is too complex to be recognized;
        the 'libg28' cell becomes a black box.  (LIBG-28)
```

### What Next

Check your library and add the functional information, in case it is absent, to the tri-statable pin. The functional model can be defined by the 'function' attribute, or ff, latch, memory, or unigen statements.

Feedback

## LIBG-29

(warning) The state group needs more statements to be\n \tmeaningful; cell becomes black-box.

### Description

This message is issued when a clocked_on or a next_state statement is missing. This message also appears when a latch does not have a minimum of two force statements (force_00, force_01, force_10, and force_11). Library Compiler treats a cell without a functional model as a black-box.

The following example shows an instance where this message occurs:

```
cell(libg29) {
            area : 12;
            pin(D) {
              direction : input;
              capacitance : 1;
            }
            pin(CK) {
              direction : input;
              capacitance : 1;
            }
            pin(Q) {
              direction : output;
            }
              state("IQ","IQN"){
                next_state : "D";       /* clocked_on : "CK"; missing */
              }
          }
```

The following is an example message:

```
Warning: Line 460, The state group needs more statements to be
        meaningful; cell becomes black-box. (LIBG-29)
```

### What Next

Check the library for a wrong model and add the missing statement of the corresponding group.

## LIBG-30

(warning) The '%s' equation on the '%s' cell evaluates\n \tto a constant '%s'. Since this implies the output will never become\n \t%s, this equation is ignored.

Feedback

### Description

The sequential library cell has both a force_00 and a force_11 declared. The translation algorithm cannot handle library cells with both of these declarations. This warning message is specific to structural models.

The following example shows an instance where this message occurs:

```
cell(libg30) {
        area : 6.50;
        pin(CLK) {
                direction : input;
                capacitance : 1.5;
        }
        pin(SET) {
                direction : input;
                capacitance : 3.0;
        }
        ff("IQ","IQN") {
                next_state : "IQ'";
                clocked_on : "CLK";
                preset     : "SET'";
        }
        pin(Q) {
                direction : output;
                function : "IQ";
        }
        pin(QN) {
                direction : output;
                function : "IQN";
                three_state : "0";
        }
      }
```

The following is an example message:

```
Warning: Line 218, The 'three_state' equation on the 'libg30' cell
 evaluates
        to a constant '0'. Since this implies the output will never
 become
        tri-stated, this equation is ignored. (LIBG-30)
```

### What Next

Declare instead either a pair of force_10 and force_01, force_10 and force_11, force_00 and force_01, force_00 and force_10, or force_01 and force_11. The Q and QN outputs of the specified sequential cell are not complementary. Possible reasons for this are: 1. The library specifies that the outputs are not complementary; 2. The functions driving the force_00 and force_11 inputs of the degenerated sequential cell are not logical zero. For purposes of verification, every sequential cell is degenerated into a primitive sequential cell. The outputs (sourcepoints) of the primitive are Q and QN. The inputs (endpoints) of the primitive are next_state, force_00, force_01, force_10, force_11, clocked_on, and

Feedback

clocked_on_also (for Master-slave cells only). The functions driving the force endpoints characterize the asynchronous properties of the primitive sequential cell. The library might also contain an attribute asserting that the outputs Q and QN are complementary. The force_01 function represents the conditions under which the Q pin can be driven to zero and the QN driven to 1 simultaneously. The other force inputs have a similar interpretation. When the attribute is not present, verification tries to determine that Q and QN are effectively complementary by establishing that both force_00 and force_11 are logically zero. The exact degeneration process depends on the library characterization of the sequential cell and is not described here.

Sequential cells should be modeled using the flip-flop and latch attributes. If you model sequential cells with state attributes, use a local function to describe the asynchronous behavior of each output pin. These functions are hard-coded as async_IQ and async_IQN. The async_IQ, async_IQN, and active functions are based on the forceXX expressions. An example is this flip-flop with Preset and Clear: state (IQ,IQN) { ... force00 : "!CD & !SD" force01 : "!CD & SD" force10 : "CD & !SD" ... } When CD is low, Q is always 0, so the VHDL library generator ORs these two expressions together; instead of (!CD & SD) | (!CD & !SD), the equation becomes !CD. The ORed expression is simplified by Boolean algorithms, which use only 1 and 0 values. If you specify an X for one of these values, the library generator might modify the results.

---

## LIBG-31

(warning) The test cell on cell '%s' does not have any function.

**Description**

This message indicates that no output port of the test cell has a function attribute.

**What Next**

An output port in the test cell must have a function attribute, a signal_type attribute, or both. At least one output port must have a function attribute. The function attribute can reflect only the nontest behavior of the cell.

---

## LIBG-32

(error) The '%s' parallel sequential cell must use bus/bundle for outputs.

**Description**

One of the requirements of the parallel sequential cells that are described by the ff_bank or latch_bank groups, is that all the outputs are busses or bundles.

This message applies to a parallel sequential cell whose outputs are not busses or bundles. However, this message does not apply to a multibit scan cell with a single-bit output pin, when the *function* attribute is not specified on the pin.

Feedback

The following example shows an instance where this message occurs:

```
cell (LSFDS) {
                area : 31.000000 ;
                dont_touch : true ;
                dont_use : true ;
                bus (D) {
                        bus_type : BUS4;
                        direction : input ;
                        capacitance : 2.000000 ;
                }
                pin (CK) {
                        direction : input ;
                        capacitance : 1.000000 ;
                }
                ff_bank (IQ,IQN,4) {
                        next_state : "D" ;
                        clocked_on : "CK" ;
                }
                bus (Q) {
                        bus_type : BUS4;
                        direction : output ;
                        function : "IQ" ;
                }
                pin (SO) {
                        direction : output ;
                        function : "IQN" ;
                }
        }
```

In this case, the pin 'SO' must be defined as a bus of type BUS4.

The following is an example message:

```
Error: Line 168, The 'libg32' parallel sequential cell must use
bus/bundle for outputs. (LIBG-32)
```

### What Next

Check your library for parallel sequential cells with single-bit output pins with the *function* attribute specified on them. Define these pins as busses or bundles.

---

## LIBG-33

(warning) In the '%s' cell, the '%s' inout pin\n \tcannot have feedback in its function. The cell becomes a black box.

### Description

The inout pin is in its own function. Library Compiler does not support combinational functions with feedback.

Feedback

The following example shows an instance where this message occurs:

```
cell (libg33) {
    pin(C) {
        direction : input;
    }
    pin(D) {
        direction : inout;
        function : "A'";
        three_state : "C";
    }
    pin(A) {
        direction : inout;
        function : "D'";
        three_state : "C"; /* It will cause loop combinational function
 loop in the design.
                           * To break the loop, change "C" into "C'" */
    }
  }
```

The following is an example message:

```
Warning: Line 64, Cell 'libg33', In the 'libg33' cell, the 'A' inout pin
        cannot have feedback in its function. The cell becomes a black
 box. (LIBG-33)
```

**What Next**

Either break the feedback or try to use a sequential definition.

---

## LIBG-34

(error) The '%s' pin in the '%s' cell has the '%s' direction\n \tThe direction must be '%s'.

**Description**

The pin on the programmable cell must have the direction specified.

**What Next**

Replace the pin direction attribute by the correct direction.

---

## LIBG-35

(warning) The 'nextstate_type' attribute on the '%s' pin/bus in\n \tthe '%s' cell is inconsistent with its function. The attribute is ignored.

**Description**

In a pin group, nextstate_type defines the type of a next_state attribute to be used in an ff group, a seq group, or a ff_bank group.

Feedback

Any pin with the *nextstate_type* attribute must be in the *nextstate* function. A consistency check is also made between the pin's nextstate_type attribute and the nextstate function. This message is generated when the attribute 'nextstate_type' does not match the functional information specified in the cell's sequential state description.

The following example shows an instance where this message occurs:

```
cell(libg35) {
        area : 6.50;
        pin(D) {
           direction : input;
           capacitance : 3.0;
        }
        pin(CLK) {
           direction : input;
           capacitance : 1.5;
        }
        pin(SET) {
           direction : input;
           nextstate_type : scan_in;
           capacitance : 3.0;
        }

        ff("IQ","IQN") {
           next_state : "D";
           clocked_on : "CLK";
           preset     : "SET'";
        }

        pin(Q) {
           direction : output;
           function : "IQ";
        }
        pin(QN) {
           direction : output;
           function : "IQN";
        }
   }
```

The following is an example message:

```
Warning: Line 153, The 'nextstate_type' attribute on the 'SET' pin/bus in
        the 'libg35' cell is inconsistent with its function. The
 attribute is ignored. (LIBG-35)
```

### What Next

Check your library for a pin with a nextstate_type attribute but not used in the next_state statement of a sequential group and correct the inconsistency.

Feedback

## LIBG-36

(warning) In the '%s' cell, the '%s' input drives more than one\n \tfunction. The cell might not be optimally used.

### Description

This warning is issued when more than one attribute inside ff, ff_bank, latch, latch_bank, or seq group share one or more inputs pin names, or when attribute function/ state_function and three_state on a port have common input pins.

The following example shows an instance where this message occurs:

```
cell(libg36) {
         area : 6.50;
         pin(D TE) {
           direction : input;
           capacitance : 3.0;
         }
        pin(CLK) {
                direction : input;
                capacitance : 1.5;
        }
        ff("IQ","IQN") {
                next_state : "D TE";
                clocked_on : "CLK";
                preset     : "D'";
                }

        pin(Q) {
                direction : output;
                function : "IQ";
        }
        pin(QN) {
                direction : output;
                function : "IQN";
        }
      }
```

In this case, the input pin 'D' is present in the next_state's string and the preset's string.

The following is an example message:

```
Warning: Line 189, In the 'libg36' cell, the 'D' input drives more than
 one function.
        The cell might not be optimally used. (LIBG-36)
```

### What Next

Check your library for shared input pins and associate a unique string to each of them.

Feedback

## LIBG-37

(warning) The '%s' pin/bus is unused in the '%s' cell.\n

### Description

This message is generated when an input pin is not used in the functionality of the cell.

The following example shows an instance where this message occurs:

```
cell(libg37) {
        area : 6.50;
        pin(CLK) {
                direction : input;
                capacitance : 1.5;
        }
        pin(SET) {
                direction : input;
                capacitance : 3.0;
        }
        ff("IQ","IQN") {
                next_state : "IQ'";
                clocked_on : "CLK";
        }
        pin(Q) {
                direction : output;
                function : "IQ";
        }
        pin(QN) {
                direction : output;
                function : "IQN";
              }
  }
```

The following is an example message:

```
Warning: Line 202, The 'SET' pin/bus is unused in the 'libg37' cell.
 (LIBG-37)
```

### What Next

Either delete the unused input pin or add it the functionality of the cell.

## LIBG-38

(warning) The '%s' enable pin with a three_state attribute in the\n \t '%s' cell should have non-unate timing arcs.

Feedback

### Description

Tri-state enable pins should have "non-unate" timing arcs because the transition depends on the data input. If incorrect timing sense is detected, a warning is issued.

### What Next

Change the timing sense to "non-unate."

---

## LIBG-39

(warning) The '%s' asynchronous input of the '%s' cell is inconsistent in active level.

### Description

The asynchronous input is inconsistent in "active" level. If "active low," the clear timing arc is *positive-unate*, the preset timing arc is *negative unate*, and the intrinsic fall in recovery constraint is *zero*.

If "active high," the clear timing arc is *negative-unate*, the preset timing arc is *positive-unate*, and the intrinsic rise in recovery constraint is *zero*.

Since the active level of asynchronous input is not explicitly specified, the warning message simply states that, due to inconsistency, the active level of the asynchronous pin cannot be determined by the set of associated timing groups. An inconsistency is found at the given line number.

This message can show up in situations when the clear/preset attribute inside ff/latch group is a complex Boolean expression of signal pins instead of a single input pin. Due to these complex Boolean values for the clear/preset attributes, user should ignore this warning message after making sure that all timing arc(s) have been modeled correctly. Even after generating this warning LC will store the user specified timing arcs as modeled in the library.

The following example shows an instance where this message occurs:

```
cell("libg39") {
  area : 5 ;
  state("IQ","IQN") {
    force_00 : "S*R" ;
    force_01 : "R*!S" ;
    force_10 : "S*!R" ;
  }
  pin("S") {
    direction : input ;
    capacitance : 0.145 ;
  }
  pin("R") {
    direction : input ;
    capacitance : 0.145 ;
```

Feedback

```
  }
  pin("Q") {
    direction : output ;
    function : "IQ" ;
    timing() {
      timing_type       : preset ;
      timing_sense      : negative_unate ;
      rise_resistance   :  1.000 ;
      related_pin       : "S" ;
    }
    timing() {
      timing_type       : clear ;
      timing_sense      : negative_unate ;
      intrinsic_fall    :  1.000 ;
      fall_resistance   :  1.000 ;
      related_pin       : "R" ;
    }
  }
  pin("QN") {
    direction : output ;
    function : "IQN" ;
    timing() {
      timing_type       : clear ;
      timing_sense      : negative_unate ;
      intrinsic_fall    :  1.000 ;
      fall_resistance   :  1.000 ;
      related_pin       : "S" ;
    }
    timing() {
      timing_type       : preset ;
      timing_sense      : positive_unate ;
      intrinsic_rise    :  1.000 ;
      rise_resistance   :  1.000 ;
      related_pin       : "R" ;
    }
  }
}
```

To fix the warning, the timing arc of S in the pin 'Q' has to be positive_unate.

The following is an example message:

```
Warning: Line 32, The 'S' asynchronous input of the 'libg39' cell is
 inconsistent in active level. (LIBG-39)
```

### What Next

Determine the asynchronous pin active level. Check the current timing group, and, if inconsistent, make the modification. Otherwise, check other timing groups associated with this asynchronous pin, and change any inconsistency.

Feedback

## LIBG-40

(warning) The '%s' clock/gate input of the '%s' cell is inconsistent in active edge/level.

### Description

A clock is consistent in triggering edge. If positive edge is triggered, only *rising_edge*, *setup_rising*, *hold_rising*, and *recovery_rising* timing types are allowed. If negative edge is triggered, only *falling_edge*, *setup_falling*, *hold_falling*, and *recovery_falling* timing types are allowed. Since the clock-pin triggering edge is not explicitly specified, the warning message simply indicates that, due to inconsistency, the triggering edge of the clock pin cannot be derived from its related timing-group data. An inconsistency is found at the given line number.

A gate is consistent in active level. If "active low," only *non_unate*, *falling_edge*, *setup_rising*, *hold_rising*, and *recovery_rising* timing types are allowed. If "active high," only *non_unate*, *rising_edge*, *setup_falling*, *hold_falling*, and *recovery_falling* are allowed. Since the gate-pin active level is not explicitly specified, the warning message simply indicates that, due to inconsistency, the active level of the gate pin cannot be derived from its related timing group data. An inconsistency is found at the given line number.

There is one exception: *rising_edge* and *falling_edge* are not checked if the library cell is a black box.

The following example shows an instance where this message occurs:

```
cell(RAM1) {
        area : 4;
        pin(D) {
        direction : input;
        capacitance : 2;
        timing() {
              timing_type : setup_rising;
              intrinsic_rise : 1.00;
              intrinsic_fall : 1.00;
              related_pin : "WR";
            }
        }
        pin(WR) {
            direction : input;
            capacitance : 1;
        }
        state ("IQ","IQN") {
            force_01 : "D' WR";
            force_10 : "D WR ";
        }
        pin(QN) {
            direction : output;
            function : "IQN";
            timing() {
```

Feedback

```
                timing_type : rising_edge;
                intrinsic_rise : 1.00;
                intrinsic_fall : 1.00;
                rise_resistance : 0.1000;
                fall_resistance : 0.1000;
                related_pin : "WR";
              }
        }
    }
```

The following is an example message:

```
Warning: Line 266, The 'WR' clock/gate input of the 'RAM1' cell is
 inconsistent in active edge/level. (LIBG-40)
```

### What Next

For a clock pin, determine the triggering edge first. Check the current timing group data, and modify timing-group data if found to be inconsistent with clock-triggering edge. Otherwise, check other timing groups that are related to the clock pin, and, if inconsistency is found, make the change.

For a gate pin, determine the active level first, check the current timing-group data, and modify timing-group data if found to be inconsistent with gate-active level. Otherwise, check other timing groups that are related to the gate pin, and, if inconsistency is found, make the change.

---

## LIBG-42

(error) The '%s' related pin of the '%s' cell has clear/preset\n \ttiming arcs but is not asynchronous.

### Description

The timing group is a clear/preset arc but the related pin is not asynchronous. The related pin must be in a force statement or in a clear/preset statement to be in a clear/preset timing arc.

The following example shows an instance where this message occurs:

```
cell (libg42)  {
        area : 8.000;
        pin(D)  {
                direction : input;
                capacitance : 0.100;
        }
        pin(CK)  {
                direction : input;
                capacitance : 0.100;
        }
        ff(IQ,IQB)  {
```

Feedback

```
                next_state : "D";
                clocked_on : "CK";
        }
        pin(QB)  {
                direction : output;
                function : "IQB" ;
                timing()  {
                        timing_type : rising_edge;
                        intrinsic_rise : 2.000;
                        intrinsic_fall : 3.000;
                        rise_resistance : 8.000;
                        fall_resistance : 6.000;
                        related_pin : "CK";
                }
        }
        pin(Q)  {
                direction : output;
                function : "IQ" ;
                timing()  {
                        timing_type : clear ;
                        timing_sense : negative_unate ;
                        intrinsic_fall : 1.000;
                        fall_resistance : 4.000;
                        related_pin : "QB";
                }
        }
}
```

The following is an example message:

```
Error: Line 270, The 'QB' related pin of the 'libg42' cell has
 clear/preset
        timing arcs but is not asynchronous. (LIBG-42)
```

**What Next**

Define the related pin as asynchronous in the functional description or remove the timing arc.

---

## LIBG-43

(error) The '%s' related pin of this timing arc is an output pin and\n \tthe timing_sense is not specified.

**Description**

If the related pin of a timing arc is an output pin, provide the *timing_sense* attribute.

The following example shows an instance where this message occurs:

```
cell(libg43) {
      area : 8.000;
```

Feedback

```
      pin(A E) {
        direction : input;
        capacitance : 1.00;
      }
      pin(B) {
        direction : output;
        function : "!A";
        timing() {
          related_pin : "A";
          intrinsic_rise : 1.0;
          intrinsic_fall : 1.0;
        }
      }
      pin(C) {
        direction : inout;
        function : "A";
        three_state : "E";
        timing() {
          related_pin : "E";
          timing_type : three_state_disable;
          intrinsic_rise : 1.0;
          intrinsic_fall : 1.0;
        }
        timing() {
          related_pin : "E";
          intrinsic_rise : 1.0;
          intrinsic_fall : 1.0;
        }
        timing() {
          related_pin : "B";
          intrinsic_rise : 1.0;
          intrinsic_fall : 1.0;
        }
      }
    }
```

In this case, a timing_sense attribute is missing in the timing group between the inout pin 'C' and the output pin 'B'.

The following is an example message:

```
Error: Line 75, The 'B' related pin of this timing arc is an output pin
 and
        the timing_sense is not specified. (LIBG-43)
```

**What Next**

Check the functionality of the cell, and provide the correct *timing_sense* to this timing arc.

Feedback

## LIBG-44

(error) The logic represented by the 'when' string (%s) in this\n \ttiming group is not mutually exclusive with the logic represented by\n \tthe 'when' string (%s) in the timing group on line %d.

### Description

The logic represented by the 'when' string of all state-dependent timing arcs between a pair of pins should be mutually exclusive. If it isn't, Library Compiler is not be able to determine which timing arc to use to propagate signal through the path when the condition causes both of them to be evaluated 'TRUE'.

The following example shows an instance where this message occurs:

```
cell(libg44) {
      area : 2;
      pin(D1 D0 S0) {
        direction : input;
        capacitance : 1;
      }
      pin(Z) {
        direction : output;
        function : "S0'D0 + S0 D1";
        timing() {
          timing_sense : non_unate;
          intrinsic_rise : 1.00;
          intrinsic_fall : 1.00;
          rise_resistance : 0.1;
          fall_resistance : 0.1;
          related_pin : " D0 D1";
        }
        timing() {
          when : "S0'D0 + S0 D1'";
          when : "D1' S0 + S0'D0 ";
          sdf_cond : "(D1 == 1'b0 && S0 == 1'b1) || (S0 == 1'b0 && D0 ==
 1'b1)";
          timing_sense : positive_unate;
          intrinsic_rise : 1.00;
          intrinsic_fall : 1.00;
          rise_resistance : 0.1;
          fall_resistance : 0.1;
          related_pin : "S0";
        }
        timing() {
          when : "D1 S0 + S0'D0";
          sdf_cond : "(D1 == 1'b1 && S0 == 1'b1) || (S0 == 1'b0 && D0 ==
 1'b0)";
          timing_sense : non_unate;
          intrinsic_rise : 0.1;
          intrinsic_fall : 0.1;
```

Feedback

```
          rise_resistance : 0.1;
          fall_resistance : 0.1;
          related_pin : "S0";
        }
        timing() { /* default */
          intrinsic_rise : 1.00;
          intrinsic_fall : 2.00;
          rise_resistance : 0.1;
          fall_resistance : 0.1;
          related_pin : "S0";
        }
      }
    }
```

In this case, the second when attribute's string has a typo. It should be "D1 S0 + S0'D0'"; where the pin D0 should be ticked according to the sdf_cond attribute's string.

The following is an example message:

```
Error: Line 258, The logic represented by the 'when' string (S0'D0 + S0
 D1') in this
        timing group is not mutually exclusive with the logic represented
 by
        the 'when' string (D1 S0 + S0'D0) in the timing group on line
 268. (LIBG-
44)
```

### What Next

Check the 'when' strings of both timing groups for wrong information and fix it.

---

## LIBG-44w

(warning) The logic represented by the 'when' string (%s) in this\n \ttiming group is not mutually exclusive with the logic represented by\n \tthe 'when' string (%s) in the timing group on line %d.

### Description

The logic represented by the 'when' string of all state-dependent timing arcs between a pair of pins should be mutually exclusive. If it isn't, Library Compiler is not be able to determine which timing arc to use to propagate signal through the path when the condition causes both of them to be evaluated 'TRUE'.

The following example shows an instance where this message occurs:

```
cell(libg44) {
      area : 2;
      pin(D1 D0 S0) {
        direction : input;
        capacitance : 1;
```

Feedback

```
      }
      pin(Z) {
        direction : output;
        function : "S0'D0 + S0 D1";
        timing() {
          timing_sense : non_unate;
          intrinsic_rise : 1.00;
          intrinsic_fall : 1.00;
          rise_resistance : 0.1;
          fall_resistance : 0.1;
          related_pin : " D0 D1";
        }
        timing() {
          when : "S0'D0 + S0 D1'";
          when : "D1' S0 + S0'D0 ";
          sdf_cond : "(D1 == 1'b0 && S0 == 1'b1) || (S0 == 1'b0 && D0 ==
1'b1)";
          timing_sense : positive_unate;
          intrinsic_rise : 1.00;
          intrinsic_fall : 1.00;
          rise_resistance : 0.1;
          fall_resistance : 0.1;
          related_pin : "S0";
        }
        timing() {
          when : "D1 S0 + S0'D0";
          sdf_cond : "(D1 == 1'b1 && S0 == 1'b1) || (S0 == 1'b0 && D0 ==
1'b0)";
          timing_sense : non_unate;
          intrinsic_rise : 0.1;
          intrinsic_fall : 0.1;
          rise_resistance : 0.1;
          fall_resistance : 0.1;
          related_pin : "S0";
        }
        timing() { /* default */
          intrinsic_rise : 1.00;
          intrinsic_fall : 2.00;
          rise_resistance : 0.1;
          fall_resistance : 0.1;
          related_pin : "S0";
        }
      }
    }
```

In this case, the second when attribute's string has a typo. It should be "D1 S0 + S0'D0'"; where the pin D0 should be ticked according to the sdf_cond attribute's string.

The following is an example message:

```
Warning: Line 258, The logic represented by the 'when' string (S0'D0 + S0
 D1') in this
```

Feedback

```
        timing group is not mutually exclusive with the logic represented
 by
        the 'when' string (D1 S0 + S0'D0) in the timing group on line
 268. (LIBG-
44w)
```

### What Next

Check the 'when' strings of both timing groups for wrong information and fix it.

---

## LIBG-45

(error) The 'timing_sense' attribute is missing. It is required\n \tfor '%s' timing arc.

### Description

For timing arcs of timing_type 'clear' or 'preset', timing_sense is a required attribute. It can be either 'positive_unate' or 'negative_unate' or 'non_unate'.

The following example shows an instance where this message occurs:

```
cell(libg45) {
      area : 9;
      pin(D CP CD) {
        direction : input;
        capacitance : 1;
      }
      state("IQ","IQN") {
        next_state : "D";
        clocked_on : "CP";
        force_01   : "CD'";
      }
      pin(Q) {
        direction : output;
        function : "IQ";
        timing() {
          timing_type : clear;  /* timing_sense missing */
          intrinsic_fall : 0.77;
          fall_resistance : 0.0523;
          related_pin : "CD";
        }
      }
      pin(QN) {
        direction : output;
        function : "IQN";
        timing() {
          timing_type : preset;
          timing_sense : negative_unate;
          intrinsic_rise : 0.87;
          rise_resistance : 0.1523;
          related_pin : "CD";
        }
```

Feedback

```
      }
    }
```

In this case, there is a missing timing_sense attribute with positive_unate for the 'clear' timing arc of the pin 'Q'.

The following is an example message:

```
Error: Line 256, The 'timing_sense' attribute is missing. It is required
        for 'clear' timing arc. (LIBG-45)
```

### What Next

Check your logic between the input and output pins and put the appropriate timing_sense in the timing group.

---

## LIBG-46

(error) The '%s' pin of the '%s' cell is not a clock/enable pin and\n \tcannot be used in the 'related_pin' of timing arc with the \n \t'%s' timing_type.

### Description

Pin in the 'related_pin' of the timing group with the timing_type rising_edge or falling_edge should be a clock/enable pin.

The following example shows an instance where this message occurs:

```
cell(libg46) {
  area : 10;
  pin(D) {
    direction : input;
    capacitance : 1;
  }
  pin(CP) {
    direction : input;
    capacitance : 1;
    min_period : 99
    timing() {
      timing_type : rising_edge
      intrinsic_rise : 1.640000
      intrinsic_fall : 1.880000
      rise_resistance : 0.182000
      fall_resistance : 0.059000
      related_pin : "D"
    }
  }
  statetable ( "D CP", "Q QB") {
    table : "L/H R  : - - : L/H H/L,\
             -   ~R : - - : N   N";
  }
```

Feedback

```
  pin(Q) {
    direction : output;
    internal_node : "Q";
    inverted_output : FALSE;
    timing() {
      timing_type : rising_edge
      intrinsic_rise : 1.640000
      intrinsic_fall : 1.880000
      rise_resistance : 0.182000
      fall_resistance : 0.059000
      related_pin : "CP"
    }
  }
  pin(QB) {
    direction : output;
    internal_node : "Q";
    inverted_output : TRUE;
    timing() {
      timing_type : rising_edge
      intrinsic_rise : 1.640000
      intrinsic_fall : 1.880000
      rise_resistance : 0.182000
      fall_resistance : 0.059000
      related_pin : "CP"
    }
  }
}
```

The following is an example message:

```
Error: Line 120, The 'D' pin of the 'libg46' cell is not a clock/enable
 pin and
        cannot be used in the 'related_pin' of timing arc with the
        'rising_edge' timing_type. (LIBG-46)
```

**What Next**

Check your library to see if you place the wrong pin in the 'related_pin' attribute or you have specified wrong 'timing_type'.

---

## LIBG-47

(warning) Cell(%s): The state table entries have been expanded.\n \tThe following %d %s of overlapping entries %s been found:\n %s\n \tFor each occurrence, the first "rule out" entry is used.

**Description**

This message informs the user that overlapping entries have been found in the state table, and that the first one encountered is being used. Overlapping entries can occur when entries (that is, rules) in the *table* attribute of the *statetable* group are expanded.

Feedback

Overlapping means that for a single input value combination, there are multiple next internal node combinations. During expansion, some entries might overlap with others; the entry highest in the table overrides those below it.

NOTE:

```
All inputs that are always don't care are not mentioned in
        the input string of occurrences.
```

The following example shows an instance where this message occurs:

```
statetable ( "D WR WRN", "Q QN") {
    table : "L/H H L : - - : L/H H/L,\\ /* rule 1 */
              -   - - : - - : N   N";   /* rule 2 */
  }
```

During the expansion of the table into an ordered table with L and H inputs and L, H, X, and N next internal nodes, rule 1 expands into

```
L H L : - - : L H /* rule 3 */
            H H L : - - : H L /* rule 4 */
```

and rule 2 expands into

```
L H L : - - : N N /* rule 5 */
            H H L : - - : N N /* rule 6 */
```

In this case, there is an overlap between rules 3 and 5 and rules 4 and 6.

The following is an example message:

```
Warning: Line 214, Cell(libg47): The state table entries have been
 expanded.
        The following 2 occurrences of overlapping entries have been
 found:
        occurrence 1: rule in: "L H L", rule out #1: "L H", rule out #2:
 "N N"
        occurrence 2: rule in: "H H L", rule out #1: "H L", rule out #2:
 "N N"
        For each occurrence, the first "rule out" entry is used.
 (LIBG-47)
```

**What Next**

You can avoid receiving this message by once explicitly listing the next internal node for every permutation of L and H for the current inputs, previous inputs, and current states.

---

## LIBG-48

(warning) Cell(%s): The following %d %s, which includes\n \tcurrent and delayed input values in the state table, %s unspecified: %s\n \tOutputs are made unknown.

Feedback

### Description

In the 'table' attribute of the 'statetable' group, some entries (rules) are unspecified. All permutations of the input nodes are not specified by the table. Therefore, the internal node is made unknown for all of the missing entries.

The following example shows an instance where this message occurs:

```
statetable ( "D CP", "Q QB") {
    table : "L R : - - : L H,\
             - ~R : - - : N   N";
  }
```

During the table expansion, the output for the input permutation of CP* (delayed), CP, and D: "L H H" is not specified. The output is set to "X X".

The following is an example message:

```
Warning: Line 134, Cell(libg48): The following 1 entry, which includes
        current and delayed input values in the state table, is
 unspecified:
        inputs: CP*, CP, D
                "011"
        Outputs are made unknown. (LIBG-48)
```

### What Next

This warning can be eliminated by carefully specifying the state table to avoid unspecified entries.

Note:

```
The input edges symbols, {"R", "~R", "F", "~F"}, represent
        permutations of L and H for the delayed input and the current
        input. Every edge-sensitive input(one with at least one
        input edge symbol) is expanded into two level-sensitive
        inputs: the current input value and the delayed input value.
        For example, the input edge symbol "R" expands to "L" for the
        delayed input value and "H" for the current input value.
```

---

## LIBG-49

(error) Cell(%s): '%s' token in "%s" field is not recognized.

### Description

The token is not a valid state table token for that field in the table. The valid tokens for input nodes are {L, H, -, L/H, H/L, R, F, ~R, and ~F}. The valid tokens for current internal nodes are {L, H, -, L/H, and H/L}. The valid tokens for next internal nodes are {L, H, L/H, H/L, X, and N}. Therefore, the table is deleted.

Feedback

### What Next

Replace the invalid token with a valid one.

---

## LIBG-50

(error) Cell(%s): Incorrect "%s" field size in table.\n \tSize found: %d, expecting: %d.

### Description

In the 'table' attribute of the 'statetable' group, either the input nodes, the current internal nodes, or the next internal nodes has an incorrect number of tokens.

The following example shows an instance where this message occurs:

```
statetable ( "D CP", "Q QB") {
    table : "L/H R  : - : L/H H/L,\
              -   ~R : - : N   N";
  }
```

The following is an example message:

```
Error: Line 134, Cell(libg50): Incorrect "current" field size in table.
        Size Found: 1, expecting: 2. (LIBG-50)
```

### What Next

Fix the number of entries in the table. For instance, The current field must have 2 tokens instead of 1. Change the state table to

```
statetable ( "D CP", "Q QB") {
    table : "L/H R  : - - : L/H H/L,\
              -   ~R : - - : N   N";
  }
```

---

## LIBG-51

(error) Cell(%s): The '%s' port has too many port names in 'input_map'.\n \tNumber of ports specified is %d, the maximum number expected is %d.

### Description

There are too many port names specified in the 'input_map' attribute of the port. The maximum number of ports in 'input_map' is equal to the number of input nodes plus the number of internal nodes plus the number of input nodes that are edge sensitive. Therefore, the table is deleted.

The following example shows an instance where this message occurs:

```
statetable ( "D CP", "Q QB") {
    table : "L/H R  : - - : L/H H/L,\
```

Feedback

```
            -   ~R : - - : N   N";
  }
  pin(Q) {
    direction : output;
    function : "IQ";
    internal_node : "Q";
    input_map : "D CP Q QB CP CP";
    ...
  }
```

The maximum number of port names expected in the input_map attribute is 5: 2 input nodes (D, CP), one delayed input (CP*), and 2 internal nodes(Q, QB).

The following is an example message:

```
Error: Line 139, Cell(libg51): The 'Q' port has too many port names in
 'input_map'.
        Number of ports specified is 6, the maximum number expected is 5.
 (LIBG-51)
```

**What Next**

Remove some of the port names.

---

## LIBG-52

(error) Cell(%s): The '%s' port requires either 'internal_node' or\n \t'state_function'.

**Description**

The output or inout port does not have either an 'internal_node' attribute or a 'state_function' attribute even though a state table is defined. If a state table is defined, every output and inout port must have either an 'internal_node' or a 'state_function'. Therefore, the table is deleted.

The following example shows an instance where this message occurs:

```
statetable ( "D CP", "Q QB") {
    table : "L/H R  : - - : L/H H/L,\
             -   ~R : - - : N   N";
  }

  pin(Q) {
    direction : output;
    function : "IQ";
    timing() {
      timing_type : rising_edge;
      intrinsic_rise : 1.34;
      intrinsic_fall : 1.54;
      rise_resistance : 0.0718;
      fall_resistance : 0.0347;
      related_pin : "CP";
```

Feedback

```
    }
  }
```

In this example, the pin Q is missing the *state_function* or the *internal_node* attribute.

The following is an example message:

```
Error: Line 139, Cell(libg52): The 'Q' port requires either
 'internal_node' or
        'state_function'. (LIBG-52)
```

**What Next**

Add either an 'internal_node' attribute or a 'state_function' attribute to the port.

---

## LIBG-53

(error) Cell(%s): The 'internal_node' in the '%s' port\n \tdoes not match any internal node name in the state table.

**Description**

The 'internal_node' attribute on the port is not equal to the name of any internal node in the state table. Every 'internal_node' attribute must match one of the internal node names. Therefore, the table is deleted.

The following example shows an instance where this message occurs:

```
statetable ( "D CP", "Q QB") {
    table : "L/H R  : - - : L/H H/L,\
              -   ~R : - - : N   N";
  }

  pin(Q) {
    direction : output;
    function : "IQ";
    internal_node : "Q1";
    timing() {
      timing_type : rising_edge;
      intrinsic_rise : 1.34;
      intrinsic_fall : 1.54;
      rise_resistance : 0.0718;
      fall_resistance : 0.0347;
      related_pin : "CP";
    }
  }
```

In this case, Q1 is not an internal node in the state table.

Feedback

The following is an example message:

```
Error: Line 139, Cell(libg53): The 'internal_node' in the 'Q' port
        does not match any internal node name in the state table.
 (LIBG-53)
```

### What Next

Change the 'internal_node' attribute to equal an internal node name, or change an internal node name to equal the 'internal_node' attribute.

---

## LIBG-54

(information) Cell(%s): The '%s' internal pin does not drive any\n \toutput or is unneeded. The pin is ignored.

### Description

This message indicates that the specified internal pin does not directly or indirectly drive any output port, or that the pin can be deleted without affecting the functionality of the cell. An internal pin that is never visible is functionally useless. Every internal pin must have a unique functional relationship with an output port. The specified internal pin is ignored.

The following example shows an instance where this message occurs:

```
cell(libg54) {
          area : 10;
          pin(D) {
            direction : input;
            capacitance : 1;
            timing() {
              timing_type : setup_rising;
              intrinsic_rise : 0.9;
              intrinsic_fall : 0.9;
              related_pin : "CP";
            }
            timing() {
              timing_type : hold_rising;
              intrinsic_rise : 0.4;
              intrinsic_fall : 0.4;
              related_pin : "CP";
            }
          }
          pin(CP) {
            direction : input;
            capacitance : 1;
          }

          statetable ( "D CP", "Q QB") {
            table : "L/H R  : - - : L/H H/L,\
                      -   ~R : - - : N   N";
```

Feedback

```
    }

    pin(n1) {
      direction : internal;
      internal_node : "Q";
      timing() {
        timing_type : rising_edge;
        intrinsic_rise : 1.34;
        intrinsic_fall : 1.54;
        rise_resistance : 0.0718;
        fall_resistance : 0.0347;
        related_pin : "CP";
      }
    }
    pin(Q) {
      direction : output;
      internal_node : "Q";
      timing() {
        timing_type : rising_edge;
        intrinsic_rise : 1.34;
        intrinsic_fall : 1.54;
        rise_resistance : 0.0718;
        fall_resistance : 0.0347;
        related_pin : "CP";
      }
    }
    pin(QB) {
      direction : output;
      internal_node : "QB";
      timing() {
        timing_type : rising_edge;
        intrinsic_rise : 1.34;
        intrinsic_fall : 1.54;
        rise_resistance : 0.0718;
        fall_resistance : 0.0347;
        related_pin : "CP";
      }
    }
  }
```

In this case, n1 is not functionally used.

The following is an example message:

```
Information: Line 139, Cell(libg54): The 'n1' internal pin does not drive
 any
        output or is unneeded. The pin is ignored. (LIBG-54)
```

**What Next**

Either delete the internal pin or give it a unique functional relationship with an output port.

Feedback

## LIBG-55

(error) Cell(%s): The 'input_map' in the '%s' port is not\n \tallowed with the 'state_function'. Deleting the table.

### Description

This message indicates that the specified port has both a *state_function* attribute and an *input_map* attribute. A port with a *state_function* attribute cannot have an *input_map* attribute. The *input_map* attribute can occur only with the *internal_node* attribute. Therefore, the table is deleted. However, the following combinations of attributes can be specified on the same port:

```
* state_function and three_state attributes
        * internal_node and input_map attributes
        * internal_node and three_state attributes
```

The following example shows an instance where this message occurs:

```
pin(Q) {
    direction : output;
    state_function : "Q";
    input_map : "D CP";
    timing() {
      timing_type : rising_edge;
      intrinsic_rise : 1.34;
      intrinsic_fall : 1.54;
      rise_resistance : 0.0718;
      fall_resistance : 0.0347;
      related_pin : "CP";
    }
  }
```

You cannot have both state_function and input_map attributes in the same pin group.

The following is an example message:

```
Error: Line 139, Cell(libg55): The 'input_map' in the 'Q' port is not
        allowed with the 'state_function'. Deleting the table. (LIBG-55)
```

### What Next

Either replace the *state_function* attribute with an *internal_node* attribute, or delete the *input_map* attribute.

## LIBG-56

(error) Cell(%s): The '%s' node must be specified in\n \tthe 'input_map' of the '%s' port. Deleting the table.

Feedback

### Description

This message indicates that a node was specified as a don't care "-" in the 'input_map' of the port, but the node is necessary for the port. Every functionally related input node and current internal node must be specified. Therefore, the table is deleted.

The following example shows an instance where this message occurs:

```
statetable ( "D CP", "Q QB") {
    table : "L/H R  : - - : L/H H/L,\
             -   ~R : - - : N   N";
  }

  pin(Q) {
    direction : output;
    internal_node : "Q";
    input_map : "D -";
    timing() {
      timing_type : rising_edge;
      intrinsic_rise : 1.34;
      intrinsic_fall : 1.54;
      rise_resistance : 0.0718;
      fall_resistance : 0.0347;
      related_pin : "CP";
    }
  }
```

The use of a dont'care, "-", in the input_map means that the inputs 'CP' and 'CP*' are not used for the outport port 'Q'. However, given the state table, the nodes 'CP' and 'CP*' (delayed) are necessary for 'Q' and must be specified in the input_map string.

The following is an example message:

```
Error: Line 139, Cell(libg56): The 'CP' node must be specified in
        the 'input_map' of the 'Q' port. Deleting the table. (LIBG-56)
```

### What Next

Investigate why the assumption that the node is functionally unrelated to the port is incorrect. Possible reasons might be that the wrong internal code is chosen, the state table is incorrectly specified, or the 'input_map' is incorrect.

---

## LIBG-57

(error) Cell(%s): Duplicate '%s' names in the 'input_map' of the '%s' port.

### Description

Two or more nodes have been mapped to the same name either implicitly or explicitly in the 'input_map' attribute of the port. Every node used in the table must be mapped to a

Feedback

unique port. Unspecified node names in the 'input_map' are implicitly mapped to the node names specified in the state table. Therefore, the table is deleted.

The following example shows an instance where this message occurs:

```
statetable ( "D CP", "Q") {
    table : "L/H R  : - - : L/H,\
             -   ~R : - - : N";
  }

  pin(Q) {
    direction : output;
    internal_node : "Q";
    input_map : "D D";
    timing() {
      timing_type : rising_edge;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "CP";
    }
  }
```

In the previous case, the port name 'D' is explicitly duplicated for nodes D and CP.

```
statetable ( "D CP", "Q") {
    table : "L/H R  : - - : L/H,\
             -   ~R : - - : N";
  }

  pin(Q) {
    direction : output;
    internal_node : "Q";
    input_map : "CP";
    timing() {
      timing_type : rising_edge;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "CP";
    }
  }
```

In the previous case, the port name 'CP' is implicitly duplicated for nodes 'D' (explicitly specified in the input_map string) and 'CP' (implicitly mapped to the node name specified in the state table).

The following is an example message:

```
Error: Line 139, Cell(libg57): Duplicate 'D' names in the 'input_map' of
 the 'Q' port. (LIBG-57)
```

Feedback

### What Next

Change the 'input_map' attribute to eliminate duplicate names.

---

## LIBG-58

(error) Cell(%s): For the '%s' port, the '%s' node is mapped to\n \t'%s', which is not a port.

### Description

After the state table node names have been explicitly and implicitly mapped to names, one of the mapped node names is not the name of a port. Every mapped node name must be the name of a port. Unspecified node names in the 'input_map' are implicitly mapped to the node names specified in the state table. Therefore, the table is deleted.

The state table is in a completely independent, isolated name space. The term "node" refers to the identifiers. The "input nodes" and the "internal nodes" can have any name made up of any character except white space and comments.

The node names are resolved to the real port names by each port with an internal_node attribute. This mapping occurs by a combination of the 'input_map' attribute and the 'internal_node' attribute. If a node name is not mapped explicitly to a real port name in the 'input_map', it automatically inherits the node name as the real port name. The internal node name specified by the 'internal_node' attribute maps implicitly to the output being defined.

After all node names have been mapped to real port names, the real port names are checked to verify that they exist and are of the right direction. That is why the error message occurs at the output port. This is when elaboration occurs.

The following example shows an instance where this message occurs:

```
statetable ( "D CP", "Q") {
    table : "L/H R  : - - : L/H,\
             -   ~R : - - : N";
  }

  pin(Q) {
    direction : output;
    internal_node : "Q";
    input_map : "data CP";
    timing() {
      timing_type : rising_edge;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "CP";
    }
  }
```

Feedback

In this case, the port 'data' does not exist in the cell.

The following is an example message:

```
Error: Line 139, Cell(libg58): For the 'Q' port, the 'D' node is mapped
 to
        'data', which is not a port. (LIBG-58)
```

### What Next

Change the *input_map* attribute.

---

## LIBG-59

(error) Cell(%s): The 'input_map' of the '%s' port maps\n \tthe '%s' input port to the '%s' internal node.

### Description

The mapped internal node name refers to an input port. An internal node must be mapped to either an output, an internal pin, or an inout port but not to an input port. An input port cannot be driven. Therefore, the table is deleted.

The following example shows an instance where this message occurs:

```
pin(CP EN) {
    direction : input;
    capacitance : 1;
  }

  statetable ( "D CP", "Q QB") {
    table : "L/H R  : L - : L/H H/L,\
             L/H R  : H - : L H,\
             -   ~R : - - : N   N";
  }

  pin(Q) {
    direction : output;
    internal_node : "Q";
    input_map : "D CP EN";
    timing() {
      timing_type : rising_edge;
      intrinsic_rise : 1.34;
      intrinsic_fall : 1.54;
      rise_resistance : 0.0718;
      fall_resistance : 0.0347;
      related_pin : "CP";
    }
  }
```

Feedback

In this case, The internal node 'Q' is mapped to an input port 'EN'. Remove 'EN' from the input_map, or change its direction.

The following is an example message:

```
Error: Line 140, Cell(libg59): The 'input_map' of the 'Q' port maps
        the 'EN' input port to the 'Q' internal node. (LIBG-59)
```

**What Next**

Change the 'input_map' attribute.

---

## LIBG-60

(warning) Cell(%s): translates into a combinational cell.\n \tThe State table information is not created.

**Description**

This message indicates that the library cell has a sequential description that actually represents a combinational function. The state table information is allowed only for describing sequential functions. This situation occurs only when reading in an old technology library DB file. Issuing this warning is controlled by the *dc_shell* or the *lc_shell* environment variable *read_db_lib_warnings*.

The following is an example message:

```
Warning: Cell(libg60): translates into a combinational cell.
        The state table information is not created. (LIBG-60)
```

**What Next**

Update the original technology library source file and recompile.

---

## LIBG-61

(error) Cell(%s): Verification failure on the '%s' port. Deleting the table.

**Description**

When a cell has both a state table and the v3.1 format, the Library Compiler automatically converts the v3.1 format to UNIGEN. The UNIGEN generated from the v3.1 format is formally compared against the UNIGEN generated from the state table. They must match.

This message indicates that the state table description for the port does not match its sequential description. If both a state table description and a sequential description (ff, latch, ff_bank, or latch_bank) are specified for the library cell, they must match each other exactly. A listing of the differences is printed out. The maximum number of differences to be listed is controlled by the *dc_shell* environment variable *libgen_max_difference*.

Feedback

The following example shows an instance where this message occurs:

```
cell(libg61) {
          area : 10;
          pin(D) {
            direction : input;
            capacitance : 1;
            timing() {
              timing_type : setup_rising;
              intrinsic_rise : 0.1;
              intrinsic_fall : 0.1;
              related_pin : "CP";
            }
            timing() {
              timing_type : hold_rising;
              intrinsic_rise : 0.1;
              intrinsic_fall : 0.1;
              related_pin : "CP";
            }
          }
          pin(CP) {
            direction : input;
            capacitance : 1;
          }

          seq( IQ, IQN) {
            clocked_on : "CP";
            next_state : "D";
          }

          statetable ( "D CP", "Q QB") {
            table : "L/H F  : - - : L/H H/L,\
                      -   ~F : - - : N   N";
          }

          pin(Q) {
            direction : output;
            function : "IQ";
            internal_node : "Q";
            timing() {
              timing_type : rising_edge;
              intrinsic_rise : 1.0;
              intrinsic_fall : 1.0;
              rise_resistance : 0.1;
              fall_resistance : 0.1;
              related_pin : "CP";
            }
          }
          pin(QB) {
            direction : output;
            function : "IQN";
            internal_node : "QB";
            timing() {
```

Feedback

```
            timing_type : rising_edge;
            intrinsic_rise : 1.0;
            intrinsic_fall : 1.0;
            rise_resistance : 0.1;
            fall_resistance : 0.1;
            related_pin : "CP";
          }
        }
      }
```

The following is an example message:

```
Error: Line 139, Cell(libg61): Verification failure on the 'Q' port.
 Deleting the table.  (LIBG-61)
  Both table inputs are: D*, CP*, CP
  List of differences between statetable vs. v3.1 table:
    001    N <=> L
    010    L <=> N
    101    N <=> H
    110    H <=> N
```

### What Next

Fix or delete the state table, or fix or delete the sequential description.

---

## LIBG-62

(warning) Cell(%s): The '%s' input node is not required by any\n \tinternal node used. The input node is ignored.

### Description

This message indicates that the specified input node is not functionally related to any internal node used. Therefore, the input node is unused. An input node must be related to at least one internal node. This warning is for informational purposes because of the possibility of a user error. This message is suppressed if the input node has always been specified with the don't care,"-", token.

The following example shows an instance where this message occurs:

```
cell(libg62) {
          area : 10;
          pin(D) {
            direction : input;
            capacitance : 1;
            timing() {
              timing_type : setup_rising;
              intrinsic_rise : 0.1;
              intrinsic_fall : 0.1;
              related_pin : "CP";
            }
```

Feedback

```
    timing() {
      timing_type : hold_rising;
      intrinsic_rise : 0.1;
      intrinsic_fall : 0.1;
      related_pin : "CP";
    }
  }
  pin(CP) {
    direction : input;
    capacitance : 1;
  }

  ff( IQ, IQN) {
    clocked_on : "CP";
    next_state : "D";
  }

  statetable ( "D CP X", "Q QB") {
    table :  "L/H R  - : - - : L/H H/L,\
              -   ~R L/H : - - : N   N";
  }

  pin(Q) {
    direction : output;
    function : "IQ";
    internal_node : "Q";
  timing() {
      timing_type : rising_edge;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "CP";
    }
  }
  pin(QB) {
    direction : output;
    function : "IQN";
    internal_node : "QB";
    timing() {
      timing_type : rising_edge;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "CP";
    }
  }
}
```

Feedback

The following is an example message:

```
Warning: Line 134, Cell(libg62): The 'X' input node is not required by
 any
        internal node used. The input node is ignored. (LIBG-62)
```

### What Next

Determine why the input node is unused, and correct the problem.

---

## LIBG-63

(error) Cell(%s): The '%s' internal node is combinational.

### Description

In the table description of the internal node, there is no hold state "N". Every internal node should have a hold state "N". Otherwise, it is not a sequential node. Every internal node should be a single sequential storage element. Therefore, the table is deleted.

The following example shows an instance where this message occurs:

```
cell(libg63) {
          area : 10;
          pin(D) {
            direction : input;
            capacitance : 1;
            timing() {
              timing_type : setup_rising;
              intrinsic_rise : 0.1;
              intrinsic_fall : 0.1;
              related_pin : "CP";
            }
            timing() {
              timing_type : hold_rising;
              intrinsic_rise : 0.1;
              intrinsic_fall : 0.1;
              related_pin : "CP";
            }
          }
          pin(CP) {
            direction : input;
            capacitance : 1;
          }

          statetable ( "D CP", "Q QB") {
            table : "L/H R  : - - : L/H H/L,\
                      -   ~R : - - : L   N";
          }

          pin(Q) {
            direction : output;
```

Feedback

```
      internal_node : "Q";
      timing() {
        timing_type : rising_edge;
        intrinsic_rise : 1.0;
        intrinsic_fall : 1.0;
        rise_resistance : 0.1;
        fall_resistance : 0.1;
        related_pin : "CP";
      }
    }
    pin(QB) {
      direction : output;
      internal_node : "QB";
      timing() {
        timing_type : rising_edge;
        intrinsic_rise : 1.0;
        intrinsic_fall : 1.0;
        rise_resistance : 0.1;
        fall_resistance : 0.1;
        related_pin : "CP";
      }
    }
  }
```

In this case, the node 'QB' has a "N" state in the table but, the node 'Q' does not have an "N" state. All internal nodes in the table require at least one "N" state and at least one "non-N", or "non-X" state.

The following is an example message:

```
Error: Line 134, Cell(libg63): The 'Q' internal node is combinational.
 (LIBG-63)
```

### What Next

Add a hold state to the internal node or delete the internal node.

---

## LIBG-64

(error) Cell(%s): The '%s' port cannot have both 'internal_node'\n \tand 'state_function' attributes.

### Description

The port has both an 'internal_node' attribute and a 'state_function' attribute. Ports can only have either an 'internal_node' attribute, which is a name of an internal node, or a 'state_function' attribute, which is the output logic. Therefore, the table is deleted.

Feedback

A port can have the following combination of attributes:

```
* state_function and three_state attributes
        * internal_node and input_map attributes
        * internal_node and three_state attributes
```

The following example shows an instance where this message occurs:

```
cell(libg64) {
          area : 10;
          pin(D) {
            direction : input;
            capacitance : 1;
            timing() {
              timing_type : setup_rising;
              intrinsic_rise : 0.1;
              intrinsic_fall : 0.1;
              related_pin : "CP";
            }
            timing() {
              timing_type : hold_rising;
              intrinsic_rise : 0.1;
              intrinsic_fall : 0.1;
              related_pin : "CP";
            }
          }
          pin(CP) {
            direction : input;
            capacitance : 1;
          }

          statetable ( "D CP", "Q QB") {
            table : "L/H R  : - - : L/H H/L,\
                      -   ~R : - - : N   N";
          }

          pin(Q) {
            direction : output;
            internal_node : "Q";
            state_function : "Q";
            timing() {
              timing_type : rising_edge;
              intrinsic_rise : 1.0;
              intrinsic_fall : 1.0;
              rise_resistance : 0.1;
              fall_resistance : 0.1;
              related_pin : "CP";
            }
          }
          pin(QB) {
            direction : output;
            internal_node : "QB";
            timing() {
```

Feedback

```
            timing_type : rising_edge;
            intrinsic_rise : 1.0;
            intrinsic_fall : 1.0;
            rise_resistance : 0.1;
            fall_resistance : 0.1;
            related_pin : "CP";
          }
        }
      }
```

The following is an example message:

```
Error: Line 139, Cell(libg64): The 'Q' port cannot have both
 'internal_node'
        and 'state_function' attributes. (LIBG-64)
```

### What Next

Remove either the 'internal_node' attribute or the 'state_function' attribute.

---

## LIBG-65

(error) Cell(%s): The '%s' internal pin cannot have a 'three_state' attribute.

### Description

The internal pin has a 'three_state' attribute. Internal pins are not allowed to have a 'three_state' attribute. The attribute 'three_state' can only be specified on output or inout ports.

The following example shows an instance where this message occurs:

```
cell(libg65) {
          area : 10;
          pin(D) {
            direction : input;
            capacitance : 1;
            timing() {
              timing_type : setup_rising;
              intrinsic_rise : 0.1;
              intrinsic_fall : 0.1;
              related_pin : "CP";
            }
            timing() {
              timing_type : hold_rising;
              intrinsic_rise : 0.1;
              intrinsic_fall : 0.1;
              related_pin : "CP";
            }
          }
          pin(CP) {
            direction : input;
```

Feedback

```
    capacitance : 1;
  }

  statetable ( "D CP", "Q QB") {
    table : "L/H R  : - - : L/H H/L,\
             -   ~R : - - : N   N";
}

pin(Q) {
    direction : internal;
    internal_node : "Q";
    three_state : "D";  /* not allowed for internal pin */
    timing() {
      timing_type : rising_edge;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "CP";
    }
  }

  pin(QB) {
  direction : output;
    internal_node : "QB";
    timing() {
      timing_type : rising_edge;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "CP";
    }
  }
}
```

The following is an example message:

```
Error: Line 139, Cell(libg65): The 'Q' internal pin cannot have a
 'three_state' attribute. (LIBG-65)
```

**What Next**

Either change the internal pin to a port or remove the 'three_state' attribute.

---

## LIBG-67

(warning) Cell(%s): The '%s' port has both the internal_node\n \tand the input_map specified. However, there is a mismatch in the mapping.\n \tThe '%s' port is mapped by the input_map instead of the '%s' port, \n \tthe name of the internal_node. The '%s' port, specified in the input_map, is used.

Feedback

### Description

The port mapped by 'input_map' to the internal node specified in internal_node is not equal to the name of the port being defined. The current state of an output port should be equal to the state of the output port, not another port. This condition should almost never be violated, but the user is given the freedom to do so with a warning.

The following example shows an instance where this message occurs:

```
cell(libg67) {
          area : 10;
          pin(D) {
            direction : input;
            capacitance : 1;
            timing() {
              timing_type : setup_rising;
              intrinsic_rise : 0.1;
              intrinsic_fall : 0.1;
              related_pin : "CP";
            }
            timing() {
              timing_type : hold_rising;
              intrinsic_rise : 0.1;
              intrinsic_fall : 0.1;
              related_pin : "CP";
            }
          }
          pin(CP) {
            direction : input;
            capacitance : 1;
          }

          statetable ( "D CP", "Q QB") {
            table : "L/H R  : L - : L/H H/L,\
                   L/H R  : H - : L H,\
                     -   ~R : - - : N   N";
          }

          pin(Q) {
            direction : output;
            internal_node : "Q";
            input_map : "D CP QB";
            timing() {
              timing_type : rising_edge;
              intrinsic_rise : 1.0;
              intrinsic_fall : 1.0;
              rise_resistance : 0.1;
              fall_resistance : 0.1;
              related_pin : "CP";
            }
          }
          pin(QB) {
```

Feedback

```
            direction : output;
            internal_node : "QB";
            timing() {
              timing_type : rising_edge;
              intrinsic_rise : 1.0;
              intrinsic_fall : 1.0;
              rise_resistance : 0.1;
              fall_resistance : 0.1;
              related_pin : "CP";
            }
          }
        }
```

The following is an example message:

```
Warning: Line 140, Cell(libg67): The 'Q' port has both the internal_node
        and the input_map specified. However, there is a mismatch in the
 mapping.
        The 'QB' port is mapped by the input_map instead of the 'Q' port,
        the name of internal_node. The 'QB' port, specified in the
 input_map, is used. (LIBG-67)
```

**What Next**

Change the input_map attribute to map the current state to the port being defined.

---

## LIBG-68

(error) Cell(%s): Mismatch in internal pins between\n \tthe v3.1 format and the state table. Deleting the table.

**Description**

UNIGEN generated from the v3.1 format is formally compared against UNIGEN generated from the state table. They must match.

This message indicates that the v3.1 format has an implied internal pin that cannot be mapped to an internal pin in the state table. There must be a one-to-one match between the internal pins of the state table and the v3.1 format.

The following example shows an instance where this message occurs:

```
cell(libg68) {
          area : 10;
          pin(D) {
            direction : input;
            capacitance : 1;
            timing() {
              timing_type : setup_rising;
              intrinsic_rise : 0.1;
              intrinsic_fall : 0.1;
              related_pin : "CP";
```

Feedback

```
      }
      timing() {
        timing_type : hold_rising;
        intrinsic_rise : 0.1;
        intrinsic_fall : 0.1;
        related_pin : "CP";
      }
    }
    pin(CP CP_ALSO) {
      direction : input;
      capacitance : 1;
    }

    ff( IQ, IQN) {
      clocked_on : "CP";
      clocked_on_also : "CP_ALSO";
      next_state : "D";
    }

    statetable ( "D CP", "Q QB") {
      table : "L/H R  : - -  : L/H H/L,\
               -   ~R : - - : N   N";
    }

    pin(QB) {
      direction : output;
      function : "IQN";
      internal_node : "QB";
      timing() {
        timing_type : rising_edge;
        intrinsic_rise : 1.0;
        intrinsic_fall : 1.0;
        rise_resistance : 0.1;
        fall_resistance : 0.1;
        related_pin : "CP";
      }
      timing() {
        timing_type : rising_edge;
        intrinsic_rise : 1.0;
        intrinsic_fall : 1.0;
        rise_resistance : 0.1;
        fall_resistance : 0.1;
        related_pin : "CP_ALSO";
      }
    }
  }
```

The following is an example message:

```
Error: Line 140, Cell(libg68): Mismatch in internal pins between
        the v3.1 format and the state table. Deleting the table.
 (LIBG-68)
```

Feedback

### What Next

Determine whether the state table is specified incorrectly or is missing an internal pin. Correct the problem.

---

## LIBG-69

(warning) Cell(%s): The '%s' input is specified as\n \tdelayed in the state table. It will be used.

### Description

Delayed input nodes are specified in the state table description, and can be specified with the asterisk (*) suffix. Only advanced users should consider specifying delayed inputs because there is a potential for error and confusion.

The following example shows an instance where this message occurs:

```
cell(libg69) {
          area : 10;
          pin(D) {
            direction : input;
            capacitance : 1;
            timing() {
              timing_type : setup_rising;
              intrinsic_rise : 0.1;
              intrinsic_fall : 0.1;
              related_pin : "CP";
            }
            timing() {
              timing_type : hold_rising;
              intrinsic_rise : 0.1;
              intrinsic_fall : 0.1;
              related_pin : "CP";
            }
          }
          pin(CP) {
            direction : input;
          capacitance : 1;
          }

          statetable ( "D CP* CP", "Q QB") {
            table : "L/H L  H  : - - : L/H H/L,\
                     -   H  - : - - : N   N,\
                     -   L  L : - - : N   N";
          }

          pin(Q) {
            direction : output;
            internal_node : "Q";
            timing() {
```

Feedback

```
            timing_type : rising_edge;
            intrinsic_rise : 1.0;
            intrinsic_fall : 1.0;
            rise_resistance : 0.1;
          fall_resistance : 0.1;
            related_pin : "CP";
          }
        }

        pin(QB) {
          direction : output;
          internal_node : "QB";
          timing() {
            timing_type : rising_edge;
            intrinsic_rise : 1.0;
            intrinsic_fall : 1.0;
            rise_resistance : 0.1;
            fall_resistance : 0.1;
            related_pin : "CP";
          }
        }
      }
```

The following is an example message:

```
Warning: Line 134, Cell(libg69): The 'CP' input is specified as
        delayed in the state table. It will be used. (LIBG-69)
```

### What Next

Try to specify the state table without referring to delayed inputs.

---

## LIBG-70

(warning) Cell(%s): Delayed inputs are specified in\n \tthe input_map of the '%s' port. They will be used.

### Description

Delayed input ports are specified in the input_map attribute, and are specified after the undelayed inputs and the internal nodes. Only advanced users should consider specifying delayed inputs because there is a potential for error and confusion.

The following example shows an instance where this message occurs:

```
cell(libg70) {
          area : 10;
          pin(D) {
            direction : input;
            capacitance : 1;
            timing() {
              timing_type : setup_rising;
```

Feedback

```
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      related_pin : "CP";
    }
    timing() {
      timing_type : hold_rising;
      intrinsic_rise : 1.0;
    intrinsic_fall : 1.0;
      related_pin : "CP";
    }
  }
  pin(CP) {
    direction : input;
    capacitance : 1;
  }

  statetable ( "D CP", "Q QB") {
    table : "L/H R  : - - : L/H H/L,\
             -   ~R : - - : N   N";
  }

pin(Q) {
    direction : output;
    inverted_output : FALSE;
    internal_node : "Q";
    input_map : "D CP Q - CP";
    timing() {
      timing_type : rising_edge;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "CP";
    }
  }

  pin(QB) {
    direction : output;
    internal_node : "QB";
    inverted_output : TRUE;
    timing() {
      timing_type : rising_edge;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "CP";
    }
  }
}
```

Feedback

The following is an example message:

```
Warning: Line 139, Cell(libg70): Delayed inputs are specified in
        the input_map of the 'Q' port. They will be used. (LIBG-70)
```

**What Next**

Try to specify the 'input_map' without mapping to delayed inputs.

---

## LIBG-71

(error) Cell(%s): The '%s' port was expected to be %s with\n \trespect to the '%s' port.

**Description**

The two ports are not the function of each other as expected. The expected behavior is derived from the v3.1 format, and the state table is expected to match. Therefore, the table is deleted.

The following example shows an instance where this message occurs:

```
cell(libg71) {
          area : 10;
          pin(D) {
            direction : input;
            capacitance : 1;
            timing() {
              timing_type : setup_rising;
              intrinsic_rise : 0.1;
              intrinsic_fall : 0.1;
              related_pin : "CP";
            }
            timing() {
              timing_type : hold_rising;
              intrinsic_rise : 0.1;
              intrinsic_fall : 0.1;
              related_pin : "CP";
            }
          }

          pin(CP) {
            direction : input;
            capacitance : 1;
          }

          ff( IQ, IQN) {
            clocked_on : "CP";
            next_state : "D";
          }

          statetable ( "D CP", "Q QB") {
            table : "L/H R  : - -  : L/H H/L,\
```

Feedback

```
                    -   ~R : - -  : N   N";
          }

          pin(Q) {
            direction : output;
            function : "IQ";
            internal_node : "Q";
            timing() {
              timing_type : rising_edge;
              intrinsic_rise : 1.0;
              intrinsic_fall : 1.0;
              rise_resistance : 0.1;
              fall_resistance : 0.1;
              related_pin : "CP";
            }
          }

          pin(QB) {
            direction : output;
            function : "IQ";            /* Wrong! function should be IQN
 */
            internal_node : "QB";
            timing() {
              timing_type : rising_edge;
              intrinsic_rise : 1.0;
              intrinsic_fall : 1.0;
              rise_resistance : 0.1;
              fall_resistance : 0.1;
              related_pin : "CP";
            }
          }
        }
```

In this case, the function descriptions for the two ports are inconsistent. From the v3.1 table, the two ports have the same function (notice the typo in the function of the pin 'QB', instead of 'IQN', the function string is 'IQ'). From the UNIGEN table they are inverted.

The following is an example message:

```
Error: Line 139, Cell(libg71): The 'Q' port was expected to be uninverted
 with
        respect to the 'QB' port. (LIBG-71)
```

**What Next**

Correct the state table or the v3.1 format.

---

## LIBG-72

(error) Cell(%s): Cannot find an implicit internal pin in\n \tthe v3.1 format to match the '%s' internal pin.

Feedback

### Description

The internal pin does not have a corresponding internal pin in the v3.1 format. The internal pins of both the v3.1 format and the state table format must match. Therefore, the table is deleted.

The following example shows an instance where this message occurs:

```
cell(libg72) {
          area : 10;
          pin(force11) {
            direction : input;
            capacitance : 1;
          }
          pin(force10) {
            direction : input;
            capacitance : 1;
          }
          pin(force01) {
            direction : input;
            capacitance : 1;
          }
          pin(force00) {
            direction : input;
            capacitance : 2;
          }
          pin(ck ck_also) {
            direction : input;
            capacitance : 2;
          }
          pin(next) {
            direction : input;
            capacitance : 2;
        }

          state("IQ","IQN") {
            next_state : "next";
            clocked_on : "ck";
            clocked_on_also : "ck_also";
            force_00   : "force00";
            force_01   : "force01";
            force_10   : "force10";
            force_11   : "force11";
          }

          statetable ("next ck force00 force01 force10 force11",\
                      "int_net") {\
             table : "-   -   H   L   L   L   : -   : L,\
                      -   -   L   H   L   L   : -   : L,\
                      -   -   L   L   H   L   : -   : H,\
                      -   -   L   L   L   H   : -   : H,\
                      -   ~R  L   L   L   L   : -   : N,\
                      L/H R   L   L   L   L   : -   : L/H";
```

Feedback

```
          }

          pin(int_net) {
            direction : internal;
            internal_node : "int_net";
            input_map : "next ck_also force00 force01 force10 force11
int_net";
        }

          pin(Q) {
            direction : output;
            function : "IQ";
            internal_node : "int_net";
            input_map : "int_net ck_also force00 force01 force10 force11
Q";
          }
        }
```

The following is an example message:

```
Error: Line 165, Cell(libg72): Cannot find an implicit internal pin in
        the v3.1 format to match the 'int_net' internal pin. (LIBG-72)
```

**What Next**

Correct the state table or the v3.1 format.

---

## LIBG-73

(warning) Cell(%s): Table inputs for the '%s' port exceed 16 inputs.

**Description**

The state table is stored with over 16 table inputs. There is an exponential cost in performance and memory as the number of table inputs increases. This warning tells the user about the excessive size of table inputs. To list the table, use the report_lib command with the *full_table* option.

The following example shows an instance where this message occurs:

```
cell(libg73) {
          area : 10;
          pin(i0 i1 i2 i3 i4 i5 i6 i7 i8 j0 j1 j2 j3 j4 j5) {
            direction : input;
            capacitance : 1;
          }
          pin(D) {
            direction : input;
            capacitance : 1;
            timing() {
              timing_type : setup_rising;
              intrinsic_rise : 0.9;
```

Feedback

```
              intrinsic_fall : 0.9;
              related_pin : "CP";
            }
            timing() {
              timing_type : hold_rising;
              intrinsic_rise : 0.4;
              intrinsic_fall : 0.4;
              related_pin : "CP";
            }
          }
          pin(CP) {
            direction : input;
            capacitance : 1;
          }

          statetable ( "D i0 i1 i2 i3 i4 i5 i6 i7 i8 j0 j1 j2 j3 j4 j5
 CP", "Q ") {
           table : "H H H H H H H H H H H H H H H H R   : - : H,\
                    - - - - - - - - - - - - - - - - R   : - : L,\
                    - - - - - - - - - - - - - - - - ~R  : - : N";
          }

          pin(Q) {
            direction : output;
            internal_node : "Q";
            timing() {
              timing_type : rising_edge;
              intrinsic_rise : 1.34;
              intrinsic_fall : 1.54;
              rise_resistance : 0.0718;
              fall_resistance : 0.0347;
              related_pin : "CP";
            }
          }
        }
```

The following is an example message:

```
Warning: Line 149, Cell(libg73): Table inputs for the 'Q' port exceed 16
 inputs.
(LIBG-73)
```

**What Next**

If possible, take advantage of the input_map attribute to reduce the size of the table.

---

## LIBG-74

(error) Cell(%s): No internal_node output pin is found.

Feedback

### Description

A state table is defined for the library cell, but none of the output pins are defined as sequential. There must be at least one output with an internal_node attribute specified.

The following example shows an instance where this message occurs:

```
cell(libg74) {
          area : 10;
          pin(D) {
            direction : input;
            capacitance : 1;
            timing() {
              timing_type : setup_rising;
              intrinsic_rise : 0.1;
              intrinsic_fall : 0.1;
              related_pin : "CP";
            }
            timing() {
              timing_type : hold_rising;
              intrinsic_rise : 0.1;
              intrinsic_fall : 0.1;
            related_pin : "CP";
            }
          }

          pin(CP) {
            direction : input;
            capacitance : 1;
          }

          statetable ( "D CP ", "Q QB") {
            table : "L/H R  : - - : L/H H/L,\
                      -   ~R : - - : N   N";
          }

          pin(Q) {
            direction : output;
            state_function : "D";
            timing() {
              timing_type : rising_edge;
              intrinsic_rise : 1.0;
              intrinsic_fall : 1.0;
              rise_resistance : 0.1;
              fall_resistance : 0.1;
              related_pin : "CP";
            }
          }

          pin(QB) {
            direction : output;
            state_function : "D";
            timing() {
```

Feedback

```
            timing_type : rising_edge;
            intrinsic_rise : 1.0;
          intrinsic_fall : 1.0;
          rise_resistance : 0.1;
            fall_resistance : 0.1;
            related_pin : "CP";
          }
        }
      }
```

In this case, both output pins 'Q' and 'QB' have state_function and no internal_node attribute.

The following is an example message:

```
Error: Line 137, Cell(libg74): no internal_node output pin is found.
 (LIBG-74)
```

### What Next

Remove the state table description or add an internal_node attribute to at least one output.

---

## LIBG-75

(error) Cell(%s): The state_function on the '%s' port has\n \tan invalid port in the expression.

### Description

A valid port in a state_function expression is either an input port, an inout port with a three_state attribute, or an output port with the internal_node attribute. This error is issued if a port in the state_function is either an output port without an internal_node attribute or an inout port without a three_state attribute. Neither of these two kinds of ports is allowed in the state_function expression.

The following example shows an instance where this message occurs:

```
cell(libg75) {
          area : 10;
          pin(D) {
            direction : input;
            capacitance : 1;
            timing() {
              timing_type : setup_rising;
              intrinsic_rise : 0.1;
              intrinsic_fall : 0.1;
              related_pin : "CP";
            }
            timing() {
              timing_type : hold_rising;
              intrinsic_rise : 0.1;
```

Feedback

```
            intrinsic_fall : 0.1;
            related_pin : "CP";
          }
        }

        pin(CP) {
          direction : input;
          capacitance : 1;
        }

        statetable ( "D CP ", "Q QB") {
          table : "L/H R  : - - : L/H H/L,\
                    -   ~R : - - : N   N";
        }

        pin(Q) {
          direction : output;
          internal_node : "Q";
          timing() {
            timing_type : rising_edge;
            intrinsic_rise : 1.0;
            intrinsic_fall : 1.0;
            rise_resistance : 0.1;
            fall_resistance : 0.1;
            related_pin : "CP";
          }
        }

        pin(QB) {
          direction : output;
          state_function : "Q";
          timing() {
            timing_type : rising_edge;
            intrinsic_rise : 1.0;
            intrinsic_fall : 1.0;
            rise_resistance : 0.1;
            fall_resistance : 0.1;
            related_pin : "CP";
          }
        }

        pin(QC) {
          direction : output;
          state_function : "QB"; /* 'QB' is an output without
internal_node*/
          timing() {
            timing_type : rising_edge;
            intrinsic_rise : 1.0;
            intrinsic_fall : 1.0;
            rise_resistance : 0.1;
            fall_resistance : 0.1;
            related_pin : "CP";
        }
```

Feedback

```
        }
      }
```

The following is an example message:

```
Error: Line 167, Cell(libg75): The state_function on the 'QC' port has
       an invalid port in the expression. (LIBG-75)
```

### What Next

Change the state_function expression, or change the invalid port.

---

## LIBG-76

(warning) The '%s' test cell pin on the '%s' cell\n \thas a timing arc. The timing arc is ignored.

### Description

The test cell pin has a timing arc, but timing arcs are only valid on the library cell.

The following example shows an instance where this message occurs:

```
cell(libg76) {
            area : 11;
            pin(D0) {
              direction : input;
              capacitance : 1;
              timing() {
                timing_type : setup_rising;
                intrinsic_rise : 1.0;
                intrinsic_fall : 1.0;
                related_pin : "CK1";
              }
              timing() {
                timing_type : hold_rising;
                intrinsic_rise : 0.1;
                intrinsic_fall : 0.1;
                related_pin : "CK1";
              }
            }
            pin(TI) {
              direction : input;
              capacitance : 1;
              timing() {
                timing_type : setup_rising;
                intrinsic_rise : 1.0;
                intrinsic_fall : 1.0;
                related_pin : "CK1";
              }
              timing() {
                timing_type : hold_rising;
```

Feedback

```
    intrinsic_rise : 0.1;
    intrinsic_fall : 0.1;
    related_pin : "CK1";
  }
}
pin(TE) {
  direction : input;
  capacitance : 1;
  prefer_tied : "0";
  timing() {
    timing_type : setup_rising;
    intrinsic_rise : 1.0;
    intrinsic_fall : 1.0;
    related_pin : "CK1";
  }
  timing() {
    timing_type : hold_rising;
    intrinsic_rise : 0.1;
    intrinsic_fall : 0.1;
    related_pin : "CK1";
  }
}
pin(CK1) {
  direction : input;
  capacitance : 1;
}
pin(CK2) {
  direction : input;
  capacitance : 1;
}
statetable("D0 TE TI CK1 CK2", "n1 Q") {
  table : "-   -  -  -  L  : L/H - : - L/H,\
           L/H L  -  L  -  : -   - : L/H -,\
           -   H L/H L  -  : - - : L/H -,\
           -   -  -  -  -  : - -  : N N";
}

pin(n1) {
  direction : internal;
  internal_node : "n1";
}

pin(Q) {
  direction : output;
  internal_node : "Q";
  timing() {
    timing_type : falling_edge;
    intrinsic_rise : 1.0;
    intrinsic_fall : 1.0;
    rise_resistance : 0.1;
    fall_resistance : 0.1;
    related_pin : "CK2";
  }
```

Feedback

```
                }
                pin(QN) {
                  direction : output;
                  state_function : "Q'";
                  timing() {
                    timing_type : falling_edge;
                    intrinsic_rise : 1.0;
                    intrinsic_fall : 1.0;
                    rise_resistance : 0.1;
                    fall_resistance : 0.1;
                    related_pin : "CK2";
                  }
                }
                test_cell() {
                  pin(D0) {
                    direction : input;
                  }
                  pin(TI) {
                    direction : input;
                    signal_type : "test_scan_in";
                    timing() {                        /* move  it to the cell
level*/
                      timing_type : setup_rising;
                      intrinsic_rise : 1.0;
                      intrinsic_fall : 1.0;
                      related_pin : "CK1";
                    }
                  }
                  pin(TE) {
                    direction : input;
                    signal_type : "test_scan_enable";
                  }
                  pin(CK1) {
                    direction : input;
                    capacitance : 1;
                  }
                  pin(CK2) {
                    direction : input;
                    capacitance : 1;
                  }
                  statetable("D0 CK1", "Q") {
                    table : "-   H : - : N,\
                             L/H L  : - : L/H";
                  }

                  pin(n1) {
                    direction : internal;
                    internal_node : "Q";
                  }
                  pin(Q) {
                    direction : output;
                    internal_node : "Q";
                    input_map : "n1 CK2";
```

Feedback

```
            signal_type : "test_scan_out";
          }
          pin(QN) {
            direction : output;
            state_function : "Q'";
            signal_type : "test_scan_out_inverted";
          }
        }
      }
```

The following is an example message:

```
Warning: Line 209, The 'TI' test cell pin on the 'libg76' cell
        has a timing arc. The timing arc is ignored. (LIBG-76)
```

### What Next

Move all timing arcs in the test cell to the top-level library cell.

---

## LIBG-77

(error) The '%s' test cell pin on the '%s' cell\n \thas a conflicting pin type.

### Description

This message indicates that the test cell has one of these invalid combinations:

```
Two pins, of which one has the signal_type attribute
           "test_scan_in" and the other has "test_scan_in_inverted";

        or  Two pins, of which one has the signal_type attribute
           "test_scan_enable" and the other has
 "test_scan_enable_inverted".
```

A test cell cannot have two pins whose *signal_type* attributes are the inversions of each other.

### What Next

Change one of the *signal_type* attributes so that the test cell has only one of the pins in the above invalid combinations.

---

## LIBG-78

(warning) The '%s' test cell pin on the '%s' cell\n \thas a conflicting pin type. Removing the test cell.

Feedback

### Description

This message indicates that the test cell has one of these invalid combinations:

```
Two pins, of which one has the signal_type attribute
           "test_scan_in" and the other has "test_scan_in_inverted";

        or  Two pins, of which one has the signal_type attribute
           "test_scan_enable" and the other has
 "test_scan_enable_inverted".
```

A test cell cannot have two pins whose *signal_type* attributes are the inversions of each other. The test cell has been removed; DFT Compiler will not use it.

### What Next

Change one of the *signal_type* attributes so that the test cell has only one of the pins in the above invalid combinations. Then re-execute.

---

## LIBG-79

(error) The '%s' test cell pin on the '%s' cell\n \thas a conflict between the function and the test type.

### Description

The test cell pin either has a signal_type attribute of "test_scan_out" and a function of the inverted output (i.e. second variable) or the pin has a signal_type attribute of "test_scan_out_inverted" and a function of the non-inverted output (i.e. first variable).

If a pin has both a signal_type attribute and a function attribute, the two attributes must be consistent. A "test_scan_out" corresponds to the non-inverted output function and a "test_scan_out_inverted" corresponds to the inverted output.

### What Next

Change either the signal_type attribute or the function.

---

## LIBG-80

(warning) The '%s' test cell pin on the '%s' cell has a\n \tconflict between the function and the test type. The test cell is removed.

### Description

The test cell pin either has a signal_type attribute of "test_scan_out" and a function of the inverted output (i.e. second variable) or the pin has a signal_type attribute of "test_scan_out_inverted" and a function of the non-inverted output (i.e. first variable).

Feedback

If a pin has both a signal_type attribute and a function attribute, the two attributes must be consistent. A "test_scan_out" corresponds to the non-inverted output function and a "test_scan_out_inverted" corresponds to the inverted output.

The test cell is removed and DFT Compiler will not use it.

**What Next**

Change either the signal_type attribute or the function.

---

## LIBG-81

(error) Cell(%s): translates into a combinational cell.

**Description**

The library cell has a sequential description, which represents a combinational function. The state table information is only allowed on sequential library cells.

The following example shows an instance where this message occurs:

```
cell(libg81) {
      area : 4;
      pin(Q) {
        max_fanout : 1.0;
        direction : output;
        function : "IQ";
        timing() {
          timing_type : rising_edge;
          intrinsic_rise : 1.0;
          intrinsic_fall : 1.0;
          rise_resistance : 0.1;
          fall_resistance : 0.1;
          related_pin : "CK";
        }
        timing() {
          timing_type : preset;
          timing_sense : "negative_unate";
          intrinsic_rise : 1.0;
          rise_resistance : 0.1;
          related_pin : "SETZ"
        }
      }
      pin(QZ) {
        direction : output;
        function : "IQZ";
        timing() {
          timing_type : rising_edge;
          intrinsic_rise : 1.0;
          intrinsic_fall : 1.0;
          rise_resistance : 0.1;
          fall_resistance : 0.1;
```

Feedback

```
    related_pin : "CK";
  }
  timing() {
    timing_type : clear;
    timing_sense : "positive_unate";
    intrinsic_fall : 1.0;
    fall_resistance : 1.0;
    related_pin : "SETZ";
  }
}
pin(CK) {
  direction : input;
  clock : true;
  min_period : 1.0;
  capacitance : 0.1;
}
pin(S) {
  direction : input;
  capacitance : 1.0;
  timing() {
    timing_type : setup_rising;
    intrinsic_rise : 1.0;
    intrinsic_fall : 1.0;
    related_pin : "CK";
  }
  timing() {
    timing_type : hold_rising;
    intrinsic_rise : 1.0;
    intrinsic_fall : 1.0;
    related_pin : "CK";
  }
}
pin_opposite("Q","QZ");
state("IQ","IQZ") {
  next_state : "(S' A) + (S B)";
  clocked_on : "CK";
  force_01   : "SETZ";
  force_10   : "SETZ'";
}
pin(A) {
  direction : input;
  capacitance : 1.0;
  timing() {
    timing_type : setup_rising;
    intrinsic_rise : 1.0;
    intrinsic_fall : 1.0;
    related_pin : "CK";
  }
  timing() {
    timing_type : hold_rising;
    intrinsic_rise : 1.0;
    intrinsic_fall : 1.0;
    related_pin : "CK";
```

Feedback

```
      }
    }
    pin(B) {
      direction : input;
      capacitance : 0.1;
      timing() {
        timing_type : setup_rising;
        intrinsic_rise : 1.0;
        intrinsic_fall : 1.0;
        related_pin : "CK";
      }
      timing() {
        timing_type : hold_rising;
        intrinsic_rise : 1.0;
        intrinsic_fall : 1.0;
        related_pin : "CK";
      }
    }
    pin(SETZ) {
      direction : input;
      capacitance : 0.1;
    }
  }
```

The following is an example message:

```
Error: Line 109, Cell(libg81): translates into a combinational cell.
 (LIBG-81)
```

**What Next**

Change the sequential description.

---

## LIBG-82

(error) The test cell on the '%s' cell is not DFT Compiler supported.

**Description**

This message indicates that the description of the specified test cell is not supported by DFT Compiler. DFT Compiler has restrictions on the non-test mode and test inputs; the test cell violates one or more of these restrictions. For details on these restrictions, refer to the *DFT Compiler Test Design Rule Checking User Guide*.

**What Next**

Revise the test cell description so that it conforms to DFT Compiler restrictions; or, remove the test cell.

Feedback

## LIBG-83

(warning) The test cell on the '%s' cell is not DFT Compiler supported.\n \tRemoving the test cell.

### Description

The test cell has a description that is not supported by DFT Compiler. DFT Compiler has restrictions on the non-test mode and test inputs; the test cell violates one or more of these restrictions. For details on these restrictions, refer to the *DFT Compiler Test Design Rule Checking User Guide*.

The test cell has been removed; DFT Compiler will not use this cell during scan insertion.

### What Next

Obtain the technology library source and revise the test cell description so that it conforms to DFT Compiler restrictions; or, remove the test cell.

## LIBG-84

(warning) The '%s' port in the '%s' cell does not\n \thave a function in the test cell descriptions. The cell is a black box.

### Description

No function for that port can be derived from the test cells. The port function must be either explicitly specified or inferable from the test cell descriptions. The cell is a black box.

This warning is issued when reading a .db library and can be switched off by setting the environment variable *read_db_lib_warnings* to FALSE (default).

The same problem in a .lib library causes an error. See the .lib example for LIBG-85.

### What Next

Change the technology library source by explicitly or implicitly providing the function for that port.

## LIBG-85

(error) The '%s' port in the '%s' cell does not\n \thave a function in the test cell descriptions.

### Description

No function for that port can be derived from the test cells. The port function must be either explicitly specified or inferable from the test cell descriptions. Some functional

Feedback

information is described but not all. Either the cell is completely specified, or it is completely unspecified.

The following example shows an instance where this message occurs:

```
cell(libg85) {
            area : 11;
            pin(D0) {
              direction : input;
              capacitance : 1;
              timing() {
                timing_type : setup_rising;
                intrinsic_rise : 1.3;
                intrinsic_fall : 1.3;
                related_pin : "CK1";
              }
              timing() {
                timing_type : hold_rising;
                intrinsic_rise : 0.3;
                intrinsic_fall : 0.3;
                related_pin : "CK1";
              }
            }
            pin(TI) {
              direction : input;
              capacitance : 1;
              timing() {
                timing_type : setup_rising;
                intrinsic_rise : 1.3;
                intrinsic_fall : 1.3;
                related_pin : "CK1";
              }
              timing() {
                timing_type : hold_rising;
                intrinsic_rise : 0.3;
                intrinsic_fall : 0.3;
                related_pin : "CK1";
              }
            }

            pin(TE) {
              direction : input;
              capacitance : 1;
              prefer_tied : "0";
              timing() {
                timing_type : setup_rising;
                intrinsic_rise : 1.3;
                intrinsic_fall : 1.3;
                related_pin : "CK1";
              }
              timing() {
                timing_type : hold_rising;
                intrinsic_rise : 0.3;
```

Feedback

```
    intrinsic_fall : 0.3;
    related_pin : "CK1";
  }
}
pin(CK1) {
  direction : input;
  capacitance : 1;
}

pin(Q) {
  direction : output;
  inverted_output : FALSE;
  timing() {
    timing_type : rising_edge;
    intrinsic_rise : 1.09;
    intrinsic_fall : 1.37;
    rise_resistance : 0.1458;
    fall_resistance : 0.0523;
    related_pin : "CK1";
  }
}

pin(QN) {
  direction : output;
  inverted_output : TRUE;
timing() {
    timing_type : rising_edge;
    intrinsic_rise : 1.59;
    intrinsic_fall : 1.57;
    rise_resistance : 0.1458;
    fall_resistance : 0.0523;
    related_pin : "CK1";
  }
}

test_cell() {
  pin(D0) {
  direction : input;
  }
  pin(TE) {
    direction : input;
    signal_type : "test_scan_enable";
  }
  pin(TI) {
    direction : input;
    signal_type : "test_scan_in";
  }
  pin(CK1) {
    direction : input;
    capacitance : 1;
  }

  statetable("D0 CK1", "Q") {
```

Feedback

```
        table : "L/H F : - : L/H,\
                 -  ~F : - : N";
      }

      pin(Q) {
        direction : output;
        internal_node : "Q";
        signal_type : "test_scan_out";
      }
      pin(QN) {
        direction : output;
        signal_type : "test_scan_out_inverted";
      }
    }
  }
```

The following is an example message:

```
Error: Line 177, The 'QN' port in the 'libg85' cell does not
        have a function in the test cell descriptions. (LIBG-85)
```

**What Next**

Change the technology library source by either describing the port function explicitly or implicitly in the test cell descriptions or removing all functional descriptions.

---

## LIBG-86

(warning) The '%s' port in the '%s' cell has a timing\n \tarc related to '%s' that is inconsistent.

**Description**

The timing type of the timing arc from the specified port to the related port is inconsistent with respect to other timing in the cell. For example, a port cannot have both a *rising_edge* timing type and a *falling_edge* timing type.

This warning occurs only when reading in a technology library DB file. Issuing this warning is controlled by the *dc_shell* or the *lc_shell* environment variable *read_db_lib_warnings*.

The same problem in a .lib library causes an error. See the .lib example for LIBG-88.

The following is an example message:

```
Warning: The 'Q' in the 'libg86' cell has a timing
        arc related to 'D' that is inconsistent. (LIBG-86)
```

**What Next**

Update the original technology library source. Make timing arcs consistent.

Feedback

## LIBG-87

(error) Cell(%s): the internal node has L/H or H/L but\n \tno input has L/H or H/L. Deleting the table.

### Description

In the *table* attribute of the *statetable* group, an internal node has either L/H or H/L specified but none of the input nodes have either L/H or H/L specified. If an internal node has either the L/H or H/L value, at least one input node must be L/H or H/L. Therefore, the table is deleted.

The following example shows an instance where this message occurs:

```
cell(libg87) {
          area : 10;
          pin(D) {
            direction : input;
            capacitance : 1;
            timing() {
              timing_type : setup_rising;
              intrinsic_rise : 0.1;
              intrinsic_fall : 0.1;
              related_pin : "CP";
            }
            timing() {
              timing_type : hold_rising;
              intrinsic_rise : 0.1;
              intrinsic_fall : 0.1;
              related_pin : "CP";
            }
          }
          pin(CP) {
            direction : input;
            capacitance : 1;
          }

          statetable ( "D CP", "Q QB") {
            table :     "L R  : - - : L/H H/L,\
                         - ~R : - - : N   N";
          }

          pin(Q) {
            direction : output;
            internal_node : "Q";
            timing() {
              timing_type : rising_edge;
              intrinsic_rise : 1.0;
              intrinsic_fall : 1.0;
              rise_resistance : 0.1;
              fall_resistance : 0.1;
              related_pin : "CP";
```

Feedback

```
        }
      }

      pin(QB) {
        direction : output;
        internal_node : "QB";
        timing() {
          timing_type : rising_edge;
          intrinsic_rise : 1.0;
          intrinsic_fall : 1.0;
          rise_resistance : 0.1;
          fall_resistance : 0.1;
          related_pin : "CP";
        }
      }
    }
```

The following is an example message:

```
Error: Line 132, Cell(libg87): the internal node has L/H or H/L but
        no input has L/H or H/L. Deleting the table. (LIBG-87)
```

**What Next**

Change the *table* attribute to replace L/H or H/L for the internal node with either L, H, X, or N, or make one of the input nodes L/H or H/L.

---

## LIBG-88

(error) The '%s' port in the '%s' cell has a timing\n \tarc related to '%s' that is inconsistent.

**Description**

The timing type of the timing arc from the specified port to the related port is inconsistent with respect to other timing in the cell. For example, a port cannot have both a *rising_edge* timing type and a *falling_edge* timing type.

The following example shows an instance where this message occurs:

```
cell(libg88) {
          area : 13;
          pin(D) {
            direction : input;
            capacitance : 1;
            timing() {
              timing_type : setup_rising;
              intrinsic_rise : 1.0;
              intrinsic_fall : 1.0;
              related_pin : "C";
            }
            timing() {
              timing_type : hold_rising;
```

Feedback

```
    intrinsic_rise : 0.1;
    intrinsic_fall : 0.1;
  related_pin : "C";
  }
}
pin(C) {
  direction : input;
  capacitance : 2;
}
pin(D2) {
  direction : input;
  capacitance : 1;
  timing() {
    timing_type : setup_rising;
  intrinsic_rise : 1.0;
  intrinsic_fall : 1.0;
    related_pin : "C2";
  }
  timing() {
    timing_type : hold_falling;     /* error */
    intrinsic_rise : 0.0;
    intrinsic_fall : 0.0;
  related_pin : "C2";
  }
}
pin(C2) {
  direction : input;
  capacitance : 2;
}
pin(Q) {
  direction : output;
  timing() {
    timing_type : rising_edge;
    intrinsic_rise : 1.0;
    intrinsic_fall : 0.1;
    rise_resistance : 0.1;
    fall_resistance : 0.1;
    related_pin : "C";
  }
  timing() {
  timing_type : rising_edge;
    intrinsic_rise : 1.0;
    intrinsic_fall : 0.1;
  rise_resistance : 0.1;
  fall_resistance : 0.1;
    related_pin : "C2";
  }
}
pin(QN) {
  direction : output;
  timing() {
    timing_type : rising_edge;
    intrinsic_rise : 1.0;
```

Feedback

```
      intrinsic_fall : 2.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "C";
    }
    timing() {
      timing_type : rising_edge;
      intrinsic_rise : 1.0;
      intrinsic_fall : 2.0;
      rise_resistance : 0.1;
    fall_resistance : 0.1;
      related_pin : "C2";
    }
  }
  test_cell() {
    pin(D) {
    direction : input;
  }
    pin(C) {
      direction : input;
  }
    pin(D2) {
      direction : input;
    signal_type : "test_scan_in";
  }
  pin(C2) {
    direction : input;
      signal_type : "test_scan_clock";
    }
  state ("IQ","IQN") {
      next_state : "D";
      clocked_on : "C";
  }
    pin(Q) {
      direction : output;
      function : "IQ";
      signal_type : "test_scan_out";
    }
    pin(QN) {
      direction : output;
      function : "IQN";
      signal_type : "test_scan_out_inverted";
    }
  }
}
```

The following is an example message:

```
Error: Line 89, The 'D2' port in the 'libg88' cell has a timing
       arc related to 'C2' that is inconsistent. (LIBG-88)
```

Feedback

### What Next

Update the technology library source to make the timing arcs consistent.

---

## LIBG-89

(warning) The '%s' port in the '%s' cell has an inconsistent\n \tfunction in the test cells.

### Description

The test_cells define different functions for the port. The port must have one unique function among all of the test cells.

When reading a .db library, Library Compiler uses the first test_cell to create the full functional description. The second test_cell is ignored.

This warning can be switched off by setting the environment variable *read_db_lib_warnings* to FALSE (default).

The same problem in a .lib library causes an error. See the .lib example for LIBG-90.

### What Next

The default behavior of selecting the first test_cell as the correct one should be correct in most cases.

Change the technology library source to make the function of the port in all of the test cells consistent.

---

## LIBG-90

(error) The '%s' port in the '%s' cell has an inconsistent\n \tfunction in the test cells.

### Description

This message indicates that the test cells define different functions for the specified port. The port must have one unique function among all test cells.

The following example shows an instance where this message occurs:

```
cell(libg90) {
            area : 21;
            pin(D1) {
              direction : input;
              capacitance : 1;
            }
            pin(C1) {
              direction : input;
              capacitance : 2;
            }
```

Feedback

```
pin(D2) {
  direction : input;
  capacitance : 1;
}
pin(C2) {
  direction : input;
  capacitance : 2;
}
pin(C3) {
  direction : input;
capacitance : 1;
}

pin(Q1) {
  direction : output;
  timing() {
  timing_type : rising_edge;
    intrinsic_rise : 1.0;
    intrinsic_fall : 0.0;
    rise_resistance : 0.1;
    fall_resistance : 0.1;
    related_pin : "C1";
  }
  timing() {
    timing_type : rising_edge;
    intrinsic_rise : 1.0;
    intrinsic_fall : 0.0;
    rise_resistance : 0.1;
    fall_resistance : 0.1;
    related_pin : "C2";
  }
}
pin(Q2) {
  direction : output;
  timing() {
    timing_type : rising_edge;
    intrinsic_rise : 1.0;
    intrinsic_fall : 1.0;
    rise_resistance : 0.1;
    fall_resistance : 0.1;
    related_pin : "C3";
  }
}
test_cell() {
  pin(D1) {
    direction : input;
  }
  pin(C1) {
    direction : input;
  }
  pin(D2) {
    direction : input;
    signal_type : "test_scan_in";
```

Feedback

```
    }
    pin(C2) {
      direction : input;
      signal_type : "test_scan_clock_a";
  }
    pin(C3) {
      direction : input;
      signal_type : "test_scan_clock_b";
    }
    state("IQ","IQN") {
      force_10 : "C1 D1";
      force_01 : "C1 D1'";
    }
    pin(Q1) {
      direction : output;
    function : "IQ";
    }
    pin(Q2) {
      direction : output;
      signal_type : "test_scan_out";
    }
  }
  test_cell() {
    pin(D1) {
      direction : input;
    }
    pin(C1) {
      direction : input;
  }
    pin(D2) {
      direction : input;
      signal_type : "test_scan_in";
    }
    pin(C2) {
      direction : input;
      signal_type : "test_scan_clock_a";
  }
    pin(C3) {
      direction : input;
    }
    seq("IQ","IQN") {
      data_in  : "D1";
      enable : "C1";
      enable_also : "C3'";
    }
    pin(Q1) {
      direction : output;
  }
    pin(Q2) {
      direction : output;
      function : "IQ";
      signal_type : "test_scan_out";
    }
```

Feedback

```
            pin(Q2N) {
              direction : output;
            function : "IQN";
              signal_type : "test_scan_out_inverted";
            }
          }
        }
```

For the pin 'Q2', the 'function' attribute is missing in the first test_cell.

The following is an example message:

```
Error: Line 175, The 'Q2' port in the 'libg90' cell has an inconsistent
        function in the test cells. (LIBG-90)
```

**What Next**

Update the technology library source so that the function of the port in all test cells is consistent.

---

## LIBG-91

(error) The '%s' attribute is required when both 'clear'\n \tand 'preset' are present in the sequential model and the %s '%s'\n \tis used in the function attribute of the output pin(s) of the cell.

**Description**

When both 'clear' and 'preset' are present in the sequential model, the *clear_preset_var1* and/or *clear_preset_var2* attributes are required, depending on which variable of the sequential model is used in the function attribute of the output pin(s) of the cell. For example, if the first variable is used, *clear_preset_var1* is required.

**What Next**

Make the appropriate correction in your library, as indicated in the error message.
For more information on the *clear_preset_var* attributes, refer to the *Library Compiler Reference Manual*, Vol. 1.

---

## LIBG-92

(warning) The '%s' port in the '%s' cell has a circular dependency.

**Description**

There is an undelayed feedback loop to the output, inout, and internal port. This circular dependency is not allowed.

This warning is issued if the current port appears in the list of expanded inputs of the port.

Feedback

This warning is issued when reading a .db library. You can switch off the warning by setting the environment variable *read_db_lib_warnings* to FALSE (default).

The same problem in a .lib library causes an error. See the .lib example for LIBG-93.

**What Next**

Break the feedback loop by delaying one of the inputs in the feedback path.

---

## LIBG-93

(error) The '%s' port in the '%s' cell has a circular dependency.

**Description**

There is an undelayed feedback loop to the output port. This circular dependency is not allowed.

This error is issued if the current port appears in the list of expanded inputs of the port.

The following example shows an instance where this message occurs:

```
cell(libg93) {
          area : 7;
          pin(D) {
            direction : input;
            capacitance : 1;
          }
          pin(R) {
            direction : input;
            capacitance : 1;
          }

          statetable("D R CP", "Q") {
            table : "H/L L  R : - : H/L,\
                       -  L ~R : - :  N,\
                       -  H -  : - :  H";
          }

          /* internal sequential state_function */
          pin(E) {
            direction : internal;
            state_function : "Q1+R";
          }

          /* slave with sequential reset */
          pin(Q1) {
            direction : internal;
            internal_node : "Q";
            input_map : "D Q2 CP";
          }
```

Feedback

```
            /* master driving Q1 */
            pin(Q2) {
              direction : output;
              internal_node : "Q"
              input_map : "D E CP";
            }

            pin(CP) {
              direction : input;
              capacitance : 1;
            }

          }
```

The following is an example message:

```
Error: Line 274, The 'Q1' port in the 'libg93' cell has a circular
 dependency. (LIBG-93)
```

### What Next

Break the feedback loop by delaying one of the inputs in the feedback path. In the example, there is a feedback loop for 'Q1'. To break the dependency, delay one of the inputs in the feedback path. The pins 'E' and 'Q2' in the cell group can be changed as follows:

```
pin(E) {
            direction : internal;
            state_function : "D+Q1";
          }
          pin(Q2) {
            direction : output;
            internal_node : "Q"
            input_map : "E R CP";
          }
```

---

## LIBG-94

(warning) The '%s' cell needs a noninverted output.

### Description

The test cell does not have any noninverted output. This particular type of test cell requires at least one noninverted output to be defined.

This warning is issued when reading a .db library. You can switch off the warning by setting the environment variable *read_db_lib_warnings* to FALSE (default).

The same problem in a .lib library causes an error. See the .lib example for LIBG-95.

Feedback

### What Next

Specify the full function or define an output to be noninverted.

---

## LIBG-95

(error) The '%s' cell needs a noninverted output.

### Description

The test cell has no noninverted output. This particular type of test cell requires at least one noninverted output to be defined.

The following example shows an instance where this message occurs:

```
cell(libg95) {
            area : 15;
            pin(D) {
              direction : input;
              capacitance : 1;
            }
            pin(C) {
              direction : input;
              capacitance : 2;
            }
            pin(TSI) {
              direction : input;
              capacitance : 1;
            }
            pin(TSE) {
              direction : input;
              capacitance : 2;
            }
            pin(QN) {
              direction : output;
              capacitance : 2;
            }
            pin(TSO TSOI) {
              direction : output;
            }
            test_cell() {
              pin(D) {
                direction : input;
              }
              pin(C) {
                direction : input;
              }
              pin(TSI) {
                direction : input;
                signal_type : "test_scan_in";
              }
              pin(TSE) {
```

Feedback

```
            direction : input;
            capacitance : 2;
            signal_type : "test_scan_enable";
          }
          statetable("D C", "QN") {
            table : "L/H H : - : H/L,\
                     -   L : - : N";
          }

          pin(TSOI) {
            direction : output;
            signal_type : "test_scan_out_inverted";
          }
          pin(TSO) {
            direction : output;
            signal_type : "test_scan_out";
          }
          pin(QN) {
            direction : output;
            internal_node : "QN";
          }
        }
      }
```

The following is an example message:

```
Error: Line 142, The 'libg95' cell needs a noninverted output. (LIBG-95)
```

**What Next**

Specify the full function or define an output to be noninverted.

In the example, the cell is a latch. To fix the problem, add another noninverted pin. For example, the following change to the cell will fix the problem.

```
cell(libg95) {
        ...
        pin(Q) {
          direction : output;
          capacitance : 2;
        }
        ...
        test_cell() {
            ...
          statetable("D C", "Q QN") {
            table : "L/H H : - - : L/H H/L,\
                     -   L : - - : N N";
          }
            ...
          pin(Q) {
            direction : output;
            internal_node : "Q";
          }
```

Feedback

```
        }
    }
```

## LIBG-96

(warning) The '%s' port in the '%s' cell is used in both test and\n \tnon test mode.

### Description

The input port has a test attribute and is also used in the non test mode. This is not allowed.

This warning is issued when reading a .db library. You can switch off the warning by setting the environment variable *read_db_lib_warnings* to FALSE (default).

The same problem in a .lib library causes an error. See the .lib example for LIBG-97.

### What Next

Specify the full function of the test cell.

## LIBG-97

(error) The '%s' port in the '%s' cell is used in both\n \ttest and non test mode.

### Description

The input port has a test attribute and is also used in the non test mode. This is not allowed.

The following example shows an instance where this message occurs:

```
cell(libg97) {
            area : 13;
            pin(D TSI) {
              direction : input;
              capacitance : 1;
            }
            pin(C TSE) {
              direction : input;
              capacitance : 2;
            }
            pin(Q TSO) {
              direction : output;
            }
            test_cell() {
              pin(D C) {
                direction : input;
              }
              pin(TSI) {
                direction : input;
```

Feedback

```
            signal_type : "test_scan_in";
          }
          pin(TSE) {
            direction : input;
            capacitance : 2;
            signal_type : "test_scan_enable";
          }
          statetable("C D TSI", "Q") {
            table : "H L/H H/L : - : L/H,\
                      L -  -  : - : N";
          }
          pin(Q) {
            direction : output;
            internal_node : "Q";
          }
          pin(TSO) {
            direction : output;
            signal_type : "test_scan_out";
    }
       }
     }
```

In this case, the input port 'TSI' has the "test_scan_in" attribute, and it is used in the state table input entry.

The following is an example message:

```
Error: Line 141, The 'TSI' port in the 'libg97' cell is used in both
        test and non test mode. (LIBG-97)
```

**What Next**

Specify the full function of the test cell.

---

## LIBG-98

(warning) The '%s' test cell has an incompatible non test mode.

**Description**

The non test mode of the test cell is incompatible with the test attributes of the test cell.

This warning appears after any of the following warning messages: LIBG-128, LIBG-129, LIBG-130, and LIBG-131. For more information, refer to the man page of the issued warning.

This warning is issued when reading a .db library. You can switch off the warning by setting the environment variable *read_db_lib_warnings* to FALSE (default).

The same problem in a .lib library causes an error. See the .lib example for LIBG-99.

Feedback

### What Next

Specify the full function of the test cell.

---

## LIBG-99

(error) The '%s' test cell has an incompatible non test mode.

### Description

This message indicates that the non test mode of the test cell is incompatible with the test attributes of the test cell.

This error is displayed for a cell in UNIGEN format after any of the following error messages: LIBG-158, LIBG-159, LIBG-160, and LIBG-161. For more information, refer to the man page of the issued error.

The following example shows an instance where this message occurs:

```
cell(libg99) {
            area : 11;
            pin(D1 D2 D3) {
              direction : input;
              capacitance : 1;
            }
            pin(CP SCK1 SCK2) {
              direction : input;
              capacitance : 1;
            }
            pin(SI) {
              direction : input;
              capacitance : 1;
              timing() {
                timing_type : setup_falling;
                intrinsic_rise : 1.0;
                intrinsic_fall : 1.0;
                related_pin : "SCK2";
              }
            }
            pin(Q) {
              direction : output;
            }
            pin(Q2) {
              direction : output;
              timing() {
                timing_type : rising_edge;
                intrinsic_rise : 1.0;
                intrinsic_fall : 1.0;
                rise_resistance : 0.1;
                fall_resistance : 0.1;
                related_pin : "CP SCK1";
```

Feedback

```
    }
  }
  test_cell() {
    pin(D1 D2 D3 CP) {
      direction : input;
    }
    pin(SCK1) {
      direction : input;
      signal_type : "test_scan_clock_b";
    }
    pin(SCK2) {
      direction : input;
      signal_type : "test_scan_clock_a";
    }
    pin(SI) {
      direction : input;
      signal_type : "test_scan_in";
    }
    statetable("C D ", "Q") {
      table :  "R  H/L : - : H/L,\
               ~R   -  : - :  N";
    }
    pin(Q) {
      direction : output;
      internal_node : "Q";
      input_map : "CP D1";
      signal_type : "test_scan_out";
    }
    pin(Q2) {
      direction : output;
      internal_node : "Q";
      input_map : "CP D2";
      signal_type : "test_scan_out";
    }
  }
}
```

In this case, the pin 'Q' generated error LIBG-161, which generated this error.

The following is a message example:

```
Error: Line 62, The 'libg99' test cell has an incompatible non test mode.
 (LIBG-99)
```

**What Next**

Update the non test mode description or the test mode description to make them consistent.

Feedback

## LIBG-100

(warning) The '%s' port in the '%s' cell has both a function\n \tand a test attribute. The test_cell will be deleted.

### Description

This message indicates that the specified port has both a function statement and a test attribute. For this particular test cell type, a port cannot have both a function statement and a test attribute.

This warning is issued when reading a .db library. You can switch off the warning by setting the environment variable *read_db_lib_warnings* to FALSE (default).

The same problem in a .lib library causes an error. See the .lib example for LIBG-101.

### What Next

Remove either the function statement or the test attribute to correctly specify the test_cell.

## LIBG-101

(error) The '%s' port in the '%s' cell has both a function\n \tand a test attribute.

### Description

This messages is issued only for single-latch LSSD and muxed-latch scan cells. The port has both a function statement and a test attribute. For the particular test cell type, a port cannot have both a function statement and a test attribute.

A function statement can be a *function*, a *state_function*, or an *internal_node* attribute.

A test attribute can be a *test_scan_in*, a *test_scan_in_inverted*, a *test_scan_enable*, a *test_scan_enable_inverted*, a *test_scan_out*, a *test_scan_out_inverted* , a *test_scan_clock*, a *test_scan_clock_a*, a *test_scan_clock_b*, or a *test_clock*.

The following example shows an instance where this message occurs:

```
cell(libg101) {
            area : 13;
            pin(D SDI) {
              direction : input;
              capacitance : 1;
            }
            pin(C SE) {
              direction : input;
              capacitance : 2;
            }
            pin(QN) {
              direction : output;
```

Feedback

```
    }
    test_cell() {
      pin(D C) {
        direction : input;
      }
      pin(SDI) {
        direction : input;
        signal_type : "test_scan_in";
      }
      pin(SE) {
        direction : input;
        capacitance : 2;
        signal_type : "test_scan_enable";
      }
      statetable("D C", "QN") {
        table : "L/H H : - : H/L,\
                  -   L : - : N";
      }
      pin(QN) {
        direction : output;
        inverted_output : TRUE;
        internal_node : "QN";
        signal_type : "test_scan_out_inverted";
      }
    }
  }
```

The following is an example of the message:

```
Error: Line 152, The 'QN' port in the 'libg101' cell has both a function
        and a test attribute. (LIBG-101)
```

In this case, the pin 'QN' has both an *internal_node* and a *test_scan_out_inverted* attributes.

**What Next**

If you can access the original technology library source file, remove either the function statement or the test attribute to correctly specify the test_cell.

---

## LIBG-102

(warning) The '%s' cell has more than one clock.

**Description**

The auxiliary clocked LSSD cell has more than one clock. This is not supported by DFT Compiler.

This warning is issued when reading a .db library. You can switch off the warning by setting the environment variable *read_db_lib_warnings* to FALSE (default).

Feedback

The same problem in a .lib library causes an error. See the .lib example for LIBG-103.

**What Next**

This type of cell is not supported by DFT Compiler, so no further action is indicated.

---

## LIBG-103

(error) The '%s' cell has more than one clock.

**Description**

The auxiliary clocked LSSD cell has more than one clock. This is not supported by DFT Compiler.

The following example shows an instance where this message occurs:

```
cell(libg103) {
            area : 12;
            pin(D) {
              direction : input;
              capacitance : 1;
            }
            pin(CK,CK2,IH) {
              direction : input;
              capacitance : 1;
            }
            pin(A,B) {
              direction : input
                capacitance : 2;
            }
            pin(SI) {
              direction : input;
              capacitance : 1;
              timing() {
                timing_type : setup_falling;
                intrinsic_rise : 1.0;
                intrinsic_fall : 1.0;
                related_pin : "A";
              }
              timing() {
                timing_type : hold_falling;
                intrinsic_rise : 1.0;
                intrinsic_fall : 1.0;
                related_pin : "A";
              }
            }
            pin(Q) {
              direction : output;
              timing() {
                timing_type : rising_edge;
                intrinsic_rise : 1.0;
```

Feedback

```
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "CK IH";
    }
    timing() {
      timing_type : falling_edge;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "B";
    }
  }
  pin(XQ) {
    direction : output;
  }
  test_cell(){
    pin(D,CK,CK2){
      direction : input;
    }
    pin(IH){
      direction : input;
      signal_type : "test_clock";
    }
    pin(SI){
      direction : input;
      signal_type : "test_scan_in";
    }
    pin(A){
      direction : input;
      signal_type : "test_scan_clock_a";
    }
    pin(B){
      direction : input;
      signal_type : "test_scan_clock_b";
    }
    ff("IQ","IQN"){
      clocked_on : "CK CK2";
      next_state : "D";
    }
    pin(Q){
      direction : output;
      function : "IQ";
      signal_type : "test_scan_out";
    }
    pin(XQ){
      direction : output;
      function : "IQN";
      signal_type : "test_scan_out_inverted";
    }
  }
}
```

Feedback

In this case, there are two clocks, 'CK' and 'CK2', which is not allowed.

The following is an example of the message:

```
Error: Line 162, The 'libg103' cell has more than one clock. (LIBG-103)
```

**What Next**

This type of cell is not supported by DFT Compiler, so no further action is indicated.

---

## LIBG-104

(warning) The '%s' pin of the '%s' design is not a clock pin\n and should not be used in the 'related_pin' of '%s' timing arc.

**Description**

By definition, a pin in the *related_pin* attribute of the timing group with the *timing_type setup_rising, setup_falling, hold_rising* or *hold_falling* should be a clock pin. It is treated as a warning only for backward compatibility.

The following example shows an instance where this message occurs:

```
cell(libg104) {
          area : 10;
          pin(CK) {
            direction : input;
            clock     : true;
            capacitance : 1.0;
          }
          pin(D) {
            direction : input ;
            capacitance : 1.0 ;
            timing() {
              related_pin : "CK";
              timing_type : setup_rising;
              intrinsic_fall : 1.0;
              intrinsic_rise : 1.0;
            }
            timing() {
              related_pin : "CK";
              timing_type : hold_rising;
              intrinsic_fall : 0.1;
              intrinsic_rise : 0.1;
            }
          }
          pin(P) {
            direction : input;
            capacitance : 1.0;
            timing() {
              related_pin : "CK";
              timing_type : setup_rising;
```

Feedback

```
            intrinsic_rise : 0.0;
          }
        }
        ff ("IQ", "IQN") {
           next_state : "D";
           clocked_on : "CK";
           preset : "P";
       }
        pin(Q) {
          direction : output;
          function  : "IQ";
          timing() {
            related_pin : "P";
            timing_type : setup_rising;
            intrinsic_rise : 1.0;
            intrinsic_fall : 1.0;
          }
        }
      }
```

The timing group of the pin 'Q' contains the pin 'P' in the related_pin attribute. This caused the warning because the timing_type is *setup_rising*

The following is an example of the message:

```
Warning: Line 153, The 'P' pin of the 'libg104' design is not a clock pin
        and should not be used in the 'related_pin' of 'setup' timing
 arc. (LIBG-104)
```

**What Next**

Check your library for a wrong pin in the *related_pin* attribute or for an incorrect *timing_type*.

---

## LIBG-105

(error) There is a missing timing arc between the '%s' and '%s'\n \tpins in the '%s' cell. A timing arc of the right timing_type\n \tis expected between the two specified pins of the cell.

**Description**

A timing arc of the right timing_type is expected between the two specified pins of the cell. For example, a timing arc with timing_type rising_edge/falling_edge is required from the clock/enable pin to the output pin. This error might be generated by a previous warning or error such as: LIBG-104.

The following example shows an instance where this message occurs:

```
cell(libg105) {
          area          : 10;
```

Feedback

```
  pin(CK) {
    direction            : input;
    clock                : true;
    capacitance          : 1.0;
  }
  pin(D) {
    direction            : input;
    capacitance          : 1.0;
    timing() {
      related_pin        : "CK";
      timing_type        : setup_rising;
      intrinsic_fall     : 1.0;
      intrinsic_rise     : 1.0 ;
    }
    timing() {
      related_pin        : "CK";
      timing_type        : hold_rising;
      intrinsic_fall     : 0.0;
      intrinsic_rise     : 0.0;
    }
  }

  pin(P) {
    direction            : input;
    capacitance          : 1.0;
    timing() {
      related_pin        : "CK";
      timing_type        : setup_rising;
      intrinsic_rise     : 0.0;
    }
  }
  ff ("IQ", "IQN") {
     next_state : "D";
     clocked_on : "CK";
     preset : "P";
 }
  pin(Q) {
    direction            : output;
    function             : "IQ";
    timing() {
      related_pin        : "CK";
      timing_type        : rising_edge;
      intrinsic_rise     : 1.0;
      intrinsic_fall     : 1.0;
    }
  }
}
```

A timing group is missing in the pin 'Q' for the pin 'P'.

Feedback

The following is an example message:

```
Error: Line 151, There is a missing timing arc between the 'P' and 'Q'
        pins in the 'libg105' cell. A timing arc of the right timing_type
        is expected between the two specified pins of the cell.
 (LIBG-105)
```

### What Next

Specify or correct the timing arc to make the cell description complete.

---

## LIBG-106

(warning) The '%s' timing type is invalid between the\n \ttwo specified pins.

### Description

The specified *timing_type* attribute is invalid between the pin that has the timing group and the pin in the *related_pin* attribute specified. It is most likely that you have specified a "combinational" timing arc from the clock, enable, clear, or preset pin to the output pin, or that the *timing_type* attribute is missing.

The following example shows an instance where this message occurs:

```
cell(libg106) {
          area           : 10;
          pin(CK) {
            direction            : input;
            clock                : true;
            capacitance          : 1.0;
          }
          pin(D) {
            direction            : input;
            capacitance          : 1.0;
            timing() {
              related_pin        : "CK";
              timing_type        : setup_rising;
              intrinsic_fall     : 1.0;
              intrinsic_rise     : 1.0;
            }
            timing() {
              related_pin        : "CK";
              timing_type        : hold_rising;
              intrinsic_fall     : 0.0;
              intrinsic_rise     : 0.0;
            }
          }
          pin(P) {
            direction            : input;
            capacitance          : 1.0;
            timing() {
              related_pin        : "CK";
```

Feedback

```
            timing_type         : setup_rising;
            intrinsic_rise      : 0.0;
          }
        }
        ff ("IQ", "IQN") {
           next_state : "D";
           clocked_on : "CK";
           preset : "P";
       }
        pin(Q) {
          direction             : output;
          function              : "IQ";
          timing() {
            related_pin         : "CK";
            timing_type         : setup_rising;
            intrinsic_rise      : 1.0;
            intrinsic_fall      : 1.0
          }
        }
      }
```

In this case, there is a 'setup_rising' attribute between the output port 'Q' and the clock 'CK'.

The following is an example of the message:

```
Warning: Line 151, The 'setup_rising' timing type is invalid between the
         two specified pins. (LIBG-106)
```

**What Next**

Specify the correct timing_type attribute.

---

## LIBG-107

(error) The complex state_function is not allowed in the '%s' test cell.

**Description**

The test cell has a port with a complex state_function. A complex state_function is any expression not equivalent to a buffer or an inverter.

Currently, complex state_functions are not supported in test cells.

The following example shows an instance where this message occurs:

```
cell(libg107) {
    area : 21;
    pin(D1 D2) {
      direction : input;
      capacitance : 1;
    }
```

Feedback

```
pin(C1 C2 C3) {
  direction : input;
  capacitance : 2;
}
pin(Q1 Q2 Q2N) {
  direction : output;
}
pin(Q1N) {
  direction : output;
  timing() {
    timing_type : rising_edge;
    intrinsic_rise : 1.0;
    intrinsic_fall : 1.0;
    rise_resistance : 0.1;
    fall_resistance : 0.1;
    related_pin : "C2";
  }
}
test_cell() {
  pin(D1 C1 C3) {
    direction : input;
  }
  pin(D2) {
    direction : input;
    signal_type : "test_scan_in";
  }
  pin(C2) {
    direction : input;
    signal_type : "test_scan_clock_a";
  }
  statetable ("C1 D1 C3", "Q1 Q2") {
    table : "H L/H - : H - : L/H -,\
               H - -   : L - : L -,\
           - -   H : L/H - : - L/H,\
             - -   - : - - : N   N";
  }

  pin(Q1) {
    direction : output;
    internal_node : "Q1";
  }
  pin(Q1N) {
    direction : output;
    state_function : "Q1'";
  }
  pin(Q2) {
    direction : output;
    internal_node : "Q2";
    signal_type : "test_scan_out";
  }
  pin(Q2N) {
    direction : output;
```

Feedback

```
        /* This state_function is too complex, change it to a
 buffer/inverter*/
        state_function : "Q2' + C1";
        signal_type : "test_scan_out_inverted";
      }
    }
  }

}
```

The following is an example message:

```
Error: Line 134, The complex state_function is not allowed in the
 'libg107' test cell. (LIBG-107)
```

**What Next**

Either remove the state_function or make it into a buffer or an inverter.

---

## LIBG-108

(error) The three_state attribute is not allowed in the '%s' test cell.

**Description**

The test cell has a port with a three_state attribute.

Currently, the three_state attribute is not supported in test cells.

The following example shows an instance where this message occurs:

```
cell(libg108) {
    area : 21;
    pin(D1 D2) {
      direction : input;
      capacitance : 1;
    }
    pin(C1 C2 C3) {
      direction : input;
      capacitance : 1;
    }
    pin(Q1 Q2 Q2N) {
      direction : output;
    }
    pin(Q1N) {
      direction : output;
      timing() {
        timing_type : rising_edge;
        intrinsic_rise : 1.0;
        intrinsic_fall : 1.0;
        rise_resistance : 0.1;
        fall_resistance : 0.1;
```

Feedback

```
        related_pin : "C2";
      }
    }
    test_cell() {
      pin(D1 C1 C3) {
        direction : input;
      }
      pin(D2) {
        direction : input;
        signal_type : "test_scan_in";
      }
      pin(C2) {
        direction : input;
        signal_type : "test_scan_clock_a";
      }
      statetable ("C1 D1 C3", "Q1 Q2") {
        table : "H L/H - : H - : L/H -,\
                 H - - : L - : L -,\
                 - -   H : L/H - : - L/H,\
                 - -   - : - - : N   N";
      }

      pin(Q1) {
        direction : output;
        internal_node : "Q1";
      }
      pin(Q1N) {
        direction : output;
        state_function : "Q1'";
      }
      pin(Q2) {
        direction : output;
        internal_node : "Q2";
        signal_type : "test_scan_out";
      }
      pin(Q2N) {
        direction : output;
        state_function : "Q2'";
        /* three_state attribute is not allowed in test_cell */
        three_state : "C1";
        signal_type : "test_scan_out_inverted";
      }
    }
  }

}
```

The following is an example message:

```
Error: Line 169, The three_state attribute is not allowed in the
 'libg108' test cell. (LIBG-108)
```

Feedback

### What Next

Remove the three_state attribute from the test cell.

---

## LIBG-109

(warning) Asynchronous specifications in the '%s' cell are\n \tincompletely specified.

### Description

There is a set of input values for asynchronous inputs that causes more than one force_* condition to be true. The force_* conditions are not mutually exclusive; the cell is unspecified when more than one force_* condition is valid.

Design Compiler cannot support this cell and might generate invalid logic if the cell is used.

This warning is issued when reading a .db library. You can switch off the warning by setting the environment variable *read_db_lib_warnings* to FALSE (default).

The same problem in a .lib library causes an error. See the .lib example for LIBG-110.

### What Next

You can put the attributes *dont_touch* and *dont_use* on the library cell to prevent this cell from being used and potentially generating invalid logic.

Alternatively, you can change the source technology library. Change the force_* expressions so they are mutually exclusive. There should be no input condition for which more than one force_* statement is valid.

---

## LIBG-110

(warning) Asynchronous specifications in the '%s' cell \n are incomplete or are not mutually exclusive.\n \tThe cell has been made a black box.

### Description

There is a set of input values for asynchronous inputs that causes more than one force_* condition to be true.

The force_* conditions are not mutually exclusive; the cell is unspecified when more than one force_* condition is valid.

Design Compiler cannot support this cell; therefore, the cell has been made a black box.

The following example shows an instance where this message occurs:

```
cell(libg110) {
  area : 10;
```

Feedback

```
    pin(force11) {
      direction : input;
      capacitance : 1;
    }
    pin(force10) {
      direction : input;
      capacitance : 1;
    }
    pin(force01) {
      direction : input;
      capacitance : 1;
    }
    pin(force00) {
      direction : input;
      capacitance : 2;
    }
    pin(ck) {
      direction : input;
      capacitance : 2;
    }
    pin(next) {
      direction : input;
      capacitance : 2;
    }
    state("IQ","IQN") {
      next_state : "next";
      clocked_on : "ck";
      force_00   : "force00";
      force_01   : "force01";
      force_10   : "force10";
      force_11   : "force11";
    }
    statetable ("next ck ck* force00 force01 force10 force11", "out") {
      table : "- - - H L L L : - : L,\
               - - - L H L L : L/H : L,\
               - - - L L H L : - : H,\
               - - - L L L H : - : H,\
               L/H H L L L L L : - : L/H,\
               - L L L L L L : - : N,\
               - L H L L L L : - : N,\
               - H H L L L L : - : N";
    }

    pin(Q) {
      function : "IQ";
      internal_node : "out";
      direction : output;
    }
  }
```

Feedback

The following is an example message:

```
Warning: Line 111, Asynchronous specifications in the 'libg110' cell
        are incomplete or are not mutually exclusive.
        The cell has been made a black box. (LIBG-110)
```

### What Next

Change the force_* expressions so they are mutually exclusive. There should be no input condition for which more than one force_* statement is valid.

---

## LIBG-111

(warning) Design Compiler does not support the values for\n \tclear_preset_var? in the '%s' cell. The cell has been made a black box.

### Description

The cell has invalid values (N, X or T) for clear_preset_var1 and clear_preset_var2.

Because of these values, Design Compiler cannot support the cell. The cell has been made a black box.

The following example shows an instance where this message occurs:

```
cell(libg111) {
        area : 11;
        pin(D) {
                direction : input;
                capacitance : 1;
        }
        pin(CP) {
                direction : input;
                capacitance : 1;
        }
        pin(CLR) {
                direction : input;
                capacitance : 1;
        }
        pin(SET) {
                direction : input;
                capacitance : 1;
        }
        ff("IQ","IQN") {
                next_state : "D";
                clocked_on : "CP";
                clear   : "CLR'";
                preset   : "SET'";
                clear_preset_var1 : X;
                clear_preset_var2 : X;
        }
        pin(Q) {
```

Feedback

```
                direction : output;
                function : "IQ";
                timing() {
                        timing_type : rising_edge;
                        intrinsic_rise : 1.0;
                        intrinsic_fall : 1.0;
                        rise_resistance : 0.1;
                        fall_resistance : 0.1;
                        related_pin : "CP";
                }
                timing() {
                        timing_type : clear;
                        timing_sense : positive_unate;
                        intrinsic_fall : 1.0;
                        fall_resistance : 0.1;
                        related_pin : "CLR";
                }
                timing() {
                        timing_type : preset;
                        timing_sense : negative_unate;
                        intrinsic_fall : 1.0;
                        fall_resistance : 0.1;
                        related_pin : "SET";
                }
        }
        pin(QN) {
                direction : output;
                function : "IQN";
                timing() {
                        timing_type : rising_edge;
                        intrinsic_rise : 1.0;
                        intrinsic_fall : 1.0;
                        rise_resistance : 0.1;
                        fall_resistance : 0.1;
                        related_pin : "CP";
                }
                timing() {
                        timing_type : preset;
                        timing_sense : negative_unate;
                        intrinsic_fall : 1.0;
                        fall_resistance : 0.1;
                        related_pin : "CLR";
                }
                timing() {
                        timing_type : clear;
                        timing_sense : positive_unate;
                        intrinsic_fall : 1.0;
                        fall_resistance : 0.1;
                        related_pin : "SET";
                }
        }
}
```

Feedback

The following is an example message:

```
Warning: Line 111, Design Compiler does not support the values for
        clear_preset_var? in the 'libg111' cell. The cell has been made a
 black box. (LIBG-111)
```

### What Next

Change the N, X or T value to a L or H value.

---

## LIBG-112

(error) The timing check condition represented in this timing\n \tgroup is not mutually exclusive with the timing check condition\n \trepresented in the timing group on line %d.

### Description

The timing check conditions represented by the 'when' string of all conditional period and pulse width timing checks should be mutually exclusive. A nonconditional timing check is assumed to have the implied '1' timing check condition (always *TRUE*). If timing check conditions are not mutually exclusive, Library Compiler cannot determine which timing check to use when the condition causes both timing check conditions to be evaluated *TRUE*.

The following example shows an instance where this message occurs:

```
cell(libg112) {
        area : 6.0;
        pin(CLK) {
                direction : input;
                capacitance : 1.0;
                minimum_period() {
                        constraint : 99;
                        when : " Q'";
                        sdf_cond : "SIG_5 == 1'b1 ";
                }
                minimum_period() {
                        constraint : 99;
                        when : " Q'"
                        sdf_cond : "SIG_5 == 1'b0 ";
                }
        }
        pin(SET) {
                direction : input;
                capacitance : 1.0;
        }
        ff("IQ","IQN") {
                next_state : "IQ'";
                clocked_on : "CLK";
                preset     : "SET'";
        }
```

Feedback

```
pin(Q) {
        direction : output;
        function : "IQ";
        timing() {
                timing_type : rising_edge;
                intrinsic_rise : 1.0;
                intrinsic_fall : 1.0;
                rise_resistance : 0.1;
                fall_resistance : 0.1;
                related_pin : "CLK";
        }
        timing() {
                timing_type : preset;
                timing_sense : negative_unate;
                intrinsic_rise : 1.0;
                rise_resistance : 0.1;
                related_pin : "SET";
        }
}
pin(QN) {
        direction : output;
        function : "IQN";
        timing() {
                timing_type : rising_edge;
                intrinsic_rise : 1.0;
                intrinsic_fall : 1.0;
                rise_resistance : 0.1;
                fall_resistance : 0.1;
                related_pin : "CLK";
        }
        timing() {
                timing_type : clear;
                timing_sense : positive_unate;
                intrinsic_fall : 1.0;
                fall_resistance : 0.1;
                related_pin : "SET";
        }
}
```

The following is an example message:

```
Error: Line 115, The timing check condition represented in this timing
       group is not mutually exclusive with the timing check condition
       represented in the timing group on line 120. (LIBG-112)
```

### What Next

Check the 'when' strings of both timing groups, and make corrections necessary to eliminate duplicate nonconditional timing checks or other erroneous information. Also, ensure that there is no nonconditional timing check defined when at least one conditional timing check of the same pins and timing type is defined.

Feedback

## LIBG-112w

(warning) The timing check condition represented in this timing\n \tgroup is not mutually exclusive with the timing check condition\n \trepresented in the timing group on line %d.

### Description

The timing check conditions represented by the 'when' string of all conditional period and pulse width timing checks should be mutually exclusive. A nonconditional timing check is assumed to have the implied '1' timing check condition (always *TRUE*). If timing check conditions are not mutually exclusive, Library Compiler cannot determine which timing check to use when the condition causes both timing check conditions to be evaluated *TRUE*.

The following example shows an instance where this message occurs:

```
cell(libg112) {
        area : 6.0;
        pin(CLK) {
                direction : input;
                capacitance : 1.0;
                minimum_period() {
                        constraint : 99;
                        when : " Q'";
                        sdf_cond : "SIG_5 == 1'b1 ";
                }
                minimum_period() {
                        constraint : 99;
                        when : " Q'"
                        sdf_cond : "SIG_5 == 1'b0 ";
                }
        }
        pin(SET) {
                direction : input;
                capacitance : 1.0;
        }
        ff("IQ","IQN") {
                next_state : "IQ'";
                clocked_on : "CLK";
                preset     : "SET'";
        }
        pin(Q) {
                direction : output;
                function : "IQ";
                timing() {
                        timing_type : rising_edge;
                        intrinsic_rise : 1.0;
                        intrinsic_fall : 1.0;
                        rise_resistance : 0.1;
                        fall_resistance : 0.1;
                        related_pin : "CLK";
```

Feedback

```
                }
                timing() {
                        timing_type : preset;
                        timing_sense : negative_unate;
                        intrinsic_rise : 1.0;
                        rise_resistance : 0.1;
                        related_pin : "SET";
                }
        }
        pin(QN) {
                direction : output;
                function : "IQN";
                timing() {
                        timing_type : rising_edge;
                        intrinsic_rise : 1.0;
                        intrinsic_fall : 1.0;
                        rise_resistance : 0.1;
                        fall_resistance : 0.1;
                        related_pin : "CLK";
                }
                timing() {
                        timing_type : clear;
                        timing_sense : positive_unate;
                        intrinsic_fall : 1.0;
                        fall_resistance : 0.1;
                        related_pin : "SET";
                }
        }
```

The following is an example message:

```
Warning: Line 115, The timing check condition represented in this timing
        group is not mutually exclusive with the timing check condition
        represented in the timing group on line 120. (LIBG-112w)
```

**What Next**

Check the 'when' strings of both timing groups, and make corrections necessary to eliminate duplicate nonconditional timing checks or other erroneous information. Also, ensure that there is no nonconditional timing check defined when at least one conditional timing check of the same pins and timing type is defined.

---

## LIBG-113

(error) The timing check starting and ending conditions\n \trepresented in this timing group are not mutually exclusive with the\n \ttiming check starting and ending conditions represented in the \n \ttiming group on line %d.

Feedback

### Description

One of the following two conditions must be satisfied between two conditional timing checks of the same pins:

1. The timing check starting conditions represented by the **when_start** or **when** string of all conditional timing checks (setup, hold, recovery, removal and skew) are mutually exclusive.
2. The timing check ending conditions represented by the **when_end** or **when** string of all conditional timing checks (setup, hold, recovery, removal and skew) are mutually exclusive.

A nonconditional timing check is assumed to have the implied '1' timing check starting and ending condition (always *TRUE*). If both starting and ending conditions are not mutually exclusive, Library Compiler cannot determine which timing check to use when the condition enables both conditional timing check conditions.

The following example shows an instance where this message occurs:

```
cell(libg113) {
    area : 10;
    pin(D) {
      direction : input;
      capacitance : 1;
      /* these two timing groups are not mutually exclusive */
      timing() {
        timing_type : setup_falling;
        intrinsic_rise : 1.0;
        intrinsic_fall : 1.0;
        when : " MQ ";
        sdf_cond : " MQ == 1'b1 ";
        related_pin : "MC";
      }
      timing() {
        timing_type : setup_falling;
        when : " MQ ";
        sdf_cond : " MQ == 1'b1 ";
        intrinsic_rise : 1.0;
        intrinsic_fall : 1.0;
        related_pin : "MC";
      }
    }
    pin(MC) {
      direction : input;
      capacitance : 2;
    }
    statetable ( "D CP", "Q ") {
        table : "H/L  R : - : H/L,\
                 -   ~R : - : N ";
    }

    pin (MQ) {
```

Feedback

```
      direction : internal;
      inverted_output : FALSE;
      internal_node : "Q";
      input_map : "D MC MQ";
    }
    pin(Q) {
      direction : output;
      internal_node : "Q";
      input_map : "MQ MC Q";
      inverted_output : FALSE;
      timing() {
        timing_type : rising_edge;
        intrinsic_rise : 1.0;
        intrinsic_fall : 1.0;
        rise_resistance : 0.1;
        fall_resistance : 0.1;
        related_pin : "MC";
      }
    }
}
```

The following is an example message:

```
Error: Line 116, The timing check starting and ending conditions
        represented in this timing group are not mutually exclusive with
 the
        timing check starting and ending conditions represented in the
        timing group on line 124. (LIBG-113)
```

**What Next**

Check the *when_start*, *when_end* and *when* strings of both timing groups and make any corrections necessary to eliminate duplicate nonconditional timing checks or other erroneous information. Also, ensure that there is no nonconditional timing check defined when at least one conditional timing check of the same pins and timing type is defined.

---

## LIBG-115

(warning) The when attribute uses '%s' pins\n \tthat cannot be found in the %s %s.

**Description**

In a sequential cell, the following pins are allowed to appear in the *when* attribute of the *state-dependent* timing arc:

1) Pins used in *enable*, *enable_also*, and *data_in* in the *latch* group

2) Pins used in *clocked_on*, *clocked_on_also*, and *next_state* in the *ff* group for *timing_arc* with *timing_type* of *rising_edge/falling_edge*

Feedback

If the timing arc's timing_type is clear/preset:

1) The related clear or preset attribute must be specified inside the sequential group.

2) The clear/preset pin must be used in related_pin or when attribute of the timing arc.

This warning informs you that one of the previous restrictions has been violated.

The following example shows an instance where this message occurs:

```
cell(libg115) {
  area : 9;
  pin(D) {
    direction : input;
    capacitance : 1;
  }
  pin(CP) {
    direction : input;
    capacitance : 1;
  }
  pin(CD) {
    direction : input;
    capacitance : 2;
  }
  ff("IQ","IQN") {
    next_state : "D";
    clocked_on : "CP";
    clear   : "CD'";
  }
  pin(Q) {
    direction : output;
    function : "IQ";
    timing() {
      when : "CD";
      sdf_cond : "D==1'b1";
      timing_type : rising_edge;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "CP";
    }
     timing() {
      timing_type : clear;
      timing_sense : positive_unate;
      intrinsic_fall : 1.0;
      fall_resistance : 0.1;
      related_pin : "CD";
    }
  }
}
```

Feedback

The following is an example message:

```
Warning: Line 134, The when attribute uses 'CD' pins
        that cannot be found in the ff/latch group . (LIBG-115)
```

**What Next**

Check your *when* attribute, and correct any wrong information in the string or in the *timing_type*.

---

## LIBG-116

(warning) No test_scan_in/test_scan_in_inverted signal \n \ttypes exist on the '%s' test_cell cell. Converting the entire cell to a black box.

**Description**

To be able to accept scan data, each test_cell must have either a *test_scan_in* or *test_scan_in_inverted* signal type.

This warning informs you that the indicated test_cell does not have the *test_scan_in* or *test_scan_in_inverted* signal types. Thus, the test_cell is not valid, and the entire cell has been converted to a black box.

There is functional information missing, and other tools will not be able to use this cell.

This warning is issued when reading a .db library. you can switch off the warning by setting the environment variable *read_db_lib_warnings* to FALSE (default).

The same problem in a .lib library causes an error. See the .lib example for LIBG-147.

**What Next**

Check the library description to ensure that each test_cell has a *test_scan_in* or *test_scan_in_inverted* signal type. These signal types can be added within the test_cell using this syntax:

```
signal_type : test_scan_in (or test_scan_in_inverted)
```

---

## LIBG-117

(warning) No test_scan_out/test_scan_out_inverted signal \n \ttypes exisit on the '%s' test_cell cell. Converting the entire cell to a black box.

**Description**

To be able to scan out data, each test_cell must have either a *test_scan_out* or a *test_scan_out_inverted* signal type.

Feedback

This warning informs you that the indicated test_cell has neither the *test_scan_out* nor the *test_scan_out_inverted* signal types. Thus, the test_cell is not valid, and the entire cell is converted to a black box.

There is functional information missing, and other tools will not be able to use this cell.

This warning is issued when reading a .db library. You can switch off the warning by setting the environment variable *read_db_lib_warnings* to FALSE (default).

The same problem in a .lib library causes an error. See the .lib example for LIBG-148.

**What Next**

Check the library description to ensure that each test_cell either has a *test_scan_out* or a *test_scan_out_inverted* signal type. These signal types can be added within the test_cell using this syntax:

```
signal_type : test_scan_out (or test_scan_out_inverted)
```

---

## LIBG-118

(warning) The clocked_on_also function on the '%s' port for\n \ttest_cell on the '%s' cell is not specified on the correct pin.\n \tConverting the entire cell to a black box.

**Description**

The *clocked_on_also* function for this test_cell is not specified on the correct pin. This function needs to be specified on the clock of the slave stage of a master-slave configuration.

This warning informs you that the test_cell pointed to has the *clocked_on_also* function specified on an incorrect pin. Thus, the test_cell is not valid, and the entire cell is converted to a black box. Other tools will not be able to use this cell if it is made into a black box.

This warning is issued when reading a .db library. You can switch off the warning by setting the environment variable *read_db_lib_warnings* to FALSE (default).

The same problem in a .lib library causes an error. See the .lib example for LIBG-149.

**What Next**

Check the library description to ensure that each test_cell has specified the *clocked_on_also* function. To determine the correct function:

```
* Create a cell with the non test mode as the functional part
          of the cell.
        * Compile this cell.
        * Issue the following command:

             report_lib -table <library name> <cell name>
```

Feedback

## LIBG-119

(warning) The test_scan_clock_b signal type on the '%s' port for\n \ttest_cell on the '%s' cell is not specified on the correct pin.\n \tConverting the entire cell to a black box.

### Description

The *test_scan_clock_b* signal type for this test_cell is not specified on the correct pin. This signal type needs to be specified on the clock of the slave stage of an LSSD configuration.

This warning informs you that the test_cell pointed to has the *test_scan_clock_b* signal type specified on an incorrect pin. Thus, the test_cell is not valid, and the entire cell is converted to a black box. Other tools will not be able to use this cell if it is made into a black box.

This warning is issued when reading a .db library. You can switch off the warning by setting the environment variable *read_db_lib_warnings* to FALSE (default).

The same problem in a .lib library causes an error. See the .lib example for LIBG-150.

### What Next

Check the library description to ensure that each test_cell has specified the *test_scan_clock_b* signal type.

## LIBG-120

(warning) The test_scan_in signal type on the '%s' port for\n \ttest_cell on the '%s' cell is not specified on the correct pin.\n \tConverting the entire cell to a black box.

### Description

The *test_scan_in* signal type for this test_cell is not specified on the correct pin. This signal type needs to be specified on the pin that takes in the scan data. All scan types and, thus, all test_cells need to have either the *test_scan_in* or *test_scan_in_inverted* signal type.

This warning informs you that the test_cell pointed to has the *test_scan_in* signal type specified on an incorrect pin. Thus, the test_cell is not valid, and the entire cell is converted to a black box . There is functional information missing, and other tools will not be able to use this cell.

This warning is issued when reading a .db library. You can switch off the warning by setting the environment variable *read_db_lib_warnings* to FALSE (default).

The same problem in a .lib library causes an error. See the .lib example for LIBG-151.

Feedback

### What Next

Check the library description to ensure that each test_cell has correctly specified the *test_scan_in* signal type.

---

## LIBG-121

(warning) The test_scan_in_inverted signal type on the '%s' port for\n \ttest_cell on the '%s' cell is not specified on the correct pin.\n \tConverting the entire cell to a black box.

### Description

The *test_scan_in_inverted* signal type for this test_cell is not specified on the correct pin. This signal type needs to be specified on the pin that takes in the scan data. All scan types and, thus, all test_cells need to have either the *test_scan_in* or *test_scan_in_inverted* signal type.

This warning informs you that the test_cell pointed to has the *test_scan_in_inverted* signal type specified on an incorrect pin. Thus, the test_cell is not valid, and the entire cell is converted to a black box. There is functional information missing, and other tools will not be able to use this cell.

This warning is issued when reading a .db library. You can switch off the warning by setting the environment variable *read_db_lib_warnings* to FALSE (default).

The same problem in a .lib library causes an error. See the .lib example for LIBG-152.

### What Next

Check the library description to ensure that each test_cell has correctly specified the *test_scan_in_inverted* signal type.

---

## LIBG-122

(warning) The test_scan_enable signal type on the '%s' port for\n \ttest_cell on the '%s' cell is not specified on the correct pin.\n \tConverting the entire cell to a black box.

### Description

The *test_scan_enable* signal type for this test_cell is not specified on the correct pin. This signal type needs to be specified on the pin that switches the input mode of the cell between test and mission mode. Only multiplexed flip-flop requires this signal type. This signal type can also be used in clocked scan cells.

This warning informs you that the test_cell pointed to has the *test_scan_enable* signal type specified on an incorrect pin. Thus, the test_cell is not valid, and the entire cell is converted to a black box. There is functional information missing, and other tools will not be able to use this cell.

Feedback

This warning is issued when reading a .db library. You can switch off the warning by setting the environment variable *read_db_lib_warnings* to FALSE (default).

The same problem in a .lib library causes an error. See the .lib example for LIBG-153.

**What Next**

Check the library description to ensure that each test_cell has correctly specified the *test_scan_enable* signal type.

---

## LIBG-123

(warning) The test_scan_enable_inverted signal type on the '%s' port for\n \ttest_cell on the '%s' cell is not specified on the correct pin.\n \tConverting the entire cell to a black box.

**Description**

The *test_scan_enable_inverted* signal type for this test_cell is not specified on the correct pin. This signal type needs to be specified on the pin that switches the input mode of the cell between test and mission mode. Only, multiplexed flip-flop requires this signal type. This signal type can also be used in clocked scan cells.

This warning informs you that the test_cell pointed to has the *test_scan_enable_inverted* signal type specified on an incorrect pin. Thus, the test_cell is not valid, and the entire cell is converted to a black box. There is functional information missing, and other tools will not be able to use this cell.

This warning is issued when reading a .db library. You can switch off the warning by setting the environment variable *read_db_lib_warnings* to FALSE (default).

The same problem in a .lib library causes an error. See the .lib example for LIBG-154.

**What Next**

Check the library description to ensure that each test_cell has correctly specified the *test_scan_enable_inverted* signal type.

---

## LIBG-124

(warning) The test_scan_clock signal type on the '%s' port for\n \ttest_cell on the '%s' cell is not specified on the correct pin.\n \tConverting the entire cell to a black box.

**Description**

The *test_scan_clock* signal type for this test_cell is not specified on the correct pin. This signal type needs to be specified on the scan clock for clocked scan cells.

This warning informs you that the test_cell pointed to has the *test_scan_clock* signal type specified on an incorrect pin. Thus, the test_cell is not valid, and the entire cell is

Feedback

converted to a black box. There is functional information missing, and other tools will not be able to use this cell.

This warning is issued when reading a .db library. You can switch off the warning by setting the environment variable *read_db_lib_warnings* to FALSE (default).

The same problem in a .lib library causes an error. See the .lib example for LIBG-155.

**What Next**

Check the library description to ensure that each test_cell has correctly specified the *test_scan_clock* signal type.

---

## LIBG-125

(warning) The test_scan_clock_a signal type on the '%s' port for\n \ttest_cell on the '%s' cell is not specified on the correct pin.\n \tConverting the entire cell to a black box.

**Description**

The *test_scan_clock_a* signal type for this test_cell is not specified on the correct pin. This signal type needs to be specified on the clock of the master stage of an LSSD configuration.

This warning informs you that the test_cell pointed to has the *test_scan_clock_a* signal type specified on an incorrect pin. Thus, the test_cell is not valid, and the entire cell is converted to a black box. There is functional information missing, and other tools will not be able to use this this cell.

This warning is issued when reading a .db library. You can switch off the warning by setting the environment variable *read_db_lib_warnings* to FALSE (default).

The same problem in a .lib library causes an error. See the .lib example for LIBG-156.

**What Next**

Check the library description to ensure that each test_cell has correctly specified the *test_scan_clock_a* signal type.

---

## LIBG-126

(warning) The test_clock signal type on the '%s' port for\n \ttest_cell on the '%s' cell is not specified on the correct pin. \n \tConverting the entire cell to a black box.

**Description**

The *test_clock* signal type for this test_cell is not specified on the correct pin. This signal type needs to be specified on the clock used for ATPG capture in auxiliary clock lssd cells.

Feedback

This warning informs you that the test_cell pointed to has the *test_clock* signal type specified on an incorrect pin. Thus, the test_cell is not valid, and the entire cell is converted to a black box. There is functional information missing, and other tools will not be able to use this cell.

This warning is issued when reading a .db library. You can switch off the warning by setting the environment variable *read_db_lib_warnings* to FALSE (default).

The same problem in a .lib library causes an error. See the .lib example for LIBG-157.

**What Next**

Check the library description to ensure that each test_cell has correctly specified the *test_clock* signal type.

---

## LIBG-127

(warning) The '%s' clocked test cell is not valid \n \tbecause it has a master-slave cell as its nontest mode.\n \tConverting the entire cell to a black box.

**Description**

A clocked test cell has the following characteristics:

```
- signal types : test_scan_in/test_scan_in_inverted, test_scan_enable/
        test_scan_enable_inverted (optional), test_scan_clock,
        test_scan_out/test_scan_out_inverted

    - nontest modes : flip-flop or latch
```

This warning informs you that the signal types identify this cell as a clocked scan test cell, but it does not have a flip-flop or latch as its nontest mode. It is currently specified as a master-slave cell. Thus, the test cell is not valid, and the entire cell is converted to a black box. There is functional information missing, and other tools cannot use this cell.

This warning is issued when reading a .db library. You can switch off the warning by setting the environment variable *read_db_lib_warnings* to FALSE (default).

The same problem in a .lib library causes an error. See the .lib example for LIBG-174.

**What Next**

Check the library description to ensure that nontest modes and signal types are correctly pointed to by the test_cell. To determine why the nontest function is not one of the previous types,

```
* Create a cell with the nontest mode as the functional part
          of the cell.
        * Compile this cell.
        * Issue the following command:
```

Feedback

```
report_lib -table <library name> <cell name>
```

In the report_lib command's report, determine which pins correspond to states (pins with an N output) by looking at their functional description.

---

## LIBG-128

(warning) Invalid nontest mode for test_cell on the '%s' cell;\n \tthere are more than 2 sequential elements. \n \tConverting the entire cell to a black box.

### Description

The nontest mode for test_cells must be one of the following:

```
- A single flip-flop
        - A single latch
        - A single master-slave latch configuration using clocked_on and
                clocked_on_also
```

This warning informs you that the nontest mode of the test_cell has more than 2 distinct sequential elements, so the test_cell does not fit the previous requirements. Thus, the test_cell is not valid, and the entire cell is converted to a black box. There is functional information missing, and other tools cannot use this cell.

This warning is issued when reading a .db library. You can switch off the warning by setting the environment variable *read_db_lib_warnings* to FALSE (default).

### What Next

Check the library description to ensure that each test_cell has properly specified the correct nontest mode. To determine why there are multiple sequential elements in this description,

```
* Create a cell with the nontest mode as the functional part
          of the cell.
        * Compile this cell.
        * Issue the following command:

             report_lib -table <library name> <cell name>
```

In the report_lib command's report, determine which pins correspond to states (pins with an N output) by looking at their functional description.

---

## LIBG-129

(warning) Invalid nontest mode for test_cell on the '%s' cell. \n \tThis mode describes a ganged cell.

Feedback

### Description

The nontest mode for test_cells must be one of the following:

```
- A single flip-flop
        - A single latch
        - A single master-slave latch configuration using clocked_on and
                clocked_on_also
```

This warning informs you that the nontest mode of the test cell is a ganged cell, and it does not fit the previous requirements. Thus, the test_cell is not valid, and the entire cell is converted to a black box. There is functional information missing, and other tools cannot use this cell.

This warning is issued when reading a .db library. You can switch off the warning by setting the environment variable *read_db_lib_warnings* to FALSE (default).

### What Next

Check the library description to ensure that each test_cell does not specify a ganged cell as its nontest mode. Ganged cells can be recognized by the keywords ff_bank or latch_bank.

---

## LIBG-130

(warning) The nontest mode for test_cell on the '%s' cell is invalid.\n \tThe '%s' slave port does not directly feed the output.

### Description

The nontest mode for test_cells must be one of the following:

```
- A single flip-flop
        - A single latch
        - A single master-slave latch configuration using clocked_on
          and clocked_on_also
```

This warning informs you that in nontest mode of the test cell, the slave does not directly feed the output. This means that it is not a master-slave latch configuration that the DFT Compiler supports. Thus, the test_cell is not valid, and the entire cell is a black box.

This warning is issued when reading a .db library. You can switch off the warning by setting the environment variable *read_db_lib_warnings* to FALSE (default).

The same problem in a .lib library causes an error. See the .lib example for LIBG-160.

Feedback

### What Next

Check the library description to ensure that the nontest mode of the test_cell is correctly specified. If you are using a master-slave configuration as the nontest mode, the only accepted description is the state description using

```
clocked_on : <master clock>
        clocked_on_also : <slave clock>
```

---

## LIBG-131

(warning) The nontest mode for test_cell on the '%s' cell is invalid.\n \tThe '%s' state pin is not a latch for a master-slave configuration.

### Description

The nontest mode for test_cells must be one of the following:

```
- A single flip-flop
        - A single latch
        - A single master-slave latch configuration using clocked_on
          and clocked_on_also
```

This warning informs you that the nontest mode of the test cell is an incorrect master-slave description. This stage of the master-slave configuration is not a latch, and it does not fit the previous requirements. Thus, the test cell is not valid, and the entire cell is converted to a black box. There is functional information missing, and other tools cannot use this cell.

This warning is issued when reading a .db library. You can switch off the warning by setting the environment variable *read_db_lib_warnings* to FALSE (default).

The same problem in a .lib library causes an error. See the .lib example for LIBG-161.

### What Next

Check the library description to ensure that the nontest mode of the test_cell is correctly specified. If you are using a master-slave configuration as the nontest mode, the only accepted description is the state description using clocked_on : <master clock> clocked_on_also : <slave clock>

---

## LIBG-132

(warning) The test cell for auxiliary clocked lssd '%s' cell is not\n \tvalid because its nontest mode is not a flip-flop.

Feedback

### Description

An auxiliary clocked lssd test_cell has the following characteristics:

```
- signal types : test_scan_in/test_scan_in_inverted,
                         test_scan_out/test_scan_out_inverted,
                         test_scan_clock_a, test_scan_clock_b, test_clock
        - nontest mode : flip-flop
```

This warning informs you that the signal types point to this auxiliary clocked lssd test_cell, but it does not have a flip-flop as its nontest mode. Thus, the test cell is not valid, and the entire cell is converted to a black box. There is functional information missing, and other tools cannot use this cell.

This warning is issued when reading a .db library. You can switch off the warning by setting the environment variable *read_db_lib_warnings* to FALSE (default).

The same problem in a .lib library causes an error. See the .lib example for LIBG-175.

### What Next

Check the library description to ensure that nontest modes and signal types are correctly pointed to by the test_cell. To determine why the nontest function is not one of the previous types,

```
* Create a cell with the nontest mode as the functional part
          of the cell.
        * Compile this cell.
        * Issue the following command:

             report_lib -table <library name> <cell name>
```

In the report_lib command's report, determine which pins correspond to states (pins with an N output) by looking at their functional description.

---

## LIBG-133

(warning) The test_cell on the '%s' cell cannot have more than one\n \ttest_scan_in/ test_scan_in_inverted signal type. This happens on \n \tthe '%s' port.

### Description

Each test_cell must have only one test_scan_in or test_scan_in_inverted signal type. This signal type is specified on the pin that is used to take scan data into the cell.

This warning informs you that the test_cell pointed to has multiple test_scan_in(test_scan_in_inverted) signal types. The message points to the second port that has this signal type. Thus, the test cell is not valid, and the entire cell is converted to a black box. There is functional information missing, and other tools cannot use this cell.

Feedback

This warning is issued when reading a .db library. You can switch off the warning by setting the environment variable *read_db_lib_warnings* to FALSE (default).

The same problem in a .lib library causes an error. See the .lib example for LIBG-162.

**What Next**

Check the library description to ensure that each test_cell has only one test_scan_in or test_scan_in_inverted signal type.

---

## LIBG-134

(warning) The '%s' port in test_cell on the '%s' cell has a\n \ttest_scan_out signal type on an inverted output.

**Description**

The test_scan_out signal type can only be specified on a noninverted output. If the output is inverted, the proper signal type should be test_scan_out_inverted. This signal type specifies the polarity of the port with respect to the state.

This warning informs you that the test_cell pointed to has the test_scan_out signal type specified on an inverted output. When reading a .db library, Library Compiler ignores the scan_out attribute and creates the functional description of the cell using the function statement. In other words, that output is inverted in the functional description.

The same problem in a .lib library causes an error. See the .lib example for LIBG-163.

**What Next**

The default behavior of selecting the function over the signal_type to determine the inversion of the output is be correct in most cases.

Check the library description to ensure that the test_scan_out signal type is only placed on noninverted outputs. To determine if this is an inverted or noninverted output,

```
* Create a cell with the nontest mode as the functional part
          of the cell.
        * Compile this cell.
        * Issue the following command:

             report_lib -table <library name> <cell name>
```

In the report_lib command's report, an inverted output is not a state if it does not have N. Instead, the output contains an inversion of the state pin.

Feedback

## LIBG-135

(warning) The '%s' port in test_cell on the '%s' cell has a\n \ttest_scan_out_inverted signal type on a noninverted output.

### Description

The test_scan_out_inverted signal type can only be specified on an inverted output. If the output is noninverted, the proper signal type should be test_scan_out. This signal type specifies the polarity of the port with respect to the state.

This warning informs you that the test_cell pointed to has the test_scan_out signal type specified on an inverted output. When reading a .db library, the Library Compiler ignores the scan_out_inverted attribute and creates the functional description of the cell using the function statement. In other words, that output is noninverted in the functional description.

The same problem in a .lib library causes an error. See the .lib example for LIBG-164.

### What Next

The default behavior of selecting the function over the signal_type to determine the inversion of the output is correct in most cases.

Check the library description to ensure that the test_scan_out_inverted signal type is only placed on inverted outputs. To determine if this is an inverted or noninverted output,

```
* Create a cell with the nontest mode as the functional part
          of the cell.
        * Compile this cell.
        * Issue the following command:

             report_lib -table <library name> <cell name>
```

In the report_lib command's report, an inverted output is not a state if it does not have N. Instead, the output contains an inversion of the state pin.

## LIBG-136

(warning) The test cell on the '%s' cell cannot have more than one\n \ttest_scan_enable or test_scan_enable_inverted signal type. This \n \thappens on the '%s' port.

### Description

Each test_cell must have only one test_scan_enable or test_scan_enable_inverted signal type. This signal type is specified on the pin that is used to select between scan and mission mode operation of the cell.

This warning informs you that the test_cell pointed to has multiple test_scan_enable(test_scan_enable_inverted) signal types. The message points to the

Feedback

second port that has this signal type. Thus, the test cell is not valid, and the entire cell is converted to a black box. There is functional information missing, and other tools cannot use this cell.

This warning is issued when reading a .db library. You can switch off the warning by setting the environment variable *read_db_lib_warnings* to FALSE (default).

The same problem in a .lib library causes an error. See the .lib example for LIBG-165.

**What Next**

Check the library description to ensure that each test_cell has only one test_scan_enable or test_scan_enable_inverted signal type.

---

## LIBG-137

(warning) The test_scan_clock '%s' port in test_cell on the '%s' cell\n \tdoes not have an active sense.

**Description**

The test_scan_clock must have an active sense in the full cell description. The Library Compiler determines the active sense by checking the function of the full cell. The test_scan_clock is the scan clock in the clocked scan type.

Because the test_cell and the full cell have a discrepancy, neither the test cell nor the full cell is recognized. Both the test_cell and the entire cell are converted to a black box. There is functional information missing, and other tools cannot use this cell.

This warning is issued when reading a .db library. You can switch off the warning by setting the environment variable *read_db_lib_warnings* to FALSE (default).

The same problem in a .lib library causes an error.

**What Next**

Check the library description to ensure that the functional descriptions in the test_cell and full cell are correct and that the timing arcs are consistent. To report on the active sense and functions of the full cell,

```
* Create a cell with the nontest mode as the functional part
          of the cell.
        * Compile this cell.
        * Issue the following command:

             report_lib -table <library name> <cell name>
```

Feedback

## LIBG-138

(warning) The test_scan_clock_a '%s' port in test_cell on the '%s' cell \n \tdoes not have an active sense.

### Description

The test_scan_clock_a must have an active sense in the full cell description. The Library Compiler determines the active sense by checking the function of the full cell. The test_scan_clock_a is the scan clock for the master stage of the LSSD scan style.

Because the test_cell and the full cell have a discrepancy, neither the test cell nor the full cell is recognized. Both the test_cell and the entire cell are converted to a black box. There is functional information missing, and other tools cannot use this cell.

This warning is issued when reading a .db library. You can switch off the warning by setting the environment variable *read_db_lib_warnings* to FALSE (default).

The same problem in a .lib library causes an error.

### What Next

Check the library description to ensure that the functional descriptions in the test_cell and full cell are correct and that the timing arcs are consistent. To report on the active sense and functions of the full cell,

```
* Create a cell with the non test mode as the functional part
          of the cell.
        * Compile this cell.
        * Issue the following command:

             report_lib -table <library name> <cell name>
```

## LIBG-139

(warning) The test_scan_clock_b '%s' port in test_cell on the '%s' cell\n \tdoes not have an active sense.

### Description

The test_scan_clock_b must have an active sense in the full cell description. The Library Compiler determines the active sense by checking the function of the full cell. The test_scan_clock_a is the scan clock for the slave stage of the LSSD scan style.

Because the test_cell and the full cell have a discrepancy, neither the test cell nor the full cell is recognized. Both the test_cell and the entire cell are converted to a black box. There is functional information missing, and other tools cannot use this cell.

Feedback

This warning is issued when reading a .db library. You can switch off the warning by setting the environment variable *read_db_lib_warnings* to FALSE (default).

The same problem in a .lib library causes an error.

**What Next**

Check the library description to ensure that the functional descriptions in the test_cell and full cell are correct and that the timing arcs are consistent. To report on the active sense and functions of the full cell,

```
* Create a cell with the nontest mode as the functional part
          of the cell.
        * Compile this cell.
        * Issue the following command:

             report_lib -table <library name> <cell name>
```

---

## LIBG-140

(warning) The test_clock '%s' port in test_cell on the '%s' cell does\n \tnot have an active sense.

**Description**

The test_clock must have an active sense in the full cell description. The Library Compiler determines the active sense by checking the function of the full cell. The test_clock is the capture clock used by ATPG in the auxiliary clocked lssd scan style.

Because the test_cell and the full cell have a discrepancy, neither the test cell nor the full cell is recognized. Both the test_cell and the entire cell are converted to a black box. There is functional information missing, and other tools cannot use this cell.

This warning is issued when reading a .db library. You can switch off the warning by setting the environment variable *read_db_lib_warnings* to FALSE (default).

The same problem in a .lib library causes an error.

**What Next**

Check the library description to ensure that the functional descriptions in the test_cell and full cell are correct and that the timing arcs are consistent. To report on the active sense and functions of the full cell,

```
* Create a cell with the nontest mode as the functional part
          of the cell.
        * Compile this cell.
        * Issue the following command:

             report_lib -table <library name> <cell name>
```

Feedback

## LIBG-141

(warning) The '%s' port in test_cell for the '%s' cell has an invalid\n \tsignal type %s.

### Description

The valid signal_types in test_cells are

```
- test_scan_in
        - test_scan_in_inverted
        - test_scan_enable
        - test_scan_enable_inverted
        - test_scan_out
        - test_scan_out_inverted
        - test_scan_clock
        - test_scan_clock_a
        - test_scan_clock_b
        - test_clock
```

This warning informs you that the port in the test_cell pointed to has a signal_type other than the ones listed previously. Thus, the test_cell is not valid, and the entire cell is converted to a black box.

### What Next

Check the test_cell pointed to, and change the invalid signal_type.

## LIBG-142

(warning) The LSSD test_cell for the '%s' cell has neither\n \ta test_scan_clock_b signal type nor a master-slave configuration as its non test mode.

### Description

LSSD scan styles must have the following test_cell characteristics:

```
clocked lssd
        - signal types : test_scan_in/test_scan_in_inverted,
 test_scan_out/
                test_scan_out_inverted, test_scan_clock_a,
 test_scan_clock_b
        - nontest mode: flip_flop
   auxiliary clock lssd
        - signal types : test_scan_in/test_scan_in_inverted,
 test_scan_out/
                test_scan_out_inverted, test_scan_clock_a,
 test_scan_clock_b,
                test_clock
        - nontest mode : flip_flop
   single-latch lssd
```

Feedback

```
       - signal types : test_scan_in/test_scan_in_inverted,
test_scan_out/
                test_scan_out_inverted, test_scan_clock_a,
test_scan_clock_b
       - nontest mode: latch
  single-latch lssd
       - signal types : test_scan_in/test_scan_in_inverted,
test_scan_out/
                test_scan_out_inverted, test_scan_clock_a,
                test_scan_clock_b (optional)
       - nontest mode: master-slave (clocked_on & clocked_on_also)
```

This warning informs you that the other signal types point to the fact that this is an lssd scan style because it has test_scan_in or test_scan_in_inverted, test_scan_out or test_scan_out_inverted, and test_scan_clock_a signal types. However, it is invalid because it has neither a test_scan_clock_b signal type nor a master-slave configuration as its nontest mode. Thus, the test cell is not valid, and the entire cell is converted to a black box. There is functional information missing, and other tools cannot use this cell.

This warning is issued when reading a .db library. You can switch off the warning by setting the environment variable *read_db_lib_warnings* to FALSE (default).

The same problem in a .lib library causes an error. See the .lib example for LIBG-171.

**What Next**

Check the library description to ensure that each test_cell has correctly specified the nontest modes and signal types in the test_cell pointed to. If you do not understand why the nontest function is not one of the previous types,

```
* Create a cell with the nontest mode as the functional part
         of the cell.
       * Compile this cell.
       * Issue the following command:

            report_lib -table <library name> <cell name>
```

In the report_lib command's report, determine which pins correspond to states (pins with an N output) by looking at their functional description.

---

## LIBG-144

(warning) The test_cell for the '%s' cell is not valid because\n \tit has missing or extra test signals.

Feedback

### Description

The supported scan styles must have the following test_cell characteristics:

```
multiplexed flip-flop
        - signal types : test_scan_in/test_scan_in_inverted,
 test_scan_enable/
                test_scan_enable_inverted,
 test_scan_out/test_scan_out_inverted
        - nontest mode : flip-flop, master slave latch pair, or latch
   clocked scan
        - signal types : test_scan_in/test_scan_in_inverted,
 test_scan_enable/
                test_scan_enable_inverted (optional), test_scan_clock,
                test_scan_out/test_scan_out_inverted
        - nontest modes : flip-flop or latch
   clocked lssd
        - signal types : test_scan_in/test_scan_in_inverted,
 test_scan_out/
                test_scan_out_inverted, test_scan_clock_a,
 test_scan_clock_b
        - nontest mode: flip-flop
   auxiliary clock lssd
        - signal types : test_scan_in/test_scan_in_inverted,
 test_scan_out/
                test_scan_out_inverted, test_scan_clock_a,
 test_scan_clock_b,
                test_clock
        - nontest mode : flip-flop
   single-latch lssd
        - signal types : test_scan_in/test_scan_in_inverted,
 test_scan_out/
                test_scan_out_inverted, test_scan_clock_a,
 test_scan_clock_b
        - nontest mode: latch
   single-latch lssd
        - signal types : test_scan_in/test_scan_in_inverted,
 test_scan_out/
                test_scan_out_inverted, test_scan_clock_a,
 test_scan_clock_b
        - nontest mode: latch
   single-latch lssd
        - signal types : test_scan_in/test_scan_in_inverted,
 test_scan_out/
                test_scan_out_inverted, test_scan_clock_a,
                test_scan_clock_b (optional)
        - nontest mode: master-slave (clocked_on & clocked_on_also)
```

This warning informs you that the other signal types do not correspond to one of the previously supported scan styles. Because there are extra or missing signal types, this test_cell is not valid, and the entire cell is a black box.

Feedback

This warning is issued when reading a .db library. You can switch off the warning by setting the environment variable *read_db_lib_warnings* to FALSE (default).

The same problem in a .lib library causes an error. See the .lib example for LIBG-172.

**What Next**

Check the library description to ensure that each test_cell has correctly specified the nontest modes and signal types in the test_cell pointed to. If you do not understand why the nontest function is not one of the previous types,

```
* Create a cell with the non test mode as the functional part
          of the cell.
        * Compile this cell.
        * Issue the following command:

             report_lib -table <library name> <cell name>
```

In the report_lib command's report, determine which pins correspond to states (pins with an N output) by checking their functional description.

---

## LIBG-145

(warning) Cannot derive full function based on test_cell\n \ton the '%s' cell in a library with DPCM vendor specific delay model.

**Description**

The Library Compiler generates a functional description of the entire cell based on the information in the test_cell. However, if the library is a DPCM library, no full function description will be generated.

This warning is not issued when reading a .db library.

**What Next**

Fill the full function with state table description.

---

## LIBG-146

(warning) Cannot derive full function based on test_cell\n \ton the '%s' cell due to previous warnings or errors.

**Description**

The Library Compiler generates a functional description of the entire cell based on the information in the test_cell. However, if there are warnings or errors in the test_cell, a functional description of the entire cell cannot be generated.

Feedback

This warning informs you that the test_cell pointed to has previous errors or warnings. Thus, the test_cell is not valid, and the entire cell is a black box. There is functional information missing, and other tools cannot use this cell.

This warning is issued when reading a .db library. You can switch off the warning by setting the environment variable *read_db_lib_warnings* to FALSE (default).

The same problem in a .lib library causes an error. See the .lib example for LIBG-173.

**What Next**

Fix the previous problems in order to recognize both the test_cell and full cell functionality. You might have to add a state table description of the entire cell.

---

## LIBG-147

(error) No test_scan_in/test_scan_in_inverted signal\n \ttypes exist on the '%s' test_cell cell.

**Description**

Each test_cell must have either a test_scan_in or test_scan_in_inverted signal type to be valid. Otherwise, this cell cannot accept scan data.

This warning informs you that the test_cell pointed to does not have the test_scan_in or test_scan_in_inverted signal types. Thus, the test_cell is not valid, and the entire cell is converted to a black box.

There is functional information missing, and other tools cannot use this cell.

The following example shows an instance where this message occurs:

```
cell(libg147) {
      area : 10;
      pin(D) {
        direction : input;
        capacitance : 1;
      }
      pin(C) {
        direction : input;
        capacitance : 2;
      }
      pin(D2) {
        direction : input;
        capacitance : 1;
      }
      pin(C2) {
        direction : input;
        capacitance : 2;
      }
      pin(C3) {
```

Feedback

```
    direction : input;
    capacitance : 1;
  }
  pin(MQ) {
    direction : internal;
    internal_node : "MQ";
    timing() {
      timing_type : rising_edge;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "C C2";
    }
  }
  pin(Q) {
    direction : output;
    internal_node : "SQ";
    timing() {
      timing_type : rising_edge;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "C C3";
    }
  }
  pin(QN) {
    direction : output;
    state_function : "(Q)'";
    timing() {
      timing_type : rising_edge;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "C C3";
    }
  }
  statetable("C D C2 C3 D2", "MQ SQ") {
    table :  "R  H/L L L  -  : -   - : H/L H/L,\
            ~R   -  - H  -  : H/L - :  -  H/L,\
            ~R   -  - L  -  : -   - :  -  N,\
             R   -  H -  -  : -   - :  X  X,\
             R   -  - H  -  : -   - :  X  X,\
            ~R   -  H - H/L : -   - : H/L -,\
            ~R   -  L -  -  : -   - :  N  -";
  }
  test_cell() {
    pin(D) {
      direction : input;
    }
    pin(C) {
```

Feedback

```
          direction : input;
          capacitance : 2;
          signal_type : "test_scan_clock";
        }
        pin(D2) {
        /* missing test_scan_in signal_type */
          direction : input;
        }
        pin(C2) {
          direction : input;
          capacitance : 2;
          signal_type : "test_scan_clock_a";
        }
        pin(C3) {
          direction : input;
          capacitance : 1;
          signal_type : "test_scan_clock_b";
        }
        ff ("IQ","IQN") {
          next_state : "D";
          clocked_on : "C";
        }
        pin(Q) {
          direction : output;
          function : "IQ";
          signal_type : "test_scan_out";
        }
        pin(QN) {
          direction : output;
          function : "IQN";
          signal_type : "test_scan_out_inverted";
        }
      }
    }
  }
```

The following is an example message:

```
Error: Line 177, No test_scan_in/test_scan_in_inverted signal
        types exist on the 'libg147' test_cell cell. (LIBG-147)
```

**What Next**

Check the library description to ensure that each test_cell has a *test_scan_in* or *test_scan_in_inverted* signal type. These signal types can be added within the test_cell using the syntax:

```
signal_type : test_scan_in [ or test_scan_in_inverted]
```

Feedback

## LIBG-148

(error) No test_scan_out/test_scan_out_inverted signal\n \ttypes exist on the '%s' test_cell cell.

### Description

Each test_cell must have either a test_scan_out or test_scan_out_inverted signal type in order to be valid. Otherwise, this cell cannot scan out data.

This warning informs you that the test_cell pointed to does not have the test_scan_out or test_scan_out_inverted signal types. Thus, the test_cell is not valid, and the entire cell is converted to a black box.

There is functional information missing, and other tools cannot use this cell.

The following example shows an instance where this message occurs:

```
cell(libg148) {
      area : 9;
      pin(D) {
        direction : input;
        capacitance : 1;
      }
      pin(C) {
        direction : input;
        capacitance : 1;
      }
      pin(SI) {
        direction : input;
        capacitance : 1;
      }
      pin(SE) {
        direction : input;
        capacitance : 2;
      }

      pin(Q) {
        direction : output;
        inverted_output : FALSE;

      }
      pin(QN) {
        direction : output;
        inverted_output : TRUE;
      }
      test_cell() {
        pin(D) {
          direction : input;
        }
        pin(C) {
          direction : input;
```

Feedback

```
        }
        pin(SI) {
          direction : input;
          signal_type : "test_scan_in";
        }
        pin(SE) {
          direction : input;
          signal_type : "test_scan_enable";
        }

        ff ("IQ","IQN") {
          next_state : "D";
          clocked_on : "C";
        }

        pin(Q) {
          direction : output;
          function : "IQ";
          /*      signal_type : "test_scan_out"; */
        }
        pin(QN) {
          direction : output;
          function : "IQN";
          /*      signal_type : "test_scan_out_inverted";*/
        }
      }
    }
```

The following is an example message:

```
Error: Line 139, No test_scan_out/test_scan_out_inverted signal
        types exist on the 'libg148' test_cell cell. (LIBG-148)
```

### What Next

Check the library description to ensure that each test_cell has a test_scan_out or test_scan_out_inverted signal type. These signal types can be added within the test_cell using the syntax:

```
signal_type : test_scan_out [ or test_scan_out_inverted]
```

---

## LIBG-149

(error) The clocked_on_also function on the '%s' port for\n \ttest_cell on the '%s' cell is not specified on the correct pin.

### Description

The *clocked_on_also* function for this test_cell is not specified on the correct pin. This function needs to be specified on the clock of the slave stage of a master-slave configuration.

Feedback

This warning informs you that the test_cell pointed to has the *clocked_on_also* function specified on an incorrect pin. Thus, the test_cell is not valid, and the entire cell is converted to a black box. There is functional information missing, and other tools cannot use this cell.

**What Next**

Check the library description to ensure that each test_cell has correctly used the *clocked_on_also* function. To determine the correct function,

```
* Create a cell with the nontest mode as the functional part
          of the cell.
        * Compile this cell.
        * Issue the following command:

             report_lib -table <library name> <cell name>
```

---

## LIBG-150

(error) The test_scan_clock_b signal type on the '%s' port for \n \ttest_cell on the '%s' cell is not specified on the correct pin.

**Description**

The test_scan_clock_b signal type for this test_cell is not specified on the correct pin. This signal type needs to be specified on the clock of the slave stage of an LSSD configuration.

This warning informs you that the test_cell pointed to has the test_scan_clock_b signal type specified on an incorrect pin. Thus, the test_cell is not valid, and the entire cell is converted to a black box. There is functional information missing, and other tools cannot use this cell.

The following example shows an instance where this message occurs:

```
cell(libg150) {
    area : 13;
    pin(D) {
      direction : input;
      capacitance : 1;
    }
    pin(C) {
      direction : input;
      capacitance : 2;
    }
    pin(D2) {
      direction : input;
      capacitance : 1;
    }
    pin(C2) {
      direction : input;
      capacitance : 2;
```

Feedback

```
}
pin(C3) {
  direction : input;
  capacitance : 1;
}
pin(MQ) {
  direction : internal;
  internal_node : "MQ";
  timing() {
    timing_type : rising_edge;
    intrinsic_rise : 1.0;
    intrinsic_fall : 0.0;
    rise_resistance : 0.1;
    fall_resistance : 0.1;
    related_pin : "C C2";
  }
}
pin(Q) {
  direction : output;
  internal_node : "SQ";
  timing() {
    timing_type : rising_edge;
    intrinsic_rise : 1.0;
    intrinsic_fall : 1.0;
    rise_resistance : 0.1;
    fall_resistance : 0.1;
    related_pin : "C C3";
  }
}
pin(QN) {
  direction : output;
  state_function : "(Q)'";
  timing() {
    timing_type : rising_edge;
    intrinsic_rise : 1.0;
    intrinsic_fall : 1.0;
    rise_resistance : 0.1;
    fall_resistance : 0.1;
    related_pin : "C C3";
  }
}
statetable("C D C2 C3 D2", "MQ SQ") {
  table :  "R  H/L L L  -  : -   - : H/L H/L,\
           ~R   -  - H  -  : H/L - :  -  H/L,\
           ~R   -  - L  -  : -   - :  -  N,\
            R   -  H -  -  : -   - :  X  X,\
            R   -  - H  -  : -   - :  X  X,\
           ~R   -  H - H/L : -   - : H/L -,\
           ~R   -  L -  -  : -   - :  N  -";
}
test_cell() {
  pin(D) {
    direction : input;
```

Feedback

```
      }
      pin(C) {
        direction : input;
        capacitance : 2;
        signal_type : "test_scan_clock_b";
      }
      pin(D2) {
        direction : input;
      }
      pin(C2) {
        direction : input;
        capacitance : 2;
        signal_type : "test_scan_clock_a";
      }
      pin(C3) {
        direction : input;
        capacitance : 1;
      }
      ff ("IQ","IQN") {
        next_state : "D";
        clocked_on : "C";
      }
      pin(Q) {
        direction : output;
        function : "IQ";
        signal_type : "test_scan_out";
      }
      pin(QN) {
        direction : output;
        function : "IQN";
        signal_type : "test_scan_out_inverted";
      }
    }
  }
```

The following is an example message:

```
Error: Line 181, The test_scan_clock_b signal type on the 'C' port for
         test_cell on the 'libg150' cell is not specified on the correct
 pin.(LIBG-150)
```

**What Next**

Check the library description to ensure that each test_cell has correctly specified the test_scan_clock_b signal type.

---

## LIBG-151

(error) The test_scan_in signal type on the '%s' port for\n \ttest_cell on the '%s' cell is not specified on the correct pin.

Feedback

### Description

The *test_scan_in* signal type for this test_cell is not specified on the correct pin. This signal type needs to be specified on the pin that takes in the scan data. All scan types, and therefore, all test_cells need to have either the *test_scan_in* or *test_scan_in_inverted* signal type.

This warning informs you that the test_cell pointed to has the *test_scan_in* signal type specified on an incorrect pin. Thus, the test_cell is not valid, and the entire cell is converted to a black box. There is functional information missing, and other tools cannot use this cell.

The following example shows an instance where this message occurs:

```
cell(libg151) {
    area : 13;
    pin(D) {
      direction : input;
      capacitance : 1;
    }
    pin(C) {
      direction : input;
      capacitance : 2;
    }
    pin(D2) {
      direction : input;
      capacitance : 1;
    }
    pin(C2) {
      direction : input;
      capacitance : 2;
    }
    pin(C3) {
      direction : input;
      capacitance : 1;
    }
    pin(MQ) {
      direction : internal;
      internal_node : "MQ";
      timing() {
        timing_type : rising_edge;
        intrinsic_rise : 1.0;
        intrinsic_fall : 1.0;
        rise_resistance : 0.1;
        fall_resistance : 0.1;
        related_pin : "C C2";
      }
    }
    pin(Q) {
      direction : output;
      internal_node : "SQ";
      timing() {
```

Feedback

```
    timing_type : rising_edge;
    intrinsic_rise : 1.0;
    intrinsic_fall : 1.0;
    rise_resistance : 0.1;
    fall_resistance : 0.1;
    related_pin : "C C3";
  }
}
pin(QN) {
  direction : output;
  state_function : "(Q)'";
  timing() {
    timing_type : rising_edge;
    intrinsic_rise : 1.0;
    intrinsic_fall : 1.0;
    rise_resistance : 0.1;
    fall_resistance : 0.1;
    related_pin : "C C3";
  }
}
statetable("C D C2 C3 D2", "MQ SQ") {
  table :  "R  H/L L L  -  : -   - : H/L H/L,\
           ~R   -  - H  -  : H/L - :  -  H/L,\
           ~R   -  - L  -  : -   - :  -  N,\
            R   -  H -  -  : -   - :  X  X,\
            R   -  - H  -  : -   - :  X  X,\
           ~R   -  H - H/L : -   - : H/L -,\
           ~R   -  L -  -  : -   - :  N  -";
}
test_cell() {
  pin(D) {
    direction : input;
  }
  pin(C) {
    direction : input;
    capacitance : 2;
    signal_type : "test_scan_in";
  }
  pin(D2) {
    direction : input;
  }
  pin(C2) {
    direction : input;
    capacitance : 2;
    signal_type : "test_scan_clock_a";
  }
  pin(C3) {
    direction : input;
    capacitance : 1;
    signal_type : "test_scan_clock_b";
  }
  ff ("IQ","IQN") {
    next_state : "D";
```

Feedback

```
        clocked_on : "C";
      }
      pin(Q) {
        direction : output;
        function : "IQ";
        signal_type : "test_scan_out";
      }
      pin(QN) {
        direction : output;
        function : "IQN";
        signal_type : "test_scan_out_inverted";
      }
    }
  }
```

In this case the pin 'C' is a clock and should not have a test_scan_in signal_type attribute. The test_scan_in should be on the pin 'D2'.

The following is an example message:

```
Error: Line 182, The test_scan_in signal type on the 'C' port for
        test_cell on the 'libg151' cell is not specified on the correct
 pin. (LIBG-151)
```

**What Next**

Check the library description to ensure that each test_cell has correctly specified the test_scan_in signal type.

---

## LIBG-152

(error) The test_scan_in_inverted signal type on the '%s' port for\n \ttest_cell on the '%s' cell is not specified on the correct pin.

**Description**

The test_scan_in_inverted signal type for this test_cell is not specified on the correct pin. This signal type needs to be specified on the pin which takes in the scan data. All scan types, and all test_cells need to have either the test_scan_in or test_scan_in_inverted signal type.

This warning informs you that the test_cell pointed to has the test_scan_in_inverted signal type specified on an incorrect pin. Thus, the test_cell is not valid, and the entire cell is converted to a black box. There is functional information missing, and other tools will not be able to use this cell.

The following example shows an instance where this message occurs:

```
cell(libg152) {
    area : 13;
    pin(D) {
```

Feedback

```
    direction : input;
    capacitance : 1;
  }
  pin(C) {
    direction : input;
    capacitance : 2;
  }
  pin(D2) {
    direction : input;
    capacitance : 1;
  }
  pin(C2) {
    direction : input;
    capacitance : 2;
  }
  pin(C3) {
    direction : input;
    capacitance : 1;
  }
  pin(MQ) {
    direction : internal;
    internal_node : "MQ";
    timing() {
      timing_type : rising_edge;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "C C2";
    }
  }
  pin(Q) {
    direction : output;
    internal_node : "SQ";
    timing() {
      timing_type : rising_edge;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "C C3";
    }
  }
  pin(QN) {
    direction : output;
    state_function : "(Q)'";
    timing() {
      timing_type : rising_edge;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "C C3";
```

Feedback

```
    }
  }
  statetable("C D C2 C3 D2", "MQ SQ") {
    table :  "R  H/L L L  -  : -   - : H/L H/L,\
             ~R   -  - H  -  : H/L - :  -  H/L,\
             ~R   -  - L  -  : -   - :  -  N,\
              R   -  H -  -  : -   - :  X  X,\
              R   -  - H  -  : -   - :  X  X,\
             ~R   -  H - H/L : -   - : H/L -,\
             ~R   -  L -  -  : -   - :  N  -";
  }
  test_cell() {
    pin(D) {
      direction : input;
    }
    pin(C) {
      direction : input;
      capacitance : 2;
      signal_type : "test_scan_in_inverted";
    }
    pin(D2) {
      direction : input;
    }
    pin(C2) {
      direction : input;
      capacitance : 2;
      signal_type : "test_scan_clock_a";
    }
    pin(C3) {
      direction : input;
      capacitance : 1;
      signal_type : "test_scan_clock_b";
    }
    ff ("IQ","IQN") {
      next_state : "D";
      clocked_on : "C";
    }
    pin(Q) {
      direction : output;
      function : "IQ";
      signal_type : "test_scan_out";
    }
    pin(QN) {
      direction : output;
      function : "IQN";
      signal_type : "test_scan_out_inverted";
    }
  }
}
```

In this case the pin 'C' is a clock and should not have a test_scan_in_inverted signal_type attribute. The test_scan_in_inverted should be on the pin 'D2'.

Feedback

The following is an example message:

```
Error: Line 181, The test_scan_in_inverted signal type on the 'C' port
 for
       test_cell on the 'libg152' cell is not specified on the correct
 pin. (LIBG-152)
```

### What Next

Check the library description to ensure that each test_cell has correctly specified the *test_scan_in_inverted* signal type.

---

## LIBG-153

(error) The test_scan_enable signal type on the '%s' port for\n \ttest_cell on the '%s' cell is not specified on the correct pin.

### Description

The test_scan_enable signal type for this test_cell is not specified on the correct pin. This signal type needs to be specified on the pin that switches the input mode of the cell between test and mission mode. Only, a multiplexed flip-flop requires this signal type. This signal type can also be used in clocked scan cells.

This warning informs you that the test_cell pointed to has the test_scan_enable signal type specified on an incorrect pin. Thus, the test_cell is not valid, and the entire cell is converted to a black box. There is functional information missing, and other tools cannot use this cell.

The following example shows an instance where this message occurs:

```
cell(libg153) {
    area : 13;
    pin(D) {
      direction : input;
      capacitance : 1;
    }
    pin(C) {
      direction : input;
      capacitance : 2;
    }
    pin(D2) {
      direction : input;
      capacitance : 1;
    }
    pin(C2) {
      direction : input;
      capacitance : 2;
    }
    pin(C3) {
      direction : input;
```

Feedback

```
    capacitance : 1;
  }
  pin(MQ) {
    direction : internal;
    internal_node : "MQ";
    timing() {
      timing_type : rising_edge;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "C C2";
    }
  }
  pin(Q) {
    direction : output;
    internal_node : "SQ";
    timing() {
      timing_type : rising_edge;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "C C3";
    }
  }
  pin(QN) {
    direction : output;
    state_function : "(Q)'";
    timing() {
      timing_type : rising_edge;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "C C3";
    }
  }
  statetable("C D C2 C3 D2", "MQ SQ") {
    table :  "R  H/L L L  -  : -   - : H/L H/L,\
             ~R   -  - H  -  : H/L - :  -  H/L,\
             ~R   -  - L  -  : -   - :  -  N,\
              R   -  H -  -  : -   - :  X  X,\
              R   -  - H  -  : -   - :  X  X,\
             ~R   -  H - H/L : -   - : H/L -,\
             ~R   -  L -  -  : -   - :  N  -";
  }
  test_cell() {
    pin(D) {
      direction : input;
    }
    pin(C) {
      direction : input;
```

Feedback

```
        capacitance : 2;
        signal_type : "test_scan_enable";
      }
      pin(D2) {
        direction : input;
      }
      pin(C2) {
        direction : input;
        capacitance : 2;
        signal_type : "test_scan_clock_a";
      }
      pin(C3) {
        direction : input;
        capacitance : 1;
        signal_type : "test_scan_clock_b";
      }
      ff ("IQ","IQN") {
        next_state : "D";
        clocked_on : "C";
      }
      pin(Q) {
        direction : output;
        function : "IQ";
        signal_type : "test_scan_out";
      }
      pin(QN) {
        direction : output;
        function : "IQN";
        signal_type : "test_scan_out_inverted";
      }
    }
  }
```

The pin 'C' is a clock and should not have a test_scan_enable signal_type attribute. The test_scan_enable should be on the pin 'D2'.

The following is an example message:

```
Error: Line 181, The test_scan_enable signal type on the 'C' port for
        test_cell on the 'libg153' cell is not specified on the correct
 pin. (LIBG-153)
```

**What Next**

Check the library description to ensure that each test_cell has correctly specified the *test_scan_enable* signal type.

---

## LIBG-154

(error) The test_scan_enable_inverted signal type on the '%s' port for\n \ttest_cell on the '%s' cell is not specified on the correct pin.

Feedback

### Description

The test_scan_enable_inverted signal type for this test_cell is not specified on the correct pin. This signal type needs to be specified on the pin that switches the input mode of the cell between test and mission mode. Only a multiplexed flip-flop requires this signal type. This signal type can also be used in clocked scan cells.

This warning informs you that the test_cell pointed to has the test_scan_enable_inverted signal type specified on an incorrect pin. Thus, the test_cell is not valid, and the entire cell is converted to a black box. There is functional information missing, and other tools cannot use this cell.

The following example shows an instance where this message occurs:

```
cell(libg154) {
    area : 13;
    pin(D) {
      direction : input;
      capacitance : 1;
    }
    pin(C) {
      direction : input;
      capacitance : 2;
    }
    pin(D2) {
      direction : input;
      capacitance : 1;
    }
    pin(C2) {
      direction : input;
      capacitance : 2;
    }
    pin(C3) {
      direction : input;
      capacitance : 1;
    }
    pin(MQ) {
      direction : internal;
      internal_node : "MQ";
      timing() {
        timing_type : rising_edge;
        intrinsic_rise : 1.0;
        intrinsic_fall : 1.0;
        rise_resistance : 0.1;
        fall_resistance : 0.1;
        related_pin : "C C2";
      }
    }
    pin(Q) {
      direction : output;
      internal_node : "SQ";
      timing() {
```

Feedback

```
    timing_type : rising_edge;
    intrinsic_rise : 1.0;
    intrinsic_fall : 1.0;
    rise_resistance : 0.1;
    fall_resistance : 0.1;
    related_pin : "C C3";
  }
}
pin(QN) {
  direction : output;
  state_function : "(Q)'";
  timing() {
    timing_type : rising_edge;
    intrinsic_rise : 1.0;
    intrinsic_fall : 1.0;
    rise_resistance : 0.1;
    fall_resistance : 0.1;
    related_pin : "C C3";
  }
}
statetable("C D C2 C3 D2", "MQ SQ") {
  table :  "R  H/L L L  -  : -   - : H/L H/L,\
           ~R   -  - H  -  : H/L - :  -  H/L,\
           ~R   -  - L  -  : -   - :  -  N,\
            R   -  H -  -  : -   - :  X  X,\
            R   -  - H  -  : -   - :  X  X,\
           ~R   -  H - H/L : -   - : H/L -,\
           ~R   -  L -  -  : -   - :  N  -";
}
test_cell() {
  pin(D) {
    direction : input;
  }
  pin(C) {
    direction : input;
    capacitance : 2;
    signal_type : "test_scan_enable_inverted";
  }
  pin(D2) {
    direction : input;
  }
  pin(C2) {
    direction : input;
    capacitance : 2;
    signal_type : "test_scan_clock_a";
  }
  pin(C3) {
    direction : input;
    capacitance : 1;
    signal_type : "test_scan_clock_b";
  }
  ff ("IQ","IQN") {
    next_state : "D";
```

Feedback

```
        clocked_on : "C";
      }
      pin(Q) {
        direction : output;
        function : "IQ";
        signal_type : "test_scan_out";
      }
      pin(QN) {
        direction : output;
        function : "IQN";
        signal_type : "test_scan_out_inverted";
      }
    }
  }
```

The pin 'C' is a clock and should not have a test_scan_enable_inverted signal_type attribute. The test_scan_enable_inverted should be on the pin 'D2.'

The following is an example message:

```
Error: Line 181, The test_scan_enable_inverted signal type on the 'C'
 port for
        test_cell on the  'libg154'cell is not specified on the correct
 pin. (LIBG-154)
```

**What Next**

Check the library description to ensure that each test_cell has correctly specified the *test_scan_enable_inverted* signal type.

---

## LIBG-155

(error) The test_scan_clock signal type on the '%s' port for\n \ttest_cell on the '%s' cell is not specified on the correct pin.

**Description**

The test_scan_clock signal type for this test_cell is not specified on the correct pin. This signal type needs to be specified on the scan clock for clocked scan cells.

This warning informs you that the test_cell pointed to has the test_scan_clock signal type specified on an incorrect pin. Thus, the test_cell is not valid, and the entire cell is converted to a black box. There is functional information missing, and other tools cannot use this cell.

The following example shows an instance where this message occurs:

```
cell(libg155) {
    area : 13;
    pin(D) {
      direction : input;
```

Feedback

```
    capacitance : 1;
  }
  pin(C) {
    direction : input;
    capacitance : 2;
  }
  pin(D2) {
    direction : input;
    capacitance : 1;
  }
  pin(C2) {
    direction : input;
    capacitance : 2;
  }
  pin(C3) {
    direction : input;
    capacitance : 1;
  }
  pin(MQ) {
    direction : internal;
    internal_node : "MQ";
    timing() {
      timing_type : rising_edge;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "C C2";
    }
  }
  pin(Q) {
    direction : output;
    internal_node : "SQ";
    timing() {
      timing_type : rising_edge;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "C C3";
    }
  }
  pin(QN) {
    direction : output;
    state_function : "(Q)'";
    timing() {
      timing_type : rising_edge;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "C C3";
    }
```

Feedback

```
    }
    statetable("C D C2 C3 D2", "MQ SQ") {
      table :  "R  H/L L L  -  : -   - : H/L H/L,\
               ~R   -  - H  -  : H/L - :  -  H/L,\
               ~R   -  - L  -  : -   - :  -  N,\
                R   -  H -  -  : -   - :  X  X,\
                R   -  - H  -  : -   - :  X  X,\
               ~R   -  H - H/L : -   - : H/L -,\
               ~R   -  L -  -  : -   - :  N  -";
    }
    test_cell() {
      pin(D) {
        direction : input;
      }
      pin(C) {
        direction : input;
        capacitance : 2;
      }
      pin(D2) {
        direction : input;
        signal_type : "test_scan_in";

      }
      pin(C2) {
        direction : input;
        capacitance : 2;
        signal_type : "test_scan_clock_a";
      }
      pin(C3) {
        direction : input;
        capacitance : 1;
        signal_type : "test_scan_clock";
      }
      ff ("IQ","IQN") {
        next_state : "D";
        clocked_on : "C";
      }
      pin(Q) {
        direction : output;
        function : "IQ";
        signal_type : "test_scan_out";
      }
      pin(QN) {
        direction : output;
        function : "IQN";
        signal_type : "test_scan_out_inverted";
      }
    }
  }
}
```

The pin 'C3' is not a clock and should not have a test_scan_clock signal_type attribute.

Feedback

The following is an example message:

```
Error: Line 195, The test_scan_clock signal type on the 'C3' port for
        test_cell on the 'libg155' cell is not specified on the correct
 pin. (LIBG-155)
```

**What Next**

Check the library description to ensure that each test_cell has correctly specified the *test_scan_clock* signal type.

---

## LIBG-156

(error) The test_scan_clock_a signal type on the '%s' port for\n \ttest_cell on the '%s' cell is not specified on the correct pin.

**Description**

The test_scan_clock_a signal type for this test_cell is not specified on the correct pin. This signal type needs to be specified on the clock of the master stage of an LSSD configuration.

This warning informs you that the test_cell pointed to has the test_scan_clock_a signal type specified on an incorrect pin. Thus, the test_cell is not valid, and the entire cell is converted to a black box. There is functional information missing, and other tools cannot use this cell.

The following example shows an instance where this message occurs:

```
cell(libg156) {
    area : 13;
    pin(D) {
      direction : input;
      capacitance : 1;
    }
    pin(C) {
      direction : input;
      capacitance : 2;
    }
    pin(D2) {
      direction : input;
      capacitance : 1;
    }
    pin(C2) {
      direction : input;
      capacitance : 2;
    }
    pin(C3) {
      direction : input;
      capacitance : 1;
    }
```

Feedback

```
pin(MQ) {
  direction : internal;
  internal_node : "MQ";
  timing() {
    timing_type : rising_edge;
    intrinsic_rise : 1.0;
    intrinsic_fall : 1.0;
    rise_resistance : 0.1;
    fall_resistance : 0.1;
    related_pin : "C C2";
  }
}
pin(Q) {
  direction : output;
  internal_node : "SQ";
  timing() {
    timing_type : rising_edge;
    intrinsic_rise : 1.0;
    intrinsic_fall : 1.0;
    rise_resistance : 0.1;
    fall_resistance : 0.1;
    related_pin : "C C3";
  }
}
pin(QN) {
  direction : output;
  state_function : "(Q)'";
  timing() {
    timing_type : rising_edge;
    intrinsic_rise : 1.0;
    intrinsic_fall : 1.0;
    rise_resistance : 0.1;
    fall_resistance : 0.1;
    related_pin : "C C3";
  }
}
statetable("C D C2 C3 D2", "MQ SQ") {
  table :  "R  H/L L L  -  : -   - : H/L H/L,\
           ~R   -  - H  -  : H/L - :  -  H/L,\
           ~R   -  - L  -  : -   - :  -  N,\
            R   -  H -  -  : -   - :  X  X,\
            R   -  - H  -  : -   - :  X  X,\
           ~R   -  H - H/L : -   - : H/L -,\
           ~R   -  L -  -  : -   - :  N  -";
}
test_cell() {
  pin(D) {
    direction : input;
  }
  pin(C) {
    direction : input;
    capacitance : 2;
    signal_type : "test_scan_clock_a";
```

Feedback

```
      }
      pin(D2) {
        direction : input;
      }
      pin(C2) {
        direction : input;
        capacitance : 2;
        signal_type : "test_scan_in";
      }
      pin(C3) {
        direction : input;
        capacitance : 1;
        signal_type : "test_scan_clock_b";
      }
      ff ("IQ","IQN") {
        next_state : "D";
        clocked_on : "C";
      }
      pin(Q) {
        direction : output;
        function : "IQ";
        signal_type : "test_scan_out";
      }
      pin(QN) {
        direction : output;
        function : "IQN";
        signal_type : "test_scan_out_inverted";
      }
    }
  }
```

The pin 'C' is a clock and should not have a test_scan_clock_a signal_type attribute.

The following is an example message:

```
Error: Line 214, The test_scan_clock_a signal type on the 'C' port for
        test_cell on the 'libg156' cell is not specified on the correct
 pin. (LIBG-156)
```

**What Next**

Check the library description to ensure that each test_cell has correctly specified the *test_scan_clock_a* signal type.

---

## LIBG-157

(error) The test_clock signal type on the '%s' port for\n \ttest_cell on the '%s' cell is not specified correctly.

Feedback

### Description

The test_clock signal type for this test_cell is not specified on the correct pin. This signal type needs to be specified on the clock used for ATPG capture in auxiliary clock lssd cells.

This warning informs you that the test_cell pointed to has the test_clock signal type specified on an incorrect pin. Thus, the test_cell is not valid, and the entire cell is converted to a black box. There is functional information missing, and other tools cannot use this cell.

The following example shows an instance where this message occurs:

```
cell(libg157) {
    area : 13;
    pin(D) {
      direction : input;
      capacitance : 1;
    }
    pin(C) {
      direction : input;
      capacitance : 2;
    }
    pin(D2) {
      direction : input;
      capacitance : 1;
    }
    pin(C2) {
      direction : input;
      capacitance : 2;
    }
    pin(C3) {
      direction : input;
      capacitance : 1;
    }
    pin(MQ) {
      direction : internal;
      internal_node : "MQ";
      timing() {
        timing_type : rising_edge;
        intrinsic_rise : 1.0;
        intrinsic_fall : 1.0;
        rise_resistance : 0.1;
        fall_resistance : 0.1;
        related_pin : "C C2";
      }
    }
    pin(Q) {
      direction : output;
      internal_node : "SQ";
      timing() {
        timing_type : rising_edge;
        intrinsic_rise : 1.0;
```

Feedback

```
    intrinsic_fall : 1.0;
    rise_resistance : 0.1;
    fall_resistance : 0.1;
    related_pin : "C C3";
  }
}
pin(QN) {
  direction : output;
  state_function : "(Q)'";
  timing() {
    timing_type : rising_edge;
    intrinsic_rise : 1.0;
    intrinsic_fall : 1.0;
    rise_resistance : 0.1;
    fall_resistance : 0.1;
    related_pin : "C C3";
  }
}
statetable("C D C2 C3 D2", "MQ SQ") {
  table :  "R  H/L L L  -  : -   - : H/L H/L,\
           ~R   -  - H  -  : H/L - :  -  H/L,\
           ~R   -  - L  -  : -   - :  -  N,\
            R   -  H -  -  : -   - :  X  X,\
            R   -  - H  -  : -   - :  X  X,\
           ~R   -  H - H/L : -   - : H/L -,\
           ~R   -  L -  -  : -   - :  N  -";
}
test_cell() {
  pin(D) {
    direction : input;
  }
  pin(C) {
    direction : input;
    capacitance : 2;
    signal_type : "test_scan_clock_a";
  }
  pin(D2) {
    direction : input;
  }
  pin(C2) {
    direction : input;
    capacitance : 2;
    signal_type : "test_clock";
  }
  pin(C3) {
    direction : input;
    capacitance : 1;
    signal_type : "test_scan_clock_b";
  }
  ff ("IQ","IQN") {
    next_state : "D";
    clocked_on : "C";
  }
```

Feedback

```
      pin(Q) {
        direction : output;
        function : "IQ";
        signal_type : "test_scan_out";
      }
      pin(QN) {
        direction : output;
        function : "IQN";
        signal_type : "test_scan_out_inverted";
      }
    }
  }
```

The pin 'C2' is not a clock and should not have a test_clock signal_type attribute.

The following is an example message:

```
Error: Line 233, The test_clock signal type on the 'C2' port for
        test_cell on the 'libg157' cell is not specified correctly.
 (LIBG-157)
```

### What Next

Check the library description to ensure that each test_cell has correctly specified the *test_clock* signal type.

---

## LIBG-158

(error) Invalid nontest mode for test_cell on the '%s' cell.\n \tThere are more than 2 sequential elements.

### Description

The nontest mode for test_cells must be one of the following:

```
- A single flip-flop
        - A single latch
        - A single master-slave latch configuration using clocked_on and
                clocked_on_also
```

This warning informs you that the nontest mode of the test cell has more than 2 distinct sequential elements. Thus, it does not fit the previous requirements. The test_cell is not valid, and the entire cell is converted to a black box. There is functional information missing, and other tools cannot use this cell.

Feedback

### What Next

Check the library description to ensure that each test_cell has properly specified the correct nontest mode. To determine the multiple sequential elements in this description,

```
* Create a cell with the nontest mode as the functional part
          of the cell.
        * Compile this cell.
        * Issue the following command:

             report_lib -table <library name> <cell name>
```

In the report_lib command's report, determine which pins correspond to states (pins with an N output) by looking at their functional description.

---

## LIBG-159

(error) Invalid nontest mode for test_cell on the '%s' cell. \n \tThis describes a ganged cell.

### Description

The nontest mode for test_cells must be one of the following:

```
- A single flip-flop
        - A single latch
        - A single master-slave latch configuration using clocked_on and
                clocked_on_also
```

This warning informs you that the nontest mode of the test cell is a ganged cell. Thus, it does not fit the previous requirements. The test_cell is not valid, and the entire cell is converted to a black box. There is functional information missing, and other tools cannot use this cell.

### What Next

Check the library description to ensure that each test_cell does not specify a ganged cell as its nontest mode. Ganged cells can be recognized by the keywords ff_bank or latch_bank.

---

## LIBG-160

(error) The nontest mode for test_cell on the '%s' cell is invalid.\n \tThe '%s' slave port does not directly feed the output.

### Description

The nontest mode for test_cells must be one of the following:

```
- A single flip-flop
        - A single latch
```

Feedback

```
        - A single master-slave latch configuration using clocked_on,
              clocked_on_also
```

This warning informs you that in nontest mode of the test cell, the slave does not directly feed the output. This means that this is not a master-slave latch configuration that DFT Compiler supports. Thus, the test_cell is not valid, and the entire cell is converted to a black box.

The following example shows an instance where this message occurs:

```
cell(libg160) {
      area : 10;
      bundle (D) {
        members(D1, D2);
        direction : input;
        capacitance : 1;
      }
      pin(G) {
        direction : input;
        capacitance : 1;
      }
      latch_bank ("IQ","IQN", 2) {
        data_in : "D";
        enable : "G";
      }
      bundle (Q) {
        members(Q1, Q2);
        direction : output;
        function : "IQ";
        timing() {
          timing_sense : positive_unate;
          intrinsic_rise : 1.0;
          intrinsic_fall : 1.0;
          rise_resistance : 0.1;
          fall_resistance : 0.1;
          related_pin : "D";
        }
        timing() {
          intrinsic_rise : 1.0;
          intrinsic_fall : 1.0;
          rise_resistance : 0.1;
          fall_resistance : 0.1;
          related_pin : "G";
        }
      }
      test_cell() {
        pin(D1) {
          direction : input;
        }
        pin(G) {
          direction : input;
        }
```

Feedback

```
        pin(D2) {
          direction : input;
          signal_type : "test_scan_in";
        }
        pin(Q1 Q2 ) {
          direction : output;
          function : "G D2 D1";
          signal_type : "test_scan_out";
        }
      }
    }
```

The following is an example message:

```
Error: Line 193, The nontest mode for test_cell on the 'libg160' cell is
 invalid.
        The 'Q1' slave port does not directly feed the output. (LIBG-160)
```

### What Next

Check the library description to ensure that the nontest mode of the test_cell is correctly specified. If you are using a master-slave as the non test mode, the only accepted description is the state description using

```
clocked_on : <master clock>
        clocked_on_also : <slave clock>
```

---

## LIBG-161

(error) The nontest mode for test_cell on the '%s' cell is invalid\n \tThe '%s' state pin is not a latch for a master-slave configuration.

### Description

The nontest mode for test_cells must be one of the following:

```
- A single flip-flop
        - A single latch
        - A single master-slave latch configuration using clocked_on and
                clocked_on_also
```

This warning informs you that the nontest mode of the test cell is an incorrect master-slave description. This stage of the master-slave configuration is not a latch, thus, it does not fit the previous requirements. The test cell is not valid, and the entire cell is converted to a black box. There is functional information missing, and other tools cannot use this cell.

The following example shows an instance where this message occurs:

```
cell(libg161) {
      area : 11;
      pin(D1 D2 D3) {
        direction : input;
```

Feedback

```
    capacitance : 1;
  }
  pin(CP) {
    direction : input;
    capacitance : 2;
  }
  pin(SCK1 SCK2) {
    direction : input;
    capacitance : 2;
  }
  pin(SI) {
    direction : input;
    capacitance : 1;
    timing() {
      timing_type : setup_falling;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      related_pin : "SCK2";
    }
    timing() {
      timing_type : hold_falling;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      related_pin : "SCK2";
    }
  }
  pin(Q) {
    direction : output;
    timing() {
      timing_type : rising_edge;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "CP SCK1";
    }
  }
  pin(Q2) {
    direction : output;
    timing() {
      timing_type : rising_edge;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "CP SCK1";
    }
  }
  pin(Q3) {
    direction : output;
    timing() {
      timing_type : rising_edge;
      intrinsic_rise : 1.0;
```

Feedback

```
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "CP SCK1";
    }
  }
  test_cell() {
    pin(D1) {
      direction : input;
    }
    pin(D2) {
      direction : input;
    }
    pin(D3) {
      direction : input;
    }
    pin(CP) {
      direction : input;
    }
    pin(SCK1) {
      direction : input;
      signal_type : "test_scan_clock_b";
    }
    pin(SCK2) {
      direction : input;
      signal_type : "test_scan_clock_a";
    }
    pin(SI) {
      direction : input;
      signal_type : "test_scan_in";
    }
    statetable("C D ", "Q") {
      table :  "R  H/L : - : H/L,\
               ~R   -  : - :  N";
    }
    pin(Q) {
      direction : output;
      internal_node : "Q";
      input_map : "CP D1";
      signal_type : "test_scan_out";
    }
    pin(Q2) {
      direction : output;
      internal_node : "Q";
      input_map : "CP D2";
      signal_type : "test_scan_out";
    }
    pin(Q3) {
      direction : output;
      internal_node : "Q";
      input_map : "CP D3";
      signal_type : "test_scan_out";
    }
```

Feedback

```
      }
    }
```

The following is an example message:

```
Error: Line 218, The nontest mode for test_cell on the 'libg161' cell is
 invalid.
        The 'Q2' state pin is not a latch for a master-slave
 configuration. (LIBG-161)
```

### What Next

Check the library description to ensure that the nontest mode of the test_cell is correctly specified. If you are using a master-slave as the nontest mode, the only accepted description is the state description using

```
clocked_on : <master clock>
        clocked_on_also : <slave clock>
```

---

## LIBG-162

(error) The test_cell on the '%s' cell cannot have more than one\n \ttest_scan_in or test_scan_in_inverted signal type. This happens on the \n \t'%s' port.

### Description

Each test_cell must have only one test_scan_in or test_scan_in_inverted signal type. This signal type is specified on the pin that is used to take scan data into the cell.

This warning informs you that the test_cell pointed to has multiple test_scan_in or test_scan_in_inverted signal types. The message points to the second port that has this signal type. Thus, the test cell is not valid, and the entire cell is converted to a black box. There is functional information missing, and other tools cannot use this cell.

The following example shows an instance where this message occurs:

```
cell(libg162) {
      area : 2;
      pin_opposite ( "Q", "NQ" );
      latch ( "QX" , "QY" ) {
        data_in   : " ((D*!SE) + (SI*SE)) ";
        enable  : " CK  ";
      }
      pin (D)    {
        direction : input;
        capacitance : 1;
      }
      pin (SI, SIN, SE) {
        direction : input;
        capacitance : 1;
      }
```

Feedback

```
      pin (CK)    {
        direction : input    ;
        capacitance :   2 ;
      }
      pin (Q)    {
        direction : output   ;
        function  : "QX" ;
      }
      pin (NQ)    {
        direction : output   ;
        function  : "QY" ;
      }
      test_cell() {
        latch ( "QX", "QY") {
          data_in : " D ";
          enable : " CK' ";
        }
        pin (D,CK) {
          direction : input;
        }
        pin (SI) {
          direction : input;
          signal_type : "test_scan_in";
        }
        pin (SIN) {
          direction : input ;
          signal_type : "test_scan_in_inverted";
        }
        pin (SE) {
          direction : input ;
          signal_type : "test_scan_enable";
        }
        pin (Q) {
          direction : output ;
          function : " QX ";
          signal_type : "test_scan_out";
        }
        pin (NQ) {
          direction : output ;
          function : " QY ";
          signal_type : "test_scan_out_inverted";
        }
      }
    }
```

The following is an example message:

```
Error: The test_cell on the 'libg162' cell cannot have more than one
       test_scan_in or test_scan_in_inverted signal type. This happens
 on
       the 'SIN' port. (LIBG-162)
```

Feedback

### What Next

Check the library description to ensure that each test_cell has only one test_scan_in or test_scan_in_inverted signal type.

---

## LIBG-163

(error) The '%s' port in test_cell on the '%s' cell has a\n \ttest_scan_out signal type on an inverted output.

### Description

The test_scan_out signal type can only be specified on a noninverted output. If the output is inverted, the proper signal type is test_scan_out_inverted. This signal type specifies the polarity of the port with respect to the state.

This warning informs you that the test_cell pointed to has the test_scan_out signal type specified on an inverted output. Thus, the test_cell is not valid, and the entire cell is converted to a black box. There is functional information missing, and other tools cannot use this cell.

The following example shows an instance where this message occurs:

```
cell(libg163) {
    area : 13;
    pin(D) {
      direction : input;
      capacitance : 1;
      timing() {
        timing_type : setup_falling;
        intrinsic_rise : 1.0;
        intrinsic_fall : 1.0;
        related_pin : "C";
      }
      timing() {
        timing_type : hold_falling;
        intrinsic_rise : 1.0;
        intrinsic_fall : 1.0;
        related_pin : "C";
      }
    }
    pin(C) {
      direction : input;
      capacitance : 2;
    }
    pin(SDI) {
      direction : input;
      capacitance : 1;
      timing() {
        timing_type : setup_rising;
        intrinsic_rise : 1.0;
```

Feedback

```
      intrinsic_fall : 1.0;
      related_pin : "C";
    }
    timing() {
      timing_type : hold_rising;
      intrinsic_rise : 0.0;
      intrinsic_fall : 0.0;
      related_pin : "C";
    }
  }
  pin(SE) {
    direction : input;
    capacitance : 2;
  }
  pin(TQ) {
    direction : output;
    timing() {
      timing_type : rising_edge;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "C";
    }
  }
  pin(SDO) {
    direction : output;
    timing() {
      timing_type : falling_edge;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "C";
    }
  }
  test_cell() {
    pin(D) {
      direction : input;
    }
    pin(C) {
      direction : input;
    }
    pin(SDI) {
      direction : input;
      signal_type : "test_scan_in";
    }
    pin(SE) {
      direction : input;
      capacitance : 2;
      signal_type : "test_scan_enable";
    }
```

Feedback

```
      statetable("C D SDI", "Q") {
        table : "H L/H H/L : - : L/H,\
                  L -  -  : - : N";
      }

      pin(TQ) {
        direction : output;
        internal_node : "Q";
      }
      pin(SDO) {
        state_function : "TQ";
        direction : output;
        signal_type : "test_scan_out";
      }
    }
  }
```

The following is an example message:

```
Error: Line 176, The 'SDO' port in test_cell on the 'libg163' cell has a
        test_scan_out signal type on an inverted output. (LIBG-163)
```

**What Next**

The default behavior of selecting the function over the signal_type to determine the inversion of the output is correct in most cases.

Check the library description to ensure that the test_scan_out signal type is only placed on noninverted outputs. To determine if this is an inverted or noninverted output

```
* Create a cell with the nontest mode as the functional part
          of the cell.
        * Compile this cell.
        * Issue the following command:

             report_lib -table <library name> <cell name>
```

In the report_lib command's report, determine which pins correspond to states (pins with an N output) by looking at their functional description.

---

## LIBG-164

(error) The '%s' port in test_cell on the '%s' cell has a\n \ttest_scan_out_inverted signal type on a noninverted output.

**Description**

The test_scan_out_inverted signal type can only be specified on an inverted output. If the output is noninverted, the proper signal type is test_scan_out. This signal type specifies the polarity of the port with respect to the state.

Feedback

This error informs you that the test_cell pointed to has the test_scan_out_inverted signal type specified on a noninverted output. Thus, the test cell is not valid, and the entire cell is converted to a black box. There is functional information missing, and other tools cannot use this cell.

The following example shows an instance where this message occurs:

```
cell(libg164) {
            area : 11;
            pin(D0) {
              direction : input;
              capacitance : 1;
              timing() {
                timing_type : setup_rising;
                intrinsic_rise : 1.0;
                intrinsic_fall : 1.0;
                related_pin : "CK1";
              }
              timing() {
                timing_type : hold_rising;
                intrinsic_rise : 1.0;
                intrinsic_fall : 1.0;
                related_pin : "CK1";
              }
            }

            pin(TI) {
              direction : input;
              capacitance : 1;
              timing() {
                timing_type : setup_rising;
                intrinsic_rise : 1.0;
                intrinsic_fall : 1.0;
                related_pin : "CK1";
              }
              timing() {
                timing_type : hold_rising;
                intrinsic_rise : 1.0;
                intrinsic_fall : 1.0;
                related_pin : "CK1";
              }
            }

            pin(TE) {
              direction : input;
              capacitance : 1;
              timing() {
                timing_type : setup_rising;
                intrinsic_rise : 1.0;
                intrinsic_fall : 1.0;
                related_pin : "CK1";
              }
```

Feedback

```
    timing() {
      timing_type : hold_rising;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      related_pin : "CK1";
    }
  }

  pin(CK1) {
    direction : input;
    capacitance : 1;
  }
  pin(Q) {
    direction : output;
    inverted_output : FALSE;
    timing() {
      timing_type : rising_edge;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "CK1";
    }
  }

  pin(QN) {
    direction : output;
    inverted_output : TRUE;
    timing() {
      timing_type : rising_edge;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "CK1";
    }
  }

  test_cell() {
    pin(D0) {
      direction : input;
    }
    pin(TE) {
      direction : input;
      signal_type : "test_scan_enable";
    }
    pin(TI) {
      direction : input;
      signal_type : "test_scan_in";
    }
    pin(CK1) {
      direction : input;
      capacitance : 1;
```

Feedback

```
            }

            statetable("D0 CK1", "Q") {
              table : "L/H F : - : L/H,\
                        -  ~F : - : N";
            }

            pin(Q) {
              direction : output;
              inverted_output : FALSE;
              internal_node : "Q";
              signal_type : "test_scan_out";
            }

            pin(QN) {
              direction : output;
              inverted_output : TRUE;
              state_function : "Q";
              signal_type : "test_scan_out_inverted";
          }
          }
        }
```

The following is an example message:

```
Error: Line 220, The 'QN' port in test_cell on the 'libg164' cell has a
        test_scan_out_inverted signal type on a noninverted output.
 (LIBG-164)
```

**What Next**

Check the library description to ensure that the test_scan_out_inverted signal type is only placed on inverted outputs.

To determine if this is an inverted or noninverted output

```
* Create a cell with the nontest mode as the functional part
          of the cell.
        * Compile this cell.
        * Issue the following command:

            report_lib -table <library name> <cell name>
```

In the report_lib command's report, an inverted output is a state only if it has N. Instead the output contains an inversion of the state pin.

---

## LIBG-165

(error) The test cell on the '%s' cell cannot have more than one\n \ttest_scan_enable or test_scan_enable_inverted signal type.\n \tThis happens on the '%s' port.

Feedback

### Description

Each test_cell must have only one test_scan_enable or test_scan_enable_inverted signal type. This signal type is specified on the pin that is used to select between scan and mission mode operation of the cell. Using the state table format, Library Compiler expects a single test_scan_enable or test_scan_enable_inverted signal type. Otherwise, Library Compiler generates a black box cell.

This warning informs you that the test_cell pointed to has multiple test_scan_enable or test_scan_enable_inverted signal types. The message points to the second port that has this signal type. Thus, the test cell is not valid, and the entire cell is converted to a black box. There is functional information missing, and other tools cannot use this cell.

The following example shows an instance where this message occurs:

```
cell(libg165) {
      area : 13;
      pin(D) {
        direction : input;
        capacitance : 1;
      }
      pin(G) {
        direction : input;
        capacitance : 2;
      }
      pin(D2) {
        direction : input;
        capacitance : 1;
      }
      pin(C2) {
        direction : input;
        capacitance : 2;
      }
      pin(SE) {
        direction : input;
        capacitance : 2;
      }
      pin(SEB) {
        direction : input;
        capacitance : 2;
      }
      pin(Q) {
        direction : output;
        timing() {
          timing_type : rising_edge;
          intrinsic_rise : 1.0;
          intrinsic_fall : 1.0;
          rise_resistance : 0.1;
          fall_resistance : 0.1;
          related_pin : "G C2";
        }
      }
```

Feedback

```
    pin(QN) {
      direction : output;
      timing() {
        timing_type : rising_edge;
        intrinsic_rise : 1.0;
        intrinsic_fall : 1.0;
        rise_resistance : 0.1;
        fall_resistance : 0.1;
        related_pin : "G C2";
      }
    }
    test_cell() {
      pin(D) {
        direction : input;
      }
      pin(G) {
        direction : input;
      }
      pin(D2) {
        direction : input;
        signal_type : "test_scan_in";
      }
      pin(C2) {
        direction : input;
        signal_type : "test_scan_clock";
      }
      pin(SE) {
        direction : input;
        capacitance : 2;
        signal_type : "test_scan_enable";
      }
      pin(SEB) {
        direction : input;
        capacitance : 2;
        signal_type : "test_scan_enable_inverted";
      }
      state ("IQ","IQN") {
        force_01 : "G D'";
        force_10 : "G D";
      }
      pin(Q) {
        direction : output;
        function : "IQ";
        signal_type : "test_scan_out";
      }
      pin(QN) {
        direction : output;
        function : "IQN";
        signal_type : "test_scan_out_inverted";
      }
    }
  }
```

Feedback

The following is an example message:

```
Error: The test cell on the 'libg165' cell cannot have more than one
        test_scan_enable or test_scan_enable_inverted signal type.
        This happens on the 'SEB' port. (LIBG-165)
```

**What Next**

Check the library description to ensure that each test_cell has only one test_scan_enable or test_scan_enable_inverted signal type. Otherwise, provide a full functional description of the non test_cell cell to bypass the multuple scan enable check.

---

## LIBG-166

(error) The '%s' test_scan_clock port in test_cell on the '%s' cell\n \tdoes not have an active sense.

**Description**

The test_scan_clock must have an active sense in the full cell description. The Library Compiler determines the active sense by checking the function of the full cell. The test_scan_clock is the scan clock in the clocked scan type.

Because the test_cell and the full cell have a discrepancy, neither the test cell nor the full cell are recognized. Both the test_cell and the entire cell are converted to a black box. There is functional information missing, and other tools cannot use this cell.

**What Next**

Check the library description to ensure that the functional descriptions in the test_cell and full cell are correct and that the timing arcs are consistent. To report on the active sense and functions of the full cell

```
* Create a cell with the non test mode as the functional part
          of the cell.
        * Compile this cell.
        * Issue the following command:

             report_lib -table <library name> <cell name>
```

---

## LIBG-167

(error) The '%s' test_scan_clock_a port in test_cell on the '%s' cell\n \tdoes not have an active sense.

Feedback

### Description

The test_scan_clock_a must have an active sense in the full cell description. The Library Compiler determines the active sense by checking the function of the full cell. The test_scan_clock_a is the scan clock for the master stage of the LSSD scan style.

Because the test_cell and the full cell have a discrepancy, neither the test cell nor the full cell are recognized. Both the test_cell and the entire cell are converted to a black box. There is functional information missing, and other tools cannot use this cell.

### What Next

Check the library description to ensure that the functional descriptions in the test_cell and full cell are correct and that the timing arcs are consistent. To report on the active sense and functions of the full cell

```
* Create a cell with the nontest mode as the functional part
          of the cell.
        * Compile this cell.
        * Issue the following command:

             report_lib -table <library name> <cell name>
```

---

## LIBG-168

(error) The '%s' test_scan_clock_b port in test_cell on the '%s' cell\n \tdoes not have an active sense.

### Description

The test_scan_clock_b must have an active sense in the full cell description. The active sense is determined by Library Compiler by checking the function of the full cell. The test_scan_clock_b is the scan clock for the slave stage of the LSSD scan style.

Because the test_cell and the full cell have a discrepancy, neither the test cell nor the full cell are recognized. Both the test_cell and the entire cell are converted to a black box. There is functional information missing, and other tools cannot use this cell.

### What Next

Check the library description to ensure that the functional descriptions in the test_cell and full cell are correct and that the timing arcs are consistent. To report on the active sense and functions of the full cell

```
* Create a cell with the nontest mode as the functional part
          of the cell.
        * Compile this cell.
        * Issue the following command:

             report_lib -table <library name> <cell name>
```

Feedback

## LIBG-169

(error) The '%s' test_clock port in test_cell on the '%s' cell does \n \tnot have an active sense.

### Description

The test_clock must have an active sense in the full cell description. Library Compiler determines the active sense by checking the function of the full cell. The test_clock is the capture clock used by ATPG in the auxiliary clocked lssd scan style.

Because the test_cell and the full cell have a discrepancy, neither the test cell nor the full cell are recognized. Both the test_cell and the entire cell are converted to a black box. There is functional information missing, and other tools cannot use this cell.

### What Next

Check the library description to ensure that the functional descriptions in the test_cell and full cell are correct and that the timing arcs are consistent. To report on the active sense and functions of the full cell

```
* Create a cell with the nontest mode as the functional part
          of the cell.
        * Compile this cell.
        * Issue the following command:

             report_lib -table <library name> <cell name>
```

## LIBG-170

(error) The '%s' port in test_cell for the '%s' cell has an invalid\n \tsignal type %s.

### Description

The valid signal_types in test_cells are

```
- test_scan_in
        - test_scan_in_inverted
        - test_scan_enable
        - test_scan_enable_inverted
        - test_scan_out
        - test_scan_out_inverted
        - test_scan_clock
        - test_scan_clock_a
        - test_scan_clock_b
        - test_clock
```

This warning informs you that the port in the test_cell pointed to has a signal_type other than the ones listed previously. Thus, the test_cell is not valid, and the entire cell is converted to a black box.

Feedback

### What Next

Check the test_cell pointed to and change the invalid signal_type.

---

## LIBG-171

(error) The LSSD test_cell for the '%s' cell has neither\n \ta test_scan_clock_b signal type nor a master-slave as its nontest mode.

### Description

LSSD scan styles must have the following test_cell characteristics:

```
clocked lssd
        - signal types : test_scan_in/test_scan_in_inverted,
 test_scan_out/
                test_scan_out_inverted, test_scan_clock_a,
 test_scan_clock_b
        - nontest mode: flip_flop
   auxiliary clock lssd
        - signal types : test_scan_in/test_scan_in_inverted,
 test_scan_out/
                test_scan_out_inverted, test_scan_clock_a,
 test_scan_clock_b,
                test_clock
        - nontest mode : flip_flop
   single-latch lssd
        - signal types : test_scan_in/test_scan_in_inverted,
 test_scan_out/
                test_scan_out_inverted, test_scan_clock_a,
 test_scan_clock_b
        - nontest mode: latch
   single-latch lssd
        - signal types : test_scan_in/test_scan_in_inverted,
 test_scan_out/
                test_scan_out_inverted, test_scan_clock_a,
                test_scan_clock_b (optional)
        - nontest mode: master-slave (clocked_on & clocked_on_also)
```

This error informs you that this cell is an lssd scan style because it has test_scan_in/ test_scan_in_inverted, test_scan_out/test_scan_out_inverted, and test_scan_clock_a signal types. However, it is invalid because it has neither a test_scan_clock_b signal type nor a master-slave as its nontest mode. Thus, the test cell is not valid, and the entire cell is converted to a black box. There is functional information missing, and other tools cannot use this cell.

The following example shows an instance where this message occurs:

```
cell(libg171) {
      area : 21;
             pin(D1) {
```

Feedback

```
  direction : input;
  capacitance : 1;
}
pin(C1) {
  direction : input;
  capacitance : 2;
}
pin(D2) {
  direction : input;
  capacitance : 1;
}
pin(C2) {
  direction : input;
  capacitance : 2;
}
pin(C3) {
  direction : input;
capacitance : 1;
}

pin(Q1) {
  direction : output;
  timing() {
  timing_type : rising_edge;
    intrinsic_rise : 1.0;
    intrinsic_fall : 0.0;
    rise_resistance : 0.1;
    fall_resistance : 0.1;
    related_pin : "C1";
  }
  timing() {
    timing_type : rising_edge;
    intrinsic_rise : 1.0;
    intrinsic_fall : 0.0;
    rise_resistance : 0.1;
    fall_resistance : 0.1;
    related_pin : "C2";
  }
}
pin(Q2) {
  direction : output;
  timing() {
    timing_type : rising_edge;
    intrinsic_rise : 1.0;
    intrinsic_fall : 1.0;
    rise_resistance : 0.1;
    fall_resistance : 0.1;
    related_pin : "C3";
  }
}
test_cell() {
  pin(D1) {
    direction : input;
```

Feedback

```
  }
  pin(C1) {
    direction : input;
  }
  pin(D2) {
    direction : input;
    signal_type : "test_scan_in";
  }
  pin(C2) {
    direction : input;
    signal_type : "test_scan_clock_a";
}
  pin(C3) {
    direction : input;
    signal_type : "test_scan_clock_b";
  }
  state("IQ","IQN") {
    force_10 : "C1 D1";
    force_01 : "C1 D1'";
  }
  pin(Q1) {
    direction : output;
  function : "IQ";
  }
  pin(Q2) {
    direction : output;
    signal_type : "test_scan_out";
  }
}
test_cell() {
  pin(D1) {
    direction : input;
  }
  pin(C1) {
    direction : input;
}
  pin(D2) {
    direction : input;
    signal_type : "test_scan_in";
  }
  pin(C2) {
    direction : input;
    signal_type : "test_scan_clock_a";
}
  pin(C3) {
    direction : input;
  }
  state("IQ","IQN") {
    force_10 : "C1 C3' D1";
    force_01 : "C1 C3' D1'";
  }
  pin(Q1) {
    direction : output;
```

Feedback

```
            }
              pin(Q2) {
                direction : output;
                function : "IQ";
                signal_type : "test_scan_out";
              }
            }
          }
```

The following is an example message:

```
Error: Line 229, The LSSD test_cell for the 'libg171' cell has neither
        a test_scan_clock_b signal type nor a master-slave as its nontest
 mode. (LIBG-171)
```

### What Next

Check the library description to ensure that each test_cell has correctly specified the nontest modes and signal types in the test_cell pointed to. If you do not understand why the nontest function is not one of the previous types,

```
* Create a cell with the nontest mode as the functional part
         of the cell.
        * Compile this cell.
        * Issue the following command:

            report_lib -table <library name> <cell name>
```

In the report_lib command's report, determine which pins correspond to states (pins with an N output) by looking at their functional description.

---

## LIBG-172

(error) The test_cell for the '%s' cell is not valid because\n \tit has missing or extra test signals.

### Description

The supported scan styles must have the following test_cell characteristics:

```
multiplexed flip-flop
        - signal types : test_scan_in/test_scan_in_inverted,
 test_scan_enable/
                test_scan_enable_inverted,
 test_scan_out/test_scan_out_inverted
        - nontest mode : flip-flop, master-slave latch pair, or latch
   clocked scan
        - signal types : test_scan_in/test_scan_in_inverted,
 test_scan_enable/
                test_scan_enable_inverted (optional), test_scan_clock,
                test_scan_out/test_scan_out_inverted
        - nontest modes : flip-flop or latch
```

Feedback

```
  clocked lssd
       - signal types : test_scan_in/test_scan_in_inverted,
test_scan_out/
                test_scan_out_inverted, test_scan_clock_a,
test_scan_clock_b
       - nontest mode: flip-flop
  auxiliary clock lssd
       - signal types : test_scan_in/test_scan_in_inverted,
test_scan_out/
                test_scan_out_inverted, test_scan_clock_a,
test_scan_clock_b,
                test_clock
       - nontest mode : flip-flop
  single-latch lssd
       - signal types : test_scan_in/test_scan_in_inverted,
test_scan_out/
                test_scan_out_inverted, test_scan_clock_a,
test_scan_clock_b
       - nontest mode: latch
  single-latch lssd
       - signal types : test_scan_in/test_scan_in_inverted,
test_scan_out/
                test_scan_out_inverted, test_scan_clock_a,
test_scan_clock_b
       - nontest mode: latch
  single-latch lssd
       - signal types : test_scan_in/test_scan_in_inverted,
test_scan_out/
                test_scan_out_inverted, test_scan_clock_a,
                test_scan_clock_b (optional)
       - nontest mode: master-slave (clocked_on & clocked_on_also)
```

This warning informs you that the other signal types do not correspond to one of the previously supported scan styles. Extra or missing signal types point to the fact that the test_cell is not valid, and the entire cell is a black box.

The following example shows an instance where this message occurs:

```
cell(libg172) {
      area : 13;
      pin(D) {
        direction : input;
        capacitance : 1;
      }
      pin(C) {
        direction : input;
        capacitance : 2;
      }
      pin(SDI) {
        direction : input;
        capacitance : 1;
      }
      pin(SDOB) {
```

Feedback

```
        direction : output;
      }
      test_cell() {
        pin(D) {
          direction : input;
        }
        pin(C) {
          direction : input;
        }
        pin(SDI) {
          direction : input;
          signal_type : "test_scan_in";
        }
        statetable("D C", "QN") {
          table : "L/H H : - : H/L,\
                  -   L : - : N";
        }

        pin(SDOB) {
          direction : output;
          inverted_output : TRUE;
          internal_node : "QN";
          signal_type : "test_scan_out_inverted";
        }
      }
    }
  }
```

The following is an example message:

```
Error: Line 160, The test_cell for the 'libg172' cell is not valid
 because
        it has missing or extra test signals. (LIBG-172)
```

**What Next**

Check the library description to ensure that each test_cell has correctly specified the nontest modes and signal types in the test_cell pointed to. If you do not understand why the nontest function is not one of the previous types,

```
* Create a cell with the nontest mode as the functional part
          of the cell.
        * Compile this cell.
        * Issue the following command:

             report_lib -table <library name> <cell name>
```

In the report_lib command's report, determine which pins correspond to states (pins with an N output) by looking at their functional description.

Feedback

## LIBG-173

(error) Cannot derive full function based on test_cell\n \ton the '%s' cell due to previous warnings or errors.

### Description

Library Compiler generates a functional description of the entire cell based on the information in the test_cell. However, if there are warnings or errors in the test_cell, a functional description of the entire cell cannot be generated.

This error informs you that the test_cell pointed to has previous errors or warnings. Thus, the test_cell is not valid, and the entire cell is a black box. There is functional information missing, and other tools cannot use this cell.

The following example shows an instance where this message occurs:

```
cell(libg173) {
            area : 11;
            pin(D0) {
              direction : input;
              capacitance : 1;
              timing() {
                timing_type : setup_rising;
                intrinsic_rise : 1.0;
                intrinsic_fall : 1.0;
                related_pin : "CK1";
              }
              timing() {
                timing_type : hold_rising;
                intrinsic_rise : 1.0;
                intrinsic_fall : 1.0;
                related_pin : "CK1";
              }
            }

            pin(TI) {
              direction : input;
              capacitance : 1;
              timing() {
                timing_type : setup_rising;
                intrinsic_rise : 1.0;
                intrinsic_fall : 1.0;
                related_pin : "CK1";
              }
              timing() {
                timing_type : hold_rising;
                intrinsic_rise : 1.0;
                intrinsic_fall : 1.0;
                related_pin : "CK1";
              }
            }
```

Feedback

```
pin(TE) {
  direction : input;
  capacitance : 1;
  timing() {
    timing_type : setup_rising;
    intrinsic_rise : 1.0;
    intrinsic_fall : 1.0;
    related_pin : "CK1";
  }
  timing() {
    timing_type : hold_rising;
    intrinsic_rise : 1.0;
    intrinsic_fall : 1.0;
    related_pin : "CK1";
  }
}

pin(CK1) {
  direction : input;
  capacitance : 1;
}
pin(Q) {
  direction : output;
  inverted_output : FALSE;
  timing() {
    timing_type : rising_edge;
    intrinsic_rise : 1.0;
    intrinsic_fall : 1.0;
    rise_resistance : 0.1;
    fall_resistance : 0.1;
    related_pin : "CK1";
  }
}

pin(QN) {
  direction : output;
  inverted_output : TRUE;
  timing() {
    timing_type : rising_edge;
    intrinsic_rise : 1.0;
    intrinsic_fall : 1.0;
    rise_resistance : 0.1;
    fall_resistance : 0.1;
    related_pin : "CK1";
  }
}

test_cell() {
  pin(D0) {
    direction : input;
  }
  pin(TE) {
```

Feedback

```
        direction : input;
        signal_type : "test_scan_enable";
      }
      pin(TI) {
        direction : input;
        signal_type : "test_scan_in";
      }
      pin(CK1) {
        direction : input;
        capacitance : 1;
      }

      statetable("D0 CK1", "Q") {
        table : "L/H F : - : L/H,\
                 -  ~F : - : N";
      }

      pin(Q) {
        direction : output;
        inverted_output : FALSE;
        internal_node : "Q";
        signal_type : "test_scan_out";
      }

      pin(QN) {
        direction : output;
        inverted_output : TRUE;
        state_function : "Q";
        signal_type : "test_scan_out_inverted";
    }
    }
  }
```

In this case, the test_cell pointed to has the test_scan_out_inverted signal type specified on a noninverted output 'QN' (LIBG-164). Thus, the test_cell is not valid, and the entire cell is converted to a black box.

The following is an example message:

```
Error: Line 189, Cannot derive full function based on test_cell
        on the 'libg173' cell due to previous warnings or errors.
 (LIBG-173)
```

**What Next**

Fix the previous problems in order to recognize both the test_cell and full cell functionality. You might have to add a state table description of the entire cell.

Feedback

## LIBG-174

(error) The '%s' clocked test cell is not valid\n \tbecause it has a master-slave configuration as its nontest mode.

### Description

A clocked test_cell has the following characteristics:

```
- signal types : test_scan_in/test_scan_in_inverted, test_scan_enable/
                 test_scan_enable_inverted (optional), test_scan_clock,
                 test_scan_out/test_scan_out_inverted
        - nontest modes : flip-flop or latch
```

This warning informs you that the signal types point to this cell as being a clocked scan test_cell, but it does not have a flip-flop or latch as its nontest mode; it is currently specified as a master-slave cell. Thus, the test cell is not valid, and the entire cell is converted to a black box. There is functional information missing, and other tools cannot use this cell.

The following example shows an instance where this message occurs:

```
cell(libg174) {
    area : 10;
    pin(D) {
      direction : input;
      capacitance : 1;
      timing() {
        timing_type : setup_falling;
        intrinsic_rise : 1.0;
        intrinsic_fall : 1.0;
        related_pin : "G";
      }
      timing() {
        timing_type : hold_falling;
        intrinsic_rise : 1.0;
        intrinsic_fall : 1.0;
        related_pin : "G";
      }
    }
    pin(G G2) {
      direction : input;
      capacitance : 1;
    }
    pin(D2) {
      direction : input;
      capacitance : 1;
      timing() {
        timing_type : setup_rising;
        intrinsic_rise : 1.0;
        intrinsic_fall : 1.0;
        related_pin : "C2";
```

Feedback

```
    }
    timing() {
      timing_type : hold_rising;
      intrinsic_rise : 0.0;
      intrinsic_fall : 0.0;
      related_pin : "C2";
    }
  }
  pin(C2) {
    direction : input;
    capacitance : 2;
  }
  pin(Q) {
    direction : output;
    timing() {
      timing_type : rising_edge;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "G C2";
    }
  }
  pin(QN) {
    direction : output;
    timing() {
      timing_type : rising_edge;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : " G2 G C2";
    }
  }
  test_cell() {
    pin(D) {
      direction : input;
    }
    pin(G G2) {
      direction : input;
    }
    pin(D2) {
      direction : input;
      signal_type : "test_scan_in";
    }
    pin(C2) {
      direction : input;
      signal_type : "test_scan_clock";
    }
    latch ("IQ","IQN") {
      data_in : "D";
      enable : "G";
      enable_also : "G2";
```

Feedback

```
      }
      pin(Q) {
        direction : output;
        function : "IQ";
        signal_type : "test_scan_out";
      }
      pin(QN) {
        direction : output;
        function : "IQN";
        signal_type : "test_scan_out_inverted";
      }
    }
  }
```

In this case, the cell is a master-slave with an unknown test cell type.

The following is an example message:

```
Error: Line 208, The 'libg174' clocked test cell is not valid
        because it has a master-slave configuration as its nontest mode.
 (LIBG-174)
fi
```

**What Next**

Check the library description to ensure that nontest modes and signal types are correctly pointed to by the test_cell. To determine why the nontest function is not one of the previous types,

```
* Create a cell with the nontest mode as the functional part
          of the cell.
        * Compile this cell.
        * Issue the following command:

             report_lib -table <library name> <cell name>
```

In the report_lib command's report, determine which pins correspond to states (pins with an N output) by looking at their functional description.

---

## LIBG-175

(error) The test cell for the '%s' auxiliary clocked lssd cell is not \n \tvalid because its nontest mode is not a flip-flop.

**Description**

An auxiliary clocked lssd test_cell has the following characteristics:

```
- signal types : test_scan_in/test_scan_in_inverted, test_scan_out/
                 test_scan_out_inverted, test_scan_clock_a,
 test_scan_clock_b,
```

Feedback

```
                test_clock
        - nontest mode : flip-flop
```

This warning informs you that the signal types point to this cell as being a auxiliary clocked lssd test_cell, but it does not have a flip-flop as its nontest mode. This makes the test_cell invalid and the entire cell is converted to a black box. There is functional information missing, and other tools cannot use this cell.

The following example shows an instance where this message occurs:

```
cell(libg103) {
      area : 12;
      pin(D) {
        direction : input;
        capacitance : 1;
      }
      pin(CK,IH, CK1) {
        direction : input;
        capacitance : 1;
      }
      pin(A,B) {
        direction : input;
        capacitance : 2;
      }
      pin(SI) {
        direction : input;
        capacitance : 1;
        timing() {
          timing_type : setup_falling;
          intrinsic_rise : 1.0;
          intrinsic_fall : 1.0;
          related_pin : "A";
        }
        timing() {
          timing_type : hold_falling;
          intrinsic_rise : 1.0;
          intrinsic_fall : 1.0;
          related_pin : "A";
        }
      }
      pin(Q) {
        direction : output;
        timing() {
          timing_type : rising_edge;
          intrinsic_rise : 1.0;
          intrinsic_fall : 1.0;
          rise_resistance : 0.1;
          fall_resistance : 0.1;
          related_pin : "CK IH CK1";
        }
        timing() {
          timing_type : falling_edge;
```

Feedback

```
        intrinsic_rise : 1.0;
        intrinsic_fall : 1.0;
        rise_resistance : 0.1;
        fall_resistance : 0.1;
        related_pin : "B";
      }
    }
    pin(XQ) {
      direction : output;
    }
    test_cell(){
      pin(D,CK,CK1){
        direction : input;
      }
      pin(IH ){
        direction : input;
        signal_type : "test_clock";
      }
      pin(SI){
        direction : input;
        signal_type : "test_scan_in";
      }
      pin(A){
        direction : input;
        signal_type : "test_scan_clock_a";
      }
      pin(B){
        direction : input;
        signal_type : "test_scan_clock_b";
      }
      ff("IQ","IQN"){
        clocked_on : "CK";
        next_state : "D";
        clocked_on_also : "CK1";
      }

      pin(Q){
        direction : output;
        function : "IQ";
        signal_type : "test_scan_out";
      }
      pin(XQ){
        direction : output;
        function : "IQN";
        signal_type : "test_scan_out_inverted";
      }
    }
  }
```

Feedback

The following is an example message:

```
Error: Line 196, The test cell for the 'libg175' auxiliary clocked lssd
 cell is not
        valid because its nontest mode is not a flip-flop. (LIBG-175)
```

**What Next**

Check the library description to ensure that nontest modes and signal types are correctly pointed to by the test_cell. To determine why the nontest function is not one of the previous types,

```
* Create a cell with the nontest mode as the functional part
          of the cell.
        * Compile this cell.
        * Issue the following command:

             report_lib -table <library name> <cell name>
```

In the report_lib command's report, determine which pins correspond to states (pins with an N output) by looking at their functional description.

---

## LIBG-176

(error) Neither the memory_read nor the memory write group\n \tis allowed on scalar ports.

**Description**

Either the memory_read group or the memory_write group is specified on a scalar port. These groups are only allowed on bus groups.

**What Next**

Remove the memory_read and memory_write group from all scalar ports.

---

## LIBG-177

(error) The '%s' cell is a ROM, and ROMs cannot have any\n \tmemory_write groups.

**Description**

The cell is defined as a ROM type in the memory group. ROM cells cannot have any memory_write groups because the memory block cannot be written to.

The following example shows an instance where this message occurs:

```
cell(libg177) {
            ...
            memory() {
              type : rom;
```

Feedback

```
            address_width : 10;
            word_width : 8;
          }
           ...
          bus (data_in) {
              bus_type   : "bus8"
              direction : input;
              capacitance : 1.46;
              fanout_load : 1.46;

              memory_write() {          /* Not allowed on ROM cells */
                address : ram_addr;
               enable : WR;
              }
          }
           ...
        }
```

The following is an example message:

```
Error: Line 22, The 'libg177' cell is a ROM, and ROMs cannot have any
       memory_write groups. (LIBG-177)
```

**What Next**

Either change the memory type to a ram type or remove the memory_write group.

---

## LIBG-178

(error) The '%s' cell is a RAM, and RAMs must have at\n \tleast one memory_read port and at least one memory_write port.

**Description**

The cell is defined as a RAM type in the cell memory group. RAM cells must have at least one memory_read port and at least one memory_write port.

The following example shows an instance where this message occurs:

```
cell (libg178) {

      area : 2300.0;
      memory() {
        type : ram;
        address_width : 10;
        word_width : 8;
      }

      bus (ram_addr) {
           bus_type  : "bus10"; /* defined in the library */
           direction : input;
           capacitance : 1.0;
```

Feedback

```
      }
      bus (data_in) {
            bus_type   : "bus8";  /* defined in the library */
            direction : input;
            capacitance : 1.0;
            memory_write() {
              address : ram_addr;
              enable : WR;
            }
      }
      pin (WR) {
        direction : input;
        capacitance : 1.0;
        clock : true;
      }

      bus(data_out){
        bus_type  : "bus8";
        direction : output;

     }
  }
```

In this case, the memory_read group is missing in the cell. Add the following group to the port 'data_out' :

```
memory_read() {
          address : ram_addr;
        }
```

The following is an example message:

```
Error: Line 22, The 'libg178' cell is a RAM, and RAMs must have at
        least one memory_read port and at least one memory_write port.
 (LIBG-178)
```

### What Next

You can implement either of the following suggestions:

```
(1) Change the memory type to rom and ensure only the memory_read
group is provided.
(2) If the model is for a RAM, make sure that both the memory_read and
memory_write groups are provided.
```

---

## LIBG-179

(error) The '%s' cell is a ROM, and ROMs must have at\n \tleast one memory_read group.

Feedback

### Description

The cell is defined as a ROM type in the cell memory group. ROM cells require at least one memory_read port.

The following example shows an instance where this message occurs:

```
cell (libg179) {

      area : 2300.0;
      memory() {
        type : rom;
        address_width : 10;
        word_width : 8;
       }

       bus (ram_addr) {
            bus_type  : "bus10"; /* defined in the library */
            direction : input;
            capacitance : 1.0;
     }
     bus (data_in) {
            bus_type   : "bus8"
            direction : input;
            capacitance : 1.0;
     }
     pin (WR) {
       direction : input;
       capacitance : 1.13;
       clock : true;
     }
     bus(data_out){
       bus_type  : "bus8";
       direction : output;
    }
 }
```

In this case, the memory_read group is missing in the cell. Add the following group to the port 'data_out' :

```
memory_read() {
         address : ram_addr;
       }
```

The following is an example message:

```
Error: Line 22, The 'libg179' cell is a ROM, and ROMs must have at
        least one memory_read group. (LIBG-179)
```

### What Next

Add the memory_read group to the data port.

Feedback

## LIBG-180

(error) The '%s' bus has a %s group, but the '%s' cell\n \thas no memory group.

### Description

The cell is missing a memory group. The bus has memory specific syntax, which is not allowed.

The following example shows an instance where this message occurs:

```
cell (libg179) {

      area : 2300.0;
      bus (ram_addr) {
           bus_type  : "bus10"; /* defined in the library */
           direction : input;
           capacitance : 1.0;
     }
     bus (data_in) {
           bus_type   : "bus8"
           direction : input;
           capacitance : 1.0;
           memory_write() {
             address : ram_addr;
             enable : WR;
           }
     }
     pin (WR) {
       direction : input;
       capacitance : 1.0;
       clock : true;
     }
     bus(data_out){
       bus_type  : "bus8";
       direction : output;

    }
 }
```

The following is an example message:

```
Error: Line 38, The 'data_in' bus has a memory_write group, but the
 'lib180' cell
        has no memory group. (LIBG-180)
```

### What Next

Either add the memory group to the cell or remove the memory_read/write groups from the ports.

Feedback

## LIBG-181

(error) In the '%s' cell, the '%s' data bus has a width of %d,\n \tbut the memory group has a word width of %d.

### Description

The size of the data bus does not match the size of the memory group specified in the cell memory group. They must be consistent.

The following example shows an instance where this message occurs:

```
cell (libg181) {
      area : 2300.0;
      memory() {
        type : ram;
        address_width : 8;
        word_width : 6;
      }

      bus (ram_addr) {
           bus_type  : "bus8"; /* defined in the library */
           direction : input;
           capacitance : 1.0;
     }
     bus (data_in) {
           bus_type   : "bus8"
           direction : input;
           capacitance : 1.0;

           memory_write() {
             address : ram_addr;
             enable : WR;
           }
     }
     pin (WR) {
       direction : input;
       capacitance : 1.0;
       clock : true;
     }

     bus(data_out){
       bus_type  : "bus8";
       direction : output;
       memory_read() {
         address : ram_addr;
       }
    }
  }
```

Feedback

In this case, the width of the bus data_out is 8 and the word width in the memory group is 6. Change either the bus_data's width to 6 or the memory group's word width to 8 to make them consistent.

The following is an example message:

```
Error: Line 54, In the 'libg181' cell, the 'data_out' data bus has a
 width of 8,
        but the memory group has a word width of 6. (LIBG-181)
```

**What Next**

Change the library to make the data bus width and the memory group word's width consistent.

---

## LIBG-182

(error) The '%s' memory_write data bus can not be an output.

**Description**

The bus is an input to the memory block because it has the memory_write group specified. Inputs to the memory block can not be an output type.

**What Next**

The direction of the bus must be changed to either input, inout, or internal.

---

## LIBG-183

(error) In the '%s' cell, the '%s' address bus is not defined.

**Description**

The address bus specified in the group is not defined in the cell.

The following example shows an instance where this message occurs:

```
cell (libg183) {
      area : 2300.0;

      memory() {
        type : ram;
        address_width : 8;
        word_width : 8;
      }
     bus (data_in) {
           bus_type   : "bus8" /* defined in the library */
           direction : output;
           capacitance : 1.0;
           memory_write() {
```

Feedback

```
            address : ram_addr;
            enable : WR;
          }
      }
      pin (WR) {
        direction : input;
        capacitance : 1.0;
        clock : true;
      }
      bus(data_out){
        bus_type  : "bus8";
        direction : output;
        memory_read() {
          address : ram_addr;
        }
     }
  }
```

In this case, define the ram_addr as a port in the cell as follows:

bus (ram_addr) { bus_type : "bus8"; /* defined in the library */ direction : input; capacitance : 1.0; }

The following is an example message:

```
Error: Line 58, In the 'libg183' cell, the 'ram_addr' address bus is not
 defined. (LIBG-183)
```

**What Next**

The address specification must refer to a bus defined in the cell.

---

## LIBG-184

(error) In the '%s' cell, the '%s' address bus has a width of %d,\n \tbut the memory group has an address width of %d.

**Description**

The size of the address bus does not match the address size of the memory group specified in the cell memory group. They must be consistent.

The following example shows an instance where this message occurs:

```
cell (libg184) {
      area : 2300.0;
      memory() {
        type : ram;
        address_width : 10;
        word_width : 8;
      }
```

Feedback

```
      bus (ram_addr) {
           bus_type  : "bus8"; /* defined in the library */
           direction : input;
           capacitance : 1.0;
     }
     bus (data_in) {
           bus_type   : "bus8"
           direction : input;
           capacitance : 1.0;

           memory_write() {
             address : ram_addr;
             enable : WR;
           }
     }
     pin (WR) {
       direction : input;
       capacitance : 1.0;
       clock : true;
     }

     bus(data_out){
       bus_type  : "bus8";
       direction : output;
       memory_read() {
         address : ram_addr;
       }
    }
  }
```

In this case, the width of the bus data_out is 8 and the address width in the memory group is 10. Either change the width of the ram_addr to 10 or change the memory group's address width to 8 to make them consistent.

The following is an example message:

```
Error: Line 58, In the 'libg184' cell, the 'ram_addr' address bus has a
 width of 8,
        but the memory group has an address width of 10. (LIBG-184)
```

**What Next**

Change the library to make the address bus width and the memory group address width consistent.

---

## LIBG-185

(error) The '%s' memory address can not be an output.

Feedback

### Description

The address bus is an input to the memory block. Inputs to the memory block can not be an output type.

### What Next

The direction of the bus must be changed to either input, inout, or internal.

---

## LIBG-186

(error) In the '%s' cell, the '%s' memory_read data bus\n \tcannot be an input.

### Description

The data bus is an output of the memory block because it has the memory_read group specified.

The following example shows an instance where this message occurs:

```
cell (libg186) {
      area : 2300.0;
      memory() {
        type : ram;
        address_width : 8;
        word_width : 8;
      }
      bus (ram_addr) {
           bus_type  : "bus8"; /* defined in the library */
           direction : output;
           capacitance : 1.0;
     }
     bus (data_in) {
           bus_type   : "bus8"
           direction : input;
           capacitance : 1.0;
           memory_write() {
             address : ram_addr;
             enable : WR;
           }
     }
     pin (WR) {
       direction : input;
       capacitance : 1.13;
       clock : true;
     }
     bus(data_out){
       bus_type  : "bus8";
       direction : input; /* should be output */
       memory_read() {
         address : ram_addr;
       }
```

Feedback

```
    }
  }
```

The following is an example message:

```
Error: Line 54, In the 'libg186' cell, the 'data_out' memory_read data
 bus
        cannot be an input. (LIBG-186)
```

**What Next**

Change the direction of the bus that has the memory_write group to either output, inout, or internal.

---

## LIBG-187

(error) In the '%s' cell, The %s '%s' attribute is not\n \tthe name of a port or bus.

**Description**

The attribute specified in the group is not defined in the cell as either a port or a bus.

You cannot specify an expression here, only a port or a bus is valid. If you need to model an expression(logic), create a pin with a state_function attribute and use that internal pin.

The following example shows an instance where this message occurs:

```
cell (libg187) {
      area : 1380.0;
      interface_timing : TRUE;

      memory() {
        type : ram;
        address_width : 10;
        word_width : 8;
      }

      bus (RA) {
        bus_type  : "bus10";
        direction : input;
        capacitance : 1.0;
      }

      bus (DI) {
        bus_type   : "bus8";
        direction : input;
        capacitance : 1.0;

        memory_write() {
          address : RA;
          enable : "WRE";
        }
```

Feedback

```
      }

      pin (WE) {
        direction : input;
        capacitance : 1.0;
        clock : true;
      }

      pin (CE) {
        direction : input;
        capacitance : 1.0;
        clock : true;
      }

      bus(DO){
        bus_type  : "bus8";
        direction : output;
        capacitance : 1.0;

        memory_read() {
          address : RA;
        }
      }
    }
```

In this case, the port 'WRE' can be defined :

```
1. as an internal pin
        2. is state_function defines the writing occurence only when
           WE and CE are active.

      pin (WRE) {
        direction : internal;
        capacitance : 1.0;
        state_function : "!(WE' * CE')";
        clock : true;
      }
```

The following is an example message:

```
Error: Line 43, In the 'libg187' cell, the enable 'WRE' attribute is not
        the name of a port or bus. (LIBG-187)
```

**What Next**

Make sure the specification refers to a port or a bus defined in the cell.

---

## LIBG-188

(error) In the '%s' cell, the %s '%s' bus has a width of %d,\n \tbut the memory group has a word width of %d.

Feedback

### Description

The size of the bus does not match the word size of the memory block specified in the cell memory group. They must be consistent.

The following example shows an instance where this message occurs:

```
cell (libg188) {
      area : 2300.0;
      memory() {
        type : ram;
        address_width : 10;
        word_width : 8;
       }

       bus (ram_addr) {
            bus_type  : "bus10"; /* defined in the library */
            direction : input;
            capacitance : 1.0;
     }
     bus (data_in) {
            bus_type   : "bus8"
            direction : input;
            capacitance : 1.0;

            memory_write() {
              address : ram_addr;
              enable : WR;
            }
     }
     bus (WR) {
       bus_type   : "bus10"
       direction : input;
       capacitance : 1.0;
       fanout_load : 1.0;
       clock : true;
     }
     bus(data_out){
       bus_type  : "bus8";
       direction : output;
       memory_read() {
         address : ram_addr;
       }
    }
 }
```

In this case, the bus type of the enable 'WR' is bus10. Change it to bus8.

The following is an example message:

```
Error: Line 42, In the 'libg188' cell, the enable 'WR' bus has a width of
 10,
        but the memory group has a word width of 8. (LIBG-188)
```

Feedback

### What Next

Change the library to make the bus width and the memory group word width consistent.

---

## LIBG-189

(error) In the '%s' cell, the memory_write group cannot\n \thave both the enable attribute and the clocked_on attribute.

### Description

The memory_write group has both the enable attribute and the clocked_on attribute specified. The Library Compiler does not allow it.

The following example shows an instance where this message occurs:

```
cell (libg189) {
      area : 2300.0;
      memory() {
        type : ram;
        address_width : 10;
        word_width : 8;
      }

      bus (ram_addr) {
           bus_type  : "bus10"; /* defined in the library */
           direction : input;
           capacitance : 1.0;
     }
     bus (data_in) {
           bus_type   : "bus8";
           direction : input;
           capacitance : 1.0;

           memory_write() {
             address : ram_addr;
             enable : WR;
             clocked_on : WR;
           }
     }
     pin (WR) {
       direction : input;
       capacitance : 1.0;
       clock : true;
     }

     bus(data_out){
       bus_type  : "bus8";
       direction : output;
       memory_read() {
         address : ram_addr;
       }
```

Feedback

```
    }
 }
```

For the bus 'data_in', you can either have the enable attribute or the clocked_on attribute defined in the memory_write, but not both.

The following is an example message:

```
Error: Line 40, In the 'libg189' cell, the memory_write group cannot
        have both the enable attribute and the clocked_on attribute.
 (LIBG-189)
```

### What Next

If the memory write process is edge triggered, use the clocked_on attribute. If the memory write process is level-sensitive triggered, use the enable attribute. You cannot specify both at once.

---

## LIBG-190

(error) In the '%s' cell, the v3.1 sequential syntax is not\n \tsupported on memory cells.

### Description

The cell has both the memory syntax and the v3.1 sequential syntax defined by either the *function* attribute or the *ff* and *latch* groups. The two syntaxes are not compatible and cannot both be specified on the same cell.

The following example shows an instance where this message occurs:

```
cell (libg190) {
      area : 2300.0;
      memory() {
        type : ram;
        address_width : 10;
        word_width : 8;
      }

      bus (ram_addr) {
           bus_type  : "bus10"; /* defined in the library */
           direction : input;
           capacitance : 1.0;
     }
     bus (data_in) {
           bus_type   : "bus8"
           direction : input;
           capacitance : 1.0;

           memory_write() {
             address : ram_addr;
             enable : WR;
```

Feedback

```
        }
      }
      pin (WR) {
        direction : input;
        capacitance : 1.0;
        clock : true;
      }

      bus(data_out){
        bus_type  : "bus8";
        direction : output;
        function : "WR + data_in";
        memory_read() {
          address : ram_addr;
        }
     }
  }
```

The function attribute is defined for the pin 'data_out'. This message is displayed for each element of the bus data_out that has the function attribute. To fix the problem, temove the function attribute from the pin.

The following is an example message:

```
Error: Line 60, In the 'libg190' cell, the v3.1 sequential syntax is not
        supported on memory cells. (LIBG-190)
```

**What Next**

Remove the v3.1 sequential syntax, and use the new state table syntax.

---

## LIBG-191

(error) The '%s' memory output cannot have a function.

**Description**

The memory output has an internal_node or a state_function attribute. A memory output has its function defined by the memory_read group and cannot also have an internal_node or a state_function attribute.

The following example shows an instance where this message occurs:

```
cell (libg191) {
      area : 2300.0;
      memory() {
        type : ram;
        address_width : 10;
        word_width : 8;
      }

      bus (ram_addr) {
```

Feedback

```
            bus_type  : "bus10"; /* defined in the library */
            direction : input;
            capacitance : 1.46;
        }
        bus (data_in) {
            bus_type   : "bus8"
            direction : input;
            capacitance : 1.46;

            memory_write() {
              address : ram_addr;
              enable : WR;
            }
        }
        pin (WR) {
          direction : input;
          capacitance : 1.13;
          clock : true;
        }

        bus(data_out){
          bus_type  : "bus8";
          direction : output;
          state_function : "WR + data_in"; /* not allowed */
          memory_read() {
            address : ram_addr;
          }
       }
    }
```

This message is displayed for each element of the bus data_out that has the state_function attribute.

The following is an example message:

```
Error: Line 60, The 'data_out[0]' memory output cannot have a function.
 (LIBG-191)
```

**What Next**

Remove the internal_node or state_function attribute.

---

## LIBG-192

(error) The '%s' port that has both memory_read and memory_write\n \tmust have the inout direction.

**Description**

Buses with both memory_read and memory_write groups must have the direction inout.

Feedback

The following example shows an instance where this message occurs:

```
cell (libg192) {
      area : 2300.0;
      memory() {
        type : ram;
        address_width : 10;
        word_width : 8;
      }

      bus (ram_addr) {
           bus_type  : "bus10"; /* defined in the library */
           direction : input;
           capacitance : 1.0;
     }
     bus (data_in) {
           bus_type   : "bus8"
           direction : input;
           capacitance : 1.0;

     }
     pin (WR) {
       direction : input;
       capacitance : 1.0;
       clock : true;
     }

     bus(data_out){
       bus_type  : "bus8";
       direction : output;      /* must be inout */
       memory_read() {
         address : ram_addr;
       }
       memory_write() {
          address : ram_addr;
          enable : WR;
       }
    }
 }
```

In this case, the direction of the bus 'data-out' is output. To fix the problem, set the direction to inout. This message is displayed for each element of the bus data_out.

The following is an example message:

```
Error: Line 62, The 'data_out[0]' port that has both memory_read and
 memory_write
        must have the inout direction. (LIBG-192)
```

**What Next**

Either change the direction of this bus to inout, or remove either the memory_read or memory_write group from this bus.

Feedback

## LIBG-193

(error) The '%s' port that has both memory_read and memory_write\n \tmust have a three_state expression.

### Description

Buses with both memory_read and memory_write groups must have the *three_state* attribute. During the memory write cycle, when the new data value is driven in from outside the cell, the port associated with the memory_read group must have a three_state attribute.

The following example shows an instance where this message occurs:

```
cell (libg193) {
      area : 2300.0;
      memory() {
        type : ram;
        address_width : 10;
        word_width : 8;
       }

      bus (ram_addr) {
           bus_type  : "bus10"; /* defined in the library */
           direction : input;
           capacitance : 1.0;
     }
     bus (data_in) {
           bus_type   : "bus8"
           direction : input;
           capacitance : 1.0;

     }
     pin (WR) {
       direction : input;
       capacitance : 1.0;
       clock : true;
     }

     bus(data_out){
       bus_type  : "bus8";
       direction : inout;       /* must have three_state */
       memory_read() {
         address : ram_addr;
       }
       memory_write() {
          address : ram_addr;
          enable : WR;
       }
    }
 }
```

Feedback

In this case, the bus 'data_out' has both memory_read and memory_write groups, but no three_state attribute. This message is displayed for each element of the bus data_out. To fix the problem, add the three_state attribute to the bus as follows:

```
pin (CE) {
       direction : input;
       capacitance : 1.0;
       clock : true;
     }
     bus(data_out){
       bus_type  : "bus8";
       direction : inout;
       three_state : "CE";       /* new statement */
       memory_read() {
         address : ram_addr;
       }
       memory_write() {
          address : ram_addr;
          enable : WR;
       }
    }
```

The following is an example message:

```
Error: Line 62, The 'data_out[0]' port that has both memory_read and
 memory_write
        must have a three_state expression. (LIBG-193)
```

**What Next**

Add a valid three_state expression to this bus, or remove either the memory_read or memory_write group from this bus.

---

## LIBG-194

(warning) The function attribute on the '%s' pin is ignored.

**Description**

This message indicates that the *function* attribute on the specified pin is ignored. If the pin is an internal pin, use *state_function* attribute. Internal pins are equal to internal nodes, which are single sequential elements. If the cell is a memory cell, it does not need a *function* attribute if the pin already has a *memory_read*, has a *state_function* attribute, or is an internal node.

The following example shows an instance where this message occurs:

```
cell(libg202) {
  area : 10;
  pin(D) {
    direction : input;
```

Feedback

```
    capacitance : 1;
  }
  pin(CP) {
    direction : input;
    capacitance : 1;
  }
  statetable ( "D CP", "Q QB") {
    table : "L/H R  : - - : L/H H/L,\
             -   ~R : - - : N   N";
  }

  pin(Q) {
    direction : output;
    inverted_output : FALSE;
    function : "Q";
    internal_node : "Q";
  }
  pin(QB) {
    direction : output;
    inverted_output : TRUE;
    internal_node : "QB";
  }
}
```

The following is an example message:

```
Warning: Line 127, the function attribute on the 'Q' pin is ignored.
 (LIBG-194)
```

**What Next**

Either change the internal pin to a output or inout port or remove the *function* attribute.

---

## LIBG-195

(warning) A bundle is not supported in the memory group.

**Description**

Currently, Library Compiler does not support memory cells with bundles.

The following example shows an instance where this message occurs:

```
cell (libg195) {
      area : 2300.0;
      memory() {
       type : ram;
       address_width : 10;
       word_width : 8;
      }

      bundle (ram_addr) {
```

Feedback

```
        members( D_1 D_2);
        direction : input;
        capacitance : 1.0;
      }
      bus (data_in) {
        bus_type   : "bus8"
        direction : input;
        capacitance : 1.0;
        memory_write() {
           address : ram_addr;
           enable : WR;
        }
      }
      pin (WR) {
        direction : input;
        capacitance : 1.0;
        clock : true;
      }
      bus(data_out){
        bus_type  : "bus8";
        direction : output;
        memory_read() {
          address : ram_addr;
        }
     }
  }
```

The following is an example message:

```
Warning: Line 94, A bundle is not supported in the memory group.
 (LIBG-195)
```

**What Next**

Modify the bundle into a bus.

---

## LIBG-196

(warning) The inverted_output attribute is not defined in the '%s' output port.

**Description**

Library Compiler uses the *inverted_output* attribute value to determine whether a sequential output pin is inverting or noninverting.

If this attribute is not defined. LC will make following assumptions: - For the output with "internal_node" == first output node in statetable, it is assumed to be noninverting; - For the output with "internal_node" == second output node in statetable, it is assumed to be inverting.

Feedback

The following example shows an instance where this message occurs:

```
cell(libg202) {
  area : 10;
  pin(D) {
    direction : input;
    capacitance : 1;
  }
  pin(CP) {
    direction : input;
    capacitance : 1;
  }
  statetable ( "D CP", "Q QB") {
    table : "L/H R  : - - : L/H H/L,\
             -   ~R : - - : N   N";
  }

  pin(Q) {
    direction : output;
/*    inverted_output : FALSE; */
    internal_node : "Q";
  }
  pin(QB) {
    direction : output;
    inverted_output : TRUE;
    internal_node : "QB";
  }
}
```

In this case, set the inverted_output value to FALSE in the port 'Q' to construct the correct logic for the Design Compiler. Without it, LC will make assumption that the port 'Q' is noninverting.

The following is an example message:

```
Warning: Line 127, The inverted_output attribute is not defined in the
 'Q' output port. (LIBG-196)
```

### What Next

Make sure each inout and output pin has its "inverted_output" attribute set.

---

## LIBG-197

(warning) In the '%s' cell, the '%s' and '%s'\n \tmemory read or memory write ports have the same address attribute.

### Description

This message indicates that the same address attribute is used for both of the specified memory read or memory write ports. Each port must have a unique address attribute.

Feedback

The following example shows an instance where this message occurs:

```
cell (libg197) {
      area : 2300.0;
      memory() {
       type : ram;
       address_width : 10;
       word_width : 8;
      }

      bus (ram_addr) {
       bus_type  : "bus10"; /* defined in the library */
       direction : input;
       capacitance : 1.0;
     }
     bus (data_in) {
       bus_type   : "bus8"
       direction : input;
       capacitance : 1.0;
       memory_write() {
          address : ram_addr;
          enable : WR;
       }
     }
     pin (WR) {
       direction : input;
       capacitance : 1.0;
       clock : true;
     }
     pin (EN) {
       direction : input;
       capacitance : 1.0;
     }

     bus(data_out){
       bus_type  : "bus8";
       direction : inout;
       three_state : "EN";
       memory_read() {
         address : ram_addr;
       }
       memory_write() {
          address : ram_addr;
          enable : WR;
       }
    }
 }
```

In this case, the ports 'data_in' and 'data_out' have the same address attribute for the memory_write group. This message is displayed for each element of the bus data_in and the bus data_out.

Feedback

The following is an example message:

```
Warning: Line 38, In the 'libg197' cell, the 'data_in[0]' and
 'data_out[0]'
        memory read or memory write ports have the same address
 attribute. (LIBG-197)
```

**What Next**

Edit the cell description so that the specified ports have unique address attributes.

---

## LIBG-198

(error) The '%s' pin of the '%s' design must be an '%s'.

**Description**

By definition, a pin in the *related_pin* attribute for timing constraints must be a input pin.

The following is an example message:

```
Error: Line 153, The 'P' pin of the 'libg198' design must be an 'input'.
 (LIBG-198)
```

**What Next**

Check your library for a wrong pin in the *related_pin* attribute or for an incorrect *timing_type*.

---

## LIBG-199

(error) The '%s' input port in the '%s' cell cannot have timing arcs.

**Description**

Input ports cannot have timing arcs. There is a possible problem in the direction of the pin, which should be output or inout.

The following is an example message:

```
Error: Line 53, The 'D' input port in the 'libg199' cell cannot have
 timing arcs. (LIBG-198)
```

**What Next**

Check your library for a pin with a wrong direction. Change the direction to output or inout.

Feedback

## LIBG-201

(error) The min_period and min_pulse_width attributes are not allowed on the '%s' port in the '%s' cell.

### Description

This message indicates that the specified port has a *min_period*, *min_pulse_width_high*, or *min_period_width_low* attribute. These attributes are allowed only on clock enable ports.

### What Next

Update the technology library source to remove the offending attributes.

## LIBG-202

(information) In the '%s' cell, the '%s' internal pin is\n \treplaced with the inverted '%s' output pin.

### Description

This message indicates that the specified internal pin is an inversion of the output pin. The internal pin is replaced by the inverted output pin. This deletion is done without affecting the functionality of the cell. An internal pin that is never visible is functionally useless. Every internal pin must have a unique functional relationship with an output port.

The following example shows an instance where this message occurs:

```
cell(libg202) {
  area : 10;
  pin(D) {
    direction : input;
    capacitance : 1;
  }
  pin(CP) {
    direction : input;
    capacitance : 1;
  }
  statetable ( "D CP", "Q QB") {
    table : "L/H R  : - - : L/H H/L,\
             -   ~R : - - : N   N";
  }

  pin(Q2) {
    direction : internal ;
    internal_node : "QB" ;
  }
  pin(Q) {
    direction : output;
```

Feedback

```
    internal_node : "Q";
  }
  pin(QB) {
    direction : output;
    state_function : "Q2'";
  }
}
```

In this case the internal node 'Q2' is an inversion of the output 'Q'.

The following is an example message:

```
Information: Line 125, In the 'libg202' cell, the 'Q2' internal pin is
        replaced with the inverted 'Q' output pin. (LIBG-202)
```

**What Next**

Either delete the internal pin or give it a unique functional relationship with an output port.

---

## LIBG-203

(warning) The '%s' test cell has no function.

**Description**

One of the test_cells has no functional description. Each test cell needs a functional description (flip-flop or latch description). This is used to determine the full functionality of the cell and is also used for the command insert_test.

When reading a .db library, Library Compiler creates a full function if there is any test_cell with a functional description. If no test_cells have a functional description and there is no full cell functional description, the cell is converted to a black box.

**What Next**

Modify the library to specify a functional description within the test_cell group. If a full cell has the function description instead, the cell is no longer a black box, but the *insert_test* command will not be able to use the functional description.

---

## LIBG-204

(error) The '%s' test cell has no function and the cell is black box.

**Description**

One of the test_cells has no functional description. Each test cell needs a functional description (ff or latch description). This is used to determine the full function of the cell and is also used for the command insert_test.

Feedback

This cell becomes a black box and read_lib cannot proceed with reading the library if this error is found.

**What Next**

Specify a functional description within the test_cell. If you specify a full cell description instead, the cell is no longer a black box but insert_test cannot use the functional description.

---

## LIBG-205

(warning) Cell(%s): The function cannot be recognized \n\tduring functional optimization by Design Compiler.

**Description**

This warning message informs you that Library Compiler encountered one of the following problems:

1. Library Compiler failed to map the state table description of the cell to the flip-flop or latch group description due to the complexity of the function.
2. Library Compiler failed to degenerate a multiple flip-flop or latch group to a single flip-flop or latch group due to the complexity of the function.

The function cannot be recognized by Design Compiler. This cell becomes a black box cell during synthesis. However, it might not be a black box cell to DFT Compiler or other tools, depending on its test_cell description, if there is one.

In the following example, the cell cannot be modeled in v3.1 format, so the warning message should be ignored:

```
cell(libg205) {
  area : 16;
  pin(D) {
    direction : input;
    capacitance : 1;
  }
  pin(CP) {
    direction : input;
    capacitance : 2;
  }
  pin (Q0) {
    direction : output;
    internal_node : "Q";
    input_map : "D CP";
    ...
  }
  pin (Q1) {
    direction : output;
```

Feedback

```
    internal_node : "Q";
    input_map : "Q0 CP";
    ...
  }
  pin (Q2) {
    direction : output;
    internal_node : "Q";
    input_map : "Q1 CP";
    ...
  }
  pin (Q3) {
    direction : output;
    internal_node : "Q";
    input_map : "Q2 CP";
    ...
  }
  statetable( "D CP", "Q QN" ) {
    table : "- ~R : - - : N N,\
             H/L R : - - : H/L L/H";
  }
}
```

This example generates the following warning message:

```
Warning: Line 21, Cell(libg205): The function cannot be
         recognized by Design Compiler. The cell becomes a black box.
         (LIBG-205)
```

In the following example, the cell cannot be simplified into one flip-flop description by giving disable values (NRESTORE = 1 and SAVE = 0):

```
cell(libg205_1) {
  latch ("p1",IQ,IQN) {
     data_in : "((D !SE) + (SE SI))";
     enable : "!p1 + SAVE";
     clear : "!CDN";
  }
  latch ("p1, p2",IQ1,IQN1) {
     data_in : "p2";
     enable : "p1 + NRESTORE";
     clear : "!CDN";
  }
  pin (inQ) {
     function : "IQ";
     reference_input : "CP";
  }
  pin (NRESTORE) {
     retention_pin("restore",1);
  }
  pin (Q) {
    function : "IQ1";
    reference_input : "CP inQ";
  }
```

Feedback

```
  pin (SAVE) {
    retention_pin("save",0);
  }
}
```

This example results in the following warning message:

```
Warning: Line 21, Cell(libg205_1): The function cannot be
         recognized by Design Compiler. The cell becomes a black box.
         (LIBG-205)
```

### What Next

For item 1 above, ignore this message if the intended function cannot be described in v3.1 flip-flop or latch syntax. Otherwise, correct the state table function.

For item 2 above, correct the multiple sequential description if it cannot be simplified to a single sequential descriptor by tied up disable values. Otherwise, ignore the message.

---

## LIBG-206

(warning) The logic represented by the 'when' string (%s) in this %s group is equivalent to logic %s.

### Description

The logic represented by the 'when' string of state-dependent internal_power and leakage_power groups should not be logically equivalent to logic 0 or logic 1.

A power table with its 'when' condition equivalent to logic 0 is meaningless because it will always be ignored during power calculation.

A power table with its 'when' condition equivalent to logic 1 is valid all states and is therefore not state dependent. It should be represented by a power table without the 'when' condition.

### What Next

Check the 'when' conditions in the internal power or leakage power group and check that that are not equivalent to logic 0 or 1. Since such condition values are typically meaningless, the power tables may have been characterized incorrecly. 'when' strings with conditons equivalent to logic 1 should be removed. Power tables with conditions equivalent to logic 0 should be removed. This warning may be promoted to an error in future releases.

Feedback

## LIBG-207

(error) The logic represented by the 'when' string (%s) in this\n \t%s group is not mutually exclusive with the logic represented by\n \tthe 'when' string (%s) in the %s group on line %d.

### Description

The logic represented by the 'when' string of all state-dependent or path-dependent internal_power groups between a pair of pins should be mutually exclusive. If it is not, Library Compiler is not be able to determine which internal_power group to use when propagating a signal through the path, if the condition causes both of them to be evaluated 'TRUE'.

The logic represented by the 'when' string of all state-dependent leakage_power groups should be mutually exclusive. If it is not, Library Compiler is not be able to determine which leakage_power group to use if the condition cause more than one condition to be evaluated 'TRUE'.

The following example shows an instance where this message occurs:

```
cell(libg44) {
      area : 2;
      pin(D1 D0 S0) {
        direction : input;
        capacitance : 1;
      }
      pin(Z) {
        direction : output;
        function : "S0'D0 + S0 D1";
        timing() {
          timing_sense : non_unate;
          intrinsic_rise : 1.00;
          intrinsic_fall : 1.00;
          rise_resistance : 0.1;
          fall_resistance : 0.1;
          related_pin : " D0 D1";
        }
        timing() {
          when : "S0'D0 + S0 D1'";
          when : "D1' S0 + S0'D0 ";
          sdf_cond : "(D1 == 1'b0 && S0 == 1'b1) || (S0 == 1'b0 && D0 ==
 1'b1)";
          timing_sense : positive_unate;
          intrinsic_rise : 1.00;
          intrinsic_fall : 1.00;
          rise_resistance : 0.1;
          fall_resistance : 0.1;
          related_pin : "S0";
        }
        timing() {
```

Feedback

```
          when : "D1 S0 + S0'D0";
          sdf_cond : "(D1 == 1'b1 && S0 == 1'b1) || (S0 == 1'b0 && D0 ==
 1'b0)";
          timing_sense : non_unate;
          intrinsic_rise : 0.1;
          intrinsic_fall : 0.1;
          rise_resistance : 0.1;
          fall_resistance : 0.1;
          related_pin : "S0";
        }
        timing() { /* default */
          intrinsic_rise : 1.00;
          intrinsic_fall : 2.00;
          rise_resistance : 0.1;
          fall_resistance : 0.1;
          related_pin : "S0";
        }
      }
    }
```

In this case, the second when attribute's string has a typographical error. It should be "D1 S0 + S0'D0'"; where the pin D0 should be ticked according to the sdf_cond attribute's string.

The following is an example message:

```
Error: Line 258, The logic represented by the 'when' string (S0'D0 + S0
 D1') in this
        timing group is not mutually exclusive with the logic represented
 by
        the 'when' string (D1 S0 + S0'D0) in the timing group on line
 268. (LIBG-
44)
```

### What Next

Check the 'when' strings of both timing groups for wrong information and fix.

---

## LIBG-207w

(warning) The logic represented by the 'when' string (%s) in this\n \t%s group is not mutually exclusive with the logic represented by\n \tthe 'when' string (%s) in the %s group on line %d.

### Description

The logic represented by the 'when' string of all state-dependent or path-dependent internal_power groups between a pair of pins should be mutually exclusive. If it is not, Library Compiler is not be able to determine which internal_power group to use when

Feedback

propagating a signal through the path, if the condition causes both of them to be evaluated 'TRUE'.

The logic represented by the 'when' string of all state-dependent leakage_power groups should be mutually exclusive. If it is not, Library Compiler is not be able to determine which leakage_power group to use if the condition cause more than one condition to be evaluated 'TRUE'.

The following example shows an instance where this message occurs:

```
cell(libg44) {
      area : 2;
      pin(D1 D0 S0) {
        direction : input;
        capacitance : 1;
      }
      pin(Z) {
        direction : output;
        function : "S0'D0 + S0 D1";
        timing() {
          timing_sense : non_unate;
          intrinsic_rise : 1.00;
          intrinsic_fall : 1.00;
          rise_resistance : 0.1;
          fall_resistance : 0.1;
          related_pin : " D0 D1";
        }
        timing() {
          when : "S0'D0 + S0 D1'";
          when : "D1' S0 + S0'D0 ";
          sdf_cond : "(D1 == 1'b0 && S0 == 1'b1) || (S0 == 1'b0 && D0 ==
 1'b1)";
          timing_sense : positive_unate;
          intrinsic_rise : 1.00;
          intrinsic_fall : 1.00;
          rise_resistance : 0.1;
          fall_resistance : 0.1;
          related_pin : "S0";
        }
        timing() {
          when : "D1 S0 + S0'D0";
          sdf_cond : "(D1 == 1'b1 && S0 == 1'b1) || (S0 == 1'b0 && D0 ==
 1'b0)";
          timing_sense : non_unate;
          intrinsic_rise : 0.1;
          intrinsic_fall : 0.1;
          rise_resistance : 0.1;
          fall_resistance : 0.1;
          related_pin : "S0";
        }
        timing() { /* default */
          intrinsic_rise : 1.00;
```

Feedback

```
          intrinsic_fall : 2.00;
          rise_resistance : 0.1;
          fall_resistance : 0.1;
          related_pin : "S0";
        }
      }
    }
```

In this case, the second when attribute's string has a typographical error. It should be "D1 S0 + S0'D0'"; where the pin D0 should be ticked according to the sdf_cond attribute's string.

The following is an example message:

```
Warning: Line 258, The logic represented by the 'when' string (S0'D0 + S0
 D1') in this
        timing group is not mutually exclusive with the logic represented
 by
        the 'when' string (D1 S0 + S0'D0) in the timing group on line
 268. (LIBG-
207w)
```

### What Next

Check the 'when' strings of both timing groups for wrong information and fix.

---

## LIBG-208

(warning) Different number of pins on the '%s' test_cell are found.\n \tThe test_cell is removed.

### Description

The cell and its associated test_cell must have the same number of pins.

This warning informs you that the test_cell pointed to has a different number of pins than its parent cell.

The following example shows an instance where this message occurs:

```
cell(lbdb128) {
      area : 13;
      pin(D) {
        direction : input;
        capacitance : 1;
      }
      pin(G) {
        direction : input;
        capacitance : 2;
      }
      pin(D2) {
        direction : input;
```

Feedback

```
    capacitance : 1;
  }
  pin(C2) {
    direction : input;
    capacitance : 2;
  }
  pin(Q) {
    direction : output;
    timing() {
      timing_type : rising_edge;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "G C2";
    }
  }
  pin(QN) {
    direction : output;
    timing() {
      timing_type : rising_edge;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "G C2";
    }
  }
  test_cell() {
    pin(D) {
      direction : input;
    }
    pin(G) {
      direction : input;
    }
    pin(D2) {
      direction : input;
      signal_type : "test_scan_in";
    }
    pin(C2) {
      direction : input;
      signal_type : "test_scan_clock";
    }
    pin(SE) {
      direction : input;
      capacitance : 2;
      signal_type : "test_scan_enable";
    }
    state ("IQ","IQN") {
      force_01 : "G D'";
      force_10 : "G D";
    }
    pin(Q) {
```

Feedback

```
          direction : output;
          function : "IQ";
          signal_type : "test_scan_out";
        }
        pin(QN) {
          direction : output;
          function : "IQN";
          signal_type : "test_scan_out_inverted";
        }
      }
    }
```

In this case, the 'SE' pin is defined in the test_cell, but it is not defined in the parent cell. To fix the problem, add the definition,

```
pin(SE) {
        direction : input;
        capacitance : 2;
      }
```

The following is an example message:

```
Warning: Line 124, Different number of pins on the 'lbdb128' cell are
 found.
        The test_cell is removed. (LIBG-208)
```

### What Next

Check the library description to ensure that each test_cell has the same number of pins as the parent cell.

---

## LIBG-209

(warning) Unable to find the '%s' pin on the '%s' test_cell.\n \tThe test_cell is removed.

### Description

This message indicates that you specified a different pin name in the cell and its associated test_cell. This might be caused by a typo. The Library Compiler ignores the associated test_cell.

The following example shows an instance where this message occurs:

```
cell(lbdb129) {
      area : 13;
      pin(D) {
        direction : input;
        capacitance : 1;
      }
      pin(G) {
        direction : input;
        capacitance : 2;
```

Feedback

```
}
pin(D2) {
  direction : input;
  capacitance : 1;
}
pin(C2) {
  direction : input;
  capacitance : 2;
}
pin(SE) {
  direction : input;
  capacitance : 2;
}
pin(Q) {
  direction : output;
  timing() {
    timing_type : rising_edge;
    intrinsic_rise : 1.0;
    intrinsic_fall : 1.0;
    rise_resistance : 0.1;
    fall_resistance : 0.1;
    related_pin : "G C2";
  }
}
pin(QN) {
  direction : output;
  timing() {
    timing_type : rising_edge;
    intrinsic_rise : 1.0;
    intrinsic_fall : 1.0;
    rise_resistance : 0.1;
    fall_resistance : 0.1;
    related_pin : "G C2";
  }
}
test_cell() {
  pin(D) {
    direction : input;
  }
  pin(G) {
    direction : input;
  }
  pin(D2) {
    direction : input;
    signal_type : "test_scan_in";
  }
  pin(C2) {
    direction : input;
    signal_type : "test_scan_clock";
  }
  pin(SEB) {
    direction : input;
    capacitance : 2;
```

Feedback

```
          signal_type : "test_scan_enable";
        }
        latch ("IQ","IQN") {
          data_in : "D";
          enable : "G";
        }
        pin(Q) {
          direction : output;
          function : "IQ";
          signal_type : "test_scan_out";
        }
        pin(QN) {
          direction : output;
          function : "IQN";
          signal_type : "test_scan_out_inverted";
        }
      }
    }
```

In this case, a pin is named 'SE' at the cell level and 'SEB' at the test_cell level. To fix the problem, give the pin the same name.

The following is an example message:

```
Warning: Line 124, Unable to find the 'SE' pin on the 'lbdb129'
 test_cell.
        The test_cell is removed. (LIBG-209)
```

**What Next**

Check the library description to ensure that each test_cell has the same number of pins as the parent cell.

---

## LIBG-210

(error) The '%s' pin of the '%s' cell has conflicting signal_types\n \tbetween the cell and the test_cell.

**Description**

This message indicates that you specified conflicting signal_type attributes for the same pin on different test_cell groups.

The following example shows an instance where this message occurs:

```
cell(lbdb131) {
      area : 13;
      pin(D) {
        direction : input;
        capacitance : 1;
      }
      pin(G) {
```

Feedback

```
    direction : input;
    capacitance : 2;
  }
  pin(D2) {
    direction : input;
    capacitance : 1;
  }
  pin(C2) {
    direction : input;
    capacitance : 2;
  }
  pin(SE) {
    direction : input;
    capacitance : 2;
  }
  pin(Q) {
    direction : output;
    timing() {
      timing_type : rising_edge;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "G C2";
    }
  }
  pin(QN) {
    direction : output;
    timing() {
      timing_type : rising_edge;
      intrinsic_rise : 1.0;
      intrinsic_fall : 1.0;
      rise_resistance : 0.1;
      fall_resistance : 0.1;
      related_pin : "G C2";
    }
  }
  test_cell() {
    pin(D) {
      direction : input;
    }
    pin(G) {
      direction : input;
    }
    pin(D2) {
      direction : input;
      signal_type : "test_scan_in";
    }
    pin(C2) {
      direction : input;
      signal_type : "test_scan_clock";
    }
    pin(SE) {
```

Feedback

```
      direction : input;
      capacitance : 2;
      signal_type : "test_scan_enable";
    }
    state ("IQ","IQN") {
      force_01 : "G D'";
      force_10 : "G D";
    }
    pin(Q) {
      direction : output;
      function : "IQ";
      signal_type : "test_scan_out";
    }
    pin(QN) {
      direction : output;
      function : "IQN";
      signal_type : "test_scan_out_inverted";
    }
  }
  test_cell() {
    pin(D) {
      direction : input;
    }
    pin(G) {
      direction : input;
    }
    pin(D2) {
      direction : input;
      signal_type : "test_scan_in";
    }
    pin(C2) {
      direction : input;
      signal_type : "test_scan_clock";
    }
    pin(SE) {
      direction : input;
      capacitance : 2;
      signal_type : "test_scan_enable_inverted";
    }
    state ("IQ","IQN") {
      force_01 : "G D'";
      force_10 : "G D";
    }
    pin(Q) {
      direction : output;
      function : "IQ";
      signal_type : "test_scan_out";
    }
    pin(QN) {
      direction : output;
      function : "IQN";
      signal_type : "test_scan_out_inverted";
    }
```

Feedback

```
      }
    }
```

In this case, The 'SE' pin has a test_scan_enable signal_type in the first test_cell group and a test_scan_enable_inverted signal_type in the second test_cell group. To fix the problem, make the signal_type consistent between the two test_cell groups.

The following is an example message:

```
Error: Line 94, The 'SE' pin of the 'lbdb131' cell has conflicting
 signal_types
       between the cell and the test_cell. (LIBG-210)
```

### What Next

Check the library description to ensure that each test_cell has no conflicting signal_type attributes on its pins.

---

## LIBG-211

(warning) Cell '%s' contains unused input pin(s).\n\tIt is labeled dont_use and dont_touch.

### Description

An input pin is unused if it does not appear in any functional statement, or state table input or input map, nor does it have any test signal_type description. Such cells may cause synthesis to generate errouneous logic because of incomplete modeling.

The following example shows an instance where this message occurs:

```
cell(acell) {
        area : 6.50;
        pin(CLK) {
                direction : input;
                capacitance : 1.5;
        }
        pin(SET) {
                direction : input;
                capacitance : 3.0;
        }
        ff("IQ","IQN") {
                next_state : "IQ'";
                clocked_on : "CLK";
        }
        pin(Q) {
                direction : output;
                function : "IQ";
        }
        pin(QN) {
                direction : output;
                function : "IQN";
```

Feedback

```
                }
  }
```

The following is an example message:

```
Warning: Line 202, Cell 'acell' contains unused input pin(s).
        It is labeled dont_use and dont_touch. (LIBG-211)
```

**What Next**

Either delete the unused input pin or add it the functionality of the cell.

---

## LIBG-212

(warning) The logic represented by the '%s' when string in this %s group\n \tdoes not sensitize one or all pins in the '%s' related_pin attribute for the %s function attribute.

**Description**

This message indicates that the when attribute within an internal_power group does not make sense given the function associated with the output pin. Library Compiler warns you that allowing these extra tables in the library, you are unnecessarily increasing the size of the library and may be characterizing unnecessary data.

The following example shows an instance where this message occurs:

```
cell(libg212) {
      area : 2;
      pin(A B) {
        direction : input;
        capacitance : 1;
      }
      pin(Z) {
        direction : output;
        function : "A & B";
        internal_power() {
          related_pin : "B";
          when : "!A'";
           ...
        }
      }
    }
```

In this case, this internal power table will never be accessed. This is because in order for B to cause Z to change, pin A must be high.

The following is an example message:

```
Warning: Line 191, The logic represented by the '!A' when string in this
 internal_power group
        does not sensitize one or all pins in the 'B ' related_pin
 attribute for the (A & B) function attribute. (LIBG-212)
```

Feedback

### What Next

Check the 'when' string of internal_power group for wrong information and fix.

---

## LIBG-213

(warning) The default internal power is not needed for the %s pin.

### Description

This message indicates that the set of 'when' attributes is complete for the internal_power and there is no need for the default internal power table.

The following example shows an instance where this message occurs:

```
cell(libg213) {
      area : 2;
      pin(A B) {
        direction : input;
        capacitance : 1;
      }

       pin ( Z ) {
         direction : output ;
         function : "A & B";
         internal_power () {
           when : "A";
           related_pin : "B";
           power(power_template) {
           values (" 1.000000, 1.000000, 1.000000, 1.000000");
           }
         }
         internal_power () {
           when : "B";
           related_pin : "A";
           power(power_template) {
           values (" 1.000000, 1.000000, 1.000000, 1.000000");
           }
         }
         internal_power () {
           related_pin : "A B";
           power(power_template) {
           values (" 1.000000, 1.000000, 1.000000, 1.000000");
           }
         }
      }
    }
```

In this case, the default internal power table is not needed. This is because in the when attribute contains all possible paths which will sensitize the related_pin B to the output. Therefore, no default internal power table is needed.

Feedback

The following is an example message:

```
Warning: Line 183, The default internal power is not needed for the A
 pin. (LIBG-213)
```

**What Next**

Check the 'when' string of internal_power groups for wrong information and fix or remove the default internal power group.

---

## LIBG-214

(warning) The default internal power is required for the '%s' pin.

**Description**

This message indicates that the set of 'when' attributes is not complete for the internal_power and there is a need for the default internal power table.

The following example shows an instance where this message occurs:

```
cell(libg214) {
      area : 2;
      pin(A B) {
        direction : input;
        capacitance : 1;
      }

       pin ( Z ) {
         direction : output ;
         function : "A & B";
         internal_power () {
           when : "A";
           related_pin : "B";
           power(power_template) {
           values (" 1.000000, 1.000000, 1.000000, 1.000000");
           }
         }
      }
    }
```

In this case, the default internal power table is needed for the related_pin A. This is because the when attribute does not contain all possible paths which will sensitize the related_pin A to the output. Therefore, you either add the power table associated with the related pin A and a when or add the default internal power table as show below:

either internal_power () { when : "B"; related_pin : "A"; power(power_template) { values (" 1.000000, 1.000000, 1.000000, 1.000000"); } } or internal_power () { related_pin : "B"; power(power_template) { values (" 1.000000, 1.000000, 1.000000, 1.000000"); } }

Feedback

The following is an example message:

```
Warning: Line 183, The default internal power is required for the 'A'
 pin. (LIBG-214)
```

### What Next

Check the 'when' string of internal_power groups for wrong information and fix or add the default internal power group.

---

## LIBG-215

(error) The '%s' when condition includes the '%s' related pin.

### Description

This message indicates that the 'when' and 'related_pin' attributes contain the same pin. This condition does not make sense because the when attribute is a static state while the related_pin attribute is the transitioning pin.

Note that the 'related_pin' of a pin includes the pin itself.

The following example shows an instance where this message occurs:

```
cell(libg215) {
      area : 2;
      pin(A B) {
        direction : input;
        capacitance : 1;
      }

       pin ( Z ) {
         direction : output ;
         function : "A & B";
         internal_power () {
           when : "A";
           related_pin : " A B";
           power(power_template) {
           values (" 1.000000, 1.000000, 1.000000, 1.000000");
           }
         }
      }
    }
```

In this case, the internal power table contains the A pin in both the when and the related_pin attributes. Either remove the A pin from the related_pin or change the when statement.

The following is an example message:

```
Error: Line 183, The 'A' when condition includes the 'A' related pin.
 (LIBG-215)
```

Feedback

### What Next

Check the 'when' and the 'related_pin' strings of the internal_power group for wrong information and fix it.

---

## LIBG-220

(warning) The '%s' equation on the '%s' cell evaluates\n \tto a '%s' constant.

### Description

This message is generated when Library Compiler encounters a three-stated pin, whose three_state attribute always evaluates to 1.

The following example shows an instance where this message occurs:

```
cell(libg220) {
        area : 6.50;
        pin(A) {
                direction : input;
                capacitance : 1.5;
        }
        pin(Y) {
                direction : output;
                function : "A";
                three_state : "1";
        }
      }
```

The following is an example message:

```
Warning: Line 218, The 'three_state' equation on the 'libg220' cell
 evaluates
        to a '1' constant. (LIBG-220)
```

### What Next

Verify the cell and check the library description of the offending cell for correctness.

---

## LIBG-221

(warning) Cell %s, referred to by cell %s, does not exist.

### Description

The missing cell is referred to by the "single_bit_degenerate" attribute from a multibit cell. This attribute will be removed from the multibit cell.

Feedback

The following example shows an instance where this message occurs:

```
cell(FDBBX4) {
    area : 18;
    single_bit_degenerate : FDBB;
    bus(D) {
      ...
    }
    ...
  }
```

In this case, cell FDBB is not in the library.

The following is an example message:

```
Warning: Line 258, Cell FDBB, referred to by cell FDBBX4, does not exist.
 (LIBG-221)
```

The missing cell is referred to by the "retention_equivalent_cell" attribute from a non retention cell.

The following example shows an instance where this message occurs:

```
cell(non_retention) {
    area : 18;
    retention_equivalent_cell : my_retention;
    pin(D) {
      ...
    }
    ...
  }
```

In this case, cell my_retention is not in the library.

The following is an example message:

```
Warning: Line 188, Cell my_retention, referred to by cell non_retention,
 does not exist. (LIBG-221)
```

### What Next

Correct the attribute value or remove the attribute, or add the required cell in the library.

---

## LIBG-222

(warning) Cell %s, referred to by cell %s, is multibit.

### Description

The offending cell is referred to by the "single_bit_degenerate" attribute from a multibit cell. This cell must be a single-bit cell.

Feedback

The following example shows an instance where this message occurs:

```
cell(FDBBX4) {
    area : 18;
    single_bit_degenerate : FDBBX2;
    bus(D) {
      ...
    }
    ...
  }
```

In this case, cell FDBBX2 is a multibit cell.

The following is an example message:

```
Warning: Line 58, Cell FDBBX2, referred to by cell FDBBX4, is multibit.
 (LIBG-221)
```

### What Next

Correct the attribute value or remove the attribute, or add the required cell in the library.

---

## LIBG-223

(warning) Cells %s and %s have different interfaces.

### Description

The first cell is referred to by the "single_bit_degenerate" attribute from the second cell. These cells must have the same interfaces.

The following example shows an instance where this message occurs:

```
cell(FDBBX4) {
    area : 18;
    single_bit_degenerate : FDBB;
    bus(D) {
      ...
    }
    pin(CP) {
    }
    bus(Q) {
    }
  }
  cell(FDBB) {
    pin(D) {
    }
    pin(clk) {
    }
    pin(Q) {
    }
  }
```

Feedback

In this case, bus D and pin D, and bus Q and pin Q are matched. But pin CP and pin clk, are not. To correct, either change clk to CP, or change CP to clk.

The following is an example message:

```
Warning: Line 58, Cells FDBB  and FDBBX4 have different interfaces.
 (LIBG-223)
```

**What Next**

Correct the interfaces.

---

## LIBG-224

(warning) Cell %s has unequal bus or bundle widths.

**Description**

To be eligible for having the "single_bit_degerate" attribute, the cell must be a multibit black box, and all buses must have the same width.

The following example shows an instance where this message occurs:

```
type (downbus3) {
    base_type : array;
    data_type : bit;
    bit_width : 3;
    bit_from : 2;
    bit_to   : 0;
    downto   : true;
  }
  type (bus4) {
    base_type : array;
    data_type : bit;
    bit_width : 4;
    bit_from : 0;
    bit_to   : 3;
    downto   : true;
  }

  cell(FDBBX4) {
    area : 18;
    single_bit_degenerate : FDBB;
    bus(D) {
      bus_type : downbus3;
      ...
    }
    pin(CP) {
    }
    bus(Q) {
      bus_type : bus4;
```

Feedback

```
    }
  }
```

In this case, bus D has pin from 3 to 0, bus bus Q has range from 0 to 3.

The following is an example message:

```
Warning: Line 224, Cell FDBBX4 has unequal bus or bundle
 widths.(LIBG-224)
```

### What Next

Correct the bus range.

---

## LIBG-225

(warning) Errors have been detected. The single_bit_degenerate\n \tattribute is removed from cell %s.

### Description

The single_bit_degenerate attribute is removed from the cell because there is an error regarding the attribute specification on this cell or related cell.

The following example shows an instance where this message occurs:

```
cell(FDBBX2) {
  area : 18;
  single_bit_degenerate : FDBB;
  bundle(D) {
    members(D0, D1); ...
  }
  pin(CP) { ...  }

  bundle(Q) {
    members(Q0, Q1); ...
  }
  bundle(QN) {
    members(Q0N, Q1N); ...
  }
}

cell(FDBB) {
  area : 18;
  pin(Data)  { ...  }
  pin(CP) { ...  }
  pin(Q)  { ...  }
  pin(QN) { ...  }
}
```

The single_bit_denerate would be ok if pin "Data" is renamed to "D".

Feedback

The following is an example message:

```
Warning: Line 225, Errors have been detected. Attribute
 single_bit_degenerate
        is removed from cell %s. (LIBG-225)
```

### What Next

To specify "single_bit_degenerate" on a cell, the cell must satisfy:

```
It is a multibit black box with bus/bundle pin
        The bus member must range from 0 to N-1 or N-1 to 0 (N is the
 width)
        The single-bit cell referred to must be a single-bit cell.
        The two cells must have identical interface naming.
```

---

## LIBG-226

(warning) Only cell %s has geometry_print %s.

### Description

It is of no use to put a geometry_print on a single cell.

The following example shows an instance where this message occurs:

```
cell(FDBBX2) {
  area : 18;
  geometry_print : GP1;
  bundle(D) {
    members(D0, D1); ...
  }
  pin(CP) { ...  }

  bundle(Q) {
    members(Q0, Q1); ...
  }
  bundle(QN) {
    members(Q0N, Q1N); ...
  }
}
```

GP1 only appears on FDBBX2.

The following is an example message:

```
Warning: Line 226, Onle cell FDBBX2 has geometry_print GP1. (LIBG-226)
```

### What Next

Remove it or put it on other cells when appropriate.

Feedback

## LIBG-228

(warning) Failed to transform the function of cell '%s' to a netlist. The cell becomes a black box for Formality.

### Description

Library Compiler fails to synthesize a netlist structure representing functionality of this cell. Library Compiler does not accept a derived statetable description when performing this netlist transformation. That is, if the full-function description is not specified in the .lib file, this warning message will be issued. Other causes of failure should have been reported by other warning messages. This netlist structure is referenced by Formality. Without this netlist, Formality can only treat this cell as a black box.

### What Next

Check the statetable description and make sure it is specified correctly.

## LIBG-229

(warning) Cell '%s' has non-disjoint input.

### Description

One or more input pins used in the statetable are used in more than one sequential function, such as clock and enable. As a result, this function cannot be transformed by Library Compiler to a netlist structure referenced by Formality.

### What Next

Check the statetable description and correct all mistakes.

## LIBG-230

(warning) Internal pin '%s' is dangling. The pin is removed.

### Description

The internal pin is not associated with any statetable, it does not have a function or a state_function, it is not a memory port, nor is it part of any timing-arc specification. It is redundant and therefore being removed.

### What Next

You can add the required element if it is accidentally missing. Or a dummy timing arc will serve the purpose of keeping the internal pin interface.

Feedback

## LIBG-231

(information) Internal pin '%s' is for dcm timing.

### Description

The internal pin is labeled for dcm timing use only.

### What Next

Make sure the pin is used for dcm timing only, not for any other purpose.

## LIBG-232

(error) The 'function' attribute on the '%s' internal Pin/bus of the\n \t'%s' cell is not allowed.

### Description

This error is issued when any internal pin of a combinational cell defines a function attribute. In a combinational cell, a function attribute can only be defined on an output pin.

The following example shows an instance where this message occurs:

```
cell(libg232) {
  area : 1 ;
  pin (A) {
        direction : input ;
        capacitance : 0 ;
  }
  pin (A_INT) {
        direction : internal ;
        function : A;
        capacitance : 0 ;
        timing() {
                related_pin     : "A" ;
                intrinsic_rise  : 1.0 ;
                rise_resistance : 0.0 ;
                intrinsic_fall  : 1.0 ;
                fall_resistance : 0.0 ;
        }
  }
  pin ("Y") {
        direction : output ;
        function : "A'" ;
        timing() {
                related_pin     : "A_INT" ;
                intrinsic_rise  : 1.0 ;
                rise_resistance : 0.0 ;
                intrinsic_fall  : 1.0 ;
                fall_resistance : 0.0 ;
```

Feedback

```
        }
  }
```

The following is an example message:

```
Error: Line 10, The 'function' attribute on the 'A_INT' internal pin/bus
 of the
        'libg232' cell is not allowed. (LIBG-232)
```

### What Next

Check your library for wrong attributes.

---

## LIBG-233

(error) The 'clock_gate_enable_pin' attribute is defined on more than one\n \tpin in the '%s' cell.

### Description

This error is issued when the clock_gate_enable_pin attribute is defined on more than one pin in a cell. In a clock gating cell, there should only be at most one pin with the clock_gate_enable_pin attribute defined.

The following example shows an instance where this message occurs:

```
cell(libg233) {
  area : 1 ;
  is_clock_gating_cell : TRUE;
  pin (A) {
        direction : input ;
        capacitance : 0 ;
        clock_gate_enable_pin : TRUE;
  }
  pin (B) {
        direction : input ;
        capacitance : 0 ;
        clock_gate_enable_pin : TRUE;
  }
  pin ("Y") {
        direction : output ;
        function : "A' B'" ;
        timing() {
                related_pin     : "A B" ;
                intrinsic_rise  : 1.0 ;
                rise_resistance : 0.0 ;
                intrinsic_fall  : 1.0 ;
                fall_resistance : 0.0 ;
        }
  }
```

Feedback

The following is an example message:

```
Error: Line 10, The 'clock_gate_enable_pin' attribute is defined on more
 than one
       pin in the 'libg233' cell. (LIBG-233)
```

**What Next**

Check your library and remove extra clock_gate_enable_pin attribute.

---

## LIBG-234

(error) The '%s' cell cannot be used as a clock gating cell.

**Description**

This error is issued when the is_clock_gating_cell attribute is defined on a cell with disallowed function.

The is_clock_gating_cell attribute can be set on

- 2-input clock gates Examples of 2-input clock gates are AND, NAND, OR, and NOR library cells that are used to gate clocks.

- 1-input cells Examples of 1-input clock gates are buffer and inverter library cells that are used in the fanin and fanout of the 2-input clock gate.

- 2-input D latches These latches can be active high or low and must have a noninverting output.

The following example shows an instance where this message occurs:

```
cell(libg234) {
  area : 1 ;
  is_clock_gating_cell : TRUE;
  pin (A) {
        direction : input ;
        capacitance : 0 ;
        clock_gate_enable_pin : TRUE;
  }
  pin (B) {
        direction : input ;
        capacitance : 0 ;
  }
  pin (C) {
        direction : input ;
        capacitance : 0 ;
  }
  pin ("Y") {
        direction : output ;
        function : "A B C" ;
        timing() {
```

Feedback

```
                related_pin     : "A B C" ;
                intrinsic_rise  : 1.0 ;
                rise_resistance : 0.0 ;
                intrinsic_fall  : 1.0 ;
                fall_resistance : 0.0 ;
        }
  }
```

The following is an example message:

```
Error: Line 10, The 'libg234' cell cannot be used as a clock gating cell.
 (LIBG-234)
```

### What Next

Check your library and correct the error.

---

## LIBG-235

(error) The '%s' attribute is either not\n \tdefined on any pin in the integrated clock gated '%s' cell\n \tor it is defined on a wrong pin.

### Description

This error is issued when the clock_gate_enable_pin or clock_gate_clock_pin or clock_gate_outp_pin attribute is either not defined on any pin in a cell or it is defined on the wrong pin. In a clock gating cell, there should only be at most one pin with the clock_gate_enable_pin or clock_gate_out_pin or clock_gate_clock_pin attribute defined.

The following example shows an instance where this message occurs:

```
cell(libg235) {
  area : 1;
  clock_gating_integrated_cell : "latch_posedge";
  dont_use : true;
  dont_touch : true;
  statetable(" CP EN ", "IQ ") {
     table : " L  L  : - : L ,\
               L  H  : - : H ,\
               H  -  : - : N ";
  }
  pin(IQ) {
    direction : internal;
    internal_node : "IQ";
  }
  pin(EN) {
    direction : input;
    capacitance : 1.0;
  }
  pin(CP) {
    direction : input;
```

Feedback

```
    capacitance : 1.0;
    min_pulse_width_low  : 0.319;
    clock_gate_clock_pin : TRUE;
  }
  pin(Z) {
    direction : output;
    state_function : "CP * IQ";
    max_capacitance : 1.0;
    clock_gate_out_pin : TRUE;
  }
```

The following is an example message:

```
Error: Line 10, The 'clock_gate_enable_pin' attribute is either not
        defined on any pin in the integrated clock gated 'libg235' cell
        or it is defined on a wrong pin. (LIBG-235)
```

### What Next

Check your library and add the missing clock gate attribute.

---

## LIBG-236

(error) The '%s' clock gate cell has the wrong number of pins.

### Description

This error is issued when the clock_gating_integrated_cell attribute is defined on a cell with disallowed function.

The clock_gating_integrated_cell attribute must have at most two inputs pins and two output pins.

The following example shows an instance where this message occurs:

```
cell(libg236) {
  area : 1;
  clock_gating_integrated_cell : "latch_posedge";
  dont_use : true;
  dont_touch : true;
  statetable(" CP1 EN ", "IQ ") {
     table : " L  L  : - : L ,\
               L  H  : - : H ,\
               H  -  : - : N ";
  }
  pin(IQ) {
    direction : internal;
    internal_node : "IQ";
  }
  pin(EN) {
    direction : input;
    clock_gate_enable_pin : true;
```

Feedback

```
    capacitance : 1.0;
  }
  pin(CP1) {
    direction : input;
    capacitance : 1.0;
    min_pulse_width_low  : 0.319;
  }

  pin(CP2) {
    direction : input;
    capacitance : 1.0;
    min_pulse_width_low  : 0.319;
  }
  pin(Z) {
    direction : output;
    state_function : "CP1 * IQ";
    max_capacitance : 1.0;
  }
```

The following is an example message:

```
Error: Line 10, The 'libg236' clock gate cell has the wrong number of
 pins. (LIBG-236)
```

### What Next

Check your library and correct the error.

---

## LIBG-237

(error) The '%s' clock gate cell has the wrong type of pins.

### Description

This error is issued when the clock_gating_integrated_cell attribute is defined on a cell with the wrong pin types

The clock_gating_integrated_cell attribute expects the following attributes on its pins: - clock_gate_enable_pin - clock_gate_obs_pin - clock_gate_test_pin

The following example shows an instance where this message occurs:

```
cell(libg237) {
  area : 1;
  clock_gating_integrated_cell : "latch_posedge_precontrol";
  dont_use : true;
  dont_touch : true;
  statetable(" CP EN CTL", "IQ ") {
     table : " L  L  L : - : L ,\
               L  L  H : - : H ,\
               L  H  L : - : H ,\
               L  H  H : - : H ,\
```

Feedback

```
                H  -  - : - : N ";
  }
  pin(IQ) {
    direction : internal;
    internal_node : "IQ";
  }
  pin(EN) {
    direction : input;
    capacitance : 0.017997;
    clock_gate_enable_pin : true;
  }
  pin(CTL) {
    direction : input;
    capacitance : 0.017997;
  }
  pin(CP) {
    direction : input;
    capacitance : 0.031419;
    min_pulse_width_low  : 0.319;
  }
  pin(Z) {
    direction : output;
    state_function : "CP * IQ";
    max_capacitance : 0.500;
  }
}
```

The attribute clock_gate_test_pin is missing. You need to add it to the 'CTL' pin.

```
clock_gate_test_pin : true;
```

The following is an example message:

```
Error: Line 10, The 'libg237' clock gate cell has the wrong type of pins.
 (LIBG-237)
```

**What Next**

Check your library and correct the error.

---

## LIBG-238

(error) The '%s' clock gate cell is not of type '%s'.

**Description**

This error is issued when the clock_gating_integrated_cell attribute is defined on a cell with disallowed function.

Check the Library Compiler User Guide for the list of the clock gating integrated cell types.

Feedback

The following example shows an instance where this message occurs:

```
cell(libg238) {
  area : 1;
  clock_gating_integrated_cell : "latch_posedge";
  dont_use : true;
  dont_touch : true;
  statetable(" CP EN ", "IQ ") {
     table : " L  L  : - : L ,\
               L  H  : - : H ,\
               H  -  : - : N ";
  }
  pin(IQ) {
    direction : internal;
    internal_node : "IQ";
  }
  pin(EN) {
    direction : input;
    capacitance : 1.0;
  }
  pin(CP) {
    direction : input;
    capacitance : 1.0;
    min_pulse_width_low  : 0.319;
  }
  pin(Z) {
    direction : output;
    state_function : "CP * IQ * EN";
    max_capacitance : 1.0;
  }
```

The following is an example message:

```
Error: Line 10, The 'libg238' clock gate cell is not of type 'latch_pos'.
 (LIBG-238)
```

### What Next

Check your library and correct the error.

---

## LIBG-238w

(warning) The '%s' clock gate cell is not of type '%s'.

### Description

This warning is issued when the clock_gating_integrated_cell attribute is defined on a cell with disallowed function.

Check the Library Compiler User Guide for the list of the clock gating integrated cell types.

Feedback

The following example shows an instance where this message occurs:

```
cell(libg238) {
  area : 1;
  clock_gating_integrated_cell : "latch_posedge";
  dont_use : true;
  dont_touch : true;
  statetable(" CP EN ", "IQ ") {
     table : " L  L  : - : L ,\
               L  H  : - : H ,\
               H  -  : - : N ";
  }
  pin(IQ) {
    direction : internal;
    internal_node : "IQ";
  }
  pin(EN) {
    direction : input;
    capacitance : 1.0;
  }
  pin(CP) {
    direction : input;
    capacitance : 1.0;
    min_pulse_width_low  : 0.319;
  }
  pin(Z) {
    direction : output;
    state_function : "CP * IQ * EN";
    max_capacitance : 1.0;
  }
```

The following is an example message:

```
Warning: Line 10, The 'libg238' clock gate cell is not of type
 'latch_pos'. (LIBG-238w)
```

### What Next

Check your library and correct the warning.

---

## LIBG-239

(information) The timing arc from '%s' to\n \t'%s' is resolved into the three_state_enable type.

### Description

The timing arc is entered as combitional/sequential. But the related pin is a fanin to the three_state function only.

Feedback

### What Next

Add "timing_type : three_state_enable;" to the timing group. Or check if this arc is intended for three_state_enable.

---

## LIBG-240

(warning) The phase relationship of the outputs of cell '%s'\n \tis different from that of its test cell at line %d.

### Description

The two outputs of the full cell are equal, but the outputs of the test cell are opposite; or vice versa.

The following example shows an instance where the message occurs:

```
cell (SCAN) {
    ff(IQ, IQN) {
      ...
    }
    pin (Q1) {
      direction : output;
      function  : IQ;
      ...
    }
    pin (Q2) {
      direction : output;
      function  : IQ;
      ...
    }
    ...
    test_cell() {
      ff(IQ, IQN) {
        ...
      }
      pin (Q1) {
        direction : output;
        function  : IQ;
        ...
      }
      pin (Q2) {
        direction : output;
        function  : IQN;
        ...
      }
      ...
    }
  }
```

Feedback

The following is an example message:

```
Warning: Line 200, The phase relationship of the outputs of cell 'SCAN'
is different from that of its test cell at line %300. (LIBG-240)
```

### What Next

If you intended that the outputs of the full cell differ from the outputs of the test cell, no action is required. Otherwise, change the specification of one of the cells so that they agree.

---

## LIBG-241

(warning) The '%s' cell has %d pins which exceeds the limit 16.\n \tThe state table is not generated.

### Description

The cell has over 16 pins. There is an exponential cose in performance and memory as the number of pins increases. This warning tells the user about the excessive size of the table that is being generated.

The following example shows an instance where this message occurs:

```
cell (libg241) {
latch(iq,iqn) {
    data_in : "((c1 *d1_b') +(c2 *d2_b') +(c3 *d3_b') +(c4 *d4_b')
 +(c5 *d5_b') +(c6 *d6_b') +(c7 *d7_b') +(c8 *d8_b') +(c9 *d9_b') +(c10
 *d10_b') +(c11 *d11_b') + (c12 * d12_b'))" ;
    enable : "(c1 +c2 +c3 +c4 +c5 +c6 +c7 +c8 +c9 +c10 +c11 +c12)" ;
    clear : "!rb";
    preset : "s";
    clear_preset_var1 : "L";
    clear_preset_var2 : "H";
}

      pin ( q ) {
         direction : output ;
         function : "iq";
        timing () {
            related_pin : "d1_b d2_b d3_b d4_b d5_b d6_b d7_b d8_b
d9_b d10_b d11_b d11_b " ;
            cell_rise ( prop1 ) {
               values ("0.102565, 0.092949, 0.117635, 0.129560");
            }
            rise_transition ( trans1 ) {
               values ("0.090691, 0.085401, 0.117467, 0.143013");
            }
            cell_fall ( prop1 ) {
               values ("0.099571, 0.101830, 0.122101, 0.129110");
            }
```

Feedback

```
            fall_transition ( trans1 ) {
               values ("0.100989, 0.083367, 0.104448, 0.185206");
            }
         }

         timing () {
            related_pin : "c1 c2 c3 c4 c5 c6 c7 c8 c9 c10 c11 c12" ;
            cell_rise ( prop1 ) {
               values ("0.101839, 0.105114, 0.140100, 0.165207");
            }
            rise_transition ( trans1 ) {
               values ("0.097529, 0.087806, 0.104830, 0.119332");
            }
            cell_fall ( prop1 ) {
               values ("0.117794, 0.121551, 0.145506, 0.159811");
            }
            fall_transition ( trans1 ) {
               values ("0.099574, 0.088423, 0.085064, 0.108336");
            }
            timing_type : rising_edge ;
         }

         timing () {
           related_pin : "s" ;
            cell_rise ( prop1 ) {
               values ("0.111990, 0.110565, 0.144589, 0.173835");
            }
            rise_transition ( trans1 ) {
               values ("0.102461, 0.097734, 0.129652, 0.175103");
            }
            timing_type : preset ;
            timing_sense : positive_unate ;
         }

         timing () {
            related_pin : "rb" ;
            cell_fall ( prop1 ) {
               values ("0.135732, 0.139003, 0.193663, 0.182614");
            }
            fall_transition ( trans1 ) {
               values ("0.090745, 0.091247, 0.107943, 0.171294");
            }
            timing_type : clear ;
            timing_sense : positive_unate ;
         }
      }

      pin(d1_b d2_b d3_b d4_b d5_b d6_b d7_b d8_b d9_b d10_b d11_b d12_b
rb s){
         direction : input ;
         capacitance : 0.1 ;

      }
```

Feedback

```
        pin ( c1 c2 c3 c4 c5 c6 c7 c8 c9 c10 c11 c12) {
          direction : input ;
          capacitance : 0.1 ;
          clock : true ;
        }
    }
```

The following is an example message:

```
Warning: The 'libg241' cell has 26 pins which exceeds the limit of 16.\n
         The state table is not generated. (LIBG-241)
```

**What Next**

If possible, take advantage of the state table group and the input_map attribute to reduce the size of the table.

---

## LIBG-242

(warning) The '%s' cell has signal pins specified with 'x_function' attribute hence\n \tit is a black box for optimization tools.

For combinational cells or FF/LATCH sequential cells with x function, the function_id, function_class and cmog_class are set to "unknown".

**What Next**

Make sure the description is correct.

**Description**

The following example shows an instance where this message occurs:

```
cell(test) {
          area : 1;
          pin(A) {
            direction : input;
            capacitance : 0;
          }
          pin(B) {
            direction : input;
            capacitance : 0;
          }
          pin(O) {
            direction : output;
            function : "A'";
            x_function : "B";
          }
        }
```

Feedback

The following is an example message:

```
Warning: Line 57, Cell 'cell1', The 'cell1' cell has signal pins
 specified with 'x_function' attribute hence
 it is a black box for optimization tools. (LIBG-242)
```

## LIBG-243

(warning) The '%s' cell is missing the 'nextstate_type' attribute\n \tfor some input pin(s) specified in 'next_state' of its ff/ff_bank group.

### Description

This warning message occurs when the attribute *nextstate_type* is not defined in the input pin included in the *next_state* attribute from the cell's sequential state description. In a pin group, *nextstate_type* defines the type of a *next_state* attribute to be used in an *ff* group, a sequential group, or an *ff_bank* group.

The following example shows an instance where this message occurs:

```
cell(libg243) {
        area : 6.50;
        pin(D) {
           direction : input;
           capacitance : 3.0;
        }
        pin(CLK) {
           direction : input;
           capacitance : 1.5;
        }
        pin(SET) {
           direction : input;
           nextstate_type : scan_in;
           capacitance : 3.0;
        }

        ff("IQ","IQN") {
           next_state : "D CLK";
           clocked_on : "CLK";
           preset     : "SET'";
        }

        pin(Q) {
           direction : output;
           function : "IQ";
        }
        pin(QN) {
           direction : output;
           function : "IQN";
        }
   }
```

Feedback

The following is an example message:

```
Warning: Line 153, The 'libg243' cell is missing the 'nextstate_type'
 attribute
        for some input pin(s) specified in 'next_state" of its ff/ff_bank
 group. (LIBG-243)
```

### What Next

This is only a warning message. No action is required.

However, if the result is not what you intended, check the library for each input pin used in the *next_state* statement of a sequential group and add the *nextstate_type* attribute if necessary.

---

## LIBG-244

(warning) The values for clear_preset_var1 and \n \tclear_preset_var2 in the '%s' cell are opposite. \n \tThis may cause incorrect DC optimization.

### Description

The following example shows an instance where this message occurs:

```
cell(libg111) {
        area : 11;
        pin(D) {
                direction : input;
                capacitance : 1;
        }
        pin(CP) {
                direction : input;
                capacitance : 1;
        }
        pin(CLR) {
                direction : input;
                capacitance : 1;
        }
        pin(SET) {
                direction : input;
                capacitance : 1;
        }
        ff("IQ","IQN") {
                next_state : "D";
                clocked_on : "CP";
                clear   : "CLR'";
                preset   : "SET'";
                clear_preset_var1 : L;
                clear_preset_var2 : H;
        }
        pin(Q) {
                direction : output;
```

Feedback

```
                function : "IQ";
                timing() {
                        timing_type : rising_edge;
                        intrinsic_rise : 1.0;
                        intrinsic_fall : 1.0;
                        rise_resistance : 0.1;
                        fall_resistance : 0.1;
                        related_pin : "CP";
                }
                timing() {
                        timing_type : clear;
                        timing_sense : positive_unate;
                        intrinsic_fall : 1.0;
                        fall_resistance : 0.1;
                        related_pin : "CLR";
                }
                timing() {
                        timing_type : preset;
                        timing_sense : negative_unate;
                        intrinsic_fall : 1.0;
                        fall_resistance : 0.1;
                        related_pin : "SET";
                }
        }
        pin(QN) {
                direction : output;
                function : "IQN";
                timing() {
                        timing_type : rising_edge;
                        intrinsic_rise : 1.0;
                        intrinsic_fall : 1.0;
                        rise_resistance : 0.1;
                        fall_resistance : 0.1;
                        related_pin : "CP";
                }
                timing() {
                        timing_type : preset;
                        timing_sense : negative_unate;
                        intrinsic_fall : 1.0;
                        fall_resistance : 0.1;
                        related_pin : "CLR";
                }
                timing() {
                        timing_type : clear;
                        timing_sense : positive_unate;
                        intrinsic_fall : 1.0;
                        fall_resistance : 0.1;
                        related_pin : "SET";
                }
        }
}
```

Change the value of clear_preset_var1 to H or the value of clear_preset_var1 to L.

Feedback

The following example shows an instance where this message occurs:

```
Warning: Line 244, The values for clear_preset_var1 and clear_preset_var2
 in the 'libg111' cell are opposite.
This may cause incorrect DC optimization. (LIBG-244)
```

**What Next**

Change the value of clear_preset_var1 or clear_preset_var2.

---

## LIBG-245

(information) Cell(%s) test scan type is %s. DFT Compiler\n \tshould be able to infer this cell during compile -scan. \n \tIt is not a black box for DFT Compiler.

**Description**

Library Compilier recognizes test scan type for the scan cell according to its test_cell description.

**What Next**

report_lib also reports scan type for test scan cell.

---

## LIBG-246

(warning) Cell(%s) test scan type is not recognized by Library Compiler. It becomes a black box for DFT Compiler.

**Description**

Library Compiler fails to recognize test scan type of the scan cell according to its test_cell description.

**What Next**

Check the signal type of each pin in test_cell description and correct it if it is wrong.

---

## LIBG-247

(error) In test_cell of Cell(%s) clocked_on pin's polarity is opposite from the cell's definition.

**Description**

The test_cell's non-test mode behavior has to be consistent with what is defined in the cell section. Library Compiler detects that the clocked_on pin in test_cell has opposite polarity than the cell's defintion.

Feedback

The following example shows an instance where the message occurs:

```
cell (TNB) {
          ff ("IQ","IQZ")  {
             next_state : " (D SCAN') + (SD SCAN) ";
             clocked_on : "CLK'";
          }
          test_cell() {
            ff ("IQ","IQZ")  {
            next_state : "D";
            clocked_on : "CLK";
          }
          ...
        }
```

In this case the cell TNB is falling-edge triggered but its test_cell describes a rising-edge triggered flip-flop.

**What Next**

Correct test_cell's clocked_on pin's polarity to be consistent with the cell.

---

## LIBG-248

(warning) The '%s' cell's %s groups for %s '%s' \n \tdo not cover all the states represented by the 'when' string, and the default \n \t%s group is not defined for %s '%s', thus 0 will be used as \n \tthe %s value for the missing states of %s '%s'.

**Description**

This message indicates the leakage_power information defined in a cell does not cover all the states for the power_level(power_rail value). For the missing states of the power_level, 0 will be used as the leakage_power value.

It is also used to check for related_pg_pin-based leakage_power information.

The following example shows an instance where this message occurs:

```
cell(AN2) {
      leakage_power () {
        power_level : "VDD1";
        when : "A";
        value : 1.0;
      }
      leakage_power () {
        power_level : "VDD1";
        when : "!A";
        value : 1.5;
      }
      leakage_power () {
        power_level : "VDD2";
```

Feedback

```
        when : "A";
        value : 1.0;
      }
      ...
  }
```

In this case, the leakage_power for the state '!A' of power_level 'VDD2' is missing, . To fix the problem, add the leakage_power for the missing state in the cell group:

```
cell(AN2) {
      leakage_power () {
        power_level : "VDD1";
        when : "A";
        value : 1.0;
      }
      leakage_power () {
        power_level : "VDD1";
        when : "!A";
        value : 1.5;
      }
      leakage_power () {
        power_level : "VDD2";
        when : "A";
        value : 1.0;
      }
      leakage_power () {
        power_level : "VDD2";
        when : "!A";
        value : 1.2;
      }
      ...
  }
```

The following is an example message:

```
Warning: Line 12, The 'AN2' cell's leakage_power groups for power_level
 'VDD2'
        do not cover all the states represented by the 'when' string, and
 the default
        leakage_power group is not defined for power_level 'VDD2', thus 0
 will be used as
        the leakage_power value for the missing states of power_level
 'VDD2'. (LIBG-248)
```

**What Next**

Add the missing leakage_power information for the power_level.

Feedback

## LIBG-249

(error) The '%s' cell does not specify the %s group \n \twith the 'when' string (%s) for the power_level '%s'.

### Description

This message indicates the leakage_power information defined in a cell does not cover the state('when' string) for the power_level(power_rail value). For the power-level specific leakage power modeling, for each 'when' statement which appears in the leakage_power groups of a cell, there must exist a leakage_power group with that 'when' statement for every modeled power_level.

The same rule applies for the internal_power groups.

The following example shows an instance where this message occurs:

```
cell(AN2) {
      leakage_power () {
        power_level : "VDD1";
        when : "AB";
        value : 1.0;
      }
      leakage_power () {
        power_level : "VDD1";
        when : "!AB";
        value : 1.5;
      }
      leakage_power () {
        power_level : "VDD2";
        when : "AB";
        value : 1.0;
      }
      ...
  }
```

In this case, the leakage_power for the state '!AB' of power_level 'VDD2' is missing, . To fix the problem, add the leakage_power for the missing state in the cell group:

```
cell(AN2) {
      leakage_power () {
        power_level : "VDD1";
        when : "AB";
        value : 1.0;
      }
      leakage_power () {
        power_level : "VDD1";
        when : "!AB";
        value : 1.5;
      }
      leakage_power () {
        power_level : "VDD2";
```

Feedback

```
        when : "AB";
        value : 1.0;
      }
      leakage_power () {
        power_level : "VDD2";
        when : "!AB";
        value : 1.2;
      }
      ...
  }
```

The following is an example message:

```
Error: Line 12, The 'AN2' cell does not specify the leakage_power group
        with the 'when' string (!AB) for the power_level 'VDD2'.
 (LIBG-249)
```

### What Next

Add the missing leakage_power information for the power_level.

---

## LIBG-250

(error) The logic represented by the 'when' string (%s) in this\n \t%s group is not equal to or mutually exclusive with the logic represented by\n \tthe 'when' string (%s) in the %s group on line %d.

### Description

The logic represented by the 'when' string of all state-dependent or path-dependent internal_power groups between a pair of pins should be either equal or mutually exclusive for different power_levels. If it is not, Power Compiler will not be able to determine correctly which internal_power group to use when propagating a signal through the path.

The following example shows an instance where this message occurs:

```
cell(libg44) {
      area : 2;
      pin(D1 D0 S0) {
        direction : input;
        capacitance : 1;
      }
      pin(Z) {
        direction : output;
        function : "S0'D0 + S0 D1";
        ...
        internal_power() {
          power_level : VDD1;
          when : "S0'D0 + S0 D1'";
          when : "D1' S0 + S0'D0 ";
          related_pin : "S0";
```

Feedback

```
          power(scalar) { values ( "1.0" ); }
        }
        internal_power() {
          power_level : VDD2;
          when : "D1 S0 + S0'D0";
          related_pin : "S0";
          power(scalar) { values ( "2.0" ); }
        }
      }
    }
```

In this case, the second when attribute's string has a typographical error. It should be "D1 S0 + S0'D0"'; where the pin D0 should be ticked according to the sdf_cond attribute's string.

The following is an example message:

```
Error: Line 258, The logic represented by the 'when' string (S0'D0 + S0
 D1') in this
        internal_power group is not equal to or mutually exclusive with
 the logic represented by
        the 'when' string (D1 S0 + S0'D0) in the internal_power group on
 line 268. (LIBG-250)
```

**What Next**

Check the 'when' strings of both internal_power groups for wrong information and fix.

---

## LIBG-251

(error) The '%s' group has conflicting 'related_pin' with the \n \t'%s' group on line %d.

**Description**

The "related_pin" in different internal_power groups with the same power_level of the same pin must not share pins. Similarly, the related_pin in different internal_power groups without power_level definition of the same pin must not share pins.

One exception is that, for the above 2 cases, if the pins specified in the "related_pin" of the first internal_power are all included in the "related_pin" of the second internal_power, then Library Compiler will issue LBDB-107 warning and the second internal_power will overwrite the first one.

The following example shows an instance where this message occurs:

```
cell(libg44) {
      area : 2;
      pin(D1 D0 S0) {
        direction : input;
        capacitance : 1;
```

Feedback

```
    }
    pin(Z) {
      direction : output;
      function : "S0'D0 + S0 D1";
      ...
      internal_power() {
        power_level : VDD1;
        related_pin : "S0";

        power(scalar) { values ( "1.0" ); }
      }
      internal_power() {
        power_level : VDD1;
        related_pin : "S0 S1";
        power(scalar) { values ( "2.0" ); }
      }
    }
  }
```

In this case, the second internal_power group's 'related_pin' conlficts with that of the first internal_power group. One fix can be that replacing the 'related_pin' of the second internal_power group from "S0 S1" to "S1".

The following is an example message:

```
Error: Line 251, The 'internal_power' group has conflicting 'related_pin'
 with the
        'internal_power' group on line 268. (LIBG-251)
```

**What Next**

Correct the 'related_pin' strings of both internal_power groups.

---

## LIBG-252

(error) The 'function' of the '%s' pin/bus on the '%s' cell\n \tis not correctly defined.

**Description**

This error is issued when the function is not correctly defined. A valid function statement in the sequential model must satisfy the following 2 conditions:

For a ff/latch, the noninverted is called teh first variable and the inverted output is called the second variable. A function statement of each primary output must include either the first variable or the second variable.

1. It must include and can only include the the first variable or the second variable or a the valid input pins defined for the cell in addition to the boolean operators.
2. All the component insider the function statement should be separated by a space or spaces.

Feedback

The following example shows an instance where this message occurs:

```
cell(libg17) {
        area : 6.50;
        pin(CLK) {
                direction : input;
                capacitance : 1.5;
        }
        pin(SET) {
                direction : input;
                capacitance : 3.0;
        }
        ff("IQ","IQN") {
                next_state : "IQ'";
                clocked_on : "CLK";
                preset     : "SET'";
        }
        pin(Q) {
                direction : output;
                function : "IQ1";       /* wrong, should be IQ */
        }
        pin(QN) {
                direction : output;
                function : "IQN";
        }
      }
```

The following is an example message:

```
Error: Line 23, The 'function' of the 'Q'  in/bus on the 'libg17' cell
        is not correctly defined. (LIBG-252)
```

**What Next**

Make the appropriate correction in your library, as indicated in the error message. For more information on sequential models, refer to the *Library Compiler Reference* and *User Guide Manuals*.

---

## LIBG-253

(error) Cell(%s): The '%s' pin is a duplicate of '%s' internal pin.

**Description**

This message indicates that the specified internal pin is a duplicate of the other relative pin. For statetable-based sequential cells, the "IQ" function can only be defined in an internal pin with internal_node : "IQ".

The following example shows an instance where this message occurs:

```
cell(libg253) {
```

Feedback

```
                ...
                statetable(" CEN CLK ", " IQ  "){
                        table : "\
                        L L : - : L , \
                        H L : - : H , \
                        - H : - : N " ;
                }
                ...
                pin(IQ){
                        direction : internal;
                        internal_node : "IQ";
                        ...
                }
                pin(Y){
                        direction : output;
                        state_function : "IQ";
                        ...
                }
                ...
        }
```

In this case, internal pin IQ and pin Y are duplicate. The correct model should be:

cell(libg253) {

... statetable(" CEN CLK ", " IQ "){ table : "\ L L : - : L , \ H L : - : H , \ - H : - : N " ; } ... pin(IQ){ direction : output; internal_node : "IQ"; ... } ... }

The following is an example message:

```
Warning: Line 139, Cell(libg253): The pin 'SOSV' is a duplicate of the
 'IQ' internal pin. (LIBG-253)
```

**What Next**

Combine the information of the two pins into the internal pin and remove the other pin.

---

## LIBG-254

(error) An invalid '%s' %s string in the '%s' cell.

**Description**

The string given is not valid for representing a boolean logic defined in Synopsys format.

The following example shows an instance where this message occurs:

```
cell(sample) {
  switch_cell_type : coarse_grain;
  area : 1 ;
  pg_pin ("VVDD") {
        pg_type : internal_power;
```

Feedback

```
        pg_function : "(IO" ;
  }
  ...
```

In this simple case, the closing parenthesis is missing in the function string.

The following is an example message:

```
Error: Line 400, An invalid '(IO' pg_function string in the 'sample'
 cell. (LIBG-254)
```

**What Next**

Refer to your Library Compiler Reference Manual to find out the valid format of the string. It is described in the "function simple attribute" under the "Cell Group Description and Syntax".

---

## LIBG-255

(error) A bad '%s' pg_pin name in the '%s' cell.

**Description**

The error gets detected during the *read_lib* and *check_library* commands. This error occurs when either the pg_pin name is not found or an incorrect pin name in one or more attributes of the pin group is detected. The pg_pin group is not recognized and the pin name is labeled as bad.

The following example shows an instance where this message occurs:

```
cell (libg3) {
    pg_pin(VSS) {
        pg_type : primary_power;
        voltage_name : VSS;
    }
    pg_pin(VDD) {
        pg_type : primary_power;
        voltage_name : VDD;
    }
    pg_pin(VVDD) {
        pg_type : internal_power;
        voltage_name : VDD1;
        pg_function : "E";       /* <-- This is wrong */
    }
 }
```

The following is an example message:

```
Error: Line 426, A bad 'E' pg_pin name in the 'libg3' cell. (LIBG-255)
```

Feedback

### What Next

Check your library for an incorrect pin name in a related attribute or for an incorrect pin group.

---

## LIBG-256

(error) The '%s' group is overlapping with the '%s' group on line %d.

### Description

This message indicates that two *dynamic_current* groups in ccs power are overlapping with each other.

If following three conditions is true, then two groups are considered to be overlapped :

1. Two groups have at least one pin specified in *related_inputs* attribtue is identical.
2. Both have same list of pins specified in *related_outputs* or both have no *related_outputs* attribtue.
3. *When* atrributes specified in both groups are not mutually exclusive, or both groups have no *when* attribute.

The following example shows an instance where this message occurs:

```
cell(libg256) {
  ...
  pin(A1) {
    direction : input;
    ...
  }
  pin(A2) {
    direction : input;
    ...
  }
  pin(A3) {
    direction : input;
    ...
  }
  pin(A4) {
    direction : input;
    ...
  }
  pin(A5) {
    direction : input;
    ...
  }
  pin(ZN1) {
    direction : output;
    ...
```

Feedback

```
}
pin(ZN2) {
  direction : output;
  ...
}
...
dynamic_current() {
  when : "A2";
  related_inputs : "A5 A4";
  related_outputs : "ZN2 ZN1";
  switching_group() {
    pg_current(V2) {
    ...
    }
    ...
  }
}
dynamic_current() {
  when : "!A1";
  related_inputs : "A4 A3";
  related_outputs : "ZN1 ZN2";
  switching_group() {
    pg_current(V2) {
    ...
    }
    ...
  }
}
...
```

In this case, "A2" and "!A1" specified in two when attributes respectively are not mutually exclusive, pin "A4" is specified in both related_inputs groups, and same list of pins ("ZN1" and "ZN2") is specified in both related_outputs groups. Two dynamic_current are overlapping. You can modify any of these attributes to fix the problem. For example : change "A4 A3" to "A2 A3".

The following is an example message:

```
Error: Line 79, The 'dynamic_current' group is overlapping with the
 'dynamic_current' group on line 88. (LIBG-256)
```

**What Next**

Check *related_inputs*, *related_outputs*, and *When* attributes for all *dynamic_current* groups to see if any of them breaks above rules, and make the necessary correction.

---

## LIBG-257

(error) The %s must be set to 0.0 or omitted.

Feedback

### Description

There are two attributes, input_high_value and input_low_value, within a gate_leakage group. The input_high_value is to specify gate leakage current on a pin when this pin is set to high. The input_low_value is to specify gate leakage current on a pin when this pin is set to low. If the 'when' statement inside leakage_current or va_leakage_current implies that the pin high never occurs, then input_high_value shall be set to 0.0 or ignored. If the 'when' statement implies that the pin low shall never occur, then input_low_value must be set to 0.0 or ignored.

The following example shows an instance where this message occurs:

```
leakage_current() {
     when : "A1 & A2 & !ZN";
     ...
     gate_leakage(A2) {
       input_high_value : 2.1;
       input_low_value : -1.7;
     }
     gate_leakage(A1) {
       input_high_value : 7.1;
       input_low_value : -8.7;
     }
  }
```

In this case, we have when condition "A1 & A2 & !ZN", which means pins A1 and A2 shall never set to low because they will never occur within this leakage_current. So, input_low_value is meaningless for both pins. We shall set both to 0.0 or comment them out.

The following is an example message:

```
Error: Line 210, The input_low_value must be set to 0.0 or omitted.
 (LIBG-257)
```

### What Next

Check *input_low_value* and *input_high_value*, and make the necessary correction.

---

## LIBG-258

(error) No %s is allowed within an intrinsic_parasitic\n \tif power_cell_type is %s.

### Description

If the power_cell_type is macro, then followings groups can't be specified within an intrinsic_parasitic : state-dependent condition, intrinsic_resistance and intrinsic_capacitance.

Feedback

If the power_cell_type is standard cell or there is no power_cell_type specified, then total_capacitance can't be specified within an intrinsic_parasitic.

The following example shows an instance where this message occurs:

```
cell (test) {
  intrinsic_parasitic() {
    when : "!A1 & A2 & !ZN";
    total_capacitance(V2) {
       value : 9.0;
    }
  }
 ...
}
```

In this case, there is no power_cell_type specified under cell "test", meaning power_cell_type is defaulted to standard cell. The total_capacitance is not allowed within an intrinsic_parasitic group.

```
cell (test) {
  power_cell_type : macro;
  intrinsic_parasitic() {
    when : "!A1 & A2 & !ZN";
    intrinsic_resistance(G1) {
       related_output : "ZN";
       value : 9.0;
    }
    intrinsic_capacitance(G1) {
       value : 8.2;
    }
  }
 ...
}
```

In this case, 'when' attribute, 'intrinsic_resistance' and 'intrinsic_capacitance' are not allowed in an 'intrinsic_parasitic' group because the 'power_cell_type' is 'macro'.

The following is an example message:

```
Error Line 272, No intrinsic_resistance is allowed within an
 intrinsic_parasitic
        if power_cell_type is macro. (LIBG-258)
```

**What Next**

Check the library source file, and make the necessary correction.

---

## LIBG-259

(error) The power_cell_type must be macro if\n \tmin_input_switching_count and max_input_switching_count are defined.

Feedback

### Description

If min_input_switching_count and max_input_switching_count are defined within switching_group, then power_cell_type defined in cell must be macro.

The following example shows an instance where this message occurs:

```
cell (test) {
 ...
 power_cell_type : stdcell;
 dynamic_current() {
   ...
   switching_group() {
      min_input_switching_count : 1;
      max_input_switching_count : 9;
      ...
   }
   ...
 }
 ...
}
```

In this case, min_input_switching_count and max_input_switching_count both are defined under one of switching_group groups within a dynamic_current, and power_cell_type is standard cell, which is wrong. We shall change "stdcell" to "macro" to avoid the error.

The following is an example message:

```
Error Line 272, The power_cell_type must be macro if
        min_input_switching_count and max_input_switching_count are
 defined. (LIBG-259)
```

### What Next

Check the library source file, and make the necessary correction.

---

## LIBG-260

(error) The logic represented by the %s string is not equal\n \tto or mutually exclusive with the logic represented by the %s string.

### Description

The logic represented by one programmable driver type string must be not equal to or mutually exclusive with the logic represented by other programmable driver types under the same pin.

The following example shows an instance where this message occurs:

```
cell(libg260) {
     pad_cell : true;
```

Feedback

```
...
 pin(A1) {
  direction : input;
  is_pad : true;
 ...}
 pin(A2) {
  direction : input;
  ...}
 pin(A3) {
   direction : input;
  ...}
 pin(ZN1) {
  direction : inout;
  max_capacitance : 0.1;
  pull_up_function : "!A1 * !A2 * !A3";
  pull_down_function : "A1 * A2 * !A3";
  bus_hold_function : "A1 * !A2 * !A3";
  open_drain_function : "!A1 * A2 * !A3";
  open_source_function : "!A1 * !A2 * A3";
  resistive_function : "A1 * A2 * A3";
  resistive_0_function : "A1 * A2";
  resistive_1_function : "!A1 * A2 * A3";
 ...}
}
```

The following are example messages:

```
Error: Line 102, The logic represented by the resistive_0_function string
 is not equal
        to or mutually exclusive with the logic represented by the
 pull_down_function string. (LIBG-260)

Error: Line 102, The logic represented by the resistive_0_function string
 is not equal
        to or mutually exclusive with the logic represented by the
 resistive_function string. (LIBG-260)
```

### What Next

```
In this case, the resistive_0_function is not mutually exclusive with
pull_down_functioni and resistive_function.  To fix the problem, you
might want to change it as following :

       resistive_0_function : "A1 * !A2 * A3";
```

---

## LIBG-261

(error) The programmable driver type can only be defined on a pad cell.

Feedback

### Description

The programmable driver type can only be defined a pad cell, which has either pad_cell or auxiliary_pad_cell attribute.

The following example shows an instance where this message occurs:

```
cell (libg261) {
    ...
    pin (A) {
      direction : input;
      pull_up_function : "p1 * p2";
      ...
    }
   ...}
```

In this case, a programmable driver type, pull_up_function, is defined on pin A, which is under a cell, ligb261. The cell is not a pad cell, which is wrong. To fix the problem, please either add pad_cell attribute under cell level or remove pull_up_function attribute from the pin.

The following is an example message:

```
Error: Line 94, The programmable driver type can only be defined on a pad
 cell. (LIBG-261)
```

### What Next

Check the library source file, and make the necessary correction.

---

## LIBG-262

(information) Cell(%s) is recognized as one-hot mux cell.

### Description

This message is issued when a cell is recognized as one-hot mux cell.

One-hot mux cells have following characteristic: 1. Two equal sized, disjoint sets of input pins, one is select pins(e.g. "S0,S1"), another set is data input pins(e.g. "D0,D1"). 2. Have cell attribute "contention_condition", e.g. Fc = S0'*S1 + S0*S1. That is there could be one and ONLY one select pin can be logic 1, others are logic 0. 3. There is only one output pin defined and its function is defined with respect to all of its inputs, e.g. Fo = S0*D0 + S1*D1;

The following example shows an instance where this message occurs:

```
cell(OHMUX2) {

  ...
  contention_condition : "( S0 S1 + S0' S1' )";
```

Feedback

```
  pin(D0) {
    direction : input;
    capacitance : 1;
  }
  pin(D1) {
    direction : input;
    capacitance : 1;
  }
  pin(S0) {
    direction : input;
    capacitance : 1;
  }
  pin(S1) {
    direction : input;
    capacitance : 1;
  }
  pin(Z) {
    direction : output;
    function : "(S0 D0 + S1 D1)";
    ...
  }
}
```

The following is an example message:

```
Warning: Line 139, Cell(OHMUX2) is recognized as one-hot mux
 cell.(LIBG-262)
```

### What Next

The one-hot mux cell will be used in DC logical synthesis.

---

## LIBG-263

(warning) Cell(%s) is not recognized as one-hot mux cell.

### Description

This message is issued when there is cell level attribute "contention_condition" defined, but it does not match with output pin function as a one-hot mux cell.

One-hot mux cells have following characteristic: 1. Two equal sized, disjoint sets of input pins, one is select pins(e.g. "S0,S1"), another set is data input pins(e.g. "D0,D1"). 2. Have cell attribute "contention_condition", e.g. Fc = S0'*S1 + S0*S1. That is there could be one and ONLY one select pin can be logic 1, others are logic 0. 3. There is only one output pin defined and its function is defined with respect to all of its inputs, e.g. Fo = S0*D0 + S1*D1;

This warning message is reported in step 3, when it find the output function with restiction of contention function does not match with the function of one-hot mux cell.

Feedback

The following example shows an instance where this message occurs:

```
cell(OHMUX2) {

  ...
  contention_condition : "( S0 S1 + S0' S1' )";
  pin(D0) {
    direction : input;
    capacitance : 1;
  }
  pin(D1) {
    direction : input;
    capacitance : 1;
  }
  pin(S0) {
    direction : input;
    capacitance : 1;
  }
  pin(S1) {
    direction : input;
    capacitance : 1;
  }
  pin(Z) {
    direction : output;
    function : "(S0 D0 + S1 D1 + D0' D1)";
    ...
  }
}
```

In this case, the contention function matches with one-hut mux cell select pin restriction, but the output pin function is not a function of one-hot mux cell.

The following is an example message:

```
Warning: Line 139, Cell(OHMUX2) is not recognized as one-hot mux
 cell.(LIBG-263)
```

**What Next**

If users intend to define the cell as one-hot mux cell, and use it later on in logical synthesis, check the cell contention_condition definition and output pin function, modify to make them match the definition of one-hot mux cell.

If users won't use the cell as one-hot mux cell, just ignore this warning.

---

## LIBG-264

(warning) ICG cell '%s' function can not be recognized during functional optimization by Power Compiler.

Feedback

### Description

A cell with attribute "clock_gating_integrated_cell" defined as "generic" is called a generic ICG cell.

Library Compiler will try to resolve ICG type of a generic ICG cell. Valid resolved types are,

latch_(posedge | negedge | posedgeactivelow | negedgeactivelow )_ (precontrol | postcontrol| *)_(obs | *)_(invgclk | *) or none | (posedge | negedge | posedgeactivelow | negedgeactivelow )_ (control | *)_(invgclk | *).

Please refer to Library Compiler User Guide for the corresponding schematics of these types.

If a generic ICG cell can't be resolved as any exact type, Library Compiler will resolve them as blackbox. It means this generic ICG will not be auto inferred by Power Compiler and understood by Formality and other client tools that need to understand the function of such cells.

FF-based ICG cell won't be supported by Power Compiler, either.

The following example shows an instance where this message occurs:

The following is an example message:

```
Warning: ICG cell 'cell1' function can not be recognized during
 functional optimization by Power Compiler. (LIBG-264)
```

### What Next

Change the cell to be one of the valid ICG types.

---

## LIBG-265

(information) Derived ICG type for cell '%s' is '%s'.

### Description

A cell with attribute "clock_gating_integrated_cell" defined as "generic" is called a generic ICG cell.

Library Compiler will try to resolve ICG type of a generic ICG cell. If successful, the value of attribute "clock_gating_integrated_cell" will be updated.

The following is an example message:

```
Information: Derived ICG type for cell 'cell1' is 'latch_posedge'.
 (LIBG-265)
```

Feedback

## LIBG-266

(warning) The 'data_in_type' attribute on the '%s' pin/bus in\n \tthe '%s' cell is inconsistent with its function. The attribute is ignored.

### Description

This warning message occurs when the *data_in_type* attribute does not match the functional information specified in the cell's sequential state description.

In a pin group, *data_in_type* defines the type of *data_in* attribute to be used in an ff group, a seq group, or a ff_bank group. Any pin with the *data_in_type* attribute must be in the *data_in* function. The tool performs a consistency check between the pin's *data_in_type* attribute and the *data_in* function and generates this warning when inconsistencies are found.

The following is an example message:

```
Warning: Line 153, The 'data_in_type' attribute on the 'SET' pin/bus in
        the 'libg35' cell is inconsistent with its function. The
 attribute is ignored. (LIBG-266)
```

### What Next

This is only a warning message. No action is required.

However, to avoid this warning, check your library for a pin that has a *data_in_type* attribute but is not used in the *data_in* statement of a sequential group. After making your changes, run the command again.

## LIBG-267

(error) The power_down_function attribute should not be specified inside the test_cell's ff/latch/statetable group.

### Description

This error message occurs because the *power_down_function* attribute must be specified in a cell's ff/latch/statetable group, not specified inside the test_cell.

The following example shows an instance where this message occurs: The following example results in this error message:

```
test_cell () {
         pin (Q) {
            direction : output;
            function : "IQ";
            signal_type : test_scan_out;
         }
         pin (CK) {
```

Feedback

```
            direction : input;
         }
         pin (D) {
            direction : input;
         }
         pin (SI) {
            direction : input;
            signal_type : test_scan_in;
         }
         pin (SE) {
            direction : input;
            signal_type : test_scan_enable;
         }
         ff (IQ,IQN) {
            clocked_on : "!(CK)";
            next_state : "D";
            power_down_function : "!VDD + VSS"; /* <-- This is wrong */
         }
      }
```

The following is an example message:

Error: Line 426, Cell 'test', The power_down_function attribute should not be specified inside the test_cell's ff/latch/statetable group. (LIBG-267)

**What Next**

Check your library to delete the *power_down_function* attribute in test_cell.

---

## LIBG-268

(error) A function cannot point to two different flip-flop or latch groups.

**Description**

This error message occurs when the Boolean expression in the *function* attribute of this pin contains more than one internal node of flip-flop or latch. Only one internal node is allowed.

**What Next**

Specify individual internal nodes separately as shown in the following example:

```
pin(temp) {
   direction :internal;
   function : "Q1";
   reference_input : "RET CK q1";
   ...
pin(Q) {
   direction : output;
   function : "Q2 + temp";
   reference_input : "RET CK q1";
```

Feedback

```
    ...
}
```

## LIBG-269

(error) The '%s' has a count of %d, which does not match\n \tthe size %d specified.

### Description

This error message occurs when the specification of the total number of reference_input is not matched to total number of internal nodes specified in ff/latch/ff_bank/latch_banck group.

### What Next

Check the library source file and correct the problem by grouping the values together in the required format.

## LIBG-270

(error) The variable in the Boolean expression is not allowed.

### Description

This error message occurs because some variables, such as IQ or IQN, in flip-flop or latch groups cannot be specified in a functional expression if the group is referred by reference_input defined in the pin.

The following example shows an instance where this message occurs: In the following example, latch 1 is referred by 2 pins, internal_1 and internal_2. That means the variables IQ_0 and IQN_0 can be referred by either the internal_1 or internal_2 pin. In latch 2, data_in is connected to IQ_0, which is not allowed because it could be either the internal_1 or internal_2 pin.

```
pin(output_1) {
    direction : "output";
    function : "IQ_1";
    reference_input : "IQ_0";
  }
  pin(internal_1) {
    direction : "internal";
    function : "IQ_0";
    reference_input : "D CP";
  }
  pin(internal_2) {
    direction : "internal";
    function : "IQ_0";
    reference_input : "D1 CP1";
  }
```

Feedback

```
  latch ("p1 p2","IQ_0","IQN_0") {    /* latch 1 */
     data_in : "p1";
     enable : "p2";
  }
  latch ("p1", "IQ_1","IQN_1") {      /* latch 2 */
     data_in : "p1";
     enable : "CP";
  }
```

The following is an example message:

```
Error: Line 160, The variable in boolean expression is not allowed.
 (LIBG-270)
```

### What Next

Check the library source file, and correct the problem by grouping the values together in the required format.

---

## LIBG-271

(warning) The disable value of the retention pin might not be valid.

### Description

This warning message occurs because the disable value that you provided might be incorrect. The tool ties up the disable values from all logic expression functions to see if the results are reasonable.

The following example shows an instance where this message occurs: The following example shows that data_in becomes logic "0" when the RESTORE pin is tied up to disable value "0". The 2 valid pins in data_in are "D" and "RESTORE". The expected data_in after tie up to the disable value is "D", not logic "0". So, in this case, the tool issues the warning message because it is possible that the disable value might be invalid.

```
pin (RESTORE) {
    direction : "input";
    retention_pin("restore",0);
    ...
  }
  pin (D) {
    direction : "input";
    ...
  }
  latch ("IQ","IQN") {
     data_in : "D * RESTORE";
     ...
  }
```

Feedback

The following is an example message:

```
Warning: Line 160, The disable value of retention pin might not be valid.
(LIBG-271)
```

**What Next**

Check the library source file to see if the disable value is valid.

---

## LIBG-272

(information) Library '%s' is being compiled with pin_number recomputation enabled.

**Description**

The pin number of the cells in this library will be assigned by a consistent rule, so the pin number can be assigned consistently when the cells are logically equal but have different function expression or pin orders in .lib are different.

The following is an example message:

```
Information:  Library 'lib_example' is being compiled with pin_number
 recomputation enabled.(LIBG-272)
```

---

## LIBG-273

(error) the logic represented by the %s string \n \t(%s) is not mutually exclusive with the logic represented by\n \tthe %s string (%s) on line %d.

**Description**

The logic represented by the clamp_*_function and illegal_clamp_condition string should be mutually exclusive. If it is not, Library Compiler is not be able to determine which value the cell will clamp to.

The following example shows an instance where this message occurs:

```
cell(libg273) {
      area : 2;
      ...
      pin(Z) {
        direction : output;
        function : "S0'D0 + S0 D1";
        clamp_0_function : "D0";
        clamp_1_function : "D0";
        clamp_latch_function : "!D0";
        ...
      }
    }
```

Feedback

In this case, the clamp_0_function string is not exclusive with clamp_1_function string.

The following is an example message:

```
Error: Line 258, Cell 'libg273', pin 'Z', the logic represented by the
clamp_0_function string (D0) is notmutually exclusive with the logic
 represented
by the clamp_1_function string (D0) on line 268. (LIBG-273)
```

**What Next**

Check the clamp_*_function and illegal_clamp_condition strings of both timing groups for wrong information and fix.

---

## LIBG-274

(error) You are not allowed to specify the function attribute on the single-bit output pin of the test_cell group of a parallel sequential cell.

**Description**

In parallel sequential cells, the single-bit output pin is only a test output pin. Therefore, do not specify the *function* attribute on this pin, in the *test_cell* group.

This message identifies a parallel sequential cell that has a single-bit output pin with the *function* attribute.

The following example shows an instance where this message occurs:

```
cell (LSFDS) {

  ...

  test_cell() {
    bus (D) {
      bus_type : BUS4;
      direction : input ;
      capacitance : 2.000000 ;
    }
    pin (CK) {
      direction : input ;
      capacitance : 1.000000 ;
    }
    ff_bank (IQ,IQN,4) {
      next_state : "D" ;
      clocked_on : "CK" ;
    }
    bus (Q) {
      bus_type : BUS4;
      direction : output ;
      function : "IQ" ;
    }
```

Feedback

```
    pin (SO) {
      direction : output ;
      signal_type : "test_scan_out";
      function : "IQ[3]" ;
    }
  }
}
```

In this example, the *function* attribute should not be defined on the pin 'SO'.

The following is an example message:

```
Error: Line 302, Cell 'test', You are not allowed to specify the function
 attribute on the
single-bit output pin of the test_cell group of a parallel sequential
 cell. (LIBG-274)
```

### What Next

Check your library, and delete the *function* attribute specified on the single-bit output pin.

---

## LIBG-275

(information) Complex function sequential cell will skip\n sequential cell timing consistency check.

### Description

This message is to infer user that a complex function seuqential cell which contains mux/xor/maj with Boolean logic will skip sequential cell timing consistency check.

The following example shows an instance where this message occurs:

```
cell (test) {

    ...

    ff(IQ,IQN) {
      clocked_on : "CK" ;
      next_state : "RETN & ((SE SI) + (!SE D))" ;
    }
}
```

The following is an example message:

```
Information: Line 118, Cell 'SDRFFQ_X1M_A12TR', Complex function
 sequential cell will skip sequential cell timing consistency check.
 (LIBG-275)
```

### What Next

The cell timing consistency need to be checked by user.

Feedback

## LIBG-276

(error) Port '%s', only one %s can be the synchronized load enable pin in the cell.

### Description

The synchronized load enable pin is the pin with attribute *nextstate_type : load*. A synchronized load enable register cell should have only one load enable pin. So only one pin/bus/bundle should be specified with this attribute.

This message identifies a sequential cell that has more than one pin/bus/bundle being specified with attribute *nextstate_type : load*.

The following example shows an instance where this message occurs:

```
cell (test) {

  ...

  ff (IQ,IQN) {
      next_state : "E*D + !E*IQ" ;
      clocked_on : "CK" ;
  }
  pin (D) {
      direction : input ;
      nextstate_type : load ;
  }
  pin (E) {
      direction : input ;
      nextstate_type : load ;
  }
  pin (CK) {
      direction : input ;
      capacitance : 1.000000 ;
  }
  bus (Q) {
      direction : output ;
      function : "IQ" ;
  }
}
```

In this example, the *nextstate_type : load* attribute should not be defined on the pin 'D', only pin 'E' should have this attribute.

The following is an example message:

```
Error: Line 302, Cell 'test', Only one pin/bus/bundle can be the
 synchronized load enable pin in the cell. (LIBG-276)
```

Feedback

### What Next

Check your library, and correct the *nextstate_type : load* attribute specified on the wrong pin.

---

## LIBG-277

(warning) The port '%s' is not working as a synchronized load enable pin,\n \tthe cell can not be degenerated into non-synchronized load enable register cell.

### Description

The synchronized load enable(SLE) pin is the pin with attribute *nextstate_type : load*. When SLE pin is active, the register allows new values to load at the data input. When SLE pin is inactive, the register output is holding unchanged.

So when the load enable pin is at inactive level, the simplified next_state should be a Boolean expression with IQ; when the load enable pin is at active level, the simplified next_state should be a Boolean expreassion without IQ, but not tie to "1" or "0".

This message identifies the SLE pin doesn't satisfy the above conditions.

The following example shows an instance where this message occurs:

```
cell (test) {

  ...

  ff (IQ,IQN) {
      next_state : "E*D + IQ" ;
      clocked_on : "CK" ;
  }
  pin (D) {
      direction : input ;
      nextstate_type : data ;
  }
  pin (E) {
      direction : input ;
      nextstate_type : load ;
  }
  pin (CK) {
      direction : input ;
      capacitance : 1.000000 ;
  }
  bus (Q) {
      direction : output ;
      function : "IQ" ;
  }
}
```

In this example, the *next_state* attribute should be correct to "E*D + !E*IQ"

Feedback

The following is an example message:

```
Warning: Line 302, Cell 'test', The port 'E' is not working as a
 synchronized load enable pin, the corresponding non-SLER degenerated
 model can't be generated. (LIBG-277)
```

### What Next

Check your library, and correct the "next_state" attribute or the SLE pin.

---

## LIBG-278

(warning) The sequential element %s variable '%s' is unused in the cell function,\n \tthe '%s' attribute should not be specified and will be removed.

### Description

This message indicates the sequential cell has both clear_preset_var1 and clear_preset_var2, but one of them is unused and will be removed from the library.

If the sequential element second var (e.g. IQN) is unused in cell function, then we don't need clear_preset_var2 to be specified. If the sequential element first var (e.g. IQ) is unused in cell function, then we don't need clear_preset_var1 to be specified.

You will get this message if you specify the unused clear_preset_var, and it will be removed in the process.

The following example shows an instance where this message occurs:

```
cell(libg278) {
        area : 11;
        pin(D) {
                direction : input;
                capacitance : 1;
        }
        pin(CP) {
                direction : input;
                capacitance : 1;
        }
        pin(CLR) {
                direction : input;
                capacitance : 1;
        }
        pin(SET) {
                direction : input;
                capacitance : 1;
        }
        ff("IQ","IQN") {
                next_state : "D";
                clocked_on : "CP";
                clear   : "CLR'";
```

Feedback

```
                preset   : "SET'";
                clear_preset_var1 : L;
                clear_preset_var2 : L;  /* this is unsed as IQN is unused
 */
        }
        pin(Q) {
                direction : output;
                function : "IQ";
                timing() {
                        timing_type : rising_edge;
                        intrinsic_rise : 1.0;
                        intrinsic_fall : 1.0;
                        rise_resistance : 0.1;
                        fall_resistance : 0.1;
                        related_pin : "CP";
                }
                timing() {
                        timing_type : clear;
                        timing_sense : positive_unate;
                        intrinsic_fall : 1.0;
                        fall_resistance : 0.1;
                        related_pin : "CLR";
                }
                timing() {
                        timing_type : preset;
                        timing_sense : negative_unate;
                        intrinsic_fall : 1.0;
                        fall_resistance : 0.1;
                        related_pin : "SET";
                }
        }
}
```

The following is an example message:

```
Warning: Line 111, The sequential element second variable 'IQN' is unused
 in the cell function,\n
\tthe 'clear_preset_var2' attribute should not be specified and will be
 removed. (LIBG-278)
```

**What Next**

Edit the .lib and remove the unused clear_preset_var1/var2

---

## LIBG-280

(information) The cell '%s' contiains memory/memory_read/memory_write groups which are deprecated, use the is_memory_cell attribute instead.

Feedback

### Description

The groups are now obsolete and are no longer used by Synopsys tools. Use is_memory_cell attribute for memory cell. For example: cell (my_memory_cell) { is_memory_cell : true ; }

By default, bypass the checking for the groups(memory/memory_read/memory_write) for the same reason. Set the gvar lc_enable_memory_group_check to true to bring back the checking.

### What Next

Remove the deprecated groups(memory/memory_read/memory_write) and add is_memory_cell attribute for the cell.

---

## LIBG-281

(warning) The when condition of '%s' has been changed to '%s', by removing output pin(s) from the Boolean expression.

### Description

The string associated with a *when* attributes for receiver_capacitance includes pins of *output* directions . Tool will remove all the output pins from when attribute and copy original when to char_when if there is no char_when attribute.

The following example shows an instance where this message occurs:

```
cell (libg281) {
  area : 4
  pin(A2) {
    direction : input
    receiver_capacitance() {
     when : "A2&!B&ZN" ;
    }
  }
  pin(B) {
    direction : input
  }
  pin (ZN) {
    direction : output
  }
  }
```

The following is an example message:

```
Error: Line 431, The when condition of 'A2&!B&ZN' has been changed to 'A2
 B'', by removing output pin(s) from the Boolean expression. (LIBG-281)
```

Feedback

**What Next**

Check your library and remove output pins from when condition

# LIBMAP

## LIBMAP-1

(error) %s Only 1 file match in libmap is allowed for RTL file '%s', but %d were found.

**Description**

More than 1 file was matched by libmap following the specified patterns. Make sure each RTL matches are unique to avoid ambiguities.

**What Next**

Re-specify the libmap so that each rule only univocally matches one RTL file.

## LIBMAP-2

(error) %s Only 1 directory match in libmap is allowed for RTL file '%s', but %d were found.

**Description**

More than 1 directory was matched by libmap following the specified patterns. Make sure each RTL matches are unique to avoid ambiguities.

**What Next**

Re-specify the libmap so that each rule only univocally matches one directory for the specified RTL file.

## LIBMAP-3

(error) %s Only 1 wildcard match in libmap is allowed for RTL file '%s', but %d were found.

**Description**

More than 1 wildcard was matched by libmap following the specified patterns. Make sure each RTL matches are unique to avoid ambiguities.

**What Next**

Re-specify the libmap so that each rule only univocally matches one wildcard.

Feedback

## LIBMAP-4

(error) %s Configuration declaration is not supported in libmap file '%s' at line %d.

**Description**

Libmap configuration is not supported.

**What Next**

Remove or comment any config/endconfig block from the provided libmap.

## LIBMAP-5

(Warning) %s Only spice library is supported in libmap file '%s' at line %d, but '%s' was given.

**Description**

Only spice libraries are supported.

**What Next**

Remove any other library types than 'spice'.

## LIBMAP-6

(error) Libmap files are not supported for VHDL.

**Description**

Libmap support is not enabled for using VHDL.

**What Next**

Remove -libmap flag.

## LIBMAP-7

(warning) Libmap file is ignored if -library option is used in analyze command.

**Description**

If both -libmap option and -library option is specified to the analyze command, -library option takes precedence and -libmap is ignored.

**What Next**

Use only one of -library or -libmap option.

Feedback

## LIBMAP-8

(error) %s Directory for library '%s' recognized in libmap file has not been declared.

**Description**

Libmap file was matched to a library that has not been declared.

**What Next**

Declare the library and its path using *define_design_lib*.

## LIBMAP-9

(error) %s Invalid import in libmap file '%s' at line %d, a semicolon is required.

**Description**

A semicolon is required after library import.

**What Next**

Add a semicolon after library import, for example, 'include libmap.map;', and try again.

## LIBMAP-10

(error) %s Corrupt libmap include chain.

**Description**

Libmap include failed because of invalid library definition.

**What Next**

Add a semicolon after each library import, and make sure each imported libmap file exists and is readable.

## LIBMAP-11

(error) Libmap provided file for -libmap argument in analyze command cannot be empty.

**Description**

The -libmap argument requires the definition of *libmap_file* in order to work. Declare the file first.

**What Next**

Remove -libmap flag, or provide a file for libmap to process.

Feedback

## LIBMAP-12

(error) %s Provided libmap file '%s' cannot be loaded. It does not exist or is not a regular file.

**Description**

Provided libmap file from -libmap argument is not a regular file, thus it cannot be processed.

**What Next**

Make sure the provided libmap file exists or it path is correctly defined.

## LIBMAP-13

(error) %s Invalid libmap library definition '%s' at line %d, a semicolon is required.

**Description**

A semicolon is required after library definition.

**What Next**

Add a semicolon after library definition, for example, 'library lib1 rtl/sub1.sv;', and try again.

## LIBMAP-14

(error) %s Missing ',' as path separator between '%s' and '%s' of file '%s' at line %d.

**Description**

Path separator for libmap definition is missing.

**What Next**

Add a comma to separate each path when describing libraries in libmap file.

# LLE

## LLE-001

(Warning) Overlay file has duplicated definition at file %s line %d for the corner, this line will be ignored.

Feedback

**Description**

LLE overlay side file definition is based on cell namd and TR number, this warning means the same definition(same cell name and same TR number) has already been defined for the same corner. This line's duplicated definition will be ignored.

**What Next**

Check and correct the lle overlay side files.

# LM

## LM-001

(warning) %d missing %s%s across all libraries.

**Description**

The *check_workspace* command found some missing arcs or checks. This is usually due to inconsistent use of arc sense (like clear_low versus clear2_low), modes (multiple arcs with separate modes versus one arc with multiple modes), or sometimes true missing arcs.

**What Next**

The *check_workspace* command used with the *-allow_missing* option can auto-correct many of these situations. There is a modes solver, a sense2 solver, and true missing arcs can sometimes be duplicated in the base library. For true missing arcs/checks (and library cells), you can also choose to ignore them. See the documentation for the *check_workspace* command.

If after *check_workspace -allow_missing* there are still errors, your library source files may need to be revisited.

## LM-002

(information) Fixed all missing %ss.

**Description**

The missing objects of the given type were all resolved.

**What Next**

No action is needed.

Feedback

## LM-003

(information) Verifying missing arcs/checks fix...

**Description**

After missing arcs/checks have been resolved, we re-verify that the libraries are all aligned. This is just an informational message to tell you that this verification has begun.

**What Next**

No action needed.

## LM-004

(error) Timing/power object verification across libs failed.

**Description**

Verification of objects used for timing (arc, check, receiver capacitance, etc.) or power (leakage, dynamic) failed. Previous messages would give you an indication of the problems encountered.

**What Next**

You may need to make some changes to your setup in order to get *check_workspace* to succeed. Passing additional arguments to *check_workspace* to allow missing objects or missing library cells might help. Changing (or removing) scaling groups is a possible fix because you cannot have missing objects in the context of scaling groups. Removing some problem library cells or libraries might also work as a last resort.

## LM-005

(information) Only one logical library loaded - internal checks only

**Description**

You entered the *check_workspace* command with only one DB library in memory. This message is just telling you that there will be no checks between logical libraries because there are no logical libraries to compare. Comparisons between logical and physical libraries will still occur.

**What Next**

No action needed.

Feedback

## LM-006

(warning) Library %s has %d duplicate %s(s).

### Description

The *check_workspace* command found duplicate DB objects such as arcs, leakage/dynamic power records, cell em records, receiver capacitance records, or ccsn records. Duplicates are defined as objects that share the same signature, which means that there is no way to tell them apart. The *check_workspace* command will try to make the objects unambiguous and match objects of this type across libraries.

### What Next

Sometimes, duplicates are allowed. Therefore, the tool will try to make the duplicates unambiguous. If this succeeds and all duplicates are resolved, no action is needed. If there are still duplicates, the check will fail. In that case, use the *check_workspace -details* with the appropriate detail value to report the specific objects that are duplicates.

## LM-007

(information) Duplicate %s: %s

### Description

The checker found a duplicate object, and *check_workspace* was call with the *-details* option to provide more verbose information. This message is emitted to show the specific object which was ambiguous.

### What Next

This is a companion message for LM-006. See the description there.

## LM-008

(information) %d libraries need their breakpoints repaired.

### Description

While comparing NLDM between related attributes in the same library, breakpoints were found that either have the same variables but different dimensions, an extra variable, or a split pair (for example, one attribute has a table which is function of Tclk and the other has Tdat). These can be automatically repaired. The message tells you that *check_workspace* will correct these issues for some number of libraries.

### What Next

No action needed.

Feedback

## LM-009

(error) Unfixable %s '%s' because\n \t%s.

### Description

While comparing two arcs/checks across libraries, a serious difference was found between the delay data for the two objects being compared. The message will detail the irreparable problem. For example, the data might exist in one library and be missing from the one being compared. This will cause check_workspace to fail, and some action (usually by changing the source library) is needed to fix it.

### What Next

Using the information from the message, you can try to adjust the .lib source. It is also possible that you have created a scaling group which tightens the restrictions on missing data across libs. You might be able to get a library created without scaling groups.

## LM-010

(error) No libraries to %s

### Description

You tried to do some operation which requires that there are some libraries in memory, and you have not yet loaded any. For example, you tried to save the library to disk and no logical or physical library files have been loaded. Or, you tried to *check_workspace* without having a base logical library to which other logical and physical libraries would be compared.

## LM-011

(error) You must check your workspace before committing

### Description

You will get this message from *commit_workspace if the workspace needs to be checked. It could be that you have not run check_workspace yet. Or, it could be that you had a successful check, then either changed the PVT configuration, or added or removed files from the workspace. Anything that changes the state of the workspace will require a check before committing.*

### What Next

The workspace must be checked and clean before committing it to disk.

Feedback

## LM-012

(error) The checker found problems which must be resolved before committing

### Description

You tried to save the library without first resolving the issues found by the checker using additional options for *check_workspace*.

### What Next

Libraries must be checked and clean before saving to disk.

## LM-013

(error) %s thresholds do not match

### Description

While creating a scaling group, some thresholds from one of the libraries did not match the first library in the group. These could be the thresholds defined at the library level or pin-specific thresholds. Each library in the group must agree for the threshold values input_threshold_pct_fall, input_threshold_pct_rise, output_threshold_pct_fall, output_threshold_pct_rise, slew_lower_threshold_pct_fall, slew_lower_threshold_pct_rise, slew_upper_threshold_pct_fall, slew_upper_threshold_pct_rise, and slew_derate_from_library, as defined in Liberty.

This will cause *create_scaling_group* to fail.

### What Next

You can use *set_thresholds* to change the thresholds for libraries and lib cell pins so that they match. However, you should to go back to the source of the DB libraries and ensure that these libraries are compatible.

### See Also

- create_scaling_group
- set_thresholds
- report_thresholds

## LM-014

(information) Library %s:%s is missing %d design(s)

Feedback

### Description

In comparison to the superset of all designs found in all logical libraries, the given library is missing some designs. This is a summary which was preceded by specific warnings (for designs missing in a downstream library) or errors (for designs missing in the base library). For example, if the base library contains the design AN2:

```
Base:     lib 1
Comparing lib 2 /u/me/libs/or2.db:or2...

Error: Design OR2 is missing in base library /u/me/libs/an2.db:an2
 (LM-091)
Information: Library /u/me/libs/an2.db:an2 has 1 missing lib cell(s)
 (LM-014)

Warning: Design AN2 is missing in library /u/me/libs/or2.db:or2 (LM-090)
Information: Library /u/me/libs/or2.db:or2 has 1 missing lib cell(s)
 (LM-014)
```

### What Next

A missing design in a downstream library is less of an issue than a missing design in the base library, but both indicate inconsistent libraries which is something you should address. Missing designs can be resolved by using the *-allow_missing* option to *check_workspace*.

### See Also

- check_workspace
- LM-090
- LM-091

---

## LM-015

(error) Check FAILED

### Description

This is just a summary message that the *check_workspace* command failed. It should be the last thing written by the command before returning to the command line.

### What Next

Examine the errors and warnings issued by *check_workspace* and determine if additional command options can correct the issues.

Feedback

## LM-016

(error) Incompatible graph found for design %s: %s.

### Description

This is a serious error. It means that two logic libraries that have the same cell and ports (by name) but they do not have the same arcs. More importantly, the graph has different nodes. This could indicate cells with a different function. The *check_workspace* command will remove the cell from the non-base library.

### What Next

To use these libraries to create a reference library, you must ensure that you read the logic library that has the superset of the graph first. To do this, remove the current library workspace, reread the logic libraries in a different order, and then use the *check_workspace -ignore_missing* command to combine the libraries without trying to create a superset of missing arcs.

Note that the library is still incorrect, and you should go back to the library source files to determine why they have two cells with substantially different arcs.

## LM-017

(error) %s '%s' has invalid thresholds.

### Description

This message indicates that invalid threshold values were specified for a library or library cell pin. This message is issued either when reading a logic library that contains invalid threshold values or when invalid values are specified in the *set_thresholds* command.

Thresholds other than the slew derating factors must be in the range of 5 percent to 95 percent, and the high and low slew thresholds must differ by at least 1 percent. In addition, the low threshold must be less than the high threshold.

You cannot write a library until you correct the values. To show the thresholds for a specific library or library cell pin, use the *report_thresholds* command.

### What Next

Check with your library vendor and correct the values before proceeding. To modify the threshold values, use the *set_thresholds* command.

## LM-018

(error) Must have at least 2 libraries for a scaling group

Feedback

### Description

A scaling group requires at least 2 libraries, and you passed in only 1 library to the command *create_scaling_group*.

### What Next

Make sure the specification of libraries to *create_scaling_group* is correct.

---

## LM-019

(error) Library %s is blocked from inclusion in scaling groups%s

### Description

There are two possible problems: either the source library is already in use in a different scaling group, or the checker found some problem earlier and marked the listed library such that it cannot be included in any scaling groups. One example problem which causes this is when duplicate arcs or checks were found in the library.

### What Next

In cases where there was an error which prevented the source library from being in a group, the libraries in question can be included in the library you write, but only as a single pane (that is, a specific unscaled characterization point). See the specific errors or warnings which occurred previously.

### See Also

- LM-006

---

## LM-020

(error) Scaling library group %u is no longer valid

### Description

You removed a library which was part of a scaling library group. By doing so, the SLG became invalid because a v,t point is missing.

### What Next

You might need to recreate other scaling library groups.

---

## LM-021

(information) Removed scaling library group %u

Feedback

**Description**

The scaling library group was removed because it became empty due to the its last library being removed.

**What Next**

No action required.

---

## LM-022

(warning) Scaling group already exists with these libraries

**Description**

You tried to create a scaling group but have already done so with the same set of libraries.

**What Next**

Check for a typo in your command.

---

## LM-023

(error) For rail requirement%s, %s characterization point %s

**Description**

While creating a scaling group, one of two things is wrong. In one case, there is a missing characterization point. For each voltage required for an arc (there might be an output voltage, an input and output voltage, or either of those along with a bias voltage), you must have each temperature, and vice-versa. So if you had 2 temperatures, two voltages for the main rail, and two voltage values for an input rail on level shifters, you would typically need eight libraries to cover those points.

In the other case, more than one library has the same characterization point. Each point can occur only once.

**What Next**

Try using a different set of libraries for your scaling group.

---

## LM-024

(information) For rail requirement%s, found%s

**Description**

While creating a scaling group, the application found missing or duplicate characterization points (PVTs). This message tells you which temperatures, output voltages (on the To pin

Feedback

of an arc), and input voltages (on the from pin of an arc, if a different rail than the to pin) were found. To get complete coverage, each combination of these values would usually have to be found in a library to allow a valid scaling group. In some cases, missing PVTs are allowed, for example, if you have multi-voltage arcs but in all cases, Vout = Vinp.

This informational message is paired with other messages, which will indicate if this is allowed or if it is an error which will prevent creating of the scaling group.

*LM-023 (n) LM-138 (n)*

---

## LM-025

(error) Could not fix all missing %s

### Description

There were missing objects of the given type, and not all could be resolved. The likely cause is more objects in one library than another, for example, an extra timing arc.

To view the missing objects, you could use *check_workspace -details* and pass in the proper value according to the information you need. For example, use *arcs_progress* for missing arcs, *leakage_progress* for leakage power records, and so on. Note that this can produce large output.

### What Next

You could try using *check_workspace -allow_missing* to allow missing objects to be passed through to the merged library. There would be no data for that object in any pane that includes the source library. Note that this will not work if you have created scaling groups, as there cannot be any missing objects. If the command option succeeds, a warning version of this message (LM-125) is output.

### See Also

- LM-125

---

## LM-026

(warning) Lib cell %s is defined in more than one physical library

### Description

Duplicate physical definitions are found for the lib cell. The application will use the first one that was loaded into the workspace, and ignore the duplicates. The lib cell's physical definitions are from LEF file or GDSII file.

You can use *get_lib_cells* command to find all of the library cells with this name.

Feedback

### What Next

Check your LEF or GDSII file and remove the duplicate, then restart the library creation with updated files.

---

## LM-027

(Information) Identified potential valid scaling groups.

### Description

The *group_libs* command analyzes the logic libraries loaded in the library workspace to check for potential scaling groups. If the command identifies one or more potential scaling groups, it issues this message.

### What Next

If you want the *group_libs* command to create the identified scaling groups, set the *lib.workspace.group_libs_create_slg* application option to *true*, and then run the *group_libs* command again.

### See Also

- group_libs

---

## LM-028

(warning) Some port differences found with %s versus base design %s as follows:

### Description

Each logical source library (from DB, for example) other than the first (known as the base) is compared to the base. Each design (library cell) is compared. Here, some differences were found with one or more ports for a specific design. This is an introductory message to show the two designs being compared. Subsequent messages will show the individual problems, and whether they were errors or were repaired (warnings and information only). Problems include missing ports, different port directions, ports out of order, and so on.

### What Next

When reading multiple logical source libraries, the library cell function, port names, port directions and many other port attributes must match those of the same cell in the base library. Libraries with this sort of error are probably out of date and should not be used. However, you can choose to remove the design from one or more libraries with *remove_lib_cell*, or in some cases, you can change port attributes. You can also have incompatible library cells removed automatically using the application option *lib.logic_model.auto_remove_incompatible_timing_designs*.

Feedback

**See Also**

- LM-029
- LM-056
- LM-123
- LM-124
- LM-129
- lib.logic_model.auto_remove_incompatible_timing_designs
- remove_lib_cell

## LM-029

(information) Successfully reordered out of position ports on design %s

**Description**

Previous messages (LM-028, LM-124) indicated that there were some ports out of position with the given design in comparison to the base DB library. The application was able to correct the problem.

## LM-030

(error) '%s' is not a valid DB/Liberty library file.

**Description**

The tool could not read the specified library file.

**What Next**

In the case of a DB file, ensure the file is a valid logic library file created with the Synopsys Library Compiler tool. In the case of a Liberty .lib file, ensure that the file conforms to the Liberty syntax.

## LM-031

(error) File '%s' is protected by a license code

**Description**

The .db file is protected by a license code. This product does not currently support reading of licensed .db files.

Feedback

### What Next

Obtain a non-licensed version of the library.

---

## LM-032

(error) Design %s is incompatible between these libraries:\n \t%s:%s\n \t%s:%s

### Description

During *check_workspace*, the first occurrence of each timing design (library cell) is compared to occurrences in all downstream libraries. If there are differences with function id, missing ports, or different port directions, the design is incompatible. Earlier messages showed the specific results of the comparison.

### What Next

When reading multiple DB or Liberty libraries, the library cell function, port names, and port directions must match those of the same cell in the base library. Libraries with this sort of error are probably out of date and should not be used. By default, the design remains. You can remove it using the *remove_lib_cell* command, then rerun the check. If you want offending library cells to be automatically removed from the offending library, you can set the application option *lib.logic_model.auto_remove_incompatible_timing_designs* to true and the designs will be automatically removed. Note that there are some cases where the design is always removed, for example, incompatible graphs.

### See Also

- check_workspace
- remove_lib_cell
- lib.logic_model.auto_remove_incompatible_timing_designs

---

## LM-033

(warning) Mismatched PG pins between the logic library and %s for block %s: %s.

### Description

The PG pins for the specified block differ between the logic and physical library source files. The tool cannot fix this mismatch.

This can happen when the logic library and physical library have more than two PG pins for a block, but the names do not match.

Feedback

### What Next

This is a warning, and it does not prevent the library from being written. However, the specified block will have all of the PG ports from both the logic and physical library source files, which might not be desirable.

Although the tool allows this mismatch, your library source files should match. You should alert the provider of your library source files about this discrepancy.

---

## LM-034

(information) Combined %d physical design%s:%s

### Description

This message, along with LM-052, summarizes the results of combining physical libraries into the workspace. In this case, there were designs from logical sources (DB), so the physical designs are combined with the logical source in the workspace. A count of unique design names is listed, followed by a line for each type of view (design, frame, layout) which had any content combined. For each view type, if any ports were reordered because they did not match the logical design, that is also listed. For example:

```
Information: Combined 20 physical designs:
      20 with a design view (14 had ports reordered);
      20 with a frame view (14 had ports reordered) (LM-034)
```

---

## LM-035

(error) Design %s does not exist in physical libraries

### Description

During *check_workspace*, when combining physical views into the workspace, neither a frame view nor a design view exists for the named library cell in any physical source library read into the application. Since the library cell does have a timing view, this means it can be timed, but not placed. This is an error and could be serious. This issue can be fixed automatically by setting the application option *lib.logic_model.auto_remove_timing_only_designs* to true, in which case you will see a similar message with the warning severity (LM-135). However, we recommend that you investigate why your DB or LIB files have designs that do not exist in any physical source.

Note that in the physical-only flow, there is a similar message which is always a warning. See LM-063.

### What Next

Verify that you are using the most up to date LEF or GDS and DB or LIB files. To continue, you would need to remove the library cell from the all DB or LIB files that contain it.

Feedback

## LM-036

(error) %s/%s was not copied from physical library

**Description**

The design was not combined into the base library. The likely cause is an attempt to associate different designs to one another (by name) that have significantly different footprints. This is an error and could be serious, but it will not block writing of the library.

**What Next**

Verify that you are using the most up to date LEF or GDS file.

## LM-037

(information) Combined physical block %s/%s (with %s port reordering).

**Description**

A verbose information message that indicates that a physical block was combined into the base library.

## LM-038

(warning) Port %s for block %s is out of order.

**Description**

When adding a physical design into the base library, the tool found a port on the specified block whose location differs from that in the logic library.

This message is issued only in verbose mode.

**What Next**

No action is required. The tool will reorder the ports using the order from the base library.

## LM-039

(error) Port %s missing on %s design %s

**Description**

When combining a physical design into the base library, a port was found to be missing on either the physical or base logical design.

Feedback

## LM-040

(error) Port %s/%s has related %s pin %s in lib %s, %s in lib %s

### Description

This is an incompatibility between the PG pin configuration between two DB files. The same cell in all DB files must have the same PG pins, and the signal pins must have the same related PG pins. In this case, a port refers to different related PG pins across libraries. This is related to LM-041, as pins with different related power pins will often have a different signal level.

### What Next

There are a few ways to correct this. Starting from a .lib file, ensure that every signal pin has related power/ground pins. It's better not to rely on Library Compiler to use a default value. If you can't modify the .lib files, then from this application, you can use attributes to fix the problem. Given that you have a cell A with signal pin X with related power pin VDD and in another library lib2 it has related power pin VDDX, and both power pins are available on cell A:

```
set_attribute [get_lib_pins {lib2/A/X}] related_power_pin_name VDD
```

Then, rerun the *check_workspace* command.

## LM-041

(error) Port %s has signal level %s in lib %s, %s in lib %s

### Description

This is an incompatibility between the PG pin configuration between two DB files. The same port in all DB files must have the same signal level.

### What Next

This is related to LM-040. See the comments there. Usually, fixing the related_power_pin_name attribute should also correct the signal_level mismatch.

## LM-042

(warning) Different number of rails between libs %s (%d) and %s (%d)

### Description

This is an incompatibility between the PG configuration between two DB files. Each DB file must have the same number of rails, and the rail names must align. This may be followed by some LM-043 messages about rail alignment. If the rails are just extra power rails and

Feedback

they are not used, you might be able to write a library. If there are different signal levels or the rail alignment is different for power rails, you will not be able to write a library.

## LM-043

(error) Different rail %s between the %s (%s) and %s (%s) libraries.

### Description

The PG configuration differs between the specified logic libraries. Each logic library must have the same number of rails, with the same names, in the same order. This message indicates that the alignment of the rails is wrong, either because one library has extra rails, or the rail names or types are simply different.

### What Next

In some cases, such as rail type differences, this error prevents the tool from writing the library and you must fix the logic library source file. If the rail names and positions differ, you can use the *rename_rail* command to make the rail names match. For example, if library A has a rail named VDD1.0 and library B has a rail named VDD1.1, use the following command to fix the mismatch:

```
rename_rail -library A -from VDD1.0 -to VDD
rename_rail -library B -from VDD1.1 -to VDD
```

When you rerun the *check_workspace* command, it issues this message again; however, it can correct the mismatch because the names now match.

Note that within a logic library, the voltage rail name is essentially a constant that is replaced with a voltage value. However, the tool uses the voltage rail names to match rails across libraries. If the rail names are inconsistent, the tool must trace back from the PG pins in the two libraries being compared to determine which rail is which, and whether they can be matched, which increases runtime.

### See Also

- LM-048
- LM-049
- rename_rail

## LM-044

(warning) Extra rail %s in lib %s

Feedback

**Description**

This is an incompatibility between the PG configuration between two DB files. Each DB file must have the same number of rails, and the rail names must align. In this case, one lib has an extra, unused rail.

---

## LM-045

(information) Fixed PG mismatch: %s name %s overrides %s.

**Description**

The PG pin names in the logic and physical library source files do not match. In this case, each design has exactly one PG pin of a type (power or ground) that does not match, so the tool will use the name in the physical library source file. Note that power and ground pins are considered separately.

**What Next**

Although the tool allows this mismatch, your library source files should match. You should alert the provider of your library source files about this discrepancy.

---

## LM-046

(information) Allowed extra internal PG pin %s from the logic library that is missing in %s.

**Description**

The PG pin names in the logic and physical library source files do not match. In this case, the block has an extra PG pin of type internal power or ground in the logic library. This might be due to an old method to emulate bias pins, or it could indicate that the logic library and physical library are out of sync.

**What Next**

Although the tool allows this mismatch, your library source files should match. You should alert the provider of your library source files about this discrepancy.

---

## LM-047

(warning) Port %s/%s changed from %s to %s.\n \tDeleting all arcs and checks at this port.

**Description**

The PG pin types in DB and LEF do not match. The most likely case is that the type is a power type in LEF and signal in DB. The default behavior is to take the type from LEF.

Feedback

Since power pins cannot have any arcs, any that exist are removed. You will also get this message if you change the port_type attribute of a pin from signal to power or ground.

**What Next**

Currently, there is no method to override this behavior.

---

## LM-048

(information) For the rail mismatches shown above, find common names, then use rename_rail commands to make them match. Then rerun check_workspace.

**Description**

When rail names don't match across DB libraries, you need to use *rename_rail* to make them match. For example, if you have library A with a rail named VDD1.0 and library B with a rail named VDD1.1:

```
rename_rail -library A -from VDD1.0 -to VDD
rename_rail -library B -from VDD1.1 -to VDD
```

That is one of many ways to get the rail names to match. During the repeat of *check_workspace*, you may see LM-043 messages again if the rails are still in different orders across libraries, but this should be followed with an indication that the problem was corrected.

**See Also**

- LM-043
- LM-049

---

## LM-049

(information) All rail names match, reordering rails...

**Description**

After the *check_workspace* command detected mismatched rail names, you used the *rename_rail* command to correct the problem. This message indicates that the names now match but the rails are still in a different order. The tool should be able to correct this problem.

**See Also**

- LM-043
- LM-048

Feedback

## LM-050

(information) Copied %d physical design%s

**Description**

A message summarizing the results of combining physical libraries into the workspace. In this case, there were no logical designs, so the physical designs are just transferred to the workspace.

## LM-051

(information) Copied %d physical-only design%s

**Description**

A message summarizing the results of combining physical libraries into the workspace. In this case, there were designs from logical sources (DB or LIB) and you are in a *keep_all_physical_cells* flow, so designs known to be physical-only were copied as well.

**What Next**

Note that this path may not be supported. In the future, it may only be possible to include physical-only cells in a workspace that only has designs from physical sources like LEF or GDS.

## LM-052

(information) Skipped %d physical design%s

**Description**

This message, along with LM-034, summarizes the results of combining physical libraries into the workspace. Some designs were not combined into the workspace. Reasons for this include a duplicate design (exists in more than one physical library), a design in physical library source that does not exist in DB, or physical-only designs without the appropriate option to *check_workspace*.

## LM-053

(error) Could not remove library cell %s/%s

**Description**

The *remove_lib_cell* command could not remove the specified library cell. This can happen if the library is in read-only mode.

Feedback

## LM-054

(information) Fixed direction mismatches for block %s.

### Description

The tool found ports on the specified block that did not match the directions in the timing view. These mismatches were detailed with NDM-032 errors. You enabled the setting that allows timing view directions to be applied to the design view to make them match.

### What Next

Although the tool allows this mismatch, your library source files should match. You should alert the provider of your library source files about this discrepancy.

## LM-055

(warning) Direction for port %s on %s is changed from unknown to %s.

### Description

The port on the specified physical design has an unknown direction. This typically indicates that the library source file for this physical design does not specify a direction for the port. If the design also has logical data, the library manager gets the port direction from the corresponding logical design. However, for physical-only designs, the library manager changes port direction to in if the *check_workspace* command did not find any problem. This message reports this change.

### What Next

Although the library manager allows an unknown port direction and changes it to the in direction, you should verify that this is desired. You should notify the provider of your library source files about this missing direction issue.

## LM-056

(error) Port %s/%s has %s '%s' in lib %s, '%s' in lib %s.

### Description

There is an incompatibility of port attributes between two logical libraries. Certain attributes must be the same for a given design port in all DB or Liberty source files. For example, port_type (signal, power, ground, etc.), direction, and various other attributes . *check_workspace* will fail if there is a mismatch for these attributes. In most cases, the user will need to fix the mismatch in the source, reload it and run *check_workspace* again. There may be some cases where the attribute can be changed in the library manager, or where an application option can change the error to a warning.

Feedback

### What Next

Starting from a Liberty source file, modify the attribute on the port, load the updated DB or Liberty source file, and rerun the *check_workspace* command.

---

## LM-057

(error) Your scaling group is very large (%1.2f %s). This is\n \t%1.1f times the size of the representation as %u unscaled panes.\n \tScaling group not created due to excessive runtime required to\n \tcreate it, and the large projected memory overhead to use it.

### Description

You are using *create_scaling_group*, and the group is going to be very large in memory and disk space, and it will also take a very long time to create. Scaling is not for everyone. You need a complete grid of logical libraries to cover the range of voltages and temperature, but that alone does not mean you will get an efficient representation. If the breakpoints (table indices) do not match across the PVTs, tables get expanded to cover the complete range of breakpoints in order to guarantee that all values from the contributing logical libraries can be retrieved exactly. That expansion costs memory, and because all of the extra points need to be filled in, the time to create the group (during *check_workspace*) can be extensive.

For example, say you have 7x7 tables and 6 contributing libraries over 2 voltages and 3 temperatures. As unscaled panes, you could assign a cost of 7x7x6, or 194, for one table. Scaling groups always have a small incremental cost over unscaled panes due to optimizations for delay calculation. So with the same breakpoints across all 6 libraries, you could assign a cost of about 250 to the scaling group. Now, in the worst case, if you have no overlap at all in the breakpoints, the final tables would be 49x49x2x3, or a cost of 14406, 75x the unscaled size rather than 1.3x. If you have more contributing libraries, the factor could be even higher.

This message is telling you that you have fallen into this situation and you should reconsider the use of scaling groups. You can force the application to create the group for you using the *-force* option to *create_scaling_group*, but this is not recommended.

### What Next

Scaling is a powerful feature which allows you to interpolate in the voltage and temperature dimensions, but your source libraries need to be designed with that in mind. Check with the provider of your logical libraries and request libraries where the breakpoints match across all PVTs.

### See Also

- create_scaling_group

Feedback

## LM-058

(warning) Duplicate physical block %s accepted only from library file '%s'.

**Description**

The *check_workspace* command found blocks with the same name in two or more physical library source files, such as LEF files. When the physical library source files contain duplicate block names, the tool uses the block from the first physical library source file read.

## LM-059

(error) There are no physical-only blocks.

**Description**

You created a physical-only library workspace with the *create_workspace -flow physical_only* command. However, the *check_workspace* command found all blocks defined in the physical libraries in one or more logic libraries; therefore, there are no physical-only blocks to include in the reference library.

**What Next**

If you want to create a library that contains only physical information for the blocks, read only the physical library source files and not the logic library source files.

## LM-060

(error) Equal/opposite relationships between %s/%s and %s are different:\n \t%s in %s, %s in %s

**Description**

Equal/opposite information, loaded from DB, is different for a design in the base library in comparison to another library.

**What Next**

In general, these libraries are incompatible, and you should go back to the source of the library and make sure these libraries were intended to be used together.

An app option lib.logic_model.relax_pin_equal_opposite_attr_check is provided to relax the consistencies check for equal/opposite attributes. It is a boolean option, default is false. When the app option set as true, the error will be treated as a warning. In that case, the LM-160 message is output.

SEE ALSO

Feedback

**LM-160(n)**
**lib.logic_model.relax_pin_equal_opposite_attr_check(3)**

---

## LM-061

(error) Switch cell %s has incorrect number of primary power/ground pins

### Description

A switch cell requires at least one primary power and one primary ground pin, and then at least one internal power and/or one internal ground pin.

### What Next

You can correct this in the library manager by changing the *pg_type* attribute on a lib pin. However, it is probably best to review the Liberty source of the library and correct it there.

---

## LM-062

(error) %s %s refers to a %s '%s' that is not on the parent library cell '%s'.

### Description

An object in a .db library file refers to a power or ground pin that does not exist on the library cell that owns the object. The .db file is corrupt.

### What Next

Recompile the library source file (.lib) into a new .db file.

---

## LM-063

(warning) DB design %s does not exist in physical libraries

### Description

During *check_workspace -flow physical_only*, neither a frame view nor a design view exists for the named library cell in any physical library read into the application. Since we are creating a physical-only library (that is, designs which exist only from physical library source), this is a warning in this flow. It would be an error in the normal flow (see LM-035).

---

## LM-064

(warning) Cell %s has more than one physical %s pin and none in DB.\n \tSome related %s pins cannot be derived.

Feedback

### Description

In the case where DB has no PG pins and the physical design has more than one power and/or ground pin, the application cannot derive the related pg pin on signal pins.

### What Next

You can use the *related_power_pin_hint* and *related_ground_pin_hint* attributes on pins in the DB file(s) to help steer the application towards a solution. In the case that you have multiple DB files, the attributes only need to be set on one instance of the pin. If you continue to get this error, use *get_attribute* or *report_lib_pins* to determine which pins are missing related pins, provide hints, and rerun *check_workspace*.

Although this is an error, it will not prevent the user from writing a library at this time; the library will simply be missing related power/ground pins for the problem cell(s). However, there may be some modes when such errors will prevent writing the file.

---

## LM-065

(error) Rail %s on port %s is defined as %s;\n \tpreviously defined as %s (on %s)

### Description

There's an inconsistency with a voltage rail within a DB file. On different cells, each has a PG port that refers to the same rail using the *voltage_name* attribute (which refers to an entry in the *voltage_map*). In one case, the pg_type of the port is power; in another, it is ground. It was likely a typo in the .lib file. Although you can have mixed types for an entry in a voltage_map, power and ground is the one combination that is not allowed.

### What Next

Someone really should follow up with the provider of the DB because this is a very serious error. There is no automatic recovery for this error. The PG ports in question will have no rail associated with them. You can use *set_attribute* to set the *rail_name* attribute to something appropriate.

---

## LM-065w

(warning) Rail %s on port %s is defined as %s;\n \tpreviously defined as %s (on %s),will add extra rail for it to continue

### Description

There's an inconsistency with a voltage rail within a DB file. On different cells, each has a PG port that refers to the same rail using the *voltage_name* attribute (which refers to an entry in the *voltage_map*). In one case, the pg_type of the port is power; in another, it is ground. It was likely a typo in the .lib file. Although you can have mixed types for an entry

Feedback

in a voltage_map, power and ground is the one combination that will be handled by adding an extra rail.

**What Next**

Someone really should follow up with the provider of the DB because this is a very serious warning. There is automatic recovery for this warning. The PG ports in question will have an extra rail associated with them.

---

## LM-066

(error) PG port %s has voltage_name %s which does not exist in the voltage_map

**Description**

Logic libraries define PG pin connections by using library-level *voltage_map* attributes and cell-level *pg_pin* groups. The *voltage_map* attributes define the power rail names. The *pg_pin* groups define the PG pin connections for a cell; they use the *voltage_name* attribute to assign one of the defined power rails to each PG pin.

The tool issues this error when the power rail specified for the *voltage_name* attribute is not defined in a library-level *voltage_map* attribute.

**What Next**

If you have access to the logic library source, you can update the library to correct this issue. Otherwise, you can use the *set_attribute* command to modify the *rail_name* attribute of the PG pin in the reference library (the *voltage_name* attribute in the logic library is stored as the *rail_name* attribute in the reference library).

To determine the power rail names defined for the library, use the *report_workspace -panes* command. If the tool deleted a power rail because it appeared to be unused, use the *create_rail* command to re-create the power rail.

---

## LM-067

(error) Attribute %s on similar %ss %s (%s) and %s (%s) must match

**Description**

There's an inconsistency with an attribute between two objects. They must match. For example, if this is a *related_power_pin_hint* attribute, if it is applied to similar lib pins in different libraries, the value must be the same, because the final library will only have one representation of the lib pin.

**What Next**

Use the *set_attribute* command to change one of the attributes.

Feedback

## LM-068

(warning) Unknown pin '%s' for attribute '%s' on '%s/%s'

### Description

You provided a hint for the checker to select the correct related power, ground, or bias pin when the DB has no PG pins and the LEF has more than one of a particular type. The hint pin name did not exist in the LEF so it does not exist on the cell in the workspace. For example, if the LEF power pins are VDD and VDDx, and you used *vdd* as the value for the *related_power_pin_hint* attribute, you would see this warning. Even though this is a warning, it will cause *check_workspace* to fail in some modes.

### What Next

Use the *set_attribute* command to change one of the attributes.

## LM-069

(error) Table template %s was not loaded from DB or .lib

### Description

There's a table in the DB or Liberty source file that refers to a template of an unsupported type. Since the template was not loaded, the table cannot be loaded either. One current example is an internal power table with a reference to a poly_power_template.

### What Next

The result is that this internal power record is ignored. It does not prevent the library from being written.

## LM-070

(error) %s is not a single cell library

### Description

You created a workspace with the ETM flow or the organize_by_pvt flow. Currently, these flows only allow single cell libraries, and the named library has more than one cell.

## LM-071

(error) Design %s already exists for PVT '%s'\n \tfrom lib %s;\n \tcannot add it from lib %s

Feedback

### Description

You created a workspace to reorganize logical DB libraries according to PVT. There can be only one combination of design name and PVT, and the application found a duplicate of a combination.

---

## LM-072

(error) Cannot reorganize by PVT

### Description

This is a summary message. See previous messages, for example, LM-070 and LM-071 for specific problems trying to reorganize the workspace by PVT.

---

## LM-073

(error) %s have the same PVT:\n \t%s

### Description

At the conclusion of *check_workspace*, the application verifies that you don't have multiple contributing libs using the same PVT. This causes ambiguity in the NDM created; the timer will be unable to distinguish panes using these libraries. Only one pane will be used. There can be several causes of a duplicate PVT, but in general, this occurs when multiple DB libraries were intended to have a different process, but the values in the libraries are the same. Prior to *check_workspace*, either use the *-process_label* option for *read_db*, or use *set_process* to adjust the process number or label in the initial libraries after reading.

The message will list each conflicting contributing DB library and the pane (in the final NDM) it is in.

### What Next

This is an error. You should be able to make adjustments to your script and rebuild the library to remove the situation.

---

## LM-074

(warning) Units for lib %s do not match the base

### Description

Libraries that come from DB files don't need to have the same units - all values are scaled appropriately when the file is read. However, there is a legacy idea of a "main" library and the units for various quantities in it. This establishes units for commands that deal with values, like timing, capacitance, and so on. This is just an advisory message to tell you

Feedback

that the units the main application will use are not necessarily all the same as the units from this particular library.

## LM-075

(warning) DB/Liberty file %s has invalid nominal voltage %g.

### Description

The nominal voltage in DB is based on the *nom_voltage* library attribute, or the voltage value in the default operating conditions. For this application, if the Liberty source has a *voltage_map*, then the nominal voltage must match a value from the map. If there is no match found, this warning message is issued. A library with an invalid nominal voltage cannot be included in any scaling group. You will get an LM-076 error message in the *create_scaling_group* command if your try to use the library with invalid nominal voltage in a scaling group.

Why is there a DB or Liberty file with an invalid nominal voltage? One possibility is that the voltage from the default operating conditions overrides the attribute, and that voltage is not in the voltage map. In general, this seems to be a typo when creating the default operating conditions in the .lib file. Why is it a problem? Capacitance and DRC values in the library are characterized to the nominal voltage. This voltage value needs to be in the voltage map because scaling is based on the nominal voltage and the values in the voltage map.

### What Next

You can modify the .lib to fix the issue listed above and regenerate the DB file. Or, you can use the *set_attribute* command in this application to change the *nom_voltage* attribute to an valid value for the loaded library. The following example shows changing nom_voltage to 0.65 for lib A.

```
set_attribute [get_libs A] nom_voltage 0.65
```

### See Also

- LM-076

## LM-076

(error) Library %s has invalid nominal voltage.

### Description

To be included in a scaling group, a library must have a valid nominal voltage. If you are getting this message, it means that the library had a voltage map, and the nominal voltage was not in it.

Feedback

Prior to this message, you should have seen an LM-075 message during *read_db*. LM-075 provides more detailed information about why the library's nominal voltage is invalid.

**What Next**

You can use the *set_attribute* command to change the nominal voltage to an valid value for the loaded library. The attribute name is "nom_voltage". The following example shows how to change the nominal voltage to 0.65 for lib A.

```
set_attribute [get_libs A] nom_voltage 0.65
```

**See Also**

- LM-075

---

## LM-077

(error) The value specified for nominal voltage (%g) is not valid; it must match a voltage rail.

**Description**

You tried to change the nominal voltage using *set_attribute*. The value specified is not a valid rail voltage, that is it was not in the voltage map in the DB. You can see the rails using *report_workspace*.

**What Next**

Use one of the voltages from the existing rails as the nominal voltage attribute value.

**See Also**

- LM-076
- LM-075

---

## LM-078

(warning) All designs were filtered out of library '%s'

**Description**

You were reading a source file into the library manager, and the combination of filtering application options lib.workspace.include_design_filters and lib.workspace.exclude_design_filters, or other command options, caused all cells to be filtered out, yielding an empty library. As an example of command options, if you use the *-views* options for the *read_ndm* command, and ask for only layout views and the file has only frame views, that would be considered filtering out all designs.

Feedback

### What Next

This is only a warning to call your attention to a possible typo in one of your filters. If it was an error, you can remove the library with *remove_lib*, adjust your filters, then re-read the file.

*read_db (2), read_lib (2), read_lef (2), read_gds (2), read_ndm (2), lib.workspace.include_design_filters (3), lib.workspace.exclude_design_filters (3).*

---

## LM-079

(error) Library %s is not compatible with the %s flow:%s

### Description

Some flows restrict the content of a Liberty/DB library. The named library has a problem construct which is not allowed in the flow, or is missing a required construct or attribute.

### What Next

Either change flows, or modify the library source.

---

## LM-080

(warning) Attribute '%s' for %s/%s does not match the base DB %s

### Description

During *check_workspace*, attributes are compared between the base logical (DB) library and each other contributing logical libraries. For certain attributes, they are checked for consistency across the DBs, and in cases where they do not match, this warning is issued. For example, if the dont_touch *attribute is true on a design in the base library but false in another library, we would report it. This alone does not cause check_workspace to fail, and if it succeeds, then for boolean attributes, the value from the base DB library is used. For string attributes, the value is removed from all libraries.*

You can use *get_attribute* to see the specific values for the given objects across different source libraries.

Depending on the particular attribute and various other settings, this message may be accompanied by the LM-081 error which will cause *check_workspace* to fail. For more information, consult the man page for LM-081.

*LM-081(n)*

Feedback

## LM-081

(error) Attribute '%s' mismatches considered as error

### Description

An attribute or construct in the base DB library did not match another DB. A warning was issued, but due to application option settings, some were upgraded to error. Examples:

- The app option *lib.logic_model.require_same_opt_attrs* is true and there were mismatches for *dont_touch*, *dont_use*, or *use_for_size_only* attributes.
- The app option *lib.logic_model.require_same_voltage_range* is true and there were mismatches for *input_voltage_range* or *output_voltage_range* constructs.

In some cases, the *set_attribute* command can be used to correct the problem. Or, you can change the value of the app option to allow *check_workspace* to succeed, in which case the value from the base DB library is used by default. In some cases, there are other choices to control how the value is computed.

*LM-080(n) LM-115(n) lib.logic_model.require_same_opt_attrs(3) lib.logic_model.require_same_voltage_range(3) set_attribute(2)*

## LM-082

(information) Some objects are missing in some logic libraries

### Description

Compared to the base logic library, some objects (arcs, leakage power records, dynamic power records, and so on) in the same relative position are missing in some logic libraries. Previous LM-001, LM-025 and other messages would be an indicator that this workspace has missing objects. Note that missing objects cannot be allowed in the presence of scaling groups, so if you are using scaling groups, you will not be able to use -allow_missing to fix this problem.

### What Next

There should not be missing objects across logic library source files. Either correct the source, or remove the cells which are causing objects to show as missing, or if using scaling groups, remove the scaling group(s). Then rerun *check_workspace*, potentially with -allow_missing if you are not removing cells or correcting the source.

## LM-083

(information) Breakpoints comparison failed

Feedback

### Description

Within a library, complimentary attributes (for example, rise delay and rise transition) for some arcs did not have identical breakpoints.

### See Also

- NLDM-003
- NLDM-004

---

## LM-084

(warning) Invalid leakage_power record for design %s: \t%s. Record ignored.

### Description

There's an inconsistency with a leakage_power record in your source file. It could be that it has a when condition which refers to a port that does not exist on the design. In another case, the record is missing a value, which probably means it was negative in the Liberty source (which is illegal, but not an error). The content of the message will tell you want the exact issue it. We disregard the record.

### What Next

This is a warning. But you should really investigate the history of the Liberty source file.

---

## LM-085

(error) Current workspace is not in exploration mode

### Description

Commands *group_libs report_lib_groups write_lib_groups* are only supported in exploration mode. Command *create_workspace* with exploration flow will create workspace in exploration mode.

---

## LM-086

(error) Lib groups are not created yet, use command group_libs to create lib groups.

### Description

Command *group_libs* will create lib groups based on the logical and physical libraries in current workspace. Then you can *report_lib_groups write_lib_groups* and *rename_lib_group* to query and edit lib groups.

Feedback

## LM-087

(error) cannot find lib group %s.

### Description

Failed to find lib group with user specified name.

### What Next

Use *report_workspace* to list all the lib groups, check whether your lib group name is valid.

## LM-088

(error) The related power or ground pin information is missing for one or more signal pins.

### Description

This is a summary message that indicates that one or more signal pins are missing information about the related power or ground pin. Previous LM-098 information messages listed the names of the signal pins that are missing the *related_power_pin* or *related_ground_pin* attribute in the logic library and the names of their owning cells.

### What Next

If the library cell has PG pins in the logic library, you can set the *related_power_pin_name* or *related_ground_pin_name* attribute directly.

If the library cell does not have PG pins in the logic library but does in the physical library, you can help the tool derive the missing related PG pins by setting the *related_power_pin_hint* and *related_ground_pin_hint* attributes of the problematic pins. If you have more than one logic library source file, you need to set the attributes only on one instance of each pin.

You can allow missing related power and ground pins by setting the *lib.workspace.allow_missing_related_pg_pins* application option to *true*. However, this is not recommended because without related PG pins, the tool that reads the library cannot determine the supply connectivity for the signal pin. This can cause issues with multivoltage constraints, especially when using alternative (non-primary) power supplies.

For example,

```
prompt> check_workspace
 ...
Information: Signal pin A in cell OR2 does not have related power pin.
 (LM-098)
Error: The related power or ground pin information is missing for one or
 more signal pins. (LM-088)
```

Feedback

```
prompt> set_attribute [get_lib_pins ao1.0/OR2/A] related_power_pin_hint
 VDD
prompt> check_workspace
 ...
Workspace check succeeded!
```

### See Also

- LM-064
- LM-098
- lib.workspace.allow_missing_related_pg_pins

---

## LM-089

(warning) Unable to find PG port %s - %s

### Description

The *check_workspace* command found a dangling reference to a PG port. This can happen in a few cases when DB or .lib files do not have PG pins.

When a DB or .lib source file has a power record (leakage_power or internal_power) which uses a power_level attribute to specify a forward reference to a PG pin (expected to be declared in a physical source file), and this PG pin cannot be found, you would get this warning, and the power record will be removed.

When a DB or .lib source file uses the rail_connection group to connect an existing power rail to a forward reference to a PG pin (expected to be declared in a physical source file), and this PG pin cannot be found, you would get this warning.

### What Next

In the case of leakage_power, this is just a warning to alert you to the situation. This could indicate an inconsistency between the DB and the physical source (LEF, GDS). In the case of rail_connection, it's a warning here, but will likely become an error because the design will probably have missing related power pins.

*lib.workspace.allow_missing_related_pg_pins(3)*

---

## LM-090

(warning) Design %s is missing in%s library %s:%s

### Description

In comparison to the union of all designs in all logical libraries, the given library is missing a design. All logical libraries in a workspace should contain the same designs. This is a

Feedback

warning because you used the *-allow_missing* option to *check_workspace* so that the union will be created in the workspace.

**What Next**

A missing design in a downstream library is less of an issue than a missing design in the base library, but both indicate inconsistent libraries which is something you should address.

**See Also**

- check_workspace
- LM-014
- LM-091

---

## LM-091

(error) Design %s is missing in%s library %s:%s

**Description**

In comparison to the union of all designs in all logical libraries, the given library is missing a design. All logical libraries in a workspace should contain the same designs. This is an error because you did not tell *check_workspace* how to resolve the problem.

**What Next**

A missing design in a downstream library is less of an issue than a missing design in the base library, but both indicate inconsistent libraries which is something you should address. Missing designs can be resolved by using the *-allow_missing* option to *check_workspace*.

**See Also**

- check_workspace
- LM-014
- LM-090

---

## LM-092

(information) Removing all timing-only designs

Feedback

### Description

You got one or more LM-035 messages because designs in DB that did not have a contribution from a physical source like LEF or GDS. The application option auto_remove_timing_only_designs has been set to true, so the tool removed these designs from the workspace.

### See Also

- LM-035

---

## LM-093

(information) Some power data was not loaded for %s/%s

### Description

Something was wrong with the internal power data from DB or .lib. Earlier messages would have indicated the problem. This just informs you that the data was ignored.

### What Next

The result is that this internal power record is ignored. It does not prevent the library from being written.

### See Also

- LM-069

---

## LM-094

(warning) One or more suspicious values found in base_curves, curve_y=%d: (%s )

### Description

A very small value (which will eventually be quantized to zero) was found in the given compact CCS base curves record from DB (or .lib). This is not expected. This will result in the failure to create a delay table, leaving a hole in the data. If this library is part of a scaling group, check_workspace will fail.

### What Next

If you are not using scaling groups, this will not cause library creation to fail, but there may be missing data for any arcs which use this base curve. If you are using scaling groups, you will not be able to get a library created unless you modify the .lib data.

Feedback

## LM-095

(warning) Bus property only exist in frame view %s/%s.

### Description

A port is defined as a bus port in frame view while in timing view, it is not a bus port. *check_workspace* will remove such bus properties automatically, when the application option *lib.workspace.remove_frame_bus_properties* is *true*.

### See Also

- lib.workspace.remove_frame_bus_properties
- LM-096

## LM-095e

(error) Bus property only exist in frame view %s/%s.

### Description

A port is defined as a bus port in frame view while in timing view, it is not a bus port. This inconsistency will bring bad result in *link_block* and *write_verilog*, which must be fixed.

Users should add the bus information into timing view through updated db file, or remove the bus property from frame view. *check_workspace* can remove such bus properties automatically, when the application option *lib.workspace.remove_frame_bus_properties* is *true*.

Users can also enable the Design Mismatch Manager (DMM) by command *set_current_mismatch_config* to repair the mismatch.

### See Also

- lib.workspace.remove_frame_bus_properties
- LM-096

## LM-096

(information) Remove bus property from %s/%s.

### Description

Remove the bus properties which only exist in frame view.

Feedback

**See Also**

- LM-095
- lib.workspace.remove_frame_bus_properties

---

## LM-097

(error) For design %s, PG pin '%s' is also used as a signal pin%s

**Description**

A cell in a DB file has a PG pin and a signal pin with the same name. Modern versions of Library Compiler do not allow this.

**What Next**

Currently, the program deletes the signal pin and continues. But someone really should follow up with the provider of the DB because this is a very serious error.

---

## LM-098

(information) Signal pin %s in cell %s does not have a related %s pin.

**Description**

This message indicates that the relationship between a signal pin and a related power or ground pin is missing.

**What Next**

In the case that the library cell in question does have PG pins, you can set the *related_power_pin_name* or *related_ground_pin_name* on pins in the library which came from DB directly. If the library cell does not have PG pins, but the physical contribution (from LEF, GDS, etc) does, then you can set the *related_power_pin_hint* or *related_ground_pin_hint* attribute on the signal pin to help the application derive the missing related power or ground pin. In the case that you have multiple DB files, the attribute only needs to be set on one instance of the pin.

These messages will be followed up with a single LM-088 error to summarize all of these informational messages. If allowing missing related power and ground pins is desirable, you can set the *lib.workspace.allow_missing_related_pg_pins* application option to true to allow signal pins to have no related power and ground pins.

Feedback

**See Also**

- lib.workspace.allow_missing_related_pg_pins
- LM-064
- LM-088

## LM-099

(information) Previous '%s' %s message(s) are considered as %s\n \tdue to '%s'

**Description**

The severity level of a message is considered to a different level. The reason will be given in the message. The reason can be application option setting, command line option, etc.

## LM-100

(error) Table size mismatch for %s attribute %s on library cell %s on an arc %s->%s: expected %u, found %u.

**Description**

The tool found an inconsistent attribute represented as a table while reading a logic library source file. The expected size of the table did not match what was actually found.

**What Next**

This is a serious error. If you see this error, it should give enough information so that you can find the problem in the .lib file.

## LM-101

(error) More than one instance of cell %s, mode %s, PVT '%s'

**Description**

For the ETM flow, in the early stages of *check_workspace*, each logical design with the same name is compared using the mode label and the PVT information from *read_db* or *read_lib*. There can be only one unique combination of logical design name, mode, and PVT. You get this message if there is a duplicate.

**What Next**

Make sure you have applied the correct mode label to each logical library read into the ETM flow workspace. Once you know which library has the wrong label, remove it, then re-read it with the correct label.

Feedback

**See Also**

- check_workspace
- read_db
- read_lib

---

## LM-102

(warning) Invalid rail_connection for %s/%s ignored: rail %s does not exist

**Description**

While reading a DB or .lib file, an invalid rail_connection was found. A rail_connection specifies a forward reference connecting an existing rail to a PG pin that will be supplied by a physical source file. If the rail is not defined in the source file, this is invalid, and the rail_connection is ignored.

**What Next**

Ensure that the logical source files that you are supplying use a consistent and complete specification for PG pins and rails. Note that rail_connection and power_supply have been superseded by the pg_pin and voltage_map constructs, and the latter is the preferred method for specifying PG pins and rails.

---

## LM-103

(error) Inconsistent rail_connection for %s/%s: %s

**Description**

The *check_workspace* command found an inconsistent reference based on a rail_connection from Liberty. A rail_connection specifies a forward reference connecting an existing rail to a PG pin that will be supplied by a physical source file. The rail_connection group was superseded by the pg_pin construct, but Library Compiler still allows use of the rail_connection. Each design in each contributing DB or .lib file for different PVTs must have the same connections, or none at all. You may also see other messages regarding problem voltage rails in the same context.

You get this message when we are comparing a downstream library to the base (first) library. If a PG pin name is found in a rail_connection in both, but the rail names different, you would see that we found a rail name that was not expected. In another case, if a PG pin name is missing in the downstream library, you would see something like PG pin missing.

Feedback

### What Next

Ensure that the logical source files that you are supplying use a consistent specification for PG pins and rails. All such files should use either pg_pin, or rail_connection plus the power_supply groups. When using rail_connection, the definitions must be the same across source files. It is possible that *rename_rail* might be able to fix the problem when the PG pin reference is the same in all libraries but the rail names are different.

---

## LM-104

(error) With mode %s, design %s has incompatible port function setup between libraries %s:%s and %s:%s

### Description

For the ETM flow, for a given mode (applied with the -mode_label option to *read_db* or *read_lib*) across all PVTs, it is required that the function information for ports match. The *function* attribute will be 0 or 1 for various ports. For each port, the attribute must be the same: either both 0, both 1, or both missing. In this case, one or more ports were missing, and *check_workspace* detected it.

### What Next

Verify that you have applied the correct mode to each library. If you have, at this time, you would need to go back to the .lib file or the creator of it and verify that the function information for the ports is consistent.

### See Also

- check_workspace
- read_db
- read_lib

---

## LM-105

(error) With mode %s, function for port %s/%s does not match: \tValue %d in library %s: %s \tValue %d in library %s:%s

### Description

For the ETM flow, for a given mode (applied with the -mode_label option to *read_db* or *read_lib*) across all PVTs, it is required that the function information for ports match. The *function* attribute will be 0 or 1 for various ports. For each port, the attribute must be the same: either both 0, both 1, or both missing. In this case, the listed port was 0 in one library and 1 in the other.

Feedback

### What Next

Verify that you have applied the correct mode to each library. If you have, at this time, you would need to go back to the .lib file or the creator of it and verify that the function information for the ports is consistent.

### See Also

- check_workspace
- read_db
- read_lib

---

## LM-106

(warning) There's no power pin or ground pin of the antenna diode for design %s

### Description

There are three types of diode cells: power, ground and power_ground which depend on the connections from diodes to VDD/VSS. Check the physical library and the logical library source.

---

## LM-107

(warning) More than one power pin or ground pin of the antenna diode for design %s

### Description

There is at most one power pin and one ground pin for the antenna diode cell. Check the physical library and the logical library source.

---

## LM-108

(information) For PG pin '%s', replaced rail '%s' with '%s'

### Description

While reading a Liberty(.lib or DB) file, an entry in the *voltage_map* is being used for multiple purposes. In this application, rails have a single type (power, ground, etc.). So, the rail is duplicated, with the new one renamed to include the type. For example, if the entry was VDD, and there are power and nwell PG pins in the file using it, you would see this message for the first PG pin which detected the need for VDD-nwell. You can see rails using *report_workspace -panes* or use *get_attribute* to see the *rail_name* attribute on your PG pins.

Feedback

### What Next

Rail names are not widely used, so there is nothing you need to do. You do have the option to create dedicated entries in the voltage_map for each type.

---

## LM-109

(warning) Copied PG pin '%s' from physical source (was missing in logical source)

### Description

In *check_workspace*, a PG pin was found in the physical source files which was not in the logical source. The application created a pin in the timing view to cover the difference.

### What Next

This could because of old DB files without pg_pin's, or something is out of sync between the physical and logical sources. If you want issues like this to force an error, you can use set_message_info to make scripts stop sometime after the issue occurs.

---

## LM-110

(warning) '%s' and '%s' is incompatible, '%s' ignored.

### Description

If the port in physical design is not consistent with timing view, the physical design will be ignored in edit flow.

---

## LM-111

(warning) Missing physical library cell, create placeholder physical library cell '%s'.

### Description

You got one or more LM-035 messages because designs in DB that did not have a contribution from a physical source like LEF or GDS. when the application option auto_remove_timing_only_designs has not been set to true, the tool will create placeholder physical cell for the missing cell. Following are basic information for a placeholder physical lib cell. Lib cell width and height (default: unit site width and height) Lib cell original (default: 0, 0) LEF site associated with the cell (default: unit site) Pin location (default: 0,0) Pin terminal layer (default: metal1 (layer maskName)) Pin terminal geometry (default: metal1 min width x min width)

Feedback

**See Also**

- LM-035
- LM-092

---

## LM-112

(warning) Missing physical library port, create placeholder physical library port '%s' for library cell '%s'.

**Description**

The tool found missing ports on the specified block compare to the ports in the timing view. These mismatches were detailed with NDM-031 errors. Following are basic information for a placeholder physical lib port. Pin location (default: 0,0) Pin terminal layer (default: metal1 (layer maskName)) Pin terminal geometry (default: metal1 min width x min width)

**See Also**

- NDM-031
- LM-111

---

## LM-113

(warning) Port '%s' on library cell '%s' has been created as a placeholder for a missing logical port.

**Description**

The tool found a missing logical port on the specified logical cell compared to the ports in the design view. The mismatch has also been reported with an NDM-031 error message. A placeholder port in the logical library has been created with the following characteristics: 1. Physical port direction and type were used for the placeholder port. 2. No timing information was created.

**See Also**

- NDM-031

---

## LM-114

(information) Changing port type from '%s' to '%s' on port '%s/%s'.

Feedback

### Description

Ports can roughly be divided into two primary classes: signal and PG. PG ports are further subdivided into power, ground, pwell, and nwell. Some physical source formats, like LEF, might classify ports with a "use" like signal and power, but also provides additional sub-types for signals, such as "clock". As the application does not have a primary port type of "clock", it converts that port type on the physical view to "signal". This message is used to convey that. The information about a port's sub-type, like clock, is carried as a dedicated attribute on the port, and that information is normally taken from the logical source (DB or .lib).

### What Next

This is just an informational message; no action is necessary.

---

## LM-115

(warning) %s %s/%s has voltage range %s in lib %s versus %s in the base lib.

### Description

Voltage range is imported from the Liberty input_voltage_range and output_voltage_range constructs. The values differ between the given library and the base. In general, these should be the same. The application option *lib.logic_model.require_same_voltage_range* controls whether you get this warning or an error, and whether the final values come from the base library (default) or from the widest range of values (the min of the lows of the range, the max of the highs).

### What Next

This is just a warning, so no action is needed unless you want it to be an error. See the man page for lib.logic_model.require_same_voltage_range for details.

### See Also

- LM-081

---

## LM-116

(Error) Cannot use read_ndm with '%s': %s

### Description

The *read_ndm* command is used to load NDM files into a workspace. This is subject to a number of rules based on the current flow and the type of NDM file. In this case, you tried to read a library which is not valid in the current context. For example, you tried to read an

Feedback

aggregate library, which is never allowed. Or, you tried to load frame views from a design library in a flow other than the etm_moded flow.

**What Next**

Check the type of the library in comparison to the flow you are using. See the man page for *read_ndm* for complete details.

---

## LM-117

(Error) Technology data in %s and workspace technology data is not compatible

**Description**

Technology data in the library which is reading and workspace technology data must be compatible, otherwise *read_ndm* will fail.

**What Next**

Try another library with compatible technology data.

---

## LM-118

(warning) Workspace technology data replaced by the technology data from '%s'

**Description**

*read_ndm* will replace the workspace technology data with technology data from the library which is reading. The replacement occurs when the two technology data are compatible or the workspace technology data is not referred by other physical libraries.

---

## LM-120

(information) List of cells with rectilinear boundaries in current workspace library: %s

**Description**

This message is for reporting cells with rectilinear boundaries in the current workspace library. All cells with rectilinear boundaries will be reported in the message's cell name list.

**What Next**

This is information only. No further action is required.

---

## LM-121

(information) Cell misaligned logical libraries are detected, and cell shadowing fixing is %s.

Feedback

### Description

*group_libs* groups logical libraries based on cell name signatures, if a set of libraries with exact same lib cells, those libraries will be in one group. Sometimes, instead having exact matching or well aligned cells among libraries, cells are partially overlapping in a set of libraries, the cell misaligned case is frequently seen in low power libraries. If use pure cell signature based algorithm for cell misaligned libraries, some cells may be created in multiple reference ndm libraries, each library only contains partial of pvt panes, this will cause cell shadowing during link designs. So special handling is needed to achieve good partition result, we call it cell shadowing fix.

You get this message because cell misaligned logical libraries are detected, and also telling whether the cell shadowing fix is turned on, it is on by default, user can set app option *lib.workspace.group_libs_fix_cell_shadowing* to turn it off.

### Examples

Library pvt1_2_cell has cell LVLH and ISO, library pvt2_1_cell has cell LVLH.

With pure signature based grouping strategy, 2 ndm ref lib will be created: ndm1: libs: pvt1_2_cell cells: 2 cells LVLH and ISO panes: 1 pvt pane pvt1

ndm2: libs: pvt2_1_cell cells: 1 cell LVLH panes: 1 pvt pane pvt2

Cell LVLH is in both ndm1 and ndm2, ndm1 only with pvt pane pvt1, ndm2 only with pvt pane pvt2. ndm1 and ndm2 will shadow each other if both are in linking path.

With cell shadowing fix on grouping strategy, 2 ndm ref lib will be created: ndm1: libs: pvt1_2_cell and pvt2_1_cell cells: 1 cell LVLH panes: 2 pvt panes, pvt1 and pvt2

ndm2: libs: pvt1_2_cell cells: 1 cell ISO panes: 1 pvt pane pvt1

Cell LVLH is in ndm1 with 2 pvt panes.

---

## LM-122

(Error) Multiple designs are detected in etm flow.

### Description

ETM flow will build a library from Extracted Timing Models (ETMs), it will merge multiple PVTs and modes for one design into a single library cell. For each session, will do the merge for only one design. You get this error message because multiple designs are detected in current session.

Feedback

### What Next

Only read DB or LIB files that contain the specific design you want to do the merge and create nlib. You can start again by reading only related DB or LIB files, or you can use *remove_lib* to remove redundant library.

---

## LM-123

(error) Port(s) out of position ports also used in the logic function: %s/%s, %s

### Description

The checker is comparing ports for a particular design to the same design from the base library. A port was found in a different position, that is, the ports are in a different order. In this case, either the port from the comparison design or the base (or both) is also used in the logic function. This is an error, and it cannot be repaired. This is an indication of two libraries are out of sync with one another.

---

## LM-124

(information) Port out of position: %s/%s aligned with %s

### Description

The checker is comparing ports for a particular design to the same design from the base library. A port was found in a different position, that is, the ports are in a different order. Often, this is reparable. If it can be repaired, and no other serious port related errors are found, an LM-029 message will be output indicating the ports were reordered. If not, other error messages will be output.

---

## LM-125

(warning) Could not fix all missing %s

### Description

There were missing objects of the given type, and not all could be resolved. The likely cause is more objects in one library than another, for example, an extra timing arc.

To view the missing objects, you could use *check_workspace -details* and pass in the proper value according to the information you need. For example, use *arcs_progress* for missing arcs, *leakage_progress* for leakage power records, and so on. Note that this can produce large output.

This message is the warning version of LM-025, which is used when the *-allow_missing* option is passed to *check_workspace* and there are no scaling groups. Consult the manpage for *check_workspace* for details on the impact of allowing missing objects.

Feedback

SEE ALSO

**LM-025(n)**

## LM-126

(Error) Some %ss failed to compare.

### Description

The *compare_db* command issues this error message to indicate that some objects, such as timing arcs or leakage power records, did not match between the base library and the comparison library. This message indicates only that errors occurred; previous messages provide the detailed information about the mismatches.

### What Next

Review the previous messages to determine the mismatched objects.

### See Also

- compare_db

## LM-127

(Error) The base library does not contain any logic library information.

### Description

The *compare_db* command issues this error message when it cannot perform the verification because the base library does not contain any logic library information.

### What Next

Ensure that you loaded the correct reference library into the verification workspace.

### See Also

- compare_db
- create_workspace

## LM-128

(Error) Could not find a matching pane in the base library.

Feedback

### Description

The *compare_db* command issues this error message when it cannot find a pane in the base library that matches the voltage, temperature, process number, and process label data of the comparison library.

### What Next

Review the voltage, temperature, process number, and process label data of the comparison library. You can determine this information either from the message issued by the *compare_db* command just prior to this message, or by using the *report_workspace -panes* command.

It is likely that the comparison library did not contribute to the building of the base library. It is also possible you either did not use the *-process_label* option with the *compare_db* command or you used it but specified an incorrect value.

### See Also

- compare_db
- report_workspace

---

## LM-129

(error) Could not correct all port differences with design %s

### Description

Previous messages, starting with LM-028, indicated that there was an issue with some ports on the given design, for example, a port had a different direction in the comparison library and the base library. The application was not able to repair the problems. This is a summary message. Consult the man page for the specific error messages above for the given design.

---

## LM-130

(warning) Shadowed design %s is %s visible from source library %s

### Description

In the library manager, you used *check_workspace* in an aggregate workspace. The check found a cell which appears in more than one library in the workspace. This is called shadowing. Only the cell in the first library in the order will be visible to the user of the aggregate. You will get this message for each library which contains the shadowed cell, indicating whether it will or won't be visible.

Feedback

Sometimes, shadowing is intentional. For example, if you have a complete library then receive an update for a single cell, you can create an aggregate with the update followed by the original. It can also be an error, for example, if you have not created a physical-only library correctly you could potentially have a library with only physical data shadowing a library that has physical and logical data.

**What Next**

If the shadowing is intentional, no action is needed. If it was not intentional, you need to remove one or more libraries and/or adjust their order to make sure the correct version of a cell is found.

---

## LM-131

(Information) Base library does not have a contributing logic library with the exact path '%s'%s.

**Description**

The *compare_db* command issues this informational message to indicate that the comparison library does not have an exact match in the base library based on the file path.

If the command finds a contributing logic library with the same file name, but a different path, the message indicates this. For example, if the comparison library path is /path/to/my.db and the base library contains a contributing library with the path /different/path/to/my.db, the end of the message indicates that the leaf file name was found.

**What Next**

This is only an informational message. If the command does not find the contributing logic library by path name, it searches for a matching pane in the base library by using the PVT information from the comparison library.

**See Also**

- compare_db

---

## LM-132

(information) Configuration change %s file %s

**Description**

In the library manager, you used *set_pvt_configuration* and the change in configuration caused one or more files to be de-activated, or re-activated. In the case of re-activation, it's possible the file was re-read.

Feedback

**What Next**

This is an informational message. No action is needed.

---

## LM-133

(warning) Did not load %s because its PVT (label='%s', P=%.7g, V=%.7g, T=%.7g) does not match the current configuration

**Description**

In the library manager, you used *read_db* or *read_lib* to read a logical library source file. The PVT of the file does not match the current PVT configuration, which you set with *set_pvt_configuration*.

Generally, when using PVT configuration, it is expected that you would attempt to load many more DB or .lib files than you need and let the application filter out the files which don't match your configuration. For example, you might select all DB files in a directory using the *glob* command to match the files. In this user model, this warning is expected.

**What Next**

If you believe that the file should have been loaded, verify that your PVT configuration is set correctly. You can use *report_workspace* to show the current configuration.

---

## LM-134

(warning) Process label '%s' from command line overrides value '%s' from Liberty source

**Description**

In the library manager, you used *read_db* or *read_lib* to read a logical library source file with the *-process_label* option. The source file already had a process label specified on the default operating conditions, and the value you specified on the command line differs from the source file.

**What Next**

Ensure that the process label you are using is correct.

---

## LM-135

(warning) Design %s does not exist in physical libraries

**Description**

During *check_workspace*, when combining physical views into the workspace, neither a frame view nor a design view exists for the named library cell in any physical

Feedback

source library read into the application. Since the library cell does have a timing view, this means it can be timed, but not placed. Because you set the application option *lib.logic_model.auto_remove_timing_only_designs* to true, the application automatically removed the timing-only designs. Without that option, the LM-035 error would have been output. However, we recommend that you investigate why your DB or LIB files have designs that do not exist in any physical source.

---

## LM-136

(information) Must-join port %s missing on %s, will copy from %s

### Description

This message indicates that a must-join port was found on physical view only. It will be copied to logical view later to keep consistency between all views of a design. A must-join port can be defined in LEF by MUSTJOIN statement in PIN section.

### What Next

This is an informational message. No action needed.

---

## LM-137

(Error) Design '%s' does not have a %s view in the base library.

### Description

The tool issues this error message when you are running a library verification command, such as *compare_db*, and the base library does not contain the requested view for a specific design.

For example, if you are using the *compare_db* command, this means that the base library does not have a timing view for the given design.

### What Next

It is rare for a design to exist in the reference library with missing views. This would mean that the design had contributions from some source libraries but not others. Some possible causes of this error are misuse of the design filter application options when building the base library or changes to the comparison library since the base library was built.

### See Also

- compare_db

Feedback

## LM-138

(Information) Allowed missing characterization points since all provided points have %s = %s

### Description

While creating a scaling group, there is a missing characterization point. For example, there is an arc which has a different voltage rail connection on the input and output, and not all combinations of Vout and Vinp have been read. However, if in all provided libraries, the Vinp and Vout values are the same, then the scaling group is allowed, and this informational message is provided in case it is not what the user expected.

## LM-139

(error) Lib %s pin %s %s %s.

### Description

This message indicates there is frequency max_cap or max_trans attribute mismatch on the specified pin between the checking lib and base lib.

The mismatches can be one of following: - attribute missing in one lib. - table index dimensions are different. - table with different index variables. - related_pin attributes on max_cap or max_trans are different.

### What Next

You have to fix the mismatches in the liberty files to pass the *check_workspace*.

## LM-141

(error) %s '%s' has inconsistent receiver capacitance thresholds comparing with base library.

### Description

This message indicates that the library has different receiver capacitance threshold values comparing with base library. All logical libraries should have exact the same number and the same value for both rise and fall of receiver capacitance thresholds.

You cannot write a library until you correct the values.

### What Next

Check with your library vendor and correct the values before proceeding.

Feedback

## LM-142

(warning) Cell %s has width of '%d' tile, not equals to size %lu in vertical_abutment_labels_top/bottom.

**Description**

This message indicates that the cell needs to occupy specified tiles in horizontal row, but the vertical_abutment_labels_top/bottom has different number of labels assigned to the cell.

You should assign exact the same number of labels to the cell as its width in tile.

**What Next**

Check the placement rule file and the cell, correct the setting before proceeding.

## LM-143

(warning) length_precision %u is not identical to target library scaling factor %u. Use target library scaling factor.

**Description**

This message indicates that the length_precision defined in the PRF is not equal to the library scaling factor, which may introduce data inconsistency. By default, library scaling factor is 10000.

**What Next**

Change the length_precision in PRF to same value as the scaling factor in target library.

## LM-144

(warning) %s(%g,%g,%g,%g) is out side of cell boundary of %s, ignored.

**Description**

This message indicates that the rectangle defined by the attribute is not within cell boundary, which will be ignored.

**What Next**

Check the rule and the cell, correct the setting before proceeding.

## LM-145

(warning) Layer %s specified in %s does not exist.

Feedback

### Description

This message indicates that the specified layer defined in ignored_pnet_layers or must_join_group does not exist in current library, it will be ignored.

### What Next

Check the layer setting, correct the setting before proceeding.

---

## LM-146

(warning) tap_rule %s specified in %s under %s group does not exist.

### Description

This message indicates that the tap_rule specified in tap_distance under one tap_boundary_wall_cell_properties group does not exist in current library, this tap_boundary_wall_cell_properties group will be ignored.

### What Next

Define the tap_rule before using it.

---

## LM-152

(Warning) The base library and comparison library have different schema versions.

### Description

The tool issues this warning message when you try to use the verification flow to compare libraries with different schema versions.

### What Next

In general, you should not compare libraries with different schema versions, as this can be very error prone.

### See Also

- compare_db

---

## LM-153

(Error) Differences found for %s '%s'.

Feedback

### Description

The *compare_db* command issues this error message when it finds mismatches between objects, such as timing arcs or leakage power records, in the base library and the comparison library.

For example, the timing arcs could have tables with different data in the two libraries or could be missing tables in one of the libraries.

### What Next

Review the messages issued by the tool following this message; these messages provide information about the objects with mismatches. Note that detailed information about the specific data that is different is not available.

### See Also

- compare_db

---

## LM-154

(Warning) %s has no %ss for %s; %s has %u-> Was this design removed?

### Description

The *compare_db* command issues this warning message when it finds objects, such as timing arcs or leakage power records, that exist in either the base library or the comparison library, but not in both.

### What Next

In many cases, this warning occurs because a design was removed for a particular pane when the reference library was built. This can be due to using the *-allow_missing* option with the *check_workspace* command when the reference library was created or due to the application option settings used when the reference library was created. If this is intended, no action is required.

This warning can also occur if changes occurred to the logic library source file after generating the reference library. In this case, you might need to regenerate the reference library.

### See Also

- compare_db
- check_workspace

Feedback

## LM-155

(Error) %s missing in the base library: '%s'.

### Description

The *compare_db* command issues this error message when an object, such as a timing arc or leakage power record, is missing in the base library.

### What Next

A missing object could be due to one of the following causes: use of the *-allow_missing* option with the *check_workspace* command, changes to the comparison library after building the base library, or other changes that were used to work around out-of-date source files.

### See Also

- compare_db

## LM-156

(Error) Comparison port %s/%s has differences.

### Description

The tool issues this error message when a library verification command, such as *compare_db*, detects a design port that has different attributes in the base library and the comparison library.

### See Also

- compare_db

## LM-157

(Error) Comparison design %s %s port '%s'.

### Description

The tool issues this error message when a library verification command, such as *compare_db*, detects a design that has a different number of ports in the base library and the comparison library.

Note that a design in a logic library might be missing PG ports that exist in the base library; this is valid and is not reported as an error.

Feedback

### See Also

- compare_db

---

## LM-158

(Warning) Comparison library is missing design '%s'.

### Description

The tool issues this warning message when a library verification command, such as *compare_db*, detects a design that exists in the base library but not in the comparison library.

### What Next

There are several reasons why the base library might contain a superset of the designs in the comparison library. For example, if the *-allow_missing* option was used with the *check_workspace* command when the base library was built.

### See Also

- compare_db

---

## LM-159

(Warning) Comparison library has extra design '%s'.

### Description

The tool issues this warning message when a library verification command, such as *compare_db*, detects a design that exists in the comparison library but not in the base library.

### What Next

This warning is likely caused by an application option setting when the base library was built, such as a design filter or the removal of timing-only designs. You should investigate the cause of the warning to determine if you should regenerate the reference library.

This warning is also likely because the base library is one of the outputs of an exploration flow. So it's possible that some designs in the comparison library are existing in other outputs of that exploration flow. But it doesn't exist in the base library.

### See Also

- compare_db

Feedback

## LM-160

(warning) Equal/opposite relationships between %s/%s and %s are different:\n \t%s in %s, %s in %s

### Description

Equal/opposite information, loaded from DB, is different for a design in the base library in comparison to another library.

### What Next

In general, these libraries are incompatible, and you should go back to the source of the library and make sure these libraries were intended to be used together. In some cases, this can be treated as an error. In that case, the LM-060 message is output.

## LM-161

(error) Layer %s (%d) in the source technology file %s does not match the layer of the same name in the destination library.

### Description

You get this error message when you try to apply a new technology file that is incompatible with the current technology in the library. The existing technology is not replaced with the new technology file due to incompatible layer that does not match to the one that already exists in the existing library.

### What Next

Update the technology file so that the layer number and layer name matches the destination library.

### See Also

- read_ndm
- create_workspace

## LM-162

(error) Layer %s (%d) is in the destination library but not in the source technology file %s.

### Description

You get this error message when you try to apply a new technology file that is missing layers that are currently used in the current technology library. The existing technology is

Feedback

not replaced with the new technology file due to missing layer that already exists in the existing library.

**What Next**

Update the technology file so that the layer number and layer name matches the destination library.

**See Also**

- read_ndm
- create_workspace

---

## LM-163

(error) Via definition %s in the source technology file %s does not match the via definition of the same name in the destination library.

**Description**

You get this error message when you try to apply a new technology file that has incompatible via definition that are currently used in the current technology library. The existing technology is not replaced with the new technology file due to incompatible via definition that does not match to the one that already exists in the existing library. Look at the via definition and compare the layer definition to that of source to ensure that they match on layer numbers and layer names before applying to the new technology file with the destination library.

**What Next**

Update the technology file so that the via defintion for layer number and layer name matches the destination library.

**See Also**

- read_ndm
- create_workspace

---

## LM-164

(error) Via definition %s is in the destination library but not in the source technology file %s.

Feedback

### Description

You get this error message when you try to apply a new technology file that has missing via definition that are currently used in the current technology library. The existing technology is not replaced with the new technology file due to missing via definition that already exists in the existing library.

### What Next

Update the technology file so that the via defintion for layer number and layer name matches the destination library.

### See Also

- read_ndm
- create_workspace

---

## LM-165

(error) Site definition %s in the source technology file %s does not match the site definition of the same name in the destination library.

### Description

You get this error message when you try to apply a new technology file that has different site width and height compared to the one in the source library. The existing technology is not replaced with the new technology file due to incompatible site definition that does not match to the one that already exists in the existing library.

### What Next

Update the technology file so that the site definition matches the destination library.

### See Also

- read_ndm
- create_workspace

---

## LM-166

(error) Site definition %s is in the destination library but not in the source technology file %s.

Feedback

### Description

You get this error message when you try to apply a new technology file that has missing sites compared to the one in the destination library. The existing technology is not replaced with the new technology file due to missing site definition that already exists in the existing library.

### What Next

Update the technology file so that the missing site definition are added to the technology file before applying the technology file to the destination library.

### See Also

- read_ndm
- create_workspace

---

## LM-167

(error) The %s library workspace contains cells with layout errors.

### Description

The *check_workspace* command issues this error message if the specified library workspace contains one or more cells with layout errors. Issues in a cell's physical layout can introduce serious DRC violations in the frame view, and therefore in the design.

When the *lib.workspace.library_developer_mode* application option is *true*, cell layout issues are identified as errors. To pass validation by the *check_workspace* command, these issues must be resolved.

### What Next

Fix the issues found in the input LEF file, and then read in the file again.

### See Also

- check_workspace
- lib.workspace.library_developer_mode

---

## LM-168

(error) The %s library contains cells with layout errors.

Feedback

### Description

The *create_frame* command issues this error message if the specified library contains one or more cells with layout errors. Issues in a cell's physical layout can introduce serious DRC violations in the frame view, and therefore in the design.

When the *physical_lib.library_developer_mode* application option is *true*, cell layout issues are identified as errors. To pass validation by the *create_frame* command, these issues must be resolved.

### What Next

Fix the issues found in the input LEF file, and then read in the file again.

### See Also

- read_lef
- create_frame

---

## LM-170

(information) create timing view for variant cell %s

### Description

This message indicates that the tool create timing view for the variant cell of a cell group. The new timing view of the variant cell is just the same as its master's timing view. For example: Users specified the below comamnd before check_workspace: create_cell_group -share_timing_view -name A [get_lib_cells "phyLib/A phyLib/A_1 phyLib/A_2 phyLib/A_3"] The command means A_1, A_2, A_3 share the same timing view as the master A's timing view. When we use A_1, A_2, A_3 in normal flow, if we can find the timing view of the master A, we will create timing view for the variant cells(A_1, A_2, A_3).

### What Next

This is an informational message. No action needed.

---

## LM-171

(information) no timing view for master cell %s in target lib %s

### Description

This message indicates that the tool cannot find the master's timing view in the target library. For example: Users specified the below command before check_workspace: create_cell_group -share_timing_view -name A [get_lib_cells "phyLib/A phyLib/A_1

Feedback

phyLib/A_2 phyLib/A_3"] The command means A_1, A_2, A_3 share the same timing view as the master A's timing view. When we use A_1, A_2, A_3 in normal flow, if we cannot find the timing view of the master A in the target library, the tool will issue LM-171.

**What Next**

This is an informational message. No action needed.

---

## LM-172

(warning) cell %s has its own timing view and it does not share timing view with the master %s

**Description**

This message indicates that the variant cell has its own timing view and it does not share its master's timing view. For example: Users specified the below comamnd before check_workspace: create_cell_group -share_timing_view -name A [get_lib_cells "phyLib/A phyLib/A_1 phyLib/A_2 phyLib/A_3"] The command means A_1, A_2, A_3 share the same timing view as the master A's timing view. If users provide .db/.lib which contains timing data for A_1, then the tool will issue LM-172 for A_1. And A_1 has its own timing view which will not share the timing view of its master A.

**What Next**

Please check the create_cell_group command for the cell of the message, if you want the variant cell share the same timing view as its master, please do not provide timing data for the cell.

---

## LM-173

(Warning) Could not find a matching pane in the base library.

**Description**

The *compare_db* command issues this warning message when it can find the a logic library contributed by a db whose name is the same as the comparison library but it cannot find a pane in the base library that matches the voltage, temperature, process number, and process label data of the comparison library. The comparison will continue using the logical library in the base lib.

The voltage name can be renamed when setting lib.logic_model.use_db_rail_names to false. The voltage values may not matched in this case when finding the matched pane. So no exactly matched pane will be found.

Feedback

### What Next

Review the voltage, temperature, process number, and process label data of the comparison library. You can determine this information either from the message issued by the *compare_db* command just prior to this message, or by using the *report_workspace -panes* command.

It is likely that the comparison library did not contribute to the building of the base library. It is also possible you either did not use the *-process_label* option with the *compare_db* command or you used it but specified an incorrect value.

### See Also

- compare_db
- report_workspace

---

## LM-174

(warning) Bus property only exist in timing view %s/%s.

### Description

A port is defined as a bus port in timing view while in frame view, it is not a bus port. The physical data need to be checked to ensure the missing is expected. *LM-095(n)*

---

## LM-175

(information) Deriving %s pin for cell %s.

### Description

Some signal pins in the cell does not connected to power and/or ground rail, the tool will try to associate the signal pin to the rail if only one exist.

The association won’t be performed on pad/memory/macro cells in PG-pin based libraries, neither for pins that are feedthrough(shorted) or unconnected.

---

## LM-176

(Warning) Failed to update non-existing site definition '%s'!

### Description

The site definition should be found in current technology library. otherwise, cannot update successfully. If you want to create site, please insure width and height should be specified. when width or height is not defined, only update existing site.

Feedback

site ("site_def_1") { width : 2; height : 3; type : pad; symmetry ("Y"); is_default : false; }

---

## LM-177

(Warning) Cell '%s' Port '%s' : The attribute '%s' value '%s' is inconsistent with its value '%s' in logical library!

**Description**

When the attribute is set in logical library and logical library has high priority, no update happens.

---

## LM-178

(Warning) Cell '%s' Port '%s' : Cannot set '%s' to '%s' on non power/ground/bias pin!

**Description**

is_secondary_pg can only be set on power/ground/bias pin!

---

## LM-179

(Warning) Failed to set attribute '%s' value to '%s' on %s.%s!

**Description**

For attribute mask_shift_layers, mask_shift_layers cannot be set when is_mask_shiftable is 'false' in the frame cell.

---

## LM-180

(Warning) Cell group '%s' ignored due to more than one cells are in the group with site id.

**Description**

For cell group with site id, only one cell can be in a cell group. Otherwise, the warning will be issued. For example: library(normal) { distance_unit : 1um; length_precision : 10000; cell(TNBUFFX2_HVT) { cell_group : "group1"; start_site_id(2, 3); } cell(TNBUFFX32_HVT) { cell_group : "group1"; start_site_id(4, 3); } }

Both cell 'TNBUFFX2_HVT' and 'TNBUFFX32_HVT' are in the same group 'group1'. You can change one of the cell_group name to another name.

Feedback

## LM-181

(Warning) The number of routing_blockage with following layer and geometry in %s.%s is inconsistent with PRF file. Ignored!

**Description**

The warning indicates the number of routing_blockage with same layer and geometry doesn't match between input PRF file and FRAME/DESIGN view in clib.

## LM-182

(Warning) Failed to set density %s on layer %s for design %s.%s!

**Description**

Following are possible reason: 1) The layer is neither a base layer which can be found in technology file nor a derived layer. 2) There is no geometry on the layer.

## LM-183

(Warning) Both C1CN and C1C2 data models exist!

**Description**

The warning indicates both C1CN and C1C2 data models exist in this library.

## LM-184

(warning) The routing direction '%s' of layer '%s' is different from the existing routing direction '%s' of the layer. Updating existing to new one.

**Description**

The specified direction is different from the existing routing direction of the layer. The tool will update with the input routing direction.

**What Next**

Please check the routing direction specified is correct.

## LM-185

(warning) Default site definition is missing in the library.

Feedback

### Description

You get this warning message when you update site definition into the library. Tool will check whether there is default site definition after updating site definition through command read_physical_rules. As there will be an error in downstream flow if there is no default site definition.

### What Next

Update physical rule file (PRF) so that the default site definition is set. site(site_def_1) { width: 2; height: 3; type: pad; symmetry (x, y, r90); is_default: true; }

### See Also

- read_physical_rules
- write_physical_rules

---

## LM-186

(warning) Overwrite existing value '%s' for attribute '%s' with new value '%s'!

### Description

You get this warning message when you update user-define attribute into the library. Tool will check whether there is existing attribute before updating.

### What Next

Please make sure you want to overwrite the value in NDM.

### See Also

- read_physical_rules
- write_physical_rules

---

## LM-187

(Error) Library cell %s has incompatible tlatch record across db files.

### Description

The tool issues this error message when check_library detects a library cell has different tlatch record across the db files. If having superset of tlatch record is desired, specify -allow_missing in check_library.

Feedback

**What Next**

Check tlatch consistency across db files. Specify -allow_missing in check_library if want to take union of the record.

---

## LM-188

(warning) Ignore invalid %s with %s.

**Description**

This message indicates that the antenna_rule or antenna_layer_rule has syntax error, which will be ignored. Please check the detail error before the message.

**What Next**

Check the rule and correct the setting before proceeding.

---

## LM-189

(information) Setting terminal class to bump on port '%s'.

**Description**

Pad pins in a macro cell must be of class bump. Sets pin class to bump for pins which are not of class bump.

**What Next**

This is just an informational message; no action is necessary.

---

## LM-190

(warning) Cannot find port '%s'/'%s'.

**Description**

The pin in the attribute does not exist.

**What Next**

Check the attribute, such as via_ladder_pin_order, make sure that each pin actually exsits in cell.

---

## LM-191

(warning) Failed to create track_pattern '%s' for %s '%s'.

Feedback

**Description**

This message indicates the track_pattern is not create successfully for the row_pattern or site.

**What Next**

Check the track_pattern and correct the track_pattern

# LMUI

## LMUI-001

(error) %s cannot be the empty string

**Description**

The given option must have at least one character.

**What Next**

Re-enter the command with a non-empty string passed to the given option.

## LMUI-002

(error) All libraries in a scaling group must have the same process information

**Description**

You passed a set of libraries to *create_scaling_group* and there's more than one process id and/or process label between them.

**What Next**

Either remove some of the libraries from the collection passed to *create_scaling_group*, or if you believe the process information is wrong in one or more libraries, use *set_process* to correct them. You can also rerun the application and pass a more consistent set of process labels to *read_db*. Or

## LMUI-003

(error) Library %s has no timing view; cannot %s

Feedback

### Description

Certain operations in the application require that the library used has a timing view. For example, the libraries passed to *create_scaling_group* or *rename_rail* must have a timing view.

---

## LMUI-004

(error) Library %s is already merged.

### Description

The libraries passed to the *create_scaling_group* command have already been merged. This means that you previously created scaling groups and wrote out a reference library, and then tried to create more scaling groups. You must create all of your scaling groups before writing the reference library.

---

## LMUI-005

(error) Cannot %s because there is no library workspace.

### Description

The first step in the library preparation flow is to create a library workspace by using the *create_workspace* command. You must have an open library workspace to perform library preparation tasks.

### What Next

Create a library workspace, and then reenter the command you used.

### See Also

- create_workspace

---

## LMUI-006

(error) You can have only one library workspace.

### Description

The first step in the library preparation flow is to create a library workspace by using the *create_workspace* command. You can have only one library workspace open at a time. The tool issues this message when you try to create a library workspace when another workspace is already open.

Feedback

### What Next

Before creating a new library workspace, either complete the library preparation flow for the current workspace and commit it to a saved reference library, which removes the workspace from memory, or remove the current workspace from memory without committing it by using the *remove_workspace* command.

---

## LMUI-007

(error) Invalid library name

### Description

The library name specified to must be compatible with the filesystem. The name specified cannot be used as a file name because it contains invalid characters, like space or slash. Further restrictions disallow certain characters that are used in the specification of a library name. For example, vertical bar is used in the full name of an aggregate member library.

### What Next

Use a different name.

---

## LMUI-008

(error) Cannot %s %s the library workspace.

### Description

You attempted an invalid action using objects in the library workspace. For example, you cannot apply the *remove_lib_cell* command to library cells in the workspace.

### What Next

Use a different object.

---

## LMUI-009

(warning) Cannot specify the %s %s in the %s flow. It is ignored.

### Description

In the library manager, you attempted to specify a command or command option that is incompatible with the current flow, so it was ignored.

For example, if you specify the option *-allow_missing* to *check_workspace* in the *physical_only* flow, you will see this message, because the option does not apply to the *physical_only* flow.

Another example is using *process_workspaces -output* in the exploration flow and more than one sub-workspace is being processed, and you are not creating an aggregate (with the application option *lib.workspace.explore_create_aggregate*).

---

## LMUI-010

(error) Cannot %s after a successful check_workspace

**Description**

In the library manager, you attempted an operation after a successful *check_workspace* which is not allowed at that point. For example, you cannot change the process label or number for a library with *set_process* after a successful check.

---

## LMUI-011

(error) Must specify %s

**Description**

In the library manager, a command requires that something is specified. For example, if a command has options -a and -b and one or more of them is required but none is specified, some commands will issue this message.

---

## LMUI-012

(warning) Cannot use %s on %s %s:\n \t%s

**Description**

In the library manager, a command was used on an invalid object. For example, if you try to use *set_process* on a physical library, it will be rejected because that library does not have a timing view where process information is stored.

---

## LMUI-013

(information) %s %s already has these settings

**Description**

In the library manager, you used a command to change settings on one or more objects, and the objects already have those settings. For example, if you used *set_process* and the library already has the process label and number that you are trying to set, you would get this informational message in case your entry was a typo.

Feedback

## LMUI-014

(error) Could not %s for library %s

**Description**

In the library manager, a problem occurred in a command which prevented it from completing. Previous messages should indicate the specific problem. This is a summary message.

## LMUI-015

(error) %s '%s' does not exist%s

**Description**

An object specified to a command does not exist, and it must for the command to function.

## LMUI-016

(error) %s '%s' already exists%s

**Description**

An object specified to a command already exists, and it must not for the command to function.

## LMUI-017

(error) Library file '%s'\n \tis already registered.

**Description**

You tried to read a file into the library manager which has already been loaded into memory.

**What Next**

If the file has changed and you want to reread it, you can remove the library using the *remove_lib* command, then reread the file.

## LMUI-018

(error) %s failed: %s.

Feedback

**Description**

Some commands require auxiliary files either in the installation root or set in application options. Something is missing, and the message will tell you which it is.

**What Next**

Ensure that the software is installed correctly and all required files in the installation and elsewhere are present.

---

## LMUI-019

(error) Read of Liberty file failed: %s

**Description**

This is a summary message at the end of *read_lib* if one or more of the files you provided failed to load.

**What Next**

The text in this message and/or other error messages from the reader should guide you to the problem in the .lib file(s) that need to be corrected.

---

## LMUI-020

(error) The %s command is not supported in fast exploration mode.

**Description**

In the fast version of the exploration flow, you create library groups and write scripts for them using *group_libs* and *write_workspace*. You cannot process (check and commit) the workspaces. The fast version (enabled with the application option *lib.workspace.fast_exploration* set to true) can be more than 50 times faster than the original version. It is functionally equivalent to process a workspace versus writing a workspace creation script and sourcing it back.

---

## LMUI-021

(error) Cannot include library '%s' in a scaling group:\n \t%s

**Description**

You used *create_scaling_group* and a library that you passed to it was disqualified from being included in a scaling group. The reason is given in the message. One reason is that the library does not have any pins. It might have cells, but they are pinless, and therefore there are no arcs, checks, or capacitances to scale.

Feedback

### What Next

Verify that you are using the correct libraries for scaling, or remove the command from your script.

---

## LMUI-022

(error) Cannot specify %s library as the seed for the edit or verification flows.

### Description

When you create a workspace with the *create_workspace* command using the edit or verification flows, you can only supply a fully committed library as the library argument. You might have specified a frame-only NDM library, or some other type of NDM library which is not allowed.

### What Next

If you want to edit a frame library, use the frame flow, and *read_ndm* to load the file into the workspace.

### See Also

- create_workspace
- read_ndm

---

## LMUI-023

(error) Command %s requires the %s flow workspace.

### Description

The command that you just used was used in a flow that does not support the command.

### What Next

Create a flow workspace that supports the command before you use the command.

### See Also

- create_workspace
- remove_workspace

---

## LMUI-024

(information) %s is ignored in the aggregate flow.

Feedback

**Description**

An aggregate library is an NDM library that contains a group of individual NDM lib-cell libraries. The command or command option you entered is invalid while creating an aggregate library in the aggregate flow.

---

## LMUI-025

(error) Cannot specify %s library as the seed for the aggregate flow.

**Description**

When creating a workspace with the aggregate flow, you can pass in a new library name to create a new aggregate, or an existing aggregate library to modify it (add new libraries, remove libraries, reorder libraries). The existing library must be an aggregate library. It cannot be any other type of NDM library.

**What Next**

Make sure you are using the correct library.

---

## LMUI-026

(error) Cannot use '%s' when there are open libraries

**Description**

You tried to use a command which requires no open libraries. For example, you cannot create a workspace while there are open libraries.

**What Next**

Close all of the open libraries.

---

## LMUI-027

(information) Library is effectively read-only.

**Description**

The *open_lib* command in the library manager shell is used to query a reference library. It is not used to open a library for the purpose of editing and saving. If you want to change the library, create a workspace with the appropriate flow to edit the type of library you are planning to change.

Feedback

## LMUI-028

(error) Cannot read logic library source file in edit flow.

**Description**

In edit flow, Library Manager cannot read logic library files.

## LMUI-029

(error) Cannot %sworkspace %s

**Description**

In the library manager, you attempted an operation on a workspace which is invalid. For example, you tried to remove a sub-workspace (a workspace created by *group_libs*). Or, you tried to use *commit_workspace*, *check_workspace*,

## LMUI-030

(warning) Without -mode_label, the library will be modeless

**Description**

In the ETM flow, you specify the mode for the Liberty library with the *-mode_label* option to *read_db* or *read_lib*. The option is not required, so we warn you just in case you wanted to specify a mode and forgot.

**What Next**

There is no way to apply a mode to the logical source library, so if that was your intention, you will need to remove the library with *remove_lib*, and re-read it, specify the *-mode_label* option.

## LMUI-031

(warning) Ignored technology file '%s' in edit flow .

**Description**

The technology file specified to *create_workspace* will be ignored in edit flow.

## LMUI-032

(warning) Can not remove lib '%s', it is associated with current workspace.

Feedback

### Description

The lib which associated with current workspace is loaded during *create_workspace* in edit flow, this lib can not be removed, otherwise the current workspace will be invalid. You can use command *remove_workspace* to remove workspace and it's associated lib.

---

## LMUI-033

(error) Option -mode_label is not consistent with previous read_db or read_lib command.

### Description

In the ETM flow, if you want to create a multiple modes library, use option -mode_label for every *read_db* and *read_lib* command. In the ETM flow, if you want to create a single mode library, you should not use option -mode_label in any *read_db* or *read_lib* command. If you specify -mode_label option in one *read_db* or *read_lib* command, you have to specify -mode_label option for every *read_db* or *read_lib* command. Otherwise, you will get this *LMUI-033* error message.

---

## LMUI-034

(warning) Without -mode_label, the library will be modeless

### Description

In the ETM flow, you specify the mode for the Liberty library with the *-mode_label* option to *read_db* or *read_lib*. The option is not required, so we warn you just in case you wanted to specify a mode and forgot. If you specify the mode for any Liberty library, you have to specify mode for all Liberty libraries in current session. You cannot have mixed moded and non-moded logical libraray in one workspace in ETM flow, LMUI-033 will be issued for such case.

### What Next

There is no way to apply mode to the logical source library, so if that was your intention, you will need to remove the library with *remove_lib*, and re-read it, specify the *-mode_label* option.

---

## LMUI-035

(error) Cannot read %s views with read_ndm in the %s flow

### Description

In some flows, *read_ndm* can only read certain views. You requested a view/flow combination that is not supported.

Feedback

## LMUI-036

(warning) Option %s is applicable to the %s command only if the flow type of the current workspace is %s. The option was ignored.

### Description

In the library manager, you attempted to specify a command option that is not applicable in the current flow; therefore, the option was ignored. For example, if you specify option *-allow_missing* to the *check_workspace* command in the *physical_only* flow, since the option does not apply to the *physical_only* flow it will be ignored.

### What Next

You can use the "get_attribute [get_workspace [current_workspace]] flow" command to show the flow type of the current workspace.

### See Also

- current_workspace

## LMUI-037

(warning) Multiple antenna layer rules defined in layer '%s' with name '%s' and mode '%d'. The latter one will overwrite the former one.

### Description

This warning is issued when multiple antenna layer rules defined in a layer with same name and same mode with command "define_antenna_layer_rule". Only the last rule will take effect.

### What Next

Remove redundant antenna layer rules or rename them.

## LMUI-038

(error) Antenna layer rule defined in layer '%s' with name '%s' and mode '%d' associates with an undefined antenna rule.

### Description

This error is issued when an antenna layer rule defined in a layer with command "define_antenna_layer_rule", But there is no antenna rule defined with the name and the mode with command "define_antenna_rule".

Feedback

**What Next**

Add antenna rule before the antenna layer rule.

---

## LMUI-039

(error) Cannot %s when app option lib.workspace.read_mw_frame is true.

**Description**

When app option lib.workspace.read_mw_frame is true, that means you want to create a library from Milkyway FRAM view directly. In this case, only a packed file for Milkyway FRAM view data can be read by read_lef. All other data cannot be read into the workspace. Moreover, only one packed file can be read by read_lef.

**What Next**

Commit or remove the current workspace and start a new library preparation flow.

---

## LMUI-040

(Error) Technology data in %s and workspace technology data is not compatible

**Description**

Technology data in the library which is reading and workspace technology data must be compatible, otherwise *create_workspace* will fail.

**What Next**

Try another libary with compatible technology data.

---

## LMUI-041

(error) Cannot invoke the %s command because %s file has already been loaded.

**Description**

The read_lef, read_gds, read_ndm, read_db, and read_lib commands cannot be invoked if the import_icc_fram command has already been invoked to successfully read an icc fram tar file, which was created by the export_icc2_frame command in the icc shell, into the current workspace, and vice versa.

**What Next**

Commit or remove the current workspace and start a new library preparation flow.

Feedback

### See Also

- import_icc_fram

---

## LMUI-042

(Warning) No workspace to %s.

### Description

There is no workspace for the specified operation. Either there is no sub-workspaces at all, or the workspace specified is an exploration mode workspace.

### What Next

Please read some logical and physical libraries, and use *group_libs* command to generate some sub-workspaces, then try this command again.

### See Also

- process_workspaces
- write_workspace

---

## LMUI-043

(warning) PVT configuration is not used in the %s flow

### Description

This is just a reminder that PVT configuration is not used by the library manager workspace flow you are in. Either you have an existing PVT configuration and you created a workspace in a flow that does not use it, or you made a change to the PVT configuration using *set_pvt_configuration* while in one of those flows.

---

## LMUI-044

(information) %d/%d workspaces %s successfully

### Description

In the library manager, while processing workspaces in the exploration flow, a step completed, and this is an informational message to tell you the results of that step.

---

## LMUI-045

(warning) Cannot use the %sworkspace as an argument to %s. It is ignored.

Feedback

**Description**

In the library manager, you attempted an operation on a workspace which is invalid. For example, you passed the root workspace in the exploration flow to *write_workspace* or *process_workspaces*. The application ignored this argument.

---

## LMUI-046

(warning) Option '%s' is ignored %s

**Description**

In the library manager, you used an invalid option which is ignored. For example, your current workspace is a sub-workspace, and you tried to pass a list of workspaces to *process_workspace* or *write_workspace*. In these cases, only the current workspace will be processed or written.

---

## LMUI-047

(warning) Must_join_group '%s' on terminal '%s' of block '%s' will be mapped to 1.

**Description**

When convert must_join_group information of a library from IC Compiler II to IC Compiler, all defined values will be mapped to 1. If the value is not 1, then it will lose information.

**What Next**

This is a warning message and tool will continue. There is no action needed if the lose of information is expected.

---

## LMUI-048

(warning) Ignored must_join_group '%s' on terminal '%s' of block '%s'.

**Description**

When convert must_join_group information of a library from IC Compiler II to IC Compiler, only terminals with rect and polygon shapes will be converted. The specified terminal is not with rect and polygon shapes so the must_join_group information of it will not be converted.

**What Next**

This is a warning message and tool will continue. There is no action needed if the lose of information is expected.

Feedback

## LMUI-049

(information) Use mode label '%s' to read db as there is %s match in the base lib.

**Description**

The *compare_db* command found the specified db file matches one in the base lib, and the one in the base lib was read using the reported mode label. So the *compare_db* command will use this mode label to read specified db file.

The match could be either a full file name match or a leaf file name match.

**What Next**

This is a informational message and no action is required.

## LMUI-050

(Warning) The %s %s is deprecated. %s

**Description**

Some aspect of the user interface (a command, an option, a value for an option, an application option, and so on) has become deprecated and will be removed in a subsequent release.

**What Next**

This is a warning. Your action will work as planned. The message text will indicate what the correct action or replacement is for the deprecated feature.

## LMUI-051

(Error) The %s %s is obsolete. %s

**Description**

Some aspect of the user interface (a command, an option, a value for an option, an application option, and so on), which was deprecated in a previous release, is now obsolete. It will be completely removed in a subsequent release.

**What Next**

The message text will indicate what the correct action or replacement is for the deprecated feature. As this is an error, you will need to change your usage to continue.

Feedback

## LMUI-052

(information) Can reuse library %s for workspace %s.

### Description

This message indicates that an existing library can be reused for the specified workspace. The processing of the workspace will be skipped.

## LMUI-053

(information) Cannot reuse library %s for workspace %s: %s.

### Description

This message indicates that no existing library can be reused for the specified workspace. The reason was given with the message. It will continue to process the workspace.

## LMUI-054

(information) The default value of %s will be '%s' in %s release.

### Description

This message indicates the default value of the specified application option will be changed in specified release. No action required in this release, but you need pay attention to the flow that will be impacted.

## LMUI-055

(warning) The size of the %s library cell does not match the site_def %s.

### Description

The *set_cell_site* command issues this warning message when the height or width of the specified library cell is not an integer multiple of the specified site_def, the command will not change the site_def of this cell.

### Example Message

```
Warning: The size of the AND library cell does not match site_def CORE.
 (LMUI-055)
```

### What Next

Change the site_def definition by using the *set_attribute* command.

Feedback

### See Also

- read_lef
- set_attribute

---

## LMUI-056

(information) The '%s' library cell is ignored.

### Description

The *set_cell_site* command issues this message because the type of the library cell is one of macro, black_box, module, pad, pad_spacer, flip_chip_driver, corner_pad. The command will not change the site_def of this cell.

### What Next

No action requried.

### See Also

- read_lef
- set_attribute

---

## LMUI-057

(error) : Failed to create cell group '%s' for cell list (%s)!

### Description

This message indicates that the cell group is not created successfully from EEQ in input LEF file, or from physcial_variant_cells in input db/lib file.

### What Next

Please check the error message printed by create_cell_group and correct them. For example, if the cell group already exists, please use command remove_cell_group to remove the cell group and redo check_workspace.

### See Also

- read_lef

---

## LMUI-058

(information) : Cell group '%s' is created successfully for cell list (%s)!

Feedback

### Description

This message indicates that the cell group is created successfully from EEQ in in input LEF file, or from physcial_variant_cells in input db/lib file.

### What Next

No action required

### See Also

- read_lef

---

## LMUI-059

(warning) Reading new physical rules, unsaved data in previous execution will be lost.

### Description

The command read_physical_rules is executed more than once, the cell level attributes in previous file will be lost if have not been applied by check_workspace.

### What Next

Check the physical rules by write_physical_rules.

---

## LMUI-060

(information) %s is only avaiable in the %s flow.

### Description

The command or command option you entered is invalid while current workspace is not in the specified flow.

---

## LMUI-061

(warning) Application option '%s' will be ignored in the %s flow.

### Description

The reported application option will be ignored in the specified flow. Please refer to the man page of the application option for details.

---

## LMUI-062

(error) Cannot rename %s to an existing library cell %s.

Feedback

**Description**

The message indicates you are trying to rename a library cell to an existing library cell in the same library, which is not allowed.

**What Next**

Correct the command line option or remove the library cell.

---

## LMUI-063

(information) Trim leading and trailing whitespaces for %s value: changed '%s' to '%s'.

**Description**

This message indicates the tool removes the leading and trailing whitespaces of the specified value.

---

## LMUI-064

(error) Invalid %s value '%s' is not used since it contains '\\n'.

**Description**

The value specified is invalid since it contains invalid characters ('\\n').

**What Next**

Remove the invalid charaters

---

## LMUI-065

(warning) The '-scale_factor' option is ignored.

**Description**

The "-scale_factor" option is ignored when the *design.allow_multiple_scale_factors* app option is set to true. If used, the specified *-scale_factor* must be same as the *scale_factor* specified with the app option *design.session_scale_factor* .

**What Next**

Nothing, it is an informational message. Please do not specify the "-scale_factor" if *design.allow_multiple_scale_factors* is set

---

## LMUI-066

(error) The specified scale factor '%d' is not usable.

Feedback

**Description**

The specified scale_factor is not usable for scale factor conversion. The scale factor specified must be larger than current scale factor of the library and must be an integral multiple of scale factor of the library being opened for edit. For example, if the library has a scale factor of 10000, then a new scale factor of 20000 is okay to use. However, a scale factor of 4000 or 8000 will result in an error.

**What Next**

Please specify a usable scale factor.

# LNK

## LNK-003

(error) Block '%s' was not successfully linked.

**Description**

A *link_block* message indicating that the link process failed for your design.

The link process will fail if there are port mismatches between modules within a design, or to lib cells. The link process will also fail if there are circular or recursive references in the module hierarchy.

**What Next**

See previous error messages for more details. Either update your Verilog source, correct the ref_libs, view switch list, or *search_path* used for this design.

**See Also**

- link_block
- read_verilog
- remove_blocks
- set_ref_libs
- set_view_switch_list
- search_path

## LNK-005

(warning) Unable to resolve reference to '%s' first referenced from module '%s'.

Feedback

### Description

The reference was not resolved to a module, block or library cell.

This warning is issued once for the first module containing a cell making this reference. Additional cells making this reference (in this module or in other modules within this block) are included in the unbound instance count at the end of *link_block*.

### What Next

Look further back in the error log to see if a previous error caused this reference to be unresolved.

If the reference is the name of a module in your Verilog source, see if the module was loaded with *get_modules module_name*. Also, check the Verilog source to verify that that module was read with the same *read_verilog* command as the module reference.

Use *get_blocks* *:*block_name*.* to see if there is a block with that name in any open library.

Check the view switch list with *get_view_switch_list* to see if the views selected by the switch list match any available views of that block. Lib cell references automatically select the timing, frame, or design views regardless of the view switch list setting.

Run *list_blocks -ref_libs -lib_cells > file* and examine the file to see if the desired view of the block is shadowed by another block with the same name in an earlier reference library.

### See Also

- get_blocks
- get_modules
- set_ref_libs

---

## LNK-008

(warning) Cannot find port '%s' on reference '%s', referenced by instance '%s'.

### Description

During the link process, an instance in the design could not be resolved because a port on the instance was not found in the design. This indicates a mismatch between the pinout of the instance and the design to which it should have resolved.

For example, if you have only these three instances of an FD1 in a design:

```
FD1 u1 ( .Q(n1), .CP(clock1), .D(data1));
  FD1 u2 ( .Q(n2), .CP(clock2), .D(data2));
  FD1 u3 ( .Q(n3), .CK(clock3), .D(data3));
```

Feedback

There is a typo in the third instance, listing "CK" instead of "CP". By default, you will get the LNK-008 message for the first instance only.

```
Warning: Cannot find port 'CK' on design 'FD1',
       referenced by instance 'u1'. (LNK-008)
```

This might seem confusing, since that instance does not have a CK port. But for a name-based reference, each instance expects at least the total of all listed ports. To see all instances which cannot be resolved, use the -verbose option to *link_block*:

```
Warning: Unable to resolve reference to 'FD1' in 'top'. (LNK-005)
Warning: Unable to resolve reference to 'FD1' in 'alu'. (LNK-005)
```

The LNK-005 messages are generated in addition to LNK-008.

**What Next**

Examine the library's ref_lib list, log messages, and use the *which* command to find out which files were loaded. This could also be due to a typo in the netlist source.

**See Also**

- link_block
- set_ref_libs
- LNK-005

---

## LNK-010

(error) In module '%s', too few ports on instance '%s' referencing '%s'.

**Description**

The specified reference to a module, block or library cell does not have enough ports. This indicates a mismatch between the library and the netlist.

**What Next**

Make sure the library is correct. Then, verify that the netlist is referencing the correct cell.

---

## LNK-011

(error) In module '%s', too many ports on instance '%s' referencing '%s'.

**Description**

The specified reference to a module, block or library cell has too many ports. This usually indicates a mismatch between the library and the netlist. It can also indicate that there are

Feedback

multiple conflicting references to the same library cell or black box. One such conflict is two references with different pin counts.

**What Next**

This error causes the link to fail. Make sure the library is correct. Then, verify that the netlist is referencing the correct cell.

---

## LNK-012

(error) In module '%s', width mismatch on port '%s' on instance '%s' referencing '%s'.

**Description**

The linker matched a bused port on an instance with a bused port on a library cell, but bus width is different between the two. This could indicate an incorrect library or a netlist which is out of date with the library.

**What Next**

Verify that your source is in sync with the library.

---

## LNK-017

(error) In module '%s', port mismatch on instance '%s' referencing '%s'.

**Description**

The reference to the module has an interface that is incompatible with the ports that are defined on that module. This indicates an inconsistency between modules within the Verilog netlist.

**What Next**

Correct the inconsistency within the Verilog netlist. Then re-read the Verilog.

---

## LNK-023

(error) Recursive hierarchy detected in block '%s': %s.

**Description**

The linker detected recursive hierarchy in the block that is being linked. This is a design error. The message will list the modules that create the recursion.

For example, if module A has an instance of B and module B has an instance of A, that is a recursive loop.

Feedback

### What Next

Remove the recursion, reread the blocks, and relink.

---

## LNK-024

(error) Cannot link flat library cell %s.

### Description

The current block is a flat library cell. Flat library cells have no hierarchy, so they cannot be linked.

### What Next

Change the current block to a design block, then link the design from there.

### See Also

- current_block
- link_block

---

## LNK-025

(error) Cannot rebind block %s before it has been linked.

### Description

The current block has been newly read in, but it has not yet been linked. A block cannot be rebound before initial link.

### What Next

Change the current block to the block that should be rebound. Then rebind that block.

### See Also

- current_block
- link_block
- rebind_block

---

## LNK-026

(warning) Block %s has conflicting link contexts: library %s and library %s.

Feedback

### Description

This multiply instantiated block is referenced by parent blocks that are linked with different hierarchical reference library lists.

When the entire block hierarchy is linked with the same reference library list, it is straight forward to decide how to resolve references within each block.

But when hierarchical reference library lists are used, a multiply instantiated block can be referenced from different libraries each with different reference library lists. In this case it is not clear which reference library list should be used to resolve references within the block. The first library's reference library list is used, but that may not be the designer's intent.

### What Next

Review the block hierarchy. Determine which blocks should be linked with which reference library lists.

If a library has the *use_hier_ref_libs* attribute incorrectly set, the setting can be removed with:

```
prompt> set_attribute [get_libs library-name] false
```

If a library is missing the *use_hier_ref_libs* attribute, the setting can be removed with:

```
prompt> set_attribute [get_libs library-name] true
```

If a multiply instantiated block really should be linked with different reference library lists, the block should be copied so that instances that are to be linked different reference library lists reside in different libraries.

```
prompt> copy_block -from_block my-block -to_block new-lib:my-block
```

### See Also

- copy_block
- link_block
- set_ref_libs

---

## LNK-031

(information) Skipping module '%s', using frozen block '%s'.

### Description

This message tells you that a stub module in the Verilog source is being skipped and the 'frozen' version of the block is being used from the reference library list. A block is

Feedback

considered frozen if it has a FRAME view. This action only occurs during the initial link of newly read Verilog.

If the app option *link.prefer_frame* is set to true, then 'frozen' blocks (that is, blocks with a FRAME view) are preferred over Verilog modules. If the app option is set to false, then Verilog modules are preferred over 'frozen' blocks.

When a 'frozen' block is used, the linker uses the view switch list to decide which view of that block to link to. This can be changed with the *set_view_switch_list* command. Individual instances can be changed to reference a different view with the *change_abstract*, *change_view* or *set_reference* commands.

The number of skipped module messages displayed for each block is controlled by the app option *link.reference_limit*. Additional messages are suppressed. When a block has skipped modules, a message is displayed giving the number of modules skipped for each block.

**What Next**

No action is required. This is just information only.

To see more skipped module messages, set the app option *link.reference_limit* to a larger number, re-read the verilog and relink the block.

```
prompt> set_app_options -global { link.reference_limit 99999 }
prompt> read_verilog ...
prompt> link_block
```

To see the names of all the modules being suppressed, set the app option *link.reference_limit* to -1, re-read the verilog and relink the block.

```
prompt> set_app_options -global { link.reference_limit -1 }
prompt> read_verilog ...
prompt> link_block
```

To not prefer a specific block over the Verilog module, remove the library containing that block from the reference library list.

```
prompt> set_ref_libs -remove frozen_blocks.nlib
```

To remove the 'frozen' state from a specific block, remove the FRAME view of that block.

```
prompt> remove_blocks -force myLib:myBlock.frame
```

To always prefer Verilog modules over 'frozen' blocks, set the *link.prefer_frame* app option to false.

```
prompt> set_app_options -global { link.prefer_frame false }
```

Feedback

### See Also

- change_abstract
- change_view
- get_view_switch_list
- link_block
- link_options
- read_verilog
- remove_blocks
- set_app_options
- set_reference
- set_view_switch_list

---

## LNK-032

(information) Skipped %d modules from block '%s'.

### Description

This message tells you that stub modules in the Verilog source have been skipped and the 'frozen' versions of those blocks are being used from the reference library list. A block is considered frozen if it has a FRAME view. This action only occurs during the initial link of newly read Verilog.

If the app option *link.prefer_frame* is set to true, then 'frozen' blocks (that is, blocks with a FRAME view) are preferred over Verilog modules. If the app option is set to false, then Verilog modules are preferred over 'frozen' blocks.

When a 'frozen' block is used, the linker uses the view switch list to decide which view of that block to link to. This can be changed with the *set_view_switch_list* command. Individual instances can be changed to reference a different view with the *change_abstract*, *change_view* or *set_reference* commands.

The number of skipped module messages displayed for each block is controlled by the app option *link.reference_limit*. Additional messages are suppressed. When a block has skipped modules, a message is displayed giving the number of modules skipped for each block.

### What Next

No action is required. This is just information only.

Feedback

To see more skipped module messages, set the app option *link.reference_limit* to a larger number, re-read the verilog and relink the block.

```
prompt> set_app_options -global { link.reference_limit 99999 }
prompt> read_verilog ...
prompt> link_block
```

To see the names of all the modules being suppressed, set the app option *link.reference_limit* to -1, re-read the verilog and relink the block.

```
prompt> set_app_options -global { link.reference_limit -1 }
prompt> read_verilog ...
prompt> link_block
```

See the manual page for LNK-031 for information on how to control which blocks are considered 'frozen'.

```
prompt> man LNK-031
```

To always prefer Verilog modules over 'frozen' blocks, set the *link.prefer_frame* app option to false.

```
prompt> set_app_options -global { link.prefer_frame false }
```

**See Also**

- change_abstract
- change_view
- get_view_switch_list
- link_block
- link_options
- read_verilog
- set_app_options
- set_reference
- set_view_switch_list

---

## LNK-040

(information) User %sunits loaded from library '%s'

Feedback

### Description

This message appears when you have not explicitly set your design units (time, capacitance, etc.) with the *set_user_units* command. The units used in your shell session will be taken from the default units in the given library.

### What Next

You can use *report_user_units* to see what units have been selected. You can use the *set_user_units* to force different units of your choice.

### See Also

- set_user_units
- report_user_units

---

## LNK-041

(warning) Cannot connect net to internal pin '%s'

### Description

Net cannot be connected to internal pins of a lib_cell. Such connections will be ignored by *link_block*.

### What Next

Please check the verilog file to correct references to internal pins.

### See Also

- link_block

---

## LNK-042

(warning) Cannot link scalar net to bus port '%s' on reference '%s', referenced by instance '%s'.

### Description

During the link process, an instance in the design could not be resolved because a bus port on the instance was being connected to a scalar net. This indicates a mismatch between the pinout of the instance and the design to which it should have resolved.

The LNK-005 messages are generated in addition to LNK-042.

Feedback

### What Next

Review your source rtl.

### See Also

- link_block
- LNK-005

---

## LNK-043

(information) Message limit (link.reference_limit %d) reached for this block, remaining reference messages will be suppressed.

### Description

The number of unresolved references in this block has exceeded the limit set in the app option *link.reference_limit*. Additional error messages about unresolved references will be suppressed.

### What Next

To see more unresolved reference messages, set the app option to a larger number and relink the block.

```
prompt> set_app_options -global { link.reference_limit 99999 }
prompt> link_block
```

To see all unresolved reference messages, set the app option to -1 and relink the block.

```
prompt> set_app_options -global { link.reference_limit -1 }
prompt> link_block
```

### See Also

- get_app_option_value
- link_block

---

## LNK-044

(info) Changed reference of instance '%s' from '%s' to '%s'.

### Description

An exact match for the reference could not be found but a match was found using case insensitive search. Based on design repair settings, the reference was changed.

Feedback

### What Next

The design has been repaired as based on the repair settings set by the user.

If the repair is not acceptable, please change the repair settings and start again.

### See Also

- get_blocks
- get_modules
- set_ref_libs

---

## LNK-045

(info) Disconnected pin '%s' of instance '%s', in module '%s'.

### Description

The pin of instance is being disconnected from its net

The design is being repaired based on the repair settings set by the user. The pin of the instance does not have a corresponding port on the reference the instance is being bound to, so the pin is being disconnected and will be deleted.

### What Next

The design has been repaired as based on the repair settings set by the user.

If the repair is not acceptable, please change the repair settings and start again.

### See Also

- get_blocks
- get_modules
- set_ref_libs

---

## LNK-046

(info) Removed pin '%s' of instance '%s', in module '%s'.

### Description

The pin of instance is being deleted.

Feedback

The design is being repaired based on the repair settings set by the user. The pin of the instance does not have a corresponding port on the reference the instance is being bound to, so the pin is being deleted.

**What Next**

The design has been repaired as based on the repair settings set by the user.

If the repair is not acceptable, please change the repair settings and start again.

**See Also**

- get_blocks
- get_modules
- set_ref_libs

---

## LNK-047

(info) Renamed pin '%s' of instance '%s', to '%s', in module '%s'.

**Description**

The pin of instance is being renamed.

The design is being repaired based on the repair settings set by the user. The pin of the instance is being mapped to a port with different name on the reference the instance is being bound to.

**What Next**

The design has been repaired as based on the repair settings set by the user. Some possibilities are that the name is being mapped by a case insensitive match or the names are declared as synonyms

If the repair is not acceptable, please change the repair settings and start again.

**See Also**

- get_blocks
- get_modules
- set_ref_libs

---

## LNK-050

(error) Unable to link design. Design exceeds the database capacity.

Feedback

### Description

The tool is not able to link this design, because the number of objects in the design exceeds the database capacity.

### What Next

Please contact Synopsys for technical help in resolving this issue.

### See Also

- link_block

---

## LNK-053

(information) Linking instances in module '%s':

### Description

The *link_block* command prints the name of each module as it links when run with the *-verbose* flag.

### What Next

Run the *link_block* command without the *-verbose* flag.

### See Also

- read_verilog
- link_block

---

## LNK-055

(warning) Referenced block '%s' does not have a physical view.

### Description

The linked block references another block that does not have a physical description defined. Without a physical definition of sub-blocks, the physical hierarchy cannot be created.

If the sub-block is a lib cell, (most likely a timing view) it is missing its frame view definition. Without a frame view, there is no physical information for that cell.

To see what views are available for what lib cells or blocks, use the *list_blocks* command.

If the sub-block is a logic-only block from the Design Planning budgeting flow, it can only be referenced by other logic-only blocks.

Feedback

### What Next

If the sub-block is a lib cell, there was a problem in creating that library. Go back and recreate that library. During library preparation, make sure that LEF exists for each of the lib cells in that library.

If the sub-block is a logic-only block, then a physical version of that sub-block is required. If needed, re-read the sub-block from Verilog, without setting the logic_only attribute on it.

Generally commands cannot handle sub-blocks without physical information. In rare cases blocks without physical sub-blocks may be processed by *link_block* by setting the *link.require_physical* app option to false. Subsequent commands may give unexpected results with these blocks.

### See Also

- link_block
- list_blocks
- read_verilog
- set_app_options
- block_attributes
- link_options

---

## LNK-056

(info) Block '%s' reference '%s' needs to be rebound.

### Description

This is a verbose message indicating which reference within the named block is causing that block to be rebound. There may be several references in the block that require updating, this message reports the first such reference discovered.

There are several reasons why a block reference would need to be rebound. A change in a reference library or the ref lib list could change which block the reference should be bound to. A change in the view switch list could change which block view is used for default view references. Or a change in the referenced block's port interface could require updating the pins on each cell referencing that block.

This message can be enabled with the *link_block -verbose* options, or by setting the *link.verbose* app option to true.

Feedback

### What Next

To disable this message, either set the *link.verbose* app option to false, or remove the *-verbose* option from the *link_block* command.

### See Also

- get_view_switch_list
- link_block
- set_app_options
- set_view_switch_list
- link_options

---

## LNK-062

(warning) Library %s reference library '%s' is not bound.

### Description

This library has reference libraries that have not been bound to the actual library named. The *link_block* command cannot resolve references to blocks or lib cells in the unbound library. Those references will be unresolved.

The specified block is in a library that has not had its reference library list set. The reference library list is used to resolve references to lib_cells and other blocks in the block hierarchy.

### What Next

Verify that the reference library list on this library is set correctly. Correct the setting with the *set_ref_libs* command. Check to see if the TCL search_path variable is set to locate that library. Verify that the library is on disk and readable. Try to explicitly open the library with the *open_lib* command.

---

## LNK-063

(warning) Library %s has no reference libraries.

### Description

The specified block is in a library that has not had its reference library list set. The reference library list is used to resolve references to lib_cells and other blocks in the block hierarchy.

Feedback

**What Next**

Set the reference library list for this library with the *set_ref_libs* command.

---

## LNK-064

(warning) In library %s, no block views exist for block %s.

**Description**

The library contained a library catalog entry for the block, but no actual design data. If the block should exist in that library, the library may be in an inconsistent state.

**What Next**

Recreate the missing block from the Verilog source, or copy the block from another library.

**See Also**

- copy_block
- read_verilog

---

## LNK-065

(warning) In library %s, block %s.%s has no top module.

**Description**

The library contains a block that does not have a top module specified. Without a top module, the block has no port interface and cannot be used to resolve block references.

**What Next**

Re-read the block Verilog specifying the top module for this block.

**See Also**

- read_verilog

---

## LNK-066

(warning) Linking block %s.%s with an empty view switch list.

Feedback

### Description

The block, library, or perhaps global view switch list is empty. If the switch list is empty no block views will be used to resolve references in this block. So the block will have no physical hierarchy.

### What Next

Set the view switch list to some of the views desired for this physical block hierarchy, and re-link the block.

### See Also

- get_view_switch_list
- link_block
- set_view_switch_list

---

## LNK-067

(Warning) Block '%s' is already linked.

### Description

The specified block has already been linked.

### What Next

Most often this means the extraneous link command may be removed from a script.

---

## LNK-068

(warning) Could not open block %s.

### Description

There was a reference to the block, but *link_block* was unable to open the block to read it from the library. References to that block will not be resolved.

### What Next

Verify that the block is in the library and not locked from use by another user. Re-build the block from Verilog, or recreate the abstract or frame view as needed.

Feedback

**See Also**

- create_abstract
- create_frame
- link_block
- read_verilog_outline
- read_verilog

---

## LNK-069

(information) Binding to reference library %s.

**Description**

This library had a reference library that was not bound to the actual library named. The library binding has been restored.

**What Next**

If the binding is now correct, there is nothing further to do. If the binding was incorrect, update the *search_path* variable and rebind the reference libraries with the *set_ref_libs* command.

**See Also**

- link_block
- set_ref_libs
- search_path

---

## LNK-070

(information) Block '%s' was not successfully auto-linked.

**Description**

A summary message indicating that the auto link process failed for your block.

**What Next**

See previous error messages for more details.

Feedback

## LNK-071

(error) Block '%s' has %d inconsistent port references.

### Description

A summary message indicating that the link process failed for your block.

Mismatched port references to a lib cell or to a module in the same block will cause *link_block* to fail. In this case, the block must be removed, the Verilog corrected, and re-read.

### What Next

See previous error messages for more details. Remove the block, fix the failed port references listed in previous error messages, then re-read the Verilog and re-link the block.

### See Also

- link_block
- read_verilog
- remove_blocks
- set_ref_libs
- set_view_switch_list
- search_path

## LNK-072

(information) Block '%s' has %d failed port references.

### Description

A *link_block* summary message indicating the number of references that failed due to port reference mismatches between blocks. These mismatches caused the link process to fail for your block.

### What Next

See previous error messages for more details. Either update your Verilog source, correct the ref_libs, view switch list, or *search_path* used for this block.

Feedback

### See Also

- link_block
- read_verilog
- remove_blocks
- set_ref_libs
- set_view_switch_list
- search_path

---

## LNK-073

(information) Block '%s' has %d unbound instances.

### Description

A *link_block* summary message indicating the number of instances that could not be bound to other blocks or library cells. These unbound instances caused the link process to fail for your block.

### What Next

See previous error messages for more details. Either update your Verilog source, correct the ref_libs, view switch list, or *search_path* used for this block.

### See Also

- link_block
- read_verilog
- remove_blocks
- set_ref_libs
- set_view_switch_list
- search_path

---

## LNK-074

(information) Block '%s' has %d unresolved references.

Feedback

### Description

A *link_block* summary message indicating the number of unresolved block references for this block. These unresolved references caused the link process to fail for your block.

### What Next

See previous error messages for more details. Either update your Verilog source, correct the ref_libs, view switch list, or *search_path* used for this block.

### See Also

- link_block
- read_verilog
- remove_blocks
- set_ref_libs
- set_view_switch_list
- search_path

---

## LNK-075

(error) In block '%s', cannot remap instance %s to block %s because pin %s would have to be deleted and it is connected to net %s.

### Description

The specified instance is now being mapped to a block with different port interface than used previously and one of the ports removed from the block corresponds to a pin which is connected to a net. So the instance cannot be remapped, leaving the instance unbound. The command *link_block* is unable to link the block.

### What Next

The instance needs to be mapped to a block with the port available. Either correct the ref_libs, view switch list, or *search_path* used for this block.

### See Also

- link_block
- remove_blocks
- set_ref_libs

Feedback

- set_view_switch_list
- search_path

---

## LNK-076

(error) Reference %s is bound to both %s and %s.

**Description**

An instance within the block is bound to two different views of two different blocks. An instance may be bound to two different views of the same block, but not different blocks.

**What Next**

The *link_block* command will attempt to repair the reference to use views from the first block, and discard references to the second block.

Verify that the block referenced is the desired block. If not, update the view switch list, or the library's ref_lib list to select the desired block.

**See Also**

- link_block
- set_ref_libs
- set_view_switch_list
- search_path

---

## LNK-077

(error) In block %s, cannot resolve references to %s and %s. Target blocks have incompatible port interfaces.

**Description**

The block is referencing two different views of a child block. But the two views of the child block have different port interfaces.

A block may not simultaneously reference two views of a child block unless the views have identical port interfaces.

**What Next**

Use the *change_view* command on those instances that reference the undesired block view. Then re-link the block.

Feedback

### See Also

- change_view
- link_block

---

## LNK-078

(error) In block %s, could not resolve reference to %s.

### Description

The block reference could not be resolved to a block because the reference expected a port that did not exist on the targeted block.

### What Next

Check the verilog source to see if the port interface on the block reference is correct.

Check the block selected by the ref_lib lib and view switch list to see if they have the expected port interface.

Correct any issues, then relink the block.

### See Also

- change_view
- link_block
- set_ref_libs
- set_view_switch_list
- search_path

---

## LNK-079

(information) Reference to '%s' has %d pin removal failures.

### Description

For this reference, some instances had net connections on pins that do not exist on the new block. Because those pins are being used, they could not be removed to make the pin interface of the instance match the port interface of the new block.

This message tells you how many cell instance pin removals were attempted but failed while rebinding that reference.

Feedback

### What Next

Verify that the instances should be bound to this view of the new block. Verify whether the pins on the instances should be connected to a net.

To see more pin removal failures for this reference use the *link_block -verbose* command.

### See Also

- link_block
- set_ref_libs
- set_view_switch_list
- search_path

---

## LNK-080

(error) Error while rebinding reference to '%s'.

### Description

Errors occurred while rebinding this reference.

### What Next

Look at the preceding error messages and resolve those issue. Then relink the block with the *link_block* command.

### See Also

- link_block

---

## LNK-081

(error) Unable to resolve reference to '%s'.

### Description

No block or lib cell was found in the ref_lib list for this library to resolve this reference.

### What Next

Check the ref_lib list for this library to see if the desired target block exists. Update the ref_lib list to include the missing library, then relink the block.

Feedback

### See Also

- link_block
- set_ref_libs

---

## LNK-082

(error) Unable to resolve reference to '%s'.

### Description

No block view was found in the view switch list to resolve this reference.

### What Next

See if you are using the global, library, or block view switch_list. See which views are selected in that list. See which views of that block are available in that library.

Update the view switch_list to include the desired view.

See which library is being used to resolve the block reference. If multiple libraries have a block with the same name, a block in an earlier library may be over-shadowing a desired block in a later library.

Update the ref_lib list so that the desired library is earlier.

### See Also

- get_view_switch_list
- link_block
- set_ref_libs
- set_view_switch_list

---

## LNK-083

(error) Unable to resolve reference to '%s.%s'.

### Description

An instance requested a specific view of this block, but the reference could not be bound to that block view.

### What Next

Verify that the instance is requesting the desired block view. If not, use the *change_view* command to cause the instance to use the desired block view.

Feedback

Verify that you're using the block from the desired library. If not, move that library earlier in the ref_lib list.

**See Also**

- change_view
- link_block
- set_ref_libs

---

## LNK-084

(error) Block '%s' has %d references with pins that could not be removed.

**Description**

This block has references that have pins that could not be removed because those pins are connected to nets. Those references could not be resolved.

**What Next**

See previous error messages for more details. Either update the pin net connections, or update the libraries or views being referenced by *link_block*.

**See Also**

- link_block
- set_ref_libs
- set_view_switch_list
- search_path

---

## LNK-085

(error) Block '%s' has %d mismatched view port interfaces.

**Description**

This block is referencing two views of a block that have different port interfaces. All views referenced by a block must have the same port interface.

**What Next**

See previous error messages for more details. Decide which view is desired for this block. Use the *change_view* command to cause instances that are requesting the incompatible block view to use the desired block view. Then run *link_block* again.

Feedback

### See Also

- change_view
- link_block
- set_ref_libs
- set_view_switch_list
- search_path

---

## LNK-086

(information) Block '%s' was not successfully rebound.

### Description

There were failures in some references when trying to rebind this block.

### What Next

See previous error messages for more details. Resolve the issues described in those error messages. Then run *link_block* again.

### See Also

- change_view
- link_block
- set_ref_libs
- set_view_switch_list
- search_path

---

## LNK-090

(warning) The 'link_path' variable is obsolete, it is no longer used.

### Description

The *link_path* application variable is obsolete. It is no longer used by the *link_block* command to resolve block or lib cell references. You may set and use this variable in your scripts, but it will not affect linking.

The *link_library* variable is an alias to the *link_path* variable. It is also obsolete.

Feedback

### What Next

Use the *set_ref_libs* command to tell the linker which libraries to use to locate blocks to resolve references and bind sub-blocks.

Use the *search_path* variable to specify which directories to search for files and libraries.

Use the *suppress_message* command to suppress this message.

### Examples

```
prompt> set_app_var link_path
 "$link_path /usr/libs/commonLib /usr/libs/leafCells"
Warning:  The 'link_path' variable is obsolete, it is no longer used.
 (LNK-090)
myLib /usr/libs/commonLib /usr/libs/leafCells

prompt> suppress_message LNK-090

prompt> set_app_var search_path "$search_path /usr/libs"
/global/libs . /usr/verilog /usr/libs
prompt> set_ref_libs {commonLib leafCells}
commonLib leafCells

prompt> link_block
Using libraries: myLib commonLib leafCells
Linking block myLib:TOP.design
Information: units loaded from library 'leafCells' (LNK-040)
Design 'TOP' was successfully linked.
1
```

### See Also

- get_app_var
- link_block
- report_app_var
- set_app_var
- set_ref_libs
- suppress_message
- search_path

---

## LNK-091

(error) The cell array instance name, %s, already exists.

Feedback

### Description

The linker cannot create a cell array instance with a name that is already used by another instance.

### What Next

Revise the verilog to make sure that the name of each cell array instance is unique.

### Examples

```
module top ( SET, A, CLOCK, Z );

  input  SET, CLOCK;
  input A;
  output Z;

  bot  \\U1[3] (.set(SET), .z(Z), .a(A), .clock(CLOCK));
  bot  U1 [3:0] (.set(SET), .z(Z), .a(A), .clock(CLOCK));
endmodule

The U1[3] name in the cell array U1[3:0] has already been used.
```

### See Also

- get_cells

---

## LNK-092

(error) Instantiated synthetic module '%s' has mismatched parameter specification.

### Description

The parameter spefication in the instantiation does not match the corresponding synthetic module. It can be due to the following reasons:

- A required parameter is missing.
- An incorrect parameter name is provided.
- The instantiation has more parameters than the syntheic module.

### What Next

Check the parameters for the synthetic module in the corresponding .sl file and make correction accordingly. If you are using DesignWare developer, you can modify the module/implementation definition in your *.sl file as appropriate.

Feedback

## LNK-093

(info) Ignoring parameters in reference to '%s' in '%s'.

### Description

The down design has no parameter information and we can't verify parameter equivalence (this can occur in an ascii read flow).

### What Next

Check the design hierarchy for correctness.

## LNK-094

(error) '%s' is an empty RTL module without an associated physical blackbox design.

### Description

This soft error denotes that an empty RTL module has been read. Presence of this error doesn't denote a link failure, but denotes that the empty logic module will be optimized away in synthesis. A physical blackbox representation for this modules is needed to preserve the module through synthesis. In absence of a physical representation or synthesis restrictions on this module it will get optimized away in compile_fusion. Users can ignore this error if a physical representation is going to be added to this cell before proceesing with the flow.

### What Next

Use report_mismatch_config output to check which action types are allowed for empty_logic_module mismatch. Use set_current_mismatch_config or set_attribute command to enable the correct mitigation.

The following will enable the repair strategy for predefined default configuration set_attribute -objects [get_mismatch_types empty_logic_module] -name action(default) -value *repair*

Repairing empty logic modules will create macro blackboxes in place of these standard cell instances. These macros will be equivalent to user defined macros and will be placed in synthesis by macro placer.

If the physical representation will be added after linking the design, change empty_logic_module's mitigation strategy to ignore. This will prevent this error from being issued by the tool. set_attribute -objects [get_mismatch_types empty_logic_module] -name action(default) -value *ignore*

Feedback

**See Also**

- report_mismatch_configs
- set_current_mismatch_config
- set_attribute
- create_blackbox

---

## LNK-095

(information) Physical representation was created for empty module '%s'.

**Description**

A physical representation for empty module was created to allow optimization to complete.

**What Next**

Use report_design_mismatch and get_mismatch_objects to identify the mismatches

**See Also**

- report_design_mismatch
- get_mismatch_objects

---

## LNK-526

(Information) Design name shortened to '%s'.

**Description**

Down stream tools have limitation for the length of design name. The design name generated by HDL compiler is shortened, then the design can be accepted by down stream tools.

**What Next**

To avoid this message, ensure that your design names do not exceed the limit.

---

## LNK-527

(error) The Width '%u' of the Net is less than the width '%u' of port '%s' of reference '%s'.

**Description**

Width mismatch of the net with the port of the the reference.

Feedback

**What Next**

Please correct the error and rerun again.

# LPFM

## LPFM-1

(information) %s

**Description**

This is a general informational message.

**What Next**

No action required.

## LPFM-2

(warning) Clib library '%s' registry file error at line %d. %s

**Description**

The registry.dat file inside a Clib library (directory) is created and updated by Library Compiler. Note that in the early phase of testing, there may be frequent incompatibility between versions. The following descriptions should help in diagnosing issues.

*Bad symbolink link.* Most likely the symbolic link points to a file no longer exists.

*First line not recognizable.* The first line is of a fixed "key" format that users must not modify.

*File type not supported.* Clib only supports a limited number of file.

*Bad data format.* Each line consists of 2 or 3 tokens (fie_type file_name checksum). This line is invalid.

*File name already registered.* This is most likely a bug as two files cannot have the same name in a given (clib2020) directory.

*File missing or not readable.* The file has been removed or is not readable.

*Bad checksum in registry.* The registry checksum in the first line no longer agrees with the checksum generated from the rest of the file. This means a user may have modified the registry.

Feedback

### What Next

If the problem is a bad symbolic link because of a moved file, update the symbolic link. In general, the clib2020 library needs to be re-checked and saved in order to update the registry. R&D.

---

## LPFM-3

(error) Clib library '%s' not qualified.

### Description

This error message occurs after the Clib checks failed either by check_clib command or internally called in create_clib or save_clib command (equivalent to check_library -logic_vs_physical). The default check is -logic_vs_physical. When it is not qualified by the default check, save_clib cannot save it as qualified clib on disk. When there is no frame file, it will print out: No physical library found.

The Clib passing criteria depend on the policy specified by set_early_data_check_policy command:

1) error policy, default mode No Error/Warning issues reported in LIBCHK-xxx.

2) tolerate policy Library data as is, and no severe errors below reported LIBCHK-212, 213 LIBCHK-300, 301, 320, 321, 322, 323, 330, 331 LIBCHK-124, 125

3) repair policy: Use the same criteria as tolerate policy, and apply check_library -auto_fix

### What Next

Check the reported issues in LIBCHK-xxx tables for detail and correct the library accordingly or apply auto fix.

---

## LPFM-4

(error) Failed to save cell library '%s': %s.

### Description

This message indicates that an operation of saving the specified cell library failed. The reason is given with the message.

### What Next

This is just a summary message. Usually there will be other error message(s) before this one to indicate the actual problem. Please refer to those error messages for how to correct the issue.

Feedback

## LPFM-5

(error) Failed to create cell library '%s': %s.

**Description**

This message indicates that an operation of creating a cell library failed. The reason is given with the message.

**What Next**

Correct the problem and try again.

## LPFM-6

(error) Failed to %s ECO file '%s': %s.

**Description**

This message indicates that an operation of specified ECO file failed. The reason is given with the message.

**What Next**

Correct the problem and try again.

## LPFM-7

(error) Failed to save frame '%s' for cell library '%s'.

**Description**

This message indicates that saving specified frame of cell library failed. The reason is given with the message.

**What Next**

Correct the problem and try again.

## LPFM-8

(error) Failed to open cell library '%s': %s.

**Description**

This message indicates that an operation of open the specified cell library failed. The reason is given with the message.

Feedback

**What Next**

Correct the problem and try again.

---

## LPFM-9

(information) ECO file '%s' was changed: %s.

**Description**

This message indicates that the specified ECO file had been changed outside Library Compiler. The actual change and tool's re-action are given with the message.

**What Next**

Verify whether the change is expected.

---

## LPFM-9w

(warning) ECO file '%s' was changed: %s.

**Description**

This message indicates that the specified ECO file had been changed outside Library Compiler. The actual change and tool's re-action are given with the message.

**What Next**

Verify whether the change is expected.

---

## LPFM-10

(error) Unable to create a temporary work directory

**Description**

Library Compiler is unable to create a temporary work directory to store data. If the environment variable TMPDIR is present, please make sure it points to a writable disk location which is not full. If TMPDIR is missing, Library Compiler uses /usr/tmp, please make sure /usr/tmp is writable and not full.

**What Next**

Fix the problem and run again.

---

## LPFM-11

(information) Clib2020 %s requires %d number of refresh check(s)

Feedback

**Description**

Refresh checks are triggered because the Clib2020 has been copied or moved imappropriately. Although totally harmless, they introduce unnecessary performance overheads. The underlying cause is that some modification time stamps are changed.

When copying using "cp", be sure to specify "-a" (for archive), rather than "-R":

cp -a clib_dir /destination

When copying using "tar", preserve high-resolution time stamps using posix format.

tar --posix -cf output.tar clib_dir

Alternatively, you may wish to refresh it using open_clib & save_clib commands in Library Compiler

---

## LPFM-12

(error) Cannot open cell library '%s' as it had been changed since last save. Please save it again using Library Compiler first.

**Description**

This message indicates that the specified cell library had been changed since last save and cannot be opened. Here, the change means any content or time stamp change of any relavent files in the cell library.

**What Next**

Save the cell library using Library Compiler and try again.

---

# LPP

---

## LPP-001

(error) No valid scenario with dynamic power found with switching power consumption.

**Description**

This message occurs when low-power placement is enabled but no dynamic power enabled scenario is found. This message indicates that low-power placement will not be used.

**What Next**

Make sure at least one scenario is active with dynamic power, so that low-power placement can work on optimizing the dynamic power consumption.

Feedback

**See Also**

- set_scenario_status
- place.coarse.low_power_placement

## LPP-002

(warning) Mismatched switching activities found for the physically equivalent nets: (1) net %s with toggle rate (r+f) = %f vs. (2) net %s with toggle rate (r+f) = %f. The toggle rate of (1) will be consistently used for low-power placement.

**Description**

This message indicates a mismatch in switching activities between physically equivalent nets during low-power placement preparation. This is likely an erroneous condition for power analysis.

**What Next**

Check the SAIF file or activities on those nets.

# LRA

## LRA-1000

(Information) Created libset %s from: %s.

**Description**

Libset was created with all the libraries provided to create_libset command.

**What Next**

No action is needed if the list of libraries is as expected.

## LRA-1001

(Error) Unable to %s DB files. Cannot continue with create_suggestion_libs command

**Description**

The db files corresponding to the reference libraries are missing. The existence of db files is needed for create_suggestion_libs command.

Feedback

**What Next**

If the location of the .db files has changed since creation, update search_path to include the new path. Or use the set_db_file_mapping command to specify the new path.

---

## LRA-1002

(Information) Successfully created the suggestion library "%s" in libset "%s", referenced from library "%s"

**Description**

The suggestion library creation needed for new lib cell creation is successful.

**What Next**

No action is needed.

---

## LRA-1003

(Error) Library set %s requires to have at least an inverter, and one of {NAND2 NOR2 AND2 OR2}

**Description**

The libraries that are passed to create_libset does not satisfy the minimum gates needed for analysis.

**What Next**

Check the arguments to create_libset command.

---

## LRA-1004

(Error) The reference library

**Description**

There can be only be on suggestion library for a given reference library in a given libset.

**What Next**

Check the arguments to create_suggestion_libs command.

---

## LRA-1005

(Error) problem in create_libset: no valid libraries were specified

Feedback

**Description**

There is no valid library passed to create_libset command.

**What Next**

Check the arguments and make sure you pass atleast one valid library name to create_libset.

---

## LRA-1006

(Error) Cannot create the suggestion library

**Description**

There is a already suggestion library with the name in this libset.

**What Next**

Check the -name_prefix argument to create_suggestion_libs command.

---

## LRA-1010

(Information) The libcell %s is marked as available for analysis

**Description**

The lib cell is marked as available for library analysis. By default all library cells except some special cells like UPF cells are available for library analysis. This message is printed if the user has changed this for certain lib cells using set_use_for_library_analysis and then reversing it by issuing set_use_for_library_analysis again with argument true.

**What Next**

No action is needed if this is the intention.

---

## LRA-1011

(Information) The libcell %s is marked as unavailable for analysis

**Description**

The given lib cell is marked as not available for library analysis. By default all library cells except some special cells like UPF cells are available for library analysis. This message is printed if the user is issuing the command set_use_for_library_analysis to change this.

**What Next**

No action is needed if this is the intention.

Feedback

## LRA-1012

(Information) The libcell %s is marked as unavailable for this session and cannot be changed

**Description**

The given lib cell is marked as not available for library analysis permanently. By default all library cells except some special cells like UPF cells are available for library analysis. This message is printed if the user is issuing the command set_use_for_library_analysis with -force option to skip analysis.

**What Next**

No action is needed if this is the intention.

## LRA-1020

(Warning) The libcell %s is ignored since it is a synthetic library cell

**Description**

When a synthetic library cell is passed to set_use_for_library_analysis command, it will be ignored. The reason is the synthetic library cell is not part of library analysis.

**What Next**

Review the arguments to set_use_for_library_analysis command.

## LRA-1021

(Warning) The libcell %s cannot be marked for analysis because it is a %s lib_cell

**Description**

Some of the lib cells like UPF lib cells and MACRO lib cells are considered during library analysis. Hence passing those lib cells to set_library_analysis will receive this warning.

**What Next**

Review the arguments to set_use_for_library_analysis command.

## LRA-1022

(Warning) Cannot use -force option with value true

Feedback

**Description**

The -force option to set_use_for_library_analysis can only be used with value false. By default all library cells are part of analysis except UPF and macro lib cells. And invoking -force with value false will cause the analysis to be skipped for those lib cells permanently.

**What Next**

Review the arguments to set_use_for_library_analysis command.

---

## LRA-1051

(Warning) Library %s is not added to libset %s since it does not have timing view.

**Description**

The library passed to create_libset command does not have the timing view. To be able to perform library analysis, you will need the timing view

**What Next**

Review the library creation step and make sure that your library has timing view.

---

## LRA-1052

(Information) Using the mapped DB file %s instead of the original DB file %s for library %s

**Description**

The original DB file used during the library creation step is not accessible during create_suggestion_lib. Hence the mapped DB file path is used. The mapped DB file path can come from search_path or from set_db_file_mapping command.

**What Next**

No action is needed if this is intentional

---

## LRA-1054

(Error) The provided input should be a library object or the library name of an opened library. Please check the specified input %s. If this is the path to a library not yet opened, please read in this library first then properly specify it.

**Description**

The provided input should be a library object or the library name of an opened library. Please check the specified input %s. If this is the path to a library not yet opened, please read in this library first then properly specify it.

Feedback

**What Next**

Use the correct library name

---

## LRA-1055

(Error) Unable to add library %s to libset %s

**Description**

The library cannot be added to libset.

**What Next**

Check the arguments to create_libset

---

## LRA-1056

(Warning) Library set has inconsistent pvt pane counts Maximum found: %u panes (eg: %s), Minimum found: %u panes (eg: %s)

**Description**

The libraries used in the libset should have same number of pvt panes.

**What Next**

Review the library preparation step

---

## LRA-1057

(Error) Db file %s is not available

**Description**

The .db file corresponding to the reference library is not available. To be able to successully run create_suggestion_lib command, the .db file should exist.

**What Next**

If the location of the .db files has changed since creation, update search_path to include the new path. Or use the set_db_file_mapping command to specify the new path.

---

## LRA-1058

(Error) Library %s is not a suggestion library

Feedback

**Description**

User reference library is used in place of libra suggestion library during create_libset

**What Next**

Check the arguments to create_libset

---

## LRA-1059

(Error) Unable to find a corresponded reference library for the given existing suggestion library %s

**Description**

The reference library is not read in before executing create_libset command.

**What Next**

Check your library setup

---

## LRA-1060

(Information) Attaching the existing %s suggestion library %s to the reference library %s.

**Description**

The association between the reference and suggestion library has been successuflly made during create_libset command.

**What Next**

No action is needed.

---

## LRA-1400

(warning) logic function %s is not canonical %s

**Description**

The specified logic function string was not able to be represented canonically. This can happen for a number of reasons. For instance, perhaps the function string is badly formed, or perhaps the function can be minimized in a way that eliminates one or more of the variables so they do not influence the output. You will see this message when that happens. This message can also happen for a more subtle reason: Library Compiler must be able to understand the function being created. For some large/complex functions, the existing Library Compiler code that recognizes and canonicalizes functions may 'give up', and this message will result.

Feedback

**What Next**

Change your function description to be correct, or less complex.

---

## LRA-1401

(warning) %s %s already exists

**Description**

You will get this message when you are creating a duplicate object in the Libra data manager, ie, one that already exists. This is usually due to a naming conflict

**What Next**

Rename the object you are creating and try again.

---

## LRA-1402

(warning) %s %s does not exist

**Description**

You will get this message when you have given an object or an object name as a command argument that doesn't exist.

**What Next**

Check the name or object that you are passing in, and make sure it is valid.

---

## LRA-1601

(Info) Successfully created library cell family %s using %s.

**Description**

A library cell family is successfully created using either library cells or sub-families.

**What Next**

Review the definition of the newly created library cell family.

---

## LRA-1602

(Error) Failed to create library cell family %s.

**Description**

Failed to create the library cell family.

Feedback

**What Next**

Check the specified library cells or sub-families, makes sure they satisfy grouping conditions (e.g. consistent functional properties)

---

## LRA-1603

(Info) Successfully created initial library cell families for library %s.

**Description**

Successfully identified and created initial library cell families for the specified library.

**What Next**

Review the newly created initial library cell families.

---

## LRA-1604

(Error) Failed to create initial library cell families for library %s.

**Description**

Failed to identify initial library cell families for the specified library.

**What Next**

Check the library content and grouping conditions.

---

## LRA-1605

(Info) %s library cell

**Description**

Successfully added a library cell to an existing library cell family.

**What Next**

Review the content of the updated library cell family.

---

## LRA-1606

(Info) %s library cell

**Description**

Successfully removed a library cell from an existing library cell family.

Feedback

**What Next**

Review the content of the updated library cell family.

---

## LRA-1607

(Info) %s library cell

**Description**

The specified library cell is not included in the target library cell family, ignored for further process.

**What Next**

Check the specified library cell and the target library cell family's content.

---

## LRA-1608

(Info) Ignoring expression

**Description**

Ignoring the family rule expression.

**What Next**

Check and correct the indicated missing portion.

---

## LRA-1609

(Info) Ignoring expression

**Description**

Ignoring the family rule expression due to syntax problem.

**What Next**

Check expression syntax.

---

## LRA-1610

(Info) Ignoring expression

**Description**

Ignoring the family rule expression, wild card is not supported.

Feedback

**What Next**

Remove wildcard expressions.

---

## LRA-1611

(Warning) The specified sub-family named %s does not exist, please provide a correct library cell family name.

**Description**

The specified sub-family name is not correct.

**What Next**

Check and provide a valid library cell family name.

---

## LRA-1612

(Info) The family rules does not apply to family creation using sub-families, the rules are ignored.

**Description**

Family rules can not be applied to a library cell family creation by grouping sub-families.

**What Next**

Don't provide a family rule.

---

## LRA-1613

(Warning) The given family

**Description**

Manually modifying an initial family's content can cause conformity problem of the initial library cell set of a library.

**What Next**

Check if the outcome of this manipulation is as needed.

---

## LRA-1614

(Warning) The given family

Feedback

**Description**

Arbitrarily manipulating initial library cell families is not allowed.

**What Next**

Avoid such manipulations.

---

## LRA-1615

(Warning) The given family

**Description**

This library cell family is created by grouping sub-families. Adding or removing library cells is not allowed.

**What Next**

Avoid such operaiton.

---

## LRA-1616

(Error) The given library cell family rule name %s does not exist.

**Description**

The specified library cell family rule is not valid.

**What Next**

Check the specified rule name.

---

## LRA-1617

(Error) The given library cell family name %s does not exist.

**Description**

The specified library cell family name is not valid.

**What Next**

Check the specified family name.

---

## LRA-1618

(Warning) The given library cell

Feedback

**Description**

The specified library cell has different logical function to the target library cell family. Ignored for further process.

**What Next**

Check the specified library cell is correct.

---

## LRA-1619

(Warning) The given library cell

**Description**

The specified library cell is a Libra suggestion library cell, ignored for further process.

**What Next**

Check the specified library cell is correct.

---

## LRA-1620

(Error) The input libset '%s' does not exist.

**Description**

The specified libset does not exist, stop further processes.

**What Next**

Check the specified libset is correct.

---

## LRA-1621

(Error) There is no library defined in the input libset '%s'.

**Description**

The specified libset is empty, it does not have any library.

**What Next**

Check the specified libset is correct.

---

## LRA-1633

(Info) Family rule name %s does not exist, please provide a correct name.

Feedback

**Description**

This family rule does not exist, skipped for further process.

**What Next**

Use an existing family rule.

---

## LRA-1634

(Warning) Family name '%s' already exists, please use another name.

**Description**

Duplicated family names are not allowed.

**What Next**

Use a different family name.

---

## LRA-1635

(Warning) No %s is provided, unable to create a family.

**Description**

Missing inputs to create a library cell family.

**What Next**

Provide valid library cells or existing families to be grouped as a new library cell family.

---

## LRA-1636

(Warning) A sub-family of a family must be defined by libcells; however, the specified sub-family %s is defined by sub-fmailies.

**Description**

A family can be defined by grouping a set of sub-families. A sub-family must be defined by grouping a set of library cells; it can not be defined by grouping sub-families.

**What Next**

Provide valid sub-families.

Feedback

## LRA-1637

(Warning) All the sub-families in a family must have same functional properties; however some of the provided sub-families (e.g. %s) are different.

**Description**

All the sub-families to be grouped as a new family must be with same functional properties.

**What Next**

Check the functional properties of the specified sub-families, make sure they are consistent.

## LRA-1638

(Warning) All the library cells in a family must have same functional properties; however some of the provided library cells (e.g. %s) are different. %s

**Description**

All the library cells to be grouped as a new family must be with same functional properties.

**What Next**

Check the functional properties of the specified library cells, make sure they are consistent.

## LRA-1639

(Warning) No library cell matches the given family rule %s.

**Description**

No library cell matches the conditions defined in the specified family rule.

**What Next**

Check the family rule definition or the input library cell set.

## LRA-1640

(Warning) Deleting library's initial families are not allowed. Only user-defined families can be deleted. Ignoring %s.

Feedback

**Description**

Library's initial families are not allowed to be arbitrarily deleted.

**What Next**

If initial families need to be updated, use the proper UIs to create initial families again.

---

## LRA-1641

(Warning) Unable to move library cells between families (Source: %s, Target: %s): %s.

**Description**

Unable to move library cells between two library cell families. Check the reason that causes this failed move.

**What Next**

Check the reason and fix it.

---

## LRA-1642

(Warning) Some library cells are not in the source family %s, ignored (%s).

**Description**

Some library cells are not in the source library cell family, ignored.

**What Next**

Check the correctness of the selected input cells.

---

## LRA-1650

(Warning) The sub-family '%s' has already included in this family.

**Description**

The target family already includes this sub-family, it will not be included twice.

**What Next**

Don't add this sub-family to the target family.

---

## LRA-1655

(Info) Deleting sub-rule %s from family rule %s

Feedback

**Description**

Deleting a sub-family from a family rule.

**What Next**

Review the family rule definition

---

## LRA-1661

(error) Target libcell %s has a non-supported height. Currently we only support single height target libcells

**Description**

The physical model for new libcell creation today only supports single-height libcells as target cells. You will see this message when you have given a non-single-height libcell as a target libcell.

**What Next**

Use a single height target libcell instead.

---

## LRA-1662

(info) Height check %s for target libcell %s

**Description**

Tells if a target libcell's height check was not able to be done, or if it succeeded.

**What Next**

If the check was not able to be done, check your physical library set up to make sure physical data are available.

---

## LRA-1671

(info) Created %lu new libcells from %s

**Description**

Library cell creation was successful, and new libcells were created.

**What Next**

Save the suggestion library containing the new libcells, and use them in optimization with FC, etc.

Feedback

## LRA-1672

(Error) Unable to create new function %s

### Description

The new libcell(s) that were requested were not created. This can be for a variety of reasons. It may be that the supplied function or design was incorrect, or it may be that the target libcell(s) were not valid, or didn't align with the supplied names, or it might be that there are new desired libcell names which conflict with existing ones.

### What Next

Review the messages that came ahead of this one and see if there are problems. Review the supplied function equation, or design, and the target libcells.

## LRA-1673

(Error) Unable to build a netlist for function %s

### Description

The new libcell(s) that were requested were not created. This can be for a variety of reasons. It may be that the supplied function or design was incorrect, or it may be that the target libcell(s) were not valid, or didn't align with the supplied names, or it might be that there are new desired libcell names which conflict with existing ones.

### What Next

Review the messages that came ahead of this one and see if there are problems. Review the supplied function equation, or design, and the target libcells.

## LRA-1674

(Info) A base name is required for each function. Please specify %d base name(s)

### Description

The name for a new libcell includes a 'base name' which is supplied by the user, in a list. This message occurs if that list is not the same size as the set of new libcells which will be created.

### What Next

Make sure that your list of target libcells is the same size as the list of names.

Feedback

## LRA-1675

(Info) An exact name is required for each lib cell. Please specify %d exact name(s)

**Description**

Users can optionally supply a set of 'exact' new libcell names for created new libcells, in a list. This message occurs if that list is not the same size as the set of new libcells which will be created.

**What Next**

Make sure that your list of target libcells is the same size as the list of names.

## LRA-1676

(Error) Found duplicate exact name '%s'

**Description**

Users can optionally supply a set of 'exact' new libcell names for created new libcells, in a list. This message occurs if that list is not unique, or not the same size as the set of new libcells that will be created.

**What Next**

Make sure that your list of target libcells is the same size as the list of names.

## LRA-1677

(Info) Model effort value %s is being ignored

**Description**

This message means that an invalid model effort value was given. "none", "low" and "high" are the valid values.

**What Next**

Revise your model effort argument and rerun.

## LRA-1678

(Error) Libcell %s is not in analysis named %s

Feedback

**Description**

The target libcell supplied has not been analyzed in the analysis set which was supplied. This means that we cannot get the data we need from the target libcell in order to create a new libcell.

**What Next**

Run library analysis on the library(ies) containing target libcells supplied to this command.

---

## LRA-1679

(Info) Found an analysis set matching libset name %s

**Description**

We were able to find an analysis set matching the libset name supplied.

**What Next**

Nothing -- this is an informational message.

---

## LRA-1680

(Error) Use static_analyze to create the analysis name %s first

**Description**

In order to use a reference libcell for new libcell creation, library analysis has to have been run on it (and its library) first. This message occurs when we cannot find a valid analysis with that libcell.

**What Next**

Run library analysis on the reference libcells that you supply to this command.

---

## LRA-1681

(Info) Analysis set name %s does not exist. Searching with libset name %s

**Description**

Reference libcells have to have been analyzed first, with library analysis. So, if a specified analysis can't be found, we also search to see if there is an analysis whose name matches that of the libset that you supplied.

Feedback

**What Next**

Nothing, unless this search fails. In that case, make sure you have analyzed a library before creating new libcells using its members as references.

---

## LRA-1682

(Info) -cost_check and -cost_percent lists need to have the same number of entries, so ignoring -cost_check and -cost_percent

**Description**

This message occurs when cost check and cost percentage arguments are different. If they are, they will be ignored, and this message will be printed.

**What Next**

Review and revise your arguments to this command.

---

## LRA-1683

(Info) Invalid weights found so setting cost weights equally

**Description**

Cost values must be numerically valid -- if they aren't you get this message (when they are all zero).

**What Next**

Revise your weights to be nonzero.

---

## LRA-1684

(Error) Target family %s %s

**Description**

The family named in the option was not found. Without it, the command will fail.

**What Next**

Either create the family, or revise your arguments to the command.

---

## LRA-1685

(Error) Libset %s does not have a new function suggestion library %s. Use create_suggestion_lib to create one first

Feedback

**Description**

This message occurs if there is a difference between the suggestion library for the given libset and the suggestion library from the command.

**What Next**

Insure that you are using the correct suggestion library.

---

## LRA-1686

(Error) Can only specify one suggestion library at a time

**Description**

Currently the command will only allow one suggestion library to be supplied. We may extend this in the future.

**What Next**

Only use one suggestion library at a time.

---

## LRA-1687

(Error) No target library found

**Description**

This command has to have at least one valid target libcell. If none is given, you get this message.

**What Next**

Review and revise your arguments to the command, so that you give at least one valid target libcell.

---

## LRA-1688

(Error) All target libcells must come from the same library

**Description**

This is a necessary limitation for now -- we don't allow you to give a set of target libcells that spans separate libraries. That may change in the future, but for now this keeps things consistent.

Feedback

**What Next**

Split your command invocation into multiple pieces, one for each reference suggestion library.

---

## LRA-1689

(Error) New function %s is invalid

**Description**

This is a 'canonicalization' error -- the idea is that we must be able to find a unique canonical form for each new libcell function. That's not always possible, which is unfortunate. But if you get this message, it also means that Library Compiler would fail as well -- in other words, even if this command *were* able to generate a new libcell using the function you have given, Library Compiler would not recognize it. So that's why this is an error.

Another possibility is that the is a functional error in your function description, or design.

**What Next**

Revise your function / design input to be correct, or smaller.

---

## LRA-1690

(Error) Failed to get the target suggestion library

**Description**

Currently this command needs 'target' libcells, which are examples to follow when building new libcells. If no target libcell was found, you get this error.

**What Next**

Add a target libcell to the command.

---

## LRA-1691

(Error) Failed to get the libset %s

**Description**

The named libset was not found, and that is an error.

**What Next**

Either create the libset you've named in this command, or change the libset name to be one that is already created.

Feedback

## LRA-1692

(Error) Libset %s does not include this target library

**Description**

The libset used must include the target library. A target library contains the example libcells that will be used to create new function libcells.

**What Next**

Use a valid libset which contains the target library, or create it prior to this command.

## LRA-1693

(Error) Problem in create_scaled_lib_cell: no valid lib_cell specified for option -ref_lib_cell_%s. This lib_cell must be in the referenced library of the specified target suggestion library

**Description**

The low or high reference libcell must be valid. If not, you will get this error.

**What Next**

Supply a valid low or high reference libcell.

## LRA-1694

(Error) Problem in create_scaled_lib_cell: only one valid lib_cell can be specified for option -ref_lib_cell_%s

**Description**

This message occurs when an invalid number of low or high reference libcells are found.

**What Next**

Supply only one libcell at a time.

## LRA-1695

(Error) Problem in create_scaled_lib_cell: unable to create a new lib_cell

**Description**

An error occurred when attempting to create the new libcell. This may be because the weightings were invalid, or because of a problem with the supplied low or high reference libcells.

Feedback

**What Next**

Review your command options, and make sure the reference libcells and weights are correct.

---

## LRA-1696

(Info) Successfully created a new lib_cell

**Description**

A new libcell was created.

**What Next**

Save the suggestion library and use the libcell in your optimization flow.

---

## LRA-1697

(Info) Equation parse %s for output %s and equation %s

**Description**

You get this message to describe what happened when parsing your supplied equation.

**What Next**

If the parsing failed, review your equation to make sure it is in correct infix format.

---

## LRA-1698

(Error) Equation string %s %s

**Description**

There was a problem in your equation specification. Here is an example of a valid equation argument:

-function "Y = (A B) + (C D)"

**What Next**

Make sure your equation string has a correct form.

---

## LRA-1699

(Error) %s for the -design option

Feedback

**Description**

There was an invalid *-design* argument used. The design needs to be a valid combinational (for now) logic design.

**What Next**

Review and revise your call to the command.

---

## LRA-1700

(Error) Design %s has an error: %s

**Description**

There is a problem with the supplied design -- it either contains a sequential element (not supported yet), or there is another problem with the functionality it describes. There are limits to the size and complexity of functions that can be made into libcells, and this message might indicate that you've hit one of them.

**What Next**

Revise the design you are supplying, and rerun.

---

## LRA-2180

(error) No target libcells were specified

**Description**

No target libcells were specified, so no new libcell can be created.

**What Next**

One or more target libcells must be supplied to *create_new_function_lib_cells*.

---

## LRA-2181

(error) Mismatch between target libcells and supplied names

**Description**

The names that are in the exact names list must match one-for-one for the new libcells to be created. If that list doesn't match, then you will see this error.

**What Next**

Inspect your use of the command and insure that the list of target libcells matches up with the list of exact names that you provided.

Feedback

## LRA-2182

(error) Libcell %s already exists, please try another name

**Description**

Exact names given by the user for new libcells must be unique.

**What Next**

Inspect your use of the command and insure that the list of exact names that you provided are unique amongst themselves and among the existing libcells.

## LRA-2183

(info) Unable to lexically classify target libcell %s, using ad-hoc name

**Description**

We try to name the new libcell intelligently, which requires that lexical classification has been run on the target library so that we know how drive variants are named. If your target library has not been lexically classified, then the name generated for the new libcells is less descriptive.

**What Next**

See if you can run lexical classification on the target library, if you want the new libcell names to be more descriptive.

## LRA-2184

(info) Cost-Based Filter for libcell %s -- %s

**Description**

This is a cost comparison message that is used when costing new potential implementations for new libcells. It explains how your cost comparisons happened, based on your input to the command.

**What Next**

This is an informational message -- make sure the cost filtering is occuring as you wish, based on these messages.

## LRA-2185

(info) %s does not have a specified naming conventioin for lexical classification. No lexical attributes could be referenced for the cell naming.

Feedback

**Description**

We try to name the new libcell intelligently, which requires that lexical classification has been run on the target library so that we know how the target library cell is named. If your target library has not been lexically classified, then the name generated for the new libcells is less descriptive.

**What Next**

See if you can run lexical classification on the target library, if you want the new libcell names to be more descriptive.

---

## LRA-2186

(info) %s has not been lexically classified. No lexical attributes could be referenced for the cell naming.

**Description**

The given object was not lexically classified. We try to name the new libcell intelligently, which requires that lexical classification has been run on the target library so that we know how the target library cell is named. If your target library has not been lexically classified, then the name generated for the new libcells is less descriptive.

**What Next**

See if you can run lexical classification on the target library, if you want the new libcell names to be more descriptive.

---

## LRA-2187

(info) Unable to supply a family for new libcell %s

**Description**

This message indicates that it was not possible to find a family of libcells to which this new libcell could be added, or it was not possible to install this libcell into that family.

**What Next**

This is an informational message -- this condition can occur when your initial or user-defined family setup is incomplete.

---

## LRA-3001

(Error) The input libset '%s' is does not exist.

Feedback

**Description**

The specified libset does not exist, unable to continue the flow.

**What Next**

Provide a valid libset.

---

## LRA-3002

(Error) There is no library defined in the input libset '%s'.

**Description**

The specified libset does not include any library.

**What Next**

Review the libset definition, provide a valid libset.

---

## LRA-3003

(Error) The input column for naming convention definition must be a positive integer.

**Description**

The column in the naming convention definition must be positive integer equal or larger than 1.

**What Next**

Review and correct the naming convention.

---

## LRA-3004

(Error) The input attribute is not valid. It must be a predefined attribute name, or an user-defined attribute name with '-is_userdef_attr' specified.

**Description**

The input attribute for naming convention definition is not correct. It must be a predefined lexical attribute name or an user-defined attribute name. If this is an user-defined attribute, specify with '-is_userdef_attr' option.

**What Next**

Review and correct the naming convention.

Feedback

## LRA-3005

(Error) The input regular expression pattern '%s' has syntax error.

**Description**

The regular expression pattern is not correct, a syntax error is detected.

**What Next**

Review and correct the regular expression specified to naming convention.

## LRA-3006

(Error) The input value for attribute '%s' must be one of {TRUE, true, True, 1, FALSE, false, False, 0}

**Description**

This attribute is with boolean type, need to specify a correct value.

**What Next**

Specify the value with correct data format.

## LRA-3007

(Error) Full library name or cell name can not be set for classification.

**Description**

The lexical attribute for the library or cell name can not be specified for classification. They are directly derived from the library cell information.

**What Next**

Do not specify these attributes for lexical classifications.

## LRA-3008

(Error) The given naming convention is empty, please provide a valid name.

**Description**

A valid name needs to be specified for this naming convention.

**What Next**

Provide a name for this naming convention.

Feedback

## LRA-3009

(Info) Added or modified column %d definition for the naming convention named '%s' %s.

**Description**

One of the column of this naming convention has been added or modified.

**What Next**

Review and check if this change is correct.

## LRA-3010

(Info) An reserved attribute keyword '%s' is detected, option '-is_userdef_attr' is ignored.

**Description**

A reserved attribute keyword is detected, it can not be used as an user-defined attribute name.

**What Next**

Review the correctness of the definition.

## LRA-3011

(Warning) Unable to find the keyword mapping set '%s'.

**Description**

The specified keyword mapping set does not exist.

**What Next**

Check if the specified keyword mapping set name is valid.

## LRA-3012

(Warning) Unable to find the keyword mapping set '%s' for deletion.

**Description**

The specified keyword mapping set does not exist, it can not be deleted.

**What Next**

Check if the specified keyword mapping set name is valid.

Feedback

## LRA-3013

(Warning) Unable to find the keyword mapping set '%s' to report, ignored.

**Description**

The specified keyword mapping set does not exist, unable to report the content.

**What Next**

Check if the specified keyword mapping set name is valid.

## LRA-3014

(Warning) Unable to remove the keyword '%s' from the keyword mapping set '%s', ignored.

**Description**

A keyword can not be removed from a keyword mapping set.

**What Next**

Check if the specified keyword is included in the keyword mapping set.

## LRA-3015

(Warning) Unable to add the mapping of keyword '%s' and predefined attribute '%s', check if the given attribute is a valid predefined attribute.

**Description**

A mapping relationship of a keyword and a specified lexical attribute can not be added.

**What Next**

Check if the given attribute is a valid predefined lexical attribute.

## LRA-3016

(Info) Classified library %s.

**Description**

The library has been successfully processed by lexical classification.

**What Next**

Review the classification results.

Feedback

## LRA-3017

(Warning) Unable to classify library %s, please check the corresponded library cell naming convention.

**Description**

The library could not be successfully processed by lexical classification.

**What Next**

Check the correctless of the specified naming convention.

## LRA-3018

(Info) Cleared naming convention of library %s.

**Description**

The library's naming convention is cleared.

**What Next**

Nothing if this is an expected process.

## LRA-3019

(Info) Cleared naming convention %s.

**Description**

The specified naming convention is cleared.

**What Next**

Nothing if this is an expected process.

## LRA-3020

(Warning) The naming convention %s does not exist.

**Description**

The specified naming convention does not exist, unable to do further process.

**What Next**

Check the correctness of the given naming convention.

Feedback

## LRA-3100

(Warning) The input attribute name '%s' is not a pre-defined name, please check the correctness

**Description**

The specified attribute name is not a pre-defined name

**What Next**

Check the correctness of the specified attribute name

## LRA-3101

(Warning) The user-defined attribute name can only be set through library cell naming convention rules.

**Description**

The user-defined attribute name can not arbitrarily changed. They are derived from cell naming conventions.

**What Next**

Change the user-defined attribute name by configuring naming conventions.

## LRA-3102

(Warning) The corresponded user-defined attribute name did not set, which has to be firstly set through library cell naming convention rules and run classifier.

**Description**

The corresponded user-defined lexical attribute name did not set.

**What Next**

Set the user-defined attribute through library cell naming convention. Then, run lexical classification.

## LRA-3103

(Warning) The lexical attribute '%s' does not exist.

**Description**

The specified lexical attribute does not exist.

Feedback

**What Next**

Check the correctness of the specified attribute name

---

## LRA-3200

(Warning) The naming convention for column '%d' in library '%s' is not properly defined.

**Description**

The naming convention for this column is not properly defined.

**What Next**

Check the correctness of the definition on the column indicated in the message.

---

## LRA-3301

(Warning) Unable to generate a valid regular expression pattern based on the input library naming convention. Please check the naming convention definitions.

**Description**

The tool reads the specified library naming convention to create a regular expression pattern for further processes. Based on the input library naming convention, a valid regular expression pattern could not be generated.

**What Next**

Check the correctness of the naming convention definitions.

---

## LRA-3302

(Warning) Library cell '%s' can not be successfully classified by the given lexical naming convention.

**Description**

This library cell can not be classified by the given lexical naming convention.

**What Next**

Check the correctness of the naming convention definitions. If the library has multiple naming conventions for different cells, try to define multiple naming convention rules and perform incremental classification.

Feedback

## LRA-3303

(Warning) The lexical classification results of library cell '%s' may not be entirely correct.

**Description**

The tool trys to perform a sanity check after the classification. This message indicates the results may not be entirely correct.

**What Next**

Review the results of this libray cell and adjust the naming convention in need.

## LRA-3304

(Warning) The keywords defined in set '%s' may not be sufficient for the library cell '%s'.

**Description**

This cell's name includes certain information that is not covered by the defined keyword set.

**What Next**

Check this cell's name and add additional keywords in need to cover it.

## LRA-4103

(error) Removing analysis set %s.

**Description**

An existing library analysis set is being removed due to modification to libraries.

**What Next**

This is an informational message, no action is required.

## LRA-4801

(warning) Argument -plot has been specified, which requires a GUI session to be initialized, please run

**Description**

A GUI session was not initialized.

Feedback

**What Next**

Start a GUI session.

---

## LRA-4802

(information) Could not find analysis set named %s.

**Description**

The analysis set of the given name could not be found, need to provide a correct name to proceed.

**What Next**

Provide a correct analysis name.

---

## LRA-4803

(warning) The specified Op Cond

**Description**

The specified Op Cond name is not coverered by the given library analysis, unable to read the analysis results for comparison.

**What Next**

Check the correctness of the Op Cond name, or provide an analysis that covers this Op Cond.

---

## LRA-4804

(warning) The specified PPVT process corner

**Description**

The specified PPVT process corner is not coverered by the given library analysis, unable to read the analysis results for comparison.

**What Next**

Check the correctness of the PPVT process corner information, or provide an analysis that covers this Op Cond.

---

## LRA-4805

(warning) The PPVT format of

Feedback

**Description**

The specified PPVT process corner format is not correct. The format is {Process Label, Process, Voltage, Temperature}, as specified for analysis environment creation. For some libraries without Process Labels, use the reserved keyword "(none)" to specify it.

**What Next**

Provide a correct format for PPVT process corner information.

---

## LRA-4806

(warning) The specified input slew point

**Description**

The specified input slew point is not coverered by the given library analysis, unable to read the analysis results for comparison.

**What Next**

Check the correctness of the input slew point.

---

## LRA-4807

(warning) The specified input slew index

**Description**

The specified input slew index is not coverered by the given library analysis, unable to read the analysis results for comparison.

**What Next**

Check the correctness of the input slew index.

---

## LRA-4808

(warning) The specified input slew index

**Description**

The input slew index is an integer equal or larger than zero, a negative value is not valid.

**What Next**

Check the correctness of the input slew index.

Feedback

## LRA-4809

(warning) The specified output load point

**Description**

The specified output load point is not coverered by the given library analysis, unable to read the analysis results for comparison.

**What Next**

Check the correctness of the output load point.

## LRA-4810

(warning) The specified output load index

**Description**

The specified output load index is not coverered by the given library analysis, unable to read the analysis results for comparison.

**What Next**

Check the correctness of the output load index.

## LRA-4811

(warning) The specified output load index

**Description**

The output load index is an integer equal or larger than zero, a negative value is not valid.

**What Next**

Check the correctness of the output load index.

## LRA-4812

(info) Op Cond / PPVT setting is not provided, default setting in the %s analysis is used (Library: %s)

**Description**

The interested Op Cond or PPVT process corner information is not provided, the tool will use the default process corner defined in the analysis to compare.

Feedback

**What Next**

Check the compared process corner, if it is not the interested process corner, specify it directly to the command.

---

## LRA-4813

(info) Input slew setting is not provided, default setting in the analysis is used. Reference library: Slew Value = %f (Index = %d), Target library: Skew Value = %f (Index = %d)

**Description**

An input slew setting is not provided, the tool uses the default setting defined in the analysis.

**What Next**

Check the compared input slew setting, if it is not the interested setting, specify it directly to the command.

---

## LRA-4814

(info) Output load setting is not provided, default setting in the analysis is used. Reference library: Load Value = %f (Index = %d), Target library: Load Value = %f (Index = %d)

**Description**

An output load setting is not provided, the tool uses the default setting defined in the analysis.

**What Next**

Check the compared output load setting, if it is not the interested setting, specify it directly to the command.

---

## LRA-4815

(warning) The input slew settings of analysis environments do not match. Environment

**Description**

The input slew settings of the analysis environments for reference and target libraries do not match.

**What Next**

Check and align the input slew settings of the anlaysis environments.

Feedback

## LRA-4816

(warning) The output load settings of analysis environments do not match. Environment

**Description**

The output load settings of the analysis environments for reference and target libraries do not match.

**What Next**

Check and align the output load settings of the anlaysis environments.

## LRA-4817

(warning) The input slew settings to be compared are not matched. Reference library: %.8f, Target library: %.8f

**Description**

The input slew settings do not match between the reference and target libraries' analysis environment. This could be caused by input slew points are different under the same input slew index.

**What Next**

Check and align the input slew settings.

## LRA-4818

(warning) The output load settings to be compared are not matched. Reference library: %.8f, Target library: %.8f

**Description**

The output load settings do not match between the reference and target libraries' analysis environment. This could be caused by output load points are different under the same output load index.

**What Next**

Check and align the output load settings.

## LRA-4819

(warning) The comparison conditions given to the reference and the target libraries are not matched, please align them.

Feedback

**Description**

The comparison conditions do not match between the reference and the target libraries. They need to be aligned for a valid comparison.

**What Next**

Check and align the comparison conditions.

---

## LRA-4820

(info) The given attribute

**Description**

The given attribute is not valid, it is ignored.

**What Next**

Check the attribute name and provide the correct name.

---

## LRA-4821

(info) No valid attribute is give, all valid attributes will be reported.

**Description**

This message informs the user that all valid attributes will be compared and reported, since no valid attribute was provided.

**What Next**

Review the comparison report.

---

## LRA-4822

(info) The given family named %s is not found.

**Description**

The provided family name is not valid.

**What Next**

Check and correct the family name.

Feedback

## LRA-4823

(info) The given family named %s is an initial family, but it does not belong to the reference library %s.

**Description**

The provided family should include a set of library cells belong to the reference library. However, the given family is an initial family defined for some other library.

**What Next**

Check and correct the family name.

## LRA-4824

(warning) The specified %s file does not exist.

**Description**

The specified file for this command does not exist.

**What Next**

Provide a valid file.

## LRA-4825

(warning) Unable to read the specified input file

**Description**

The specified input file can not be successfully read in.

**What Next**

Provide a valid file.

## LRA-4826

(info) The timing units of the reference library and the target library are not matched, auto-scaling will be applied if it is applicable.

**Description**

The timing units are not consistent between the reference and the target libraries. The tool will try to do auto-scaling for a valid comparison.

Feedback

**What Next**

Review the comparison results.

---

## LRA-4827

(info) The power units of the reference library and the target library are not matched, auto-scaling will be applied if it is applicable.

**Description**

The power units are not consistent between the reference and the target libraries. The tool will try to do auto-scaling for a valid comparison.

**What Next**

Review the comparison results.

---

## LRA-4828

(info) The capacitance units of the reference library and the target library are not matched, auto-scaling will be applied if it is applicable.

**Description**

The capacitance units are not consistent between the reference and the target libraries. The tool will try to do auto-scaling for a valid comparison.

**What Next**

Review the comparison results.

---

## LRA-4829

(info) A line must be with the format

**Description**

The information specified on the indicated line in the given custom cell pair mapping file is not correct. This line will be ignored. The correct format is "A_Cell_in_Ref_Lib A_Cell_in_Target_Lib" (e.g. INV_ABC INV_XYZ).

**What Next**

Check the correctness of the line indicated in this message.

Feedback

## LRA-4830

(info) The reference library cell %s defined in this line

**Description**

A duplicated reference cell definition is detected in the given custom cell pair mapping file. This line will be ignored.

**What Next**

Check the correctness of the line indicated in this message.

## LRA-4831

(info) A line must be with the format

**Description**

The information specified on the indicated line in the given lexical attribute values mapping file is not correct. This line will be ignored. The correct format is "A_Lexical_Attribute Value_in_Ref_Lib Valie_in_Target_Lib

**What Next**

Check the correctness of the line indicated in this message.

## LRA-4832

(info) The attribute value of the reference library %s defined in this line

**Description**

A duplicated lexical attribute value is detected in the given lexical attribute values mapping file. This line will be ignored.

**What Next**

Check the correctness of the line indicated in this message.

## LRA-4833

(warning) Unable to write to the specified output file

**Description**

The tool is not able to write to the specified output file.

Feedback

**What Next**

Check the write permission of the specified file.

---

## LRA-4834

(info) %s library cell %s does not exist.

**Description**

The message indicates the specified reference or target library cell does not exist.

**What Next**

Check and provide a valid cell name.

---

## LRA-4835

(warning) Detected different library vendors, stop comparison.

**Description**

The tool detects the given reference and target libraries are created by different library vendors. The comparison will be stopped.

**What Next**

Provide libraries from the same vendor.

---

## LRA-4836

(info) The given family named %s is an user-defined family. The library cells included in this family but not belong to the reference library %s will be ignored.

**Description**

The given family is an user-defined family. ly may include library cells from different libraries. The library cells not belong to the reference library will be ignored.

**What Next**

Review the compared library cells.

---

## LRA-4837

(info) The given lexical attribute mapping definition will be ignored since a custom comparison pair definition file is specified.

Feedback

**Description**

The custom comparison pair mapping definition has higher priority. The given lexical attribute mapping definitions will be ignored.

**What Next**

Review the compared library cells.

---

## LRA-4838

(warning) Unable to find an exactly matched input slew point

**Description**

The tool is not able to find an exactly matched input slew point from the library analysis by the specified value. A closest available input slew point will be used as the comparison condition.

**What Next**

Review the input slew comparison condition.

---

## LRA-4839

(warning) Unable to find an exactly matched output load point

**Description**

The tool is not able to find an exactly matched output load point from the library analysis by the specified value. A closest available output load point will be used as the comparison condition.

**What Next**

Review the output load comparison condition.

---

## LRA-4840

(info) Set the number of significant digits to 5, the default value

**Description**

The tool will report the absolute values with the number of significant digits equals to 5. This is the default settings of the tool.

**What Next**

Adjust the number of significant digits if there is a need.

Feedback

## LRA-5701

(information) Creating lib cell %s ...

**Description**

Libra command *create_new_function_lib_cells* is creating the lib cell.

**What Next**

This is an informational message, no action is required.

## LRA-5702

(error) %s executable is missing. %s

**Description**

Libra command *create_new_function_lib_cells* needs supports from PrimeTime and Library Compiler. This error message indicates the tool is not able to find the executable of PT or LC.

**What Next**

Try to set the app option and environment variable according to the error message.

## LRA-5703

(error) Libra create_new_function_lib_cells encounters an internal error. %s

**Description**

Libra command *create_new_function_lib_cells* encounters an internal error and not able to resolve by itself. The lib cell is not able to be created.

**What Next**

Set the app option to decrypt log file and search for the error.

## LRA-5704

(error) Lib cell %s cannot be created. %s

**Description**

Libra command *create_new_function_lib_cells* cannot create the lib cell. Currently Libra can only create the new function lib cell which has only one output pin with at least one input pin.

Feedback

**What Next**

Please check the design meet the limitation and try again.

---

## LRA-5705

(error) Library Compiler executable is missing.

**Description**

Libra command *save_suggestion_libs* needs supports from Library Compiler. This error message indicates the tool is not able to find the executable of LC.

**What Next**

Try to set the environment variable SYNOPSYS_LC_ROOT to a proper location.

---

## LRA-5706

(error) Can not find the original DB file of %s

**Description**

Libra command *save_suggestion_libs* needs original DB file of the reference library. This error message indicates the tool is not able to find the original DB file.

**What Next**

Put the original DB file back to the location. User might be able to use *report_lib* to find out the location.

---

## LRA-5707

(error) Library Compiler encounters an error when writing out the library '%s'

**Description**

Libra command *save_suggestion_libs* needs support from LC. This error message indicates the LC encoutners an error while writing out the library.

**What Next**

Check if LC can be launched successfully outside Libra session. Check if the environment variable SYNOPSYS_LC_ROOT is set properly.

---

## LRA-5708

(error) Tool doesn't have write permission.

Feedback

**Description**

Libra command *save_suggestion_libs* does not have the permission to write out suggestion library.

**What Next**

Grant user the permission to write file into output location.

---

## LRA-5709

(error) save_suggestion_libs is not able to write the library '%s'. %s

**Description**

The format or the flow that reference library of the library user wants to write out is not supported by *save_suggestion_libs*.

**What Next**

If there is any library refer to a non-pg-pin-based library, filter out that library and issue the command again. If the reference library is read in thru CLIB flow, close it, and use DB + FRAME (*create_ref_lib*) flow to read in again.

---

## LRA-5710

(warning) *save_suggestion_libs*: There is no library be specified to write out

**Description**

*save_suggestion_libs* doesn't recognize any input of library that user wants to save out.

**What Next**

Check the input argument of *save_suggestion_libs* again. Make sure the library could be queried by *get_libs* command.

---

## LRA-5711

(error) *save_suggestion_libs*: %s libset does not exist

**Description**

*save_suggestion_libs* doesn't recognize the input of libset.

**What Next**

Check if the input libset exists in the session.

Feedback

## LRA-5712

(error) *save_suggestion_libs*: Pane ID %d invalid. Library %s has max Pane ID of %d

**Description**

*save_suggestion_libs* doesn't recognize the input of pane ID.

**What Next**

Specify a correct pane ID. User can use *report_libs* to look for the pane ID.

## LRA-5713

(warning) *save_suggestion_libs*: Missing frame for lib_cell %s.

**Description**

lib_cell doesn't have frame view in memory. The frame view of the lib cell would not be written out.

**What Next**

When *create_ref_libs*, make sure to include the frame file which contains all lib_cells frame view. After that, user has to *create_suggestion_libs* again if user wants to have lib cell frame view be written out.

## LRA-6000

(error) No current design supplied, exiting.

**Description**

The *run_design_driven_analysis* command needs a current design for analysis, and no current design was found.

**What Next**

Read in a design, link it, and rerun the command.

## LRA-6001

(error) Library error for design-driven analysis on design %s, exiting.

Feedback

**Description**

In addition to a design, the *run_design_driven_analysis* command needs a collection of library cells supplied within a libset in order to generate suggestions for new libcells. There was an error in getting that library information.

**What Next**

Supply a libset for the command to use when invoking the command.

---

## LRA-6002

(error) Invalid -num_suggest value of %d (must be a positive integer). Exiting

**Description**

The limit on the number of new libcell functions to suggest must be a positive integer.

**What Next**

Supply a positive integer value for *-num_suggest* and rerun.

---

## LRA-6003

(error) No suggested libcells were found for design %s.

**Description**

There were no potential new cell functions identified for this design.

**What Next**

Make sure that you have supplied a valid design and library as input for the command.

---

## LRA-6004

(error) Cannot locate libset %s

**Description**

The given libset was not available.

**What Next**

Make sure that you have supplied a valid libset as input for the command.

---

## LRA-7001

(Error) Library cell family '%s' not found.

Feedback

**Description**

The specified library cell family name could not be found.

**What Next**

Check the specified family name.

---

## LRA-7002

(Error) Please specify a valid %s attribute name.

**Description**

The specified attribute name is invalide.

**What Next**

Check the specified attribute name.

---

## LRA-7003

(Error) Please specify a valid library analysis name.

**Description**

The specified analysis name could not be found.

**What Next**

Check the specified analysis name.

---

## LRA-7004

(Error) Please run library analysis.

**Description**

Library analysis could not be run.

**What Next**

Check the analysis name or a current analysis has been specified.

---

## LRA-7005

(Info) x attribute for bar chart plot is library cell name.

Feedback

**Description**

Using library cell name for x attribute.

**What Next**

Review the content of the updated x attribute name.

---

## LRA-7006

(Error) Library set '%s' not found.

**Description**

The specified libset could not be found.

**What Next**

Check the specified libset name.

---

## LRA-7007

(Info) No '%s' attribute value for library cell '%s'.

**Description**

Attribute value does not exist for the library cell.

**What Next**

Review the attribute values for the library cell.

---

## LRA-7008

(Error) Please run start_gui.

**Description**

GUI has not been started, please run start_gui.

**What Next**

Run start_gui.

---

## LRA-7009

(Error) Please specify value in range (0 to %u) for '%s'.

Feedback

**Description**

The specified value was not within a valid range.

**What Next**

Specify a value withing the valid range.

---

## LRA-7010

(Info) No '%s' attribute value for timing arc object.

**Description**

Attribute value does not exist for the timing arc object.

**What Next**

Review the attribute values for the timing arc object.

---

## LRA-7011

(Info) No '%s' attribute value for all library cells.

**Description**

Attribute value does not exist for any of the library cells.

**What Next**

Review the attribute values for the library cells.

---

## LRA-7012

(Info) No %s to plot.

**Description**

There are no library cells or timing arcs to plot.

**What Next**

Review your library cells or timing arcs.

Feedback

# LRO

## LRO-0005

(warning) Could not find the name %s

**Description**

This message indicates that the name could not be found.

**What Next**

Provide a correct name for removal

## LRO-0006

(error) Gap name %s not found

**Description**

Provided gap name could not be found, need to provide a correct name to proceed.

**What Next**

Provide the correct gap name in gap_names option

## LRO-0007

(error) Skipping creation of gap '%s' because it is missing one or more lib cell names

**Description**

This message indicates that the creation of the gap is skipped due to missing some names in the lib cell names option. The number of the lib cell names must match with the number of the gaps. For example, you can have g0 with gap size of 2 and g1 with gap size of 1. In this case, the command options must be used as *"-gap_names {g0 g1} -lib_cell_names { {name1 name2} {name3} }"*. The first new gap lib cell in g0 will be named as "name1", and the second new gap lib cell in g0 will be named as "name2". The new gap lib cell in g1 will be named as "name3".

**What Next**

Provide a correct list of lib cell names or gap names to be able to create this gap

Feedback

## LRO-0008

(warning) Skipping gap %s because it is already implemented

**Description**

This message indicates that the gap was already implemented and will not be created again

**What Next**

Remove this gap from the gap names list to avoid this warning

## LRO-0009

(error) Could not create gap lib cell '%s'

**Description**

This message indicates that the gap cell creation is not successful

**What Next**

Please check for previous errors that cause the gap cell creation to fail

## LRO-0010

(error) Please use a percentage coefficient between 0 and 1

**Description**

This message indicates that provided percentage coefficient value is not within the acceptable range.

**What Next**

Provide a valid percentage coefficient value to proceed.

## LRO-0011

(warning) No analysis data can be found for lib cell '%s'

**Description**

This message indicates that the library analysis data could not be found for the lib cell, and it will be skipped.

Feedback

**What Next**

Please check to see for possible errors during run_library_analysis for this lib cell

---

## LRO-0012

(warning) Skipping gap %s because no suggestion library found for reference library '%s'

**Description**

This message indicates that the gap is skipped due to missing a suggestion library for the reference library

**What Next**

Use create_suggestion_libs command to create a suggestion library for the reference library to be able to create this gap

---

## LRO-0014

(error) Could not find the libset named %s

**Description**

The libset of the given name could not be found, need to provide a correct name to proceed.

**What Next**

Provide a correct libset name.

---

## LRO-0015

(error) Found a mismatch between gap names and lib cell names

**Description**

The number of the lib cell names must match with the number of the gaps. For example, you can have g0 with gap size of 2 and g1 with gap size of 1. In this case, the command options must be used as *"-gap_names {g0 g1} -lib_cell_names { {name1 name2} {name3} }"*. The first new gap lib cell in g0 will be named as "name1", and the second new gap lib cell in g0 will be named as "name2". The new gap lib cell in g1 will be named as "name3".

**What Next**

Provide a correct list of lib cell names or gap names to proceed.

Feedback

## LRO-0016

(warning) Less than two active costs. No tradeoff to perform

**Description**

This message indicates that there is less than two active costs enabled. Therefore there is no tradeoff to perform.

**What Next**

Active more than two costs to proceed.

## LRO-1000

(error) Could not find analysis set named %s

**Description**

The analysis set of the given name could not be found, need to provide a correct name to proceed.

**What Next**

Provide a correct analysis name or run "run_library_analysis" command to generate an analysis set.

## LRO-1002

(warning) Libcell %s is a redundant variant of %s. Redundant variants will not be listed as dont_use in the present flow

**Description**

This message indicates that the libcell is a redundant physical variant of a libcell. This libcell will not be listed in the dont_use list.

## LRO-1003

(information) Generated %d library cells from gap information

**Description**

This message shows the total number of generated library cells from gap information

**What Next**

This is an informational message, no action is required.

Feedback

## LRO-1005

(information) Removed %d redundancies, outliers, and equivalences

**Description**

This message shows the total number of removed redundancies, outliers, and equivalences.

**What Next**

This is an informational message, no action is required.

## LRO-1006

(warning) Could not find library cell family named %s

**Description**

The given library cell family name could not be found, need to provide a correct name to proceed.

**What Next**

Provide a correct library cell family name.

## LRO-1007

(error) Could not find any library cell families

**Description**

This message indicates that no library cell families can be found from the provided family list.

**What Next**

Provide valid library cell family names.

## LRO-1008

(warning) Family %s was not provided in the family list and will be ignored for keeping it separate

**Description**

This message indicates that the library cell family is not provided in the family list and will be ignored for keeping it seperate.

Feedback

**What Next**

Provide the library cell family name in the family list.

---

## LRO-1009

(information) Removed %d gaps

**Description**

This message shows the total number of removed gaps.

**What Next**

This is an informational message, no action is required.

---

## LRO-2000

(information) Created %d gap cells from gap %s

**Description**

This message shows the total number of implemented gap cells from the gap information

**What Next**

This is an informational message, no action is required.

---

## LRO-4000

(information) Found %d libcells equivalent to other libcells across all PVTs

**Description**

This message shows the total number of equivalent lib cells that are found during analysis across all PVT indices.

**What Next**

This is an informational message, no action is required.

---

## LRO-4001

(information) Found %d libcells equivalent to other libcells at some PVTs

**Description**

This message shows the total number of equivalent lib cells that are found during analysis at some PVT indices.

Feedback

**What Next**

This is an informational message, no action is required.

---

## LRO-4100

(information) Found %d libcells redundant across all PVTs

**Description**

This message shows the total number of redundant lib cells that are found during analysis across all PVT indices.

**What Next**

This is an informational message, no action is required.

---

## LRO-4101

(information) Found %d libcells redundant at some PVTs

**Description**

This message shows the total number of redundant lib cells that are found during analysis at some PVT indices.

**What Next**

This is an informational message, no action is required.

---

## LRO-4200

(information) Found %d libcells outlier across all PVTs

**Description**

This message shows the total number of outlier lib cells that are found during analysis across all PVT indices.

**What Next**

This is an informational message, no action is required.

---

## LRO-4201

(information) Found %d libcells outlier at some PVTs

Feedback

**Description**

This message shows the total number of outlier lib cells that are found during analysis at some PVT indices.

**What Next**

This is an informational message, no action is required.

---

## LRO-4300

(information) In analysis %s :

**Description**

This message shows the analysis name that is used for gap, redundancy, outlier, and equivalent lib cell analysis

**What Next**

This is an informational message, no action is required.

---

## LRO-4301

(information) In families %s :

**Description**

This message shows the family names that are used for gap, redundancy, outlier, and equivalent lib cell analysis

**What Next**

This is an informational message, no action is required.

---

# MBIT

---

## MBIT-001

(Error) %s. Banking/Debanking aborted.

**Description**

This message indicates that either banking or debanking is aborted due to some reason.

**What Next**

Users need to fix the reason.

Feedback

**See Also**

- create_multibit
- split_multibit
- identify_multibit

---

## MBIT-003

(Error) The specified cell (%s) is invalid because the cell is locked.

**Description**

This message indicates that the specified register cell is invalid for create_multibit or split_multibit, because the cell is locked.

**What Next**

Users need to check the specified cell.

**See Also**

- create_multibit
- split_multibit

---

## MBIT-004

(Info) The specified cell (%s) is invalid because of dont_touch attribute.

**Description**

This message indicates that the specified register cell is invalid during banking or debanking, because the cell has dont_touch attribute.

**What Next**

Users need to check the specified cell.

**See Also**

- identify_multibit
- create_multibit
- split_multibit
- place_opt

Feedback

## MBIT-005

(Error) Invalid specified cell (%s). The cell must be sequential cell.

### Description

This message indicates that the specified register cell is invalid for create_multibit or split_multibit, because the cell is not sequential cell.

### What Next

Users need to check the specified cell.

### See Also

- create_multibit
- split_multibit

## MBIT-006

(Error) The specified cell (%s) is invalid because its location is invalid.

### Description

This message indicates that the specified register cell is invalid for create_multibit or split_multibit, because the cell's location is invalid.

### What Next

Users need to check the specified cell.

### See Also

- create_multibit
- split_multibit

## MBIT-007

(Error) The specified cell (%s) is invalid because its reference library cell is different with other candidate cells.

### Description

This message indicates that the specified register cell is invalid for create_multibit, because the cell's reference library cell is different with other candidate cells.

Feedback

### What Next

Users need to remove the specified register from the candidate registers list.

### See Also

- create_multibit
- split_multibit

---

## MBIT-008

(Error) The specified cell (%s) is invalid because its hierarchy is different with other candidate cells.

### Description

This message indicates that the specified register cell is invalid for create_multibit or split_multibit, because the cell's hierarchy is different with other candidate cells.

### What Next

Users need to check the specified cell.

### See Also

- create_multibit
- split_multibit

---

## MBIT-009

(Error) The specified cell (%s) is invalid because of inconsistent size_only attribute.

### Description

This message indicates that the specified register cell is invalid for create_multibit or split_multibit, because the cell's size_only attribute is inconsistent with other candidate cells.

### What Next

Users need to check the specified cell.

### See Also

- create_multibit
- split_multibit

Feedback

## MBIT-010

(Error) The specified cell (%s) is invalid because timing constraint is not removable.

### Description

This message indicates that the specified register cell is invalid for create_multibit or split_multibit, because cell has un-removable timing constraints.

### What Next

Users need to check the timing constraints on specified cell.

### See Also

- create_multibit
- split_multibit

## MBIT-011

(Error) Failed to create multibit cell.

### Description

This message indicates that tool failed to create a multibit cell.

### What Next

Users need to double check design setup.

### See Also

- create_multibit
- split_multibit

## MBIT-012

(Error) Failed to set location on cell (%s).

### Description

This message indicates that tool failed to set location on cell.

### What Next

Users need to double check the design setup.

Feedback

**See Also**

- create_multibit
- split_multibit

---

## MBIT-013

(Warning) Failed to find %s pins on cell %s .

**Description**

This message indicates that tool failed to find related pins on cell.

**What Next**

Users need to double check the reference library cell.

**See Also**

- create_multibit
- split_multibit

---

## MBIT-014

(Error) Inconsistent control pin connections on candidate flops (%s and %s)

**Description**

This message indicates that the control pin connection on specified candidate flops are not electronically equivalent.

**What Next**

Users need to check the specified cells' connection.

**See Also**

- create_multibit
- split_multibit

---

## MBIT-015

(Warning) SCANDEF update fails because %s.

Feedback

**Description**

This message occurs when SCANDEF update failes due to the described reason.

**What Next**

Call check_scan_chain beforehand to ensure all the scan chains are valid.

**See Also**

- check_scan_chain
- create_multibit
- split_multibit

---

## MBIT-016

(Error) %s at or near line number %d.

**Description**

Syntax error at or near the reported line number.

**What Next**

Correct the syntax error and rerun the command.

---

## MBIT-017

(Error) Insufficient bit width for banking or splitting.

**Description**

This message indicates that specified candidates has insufficient bit width for banking or splitting. For create_multibit, the candidate flops' count should be less than or equal to bank's bit width. For split_multibit, the sub-bank's width should be equal to original bank's width.

**What Next**

Users need to double check the specified banking or splitting candidates.

**See Also**

- create_multibit
- split_multibit

Feedback

## MBIT-018

(Info) create_multibit cannot continue because %s.

### Description

This message occurs when the single-bit flops to be banked are failed due to printed reason.

### What Next

Use write_def to dump out the SCANDEF file to check the issue, and specify the single-bit flops to meet the scan chain constraints. All the single-bit scan flops to be banked are either ORDERED list or from FLOATING list of scanchain or not present in scan chain.

### See Also

- write_def
- create_multibit

## MBIT-019

(Error) Inconsistent data input / output pin connection.

### Description

This message indicates that tool found the connected input data pin count is not equal to connected output pin count. Since the connection count is inconsistent, tool skipped the splitting.

### What Next

Users need to double check the specified banking or splitting candidates.

### See Also

- create_multibit
- split_multibit

## MBIT-020

(Error) The specified cell (%s) is invalid because of legalize_only attribute.

Feedback

### Description

This message indicates that the specified register cell is invalid for create_multibit or split_multibit, because the cell has legalize_only attribute.

### What Next

Users need to check the specified cell.

### See Also

- create_multibit
- split_multibit

---

## MBIT-021

(Warning) Could not find the lib cell %s.

### Description

Reported library cell could not be found.

### What Next

Users need to check the specified library cell.

### See Also

- identify_multibit

---

## MBIT-022

(Warning) Bit width of library cell %s will be treated as 1.

### Description

Could not determine the bit_width of the reported library cell; bit_width will be treated as 1.

### What Next

Users need to check the specified library cell.

### See Also

- identify_multibit

Feedback

## MBIT-023

(Warning) Number of register should be greater than 0.

**Description**

Number of register should be greater than 0.

**What Next**

Correct the number and rerun the command.

**See Also**

- identify_multibit

## MBIT-024

(Warning) Input bit %d cannot be less than 2.

**Description**

Input bit cannot be less than 2. Specific entry will be ignored.

**What Next**

Correct the mapping entry and rerun the command; otherwise specific entry will be ignored.

**See Also**

- identify_multibit

## MBIT-025

(Warning) Entry for %d bits is not proper, calculated bit is %d

**Description**

Calculated total number of bits is more than its entry bit. Entry will be ignored.

**See Also**

- identify_multibit

## MBIT-026

(Warning) Found duplicate entry for %d bit in functional group %s.

Feedback

### Description

Duplicate entry found. The entry will be ignored.

### See Also

- identify_multibit

---

## MBIT-027

(Warning) Functional group with name %s already exists.

### Description

Reported functional group already exists. You can continue but this group will be ignored. Alternatively rename the group and rerun the command.

### See Also

- identify_multibit

---

## MBIT-028

(Error) The specified cell (%s) is invalid because its bound is different with other candidate cells.

### Description

This message indicates that the specified register cell is invalid for create_multibit, because the cell's bound is different with other candidate cells.

### What Next

Users need to check the specified cell.

### See Also

- create_multibit
- split_multibit

---

## MBIT-029

(Error) The specified cell (%s) is invalid because its voltage area is different with other candidate cells.

Feedback

### Description

This message indicates that the specified register cell is invalid for create_multibit, because the cell's voltage area is different with other candidate cells.

### What Next

Users need to check the specified cell.

### See Also

- create_multibit
- split_multibit

---

## MBIT-030

(Warning) Single-bit lib cell %s has already been added.

### Description

Reported single-bit lib cell was already added to some functional group. It cannot be added again. Present occurrence of lib cell will be ignored.

### See Also

- identify_multibit

---

## MBIT-031

(Error) The specified cell (%s) is invalid because it belongs to RP group.

### Description

This message indicates that the specified register cell is invalid for create_multibit or split_multibit, because the cell belongs to RP group.

### What Next

Users need to check the specified cell.

### See Also

- create_multibit
- split_multibit

Feedback

## MBIT-032

(Warning) WNS threshold for %s has already been set. Value specified at line number %d is ignored.

### Description

WNS threshold value for the reported group is already set. You are trying to set it once more and hence the warning.

### See Also

- identify_multibit

## MBIT-033

(Warning) No mapping entry found for functional group %s.

### Description

For the reported functional group there is no mapping entry. This group will be ignored during banking.

### See Also

- identify_multibit

## MBIT-034

(Warning) Mapping entry for bit %d not found for functional group %s.

### Description

Reported bit entry for the group is not found. This may result in lower banking ratio.

### See Also

- identify_multibit

## MBIT-035

(Error) The specified reference lib cell is dont-use: %s.

### Description

Tool could not create or split cell because the specified reference library cell is marked as dont-use.

Feedback

### What Next

Users need to check the specified library cell.

### See Also

- identify_multibit

---

## MBIT-036

(Warning) The reference lib cell %s is dont-use.

### Description

The reported reference lib cell is ignored.

### What Next

Users need to check the specified library cell.

---

## MBIT-037

(Error) No proper functional group defined in input file %s.

### Description

You are getting this message as the tool didn't find a single valid functional group in input_map_file.

### What Next

Provide proper functional group information through input_map_file. If input_map_file contains functional group information, check for error warning to know why they are ignored.

### See Also

- identify_multibit

---

## MBIT-038

(Warning) There is no scandef info in current design

### Description

No scandef info is in current design

Feedback

### What Next

Users need to check whether the missing scandef is intended

### See Also

- check_scan_chain

---

## MBIT-039

(Error) Inconsistent timing constraints on candidate flops (%s and %s)

### Description

This message indicates that the timing constraints on specified candidate flops are not consistent.

### What Next

Users need to check the specified cells' timing constraints.

### See Also

- create_multibit
- split_multibit

---

## MBIT-040

(Error) Failed to deduce library-cells from specified multi-bit cell.

### Description

Tool failed to deduce library-cells from specified multi-bit cell. User needs to manually specify '-lib_cells' option.

### What Next

User needs to manually specify '-lib_cells' option.

### See Also

- identify_multibit

---

## MBIT-041

(Warning) No mbit lib cell found for strategy %s to replace %lu cells.

Feedback

### Description

No mbit lib cell is mapped to the strategy mentioned.

### What Next

Map level_shifter_cell or isolation_cell as required and re run the command.

### See Also

- map_isolation_cell
- map_level_shifter_cell

---

## MBIT-042

(Error) The specified reference lib cell is invalid: %s.

### Description

Tool could not create or split cell because the specified reference library cell is invalid.

### What Next

Users need to check the specified library cell.

### See Also

- identify_multibit

---

## MBIT-043

(Error) The app option required for banking of multibit cell '%s' has not been specified.

### Description

This error message occurs when the tool detects that you have specified a multibit cell for banking during create_multibit/identify_multibit command without using the app option for banking multibit cells.

### What Next

Please specify the following app option: mbit.enable_banking_of_multibit_cell and run the command again.

Feedback

**See Also**

- identify_multibit
- create_multibit

---

## MBIT-045

(Error) %s cell specified for the command option '-cells'.

**Description**

This error message occurs when the tool detects that you have specified invalid list of cells for banking/de-banking during identify_multibit/split_multibit command with cells option.

**What Next**

Please specify valid list of cells and run the command again.

**See Also**

- identify_multibit
- split_multibit

---

## MBIT-046

(Error) The specified cell (%s) is non considered as it has size_only attribute.

**Description**

This message indicates that the specified cell is invalid for create_multibit or split_multibit or identify_multibit, because size_only attribute is set on it.

**What Next**

Users need to check the specified cell.

**See Also**

- create_multibit
- split_multibit
- identify_multibit

---

## MBIT-047

(Error) SDC constraint is set on the cell (%s)

Feedback

### Description

This message indicates that the SDC constraints is specified on candidate flops because of which it is ignored.

### What Next

Users need to check the SDC constraint applied.

---

## MBIT-048

(Error) %s Lib cell does not belong to strategy %s.

### Description

The specified reference library cell does not belong to proper strategy.

### What Next

Please use the lib-cell from proper power strategy.

### See Also

- identify_multibit

---

## MBIT-049

(Error) The specified cell (%s) is invalid because of retention cell criteria.

### Description

This message indicates that the specified register cell is either not a retention cell or it retention strategy is not matching with other cells.

### What Next

Users need to check the specified cell.

### See Also

- create_multibit
- split_multibit

---

## MBIT-050

(Warning) The specified cell (%s) is invalid because %s.

Feedback

**Description**

This message indicates that the specified register cell is invalid for create_multibit or split_multibit.

**What Next**

Users need to check the specified cell.

**See Also**

- create_multibit
- split_multibit

---

## MBIT-051

(Info) No multibit banking performed as no valid multibit lib-cell found in the library.

**Description**

This message indicates that no valid multibit lib-cell found in the library for multibit banking.

**What Next**

Users need to check library for valid multibit lib-cells.

**See Also**

- create_multibit
- split_multibit

---

## MBIT-052

(Error) Cells %s and %s does not having matching constraints.

**Description**

This message indicates that cells which are expected to be banked are ignored as they don't having matching constraints on them.

**What Next**

cells are expected to have matching constraint

Feedback

## MBIT-053

(Error) Cells to be banked does not having matching restrictions.

### Description

This message indicates that cells which are expected to be banked are ignored as they are not allowed by restriction manager.

### What Next

cells are expected to have matching restriction

## MBIT-054

(Error) Cell %s to be debanked does not passes restrictions check.

### Description

This message indicates that cell %s which is expected to be debanked is ignored as it does not passes restriction manager check.

### What Next

cell is expected to pass restriction check.

## MBIT-055

(Warning) Design has scan stitched cells but no scandef found.

### Description

This message indicates that scandef is not provided but cells have valid scan connection. During multibit banking such cells will be banked if they are found consecutive and that may reduce banking ratio.

### What Next

Read scandef before multibit banking.

### See Also

- identify_multibit
- split_multibit

Feedback

## MBIT-056

(Warning) The multibit cell %s may not be used during banking becuase the delay of arc from clock pin to output pin could not be computed.

### Description

The multibit cell %s may not be used during banking becuase the delay of arc from clock pin to output pin could not be computed and Please check if libcell is not corrupted.

### What Next

Please check if libcell is corrupted or not.

### See Also

- identify_multibit

## MBIT-057

(Warning) Lib cell %s does not match cell %s.

### Description

The specified reference library cell cannot be used to replace specified cell.

### What Next

Please use the lib-cell match with cell.

### See Also

- identify_multibit
- split_multibit
- create_multibit

## MBIT-058

(Warning) Bus based banking is not supported for MV cells

### Description

Physical aware multibit engine does not support bus based banking for MV cells. As of now bus based banking is only supported for register cells.

Feedback

### What Next

Use register cells for bus based banking.

### See Also

- identify_multibit
- split_multibit
- create_multibit

---

## MBIT-059

(Warning) Cell %s is not debanked as its ref lib cell has dont_use attribute true.

### Description

Multibit cell instance having its ref lib cell has dont_use attribute true cannot be debanked.

### What Next

Remove dont_use attribute on lib cell and then debank the cells.

### See Also

- identify_multibit
- split_multibit
- create_multibit

---

## MBIT-060

(Warning) set_target_library_subset will not be honored.

### Description

Set_target_library_subset in split_multibit will not be honored if we don't use either -cells or -lib_cells option with split_multibit.

### What Next

Use -cells or -lib_cells option with split_multibit to honor set_target_library_subset.

### See Also

- split_multibit

Feedback

## MBIT-061

(Error) Invalid cell list specified for the command option '-exclude_instance'.

### Description

This error message occurs when the tool detects that you have specified invalid list of cells for banking/de-banking during identify_multibit/split_multibit command with exclude_instance option.

### What Next

Please specify valid list of cells and run the command again.

### See Also

- identify_multibit
- split_multibit

## MBIT-062

(Warning) Invalid specified cell (%s). The cell specified by the command option '-exclude_instance' must be a sequential cell.

### Description

This message indicates that the specified register cell is invalid for identify_multibit/ split_multibit, because the cell specified by the command option '-exclude_instance' is not a sequential cell.

### What Next

Users need to check the specified cell for the command option '-exclude_instance'.

### See Also

- identify_multibit
- split_multibit

## MBIT-063

(Error) Please specify more than one lib cell for the command option '-lib_cells' in debanking.

Feedback

### Description

This message indicates that the number of specified lib cells is not sufficient.

### What Next

Users should make sure to specify more than one lib cell and the sum of the bit width of all lib cells should be equal to bit width of multibit cell be debanked.

### See Also

- split_multibit

---

## MBIT-064

(Error) The width of specified lib cells for debanking is higher than the width of multlibit cell.

### Description

This message indicates that the width of specified lib cells for debanking is higher than the width of multlibit cell. The sub-bank's width should be equal to original bank's width.

### What Next

Users should make sure to specify more than one lib cell and the sum of the bit width of all lib cells should be equal to bit width of multibit cell be debanked.

### See Also

- split_multibit

---

## MBIT-065

(Error) No -lib_cell is expected with the provided list of unmapped cells.

### Description

The list of provided unmapped cells will be added to a bus instance and will be candidates for banking by the sequential RTL mapping and banking engine. Therefore, no -lib_cell is expected.

### What Next

Please remove the -lib_cell option from the create_multibit command and try again.

### See Also

- create_multibit

Feedback

## MBIT-066

(Error) cell_list does not support a mix of mapped and unmapped cells.

**Description**

create_multibit does not support a list of mapped and unmapped cells.

**What Next**

Please check the list of provided cells and try the create_multibit command again.

**See Also**

- create_multibit

## MBIT-067

(Error) Provided list of cells has a mix of sequential and non-sequential cells.

**Description**

Non-sequential cells are not supported in the list of provided cells for create_multibit.

**What Next**

Please check the list of cells provided to the create_multibit command.

**See Also**

- create_multibit

## MBIT-068

(Error) create_multibit command failed: multibit flow is disabled.

**Description**

create_multibit could not create the bus instance because of multibit flow is disabled.

**What Next**

Please check the app-option setting for enabling the multibit flow.

**See Also**

- create_multibit

Feedback

## MBIT-069

(Error) create_multibit command failed: provided list of cells do not all belong to the same hierarchy.

### Description

create_multibit could not create the bus instance because the provided list of cells do not belong to the same hierarchy.

### What Next

Please check the list of cells provided to the create_multibit command.

### See Also

- create_multibit

## MBIT-070

(Error) create_multibit command failed: one or more cells already belong to a bus instance.

### Description

create_multibit did not create the bus instance because one or more cells already belong to a bus instance and are candidates for banking for the sequential mapping engine.

### What Next

Please check the list of cells provided to the create_multibit command.

### See Also

- create_multibit

## MBIT-071

(Error) Failed to create bus instance because single-bit cells do not share control signals or are functionally not equivalent.

### Description

create_multibit expects the single-bit cells to be functionally equivalent for them to be added to a bus instance.

Feedback

### What Next

Please check the list of cells provided to the create_multibit command.

### See Also

- create_multibit

---

## MBIT-072

(Error) Banking step is skipped because of no setup scenario.

### Description

This message indicates that the banking step is skipped because of no setup scenario.

### What Next

Users should specify at least one setup scenario.

### See Also

- identify_multibit
- place_opt

---

## MBIT-073

(Warning) Cannot identify the correct multibit_width of lib cell (%s). The number of input data pins (%u) and output data pins (%u) are mismatched.

### Description

This message indicates that the report_multibit command may not get the correct multibit_width attribute of the lib cell. The multibit_width is used to identify the bit width of the lib cell, and if this attribute is missing, the command uses the number of data pins to identify the bit width.

### What Next

Users need to check the multibit_width and pin attribute of the lib cell.

### See Also

- report_multibit

Feedback

# MDXF

## MDXF-001

(error) Could not parse '%s' with value string '%s' inside '%s'

**Description**

This message indicates either the given rule name or value string is syntactically wrong.

**What Next**

Please refer to the MDXF Reference Manual and provide correct rule name/value string.

## MDXF-002

(error) Multiple '%s' provided

**Description**

Multiple section(s)/attribute(s) provided whose occurrence number one or at most one.

**What Next**

Please refer to the MDXF Reference Manual and remove repetitive occurrences.

## MDXF-003

(error) Wrong %s provided

**Description**

If file version is mentioned in the error, the given MDXF file version number is greater than reader's version number or non-numeric characters have been mentioned.

If the unit is mentioned in the error, not supported distance unit has been provided.

**What Next**

Please refer to the MDXF Reference Manual and provide correct distance unit or file version.

## MDXF-004

(error) Encountered while converting the data to %s

Feedback

**Description**

This message indicates an internal data model error.

**What Next**

Send the log to Synopsys.

---

## MDXF-005

(error) Required %s not provided

**Description**

If file version is mentioned in the error, the required mdxf file version number is not provided in the mdxf file.

If the unit is mentioned in the error, the required attribute distance unit is not provided in the mdxf file.

**What Next**

Please refer to the MDXF Reference Manual and provide attribute distance unit or file version.

---

## MDXF-006

(error) Syntax error in the file : '%s'

**Description**

Syntax error related to XML formatting was found in the specified MDXF file in the error message.

The line number of the syntax error and additional details will also be printed in case of missing operands like quote("), equal sign(=) and end tag(>).

In some other cases of a missing slash tag (/) or start tag(<) etc., the line number corresponding to end of file will be printed, along with error details like 'unexpected end of data' or 'expected element name'

**What Next**

Please fix the syntax error(s) and rerun the command.

---

## MDXF-007

(error) Could not parse mdxf file '%s'.

Feedback

**Description**

There was a major syntax error in the specified mdxf file. It should be a single text file containing 3DIC information. Please refer to messages printed before this message.

**What Next**

Please correct the input mdxf file and rerun the command

---

## MDXF-008

(error) Unsupported %s file '%s' format '%s'.

**Description**

The input/include file format specified is not supported. Hence the command cannot parse the text file.

**What Next**

Provide a valid format, e.g. "mdxf".

---

## MDXF-009

(warning) Ignoring unknown %s : '%s'

**Description**

This message indicates the given section/attribute name is syntactically wrong.

**What Next**

Please refer to the MDXF Reference Manual and provide correct name.

---

## MDXF-010

(error) File '%s' included multiple times.

**Description**

The file specified is either included multiple times or having a cyclic dependency.

**What Next**

Please include the file only once. If file is included only once then check and remove the cyclic dependency.

Feedback

## MDXF-011

(error) Rule '%s' of rule type '%s' is not supported in '%s' section.

**Description**

This rule will not be considered in the mentioned section.

**What Next**

Please refer to the MDXF Reference Manual and provide only supported rule types in the corresponding owner section.

## MDXF-012

(error) Section '%s' is not supported in '%s' section.

**Description**

This section will not be considered in the mentioned owner section.

**What Next**

Please refer to the MDXF Reference Manual and provide only supported section types in the corresponding owner section.

## MDXF-013

(warning) Skipping rule '%s' because of wrong rule name inside '%s'

**Description**

This message indicates the given rule name is syntactically wrong.

**What Next**

Please refer to the MDXF Reference Manual and provide correct rule name.

## MDXF-014

(warning) Skipping rule '%s' because multiple rules with same name provided inside '%s'

**Description**

This message indicates multiple rules with the same rule name has been specified inside owner section mentioned in the message.

Feedback

**What Next**

Please remove the multiple rules specified with the same name.

---

## MDXF-015

(warning) No floorplan rule in current design.

**Description**

This message indicates that there is no floorplan rule in current design.

**What Next**

Please ensure you have opened the proper design or have already read MDXF data.

---

## MDXF-016

(warning) Rule '%s' has been provided with one value '%s' inside '%s', expected two values.

**Description**

This message indicates the given rule name is expected two values but has been provided with one. The rule has been created assuming the two values are equal to the provided value.

**What Next**

Please refer to the MDXF Reference Manual and provide correct number of values.

---

## MDXF-017

(warning) The min and max values of the rule '%s' are given in the wrong order as '%s' inside '%s'.

**Description**

This message indicates the minium value provide is greater than the maximum value i.e. the values has been provided in the wrong order. The values have been swapped while creating the rule i.e. the greater of the two values considered as max and min will be the lesser of the two.

**What Next**

Please provide the values in the correct order.

Feedback

## MDXF-018

(warning) Skipping element '%s' because multiple elements with same name provided inside '%s'

**Description**

This message indicates same element has been specified multiple times inside owner section mentioned in the message. The data of first one will be used.

**What Next**

Please remove the multiple element and specify the correct data.

## MDXF-019

(error) Wrong %s : '%s' provided %s

**Description**

The element data provided is not valid.

Example: If the element data is expecting name as the value, then the name should not contain any special char (i.e. '*', '?', '^', '$'). If the element data is expecting orientation as the value, the orientation value should be one of the valid orientation.

**What Next**

Please refer to the MDXF Reference Manual and provide valid data.

## MDXF-020

(warning) Inefficient usage: Rule '%s' has been provided with overlapped range values as '%s' inside '%s'.

**Description**

This message indicates that among the multiple ranges which are specified, there is a range which is overlapping with another range which leads to inefficient usage.

**What Next**

Please provide the non-overlapping ranges.

## MDXF-021

(error) Error encountered in encrypted sections of the input. Detailed information suppressed to protect encrypted data.

Feedback

**Description**

This message indicates that an error was encountered in the encrypted sections of the input. More detailed information is suppressed to protect encrypted data.

**What Next**

Please refer to the MDXF Reference Manual and correct the encrypted sections of the input.

---

## MDXF-022

(warning) Warning encountered in encrypted sections of the input. Detailed information suppressed to protect encrypted data.

**Description**

This message indicates that a warning was encountered in the encrypted sections of the input. More detailed information is suppressed to protect encrypted data.

**What Next**

Please refer to the MDXF Reference Manual and correct the encrypted sections of the input.

---

## MDXF-023

(warning) Could not determine the z_level of die '%s' : %s

**Description**

This message indicates that there can be a cyclic dependency among the dies or the bottom most die was not found, due to which z_level of die could not be determined.

**What Next**

Please refer to the MDXF Reference Manual and provide correct dependency.

---

## MDXF-024

(error) %s must be the first element inside %s section, if specified.

**Description**

The element should be first element inside the mentioned section.

Feedback

**What Next**

Please refer to the MDXF Reference Manual and provide the elements in the required order.

---

## MDXF-025

(warning) Skipping rule '%s' because of wrong object type inside '%s'

**Description**

This message indicates the given rule object type is syntactically wrong.

**What Next**

Please refer to the MDXF Reference Manual and provide correct rule name.

---

## MDXF-026

(error) Cannot find reference or cell name mentioned in '%s' with value string '%s' inside '%s' section

**Description**

This message indicates that the mentioned reference or cell name in the value string was not found. Reference cell or lib cell with this name was not found in the reference libraries of the current lib. Also, cell instance with this name is not present in the current physical hierarchy of the design.

**What Next**

Please refer to the MDXF Reference Manual and provide correct rule name/value string.

---

## MDXF-027

(error) Cannot find attribute name mentioned in '%s' with value string '%s' inside '%s' section

**Description**

This message indicates that the mentioned attribute name in the value string was not found.

**What Next**

This usually means that this attribute was not defined in this section earlier using the DESIGN_TAG entry. Please refer to the MDXF Reference Manual and provide correct rule name/value string.

Feedback

## MDXF-028

(warning) Skipping rule '%s' because multiple attributes with same name '%s' are provided inside '%s'

**Description**

This message indicates multiple attribute with the same name in same rule has been specified inside owner section mentioned in the message.

**What Next**

Please remove or modify the lines with the same rule specified with the same attribute name.

## MDXF-029

(error) Cannot find reference block '%s' mentioned in '%s' inside '%s' section

**Description**

This message indicates that the mentioned reference block in the value string was not found in the reference libraries of the current lib.

**What Next**

Please refer to the MDXF Reference Manual and provide correct value string.

## MDXF-030

(warning) Rule '%s' is deprecated and will be removed in a future release. Please consider removing it from your file.

**Description**

This message indicates this particular rule has been deprecated and will be removed in a future release.

**What Next**

Please remove the lines with deprecated rules.

## MDXF-031

(warning) The value '%s' mentioned in rule '%s' does not match with '%s'.

Feedback

**Description**

This message indicates the value mentioned in this rule does not match with actual data.

**What Next**

Please provide correct values.

---

## MDXF-032

(warning) At least one '%s' in the value string '%s' mentioned in '%s' is not found.

**Description**

This message indicates that at least one die in the mentioned value string was not found in the current design.

**What Next**

Please modify the MDXF file with correct die names and run again.

---

## MDXF-033

(warning) Section '%s' is deprecated and will be removed in a future release. Please consider removing it from your file and replacing it with '%s'.

**Description**

This message indicates this particular section has been deprecated and will be removed in a future release.

**What Next**

Please remove the lines with deprecated sections and replace them with new section names.

---

## MDXF-034

(warning) Some '%s' of '%s' were not written out because '%s' was not found.

**Description**

This message indicates few sections were not written out since the corresponding refrence was not found.

---

## MDXF-035

(warning) Section '%s' is deprecated and will be removed in a future release.

Feedback

**Description**

This message indicates this particular section has been deprecated and will be removed in a future release.

**What Next**

Please remove the lines with deprecated sections.

# MENU

## MENU-001

(Error) Menu name is empty.

**Description**

An empty string was given as the menu name option value.

**What Next**

Provide a non-empty menu name.

## MENU-002

(Error) The menu name needs to be a full hierarchical menu name.

**Description**

A full hierarchical menu name needs to be specified for the menu name. An example: "&File->Exit".

**What Next**

Check the option value of the -menu option.

## MENU-003

(Error) The -anchor_offset option value needs to be a positive/negative integer and not zero.

**What Next**

Correct the option value of the -anchor_offset option.

Feedback

## MENU-004

(Error) -anchor_offset option needs to be specified when -anchor_item option is present.

## MENU-005

(Error) %s option value is specified incorrectly.

**What Next**

Check and correct the option value for the specified option.

## MENU-006

(Error) -anchor_item option needs to be specified when -anchor_offset option is present.

## MENU-007

(Error) Anchor menu item '%s' not found.

**What Next**

Check the spelling of the option value of the -anchor_menu option. Make sure that the anchor menu item already existed before using it as anchor_menu option value in the gui_create_menu command call.

## MENU-008

(Error) Menu bar items must be popup menus.

**Description**

First-level items on the menu bar must be popup menus.

**What Next**

Check and correct the option value of the -menu option.

## MENU-012

(Error) Menu root: %s not found.

**What Next**

Check and correct the option value of the -menu_root option.

Feedback

## MENU-013

(Warning) Hotkey string: %s is either not supported or reserved by system for other use.

**What Next**

Please check and correct the option value of the -hot_key option. This hotkey key combination may not be supported or has been reserved by the application for some other use.

## MENU-014

(Error) Duplicate menu item: %s .

**Description**

Another existiing menu item already has this menu name.

## MENU-015

(Error) %s option is specified with invalid value: %s.

**What Next**

Check and correct the option value of the specified option.

## MENU-016

(Error) Cannot specify both -anchor_offset and -position options.

## MENU-017

(Error) -position option value must not be a negative integer.

**Description**

Position option value must be zero or a positive integer. It specifies the position of the menu item within the parent popup menu.

## MENU-018

(Warning) Specified hotkey %s for this menu item %s will be ignored. Hotkey is already in use for %s menu item.

Feedback

**Description**

This hotkey has already been mapped to another existing menu item. It will be ignored and not set for this menu item.

**What Next**

Change the -hotkey option value to specify another hotkey or remove this option.

---

## MENU-019

(Error) Menu item '%s' not found.

**Description**

The specified menu item was not found in the currently active window. It could not be executed.

**What Next**

Check the menu name to make sure it matches an existing menu item in the active window.

---

## MENU-020

(Error) There is no active main window.

**Description**

No active main window was found.

**What Next**

Activate a main window before executing this command.

---

## MENU-021

(Error) Menu %s is not a popup menu item.

**What Next**

Add menu items to a popup menu, not to an item with associated command or action. Popup menus are created implicitly by specifying a full hierarchical menu name.

---

## MENU-022

(Warning) Menu bar items must be popup menus. %s not added.

Feedback

**Description**

First-level items on the menu bar must be popup menus.

**What Next**

Do not explicitly create popup menus in hierarchical menus. Pop-up menus are created implicitly by creating command menu items with full hierarhcical names. For example, creating a menu with the menu name "&File->&Open..." creates the "File" pop-up menu in the menu bar.

---

## MENU-023

(Error) Window named %s provided for the -window option was not found.

**Description**

The provided window name is not valid.

**What Next**

Check the spelling of the window name. If you omit the -window option, the currently active top-level main window will be used.

---

## MENU-024

(Error) Window named was not provided and there is no active main window.

**Description**

When no window name is provided for the -window option, the currently active top-level window is used. However, no active top-level main window eixsts.

**What Next**

Create an instance of the top-level main window on which you wish to operate, then provide its name with the -window option or activate the window and omit the -window option.

---

## MENU-025

(Warning) There are no toolbars in the specified window.

**Description**

Since there are no toolbars in the specified window, the operation resulted in no action.

Feedback

## MENU-026

(Error) There is no toolbar with the name %s.

### Description

A toolbar with the given name could not be found. Therefore, the operation resulted in no action.

## MENU-028

(Error) Options %s are not supported with the -window option.

### Description

Supported options for menu attribute setting in a specific window are: -menu_text, -tooltip, and -help_string.

# MIB

## MIB-001

(warning) There are either no MIBs in the design or all MIB blocks are marked un-editable.

### Description

You receive this message when there are no MIBs in the design. This command should only be used in MIB-based designs. You also receive this message when all the MIB instances in the design are disabled for planning or marked read-only using the hierarchical edit control command set_editability.

### What Next

Check for MIB instances in the design. Also, report the hierarchical edit-ability control set on the MIB blocks in the design. Mark the MIB blocks editable if you wish to report perform MIB alignment check on them.

### See Also

- report_mibs
- report_editability

Feedback

## MIB-002

(error) -layers option must be specified only with -wire_tracks option or -pg option.

**Description**

This option is to specify the layer on which wire tracks and/or pg straps will be checked for alignment. It is invalid if -wire_tracks or -pg option is not present.

**What Next**

Specify -wire_tracks or -pg option.

## MIB-003

(warning) There are no P/G wires that intersect MIB blocks and MIB alignment can not be determined.

**Description**

No P/G wires intersect MIB blocks. Alignment check can not be determined.

**What Next**

Create P/G wires in the design first before checking MIB alignment.

## MIB-004

(warning) No site row found in instance %s.

**Description**

No site rows are crossing instance %s. This could be due to the instance is not placed in the core area.

**What Next**

Check the instance and place it properly.

## MIB-005

(error) MIB instance %s's orientation %s is not valid.

**Description**

MIB blocks are not allowed to rotate by 90 or 270 degrees due to alignment issue.

Feedback

**What Next**

Check the MIB rotation.

---

## MIB-006

(error) No grid name.

**Description**

Please specify a non-empty name for the block grid that you want the tool to work on.

**What Next**

Run the command again with a valid grid name.

---

## MIB-007

(error) Invalid value %f for option %s.

**Description**

The options -x_step and -y_step require a positive value. The options -x_offset and -y_offset require a non-negative value. The specified value is not valid.

**What Next**

Correct the value and re-issue the command.

---

## MIB-008

(error) Invalid combination of options.

**Description**

The options -x_step/-y_step/-x_offset/-y_offset/-orientations are for user defined block grid.

The options -pg_strategy/-layers/-site_rows are for automatically derived block grid.

An block grid can either be defined by the user or derived automatically from PG strategies and/or site rows. But it cannot be both. The above two sets of options are mutually exclusive with each other.

**What Next**

Correct the options and re-issue the command.

Feedback

## MIB-009

(error) Options for %s grid is not allowed because block grid %s was previously set to be a %s grid.

**Description**

The options -x_step/-y_step/-x_offset/-y_offset/-orientations are for user defined (manual) block grid.

The options -pg_strategy/-layers/-site_rows are for automatically derived (auto) block grid.

An block grid can either be defined by the user or derived automatically from PG strategies and/or site rows. But it cannot be both.

You are trying to set contradicting options for the specified block grid. It is not allowed.

**What Next**

You must be calling set_grid without -reset option.

If you want the new options to overwrite any existing option on the grid that was previously set, please call the command with the -reset option.

If you want to create a new block grid with the specified options, please call create_grid command.

## MIB-010

(error) No block grid is specified.

**Description**

You must specify an existing block grid for set_grid command.

**What Next**

Check whether the block grid you want to change was created successfully.

## MIB-011

(error) Please specify at least one of or both -x_step and -y_step.

**Description**

To define a manual MIB grid, you need to provide the grid pitch in at least one direction.

**What Next**

Specify -x_step or -y_step or both with command set_grid.

Feedback

## MIB-012

(error) No valid orientation specified.

**Description**

Option -orientations specifies which orientations are allowed for MIB instances that might be associated with the MIB grid in the future.

This option is for user defined MIB grid. The valid orientation values are: R0 (or N), MX (or FS), MY (or FN), R180 (or S). If the option is not in the command line, all four orientations are allowed. If the option is specified, it must be followed with at least one valid orientation value.

**What Next**

Correct the command line and re-issue it.

## MIB-013

(error) Block grid %s already exists.

**Description**

You get this error message because create_grid is called to create a block grid with the specified name. However there is an existing block grid with the same name.

**What Next**

If you want to change the existing block grid, please use set_grid command. If you want to create a new block grid in addition to the existing one, please choose another name for the new block grid.

## MIB-014

(error) Failed to create block grid %s.

**Description**

You get this error message because create_grid is called to create block grid with the specified name, but the tool is unable to create it.

**What Next**

Please check if there are other error or warning messages that indicates the nature of the problem.

Feedback

## MIB-015

(error) Rows with different site definition are not supported.

**Description**

You get this error message because you asked the tool to derive block grid from site rows with different site definitions. This is not supported.

**What Next**

Please specify rows with the same site definition.

## MIB-016

(error) Rows with different rotations are not supported.

**Description**

You get this error message because you asked the tool to derive block grid from horizontal and vertical site rows. This is not supported.

**What Next**

Please specify all horizontal rows or all vertical rows.

## MIB-017

(warning) The new grid %s step size (%.3f) is too large.

**Description**

You get this message because block grid derived from the specified options has a step size that is too big. It would not be possible for applications in the tool, such as shape_blocks, to use this block grid.

**What Next**

Please review all the options you specified for this block grid and consider removing some of them.

## MIB-018

(error) Cell instance %s is not on block grid.

Feedback

### Description

You get this error message because the specified cell instance's snap-point location is not on its associated block grid.

### What Next

Please check the location of the cell. You can use snap_cells_to_block_grid to move the cell to be on grid.

---

## MIB-019

(error) The current orientation of cell instance %s is not allowed by its associated block grid.

### Description

You get this error message because the specified cell instance's orientation is not among the allowed orientations of its associated block grid.

### What Next

Please check the orientation of the cell. Either correct the cell's orientation, or change the block grid.

---

## MIB-020

(error) Please specify -all or the grid to remove.

### Description

You get this error message because remove_grids is called without any option.

### What Next

If you want to remove all block grids, please use the option -all. If you want to remove a specific block grid, please specify the grid.

---

## MIB-021

(error) Cannot specify '-adjust_snap_point_of_cell' with '-designs' or '-grid' or '-snap_point' or '-reset'.

### Description

You get this error message because *set_block_grid_references* is called with option *-adjust_snap_point_of_cell* and at least one other option which is mutually exclusive with *-adjust_snap_point_of_cell*.

Feedback

**What Next**

If you want to adjust the snap point of a cell based on the current location of the cell, please use the option *-adjust_snap_point_of_cell* alone.

If you want to use other options of the command, please call them without *-adjust_snap_point_of_cell*.

---

## MIB-022

(error) Cannot specify '-reset' with '-adjust_snap_point_of_cell' or '-grid' or '-snap_point'.

**Description**

You get this error message because *set_block_grid_references* is called with option *-reset* and at least one other option which is mutually exclusive with *-reset*.

**What Next**

If you want to reset the block grid information of a block reference, please use the option *-reset* alone.

If you want to use other options of the command, please call them without *-reset*.

---

## MIB-023

(error) Must specify '-designs' with '-reset'.

**Description**

You get this error message because *set_block_grid_references* is called with option *-reset* but option *-designs* is not specified.

**What Next**

If you want to reset the block grid information of a block reference, please use the option *-designs* to tell the tool which block reference you want to reset.

---

## MIB-024

(error) Must specify at least one of '-reset', '-grid' or '-snap_point' with '-designs'.

**Description**

You get this error message because *set_block_grid_references* is called with option *-designs* alone.

Feedback

**What Next**

If you want to set the block grid information of a block reference, please use options *-grid* and/or *-snap_point* together with *-designs*.

If you want to reset the block grid information of a block reference, please use option *-reset* together with *-designs*.

---

## MIB-025

(error) Must specify '-designs' with option '%s'.

**Description**

You get this error message because *set_block_grid_references* is called with option *-grid* or *-snap_point* but without option *-designs*.

**What Next**

If you want to set the block grid information of a block reference, please use options *-grid* and/or *-snap_point* together with *-designs*.

---

## MIB-026

(error) Cannot set snap point for block reference %s. The block has no associated block grid.

**Description**

You get this error message because you call *set_block_grid_references* to set snap-point on the specified block reference when the block is not associated with any block grid.

**What Next**

Please use options *-grid* to set up the association to a block grid first.

---

## MIB-027

(error) incorrect type %s. for the option -print.

**Description**

Only the following types are allowed: cell_rows, pg_straps, and wire_tracks.

**What Next**

Please use correct type to print out information in detail.

Feedback

## MIB-028

(warning) MIB instance %s does not contain PG straps for the specified layer.

**Description**

To perform the PG alignment checking, the checked MIB instance need to have PG straps with the specified layer.

**What Next**

Please add PG straps of the specified layer for alignment checking.

## MIB-029

(warning) Cell %s is not associated with any block grid.

**Description**

You get this message when you call *check_dp_design -pre_block_shaping*.

This could happen if the cell is not a block (aka, soft macro), or, if the cell's reference design has not been associated with any block grid.

A block grid can only be associated with a block (aka, soft macro). To associate a block with a block grid, use the command *set_block_grid_references*. Please see the man page of the command for details.

**What Next**

Check if the cell is a block. If it is a block, associate the block reference to the block grid you want.

If you want to snap a cell to an unassociated block grid, please specify the grid using option *-grid* with option *-cells* in command *snap_cells_to_block_grid*.

## MIB-030

(error) No block grid is specified.

**Description**

You must specify corresponding block grid if you have MIB in the design before block shaping.

**What Next**

Check whether the block grid has been set correctly in the design.

Feedback

## MIB-031

(warning) MIB instances that are part of the MIB group of reference-block %s do not contain PG straps.

**Description**

To perform the PG alignment checking, the checked MIB instance need to have PG straps.

**What Next**

Please add PG straps for alignment checking.

## MIB-032

(error) The specified %s in option %s is not correct.

**Description**

This option is to specify objects to be checked.

**What Next**

Specifies correct option value.

## MIB-201

(warning) Misalignment on P/G straps between MIB instances %s and %s.

**Description**

Power and/or ground straps in two MIB instances do not match. This will create errors in MIB block implementation.

**What Next**

Move or re-shape the MIB blocks with correct dimension and offsets, or update P/G mesh in top-level design.

## MIB-202

(warning) Misalignment at the site row between MIB instances %s and %s.

**Description**

Site rows in two MIB instances do not match. This will create errors in MIB block implementation.

Feedback

**What Next**

Move or re-shape the MIB blocks with correct dimension and offsets.

---

## MIB-203

(warning) Misalignment at track %s on layer %s between MIB instances %s and %s. Track offset of %s is %s, and track offset of %s is %s.

**Description**

Wire tracks %s on layer %s in two MIB instances do not match. This will create errors in MIB block implementation.

**What Next**

Move or re-shape the MIB blocks with correct dimension and offsets.

---

## MIB-204

(warning) Option -layer is ignored because -pg_strategy is not specified.

**Description**

Option -layer specifies which layer or layers in the specified PG strategy to consider when creating the grid. It is meaningless to specify -layer without -pg_strategy.

**What Next**

Check the command line, see if you really want to specify -layer but forgot to specify -pg_strategy, or if -layer was given mistakenly.

If -strategy is needed, please add -pg_strategy and re-issue the command. If -layer was given mistakenly and is not needed, no action is needed.

---

## MIB-205

(warning) Invalid orientation %s is ignored.

**Description**

Option -orientations specifies which orientations are allowed for MIB instances that might be associated with the MIB grid in the future.

For a MIB grid that is derived from PG strategy and/or site rows, the allowed orientations are derived automatically by the tool.

This option is for user defined MIB grid only. The valid orientation values are: R0 (or N), MX (or FS), MY (or FN), R180 (or S).

Feedback

**What Next**

If ignoring the said invalid orientation name is ok, no action is needed. If there is a typo and you want to correct it and set it again, please correct the command line and re-issue the command with -add option.

---

## MIB-206

(warning) Cannot find MIB grid %s.

**Description**

The tool is unable to find the MIB grid with the specified name in the current design.

**What Next**

Check if the grid was created successfully, or whether the grid name was spelled correctly.

---

## MIB-207

(warning) More than 2 numbers specified for -offset.

**Description**

The tool expects user to specify two numbers for the -offset option. One for X offset, and the other one for Y offset. If more than 2 numbers are specified, only the first two are taken, the rest are ignored.

**What Next**

No action required, unless the first 2 numbers are incorrect.

---

## MIB-208

(warning) Less than 2 numbers specified for -offset.

**Description**

The tool expects user to specify two numbers for the -offset option. One for X offset, and the other one for Y offset. If less than 2 numbers are specified, there is not enough information to set up the MIB module's offset to its MIB grid.

**What Next**

Please specify the correct offsets. If you want to remove the offset that was set previously, you can call set_grid command with the -grid option again, but without the -offset option.

Feedback

## MIB-209

(warning) Cell %s is not associated with any block grid.

**Description**

You get this message when you call *set_block_grid_references -adjust_snap_point_of_cell* or *snap_cells_to_block_grid -cells <cell_list>* with a cell that has no associated block grid.

This could happen if the cell is not a block (aka, soft macro), or, if the cell's reference design has not been associated with any block grid.

A block grid can only be associated with a block (aka, soft macro). To associate a block with a block grid, use the command *set_block_grid_references*. Please see the man page of the command for details.

**What Next**

Check if the cell is a block. If it is a block, associate the block reference to the block grid you want.

If you want to snap a cell to an unassociated block grid, please specify the grid using option *-grid* with option *-cells* in command *snap_cells_to_block_grid*.

## MIB-210

(warning) Attribute %s of block grid %s is not settable through set_attribute. Because the grid is auto-derived.

**Description**

You get this message when you call *set_attribute* to change the specified attribute of the specified block grid.

Since the block grid is auto-derived from PG strategy and/or site rows, its attributes are not modifiable.

**What Next**

If you want to change the block grid to use a certain pitches and offsets, please use *set_grid* to reset the grid first, then use either *set_attribute* or *set_grid* to set the desired X step, Y step, X offset and Y offset.

## MIB-211

(warning) Cell %s is not associated with block grid %s but is being snapped on to the grid.

Feedback

### Description

You get this message when you call *snap_cells_to_block_grid* to move a cell to a block grid that is different from the one that the cell is associated to.

### What Next

Check and make sure that this operation is intended.

---

## MIB-212

(warning) Cannot find an offset point to allow flipping around the %s axis. Limiting the allowed orientations.

### Description

You get this message when you call *create_grid* to create a block grid by combining PG grid with site row grid. Both the PG grid and site row grid allow the specified flipping. But the tool is unable to find a common offset point that the two grids can both flip around. So the allowed orientations for the combined grid is limited.

For horizontal site rows, the site row grid allows flipping around the Y axis with the grid points either on the edge of a site unit or in the middle of a site unit. If the rows are placed back-to-back flipped, the site row grid would allow flipping around the X axis with the grid points in the middle of two neighboring rows. If the rows are all placed with the same orientation, the flipping around the X axis is not allowed.

Similar rules apply to verical site rows.

### What Next

Check the steps and offsets of the PG grid and site rows reported by *create_grid*. If the report is correct and expected, no action is required. If the report is correct but not expected, you may want to adjust either the PG grid or site row positions to fix the problem. If the report is incorrect, please file a bug report.

---

## MIB-213

(warning) MIB instances %s and %s are not aligned on power/ground strap. Found misaligned horizontal %s strap on metal layer %s at bbox {{%.3f %.3f} {%.3f %.3f}} in instance %s.

### Description

The MIB instance has misaligned horizontal strap.

Feedback

**What Next**

Move or re-shape the MIB blocks with correct dimension and offsets, or update P/G mesh in top-level design.

---

## MIB-214

(warning) MIB instances %s and %s are not aligned on power/ground strap. Found misaligned vertical %s strap on metal layer %s at bbox {{%.3f %.3f} {%.3f %.3f}} in instance %s.

**Description**

The MIB instance has misaligned vertical strap.

**What Next**

Move or re-shape the MIB blocks with correct dimension and offsets, or update P/G mesh in top-level design.

---

## MIB-215

(warning) MIB instance %s does not have horizontal straps, but MIB instance %s contain horizontal PG straps.

**Description**

One MIB instance does not contain horizontal PG straps, but the other has horizontal PG straps.

---

## MIB-216

(warning) MIB instance %s does not have vertical straps, but MIB instance %s contain vertical PG straps.

**Description**

One MIB instance does not contain vertical PG straps, but the other has vertical PG straps.

---

## MIB-217

(warning) MIB instances %s and %s are not aligned on cell row. Found misaligned cell row %s at %s %.3f in instance %s.

Feedback

**Description**

Site rows in two MIB instances do not match. This will create errors in MIB block implementation.

**What Next**

Move or re-shape the MIB blocks with correct dimension and offsets.

---

## MIB-218

(warning) MIB instances %s and %s are not aligned on cell site. Found misaligned cell site in cell row %s at %s %.3f in instance %s.

**Description**

Cell site in two MIB instances do not match. This will create errors in MIB block implementation.

**What Next**

Move or re-shape the MIB blocks with correct dimension and offsets.

---

## MIB-219

(error) There is no cell site in cell row %s at %s %.3f in instance %s.

**Description**

There is no cell site in cell row in MIB instance. This will create errors in MIB block implementation.

**What Next**

Create cell row with cell site or not remove cell site in cell row.

---

## MIB-220

(warning) Track %s's direction is not same as prefered direction %s.

**Description**

Wire track do not follow the prefered direction.

**What Next**

Check the prefered direction.

Feedback

## MIB-221

(warning) MIB instances %s and %s are not aligned on wire track. There is track %s that mask is %s on layer %s at location (%s, %s) in %s direction to go through MIB instance %s's boundary edge %s, while there is no same mask track on layer %s at location (%s, %s) in %s direction to go through MIB instance %s's boundary edge %s.

**Description**

Wire tracks in two MIB instances do not match. This will create errors in MIB block implementation.

**What Next**

Move or re-shape the MIB blocks with correct dimension and offsets.

## MIB-301

(info) Reference design %s has no associated block grid.

**Description**

There is no block grid associated with the specified reference design.

**What Next**

Please check if set_block_grid_references was run successfully to associate the reference design with any block grid.

## MIB-302

(info) The PG patterns are not symmetric with respect to %s axis which restricts the grid's allowed orientations.

**Description**

This is an informational message.

**What Next**

If the situation is expected, nothing needs to be done. If you expect symmetric PG pattern, please check your PG pattern/strategy settings.

## MIB-303

(info) The site rows are not symmetric with respect to %s axis which restricts the grid's allowed orientations.

Feedback

**Description**

This is an informational message. When a neighboring row is flipped, it makes the site rows symmetric. When all the rows are of the same orientation, it is not symmetric.

**What Next**

If the situation is expected, nothing needs to be done. If you expect symmetric site rows, please check your site row settings.

# ML

## ML-001

(error) Too few clustered rDRC grids in label extraction; training will fail.

**Description**

estimate_routability found too few clustered rDRC grids in label extraction; training will fail.

**What Next**

Examine the drawing file in mlpao/train/tLabel.draw.<stage>.<design>.nr.csv Pick designs that show more than 5 clustered rDRC grids in the drawing file (this can be told by #lines with non-negative cluster indices in the last column).

# MLUTIL

## MLUTIL-001

(error) Failed to '%s'.

**Description**

You receive this error message when something is wrong in MLUTIL.

**What Next**

Fix your input and re-run.

## MLUTIL-002

(error) Failed to initialize IPC service %s

Feedback

**Description**

You receive this error message when IPC service cannot be initialized.

---

## MLUTIL-003

(error) Cannot create log directory in %s! Please check permission.

**Description**

You receive this error message when directory cannot be created.

**What Next**

Check working directory writing privilege

---

## MLUTIL-004

(error) Python file %s is not found!

**Description**

You receive this error message when Python file cannot be found.

**What Next**

Provide correct path to Python file.

---

## MLUTIL-005

(error) ML is not available in this platform!

**Description**

You receive this error message when your are running on a platform that does not have machine learning capability.

**What Next**

Run in a machine that has Python and machine learning packages.

---

## MLUTIL-006

(error) Please specify the python settings!

**Description**

You receive this message due to the lack of the python settings in mlutil module.

Feedback

**What Next**

Please Specify the python settings.

---

## MLUTIL-007

(warning) The path of shared library is missing.

**Description**

You should specify the shared libary path in product setup.

**What Next**

Please Specify the shared library path.

---

## MLUTIL-008

(error) The cmlp_root path setting is not correct. Please set it by environment variable SYNOPSYS_CMLP_INSTALL_ROOT.

**Description**

The CMLP root path is missing.

**What Next**

Please Specify the CMLP root path by environment variable SYNOPSYS_CMLP_INSTALL_ROOT.

---

## MLUTIL-009

(error) The remote host %s is not reachable!

**Description**

For RSH protocol, the correct host name is needed.

**What Next**

Please check if the remote host can be reached.

---

# MP

---

## MP-001

(info) Pulling level %d cell %s from (%d, %d) to (%d, %d) with orientation %s.

Feedback

**Description**

The message shows that magnet_placement is pulling current level cell towards the magnet object with new location and orientation info.

**What Next**

legalize_placement, check_legality

---

## MP-002

(warning) Tracing stops at %s %s since %s.

**Description**

The message shows that magnet_placement tracing stops at the cell or the net due to the given reason.

**What Next**

report_cell, report_net, report_attributes, get_attribute

---

## MP-003

magnet_placement terminates because %s.

**Description**

The message shows that magnet_placement stops due to the reported reason.

**What Next**

report_cell, report_net, report_attributes, get_attribute

---

# MSE

---

## MSE-001

(error) Hierarchical path name '%s' is not allowed in module scoped editing.

**Description**

While directly editing the module during module scoped editing, hierarchical path names like A/B or A/B/C are not allowed as inputs for cells, nets, ports, terminals. Only the direct hierarchy within the module is editable.

Feedback

### MSE-002

(error) Can't remove module '%s' which is currently being edited.

**Description**

During module scoped editing, the module being edited can't be removed using remove_modules within the editing command's scope.

## MSG

### MSG-1445

(error) Bad filter pattern: '%s'

**Description**

The specified pattern length is too small.

**What Next**

Please specify pattern whose length is more than 3.

### MSG-1446

(error) Maximum count of suppressed patterns (%d) exceeded for tag '%s'

**Description**

The maximum count of suppressed patterns has been reached for the specified tag.

**What Next**

Please remove some patterns before specifying more suppressed pattern.

### MSG-2446

(warning) Filter pattern '%s' does not exist for message tag '%s'

**Description**

The specified pattern does not match with the suppressed patterns for specified tag.

**What Next**

Please refer to print_suppressed_messages to check and specify the correct pattern to be unsuppressed for the specified tag.

Feedback

## MSG-3026

(information) #prnt #trgr #lmt Tag Level Format (or last printed message)

### Description

Prints the header for the message data printed by MSG-3032 message.

```
#prnt means number of times message printed.
#trgr means number of times message triggered for printing.
#lmt means message limit.
Tag means message tag.
Level means message level.
Format means message format.
```

### See Also

- MSG-3032

## MSG-3032

(information) %5d %5d %2d %s

### Description

Prints all data on messages which are triggered or printed from the start of the session. Message data contains number of times message printed, number of times message triggered for printing, message limit, message tag, message level, message format (or last printed message).

### See Also

- MSG-3026

## MSG-3121

(info) set_msg: changed print limit per command of %d message%s matching %s to %d

### Description

Prints the information of the new per command limit of the message(s).

## MSG-3296

(info) Using new severity_first format for messages. e.g. 'Warning: <tag> ... '

Feedback

**Description**

The messaging format : 'Warning: <tag> ... ' will be used to print the messages. Means Severity of the message will be printed first followed by message tag and then message contents.

Example: Message CMD-005 will be printed as below: Error: CMD-005 unknown command 'incorrect_command'

---

## MSG-3446

(info) Filtering out any '%s' tagged output containing pattern '%s'

**Description**

The specified tag's output messages containing specified pattern will be suppressed.

**What Next**

Please use unsuppress_message command if output message containing specified pattern needs to be printed.

---

## MSG-3913

(info) %d out of %d %s messages were not printed due to limit %d %s

**Description**

This message prints the summary of the suppressed messages at the end of every source or mega command. This message will also be printed when we change the limit of a message which was suppressed.

The file information will be printed if this message is triggered before the end of the source file. The message will be printed as below:

3 out of 5 NDMUI-669 messages were not printed due to limit 2 (after 'set_message_info' at test.tcl:21) (MSG-3913) 2 out of 4 ATR-012 messages were not printed due to limit 2 (after 'set_message_info' at test.tcl:22) (MSG-3913)

And if this message is triggered after the completion of the source file. It will be printed as below:

1 out of 3 ATR-012 messages were not printed due to limit 2 (MSG-3913) 1 out of 3 NDMUI-669 messages were not printed due to limit 2 (MSG-3913)

---

## MSG-3915

(info) %d out of %d %s messages were not printed due to per command limit %d %s

Feedback

**Description**

This message gets printed if the per command limit is hit and global limit doesn't.

This message prints the summary of the suppressed messages at the end of every source or mega command. This message will also be printed when we change the limit of a message which was suppressed.

The file information will be printed if this message is triggered before the end of the source file. The message will be printed as below:

3 out of 5 NDMUI-669 messages were not printed due to per command limit 2 (after 'set_msg' at test.tcl:21) (MSG-3915) 2 out of 4 ATR-012 messages were not printed due to per command limit 2 (after 'set_msg' at test.tcl:22) (MSG-3915)

And if this message is triggered after the completion of the source file. It will be printed as below:

1 out of 3 ATR-012 messages were not printed due to per command limit 2 (MSG-3915) 1 out of 3 NDMUI-669 messages were not printed due to per command limit 2 (MSG-3915)

### MSG-3995

(information) %s

**Description**

Prints the machine information of the current run.

## MUX

### MUX-6091

(info) Identified %lu crossbars in module '%s'. %lu out of %lu potentially congested crossbars amongst these are candidates for mux tree implementation.

**Description**

This message signifies that the tool was able to identify some crossbar structures in a given module. A crossbar is a set of SELECT_OPs that have a set of common data inputs. Crossbar structures typically lead to congestion in circuits and can be synthesized to mux trees in congested designs to alleviate such congestion with some trade-off with area.

Feedback

# MUXEO

## MUXEO-001

(warning) %s:%d: One-hot signals not matched to AND/OR.

### Description

This message explains why a one-hot pragma in your RTL is not converted into a tree of MUX standard cells. To be matched, each one-hot signal must be ANDed with a data signal, and the output of the ANDs must be ORed together. The following shows RTL that will match.

//synopsys $onehot(one_hot) assign out = (({64{one_hot[0]}} & data0) | ({64{one_hot[1]}} & data1) | ({64{one_hot[2]}} & data2) | ({64{one_hot[3]}} & data3));

### What Next

You can safely ignore this warning because the RTL will be correctly implemented with AND/OR standard cells. If you want it converted into a tree of MUX standard cells, then rewrite the RTL so that each one-hot signal is ANDed with a data signal, and the output of ANDs are ORed.

# MV

## MV-001

(Error) Top power domain is not created yet.

### Description

This error occurs doing commit_upf if a top scoped power domain has not been created in the UPF constraints. All MV designs required a top power domain (a domain including the top scope /).

### What Next

Use 'set_scope /' to set the current scope to the topmost scope, and create a power domain with create_power_domain at that scope. Alternately, use 'create_power_domain -scope /' to create a power domain at the top scope.

### See Also

- create_power_domain
- set_scope

Feedback

## MV-002

(Error) Primary %s of domain '%s' is not specified.

**Description**

Primary power/ground of power domain is not specified, but is required.

**What Next**

Specify the primary power/ground for all the power domains.

**See Also**

- create_power_domain
- set_domain_supply_net

## MV-003

(Error) %s of explicit supply set '%s' is not resolved to supply net.

**Description**

Power/ground/nwell/pwell of explicit supply set is not resolved to supply net.

**What Next**

Resolve power/ground/nwell/pwell of all explicit supply sets to supply nets. Check if the supply nets are available in all the expected power domains. If this error message is about a pwell supply net, make sure that you do not incorrectly have the mv.upf.enable_nwell_only_support app-option set. If you did intend to use mv.upf.enable_nwell_only_support, no pwell supplies should be referenced in the UPF.

**See Also**

- set_domain_supply_net
- create_supply_set
- connect_supply_net

## MV-004

(error) Design's multi-voltage data is not ready yet. Please run commit_upf and make sure it completes successfully.

Feedback

### Description

This error occurs when placement or optimization is invoked, before UPF data has been successfully processed by commit_upf. IC Compiler II requires that commit_upf is run after all UPF information have been loaded to signal completion and verification of UPF intent.

### What Next

Execute commit_upf and clean up any errors it reports before proceeding.

### See Also

- commit_upf

---

## MV-005

(warning) PG pin '%s' is unconnected.

### Description

This message occurs when the specified PG pin of mapped cell is unconnected. One possible cause is that the supply net connection cannot be determined in power intent.

### What Next

Invoke 'connect_pg_net" command to connect the power and ground pin. Also, check the supply net connection of the PG pin.

### See Also

- connect_pg_net
- report_cell
- MV-009

---

## MV-006

(error) The %s pin '%s' is connected to %s net '%s'.

### Description

This message occurs when the specified PG pin is connected to a net of conflicting power and ground net type. One possible cause is that the PG pin is connected to a supply net of conflicting power and ground type in power intent.

### What Next

Check the pin connection. Also check the supply net connection in power intent.

Feedback

**See Also**

- report_cell
- report_net
- MV-043

---

## MV-007

(error) PG net '%s' has no valid PG source(s) or driver(s).

**Description**

This message occurs when the specified PG net is not connected to any valid pg source(s) or driver(s) in the netlist. One possible cause is that the supply net does not have a valid supply source or driver in power intent and so the resulting PG netlist created by create_pg_net also does not have any valid PG source(s) or driver(s).

**What Next**

Check and make sure the power and ground drivers exist in the PG netlist. If PG net drivers already exist, please run the 'connect_pg_net' command to connect them. Also, check the supply net connection of the PG drivers to make sure they are connected to the correct supply net(s) in power intent.

**See Also**

- report_net
- connect_pg_net
- MV-011

---

## MV-008

(error) PG net '%s' is not defined in UPF as a supply net.

**Description**

This message occurs when the corresponding supply net is not defined in UPF for the power and ground net.

**What Next**

Check the supply nets defined in the correct domain and scope in the power intent.

Feedback

**See Also**

- report_supply_nets
- create_supply_net

---

## MV-009

(error) Supply net connection cannot be determined for PG pin '%s'.

**Description**

This message occurs when the tool cannot determine the supply net connection of the power and ground pin.

The power and ground connection of the PG pin is required to properly power up the cell. Without the supply net connection of the PG pin, the tool will not able to connect the proper PG net to the PG pin. Also, certain operations, including optimization, implementation of tie-off connections etc..., will be skipped for any signal pins with unknown related supply net connection.

**What Next**

Specify additional power intent using the connect_supply_net, set_isolation, set_retention, and create_power_switch commands.

**See Also**

- report_cell
- connect_supply_net
- set_isolation
- set_retention
- create_power_switch

---

## MV-010

(error) PG pin '%s' is connected to PG net '%s' instead of supply net '%s' as specified in power intent.

**Description**

This message occurs when the PG pin is connected to a PG net different from what is specified in power intent.

Feedback

### What Next

Check the supply net and PG net connections of the PG pin. Invoke resolve_pg_nets command to resolve the conflicts between power intent and PG netlist.

### See Also

- report_net
- report_power_domain
- resolve_pg_nets

---

## MV-011

(error) Supply net '%s' does not have any valid supply source(s) or driver(s) in power intent.

### Description

This message occurs when the specified supply net is not connected to any valid supply source(s) or driver(s). A valid supply source or driver can be a supply port or leaf PG pin that drives the supply net.

### What Next

Check the supply net connections in power intent.

### See Also

- connect_supply_net
- report_supply_nets

---

## MV-012

(warning) %s '%s' (supply net '%s' [%s]) cannot drive load pin '%s' (supply net '%s' [%s]) due to voltage difference.

### Description

This message occurs when the voltage difference of driver supply net and load supply net is larger than the voltage threshold. This can happen when the driver supply net and the load supply net have different voltage values in power state table or the port states definitions are incomplete.

Feedback

### What Next

- Check the voltage values of the two supply nets.
- Also, consider to change the voltage threshold if the voltage difference is within acceptable tolerance.
- Check *analyze_mv_design -level_shifter* to report issues that prevent Level Shifter insertion.
- Check *analyze_mv_feasibility* to know why Enabled Level Shifter cells cannot be use for insertion.
- Please check Level Shifter supply availability in Power Domains where cell is expected to be inserted.
- Check if there is a dont_touch present on the net requiring a Level Shifter.
- To debug Enabled Level Shifter issues, check *report_mv_path* for any Isolation or Level Shifter strategy that may apply on the path.

### See Also

- analyze_mv_design
- analyze_mv_feasibility
- report_mv_cells
- report_mv_lib_cells
- report_mv_path
- report_net
- report_pst
- report_supply_nets
- set_level_shifter

---

## MV-013

(warning) %s '%s' [supply net '%s', power state '%s'] is less always-on than the load pin '%s' [supply net '%s', power state '%s'].

### Description

This message occurs when the supply net of the specified driver pin or PG net is less always-on than the related supply net of the load pin and an Isolation Cell is missing. An

Feedback

Isolation Cell is required to isolate the signal path if the driver of the net is less always-on or unrelated to the load pin.

**What Next**

- Check the power states of the supply nets with *report_pst*.
- Use *report_mv_path* to report any Isolation strategy that may apply on the path.
- Check *analyze_mv_feasibility* for Isolation Cell and Enabled Level Shifter cell mapping related issues.
- Check *report_mv_lib_cells* for availability of a suitable Isolation Cells.
- Run *report_mv_path -net -isolation* to check if there is any strategy applied but not implemented. If there is no strategy applied, please add necessary Isolation strategy.

**See Also**

- report_net
- report_pst
- report_mv_path
- analyze_mv_feasibility
- report_mv_lib_cells

---

## MV-014

(error) Enable pin '%s' of isolation cell '%s' is not connected to any driver(s).

**Description**

This message occurs when the enable pin of the isolation cell is not connected to a net or the net has no driver(s).

**What Next**

Check the pin connection.

**See Also**

- report_cell
- report_net

Feedback

## MV-015

(warning) Found redundant level shifter '%s'%s%s.

### Description

This message occurs when the level shifter exists but it is not required. This can happen when the driver(s) and the load(s) of the level shifter have no voltage difference or the input supply net and output supply net of a dual-rail level shifter have no voltage difference. A level shifter is also not required when it is connected to ground net or tie-low cell. Presence of level shifter cells triggers design constraining, and this may result in QOR loss in the final netlist.

### What Next

Check the supply net connections and their voltage values.

### See Also

- get_related_supply_nets
- report_pst
- report_cell
- report_net

## MV-016

(error) Save and restore pin '%s' of retention cell '%s' is not connected to any driver(s).

### Description

This error occurs when the save and restore pin of the specified retention cell is not connected to a net or a net that has a driver(s).

### What Next

Check the pin connection for a driver manually using the report_nets -connections <net_name> command or try using the all_fanin -flat -to <net_name> command to help locate the source of the subcircuit.

### See Also

- report_cells
- report_nets
- all_fanin

Feedback

## MV-019

(error) Power domain '%s' does not have any primary voltage area.

### Description

This message occurs when primary voltage area of the specified power domain is not defined. The power and ground supplies of primary voltage area of a power domain must match the primary power and ground supplies of the domain.

### What Next

Check the voltage area(s) of the power domain and their power and ground supplies.

### See Also

- report_power_domain
- report_voltage_areas

## MV-020

(error) The backup supply net '%s' of %s cell '%s' is not more always-on than the primary supply net '%s' of the cell.

### Description

This message occurs when the supply net of backup PG pin of the isolation cell or retention cell is not more always-on than the supply net of the primary PG pin. For isolation and state retention, the backup supply net should be more always-on than the primary supply net.

### What Next

Check the supply net connections of the PG pins. Also check the power states of the two supply nets.

### See Also

- report_cell

## MV-021

(information) Total %d %s cell(s) in the design.

Feedback

### Description

This message is printed by commit_upf and associate_mv_cells, and shows the total number of different categories of power management cells, like isolation or level-shifters or repeater, that have been detected in the design.

### What Next

This is an information-only message. No action is required.

### See Also

- commit_upf
- associate_mv_cells

---

## MV-022

(warning) %s of supply port '%s' is unconnected in power intent.

### Description

This message occurs when the inside or outside connection of the supply port is open in power intent. Unless the supply port is not intended to be implemented, a supply port should be properly connected to supply net(s).

### What Next

Check the supply net connections of the supply port in power intent.

### See Also

- connect_supply_net
- report_supply_ports

---

## MV-023

(warning) Found redundant isolation cell '%s'%s.

### Description

This message is printed when an isolation cell exists but it is not required. This can happen when the isolation cell is connected to its driver(s) and relatively power down fanout(s). Presence of isolation cells triggers design constraining, and this may result in QOR loss in the final netlist.

Feedback

### What Next

Please check always-on relationship between the driver(s) and fanout(s) of the isolation cell.

### See Also

- report_net
- report_pst

---

## MV-024

(error) The control pin '%s' of switch cell '%s' is not connected to any driver(s).

### Description

This message occurs when the control pin of the specified switch cell is not connected to a net or the net has no driver(s).

### What Next

Check the pin connection.

### See Also

- report_cell
- report_net

---

## MV-025

(error) The output supply net '%s' of the switch cell '%s' is same as its input supply net.

### Description

This message occurs when the input and output supply net of the specified switch cell is identical. The output supply net of a switch cell should be a switchable supply, and it should be less always-on than the input supply net.

### What Next

Check the supply net connections of the switch cell.

### See Also

- report_cell

Feedback

## MV-026

(error) The output supply net '%s' of the switch cell '%s' is not less always-on than its input supply net '%s'.

### Description

This message occurs when the output supply net of the specified switch cell is not less always-on than its input supply net. The output supply net of a switch cell should be a switchable supply, and it should be less always-on than the input supply net.

### What Next

Check the supply net connections of the switch cell.

### See Also

- report_cell
- report_pst

## MV-027

(warning) The tie-off connection '%s' has not been implemented.

### Description

This message occurs when 1'b1 or 1'b0 has not been implemented as a tie cell or PG net. Constant nets that are floating or connected to unmapped pins only are excluded.

### What Next

Ensure that related supply net of constant pins can be determined. Use connect_pg_net to connect tie-off pins in either automatic or manual mode.

### See Also

- report_net
- connect_pg_net

## MV-028

(information) Connecting source '%s' to '%d' cells in '%s' mode.

Feedback

### Description

An informational message from connect_power_switch, indicating how many connections have been made from the source of the power switch enable signal to the power switches. It also indicates which style of connection was used (HFN, daisy-chain, clustering or fishbone).

### What Next

Optionally check that the expected number of power switch connections have been made by connect_power_switch.

### See Also

- connect_power_switch
- create_power_switch
- create_power_switch_array
- create_power_switch_ring

---

## MV-029

(Error) The argument '%s' of option -source is not a valid '%s' pin.

### Description

The pin or port specified as the source of this switch network is not of correct type or direction. For example, an input pin of a leaf-level cell cannot be the source for the control signal of the power switch network.In case of clustering mode, the command errors out for hierarchical pins with no location, input pins and output ports.

### What Next

Please re-issue the command with a valid pin or port as source.

### See Also

- connect_power_switch

---

## MV-030

(warning) Skipping cell %s, %s.

Feedback

### Description

This warning message occurs if a given cell:

0

is not a switch cell.

0

is same as the cell of the source pin.

0

has multiple sleep/ack pins but none defined using -lib_pin.

0

has no valid sleep/ack pins.

0

has sleep/ack pins already connected to some other nets.

### What Next

Please check the cell for the above conditions and re-issue the command.

### See Also

- connect_power_switch

---

## MV-031

(error) Error in argument '%s'. '%s'.

### Description

This is an error message indicating syntax errors using the connect_power_switch command. For example, it may be issued if an enumerated option like -mode or -direction is giving an illegal value. It may also be issued if certain options are used in combination that are disallowed together. It is also issued if the same pin is given for the -source and -ack_out options, as well as many other situations.

### What Next

Review the specific information given by the MV-031 error message and refer to the connect_power_switch man page for complete information about using the command.

### See Also

- connect_power_switch

Feedback

## MV-032

(error) None of the cells in the object_list have a valid switch pin of coarse-grain power switch cell.

### Description

This error occurs when connect_power_switch command is not able to find a valid switch pin of coarse-grain power switch cells to connect to.

### What Next

Check the -object_list. The list should only contain a valid list of coarse-grain power switch cells with sleep pins and (or) ack pins.

The sleep pin (or enable pin) of a coarse-grain switch cell will have a switch_pin attribute in liberty (.lib file).

### See Also

- connect_power_switch

## MV-033

(error) Primary PG pin '%s' and backup pin '%s' of the cell '%s' are connected to the same PG net '%s'.

### Description

This message occurs when the primary PG and backup PG pins of the specified cell are connected to the same power and ground net. One possible cause is that the PG pins are connected to the same supply net in power intent.

### What Next

Check the net connection of the PG pins. Also check the supply net connection of the pins in power intent.

### See Also

- report_cell
- MV-101

## MV-034

(error) Cannot find power switch %s.

Feedback

### Description

This message occurs when either the create_power_switch_array or create_power_switch_ring commands are used and the tool cannot find the specified power switch strategy. Power switch strategy is created by UPF command create_power_switch.

### What Next

Check that the create power switch strategy defined in the UPF matches that being used in the create_power_switch command.

### See Also

- create_power_switch
- create_power_switch_array
- create_power_switch_ring

---

## MV-035

(error) Must specify -power_switch.

### Description

This message occurs when the option -power_switch is not specified for create_power_switch_array or create_power_switch_ring. -power_switch is required to insert and associate switch cells.

### What Next

Specify the power switch strategy.

### See Also

- create_power_switch_array
- create_power_switch_ring

---

## MV-036

(error) Cannot specify negative pitch value.

### Description

The pitch value for power switch array must be non-negative as it specifies the distance between centers of power switches or power switch patterns. When they are zero, it indicates one row/column.

Feedback

### What Next

Specify positive pitch value.

### See Also

- create_power_switch_array

---

## MV-037

(error) Cannot specify more than one voltage area.

### Description

User can only specify one voltage area at a time when creating a power switch array or ring.

### What Next

Specify only one voltage area as input.

### See Also

- create_power_switch_array
- create_power_switch_ring

---

## MV-038

(error) Cannot find any lib cell mapped to power switch %s

### Description

This error happens when tool cannot find the corresponding physical lib cells that are mapped to the specified power switch. Usually, one power switch should be mapped to at least (can be more) than one lib cells in the design by the *map_power_switch* command.

### What Next

Check your UPF setting.

### See Also

- create_power_switch_array
- create_power_switch_ring
- map_power_switch

Feedback

## MV-039

(error) Cannot specify more than one lib cell

### Description

This error happens when user specifies more than one lib cell to the -lib_cell option of *create_power_switch_ring* or *create_power_switch_array* command. This option can only take one lib cell at a time.

### What Next

Specify only one lib cell to the -lib_cell option of the command.

### See Also

- create_power_switch_ring
- create_power_switch_array

## MV-040

(error) Voltage area %s does not define power switch strategy %s.

### Description

This warning message occurs in the *create_power_switch_ring* or *create_power_switch_array* command when the power domain of user-specified voltage does not define user-specified power switch strategy.

### What Next

Check your command option and UPF file to ensure the power switch strategy defined in the UPF matches that being passed to the create_power_switch_array or create_power_switch_ring command that is being used.

### See Also

- create_power_switch
- create_power_switch_array
- create_power_switch_ring

## MV-041

(Error) Failed to get load pin/ports.

Feedback

### Description

You need to specify at least one load pin or port.

### What Next

Specify at least on load pin or port with the -loads option.

---

## MV-042

(warning) Unable to derive power and ground type for supply net %s.

### Description

This warning message occurs when the tool cannot determine if the specified supply net and its connected supply_nets through supply ports or power switches are used as power supply nets or ground supply nets.

A supply net and its connected supply nets should be connected to power domains or domain elements and used for one of the following connections:

- `Primary power or ground net of power domain(s)`
- `Explicit connections on power or ground pins of cell(s)`
- `Related supply nets of power or ground pins of cell(s)`
- `Isolation power or ground supply`
- `Retention power or ground supply`
- `Power or ground function of a supply set`

### What Next

Check the UPF constraints and make sure the supply net or its connected supply nets are connected to a domain or domain elements.

### See Also

- set_domain_supply_net
- connect_supply_net
- set_related_supply_net
- set_isolation
- set_retention

Feedback

- create_supply_set
- create_power_switch

---

## MV-043

(error) Supply net %s and its connected supply net(s) are used as both power and ground supply net.

### Description

This error occurs when a supply net and its connected supply_nets through supply ports or power switches are used as power supply net for some power domain(s) or domain element(s), but also as ground supply net for some other power domain(s) or domain element(s).

### What Next

Check all UPF connections and constraints using this supply net and its connected supply nets, and make sure all of them are used only as power supply net or ground supply net.

### See Also

- set_domain_supply_net
- connect_supply_net
- set_related_supply_net
- set_isolation
- set_retention
- create_supply_set
- create_power_switch

---

## MV-044

(Error) Missing required options for check_bufferability.

### Description

check_bufferability can be called in two ways: using both the -net and -voltage_area options together, or alternately using all three of the -driver, -loads, and -voltage_area options together.

Optionally, -verbose can be used with either combination to get additional detailed information about the bufferability.

Feedback

### What Next

Specify all required options to avoid this error.

### See Also

- check_bufferability

---

## MV-045

(error) The input supply net '%s' of the level shifter '%s' is less always-on than its output supply net '%s'.

### Description

This message occurs when the input supply net of the level shifter is less always-on than its output supply net.

### What Next

Check the power state table for the referenced supply nets. You can use the -supplies option with report_pst command to focus only on the referenced supply nets. If the supplies are correct for the net crossing the power domain, than an isolation cell should precede the level shifter to protect the more always-on logic. The commands create_mv_cells -isolation can be used to implement isolation cells based on define isolation strategies in the UPF.

### See Also

- report_cells
- report_pst

---

## MV-046

(error) Level shifter '%s'%s of level shifter type '%s' is incompatible with the voltage conversion from %s to %s.

### Description

This message occurs when the type of level shifter library cell is incompatible with the voltage conversion performed by the specified level shifter.

### What Next

Check the supply nets of the specified level shifter. Also check the level shifter type of the library cell.

Feedback

### See Also

- report_cell

---

## MV-047

(warning) %s voltage %s of %s '%s'%s does not satisfy the voltage range %s defined in the library.

### Description

This message occurs when the level shifter library cell or macro input pin with level shifter inside cannot satisfy the voltage shifting range.

### What Next

Check the supply nets of the specified level shifter. Also, check the voltage range of the library cell.

### See Also

- report_cell

---

## MV-048

(error) Neither driver nor load supply can be determined. Failed to analyze bufferability.

### Description

Errors have occurs during check_bufferability, as supply of driver and/or loads cannot be determined.

### What Next

Please make sure that you can get related supply information from both driver and loads. Ensure you specified a valid leaf signal driver pin, signal driving port, set of leaf signal load pins or set of leaf signal load ports.

### See Also

- all_fanin
- all_fanout
- all_connected

Feedback

## MV-049

(Error) The argument '%s' of option -ack_out is not a valid '%s' pin.

### Description

The pin or port specified as the acknowledge out of this switch network is not of correct type or direction. For example, an output pin of a leaf-level cell may have been specified, which cannot be connected to the output ack_out pin of a power switch cell.

### What Next

Please re-issue the command with valid pin or port connected to ack_out.

### See Also

- connect_power_switch

## MV-050

(error) %s related supply net(s) of signal pin '%s' cannot be determined.

### Description

This message occurs when the tool cannot determine the related supply net(s) of the specified signal pin.

The related supply nets of a signal pin are derived from the supply net connections of its related PG pins. If the related PG pins are not properly defined for the signal pin or their supply net connections in power intent cannot be determined, then the tool will not be able to determine the related supply net connections for the signal pin.

### What Next

Check the related power and ground pins of the specified signal pin. Also check the supply net connections of its related PG pins.

### See Also

- report_cell
- set_domain_supply_net
- connect_supply_net
- MV-009
- MV-062

Feedback

## MV-051

(warning) '%s' is already connected to net '%s'. It might be disconnected.

### Description

This message is printed when a port or pin specified for sleep or ack out is already connected to a net. In this case, the tool might disconnect the original net and reconnect the port or the pin to the new net.

### What Next

Typically this message occurs in relation to ack ports which are usually hierarchical. When connecting these ack ports/nets they can get disconnected or reused in some cases. Please check the connections of the specified port or pin in this case.

### See Also

- connect_power_switch

## MV-052

(error) -net cannot be specified together with -driver or -loads option.

### Description

This error occurred because you have specified incompatible options with the command check_bufferability.

### What Next

Please refer to the manpage of check_bufferability command for usage information and examples.

## MV-053

(warning) Retention cell association skipped the element '%s' of strategy '%s' in power domain '%s' because %s.

### Description

This message indicates that the element of retention strategy was skipped by retention cell association.

### What Next

Check the retention strategy.

Feedback

**See Also**

- associate_mv_cells
- set_retention
- set_retention_control
- map_retention_cell

---

## MV-054a

(information) Total %d level shifter cells have been %s incrementally.

**Description**

This message shows the number of level shifter cells that have been inserted or removed incrementally during optimization flow.

**What Next**

This is an information-only message. No action is required.

---

## MV-054

(information) Total %d %s cells have been inserted.

**Description**

This message shows the number of each type of mv cells (eg, level shifters, isolation cells, repeater cells, etc) that have been inserted with the defined strategies.

**What Next**

This is an information-only message. No action is required. Ensure the expected number of each type was inserted. Otherwise, the strategies might have to be adjusted in the UPF.

**See Also**

- create_mv_cells

---

## MV-055

(information) Total %d %s%s%s been converted from %s-rail to %s-rail.

**Description**

This message shows the number of buffers or inverters that have been converted from single-rail to dual-rail, and vice-versa.

Feedback

### What Next

This is an information-only message. No action is required.

### See Also

- create_mv_cells

---

## MV-057

(information) Total %d enable level shifter cells have been inserted to replace isolation cells.

### Description

This message shows the number of enabled level shifters that have been inserted during Level Shifter insertion, with the same number of isolation cells being replaced by these enabled level shifters.

### What Next

This is an information-only message. No action is required.

### See Also

- create_mv_cells

---

## MV-058a

(error) There are some instantiated single rail cells inside an AO Block Domain without any connect_supply_net, and the primary PG pin connection of the cells is different from the domain main rail supply.

### Description

This message occurs when a single rail cell without explicit connect_supply_net is in the AO Block Domain. If a user expects to connect the primary PG pin with the main rail of the AO Block Domain, the user needs to specify connect_supply_net command explicitly. If a user expects to connect the primary PG pin with the primary supply of the AO Block Domain, the user needs to re-map the single rail cell to the dual rail cell.

### What Next

Check the supply net of cell's primary power and ground pin. Also check the main rail and primary of the power domain. If a user expects to connect the primary PG pin with the primary supply of the AO Block Domain, use "create_mv_cells -map_ao_block" to fix.

Feedback

### See Also

- report_cell
- report_power_domains
- connect_supply_net
- create_mv_cells

---

## MV-058b

(error) Primary PG pin '%s' is implicitly connected to supply net '%s' that is different from the main rail supply net '%s' of the AO Block Domain.

### Description

This message occurs when a single rail cell without explicit connect_supply_net is in the AO Block Domain. If a user expects to connect the primary PG pin with the main rail of the AO Block Domain, the user needs to specify connect_supply_net command explicitly. If a user expects to connect the primary PG pin with the primary supply of the AO Block Domain, the user needs to re-map the single rail cell to the dual rail cell.

### What Next

Check the supply net of cell's primary power and ground pin. Also check the main rail and primary of the power domain. If a user expects to connect the primary PG pin with the primary supply of the AO Block Domain, use "create_mv_cells -map_ao_block" to fix.

### See Also

- report_cell
- report_power_domains
- connect_supply_net
- create_mv_cells

---

## MV-058

(warning) Primary PG pin '%s' is connected to supply net '%s' that is different from the primary supply net '%s' of the voltage area.

### Description

This message occurs when the specified primary power and ground pin is connected to a supply net that is different from the primary supply net of the voltage area.

Feedback

### What Next

Check the supply net of primary power and ground pin. Also check the voltage area of the cell.

### See Also

- report_cell
- report_voltage_areas

---

## MV-059

(warning) Isolation cell '%s'%s has been inserted and associated with isolation strategy '%s' but it is not required.

### Description

This message is printed when an Isolation Cell exists due to the specified isolation strategy, but it is not required. This can happen when the Isolation Cell is connected to its driver(s) and relatively power down fanout(s). Presence of isolation cells triggers design constraining, and this may result in QOR loss in the final netlist.

### What Next

- Please check always-on relationship between the driver(s) and fanout(s) of the Isolation Cells.
- Check the isolation strategy associated with the cell.
- Please run *report_mv_path* to report strategies that apply to the path.
- As Tool inserts Isolation Cells according to the defined strategy, modify strategy if an Isolation Cell is not required for certain paths.

### See Also

- report_power_domain
- report_net
- report_pst
- report_mv_path
- report_mv_cells

Feedback

## MV-060

(warning) %s '%s' [supply net '%s', power state '%s'] is less always-on than the load pin '%s' [supply net '%s', power state '%s'] while "-no_isolation" of isolation strategy is specified.

### Description

This message occurs when the supply net of the specified driver pin or PG net is less always-on than the related supply net of the load pin and an isolation cell is missing because "-no_isolation" option of isolation strategy is specified. An isolation cell is required to isolate the signal path if the driver of the net is less always on or unrelated to the load pin.

### What Next

Check the power states of the supply nets. Also, check the isolation strategy.

### See Also

- report_power_domain
- report_net
- report_pst

## MV-062

(warning) Supply net connection cannot be determined for %s PG pin '%s'.

### Description

This message occurs when the tool cannot determine the supply net connection of the power and ground pin.

The power and ground connection of the PG pin may not be required to properly power up the cell. However, if the specified PG pin is the related power or ground pin of any signal pins of the cell, the tool will not be able to determine the related supply net of those signal pins. As a result, certain operations, including optimization, implementation of tie-off connections etc..., will be skipped for those signal pins.

### What Next

Specify additional power intent using the connect_supply_net, set_isolation, set_retention, and create_power_switch commands.

Feedback

### See Also

- report_cell
- connect_supply_net
- set_isolation
- set_retention
- create_power_switch

---

## MV-063

(warning) Diode cell '%s' (%s type) is connected to supply net '%s' that is less always-on than the related supply net '%s' of the driver pin '%s'.

### Description

This message is printed when the diode cell is connected to the supply net that is less always-on than the related supply net of the driver pin. It is possible that a sneaky leakage path through the diode cell may exist.

### What Next

Check the connections of the drivers, loads and the diode cell(s). Also check the power an d ground connections of the cells.

### See Also

- report_net

---

## MV-064

(warning) Diode cell '%s' (%s type) is connected to supply net '%s' [%s] which has lower voltage value than the related supply net '%s' [%s] of the driver pin '%s'.

### Description

This message is printed when the diode cell is connected to the supply net whose voltage value is lower than the voltage of the related supply net of the driver pin. It is possible that a sneaky leakage path through the diode cell may exist.

### What Next

Check the connections of the drivers, loads and the diode cell(s). Also check the power an d ground connections of the cells.

Feedback

### See Also

- report_net

---

## MV-065

(information) Total %d supply connections have been inferred from PG netlist.

### Description

This message shows the number of supply connections that have been inferred from PG netlist.

### What Next

This is an information-only message. No action is required.

### See Also

- commit_upf

---

## MV-066

(error) The propagated clamp value of isolation cell '%s'%s to power domain boundary pin '%s' is '%s' but it differs from the clamp value '%s' specified in the isolation strategy '%s' of power domain '%s' at that power domain boundary pin.

### Description

This message is printed when the propagated clamp value at the power domain boundary pin is different from the clamp value specified in the isolation strategy applied to the specified power domain boundary pin.

When isolation is enabled, the clamp value of the isolation cell is propagated to the power domain boundary through any buffers and/or inverters; the propagated logic value from the isolation cell to the power domain boundary pin should match the clamp value specified in the isolation strategy.

### What Next

Please check the clamp value of both the isolation cell and the isolation strategy. Also, please check the connections of the isolation cell to the power domain boundary pin and make sure the logic value is not inverted unintentionally.

Feedback

### See Also

- report_power_domain
- report_cell
- report_net

---

## MV-067

(warning) The clamp value '%s' of isolation cell '%s' is different from the logic constant '%s' of its input pin '%s'.

### Description

This message is printed when the clamp value of the isolation cell or enable level shifter is different from the logic constant connected to the data input pin of the cell. As a result, the output pin will have different logic values depending on whether isolation is enabled or not.

### What Next

Please check it is intentional that the clamp value is different from the logic constant when isolation is disabled. Also, please check the clamp value of the isolation cell and the connection of its data input pin.

### See Also

- report_cell
- report_net

---

## MV-068

(warning) There are no isolation cells implemented and associated with isolation strategy '%s' in power domain '%s'.

### Description

This message is printed when isolation cells exist in the design but there are no isolation cells implemented and associated with the specified isolation strategy.

### What Next

Check the isolation strategy and the association of isolation cells.

Feedback

### See Also

- associate_mv_cells
- report_power_domains
- report_mv_path
- report_cell

---

## MV-069

(warning) There are no retention cells implemented and associated with retention strategy '%s' in power domain '%s'.

### Description

This message is printed when retention cells exist in the design but there are no retention cells implemented and associated with the specified retention strategy.

### What Next

Check the retention strategy and the association of retention cells.

### See Also

- associate_mv_cells
- report_power_domains
- report_cell

---

## MV-070

(warning) There are no power switch cells implemented and associated with power switch strategy '%s' in power domain '%s'.

### Description

This message is printed when power switch cells exist in the design but there are no power switch cells implemented and associated with the specified power switch strategy.

### What Next

Check the power switch strategy and the association of power switch cells.

Feedback

**See Also**

- associate_mv_cells
- report_power_domains
- report_cell

---

## MV-071

(warning) The isolation cell '%s' is not associated with any isolation strategies.

**Description**

This message is printed when the specified Isolation cell is not associated with any Isolation strategies.

The tool cannot associate Isolation Cells with Isolation strategies when there is any discrepancy between them. In such case, supply net connections of the Isolation Cells cannot be automatically derived by the tool. While it is sufficient to continue the implementation flow if supply net connections are explicitly specified, Tool will not be able to check and ensure that the Isolation Cells are properly inserted and consistent with the Isolation strategies intended for the cells.

**What Next**

- Run *analyze_mv_feasibility -isolation -format html* to report detailed reasons for failed association.
- Check the related Isolation strategy and the association of the Isolation Cell.
- Check *report_mv_path* to report supply availability, inputs, outputs, drivers, sinks, associated strategies and corresponding power domain for MV cells.

**See Also**

- analyze_mv_feasibility
- associate_mv_cells
- report_cell
- report_mv_path
- report_power_domains

---

## MV-072

(information) The level shifter '%s' is not associated with any level shifter strategies.

Feedback

### Description

This message is printed when the specified level shifter is not associated with any level shifter strategies.

When the tool cannot associate level shifter cells with level shifter strategies, it might be caused by one of these reasons: level shifter cell is not inserted based on any strategy, it might purely be inserted to fix the voltage shifting automatically; or the level shifter is inserted base on certain strategy, but the nestlist got changed without introducing any extra violation. In any case, existing power/ground supply nets of the cell are maintained. No association will pose no problem in downstream tools.

### What Next

This is an information-only message. No action is required.

### See Also

- associate_mv_cells
- report_power_domains
- report_mv_path
- report_cell

---

## MV-073

(warning) The retention cell '%s' is not associated with any retention strategies.

### Description

This message is printed when the specified retention cell is not associated with any retention strategies.

The tool cannot associate retention cells with retention strategies if there are any discrepancies between them. In such a case, supply net connections of the retention cells cannot be automatically derived by the tool. While it is sufficient to continue the implementation flow if supply net connections are explicitly specified, the tool will not be able to check and ensure that the retention cells are properly inserted and consistent with the retention strategies intended for the cells.

### What Next

Check the related retention strategy and the association of the retention cell.

Feedback

**See Also**

- associate_mv_cells
- report_power_domains
- report_cell

## MV-074

(warning) The power switch cell '%s' is not associated with any power switch strategies.

**Description**

This message is printed when the specified power switch cell is not associated with any power switch strategies.

The tool cannot associate power switch cells with power switch strategies if there are any discrepancies between them. In such a case, supply net connections of the power switch cells cannot be automatically derived by the tool. While it is sufficient to continue the implementation flow if supply net connections are explicitly specified, the tool will not be able to check and ensure that the power switch cells are properly inserted and consistent with the power switch strategies intended for the cells.

**What Next**

Check the related power switch strategy and the association of the power switch cell.

**See Also**

- associate_mv_cells
- report_power_domains
- report_cell

## MV-075

(information) Total %d sequential cells have been converted to %sretention cells.

**Description**

This information message indicates that the reported number of flip-flops and/or latches which are targets of retention strategy (retention candidates), have been converted to either generic or technology specific retention cells. When the candidate cells are generic sequential elements, they will be converted generic retention elements. If the candidate cells are already from technology library, they will converted to respective retention equivalent cell in the technology library.

Feedback

### What Next

This is just an information message. No action is required.

### See Also

- create_mv_cells
- set_retention
- set_retention_control
- map_retention_cell

---

## MV-076

(warning) Sequential cell %s %s converted to retention cell.

### Description

This warning message indicates that the technology specific sequential cell cannot be converted to its retention counterpart. This can happen during *create_mv_cells* when *-all* option is used.

### What Next

Invoke sequential mapping step/command to get the sequential cell converted to its retention counterpart.

### See Also

- create_mv_cells

---

## MV-077

(warning) Supply net '%s' of PG pin '%s' is more always-on than the supply net '%s' of its related bias pin '%s'.

### Description

This message occurs when the supply net of a power and ground pin is more always-on than the supply net of its related bias pin.

### What Next

Check the supply net connections of the PG and bias pins. Also check the power states of the supply nets.

Feedback

### See Also

- report_cell
- report_pst

---

## MV-078

(warning) Supply net '%s' of bias pin '%s' is different from the default %s supply net '%s' of its voltage area.

### Description

This message occurs when the bias pins in a voltage area do not share the same bias supply net.

### What Next

If this message is reported for cells in an AO Block Synthesis Domain, it may be ignored. Otherwise, check the supply net connections of the bias pin(s).

### See Also

- report_cell

---

## MV-079

(information) Message '%s' limit (%d) exceeded. Remainder will be suppressed. The limit can be changed with '-max_message_count' option.

### Description

This message indicates that the message limit has been exceeded and the remainder will be suppressed.

### What Next

Run check_mv_design with different value for the -max_message_count option.

### See Also

- check_mv_design

---

## MV-080a

(information) Total %d %s violations.

Feedback

### Description

This message, printed by MV design critical check, gives the total number of a specific type of MV violation.

### What Next

No further action is required.

### See Also

- MV-009

---

## MV-080

(information) Total %d %s violations.

### NAME

*MV-080* (warning) Total %d %s violations.

### Description

This message, printed by check_mv_design, gives the total number of a specific type of MV violation. All violations may not be printed in the report. By default, only 20 of each type of violation are printed. This number can be controlled by the check_mv_design -max_message_count option. To see details of every violation, use check_mv_design -max_message_count all.

### What Next

No further action is required. Optionally, increase the number of printed violations to see more details by using check_mv_design -max_message_count.

### See Also

- check_mv_design

---

## MV-081

(information) Total %d out of %d %s cells have been associated with %s strategy(s).

### Description

This message shows the number of mv cells out of total same type cells in design that have been associated with the strategy. Association is the process of pairing UPF power management strategies (i.e. level shifting, isolation, repeater, etc.) with existing power management cells in the design.

Feedback

### What Next

This is an information-only message. No action is required if the number of associated cells is expected. If some cells are not associated, it is an indication that some part of the defined strategy is not matching with the instantiated power management cell. For example, if the related supplies of a level shifter input and output pin are electrically equivalent, the level shifter will not get associated because voltage differential is needed. The related supplies of a pin can be checked with the get_related_supply_nets command. In order for any cell to get automatically associated, all of the defined UPF constrains associated with that strategy need to match with the implementation of the power management cell. Any discrepancy can lead to an unassociated cell. The following is a partial list of checks performed during automatic association of power management cells:

Level shifters -map cells against instance lib cell -physical location vs. strategy location -related supplies of load and driver -voltage differential against threshold and level shifter rules -included and excluded element list associated with strategy

Isolation cells -map cells vs. instance lib cell -physical location vs. strategy location -related supplies of load and driver and enable pin vs. strategy -connection to the control signal -included and excluded element list associated with strategy

Retention registers -map cells vs. instance lib cell -location vs. power domain -connection to the save and restore signals -associated supply sets -included and excluded element list associated with strategy

Power switches -map cells vs. instance lib cell -location vs. power domain (i.e. voltage area) -connection to sleep and acknowledge pins -related supplies of input, output and control ports -associated supply sets -instances associated with strategy

Repeater cells -location vs. power domain -related supplies of output vs. strategy -included and excluded element list associated with strategy

Association can be manually performed using the set_power_strategy_attribute command. If automatic association fails and manual association is used, check_mv_design report should be used to validate the newly associated cell is not failing multi-voltage checks. If a power management cell is associated with a UPF strategy, the association will be registered under cell attribute power_strategy.

Some useful commands for debugging association issues are: -get_related_supply_net for checking driver and load related supplies -all_fanin or all_fanout for tracing pin connectivity to control signals -report_cell -power for seeing PG pin connectivity of a cell -report_mv_path -full for understanding the connectivity of the multi-voltage path

### See Also

- associate_mv_cells
- report_mv_path

Feedback

- set_power_strategy_attribute
- check_mv_design

---

## MV-082

(information) Total %d error(s) and %d warning(s) from check_mv_design.

**Description**

This summary message shows the total number of error(s) and warning(s) reported by check_mv_design. Details for some of these messages may have been suppressed. Use check_mv_design -max_message_count to specify the count of detailed outputs per message tag. Use check_mv_design -max_message_count all to report all details.

**What Next**

Check the details of the reported violations in the check_mv_design report.

**See Also**

- check_mv_design

---

## MV-083

(warning) Cell '%s' in bias power domain '%s' is mapped to non-bias library cell "%s".

**Description**

This warning message occurs when a cell in a bias power domain is mapped to a non-bias library cell.

**What Next**

Run report_mv_lib_cells to check if the lib-cell has bias pins modeled in the library. Check the library of the lib-cell.

**See Also**

- report_cell

---

## MV-084

(error) Hierarchical pin %s is passed in as either driver or load. Failed to perform check_bufferability.

Feedback

### Description

A hierarchical pin was incorrectly given as input to check_bufferability.

### What Next

Please make sure that you pass only leaf pins or top ports as driver or loads.

---

## MV-085

(warning) The %s supply of supply set '%s' is less always-on than its %s supply.

### Description

is less always-on than its power or ground supply. This warning message occurs because the bias function of the specified supply set is less always-on than its power or ground function.

### What Next

Ensure the bias function of the specified supply set is more or equally always-on than its power or ground function.

### See Also

- report_supply_net
- report_pst

---

## MV-086

(error) The related supply net '%s' of the enable pin of level shifter '%s' is less always-on than its output supply net '%s'.

### Description

This message occurs when the related supply net of level shifter enable pin is less always-on than output supply net of the level shifter.

### What Next

Check the power states of the supply nets.

### See Also

- report_cell
- report_pst

Feedback

## MV-087

(error) There are no level shifter implemented and associated with level shifter strategy '%s' in power domain '%s' which is specified "-force_shift".

### Description

This message occurs when level shifter strategy is specified "-force_shift" but no level shifter inserted.

### What Next

Check the level shifter strategy.

### See Also

- associate_mv_cells
- get_related_supply_nets
- report_mv_path
- set_level_shifter

## MV-088

(error) The Driver %s '%s' [supply net '%s', power state '%s'] of enable signal of isolation cell '%s' is less always-on than the load pin '%s' [supply net '%s', power state '%s'] of the cell.

### Description

This message occurs when the related supply net of the enable signal is less always-on than the related supply net of the load pin of an isolation cell. An isolation cell is required to isolate the signal path if the signal is less always on or unrelated to the load pin.

### What Next

Check the power states of the supply nets.

### See Also

- report_net
- report_pst

Feedback

## MV-089

(error) The Driver %s '%s' [supply net '%s', power state '%s'] of data input signal of %s isolation cell '%s' is less always-on than the supply net '%s' [power state '%s'] of the cell.

### Description

This message occurs when the related supply net of data input signal is less always-on than the supply net of the nor/nand style isolation cell. This will cause electrical violation due to power-ground short.

### What Next

Check the power states of the supply nets.

### See Also

- report_net
- report_pst

## MV-096

(error) The %s pin '%s' is tied to the primary %s pin '%s' of the cell but is instantiated in power domain '%s' where the primary %s and primary %s supply nets are different.

### Description

This message occurs when a bias pin is tied to the primary power or ground pin but is instantiated in a power domain where the primary nwell/pwell supply net is different from the primary power or ground supply net.

### What Next

Check the primary power/ground and nwell/pwell supply nets of the power domain. Also, check the supply net connection of the pin.

### See Also

- report_cell
- report_power_domain

## MV-097

(error) There is a forward/reverse bias violation between the backup power pin '%s' [supply net '%s'] and its related bias pin '%s' [supply net '%s'] of a shared-well cell.

Feedback

### Description

This message occurs when there is a voltage difference between the backup supply net and its related bias supply net of a shared-well cell. The voltage difference causes forward/reverse bias violation between the two pins.

### What Next

Check the supply net connections and supply voltages of the backup PG pin and its related bias pin.

### See Also

- report_cell
- report_pst

---

## MV-099

(warning) Supply state(s) has not been defined on the %s '%s'.

### Description

This message occurs when supply states are not defined on the specified supply net or supply port. The tool will assume that any possible combinations of states are possible, and isolation violations or level shifter violations will be flagged between signal pins that are powered by supply nets/ports without any defined supply states.

### What Next

Please check connectivity of the supply net/supply port or define its legal supply states.

### See Also

- add_port_state
- add_power_state
- report_power_domains
- report_pst

---

## MV-100

(error) cannot find voltage area %s.

### Description

This message occurs when the voltage area specified cannot not be found.

Feedback

### What Next

Check the name of the voltage area.

### See Also

- create_voltage_area

---

## MV-101

(warning) Primary PG pin '%s' and backup pin '%s' of the cell '%s' are connected to the same supply net '%s'.

### Description

This message occurs when the primary and backup PG pins of the specified cell are connected to the same supply net.

Typically, dual-rail cells are required when a cell is powered by the backup supply that is different from its primary supply net. If the primary and backup supply of a cell is the same, a dual-rail cell is not required and it should be swapped to a single-rail cell to avoid QOR loss in the final netlist.

### What Next

Check the supply net connections of the PG pins. If the dual-rail cell is not required, swap it to a single-rail cell.

### See Also

- report_cell

---

## MV-102

(warning) The backup supply net '%s' of the dual-rail cell '%s' is PST-equivalent to the primary supply net '%s' of the cell.

### Description

This message occurs when the supply net of backup PG pin of the dual-rail cell is PST-equivalent to the supply net of the primary PG pin.

Typically, dual-rail cells are required when a cell is powered by the backup supply that is different from its primary supply net. If the primary and backup supply of a cell is PST-equivalent, a dual-rail cell is not required and it should be swapped to a single-rail cell to avoid QOR loss in the final netlist.

Feedback

### What Next

Check the supply net connections of the PG pins and the power states of the supply nets. If the dual-rail cell is not required, swap it to a single-rail cell.

### See Also

- report_cell
- report_pst

---

## MV-103

(warning) The sequential element '%s' belonging to retention strategy '%s' in power domain '%s' is not mapped to a retention library cell.

### Description

This warning message indicates that the sequential element is not mapped to retention library cells despite being part of a retention strategy. This results in verification/simulation failure.

Sequential element can be a mapped/unmapped register, ETM or hard-macro with sequential timing arcs.

### What Next

Check the retention strategy commands for the correctness and/or make sure the library has required retention cells and run the command to map the netlist. If the reported sequential element is an ETM or hard-macro and you didn't intend replace it with retention equivalent, you can ignore this message for such elements.

### See Also

- set_retention
- set_retention_control
- map_retention_cell
- check_mv_design

---

## MV-104

(error) Pattern '%s' could not be created.

Feedback

### Description

This messages is printed when the connect_power_switch command failed to create the specified power switch pattern.

### What Next

Check if the pattern name specified by -name already exists.

### See Also

- connect_power_switch

---

## MV-105

(information) Pattern '%s' successfully created.

### Description

This messages is printed when connect_power_switch -pattern_list successfully connects the control and ack_out pins of the power switches in the specified pattern.

### See Also

- create_power_switch
- create_power_switch_array
- create_power_switch_ring
- connect_power_switch
- set_power_switch_placement_pattern

---

## MV-106

(warning) There are no repeater cells implemented and associated with port(pin) '%s' for repeater %s.

### Description

This message is printed when there are no repeater cells implemented and associated with the specified port(pin).

This message is reported even when a repeater was actually inserted but is gobbled due to constant propagation.

Feedback

### What Next

Check the port attributes/strategy of repeater. If the port(pin) is driven by a constant and there is no MV violation, you can ignore this message.

### See Also

- set_port_attributes
- set_repeater

---

## MV-108

(warning) Pin/Port %s is constrained with attribute clamp_value '%s', but the isolation strategy %s from power domain %s that applies on this Pin/Port has the conflicting clamp_value '%s'. Clamp value '%s' will be used while implementing the isolation cell.

### Description

This warning message occurs when the tool encounters design ports or pins that have clamp value settings that do not match the explicit clamp value of the isolation strategy that applies to them. During implementation the tool will honor clamp value settings on the pin being isolated over the clamp value specified for the isolation strategy being implemented.

Clamp value settings for pins or ports are set using the *set_port_attributes* command with the *-clamp_value* option.

Explicit clamp value settings for isolation strategies are managed using the *-clamp_value* option of the *set_isolation command*.

### What Next

Review *set_isolation -clamp_value instances* to fix any clamp value mismatches.

### See Also

- set_port_attributes
- set_isolation

---

## MV-109

(warning) Supply net connection is not specified and related supply of signal pins will be derived from on PG function '%s' of PG pin '%s'.

Feedback

### Description

This message occurs when the tool cannot determine the supply net connection of the internal power and ground pin of a macro.

In the case where supply net connection to internal PG pin is not specified explicitly by connect_supply_net, for related supply of INPUT signal pins related to the internal PG pin, tool will try to get the supply from the pg_function of the internal PG pin. The feature is controlled by app option "mv.upf.use_preswitched_supply_for_macro_pins".

### What Next

Specify additional power intent using the connect_supply_net commands if you do not want to tool to use the supply from PG function.

### See Also

- connect_supply_net
- MV-062
- mv.upf.use_preswitched_supply_for_macro_pins

---

## MV-110

(Error) Power-domain-on-instance element %s is specified in the user (non-derived) section of the UPF constraints for power domain %s. Legal crosstool UPF cannot be generated.

### Description

This message occurs when these conditions are true:

- Power-domain-on-instance (PDOI) repeater elements are present in *create_power_domain* statements in the user section of the UPF file
- The app option *mv.upf.write_crosstool_wrappers* is set true
- The app option *mv.upf.write_crosstool_no_upf_update_command* is set false

Power-domain-on-instance (PDOI) elements are leaf cells as elements in UPF *create_power_domain* statements. This message is specifically about buffers and inverters as leaf cell elements. These are called "physical feedthrough buffers", and are added by the tool automatically with the app option *opt.common.allow_physical_feedthrough* set true. These constraints allow the repeater cell to be placed in a voltage area of a different hierarchy than it logically belongs. When the tool creates UPF changes for these PDOI repeaters, it adds them in the derived section at the end of the UPF file. *create_power_domain -update* statements are created in that section that add these cells to the power domain. This error occurs when *create_power_domain* statements

Feedback

referencing PDOI repeaters are present in the user section of the UPF file. This should not be done by any optimization command in the tool, and if this occurs, it typically means that the user had added these constraints manually to the UPF file. This might have been done by another tool or by manual ECO commands in the tool.

When *mv.upf.write_crosstool_wrappers* is set true, these PDOI leaf instances are removed from the UPF, and replaced by wrapper hierarchies, to support UPF export to other tools that do not support PDOI elements in the UPF. But, the tool is also normally only allowed to modify the derived section of the UPF. So if these PDOI repeaters exist in the user section, they cannot be modified and this error occurs.

**What Next**

There are two approaches to resolve this error:

- Move any create_power_domain statements referencing PDOI repeaters from the user UPF section into the derived section. Typically, the beginning of the UPF file is the user section, and the derived section is at the end. Look for the following, and move these constraints inside. For example:

```
set derived_upf true
if ($derived_upf) {

create_power_domain -elements ... -update
}
```

- Set *mv.upf.write_crosstool_no_upf_update_command* to true. With this setting, the crosstool UPF exported by having *mv.upf.write_crosstool_wrappers* set to true is modified to not include any *-update* options in the UPF commands. This is to support external tools that don't support this newer UPF syntax. To make these changes, the tool has to override it's typical behavior of not modifying the user UPF section. A side effect of this is that the tool is now also free to remove the PDOI leaf instances and replace them with their wrapper hierarchies, even if they are defined in the user section. So setting this app option does have the effect of avoiding the MV-110 error.

**See Also**

- create_power_domain

---

## MV-111

(information) DFT %s %s belonging to retention strategy %s of domain %s is skipped from retention mapping.

**Description**

This information message indicates that the reported sequential element is created by DFT engine and is excluded from retention mapping.

Feedback

### What Next

This is just an information message. No action is required. However, if this behavior is not desired, use the following app option to get those DFT sequential elements mapped to retention. *set_app_options -list {mv.upf.skip_retention_on_dft_registers false}* This app option doesn't not apply to lockup latches. They will always be excluded from retention mapping.

### See Also

- create_mv_cells
- set_retention

---

## MV-112

(warning) map_retention_cell of retention strategy %s for power domain %s specifies unusable library cells : %s.

### Description

This warning message indicates that some library cells specified in *map_retention_cell* are not usable for retention mapping since they either have too strict lib_cell_purpose or dont_touch attribute set on them.

### What Next

Use set_lib_cell_purpose to include the reported library cells optimization. Also use set_dont_touch command to remove dont_touch attribute from the report library cells.

### See Also

- set_lib_cell_purpose
- set_dont_touch
- commit_upf

---

## MV-113

(error) The related supply net '%s' [power state '%s'] of driver '%s' pin '%s' is less always-on than the related supply net '%s' [power state '%s'] of load '%s' pin '%s'.

### Description

This message occurs when the related supply net of the driver is less always-on than the related supply net of the load pin and an isolation cell is missing and the driver or load are

Feedback

of direction inout. An isolation cell is required to isolate the signal path if the driver of the net is less always on or unrelated to the load pin.

**What Next**

Check the power states of the supply nets.

**See Also**

- report_net
- report_pst

---

## MV-114

(Information) UPF command format and object tracking feature %s.

**Description**

This information message occurs when UPF object and command order tracking engine gets disabled by the tool.

**What Next**

This is just an information message.

---

## MV-115

(warning) Abutted pins '%s' [related supply net '%s'] and '%s' [related supply net '%s'] have different related supply nets.

**Description**

This message occurs when two physical block pins are abutted together but have different related supply nets. When the hierarchical block pins are abutted together, there is no space outside the sub-blocks to fix any multi-voltage related violations. As a result, the abutted pins should have equivalent related supply net(s) so that any possible multi-voltage violations (e.g. level shifting violations) would only occur and get resolved within the sub-block(s).

**What Next**

Check the related supply nets of the specified pins and consider to assign equivalent related supply nets in the sub-blocks.

Feedback

**See Also**

- get_related_supply_nets
- check_supply_equivalence
- set_related_supply_net
- set_port_attributes

---

## MV-116

(error) Retention register %s belonging to strategy %s of power domain %s does not conform with the mapping directive.

**Description**

This error message indicates one of the following possibilities. - the reported register instance is mapped to a library cell that is not in the list specified by *map_retention_cell* command for the strategy. - retention strategy has different save and restore control signals, but the reported register instance is mapped to a 1-pin or a 0-pin retention library cell.

This might result in strategy association failure during *commit_upf*.

**What Next**

Tool has either mapped or sized this register instance incorrectly. This must be fixed.

**See Also**

- set_retention
- map_retention_cell
- check_mv_design

---

## MV-117

(warning) Net %s connected to PG pins does not have a matching UPF supply net.

**Description**

The PG net listed above does not have a corresponding supply net with the same name in the power intent.

**What Next**

Check the net connection of the PG pins.

Feedback

For inference of PG connections in the power intent, the power intent must have a supply net of the same name as this net. This net can be created using 'create_supply_net' if it does not already exist. If this net is created and inference of supply net connections in the UPF is not disabled, infer_supply_from_pg_net will create connections using 'connect_supply_net' in the power intent, based on the net connections in the RTLPG.

**See Also**

- infer_supply_from_pg_net
- create_supply_net
- connect_supply_net

---

## MV-118

(warning) %d net(s) do not have matching supply nets in the UPF.

**Description**

These PG nets do not have corresponding supply nets of the same names in the power intent.

**What Next**

Please run check_mv_design -power_connectivity for details on which of the nets do not have matching UPF supply nets. For inference of PG connections in the power intent, the UPF must have supply nets with the same names as the PG nets in the design. These supply nets can be created using create_supply_net if they do not exist. If they are created and inference of supply net connections in the UPF is not disabled, commit_upf will create connections using 'connect_supply_net' in the power intent, based on the net connections in the RTLPG.

**See Also**

- commit_upf
- create_supply_net
- connect_supply_net

---

## MV-119

(warning) Power domain '%s' does not have any primary voltage area.

Feedback

### Description

This message occurs when primary voltage area of the specified power domain is not defined. The power and ground supplies of primary voltage area of a power domain must match the primary power and ground supplies of the domain.

### What Next

Check the voltage area(s) of the power domain and their power and ground supplies.

### See Also

- report_power_domain
- report_voltage_areas

---

## MV-120

(warning) Net %s is missing %s. Skipping level shifter insertion evaluation for the net.

### Description

This message occurs during level shifter insertion. The net is missing a valid driver or load. It will be skipped during evaluation.

### What Next

Check the net for connectivity problems if you believe it should have a level shifter inserted on it. Otherwise, you can ignore this message.

### See Also

- all_fanin
- all_fanout
- all_connected

---

## MV-121

(warning) Net %s %s %s is missing power supply connection, skip LS insertion/association evaluation of the net.

### Description

This message occurs during level shifter insertion/association, the net driver pin/port is missing power supply definition, it will be skipped during insertion evaluation or level shifter cell association. This may cause the voltage shifting issue or level shifter cell unassociation issue.

Feedback

### What Next

Check the supply connection for the pin/port.

### See Also

- connect_supply_net

---

## MV-122

(Error) No valid library cell available for LS insertion, LS insertion on net %s failed.

### Description

This message occurs during level shifter insertion, there is no valid level shifter reference design can be found from library matching all the conditions, level shifter insertion on this net failed.

### What Next

Check the library setup

---

## MV-123

(Warning) Net has multi-supply loads which cannot be split based on the level shifter strategy setting, and level shifter insertion on net %s failed.

### Description

This message occurs during level shifter insertion in cases where the net has multi-supply loads, but the strategy setting has conflict with splitting the loads so level shifter insertion on this net failed. For example, the strategy defines the inserting location is parent, all the loads inside the domain have different supply settings compared to those outside. In this case then, splitting cannot be done in parent domain.

### What Next

Check the UPF strategy setting.

### See Also

- set_level_shifter

---

## MV-124

(warning) %s supply net %s does not have valid voltage defined. LS insertion will skip evaluating the connected net.

Feedback

### Description

This message occurs during level shifter insertion, supply net does not have valid voltage defined, insertion will skip evaluating the connected net.

### What Next

Check the voltage setting of the supply net, check the port state setting from UPF.

### See Also

- add_port_state

---

## MV-125

(warning) %s supply net %s is not available in domain %s, level shifter cell insertion for strategy %s on net %s failed.

### Description

This message occurs during level shifter insertion in the create_mv_cells command. A level shifter cell could not be inserted because the related supply for either the input or output of the level shifter is not available in the power domain. In this case, the level shifter cell is not inserted.

### What Next

To further debug, use report_mv_path -net <net>, where <net> is the net name from the MV-125 warning. This report gives detailed information about the logic connected to that net, the related supplies of that logic, and any power domain boundaries on that net. The related supplies of the connected logic determine the necessary supplies for the level shifter.

A common reason for level shifter insertion failure is that the required supply is not available in the power domain, in which case it may need to be added using the create_supply_port and create_supply_net commands. Alternately, the level shifter may have been inserted in the wrong domain which doesn't include the required supplies, in which case the set_level_shifter -location option can be modified to change whether the level shifter is inserted in the self, parent, or fanout domain hierarchy.

After correcting the UPF constraints, rerun that part of the flow through commit_upf and create_mv_cells and ensure that the MV-125 warnings go away.

Feedback

**See Also**

- create_mv_cells
- report_mv_path
- set_level_shifter

---

## MV-126

(Error) Command line option strategy_output can be used only for -generate_strategy.

**Description**

This message occurs in create_mv_cells command with -strategy_output option but no -generate_strategy option

**What Next**

Specify -generate_strategy to generate the strategy output file.

---

## MV-127

(Error) Wrong input '%s' for -generate_strategy option, only 'level_shifter' is accepted.

**Description**

This message occurs in create_mv_cells command with -generate_strategy option but not specifying level_shifter. In current version, only level_shifter is supported.

**What Next**

Specify -generate_strategy level_shifter to generate level shifter strategies.

---

## MV-128

(error) Option -keep_order should be specified together with option -object_list.

**Description**

This error occurred because connect_power_switch option -keep_order was not specified with -object_list option.

**What Next**

Please refer to the manpage of connect_power_switch command for usage information and examples.

Feedback

### See Also

- connect_power_switch

---

## MV-129

(Error) Option -keep_order should be specified for -mode daisy.

### Description

This message occurs when the connect_power_switch command is issued with the -keep_order option, but -mode option is not daisy.

### What Next

Specify the -keep_order option only with -mode daisy.

### See Also

- connect_power_switch

---

## MV-130

(information) Using pin(port) '%s' of net '%s' for connections.

### Description

This message shows the port or pin of the specified net that is being used. The -source and -ack_out options use the driver for connections. If the net does not have a driver, will use the load instead.

### What Next

This is an information-only message. No action is required.

### See Also

- connect_power_switch

---

## MV-131

(warning) Cannot insert repeater cell for pin(port) '%s', because some isolation or level shifter cells already exist on the path.

### Description

This message occurs during repeater insertion. Since the path has already inserted isolation or level shifter cells, the repeater cells will not be inserted.

Feedback

### What Next

Remove isolation or level shifter cells of the path and create_mv_cells again.

---

## MV-132

(Error) Repeater supply ('%s','%s') at pin '%s' is not available to domain '%s'.

### Description

This error message occurs when the repeater supply attribute specified for a port/pin or an element is not found available to the domain where the repeater will be inserted.

### What Next

Make sure the supplies are appropriately brought in to the domain. If the domain is restricted using extra_supplies handle, make the repeater supply available using extra_supplies_# handle. If the supplies are domain dependent, you may need to use -reuse switch to make it available to the domain.

### See Also

- create_power_domain
- create_supply_net

---

## MV-133

(information) UPF is not committed yet.

### Description

This message indicates that UPF has not been committed yet. Therefore, the result of the last executed command may not be accurate.

### What Next

Run commit_upf and re-run the previous command to get the results after UPF information is committed.

### See Also

- commit_upf

---

## MV-134

(information) Committing UPF.

Feedback

### Description

Some commands require UPF to be committed before they get executed. This message indicates that UPF is being committed automatically for such a command.

### What Next

N/A

### See Also

- commit_upf

---

## MV-135a

(Information) No Level Shifter library cell available due to %s.

### Description

This message is from level-shifter insertion engine, only create_mv_cell is specified with -verbose.

No level-shifter cell can be inserted due to a problem with lib_cell availability. And the reasons might be:

1. Library cell is has don't_touch attribute set.
2. Voltage range mismatch: a library cell's voltage range defined in library should match with driver/load supply's voltage range before it can be inserted.
3. Library cell is not overdriven while the path needs overdriven Level Shifter.
4. Library cell is not single ground while single ground library cell is required.
5. Strategy rule mismatch: a library cell's level-shifter type should match with the voltage shifting direction on the path.
6. Standard cell main-rail mismatch: a library cell's main-rail supply should match with the domain primary supply.
7. Library cell has pvt mismatch.
8. Library cell (Enable Level Shifter) clamp has mismatch or cannot perform automatic phase correction due to clamp mismatch
9. Library cell (Enable Level Shifter) sense has mismatch or cannot perform automatic phase correction due to sense mismatch
10. Library cell (Enable Level Shifter) does not have enable pin.

Feedback

11. Library cell is Combo Enable Level Shifter, but has data mismatch.
12. No library cell can be selected due to voltage violation between input and output supply net.

**What Next**

Identify specific lib_cells you expect should be available, check lib_cell information by report_lib_cell and check the PST table definition in UPF file.

**See Also**

- add_power_state
- create_pst
- report_mv_lib_cells
- set_level_shifter

---

## MV-135

(information) No level shifter library cell available due to voltage range mis-matched.

**Description**

This message is from level-shifter insertion engine, only create_mv_cell is specified with -verbose.

No level-shifter cell can be inserted due to a problem with lib_cell availability. And the reasons might be:

1. Strategy rule mismatch: a library cell's level-shifter type should match with the voltage shifting direction on the path.
2. Standard cell main-rail mismatch: a library cell's main-rail supply should match with the domain primary supply.
3. Voltage range mismatch: a library cell's voltage range defined in library should match with driver/load supply's voltage range before it can be inserted.

this message will be printed for the reason #3.

**What Next**

Identify specific lib_cells you expect should be available, check lib_cell information by report_lib_cell and check the PST table definition in UPF file.

Feedback

### See Also

- add_power_state
- create_pst
- report_mv_lib_cells

---

## MV-136

(information) No level shifter library cell available due to standard cell main-rail mismatch.

### Description

This message is from level-shifter insertion engine, only create_mv_cell is specified with -verbose.

No level-shifter cell can be inserted due to a problem with lib_cell availability. And the reasons might be:

1. Strategy rule mismatch: a library cell's level-shifter type should match with the voltage shifting direction on the path.
2. Standard cell main-rail mismatch: a library cell's main-rail supply should match with the domain primary supply.
3. Voltage range mismatch: a library cell's voltage range defined in library should match with driver/load supply's voltage range before it can be inserted.

this message will be printed for the reason #2.

### What Next

Identify specific lib_cells you expect should be available and check the location of level-shifter strategy which is specified in set_level_shifter command.

### See Also

- report_mv_lib_cells
- set_level_shifter

---

## MV-137

(information) No level shifter library cell available due to strategy rule mismatch.

### Description

This message is from level-shifter insertion engine, only create_mv_cell is specified with -verbose.

Feedback

No level-shifter cell can be inserted due to a problem with lib_cell availability. And the reasons might be:

1. Strategy rule mismatch: a library cell's level-shifter type should match with the voltage shifting direction on the path.
2. Standard cell main-rail mismatch: a library cell's main-rail supply should match with the domain primary supply.
3. Voltage range mismatch: a library cell's voltage range defined in library should match with driver/load supply's voltage range before it can be inserted.

this message will be printed for the reason #1.

**What Next**

Identify specific lib_cells you expect should be available. Check lib_cell information by report_lib_cell.

**See Also**

- report_mv_lib_cells

---

## MV-138

(Warning) Isolation cell %s cannot be used to clamp retention cell pin %s (strategy %s of power domain %s). %s mismatch.

**Description**

This warning message occurs when the isolation cell cannot be used to clamp the load retention cell. It could be caused by clamp value mismatch, retention supply mismatch or retention control signal mismatch.

**What Next**

Check the related retention strategy and the association of the isolation cell.

**See Also**

- set_retention
- set_retention_control
- map_retention_cell

---

## MV-139

(Warning) Retention cell pin %s (strategy %s of power domain %s) is not clamped.

Feedback

### Description

This warning message occurs when a zero-pin retention cell pin is not clamped.

### What Next

Check the related retention strategy and zero-pin retention library cell.

### See Also

- set_retention
- set_retention_control
- map_retention_cell

---

## MV-140

(information) MV cell with %s strategy %s cannot be inserted on port %s: standard cells may not be inserted between a pad and the outside world.

### Description

This message occurs when there is an isolation, level shifter or repeater strategy defined on a top-level port that is connected to a pad cell.

### What Next

This is an information-only message. No action is required. However, you should check the power intent to ensure the strategies are specified correctly.

### See Also

- commit_upf
- create_mv_cells

---

## MV-141

(warning) Instance %s connected to top-level port %s has is_pad set to true, while instance %s connected to the same port has 'is_pad' set to false. This should not be implemented as mixture of pad cells and standard cells on the same port is uncommon. Supply at top-level port is set from pad cell supply.

### Description

This message occurs when some pins connected to a top-level port have is_pad : true, and some don't. This may not be physically implementable.

Feedback

### What Next

Check the net connections between the top-level port and pad cell.

### See Also

- commit_upf

---

## MV-142

(warning) Top-level port %s is connected to multiple pad cells, but the supplies are inconsistent. Pad cell %s connected the port with supply net '%s' while pad cell %s connected it with supply net '%s'. UPF supply for the top-level port will be applied.

### Description

This message occurs when more than one cell pin connected to the top-level port has is_pad : true, and the supply of the pins is inconsistent.

### What Next

Check the net connections between the top-level port and the pad cell. Fix the design so only one pad cell is connected to the port, or so that the pad cells connected to the same net have the same related supply.

### See Also

- commit_upf
- report_pst

---

## MV-143

(warning) Pre-existing standard cell %s cannot be connected to pad cell %s: standard cells may not be inserted between a pad and the outside world. is_pad attribute on pad pin %s will be ignored.

### Description

This message occurs when a pin with is_pad : true is not directly connected to top-level port.

### What Next

Check the net connections between top-level port and pad cell. Cell is not allowed to be inserted between a top-level port and a pad cell.

Feedback

### See Also

- commit_upf

---

## MV-144

(information) User has specified '%s' on top-level port %s. The supply net %s derived from pad cells connected with the port is inconsistent with user setting. Supply at top-level port is set from pad cell supply; UPF supply for the port is not applied.

### Description

This message occurs when user has specified UPF constraints like 'set_port_attribute -driver/-receiver supply' or 'set_related_supply_net' on top-level port that is connected to a pad cell. By default, the supply derived from pad cell will be honored.

### What Next

Check the net connections between the top-level port and pad cell.

### See Also

- commit_upf
- mv.upf.use_driver_receiver_for_io_supplies

---

## MV-145

(error) A leaf cell %s is found for option -hierarchy. Specify only logical hierarchy cells with this option.

### Description

The *check_bufferability -hierarchy* option is required when using the *-driver* option. Without it, accurate checking cannot be performed for bufferability issues related to logical hierarchies such as target library subset constraints and bias optimization.

This option can only take a logical hierarchy cell as an argument, not leaf cells.

### What Next

Change the argument to the *check_bufferability -hierarchy* option to a logical hierarchy cell. It may be sufficient to specify the logical hierarchy owner of the specified leaf cell. For example, if a leaf cell *a/b/buffer1* is specified, instead specify just *a/b*.

The correct logical hierarchy cell to specify depends on what you are trying to learn from the *check_bufferability* call, though. See the *check_bufferability* manpage for more details.

Feedback

**See Also**

- check_bufferability

---

## MV-146

(error) Missing required option(s) %s for check_bufferability.

**Description**

The *check_bufferability* command has two use models:

1. *check_bufferability -nets [-voltage_area]*
2. *check_bufferability -driver [-loads] -hierarchy [-voltage_area]*

The *-voltage_area* option is optional. And either *-nets* or two options *-driver -hierarchy* must be used. For the second usage, *-voltage_area* is optional. This error message indicates that the required combination of command options has not been specified.

**What Next**

Correct the options on *check_bufferability* as described in the error message. Refer to the *check_bufferability* manpage for a detailed description of the usage of this command.

**See Also**

- check_bufferability

---

## MV-147

(error) A hierarchical pin %s is found for option -driver. Specify only a leaf cell pin or a current design port with this option.

**Description**

The *check_bufferability -driver* option can only accept a physical driver pin or port, whether that is a leaf cell pin or a current design port.

**What Next**

Change the *check_bufferability -driver* option to specify a valid physical driver pin or port.

**See Also**

- check_bufferability

Feedback

## MV-148

(error) A wrong direction pin or port %s is found for option -driver. Do not use a leaf cell input pin or a current design output port with this option.

### Description

The *check_bufferability -driver* option can only accept a physical driver pin or port, whether that is a leaf cell output pin or a current design input port. A bidirection (inout) pin or port is also accepted.

### What Next

Change the *check_bufferability -driver* option to specify a valid physical driver pin or port.

### See Also

- check_bufferability

## MV-149

(warning) Option %s is specified together with option(s) %s. The command will only take option %s and ignore the rest.

### Description

The *check_bufferability* command has two usage models:

1. *check_bufferability -nets [-voltage_area]*
2. *check_bufferability -driver [-loads] -hierarchy [-voltage_area]*

This warning is issued if any of the *-driver -loads -hierarchy* options are specified together with *-nets*. In this case, they are ignored and only *-nets* is honored.

### What Next

This is just a warning, and can be ignored if the resulting options accepted by *check_bufferability* still perform the check you want. Otherwise, change the specified options to perform the correct check. See the *check_bufferability* manpage for more details about correct usage of this command.

### See Also

- check_bufferability

Feedback

## MV-150

(warning) Multiple %s type objects are found for option %s. The command will only take the first object and ignore the rest.

### Description

For the options *-net, -driver, -hierarchy,* and *-voltage_area*, they can take either a single name or single object. If more than one name or object is passed to those options, then this warning will occur.

### What Next

You might need to split your query into multiple *check_bufferability* calls to check all of the nets, drivers, hierarchies, or voltage areas you are trying to check.

### See Also

- check_bufferability

## MV-151

(warning) A hierarchical pin %s is found for option -loads. It will be ignored.

### Description

The *check_bufferability -loads* option can only accept physical load pins or ports, whether those are leaf cell pins or current design ports. Any hierarchical pins specified with this option are ignored.

### What Next

The warning can be ignored, as long as it is understood that the specified hierarchical pins will not be used by *check_bufferability*.

### See Also

- check_bufferability

## MV-152

(warning) A wrong direction pin or port %s is found for option -loads. It will be ignored.

### Description

The *check_bufferability -loads* option can only accept physical load pins or ports, whether those are leaf cell input pins or current design output ports. Bidirection (inout) pins or ports are also accepted. Any pins or ports violating this requirement are ignored.

Feedback

### What Next

The warning can be ignored, as long as it is understood that the wrong direction pins or ports are not used by *check_bufferability*.

### See Also

- check_bufferability

---

## MV-153

(warning) The specified -driver, -loads and -hierarchy options do not correspond to a real net segment. Net dont_touch and net is_ideal checks cannot be performed.

### Description

The *check_bufferability* command has two usage models:

1. *check_bufferability -net -voltage_area*
2. *check_bufferability -driver -loads -hierarchy -voltage_area*

For the first, an actual net segment in the design is always identified and queried for dont_touch and is_ideal properties, when checking non-MV reasons that a net cannot be buffered.

For the second, a real net segment can be inferred if the driver and loads all connect to the same net, and at least one segment of that net belongs to the specified hierarchy. Otherwise, this warning message is printed. In this case, dont_touch and is_ideal checks cannot be performed since no actual net segment in the design could be identified corresponding to the specified *-driver -loads -hierarchy* options.

### What Next

In most cases, *check_bufferability* is used to check real net segments. If you are trying to check a real net segment, then this warning is unexpected. The limitations on checking dont_touch and is_ideal attributes will occur, but additionally the rest of the bufferability results may be wrong. Revisit the arguments passed to *-drivers -loads -hierarchy* and correct them so the real net segment can be found. Alternately, just pass that net segment name directly to *check_bufferability -net* instead of using the *-driver -loads -hierarchy* options.

*check_bufferability* can also be used to make hypothetical bufferability queries, with drivers and loads that aren't connected to the same net, or to query an unrelated hierarchy. If you are trying to make a hypothetical query, then this warning is expected. Just be aware that dont_touch and is_ideal bufferability checks cannot be performed.

Feedback

**See Also**

- check_bufferability

---

## MV-154

(warning) Cannot find any loads for net %s. Supply net availability and lib cell availability checks cannot be performed.

**Description**

The *check_bufferability* command performs three categories of checks:

1. Supply availability, to determine if single rail or dual rail buffers are needed, and the buffering supply
2. lib_cell availability, to determine if any lib_cells are available matching the supply requirements
3. Non-MV bufferability checks, such as dont_touch or is_ideal net attributes, and many others

This warning message indicates that the first two categories of checks, supply and lib_cell availability, cannot be performed because the net has no loads. Only the non-MV bufferability checks are performed.

**What Next**

No further action is required.

**See Also**

- check_bufferability

---

## MV-155

(warning) Cannot find any driver for net %s. Supply net availability and lib cell availability checks cannot be performed.

Feedback

### Description

The *check_bufferability* command performs three categories of checks:

1. Supply availability, to determine if single rail or dual rail buffers are needed, and the buffering supply
2. lib_cell availability, to determine if any lib_cells are available matching the supply requirements
3. Non-MV bufferability checks, such as dont_touch or is_ideal net attributes, and many others

This warning message indicates that the first two categories of checks, supply and lib_cell availability, cannot be performed because the net has no driver. Only the non-MV bufferability checks are performed.

### What Next

No further action is required. Once the design problem of "no net driver" is resolved, *check_bufferability* can be rerun to do more complete checking on the net.

### See Also

- check_bufferability

---

## MV-156

(warning) Net %s is not bufferable because it is driven by constant logic 0 or 1. The command assumes the driver will be implemented with normal tie cell to analyze supply availability.

### Description

Since the driver is constant 0 or 1, the net is not bufferable. However, *check_bufferability* will still analyze supply availability by assuming the driver is implemented with normal tie cell.

### What Next

Implement the constant logic 0 or 1 with proper tie cell prior to running *check_bufferability*.

### See Also

- check_bufferability

Feedback

## MV-157

(warning) Multiple drivers are found for net %s. The command will only take the %s as unique driver to perform supply availability and lib cell checks.

### Description

The standard data and CTS buffering engines do not build multi-driven nets that have multiple drivers. *check_bufferability* will report this as one of the non-MV bufferability problems: "The net has multiple drivers". Despite this, *check_bufferability* will still perform supply availability and lib_cell availability checks for buffering this net, by ignoring all drivers except the one listed in this warning message.

### What Next

No further action is required. Be aware that the standard buffering engines will not build this net because of multiple drivers. But for the purposes of checking supply and lib_cell availability for a different buffering solution, like multisource CTS, *check_bufferability* is still useful on such a net.

### See Also

- check_bufferability

## MV-158

(information) Found net %s corresponding to the specified -driver, -loads, and -hierarchy options. It will be used to perform net dont_touch and net is_ideal checks.

### Description

The *check_bufferability* command has two usage models:

1. *check_bufferability -net -voltage_area*
2. *check_bufferability -driver -loads -hierarchy -voltage_area*

For the first, an actual net segment in the design is always identified and queried for dont_touch and is_ideal properties, when checking non-MV reasons that a net cannot be buffered. For the second, a real net segment can be inferred if the driver and loads all connect to the same net, and at least one segment of that net belongs to the specified hierarchy. This information message indicates the an actual net segment was identified. As a result, the complete set of bufferability checks can be performed.

### What Next

No further action is required.

Feedback

### See Also

- check_bufferability

---

## MV-159

(warning) Could not determine a buffering supply. Lib cell availability checks cannot be performed.

### Description

The *check_bufferability* command performs three categories of checks:

1. Supply availability, to determine if single rail or dual rail buffers are needed, and the buffering supply
2. lib_cell availability, to determine if any lib_cells are available matching the supply requirements
3. Non-MV bufferability checks, such as dont_touch or is_ideal net attributes, and many others

In the case that the supply availability check fails to identify a valid buffering supply, the lib_cell availability check cannot be performed. So, the complete set of *check_bufferability* checks is not done on this net.

### What Next

No further action is required. Be aware that if the supply availability problem is resolved, and *check_bufferability* is rerun on this net, it is possible that lib_cell availability failures will now be reported that could not previously be identified when the supply availability issue existed.

### See Also

- check_bufferability

---

## MV-160

(information) No enabled level shifter library cell available due to sense or clamp value mis-matched.

### Description

This message is from level-shifter insertion engine, only create_mv_cell is specified with -verbose.

Feedback

No enabled level-shifter cell can be inserted due to a problem with lib_cell availability. And the reasons might be:

1. Isolation type mismatch: the replaced isolation cell and the enabled level shifter cell should has the same isolation type.
2. Strategy rule mismatch: a library cell's level-shifter type should match with the voltage shifting direction on the path.
3. Voltage range mismatch: a library cell's voltage range defined in library should match with driver/load supply's voltage range before it can be inserted.
4. Sense or clamp value mismatch: the replaced isolation cell and the enabled level shifter cell should has the same sense and clamp value.

this message will be printed for the reason #4.

**What Next**

Identify specific lib_cells you expect should be available and check the options -clamp_value and -sense from the set_isolation command.

**See Also**

- report_mv_lib_cells
- set_isolation

---

## MV-161

(information) No enabled level shifter library cell available due to isolation type mismatched.

**Description**

This message is from level-shifter insertion engine, only create_mv_cell is specified with -verbose.

No level-shifter cell can be inserted due to a problem with lib_cell availability. And the reasons might be:

1. Isolation type mismatch: the replaced isolation cell and the enabled level shifter cell should has the same isolation type.
2. Strategy rule mismatch: a library cell's level-shifter type should match with the voltage shifting direction on the path.

Feedback

3. Voltage range mismatch: a library cell's voltage range defined in library should match with driver/load supply's voltage range before it can be inserted.
4. Sense or clamp value mismatch: the replaced isolation cell and the enabled level shifter cell should has the same sense and clamp value.

this message will be printed for the reason #1.

**What Next**

Identify specific lib_cells you expect should be available.

**See Also**

- report_mv_lib_cells

---

## MV-162

(warning) The timing data is not up-to-date. %s

**Description**

This warning occurs when the timer is not fully updated prior to running *check_bufferability*. *check_bufferability* still runs, because most bufferability checking can be done without the timer. But, the bufferability checks that rely on timing information will be skipped. These can include:

- Dont touch status of a net
- Ideal network status of a net
- Freeze port status of a logical feedthrough hierarchy
- The type of net being buffered (clock, or data, or mixed clock and data)

The MV-162 message indicates which of these checks will be skipped, which can vary depending on the exact timer state.

For dont touch and ideal network constraints, the checking is skipped. It is possible that the net cannot be buffered because of one of these constraints, but that *check_bufferability* reports the net as bufferable, if run without updated timing.

For freeze port constraints, the checking is skipped. It is possible that the net cannot be buffered in logical feedthrough hierarchy because the hierarchy has freeze port constraints, but that *check_bufferability* reports the net as bufferable, if run without updated timing.

Normally *check_bufferability* determines the type of net being buffered, and reports repeater lib cell available with the correct lib cell purpose, usually 'cts' purpose for clock nets and 'optimization' purpose for data nets. If the net type cannot be determined due to

Feedback

the timer state, then *check_bufferability* will query lib cell availability with a purpose of 'cts' OR 'optimization'.

**What Next**

To ensure that all checks are performed, and make the MV-162 warning go away, explicitly run update_timing prior to running *check_bufferability*.

**See Also**

- check_bufferability

---

## MV-163

(warning) The isolation cell '%s' is associated with isolation strategy '%s' by name, but %s of the cell is not match with the strategy.

**Description**

This message is printed when the specified isolation cell is naming match with the strategy but upf intent is inconsistent.

The tool can associate isolation cells with isolation strategies by name even there is UPF intent mismatch.

**What Next**

Check the upf intent of related isolation strategy with the isolation cell.

**See Also**

- associate_mv_cells
- check_mv_design
- mv.cells.use_name_based_iso_matching

---

## MV-164

(warning) The net %s is connected to PG pin %s, which has a direction other than "in" or "out".

**Description**

The PG Pin that this net is connected to has a direction other than "in" or "out". When commit_upf tries to translate all of the RTLPG to UPF, it is unable to create a supply port in this case, since the port direction cannot be determined.

Feedback

### What Next

Please check your RTL.

### See Also

- commit_upf
- create_supply_net
- connect_supply_net

---

## MV-165

(warning) Net %s, which is connected to pin %s, does not have a matching supply net in the UPF.

### Description

This PG Net does not have a matching supply net in the UPF. For connections in the RTL PG to be translated correctly to UPF, the UPF must have a matching supply net, i.e. either a supply net of the same name as the RTL net which is available in the same scope, or a supply net with the same name as the topmost net segment of the net, if net segments have different names.

### What Next

Please check your UPF and create the appropriate supply nets.

### See Also

- commit_upf
- create_supply_net
- connect_supply_net

---

## MV-166

(warning) Net is bufferable. But the %s violation on the net will be preserved.

### Description

This warning occurred because the net has Isolation and (or) Level Shifter violation. Some segments of the net might have limited bufferability.

Feedback

### What Next

- Check whether the existing violation is expected or not. If the existing violation is not expected, fix before continuing. Otherwise it can lead to wrong buffer selection for buffering.
- Use *report_mv_path -net* to identify existing violation and resolve it.

### See Also

- check_mv_design
- report_mv_path
- analyze_mv_design

---

## MV-167

(error) Detected unsupported command(%d) and option(%d) in design UPF file for block '%s'. All the original UPF power constraints aren't applied in the design '%s'.

### Description

UPF file is not forward compatible. When the opened design from future release contains design UPF file, which has unsupported UPF command or option, the NDM design would be treated as UPF incompatible. As a result, all the original UPF power constraints aren't applied.

### What Next

You can use *save_upf* to write out the original design UPF file, and remove incompatible UPF commands and options from the design UPF file.

You need to do *reset_upf* and load a new UPF before you use any UPF power constraints in the design.

### See Also

- load_upf
- reset_upf
- save_upf

---

## MV-168

(error) Incompatible design UPF file was detected. This command '%s' is disabled.

Feedback

### Description

UPF file is not forward compatible. When the opened design from future release contains design UPF file, which has unsupported UPF command or option, the NDM design would be treated as UPF incompatible. As a result, this command is disabled.

### What Next

You need to do *reset_upf* before executing this command.

### See Also

- load_upf
- reset_upf
- MV-167

---

## MV-169

(warning) Incompatible design UPF file was saved.

### Description

UPF file is not forward compatible. When the opened design from future release contains design UPF file, which has unsupported UPF command or option, the NDM design would be treated as UPF incompatible. As a result, all the original UPF power constraints aren't applied. But by save_upf or save_block, the original incompatible design UPF would be saved.

### What Next

You can write out the incompatible design UPF by save_upf and remove incompatible UPF commands and options from it, then re-load the compatible design UPF file.

### See Also

- copy_block
- load_upf
- reset_upf
- save_block
- save_upf
- MV-167

Feedback

## MV-170

(warning) The clamp value '%s' of isolation cell '%s' is inconsistent with the clamp value '%s' defined by associated strategy '%s'.

### Description

This message occurs when the isolation cell is associated with a strategy but the clamp value is inconsistent with the strategy. This is possible that netlist has been changed but association do not update.

### What Next

Try to use associate_mv_cells to re-association isolation cells. Check if there is port attribute -clamp_value defined on boundary port which has higher precedence than strategy.

### See Also

- associate_mv_cells
- report_cell

## MV-171

(warning) The control signal of isolation cell '%s' is inconsistent with associated strategy '%s'.

### Description

This message occurs when the isolation cell is associated with a strategy but the control signal is inconsistent with the strategy. This is possible that netlist has been changed but association do not update.

### What Next

Try to use associate_mv_cells to re-association isolation cells. Check the control signal path from isolation cell enable pin. Check if the signal source port/net name and sense match with the strategy.

### See Also

- associate_mv_cells
- report_cell
- report_net

Feedback

## MV-172

(warning) The supply net of isolation cell '%s' is inconsistent with associated strategy '%s'.

### Description

This message occurs when the isolation cell is associated with a strategy but the supply net is inconsistent with the strategy. This is possible that netlist has been changed but association do not update.

### What Next

Try to use associate_mv_cells to re-association isolation cells. Check the power and ground net of isolation cell and the supply net specified by the strategy -isolation_power_net/-isolation_ground_net/-isolation_supply_set.

### See Also

- associate_mv_cells
- report_cell
- report_net

## MV-173

(warning) The location of isolation cell '%s' is inconsistent with associated strategy '%s'.

### Description

This message occurs when the isolation cell is associated with a strategy but logical location is inconsistent with the strategy. This is possible that netlist has been changed but association do not update.

### What Next

Try to use associate_mv_cells to re-association isolation cells. Check if location of isolation cell with the strategy domain boundary match with -applies_to and -location defined in strategy. If the strategy is expected, the isolation cell may need to removed and reinserted. If the isolation cell is expected, the strategy should be updated.

### See Also

- associate_mv_cells
- report_mv_path

Feedback

## MV-174

(warning) The driver or receiver supply sets of isolation cell '%s' is inconsistent with associated strategy '%s'.

### Description

This message occurs when the isolation cell is associated with a strategy but driver or receiver supply sets is inconsistent with the strategy. This is possible that when isolation cell is associated by name, upf intent check will be relaxed.

### What Next

Check if the supply set of driver and receiver of the isolation cell on mv path match with -source/-sink/-diff_supply_only defined in strategy.

### See Also

- associate_mv_cells
- report_mv_path

## MV-175

(warning) The name %s of isolation cell '%s' is inconsistent with associated strategy '%s'.

### Description

This message occurs when the isolation cell is associated with a strategy but name prefix or suffix is inconsistent with the strategy. This is possible that when isolation cell is associated by name, upf intent check will be relaxed.

### What Next

Check if there is strategy change after the isolation cell inserted and associated. The isolation cell may need to removed and reinserted to match the strategy name prefix/suffix.

### See Also

- associate_mv_cells
- report_cell

## MV-176

(warning) The reference library cell '%s' of isolation cell '%s' is inconsistent with associated strategy '%s'.

Feedback

### Description

This message occurs when the isolation cell is associated with a strategy but reference library cell is inconsistent with the strategy. This is possible that when isolation cell is associated by name, upf intent check will be relaxed.

### What Next

Check the reference library cell of the isolation cell with the mapping library defined by map_isolation.

### See Also

- associate_mv_cells
- report_cell

---

## MV-178

(error) specified improper library cell pair: %s and %s.

### Description

This message occurs when user specifies improper library cell pair in the command heal_mv_path for cell swapping. The possible reasons can be:

- `incorrect library cell type: only buffer or inverter is accepted.`

- `library cell type mismatched: it's not accepted that one is buffer while another is inverter.`

- `incorrect rail count: should be single-rail versus dual-rail.`

### What Next

Please make sure to specify the right library cell pair in the option -user_mapping in the command heal_mv_path.

---

## MV-179

(error) Isolation strategy '%s' in power domain '%s' has self dependency issue on element '%s' and control signal '%s'.

Feedback

### Description

This message occurs when the element defined in an isolation strategy will also be used as control signal in the same isolation strategy. The element will be dropped from the isolation strategy.

### What Next

Review the issued strategy for the self dependency definition.

### See Also

- commit_upf
- create_mv_cells

---

## MV-180

(error) Circular dependencies are found among isolation control signals "%s".

### Description

Isolation process has dependency that elements specified as the isolation signal of an isolation strategy should be implemented after other elements.

This message occurs when there is circular dependencies defined multiple isolation strategies. Isolation control signal processing order can not be resolved. For example:

set_isolation iso1 -elements {ctrl1} -isolation_signal ctrl2 set_isolation iso2 -elements {ctrl2} -isolation signal ctrl1 set_isolation iso3 -elements {ctrl3} -isolation_signal ctrl3

### What Next

Check the strategies using issued isolation control signals and fix the circular dependency.

### See Also

- commit_upf
- create_mv_cells

---

## MV-181

(error) %s pin %s of retention register %s belonging to strategy %s of power domain %s has incorrect connection to %s signal.

Feedback

### Description

This error message indicates that the reported register instance has a wrong connection to the save/restore signal of the applicable retention strategy.

### What Next

If you have instantiated the retention register, make sure the save/restore connection matches the save/restore signal and sense specified by the applicable strategy.

### See Also

- set_retention
- check_mv_design

---

## MV-182

(error) Retention register %s belonging to strategy %s of power domain %s has incorrect backup %s supply connection.

### Description

This error message indicates that the reported register instance has a wrong backup retention supply connection which doesn't match the one specified by the applicable retention strategy.

### What Next

Make sure the backup power/ground pins have the same supply as that of the applicable retention strategy.

### See Also

- set_retention
- check_mv_design

---

## MV-183

(information) Adding inverter %s before enable pin of isolation cell %s (isolation strategy = %s).

### Description

We allow to use isolation library cell that has inverse sense than isolation strategy when there is no isolation library cell with correct sense available. Inverter cell will be automatically inserted to make the isolation control sense logic correct.

Feedback

This message occurs when inverter cell gets inserted before isolation cell enable pin.

**What Next**

This is an information-only message. No action is required.

**See Also**

- create_mv_cells

---

## MV-184

(Information) Tool has found no MV Violation to fix for buffer tree driven by '%s'.

**Description**

This message is printed when *fix_mv_design -verbose* runs but it does not find any MV Violation to fix, therefore the command finished without making changes to the specified buffer tree.

**See Also**

- fix_mv_design

---

## MV-185

(Warning) Attribute *is_upf_mim_primary_instance* can only set to true.

**Description**

The message is reported when setting the attribute *is_upf_mim_primary_instance*. You can only set attribute to value *true*.

**What Next**

- Instead of setting the attribute to false, you can remove the attribute by using *remove_attributes -objects [get_cells <instance_name>] -name "is_upf_mim_primary_instance"*.

**See Also**

- set_attribute
- remove_attributes

Feedback

## MV-186

(Warning) Attribute *is_upf_mim_primary_instance* is already set on instance '%s'.

### Description

The message is reported when setting the attribute *is_upf_mim_primary_instance*. You can only set the attribute to one instance of each module.

### What Next

- Make sure which instance should have defined the attribute.
- If the attribute is no longer required for the mentioned instance, you can remove the attribute by using *remove_attributes -objects [get_cells <instance_name>] -name "is_upf_mim_primary_instance"*.

### See Also

- set_attribute
- remove_attributes

## MV-187

(warning) For pin %s, power supply net %s has less voltage than ground supply net %s voltage.

### Description

This message is printed when there is a pin, whose power supply voltage is less than its ground supply voltage. The supply voltages from PST states need to be checked. Pins from power management cells, standard cells, and TIE cells are checked.

### What Next

Check port states on supply nets, and also check how sup- ply nets connect to pins.

### See Also

- add_port_state
- add_power_state
- connect_supply_net

Feedback

## MV-188

(warning) Repeater is not inserted at pin %s as it does not have any %s.

### Description

This message shows the port where no repeater is inserted due to the absence of driver or load on the port.

### What Next

This is a warning message. No action is required.

### See Also

- set_port_attributes
- set_repeater

## MV-189

(warning) For Domain %s, primary power supply net %s has less voltage than primary ground supply net %s voltage.

### Description

This message is printed when, for a given domain, its power supply voltage is less than its ground supply voltage. The supply voltages from PST states need to be checked.

### What Next

Check port states on supply nets, and also check how sup- ply nets connect to pins.

### See Also

- add_port_state
- add_power_state
- connect_supply_net

## MV-190

(information) Total %d out of %d cells missing supply net connections have been inferred with connect_supply_net.

Feedback

### Description

This message shows the number of cells inferred with connect_supply_net by the tool in DMM prototype mode.

### What Next

This is an information-only message. No action is required if the number of cells missing supply net is expected.

### See Also

- commit_upf

---

## MV-192

(warning) Repeater cell %s found for the port(pin) %s matching the repeater strategy %s, but not associated to the strategy.

### Description

This warning message appears when a cell matches the repeater strategy on the port(pin), but that cell is not associated as a repeater to the strategy.

### What Next

Run commit_upf to trigger association.

### See Also

- commit_upf
- set_repeater

---

## MV-193

(error) Pre-instantiated cell %s is situated in an AO Block Domain and cannot be remapped to a physically legal dual-rail cell.

### Description

This message indicates that the given cell, situated in an AO Block Domain (with either no explicit CSN or with a CSN to the primary supply of the AO Block Domain), cannot be remapped by the tool to a physically legal dual-rail cell.

Feedback

### What Next

If the intention is to let the tool proceed and ignore these errors, app-option mv.cells.ignore_remapping_error_for_pre_instantiated_cells_in_ao_block can be set to true. If not, please check the CSN connections of this cell.

### See Also

- create_mv_cells

---

## MV-194

(warning) There are unmapped %s in the design.

### Description

Generally, connect_pg_net requires all cells to be mapped, but if -create_nets_only option is specified or there are only unmapped power switch placeholders, it will still run. This warning just mentions that there are some unmapped cells or power switch placeholders in the design.

### What Next

Check or remove the unmapped cells or power switches in the design.

### See Also

- create_power_switch
- map_power_switch
- commit_upf

---

## MV-195

(error) The design contains at least one unmapped cell: %s.

### Description

Generally, connect_pg_net requires all cells to be mapped, but if there are unmapped power switch placeholders, it will still run. In this case, there is an unmapped cell in the design which is not a power switch placholder, and therefore, connect_pg_net cannot complete.

### What Next

Make sure all cells are mapped to library cells by running compile command.

Feedback

## MV-196

(error) Unknown option detected from UPF command/block: \n\ %s.

### Description

During UPF loading by linking, when unknown option detects, the tool will mark UPF incompatible and print this message.

### What Next

You can check log file to find the unknown option name, and follow the instructions from *MV-167* to fix the incompatible UPF.

### See Also

- CMD-005
- CMD-010
- MV-167

## MV-197

(error) Cannot use %s options at the same time for command %s.

### Description

This message occurs when the command is called with the specified options. Only one of the specified options can be used at a time.

### What Next

Choose only one of the specified options for the command.

## MV-198

(Error) Supply (%s, %s) of terminal boundary pin %s does not match supply (%s, %s) of %s pin %s.

### Description

This error message occurs on terminal boundary port in either of the following conditions:

1. For input port, driver supply defined in 'set_port_attribute -driver_supply' does not match the supply of the driver pin it connects.
2. For output port, receiver supply defined in 'set_port_attribute -receiver_supply' does not match the supply of the load pins it connects.

Feedback

### What Next

Please make sure to avoid the above conditions.

---

## MV-199

(Information) The name of the %s cell is being changed from %s to %s.

### Description

This message occurs when any repeater or isolation cell name is being changed. Changing of repeater/isolation cell name occurs during commit_upf when the existing cell name fails to obey the repeater/isolation naming convention. The changed name satisfies the naming convention.

set mv.cells.rename_isolation_cell_with_formal_name to false can avoid renaming isolation cells at commit_upf.

### What Next

This is just an information message. No action is required.

### See Also

- commit_upf

---

## MV-200

(Error) Always_on_strategy '%s' is not allowed.

### Description

This error message occurs when an invalid always_on_strategy is specified for command 'set_power_domain_constraints'. 'single_power' and 'dual_power' are the only valid candidates.

### What Next

Please input correct value for option '-always_on_strategy' of command 'set_power_domain_constraints'.

### See Also

- set_power_domain_constraints

Feedback

## MV-201

(warning) Repeater at port %s with repeater supply (%s, %s) is not inserted due to non-availability of matching %s in the libraries.

### Description

This message is issued when the tool cannot insert repeaters due to its non availability in the libraries specified.

Either all the cells of required type has been constrained with lib_cell_purpose or the libraries don't have the buffers/inverters.

### What Next

Check for the presence of the required cells in the library using get_lib_cells. If not present, use proper libraries that have the required lib-cells. If present, check for any constraint on the library cells set by set_lib_cell_purpose. If found, remove the constraints.

### See Also

- set_lib_cell_purpose
- get_lib_cells
- set_dont_touch
- create_mv_cells

## MV-202

(Warning) Cell intance '%s' belongs to a power domain in voltage area '%s', but is placed in voltage area '%s'.

### Description

This warning message is printed when a cell is placed in a voltage area different from the one to which its power domain belongs.

### See Also

- create_voltage_area

## MV-203

(Information) Hierarchy cell '%s' is DesignWare component, %s.

Feedback

### Description

This information message is printed during manual or automatic PG connection. Manual PG connection will be aborted if the connection will cross DesignWare component or SynthRef hierarchy. Auto PG connection will skip connection inside DesignWare component or SynthRef hierarchy.

### See Also

- connect_pg_net

---

## MV-204

(information) Supply net '%s' is inferred as '%s' by name.

### Description

This message is reported for a supply net that cannot be inferred as power or ground by supply set, supply connection, isolation or retention strategy, design attribute user PG type and power switch strategy. The PG net type is inferred according to the net name.

### See Also

- MV-205

---

## MV-205

(information) Supply net '%s' is inferred as '%s' by voltage.

### Description

This message is reported for a supply net that cannot be inferred as power or ground by supply set, supply connection, isolation or retention strategy, design attribute user PG type and power switch strategy. The PG net type is inferred according to the voltage.

### See Also

- MV-204

---

## MV-206

(Warning) Supply net '%s' has been assigned as PG type ground because no known type has been assigned.

Feedback

### Description

This message is reported for a supply net that cannot be inferred as power or ground by supply set, supply connection, isolation or retention strategy, design attribute user PG type and power switch strategy. The PG net type is assigned as ground.

---

## MV-207

(warning) %s %s cell on net %s is skipped because this net is marked as analog.

### Description

Isolation cells, repeaters, and level shifters are not inserted on analog nets.

### What Next

Find out why these nets are marked as analogs. A net becomes analog under the following conditions: 1. If the net is connected to a pin that has "is_analog" liberty attribute 2. If the net is connected to a pin, and that pin is set to analog by "set_port_attributes -is_analog" command.

A net becomes analog if it is connected to at least one analog pin.

### See Also

- set_port_attributes

---

## MV-208

(Warning) Pin(s) '%s' of library cell '%s' is(are) missing the "is_unconnected" attribute

### Description

This warning message appears when the tool identifies pins of a library cell where the "is_unconnected" attribute is expected but missing.

The "is_unconnected" attribute is expected on library pins with the following characteristics:

- The pin has no related power/ground pin information
- The pin is not part of a shorted ("short") path in the cell model

Such pins are treated as internally unconnected by the tool and the "is_unconnected" attribute is used as way to confirm that this is the intent of the library creator.

Feedback

### What Next

If you agree with the identified pin(s) being treated as internally unconnected, you can safely ignore this warning. Otherwise, you can use the set_related_supply_net command on each instance of the cell as an override.

### See Also

- set_port_attributes
- set_related_supply_net

---

## MV-209

(Error) Pin(s) '%s' of library cell '%s' have conflictive "is_unconnected" attribute setting

### Description

This error message appears when the tool identifies pins of a library cell where the "is_unconnected" attribute is conflicting with the rest of the pin model.

The "is_unconnected" attribute is expected on library pins with the following characteristics:

- The pin has no related power/ground pin information
- The pin is not part of a shorted ("short") path in the cell model

Such pins are treated as internally unconnected by the tool and the "is_unconnected" attribute is used as way to confirm that this is the intent of the library creator. A library pin with the "is_unconnected" attribute that does not match the above description is considered a modeling error

### What Next

Review the conflictive library cell

---

## MV-210

(warning) %s strategy %s from power domain %s is specified on Pin/Port %s which is on an analog net.

### Description

This warning message is printed when isolation, level shifter or repeater strategy is specified on analog net. Power management cells will not be inserted on analog net.

Feedback

### What Next

Check the power strategies and the analog net. Add *set_isolation -no_isolation* to avoid apply isolation strategy on analog path.

Add *set_level_shifter -no_shift\P to avoid applying level shifter strategy on analog path.*

Use *set_repeater -exclude_elements* to avoid apply repeater strategy on analog path.

### See Also

- set_port_attributes
- set_isolation
- set_level_shifter
- set_repeater

---

## MV-211

(warning) Pin(s) %sis(are) defined as analog on instance %s of module %s, but same pin(s) are not defined as analog on instance(s) %s of module %s.

### Description

This warning message is printed when ports/pins of different instances of same module differ in analog setting. This happens when user has not used -model option to set -is_analog attribute with set_port_attributes command.

### What Next

Check if is_analog attribute can be set with set_port_attributes -is_analog -model so that all instances of a module have same analog settings.

### See Also

- set_port_attributes

---

## MV-212

(warning) %s '%s' is connected with analog net '%s'.

### Description

This warning message is printed when there is an isolation cell or level shifter connected with analog net. Isolation cell or level shifter should not be inserted on analog net.

Feedback

### What Next

Check if the ISO/LS already exists before apply the is_analog attribute. Remove the ISO/LS if the net is expected to be an analog net.

### See Also

- set_port_attributes
- report_nets

---

## MV-213

(warning) Found net %s connected to PG pin %s, while %s UPF constraint use supply net %s to connected with the same pin. UPF constraint will be given precedence.

### Description

This warning message is printed by infer_supply_from_pg_net verbose mode. if UPF CSN conflict with PG netlist, this warning will be reported for each conflict in verbose mode when app option mv.pg.continue_infer_pg_with_conflict is on. UPF CSN will be preserved, and later resolve_pg_net/connect_pg_net will update PG netlist to be consistent as UPF.

### What Next

Check the existing pg netlist.

### See Also

- infer_supply_from_pg_net
- connect_pg_net
- mv.pg.continue_infer_pg_with_conflict

---

## MV-214

(warning) Buffer cell '%s' is connected with analog net '%s'.

### Description

This warning message is printed when there is a buffer or inverter cell connected with analog net. Buffer or inverter cell should not be inserted on analog net.

### What Next

Check if the buffer cell already exists before apply the is_analog attribute. Remove the buffer cell if the net is expected to be an analog net.

Feedback

**See Also**

- set_port_attributes
- report_nets
- remove_buffers

---

## MV-215

(information) Total %d logical feedthrough buffers have been committed.

**Description**

This message shows the number of logical feedthrough buffers that have been committed.

**What Next**

This is an information-only message. No action is required.

---

## MV-216

(information) Total %d hierarchical ports have been created for logical feedthrough buffering.

**Description**

This message shows the number of hierarchical ports that have been created for logical feedthrough buffering.

**What Next**

This is an information-only message. No action is required.

---

## MV-217

(warning) Analog driver '%s' voltage is %s which is different than the analog voltage of the load '%s' at %s.

**Description**

Two analog pins that are connected should have the same voltage.

**What Next**

Find out if these pins are correctly marked as analog. A pin is set to analog either in .lib file or by using set_ports_attributes command.

Feedback

## MV-218

(warning) A mix of analog and digital pins are connected to each other. Driver pins are {%s}, and load pins are {%s}.

### Description

In general analog and digital pins should not be mixed and connected to each other.

### What Next

Find out if these pins are correctly marked as analog. A pin is set to analog either in .lib file or by using set_ports_attributes command.

## MV-219

(warning) Multiple drivers on unresolved supply net '%s'

### Description

This message indicates that multiple power sources are connected to a net specified as unresolved with the create_supply_net command -resolve option.

### What Next

Supply nets specified as unresolved may receive only one driver. Either disconnect other sources or specify the supply net as parallel.

### See Also

- create_supply_net

## MV-220

(Error) There are uncommitted logical feedthrough buffers in the design. Please run commit_logical_feedthrough before saving it.

### Description

This message indicates that there are uncommitted logical feedthrough buffers in the design. Please perform commit_logical_feedthrough to commit the logical feedthrough buffers.

### What Next

Perform commit_logical_feedthrough.

Feedback

## MV-221

(warning) Missing supply net information on pin %s with related power pin %s and related ground pin %s of cell %s

### NAME

*MV-221* (error) Missing supply net information on pin %s with related power pin %s and related ground pin %s of cell %s

### Description

This message occurs when the tool fails to derive the supply net information from pin.

### What Next

Set the related supply net information on pin manually or do connect_supply_net on the pg pins that this pin is related to.

### See Also

- connect_supply_net

## MV-222

(error) Missing supply net information on port %s

### Description

This message occurs when the tool fails to derive the supply net information from the port.

### What Next

Manually set the related supply net information using the set_related_supply_net command.

### See Also

- set_related_supply_net
- get_related_supply_nets

## MV-223

(error) Missing supply net information on pin %s of cell %s, this pin is missing related power/ground pin information

Feedback

**Description**

This message occurs when the tool fails to derive the supply net information from pin.

**What Next**

Need to associate related power/ground pin information on the signal pins that are missing supply information.

---

## MV-224

(error) Missing supply net information on macro pin %s of cell %s

**Description**

This message occurs when the tool fails to derive the supply net information from macro.

**What Next**

Set the related supply net information on pin manually or do connect_supply_net on the pg pins that this pin is related to.

---

## MV-225

(warning) %s supply net same voltage as %s with same AOness with %s is not available in domain %s, level shifter cell insertion for strategy %s on net %s failed.

**Description**

This message occurs during level shifter insertion in the create_mv_cells command. A level shifter cell could not be inserted because the related supply output of the level shifter is not available in the power domain. Inserted Level Shifter output supply will use the same voltage of load supply but maintain the same AO status as driver supply so avoid introducing ISO issue. In this case, the level shifter cell is not inserted.

**What Next**

To further debug, use report_mv_path -net <net>, where <net> is the net name from the MV-225 warning. This report gives detailed information about the logic connected to that net, the related supplies of that logic, and any power domain boundaries on that net. The related supplies of the connected logic determine the necessary supplies for the level shifter.

A common reason for level shifter insertion failure is that the required supply is not available in the power domain, in which case it may need to be added using the create_supply_port and create_supply_net commands. Alternately, the level shifter may have been inserted in the wrong domain which doesn't include the required supplies, in

Feedback

which case the set_level_shifter -location option can be modified to change whether the level shifter is inserted in the self, parent, or fanout domain hierarchy.

After correcting the UPF constraints, rerun that part of the flow through commit_upf and create_mv_cells and ensure that the MV-225 warnings go away.

### See Also

- create_mv_cells
- report_mv_path
- set_level_shifter

---

## MV-226

(warning) Failed to insert Level Shifter Cell for strategy %s applied on port '%s' (net '%s') with location %s.

### Description

Tool tries to follow user defined Level Shifter strategy but it cannot find useable Level Shifter for current operation conditions in the specified insertion point.

### What Next

- Use *analyze_mv_design -level_shifter -through* on the specific insertion point to report reasons for Level Shifter insertion failure.
- Use *report_mv_path -pin* on the specified port to get information of each insertion point in the path (such operational conditions and applied strategies).
- Check for library cell availability using *report_mv_lib_cells -level_shifter* for appropriate Level Shifter strategy.
- Use *check_mv_design* to get additional warning/error messages that could help to find the root cause.

### See Also

- analyze_mv_design
- map_level_shifter_cell
- report_mv_lib_cells
- report_mv_path
- MV-135

Feedback

- MV-135a
- MV-136
- MV-137
- MV-160

---

## MV-227

(warning) Failed to insert isolation cell for strategy %s of domain %s, which is applied on boundary port '%s' with location %s to isolate load pin '%s'.

### Description

This message is from isolation insertion engine. This warning message indicates that isolation insertion engine can't insert isolation cell due to location.

### What Next

If the specified point has MV violation reported by check_mv_design, you need to check further informational messages to find the root cause.

### See Also

- report_mv_path

---

## MV-228

(Warning) There are uncommitted logical feedthrough buffers in the design. Please run commit_logical_feedthrough before saving it.

### Description

This message indicates that there are uncommitted logical feedthrough buffers in the design. Please perform commit_logical_feedthrough to commit the logical feedthrough buffers.

### What Next

Perform commit_logical_feedthrough.

---

## MV-229

(information) The file extention %s has been appended to the file name %s.

Feedback

### Description

This information message is reported when the output report has been assigned with a file extention.

---

## MV-231

(information) The power domain %s is associated to the voltage area %s by enable_missing_voltage_area.

### Description

This message indicates that when the app option "mv.upf.enable_missing_voltage_area" is true, the tool automatically assigns DEFAULT_VA to power domains which do not yet have associated voltage area.

### What Next

This is just an information message.

### See Also

- mv.upf.enable_missing_voltage_area
- create_voltage_area
- set_voltage_area

---

## MV-232

(error) Failed to set always_on_strategy as single_power on the power domain '%s' because it is an AO Block Domain.

### Description

This warning message occurs when an AO Block Domain is specified in command 'set_power_domain_constraints' and the *-always_on_strategy* is specified with the value *single_power*.

### What Next

No further action needed.

### See Also

- create_power_domain
- set_power_domain_constraints

Feedback

## MV-233

(information) Isolation strategy %s(%s) can't be implemented at domain boundary %s with location %s to isolate path to pin(s)/port(s)%s.

### Description

Isolation cell can't be inserted at location specified in strategy due to hetero-fanout.

### What Next

This is an information-only message. No action is required.

### See Also

- create_mv_cells

## MV-234

(warning) Caution with argument '%s'. '%s'.

### Description

This is a warning message for connect_power_switch command. Among other cases, it may be issued if certain options that are unnecessary for certain modes are entered but being ignored.

### What Next

Review the specific information given by the MV-234 error message and refer to the connect_power_switch man page for complete information about using the command.

### See Also

- connect_power_switch

## MV-235

(error) Either -layers or -region has to be specified.

### Description

This message occurs when neither -layers nor -region option is specified for create_secondary_pg_placement_constraints command.

### What Next

Specify either -layers or -region.

Feedback

### See Also

- create_secondary_pg_placement_constraints

---

## MV-236

(error) secondary_pg_constraints '%s' has been created already.

### Description

This message occurs when the name specified for create_secondary_pg_placement_constraints command conflicts with existing secondary_pg_placement_constraints.

### What Next

Specify differnt name for create_secondary_pg_placement_constraints.

### See Also

- create_secondary_pg_placement_constraints

---

## MV-237

(error) Related supply of data output pin of the retention cell '%s' belonging to strategy '%s' of power domain '%s' doesn't match -use_retention_as_primary requirement of the strategy.

### Description

This error message occurs when user has applied a retention strategy specified with -use_retention_as_primary option to a retention cell which is mapped to a library cell that doesn't have its data output pin related to the backup PG pin.

### What Next

Specify correct library cells or library cell types in map_retention_cell, that would match -use_retention_as_primary requirement. You could also remove -use_retention_as_primary option from the strategy if library doesn't have library cells matching this requirement.

### See Also

- set_retention
- map_retention_cell

Feedback

## MV-238

(warning) Retention library cell %s is unsuitable for retention strategy %s of power domain %s.

### Description

This warning indicates that retention library cell specified in map_retention_cell command is not suitable to use for the retention strategy. This message appears when retention strategy specifies two different signals for save and restore (which indicates 2-pin retention library cell to be used for mapping) and map_retention_cell has 1-pin or 0-pin retention library cells. Such unsuitable library cells will not be considered for mapping.

### What Next

No action is needed if commit_upf doesn't report any error. But if you don't see the expected retention library cell in the netlist or if all library cells are unsuitable, revisit your map_retention_cell and give library cells that are suitable for the strategy.

### See Also

- set_retention
- map_retention_cell
- commit_upf

## MV-239

(warning) PG-MUX Cells %s and %s are PGMUX with equivalent internal/output supply nets %s and %s, but the tool was not able to establish the equivalence of their %s pins

### Description

This message indicates two PG-MUX cells have equivalent internal/output supply nets but the tool was not able to establish equivalence of their input control or PG pins.

### What Next

Find out if the internal/output supply pins of these two PG-MUX cells should be connected. If so, check why their control pins or their input PG pins are not connected to the same net.

### See Also

- connect_supply_net

Feedback

## MV-240

(warning) map_retention_cell is not specified for the retention strategy %s in power domain %s.

### Description

This warning indicates that mapping constraints have not been defined on the retention strategy for the specified power domain using the map_retention_cell command. If registers specified in the set_retention command are SEQGENs, they will be mapped to non-retention library cells and if registers are mapped retention registers, they will be preserved as retention.

### What Next

Define the mapping constraints for the retention strategy using the map_retention_cell command.

### See Also

- set_retention
- map_retention_cell
- report_power_domain

## MV-241

(error) Tool tried to perform auto commit_upf but it failed.

### Description

This error occurs when tool tries to perform commit_upf under the hood but it fails. This error is seen while executing commands that require the UPF intent to be ready.

### What Next

Execute commit_upf and clean up all the errors it reports before proceeding.

### See Also

- commit_upf

## MV-242

(error) Voltage area setup is not ready for physical implementation.

Feedback

### Description

This message occurs when physical implementation commands, such as create_power_switch_ring, create_power_switch_array, create_placement, place_opt etc. are called but the voltage areas have not been created for all power domains. Note, that voltage areas are required for all power domains, except power domains containing only macros and/or I/O pads.

### What Next

If the UPF has been successfully committed then create voltage areas for all power domains that require a voltage area (all power domains except those containing only macro and/or IO pads, including voltage areas in abstracts). You can use the 'check_mv_design' command to identify any power domains that are missing a voltage area, such power domains are reported under the "Power domain rule" section.

Then you can use "create_voltage_area" command to assign voltage areas to power domains, or turn on the "mv.incomplete_upf.enable" to enable incomplete UPF flow to bypass this check with default VA assigned to those "missing VA" domains.

### See Also

- load_upf
- commit_upf
- check_mv_design
- mv.incomplete_upf.enable
- mv.upf.enable_missing_voltage_area
- MV-511

---

## MV-243

(error) Tool tried to perform auto commit_upf but it cannot be done since the design has unpropagated physical blocks. Explicit commit_upf is required for constraints propagation.

### Description

This error occurs when tool tries to perform commit_upf under the hood but it cannot be performed as the design has unpropagated physical blocks.

### What Next

Execute commit_upf and and make sure constraints propagation is done for all the physical blocks.

Feedback

**See Also**

- commit_upf

---

## MV-244

(warning) Cell %s is partially outside the VA-shape(s) for secondary_pg_placement constraint of supply %s

**Description**

This message indicates the cell is not placed inside the VA-shape(s) completely for secondary_pg_placement constraint of the supply.

**What Next**

Run legalize_placement to have the cell placed in the correct VA-shape.

---

## MV-245

(error) Cell %s violation secondary_pg_placement constraint for supply %s

**Description**

This message indicates the cell is not placed in the VA-shape for secondary_pg_placement constraint of the supply.

**What Next**

Run legalize_placement to have the cell placed in the correct VA-shape.

---

## MV-246

(error) Found net %s connected to PG pin %s, it is conflict with supply net %s defined in %s connect_supply_net constraint in UPF.

**Description**

This error message is printed by infer_supply_from_pg_net. if UPF CSN conflict with PG netlist, by default this error will be reported for each conflict.

**What Next**

Check the existing pg netlist.

Feedback

### See Also

- infer_supply_from_pg_net
- mv.pg.continue_infer_pg_with_conflict
- MV-247

---

## MV-247

(error) Total %d conflict(s) are found between existing PG connection in netlist and UPF constraint.

### Description

This error message is printed by infer_supply_from_pg_net. if UPF CSN conflict with PG netlist, by default this error will be reported for conflict summary.

### What Next

User infer_supply_from_pg_net verbose mode to check each conflict.

### See Also

- infer_supply_from_pg_net
- mv.pg.continue_infer_pg_with_conflict
- MV-246

---

## MV-248

(error) Advanced Legalizer has not been turned on.

### Description

This message shows when user trying to do commit_secondary_pg_placement_constraints, but advanced legalizer has not been turned on. Currently only advanced legalizer can support secondary_pg_placement constraints.

### What Next

Please check with legalization team if advanced legalizer can be applied to this design. If yes, you have to turn it on by setting place.legalize.enable_advanced_legalizer to true. If not, this design cannot have secondary_pg_placement constraints.

Feedback

### See Also

- commit_secondary_pg_placement_constraints

---

## MV-249

(error) Power switch insertion/connection cannot be implemented for power switch strategy %s : %s.

### Description

This error message is printed when the power switch insertion or connection cannot be implemented when multiple power domains share a single voltage area. In such a case, power switch strategy from the master domain should be implemented.

### What Next

Please check the power switch strategy, power domain of the strategy and the voltage area that is shared by this power domain.

### See Also

- report_power_domains
- report_voltage_areas

---

## MV-250a

(warning) Power domains %s cannot be assigned to default voltage area due to shared_voltage_area attribute conflict.

### Description

This warning message is printed only when missing va flow and shared va flow are both on. It is issued when multiple power domains are specified in UPF design attribute "shared_voltage_area", but they are conflicting with missing VA domain list which will be assigned to default voltage area by the tool.

### What Next

You can use the 'check_mv_design' command to identify any power domains that are missing a voltage area, such power domains are reported under the "Power domain rule" section.

Then you can use "create_voltage_area" command to assign voltage areas to power domains.

Feedback

**See Also**

- check_mv_design
- report_power_domains
- report_voltage_areas
- mv.upf.enable_missing_voltage_area

---

## MV-250

(error) Power domains of the voltage area and shared_voltage_area attribute are different : power domain '%s' is not specified as part of the %s.

**Description**

This error message is printed when multiple power domains share a single voltage area, but power domains of the voltage area are different from the power domain list of UPF design attribute 'shared_voltage_area'.

**What Next**

Please check the power domains of the voltage area and shared_voltage_area attribute.

**See Also**

- report_voltage_areas

---

## MV-251

(error) Voltage area '%s' is not completely in current physical block '%s'.

**Description**

This message shows when voltage area specified is not completely in current physical block for create_secondary_pg_placement_constraints.

**What Next**

Please only specify voltage area which is completely in currently physical block (doesn't have any portion in linked physical block).

**See Also**

- create_secondary_pg_placement_constraints

Feedback

## MV-252

(error) Supply '%s' is domain primary of voltage area '%s'.

### Description

This message shows when the supply specified is domain primary of the voltage area for create_secondary_pg_placement_constraints.

### What Next

Please only specify voltage area which the supply is non-domain-primary.

### See Also

- create_secondary_pg_placement_constraints

## MV-253

(error) Supply '%s' is not available in the power domain of voltage area '%s'.

### Description

This message shows when the supply specified is not available in the power domain of the voltage area for create_secondary_pg_placement_constraints.

### What Next

Please remove the voltage area if the supply is really not available in it, or update the UPF to make the supply available.

### See Also

- create_secondary_pg_placement_constraints

## MV-254

(error) Cannot find qualified voltage area for constraint '%s'.

### Description

This message shows when tool cannot find any qualified (non-domain-primary and is available in the domain) for the secondary_pg_placement constraint.

### What Next

Please remove this constraint if it's not needed, or recreate the secondary_pg_placement constraint with correct supply and/or voltage_area.

Feedback

**See Also**

- create_secondary_pg_placement_constraints

---

## MV-255

(error) Cannot find PG strap in voltage area '%s' for constraint '%s'.

**Description**

This message shows when tool cannot find any PG strap for the voltage area specified for the secondary_pg_placement constraint (with -layers).

**What Next**

It's possible that the voltage_area is wrongly specifed, or the PG strap hasn't been implemented. If the PG strap hasn't been implemented yet, user should use -region for creating secondary_pg_placement constraint.

**See Also**

- create_secondary_pg_placement_constraints

---

## MV-256

(error) Cannot derive any valid shape for constraint '%s'.

**Description**

This message shows when tool cannot derive any valid shape the secondary_pg_placement constraint.

**What Next**

Remove the constraint if it's not needed, or recreate it with correct options.

**See Also**

- create_secondary_pg_placement_constraints
- commit_secondary_pg_placement_constraints

---

## MV-257

(warning) If set_power_domain_constraints -prefer_tie_off_nwell/-prefer_tie_off_pwell is used, and mv.upf.auto_enable_type2n3_selection is set, set_power_domain_constraints gets precedence.

Feedback

### Description

If mv.upf.auto_enable_type2n3_selection is already set, and -prefer_tie_off_nwell/ -prefer_tie_off_pwell options of set_power_domain_constraints are used, the set_power_domain_constraints command takes precedence. Insulated tied-off cells will be 'preferred' only in those power domains specified with the -domains option of set_power_domain_constraints.

### What Next

The app-option mv.upf.auto_enable_type2n3_selection will be deprecated in future releases. Switch to using the new command options of set_power_domain_constraints command

---

## MV-258

(warning) If set_power_domain_constraints -disallow_forward_biasing/-disallow_reverse_biasing is used, and either of mv.cells.disallow_forward_biasing/ mv.cells.disallow_forward_reverse_biasing/mv.cells.disallow_reverse_biasing are set, set_power_domain_constraints gets precedence.

### Description

If either of mv.cells.disallow_forward_biasing/mv.cells.disallow_forward_reverse_biasing/ mv.cells.disallow_reverse_biasing is already set, and -disallow_forward_biasing/-disallow_reverse_biasing options of set_power_domain_constraints are used, the set_power_domain_constraints command takes precedence.

### What Next

App-options mv.cells.disallow_forward_biasing/mv.cells.disallow_forward_reverse_biasing/ mv.cells.disallow_reverse_biasing will be made obsolete in future releases. Switch to using the new command options of set_power_domain_constraints command

---

## MV-259

(warning) %s %s cell on net %s is skipped because this net is connected with leaf pin related with relative PG pin.

### Description

Isolation cells, repeaters, and level shifters are not inserted on relative signal nets.

### What Next

Find out why these nets are marked as relative signal nets. A net becomes relative signal net when it connected with a leaf signal pin which use relative PG pin as related power pin or ground pin.

Feedback

A leaf PG pin, it will be called as relative PG pin, when it used as 'related power pin' for one signal pin and 'related ground pin' for another signal pin. Relative PG pin can be queried by: 1. "relative PG" liberty attribute that derived by liberty. 2. on-the-fly marking that generated by "set_port_attributes -relative_pg" command.

**See Also**

- set_port_attributes

---

## MV-260a

(Warning) Name based matching has failed for Isolation cell %s since it can match strategy (%s,%s) on pin %s or strategy (%s,%s) on pin %s.

**Description**

This message occurs during isolation cell matching when multiple isolation strategies can be matched to the same isolation cell. Due to the ambiguity, name based isolation cell matching has failed. Topology based matching will still attempt to match and associate the isolation cell.

**What Next**

Use unique power domain and or isolation strategy name in order to distinguish isolation cells.

---

## MV-260

(Error) Isolation cell %s matches both strategy (%s,%s) on pin %s and (%s,%s) on pin %s, matching failed.

**Description**

This message occurs during isolation cell matching when multiple isolation strategies can be matched to the same isolation cell. Due to the ambiguity, isolation cell match fails.

**What Next**

Use unique power domain and or isolation strategy name in order to distinguish isolation cells.

---

## MV-261

(information) Inferring supply port %s for RTL PG port %s.

Feedback

### Description

This message is printed by infer_supply_from_pg_net verbose mode when a new create_supply_port UPF constraint get inferred from PG netlist.

### What Next

This is an information-only message. No action is required.

### See Also

- infer_supply_from_pg_net
- create_supply_port

---

## MV-262

(information) Inferring supply net %s connection to %s port %s.

### Description

This message is printed by infer_supply_from_pg_net verbose mode when a new connect_supply_net UPF constraint get inferred from PG netlist.

### What Next

This is an information-only message. No action is required.

### See Also

- infer_supply_from_pg_net
- connect_supply_net

---

## MV-263

(warning) Driver '%s' (supply set (%s, %s)) cannot drive load pin '%s' (supply set (%s, %s)) due to voltage difference on relative signal path.

### Description

Two pins that are connected by relative signal net should have the same voltage. A net is recoginized as relative signal net if the net is connected with leaf pin which is related with relative PG pin.

It will compare driver power supply with load power supply and compare driver ground supply with load ground supply, violation will be reported if one of the comparison find voltage difference.

Feedback

### What Next

Find out if related power/ground pin of these pins are correctly marked as relative PG pin. A pin is set to relative PG pin either in .lib file or by using set_ports_attributes command.

---

## MV-264

(warning) The option -direction should not be used to speicfy ring connection, please use -ring_connection instead.

### Description

This is a warning message for connect_power_switch command. When the switch cells are placed as a ring, and user has specified -direction to connect them.

### What Next

Suggest to use -ring_direction to specify clockwise or counter_clockwise. It is now a warning to keep backward compatability with old script.

### See Also

- connect_power_switch

---

## MV-265

(warning) Pin '%s' of %s cell '%s' is connected with relative signal net '%s'.

### Description

This warning message is printed when there is an existing cell connected with a net which is connected to leaf pin related with relative PG pin. Only analog pin and cell contains relative PG pin allow to connected with such net.

### What Next

Remove the existing cell if the net is expected to be a relative signal net.

---

## MV-266

(warning) Unknown type %s is skipped.

### Description

Option -type must specify a predefined connection type. This warning message indicates the unknown type is skipped.

Feedback

**What Next**

Make sure predefined type is specified.

---

## MV-267

(warning) Strategy %s is ignored for %s because strategy %s has already been specified.

**Description**

There shall be only one strategy specified for a given setting of option type and/or dft_target_domain/dft_source_supply. The strategy is ignored if tool detects a strategy has been defined.

**What Next**

Make sure the setting of option type and/or dft_target_domain/dft_source_supply is correct.

---

## MV-268

(warning) Isolation cell '%s' pin '%s' is connected with relative supply net %s.

**Description**

This warning message occurs when the isolation cell use relative supply net as primary or backup supply.

**What Next**

Remove the existing isolation cell if it is unexpected.

---

## MV-269

(warning) Level shifter cell '%s' pin '%s' is connected with relative supply net %s.

**Description**

This warning message occurs when the level shifter cell use relative supply net as primary or backup supply.

**What Next**

Remove the existing level shifter cell if it is unexpected.

---

## MV-270

(warning) Retention cell '%s' pin '%s' is connected with relative supply net %s.

Feedback

### Description

This warning message occurs when the retention cell use relative supply net as primary or backup supply.

### What Next

Remove the existing retention cell if it is unexpected.

---

## MV-271

(warning) Power switch cell '%s' use relative supply net %s as %s supply.

### Description

This warning message occurs when the power switch instance use relative supply net as input or output supply.

### What Next

Remove the existing power switch instance if it is unexpected.

---

## MV-272

(warning) Sequential element %s belonging to strategy %s of domain %s has incorrect retention_equivalent attribute set on its library cell. Reason : %s

### Description

This warning message indicates that the reported register instance is a complex non-retention library cell with no functional information and the instance could not be remapped to complex retention library cell based on the *retention_equivalent* attribute specified on the complex non-retention library cell. Reasons will be one of the following. - No complex retention library cell was found in the reference library with the name specified in retention_equivalent attribute - Complex retention library cell was not found in *map_retention_cell* command for the retention strategy - Complex retention library cell was not found suitable for the PVT requirements - Syntax used to specify pin map in retention_equivalent is wrong - Complex retention library cell didn't have all the pins of its complex non-retention counterpart - Not all the pins specified in the pin map were found either on non-retention or on retention library cell

### What Next

Make sure retention_equivalent attribute on the complex non-retention library cell is of the syntax {complex_retention_library_cell_name {old_pin_name new_pin_name}*}. and reference library has a retention cell with the name used in the attribute and it has all the pins of the non-retention variant. Also, pins specified in the pin map of retention_equivalent attribute must all be present on the respective library cells.

Feedback

### See Also

- define_user_attribute
- set_attribute
- check_mv_design
- mv.cells.infer_complex_retention_cells

---

## MV-273

(warning) Power domain '%s' (supply: '%s') and its primary voltage area '%s' (supply: '%s') have different primary %s supplies.

### Description

This message occurs when the primary supply of the specified power domain is different from the supply net of primary voltage area of the power domain. This may potentially cause bad logic as the primary pg pins of the core cells in the voltage area by default are physically connected to and powered by the primary supply of the voltage area.

### What Next

Check the voltage area(s) of the power domain and their power and ground supplies. If the power domain should not belong to the specified voltage area, please consider to add the power domain to a different voltage area.

### See Also

- create_power_domain
- set_voltage_area
- report_power_domain
- report_voltage_areas
- mv.upf.enable_missing_voltage_area

---

## MV-274

(error) region specified for constraint '%s' overlaps with non-qualified voltage_area '%s' (%s).

### Description

This message shows when the region specified for command create_secondary_pg_placement_constraints overlaps with non-qualified voltage_area

Feedback

(a gas-station type of voltage_area or the supply is domain primary or the supply is not available in this voltage_area).

**What Next**

Adjust the region to cover only qualified voltage_area(s).

**See Also**

- create_secondary_pg_placement_constraints

---

## MV-275

(error) The primary %s supply nets '%s' of power domain '%s' and '%s' of power domain '%s' in the same shared voltage area list have different driver types.

**Description**

This error message is printed when multiple power domains are part of a shared voltage area list specified by the 'shared_voltage_area' design attribute, but the primary supply nets of those power domains have different driver types.

**What Next**

Please check the primary supply nets of the power domains of the shared voltage area and their drivers.

**See Also**

- report_power_domains
- report_voltage_areas

---

## MV-276

(error) The primary %s supply '%s' of power domain '%s' is same as primary %s supply of power domain '%s' that is not in the same shared voltage area list.

**Description**

This error message is printed when multiple power domains are part of a shared voltage area list specified by the 'shared_voltage_area' design attribute, but the primary supply net of the specified power domain is same as the primary supply net of the power domain that are not the master power domain. As a result, the primary supply net will not be physically implemented.

Feedback

### What Next

Please check the primary supply nets of the power domains and the power domains of the shared voltage area.

### See Also

- report_power_domains

---

## MV-277

(error) Power switch strategy '%s' is not valid for power switch insertion/implementation.

### Description

This error happens when the specified power switch strategy is invalid for switch cell insertion and physical implementation. Possible causes include undefined output supply net of the strategy, unknown PG type of the supply net etc...

### What Next

Check the power switch strategy.

### See Also

- create_power_switch_ring
- create_power_switch_array
- create_power_switch

---

## MV-278

(warning) Level shifter cell %s belonging to strategy %s of power domain %s does not conform with the mapping directive.

### Description

This error message indicates that the reported level shifter instance is mapped to a library cell that is not in the list specified by the *map_level_shifter_cell* command for the strategy.

This might result in strategy association failure during *commit_upf*.

### What Next

Tool has either mapped or sized this level shifter instance incorrectly. This needs to be fixed.

Feedback

### See Also

- set_level_shifter
- map_level_shifter_cell
- check_mv_design

---

## MV-279

(error) Invalid value specified for one or more command options of set_power_domain_constraints.

### Description

This error message indicates that one or more of the bias related command options of set_power_domain_constraints have invalid values specified. These options take only boolean true/false values. Specifying any other value other than "true" or "false" is an error.

### What Next

check your set_power_domain_constraints command to see if set_power_domain_constraints is being used correctly.

### See Also

- report_power_domain

---

## MV-281

(Information) Query rules have been created for renamed isolation cells. Query rules are used during read_def if app option "file.def.rule_based_name_matching" is set to true and get_* commands if app option "design.enable_rule_based_query" is set to true.

### Description

This information message is printed during commit_upf. When pre-instantiated isolation cells are renamed to conform with Synopsys naming rules, query rules will be created for read_def and get_* commands purpose.

### See Also

- set_query_rules

Feedback

## MV-282

(Warning) The switch cell '%s' is skipped for ring connection when it's specified by option '%s'.

### Description

This warning message is reported only when the switch cell is specified in -object_list/ -voltage_area and power switch ring connection skips it because it's not closest to the voltage area edge or other reasons which have been reported - source pin is same as ack pin or sleep pin is same as ack out.

We only report at most 5 switch cells to avoid bogus message.

### See Also

- connect_power_switch
- MV-030

## MV-283

(error) Retention strategy %s of power domain %s applies to safety register %s.

### Description

This error message indicates that safety registers (TMR candidates and Failsafe FSM registers) are part of retention strategy. Since safety registers are not expected to shut down, retention strategies are not allowed to be defined on safety registers.

### What Next

Refine the UPF to exclude safety registers from being target of *set_retention* commands.

### See Also

- set_retention
- create_safety_register_group
- create_safety_register_rule
- set_safety_register_rule

## MV-284

(warning) Options -layers/-margin/-voltage_areas are specified together with -region. Options -layers/-margin/-voltage_areas will be ignored.

Feedback

### Description

This message occurs when -layers/-margin/-voltage_areas are specified together with -region for create_secondary_pg_placement_constraints command. Only -region will be considered, -layers/-margin/-voltage_areas will be ignored.

### What Next

Specify either -region or -layers/-margin/-voltage_areas.

### See Also

- create_secondary_pg_placement_constraints

---

## MV-285

(error) The driver %s '%s' [supply net '%s', power state '%s'] of enable signal of NOR/NAND isolation cell implemented on the output port %s of MACRO instance %s is less always-on than the load pin '%s' [supply net '%s', power state '%s'] of the cell.

### Description

This message occurs when the related supply net of the enable signal is less always-on than the related supply net of the load pin of an NOR isolation cell implemented on a output port of a MACRO instance.

For single enable NOR isolation inside MACRO, check is performed between the driver of the enable pin to load pin. In the case of multiple enable pins, as they do not drive the output pin of NOR directly, the ON-ness check is performed with the enable pins.

### What Next

Check the power states of the supply nets.

### See Also

- report_net
- report_pst

---

## MV-286

(error) The driver %s '%s' [supply net '%s', power state '%s'] of data input pin (having alive_during_power_up attribute) of NOR/NAND isolation cell implemented in the MACRO instance %s is less always-on than the data pin '%s' [supply net '%s', power state '%s'] of the cell.

Feedback

### Description

This message occurs when the related supply net of the driver of data input pin is less always-on than the related supply net of the data pin with alive_during_power_up attibute of a NOR isolation cell implemented at an input port of a MACRO instance. This can cause subthreshold leakage.

### What Next

Check the power states of the supply nets.

### See Also

- report_net
- report_pst

---

## MV-287

(error) The %s supply [supply net '%s', power state '%s'] of output pin of %s isolation cell implemented in the output port %s of MACRO instance %s is less always-on than the load pin '%s' [supply net '%s', power state '%s'] of the cell.

### Description

This message occurs when the related ground(NOR)/power(NAND) supply net of the output pin is less always-on than the related ground/power supply net of the load pin of the NOR/NAND isolation cell implemented in a output port of a MACRO instance.

### What Next

Check the power states of the supply nets.

### See Also

- report_net
- report_pst

---

## MV-288

(error) The related %s pin [supply net '%s'] of enable pin %s of %s isolation cell implemented in the output port %s of MACRO instance %s is different than the related %s pin of the output pin [supply net '%s'].

Feedback

### Description

This error message occurs when there is a NOR/NAND isolation cell in MACRO output which is driven by single enable pin and the related supply net of the enable pin is different than the related supply net of the output pin.

### What Next

Check the power states of the supply nets.

### See Also

- report_net
- report_pst

---

## MV-289

(warning) Supply net %s is available in power domain %s according to UPF, but not physically available due to secondary pg constraint %s under block %s .

### Description

This message occurs when there is a supply net defined as available in UPF, but is excluded from supply availability due to secondary pg constraints.

### What Next

Check secondary pg constraints

### See Also

- create_secondary_pg_placement_constraints
- remove_secondary_pg_placement_constraints

---

## MV-290

(error)Value of app option mv.upf.save_upf_line_width or mv.upf.save_upf_line_indent is negative.

### Description

This message occurs when value of app option mv.upf.save_upf_line_width or mv.upf.save_upf_line_indent is negative. Both of them should be set as non-negative.

### What Next

Specify a non-negative value by set_app_options.

Feedback

**See Also**

- mv.upf.save_upf_line_width
- mv.upf.save_upf_line_indent

---

## MV-291

(error) The value of app option mv.upf.save_upf_line_width is not bigger than that of mv.upf.save_upf_line_indent.

**Description**

This message occurs when the value of app option mv.upf.save_upf_line_width is less than that of mv.upf.save_upf_line_indent, which will result in the continuing line cannot be output correctly for a command in multiple lines.

The value of app option mv.upf.save_upf_line_width should be bigger than that of mv.upf.save_upf_line_indent

**What Next**

Specify the value of app option mv.upf.save_upf_line_width bigger than that of mv.upf.save_upf_line_indent by set_app_options.

**See Also**

- mv.upf.save_upf_line_width
- mv.upf.save_upf_line_indent

---

## MV-292

(Warning) Always-on strategy cannot be set on a power domain in child design.

**Description**

This message is reported for users try to set always-on strategy on a power domain in child design. Only power domains on current top design can be set successfully.

---

## MV-293

(error) Supply '%s' is unresolved implcit supply net.

**Description**

This message shows when the supply specified for create_secondary_pg_placement_constraints is an unresolved implicit supply net

Feedback

### What Next

Please only specify either explicit supply net or implicit supply net but being resolved already.

### See Also

- create_secondary_pg_placement_constraints

---

## MV-294

(error) Either -supply or -exclude_supply should be specified.

### Description

This message shows when neither -supply nor -exclude_supply is specified for create_secondary_pg_placement_constraints or both -supply and -exclude_supply are specified.

### What Next

Please specify either -supply or -exclude_supply for create_secondary_pg_placement_constraints.

### See Also

- create_secondary_pg_placement_constraints

---

## MV-295

(error) -layers and -region should not be specified together with -exclude_supply.

### Description

This message shows when either -layers or -region is specified together with -exclude_supply for create_secondary_pg_placement_constraits

### What Next

Please do not specify either -layers or -region when -exclude_supply is specified for create_secondary_pg_placement_constraints.

### See Also

- create_secondary_pg_placement_constraints

Feedback

## MV-296

(error) Supply '%s' or its equivalent supply from secondary pg constraint %s cannot be specified as physically unavailable for voltage area '%s', because it is already used by pin or port %s'.

### Description

This message shows when the supply specified in create_secondary_pg_placement_constraint -exclude_supply is already used in one of the existing cells in the netlist, or one of the top level port. When this happens, the secondary pg constraint with -exclude_supply will be ignored

### What Next

Please remove the secondary pg constraint that has primary supply specified in -exclude_supply.

### See Also

- create_secondary_pg_placement_constraints
- remove_secondary_pg_placement_constraints

## MV-297

(error) Supply '%s' cannot be specified as physically unavailable, because it is a supply that required by power management cells in domain %s for power strategy: %s.

### Description

This message shows when the supply specified in create_secondary_pg_placement_constraint -exclude_supply is a supply that might be used by power management cells. When this happens, the secondary pg constraint with -exclude_supply will be ignored

### What Next

Please remove the secondary pg constraint that has primary supply specified in -exclude_supply.

### See Also

- create_secondary_pg_placement_constraints
- remove_secondary_pg_placement_constraints

Feedback

## MV-298

(warning) Supply '%s' cannot be specified as physically unavailable for voltage area '%s', because there are already explicit secondary pg constraints indicating this supply is physically available for the same VA.

### Description

This message shows when the supply specified in create_secondary_pg_placement_constraints -exclude_supply is already a physically available supply derived by other create_secondary_pg_placement_constraints -supply

### See Also

- create_secondary_pg_placement_constraints
- remove_secondary_pg_placement_constraints

## MV-299

(error) Supply '%s' is not available in VA '%s' from UPF standpoitn, but there are '%s' straps available within VA '%s'.

### Description

This message occurs during check_secondary_pg_placement_constraints. It shows that even though a certain supply is not available in VA from UPF standpoint, there are straps available in area which belongs to the VA. This would be a waste of resource, as there will not be cells trying to use the straps.

### What Next

Fix the inconsistency between UPF and the PG network. Either remove the unnecessary straps, or add the secondary supply to UPF.

### See Also

- remove_shapes

## MV-300

(warning) voltage value (corner "%s", %s) of '%s' is invalid.

Feedback

### Description

This warning message is printed when specified voltage value is invalid or uninitialized. As a result, the voltage value in the voltage_map of the budget shell will be incorrect and may cause potential error(s) when generating the budget shell library.

### What Next

Check the voltage values of the specified object.

### See Also

- set_voltage
- report_supply_nets

---

## MV-302

(Information) There is at least one library cell with is_soi attribute and PG connection checks for bias will be performed accordingly.

### Description

This message is to inform users that there are library cells with is_soi attribute and bias PG connection checks will be performed accordingly.

---

## MV-303

(error) Strategy %s is not element based.

### Description

Command set_dft_isolation only accepts element based isolation strategy.

### What Next

Make sure an element-based strategy is specified.

### See Also

- set_isolation

---

## MV-304

(warning) UPF update is not supported when compile_fusion starts from stage %s.

Feedback

### Description

When compile_fusion starts from a stage other than initial_map, UPF update is only supported for gate-level insert_dft flow where UPF update happens between gate-level insert_dft and the successive compile_fusion session which starts from stage initial_place, initial_drc or initial_opto.

### What Next

Check if the UPF update is intended.

---

## MV-305

(error) The isolation enable signal "%s" in strategy "%s" applied on port "%s", cannot isolate the load "%s" from driver "%s".

### Description

The message is given when the constrained value of the isolation signal does not assert the clamp condition on the isolation cell causing an "X" to propagate to the output loads from the driver.

The message is only issued for the enable signal of a isolation strategy which is constrained to a value '0' or '1' using the "-logic_expr" in an add_power_state command.

### What Next

Check if the isolation signal is the correct signal to be isolating the specified driver and load pair and if it's value is correctly constrained in the add_power_state command.

---

## MV-306

(warning) The logic value of the isolation enable signal "%s" in strategy "%s" is not specified or constrained in any system power state. The tool will not perform corruption propagation checks for this strategy.

### Description

The message is seen because only some of the isolation enable signals of the design are specified or constrained in the system power states and others are not.

The message is issued for all the isolation signals which are not specified or constrained in the system power states.

### What Next

Please check if the all the necessary isolation enable signals have been specified in the system power states.

Feedback

## MV-307

(error) cannot use -voltage_area_shape option for check_bufferability when secondary pg placement constraints are not up to date.

### Description

The error is seen when user tries to use -voltage_area_shape option in check_bufferability when secondary pg placement constraints are not committed and up to date.

### What Next

Please do commit_secondary_pg_placement_constraints manually and specify the up to date voltage area shape name with -voltage_area_shape option

### See Also

- commit_secondary_pg_placement_constraints
- create_secondary_pg_placement_constraints

## MV-308

(error) Unable to connect the sleep and ack of the shared power domains: %s.

### Description

This error message is printed when '-connect_sleep_and_ack_in_shared_domains' is specified but the command failed to connect the sleep and ack signals of the shared power domains due to the specified reason.

### What Next

Please check the reason that caused the connection to fail.

### See Also

- connect_power_switch

## MV-309

(error) Invalid %s is specified by '-fine_grained_lib_pins' command option: %s.

### Description

This error message is printed when '-fine_grained_lib_pins' is specified but the specified pairs of sleep and ack library pins of fine-grained macro are invalid. An example of

Feedback

valid option value is : -fine_grained_lib_pins {{macroLib/macroLibCellA/sw macroLib/macroLibCellA/ack} {macroLib/macroLibCellB/sw macroLib/macroLibCellB/ack}}

**What Next**

Please make sure that the sleep and ack library pin pairs of fine-grained macro are valid.

**See Also**

- connect_power_switch

---

## MV-310

(Error) %s supply (%s, %s) on soft macro pin %s is %s on than supply (%s, %s) on its %s pin %s.

**Description**

This error message occurs on soft macro boundary pins in either of the following situations:

1. The driver supply specified in *set_port_attribute -driver_supply* on the pin is more on than the supply on the actual driver pin to which is connected.
2. The receiver supply specified in *set_port_attribute -receiver_supply* on the pin is less on than the supply on the actual load pins to which is connected.

Isolation insertion based on the port attributes on the soft macro boundary pin might not be sufficient for correcting electrical violations on the path.

**What Next**

Correct the supplies set on soft macro boundaries to match the actual supplies in the design.

---

## MV-311

(Warning) %s supply (%s, %s) on soft macro pin %s is %s on than supply (%s, %s) on its %s pin %s.

Feedback

**Description**

This error message occurs on soft macro boundary pins in either of the following situations:

1. The driver supply specified in *set_port_attribute -driver_supply* on the pin is less on than the supply on the actual driver pin to which is connected.
2. The receiver supply specified in *set_port_attribute -receiver_supply* on the pin is more on than the supply on the actual load pins to which is connected.

**What Next**

This is a warning message only. No action is required, but consider correcting the supplies set on soft macro boundaries to match the actual supplies in the design.

---

## MV-312

(Error) Voltage value on %s supply (%s, %s) on soft macro pin %s is different from that of supply (%s, %s) on its %s pin %s.

**Description**

This error message occurs on soft macro boundary pins in either of the following situations:

1. The driver supply specified in *set_port_attribute -driver_supply* on the pin has a different voltage value from that of the supply on the actual driver pin to which is connected.
2. The receiver supply specified in *set_port_attribute -receiver_supply* on the pin has a different voltage value from that of the supply on the actual load pins to which is connected.

Level shifters insertion based on the port attributes on the soft macro boundary pin might not be sufficient for correcting voltage violations on the path.

**What Next**

Correct the supplies set on soft macro boundaries to match the actual supplies in the design.

---

## MV-313

(error) exclude supply net specified in constraint '%s' is gas station '%s' '%s', this supply cannot be excluded.

Feedback

### Description

This message shows when -exclude_supply is specified for a supply that cannot be excluded from the given gas station. The disqualified supply includes gas station power supply, gas station ground supply, gas station nwell supply and gas station pwell supply.

### What Next

remove this secondary pg placement constraint or specify different power/ground/nwell/pwell supply for gas station.

### See Also

- create_secondary_pg_placement_constraints
- create_voltage_area
- set_voltage_area

---

## MV-314

(warning) PG pin connections of some cells have changed. Please run connect_pg_net to reconnect PG nets.

### Description

The *fix_mv_buffer_trees* command has changed PG pin connections of some buffers, inverters, level shifters, or enable level shifters. Existing PG nets connecting to these cells are removed. Please run the *connect_pg_net* command to reconnect these PG pins.

### See Also

- connect_pg_net

---

## MV-315a

(information) Total %d %s cells not mapped were removed(not inserted) in design.

### Description

This message shows the number of each type of mv cells (eg, level shifters, isolation cells, repeater cells, etc) that cannot be mapped successfully with the library cells are removed from the design, they are not finally get inserted in the design.

### What Next

This is an information-only message. No action is required. The unmapped report will be generated to show some failure reason for mapping .

Feedback

### See Also

- create_mv_cells

---

## MV-315

(information) The %s cell %s was not inserted because a matching library cell cannot be found in user technology library.

### Description

This message warning message occurs in create_mv_cells command if a multi-voltage cell is needed but not inserted because a valid reference for the cell is not found in the user provided technology library.

### What Next

Please provide the required technology library cells to avoid multi voltage violations.

### See Also

- create_mv_cells

---

## MV-316

(Error) %s options should be specified together.

### Description

This error message occurs in create_mv_cells command if the option -multibit is specified without the -mapped option.

### What Next

Plase specify the option -multibit along with the option -mapped in create_mv_cells.

### See Also

- create_mv_cells

---

## MV-318

(information) Message '%s' limit (%d) exceeded. Remainder will be suppressed. To print all the messages use '-verbose' option.

Feedback

### Description

This message indicates that the message limit has been exceeded and the remainder will be suppressed.

### What Next

Run create_mv_cells with the -verbose option.

### See Also

- create_mv_cells

---

## MV-319

(Error) Neither option -buffer nor option -diodes is specified in fix_mv_design command.

### Description

This error message occurs when neither option -buffer nor option -diodes is specified in fix_mv_design command. fix_mv_design needs requires user to specify either -buffer or -diodes.

### What Next

Specify either -buffer or -diodes in fix_mv_design command

### See Also

- fix_mv_design

---

## MV-320

(Error) Option -from/-lib_cells/-minimize_dual_rail/-level_shifter/-silent/-debug/-html/-csv/-no_split/-report_all/-mapped/-output cannot be specified together with option -diodes in fix_mv_design command.

### Description

This error message occurs when one of more of the options in the following option lists {-from, -lib_cells, -min_dual_rail, -level_shifter, -silent, -debug, -html, -cvs, -no_split, -report_all, -mapped, -output} is specified together with option -diodes in command fix_mv_design.

### What Next

When specify -diodes in fix_mv_design, do not specify option -from/-lib_cells/-min_dual_rail/-level_shifter/-silent/-debug/-html/-cvs/-no_split/-report_all/-mapped/-output

Feedback

### See Also

- fix_mv_design

---

## MV-321

(information) %d connections are made to connect sleep and ack signals of shared power domains.

### Description

An informational message from connect_power_switch, indicating how many connections have been made to connect the sleep and ack signals of shared power domains from the ack_out pin that is specified as command option.

### What Next

Optionally check that the expected number of connections have been made by connect_power_switch.

### See Also

- connect_power_switch
- create_power_switch

---

## MV-322

(Information) Derive secondary pg constraints based on pg straps. All prior existed tool derived secondary pg constraints will be removed.

### Description

This information message occurs user issues derive_secondary_pg_placement_constraints. All prior existed tool derived secondary pg constraints will be removed first.

### What Next

This is just an information message.

---

## MV-323

(error) MV failed to process multi bit buffer chain.

Feedback

### Description

This message occurs when MV fails to process the multi bit buffer chain during set_eco_power_intent. The failure could be due to input/output not having the same supplies; input being a tree of buffers/inverters other than chain; buffer chain having a mix of multi bit buffers/inverter and normal/single bit buffer/inverters and many others.

### What Next

Please check to make sure the input is correct.

---

## MV-324

(error) MV failed to annotate MV information on buffer/inverters in the multi bit buffer chain.

### Description

This message occurs when MV fails to annotate supply or power domain information on the buffer/inverters in the multi bit buffer chain during set_eco_power_intent. The failure could be due to required supply being not available; PD-VA mismatch being unsolvable; lib cell type being incorrect and many others.

### What Next

Please check the upf to make sure the required supply is available or change the lib cell type so that we can annotate the supply information correctly. If its because PD-VA mismatch issue, please check the PFT setting.

---

## MV-325

(error) Supply '%s' in secondary pg constraint '%s' does not have physical straps in the region specified in the constraint.

### Description

This message occurs during check_secondary_pg_placement_constraints. The secondary pg constraint is trying to specify certain supply being available in a given region. However, there is no physical straps existed in that region.

### What Next

Specify the secondary pg constraint correctly so that it matches the physical straps availability, or re-compile pg.

Feedback

### See Also

- remove_secondary_pg_placement_constraints
- create_secondary_pg_placement_constraints
- compile_pg

---

## MV-326

(warning) Supply '%s' in secondary pg constraint '%s' does not have physical straps in VA '%s' which is specified in the constraint.

### Description

This message occurs during check_secondary_pg_placement_constraints. The secondary pg constraint is trying to specify certain supply being available in a voltage area. However, there is no physical straps existed in that voltage area.

### What Next

Specify the secondary pg constraint correctly so that it matches the physical straps availability, or re-compile pg.

### See Also

- remove_secondary_pg_placement_constraints
- create_secondary_pg_placement_constraints
- compile_pg

---

## MV-327

(warning) Exclude supply '%s' in secondary pg constraint '%s' does have physical straps in VA '%s' which is specified in the constraint.

### Description

This message occurs during check_secondary_pg_placement_constraints. The secondary pg constraint is trying to exclude certain supply being used in a voltage area. However, there is physical straps for that supply existed in that voltage area.

### What Next

Specify the secondary pg constraint correctly so that it matches the physical straps availability, or re-compile pg.

Feedback

**See Also**

- remove_secondary_pg_placement_constraints
- create_secondary_pg_placement_constraints
- compile_pg

---

## MV-328

(error) Tool derived secondary pg constraint '%s' is conflicting with the following user specified secondary pg constraints '%s'.

**Description**

This message occurs during check_secondary_pg_placement_constraints or commit_secondary_pg_constraint. There is a tool derived secondary pg constraint for certain supply. However, there are also user specified secondary pg constraints on the same or equivalent supply. The user specified pg constraints will be honored. The tool derived constraints will be removed automatically during commit_secondary_pg_placement_constraints.

**What Next**

Specify the secondary pg constraint correctly so that there will be no conflicts

**See Also**

- remove_secondary_pg_placement_constraints
- create_secondary_pg_placement_constraints
- check_secondary_pg_placement_constraints
- commit_secondary_pg_placement_constraints

---

## MV-329

(error) Invalid region. Region is outside of the current physical block boundary.

**Description**

You receive this error when you are trying to specify a region which is outside of the current physical block boundary during create_secondary_pg_placement_constraints command. The region you specified should always be within the current physical block boundary.

Feedback

### What Next

Specify a region inside the current physical block boundary and try this command again.

### See Also

- create_secondary_pg_placement_constraints

---

## MV-330

(error) Physical hierarchy '%s' is not ready in terms of secondary pg constraints.

### Description

This message occurs during top level commit secondary pg constraints. It is required that if there are secondary pg constraints in subblocks, there should be commit happened at the block level prior to the blocks being linked back to the top. The tool will error if the blocks are not committed yet when commit is happening at the top.

### What Next

User can do commit_secondary_pg_placement_constraints -commit_subblock or he/she can open the specific subblock and commit secondary pg at the block level.

### See Also

- create_secondary_pg_placement_constraints
- commit_secondary_pg_placement_constraints

---

## MV-331

(error) Margin specified cannot be negative value.

### Description

This message occurs when the margin you specified is less than zero. The tool allows user to specify margin with -margin option in create_secondary_pg_placement_constraints and derive_secondary_pg_placement_constraints. The margin you specified has to be value equal or greater than zero.

### What Next

User can respecify -margin in create_secondary_pg_placement_constraints command or derive_secondary_pg_placement_constraints command.If you do not know the margin value, you do not have to specify any value.

Feedback

### See Also

- create_secondary_pg_placement_constraints
- derive_secondary_pg_placement_constraints

---

## MV-332

(error) Incorrect layer name '%s' specified.

### Description

This message occurs when the layer name you specified is incorrect.
You can only specify interconnect layer names when you specify
-layers in create_secondary_pg_placement_constraints command and
derive_secondary_pg_placement_constraints command.

### What Next

User can respecify -layers in create_secondary_pg_placement_constraints command or derive_secondary_pg_placement_constraints command. You can get a list of interconnect layers by doing get_layers -filter {layer_type==interconnect}

### See Also

- create_secondary_pg_placement_constraints
- derive_secondary_pg_placement_constraints
- get_layers

---

## MV-333

(error) Missing required option(s) -net or -pin for generate_mv_feedback.

### Description

One of -net or -pin option must be specified for command generate_mv_feedback.

### What Next

Correct the options on *generate_mv_feedback* as described in the error message. Refer to the *generate_mv_feedback* manpage for a detailed description of the usage of this command.

### See Also

- generate_mv_feedback

Feedback

## MV-334

(Information) Total number of MV cells in the design.

### Description

This information message is printed during MV cell insertion compile flow step.

### What Next

This is just an information message.

## MV-335

(information) power domain root cell '%s' is being deleted.

### Description

This message occurs during the root cell of a power domain is being deleted since it is an empty hierarchy.

### What Next

To prevent a power domain root cell from being deleted by optimization, app-option mv.upf.preserve_power_domain_root_cells can be set to true.

## MV-336

(warning) The 'set_port_attributes %s %s' setting on macro pin '%s' would result in an isolation violation if used as the %s supply of the macro pin.

### Description

This warning message lets the user know that if the tool were to perform implementation using the mentioned set_port_attribute setting instead of the real driver/load supply an isolation violation would occur.

### What Next

Review the UPF to make sure the setting is correct.

### See Also

- set_port_attributes

## MV-337

(warning) App option %s is not applicable with command %s, and it's setting is ignored.

Feedback

### Description

This message occurs during execution of above specified command where certain app option setting is not applicable. The value set on the app option is ignored by the above command.

### What Next

Check the app option setting is intended. You can ignore this message if app option setting is not intended for this command.

---

## MV-338

(warning) Swapping of complex sequential instance %s to its retention equivalent failed. %s.

### Description

Fusion Complier does automatic swapping of complex sequential instances when *mv.cells.infer_complex_retention_cells* is set to true. When any of the conditions required for the successful swapping is not met, tool issues this warning, along with the reason.

### What Next

Use *check_mv_design* to know the detailed reason(s) for the swapping failure.

### See Also

- set_retention
- map_retention_cell
- check_mv_design
- mv.cells.infer_complex_retention_cells

---

## MV-339

(Warning) Can't find a power domain using (%s, %s) as primary supply nets for %s '%s'.

### Description

This warning message occurs when *-update_power_domain {}* is specifed to *fix_mv_design* command. Because no specific power domain is specified in the option, tool tries to find a power domain using the specified driver supply nets as domain primary supply but fails to find one. Tool will use the current power domain for the specified buffer or inverter.

Feedback

### What Next

Please add a power domain using the specified supply nets as domain primary supply and specify it in *-update_power_domain* option.

### See Also

- fix_mv_design

---

## MV-340

(Warning) Found %d power domains using (%s, %s) as primary supply nets: fix_mv_design chooses %s as power domain for %s '%s'.

Found %d power domains available for %s '%s' in -update_power_domain option: power domain '%s' will be used.

### Description

This warning message occurs when *-update_power_domain {}* is specifed to *fix_mv_design* command. Because no specific power domain is specified in the option, tool tries to find a power domain using the specified driver supply nets as domain primary supply but it found multiple matching power domains. Tool will use the first matching power domain it found for the specified buffer or inverter.

### What Next

Please add a power domain using the specified supply nets as domain primary supply and specify it in *-update_power_domain* option.

### See Also

- fix_mv_design

---

## MV-341

(information) Power and ground nets will be connected to mapped cells only. Unmapped instances and switch placeholder cells in the design are skipped.

### Description

connect_pg_net makes power and ground connections to mapped cells only, and unmapped instances and switch placeholder cells in the design will be skipped.

### What Next

This message is informational. No action is required.

Feedback

### See Also

- connect_pg_net

---

## MV-342

(warning) Pin/Port %s is constrained with attribute async_clamp_value '%s', but the isolation strategy %s from power domain %s that applies on this Pin/Port has the conflicting async_clamp_value '%s'. Async clamp value '%s' will be used while implementing the isolation cell.

### Description

This warning message occurs when the tool encounters design ports or pins that have async clamp value settings that do not match the explicit async clamp value of the isolation strategy that applies to them. During implementation the tool will honor async clamp value settings on the pin being isolated over the async clamp value specified for the isolation strategy being implemented.

Async clamp value settings for pins or ports are set using the *set_port_attributes* command with the *-UPF_async_clamp_value* option.

Explicit async clamp value settings for isolation strategies are managed using the *-UPF_async_clamp_value* option of the *set_isolation command*.

### What Next

Review *set_isolation -async_clamp_value instances* to fix any async clamp value mismatches.

### See Also

- set_port_attributes
- set_isolation

---

## MV-343

(warning) Isolation strategy %s from power domain %s that applies to pin/port %s does not have async_clamp_value specified in set_isolation, nor UPF_async_clamp_value on this pin/port, -async_set_reset option will be ignored.

### Description

This warning message occurs when the isolation strategy is specified with -async_set_reset option, but there is no -async_clamp_value specified in the set_isolation or through set_port_attributes.

Feedback

Async clamp value settings for pins or ports are set using the *set_port_attributes* command with the *-UPF_async_clamp_value* option.

Explicit async clamp value settings for isolation strategies are managed using the *-async_clamp_value* option of the *set_isolation command*.

**What Next**

Review *set_isolation -async_clamp_value* to fix any async clamp value mismatches.

**See Also**

- set_port_attributes
- set_isolation

---

## MV-344

(warning) Pin/Port %s is constrained with attribute clamp_value '%s', "-async_set_reset" and "-async_clamp_value" option of the isolation strategy %s from power domain %s that applies on this Pin/Port will be ignored. Clamp value '%s' will be used while implementing the isolation cell.

**Description**

This warning message occurs when the tool encounters design ports or pins that have clamp value settings which is not latch. "-async_set_reset" and "-async_clamp_value" option will be ignore. During implementation the tool will honor clamp value settings on the pin being isolated.

Clamp value settings for pins or ports are set using the *set_port_attributes* command with the *-clamp_value* option.

**What Next**

Review *set_isolation -clamp_value instances* to fix any clamp value mismatches.

**See Also**

- set_port_attributes
- set_isolation

---

## MV-345

(Error) No power switch cells found in the design.

Feedback

### Description

This error message occurs when no power switch cells can be found in the design when the tool attempts to run connect_power_switch.

### What Next

Please check the design and libraries to see if power switch cells are set up correctly.

### See Also

- connect_power_switch

---

## MV-346

(warning) Isolation signal '%s' of isolation strategy '%s' in power domain '%s' is connected to power/ground pin.

### Description

This message occurs when isolation signal of isolation strategy is driven by power/ground pin.

### What Next

Review isolation signal net connections.

### See Also

- commit_upf
- create_mv_cells

---

## MV-347

(warning) Supply '%s' or its equivalent supply from secondary pg constraint %s is specified as exclude supply, for voltage area '%s', please note that this supply is already used by top port/macro pin/fixed std cell pin %s'.

### Description

This message shows a warning when the supply specified in create_secondary_pg_placement_constraint -exclude_supply is already used by one of the top level port or macro pin or fixed standard cell pin. -exclude_supply is still honored. The specified supply is excluded from further usage by buffering.

Feedback

**See Also**

- create_secondary_pg_placement_constraints
- remove_secondary_pg_placement_constraints

---

## MV-348

(warning) Strategy '%s' specified by command '%s' does not exist in power domain '%s'.

**Description**

This message occurs when the strategy cannot be found in the power domain. Strategy will be dropped from map_retention_clamp_cell command.

**What Next**

Check the domain name or the strategy name to make sure the strategy is created before this command is issued.

**See Also**

- map_retention_clamp_cell

---

## MV-349

(Error) None of retention strategies specified in '%s' exists in power domain '%s'.

**Description**

This warning message occurs when all strategy names specified with map_retention_clamp_cell command are illegal.

**See Also**

- map_retention_clamp_cell

---

## MV-350

(warning) Cannot apply %s on retention strategy '%s' of power domain '%s' because -no_retention is specified.

**Description**

This message occurs when map_retention_clamp_cell is specified for the retention strategy with -no_retention. Strategy will be dropped from map_retention_clamp_cell command.

Feedback

### What Next

Check the specified retention strategy.

### See Also

- map_retention_clamp_cell

---

## MV-351

(error) Library cell specified with %s in map_retention_clamp_cell is already specified.

### Description

This error message occurs when library cells is specified in map_retention_clamp_cell with duplicated "UPF_GENERIC_CLOCK" or "UPF_GENERIC_ASYNC_LOAD".

### See Also

- map_retention_clamp_cell

---

## MV-352

(warning) The reference library cell '%s' of isolation cell '%s' is specified in map_retention_clamp_cell command. It should not be used for isolation cell mapping.

### Description

This message occurs when the reference library cell of isolation cell is specified in map_retention_clamp_cell, which can only be used for retention clamp cell unless it is also specified in map_isolation_cells command.

### What Next

Check the reference library cell of the isolation cell with the mapping library defined by map_isolation_cells and map_retention_clamp_cells.

### See Also

- report_cell

---

## MV-353

(warning) The reference library cell '%s' of retention clamp cell '%s' is inconsistent with associated strategy '%s'.

Feedback

### Description

This message occurs when the retention clamp cell is associated with a strategy but reference library cell is inconsistent with the strategy.

### What Next

Check the reference library cell of the retention clamp cell with the mapping library defined by map_retention_clamp_cell.

### See Also

- associate_mv_cells
- report_cell
- map_retention_clamp_cell

---

## MV-354

(warning) Skipped deriving supply net connection for %s '%s' : %s.

### Description

The command failed to derive UPF supply net connection for the specified object and detailed reason for the failure is given.

### What Next

Check and resolve the issue and reason that causes the failure.

### See Also

- create_left_right_filler_cells

---

## MV-355

(warning) Related %s supply '%s' of bump pin '%s' is different from related supply '%s' of port '%s'.

### Description

This message occurs when related supply of bump pin and port are different. Related supply net of bump pin will be ignored.

### What Next

Check the related supply net of top-level port and bump cell.

Feedback

### See Also

- commit_upf

---

## MV-356

(error) Bump pin '%s' is connected to standard cell '%s' but does not have connection to any ports.

### Description

This message occurs when the specified bump pin is connected to standard cell(s) but does not have any connection to any ports. A bump cell is the physical implementation of the logical port. Unless the bump cell is not connected or used, bump cell should always be connected to a port.

### What Next

Check the net connections of top-level port and bump cell.

### See Also

- commit_upf

---

## MV-357

(information) Net '%s' has been disconnected from %s '%s' to resolve conflicts between power intent and PG netlist.

### Description

This is a informational message indicating the net has been disconnected from the pin/port to resolve conflicts between power intent and PG netlist.

### What Next

Check the net connection and power intent of the specified port/pin.

### See Also

- resolve_pg_nets

---

## MV-358

(information) %s '%s' is already connected to net '%s' and will be skipped.

Feedback

### Description

This is an informational message indicating the specified pin/port is skipped because it has an existing net connection while reconnection is disallowed.

### What Next

Check the net connection of the specified port/pin. If reconnection is desired, rerun the command and specify that reconnection is allowed with the "-reconnect" command option.

### See Also

- connect_pg_net

---

## MV-359

(information) -lib_cells will be obsolete in a near future release. Please replace with -clock_clamp_lib_cells or -async_clamp_lib_cells.

### Description

map_retention_clamp_cell -lib_cells will be obsolete in a near future release, customer should use -clock_clamp_lib_cells or -async_clamp_lib_cells to specify lib cells instead.

### What Next

```
stop using -lib_cells for map_retention_clamp_cell command.
```

---

## MV-360

(warning) Connection to pg pin '%s' is skipped due to illegal supply net connection to supply net '%s'.

### Description

This message occurs when the specified supply net connection to the pg pin is illegal and the supply net is not visible in the scope of the pg pin.

### What Next

Check the supply net connection of the pg pin.

### See Also

- connect_pg_net
- connect_supply_net

Feedback

## MV-361

(warning) Connection to pg pin '%s' is skipped due to name conflict during port punching across different scopes.

### Description

This message occurs when the net connection to the pg pin is skipped because port punching from the scope of supply net to the specified pg pin may cause potential name conflict(s) in the resulting pg netlist.

### What Next

Check the supply net connection of the pg pin.

### See Also

- connect_pg_net
- connect_supply_net
- create_supply_net

## MV-362

(information) Supply net '%s' has been created to resolve name conflict during port punching across different scopes.

### Description

This is an informational message indicating that the named supply net has been created in UPF intent to resolve name conflict during port punching across different scopes.

### What Next

This message is informational. No action is required.

### See Also

- connect_pg_net
- resolve_pg_nets
- MV-361

Feedback

## MV-363

(information) Supply port '%s' has been created to resolve name conflict during port punching across different scopes.

### Description

This is an informational message indicating that the named supply port has been created in UPF intent to resolve name conflict during port punching across different scopes.

### What Next

This message is informational. No action is required.

### See Also

- connect_pg_net
- resolve_pg_nets
- MV-361

## MV-364

(information) PG pin '%s' is reconnected to supply net '%s' to resolve name conflict during port punching across different scopes.

### Description

This is an informational message indicating that the PG pin is reconnected to the specified supply net in UPF intent to resolve name conflict during port punching across different scopes.

### What Next

This message is informational. No action is required.

### See Also

- connect_pg_net
- resolve_pg_nets
- MV-361

Feedback

## MV-365

(information) %d name conflict(s) during port punching across different scopes are resolved by updating to different UPF supply net connection(s).

### Description

The informational message shows the number of name conflict(s) during port punching across different scopes that are resolved by updating to different UPF supply net connection(s).

### What Next

This message is informational. No action is required.

### See Also

- connect_pg_net
- resolve_pg_nets
- MV-361

## MV-366

(warning) Given design is detected to be an SOI design and enable_bias derived will proceed to derive nwell to power and pwell to ground where applicable.

### Description

This message indicates that given design is an SOI design and enable_bias attribute is set to 'derived'. Tool will proceed with auto-derivation of bias supplies similar to non-SOI designs where nwell is derived from power and pwell is derived from ground where applicable.

### What Next

Check enable_bias attribute value setting and refine the UPF if needed.

### See Also

- set_design_attributes
- MV-302

Feedback

## MV-367

(error) Root cells of power domain %s must either be all bias-enabled blocks or all bias-disabled blocks.

### Description

This message indicates that given power domain has some elements with bias enabled and some with bias disabled. Mixing of bias and non-bias blocks in the same power domain is not supported.

### What Next

Check power intent.

### See Also

- set_design_attributes

## MV-368

(error) Setting does not match the current matching type being used.

### Description

It is not allowed to mix the -dft_target_domain/-dft_source_domain options with the -dft_source_supply/-dft_sink_supply options across set_dft_isolation setting. All defined strategies must use one or the other.

### What Next

Make sure that all defined set_isolation_isolation strategy that require source/sink matching use either the -dft_taget_domain and -dft_source_domain options, or the -dft_sink_supply and -dft_source_supply options.

## MV-369

(information) '%s' is determined as supply net which is available in power domain '%s' to drive tie-off pin '%s' with heterogeneous supplies %s.

### Description

The specified supply net is derived as supply net that is available in power domain of the tie-off net to drive the constant pin with heterogeneous supplies.

### What Next

This message is informational. No action is required.

Feedback

### See Also

- connect_pg_net

---

## MV-370

(Error) '-suggest_domain_based' cannot be used due to existing set_dft_isolation strategy with supply-based filtering option.

### Description

The '-suggest_domain_based' argument of the 'check_isolation_coverage' command cannot be used when there are 'set_dft_isolation' strategies present that use the '-dft_source_supply' and/or '-dft_sink_supply' options.

### What Next

Remove the '-suggest_domain_based' argument. Tool will generate 'set_dft_isolation' strategy suggestions with a matching style that follows existing 'set_dft_isolation' strategies

---

## MV-371

(Warning) Cell/Library Cell %s is not supported with this command.

### Description

The set_non_bias_approved_list command can be used to suppress bias checks related to non-bias cell being inserted in a bias domain. However, this check cannot be suppressed for any types of Multi-voltage cells including Isolation, Level-shifter, Enable-Level-Shifter, Repeater, Retention and Always-on buffer/inverter cells.

### What Next

Check and refine the options of set_non_bias_approved_list command.

### See Also

- set_non_bias_approved_list

---

## MV-372

(error) Virtual %s pin '%s' of tap cell '%s' is instantiated in power domain '%s' while its bias supply net '%s' and primary supply net '%s' are different and cause a short.

Feedback

### Description

This message occurs when the virtual bias pin of tap cell is instantiated in a bias power domain and the different bias and primary supply nets cause a short.

### What Next

Check the primary and bias supply nets of the tap cell. Also check whether this tap cell with virtual pg pin(s) should be used in the specified power domain.

### See Also

- report_cell
- report_power_domain

---

## MV-373

(error) -force can not be specified with -supply. -force can only be specified with -exclude_supply.

### Description

This message occurs when -force is specified together with -supply.

### What Next

Please specify -force with -exclude_supply

### See Also

- create_secondary_pg_placement_constraints

---

## MV-374

(error) invalid lib cell type is specified with -lib_cell_types.

### Description

This message occurs when invalid lib cell type is specified with -lib_cell_types. The valid lib cell types are: isolation_cells, level_shifters, enable_level_shifters, retention_registers, repeaters and tie_cells.

### What Next

Please specify valid lib cell types with -lib_cell_types

Feedback

### See Also

- create_secondary_pg_placement_constraints

---

## MV-375

(Warning) Hierarchical cells are not supported with this command. Cell %s will be ignored.

### Description

The set_non_bias_approved_list command can be used to suppress bias checks related to non-bias cell being inserted in a bias domain, however this command does not support hierarchical cells specified with -cells.

### What Next

Check and refine the options of set_non_bias_approved_list command.

### See Also

- set_non_bias_approved_list

---

## MV-376

(Error) Option -from/-tie_cell/-minimize_dual_rail/-level_shifter/-move_within_x/-move_within_y/-mapped/-update_power_domain/-diode cannot be specified together with option -eco in fix_mv_design command.

### Description

This error message occurs when one of more of the options in the following option lists {-from, -min_dual_rail, -level_shifter, -mapped, -tie_cell, -update_power_domain, -move_within_x, -move_within_y, -diode} is specified together with option -eco in command fix_mv_design.

### See Also

- fix_mv_design

---

## MV-377

(information) Total %d DFT ports are added to isolation strategy.

### Description

This message shows the number of DFT ports are added to isolation strategy with -update to address isolation violation on DFT paths.

Feedback

### What Next

This is an information-only message. No action is required.

### See Also

- generate_upf_strategies

---

## MV-378

(warning) Total %d DFT paths with isolation violation not fixed.

### Description

This message shows the number of DFT paths still have isolation violation not fixed.

### What Next

This is an information-only message. To fix the violations, user can use define_upf_isolation_rule to create a new strategy for the violating path.

### See Also

- generate_upf_strategies
- define_upf_isolation_rule

---

## MV-379

(information) DFT port "%s" is added to isolation strategy "%s" to fix iso violation from (%s, %s) to (%s, %s).

### Description

This message occurs when DFT port is added to isolation strategy with -update to address isolation violation on path.

### What Next

This is an information-only message. No action is required.

### See Also

- generate_upf_strategies

---

## MV-380

(warning) Isolation violation on DFT path from "%s" (%s, %s) to "%s" (%s, %s) is not fixed.

Feedback

### Description

This message occurs isolation violation on DFT path can't be auto fixed with generate_upf_strategies.

### See Also

- generate_upf_strategies

---

## MV-381

(warning) Isolation violation on DFT path from "%s" to "%s" is not fixed, as path is hetero-fanout.

### Description

This message occurs when isolation violation on DFT path can't be fixed with generate_upf_strategies.

### See Also

- generate_upf_strategies

---

## MV-382

(info) Connections of %d power/ground pin(s) are created or changed.

### Description

This message occurs when user executes the *connect_pg_net* command. It shows the number of connections of pin(s) that are created or changed.

### See Also

- connect_pg_net

---

## MV-383

(information) Can insert single rail tie %s cell (%d lib cells) powered by supply nets (%s).

### Description

This is an informational messages printed by *check_tie_cell_insertion* command to shows the number of single rail cell were powered by a supply net.

Feedback

## MV-384

(information) Cannot insert single rail tie high cell because there is no appropriate tie cells in the library.

### Description

This is an informational messages printed by *check_tie_cell_insertion* command. It tells if there is any single rail that cannot be found in the library.

### What Next

Verify if you have proper lib-cells available to be used. Also check if their lib-cell purpose is correctly set.

- To check if there are tie high cells on the library run get_lib_cells */* -filter "@function_id=~a0.0"
- To check the purpose of the lib-cell run: get_attribute -objects <lib_name>/<cell_name> -name valid_purposes

e.g: get_attribute -objects bias_std_cells.tied_to/TIEHHVT_tied_to -name valid_purposes

### See Also

- get_lib_cells

## MV-385

(information) Can insert ao tie %s cell (%d lib cells) powered by supply nets (%s).

### Description

This is an informational messages printed by *check_tie_cell_insertion* command to shows the number of AO cell were powered by a supply net.

## MV-386

(information) Cannot insert ao tie high cell because there is no appropriate tie cells in the library.

### Description

This message is reported by *check_tie_cell_insertion* command when the user library has no AO tie high cells.

Feedback

### What Next

Verify if you have proper lib-cells available to be used. Also check if their lib-cell purpose is correctly set.

- To check if there are tie high cells on the library run get_lib_cells */* -filter "@function_id=~a0.0 && @always_on==true"
- To check the purpose of the lib-cell run: get_attribute -objects <lib_name>/<cell_name> -name valid_purposes

e.g: get_attribute -objects bias_std_cells.tied_to/TIEHHVT_tied_to -name valid_purposes

### See Also

- get_lib_cells

---

## MV-387

(information) Can insert insulated ao tie low cell (%d lib cells) powered by supply nets (%s).

### Description

This is an informational messages printed by *check_tie_cell_insertion* command to shows the number of insulated AO cells which were powered by the specified supply net.

---

## MV-388

(information) Cannot derive supply required to power tie high/low cells.

### Description

This message is reported by *check_tie_cell_insertion* when there is no supply nets to power tie cells.

### What Next

Verify if you declare all the neccessary supply nets in the UPF.

---

## MV-389

(error) Isolation rule "%s" with source/sink supply (%s, %s) is functional equivalent to isolation rule "%s" with (%s, %s) defined earlier. Second rule will be dropped.

Feedback

### Description

It is not allowed to have multiple isolation rules with functional equivalent source and sink supply. Second rule will be dropped.

### What Next

Make sure that isolation rule defined have different source/sink.

### See Also

- define_upf_isolation_rule

---

## MV-390

(error) Isolation rule with name "%s" is defined earlier. Second rule will be dropped.

### Description

It is not allowed to have multiple isolation rules with same name. Second rule will be dropped.

### What Next

Make sure that isolation rule defined have different name.

### See Also

- define_upf_isolation_rule

---

## MV-391

(information) Isolation rule "%s" can't be used for boundary port "%s" in domain "%s", as isolation supply "%s" is not available in this domian.

### Description

Isolation rule can't be used to generate isolation strategy in domain if isolation supply is not available.

### What Next

Make isolation supply available to the domain before running generate_upf_strategies command.

Feedback

### See Also

- define_upf_isolation_rule
- generate_upf_strategies
- MV-392
- MV-393

---

## MV-392

(information) Isolation rule "%s" can't be used for boundary port "%s" in domain "%s", which is either not specified in -domains or specified in -exclude_domains.

### Description

Isolation rule can't be used to generate isolation strategy in domain if domain is not specified in -domains or specified in -exclude_domains.

### What Next

Make isolation supply available to the domain before running generate_upf_strategies command.

### See Also

- define_upf_isolation_rule
- generate_upf_strategies
- MV-391
- MV-393

---

## MV-393

(information) Isolation rule "%s" can't be used for boundary port "%s" in domain "%s", as location "%s" can't be implemented.

### Description

Isolation rule can't be used to generate isolation strategy in domain if isolation location can't be implemented.

Feedback

### See Also

- define_upf_isolation_rule
- generate_upf_strategies
- MV-391
- MV-392

---

## MV-394

(information) Isolation rule "%s" can't be used for boundary port "%s" in domain "%s", as path's driver supply (%s, %s) and load supply (%s, %s) does not match isolation rule source and sink supply.

### Description

Isolation rule can't be used to generate isolation strategy for the path if driver supply and load supply do not match with isolation rule source and sink supply.

### See Also

- define_upf_isolation_rule

---

## MV-395

(information) Isolation rule "%s" can't be used for boundary port "%s" in domain "%s" with target location "%s".

### Description

Isolation rule can't be used to generate isolation strategy in domain if target location does not match.

### See Also

- define_upf_isolation_rule
- generate_upf_strategies
- MV-391
- MV-393

---

## MV-451

(information) The net %s can be buffered with single rail buffers/inverters.

Feedback

### Description

The check_bufferability command analyzed the bufferability on this net and concluded that this net can be buffered with single rail buffers or inverters.

### What Next

If you believe a single-rail repeater is acceptable on this net, no further action is required. However, if you believe otherwise, please check the related supply nets of the driver and loads.

---

## MV-452

(information) Can buffer with single rail buffers ( number of buffers %d ) and inverters ( number of inverters %d ) powered by ( power %s, ground %s ).

### Description

check_bufferability command analyzes the bufferability and concludes that the tool can buffer between the driver and loads in the given voltage area with this number of single rail buffers and this number of single rail inverters with the relevant power and ground.

### What Next

This is an information-only message. No action is required.

### See Also

- check_bufferability

---

## MV-453

(information) Can insert single rail buffers (%d lib_cells) and inverters (%d lib_cells) with supply nets (%s).

### Description

The *check_bufferability* command found a valid buffering supply and available lib_cells for single rail buffering. No buffering problems are detected.

### What Next

The net is bufferable, no further action is required. For further information about how the buffering supply is determined, or which lib_cells are available for buffering, use *check_bufferability -verbose*.

Feedback

### See Also

- check_bufferability

---

## MV-454

(information) Can insert dual rail buffers (%d lib_cells) and inverters (%d lib_cells) with supply nets (%s).

### Description

The *check_bufferability* command found a valid buffering supply and available lib_cells for dual rail buffering. No buffering problems are detected.

### What Next

The net is bufferable, no further action is required. For further information about how the buffering supply is determined, or which lib_cells are available for buffering, use *check_bufferability -verbose*.

### See Also

- check_bufferability

---

## MV-455

(information) Can insert insulated dual rail buffers (%d lib_cells) and inverters (%d lib_cells) with supply nets (%s).

### Description

The *check_bufferability* command found a valid buffering supply and available lib_cells for insulated dual rail buffering. No buffering problems are detected.

### What Next

The net is bufferable, no further action is required. For further information about how the buffering supply is determined, or which lib_cells are available for buffering, use *check_bufferability -verbose*. Insulated dual rail buffers are only used in bias designs, in cases where the power domain primary nwell bias supply is less-on than the required backup supply for the dual rail cell. If a standard dual rail buffer is expected, use *report_pst* to examine the power state table and see why an insulated cell is required.

### See Also

- check_bufferability
- report_pst

Feedback

## MV-456

(information) Cannot insert single rail buffers or inverters because of no available lib_cells. But required supply nets (%s) are available.

### Description

The *check_bufferability* command found a valid supply for single rail buffering, but no lib_cells are available to use. No other buffering problems are detected.

### What Next

The net is not bufferable due to a problem with lib_cell availability. Individual lib_cells may be unavailable for one of the following reasons:

1. An incompatible *set_lib_cell_purpose*
2. A *set_dont_touch* constraint on the lib_cell
3. A *set_target_library_subset* constraint excludes this lib_cell in this hierarchy
4. A mismatch in site_def between the lib_cell and the available site_rows in the voltage area
5. Incompatible supply rails: for example, a backup power pin when a single rail cell is expected
6. Bias mismatch: for example, a lib_cell with nwell/pwell pg_pins when none are expected
7. Voltage mismatch: the lib_cell is not characterized for the supply voltage

Identify specific lib_cells you expect should be available for buffering this net, and use *check_bufferability -verbose* to get a list of all unavailable lib_cells with the reason they are unavailable.

### See Also

- check_bufferability
- set_lib_cell_purpose
- set_dont_touch
- set_target_library_subset

## MV-457

(information) Cannot insert dual rail buffers or inverters because of no available lib_cells. But required supply nets (%s) are available.

Feedback

### Description

The *check_bufferability* command found a valid supply for dual rail buffering, but no lib_cells are available to use. No other buffering problems are detected.

### What Next

The net is not bufferable due to a problem with lib_cell availability. Individual lib_cells may be unavailable for one of the following reasons:

1. An incompatible *set_lib_cell_purpose*
2. A *set_dont_touch* constraint on the lib_cell
3. A *set_target_library_subset* constraint excludes this lib_cell in this hierarchy
4. A mismatch in site_def between the lib_cell and the available site_rows in the voltage area
5. Incompatible supply rails: for example, a backup power pin when a single rail cell is expected
6. Bias mismatch: for example, a lib_cell with nwell/pwell pg_pins when none are expected
7. Voltage mismatch: the lib_cell is not characterized for the supply voltage

Identify specific lib_cells you expect should be available for buffering this net, and use *check_bufferability -verbose* to get a list of all unavailable lib_cells with the reason they are unavailable.

### See Also

- check_bufferability
- set_lib_cell_purpose
- set_dont_touch
- set_target_library_subset

---

## MV-458

(information) Cannot insert insulated dual rail buffers or inverters because of no available lib_cells. But required supply nets (%s) are available.

### Description

The *check_bufferability* command found a valid supply for insulated dual rail buffering, but no lib_cells are available to use. No other buffering problems are detected.

Feedback

### What Next

The net is not bufferable due to a problem with lib_cell availability. Individual lib_cells may be unavailable for one of the following reasons:

1. An incompatible *set_lib_cell_purpose*
2. A *set_dont_touch* constraint on the lib_cell
3. A *set_target_library_subset* constraint excludes this lib_cell in this hierarchy
4. A mismatch in site_def between the lib_cell and the available site_rows in the voltage area
5. Incompatible supply rails: for example, a backup power pin when a single rail cell is expected
6. Bias mismatch: for example, a lib_cell with nwell/pwell pg_pins when none are expected
7. Voltage mismatch: the lib_cell is not characterized for the supply voltage

Identify specific lib_cells you expect should be available for buffering this net, and use *check_bufferability -verbose* to get a list of all unavailable lib_cells with the reason they are unavailable.

### See Also

- check_bufferability
- set_lib_cell_purpose
- set_dont_touch
- set_target_library_subset

---

## MV-459

(information) Cannot insert buffers or inverters because no valid supply for buffering.

### Description

The *check_bufferability* command did not find a valid supply for buffering, so the net segment cannot be buffered. No other bufferability problems were detected, though lib_cell availability checking is skipped, because it cannot be performed without a valid buffering supply.

Feedback

### What Next

For most nets, the buffering supply is determined by first considering the related supply of all drivers and loads being considered. Find the related supplies of the drivers and loads from the output of *check_bufferability -verbose*, or by using the *get_related_supply_nets* command on the driver and load pins. From the set of related supplies of drivers and loads, a valid buffering supply must satisfy two criteria:

1. It must be available in the voltage area being buffered, or a PST equivalent supply must be available. Supply availability can be determined from the *check_bufferability -verbose* output, or from *report_power_domain*. PST equivalence of supplies can be determined from the *report_pst* PST tables, or using the *check_supply_equivalence* command.
2. It must not cause an MV violation (isolation or voltage shifting violation). This means the supply must be of a compatible voltage with all driver and load related supplies. It also must be less-than-or-equal-on as the driver related supply, and more-than-or-equal-on than all load related supplies. The voltages and on-ness are determined from the supply states as defined by either the *add_port_state* or *add_power_state* command.

For special nets that have loads which are isolation cells, enable level shifter cells, or overdriven level shifter cells, the buffering supply can only be the driver related supply or a PST equivalent supply. The load related supplies are not considered for these nets.

Use this information about buffering supplies to understand why no supply was available, and what can be done to resolve the problem. Possibilities include:

o Make sure the correct voltage area is specified to *check_bufferability* for analysis.

o This net segment may not be bufferable, but the net could still be buffered if other net segments are bufferable. Use *check_bufferability* to investigate other segments of the same net for bufferability.

o Some supplies that are restricted from use in the power domain of the voltage area may need to be made available through the UPF constraints for buffering to occur. By default, all supplies are available in all power domains, as long as they are in the scope of the power domain. Restrictions may have been in the UPF by specifying *create_supply_net -domain* to define a domain specific supply. Alternately, *create_power_domain -supply* can be used to restrict extra supplies available in a power domain.

o Floorplan changes may be necessary if a net must cross a wide unbufferable voltage area.

### See Also

- check_bufferability
- get_related_supply_nets

Feedback

- report_power_domain
- report_pst
- check_supply_equivalence
- add_port_state
- add_power_state
- create_supply_net
- create_power_domain

---

## MV-461

(information) The net %s can be buffered with dual rail buffers/inverters.

### Description

The check_bufferability command analyzed the bufferability on this net and concluded that this net can be buffered with dual rail buffers or inverters.

### What Next

If you believe a dual-rail repeater is acceptable on this net, no further action is required. However, if you believe otherwise, please check the related supply nets of the driver and loads.

---

## MV-462

(information) Can buffer with dual rail buffers (number of buffers %d ) and inverters ( number of inverters %d ) powered by ( power %s, ground %s )

### Description

check_bufferability command analyzes the bufferability and concludes that the tool can buffer between the driver and loads in the given voltage area with this number of dual rail buffers and this number of dual rail inverters.

### What Next

This is an information-only message. No action is required.

### See Also

- check_bufferability

Feedback

## MV-463

(information) Single rail buffers (%d lib_cells) and inverters (%d lib_cells) with supply nets (%s) are available, but there are other issues that will prevent or restrict buffering.

### Description

The *check_bufferability* command found a valid buffering supply and available lib_cells for single rail buffering. But, buffering is prevented or restricted by other reasons.

### What Next

The bufferability problems will be listed in the 'Bufferability Problems' section of the report. They may include:

- The net has no driver
- The net has multiple drivers
- The net is dont_touch
- The net is ideal
- The net is tri-state
- The net belongs to a read only block
- The net is not in the block of the given voltage area
- The net's hierarchy is not in the domain scope of the given voltage area
- Physical feedthrough buffering must be enabled to buffer the net in the given voltage area

### See Also

- check_bufferability

## MV-464

(information) Dual rail buffers (%d lib_cells) and inverters (%d lib_cells) with supply nets (%s) are available, but there are other issues that will prevent or restrict buffering.

### Description

The *check_bufferability* command found a valid buffering supply and available lib_cells for dual rail buffering. But, buffering is prevented or restricted by other reasons.

Feedback

### What Next

The bufferability problems will be listed in the 'Bufferability Problems' section of the report. They may include:

- The net has no driver
- The net has multiple drivers
- The net is dont_touch
- The net is ideal
- The net is tri-state
- The net belongs to a read only block
- The net is not in the block of the given voltage area
- The net's hierarchy is not in the domain scope of the given voltage area
- Physical feedthrough buffering must be enabled to buffer the net in the given voltage area

### See Also

- check_bufferability

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## MV-465

(information) Insulated dual rail buffers (%d lib_cells) and inverters (%d lib_cells) with supply nets (%s) are available, but there are other issues that will prevent or restrict buffering.

### Description

The *check_bufferability* command found a valid buffering supply and available lib_cells for insulated dual rail buffering. But, buffering is prevented or restricted by other reasons.

### What Next

The bufferability problems will be listed in the 'Bufferability Problems' section of the report. They may include:

- The net has no driver
- The net has multiple drivers
- The net is dont_touch
- The net is ideal

Feedback

- The net is tri-state
- The net belongs to a read only block
- The net is not in the block of the given voltage area
- The net's hierarchy is not in the domain scope of the given voltage area
- Physical feedthrough buffering must be enabled to buffer the net in the given voltage area

### See Also

- check_bufferability

---

## MV-466

(information) Supply nets (%s) are available for single rail buffering, but cannot insert buffers or inverters because of no available lib_cells. There are also other issues that will prevent or restrict buffering.

### Description

The *check_bufferability* command found a valid buffering supply for single rail buffering. But, there are no lib_cells that can be used to implement that buffering. In addition, buffering is prevented or restricted by other reasons.

### What Next

The net is not bufferable in part due to a problem with lib_cell availability. Individual lib_cells may be unavailable for one of the following reasons:

1. An incompatible *set_lib_cell_purpose*
2. A *set_dont_touch* constraint on the lib_cell
3. A *set_target_library_subset*constraint excludes this lib_cell in this hierarchy
4. A mismatch in site_def between the lib_cell and the available site_rows in the voltage area
5. Incompatible supply rails: for example, a backup power pin when a single rail cell is expected
6. Bias mismatch: for example, a lib_cell with nwell/pwell pg_pins when none are expected
7. Voltage mismatch: the lib_cell is not characterized for the supply voltage

Feedback

Identify specific lib_cells you expect should be available for buffering this net, and use *check_bufferability -verbose* to get a list of all unavailable lib_cells with the reason they are unavailable.

The other bufferability problems will be listed in the 'Bufferability Problems' section of the report. They may include:

- The net has no driver
- The net has multiple drivers
- The net is dont_touch
- The net is ideal
- The net is tri-state
- The net belongs to a read only block
- The net is not in the block of the given voltage area
- The net's hierarchy is not in the domain scope of the given voltage area
- Physical feedthrough buffering must be enabled to buffer the net in the given voltage area

**See Also**

- check_bufferability

---

## MV-467

(information) Supply nets (%s) are available for dual rail buffering, but cannot insert buffers or inverters because of no available lib_cells. There are also other issues that will prevent or restrict buffering.

**Description**

The *check_bufferability* command found a valid buffering supply for dual rail buffering. But, there are no lib_cells that can be used to implement that buffering. In addition, buffering is prevented or restricted by other reasons.

**What Next**

The net is not bufferable in part due to a problem with lib_cell availability. Individual lib_cells may be unavailable for one of the following reasons:

1. An incompatible *set_lib_cell_purpose*
2. A *set_dont_touch* constraint on the lib_cell

Feedback

3. A *set_target_library_subset*constraint excludes this lib_cell in this hierarchy
4. A mismatch in site_def between the lib_cell and the available site_rows in the voltage area
5. Incompatible supply rails: for example, a backup power pin when a single rail cell is expected
6. Bias mismatch: for example, a lib_cell with nwell/pwell pg_pins when none are expected
7. Voltage mismatch: the lib_cell is not characterized for the supply voltage

Identify specific lib_cells you expect should be available for buffering this net, and use *check_bufferability -verbose* to get a list of all unavailable lib_cells with the reason they are unavailable.

The other bufferability problems will be listed in the 'Bufferability Problems' section of the report. They may include:

- The net has no driver
- The net has multiple drivers
- The net is dont_touch
- The net is ideal
- The net is tri-state
- The net belongs to a read only block
- The net is not in the block of the given voltage area
- The net's hierarchy is not in the domain scope of the given voltage area
- Physical feedthrough buffering must be enabled to buffer the net in the given voltage area

**See Also**

- check_bufferability

---

## MV-468

(information) Supply nets (%s) are available for insulated dual rail buffering, but cannot insert buffers or inverters because of no available lib_cells. There are also other issues that will prevent or restrict buffering.

Feedback

### Description

The *check_bufferability* command found a valid buffering supply for insulated dual rail buffering. But, there are no lib_cells that can be used to implement that buffering. In addition, buffering is prevented or restricted by other reasons.

### What Next

The net is not bufferable in part due to a problem with lib_cell availability. Individual lib_cells may be unavailable for one of the following reasons:

1. An incompatible *set_lib_cell_purpose*
2. A *set_dont_touch* constraint on the lib_cell
3. A *set_target_library_subset*constraint excludes this lib_cell in this hierarchy
4. A mismatch in site_def between the lib_cell and the available site_rows in the voltage area
5. Incompatible supply rails: for example, a backup power pin when a single rail cell is expected
6. Bias mismatch: for example, a lib_cell with nwell/pwell pg_pins when none are expected
7. Voltage mismatch: the lib_cell is not characterized for the supply voltage

Identify specific lib_cells you expect should be available for buffering this net, and use *check_bufferability -verbose* to get a list of all unavailable lib_cells with the reason they are unavailable.

The other bufferability problems will be listed in the 'Bufferability Problems' section of the report. They may include:

- The net has no driver
- The net has multiple drivers
- The net is dont_touch
- The net is ideal
- The net is tri-state
- The net belongs to a read only block
- The net is not in the block of the given voltage area

Feedback

- The net's hierarchy is not in the domain scope of the given voltage area
- Physical feedthrough buffering must be enabled to buffer the net in the given voltage area

### See Also

- check_bufferability

---

## MV-469

(information) Cannot insert buffers or inverters because no valid supply for buffering, and/or other issues that will prevent or restrict buffering.

### Description

The *check_bufferability* command did not find a valid supply for buffering, so the net segment cannot be buffered. In addition, there are other reasons that will prevent or restrict buffering on the net. In this case, lib_cell availability is inconclusive. Checking is skipped, because it cannot be performed without a valid buffering supply.

### What Next

For most nets, the buffering supply is determined by first considering the related supply of all drivers and loads being considered. Find the related supplies of the drivers and loads from the output of *check_bufferability -verbose*, or by using the *get_related_supply_nets* command on the driver and load pins. From the set of related supplies of drivers and loads, a valid buffering supply must satisfy two criteria:

1. It must be available in the voltage area being buffered, or a PST equivalent supply must be available. Supply availability can be determined from the *check_bufferability -verbose* output, or from *report_power_domain*. PST equivalence of supplies can be determined from the *report_pst* PST tables, or using the *check_supply_equivalence* command.
2. It must not cause an MV violation (isolation or voltage shifting violation). This means the supply must be of a compatible voltage with all driver and load related supplies. It also must be less-than-or-equal-on as the driver related supply, and more-than-or-equal-on than all load related supplies. The voltages and on-ness are determined from the supply states as defined by either the *add_port_state* or *add_power_state* command.

For special nets that have loads which are isolation cells, enable level shifter cells, or overdrive level shifter cells, the buffering supply can only be the driver related supply or a PST equivalent supply. The load related supplies are not considered for these nets. Use this information about buffering supplies to understand why no supply was available, and what can be done to resolve the problem.

Feedback

The other bufferability problems will be listed in the 'Bufferability Problems' section of the report. They may include:

- The net has no driver
- The net has multiple drivers
- The net is dont_touch
- The net is ideal
- The net is tri-state
- The net belongs to a read only block
- The net is not in the block of the given voltage area
- The net's hierarchy is not in the domain scope of the given voltage area
- Freeze ports are found and not able to find LFT hierarchy
- No LFT buffer insertion is allowed for the given voltage area, please check its voltage area routing rule.
- Physical feedthrough buffering must be enabled to buffer the net in the given voltage area

**See Also**

- check_bufferability

---

## MV-471

(information) The net %s cannot be buffered

**Description**

Analyze_bufferability command analyzes the bufferability on this net and concludes that this net can not be buffered.

**What Next**

You need to check for the availability of both the required supply net and the appropriate dual rail buffers.

---

## MV-472

(information) Cannot insert buffer/inverter between the given driver and loads in the given voltage area.

Feedback

### Description

check_bufferability command analyzes the bufferability and concludes that the tool cannot buffer between the driver and loads in the given voltage because of various reasons. Please use -verbose to find detailed information.

### What Next

You need to check for the availability of the supplies in the given voltage area

### See Also

- check_bufferability

---

## MV-491

(information) Required supply is ( power %s, ground %s ), cannot find appropriate single rail buffers/inverters in the library.

### Description

This message occurs when there is no required single rail buffers/inverters in the libraries

### What Next

Please make sure to link the required logical and physical libraries during library preparation stage.

---

## MV-492

(information) Required supply is ( power %s, ground %s ), cannot find appropriate dual rail buffers/inverters in the library.

### Description

This message occurs when there is no required dual rail buffers/inverters in the libraries

### What Next

Please make sure to link the required logical and physical libraries during library preparation stage.

---

## MV-493

(Error) Failed to create ISO/LS cell instance %s due to system problem.

Feedback

### Description

This is a low level system error occurring during create_mv_cells. This command is attempting to insert an isolation or level shifter cell, but is unable to due to a system problem. The error is not related to MV criteria for cell insertion. It can be caused by memory or other machine issues on the icc2_shell host machine.

### What Next

This is a serious error impacting the entire icc2_shell session, which may become unstable or crash due to a system issue. Check the host machine to ensure enough memory is available for this job. Restart your job with more memory or on a new host machine to try to resolve this error.

---

## MV-498

(error) Found %d power domain%s without valid voltage area:%s.

### Description

This message occurs during physical implementation when tool found power domains which do not have valid voltage areas. Note that voltage areas should be created for all power domains during physical implementation (except power domain containing only macros and/or I/O pads).

### What Next

Create voltage areas for all listed power domains. You can run *check_mv_design* to verify that all power domains have a valid voltage area.

### See Also

- check_mv_design

---

## MV-501

(error) PG net %s is not match with the input supply %s of power switch strategy %s of domain %s.

### Description

This error occurs when the specified PG net associated with the pg_strategy or pg_strap does not match with the input supply specified by the UPF power switch strategy. Switch insertion fails because inserting the power switch would be a violation of the power switch strategy.

Feedback

### What Next

You could fix the mismatch in two ways:

1. Changing the UPF power switch requirements
2. Specifying vertical straps or pg_strategy that matches the required supply of the UPF power switch strategy.

Use report_power_domain or save_upf commands to view the specified power switch strategy and related supply. To view available pg_strategy definitions and associated PG nets, use the report_pg_strategies command.

### See Also

- create_power_switch_array
- create_power_switch
- report_pg_strategies
- report_power_domains

---

## MV-502

(warning) Voltage area region %s is not effective.

### Description

This message occurs during create_power_switch_array/ring when the voltage area region is abutted or over-lapped with other voltage area regions of the same voltage area.

### What Next

Please recreate your voltage area or use set_voltage_area -merge_regions to merge all abutted and overlapping shapes of this voltage area.

### See Also

- create_power_switch_array
- create_power_switch_ring
- set_voltage_area

---

## MV-503

(error) Cannot specify more than one voltage area shape.

Feedback

### Description

User can only specify one voltage area shape at a time when creating power switch array or ring.

### What Next

Specify only one voltage area shape as input.

### See Also

- create_power_switch_array
- create_power_switch_ring

---

## MV-504

(error) Invalid -align_pg_mesh argument. Please specify { pg_xPitch pg_yPitch pg_xOffset pg_yOffset }

### Description

This message occurs when -align_pg_mesh argument specified by users is invalid. Users need to provide four numbers representing xPitch, yPitch, xOffset, yOffset of the power mesh.

### What Next

Please provide valid pitch and offset of the power mesh to be aligned.

### See Also

- create_power_switch_array

---

## MV-505

(error) Power switch cell %s has multiple input or output supply pins.

### Description

This error occurred because incompatible switch cells have been used. Tool only supports power switch cell with one input supply pin and one output supply pin.

### What Next

You can either replace the power switch cells with switch cells that are supported by the tool, or manually connect the PG pin of the existing power switch cells using connect_pg_net command.

Feedback

Use report_cells -power with the specified cell to report the related PG pin information for the cell to verify the extra input or output supply pins exist.

**See Also**

- create_power_switch_array
- create_power_switch_ring
- create_power_switch
- connect_pg_net
- report_cells

---

## MV-506

(warning) User specified %s is adjusted to %.3f to make it compatible with finfet grid.

**Description**

This message occurs when user specified pitch is not multiple of finfet grid pitch. The value is rounded to be multiple of the finfet grid pitch.

**See Also**

- create_power_switch_ring
- create_power_switch_array

---

## MV-507

(Error) User specified power switch cell dimension is not compatible with finfet grid.

**Description**

This message occurs when user specified power switch cell height/width is not multiple of finfet grid pitch.

**What Next**

Please check finfet grid settings or change power switch cell.

Feedback

### See Also

- create_power_switch_ring
- create_power_switch_array
- check_finfet_grid

---

## MV-508

(warning) The primary voltage area of power domain '%s' is still assigned to default voltage area by the tool.

### Description

This message occurs when the app option 'mv.upf.enable_missing_voltage_area' is set to false or missing VA flow is turned off, however the primary voltage area of the specified power domain is still assigned to default voltage area by the tool. You also can use 'check_mv_design' command to identify any power domain not assigned to a user-specified voltage area, such power domains are reported under "Power domain rule" section.

### What Next

Please use the "create_voltage_area" command to assign voltage areas to those power domains for missing VA off flow.

### See Also

- mv.upf.enable_missing_voltage_area
- MV-119
- check_mv_design

---

## MV-509

(error) No valid driver cell is specified.

### Description

This message occurs when -placement_type is array and no valid driver cell is specified. A valid driver power switch cell must be specified with -driver option for an array based power switch placement pattern.

### What Next

Please specify a valid driver cell.

Feedback

### See Also

- set_power_switch_placement_pattern

---

## MV-510

(warning) MV data or voltage area setup is not ready for physical implementation.

### Description

This message occurs when physical implementation commands, such as create_power_switch_ring, create_power_switch_array, create_placement, place_opt etc. are called but the MV data is not ready. There are two main reasons that cause this message to be issued.

1. commit_upf has not been ran successfully 2. voltage areas have not been created for all power domains. Note, that voltage areas are required for all power domains, except power domains containing only macros and/or I/O pads.

### What Next

If commit_upf has not been ran successfully, invoke the 'commit_upf' command and make sure the power intent is committed, resolve any errors that are reported until commit_upf succeeds.

If the UPF has been successfully committed then create voltage areas for all power domains that require a voltage area (all power domains except those containing only macro and/or IO pads, including voltage areas in abstracts). You can use the 'check_mv_design' command to identify any power domain not assigned to a voltage area, such power domains are reported under the "Power domain rule" section. Then you can use the "create_voltage_area" command to assign voltage areas to power domains, or turn on the "mv.incomplete_upf.enable" app option to enable the incomplete UPF flow to bypass this check.

### See Also

- load_upf
- commit_upf
- check_mv_design
- mv.incomplete_upf.enable
- mv.upf.enable_missing_voltage_area
- MV-242

Feedback

## MV-511a

(error) MV voltage area setup is not ready for physical implementation.

### Description

This message occurs when physical implementation commands, such as place_opt, route_opt, clock_opt, refine_opt etc. are called but there are some power domains which don't have user-specified voltage area and are still being assigned the default voltage area by the tool.

Note, that voltage areas are required for all power domains, except power domains containing only macros and/or I/O pads, including voltage areas in abstracts.

### What Next

You can use the 'check_mv_design' command to identify any power domain not assigned to a user-specified voltage area, such power domains are reported under the "Power domain rule" section. Then you can use the "create_voltage_area" command to assign voltage areas to power domains, or turn on the "mv.incomplete_upf.enable" app option to enable the incomplete UPF flow to bypass this check.

### See Also

- load_upf
- commit_upf
- check_mv_design
- mv.incomplete_upf.enable
- mv.upf.enable_missing_voltage_area
- MV-511
- MV-242

## MV-511

(error) MV data or voltage area setup is not ready for physical implementation.

### Description

This message occurs when physical implementation commands, such as create_power_switch_ring, create_power_switch_array, create_placement, place_opt etc. are called but the MV data is not ready. There are two main reasons that cause this message to be issued.

Feedback

1. commit_upf has not been ran successfully 2. voltage areas have not been created for all power domains. Note, that voltage areas are required for all power domains, except power domains containing only macros and/or I/O pads.

**What Next**

If commit_upf has not been ran successfully, invoke the 'commit_upf' command and make sure the power intent is committed, resolve any errors that are reported until commit_upf succeeds.

If the UPF has been successfully committed then create voltage areas for all power domains that require a voltage area (all power domains except those containing only macro and/or IO pads, including voltage areas in abstracts). You can use the 'check_mv_design' command to identify any power domain not assigned to a voltage area, such power domains are reported under the "Power domain rule" section. Then you can use the "create_voltage_area" command to assign voltage areas to power domains, or turn on the "mv.incomplete_upf.enable" app option to enable the incomplete UPF flow to bypass this check.

**See Also**

- load_upf
- commit_upf
- check_mv_design
- mv.incomplete_upf.enable
- mv.upf.enable_missing_voltage_area
- MV-242

---

## MV-512

(error) Lib cell %s is not mapped to power switch %s.

**Description**

This error happens when the specified lib cell is not mapped to the specified power switch.

**What Next**

Check your UPF setting.

Feedback

### See Also

- create_power_switch_ring
- create_power_switch_array
- map_power_switch

---

## MV-513

(warning) Specified %s point (%.4f %.4f) is projected to the closest point (%.4f %.4f) on the boundary of power switch insertion region.

### Description

This message occurs when -start_point or -end_point specified by users is not on the boundary of power switch insertion region. The tool automatically projects the specified point to the boundary.

### What Next

Please check if the specified/projected point is correct.

### See Also

- create_power_switch_ring

---

## MV-514

(warning) Switch cells are dropped along edge (%.3f %.3f) (%.3f %.3f).

### Description

This message occurs when switch cells cannot be inserted along certain edge. Switch cells may be dropped because they are outside the core area, or blocked by blockages, including macros, keep-out margin, placement blockages, other voltage areas and existing switch cells.

### What Next

If users want to have switch cells outside the core area, please set app option "plan.mtcmos.allow_power_switch_outside_core", i.e. set_app_options -name plan.mtcmos.allow_power_switch_outside_core -value true.

### See Also

- create_power_switch_ring

Feedback

## MV-515

(warning) No power switches are inserted because there is no overlapping between voltage area regions and user specified boundary.

### Description

This message occurs when user specified boundary is not overlapping with voltage area regions.

### What Next

Please check the boundary of power switch insertion region

### See Also

- create_power_switch_array

## MV-516

(error) report_mv_path cannot report %s simultaneously.

### Description

This message occurs when two or more of -isolation, -level_shifter or -repeater options are specified with single report_mv_path command. Since the path driving and loading conditions are processed differently for Isolation, Level Shifter and Repeater, only one type of the report can be done. If need to report more than one, please run the command separately with -isolation, -level_shifter or -repeater option. Default is -isolation.

### See Also

- report_mv_path
- create_mv_cells
- commit_upf
- load_upf

## MV-517

(warning) Power switches are inserted along the outer boundary of donut shape VA only.

Feedback

### Description

This message occurs when users create power switch ring for donut shape VA. Power switches are only inserted along the outer boundary. No power switches are inserted along the holes.

### See Also

- create_power_switch_ring

---

## MV-518

(warning) Start point location (%.4f %.4f) and end point location (%.4f %.4f) are the same. A full power switch ring will be created.

### Description

This warning occurs when the options -start_point and -end_point are used for creating a partial switch ring, but the specified (or projected) points overlap, resulting in a full switch ring.

### What Next

The warning is a notification that a full ring is created, despite the use of -start_point and -end_point options intended for creating a partial switch ring. If a partial ring is expected, validate and update the specified start and end point values.

### See Also

- create_power_switch_ring

---

## MV-519

(error) Invalid argument for option -pg_strategy. The argument should be a triplet {pg_strategy_name net_name layer_name}.

### Description

This message occurs when the argument of -pg_strategy is not valid.

### What Next

Please check -pg_strategy argument.

### See Also

- create_power_switch_array

Feedback

## MV-520

(warning) Cell instance %s is not Isolation, Level Shifter or Repeater cell.

### Description

This message occurs when non-MV cell is input for -cell option. No MV report will be done for such cell.

### What Next

Correct the cell or use report_cell command to report normal cell.

### See Also

- report_cell

## MV-521

(error) None vertical straps belong to the input supply net of power switch %s. Power switches cannot be inserted.

### Description

This error occurs when the -pg_straps option is used to align the power switches with vertical straps, but the specified collection of straps do not include any vertical supply straps. Power switches can only be inserted under vertical straps.

### What Next

Evaluate the collection of power straps to ensure at least one vertical input supply strap exists.

### See Also

- create_power_switch_array

## MV-522

(error) Options -boundary, -voltage_area and -voltage_area_shape are mutually exclusive for create_power_switch_ring.

### Description

This message occurs when -boundary, -voltage_area and -voltage_area_shape are specified together for create_power_switch_ring.

Feedback

### What Next

Specify either voltage area or voltage area shape or polygon.

### See Also

- create_power_switch_ring

---

## MV-523

(error) Lib cell %s is not a valid switch cell.

### Description

This error happens when the specified lib cell does not have valid sleep/acknowledge pins.

### What Next

Check your library cell.

### See Also

- create_power_switch_ring
- create_power_switch_array
- map_power_switch
- create_power_switch

---

## MV-524

(error) Cannot find library cell %s.

### Description

This message occurs when specified library cell is not found in the design referenced library.

### What Next

Only library cells in the design referenced libraries can be used with -lib_cell option. Check for the specified cell reference using the get_lib_cells command. You can also use the get_ref_libs command to see a list of libraries referenced by the current design.

Feedback

**See Also**

- get_lib_cells
- report_ref_libs
- create_power_switch_array
- create_power_switch_ring
- set_power_switch_placement_pattern

---

## MV-525

(error) Power switch placement pattern %s does not exist.

**Description**

This error occurs when a power switch placement pattern is specified using the -pattern option, but the pattern does not exist.

**What Next**

The -pattern option can only be used to specify a predefined placement pattern. You can use the command report_power_switch_placement_patterns to view a list of existing placement patterns defined in the design. You can create a new placement pattern with the set_power_switch_placement_pattern command.

**See Also**

- set_power_switch_placement_pattern
- report_power_switch_placement_patterns
- create_power_switch_ring
- create_power_switch_array

---

## MV-526

(warning) User specified %s value is smaller than the %s of the %s, reset %s to the %s of the %s.

**Description**

This message occurs when user specified pitch is smaller than the width/height of power switch cell/pattern. Tool automatically resets the pitch to the width/height of power switch cell/pattern.

Feedback

### What Next

If using the set_power_switch_placement_pattern check the pitch specified to create_power_switch_array, this should be greater than the pattern given. For individual cell insertion please check the pitch given is greater than the width/height of the cell as relevant. Please double check the pitch value.

### See Also

- set_power_switch_placement_pattern
- create_power_switch_array

---

## MV-527

(error) User specified collection of wires is not legal. Please specify a collection of wires of same PG net.

### Description

This message occurs when user specified wires for -pg_straps option are not legal for power switch insertion. The reason could be 1: there are non-PG wires in the collection. 2: there are multiple nets in the collection.

### What Next

Please specify a collection of wires of same PG net.

### See Also

- create_power_switch_array

---

## MV-528

(error) Power switch placement pattern syntax error: %s.

### Description

This message occurs when the syntax for the set_power_switch_placement_pattern command is not correct.

### What Next

Review the error message details and the manpage for set_power_switch_placement_pattern to correct the syntax.

Feedback

### See Also

- set_power_switch_placement_pattern

---

## MV-529

(warning) Found unused isolation cell '%s'%s.

### Description

This message is printed when an isolation cell exists but it is not used. This can happen when the isolation cell does not have any drivers or it does not have any loads. Presence of isolation cells triggers design constraining, and this may result in QOR loss in the final netlist.

### What Next

Remove cell from the design

---

## MV-530

(error) Net %s does not have a non-PG real driver. A multi-voltage path report cannot be generated.

### Description

This error occurs when the specified pin or net on a multi-voltage path is traced, but a drive is not found for the subcircuit. report_mv_path cannot generate a report for a path without a driver.

### What Next

Check the path for a driver manually using the report_nets -connections <net_name> command or try using the all_fanin -flat -to <net_name> command to help locate the source of the subcircuit.

### See Also

- all_fanin
- report_net

---

## MV-531

(error) Setup of Voltage Area '%s' with multiple Power Domains is invalid: %s.

Feedback

### Description

This error message occurs when the specified Voltage Area with multiple Power Domains is not set up properly. Power Domains must be electrical-equivalent in order to be implemented together as one Voltage Area.

### What Next

- Check the Power Domains of the Voltage Area. • Check the UPF intent of each power domain.

### See Also

- create_voltage_area
- remove_voltage_areas
- set_voltage_area
- create_power_domain
- report_app_options
- mv.upf.multi_pd_single_va_strict_check

---

## MV-532

(error) Unable to re-associate voltage area '%s' with power domains: %s.

### Description

This error message occurs when the tool fails to re-associate voltage area (retained after reset_upf) with power domain(s) and supply net(s) in commit_upf after new UPF is loaded.

### What Next

Delete the voltage area if it is no longer needed or correct. Otherwise check and modify the UPF intent accordingly.

### See Also

- load_upf
- reset_upf
- commit_upf
- create_voltage_area
- remove_voltage_areas

Feedback

## MV-533

(warning) The default %s with name %s already exists.

### Description

The app option doesn't have any effect since the default supply net or supply port is already created.

### What Next

Set the app option prior the default supply net or supply port is created.

### See Also

- report_app_options
- set_app_options

## MV-534

(warning) Unable to re-associate default voltage area with power domains: %s.

### Description

This warning message occurs when the tool fails to re-associate the default voltage area with the specified power domain(s) in commit_upf after new UPF is loaded. The specified power domain(s) will be removed from the default voltage area.

### What Next

Check and modify the UPF intent accordingly.

### See Also

- load_upf
- reset_upf
- commit_upf
- create_voltage_area

## MV-535

(warning) Isolation cell '%s'%s has been associated with isolation strategy '%s' but it is unused.

Feedback

### Description

This message is printed when an isolation cell exists due to the specified isolation strategy but it does not have any load or it does not have any driver. Presence of isolation cells triggers design constraining, and this may result in QOR loss in the final netlist.

### What Next

Review your isolation strategy and remove cell from the design

### See Also

- report_power_domain

---

## MV-536

(error) Please specify either "-power_switch" or "-lib_cell" option

### Description

This error happens when user does not specify either -lib_cell or -power_switch option for *create_power_switch_ring* or *create_power_switch_array* command. One of the two options is required for these commands to proceed. These options are used by the command to find the library cell to be instantiated.

### What Next

Specify either -power_switch or -lib_cell options with correct values.

### See Also

- create_power_switch_ring
- create_power_switch_array
- create_power_switch

---

## MV-537

(error) Please specify one of the following options: "-power_switch","-voltage_area" or "-voltage_area_shape"

### Description

This error happens when user does not specify one the following options for *create_power_switch_ring* or *create_power_switch_array* commands: -power_switch, -voltage_area or -voltage_area_shape. One of the these options is required for these

Feedback

commands to proceed. These options are used by the command to determine the hierarchy where library cells are to be inserted.

**What Next**

Specify one of these options : -power_switch, -voltage_area or -voltage_area_shape with correct values.

**See Also**

- create_power_switch_ring
- create_power_switch_array
- create_power_switch

---

## MV-538

(error) Please specify "-power_switch" if "-pg_strategy" is specified.

**Description**

This error happens when user does not specify -power_switch for *create_power_switch_ring* or *create_power_switch_array*. The options is required for these commands to proceed. These options are used by the command to determine how power switch should be aligned to pg strategy.

**What Next**

Specify -power_switch with correct values.

**See Also**

- create_power_switch_ring
- create_power_switch_array
- create_power_switch

---

## MV-540

(warning) %s related supply net(s) of switch cell '%s' does not match supply net '%s' specified in UPF.

**Description**

This message occurs when the related supply net for control or ack pins of a given switch cell does not match the supply nets specified with the -supply_set option of

Feedback

create_power_switch command. The related supply nets of a control or ack pin are derived from the supply net connections of its related PG pins.

**What Next**

Check the supply set specified by the -supply_set option of the create_power_switch command.

**See Also**

- report_cell
- set_domain_supply_net
- connect_supply_net
- create_power_switch

---

## MV-541

(error) Lib cell %s is skipped as it conflicts with the power switch strategy %s.

**Description**

This error happens when the specified lib cell conflicts with the power strategy it is mapped to. This happens when the strategy specifies a header cell and library cell is a footer cell or vice versa.

**What Next**

Check the library cell list specified for map_power_switch.

**See Also**

- create_power_switch_ring
- create_power_switch_array
- map_power_switch

---

## MV-542

(error) Specified options "-orient" if "-snap_to_site_row" are conflict.

**Description**

This error happens when user specifies -orient R90, R270, MXR90 or MYR90. Those orientation cannot snap power switch to siterow.

Feedback

### What Next

Please adjust value of -orient or set -snap_to_site_row to falses.

### See Also

- create_power_switch_ring
- create_power_switch_array
- create_power_switch

---

## MV-543

(error) Voltage area setup is not ready for design implementation with missing va.

### Description

This message occurs when voltage area setup is not ready for design implementation because some power domains do not have a user-specified voltage area and cannot be automatically assigned to default voltage area by the tool. Note, that voltage areas are required for all power domains, except power domains containing only macros and/or I/O pads.

### What Next

You can use the 'check_mv_design' command to identify any power domains that are missing a voltage area, such power domains are reported under the "Power domain rule" section. Also, check why such power domain(s) cannot be assigned to default voltage area automatically.

Then you can use "create_voltage_area" command to assign voltage areas to power domains.

### See Also

- check_mv_design
- report_power_domains
- report_voltage_areas
- MV-250a
- mv.upf.enable_missing_voltage_area

Feedback

## MV-545

(information) Support for always-on library cells without primary power pin is enabled: regular dual-rail library cells will not be used.

### Description

This message occurs when app option *mv.cells.ao_cells_without_primary_pg_pin* is set to true. This will enable the support of always-on library cells without primary power pin in the tool. Whenever there is a need for dual-rail buffers or inverters, only always-on library cells without primary power pin will be considered. Regular dual-rail library cells will be skipped.

### What Next

No action required.

### See Also

- set_app_options
- report_app_options

## MV-546

(error) Regular dual-rail library cell '%s' cannot be used for cell '%s' because it has primary power pin.

### Description

This message occurs when app option *mv.cells.ao_cells_without_primary_pg_pin* is set to true but a regular dual-rail library cell is used by the design. This is illegal because only always-on library cells without primary power pin can be used in dual-rail situations when the app option is enabled.

### What Next

If the library cell is a buffer or inverter, you can run *fix_mv_design -buffer* to correct the problem. If there is any dont_touch attribute on the cell, please remove it so that tool can swap the library cell into an always-on library cell without primary power pin.

### See Also

- check_mv_design
- fix_mv_design
- report_cell

Feedback

## MV-547

(warning) Backup power pin '%s' of always-on library cell '%s' which has no primary power pin is connected to domain primary power '%s'.

### Description

This message occurs when app option *mv.cells.ao_cells_without_primary_pg_pin* is set to true and the backup power pin of an always-on library cells without primary power pin is connected to the same supply net as the domain primary power. This is illegal because the backup power pin of such cell should be connected to a supply net which is different from the domain primary power.

### What Next

If the library cell is a buffer or inverter, you can run *fix_mv_design -buffer* to correct the problem. If there is any dont_touch attribute on the cell, please remove it so that tool can swap the library cell into an always-on library cells with no primary power pin.

### See Also

- check_mv_design
- fix_mv_design
- report_cell

## MV-548

(warning) The backup supply net '%s' of always-on cell '%s' which has no primary power pin is PST-equivalent to domain primary power '%s'.

### Description

This message occurs when app option *mv.cells.ao_cells_without_primary_pg_pin* is set to true and the supply net of backup PG pin of an always-on library cell without primary power pin is PST-equivalent to the domain primary supply net.

Typically, an always-on library cell without primary power pin is required when a cell is powered by a backup supply that is different from its primary supply net. If the primary and backup supply of a cell is PST-equivalent, always-on library cell without primary power pin is not required and it should be swapped to a single-rail cell to avoid QOR loss in the final netlist.

### What Next

Check the supply net connections of the PG pins and the power states of the supply nets. If the always-on library cell without primary power pin not required, please swap it to a

Feedback

single-rail cell. If the library cell is a buffer or inverter, you can run *fix_mv_design -buffer* to correct the problem.

**See Also**

- fix_mv_design
- report_cell
- report_pst

---

## MV-550

(warning) Net %s has multiple drivers, skip LS insertion/association evaluation of the net.

**Description**

This message occurs during level shifter insertion/association, the net has multiple drivers, cannot determine real driver of the path for supply voltage evaluation.

**What Next**

Check the driver of the path.

---

## MV-560

(error) Inserting mapped MV cells in Wordview flow design is not supported.

**Description**

This error happens when user specified -mapped in wordview flow for create_mv_cells to insert mapped PM cells.

**What Next**

Insert gtech PM cells in wordview flow.

---

## MV-562

(warning) Always-On legalization failed in buffer trees. Design may have voltage, isolation, or other MV errors.

**Description**

Tool fails to fix MV violations in buffer trees in the design. These violations may include: - Voltage violation - Isolation violation - PVT mismatch - Bias mismatch - Target library subset violation - Lib cell subset violation - Lib cell purpose violation

Feedback

### What Next

Please run check_mv_design to identify any MV violations existing in the design.

### See Also

- check_mv_design
- fix_mv_design
- set_target_library_subset
- set_lib_cell_purpose

---

## MV-611

(warning) %d nets required level shifters but matching level shifters were not found.

### Description

This warning message shows the total number of global nets which requires Level Shifters but no suitable non-GTECH Level Shifters can be found in user library. In general, to successfully insert Level Shifters, all conditionals including Level Shifters type, operating conditions, Standard Cell Main Rail setting, input/output voltage range, and availability of supply nets should be satisfied.

### What Next

- Run *analyze_mv_design -level_shifter -global_report -verbose* to know which error affect each global net. For each global net (driver pin), run *analyze_mv_design -level_shifter -through* to obtain specific failure reasons and try to solve those issues before continuing.
- Use *create_mv_cells -verbose* to display possible reasons why required Level Shifters are not inserted in each net.
- Use *report_mv_path to report strategies applied to path.*
- Check *analyze_mv_feasibility* to report mapping failures for Enabled Level Shifters.

### See Also

- analyze_mv_design
- analyze_mv_feasibility
- create_mv_cells

Feedback

- report_mv_path
- report_mv_lib_cells

## MV-612

(information) report all heterogeneous paths only with -all_heterogeneous_paths option, any other option will be ignored.

### Description

This information message just to inform the report_mv_path with -all_heterogeneous_paths option will report all heterogeneous paths in the design. It will ignore all other options.

## MV-616

(Warning) Level shifter '%s'%s of level shifter type '%s' is incompatible with the voltage conversion from %s to %s, but user level shifter strategy '%s' with rule '%s' is set.

### Description

This message occurs when the type of level shifter library cell is incompatible with the voltage conversion performed by the specified level shifter and a user level shifter strategy with a different rule of high_to_low/low_to_high is set.

### What Next

Check the supply nets of the specified level shifter. Also check the level shifter type of the library cell and the rule of the strategy. But if the level shifter strategy is set as intended, this warning can be ignored.

### See Also

- report_cell

## MV-617

(Warning) %s '%s' (supply net '%s' [%s]) cannot drive load pin '%s' (supply net '%s' [%s]) due to voltage difference. Level shifter insertion is prevented due to a no_shift level shifter strategy.

### Description

This message occurs when there is voltage difference between a driver supply net and load supply net in a path with a level shifter strategy with -no_shift.

Feedback

### What Next

Check the voltage values of the two supply nets or the level shifter strategy. If the strategy is done with this intent, the message can be ignored.

### See Also

- report_net
- report_supply_nets
- report_pst
- set_level_shifter

---

## MV-654

(information) Total %d %s cells have been mapped successfully in design.

### Description

This message shows the number of each type of mv cells (eg, level shifters, isolation cells, repeater cells, etc) that have been mapped successfully with the library cells.

### What Next

This is an information-only message. No action is required. Ensure the expected number of each type was mapped. Otherwise, the unmapped report will be generated to show some failure reason for mapping .

### See Also

- create_mv_cells

---

## MV-655

(error) analyze_mv_feasibility -element can be used for only one type of cell report.

### Description

analyze_mv_feasibility command -element option can be used for only one type of cell report, don't specify multiple cells report in the meantime, different type of cell report accept different type of element.

### What Next

correct the command option

Feedback

### See Also

- analyze_mv_feasibility

---

## MV-660

(error) Assign_feedthrough_supply fails due to domain boundary ports on MIM instances have different level shifter strategy, %s(%s) v.s. %s(%s).

### Description

Assign_feedthrough_supply requires domain boundary ports on all the paths go through the same MIM design to have the same level shifter strategy. No boundary supply will be assigned on the path when inconsistent level shifter stratgy is found.

### What Next

correct the design

### See Also

- assign_feedthrough_supply

---

## MV-1030

(warning) PVT mismatch found in %d %s%s.

### Description

This message occurs when PVT mismatch is found in the specified type of cells in the design. For example:

```
Warning: PVT mismatch found in 2 level shifters. (MV-1030)
Warning: PVT mismatch found in 4 isolation cells. (MV-1030)
Warning: PVT mismatch found in 1 retention register. (MV-1030)
```

Possible reasons for mismatch can be:

```
• Cell is powered by a supply net whose voltage does not match the
supported voltage range of the library cell.

• Cell's hierarchy has a temperature setting which does not match
the supported temperature range of the library cell.

• Cell's hierarchy has a process number setting which does not match
that of the library cell.

• Cell's hierarchy has a process label setting which does not match
that of the library cell.
```

Feedback

```
• Library cell does not match set_target_library_subset constraints
applied to the hierarchy.

• Library cell does not match site def of the voltage area.

• Library cell does not have a valid lib cell purpose.

• Library cell has a user don't touch setting.

• Library cell has incorrect bias settings.

• Library cell has a subset_name attribute which does not match
the cell.

• Library cell has a threshold_voltage_group attribute which does
not match threshold voltage group type required by tool.
```

### What Next

Please run *check_mv_design* to find out if there is any issues with the specified type of cells in the design.

### See Also

- check_mv_design
- set_process_label
- set_process_number
- set_temperature
- set_voltage
- report_dont_touch
- report_target_library_subset
- report_threshold_voltage_groups

---

## MV-1031

(warning) PVT mismatch found in %s (%s).

### Description

This message occurs when PVT mismatch is found in the specified cell. This message is printed if app option *mv.cells.report_pvt_verbose* is set to true. For example:

```
Warning: PVT mismatch found in MID1/in_UPF_ISO (LIB1/iso). (MV-1031)
Warning: Can't use LIB1/ISO because it doesn't match isolation power
supply net voltage (VDD2 = 0.500V) in corner 'default'. (PVT-1201)
```

Feedback

```
Warning: PVT mismatch found in MID2/in_UPF_ISO (LIB2/iso). (MV-1031)
Warning: Can't use LIB2/ISO because it doesn't match process number
 (2.000)
in corner 'default'. (PVT-1201)
Warning: PVT mismatch found in 2 isolation cells. (MV-1030)
```

Possible reasons for mismatch can be:

- Cell is powered by a supply net whose voltage does not match the supported voltage range of the library cell.
- Cell's hierarchy has a temperature setting which does not match the supported temperature range of the library cell.
- Cell's hierarchy has a process number setting which does not match that of the library cell.
- Cell's hierarchy has a process label setting which does not match that of the library cell.
- Library cell does not match set_target_library_subset constraints applied to the hierarchy.
- Library cell does not match site def of the voltage area.
- Library cell does not have a valid lib cell purpose.
- Library cell has a user don't touch setting.
- Library cell has incorrect bias settings.
- Library cell has a subset_name attribute which does not match the cell.
- Library cell has a threshold_voltage_group attribute which does not match threshold voltage group type required by tool.

### What Next

Please run *check_mv_design* to find out if there is any issues with the specified type of cells in the design.

### See Also

- check_mv_design
- set_process_label
- set_process_number
- set_temperature
- set_voltage

Feedback

- report_dont_touch
- report_target_library_subset
- report_threshold_voltage_groups
- mv.cells.report_pvt_exact
- mv.cells.report_pvt_verbose

---

## MV-1055

(Information) Related supply assignment details reported in %s.

**Description**

Indicates the name of the output report of *assign_feedthrough_supply*.

**See Also**

- assign_feedthrough_supply

---

## MV-1056

(Information) Related Supplies assigned for %d nets.

**Description**

Indicates the number of nets which *assign_feedthrough_supply* successfuly propagate.

**See Also**

- assign_feedthrough_supply

---

## MV-1057

(Information) Related Supplies cannot be assigned for %d nets.

**Description**

Indicates the number of nets which *assign_feedthrough_supply* failed to propagate.

**See Also**

- assign_feedthrough_supply

Feedback

## MV-1058

(Information) Net driven by '%s' cannot be processed by assign_feedthrough_supply because it has bidirectional pins.

### Description

This message occurs when *assign_feedthrough_supply* cannot process the specified net because Tool cannot work with bidirectional pins.

### What Next

- Check which pin in the net is bidirectional in the net with *get_attribute -name direction -objects pin*. If the returned value is *inout* it means it's bidirectional.

### See Also

- get_attribute
- assign_feedthrough_supply

## MV-1059

(Information) Net driven by '%s' is a clock net and cannot be processed by assign_feedthrough_supply as automatic Level Shifter insertion is disabled on clock nets.

### Description

When the application option *mv.upf.auto_ls_clock_nets* has the default value of *"*"*, *assign_feedthrough_supply* can run on any clock net. If the value of *mv.upf.auto_ls_clock_nets* is different than "*", you must add the specified net to the list of permitted clock nets in order for *assign_feedthrough_supply* to process it.

### What Next

- Check why *mv.upf.auto_ls_clock_nets* has non-default value. If this is intended, add the specified clock nets to the list of permitted clock nets using the application option *mv.upf.auto_ls_clock_nets*.

### See Also

- set_app_options
- mv.upf.auto_ls_clock_nets
- assign_feedthrough_supply

Feedback

## MV-1060

(Information) Net driven by '%s' cannot be processed by assign_feedthrough_supply because it is an analog net.

### Description

This message occurs when *assign_feedthrough_supply* cannot process the specified net because Tool cannot work with analog nets.

### What Next

- Check for a call to command *set_port_attributes -is_analog true*. When this property is assigned to a port, it's also propagated to the whole net.

### See Also

- assign_feedthrough_supply
- set_port_attributes

## MV-1061

(Error) Net driven by '%s' has Level Shifter Violation. Please, fix the violation before continuing.

### Description

This message occurs when *assign_feedthrough_supply* cannot process the specified net because it has Level Shifter Violation.

### What Next

- Run *analyze_mv_feasibility -level_shifter* to get detailed information about Level Shifter Violation.
- Run *analyze_mv_design -level_shifter -through* on the specified net to know the failure reasons.
- Run *report_mv_path -level_shifter -pin pin* to analyze Level Shifter Violation along a specified path.

### See Also

- assign_feedthrough_supply

Feedback

## MV-1062

(Information) Net driven by '%s' cannot be processed by assign_feedthrough_supply because it is driven by bump cell.

### Description

This message occurs when *assign_feedthrough_supply* cannot process the specified net because Tool cannot work with nets driven by a bump cell.

### What Next

- Check why a bump cell drives the net.

### See Also

- assign_feedthrough_supply

## MV-1063

(Information) Net driven by '%s' cannot be processed by assign_feedthrough_supply because net does not cross any physical boundary.

### Description

This message occurs when *assign_feedthrough_supply* cannot process the specified net because it does not cross any physical boundary.

### What Next

- Make sure the net cross a physical boundary to run *assign_feedthrough_supply*
- Check if *place_pins* finishes correctly. If a cell remains unplaced it could have their pins assigned in the wrong direction.

### See Also

- assign_feedthrough_supply

## MV-1064

(Information) Net driven by '%s' cannot be processed by assign_feedthrough_supply because it's undriven.

### Description

This message occurs when *assign_feedthrough_supply* cannot process the specified net because the net has no driver (undriven).

Feedback

### What Next

- Please, check why the net is undriven.

### See Also

- assign_feedthrough_supply

---

## MV-1090

(error) Cannot use %s options at the same time.

### Description

This message occurs when the create_mv_cells command is called with the specified options. Only one of the specified options can be used at a time.

### What Next

Choose only one of the specified options for the create_mv_cells command.

### See Also

- create_mv_cells

---

## MV-1091

(info) User library has %d level shifter%s and %d enabled level shifter%s available to be used.

### Description

This message occurs when user executes the *create_mv_cells* command with *-verbose* option. It shows the number of level shifter and enabled level shifter library cells that are available to the tool. Notice that if a library cell has the dont_touch attribute set to true, it cannot be used by the tool.

### What Next

No action is required.

### See Also

- set_level_shifter
- create_mv_cells
- set_dont_touch

Feedback

## MV-1092

(info) User library has %d level shifter%s and %d enabled level shifter%s with dont_touch attribute.

### Description

This message occurs when user executes the *create_mv_cells* command with *-verbose* option. It shows the number of level shifter and enabled level shifter library cells that have the dont_touch attribute set to true. If a library cell has the dont_touch attribute set to true, it cannot be used by the tool.

### What Next

If user design needs level shifters or enabled level shifters to fix voltage violations, make sure that the required level shifter or enabled level shifter library cells do not have the dont_touch attribute set to true.

### See Also

- set_level_shifter
- create_mv_cells
- set_dont_touch

## MV-1093

(info) User library has %d ISO-LS type of enabled level shifter%s.

### Description

This message occurs when user executes the *create_mv_cells* command with *-verbose* option. It shows the number of ISO-LS type of enabled level shifters in the user library.

### What Next

No action is required.

### See Also

- set_level_shifter
- create_mv_cells

## MV-1094

(warning) Total %d supply net%s not found in PST: %s

Feedback

### Description

This message occurs when supply nets are defined but not included in PST or the supply nets only have OFF power states. These missing supply nets will not be considered in voltage violation and level shifter insertion. Required level shifters may not be inserted.

### What Next

Add the missing supply nets to PST.

### See Also

- create_pst
- add_pst_state
- add_port_state
- add_power_state
- set_level_shifter
- create_mv_cells

---

## MV-1095

(warning) User library has no level shifter or enabled level shifter available to be used.

### Description

This message occurs when user executes the *create_mv_cells* command with *-verbose* option. There are no level shifter or enabled level shifters available in the user library. Tool will not be able to insert level shifters to fix voltage violation. Notice that if a library cell has the dont_touch attribute set to true, it cannot be used by the tool.

### What Next

Make sure that user library has available level shifters or enabled level shifter to be used. Use *set_dont_touch* to make level shifter library cells available to be used by the tool.

### See Also

- set_level_shifter
- create_mv_cells
- set_dont_touch

Feedback

## MV-1096

(warning) fix_mv_design -move_within_x or -move_within_y is specified but both physical and logical feedthrough are not enabled in voltage area '%s'.

### Description

This message occurs when user executes the *fix_mv_design* command with *-move_within_x* or *-move_within_y* option so that the command can move a buffer or inverter to a nearby voltage area. However, the command cannot do so if both physical and logical feedthrough are not enabled in the target voltage area.

### What Next

Enable physical or logical feedthrough or both in *create_voltage_area_rule* command.

```
Example:
create_voltage_area_rule -default_rule \\
  -allow_physical_feedthrough true \\
  -allow_logical_feedthrough true
```

### See Also

- fix_mv_design
- create_voltage_area_rule

## MV-1100

(error) Cannot insert level shifter for global net driven by driver pin '%s' because the global net does not cross any power domain boundary.

### Description

This message occurs when the global net driven by the specified driver pin has a voltage violation but the global net does not cross any power domain boundary. As a result, there is no available power domain boundary pin to insert a level shifter.

### What Next

Change the design so that the specified global net crosses at least one power domain boundary.

### See Also

- create_mv_cells
- set_level_shifter

Feedback

## MV-1101

(error) Cannot insert level shifter in the path from driver pin '%s' (related supply net: %s [%s]) to load pin '%s' (related supply net: %s [%s]) because the path does not cross any power domain boundary.

### Description

This message occurs when the specified driver pin has a voltage violation to the specified load pin but the signal path does not cross any power domain boundary. As a result, there is no available power domain boundary pin to insert a level shifter.

### What Next

Change the design so that the specified signal path crosses at least one power domain boundary.

### See Also

- create_mv_cells
- set_level_shifter

## MV-1102

(error) Cannot insert enabled level shifter in the path from driver pin '%s' (related supply net: %s [%s]) to load pin '%s' (related supply net: %s [%s]) because there is no isolation cell on the path.

### Description

This message occurs when the specified driver pin has both voltage and isolation violations driving the specified load pin. An enabled level shifter is required to fix both the voltage and isolation violations. However, there is no isolation cell on the path which is needed for the insertion of enabled level shifter.

### What Next

Add an isolation strategy on a domain boundary pin along the path.

### See Also

- set_isolation
- set_level_shifter
- create_mv_cells

Feedback

## MV-1103

(error) Cannot convert isolation cell '%s' into enabled level shifter because of no_shift level shifter strategy '%s'.

### Description

This message occurs when the specified isolation cell needs to be converted into an enabled level shifter to fix a voltage violation but a user no_shift level shifter strategy prevents the conversion.

### What Next

Remove the no_shift level shifter strategy

### See Also

- set_level_shifter
- set_isolation
- create_mv_cells

## MV-1104

(error) Cannot fulfill level shifter strategy '%s' at pin '%s' because driver power supply (%s) is not available in pin's power domain (%s).

### Description

This message occurs when user specifies a level shifter strategy at the specified pin but power supply of the driver pin is not available in the specified pin's power domain. As a result, tool cannot insert a level shifter at the specified pin.

### What Next

Add the driver supply to the available supply list of the specified pin's power domain or apply the level shifter strategy to another pin in a power domain in which the driver power is available.

### See Also

- set_level_shifter
- create_power_domain
- create_mv_cells

Feedback

## MV-1105

(error) Cannot fulfill level shifter strategy '%s' at pin '%s' because all load power supplies (%s) are not available in pin's power domain (%s).

### Description

This message occurs when user specifies a level shifter strategy at the specified pin but power supplies of all load pins are not available in the specified pin's power domain. As a result, tool cannot insert a level shifter at the specified pin.

### What Next

Add the load power supplies to the available supply list of the specified pin's power domain or apply the level shifter strategy to another pin in a power domain in which the load power supplies are available.

### See Also

- set_level_shifter
- create_power_domain
- create_mv_cells

## MV-1106

(error) Cannot fulfill level shifter strategy '%s' at pin '%s' in power domain '%s' because the primary power supply (%s) cannot drive any load power supply.

### Description

This message occurs when:

- User specifies a level shifter strategy at the specified pin; and
- The specified pin's domain primary supply cannot drive any power supplies of the load pins without voltage violations; and
- All level shifter library cells available to the design or specified in map_level_shifter_cell/map_isolation_cell command use output rail as standard cell main rail (i.e., output supply of these level shifter library cells need to be connected to the domain primary supply).

As a result, tool cannot insert a level shifter at the specified pin because it will result in a standard cell main rail violation.

Feedback

### What Next

You can do one of the followings:

- Apply the level shifter strategy to another domain boundary pin in a power domain in which the primary supply can drive the load supplies without voltage violations; or
- Add level shifter library cells which use input rail as standard cell main rail to the design or applicable map_level_shifter_cell/map_isolation_cell command; or
- Ignore standard cell main rail checking in level shifter insertion by:

```
set_app_options -as_user_default -list {mv.upf.ignore_ls_main_rail
 true}
```

### See Also

- set_level_shifter
- create_mv_cells

---

## MV-1107

(error) Cannot insert level shifter in the path from driver %s '%s' (related supply net: %s [%s]) to load pin '%s' (related supply net: %s [%s]) because no domain primary supply along the path can drive the load supply.

### Description

This message occurs when:

- the driver pin has a voltage violation when driving the load pin; and
- none of the primary supplies of power domains along the path from the driver pin to the load pin can drive the load supply without a voltage violation; and
- all level shifter library cells available to the design or specified in map_level_shifter_cell/map_isolation_cell command use output rail as standard cell main rail (i.e., output supply of these level shifter library cells need to be connected to the domain primary supply).

As a result, tool cannot insert a level shifter at any domain boundary pins along the path from the driver pin to the load pin because it will result in a standard cell main rail violation.

Feedback

### What Next

You can do one of the followings:

- Add level shifter library cells which use input rail as standard cell main rail to the design or applicable map_level_shifter_cell/map_isolation_cell command; or
- Ignore standard cell main rail checking in level shifter insertion by: set_app_options -global {upf.ignore_ls_main_rail true}

### See Also

- create_mv_cells

---

## MV-1108

(error) Cannot insert level shifter in the path from driver %s '%s' (related supply net: %s [%s]) to load pin '%s' (related supply net: %s [%s]) because the path does not have an effective level shifter strategy. %s

### Description

This message occurs when user specifies the app option upf.insert_ls_on_user_cstr_only to be true (i.e. tool can only insert a level shifter at a domain boundary pin with user level shifter strategy specified) but there is no effective level shifter strategy found from the driver pin to the load pin. An effective level shifter strategy is defined as one which does not have -no_shift option specified and the applied domain boundary pin is not on a don't touch net or don't touch network.

### What Next

Add a level shifter strategy which does not have -no_shift option specified to one of the domain boundary pins between the driver pin and the load pin. Make sure that the applied domain boundary pin is not on a don't touch net or don't touch network.

### See Also

- set_level_shifter
- set_dont_touch
- set_dont_touch_network
- create_mv_cells

Feedback

## MV-1109

(error) Cannot convert isolation cell '%s' into enabled level shifter because the path has %s voltage transition but no %s enabled level shifter is included in %s.

### Description

This message occurs when there is high-to-low, low-to-high, or both types of voltage transitions going through the specified isolation cell. The isolation cell needs to be converted into an enabled level shifter in order to fix the violation violation. However, enabled level shifters which can perform this type of voltage transition are not available in the user library or user isolation strategy. Therefore the tool cannot fix the voltage violation going through the specified isolation cell.

### What Next

Add the required type of enabled level shifters to user library or isolation strategy.

### See Also

- set_isolation
- set_level_shifter
- create_mv_cells

## MV-1110

(error) Cannot insert level shifter in the path from driver %s '%s' (related supply net: %s [%s]) to load pin '%s' (related supply net: %s [%s]) because driver supply net %s is not available in any power domain along the path.

### Description

This message occurs when:

- the driver pin has a voltage violation; and
- the driver supply net is not available in any power domain along the path from the driver pin to the load pin; or if the driver supply net is available, the domain boundary pin is not available for level shifter insertion because there is a -no_shift strategy, a conflicting -location strategy, or other dont_touch reasons.

  As a result, tool cannot insert a level shifter at any domain boundary pins along the path from the driver pin to the load pin because it cannot power the input side of a level shifter with the driver supply net.

### What Next

Please make the driver supply net available in at least one of the power domains along the path from the driver pin to the load pin.

### See Also

- create_mv_cells

---

## MV-1111

(error) Cannot convert isolation cell '%s' with bias supplies (nwell=%s, pwell=%s) into enabled level shifter because no enabled level shifter with %sbias support is included in %s.

### Description

This message occurs when there is a voltage violation going through an isolation cell and the applied isolation strategy has specified an isolation supply set with bias supplies. However, tool cannot convert the isolation cell into enabled level shifter because user library or map_isolation_cell command does not include such lib cells which can support bias (or insulated bias).

### What Next

Add enabled level shifters with bias (or insulated bias) support in the user library or map_isolation_command.

### See Also

- set_isolation
- set_level_shifter
- create_mv_cells

---

## MV-1112

(error) Cannot insert level shifter for global net driven by driver pin '%s' because no level shifter with bias support is included in %s.

### Description

This message occurs when there is a voltage violation in the global net driven by the specified driver pin. The global net goes through power domains which have bias supplies specified. Therefore tool needs to insert level shifters with bias support in these power domains. However, user library or map_level_shifter_cell command does not include level shifters with bias support.

Feedback

### What Next

Add level shifters with bias support in the user library or map_level_shifter_cell command.

### See Also

- set_isolation
- set_level_shifter
- create_mv_cells

---

## MV-1113

(error) Found level shifter strategy '%s' with -location parent on pin '%s' but tool cannot insert level shifter in the parent power domain because net '%s' has a user don't touch attribute.

### Description

This message occurs when there is a level shifter strategy with -location parent set on a power domain boundary pin. However, there is a user dont touch attribute set on the net connecting to the pin in the parent power domain. Therefore tool cannot insert level shifter in the parent power domain of the pin.

### What Next

Remove the dont touch attribute set on the net in the parent power domain or change the level shifter strategy to use -location self.

### See Also

- create_mv_cells
- report_dont_touch
- set_dont_touch
- set_level_shifter

---

## MV-1114

(error) Found level shifter strategy '%s' with -location self on pin '%s' but tool cannot insert level shifter in the power domain because net '%s' has a user don't touch attribute.

### Description

This message occurs when there is a level shifter strategy with -location self set on a domain boundary pin. However, there is a user dont touch attribute set on the net

Feedback

connecting to the pin in its power domain. Therefore tool cannot insert level shifter in the power domain of the pin.

**What Next**

Remove the dont touch attribute set on the net in the power domain or change the level shifter strategy to use -location parent.

**See Also**

- create_mv_cells
- report_dont_touch
- set_dont_touch
- set_level_shifter

---

## MV-1115

(error) Cannot convert isolation cell '%s' into %s enabled level shifter because power supplies (%s, %s) of its driver pin '%s' cannot drive domain primary supplies (%s, %s) of the isolation cell.

**Description**

This message occurs when:

- There is a voltage violation from the specified driver pin to the data input pin of an isolation cell; and
- The supply of the specified driver pin cannot drive the domain primary supply of the isolation cell because of voltage or isolation violation; and
- All high-to-low (or low-to-high) enabled level shifter library cells available to the design or specified in map_level_shifter_cell/map_isolation_cell command use input rail as standard cell main rail (i.e., input supply of these enabled level shifter library cells needs to be connected to the domain primary supply).

As a result, tool cannot convert the specified isolation cell into a high-to-low (or low-to-high) enabled level shifter because it will result in a standard cell main rail violation.

Feedback

### What Next

You can do one of the followings:

- Add high-to-low (or low-to-high) enabled level shifter library cells which use output rail as standard cell main rail to the design or applicable map_level_shifter_cell/ map_isolation_cell command; or
- Ignore standard cell main rail checking in level shifter insertion by:

```
set_app_options -as_user_default -list {mv.upf.ignore_ls_main_rail
 true}
```

### See Also

- set_level_shifter
- create_mv_cells

---

## MV-1116

(error) Cannot convert isolation cell '%s' into %s enabled level shifter because its domain primary supplies (%s, %s) cannot drive any of its loads.

### Description

This message occurs when:

- There is a voltage violation from the output pin of an isolation cell to its loads; and
- The domain primary supply of the isolation cell cannot drive any of its loads because of voltage or isolation violation; and
- All high-to-low (or low-to-high) enabled level shifter library cells available to the design or specified in map_level_shifter_cell/map_isolation_cell command use output rail as standard cell main rail (i.e., output supply of these enabled level shifter library cells needs to be connected to the domain primary supply).

As a result, tool cannot convert the specified isolation cell into a high-to-low (or low-to-high) enabled level shifter because it will result in a standard cell main rail violation.

Feedback

### What Next

You can do one of the followings:

- Add high-to-low (or low-to-high) enabled level shifter library cells which use input rail as standard cell main rail to the design or applicable map_level_shifter_cell/ map_isolation_cell command; or
- Ignore standard cell main rail checking in level shifter insertion by:

```
set_app_options -as_user_default -list {mv.upf.ignore_ls_main_rail
 true}
```

### See Also

- set_level_shifter
- create_mv_cells

---

## MV-1117

(error) Cannot use driver supply net %s (of driver pin '%s') to power %s '%s' because the supply net is not available in the %s's power domain (%s).

### Description

This message occurs when the driver supply net is not available in a Level Shifter's, Enable Level Shifter's or Isolation Cell's power domain. As a result, tool cannot use the driver supply net as the input supply net of the Level Shifter, Enable Level Shifter or Isolation Cell.

### What Next

Please make the driver supply net available in the power domain of the Level Shifter, Enable Level Shifter or Isolation Cell.

### See Also

- create_mv_cells

---

## MV-1118

(error) Tool cannot insert level shifter in net '%s' because has a don't touch attribute.

### Description

This message occurs when the net in wich a level shifter should be inserted has don't touch attribute.

Feedback

### What Next

Remove the dont touch attribute set on the net

### See Also

- create_mv_cells
- report_dont_touch
- set_dont_touch
- set_level_shifter
- analyze_mv_design

---

## MV-1119

(Information) Tool cannot insert Level Shifter because there is no usable lib cell for this insertion point. The initial %s lib cells where filtered by: %s.

### Description

This message occurs when there is no lib cell that can be used to insert a Level Shifter at this insertion point. Reasons might be:

1. Library cell is has dont_touch attribute set.
2. Voltage range mismatch: a library cell's voltage range defined in library should match with driver/load supply's voltage range before it can be inserted.
3. Library cell is not overdriven while the path needs overdriven Level Shifter.
4. Library cell is not single ground while single ground library cell is required.
5. Strategy rule mismatch: a library cell's level-shifter type should match with the voltage shifting direction on the path.
6. Standard cell main-rail mismatch: a library cell's main-rail supply should match with the domain primary supply.
7. Library cell has pvt mismatch.
8. Library cell (Enable Level Shifter) clamp has mismatch or cannot perform automatic phase correction due to clamp mismatch
9. Library cell (Enable Level Shifter) sense has mismatch or cannot perform automatic phase correction due to sense mismatch
10. Library cell (Enable Level Shifter) does not have enable pin.

Feedback

11. Library cell is Combo Enable Level Shifter, but has data mismatch.
12. No library cell can be selected due to voltage violation between input and output supply net.

**What Next**

Check the *map_level_shifter_cell* command associated with this insertion point if any. Check the operating conditions for the Level Shifter lib cells available for use with *report_mv_lib_cells -level_shifter*. If current insertion point has Isolation (or Isolation strategy) use *analyze_mv_feasibility* to see a detailed information for unmap Enabled Level Shifter cells.

**See Also**

- add_power_state
- create_pst
- report_mv_lib_cells
- set_level_shifter
- analyze_mv_feasibility
- map_level_shifter_cell

---

## MV-1120

(error) Cannot insert level shifter in the path from driver %s '%s' (related supply net: %s [%s]) to load pin '%s' (related supply net: %s [%s]) because load supply net %s is not available in any power domain along the path.

**Description**

This message occurs when:

- there is a voltage violation; and
- the load supply net, required to implement a level shifter, is not available in any power domain along the path from the driver pin to the load pin; or if the load supply net is available, the domain boundary pin is not available for level shifter insertion because there is a -no_shift strategy, a conflicting -location strategy, or other dont_touch reasons.

  As a result, tool cannot insert a level shifter at any insertion point along the path from the driver pin to the load pin, because it cannot power the level shifter with the driver supply net.

Feedback

### What Next

Please make the load supply net available in at least one of the power domains along the path from the driver pin to the load pin.

### See Also

- create_mv_cells

---

## MV-1121

(Error) User library has no level shifter cells of type '%s' available to be used.

### Description

Tool detects one or more voltage violations that need level shifter cells of type 'high-to-low' (HL), 'low-to-high' (LH) or both (HL_LH) but user library does not contain level shifter cells of the specified type. Tool is not be able to perform level shifter insertion to solve voltage shift violations.

### What Next

o Check if user library contains level shifters of the specified type (HL, LH or HL_LH) using *report_mv_lib_cells -level_shifter*.

o Also check if lib cells have *dont_touch attribute* (dt) in the report of *report_mv_lib_cells -level_shifter*. Lib cells with 'dont_touch attribute' cannot be used by insertion engine. Use *set_dont_touch* to make level shifter library cells available to be used by the tool.

### See Also

- set_level_shifter
- create_mv_cells
- set_dont_touch
- report_mv_lib_cells

---

# NDM

---

## NDM-001

(error) %s

Feedback

**Description**

This message indicates an internal data model error.

**What Next**

Send the log to Synopsys.

---

## NDM-002

(error) %s

**Description**

Permission denied.

**What Next**

Please set proper access right and rerun.

---

## NDM-003

(error) %s

**Description**

The object already exists.

**What Next**

Please don't create an object with name/id of an existing object.

---

## NDM-004

(error) %s

**Description**

Input has invalid argument.

**What Next**

Please send the log to Synopsys.

---

## NDM-005

(error) %s

Feedback

**Description**

Operation interrupted.

**What Next**

Please send the log to Synopsys.

---

## NDM-006

(error) %s

**Description**

Is a directory.

**What Next**

Please check and use the right file name.

---

## NDM-007

(error) %s

**Description**

No such file or directory.

**What Next**

Please check and use the right file/directory name.

---

## NDM-008

(error) %s

**Description**

Out of memory.

**What Next**

Please pick a machine with large enough memory and rerun.

---

## NDM-009

(error) %s

Feedback

**Description**

Not a file.

**What Next**

Please check and use the right file name.

---

## NDM-010

(error) %s

**Description**

Not a directory.

**What Next**

Please check and use the right directory name.

---

## NDM-011

(error) %s

**Description**

Not (yet) implemented.

**What Next**

Please send the log to Synopsys.

---

## NDM-012

(error) %s

**Description**

Operation timed out.

**What Next**

Please check and retry.

---

## NDM-013

(error) %s

Feedback

**Description**

File open/lock/read/write error

**What Next**

This is an internal error. Please report to Synopsys.

---

## NDM-014

(error) %s

**Description**

Object not exists

**What Next**

This is an internal error. Please report to Synopsys.

---

## NDM-015

(error) %s

**Description**

Invalid Object

**What Next**

Internal message, please report to Synopsys.

---

## NDM-016

(error) NDM file version %s is no longer supported. Current compatible version is %s.

**Description**

This version of the NDM file is not supported.

**What Next**

Re-generate design and library data with the latest binary.

---

## NDM-017

(error) Failed to read %s object from file %s.

Feedback

**Description**

The specified object could not be read correctly read from file because of an I/O error.

**What Next**

Re-create the file with the latest executable.

---

## NDM-018

(error) Failed to restore %s object from file %s.

**Description**

The object cannot be correctly restored from file because of an I/O error.

**What Next**

Re-create the file with the latest executable.

---

## NDM-019

(error) Cannot save the library or block because it is in read-only mode.

**Description**

The library or block cannot be saved to disk because it was opened in read mode.

**What Next**

Close the library or block and reopen it in edit mode.

---

## NDM-020

(error) '%s' is not a valid NDM binary file

**Description**

The file cannot be read because not in valid NDM binary format.

**What Next**

Verify the file location is correct or re-create it with the latest binary.

---

## NDM-021

(warning) re-binding failed for %s '%s'.

Feedback

**Description**

Re-binding a cell to a valid reference block failed during *open_block*.

Binding failed either because the target reference block could not be found, or because the target reference block port interface has changed since the last time this cell was bound.

**What Next**

If the target block was not found, verify that the search_path location or reference library list setting correctly identifies the target block and try again.

If the target block port has changed, use the *link_block* command to update the cell's pin interface to match the referenced block's new port interface.

---

## NDM-022

(warning) '%s' is logical only. Placement status is not available.

**Description**

Placement status cannot be set on a logical only cell/port.

---

## NDM-023

(error) Placement status cannot be set on object type '%s'.

**Description**

Placement status can only be set on cells and ports.

---

## NDM-024

(error) Invalid %s placement status '%s'.

**Description**

For ports, valid status values are unplaced, placed, fixed, and locked. For cells, valid status values are unplaced, placed, legalize_only, application_fixed, fixed, and locked.

**What Next**

Specify a valid placement status.

---

## NDM-025

(error) The value %f is too %s causing an arithmetic overflow error.

Feedback

**Description**

The specified value cannot be represented within the database. It is either too large or too small, causing an arithmetic overflow error when converted into internal database units.

**What Next**

Check the technology units and specify a valid value.

---

## NDM-026

(error) Can not create hier_boundary as the cell already has a hier_boundary.

**Description**

The hierarchical cell already has a hier_boundary associated with it, therefore hier_boundary creation has failed.

**What Next**

Please provide proper input to command.

---

## NDM-027

(error) Can not create partition as the cell is already a partition.

**Description**

The hierarchical cell is already a partition, therefore partition creation has failed.

**What Next**

Please provide proper input to command.

---

## NDM-028

(warning) Block '%s.%s' is out of sync with the '%s' view. '%s' view has changed since last synced.

**Description**

The specified block view has changed since it was set to be in-sync with its source view.

**What Next**

please re-sync the (derived) block with its source view.

Feedback

## NDM-029

(error) '%s' failed. Block '%s' is being edited by process %d %s@%s.

**Description**

The specified operation failed because write acccess is held by another process.

**What Next**

Please coordinate with the user holding the block for edit about saving and releasing the write access.

## NDM-030

(error) %s for design %s does not match design %s: found '%s', expected '%s'

**Description**

While comparing two designs, the function id does not match.

**What Next**

If running the library manager, this design will be removed from the library in question. However, libraries with this sort of error are probably out of date and should not be used.

## NDM-031

(error) Port %s missing on %s

**Description**

While comparing two designs, a port was found on one but not the other.

**What Next**

If running the library manager, this design will be removed from the library in question. However, libraries with this sort of error are probably out of date and should not be used.

## NDM-032

(error) Direction '%s' for port %s on %s\n \tdoes not match expected value '%s' from %s

**Description**

While comparing two designs, a port has a different direction in each design.

Feedback

### What Next

If running the library manager, this design will be removed from the library in question. However, libraries with this sort of error are probably out of date and should not be used.

---

## NDM-033

(Error) %s '%s' for pin %s on %s does not match the expected value '%s' from %s.

### Description

The tool issues this warning message if it detects a pin that has a different *port_type* or *pg_type* attribute when comparing two designs. The *port_type* attribute identifies the type of the library cell pin, such as *signal*, *power*, or *ground*. The *pg_type* attribute identifies the type of a PG pin, such as *backup*, *internal*, or *primary*.

- The *check_workspace* command issues this message for *pg_type* mismatches between two designs in the library workspace.
- The *compare_db* command issues this message for both *port_type* and *pg_type* mismatches between the design in the base library and the comparison library.

### What Next

If you get this message during library preparation, by default, the design is removed from the reference library. To modify this behavior, set a mismatch configuration.

If you get this message during library validation, the library is probably out-of-date and you not should use it.

### See Also

- check_workspace
- create_mismatch_config
- set_current_mismatch_config
- compare_db

---

## NDM-034

(error) Cannot locate block %s for instance %s.

### Description

The instance is attempting to bind to a physical block, but the requested view of the physical block cannot be located, so the instance is unbound.

Feedback

**What Next**

A number of things can go wrong when looking for a view of a block.

Verify that the block exists, and is in a library that is loaded into the session.

Verify that the library is on the reference library list of the library containing the top block.

The desired view may be obscured by another block in an earlier library. Look at the viewtype requested for that instance. Verify that the desired view (DESIGN ABSTRACT OUTLINE) exists for that block in the first reference library that contains a block of that name.

---

## NDM-035

(error) NDM_LOCK_FAIL_ACTION `%s' not one of: `off', `complain', or `die'. Turning off.

**Description**

This is a internal error. Please report to Synopsys.

**What Next**

Internal error, please report to Synopsys.

---

## NDM-036

(info) %s lock check failed!.

**Description**

This is a internal message. Please report to Synopsys.

**What Next**

Internal message, please report to Synopsys.

---

## NDM-037

(warning) Conn count of instance '%s' does not match reference module '%s'

**Description**

The conn count of the specified instance '%s' does not match its reference module. This could be caused by a change in the port interface after the instance was created.

**What Next**

Verify the port interface of the reference module matches the instance.

Feedback

## NDM-038

(warning) power_domain '%s' is already associated with a voltage_area.

**Description**

The specified power_domain is already associated with a voltage_area.

**What Next**

Specify a different power_domain and try this command again.

## NDM-039

(warning) Top-level power_domain '%s' is reserved for default voltage_area.

**Description**

The specified top-level power_domain is reserved for the default voltage_area.

**What Next**

Specify a different power_domain and try this command again.

## NDM-040

(information) NDM file '%s', version %s does not match current version %s.

**Description**

The NDM file is of an earlier version than the current version.

## NDM-041

(info) Net '%s' has been created.

**Description**

This is a verbose message indicating the named net has been created.

## NDM-042

(info) Port '%s' has been created.

**Description**

This is a verbose message indicating the named port has been created.

Feedback

## NDM-043

(info) Net '%s' has been connected to %s '%s'.

**Description**

This is a verbose message indicating the net has been connected to the pin/port.

## NDM-044

(info) Net '%s' has been disconnected from %s '%s'.

**Description**

This is a verbose message indicating the net has been disconnected from the pin/port.

## NDM-045

(error) Unable to create net '%s'. A net of that name already exists.

**Description**

This is an internal error. Please report to Synopsys.

## NDM-046

(warning) Connection from net '%s' to %s '%s' is not consistent with supply net '%s'.

**Description**

The intended connection is not consistent with supply net, but the manually specified connection has been made.

## NDM-047

(warning) Unable to create net '%s'. A net of that name already exists. Net is renamed to '%s'.

**Description**

Net has been renamed in order to make specified connection.

## NDM-048

(error) MV data is not ready.

Feedback

**Description**

MV data is not ready to run this command. Please run commit_upf.

---

## NDM-049

(error) The conflicts between PG network (%s'%s') and UPF (%s'%s') have not been resolved.

**Description**

The conflicts between PG network and UPF must be resolved before running this command.

---

## NDM-050

(error) Unable to create port '%s'. A port of that name already exists.

**Description**

This is an internal error. Please report to Synopsys.

---

## NDM-051

(error) Unable to determine the supply net to connect to the tie net '%s'.

**Description**

This is an internal error. Please report to Synopsys.

---

## NDM-052

(error) Cannot push into partition '%s'.

**Description**

The partition cannot become current block.

---

## NDM-053

(error) Unable to determine supply net to connect to %s '%s'.

**Description**

Supply net cannot be determined to make tie connection.

Feedback

### What Next

Fix your power constraints and try again.

---

## NDM-054

(warning) Unable to determine supply net to connect to %s '%s'. %s

### Description

Supply net cannot be determined to make tie connection. This could be due to related power/ground pin is not defined for the signal pin in library or supply net connection cannot be determined for the related power/ground pin by UPF.

### What Next

Check whether related power/ground pin is defined for the signal pin in library or supply net connection for the related power/ground pin is defined in UPF. If the pin is a macro input pin that is related to an internal pg pin, consider to set the app option "mv.upf.use_preswitched_supply_for_macro_pins" to use the pre-switched supply to determine the supply net for the tie connection.

### See Also

- mv.upf.use_preswitched_supply_for_macro_pins

---

## NDM-055

(info) In block '%s', uniquifying module '%s' from '%s' for cell '%s'.

### Description

This is a verbose message indicating the multiply instantiated module (MIM) has been automatically split into two modules so that changes to that instance of the module for that logical cell will not be replicated into all cells that reference that module.

A cell's module can be auto-uniquified whenever the contents of the cell are changed. Only multiply instantiated cell modules (MIMs) will be auto-uniquified. Physical blocks and lib cells are never uniquified.

### What Next

This message can be enabled or disabled by setting the *design.verbose_uniquify* app option.

You can get a cell's reference module with the *ref_name* attribute.

You can get all the modules in a block with the *get_modules* command.

Feedback

**See Also**

- get_attribute
- get_modules
- set_app_options
- uniquify
- cell_attributes
- design_options

---

## NDM-056

(info) Port '%s' is removed.

**Description**

This is a verbose message indicating the port is removed.

---

## NDM-057

(info) Net '%s' type is set to be '%s'.

**Description**

This is a verbose message indicating the net type is set.

---

## NDM-058

(info) Net '%s' name is set to be '%s'.

**Description**

This is a verbose message indicating the net name is set.

---

## NDM-059

(info) Net '%s' is removed.

**Description**

This is a informational message indicating the net is removed when resolve_pg_nets resolves the conflicts between power intent (based on UPF) and pg net connections (from DEF or PG netlist).

Feedback

### What Next

Please check the power intent.

### See Also

- commit_upf
- resolve_pg_nets

---

## NDM-060

(info) Port '%s' direction is set to '%s'.

### Description

This is a verbose message indicating the port direction is set.

---

## NDM-061

(info) %s '%s' cannot be created because the %s is not PG type.

### Description

This is a verbose message indicating that the named port/net cannot be created since the supply port/net is not PG type.

---

## NDM-062

(warning) Block '%s.%s' is out of sync with the '%s' view. Both views have changed since last synced.

### Description

The specified block views have changed since they were set to be in-sync.

### What Next

please re-sync the block views.

---

## NDM-063

(information) Block '%s' is not being saved because it is in read mode.

### Description

The specified block view is not saved because it is open in read mode.

Feedback

### What Next

Open the block in edit mode and try again.

---

## NDM-064

(info) Creating block '%s', mode '%s'.

### Description

This is a verbose message indicating the named block has been created with the named open mode.

A block can have one of three open modes: A new, unsaved block has an open mode of 'new'. A block open for modification has an open mode of 'edit'. A block open read-only has an open mode of 'read'.

### What Next

This message can be enabled or disabled by setting the *design.verbose_io* app option.

You can get a block's open mode with the *open_mode* attribute.

### See Also

- create_block
- get_attribute
- set_app_options
- design_options

---

## NDM-065

(info) Opening block '%s', mode '%s'.

### Description

This is a verbose message indicating the named block has been read in from its library with the named open mode.

### What Next

This message can be enabled or disabled by setting the *design.verbose_io* app option.

You can get a block's open mode with the *open_mode* attribute.

Feedback

**See Also**

- get_attribute
- open_block
- set_app_options
- design_options

---

## NDM-066

(info) Saving block '%s'.

**Description**

This is a verbose message indicating the named block has been written to its library.

**What Next**

This message can be enabled or disabled by setting the *design.verbose_io* app option.

**See Also**

- save_block
- set_app_options
- design_options

---

## NDM-067

(info) Closing block '%s'.

**Description**

This is a verbose message indicating the named block has been closed and removed from memory.

**What Next**

This message can be enabled or disabled by setting the *design.verbose_io* app option.

**See Also**

- remove_blocks
- set_app_options
- design_options

Feedback

## NDM-068

(info) Changing block '%s' mode from '%s' to '%s'.

### Description

This is a verbose message indicating the named block's open mode has been changed from its previous mode to a new mode.

A block can have one of three open modes: A new, unsaved block has an open mode of 'new'. A block open for modification has an open mode of 'edit'. A block open read-only has an open mode of 'read'.

### What Next

This message can be enabled or disabled by setting the *design.verbose_io* app option.

You can get a block's open mode with the *open_mode* attribute.

### See Also

- get_attribute
- reopen_block
- set_app_options
- design_options

## NDM-069

(error) Cannot include read-only block '%s' as an editable part of abstract '%s'.

### Description

When creating an editable abstract, all of the editable sub-blocks of the abstract must first be opened in edit mode. It is possible the the read-only block in question is already part of an existing abstract.

The new abstract cannot be created.

### What Next

Use the reopen_block command to obtain edit permission for the given block. Alternatively, you could create a read-only abstract.

Feedback

### See Also

- reopen_block
- create_abstract

---

## NDM-070

(error) Cannot include '%s' as a part of new abstract '%s'. Existing abstract '%s' must be merged or removed first.

### Description

When creating a new abstract, none of the sub-blocks of the abstract may be part of an existing editable abstract. The abstract cannot be created.

### What Next

You can merge the changes in the existing editable abstract by using the merge_abstract command. After running that command, the abstract will no longer be editable, and new abstracts can be created.

Alternatively, you can remove the existing editable abstract by using the remove_abstract command.

### See Also

- merge_abstract
- remove_abstract

---

## NDM-071

(information) Saving '%s' and adding it as a part of editable abstract '%s'.

### Description

The given block is now part of an editable abstract. The block has been written to disk, and has been set to read-only. The block will remain read only while the editable abstract still exists.

### What Next

You might see future information messages that the read-only block cannot be saved. This is OK, because it was saved in this step.

In order to make your block editable again, you must either merge the changes in the editable abstract by using merge_abstract, or you can remove the existing editable abstract by using the remove_abstract command.

Feedback

### See Also

- merge_abstract
- remove_abstract

---

## NDM-072

(warning) Non-editable abstract '%s' can only be opened as read-only

### Description

An attempt was made to open a read-only abstract in edit mode. The abstract has been opened in read-only mode instead.

### What Next

If you want to edit or optimize the logic inside of an abstract, you must create an editable abstract. Refer to the create_abstract man page for details.

### See Also

- create_abstract

---

## NDM-073

(information) %s view of block '%s' is not being saved because it is in read mode.

### Description

The specified block view is not saved because it is open in read mode.

### What Next

Open the block in edit mode and try again.

---

## NDM-074

(warning) Block '%s' can only be opened as read-only because it is part of an editable abstract '%s'.

### Description

When a block is included in an editable abstract, it becomes read-only. The abstract must be processed or deleted before the block may be edited again.

Feedback

### What Next

In order to make your block editable again, you must remove the existing editable abstract by using the remove_abstract command.

Alternatively, you may force the block to be editable by using "reopen_block -edit". If you do this, the editable abstract will be rendered unusable.

Changes made in the editable abstract view since its creation are automatically merged to the design view when the design view is loaded into memory.

### See Also

- merge_abstract
- remove_abstract

---

## NDM-075

(warning) Block '%s' was part of editable abstract '%s'. The abstract has been downgraded to read-only.

### Description

When a block is included in an editable abstract, it becomes read-only. The abstract must be processed or deleted before the block may be edited again.

By using the reopen_block command, you have forced the block to be editable. This means that the editable abstract is no longer usable. You will not be able to perform the merge_abstract operation.

### What Next

If you need a new editable abstract, you need to rerun the create_abstract command.

### See Also

- create_abstract
- remove_abstract

---

## NDM-076

(error) The abstract for block '%s' is not editable. The merge_abstract operation is not possible.

### Description

Only editable abstracts can be merged back into their parent block.

Feedback

### What Next

Please refer to the create_abstract man page for details about creating editable abstracts.

### See Also

- create_abstract

---

## NDM-077

(error) The abstract for '%s' depends on the design view of block '%s:%s', but the design view has changed or is missing. The merge_abstract operation is not possible.

### Description

When using editable abstracts, it is important to preserve the original design view block hierarchy on which the abstract is based. It is not possible to merge changes from the abstract back to the design view, if the block has changed or you are using a different copy of the block.

### What Next

It is possible that you changed the original design view hierarchy by using "reopen_block" or "remove_abstract". If the design view has changed, then you cannot run merge_abstract. Use remove_abstract on your top-level block to return it to a writable state.

It is also possible that you did not change your original design view hierarchy, but you are instead using a modified *copy* of your original design view. Try to find the copy of the block that was used to create the abstract.

### See Also

- create_abstract
- merge_abstract

---

## NDM-078

(error) Cannot include view '%s' of block '%s:%s' in an abstract.

### Description

When creating an abstract, the subblocks of the design must either be a "design" view or an "abstract" view.

Feedback

### What Next

Use the change_link command to include the proper view for the subblock.

### See Also

- change_link

---

## NDM-079

(information) %s view of block '%s:%s' has been restored to edit mode.

### Description

After merging or deleting an editable abstract, the blocks on which the abstract was based are reverted to edit mode.

### What Next

No action is required.

### See Also

- merge_abstract
- remove_abstract

---

## NDM-080

(warning) Orientation %s of cell %s is not allowed for grid %s.

### Description

A snap to grid operation is called trying to move the specified cell instance to be on the specified grid. However, the current orientation of the cell instance is not allowed for the specified grid. The cell is not moved.

### What Next

Check the cell's orientation, see if it needed to be rotated. Also check the association of the cell and the grid, see if the cell needs to be using another grid.

### See Also

- set_block_grid_references

Feedback

## NDM-081

(warning) Cell %s is not associated with grid %s.

### Description

A snap to grid operation is called trying to move the specified cell instance to be on the specified grid. However, the cell instance is not associated with the specified grid. The cell is not moved.

### What Next

Check the association of the cell and the grid.

### See Also

- set_block_grid_references

## NDM-082

(warning) Problem exporting %s %s : %s.

### Description

A problem was encountered when exporting the specified object to the specified format, likely due to differences between the object's configuration in the current database versus the configurability of the object in the target format.

## NDM-083

(error) Write-access request for %s failed. Library '%s' is being edited by process %d %s@ %s.

### Description

Write ccess requested by the specified operation failed because the library is being edited by another process.

### What Next

Please either try again (write-access on libraries are only held for a short time during actual write-to-disk and read-from-disk operation), or coordinate with the user holding the library for edit about releasing it.

The maximum wait time is controlled by the app_option "lib.setting.max_wait_time". The default value is 3600 seconds.

Feedback

## NDM-084

(error) '%s' failed. Another process has modified and saved library '%s' while this process was running, or a library with the same name has been copied, renamed or unpacked into this save location.

### Description

The specified operation failed because the library was modified and saved by another process while this process was running, or a library with the same name has been copied, renamed or unpacked into this save location.

### What Next

Please use save_lib -as to save your work to another library, or move/remove the conflicting on-disk library.

## NDM-085

(error) Cannot %s %s %s because %s.

### Description

The specified remove, modify, or create operation for the named type of the object given at full path has failed for the reason stated. or a library with the same name has been copied, renamed or unpacked into this save location.

### Examples

ERROR: Cannot remove net Reset because net is marked don't touch. (NDM-085)

### What Next

Obey the restriction, which is meant to keep the user from making unsafe changes that will have side effects in the flow. If this is not practical, set the app option design.eco_restrictions to false, and the operation will be permitted. Note that in the case of permitted unsafe changes, the performance may suffer significantly due to the requirements for more detailed checking and updating or recreation of many parts of the system data.

## NDM-086

(error) NDM file version %s is newer than the currently supported version %s. Use '%s <new_name>' in your original %s program to create an NDM library which can be read by this version of %s. See the man page for '%s' for further details.

Feedback

### Description

The library version is newer than the currently supported version.

### What Next

Use the original program that creates the library and save it to a supported version:

```
On IC Compiler II:

prompt> open_lib <lib_name>
prompt> save_lib -version J-2014.12 -as <new_name>

On IC Compiler II Library Manager:

prompt> create_workspace <workspace_name>
prompt> open_lib <lib_name>
prompt> commit_workspace -version J-2014.12
```

### See Also

- save_lib
- commit_workspace

---

## NDM-087

(warning) Block %s not found in current design, skipping.

### Description

A block listed in the connect_pg_net -block option could not be found instantiated in the current design. Connect_pg_net will skip missing blocks.

### What Next

To eliminate warnings, update -block list to include only blocks which are instantiated directly in current design.

---

## NDM-088

(info) Attempting to acquire write access for library (max wait until timeout is %d seconds).

### Description

Waiting to acquire write access on library which is held by another process.

Feedback

## NDM-089

(error) Another process has modified attributes and saved library '%s' while this process was running.

**Description**

The specified operation failed because attributes in this library were modified and saved by another process while this process was running.

**What Next**

Please use save_lib -as to save your work to another library, or move/remove the conflicting on-disk library.

## NDM-090

(error) Multiple processes have simultaneously attempted to create the same design '%s' and save to library '%s'.

**Description**

The specified operation failed because multiple processes attempted to simultaneously create and save the same design to a library.

**What Next**

Please use save_lib -as to save your work to another library, or move/remove the conflicting on-disk library.

## NDM-091

(error) Another processes attempted to delete the open design '%s' (detected while saving to library '%s').

**Description**

The specified operation failed because another processes attempted to delete a design that is open in the current process.

**What Next**

Please use save_lib -as to save your work to another library, or move/remove the conflicting on-disk library.

## NDM-092

(error) Unable to create %s '%s'. A %s already exists%s.

Feedback

**Description**

The specified operation failed because an object with the given name/attribute already exists.

**What Next**

Choose a different name/attribute for this new object.

---

## NDM-093

(error) '%s' has an experimental schema version '%s' incompatible with this production binary. The binary schema version is '%s'.

**Description**

The file has an experimental schema version. It is meant only for testing and it can only be opened with binaries from the same experimental branch. This schema version is incompatible with regular release versions and cannot be opened with production binaries.

**What Next**

Try again with a binary from the same experimental branch.

---

## NDM-094

(error) '%s' of experimental schema version '%s' is incompatible with this experimental binary from a different experimental branch. The binary schema version is '%s'.

**Description**

The file has an experimental schema version. It is meant for testing only and can be opened with binaries from the same experimental branch. This schema version is incompatible with regular release binaries or binaries with a different experimental suffix (The last portion of the schema version after .X).

**What Next**

Try again with a binary from the same experimental branch.

---

## NDM-095

(warning) Could not restore binding for reference '%s'.

**Description**

During *open_block* a block or lib cell reference could not be restored.

Feedback

The block being opened contains a reference to another block or lib cell. A block or lib_cell of that name could not be found in that library. The binding to that referenced block or lib cell could not be restored.

**What Next**

Verify that the requested library is in the ref_lib list and is open.

Check the named library to verify that a block with that name exists.

Use the *link_block* command to restore the bindings of the references in this block.

**See Also**

- link_block
- open_block
- set_ref_libs
- search_path

---

## NDM-096

(warning) Could not restore binding. Cannot locate library for reference '%s'.

**Description**

During *open_block* a block or lib cell reference could not be restored.

The block being opened contains a reference to another block or lib cell. The binding to that referenced block or lib cell could not be restored because the referenced library could not be found in the current ref_lib list.

**What Next**

Verify that the search_path location or reference library list setting correctly identifies the target block and try again.

If the reference libraries have moved but are still in the search_path, update the path to the ref_libs and re-link the design:

```
prompt> set_ref_libs -rebind
prompt> link_block
```

If the search_path to ref_lib list needs to be updated, do so, then re-link the design:

```
prompt> set_ref_libs -ref_libs { ... }
prompt> link_block
```

Feedback

### See Also

- link_block
- open_block
- set_ref_libs
- search_path

---

## NDM-097

(warning) Could not restore binding. Cannot locate block view '%s' for reference '%s'.

### Description

During *open_block* a block or lib cell reference could not be restored.

The block being opened contains a reference to another block or lib cell. The binding to that referenced block or lib cell could not be restored because the referenced block view could not be found in that library.

### What Next

Check the referenced library to see what views exist of that block:

```
prompt> get_blocks -all -lib_cells LIBRARY-NAME:BLOCK-NAME.*
```

If the desired block view is in another library, make that library part of the reference library list with the *set_ref_libs* command. Then re-link the design with the *link_block* command.

If a different block view is desired, change the cell's reference with the *change_view* or *set_reference* command.

### See Also

- change_view
- get_blocks
- link_block
- open_block
- set_ref_libs
- set_reference
- search_path

Feedback

## NDM-098

(warning) Could not restore binding. Port interface of block '%s' does not match reference '%s'.

### Description

During *open_block* a block or lib cell reference could not be restored.

The block being opened contains a reference to another block or lib cell. The binding to that referenced block or lib cell could not be restored because the port interface of the referenced block has changed since the block was last saved.

### What Next

Update the cell's pin interface to match the referenced block's new port interface using the *link_block* command:

```
prompt> link_block
```

### See Also

- link_block
- open_block

## NDM-099

(info) Running auto PG connection.

### Description

This is a message indicating auto PG connection is running.

## NDM-100

(error) Another process deleted design '%s' (detected while syncing to library '%s').

### Description

The specified operation failed because another process deleted a design that is not yet opened in the current process.

### What Next

The specified design is no longer available. Check the flow or concurrent library environment.

Feedback

## NDM-101

(error) Cannot open block '%s'. Binding to technology library is missing.

### Description

The reference technology library for this block is required but is missing. The block cannot be opened until this reference is resolved. Check the refLib list on the library and verify all refLibs are bound.

### What Next

Use report_ref_libs to verify all refLibs are bound. Unbound refLibs are those refLibs whose name or location are not available. If there are unbound refLibs, modify the search_path so that the refLib paths are reachable and use set_ref_libs -rebind to restore the binding.

```
prompt> report_ref_libs
****************************************
Report : Reference Library Report
Library: 1
****************************************

   Name Path      Location
   -----------------------
   -    tech_lib  -
    "-" = Name and location not available
1

prompt> set_ref_libs -rebind
```

### See Also

- report_ref_libs
- set_ref_libs

## NDM-102

(warning) Technology '%s' used for frame-view creation in library '%s', is inconsistent with the current technology '%s' of library '%s'. Please run derive_design_level_via_regions to generate up-do-date via region in the design library.

### Description

The technology of the reference library is inconsistent with technology of the design library. Need to run derive_design_level_via_regions to generate up-do-date via region in the design library.

Feedback

**What Next**

Update the library with consistent technology and try again. Or run derive_design_level_via_regions to generate up-do-date via region in the design library.

**See Also**

- create_lib
- open_lib
- read_tech_file
- derive_design_level_via_regions

---

## NDM-103

(info) %s Sparse View library '%s' %s base library '%s'.

**Description**

This is a message indicating set/update of the base library attribute of a Sparse View library.

---

## NDM-104

(info) Updating %s of Sparse View library '%s' from base library '%s'.

**Description**

This is a message indicating synchronization of a Sparse View library to pull-in updates from its base library.

---

## NDM-105

(info) Adding %s '%s' to Sparse View library '%s'.

**Description**

This is a synchronization message indicating addition of remote content to the Sparse View library from the local on-disk content of the base library.

---

## NDM-106

(info) Removing %s '%s' from Sparse View library '%s'.

Feedback

**Description**

This is a synchronization message indicating removal of remote content from the Sparse View library because its local on-disk content doesn't exist in the base library.

---

## NDM-107

(error) Cannot open '%s'. It is an internal file within a directory format library.

**Description**

The specified path is an internal data file within a directory format library. It cannot be opened directly.

**What Next**

Find out the name of the library and use open_lib to open it:

prompt> *open_lib <lib_path>*

**See Also**

- open_lib
- open_block

---

## NDM-108

(error) The given drc error file does not exist.

**Description**

This error message occurs when the specified file does not exist in the directory. It may be either misspelled or an empty string.

**What Next**

Verify that you specified the correct path for the file and the name string.

---

## NDM-109

(error) Cannot restore site name on '%s'. Site definition '%s' is missing from technology.

**Description**

The specified site definition is missing. Add the missing site definition to the technology and try again.

Feedback

### See Also

- open_block
- create_site_def
- get_site_defs

---

## NDM-110

(information) Recycled %d file descriptors.

### Description

The specified number of file descriptors have been recycled as the user-specified open file limit has been reached.

### See Also

- open_lib

---

## NDM-111

(warning) Frame views in library '%s' is inconsistent with technology '%s'.

### Description

The technology of this library has changed after frame creation. Frame views in this library are out-of-sync with the technology and should be re-created.

### What Next

Re-generate the frame views using a consistent technology.

### See Also

- open_lib
- create_frame
- create_workspace

---

## NDM-112

(Information) Opening block '%s' in read-only mode.

Feedback

**Description**

The block is being opened in read-only mode because system resources have been exhausted.

---

## NDM-113

(Information) Changing block '%s' to read-only mode.

**Description**

The block is being changed to read-only mode because system resources have been exhausted.

---

## NDM-114

(error) Invalid port to be %s the port bus.

**Description**

Ports which have the same base name as port bus and are part of vector can be added or removed from the port bus.

**What Next**

Verify that you specified the correct port for the bus port.

---

## NDM-115

(error) Port %s will create a hole in the port bus.

**Description**

Ports addition/removal should be done from the sides of the port bus. Post update there should not be any hole in the port bus.

**What Next**

Verify that you specified the correct port for the bus port.

---

## NDM-116

(error) Port can not be removed from the port bus having single port.

**Description**

There should be non empty port bus post port removal.

Feedback

**What Next**

Verify that bus port should not become empty post removal.

---

## NDM-117

(Information) The specified row/column should not less than cut row/column number '%u/%u' of base vias inherited from the via def '%s'.

**Description**

The cut row/column number isn't specified or the specified number is too small resulting that the number less than the cut row/columhn inherited from the via def. The row/column change doesn't succeed.

---

## NDM-118

(error) cannot change %s for the custom via def.

**Description**

This command cannot change row/column/XPitch/YPitch of the specified base via. Base via must be simple base via, and simple base via cannot be changed to custom base via.

**What Next**

Please provide the correct input for simple base via only.

---

## NDM-119

(error) Invalid net to be %s the net bus.

**Description**

Nets which have the same base name as net bus and are part of vector can be added or removed from the net bus.

**What Next**

Verify that you specified the correct net for the bus net.

---

## NDM-120

(error) Net %s will create a hole in the net bus.

Feedback

**Description**

Nets addition/removal should be done from the sides of the net bus. Post update there should not be any hole in the net bus.

**What Next**

Verify that you specified the correct net for the bus net.

---

## NDM-121

(error) Net can not be removed from the net bus having single net.

**Description**

There should be non empty net bus post net removal.

**What Next**

Verify that bus net should not become empty post removal.

---

## NDM-122

(information) Loading M-2016.12 library '%s'.

**Description**

This message is to inform the user that a forward version library, created from M-2016.12 binary is being loaded. The library and its design data will be saved in M-2016.12 format by default, or can be down converted with the save_lib -as <lib_name> -version L-2016.03 command

---

## NDM-123

(error) Can't create supernet bundle '%s' as another bundle exists by same name.

**Description**

Data was saved using a version that created supernet bundles and then was taken back into an older version with same schema that doesn't know about supernet bundles wherein new bundle was created with the same name as that of the supernet bundle. The data was saved in this older version and then brought back into current version.

**What Next**

Please use a different name than the name of supernet bundle while creating bundle in the older version.

Feedback

## NDM-124

(error) %s '%s' as an object of supernet bundle '%s' doesn't exist.

### Description

Supernet bundle contains supernets and/or bundles of supernets. Data was saved using a version that created supernet bundles and then was taken back into an older version with same schema that doesn't know about supernet bundles.

If the message refers to another bundle of supernets then a new bundle was created in the older version with the same name as that of a supernet bundle. If message refers to a supernet then the supernet was removed in the older version. The data was saved in this older version and then brought back into current version.

### What Next

If the message refers to another bundle of supernets then please use a different name than the name of supernet bundle while creating bundle in the older version. If message refers to a supernet then it is fine as it was removed in the older version by intent.

## NDM-125

(error)Specified row and column(%u %u) exceed via matrix size %u and %u for %s.

### Description

The Specified row and column should be non negative and less than via matrix size.

### What Next

Specify the row and column between 0 and via matrix size.

## NDM-126

(warning) Hierarchical edit-control of uneditable (read-only) block %s is violated for modification of %s object %s(%d)%s.

### Description

This message is issued when a read-only block is being modified. The hierarchical edit-control of the block doesn't allow edit but the reported object of such uneditable (read-only) block is being modified.

### What Next

Report this issue to Synopsys.

Feedback

## NDM-127

(warning) Saving uneditable (read-only) block %s because it has already been modified.

**Description**

This message is issued when a read-only block is being saved. The hierarchical edit-control of the reported block doesn't allow edit but the uneditable (read-only) block has already been modified and requires save.

**What Next**

Report this issue to Synopsys.

## NDM-128

(information) Uneditable (read-only) block %s is not being saved.

**Description**

This message is issued when a read-only block is not being saved. The hierarchical edit-control of the reported block doesn't allow edit and hence the block doesn't require save.

## NDM-129

(information) The hierarchical edit-control of committed block %s is set editable in the top-level context of %s.

**Description**

This message is issued when a physical hierarchy is being committed as a new block. The hierarchical edit-control of the reported block is set to allow edits.

## NDM-130

(information) The hierarchical edit-control of uncommitted block %s is cleared from the top-level context of %s.

**Description**

This message is issued when a physical hierarchy is being uncommitted and its block disappears. The hierarchical edit-control of the reported block is cleared from its top-level context.

Feedback

## NDM-131

(information) Master lock use for library "%s" changed to "%s" due to detection of %s.

**Description**

This message is issued when the lock mechanism is changed from the user-defined value. This change is due to the detection of another lock mechanism in use for the lib.

**What Next**

If the resulting change causes an error, close the library and notify other session that are using the lib to close the library and either let you open the library first, or change their design.master_lock to the same value you are using.

## NDM-132

(warning) Resulting %s bound has %d elements, suggested limit is %s.

**Description**

This message is issued when the given type of bound has more than the suggested number of elements. The suggested number usually indicates a good threshold for maintaining good performance during placement operations.

**What Next**

If resulting placement operations slow down significantly consider reducing the number of elements in this bound.

## NDM-133

(error) Repelling bound requires argument -dimension.

**Description**

A repelling bound must have dimensions set upon creation.

**What Next**

Create the bound with -dimension argument specified.

## NDM-134

(error) Resulting repelling bound contains only port objects.

Feedback

**Description**

A repelling bound cannot consist of only port objects as ports are fixed. As such a repelling bound containing only ports cannot respect a given repulsion dimension.

**What Next**

Try including at least one cell in the bound.

---

## NDM-135

(information) Overwriting design '%s' in library '%s' created by another process.

**Description**

This info message is to alert the user that a design created by another process is being overwritten.

---

## NDM-136

(warning) The Path '%s' is an all angle path. The the boundary coordinates of the path will be displayed similar to 45 degree path.

**Description**

This message is issued when the input path shape is an all angle path. The polygon path will be dispalyed similar to all angle paths with 45 degree points.

**What Next**

If the returned path points are not desirable, the centre point coordinates can be modified to make it a 45 degree path.

---

## NDM-137

(information) Setting blockage type to '%s' requires blocked percentage to be set as well. Blocked percentage is set to 100.

**Description**

Setting blockage type to above requires blocked percentage to be set as well. Blocked percentage is set to 100.

**What Next**

Please set blockage percentage using "set_attribute -object 'foo' -name blocked_percentage -value 'bar'"

Feedback

## NDM-138

(information) Setting blockage type to '%s' requires category name to be set as well. Category name is set to "".

### Description

Setting blockage type to above requires category name to be set as well. Category name is set to "".

### What Next

Please set category name using "set_attribute -object 'foo' -name category -value 'bar'"

## NDM-139

(error) Can't create shield overhang '%s' as another constraint group exists by same name.

### Description

Data was saved using a version that created shield overhangs and then was taken back into an older version with same schema that doesn't know about shield overhangs wherein new constraint group was created with the same name as that of the shield overhang. The data was saved in this older version and then brought back into current version.

### What Next

Please use a different name than the name of shield overhang while creating constraint group in the older version.

## NDM-140

(error) %s '%s' as an object of shield overhang '%s' doesn't exist.

### Description

Shield overhang contains objects of types nets, bundles or topology edge. Data was saved using a version that created shield overhangs and then was taken back into an older version with same schema that doesn't know about shield overhangs.

The mentioned object in the message was removed in the older version. The data was saved in this older version and then brought back into current version.

### What Next

The mentioned object was removed in the older version by intent. It is no longer possible to have that object in the shield overhang constraint group.

Feedback

## NDM-141

(error) Unable to open_block '%s' due to license check failure. The license key '%s' is required.

**Description**

Ths block is license protected. Opening it requires certain licenses to be available. The tool was not able to check out the required licenses, so the block cannot be opened.

**What Next**

Please make sure the required licenses are available and can be checked out.

## NDM-142

(error) Cannot add shape '%s' to via ladder because it is connected to a different net than other shapes in this via ladder.

**Description**

The specified shape cannot be added to the via ladder because it is connetced to a different net than other shapes in this via ladder. All the shapes in a via ladder should be connected to the same net.

**What Next**

Pass a collection of shapes to the createViaLadder api command which belong to the same net and try again.

## NDM-143

(error) Shape '%s' is a part of an existing via ladder object.

**Description**

A shape can only be a part of exactly one via ladder object.

**What Next**

Please remove this shape from the list of shapes for the current via ladder.

## NDM-144

(error) Cannot delete rule '%s' since it is being referenced by group '%s' .

Feedback

**Description**

There is some problem in remove. This could happen when object is getting referenced by some other object.

**What Next**

First remove the rule from group then remove rule.

---

## NDM-145

(warning) The pin %s is not of any of the types specified in the rule.

**Description**

There is some problem in pin(s). This could happen when type of pin(s) does not match with the type(s) specified in the rule.

**What Next**

Specify the pin matching with the type specified in rule.

---

## NDM-146

(information) Layer %s (%d) in the source technology file %s does not match the layer of the same name in the destination library.

**Description**

You get this message when you apply a new technology file which has layer number and layer name different from current technology.

**What Next**

Make sure that the differences are expected or replace with correct technology file. If differences are expected then update various layer mapping files also e.g. related with read_parasitic_tech etc. as per layers in new technology file.

---

## NDM-147

(information) Layer %s (%d) is in the destination library but not in the source technology file %s.

**Description**

You get this message when you apply a new technology file which is missing layer(s) which are present in the current technology.

Feedback

**What Next**

Make sure that the differences are expected or replace with correct technology file. If differences are expected then update various layer mapping files also e.g. related with read_parasitic_tech etc. as per layers in new technology file.

---

## NDM-148

(error) Resulting %s bound has %d cores, while the limit is %d.

**Description**

This message is issued when the given type of bound has more than the allowed number of cores. Cores are modelled as either hierarchical cells or repelling bound groups.

**What Next**

Reduce the number of cores in this bound.

---

## NDM-149

(error) Failed to open the cell-rename map file '%s'.

**Description**

The cell-rename map file could not be opened. Verify the file-permissions and retry.

---

## NDM-150

(error) Required section '%s' was not found before section '%s'.

**Description**

The name mapping for mask-shifted cells comprises three sections: MASKSHIFTLAYER, MASKSHIFTPATTERN and CELLMAP. You need to define the sections in the order above.

**What Next**

Supply the required section.

---

## NDM-151

(error) Invalid mask shift layer '%s'%s.

**Description**

This error message occurs when the specified mask shift layer is invalid, it must meet the following conditions. 1. Multiple cell mapping files should have the same mask shift layer,

Feedback

whether the number or the order of layer must be the same. 2. The layers must exist in the current library.

**What Next**

Please check and correct the mask shift layer, then try again.

---

## NDM-152

(warning) Incorrect syntax '%s' specified for cell-rename entry. The entry will be ignored.

**Description**

The syntax for the cell-rename entry is not a valid one. Refer the man-page of write_verilog/write_gds for the appropriate syntax.

---

## NDM-153

(error) Invalid mask shift pattern '%s'%s.

**Description**

This error message occurs when the specified mask shift pattern is invalid, it must meet the following conditions. 1. Multiple cell mapping files should have the same mask shift pattern. 2. The number of the mask shift combination is equal to or less than pow(m, n), where n is the number of layers specified in MASKSHIFTLAYER and m is the count of the shift values. For 2-mask layers, valid shift values are 0 and 1, so m is 2. For 3-mask layers, valid shift values are 0, 1, 3, 4 and 5, so m is 5. When the shift value is 3, 4 or 5, the shifted layer must have fixed mask 1, 2 or 3, repectively and its number_of_masks must be more than 2. 3. MASKSHIFTPATTERN must contain the mask shift combination for the cell without any mask shifting.

**What Next**

Please check and correct the mask shift pattern, then try again.

---

## NDM-154

(error) Invalid cell mapping value '%s'.

**Description**

The number of the cell mapping value is not consistent with the number of the specified mask shift pattern.

**What Next**

Please check and correct the cell mapping value, then try again.

Feedback

## NDM-155

(error) Mask shifted cell '%s' does not have its corresponding mapping in the map file.

**Description**

There is not a corresponding mapping for the mask shifted cell in the map file.

**What Next**

Please add the corresponding mapping to the map file and then try again.

## NDM-156

(error) Mask shift property mismatch on cell '%s'.

**Description**

There is mask shift property mismatch between different map files.

**What Next**

Please correct the mask shift property and then try again.

## NDM-157

(error) Fail to find the pattern '%s' for cell '%s'.

**Description**

The pattern is not specified in MASKSHIFTPATTERN section.

**What Next**

Please check and correct it, and then try again.

## NDM-158

(warning) The design '%s' with the view '%s' is not found in Library '%s'. But different views of the design exists in the library.

**Description**

The design with the specified name and view is not found in the searched library. But other views of the design exists in the Library.

**What Next**

Please specify the correct Library for the search.

Feedback

## NDM-159

(Info) Don't support alternating mask pattern.

**Description**

Don't support 'alternating' mask pattern. The valid mask pattern is one of uniform, arbitrary, alternate_rows, alternate_columns.

**What Next**

Please specify the valid mask pattern.

## NDM-160

(warning) Cannot edit %s because it is part of via_ladder.

**Description**

The specified object cannot be edited or deleted because it is member of a via ladder and via ladder protection is enabled.

**What Next**

Disable via ladder protection using app option design.via_ladder_protection or remove the object from its via ladder.

## NDM-161

(error) Attempt to write '%s' failed due to '%s'.

**Description**

Unable to write a file.

## NDM-162

(error) Cannot delete fsm '%s' since it is being referenced by rule or group '%s' .

**Description**

There is some problem in remove. This could happen when object is getting referenced by some other object.

**What Next**

First remove the rule or group then remove fsm.

Feedback

## NDM-163

(error) Invalid '%s' option usgae : It can only be used with '%s' type objects in the list.

**Description**

Command option '-original_parent' can only be used with cell type of objects in the list and not with port or module/block objects.

**What Next**

Please provide cell type of objects with '-original_parent' option otherwise don't use this option for other type of objects in the list.

## NDM-164

(error) Reading from '%s' failed due to '%s'.

**Description**

Error at reading a file

## NDM-165

(information) Waiting to acquire write access for %s. Library '%s' is being edited by process %d %s@%s.

**Description**

Unable to acquire write access to the library because write access is held by another process. The specified operation will be retried until the maximum wait time is reached.

The maximum wait time is controlled by the app_option "lib.setting.max_wait_time". The default value is 3600 seconds.

## NDM-166

(warning) Rail index %d is invalid on port %s.

**Description**

The rail index on the specified port is invalid, since it is corrupted data, the rail attribute on the port will be dropped when loading the reference library. To be specific, the input_signal_level and output_signal_level attributes on signal port will be lost, and rail attribute on power or ground port will be lost . There is a known issue in Library Preparation on 2019.12-SP2 release which leads to this error. The problem was fixed in 2020.09-SP6 release. The missing rail attributes on port could impact the result of

Feedback

downstream tools, so it is highly recommended to regenerate the reference libraries using 2020.09-SP6 or later releases.

**What Next**

Regenerate the reference libraries using 2020.09-SP6 or later releases.

---

## NDM-167

(Error) The current shell cannot open designs of type '%s'.

**Description**

Some design types may only be opened in specific applications. For example designs specific to 3dic systems (3dic, package, substrate, bridge and interposer) may only be opened by the 3dic shell and not by Fusion Compiler and IC Compiler II.

**What Next**

Open the design in the application used to create the design.

---

## NDM-168

(Warning) This type of repelling group bounds for functional safety is deprecated and replaced by safety_core_groups.

**Description**

The use of repelling bounds for specifying dual-core lockstep for functional safety is deprecated and has been replaced by safety_core_groups.

**What Next**

Use create_safety_core_rule and create_safety_core_group commands to define dual-core lockstep (DCLS) functionality.

---

## NDM-169

(warning) Cannot change net_type from %s to %s on net '%s'.

**Description**

The segment net is "*Logic0*" or "*Logic1*". These are reserved by Synopsys for the tie-low and tie-high constant nets. Their types cannot be changed.

**What Next**

No action is required.

Feedback

### NDM-170

(information) Restored app option '%s' with value '%s'.

**Description**

This info message is to alert the user that app option is restored.

### NDM-171

(error) Could not find a system startup file '%s' in '%s' .

**Description**

There is some problem in sourcing system startup files. This could happen when the system startup file is not available or permission is not proper.

**What Next**

Please check if the system startup file is available or if the file permissions are proper.

## NDMIO

### NDMIO-010

(warning) App option '%s' of design '%s' is not registered in current tool version, hence not effective. This setting will be stored internally for compatibility purpose.

**Description**

The specified app option is not registered. This could happen when an app option is removed after the app option data was saved.

**What Next**

Use get_app_options to verify that the app option is available. Also can use 'report_app_options -unknown' to report the unknown app option values.

### NDMIO-011

(warning) The value for the app option '%s' of design '%s' could not be set. This setting will be stored internally for compatibility purpose.

Feedback

**Description**

The specified app option value couldnot be set. This could happen when an app option value type is modified after the app option data was saved.

**What Next**

Use get_app_options to verify that the app option is available. Also can use 'report_app_options -unknown' to report the unknown app option values.

# NDMUI

## NDMUI-001

(warning) Boundary option is ignored when hierarchy_type being changed to normal.

**Description**

The boundary option is ignored in the command *set_cell_hierarchy_type* when the new *hierarchy_type* is *normal*.

**What Next**

Nothing is needed, boundary specified is ignored

## NDMUI-002

(error) Only one cell can be specified as input%s

**Description**

This command accepts only one object

**What Next**

Please provide the correct input.

## NDMUI-003

(error) The command '%s' can only be used for resolved hierarchical cells. Cell '%s'.

**Description**

Only hierarchical cells can be used in this command. The cell must also be resolved during link.

Feedback

**What Next**

Use hierarchical cells instead of leaf cells

---

## NDMUI-004

(error) The hierarchical cell belongs to different physical hierarchy.

**Description**

The cell used as input to command *set_cell_hierarchy_type* must belong to the top physical hierarchy.

**What Next**

Please provide the correct input.

---

## NDMUI-005

(error) Failed to create %s.

**Description**

hier_boundary or partition could not be created for the specified cell.

**What Next**

Please check the errors issued in the command.

---

## NDMUI-006

(error) Only hierarchical cells which are physical hierarchies can be used.

**Description**

Only a cell which is a physical hierarchy can be specified. The cell must also be fully resolved. Unresolved cells cannot be accepted as input to this command.

**What Next**

Please provide the correct input.

---

## NDMUI-007

(error) Only rectangles are supported.

**Description**

The option *-coordinate* must specify a rectangle for this command.

Feedback

**What Next**

Please provide the correct input.

---

## NDMUI-008

(error) Failed to create routing guide.

**Description**

The routing guide could not be created.

**What Next**

Please check any previous errors and provide correct input.

---

## NDMUI-009

(error) Design with name %s already exists in current library.

**Description**

The current library already has a design with the specified name, therefore a new design with the same name can not be created.

**What Next**

Please specify a name which does not conflict with existing design names

---

## NDMUI-010

(warning) The '%s' command cannot be used on library cell '%s' which has no logical model.

**Description**

The command is valid only for library cells with a logical/timing model from .lib/.db.

**What Next**

If there is a .db library file for this library, perform library data preparation again with this data included. If not, re-enter the command with library cells that do have a logical model.

---

## NDMUI-011

(error) cannot find via definition '%s'.

Feedback

**Description**

Specified via definition cannot be found.

**What Next**

Please provide the correct via definition name.

---

## NDMUI-012

(error) value '%d' for '%s' is invalid; must be an integer>=1.

**Description**

Specified value is invalid; The value must be an integer>=1;

**What Next**

Please provide the correct input.

---

## NDMUI-013

(error) Cannot instantiate current_design within itself.

**Description**

The *create_cell* command was used to try to create a cell in the current design that referenced the current design. This would create a cycle in the design hierarchy, which is not allowed.

**What Next**

Choose a different *reference* for the *create_cell* command.

---

## NDMUI-014

(error) Cell %s already exists.

**Description**

The *create_cell* command was used to try to create a cell, but a cell with that name already exists.

**What Next**

Choose a different *cell name* for the *create_cell* command.

Feedback

## NDMUI-015

(error) Invalid reference specified.

**Description**

The *create_cell* can only accept library cells or physical blocks as reference. Any other object type is invalid.

**What Next**

Please specify a lib cell or block as reference.

## NDMUI-016

(error) Cannot set origin of cell '%s'.

**Description**

The *create_cell* command was used to create an instance of logical hierarchy, and the origin was specified. That origin could not be set on the newly created cell.

Instances of logical hierarchy do not have locations. Only instances of physical designs or lib_cell designs can have locations.

**What Next**

Do not specify *-origin* for logical hierarchy cells. Only specify *-origin* for instances of lib_cells or physical designs.

## NDMUI-017

(error) Cannot create cell '%s'.

**Description**

The *create_cell* command was unable to create the cell.

**What Next**

Try using a different name for the cell. Or try using a different reference.

## NDMUI-018

(error) Cannot remove cell '%s'.

**Description**

The *remove_cell* command was unable to remove the specified cell.

Some hierarchical cells cannot be removed while their contents are active.

**What Next**

Try removing the contents of a hierarchical cell before removing the parent cell. Try removing a different cell.

---

## NDMUI-019

(error) Cannot remove net '%s'.

**Description**

The *remove_nets* command was unable to remove the specified net.

**What Next**

Try removing a different net.

---

## NDMUI-020

(error) Cannot remove port '%s'.

**Description**

The *remove_ports* command was unable to remove the specified port.

**What Next**

Try removing a different port.

---

## NDMUI-021

(error) Cannot remove pin '%s'.

**Description**

The *remove_pins* command was unable to remove the specified pin.

**What Next**

Try removing a different pin.

Feedback

## NDMUI-022

(error) Cannot connect driver pin '%s' to pins:

### Description

The *connect_pins* command was unable to connect the given driver pin to the given load pins. Some levels of hierarchy could not have ports created in them.

### What Next

Try removing some of the load pins from the connect_pins command to discover which part of the hierarchy could not be connected through.

## NDMUI-023

(error) Cannot size cell '%s' because the new lib_cell '%s' not compatible with current lib_cell '%s'.

### Description

The *size_cell* command was unable to change the lib_cell used for the specified cell. The new lib_cell either has a different port interface, or a different function from the current lib_cell.

The *size_cell* command cannot be used to change the functionality or the connectivity of the net list.

### What Next

Try using a different lib_cell when sizing this cell.

## NDMUI-024

(error) Cannot size cell '%s', the new reference design %s has a different port interface.

### Description

The *size_cell* command was unable to set the cell to the new reference the given design. The new reference design has a different port interface from the current design.

The *size_cell* command cannot be used to change the connectivity of the net list.

### What Next

Try using a different lib_cell when sizing this cell.

Feedback

## NDMUI-025

(error) Cannot remove buffer '%s' because it has %u pins.

**Description**

The *remove_buffers* command can only removes buffer that have 2 pins. Buffers with more or less than 2 pins cannot be removed with the *remove_buffers* command.

The *remove_buffers* command doesn't know how to restore the net connections for cells that have other than 2 pins.

**What Next**

Manually remove the cell and restore the net connections with the *remove_cell* and *connect_net* commands.

## NDMUI-026

(error) Cannot remove cell '%s' because has no %s connection.

**Description**

The *remove_buffers* command can only remove buffers that have input and output net connections. Without both an input and output, the net connection cannot be restored after the buffer is removed.

**What Next**

Manually remove the cell and restore the net connections with the *remove_cell* and *connect_net* commands.

## NDMUI-027

(error) Cannot remove buffer cell '%s'.

**Description**

The *remove_buffers* command was unable to remove the specified buffer.

Perhaps the cell is not a buffer, the cell could not be removed, or is not connected to nets.

**What Next**

Manually remove the cell and restore the net connections with the *remove_cell* and *connect_net* commands.

Feedback

## NDMUI-028

(error) Module '%s' already exists in design '%s'.

**Description**

A module with the given name already exists in this design. Or this design already references another design with this name.

**What Next**

Choose a different name for this new module.

## NDMUI-029

(error) Cannot create module '%s' because a module with that name already exists in design '%s'.

**Description**

A module with the given name already exists in this design. Or this design already references another design with this name.

**What Next**

Choose a different name for this new module.

## NDMUI-030

(error) Cannot remove top module '%s' from design '%s'.

**Description**

The *remove_modules* command attempted to remove the top module from this design. Only unused hierarchical children modules may be removed from a design.

**What Next**

Remove the entire design with the *remove_block* command.

## NDMUI-031

(error) Cannot remove module '%s' used in design '%s'.

**Description**

The *remove_modules* command attempted to remove a module that is currently used as a hierarchical cell instance in the design.

Feedback

The module cannot be removed before all instances of that module are removed.

**What Next**

Locate all cells that reference the module by filtering the result of the *get_cells* command with the *ref_module_name* attribute. Then remove those cells with the *remove_cell* command.

---

## NDMUI-032

(error) Cannot change view of cell '%s' because it is not an instance of a lib_cell or block.

**Description**

The *change_view* command attempted to change the referenced view of a logical hierarchy cell. Views can only be changed on cells that instantiate physical designs, or lib_cells.

**What Next**

Choose a leaf cell, or a cell that instantiates a physical design.

---

## NDMUI-033

(error) Min (%d) can not be greater than max (%d).

**Description**

The *min_routing_layer* specified can not be greater than the *max_routing_layer* specified.

**What Next**

Provide the correct inputs and rerun the command.

---

## NDMUI-034

(info) Update of ignored layers for design %s %s.

**Description**

Informational message showing whether the update succeeded or failed.

**What Next**

This is informational message only. To turn it off, do not use the verbose option to the command

Feedback

## NDMUI-035

(error) Data type '%s' may be included only with format '%s'.

**Description**

This data type cannot be used with the currently specified format. It is only supported format indicated in the error message.

**What Next**

To continue including this data type, specify the supported format with the *-format* option. Or to continue using this format, remove this data type from the *-include* option, or explicitly exclude this data type using the *-exclude* option.

## NDMUI-036

(error) The specified object '%s' is in a different physical design.

**Description**

The input to this command must be a object in the current physical design. The specified object belongs to a different physical design.

**What Next**

Use the command 'current_design' to switch to the design which actually owns the object and rerun the command with proper input.

## NDMUI-037

(warning) voltage_area_shape '%s' is already at the top of the stack.

**Description**

The specified voltage_area_shape is already at the top of the position stack. set_voltage_area_shape will be a no-op.

## NDMUI-038

(warning) voltage_area_shape '%s' is already at the bottom of the stack.

**Description**

The specified voltage_area_shape is already at the bottom of the position stack. set_voltage_area_shape will be a no-op.

Feedback

## NDMUI-039

(warning) voltage_area_shape '%s' is already directly %s voltage_area_shape '%s'.

**Description**

The voltage_area_shape is already directly above or below the specified shape. set_voltage_area_shape will be a no-op.

## NDMUI-040

(error) A voltage_area_shape cannot be placed below the default voltage_area_shape.

**Description**

The first shape of the default voltage_area corresponds to the core area and must remain at the bottom of the position stack.

## NDMUI-041

(error) voltage_area_shape cannot be moved below or above itself.

**Description**

It is an error to move a voltage_area_shape relative to itself.

## NDMUI-042

(error) voltage_area_shape '%s' belong to a different design '%s'.

**Description**

Each each has a separate voltage_area_shape stack. It is an error to move a voltage_area_shape from one design to another.

## NDMUI-043

(error) Unable to apply connect_pg_net to '%s' net '%s'. Only power and ground nets are allowed.

**Description**

Only power and ground nets can be specified in this command.

**What Next**

Specify a power or ground net.

Feedback

## NDMUI-044

(error) Default voltage_area_shape cannot be moved.

**Description**

The first shape of the default voltage_area corresponds to the core area and must remain at the bottom of the position stack.

## NDMUI-045

(information) Command %s will be obsoleted, please use %s instead.

**Description**

This message informs you that the command specified in the information message will be obsolete. You are advised to use the specified replacement command instead.

**What Next**

Use the replacement command instead.

## NDMUI-046

(error) Option '%s' must be used when using options '%s' and '%s' together in this command.

**Description**

The option '-physical_context' must be used when 'get_nets' is invoked with '-hierarchical' and '-of_objects'. Further, the objects specified as input to '-of_objects', must be a net. No other use of '-hierarchical' and '-of_objects', is permitted for 'get_nets'.

**What Next**

Please provide the correct inputs.

## NDMUI-047

(error) May specify no more than one of the -design, -library, and -global arguments.

**Description**

This command applies to the view switch list of either a design, library, or the global session. Therefore only one of the -design, -library, or -global arguments may be specified.

Feedback

**What Next**

Change the arguments to include no more than one of -design, -library, or -global. Note that removing them all is equivalent to using the -global switch.

---

## NDMUI-048

(error) Duplicate view '%s' specified in view switch list.

**Description**

The view switch list argument of this command must be a list of unique view names.

**What Next**

Remove the duplicate view names from the view switch list argument.

---

## NDMUI-049

(error) View of the current design and view of the design being uncommitted are different.

**Description**

The view of the current design and the view of the design being uncommitted do not match. The design associated with the physical hierarchical cell being uncommitted must be of the same view as the current design.

**What Next**

Please provide the correct inputs.

---

## NDMUI-050

(error) Value '%s' for routing corridor shape is invalid. It must be in rectangle, rectilinear or path format.

**Description**

The value of points must specify a legal shape of routing corridor. The legal shape must be a rectangle, or rectilinear or path.

**What Next**

Please provide the correct input representing legal routing corridor shapes.

---

## NDMUI-051

(error) Value '%s' for routing corridor min_layer (max_layer) is invalid.

Feedback

**Description**

Min layer and max layer must be legal routing layers. For specified min routing layer, it must be lower than max routing layer. For specified max routing layer, it must be higher than min routing layer.

**What Next**

Please provide the correct inputs.

---

## NDMUI-052

(error) Unable to create routing corridor '%s'. A corridor of that name already exists.

**Description**

The *create_routing_corridor* command was used to try to create a routing corridor, but a routing corridor with that name already exists.

**What Next**

Choose a different *routing corridor name* for the *create_routing_corridor* command.

---

## NDMUI-053

(warning) Object of name '%s' already exists in the routing %s '%s'. It will be ignored.

**Description**

The specified net has already been associated with this routing-corridor or routing-blockage object.

**What Next**

There is no need to specify a net already associated with this routing-corridor or routing-blockage object.

---

## NDMUI-054

(warning) Object of name '%s' does not exist in the routing %s '%s'. It will be ignored.

**Description**

The specified net has not been associated with this routing-corridor or routing-blockage object.

**What Next**

Please provide the correct input.

Feedback

## NDMUI-055

(error) Failed to add object '%s' to the given bundle

**Description**

The specified object could not be added to the given bundle.

**What Next**

Please provide the correct inputs.

## NDMUI-056

(error) Failed to remove object '%s' from the given bundle

**Description**

The specified object could not be removed from the given bundle.

**What Next**

Please provide the correct inputs.

## NDMUI-057

(error) Failed to remove bundle '%s' from the design

**Description**

The specified bundle could not be removed.

**What Next**

Please provide the correct inputs.

## NDMUI-058

(error) A bundle with the name '%s' already exists

**Description**

The design already contains a bundle with the name specified, for creating a new bundle.

**What Next**

Please provide the correct inputs.

Feedback

## NDMUI-059

(error) Only nets, bundles or supernets can be added to bundles

**Description**

As of now, only nets, bundles or supernets can be added to bundle objects. Any other object is not supported.

**What Next**

Please provide the correct inputs.

## NDMUI-060

(error) Only objects in the current design can be added to bundles

**Description**

Only those objects which are physically present in the design, can be added to a bundle.

**What Next**

Please provide the correct inputs.

## NDMUI-061

(error) %s be specified

**Description**

The number of bundles specified is invalid.

**What Next**

Please provide the correct inputs.

## NDMUI-062

(error) Cell '%s' cannot be committed.

**Description**

The cell is either a leaf cell, an instance of another block, or an unbound instance. In these cases, the cell cannot be committed to form a new block.

Feedback

**What Next**

Check the cell to verify that it is an instance of logic hierarchy within the same block, and not a reference to another block.

---

## NDMUI-063

(warning) %s '%s' is not physical.

**Description**

The indicated object is not physical. Only physical objects may be added to a routing corridor. This object is skipped and not added.

**What Next**

Check that the correct physical objects are added.

---

## NDMUI-064

(information) Removed design '%s' from memory and disk.

**Description**

The remove_blocks command removes a design from memory, and from disk as well if the design has an on-disk copy in a directory-format library. Please use with caution.

---

## NDMUI-065

(error) The boundary is not valid.

**Description**

The specified boundary is not rectilinear or its area is zero. The boundary must be a rectilinear polygon or rectangle. In addition, its area must be greater than zero.

**What Next**

Correct the boundary.

---

## NDMUI-066

(error) %sNet '%s' has already been added to routing corridor '%s'.

**Description**

The specified net or super net has already been added to another routing corridor. Nets or super nets cannot be added to more than one routing corridor.

Feedback

**What Next**

Use the remove_from_routing_corridor command to remove the net or super net from its current routing corridor, and then call this command again.

---

## NDMUI-067

(error) Cannot remove object '%s' (type %s).

**Description**

The *remove_objects* command was unable to remove the specified object.

**What Next**

Modify the object's configuration so that it may be removed.

---

## NDMUI-068

(warning) '-all' option is ignored, as the reference of the specified cell has no other instances.

**Description**

The "-all" option in this command makes sense only when the operation is meant to be performed for "all" other instances of the reference, as the given cell. In this case, the given cell is the only instance of the reference, hence the option does not have any meaning.

**What Next**

Next time use the command without the "-all" option.

---

## NDMUI-069

(error) The specified polygon or path as corridor shape is not rectilinear.

**Description**

The shape specified must be a rectilinear polygon or path. In addition, its area must be greater than zero.

**What Next**

Correct the points.

Feedback

## NDMUI-070

(error) All the specified cells do not have the same reference.

**Description**

All the cells specified as input to this command, must have the same reference. Cells with different references can not be processed in one run of the command.

**What Next**

Only specify the cells with same reference.

## NDMUI-071

(Warning) Specified cell '%s' could not be uniquified.

**Description**

This warning is displayed during "uniquify" and "uniquify_block" commands, when some error happens while uniquifying mentioned cell.

## NDMUI-072

(Warning) Skipping cell '%s' because it does not have reference object of type "%s".

**Description**

Cell specified does not have reference of required type. For uniquify command, reference should be of type "logical module". For uniquify_block command, reference should of type "block". For both the commands, cell of a library cell will be invalid input.

**What Next**

Specify cell with reference of required type for the respective command.

## NDMUI-073

(error) All the specified cells are not bound to the same view.

**Description**

All the cells specified as input to this command, must be bound to the same view of the reference design. Cells bound to different views of the reference can not be processed in one run of the command.

Feedback

**What Next**

Only specify the cells with bound to the same view of the reference.

---

## NDMUI-074

(error) Module with name '%s' not found in the current design.

**Description**

A module with the specified name must already exist in the current design, for the command to work properly.

**What Next**

Specify a module which already exists in the design.

---

## NDMUI-075

(error) Port interface of the new %s is not same as that of the existing %s

**Description**

The reference or view of cells can be changed only with a reference or view with a identical port interface.

**What Next**

Specify a reference or view with a port interface identical to the current one.

---

## NDMUI-076

(error) The type of current reference ('%s') and new reference ('%s') do not match.

**Description**

The reference of cells can be changed only with a reference of the same type. A cell whose current reference is a module, must have a module as new reference. A cell whose current reference is a design, must have a design as new reference.

**What Next**

Specify a reference of the same type as the current reference of the cells

---

## NDMUI-077

(warning) '%s' is not physical net.

Feedback

### Description

The indicated net is not physical. Only physical nets are supported. This net is skipped and will not be added to or removed from the corridor.

### What Next

Check that the correct physical nets are added to or removed from the routing corridor.

---

## NDMUI-078

(error) The specified cell '%s' is multiply instantiated.

### Description

The input to this command must either be a uniquely instantiated hierarchical cell or the "-all" option must be used. For multiply instantiated cells, either use the option "-all" to commit all instances of the same master, or use the command "uniquify" to create a unique master.

### What Next

Please provide the correct input.

---

## NDMUI-079

(error) Design '%s' is not in the reference library list of library '%s'.

### Description

The referenced design must belong to the current library containing the current design, or a reference library of the current library.

### What Next

Either choose a different library for the reference design, or add the desired library to the current design's library reference library list.

To see what library a design is in, use the *get_attribute* command:

```
get_attribute [get_designs MID] lib
  RefLibB
```

Or use the *get_libs* command:

```
get_libs -of [get_designs MID]
  {RefLibB}
```

Feedback

To see what libraries are in the current library's ref_lib list, use the *get_attribute* command:

```
get_attribute [current_lib] ref_libs
  RefLibA RefLibC leafCellLib
```

To add a library to the current library's ref_lib list, use the *set_ref_libs* command:

```
set_ref_libs
  RefLibA RefLibC leafCellLib
  set_ref_libs -ref_libs {RefLibA RefLibB RefLibC leafCellLib}
  RefLibA RefLibB RefLibC leafCellLib
```

Then re-run the command.

---

## NDMUI-080

(error) '%s' cannot be used in user specified names for the requested object.

**Description**

The specified character is a reserved for internally generated names and can not be used in user specified names for the requested object.

**What Next**

Please provide the correct inputs.

---

## NDMUI-081

(error) No objects found to add to the bundle.

**Description**

No valid objects could be found in the specified objects to add to the bundle.

**What Next**

Please provide the correct inputs.

---

## NDMUI-082

(error) The object before which new objects are to be added, does not exist in the bundle.

**Description**

The object specified as input to option '-beforeObject' does not exist in the specified bundle. Therefore new objects cannot be added to the bundle.

Feedback

**What Next**

Please provide the correct inputs.

---

## NDMUI-083

(error) Only normal hierarchy type can be used for uncommitting Multiple-Instantiated Modules (MIMs).

**Description**

Only normal hierarchy type can be used for uncommitting Multiple-Instantiated Modules (MIMs).

**What Next**

Please specify hierarchy type normal.

---

## NDMUI-084

(error) The library %s is read only.

**Description**

*commit_cell* command needs to modify the library %s. If the library is read only, *commit_cell* command cannot be completed.

**What Next**

Please change the write permission of the library.

---

## NDMUI-085

(information) The design does not have any cells referencing block '%s'.

**Description**

All cells referencing the given block were specified as input to this command. But the design does not contain any cells that reference the given block. So no cells were specified.

**What Next**

Check that the correct block was specified.

Check that the current block is the expected current block.

Feedback

## NDMUI-086

(error) Reference block does not exist.

### Description

The specified reference does not exist. The command failed because a cell could not be created without a valid reference block.

### What Next

Please provide the correct inputs.

## NDMUI-087

(error) Cells must reference a block or lib cell.

### Description

All the cells given are logic-only module references. The *set_reference* command failed because it requires cells that are references to either lib cells or another block of physical hierarchy.

### What Next

Choose only physical cells for the *set_reference* command. To convert a logical cell into a physical cell, see the *commit_block* command.

### See Also

- commit_block
- set_reference

## NDMUI-088

(error) Block '%s' port interface is different from '%s'.

### Description

The port interface on the new block is different from the pin interface on the specified cells and the *-pin_rebind none* option was given which requires that all mappings be made without changes to the cell pin interface.

The *set_reference* command cannot set the reference without changing the cell's pin interface, so the *set_reference* command failed.

Feedback

### What Next

Run the *set_reference* command with the *-pin_verbose* option for more information on what pin changes the *set_reference* command would make with this block and this cell. Use the *-pin_rebind test* option to get the pin rebind mapping information without making any changes to the cells.

Verify that the new block is the correct block to be used.

Verify that the correct cells to be changed were specified.

Check the new block's port interface to verify that required ports are present on the block.

```
prompt> current_block newBlock
  {myLib:newBlock.design}
  prompt> get_ports
  {i1 i2 i3 i4 i5 i6 i7 ...}
```

### See Also

- current_block
- get_blocks
- get_ports
- set_reference

---

## NDMUI-089

(warning) Skipping cell '%s' to avoid dropping cell pin net connection.

### Description

The *set_reference* command is skipping that cell because changing the cell's reference would drop a pin on the cell that is connected to a net.

The *set_reference* command is using *-pin_rebind safe* which will not change the reference of the cell if that requires dropping net connections for that cell.

### What Next

Run the *set_reference* command with the *-pin_verbose* option for more information on what pin changes the *set_reference* command would make with this block and this cell. Use the *-pin_rebind test* option to get the pin rebind mapping information without making any changes to the cells.

Verify that the new block is the correct block to be used.

Verify that the correct cells to be changed were specified.

Feedback

Check the new block's port interface to verify that required ports are present on the block.

```
prompt> current_block newBlock
  {myLib:newBlock.design}
  prompt> get_ports
  {i1 i2 i3 i4 i5 i6 i7 ...}
```

Use the *set_reference* option *-pin_rebind force* to force *set_reference* to disconnect nets from pins that will be removed from the cell.

**See Also**

- current_block
- get_blocks
- get_ports
- set_reference

## NDMUI-090

(warning) Bound created without any cells or ports.

**Description**

Only cells and ports may be assigned to the bound.

**What Next**

Use the *add_to_bound* command to assign cells and ports to the bound.

## NDMUI-091

(error) No cells or ports are specified.

**Description**

Only cells and ports may be assigned to the bound.

**What Next**

Specify cells and ports in the *cell_and_port_list* option.

## NDMUI-092

(warning) Via mapping from '%s' to '%s' already exists. Duplicated definition is ignored.

Feedback

**Description**

A via mapping with the given new 'from' and 'to' via pattern already exists in the via mapping table, therefore a new mapping with the same contents can not be created. The new mapping must not already be used in the design.

**What Next**

Specified a new via mapping.

---

## NDMUI-093

(error) The number of contacts '%s' in the '%s' direction of via mapping is invalid.

**Description**

The number of contact must be an integer>=1.

**What Next**

Specified the correct number.

---

## NDMUI-094

(error) The -all option cannot be used with -from or -to options.

**Description**

-all option cannot be used with -from ot -to options.

**What Next**

Specified the correct option.

---

## NDMUI-095

(error) Via pattern '%s' not found in via mapping table.

**Description**

Specified via definition name cannot be found in via mapping table.

**What Next**

Specified the valid via definition name.

Feedback

## NDMUI-096

(warning) Unable to determine primary power because multiple power or ground nets exist in the design and UPF has not been committed. Auto PG connection is skipped.

### Description

This message is issued when running connect_pg_net automatic mode on a design with multiple power or ground nets, and the design has no UPF committed. IC Compiler II supports auto PG connection for designs with only one pair power and ground nets.

If no UPF is loaded and committed and the pair of power and ground nets do not exist in design, tool uses app options mv.pg.default_power_supply_net_name and mv.pg.default_ground_supply_net_name to determine the power and ground nets name and connects all PG pins. If the pair power and ground nets exist in the design, then tool uses the existing power and ground nets and connects all PG pins to them.

Unexpected power and ground nets may be coming from analog blocks or special hard macros. To see what power & ground nets are being used in the design use the get_supply_nets command.

For designs with multiple power and ground nets but no user UPF committed, auto mode connect_pg_net does not work.

### What Next

To get PG nets, 'get_nets -hier -filter "net_type==power||net_type==ground"' can be used. There are two possible methods to resolve this issue:

1. Add user power intent by loading UPF using load_upf command followed by commit_upf. With UPF loaded, the PG connections will be made automatically.

-or-

2. Perform manual PG net connections with connect_pg_nets command, specifying the supply net with the -net option and providing a complete list of pins and port it needs to connect to.

## NDMUI-097

(information) Disconnecting net '%s' from pin '%s'.

### Description

The *set_reference* command is removing the pin from the net because that port does not exist on the new block so the pin will be dropped from the cell.

The *set_reference* command is using *-pin_rebind force* which will drop net connections as needed when removing pins from cells.

Feedback

### What Next

Run the *set_reference* command with the *-pin_verbose* option for more information on what pin changes the *set_reference* command would make with this block and this cell. Use the *-pin_rebind test* option to get the pin rebind mapping information without making any changes to the cells.

Verify that the new block is the correct block to be used.

Verify that the correct cells to be changed were specified.

Check the new block's port interface to verify that required ports are present on the block.

```
prompt> current_block newBlock
  {myLib:newBlock.design}
  prompt> get_ports
  {i1 i2 i3 i4 i5 i6 i7 ...}
```

Use the *set_reference* option *-pin_rebind safe* to keep *set_reference* from changing a cell's reference if that means disconnecting nets from pins that will be removed from the cell.

To restore the previous cell references, cell pins and net connections, use the *undo* command.

### See Also

- current_block
- get_blocks
- get_ports
- set_reference
- undo

---

## NDMUI-098

(error) '%s' layer of via mapping is invalid.

### Description

The specified lower, upper, or cut layer in via mapping must be consistent.

### What Next

Specified the correct contact to map.

Feedback

## NDMUI-099

(information) Disconnecting net '%s' from pin '%s' for cell '%s' and all instances of module '%s' from block '%s'.

### Description

The *set_reference* command is removing the pin from the net because that port does not exist on the new block so the pin will be dropped from the cell.

The *set_reference* command is using *-pin_rebind force* which will drop net connections as needed when removing pins from cells.

If the cell is a multiply instantiated module (MIM) or block (MIB), the net will be disconnected from all instances of the cell in the design hierarchy.

### What Next

Run the *set_reference* command with the *-pin_verbose* option for more information on what pin changes the *set_reference* command would make with this block and this cell. Use the *-pin_rebind test* option to get the pin rebind mapping information without making any changes to the cells.

Verify that the new block is the correct block to be used.

Verify that the correct cells to be changed were specified.

Check the new block's port interface to verify that required ports are present on the block.

```
prompt> current_block newBlock
  {myLib:newBlock.design}
  prompt> get_ports
  {i1 i2 i3 i4 i5 i6 i7 ...}
```

Use the *set_reference* option *-pin_rebind safe* to keep *set_reference* from changing a cell's reference if that means disconnecting nets from pins that will be removed from the cell.

To restore the previous cell references, cell pins and net connections, use the *undo* command.

### See Also

- current_block
- get_blocks
- get_ports

Feedback

- set_reference
- undo

---

## NDMUI-100

(error) Invalid coordinates '%s' in list.

**Description**

A coordinate list argument is expected to contain a list of coordinates. The specified value is not a valid coordinate pair such as { x y }.

**What Next**

Fix the offending list element.

---

## NDMUI-101

(error) Invalid coordinate list '%s'; %s.

**Description**

A coordinate list argument is expected to contain a list of coordinates. The specified list is not a valid coordinate list for the specified reason.

**What Next**

Fix the offending list.

---

## NDMUI-102

(error) Regions in routing corridor '%s' are not connected.

**Description**

The regions in the routing corridor are not connected. A legal routing corridor must contain connected regions.

**What Next**

Fix the routing corridor with connected corridor regions.

---

## NDMUI-103

(error) Routing corridor '%s' does not cover %s '%s'.

Feedback

**Description**

The routing corridor does not cover the port or pin. A legal routing corridor must cover all ports and pins its member nets connected.

**What Next**

Fix the routing corridor to cover all ports and pins.

---

## NDMUI-104

(error) The reference of the specified cell is a internal module.

**Description**

The cell specified as input to the command create_blackbox, must have an unresolved design as its reference.

**What Next**

Please provide correct input.

---

## NDMUI-105

(error) The reference of the specified cell is already resolved.

**Description**

The cell specified as input to the command create_blackbox, must have an unresolved design as its reference.

**What Next**

Please provide correct input.

---

## NDMUI-106

(error) The library '%s' specified to create the new design in, is not a reference library of the current design

**Description**

The library specified as input to create the blackbox design in, must be a reference library for the library containing the current design. Otherwise, the design can not be linked properly.

**What Next**

Please provide correct input.

## NDMUI-107

(error) The library '%s' specified to create the new design in, is invalid.

**Description**

The library specified as input to create the blackbox design in, must be a design cell library and must be opened in edit mode. The specified library does not meet this one or more of these criteria and can not be used.

**What Next**

Please provide correct input.

## NDMUI-108

(error) Invalid orientation '%s' in list.

**Description**

The specified orientation is invalid. The orientation must be one of these values: R0, R90, R180, R270, MX, MXR90, MY, MYR90

**What Next**

Fix the offending orientation.

## NDMUI-109

(error) Only module hierarchy type can be used for uncommitting Multiple-Instantiated Blocks (MIBs).

**Description**

Only module hierarchy type can be used for uncommitting Multiple-Instantiated Blocks (MIBs).

**What Next**

Please specify hierarchy type module for MIB blocks.

## NDMUI-110

(error) The required option %s is not specified for the given type of routing guide.

**Description**

This option is a required option for the type of routing guide.

Feedback

**What Next**

Check the command man page and specify the required option.

---

## NDMUI-111

(error) Value '%s' for option '%s' does not specify a rectangle. Rectangle syntax is { { llx lly } { urx ury } }.

**Description**

The reported option must specify a rectangle. A rectangle is defined by its lower-left and upper-right vectices. The syntax is: { { llx lly } { urx ury } }

**What Next**

Provide the correct value to the reported option.

---

## NDMUI-112

(error) Value '%s' for option '%s' does not specify a polygon. Polygon syntax is { x1 y1 } { x2 y2 } ... { xn yn } }.

**Description**

The reported option must specify a polygon. A polygon is defined by a list of 3 or more points. The syntax is: { { x1 y1 } { x2 y2 } ... { xn yn } } in which n is 3 or more.

**What Next**

Provide the correct value to the reported option.

---

## NDMUI-113

(error) Value '%s' for option '-path' does not specify a 2-point path. Path syntax is { { x1 y1 } { x2 y2 } }.

**Description**

The -path option must specify a 2-point path when used with the '-shape_type path' option. A 2-point path is defined by a list of exactly two points. The syntax is: { { x1 y1 } { x2 y2 } }

**What Next**

Provide the correct value to the -path option.

Feedback

## NDMUI-114

(error) Must specify option '%s' when a variable endcap is specified for option '%s'.

**Description**

When specifying a variable endcap on the path, the extension value must be specified. The extension is a non-negative distance value. The start extension indicates how far passed the first path coordinate the endcap extends. The end extension indicates how far passed the last path coordinate the endcap extends.

**What Next**

Provide the extension value.

## NDMUI-115

(error) Cannot specify option '%s' when a %s endcap is specified for option '%s'.

**Description**

When specifying a flush, full_width, half_width, or octagon endcap on the path, the extension value cannot be specified. The extension value is automatically derived for these endcap styles.

**What Next**

Remove the extension option.

## NDMUI-116

(error) Cannot specify option '%s' when a custom via definition is specified for option '%s'.

**Description**

When creating an instance of a custom via definition, the cut array size and pitch may not be specified. Custom via geometries are predefined and may not be modified.

The cut array pitch and size may be specified for instances of simple via definitions.

**What Next**

Remove the indicated option or specify a simple via definition.

## NDMUI-117

(warning) Ignoring option '%s'; the via contains a single cut.

Feedback

**Description**

Single cut vias do not have pitch attributes. If a pitch is specified, it will be ignored.

**What Next**

Remove the indicated option.

---

## NDMUI-118

(error) %s '%s' cannot be added to net '%s'; it currently belongs to net '%s'. No objects were added to the net.

**Description**

The specified shape or via is owned already by another net as indicated in the message. Each shape or via may belong to at most one net.

The command exited without adding any shapes or vias to the net.

**What Next**

Use the remove_from_net command to remove the shape or via from its owner net, and then call this command again.

---

## NDMUI-119

(error) %s '%s' cannot be added to net '%s'; it belongs to terminal '%s' of port '%s'. No objects were added to the net.

**Description**

The specified shape or via is owned by a terminal as indicated in the message. Each shape or via may belong to a net or terminal, but not both.

The command exited without adding any shapes or vias to the net.

**What Next**

Remove the shape from the terminal using *remove_terminals*, then call this command again.

---

## NDMUI-120

(error) Space value cannot be zero.

**Description**

You receive this error because the specified space value is zero.

Feedback

### What Next

Please use the non-zero space value.

### See Also

- create_track
- get_tracks
- remove_tracks
- report_tracks

---

## NDMUI-121

(error) Couldn't get the pitch value of the layer '%s' from the library.

### Description

You receive this error because the specified layer's pitch value cannot be fetched from the edit library.

### What Next

Please check the technology information of the edit library.

### See Also

- create_track
- get_tracks
- remove_tracks
- report_tracks

---

## NDMUI-122

(warning) Coordinate value is not specified. Using coordinate that is half of the space distance inside of the die area.

### Description

You receive this warning because the value of the start of the first track is not specified. If the track's boundary box is given, the tool is able to derive the start coordinate from the boundary box. However, in this case, the boundary box is not specified either. When the coordinate value is not specified and boundary ix is also not specified, the tool attempts to use half of the space from the left(bottom) coordinate of the die area.

### What Next

No action is required on your part. You can see the man page for the command that caused the warning for detailed information on specifying optional and required parameters.

### See Also

- create_track
- get_tracks
- remove_tracks
- report_tracks

---

## NDMUI-123

(error) specified count value is not consistent with the specified bounding box, based on its space value.

### Description

You receive this error message because the number of tracks to be created based on the given bounding box is not the same as the number of tracks to be created based on given count.

### What Next

Specify only one of bounding box and count, but not both. If both are specified, count must be consistent with number of tracks the given bounding box can have.

### See Also

- create_track
- get_tracks
- remove_tracks
- report_tracks

---

## NDMUI-124

(warning) Space is not specified. Using pitch.

Feedback

### Description

You receive this warning because the space between the tracks is not provided. The tool uses the default value, which is the pitch value provided by the library.

### What Next

No action is required on your part. You can see the man page for the command that caused the warning for detailed information on specifying optional and required parameters.

### See Also

- create_track
- remove_tracks
- report_tracks

---

## NDMUI-125

(warning) Direction is not specified. Using the layer preferred direction.

### Description

You receive this warning because the direction is not specified in the command invoked, so the tool uses the layer preferred direction.

### What Next

No action is required on your part unless you want to specify a direction. You can examine the design to see the layer preferred direction and the man page for the command that caused the warning for detailed information on specifying a direction value.

### See Also

- create_track
- remove_tracks
- report_tracks

---

## NDMUI-126

(warning) Coordinate value is not specified. Defaulting to the left or bottom coordinate of the track's bounding box.

Feedback

### Description

You receive this warning because the value of the start of the first track is not specified. When the coordinate value is not specified, the tool attempts to use the left(bottom) coordinate of the track's bounding box.

### What Next

No action is required on your part. You can see the man page for the command that caused the warning for detailed information on specifying optional and required parameters.

### See Also

- create_track
- remove_tracks
- report_tracks

---

## NDMUI-127

(error) specified start value is not consistent with the specified bounding box.

### Description

You receive this error message because the given start value does not match the given track bounding box. For horizontal track, the start value must match the Y coordinate of the lower left corner of given bounding box. For vertical track, the start value must match the X coordinate of the lower left corner of given bounding box.

### What Next

Specify only one of bounding box and start, but not both. If both are specified, correct the start or bounding box values.

### See Also

- create_track
- remove_tracks
- report_tracks

---

## NDMUI-128

(error) Could not find pin '%s' on reference '%s'.

Feedback

### Description

The *set_reference* command option *-pin_map* could not find a pin on the source reference or a port on the destination reference with the given name.

### What Next

Verify that the name was entered correctly and that the source and destination pin names were not accidentally swapped.

Verify that the new block is the correct block to be used.

Verify that the correct cells to be changed were specified.

Check that the source reference has a pin with the given name.

```
prompt> get_pins u15/u23/selectPin4
  {u15/u23/selectPin4}
```

Check that the destination reference has a port with the given name.

```
prompt> current_block newBlock
  {myLib:newBlock.design}
  prompt> get_ports driverPin1
  {driverPin1}
```

Run the *set_reference* command with the options *-pin_verbose -pin_rebind test* to see the pin mappings on the source cell and destination block reference.

```
prompt> set_reference -block newBlock u15/u23 \\
    -pin_verbose -pin_rebind test
  ...
  Pin mapping from 'myLib:oldBlock.design' to 'myLib:newBlock.design':
    Pin    1 selectPin4 stays at pin    1
  ...
```

### See Also

- current_block
- get_blocks
- get_pins
- get_ports
- set_reference

---

## NDMUI-129

(warning) Cannot remove %s '%s' because it is not in the current design.

Feedback

**Description**

The specified object cannot be removed because it belongs to another design or lib_cell. This command may only remove objects from the current design.

**What Next**

Set current_design to the design that contains the object, and then remove it.

---

## NDMUI-130

(warning) %s '%s' cannot be removed because it does not belong to net '%s'; %s.

**Description**

The specified object is not owned by the net, and so it cannot be removed from the net. Each shape or via may be removed only from its owner net or terminal.

The command successfully removed the rest of the objects from the net.

**What Next**

Check the option values and verify that you are removing the correct shapes from the correct owner.

---

## NDMUI-131

(error) Terminal cannot be created with %s '%s'; the %s currently belongs to terminal '%s'.

**Description**

The terminal cannot be created with the specified shape or via, because the object is already owned by another terminal. Each shape or via may belong to at most one terminal.

**What Next**

Remove the shape or via from its current terminal using the remove_terminals command, or specify another shape or via that is not already owned by a terminal.

---

## NDMUI-132

(error) Terminal cannot be created on port '%s'; the %s '%s' belongs to net '%s'.

**Description**

The terminal cannot be created with the specified shape or via, because the shape or via is owned by a net. Each shape or via may belong to a net or terminal, but not both.

Feedback

**What Next**

Remove the shape from the net using *remove_from_net*, then call this command again.

---

## NDMUI-133

(error) Terminal cannot be created with %s '%s'; terminals cannot be created with %s shapes.

**Description**

A terminal cannot be created with a text shape. They may only be created with rects, polygons, paths, and vias.

**What Next**

Do not specify a text. Specify only rects, polygons, paths, or vias.

---

## NDMUI-134

(error) %s '%s' cannot be added to net '%s'; nets cannot contain any %s shapes. No objects were added to the net.

**Description**

A net cannot own text shapes. It may only own rects, polygons, paths, and vias.

**What Next**

Do not specify any text. Specify only rects, polygons, paths, or vias.

---

## NDMUI-135

(error) %s value '%s' cannot be used with other values in list.

**Description**

The specified value cannot be used with other values in the list. This value is mutually exclusive with all other possible values.

**What Next**

Specify only this value in the list.

---

## NDMUI-136

(warning) Count value is not specified. Covering the bounding box of track, depending on its space value.

Feedback

### Description

You receive this warning because the number of tracks to be created is not specified. When the count value is not specified, the tool attempts to calculate the count with the width of track's bounding box divided by track's space value.

### What Next

No action is required on your part. You can see the man page for the command that caused the warning for detailed information on specifying optional and required parameters.

### See Also

- create_track
- get_tracks
- remove_tracks
- report_tracks

---

## NDMUI-137

(error) power_domains '%s' and '%s' do not share equivalent primary power and ground supply_net.

### Description

When associating multiple power_domains to a voltage_area, the primary supplies of the power_domains must be electrically equivalent, meaning that their supply_nets must either be the same, or connected via supply_ports.

---

## NDMUI-138

(error) Cell '%s' is already associated with voltage_area '%s'.

### Description

Each cell must belong to a single voltage_area. It is an error if the cell to be added already explicitly or implicitly associated with another voltage_area.

### What Next

Verify the cell to be added does not belong to a voltage_area with the get_voltage_areas command

Feedback

## NDMUI-139

(error) voltage_area '%s' already exists in this design.

**Description**

Each voltage_area must have a unique name. It is an error if there is an existing voltage_area of the specified name.

**What Next**

Please use a different name for this voltage_area.

## NDMUI-140

(error) Please specify a name for this voltage_area with the -name option.

**Description**

A unique name must be provided when creating a voltage_area with multiple power_domains, or with a single power_domain that already has a primary voltage_area.

**What Next**

Please provide a name for this voltage_area.

## NDMUI-141

(warning) Logical hierarchical cell '%s' is not added to %s.

**Description**

Only physical cells can be added to a voltage_area/voltage_area shape. When adding a hierarchical cell, only the physical cells residing under it are added. The hierarchical cell itself is not added. In the case when the hierarchical cell is purely logical and contains no physical cells, no cells will be added to the voltage_area/voltage_area_shape.

**What Next**

Verify the cell is physical by checking the "is_physical" attribute or that the cell contains at least one physical cell with the get_cells -of <hier_cell> -filter "is_physical == true" option.

## NDMUI-142

(warning) power_domain '%s' is already associated with voltage_area '%s'.

Feedback

**Description**

The specified power_domain is already associated with the voltage_area. Adding it to the voltage_area again will be a no op.

**What Next**

Use the get_voltage_areas -of <power_domain> command to see if this power_domain and voltage_area is already associated.

---

## NDMUI-143

(error) Option '-use_power_domain_cells' is not allowed if the voltage_area already contains explicitly assigned cells.

**Description**

It is illegal to specify the '-use_power_domain_cells' option when creating or updating a voltage_area, if the voltage_area or any of its voltage_area_shape already contain explicitly assigned cells.

---

## NDMUI-144

(error) The '-is_fixed' option is not allowed if the voltage_area do not have any voltage_area_shapes.

**Description**

It is illegal to specify the 'is_fixed" option when creating or updating a voltage_area unless the voltage_area has at least one voltage_area_shape.

---

## NDMUI-145

(warning) power_domain '%s' is not associated with voltage_area '%s'.

**Description**

The specified power_domain is not associated with this voltage_area. Removing it from the voltage_area will be a no op.

---

## NDMUI-146

(warning) Cell '%s' is not associated with this %s.

Feedback

**Description**

To remove a cell for a voltage_area or a voltage_area_shape, the cell must have been explicitly assigned to this voltage_area or voltage_area_shape, otherwise the cell is ignored.

**What Next**

Verify the cell does belong to this voltage_area/voltage_area_shape with the get_voltage_areas/get_voltage_area_shapes command.

---

## NDMUI-147

(error) Cell '%s' is already associated with voltage_area_shape '%s'.

**Description**

Each cell can only be assigned to one voltage_area_shape. It is an error if the cell to be added is already explicitly assigned to another voltage_area_shape.

**What Next**

Verify the cell to be added does not belong to a voltage_area_shape with the get_voltage_area_shapes command.

---

## NDMUI-148

(error) Option '-use_power_domain_cells' is not allowed on voltage_area with no power_domain association.

**Description**

It is illegal to specify the '-use_power_domain_cells' option when creating or updating a voltage_area unless the voltage_area has at least one power_domain association.

---

## NDMUI-149

(warning) Cell '%s' was already added to this %s.

**Description**

A cell can only be associated with a voltage_area/voltage_area_shape once. Adding a cell to the same voltage_area/voltage_area_shape a second time will be a no-op.

**What Next**

Use the get_voltage_areas/get_voltage_area_shapes -of <cell> command to check if the cell is already associated with this voltage_area/voltage_area_shape.

Feedback

## NDMUI-150

(error) power_domain '%s' already has a voltage_area '%s' with implicit cell membership.

**Description**

The specified power_domain is already associated with the voltage_area with implicit cell membership. Each power_domain can have at most one voltage_area created with the -use_power_domain_cells option.

**What Next**

Create a voltage_area with explicit cell assignment without the -use_power_domain_cells option.

## NDMUI-151

(error) Value '%s' for option '-region' does not specify a single rectangle or polygon.

**Description**

The -region option must specify a single rectangle or polygon. For backward script compatibility, the app_option "voltage_area.disable_region_list" can be set to false to allow the -region option to continue accepting a list of rectangles/polygons.

set_app_options -global {voltage_area.disable_region_list false}

**What Next**

Provide a single rectangle or polygon for this option.

## NDMUI-152

(information) The '-merge_regions' option is an no-op. The voltage_area needs at least 2 shapes for a merge to happen.

**Description**

The "-merge_regions" option is an no operation when the voltage_area contains zero or 1 shape.

## NDMUI-153

(information) Merging existing shapes that abuts or overlaps with the new shape in voltage_area '%s'.

Feedback

**Description**

The "-merge_regions" will merge the input shape with existing shapes within the voltage_area that abuts or overlaps with it.

---

## NDMUI-154

(information) Merging abutted and overlapping shapes in voltage_area '%s'.

**Description**

The "-merge_regions" option will merge the existing abutted and overlapping shapes in the voltage_area into a minimum set of disjoint shapes.

**What Next**

create_voltage_area create_voltage_area_shape

---

## NDMUI-155

(error) Cell '%s' does not belong to a power_domain associated with this voltage_area.

**Description**

To add a cell to a voltage_area, the cell must belong to one of the power_domains associated with the voltage_area.

**What Next**

Get the power_domain of this cell:

get_power_domains -of [get_cells xxx]

Verify this power_domain is associated with this voltage_area:

get_voltage_areas -of [get_power_domains -of [get_cells xxx]]

---

## NDMUI-156

(error) Voltage Area cannot be created on logical hierarchical cell '%s'.

**Description**

Voltage area can only be created on physical hierarchical. It is an error to specify a logical hierarchical cell with the -cell option.

Feedback

### What Next

Verify the cell is physical by checking the "is_physical" attribute or that the cell contains at least one physical cell with the get_cells -of <hier_cell> -filter "is_physical == true" option.

---

## NDMUI-157

(error) The specified supply_net '%s' does not match any of the available supply_nets on the power_domain.

### Description

When specifying a power or ground supply net on a voltage_area, the supply net must be available on the power_domain.

---

## NDMUI-158

(error) Pins '%s' and '%s' mapped to same target port '%s' on block '%s'.

### Description

The *set_reference* command option *-pin_map* attempted to map two source pins to the same target port.

The *set_reference* command can add, remove, or move cell pins. It cannot merge two pins, because that would require merging two or more nets for each cell.

### What Next

Verify that the correct source and target name pairs were passed as to the *-pin_map* options.

Manually update the net connections for each cell, and drop the unneeded pins.

### See Also

- current_block
- disconnect_net
- get_blocks
- get_cells
- get_nets
- get_pins
- set_reference

Feedback

## NDMUI-159

(error) Invalid shapes specification '%s'; %s.

**Description**

A shape specification is expected to contain a layer name and a list of coordinates. The specified list is not a valid shape specification for the given reason.

**What Next**

Fix the offending list.

## NDMUI-160

(error) Design '%s' is an outline view; run the 'expand_outline' command before calling '%s'.

**Description**

This command cannot be executed on an outline view.

**What Next**

Open or create the design view of this design. To create the design view from the outline view use *expand_outline*.

## NDMUI-161

(error) Cell instance '%s' cannot be committed. It is already a physical block.

**Description**

The cell has already been committed, and exists in the top cell as a block instance already. It cannot be re-committed.

**What Next**

Be sure to attempt to commit a cell that has not already been committed.

## NDMUI-162

(error) The relative placement group '%s' already exists in this design.

**Description**

Each relative placement group must have a unique name. It is an error if there is an existing relative placement group of the specified name.

Feedback

### What Next

Please use a different name for this relative placement group.

---

## NDMUI-163

(error) The specified row %d is out of limit for RP group '%s', which has only %d rows.

### Description

This error message occurs when the *add_to_rp_group* command detects that you tried to add an object to a Relative Placement group in a row that is out of range of the rows for that group.

### What Next

Choose a new row that is within the rows of the relative placement group. After making any changes, rerun the command.

### See Also

- add_to_rp_group
- create_rp_group
- remove_from_rp_group

---

## NDMUI-164

(error) The specified column %d is out of limit for RP group '%s', which has only %d columns.

### Description

This error message occurs when the *add_to_rp_group* command detects that you tried to add an object to a Relative Placement group in a column that is out of range of the columns for that group.

### What Next

Choose a new column that is within the columns of the relative placement group. After making any changes, rerun the command.

Feedback

### See Also

- add_to_rp_group
- create_rp_group
- remove_from_rp_group

---

## NDMUI-165

(error) The specified %s '%s' already belongs to relative placement group '%s'.

### Description

This error message occurs when the *add_to_rp_group* command detects that you tried to add an object, which is already part of a relative placement group. An object can be added only once in any relative placement group.

### What Next

Either remove object from its previous location or skip this addition. After making any changes, rerun the command.

### See Also

- add_to_rp_group
- create_rp_group
- remove_from_rp_group

---

## NDMUI-166

(error) The specified column %d and row %d of relative placement group '%s' is already occupied.

### Description

This error message occurs when the *add_to_rp_group* command detects that you tried to add an object to a column and row position that is already occupied by another object. Each lattice position of a Relative Placement (RP) group can only contain one object.

### What Next

Either remove the object already located at the given lattice position before inserting another object at that location, or choose another location that is empty. After making any changes, rerun the command.

Feedback

### See Also

- add_to_rp_group
- create_rp_group
- remove_from_rp_group

---

## NDMUI-167

(error) The specified cell '%s' can not be added to relative placement group '%s', because it is hierarchical.

### Description

This error message occurs when the *add_to_rp_group* command detects that you tried to add a cell, which is hierarchical. Only leaf cells can be added to relative placement group.

### What Next

The leaf cells within hierarchical cell can be added in relative placement group.

### See Also

- add_to_rp_group
- create_rp_group
- remove_from_rp_group

---

## NDMUI-168

(Warning) The first cell of the given list will be added to the relative placement group.

### Description

This message occurs when the *add_to_rp_group* command detects that you tried to add multiple cells in a single location of relative placement group. First cell of the given multiple cells will be added to relative placement group.

### What Next

Please add other cells of the list in other empty locations or skip adding them.

Feedback

**See Also**

- add_to_rp_group
- create_rp_group
- remove_from_rp_group

---

## NDMUI-169

(error) Relative Placement Blockage name cannot be empty.

**Description**

Relative Placement Blockage name must not be empty.

**What Next**

Please specify a non-empty blockage name.

---

## NDMUI-170

(error) The specified %s '%s' does not belong to relative placement group '%s'.

**Description**

This error message occurs when the *remove_from_rp_group* command detects that you tried to remove an object, which does not belong to provided relative placement group. The provided object to be removed either does not belong to any relative placement group or is part of some other group.

**What Next**

Provide correct object to be removed.

**See Also**

- add_to_rp_group
- create_rp_group
- remove_from_rp_group

---

## NDMUI-171

(error) The specified object '%s' has not been added to any relative placement group yet.

Feedback

### Description

This error message occurs when the *get_rp_group_object_position* command detects that you tried to get position of an object which is not part of any relative placement group.

### See Also

- add_to_rp_group
- create_rp_group
- remove_from_rp_group

---

## NDMUI-172

(error) The relative placement group '%s' could not be removed.

### Description

This error message occurs when the *remove_rp_groups* command could not delete the specified relative placement group. Please note that error will be issued only for top level group, if *-hierarchical* option is given.

### See Also

- add_to_rp_group
- create_rp_group
- remove_from_rp_group

---

## NDMUI-173

(info) There are no relative placement groups in the design.

### Description

This info message occurs when relative placement command detects that there are no any relative placement groups in the design.

### See Also

- get_rp_groups

Feedback

## NDMUI-174

(Information) The object for only first relative placement group of the given list will be collected.

### Description

This error message occurs when the *get_rp_group_objects* command detects that you tried to find object at a location using *-at* option for multiple relative placement groups. The object will be collected only for the first RP group in the list.

### See Also

- add_to_rp_group
- create_rp_group
- remove_from_rp_group
- get_rp_groups

## NDMUI-175

(error) View '%s' of referenced block %s does not exist.

### Description

The block does not have indicated view.

### What Next

Look at the referenced block to see what views are available to *change_view* to. Or use the *-force* option to change to a non-existent view.

## NDMUI-176

(error) No references to %s found in block %s.

### Description

This block does not contain references to the named block.

### What Next

Check that the name of the referenced block is spelled exactly without any wild card characters.

Feedback

## NDMUI-177

(error) Cannot change view for reference '%s'.

### Description

The view of specified reference cannot be changed.

### What Next

Check if the indicated view exists.

## NDMUI-178

(error) The specified row %d and num_rows %d trying to add cell out of limit for RP group '%s', which has only %d rows.

### Description

This error message occurs when the *add_to_rp_group* command detects that you tried to add an object to a relative placement group in multiple rows, and one or more of the specified rows are out of range of the rows for that group.

### What Next

Choose a new starting row using option *-row* or change number of multiple rows using option *-num_rows*. After making any changes, rerun the command.

### See Also

- add_to_rp_group
- create_rp_group
- remove_from_rp_group

## NDMUI-179

(error) The specified column %d and num_columns %d trying to add cell out of limit for RP group '%s', which has only %d columns.

### Description

This error message occurs when the *add_to_rp_group* command detects that you tried to add an object to a relative placement group in multiple columns and one or more of the specified columns are out of range of the columns for that group.

Feedback

### What Next

Choose a new starting column using option *-column* or change number of multiple columns using option *-num_columns*. After making any changes, rerun the command.

### See Also

- add_to_rp_group
- create_rp_group
- remove_from_rp_group

---

## NDMUI-180

(error) Matching type '%s' already exists in the design.

### Description

A matching type with the specified name already exists in the design.

### What Next

Specify a different matching type name, or use the remove_matching_types command to remove the existing matching type and then run this command again.

---

## NDMUI-181

(error) %s '%s' cannot be added to the matching type; it is not physical.

### Description

Only physical cells and pins can be added to a matching type. Logical-only cells and pins are not allowed.

### What Next

Verify that the cell or pin is physical. To verify a cell, check its is_physical attribute. To verify a pin, check its cell's is_physical attribute.

---

## NDMUI-182

(error) %s '%s' cannot be added to the matching type; %s is already in matching type '%s'.

Feedback

**Description**

The object cannot be added to the matching type for the indicated reason. An object cannot be in multiple matching types. If it is already in a matching type, it must be removed from its current matching type before adding it to a new matching type.

A pin and its terminals cannot both be added to matching types. Either assign the pin to a matching type, or assign its terminals to matching types, but not both. If a pin is in a matching type, the pin must be removed from its matching type before adding its terminals to matching types. If a terminal is in a matching type, the terminal must be removed from its matching type before adding its pin to a matching type.

**What Next**

Use the remove_from_matching_type command to remove the object from the indicated matching type, and then call this command again.

---

## NDMUI-183

(warning) %s '%s' cannot be removed from the matching type; %s.

**Description**

The specified object cannot be removed from the matching type. Objects may only be removed from the matching type in which it is in.

**What Next**

If the object is in another matching type, call this command with the correct matching type.

---

## NDMUI-184

(error) Port terminal '%s' cannot be added to the matching type; it is not a pin terminal.

**Description**

Only pin terminals can be added to a matching type. Top-level port terminals are not allowed in a matching type.

**What Next**

Remove the port terminal from the object list and run this command again..

---

## NDMUI-185

(error) The option '-pin_name' is missing with pin alignment specified using option '-alignment'.

Feedback

### Description

This error message occurs when the *set_rp_group_options* command detects that you tried to set pin alignment on relative placement group using option *-alignment*, but did not provide pin name on which to do alignment using *-pin_name* option.

### What Next

Please provide option *-pin_name* along with *-alignment pin*.

### See Also

- create_rp_group
- remove_rp_groups

---

## NDMUI-186

(error) The option -pin_name can not be set on cell '%s' because its parent relative placement group does not have pin alignment.

### Description

This error message occurs when the *add_to_rp_group* command detects that you tried to set pin name on a cell using option *-pin_name*, while adding it in a relative placement group not having pin alignment set on it.

### What Next

Please set pin alignment on relative placement group using *-pin_name -alignment pin* or do not provide option *-pin_name* when adding a cell.

### See Also

- create_rp_group
- remove_rp_groups
- add_to_rp_group
- remove_from_rp_group

---

## NDMUI-187

(error) The object list '%s' is not homogeneous.

Feedback

**Description**

The specified object list contains objects of different types. The object list may only contain objects of one type.

**What Next**

Specify objects of one type only.

---

## NDMUI-188

(error) Invalid polygon operation

**Description**

The specified polygon operation is not valid. Valid polygon operations are "AND", "OR", "XOR", and "NOT".

**What Next**

Specify a valid polygon operation.

---

## NDMUI-189

(error) Invalid value for "size" parameter.

**Description**

The "size" parameter of the resize_polygons command must be a list of either 1, 2, or 4 numerical values.

**What Next**

Specify a list of 1, 2, or 4 numbers as the "size" parameter value.

---

## NDMUI-190

(error) Polygon is empty and cannot be resized.

**Description**

The resize_polygon command was called on an empty polygon. Empty polygons cannot be resized.

**What Next**

Specify a non-empty polygon.

Feedback

## NDMUI-191

(error) Invalid value for "-output" argument.

**Description**

An invalid value was specified for the "-output" argument of the split_polygons command. Valid values are "geo_mask" and "poly_rect".

**What Next**

Specify either "geo_mask" or "poly_rect" for the "-output" argument.

## NDMUI-192

(Error) Technology information of the layer '%s', differs with that of the design '%s'.

**Description**

The specified layer is associated with a different technology than that of the current design.

**What Next**

Please provide a layer whose technology information matches with that of the current design.

## NDMUI-193

(warning) Disassociating shape '%s' from its current environment.

**Description**

Removes the pin/net connections of a shape and gives information to user as warning.

## NDMUI-194

(error) The value for option -group_orientation is not valid. Specify one of: R0, MX, MY, R180.

**Description**

This error message occurs when the *set_rp_group_options* command detects that you tried to set an invalid orientation on relative placement group.

**What Next**

Please provide one of R0, MX, MY, R180 as -group_orientation for RP group.

Feedback

## NDMUI-195

(Error) Invalid options specified for user grid.

**Description**

This error message occurs when an option other than x_step, y_step, x_offset and y_offset is specified for a user grid.

**What Next**

Please use only x_step, y_step, x_offset and y_offset options for a user grid.

## NDMUI-196

(Error) No grid specified.

**Description**

This error message occurs when no grid object is specified as input.

**What Next**

Please specify a valid grid object to operate on.

## NDMUI-197

(Error) A %s grid by name '%s' already exists.

**Description**

This error occurs when a grid with a same name already exists in the block.

**What Next**

Please use a different name for the grid.

## NDMUI-198

(Error) %s grid associated with the block.

**Description**

Either there are no grids, or more than one grid is, associated with the block

**What Next**

Please provide a valid grid name.

Feedback

## NDMUI-199

(Error) No objects matched grid type '%s'.

### Description

This error occurs when none of the objects from the resultant collection is of the requested grid type.

## NDMUI-200

(Error) %s grid object specified.

### Description

This error occurs when an object of grid other than block-grid is specified.

### What Next

Please provide a valid object of block grid.

## NDMUI-201

(Error) The pin named '%s' does not exist on cell '%s'.

### Description

This error message occurs when the *add_to_rp_group* command detects that the pin provided using option *-pin_name* or using *set_attribute* command does not exist on cell.

### What Next

Please provide the name of a pin on the given cell.

### See Also

- create_rp_group
- add_to_rp_group
- remove_from_rp_group
- set_attribute

## NDMUI-202

(Error) The option -pin_name is applicable for pin alignment only and can not be specified with '%s' alignment.

Feedback

### Description

This error message occurs when the *set_rp_group_options* command detects that the option *-pin_name* is provided with left or right alignment using option *-alignment*.

### What Next

Please set pin alignment on relative placement group using *-pin_name -alignment pin* or do not provide option *-pin_name*.

### See Also

- create_rp_group
- remove_rp_groups
- add_to_rp_group
- remove_from_rp_group

---

## NDMUI-203

(error) The specified '%s' '%s' cannot be added to relative placement group '%s', because its voltage area is not same as relative placement group.

### Description

This error message occurs when the *add_to_rp_group* command detects that you tried to add a relative placement cell or a group which has a voltage area different from the relative placement group. The voltage area of the relative placement group is the voltage area of its cells.

### See Also

- create_rp_group
- remove_from_rp_group
- create_voltage_area

---

## NDMUI-204

(error) The specified cells can not be added to voltage area '%s'.

### Description

This error message occurs when the *create_voltage_area* or *set_voltage_area* command detects that you tried to add a cell, which belongs to a relative placement group. The cell

Feedback

was not allowed to be added because voltage area of all the cells of top relative placement group must be same.

**What Next**

Please add all the cells of top relative placement group together.

**See Also**

- create_voltage_area
- set_voltage_area
- add_to_rp_group

---

## NDMUI-205

(error) The specified cell '%s' can not be added to group bound '%s', because it is a relative placement cell.

**Description**

This error message occurs when the *create_bound* command detects that you tried to add a relative placement cell to a group bound. A cell cannot be part of both Group Bound and relative placement group.

**See Also**

- create_bound
- add_to_bound
- add_to_rp_group

---

## NDMUI-206

(error) The specified cells can not be added to bound '%s'.

**Description**

*create_bound* or *add_to_bound* does not allow cells of relative placement group to be added unless all cells of relative placement group are added. All the cells of relative placement group and its hierarchies must share the same move bound and group bound.

**What Next**

Please add all the cells of top relative placement group together. The following example adds all cells from the top relative placement group of cell "cell1" to bound "bound1".

Feedback

add_to_bound bound1 [get_cells -of_objects -hierarchical [get_rp_groups -top -of_objects cell1]]

**See Also**

- create_bound
- add_to_bound
- add_to_rp_group
- get_cells

## NDMUI-207

(error) The specified %s '%s' can not be added to relative placement group '%s', because its bounds are not same as relative placement group's bounds.

**Description**

This error message occurs when the *add_to_rp_group* command detects that you tried to add a cell or a child relative placement group which does not have the same named move or group bound(s) as the parent relative placement groups's bound(s). The bound of the relative placement group is the bound of its cells.

**See Also**

- add_to_rp_group
- create_rp_group
- remove_from_rp_group
- create_bound

## NDMUI-208

(warning) Option *-new_name* is ignored for blackbox cell %s.

**Description**

This warning message occurs when option *-new_name* is specified with command *create_blackbox* on a successfully bound logical hierarchy cell.

The tool would commit the specified logical hierarchy cell into a physical block which is a black_box typed design. This operation, similar to *commit_block*, uses the current reference name of the cell as the committed block's name. It does not accept user specified new name.

Feedback

### See Also

- commit_block

---

## NDMUI-209

(error) The specified cell '%s' can not be added to relative placement group '%s', because it is part of a group bound.

### Description

This error message occurs when the *add_to_rp_group* command detects that you tried to add a cell which belong to a group bound. into relative placement group. A cell cannot be part of both Group Bound and relative placement group.

### See Also

- create_bound
- add_to_bound
- add_to_rp_group

---

## NDMUI-210

(error) Anchor corner options are not allowed for non-top relative placement group %s.

### Description

This error message occurs when the *set_rp_group_options* command is given anchor corner options for non-top relative placement group. The options *-x_offset*, *-y_offset* and *-anchor_corner* are anchor corner options.

### See Also

- create_rp_group
- add_to_rp_group

---

## NDMUI-211

(error) Both anchor corner and offset values has to be provided for relative placement group %s.

Feedback

### Description

This error message occurs when the *set_rp_group_options* command tries to set only offset value (*-x_offset* or *-y_offset*) or *-anchor_corner*.

### What Next

Please set both anchor corner and offset values for the relative placement group.

### See Also

- create_bound
- add_to_bound
- add_to_rp_group

---

## NDMUI-212

(error) Allowable orientation of a cell cannot be more than eight.

### Description

This error message occurs when the *add_to_rp_group* command tries to set more than eight allowable orientation on relative placement cell.

### What Next

Please remove any duplicate orientation value from the list of orientation and rerun the command.

### See Also

- add_to_rp_group

---

## NDMUI-213

(error) Specify pins of the same instance.

### Description

This warning message occurs when the user specifies a pair of pins in the *-pins* option and the pins do not belong to the same instance.

### What Next

Please specify one pair of pins belonging to the same instance and rerun the command.

Feedback

**See Also**

- set_supernet_exceptions
- remove_supernet_exceptions

## NDMUI-214

(warning) Cannot disable non-repeater cell with no supernet transparent pins.

**Description**

This warning message occurs when the user specifies a cell to disable using the *-disable_cells* option which does not have any supernet transparent pins defined.

**What Next**

Please specify a cell which has supernet transparent pins defined to disable.

**See Also**

- set_supernet_exceptions

## NDMUI-215

(warning) Repeater does not have disable_cell settings.

**Description**

This warning message occurs when the user tries to enable a repeater cell as supernet transparent using the *-cells* option of the *remove_supernet_exceptions* command which was not disabled using the *-disable_cells* option of *set_supernet_exceptions* command.

**What Next**

Please specify a cell which has been disabled and rerun the command.

**See Also**

- set_supernet_exceptions
- remove_supernet_exceptions

## NDMUI-216

(warning) Non-repeater cell has no supernet transparent pins to remove.

Feedback

### Description

This warning message occurs when the user tries to remove the supernet transparent pins of a cell using the option *-cells* which does not have any supernet transparent pins set.

### What Next

Please specify a cell which has supernet transparent pins and rerun the command.

### See Also

- remove_supernet_exceptions

---

## NDMUI-217

(warning) Cannot delete non supernet transparent pins.

### Description

This warning message occurs when the user specifies a pair of pins using the *-pins* option which were not set as supernet transparent.

### What Next

Please specify pins which are supernet transparent and rerun the command.

### See Also

- remove_supernet_exceptions

---

## NDMUI-218

(error) The specified '%s' '%s' cannot be added to relative placement group '%s', because its power domain is not same as relative placement group.

### Description

This error message occurs when the *add_to_rp_group* command detects that you tried to add a relative placement cell or a group which has a power domain different from the relative placement group. The power domain of the relative placement group is the power domain of its cells.

### See Also

- create_rp_group
- remove_from_rp_group
- create_voltage_area

Feedback

## NDMUI-219

(warning) The block %s is not in the instance tree of the Physical object %s.

**Description**

This block option is not in the instance tree of the current physical object.

**What Next**

Check the command man page and specify the correct block option.

## NDMUI-220

(error) The name subscripting app_options 'design.bus_delimiters' and 'design.name_expansion_delimiters' cannot be set to the same value.

**Description**

The name subscripting delimiters cannot be set to the same value, because this creates ambiguous object name searches.

**What Next**

Change one of the values using the *set_app_options* command, so that it is different from the other value.

## NDMUI-221

(error) Width %s specified for layer '%s' is less than the layer minimum width %s.

**Description**

The indicated routing_rule width is invalid, because it is less than the layer minimum width. The routing_rule width specified for each layer must be greater than or equal to the corresponding layer's minimum width in the technology file.

**What Next**

Change the indicated routing_rule width, so that it is at least the layer minimum width.

## NDMUI-222

(warning) The objects passed to the option %s is inappropriate as the input is of type %s.

**Description**

The objects passed as input to the given option is not a valid input.

Feedback

### What Next

Change the objects(s) or collection passed to input for the option.

---

## NDMUI-223

(warning) The object passed to -of_objects is of type Lib Pin. The object will be ignored.

### Description

The objects passed is not a valid input for the current command. The object will be ignored.

### What Next

Change the objects(s) or collection passed to -of_objects.

---

## NDMUI-224

(error) Cannot remove fill_cell '%s'.

### Description

The *remove_fill_cells* command was unable to remove the specified fill_cell.

### What Next

Please make sure that the specified fill_cell belongs to the current_block. If all arguments are correct and the command still fails, it indicates an internal error. Please report to Synopsys.

---

## NDMUI-225

(error) Provided %s is out of core area.

### Description

This error message occurs when provided *-x_offset* or *-y_offset* for *set_rp_group_options* command is outside of core area.

### See Also

- create_rp_group
- add_to_rp_group

Feedback

## NDMUI-226

(error) Input region specified for option (%s) is of zero area.

### Description

The get_objects_by_location command was called on an empty region. Empty region cannot be specified for physical search.

### What Next

Specify a non-empty region.

## NDMUI-227

(error) Cells must all be in the same parent module.

### Description

The *group_cells* command requires that all of the cells being grouped are in the same module.

### What Next

Specify cells that are all in the same module.

### See Also

- group_cells

## NDMUI-228

(error) Unable to group cells.

### Description

The *group_cells* command is unable to group the specified cells due to problems that were detected while checking the cell list. These problems are reported separately.

### What Next

Examine the log file and correct any reported errors, then try the command again.

### See Also

- group_cells

Feedback

## NDMUI-229

(error) cell_list or -all must be specified.

### Description

You must specify either a list of cells or the *-all* option with the *ungroup_cells* command.

### What Next

Specify the list of cells or use the *-all* option to indicate that all hierarchical cells in the current instance should be ungrouped.

### See Also

- ungroup_cells

## NDMUI-230

(warning) Design has no hierarchy. No cells can be ungrouped.

### Description

You specified the *-all* option with the *ungroup_cells* command, but the current module contains no hierarchy, so there is nothing that can be ungrouped.

### See Also

- ungroup_cells

## NDMUI-231

(warning) Cannot ungroup %s cell %s.

### Description

The *ungroup_cells* command operates on logic hierarchical cells. You will see this message if you specified a cell that is not hierarchical or it is an instance of a block or an instance of a synthetic module.

A synthetic module, represents a tool generated module and is not allowed to be manipulated by the user.

### What Next

The command continued after excluding any leaf or block instance cells from the list of cells to ungroup. If the results are what you want, no action is required.

Feedback

### See Also

- ungroup_cells

---

## NDMUI-232

(error) Unable to ungroup cells.

### Description

The *ungroup_cells* command is unable to ungroup the specified cells due to problems that were detected while checking the cell list. These problems are reported separately.

### What Next

Examine the log file and correct any reported errors, then try the command again.

### See Also

- ungroup_cells

---

## NDMUI-233

(error) The specified %s %d is out of limit for relative placement group '%s', which has only %d %s.

### Description

This error message occurs when relative placement group commands detect that the row/column value given in input is out of range of the rows/columns for the specified relative placement group.

### What Next

Choose a new row/column that is within the total number of rows/columns of the relative placement group. Please rerun the command after making any changes.

### See Also

- add_to_rp_group
- create_rp_group
- remove_from_rp_group

Feedback

## NDMUI-234

(error) All locations of multiple location cell '%s' do not belong to specified %s of the relative placement group %s.

### Description

This error message occurs when the *modify_rp_groups* command detects that you tried to modify a relative placement group's rows/columns where all locations of multiple location cell do not belong to specified rows/columns of the relative placement group.

### See Also

- add_to_rp_group
- create_rp_group
- remove_from_rp_group

## NDMUI-235

(error) The specified number of %s to be deleted %d at %s %d is out of limit for relative placement group '%s', which has only %d %s.

### Description

This error message occurs when the *modify_rp_groups* command detects that you tried to delete number of rows/columns at a row/column of a relative placement group that is out of range of the rows/columns for that group.

### See Also

- add_to_rp_group
- create_rp_group
- remove_from_rp_group

## NDMUI-236

(error) The specified %s %d and %d is same so swap will not be performed for relative placement group.

### Description

This error message occurs when the *modify_rp_groups* command detects that you tried to swap a row/column with itself.

Feedback

**See Also**

- add_to_rp_group
- create_rp_group
- remove_from_rp_group

---

## NDMUI-237

(error) All %s of the relative placement group cannot be removed.

**Description**

This error message occurs when the *modify_rp_groups* command detects that you tried to delete all rows/columns of the relative placement group.

**See Also**

- add_to_rp_group
- create_rp_group
- remove_from_rp_group

---

## NDMUI-238

(error) The second argument of %s option cannot be 0.

**Description**

This error message occurs when the *modify_rp_groups* command detects that you tried to add/remove the rows/columns of the relative placement group with the second argument as 0.

**See Also**

- add_to_rp_group
- create_rp_group
- remove_from_rp_group

---

## NDMUI-239

(error) Invalid shapes specification '%s'.

Feedback

**Description**

A shape specification is expected to contain a layer name and a list of coordinates. The specified value is not a list of valid shape specifications such as { { layer {x1 y1} {x2 y2} ...} ... }.

**What Next**

Fix the offending list.

---

## NDMUI-240

(error) Invalid track color '%s' in list.

**Description**

The specified track color is invalid. The track color must be one of these values: mask_one, mask_two, mask_three

**What Next**

If you're attempting to clear the track color list, use an empty string. Otherwise, fix the offending track color.

---

## NDMUI-241

(error) Too many track color values (%d) in list.

**Description**

The track color specification is invalid because it contains too many entries. The maximum number allowed for the repeating pattern is 5.

**What Next**

Reduce the repeating track color pattern to contain no more than five values.

---

## NDMUI-242

(error) Invalid layer: '%s'.

**Description**

A shape specification is expected to contain a valid layer name and a list of coordinates. The specified layer could not be found.

**What Next**

Provide the name of a valid layer in the relevant tech.

Feedback

## NDMUI-243

(error) Cannot specify current block as reference.

**Description**

The current block cannot be specified as the reference for this command. Only blocks or library cells instantiated in the current block can be specified.

**What Next**

Please specify a block which has a interface mismatch with its instance.

## NDMUI-244

(error) Specified reference '%s' is not bound to current block.

**Description**

The reference specified as input is not bound to any instance in the current block. Therefore it cannot be used. Only blocks or library cells instantiated in the current block can be specified.

**What Next**

Please specify a block which has a interface mismatch with its instance.

## NDMUI-245

(information) Specified reference '%s' does not have any port mismatch.

**Description**

The reference specified as input has the same interface as the instances referring it.

**What Next**

Nothing to be done, this reference is not mismatched. This is just an information message

## NDMUI-246

(Error) No objects matched keepout type '%s'.

**Description**

This error occurs when none of the objects from the resultant collection is of the requested keepout type.

Feedback

## NDMUI-247

(Warning) Cannot set outer keepout margins on design '%s', as there are no instances which refer to this design.

### Description

Outer keepout margin is set on instances of the particular design in all the parent designs, where it is used as reference.

### What Next

Do not set outer keepout for the design.

## NDMUI-248

(warning) The specified cell '%s' cannot be added to relative placement group because it is fixed.

### Description

This warning message occurs when the *add_to_rp_group* command detects that you tried to add a cell, whose status is fixed. A cell can be added to relative placement group only if it is not fixed.

### See Also

- add_to_rp_group
- create_rp_group
- remove_from_rp_group

## NDMUI-249

(Error) Cell list must contain exactly one cell if -phys_wrapper is specified.

### Description

The *group_cells* command's *-phys_wrapper* option is used to create a new hierarchy around a single, hierarchical cell. You will see this message if you used the *-phys_wrapper* option and specified more than one cell for the cell list.

### What Next

Use the *-phys_wrapper* option with a single, hierarchical cell.

Feedback

### See Also

- group_cells

---

## NDMUI-250

(Error) The specified cell must be hierarchical if -phys_wrapper is specified.

### Description

The *group_cells* command's *-phys_wrapper* option is used to create a new hierarchy around a single, hierarchical cell. You will see this message if you used the *-phys_wrapper* option with a non-hierarchical cell.

### What Next

Use the *-phys_wrapper* option with a single, hierarchical cell.

### See Also

- group_cells

---

## NDMUI-251

(error) One or more objects, their parents, and/or their children have locked status.

### Description

This error message occurs when a command tries to transform or remove an object or a collection of objects, at least one of which either has locked status or is a parent/child of an object with locked status.

For example, a net may not be locked, but if it has a shape that is locked, then it will produce this error. This works the other way around as well. A shape may not be locked, but if it is owned by a net that is locked then it will produce this error.

Use "-force" option with the command to ignore locked status of the objects.

### What Next

If you see this error then one of three cases has occurred. One, one of the objects in the list has locked status. Two, a parent (at any level of the hierarchy) of one of the objects in the list has locked status. Three, a child (at any level of the hierarchy) of one of the objects in the list has locked status.

To resolve the error check the attributes of the objects, their parents, and their children to determine which objects need to be unlocked.

Feedback

**See Also**

- remove_shapes
- move_objects
- set_object_layer

---

## NDMUI-252

(warning) No port mismatches were fixed.

**Description**

The command was unable to fix any port mismatches in the design. This could happen due to one of the following reasons. a) There are no instances that needed to be fixed b) Instances with port mismatch were found but different instances for the same reference were bound to different views c) Fixing the mismatches would have caused some pins on instances to be deleted, but those pins were connected to a net

**What Next**

If the reason was "b" above, please run this command for the particular reference, specifying the reference you want to keep. If the reason was "c" above, please use "-force" option.

---

## NDMUI-253

(error) Invalid custom via_def shape: '%s'. Custom via_defs can contain only non-zero-sized Manhattan polygons.

**Description**

A custom via_def shape must be a non-zero size Manhattan (right-angled) polygon. The given shape is either zero size or contains non-right angles.

**What Next**

Provide a valid Manhattan shape specification.

---

## NDMUI-254

(error) Attribute '%s' cannot be set on via_def '%s' because it is not defined for custom via_defs. It is defined for simple via_defs only.

Feedback

**Description**

A custom via_def is defined by a collection of shapes. Some via_def attributes are not applicable to custom via_defs.

**What Next**

Use 'report_attributes' to review the set of attributes defined on the given via_def.

---

## NDMUI-255

(error) Attribute '%s' cannot be set on via_def '%s' because it is a custom via_def. This attribute is settable for simple via_defs only.

**Description**

A custom via_def is defined by a collection of shapes. Some attributes of a custom via_def are derived from its shape collection and are therefore not settable.

**What Next**

Modify the custom via_def's collection of shapes instead to achieve the desired attribute value.

---

## NDMUI-256

(error) Exactly two pins must be specified as input to "-pins" option.

**Description**

The "-pins" option for commands "set_supernet_exceptions" and "remove_supernet_exceptions" expects a list or collection with exactly two pins.

**What Next**

Please provide the correct input. If you're using a collection, make sure the collection has exactly two pins.

---

## NDMUI-257

(error) Attribute 'is_default' cannot be set on custom via_def '%s'. It is settable for simple via_defs only.

**Description**

A custom via_def is defined by a collection of shapes. A custom via_def cannot be a default via_def for a cut layer. .

Feedback

### What Next

Use 'report_via_defs' to review the set of via_defs. Choose one or more simple via_defs as default via_defs for this cut layer.

---

## NDMUI-258

(error) Term '%s' is already an anchor for a supernet.

### Description

The specified pin or port is already used an anchor for a supernet. A given pin or port can only be used as anchor for only one supernet.

### What Next

Please specify a pin or port which is not an anchor for a supernet. You can also use the existing supernet on the specified pin or port.

---

## NDMUI-259

(error) Name cannot be specified, when multiple anchors are provided.

### Description

A supernet is created for each specified anchor point. Each supernet must have a different name. When multiple anchors are specified, the name for the supernets cannot be given.

### What Next

Please rerun the command either without a name or with exactly one anchor.

---

## NDMUI-260

(Information) Query rules are not defined in block '%s'.

### Description

Query rules have not been defined in the specified block.

### What Next

Read a name map file, containing query rules, using the *read_name_map* command.

### See Also

- read_name_map
- write_name_map

Feedback

## NDMUI-261

(Error) Map application name '%s' is not supported.

### Description

In the name map file's *define_name_maps* command call, the specified application name is not supported. The application name must be 'golden_upf'.

### What Next

Correct the map application name in the name map file, then reload it using the *read_name_map* command.

### See Also

- read_name_map
- write_name_map

## NDMUI-262

(Error) Map design name '%s' does not match the current block name '%s'.

### Description

In the name map file's *define_name_maps* command call, the specified design name does not match the current block name. Name maps are loaded into the current block, so the intended destination design name must match the current block name.

### What Next

Correct the map design name in the name map file, then reload it using the *read_name_map* command.

### See Also

- read_name_map
- write_name_map

## NDMUI-263

(Error) Map column names must be '{class pattern options names}'.

Feedback

### Description

In the name map file's *define_name_maps* command call, the specified column names are not supported. The column names must match the predefined column names for the 'golden_upf' application. It must be the list {class, pattern, options, names}.

### What Next

Correct the map column names in the name map file, then reload it using the *read_name_map* command.

### See Also

- read_name_map
- write_name_map

---

## NDMUI-264

(Error) Map class name '%s' is not supported.

### Description

In the name map file's *define_name_maps* command call, the specified class name is not supported. The supported classes are cell, port, pin, net, power_domain, supply_net, supply_set, supply_port, and power_switch.

### What Next

Correct the map class name in the name map file, then reload it using the *read_name_map* command.

### See Also

- read_name_map
- write_name_map

---

## NDMUI-265

(Error) Map option name '%s' is not supported.

### Description

In the name map file's *define_name_maps* command call, the specified option name is not supported. The supported options are transitive, nocase, leaf, noleaf, in, out, and inout.

Feedback

### What Next

Correct the map option name in the name map file, then reload it using the *read_name_map* command.

### See Also

- read_name_map
- write_name_map

---

## NDMUI-266

(Error) Map entry '%s' is invalid; %s.

### Description

In the name map file's *define_name_maps* command call, the specified entry is invalid. Each entry must contain 4 elements (i.e., class, pattern, options, names). The list of names cannot be empty.

### What Next

Correct the map entry in the name map file, then reload it using the *read_name_map* command.

### See Also

- read_name_map
- write_name_map

---

## NDMUI-267

(Error) Query rule regsub '%s' is invalid; %s.

### Description

In the name map file's *set_query_rules* command call, the specified regsub is invalid. Each regsub must contain 3 elements (i.e., switches, match_regexp, substitution_exp).

### What Next

Correct the query rule regsub in the name map file, then reload it using the *read_name_map* command.

Feedback

### See Also

- read_name_map
- write_name_map

---

## NDMUI-268

(error) The specified relative placement group '%s' cannot be added to itself or one of its hierarchical relative placement groups.

### Description

This error message occurs when the *add_to_rp_group* command detects that you tried to add a relative placement group to itself or one of its hierarchical relative placement groups.

### See Also

- add_to_rp_group
- create_rp_group
- remove_from_rp_group

---

## NDMUI-269

(error) Value '%s' for option '%s' does not specify at least 2 points. The syntax is { { x1 y1 } { x2 y2 } ... }.

### Description

The reported option must specify points which form a path. A path is defined by a list of at least two points. The syntax is: { { x1 y1 } { x2 y2 } ... }

### What Next

Provide the correct value to the reported option.

---

## NDMUI-270

(error) The command '%s' is prohibited in freeze-silicon mode.

### Description

The design is in freeze_silicon mode. Only a small set of commands that modify the database can actually run properly in this mode. This command cannot run properly in freeze_silicon mode and is therefore prohibited.

Feedback

**What Next**

If you need to run this command please set the app option design.eco_allow_all_commands to enable all commands to run in this mode

---

## NDMUI-271

(error) Value '%s' for option -cuts is invalid; %s.

**Description**

This option requires a list in the format {{cut_layer_name {cut_name number_of_cuts} ...} ...}.

The cut_layer_name must correspond to a cut layer defined in the technology file with defined fat tables (i.e., fatTblThreshold, fatTblFatContactNumber, fatTblFatContactMinCuts) and cut tables (i.e., cutNameTbl, cutWidthTbl, cutHeightTbl).

The cut_name must correspond to a cut name in the cut layer's cut tables (i.e., cutNameTbl).

The number_of_cuts must be an integer greater than 0.

**What Next**

Provide the correct value to the option.

---

## NDMUI-272

(Error) The height '%f' of site def '%s' doesn't match the FinFET grid's Y pitch '%f'.

**Description**

This message was issued because the height of the Site def to be used doesn't match FinFET grid's Y pitch.

**What Next**

1.Make sure set_technology is set.

2.If 1 not work, change the Site def to be used, so that it matches the FinFET Grid.

---

## NDMUI-273

(Error) The width '%f' of site def '%s' doesn't match the FinFET grid's X pitch '%f'.

Feedback

**Description**

This message was issued because the width of the Site def to be used doesn't match FinFET grid's X pitch.

**What Next**

Change the Site def to be used, so that it matches the FinFET Grid.

---

## NDMUI-274

(Error) The height '%f' of site def '%s' with orientation '%s' doesn't match the FinFET grid's Y pitch '%f'.

**Description**

This message was issued because the effective height of the Site def for a user specified orientation doesn't match FinFET grid's Y pitch.

**What Next**

Change the Site def to be used, so that it matches the FinFET Grid.

---

## NDMUI-275

(Error) The width '%f' of site def '%s' with orientation '%s' doesn't match the FinFET grid's X pitch '%f'.

**Description**

This message was issued because the effective width of the Site def for a user specified orientation doesn't match FinFET grid's X pitch.

**What Next**

Change the Site def to be used, so that it matches the FinFET Grid.

---

## NDMUI-276

(Warning) The length '%f' of '%s' is not an integral multiple of the FinFET grid's Y pitch '%f'.

**Description**

This message was issued because the Y direction length is not an integral multiple of FinFET grid's Y pitch and hence not falling into the FinFET grid.

**What Next**

May need to change the length, so that it falls into the FinFET Grid.

Feedback

## NDMUI-277

(Warning) The length '%f' of '%s' is not an integral multiple of the FinFET grid's X pitch '%f'.

**Description**

This message was issued because the X direction length is not an integral multiple of FinFET grid's X pitch and hence not falling into the FinFET grid.

**What Next**

May need to change the length, so that it falls into the FinFET Grid.

## NDMUI-278

(Warning) The user specified coordinate '(%f, %f)' does not align with the FinFET grid.

**Description**

This message was issued because the user specified coordinate doesn't align with the FinFET grid and hence is violating the condition that all Fins should align with the FinFET grid.

**What Next**

May need to change the user specified coordinate, so that it aligns with the FinFET Grid.

## NDMUI-279

(warning) Via spacing %s is less than the minimum spacing %s of via layer '%s'.

**Description**

The indicated routing_rule via spacing is invalid, because it is less than the via layer minimum spacing. The router will use the via layer minimum spacing instead of the routing_rule via spacing.

**What Next**

Change the indicated routing_rule via spacing, so that it is at least the via layer minimum spacing.

## NDMUI-280

(error) Defining multiple via spacings between layers '%s' and '%s' is not allowed.

Feedback

**Description**

Defining multiple via spacings between layers is not allowed.

**What Next**

Specify only one via spacing between the two via layers.

---

## NDMUI-281

(error) Value for option -port_contact_selections is invalid; %s.

**Description**

This option requires a list in the format {{portName {contactCodeName ...}} ...}.

The contactCodeName must correspond to a ContactCode in technology file.

**What Next**

Provide the correct value to the option.

---

## NDMUI-282

(warning) The create_routing_rule %s option is deprecated.%s

**Description**

The indicated *create_routing_rule* option is deprecated and no longer supported.

The *-taper_distances* option is now supported by the *-taper_distance* option, which takes a single distance value, instead of a list of distance values.

The *-driver_taper_distances* option is now supported by the *-driver_taper_distance* option, which takes a single distance value, instead of a list of distance values.

The *-spacing_weights* option is now supported by the *-spacing_weight_levels* option, which takes weight levels, instead of weight float values.

The *-min_routing_layer* and *-max_routing_layer* options are now supported by the *set_routing_rule* command. The minimum and maximum routing layers are applied to each net, instead of each routing rule.

These options have no replacement: *-routing_effort_level*, *-spacing_efforts*, *-taper_level*, *-taper_modes*,

**What Next**

Remove the indicated option and use the replacement option if any.

Feedback

## NDMUI-283

(error) Attribute '%s' cannot be set on via_def '%s' because it is derived from the shapes of a custom via_def. It is settable for simple via_defs only.

**Description**

A custom via_def is defined by a collection of shapes. Some via_def attributes are not settable on custom via_defs.

**What Next**

Use 'report_attributes' to review the set of attributes that are settable on the given via_def.

## NDMUI-284

(error) Layer '%s' is not a valid cut layer. The cut layer of a simple via_def must be of type VIA_CUT or LOCAL_VIA_CUT.

**Description**

The cut layer of a simple via_def can only be of type VIA_CUT or LOCAL_VIA_CUT. Other types, such as INTERCONNECT or LOCAL_INTERCONNECT are not valid types for the cut layer.

**What Next**

Check the layers available in the tech for appropriate candidates of the allowed types.

## NDMUI-285

(Error) The height '%f' of '%s' is not an integral multiple of the FinFET grid's Y pitch '%f'.

**Description**

This message was issued because the Y direction length is not an integral multiple of FinFET grid's Y pitch and hence not falling into the FinFET grid.

**What Next**

Change the height of the object, so that it falls into the FinFET Grid.

## NDMUI-286

(Error) The width '%f' of '%s' is not an integral multiple of the FinFET grid's X pitch '%f'.

Feedback

### Description

This message was issued because the X direction length is not an integral multiple of FinFET grid's X pitch and hence not falling into the FinFET grid.

### What Next

Change the Width of the object, so that it falls into the FinFET Grid.

---

## NDMUI-287

(error) The specified utilization '0' for the relative placement group is not allowed.

### Description

This error message occurs when the *set_rp_group_options* command detects that you tried to set the utilization of a relative placement group to '0'. The utilization of a relative placement group should be greater than '0' and less than or equal to '1.0'.

### See Also

- add_to_rp_group
- create_rp_group
- remove_from_rp_group

---

## NDMUI-288

(Error) Invalid symmetry option '%s', please specify a set composed with strings "X", "Y" and "R90".

### Description

This message indicates that an invalid symmetry value is specified. The acceptable value of this option is a set composed with the strings "X", "Y" and "R90".

### What Next

Change the symmetry value to a set composed with valid strings - "X", "Y" and "R90". Example: { X } or { Y R90 } or { X Y R90 } etc.

---

## NDMUI-289

(error) The specified cell '%s' is a spare cell. This command cannot be run for this cell.

Feedback

**Description**

The design is in freeze_silicon mode. The command (change_link or set_reference) can only be used for regular cells in freeze_silicon mode.

**What Next**

Please specify regular cells to change the reference.

---

## NDMUI-290

(error) Cells may only be set to a via_def of type "through_silicon_via_def".

**Description**

Cells may be set to reference via_defs, but only those of type "through_silicon_via_def".

**What Next**

Please specify a through-silicon via_def in the "-via_def" argument of the command when using the "-cells" argument.

---

## NDMUI-291

(error) One or more TSV cells could not be created.

**Description**

One or more through-silicon via ("TSV") cells could not be created, likely due to a name collision.

**What Next**

Please specify cell names that are unique in the current block.

---

## NDMUI-292

(error) The layers list must have the same length as the boundary list.

**Description**

If the "-layers" option is specified, then the list of layers must have the same length as the boundary list, so that a one-to-one correspondence is formed between the layers and poly_rects with the associated boundaries.

**What Next**

Modify the layers list to have a layer corresponding to each respective boundary in the boundary list.

Feedback

## NDMUI-293

(error) Attribute 'shapes' cannot be set on simple via_def '%s'. It is settable for custom via_defs only.

### Description

A simple via_def is defined by its layer and geometrical parameters. The 'shapes' attribute is derived from those and cannot be set directly.

### What Next

The 'shapes' attribute of a simple via_def can be set indirectly by setting the layer, height and width parameters for each layer of the via_def. Use 'list_attributes' to review the set of parameters that define a simple via_def, and 'set_attribute' to set them.

## NDMUI-294

(error) Attribute 'shapes' could not be set on custom via_def '%s'. One or more objects in the given collection was not a poly_rect.

### Description

The 'shapes' attribute of a custom via_def is defined by a collection of poly_rects. When 'set_attribute' is used to set the 'shapes' of a custom via_def, all objects in the given collection must be poly_rects.

### What Next

Remove the non-poly_rects from the input collection. Consider using 'create_poly_rect' to create a homogeneous collection of poly_rects representing the 'shapes' of the custom via_def.

## NDMUI-295

(Error) Application option '%s' is not enabled for subscripted access/modifications.

### Description

The application option was registered to not allow subscripted access/modifications. The application option has to be of the form {name value} and needs to be enabled for subscripted access/modifications.

## NDMUI-296

(Information) Application option '%s' is global scoped. Ignoring the '%s' option.

Feedback

**Description**

Global scoped app-options cannot be set on a block.

---

## NDMUI-297

(Error) Subscript '%s' is not set on application option '%s'.

**Description**

The requested subscript is not set on the application option.

**What Next**

Try "get_app_option_value -name <applicationOptionName>" to look at all the subscripts that are set.

---

## NDMUI-298

(Information) Cell is already uniquified. Child hierarchy will be uniquified if not already uniquified.

**Description**

For "uniquify" and "uniquify_block" commands, if an already uniquified cell is provided along with "-uniquify_children" option, then the commands will uniquify children hierarchy of cell's reference.

---

## NDMUI-299

(Error) Cannot %s design scoped application option '%s', as the design is not specified.

**Description**

The current block is not set and the '-block' option is not specified. The design on which the design scoped application option is to be set could not be determined.

**What Next**

Either set the current block or specify the design to be used using the '-block' option.

---

## NDMUI-300

(Warning) Cannot use unordered layer '%s' with negative mask_order as reference_layer in '%s' option.

Feedback

**Description**

The relative mask_order cannot be determined from an unordered reference layer.

**What Next**

The mask_order would be set to its default value -1.

---

## NDMUI-301

(Warning) Blockage '%s' isn't a routing-blockage.

**Description**

The specified blockage is NOT a routing- blockage.

**What Next**

Please specify a routing-blockage.

---

## NDMUI-302

(Warning) Routing blockage '%s' cannot be used for both net and net_type specific association simultaneously.

**Description**

A routing-blockage applies to either all nets or only the specified nets. Nets can be specified by associating EITHER individual specific nets (blocks only the specified nets) OR specific net_type (example -net_types clock blocks all clock nets).

**What Next**

Please use any one of net or net_type specific association for the routing-blockage.

---

## NDMUI-303

(Error) Cannot specify the side-numbers for the given start/end object. Side numbering is to be used only when the start/end object is an object of '%s' or '%s' or '%s.

**Description**

The side numbers to be used as a topological pin feedthrough constraint is applicable only for objects of certain types.

---

## NDMUI-304

(Error) The offset value '%f' should be less than the maximum side length '%f'.

Feedback

**Description**

The offset value is limited by the maximum side length of the given start/end object.

**What Next**

Use a value within the limit specified

---

## NDMUI-305

(Error) The offset range '%s' should be less than the offset '%s'.

**Description**

The offset-range represents a distance from the offset that a pin may be placed. The range should be less than the current offset value.

**What Next**

Use a value smaller than the current offset.

---

## NDMUI-306

(Error) %s '%s' must be logically connected to the owner.

**Description**

The port or the pin object that is used as a start/end object for a particular topological pin feedthrough constraint should be logically connected to the owner of the constraint.

**What Next**

Use a port/pin that is logically connected to the owner.

---

## NDMUI-307

(Error) %s '%s' is not logically connected to the owner %s '%s'.

**Description**

The port or the pin object that is used as a start/end object for a particular topological pin feedthrough constraint should be logically connected to the owner of the constraint.

**What Next**

Use a port/pin that is logically connected to the owner.

Feedback

## NDMUI-308

(Error) The %s option cannot be specified for %s object of type '%s'.

**Description**

The option to specify side-numbers can only be used for objects of type cell, voltage-area or voltage-area-shape.

## NDMUI-309

(Error) Current block does not have the physical information for %s '%s'.

**Description**

The physical representation of the start/end object (port, pin and cell) should exist in the current block. Topological pin feedthrough constraints cannot be created across physical hierarchies.

## NDMUI-310

(Error) Invalid type '%s' specified. Supported object types are '%s'.

**Description**

The object specified as the start/end object for the constraint belongs to an unsupported type. The valid types are specified in the output.

**What Next**

Use the supported object type as a start/end object.

## NDMUI-311

(Error) Cannot specify an object of '%s' type for both the %s and %s options.

**Description**

The start/end object of a topological pin feedthrough constraint cannot be of the same type unless it is a cell.

**What Next**

Use objects of different types for the start/end object.

Feedback

## NDMUI-312

(Error) Object '%s' is in design '%s' which is not the same as the constraint's design '%s'.

**Description**

The start and end objects of a topological pin feedthrough constraint must be in the design of the constraint. It could be either in the same physical hierarchy as the constraint or a lower level of physical hierarchy.

**What Next**

Use objects in the constraint's design as the start and end objects. When specifying objects in the lower physical hierarchy, be sure to get it through the hierarchy by specifying the full hierarchical path name.

## NDMUI-313

(Error) Existing topological pin feedthrough constraint '%s' has similar configurations.

**Description**

The topological pin feedthrough constraints in a block should be unique. The tool does not allow creation of duplicate topological pin feed through constraints.

**What Next**

Use the existing topological pin feedthrough constraint or create a new one with different configuration.

## NDMUI-314

(Error) Failed to create topological pin feedthrough constraint for owner object '%s'.

**Description**

The topological pin feedthrough constraint creation command failed to create a topological constraint on the current block.

## NDMUI-315

(Error) The '%s' attribute cannot be set for %s object of type '%s'.

**Description**

The side numbers to be used as a topological pin feedthrough constraint is applicable only for objects of certain types.

Feedback

## NDMUI-316

(Error) Attribute '%s' is applicable only for start/end objects of type '%s' or '%s' or '%s'.

**Description**

The side numbers to be used as a topological pin feedthrough constraint is applicable only for objects of certain types. Subsequently the offset attribute can be specified only on objects for which the side-number attribute is applicable.

## NDMUI-317

(Error) The topological pin feed through constraint '%s' does not have the %s %s set. Cannot set the %s attribute.

**Description**

The offset and offset-range attribute can be specified only if the side numbers are set on the topological pin-feedthrough constraint.

## NDMUI-318

(Error) Valid side-number range for the object %s is (1 - %u).

**Description**

The sides numbers provided is out of the valid range for the start/end object.

**What Next**

Use the side-numbers in the range displayed.

## NDMUI-319

(Error) Voltage area '%s' has more than one (%u) regions. Cannot use the side-numbers for this voltage-area.

**Description**

If the voltage-area is used as a start/end object and if it has more-than one voltage-area-regions then one specific voltage-area-region should be used as a start/end object. The side-numbers will then correspond to the chosen voltage-area-region.

**What Next**

Use a voltage-area-region as the start/end object.

Feedback

## NDMUI-320

(Error) Site row '%s' belongs to site array '%s' and can't be removed independently.

### Description

The site row to be removed is part of a site array. Constituent site rows of site array can't be manipulated independently. Site array takes care of handling its constituent site rows.

### What Next

The site row can't be removed. If it is part of a collection specified for removal then remove this site row from the collection so that other site rows can be removed.

## NDMUI-321

(warning) The requested module name %s is not available. The new module's name is %s.

### Description

The module name requested by *group_cells -module_name* is not available, so the tool used an automatically-generated name.

### What Next

Specify a module name that is not in use or omit the *-module_name* option to the *group_cells* command.

### See Also

- group_cells

## NDMUI-322

(warning) The requested cell name %s is not available. The new cell's name is %s.

### Description

The cell name requested by *group_cells -cell_name* is not available in the parent module, so the tool used an automatically-generated name.

### What Next

Specify a cell name that is not in use or omit the *-cell_name* option to the *group_cells* command.

Feedback

**See Also**

- group_cells

---

## NDMUI-323

(Error) Specified pin '%s' does not have physical context or belong to current block.

**Description**

This error message is printed during commands *set_via_ladder_constraints*, *remove_via_ladder_constraints* and *remove_via_ladder_constraints*, when a pin specified does not have block context or belong to current block.

**What Next**

Use "get_pins -physical_context -design [current_block]" to get list of valid pins.

**See Also**

- set_via_ladder_constraints
- remove_via_ladder_constraints
- report_via_ladder_constraints

---

## NDMUI-324

(Error) Specified via ladder '%s' does not exist in technology file. Specify from:\n\t%s

**Description**

This error message is printed when a via ladder template name is specified that is not present in technology file. Along with this message, valid list of names are printed out.

**What Next**

Specify valid via ladder template names.

**See Also**

- set_via_ladder_constraints
- remove_via_ladder_constraints
- report_via_ladder_constraints

Feedback

## NDMUI-325

(Warning) Specified pin '%s' does not have via ladder constraint set.

### Description

This warning message is printed during commands "*remove_via_ladder_constraints* and *report_via_ladder_constraints* when no via ladder constraint is set for specified pin.

### What Next

Please specify pin for which via ladder constraint exists.

### See Also

- set_via_ladder_constraints
- remove_via_ladder_constraints
- report_via_ladder_constraints

## NDMUI-326

(error) The via_def specified as the cell reference must be of type 'through_silicon_via_def'.

### Description

When specifying a via_def as the 'reference' argument of the create_cell command, the via_def must be of type 'through_silicon_via_def'.

### What Next

Please specify a via_def of the correct type.

## NDMUI-327

(error) The specified %s '%s' can not be added to relative placement group '%s', because its site_height is different.

### Description

This error message occurs when the *add_to_rp_group* command detects that you tried to add a cell or a hierarchical relative placement group to another relative placement group which has a different site_height. The site_height of the relative placement group is the site_height of its cells.

Feedback

**See Also**

- add_to_rp_group
- create_rp_group
- remove_from_rp_group
- create_bound

---

## NDMUI-328

(error) Value '%s' is an invalid layer mask shift list.

**Description**

The layer mask shift list is invalid. It must be a list, { { layer_name shift } ... }, in which layer_name must be a valid layer name and shift must be 0 or 1. 0 indicates no shift on the layer. 1 indicates shift by 1 on the layer.

**What Next**

Call *set_attribute* with the corrected layer mask shift list.

---

## NDMUI-329

(error) Layer mask shift specification '%s' is invalid; %s.

**Description**

The layer mask shift specification is invalid. It must be a string "layer_name shift" in which layer_name must be a valid layer name and shift must be 0 or 1. 0 indicates no shift on the layer. 1 indicates shift by 1 on the layer.

**What Next**

Call *set_attribute* with the corrected layer mask shift specification.

---

## NDMUI-330

(error) Cell instances of block '%s' are not mask shiftable.

**Description**

The cell *mask_shift attribute cannot be set, because its reference block is not mask shiftable. The block is not mask shiftable if its is_mask_shiftable attribute value is false or if its parent library's is_mask_shiftable attribute value is false.*

Feedback

### What Next

Check the reference block's and its parent library's *is_mask_shiftable* attribute value. If not correct, correct the values. Then set the cell's *mask_shift* attribute.

---

## NDMUI-331

(warning) Shift value on layer '%s' has no effect, because the layer number_of_masks attribute value is '%d'.

### Description

The specified shift will have no effect on the indicated layer, because the layer's number_of_masks attribute is less than 2. For shifting to have any effect, the layer's number_of_masks attribute must be 2 or greater.

### What Next

Correct the layer's number_of_masks attribute value so it is at least 2.

---

## NDMUI-332

(warning) The option '%s' has been deprecated and is scheduled to be removed in a future release. %s

### Description

The given command option has been deprecated. Please see the man-page of the command for migration assistance.

---

## NDMUI-333

(error) Value '%s' for option %s is invalid; %s.

### Description

This option requires a list in the reported format.

If layer_name is expected in the reported format, it must correspond to a routing layer defined in the technology file. The mask_name must be one of the acceptable values. If the mask is fixed, then the keyword, fixed, must be specified. If the mask is not fixed, do not specify the keyword.

### What Next

Provide the correct value to the option.

Feedback

## NDMUI-334

(error) All values in '%s' must be greater than zero.

**Description**

This option requires a list of values, all greater than zero.

**What Next**

Provide the correct value to the option.

## NDMUI-335

(error) Value for '%s' must be less than sum of all values for '%s'.

**Description**

First option requires value which is less than sum of all the values provided in second option.

**What Next**

Provide the correct value to the option.

## NDMUI-336

(warning) Specification of a {name_value_pair list} without either the '%s' or '%s' option has been deprecated and is scheduled to be removed in a future release. %s

**Description**

The given command option has been deprecated. Please see the man-page of the command for the new syntax.

## NDMUI-337

(Error) Empty or invalid geometry specified for the '%s' argument of the command.

**Description**

The specified argument of the command requires a valid geometrical region with positive area.

## NDMUI-338

(error) Antenna rule with the given name '%s' does not exist in the library '%s'.

Feedback

**Description**

The specified antenna rule does not exist in the given library.

**What Next**

Use get_antenna_rule_names(2) to list the rule-names in a given library.

---

## NDMUI-339

(error) Layer '%s' of routing blockage '%s' is not a via cut layer.

**Description**

The 'is_design_rule_blockage' attribute is settable only if the blockage's layer type is VIA_CUT or LOCAL_VIA_CUT.

---

## NDMUI-340

(information) Setting 'is_design_rule_blockage' to false. Layer '%s' of routing blockage '%s' is not a via cut layer.

**Description**

The 'is_design_rule_blockage' attribute can be true only if the layer type is VIA_CUT or LOCAL_VIA_CUT.

---

## NDMUI-341

(Error) Cannot specify the '%s' argument without the '%s' or '%s' arguments.

**Description**

The specified argument of the command requires one of the two other specified arguments.

---

## NDMUI-342

(Error) Must specify either the '%s' argument or the '%s' argument.

**Description**

The command requires that one of the specified arguments is provided.

---

## NDMUI-343

(Error) Multiple polygons specified for the '%s' argument of the command.

Feedback

**Description**

The specified argument of the command requires an input geometry equivalent to a single, connected polygon. However, the specified geometry resulted in multiple, non-connected polygons.

---

## NDMUI-344

(Error) Cannot set target library subset on %s '%s'; %s.

**Description**

The target library subset cannot be set on the object. Target library subsets may only be assigned to objects in the current block. Additionally cells must reference hierarchies within the current block (i.e., they cannot reference other blocks).

**What Next**

Remove the specified object from the *-objects* option list.

---

## NDMUI-345

(Warning) Skipping %s '%s'; %s.

**Description**

The target library subset cannot be removed or reported from the specified object. The object is skipped. Target library subsets are assigned only to objects in the current block.

**What Next**

Remove the specified object from the *-objects* option list.

---

## NDMUI-346

(Error) Library '%s' is not a valid target library; %s.

**Description**

The specified library is not a valid target library. Target libraries must be lib_cell libraries.

**What Next**

Remove or replace the specified library in the *target_libs* option list.

---

## NDMUI-347

(Error) No valid target libraries or lib_cells were specified.

Feedback

**Description**

The specified target libraries and lib_cells are not valid.

**What Next**

Specify valid values for the *target_libs*, *-dont_use*, or *-only_here* options.

---

## NDMUI-348

(error) The working design is not from frame view.

**Description**

The block does not have indicated view.

**What Next**

The command is only for frame view.

---

## NDMUI-349

(error) via_def '%s' is invalid. via_def must be a simple via definition when -size is not {1 1} or -pitch is specified.

**Description**

The input via_def must be a simple via definition when -size is not {1 1} or -pitch is specified.

**What Next**

Provide a valid simple via definition.

---

## NDMUI-350

(error) input via_list contains %d custom vias, '%s'. Vias must be simple via or simple array via when -size is not {1 1} or -pitch is specified.

**Description**

The input vias must be simple via or simple array via when -size is not {1 1} or -pitch is specified.

**What Next**

Provide valid input via_list containing simple vias and simple array vias only.

Feedback

## NDMUI-351

(warning) Invalid reference design name(%s) for power constraints.

**Description**

The format of reference design name is invalid for Power constraints.

**What Next**

Provide provide the design name as libName:designName.view.

## NDMUI-352

(error) at lease one of -via_def, -pitch, and -size options must be specified.

**Description**

At least one of -via_def, -pitch, and -size options must be specified.

**What Next**

Provide at lease one of -via_def, -pitch, and -size options.

## NDMUI-353

(error) Exactly one via def must be specified as input.

**Description**

This command accepts exactly one via def.

**What Next**

Please provide the correct input.

## NDMUI-354

(error) cannot change %s of via '%s' to %s.

**Description**

This command cannot change via def from simple via def to custom via def or from custom via def to simple via def. To change via def type, please use set_via_def.

**What Next**

Please provide the correct input.

Feedback

## NDMUI-355

(error) cannot change array size of %s '%s'%s.

**Description**

This command cannot change array size of the specified via. Via must be simple array via, and simple array via cannot be changed to simple via with array size {1 1}.

**What Next**

Please provide the correct input. To change simple via to simple array via or simple array via to simplevia, please use tcl command set_via_def.

## NDMUI-356

(error) invalid array size value '%s' in list. It must be {number_of_rows number_of_columns}.

**Description**

Invalid array size input. It must be {number_of_rows number_of_columns}.

**What Next**

Please provide the correct array size.

## NDMUI-357

(error) power domains '%s' and '%s' do not share equivalent default pwell and nwell supply net(s).

**Description**

When associating multiple power domains to a voltage area, the default pwell and nwell supplies of the power domains must be electrically equivalent, meaning that their supply nets must either be the same, or connected via supply ports.

## NDMUI-358

(error) power domains '%s' and '%s' do not share equivalent available supply nets.

**Description**

When associating multiple power domains to a voltage area, the available supply nets of the power domains must be electrically equivalent, meaning that their supply nets must either be the same, or connected via supply ports.

Feedback

## NDMUI-359

(warning) Unable to obtain mask value for surround on lower metal layer of via '%s'.

**Description**

This warning message is issued since the surround on lower metal layer of the via cannot touch any pin in the design. This could be a via that only connects two layers but is not on a pin.

**What Next**

Please check the via and the design data if needed.

## NDMUI-360

(warning) The view type current block is not 'frame'. Via regions should exist only in frame views.

**Description**

The view type current block should be frame view.

**What Next**

The command is only for frame view.

## NDMUI-361

(error) '-incremental' option can only be specified when '-automatic' option is also specified.

**Description**

'-incremental' option can only be specified when '-automatic' option is also specified.

**What Next**

Run connect_pg_net with both '-incremental' and '-automatic' options or remove '-incremental' option.

## NDMUI-362

(error) Invalid 'base_lib'. Specified base library must be %s.

**Description**

Specified library cannot be used as the base for Sparse View library.

Feedback

**What Next**

Please specify a valid design library on-disk as the base of a Sparse View library.

---

## NDMUI-363

(error) Number of Sparse View libraries in the daisy chain of base library '%s' is '%d' that exceeds the maximum limit '%d'.

**Description**

Specified library cannot be used as the base for Sparse View library.

**What Next**

Please specify a different design library as the base library.

---

## NDMUI-364

(error) Design %s is open. It must be closed before %s.

**Description**

Reported design prevents the intended operation in Sparse View library because it is open.

**What Next**

Please close the reported design to proceed.

---

## NDMUI-365

(info) Reverted design '%s' and removed it from disk.

**Description**

This is a message indicating that the reported design has been reverted the in Sparse View library and its on-disk content has been removed.

---

## NDMUI-366

(error) %s failed. Library '%s' is %sSparse View library.

**Description**

Reported operation is not permitted in the specified library.

Feedback

**What Next**

Please operate on correct library.

---

## NDMUI-367

(error) %s failed. Design '%s' is %sremote design.

**Description**

Reported operation is not permitted in the specified design.

**What Next**

Please operate on correct design.

---

## NDMUI-368

(error) Via region %s is a system via region.

**Description**

System via region should not be editable, but user via region is editable.

---

## NDMUI-369

(error) Invalid input layer. Input layers must be metal layers.

**Description**

When a min area type routing guide is created, all input layers must be metal layers.

**What Next**

Provide all metal layer input.

---

## NDMUI-370

(error) '%s' option not specified and no default %s to use.

**Description**

This command requires a specific kind of object to operate on. However, neither an option specifying the object is provided nor a default object exists.

**What Next**

Either provide an object explicitly, or make sure that a default object exists.

Feedback

## NDMUI-371

(error) A track pattern for %s '%s' and layer '%s' already exists.

**Description**

A track pattern with the specified site_def/row_pattern and layer already exists.

**What Next**

Please provide the correct inputs. To change an existing track pattern, please remove the existing one using the 'remove_track_patterns' command first, then create new one.

## NDMUI-372

(warning) The direction '%s' specified by '%s' option is different from the preferred routing direction '%s' of the layer '%s'.

**Description**

The specified direction is different from the preferred routing direction of the layer. The tool will use the specified direction.

**What Next**

Please check the direction specified is correct.

## NDMUI-373

(error) '%s' option not specified and preferred routing direction '%s' of the layer '%s' is not supported.

**Description**

This command requires a direction input. However, neither the direction explicitly provided nor a valid preferred routing direction for this command exists.

**What Next**

Either provide a direction explicitly, or make sure that the preferred routing direction is valid for this command.

## NDMUI-374

(error) Cannot specify option '%s' when %s.

Feedback

**Description**

Cannot specify the option in given condition.

**What Next**

Please provide correct input.

---

## NDMUI-375

(error) '%s' not specified and pitch %s of the layer '%s' is not valid.

**Description**

This command requires a spacing input. However, neither an option explicitly provided, nor the pitch value of the layer is valid.

**What Next**

Please provide a spacing input, or make sure that the pitch of the layer is valid.

---

## NDMUI-376

(error) Must specify option '%s' when %s.

**Description**

The option is a mandatory option in given condition.

**What Next**

Please provide correct input.

---

## NDMUI-377

(error) Width %s specified for '%s' option is less than the minimum width %s of the layer '%s'.

**Description**

The width value specified is invalid, because it is less than the layer minimum width.

**What Next**

Change the indicated width value, so that it is at least the layer minimum width.

---

## NDMUI-378

(error) '%s' option should have size %s when %s.

Feedback

**Description**

The specified list option should have specified size, or same size as another option in given condition.

**What Next**

Provide correct input.

---

## NDMUI-379

(warning) '%s' option not specified. Using %s %s.

**Description**

You receive this warning because an option is not provided. The tool uses the indicated value.

**What Next**

No action is required on your part. You can see the man page for the command that caused the warning for detailed information on specifying optional and required parameters.

---

## NDMUI-380

(error) All values in '%s' must be %s than %s.

**Description**

This option requires values with indicated constraint.

**What Next**

Provide the correct value to the option.

---

## NDMUI-381

(error) More than one module can't be specified with '-instance_names' option.

**Description**

Generally many modules can be created simultaneously. But when instances are created, they can be created for one module at a time.

**What Next**

Please specify a single module whose instances are to be created.

Feedback

## NDMUI-382

(error) No valid '%s' commands in specified file '%s'.

**Description**

The specified file must contain at least one valid command as indicated.

**What Next**

Check the specified file to make sure it has valid content.

## NDMUI-383

(error) Exactly one net must be specified for topology_edges and topology_nodes.

**Description**

This command accepts exactly one net to associate with the specified topology object.

**What Next**

Please specify one net.

## NDMUI-384

(error) No more than one object may be specified for a topology_edge endpoint.

**Description**

Only one object of type topology_node may be specified as a topology_edge endpoint.

**What Next**

Please specify one endpoint object.

## NDMUI-385

(error) Invalid object type specified for a topology_edge endpoint.

**Description**

Only one object of type topology_node may be specified as a topology_edge endpoint.

**What Next**

Please specify an endpoint object of type topology_node.

Feedback

## NDMUI-386

(error) unable to determine PST-equivalence for %s supply nets of power_domains '%s' and '%s' before power intent has been committed successfully.

### Description

When associating multiple power domains to a voltage area, the primary supplies of power domains should be equivalent. However, until the power intent has been finalized and committed successfully, the tool cannot determine the supply nets are equivalent based on the power state tables.

### What Next

Invoke 'commit_upf' command and make sure power intent is committed successfully.

### See Also

- commit_upf

## NDMUI-387

(error) Cannot set both topology_edge endpoints to the same object.

### Description

Topology_edge endpoints must be distinct.

### What Next

Please specify different values for topology_edge endpoints.

## NDMUI-388

(Error) -library option is not supported in library shell.

### Description

-library option is not supported in library shell.

### What Next

Remove -library option and the operation will be done on current open library.

## NDMUI-389

(warning) List of cells missing antenna property.

Feedback

**Description**

This message is for reporting missing antenna property for cells in physical library. All the cells that are missing antenna property will be reported in this table. In this table, it lists cell names and pin names missing the antenna property and the missing property type. If an input pin is missing the gate size, it is counted as missing the property. If an output pin is missing diode protection, it is counted as missing the property. If a macro is missing the hierarchical antenna property, then it is counted as missing the property. It is considered good practice to specify hierarchical antenna properties for macros.

**What Next**

Please check and load the antenna properties within workspace.

---

## NDMUI-390

(error) Translation was not completed successfully.

**Description**

The tool has failed to translate the given drc error file.

**What Next**

Please check to make sure that the input file matches the format that was given by the -drc_type option.

---

## NDMUI-391

(error) Failed to create error cell from drc error file.

**Description**

The tool encountered an error while trying to create an error cell from the given drc error file.

**What Next**

Please check to make sure that the input file matches the format that was given by the -drc_type option.

---

## NDMUI-392

(error) The specified drc type is not supported.

**Description**

The value specified for -drc_type is not valid. Valid values are: cal- ibre

Feedback

**What Next**

Rerun the command with a valid drc type.

---

## NDMUI-393

(error) Specified view name not supported.

**Description**

The specified error cell name contains a '.' character and does not end with ".err".

**What Next**

Rerun the command without specifying a view for the error cell.

---

## NDMUI-394

(error) '.' cannot be the first character in the error cell. name.

**Description**

The specified error cell name is invalid, it does not contain a design name.

**What Next**

Rerun the command with a valid error cell name.

---

## NDMUI-395

(error) Value for option -mask_contact_selections is invalid; %s.

**Description**

This option requires a list in the format {{layerName maskConstraint {contactCodeName ...}} ...}.

The contactCodeName must correspond to a ContactCode in technology file.

**What Next**

Provide the correct value to the option.

---

## NDMUI-396

(warning) Options -mask_contact_selections and -port_contact_selections are both set. The port_contact_selections setting will have higher priority when both setting are applicable for a terminal.

Feedback

**Description**

When the setting of -mask_contact_selections and -port_contact_selections are both applicable for creating via region for a terminal, tool will use the setting of -mask_contact_selections to create via region.

**What Next**

Check if the settings are correct.

---

## NDMUI-397

(warning) Exactly two pins must be specified for each cell. Skipping cell '%s'.

**Description**

The "-pins" option for command "set_supernet_exceptions" accepts a collection of pins, however, the collection must have exactly two pins for each cell the pins belong to. A pair of in & out pins must be identified for each cell for the command to work correctly for that cell. The cell with 1 or more than 2 pins is skipped.

**What Next**

Please provide the correct input. If you're using a collection, make sure the collection has exactly two pins for each cell.

---

## NDMUI-398

(warning) One input and one output pin must be specified for each cell. Skipping cell '%s'.

**Description**

The "-pins" option for command "set_supernet_exceptions" accepts a collection of pins, however, the collection must have exactly two pins for each cell the pins belong to. A pair of in & out pins must be identified for each cell for the command to work correctly for that cell. Pins with direction "inout" are not accepted.

**What Next**

Please provide the correct input. If you're using a collection, make sure the collection has exactly two pins for each cell.

---

## NDMUI-399

(error) The specified number of %s (%d) exceeds the supported maximum (%d).

Feedback

**Description**

The specified number of rows or columns exceeds the maximum supported by the via array. In each dimension, the number of cuts must be less than the supported maximum.

**What Next**

Specify the number of rows and columns, such that each is less than the supported maximum.

---

## NDMUI-400

(error) Cannot create shape with shape_use other than area_fill, in a fill cell.

**Description**

Fill cells can only have fill shapes, which means the shape_use, if specified must be area_fill. If shape is being created in a fill cell and shape_use is not specified, the shape_use will be set as area_fill.

**What Next**

Please provide the correct input. There's no need to specify shape_use if shape is being created in a fill cell

---

## NDMUI-401

(error) Cannot find a fill cell to create fill shape in.

**Description**

Fill shapes will be created in fill cells. No fill cell was specified and the top level fill cell does not exist in the design.

**What Next**

Please provide the correct input. Please specify the fill cell to create the shape in or create fill cells first.

---

## NDMUI-402

(error) Unable to uniquify the specified fill cell '%s'.

**Description**

The reference of the specified fill cell is multiply instantiated but it cannot be uniquified. Therefore, a new shape cannot be added to it. This can happen if the reference for the fill cell is a external reference.

Feedback

**What Next**

Please provide the correct input. If possible, uniquify the reference and run the command again.

---

## NDMUI-403

(error) Layer name specified is invalid.

**Description**

The layer name provided as input cannot be found in the associated tech.

**What Next**

Please provide the correct input and run the command again.

---

## NDMUI-404

(Error) Library has no associated tech.

**Description**

This command needs a library with a valid tech to run correctly. The specified library does not have a associated tech.

**What Next**

Please provide the correct input and run the command again.

---

## NDMUI-405

(Error) A tech object is not available.

**Description**

This command needs a valid tech object to run correctly. No tech object is available at this point.

**What Next**

Please load the tech object and run the command again.

---

## NDMUI-406

(warning) %s.frame does not exist.

Feedback

**Description**

This message is for missing frame cells in physical library when reporting pin information.

**What Next**

Please check the designs in the library.

---

## NDMUI-407

(warning) Failed to set property '%s' for DensityRule

**Description**

The specified property could not be set for the density rule because it does not meet the requirements

**What Next**

Please provide correct value for the property

---

## NDMUI-408

(error) Duplicate via-def '%s' can't override a via-def of same name that already has %u via usages.

**Description**

A via-def having via usages can't be overridden.

**What Next**

Please provide a different name for the via-def.

---

## NDMUI-409

(error) Only one '%s' can be specified at a time.

**Description**

This command can only process one object at a time.

**What Next**

Please provide one input at a time.

---

## NDMUI-410

(error) input bus name must be in bus_name[start:end] format.

Feedback

**Description**

Input bus name must be in bus_name[start:end] format.

**What Next**

Fix input bus name and try again.

---

## NDMUI-411

(error) bus members '%s' do not exist.

**Description**

All bus members must exist without -create_ports option.

**What Next**

Create port bus members before running this command or use -create_ports option.

---

## NDMUI-412

(error) cannot create port bus '%s'. Port bus name is already taken.

**Description**

Input bus port name is taken.

**What Next**

Use a different name or remove existing port or port bus.

---

## NDMUI-413

(error) Options -power or -ground is not allowed for primary Voltage Area.

**Description**

Options -power or -ground can't be specified for primary Voltage Area.

**What Next**

Please don't give options -power or -ground for primary Voltage Area.

---

## NDMUI-414

(error) Option -name is not allowed for primary Voltage Area with single power domain.

Feedback

**Description**

In case of single power domain, option -name can't be specified for primary Voltage Area.

**What Next**

Please don't specify option -name for primary Voltage Area with single power domain.

---

## NDMUI-415

(error) Options -guard_band or -target_utilization is not allowed for gas-station Voltage Area.

**Description**

Options -guard_band or -target_utilization can't be specified for gas-station Voltage Area.

**What Next**

Please don't specify options -guard_band or -target_utilization for gas-station Voltage Area.

---

## NDMUI-416

(error) Options -power or -ground is required for gas-station Voltage Area.

**Description**

Options -power or -ground should be specified for gas-station Voltage Area.

**What Next**

Please do specify options -power or -ground for gas-station Voltage Area.

---

## NDMUI-417

(error) Options -region is required for gas-station Voltage Area.

**Description**

Options -region should be specified for gas-station Voltage Area.

**What Next**

Please do specify options -region for gas-station Voltage Area.

Feedback

## NDMUI-418

(error) Replacement of '%s' via_def '%s' by a duplicate '%s' via-def of same name is not supported.

**Description**

Replacement via-def must be of the same type as the original via_def.

**What Next**

Please provide a different name for the via-def or create a via_def of different type.

## NDMUI-419

(error) %s '%s' cannot be assigned to terminal '%s'; it is currently assigned to terminal '%s'.

**Description**

The specified shape or via is owned by another terminal as indicated in the message. Each shape or via may belong to at most one terminal.

The command exited without modifying any data.

**What Next**

Remove the shape from its current terminal using *remove_terminals*, then call this command again.

## NDMUI-420

(error) Port bus name '%s' is invalid. Name must be the concatenation of the bus name '%s' and bus delimiter enclosed index.

**Description**

The specified port is a bus, so its name must adhere to the bus port naming convention. Port bus names must be the concatenation of the bus name, opening bus delimiter, bus index, and closing bus delimiter. The bus delimiters are specified by the app_option, design.bus_delimiters.

**What Next**

Provide a valid name for the bus port.

Feedback

## NDMUI-421

(warning) %s object '%s' is in input %s list. It is ignored.

**Description**

There is a net object in input pin/port list or there is a pin/port object in input net list. It is ignored when searching for all connected.

**What Next**

Provide input object list containing all nets or all pins/ports.

## NDMUI-422

(info) No internal database errors were detected in module '%s'.

**Description**

This message indicates that the data-model integrity checker didn't find any internal database errors for the reported module.

## NDMUI-423

(error) The database has internal errors. You must revert to an earlier version of this database to proceed. Please contact your Synopsys representative to report this issue.

**Description**

This message indicates that the data-model integrity checker has found internal database errors.

**What Next**

It is recommended that you revert to an earlier version of this database to proceed.

## NDMUI-424

(warning) Removal of %s '%s' invalidates voltage_area_routing_rule '%s'. The rule has been removed.

**Description**

The removed voltage_area or voltage_area_shape was associated with the specified voltage_area_routing_rule. After the object was remove, the voltage_area_routing_rule was left in an invalid state (i.e., the rule was left without any starting, ending, or pass-through objects). So the rule was removed from the design.

Feedback

### What Next

Undo the remove command if you did not intend to remove these objects.

---

## NDMUI-425

(error) %s '%s' is an invalid %s object of the voltage_area_routing_rule; %s.

### Description

The starting and ending objects of a voltage_area_routing_rule must be a voltage_area, macro cell, pad cell, or macro cell pin. No other object types are allowed.

### What Next

Check the object types that are being set as the starting objects of the voltage_area_routing_rule. Remove any objects from your list which are not an allowable type.

---

## NDMUI-426

(error) Cannot append to existing rule '%s', because the specified start and end objects do not match the existing rule's start and end objects.

### Description

When a rule already exists in the design with the given name, the create_voltage_area_routing_rule command appends the specified options onto the existing rule if and only if the start and end objects matching the existing rule's start and end objects.

### What Next

Specified another rule name when calling the create_voltage_area_routing_rule command. Or if you want to append to the existing rule, specify the same start and end objects as the existing rule's start and end objects in the -from and -to command options.

---

## NDMUI-427

(warning) %s '%s' does not belong to current block. It is ignored.

### Description

This message indicates that input object does not belong to current block. It is ignored.

### What Next

Please only input object belonging to current block.

Feedback

## NDMUI-428

(warning) %s '%s' does not have corresponding physical net. It is ignored.

**Description**

This message indicates that input net does not have corresponding physical net. It is ignored.

**What Next**

Please only input net with corresponding physical net.

## NDMUI-429

(error) one of -shapes and -nets options is required for this command.

**Description**

This message indicates that none of -shapes and -nets options is specified.

**What Next**

Please input at least one of -shapes and -nets options.

## NDMUI-430

(error) shielding shape '%s' must be PG related.

**Description**

This message indicates that input shielding shape is not PG related.

**What Next**

Please input only shielding shape that is PG related.

## NDMUI-431

(error) shielded net '%s' must not be PG net.

**Description**

This message indicates that input shielded net is PG net.

**What Next**

Please input only net that is not PG net.

Feedback

## NDMUI-432

(error) shielding shape '%s' shape_use must be route_shield and physical_status must be minor_change.

**Description**

This message indicates that input shielding shape shape_use is not route_shield or physical_status is not minor_change.

**What Next**

Please input only appropriate shapes.

## NDMUI-433

(error) Command '%s' is not supported on %s view.

**Description**

This message indicates that specified command is unsupported on the reported view.

**What Next**

Please use different view for the specified command.

## NDMUI-434

(warning) Name %s is reserved for auto generated names. Ignored -name.

**Description**

This message indicates that given name is reserved for auto generated name for unnamed blockages. -name is ignored so system will generate the name.

**What Next**

Provide the name which is not reserved for auto generated names.

## NDMUI-435

(error) The specified supply_net '%s' is not available on sub-block '%s'.

**Description**

This error shows up when creating a voltage_area in a sub-block through the -cell option, when the specified supply_net exists only on the top-level block but not on the sub-block.

Feedback

### What Next

Verify the the supply_net is available on the sub-block.

---

## NDMUI-436

(error) Cell '%s' cannot be assigned to the voltage_area; it is in a different block than the voltage_area

### Description

The specified object is in a different block than the voltage_area. The voltage_area and its containing cells must be in the same physical block.

### What Next

Specify cells that are in the same block as the voltage_area.

---

## NDMUI-437

(Error) Unable to assign routing_rule '%s' to net '%s' because a different routing_rule with the name '%s' already exists in the net's block '%s'.

### Description

The specified routing_rule and net are in different blocks. In order to assign the routing_rule to the ent, the routing_rule and net must be in the same block. The command attempted to copy the routing_rule into the block of the net. However because a routing_rule already exists with the same name but different rules in the net's block, the copy was unsuccessful. The routing_rule is not assigned to the net.

### What Next

Make sure the routing_rule definitions are consistent in each block and call this command again.

---

## NDMUI-438

(Error) The specified reference design is of type 'tsv' but is not a valid TSV reference design: %s

### Description

A design of type 'tsv' was specified as the cell reference in the command, however the design fails to meet one or more requirements for use as a TSV reference design. Please see the *create_cell* man page's documentation on its *reference* argument for more information on TSV reference design criteria.

Feedback

### What Next

Please correct the design or use a different, valid TSV design as the reference.

---

## NDMUI-439

(Error) application option <%s> is obsolete. See the man page for this app_option for migration assistance.

### Description

The specific application option has been made obsolete and hence the user script needs to be migrated. The man page of the particular application option will have details about the required migration activity.

### What Next

See the man page for this application option for migration assistance.

---

## NDMUI-440

(error) Cannot change_link for cell '%s' because the port interface of design '%s' does not match port interface of current lib_cell '%s'.

### Description

The *change_link* command was unable to change the lib_cell bound to the specified cell. The order in which ports occur in the new lib_cell must be identical to the order of ports in the lib_cell the cell is currently bound to.

### What Next

Try using a different design when change_link for this cell.

---

## NDMUI-441

(warning) application option <%s> is deprecated and scheduled for removal in a future release. See the man page for this application option for migration assistance.

### Description

The specific application option has been made deprecated and will hence be removed in a future release. The man page of the particular application option will have details about the required migration activity.

### What Next

See the man page for this application option for migration assistance.

Feedback

## NDMUI-442

(Error) The application option <%s> is obsolete.

**Description**

The specific application option has been made obsolete and hence the user script needs to be migrated to use alternative application option.

**What Next**

This application option has been made obsolete. The user script will have to be replaced with the alternative application option.

## NDMUI-443

(warning) The application option <%s> is deprecated and scheduled for removal in a future release.

**Description**

The specific application option has been made deprecated and will hence be removed in a future release. Hence the user script needs to be migrated to use alternative application option.

**What Next**

This application option has been made deprecated. The user script will have to be migrated to use the alternative application option.

## NDMUI-444

(error) The %s of the default voltage_area cannot be changed.

**Description**

The default voltage_area (DEFAULT_VA) is auto-generated. Its name or supply nets cannot be changed.

## NDMUI-445

(error) Option '-class' can't be specified when all input objects are collections.

**Description**

When objects specified for get_attribute and set_attribute are collections only then -class option can't be used. Objects in the collections are already of a certain class type and so using -class option on top of it is improper.

Feedback

### What Next

Don't use -class option when input objects to get_attribute and set_attribute are through collections. Use -class option only when input object is specified as string that can't be implicitly found.

---

## NDMUI-446

(error) Option '%s' cannot be specified when creating a single-cut via.

### Description

The indicated option is allowed only on array vias. It cannot be specified if the via has only one cut.

### What Next

Remove the indicated option or specify more than one cut in the *-size* option.

---

## NDMUI-447

(error) Pattern cannot be an empty string.

### Description

The pattern cannot be an empty string. The pattern must be a space delimited string of equal length substrings (e.g., "1010 0101 1011" is a 3x4 pattern). Each substring must consist of only 1s and 0s, representing a row in the MxN pattern, starting with the bottom row in the pattern.

### What Next

Specify a non-empty pattern string.

---

## NDMUI-448

(error) Substrings in pattern '%s' do not have equal lengths.

### Description

The pattern substrings must have equal lengths. The substring length is equal to the number of columns (i.e., N) in the MxN pattern.

The pattern is a space delimited string of equal length substrings (e.g., "1010 0101 1011" is a 3x4 pattern). Each substring must consist of only 1s and 0s, representing a row in the MxN pattern, starting with the bottom row in the pattern.

Feedback

**What Next**

Correct the substrings in the pattern so that they all have the same length.

---

## NDMUI-449

(error) Pattern substring '%s' contains invalid characters.

**Description**

The pattern substring must contain only 1s and 0s. No other characters are allowed.

The pattern is a space delimited string of equal length substrings (e.g., "1010 0101 1011" is a 3x4 pattern). Each substring must consist of only 1s and 0s, representing a row in the MxN pattern, starting with the bottom row in the pattern.

**What Next**

Correct the specified pattern substring so that it contains only 1s and 0s.

---

## NDMUI-450

(error) Option '%s' cannot be used if option '%s' is specified with a non-zero value.

**Description**

Via distance thresholds cannot be used with taper rules. All taper rule options must be unspecified or set to zero.

**What Next**

Remove the taper option or set the taper option value to zero.

---

## NDMUI-451

(error) Layer '%s' is not defined with the rectangle only rule in the technology file and therefore cannot be used in option '%s'.

**Description**

The via distance thresholds can be specified only on layers which are restricted to rectangle only shapes. These layers are defined in the technology file with the 'hasRectangleOnly = 1' statement.

**What Next**

Remove the layer from the *-via_distance_thresholds* option or update the technology to include the rectangle only rule on the layer.

Feedback

## NDMUI-452

(error) Via pattern '%s' cannot be used with the specified via distance thresholds; %s.

**Description**

The via pattern cannot be used with the via distance thresholds specified through the *-via_distance_thresholds* option. When the *-via_distance_thresholds* option is used, the via pattern must meet these requirements:

1) The via_def name must correspond to a simple via_def.

2) If either of the via_def enclosure layers are assigned a via distance threshold, the via_def must be symmetric.

3) If the via_def lower enclosure layer is assigned a via distance threshold, the layer must be defined with 'hasRectangleOnly = 1' in the technology file, and the lower enclosure must be smaller than or equal to the routing_rule width.

4) If the via_def upper enclosure layer is assigned a via distance threshold, the layer must be defined with 'hasRectangleOnly = 1' in the technology file, and the upper enclosure must be smaller than or equal to the routing_rule width.

**What Next**

Remove the via pattern from the *-vias* option, or update the options to meet the above requirements.

## NDMUI-453

(warning) Name %s is reserved for auto generated names. Ignored -name.

**Description**

This message indicates that given name is reserved for auto generated name for unnamed groups. -name is ignored so system will generate the name.

**What Next**

Provide the name which is not reserved for auto generated names.

## NDMUI-454

(error) Group '%s' already exists in the design.

**Description**

A group with the specified name already exists in the design.

Feedback

**What Next**

Specify a different group name, or use the remove_groups command to remove the existing group. Alternatively either you may omit the name option, and the system will automatically select a unique name for the group or remove the existing group and create both the group with -allow_duplicate_name option.

---

## NDMUI-455

(warning) %s can not be used without %s. Ignored %s.

**Description**

This message indicates that specified option must be used with other missing option. Specified option is ignored.

**What Next**

Specify missing option along with the specified option.

---

## NDMUI-456

(error) Group %s created successfully but error occurred during member addition.

**Description**

This message indicates that group is created but error occued during object addition.

**What Next**

Multiple time addition of same object in a "set" type group is not allowed. Check if any object is being added multiple times.

---

## NDMUI-457

(error) Member %s failed in group %s.

**Description**

This message indicates that error occued during member addition or removal.

**What Next**

Multiple time addition of same object in a "set" type group is not allowed. Check if any object is being added multiple times. Check if the member belong to the same design as of group. Make sure that member addition does not form any cycle. Removal of non-member object from a group is not allowed. Make sure that object being removed is part of the group.

Feedback

## NDMUI-458

(error) Cannot create new shape in arrayed fill cell '%s'.

**Description**

The specified fill cell is arrayed. New shapes cannot be created in arrayed fill cells.

**What Next**

Please provide the correct input.

## NDMUI-459

(error) The minimum and maximum routing layer cannot be the same layer '%s'.

**Description**

The minimum and maximum routing layers must be set to two different routing layers.

**What Next**

Correct the layer specified to the *-min_routing_layer* or *-max_routing_layer* option.

## NDMUI-460

(error) Cannot specify mask value to the shape of layer '%s', because the layer number_of_masks attribute value is '%d'.

**Description**

The mask value can be specified only when the layer's number_of_masks attribute is 2 or greater.

**What Next**

Correct the layer's number_of_masks attribute value so it is at least 2.

## NDMUI-461

(error) The application option <%s> is for R&D debug and can't be used outside Synopsys.

**Description**

The specific application option is for R&D debug. It cannot be used outside Synopsys and will not be retrieved or set.

Feedback

### What Next

The user script should be modified to remove the internal app option.

---

## NDMUI-462

(Warning) The given value for '%s' is not being set for hard macro '%s'.

### Description

This error message occurs when the *add_to_rp_group* command detects that you tried to set an option or attribute which is not supported on hard macro cell. Currently pin alignment and cell orientation features are not supported for hard macros cells.

### See Also

- add_to_rp_group
- create_rp_group
- remove_from_rp_group

---

## NDMUI-463

(error) Options -net_types or -zero_spacing or -blockage_group_id or -auto_blockage_group_id are not allowed for block boundary blockages.

### Description

Options -net_types or -zero_spacing or -blockage_group_id or -auto_blockage_group_id can't be specified for block boundary blockages, as indicated with use of -boundary_internal or -boundary_external options.

### What Next

Please don't specify options -net_types or -zero_spacing or -blockage_group_id or -auto_blockage_group_id for block boundary blockages.

---

## NDMUI-464

(error) Source view '%s' is not same as specified destination view '%s'.

### Description

The source block view and destination block view must match while copying, moving or renaming a block. Any particular view can't be changed to any other view as content wise it will be incorrect.

Feedback

**What Next**

Maintain the same view for both source and destination blocks.

---

## NDMUI-465

(error) Same port can not be %s more than one port bus.

**Description**

Each port bus contains unique ports. Same port can not be added to more than one port bus.

**What Next**

Maintain the uniqueness of ports.

---

## NDMUI-466

(error) Multiple ports can not be %s the port bus at a time.

**Description**

Port bus updation is allowed one by one.

**What Next**

Remove the multiple ports and do it sequentially.

---

## NDMUI-467

(error) Port can be added to parent module's port bus only.

**Description**

Port's and port bus are in different module.

**What Next**

Port's and port bus should be in same module.

---

## NDMUI-468

(error) Port bus does not have the port to be removed.

**Description**

Only existing port can be removed from the port bus.

Feedback

**What Next**

Make sure that port to be removed is part of the port bus.

---

## NDMUI-469

(error) %s cannot be less than the corresponding minimum cut spacings %s for x_pitch and %s for y_pitch.

**Description**

A base pitch cannot be less than the corresponding minimum cut spacings.

**What Next**

Please provide the correct base pitch not less than the corresponding via_def minimum cut spacings.

---

## NDMUI-470

(error) The base_x_pitch and base_y_pitch values of '%s' cannot be less than the corresponding minimum cut spacings of %s and %s on the cut layer specified by the given via_def.

**Description**

A via_def has been specified on which the cut layer has a minimum cut spacing that is larger than the cut spacing specified for the cuts of this array via.

**What Next**

Check the cut spacing of this array via and provide a value greater than or equal to the minimum cut spacing of the cut layer specified in the via_def.

---

## NDMUI-471

(error) Value '%s' for %s option. The mask_constraints is invalid; %s.

**Description**

This option requires a list in the format {{row column mask_name [fixed]} ...}.

The mask_name must be mask_one, mask_two, or same_mask for upper/lower masks, in addition to these mask names, the cut masks can be no_mask, mask_three, mask_four, mask_five, mask_six, mask_seven, mask_eight, mask_nine, mask_ten, mask_eleven, mask_twelve, mask_thirteen, mask_fourteen, mask_fifteen. If the mask is fixed, then the keyword, fixed, must be specified. If the mask is not fixed, do not specify the keyword.

Feedback

**What Next**

Provide the correct value to the option.

---

## NDMUI-472

(error) Port is already a member of port bus %s.

**Description**

Port is already a member of the port bus.

**What Next**

Verify that you specified the correct port and port bus.

---

## NDMUI-473

(error) %s.

**Description**

Log file for individual commands within parallel_execute is not specified correctly. If -commands_only option is used, then no log file should be specified for individual commands. Similarly a log file must be specified when not using -commands_only. Also log file for an individual command can't be multiple strings.

**What Next**

Verify that if a log file is specified, when not using -commands_only, then it is a single string. If -commands_only option is used, then log file must not be used.

---

## NDMUI-474

(error) %s.

**Description**

Individual command specified within parallel_execute is either unknown or not a read-only command or parallel_execute itself.

**What Next**

Verify that each command specified within parallel_execute is a valid ICC2 command and also a read-only command like, for example, reporting commands. Any command that modifies design data like, for example, create commands are not allowed to be executed within parallel_execute. Also nested parallel_execute is not allowed.

Feedback

## NDMUI-475

(warning) The design '%s' is read only.

**Description**

*create_terminal* command needs to modify the library %s. If the design is read only, *create_terminal* command cannot be completed.

**What Next**

The design may be a read only lib cell. Otherwise, the write permission of the design is needed to be enabled.

## NDMUI-476

(warning) The shape '%s' is not a net shape.

**Description**

*create_terminal* command needs the layer shapes and vias belonging to nets to be duplicated as terminals of the port.

**What Next**

The command create_terminal needs net shapes to be provided as input for the "-of_objects" option.

## NDMUI-477

(warning) The option '%s' is ignored when specified with option '%s'.

**Description**

*create_terminal* command ignores the specified option when this is used with the option "-of_objects".

**What Next**

The above option is ignored when specified with "-of_objects".

## NDMUI-478

(warning) The shape '%s' already belongs to a terminal.

Feedback

### Description

*create_terminal* command needs the layer shapes and vias belonging to nets to be duplicated as terminals of the port. The input shapes already is associated with a terminal. The shape will be ignored.

### What Next

The command create_terminal needs net shapes to be provided as input for the "-of_objects" option.

---

## NDMUI-479

(warning) The shape '%s' is physically overlapping with multiple ports. The shape will be ignored.

### Description

*create_terminal* command needs the layer shapes and vias belonging to nets to be not physically overlapping with multiple ports. These shapes cannot be used for further processing and will be ignored.

### What Next

The command create_terminal needs net shapes to be physically overlapping with atmost one port.

---

## NDMUI-480

(error) Cell '%s' is bound to non-design view '%s' which, if modified, can lead to inconsistencies.

### Description

The design view is primarily the source of all other view derivation and for example, design view itself can't be extracted from say, abstract or frame view. If a non-design view is modified by *change_names* then it will cause this non-design view to be out of sync with its design view leading to issues like inconsistency in design data with its associated timing constraints. On the other hand if a design view is modified then other views can be derived from this modified design view and all views will be in sync.

### What Next

This situation can happen only when *change_names* is applied hierarchically across multiple physical hierarchy including sub-blocks. Make sure that only design view is used for sub-blocks to link the top block before applying *change_names* or re-link with design views in order to consistently rename all the design data and associated timing constraints.

Feedback

## NDMUI-481

(error) Same net can not be %s more than one net bus.

**Description**

Each net bus contains unique nets. Same net can not be added to more than one net bus.

**What Next**

Maintain the uniqueness of nets.

## NDMUI-482

(error) Multiple nets can not be %s the net bus at a time.

**Description**

Port bus updation is allowed one by one.

**What Next**

Remove the multiple nets and do it sequentially.

## NDMUI-483

(error) Net can be added to parent module's net bus only.

**Description**

Net and net bus are in different module.

**What Next**

Net and net bus should be in same module.

## NDMUI-484

(error) Net bus does not have the net to be removed.

**Description**

Only existing net can be removed from the net bus.

**What Next**

Make sure that net to be removed is part of the net bus.

Feedback

## NDMUI-485

(error) Net is already a member of net bus %s.

**Description**

Net is already a member of the net bus.

**What Next**

Verify that you specified the correct net and net bus.

## NDMUI-486

(error) bus members '%s' do not exist.

**Description**

All bus members must exist without -create_nets option.

**What Next**

Create net bus members before running this command or use -create_nets option.

## NDMUI-487

(error) cannot create net bus '%s'. Net bus name is already taken.

**Description**

Input bus net name is taken.

**What Next**

Use a different name or remove existing net bus.

## NDMUI-488

(error) Heterogeneous bundles are not allowed. Bundle components should be of same type.

**Description**

Bundles should be homogeneous. They can only consist of same type of objects but not a mix of different types.

**What Next**

Make sure that the object list consists of only one type of objects.

Feedback

## NDMUI-489

(warning) Skipping bundle '%s' consisting of supernets which is not supported for some applications.

### Description

Currently some applications don't support bundles of supernets. Till the time the applications are ready to support bundles of supernets, such bundles can't be used in those applications.

### What Next

Consider creating a bundle of nets that are part of the supernets.

## NDMUI-490

(warning) Ignoring the -base_pitch option because the base via is a single cut via.

### Description

When the base via is defined as single cut, the value specified by -base_pitch doesn't take effect.

### What Next

Delete the "-base_pitch" option to create a via matrix with a single cut base via or Consider creating a via matrix with multiple cut base via.

### See Also

- create_via_matrix

## NDMUI-491

(information) No net shape or via geomtries is added as a terminal shape.

### Description

The command 'create_terminal' didnot add any net shape or via as terminals. This may be because there is no physical overlap between the input net shapes and the existing terminal shapes.

## NDMUI-492

(error) The specified net %s isn't power/ground. A %s can connect with a power/ground net only.

Feedback

**Description**

Only a power/ground net can connect with the reported object_class.

**What Next**

Specify a power/ground net to connect with the reported object_class.

---

## NDMUI-493

(information) The specified %s mask pattern is uniform, only the responsible mask constraint of lower left base via can be honored.

**Description**

The specified mask_pattern is uniform, only the responsible mask constraint of lower left base via can be honored. specify the mask pattern to arbitrary to honor every mask constraint.

---

## NDMUI-494

(warning) Cannot delete auto-generated port terminal '%s'; its shape is owned by a pad cell.

**Description**

The terminal was auto-generated by the application, because its port is connected to a pad pin. The terminal attribute, *is_owned_by_pad*, is true. Its shape is really owned by the pad pin. The terminal cannot be removed by this command.

**What Next**

This terminal cannot be directly removed. However if you manually create a terminal on the port or disconnect the port from the pad pin, this terminal will be removed automatically.

---

## NDMUI-495

(warning) ignore option “-base_pitch”, because the base via is defined as single cut one.

**Description**

When the base via is defined as single cut, the value specified by -base_pitch doesn't take effect.

Feedback

**What Next**

Delete the "-base_pitch" option to create a via matrix with a single cut base via or Consider creating a via matrix with multiploe cut base via.

---

## NDMUI-496

(warning) The mask_pattern is set to uniform. Only mask constrain in lower left base via is honored. Others will be omitted.

**Description**

Mask constrains other than lower left base via will be honored only when the mask_pattern isn't uniform.

**What Next**

Consider mask pattern other than uniform to honor all maks constrains.

---

## NDMUI-497

(error) Rule type '%s' cannot be set on %s '%s' because %s.

**Description**

When setting the rule_type attribute to "net_non_default", the object must be owned by a net and the net must be constrained by a routing rule.

**What Next**

Assign the object to a net or assign a routing rule to the net as necessary. Then set the rule_type attribute.

---

## NDMUI-498

(error) Voltage area '%s' is not in the same block '%s' as the voltage area rule being created.

**Description**

The specified voltage area is not in the same block as the voltage area rule. The voltage area and its assigned rule must be in the same block.

**What Next**

Create the voltage area rule in the same block as the voltage area.

Feedback

## NDMUI-499

(error) Failed to create %s voltage area rule. Design contains %s rules created through the %s command. Design cannot contain both new and deprecated rules.

### Description

Voltage area rules may be created with either the new *create_voltage_area_rule* or deprecated *create_voltage_area_routing_rule* command, but not both. The design cannot contain both new and deprecated rules at the same time.

### What Next

Remove all existing voltage area rules from the design, and then create the voltage area rules.

## NDMUI-500

(error) Failed to create the default rule, because %s.

### Description

The command failed to create the default voltage area rule for the indicated reason. Only one default rule may exist in each design. The tool automatically names the default rule as "DEFAULT_RULE".

### What Next

If the default rule already exists and you want to modify it, use the *set_attribute* command to modify its attributes.

## NDMUI-501

(error) Physical feedthrough cannot be allowed when pass through is disallowed.

### Description

Physical feedthrough cannot be allowed when pass through is disallowed. If you want to allow physical feedthrough, you must also allow pass through.

Only these option combinations are supported:

```
-allow_pass_through false -allow_physical_feedthrough false
-allow_pass_through true -allow_physical_feedthrough false
-allow_pass_through true -allow_physical_feedthrough true
```

### What Next

Change the *-allow_pass_through* and *-allow_physical_feedthrough* option values to one of the supported combinations.

---

## NDMUI-502

(error) Cannot update existing voltage area rule '%s', because the specified -allow_pass_through, -allow_physical_feedthrough, and -allow_logical_feedthrough option values do not match the existing rule.

### Description

When a rule already exists in the design with the given name, the create_voltage_area_rule command adds the specified voltage areas to the existing rule if and only if the constraints match.

### What Next

Specified another rule name when calling the create_voltage_area_rule command.
Or if you want to append to the existing rule, specify the same *-allow_pass_through*, *-allow_physical_feedthrough*, and *-allow_logical_feedthrough* option values as the existing rule.

---

## NDMUI-503

(warning) Removing voltage area "%s" from currently assigned voltage area rule "%s". It will be constrained by the new rule.

### Description

A voltage area may be assigned to at most one voltage area rule. The voltage area has been removed from its existing rule and reassigned to the new rule.

### What Next

Verify that you intended to reassign the voltage area to the new rule.

---

## NDMUI-504

(warning) Removing voltage area rule "%s" because it is no longer constraining any voltage area.

### Description

Voltage area rules cannot exist in the design without any voltage areas to constrain. When creating a new rule on voltage areas, the voltage areas are removed from their existing

Feedback

rules first. This process may cause rules to be orphaned, left behind without any voltage areas to constrain. Such rules are removed automatically by the tool.

**What Next**

Verify that you intended to reassign the voltage areas to the new rule, causing their original rules to be orphaned and removed.

---

## NDMUI-505

(error) Length of list values specified for options '%s' must be same.

**Description**

Length of list values sepecified for these options must be same.

**What Next**

Make sure that length of each option value is same.

---

## NDMUI-506

(error) Unable to create %s as the specified input %s have different %s.

**Description**

The specified input objects must have matching attributes for conversion to the composite object of the reported output object_class.

**What Next**

Please remove the input objects with different attributes.

---

## NDMUI-507

(warning) Net shape '%s' has no port .

**Description**

The command 'create_terminal' didnot add any terminal for this net shape. This is because there is no port been detected for this terminal shape.

---

## NDMUI-508

(error) All cells must have the same 'is_tsv' attribute value.

Feedback

**Description**

Either all or none of the cells specified in the set_reference command may be TSV cells - a mixture of TSVs and non-TSVs is not permitted.

**What Next**

Please specify a cell list that is homogeneous with respect to their is_tsv attribute values.

---

## NDMUI-509

(warn) Creating synonym '%s',%s

SH DESCRIPTION The pin name synonym is not setting correctly.

**See Also**

- remove_pin_name_synonym
- report_pin_name_synonym

---

## NDMUI-510

(error) Creating synonym: '%s',%s

**Description**

The printed constraint value is not a valid value.

**See Also**

- remove_pin_name_synonym
- report_pin_name_synonym

---

## NDMUI-511

(error) Owner %s '%s' is not in the same block as the topological constraint.

**Description**

The owner net or bundle must be in the same block as the topological constraint being created. The topological constraint cannot be owned by objects in other physical blocks.

**What Next**

Specify an owner net or bundle that is in the same block as the topological constraint.

Feedback

## NDMUI-512

(error) Pitch specified is not enough to create required sadp_tracks (minimum pitch is '%s').

### Description

The command 'generate_sadp_tracks' will create sadp_tracks only if specified pitch is enough to create required sadp_tracks.

## NDMUI-513

(error) Track rule '%s' doesn't exist in current library.

### Description

The command 'generate_sadp_tracks' will create sadp_tracks based on track rule only if track rule exist.

## NDMUI-514

(information) Start Point of sadp_tracks is shifted to %s = %s.

### Description

The command 'generate_sadp_tracks' will try to find friendly start point when *-exact* is not specified.

## NDMUI-515

(error) Space between start point and end point is not enough to create required sadp_tracks.

### Description

When *-exact* is specified sadp_tracks should start from specified start point and should end before the specified specified end point but there is no enough space between start and end point to create required sadp_tracks.

### What Next

Option *-exact* can be removed so the command *generate_sadp_tracks* can find the friendly nearest start point for the required sadp_tracks.

Feedback

## NDMUI-516

(information) Tracks '%s' and '%s' overlap at point %s = %s.

**Description**

Sadp_tracks should not be overlapping with each other.

## NDMUI-517

(information) Track '%s' doesn't meet sadp spacing requirement.

**Description**

Every sadp_track should meet sadp spacing requirement.

## NDMUI-518

(error) Net '%s' doesn't exist in the current design.

**Description**

The command 'create_sadp_track_rule' will create track rule with net constrainst on track only if net exists before the creation of track rule.

## NDMUI-519

(error) Unsupported track pattern.

**Description**

The command 'create_sadp_track_rule' will create track rule only if pattern format is correct.

## NDMUI-520

(error) Unable to create required sadp_tracks as there is no enough free space on the specified layer.

**Description**

The command 'generate_sadp_tracks' will create sadp_tracks only if there is enough space on the layer.

Feedback

## NDMUI-521

(error) Cannot specify '-pitch' when count is less than two.

**Description**

Pitch is required only when count is two or more than two.

## NDMUI-522

(error) Mask value '%s' for '%s' is not valid. The value is valid only for custom vias.

**Description**

The mask value, inherit_mask, may be specified only for the mask constraint attributes on custom vias. It is not used for simple vias.

**What Next**

Specify a valid mask (i.e., a value other than inherit_mask) for the simple via.

## NDMUI-523

(error) '%s' is not a valid name, because it is reserved for the default rule.

**Description**

The specified name is not valid when creating a voltage area specific rule. "The name is reserved for the default rule and cannot be used for any other voltage area rule.

**What Next**

Specify another name for the voltage area rule.

## NDMUI-524

(Warning) The command '%s' is not supported when Golden UPF mode is not turned on.

**Description**

Name map only works with golden upf mode.

**What Next**

Ensure to turn on golden upf mode if want to use name map.

Feedback

**See Also**

- read_name_map
- mv.upf.enable_golden_upf

---

## NDMUI-525

(error) A '%s' floorplan rule by name '%s' already exists.

**Description**

While trying to create a floorplan rule using any of the *set_floorplan_** commands, the name specified for the new rule matches one of the existing floorplan rule's name.

**What Next**

Either provide a different name for new floorplan rule to be created or remove existing floorplan rule with conflicting name if seems appropriate.

---

## NDMUI-526

(error) Minimum value '%d' can not be greater than maximum value '%d' specified in '%s'.

**Description**

While trying to create a floorplan rule using any of the *set_floorplan_** commands, either values specified with *-min/-max* or *-forbidden_range* or *-valid_range* doesn't follow the min less than max criterion.

**What Next**

Please provide values such that min less than max criterion is satisfied.

---

## NDMUI-527

(error) Please only specify one layer for the -min_pin_layer option.

**Description**

The option *-min_pin_layer* is to set the minmum layer to have terminals on it. Please only specify one layer for this option.

**What Next**

Please see the man page of *report_lib* command for more infomation. report_lib(2)

Feedback

## NDMUI-528

(Error) The command '%s' is not supported when name map version is version 1.0 in Golden UPF mode.

### Description

This message occurs when reading a name map file with name map version v1.0 and related command is not supported.

### What Next

Set the name map version to 2.0 if want to use the name mapping defined by this command.

### See Also

- read_name_map
- mv.upf.enable_golden_upf

## NDMUI-529

(information) Open block '%s' is marked unbound.

### Description

The open block of this library has been marked unbound.

## NDMUI-530

(error) Unable to open reference block '%s'.

### Description

The block specified as the new reference cannot be opened and therefore cannot be used as reference.

### What Next

Please check the previous error messages and specify a compatible reference block.

## NDMUI-531

(warning) Could not set the via_def on one or more non-TSV cells.

Feedback

**Description**

Cells passed to the set_via_def command using the '-cells' argument must all have their 'is_tsv' attribute equal to 'true'. Only TSV cells may reference a via_def, and the via_def must be of type 'through_silicon_via_def'.

**What Next**

Please pass exclusively TSV cells to the '-cells' argument of the set_via_def command.

---

## NDMUI-532

(error) Only TSV cells can reference designs of type 'tsv'.

**Description**

When using the set_reference command to set the reference to a design of type 'tsv', the specified cells must all be TSV cells - that is, they must all have their 'is_tsv' attribute equal to 'true'.

**What Next**

Please pass TSV cells exclusively to the command when setting the reference to TSV designs.

---

## NDMUI-533

(error) TSV cells may only reference designs of type 'tsv'.

**Description**

When using the set_reference command to set the reference of TSV cells - that is, cells with their 'is_tsv' attribute equal to 'true' - the reference design must be of type 'tsv'.

**What Next**

Please specify a valid TSV design when setting the reference of TSV cells.

---

## NDMUI-534

(error) Logical feedthrough cannot be allowed when pass through is disallowed.

**Description**

Logical feedthrough cannot be allowed when pass through is disallowed. If you want to allow logical feedthrough, you must also allow pass through.

Feedback

Only these option combinations are supported:

```
-allow_pass_through false -allow_logical_feedthrough false
-allow_pass_through true -allow_logical_feedthrough false
-allow_pass_through true -allow_logical_feedthrough true
```

**What Next**

Change the *-allow_pass_through* and *-allow_logical_feedthrough* option values to one of the supported combinations.

---

## NDMUI-535

(information) The port die_side attribute has changed from '%s' to '%s' to match the specified preferred pin.

**Description**

When the preferred pin is set, the die side attribute is updated to match the preferred pin's location on the die.

**What Next**

Check the port die_side attribute.

---

## NDMUI-536

(error) Command argument '%s' must be a collection of %s of type %s.

**Description**

The value of the specified command argument must be a collection with the specified size and type.

**What Next**

Provide an argument value of the specified collection size and type.

---

## NDMUI-537

(error) Neither frame nor design view present for '%s'

**Description**

Frame view is preferred for lib cells specified in floorplan rules. For soft macro, the design view can be specified if frame view doesn't exist. But either frame or design view is needed.

Feedback

**What Next**

Please make sure that either frame or design view exists for specified lib cells in floorplan rules.

---

## NDMUI-538

(warning) The origin of topology_node '%s' could not be computed: %s.

**Description**

The origin of the specified topology_node could not be computed and set due to the specified reason.

**What Next**

Please adjust the topology_node's constraint attributes to a legal and unambiguous configuration.

---

## NDMUI-539

(error) Bounds may not be created in lib cells.

**Description**

Lib cells blocks may not contain bounds.

**What Next**

Please specify a valid block in which to create the bound.

---

## NDMUI-540

(error) One voltage area must be specified through the option '%s' if option '%s' is specified.

**Description**

When creating a voltage area rule on specific logical hierarchies, exactly one voltage area must be specified in the *-voltage_areas* command option. The voltage area must be associated with the power domain of the logical hierarchies.

**What Next**

Specify exactly one voltage area in the *-voltage_areas* command option.

Feedback

## NDMUI-541

(error) Cannot update existing voltage area rule '%s', because %s.

### Description

When a rule already exists in the design with the given name, the create_voltage _area_rule command cannot update the existing rule for the stated reason. If logical hierarchies are specified for a specific voltage area, but the existing rule constrains a different voltage area, the existing rule cannot be updated. If multiple voltage areas or a different voltage area is specified, but the existing rule constrains specific logical hierarchies of its associated voltage area, the existing rule cannot be updated.

### What Next

Remove the existing rule with *remove_voltage_area_rules* command option, and then create a new rule.

## NDMUI-542

(error) The '.' character is specified but the top hierarchy is not in the power domain of the specified voltage area '%s'.

### Description

When creating a voltage area rule controlling whether logical feedthrough buffering is allowed on the top hierarchy, the top hierarchy must be a root of the specified voltage area's power domain.

### What Next

Correct the *-voltage_areas* option value, so that it specifies the correct voltage area.

## NDMUI-543

(error) Cell '%s' is not a level of logical hierarchy in the top block.

### Description

When creating a voltage area rule controlling whether logical feedthrough buffering is allowed on specific logical hierarchies, the specified cells must indicate logical hierarchies in the same block as the rule.

### What Next

Correct the list of cells, so that they are logical hierarchies within the same block as the rule.

Feedback

## NDMUI-544

(error) Cell '%s' is not a power domain root cell of the specified voltage area '%s'.

**Description**

When creating a voltage area rule that controls whether logical feedthrough buffering is allowed on specific logical hierarchies, the specified cells must be root cells of the specified voltage area's power domain.

**What Next**

Correct the *-voltage_areas* option value, so that it specifies the correct voltage area. Or correct the list of cells, so that they are root cells of the voltage area's power domain.

## NDMUI-545

(error) Attributes '%s' and '%s' must be cleared before setting attribute '%s'.

**Description**

A voltage area rule may contain a list of included hierarchies or a list of excluded hierarchies, but not both at the same time. To switch from one to another, the indicated attributes must be cleared before setting the new attribute.

**What Next**

Clear the indicated attributes with the *remove_attributes*command, then set the new attribute.

## NDMUI-546

(error) Top hierarchy is not in the power domain of the rule's voltage area '%s'.

**Description**

When setting a voltage area rule to control logical feedthrough buffering on the top hierarchy, the top hierarchy must be a root of the rule's voltage area's power domain.

**What Next**

Specify the correct voltage area rule or create a new one on the appropriate voltage area.

## NDMUI-547

(warning) Couldn't include %s object '%s' when setting the '%s' collection.

Feedback

**Description**

The specified object couldn't be included when setting the specified collection of the topology object.

**What Next**

Check that the object is of the correct type, and that its handle has a hierarchical context consistent with the current block.

---

## NDMUI-548

(error) No topology_plan specified and no current topology_plan.

**Description**

The command requires that either a topology_plan be specified using the '-plan' argument, or that a current topology_plan exists.

**What Next**

Please specify a topology_plan or set the current topology_plan.

---

## NDMUI-549

(warning) The shape_layers of the topology_edge could not be set: %s.

**Description**

The shape_layers of the topology_edge could not be set, due to the specified reason.

**What Next**

Please provide a valid value for the topology_edge's shape_layers.

---

## NDMUI-550

(error) Cannot map from the selected via to the provided custom via.

**Description**

Command failed to map from the selected via to the custom via because mapping to custom via is not allowed unless the target MxN dimensions are 1x1.

**What Next**

Map to a custom via with dimension 1x1

Feedback

## NDMUI-551

(error) Cannot map from custom via.

**Description**

Mapping from custom vias is not allowed.

**What Next**

Map from simple vias or array vias

## NDMUI-552

(Error) Cannot %s site row '%s' as it is %s%s

**Description**

Site rows belonging to a site array can neither be deleted nor modified independently while unbound site rows can be deleted but not modified. Modification of a site row requires parameters of its site definition which is missing in case of an unbound site row.

**What Next**

Site rows belonging to a site array can be modified by modifying the site array parameters.

## NDMUI-553

(error) The reference design %s and the owner design %s are same.

**Description**

The reference design and the owner design are same. This will create a self reference.

## NDMUI-554

(warning) The input tech layer '%s' is not a marker layer.

**Description**

The layer passed to the command 'create_marker_layers' is not a marker layer and hence will be ignored.

## NDMUI-555

(warning) The input tech layer with name '%s' is not defined.

Feedback

### Description

The layer passed to the command 'create_marker_layers' is not a valid tech layer and hence will be ignored.

---

## NDMUI-556

(warning) Removing reference library '%s' from target library subset, because it is not in the reference library list of the destination library.

### Description

The block has a target library subset containing a reference library which does not exist in the destination library's reference library list. The target library subset must be a subset of the library's reference library list. The reference library has been removed from the block's target library subset in the destination library.

### What Next

If the reference library needs to be in the target library subset, call *set_ref_libs* to add it to the reference library list and then call *set_target_library_subset* to add it to the target library subset.

---

## NDMUI-557

(error) Cannot create cell '%s' due to overlapping cell '%s'.

### Description

The *create_relative_cell* command was unable to create the cell because *-check_placement* was specified and an overlapping cell was found.

### What Next

Try using a different offset for the cell creation. Or try removing the *-check_placement* option.

---

## NDMUI-558

(warning) Removing existing instance specific mask constraints from cell '%s', because they do not correspond to the current geometry in the reference block.

### Description

The indicated cell has existing instance specific mask constraints in the design. However these mask constraints do not correspond to the current geometries in the cell's reference block/lib_cell. This happens if the reference block/lib_cell has been modified or if the

Feedback

reference library has been swapped out with versions that have different geometries. In order to proceed, this command must remove the existing mask constraints and then assign new mask constraints.

**What Next**

Verify that the reference block/lib_cell modification was intentional.

---

## NDMUI-559

(warning) Could not color %s '%s' without violating spacing margin.

**Description**

The command could not set the specified object's mask_constraint or cut_mask_constraint to an allowable color that does not violate the specified spacing margin. The reason could be that there are other objects on the same layer nearby that span the allowable set of colors, leaving no color left for this object.

**What Next**

Correct the color congestion around the specified object, or decrease the spacing margin to avoid this warning.

---

## NDMUI-560

(warning) Setting DB file name from '%s' to '%s' is unsuccessful.

**Description**

While trying to set the DB file name from source file name to destination file name, the operation is unsuccessful.

**What Next**

Please check if the input DB file name exists in the lib-cell library provided. If the library is design library, check if the reference libraries has the DB file information.

---

## NDMUI-561

(error) Cannot set_reference on lib-cell because the design '%s' not compatible with current lib_cell '%s'.

**Description**

The *set_reference* command was unable to change the lib_cell used for the specified cell. The new lib_cell either has a different port interface, or a different function from the current lib_cell.

Feedback

The *set_reference* command cannot be used to change the functionality or the connectivity of the net list.

**What Next**

Try using a different design when set_reference for this cell.

---

## NDMUI-562

(error) The '-pattern' argument of the command cannot be empty.

**Description**

The value specified for the *-pattern* argument of the command must contain at least one mask_constraint value.

**What Next**

Specify at least one mask_constraint value for the *-pattern* argument of the command.

---

## NDMUI-563

(error) Syntax error in '-pattern' argument of the command.

**Description**

There was a syntax error in the *-pattern* argument of the command. Each list element of the argument value must one of two forms: <mask_constraint_name> or {<layer_name> {<mask_constraint_name0> <mask_constraint_name1> ...}}.

**What Next**

Edit the argument value to be a list of elements with the correct format.

---

## NDMUI-564

(error) Invalid mask_constraint name '%s' specified - valid values are: %s

**Description**

An invalid mask_constraint name was specified in the *-pattern* argument of the command.

**What Next**

Use a valid mask_constraint name.

Feedback

## NDMUI-565

(error) mask_constraint value of mask_four must be specified explicitly via cut layers only.

### Description

The mask_constraint value of mask_four can only be specified for via cut layers using the layer-specific format of the command's "-pattern" argument.

### What Next

Only specify mask_four as a mask_constraint value explicitly for via cut layers.

## NDMUI-566

(warning) The shape use type will set to detail_route

### Description

When the shape purpose is metal_cut, special_router_extension or active_fill , the shape use type should be set. If not specified, the default will be detail_route.

### What Next

Please set the shape use type when specifying metal_cut as the shape purpose.

## NDMUI-567

(warning) The command 'link_block -force' needs to be called after all the invocation of commmand "%s" are executed.

### Description

The command 'link_block -force' needs to be called after all the invocation of "%s" is executed.

### What Next

Please call command 'link_block -force'

### See Also

- link_block

## NDMUI-568

(error) XPitch and YPitch cannot be zero at the same time.

Feedback

### Description

A via matrix cannot have XPitch and YPitch both be zero at the same time. Please create a general via when the both XPitch and YPitch are expected to zero.

### What Next

Please set the right pitch value. They should not less than zero and cannot both be zero at the same time.

### See Also

- create_via_matrix

---

## NDMUI-569

(error) The -effort argument may only be specified for soft group bounds and soft move bounds.

### Description

The effort setting of a bound only applies for group bounds and soft move bounds. When creating any other type of bound (e.g. repelling, diamond, exclusive, type hard, etc), the effort level does not apply so trying to set it is an error.

### What Next

Do not specify an effort unless creating a group bound or soft move bound.

---

## NDMUI-570

(Error) Cannot specify '-remove' with '-return_as_collection'.

### Description

When 'check_duplicates' specify '-remove' with '-return_as_collection'.

### What Next

Please use -remove and -return_as_collection separately.

---

## NDMUI-571

(Warning) Active fill is older than design data in "%s".

### Description

Active fill is outdated, since the design was already updated.

Feedback

**What Next**

Please use signoff commands to update the active fill data.

---

## NDMUI-572

(error) Reference library '%s' of source library '%s' must also be part of reference library of destination library '%s'.

**Description**

While copying module from a design to another design that belongs to a different library than the source design's, the set of reference libraries of source library must be a subset of reference libraries of destination library. This is to ensure that copied module in destination design can have its leaf instances referring correctly.

**What Next**

Please make sure source library's reference libraries is a subset of destination library's reference libraries.

---

## NDMUI-573

(error) Duplicate layer number '%d' and purpose number '%d'

**Description**

Tech layer with layer number and purpose number is already existing in the tech.

**What Next**

Please specify correct values for layer number/purpose number.

---

## NDMUI-574

(error) max_layer '%s' should not be lesser than min_layer '%s'.

**Description**

While creating object, max_layer's layer_number should not be lesser than min_layer's layer_number. If any of the min_layer/max_layer option is not specified then it picks the default values and checks for the rule validity.

**What Next**

Please specify valid values for min_layer/max_layer.

Feedback

## NDMUI-575

(error) '%s' objects can only be set on layers of type '%s'.

**Description**

Certain objects are restricted in the layers to which they apply. For example, a track constraint can only be set on an INTERCONNECT layer as tracks do not exist on other layers of other types.

**What Next**

Select a valid layer based on the type of object being set.

## NDMUI-576

(error) Track constraint offset list must be given in increasing order.

**Description**

The list of offset values for a track constraint should be given in increasing order.

**What Next**

Order the offset values to be set in increasing order.

## NDMUI-577

(error) Label '%s' already defines track constraints.

**Description**

Once a given label has been assigned to a track constraint definition it cannot be re-used.

**What Next**

In order to set a new track constraint definition, choose a label that does not exist in the design. In order to re-use the label, first call unset_track_constraint to remove the existing constraint definition with the desired label.

## NDMUI-578

(error) Cannot set track constraint on block '%s'. Block boundary is not defined.

**Description**

Track constraints require block boundary data as they are defined in reference to either the block boundary or the block core area.

Feedback

**What Next**

Set the boundary data for the block and then set the desired track constraints.

---

## NDMUI-579

(error) Duplicate layer name '%s' and purpose name '%s'

**Description**

Tech layer with layer name and purpose name is already existing in the tech.

**What Next**

Please specify correct values for layer name/purpose name.

---

## NDMUI-580

(error) Cannot change reference to '%s' as it creates a self reference or cycle in the hierarchy.

**Description**

The target reference been provided to the command introduces a self refernce or cycle. This is not allowed.

**What Next**

Check the target reference been provided in the command and correct it.

---

## NDMUI-581

(error) Cannot change module reference to '%s' as the cell been passed is an unbound instance.

**Description**

The target module been provided to the command through '-to_module' option cannot be used to change the reference of the cell because the cell is initailly unbound

**What Next**

Check the unbound cell been input to the command and correct it.

---

## NDMUI-582

(error) Unable to get fill_cells. Size of fill_cells exceeds database capacity.

Feedback

**Description**

The tool is not able to collect fill_cells in the design as the number of fill_cells exceeds database capacity.

**What Next**

Please contact Synopsys for techincal help in resolving this issue.

---

## NDMUI-583

(warning) Couldn't join edges of topology_node %s.

**Description**

While removing the specified topology_node, the topology_edges with the node as an endpoint could not be joined.

**What Next**

When the "-join_edges" argument of the remove_topology_nodes command is used, the topology_nodes specified for removal must have exactly one driving edge, such that the node is the end_node of the edge, and all remaining edges the node is an endpoint of must be driven by the node, such that the node is the start_node for the edges. This restriction ensures that the join operation is well-defined.

---

## NDMUI-584

(warning) The topology_nodes must reside in the same topology_plan as the edges.

**Description**

When using the "-nodes" argument of the create_topology_edge command, the specified topology_nodes must be owned by the same topology_plan as is specified in the command, or by the current topology_plan if no plan is specified.

**What Next**

Please specify a node list of nodes in the target topology_plan.

---

## NDMUI-585

(information) -incremental option will be deprecated in future releases.

**Description**

Make the operation incremental. The driver pin must be connected to a net otherwise the option does not have any effect. Any existing loads on the net connected to the driver pin

Feedback

are preserved. Please note that connect_pins behavior defaults to -incremental usage, therefore explicit use of -incremental is not required.

**See Also**

- connect_pins

---

## NDMUI-586

(error) Value '%s' specified for option -%s is invalid; %s.

**Description**

This option requires a list in the format {min_layer max_layer area_threshold}. Max_layer's layer_number should not be lesser than min_layer's layer_mumber. area_threshold should be a valid area.

**What Next**

Provide the correct value to the option.

---

## NDMUI-587

(warning) Could not set the ref_node.

**Description**

When creating the new topology_node or setting the ref_node attribute of an existing node, its ref_node could not be set.

**What Next**

Check that the specified ref_node reside in the new node's design hierarchy, and that it will not result in a reference cycle.

---

## NDMUI-588

(error) The current %s must be in the current block.

**Description**

The current typed object may only be set to an object within the current block.

**What Next**

Specify an object in the current block.

Feedback

## NDMUI-589

Error: Design Rule '%s' already exist between layers '%s' and '%s'.

**Description**

Design rule already exist between layers. User can set new properties or update existing properties in existing desing rule.

## NDMUI-590

(warning) The option "-design" or "-library" or "-tech" is not specified. The via rule will be created in block scope instead of tech scope.

**Description**

The command creates the via rule in the block scope by default . The options "-design" or "-library" or "-tech" can be chosen to create the via rule in the correct scope.

**What Next**

Please use the option "-design" or "-library" or "-tech" to create it in correct scope.

## NDMUI-591

(information) The via_rule '%s' is created in the block. Please use '-tech' or '-library' if you intend to create this via rule into the library tech data.

**Description**

The command creates the via rule in the block scope by default or when '-design' is specified. The options "-library" or "-tech" can be chosen to create the via rule in the tech scope.

**What Next**

Please use '-tech' or '-library' if you intend to create this via rule into the library tech data.

## NDMUI-592

(warning) The '%s' command cannot be used on synthetic library cell '%s'.

**Description**

The command is valid only for non-synthetic library cells with a logical/timing model from .lib/.db.

Feedback

## NDMUI-593

(error) Cannot change net_type from %s to %s on net '%s'.

### Description

Nets "*Logic0*" and "*Logic1*" are reserved by Synopsys for the tie-low and tie-high constant nets. Their types cannot be changed.

### What Next

No action is required.

## NDMUI-594

(warning) No cells found to ungroup.

### Description

The *ungroup_cells* command is not able to find any cells to ungroup. If start_level *option was used, it implies that there are no hierarchical cells at a level deeper than the lvel specified.*

### What Next

Check that hierarchical cells exist at the level specified or use a different value.

### See Also

- ungroup_cells

## NDMUI-595

(error) Only the following position-related option combinations are valid for topology_node constraint_type '%s': %s

### Description

The specified constraint_type of the topology_node only supports the specified combinations of position-related options.

### What Next

Specify a valid set of options for the command.

Feedback

## NDMUI-596

(error) The '%s' option of the command cannot be specified unless the '%s' option is specified and set to %s

### Description

The legality of specifying the option depends on the value of another option.

### What Next

Specify a valid set of options for the command.

## NDMUI-597

(warning) Could not include internal pin '%s' in the collection.

### Description

Internal pins cannot be included in this collection because they do not have a physical aspect.

### What Next

To avoid this message, please specify a collection that does not include internal pins.

## NDMUI-598

(error) Pin rebinding is not supported for cells with synthetic library cells as reference.

### Description

Pin rebinding is not supported for cells with synthetic library cells as reference. The pins of the new reference must match exactly and *-pin_rebind none* must be used for *set_reference* to continue.

### See Also

- set_reference

## NDMUI-599

(error) Cell has physical status disallowing changes.

### Description

The physical status of the cell does not allow any changes to be made.

Feedback

**What Next**

If the physical status is *fixed*, you can use *-ignore_fixed* option to override. Otherwise, please change the physical status of the cell before running the command.

---

## NDMUI-600

(warning) The directions donot match. The direction of reference port '%s' do not match block port '%s'.

**Description**

The direction of the port of the reference design doesnot matches with the target design.

**What Next**

Please verify that the target design provided to the command are correct.

---

## NDMUI-601

(warning) The port type donot match. The port type of reference port '%s' do not match block port '%s'.

**Description**

The port type of the port of the reference design doesnot matches with the target design.

**What Next**

Please verify that the target design provided to the command are correct.

---

## NDMUI-602

(warning) The port '%s' of new design is missing in the reference design '%s'.

**Description**

The port of the new target design is missing in the old reference design.

**What Next**

Please verify that the target design provided to the command are correct.

---

## NDMUI-603

(warning) layer '%s' referenced by '%s' is undefined. This rule will be ignored.

Feedback

**Description**

The referenced layer hasn't been defined. The rule will be ignored.

**What Next**

Add the missing layer or remove the reference.

---

## NDMUI-604

(error) due to property '%s' not yet defined, property '%s' has unknown table size and will be ignored.

**Description**

The property is a table but its table size is not yet defined. The property will be ignored.

**What Next**

Add related table size property before this property.

---

## NDMUI-605

(error) the table size defined by property '%s' is '%d'. However, the actual table size of property '%s' is '%d'. This property update will be ignored.

**Description**

The property table size differs from the value of the property that defines this table size. The property will be ignored.

**What Next**

Correct the table size.

---

## NDMUI-606

(error) '%s' is not a valid value for property '%s'. It must be '%s'.

**Description**

The input value is not valid for the property.

**What Next**

Input a valid value.

Feedback

## NDMUI-607

(warning) %s '%s' '%s' and '%s' coexist.

### Description

This message occurs when two conflicting rules/attributes coexist.

For example, if uShapeMinSpacing attribute and uShapeExcludedLengthRange coexist, router does not know which rule to apply.

The following is an example of the warning message:

```
Warning: Layer 'M1' 'uShapeExcludedLengthRange' and 'uShapeMinSpacing'
 coexist. (TECH-607)
```

### What Next

It is recommended that you choose the rule/attribute that you want and remove the other.

## NDMUI-608

(warning) %s '%s' is missing the attribute '%s'.

### Description

This warning message occurs when the specified object of the specified section is missing an important optional attribute, or partial attributes of a rule are specified. If the attribute is not specified in the technology file, the attribute has a default value of 0 or an empty string.

The following is an example of the warning message:

```
Warning : Layer 'M1' is missing the attribute
 'uShapeMinLength'.(TECH-608)
```

### What Next

This is only a warning message. No action is required.

However, if you do not want to the use the default for the attribute, add the attribute with the desired value to the specified section of the technology file and then rerun the command.

## NDMUI-609

(warning) %s '%s' has invalid table values '%s'.

Feedback

### Description

This warning message occurs when the specified section contains an invalid table value list, e.g. thresholds. The value list must meet the following requirements:

- The number of the values must match the table dimension.
- All values must be nonnegative numbers and listed in ascending order.
- The first threshold value must be between 0 and minWidth inclusively.
- The second threshold value must be greater than the layer minWidth. • The first spacing value must not be smaller than minSpacing, and the first maxSpacingCheckRange value must not be smaller than minSpacing * 2 + minWidth.
- In 2-D spacing table between the same layer, the values must be symmetrical.

  In minCutsTbl, if area is specified, extension should be specified. In enclosureTbl, if no lower/upper width is specified or an equivalent ContactCode to the specified enclosures is not found, this message will occur.

In generalized cut spacing rule with parallel length, if minSpacingDiagonalCheckTbl[i][j] = 1, at least one of minSpacingXParallelLengthThresholdTbl[i][j] and minSpacingYParallelLengthThresholdTbl[i][j] should be < 0; otherwise, TECH-609 will be issued to ignore minSpacingDiagonalCheckTbl[i][j]. If minSpacingXParallelLengthThresholdTbl or minSpacingYParallelLengthThresholdTbl is defined, orthoSpacingExcludeCornerTbl[i][j] should be 1; otherwise, TECH-609 will be issued to warn users that the corners will be checked by x/y spacing.

For the table of fatTblExtensionRange, if fatTblExtensionRangeDimension defined, the value list must meet the following requirements:

- The number of the values must match the dimension.
- All values must be larger than 0 and listed in ascending order.

  For fatWireViaEnc rule, fatWireViaEncMinWidthThresholdTbl is recommended to be in ascending order. However, the tool accepts this table even it is not in ascending order as long as the min and max thresholds are paired in fatWireViaEncMinWidthThresholdTbl and fatWireViaEncMaxWidthThresholdTbl. Furthermore, if missing a width range between fatWireViaEncMinWidthThresholdTbl and fatWireViaEncMaxWidthThresholdTbl, this message will be issued.

  For maxStackLevel rule, the value in maxStackLevelCutAsViaArrayTbl for the smallest squared cut should be 0.

  For separatedCut spacing rule, if defined between the same cut layer, the following should hold: 1) separatedCut1NameTbl=separatedCut2NameTbl 2) separatedCut1ToUpperMetalExcludedSpacingTbl=separatedCut2ToUpperMetalExcludedSpacingTbl

Feedback

3) the 2D spacing tables separatedCutXMinSpacingTbl and separatedCutYMinSpacingTbl should be symmetrical.

x|yLegalDimensionTbl table values should be in ascending order. voltageAreaThresholdTbl values should be in ascending order.

For fatTblSpacing Layer property, each of its values must be greater than or equal to minSpacing value.

The following is an example of the warning message:

```
Warning : Layer 'METAL1' has invalid table values '(0, 0,
1.505, 3.005, 4.505, 7.505)'.(TECH-609)
```

**What Next**

Please change the value list of the specified section to meet the above requirements. After making your changes, run the command again.

---

## NDMUI-610

(warning) DesignRule is defined with non-consecutive layers '%s' and '%s'.

**Description**

This warning message occurs when the specified DesignRule section contains two metal layers or two via layers that do not have consecutive mask names. For example, the mask names via1 and via2 are valid consecutive mask names. The mask names via2 and via4 are not consecutive and cause the warning message to occur and the design rule to be ignored. For cut to metal spacing rule with minSpacingUpperMetalWidthThresholdTbl, the rule should be defined in DesignRule section between cut and lower metal layers (via_x and metal_x). For forbiddenCut and separatedCut spacing rule, the attributes should be defined in DesignRule section between same cut layer; otherwise, TECH-610 will be issued.

The following is an example of the warning message:

```
Warning : DesignRule is defined with non-consecutive layers 'VIA2'
and 'VIA4'.
```

**What Next**

This is only a warning message. No action is required.

However, to avoid this warning message, change the layers of the DesignRule section so that the mask names of the layers are consecutive. After making your changes, run the command again.

Feedback

## NDMUI-611

(warning) DesignRule is defined with invalid metal-to-via enclosure layers '%s' and '%s'.

### Description

This warning message occurs when the DesignRule section specifies either a minimum or an end-of-line enclosure rule, for which one layer is not a via layer and/or the other layer is not a metal layer. The application ignores the design rule. Enclosure rules are only valid between a metal and via layer.

The following is an example of the warning message:

```
Warning : DesignRule is defined with invalid metal-to-via
enclosure layers 'via2' and 'via1'. (TECH-611)
```

### What Next

This is only a warning message. No action is required.

However, if the result is not what you intended, change the layers of the DesignRule section so that one is a via layer and the other is a metal layer.

## NDMUI-612

(error) Layers '%s' and '%s' are not adjacent.

### Description

The error message indicates that layer1 and layer2 of a DesignRule section that can only be defined between two adjacent layers are not adjacent.

### What Next

Correct the erroneous layer specifications in the tech file and run the command again.

## NDMUI-613

(warning) Valid SAV%s is missing in viaRule section for via layer '%s' of DesignRule '%s'.

Feedback

### Description

This warning message occurs when a SAV/SAV cluster is missing in the viaRule section meets one of following criteria:

- DesignRule defined attribute 'diffNetNonSavCenterMinSpacingTbl' or 'nonSelfAlignedViaCenterMinSpacingTbl', valid SAV via on layer1 should be defined in ViaRule section.
- DesignRule defined attribute 'selfAlignedViaClusterCenterMinSpacingTbl', valid SAV cluster via on layer1 should be defined in ViaRule section.

Reference example:

```
ViaRule SAV1 {
 cutLayerNameTblSize = 1
 cutLayerNameTbl = (V1) # via layer name
 cutNameTbl = (Vsm) # cut name
 selfAlignedViaUpperLayerEncTblSize = 2 # N
 selfAlignedViaUpperLayerEncTbl = (0.00, 0.01)
}
DesignRule {
  layer1 = "V1"
  layer2 = "V1"
  cut1TblSize = 1
  cut2TblSize = 1
  cut1NameTbl = (Vsm)
  cut2NameTbl = (Vsm)
  nonSelfAlignedViaCenterMinSpacingTbl = (S)
}
```

If the 'SAV1' SAV via is not defined in viaRule section, the tool issues a warning message NDMUI-613 will be issued for designRule 'V1-V1'.

The following is an example of the warning message:

```
Warning: Valid SAV is missing in viaRule section for via layer 'V1' of
DesignRule 'V1-V1'. (NDMUI-613)
```

### What Next

Define or correct the invalid SAV/SAV cluster in viaRule section for the specified via layer of the DesignRule and then run the command again.

---

## NDMUI-614

(warning) Cut name %s in DesignRule cut name table %s is not in the cut name table of layer %s.

Feedback

### Description

Cut names specified in the cut1NameTbl attribute of a DesignRule must be defined in cutNameTbl of the cut layer that is specified by the layer1 attribute of the DesignRule and similarly cut names specified in cut2NameTbl of the DesignRule must be in cutNameTbl of the cut layer specified by layer2.

### What Next

Correct the problematic cut name in the tech file and then run the command again.

---

## NDMUI-615

(warning) DesignRule is defined with different %s layers '%s' and '%s'.

### Description

This warning message occurs when the specified DesignRule section must contain the same via layers or the same metal layers.

The following is an example of the warning message:

```
Warning : DesignRule is defined with different via layers 'VIA2'
and 'VIA4'. (TECH-615)
```

### What Next

If the result is not what you intended, Change the layers of the DesignRule section to make them the same.

---

## NDMUI-616

(error) Layer2 of the DesignRule is not lower than Layer1.

### Description

This message indicates that the layer specified by layer2 of the DesignRule section that generated this message must be lower than the layer specified by layer1. However, the definition of this DesignRule does not satisfy this condition.

### What Next

Correct the error in the tech file and then run the command again.

---

## NDMUI-617

(error) Attribute %s is not specified in a DesignRule section whose metal layer is a "hasRectangleOnly" layer.

Feedback

**Description**

This message indicates that the attribute named in the message can only be specified in metal-to-cut DesignRule sections and the "hasRectangleOnly" attribute of the metal layer must be 1. However, the definition of the DesignRule does not satisfy the condition.

**What Next**

Correct the error in the tech file and then run the command again.

---

## NDMUI-618

(error) Attribute %s is specified in a DesignRule section that is not defined for two consecutive via layers.

**Description**

This message indicates that the attribute named in the message can only be specified in DesignRule sections that are defined for two consecutive via layers. However, the definition of the DesignRule that contains the attribute does not satisfy the condition.

**What Next**

Correct the error in the tech file and then run the command again.

---

## NDMUI-619

(error) Diode mode 19 for antenna rules is supported only from 2017.09.

**Description**

The well dependent antenna rules associated with diode mode 19 is supported only from 2017.09.

**What Next**

Diode mode 19 can't be used. Use a different diode mode.

---

## NDMUI-620

(warning) The runtime of the command can increase significantly if the "-include_blockage_shapes" option is specified.

**Description**

Using the "-include_blockage_shapes" option may increase the runtime of the "set_instance_pg_via_mask_constraints" command.

Feedback

### What Next

To improve runtime, avoid using the option by representing FRAME via routing blockages as via shapes on the PG terminals.

---

## NDMUI-621

(Information) Setting block '%s' %s, this conflicts with existing %s setting of its instances: %s.

### Description

This message indicates that there are multiple instantiations of the concerned block at different levels and at some level its instance already has an edit-control that conflicts with the newly specified value. The editability of all instances has been changed to the specified value and user should review this change. The hierarchical edit-controls apply on the reference block in the current_block top-level context and all instances of the same block share the same editability.

### What Next

The *-blocks* option of *set_editability* command may be used to specify the correct editability value of a reference block. The specified hierarchical edit-controls apply to all instances of the specified reference blocks.

The following example sets the hierarchical edit control of multiple reference blocks to mark them as uneditable (read-only):

```
prompt> set_editability -blocks [get_blocks {blk1 blk2 blk3}] -value
 false -verbose
Block lib:blk1.design is set uneditable (read-only) in the top-level
 context of top.
Block lib:blk2.design is set uneditable (read-only) in the top-level
 context of top.
Block lib:blk3.design is set uneditable (read-only) in the top-level
 context of top.
3
```

### See Also

- set_editability
- get_editability
- report_editability

Feedback

## NDMUI-622

(information) The application option <%s> is an alias and is either been deprecated or has been made obsolete and the Primary application option for this is <%s>.

**Description**

The specific application option is an alias and is either been deprecated or has been made obsolete.

**What Next**

The user script should be modified to remove the deprecated or obsolete application option.

## NDMUI-623

(information) In design '%s', renaming module '%s' to '%s'.

**Description**

A module has been renamed.

## NDMUI-624

(error) Cannot specify more than 1 module in 'module_list' when using 'target_name'.

**Description**

When specifying a 'target_name', only one module is allowed.

**What Next**

If the intent is to rename one module, please ensure that only 1 module is in 'module_list'

## NDMUI-625

(error) Routing blockage cannot be created on layer '%s' because %s.

**Description**

Routing blockages cannot be created on operated layers. These are layers defined in the tech file with attribute "isOperatedLayer = 1" and are considered to be virtual layers. No geometry may be created on these layers.

**What Next**

Specify a valid layer to the command.

Feedback

## NDMUI-626

(error) %s feedthrough buffering cannot be allowed when buffering is disallowed.

### Description

Logical and physical feedthrough buffering cannot be allowed when buffering is disallowed. If you want to allow logical or physical feedthrough buffering, you must allow buffering.

Only these option combinations are supported:

```
-allow_buffering false -allow_logical_feedthrough false
    -allow_physical_feedthrough false
-allow_buffering true -allow_logical_feedthrough true
    -allow_physical_feedthrough false
-allow_buffering true -allow_logical_feedthrough false
    -allow_physical_feedthrough true
-allow_buffering true -allow_logical_feedthrough true
    -allow_physical_feedthrough true
```

### What Next

Change the *-allow_buffering*, *-allow_logical_feedthrough*, and *-allow_physical_feedthrough* option values to one of the supported combinations.

## NDMUI-627

(error) Could not set the specified group_master of the topology_edge: %s

### Description

The topology_edge's group_master attribute could not be set to the specified topology_edge for the specified reason.

### What Next

Adjust the topology_edge's attributes so that it is compatible with those of the group_master, or choose a different group_master edge.

## NDMUI-628

(error) The "-nodes" option of the create_topology_edge command cannot contain more than two topology_nodes when the "-group_master" option is specified.

Feedback

**Description**

Since a topology_edge must have the same endpoints as its group_master edge, it is not possible to create topology_edges over a span of node endpoints that all have the same group_master edge.

**What Next**

Specify only two nodes in the "-nodes" option that match those of the specified group_master edge when using the "-group_master" option. Alternatively, you may leave the "-nodes" option unspecified, in which case the new edge will get the same node endpoints as the group_master edge.

---

## NDMUI-629

(error) When the topology_node has constraint_type '%s', its 'objects' collection %s

**Description**

The 'objects' collection of the topology_node must meet the specified conditions when it has the specified constraint_type.

**What Next**

Specify a valid objects collection for the constraint_type.

---

## NDMUI-630

(error) Cannot specify argument(s) '%s' when creating shaping constraints of type '%s'.

**Description**

The shaping_constraint object models all types of shaping constraints, but each specific type only uses a certain subset of the data that pertains to it. Therefore, the value given to the '-type' argument determines which of the data value arguments may be specified.

**What Next**

Select the appropriate data value arguments for the desired type. For more information on which data value arguments correspond to which type please see the create_shaping_constraint man page.

---

## NDMUI-631

(error) Must specify at least one of arguments '%s' when creating shaping constraints of type '%s'.

Feedback

**Description**

The shaping_constraint object models all types of shaping constraints, and each different type requires some data value(s) upon creation.

**What Next**

Provide one of the necessary data value arguments for the given type of shaping constraint. For more information on which data value arguments correspond to which type please see the create_shaping_constraint man page.

---

## NDMUI-632

(error) Must specify all of arguments '%s' when creating shaping constraints of type '%s'.

**Description**

The shaping_constraint object models all types of shaping constraints, and some types require multiple data values upon creation.

**What Next**

Provide all the necessary data value arguments for the given type of shaping constraint. For more information on which data value arguments correspond to which type please see the create_shaping_constraint man page.

---

## NDMUI-633

(error) Attribute '%s' does not apply for shaping_constraint of type '%s'.

**Description**

The shaping_constraint object models all types of shaping constraints, but each type of shaping constraint only uses a certain subset of all the data values. Therefore, given a constraint of a certain type, some attributes cannot be set because they do not apply to that type.

**What Next**

For more information on which attributes correspond to which type of shaping constraint please see the shaping_constraint object man page.

---

## NDMUI-634

(error) Shaping channel '%s' already assigned to shaping constraint object.

Feedback

**Description**

A given shaping channel may belong to at most one shaping constraint object.

**What Next**

Either remove the existing shaping channel association by removing the assigned to shaping constraint or create a new shaping channel to use.

---

## NDMUI-635

(warning) Cannot remove derived shaping constraint object '%s'.

**Description**

Shaping constraint objects of type boundary_status may not be explicitly created or removed. The boundary_status constraint is derived from the constrained object's physical status and is_rigid_boundary constraint, and is created/removed as necessary to reflect the current status of the object.

**What Next**

Please see the shaping_constraint man page for more information about the boundary_status constraint type.

---

## NDMUI-636

(warning) Setting value '%s' for '%s' is '%s' than value '%s' for '%s'. Value should be '%s' than or equal to value for '%s'.

**Description**

Some attribute values define bounds for a value range. For example an object may have attributes for its minimum width (lower bound) and maximum width (upper bound). Some objects do not enforce strict range adherance for these attributes, meaning that an attribute defining a lower range may be set to a value above another attribute defining a higher range (i.e. a "min" may be set to larger than a "max").

---

## NDMUI-637

(Error) Setting value '%s' for '%s' is '%s' than value for '%s'.

**Description**

Some attribute values define bounds for a value range. For example an object may have attributes for its minimum width (lower bound) and maximum width (upper bound). These range boundary values must be consistent, meaning that each given boundary value must

Feedback

be greater than or equal to the next "lower" boundary value and less than or equal to the next "higher" boundary value. So an object with minimum and maximum width attributes cannot have a minimum width setting greater than its maximum width.

---

## NDMUI-638

(Error) The reference object for a parent_object shaping constraint may only be voltage area or move bound.

### Description

The parent_object constraint further restricts the reference object type to only allow voltage area and move bound objects.

### What Next

Provide a voltage area or move bound when specifying a parent_object constraint.

---

## NDMUI-639

(error) The '%s' attribute of the topology_edge cannot be set when in a topology edge group.

### Description

Certain attributes of topology_edge cannot be set when the edge is in an edge group, as those attribute values are tied to those of the group_master edge. A topology_edge is in an edge group is its 'group_master' attribute is set.

### What Next

To change this attribute, either set the same attribute on the group_master edge, or remove this edge from the edge group by removing its group_master attribute.

---

## NDMUI-640

(error) Cannot replace existing reference library '%s' with the new reference library '%s' because there are references to block '%s' in the existing reference library.

### Description

The new reference library has the same name as an existing reference library and will replace the existing reference library. However doing so will cause cells, which are still referencing the existing reference library, to become unbound. All references to the existing reference library must be changed simultaneously by specifying them to the *-references* option.

Feedback

**What Next**

Specify the names of all reference blocks, which are in the existing reference library, to the *-reference* option.

---

## NDMUI-641

(error) %s cannot be allowed when new cells are disallowed.

**Description**

Buffering is allowed if and only if new cells are allowed, since buffering requires the addition of new cells in the voltage area. If you want to allow buffering through the *-allow_buffering*, *-allow_logical_feedthrough*, or *-allow_physical_feedthrough* option, you must allow new cells through the *-allow_new_cells* option.

**What Next**

If you want to allow buffering, change the *-allow_new_cells* option value to true.

---

## NDMUI-642

(information) Disallowing new cells and all buffering in voltage_area_rule '%s'.

**Description**

Disallowing new cells also disallows all buffering, because buffering requires the ability to add new cells. So since you are setting the *is_new_cell_allowed* attribute to false, the *is_buffering_allowed*, *is_logical_feedthrough_allowed*, and *is_physical_feedthrough_allowed* attributes are automatically set to false as well.

**What Next**

If you mistakenly disallowed buffering, change the *is_buffering_allowed*, *is_logical_feedthrough_allowed*, and/or *is_physical_feedthrough_allowed* attributes back to true. Note that this action will automatically set the *is_new_cell_allowed* attribute to true.

---

## NDMUI-643

(error) Cannot set multiple objects on %s's '%s' attribute.

**Description**

The specified attribute must be a collection of no more than one object.

**What Next**

Please set the attribute value to a collection of a single attribute.

Feedback

## NDMUI-644

(error) Cannot set the anchor_object of an annotation_point of type '%s'

**Description**

Only annotation_points of type 'anchor' may have their anchor_object set.

**What Next**

Please set the anchor_object of annotation_points of type 'anchor' exclusively.

## NDMUI-645

(error) Cannot set the %s of an annotation_point of type '%s'

**Description**

Only annotation_points of type 'absolute' may have their location directly set, otherwise the location is automatically computed. Likewise, annotation_point attributes 'offset' and 'position_type' are only applicable when the annotation_point type is 'anchor'.

**What Next**

Please only attempt to modify attribute values that apply to that annotation_point's type.

## NDMUI-646

(error) Annotation_shape '%s' already exists.

**Description**

An annotation_shape with the specified name already exists. Annotation_shape names must be unique in the block.

**What Next**

Please specify a unique name, or let the tool assign an auto-generated name by leaving the "-name" option unspecified.

## NDMUI-647

(error) May only set the annotation_points of a single annotation_shape at a time

**Description**

Multiple annotation_shapes were specified in the command, but the command may only set the annotation_points of a single annotation_shape.

Feedback

### What Next

Please specify a single annotation_shape to set the annotation_points of.

---

## NDMUI-648

(error) Invalid annotation_points list: %s.

### Description

The specified list input for the "-annotation_points" option is invalid for the specified reason. The format of the input for the "-annotation_points" option is a list, where each list element represents an annotation_point to be added to the annotation_shape and must have one of the following formats:

```
(1) <coord_x> <coord_y>
- a fixed, absolute {x y} point

(2) <anchor>
- a point that follows an anchor object

(3) <anchor> <position> <offset_x> <offset_y>
- a point that follows an anchor object at the specified position and
 offset

(4) <offset_x><'+'/'-'> <offset_y><'+'/'-'>
- a point that follows the location of the previous point in the list, at
 the specified offset
```

### Data Types

```
coord_x    distance
coord_y    distance
anchor     collection of 1 object
position   string - same valid values as annotation_point's position_type
 attribute
offset_x   distance or percentage (when '%' is appended)
offset_y   distance or percentage (when '%' is appended)
'+'/'-'    the literal character '+' or '-'
```

### Notes

```
- The number of annotation_points required by an annotation_shape depends
 on the annotation_shape's type:


tab(;) allbox;
l,al.
Type;Number of annotation_points
symbol, text; 1
rectangle, circle; 2
ruler, line; 2+
```

Feedback

```
polygon; 3+


- Format (1) creates an annotation_point of type "absolute", the others
 of type "anchor"

- Anchor object must have physical extent and must reside at or below the
 current hierarchical level

- Format (4) may not be used for the first annotation_point in the list,
 as there's no "previous" point in that case

- When a percentage value is used for offset_x or offset_y, it is a
 percentage of the anchor object's bounding box dimensions in that
 direction
```

### What Next

Please correct the option input to match the valid format.

---

## NDMUI-649

(info) Active collections are potentially impacting runtime of your scripts.

### Description

Active collections are potentially impacting runtime of your scripts. Any active collection needs to be managed to make sure the objects in the collection are valid. Having big collections active, can add to the runtime of scripts.

### What Next

Please review your scripts to see if there are any TCL variables pointing to collections. If those collections are not needed anymore, you can remove them by unsetting the TCL variables. For example, the following command will unset a TCL variable named A

prompt> unset A

You can also use a TCL procedure like this, to find all TCL variables pointing to collections.

proc getClctVars {} { set clct {} foreach v [info globals] { upvar #0 $v var if {![catch {set var} val]} { # Skip array... if {[as_collection -check -- $val]} { lappend clct $v } } } return $clct }

Additionally, bigger collections are likely to have a bigger impact on runtime. You can check the size of collections by using the sizeof_collection command. For example, the following command returns the size of collection A.

prompt> sizeof_collection $A

Feedback

## NDMUI-650

(error) Cannot specify more than one reference.

**Description**

Members of *cell_bus* should have same reference.

**What Next**

Specify only one reference.

## NDMUI-651

(error) Inappropriate reference specifed.

**Description**

Members of *cell_bus* cannot have reference same as that *-block*, *-cell* or *-design*.

**What Next**

Specify appropriate reference.

## NDMUI-652

(error) cannot create cell bus '%s'.Cell bus name is already taken.

**Description**

Input bus cell name is taken.

**What Next**

Use a different name or remove existing cell bus/cell.

## NDMUI-653

(error) Members '%s' of cell bus have different reference.

**Description**

Members of *cell_bus* should have same reference.

**What Next**

Make sure all cell members of cell_bus have same reference.

Feedback

## NDMUI-654

(error) bus members '%s' do not exist.

**Description**

All bus members must exist without -create_cells option.

**What Next**

Create cell bus members before running this command or use -create_cells option.

## NDMUI-655

(error) Pin '%s' on cells '%s' have different direction.

**Description**

Members of *pin_bus* should have same direction.

**What Next**

Make sure all pin members of pin_bus have same direction.

## NDMUI-656

(error) Pin '%s/ on cells '%s' do not exist.

**Description**

All bus members must exist without -create_pins option.

**What Next**

Create pin bus members on cell bus members before running this command or use -create_pins option.

## NDMUI-657

(error) input bus name doesn't support hierarchical name.

**Description**

Input bus name shouldn't contain hierarchical seperator.

**What Next**

Use -cell option to create bus inside a cell.

Feedback

## NDMUI-658

(error) input bus name doesn't support hierarchical name.

**Description**

pin bus can be created on cell bus only.

**What Next**

To create pin bus on hierarchical cell bus, provide hierarchical cell bus name to -cell_bus option.

## NDMUI-659

(error) Cannot provide range for the creation of pin bus.

**Description**

pin bus can be created on cell bus only. There cannot be width mismatch between cell bus and pin bus.

**What Next**

Provide only pin bus name without range.

## NDMUI-660

(error) cannot create pin bus '%s'.Pin bus name is already taken.

**Description**

Input bus pin name is taken.

**What Next**

Use a different name or remove existing pin bus.

## NDMUI-661

(error) Cannot set shaping constraint '%s' because it violates rule '%s'.

**Description**

Some shaping constraints have certain rules governing the cases in which they may be set. The list of shaping constraint rules is as follows: 1) Members of a shaping group that has a set array_layout constraint may not have set target_location and/or alignment_point constraints.

Feedback

### What Next

Change the design state to correct the rule violation.

---

## NDMUI-662

(error) Cannot change reference to '%s' as the cell passed has an unresolved reference.

### Description

The target reference provided to the command cannot be used because the cell passed has an unresolved reference.

### What Next

Check the cell passed to the command and resolve its reference.

---

## NDMUI-663

(information) Allowing %s and new cells in voltage_area_rule '%s'.

### Description

Allowing buffering also allows new cells, because buffering requires the ability to add new cells. So since you are setting the *is_buffering_allowed*, *is_logical_feedthrough_allowed*, or *is_physical_feedthrough_allowed* attribute to true, the *is_new_cell_allowed* attribute is automatically set to true as well.

### What Next

If you mistakenly allowed new cells, change the *is_new_cell_allowed* attribute back to false. Note that this action will automatically set the *is_buffering_allowed*, *is_logical_feedthrough_allowed*, and *is_physical_feedthrough_allowed* attributes to false.

---

## NDMUI-664

(warning) Attribute '%s' already exists for the %s object class with different data type so unable to get/set or remove attribute.

### Description

In a session, there can not be more than one user defined attribute with same name and on same class with different data type. User can undefine one of the attribute type by using 'undefine_user_attribute' command.

Feedback

**What Next**

Please change the attribute name or type in one of the library or use 'undefine_user_attribute' command to undefine one of the attribute type.

---

## NDMUI-665

(error) Sum of values in %s must be %s than %s.

**Description**

This option requires values with indicated constraint.

**What Next**

Provide the correct value to the option.

---

## NDMUI-666

(warning) All domains specified as part of this create_voltage_area command are AO Block Domains and no regular domain is specified.

**Description**

All the power domains specified as part of this voltage are AO Block domains. No regular domain (with primary supply equal to the main_rail supply of the AO Block Domain(s)) is specified.

**What Next**

Please add the regular power-domain(s) that are intended to share their voltage area with the AO Block domain(s) to this create_voltage_area command.

---

## NDMUI-667

(error) Specified reference does not exist in the current design or reference library.

**Description**

The specified reference does not exist. The command failed because a cell could not be created without a valid reference.

**What Next**

Please provide the correct inputs.

Feedback

## NDMUI-668

(error) Specified reference does not have any ports.

**Description**

Reference should have atleast one port to create valid cell bus.

**What Next**

Please provide the correct inputs.

## NDMUI-669

(warning) All user-defined attributes are persistent by default from 2018.06 release. Please use the '-non_persistent' if non-persistent user attribute is preferred.

**Description**

Earlier user-defined attributes were non-persistent by default, user had to specify -persistent option to make them persistent. But from 2018.06 release, all user-defined attributes are persistent by default.

**What Next**

New option '-non_persistent' has been added in next 2018.06 release. The user script should be modified if user-defined attribute has to be non-persistent only.

## NDMUI-670

(error) Invalid '%s' object. For object class '%s', this argument requires an object type of '%s'.

**Description**

Certain commands take a database object as an argument. Often these arguments are limited to a subset of TCL object classes. In some cases however, the argument requires that objects be of a certain type as well as a certain class. For example, an argument that takes a move bound will require an object with the TCL class 'bound' and ...

**What Next**

Select an object of the appropriate type for the given argument.

## NDMUI-671

(error) Constraint '%s' is not valid for constrained object of type '%s'.

Feedback

**Description**

Each constrainable object has a particular set of shaping constraints that are valid for it. If an invalid constraint is set then this error is reported.

**What Next**

Select a valid constraint to set on the object.

---

## NDMUI-672

(error) Constraint '%s' cannot be set because one of '%s' is already set on this object.

**Description**

Some constraints form mutually exclusive groups. If any constraint in a mutually exclusive group is set then no other constraints in that group may also be set.

**What Next**

If the constraint should be set try unsetting the mutually exclusive constraint first.

---

## NDMUI-673

(error) Cannot set shaping constraint '%s' because '%s'.

**Description**

Most shaping constraints have some restrictions placed on their allowed values. If the current set operation violates one of these restrictions then this error is reported.

**What Next**

Choose valid value for the given shaping constraint.

---

## NDMUI-674

(error) Object '%s' cannot be added to shaping group '%s' because '%s'.

**Description**

Shaping groups require that any object-to-be-added meets the following requirements. 1) All objects being added to a shaping group must be shapeable, 2) all objects being added to a shaping group must share the same physical (block) context as the group, 3) all objects being added to a shaping group must not already be members of another shaping group, 4) if any of the objects are shaping groups, then they must not introduce a circular membership loop in the group membership structure, 5) Members of a shaping group that

Feedback

has a set array_layout constraint may not have set target_location and/or alignment_point constraints.

**What Next**

Select an object that meets the requirements of the shaping group.

---

## NDMUI-675

(warning) application option <%s> has been replaced with <%s>, please update your run script accordingly.

**Description**

The specific application option has been replaced with a new application option and will hence be removed in a future release.

**What Next**

Update the Tcl script to replace the application option with the new application option.

---

## NDMUI-676

(error) Cannot re-order shaping group '%s' because '%s'.

**Description**

Re-ordering operations must result in a valid group. If the specified re-ordering operation would lead to an invalid group then this error is reported.

**What Next**

Correct the operation to result in a valid group.

---

## NDMUI-677

(error) Argument '%s' requires both argument '%s' and argument '%s'.

**Description**

Some commands have arguments that are optional unless a certain other argument is given in which case they become required.

**What Next**

Provide the required argument.

Feedback

## NDMUI-678

(information) Ignoring argument '%s', not used when argument '%s' is given.

**Description**

Some arguments are only used in conjunction with others. In the absence of their "dependent" argument, these arguments are ignored.

**What Next**

Nothing.

## NDMUI-679

(error) Specified lib_pin '%s' is from lib_cell '%s'. It must be from lib_cell '%s'.

**Description**

The specified lib_pin is not from the lib_cell. The via_ladder_pin_order attribute must be a collection of lib_pins from the lib_cell.

**What Next**

Specify a collection of lib_pins from the lib_cell.

## NDMUI-680

(error) Maximum number of lib_pins (%d) in the via ladder pin order has been exceeded.

**Description**

The via_ladder_pin_order attribute must be a collection of 31 or less lib_pins. It is not possible to order more than 31 lib_pins per lib_cell.

**What Next**

Specify a collection of 31 or less lib_pins.

## NDMUI-681

(error) The '-create_marker_layers' argument of the command cannot be empty.

**Description**

The value specified for the *-create_marker_layers* argument of the command must contain at least one layer name, and must match the contents of the *-pattern* argument of the

Feedback

command such that the there is a clear correspondence between the specified colors and the layers that should get the marker shapes for those colors.

**What Next**

Specify at least one layer name for the *-create_marker_layers* argument of the command.

---

## NDMUI-682

(error) Syntax error in '-create_marker_layers' argument of the command.

**Description**

There was a syntax error in the *-create_marker_layers* argument of the command. Each list element of the argument value must one of two forms: <layer_name> or {<layer_name> {<layer_name0> <layer_name1> ...}}. The argument value must match the contents of the *-pattern* argument of the command such that the there is a clear correspondence between the specified colors and the layers that should get the marker shapes for those colors.

**What Next**

Edit the argument value to be a list of elements with the correct format.

---

## NDMUI-683

(error) Cannot set constraint '%s' as it is mutually exclusive with '%s'.

**Description**

Certain groups of shaping constraints may not be set on a given object at the same time. Specifying a set of constraints which violates one of these mutually exclusive groups results in this error.

**What Next**

Select a different set of constraints to set that do not violate any mutual exclusion requirements.

---

## NDMUI-684

(warning) The application option <%s> is disabled and will be ignored.

**Description**

The specific application option has been made disabled and hence will be ignored.

Feedback

**What Next**

See the man page for this application option for details.

---

## NDMUI-685

(warning)The name of the object '%s' of type '%s' already exists in destination design '%s'. This will be skipped.

**Description**

The name of the object is conflicting with existing same name object in command "copy_floorplan". The copy for the same will be skipped.

**What Next**

The option "-force" can be used to overide this.

---

## NDMUI-686

(Information) The name of the object '%s' of type '%s' will be renamed to '%s' in destination design '%s'.

**Description**

The name of the object is conflicting with existing same name object in command "copy_floorplan". The object will be renamed with another name.

**What Next**

The option "-force" shouldn't be used to prevent this.

---

## NDMUI-687

(error) The specified design label '%s' is not found.

**Description**

The specified design label name specified by the user is not found.

**What Next**

Specify the correct design label name.

---

## NDMUI-688

(error) The specified design lavel '%s' does not have a design view.

Feedback

**Description**

The specified design label name specified by the user doesn't have a design view.

**What Next**

Specify the correct design label name.

---

## NDMUI-689

(error) The current design '%s' is not a design view.

**Description**

The current design is not a design view.

**What Next**

The user can change the current design using "current_block" and execute the "copy_floorplan" command.

---

## NDMUI-690

(error) Cannot pack cell library '%s'.

**Description**

write_lib_package can't package on standard/lib-cell library or another write_lib_package is active.

**What Next**

Use write_lib_package on appropriate library only.

---

## NDMUI-691

(error) Unable to pack library '%s'.

**Description**

The tool could not create the temporary file used to package the library. Without the temporary file, *write_lib_package* command fails.

This is most likely caused by a file permission problem, but there are a number of other reasons why the tool would not be able to create the temporary file: the path name might be invalid; there might be a file with the same name that could not be overwritten; you might not have permission to write to that directory; you might have exceeded your disk quota on that file system; the file system might be read-only; or the file system might be full.

Feedback

### What Next

Verify that you can create that file in that directory and that text can be written to that file. Verify that there is sufficient free space on the disk.

To see the size of a file system, use the UNIX *df* command:

```
unix% df /tmp/.
```

Change the *shell.common.tmp_dir_path* application option setting to use a different path or file system to store store the temporary file. To change the temporary directory to /SCRATCH, use the *set_app_options* command:

```
prompt> set_app_options -list { shell.common.tmp_dir_path /SCRATCH }
```

### See Also

- expand_outline
- set_app_options
- shell.common.tmp_dir_path

---

## NDMUI-692

(error) Unable to exclude design views for specified ref_libs.

### Description

The ref_libs might not available to exclude its design views.

### What Next

Mask sure that specified ref_libs exists.

---

## NDMUI-693

(error) %s.

### Description

This message occurs when a process was unable to be forked or forked process has failed/errored-out.

---

## NDMUI-694

(error) Unable to unpack library package '%s'.

Feedback

### Description

The tool could not create/find the temporary file used to unpack the library package. Without the temporary file, *read_lib_package* command fails.

This is most likely caused by a file permission problem, but there are a number of other reasons why the tool would not be able to create the temporary file: the path name might be invalid; there might be a file with the same name that could not be overwritten; you might not have permission to write to that directory; you might have exceeded your disk quota on that file system; the file system might be read-only; the file system might be full; or library package might invalid as well.

### What Next

Verify that you can create that file in that directory and that text can be written to that file. Verify that there is sufficient free space on the disk.

To see the size of a file system, use the UNIX *df* command:

```
unix% df /tmp/.
```

Change the *shell.common.tmp_dir_path* application option setting to use a different path or file system to store store the temporary file. To change the temporary directory to /SCRATCH, use the *set_app_options* command:

```
prompt> set_app_options -list { shell.common.tmp_dir_path /SCRATCH }
```

### See Also

- expand_outline
- set_app_options
- shell.common.tmp_dir_path

---

## NDMUI-695

(error) Unable to unpack library package '%s'. Library '%s' already exists in the current directory. Please try again with the -overwrite option.

### Description

*read_lib_package* unable to create/open unpacked library in the destination directory (default is current directory).

### What Next

This error is mainly because the destination directory already has an library with the same name as the one being unpacked. Use *-overwrite* to overwrite unpacked library in the destination directory. If command still fails, it means previous unpacked

Feedback

library package has non-removable (i.e, no write permission) files so manually remove the previous unpacked directory after providing write access ( using chmod u+w -R <unpacked_directory> ) or change destination directory through *-destination* option. This error is mainly because the destination directory already has a library with the same name as the one being unpacked.

---

## NDMUI-696

(error) Reference library '%s' of the library package already opened.

**Description**

*read_lib_package* failed as reference library of the library package is already opened.

**What Next**

Try specifying *-overwrite* to open reference library from the library package by closing opened reference library.

---

## NDMUI-697

(error) Unable to open library '%s'.

**Description**

read_lib_package unable to open library.

**What Next**

Try changing destination directory through *-destination*.

---

## NDMUI-698

(error) z-offset value specified is not valid.

**Description**

z-offset can be greater than or eqal to zero but less than 32767.

---

## NDMUI-699

(error) Cannot set z-offset on cell '%s' which is not from top-level of 3d design.

**Description**

z-offset can be set only on cells of top-level 3d design.

Feedback

## NDMUI-700

(error) Persistency for user-defined attributes of types %s are not supported for all object types.

### Description

Persistency for user-defined attributes of types one_of/range_min/range_max are not supported for all object types. Earlier user-defined attributes were non-persistent by default, user had to specify -persistent option to make them persistent. But from 2018.06 release, all user-defined attributes are persistent by default.

### What Next

Please use '-non_persistent' option with user-defined attributes of types one_of/ range_min/range_max for class objects which don't have support of persistency.

## NDMUI-701

(error) Value '%s' specified for option -blocks is invalid; %s.

### Description

This option requires a list in the format { { die1[:num_layers] die2[:num_layers] } ...}.

The die names must correspond to cell names in the design. The number of layers must be equal or less than the number of layers in the die block.

### What Next

Provide the correct value to the option.

## NDMUI-702

(error) Failed to remove virtual interface block; %s.

### Description

The command failed to remove the virtual interface block for the given reason. If a single block is specified, it must be a virtual interface block. If a single cell is specified, it must be the instance of the virtual interface block. If two cells are specified, they must be the two dies used to create the virtual interface block.

### What Next

Call the command with either the virtual interface block, the cell instance of the virtual interface block, or the pair of dies used to create the virtual interface block.

Feedback

## NDMUI-703

(information) Created virtual interface block '%s' between dies '%s' and '%s'.

**Description**

The command created the indicated virtual interface block.

**What Next**

This is an informational message only.

## NDMUI-704

(information) Removed virtual interface block '%s'.

**Description**

The command removed the indicated virtual interface block.

**What Next**

This is an informational message only.

## NDMUI-705

(error) The layer specified in the '%s' argument of the command is invalid for the requested via_def type.

**Description**

The specified layer is incompatible with the type of via_def being created. For simple via_defs, the lower and upper layers must be the nearest routable upper and lower metal layers to the specified cut_layer. For custom via_defs, the lower and upper layers must correspond to the lowest and highest layers, respectively, of the specified custom shapes.

**What Next**

Please specify compatible lower and upper layers, or simply omit the layer arguments for simple and custom via_defs, in which case they will be set automatically to the correct layers.

## NDMUI-706

(error) Net %s is already a part of bus %s.

Feedback

**Description**

The specified net is already part of the existing bus. If bus base name and net name are different then it may be due to inconsistent DB.

**What Next**

Try again after running "change_names -hierarchy -rules verilog " to fix the net/bus association.

---

## NDMUI-707

(Error) Only bounds of type repelling may contain group members.

**Description**

Currently group objects (more specifically repelling bound group objects) are only valid in repelling bounds.

**What Next**

Try running the command with a repelling bound.

---

## NDMUI-708

(warning) Group '%s' cannot be added to bound because '%s'.

**Description**

Group membership in bounds must follow certain criteria. Currently only repelling bound groups are allowed in bounds, and only repelling bounds may contain repelling bound groups. Furthermore repelling bound groups may not be in multiple repelling bounds.

**What Next**

Correct any violations to the rules described above and re-try.

---

## NDMUI-709

(warning) Object '%s' cannot be added to repelling bound group because '%s'.

**Description**

The membership rules for objects in a repelling bound group follow the membership rules for repelling bounds, with the exception of repelling bounds groups (groups may not be members of repelling bounds groups). Valid members are cells and ports.

Feedback

**What Next**

Correct any violations to the rules described above and re-try.

---

## NDMUI-710

(information) The object '%s' of type '%s' is copied to the destination design '%s'.

**Description**

The object is copied to the destination design in the command "copy_floorplan".

**What Next**

This is an informational message only.

---

## NDMUI-711

(Error) %s group %s can't be modified or removed.

**Description**

HDL or FSM groups can't be removed or edited and even members can't be added or removed from the HDL groups.

---

## NDMUI-712

(Information) Skipping group %s as it is non-HDL group.

**Description**

Can't specify non-HDL groups with -module option.

---

## NDMUI-713

(Error) Hierarchical patterns are not supported for non-HDL group.

**Description**

Hierarchical pattern are supported for HDL groups to limit the scope of query.

---

## NDMUI-714

(warning) Skipping given clip region as it is not rectilinear.

Feedback

**Description**

Clip regions must be rectilinear polygons.

---

## NDMUI-715

(error) Given clip region overlaps another.

**Description**

All active clip regions must be non-overlapping.

---

## NDMUI-716

(error) Specified clip block '%s' invalid because '%s'.

**Description**

Clip blocks may only merge back into their source blocks from a given clip library.

---

## NDMUI-717

(Error) Cannot create an empty repelling bound group.

**Description**

Repelling bound groups may not be empty. If the command receives no valid objects then it will error.

**What Next**

Try running the command with at least one valid bound member.

---

## NDMUI-718

(Error) Repelling bounds have a hard limit of at most 16 members.

**Description**

Repelling bounds repel all objects within them. The intended use case involves repelling a small number of cells or groups (cores) away from one another (or one or more ports). To help guide and support the intended use the membership limit caps at 16.

**What Next**

Try restricting the number of objects being added to the bound.

Feedback

## NDMUI-719

(Error) Repelling bounds cannot contain both groups and cell/ports.

**Description**

Repelling bounds allow two membership categories. The original use case with cells (leaf and hierarchical) and ports, and the new use case with groups. Mixing the two membership categories is an error.

**What Next**

Adjust the object list to match one of the two membership categories.

## NDMUI-720

(error) Failed to create virtual interface block because design '%s' contains unresolved cell references.

**Description**

Virtual interface blocks cannot be created in a design containing unresolved cell references. Check the link_block results, specifically the LNK-081 errors, to understand and resolve the problem before running *create_3d_virtual_blocks*.

**What Next**

Configure the design correctly.

## NDMUI-721

(error) bond_pad terminal already exist with the name '%s'.

**Description**

Name of the bond_pad terminal should be unique in the specified block/port.

**What Next**

Specify a different name for bond_pad terminal.

## NDMUI-722

(warning) Via ladder %s has a shape with physical status locked or fixed, hence we cannot remove this via ladder from the design.

Feedback

**Description**

This via ladder has a shape with physical status locked or fixed, hence we cannot remove this via ladder from the design.

**What Next**

Change the physical status of the locked or fixed shape and try again.

---

## NDMUI-723

(warning) Name changes are not performed on %s bus "%s", as it converts regular bus to irregular.

**Description**

change_names does not allow conversion from regular bus to irregular bus.

**What Next**

Ensure definition of name rule does not imply generation of irregular bus.

---

## NDMUI-724

(error) '.' as current_design is not supported anymore in file_name "%s".

**Description**

Support of '.' as current_design is obsolete.

**What Next**

Revise run script to specify current_design properly.

---

## NDMUI-725

(warning) %s name change from '%s' to '%s' %s.

**Description**

Name change cannot be performed as explained in the specified reason at end of message. Following are possible reasons:

in design "designName" not performed, given new cell name repeated in design "designName" not performed, given new net name repeated in design "designName" not performed, given new port name repeated

Feedback

not performed, given cell not present in design "designName" not performed, given net not present in design "designName" not performed, given port not presented in design "designName"

not performed, given new cell name already present in design "designName" not performed, given new net name already present in design "designName" not performed, given new port name already present in design "designName"

not performed not performed due to duplicate entry in names file

**What Next**

Check name rule definition and names file.

---

## NDMUI-726

(warning) %s '%s' provided in the names file is not valid.

**Description**

Name specified in the names file is not valid.

**What Next**

Check reported name against the design data to ensure it is valid.

---

## NDMUI-727

(warning) Invalid type '%s' in name_file '%s'; The type must be port, cell, or net.

**Description**

Invalid object type found in name_file.

**What Next**

Ensure object type is one of port, cell, or net.

---

## NDMUI-728

(warning) No change in %s bus '%s'%s.

**Description**

Net/Port bus is not changed since there is another bus with the same name in the design already.

Feedback

**What Next**

Validate the reported collision with design.

---

## NDMUI-729

(warning) %s bus '%s' is bit blasted.

**Description**

Net/Port bus is bit blasted.

**What Next**

Check result design to ensure the bit blasted is intended.

---

## NDMUI-730

(info) In module '%s', net '%s' was renamed to '%s' as required to meet rule -equal_ports_nets.

**Description**

NDMUI-730 is a specific case where suppose net n1 is to be renamed to abc as it is connected to port abc but a different net by name abc already exists. Then we try to rename abc to abc_0 so that n1 can be renamed to abc and that is when we issue NDMUI-730 for abc and not for n1.

---

## NDMUI-731

(warning) While applying rule max_length, tool could not find a unique name for %s %s.

**Description**

While applying rule max_length, tool could not find a unique name for one of the following types: cell, port, bus port, net, and bus net.

**What Next**

Ensure name rule is valid with the max_length

---

## NDMUI-732

(info) The first restricted character list conflicts with the name prefix(%s). The name prefix will still be used in change names.

Feedback

**Description**

The first restricted character list conflicts with the reported name prefix. The name prefix will still be used in change names.

---

## NDMUI-733

(warning) Could not change name for feedthrough net '%s' to '%s'.

**Description**

Could not change name for feedthrough net. The failure in name change may be due to collision of names in the name space.

**What Next**

Check for collision in names.

---

## NDMUI-734

(error) The %s '%s' cannot be anchored - it is in edit_group '%s'.

**Description**

The specified object is in an edit_group, however an object in an edit_group may not be the dependent_object of an anchor.

**What Next**

Remove the object from the edit_group, or choose a different object to be anchored.

---

## NDMUI-735

(warning) To change names in module '%s' use -hierarchy option.

**Description**

Needs to use -hierarchy option if changing names of non-top level modules. Note that this message is only issued when option -names_file is used.

**What Next**

Use -hierarchy option if changing names of non-top level modules is intended.

---

## NDMUI-736

(info) The specified replacement character %s is conflicting with the specified restricted character.

Feedback

**Description**

In change_names, the specified replacement character is conflicting with the specified restricted character.

---

## NDMUI-737

(warning) All %s in the %s bus '%s' are not present in the names file.

**Description**

All members of net/port bus are not in the names file.

**What Next**

Ensure names file to verify members of net/port bus are presented in the names file.

---

## NDMUI-738

(error) Shape '%s' is a part of an existing via ladder object.

**Description**

A shape can only be a part of exactly one via ladder object.

**What Next**

Please remove this shape from the list of shapes for the current via ladder.

---

## NDMUI-739

(error) Cannot add %s '%s' to via ladder because it is not in the current design.

**Description**

The specified object cannot be added to the via ladder because it belongs to another design or lib_cell. This command may only add objects to the via ladder from the current design.

**What Next**

Set current_design to the design that contains the object, and then add it to the via ladder.

---

## NDMUI-740

(error) Cannot add shape '%s' to via ladder because it is connected to a different net than other shapes in this via ladder.

Feedback

**Description**

The specified shape cannot be added to the via ladder because it is connetced to a different net than other shapes in this via ladder. All the shapes in a via ladder should be connected to the same net.

**What Next**

Pass a collection of shapes to the create_via_ladder command which belong to the same net and try again.

---

## NDMUI-741

(error) Tech purpose with name 'drawing' and/or number '0' can not be created.

**Description**

Tech purpose with number '0' is system default purpose and reserved for 'drawing'.

**What Next**

Please change option values and try again.

---

## NDMUI-742

(warning) Tech purpose with name 'drawing' and number '0' can not be removed.

**Description**

'drawing' is system default tech purpose. It can not be removed.

**What Next**

Specify tech purpose(s) other than 'drawing'.

---

## NDMUI-743

(error) Failed to remove tech purpose %s.

**Description**

Failed to remove the tech_purpose.

**What Next**

Specfied tech purpose can not be removed.

Feedback

## NDMUI-744

(error) %s can not be created on non-default purpose.

### Description

The object can be created only on default purposes 'drawing'.

### What Next

Specify correct layer.

## NDMUI-745

(error) Cannot modify mutable lib-cell %s data in this shell.

### Description

It is not currently possible to change the named lib-cell data from inside this shell. There are three types of lib-cell data. Some cannot be changed after the lib-cell library is built. Others cannot be changed outside of the Library Manager shell, because the checks which make sure the value is changed properly only exist within the Library Manager. Still others can be changed from inside this shell, but only when the app option design.enable_lib_cell_editing is set to "mutable".

### What Next

If the data must be changed, set the app option design.enable_lib_cell_editing to "mutable". Optionally, you can return to the Library Manager and use the edit flow to change this data. Additionally, you can return to the original setup used to create the library and change the data before the workspace is checked and committed.

## NDMUI-746

(information) Removing library '%s' from the target library subsets in block '%s', because the library is no longer in the reference library list.

### Description

The specified block has one or more target library subsets containing a library which is no longer in the reference library list. Since every target library subset must be a subset of the libraries in the reference library list, the specified library has been removed from the specified target library subsets.

### What Next

If the library needs to remain in the target library subsets, add the library back into the reference library list and target library subsets.

Feedback

**See Also**

- set_target_library_subset
- report_target_library_subset

---

## NDMUI-747

(warning) Cannot modify read-only application option '%s'.

**Description**

This application option is read-only. You cannot change its value.

**What Next**

Some application options can be set only in the .synopsys_icc2.setup file, and become read-only after the file has been read. Determine if that is the case for this option.

---

## NDMUI-748

(warning) Invalid value '%s' specified for app option '%s'.

**Description**

This message indicates that the specified value for app option is invalid.

**What Next**

Provided a correct value for the app option.

---

## NDMUI-749

(error) The %s '%s' cannot be added to the edit_group - it is anchored by anchor '%s'.

**Description**

The specified object is the dependent_object of an anchor, however anchored objects may not be members of edit_groups.

**What Next**

Un-anchor the object, or choose a different object to add to the edit_group.

---

## NDMUI-750

(warning) Port count of the new %s is not same as that of the existing %s

Feedback

**Description**

The reference or view of cells can be changed only with a reference or view with a identical port interface.

**What Next**

Specify a reference or view with a port interface identical to the current one.

---

## NDMUI-751

(warning) Port name '%s' of the old reference '%s' is not found in the new reference '%s'.

**Description**

The reference or view of cells can be changed only with a reference or view with a identical port interface.

**What Next**

Specify a reference or view with a port interface identical to the current one.

---

## NDMUI-752

(error) Pin '%s' is not a physical pin.

**Description**

The pin which has a via ladder must be always be a physical pin.

**What Next**

Please use a physical pin and try again.

---

## NDMUI-753

(error) Pin group id '%u' cannot be specified in the %s option, because %s.

**Description**

The specified pin group id cannot be specified for the indicated reason. The pin group id must have been assigned to a pin, shape, or via in the net's block prior to calling this command. Additionally the specified pin group id cannot be 0, because it is the default pin group's id. Pins in the default pin group are always routed using the default rule.

**What Next**

Remove the specified pin group id from the command call.

Feedback

## NDMUI-754

(error) Pin '%s' cannot be assigned to pin group id '%u', because the pin is unconnected.

**Description**

The pin must be connected to a net before being added to a pin group.

**What Next**

Connect the pin to a net and then set the pin group id.

## NDMUI-755

(information) The object '%s' of type '%s' of design '%s' is skipped.

**Description**

The object is skipped in the command "copy_floorplan".

**What Next**

This is an informational message only.

## NDMUI-756

(error) '-new_name' is not allowed for multiple cells.

**Description**

The option '-new_name' is not allowed when multiple cells are specified in the cell_list argument.

**What Next**

Please specify single cell in the cell_list argument.

**See Also**

- uniquify

## NDMUI-757

(error) The value of the '%s' option or attribute must be a list of one or more positive distance values.

Feedback

**Description**

The specified option or attribute value must be a list of one or more distance values, all of which must be greater than zero.

**What Next**

Please specify a list of positive distances for the specified value.

---

## NDMUI-758

(error) The value of the '%s' option or attribute must be a %s distance value.

**Description**

The specified option or attribute value must be a distance value in the specified range.

**What Next**

Please specify a valid distance for the specified value.

---

## NDMUI-759

(warning) Could not remove bump_region_pattern '%s' - it is referenced by one or more bump_regions.

**Description**

A bump_region_pattern cannot be removed if it is referenced by one or more bump_regions - that is, one or more bump_regions has the bump_region_pattern as their *pattern* attribute. You can query which bump_regions these are using the *get_bump_regions -of_objects <collection of the bump_region_pattern>* command.

**What Next**

To remove the bump_region_pattern, first remove the referencing bump_regions, or change the bump_regions' *pattern* attributes to reference other bump_region_patterns.

---

## NDMUI-760

(error) module '%s' has no HDL Groups

**Description**

Module has no HDL Groups defined.

Feedback

**See Also**

- group_cells

---

## NDMUI-761

(error) No HDL Groups matched pattern '%s' for module '%s'

**Description**

The pattern which you specified did not match any HDL Groups for the module.

**See Also**

- group_cells

---

## NDMUI-762

(error) No cells found in HDL Groups

**Description**

No valid cells found in HDL Groups

**See Also**

- group_cells

---

## NDMUI-763

(error) value specified for '-new_name' is an empty string

**Description**

Argument of -new_name should be a non-empty string.

**What Next**

Please specify a non-empty string value

**See Also**

- uniquify

Feedback

## NDMUI-764

(info) Target bus naming style '%s' is applied to all modules of design '%s' even though -hierarchy is not specified.

**Description**

Value of -target_bus_naming_style is always applied to all modules in the design, regardless of the value of option -hierarchy for change_names.

## NDMUI-765

(error) Design '%s' Technology has fewer routing layers than the specified number of layers.

**Description**

Technology has fewer routing layers than the specified number of layers.

**What Next**

Make sure that the technology has sufficient routing layers.

## NDMUI-766

(error) %s.

**Description**

Individual command specified within redirect -bg is either unknown or a command that is not allowed with redirect -bg or redirect -bg itself.

**What Next**

Verify that each command specified within redirect -bg is a valid ICCII or Descartes command and also one of the supported commands within redirect -bg, for example, reporting commands. Most of the command that modifies design data like, for example, create commands are not allowed to be executed within redirect -bg. Also nested redirect -bg is not allowed.

## NDMUI-767

(warning) Parallel_execute within redirect -bg will run with %d cores instead of %d cores.

**Description**

The maximum cores allowed in parallel_execute when invoked within redirect -bg will be limited by the max_cores specified within redirect -bg.

Feedback

**What Next**

If you want run parallel_execute with more cores, please increase the value of max_cores specified to redirect -bg.

---

## NDMUI-768

(warning) redirect -bg will run with a max cores of %d instead of %d.

**Description**

The maximum cores allowed in redirect -bg will be limited by the maximum cores available in the parent.

**What Next**

In the main process, always specify max_cores to be atleast one more than what you want to make use of in redirect -bg.

---

## NDMUI-769

(error) redirect -bg cannot be used in single core.

**Description**

The main process should specify a max core value which is greater than 1.

**What Next**

Use set_host_options -max_cores to change the maximum allowed cores in main process.

---

## NDMUI-770

(error) '%s' is not a valid library package. Please run md5sum to confirm the checksum after transfer from customer site.

**Description**

*read_lib_package* cannot read library package generated by the *write_lib_package command.*

**What Next**

The lib package file provided isn't in the proper format. If you are transferring this data from customer site, please run md5sum on the lib_package file before and after file transfer to confirm the checksum remains the same.

Feedback

## NDMUI-771

(error) Specified output library package name '%s' is not writable in the current working directory.

**Description**

*write_lib_package* can create a library package only if the specified output library package name is writable in the current working directory.

**What Next**

Please provide the valid library package name

## NDMUI-772

(error) Cannot raise or lower cells in z level when multiple specified cells are distributed in different z levels as well as different stacks.

**Description**

Cells can be reordered by one position in z level only when all specified cells are in a same z level or same stack.

## NDMUI-773

(error) Cannot raise or lower cells in z level when specify multiple cells whose z_offsets are not consecutive in a stack.

**Description**

Cells in a stack can be reordered when all specified cells' z_offsets are consecutive.

## NDMUI-774

(error) The specified preferred pins are invalid for port '%s' because %s.

**Description**

The preferred pins must be all frontside pins or all backside pins. A mix of frontside and backside pins is not supported on a single port. The preferred pins' die side is used to determine the port's die side, which is either front or back but not both.

**What Next**

Specify preferred pins that are all on the same die side.

Feedback

## NDMUI-775

(warning) Could not add topology_plan '%s' to the new topology_group - the plan must be in the same block as the group

**Description**

Could not add the specified topology_plan to the newly-created topology_group - the plans of a group must reside in the same block as the group.

**What Next**

Specify plans exclusively from the groups's block.

## NDMUI-776

(error) Invalid sequential_repeaters value specified

**Description**

The value specified for the topology_plan 'sequential_repeaters' attribute is invalid - must be a string of the form "<operator> <operand>". '<operator>' must be one of the following values: <, <=, =, >=, >, less, less_or_equal, equal, greater_or_equal, or greater. '<operand>' must be either the name of a topology_plan, which may include wildcards, or a non-negative integer.

**What Next**

Please specify a valid sequential_repeaters value.

## NDMUI-777

(warning) Could not remove default topology_group '%s' - it contains topology_plans.

**Description**

The default topology_group cannot be removed if it contains topology_plans, unless the '-remove_plans' argument is specified.

**What Next**

To remove the default topology_group, please remove the plans first, or use the '-remove_plans' argument of the remove_topology_groups command.

## NDMUI-778

(error) Could not set the group of the topology_plan.

Feedback

**Description**

Could not set the topology_plan's group to the specified topology_group - the group must be in the same block as the plan.

**What Next**

Specify a valid topology_group from the plan's block.

---

## NDMUI-779

(error) Invalid color value specified

**Description**

The specified color value is not valid. Please specify a hexadecimal 24-bit rgb value preceded by '#', where the digits correspond to the format #rrggbb or #aarrggbb, for example: "#543cb1".

**What Next**

Please specify a valid color value.

---

## NDMUI-780

(warning) The core area cannot be set.

**Description**

The setting of the core area is skipped in the command "copy_floorplan".

**What Next**

The core area cannot be set because there may be existing site array or site row in the design.

---

## NDMUI-781

(warning) The boundary cannot be set.

**Description**

The setting of the boundary is skipped in the command "copy_floorplan".

**What Next**

The boundary cannot be set because it may be invalid.

Feedback

## NDMUI-782

(info) %s for first site row of site_array: %s.

**Description**

It is not necessary that first row of site array will have first value of cycle and offset provided by user in *create_site_array* command. Cycle and Offset will be aligned on the basis of virtual grid.

If user wants to have cycle and offset provided as part of *create_site_array* command to be aligned with first site row of site array then user can can create un-aligned site array.

## NDMUI-783

(error) '%s' already has reliability verification constraint '%s' with layer %s. It can't have both layer lines and layer widths.

**Description**

One object can't have reliability verification constraints with both layer lines and layer widths. It can either have reliability verification constraints with layer lines or reliability verification constraints with layer widths.

**What Next**

Review the objects listed in error message and decide which reliability verification constraints is needed for that object and then apply constraints accordingly.

## NDMUI-784

(error) Could not set the anchor's %s to '%s' - the object must be in the same or lower level of hierarchy as the dependent object.

**Description**

Could not set the specified object as an anchor object of the anchor, because it resides in an incompatible level of hierarchy.

**What Next**

Specify an object from the same or a directly descendant hierarchy as the dependent_object.

## NDMUI-785

(error) Cannot add hierarchical %s object '%s' to the edit_group: %s

Feedback

**Description**

Could not add the specified hierarchical object to the edit_group: only hierarchical objects of class edit_group may be added to an edit_group, and only those that are from a unique block, and that don't already belong to another edit_group in the top edit_group's block.

**What Next**

When adding a hierarchical object to a top edit_group, ensure that the hierarchical object is an edit_group that does not already belong to another edit_group in the top edit_group's block, and that the top edit_group does not already contain a hier_member from the same block as the hierarchial object.

---

## NDMUI-786

(error) %s.

**Description**

redirect with -bg option is not allowed within redirect -bg.

**What Next**

Verify that you do not use -bg option to redirect when using within redirect -bg option.

---

## NDMUI-787

(error) Value '%s' for option '%s' must form a rectangle.

**Description**

The reported option must specify points which form a rectangle.

**What Next**

Provide the correct value to the reported option.

---

## NDMUI-788

(warning) Duplicate label '%s' specified in label list.

**Description**

The label list argument of this command must be a list of unique label names.

**What Next**

Remove the duplicate label names from the view label list argument.

Feedback

## NDMUI-789

(warning) Empty reference list specified.

**Description**

The reference list argument of this command must be a non empty list of unique reference names.

**What Next**

Indicate one or more references to add a label link map to.

## NDMUI-790

(warning) Could not remove pseudo_tsv_def '%s' - it is referenced by one or more pseudo_tsvs or bump_regions.

**Description**

A pseudo_tsv_def cannot be removed if it is referenced by one or more pseudo_tsvs - that is, one or more pseudo_tsvs has the pseudo_tsv_def as their *tsv_def* attribute. Additionally, a pseudo_tsv_def cannot be removed if one or more bump_regions has this pseudo_tsv_def specified in its *tsv_pattern* attribute.

**What Next**

To remove the pseudo_tsv_def, ensure that it is not referenced by any pseudo_tsv or bump_region.

## NDMUI-791

(error) Block boundary is not defined for block '%s'.

**Description**

Block boundary is not defined so track cannot be reported with respect to block boundary.

**What Next**

Set the boundary data for the block and then report track relative to block boundary.

## NDMUI-792

(error) Core area is not defined for block '%s'.

Feedback

**Description**

Core area is not defined so track cannot be reported with respect to core area.

**What Next**

Define core area for block and then report track relative to core area.

---

## NDMUI-793

(error) Start coordinate provided is outside block boundary. Please provide '%s' or '%s' also.

**Description**

Start coordinate of track is outside block bounadry so user must provide '%s' or '%s' so that track can be created in a definite range.

**What Next**

Please provide '%s' or '%s'.

---

## NDMUI-794

(error) Exactly one pin or port must be specified as input of '%s' option.

**Description**

This command accepts exactly one pin or port as part of '%s' option.

**What Next**

Please provide the correct input.

---

## NDMUI-795

(warning) Command 'report_utilization' doesn't support Hybrid site-array i.e. '%s'.

**Description**

For now, command 'report_utilization' doesn't support Hybrid site-array. In future, it may be supported. Hybrid site-array is a site-array which has site-rows of multiple site names or site-defs.

---

## NDMUI-796

(error) The column and row counts for the '%s' option must be positive values.

Feedback

**Description**

The value of the specified option or attribute must be a pair of positive integer values, specifying the desired pseudo_bump array size of the bump_region. If you want to unconstrain the array size of the bump_region, you may change the bump_region's 'boundary_mode' attribute to 'boundary', or setting the 'boundary' attribute directly will also change the 'boundary_mode'.

**What Next**

Please specify a valid value.

---

## NDMUI-797

(error) The name '%s' is not in correct format.

**Description**

The name of the internal design is not in correct format.

**What Next**

Please specify the name in correct format

---

## NDMUI-798

(error) The library containing the internal design is not open.

**Description**

The library containing the internal design is not open.

**What Next**

Please open the library and run the command again.

---

## NDMUI-799

(error) The specified internal design cannot be found.

**Description**

The specified internal design cannot be found.

**What Next**

Please specify a valid internal design as input.

Feedback

## NDMUI-800

(error) Cannot specify name '%s' as it overlaps with the system-generated names for '%s' objects.

### Description

Many objects have optional names the user may set. In the absence of a user specified name the objects have a system defined name, which is usually the name of the object type followed by a number. User names may not match the syntax of a system-generated name.

### What Next

Try choosing a different name that does not conflict with a system-generated name for the object.

## NDMUI-801

(error) Name '%s' is not unique among '%s' objects.

### Description

Many objects of the same type require unique naming. If a given name already exists for another object of the same type it may not be used again.

### What Next

Try choosing a different name for the object that does not match the name of an existing object of the same type.

## NDMUI-802

(error) Cell '%s' is not in the same hierarchy as the placement attraction.

### Description

Placement attractions only allow cells that exist in the same hierarchy. The cells that the user may add must exist in the same design context as the placement attraction.

### What Next

Try selecting a different set of cells, all of which have the same design context as the placement attraction.

## NDMUI-803

(error) Cell '%s' is not at the same level of physical hierarchy as the placement attraction.

Feedback

**Description**

Placement attractions only allow cells that exist at the same level of physical hierarchy.

**What Next**

Try selecting a different set of cells, all of which exist at the placement attraction's level of hierarchy.

---

## NDMUI-804

(error) Placement attraction regions may only be defined by at most two unique coordinates.

**Description**

A placement attraction's region may be one of a point, (axis-parallel) line, or rectangle. Therefore any desired region may be specified with no more than 2 unique coordinates.

**What Next**

Identify the desired region. If the region is a point, provide the same coordinate value twice. If the region is a line, provide the start and end coordinates. If the region if a rectangle, provide the coordinates of the lower-left and upper-right corners.

---

## NDMUI-805

(warning) Cell '%s' is already a member of placement_attraction '%s', skipping.

**Description**

A placement attraction may only contain one instance of a given cell.

**What Next**

The cell is already in the placement attraction.

---

## NDMUI-806

(warning) Cell '%s' is not a member of placement_attraction '%s', skipping.

**Description**

A cell cannot be removed from a placement attraction of which it is not a member.

**What Next**

The cell is already not a member of the placement attraction.

Feedback

## NDMUI-807

(error) Cell '%s' is not a physical cell.

**Description**

Only physical cells may be members of placement attractions.

**What Next**

Remove cells that do not have physical implementation from the cell list.

## NDMUI-808

(error) Invalid mask constraint list '%s'; %s.

**Description**

A mask constraint list argument is expected to contain a list of mask constraints. The specified list is not a valid mask constraint list for the specified reason.

**What Next**

Fix the offending list.

## NDMUI-809

(error) Invalid %s '%s' specified for delta displacement.

**Description**

The distance specified for delta displacement cannot be negative. The supported directions are east, north, west, south, northeast, northwest, southwest, southeast.

**What Next**

Correct the delta displacement.

## NDMUI-810

(error) Invalid delta '%s'.

**Description**

The format of delta is "distance:direction | {x-distance:x-direction y-distance:y-direction}". In the first form, any value of direction suffix from the set {east north west south northeast northwest southwest southeast} may be added to the numeric distance to specify the delta spacing for the horizontal, vertical, or 45-degree diagonal displacement, eg.

Feedback

100.5000:east. In the second form, a general {x y} displacement may be specified for any arbitrary direction using a pair of deltas where the first delta must have either :east or :west direction suffix and the second delta must have either :north or :south direction suffix. Eg. {100.5000:east 200.5000:south}.

**What Next**

Correct the errors in delta displacement(s).

---

## NDMUI-811

(error) %s specified for%s the displacement.

**Description**

At-least one delta must be specified for the first form of delta. At-most a pair of deltas may be specified for the second form of delta also known as g-delta with the horizontal and vertical components of the displacement for any arbitrary direction. The distance specified for delta displacement cannot be negative. The supported directions are east, north, west, south, northeast, northwest, southwest, southeast.

**What Next**

Correct the delta displacement.

---

## NDMUI-812

(error) Must specify %s in the specified delta displacement '%s'.

**Description**

The distance and direction both must be specified in the input delta. The distance specified for delta displacement cannot be negative. The supported directions are east, north, west, south, northeast, northwest, southwest, southeast.

**What Next**

Correct the delta displacement.

---

## NDMUI-813

(error) The %s delta must be %s with %s.

**Description**

The format of second form of delta also known as g-delta is "{x-distance:x-direction y-distance:y-direction}". In this form, a general {x y} displacement may be specified for any arbitrary direction using a pair of deltas where the first delta must have either :east

Feedback

or :west direction suffix and the second delta must have either :north or :south direction suffix. Eg. {100.5000:east 200.5000:south}.

**What Next**

Correct the errors in g-delta displacement(s).

---

## NDMUI-814

(error) Invalid items in the delta_list '%s'.

**Description**

The delta_list must be of the form "{ delta1 delta2 ... }". The format of delta is "distance:direction | {x-distance:x-direction y-distance:y-direction}". In the first form, any value of direction suffix from the set {east north west south northeast northwest southwest southeast} may be added to the numeric distance to specify the delta spacing for the horizontal, vertical, or 45-degree diagonal displacement, eg. 100.5000:east. In the second form, a general {x y} displacement may be specified for any arbitrary direction using a pair of deltas where the first delta must have either :east or :west direction suffix and the second delta must have either :north or :south direction suffix. Eg. {100.5000:east 200.5000:south}. The delta_list comprises of any arbitrary spequence of deltas in first or second form. Eg. { 200.5000:north {100.5000:east 200.5000:north} 100.5000:south ...}.

**What Next**

Correct the errors in the reported items of the delta_list.

---

## NDMUI-815

(error) The %s cannot be %s when %s.

**Description**

This error is reported when there is a dependency amongst multiple attributes.

**What Next**

Correct the reported attribute(s).

---

## NDMUI-816

(error) The specified %s must have exactly %u delta spacing values.

**Description**

In case of shape_pattern with non-uniform spacing exactly (m_dimension * n_dimension - 1) spacings must be specified in the input delta_list.

Feedback

**What Next**

Correct the number of deltas in the specified delta_list.

---

## NDMUI-817

(error) The '%s' and '%s' of %s cannot be both less than %u.

**Description**

This error is reported when there is a dependency amongst multiple attributes.

**What Next**

Correct the reported attribute(s).

---

## NDMUI-818

(warning) The edit_group '%s' cannot be modified because it is locked.

**Description**

The specified edit_group either has its 'is_locked' attribute set to true or is itself a member of a locked edit_group, so it cannot be modified or removed, nor can any of the member objects in its objects collection be modified or removed.

**What Next**

To edit the edit_group or any of its member objects, the edit_group must be unlocked by setting its 'is_locked' attribute to 'false', and it must also not be a member of a locked edit_group.

---

## NDMUI-819

(warning) The %s '%s' cannot be modified or removed, because it is a member of a locked edit_group.

**Description**

The specified object is a member of a locked edit_group, which is an edit_group with its 'is_locked' attribute set to 'true'. When in a locked edit_group, an object can neither be modified nor removed.

**What Next**

To modify or remove the object, the edit_group that contains the object must be unlocked by setting its 'is_locked' attribute to 'false'.

Feedback

## NDMUI-820

(warning) Cannot consider %s '%s' since it is %s.

### Description

The specified cell, lib_cell or pin cannot be considered as it is not valid. This could happen when command is expecting a cell of type register and you have specified a non-register, or vice-versa. Another possibility is that the cell or pin is not from the same block as current block. In case of lib_cell, this could happen when command is expecting a fault-tolerant lib_cell and you have specified a module, or vice-versa.

In case of safety cores, this could happen when command is expecting a logical hierarchical cell and user has specified a leaf level cell or cell with physical view or cell in different hierarchy. The same is applicable for setting an SPFM target on a cell.

### What Next

Ensure that the object is from the current block and is of the correct type.

## NDMUI-821

(warning) The %s '%s' is not of type '%s', but will still consider it

### Description

The specified cell, lib_cell or pin is not of the correct design type. But the command will still accept it. This could happen when a cell or lib_cell is specified but its design type is not well_tap.

### What Next

Ensure that the tap cell or lib_cell is of the correct design type

## NDMUI-822

(warning) Changing the register_type value might make the safety_register_rule data inconsistent.

### Description

The different attributes of a safety_register_rule must be consistent with each other. Otherwise the safety_register_rule is not valid. For example, if you change the register_type from fault_tolerant to group, you must also change the map_ref_name to specify a module instead of the earlier lib-cell.

Feedback

**What Next**

Please change the other attribute values accordingly.

---

## NDMUI-823

(error) Could not find %s with name '%s'.

**Description**

There is no module or lib_cell existing with the given name. Lib cell name must be specified in the form of lib_name/libcell_name. If you are specifying a module name, that module must exist in the current design.

**What Next**

Provide a valid module/lib_cell name

---

## NDMUI-824

(error) No argument specified.

**Description**

There is no argument specified.

**What Next**

Provide argument in command.

---

## NDMUI-825

(warning) The number of registers in the group (%d) does not match the redundancy mode (%d) of the rule.

**Description**

When creating a safety register group, or modifying the grouped registers via attribute, it must be ensured that the group has the same number of registers as specified by the redundancy mode of the rule which is applied on the group. This could also happen if one or more registered specified is not a valid candidate for being a part of the group.

**What Next**

Provide the corrent number of valid registers for the safety register group

Feedback

## NDMUI-826

(warning) Could not remove %s '%s'.

**Description**

There is some problem in remove. This could happen when object is getting used by some other object, or does not exists or not specified the valid value in command.

**What Next**

Provide valid argument in command.

## NDMUI-827

(error) Value provided with %s is either too large or too small.

**Description**

Value provided with argument is either too large or too small. It must be a positive value and also lesser than the block dimensions.

**What Next**

Provide the argument value in range.

## NDMUI-828

(error) %s values can be provided with %s.

**Description**

The list has too many or too few elements.

**What Next**

Provide a valid number of elements.

## NDMUI-829

(error) Invalid value %s for %s.

**Description**

Values provided with argument is not valid.

**What Next**

Provide the valid value for argument.

Feedback

## NDMUI-830

(warning) %s cannot be added in '%s' list.

**Description**

This message indicates that input objects provided cannot be added in the list.

**What Next**

Please exclude these objects from the input.

## NDMUI-831

(error) Cannot create a group as the rule is of type 'fault_tolerant'.

**Description**

The specified safety register group could not be created via create_safety_register_group command because the rule specified is of type 'fault_tolerant' instead of 'group'.

**What Next**

Please specify another rule with the type 'group'.

## NDMUI-832

(error) Cannot associate cell %s with rule %s as this cell is already associated with rule %s .

**Description**

The specified cell is already associated with some other rule.

**What Next**

Please specify another cell with the this rule.

## NDMUI-833

(warning) The %s output is not specified in the safety logic port map for '%s'.

**Description**

The safety logic port map for the specified module or lib-cell has not been set up correctly. For rules of type dual_mode, the error output must be specified prior to running this command. Similarly, for rules of type triple_mode, the voting outout must be specified before hand.

Feedback

**What Next**

Use set_safety_logic_port_map command to set the port mappings correctly for the given module or lib-cell.

---

## NDMUI-834

(error) Argument %s doesn't have any valid object.

**Description**

No valid objects could be found in the specified objects. This might be because of objects are ignored as they are not valid for that argument. There must be at least one valid object for the argument.

**What Next**

Please provide the correct inputs.

---

## NDMUI-835

(warning) Attribute 'max_num_stagger_tracks_table' cannot be set for via_rule %s. Value must be between 0 and 9.

**Description**

Value for the 'max_num_stagger_tracks_table' attribute must be between 0 and 9.

---

## NDMUI-836

(error) The customized-mask base via ratio is greater than 20 percent.

**Description**

The customized-mask base via ratio is greater than 20 percent. The capacity is worst than the situation in arbitray mask_pattern.

**What Next**

Update the mask_pattern as arbitray, then set the mask.

---

## NDMUI-837

(error) '%s' option is not supported when RTL restructuring mode is enabled.

**Description**

Command option specified is not supported when RTL restructuring mode in enabled.

Feedback

**What Next**

Verify that each option specified with command in RTL restructuring mode *set_app_options -list {rtlfp.flow.rtl_restructuring_mode true}* is supported. for example, *-hdl_block*|*-hdl_bussed*|*-hdl_all_blocks* options of *group_cells* and *-flatten*|*-force*|*-start_level* options of *ungroup_cells* are not supported.

---

## NDMUI-838

(error) '%s' command is disabled when RTL restructuring mode is enabled.

**Description**

Command specified is not allowed when RTL restructuring mode in enabled.

**What Next**

Verify that each command specified after enabling RTL restructuring mode *set_app_options -list {rtlfp.flow.rtl_restructuring_mode true}* is one of the supported commands within RTL restructuring mode. for example, restructuring commands *group_cells, ungroup_cells, write_restructured_rtl* etc. Most of the commands that modifies design data like, for example, *create*|*change*|*set*|*rename*|*remove*|*edit* commands are not allowed to be executed within RTL restructuring mode.

---

## NDMUI-839

(error) The number of cores in the group (%d) does not match the number of cores (%d) in the rule.

**Description**

When creating a safety core group, or modifying the cores via attribute, it must be ensured that the group has the same number of cores as specified by the number of cores of the rule which is applied on the group. This could also happen if one or more core specified is not a valid candidate for being a part of the group.

**What Next**

Provide the correct number of valid cores for the safety core group

---

## NDMUI-840

(warning) Cannot %s macro group move bound : %s

Feedback

**Description**

Bound shapes can only be added to a macro group move bound while creating the move bound using *create_bound* command.

Bound shapes can not be added or removed from a macro group move bound using *create_bound_shape* or *remove_bound_shapes*.

**What Next**

For adding or removing the bound shapes to a macro group move bound, use *remove_bounds* command to delete the macro group move bound and then use *create_bound* to create the bound with the desired bound shapes.

**See Also**

- create_bound
- remove_bounds
- create_bound_shape
- remove_bound_shapes

---

## NDMUI-841

(warning) macro_group move bound 'type' cannot be changed

**Description**

The type of the macro_group cannot be changed.

---

## NDMUI-842

(info) Skipped pin_pair {%s %s} as it already exists in supernet exceptions.

**Description**

The specified pin_pair already exists in supernet exceptions.

---

## NDMUI-843

(error) Specified start and stop anchors are not on same physical hierarchy.

**Description**

Specify start and stop anchors from same physical hierarchy only.

Feedback

## NDMUI-844

(error) No path exists between specified start and stop anchors.

**Description**

No path exists between start and stop anchors.

## NDMUI-845

(warning) %s must be provided if number of valid cores is 2

**Description**

When creating a safety core group or setting cores in safety core rule, error signal must be provided if number of valid cores is 2 to handle the error.

**What Next**

Provide the correct number of valid cores for the safety core group or rule.

## NDMUI-846

(error) Different specification formats for placement separation and routing separation.

**Description**

When creating a safety core group, the routing guardband and distance must be specified in the same format. Either both should be a single radial value {r}, or both should be in terms of {x,y} pairs.

**What Next**

Provide routing_guardband in the same format as distance.

## NDMUI-847

(error) Cannot create failsafe fsm group as specified register %s is duplicated or there is overlap between state, parity and sync registers.

**Description**

Register(s) specified as state, parity and sync should be mutually exclusive. There should not be any overlap between state, parity and sync registers.

Feedback

### What Next

Ensure that the register(s) specified as part of state, parity and sync are mutually exclusive.

---

## NDMUI-848

(error) Cannot create safety core group as specified %s '%s' is already part of safety core group '%s'.

### Description

One of the specified voting / logic cell is already part of another safety core group as voting / logic cell or one of the specified safety cores is already part of another safety core group as safety core or One of the specified delay cell is already part of another safety core group as delay cell. Overlapping of safety core / voting / logic cell(s) is not allowed.

### What Next

Ensure that the safety core / voting / logic / delay cell is part of one safety core group only.

---

## NDMUI-849

(error) The topology_node's 'origin' attribute cannot be modified unless its constraint_type is 'origin'.

### Description

The topology_node's 'origin' attribute cannot be directly modified unless its 'constraint_type' attribute value is 'origin', because the value of 'origin' is automatically calculated for all other constraint modes.

### What Next

Only set the 'origin' attribute value directly when the topology_node's 'constraint_type' attribute value is 'origin'.

---

## NDMUI-850

(error) Unable to set 'offset' relative to core area as 'cycle' is not set for site-row '%s'.

### Description

'cycle' must be set before setting 'offset' for a site-row while setting offset relative to core area. Setting offset relative to core area uses cycle value so it must be set before setting offset.

Feedback

### What Next

Set the cycle first and then try to set offset relative to core area.

---

## NDMUI-851

(error) Unable to create via rule as -cut_rows and -cuts_per_row are inconsistent to each other.

### Description

The options -cut_rows and -cuts_per_row are expected to be consitent with each other. The correct format for -cut_rows is {A1 A2 …. An 1} and for -cuts_per_row is {1 A1 A2 … An}. Any format other than this will lead to a corrupted via rule and creation of via rule will be aborted.

### What Next

Make sure options -cut_rows and -cuts_per_row are in a proper format to create the via rule.

---

## NDMUI-852

(information) Setting the bump_region's pattern attribute to the default bump_region_pattern '%s'.

### Description

The bump_region you are creating will have its *pattern* attribute automatically set to the specified bump_region_pattern, which is the current default bump_region_pattern in the block. The default bump_region_pattern is automatically used when the *-pattern* option of the command is unspecified.

### What Next

No action needed.

### See Also

- create_bump_region
- create_bump_region_pattern

---

## NDMUI-853

(error) Missing '-pattern' option and no default pattern exists.

Feedback

**Description**

Could not create the bump_region object - a bump_region_pattern object is required as the reference pattern, however there was none specified, and no default bump_region_pattern exists in the block.

**What Next**

Please specify a bump_region_pattern using the command's *-pattern* option, or set an existing bump_region_pattern as the default pattern by setting its *is_default* attribute to true, or create a new bump_region_pattern object using the *-default* option of the create command.

---

## NDMUI-854

(error) Problem reading %s: %s.

**Description**

Could not read the attribute value for the specified reason.

**What Next**

Please specify a valid attribute value. A description of the attribute format can be found in the man page of the *create_bump_region* command.

**See Also**

- create_bump_region

---

## NDMUI-855

(error) Problem resolving %s '%s': %s.

**Description**

Could not find or use the object with the specified name for the specified reason.

**What Next**

Please specify the name of an existing object in the correct block.

---

## NDMUI-856

(error) The '%s' option value must be a distance list of length 2.

Feedback

**Description**

The specified option requires a value that is a distance list of length 2, where the first element is the x offset and the second element is the y offset.

**What Next**

Please specify a distance vector of length 2.

---

## NDMUI-857

(error) Could not set the %s's %s to '%s' - the %s must be in the same block.

**Description**

Could not set the specified option or attribute to the specified object, because the object is in a different block.

**What Next**

Specify an object from the same block.

---

## NDMUI-858

(error) The bump_region's '%s' option value must be a distance list of length 1, 2, or 4.

**Description**

The specified option requires a value that is a distance list of length 1, 2, or 4. This option specifies a margin between the bump_region's boundary and the pseudo_bumps it contains. If the length is 1, then the value imposes that margin on all sides. If the length is 2, then the first value is the margin for the left and right, and the second value is the margin for the top and bottom. If the length is 4, then the values are the margins for the left, bottom, right, and top, respectively.

**What Next**

Please specify a distance vector of length 1, 2, or 4.

---

## NDMUI-859

(error) Invalid value for '-index_range' option

Feedback

### Description

An invalid value was specified for the '-index_range' option of the 'get_pseudo_bumps' command. The value must have one of the following two formats:

"{<column_index> <row_index>}" to specify a single index value, where <column_index> and <row_index> are integer values within the range of 16-bit signed integer values, or

"{{<column_index_ll> <row_index_ll>} {<column_index_ur> <row_index_ur>}}" to specify an inclusive range of indices, where <column_index_ll>, <row_index_ll>, <column_index_ur>, and <row_index_ur> are integer values within the range of 16-bit signed integer values, and the following relationships are true: <column_index_ll> <= <column_index_ur> and <row_index_ll> <= <row_index_ur>

### What Next

Specify a value index range value.

---

## NDMUI-860

(warning) The '%s' option or attribute is deprecated - please use the 'anchor' object to implement bump_region anchoring.

### Description

The specified anchoring-related command option or attribute is deprecated and non-functional. Please use the "anchor" object to anchor the bump_region to another object.

### What Next

Use the "anchor" object for anchoring.

### See Also

- create_anchor

---

## NDMUI-861

(error) Could not set the anchor's dependent object to '%s' - it is already the dependent object of anchor '%s'.

### Description

Could not set the specified object as the dependent object of the anchor, because the object is already the dependent object of another anchor. An object may only be the dependent object of up to one anchor.

Feedback

**What Next**

Specify a different dependent object, or remove the object as the dependent object of the other anchor.

---

## NDMUI-862

(error) Could not set the anchor's dependent object to '%s' - the object must be in the same block as the anchor.

**Description**

Could not set the specified object as the dependent object of the anchor, because the object is in a different block.

**What Next**

Specify an object from the anchor's block.

---

## NDMUI-863

(error) Anchor's %s command option must be set when its type is '%s'.

**Description**

Certain anchor command options are required based on the anchor's 'type' attribute value.

**What Next**

Please specify the command options required for the anchor's type.

---

## NDMUI-864

(error) Anchor's '%s' %s cannot be set unless its type is '%s'.

**Description**

The settability of certain anchor attributes or command options depends on the anchor's 'type' attribute value. In this case, you attempted to set the specified attribute or command option even though the anchor has the wrong type for that attribute or command option.

**What Next**

Only set anchor attributes and command options appropriate to the anchor's type.

Feedback

## NDMUI-865

(warning) Could not implement pseudo_bumps for bump_region '%s' - the number of pseudo_bumps would exceed the system limit of %s.

### Description

The current configuration of the specified bump_region and the bump_region_pattern it references results in a pseudo_bump count that exceeds the specified system limit, therefore the application could not populate the bump_region with pseudo_bumps.

### What Next

Change the configuration of the bump_region and/or its bump_region_pattern such that a reasonable number of pseudo_bumps would be created, or modify the system limit by increasing the value of the *design.bump_region_pseudo_bump_count_limit* app option. Note that increasing the system limit may result in a significant increase in runtime and memory usage.

## NDMUI-866

(error) The topology_node's intrinsic_delay must be a list with no more than two values

### Description

The 'intrinsic_delay' attribute of the topology_node must be a list of one or two values, where each value can be either a distance or a percentage. Percentage values must be appended with the '%%' symbol.

### What Next

Specify a list of values no longer than two.

## NDMUI-867

(error) Could not create TSV cell '%s': %s

### Description

Could not create the TSV cell due to the specified reason(s).

### What Next

Please correct the specified problem(s).

Feedback

## NDMUI-868

(error) %s %s are in un-editable hierarchy and hence the command is not executed, few of which are %s.

**Description**

Unable to make modification in un-editable hierarchy.

## NDMUI-869

(error) %s %s is un-editable and hence the command is not executed.

**Description**

Design is in un-editable hierarchy.

## NDMUI-870

(error) Auto-saving is already in active state.

**Description**

This indicates auto-saving has already enabled and to suspend auto-saving please use stop_auto_save

**What Next**

To change auto-save settings, please do stop_auto_save and then do start_auto_save with the desired settings.

**See Also**

- start_auto_save
- stop_auto_save

## NDMUI-871

(error) Frequency must be initialized.

**Description**

Frequency must be set atleast once and doesn't required to be set again if auto-saving is stopped and restarted unless the frequency needs to be changed.

Feedback

### See Also

- start_auto_save

---

## NDMUI-872

(error) Frequency must more than or equal to "%s" seconds.

### Description

Minimum time interval between two auto-saves must more than or equal to 1800 seconds (30 minutes).

### See Also

- start_auto_save

---

## NDMUI-873

(error) Unable to create "%s" specified auto-save directory.

### Description

Check whether specified directory already exists and doesn't have sufficient permission to both read and write to the directory.

### WHAT'S NEXT

Try specifying another directory path.

### See Also

- start_auto_save

---

## NDMUI-874

(error) Size specification "%s" didn't match supported specification.

### Description

Units for "-allowed_disk_space" option is GigaBytes or MegaBytes. Size specification should have size (integral-part) followed by tags G, g, GB, Gb, gb, M, m, MB, Mb or mb.

### WHAT'S NEXT

Specify valid size specification.

Feedback

**See Also**

- start_auto_save

---

## NDMUI-875

(error) Specified size should be more than zero.

**Description**

Size should more than zero but make sure atleast it is more than current library size.

**See Also**

- start_auto_save

---

## NDMUI-876

(error) Number of auto-save snapshots should be more than one.

**Description**

Number of the auto-save snapshots should be more than one if auto-saving is being started.

**See Also**

- start_auto_save

---

## NDMUI-877

(information) Auto-saving started for the current session.

**Description**

Any modifications to the current library beyond this point will be auto-saved given all the prerequisite auto-save settings has met like frequency etc.

**See Also**

- start_auto_save

---

## NDMUI-878

(error) Auto-saving is not in active state.

Feedback

### Description

Auto-saving can be stopped only if auto-saving is active previously.

### See Also

- stop_auto_save
- start_auto_save

---

## NDMUI-879

(information) Auto-saving stopped for the current session.

### Description

Any modifications to the current library beyond this point wont be auto-saved.

### See Also

- stop_auto_save

---

## NDMUI-880

(error) Current library must be defined or "-library" must be specified.

### Description

Set current library or specify library name (for opened library)/ full library path (for unopened library) using -library option.

### See Also

- report_auto_save
- remove_auto_save

---

## NDMUI-881

(error) Specified library '%s' doesn't exist.

### Description

Set current library or specify library name (for opened library)/ full library path (for unopened library). using -library option.

### See Also

- report_auto_save
- remove_auto_save

---

## NDMUI-882

(error) The command %s was not executed because of an ECO policy violation.

### Description

The design.eco_restrictions app_option controls the behavior of the reported command. The app-option when true checks if a netlist change is safe before performing it. when the app-option is true command violating eco policy cannot be performed.

### What Next

If you are sure that the netlist change is safe, you need to set the app_option design.eco_restrictions to false.

### See Also

- disconnect_net
- remove_nets
- create_pin
- remove_pins
- connect_pins
- change_reference
- set_reference
- connect_net
- create_port
- create_net
- remove_cells
- change_link
- design.eco_restrictions

Feedback

## NDMUI-883

(error) In command %s, Custom via def %s cannot be used to create a routing rule. Only simple via defs may be specified.

### Description

This error message indicates that the specified NDR via is an special via.The current NDR engine in the router can only handle NDR vias with only one via enclosure on each layer.Zroute NDR routing doesn't support special via(vias with more than one via enclosures on one metal layer) as NDR vias.

### What Next

Please specify the NDR via using simple_via_def.

### See Also

- create_routing_rule

## NDMUI-884

(information) Attribute is applicable to objects of type %s.

### Description

The given attribute is applicable to objects of reported type created by the command create_pin_constraint. Please ensure that the attribute is being set for the correct object type.

### What Next

Please specify objects of reported type.

### See Also

- create_pin_constraint
- get_pin_constraints

## NDMUI-885

(information) No %s option specified. Constraints will not be set for any object.

### Description

Any individual pin constraint must be set for either a pin, port, or net. The given pin constraint doesn't have any of these options set.

Feedback

**What Next**

Please ensure you provide a pin, port, or net option.

**See Also**

- create_pin_constraint

---

## NDMUI-886

(error) Pin constraint %s should have only one side set.

**Description**

Trying to set pin placement offset for a pin constrait that has more than one side set.

**What Next**

Please ensure only one side is set before trying to set offset.

**See Also**

- create_pin_constraint

---

## NDMUI-887

(warning) No physical data defined.

**Description**

The tool cannot find the physical data from the database. Please check the design input.

**What Next**

Please check the design input and make sure that the physical data exists before running this command.

**See Also**

- create_pin_constraint
- write_pin_constraints

---

## NDMUI-888

(error) Cannot parse the value

Feedback

### Description

The value specified for the argument is not in the correct format.

### What Next

Please check that the values are specified in the correct format. The attribute expects either a single size value (eg., 10) or white space seperated layer-name and size pairs (eg., {METAL5 10}), possibly surrounded by {}.

### See Also

- create_pin_constraint

---

## NDMUI-889

(error) Offset range number must be in same sign.

### Description

The offset range specified should be both +ve or both -ve.

### What Next

Please ensure that the value specified have the same sign. Run the command again with both values either +ve or -ve.

### See Also

- create_pin_constraint
- get_attribute
- set_attribute

---

## NDMUI-890

(error) Range specified must be different.

### Description

The values specified for range must be different.

### What Next

Please ensure that the value specified are different. Run the command again with both values different specifying a range.

Feedback

**See Also**

- create_pin_constraint

---

## NDMUI-891

(warning) The object %s is not appropriate for option %s.

**Description**

This option requires objects of a certain type, but the given object is not that type. This object will be ignored.

**What Next**

Check the command man page to see correct objects types for option.

**See Also**

- create_pin_constraint

---

## NDMUI-892

(error) Object %s not in the top-level physical hierarchy of the current design.

**Description**

The object should be in the top-level physical hierarchy of the current design. This command cannot manipulate constraints on objects in sub-blocks.

**What Next**

Change the current_design to the physical hierarchy directly enclosing the object.

**See Also**

- create_pin_constraint

---

## NDMUI-893

(warning) Duplicate reference module %s detected in option %s. Instance %s ignored.

**Description**

This command option requires unique reference modules. In other words, you can specify only 1 cell instance from an MIB (multiply instantiated block) group. You have specified 2 or more cell instances with the stated reference module. The stated instance will be ignored.

Feedback

### What Next

Specify only 1 cell instance in an MIB group. It does not matter which one.

### See Also

- create_pin_constraint

---

## NDMUI-894

(warning) Skipping power or ground net %s. Power or ground nets are excluded for %s.

### Description

Creation of pin constraint on nets which are power or ground is not allowed.

### What Next

Runn the command again without power or ground nets.

### See Also

- create_pin_constraint

---

## NDMUI-895

(warning) Net %s is tied to power or ground net. The individual pin constraint is applied to the connected %s %s instead %s.

### Description

The reported net is a power or ground net. Individual pin constraints cannot be applied to power or ground net. Therefore, the tool is applying the same constraint to the applicable pins or ports instead.

### What Next

Check if the signal net tied to power or ground is user's intention.

### See Also

- create_pin_constraint

---

## NDMUI-896

(warning) Pin offset/range or start/end specified for side %d of cell %s is outside the block boundary.

Feedback

### Description

Offset/range or start/end values specified for pins/port lies outside the block boundary and hence will be ignored.

### What Next

Input a valid offset/range or start/end value which is within the limits of the associated block edge's perimeter.

### See Also

- create_pin_constraint

---

## NDMUI-897

(warning) Constraints modified on an uneditable(read-only) block %s and it will be ignored.

### Description

This message is issued when a read-only block is being modified.

### What Next

Check to see if this block is reasonable to be modified.

### See Also

- create_pin_constraint

---

## NDMUI-898

(warning) The value specified for range is incorrect.

### Description

The values specified for range must represent a span across which the pins can be placed.

### What Next

Provide two distinct values representing the range indicating a span across a side.

### See Also

- create_pin_constraint

Feedback

## NDMUI-899

(warning) Side number %d is larger than number of sides for block %s, side constraint will be ignored.

### Description

A side number is specified for side or exclude_side constraints, but this number is larger than the total number of side on selected block.

### What Next

Correct the side number then reset the constraint again.

### See Also

- create_pin_constraint

## NDMUI-900

(warning) %s. MIM grouping is disabled, continuing with group_cells only.

### Description

The tool cannot continue with *MIM grouping* because of the reason(s) specified in this message. *MIM grouping* is disabled, it will continue with normal grouping.

### What Next

Fix the reported issue that caused disabling of *MIM grouping* and try again.

### See Also

- group_cells

## NDMUI-901

(error) MIM grouping failed because '%s'.

### Description

*MIM grouping* can fail for various reasons. This error message points out one of the possible reasons why *MIM grouping* failed.

### What Next

Fix the reported issue that caused failure of *MIM grouping*.

Feedback

**See Also**

- group_cells

---

## NDMUI-902

(warning) Exclude-side constraint set for %s %s includes all the sides of it's parent block.

**Description**

All sides of the block are disallowed for pin placement. This could have bad consequences on place_pins outcome and resulting QoR.

**What Next**

Update the exclude sides constraints.

**See Also**

- remove_pin_constraints

---

## NDMUI-903

(error) Pin location coordinate for %s %s is outside of block cell boundary and will be ignored.

**Description**

If pin location is not inside block boundary, the set_pin_constraints will ignore all setting and error out.

**What Next**

Please check pin location coordinate and make it inside block boundary

---

## NDMUI-904

(warning) Net %s does not connect to cell %s. The constraint is skipped. %sPlease consider using topological constraints.

**Description**

The net and cell are not connected. The constraint is skipped. This case should be constrainted with topological constraints instead of physical pin constraints.

**What Next**

Please consider using topological constraints to constrain this case.

Feedback

**See Also**

- read_pin_constraints
- create_topological_constraint

---

## NDMUI-905

(information) The following blocks are disabled for planning:%s.

**Description**

This message reports the blocks that are disabled for planning in the current design.

**What Next**

Please ensure that you run set_editability for the block to allow changes to the block.

**See Also**

- set_editability

---

## NDMUI-906

(information) %s pin constraints associated with %s are removed from block %s.

**Description**

This message displays information about individual pin constraints that are removed from blocks.

**See Also**

- remove_pin_constraints

---

## NDMUI-907

(information) The following blocks do not have associated technology files:%s.

**Description**

This message reports blocks that do not have associated technology files. If a block does not have a technology file with it, the tool cannot know physical data needed, e.g. minimum width allowed on a certain layer. Therefore, any operations or checks that are related to physical data cannot be performed.

Feedback

### What Next

Check that the technology file is correctly set.

### See Also

- create_lib
- set_ref_libs

---

## NDMUI-908

(warning) Exclude side is set for the pin constraint.

### Description

The pin constraint has exclude side set. You cannot have both side and exclude side attribute set for a pin constraint.

### See Also

- create_pin_constraint

---

## NDMUI-909

(error) Pin constraint %s has exclude side set.

### Description

Trying to set pin placement offset for a constraint that has exclude side set.

### What Next

Please ensure that the pin constraint has no sides excluded before setting offset.

### See Also

- create_pin_constraint

---

## NDMUI-910

(warning) Specified width/length for layer %s is less than the layer minimum width/length %s.

Feedback

### Description

The user defined pin width or length constraint, or the non default routing rule width associated with net on the pin, is less than the minimum value defined by the technology rule. As a result, the user defined constraint is ignored for pin placement or pin check.

### What Next

Please double check the technology rule to make sure that user defined width or length constraint, or the non default routing rule, does not violate those defined in technology rule.

---

## NDMUI-911

(warning) %s constraint applied on net %s will be ignored.

### Description

Location/off_edge constraints are not allowed on nets. Therefore, such constraints specified by the user for a net will be ignored.

### What Next

Instead, apply the location/off-edge constraint on the pin that connects to the corresponding net.

### See Also

- create_pin_constraint

---

## NDMUI-912

(warning) Command %s can't find a logical hierarchy for reference block %s.

### Description

During a hierarchical command, the command failed to find a logical hierarchy for a physical reference block. Sometimes, when there is a problem with linking, the ICC2 database may not get correctly updated to include this logical hierachy to physical reference block mapping.

### What Next

Make sure that your design includes all necessary reference libraries in the library search path.

Feedback

## NDMUI-913

(information) %s not set for given constraint.

**Description**

The mentioned attribute is not set for the given pin constraint.

**What Next**

Please ensure the mentioned attribute is set. To set the value of a settable attribute please use the set_attribute command to set it before querying it.

**See Also**

- create_pin_constraint
- get_attribute
- set_attribute

## NDMUI-914

(Error) Cannot set attribute '%s' for rows belonging to an array.

**Description**

Site rows belonging to a site array cannot be modified independently w.r.t attributes-'name', 'orientation', 'origin', 'site_count', 'site_orientation', 'site_space'.

**What Next**

Derived site rows can be modified by modifying site array attributes.

## NDMUI-915

(information) cell %s is unmapped.

**Description**

The cell is unmapped in the design.

**What Next**

If the design has not undergone synthesis, this information can be ignored. Otherwise, you should examine the cell that is reported to see why it is still unmapped.

Feedback

## NDMUI-916

(warning) Object '%s' is not connected to net '%s'.

**Description**

The object is not conneted to the net.

**What Next**

This is only a warning message. No action is required.

## NDMUI-917

(error) cannot create port bus '%s' as existing port '%s' has different direction from other existing ports which can be part of port bus.

**Description**

Existing ports which will be part of port bus are not having same direction so unable to create port bus. All ports of port bus should have same direction. Please use command *get_attribute [get_ports portBusName*] direction* to check direction of ports.

**What Next**

Change direction of existing ports which can be part of port bus and make it same for all ports.

## NDMUI-918

(error) cannot create port bus '%s' as existing port(s) don't have same direction as provided in command option.

**Description**

Existing ports which will be part of port bus are not having same direction as requested in command option so unable to create port bus. All ports of port bus should have same direction. Please use command *get_attribute [get_ports portBusName*] direction* to check direction of ports.

**What Next**

Change direction of existing ports which can be part of port bus or change command option value.

## NDMUI-919

(warning) The requested array_size for the bump_region could not be achieved.

Feedback

**Description**

Due to the geometry of the bump_region's boundary or other configurations, the requested pseudo_bump array_size could not be implemented. Note that the actual array_size is computed based on the highest column and row indicies over all existing pseudo_bumps. This message will appear if a particular array_size was requested, but the resulting pseudo_bumps do not span that array_size range.

**What Next**

Adjustments to the bump_region's boundary, pattern configuration, or other attributes should allow the requested array_size to be achieved.

---

## NDMUI-920

(Warning) Cannot change net_type from '%s' to '%s' on net '%s' as it is overridden.

**Description**

If the net type of a net is overridden by some other segment of that net of type power or ground or constant, then the type of the net can't be modified using the command "set_attribute". "set_net_type" command can be used to change the type.

**What Next**

Use the command "set_net_type" to change the type of the net.

---

## NDMUI-921

(error) site-def '%s' doesn't exist.

**Description**

site-def with name provided as input doesn't exist tech file.

**What Next**

Rectify the input data as per details in error message.

---

## NDMUI-922

(error) incorrect input format for option '%s', should be {composite_site_def_A {elementary_site_def_1 elementary_site_def_2 ...} composite_site_def_B {elementary_site_def_3 elementary_site_def_4 ...} ...}

Feedback

**Description**

Input should have one or more pair of composite site_def, followed by two or more elementary site_defs.

**What Next**

Please correct the input format.

---

## NDMUI-923

(error) Incorrect value specified. Value range is [0-15] inclusive.

**Description**

Input should be within the range 0 to 15 inclusive.

**What Next**

Please specify a valid range for the mentioned value.

---

## NDMUI-924

(error) Invalid check %s.

**Description**

This check is not valid because it is not in the list of supported checks.

**What Next**

Execute the *report_early_data_checks -policy -hierarchical* command to get a list of valid checks. Specify a check from this list.

---

## NDMUI-925

(error) Configuration not set.

**Description**

The early data checks configuration is not set.

**What Next**

Set the early data checks configuration using the *set_early_data_check_policy -policy <policy_name>* command.

Feedback

## NDMUI-926

(error) Invalid strategy %s.

**Description**

This strategy is not valid because it is not in the list of supported strategies.

**What Next**

Execute the *report_early_data_checks -checks <check_name>* command to get a list of valid strategies for the checks. Specify a strategy from this list.

## NDMUI-927

(warning) Cell %s does not exist.

**Description**

Cells are moved in the same order as specified in the command. There are chances that due to movement of the previous cells, current cell does not exist.

**What Next**

Please check for cell being moved and also the order of the cells being moved if there are more than one cells.

## NDMUI-928

(warning) No cell to be moved.

**Description**

There are no cells to be moved. This may be due to restriction of the command or incorrect argument.

**What Next**

Please check the warning messages before this one from the command.

## NDMUI-929

(warning) %s %s cannot be uniquified.

**Description**

Hierarchical cell needs to be uniquified for reparenting of the specified cell. But due to some restriction it cannot be uniquified.

Feedback

### What Next

Please check for restriction on uniquification.

---

## NDMUI-930

(warning) Unable to ungroup cells.

### Description

The *ungroup_cells* command is unable to ungroup the specified cells due to problems that were detected while checking the cell list. These problems are reported separately.

### What Next

Examine the log file and correct any reported errors, then try the command again.

### See Also

- ungroup_cells

---

## NDMUI-931

(warning) %s %s not in current design %s.

### Description

Reparenting is allowed only in the current design. Cell being moved and the new specified parent should be in the current design.

### What Next

Please check if the cell being moved and the new parent are in current design.

---

## NDMUI-932

(warning) Cell %s cannot move to %s.

### Description

Cell being moved is either parent cell or ancestor cell of the new specified parent.

### What Next

Please check cell being moved and the new specified parent.

Feedback

## NDMUI-933

(Error) Current block does not have the physical information for %s '%s'.

**Description**

The physical representation of the scan-in and scan-out pin should exist in the current block. Element's scan-in and scan-out pins cannot be created across physical hierarchies.

## NDMUI-934

(Error) The scan-in pin '%s' and scan-out pin '%s' must be from the same scan cell.

**Description**

The scan-in and scan-out pin of each triplet must be from the same scan cell.

## NDMUI-935

(Error) %s pin '%s' is part of an existing stub chain '%s'.

**Description**

Neither scan-in nor scan-out pin of any stub chain element can refer to an existing stub chain element.

## NDMUI-936

(error) Scan Chain with name '%s' already exists.

**Description**

A Scan Chain with the given name already exists. Scan Chain name must be unique.

**What Next**

Please specify another value which is unique for the scan chain.

## NDMUI-937

(error) Stub Chain with name '%s' already exists.

**Description**

A Stub Chain with the given name already exists. Stub Chain name must be unique.

Feedback

**What Next**

Specify unique value for the stub chain name.

---

## NDMUI-938

(error) Term '%s' is already part of an existing stub chain '%s'.

**Description**

Startpoint or Endpoint term cannot be part of an existing Stub Chain. unique.

**What Next**

Remove the term from the existing stub chain or provide another term.

---

## NDMUI-939

(error) The element list must be of the form {{{scan_in_pin_name scan_out_pin_name bit_length} ...} ...}

**Description**

Required list format for floating/ordered elements is {{{scan_in_pin_name scan_out_pin_name bit_length} ...} ...}

**What Next**

Provide element list in the specified format.

---

## NDMUI-940

(warning) Failed to add stub chain '%s' to scan chain '%s'.

**Description**

Stub Chain addition to the scan chain failed.

**What Next**

Provide appropriate value in -at option.

---

## NDMUI-941

(error) -at option value is out of range for stub chain '%s' having length '%s'.

Feedback

**Description**

Value provided in -at option cannot be greater or equal to the length of any stub chain to be updated with -remove option or greater than the length of any stub_chain to be updated with -add_floating_elements/ -add_ordered_elements options.

**What Next**

Provide appropriate value in -at option.

---

## NDMUI-942

(warning) Cannot remove stub chain '%s' from scan chain '%s'.

**Description**

Stub Chain removal from the scan chain failed. A stub chain must be part of scan chain to be removed successfully.

---

## NDMUI-943

(error) Option %s cannot be empty.

**Description**

Value specified for the option is empty. Non empty value is required.

**What Next**

Specify non-empty value.

---

## NDMUI-944

(warning) Cannot remove %s '%s'.

**Description**

Delete operation failed.

---

## NDMUI-945

(info) Removed %s '%s'.

**Description**

Delete operation performed successfully.

Feedback

## NDMUI-946

(error) Failed to create stub chain element '{%s %s %s}'.

**Description**

Stub chain element creation failed.

## NDMUI-947

(error) Bit length of element '{%s %s %d}' is out of range [0, %d].

**Description**

Each stub chain element must have a bit length within the specified range.

**What Next**

Specify a valid bit length for the stub chain element.

## NDMUI-948

(error) -at option value is out of range for scan chain '%s' having length '%s'.

**Description**

Value provided in -at option value must be less than the length of the scan chain.

**What Next**

Provide appropriate value in -at option.

## NDMUI-949

(error) %s i.e. '%s' provided as part of option %s is not valid. %s

**Description**

One of the value in specified option is not valid or doesn't exist. Specified option takes list of element_extension where one element_extension is index (unsigned integer), direction and distance (separated by space) as one value like below

{{0 bottom 1.1} {1 horizontal 11.1}}

In this index should be less than number of object types or lib_cells specified. Direction should be valid direction and distance should be greater than equal to 0.

Feedback

**What Next**

Please check the option value specified is correct.

---

## NDMUI-950

(error) The option '%s' value '%s' is not in correct format.

**Description**

The specified option value is not in correct format. User has to provide direction and distance (separated by space) as one value i.e. direction_extension. User can provide list of direction_extension values like below {{bottom 1.1} {horizontal 11.1}}

**What Next**

Please check the option value specified is correct.

---

## NDMUI-951

(Error) Attribute 'row_offsets' can not be set as 'row_cycles' doesn't exist for site-array '%s'.

**Description**

This message indicates that site-array doesn't have row_cycles set. For setting row_offset, row_cycles must be set on site-array as without cycle attribute, offset doesn't have any significance.

**What Next**

Please set the row_cycles attribute on site-array first.

---

## NDMUI-952

(warning) Design %s is linked and will be packed even though provided in exclusion list for labels or views.

**Description**

Input design is linked and will be packaged.

---

## NDMUI-953

(warning) The associated object '%s' of type '%s' to '%s' of type '%s', is not presetn in design '%s' and association skipped.

Feedback

**Description**

The associated object is not present in destination design and association skipped in the command "copy_floorplan".

**What Next**

Include associated object type in copy_floorplan to copy if suppurted by copy_floorplan. check help of copy_floorplan

---

## NDMUI-954

(warning) The force option not applicable for '%s'.

**Description**

The object can not be copied forcefully in destination design.

**What Next**

This is an informational message only.

---

## NDMUI-955

(warning) The associated object '%s' of type '%s' to '%s' of type '%s', is not presetn in design '%s' and object copy skipped.

**Description**

The associated object is not present in destination design and object creation not possible.

**What Next**

Include associated object type in copy_floorplan to copy if supported by copy_floorplan. check help of copy_floorplan

---

## NDMUI-956

(warning) Design '%s' is not present in Lib '%s'

**Description**

Source design is not present in destination Lib.

**What Next**

Copy design in current Lib.

Feedback

## NDMUI-957

(warning) The via def '%s' specified is of type custom via def.

**Description**

The via def specified should be of type simple via def and specifying custom via def is not allowed.

**What Next**

Please specify a valid via def of type simple via def.

## NDMUI-958

(Error) Layer '%s' is not from the tech of the current lib '%s'.

**Description**

The specified layer is not from the tech database of the library containing the block in which the shape is being created.

**What Next**

Use a layer object from the current Lib.

## NDMUI-959

(information) Auto-saving of lib is still under process in background.

**Description**

Auto-saving of lib is still under process in background.

## NDMUI-960

(information) No Auto-saved library is present for the library '%s'.

**Description**

No auto-saved library is present for the library, original library will be opened.

## NDMUI-961

(warning) CLIB flow is not recommended for %s. It is preffered to switch to Cache-free flow.

Feedback

**Description**

CLIB does not hold enough information to execute the specified command.

**What Next**

Use create_ref_libs to generate reference liibraries.

---

## NDMUI-962

(error) One of the value in option '%s' i.e. '%s' is not in correct format.

**Description**

The specified option value is not in correct format. User has to provide object type or lib-cell with 'lib_cell' keyword like below. {object_type {lib_cell A} {lib_cell B} object_type}

**What Next**

Please check the option value specified is correct.

---

## NDMUI-963

(error) Could not open file %s for writing.

**Description**

Could not open the specified file for writing. This could be because of one of the following reasons:

- The file path is invalid.
- You do not have read and write permissions for the specified file and directory.

**What Next**

Ensure the following:

- The file path is valid.
- You have read and write permissions for the specified file and directory.

---

## NDMUI-964

(error) Policy %s is not supported for all checks.

**Description**

The specified policy is not supported for all checks.

Feedback

**What Next**

If you specify both the -checks and the -policies options, the configuration supports strict, error, tolerate, repair and lenient policies. If you specify only the -policy option and not the -checks option, the configuration supports only the strict and lenient policies. Ensure that you specify a supported policy.

---

## NDMUI-965

(error) %s is not specified in the command.

**Description**

A valid argument is not specified in the command.

.SH WHAT NEXT Specify a valid argument in the command.

---

## NDMUI-966

(error) Incorrect usage of the -references option. Setting of the current block configuration using the -references option is not allowed.

**Description**

Incorrect usage of the -references option. Setting of the current block configuration using the -references option is not allowed.

**What Next**

Specify only reference blocks in the -references option.

---

## NDMUI-967

(error) Specified policy or strategy is not supported for this check.

**Description**

Specified policy or strategy is not supported for this check.

**What Next**

Execute the *report_early_data_checks -checks <check_name>* command to get a list of supported policies and strategies. Specify a policy or strategy from this list.

---

## NDMUI-968

(warning) The option '%s' value %s is different from %s %s of the layer '%s'.

Feedback

**Description**

The specified option value is different from the value defined in technology file. The tool will use the specified option value.

**What Next**

Please check the option value specified is correct.

---

## NDMUI-973

(error) Library cell '%s' is not mask shiftable.

**Description**

The *mask_shift_layers* attribute cannot be set or removed because the specified lib_cell is not mask shiftable. The lib_cell is not mask shiftable if its *is_mask_shiftable* attribute value is false.

**What Next**

Check the lib_cell's *is_mask_shiftable* attribute value. If not correct, correct the value. Then set the lib_cell's *mask_shift_layers* attribute.

---

## NDMUI-974

(Error) Purpose '%s' is not from the tech of the current lib '%s'.

**Description**

The specified purpose is not from the tech database of the library containing the block in which the shape is being created.

**What Next**

Use a purpose object from the current Lib.

---

## NDMUI-975

(Error) Site_def '%s' is not from the tech of the current lib '%s'.

**Description**

The specified site_def is not from the tech database of the library containing the block in which the shape is being created.

**What Next**

Use a site_def object from the current Lib.

Feedback

## NDMUI-976

(Error) Via_def '%s' is not from the tech of the current lib '%s'.

**Description**

The specified via_def is not from the tech database of the library containing the block in which the shape is being created.

**What Next**

Use a via_def object from the current Lib.

## NDMUI-977

(Error) Via_rule '%s' is not from the tech of the current lib '%s'.

**Description**

The specified via_rule is not from the tech database of the library containing the block in which the shape is being created.

**What Next**

Use a via_rule object from the current Lib.

## NDMUI-978

(Error) Density_rule '%s' is not from the tech of the current lib '%s'.

**Description**

The specified density_rule is not from the tech database of the library containing the block in which the shape is being created.

**What Next**

Use a density_rule object from the current Lib.

## NDMUI-979

(Error) Design_rule '%s' is not from the tech of the current lib '%s'.

**Description**

The specified design_rule is not from the tech database of the library containing the block in which the shape is being created.

Feedback

**What Next**

Use a design_rule object from the current Lib.

---

## NDMUI-980

(Error) PR_rule '%s' is not from the tech of the current lib '%s'.

**Description**

The specified pr_rule is not from the tech database of the library containing the block in which the shape is being created.

**What Next**

Use a pr_rule object from the current Lib.

---

## NDMUI-981

(error) Values provided for the -references option are either not linked or do not exist.

**Description**

Values provided for the -references option are either not linked or do not exist.

**What Next**

Specify valid values for the -references option. Ensure that the specified designs are linked. Execute the *link_block -force* command to link the design.

---

## NDMUI-982

(error) The option '%s' value '%s' is not in correct format.

**Description**

The specified option value is not in correct format. User has to provide index (unsigned integer), direction and distance (separated by space) as one value i.e. element_extension. User can provide list of element_extension values like below {{0 bottom 1.1} {1 horizontal 11.1}}

**What Next**

Please check the option value specified is correct.

Feedback

## NDMUI-983

(error) Cannot specify tech-scoped via def '%s' when block-scoped via def of the same name exists.

### Description

Block-scoped via defs take precedence over tech-scoped via defs. In the case where a block-scoped and a tech-scoped via def share the same name, only the block-scoped via def is "visible". In such cases, the tech-scoped via def cannot be used for via def setting.

### What Next

If the block-scoped via def is applicable then use it for via def setting. If the tech-scoped via def is desired then first remove the block-scoped via def of the same name. After the block-scoped via def is removed the tech-scoped via def may be used for via def setting.

## NDMUI-984

(error) Check %s not applicable to this design.

### Description

This check is valid, but is not applicable to this design.

### What Next

Execute the *report_early_data_checks -checks -hierarchical* command to get a list of checks that are applicable to the design. Specify a check from this list.

## NDMUI-985

(information) Total %d mismatches are found on block '%s'.

### Description

If there exists any inconsistency between design and reference data, they are regarded as a mismatch in the flow. These mismatches can be related to a missing reference, port name case, missing routing direction on a metal layer etc.

When such a mismatch is detected, it is collected and recorded on design. These mismatches are captured during specific flows like linking and routability checks.

### What Next

Check output of report_early_data_checks to find more details about mismatches.

Feedback

**See Also**

- report_early_data_checks

---

## NDMUI-986

(warning) Port name '%s' of the new reference '%s' is not found in the old reference '%s'.

**Description**

The reference or view of cells can be changed only with a reference or view with a identical port interface.

**What Next**

Specify a reference or view with a port interface identical to the current one.

---

## NDMUI-987

(error) Type of rvalue is invalid.

**Description**

Rvalue should be of type "rtlVariable, rtlExpression, rtlLiteral, rtlFunction, rtlList, port, net, busPort or busNet".

**What Next**

provide the valid statement.

**See Also**

- create_assignment_statement

---

## NDMUI-988

(error) Sensitivity list must be of expression type.

**Description**

The sensitivity list is expecting the rtl statement of expression type.

**What Next**

provide the statement of expression type as a sensitivity list.

**See Also**

- create_always_statement

Feedback

## NDMUI-989

(error) All passed statements must be in the current hier.

### Description

This command requires that all of the statements being passed are in the same hier in which you are working.

### What Next

Specify statements that are all in the current hier.

### See Also

- create_always_statement
- create_assignment_statement

## NDMUI-990

(error) All statements to be used in creation of a new statement, must belong to the same '%s' scope

### Description

This command requires that all of the objects being passed are in the same required scope in which you are working or if you have passed a particular scope , then all objects must exist in the same given scope.

### What Next

Specify objects that are all in the same required scope.

### See Also

- create_always_statement
- create_assignment_statement
- create_for_statement
- create_foreach_statement
- create_if_statement
- create_while_statement
- create_repeat_statement

Feedback

- create_parameter_statement
- create_case_statement

## NDMUI-991

(error) Type of lvalue is invalid.

**Description**

Lvalue should be of type "rtlVariable, port, busPort , net , busNet and rtlList".

**What Next**

provide the valid statement.

**See Also**

- create_assignment_statement

## NDMUI-992

(error) start is not an driver term.

**Description**

Specify a driver term in -start.

## NDMUI-993

(error) stop is not an load term.

**Description**

Specify a load term in -stop.

## NDMUI-994

(error) next statement must be of assignment type.

**Description**

The next statement is expecting the rtl statement of assignment type.

**What Next**

provide the statement of assignment type as a next statement .

Feedback

**See Also**

- create_for_statement

---

## NDMUI-995

(error) count_expression must be of expression type.

**Description**

The count_expression is expecting the rtl statement of expression type.

**What Next**

provide the statement of expression type as count_expression .

**See Also**

- create_repeat_statement

---

## NDMUI-996

(error) Cannot set failsafe fsm rule or create failsafe fsm group as no existing fsm found with specified registers.

**Description**

Failsafe fsm rule can be set or failsafe fsm group can be created with specified registers only if there is an existing fsm found with similar registers.

**What Next**

Try selecting a list of registers as specified in one of the fsm.

---

## NDMUI-997

(error) Cannot create failsafe fsm group as specified %s is already part of failsafe fsm group '%s'.

**Description**

One of the register specified as state, parity or sync register(s) is already part of an existing failsafe fsm group. Registers specified in any of the failsafe fsm group should be mutually exclusive with registers specified in other failsafe fsm groups.

**What Next**

Ensure that the register(s) are mutually exclusive with other failsafe fsm group(s).

Feedback

## NDMUI-998

(error) Variable statement with name '%s' already exists.

### Description

A variable statement with the given name already exists. Variable statement name must be unique.

### What Next

Specify unique value for the variable statement name.

## NDMUI-999

(error) invalid type of expression.

### Description

This command requires that expression must be one of the type: variable, expression,literal, function, list, port, net, port_bus or net_bus.

### What Next

Specify valid type of expression.

### See Also

- create_case_statement
- create_while_statement
- create_for_statement
- create_if_statement
- create_assignment_statement

## NDMUI-1000

(warning) %s name change cannot be performed on '%s', as it is included in don't touch.

### Description

As the object is in the dont_touch list, change_names is not operational on this object. If a part of bus is included in dont_touch the bus and bus member’s name won't be changed.

Feedback

## NDMUI-1001

(error) Cannot set failsafe fsm rule as rule '%s' is already applied on set of registers specified.

### Description

Failsafe fsm rule can be set on set of registers if there is no rule specified on those set of registers.

### What Next

Try selecting a list of registers as specified in one of the fsm and don't have any rule applied on that.

## NDMUI-1002

(error) %s statement must be of %s type.

### Description

This command requires that the statement being passed must be of required type.

### What Next

Specify object of required type.

### See Also

- create_package_statement
- create_begin_statement

## NDMUI-1003

(error) %s statement already present inside %s statement.

### Description

This command requires that the statement being passed must be present inside the begin or package statement.

### What Next

Specify statement which will present inside the current begin or package statement.

Feedback

**See Also**

- create_package_statement
- create_begin_statement

---

## NDMUI-1004

(warning) Ignoring -as_user_default option since application option <%s> is global scoped.

**Description**

The specific application option is global scoped. The global scoped application option only has one value that can be set by user.

---

## NDMUI-1005

(error) %s Statement must be present inside %s statement

**Description**

This command requires that the statement being passed must be present inside the begin or package statement.

**What Next**

Specify statement which will present inside the current begin or package statement.

**See Also**

- create_package_statement
- create_begin_statement

---

## NDMUI-1006

(error) initialization statement must be of assignment type.

**Description**

The initialization statement is expecting the rtl statement of assignment type.

**What Next**

provide the statement of assignment type as an initialization statement .

**See Also**

- create_for_statement

Feedback

## NDMUI-1007

(error) Case statement already contains this expression.

### Description

This command requires that the expression statement being passed must not be present in any case condition of the given case statement.

### What Next

Specify expression which is not already present.

### See Also

- add_to_case_statement

## NDMUI-1008

(warning) Failed to remove statement '%s'.

### Description

The specified statement(s) could not be removed. Some statements are required part of other statements in the design and therefore cannot be removed directly. To remove these statements, the statement containing them must be removed instead. For example, the if-else statement has an expression as part of the statement. So, removing only the expression of the if-else statement is not allowed.

### What Next

Modify the input to the command to contain only statements that can be removed.

### See Also

- get_statements

## NDMUI-1009

(error) Cannot remove btp_pattern '%s'.

### Description

The *remove_btp_pattern* command was unable to remove the specified btp_pattern.

Feedback

**What Next**

Please make sure that the specified btp_pattern belongs to the current_block. If all arguments are correct and the command still fails, it indicates an internal error. Please report to Synopsys.

---

## NDMUI-1010

(error) Cannot remove btc_region '%s'.

**Description**

The *remove_btc_region* command was unable to remove the specified btc_region.

**What Next**

Please make sure that the specified btc_region belongs to the current_block. If all arguments are correct and the command still fails, it indicates an internal error. Please report to Synopsys.

---

## NDMUI-1011

(warning) All values in '%s' must be %s than %s.

**Description**

This option requires values with indicated constraint.

**What Next**

Provide the correct value to the option.

---

## NDMUI-1012

(error) Unable to connect net across hierarchies.

**Description**

The connect_net command was not able to connect the net with the specified pin(s). This happens because one or more of the specified pins belong to a internally generated module, which does not allow new ports to be created on them. These modules are generated during compile.

**What Next**

The operation can be performed once compile is finished and a fully mapped netlist is created. You can also make local connections within the internally generated modules, as long as the port interface is not modified.

Feedback

## NDMUI-1013

(error) Cannot create safety error code as specified pin or port %s is duplicated or there is overlap between data and checkbits pin(s) or port(s).

### Description

Pin(s) or port(s) specified as data and checkbits should be mutually exclusive. There should not be any overlap between data and checkbits pins or ports.

### What Next

Ensure that the pin(s) or port(s) specified as part of data and checkbits are mutually exclusive.

## NDMUI-1014

(error) Cannot create safety error code group as specified %s : '%s' is already part of safety error code group '%s'.

### Description

One of the pin or port specified as data or checkbits pin(s) or port(s) is already part of an existing safety error code group. Data and checkbits pins or ports specified in any of the safety error code group should be mutually exclusive with data and checkbits pins or ports specified in other safety error code groups.

### What Next

Ensure that the data and checkbits pin(s) or port(s) are mutually exclusive with other safety error code group(s).

## NDMUI-1015

(error) Cannot set safety error code rule as specified %s : '%s' is already part of safety error code rule setting.

### Description

One of the pin or port specified as data or checkbits pin(s) or port(s) is already part of a safety error code rule setting. Data and checkbits pins or ports specified in any of the previously called 'set_safety_error_code_rule' command should be mutually exclusive with data and checkbits pins or ports specified in other calls of the command.

### What Next

Ensure that the data and checkbits pin(s) or port(s) are mutually exclusive with other safety error code rules setting.

Feedback

## NDMUI-1016

(error) Invalid %s pin or port : '%s' provided.

**Description**

If pin or port is of data or checkbits, the specified pin(s) or port(s) is already part of an existing safety error code group or safety error code rule setting as one of the following type: clock, enable, correction or error_signal.

Data and checkbits pins or ports specified in any of the safety error code group or safety error code rule setting should be mutually exclusive with clock, enable, correction, error_signal pin or port specified in other safety error code groups or safety error code rule setting.

If pin or port is one of the type: clock, enable, correction, error_signal, the specified pin(s) or port(s) is already part of an existing safety error code group or safety error code rule setting as different type of pin or port. Example: Pin or port of clock is already part of existing group or rule setting as pin or port of error_signal.

Clock, enable, correction, error_signal pin or port type specified in any of the safety error code group or safety error code rule setting should be mutually exclusive with clock, enable, correction, error_signal pin or port types specified in other safety error code groups or safety error code rule setting.

**What Next**

Ensure that the data and checkbits pin(s) or port(s) are mutually exclusive with other safety error code group's or safety error code rule setting's clock, enable, correction, error_signal pins or ports.

Ensure that the Clock, enable, correction, error_signal pin or port type are mutually exclusive with other safety error code group's or safety error code rule setting's clock, enable, correction, error_signal pin or port types.

## NDMUI-1017

(error) Cannot update rule '%s' since it is being referenced by group '%s'.

**Description**

The rule is referenced by group, hence its contents cannot be updated.

**What Next**

First remove the rule from group then update the rule.

Feedback

## NDMUI-1018

(error) Attribute 'port' on terminal '%s' cannot be updated because %s.

**Description**

The terminal's port attribute cannot be updated for the indicated reason. The port attribute may be updated only on bond pad terminals. The specified port must be a physical port in the same block as the terminal.

**What Next**

Correct the indicated problem and try the command again.

## NDMUI-1019

(error) Low disk space. projected disk space required: '%s' bytes, available space: '%s' bytes.

**Description**

Disk space may not be sufficient to unpack library.

**What Next**

Unpack library to other location, which has sufficient space.

## NDMUI-1020

(Error) %s statement is already present inside %s scope.

**Description**

The specified statement is already present inside the specified scope statement.

**What Next**

Use another statement which is not present in any scope.

## NDMUI-1021

(error) Input must be of variable type.

**Description**

This command requires that the inputs being passed must be of variable type.

Feedback

**What Next**

provide the statement of variable type as input.

**See Also**

- create_function

---

## NDMUI-1022

(information) Unable to %s for fusion library %s.

**Description**

Currently "-exclude_design_view" and "-exclude_unused_libcells" option are not supported for fusion libraries.

---

## NDMUI-1023

(error) The route guide boudary is not valid.

**Description**

The route guide boundary is not a rectangle or not covering whole chip in the preferred direction.

**What Next**

Check man page of create_routing_guide to understand requirement's of boundary for -rectangle_only option

---

## NDMUI-1024

(error) The layer do not have valid routing direction.

**Description**

The layer can have either horizontal or vertical routing direction for -rectangle_only route guide.

**What Next**

Check man page of create_routing_guide for -rectangle_only option

---

## NDMUI-1025

(error) All layers must have same routing direction.

Feedback

**Description**

All layers must have same routing direction.

**What Next**

Check the man page of create_routing_guide for more detail of -rectangle_only option

---

## NDMUI-1026

(error) Cannot specify a mix of different object types for option %s.

**Description**

The collection of the objects specified must have the same object type. For example, all of them must be either pins or ports. A collection of mixed object types cannot be specified.

**What Next**

Ensure that all objects are of the same object type.

---

## NDMUI-1027

(warning) Could not determine output pin of register %s.

**Description**

The positive unate output pin of the specified register cell could not be determined. This could happen if the cell does not have a timing view or if it is not a sequential cell.

**What Next**

Please specify a valid flop or a latch and ensure that correct reference libraries were successfully used during linking of the design.

---

## NDMUI-1028

(information) Copying db files for the ref lib '%s'.

**Description**

db files are copied for ref libs when '-include_db_files' option is used.

---

## NDMUI-1029

(warning) Cannot copy db file for the ref lib '%s'.

Feedback

**Description**

Unable to copy db file, possibly db files not present at specified path.

---

## NDMUI-1030

(error) Cannot update rule '%s' since it is associated with pin(s) or port(s).

**Description**

The rule is associated with pin(s) or port(s) of data bits and check bits, hence its contents cannot be updated.

**What Next**

First remove the association of the rule with pin(s) or port(s) then update the rule.

---

## NDMUI-1031

(warning) RTL name does not exist for cell %s.

**Description**

The RTL name for the specified cell instance does not exist or is blank. Tool cannot process the safety related intent for this case.

**What Next**

This particular cell will be skipped.

---

## NDMUI-1032

(warning) %s is not an integral multiple of %s.

**Description**

The first value is not an integral multiple of the second value. In case of safety error codes for functional safety implementation, the number of data bits must be an integral multiple of the slice size of the specified rule. If this is not followed, there might be issues with the safety error code implementation by the tool.

**What Next**

In case of safety error codes for functional safety implementation, choose the data bits and slice size so that the size of the former is an integral multiple of the latter.

Feedback

## NDMUI-1033

(error) '%s' shouldnot be a negative integer value.

**Description**

The value specified is a negative integer value. The value provided should be a positive integer value.

**What Next**

The user should provide a positive integer value.

## NDMUI-1034

(error) Invalid input layer. Input layers must be %s layers.

**Description**

Option -layer_area_threshold in command define_antenna_rule only accept metal layers.
Option -via_layer_area_threshold in command define_antenna_rule only accept via layers.

**What Next**

Provide correct metal/via layer input.

## NDMUI-1035

(error) Required argument %s not specified.

**Description**

The listed argument is required argument for the condition specified in the message.

**What Next**

Specify the required argument.

## NDMUI-1036

(error) '%s' shouldnot be a less than '%s' value.

**Description**

The value specified is a positive integer value. The value provided should be a greater positive integer value.

Feedback

### What Next

The user should provide a positive integer value greater than the specified value .

---

## NDMUI-1037

(warning) Value '%s' for command option '%s' is too small.

### Description

The SPFM value specified is too small. Please note that the tool expects the value in percentage, and the valid range is from 0.0 to 100.0

### What Next

Please check whether the specified SPFM value is correct. If you have specified the value as a fraction, please multiply it by 100 to get the percentage before passing it to the command.

---

## NDMUI-1038

(warning) Duplicate track is detected for option '%s' with value '%s'.

### Description

The value provided for option offset could lead to duplicate track. Tool will caculate the effective pitch for 1) Vertical track: effective pitch = std::ceil (max_offset / siteWidth) * siteWidth) 2) Horizontal track: effective pitch = std::ceil (max_offset / siteHeight) * siteHeight) Note: max_offset is max value specified in offset option. Then subtract the n *effective pitch from the value provided by user and make sure the value is larger than effective pitch. for Exmaple, If the effctive pitch is 0.35 and offsets(0.0, 0.2, 0.35, 0.5, 0.55) For 0.35, 0.35-0.35 = 0.0 which is equal to the first value in the offsets 0.0 For 0.55, 0.55-0.35 = 0.2 which is equal to the second value in the offsets 0.2. So the warning message will be issued.

### What Next

The user should correct the value.

---

## NDMUI-1039

(error) Duplicate %s name '%s' specified.

### Description

Objects with name specified as the values of command options should be unique. There should not be any overlap.

Feedback

**What Next**

Ensure that the objects specified as command options values are mutually exclusive.

---

## NDMUI-1040

(information) Some tcl variables and app options are not packed and unpacked during write_lib_package/read_lib_package.

**Description**

Following tcl variables will not be packed and restored :- env auto_index auto_path tcl_platform tk_platform cmd errorInfo errorCode err sh_arch sh_product_version sh_dev_null sh_tcllib_app_dirname tclDefaultLibrary tcl_patchLevel tcl_library tcl_pkgPath tcl_version tcl_interactive tcl_traceExec tkDefaultLibrary tk_patchLevel tk_library tk_pkgPath tk_version tk_interactive tk_traceExec link_path link_library synopsys_program_name synopsys_root tix _d_u_m_m_y_ sh_command_log_file sh_output_log_file

Following app options will not be packed and restored :- gui.batch_x_display

**What Next**

This is only a information message. No action is required.

---

## NDMUI-1041

(error) The option '%s' has invalid value or has invalid radius specified.

**Description**

The value specified for the curved poly rect is invalid. The radius of the center from the start or end may not be same or may be Zero.

**What Next**

The user should check the point list specified for the curved poly rect.

---

## NDMUI-1042

(error) Cannot set the %s of an annotation_shape of type '%s'

**Description**

The specified attribute does not belong to the reported type of annotation_shape.

**What Next**

Please only attempt to modify attribute values that apply to that annotation_shape's type.

Feedback

## NDMUI-1043

(error) Cannot set the %s of an annotation_shape of type '%s' with annotation_point of type '%s'.

**Description**

annotation_point of an annotation_shape should not be a reported type to set reported attribute

**What Next**

Please only attempt to modify attribute values that apply to that annotation_shape's type and it's annotation_point's type.

## NDMUI-1044

(error) The specified %s '%s' is not in the same block '%s' as the scan signal.

**Description**

The user specified port or hookup pin is in a different block from the scan signal. The scan signal and its port and hookup pin must be from the same block.

**What Next**

Specify a port or hookup pin from the same block as the scan signal.

## NDMUI-1045

(error) The element '%s' specified for option '%s' is not in the scan_chain '%s'.

**Description**

The option *-before* or *-after* was specified with an element that does not exist in the specified scan_chain. In order to create the new element after or before a specific element in the scan_chain, the specified element must already exist in the scan_chain.

**What Next**

Specify an element that is in the scan_chain, or omit the options *-before* and *-after* to create the new element at the end of the scan_chain.

## NDMUI-1046

(error) The pin '%s' specified for option '%s' is invalid because %s.

Feedback

**Description**

The specified pin is invalid for the indicated reason and cannot be used to create the scan chain element. All pins used to create a scan chain element must be on the same cell. That cell must be in the same block as the scan chain element being created. And the pins must not already be part of another scan chain element.

**What Next**

Specify a valid pin to the option.

---

## NDMUI-1047

(error) The specified %s scan_signal '%s' is invalid because %s.

**Description**

The specified scan_signal is invalid for the indicated reason and cannot be used for the scan chain element. A scan_signal used for a scan chain element must be from the same scan_test_mode as the scan chain element. If the scan chain element is in multiple scan_test_modes, then the scan_signal just needs to be from one of those scan_test_modes.

**What Next**

Specify a scan_signal that is from the scan chain element's scan_test_mode.

---

## NDMUI-1048

(error) The cell '%s' specified for option '%s' is invalid because %s.

**Description**

The specified cell is invalid for the indicated reason and cannot be used to create the scan chain element. The cell must be in the same block as the scan chain element being created. And the cell must not already be part of another scan chain element.

**What Next**

Specify a valid cell to the option.

---

## NDMUI-1049

(warning) Failed to add %s '%s' to scan_chain '%s' because %s.

Feedback

**Description**

The specified scan chain element cannot be added to the scan_chain. The scan chain element must be in the same block and scan_partition as the scan_chain.

**What Next**

Specify a valid scan chain element that is in the same block and scan_partition as the scan_chain.

---

## NDMUI-1050

(error) The %s '%s' cannot be the %s of the scan_chain because %s.

**Description**

The specified port or pin is invalid for the indicated reason and cannot be assigned to the scan_chain. The port or pin must be in the same block as the scan_chain. And it must not be assigned already as the scan input, scan output, scan clock A, or scan clock B of a scan_chain.

**What Next**

Specify a valid port or pin.

---

## NDMUI-1051

(error) A '%s' floorplan group by name '%s' already exists.

**Description**

While trying to create a floorplan group using any of the *create_floorplan_** commands, the name specified for the new group matches one of the existing floorplan group's name.

**What Next**

Either provide a different name for new floorplan group to be created or remove existing floorplan group with conflicting name if seems appropriate.

---

## NDMUI-1052

(error) A %s floorplan group by name '%s' does not exist.

**Description**

While trying to add / remove member(s) in floorplan group using any of the *add_to_floorplan_** or *remove_floorplan_group_elements* or setting project cell group on

Feedback

floorplan rules by *set_floorplan_*_rules* commands, the specified group with group type doesn't exist.

**What Next**

Provide name for existing floorplan group of same group type in which element(s) need to be added or existing floorplan group which need to be removed or need to be set on floorplan rule.

---

## NDMUI-1053

(error) A '%s' floorplan group rule by name '%s' already exists.

**Description**

While trying to create a floorplan group rule using any of the *create_floorplan_group** commands, the name specified for the new group rule matches one of the existing floorplan group rule's name.

**What Next**

Either provide a different name for new floorplan group rule to be created or remove existing floorplan group rule with conflicting name if seems appropriate.

---

## NDMUI-1054

(error) A %s floorplan group rule by name '%s' does not exist.

**Description**

While trying to set group rule on a group member using any of the *add_to_floorplan_** commands or trying to remove group rule by *remove_floorplan_group_rules*, the specified group rule doesn't exist.

**What Next**

Provide name for existing floorplan group rule which need to be set on member(s) or need to be removed.

---

## NDMUI-1055

(error) Cannot specify option '%s' or '%s' with edge_type '%s'.

**Description**

When creating an edge sealing group, user can provide core_sp_x or macro_sp_x with edge type *horizontal* or *all*. Similarly, core_sp_y or macro_sp_y can be provided with edge type *vertical* or *all* only.

Feedback

**What Next**

Provide option(s) with supported edge type value.

---

## NDMUI-1056

(error) Value for options '%s' cannot be negative.

**Description**

The value for these options must be greater than or equal to zero.

**What Next**

Enter the command again with a valid option value.

---

## NDMUI-1057

(warning) Library cell '%s' is not mask shiftable when the library is not mask shiftable.

**Description**

The lib_cell/library is not mask shiftable if its *is_mask_shiftable* attribute value is false. The *is_mask_shiftable* attribute for the library cell cannot be set to true when the library is not mask shiftable.

**What Next**

Check the library's *is_mask_shiftable* attribute value must be true. If not correct, correct the value. Then set the lib_cell's *is_mask_shiftable* attribute.

---

## NDMUI-1058

(error) Invalid block name specified in the '-to' option of the command.

**Description**

The block name specified in the '-to" option of the command must be non-empty, unique, and cannot be the name of an existing block. If you want to apply the mirror operation to an existing block in-place instead of making a copy of the block and applying the operation to the copy, please use the '-no_copy' option of the command.

**What Next**

Please specify a unique destination block name for the '-to' option, or use the '-no_copy' option instead.

Feedback

## NDMUI-1059

(information) Data map '%s' removed from mirrored block '%s'.

**Description**

Rotation is not supported for data_maps, therefore blocks containing data_maps may not be mirrored. This message is a notification that the specified data_map was removed from the specified block to allow it to be mirrored.

**What Next**

No action necessary.

## NDMUI-1060

(error) Data map '%s' could not be removed from mirrored block '%s'.

**Description**

Rotation is not supported for data_maps, therefore blocks containing data_maps may not be mirrored. However, the tool failed to remove the specified data_map.

**What Next**

Please check the permissions on the specified data_map and on the block, and correct the issue.

## NDMUI-1061

(error) Block '%s' could not be copied for the block mirroring operation.

**Description**

The mirror_block command attempted to copy the specified block, but could not complete the operation.

**What Next**

Please check that the block exists and can be opened.

## NDMUI-1062

(error) The scan_signal '%s' cannot be a scan enable signal of the scan_chain because %s.

Feedback

**Description**

The specified scan_signal is invalid for the indicated reason and cannot be assigned to the scan_chain. The scan_signal must be in the same scan_test_mode as the scan_chain.

**What Next**

Specify a valid scan_signal.

---

## NDMUI-1063

(warning)The attribute '%s' of attribute group '%s' will be skipped.

**Description**

The attribute has different attribute type at ddestination object. The copy for the same will be skipped.

**What Next**

To overwrite , remove the attribute from destination object.

---

## NDMUI-1064

(warning) Current design '%s' is invalid for report_supernet_exceptions command.

**Description**

report_supernet_exceptions command do not work on leaf/macro design.

**What Next**

Please check current design.

---

## NDMUI-1065

(error) Invalid mask constraint value %s is specified.

**Description**

Given mask constraint value cannot be set on shape, shape_pattern, via, via_matrix or routing blockage. Value of mask constraint can be no_mask only when the number_of_masks value on the layer is zero or one. When the number_of_masks is greater than one, mask constraint value must be less than number_of_masks and cannot be no_mask.

Feedback

**What Next**

Check the number_of_masks attribute value of layer and use correct mask constraint value.

---

## NDMUI-1066

(error) Position '%d' specified for the -at option is invalid. The position must be %s, because %s in the scan_test_mode.

**Description**

The position value specified for the -at option is invalid. The position, at which to create the scan_signal within the scan_test_mode, must be between 0 and the number of scan_signals already in the scan_test_mode. If 0 is specified, the scan_signal is created at the front of the scan_signal list in the scan_test_mode.

**What Next**

Specify a valid position value for the -at option.

---

## NDMUI-1067

(error) The %s of the default %s cannot be changed.

**Description**

The default scan_partition and default scan_test_modes are auto-generated. Their names cannot be changed.

---

## NDMUI-1068

(error) Mask value '%s' is invalid as number_of_masks attribute value of layer '%s' is '%d'.

**Description**

The mask value cannot be greater than the layer's number_of_masks attribute. If layer's number_of_masks attribute value is 0 or 1, "no_mask" is the only supported mask value. If layer's number_of_masks attribute value is 2 or greater, mask value must be less than or equal to the number_of_masks attribute value.

**What Next**

Correct the specified mask value to make it less than or equal to the layer's number_of_masks attribute value.

Feedback

## NDMUI-1069

(error) The 'is_bond_pad' attribute of terminal '%s' cannot be changed to false.

**Description**

The terminal 'is_bond_pad' attribute cannot be changed from true to false. In order to change this attribute from true to false, the terminal must be owned by a port and must not be in a matching type or bump pin constraint. These constraints require that their terminals must be bond pads. This only applies to constraints and terminals in the same block.

**What Next**

Ensure the terminal is owned by a port and is not in any matching type or bump pin constraint. Then modify the 'is_bond_pad' attribute.

## NDMUI-1070

(warning) Command '%s' reads and ignores the data.

**Description**

The specified command is a dummy command which reads and ignores the data provided.

The dummy command is added so that the tool does not throw error while loading the SSF script written by other tools.

**What Next**

Please note that the data will be lost, hence will not be written by the tool.

## NDMUI-1071

(warning) The ref_cell for supernet exception can not be set: %s.

**Description**

The ref_cell for supernet exception could not be set, due to the specified reason.

**What Next**

Please provide a valid value for the ref_cell.

## NDMUI-1072

(warning) The excluded characters for bump name can not be set: %s.

Feedback

**Description**

The excluded characters for bump name could not be set, due to the specified reason.

**What Next**

Please provide value in valid format for the excluded characters.

---

## NDMUI-1073

(Information) Trying to create a secondary Voltage Area.

**Description**

This message is reported whenever you are trying to create a secondary Voltage Area.

**What Next**

- Make sure the creation of a gas station is what you intent to do. If not, check if you already had created a primary Voltage Area for the specified Power Domain. Only a single primary Voltage Area could exist.

---

## NDMUI-1074

(warning) The port type(s) '%s' are relevant only for module type '%s'.

**Description**

The specified port type(s) are not relevant to the specified module type.

**What Next**

Please check the relevant module type for the specified port type(s).

---

## NDMUI-1075

(error) '%s' is not a valid library package. '%s' file missing.

**Description**

*read_lib_package* cannot read library package generated by the *write_lib_package command.*

**What Next**

The lib package file provided isn't in the proper format. If you are transferring this data from customer site, please check all the required files are copied properly.

Feedback

## NDMUI-1076

(warning) Terminal %s's boundary has been updated which may be different from bond_pad_def %s's boundary.

**Description**

The boundary attribute of the terminal mentioned has been changed which was set using the boundary of the bond_pad_def. The new boundary may be different from the bond_pad_def's boundary.

**What Next**

Please use the bond_pad_def's boundary in place of terminal boundary carefully.

## NDMUI-1077

(error) Port '%s' and pin '%s' are in different blocks.

**Description**

The specified port and pad pin are not in the same block. A pad pin can be used to create a terminal only on the port of the block that contains the pad pin.

To create a top port terminal so that it has the shape of this pad pin, call create_terminal with this pad pin and the port on the subblock containing this pad pin. Ensure that the subblock's port *is_pad* attribute is true. Then logically connect the subblock's port to the top port.

**What Next**

Specify a port and pad pin that in the same block.

## NDMUI-1078

(error) Pin '%s' is not a pad pin.

**Description**

The specified pin must be a pad pin in order to create a terminal using its pad pin shape. A pin is a pad pin if and only if its reference port *is_pad* attribute is true.

**What Next**

Update the specified pin's reference port *is_pad* attribute to true, or specify a different pin that is already marked as a pad pin.

Feedback

## NDMUI-1079

(error) Pin '%s' has a terminal with an unassigned name on layer '%s'.

**Description**

The specified pad pin contains a terminal with a system generated name on the specified layer. The pad pin's terminals must have explicitly assigned names in order to create terminal using its shapes.

**What Next**

Assign names to the pad pin's terminals.

## NDMUI-1080

(error) A non-hole pseudo_tsv must reference a pseudo_tsv_def.

**Description**

Every non-hole pseudo_tsv must reference a pseudo_tsv_def. Therefore a pseudo_tsv's *tsv_def* and *tsv_def_name* attributes must be non-empty when it is not a hole, specifically when its *is_hole* attribute is 'false'. Similarly, a pseudo_bump's *tsv_def_name* attribute must be non-empty when its corresponding pseudo_tsv is not a hole.

**What Next**

Please specify a valid pseudo_tsv_def for the pseudo_tsv, or set its *is_hole* attribute to 'true' to clear the pseudo_tsv_def.

## NDMUI-1082

(warning) forbidSignalRoute is defined for layer '%s' which is in between of min ('%s') and max ('%s') routing layer.

**Description**

Property 'forbidSignalRoute' has been defined for a layer which is in between of min and max routing layer (both inclusive). If min or max has not been specified, then it will consider min or max Interconnect routing layer as min or max.

## NDMUI-1083

(warning) Cell '%s' is not a physical cell.

**Description**

Only physical cells may have their location set.

Feedback

### What Next

Remove cells that do not have physical implementation from the cell list.

---

## NDMUI-1084

(error) There is no default drawing layer with layer number %d so it cannot be used with -purpose_number option.

### Description

Default drawing layer with given layer number must exist before using -purpose_number option.

### What Next

Create default layer with layer number mentioned to be able to create tech layerPurpose along with -purpose_number option.

---

## NDMUI-1085

(error) A control group by name '%s' already exists.

### Description

While trying to create a control group using any of the *create_control_group** commands, the name specified for the new group matches one of the existing control group name.

### What Next

Either provide a different name for new control group to be created or remove existing control group with conflicting name if seems appropriate.

---

## NDMUI-1086

(error) The specified purpose '%s' already exists and cannot be used with the -purpose_number option.

### Description

The design already contains the named purpose that has been specified using purpose option corresponding to purpose_number option. Each purpose number must have an unique purpose name associated with it.

### What Next

Specify unique name for purpose option which is not already present in the design.

Feedback

## NDMUI-1087

(error) Pin '%s' does not contain any terminals on layer '%s'.

**Description**

The specified pin does not have any terminals on the specified layer. The pin must contain at least one terminal on the specified layer in order to create terminals using its shapes.

**What Next**

Specify a layer that has terminals in the pin, or update the pin so that it has at least one terminal on the specified layer.

## NDMUI-1088

(error) Cannot update shape_use for shape '%s'

**Description**

Shapes with shape_use of type 'partial_parallel_route' can't be updated through UI OR shape_use for any shape can't be updated to 'partial_parallel_route'.

**What Next**

Please provide the correct input.

## NDMUI-1090

(warning) The user specified layer '%s' in '%s' does not match the auto-derived layer value '%s'.

**Description**

The specified layer is not recommended for the type of via_def being created. For simple via_defs, the lower and upper layers should be the nearest routable upper and lower metal layers to the specified cut layer. For custom via_defs, the lower and upper layers must correspond to the lowest and highest layers of the specified custom shapes.

**What Next**

The user specified layer will take precedence over the auto-derived layer determined by the tool. If the user wants to set a different layer than the tool's suggestion then no further action is needed.

Feedback

## NDMUI-1091

(information) Library '%s' was successfully recovered. Recovered library was created after '%s' command.

### Description

Library was successfully recovered.

### See Also

- report_auto_save
- remove_auto_save
- recover_auto_save
- stop_auto_save
- start_auto_save

## NDMUI-1092

(error) Special character '%s' can't be used in name for Lib package.

### Description

Special characters can't be used in name for Lib package.

### What Next

Please remove special characters from Lib package name.

## NDMUI-1093

(warning) Cannot change net_type from %s to %s on net '%s'.

### Description

One of the segment net is "*Logic0*" or "*Logic1*". These are reserved by Synopsys for the tie-low and tie-high constant nets. Their types cannot be changed. Verify segment nets of input net.

### What Next

No action is required.

Feedback

## NDMUI-1094

(error) Unable to generate row-based grid steps because %s.

### Description

The row-based grid steps cannot be generated from the specified *-row_end_grid_low_steps* and *-row_end_group_high_steps* options. To use these options, the block must have horizontal or vertical site rows but not both. Each site row must be bound to a site_def that is specified in these options.

### What Next

Check the *-row_end_grid_low_steps* and *-row_end_group_high_steps* option values and the corresponding site_rows and site_defs.

## NDMUI-1095

(error) Duplicate steps specified for site_def '%s' in the %s option.

### Description

The site_def was specified multiple times in the indicated command option. Only one list of steps may be specified for each site_def.

### What Next

Check the indicated option and specify only one list of steps for the site_def.

### See Also

- create_track

## NDMUI-1096

(error) The file format is not correct. Please specify a file output by command '%s'.

### Description

The file format is not correct. Command is unable to process given format. Please provide file in correct format or dump the output from corresponding write command and use that file.

### What Next

Please check corresponding write command if supported.

Feedback

## NDMUI-1097

(warning) Max core is set to 1. Command '%s' will run sequentially.

**Description**

Number of cores are set to 1. Reported command can not run with multiple threadeds

**What Next**

Change number of core setting to run command non sequencially.

## NDMUI-1098

(info) No connectivity errors were detected in module '%s'.

**Description**

This message indicates that the connectivity checker didn't find any connectivity errors for the reported module.

## NDMUI-1099

(error) The database has internal connectivity errors. You must revert to an earlier version of this database to proceed. Please contact your Synopsys representative to report this issue.

**Description**

This message indicates that the data-model connectivity checker has found internal connectivity errors.

**What Next**

It is recommended that you revert to an earlier version of this database to proceed.

## NDMUI-1100

(error) seal_ring or assembly_die keepout cannot be overlapped

**Description**

The -can_overlap option cannot be specified for seal_ring and assembly_die type keepout.

**What Next**

Dont use the -can_overlap option to create seal_ring or assembly_die keepout.

Feedback

## NDMUI-1101

(error) seal_ring or assembly_die keepout can only be set as outer keepout.

**Description**

seal_ring and assembly_die represent the external seal_ring and grinding margin outside of a die so it can only be set as outer keepout.

**What Next**

Specify -outer keepout for seal_ring and assembly_die

## NDMUI-1102

(error) seal_ring or assembly_die keepout can only be set on a block.

**Description**

You cannot specify seal_ring and assembly_die keepout on a cell. seal_ring and assembly_die represent the external seal_ring and grinding margin outside of a die so it can only be set as outer keepout on a block.

**What Next**

Specify seal_ring and assembly_die on a block.

## NDMUI-1103

(error) Cannot %s safety error code %s as inverted pin '%s' of specified pin '%s' is already part of safety error code %s.

**Description**

Either the inverted pin of one of the pin(s) specified as data or checkbits pin(s) is already part of an existing safety error code group, or else the inverted pin of one of the pin(s) specified as data or checkbits pin(s) is already part of a safety error code rule setting.

The inverted pin of data and checkbits pins specified in any of the safety error code group should be mutually exclusive with data and checkbits pins specified in other safety error code groups. Similarly, the inverted pin of data and checkbits pins specified in any of the previously called 'set_safety_error_code_rule' command should be mutually exclusive with data and checkbits pins specified in other calls of the command.

**What Next**

Ensure that the inverted pin of pin(s) specified as part of data and checkbits are mutually exclusive with other safety error code group(s) and safety error code rule setting(s).

Feedback

## NDMUI-1104

(error) Cannot %s safety error code %s as specified pin '%s' and its inverted pin have been specified together.

### Description

A pin and its inverter pin cannot be specified as data or checkbits together. There should not be any overlap between data and checkbits pins and their inverter pins as data or checkbits.

### What Next

Ensure that the inverter pin of the pin(s) specified as part of data and checkbits should not be specified in the command as data or checkbits pins.

## NDMUI-1105

(error) Must specify a combination of these options: -from_object_types/-from_lib_cells and -to_object_types/-to_lib_cells.

### Description

Must specify a combination of these options: -from_object_types/-from_lib_cells and -to_object_types/-to_lib_cells.

### What Next

Please specify a combination of these options: -from_object_types/-from_lib_cells and -to_object_types/-to_lib_cells.

## NDMUI-1106

(error) -from_object_types and -from_lib_cells or -to_object_types and -to_lib_cells cannot be specified together without -merge_object option.

### Description

-from_object_types and -from_lib_cells or -to_object_types and -to_lib_cells cannot be specified together without -merge_object option.

### What Next

Please specify -merge_object option to use -from_object_types and -from_lib_cells or -to_object_types and -to_lib_cells together.

Feedback

## NDMUI-1107

(error) -object_types and -lib_cells cannot be specified together without -merge_object.

**Description**

-object_types and -lib_cells cannot be specified together without -merge_object.

**What Next**

please specify -merge_object to use -object_types and -lib_cells together.

## NDMUI-1108

(error) Multiple subset names support for lib_cell is disabled.

**Description**

Attribute subset name doesn't allow multiple subset names as per current app-option setting.

**What Next**

Please specify single subset name. To enable the suppport for multiple subset names, enable app-option allow_multiple_subset_names as below. set_app_options -name design.allow_multiple_subset_names -value true

## NDMUI-1109

(error) Number of subset_name values provided is '%s' which is greater than the maximum allowed limit of '%s'.

**Description**

Specified subset_names size cannot exceed maximum allowed limit.

**What Next**

Please specify the attribute value with size less than or equal to the max allowed limit.

## NDMUI-1110

(error) Cannot use the '-id' option if multiple layers are specified using the '-layers' option, or if multiple shapes are specified.

Feedback

**Description**

The '-id' option cannot be used when the options specify that multiple blockages be created.

**What Next**

Only use the '-id' option if a single blockage will be created.

---

## NDMUI-1111

(error) Cannot create safety peer pin group which contains number of pins less than %s.

**Description**

Safety peer pin group cannot be created with the specified number of pins.

**What Next**

Ensure that the number of pins specified should be equal to or greater than the required number.

---

## NDMUI-1112

(error) Specified pin '%s' is part of an existing safety peer pin group.

**Description**

Specified pin is already part of an existing safety peer pin group. Pins should be mutually exclusive with other safety peer pin group pins.

**What Next**

Ensure that the pins are mutually exclusive with other safety peer pin group pins.

---

## NDMUI-1113

(error) %s '%s' and '%s' are not mutually exclusive in terms of hierarchy.

**Description**

Specified objects must be mutually exclusive, i.e. separated out in terms of their hierarchies. In other words, first object must not lie anywhere inside the hierarchy of the second object at any level, and vice versa.

**What Next**

Ensure that the objects specified are mutually exclusive and are in separate hierarchies.

Feedback

## NDMUI-1118

(error) The '%s' option or attribute of the bump_region may not be specified when the pattern is of type '%s'.

**Description**

The setting of certain option or attributes may be disallowed depending on the *type* attribute of the bump_region_pattern the bump_region references.

**What Next**

Only specify option or attribute values that are applicable given the referenced bump_region_pattern's type.

## NDMUI-1119

(error) The bump_region specified in the '-region' option must be freeform.

**Description**

Only pseudo_bumps that belong to freeform bump_regions may be created or destroyed.

**What Next**

When creating a new pseudo_bump, ensure that the specified bump_region has *is_freeform* attribute value "true".

## NDMUI-1120

(error) Cannot replace pattern of type '%s' with pattern of type '%s' - new pattern must have the same type.

**Description**

A bump_region must reference the same type of bump_region_pattern over its lifetime.

**What Next**

Ensure the specified bump_region_pattern has the same *type* attribute value as the pattern it is replacing in the bump_region's *pattern* attribute.

## NDMUI-1121

(error) Cannot remove pseudo_bump %s.

Feedback

**Description**

Only pseudo_bumps that belong to freeform bump_regions may be created or destroyed.

**What Next**

Ensure that the pseudo_bumps specified for removal belong to bump_regions with *is_freeform* attribute value "true".

---

## NDMUI-1122

(error) %s is not a valid cell.

**Description**

The specified cell is not a physical cell or not in the same physical hierarchy.

**What Next**

Please provide a valid physical cell.

---

## NDMUI-1123

(error) Cannot set 'pattern_must_join' attribute on port or lib_pin '%s' having input direction.

**Description**

The 'pattern_must_join' attribute requires port or lib_pin having output or inout direction.

**What Next**

Please specify port or lib_pin having output or inout direction.

---

## NDMUI-1124

(error) Cannot set 'pattern_must_join' attribute on port or lib_pin '%s' having less than two terminals on top-layer.

**Description**

The 'pattern_must_join' attribute requires port or lib_pin having at-least two terminals on top-layer.

**What Next**

Please specify port or lib_pin having at-least two terminals on top-layer.

Feedback

## NDMUI-1125

(error) The first contact_code in cut_layer_name_table of via ladder rule '%s' does not connect to top-layer of %s '%s'.

**Description**

The first contact_code in cut_layer_name_table of reported via ladder rule must connect to the top-layer of reported pin or lib_pin.

**What Next**

Please specify correct via ladder rule for the reported pin or lib_pin.

## NDMUI-1126

(error) The first value in num_cuts_per_row_table of via ladder rule '%s' is smaller than the number of top-layer terminals of “pattern_must_join” or “is_em_via_ladder_required” %s '%s'.

**Description**

The first value in num_cuts_per_row_table of reported via ladder rule must match the number of top-layer terminals of reported pin or lib_pin having either “pattern_must_join” or “is_em_via_ladder_required” attribute set true.

**What Next**

Please specify correct via ladder rule for the reported pin or lib_pin.

## NDMUI-1127

(warning) The first value in num_cuts_per_row_table of via ladder rule '%s' is smaller than the number of top-layer terminals of %s '%s'.

**Description**

The first value in num_cuts_per_row_table of reported via ladder rule must match the number of top-layer terminals of reported pin or lib_pin. This is a warning when the pin or lib_pin does not have either “pattern_must_join” or “is_em_via_ladder_required” attribute set true but user must review if a smaller via ladder is required on purpose.

**What Next**

Please review if a smaller via ladder is required on purpose else specify correct via ladder rule for the reported pin or lib_pin.

Feedback

## NDMUI-1128

(warning) Unknown or unsupported routing_direction or pitch in top-layer of %s '%s' prevents track count estimation and large via ladder rule sanity pre-check of '%s'.

**Description**

The routing_direction and pitch of top-layer of reported pin or lib-pin are required to estimate the number of tracks for large via ladder rule sanity pre-check of reported via ladder rule.

**What Next**

Please correct routing_direction and pitch of top-layer for the reported pin or lib_pin.

## NDMUI-1129

(error) The first value in num_cuts_per_row_table of via ladder rule '%s' is larger than the estimated number of tracks for top-layer terminals of %s '%s'.

**Description**

The first value in num_cuts_per_row_table of reported via ladder rule must match the estimated number of tracks for top-layer terminals of reported pin or lib_pin.

**What Next**

Please specify correct via ladder rule for the reported pin or lib_pin.

## NDMUI-1130

(error) The specified bump_region must be structured.

**Description**

The bump_region specified in the *-region* argument of the command must be a structured bump_region, meaning that the bump_region_pattern it references must be of type "structured". Equivalently, the bump_region's *is_freeform* attribute must be "false".

**What Next**

Please specify a structured bump_region as the *-region* argument value of the command.

## NDMUI-1131

(error) A bump_region_pattern with the name '%s' already exists.

Feedback

**Description**

The command must create a new bump_region_pattern of type "freeform" to replace the bump_region's existing "structured" bump_region_pattern. The *-freeform_pattern_name* argument of the command specifies the name of the new bump_region_pattern. However, a bump_region_pattern with the specified name already exists, preventing the command from creating the required new one.

**What Next**

Please specify a name for the new bump_region_pattern that does not conflict with the name of any existing bump_region_patterns. Alternatively, you may omit this argument and the system will generate a unique name for the new bump_region_pattern.

---

## NDMUI-1132

(error) The specified bump_region may not have pseudo_tsvs.

**Description**

The bump_region specified in the *-region* argument of the command may not have pseudo_tsvs. Equivalently, the bump_region's *has_pseudo_tsvs* attribute must be "false".

**What Next**

Please specify a bump_region without pseudo_tsvs as the *-region* argument value of the command.

---

## NDMUI-1135

(warning) One or more objects in the collection specified for the '%s' argument of the command lack a valid bounding box or geometrical area.

**Description**

The specified collection is intended to define a combined geometrical area, however one or more objects in the collection lack a valid geometrical area. These objects have no effect on the resulting geometry.

---

## NDMUI-1136

(error) '%s' is set to backside layer '%s'. This setting may be ignored by the router. Please use the '%s' option for backside routing control.

Feedback

**Description**

Old UI options i.e. -min_routing_layer and -max_routing_layer should be disallowed from specifying a layer range that spans both backside and frontside because this setting may be ignored by the router. Instead of that use new UI options.

**What Next**

Please use new UI options -backside_min_routing_layer and -backside_max_routing_layer for backside along with options -frontside_min_routing_layer and -frontside_max_routing_layer for frontside routing control.

---

## NDMUI-1137

(error) Must specify any one of these options: -to_object_types/-to_lib_cells.

**Description**

Must specify any one of these options: -to_object_types/-to_lib_cells.

**What Next**

Please specify any one of these options: -to_object_types/-to_lib_cells.

---

## NDMUI-1138

(error) Must specify a valid floorplan density rule with -rules option.

**Description**

Must specify a valid floorplan density rule with -rules option.

**What Next**

Please specify a valid floorplan density rule with -rules option.

---

## NDMUI-1141

(warning) Layer %s is not suitable for bond pads.

**Description**

Layer is not suitable for creating bond pads. It must be either a top-most metal layer for creating frontside terminals, or bottom-most layer for creating backside terminals.

**What Next**

Specify an HBL layer, or a top-most or bottom-most metal layer.

Feedback

# NDMX

## NDMX-001

(error) Failed to open the cell-rename map file '%s'.

### Description

This error message occurs when the cell-rename map file could not be opened. The cell-rename map file is instroduced in the man page of message NDMX-009, you can refer to it if necessary.

### What Next

Verify the file-permissions and retry.

## NDMX-002

(error) Required section '%s' was not found before section '%s' in the cell-rename map file '%s'.

### Description

This message occurs when the required section is missed or the required sections are not specified in the required order.

The mask-shifted cell-rename map file comprises three sections: MASKSHIFTLAYER, MASKSHIFTPATTERN and CELLMAP. You need to define the sections in the order above.

Here is an example about the map file.

```
<MASKSHIFTLAYER>
M1 M2
</MASKSHIFTLAYER>
<MASKSHIFTPATTERN>
00 01 10 11
</MASKSHIFTPATTERN>
<CELLMAP>
A AM2B NONE AM1BM2B
</CELLMAP>
```

### What Next

Refer to the detail about cell-rename map file in the man page of message NDMX-009 and supply the required section.

Feedback

## NDMX-003

(error) Invalid mask shift layer '%s'%s in the cell-rename map file.

### Description

This error message occurs when the specified mask shift layer in the cell-rename map file is invalid, it must meet the following conditions. 1. If multiple cell mapping files are specified, the following type of files are valid: a) files have the same mask shift layer name and the order, e.g. file1: M1 M2; file2: M1 M2 b) each mask shift layer in each file is different, e.g. file1: M1 M2; file2: M3 M4 2. The layers must exist in the current library.

For example, assuming that the following map files are specified, and file1 has mask shift layers M1 and M2, but file2 has mask shift layers M1 and M3, they are partially different, will trigger this message.

```
file1:
<MASKSHIFTLAYER>
M1 M2
</MASKSHIFTLAYER>
<MASKSHIFTPATTERN>
00 01 10 11
</MASKSHIFTPATTERN>
<CELLMAP>
A AM2B NONE AM1BM2B
</CELLMAP>

file2:
<MASKSHIFTLAYER>
M1 M3
</MASKSHIFTLAYER>
<MASKSHIFTPATTERN>
00 01 10 11
</MASKSHIFTPATTERN>
<CELLMAP>
A AM2B NONE AM1BM2B
</CELLMAP>
```

Multiple cell mapping files with completely different mask shift layers will be treat as followed

```
file1:                  file2:
<MASKSHIFTLAYER>        <MASKSHIFTLAYER>
M1                      M2
</MASKSHIFTLAYER>       </MASKSHIFTLAYER>
<MASKSHIFTPATTERN>      <MASKSHIFTPATTERN>
0 1                     0 1
</MASKSHIFTPATTERN>     </MASKSHIFTPATTERN>
<CELLMAP>               <CELLMAP>
A AM1B                  B BM2B
</CELLMAP>
```

Feedback

```
Treated as:

<MASKSHIFTLAYER>
M1 M2
</MASKSHIFTLAYER>
<MASKSHIFTPATTERN>
00 01 10 11
</MASKSHIFTPATTERN>
<CELLMAP>
A NONE AM1B NONE
B BM2B NONE NONE
</CELLMAP>
```

The cell-rename map file is introduced in the man page of message NDMX-009, you can refer to it if necessary.

**What Next**

Please check and correct the mask shift layer, then try again.

---

## NDMX-004

(warning) Incorrect syntax '%s' specified for cell-rename entry. The entry will be ignored.

**Description**

This warning message occurs when the syntax for the cell-rename entry is not a valid one.

For example, assuming that the following map file is specified, and there is a HI section in the map file, which is not supported and will trigger this message.

```
<MASKSHIFTLAYER>
M1 M2
</MASKSHIFTLAYER>
<HI>
00 01 10 11
</HI>
<CELLMAP>
A AM2B NONE AM1BM2B
</CELLMAP>
```

**What Next**

Please refer to the detail about the cell-rename map file in the man page of message NDMX-009 to correct it.

---

## NDMX-005

(error) Invalid mask shift pattern '%s'%s.

Feedback

### Description

This error message occurs when the specified mask shift pattern is invalid in the cell-rename map file.

The mask shift pattern must meet the following conditions. 1. If multiple cell mapping files have the same mask shift layers and order, they should have the same number of mask shift pattern. 2. The number of the mask shift combination is equal to or less than pow(m, n), where n is the number of layers specified in MASKSHIFTLAYER and m is the count of the shift values. For 2-mask layers, valid shift values are 0 and 1, so m is 2. For 3-mask layers, valid shift values are 0, 1, 3, 4 and 5, so m is 5. When the shift value is 3, 4 or 5, the shifted layer must have fixed mask 1, 2 or 3, repectively and its number_of_masks must be more than 2. 3. MASKSHIFTPATTERN must contain the mask shift combination for the cell without any mask shifting.

The cell-rename map file is instroduced in the man page of message NDMX-009, you can refer to it if necessary.

### What Next

Please check and correct the mask shift pattern, then try again.

---

## NDMX-006

(error) Invalid cell mapping value '%s' in the cell-rename map file.

### Description

This error message occurs when the number of the cell mapping value is not consistent with the number of the specified mask shift pattern.

For example, assuming that the following map file is specified, the number of the cell mapping values in CELLMAP section is 5, but the number of the mask shift pattern in MASKSHIFTPATTERN is 4, they are inconsistent, this is why it will trigger this message.

```
<MASKSHIFTLAYER>
M1 M2
</MASKSHIFTLAYER>
<MASKSHIFTPATTERN>
00 01 10 11
</MASKSHIFTPATTERN>
<CELLMAP>
A AM2B NONE AM1BM2B AM1B
</CELLMAP>
```

The cell-rename map file is instroduced in the man page of message NDMX-009, you can refer to it if necessary.

Feedback

### What Next

Please check and correct the cell mapping value, then try again.

---

## NDMX-007

(error) Mask shifted cell '%s' does not have its corresponding mapping in the cell-rename map file.

### Description

This error message occurs when there is not a corresponding mapping for the mask shifted cell in the cell-rename map file.

For example, assuming that the following map file is specified, and there is a cell in the database which references B and has mask shift 1 on layer M1, it will trigger this message, because B is not specified in the CELLMAP section.

```
<MASKSHIFTLAYER>
M1 M2
</MASKSHIFTLAYER>
<MASKSHIFTPATTERN>
00 01 10 11
</MASKSHIFTPATTERN>
<CELLMAP>
A AM2B NONE AM1BM2B
</CELLMAP>
```

The cell-rename map file is introduced in the man page of message NDMX-009, you can refer to it if necessary.

### What Next

Please add the corresponding mapping to the map file and then try again.

---

## NDMX-008

(error) Mask shift property mismatch on cell '%s'.

### Description

This error message occurs when there is mask shift property mismatch between different cell-rename map files.

For example, assuming that the following map files are specified, the new names corresponding to the pattern 01 in CELLMAP section are inconsistent, which will trigger this message.

```
file1:
<MASKSHIFTLAYER>
```

Feedback

```
M1 M2
</MASKSHIFTLAYER>
<MASKSHIFTPATTERN>
00 01 10 11
</MASKSHIFTPATTERN>
<CELLMAP>
A AM2B NONE AM1BM2B
</CELLMAP>

file2:
<MASKSHIFTLAYER>
M1 M2
</MASKSHIFTLAYER>
<MASKSHIFTPATTERN>
00 01 10 11
</MASKSHIFTPATTERN>
<CELLMAP>
A AM2BBC NONE AM1BM2B
</CELLMAP>
```

The cell-rename map file is instroduced in the man page of message NDMX-009, you can refer to it if necessary.

**What Next**

Please correct the mask shift property and then try again.

---

## NDMX-009

(error) Fail to find the mask shift pattern '%s' for cell '%s' in cell-rename map file(s).

**Description**

This error message occurs when the mask shift pattern is not specified in MASKSHIFTPATTERN section of the cell-rename map file(s).

For example, assuming the following map file to rename the cell A with mask shifting on layer M1 or/and M2 is specified.

```
<MASKSHIFTLAYER>
M1 M2
</MASKSHIFTLAYER>
<MASKSHIFTPATTERN>
00 01 10
</MASKSHIFTPATTERN>
<CELLMAP>
A AM2B NONE
</CELLMAP>
```

Feedback

If there is a cell in the database which references A and has mask shift 1 on both layer M1 and layer M2, it will trigger this message, because 11 is not specified in MASKSHIFTPATTERN section of the map file above.

Normally, all the pattern should be specified in the map file.

The following information is the detail about the map file, you can ignore it if you have known it.

The cell-rename map file is to rename the mask shift cells and comprises three sections: MASKSHIFTLAYER, MASKSHIFTPATTERN and CELLMAP.

For example, to rename the cell A with mask shifting on layer M1 or/and M2, enter the following:

```
<MASKSHIFTLAYER>
M1 M2
</MASKSHIFTLAYER>
<MASKSHIFTPATTERN>
00 01 10 11
</MASKSHIFTPATTERN>
<CELLMAP>
A AM2B NONE AM1BM2B
</CELLMAP>
```

MASKSHIFTLAYER specifies the relevant metal layers, in order, on a single row.

MASKSHIFTPATTERN specifies the mask shift combinations to apply to the metal layers. The digits in each combination correspond to the metal layers listed in MASKSHIFTLAYER, respectively. For example, the 00 combination represents a shift value of 0 on layers M1 and M2; the 01 combination represents a shift value of 0 on layer M1 and 1 on layer M2; the 10 combination represents a shift value of 1 on layer M1 and 0 on layer M2; and so on.

The number of mask shift combinations is less than or equal to pow(m, n), where n is the number of layers specified in MASKSHIFTLAYER and m is the count of the shift values. For 2-mask layers, valid shift values are 0 and 1, so m is 2. In the example above, two metal layers are specified (M1 M2), so there are four mask shift combinations (00 01 10 11).

CELLMAP specifies the cell mappings that correspond to each mask shift pattern for a given cell. (Each row specifies the mappings for a single cell, so the number of mappings in each row corresponds to the number of MASKSHIFTPATTERN combinations.)

In the example above, cell A is mapped to A, AM2B, NONE, and AM1BM2B when its mask_shift attribute is set to 0 on both layers (00), set to 1 on layer M2 (01), set to 1 on layer M1 (10), and set to 1 on both layers (11), respectively. When a mask-shifted cell is not expected for a particular mask shift combination, the cell name is mapped to NONE.

Feedback

**What Next**

Please check and correct it, and then try again.

---

## NDMX-010

(error) Invalid value for "-write_overlap_shape_of_via".

**Description**

This message occurs when "-write_overlap_shape_of_via" is specified to write_gds/oasis and the value is not expected. The value must be of the form { { viaLayerName CRLayeName resizeValue } ...}

**What Next**

Please check and correct the option value.

---

## NDMX-011

(information) CBCM engine is running in single thread mode.

**Description**

CBCM engine has two modes: single thread mode and multiple thread mode, multiple thread mode will be automatically turned on when max cores is more than 2. e.g. set_host_options -max_cores 4

**What Next**

Please set max cores if you'd like to turn on multiple thread mode.

---

# NEX

---

## NEX-001

(warning) Technology layer '%s' setting '%s' is not valid

**Description**

The specified parameter of the technology layer is not valid, it is recommanded to check technology setup first.

**What Next**

Check technology data and design setup.

Feedback

## NEX-002

(warning) Global NDR rule '%s' setting '%s' is not valid

**Description**

The specified Global NDR rule is not valid, it is recommanded to check TCL script file.

**What Next**

Check Global NDR rule in TCL script and design setup.

## NEX-003

(info) parasitic model has %i cap and %i res corners

**Description**

Extraction is setup with multiple Parasitic tech libraries(tlu+), and with multi temperatures. Num of cap corners and different(or same) num of res corners was checked during extraction setup.

**What Next**

Multi corner Multi mode rc values.

## NEX-004

(info) HNSCALE = %1.3f

**Description**

HFN(Half Node process Scaling) factor, it is a geometry scaling factor that real silicon geoemtry size equals to layout size multiplied by HFN.

**What Next**

Half node process

## NEX-005

(warning) Ignore different TLUPLus HNSCALE = %1.3f in corner '%s'

**Description**

Extraction only support 1 HNSCALE (Half Node Scale) value, All MCMM parasitic tech lib should carry same HNSCALE. Different HNSCALE will be ignored.

Feedback

**What Next**

Check and compare HNSCALE in parasitic tech libraries.

---

## NEX-006

(error) Can not find tech layer '%s' in Parasitic Tech Library

**Description**

Extraction can not model RC for specified tech layer becasue parasitic tech library does not have specified tech layer.

**What Next**

Check technology file and parasitic tech lib(tlu+) for specified layer

---

## NEX-007

(warning) Parasitic Tech Library layer '%s' parameter '%s' = %2.3f does not match %2.3f of technology file

**Description**

There will be accuracy concern becasue specified parasitic tech library parameter doesnt match setting in techlogy file.

**What Next**

Check technology file and parasitic tech lib(tlu+) for specified layer and parameter

---

## NEX-008

(error) Parasitic tech lib via '%s' FROM '%s' To '%s' dont match technology '%s' or '%s'

**Description**

This is a setup error, parasitic tech lib via FROM/TO layer doesn't match technology setting

**What Next**

Check technology file and parasitic tech lib(tlu+) for specified via

---

## NEX-009

(info) r = %f ohm/um, c = %f ff/um, cc = %f ff/um (%s %s)

Feedback

**Description**

This is a information message of average rc value per um. Each metal layer and each routing direction may have seperate average rc value.

The average rc value is based on parasitic tech libraries and pre-defined default utilization rate.

The purpose of average rc is to check parasitic tech lib(tlu+) and to calculate per net rc roughly.

**What Next**

n/a

---

## NEX-010

(info) r = %f ohm, c = %f ff (%s)

**Description**

This is a information message of rc value per via. RC value is based on parasitic tech library.

The purpose of average rc is to check parasitic tech lib(tlu+) and to calculate per net rc roughly.

**What Next**

n/a

---

## NEX-011

(info) %s

**Description**

This message tell extraction progress, the specified step begin to run.

**What Next**

n/a

---

## NEX-012

(info) %s successfully

Feedback

**Description**

This message tell extraction progress, the specified step complete successfully.

**What Next**

n/a

---

## NEX-013

(error) %s failed

**Description**

This message tell extraction progress, the specified step failed.

**What Next**

n/a

---

## NEX-014

(info) %s with %i threads

**Description**

Extraction uses multi thread technology to speed up. This message tells current specified step will run with how many num of threads.

**What Next**

Check extraction elapsed run time improvement.

---

## NEX-015

(info) %s takes %2.2f seconds, elapse %2.2f seconds

**Description**

This is infomation message to tell run time usage of specified extraction step, it will print out cpu time and elapsed times

**What Next**

n/a

---

## NEX-016

(error) term %s bbox not valid

Feedback

**Description**

The specified term doesn't have a valid boundary box

**What Next**

check design set data of specified term

---

## NEX-017

(info) r = %f ohm/um, via_r = %f ohm/cut, c = %f ff/um, cc = %f ff/um (%s %s)

**Description**

This is a information message of average rc value per um. Each routing direction may have seperate average rc value.

The average rc value is based on parasitic tech libraries and current whole design congestion average rate.

The purpose of average rc is to check rc value roughly.

**What Next**

n/a

---

## NEX-018

(warning) no valid parasitic for %s corner(%s)

**Description**

There is no valid parasitic corner, usually because constraint corner has no parasitic tech setup.

**What Next**

Check constraint corner's parasitic tech setup in run script. Run report_parasitic_parameters to make sure all corners are correctly set.

---

## NEX-019

(error) rc scaling parameter %s %0.2f is beyond valid range

**Description**

the rc scaling parameter allowed range is 0~inf

Feedback

### What Next

Check rc scaling parameter in set_extraction_options command

---

## NEX-020

(warning) Net '%s' is exceeding threshold (over %d pins) and will be skipped.

### Description

This net has a very large number of pins so it will not be extracted. This net was probably not meant to be detail routed. There will be no parasitics and no back-annotated delay computed for this net. Timing analysis will use the wire load model with the high fanout threshold assumptions. The current default for the max number of pins is 1000. You can use the option *extract.high_fanout_threshold* to adjust it.

### What Next

Buffer this net and detail route it to be able to extract it.

---

## NEX-021

(warning) Net '%s' is not fully connected and will be estimated.

### Description

Two route segments that you specified are not connected and the tool will estimate the resistance and capacitance of the net.

The tool performs extraction only on completely routed nets. For partially routed nets, the tool uses virtual routing to estimate the resistance and capacitance values. The tool outputs the RC results for all the nets.

As default, only first 100 open nets will have detail info. And total number of open nets will be printed after the extraction done.

### What Next

Perform routing verification to identify all open nets.

---

## NEX-022

(information) The RC mode used is %s.

### Description

This message advises you of the RC model that the tool will use to compute RC value.

Feedback

The valid RC modes and their descriptions are shown below, starting with the highest priority and ending with the lowest priority based on design route status.

- DR(detail routed), if 60% of all nets are detail routed
- GR(global routed), if 60% of all nets are global or detail routed
- CTO(clock routed), if 1 or more nets are global or detail routed
- VR(virtual routed), can be done with/without routed net depending on timer's control
- RDE(route driven estimation), can be done with/without routed net depending on timer's control. Estimation is done thru captured data learned from routing.

---

## NEX-023

(information) The mixed mode parasitic extraction setting is ON.

**Description**

Some nets are routed and some nets are not. CTO/GR/DR extraction will be performed on the fully connected nets. For partially routed or unrouted nets, VR estimation will be performed.

---

## NEX-024

(info) Design %s has %i nets, %i global routed, %i detail routed.

**Description**

Message about route status, which will decide extraction mode.

**What Next**

NEX-022 will be issued based on the ratio of routed/totalNets.

---

## NEX-025

(warning) Design is not placed.

**Description**

Message about design is not placed, not suitable to do parasitic extraction

**What Next**

run placement and the problem will be solved

Feedback

## NEX-026

(warning) Ignore corner %s %s vr scaling, will use corner %s %s vr scaling

**Description**

extractor does not allow different vr scaling for same parasitic tech and temperature. App options extractor.corner_for_vr_scaling is used to control this feature.

**What Next**

Check corner vr scaling setup.

## NEX-027

(warning) Design core area is invalid.

**Description**

Message about design with invalid core area, not suitable to do parasitic extraction

**What Next**

check design and make sure block is correct.

## NEX-028

(info) %i nets are successfully extracted.

**Description**

Message about extraction status, extracted means the net has content in rcCache.

**What Next**

Timer/SPEF output can use RC in rcCache.

## NEX-029

(info) coupling capacitance is created explicitly.

**Description**

Message about coupling capacitance is extracted.

**What Next**

Timer SI can work with coupling capacitance.

Feedback

## NEX-030

(info) coupling capacitance is lumped to ground.

**Description**

Message about coupling capacitance is not extracted.

**What Next**

Timer SI does not work with non-CC extraction.

## NEX-031

(error) No reference direction set in either TLU+ or set_extraction_options

**Description**

TLU+ has orientation ETCH tables, but there is no reference direction setting in either TLU + or using *set_extraction_options*.

**What Next**

Please set reference direction using *set_extraction_options*.

## NEX-032

(information) Orientation ETCH is using reference direction: %s.

**Description**

This message advises you of the reference direction used in orientation ETCH. It can be from either TLU+ or *set_extraction_options*.

**What Next**

This is for your information only.

## NEX-033

(warning) Reference direction is not consistent among TLU+ files, will use direction set in set_extraction_options.

**Description**

This message advises you of the reference direction setting issue in orientation ETCH. ICC2 finds that reference direction is not consistent among TLU+ files. ICC will use the

Feedback

reference direction set in *set_extraction_options*. The final reference direction will be choosen by ICC2 and reported in the log.

**What Next**

Please set desired reference direction using *set_extraction_options*.

---

## NEX-034

(warning) Failed to write SPEF file.

**Description**

This message advises you of empty or incomplete SPEF file.

**What Next**

Please make sure that the SPEF file you specified has enough free disk space.

---

## NEX-035

(warning) Number of mask colors is different among TLU+ files.

**Description**

This message advises you that different number of mask colors has been specified in TLU + files. This will affect accuracy.

**What Next**

Please make sure that the same number of mask colors is specified in all TLU+ files.

---

## NEX-036

(error) read_parasitics failed: %s.

**Description**

This message advises you of the failure of read_parasitics command.

---

## NEX-037

(error) read_parasitics failed: %s.

**Description**

This message advises you of the failure of read_parasitics command when design and spef files are not matched.

Feedback

**What Next**

Please check the consistency among design and the input spef files.

---

## NEX-038

(warning) read_parasitics: %s.

**Description**

This message advises you of the warning of read_parasitics command.

**What Next**

Please check command input or the consistency among design and the input spef files.

---

## NEX-039

(warning) cut loop %s.

**Description**

This message advises you of the warning of some nets defined in spef file have the loop, and the loops have to be cut in reading in the spef files, due to ICC2 timer cannnot resolve the RC grap with loop. By cutting loop, the P2P resistance of this net is not consistent as the original input in spef files, and it would impact the result of timer commands and write_parasitics.

**What Next**

Please provide the input spef files without loops, in order to have accurate RC values annonated.

---

## NEX-040

(error) No rule match is found in RPV_VS_COVERAGE AND no RPV_VS_AREA is specified

**Description**

TLU+ has RPV_VS_COVERAGE table, but no coverage rule is found in the table that matches the layout via coverage. In this case, the tool will use RPV_VS_COVERAGE table to compute via resistance. However, if RPV_VS_AREA table is not specified, this error will be issued. No single RPV value will be used. This is to be consistent with RCXT.

**What Next**

Please specify RPV_VS_AREA table along with RPV_VS_COVERAGE.

Feedback

## NEX-041

(error) RPV_VS_COVERAGE cannot be applied to via array AND no RPV_VS_AREA is specified

**Description**

TLU+ has RPV_VS_COVERAGE table, but it doens't apply to via array. In this case, the tool will use RPV_VS_COVERAGE table to compute via resistance. However, if RPV_VS_AREA table is not specified, this error will be issued. No single RPV will used. This is to be consistent with RCXT.

**What Next**

Please specify RPV_VS_AREA table along with RPV_VS_COVERAGE.

## NEX-042

(error) Number of MASK in TLUPlus exceeds limit of 6.

**Description**

TLU+ has MASK defined and the nubmer of MASKs is over the set limit of 6.

**What Next**

Please correct ITF and regenerate TLUPlus.

## NEX-043

(warning) Both target utilization ratio and fill ratio are set. Target utilzation ration will be used.

**Description**

If both target utilization ratio and fill ratio are set, target utilzation ration will be used. Fill ration from TLUPlus will be ignored.

**What Next**

Please correct your settings.

## NEX-044

(warning) Target utilization ratio cannot be used in real metal fill extraction.

Feedback

**Description**

When real metal fill extraction is set, target utilization ratio will be ignored.

**What Next**

Please correct your settings.

---

## NEX-045

(warning) Missing layer mapping for VIA. Please check layer mapping file.

**Description**

Layer mapping for a via is missing, please check layer mapping file, ITF and TF.

**What Next**

Please correct layer mapping for via.

---

## NEX-046

(warning) Net '%s' has too many geometries for R-network builder and will be extracted with VR mode.

**Description**

Timer/detaly calculator have limitation to number of nodes/edges from R-network. It cannot use node/edge with id >= 65536. So R-tree builder will exit beforehand if there're already more geometries than expected.

**What Next**

VR extraction mode will be applied to this net.

---

## NEX-047

(Error) No valid design block.

**Description**

This message occurs when there is no valid design block for extraction.

**What Next**

Please check design setup.

Feedback

## NEX-048

(Error) Port shape or size is abnormal for port '%portName'.

**Description**

This message occurs when a port has abnormal size.

**What Next**

Please check design setup.

## NEX-049

(Error) Abnormal temperature setting for parasitic corner '%cornerName'.

**Description**

This message occurs when a parasitic corner temperature setting is out of range.

**What Next**

Invoke 'report_parasitic_parameters' command to check parasitic settings.

## NEX-050

(Error) Less than 1 net or 1 instance in design '%designName'.

**Description**

This message occurs when there is less than 1 net or 1 instance in design, extraction cannot continue.

**What Next**

Please check design setup.

## NEX-051

(Error) No valid parasitic corner in design '%designName'.

**Description**

This message occurs when there is no constraint corner with valid parasitic parameter.

**What Next**

Invoke 'set_parasitic_parameters' command to set parasitic tech.

Feedback

## NEX-052

(Warning) Layout width is smaller than TF minimum width. ICC2 will use TF minimum width instead.

**Description**

This message occurs when layout width is found to be smaller than Technology File defined minimum width for corresponding metal layers. This indicates a DRC violation found in the layout.

**What Next**

Please clean up your design.

## NEX-053

(information) read_parasitics: %s.

**Description**

This message advises you how many nets are annotated from spef file.

**What Next**

Please check the consistency among design and the input spef files.

## NEX-054

(Error) starrc_extract_rc failed: %s. This can be ignored if using fusion extraction.

**Description**

This message advises you of the failure in starrc_extract_rc when running StarRCInDesign or PT ECO fusion. It's used by fusion extraction too. The error about non-exist starRC executable image can be ignored if fusion extraction is used in later flow, as fusion extraction integrated StarRC API in FC image and no need to run StarRC.

**What Next**

Invoke 'set_starrc_in_design'

## NEX-055

(warning) starrc_extract_rc: %s.

Feedback

**Description**

This message advises you of the warning in starrc_extract_rc when running StarRCInDesign or PT ECO fusion.

**What Next**

Invoke 'set_starrc_in_design'

---

## NEX-056

(info) %s

**Description**

This message about StarRC in Design progress information

**What Next**

n/a

---

## NEX-057

(error) Net %s has non-TLU+ layer geometry from layer# %d.

**Description**

This message tells there's non-tluplus layer geometry from problem net. RC extraction may be inacccurate.

**What Next**

User need check TLU+/mapping setting to make sure the correct TLU+ is used. For non-TLU+ layer, user should set it as ignored layer before routing starts. Otherwise RC value from the non-TLU+ layer will be 0, which may not be expected.

---

## NEX-058

(warning) The %s block contains invalid active fill.

**Description**

This message occurs when RC extraction detects invalid active fill in a block. Invalid active fill can cause inaccurate RC extraction results.

**What Next**

None.

Feedback

## NEX-059

(error) Empty TLU+ table: %s.

**Description**

This message tells cap table in TLU+/mapping is incorrect. RC extractor cannot be initialized. The table's name will appear as: 8(1)_0_9;Mask#0_0, and the meaning of each number is: 8: ITF/TLU+ sequence number 1: metal1(in mapping file) as working layer 0: 1 layer below working layer(should be ground) 9: 10 layers above working layer(should be metal11) Last 2 numbers are mask ID.

**What Next**

User need check TLU+/mapping setting to make sure the correct TLU+ is used.

## NEX-060

(error) RDE is not enabled %s.

**Description**

This message tells RDE is not setup to be used

**What Next**

User need check opt.common.enable_rde is enabled.

## NEX-061

(error) No valid parasitic for %s.

**Description**

This message tells that there is no parasitic corner defined for a given module.

**What Next**

User needs check corners are well defined in a current module for RC extraction.

## NEX-062

(error) Core area is not valid %s.

**Description**

This message tells core are is not valid for a given design. It appears when core area itself is not valid or height is not larger than 0 or width is not larger than 0.

Feedback

## NEX-063

(warning) StarRCInDesign: %s.

**Description**

This message advises you of the warning of StarRCInDesign command to indicate some nets which terminal has the imcompleted RC network from StarRC.

**What Next**

Please check these terminals/pins in the input design, and these pins might not be placed. Or some issues in StarRC. Please use standalone StarRC to verify the design for these nets.

## NEX-064

(warning) RDE isn't sync up with an app option (extract.rde_via_model).

**Description**

This message tells RDE is not captured with current via model setup (extract.rde_via_model (2|3)).

**What Next**

RDE behavior will do based on captured via model enahancement but we recommend to run the flow from the beginning with RDE via model enhancement.

## NEX-065

(warning) RDE has via model enhancement(2|3) but app option is too old setup(extract.rde_via_model 0|1).

**Description**

This message tells RDE is captured with via model enhancement(2|3) which is OBD but extract.rde_via_model is 0|1 which is not sync up with RDE.

**What Next**

RDE bahavior will do with via model enhancement but we recommend to change RDE setup to turn on rde via enhancement by setting extract.rde_via_model to 2|3 to sync up with RDE.

Feedback

## NEX-066

(error) write_parasitics -attach_gpd failed: ndm design %s does not have permission to attach GPD .

**Description**

In order to attach GPD to ndm, the opened ndm lib and design need to have the written permission.

## NEX-067

(information) metal fill emulation.

**Description**

parasitics technology has metal fill emulation.

**What Next**

DO NOT DO RC correlation with StarRC with metal fill emulation.

## NEX-068

(information) StarRC-in-design is not invoked because no config file is specified in set_starrc_in_design. Native ICC2 extractor is used.

**Description**

StarRC-in-design is not invoked because no config file is specified in set_starrc_in_design. Native ICC2 extractor is used.

**What Next**

please setup set_starrc_in_design.

## NEX-069

(warning) StarRC-in-design is not invoked because no config file is specified in set_starrc_in_design. Native ICC2 extractor is used.

**Description**

StarRC-in-design is not invoked because no config file is specified in set_starrc_in_design. Native ICC2 extractor is used.

Feedback

**What Next**

please setup set_starrc_in_design.

---

## NEX-070

(warning) "Missing the StarRC command specified in COMMAND_FILE in config file for starrc_centric mode. icc2_centric mode is used."

**Description**

If set -mode starrc_centric in set_starrc_in_design to user the RC setting from StarRC, users need to specify StarRC command file in COMMAND_FILE in config file. Otherwise, icc2_centric mode is used in StarRCInDesign to set RC setting from ICC2 settting.

**What Next**

please refer to set_starrc_in_design -mode.

---

## NEX-071

(error) "RDE Data doesn't match with Design Technology. Layer mismatch RDE( %d ) Design( %d )"

**Description**

When importing RDE it errors out if layer number mismatch between RDE and Design.

---

## NEX-072

(error) "Virtual Metall Fill parameter file %s is not specified or does not exist, or cannot be open/read."

**Description**

Virtual metal fill application needs to specify the paramter file in the command set_extraction_options -virtual_metalfill_parameter_file fileName

**What Next**

Invoke 'set_extraction_options'

---

## NEX-073

(information) "Virtual Metall Fill parameter file uses the starrc metalfill parameter file %s."

Feedback

**Description**

If the parameter file is not specified in -virtual_metalfill_parameter_file of the command set_extraction_options and starrc virutal metalfill parameter file is specified in -virtual_metalfill_starrc_parameter_file , then the starrc virtual metalfill parameter file would be used.

**What Next**

Invoke 'set_extraction_options'

---

## NEX-074

(Error) "Virtual Metall Fill parameter file %s is not valid: %s."

**Description**

The virtual metalfill parameter file is specifed with fill width and fill space on the routing layers.

**What Next**

Invoke 'set_extraction_options'

---

## NEX-075

(warning) "RDE Data is not loaded because of design size change. Please re-generate RDE or it goes with VR."

**Description**

When design size changes as like by pin or port change, RDE is not loaded because congestion map mismatches by sizing change. Normally, RDE congestion Map is registered to RDE itself when it is generated. But RDE generated by old version is not handled.

**What Next**

User is supposed to re-generate RDE or it would go with VR.

---

## NEX-076

(information) "Virtual Metall Fill in fusion extraction to use the starrc metalfill options file %s."

Feedback

**Description**

If the options file is specified in -virtual_metalfill_starrc_options_file of the command set_extraction_options, then the starrc virtual metalfill options file would be used in Fusion extraction.

**What Next**

Invoke 'set_extraction_options'

---

## NEX-079

(warning) "Failed to to capture RDE for a design %s due to no routing shape."

**Description**

There is no routing shapes which are essential for RDE capturing.

Design is supposed to have routed shape with GR/DR for RDE and RDE calls a global router to generate routing shapes by default unless it is forced not to call a global router. LighRDE(FC) doesn't call a global router because it uses glinks from place_mode GR from previous stage.

But it seems RDE capturing is done on unrouted design forcing not to call a global router or global router failed to provide glinks to RDE for some reason.

**What Next**

User is supposed to re-generate RDE or it would go with VR.

---

## NEX-080

(Error) "IVM file parsing parameter failed. %s Table will be ignored."

**Description**

IVM file is specified with right to its own syntax.

**What Next**

Invoke 'read_ivm'

---

## NEX-081

(Error) "Via Variation setup failed. %s"

**Description**

There is associating process to bind ivm file to parasitic corners.

Feedback

**What Next**

Invoke 'read_ivm'

---

## NEX-082

(Error) "IVM file name needs to be specified."

**Description**

IVM file name needs to be specified for the command read_ivm / write_ivm

**What Next**

Please refer to 'read_ivm' and 'write_ivm'

---

## NEX-083

(Error) "No valid corner list."

**Description**

Given corner name should be valid for ivm file UI commands.

**What Next**

Invokd 'read_ivm', 'write_ivm', 'report_ivm'

---

## NEX-084

(Error) "No via variation table exists."

**Description**

There is no via variation table for a given corner ivm UI commands.

**What Next**

Invokd 'read_ivm'.

---

## NEX-085

(Warning) "NEX: fusion cannot accept: (%s) from StarRC command file, please use (set_starrc_in_design -reduction) to provide the REDUCTION setting."

Feedback

**Description**

REDUCTION setting from StarRC command file is ignored, user should use set_starrc_in_design -reduction to provide the REDUCTION setting used by fusion StarRC engine.

**What Next**

Invoke 'set_starrc_in_design'

---

## NEX-089

(information) "TLUPlus is fill emulated."

**Description**

TLUPlus is fill emulated

**What Next**

Invoke 'read_parasitic_tech'

---

## NEX-090

(error) Conducing layer %s doesn't have mask aware capacitance table.

**Description**

conducting layers doesn't have etch cap table during read_parasitic_tech it.

---

## NEX-091

(information) "NEXML %s"

**Description**

This message indicates the local customized ML train models are used.

---

## NEX-092

(Warning) "NEX: TLU+ file doesnot contain nxtgrd for fusion or some corner nxtgrd is not set, back to native extractor."

**Description**

To run fusion extraction needs nxtgrd mapping file in starrc config file. If there is no nxtgrd mapping file, then TLU+ is triggered and run.

Feedback

### NEX-093

(Error) "write_rde command failed, %s"

**Description**

write_rde failed due to no RDE exists.

**What Next**

Invoke 'create_rde' to create RDE before 'write_rde'.

## NLDM

### NLDM-001

(error) Discarded NLDM for %s on lib cell %s for arc:\n \t'%s'.\n \tReason: %s

**Description**

While reading a db file and converting NLDM data from timing arcs, an unsupported construct was found which caused the NLDM for this arc to be discarded. For example, if a rise delay table contains an invalid value (NaN, infinity, or -infinity) which could later cause a floating point exception, we discard the table. The message will indicate which arc is causing the problem. Note that Library Compiler does issue warnings for the use of such values. We not only discard the table with invalid value, but also discard it's complimentary table, so you wouldn't get NLDM-008 during *check_workspace*. For example, if cell_rise data was discarded, it's complimentary rise_transition data was also discarded.

**What Next**

Keep in mind that by removing NLDM data for part of an arc will likely cause additional errors during *check_workspace*. NLDM-008 will tell you that cell_rise is missing for the same arc you saw in this message, and NLDM-003 will summarize that there were irreparable NLDM attributes. Your only recourse within the application is to remove the offending lib cell with *remove_lib_cell*. Externally, you can look at the Liberty (.lib) file and fix the erroneous value(s).

**See Also**

- NLDM-008
- NLDM-003
- remove_lib_cell

Feedback

## NLDM-002

(warning) Missing %s_transition for the following arc: \t%s

### Description

The listed timing arc has a delay attribute without the complimentary transition (slew) attribute. For example, a cell_rise was found without rise_transition. Modern versions of Library Compiler do not allow this however this situation can be found in older DB libraries. Prior to this message you would see an indication of the library being checked, for example:

```
\&...verifying arcs for file1.db
```

That is the problem library. The arc description will show the cell, from and two pins, sense, when condition and sdf condition (if applicable) in order to make it as clear as possible which arc is at fault.

### What Next

This is a warning. The application will correct this for you by adding an identity transition table so that output transition time equals input transition. But if you have the source of the DB file, you should consider updating it.

## NLDM-003

(error) Found irreparable NLDM attributes.

### Description

Within a logic library, complimentary attributes (for example, rise delay and rise transition) for some arcs did not have identical breakpoints, and there was some issue preventing the breakpoints from being homogenized.

### What Next

This issue causes the *check_workspace* command to fail. You must run the *check_workspace* command with the *-details arcs_progress* option to determine the specific problem arcs. After identifying the problem arcs, update the logic library and redo library preparation.

## NLDM-004

(information) Found some variation in breakpoints between complementary\n \tattributes - %d can be repaired.

Feedback

### Description

Within a library, complementary attributes (for example, rise delay and rise transition) for some arcs did not have identical breakpoints, but had a type of variation which can be repaired. Currently, there are only automated repair methods. If the difference is that the attributes are a function of the same variables but have different indices, the union of the table indices will be used with no loss of accuracy.

In cases where one attribute has an extra variable, the lesser breakpoints will pick up the missing variable with no loss of accuracy.

In cases where one attribute is a function of one variable (like Tclk) and the other is a function of a different but paired variable (Tdat in this example), each breakpoints is expanded to include the missing variable, with no loss of accuracy.

In cases where the arc has a delay attribute but transition is missing, the application will correct this for you by adding an identity transition table so that output transition time equals input transition. This case should be very rare as the current Library Compiler does not allow creation of such an arc.

### What Next

At this time, there is no user intervention available. The *check_workspace* command will fix these issues.

---

## NLDM-005

(error) Different number of %s and %s attributes.

### Description

Delay arcs in the logic library must have both delay and slew attributes. To quickly verify that the library meets this requirement, the tool verifies that there are the same total number of each attribute, for example, rise delay and rise slew. The tool issues this message if the attribute counts do not match.

### What Next

Analyze the logic library source file to determine why some arcs are missing the delay or slew attribute.

---

## NLDM-006

(error) %s data missing for some of these libraries

Feedback

### Description

The data from Liberty needs to exist in all or none of the libraries in a scaling group. The libraries in your proposed scaling group do not fit that description. You cannot create a scaling group with these libraries.

---

## NLDM-007

(error) Failed to write to on-disk cache in directory %s. \tReverting to in-memory only.

### Description

NLDM data is cached on disk and is loaded on-demand, per-arc, when some part of the application requests the data. The path to this cache starts with the value of the application option *shell.tmp_dir_path*. This message is similar to NLDM-006, but slightly different. This one indicates that the application was able to create the disk cache, but sometime later, it was unable to write to it. The most likely causes are a full disk. In this situation, the application falls back to an in-memory only representation. This can dramatically increase memory usage.

### What Next

The application will function as before, albeit with a much larger memory footprint. You can also stop and restart after setting the application option *shell.tmp_dir_path* to a different disk with more space.

---

## NLDM-008

(error) Missing cell_%s \t'%s'

### Description

The listed timing arc has a transition (slew) attribute without the complimentary delay attribute. For example, a rise_transition was found without cell_rise. As modern versions of Library Compiler do not allow this, either the DB is an old version, or the source might be using rise/fall_propagation groups, which this application does not support. Prior to this message you would see an indication of the library being checked, for example:

```
\&...verifying arcs for file1.db
```

That is the problem library. The arc description will show the cell, from and two pins, sense, when condition (if applicable) in order to make it as clear as possible which arc is at fault.

It is also possible that the cell_rise/fall for this arc had invalid values. If this is the case, you would have seen NLDM-001 messages when you read the source file.

Feedback

### What Next

This is an error. If you have the Liberty source file, you need to express delay in terms of cell_rise and cell_fall, and these constructs must have valid numerical values. Otherwise, the only recourse is to remove the offending cell using *remove_lib_cell* on the contributing library.

### See Also

- NLDM-001
- remove_lib_cell

# NP

## NP-001

(Warning) No physical information exists for design '%s'. Zero interconnect delay is used in timing analysis.

### Description

Either the physical information for the design is not loaded or it is incorrectly initialized. The net parasitics can not be calculated and the timer will use the zero interconnect delay for net estimation.

### What Next

Check the set up for the design. Check if the design was placed correctly and has valid core box defined.

## NP-002

(Warning) The extractor can not be initialized for design '%s'. Zero interconnect delay is used in timing analysis.

### Description

Either the physical information for the design is not loaded or it is incorrectly initialized. The net parasitics can not be calculated and the timer will use the zero interconnect delay for net estimation.

### What Next

Check the set up for the design. Check constraint corner's parasitic tech setup in run script. Run report_parasitic_parameters to make sure all corners are correctly set.

Feedback

### NP-003

(Warning) The extractor can not be initialized for block '%s'.

**Description**

Either the physical information for the block is not loaded or it is incorrectly initialized. The extractor for the top block will be used to estimate the nets within the block.

**What Next**

Check the set up for the block. Check if the design was placed correctly and has valid core box defined.

### NP-004

(Information) Coupling caps annotated for %u nets in block '%s' (Mem = %.4f MB).

**Description**

Information about the coupling caps annotated from an abstract block for the nets retained in the abstract netlist. The annotated coupling caps are used in signal integrity (SI) analysis during the timing update. The message also prints the total memory requirement for the annotated coupling caps.

## NPLDRC

### NPLDRC-001

(info) Analyzing PG structure for %ld layers using %d cores.

**Description**

The tool is analyzing the PG and user route shapes and vias so it can check for DRC violations between placed lib_cells and the PG grid.

The result of the PG structure analysis will be saved with the design in its NDM library so it can be re-used in later runs.

In future runs if the PG structures have changed then the PG structure analysis will be re-run on the updated PG structures.

**What Next**

If the design has been freshly read and not yet been saved to an NDM library, there is no place for the analysis to be saved. The analysis will re-run each time PG DRC checking is done.

Feedback

Try saving the design to its NDM library after reading the design:

```
prompt> save_block
```

**See Also**

- legalize_placement
- save_block

## NPLDRC-002

(info) Saving PG structure for '%s'.

**Description**

The tool is saving the PG structure analysis in the design's NDM library so it can be re-used in later runs.

In future runs if the PG structures have changed then the PG structure analysis will be re-run on the updated PG structures.

**What Next**

The tool can now check for DRC violations between placed lib_cells and the PG grid.

**See Also**

- legalize_placement

## NPLDRC-003

(info) Reading PG structure for '%s'.

**Description**

The tool is reading the PG structure analysis saved in the design's NDM library from a previous run.

Since the power structures have not changed since the previous run the PG structure analysis results can be re-used.

**What Next**

The tool can now check for DRC violations between placed lib_cells and the PG grid.

**See Also**

- legalize_placement

Feedback

## NPLDRC-004

(info) Updating unknown PG structure for '%s'.

### Description

The tool could not read the saved PG structure analysis for this design from the NDM library because the file format was not recognized.

The tool will re-analyze the PG and user route shapes and vias so it can check for DRC violations between placed lib_cells and the PG grid.

The result of the PG structure analysis will be saved with the design in its NDM library so it can be re-used in later runs.

### What Next

The tool will automatically re-run the PG structure analysis for this design.

### See Also

- legalize_placement

## NPLDRC-005

(info) Updating PG structure for '%s'.

### Description

The tool could not re-use the saved PG structure analysis for this design from the NDM library because the PG structures have changed since that run.

Changing the PG grid or user route shapes causes the tool to re-run the PG structure analysis before checking for DRC violations.

The tool will re-analyze the PG and user route shapes and vias so it can check for DRC violations between placed lib_cells and the PG grid.

The result of the PG structure analysis will be saved with the design in its NDM library so it can be re-used in later runs.

### What Next

To avoid unneeded PG structure analysis, make all updates to the PG grid and setting clock tree and other user routes at once instead of incrementally.

The tool will automatically re-run the PG structure analysis for this design.

Feedback

**See Also**

- legalize_placement

---

## NPLDRC-006

(warning) Could not read PG structure for '%s'.

**Description**

The tool could not read the saved PG structure analysis for this design from the NDM library because the file contents were not recognized.

The tool will re-analyze the PG and user route shapes and vias so it can check for DRC violations between placed lib_cells and the PG grid.

The result of the PG structure analysis will be saved with the design in its NDM library so it can be re-used in later runs.

**What Next**

The tool will automatically re-run the PG structure analysis for this design.

**See Also**

- legalize_placement

---

## NPLDRC-007

(warning) Detected a sector %s on layer %s with %d metal and/or via shapes.

**Description**

Unusual PG grid definition consisting of combinations of small straps and inlaid vias or vias with abutting enclosures create huge sectors that are usually long and consist of many shapes. This might cause long runtime because the tool must merge these shapes to perform legality checks.

**What Next**

Check the PG grid definition and reformat if possible.

---

## NPLDRC-008

(warning) place.legalize.auto_adjust_drc_rules_for_short_pin disables %s rule in %s layer..

Feedback

**Description**

The tool detects short pins in the design and automatically disables color_based_end_of_line_alignment rule in M1.

**What Next**

The tool would not report Color-Based End-of-line Alignment rule on M1 short pins which can be patched or extended during detail_route.

**See Also**

- legalize_placement

---

## NPLDRC-009

(Warning) Net name changed from %s to %s in cache.

**Description**

Net name cached by tool has been changed.

**What Next**

This is legitimate. Nothing needs to be done.

**See Also**

- legalize_placement

---

## NPLDRC-010

(Warning) Net %s removed from cache.

**Description**

Net cached by tool has been removed.

**What Next**

This is legitimate. Nothing needs to be done.

**See Also**

- legalize_placement

Feedback

### NPLDRC-011

(Warning) Technology file error. Via layer %s cut definition properties, cutNameTbl(%d), cutWidthTbl(%d), cutHeightTbl(%d), do not match size. Skipping processing of via layer.

**Description**

Error in technology file via layer cut definition.

**What Next**

Fix technology file.

**See Also**

- legalize_placement

## OASIS

### OASIS-001

(error) Library %s is not Found.

**Description**

The Given library is not Found.

### OASIS-002

(error) Design %s with view type %s is not Found in the Library.

**Description**

The Given Design name with the specified viewtype is not Found in the Specified Library.

### OASIS-003

(error) Technology Defnition for Layer Number %d and Purpose %d not defined.

**Description**

The Given Design doesn't have a LAYER Definition for the Layer in question in the Associated Technology Library.

Feedback

## OASIS-004

(information) Text %s (%d, %d)is creating a short with already detected Pin %s.The current text will be ignored.

**Description**

The Text is having a coordinate which overlaps with a geometry with has been already detected as another pin. So this is creating a short and will be ignored.

## OASIS-005

(error) Current Design %s is not Found.

**Description**

The current Design is not Found.

## OASIS-006

(error) Current Design is not of LAYOUT type.

**Description**

The current Design loaded is not of Layout type.

## OASIS-007

(information) No Design has been Added or Modified in the current read_oasis. So no tracing will happen.

**Description**

The current read_oasis has not modified any Designs, so pin tracing won't happen as part of read_oasis.

## OASIS-008

(warn) Multiple Boundary definition found for Module %s, keeping the original definition.

**Description**

The current OASIS file has multiple boundary definition for a module.

Feedback

## OASIS-009

(error) Reading Directory is not supported. File %s is a directory.

**Description**

The input file is a directory.

## OASIS-010

(warn) Layer mapping rule for "%d:%d" is defined multiple times. Only the last instance will be effective.

**Description**

Multiple Layer Mapping rule is not supported in OASIS read/write utility.

## OASIS-011

(error) OASIS Parsing Error: %s

**Description**

OASIS file has problem.

## OASIS-012

(warn) Technology Defnition for Layer Number %d and Purpose %d not defined.

**Description**

The Given Design doesn't have a LAYER Definition for the Layer in question in the Associated Technology Library.

## OASIS-013

(warn) Layer Mapping for OASIS LAYER %d and DATATYPE %d not specified.

**Description**

For the OASIS LAYER and DATATYPE, layer mapping is not specified in the layer map file.

## OASIS-014

(error) Library name has spaces.

Feedback

**Description**

A library name had spaces.

**What Next**

Check that the given library name is valid simple path.

---

## OASIS-015

(error) Library name specified %s cannot be same as workspace library.

**Description**

The Given library name is not allowed.

---

## OASIS-016

(error) Wrong Block Map file Syntax.

**Description**

The syntax of Block Map file is not allowed.

---

## OASIS-017

(error) Wrong layer map file syntax at line %lu.

**Description**

The syntax of Layer Map file at the specified line number is not allowed.

**What Next**

Please check the man-page of the read_oasis(2), write_oasis(2) for the correct syntax.

---

## OASIS-018

(warning) FIXEDMASK defined, geometry without MASK will be set as same_mask type.

**Description**

The module has the fixed mask attribute set but the geometry doesn't have a mask attribute defined. The geometry will be set as same_mask type.

Feedback

## OASIS-019

(error) The mask number exceeds the number of masks defined for the tech layer %d.

**Description**

The Mask number should not exceed the number of masks defined in the Tech layer.

## OASIS-020

(error) Library '%s' is not a physical lib_cell library.

**Description**

The library specified for read OASIS must be a physical lib_cell library.

**What Next**

Choose a physical lib_cell library to read OASIS in.

## OASIS-021

(information) Read OASIS into %s library '%s'.

**Description**

This message reports the name of the library to accomodate OASIS data. It also indicates whether the library is an existing library or a new library.

**What Next**

No action required.

## OASIS-022

(warn) The Application option "oasis.port_type_map" is not defined before "read_oasis" command.

**Description**

The Application option "oasis.port_type_map" is not defined before "read_oasis" command. OASIS generated libraries will not generate any PG pins. This may create problem in final library generation. Please check your setup.

Feedback

## OASIS-023

(warn) The Module '%s' doesnot have any port matching the PG names provided by "oasis.port_type_map" application option

### Description

The Module doesnot have any port matching the PG names provided by "oasis.port_type_map" application option. OASIS generated Module will not have any PG pins. This may create problem in final library generation. Please check your setup.

## OASIS-024

(warn) Text '%s' is not on routing layer. Tracing for the given text may result in identification of wrong pins.

### Description

Text '%s' is not on routing layer. Tracing for this text has been enabled on all routing layers by setting the application option "oasis.trace_unmapped_text".Also, routing layer to text layer mapping is not specified using "oasis.text_layer_map" application option. Connectivity tracing for the text will be done on all layers. This may lead to errornous result. Please specify routing layer to text layer mapping using "oasis.text_layer_map". Please check your setup.

### What Next

```
oasis.port_type_map(3),
oasis.trace_unmapped_text(3),
```

## OASIS-025

(error) The %d is outside the alowed valid range for %s in layer map. Valid range is between %d and %d.

### Description

Value specified outside valid range.

## OASIS-026

(error) The special charecter %s is not allowed in %s field of layer map.

### Description

The special character is not allowed in layer map field.

Feedback

## OASIS-027

(error) The input OASIS file name and the output ascii file name can't be same.

**Description**

OASIS file name and Ascii file name are same.

## OASIS-028

(warning) The string '%s' is longer than %lu characters. Please use '-long_names' for proper output.

**Description**

The length of the string name to be written in the OASIS file is longer than the supported length. The name will trucated.

**What Next**

The command "write_oasis" needs to be run with "-long_names" option.

## OASIS-029

(Information) %s '%s' already exists (will %s).

**Description**

This message indicates that specified object already exists when reading a OASIS file into an existing library. The library manager shell, lm_shell, will take action according to the merge settings.

**What Next**

No action required.

## OASIS-030

(information) Change %s name '%s' from OASIS file to '%s' to resolve a conflict.

**Description**

When the object to be created by read OASIS conflict with an existing object, there is an attempt to change the name of the object to be created to a unique name to resolve the conflict.

Feedback

**What Next**

No action required.

---

## OASIS-031

(warning) Merge action in '%s' will be ignored when read OASIS into a new library.

**Description**

When read OASIS into a new library, merge settings will be ignored.

**What Next**

Do not specify merge related options when read OASIS into a new library.

---

## OASIS-032

(warning) The width of the boundary %d is more than width %d of site width.

**Description**

The width of the boundary is more than width of the site def.

**What Next**

May need to check the boundary or the site def of the lib-cell.

---

## OASIS-033

(warning) The height of the boundary %d is more than height %d of site def.

**Description**

The height of the boundary is more than height of the site def.

**What Next**

May need to check the boundary or the site def of the lib-cell.

---

## OASIS-034

(warning) The user specified view '%s' for lib-cell '%s' is not available.

**Description**

The user specified view for lib-cells is not available for this lib-cell design. The definition for the current lib-cell design will not be written to the OASIS file.

Feedback

### What Next

The user may need to change the view to be written to OASIS file for the lib-cell design using option "-lib_cell_view".

---

## OASIS-035

(warning) The design '%s' is found to have mismatches. Use allow_design_mismatch option to write OASIS file.

### Description

The design is found to have mismatches due to which allow_design_mismatch option needsto be used to write OASIS file. Please use report_design_mismatch user command initially to review all the mismatches. If the design mistmatches are found to be fine, use the option allow_design_miismatch to write the OASIS file.

### What Next

Use the command report_design_mismatch to review all the design mismatches. The user may then use allow_design_mismatch option to write the OASIS file.

---

## OASIS-036

(error) The mask number(%s) for the tech layer %d doesn't belong to a VIA Cut Layer.

### Description

The Mask number should be asscoiated with a VIA cut layer. The input layer number is not a VIA cut layer.

### What Next

Please check the OASIS layer map.

---

## OASIS-037

(warning) Unable to obtain mask value for surround on lower metal layer of via '%s'.

### Description

This warning message is issued since the surround on lower metal layer of the via cannot touch any pin in the design. This could be a via that only connects two layers but is not on a pin.

### What Next

Please check the via and the design data if needed.

Feedback

## OASIS-038

(warn) Zero width spur in the input point list {%s}.

**Description**

The Input point list has zero width spur. The polygon will not be added.

**What Next**

Please check the point list been passed.

## OASIS-039

(error) Invalid layer-name '%s' specified at line %lu in the input layer-map file.

**Description**

A layer by the specified name does not exist. Please check the layers present in the associated technology file.

## OASIS-040

(error) Invalid value '%s' specified for '%s', at line %lu in the input layer-map file.

**Description**

The value specified for the flag is not a valid value. Please check the man-page of read_oasis(2), write_oasis(2) for the allowed values. The stream file will still be written out but ignoring the line with invalid value in the layer mapping file. This error will be reported in several scenarios. For example, the input value cannot be recognized due to type error or the use_type in newer version of the tool is not supported in the older version.

**What Next**

Please modify the layer mapping file for the invalid value or use the latest version of the tool.

## OASIS-041

(warning) Ignoring the invalid layer map file syntax at line %lu.

**Description**

The syntax of Layer Map file at the specified line number is not allowed.

Feedback

**What Next**

Please check the man-page of the read_oasis(2), write_oasis(2) for the correct syntax.

---

## OASIS-042

(error) Line %lu: multi-pattern map entry with wild-card cannot co-exist with multi-pattern map entry with specified layer for same mask type.

**Description**

This message occurs when there is invalid line in the layer mapping file. Multi-pattern map entry with wildcard cannot co-exist with Multi-pattern map entry with specified layer for same mask type. For example, M1 * *:108 M1 12 12:108

**What Next**

Remove Multi-pattern map with wildcard or Multi-pattern map with specified layer.

---

## OASIS-043

(error) Line %lu: same OASISLayer:OASISDataType appears in both regular mapping and multi-pattern mapping.

**Description**

This message occurs when there are invalid lines in the layer mapping file. Same OASISLayer:OASISDataType cannot appear in both regular mapping and multi-pattern mapping. For Example, A 12 13:121 M1 12 13:121

**What Next**

If in multiple-pattern technology process, please remove the regular mapping. If not, remove the multi-pattern mapping.

---

## OASIS-044

(error) The ICC2 layer number '%d', specified at line %lu is not a valid value for %s layer map format.

**Description**

The ICC2 layer number specified should follow the limits below: icc2 : 0 - 32767 icc default : 0 - 255 icc extended : 0 - 4095

**What Next**

Use layer-numbers within the specified limits.

Feedback

## OASIS-045

(warning) Technology layer for layer number %d, does not exist. Ignoring the layer-map entry at line %lu.

**Description**

The technology file associated with the current library does not have a layer defined with the given layer-number.

## OASIS-046

(warning) The magnification %f is different than the default value of 1.0. This will be ignored.

**Description**

The magnification specified is different from 1.0 and hence is ignored. Only magnification of 1.0 is supported now.

## OASIS-047

(error) Invalid purpose-name '%s' specified at line %lu in the input layer-map file.

**Description**

A tech-layer purpose by the specified name does not exist. Please check the tech-layer purposes present in the associated technology file.

## OASIS-048

(error) The simple via def '%s' doesnot have the lower overhang metal layer set.

**Description**

A simple via def has both the lower and upper metal overhang defined. A simple via_def cannot have either of them not defined .

**What Next**

Use 'report_via_defs' to review the set of via_defs.

## OASIS-049

(error) The simple via def '%s' doesnot have the upper overhang metal layer set.

Feedback

**Description**

A simple via def has both the lower and upper metal overhang defined. A simple via_def cannot have either of them not defined .

**What Next**

Use 'report_via_defs' to review the set of via_defs.

---

## OASIS-050

(information) Translating cell '%s' as cell '%s/%s'.

**Description**

The OASIS cell being read is translated to the given cell name.

---

## OASIS-051

(error) Input file '%s' is neither a GDSII nor an OASIS file.

**Description**

The input file can be either a GDSII file or an OASIS file.

---

## OASIS-053

(error) Input file '%s' is not a flat structured OASIS file.

**Description**

Only flat structured OASIS file can be given as input file when merge_action is replace_layer and append.

---

## OASIS-054

(error) Input file '%s' is not a appropriate OASIS file.

**Description**

Illegal usage of the command.

---

## OASIS-055

(error) Reading of input OASIS file failed.

Feedback

**Description**

When reading of OASIS file fails, design will be restored to it's previous state.

---

## OASIS-056

(information) no mapping information for system purpose %s in layer mapping file. Mapped to default data type.

**Description**

When there is explicit mapping of system purpose (-1, -2, -3 and -4) in layer mapping file, the tool will use that mapping; for example, with the mapping "all 32:-1 32:55", the tool maps database layer "32:-1" to GDS layer 32:55; if no mapping for these system purposes, the tool will follow the mapping of the purpose 0 of the same layer. For example, if the mapping has "all 32:0 32:70", system purpose "-1" will be mapped to GDS layer 32:70. Please note that GDS datatype can't be any data less than 0, the usage like "all 32:-1 132:-1" is invalid for GDS language, and the datatype will be taken as 0 like "A 32:-1 132 0".

**What Next**

If the implicit mapping as described above does not meet the requirement, you can explicitly map the system purpose (-1, -2, -3 and -4) in layer mapping file.

---

## OASIS-057

(warning) Found child design '%s' for option "-bbox_list".

**Description**

Option "-bbox_list" is only for flatten design, otherwise, it might cause OASIS file inconsistency.

**What Next**

To flatten the design and try the option "-bbox_list" again.

---

## OASIS-058

(error) Block %s is not Found in the Library.

**Description**

The given VIB block name is not Found in the current design of the Specified Library.

Feedback

## OASIS-059

(information) Outputting hierarchy '%s' from '%s' view.

**Description**

write_oasis is generating the structure for the hierarchy.

## OASIS-060

(error) command failed at outputting structure '%s'.

**Description**

write_oasis failed at generating the content for the structure.

**What Next**

Please check what's the exact data caused the failure.

## OASIS-061

(warning) The user specified view '%s' for cell '%s' is not available.

**Description**

The user specified view for cells is not available for this cell. If all specified views cannot be found for the cell, design view will be output, if any. The definition for the cell will not be written to the GDSII file, if all specified views and design view cannot be found.

**What Next**

The user may need to change the view to be written to OASIS file for the cell using option "-switch_view_list".

## OASIS-062

(information) Cell instance '%s' referred to reference design '%s' is retained and will refer to designs in reference library.

**Description**

It means the block to which the cell instance referred is retained in block mapping file, and the CELL record in OASIS file is skipped, the cell instance will be bound to corresponding reference design in reference library.

Feedback

**What Next**

No action required.

---

## OASIS-064

(error) Input file '%s' is GDSII format file instead of OASIS.

**Description**

Command 'read_oasis' only support OASIS format file.

**What Next**

Please use 'read_gds' command to read in this file.

---

## OASIS-065

(error) Invalid value for option -mask_shift_layers.

**Description**

There can not be duplicated layers in the specified value.

---

## OASIS-066

(error) Mask shifted cell '%s' does not have its corresponding mapping in the map file.

**Description**

There is not a corresponding mapping for the mask shifted cell in the map file.

**What Next**

Please add the corresponding mapping to the map file and then try again.

---

## OASIS-067

(error) Mask shift property mismatch on cell '%s'.

**Description**

There is mask shift property mismatch between different map files.

**What Next**

Please correct the mask shift property and then try again.

Feedback

## OASIS-068

(error) Fail to remove the file '%s'.

**Description**

The oasis file is removed if failure occurs.

**What Next**

Please ignore the file.

## OASIS-069

(warning) Found uncolored multi-patterning shapes in the database which won't be output.

**Description**

This warning message occurs when there are some uncolored multi-patterning shapes in the database the check is controlled by the app option file.oasis.ignore_uncolored_shapes or the syntax forbidNoMaskConstraint in technology file. write_oasis will continue to write out all other shapes except the uncolored ones.

**What Next**

Please use command check_shapes -uncolored to find out all these shapes and color them, or turn off the app option mentioned above to output uncolored shapes. Note, if the syntax forbidNoMaskConstraint is 1, the value of the app option will be ignored.

## OASIS-070

(error) Unsupported option '%s' to new engine of write_oasis.

**Description**

It means the command option or app option isn't supported at current stage.

**What Next**

To achieve the option feature, turn off the file.oasis.enable_new_writer to output OASIS layout file.

## OASIS-071

(warning) The deprecated option '%s' is ignored when the new writer app option is enabled.

Feedback

**Description**

It means that the command option or app option is deprecated and ignored when turning on the app option "file.oasis.enable_new_writer".

**What Next**

Remove the option setting

---

## OASIS-075

(information) Output mask-shifted cell name according to rename files.

**Description**

This message occurs when -rename_cell is specified to write_oasis and the rename files are only for mask-shifted cells.

**What Next**

No action required.

---

## OASIS-077

(warning) Ignoring %s, because there is another design with the same name %s.

**Description**

This warning message occurs when app option "file.oasis.output_first_same_name_cell" is on, write_oasis only writes the first encountered cell.

**What Next**

Please turn off the app option "file.oasis.output_first_same_name_cell" if you expect that write_oasis renames the cells with the same name and writes them as different structures in the output oasis file.

---

## OASIS-078

(warning) Found uncolored shapes on multi-patterning layer in the database which won't be output.

**Description**

This warning message occurs when there are some uncolored shapes on multi-patterning layer in the database, the check is controlled by the app option file.oasis.check_mask_constraints, value "loose". write_oasis will continue to write out all other shapes except the uncolored ones.

Feedback

**What Next**

Please use command check_shapes to find out all these shapes and color them, or turn off the app option mentioned above to output uncolored shapes.

---

## OASIS-079

(warning) Found colored shapes on uni-patterning layer in the database.

**Description**

This warning message occurs when there are some colored shapes on uni-patterning layer in the database, the check is controlled by the app option file.oasis.check_mask_constraints, value "loose". write_oasis will continue to write out all shapes.

**What Next**

Please use command check_shapes to find out all these shapes.

---

## OASIS-080

(error) Found mismatch on shapes and layers for color check.

**Description**

This error message occurs when there is mismatch on shapes and layers for color check in the database, the check is controlled by the app option file.oasis.check_mask_constraints, value "strict".

**What Next**

Please use command check_shapes to find out all these shapes and fix them.

---

## OASIS-081

(error) The layer '%s' provided by "file.oasis.text_layer_map" application option was not found.

**Description**

A layer by the specified name in "file.oasis.text_layer_map" application option does not exist. Please check the layers present in the associated technology file.

**What Next**

```
file.oasis.text_layer_map(3)
```

Feedback

## OASIS-082

(warning) Ignoring invalid value '%s' provided for app option file.oasis.shape_use_for_pg_mesh.

**Description**

Value provided for app option file.oasis.shape_use_for_pg_mesh is invalid. Valid values are case-insensitive "stripe", "ring", "lib_cell_pin_connect" or "macro_pin_connect".

**What Next**

Specify correct values for app option.

## OASIS-083

(error) The design %s is not of the 3dic_top type.

**Description**

The design %s is not of the 3dic_top type

**What Next**

Make sure the design is 3dic_top type and run the command again

## OASIS-100

(warning) Ignoring unsupported magnification value '%f'.

**Description**

This message occurs when the input OASIS file contains magnification information and the magnification value is not equal to 1.0. Magnified cell instance is not supported, so ignore it and use the default value 1.0.

## OASIS-101

(Error) Fail to write out PLACEMENT record for cell instance %s at (%d %d).

**Description**

Fail to write out PLACEMENT record in OASIS file for the cell instance at specified location.

Feedback

## OASIS-102

(Error) Fail to write out RECTANGLE record.

**Description**

Fail to write out RECTANGLE record in OASIS file.

## OASIS-103

(Error) Fail to write out POLYGON record.

**Description**

Fail to write out POLYGON record in OASIS file.

## OASIS-104

(Error) Fail to write out PATH record.

**Description**

Fail to write out PATH record in OASIS file.

## OASIS-105

(Error) Fail to write out TEXT record.

**Description**

Fail to write out TEXT record in OASIS file.

## OASIS-106

(error) Failed to write out Path '%s'.

**Description**

The path is irregular, please check its coordinates.

## OASIS-107

(error) %s "%s" has an angle of %.3f degrees which is not supported currently.

Feedback

**Description**

This message occurs when sref or aref specified an unsupported ANGLE value. Supported values are: 0/90/180/270 degrees.

**What Next**

Check the OASIS file to find the "abnormal" object, change the ANGLE value of it, or delete the object.

---

## OASIS-108

(warning) Ignoring unsupported circle.

**Description**

This message occurs when the input OASIS file contains circle record.

---

## OASIS-109

(warning) Ignoring unsupported abstract for '%s'.

**Description**

This message occurs when the -abstract and -type die/interposer are specified together, read_oasis will create layout view when tracing is not enabled, otherwise, design view is created.

---

## OASIS-110

(warning) Unplaced cell '%s' is written to the OASIS file.

**Description**

This warning is displayed upto 20 times when unplaced cells are present in the design. By default unplaced cells will be outputted to OASIS file.

**What Next**

Please use option -skip_unplaced_cells if you do not need unplaced cells.

---

## OASIS-111

(warning) Ignoring line %d in file '%s' for the reason: '%s'.

Feedback

### Description

This warning message occurs when there is something wrong in the marker layer map file, which was instroduced to duplicate routing data in database to specified GDSII/OASIS marker layer.

There are 4 types (layer, region, net and cell) in the marker layer map file. Each line indicates a mapping, the mapping starts with marker and ends with ";", any text after a semicolon "#" in each line is a comment.

For all the types, -FBlayer, -cad_layer and -data_type are required options. When the type value is not cell, the -layer option is also required option. When the type value is region, the -bbox is also required option. When the type value is net, the -net is also required options. When the type value is cell, the -cell is also required option.

The following statements are the example for all kinds of type. marker -type layer -layer METAL -FBlayer FB9 -cad_layer 135 -data_type 1999 ; #layer based marker -type region -layer METAL -FBlayer FB9 -bbox {{1294 479} {1415 507}} -cad_layer 222 -data_type 1999 ; #region based marker -type net -net VDD -layer METAL3 -from U458/ZN -to H1/H3/B1 -FBlayer FB6 -cad_layer 333 -data_type 1999 ; #net based marker -type cell -cell inst2 -FBlayer FB7 -cad_layer 444 -data_type 1999 ; #cell based

### What Next

Please check and correct the marker layer map file according to the message about the warning reason.

# OCVM

## OCVM-001

(error) Unrecognized specifier '%s' at line %d of %s.

### Description

This message is issued when the *read_ocvm* command cannot find a valid specifier name.

### What Next

Please check the PrimeTime reference manual for the correct AOCVM syntax.

### See Also

- read_ocvm

Feedback

## OCVM-002

(error) Multiple entries for specifier '%s' at line %d of %s.

### Description

This message is issued when the *read_ocvm* command encounters the same specifier multiple times for a given table. The *version* should be specifed once per file.

### What Next

Please make sure that there is only one entry for a given specifier for a table. If it is an optional specifier (like *path_type*), please make sure that there is at least one required specifier (like *derate_type*) after this specifier for the given table.

### See Also

- read_ocvm

## OCVM-003

(error) '%s' at line %d of %s is not supported for the given file version.

### Description

This message is issued when the given specifier is not supported for the version given in the file.

### What Next

Please check the PrimeTime reference manual for the correct AOCVM syntax.

### See Also

- read_ocvm

## OCVM-004

(error) Unrecognized entry for '%s' at line %d of %s.

### Description

This message is issued when the parser cannot recognize the entries for the given specifier. This message is also issued if the number of entries specified is more or less than expected.

Feedback

### What Next

Please check the PrimeTime reference manual for the correct AOCVM syntax.

### See Also

- read_ocvm

---

## OCVM-005

(error) Unsupported specifier '%s' at line %d of %s.

### Description

This message is issued when the parser encounters a specifier that is not supported. The specifier *group_name* is not supported in the tool.

### What Next

Remove the unsupported specifier.

### See Also

- read_ocvm

---

## OCVM-006

(warning) Ignoring specifier '%s' at line %d of %s.

### Description

This message is issued when the parser encounters a specifier that is not supported. The specifier *voltage* is not supported in the tool. This line will be ignored and the rest of the table will be parsed.

### What Next

Ignoring the specifier may mean that the table may be overriding already specified table. For instance, a table for libcell A may have been already read when another table for libcell A with voltage 1.0 is read. In this case, the voltage line will be ignored and the new table will override the existing table. Please use the *report_ocvm* command to see the annotated tables.

### See Also

- read_ocvm
- report_ocvm

Feedback

## OCVM-007

(error) Multiple entries for '%s' at line %d of %s.

### Description

This message is issued when mutiple values are specifed for a specifier when only one entry is expected.

### What Next

Please check the PrimeTime reference manual for the correct AOCVM syntax.

### See Also

- read_ocvm

## OCVM-008

(Warning) Ignoring 'object_spec' as 'object_type' is 'design' (line: %d of file: %s).

### Description

The *object_spec* entry should be empty when *object_type* is *design*. The tool will ignore the object_spec and apply the table on top level design.

### What Next

Please check the PrimeTime reference manual for the correct AOCVM syntax.

### See Also

- read_ocvm

## OCVM-009

(error) 'object_type' is not known for the given 'object_spec' (line: %d of file: %s).

### Description

This error message is issued when the parser does not know the type of objects.

### What Next

Please make sure that there is an entry for *object_type* before *object_spec*.

### See Also

- read_ocvm

Feedback

## OCVM-010

(error) '%s' can only be '%s' or '%s'.

**Description**

This error message is issued when the entry for a specifier is not valid.

**What Next**

Please check the PrimeTime reference manual for the correct AOCVM syntax.

**See Also**

- read_ocvm

## OCVM-011

(error) Invalid float value at line %d of %s.

**Description**

This message is issued when the *read_ocvm* command cannot find a valid float value.

**What Next**

Please check the line for bad float entries.

**See Also**

- read_ocvm

## OCVM-012

(error) Incomplete table at line %d of %s.

**Description**

This message is issued when the *read_ocvm* command reaches end of file without getting a complete table.

**What Next**

Please add all the required specifiers for the table.

**See Also**

- read_ocvm

Feedback

## OCVM-013

(error) '%s' specification is not monotonic at index %d.

**Description**

The depth or distance specification is expected to be strictly monotonic.

**What Next**

Please make sure that the entries for depth or distance are strictly monotonic. Please note that the index starts at 0.

**See Also**

- read_ocvm

## OCVM-014

(error) 'table' is not monotonic for '%s' at index %d.

**Description**

The table specification is expected to be monotonic for depth or distance. It is expected that any row or column of the table is monotonic.

**What Next**

Please make sure that the entries in any row or columns are monotonic. Please note that the index starts at 0.

**See Also**

- read_ocvm

## OCVM-015

(error) Cannot specify '%s' when '%s' is '%s'.

**Description**

This message is reported when parser encounters an illegal combination of specifiers.

**What Next**

Please check the PrimeTime reference manual for the correct AOCVM syntax.

Feedback

**See Also**

- read_ocvm

---

## OCVM-016

(error) Sanity check failed for table at lines: %d-%d of %s.

**Description**

This message is reported when parser encounters an error in a table. An additional error message printed before this one would describe the actual problem.

**What Next**

Look at the error message printed just before this message to resolve the problem.

**See Also**

- read_ocvm

---

## OCVM-017

(error) Number of 'table' entries(%d) is not consistent with depth(%d) and distance(%d).

**Description**

The number of entries in the *table* specifier is expected be a product of number of entries in *depth* and *distance\FP specifications. An empty depth or distance specification is considered to have one entry.*

**What Next**

Check the specifications to make sure that the table has entry for all rows and columns.

**See Also**

- read_ocvm

---

## OCVM-018

(error) '%s' montonicity does not match table montonicity.

**Description**

This message is printed when the parser encounters inconsistency between the entries for *depth* or *distance* and *table* specifications. The *table* entries are expected to increase

Feedback

with depth and decrease with distance when the *derate_type* is *early*. The *table* entries are expected to decrease with depth and increase with distance when the *derate_type* is *late*.

**What Next**

Please check the tables for monotonicity.

**See Also**

- read_ocvm

---

## OCVM-019

(error) Derate value ('%f') is out of range.

**Description**

This message is printed when the parser encounters a derate value which is out of range. The expected range is 0 to 6.55.

**What Next**

Please check the derate values in the table.

**See Also**

- read_ocvm

---

## OCVM-020

(Warning) No object_spec specified (line: %d of file: %s).

**Description**

This message is emitted when the object_spec specification does not have any entry for libcell or cell objects.

**What Next**

Please fix the object_spec specification in the table.

**See Also**

- read_ocvm

---

## OCVM-021

(Warning) No OCV table specified for corner %s.

Feedback

**Description**

This message is emitted when there is no AOCV table specified for the given corner.

**What Next**

Please provide the table for the given corner if AOCV analysis is needed.

**See Also**

- read_ocvm

---

## OCVM-022

(error) Negative values are not allowed.

**Description**

Negative values are not allowed for the distance, depth arrays or in the 1D/2D tables.

**What Next**

Remove negative values from the table file.

**See Also**

- read_ocvm

---

## OCVM-023

(warning) Cannot associate table with leaf cell '%s'.

**Description**

AOCVM/POCVM does not support the association of AOCVM/POCVM data with leaf cells.

**What Next**

Associate AOCVM/POCVM tables with hierarchical cells, library cells and designs only.

---

## OCVM-030

(Error) Expected to find a valid field name but found '%s'.

**Description**

This error occurs when the parser cannot find a valid field name.

Feedback

**What Next**

Correct the COCV CPODE syntax by ensuring that each field is specified in the format name: data where name is one of version, table, distance, depth, object_spec, derate_type, delay_type, rf_type, object_type.

---

## OCVM-031

(error) The version '%s' is not valid. Allowable versions are '%s'

**Description**

An invalid version number has been specified.

**What Next**

Check the version number.

---

## OCVM-032

(error) The current table is missing one or more fields

**Description**

This error occurs when a table is found at the end of the AOCV file without the required number of fields.

**What Next**

Correct the AOCV syntax by adding the missing fields to the file.

---

## OCVM-033

(error) Incorrect table value.

**Description**

The bottom right value of the 2D COCV table must be zero.

**What Next**

Correct the table values.

---

## OCVM-037

(error) COCVM CPODE derate bin label "%s" already defined.

Feedback

**Description**

A COCVM CPODE derate bin label can be associated with only one bin.

**What Next**

Specify unique labels across derate bins.

---

## OCVM-038

(error) Number of bin values is different from number of derate columns.

**Description**

Number of derate values in every bin must be same as the number of entries specified as DERATE_COLUMNS field of COCVM CPODE derate side file.

**What Next**

Provide proper number of derate values.

---

## OCVM-039

(error) "default" bin must be specified and must be the last bin.

**Description**

Default bin definition missing in the COCVM CPODE dertate side file.

**What Next**

Include default bin definition.

---

## OCVM-040

(error) Could not find the specified AOCV file

**Description**

The AOCV file could not be found.

**What Next**

Correct the name and path of the AOCV file.

---

## OCVM-041

(warning) Unsupported derate type, %s skipped.

Feedback

**Description**

The derate type specified is not supported in cpode derate file. Corresponding derates are read and skipped.

**What Next**

Either correct the table derate type to correct type or ignore the warning if want to proceed with ignoring derate values.

---

## OCVM-042

(error) This command applies to hierarchical cells only, '%s' is a leaf cell.

**Description**

The specified cell is a leaf cell, and the command you executed applies only to hierarchical cells.

**What Next**

Run the command again, and ensure that you do not specify a leaf cell.

---

## OCVM-043

(warning) There is no aocvm named table '%s'.Ensure that it is loaded before update timing.

**Description**

This message is issued when a named table set is assigned to a design or hierarchy but no table set with this name has been read yet. Since set_ocvm_table_group is always before corresponding read_ocvm commands. this is only a warning because it is possible that the named table set will still be read after the set_ocvm_table_group command.

**What Next**

Either correct the table derate type to correct type or ignore the warning if want to proceed with ignoring derate values.

---

# ODB

---

## ODB-001

(error) Invalid library name '%s' specified in '%s' command option of read_odb.

Feedback

**Description**

The library name specified is invalid because it contains spaces.

**What Next**

Ensure that there are no spaces in library name specified.

---

## ODB-002

(error) Library '%s' specified in '%s' command option of read_odb already exists!

**Description**

There is/are existing librar(y/ies) with library name specified. Use a fresh library name to create a new library with design from ODB++ files.

**What Next**

Ensure that there are no existing libraries in the current session with the name specified.

---

## ODB-003

(warning) NDM expects only two solder mask layers. Skipping layer %s.

**Description**

NDM Technology expects only two solder mask layers - one with mask_name "solder_mask_top" and another with mask_name "solder_mask_bottom". Remaining solder mask layers are skipped.

**What Next**

Ensure that there are not more than two solder mask layers in ODB++ design being read.

---

## ODB-004

(warning) Unsupported ODB++ layer type. Skipping layer %s.

**Description**

NDM Technology only supports ODB++ layers of type SIGNAL, POWER_GROUND, MIXED, DIELECTRIC, DRILL and SOLDER_MASK. Remaining type of layers are skipped.

**What Next**

Not applicable.

Feedback

## ODB-005

(warning) Layer '%s' does not have a valid %s layer.

**Description**

The specified dielectric layer does not have a metal layer above/below it.

**What Next**

Confirm that the ODB++ layer stack is okay.

## ODB-006

(error) read_odb Automatic Technology Generation failed.

**Description**

The automatic technology generation feature of read_odb failed.

**What Next**

Check the previous error and warning messages to detect the issue.

## ODB-007

(warning) read_odb finds more than one steps in ODB++ design, will only read the first one.

**Description**

read_odb only support one step.

**What Next**

Check the ODB++ design whether has multiple steps or not.

## ODB-008

(info) Skip bumps in component layer %s with unsupported angle value(%d).

**Description**

Only support angle value 0/90/180/270 for bumps.

**What Next**

Check the ODB++ design, find another way to describe the feature shapes instead of unsupported angle value.

Feedback

## ODB-009

(warning) Syntax error for SURFACE records. First polygon in surface is a hole. Skip it.

**Description**

SURFACE must start with an island.

**What Next**

Check the ODB++ design and fix the error.

## ODB-010

(warning) Unsupported SURFACE records syntax: an island is contained in a hole. Skip it.

**Description**

For SURFACE records, read_odb only support an island contains multiple holes.

**What Next**

This is a limitation.

## ODB-011

(info) Design source is not from ODB++, cannot dump out components info.

**Description**

The current package design source is not from ODB++, that means it does not created by read_odb. In this situation, write_odb will be not able to dump out ODB++ specified info such as components.

**What Next**

Check the design source, if it is expected, ignore the info message.

## ODB-014

(error) The command '%s' failed because '%s'.

**Description**

The given command is failed to execute successfully.

**What Next**

Check the cause of error and take necessary course of action.

Feedback

### ODB-015

(info) %s.

**Description**

This is the information provided by the read_odb/write_odb command.

**What Next**

Validate that ODB++ file and the design created meet the requirements.

## OPT

### OPT-001

(error) There are undefined voltages in setup. Optimization aborted.

**Description**

The MV design has undefined voltages in setup optimization. As a result, optimization does not have sufficient information to correctly optimize the design. Therefore, optimization aborted upon detecting this situation.

**What Next**

Check the voltages defined for every supply net and make sure they are correct. Try using the report_supply_nets command to see the current settings.

### OPT-002

(error) Error: Design is MV design, but design is not MV ready. Perhaps, commit_upf has not been run.

**Description**

The design has been detected as an MV design, but the MV data is not ready. Most likely commit_upf command has not been run.

**What Next**

Check the commit_upf has been run. If not, please run it before running optimization.

Feedback

## OPT-003

(error) Design is MV design, but design is not ready for implementation. Please check UPF.

### Description

The design has been detected as an MV design, but the MV data is not ready for implementation. This indicates that there are errors in the UPF data.

### What Next

Check the UPF data is correct and fix it before running optimization.

## OPT-004

(error) The design of block "%s" is opened in read mode, failed to enable optimization on the block.

### Description

The specified operation failed to enable optimization on the block as the design of the block or the design library is opened in read mode.

There can be multiple reasons why a block design is opened in read mode. Few possible reasons:

- If the block is coming from a different library than Top library, then "-ref_libs_for_edit" should be used while opening the Top library/design; otherwise the block will be implicitly in read mode.
- If the block is linked to abstract view and the abstract was created with "-read_only" option, then it will be opened in read mode only. Check for "NDM-072" messages which tells if the block is being opened in read mode because of non-editable abstract.
- If the block is linked to design view and there exists an editable abstract view for it, then the block will be implicitly in read mode. Check for "NDM-074" messages which tells if the block is being opened in read mode due to the presense of editable abstract views.

### What Next

Please re-open design in edit mode and try again.

## OPT-005

(warning) Cell "%s" is bound to an abstract view, restricting optimization to interface logic only.

Feedback

**Description**

The specified operation tried to enable optimization on entire logic of the cell, but the cell is bound to an abstract view. Optimization is restricted to interface logic only.

**What Next**

Please use an optimizable design view for complete logic optimization.

---

## OPT-006

(warning) Design power is incorrectly set (%s) (0.0/inf/nan).

**Description**

The power for the design has invalid values for certain scenarios.

**What Next**

Check report_corner/report_supply_net.

---

## OPT-007

(warning) Net %s has MV violations. Buffering will be limited.

**Description**

The given net has MV violations. Examples including missing supply net information on the net pins, level shifter violations, etc. Please check the UPF to make sure that it is correct.

When a net has MV violations, buffering will be limited on this net.

**What Next**

Use check_mv_design to see the violations in the design.

---

## OPT-008

(warning) Libcell %s is dont-touch but is qualified for purpose: %s.

**Description**

When a dont-touch cell is introduced into the design, it cannot be altered later. However, dont-touch semantics do not forbid such cells from being used by specified purpose(s).

**What Next**

report_dont_touch(2), get_attribute(2), report_lib(2), set_lib_cell_purpose(2)

Feedback

~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~

## OPT-009

(Information) Net %s with %s violation is a dont_touch net.

**Description**

The net is violated with some design rule constraint. But it could not be fixed since the net is set as dont_touch.

**What Next**

User can check whether the setting is reasonable.

~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~

## OPT-010

(Information) Net %s with %s violation is highly blocked.

**Description**

The net is violated with some design rule constraint. But It is hard to be fixed since the net is highly blocked.

**What Next**

User can check whether the placement constraint is too tight.

~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~

## OPT-011

(Information) Net %s with %s violation is in high density areas.

**Description**

The net is violated with some design rule constraint. But It is hard to be fixed since the local density is too high.

**What Next**

User need to check whether the placement is reasonable.

~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~

Feedback

## OPT-012

(Information) Net %s with %s violation is partially blocked.

**Description**

The net is violated with some design rule constraint. But It is hard to be fixed since there are some pins in the net are blocked.

**What Next**

User can check whether the placement constraint is too tight.

~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~

## OPT-013

(Error) The library cell %s does not have site row definition.

**Description**

The library cell does not have site row definition.

**What Next**

User need to check if the library cell has site definition.

## OPT-014

(Warning) No default buffer/inverter available for voltage areas %s.

**Description**

There are no available buffers or inverters for the optimization inside voltage areas.

**What Next**

User need to check whether the target library setting and UPF setting is correct.

## OPT-015

(Error) No legal location can be found for new diode cell %s.

**Description**

The tool cannot find the legal locaiton for the new added diode cell.

Feedback

**What Next**

User need to check whether the design has avialable space for the new added diode cell first.

---

## OPT-016

(Warning) Can not find available buffer/inverter lib cells for the site %s.

**Description**

Tool can not find any valid library buffer or inverter that matches the specific site row. For each site row definition in the design, at least one valid library buffer or inverter are needed.

**What Next**

User need to add library that matches the site.

---

## OPT-017

(Warning) There are buffers/inverters library cells with dont_use attribute for the site %s.

**Description**

Tool finds library buffers or inverters with dont_use that match the specific site row.

**What Next**

User need check if the library buffers with dont_use can be reset so that tool can use them for optimization.

---

## OPT-018

(information) Long chain arithmetic units detected.

**Description**

Long chain of arithmetic units is found to be aligned with critical paths. Check the following line for the gate that is involved.

**What Next**

Check if the chain is still in the critical path in after place_opt/clock_opt netlist. If so, please review the architectures used for the corresponding operators in RTL design.

Feedback

## OPT-019

(Warning) Cannot find buffers. Net %s cannot be buffered.

### Description

This means that the tool cannot find any suitable buffers or inverters to use to correctly (power-state wise) buffer the net in all voltage areas. The UPF needs to be corrected before this net can be buffered.

### What Next

Use report_pst to see the state of the supply nets. The supply net of the driver needs to be at least as on as that of the load(s) it is driving, and the buffer to be inserted needs to be powered by a supply net that is no more on than the driver and no less on than the sinks it is driving. The user can use report_power_domains to see what supplies are available in each power domain. The user can also use check_bufferability to see the buffers, if any, that can be used to buffer the net in a given voltage area.

## OPT-020

(Information) %s (Reference: %s) is treated as %s

### Description

This message indicates whether sub-blocks/top-level design are modifiable or non-modifiable, along with reason, during Toplevel implementation flow.

## OPT-021

(Error) MV initialization during optimization failed. Optimization aborted.

### Description

The optimization tool was not able to initialize the MV information properly. As result, optimization cannot continue and will be aborted.

### What Next

User need to ensure that the MV information is correct. Please review the UPF for correctness with the help of check_mv_design command.

### See Also

- check_mv_design

Feedback

## OPT-022

(Error) No voltage area detected. Optimization aborted.

### Description

The optimization tool was not able to see any voltage area. In the case where the design is non-MV and does not have any UPF, then the tool automatically defines the default VA. However, if the user specifies an incomplete UPF but did not define any voltage, then the tool does not define a default VA. As result, optimization cannot continue and will be aborted.

### What Next

User need to ensure that the VA information is correct. Please review the UPF for correctness.

### See Also

- check_mv_design
- report_voltage_areas

## OPT-023

(Warning) Voltage Area %s does not have associated power domain.

### Description

The given voltage area does not have an associated power domain. Thus, there VA is not usuable.

### What Next

User need to ensure that the MV information is correct. Please review the UPF for correctness.

### See Also

- check_mv_design
- report_voltage_areas

## OPT-024

(Warning) Power Domain %s does not have a primary supply.

Feedback

### Description

The optimization tool detected that the given power domain did not have a primary supply.

### What Next

User need to ensure that the MV information is correct. Please review the UPF for correctness.

### See Also

- check_mv_design
- report_power_domains

---

## OPT-025

(Error) There are more than one default voltage area detected. Optimization aborted.

### Description

The optimization tool detected that there are more than one default voltage area in the design. As result, optimization cannot continue and will be aborted.

### What Next

User need to ensure that the MV information is correct. Please review the UPF for correctness.

### See Also

- check_mv_design
- report_voltage_areas

---

## OPT-026

(Warning) Found long segment of %d um for net %s between (%d, %d) and (%d, %d) that is not bufferable. This segment crosses the following VAs: %s.

### Description

The optimization tool detected that there are long segments that cannot be buffered. The reasons for non-bufferability may be due to 1) blockages, 2) macros, 3) lack of proper row sites, 4) required supply nets not available, 5) improper target library subset setting on module based movebounds, etc. Please check design to determine that none of the factors are affecting bufferability.

Feedback

### What Next

User need to the areas indicated by the message and make sure they are bufferable.

### See Also

- check_mv_design
- report_voltage_areas
- set_target_library_subset

---

## OPT-027

(Error) Support name synonym for ports with matching names.

### Description

Optimization cannot proceed because one or more port having matching name with other port.

### What Next

Run report_design_mismatch to find the port name which is having same matching name. Change the name of any of port which are causing this port_name_synonym case. Then run place_opt.

---

## OPT-028

(Error) Missing port %s in the design.

### Description

Optimization cannot proceed for an unbound pin found, unable to find it's corresponding port in the design master.

### What Next

Run report_design_mismatch to find the unbound pin. Then add the missing port or remove the pin for which port is missing.

---

## OPT-029

(Error) Port name case mismatch for block port %s in the design.

Feedback

### Description

Optimization cannot proceed for port name case mismatch happened, name of the port and the corresponding pin don't match.

### What Next

Run report_design_mismatch to find the block port for which port name case mismatch happened. Change the name of the block port for which port name case mismatch happened.

---

## OPT-030

(error) place_opt failed during trial CTS.

### Description

This message occurs when trial clock tree synthesis is enabled but is not successful inside place_opt. With this condition place_opt immediately exits.

### What Next

Make sure clock setups are correct.

### See Also

- place_opt
- place_opt.flow.trial_clock_tree

---

## OPT-031

(warning) optimize_icgs will optimize the ICGs in the current_block only, for this top-level place_opt invocation.

### Description

This message indicates that ICG optimization flow is turned on for top-level place_opt. ICG optimization will be limited to current_block. place_opt will immediately exit with an Error, unless all sub-blocks are read-only.

### See Also

- place_opt
- place_opt.flow.optimize_icgs

Feedback

## OPT-032

(error) optimize_icgs does not support top-level place_opt with modifiable sub-blocks.

### Description

This message indicates that ICG optimization flow is turned on for top-level place_opt, but modifiable sub-blocks are dectected. optimize_icgs does not support top-level place_opt with modifiable sub-blocks.

### What Next

Turn off optimize_icgs for place_opt or make all sub-blocks read-only

### See Also

- place_opt
- place_opt.flow.optimize_icgs

## OPT-033

(Information) Terminating the integrated optimization with useful skew.

### Description

This message indicates that the useful skew computation was unable to find predictable timing improvement, so the useful skew based optimization is terminating early. The last successful useful skew commitment is still effective, and tool can continue optimization.

### See Also

- place_opt
- clock_opt

## OPT-034

(Warning) The hierarchy port %s is defined as %s but used as %s.

### Description

The specified hierarchy port is either defined as input but used as output or defined as output but used as input. The mismatch can cause some internal data invalid. Sometimes it results in fatal.

Feedback

### What Next

User needs to correct the port definition.

---

## OPT-035

(warning) Power app option %s has been deprecated. Please remove it from your script.

### Description

App options opt.power.mode and opt.power.effort have been deprecated. Please remove them from your script.

### What Next

Switch to compile.flow.enable_power, place_opt.flow.enable_power and clock_opt.flow.enable_power.

### See Also

- set_qor_strategy
- place_opt.flow.enable_power
- clock_opt.flow.enable_power

---

## OPT-040

(warning) Net %s is skipped from optimization since it has an unplaced pin %s.

### Description

This message indicates that the nets with unplaced pin(s) are skipped during optimization.

### What Next

Make sure all optimizable nets have placed pins.

~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~

---

## OPT-041

(error) Scenario Manager creation failed. Optimization aborted.

### Description

Without the scenario manager the optimization cannot proceed. Therefore, optimization aborted upon detecting this situation.

Feedback

**What Next**

1> Parasitics may be missing at the corners. Please check the parasitics setup.

---

## OPT-042

(Error) Failed to create routing topology for driver %s as one or more hierarchical pin placed outside it's block boundary.

**Description**

During topology creation tool expect all port and pin inside the block should remain inside the block bounding box. Topology generator found pin placed beyond block bounding box.

**What Next**

Run place_pin for the block. And make sure that succeed.

---

## OPT-043

(Warning) Cannot find default buffer/inverter for VA %s with Block Hierarchy %s.

**Description**

This means the tool cannot find any usable buffers/inverters for the combination of the specified voltage area and the block hierarchy. It can lead to nets lying under the block hierarchy to remain unbuffered inside this VA. The UPF needs to be corrected.

**What Next**

User needs to check whether the target library setting or UPF setting is correct. User can also check log file for OPT-019 messages to see which all nets are unbuffered in the design.

---

## OPT-044

(Warning) Block hierarchy %s is read-only hierarchy.

**Description**

This means the tool will not be able to optimize nets lying under this block hierarchy.

**What Next**

User needs to ensure if the design settings are correct for this block.

Feedback

## OPT-045

(Error) Cannot find usable buffers or inverters.

**Description**

This means that the tool cannot find any usable buffers or inverters for all the voltage areas in the design. This could be due to multiple reasons a) All blocks passed are non-modifiable in the design. b) Target library settins are not correct in the design. c) UPF setting are not correct for the design.

User needs to ensure that these settings are correct before passing the design for the optimization.

**What Next**

User should look for if all the blocks are set as non-modifiable by the user or UPF or target library settings are correct or nt. User should look for "OPT-043" and "OPT-044" messages in the log for the additional info.

## OPT-046

(Warning) Can not find available TIE lib cells for the site %s.

**Description**

Tool can not find any valid library TIE cell that matches the specific site. It will still add TIE cell with other lib cells. The TIE cell will be far away from its load.

**What Next**

User need to add library that matches the site.

## OPT-049

(error) Block %s has no physical data.\n

**Description**

There are physical hierarchical blocks which have no physical data defined. Optimization flow cannot run on such designs

**What Next**

Ensure that all blocks have physical data

Feedback

## OPT-050

(Error) Unbound via, site_row or site_array or instances in the block.

**Description**

Optimization cannot proceed because one or more objects in the block is unbound, i.e., is referencing a missing definition in the tech. For example, a via can reference a missing via_def, or a site_row can reference a missing site_def.

**What Next**

Run report_unbound to identify the unbound objects. Then remove the unbound objects from the block or add the missing definitions to the tech to resolve the problem.

## OPT-051

(Error) Incorrect setting of physical feedthrough option in create_voltage_area_rule.Optimization aborted.

**Description**

Option -allow_physical_feedthrough set to true in create_voltage_area_rule command is not supported yet for VA specific rule. As result, optimization cannot continue and will be aborted.

**What Next**

User need to ensure that the -allow_physical_feedthrough option is set to false for VA specific create_voltage_area_rule rules. Currently user can use the -all_voltage_areas option with value true for all voltage areas in the design.

## OPT-052

(Warning) Net %s will route through VA (name=%s,vaIndex=%d) even though the VA is passthrough disabled.

**Description**

Due to no other routing space, route topology generator is forced to route the net through this VA even it is passthrough disabled for the given net.

**What Next**

Consider updating the va-feedthru rule for this VA.

Feedback

## OPT-053

(Warning) VA (name=%s, vaIndex=%d) is made passthrough enabled for net %s due to its hierarchy %s.

### Description

This voltage area was originally passthrough disabled for this net. However, it is made passthrough enabled internally in order to allow routing of net as net has hierarchy in this voltage area.

### What Next

Consider updating the va-feedthru rule for this VA.

## OPT-054

(Warning) VA (name=%s, vaIndex=%d) is made passthrough enabled for net %s due to its pin %s.

### Description

This voltage area was originally passthrough disabled for this net. However, it is made passthrough enabled internally in order to allow routing of net as net has pin in this voltage area

### What Next

Consider updating the va-feedthru rule for this VA.

## OPT-055

(Information) Current block utilization is '%.5f', effective utilization is '%.5f'.

### Description

This message provides the current block utilization numbers during optimization. In some designs, density PG structure at low metal layer may block some sites for placement. Also, existence of multi-row or inbound cells can cause unusable sites adjacent to these cells. There are also color-based cell spacing rules that can cause unusable sites. The effective utilization number is calculated by treating these unusable sites as used area. Therefore, this is a more accurate representation of the cell density seen by placement and optimization engines.

The estimation for the number of unusable sites adjacent to multi-row or inbound cells is controlled by place.utilization.multi_row_penalty and place.utilization.inbound_penalty options.

Feedback

The estimation for the number of unusable sites under low metal layer PG nets is controlled by place.coarse.pnet_aware_layers, place.coarse.pnet_aware_min_width and place.coarse.pnet_aware_density options.

Keepout margins are not included in block utilization but the sites covered by keepout margins are not available for optimization usage. Therefore, keepout margins will also increase the effective utilization number.

**What Next**

If effective utilization is much larger than utilization, check the low metal layer PG structure or use "analyze_lib_cell_placement -lib_cells [all_lib_cells]" to see how many library cells became unusable due to PG structure. Also, check to see if the utilization penalty for multi-row and inbound cells are setup properly.

**See Also**

- report_utilization
- analyze_lib_cell_placement
- report_keepout_margins
- place.utilization.multi_row_penalty

---

## OPT-056

(Warning) opt.common.allow_physical_feedthrough is going to be retired in 18.06 release. Please plan to migrate to create_voltage_area_rule in order to set physical feedthrough option on different voltage areas.

**Description**

This message informs user that app option opt.common.allow_physical_feedthrough is going to be retired in 18.06 release. Please switch to the create_voltage_area_rule in order to set physical feedthrough on differnt voltage areas in the design. Please visit the man page of create_voltage_are_rule for details on how to create a voltage area rule.

---

## OPT-057

(Error) Timer is in Zero Interconnect Delay mode for optimization.

**Description**

This message informs user that timer is using Zero Interconnect Delay model while performing optimization. It can lead to incorrect timing delay optimization, hence shold be avoided. It could be due to either users has set time.delay_calculation_style or design has no physical data.

Feedback

## OPT-060

(Warning) Specified Object %s is not a tie-cell

**Description**

Non tie-cell objects will be skipped during tie-cell removal.

**What Next**

User needs to ensure that the list of tie-cells provided to remove_tie_cells are indeed tie-cells.

## OPT-061

(error) place_opt failed during multisource CTS.

**Description**

This message occurs when multisource clock tree synthesis is enabled but is not successful inside place_opt. With this condition place_opt immediately exits.

**What Next**

Make sure multisource clock setups are correct.

**See Also**

- place_opt
- place_opt.flow.enable_multisource_clock_trees

## OPT-062

(error) %s bailed out due to unplaced cells.

**Description**

This message occurs when physical optimization starts from a netlist with unplaced cells.

**What Next**

Make sure design is placed before running optimization.

**See Also**

- place_opt

Feedback

## OPT-063

(warning) Constant net %s is a dont_touch net.

**Description**

The constant/power/ground net is marked as dont_touch. Tie-cell insertion cannot be done for loads of this net.

**What Next**

User can check whether the dont_touch setting is reasonable.

~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~

## OPT-064

(Error) all the specified ports are unplaced.

**Description**

All the specified ports are unplaced. The command only adds protection diode for the placed ports.

**What Next**

Place the ports first and then use the command for protection diode insertion.

## OPT-065

(Warning) Instance %s has invalid layer attribute.

**Description**

This message indicates that the layer attribute is missing or is illegal when attempting to get the port location of the instance, which might cause tool crash in the later stage.

**What Next**

Check the layer attribute of the instance, and make sure it is correct.

## OPT-066

(warning) High fanout net "%s" could be partially buffered because tool finds some unbufferable segments on it.

Feedback

**Description**

The high fanout net may be partially buffered because tool finds some segments are unbufferable. The reason of unbufferable could be dont_touch logical net, or some segments physically are blocked.

**What Next**

To remove dont_touch or placement blockages on the net.

---

## OPT-067

(info) Pin %s is on clock network. Skipping.

**Description**

The given pin is on a clock network, but the net connecting the pin is a signal net. So, buffering engine will skip this net during optimization.

**What Next**

Make sure the signal net is not on a clock network.

~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~

---

## OPT-068

(info) Disabling tracking of voltage threshold-based lib-cells during tieopt

**Description**

Tracking of low_vt usage is not needed for tieopt engine as costing is not involved here.

**What Next**

There may be a change in low-vt libcell count usage post tieopt. These will be addressed in subsequent stages.

~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~

---

## OPT-069

(warning) Skipping pin %s. Reason - %s.

**Description**

Tie-cell insertion is skipped for a pin due to specified reason.

Feedback

### What Next

User can check whether the reason mentioned is reasonable.

~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~

---

## OPT-070

(Warning) Cannot find any max transition constraint on the design.

### Description

Without max transition constraint on design active scenarios, nets on the critical timing path with large transitions might not be able to be selected for max transition fixing, and thus leave large transitions on the timing path after the flow. It could be helpful to have a proper max transition constraint set on the design for improving timing.

### What Next

It suggested to use *set_max_transition* command to set appropriate max transition constraints on design.

### See Also

- set_max_transition

---

## OPT-071

(Warning) Critical MV check failure on the design.

### Description

This message occurs when some critical MV setting is not ready for place_opt clock_opt and route_opt, such as data or voltage area setup, design's multi-voltage data, etc. That may lead to unpredictable effects on the subsequent flow.

### What Next

Please check whether the messages of *MV-510*,*MV-058a*,*MV-511*,*MV-511a*, and *MV-004* in the log and follow the instructions of theses man pages.

### See Also

- MV-510
- MV-058a
- MV-511

Feedback

- MV-511a
- MV-004

---

## OPT-072

(Error) Design has no active hold scenarios with CCD enabled.

**Description**

Place or route optimization cannot use designs without any hold analyses when CCD enabled. Please use report_scenarios to check that at least one active scenario has hold analysis enabled.

**What Next**

User can solve the error by setting hold analyses to true in one or more active scenarios using set_scenario_status {<scenario list>} -hold true.

**See Also**

- report_scenarios
- set_scenario_status
- place_opt
- clock_opt

---

## OPT-073

(error) Failed to find any usable buffers. Please check library and user setup.

**Description**

This message occurs when there is no usable buffers when running optimization.

**What Next**

Check library and user setup.

---

## OPT-074

(error) Failed to find any usable inverters. Please check library and user setup.

**Description**

This message occurs when there is no usable inverters when running optimization.

Feedback

### What Next

Check library and user setup.

---

## OPT-075

(error) Found %d buffers in design that are not dont_touch. Please clean up buffers or refer to the man page of flow.common.inverter_only for alternate solution.

### Description

This message occurs when there are buffers in designs that are not dont_touch in inverter only flow.

### What Next

Clean up buffers or refer to the man page of flow.common.inverter_only for alternate solution.

---

## OPT-076

(warning) There are %d available buffer libcells (including %d opto %d hold %d power %d cts purpose) incoming to inverter only flow.

### Description

This message occurs when there are usable buffer libcells in library in inverter only flow.

### What Next

Use set_lib_cell_purpose to disable all buffer libcells in the library in inverter-only flow.

### See Also

- set_lib_cell_purpose

---

## OPT-077

(warning) Unable to fix violating net %s because it is not optimizable.

### Description

The net with attribute such as dont_touch is not optimizable during buffering.

### What Next

Check whether the net has any unexpected restrictions.

Feedback

## OPT-080

(warning) Adjusted max transition constraint "%f" ns has been applied to "%s".

**Description**

Adjusted constraint specified has been appled to the pin (initially pin has had an infinite transiton constraint. The reason of unbufferable could be dont_touch logical net, or some segments physically are blocked.

## OPT-100

(Warning) Constraints are too tight for corner %s and libcell %s,expect over-buffering.

**Description**

The constraints for one of the corners is too tight that buffer can not even drive itself.

**What Next**

Consider lifting the constraints.

## OPT-200

(warning) No tie cell is available for constant fixing.

**Description**

There is no tie cell in the library that is available for constant (1'b0 or 1'b1) fixing. This could be because all tie cells have a *dont_touch* attribute, the library cell purposes for the tie cells do not include optimization, or the library cell is associated with a site that is not defined in the current design.

In some design styles, constant pins are directly tied to power or ground nets using the *connect_pg_net* command after optimization is done.

**What Next**

To insert tie cells during optimization, verify the site definitions and the tie cell settings for the *dont_touch* attribute and included library cell purposes. If necessary, update the *dont_touch* attributes using the *set_attribute* command and the library cell purposes using the *set_lib_cell_purpose* command. After making the fixes, restart the optimization.

Otherwise, tie the constant pins to power or ground nets using the *connect_pg_net* command after optimization.

Feedback

## OPT-201

(warning) Corner %s is similar to %s but has a different process label.

### Description

The two corners are similar in voltage, parasitics and derates, but differ in temperature and process label. It is likely that the two corners actually share the same process, but there is an error during library preparation. A common mistake is that multiple reference libraries were built for the same process but with different process labels. For example, two reference libraries for 'SS' process were built with process labels 'SS_125' and 'SS_m40'. This caused the timing information for different temperatures of the same process to be put in different panes and prvents synthetic corners from being created to achieve faster optimization runtime.

### What Next

If this is indeed a process label issue, the solution is to redo library preparation with the correct process label. If there is no other reference library with the same voltage and temperature for a different process, a workaround is to remove the process label restriction in the constraint corners using *set_process_label -clear.*

### See Also

- opt.mcmm.use_synthetic_corners
- report_pvt

## OPT-202

(warning) No legal tie cell is available to drive '%s' pin.

### Description

There is no tie cell in the library that is available for constant (1'b0 / 1'b1) fixing, given the constraints for the reported pin. This could be because of target library subset selection, power constraints etc.

### What Next

Check and update power constraints or targer library subset constraints or provide the library with additional cells.

## OPT-204

(Warning) The site row %s of lib cell %s/%s is illegal as it does not match any design site row.

Feedback

**Description**

The site row of given library cells does not match any site row of design.

**What Next**

User need to check if the site row definition in technique file is correct for the issue lib cell.

---

## OPT-205

(Warning) The site row %s of lib cell %s/%s is illegal as it is not integral multiple of design site.

**Description**

If the lib cell site row is not integral multiple design site row height, legalization will report error since it can not find legal location for this lib cell.

**What Next**

User need to check if the site row definition in technique file is correct for the issue lib cell.

---

## OPT-206

(Error) There is no default lib buffers/inverts found for the design %s.

**Description**

Tool can not find valid library buffers/inverters for optimization.

**What Next**

User need check design setting to ensure valid library buffers/inverters for optimization.

---

## OPT-207

(Error) VA %s has no Power Domain.

**Description**

Tool can not find power domain definition for VA.

**What Next**

User need to generate UPF power domain definition for each VA.

---

## OPT-208

(Error) Power domain %s has no primary supply net.

Feedback

**Description**

Tool can not find power supply definition for power domain.

**What Next**

User need to check UPF file to create power supply for each power domain.

---

## OPT-209

(Warning) Design includes unreasonable large hold violation(s).

**Description**

Unreasonable hold violation here means the hold violation is so large that it costs more than 20 max-delay buffers to fix. However, fixing such hold violations usually leads to density and QoR issues, so it is now skipped in the hold fixing engine by default.

**What Next**

Please check those large hold violations and try to resolve them in advance before rerunning the optimization flow. If you still want the tool to fix these large violations with current library setup, you can use *set_early_data_check_policy -policy tolerate -checks opt.sanity_check.large_hold -strategy report_only* , before clock_opt or route_opt.

**See Also**

- report_qor
- report_units

---

## OPT-210

(error) Scenario %s has an unreasonbly tight max transition constraint of %f.

**Description**

The indicated scenario has an unreasonbly tight max transition constraint that cannot be satisfied. This most likely will lead to area and Qor problems. This is usually because the same max transition constraint for fast corners/scenarios is also applied to slow corners/scenarios or it could be due to unit problem.

**What Next**

Please update the max transition constraint for this scenario and rerun optimization.

Feedback

**See Also**

- set_max_transition
- report_units

---

## OPT-211

(Warning) Input cell %s with library module %s is from wrong target library.

**Description**

The library module of the indicated cell in the input design is outside the scope of the permissible library cells allowed in accordance with the target library subset set on the cell's hierarchy.

**What Next**

Please correct the input design to have the correct library module for the indicated cell.

---

## OPT-212

(error) There are problems in end points specification. Optimization aborted.

**Description**

The list does not have any valid end points. Therefore, optimization aborted upon detecting this situation.

**What Next**

Check the end point list and make sure they are correct.

---

## OPT-213

(error) There are problems in nets specification. Optimization aborted.

**Description**

The list does not have any valid nets. Therefore, optimization aborted upon detecting this situation.

**What Next**

Check the net list and make sure they are correct.

Feedback

## OPT-214

(error) Exclusive flow and current command line argument cannot be honored at the same time. Optimization aborted.

### Description

The command line argument requires subset flow (i.e. -from or -to) or path_opt only flow (i.e. -end_points, -path_groups or -nets). However, there is an app option (i.e. refine_opt.flow.exclusive) requires exclusive flow, and the only supported combination is *hold* exclusive and *-end_points*. Therefore, optimization aborted upon detecting this situation.

### What Next

Check the command line and app options and make sure they are not conflicting.

### See Also

- refine_opt.flow.exclusive

## OPT-215

(Warning) the app option 'time.si_enable_analysis' should not be set to be true in pre-route stage, enforce it to be false temporarily.

### Description

The app option 'time.si_enable_analysis' is only supported in post route optimization stage now, and not in pre-route stage. It is now set to be true by the user in pre-route stage, and the tool sets it back to false temporarily in pre-route stage to avoid the potential issue.

### What Next

Please correct the app option setting and continue with the flow.

## OPT-216

(Error) Net %s with %d loads is not properly buffered.

### Description

The design has unbuffered high-fanout nets. Most likely *place_opt* has not been run, or has been run but failed to buffer this net. The *place_opt* initial_drc stage does HFS/DRC buffering, which is responsible for building the initial high fanout buffer trees.

Feedback

### What Next

Make sure that *place_opt* initial_drc was previously run. If it was run, check for reasons that the net may not have been buffered:

- Ensure the net does not have an MV violation, by looking in the *place_opt* logfile for OPT-007 errors, or by using *check_mv_design*
- Check for many bufferabilty problems by using the *check_bufferability* command. This command works on an individual net segment, and a high fanout net may consist of many segments, so use *check_bufferability* on each segment of the net to get a complete understanding.
- Floorplan issues like blockages, high cell density, and high congestion can also cause buffering problems. Use the GUI layout window to check for these issues.

### See Also

- check_bufferability
- check_mv_design
- place_opt
- OPT-007

---

## OPT-217

(error) There are problems in path group specification. Optimization aborted.

### Description

The list does not have any valid path groups. Therefore, optimization aborted upon detecting this situation.

### What Next

Check the path group list and make sure they are correct.

---

## OPT-218

(error) This is a top level design which is not supported by route-based refine_opt. Optimization aborted.

### Description

Refine_opt does not support top level designs in route-based optimization flow. Therefore, otimization aborted upon detecting this situation.

Feedback

**What Next**

Run the virtual routed based refine_opt.

---

## OPT-219

(warning) Library cell %s is useless due to zero input capacitance.

**Description**

This particular library buffer or inverter cell is with zero input capacitance, and cannot be used for optimization. If the number of this kind of library buffer or inverter cell is large, the QoR will be affected a lot due to few number of useful library buffer or inverter cell.

**What Next**

Check the library to provide enough usefull library buffer or inverter cells.

---

## OPT-220

(error) Worst hold violation exceeds 2000 times nominal buffer delay. This usually indicates a design setup issue. Hold optimization aborted.

**Description**

To prevent the hold fixing engine from working on unreasonable hold violations. The tool will bail out when the initial hold violation is too large.

**What Next**

Inpect the hold constraints and CTS result.

---

## OPT-221

(warning) Unplaced RP Groups detected.

**Description**

The design has unplaced RP Groups. Most likely place_opt has not been run. DTDP stage included in the place_opt mega-command is needed to process these RP Groups.

**What Next**

Please run place_opt first.

---

## OPT-222

(error) Extraction failed. place_opt is running with ZWD settings.

Feedback

### Description

Extraction failed. place_opt is running under ZWD settings, hence erring out.

### What Next

Please run place_opt with proper ZWD settings.

---

## OPT-223

(warning) Current design contains soft blockages, and optimization may add cells onto soft blockages.

### Description

Current design contains soft blockages, and optimization may add cells onto soft blockages. However, if users run stand-alone commands to do incremental placement after optimization, those cells will be forcefully moved out, because neither *place.coarse.fix_cells_on_soft_blockages* nor *place.coarse.enable_enhanced_soft_blockages* was set.

Incremental placement commands include '*create_placement -incremental*', '*create_placement -use_seed_locs*' and '*refine_placement*'.

### What Next

Setting *place.coarse.fix_cells_on_soft_blockages* to *true* will let incremental placement not move cells on soft blockages.

Setting *place.coarse.enable_enhanced_soft_blockages* to *true* will let incremental placement move cells on soft blockages only inside their soft blockages.

The two app options are mutually exclusive.

### See Also

- create_placement
- refine_placement
- place.coarse.fix_cells_on_soft_blockages
- place.coarse.enable_enhanced_soft_blockages

---

## OPT-224

(Warning) Removing buffer %s will lead to logical feedthrough.

Feedback

### Description

Removing this buffer will lead to logical feedthrough. A logical feedthrough occurs when a netlist logical module has a connection between two or more ports with no leaf logic in between. Logical feedthroughs result in assign statements in the exported Verilog netlist. While logical feedthroughs cause no problem for IC Compiler II, some verification tools cannot handle them.

### What Next

No action is needed if logical feedthroughs are acceptable in your flow. To prevent the creation of logical feedthroughs by remove_buffer_trees as well as all other commands in the ICC II flow, you can set the app option *opt.port.eliminate_verilog_assign* to the *true* value. The *add_feedthrough_buffers* command can also be used to insert buffers to break any existing logical feedthroughs in the design.

### See Also

- remove_buffer_trees
- add_feedthrough_buffers
- opt.port.eliminate_verilog_assign
- OPT-225

---

## OPT-225

(Warning) There are %d removed buffers which will lead to logical feedthroughs.

### Description

Buffer removal by the *remove_buffer_trees* command has introduced one or more logical feedthroughs in the netlist. A logical feedthrough occurs when a netlist logical module has a connection between two or more ports with no leaf logic in between. Logical feedthroughs result in assign statements in the exported Verilog netlist. While logical feedthroughs cause no problem for IC Compiler II, some verification tools cannot handle them.

### What Next

To see the specific buffer cells whose removal caused logical feedthroughs, specify the *-verbose* option to list them and look for *OPT-224* messages in the logfile.

No action is needed if logical feedthroughs are acceptable in your flow. To prevent the creation of logical feedthroughs by *remove_buffer_trees* as well as all other commands in the ICC II flow, you can set the app option *opt.port.eliminate_verilog_assign* to the *true* value. The *add_feedthrough_buffers* command can also be used to insert buffers to break any existing logical feedthroughs in the design.

Feedback

**See Also**

- remove_buffer_trees
- add_feedthrough_buffers
- opt.port.eliminate_verilog_assign
- OPT-224

---

## OPT-230

(Warning) There are cells placed illegally.

**Description**

The final path_opt stage expects input to be legalized. Please use *check_legality* to view details about these violations. Run *legalize_placement* and invoke final path_opt by *refine_opt -from final_path_opt* after legalization violations are cleaned.

**What Next**

Make sure the design is legalized before running the final path_opt stage.

- Check legalization violations by *check_legality*
- Run legalization by *legalize_placement*
- Run final path_opt stage by *refine_opt -from final_path_opt*

**See Also**

- check_legality
- legalize_placement

---

## OPT-231

(error) This is a detailed routed design which is not supported by %s command. Optimization aborted.

**Description**

The command does not support detailed routed designs in flow. Therefore, otimization aborted upon detecting this situation.

**What Next**

Run command before design is detailed routed.

Feedback

## OPT-232

(error) Cannot run %s on a design that has already run %s as they run different pre-route flows.

### Description

The place_clock_opt command runs a different pre-route flow than the traditional place_opt / clock_opt flow, with different assumptions, extraction models, etc. Taking the input from one flow into another can result in unexpected results and is not recommended.

### What Next

Use place_opt / clock_opt to continue with traditional pre-route flow, and only use place_clock_opt for the new pre-route flow.

### See Also

- place_opt
- clock_opt

## OPT-233

(Warning) true_footprint mode is not supported with CCD on -- turning off CCD.

### Description

CCD doesn't support true_footprint mode, therefore CCD must be off when running true footprint based size only. App options affected: route_opt.flow.enable_ccd set to false route_opt.flow.enable_ccd_clock_drc_fixing set to 'auto' route_opt.flow.enable_clock_power_recovery set to 'auto'

### What Next

User can either turn off CCD or set a different size only mode.

### See Also

- route_opt
- report_timing

## OPT-500

(Warning) Advanced HFN buffering will not be used for initial HFS and DRC step as global route based initial drc is turned on.

Feedback

### Description

Global route based initial drc does not work with advanced HFN buffering. Hence, when it is turned on, advanced HFN buffering will not be used.

### What Next

Make sure that the desired behavior is specified to *place_opt*

### See Also

- place_opt

---

## OPT-501

(Warning) Advanced HFN buffering will not be used for initial HFS and DRC step as GRLB is turned on.

### Description

GRLB does not work with advanced HFN buffering. Hence, when it is turned on, advanced HFN buffering will not be used.

### What Next

Make sure that the desired behavior is specified to *place_opt*

### See Also

- place_opt

---

## OPT-502

(Warning) Library cell %s has zero or negligible value specified for input capacitance.

### Description

Logical effort computation will not be possible for zero input capacitance of this library cell. Hence this cell will not be used for buffering. But if the cell is already in the netlist, then either a small value will be set if capacitance is specified as zero for all corners, or an average of the non-zero values from other corners will be set for the corner where it was specified as zero.

### What Next

Make sure that the library cell capacitance values are specified correctly.

Feedback

## OPT-503

(Warning) Routing blockage %s of block % is ignored because it has an invalid layer specification.

### Description

The virtual routing topology engine takes routing blockage into conisideration when building a map of blocked areas. This is done based on the layer to which the routing blockage is assigned to. In case this information is no valid the routing blockage cannot be mapped into the blockage map.

### What Next

Investigate the layer information of the reported routing blockage by checking the layer_name or layer attribute. If the routing blockage is required adjust the attributes accordingly.

### See Also

- get_routing_blockages
- report_attributes
- get_attribute
- set_attribute

## OPT-504

(information) It is recommended to use 'set_app_options -name opt.buffering.min_skinny_blockage_size -value %.4f'.

### Description

The design has several long but thin obstacles creating large number of extra detour points for the virtual routing engine. This can cause unnecessary high runtime. By using the recommended setting this penalty can be avoided. It may come with a slight QoR impact.

### What Next

Set the recommended app option and re-run. Check the QoR at the end of the run.

### See Also

- opt.buffering.check_for_skinny_blockages
- opt.buffering.min_skinny_blockage_size

Feedback

## OPT-505

(information) The VR topology generator is working as if 'set_app_options -name opt.buffering.min_skinny_blockage_size -value %.4f' had been set.

### Description

The evaluation of the obstacles found in the floorplan of the given design has found several long but skinny obstacles. Only a small number of non-fixed standard cells overlap with those obstacles. As a consequence these obstacles are ignored as objects for detouring. More details can be found in the man page for app option *opt.buffering.min_skinny_blockage_size*.

### What Next

If the QoR is not as expected re-run with setting app option *opt.buffering.check_for_skinny_blockages* to false.

### See Also

- opt.buffering.check_for_skinny_blockages
- opt.buffering.min_skinny_blockage_size

## OPT-700

(Error) Feedthrough buffer cannot be inserted on multiport net %s because no legal placement location within %d um of the net.

### Description

When the *opt.port.eliminate_verilog_assign* application option is set to *true,* optimization inserts feedthrough buffers on some nets to eliminate assign statements in the exported Verilog netlist. No nearby legal placement location could be found for a buffer on this net. As a result, optimization skipped insertion of a feedthrough buffer on this net to avoid possible QOR degradation.

### What Next

The floorplan may not permit a buffer to be inserted near this net, due to placement blockages, macros, site row availability, or other factors. Either modify the floorplan to correct this issue, or set the app option *opt.buffering.feedthrough_ignore_distance_threshold* to *true*. Setting the app option true will prevent the errors and force insertion of the feedthrough buffers, at a possible cost to timing or logical DRCs.

Feedback

**See Also**

- place_opt
- opt.port.eliminate_verilog_assign

---

## OPT-800

(Information) Design Fusion Restructuring targeted %u cells out of which %u are not optimizable.

**Description**

Logic restructuring cannot optimize cells with attribute such as size_only/dont_touch/fixed/locked. If majority of the cells are not optimizable, improvement from logic restructuring will be minimal.

**What Next**

User can check whether the data path cells have any unexpected restrictions.

---

## OPT-801

(Warning) Skipping Design Fusion Restructuring for total power because no dynamic/leakage power scenarios were found. Doing Design Fusion Restructuring for area instead.

**Description**

There is no scenario available for power optimization. Hence, we run area optimization.

**What Next**

User can check whether there is power scenario available.

---

## OPT-802

(Warning) No dynamic power scenarios were found. Doing Design Fusion Restructuring for leakage power only.

**Description**

There is no dynmaic scenario available for dynamic power optimization. Hence, we run leakage power optimization.

**What Next**

User can check whether there is dynamic power scenario available.

Feedback

## OPT-900

(Error) Secondary-PG constraints will not be honored for the Refine-Opt command

### Description

Secondary-PG constraints are only supported inside Place-opt/Clock-opt command. These constraints are not supported inside Refine-opt command and therefore will not be honored when Refine-opt command is run. This can lead to some QoR degradation.

### What Next

User need to check whether the design needs secondary-PG related constraints in Refine-opt command or if they need to run Refine-opt command anymore.

## OPT-901

(error) Power flow is enabled but neither leakage power nor dynamic power analysis is enabled.

### Description

With Power flow enabled, Pre-route optimization can use designs only if either leakage power or dynamic power or both are enabled. Please use report_scenarios to check that at least one active scenario has leakage analysis enabled when using power flow.

### What Next

User can solve the error by setting leakage analysis to true in one or more active scenarios using set_scenario_status {<scenario list>} -leakage true.

### See Also

- report_scenarios
- set_scenario_status
- place_opt
- clock_opt

## OPT-902

(warning) No clock routing rules were specified. Use set_clock_routing_rules to specify routing rules for accurate clock latency estimation.

Feedback

### Description

Clock latency estimation functionality was called without having specified non-default clock routing rules. Typically, this points to an incomplete setup of the clock tree synthesis (CTS) constraints. Clock latency estimation may be performed in pre-CTS stages of the flow to estimate the effect of clock tree topology on timing, e.g. for ICG enable timing paths. In order to have an accurate pre-CTS clock latency estimation, CTS constraints like CTS routing rules and CTS repeater selection should be setup before compile_fusion or place_opt.

### What Next

Specify the clock routing rules using set_clock_routing_rules before compile_fusion or place_opt.

### See Also

- set_clock_routing_rules
- place_opt

---

## OPT-903

(warning) No clock repeaters were configured. Use set_lib_cell_purpose to configure CTS repeaters for accurate clock latency estimation.

### Description

Clock latency estimation functionality was called without having specified the repeaters (buffers and/or inverters) to be used for clock tree synthesis (CTS). Typically, this points to an incomplete setup of the CTS constraints. Clock latency estimation may be performed in pre-CTS stages of the flow to estimate the effect of clock tree topology on timing, e.g. for ICG enable timing paths. In order to have an accurate pre-CTS clock latency estimation, CTS constraints like CTS routing rules and CTS repeater selection should be setup before compile_fusion or place_opt.

### What Next

Specify the repeaters that will be used during CTS using the set_lib_cell_purpose command.

### See Also

- set_lib_cell_purpose
- place_opt

Feedback

## OPT-904

(info) Created %d clock network latencies at the output of clock gates in scenario '%s'.

### Description

This message indicates that the tool created SDC set_clock_latency at the output of clock gates. This is to prepare for latency estimation at the input of the clock gates as imposed by app option 'place.coarse.clock_gate_latency_aware'. In order to avoid that estimated latencies at the input of a clock gate impact the ideal clock network latency in the fanout of the clock gate, the ideal clock network latency at the input is set to the output of the clock gate. This will only happen if the output of the clock gate does not have an existing SDC clock network latency specified.

Note that a clock gate in this context not only refers to integrated clock gates (ICGs) but also to discrete clock gates and clock dividers. Basically it refers to all clock cell that have a timing-checked data input.

### See Also

- place.coarse.clock_gate_latency_aware

## OPT-905

(info) Removed %d clock network latencies at clock gate inputs in scenario '%s'. New clock gate input latencies will be estimated.

### Description

This message indicates that user- or tool-specified latencies at clock gate inputs will be ignored (removed) and that new latencies will be applied based on internal clock tree latency estimation. This will happen if app option 'place.coarse.clock_gate_latency_aware' is set to 'true'. The message indicates how many latency settings are removed/ignored. Removing latencies at clock gate inputs may imply that latency settings have to be moved (copied) to the clock gate output to preserve the ideal latency in the fanout of the clock gate. If this is required, the tool will automatically take care of that.

Note that a clock gate in this context not only refers to integrated clock gates (ICGs) but also to discrete clock gates and clock dividers. Basically it refers to all clock cell that have a timing-checked data input.

### See Also

- place.coarse.clock_gate_latency_aware

Feedback

## OPT-906

(error) Design has atleast one corner with no Parasitic information.

### Description

Optimization cannot use designs with corners without Prasitic information. Please use report_parasitic_parameters to check the parasitic informatin for all the corners.

### What Next

User can solve the error by setting parasitic using set_parasitic_parameters or disabling the corresponding scenario of the corner without parasitics.

### See Also

- report_scenarios
- set_parasitic_parameters
- report_parasitic_parameters

## OPT-907

(error) '%s' failed during congestion-mode timing-driven placement.

### Description

Error is caused due to failure during congestion-mode timing-driven placement.

### See Also

- report_scenarios

## OPT-908

(error) place_opt failed during timing-driven placement.

### Description

Error is caused since, place_opt failed during timing-driven placement

### See Also

- report_scenarios

Feedback

## OPT-909

(info) Estimating clock gate latencies for scenario '%s'.

### Description

This informative message indicates that clock latencies are estimated for the specified scenario during pre-CTS optimization in the place_opt or compile_fusion flow. The estimated latencies are annotated on the design as an offset on top of user-annotated latencies using 'set_clock_latency -offset'. Note that these are not SDC constraints and will not be reported when 'write_sdc' is called. To inspect the latency offsets, please use the 'write_script' command.

### See Also

- write_sdc
- write_script
- create_placement
- place_opt

## OPT-910

(warning) Skipping clock gate latency estimation for scenario '%s' with propagated clocks.

### Description

Although clock gate latency estimation was called, it is skipped for the specified scenario because that scenario has at least one propagated clock. Clock gate latency estimation is only supported for scenarios where all clocks are in ideal mode, typically during pre-CTS stages of the flow.

### See Also

- place_opt
- create_placement

## OPT-911

(info) Skipped clock gate latency estimation for %d pin%s.

### Description

During clock gate latency estimation, a number of clock gate pins were excluded from estimation. Latency may not estimated for clock gate pins inside abstracts, for clock gate

Feedback

pins inside subtrees of a structural Multi-Source clock network or for pins that have the attribute *dont_estimate_clock_latency* on it.

**What Next**

To list all pins that were skipped and the reason why they were skipped, set application option *opt.clock_latency_estimation.report_skipped_pins* to *true* and rerun.

**See Also**

- opt.clock_latency_estimation.report_skipped_pins

---

## OPT-912

(warning) The design timing appears to be over-constrained (dirty). Optimization effort will be reduced to avoid excessive runtime.

**Description**

Based on analyzing various design timing metrics, the tool has determined that the design's timing is over-constrained, possibly because it is early in the design cycle. To avoid excessive runtime spent in fixing every setup timing violation, the tool will reduce the optimization effort and turn off runtime intensive features for the current session as well as future sessions that start from the same netlist.

**What Next**

If the design timing is dirty, reduced effort can still provide a reasonable estimate of power-performance-area (PPA) without significant runtime. However, if design is correctly constrained, the dirty timing detection can be disabled by setting the application option compile.physical.dirty_timing_handling to value “off” and rerunning compile_fusion.

---

## OPT-913

(info) Clock gate latency estimation for pin %s was skipped because the pin has attribute 'dont_estimate_clock_latency' on it.

**Description**

Automatic clock gate latency estimation skips clock pins that have the attribute 'dont_estimate_latency' on it.

The message is only generated if application option *opt.clock_latency_estimation.report_skipped_pins* was set to *true*.

**What Next**

Check whether this pin should really be excluded from clock gate latency estimation.

Feedback

### See Also

- opt.clock_latency_estimation.report_skipped_pins

---

## OPT-914

(info) Clock gate latency estimation for pin %s was skipped because the pin is inside an abstracted block.

### Description

Automatic clock gate latency estimation skips clock pins that are inside an abstracted block. Since the netlist inside an abstracted block is incomplete, no accurate latency estimation can be performed.

The message is only generated if application option *opt.clock_latency_estimation.report_skipped_pins* was set to *true*.

### See Also

- opt.clock_latency_estimation.report_skipped_pins

---

## OPT-915

(info) Clock gate latency estimation for pin %s was skipped because it is part of a structural Multi-Source clock network.

### Description

Automatic clock gate latency estimation may skip clock pins that are inside sub trees of a structural Multi-Source clock network. For these clock gates, a manual clock latency annotation should be done.

The message is only generated if application option *opt.clock_latency_estimation.report_skipped_pins* was set to *true*.

### What Next

Make sure that the clock gates inside these sub trees have a proper clock latency annotated.

### See Also

- opt.clock_latency_estimation.report_skipped_pins

Feedback

## OPT-916

(warning) Design timing is over-constrained. Turning off dirty timing handling will cause optimization to run with regular effort with potentially high runtime.

**Description**

This message is printed when compile.physical.dirty_timing_handling is set to off while the design is detected to have over-constrained timing. The default optimization will still be run despite over-constrained timing, and increased runtime may be expected.

## OPT-921

(warning) Skipping clock gate latency estimation because design setup is incomplete/invalid.

**Description**

Skipping clock gate latency estimation because design setup is incomplete/invalid.

## OPT-922

(warning) Cannot determine layer RC values for buffer characterization in corner %s.

**Description**

Cannot determine layer RC values for buffer characterization in corner %s.

## OPT-1006

(error) Library analysis failure - cannot find minimum required gates i.e buffer, inverter and a basic gate

## OPT-1007

(error) Library analysis failure - No technology data

## OPT-2001

(error) Could not initialize scenarios settings

## OPT-3000

(error) Design is not Linked or has link issues

Feedback

## OPT-3001

(error) Cell '%s' is not resolved after linking

## OPT-3002

(error) Cell '%s' is Physical hierarchy and not allowed in compile

## OPT-3003

(error) Cell '%s' is Logical hierarchy and not allowed in compile

## OPT-3005

(error) Design is not present

## OPT-4000

(error) Design does not have a floorplan or a valid floorplan

## OPT-4001

(error) Design does not have a valid die area

## OPT-4002

(error) Macro '%s' does not have placement information

## OPT-4003

(error) Design has '%d' unplaced Macros

## OPT-4004

(error) Design port %s does not have placement information

## OPT-4005

(error) Design has '%d' unplaced ports

Feedback

## OPT-4008

(error) Design has no site rows defined

## OPT-4009

(error) Design has no tracks defined

## OPT-4010

(error) Pad '%s' connected to port '%s' does not have placement information

## OPT-4011

(Warning) Could not find a legal location or the location is too far to instantiate isolation buffer for port %s

## OPT-4013

(warning) The port named '%s' cannot be isolated because it is on clock network.

### See Also

- set_isolate_ports

## OPT-4014

(warning) The port named '%s' cannot be isolated because it is not connected to a net.

### See Also

- set_isolate_ports

## OPT-4015

(warning) The port named '%s' cannot be isolated because it is of INOUT direction.

### See Also

- set_isolate_ports

Feedback

## OPT-4016

(warning) The port named '%s' cannot be isolated because it is not driven

**See Also**

- set_isolate_ports

## OPT-4017

(warning) The port named '%s' cannot be isolated because it is connected to don't touch/ constant/pad net.

**See Also**

- set_isolate_ports

## OPT-4018

(warning) The port named '%s' cannot be isolated because it is a clock port.

**See Also**

- set_isolate_ports

## OPT-4020

(error) The design contains safety registers. Please set the app-option 'compile.flow.enable_safety_register_groups' to true to use the safety features.

## OPT-4021

(error) Exclusive Movebound '%s' is completely covered by blockages

## OPT-4022

(error) Voltage Area '%s' is completely covered by blockages

## OPT-4030

(warning) Netlist contains routing shapes on optimizable signal nets. Please run 'compile_fusion -check_only' for more details.

Feedback

**Description**

Netlist contains detail route shapes on optimizable nets.

**What Next**

Run 'compile_fusion -check_only' command for more details of these detail route shapes. Set app option compile.flow.remove_dr_shapes to true will remove these route shapes during compile_fusion flow.

**See Also**

- set_app_options

---

## OPT-4031

(warning) Netlist contains %d routing shapes on '%s' optimizable signal net.

**Description**

Netlist contains no.of detail route shapes on individual optimizable net.

**What Next**

Set app option compile.flow.remove_dr_shapes to true will remove these route shapes during compile_fusion flow.

**See Also**

- set_app_options

---

## OPT-4032

(warning) Netlist contains %d routing shapes on %d optimizable signal net(s).

**Description**

Netlist contains no.of detail route shapes on optimizable nets.

**What Next**

Set app option compile.flow.remove_dr_shapes to true will remove these route shapes during compile_fusion flow.

**See Also**

- set_app_options

Feedback

## OPT-4033

(info) Netlist contains %d routing shapes on '%s' optimizable signal net has been removed.

### Description

Netlist contains no.of detail route shapes on individual optimizable net.

### See Also

- set_app_options

## OPT-4034

(info) Netlist contains %d routing shapes on %d optimizable signal net(s). These will be removed

### Description

Netlist contains no.of detail route shapes on optimizable nets.

### See Also

- set_app_options

## OPT-5003

(error) Failure in hierarchy checks. Run check_hier_design -stage pre_compile for details

### Description

Reported some errors related to design checks in the top-level closure. Please fix the hierarchical checks which are reported as TL errors.

## OPT-6000

(warning) Physical status of unmapped cell %s is converted from fixed to unplaced.

### Description

Physical status of unmapped cell is marked as unplaced and it will be treated as regular standard cell. This cell will be part of optimization and might get gobbled. If the cell needs to be preserved then compile_fusion can change the status of this cell to legalize_only and allow mapping/limited replacement on this cell. Set app option compile.flow.legalize_unmapped_fixed to change fixed status to legalize_only.

Feedback

**See Also**

- set_app_options

---

## OPT-6001

(warning) Physical status of unmapped cell %s is converted from fixed to legalize_only.

**Description**

Physical status of unmapped cell is marked as legalize_only. This cell will be mapped in compile_fusion and will be part of optimization. Out of Unmapped Syntheic cells only SeqGen's and WordView Gtech's are converted to legalize_only, rest other type of synthetic cells will be marked unplaced. Set app option compile.flow.legalize_unmapped_fixed false to treat these unmapped cells as regular standard cells.

**See Also**

- set_app_options

---

## OPT-6003

(warning) New cell %s was added to module %s with no new cells allowed.

**Description**

New cell is added to module/cell having attribute is_new_cell_allowed false.

**What Next**

Either update the constraint using set_allow_new_cells or don't create the cell inside restricted module/cell. One should start by looking at the first occurring OPT-6003 or OPT-6006 messages in the flow. Subsequent messages might occur because of optimization of new cells added earlier in the flow and this is expected. If you think first warning message is application issue, then contact Synopsys.

---

## OPT-6004

(error) Can not be updated for %s %s.

**Description**

set_allow_new_cells/is_new_cell_allowed attribute updation is not allowed due to the given reason.

Feedback

**What Next**

Avoid updating set_allow_new_cells/is_new_cell_allowed attribute.

---

## OPT-6005

(warning) %s %s already contains leaf cells. New cell addition cannot be restricted.

**Description**

This hierarchy already contains leaf cell so various optimization can still create new cells in this hierarchy even if this is restricted. Soft/hard marco are not considered as leaf cell and this warning will not be issued for macros.

**What Next**

Either remove all leaf cells from this hierarchy else use other restrictions to disallow new cell creation.

---

## OPT-6006

(warning) New cell %s was added to voltage area %s with no new cells allowed.

**Description**

New cell is added to voltage area having attribute is_new_cell_allowed false.

**What Next**

Either update the constraint using create_voltage_area_rule -allow_new_cells or don't create the cell inside restricted VA. One should start by looking at the first occurring OPT-6003 or OPT-6006 messages in the flow. Subsequent messages might occur because of optimization of new cells added earlier in the flow and this is expected. If you think first warning message is application issue, then contact Synopsys.

One may also encounter OPT-6006 for transitional violations e.g. cell was temporarily created in restricted VA and then immediately moved to proper VA. One can ignore such violations safely as long as they are not present at the end of the command. These tranistional violations will not cause any real violations.

---

# PAC

---

## PAC-001

(error) Current library is not created by command create_pin_check_lib, cannot run pin access checking command for it.

Feedback

**Description**

The tool perfoms pin access checking only on the library created by command create_pin_check_lib.

**What Next**

Please run create_pin_check_lib to create the pin check library.

---

## PAC-002

(warning) Run tcl command failed. Command: %s

**Description**

Run tcl command and the command is failure.

**What Next**

Please check the tcl command.

---

## PAC-003

(error) Empty value in app_option %s.

**Description**

The tool cannot accept empty value of this app_option.

**What Next**

Please input valid app_option value.

---

## PAC-004

(error) Cannot use app_option %s in current checking -mode %s. %s

**Description**

The tool cannot support this app_option in current -mode.

**What Next**

Please check and modify the app_option value.

---

## PAC-005

(error) Require app_option %s when %s.

Feedback

**Description**

The tool needs the prerequisite app_option.

**What Next**

Please add the prerequisite app_option.

---

## PAC-006

(error) -to stage cannot be eariler than -from stage. The stage order is prepare, place, report_place, route, report_route.

**Description**

The command option -from and -to must be in correct order. -from stage should be eariler than -to stage.

**What Next**

Please correct the value of command option -from and -to.

---

## PAC-007

(error) No libcell found. Please check pin_check.input.cells and pin_check.input.exclude_cells.

**Description**

No libcell found when preparing pin_check database.

**What Next**

Please check the pin_check.input.* and the libcell names in ref_libs.

---

## PAC-008

(error) Create cell_pin_access block failed.

**Description**

Something wrong when running for -mode analyze_lib_cell or analyze_lib_pin.

**What Next**

Please check the message.

Feedback

## PAC-009

(warning) Cannot find cell %s.frame in reference library. %s

**Description**

Cannot find the target cell (.frame) in reference library.

**What Next**

Please check the input cell name in corresponding options.

## PAC-010

(information) No against cells in pin_check.input.include_for_checking, automatically copy from pin_check.input.cells.

**Description**

The tool default uses input cells as against cells when the option is not set.

**What Next**

NA

## PAC-011

(warning) Cannot set pin threshold to 0 in %s mode by app_option pin_check.input.filter_cells_by_pin_count. Use default pin threshold 1 instead.

**Description**

The tool does not support 0 pin threshold in current checking mode. Use the default value instead.

**What Next**

Please set non zero value in pin_check.input.filter_cells_by_pin_count.

## PAC-012

(error) The pin threshold exceeds the maximum pin count %d. Please set app_option pin_check.input.filter_cells_by_pin_count with a smaller value.

**Description**

The pin threshold value is over the maximum pin count. No cell can be selected for checking.

Feedback

**What Next**

Please set a smaller value in pin_check.input.filter_cells_by_pin_count.

---

## PAC-013

(error) report_cell_pin_access failed; %s

**Description**

Something wrong in -mode analyze_lib_pin.

**What Next**

Please check the error message.

---

## PAC-014

(warning) Invalid layer value "%s" in option %s.

**Description**

The input option value has invalid layers.

**What Next**

Please correct the layer name.

---

## PAC-015

(warning) Cell %s is design_type %s, which is only supported by single_cell mode. Skip pin access checking for this cell in current -mode %s.

**Description**

The following design_type is only supported in single_cell mode: macro, pad, analog, black_box, flip_chip_pad, flip_chip_driver, abstract, pad_spacer, module.

**What Next**

Please check the reference library.

---

## PAC-016

(information) Directory %s is overwritten automatically.

**Description**

The tool overwrites the directory when it already exists.

Feedback

**What Next**

Please backup the file if you want to keep the old file.

---

## PAC-017

(warning) Skip placing %s because it is unprepared.

**Description**

The tool can only place a well prepared design.

**What Next**

Please run check_libcell_pin_access -from "prepare" in advance for the cell.

---

## PAC-018

(warning) Skip routing %s because it is unplaced.

**Description**

The tool can only route a well placed design.

**What Next**

Please run check_libcell_pin_access -to "route" in advance for the cell.

---

## PAC-019

(error) SiteDef information is not initialized.

**Description**

The tool needs site width / height information for the checking process.

**What Next**

Please check the site_def information in technology.

---

## PAC-020

(error) Fail to create %s.

**Description**

The block is required in the checking process.

Feedback

**What Next**

Please check the pin_check library.

---

## PAC-021

(error) Fail to create %s due to read_verilog %s failed.

**Description**

The tool cannot create the block due to read_verilog failed.

**What Next**

Please check the verilog file content.

---

## PAC-022

(warning) Cannot find legal location for %s. %s

**Description**

The tool cannot find legal location for the cell.

**What Next**

Please chech the place options.

---

## PAC-023

(warning) Cannot find target cells, please check the option setting. %s

**Description**

The tool cannot find target cells in current process.

**What Next**

Please check the input library or option setting.

---

## PAC-024

(warning) Cannot find against cells, please check the option setting. %s

**Description**

The tool cannot find agsint cells in current process.

Feedback

**What Next**

Please check the input library or option setting.

---

## PAC-025

(warning) Cannot find the ICV executable for signoff_check_drc.

**Description**

The tool cannot find the ICV executable and skip the signoff_check_drc process.

**What Next**

Please set correct path by app_option pin_check.report.icv_home_dir or env ICV_HOME_DIR.

---

## PAC-026

(warning) The preferred routing direction is not defined for layer {%s}. Please set routing_direction attribute as horizontal or vertical in the preplaced option file.

**Description**

The routing_direction attribute of the routing layer is unknown. Please set the attribute in the preplaced option file.

**What Next**

Please set routing_direction attribute in the file of pin_check.place.preplaced_option_file. For example: set_attribute -objects [get_layers M1] \\ -name routing_direction \\ -value horizontal|vertical

---

## PAC-027

(warning) Number of selected cells (%d) is less than the number of each run (%d). The command will still perform pin checking in single run.

**Description**

According to current multiple run setting, the target cells number is less than the required testing cell number of each run. The tool will not divide the pin checking into multiple runs.

**What Next**

Please check the setting in pin_check.common.num_cells_per_run or pin_check.common.num_jobs.

Feedback

## PAC-028

(warning) Something wrong in job %d, please check the log file %s for the core dump or other information.

**Description**

Something wrong in the job of multiple runs. Please check the log file for detail information.

**What Next**

Please check the log file for detail information.

## PAC-029

(error) Fail to submit job for file %s. %s

**Description**

Fail to submit job for multiple runs pin checking flow.

**What Next**

Please check the log file or correct the grd command.

## PAC-030

(warning) The main flow is terminated, but %d jobs may be still running in multiple runs. Please check and delete the jobs manually.

**Description**

The main flow is terminated, please check and delete the jobs in multiple runs if you don't need them.

**What Next**

Please terminate the jobs in multiple runs, and check the log for error information.

## PAC-031

(warning) Missing row_pattern specification in PRF when pin_check.place.row_pattern app_option is specified.

Feedback

**Description**

Can't find row_pattern specification in PRF. The design will be created with regular row. Please check PRF file.

**What Next**

Please check PRF file or specify correct row_pattern.

---

## PAC-032

(error) No test design found for current -mode %s. Please correct the -mode or run with "-from prepare".

**Description**

No test design found when preparing pin_check database. The checking mode may be inconsistent with previous checking.

**What Next**

Please correct the command -mode option or start from "prepare" step.

---

## PAC-033

(warning) Cell %s is design_type %s, which is not supported in current command. Skip the cell specified in option %s.

**Description**

The following design types are not supported in check_libcell_pin_access: macro, pad, analog, black_box, flip_chip_pad, flip_chip_driver, abstract, pad_spacer, module.

**What Next**

Please correct the related application option.

---

## PAC-034

(warning) Disable set_host_options command in the sub job of parallel flow.

**Description**

Disable set_host_options command in the sub job of parallel flow to prevent the threading resource issue.

The set_host_options command in the Tcl files sourced by pin accessibility checking will become invalid.

Feedback

### What Next

Please review the Tcl files used in the pin accessibility checking command.

---

## PAC-035

(warning) Cannot launch %d jobs in non-GRD mode, which exceeds pin_check.common.max_cores %d. The tool ignores multiple runs setting and performs pin accessibility checking in single run.

### Description

The max number of thread is not enough for the jobs in parallel flow. The tool will ignore pin_check.common.num_jobs or pin_check.common.num_cells_per_run and keep single run flow.

### What Next

Please review the number of thread setting in pin_check.common.max_cores.

Please set a smaller value for pin_check.common.num_jobs or a larger value for pin_check.common.num_cells_per_run. Or enable pin_check.common.grd to submit jobs in different machines.

---

## PAC-036

(information) Automatically set pin_check.common.max_cores as %d for each sub job in non-GRD parallel flow.

### Description

In non-GRD parallel flow, automatically calculate and set the max_cores for each sub job.

### What Next

Please review pin_check.common.max_cores and parallel flow options.

---

## PAC-037

(error) The reference library %s is not a libcell library, which is unsupported in create_pin_check_lib.

### Description

The tool perfoms pin access checking only for libcell type reference libraries.

### What Next

Please check the reference libraries.

Feedback

## PAC-038

(warning) There is no stage control in current checking mode (%s). Ignore option -from/-to and run full flow.

**Description**

There is no stage control in current pin access checking mode. The tool runs full checking flow.

## PAC-039

(warning) The value of both %s and %s are not empty, use %s for the composite_cell mode.

**Description**

In composite_cell mode, can only set either composite_inst_count or composite_cell_area_ratio.

**What Next**

Please set either composite_inst_count or composite_cell_area_ratio if you do not want to see the warning message.

## PAC-040

(error) Invalid %s value %f, please specify a number in um between 5 x placement_site_width to the length of shorter core side.

**Description**

In tool cannot support invalid value of this app_option.

**What Next**

Please specify a valid app_option value.

## PAC-041

(warning) Option %s is only for check_libcell_pin_access, which has no effect in check_macro_pin_access.

**Description**

The application option is only for command check_libcell_pin_access. The option setting has no effect when executing check_macro_pin_access.

Feedback

**What Next**

Please check the usage of the application option.

---

## PAC-042

(warning) Option %s is only for check_macro_pin_access, which has no effect in check_libcell_pin_access.

**Description**

The application option is only for command check_macro_pin_access. The option setting has no effect when executing check_libcell_pin_access.

**What Next**

Please check the usage of the application option.

---

## PAC-043

(error) The utilization rate is higher than 0.2. There may be no enough space to place all cells.

**Description**

In tool cannot support the design which utilization rate is too high. To consider hybrid rows, PG structure ..., it may need more space to place all cells.

**What Next**

Please specify a correct design or use less test cells.

---

# PARSE

---

## PARSE-1

(error) %s: %s on line %d at or near '%s'.

---

## PARSE-2

(error) %s on line %d at or near '%s'.

Feedback

# PDC

## PDC-001

(Warning) Reference '%s' pin '%s' has no via regions. Using pin shape.

**Description**

The pin specified does not contain via regions generated in the frame. Pin shape is used instead for legality checking.

**What Next**

Check your library preparation procedure for correctness.

## PDC-002

(Error) Design '%s' has no associated Technology.

**Description**

The design specified is missing Technology file information. Legalization is thus unable to proceed.

**What Next**

Check your data preparation procedure for correctness.

## PDC-003

(Warning) Routing direction of metal layer %s is neither "horizontal" nor "vertical". PDC checks will not be performed on this layer.

**Description**

This warning message indicates that no routing direction has been set on the metal layer. This may result in DRC violations due to illegal placement.

**What Next**

set_attribute [get_layer <layer>] routing_direction <horizontal/vertical>

## PDC-004

(Warning) Technology file error detected. Excluded Spacing Range defined but corresponding Min Spacing Table and/or Parallel Length Table is not.

Feedback

**Description**

This warning message indicates that a technology file error has been detected. This may result in DRC violations due to illegal placement.

**What Next**

Check and fix technology file.

---

## PDC-005

(Error) Power/Ground pin %s/%s shorts with unconnected metal shape %s.

**Description**

This error message indicates that a short between a cell's power/ground pin and an unconnected metal shape has been found which likely indicates a failure to properly connect the Power/Ground rails to their nets. This will result in legalization failure.

**What Next**

Fix Power/Ground rail connectivity.

---

## PDC-006

(Warning) Technology file via layer %s definition has conflicting sizes of cut name, width, height tables.

**Description**

This warning message indicates that technology file fields "cutNameTbl", "cutWidthTbl" and "cutHeightTbl" in the given via layer definition are inconsistent with each other. Via spacing checking on this layer will be skipped.

**What Next**

Fix technology file.

---

## PDC-007

(Warning) No tracks defined on metal layer %s. Pin access checking disabled for this layer.

**Description**

This warning message indicates that no track definition has been found in the NDM on the given metal layer. Pin access checking on this layer will be skipped.

Feedback

**What Next**

Fix floorplan.

---

## PDC-008

(Warning) Command set_legalizer_preroute_keepout encountered error in setting keepout on provided pins or lib_pins.

**Description**

This warning message indicates the keepout could not be set on one or more of the named pins/lib_pins.

**What Next**

Review the pin/lib_pin list.

---

## PDC-009

(Warning) Command set_legalizer_access_track_constraint encountered error in setting the constraint on provided pins or lib_pins.

**Description**

This warning message indicates the constraint could not be set on one or more of the named pins/lib_pins.

**What Next**

Review the pin/lib_pin list.

---

## PDC-010

(Error) No site_defs found in top block.

**Description**

This error message indicates that there are no site_defs defined in the design block. Pre-routed net checking is disabled.

Feedback

**What Next**

Review the floorplanning steps,

---

## PDC-011

(Warning) Technology file field cut%dNameTbl in Design Rule for layers %s, %s has undefined cut name %s.

**Description**

This warning message indicates that the technology file definition for given Design Rule lists a cut name that is not defined in the Via layer section. Legalization will not be able to honor the associated rules.

**What Next**

Fix the technology file.

---

## PDC-012

(Warning) Pin %s of RefLib %s has different direction from preferred track.

**Description**

Pin/track alignment requires pin and track are always in the same direction. This message indicates some pin direction do not follow this rule.

**What Next**

Check input design to see if pin/track direction is set correctly.

---

## PDC-013

(Warning) Grid %s is not found. Via0 alignment is disabled.

**Description**

The named via0 alignment grid is not found in NDM. User must provide valid via0 grid name to app_option "place.legalize.via0_alignment_grid_name" to enable the alignment to the via0 grid. The grid object is created by executing command "create_grid".

**What Next**

Check if command "create_grid" has been executed to create the via0 grid object.

Feedback

## PDC-014

(Warning) Pin-track color alignment checking skipped for layer %s.

**Description**

Pin-track color alignment is not defined for the named metal layer. Only metal layers with tech file field "hasRectangleOnly = 1" are appropriate for Pin-color alignment checking.

**What Next**

Check application options and technology file.

## PDC-015

(Error) One or more metal/via layer has been deleted. Pre-routed net checking is disabled.

**Description**

This error message indicates that there is at least one deleted metal or via layer from the database. Pre-routed net checking is disabled. Any legalization or check_legality results are unreliable.

**What Next**

Review the floorplanning steps,

## PDC-016

(Warning) On layer %s found %d identical %s %s at location %s. Check for duplicate power grid.

**Description**

Multiple identical vias or metal shapes were found at the same location. This could mean that this design's power grid was created or loaded multiple times. A duplicated power grid can greatly increase run-time, and should be corrected.

**What Next**

Check the vias or metal shapes at the location. If there are multiple identical shapes, the power grid has been duplicated. Remove the power grid from the affected layers and reload or recreate it once for those layers. Check your design loading script for duplication.

## PDC-017

(Warning) Cannot save PG structure analysis in block '%s' for reuse. Later runs must recompute PG data.

Feedback

### Description

After PG structure analysis the tool could not save the PG structure data to the block for reuse. Later runs of the tool cannot reuse the saved data, the PG structure analysis must be recomputed.

The PG structure analysis can take considerable time and memory. The tool saves the results in the block so that the PG structure can be quickly re-read and reused. If the data cannot be saved it must recomputed, sometimes several times in the same flow. This can take considerable time and memory.

### What Next

The most common reason why the PG structure analysis cannot be saved in the block is that the block has not yet been saved to disk. Either the block was just read in, or the block was copied but not yet saved to its library. Use *save_block* to save the current block to its library on disk.

Another reason why the PG structure analysis cannot be saved is that the current user does not have write permission to the library containing the block. Use *copy_block* to copy the block to a library with write permissions. Then use *save_block* to save the copied block to disk.

If the library Unix directories and files have lost write permissions, the permissions can be altered with the Unix *chmod* command. Use *chmod -R u+w /path/to/library* to add user write permissions to the library directories.

### See Also

- copy_block
- open_block
- save_block
- save_lib

---

## PDC-018

(Warning) Cannot save %s cache to block '%s' for reuse. Later runs must recompute cache data.

### Description

After completing legalization design rule checking, the tool could not save the DRC caches to the block for reuse. Later runs of the tool cannot reuse the saved data, those checks must be re-run.

Feedback

There can be millions of design rule checks and each one takes time. The tool caches the result of each check so it does not have to re-run each check for each lib cell. At the end of a run, the DRC results are saved to the block. At the start of the next run the DRC cache is reloaded from the block to save the time of re-running those design rule checks.

**What Next**

The most common reason why the DRC cache cannot be saved in the block is that the block has not yet been saved to disk. Either the block was just read in, or the block was copied but not yet saved to its library. Use *save_block* to save the current block to its library on disk.

Another reason why the DRC cache cannot be saved is that the current user does not have write permission to the library containing the block. Use *copy_block* to copy the block to a library with write permissions. Then use *save_block* to save the copied block to disk.

If the library Unix directories and files have lost write permissions, the permissions can be altered with the Unix *chmod* command. Use *chmod -R u+w /path/to/library* to add user write permissions to the library directories.

**See Also**

- copy_block
- open_block
- save_block
- save_lib

---

## PDC-019

(warning) Using 'ignored_pnet_shape_layers' from the design library instead of from the reference library for lib-cell '%s'.

**Description**

The tool picks up the ignored_pnet_shape_layers rules (see PRF flow) first from the design library, and then from the reference libraries. This message indicates that the layer definitions were found in the design library as well as in the reference libary of the given lib-cell. The ones from the reference library are ignored.

Feedback

# PGR

## PGR-001

(error) The required option %s is not specified.

**Description**

This option is a required option.

**What Next**

Check the command man page and specify the required option.

## PGR-002

(error) For option %s, the specified value %s is invalid.

**Description**

This option value is invalid.

**What Next**

Check the command man page and specify the correct value.

## PGR-003

(error) Some variables have been defined multiple times.

**Description**

Variables should be only defined once in -parameters option which each variable can be used multiple times by other options.

**What Next**

Remove the redundant variables.

## PGR-004

(error) Some used parameters are not defined in -parameters option.

**Description**

Parameters used by any options need to be defined in -parameters option.

Feedback

**What Next**

Define the missing variables in -parameters option.

---

## PGR-005

(error) In option %s, the specified value %s for keyword %s is invalid.

**Description**

This value is invalid.

**What Next**

Check the command man page and specify the correct value.

---

## PGR-006

(error) In the pattern expression %s, the specified string %s is unknown.

**Description**

This string including keyword and data is unknown.

**What Next**

Check the command man page and specify the correct string.

---

## PGR-007

(error) In pattern expression %s, the required keyword %s is missing.

**Description**

Missing required keyword.

**What Next**

Check the command man page and specify the required keyword.

---

## PGR-008

(error) In pattern expression %s, the value for keyword %s is missing.

**Description**

This value is not specified.

Feedback

**What Next**

Check the command man page and specify the correct value.

---

## PGR-009

(error) In pattern expression %s, the following keywords %s have been specified more than once.

**Description**

Some keywords have been specified more than once in the pattern expression.

**What Next**

Remove the redundant keywords.

---

## PGR-010

(error) In composite pattern, some nets used in sub-pattern expressions are not defined in "-nets" option.

**Description**

Nets used in sub-patterns need be be defined in "-nets" options in a composite pattern;

**What Next**

Correct the "-nets" option or nets entry in sub-pattern expression.

---

## PGR-011

(error) In composite pattern, some sub patterns are neither wire nor composite pattern types.

**Description**

Composite pattern can only contain wire or composite patterns.

**What Next**

Correct the pattern expression in the composite pattern.

---

## PGR-012

(error) Parameter interface is inconsistent between the composite pattern and some child patterns.

Feedback

**Description**

The parameter interface need to be matched.

**What Next**

Correct the pattern expression in the composite pattern.

---

## PGR-013

(error) Composite pattern can only contain other composite patterns or wire patterns as child patterns.

**Description**

Composite pattern can only contain other composite patterns or wire patterns as child patterns.

**What Next**

Correct the pattern expression in the composite pattern.

---

## PGR-014

(error) Composite pattern cannot be defined recursively.

**Description**

Composite pattern cannot contain itself as a child or descendant pattern.

**What Next**

Correct the pattern expression in the composite pattern.

---

## PGR-015

(error) In via rule %s, the %s type rule has been specified several times.

**Description**

Some information such as intersection rule, filtering rule or via definition rule have been specified more than once or twice in one via rule.

**What Next**

Correct the via rule expression string, i.e. no more than one intersection rule, two filtering rules or one via definition rule in one via rule.

Feedback

## PGR-016

(error) In via rule %s, multiple types of rules have been group in one brace pair.

**Description**

Some information such as intersection rule, filtering rule or via definition rule have been group within one brace pair.

**What Next**

Correct the via rule expression string, i.e. the brace pair may not be wrong.

## PGR-017

(error) In the via filtering rule %s, the specified string %s is unknown.

**Description**

This string including keyword and data is unknown.

**What Next**

Check the command man page and specify the correct string.

## PGR-018

(error) In via filtering rule %s, the following keywords %s have been specified more than once.

**Description**

Some keywords have been specified more than once in the via rule.

**What Next**

Remove the redundant keywords.

## PGR-019

(error) In the strategy extension %s, the specified string %s is unknown.

**Description**

This string including keyword and data is unknown.

**What Next**

Check the command man page and specify the correct string.

Feedback

## PGR-020

(error) In the strategy blockage %s, the specified string %s is unknown.

**Description**

This string including keyword and data is unknown.

**What Next**

Check the command man page and specify the correct string.

## PGR-021

(error) Strategy %s is obsolete and needs to be redefined.

**Description**

Some pattern used by this strategy may be removed or redefined after the strategy is defined. The strategy needs to be redefined using the up-to-date patterns.

**What Next**

Redefine the strategy using set_pg_strategy command.

## PGR-022

(error) Name of %s is missing.

**Description**

To use -tcl option, the name of reported pattern/strategy/via_rule/mask_constraint needs to be specified.

**What Next**

Please specify the name of the reported pattern/strategy/via_rule/mask_constraint.

## PGR-023

(error) The region polygon is self-intersected.

**Description**

A self-intersected polygon can be divided into multiple polygons. PG router does not know PG area should include or exclude certain polygons.

Feedback

**What Next**

Please modify the region to non-self-intersected and specify blockage using the option -blockage of the command set_pg_strategy to guide PG router to exclude certain areas.

---

## PGR-024

(error) Failed to load technology library from the design.

**Description**

PG router cannot continue without routing layer and design rule info from technology file.

**What Next**

Please check if a valid tech file is loaded for the current block.

---

## PGR-025

(error) Fail to find PG objects in specified regions.

**Description**

PG router cannot find PG shapes of nets (specified by -nets) in the regions (specifed by -within_bbox).

**What Next**

Please check if -nets and -within_bbox are correctly specified.

---

## PGR-026

(error) Failed to create pin at %s for net %s because: %s

**Description**

Failed to create pins at the boundary.

**What Next**

Follow the error message instruction.

---

## PGR-027

(error) Failed to create directory %s to write.

**Description**

Failed to write PG constraint files to directory.

Feedback

**What Next**

Please grant write permission to the working directory.

---

## PGR-028

(error) Failed to extend %s end of wire at %s to %s due to DRC errors.

**Description**

Failed to extend wires.

**What Next**

Check the DRC issues. You may re-run the command with DRC checking turned off and then use check_pg_drc to find out details of DRC violations.

---

## PGR-029

(error) Strap with center %s and width %s is outside design boundary.

**Description**

Failed to create pg strap because it is outside design boundary.

**What Next**

Please check the center locations for PG strap to be created.

---

## PGR-030

(error) Failed to open file %s.

**Description**

Failed to open script/side/mapping files to read or write.

**What Next**

Please check if the file exist and grant read/write permission to the working directory.

---

## PGR-031

(error) Failed to find interconnect layer with %s direction.

**Description**

The command failed because there is no interconnect layer with vertical or horizontal direction in the tech file.

Feedback

**What Next**

Please examine the tech file to check interconnect layer settings.

---

## PGR-032

(error) Cannot find layer %s in the library.

**Description**

Cannot find routing layer in the design library

**What Next**

Please check if the layer name is correct.

---

## PGR-033

(error) %s is undefined.

**Description**

The keyword is not defined in the design.

**What Next**

Please define the required info.

---

## PGR-034

(error) %s is not specified in %s.

**Description**

The keyword is not specifed in user input.

**What Next**

Please specify the keyword.

---

## PGR-035

(error) Non-numerical value is specified for %s.

**Description**

The keyword is specifed with a non-numerical value.

Feedback

**What Next**

Please specify numerical value for the keyword.

---

## PGR-036

(error) Non-positive value is specified for %s.

**Description**

The keyword is specifed with a non-positive value.

**What Next**

Please specify positive value for the keyword.

---

## PGR-037

(error) Failed to load the design core area.

**Description**

Failed to load the design core area.

**What Next**

Please check if the core area is defined for the design

---

## PGR-038

(error) Failed to load the design boundary.

**Description**

Failed to load the design boundary.

**What Next**

Please check if the design boundary is defined

---

## PGR-039

(error) Invalid value %s is specified for %s.

**Description**

Invalid argument is specified for the keyword.

Feedback

**What Next**

Please check the man page to see acceptable value for the keyword

---

## PGR-040

(error) User specified routing layers are not adjacent.

**Description**

The error is issued when -from_layer and -to_layer are not adjacent for create_pg_stapling_vias. Currently, this command is limited to create stapling vias for parallel wires on adjacent metal layers

**What Next**

Please specify adjacent routing layers for -from_layer and -to_layer.

---

## PGR-041

(error) Specified contact code does not match specified metal layers.

**Description**

The error is issued when -contact_code does not match -from_layer and -to_layer for create_pg_stapling_vias.

**What Next**

Please specify matching -contact_code for -from_layer and -to_layer.

---

## PGR-042

(error) Failed to assign mask constraint for PG shape at %s, layer %s.

**Description**

The error is issued when the command failed to assign mask constraint for the shape. This error could be caused by the following reasons:

1. The shape is not center-aligned with any routing tracks. 2. The center-aligned routing track does not have mask constraint itself. 3. The shape is touching with other shapes which have conflict mask constraints.

**What Next**

Please check which reason causes the failure and fix accordingly.

Feedback

## PGR-043

(error) Keyword %s has been specified multiple times in %s.

**Description**

The keyword has been specified multiple times in the spec.

**What Next**

Please correct the spec.

## PGR-044

(error) Unknown string %s inside the specification %s.

**Description**

There are unknown keyword defined in the specification.

**What Next**

Please correct the specification.

## PGR-045

(error) Dummy net name

**Description**

The error is issued when users try to specify dummy net name for ring, rail, macro connection, or other special PG patterns.

**What Next**

Please specify valid net name for the pattern.

## PGR-046

(error) Cannot find net %s in the design.

**Description**

The error is issued when pg router cannot find the specified net in the current block.

**What Next**

Please check if the specified net exists.

Feedback

## PGR-047

(error) Cannot find contact code or via rule %s in the design.

**Description**

The error is issued when pg router cannot find the specified contact code in the tech file or the pg via rule defined for the current block.

**What Next**

Please specify correct contact code or via rule.

## PGR-048

(error) Layer %s is not a wire routing layer.

**Description**

This error is issued when user specified layer is not a wire routing layer

**What Next**

Please specify valid routing layer.

## PGR-049

(error) Failed to find intersecting PG objects for specified nets in specified regions.

**Description**

This error is issued when there is no intersecting PG shapes in the specified regions. As a result, no vias could be inserted for create_pg_vias command.

**What Next**

Please check if PG shapes are properly created in that region.

## PGR-050

(error) Existing vias prevented new vias to be inserted.

**Description**

This error is issued when there are existing vias at the intersections where new vias intended to be inserted by create_pg_vias command.

Feedback

**What Next**

Please check if extra vias are needed for intersections already having vias. User may specify option -insert_additional_vias to allow extra vias to be created.

---

## PGR-051

(error) DRC violations prevented new vias to be inserted.

**Description**

This error is issued when vias could not be created due to DRC violations for create_pg_vias command.

**What Next**

User may specify -drc no_check to skip DRC checking. After that, user may use check_pg_drc command to debug what DRC rules are preventing vias to be inserted.

---

## PGR-052

(error) Failed to apply via rule to insert vias or intersection ratio is smaller than via site threshold.

**Description**

This error is issued when create_pg_vias command failed to apply PG via rules specified by -via_masters options. The intersection ratio in vertical/horizontal is smaller via site threshold.

**What Next**

Please check if the specified via rule is applicable and via site thresold value.

---

## PGR-053

(error) Incorrect usage of snapping reference point.

**Description**

Snapping reference point should be used together with allow_multiple, but mutually exclusive with offset and offset start, as well as via site ratio. This error is issued when the usage is not correct.

Feedback

**What Next**

Please check man page of set_pg_via_master_rule command to understand the usage of snapping reference point. Specify it with allow_multiple option but not offset, offset_start, or via site ratio.

---

## PGR-054

(error) Failed to parse special pattern for %s.

**Description**

The error is issued when user specified special pattern string cannot be correctly parsed.

**What Next**

Please check man page of create_pg_special_pattern to correct the special pattern specification.

---

## PGR-055

(error) Fail to set pg via master rule: %s.

**Description**

The error is issued when user cannot create via master rule correctly. It can be incorrect input or lack of technology data.

**What Next**

Please check man page of set_pg_via_master_rule to correct the issue.

---

## PGR-056

(error) Cannot find via master rule %s in the design.

**Description**

The error is issued when pg router cannot find the specified pg via master rule for the current block.

**What Next**

Please define via master rule using set_pg_via_master_rule.

Feedback

## PGR-057

(error) Number of values specified for %s does not match number of values specified for -nets.

**Description**

The error is issued when the number of values specified for -start option is not equal to the number of nets specified by -nets option, or the number of values specified for -pitch option is more than the number of nets specified by -nets option.

**What Next**

Please specify equal number of arguments for -start and -nets, and no more arguments for -pitch than -nets.

## PGR-058

(error) Invalid pitch value for create_pg_stapling_vias.

**Description**

The error is issued when pitch value is not specified or cannot be derived automatically when the specified contact code is custom via def. Pitch value can be either specified using -pitch option, or obtained from specified pg via master rule or derived for a simple via def.

**What Next**

Please specify valid -pitch value for custom via def.

## PGR-059

(error) More than one %s are specified together with -locations for create_pg_vias .

**Description**

The error is issued when more than one net/from_layers/to_layers are specified together with -locations for create_pg_vias. Only one net, one from layer and one to layer is allowed for coordinate based via creation.

**What Next**

Please specify one value for -net/-from_layers/-to_layers.

## PGR-060

(error) One of -pins and -cells must be specified.

Feedback

**Description**

The error is issued when neither -pins nor -cells are specified for connect_pg_via_ladders command.

**What Next**

Please specify either -pins or -cells and -nets.

---

## PGR-061

(error) PG router doesn't support unbound %s.

**Description**

The error is issued when there is unbound data in design. For example, shape, via def, tech layer and etc.

**What Next**

Please check if any required information is removed or modified.

---

## PGR-062

(error) There are 2nd pins of %s in the specified region. Please set associate pattern to the net: %s

**Description**

The error happens when the 2nd pins should have associated pattern with net for wire creation.

**What Next**

Please set required pattern information for -pattern_net_list.

---

## PGR-063

(error) There is no PG region can be generated from create_secondary_pg_trunk.

**Description**

It can be no PG wire in user specified -horizontal_layer or -vertical_layer.

**What Next**

Check the command man page and specify the required options.

Feedback

## PGR-064

(error) Specified custom via def is not supported.

**Description**

The error is issued when user specified custom via def contains more than one cut layers.

**What Next**

Please specify via def which contains only upper/cut/lower layers.

## PGR-065

(error) There are no 2nd PG pins in the design.

**Description**

The error happens when there are no dual rail cells in the design or no backup type PG pin in dual rail cell which means there is no 2nd pins for connecting.

**What Next**

Please check the design.

## PGR-066

(error) The specified pattern %s is undefined.

**Description**

The specified PG pattern is not defined before the command.

**What Next**

Please check the specified pattern name is correct or create pg pattern first.

## PGR-067

(error) cannot create the specified directory.

**Description**

The error is issued when training data in the specified directory do not exist.

**What Next**

Please specify correct directory where the ML training data are located.

Feedback

## PGR-068

(error) %d output directories are specified. Only one output directory can be specified.

**Description**

Only one output data is allowed to store created training data.

**What Next**

Please specify only one output directory.

## PGR-069

(error) Invalid wire width %s. It is smaller than the minimum layer width %s.

**Description**

This message is issued when user specified wire width is smaller than the minimum width in the tech file for that layer.

**What Next**

Please specify width equal or greater than minimum layer width.

## PGR-070

(error) Invalid wire width %s. It is greater than the maximum layer width %s.

**Description**

This message is issued when user specified wire width is greater than the maximum width in the tech file for that layer.

**What Next**

Please specify width equal or smaller than maximum layer width.

## PGR-071

(error) %s is not defined in config %s.

**Description**

The required information is not defined.

**What Next**

Please define the required information.

Feedback

## PGR-072

(error) %s is not defined in layer %s of config %s.

**Description**

The required information is not defined.

**What Next**

Please define the required information.

## PGR-073

(error) incorrect file.

**Description**

The required information is not defined.

**What Next**

Please define the required information.

## PGR-074

(error) incorrect layer name %s is specified in config %s.

**Description**

The specified layer name does not existed.

**What Next**

Please specify correct layer name.

## PGR-075

(error) category %s in config %s is not supported.

**Description**

The specified category is not supported.

**What Next**

Please specify correct category.

Feedback

## PGR-076

(error) %s is not defined in via rule %s.

**Description**

The required information is not defined.

**What Next**

Please define the required information.

## PGR-077

(error) category %s in viaRule %s is not supported.

**Description**

The specified category is not supported.

**What Next**

Please specify correct category.

## PGR-078

(error) no viaRule defined in ViaConfig %s.

**Description**

viaRule is required in ViaConfig. The specified category is not supported.

**What Next**

Please define viaRule in ViaConfig.

## PGR-079

(error) pintype %s in config %s is not supported.

**Description**

The specified pintype is not supported.

**What Next**

Please specify correct pintype: ring_pin, long_pin, or scattered_pin.

Feedback

## PGR-080

(error) %s is not defined in layer %s of strapConfig %s.

**Description**

The required information is not defined.

**What Next**

Please define the required information.

## PGR-081

(error) Failed to get pin bbox from net %s for rail creation.

**Description**

Rail creation needs to get width information from standard cell PG pin. The error happens when 1. There are siterows with different heights in the creation region 2. There is no standard cell in the creation region

**What Next**

Please check if rail creation region is well-defined for rail creation.

## PGR-082

(error) Failed to perform undo since %s

**Description**

Compile_pg -undo fails so no shapes will be remove in the command. It fails when shapes are moved or deleted. It may also reach maximum undo step.

**What Next**

Please check log if there is any command cause undo fails.

## PGR-083

(error) Please specify only one of the following options: -core, -design_boundary, -voltage_area, -polygon, -macros, -blocks and -pg_regions.

**Description**

set_pg_strategy only accepts one option from the list above. User cannot combine it in one strategy.

Feedback

**What Next**

Please check if the region option is well-specified.

---

## PGR-084

(error) Distance required by number and pitch setting in %s cell alignment pattern exceeds pin width.

**Description**

The error is issued when distance required by number and pitch setting in switch/physical cell alignment pattern exceeds pin width.

**What Next**

Please check that the distance required by number and pitch setting in switch/physical cell alignment pattern must not exceeds pin width.

---

## PGR-085

(error) No legal mask can be assigned for PG via cut %s.

**Description**

The error is issued when no legal mask can be assigned for given PG via cut.

**What Next**

Please check if the spacing between via cut is correct.

---

## PGR-086

(error) Can not find valid site row for power switch insertion.

**Description**

The error is issued when tool can not find valid site row for power switch insertion.

**What Next**

Please check if site row is created correctly.

---

## PGR-087

(error) Your specified pitch %s on layer %s in pattern %s is too small to accommodate all straps.

Feedback

**Description**

The tool issues this error when the value specified in pitch in a pattern is too small to fit all straps. The minimum pitch is based on the widths specified in the pattern and the required spacing between wires.

**What Next**

Increase the value of pitch or decrease the value of widths specified in the pattern.

---

## PGR-088

(error) plan.pgroute.support_polygon_wires only support adjacent layers creation

**Description**

The error is issued when the app_option is set to true and user hasn't specified adjacent layers in -from_layers and -to_layers in create_pg_vias. Non-adjacent layers is not supported. The application option is specialized for complex intersection handling.

**What Next**

Please check if the setting is intended.

---

## PGR-089

(info) Option "%s" is not set, automatically load it %s for backside ring creation.

**Description**

When set_pg_strategy -backside, and some of the options are not set, then the missing values are automatically set.

Including:

Set command create_pg_ring_pattern option "-horizontal_layer" and "-vertical_layer" as the value of application option "plan.backside.ring_layer".

Set command create_pg_ring_pattern option "-horizontal_width" and "-vertical_width" as the value of application option "plan.backside.ring_width".

Set command set_pg_strategy option "offset" as minus half the value of application option "plan.backside.ring_width".

Set command set_pg_strategy option "side_offset: {side }" as "T B".

Set command set_pg_strategy option "side_offset: {offset }" as the value of application option "plan.backside.alignmnet".

Feedback

### What Next

Check if the automatically set value meet the requirement. If not, input the value from the related commands.

---

## PGR-090

(warn) In region %s, failed to smoothen the merged boundary of macros and blockages for ring creation. Use the unsmooth boundary to create ring.

### Description

For ring pattern set with {merge: placement_blockage}, the macros and placement blockages are merged to form a region to create ring. Small slivers and holes in the merged boundary caused by unaligned placement blockage is smoothen before ring creation. The smoothen process might failed when the unaligned condition is serious. Then the ring is created on the unsmooth boundary.

### What Next

Aligned the placement blockage to the macros manually using the GUI tools and create the PG ring again.

---

## PGR-091

(warn) %s

### Description

This message reports a failed attempt to create wire or via using the compile_pg command. The reason for failure can be either wrong parameters in the compile_pg command or DRC violations. Perform the following steps to ascertain the actual cause of failure. 1) Check for invalid parameter in the compile_pg command using <-ignore_drc> option. If the compile_pg command fails to create wires and/or vias with the <-ignore_drc> option, it means that some invalid parameter has been provided to the compile_pg command. Use the details provided in the compile_pg command page to know more about valid parameters that you can use with this command. Try running the command again with correct parameters. If the command succeeds in creating wires and vias, proceed to the next step. 2) Run the compile_pg command without the <-ignore_drc> option but add the <-show_phantom> option. If the command fails to create some wires and vias then some DRC violations have occurred. To view the details of the DRC violations, use View->Error Browser in the tool GUI.

### What Next

On finding any DRC rule violation you can look at the relevant DRC manuals to rectify any related issues in the design. The DRC manuals can be found in Solvnetplus.

Feedback

## PGR-501

(warning) Some defined variables in -parameters option are not used by other options.

**Description**

Some defined variables are not used.

**What Next**

Remove the unused variables from -parameters option.

## PGR-502

(warning) Option %s is incompatible with other option %s. The first option is therefore ignored.

**Description**

Ignore some incompatible option.

**What Next**

Remove the incompatible option when specifying the command.

## PGR-503

(warning) For option %s, the specified value %s is invalid and ignored.

**Description**

This option value is invalid.

**What Next**

Check the command man page and specify the correct value.

## PGR-504

(warning) Via master rule %s has been removed and will be ignored.

**Description**

This via master rule is removed and will be ignored in the corresponding strategy via rules or patterns.

**What Next**

Redefine the removed via master rule using set_pg_via_master_rule command.

Feedback

## PGR-505

(warning) Strategy %s in strategy via rule is undefined and the rule will be ignored.

**Description**

PG route adopts a methodology called lazy evaluation, which allow users to specify constraints like set_pg_strategy even though certain contents (like strategy) does not exist. The corresponding strategy via rule will be ignored by compile_pg if the missing strategy is not specified.

**What Next**

Define the corresponding PG strategy before invoking compile_pg.

## PGR-506

(warning) %s is undefined.

**Description**

The keyword is not defined in the design.

**What Next**

Please define the required info.

## PGR-507

(warning) The specified net %s is not P/G net. Create_pg_vias may not generate vias successfully.

**Description**

App_option plan.pgroute.honor_signal_route_drc default vaule is false. Create_pg_vias will not check signal route drc when the app option is off.

**What Next**

Set the app option plan.pgroute.honor_signal_route_drc true or check if the net name is correct.

## PGR-508

(warning) Metal or via layer is missing in tech file.

Feedback

**Description**

Metal or via layer is missing, PG creation command will error out.

**What Next**

Check if there is any metal or via layer is missing in tech file.

---

## PGR-509

(warning) Cannot find default via between layer %s and layer %s. %s

**Description**

Default via def cannot be found in the tech file.

**What Next**

Check if the current tech file is correct for via creation. It may need to be replaced.

---

## PGR-510

(warning) The %s is deprecated. It will be retired on %s release.

**Description**

The feature is going to be retired soon.

**What Next**

Check if the feature is used intended.

---

## PGR-599

(warning) %s

**Description**

General warning message.

**What Next**

Follow the warning message instruction.

---

## PGR-600

(error) %s

Feedback

**Description**

Error in command line options for *check_pg_drc*.

**What Next**

Correct the command line options, please consult the manpage of check_pg_drc.

---

## PGR-601

(warning) %s

**Description**

Warning in command line options for *check_pg_drc*.

**What Next**

Consult the manpage of check_pg_drc for correcting the warning issues.

---

## PGR-610

(error) Wrong Contact Code Violation for rule %s in the design. Required Contact code %d, and actual is %d. So we are dropping these via. Change Via Def Name %s and add correct via defination.

**Description**

The error is issued when pg router cannot find the specified contact code in the tech file or the pg via rule defined for the current block.

**What Next**

Please specify correct contact code or via rule.

---

## PGR-999

(error) %s

**Description**

General error.

**What Next**

Follow the error message instruction.

Feedback

# PI

## PI-001

(error) An error prevented PI optimization. %s

**Description**

Something in the design or the configuration prevented the successful run of a Power Integrity optimization.

**What Next**

It may be necessary to alter the design, configuration or application options to complete the information required for the PI optimization.

# PLACE

## PLACE-001

(warning) Infeasible tiling in MVDD placer.

**Description**

Infeasible tiling in MVDD placer. Please check your bound creation.

## PLACE-002

(warning) Infeasible tiling in MVDD placer: %s

**Description**

Infeasible tiling in MVDD placer. Please check your bound creation.

## PLACE-003

(error) Error in placement tiling. Not enough room available for the following cells: %s

**Description**

Infeasible tiling in placer.

**What Next**

Fix the amount of space available for these cells and run placement again.

Feedback

## PLACE-004

(error) Fail in placement: %s

**Description**

Placement failure

**What Next**

Fix the error and run placement again.

## PLACE-005

(warning) Fail in placement: %s

**Description**

Fixed area greater or equal than chip area. Total moveable area is zero.

**What Next**

Fix the error and run placement again.

## PLACE-006

%s

**Description**

Placement message

**What Next**

Informational message - may require analysis to see if placement needs to be run again.

## PLACE-007

(Error) Library cell %s/%s has Vth implant on layer %d which does not correspond to any Vth implant layer defined in the technology file. Advanced rule checking will not be performed for this library cell.

**Description**

This error message indicates an inconsistency between reference library technology file and design library technology file. This error will likely result in incorrect placement legalization and legality checking results.

Feedback

**What Next**

Revisit the design preparation process.

---

## PLACE-008

(Error) Library cell %s/%s has TrimPoly implant on layer %d which does not correspond to any TrimPoly implant layer defined in the technology file. Advanced rule checking will not be performed for this library cell.

**Description**

This error message indicates an inconsistency between reference library technology file and design library technology file. This error will likely result in incorrect placement legalization and legality checking results.

**What Next**

Revisit the design preparation process.

---

## PLACE-009

(Warning) User set forbidden pair "%d - %d" is not legal. It will be ignored.

**Description**

This error message indicates that illegal values for "forbidden pair" have been set by the user with the app_option "place.legalize.vertical_abutment_forbidden_pairs". Values of forbidden indexes must be greater than 0. As a result, the given pair will be skipped. This user error may result in incorrect placement legalization and legality checking results.

**What Next**

Revisit the design preparation process.

---

## PLACE-010

(Warning) %d cells are not placed.

**Description**

This error message indicates that legalization engine found some unplaced cells in the design. It is expected that user run coarse placement prior to legalization. A design may contain a small number of unplaced cells, but a large number usually indicates a flow/script problem.

Feedback

**What Next**

Revisit the flow script.

---

## PLACE-011

(Warning) Overlapping site rows found in block %s.

**Description**

This warning message indicates that legalization engine found overlapping site rows of different siteDefs in the placement block. This design style is not completely supported. This error may result in failure of legalization and/or legality checking.

**What Next**

Revisit the design floorplan.

---

## PLACE-012

(Warning) Reference cell %s has site dimensions (%d, %d) that do not equal top block site dimensions (%d, %d).

**Description**

This error message indicates that legalization engine found the given reference cell is associated with site def dimensions that are different then those used by the top design. This usually indicates a library data preparation problem. The reference library was likely created with a different version of the technology file than the top design.

**What Next**

Revisit the design and refererence library preparation procedures.

---

## PLACE-013

(Warning) Invalid capacitance value detected. It will be ignored.

**Description**

This error message indicates that an invalid capacitance value was encountered. The value will be changed to zero.

**What Next**

This is an internal error. Please report to Synopsys.

Feedback

## PLACE-014

(Warning) Reference cell %s has width %d that is not multiple of its site def %d.

**Description**

This error message indicates that legalization engine found the given reference cell is associated with site def with dimensions that are not compatible with its size. This usually indicates a library data preparation problem. The reference library was likely created with a different version of the technology file than the top design.

**What Next**

Revisit the design and refererence library preparation procedures.

## PLACE-015

(Warning) Circuit has no valid timing endpoints

**Description**

This error message indicates that the circuit has no valid timing endpoints. Placement will continue.

**What Next**

Revisit the timing information

## PLACE-016

(Error) Placer has detected a timing graph with too many logic levels. This may indicate a loop in the timing graph.

**Description**

This error message indicates that the timnig graph either has too many logic levels or has a loop. Placement cannot continue.

**What Next**

Revisit the timing information

## PLACE-017

(error) Error in placement constraints.

Feedback

**Description**

Placement bound constraints are infeasible.

**What Next**

Fix the placement bound constraints and run placement again. To check the constraints before running placement, use check_placement_constraints.

---

## PLACE-018

(error) No valid core area defined.

**Description**

Coarse placement cannot be run without a valid core area..

**What Next**

Use *initialize_floorplan* to define your core area.

---

## PLACE-019

(error) The region defined by the constraints %s has zero area and %ld cells.

**Description**

Infeasible constraints in coarse placement.

**What Next**

Fix the constraints and run the placement again.

---

## PLACE-020

(error) The region defined by the constraints %s is not feasible.

**Description**

Infeasible constraints in coarse placement.

**What Next**

Fix the constraints and run the placement again.

---

## PLACE-021

(Error) Library cell %s/%s has TrimPoly implant width which is wider than cell width. Advanced rule checking will not be performed for this library cell.

Feedback

**Description**

This error message indicates an error in library preparation for this cell. This error will likely result in incorrect placement legalization and legality checking results if this cell is instantiated.

**What Next**

Revisit the library/design preparation process.

---

## PLACE-022

(Info) %d placement spacing labels and %d placement spacing rules are stored in design NDM.

**Description**

This message notifies user that user placement spacing rules and labels are stored in the design NDM and will be loaded the next time the design is opened.

---

## PLACE-023

(Error) Layer %s has non-positive RC values.

**Description**

This warning message indicates that the RC values coming from extraction are either negative, zero, or infinity for the given layer. RC values could be zero for superconducting layers. If certain options, such as buffer aware placement are enabled, this will cause the placer to exit.

**What Next**

Verify that the RC values in the TLUplus for the given layer are correct. Check if this layer should be specified as an ignored layer.

---

## PLACE-024

(Error) Placement of Power/Ground rails conflicts with NDM row definition.

**Description**

This error message indicates that placement of Power/Ground rails is such that no legal placement exists for any cell. Legalization is aborted. This error indicates a problem with the floorplan, either in the PG grid generation, or in the site row definition.

Feedback

### What Next

Please review your floorplanning steps. The check that produces this error may be switched off with application option "place.legalize.enable_pg_rail_sanity_check" set to false. Be aware that doing so, will result in a placement that cannot be legally taped out.

---

## PLACE-025

(Info) Instance %s dimensions are not a multiple of unit site def.

### Description

This message indicates given standard cell instance dimensions are not a multiple of the unit site def. This notice may be safely ignored if the cell's intended row is not a unit site row.

### What Next

Please review your floorplanning and library prep steps, if necessary.

---

## PLACE-026

(error) There are unfixed macros in design. Use create_placement -floorplan to place macros or fix all macros with the app option place.coarse.fix_hard_macros.

### Description

You are getting this message because the current design has unfixed macros. If you want to place these macros then you need to use the command create_placement -floorplan which will invoke the macro placement engine. If you already have a macro placement and want to fix all the macros then you can set the app option place.coarse.fix_hard_macros to true.

### What Next

Based on what you want to do action is described above.

---

## PLACE-027

(info) Automatic density control has selected the following settings: max_density %.2f, congestion_driven_max_util %.2f.

### Description

This message indicates that automatic density control is enabled for coarse placement, and reports the selected settings.

Feedback

## PLACE-028

(Warning) Characterization of reference cell %s VT ignored.

### Description

This message indicates that issues were encountered in characterization of the given reference cell's VT parameters. This may lead to incorrect placement of instances of the given reference cell.

### What Next

Please review your library prep steps, if necessary.

## PLACE-029

(Warning) There is insufficient area available to achieve the target routing density of %s %s. Using %s of %s instead.

### Description

This message indicates that the congestion reduction step of coarse placement cannot reduce the cell density enough in congested areas to achieve the target routing density, because there is not enough area available for that. This can happen because there is too much area that is congested, or because the cell density in uncongested areas cannot be increased further. The message also indicates which routing density value is used by the congestion reduction step instead.

### What Next

Check the value of application option "place.coarse.congestion_driven_max_util" to make sure that it is not set too low (in particular a value below 0.8). Review the global route congestion map and the cell density map, to determine whether the main issue is the amount of congestion or the cell density in the uncongested areas. In the latter case, check whether the cell area is reduced by the final_opto step of place_opt. If that is the case, running "place_opt -from final_place" or "refine_opt" after "place_opt" may help with reducing congestion further.

## PLACE-030

(info) Reducing cell density for %s%% of the movable cells to alleviate congestion. This changes the average cell density in non-congested areas from %s to %s.

### Description

This message reports several statistics for the congestion reduction step of coarse placement: It reports the percentage of movable cells that are in areas where congestion

Feedback

removal is reducing the cell density. It also reports how that increases cell density for cells in non-congested areas.

## PLACE-032

(Error) Unequal vertical abutment pattern sizes found for top (=%d bits) and bottom (=%ld bits) of reference %s. Check database consistency.

**Description**

This message notifies user that the vertical abutment pattern on the for top and bottom edges are of unequal length.

## PLACE-033

(Error) Reference width (=%u bits) is inconsistent with vertical abutment pattern size (=%u bits) for reference %s. Check database inconsistency.

**Description**

This message notifies user that the length of vertical abutment pattern set on a reference is not equal to the width of that reference.

## PLACE-034

(Error) PNET_AWARE: Unable to find core site. The app_options

**Description**

The pnet aware functionality will be disabled if there are no core sites.

**What Next**

Make sure the design has core sites.

## PLACE-035

(Info) PNET_AWARE: %d power stripes treated as placement blockages.

**Description**

Report how many power stripes are treated as placement blockages.

**What Next**

Please review the number of power stripes treated as placement blockage is reasonable.

Feedback

## PLACE-036

(Warning) The user setting for the target routing density of %.2f is outside the range from 0.8 to 1.0. Using a value of %.2f instead.

**Description**

To avoid excessive or insufficient congestion reduction, the coarse placer only supports target routing density values in the range from 0.8 to 1.0. If the user setting is outside the range, the nearest value within the range is used instead.

**What Next**

Update the value of application option "place.coarse.target_routing_density" or restore the default value.

## PLACE-037

(Info) PNET_AWARE: The app_option

**Description**

Use setting of the minimal width.

**What Next**

Please review the setting is intended or not.

## PLACE-038

(Info) PNET_AWARE: The app_option

**Description**

Print out the internally computed the default minimal width.

**What Next**

Please review the setting is intended or not.

## PLACE-039

(Info) PNET_AWARE: The minimal height for horizontal power stripes to be treated as a placement blockage is set to %lg microns (half of row-height). (This does not affect vertical power stripes).

Feedback

**Description**

Print out the internally decided the minimal height for horizonal power stripes.

**What Next**

Please review the setting is intended or not.

---

## PLACE-040

(Error) PNET_AWARE: Unable to find layer

**Description**

The layer name specified in app_option

**What Next**

Please check the layer name.

---

## PLACE-041

(Error) The option %s is obsolete. See the man page for %s for migration assistance.

**Description**

This message notifies that the option %s to the command %s is obsolete.

---

## PLACE-042

(error) No valid scan def found.

**Description**

This error message indicates that scan def has not been loaded for this design.

**What Next**

To run placement without scan def please use: set_app_options -name place.coarse.continue_on_missing_scandef -value true

---

## PLACE-043

(Warning) Fixed cell %s (%s) has illegal site_name

Feedback

**Description**

This warning message indicates that a fixed cell is associated with an invalid site def. It will be trated as a blockage. This may result in placement violations if there are any placement rules associated with the cell. This message is often expected behavior for certain flows.

**What Next**

Verify that this is expected behavior.

---

## PLACE-046

(warning) Soft move bound %s is present inside exclusive move bound %s.

**Description**

The reported soft move bound does not make sense because it is covered by an exclusive move bound. Please check your bound creation.

---

## PLACE-047

(warning) Overlap between move bounds %s and %s.

**Description**

The reported overlapping move bounds may fail the create_placement because one or both of them are exclusive. Please check your bound creation.

---

## PLACE-048

(warning) Parent move bound %s is not covering child move bound %s.

**Description**

The reported parent move bound is expected to cover the child move bound because they are hierarchially related(ancestor-descendant relationship). Please check bound creation.

---

## PLACE-049

(error) Exclusive move bound %s covers hard/exclusive move bound %s.

**Description**

The reported hard or exclusive move bound must not be covered by the exclusive move bound because during placement it will have zero area for cells present inside it. Infeasible move bounds definitions.

Feedback

**What Next**

Fix the constraints and check placement again.

---

## PLACE-050

(error) Hierarchically related disjoint Hard/Exclusive move bounds %s and %s.

**Description**

The ancestor move bound should cover the descendant move bound because they are both of type hard or exclusive.

**What Next**

Fix the constraints and check placement again.

---

## PLACE-051

(warning) Unplaced cell %s is not placed %s

**Description**

Coarse placement will not place the cell which has CTS size_only flag or is the isolation cell. If such a cell is unplaced, it will keep the unplaced physical status after placement, causing ZRT-064 error in router.

**What Next**

Review each unplaced cell and mark it as placed (or fixed if the cell is not allowed to be moved in the following optimization steps) if the cell location is still valid. Otherwise go back to steps in which such a cell is placed.

---

## PLACE-052

(error) The app options %s %s are not compatible.

**Description**

The two given app options are incompatible and cannot be used together.

**What Next**

Decide which of the 2 app options you want to use and use only that one.

Feedback

## PLACE-053

(error) IR drop aware placement disabled. Unable to find a scenario with power analysis enabled.

### Description

To run IR drop aware placement there needs to be at least 1 scenario with power analysis enabled. This is needed to compute the power usage of each cell.

### What Next

Check your scenarios using report_scenarios and turn on leakage power and/or dynamic power analysis using set_scenario_status.

## PLACE-054

(error) Power density aware placement disabled. Unable to determine a maximum power density target.

### Description

To run power density aware placement there needs to be a maximum power density target. By default the tool will attempt to calculate a maximum power density target which is 1.4 times the average power density. However, this message indicates that the calculation has failed some some reason.

### What Next

Set a maximum power density manually by using the app option place.coarse.max_power_per_area. One way to get a reasonable value is to look at the power density map of the placed design in the GUI. Set a value somewhat less than the maximum power density value shown.

## PLACE-055

### NAM5

*PLACE-055* (info) %s aware placement disabled. Reason: Unable to load %s data.

### Description

To run %s aware placement the %s data must be loaded into the design before placement is run.

### What Next

Load the data.

Feedback

## PLACE-056

(info) Because automatic density control is enabled, congestion analysis effort high is used and the value of application option place.coarse.congestion_analysis_effort is ignored.

### Description

This message indicates that effort high is used for congestion analysis, instead of the effort level selected by application option place.coarse.congestion_analysis_effort. The reason for using effort high is that automatic density control is enabled, which requires high effort congestion analysis.

## PLACE-057

(error) Cannot run timing driven placement, because zero interconnect delay is used in timing analysis.

### Description

Timing analysis uses zero interconnect delay for the current design, which is not supported by timing driven placement.

### What Next

Change the design setup such that the interconnect delays are not zero, or run coarse placement with timing driven placement disabled.

## PLACE-058

### NAM5

*PLACE-058* (error) Effective resistance placement disabled. Unable to load effective resistance data.

### Description

To run effective resistance placement the effective resistance data must be loaded into the design before placement is run.

### What Next

Load the effective resistance data.

## PLACE-060

(warning) MVDD placer. Tile %s, does not have enough area.

Feedback

**Description**

The Voltage Area doesn't have enough area to accommodate all the cells.

---

## PLACE-061

(warning) Ignoring %s bound on cell %s due to placement obstructions.

**Description**

The movebound on the reported cell is completely blocked by placement obstructions. The bound will be ignored.

**What Next**

Check the placement obstructions and bounds and modify the placement constraints to resolve the conflict.

---

## PLACE-062

(warning) Cell %s cannot satisfy placement constraints.

**Description**

This error message occurs when a cell cannot be legally placed anywhere in the design. Commonly, this is caused by conflicting hard move bounds and blockages.

**What Next**

Analyze the placement constraints of the indicated cell to identify sets of conflicting constraints such as hard move bounds and placement blockages. Experiment with deleting and adding subsets of these bounds and blockages to find a set that does not overconstrain the cell.

---

## PLACE-063

(warning) Placement timed out on user-defined macros.

**Description**

The number of macros increases in each pass, increasing the runtime for overlap removal for a plane and hence placement.

**What Next**

Reduce the number of macros and run again.

Feedback

## PLACE-064

(warning) %s cells have hard move bounds.

**Description**

Specified cells are forced to be places inside the hard MBs. With increase in number of hars mbs, optimization of the circuit degrades due to increase in congestion and poor quality of placement.

**What Next**

Reduce the number of move bounds and run placement again.

## PLACE-065

(warning) Infeasibity during density aware blockage shoving due to movebounds.

**Description**

Presence of hard/exclusive MBs on blockages would force the cells inside them to be placed over blockages. Shoving won't be possible for such cells.

**What Next**

Re-check the movebounds definitions and run placement again.

## PLACE-066

(warning) Constraints for macro cells present on logic %s are unsatisfied.

**Description**

Overlap removal is given up on logic containing macro cells having auto keepouts around them.

**What Next**

Check placement constraints

## PLACE-067

(error) Placer detected conflicting constraints on the following cells: %s.

**Description**

Cells violating macro constraints are present in the logic. Placement/Parameter Calculation will be aborted.

Feedback

**What Next**

Recheck placement constraints and run again.

---

## PLACE-068

(warning) Placer detected conflicting constraints on the following bounds : %s.

**Description**

Bounds conflicting placement constraints found. Differernt bounds might be containing same cells, or they might be overlapping. Either such groupbounds are removed and conflicting quadratic groupbounds may be converted to linear bounds, or placement will be aborted.

**What Next**

Recheck placement constraints and run again.

---

## PLACE-069

(warning) Conflicting constraints detected on bounds. Too many quadratic groupbounds present.

**Description**

Conflicting bounds found. All groupbounds with more than two cells are converted to linear.

**What Next**

Recheck constraints and run placement again.

---

## PLACE-070

(warning) Floating point stack is corrupt.

**Description**

Errors have occurred in floating point stack calculation.

**What Next**

Clean up the stack and continue.

Feedback

## PLACE-071

(info) The cpu mem report : %s CPU: %s s (%s hr) ELAPSE: %6s s (%s hr) MEM-PEAK: %s Mb.

**Description**

Runtime and memory information.

## PLACE-072

(info) The final rpsa process memory report : %s CPU: %s s (%s hr) ELAPSE: %6s s (%s hr) MEM-PEAK: %s Mb.

**Description**

The final runtime and memory information.

## PLACE-073

(warning) Highest utilization value of a plane for the circuit is %s.

**Description**

Plane/s in the circuit is/are over-utilized, resulting in increase in congestion and insufficient placement area.

**What Next**

Check design constraints and run placement again.

## PLACE-074

(warning) Utilization of %s is %s.

**Description**

Plane/chip over-utilized. Insufficient available area for movable cells. Load file

**What Next**

Recheck

## PLACE-075

(warning) The circuit has a large number of planes present.

Feedback

**Description**

Circuit has a total of %s exclusive bounds, plan groups, voltage areas, and/or category blockages. Large coarse placement runtime / memory usage expected.

**What Next**

Reduce the number of planes and run again.

---

## PLACE-076

(error) Placer has detected a timing graph in excess of %s logic levels in depth. Aborting.

**Description**

Excess graph depth while seting up the graph after the nodes etc have been added.

---

## PLACE-077

(warning) Cache circuit count is inconsistent.

**Description**

Overlap area cache memory has been modified, but the memory still contains the old data.

**What Next**

Clean up the memory and run placement again.

---

## PLACE-078

(info) Constraint %s has utilization %s. Adjusting max density from %s to %s for this area since soft_max_density is enabled.

**Description**

This message reports the change in max density when the soft_max_density app option is enabled.

---

## PLACE-079

(info) Placer used soft congestion map.

**Description**

This message indicates that the placer is driven by ZGR's soft congestion map.

Feedback

## PLACE-080

(warning) Disabling clock gate latency aware placement because trial clock tree construction was enabled.

### Description

Latency estimation during coarse placement only works for non-propagated clocks. This is incompatible with trial tree construction and therefor clock gate latency aware placement was disabled.

Clock gate latency aware placement is enabled by setting place.coarse.clock_gate_latency_aware to "true", or by setting place.coarse.clock_gate_latency_aware to "auto" and enabling clock gate latency estimation at the flow level by seting one of app option place_opt|refine_opt|compile.flow.estimate_clock_gate_latency to "true".

### What Next

Decide whether you want trial clock tree construction OR clock gate latency aware placement for latency estimation. If the latter, switch off trial_clock_tree and/or optimize_icgs. Note that optimize_icgs implicitly enables trial clock tree construction.

### See Also

- place.coarse.clock_gate_latency_aware
- place_opt.flow.trial_clock_tree
- place_opt.flow.optimize_icgs
- place_opt.flow.estimate_clock_gate_latency

## PLACE-081

(info) Removed %d clock network latencies at clock gate inputs in scenario '%s'. New clock gate input latencies will be estimated.

### Description

This message indicates that user- or tool-specified latencies at clock gate inputs will be ignored (removed) and that new latencies will be applied based on internal clock tree latency estimation. This will happen if app option 'place.coarse.clock_gate_latency_aware' is set to 'true'. The message indicates how many latency settings are removed/ignored. Removing latencies at clock gate inputs may imply that latency settings have to be moved (copied) to the clock gate output to preserve the ideal latency in the fanout of the clock gate. If this is required, the tool will automatically take care of that.

Feedback

Note that a clock gate in this context not only refers to integrated clock gates (ICGs) but also to discrete clock gates and clock dividers. Basically it refers to all clock cell that have a timing-checked data input.

**See Also**

- place.coarse.clock_gate_latency_aware

---

## PLACE-082

(info) Created %d clock network latencies at the output of clock gates in scenario '%s'.

**Description**

This message indicates that the tool created SDC set_clock_latency at the output of clock gates. This is to prepare for latency estimation at the input of the clock gates as imposed by app option 'place.coarse.clock_gate_latency_aware'. In order to avoid that estimated latencies at the input of a clock gate impact the ideal clock network latency in the fanout of the clock gate, the ideal clock network latency at the input is set to the output of the clock gate. This will only happen if the output of the clock gate does not have an existing SDC clock network latency specified.

Note that a clock gate in this context not only refers to integrated clock gates (ICGs) but also to discrete clock gates and clock dividers. Basically it refers to all clock cell that have a timing-checked data input.

**See Also**

- place.coarse.clock_gate_latency_aware

---

## PLACE-083

(warning) Placement continues with over utilized regions in the design.

**Description**

Placement is forced to continue eventhough there are overutilized exclusive movebounds or voltage areas present in design or whole design is overutilized.

**What Next**

Please check the PLACE-074 warnings printed before this warning. Increase the available region or modify blockages to remove the PLACE-074 warnings. These warnings need to be resolve before running normal place_opt and clock_opt flows. Please be aware that this is likely to cause high density areas.

Feedback

## PLACE-084

(warning) Overutilized regions for %s are modified by %s.

### Description

Insufficient available area for movable cells. Early data flow modifies blockages and fix cells to decrease utilization.

### What Next

Increase the available region or modify blockages to remove the warning. These warnings need to be resolve before running normal place_opt and clock_opt flows.

## PLACE-085

(info) Clock gate latency aware placement is enabled.

### Description

Informs that timing-driven placement will be performed with consideration of (estimated) clock network timing. This will have an impact on the timing budget of clock gate enable paths. At the end of placement, the estimated clock network timing will be annotated as set_clock_latency on the clock gates.

### See Also

- place.coarse.clock_gate_latency_aware
- place_opt.flow.estimate_clock_gate_latency

## PLACE-086

(info) Auto congestion effort reduced congestion effort from high to medium.

### Description

Informs that enable_auto_congestion_effort reduced congestion effort to medium because congestion is low.

### See Also

- place_opt.place.congestion_effort

## PLACE-087

(warning) %s Exiting wide-cell placement mode.

Feedback

**Description**

This message indicates that the placement model is unable to return information about the PG pitch and width, which is used by the wide-cell plane to determine how to handle the densities of wide-cells.

**What Next**

Check if the PG structure is set up correctly. The feature place.coarse.wide_cell_use_model should be used on N7 designs with narrow M1 PG straps.

---

## PLACE-088

(warning) A manual setting has been set when a value could be auto-derived from the app_option '%s'.

**Description**

This message indicates that user has set a value which could have been auto-derived.

**What Next**

The user may want to use the auto-derived value by using the app_option mentioned in the warning message.

---

## PLACE-089

(warning) Utilization of %s is %s.

**Description**

Reports Movebound/Voltage region having utilization more than a threshold. The threshold is controlled by app option place.coarse.utilization_warning_threshold. Default is 90%. In case a movebound is intersecting with one or more exclusive movebound or voltage areas, then utilization of each overlapping region of the movebound is calculated and worst region is reported if it is more than the threshold. High utilization could cause longer runtime.

**What Next**

Resize the Movebounds or Voltage areas to decrease utilization.

---

## PLACE-090

(info) Found %d sequential and %d combinational multibit cells in the design.

Feedback

### Description

There are sequential/combinational multibit cells that could impact the quality of placement results.

### What Next

Debank multibit cells with the *split_multibit* command, and call banking after initial_opto.

---

## PLACE-091

(warning) Value 'true' for app option place.coarse.clock_gate_latency_aware is obsolete and will be interpreted as 'auto'.

### Description

The value 'true' for app option place.coarse.clock_gate_latency_aware is obsolete. The tool will interpret the value of 'true' as 'auto'.

### What Next

Modify your scripts to set the value to 'auto'. See the manpage of place.coarse.clock_gate_latency_aware for more information on app option values.

### See Also

- place.coarse.clock_gate_latency_aware

---

## PLACE-092

(warning) Skipping %s command as global_route_opt has been run.

### Description

Message indicates that the tool detected a situation where global_route_opt has already been run. Therefore, this command will be skipped. Application option place.coarse.force_rerun_after_global_route_opt can be used to override this behavior.

### What Next

No action necessary.

---

## PLACE-093

(info) Forcing rerun of %s command after global_route_opt due to application option place.coarse.force_rerun_after_global_route_opt being set.

### Description

Message indicates that the tool detected place.coarse.force_rerun_after_global_route_opt as set. In this case, placer will be forced to rerun after global_route_opt.

### What Next

No action necessary.

---

## PLACE-094

(error) Early data check failures detected. Placement failed.

### Description

During the coarse of placement, one or more early data checks failed. At least one of those failing checks was configured to be an error.

### What Next

Use 'report_early_data_checks' to inspect the failling check(s) after placement. For more detail, search in the log file for EDC-001 early data check messages. Fix the issue(s) or use 'set_early_data_check_policy' to relax the failing check(s).

---

## PLACE-095

(error) Value '%s' is not valid. Specify one of 'false' or 'auto'.

### Description

This app option only supports values 'false' and 'auto'.

### What Next

Specify one of the supported values. See manpage of place.coarse.clock_gate_latency_aware for details.

### See Also

- place.coarse.clock_gate_latency_aware

---

## PLACE-096

(Info) Technology node %s has been set by set_technology command. Technology-specific rules may not be fully supported by export_advanced_technology_rules command. Please consider using PrimeECO flow instead.

Feedback

**Description**

Message indicates that the tool detected the technology node set by user. Some advanced technology node rules are not supported. Please use PrimeECO flow instead.

**What Next**

Use PrimeECO flow.

---

## PLACE-097

(Warning) Enhanced Legalization Driven Placement technology is skipped for a hybrid row design.

**Description**

This warning message indicates that eLDP (place.legalize.enable_advanced_legalizer) is kipped for a hybrid-row design. Placement will continue.

**What Next**

Disable eLDP in the hybrid-row design flow.

---

## PLACE-098

(info) %sAuto Density Control is %s.

**Description**

This message indicates the status of automatic density control for the coarse placer.

**See Also**

- place.coarse.auto_density_control

---

## PLACE-099

(info) Scan connections are present in the design. More than %d%% of sequential cells are stitched. Valid scan def is required.

**Description**

This message indicates that scan def is required for this design as design is stitched.

**What Next**

Provide a valid scan def to run placement.

Feedback

## PLACE-100

(warning) Running placement without scan def on stitched design can result in suboptimal placement.

**Description**

This message indicates that scan def has not been loaded for this design. Placement result may be suboptimal.

**What Next**

Provide a valid scan def to avoid possible suboptimal placement.

## PLACE-101

(Warning) %s attribute not created.

**Description**

This message notifies user that the specified attribute is failed to be created in ndm and feature will be no-op.

## PLACE-102

(warning) Placer may continue without valid scan def.

**Description**

This warning message indicates that scan def has not been loaded for this design and the placer will continue.

**What Next**

A valid scan def is needed.

## PLACE-103

(error) Invalid power setup detected.

**Description**

This error message indicates that the power setup is invalid.

**What Next**

To check the UPF file and make sure it is complete.

Feedback

## PLACE-104

(warning) Placer may continue with invalid power setup.

**Description**

This waring message indicates that the placer will run with invalid power setup.

**What Next**

To check the UPF file.

## PLACE-105

(error) Invalid IR drop setup detected%s.

**Description**

This error message indicates that the IR drop setup is invalid.

**What Next**

To check the IR drop setup for print out message.

## PLACE-106

(warning) Skipping IR drop aware placement because setup is invalid%s.

**Description**

This waring message indicates that the placer will run with skipping IR drop aware placement.

**What Next**

To check the IR drop with print out message.

## PLACE-107

(warning) Placer may continue with infeasible constraints %s.

**Description**

This warning message indicates that the placer may continue with infeasible constraints.

**What Next**

To check the multi-voltage data and its placement constraints.

Feedback

## PLACE-108

(error) Placer will abort due to invalid region(s).

**Description**

This error message indicates that the physical region infered from voltage_areas and/or placement bounds have issues.

**What Next**

To check the voltage areas or bounds have invalid region.

## PLACE-109

(warning) Placer may continue with invalid region(s). Ignoring movebounds and voltage areas.

**Description**

This waring message indicates that the placer will run with ignoring movebounds and voltage areas.

**What Next**

To check the movebounds or voltage areas have valid region or not.

## PLACE-110

(error) Issues during timing analysis detected%s.

**Description**

This error message indicates that the timing analysis for timing-driven placement has issues.

**What Next**

To check the timing analysis for print out message.

## PLACE-111

(warning) Issues during timing analysis detected%s.

**Description**

This warning message indicates that the timing analysis for timing-driven placement has issues.

Feedback

**What Next**

To check the timing analysis for print out message.

---

## PLACE-112

(error) Data for timing-driven placement is missing.

**Description**

This error message indicates that the required data for timing-driven placement is not complete or not correct.

**What Next**

To check the timing data for timing-driven placement.

---

## PLACE-113

(warning) Data for timing-driven placement is missing.

**Description**

This warning message indicates that the data for the required data for timing-driven placement is complete and correct.

**What Next**

To check the required data for timing-driven placement.

---

## PLACE-114

(error) Macro %s should have physical_status fixed.

**Description**

This error message indicates that the macro is not fixed during placement.

**What Next**

To fix the macro or run with command create_placement -floorplan

---

## PLACE-115

(warning) Macro %s should have physical_status fixed.

Feedback

**Description**

This warning message indicates that the macro is not fixed during placement and it will be mark as fixed.

---

## PLACE-116

(error) Spare cell placement will abort due to infeasible spare cell constraint(s).

**Description**

This error message indicates that the spare cells constraint is infeasible.

**What Next**

To check with the constraint of the spare cells, like voltage area or valid region.

---

## PLACE-117

(warning) Spare cell placement may continue with infeasible spare cell constraint(s).

**Description**

This warning message indicates that the spare cells constraint is infeasible.

**What Next**

To r-name place.coarse.continue_on_missing_scandef -value true To check with the constraint of the spare cells, like voltage area or valid region.

---

## PLACE-118

(error) There are unfixed physical only cells in the design. Change the early data check policy unfixed_physical_cells to tolerate or fix the unfixed physical only cell.

**Description**

This error message indicates that there are unfixed physical only cells and may lead to bad runtime for placer.

**What Next**

To fix the unfixed physical only cell or change the policy to tolerate.

---

## PLACE-119

(warning) Unfixed physical cells are fixed.

Feedback

**Description**

This warning message indicates that there are unfixed physical only cells and the placer has marked it as fixed for itself.

---

## PLACE-120

(Info) %s

**Description**

This message is a generic informational message for coarse placer.

---

## PLACE-121

(Info) Adjusting congestion_driven_max_util to %s. This reserve area (%s) for %d cells.

**Description**

This message indicates that congestion_driven_max_util was adjusted to handle resereved area for certain cells. (legalization and/or timing)

---

# PNA

---

## PNA-001

(error) No power or ground net is specified

**Description**

User should specify one or multiple power or ground nets for analysis

**What Next**

Please specify one or a list of power and/or ground nets.

---

## PNA-002

(error) PNA is running out of memory.

**Description**

There is no enough memory for PNA to solve matrix during IR drop simulation.

Feedback

### What Next

Please use app_option plan.pna.ignore_objects_below_layer to ignore lower layer objects. It can help reducing memory usage but degrading accuracy of IR drop.

### See Also

- plan.pna.ignore_objects_below_layer

---

## PNA-003

(error) Net %s is not a power or ground net.

### Description

The net type of the specified net is neither Power nor Ground. PNA can only be performed on Power or Ground nets.

### What Next

Please update the net type or specify correct P/G nets.

---

## PNA-004

(error) No power pad can be found for net %s in the design.

### Description

PNA requires at least one power pad to be specified for the P/G net to be analyzed.

### What Next

Please specify power pad using *set_virtual_pad* command.

### See Also

- set_virtual_pad

---

## PNA-005

(error) Power pads are not connected to net %s.

### Description

PNA cannot identify any power pads or virtual pads that is physically connected to the %s P/G network; nor can PNA create virtual connections to the pads possibly due to route blockage.

Feedback

### What Next

Please change the location of P/G pads or virtual pads, or disable the blockage for virtual connection to pads.

---

## PNA-006

(error) No cell instance is connected to the %s power network or all the cell instances cannot reach the power pads.

### Description

PNA cannot find any cell instance that is connected to the power pads through the %s P/G network. It could be that none of the cells is logically connected to net %s; or they are floating.

### What Next

Please check the logic connection of cells in the design or check floating cells.

---

## PNA-007

(error) Power tap type is not defined.

### Description

User must specify one of the options '-virtual', '-pad_lib_cells' and '-pad_cells' to define power taps.

### What Next

Please specify the correct option.

### See Also

- analyze_power_plan

---

## PNA-008

(error) Cannot specify the option '-layer'.

### Description

The '-layer' option is valid only when virtual power taps are specified.

### What Next

Please specify '-layer' option together with '-virtual'.

Feedback

**See Also**

- analyze_power_plan

---

## PNA-009

(error) Power network analysis failed because no instance is connected to net %s.

**Description**

At least one cell in the design need to be logically connected to the P/G net being analyzed before power network analysis is performed.

**What Next**

Please check if cells' ports are connected to net %s. If not, please use *connect_pg_net* command to connect them.

**See Also**

- connect_pg_net

---

## PNA-010

(error) Cannot open file %s for read.

**Description**

Power network analysis may need the information from file %s.

**What Next**

Please check the access to file %s.

---

## PNA-011

(error) Cannot open file %s for write.

**Description**

Power network analysis may need write out information to file %s. It could be due to not sufficient disk space or the file name is incorrect.

**What Next**

Please check the access to file %s.

Feedback

## PNA-012

(error) Cannot specify the option '-coordinate'.

**Description**

The '-coordinate' option is valid only when virtual power taps are specified.

**What Next**

Please specify '-coordinate' option together with '-virtual'.

**See Also**

- analyze_power_plan

## PNA-013

(error) No horizontal or vertical straps.

**Description**

Both horizonatl and vertical straps need to exist before invoking analyze_rail.

**What Next**

Please create both horizonatl and vertical straps.

**See Also**

- analyze_power_plan

## PNA-014

(error) Fail to get parasitic model and corner to perform resistance extraction of P/G wires.

**Description**

The extraction engine of power network analysis uses parasitic model and corner to extract resistance of P/G wires and build network for IR drop analysis. The extractor failed to get defined parasitic model, and/or corner with associated parasitic model.

**What Next**

Please define parasitic model using "read_parasitic_tech", create a corner using "create_corner", and set corner & parasitic model association using "set_parasitic_parameters".

Feedback

**See Also**

- read_parasitic_tech
- create_corner
- set_parasitic_parameters

---

## PNA-015

(error) The required option -net is missing.

**Description**

For the command remove_virtual_pads, user can either use -all option to remove all existing virtual pads, or use -net and -coordinate to remove a specific virtual pad.

**What Next**

Please specify both -net and -coordinate options.

**See Also**

- remove_virtual_pads

---

## PNA-016

(error) The required option -coordinate is missing.

**Description**

For the command remove_virtual_pads, user can either use -all option to remove all existing virtual pads, or use -net and -coordinate to remove a specific virtual pad.

**What Next**

Please specify both -net and -coordinate options.

**See Also**

- remove_virtual_pads

---

## PNA-017

(error) Neither power budget nor power file is specified.

Feedback

**Description**

To perform power estimation for IR drop analysis, either -power_budget or -read_power_file needs to be specified.

**What Next**

Please specify -power_budget or -read_power_file option.

**See Also**

- analyze_power_plan

---

## PNA-018

(error) Syntax error in via rule file %s on line %d: %s.

**Description**

This message occurs when the specified via rule file has syntax errors.

**What Next**

Please refer to man page and correct the syntax errors.

**See Also**

- check_pg_missing_vias

---

## PNA-019

(error) Unrecognized power file format.

**Description**

This message occurs if the format of the specified power file is not default, or PT-PX.

**What Next**

Please apply default power file or PT-PX power report file.

---

## PNA-020

(error) Unable to find %s wire for virtual pad connection at %s on layer %s.

**Description**

This message occurs when there is no wire to be connected for the specified virtual pad.

Feedback

**What Next**

Please apply correct virtual pad.

---

## PNA-021

(error) Power analysis failed for net %s.

**Description**

This message occurs when calculating cell instance power fails.

**What Next**

Please check if power budget is set or power file exists.

---

## PNA-022

(error) fail to extract resistance for wire segment (%.3f %.3f %.3f %.3f) on layer %d .

**Description**

This message occurs when extracting resistance fails. The reason could be lack of RC information in this layer

**What Next**

Please check if tech file and corner are porperly set.

---

## PNA-023

(error) fail to extract resistance for via at (%.3f %.3f).

**Description**

This message occurs when extracting resistance fails. The reason could be lack of RC information in this layer

**What Next**

Please check if tech file and corner is porperly set.

---

## PNA-024

(error) Cannot get technology data for NDM.

**Description**

Power network analysis needs technology file information.

Feedback

**What Next**

Please check the access to technology file is correct.

---

## PNA-025

(error) Cannot find net %s in the design. Virtual pads cannot be created.

**Description**

Virtual pad can be created only if the desired net exists.

**What Next**

Please generate desired power nets and then use this command.

**See Also**

- read_virtual_pad_file

---

## PNA-026

(error) It is an unbound design.

**Description**

Power network analysis needs via definition in the design.

**What Next**

Please check if any required information is removed.

---

## PNA-027

(error) Cannot find the cut layer name in group_stapling_vias: %s

**Description**

User has to specify one correct cut layer name in group_stapling_vias.

**What Next**

Please specify one cut layer name in group_stapling_vias option. i.e. -group_stapling_vias VIA1

---

## PNA-028

(error) Neither power net nor report_track_utilization is specified.

Feedback

### Description

To perform power estimation for IR drop analysis or track utilization, either -nets or -report_track_utilization needs to be specified.

### What Next

Please specify -nets or -report_track_utilization option.

### See Also

- analyze_power_plan

---

## PNA-029

(error) The input network is too large to solve. Please use the app option: plan.pna.pna_auto_simplify_network -value true.

### Description

The input network is too large to solve. The size is over the capacity of matrix solver.

### What Next

Please try the app option plan.pna.pna_auto_simplify_network -value true.

### See Also

- analyze_power_plan

---

## PNA-030

(error) Current scenario doesn't exist. Please create a scenario.

### Description

Analyze_power_plan needs current scenario when using -analyze_power option.

### What Next

Please create a scenario.

### See Also

- analyze_power_plan
- create_scenario

Feedback

## PNA-501

(warning) unable to remove the specified virutal pad.

**Description**

The specified virutal pad to be removed does not exist.

**What Next**

Please specify correct net, coordinate, layer for the virtual pad to be removed.

## PNA-502

(warning) unable to create the specified virutal pad.

**Description**

The specified virutal pad exists in the design.

**What Next**

Please specify correct net, coordinate, layer for the virtual pad to be created.

## PNA-503

(warning) cannot find instance %s in the design.

**Description**

The instance defined in power file does not exist in the design.

**What Next**

Please specify correct instance in the power file.

## PNA-504

(warning) Specified power net %s is removed from net vector.

**Description**

The specified power net is secondary net, and will not be analyzed.

**What Next**

Please specify primary net, or ignore this warning.

Feedback

## PNA-505

(warning) Macro %s has no PG port.

**Description**

The macro has no PG port.

**What Next**

Create PG port for the macro, or ignore this warning.

## PNA-506

(warning) Power switch cell %s resistance is not valid, and is ignored.

**Description**

PNA expects a positive resistance value for a power switch cell.

**What Next**

Please specify a valid power switch cell resistance by set_app_options/set_attr command, or ignore this warning.

## PNA-507

(warning) The power_switch_resistance attribute of %s (%f) is not valid, and will be ignored.

**Description**

The specified power switch cell resistance by set_attr/set_attribute command is equal to or smaller than zero. PNA expects a positive resistance value for a power switch cell.

**What Next**

Please specify a valid power switch cell resistance by set_app_options/set_attr command, or ignore this warning.

## PNA-508

(warning) This virtual pad is outside of block boundary. Coordinate of the virtual pad :(%.3f, %.3f).

Feedback

**Description**

The virtual pad is out of current design boundary, but it will still be created and analyzed in PNA.

**What Next**

Please make sure the virtual power pad is in desired location, or ignore this warning.

---

## PNA-509

(warning) This virtual pad cannot connect to any PG wire. Coordinate of the virtual pad : (%.3f, %.3f).

**Description**

The virtual pad cannot create a virtual straight wire that connect to any PG wire. PNA will ignore the virtual pad.

**What Next**

Please change coordinates of the virtual power pad, or ignore this warning.

---

## PNA-510

(warning) There are standard cells/macros unplaced. PNA will distributed the power uniformly to the design.

**Description**

The power of unplaced standard cells/macros will be distributed uniformly to the design. The result will be different after placement/legalization.

**What Next**

Please place standard cells/macros and running analyze_power_plan again to achieve better result.

---

## PNA-511

(warning) For option %s, the specified value %s is invalid and ignored.

**Description**

This option value is invalid.

**What Next**

Check the command man page and specify the correct value.

Feedback

### PNA-512

(warning) User specified %s in %s is duplicated. It will ignore duplicated input.

**Description**

There are duplicated inputs in option. The command will only ignore duplicated input.

**What Next**

Please check if the input is intended.

### PNA-513

(warning) The value specified in %s is ignored. %s will take the value from %s only.

**Description**

There are exclusive control. Tool will only accept one of the setting

**What Next**

Please check if the input is intended.

## POW

### POW-001

(information) Propagated activity information for %s has been invalidated.

**Description**

There was propagated activity information available for certain modes or scenarios, but due to a design change it has been invalidated. It will be re-computed automatically next time power analysis is run.

**What Next**

No action required.

### POW-002

(warning) Design context is not valid.

**Description**

This message occurs when wrong design context is specified.

Feedback

### What Next

Specify correct design context.

### See Also

- report_power

---

## POW-003

(warning) Constraint design is not valid.

### Description

This message occurs when wrong constraint design is specified.

### What Next

Specify correct constraint design.

### See Also

- report_power

---

## POW-004

(warning) Design is not valid.

### Description

This message occurs when wrong design is specified.

### What Next

Specify correct design.

### See Also

- report_power

---

## POW-005

(information) Activity for scenario %s was cached, no propagation required.

### Description

Switching activity for a scenario was already calculated earlier. Therefore switching activity information will not be re-calculated.

Feedback

### What Next

No action is required.

---

## POW-006

(information) Propagated activity on scenario %s identical to that on %s

### Description

During activity propagation, a scenario with propagated data was detected that already contains the propagated activities. This data is being reused.

### What Next

No action required.

---

## POW-007

(warning) Pin %s is not a power pg pin.

### Description

This message occurs when specified pin is not a power pg pin.

### What Next

Specify a power pg pin.

---

## POW-008

(warning) No valid %sscenarios found in the user specified mode/corner/scenario list in design '%s'%s.

### Description

This message occurs when no valid scenarios are found in the mode/corner/scenario list specified by the user.

### What Next

Specify a valid scenario in the mode/corner/scenario list.

### See Also

- report_power
- report_switching_activity

Feedback

- set_switching_activity
- get_switching_activity
- reset_switching_activity
- propagate_switching_activity
- read_saif

## POW-009

(warning) Power analysis is disabled %s as leakage/swcap/internal mode power app options can be off and/or leakage_power/dynamic_power scenario flags are set to false.

**Description**

This message occurs when the scenario power flags are false and/or leakage/swcap/internal app options are 'off'.

**What Next**

Set the required power analysis app option to 'on' and enable the scenario power flags.

**See Also**

- report_power
- report_scenarios
- power.leakage_mode
- power.swcap_mode
- power.internal_mode

## POW-010

(warning) Ignoring invalid %s '%s' for %s-dependent %s specifications for the %s '%s'.

**Description**

state-condition or path-source specified for annotating sdpd data is not valid.

**What Next**

Specify a valid state-condition and/or path-source.

Feedback

## POW-011

(warning) Object for '%s' not found.

### Description

This message occurs when the object (instance, net, port, iopath or leakage) specified in the SAIF file is not found in the design.

### What Next

Check the name and pathname of the object and specify the correct object name which is present in the design.

### See Also

- read_saif

## POW-012

(error) Wrong format in line no %d of SAIF file.

### Description

This message occurs when the SAIF file contains a line which has incorrect format.

### What Next

Check the incorrect line number and re-write it according to the SAIF format.

### See Also

- read_saif

## POW-013

(warning) Unable to find pin '%s' specified as path source for the state condition '%s'.

### Description

This message occurs when an improper state dependent path dependent directive is specified in the SAIF file.

### What Next

Ensure that the pins specified in the state condition and the path source for the concerned object in the SAIF file actually exist in the design.

Feedback

### See Also

- read_saif

---

## POW-014

(warning) A value of '0.0' has been specified in the '-period' option. The tool will set it to the default value of '1.0' for activity computation.

### Description

This message occurs when a value of '0.0' is specified for the '-period' option.

### What Next

Check the value specified in the '-period' option to confirm if it should indeed be '0.0'.

### See Also

- get_switching_activity
- reset_switching_activity
- report_switching_activity

---

## POW-015

(warning) Cell '%s' is hierarchical, specify a non-hierarchical cell.

### Description

Command needs a non-hierarchical cell.

### What Next

Specify a non-hierarchical cell.

---

## POW-016

(info) No activity data found %s for %s '%s' scenario '%s'.

### Description

The message occurs when no activity data is found for specified state-condition and/or path-source.

### What Next

No action required.

Feedback

## POW-017

(warning) Clock '%s' specified as base clock is not a valid clock. It will be ignored for setting activities.

### Description

This message occurs when an invalid base clock is specified in the 'set_switching_activity' command.

### What Next

Check the base clock name specified in the 'set_switching_activity' command.

### See Also

- get_switching_activity
- reset_switching_activity
- report_switching_activity

## POW-018

(error) 'object_list' must contain valid objects.

### Description

This message occurs when object_list contains invalid elements or is empty.

### What Next

Specify at least one element, and make sure that all objects are of the valid type(s).

### See Also

- report_power
- set_switching_activity
- reset_switching_activity
- report_threshold_voltage_groups

## POW-019

(error) No valid %s specified.

Feedback

### Description

This message occurs when -cell_power and/or -net_power is specified without specifying a valid cell and/or net.

### What Next

Specify atleast one valid cell and/or net.

### See Also

- report_power

---

## POW-020

(error) The %s option cannot be used without -net_power or -cell_power.

### Description

This message occurs when 'objects' or '-leaf' option is specified without specifying -net_power or -cell_power.

### What Next

Specify -net_power and/or -cell_power.

### See Also

- report_power

---

## POW-021

(error) The -leaf option cannot be used when specifying objects.

### Description

This message occurs when objects are specified with -leaf option.

### What Next

Specify -leaf without specifying objects.

### See Also

- report_power

Feedback

## POW-022

(warning) Power group '%s' does not exist.

**Description**

This message occurs when invalid power group is specified.

**What Next**

Specify a valid power group.

## POW-023

(warning) Cannot determine rise/fall transition for unconnected pin '%s' for scenario '%s'.

**Description**

The transition on this pin cannot be determined as pin is unconnected. update_timing could not set transition for this pin.

**What Next**

connect the unconnected pin or set the transition attribute on the unconnected pin.

## POW-024

(information) Doing activity propagation for mode '%s' and corner '%s' with effort level '%s'.

**Description**

Specifies that activity propagation is being done for a given mode and corner.

**What Next**

Activity propagation should proceed for a valid combination of mode and corner.

## POW-025

(warning) Invalid switching activity annotation on constant object '%s' is being ignored.

**Description**

This message occurs when activity is being annotated on either a power or a ground object or a logic constant in the design. If you are annotating a power or logic1 object, the probability can only be specified as 1.0. If you are annotating a ground or logic0 object, the probability can only be specified as 0.0. The toggle rates in such cases can only be specified as 0.0.

Feedback

**What Next**

Check whether you really intended to apply those activities on constant objects.

**See Also**

- get_switching_activity

---

## POW-026

(information) Activity is being annotated on an unconnected net '%s'.

**Description**

This message occurs when activity is being annotated on an unconnected net.

**What Next**

Check if activity annotated on this unconnected net is useful.

**See Also**

- get_switching_activity

---

## POW-027

(warning) Invalid switching activity type '%s' is being %s

**Description**

This message occurs when wrong activity type is specified.

**What Next**

Specify a valid activity type and re-run the command.

**See Also**

- report_switching_activity

---

## POW-028

(error) File is not a backward annotated SAIF file.

**Description**

This message occurs when the string in the DIRECTION construct of the file is not backward.

Feedback

**What Next**

Specify the DIRECTION string as backward and re-run the command.

**See Also**

- read_saif

---

## POW-029

(error) Wrong scaling factor in line number %s.

**Description**

This message occurs when the scaling factor in TIMESCALE construct is neither 1, 10 or 100.

**What Next**

Specify the scaling factor as either 1, 10 or 100 and re-run the command.

**See Also**

- read_saif

---

## POW-030

(warning) %s

**Description**

The command action returned the specified information message.

**What Next**

n/a.

---

## POW-031

(error) Wrong time unit in line number %s.

**Description**

This message occurs when the time unit in the TIMESCALE construct is neither of s, ms, us, ns, ps or fs.

**What Next**

Specify the time unit as one of allowed types and re-run the command.

Feedback

**See Also**

- read_saif

---

## POW-032

(info) Setting toggle rate to 0.0 since probability has been specified as '%f'.

**Description**

The toggle rate can only be 0.0 if probability has been set to either 1.0 or 0.0. The tool will automatically reset the user specified toggle rate to 0.0 if anything other than 0.0 is provided.

**What Next**

Check whether you really intended to specify this activity for the object under consideration.

**See Also**

- get_switching_activity

---

## POW-033

(warning) Object '%s' does not have switching activity specified %s %s %s %s.

**Description**

The specified path_source/state_condition is not valid or switching activity has not been set for this path_source/state_condition.

**What Next**

Check whether specified path_source/state_condition is correct and switching activity has been set for this path_source/state_condition.

**See Also**

- set_switching_activity

---

## POW-034

(warning) No valid clocks available in mode '%s'. Setting clock frequency to %s GHz.

Feedback

### Description

Specified mode does not have any valid clocks. The simulation resolution needed to determine random vector generation during switching activity propagation is based on the fastest clock.

### What Next

Ensure there are valid clocks defined for the mode.

### See Also

- set_switching_activity
- get_switching_activity

---

## POW-035

(info) Setting the total toggle rate for net '%s' to 0.0 since probability has been set as '%f'.

### Description

The total toggle rate can only be 0.0 if probability has been set to either 1.0 or 0.0 during switching activity propagation.

### What Next

N/A.

### See Also

- set_switching_activity
- get_switching_activity

---

## POW-036

(warning) The toggle rate value of '%0.7f' for object '%s' exceeds the maximum toggle rate value of '%0.7f' of the fastest clock in the design.

### Description

The toggle rate of the object under consideration exceeds the maximum toggle rate possible during switching activity propagation. No data signal in the design should exceed the toggle rate of the fastest clock.

Feedback

### What Next

Ensure that any user activity annotations done through the 'set_switching_activity' command or activities enforced through reading SAIF files are valid.

### See Also

- set_switching_activity
- get_switching_activity

---

## POW-037

(warning) Cannot determine rise/fall transition for pin '%s' for scenario '%s'.

### Description

The transition on this pin cannot be determined and needs to be used for power estimation. During power estimation the 0.0 is used for the transition. This might cause extrapolation of the power tables in the library.

### What Next

The reason could be that it is driven by a primary input and the transitions as not been set. Please specify the transition. Another reason could be that the pin is driven by a constant, but that the toggle rate is non-zero. Please remove the (annotated) activity on the pin.

---

## POW-039

(warning) While computing leakage for instance '%s', mismatches in operating condition settings have been found in the design.

### Description

The operating condition mismatches may be triggered due to voltage, temperature or process label mismatches.

### What Next

Check the specifications in 'set_process', 'set_process_label', 'set_voltage' and 'set_temperature' commands.

### See Also

- report_power

Feedback

## POW-040

(warning) There are more than one instances present in the design for the saif name %s. Only one instance will be considered for annotating the switching activity.

### Description

The command action returned the specified information message.

### What Next

n/a.

### See Also

- read_saif

## POW-041

(warning) report_power can not generate net report for option '-net' as either power.swcap_mode app option is 'off' or its scenario is off for dynamic power.

### Description

Option '-net' of report_power command requires both power.swcap_mode app option 'on' and its scenario turned on for dynamic power.

### What Next

Turn ON the power.swcap_mode app option and/or turn ON the dynamic power for the scenario.

### See Also

- report_power
- power.swcap_mode

## POW-042

(error) Multiple objects cannot be specified if '%s' option is used in the 'report_power_calculation' command.

### Description

For virtual internal power queries using any of the '-capacitance', '-toggle_rate', '-probability' or '-transition' options, only a single object can be specified. The {pin, value}

Feedback

pairs given in these options will apply to the pins or ports of the specified instance or the module.

**What Next**

Specify a single object for the query.

**See Also**

- report_power

---

## POW-043

(error) The object '%s' specified can only be either an instance or a module if '%s' option is used in the 'report_power_calculation' command.

**Description**

For virtual internal power queries using any of the '-capacitance', '-toggle_rate', '-probability' or '-transition' options, only a single object of type either instance or module can be specified. The {pin, value} pairs given in these options will apply to the pins or ports of the specified instance or the module.

**What Next**

Specify either a module or an instance object for the query.

**See Also**

- report_power

---

## POW-044

(error) The object '%s' specified in the '%s' option in the 'report_power_calculation' command is not a pin or port in the instance or module being queried.

**Description**

For virtual internal power queries using any of the '-capacitance', '-toggle_rate', '-probability' or '-transition' options, the ports or pins specified in those options should be part of the instance or module object being queried.

**What Next**

Specify pins or ports in the '-capacitance', '-toggle_rate', '-probability' or '-transition' options that correspond to the module or instance being queried.

Feedback

**See Also**

- report_power

---

## POW-045

(info) %s

**Description**

The command action returned the specified information message.

**What Next**

n/a.

---

## POW-046

(warning) Power table extrapolation (%s mode) for port %s on %s for parameter %s. Lowest table value = %f, highest table value = %f, value = %f

**Description**

This message is given when a parameter during power table access is outside of the table range. Based on the mode, it will be extrapolated or clipped to the range.

**What Next**

The extrapolation mode can be set by app option power.table_extrapolation

**See Also**

- report_power

---

## POW-047

(warning) Power table estimation for parameter %s reduced to %f. Resulting power number will be multiplied with factor %f.

**Description**

This message is given when a parameter during power table access is outside of the table range and extrapolation mode is estimation. This will only be done for capacitive load parameters exceeding the highest power table index. In that case the power will be computed as if the load was driven by factor parallel drivers. So the load is reduced by 1/factor, the power number will be multiplied with factor for the given reduced load.

Feedback

### What Next

The extrapolation mode can be set by app option power.table_extrapolation.

### See Also

- report_power

---

## POW-048

(error) Timing model information not found for libcell '%s'.

### Description

This message occurs when timing model information for a libcell is not found.

### What Next

Check why timing model information is not found.

---

## POW-049

(warning) Option -tables not supported for libcell %s

### Description

The detailed library format is not available because the .lib data has been marked as confidential.

### What Next

Issuing report_power_library with out -tables will show a summary of what ta is available, but not the precise numerical values. The latter cannot be obtained from this library.

---

## POW-050

(error) '%s' is not a floating point number.

### Description

This message occurs when command expects a floating point number but user has not specified a valid floating point number.

### What Next

Specify valid floating point number.

Feedback

## POW-051

(information) Timer-derived activity data cached on scenario %s is invalidated.

**Description**

There was timer-derived activity information available for certain modes or scenarios, but due to a design change it has been invalidated. It will be re-computed automatically on demand.

**What Next**

No action required.

## POW-052

(information) Timer-derived activity data is cached on scenario %s

**Description**

Timer-derived activity informations been calculated and cached for the given scenario. It will be invalidated and re-calculated automatically on demand.

**What Next**

No action required.

## POW-053

(error) '%s' is not a '{pin_name value}' pair.

**Description**

This message occurs when command expects a pair but user has not specified a valid pair.

**What Next**

Specify valid pair.

## POW-054

(information) Activity cache is invalidated for scenario '%s'. Do activity propagation for this scenario to get updated activity.

Feedback

**Description**

There was cached activity information available for the scenario but it has been invalidated.

**What Next**

Re-run activity propagation for the scenario.

---

## POW-055

(error) Only leaf cells, black box cells and physical blocks can be annotated for power. Cell '%s' is neither.

**Description**

The set_annotated_power command only accepts leaf cells, black box cells or physical blocks for annotation.

**What Next**

Inspect the passed cell list for hierarchical cells that are not physical blocks or black box cells.

---

## POW-056

(warning) %s '%s' present in activity side file %s is not present in %s '%s', ignoring this object data.

**Description**

This happens when specified object in activity side file in not present in that hierarchy in the design.

**What Next**

Check if netlist has changed after writing the activity side file, write the activity side file again.

---

## POW-057

(error) value '%f' for option '%s' is invalid: must be > 0.

**Description**

This message occurs when the command option value specified is less than or equal to zero.

Feedback

**What Next**

Specify a non-zero non-negative number for this command option and re-run the command.

---

## POW-058

(error) %s

**Description**

The command action returned the specified error message.

---

## POW-059

(warning) Static probability of %d objects has been truncated to '1', to avoid this set option normalize to true.

**Description**

The message appears when option normalize is set to false and total weight applied is greater than 1 which resulted merged probablity of an object to be greater than 1.

**What Next**

Set option normalize to true.

---

## POW-060

(error) The activity group '%s' does not exist.

**Description**

The message appears when an activity group is specified that does not exist.

**What Next**

Specify an activity group that does exists. The 'report_switching_activity -essential' will give a report with all essential groups listed.

---

## POW-061

(error) Cannot assign to a non-user defined essential activity group.

**Description**

The message appears when the user wants to mark an activity point as a non-user defined essential activity point. Non-user essential activity point are automatically

Feedback

extracted by the tool and cannot be assigned by the user. The user is allowed to mark user-defined essential activity points.

**What Next**

Specify an user-defined activity group.

---

## POW-062

(error) Cannot set essential group on hierarchical port or pin named '%s'.

**Description**

The message appears when the user wants to set an essential group on a hierarchical port or pin.

**What Next**

Specify a leaf cell pin or port.

---

## POW-063

(error) Object must be pin or port.

**Description**

The message appears when the user specifies an object that is not a port nor pin.

**What Next**

Specify a leaf cell pin or port.

---

## POW-064

(error) Object must be pin, port or net.

**Description**

The message appears when the user specifies an object that is not a port nor pin nor net.

**What Next**

Specify a leaf cell pin, port or net.

---

## POW-065

(information) Essential activity is out-of-date. Do 'report_switching_activity -essential' to update essential activity points.

Feedback

### Description

Essential activity points have been queried, but the essential activity points are not up-to-date.

### What Next

Do 'report_switching_activity -essential' to update essential activity points.

---

## POW-066

(warning) Clock '%s' for mode '%s' has period 0.0 and will be ignored during power analysis.

### Description

This message occurs when a clock with period 0.0 has been defined.

### What Next

Check the clock definition with report_clock.

### See Also

- report_clock

---

## POW-067

(error) Invalid hierarchy selection '%s' specified.

### Description

The hierarchy option defines the scope for finding nets. Valid values are current, physical or all. In current hierarchy, net selection will take place within the logical hierarchy of the objects in the object list, or in the top module when no objects were defined. It means that no objects are reported that are not in the logical hierarchy of the objects (or top module). In physical hierarchy, nets are selected in the block scope (physical scope) of the objects. When the hierarchy is set to all, then all nets in the full hierarchy may be selected.

---

## POW-068

(warning) Cell(s) "%s" are non-hierarchical, hence ignoring them.

### Description

Command needs only hierarchical cells.

Feedback

**What Next**

Specify hierarchical cells.

---

## POW-069

(warning) %s toggles on pin %s are impossible given input states; converted to %s toggles.

**Description**

Power calculation deteected a situation where there are toggles of a certain polarity (rise/fall) on a pin whereas from the input states it would appear such toggles are impossible. In this case, the toggles are interpreted as the opposite type (i.e. rise toggles as fall toggles, and vice versa).

This situation can occur when inconsistent activities have been annotated on a cell.

**What Next**

You may want to review the activities annotated on this cell.

---

## POW-070

(information) Multi driven net '%s' is %s driven.

**Description**

This message occurs when a net is multi-driven by same type of drivers.

**What Next**

No action required.

---

## POW-071

(information) Multi driven net '%s' drivers %s and %s have different %s.

**Description**

This message occurs when a net is multi-driven by same type of drivers..

**What Next**

No action required.

Feedback

## POW-072

(warning) Multiple user power groups '%s' present for object '%s' and 'power.report_user_power_groups' is set as 'exclusive'. Considering group '%s' for this object.

**Description**

This message occurs when mulitple user power groups are found for an object and 'power.report_user_power_groups' is set as 'exclusive'.

**What Next**

'power.report_user_power_groups' can be set as 'inclusive'.

## POW-073

(information) Reading file '%s'.

**Description**

This message occurs when a saif file is read.

**What Next**

No action required.

## POW-074

(information) Power analysis being done for the %sactive scenario '%s'%s

**Description**

This message occurs when -force option is specified with report_power command to generate the power report for an inactive scenario or to override the leakage_power/ dynamic_power flags of a scenario.

**What Next**

No action required.

## POW-075

(warning) '%s' present in saif_map is not found in the current design.

Feedback

### Description

This warning message will be given when an object is not found in the design when reading a saif_map. After giving this warning, the parsing of saif_map will continue to give exhaustive list of objects not found.

### Examples

If an instance is not found in the design, then following error is reported - INSTANCE (full_cell_name) present in saif_map is not found in the current design.

If a net is not found in the design, then following error is reported - NET (full_net_name) present in saif_map is not found in the current design.

If a port is not found in the design, then following error is reported - PORT (full_port_name) present in saif_map is not found in the current design.

If a pin is not found in the design, then following error is reported - PIN (full_pin_name) present in saif_map is not found in the current design.

---

## POW-076

(warning) Simulated activity at netlist object '%s' was inconsistently redefined %d times. Last inconsistent SAIF definitions are on object %s and object %s.

### Description

This warning message indicates that inconsistent activities are present in the SAIF activity file. Activities are called inconsistent if the resulting probability and/or toggle rate are different. If two inconsistent SAIF activities refer to the same netlist object, this warning will be issued. This will for example happen when two pins that are connected to the same net have different activities in the SAIF file.

### What Next

Check the SAIF file or the tool that generated the SAIF file for issues.

---

## POW-077

(warning) No netlist object found to annotate simulated activity of SAIF object %s.

### Description

This warning message indicates that a specified activity in the SAIF activity file cannot be annotated in the netlist. This can happen when the original object that existed at SAIF generation time was deleted or completely remapped.

Feedback

**What Next**

If the SAIF file belongs to the current status of the netlist, this warning is not expected and should be investigated. Possibly the top level instance in the SAIF file does not correspond to the design. However, when the netlist has undergone optimizations, objects may have been deleted or remapped and this message can typically be ignored.

---

## POW-078

(warning) SDPD leakage data in the activity side file is incorrect. Skipping the SDPD leakage data of the design.

**Description**

This happens when the activity side file has improper SDPD leakage data.

**What Next**

Generate the activity side file again.

---

## POW-079

(error) Design '%s' present in saif_map is not same as the current top module '%s'.

**Description**

This error message will be shown when the current top module is not same as the design in the saif_map.

---

## POW-080

(warning) No default %s defined in the design, taking it as zero for power calculations.

**Description**

This message occurs when default voltage/temperature is not defined in the design.

**What Next**

Specify voltage/temperature using 'set_voltage/set_temperature'.

---

## POW-081

(information) Switching and internal power cannot be computed for the module '%s' because it does not have any ports.%s

Feedback

### Description

If the specified module does not have any ports, only leakage power can be calculated for it.

### What Next

Specify a module having ports for the query.

### See Also

- report_power

---

## POW-082

(warning) Simulated activity on object %s is changed by object %s at line:%d

### Description

This message occurs when we overwrite the activity on an object that has already been annotated in the same SAIF file.

### What Next

'power.only_use_unconditional_saif_information_for_nets' can be set to false.

---

## POW-083

(warning) The setting of ratio to 1 for the generated clock '%s' may not be equivalent to reset. Usage of reset_power_clock_scaling is recommended.

### Description

This message occurs when we set the scaling ratio for a generated clock to 1.

### What Next

Use reset_power_clock_scaling.

---

## POW-084

(warning) 'HighVT_select_fail_ratio' is not specified in file '%s'. Default value 1.0 will be applied for estimate context leakage calculation.

### Description

Leakage file must have HighVT_select_fail_ratio. It is used for estimating context leakage calculation.

Feedback

**What Next**

Provide HighVT_select_fail_ratio in leakage file.

---

## POW-085

(warning) Cell '%s' is a gtech cell which will limit activity propagation and power analysis results.

**Description**

Activity propagation and power analysis results for designs containing gtech cells can be limited.

**What Next**

Please compile to std cells to remove limitations.

---

## POW-086

(warning) Generic tech cell '%s' is too complex and will be modeled as a black box cell.

**Description**

Generic tech cell is too complex for activity propagation and power analysis. It will be modeled as a black box cell. Activity propagation and power analysis for this cell will be limited.

**What Next**

Compile the design to std cells.

---

## POW-087

(info) The activity type of object '%s' is '%s' so it is associated with either a supply or a constant. The activity of these kind of objects cannot be changed.

**Description**

The object is associated with a supply (power or ground) or a logic constant. The activity of these kind of objects are given and cannot be changed or set. If you have the intent to change the probability that a block/net is switched on for power scaling of switched blocks, then you should use the command *set_supply_net_probability*.

**What Next**

Dont annotate activity on supplies or constants.

Feedback

## POW-088

(information) Activity annotation on %s %s is %s to %s %s.

**Description**

Activity annotation is moved/copied from inside the repeater tree to root of the repeater tree if power.annotate_repeater_tree is enabled.

## POW-089

(error) Invalid state_condition '%s' specified in command %s_switching_activity.

**Description**

state-condition for sdpd data is not valid.

**What Next**

Specify a valid state-condition.

## POW-090

(warning) Command 'report_switching_activity' is deprecated, please use 'report_activity' instead.

**Description**

Command 'report_switching_activity' is deprecated.

**What Next**

Please use 'report_activity' instead.

## POW-091

(warning) %s

**Description**

The command action returned the specified warning message.

## POW-092

(warning) %s

Feedback

**Description**

The command action returned the specified warning message.

---

## POW-093

(error) %s

**Description**

The command action returned the specified error message.

---

## POW-094

(warning) report_power_calculation not enabled in the library '%s'.

**Description**

'report_power_calculation' library_feature is not enabled in this library, hence users do not have access to the power table values used for power calculation.

---

## POW-095

(warning) '%s' is not a valid vtGroup.

**Description**

This message occurs when the specified vtGroup name is not a valid vtGroup.

**What Next**

Specify a valid vtGroup name for this command option and re-run the command.

---

## POW-096

(error) value '%d' for option '%s' is invalid: must be > 0.

**Description**

This message occurs when the command option value specified is less than or equal to zero.

**What Next**

Specify a non-zero non-negative number for this command option and re-run the command.

Feedback

## POW-097

(error) Calculated '%s' power value is NaN for '%s'. Please check its input data, such as slew, load, and activity.

**Description**

This message occurs when calculated power value is NaN. It could be due to invalid slew value, load value, or activity toggle rate and static probability value.

**What Next**

Check slew, load, and activity values on this pin.

## POW-098

(warning) '%s' is a leaf cell, hence ignoring it in -hierarchy report.

**Description**

This message occurs when the specified cell is a leaf cell.

**What Next**

Specify a hierarchical cell and re-run the command.

## POW-099

(information) Net '%s' is being mapped to '%s' driven by pin '%s'.

**Description**

This message occurs when combinational logic driven by sequential cell is modified outside Fusion compiler. Hence we need to map old logic to modified logic and update saif_map database.

## POW-100

(warning) Value of '%s' for object '%s' should be non-negative, ignoring it.

**Description**

This message occurs when specified value is negative.

**What Next**

Specify a non-negative value for annotation.

Feedback

## POW-101

(error) Saif file should have DURATION specified and it should be greater than 0.

**Description**

This message occurs when duration is unspecified or less than or equal to 0.

**What Next**

Specify positive duration value.

## POW-102

(warning) Ignoring activity propagation as none of the flags(power or em) are set for scenario '%s'.

**Description**

Activity propagation is not done for the scenario if none of the flags(power or em) are set.

**What Next**

Enable the power/em flags for the scenario to propagate activity.

## POW-103

(warning) Ignoring boundary leakage for cell '%s' as it has not been legalized.

**Description**

Boundary leakage is not computed for cells that have not been legalized.

**What Next**

Please run create_placement and legalize_placement.

## POW-104

(error) Indesign PrimePower flow is unable to generate gate level saif.

**Description**

This could be due to issues reading the fsdb file in PrimePower.

**What Next**

Please check the input options '-strip_path', '-path', '-time', '-format' set in set_indesign_primepower_options command. Also ensure that the input fsdb file is correct.

Feedback

**See Also**

- set_indesign_primepower_options

---

## POW-105

(error) The SAIF map database for the '%s' block is corrupt.

**Description**

The SAIF map database could not be read successfully as it is corrupt.

**What Next**

Contact Synopsys Support and provide a test case with the implementation step used to create the specified block.

---

## POW-106

(information) C1Cn pin cap is not available for lib cell pin '%s', fall back.

**Description**

This message occurs when app option use_c1cn_pin_capacitance is enabled while the c1cn library data is not available for the cell pin.

---

## POW-107

(warning) Flat net for Net '%s' is not valid, cannot perform power analysis for design '%s'.

**Description**

This message occurs when any flat net for a design is not valid, and hence power analysis will not be performed for design.

---

## POW-108

(warning) Option '-write_fsdb' is not compatible with PrimePower C-CAPP time-based analysis.

**Description**

Option '-write_fsdb' is not compatible with PrimePower C-CAPP time-based analysis. It can only work with CAPP mode.

Feedback

**What Next**

Set power.indesign_use_capp to true to switch to CAPP mode, if gate level fsdb needs to be written.

---

## POW-109

(warning) The estimated activity has calculated toggle rate being inf or nan or negative number. Tool will use default probability and toggle rate as estimated activity value.

**Description**

The calculated toggle rate during activity estimation is found to be negative value or not a valid number. Tool will use default propability and toggle rate as estimated activity value. Please check if design fastest clock frequency is too low comparing to annotated toggle rate. Usually it happens when user speicified or SAIF specified activities have much higher frequency than design's SDC clock.

**What Next**

Please check clock frequency in SAIF, and sdc clock setup in the design. If inconsisitent, please use set_power_clock_scaling to make them consistent.

---

## POW-110

(error) Cannot create saif map temp file.

**Description**

This message occurs when shell.common.tmp_dir_path points to an invalid directory or the directory has incorrect read/write permissions.

**What Next**

Check shell.common.tmp_dir_path points to a valid directory with proper read/write permissions.

---

## POW-111

(warning) Command is ignored as saif map is not started.

**Description**

This message occurs during saif_map -write_map when saif_map is not started.

**What Next**

You needs to call saif_map -start before executing saif map command.

Feedback

# PPD

## PPD-001

(error) %s

**Description**

Persistent path data internal error.

**What Next**

Please send the log to Synopsys.

## PPD-002

(error) File format not supported.

**Description**

The format of the persistent path data file is not supported.

**What Next**

Please use gui_write_timing_paths command to generate valid timing path files.

**See Also**

- gui_write_timing_paths
- gui_read_timing_paths

## PPD-003

(error) File version %s not supported.

**Description**

The version of the persistent path data file is not supported.

**What Next**

Please use the correct version of the tool to generate timing path data.

Feedback

**See Also**

- gui_write_timing_paths
- gui_read_timing_paths

## PPD-004

(warning) Some timing paths will be skipped during reading. %s

**Description**

Some timing paths in the persistent path data file could not be properly read back in. This may be due to the fact that the associated design, scenario, mode, corner, pathgroup, clocks, and/or ports/pins etc... are not defined.

**What Next**

Please use the correct version of the design before reading timing path data.

**See Also**

- gui_write_timing_paths
- gui_read_timing_paths

# PRG

## PRG-002

(Information) %s

**Description**

The is an informational progress message. Messages of this type have a medium verbosity level.

**What Next**

No action is needed.

Feedback

# PT

## PT-001

(fatal) %s is not enabled.

**Description**

The application tried to reserve the specified license, but it was not available.

**What Next**

Verify that the key file is up to date and that sufficient licenses are available.

## PT-002

(error) Unrecognized feature name '%s'.

**Description**

You tried to get or remove a license that does not exist in the key file.

**What Next**

Verify that the key file is up to date and that the feature name is spelled correctly.

## PT-003

(Information) There are already %d licenses checked out for feature %s.

**Description**

This error occurred because you specified the *-quantity* option with the *get_license command* command and the specified quantity is smaller than the number of licenses that are already checked out for the specified feature.

The value specified for *-quantity* is the total number of licenses that should be checked out after the command has completed, which might not be the same as the incremental number of licenses checked out by the command.

**What Next**

Verify that the feature name is spelled correctly in case it was a different feature that you wanted to get. Specify a quantity that is equal to or greater than the number of licenses that have already been obtained.

Feedback

**See Also**

- get_license
- list_licenses

---

## PT-004

(error) You don't have a '%s' license to remove.

**Description**

You tried to remove a license that is not checked out.

**What Next**

Verify that the feature name is spelled correctly in case it was a different feature that you wanted to remove.

---

## PT-005

(error) Can't remove your '%s' license: %s.

**Description**

You tried to remove a license that is locked by the application. For example, this might be the license that is required to launch the application, or the license might be required by a design that is in memory. The message will indicate the condition that triggered it.

**What Next**

Action based on the message text.

---

## PT-006

(error) %s is not supported on '%s' platform.

**Description**

This is an error message indicating the command or option is not supported on the specified platform.

**What Next**

Please check the command manpage or product manuals for supported platforms. Make sure you run this command on the platforms that are supported.

Feedback

## PT-007

(warning) %s is deprecated; it will be obsoleted and removed starting with the %s release of PrimeTime.

**Description**

This is a warning message indicating the specified command, option, or variable is currently deprecated, and it will become obsoleted and removed in a future release of PrimeTime.

A "deprecated" feature is a feature that has not yet been removed but is planned to be removed in a future release. Thus, its use should be discontinued or transitioned to another feature.

An "obsoleted" feature is a feature that has been removed from the tool.

**What Next**

Please check the command/variable manpages, product manuals, or release notes for more details about the deprecation, and whether there are alternative approaches or replacements for this deprecated feature.

## PT-008

(error) %s was obsoleted starting with the %s release of PrimeTime; it is no longer supported.

**Description**

This is an error message indicating that the specified command, option, or variable has been obsoleted and is no longer supported.

**What Next**

Please check the command/variable manpages, product manuals, or release notes for more details about the obsolescence, and whether there are alternative approaches or replacements for this obsoleted feature.

## PT-009

(warning) The BC_WC analysis mode will be phased out in future releases.

**Description**

The analysis mode you are setting, best case - worst case, is potentially inaccurate and should be avoided for sign-off. It will be obsoleted in a future release of PrimeTime.

Feedback

For a thorough analysis, both setup and hold times should be checked at each corner of interest using the on_chip_variation analysis mode. In the BC_WC analysis mode, two corners are analyzed simultaneously. Hold times are checked at the fast (min) corner, and setup times are checked at the slow (max) corner. In the fast corner, fast slews are propagated along both the launch and capture sides of hold timing paths. As a result, worst-case slow timing cannot be guaranteed on the hold capture paths. In the slow corner, slow slews are propagated along both the launch and capture sides of setup timing paths. Again, worst-case fast timing cannot be guaranteed on the setup capture paths. In addition, the BC_WC analysis is incomplete because setup times are not cross-checked in the fast corner, and hold times are not cross-checked in the slow corner.

**What Next**

Migrate flows and scripts to avoid setting BC_WC analysis mode.

---

## PT-010

(error) %s is not enabled.

**Description**

The application tried to reserve the specified license, but it was not available.

**What Next**

Verify that the key file is up to date and that sufficient licenses are available.

---

## PT-011

(Warning) Obtained %d of requested %d licenses for feature %s.

**Description**

This warning is printed out because you specified the *-quantity* option with the *get_license command* and the number of licenses that was obtained is less than the requested quantity of licenses the specified feature.

At least one license of the specified feature was obtained.

**What Next**

No action is necessary if it is acceptable to run the tool with less than requested number of licenses (this may have tool performance implications). Otherwise, check the number of licenses of the specified feature and their usage by other processes and re-run the command when there are enough licenses available.

Feedback

### See Also

- get_license
- list_licenses

---

## PT-012

(Error) Only %d licenses are checked out for feature %s.

### Description

This error occurred because you specified the *-keep* option with the *remove_license command and you specified a quantity that is greater than the number of licenses that have already been obtained for the specified feature.*

### What Next

Specify a quantity for the *-keep* option that is less than or equal to the number of licenses that are already checked out for the specified feature.

### See Also

- get_license
- list_licenses

---

## PT-014

(information) Successfully checked out feature '%s'.

### Description

You receive this message because you have enabled licensing queuing by setting environment variable SNPSLMD_QUEUE to TRUE.

### What Next

This is an informational message only. No action is required on your part.

### See Also

- PT-018

---

## PT-015

(information) Started queuing for feature '%s'.

Feedback

### Description

You receive this message because you have enabled licensing queuing by setting environment variable SNPSLMD_QUEUE to TRUE.

### What Next

This is an informational message only. No action is required on your part.

### See Also

- PT-018

---

## PT-016

(information) Still waiting for feature '%s'.

### Description

You receive this message because you have enabled licensing queuing by setting environment variable SNPSLMD_QUEUE to TRUE.

### What Next

This is an informational message only. No action is required on your part.

### See Also

- PT-018

---

## PT-017

(information) Timeout while waiting for feature '%s'.

### Description

You receive this message because you have enabled licensing queuing by setting environment variable SNPSLMD_QUEUE to TRUE in pt_shell.

### What Next

This is an informational message only. No action is required on your part.

### See Also

- PT-018

Feedback

## PT-018

(Information) License queuing is enabled.

**Description**

You receive this message because you have enabled license queuing.

The license queuing is enabled by setenv SNPSLMD_QUEUE TRUE

When enabled the following timeouts can be adjusted:

Timeout for the initial license: setenv SNPS_MAX_WAITTIME <number of seconds>

Timeout for all subsequent licenses: setenv SNPS_MAX_QUEUETIME <number of seconds>

Defaults are equivalent to setenv SNPS_MAX_WAITTIME 259200 setenv SNPS_MAX_QUEUETIME 28800

**What Next**

This is an informational message only. No action is required on your part.

## PT-019

(Information) Checked out license '%s'

**Description**

The application checked out 1 license of the feature specified.

## PT-020

(Information) Checked in license '%s'

**Description**

The application checked in 1 license of the feature specified.

## PT-021

(information) Setting license limit for feature '%s' to %d.

**Description**

The limit on number of licenses for the specified feature has been set.

Feedback

### What Next

This is an informational message only. No action is required on your part.

### See Also

- get_license
- remove_license

---

## PT-022

(warning) Failed to set license limit for feature '%s': existing license count is higher than limit (%d > %d).

### Description

Failed to set the limit on number of licenses for the specified feature because the number of currently used licenses for the feature is higher than the specified limit.

### What Next

Reduce the license count for the specified feature to a number lower or equal of the given limit.

---

## PT-023

(Warning) Requested '%d' licenses for feature '%s', '%d' checked out.

### Description

Failed to checkout the requested number of licenses for the specified feature but did manage to check out some.

If the number of licenses requested is higher than the number already checked out, this occurs when the license server cannot fulfil the requests for the number of licenses due to a lack of licenses or because some of its licenses are already in use. The number of license indicated as checked out are all the license that can be acquired at this time.

If the number of licenses requested is lower than the number already checked out, this occurs because incremental licensing is not enabled.

PrimeTime can only check out licenses for a single feature from a single license server at a time. If the licenses for a feature are split across multiple license servers, the maximum number of licenses of that feature that PrimeTime can check out is determined by the license server with the largest number of licenses. PrimeTime cannot span its license check outs for a single feature across multiple license servers at this time. However, it can check out licenses for different features from different license servers.

Feedback

**What Next**

If the warning is due to a failure to increase the number of licenses then increase the number of licenses available on the license server so that subsequent attempts to checkout licenses succeed.

---

## PT-024

(information) Removing license limit for feature %s.

**Description**

The limit on number of licenses for the specified feature has been removed.

**What Next**

This is an informational message only. No action is required on your part.

---

## PT-025

(error) Cannot modify this variable after the first timing update.

**Description**

This variable can only be set prior to the initial timing update of the current design.

**What Next**

Please reconsider when to set this variable in your script or interactive run.

---

## PT-026

(warning) %s has been made obsolete since the %s release of PrimeTime and is no longer supported.

**Description**

This is an error message indicating that the specified command, option or variable has been obsoleted and is no longer supported.

**What Next**

Please check the command/variable manpages, product manuals or release notes for more details about the obsolescence, and whether there are alternative approaches or replacements for this feature being obsleted.

Feedback

## PT-027

(error) Cannot set the license limit for the crucial feature '%s' to '%d'

### Description

The limit on the number of licenses of the specified crucial feature cannot be set to the level specified. Typically this occurs when setting the limit to '0' which is not allowed as at least 1 license key of a crucial feature must remained checked out at all times during a session.

### What Next

Specify a higher limit on the number of licenses.

## PT-029

(information) Failed to checkout '%s' license ... exiting!

### Description

PrimeTime exited after failing to checkout a license for a feature required by commands in the users script. Typically this occurs when all the license keys for the required feature are in use.

Exiting under these conditions is a user selected behaviour, by default PrimeTime does not exit when a feature license is not available it continues execution without the commands/ capabilities enabled.

### What Next

In order to prevent PrimeTime from exiting immediately when the feature license cannot be checked out, turn on license queuing to have the tool wait for a period of time for a license to become available. License queuing can be turned on by applying the following in the UNIX shell prior to launching PrimeTime.

setenv SNPSLMD_QUEUE TRUE

By default, the queuing timeout is 28800 seconds.

If license queuing is already enabled then the queuing time may be too short and can be set by applying the following in the UNIX shell prior to launching PrimeTime along with enabling license queuing.

setenv SNPS_MAX_QUEUETIME <number of seconds>

Feedback

## PT-030

(error) The '%s' license cannot be removed, it is a pre-requisite of the '%s' license.

**Description**

The license cannot be removed as it is a pre-requisite of another license which is currently checked out.

**What Next**

Remove the license that depends on this license before removing this license.

## PT-031

(error) the '%s' license cannot be removed, it is required by the '%s' variable setting.

**Description**

The license cannot be removed as it is required by the analysis type enabled by the specified variable.

**What Next**

Disable the analysis type specified by the variable and then remove the license.

## PT-032

(error) The '%s' license limit cannot be set to '%d' '%s' is a prerequisite of the '%s' feature which already has '%d' licenses checked out.

**Description**

Where a feature is a prerequisite of another feature, the limit on the prerequiste feature cannot be less than the number of licenses already checked out for the feature with the license dependency.

**What Next**

Reduce the number of licenes checked out for the feature with the prerequisite dependency before reducing the limit on the prerequisite feature.

## PT-033

(error) The '%s' license limit cannot be set to '%d' '%s' is a prerequisite of the '%s' feature whose limit is already set to '%d'

Feedback

### Description

Where a feature is a prerequisite of another feature, its limit cannot be set less than the limit set for the feature with the prerequisite dependency.

### What Next

Reduce the limit on the feature with the prerequisite dependency before reducing the limit for the prerequisite feature.

---

## PT-034

(Information) License filtering is enabled.

### Description

You receive this message because you have enabled licensing filtering.

The license filtering is enabled by setting the following UNIX environmental variable before starting PrimeTime.

setenv SNPSLMD_DISABLE_NTA_LICENSE_CHECKS 1

When enabled, this indicates to PrimeTime that the user is not interested in any capabilities associated with the PrimeTime New Technology feature licenses. As a result PrimeTime will not even query the license server for the existence of these licenses, it will be as if the feature keys are not present in the license server and any associated capabiliteis will remain disabled.

### What Next

This is an informational message only. No action is required on your part.

---

## PT-035

(error) Cannot reduce %s license count to %d because %d licenses of this feature are required for feature %s.

### Description

This error occurred because you specified the *-keep* option with the *remove_license command, but the number of licenses for the specifed feature cannot be reduced because it is a pre-requisite of another feature.*

Where a feature is a prerequisite of another feature, its license count cannot be set to a lower number than the license count of the feature with the prerequisite dependency.

Feedback

### What Next

Reduce the license count on the feature with the prerequisite dependency before reducing the license count for the prerequisite feature.

### See Also

- remove_license
- get_license
- list_licenses

---

## PT-036

(error) Feature %s is limited to %d licenses.

### Description

This error occurred because you specified the *-quantity* option with the *get_license command, but the number of licenses for the specifed feature is limited to a lower number than the requested number of licenses.*

The get_license *command will attempt to obtain licenses up to the limit.*

### What Next

Set higher limit or remove the limit for the specified feature using the set_license_limit *command.*

### See Also

- get_license
- list_licenses

---

## PT-037

(error) Cannot get more than %d licenses for feature %s due to pre-requisite limits.

### Description

This error occurred because you specified the *-quantity* option with the *get_license command, but the number of licenses for the specifed feature cannot be set to a higher number than the license limit set on a feature the specified feature depends on.*

The get_license *command will attempt to obtain licenses up to the maximum allowable number.*

Feedback

### What Next

Set higher limit or remove the limit for the feature the specified feature depends on using the set_license_limit *command.*

### See Also

- get_license
- list_licenses

---

## PT-040

(information) Advanced waveform propagation is enabled.

### Description

You receive this message because you have enabled the advanced waveform propagation mode.

### What Next

This is an informational message only. No action is required on your part.

---

## PT-041

(information) Message '%s' %s limit (%d) exceeded. Remainder will be suppressed.

### Description

You receive this message if the total number of warning messages of a particular message id exceeds a default (or user set) limit.

### What Next

This is an informational message only. To avoid receiving this message, you can increase the default limit using the variable *sh_message_limit* or the command *set_message_info*.

### See Also

- PT-041

---

## PT-042

(information) No license limit set for feature %s.

Feedback

**Description**

You are trying to remove the limit on the number of licenses set for a feature shown in the informational message, but the limit cannot be removed because it was not previously set.

**What Next**

This is an informational message only. No action is required on your part.

---

## PT-043

(information) No limits are set on feature licenses.

**Description**

There are no limits on the number of licenses set for any feature.

**What Next**

This is an informational message only. No action is required on your part.

---

## PT-044

(Warning) Only one %s license can be checked out at a worker process.

**Description**

You have issued the *get_license* command with the *-quantity* option, requesting check out of more than one license for the given feature at a worker process. There is no need for worker processes to use more than one license of a feature, so *get_license* will attempt to check out only one of the requested feature licenses.

**What Next**

Check out the required number of licenses at the manager process instead of checking out licenses at worker processes. Alternatively, modify your scripts to check out no more than one license of the feature by a worker process.

---

## PT-045

(Error) PrimeTime New Technology license %s cannot be checked out.

**Description**

You have issued the *get_license* command requesting check out of a PrimeTime New Technology license. PrimeTime does not require check out of New Technology feature licenses.

Feedback

**What Next**

Remove the request to check out a PrimeTime New Technology license from your script.

---

## PT-046

(Error) the '%s' feature license cannot be checked out.

**Description**

You have issued the *get_license* command requesting check out of the debug feature indicated, however the UNIX environmental variable *SNPSLMD_DISABLE_DEBUG_LICENSE_CHECKS* is set to 1 indicating that all debug licenses should be skipped for access.

**What Next**

Remove the request to check out the indicated feature from your script or unset the *SNPSLMD_DISABLE_DEBUG_LICENSE_CHECKS* UNIX environmental variable.

---

## PT-047

(error) Cannot set limit for dependent feature %s to %d which is higher than the limit %d set on a pre-requisite feature.

**Description**

This error occurred because you specified the *-quantity* option with the *set_license_limit command, but the limit of licenses for the specifed feature cannot be set to a higher number than the license limit set on a feature the specified feature depends on.*

**What Next**

Set higher limit or remove the limit for the feature the specified feature depends on using the set_license_limit *command.*

---

## PT-048

(error) Checkout of these licensed features has failed: %s.

**Description**

The application tried to reserve the specified list of licensed features, but one or more of them was not available.

**What Next**

Verify that the key file is up to date and that sufficient licenses are available.

Feedback

## PT-049

(error) Cannot reduce the %s license count from %d to %d, %d licenses are in use for %s.

### Description

This error occurred because you tried to reduce the number of licenses checked out, for the specified feature, to below the number of licenses currently in use for the analysis indicated.

### See Also

- remove_license
- get_license
- list_licenses

## PT-050

(error) cannot checkout '%s' because it is suppressed.

### Description

The feature license cannot be checked out because it has been suppressed for checkout by the user.

### What Next

If the feature license is required, do not suppress it. Remove whatever suppression method has been applied.

## PT-051

(error) cannot checkout '%s' because its prerequisite '%s' is suppressed.

### Description

The feature license cannot be checked out because it one or more of its prerequisite features has been suppressed.

### What Next

If the feature license is required, do not suppress any of its prerequisites. Remove whatever suppression method has been applied.

Feedback

## PT-052

(Information) License queuing is disabled.

**Description**

You receive this message because you have explicity disabled license queuing.

The license queuing is disabled by setenv SNPSLMD_QUEUE FALSE

**What Next**

This is an informational message only. No action is required on your part.

## PT-054

(Information) Changing the license checkout maximum wait time from '%d' seconds to '%d' seconds.

**Description**

You receive this message because you requested the maximum wait time for the initial license checkout to be changed to the specifed value.

**What Next**

This is an informational message only. No action is required on your part.

## PT-055

(Information) Changing the license checkout maximum queue time from '%d' seconds to '%d' seconds.

**Description**

You receive this message because you requested the maximum queue time for license checkouts to be changed to the specifed value.

Where the process is a PrimeTime DMSA manager, it is recommended to set this value to a reaonably small value (< 300s) because where there are insufficient licenses available, each distributed command executed at the manager will incur queuing for licenses. The queuing will occur for every command until all the required licenes have been acquired impacting performance.

**What Next**

This is an informational message only. No action is required on your part.

Feedback

## PT-056

(Error) Failed to enable incremental licensing.

**Description**

You receive this message because the tool tried to enable incremental licensing internally and failed.

**What Next**

Contact the tool support team.

## PT-058

(Error) Failed to initialize critical licensing component.

**Description**

You receive this message because a critical licensing component failed to initialise and the tool cannot start up.

**What Next**

Please open a star with Synopsys specifying this error message.

## PT-059

(Warning) could not load Itcl package.

**Description**

You receive this message because the Incremental Tcl (Itcl) package supplied as part of the PrimeTime installation cannot be loaded.

**What Next**

Please check your Primetime installation directories include the itcl package library.

## PT-060

(Warning) Background process invalidates the -baseline_delta option of mem command

**Description**

The -set_baseline option of the mem command sets a baseline of the current memory used by PrimeTime. Subsequently, mem -baseline_delta returns the peak memory since the last baseline. If a background process is launched by the tool (e.g. by parallel_execute

Feedback

or redirect -background) after a baseline has been set, the memory used by the background proces will not be accounted in the result of mem -baseline_delta.

**See Also**

- mem

---

## PT-061

(Error) Failed to initialize the licensing call-home functionality.

**Description**

You receive this message because the licensing infrastructure failed to initialize a critical component necessary for checking out licenses.

**What Next**

Please file a STAR reporting the circumstances under which this occurred.

---

## PT-063

(Error) Library Compiler executable path is not set.

**Description**

When you install the PrimeTime tool, you need to specify the path to the Library Compiler executable, which PrimeTime tool uses for reading Liberty (.lib) files. The Library Compiler executable cannot be found because no path was specified.

**What Next**

Do one or both of the following:

- Add the Library Compiler executable path to the existing PrimeTime installation by running the following script:

  ```
  $synopsys/admin/install/syn/bin/install_pt
  ```

  The script prompts you for the location of the Library Compiler root directory:

  ```
  Synopsys Library Compiler root directory:
  ```

  At the prompt, enter the root installation directory of the Library Compiler tool.

- Set the SYNOPSYS_LC_ROOT environment variable to the path to the Library Compiler executable before you invoke the PrimeTime shell.

Feedback

## PT-064

(Error) Incorrect argument %s passed to load : %s.

**Description**

You receive this message because an option passed to the *load* command is incorrect.

**What Next**

The the arguments passed to the *load* command.

## PT-065

(Error) Loading command %s failed limitations checking: %s.

**Description**

You receive this message because the *load* tried to define commands which do not conform to PrimeTime's requirements.

**What Next**

Modify the source code of the loaded shared library to make it compatible with Primetime's requirements.

## PT-066

(fatal) Existing command %s was overwritten by load.

**Description**

You receive this message because the *load* command tried to overwrite one the existing PrimeTime commands.

**What Next**

Modify the source code of the loaded shared library to use command names different to the names of existing PrimeTime commands.

## PT-067

(fatal) Failure to open file due to system error "%s".

**Description**

A file could not be opened because of operating system error.

Feedback

### What Next

Depending on the system error, take the appropriate action on your OS or in your scripts. For example, an error of "Too many open files" may indicate that file handles are being leaked (opened but not closed) in a tcl script. See the linux man page for 'errno' for more information.

---

## PT-068

(warning) The dynamically loaded library was not linked against the collections library.

### Description

The dynamically linked library was not linked against the Synopsys supplied Collections API library. The load may still proceed, but this native Collections API functionality will be unavailable.

### What Next

If you wish to access the Collections API functionality, please ensure you are following Synopsys supplied instructions for this, or contact Synopsys.

---

## PT-072

(error) License checkout is not allowed in subprocesses.

### Description

Cross command parallel execution directives *redirect -bg* and *parallel_execute* launch subprocesses to perform the commands in parallel. It is required that these subprocesses do not try to checkout additional license while performing their work. Therefore, commands and variables which implicitly try to checkout licenses are not executed and error is issued.

### What Next

Remove the license checkout command from subprocesses running in parallel, and move them to the main parent process so they are executed in serial mode. The license checked-out by the parent process will be avilable for use to all the subprocesses.

---

## PT-073

(error) Cannot request %d keys of %s in the In-Design Mode.

Feedback

**Description**

In the In-Design mode an external application instructs the tool such as PrimeTime to manage licenses in the pre-programmed way. This error occurs if the number of license keys requested by the tool is larger than 1.

**What Next**

Remove the license checkout command from the script passed to the tool (e.g. PrimeTime) which requests multiple license keys. Alternatively, multiple keys may be requested internally by the tool due to enabling of some capability, in that case rewrite the script to request only a single key of the license.

---

## PT-074

(error) The license %s is not available in the In-Design Mode.

**Description**

In the In-Design mode an external application instructs the tool such as PrimeTime to manage licenses in the pre-programmed way. This error occurs if the reported license is not allowed to be checked out by the tool.

**What Next**

Remove the license checkout command from the script passed to the tool (e.g. PrimeTime) which requests the license.

---

## PT-075

(error) %s or %s is not enabled.

**Description**

The application tried to reserve the specified license, but either one or both of them were not available.

**What Next**

Verify that the key file is up to date and that sufficient licenses are available.

---

# PTLIB

---

## PTLIB-001

(error) The input of option -index_1 is invalid.

Feedback

### Description

The command *set_lib_max_cap_table* issues this error message when option -index_1 is specified with a invalid list of values.

### What Next

Make sure the values in the list are monotonically increasing.

---

## PTLIB-002

(error) The length of the input option -index_1 and -value should be the same.

### Description

The command *set_lib_max_cap_table* issue this message when the input frequency indices and capacitance values are of two different length input lists.

### What Next

Make sure the the input list of options -index_1 and -value have same length.

---

## PTLIB-003

(error) No max cap table on the specified library pin '%s', failed to set max cap table.

### Description

The *set_lib_max_cap_table* command issues this message when there is no max cap table on the specified library pin.

### What Next

Make sure the specified library pin contains max cap table or use -force option.

---

## PTLIB-004

(error) The desired table on the specified library pin '%s' is not a 1-d frequency based max_cap table, failed to set max cap table.

### Description

The *set_lib_max_cap_table* command issue this message when the desired table on the specified library pin is not a 1-d frequency based max_cap table. The *set_lib_max_cap_tablecommand supports only to override the 1-d frequency based max_cap table.*

Feedback

**What Next**

Make sure the desired table on the specified library pin is a 1-d frequency based max_cap table.

---

## PTLIB-005

(error) The input list of options -index or -value are empty.

**Description**

The command *set_lib_max_cap_table* issues this error message when the input list of options -index or -value are empty.

**What Next**

Make sure to provide values.

---

## PTLIB-006

(error) Invalid frequency index value '%f', failed to override frequency index on the library pin '%s'.

**Description**

The *set_lib_max_cap_table* command issues this message when the specified -index_1 option is not in the table. Allowed only to override the existing table index

**What Next**

Please provide the existing max cap table frequency index.

---

## PTLIB-007

(error) Trying to set max cap table on library pin '%s' of directon '%s', failed to set max cap table.

**Description**

The *set_lib_max_cap_table* command issues this message when try to set max cap table on library pin which is not of direction Output or Inout.

**What Next**

Please provide a library pin of direction Output or Inout.

Feedback

## PTLIB-008

(error) Slew derate inconsistency found in library %s for %s.

### Description

This message is issued when slew derate inconsistency is found for the specified library. The inconsistency happens when slew derate times the difference between upper and lower thresholds for pct rice/fall is greater than 1.

### What Next

Please verify the values for slew_lower[upper]_threshold_pct_rise[fall] attributes in the library.

## PTLIB-009

(Error) %s CCS noise model is invalidated because: %s

### Description

This error message indicates that an issue was found with an arc or pin CCS noise model during library read. As a result, the model is invalidated and not used in delay calculation.

These checks are always performed, but they are reported only when the *rc_ccsn_enable_library_error_reporting* variable is set to *true*.

If library scaling groups are being used, invalidated CCS noise models can cause scaling group inconsistencies, resulting in SLG-403 errors.

The limit value for the maximum current check can be set with the following variable:

```
pt_shell> set_app_var rc_ccsn_dc_current_limit_in_ma ...
```

### What Next

Investigate and fix the CCS noise model issue in the library.

### See Also

- SLG-403

## PTLIB-010

(Warning) CCS receiver model of %s : %s

Feedback

**Description**

You receive this warning because PrimeTime find a library CCS receiver model issue that may affect delay calculation accuracy. The message shows only when rc_receiver_enable_library_error_reporting is set to TRUE.

Types of warning

Negative capacitance value detected: PrimeTime does not support negative capacitance value in delay calculation.

Capacitance glitch detected: large fluctuation in C1Cn cap value may cause delay calculation inaccuracy.

The following example shows when negative value is detected:

Warning: CCS receiver model INV_4X: arc I->ZN : negative capacitance value detected. (PTLIB-010)

**What Next**

This is an informational message and no action required.

---

## PTLIB-011

(Warning) Extrapolation is applied bounding to 10% of the limit.

**Description**

Extrapolation is applied to estimate the value of the capacitance for the respective frequency value and is bounded to 10% of the limit,if the freq lies in a range outside of the given data set.

This warning message will be thrown when user has overriden the freq-cap data from library i.e. when *set_lib_max_cap_table* is used to override the freq-cap data for the lpin.

**What Next**

This is an informational message and no action required.

---

# PVT

---

## PVT-010

(warning) Multiply-defined voltage for rail %u from supplyNet %s connected to pin %s: %g vs %g in corner %s.

Feedback

### Description

This message is issued when there are conflicting voltage settings for a power rail of some cell. This could happen if the rail has multiple power pins with different supply net connections, or if two supply nets have different voltages (from set_voltage), and are both connected to the rail.

### What Next

Review the UPF data for an incorrect supply net configuration. Review the constraints for duplicate or incorrect set_voltage commands.

---

## PVT-011

(warning) No voltage defined for supplyNet %s of pin %s, rail %u in corner %s.

### Description

This message is issued when the voltage value of a supply net connected a cell's power pin is not known. This could happen if no set_voltage command has been given to set the voltage on a supply network, or if a different supply net has the correct voltage, but there is no connection (via supply_ports) between the powered supply net and the one actually connected to the cell.

### What Next

Review the UPF data for an incorrect or incomplete supply net configuration. Review the constraints for missing set_voltage commands.

---

## PVT-012

(warning) Using default primary voltage %g.

### Description

This message is issued (usually after a PVT-011 or PVT-013 message) when the voltage value of a cell's power pin is not known, and the corner's default primary supply voltage is used instead.

### What Next

Review the UPF data for an incorrect or incomplete supply net configuration. Review the constraints for missing set_voltage commands.

Feedback

## PVT-013

(warning) No supply net for power pin %s with power function %s and pg type %s in corner %s.

### Description

This message is issued when there is no supply net connected a cell's power pin, and thus the pin's supply voltage is not known. This could happen if there is no primary supply set or no connect_supply_net command to implicitly or explicitly connect a UPF supply net to the cell's power pin. If no UPF commands have been given, the tool will automatically create default primary supply nets to implicitly connect all cells' primary power pins. In this case, this message would be seen only for non-primary power pins.

### What Next

Review the UPF data for an incorrect or incomplete supply net configuration.

## PVT-014

(warning) Network of supply net %s has no %s voltage defined in corner %s.

### Description

This message is issued when the early or late voltage value of a supply net cannot be determined for some corner. This could happen if no set_voltage command has been given to set the voltage on the supply net's network, or if the supply net has not been properly connected to the rest of its network.

### What Next

Review the constraints for missing set_voltage commands. Review the UPF data for an incorrect or incomplete supply net configuration.

## PVT-020

(warning) Mismatched voltage in corner %s.

### Description

This message is issued when a specified voltage value for some cells in a particular corner does not match any of the voltages supported by the cells' library. Additional information will be printed identifying the relevant cell library, the number of affected cells, and the voltage that will actually be used for timing analysis.

Feedback

### What Next

Review the P/V/T operating conditions supported by the relevant cell library. Review the constraints for missing or incorrect set_voltage commands. Review the UPF data for an incorrect or incomplete supply net configuration.

---

## PVT-021

(warning) Mismatched temperature in corner %s.

### Description

This message is issued when the specified temperature value for some cells in a particular corner does not match any of the temperatures supported by the cells' library. Additional information will be printed identifying the relevant cell library, the number of affected cells, and the temperature that will actually be used for timing analysis.

### What Next

Review the P/V/T operating conditions supported by the relevant cell library. Review the constraints for missing or incorrect set_temperature commands. Review the constraints for missing or incorrect set_operating_conditions commands.

---

## PVT-022

(warning) Mismatched process number in corner %s.

### Description

This message is issued when the specified process number for some cells in a particular corner does not match any of the process numbers supported by the cells' library. Additional information will be printed identifying the relevant cell library, the number of affected cells, and the process number that will actually be used for timing analysis.

### What Next

Review the P/V/T operating conditions supported by the relevant cell library. Review the constraints for missing or incorrect set_process_number commands. Review the constraints for missing or incorrect set_operating_conditions commands.

---

## PVT-023

(warning) Mismatched process label in corner %s.

Feedback

### Description

This message is issued when the specified process label for some cells in a particular corner does not match any of the process labels supported by the cells' library.

Additional information is printed identifying the relevant cell library, the number of affected cells, and the process label that is actually used for timing analysis.

Remember that process labels are optional, and are needed only to distinguish between operating conditions that have identical voltages, temperatures, and process numbers.

### What Next

Review the PVT operating conditions supported by the relevant cell library. Review the constraints for missing or incorrect *set_process_label* commands.

---

## PVT-030

(warning) Corner %s: %d process number, %d process label, %d voltage, and %d temperature mismatches.

### Description

This message is issued when there are mismatches between the specified and effective values of the process number, process label, voltage, or temperature operating conditions of a particular corner. The *report_pvt* command will print information and warnings identifying the mismatched values, the relevant cell libraries, the number of affected cells, and the operating conditions that will actually be used for timing analysis.

### What Next

Review the P/V/T operating conditions supported by the relevant cell libraries. Review whether specified P/V/T is missing some library cells. Review the constraints for missing or incorrect set_voltage, set_temperature, set_process_number, set_process_label, or set_operating_conditions commands. Review the UPF data for an incorrect or incomplete supply net configuration. The *report_pvt* command will give more information about this situation.

---

## PVT-031

(warning) %d cells affected for early, %d for late.

### Description

This message is issued when there are mismatches between the specified and effective values of the process number, process label, voltage, or temperature operating conditions of a particular corner. It describes the number of cells affected by the mismatched PVT values. It will be preceded by a PVT-030 warning, which lists the number of mismatches.

Feedback

**What Next**

Review the P/V/T operating conditions supported by the relevant cell libraries. Review the constraints for missing or incorrect set_voltage, set_temperature, set_process_number, set_process_label, or set_operating_conditions commands. Review the UPF data for an incorrect or incomplete supply net configuration. The *report_pvt* command will give more information about this situation.

---

## PVT-032

(info) Corner %s: no PVT mismatches.

**Description**

This message is issued when there are no mismatches between the specified and effective values of the process number, process label, voltage, or temperature operating conditions of a particular corner.

**What Next**

No action is required. This is the desired situation.

---

## PVT-033

(warning) %d voltage and %d temperature extrapolations.

**Description**

This message is issued when voltage or temperature scaling is happening, and the specified voltage or temperature values for some part of the design are outside the ranges supported by a cell library. This situation can degrate the accuracy of the timer's delay and slew calculations.

**What Next**

Review the V/T operating conditions supported by the relevant cell libraries.
Review the constraints for missing or incorrect set_voltage, set_temperature, or set_operating_conditions commands. Review the UPF data for an incorrect or incomplete supply net configuration. The *report_pvt* command will give more information about this situation.

---

## PVT-034

(warning) %d port driving_cells affected for early, %d for late.

Feedback

### Description

This message is issued when there are mismatches between the specified and effective values of the process number, process label, voltage, or temperature operating conditions of a particular corner. It describes the number of port driving_cells affected by the mismatched PVT values. It will be preceded by a PVT-030 warning, which lists the number of mismatches.

### What Next

Review the P/V/T operating conditions supported by the relevant cell libraries. Review the constraints for missing or incorrect set_voltage, set_temperature, set_process_number, set_process_label, or set_operating_conditions commands applied to the port or the top level of the design. The *report_pvt* command will give more information about this situation for a particular port.

---

## PVT-073

(warning) %s have the same PVT: %s.

### Description

The *report_lib* command found a reference library with multiple panes that use the same PVT values. These panes are indistinguishable, so one of them will never be used. This problem originates in the library preparation process. For every PVT-073 warning, there would have been a LM-073 warning when the library was built.

### What Next

This is just a warning to alert you to the situation. You should be able to make adjustments to your library preparation script and rebuild the library to remove the duplicates.

### See Also

- LM-073

---

## PVT-080

(warning) Incorrect PVT value(s) specified for %s, %s. Pane:%d Voltage:%f Temperature: %f

### Description

This message is generated when invalid PVT value is encountered for a lib_cell or lib_pin. The PVT value used may be user specified or tool generated.

A library with multiple panes can trigger this warning for an out of range pane value or invalid voltage or temperature values. Please use report_lib command to check the details.

Feedback

### What Next

Please check for valid pane, voltage and temperature values for the library. If the PVT values are user specified then fix the script(s) to specify the correct PVT value(s).

If the message is generated by optimization command then the PVT value(s) may be tool generated. Please check all the instances of the lib_cell for correct PVT value(s) using if the reported values are correct. As an added measure, please file a star for the command.

---

## PVT-081

(warning) Incorrect PVT value(s) passed for %s %s; LibCell (lib:%s, pane:%d) but PVT data is (lib:%s pane:%d)

### Description

This message is generated when invalid PVT value is encountered for a lib_cell or lib_pin. The PVT value used may be user specified or tool generated.

A library with multiple panes can trigger this warning for an out of range pane value or invalid voltage or temperature values. Please use report_lib command to check the details.

### What Next

Please check for valid pane, voltage and temperature values for the library. If the PVT values are user specified then fix the script(s) to specify the correct PVT value(s).

If the message is generated by optimization command then the PVT value(s) may be tool generated. Please check all the instances of the lib_cell for correct PVT value(s) using if the reported values are correct. As an added measure, please file a star for the command.

---

## PVT-085

(Warning) Rails (count=%d) in PVT value (LibId: %d Pane: %d) mismatches with specified timing library %s (count=%d)

### Description

This message is generated when the rail count ( number of effective voltages ) is less than those found in the timing library for the cell. Tool will use the larger rail count always to determine the effective voltages.

### What Next

Please check that NDM does not have any unresolved cells after linking. Please check if there exists any hierachical insts with a specified library.

Feedback

If the message is generated by optimization command then the PVT value(s) may be tool generated. Please check all the instances of the lib_cell for correct PVT value(s) using if the reported values are correct. As an added measure, please file a star for the command.

---

## PVT-1201

(warning) %s.

### Description

*PVT-1201* is a generic warning message which gives the reason why a specific library cell cannot be used.

The following messages complain that a library cell does not match PVT requirement of a corner. Please use *report_mv_lib_cell* to display PVT settings of the library cell and *report_pvt* to display PVT requirements of the corner.

- Can't use LIB_1/CELL_1 because it doesn't match PVT setting in corner 'default'. (PVT-1201)
- Can't use LIB_1/CELL_1 because it doesn't match process label/number (L1/1.000) in corner 'max'. (PVT-1201)
- Can't use LIB_1/CELL_1 because it doesn't match power supply net voltage (VDD = 0.72V) in corner 'max'. (PVT-1201)
- Can't use LIB_1/CELL_1 because it match temperature (-40C) in corner 'fast'. (PVT-1201)

The following message complains that a library cell does not have the correct lib cell purpose required by tool. Please use *set_lib_cell_purpose* to update lib cell purpose of the library cell.

- Can't use LIB_1/CELL_1 because it has a lib cell purpose of 'none' but 'opto' is required. (PVT-1201)

The following message complains that a library cell is not in the correct threshold voltage group as required by tool. Please use *report_threshold_voltage_groups* to display threshold voltage group specified in the design.

- Can't use LIB_1/CELL_1 because it is in hvt threshold voltage group but lvt group is needed. (PVT-1201) "

The following message complains that a library cell has a site def id which is different than the one specified for the design.

- Can't use LIB_1/CELL_1 because it has a site def id which is incompatible with current top module. (PVT-1201)

Feedback

The following message complains that a library cell is in a different library cell subset from that required by an instance. Please use *report_libcell_subset* to display library subsets specified to the instance.

- Can't use LIB_1/CELL_1 because it has a lib cell subset LIB_SET_A but LIB_SET_B is needed. (PVT-1201)

The following messages complain that a library cell has a conflict with the target library subset specified in a hierarchy. Please use *report_target_library_subset* to display target library subsets specified in the hierarchy.

- Can't use LIB_1/CELL_1 in %s because it is not in hierarchy's set_target_library_subset. (PVT-1201)
- Can't use LIB_1/CELL_1 in MID1 because it is in another hierarchy's set_target_library_subset -only_here list. (PVT-1201)
- Can't use LIB_1/CELL_1 in MID/BOT because it is in hierarchy's set_target_library_subset -dont_use list. (PVT-1201)

The following message complains that an isolation library cell has an isolation clamp value which is different from that specified in *set_isolation*. Please use *report_mv_lib_cells -isolation -verbose* to display isolation cell properties of the library cell.

- Can't use LIB_1/CELL_1 because of isolation clamp value mismatch. (PVT-1201)

The following message complains that a retention library cell is not a zero-pin retention library cell but one is needed. Please use *report_mv_lib_cells -retention -verbose* to display retention cell properties of the library cell.

- Can't use LIB_1/CELL_1 because it is not a zero-pin retention lib cell. (PVT-1201)

The following message complains that a library cell does not have bias PG pins but the design requires to use one with bias PG pins. Please use *report_mv_lib_cells -verbose* to display bias PG pin information of the library cell.

- Can't use LIB_1/CELL_1 because it does not have bias PG pins". (PVT-1201)

**What Next**

Please correct settings on the specified library cell to address issues described in the message.

---

## PVT-1202

(warning) %u out of %u %s%slibrary cell%s %s%s.

Feedback

### Description

*PVT-1202* is issued when tool cannot find a legal library cell of a specified type to be used from user libraries. This usually happens during library analysis phrase when tools try to confirm that certain type of library cells with correct properties exist in user libraries.

The following messages complain that no buffers or inverters cannot be used because all of them have site def id's which are different from that specified for the design. Please use *report_site_defs* to display site def id's specified in the design. Please use *get_attribute -name site_def <lib_cell_name>* to display site def id specified to a library cell. Please use *get_lib_cells -filter site_name==<site_def_name>* to collect library cells with a specified site def name.

- Warning: 957 out of 957 buffer library cells have incompatible site def id (PVT-1202)
- Warning: 869 out of 869 inverter library cells have incompatible site def id (PVT-1202)
- Error: Library analysis failure - cannot find buffers and inverters (OPT-1006)

The following messages complain that no buffer or inverter library cells can be used in the top hierarchy because all of them do not have the required bias properties specified in the design. Please use *report_mv_lib_cells -standard -verbose* to display bias PG pin information and nwell/pwell voltages specified by the library cells. Please check that their nwell and pwell voltages match that specified in nwell and pwell supply functions of top power domain's primary supply set.

- Warning: 47 out of 47 buffer library cells have bias property mismatch in top hierarchy (PVT-1202)
- Warning: 55 out of 55 inverter library cells have bias property mismatch in top hierarchy (PVT-1202)

The following messages complain that no two-input NAND library cells can be used in hierarchy MID/BOT because all of them have voltage mismatch with that of the power domain primary supply net. Please use *report_mv_lib_cell -standard* to display PVT settings of the library cells and *report_supply_nets* to display operating voltages specified on supply nets.

- Warning: 45 out of 45 two-input NAND library cells have supply net voltage mismatch in MID/BOT (PVT-1202)
- Error: Insufficient library cells to generate synthetic parts. Need either one of nand2/ nor2 or an inverter and one of and/or. (DWS-0103)

The following messages are issued from *create_mv_cells -verbose* and complain that no buffer or inverter library cells can be used to implement a repeater specified in *set_repeater* or *set_port_attributes -repeater_supply*. Some have voltage mismatch with that specified in the top power domain's primary supply net. Some have conflicts with

Feedback

target library subset specified in the top hierarchy. Please use *report_target_library_subset* to display target library subsets specified in the top hierarchy.

- Warning: 50 out of 143 buffer library cells have supply net voltage mismatch in top hierarchy (PVT-1202)
- Warning: 93 out of 143 buffer library cells have incompatible target library subset (PVT-1202)
- Warning: 40 out of 118 inverter library cells have supply net voltage mismatch in top hierarchy (PVT-1202)
- Warning: 78 out of 118 inverter library cells have incompatible target library subset (PVT-1202)
- Warning: Repeater at port MID/Y with repeater supply (VDD, VSS) is not inserted due to non-availability of matching buffers and inverters in the libraries. (MV-201)

**What Next**

Please correct settings on the specified type of library cells to address issues described in the message.

# QTCL

## QTCL-001

(error) Cannot find widget '%s'.

**Description**

The widget with the specified hierarchical name cannot bre found. The listed command options are exclusive. Only one of them can be specified.

**What Next**

Check whether the hierarchical path of the widget name is correct or whether the widget is already created.

## QTCL-002

(error) At least one of the following options need to be specified: '%s'.

**Description**

Not all required arguments are specified.

Feedback

**What Next**

Check the correct syntax of the command and add all required arguments.

---

## QTCL-003

(error) No widget name specified (option: %s).

**Description**

The new name of the widget is not specified.

**What Next**

Add the widget name with the described option.

---

## QTCL-004

(error) Sender widget '%s' not found.

**Description**

The sender widget is not found with the specified hierarchical name.

**What Next**

Check whether the hierarchical path of the widget name is correct or whether the widget is already created.

---

## QTCL-005

(error) Receiver widget '%s' not found.

**Description**

The receiver widget is not found with the specified hierarchical name.

**What Next**

Check whether the hierarchical path of the widget name is correct or whether the widget is already created.

---

## QTCL-006

(error) Parent Widget '%s' not of class type %s.

**Description**

The parent of the new widget is not of the required class type.

Feedback

**What Next**

Check which parent class types are allowed for the new widget and correct the appropriate option.

---

## QTCL-007

(error) Widget with name '%s' already exists.

**Description**

A widget with the specified hierarchical name already exists. You cannot create widgets with identical hirarchical names.

**What Next**

Choose a unique hirarchical name for the new widget.

---

## QTCL-008

(error) Property '%s' of widget '%s' is not valid or not readable.

**Description**

A widget property with the specified name does not exist or is not readable.

**What Next**

Check the existence or readability of the property or correct the name.

---

## QTCL-009

(error) Property '%s' of widget '%s' not found.

**Description**

The specified widget property does not exist.

**What Next**

Check whether the specified widget property exists or correct the property name.

---

## QTCL-010

(error) Widget '%s' of class type '%s' does not support list of items.

Feedback

**Description**

The class type of the specified widget does not support list of items. The current tcl function cannot be apllied on this widget.

**What Next**

Correct the class type or widget name.

---

## QTCL-011

(error) Setting value '%s' for property '%s' of widget '%s' failed.

**Description**

Cannot set the property value of specified name.

**What Next**

Check in the widget documentation whether the property is not writable. Check the correctness of the hierarchical widget name and property name.

---

## QTCL-012

(error) Cast of value '%s' for property '%s' of widget '%s' to type '%s' failed.

**Description**

Cannot convert the value of the specified widget value to the data type of the property.

**What Next**

Check whether this data type is supported. Check the correctness of the hierarchical widget name and the property name.

---

## QTCL-013

(error) Cannot cast value '%s' for property '%s' of widget '%s' to type '%s'.

**Description**

Cannot convert the value of the specified widget value to the data type of the property.

**What Next**

Check whether this data type is supported. Check the correctness of the hierarchical widget name and the property name.

Feedback

## QTCL-014

(error) Widget of class '%s' not supported.

**Description**

The specified class type is not yet supported.

**What Next**

Choose one of the supported widget class types.

## QTCL-015

(error) Value '%d' of option '%s' is out of range ( %d <= index <= %d ).

**Description**

The specified option value is out of the allowed range.

**What Next**

Correct the value according to the described option value range.

## QTCL-016

(error) Indices list '%s' is not correct. Syntax: {i1 i2 ... in}.

**Description**

The syntax of the index list is not correct.

**What Next**

Correct the syntax according to the described syntax.

## QTCL-017

(error) Specified ui file '%s' does not exist.

**Description**

The specified ui file does not exist.

**What Next**

Correct the path name and filename and check the existence of the file.

Feedback

## QTCL-018

(error) Value '%d' of option '%s' is out of range (must be %s).

**Description**

The option value os out the the allowed described range.

**What Next**

Correct the option value according to the described range rules.

## QTCL-019

(error) Cannot connect signal '%s' of sender '%s'\nto slot '%s' of receiver '%s'.

**Description**

Cannot setup a connection between sender and reciever widget.

**What Next**

CHeck for the correct hierarchical names of the sender and receiver widgets. Check the correct names and arguments for signal and slot functions.

## QTCL-020

(error) Cannot connect signal '%s' of sender '%s'\nto tcl command '%s'.

**Description**

Cannot setup a connection between sender widget and a tcl clallback function.

**What Next**

CHeck for the correct hierarchical name of the sender widget. Check the correct name and arguments for signal and tcl function.

## QTCL-021

(error) The star '*' is not allowed as last name in hierarchical widget name %s'.

**Description**

The charecter '*' cannot be used as the last name in the specifier of a hierarchical widget name.

Feedback

**What Next**

Use this character only for levels in between.

---

## QTCL-022

(error) Widget '%s' has already specified %sproperty '%s'.

**Description**

The widget has already a property or extra property of the specified name.

**What Next**

Check for the correct hierarchical name of the sender widget. Check the correct name of the extra widget property and make sure that no other extra property is already attached to the widget.

---

## QTCL-023

(error) Incorrect value '%s' for enum property '%s' of widget '%s'.

**Description**

The property with a enum datatype cannot be initialized with the specified value.

**What Next**

Make sure that you only use supported values for the enum datatype.

---

## QTCL-024

(error) Object '%s' is not of type 'widget'.

**Description**

The specified object is not derived from QWidget. In order to work properly is needs to be derived from QWidget or from a class derived frm QWidget.

**What Next**

Derive the object from QWidget or from a class derived from QWidget.

---

## QTCL-025

(error) Parent object '%s' is not derived from QWidget

Feedback

**Description**

The specified parent object is not derived from QWidget.

**What Next**

Derive the parent object from QWidget or from a class derived from QWidget.

---

## QTCL-026

(error) The specified file '%' does not exist or is not readable

**Description**

The specified file does not exist or is not readable.

**What Next**

Check the existence and the read permissions of the specified file.

---

## QTCL-027

(error) Cannot find parent '%s'.

**Description**

The object with the specified hierarchical name specified as parent cannot be found.

**What Next**

Check whether the hierarchical path of the widget name is correct.

---

## QTCL-028

(error) Widget '%s' is not inherited from QDialog.

**Description**

The specified widget needs to be inherited from QDialog in order to provide the correct functionality.

**What Next**

Derive specified widget from QDialog or widget derived from QDialog.

---

## QTCL-029

(error) Cannot find extension widget '%s'.

Feedback

**Description**

The specified dialog extension widget does not exist.

**What Next**

Check correct name and creation of dialog extension widget.

---

## QTCL-030

(error) Dialog extension object '%s'is not a widget.

**Description**

The specified dialog extension widget is not a widget.

**What Next**

Derive the dialog extension object from QWidget or an object derived from QWidget.

---

## QTCL-031

(error) Syntax of string list argument value '%s' incorrect.

**Description**

The syntax of the value of a string list argument is incorrect.

**What Next**

Correct the value of the string list argument.

---

## QTCL-032

(error) Cannot find menu item '%s'.

**Description**

The specified menu item does not esist.

**What Next**

Check the existence and specification of the menu item.

---

## QTCL-033

(error) Object alias '%s' already in use.

Feedback

**Description**

The object specified object alais is already in use by another object.

**What Next**

Choose another object alias name.

---

## QTCL-034

(error) Cannot disconnect signal '%s' of sender '%s'\nto slot '%s' of receiver '%s'.

**Description**

Cannot disconnect the specified signal from specified sender to receiver.

**What Next**

Check correct names of signal, sender, receiver and slot.

---

## QTCL-035

(error) Cannot disconnect signal '%s' of sender '%s'\nto tcl command '%s'.

**Description**

Cannot disconnect the specified signal from specified sender to tcl command.

**What Next**

Check correct names of signal, sender and tcl command name.

---

## QTCL-036

(error) Property '%s' of object '%s' is ReadOnly

**Description**

Cannot set the value of specified property since it is a 'ReadOnly' property.

---

## QTCL-037

(error) Value list has more items than signal or slot argument definition

**Description**

The number of values passed to a signal or slot function does not match the number of arguments of the signal or slot function.

Feedback

**What Next**

Check the number of arguments of signal or slot function and the number of values.

---

## QTCL-038

(error) Cannot activate slot '%s' of receiver object '%s'

**Description**

Cannot find the name of the slot function for the specified receiver object

**What Next**

Check correct names of the slot function and the receiver object.

---

## QTCL-099

(error) '%s'

**Description**

Generic error

**What Next**

Fix issue or request specific error

---

# RAIL

---

## RAIL-200

(error) Cannot find layer of layer number %d.

**Description**

The layer of specified layer number cannot be found. One possible cause is that the technology for creating the rail result files is different with current technology.

**What Next**

Fix the incorrect data and try again.

---

## RAIL-300

(error) Must specify option '%s' first. Use comamnd '%s' to set the relevant option.

Feedback

**Description**

The invoked command requires some option to be set before running the command.

**What Next**

Use recommended command to set the relevant option then invoke the command again.

---

## RAIL-301

(error) Option %s is not supported for %s. Specify %s for the option instead.

**Description**

The specified option is not supported for the flow.

**What Next**

Specify the option according to the message.

---

## RAIL-302

(warning) No previous ICC2 generated %s to reuse!

**Description**

Reuse not possible on first run.

**What Next**

Run analyze_rail and try again.

---

## RAIL-303

(info) Writing %s..

**Description**

A resource file is being written.

**What Next**

n/a

---

## RAIL-304

(info) Generated %s: %s

Feedback

**Description**

A resource file has been generated at a location.

**What Next**

n/a

---

## RAIL-305

(info) Skip write %s..

**Description**

The write step for a resource file is being skipped.

**What Next**

n/a

---

## RAIL-306

(info) Using user-specified %s for %s: %s

**Description**

Using a user-specified resource file for a purpose.

**What Next**

n/a

---

## RAIL-307

(info) Captured %s log.\n\tSee "%s" for details

**Description**

The output of a process has been captured.

**What Next**

n/a

---

## RAIL-308

(info) To specify %s, use: set_app_options -name %s -value %s

Feedback

**Description**

Suggesting the use of an application option to override the write step for a resource file.

**What Next**

n/a

---

## RAIL-309

(info) Script file for %s: %s

**Description**

Indicating the location of a script file for a purpose.

**What Next**

n/a

---

## RAIL-310

(info) Reusing previous ICC2 generated %s for %s.

**Description**

Indicating reuse of a previous ICC2 generated resource.

**What Next**

n/a

---

## RAIL-400

(error) Rail result with name '%s' may be modified and thus cannot be opened.

**Description**

The rail result generated from previous analyze_rail command may have been modified. Common mistakes are: result directory name is modified; or some files under the result directory have been removed or renamed.

**What Next**

Rail result directory once created cannot be renamed. Result files underneath cannot be modifiedi or deleted. Run analyze_rail command again to re-generate the result.

Feedback

## RAIL-401

(warning) Rail result is opened. It will be closed before a new rail result is generated.

**Description**

There is one rail result opened from previous analyze_rail or open_rail_result command. This result will be closed first, so a new rail result can be generated.

**What Next**

None.

## RAIL-402

(error) Rail result with name '%s' is already opened.

**Description**

This occurs when user wants to open a rail result which is already opened through previous open_rail_result command or analyze_rail command.

**What Next**

Close the rail result using close_rail_result command and then open a rail result.

## RAIL-403

(error) Rail result with name '%s' is not generated from Redhawk Analysis Fusion flow.

**Description**

This occurs when Redhawk fails to generate the result files back to ICC2 to read.

**What Next**

Check Redhawk log file to see if fusion flow is successful or not.

## RAIL-404

(error) Rail database path cannot be changed while current rail result is opened.

**Description**

This occurs when set_app_options -name rail.database is used to change rail database path while a current rail result is still opened.

Feedback

**What Next**

Close currently opened rail result, then change the rail database path.

---

## RAIL-405

(error) Rail result cannot be opened before open_block.

**Description**

This occurs when open_rail_result is invoked before open_block.

**What Next**

Invoke open_lib/open_block command first, then run open_rail_result.

---

## RAIL-406

(error) Rail result file is generated from a Redhawk version which is not compatible with the current product version. %s.

**Description**

This occurs when an incompatible Redhawk binary is used to generate the rail result files for the Fusion Analysis flow. Those files cannot be read back into the current product.

**What Next**

Use a compatible Redhawk version to rerun the flow.

---

## RAIL-407

(error) Rail result file is out of date.

**Description**

This occurs when an old rail result which is saved by an old ICC2 version is opened in a newer ICC2 version. Between those two versions of ICC2, rail result file format is changed. As a result, those rail result files cannot be opened in the new version of ICC2.

**What Next**

Use an older ICC2 version to open the rail result, or re-run the analysis flow to re-generate result files.

---

## RAIL-408

(warning) No rail result can be closed since there is no rail result currently opened.

Feedback

**Description**

This occurs when close_rail_result is invoked on a design which has no rail result currently opened. For example, once analyze_rail finishes, if close_rail_result is invoked twice consecutively, the second close_rail_result will trigger this warning message.

**What Next**

None.

---

## RAIL-409

(error) No rail result is currently opened for %s

**Description**

This occurs when there is no rail result opened in the current session; or the currently opened rail result does not contain the net specified.

**What Next**

Invoke analyze_rail to generate a rail result or use open_rail_result to open a rail result generated from previous sessions.

---

## RAIL-410

(warning) Rail result is not consistent with current design. Possible reason: %s.

**Description**

This occurs when design is changed after rail analysis is run. Design change may include design name change, or netlist change, and thus caused inconsistency between current design and saved rail result.

**What Next**

Recommend to invoke analyze_rail again to generate a new rail result for current design. If user really wants to reuse previous rail result, use *open_rail_result -back_annotate*.

---

## RAIL-411

(error) Rail result is not consistent with current design. Possible reason: %s.

**Description**

This occurs when design is changed after rail analysis is run. Design change may include design name change, or netlist change, and thus caused inconsistency between current design and saved rail result.

Feedback

### What Next

Recommend to invoke analyze_rail again to generate a new rail result for current design. If user really wants to reuse previous rail result, use *open_rail_result -back_annotate*.

---

## RAIL-412

(error) Fail to annotate rail result on %s %s. This may be due to design is changed and thus not consistent with currently opened rail result.

### Description

This occurs when design is changed after rail analysis is run. Design change may include netlist change, for example, cells or PG pins are added or removed. This will introduce inconsistency between current design and current rail result.

### What Next

Recommend to invoke analyze_rail again to generate a new rail result for current design. Use *open_rail_result -back_annotate* may not solve the problem. For newly added cells or PG pins, there will be no rail result to be annotated, and thus the same error may occur.

---

## RAIL-500

(error) In voltage drop hot spot area (grid index %d), current is dominated by one instance %s (%s). Cell movement technique may not be effective.

### Description

In voltage drop hot spot area or grid which is indexed by an id, its current is dominated by one instance. Therefore cell movement technique is not effective in this area, because otherwise the voltage drop hot spot could just be moved from one area to another different area.

### What Next

Use report_hot_spots -index id to get more detail on how current is distributed cross different cells. To reduce voltage drop in this hot spot area, techniques other than cell movement may be applied.

---

## RAIL-600

(error) %s is not a supply net.

Feedback

**Description**

The specified net name is not a supply net in the design. Rail analysis requires net names specified to be supply nets.

**What Next**

Use command create_supply_net to create relevant supply nets.

---

## RAIL-601

(warning) Supply voltage is not defined for supply net %s.

**Description**

There is no supply voltage defined for the specified supply net. Default 1.0 will be used for power nets, and 0.0 for ground nets.

**What Next**

Use command set_voltage to define supply voltage.

---

## RAIL-900

(error) %s

**Description**

Basic input error checking.

**What Next**

Please re-check the command inputs and use '-help' option to verify if the input is correct.

---

## RAIL-903

(error) Required argument for names of the power or ground nets is not found.

**Description**

There is a user settings problem or preconditions for an operation are not met.

**What Next**

Correct the user settings problem or satisfy the preconditions for the operation and try again.

Feedback

**See Also**

- analyze_rail

---

## RAIL-904

(error) -primerail_script_file and -script_only options can't enable simultaneously.

**Description**

There is a user settings problem or preconditions for an operation are not met.

**What Next**

Correct the user settings problem or satisfy the preconditions for the operation and try again.

**See Also**

- analyze_rail

---

## RAIL-905

(error) No analysis options specified. Please use -voltage_drop, -electromigration, -decap or -integrity option.

**Description**

There is a user settings problem or preconditions for an operation are not met.

**What Next**

Correct the user settings problem or satisfy the preconditions for the operation and try again.

**See Also**

- analyze_rail

---

## RAIL-909

(error) Cannot allocate memory.

**Description**

Tool fails to allocate memory for the operation.

Feedback

### What Next

Please rerun on a machine with larger memory.

### See Also

- analyze_rail

---

## RAIL-931

(error) Net '%s' is not a valid net.

### Description

You receive this message because an error occurs when the operation is scanning through the input files. It has found either a syntax error or an unsupported construct.

### What Next

Fix the syntax error or work around the unsupported construct.

### See Also

- analyze_rail

---

## RAIL-932

(error) Value string '%s' is not a valid number.

### Description

You receive this message because an error occurs when the operation is scanning through the input files. It has found either a syntax error or an unsupported construct.

### What Next

Fix the syntax error or work around the unsupported construct.

### See Also

- analyze_rail

---

## RAIL-934

(error) Cannot find the pairing voltage value to net '%s'.

Feedback

### Description

You receive this message because an error occurs when the operation is scanning through the input files. It has found either a syntax error or an unsupported construct.

### What Next

Fix the syntax error or work around the unsupported construct.

### See Also

- analyze_rail

---

## RAIL-936

(error) Invalid supply voltage in the net name and voltage pair {%s %s}.

### Description

You receive this message because an error occurs when the operation is scanning through the input files. It has found either a syntax error or an unsupported construct.

### What Next

Fix the syntax error or work around the unsupported construct.

### See Also

- analyze_rail

---

## RAIL-937

(error) Please assign the supply voltage for the internal PG net "%s", for example {%s 1.08}.

### Description

There is a user settings problem or preconditions for an operation are not met.

### What Next

Correct the user settings problem or satisfy the preconditions for the operation and try again.

### See Also

- analyze_rail

Feedback

## RAIL-938

(error) Main power or ground net name is missing.

**Description**

An unexpected failure has occurred.

**What Next**

This is an internal error.

**See Also**

- analyze_rail

## RAIL-939

(warning) Cannot duplicate stdout.

**Description**

An unexpected failure has occurred.

**What Next**

This is an internal error.

**See Also**

- analyze_rail

## RAIL-942

(error) Get current operating condition failed. Please use set_operating_conditions to set the current operating condition.

**Description**

An unexpected failure has occurred.

**What Next**

This is an internal error.

**See Also**

- analyze_rail

Feedback

## RAIL-952

(error) Cannot get current cell id.

**Description**

An unexpected failure has occurred.

**What Next**

This is an internal error.

## RAIL-954

(error) Can not set "-reuse" option, since no analyze_rail run data found in the RAIL view.

**Description**

An unexpected failure has occurred.

**What Next**

This is an internal error.

## RAIL-959

(error) PrimeTime version has to be newer than or equal to 2009.06!

**Description**

You are seeing this error because you tried to perform an operation or specify an option that is not supported.

**What Next**

There is nothing to do.

**See Also**

- analyze_rail

## RAIL-961

(error) PrimeRail version check failed. 2009.06 or later release is required.

**Description**

You are seeing this error because you tried to perform an operation or specify an option that is not supported.

Feedback

### What Next

There is nothing to do.

### See Also

- analyze_rail

---

## RAIL-963

(error) PrimeRail license check failed.

### Description

You receive this message because a license feature is not available.

### See Also

- analyze_rail

---

## RAIL-964

(error) -assign_power_only cannot co-exist with -reuse {power}.

### Description

There is a user settings problem or preconditions for an operation are not met.

### What Next

Correct the user settings problem or satisfy the preconditions for the operation and try again.

### See Also

- analyze_rail

---

## RAIL-965

(error) Cannot get library id.

### Description

An unexpected failure has occurred.

### What Next

This is an internal error.

Feedback

**See Also**

- analyze_rail

---

## RAIL-966

(error) Cannot open rail DB of cell %s.

**Description**

An unexpected failure has occurred.

**What Next**

This is an internal error.

**See Also**

- analyze_rail

---

## RAIL-967

(error) No Power DB exist for reuse power.

**Description**

An unexpected failure has occurred.

**What Next**

This is an internal error.

**See Also**

- analyze_rail

---

## RAIL-968

(error) Cannot change output_dir when use -reuse option.

**Description**

There is a user settings problem or preconditions for an operation are not met.

**What Next**

Correct the user settings problem or satisfy the preconditions for the operation and try again.

Feedback

**See Also**

- analyze_rail

---

## RAIL-969

(error) -top_level_only is set to %s, which is inconsistent to the rail DB generated in the previous run, therefore the result of the previous run cannot be reused.

**Description**

There is a user settings problem or preconditions for an operation are not met.

**What Next**

Correct the user settings problem or satisfy the preconditions for the operation and try again.

**See Also**

- analyze_rail

---

## RAIL-971

(warning) Cannot run /bin/uname on host '%s'.

**Description**

An unexpected failure has occurred.

**What Next**

This is an internal error.

**See Also**

- analyze_rail

---

## RAIL-979

(error) The platform of the assigned host '%s'(%s) is %s, while the platform of the current machine '%s' is %s. Please choose another host whose platform is %s.

**Description**

An unexpected failure has occurred.

Feedback

### What Next

This is an internal error.

### See Also

- analyze_rail

---

## RAIL-980

(error) Net pattern did not find any nets.

### Description

An unexpected failure has occurred.

### What Next

This is an internal error.

---

## RAIL-984

(error) Cannot get current cell name.

### Description

An unexpected failure has occurred.

### What Next

This is an internal error.

---

## RAIL-991

(error) Power plan region pattern did not find any power plan regions.

### Description

An unexpected failure has occurred.

### What Next

This is an internal error.

---

## RAIL-997

(error) rail child server standing by already can not start again.

Feedback

**Description**

An unexpected failure has occurred.

**What Next**

This is an internal error.

**See Also**

- analyze_rail

---

## RAIL-998

(error) Failed to start rail child server.

**Description**

An unexpected failure has occurred.

**What Next**

This is an internal error.

**See Also**

- analyze_rail

---

## RAIL-999

(error) Executing remote command failed.

**Description**

An unexpected failure has occurred.

**What Next**

This is an internal error.

**See Also**

- analyze_rail

---

## RAIL-1000

(error) No server mode PrimeRail standing by.

Feedback

### Description

An unexpected failure has occurred.

### What Next

This is an internal error.

### See Also

- analyze_rail

---

## RAIL-1002

(error) Cannot retrieve data (top cell: %s temperature: %s net: %s) from a previous PrimeRail run since no rail database is found under the output directory "%s". Please reset "-reuse" option of set_rail_options and rerun analyze_rail to generate rail database.

### Description

An unexpected failure has occurred.

### What Next

This is an internal error.

### See Also

- analyze_rail

---

## RAIL-1003

(error) The net-voltage pair '%s' which is set in the value of the -supply_voltage option is not a valid pair containing a net name and a voltage value.

### Description

There is a user settings problem or preconditions for an operation are not met.

### What Next

Correct the user settings problem or satisfy the preconditions for the operation and try again.

### See Also

- analyze_rail

Feedback

## RAIL-1005

(error) The net '%s' which is set in the value of the -supply_voltage option is not a valid power or ground net.

**Description**

There is a user settings problem or preconditions for an operation are not met.

**What Next**

Correct the user settings problem or satisfy the preconditions for the operation and try again.

**See Also**

- analyze_rail

## RAIL-1010

(error) Option -assign_power_only need -config_file to work together.

**Description**

There is a user settings problem or preconditions for an operation are not met.

**What Next**

Correct the user settings problem or satisfy the preconditions for the operation and try again.

## RAIL-1011

(error) Unable to create directory %s.

**Description**

The file cannot be found in the given path.

**What Next**

Check that the file is correctly set and spelled in the project file and that the *search_path* variable is correctly set to find that file.

## RAIL-1012

(error) end_time must be greater than or equal to start_time!

Feedback

**Description**

There is a user settings problem or preconditions for an operation are not met.

**What Next**

Correct the user settings problem or satisfy the preconditions for the operation and try again.

---

## RAIL-1013

(error) Log file name assigned to -primerail_log_file option cannot be an empty string.

**Description**

There is a user settings problem or preconditions for an operation are not met.

**What Next**

Correct the user settings problem or satisfy the preconditions for the operation and try again.

**See Also**

- analyze_rail

---

## RAIL-1014

(error) Cmd file name assigned to -primerail_cmd_file option cannot be an empty string.

**Description**

There is a user settings problem or preconditions for an operation are not met.

**What Next**

Correct the user settings problem or satisfy the preconditions for the operation and try again.

**See Also**

- analyze_rail

---

## RAIL-1016

(error) Inrush results cannot be reused.

Feedback

### Description

An unexpected failure has occurred.

### What Next

This is an internal error.

### See Also

- analyze_rail

---

## RAIL-1020

(error) Unexpected output by PrimeRail report_db_units. Aborting.

### Description

An unexpected failure has occurred.

### What Next

This is an internal error.

### See Also

- analyze_rail

---

## RAIL-1035

(error) Must issue set_tlu_plus_files.

### Description

There is a user settings problem or preconditions for an operation are not met.

### What Next

Correct the user settings problem or satisfy the preconditions for the operation and try again.

### See Also

- analyze_rail

Feedback

## RAIL-1036

(error) Taps are not specified since option use_pins_as_pads is false and none of the following options are specified: pad_master_file, pad_instance_file, user_defined_tap_file, virtual_pads.

### Description

An unexpected failure has occurred.

### What Next

This is an internal error.

### See Also

- analyze_rail

## RAIL-1041

(warning) Syntax error in UDE file "%s" on line "%s". Correct syntax is "R <netName> <layerNum1> <x1> <y1> <layerNum2> <x2> <y2> <value>".

### Description

You receive this message because an error occurs when the operation is scanning through the input files. It has found either a syntax error or an unsupported construct.

### What Next

Fix the syntax error or work around the unsupported construct.

### See Also

- analyze_rail

## RAIL-1053

(warning) PORT_INST_CURRENT_WAVEFORM only works when analysis_mode is "dynamic". Aborting.

### Description

An unexpected failure has occurred.

### What Next

This is an internal error.

Feedback

**See Also**

- analyze_rail

---

## RAIL-1065

(error) Cannot perform dynamic rail analysis without VCD switching activity input. If you want to perform dynamic VFRA analysis, please use "set_rail_options dynamic_vectorfree".

**Description**

An unexpected failure has occurred.

**What Next**

This is an internal error.

**See Also**

- analyze_rail

---

## RAIL-1066

(error) Inrush only preconditions not met.

**Description**

An unexpected failure has occurred.

**What Next**

This is an internal error.

**See Also**

- analyze_rail

---

## RAIL-1068

(error) No integrity checking strategy is set.

**Description**

There is a user settings problem or preconditions for an operation are not met.

Feedback

### What Next

Correct the user settings problem or satisfy the preconditions for the operation and try again.

---

## RAIL-1074

(error) No rail option is set.

### Description

There is a user settings problem or preconditions for an operation are not met.

### What Next

Correct the user settings problem or satisfy the preconditions for the operation and try again.

---

## RAIL-1079

(error) Cannot find %s executable.

### Description

The file cannot be found in the given path.

### What Next

Check that the file is correctly set and spelled in the project file and that the *search_path* variable is correctly set to find that file.

### See Also

- analyze_rail

---

## RAIL-1080

(error) Cannot find file '%s'.

### Description

The file cannot be found in the given path.

### What Next

Check that the file is correctly set and spelled in the project file and that the *search_path* variable is correctly set to find that file.

Feedback

**See Also**

- analyze_rail

---

## RAIL-1081

(warning) Cannot get %s.

**Description**

An unexpected failure has occurred.

**What Next**

This is an internal error.

**See Also**

- analyze_rail

---

## RAIL-1082

(error) Cannot open file '%s' for %s.

**Description**

The file cannot be found in the given path.

**What Next**

Check that the file is correctly set and spelled in the project file and that the *search_path* variable is correctly set to find that file.

**See Also**

- analyze_rail

---

## RAIL-1083

(error) %s is not supported in this version of PrimeRail. Please set rail_indesign_shell_mode variable to '%s'.

**Description**

You are seeing this error because you tried to perform an operation or specify an option that is not supported.

Feedback

### What Next

There is nothing to do.

### See Also

- analyze_rail

---

## RAIL-1084

(error) db %s unit size cannot be zero. Aborting.

### Description

There is a user settings problem or preconditions for an operation are not met.

### What Next

Correct the user settings problem or satisfy the preconditions for the operation and try again.

### See Also

- analyze_rail

---

## RAIL-1085

(error) %s is not supported in this version of PrimeRail.

### Description

You are seeing this error because you tried to perform an operation or specify an option that is not supported.

### What Next

There is nothing to do.

### See Also

- analyze_rail

---

## RAIL-1086

(error) Decap insertion: %s not found. Please set by set_rail_options -%s lib_cell_list.

Feedback

### Description

There is a user settings problem or preconditions for an operation are not met.

### What Next

Correct the user settings problem or satisfy the preconditions for the operation and try again.

### See Also

- analyze_rail

---

## RAIL-1087

(error) Error cell

### Description

The file cannot be found in the given path.

### What Next

Check that the file is correctly set and spelled in the project file and that the *search_path* variable is correctly set to find that file.

### See Also

- analyze_rail

---

## RAIL-1088

(error) Parsing error on line '%s' in PrimeRail config file '%s'. %s.

### Description

You receive this message because an error occurs when the operation is scanning through the input files. It has found either a syntax error or an unsupported construct.

### What Next

Fix the syntax error or work around the unsupported construct.

### See Also

- analyze_rail

Feedback

## RAIL-1089

(error) %s is not set.

**Description**

There is a user settings problem or preconditions for an operation are not met.

**What Next**

Correct the user settings problem or satisfy the preconditions for the operation and try again.

**See Also**

- analyze_rail

## RAIL-1090

(error) %s file

**Description**

An unexpected failure has occurred.

**What Next**

This is an internal error.

**See Also**

- analyze_rail

## RAIL-1091

(error) Strategy '%s' is not %s since it is not set.

**Description**

An unexpected failure has occurred.

**What Next**

This is an internal error.

**See Also**

- analyze_rail

Feedback

## RAIL-1092

(error) The %s value '%s' which is set to the net '%s' in the value of the -supply_voltage option is not a valid float point value.

### Description

There is a user settings problem or preconditions for an operation are not met.

### What Next

Correct the user settings problem or satisfy the preconditions for the operation and try again.

### See Also

- analyze_rail

## RAIL-1093

(error) This line in %s file has incorrect %s ==>

### Description

You receive this message because an error occurs when the operation is scanning through the input files. It has found either a syntax error or an unsupported construct.

### What Next

Fix the syntax error or work around the unsupported construct.

### See Also

- analyze_rail

## RAIL-1094

(error) %s only works when analysis_mode is

### Description

An unexpected failure has occurred.

### What Next

This is an internal error.

Feedback

**See Also**

- analyze_rail

---

## RAIL-1095

(error) Unable to calculate %s due to missing db %s unit.

**Description**

An unexpected failure has occurred.

**What Next**

This is an internal error.

**See Also**

- analyze_rail

---

## RAIL-1096

(error) Unable to load db %s unit. Aborting.

**Description**

An unexpected failure has occurred.

**What Next**

This is an internal error.

**See Also**

- analyze_rail

---

## RAIL-1097

(error) Unable to verify %s file

**Description**

An unexpected failure has occurred.

**What Next**

This is an internal error.

Feedback

**See Also**

- analyze_rail

---

## RAIL-1098

(warning) Unknown %s.

**Description**

An unexpected failure has occurred.

**What Next**

This is an internal error.

**See Also**

- analyze_rail

---

## RAIL-1099

(error) Unrecognized %s.

**Description**

An unexpected failure has occurred.

**What Next**

This is an internal error.

**See Also**

- analyze_rail

---

## RAIL-1100

(error) Failed to run analyze_rail! -- %s.

**Description**

RedHawk cannot be launched successfully.

**What Next**

Check the setting and the error message after dash line.

Feedback

**See Also**

- analyze_rail

---

## RAIL-1101

(error) Command %s is required for option %s.

**Description**

Command setting is wrong.

**What Next**

Add the required command setting.

---

## RAIL-1102

(error) Failed to check set_host_options! -- %s.

**Description**

Failed to submit a job by using this set_host_options setting.

**What Next**

Check the set_host_options setting and the error message after dash line.

---

## RAIL-1103

(error) %s does not support submitting jobs to specific machines.

**Description**

This feature does not support submitting jobs to specific machines. Please try to submit jobs to farms, e.g., LSF or GRD.

**What Next**

Set set_host_options to farms like LSF or GRD.

---

## RAIL-1104

(error) Accessing this attribute requires RedHawk sign-off license!

**Description**

You receive this message because a RedHawk sign-off license feature is not available.

Feedback

**See Also**

- analyze_rail

## RAIL-1105

(error) Cell %s is not physically connected to net %s.

**Description**

report_rail_minimum_path could not trace a minimum resistive path from a cell to the specified net if the cell is not physically connected to the net, or the cell is not included in rail analysis (for example, filler cells).

**See Also**

- report_rail_minimum_path
- analyze_rail

## RAIL-1106

(info) Session taps changed!

**Description**

A change to session taps caused by *open_rail_result* or *analyze_rail* is detected. *open_rail_result* and *analyze_rail* re-populate session taps from rail results.

**See Also**

- open_rail_result
- analyze_rail

## RAIL-1107

(error) Operation not allowed when background job is running.

**Description**

Following operations aren't allowed when background job is running: 1. analyze_rail 2. change rail database 3. open_rail_result

Feedback

**See Also**

- analyze_rail
- open_rail_result

---

## RAIL-1108

(warning) Warning: Check missing via rule with valid threshold dropped because voltage drop analysis not performed!

**Description**

*set_missing_via_check_options* settings with unspecified or positive threshold require voltage drop analysis.

*set_missing_via_check_options* settings with negative threshold do not require voltage drop analysis.

You are seeing this warning because there are *set_missing_via_check_options* settings with unspecified or positive threshold, but voltage drop analysis is not being performed.

**NEXT STEPS**

1. Re-issue *set_missing_via_check_options* with negative threshold. This will return all missing via errors without the voltage drop analysis dependency.
2. Re-issue *analyze_rail* with *-voltage_drop*. This will take into account the voltage drop analysis result but will take more time to generate. Also, this may result in fewer missing via errors due to filtering.

The default threshold 1mV is used if *set_missing_via_check_options* is called without specifying *-threshold*.

The default threshold is positive, hence, will require voltage drop analysis.

If *analyze_rail* is invoked with *-check_missing_via*, but without *set_missing_via_check_options*, a minimal *mesh vias -report_missing* will still be generated in the run script.

**See Also**

- analyze_rail
- set_missing_via_check_options
- report_missing_via_check_options

Feedback

## RAIL-1109

(error) %s is not a valid net for app-option %s.

**Description**

The net specified for the aforementioned app-option does not exist in the database.

**What Next**

Please check the net name and it's validity in the database.

## RAIL-1110

(error) %s is not a valid die name for app-option %s.

**Description**

The die name specified for the aforementioned app-option does not exist in the database.

**What Next**

Please check the die name and it's validity in the database.

## RAIL-1111

(error) %s is not a valid voltage value for app-option %s.

**Description**

The voltage value specified for the aforementioned app-option is not valid.

**What Next**

Please check the voltage value.

## RAIL-1112

(error) %s is not a valid die.

**Description**

The die specified does not exist in the database.

**What Next**

Please check the die name.

Feedback

## RAIL-1113

(error) %s.

**Description**

Errors are found during the processing of the command options.

**What Next**

Please check the manpage of the command to correct any errors related to option settings.

## RAIL-1114

(error) failed to parse %s.

**Description**

The specified PRE_ANA_SUPPLY_LOG file either doesn't exist or is of an incorrect format.

Expected format: <DIE_INSTANCE> <DIE_MASTER> <RAIL_DATABASE> <RAIL_RESULT>

**What Next**

Point app-option rail.database toward RAIL_DATABASE with PRE_ANA_SUPPLY_LOG file with the expected format.

## RAIL-1115

(warning) %s(%s) cannot find voltage waveform.

**Description**

Cannot find voltage waveform for reported pin in boundary_condition file.

**What Next**

Check if RAIL_DATABASE contains file matching pattern *boundary_condition* and it contains voltage waveform for reported pin.

## RAIL-1116

(error) Found %d tokens, but expect %d tokens in %s.

Feedback

**Description**

Self-heat result file has an incorrect number of tokens.

**What Next**

Set app-option rail.technology to the correct technology node.

---

## RAIL-1117

(error) %s is required if %s in the command set_host_options for distributed mode.

**Description**

Command setting is wrong for distributed mode.

**What Next**

Add the required command setting for distributed mode.

---

## RAIL-1118

(warning) Nonzero sum current in region: Region: %s, Current: %f

**Description**

set_block_power per-net current specifications do not sum to zero.

**What Next**

Check if set_block_power per-net current specifications sum to zero for each region (power nets having positive current and ground nets having negative current). Also check if nominal voltages are available from top. If not, it may be necessary to specify app-option rail.supply_net_voltage.

---

## RAIL-1119

(warning) rail.enable_save doesn't work when rail.allow_redhawk_license_checkout set to %s.

**Description**

rail.enable_save requires rail.allow_redhawk_license_checkout set to true.

**What Next**

Set rail.allow_redhawk_license_checkout to true and try again.

Feedback

## RAIL-1120

(error) %s might be disconnected to ideal voltage.

**Description**

For effective voltage drop, std-cells must have valid PG arc. At least one power pin and one ground pin connect to ideal voltage.

**What Next**

Check PG pins of cells if they have been connected well.

## RAIL-1121

(error) Cannot specify %s without %s.

**Description**

One command option requires another.

**What Next**

Refer to the manual page for this command for detailed information on valid options.

## RAIL-1122

(error) invalid option settings for "set_power_switch_resistance". %s.

**Description**

Errors are found during the processing of the command "set_power_switch_resistance".

**What Next**

Please check the manpage of the command to correct any errors related to option settings.

## RAIL-1123

(error) invalid option settings for "size_power_switches". %s.

**Description**

Errors are found during the processing of the command "size_power_switches".

Feedback

**What Next**

Please check the manpage of the command to correct any errors related to option settings.

---

## RAIL-1124

(error) invalid option settings for "reset_power_switch_resistance". %s.

**Description**

Errors are found during the processing of the command "reset_power_switch_resistance".

**What Next**

Please check the manpage of the command to correct any errors related to option settings.

---

# RESYN

---

## RESYN-001

(Error) Missing resynthesize setting for %s .

**Description**

This message indicates that the resynthesize setting is not valid. User needs to perform set_resynthesize_netlist_option first.

**What Next**

User needs to perform set_resynthesize_netlist_option

---

## RESYN-002

(Error) can't find specified shell %s.

**Description**

This message indicates that the shell is not executable.

**What Next**

User needs to double check set_resynthesize_netlist_option

Feedback

## RESYN-003

(Error) The specified shell %s is not recognizable.

**Description**

This message indicates that the shell is not valid synopsys tool. User needs to provide valid ICC/ICC2/DC/ICC2_LM shell.

**What Next**

User needs to double check set_resynthesize_netlist_option

## RESYN-004

(Error) Directory %s already exists. Please remove the directory and rerun.

**Description**

This message indicates that the working directory already exists. User needs to remove directory firstly.

**What Next**

User needs to remove resynthesize working directory

## RESYN-005

(Error) Design with physical hierarchy is not supported for resynthesize flow.

**Description**

Tool found design has physical hierarchy, which is not supported in resynthesize flow.

**What Next**

User needs to double check the design setting

## RESYN-006

(Error) Design with dummy power domain is not supported for resynthesize flow.

**Description**

Tool found design has dummy power domain, which is not supported in resynthesize flow.

**What Next**

User needs to double check the design setting

Feedback

## RESYN-007

(Error) Design has MV violations, abort resynthesize flow.

**Description**

Tool found design has MV violation.

**What Next**

User needs to clean up MV violation before resynthesize flow.

## RESYN-008

(Error) Could not find TLU+ %s, Consider adding -search_path switch

**Description**

Tool could not find valid TLU+ file in search path.

**What Next**

User needs to provide valid TLU+ file in search path.

## RESYN-009

(Error) Could not find TLU+ mapping file %s, Consider adding -search_path switch

**Description**

Tool could not find valid TLU+ mapping file in search path.

**What Next**

User needs to provide valid TLU+ mapping file in search path.

## RESYN-010

(Error) Could not find db file %s, Consider adding -search_path switch

**Description**

Tool could not find valid db file in search path.

**What Next**

User needs to provide valid db file in search path.

Feedback

## RESYN-011

(Error) Working directory %s does not exist.

**Description**

Tool could not find valid working directory.

**What Next**

User needs to provide valid working directory.

## RESYN-012

(Error) Script %s does not exist.

**Description**

Tool could not find valid resynthesize script.

**What Next**

User needs to provide valid resynthesize script.

## RESYN-013

(Error) Input file not found.

**Description**

Tool could not find valid input file.

**What Next**

User needs to provide valid input file.

## RESYN-014

(Error) Please create output directory first.

**Description**

Tool could not find valid output directory.

**What Next**

User needs to provide valid output directory.

Feedback

### RESYN-015

(Error) This .tbc can not be used outside DC or ICCII

**Description**

This .tbc can not be used outside DC or ICCII.

**What Next**

User needs to provide valid running shell.

### RESYN-016

(Error) Resynthesize flow failed at %s stage.

**Description**

This message indicates that the resynthesize flow failed at specific stage. User needs to double check setting and re-run this stage.

**What Next**

User needs to double check design setting and re-run the stage.

## ROPT

### ROPT-001

(error) Block %s has no physical data. route_opt will exit now!\n

**Description**

There are physical hierarchical blocks which have no physical data defined route_opt cannot run on such designs

**What Next**

Ensure that all blocks have physical data before running route_opt

### ROPT-002

(warning) SI analysis is not enabled for post-route optimization.

Feedback

### Description

SI analysis is usually enabled for post-route optimization for best correlation with signoff timing analysis. Please consider enabling it.

### What Next

Consider enabling SI before route_opt by doing "set_app_options -name time.si_enable_analysis -value true"

### See Also

- route_opt

---

## ROPT-003

(warning) repair shorts over macro is enabled after initial routing.

### Description

This is not recommended in route_opt because it may lead to increased runtime and significant detours on critical nets. Please consider disabling it.

### What Next

Consider disabling app option before route_opt by doing "set_app_options -name route.detail.repair_shorts_over_macros_effort_level -value off"

### See Also

- route_opt

---

## ROPT-004

(warning) Total %zu nets have their routing open (data: %zu nets, clock: %zu nets). This will cause problems for extraction, timing and post-route optimization. Some examples of open nets are listed below.\n

### Description

Open nets will not provide precise routing information to route_opt. If there are too many opened nets, the optimization may fail or give incorrect result.

### What Next

Review the design and use "route_eco" to reroute the opennets.

Feedback

**See Also**

- route_opt
- route_eco

---

## ROPT-005

(warning) No parasitic data is present for any scenarios.

**Description**

Parasitic data is required for post-route optimization. route_opt cannot be proceed without parasitics.

**What Next**

Configure the design correctly.

**See Also**

- route_opt

---

## ROPT-006

(warning) Many nets are not fully detailed routed. Some examples are listed below.

**Description**

route_opt expects a fully detail routed design. More than 15% of nets are not fully routed.

**What Next**

Please make the design fully detailed routed using command such as route_auto.

**See Also**

- route_opt
- route_auto

---

## ROPT-007

(error) Design has unresolved cell references..

Feedback

### Description

Post-route optimization can not take designs with unresolved cell references. Please check link_block message LNK-081 to understand problem and solve it before run route_opt.

### What Next

Configure the design correctly.

### See Also

- route_opt

---

## ROPT-008

(error) Design has unbound sites or vias.

### Description

Postroute optimization can not take designs with unbound sites or vias. Please check previous design logs, especially loading/linking message, to understand problem and solve it before run route_opt.

### What Next

Use the *report_unbound* command to identify the unbound sites and vias. To resolve the issue, either delete the unbound objects or bind the objects by replacing the technology file or using the *create_site_def* or *create_via_def* commands.

### See Also

- create_via_def
- create_site_def
- route_opt

---

## ROPT-009

(error) Design has no active scenarios.

### Description

Post-route optimization cannot use designs without any active scenario. Please use report_scenarios to check that at least one scenario is active, and at least one active scenario has either setup or hold analysis enabled.

Feedback

### What Next

User can solve the error by making one or more scenarios active using set_scenario_status {<scenario list>} -active true.

### See Also

- report_scenarios
- set_scenario_status
- route_opt

---

## ROPT-010

(error) Design has no active scenarios with setup or hold analysis enabled.

### Description

Post-route optimization cannot use designs without any setup or hold analyses enabled. Please use report_scenarios to check that at least one active scenario has either setup or hold analysis enabled.

### What Next

User can solve the error by setting setup and/or hold analyses to true in one or more active scenarios using set_scenario_status {<scenario list>} -setup true, and/or set_scenario_status {<scenario list>} -hold true.

### See Also

- report_scenarios
- set_scenario_status
- route_opt

---

## ROPT-011

(error) Power flow is enabled but no leakage power analysis is enabled.

### Description

Post-route optimization cannot use designs without leakage analysis enabled when power flow is enabled. Please use report_scenarios to check that at least one active scenario has leakage analysis enabled when using power flow.

Feedback

### What Next

User can solve the error by setting leakage analysis to true in one or more active scenarios using set_scenario_status {<scenario list>} -leakage true.

### See Also

- report_scenarios
- set_scenario_status
- route_opt

---

## ROPT-012

(error) Terminate the command since previous command has failed.

### Description

The command status is tracked during running some mega commands. There are failed commands before you see the message. It is unnecessary to continue.

### What Next

User can find messages telling which command failed like "XXX-opt command failed". User need to fix the failure and then continue.

### See Also

- route_opt
- clock_opt
- place_opt

---

## ROPT-013

(error) Design has no routing layer.

### Description

Post-route optimization cannot continue without routing layer info.

### What Next

User can solve the error by ensuring routing layer existence in the design.

Feedback

### See Also

- route_opt

---

## ROPT-014

(error) Metal fill is enabled within route_opt, but ICV executable path is not set properly.

### Description

When ropt.enable_metal_fill_within_ropt is on, metal fill will be enabled within route_opt, so ICV executable path must be set properly before route_opt command.

### What Next

User can solve the error by adding the ICV executable in the PATH.

### See Also

- signoff_create_metal_fill
- route_opt

---

## ROPT-015

(warning) The LVT percentage of the incoming netlist (%f%%) is already higher than the constraint value (%f%%). The LVT percentage in optimized netlist may not meet the constraint.

### Description

The LVT percentage of the incoming netlist is already higher than the LVT percentage constraint value. This command reduces LVT percentage without seriously degrading other QoR metrics. Therefore, result LVT percentage may still be higher than the constraint.

### What Next

Adjust LVT percentage constraint before running this command based on result LVT percentage, timing and power if needed. Or reduce the LVT percentage of the incoming netlist by tightening LVT percentage constraint for previous optimization commands.

### See Also

- set_multi_vth_constraint
- reset_multi_vth_constraint
- report_multi_vth_constraint

Feedback

## ROPT-016

(warning) Sanity check found an endpoint in hold violations with large clock skew.

### Description

Sanity check found an endpoint in hold violations with large clock skew. This check collects the worst path in each endpoint and checks if the ratio of clock skew to clock period exceeds the threshold. This ratio is used to determine whether there are potential problems with the current endpoint. If there are too many such endpoints, it will lead to excessive runtime or the hold not fixing.

### What Next

This warning message does not suppress the subsequent flow. If you suspect a problem with the hold timing clock skew related, we recommend that you combine the later provided details table, using *analyze_design_violations -type hold* and *report_timing -to -scenario* for further information. If proven to be caused by a clock skew issue, it is suggested checking the *synthesize_clock_trees* or *clock_opt* stage.

### See Also

- synthesize_clock_trees
- clock_opt
- report_timing

# RPGP

## RPGP-001

(warning) The hierarchical relative placement group '%s' will not be processed.

### Description

The command processes top level relative placement groups only so hierarchical relative placement groups given in input will not be processed.

### What Next

Please provide top level relative placement group of given group.

Feedback

### See Also

- create_rp_group
- add_to_rp_group
- report_rp_groups
- legalize_rp_groups

---

## RPGP-002

(error) Possible placement failure of relative placement group.

### Description

No enough area for the entire relative placement group to be placed together because of overlapping with fixed cells or placement obstruction. Note that the search region for the relative placement group in the core area is based on the value that you specify to the -move_effort option. The relative placement cells will be placed around blockages if possible. The relative placement group constraints might not be respected. Also, if an existing group is modified by adding or deleting objects, this message will be issued and a placement command or legalization command needs to be run.

### What Next

Bring up the GUI and check the placement results. Use the command *report_rp_groups* for finding any relative placement constraint violations.Re run any placement command to re place modified groups. If the failure is due to modification then either legalize_rp_groups or legalize_placement or any placement command needs to be run to replace the modified RP groups.

### See Also

- report_rp_groups
- create_rp_group
- set_rp_group_options

---

## RPGP-003

(warning) The relative placement group has not been placed yet.

### Description

Placement of relative placement group has not been done yet.

Feedback

### What Next

The relative placement group is either newly created or was ignored during placement. Please look at log for any ignore messages.

### See Also

- report_rp_groups
- create_rp_group
- set_rp_group_options

---

## RPGP-004

(warning) The '%s' alignment has not been respected for relative placement group.

### Description

The warning indicates failure of column alignment or row alignment of RP groups. This warning occurs when alignment specified for an RP group or any of its hierarchical RP groups has not been met. Pleas use -verbose option of *report_rp_groups* command for details of the alignment failure.

### What Next

You may open up the GUI and check the alignment of the RP groups. The specified alignment may not be met after placement because of overlap with fixed placement objects.

### See Also

- report_rp_groups
- create_rp_group
- set_rp_group_options

---

## RPGP-005

(error) The bbox set on relative placement group %s does not match with calculated bbox.

### Description

This error occurs when bbox calculated by *report_rp_groups* command does not match with bbox set on relative placement group.

Feedback

### What Next

This can happen when there is some BUG in tiling code. It can also happen where RP cells are getting optimized, but RP tiling is not being done. Please analyze and fix the issue.

### See Also

- legalize_rp_groups

---

## RPGP-006

(warning) The x_offset specified for the RP group %s has not been satisfied. The movement from the specified location is '%s'.

### Description

This warning message occurs because while processing a Relative Placement (RP) group, an *x_offset* was present, and the RP group could not be placed at the requested location.

### What Next

This is only a warning message. The movement may not be significant. If it is significant then it may be caused by obstructions/blockages around that area. In such a case please specify appropriate x_offset.

### See Also

- set_rp_group_options
- remove_rp_group_options

---

## RPGP-007

(warning) The y_offset specified for the RP group %s has not been satisfied. The movement from the specified location is '%s'.

### Description

This warning message occurs because while processing a Relative Placement (RP) group, an *y_offset* was present, and the RP group could not be placed at the requested location.

Feedback

### What Next

This is only a warning message. The movement may not be significant. If it is significant then it may be caused by obstructions/blockages around that area. In such a case please specify appropriate y_offset.

### See Also

- set_rp_group_options
- remove_rp_group_options

---

## RPGP-008

(error) Relative placement group '%s' is ignored during placement.

### Description

You receive this message because the relative placement group's estimated height or width (or both) is greater than that of core area or you anchored the group such that it cannot be placed inside the core area.

### What Next

Redo the floorplan to increase the core area or shrink the RP group.

### See Also

- remove_from_rp_group
- remove_rp_groups
- report_rp_groups
- create_rp_group

---

## RPGP-009

(warning) The option '%s' can not be set on hierarchical relative placement group '%s'.

### Description

The given option can be set on only top level relative placement group.

### What Next

Please provide top level relative placement group of given group.

Feedback

**See Also**

- create_rp_group
- set_rp_group_options
- report_rp_groups
- legalize_rp_groups

---

## RPGP-010

(warning) The keepout margin of relative placement group '%s' is overlapped with other relative placement group(s).

**Description**

This warning occurs when rp only keepout margin specified for an relative placement group has been overlapped with other relative placement group(s). The overlap of keepout margins is also considered a failure. Please use -verbose option of *report_rp_groups* command for details of the keepout margin failure for report on overlapped relative placement group(s).

**See Also**

- report_rp_groups
- create_rp_group
- set_rp_group_options

---

## RPGP-011

(info) The option '%s' will be ignored for hierarchical relative placement group '%s'.

**Description**

The given option is applicable for top level relative placement group only, so will be ignored for the hierarchical relative placement group being added.

**See Also**

- create_rp_group
- set_rp_group_options
- report_rp_groups
- legalize_rp_groups

Feedback

## RPGP-013

(warning) %s cells of the relative placement group %s are fixed.

### Description

This warning occurs because the physical status of some or all cells of the specified relative placement group is fixed. The partially fixed relative placement group will be ignored during placement/legalization and fully fixed relative placement group will not be touched during placement/legalization. Please use -verbose option of *check_rp_constraints* command to get the list of fixed cells.

### What Next

Please remove the fixed cells from relative placement group or remove the fixed physical status from the cells and rerun the command.

### See Also

- report_rp_groups
- create_rp_group
- set_rp_group_options

## RPGP-014

(warning) The even height cell of relative placement group '%s' is getting placed after multi column cell '%s' of next row.

### Description

This warning occurs because the even height cell of current row is getting placed after multi column cell of next row. The multi column cell is added in one of the previous columns and is part of current column too. The even height cell is illegal at original location so has to be moved up, but it overlaps with multi column cell at that location. To avoid this possible overlap, the even height cell moves after the multi column cell to get placed legally. There may be several reasons for even height cell ending up at illegal site row and one of them is conflicting location relative to some other even height cell of relative placement group. Please check man page for RPGP-022 for more details about conflicting location.

Feedback

**See Also**

- report_rp_groups
- create_rp_group
- set_rp_group_options

---

## RPGP-015

(warning) The anchored relative placement group '%s' overlaps with fixed object(s).

**Description**

This warning occurs because the specified anchored relative placement group may overlap with macro(s) or placement blockage(s) or other anchored relative placement group(s). This may lead to critical or non-critical failure of relative placement group. Please change the anchor location of the specified relative placement group. Please use -verbose option of *check_rp_constraints* command to get the list of overlapped figures.

**See Also**

- report_rp_groups
- create_rp_group
- set_rp_group_options

---

## RPGP-016

(warning) The relative placement group '%s' has only physical only cells.

**Description**

This warning occurs because the specified relative placement group has only physical only cells in it. This may lead to failure of relative placement group after placement.

**See Also**

- report_rp_groups
- create_rp_group
- set_rp_group_options

---

## RPGP-017

(warning) Relative placement group '%s' contains no cell.

Feedback

### Description

This warning occurs because the specified relative placement group has no cells in it and it will not be placed even if blockages are there in it.

### See Also

- report_rp_groups
- create_rp_group
- set_rp_group_options

---

## RPGP-018

(warning) The height of relative placement group '%s' is more than the height of voltage area or exclusive move bound.

### Description

This warning occurs because the height of specified relative placement group is greater than the height of the voltage area or exclusive move bound. Please use -verbose option of *check_rp_constraints* command to get the voltage area or exclusive move bound.

### See Also

- report_rp_groups
- create_rp_group
- set_rp_group_options

---

## RPGP-019

(warning) The width of relative placement group '%s' is more than the width of voltage area or exclusive move bound.

### Description

This warning occurs because the width of specified relative placement group is greater than the width of the voltage area or exclusive move bound. Please use -verbose option of *check_rp_constraints* command to get the voltage area or exclusive move bound.

Feedback

**See Also**

- report_rp_groups
- create_rp_group
- set_rp_group_options

---

## RPGP-020

(warning) The bounding box of the anchored relative placement group '%s' is going outside the voltage area or exclusive move bound.

**Description**

This warning occurs because the specified anchored relative placement group's bounding box is going outside the voltage area or exclusive Move Bound. This may lead to critical failure of RP group after placement. Please use -verbose option of *check_rp_constraints* command to get the voltage area or exclusive move bound.

**See Also**

- report_rp_groups
- create_rp_group
- set_rp_group_options

---

## RPGP-022

(warning) The even height cell '%s' in specified relative placement group '%s' has conflicting location.

**Description**

This warning occurs because the specified relative placement group contains double height (or even height) cell(s) in conflicting location(s). One of these even height cells in conflicting location may move up leading to row alignment failure.

**See Also**

- report_rp_groups
- create_rp_group
- set_rp_group_options

Feedback

## RPGP-023

(warning) The utilization of specified relative placement group '%s' is not honored.

### Description

This warning occurs because the utilization constraint of specified relative placement group is not honored after the placement. This can occur if there is any fixed cell inside the relative placement group area forcing the relative placement objects in a column to move to avoid overlap.

### See Also

- report_rp_groups
- create_rp_group
- set_rp_group_options

## RPGP-024

(warning) The specified option '%s' of relative placement group '%s' is not set.

### Description

This warning occurs because user is attempting to remove a constraint which is not set on the specified relative placement group.

### See Also

- report_rp_groups
- create_rp_group
- set_rp_group_options

## RPGP-025

(info) The estimated height is '%s' and estimated width is '%s' for relative placement group '%s".

### Description

This message prints the estimated height and width of the relative placement group during coarse placement.

Feedback

**See Also**

- create_rp_group
- set_rp_group_options
- report_rp_groups
- legalize_rp_groups

---

## RPGP-026

(warning) The cell of relative placement group '%s' is getting placed after multi row cell '%s' during horizontal compression.

**Description**

This warning occurs because the cell of relative placement group is getting placed after multi row cell of subsequent column. The multi row cell is added in one of the previous rows and is part of current row too. This happens when width of relative placement objects before multi row cell in current row is more compared to placed row of multi row cell.

**See Also**

- report_rp_groups
- create_rp_group
- set_rp_group_options

---

## RPGP-027

(warning) The '%s' is not supported with horizontal compression for the specified relative placement group '%s'.

**Description**

This message specifies that the utilization or alignment type are not supported with horizontal compression. They are supported only with bit slice tiling type.

**See Also**

- create_rp_group
- set_rp_group_options
- report_rp_groups
- legalize_rp_groups

Feedback

## RPGP-028

(error) The relative placement object(s) of the specified relative placement group are not placed in its expected location.

### Description

This message specifies that the relative placement object(s) in specified relative placement group are not placed in expected location after tiling. This error occurs if the specified relative placement object(s) is moved before/below the previous relative placement object(s) in its row/column. This error will be issued if the previous relative placement object(s) in the row/column is moved next/above the specified relative placement object(s). This error will also be reported if two or more relative placement object(s) overlap with each other. Please use -verbose option of report_rp_groups command for details of the location movement failure.

### See Also

- create_rp_group
- set_rp_group_options
- report_rp_groups
- legalize_rp_groups

## RPGP-029

(warning) The specified %s '%s's block '%s' and the relative placement group '%s's block '%s' are not same.

### Description

This error message occurs when the *add_to_rp_group* command detects that you tried to add an object, whose block is not same as the relative placement group's block. An object can be added to the relative placement group only if its block and relative placement group's block are same.

### See Also

- add_to_rp_group
- create_rp_group
- remove_from_rp_group

Feedback

## RPGP-030

(error) The right override alignment of relative placement object is not supported since its relative placement group '%s''s alignment type is pin.

### Description

This error message occurs when the *add_to_rp_group* command detects that you tried to set right override alignment for the relative placement object when the alignment type of relative placement group is pin.

### See Also

- add_to_rp_group
- create_rp_group
- remove_from_rp_group

## RPGP-031

(Warning) The cell '%s' is not placed.

### Description

This error message occurs when the *create_clock_rp_groups* command detects that the sequential cell is not placed. The command will not be able to create relative placement groups for clock tree sinks if all sequential cells are not placed.

### See Also

- create_rp_group
- create_placement

## RPGP-032

(info) The %s '%s' will be renamed to '%s' because of the naming conflict.

### Description

This information message specifies that there is a relative placement group/blockage name conflict during uncommitting of block. The relative placement group/blockage name will be prefixed with the uncommitted block's name.

Feedback

### See Also

- create_rp_group
- set_rp_group_options
- report_rp_groups
- legalize_rp_groups

---

## RPGP-033

(warning) The relative placement group '%s' in the block '%s' will not be committed because it has partial cells from hierarchical instance '%s'.

### Description

The message specifies that the relative placement group will not be copied to the committed block because it has only partial cells belonging to the committed block's hierarchical instance. The relative placement group will stay in its current block excluding the relative placement cells committed to the physical hierarchy. For a relative placement group to get fully committed to committed block, all the cells of this relative placement group should belong to the committed physical block.

### See Also

- add_to_rp_group
- create_rp_group
- remove_from_rp_group

---

## RPGP-034

(warning) The relative placement group %s is not completely inside of hierBoundary for module instance %s, but it has members that belong to that module instance.

### Description

During the command commit_blocks, a relative placement group was encountered which did not lie completely inside a hierBoundary or partition, but had cell members that belonged to that module instance. Please note that the placement information of this relative placement group is removed in the committed block.

Feedback

### See Also

- add_to_rp_group
- create_rp_group
- remove_from_rp_group

---

## RPGP-035

(warning) The '%s' of anchored relative placement group %s is completely outside the hierBoundary for module instance %s, but it has members that belong to that module instance.

### Description

During the command commit_block, the anchored relative placement group was encountered which has x_offset/y_offset that does not lie inside the hierBoundary or partition, but had cell members that belonged to that module instance. Please note that the placement information of this anchored relative placement group is removed in the committed block.

### What Next

Please set the anchor point of relative placement group inside hierBoundary/partition.

### See Also

- add_to_rp_group
- create_rp_group
- remove_from_rp_group

---

## RPGP-036

(warning) The right override alignment cannot be set on multi column relative placement cell '%s'.

### Description

This error message occurs when the *add_to_rp_group* or *set_attribute* command detects that you tried to set right override alignment on the multi column relative placement cell. The right alignment is not supported for multi column relative placement cell.

Feedback

**See Also**

- add_to_rp_group
- create_rp_group
- remove_from_rp_group

---

## RPGP-037

(warning) The Relative Placement group '%s' is already placed.

**Description**

This error message occurs when the *recover_rp_placement* command detects that you tried to recover the placement information of the Relative Placement group which is already placed. We can recover the placement information of the Relative placement group only when it is not placed and relatively placed.

**See Also**

- add_to_rp_group
- create_rp_group
- remove_from_rp_group

---

## RPGP-038

(Warning) The retrieved placement information for relative placement group '%s' may be diferent from original placement.

**Description**

This information message occurs when the *recover_rp_placement* command detects that you tried to recover the placement information of the relative placement group whose relative placement objects are not relatively placed. Hence the retrieved placement information may not be exactly same as it was in *Design Compiler* or *IC Compiler*.

**See Also**

- add_to_rp_group
- create_rp_group
- remove_from_rp_group

Feedback

## RPGP-039

(warning) The relative placement group '%s' is not being legalized because it does not belong to the current block.

### Description

This error message occurs when the *legalize_rp_groups* command detects that the relative placement group being legalized is present in a different block. The relative placement group(s) can be legalized only from their block.

### See Also

- add_to_rp_group
- create_rp_group
- remove_from_rp_group

## RPGP-040

(warning) The duplicate entry of %s '%s' is not being processed.

### Description

This error message occurs when the command detects that the duplicate entry of relative placement group or relative placement object is provided in the input. The relative placement groups from multiply instantiated blocks are also not allowed.

### See Also

- add_to_rp_group
- create_rp_group
- remove_from_rp_group

## RPGP-041

(warning) The anchor_corner set on the relative placement group '%s' may lie outside the core area when it's defined.

### Description

This warning message occurs when the *set_rp_group_options* command detects that the user tried to set the anchor_corner on the relative placement group, whose core area has not been defined yet.

Feedback

**See Also**

- add_to_rp_group
- create_rp_group
- remove_from_rp_group

---

## RPGP-042

(error) The values of both anchor_row and anchor_column of the relative placement group should not be zero.

**Description**

This error message occurs when the *set_rp_group_options* command detects that the values provided for both *anchor_row* and *anchor_column* are zero for *-anchor_corner* rp_location.

**What Next**

Please change the *anchor_corner* to bottom_left or change the value of either *anchor_row* or *anchor_column*.

**See Also**

- add_to_rp_group
- create_rp_group
- remove_from_rp_group

---

## RPGP-043

(error) The anchor_corner rp_location specified for the relative placement group should be provided with both anchor_row and anchor_column options.

**Description**

This error message occurs when the *set_rp_group_options* command detects that both *anchor_row* and *anchor_column* options are not provided for the relative placement group with *anchor_corner* rp_location.

Feedback

**See Also**

- add_to_rp_group
- create_rp_group
- remove_from_rp_group

---

## RPGP-044

(error) The anchor_row or anchor_column option(s) are not allowed for non rp_location anchor_corner of the relative placement group.

**Description**

This error message occurs when the *set_rp_group_options* command detects that the user has specified the *anchor_row* or *anchor_column* option(s) for the non rp_location *anchor_corner* of the relative placement group.

**See Also**

- add_to_rp_group
- create_rp_group
- remove_from_rp_group

---

## RPGP-045

(error) The anchored relative placement location for rp_location anchor_corner specified using anchor_row and anchor_column is empty for relative placement group '%s'.

**Description**

This error message occurs when the *set_rp_group_options* command detects that the anchored relative placement location is empty for rp_location *anchor_corner*. The relative placement location specified using *anchor_row* and *anchor_column* cannot be empty for anchoring the relative placement group.

**See Also**

- add_to_rp_group
- create_rp_group
- remove_from_rp_group

Feedback

## RPGP-046

(info) The anchor_corner for relative placement group '%s' is removed because the anchored relative placement location for rp_location anchor_corner is empty.

### Description

This error message occurs when the *modify_rp_groups* command detects that after modification of relative placement group structure, the anchored relative placement location is empty for rp_location *anchor_corner*. The *anchor_corner* of the relative placement group is removed because the relative placement location specified in command *set_rp_group_options* using *anchor_row* and *anchor_column* options cannot be empty for anchoring the relative placement group.

### See Also

- add_to_rp_group
- create_rp_group
- remove_from_rp_group

## RPGP-047

(Error) The relative placement group '%s' cannot be created because given design is read only.

### Description

This error message occurs when the command *create_rp_group* detects that given design through option *-design* is read only and relative placement group can not be created.

### See Also

- create_rp_group
- add_to_rp_group

## RPGP-048

(Error) The relative placement group '%s' cannot be created because design view is not loaded.

Feedback

### Description

This error message occurs when the command *create_rp_group* detects that given design through option *-design* is not loaded as design view, so relative placement group can not be created. The other view types are abstract view, frame view etc.

### See Also

- create_rp_group
- add_to_rp_group

---

## RPGP-049

(Warning) The relative placement group '%s' is not placed.

### Description

This error message occurs when unplaced relative placement group is detected during congestion aware coarse placement. The congestion aware coarse placement is done by *refine_placement* command or by *create_placement -congestion*. Please note that cells of such relative placement groups may move large distance.

### What Next

Please check why relative placement group was unplaced and getting rid of reason may make the group placed. If unplaced realative placement group is getting placed during congestion aware placement then please make sure desired placement is being achieved.

### See Also

- create_rp_group
- create_placement
- refine_placement

---

## RPGP-050

(Error) The number of given min_sinks is more than number of given max_sinks.

### Description

This error message occurs when the *create_clock_rp_groups* command detects that the number of minimum sinks given using option *-min_sinks* is more than max sinks given using *-max_sinks*.

Feedback

**See Also**

- create_rp_group
- create_placement

---

## RPGP-051

(Warning) The cell '%s' is being ignored because it is '%s'.

**Description**

This error message occurs when the *create_clock_rp_groups* command ignores some cell and does not consider it for clustering (i.e. fixed cell). If the cell is a driver then all its sinks are also not clustered.

**See Also**

- create_rp_group
- create_placement

---

## RPGP-052

(info) The attribute '%s' is not applicable for top level relative placement group '%s'.

**Description**

The given attribute is applicable for hierarhical relative placement group only, so will not be processed for top level relative placement group.

**See Also**

- create_rp_group
- set_rp_group_options
- report_rp_groups
- legalize_rp_groups

---

## RPGP-053

(Warning) The relative placement group '%s' is not being processed because it is unplaced and has hard macros.

Feedback

### Description

This warning message occurs when relative placement group is found to be unplaced during coarse placement or legalization and it has added hard macro cells.

### What Next

The relative placement group should be placed during floorplanning using *create_placement -floorplan* command.

### See Also

- create_rp_group
- create_placement
- legalize_placement
- legalize_rp_groups

---

## RPGP-054

(Warning) The hard macro cell '%s' is not fixed.

### Description

This warning message occurs when hard macro cell is detected as unfixed during coarse placement or legalization. This warning is given for hard macro cells which are part of relative placement group.

### What Next

The hard macros which are part of relative placement group are expected to be fixed after design planning. Please mark all relative placement hard macros fixed at start of placement and routing flow.

### See Also

- create_rp_group
- create_placement
- legalize_placement
- legalize_rp_groups

---

## RPGP-055

(error) The site def information for '%s' '%s' is missing.

Feedback

### Description

This error message indicates that the site def information is not available for the reported cell reference or site row. This error message can occur either during placement, legalization or design planning.

### What Next

You can check the site def information using either of the commands *get_site_defs* or *report_site_defs*. If no site def is present, you can add site def information and rerun the command.

### See Also

- get_site_defs
- report_site_defs

---

## RPGP-056

(error) The '%s' attribute cannot be modified for the relative placement free placement cell '%s'.

### Description

This error occurs when you try to modify either the *rp_override_alignment* or *rp_pin_name* attribute on a relative placement free placement cell. These attributes cannot be set or removed from a relative placement free placement cell.

### See Also

- add_to_rp_group
- remove_from_rp_group

---

## RPGP-057

(error) The '%s' option can be used only with '-free_placement' or '-blockage' option.

### Description

This error message occurs when the *add_to_rp_group* command detects that you used the *-height* or *-width* option without either '-free_placement' or '-blockage'. The *-height* and *-width* options are not valid with *-rp_group* option. They are also not valid with *-cells* if '-free_placement' option is not provided.

Feedback

### What Next

Please see the man page of *add_to_rp_group* command for the correct combination of options.

### See Also

- add_to_rp_group
- remove_from_rp_group

---

## RPGP-058

(error) The area provided for the relative placement free placement cells is not sufficient to place all the specified Cells.

### Description

This error message occurs when the *add_to_rp_group* command detects that the area provided by you for the relative placement free placement cells not sufficient to place the specified relative placement free placement cells.

### What Next

Please increase the height or width of the free placement region provided for the relative placement free placement cells or remove some of the cells from the list and rerun the command.

### See Also

- add_to_rp_group
- remove_from_rp_group

---

## RPGP-059

(warning) The '%s' '%s' cell cannot be added as a relative placement free placement cell.

### Description

This error message occurs when the *add_to_rp_group* command detects that you are adding a hard macro or a physical only cell to relative placement free placement region.

### What Next

Please remove any hard macro or physical only cells from the list of specified free placement cells.

Feedback

### See Also

- add_to_rp_group
- remove_from_rp_group

---

## RPGP-060

(warning) Some of the relative placement free placement cells in the relative placement group are not placed in the free placement region.

### Description

This message occurs when the *report_rp_groups* command detects that relative placement free placement cells of the either the top or hierarchical relative placement group have been placed outside the free placement region defined for them. Pleas use -verbose option of *report_rp_groups* command for details of this failure. This failure may happen because of overlap with fixed placement objects.

### What Next

Please increase the height or width of the free placement region provided for the relative placement free placement cells and rerun the create_placement command.

### See Also

- add_to_rp_group
- report_rp_groups
- remove_from_rp_group

---

## RPGP-061

(error) The anchored relative placement location for rp_location anchor_corner specified using anchor_row and anchor_column contains relative placement free placement cells.

### Description

This error message occurs when the *set_rp_group_options* command detects that the anchored relative placement location for rp_location *anchor_corner* contains relative placement free placement cells. The relative placement location specified using *anchor_row* and *anchor_column* cannot have relative placement free placement cells.

### What Next

Pleae remove the anchor_corner from the relative placement locations containing relative placement free placement cells and rerun the command.

Feedback

### See Also

- set_rp_group_options
- add_to_rp_group
- remove_from_rp_group

---

## RPGP-062

(warning) The '%s' provided for the relative placement free placement cells is less than the maximum '%s' of relative placement free placement cells.

### Description

This warning message occurs when the *add_to_rp_group* command detects that the height or width provided by you for the relative placement free placement cells is less than the maximum height or width of the specified free placement cells.

### What Next

Please increase the height or width of the free placement region provided for the relative placement free placement cells.

### See Also

- add_to_rp_group
- remove_from_rp_group

---

## RPGP-063

(error) The '-pin_name', '-override_alignment' and '-orientations' options cannot be used with '-free_placement' option.

### Description

This error message occurs when the *add_to_rp_group* command detects that you used the one of *-pin_name* or *-override_alignment* or *-orientations* options with '-free_placement'. These options cannot be used while adding relative placement free placement cells. not provided.

### What Next

Please see the man page of *add_to_rp_group* command for the correct combination of options.

Feedback

### See Also

- add_to_rp_group
- remove_from_rp_group

---

## RPGP-064

(warning) The cell of relative placement group '%s' is getting placed after multi row free placement cells during horizontal compression.

### Description

This warning occurs because the cell of relative placement group is getting placed after multi row free placement cells of subsequent column. The multi row free placement cells are added in one of the previous rows and is part of current row too. This happens when width of relative placement objects before multi row free placement cells in current row is more compared to placed row of multi row free placement cells.

### See Also

- report_rp_groups
- create_rp_group
- set_rp_group_options

---

## RPGP-065

(warning) The cell '%s' will not be legalized because all the cells of the relative placement group are not provided.

### Description

This warning message indicates that a cell was not legalized because it is relative placement cell and all the cells of the relative placement group are not provided to be legalized. The relative placement group moves as a whole unit and individual cells should not be legalized.

### What Next

Please provide all cells of the relative placement group. If you want to legalize individual cells only then please remove the relative placement group.

Feedback

### See Also

- create_placement
- legalize_rp_groups
- legalize_placement

---

## RPGP-066

(warning) The cell of relative placement group '%s' is getting placed after multi row free placement cells at row '%s' column '%s' during horizontal compression.

### Description

This warning occurs because the cell of relative placement group is getting placed after multi row free placement cells of subsequent column. The multi row free placement cells were added in one of the previous rows and are part of current row too. This happens when width of relative placement objects before multi row free placement cells in current row is more compared to placed row of multi row free placement cells.

### See Also

- report_rp_groups
- create_rp_group
- set_rp_group_options

---

## RPGP-067

(warning) The cell of relative placement group '%s' is getting placed after multi column cell '%s' during vertical compression.

### Description

This warning occurs because the cell of relative placement group is getting placed after multi column cell of subsequent row. The multi column cell is added in one of the previous columns and is part of current column too. This happens when height of relative placement objects before the multi column cell in current column is more compared to placed column of multi column cell.

Feedback

**See Also**

- report_rp_groups
- create_rp_group
- set_rp_group_options

---

## RPGP-068

(warning) The vertical compression tiling has not been respected for relative placement group '%s'.

**Description**

This warning occurs when vertical compression tiling has not been respected for the relative placement group. The gap between two objects in consecutive rows of a relative placement groups is considered as vertical compression failure.

**See Also**

- check_rp_constraints
- report_rp_groups
- set_rp_group_options

---

## RPGP-069

(warning) The cell of relative placement group '%s' is getting placed after multi column free placement cells at row '%s' column '%s' during vertical compression.

**Description**

This warning occurs when the cell of relative placement group is getting placed after multi column free placement cells of subsequent row. The multi column free placement cells were added in one of the previous columns and are part of current column too. This happens when height of relative placement object is higher than the object in same row of previous column.

**See Also**

- report_rp_groups
- create_rp_group
- set_rp_group_options

Feedback

## RPGP-070

(error) The region provided is not sufficient to perform the placement analysis for relative placement group '%s'.

### Description

This error message occurs during check_rp_constraints command when the size of the region provided by user is not large enough to perform placement analysis. The size of the region must be at least greater than the size of relative placement group.

### What Next

Please increase the size of the region provided.

### See Also

- check_rp_constraints
- report_rp_groups

## RPGP-071

(warning) The cell '%s' is illegal everywhere. The relative placement group '%s' may not be placed.

### Description

This warning message indicates that a cell is not legal on any site. The placement of parent relative placement group of this cell may fail.

### What Next

Please modify the library cell of the problematic cell to make it legal.

### See Also

- create_placement
- legalize_rp_groups
- legalize_placement

## RPGP-072

(error) The anchor_corner specified for the relative placement group '%s' is outside it's physical block '%s'.

Feedback

### Description

This error message occurs when the *set_rp_group_options* command detects that the user tried to set the anchor_corner on the relative placement group present in a physical block, outside of the defined core_area of the block.

### See Also

- add_to_rp_group
- create_rp_group
- remove_from_rp_group

---

## RPGP-073

(warning) The specified cell '%s' could not be added to the relative placement group '%s' because its reference library cell is missing.

### Description

This warning message occurs when the *add_to_rp_group* command detects that you tried to add a cell whose reference cell is not present in the library. A cell with missing reference library cell is not allowed to be added to a relative placement group. Please check the library for the missing reference library cell ('LNK-005' messages during library linking).

### See Also

- create_rp_group
- remove_from_rp_group

---

## RPGP-074

(Information) Checking the legality of '%s' relative placement cells which have been swapped during relative placmeent variant-aware legalization.

### Description

This information message occurs during relative placement variant-aware legalization when we check the legality of relative placement cells which have been swapped.

### See Also

- legalize_placement
- legalize_rp_groups

Feedback

## RPGP-075

(Information) Starting normal relative placement legalization to fix violations after variant-aware legalization.

### Description

This information message occurs when all the cells are not legal after relative placement variant-aware legalization and normal relative placement legalization is being run to fix them.

### See Also

- legalize_placement
- legalize_rp_groups

## RPGP-076

(Information) Relative Placement legalization is not supported for advanced legalization flow.

### Description

This information message occurs when legalize_rp_groups command is run with advanced legalization flow enabled. Relative Placement groups are not supported in advanced legalization flow.

### See Also

- legalize_placement

## RPGP-077

(Warning) RP cell %s have either top, bottom or both the keepouts which are not supported.

### Description

This warning message occurs when relative placement cell have either top, bottom or both the keepout as RP technology does not support these keepouts on RP cell.

## RPGP-078

(warning) Both anchor corner and site row values cannot be set on relative placement group %s.

Feedback

### Description

This warning message occurs when the *set_rp_group_options* command tries to set *-anchor_corner* and *-site_row* value on relative placement group.

### What Next

Please set both anchor corner and offset values for the relative placement group.

### See Also

- set_rp_group_options
- add_to_rp_group

# RPT

## RPT-001

(Warning) No paths found. Design has unmapped elements. This command works on mapped netlists only.

### Description

This message occurs while running the report_logic_levels command, when there are no paths found for analysis and the design has some unmapped elements. The command works only on mapped netlist.

### What Next

Please run compile command to map all the elements. If the command is already run, then try passing differnt arguments to the reporting command so that paths are found for analysis.

## RPT-002

(Warning) No paths found. Please supply slack options to include positive slack paths, if desired.

### Description

This message occurs while running the report_logic_levels command, when there are no paths found for analysis. By default, this command analyses only the -ve slack path paths.

Feedback

**What Next**

Invoke the reporting command with -slack_lesser_than & -slack_greater_than options to include +ve slack paths as well for analysis.

---

## RPT-003

(Warning) No paths matched analysis criterion. Please specify different set of options.

**Description**

This message occurs while running the report_logic_levels command, when there are no paths found for analysis.

**What Next**

Invoke the command with different set of options.

---

## RPT-004

(error) One of threshold value or reset flag must be specified.

**Description**

This message occurs while running the set_analyze_rtl_logic_level_threshold command without any of the required options.

**What Next**

Either specify the threshold value or the -reset option.

---

## RPT-005

(error) Both threshold value and reset flag cannot be specified.

**Description**

This message occurs while running the set_analyze_rtl_logic_level_threshold command, both threshold value and -reset flag is specified.

**What Next**

Specify only one of the the threshold value or the -reset option.

---

## RPT-006

(error) Invalid path group name %s supplied with -group option.

Feedback

### Description

This error message occurs if you specify a non-existent path group name when using the *-group* option with the *report_logic_levels* command.

### What Next

Specify a valid (existing) path group name when using the *-group* option and then run the *report_logic_levels* command again.

Path groups are created implicitly according to the clock of the destination register of a path. You can also define a path group with via the *group_path* command. To see a list of all valid path groups, issue the *report_path_group* command. Any of the path group names listed can be supplied with the *-group* option in the *report_logic_levels* command.

### See Also

- report_path_group

---

## RPT-007

(warning) value '%s' for app option '%s' is not valid. \n Valid values are '%s' . \n Setting value to '%s' :

### Description

The value given for the listed argument is not one of the limited allowable strings. Only valid strings are set for this app option. This messages lists all of the appropriate values.

### What Next

Supply a compatible value for the argument.

---

## RPT-008

(warning) Option

### Description

The option is in conflict with another one and therefore is ignored.

---

## RPT-181

(error) Errors occurred during write_qor_data while running the %s report.

Feedback

**Description**

*write_qor_data* captures data from tool reports (or other sources) into a structured table format. *compare_qor_data* later uses this table data to create the QORsum report. This error occurs when an unexpected problem occurs while *write_qor_data* was running or extracting data from the report named in the error message. In some cases, you may find clues to what went wrong by looking at the report file created in the *write_qor_data -output* dir: <output>/<label>/reports/<report_name>.rpt.

**What Next**

Report this problem to your Synopsys support team.

# RT

## RT-022

(error) %s has not been specified.

**Description**

The command failed since you did not specify necessary inputs. You may specify the inputs by the options of this command, or via another command prior to executing this command.

**What Next**

Specify necessary inputs and then execute this command again. Please refer to the man page of the command for more information.

## RT-036

(error) Layer '%s' does not exist in design.

**Description**

You receive this message because you used a layer that does not exist in design.

**What Next**

Check the layers from the design and use the right routing layer name.

## RT-048

(info) No DRC error is reported!

Feedback

**Description**

This information message notifies you that there is no DRC violation after running this command.

**What Next**

This is an informational message only. No action is required on your part.

---

## RT-050

(error) Failed in executing %s.

**Description**

Command has failed.

**What Next**

Please check the log file and correct any errors reported.

---

## RT-066

(warning) The '%s' via enclosure width is less than NDR width constraint of '%s' layer.

**Description**

You receive this message because the via enclosure width is less than NDR width constraint of the layer. Zroute may report "Less than NDR width" violation for these via enclosure.

**What Next**

Please increase NDR via enclosure to meet metal layer NDR width.

---

## RT-080

(info) %s

**Description**

As described in the message.

**What Next**

This is for your information.

Feedback

## RT-113

(warn) Net pattern did not match any nets.

**Description**

The net pattern specified did not match with any nets.

**What Next**

Verify that the net pattern specified is correct.

## RT-115

(warn) %s

**Description**

Basic input error checking.

**What Next**

Please recheck the command inputs and use the -help option to verify the input is correct.

## RT-116

(warn) Please set flip_chip.route.design_style to %s style. %s is auto-selected for the current command.

**Description**

flip_chip.route.design_style value does not match the actual design style. Please set the correct value so that the following push_rdl_routes, optimize_rdl_routes or route_3d_rdl command works properly.

**What Next**

For route_rdl_flip_chip, please set flip_chip.route.design_style to flip_chip_peripheral_io. For route_3d_rdl, please set flip_chip.route.design_style to 3dic_single_die.

## RT-119

(error) %s

**Description**

An unexpected error has occured.

Feedback

**What Next**

Please check the log file and correct any errors reported.

---

## RT-120

(error) %s

**Description**

Basic input error checking.

**What Next**

Please recheck the command inputs and correct any errors reported in the log file.

---

## RT-201

(error) Cells are not placed.

**Description**

This error message indicates that check_lvs stops checking because many cells are not been legally placed.

**What Next**

Please make sure all the cells are placed before running check_lvs.

---

## RT-202

(warn) Pin shape %s is skipped by check_lvs because its area is 0. Skip this pin.

**Description**

This warning message indicates that pin shape is skipped by check_lvs because its area is 0.

**What Next**

Please check your design.

---

## RT-203

(warn) Port %s have no valid pin shapes. Skip this port.

**Description**

This warning message indicates that port has no valid pin shapes. Skip this port.

Feedback

**What Next**

Please check your design.

---

## RT-204

(warn) Net %s has less than 2 valid port. Skip open checking for this net.

---

## RT-205

(warn) Via %s has no via_def. Skip this via.

**Description**

This warning message indicates that via has no via_def. Skip this via.

**What Next**

Please delete the problematic via or provide a replacement tech file. Please refer to create_via_def.

---

## RT-301

(error) The input spacing "%.3f" of layer "%s" is smaller than the minimum spacing "%.3f" defined in techfile.

**Description**

The input spacing value cannot violate the minimum spacing rule defined in techfile.

**What Next**

Please correct your input.

---

## RT-302

(error) This command only support design with rectangular boundary.

**Description**

Rectlinear or polygon design boundary is not supported in this command.

**What Next**

Please use create_routing_guide -rdl_routing to create the route guides

Feedback

## RT-303

(error) "nonDefaultRDLRegionMinSpacing" value is not defined for layer "%s" in the techfile. Please provide the value by "-spacing" option or define it in the techfile.

**Description**

For RDL routing guides, the spacing value comes either from the "nonDefaultRDLRegionMinSpacing" defined in the layer section of the techfile, or input by the "-spacing" options. Please use either way to specify the spacing value of the routing guides.

**What Next**

Please specify the spacing vaule.

## RT-304

(error) Input %s vaule %.3f is larger than half of the block width or half of the block height. Unable to create rdl routing guides using this value.

**Description**

For RDL routing guides, the -boundary_distance and -corner_distance are measured from for side of the block boundary. So the vaules need to be smaller than half of the block width and half ot the block height to create vaild rdl routing guide.

**What Next**

Please check your input and modify it.

## RT-305

(info) Remove existing RDL routing guide before creating new one.

**Description**

Remove existing RDL routing guides before creating new one to avoid ambiguty to the RDL router.

## RT-306

(warning) RDL routing guide "%s" has non-Manhattan and non-45-degree edge, which is not supported by RDL router. Ignore this routing guide.

Feedback

**Description**

RDL router only support routing guides with manhanttan or 45-degree edges. Routing guides with any angle edges are ignored during routing.

**What Next**

Please modify the routing guide.

---

## RT-307

(warning) RDL routing guide "%s" overlaps with RDL routing guide "%s" on layer "%s" with different associated spacing values. Ignore this routing guide to avoid confusion during routing.

**Description**

RDL routing guides with different associated spacing values should not overlap with each others. Such overllaped routing guides are ignored to avoid confusion during routing.

**What Next**

Please modify the region of the routing guides to avoid overlap issue.

---

## RT-308

(error) The input spacing "%.3f" of layer "%s" is smaller than the nonDefaultRDLRegionMinSpacing spacing "%.3f" defined in techfile.

**Description**

The input spacing value cannot violate the nonDefaultRDLRegionMinSpacing spacing rule defined in techfile.

**What Next**

Please correct your input.

---

## RT-309

(error) The input -boundary_distance vaule "%.3f" is larger than the -corner_distance vaule "%.3f", which is an invalid input.

**Description**

To form a vaild rdl routing guide, the -boundary_distance vaule has to be smaller or equal to -corner_distance value.

Feedback

**What Next**

Please correct your input.

---

## RT-310

(warning) The input "%s" vaule is not the multiple of minimum grid. It is snap up to the nearest on grid value.

**Description**

The side length of the routing guides needs to be the multiple of minimum grid. Otherwise, the off-grid value might cause issue during routing. Tools automatically snap up off-gird value to the nearest on-grid number.

**What Next**

Please check your input.

---

## RT-311

(Warning) Pin %s can not be found, skipping it.

**Description**

Please check if pin setting correctly.

---

## RT-502

(warn) %s

**What Next**

Please recheck the command inputs and make sure the design is verified.

---

## RT-505

(error) Cannot find contact information for the %d layer.

**Description**

An error related to contact information occurred in the design.

**What Next**

Check the via layer or contact code section in the technology file to make sure that the vias and contact codes are correctly defined.

Feedback

## RT-511

(warn) Net %s is not a valid net for this router. Skip this net.

**Description**

This net is not identified as an RDL net.

**What Next**

Please check if the flip-chip IO pad cell design type is set as 'pad' or 'flip_chip_driver' Please check if the bump cell design type is set as 'flip_chip_pad' If the net is a multiple IO pad pins to multiple bumps net, user needs to run place_io first to determine the IO pad pin to bump connections If one IO pad pin is connected to more than 4 bumps, user needs to set app option flip_chip.route.routing_style to spanning_tree style to route this IO pad pin

## RT-513

(warning) Net %s {(%.3f %.3f), (%.3f %.3f)}, corresponding cells {%s}, would have potential cross with %d nets, {%s}.

**Description**

The net would have potential cross with current net assignment.

**What Next**

Please fix the problem by swap cell before routing.

## RT-516

(warn) No routable RDL nets found.

**Description**

Tool cannot find routable RDL nets in the design.

**What Next**

Please check the following things: 1. Please check if the given net names are correct. 2. Please check if the flip-chip IO pad cell design type is set as 'pad' or 'flip_chip_driver' 3. Please check if the bump cell design type is set as 'flip_chip_pad' 4. If the net is a multiple IO pad pins to multiple bumps net, user needs to run place_io first to determine the IO pad pin to bump connections 5. If one IO pad pin is connected to more than 4 bumps, user needs to set app option flip_chip.route.routing_style to spanning_tree style to route this IO pad pin

Feedback

## RT-518

(warn) %s on net '%s' is blocked at (%.3f %.3f), (%.3f %.3f). Net cannot be routed.

**Description**

Routing point of the net is blocked at the given location. Router cannot route the net.

**What Next**

Please check the blocked net at given location.

## RT-519

(warn) Multi-pin net

**Description**

Multi-pin nets will not be routed with this command.

**What Next**

Multi-pin nets will not be routed with this command. Please disconnect unnecessary ports or use route_tsv_nets to route back-side bump-to-TSV power/ground nets, and Template-based PNS to create power/ground mesh for each power domain.

## RT-520

(warn) Net %s contains more than one bump cell set as connect-only. Router skips the net.

**Description**

There are more than one bump cell set as connect-only in the net. Router will skip routing for this net.

**What Next**

Please use report_flip_chip_bump_attributes to see the settings on the bump cells. Then modify your design by set_flip_chip_bump_attributes -connect_only (true|false) to set only one bump cell as connect-only in one net.

## RT-521

(error) Too many routing layers.

Feedback

**Description**

With -auto_via true, there should be no more than 2 routing layers.

**What Next**

Please specify only 1 or 2 routing layers or set -auto_via false.

---

## RT-522

(error) Routing layers are not adjacent.

**Description**

With -auto_via true, the routing layers should be adjacent.

**What Next**

Please specify adjacent routing layers or set -auto_via false.

---

## RT-523

(warn) No existing global routes on the design.

**What Next**

Please run global routing first.

---

## RT-524

(error) Failed to open child cell %s.%s.

**Description**

This warning message indicates that the router was not able to open the child cell for read. This could be because the reference library where the child cell resides does not have read permission or is missing.

**What Next**

Please make sure the reference library is linked and readable to the current design.

---

## RT-525

(warn) Cannot push, optimize or split an open net %s.

**Description**

The net to push, optimize or split is not detail routed.

Feedback

## RT-526

(error) Cannot push EEQ or multi-terminal nets with -mode neighbor without specifying -terminal_names.

**Description**

The net to be pushed or optimized is an EEQ or multi-terminal net. Please specify -terminal_names to selectively push the routes connected to the specified driver pins or EEQ pins.

## RT-527

(error) Cannot push more than one net with -mode neighbor.

## RT-528

(warn) Delete global routes in the design.

**Description**

All global routes in the design are deleted after command optimize_rdl_route, push_rdl_route and split_rdl_route.

## RT-529

(error) Route shape of net %s could not be split. Net width is %.3f, but it is requested to split into %d nets with width of %.3f and spacing %s in between.

**Description**

Fail to spilt route shape since the width is not wide enough.

**What Next**

Please check your width and spacing settings, and modify it by set_route_flip_chip_options. Please make sure that (width before spilt) >= 2*(width after split) + (spacing).

## RT-532

(warning) Net %s contains off-grid segments.

**Description**

Please check the segments of this net.

Feedback

**What Next**

Please fix all off-grid segments.

---

## RT-533

(warn) NDR via %d setting is useless and will be ignored.

**Description**

NDR via setting is useless and will be ignored for the follwing reason 1. This via can not be found in technology file. 2. This via array size is incorrect.

**What Next**

Please check your setting if it is correct.

---

## RT-535

(warn) The via of net %s has a spacing violation with a blockage.

**Description**

The via has a spacing violation with a metal/cut blockage or route guide.

**What Next**

Check the design and follow these instructions to correct any errors:

1. Check the net spacing setting in the technology file.
2. Check the net spacing rule in the nondefault routing rule.
3. Set the view level to 1 and turn on the visibility of route guides and system blockages. Make sure that nothing blocks the resource.

---

## RT-536

(warn) not able to create stack via for net %s

**Description**

There is no routing resource to create stack via for the target net.

Feedback

**What Next**

Check the design and follow these instructions to correct any errors:

1. Check the net spacing setting in the technology file.
2. Check the net spacing rule in the nondefault routing rule.
3. Set the view level to 1 and turn on the visibility of metal, pin, route guides and system blockages. Make sure that nothing blocks the resource.

---

## RT-537

(error) No shielding net is found.

**Description**

There is no shielding net in design.

**What Next**

Use set_route_rdl_options -shielding_net to specify shielding net.

---

## RT-538

(warn) Routes of net %s contain shapes of different width. Skip the route for splitting.

**Description**

Routes that contain shapes of different width cannot be split in order to prevent open nets. These routes are skip by the splitting command.

**What Next**

Please check if the net has tapering setting that makes the shapes vary in width.

---

## RT-539

(error) Please specify no more than %d nets.

**Description**

Too many nets are specified.

**What Next**

Please specify less nets.

Feedback

## RT-540

(warning) Net %s is previously shielded on layer %s. Skip creating shields on this net.

## RT-541

(warning) Net %s has no shielding rules defined or its shielding width defined on layer %s is 0. Skip this net.

## RT-542

(warning) Net %s has 45 degree routed shapes, which cannot be pushed, optimized or split in 90_degree mode. %s

## RT-545

(warning) Differential group parallel mode routing failed due to insufficient routing space or bad net assignment.

**Description**

Tool requires sufficient routing space and good net assignment to perform differential group parallel mode routing.

Please check that there is sufficient routing space, especially between bump pin and bump pin. For net assignment, please check that there is no flyline crossing before routing and the pins are not far away from each others.

**What Next**

Please make sure that there is sufficient routing space or re-assign the nets.

## RT-546

(error) Layer %s: The orginal width of net %s is %.3f. The route cannot be split into routes of required width and spacing.

**Description**

When -widths and -spacings options are both given, the summation of widths and spacings should equal to the oringal route width.

**What Next**

Please adjust the value of -widths and -spacings options.

Feedback

## RT-547

(warn) Cannot split net %s, since it is already split by another mode.

**Description**

A net which is already split by same_layer mode cannot be split by adjacent_layer mode, and vice versa.

## RT-548

(warn) For net %s, the width of split wire on layer %s is smaller than the rule width on layer %s.

## RT-549

(warn) Cannot split via for net %s.

**Description**

Cannot split via due to failing to find suitable small via/via array to drop on the split route.

**What Next**

Please check the split route setting and the via sizes aviable.

## RT-550

(warn) Net %s contains split vias which are marked as User Enter type, which might cause incorrect result.

**Description**

Net %s contains vias which are split by split_rdl_route -split_via command before. But now these vias are marked as UserEnter type, which might cause incorrect result.

**What Next**

If the result is incorrect, undo the command if possible and mark the vias as Signal Route. Otherwise, mark the vias as Signal Route, rip-up reroute the net and redo the split command. Then redo your command.

## RT-551

(warn) place_io guidance for net %s does not exist.

Feedback

**Description**

The tool cannot honor place_io guidance because it does not exist.

**What Next**

Make sure place_io is applied before routing, or

apply the routing command without "-honor_place_io_guidance".

---

## RT-553

(warning) No shielding rules are set to RDL nets.

---

## RT-558

(warn) Remove extension shapes with violations in the area{{%.3f %.3f} {%.3f %.3f}}.

**Description**

Remove extension shapes.

**What Next**

Please check the log file and correct any errors reported.

---

## RT-562

(warn) Tapering setting does not work for net %s on layer %s.

**Description**

Taper distance must be larger than half of NDR width of Net.

**What Next**

Please check if the net has correct tapering setting.

---

## RT-563

(warn) extend_vias_on_pin setting does not work for tappering net %s.

**Description**

extend_vias_on_pin setting does not work for tappering net.

**What Next**

Please check if the net has tapering setting.

Feedback

## RT-564

(warn) extend_vias_on_pin setting does not work for connect_pad_without_endcap net %s.

**Description**

extend_vias_on_pin setting does not work for connect_pad_without_endcap option.

**What Next**

Please check the option connect_pad_without_endcap setting.

## RT-567

(warn) NDR via {%s %dx%d %s} does not satisfy fat metal contact rule.

**Description**

You receive the message because the NDR via does not satisfy fat metal contact rule. Zroute may have unsolvable DRCs while using the NDR via.

**What Next**

Check fat metal contact rule and remove the NDR via from -via_cuts or -cuts option in non-default routing rule setting.

## RT-568

(warn) Frame or design view for block {%s} not found.

**Description**

You receive the message because the frame or design view for the child block can not be found. The router will skip this child block.

**What Next**

Make sure $search_path includes the directory of the reference library. Make sure the frame or design view is in the reference library. If not, regenerate the reference library.

## RT-569

(warn) Gui is not running. Option -display does not work.

**Description**

The option -display will sjohow flyline on GUI. The application GUI must start.

Feedback

**What Next**

Please start the application GUI.

---

## RT-570

(error) Net %s not found.

**Description**

Failed to find the net in the current design.

**What Next**

Check the spelling of the net name. Make sure net of that name exists.

---

## RT-571

(warning) Multi-bump connection on net %s is not supported for length matching routing.

**Description**

Multi-bump connection is not supported for length matching routing. Normal routing will be performed on the connection.

**What Next**

Modify the net assignment so that every connection contains only one bump.

---

## RT-572

(warning) Unable to honor connect_edge_center on bump %s.

---

## RT-584

(warning) %s.frame does not exist.

**Description**

This message indicates that check_lvs cannot find frame cell for a design in physical library.

**What Next**

If not having frame view for this design is intended, ignore this warning. If frame view is expected for this design, please check the designs in the library and man page of LNK-055 for more details.

Feedback

## RT-585

(info) Detected open violation for Net %s. %s

**Description**

As described in the message.

**What Next**

This is for your information.

## RT-586

(info) Detected short violation. %s. %s. %s

**Description**

As described in the message.

**What Next**

This is for your information.

## RT-587

(info) Detected floating route violation for Net %s. %s

**Description**

As described in the message.

**What Next**

This is for your information.

## RT-588

(info) Detected floating pin violation for Net %s. %s

**Description**

As described in the message.

**What Next**

This is for your information.

Feedback

## RT-589

(Warn) Does not meet the value of flip_chip.route.number_of_ring_connections due to space limitation. (Bump %s)

**Description**

As described in the message.

**What Next**

This is for your information.

## RT-590

(Info) Auto computed river route bounding box {%.3f %.3f} {%.3f %.3f}.

**Description**

If the river routing coordinates are not given from input, tool automatically computes the routing bounding box by all the pins/vias of the target nets on the routing layer. Note that for river routing, all pins/vias need to fall on one of the routing bounding box edge. If any pin/via lie inside the routing bounding box, then it becomes floating and cannot be routed using the river routing methodology.

**What Next**

This is for your information.

## RT-591

(error) The input coordinates form disjoint regions which are not suitable for river routing. Please correct your input.

**Description**

River routing need to be perform in a single routing region with all pins/vias lying on the region boundaries. If the the input regions are disjoint, river routing cannot be performed.

**What Next**

Please correct your input.

## RT-592

(warn) Net '%s' has pin/via at (%.3f, %.3f) which is not lying on the routing boundary. This pin/via cannot be routed using river route.

Feedback

**Description**

River routing needs to be perform in a single routing region with all target pins/vias lying on the region boundaries. Please either adjust the vias position to make them lie on the boundary, or provide a routing boundary which touches all the target pins/vias by the option -coordinates.

**What Next**

Please correct your input.

---

## RT-593

(warn) There are %d stripe shape parts of net %s in the bbox (%.3f %.3f) (%.3f %.3f). Stripe to stripe routing is skipped for the bbox.

**Description**

The tool finds the intersection parts of the stripe shapes and the bbox as routing targets. And stripe-to-stripe routing can only connect two stripe shape parts. If there are more than two intersection parts, then RDL router will skip routing in this bbox.

**What Next**

Please correct your input bbox to overlap with only two stripe shapes.

---

## RT-594

(error) There are invalid input nets for this router. Stop command under -strict mode.

**Description**

Under -strict mode, the router will not route if any invalid input nets exist.

**What Next**

Please see RT-511 for valid RDL net definition. If you want to skip the invalid nets and route the valid nets, please set option '-strict' as 'false'.

---

## RT-595

(warning) flip_chip.route.bump_via_landing_mode is ignored in non-3DIC style routing.

**Description**

flip_chip.route.bump_via_landing_mode is designed only for 3DIC flow.

Feedback

**What Next**

Please set flip_chip.route.design_style to 3DIC-related value (3dic_single_die or 3dic_interposer).

---

## RT-596

(warning) Net '%s' is not found in P2P file '%s'. Skipped.

**Description**

The P2P file should contain at least one entry for each target net, or the net is skipped.

**What Next**

Please check the P2P file and ensure there is at least one entry for the target net.

---

## RT-597

(warning) Net %s is previously shielded on layer %s within the region specified in -coordinates or -create_shield_coordinates. Skip creating shields on this net.

---

## RT-598

(warning) Ignore option -offset in side-wall shield mode.

---

## RT-599

(warning) keep_existing_route option only works with bump_via_landing_mode. Ignore the setting.

**Description**

The "keep_existing_route" option only works when "bump_via_landing_mode" option is also set as true. Otherwise, the "keep_existing_route" setting will be ignored and use the default value.

**What Next**

Please check "report_app_option flip_chip.route.bump_via_landing_mode".

---

## RT-600

(warning) Ignore RDL routing guides in this command.

Feedback

**Description**

RDL routing guide is not supported in this comamnd. It is only supported in route_rdl_flip_chip, push_rdl_routes, optimize_rdl_routes.

---

## RT-601

(warning) Point (%s, %s) is not on grid. Skip routing point pair (%s, %s) - (%s, %s) for net %s on layer %s.

**Description**

For point-to-point routing, all the specified points in the input files need to be on grid. Router can only route to on-grid points. All the off-grid points will be skiped in point-to-point routing.

**What Next**

Please correct your input.

---

## RT-602

(warning) Point (%.3f, %.3f) is neither a wire end point nor falls inside a pin/via shape of the same net. Skip routing point pair (%.3f, %.3f) - (%.3f, %.3f) for net %s on layer %s.

**Description**

For point-to-point routing, all the specified points are either wire end points or fall inside pin/via shapes of the same net. All the points not following the rule above will be skiped in point-to-point routing. Please note that the "shape_use" of the pre-placed wires need to be set as "user_route" before calling the router. Otherwise, they will be removed.

**What Next**

Please correct your input.

---

## RT-603

(warning) Cannot find same net via of bump pin '%s'. No routing for this bump.

**Description**

For bump_via_landing_mode routing, a pre-placed same net via is required. Please set "shape_use" as "user_route". Otherwise, router will treat it as dangling shape and remove it before routing.

Feedback

**What Next**

Please pre-place landing via for this bump pin.

---

## RT-604

(warning) Point (%.3f, %.3f) is too close too another same net shape. Skip routing point pair (%.3f, %.3f) - (%.3f, %.3f) for net %s on layer %s.

**Description**

For point-to-point routing, if the point is a wire end-point and very close to another same net shape so that putting a unit wire on this point will directly touch that shape, then the routing to this point is disabled.

**What Next**

Please extend the length of this wire shape to make the routing point away from the other shapes.

---

## RT-605

(warning) Ignore option -apply_new_sequence when point_to_point_connection_file_name is not specified"

**Description**

Option -apply_new_sequence only works for point-to-point routing, Ignore this optoin when it is not point to point routing.

---

## RT-606

(warning) Ignore option -half_shield for coaxial shield mode"

**Description**

Option -half_shield is not supported in coaxial shield mode.

---

## RT-607

(warning) Ignore app option flip_chip.route.route_to_user_route_shapes when flip_chip.route.point_to_point_connection_file_name is set.

Feedback

## RT-608

(Info) Skip creating shields for net %s since the net direction does not match -half_shield direction.

## RT-609

(warning) Ignore app option "flip_chip.route.auto_coordinate_for_bump_via_landing_mode" when command option "-coordinates" is set.

**Description**

Command option -coordinates has higher priority than app option "flip_chip.route.auto_coordinate_for_bump_via_landing_mode".

## RT-610

(warning) App option "flip_chip.route.auto_coordinate_for_bump_via_landing_mode" only goes with app option "flip_chip.route.bump_via_landing_mode".

**Description**

The auto coordinate calculation only works on bump-via landing mode.

**What Next**

Please set app option "flip_chip.route.bump_via_landing_mode" as true first.

## RT-611

(warning) App option "flip_chip.route.auto_coordinate_for_bump_via_landing_mode" is not applicable for this command.

**Description**

The auto coordinate calculation is only works on route_rdl_flip_chip, route_3d_rdl, optimize_rdl_routes commands.

## RT-612

(info) App option "flip_chip.route.design_style" is automatically set to "%s" to match the RDL routing command %s.

Feedback

**Description**

flip_chip.route.design_style is automatically set to match the RDL routing command. The correct value need to be set so that the following push_rdl_routes, optimize_rdl_routes command works properly.

**What Next**

If the automatically set design style does not match current design, please set the app option "flip_chip.route.design_style" correctly before calling RDL routing command.

---

## RT-613

(warn) Routing pair (%.3f %.3f) (%.3f %.3f) of net '%s' on layer '%s' is not inside routing region specified by -coordinates option. Skip this pair.

**Description**

With -coordinates option specified, routing is only allowed in the specified region. Thus, any points in the point-to-point routing file that are not inside the specified routing region are not reachable for the router. These point pairs are skipped.

**What Next**

Resolve the conflict between -coordinates option and the content of point-to-point routing file from the app option "flip_chip.route.point_to_point_connection_file_name".

---

## RT-614

(warning) flip_chip.route.extend_spanning_tree_style_routing option only works with flip_chip.route.routing_style as "spanning_tree". Ignore the setting.

**Description**

The "extend_spanning_tree_style_routing" option only works when "routing_style" option is set as "spanning_tree". Otherwise, the "extend_spanning_tree_style_routing" setting will be ignored.

**What Next**

Please check "report_app_option flip_chip.route.routing_style".

---

## RT-615

(warning) The origin (%.4f %.4f) of cell %s is not on minimum grid.

Feedback

**Description**

Off minimum grid shapes cause unexpected issue in RDL routing, thus RDL router snaps the off-gird shapes to gird to avoid issues. The snap on min-grid process is to expand the oringinal shape to fit to the minimum grid. The expanded shapes occupy more routing resource, thus the routing might fail in region where the routing resources are tight.

**What Next**

Please fix the off-grid issue manually to avoid routing issues.

---

## RT-616

(warning) Library cell %s has geometry on layer %s which is not on minGrid..

**Description**

Off minimum grid shapes cause unexpected issue in RDL routing, thus RDL router snaps the off-gird shapes to gird to avoid issues. The snap on min-grid process is to expand the oringinal shape to fit to the minimum grid. The expanded shapes occupy more routing resource, thus the routing might fail in region where the routing resources are tight.

**What Next**

Please fix the off-grid issue manually to avoid routing issues.

---

## RT-617

(warning) Option -post_shield_fill_empty_space and -extend_to_pg_structure are ignored in non-3DIC style routing.

**Description**

Option -post_shield_fill_empty_space and -extend_to_pg_structure are designed only for 3DIC flow.

---

## RT-618

(warning) Option flip_chip.route.secondary_shielding_net is ignored in non-3DIC style routing.

**Description**

Option flip_chip.route.secondary_shielding_net is designed only for 3DIC flow.

Feedback

## RT-619

(error) The routing region (from -coordinates option) is not allowed to be smaller than the desgin data read-in region (from application option flip_chip.route.data_in_coordinates). Please correct your input.

**Description**

The routing region, input from -coordinates options, needs to be smaller than the desgin data read-in region, input from application option flip_chip.route.data_in_coordinates. Otherwise, design rule violations cannot be detected correctly.

**What Next**

Please correct either the -coordinates option value or the application option flip_chip.route.data_in_coordinates value.

## RT-620

(warning) Ignore user-defined group setting on port %s since must_join_group is also set.

## RT-621

(warning) Terminal %s is not assigned to one and only one user-defined group. check_lvs will ignore user-defined group setting for port %s.

## RT-622

(Error) No valid pins to be routed.

**Description**

Among all the input pins, there is no any valid PG pins that can be routed.

**What Next**

Check the pin list to ensure there is at least one PG pin.

## RT-623

(Error) stripe layers are not specified or valid.

**Description**

The stripe layer is either not specified or not routable.

Feedback

**What Next**

Check the stripe layer setting.

---

## RT-624

(Error) No valid routing layer combinations.

**Description**

The routing layer combination is not routable given the stripe layer.

**What Next**

Check the routing layer setting, and make sure that the routing direction of stripe layer, main-trunk layer, sub-trunk layer is either horizontal, vertical, horizontal or vertical, horizontal, vertical.

---

## RT-625

(Warning) App option %s is disabled in RDL router.

**Description**

App option is disabled in RDL router when one of minWidthForConcaveCorner, concaveConvexMinEdgeLength, convexConvexMinEdgeLength, and maxLength is set in tech file for the routing layer.

---

## RT-626

(Warning) Ignore minWidthForConcaveCorner, concaveConvexMinEdgeLength, convexConvexMinEdgeLength, and maxLength rules in this command.

---

## RT-627

(Warning) Detected bad input pattern for optimize_rdl_routes in net %s. The result might be not correct after optimization.

---

## RT-628

(Warning) Ignore app option flip_chip.route.bumps_to_honor_bump_spacing_only on layer %s, since app option flip_chip.route.layer_bump_spacings is not specified.

Feedback

**Description**

Please specify flip_chip.route.layer_bump_spacings if you would like to honor bump spacing.

---

## RT-629

(Warning) Ignore bumpRegionRDLDistance, bumpRegionRDLAllowedSpacing, bumpRegionRDLAllowedWidth, and bumpRegionRDLMaxAdjRouteCount rules on layer %s since one of them is not specified.

**Description**

Please review your tech file settings.

---

## RT-630

(Warning) Ignore bumpRegionRDLDistance, bumpRegionRDLAllowedSpacing, bumpRegionRDLAllowedWidth, and bumpRegionRDLMaxAdjRouteCount rules on layer %s since routing angle is 90-degree.

**Description**

Please review the routing angle settings.

---

## RT-631

(Warning) Ignore bumpRegionRDLDistance, bumpRegionRDLAllowedSpacing, bumpRegionRDLAllowedWidth, and bumpRegionRDLMaxAdjRouteCount rules on layer %s due to invalid bump region settings.

**Description**

The invalid bump region might be caused by smaller bumpRegionRDLDistance or disjoint bump regions. Please review your tech file settings or design.

---

## RT-632

(Warning) Ignore bumpRegionRDLDistance, bumpRegionRDLAllowedSpacing, bumpRegionRDLAllowedWidth, and bumpRegionRDLMaxAdjRouteCount rules on layer %s because there are some shapes inside bump region.

**Description**

Please clean up the bump region before routing.

Feedback

## RT-633

(Warning) Skip contact %s for incorrect lower_layer_name and upper_layer_name.

**Description**

Please check your tech file.

## RT-634

(Error) Skip this command since bumpRegionRDLDistance, bumpRegionRDLAllowedSpacing, bumpRegionRDLAllowedWidth, and bumpRegionRDLMaxAdjRouteCount rules are set.

**Description**

Please check if the tech file is correct set.

## RT-635

(Warning) Ignore bumpRegionRDLDistance, bumpRegionRDLAllowedSpacing, bumpRegionRDLAllowedWidth, and bumpRegionRDLMaxAdjRouteCount rules on layer %s because rectilinear bumps are detected.

**Description**

Only round / octaongal bump are supported to bumpRegionRDLDistance, bumpRegionRDLAllowedSpacing, bumpRegionRDLAllowedWidth, and bumpRegionRDLMaxAdjRouteCount rules.

## RT-636

(Warning) Ignore bumpRegionRDLDistance, bumpRegionRDLAllowedSpacing, bumpRegionRDLAllowedWidth, and bumpRegionRDLMaxAdjRouteCount rules on layer %s because the channel width between bump to bump is too large.

**Description**

The channel width between bump to bump should be only allowed bumpRegionRDLMaxAdjRouteCount routes to pass.

## RT-637

(Warning) Ignore bumpRegionRDLDistance, bumpRegionRDLAllowedSpacing, bumpRegionRDLAllowedWidth, and bumpRegionRDLMaxAdjRouteCount rules on layer

Feedback

%s because bumpRegionRDLAllowedSpacing/bumpRegionRDLAllowedWidth is larger than minSpacing/minWidth.

**Description**

bumpRegionRDLAllowedSpacing/bumpRegionRDLAllowedWidth should be less than minSpacing/minWidth. Please review your tech file.

---

## RT-638

(Warning) Routing width of net %s on layer %s does not meet maxWidth/minWidth constraints.

**Description**

Please check if the tech file is correct set.

---

## RT-639

(Warning) Invalid pin %s(%s) for routing layer %s.

**Description**

Please check if routing layers are set correctly or the pin is a valid routing pin.

---

## RT-640

(Warning) Pin %s does not belong to a PG Net, skipping it.

**Description**

Please check if pin setting correctly.

---

## RT-641

(warn) Net '%s' specified in option '%s' is not found and is ignored.

**Description**

Failed to find the net in the current design.

**What Next**

Check the spelling of the net name. Make sure net of that name exists.

Feedback

## RT-642

(info) There is only one group for route_rdl_differential pattern match mode. Use normal routing flow in which the tool only routes one group, but more than two nets can be in the group..

**Description**

For single differential group, route_rdl_differential pattern match mode uses the normal routing flow. The normal routing flow tries more solution, and the routing pattern might differ from the pattern using group routing flow.

**What Next**

Check the routing result.

## RT-643

(info) There are %d groups for route_rdl_differential pattern match mode. Use group routing flow in which the tool uses the group routing flow to route the nets.

**Description**

For multiple differential group, route_rdl_differential pattern match mode uses the group routing flow. This flow routes all groups together, and the routing pattern might differ from the pattern using normal routing flow.

**What Next**

Check the routing result

## RT-644

(warning) More than %d nets are specified in group '%s'. This group is skipped in routing.

**Description**

Too many nets are specified in the group.

**What Next**

Specify less nets in a group.

## RT-645

(warning) Only %d nets are specified in group '%s'. This group is skipped in routing.

Feedback

**Description**

Too less valid nets are specified. The minimum is two nets.

**What Next**

Specify more nets in a group.

---

## RT-646

(warning) Fail to create parallel points for %s.

**Description**

The parallel points are created at the middle point of the pins of the net group. If there is no sufficient space at the middle point, for example, creating the parallel point causes DRC violation, or there is no feasible solution that can be found. In such cases, the parallel points cannot be created.

**What Next**

Check the routing resource at the middle point of the pins of the net group. You can do this by turning on the GUI and measure the distance between bumps/pins/blockages/existing routes, and check the width/spacing setting, to understand that the routing resource is not enough at the middle point to create parallel points.

---

## RT-647

(warning) %s: fail to route from parallel start to endpoints.

**Description**

Fail to route the parallel part for the group. Each parallel group nets have a parallel startpoint and a parallel endpoint. The routing goes in parallel from the parallel startpoint to the parallel endpoint.

**What Next**

Check the routing resource and net assignment.

---

## RT-648

(warning) Group %d: net %s fail to route from parallel point to pin.

**Description**

Fail to route from parallel point to pin.

Feedback

**What Next**

Check the routing resource and net assignment. You can do this by turning on the GUI and measure the distance between bumps/pins/blockages/existing routes, and check the width/spacing setting, to understand that the routing resource is not enough at the middle point to create parallel points.

---

## RT-650

(warning) Tried to create parallel start/end point at (%.4f %.4f) but failed due to DRC violation.

**Description**

Creation of parallel start/end points failed due to DRC violation. Please check if there are any other shapes (including shapes in frame view), blockages at or too close to the creation location. Only limited locations are tried when creating the parallel start/end points.

---

## RT-651

(warning) There is only %d valid net in group '%s'. This group is skipped in routing.

**Description**

Please check if the other input nets in this group are valid nets or duplicate in other groups.

**What Next**

Correct the group input.

---

## RT-652

(warn) No routable differential RDL nets found.

**Description**

Tool cannot find routable differential RDL nets.

**What Next**

Set the routing nets by: 1. Specified by command option -nets or -nets_in_file 2. Specified by app option flip_chip.route.differential_groups

If nets are specified by above method, check if the input net is valid by 1. Check if the given net names are correct. 2. Check if the flip-chip IO pad cell design type is set as 'pad' or 'flip_chip_driver' 3. Check if the bump cell design type is set as 'flip_chip_pad' 4. If

Feedback

the net is a multiple IO pad pins to multiple bumps net, user needs to run place_io first to determine the IO pad pin to bump connections

---

## RT-653

(Warning) Number of %s density violations exceeds 10000 limitation.

**Description**

This message indicates that number of min/max density violations exceeds 10000 limitation.

**What Next**

Use check_rdl_route -check_window_density_mode (all|max|min|none) command to skip density violations checking.

---

## RT-654

(Warning) Option flip_chip.route.bump_via_landing_mode true is incompatible with -stripe_to_stripe_bboxs. Turn off this option for the current command.

---

## RT-655

(Warning) Discard a min cut rule in tech file for not supporting area and extension contact request.

---

## RT-656

(Warning) add_large_vias_from_techfile command does not support such library.

**Description**

This message indicates the tech library cannot be modified by the command.

---

## RT-657

(Warning) Skipping add_large_vias_from_techfile. The value of app option route.common.auto_dfm_redundant_via_tech_params is empty.

**Description**

This message indicates the app option route.common.auto_dfm_redundant_via_tech_params is empty.

Feedback

**What Next**

Use set_technology to setup app option value.

---

## RT-658

(Warning) Number of user_route shapes exceeds 1000000 limitation.

**Description**

This message indicates that number of user_route shapes exceeds 1000000 limitation.

**What Next**

Turn off check_from_user_shapes for check_rdl_routes.

---

# RTLA

---

## RTLA-001

(error) User script file %s does not exist.

**Description**

User script file specified by set_explore_design_options does not exist.

**What Next**

1. Create the script file, or 2. Use set_explore_design_options to reset script file.

---

## RTLA-002

(warning) %s file %s does not exist, set it as empty.

**Description**

The script file specified by set_explore_design_options does not exist, set it as empty so the flow can continue.

**What Next**

Create the script file reset script file using set_explore_design_options command.

---

## RTLA-003

(error) Power script %s does not exist.

Feedback

**Description**

Power shell set by command set_rtl_power_analysis_options does not exist.

**What Next**

Use command set_rtl_power_analysis_options to set correct power shell.

---

## RTLA-004

(warning) Power script %s does not exist,replaced by %s.

**Description**

Power shell set by command set_rtl_power_analysis_options does not exist, replaced it by pt_shell in Synopsys installation directory so the flow can continue.

**What Next**

Use command set_rtl_power_analysis_options to set correct power shell.

---

## RTLA-005

(error) Parameter -rtl_line has syntax error.

**Description**

The correct syntax for -rtl_line is {fileName lineNumber}, for example, get_metrics -rtl_line {/foo/cpu.v 121} -metric metrics_cong_number_cells Note, line number must be positive.

**What Next**

Correct the -rtl_line parameter.

---

## RTLA-006

(error) Cannot find RTL source file %s, line %d for %s metrics.

**Description**

Cannot find the source file with line number. There are two possible reasons: 1. The file and line number does not exist, or 2. The file and line number exists, but there is no timing or congestion metrics data for that line.

---

## RTLA-007

(error) Metric %s is not found, check if it is a supported metric.

Feedback

**Description**

The tool cannot find the specified metric as being a supported metric for the command.

**What Next**

Check if spelling of metric exactly matches the metric name. To see a list of metrics look at man page for *get_metrics*.

One special situation is that certain metrics do not have support for option *-local*. For congestion, the following metrics do not support option *-local*: metrics_cong_utilization, metrics_cong_shape_distortion, and metrics_cong_shape_dispersion.

For power metrics only the following power metrics are supported with *-local*: total_power, internal_power, switching_power, leakage_power, and glitch_power.

---

## RTLA-008

(error) Block %s has no boundary.

**Description**

This combination of command and options requires that the block have a boundary.

**What Next**

Use command initialize_floorplan to create a boundary.

---

## RTLA-009

(error) Memory configue file %s does not exist.

**Description**

Memory configure file specified by set_explore_design_options does not exist.

**What Next**

Use command set_explore_design_options to set correct memory configure file.

---

## RTLA-010

(error) Cannot write script %s because missing data in section: %s sub-section: %s key: %s.

**Description**

The memory configuration data is missing information needed to proceed. The key which is missing data is reported.

Feedback

### What Next

Use the command set_memory_configuration to add the needed information. Usually there are 2 ways to add the information. Either add the information directly into the key as specified in message, or set a default value. To set a default value, put the key/data into the section: memory_compiler_section.

---

## RTLA-011

(error) Failed to find required files from gui.verdi_indesign.verdi_home app option

### Description

To application failed to find the required files in the Verdi installation home directory.

The home directory should be specified in the gui.verdi_indesign.verdi_home app option or the VERDI_HOME environment variable.

### What Next

Check the value of the gui.verdi_indesign.verdi_home app option, or VERDI_HOME environment variable, points to a valid Verdi installation home directory.

### See Also

- gui.verdi_indesign.verdi_home

---

## RTLA-012

(error) Failed to find required files from gui.verdi_indesign.vcs_home app option

### Description

To application failed to find the required files in the VCS installation home directory.

The home directory should be specified in the gui.verdi_indesign.vcs_home app option or the VCS_HOME environment variable.

### What Next

Check the value of the gui.verdi_indesign.vcs_home app option, or VCS_HOME environment variable, points to a valid VCS installation home directory.

### See Also

- gui.verdi_indesign.vcs_home

Feedback

## RTLA-013

(error) Metric %s is not supported by RTL%s.

**Description**

The tool does not support usage of the metric passed for RTL (RTL cells, RTL modules and RTL constructs).

**What Next**

The following metrics are supported in this mode: %s

## RTLA-020

(Information) Register '%s' is a potential constant '%s' register.

**Description**

The mentioned register is detected to be a potential constant register.

**What Next**

Check the logic cone of the register.

## RTLA-021

(Information) Cell '%s' is unmapped because of reason code(s): '%s'.

**Description**

The mentioned cell is unmapped because of the given reason codes.

**What Next**

Check for the description of the reason codes in the man page of the command *report_rtl_unmapped_cells*.

## RTLA-300

(error) Option -inst_name is not valid in RTLA.

**Description**

Option -inst_name is not valid in RTLA.

**What Next**

Don't use this option.

Feedback

## RTLA-301

(error) Cannot find the result '%s' in the thermal database.

**Description**

The specified result name cannot be found in the current thermal database.

**What Next**

Specify the correct result name in the thermal database.

## RTLA-302

(Warning) There is no Power Profile to be removed.

**Description**

There is no Power Profile to be removed.

**What Next**

Check the Power Profile exists or not.

## RTLA-303

(Warning) There is no Thermal Profile to be removed.

**Description**

There is no Thermal Profile to be removed.

**What Next**

Check the Thermal Profile exists or not.

## RTLA-600

(warning) Could not determine a boundary for hierarchy %s.

**Description**

To compute certain metrics, it is necessary to have a boundary for a hierarchy. This boundary will encompass the cells in the hierarchy. For the given hierarchy, a boundary could not be determined. The most likely reason for this is that the hierarchy has no cells in it.

Feedback

### What Next

If having no cells is expected, then there is nothing to do. If it is expected that this hierarchy should have cells, then check to see why this hierarchy has no cells by looking at the verilog for this hierarchy.

---

## RTLA-601

(warning) Not able to get dispersion attribute on %s.

### Description

The computation to compute the dispersion metric on given hierarchy has failed and thus there is no dispersion attribute. The most likely cause of this is that the hierarchy has no cells in it.

### What Next

If having no cells is expected, then there is nothing to do. If it is expected that this hierarchy should have cells, then check to see why this hierarchy has no cells by looking at the verilog for this hierarchy.

If this hierarchy has cells, then look at log file to see if there are other messages with information on the failure.

---

## RTLA-602

(warning) Not able to get utilization attribute on %s.

### Description

The computation to compute the utilization metric on given hierarchy has failed and thus there is no utilization attribute. The most likely cause of this is that the hierarchy has no cells in it.

### What Next

If having no cells is expected, then there is nothing to do. If it is expected that this hierarchy should have cells, then check to see why this hierarchy has no cells by looking at the verilog for this hierarchy.

If this hierarchy has cells, then look at log file to see if there are other messages with information on the failure.

Feedback

## RTLA-603

(warning) Loading old DB version %d.%d. Tool version is %d.%d. Possible inconsistent data.

### Description

Metrics data is saved to the disk so that it can be quickly loaded back. However if the metrics data database (DB) is an older version than the tool version, this can lead to data inconsistency problems. Ideally the tool version and the metrics DB version should be identical.

For example in version 6.3 of metrics DB, support for hierarchical constructs was added. Previous versions (< 6.3) do not have this data. So if the tool supports version 6.3 or later, it will load earlier metrics DB but will not have correct hierarchical construct data.

A further problem is if this inconsistent data is then saved, the inconsistent data would then be saved with a new version number, which would then hide the issue and lead to hard to discover issues in the future.

### What Next

It is best practice to rerun the design from scratch to ensure that the metrics DB and the tool are in sync. In particular, the version 6.3 has a large change to the metrics DB so any previous version than 6.3 should be rerun from scratch.

## RTLA-604

(warning) Using an old DB version %d.%d. Tool version is %d.%d. Possible need to rerun from scratch.

### Description

It is detected that an old DB version is loaded into a newer tool version. It may be the case that running compute_metrics will clean up this issue, but not in all cases. Here are some situations where compute_metrics will not clear up the problem.

In version 6.3 of metrics DB, support for hierarchical constructs was added. Previous versions (< 6.3) do not have this data. For this particular enhancement, the flow needs to be run from scratch. This is because the hierarchical construct tree originates from the RTL and needs to be created at the begining of flow and carried forward.

A further warning is if this old database is saved, the inconsistent data would then be saved with a new version number, which would then hide the issue and may lead to hard to discover issues in the future.

Feedback

**What Next**

It is best practice to rerun the design from scratch to ensure that the metrics DB version and the tool version are in sync.

---

## RTLA-700

(error) Cannot open file '%s' for write. Paths cannot be saved.

**Description**

Tool was unable to access neither the current working directory nor the temporary directory specified through app option 'shell.common.tmp_dir_path' for writing the internal temporary files which are needed to save the timing path information.

**What Next**

Check the write permission of the current working directory, or the directory specified by app option 'shell.common.tmp_dir_path'.

---

## RTLA-701

(error) Metrics data not found. Compute metrics with the current release.

**Description**

Tool was unable to access the metrics data. Re-compute the metrics data with the current release.

**What Next**

Re-compute the metrics data with the current release.

---

## RTLA-705

(error) RTLA non-physical flow is disabled for hierarchical designs.

**Description**

Command rtl_opt currently does not support non-physical flow for hierarchical designs.

---

## RTLA-706

(error) Congestion metrics computation is disabled on non-physical designs. Non-physical designs found: '%s'

Feedback

**Description**

This combination of command and options requires the blocks to have physical data. Computation of congestion metrics is disabled on non-physical blocks.

**What Next**

Rerun with designs that include physical data.

---

## RTLA-800

(warning) app_option %s will be set as %s during the rtl_opt.

**Description**

The app option %s is being set to the value %s during rtl_opt.

**What Next**

Please check for usage of the specified app option.

---

# RTLRS

---

## RTLRS-001

(warning) Please ensure that this app-option is set before RTL is analyzed/elaborated and top-level module is linked.

**Description**

The recommended flow to perform RTL restructuring is as follows:

Step-1: Enable RTL Restructuring capabilities of Fusion Compiler

set_app_options -list {rtl_restructuring.enable_restructured_rtl_generation true}
set_app_options -list {rtl_restructuring.enable_formality_script_generation true}

Step-2: Read and Elaborate RTL

analyze -format verilog {*.v} elaborate top

Step-3: Link Design and initialize RTL restructuring engine

set_top_module top

Note: set_top_module will initialize RTL restructuring engine under the hood if the app-option "rtl_restructuring.enable_restructured_rtl_generation" is set to true before analyze and elaborate.

Feedback

Step-4: Load UPF and Timing Constraints

Step-5: Run RTL Restructuring functions like group and ungroup.

Step-6: Generate modified UPF and Constraints

Step-7: Generate modified RTL and formality mapping file

write_restructured_rtl

**What Next**

Update your scripts to set the RTL restructuring mode app-option to true prior to RTL analyze step.

---

## RTLRS-002

(error) Failed to initialize RTL restructuring engine. Please set the app-option which enables RTL restructuring mode before RTL analyze step.

**Description**

The recommended flow to initialize RTL restructuring engine and perform RTL exploration is as follows:

Step-1: Enable RTL Restructuring capabilities of Fusion Compiler

set_app_options -list {rtl_restructuring.enable_restructured_rtl_generation true}
set_app_options -list {rtl_restructuring.enable_formality_script_generation true}

Step-2: Read and Elaborate RTL

analyze -format verilog {*.v} elaborate top

Step-3: Link Design and initialize RTL restructuring engine

set_top_module top

Note: set_top_module will initialize RTL restructuring engine under the hood if the app-option "rtl_restructuring.enable_restructured_rtl_generation" is set to true before analyze and elaborate.

Step-4: Load UPF and Timing Constraints

Step-5: Run RTL Restructuring functions like group and ungroup.

Step-6: Generate modified UPF and Constraints

Step-7: Generate modified RTL and formality mapping file

write_restructured_rtl

Feedback

**What Next**

Update your scripts to set the RTL restructuring mode app-option to true prior to RTL analyze step.

---

## RTLRS-003

(error) Failed to generate restructured RTL since the RTL restructuring engine has not been activated.

**Description**

The recommended flow to initialize RTL restructuring engine and perform RTL exploration is as follows:

Step-1: Enable RTL Restructuring capabilities of Fusion Compiler

set_app_options -list {rtl_restructuring.enable_restructured_rtl_generation true}
set_app_options -list {rtl_restructuring.enable_formality_script_generation true}

Step-2: Read and Elaborate RTL

analyze -format verilog {*.v} elaborate top

Step-3: Link Design and initialize RTL restructuring engine

set_top_module top

Note: set_top_module will initialize RTL restructuring engine under the hood if the app-option "rtl_restructuring.enable_restructured_rtl_generation" is set to true before analyze and elaborate.

Step-4: Load UPF and Timing Constraints

Step-5: Run RTL Restructuring functions like group and ungroup.

Step-6: Generate modified UPF and Constraints

Step-7: Generate modified RTL and formality mapping file

write_restructured_rtl

**What Next**

Update your scripts to set the RTL restructuring mode app-option to true prior to RTL analyze step.

Feedback

## RTLRS-004

(warning) RTL restructuring mode can't be disabled at this stage of the flow. Please refer to the recommended flow described in the man page.

### Description

Due to current tool limitations, once the RTL restructuring engine has been initialized, it can't be disabled in the same session to continue with the synthesis flow. The recommended flow when performing RTL restructuring is as follows:

Step-1: Initialize RTL restructuring engine (as described in the man page of the app-option rtl_restructuring.enable_restructured_rtl_generation).

Step-2: Run RTL Restructuring functions like group and ungroup.

Step-3: Generate modified UPF and Constraints.

Step-4: Generate modified RTL using write_restructured_rtl.

Step-5: Exit the shell.

Step-6: Read and Elaborate restructured RTL.

Step-7: Load modified UPF and Timing Constraints.

Step-8: Continue with synthesis flow.

Even if none of the RTL restructuring functions were performed, the RTL restructuring mode can't be disabled and user would have to exit the shell and re-elaborate the original RTL.

### What Next

Please use the recommended flow until the limitations are addressed.

## RTLRS-005

(error) Failed to generate modified RTL since the restructuring engine has not been initialized.

### Description

In order to perform RTL restructuring functions and generate modified RTL, the restructuring engine must be initialized during top-level module link by enabling the app-option "rtl_restructuring.enable_restructured_rtl_generation" before reading and elaborating RTL as described in the man page of the app-option.

Feedback

**What Next**

Please follow the recommended flow highlighted in the man page of the app-option 'rtl_restructuring.enable_restructured_rtl_generation'.

---

## RTLRS-007

(information) Generating restructured RTL. The modified RTL files and file-list info will be written to directory named %s

**Description**

The command will selectively generate RTL for modified modules only and write the updated modules in the specified format or by maintaining the source format and also preserving the original RTL source file names in the specified directory. If the output directory name isn't specified, the command will by default write the restructured RTL files to a directory named 'output'.

The command will also print a file-list that contains information about both the modified RTL files generated as well as the unchanged modules that were imported.

---

## RTLRS-008

(information) Generating formality mapping file for design %s.

**Description**

The command will generate a supplementary formality mapping file in the current working directory which is required to perform formal verification to verify the correctness of the new generated RTL which contains modified hierarchies after the design was restructured.

The generated formality mapping file must be passed along with original and new RTL files for verification with Formality.

---

## RTLRS-009

(warning) Failed to create '%s' output directory. RTL will be written in default directory output/Verilog.

**Description**

The command will generate RTL which contains modified hierarchies after the design was restructured.

If the output directory name isn't specified, the command will by default write the generated RTL files to a directory named 'output/Verilog'.

Feedback

## RTLRS-010

(error) Currently supported formats are verilog and systemverilog.

**Description**

The command will selectively generate RTL for modified modules only and write the updated modules in the specified format or by maintaining the source format and also preserving the original RTL source file names in the specified directory. If the output directory name isn't specified, the command will by default write the restructured RTL files to a directory named 'output'.

The command will also print a file-list that contains information about both the modified RTL files generated as well as the unchanged modules that were imported.

## RTLRS-011

(error) Failed to initialize rtl restructuring engine as top design '%s' has been auto black-boxed.

**Description**

If the top design is black-boxed for some reasons( e.g. due to the presence of glue-block in it), rtl restructuring engine initialization fails.

## RTLRS-012

(warning) Design '%s' is auto black-boxed as %s

**Description**

The design will be automatically black-boxed if it has always block/glue block/generate statement/multiple view present in it.

## RTLRS-013

(warning) Editing connection of instance '%s' is not allowed in rtl restructuring flow as it is a glue logic cell. This change may not reflect in Output RTLs.

**Description**

If rtl_restructuring.enable_restructured_rtl_generation is set to true, editing connection of a glue logic cell is not allowed. This changes may not reflect in Output RTLs.

Feedback

## RTLRS-014

(warning) Ungrouping the Glue logic is not permitted in rtl restructuring mode.

**Description**

If rtl_restructuring.enable_restructured_rtl_generation is set to true, Ungrouping of a glue logic is not permitted.

## RTLRS-015

(error) Command remove_port %s: Partial port removal from a bus is not supported in restructuring mode. Specify all scalar ports of a bus or bus name for successful bus port removal.

**Description**

Removal of partial bits from a bussed port is not supported. Specify all scalar ports of bus or bus name for successful bus port removal.

## RTLRS-016

(error) Command remove_pin %s: Partial pin removal from a bus is not supported in restructuring mode. Specify all scalar pins of a bus or bus name for successful bus pin removal.

**Description**

Removal of partial pin bits from a bussed pin is not supported. Specify all scalar pins present on bus or bus name for successful bus pin removal.

## RTLRS-017

(warning) Failed to create '%s' directory. RTL restructuring related collaterals will be written in cwd.

**Description**

The command will generate rtl_restructuring_commands.tcl, rtl_restructuring_sources.f files which contains modified hierarchies after restructring.

If the output_dir app option isn't specified, the command will by default write the generated rtl_restructuring_commands.tcl, rtl_restructuring_sources.f, report directory and rtl_restructuring_log directory in cwd.

Feedback

# RTM

## RTM-1000

(info) Retiming results for instance '%s' (module '%s'): Clock period (Period - Uncertainty): %f, Typical Clock2q = %f, Typical Setup = %f, Netlist modified: %s, Period before: %f, Period after: %f, Registers before: %d, Registers after: %d

### Description

This message shows the result of retiming. It shows the clock period, typical register clock2q and setup considered during retiming. 'Netlist modified' is a boolean showing whether any retiming happened or not. It also shows the achieved clock period and registers before and after retiming.

### What Next

If results are not as expected, then enable other RTM messages to debug further

## RTM-1001

(warning) No retimable registers found for instance '%s' (module '%s'). If this module contains registers, re-run by enabling RTM-1004, RTM-1005 and RTM-1013 messages to know the reason

### Description

This warning shows up if a hierarchical instance with set_optimize_registers specification does not have any retimable registers.

If this module contains registers that were not identified as retimeable then re-run with enabling verbose retiming messages to see more details on the reason. Use following commands to enable these messages:

set_msg -level info RTM-1004

set_msg -level info RTM-1005

set_msg -level info RTM-1013

### What Next

Check the output of the verbose retiming messages and confirm or modify the design setup accordingly for the detailed reasons.

Feedback

## RTM-1002

(info) %s

**Description**

This message print the 'critical loop' for retiming. For the critical loop, it shows the placement of registers before and after retiming. This is typically used for debugging the quality of results during retiming.

**What Next**

Analyze if the retiming is reasonable

## RTM-1004

(info) %s is not retimed because : %s

**Description**

Retiming does not retime a register in certain situations. This message prints the reason for no retiming.

**What Next**

Follow the instructions in the message

## RTM-1005

(info) %s : %s

**Description**

Most commonly, this message prints the registers which has an 'invalid' clock. This usually happens when the clock constraint is not correct. In rare situations, it prints a message for empty module.

**What Next**

Follow the instructions in the message

## RTM-1013

(info) %s is not retimed due to one of these size_only reasons : %s

Feedback

### Description

Retiming tolerates a few types of size_only restrictions on registers, retimes them and propagates the restrictions to the new registers. When intolerable size_only restriction is encountered on a register, this message prints the intolerable size_only reasons.

### What Next

Check the constraints that caused this intolerable size_only

---

## RTM-1016

(info) Instance '%s' (module '%s') is retimed in place of user specified retimed instances %s because of chaining

### Description

Two retiming instances (modules) are considered to be chained if there exist combinational paths without registers between them. In this case, retiming them together is beneficial for timing QoR between these instances. Hence the least common parent hierarchy of the chained instances is retimed. Any other non-retime hierarchies or registers inside the parent hierarchy are prevented from getting retimed.

### What Next

Retiming the parent hierarchy usually helps QoR. Otherwise, this can be disabled by setting the app option compile.retiming.group_chained_retiming_modules to false.

---

# RULE

---

## RULE-001

(error) Value for message list must have %d elements.

### Description

The number of elements in the rule message list cannot be changed. The new message list must have the same number of elements as the previous message list value.

### What Next

Specify the correct number of elements for the message list.

---

## RULE-002

(error) Rule '%s' is already defined.

Feedback

**Description**

The rule name is in use by an existing rule.

**What Next**

Specify a valid rule name that is not already in use.

---

## RULE-004

(error) Rule '%s' is not defined.

**Description**

The specified rule cannot be found.

**What Next**

Specify a valid rule name.

---

## RULE-005

(error) Rule '%s' is permanent.

**Description**

This command works only on user-defined rules.

**What Next**

Specify a valid list of rules.

---

## RULE-006

(error) User-defined rule name '%s' does not begin with 'UDEF'.

**Description**

User-defined rules must have names beginning with 'UDEF'. For example, 'UDEF-0345'.

**What Next**

Specify a valid name for the user-defined rule.

---

## RULE-009

(error) Property '%s' does not exist on built-in rule '%s'.

Feedback

**Description**

Each built-in rule supports zero or more property names. Only valid property names can be set on built-in rules.

**What Next**

Specify a valid property name for the rule.

---

## RULE-010

(error) Property '%s' does not exist on rule '%s'.

**Description**

The rule does not have a property of this name.

**What Next**

Specify a valid property name for the rule.

---

## RULE-012

(error) Checker procedure '%s' does not exist.

**Description**

The checker procedure associated with a user-defined rule must be a valid Tcl procedure.

**What Next**

Specify the correct checker procedure.

---

# SCP

---

## SCP-001

(error) Unable to add library %s in session checkpoint

**Description**

Unable to add a library in session checkpoint

**What Next**

Please check for the write permission in the session checkpoint directory.

Feedback

## SCP-002

(error) Unable to load library %s

**Description**

Unable to load a library

**What Next**

Please check if the library exists

## SCP-003

(error) Unable to find current library

**Description**

There is no current library set.

**What Next**

Please set the current library in the session.

## SCP-004

(error) Unable to link design %s

**Description**

Unable to link the design.

## SCP-005

(error) Unable to find design %s

**Description**

Unable to find the design.

## SCP-006

(error) Unable to open design %s

**Description**

Unable to open the design

Feedback

## SCP-007

(error) Unable to set App option {%s %s}

**Description**

Unable to set App option

## SCP-008

(error) Unable to set %d App options

**Description**

Provides number of app options that were not set.

## SCP-009

(error) Session name not given

**Description**

The name of the session checkpoint is not given

**What Next**

Please provide the name of the session checkpoint.

## SCP-010

(error) Top module not found

**Description**

Unable to find the top module

## SCP-011

(error) Unable to open session checkpoint %s for write

**Description**

The session checkpoint directory cannot be opened for write.

**What Next**

Please check for the write permission in the destination directory.

Feedback

## SCP-012

(error) Incorrect session checkpoint structure

**Description**

This message indicates that the session checkpoint is corrupt.

**What Next**

Please regenerate the session checkpoint

## SCP-013

(error) Opening session was not entirely successful

**Description**

There were errors while opening session checkpoint

**What Next**

Please review earlier error messages

## SCP-014

(error) Unable to make directory %s

**Description**

Unable to make directory

**What Next**

Please check for the write permission in the destination directory.

## SCP-015

(error) Unable to open session checkpoint %s for read

**Description**

Unable to open session checkpoint for read

**What Next**

Please check if the given session checkpoint is a valid session checkpoint

Feedback

## SCP-016

(error) Unable to open session catalog file %s with error : %s

**Description**

Unable to open session catalog file

## SCP-017

(error) Inconsistency in session catalog

**Description**

Inconsistency in session catalog

**What Next**

Session checkpoint may be corrupt, please regenerate the session checkpoint

## SCP-018

(error) Unable to create session checkpoint catalog successfully

**Description**

Unable to create session checkpoint catalog successfully

## SCP-019

(error) Unable to read session checkpoint

**Description**

Unable to read session checkpoint

## SCP-020

(warning) The library name %s is not unique

**Description**

The library name is not unique

## SCP-021

(warning) Unable to on demand load library %s

Feedback

**Description**

Unable to on demand load library

---

## SCP-022

(warning) Overwriting existing session checkpoint %s

**Description**

Overwriting existing session checkpoint

---

## SCP-023

(warning) Unable to remove old session checkpoint %s, renaming it into %s

**Description**

When unable to remove old session checkpoint, it is renamed

---

## SCP-024

(warning) Unable to rename session checkpoint %s

**Description**

Unable to rename session checkpoint

---

## SCP-025

(warning) Compression value %d not recognized

**Description**

Compression value not recognized

---

## SCP-026

(warning) Unable to read catalog successfully

**Description**

Unable to read catalog successfully

Feedback

# SDC

## SDC-1

(information) Setting sdc_version outside of an SDC file has no effect

**Description**

You set the *sdc_version* variable outside of the context of an SDC file. In that context, changing the variable has no effect.

**What Next**

No action necessary.

## SDC-2

(warning) SDC version in file (%s) does not match the version you requested\n \tfrom read_sdc (%s). Some constraints and options may not function.

**Description**

The version indicated by the setting of the *sdc_version* variable in your SDC file does not match the version requested by the *read_sdc* command.

**What Next**

Ensure that you select the correct version when issuing the *read_sdc* command.

## SDC-3

(warning) Constraint '%s' is not supported by %s.

**Description**

Not all Synopsys Design Constraints are supported by all applications. The specified constraint is not supported by the current application, and it is ignored. For example, test constraints are not recognized by PrimeTime. One SDC-3 message is issued per instance of the constraint which is ignored. Then, after *read_sdc* completes, an SDC-4 summary message will tell you how many of each constraint was ignored.

## SDC-4

(information) Ignored %d unsupported '%s' constraint%s.

Feedback

**Description**

This is a summary message indicating how many instances of a particular constraint were ignored by *read_sdc* because the constraint is unsupported.

---

## SDC-5

(error) Errors reading SDC file:\n \t%s.\n \tUse error_info for more info.

**Description**

This message is generated by *read_sdc* when an syntax error occurs during the read. The specific error is shown in the text of the message.

**What Next**

You can use *error_info* to help trace the cause of the error. For example, it might show the file and line number of the syntax error.

---

# SDXF

---

## SDXF-001

(error) Rule '%s' has wrong %s '%s'

**Description**

The value(s) provided to given rule name is wrong.

**What Next**

Please refer to the SDXF Reference Manual and provide correct value given rule name.

---

## SDXF-002

(error) Multiple '%s' provided

**Description**

Multiple section(s)/attribute(s) provided whose occurrence number one or at most one.

**What Next**

Please refer to the SDXF Reference Manual and remove repetitive occurrences.

Feedback

## SDXF-003

(error) Wrong %s provided

**Description**

If file version is mentioned in the error, the given SDXF file version number is greater than reader's version number or non-numeric characters have been mentioned.

If the unit is mentioned in the error, not supported distance unit has been provided.

**What Next**

Please refer to the SDXF Reference Manual and provide correct distance unit or file version.

## SDXF-004

(error) Encountered while converting the data to %s

**Description**

This message indicates an internal data model error.

**What Next**

Send the log to Synopsys.

## SDXF-005

(error) Required %s not provided

**Description**

If file version is mentioned in the error, the required sdxf file version number is not provided in the sdxf file.

If the unit is mentioned in the error, the required attribute distance unit is not provided in the sdxf file.

**What Next**

Please refer to the SDXF Reference Manual and provide attribute distance unit or file version.

## SDXF-006

(error) Syntax error in the file : '%s'

Feedback

**Description**

Syntax error related to XML formatting was found in the specified SDXF file in the error message.

The line number of the syntax error and additional details will also be printed in case of missing operands like quote("), equal sign(=) and end tag(>).

In some other cases of a missing slash tag (/) or start tag(<) etc., the line number corresponding to end of file will be printed, along with error details like 'unexpected end of data' or 'expected element name'

**What Next**

Please fix the syntax error(s) and rerun the command.

---

## SDXF-007

(error) Could not parse sdxf file '%s'.

**Description**

There was a major syntax error in the specified sdxf file. It should be a single text file containing 3DIC information. Please refer to messages printed before this message.

**What Next**

Please correct the input sdxf file and rerun the command

---

## SDXF-008

(error) Unsupported %s file '%s' format '%s'.

**Description**

The input/include file format specified is not supported. Hence the command cannot parse the text file.

**What Next**

Provide a valid format, e.g. "sdxf".

---

## SDXF-009

(warning) Skipping rule '%s' because of wrong rule name

**Description**

This message indicates the given rule name is syntactically wrong.

Feedback

**What Next**

Please refer to the SDXF Reference Manual and provide correct rule name.

---

## SDXF-010

(warning) Skipping rule '%s' because %s

**Description**

This message indicates the given rule name is already exist or multiple rules with the same rule name has been provided in a constraint section.

**What Next**

Please check if the multiple rules are provided with the same name, if yes remove the repetitive rule or check if the rule already exists, if yes then give different rule name or remove the existing rule first.

---

## SDXF-011

(warning) Ignoring unknown %s : '%s'

**Description**

This message indicates the given section/attribute name is syntactically wrong.

**What Next**

Please refer to the SDXF Reference Manual and provide correct name.

---

## SDXF-012

(warning) There is no SDXF data in current lib.

**Description**

This message indicates that there is no SDXF data in current lib. So write_sdxf will do nothing but quit.

**What Next**

Please import SDXF data by "read_sdxf" or open a lib which contains SDXF data.

---

## SDXF-013

(error) File '%s' included multiple times.

Feedback

**Description**

The file specified is either included multiple times or having a cyclic dependency.

**What Next**

Please include the file only once. If file is included only once then check and remove the cyclic dependency.

---

## SDXF-014

(error) Rule '%s' of rule type '%s' is not supported in '%s_CONSTRAINT' section.

**Description**

This rule will not be considered in the mentioned constraints section.

**What Next**

Please refer to the SDXF Reference Manual and provide only supported rule types in the corresponding constraints section.

---

## SDXF-015

(error) Value '%s' is too large.

**Description**

The length value specified in the rule is too large to be supported by the tool.

The name of the rule for which this value error occurred is printed in the subsequent message.

**What Next**

Please check the value and provide the appropriate value.

---

# SEC

---

## SEC-0

(error) Software is not licensed for this machine.

**Description**

Could not find a valid matching key in the key file for this feature

Feedback

**What Next**

Check to see that the machine hostid, using the machine_id utility provided by Synopsys, matches the hostid in the key file for the feature.

---

## SEC-1

(error) %s

**Description**

The key file with all of the license information cannot be opened.

**What Next**

Check directory and file read permissions.

---

## SEC-21

(error) Failed to checkout license for feature(s): %s

**Description**

The application failed to check out the licenses required to enable this product. It may be that all the licenses are in use or the site is not licensed to use this product.

**What Next**

Make sure that the required features are in the key file.

Use lmstat to make sure that the server has enabled the required features and to find the current users of the features.

---

## SEC-23

(information) Waiting for required feature(s). (%s)

**Description**

This is an informational message that is displayed periodically when the process is queued for required set of features. You can find out the current users of all the required features by using the lmstat -A command.

**What Next**

Request the current users to release the licenses, if possible.

Let the process wait in queue for indicated period of time to acquire the license.

Feedback

## SEC-51

(error) This site is not licensed for '%s'.

**Description**

A possible cause is that no such feature exists. The feature could not be found in the license file.

A possible problem is that the version is not supported at the server end. The version specified in the checkout request is greater than the highest version number the daemon supports.

A possible problem is that the server does not support this feature or the feature has expired or has not yet started (on the server node).

**What Next**

Check to make sure that the license file supports this version.

## SEC-55

(error) Can't remove your '%s' license. You must always have one or more of the following license(s): %s.

**Description**

A request was received to remove a license that is necessary for the application to be run. At least one of these licenses is required for the application to be running.

**What Next**

Make sure that you are removing the right license.

## SEC-57

(error) License daemon version is older than application's FLEXlm version.

**Description**

This is an error message that is displayed if the FLEXlm version of the application is higher than than the license daemon version.

**What Next**

Please upgrade the license server to latest version.

Feedback

## SEC-71

(error) License checkout unsuccessful. Command '%s' cannot run.

**Description**

The command that you are trying to run needs some license(s) to run successfully. The required set of licenses is not available and hence the command cannot run.

**What Next**

Please ensure that you have all the required license keys on your license server. Check for any other messages preceding this message for more details.

## SEC-72

(error) Following set of licenses does not exist on server : %s. Combination of %s keys must exist.

**Description**

The tool is checking for the existence of a particular set of license keys on the license server. The required combination of licenses is mentioned in the message. If you are seeing this error, it means that the license requirement is not met.

**What Next**

Please ensure that you have all the required license keys on your license server.

## SEC-73

(information) Waiting for license key(s) was successful.

**Description**

The tool was waiting in queue for checking out a particular set of license keys. The waiting is over since all the required keys were successfully checked out.

**What Next**

The tool will proceed with the run.

## SEC-74

(information) Waiting in queue for the following license keys : %s

Feedback

### Description

The command that you are trying to run needs to check-out a set of license keys in order to run successfully. Some of these keys (mentioned in the message) are not available right now, and hence the tool is waiting in queue on the license server in order to obtain these keys. Until then, the tool cannot proceed with the run.

If the license keys are unavailable even after a particular timeout period, the tool will stop waiting and exit the queue. You can control this timeout period by setting the environment variable SNPS_MAX_QUEUETIME with the desired value (in seconds). If this variable is not set, the default timeout period will be 8 hours.

### What Next

Wait for the license keys to be available.

---

## SEC-75

(error) Timeout while waiting for license keys.

### Description

The tool was waiting in queue for checking out a particular set of license keys. The timeout period got over and still the license keys were not available. Hence the tool cannot run the command successfully.

You can control the timeout period by setting the environment variable SNPS_MAX_QUEUETIME with the desired value (in seconds). If this variable is not set, the default timeout period will be 8 hours.

### What Next

Please ensure that you have all the required license keys on your license server.
If you want to wait for a longer period of time, set the environment variable SNPS_MAX_QUEUETIME appropriately before the next run.

---

## SEC-76

(warning) Availability of the command '%s' is deprecated in this shell and it will be disallowed from 2020.09. Please use single-shell instead.

### Description

The command will not be available in this shell in future releases. However it is available in the single-shell. If you need to use it in your flow, you should switch to the single-shell where this command is available along with the current shell's other capabilities.

Feedback

**What Next**

Please use single-shell if you need to use this command.

---

## SEC-78

(error) This command '%s' is not allowed to run in a child process.

**Description**

This message occurs in a child process which gets launched by a command which uses distributed processing. It indicates that the run script which was automatically generated by the master to be run on this child process has been modified and the tool does not allow these modifications to be run on the child process.

**What Next**

Start a new run and do not modify the run script of the child process.

---

## SEC-81

(error) Unrecognized feature name '%s'

**Description**

The specified name is not a valid license feature for this product.

**What Next**

Check the feature name and rerun the command with a valid feature name.

---

## SEC-82

(information) Feature '%s' is not checked out.

**Description**

You are trying to check in a license feature which has not been checked out right now. Hence the *remove_license* command will not do anything.

**What Next**

Check the feature name and rerun the command with a valid feature that has been checked out.

---

## SEC-83

(error) Only %d keys of '%s' are checked out.

Feedback

### Description

The number of keys of the given feauture that you are trying to check-in is more number than the number of keys that have been checked out. Hence the *remove_license* command will not do anything.

### What Next

Ensure that you check-in the same or lesser number of keys than are currently checked out.

---

## SEC-84

(error) Cannot check out '%s' as another key of a different core-type is currently checked out.

### Description

The tool supports different kinds of license packages, with each package typically enabling a different multi-core capability. However, it does not support the simultaneous check out of keys from different packages. For example, if any 8-core key has been already checked out, the tool will not allow any 16-core key to be checked out.

If you are seeing this message, it means that at least one key of a different core-type has already been checked out in this session and hence the current key cannot be checked out. This can be verified by running the *list_licenses* command. Note that any key which is common between packages does not fall under a specific core-type and hence this rule does not apply to such keys.

### What Next

First check in all keys of all other core-types and then try to check out the current one.

---

## SEC-90

(error) Cannot check in '%s' as design '%s' is open and needs this license.

### Description

If you are seeing this message, it means that there is at least one open design which needs this key.

### What Next

If you want to remove this license, please close the design and then try to check-in.

Feedback

## SEC-91

(error) Cannot remove license when background jobs using redirect -bg are running.

**Description**

It is not allowed to remove license when background jobs using redirect -bg are running.

**What Next**

Wait for the redirect -bg jobs to complete and then issue the remove_license command

**See Also**

- report_background_jobs

## SEC-92

(error) Command %s cannot run as this functionality is disabled in %s when invoked with %s.

**Description**

The command is checking for particular license keys which are not allowed to run in the invoked shell and its mode.

**What Next**

Please ensure to invoke correct shell and mode to run this functionality.

# SEL

## SEL-001

(error) No such collection '%s'

**Description**

The collection which you specified does not exist.

**What Next**

Verify that the collection is the one you want. It is possible that it existed, but was transient. Transient collections are automatically garbage-collected and cannot be relied upon

Feedback

across command boundaries. In order to make a collection persistent, set it to a variable, and then use it. For example:

```
set uPorts [get_ports U*]
    command_for_ports $uPorts
```

## SEL-002

(warning) Collection '%s' has inappropriate type (%s).

**Description**

The collection which you specified contains objects which are not acceptable for this command. Either the data type of the objects is incorrect, or the objects are out of context (for example, they are not in the current design).

**What Next**

Check the command to determine the allowable object types for it, or specify objects that are in the correct context.

## SEL-003

(warning) Nothing implicitly matched '%s'

**Description**

The pattern which you specified did not match any objects of the classes acceptable for this command.

**What Next**

Check the pattern to see if it is what you expected.

## SEL-004

(warning) No %s objects matched '%s'

**Description**

The pattern which you specified did not match any objects of the class acceptable for this command.

**What Next**

Check the pattern to see if it is what you expected.

Feedback

## SEL-005

(error) Nothing matched for %s

**Description**

The pattern(s) which you specified did not match any objects.

**What Next**

Check the values which you entered.

## SEL-006

(error) More than one object matched for '%s'.

**Description**

The pattern(s) which you specified matched more than one object. This command option accepts only a single object.

**What Next**

Check the values which you entered.

## SEL-007

(error) Invalid index %d for collection %s

**Description**

During an iteration over a collection (with foreach_in_collection), an invalid index was generated.

**What Next**

Contact your application consultant.

## SEL-008

(warning) Collection/attribute class '%s' has not been defined

**Description**

The collection class which you specified does not exist. Classes of objects include designs, cells, etc.

Feedback

**What Next**

Verify that the class name is spelled correctly, or that the class of objects is applicable for this product.

---

## SEL-009

(warning) Collection class '%s' cannot be %s

**Description**

The collection class which you specified cannot be used for the operation you attempted. Some collection classes cannot be queried, indexed, or copied, so they cannnot be used as an argument to query_objects, index_collection, or copy_collection.

**What Next**

Only use collections of this class as arguments to appropriate commands.

---

## SEL-010

(warning) %s objects from '%s' were of the %scorrect class.

**Description**

A heterogeneous collection was passed to another command. This collection contained some objects that were of a class which is not accepted by the command. The message will indicate whether some objects or no objects were accepted by the command.

**What Next**

Some commands continue to operate when only a subset of the patterns match. Other commands only perform their action when all patterns match something. So, verify that the command was applied to the objects which were expected.

---

## SEL-011

(warning) Some objects (%s) could not be queried.

**Description**

A heterogeneous collection was passed into query_objects. This collection contained some objects that were of a class which cannot be the target of a query.

**What Next**

There is no adverse affect of this situation.

Feedback

## SEL-012

(information) Iteration for collection %s was terminated\n \tbecause the collection was modified or deleted.

**Description**

Commands in the body of a *foreach_in_collection* can affect the collection which is currently in iteration. Some commands can cause objects to be removed from the collection, and others can cause the collection to be deleted. When such events occur, the iterator is modified and in some cases will terminate. This message advises you of that event.

For example:

foreach_in_collection itr [get_cells *] { remove_design [current_design] }

would cause the collection of cells to be deleted, and the iteration would be terminated.

**What Next**

No action is required.

## SEL-013

(error) Regular expression error: %s.

**Description**

While using a regular expression with a collection command, you entered an invalid regular expression. For example, use of the * (zero or more) or + (one or more) operators alone always yields the empty set; therefore, ".*" or ".+" would be appropriate. Other errors such as unmatched parens or invalid characters within square braces will also cause this error.

**What Next**

Take action based on the error that occurred.

## SEL-014

(error) At least one %scollection required for argument '%s'%s

**Description**

Some commands do not allow heterogeneous collections as arguments, whereas others allow them only in some contexts. Other commands require at least one collection as an

Feedback

argument. You entered a variation of a command which requires at least one collection (either homogeneous, or of either type) for the named argument.

**What Next**

Consult the man page from the command which failed for further information.

---

## SEL-015

(warning) Ignored all implicit elements in argument '%s'%s

**Description**

Many commands allow implicit searches for objects - an argument can be a list of collections or patterns which are searched for in a documented set of object classes. However, in some cases, it is not possible to determine any object classes in which to search for an implicit pattern. For example, attempting to add an implicit pattern to a heterogeneous collection with *add_to_collection* would cause this warning.

**What Next**

Consult the man page from the command which failed for further information.

---

## SEL-016

(error) Name patterns are not allowed in this argument context - the pattern %s will be skipped/ignored; use only collections in this argument context.

**Description**

Many commands allow implicit searches for objects - an argument can be a list of collections or name patterns which are searched for in a documented set of object classes. However, in some cases, when there is more than one object class to be searched, it is not allowed to include name patterns, and only collections should be included in such an argument.

**What Next**

In the argument context in which this error happened, remove all name patterns from the offending argument, and make sure that only collections are included in the argument. You can typically use a "get" command to convert a name pattern to a collection.

---

## SEL-017

(error) %s index %s for collection %s

Feedback

**Description**

An invalid or out of range index was supplied to a command.

**What Next**

Contact your application consultant.

# SFT

## SFT-1000

(info) Register %s is restricted. The restriction is ignored while doing automatic safety register implementation.

**Description**

This message shows that there is some restriction on the register which is ignored for automatic safety register implementation.

## SFT-1001

(error) Ignoring set_safety_register_rule (and hence automatic replication) on register %s as %s.

**Description**

This message shows why a register marked as safety can not be implemented using any safety mechanism.

## SFT-1004

(warning) Register %s can not be mapped to a scan flop due to user constraint. So mapping to non-scan flop.

**Description**

This message says that due to some user constraints mapping to scan flop is not possible, so mapping it to non-scan flop. This message comes when some register is part of shift register chain and is considered for safety register inference, and during breaking of shift-register, this register should be mapped to scan flops but it cannot be mapped to scan flop.

## SFT-1005

(error) Could not map some instances during fault-tolerant library cell inference.

Feedback

**Description**

This message says that during inference of fault-tolerant library cell, some of the pulled-out logic could not be mapped.

---

## SFT-1006

(error) Ignoring error port connection/error detection as error signal %s provided with '-error_signal' option of 'set_safety_register_rule' is invalid. So valid error signal would be of type input pin, output pin, output port.

**Description**

This message says why an error hookup point for DMR/TMR is invalid.

---

## SFT-1007

(info) Removing empty safety group %s as its registers or register-pins are gobbled.

**Description**

This message comes when removing an empty safety group because its registers or register-pins are gobbled.

---

## SFT-1009

(error) Ignoring library cell %s provided with '-logic_mapping' option of 'create_safety_register_rule' as it %s.

**Description**

This error message comes when option '-logic_mapping' of 'create_safety_register_rule' is ignored due to some reason (for example, if there is no error port mapping provided).

---

## SFT-1010

(warning) 'set_safety_register_rule -target ...' exists, but set_spfm_target does not. Safety register inference is skipped.

**Description**

This warning message comes when safety inference is skipped as 'set_spfm_target' is not provided.

Feedback

## SFT-1011

(warning) Some registers (Eg., %s) selected for safety do not have a target rule. You can use 'set_safety_register_rule -target ...' to specify a common rule for all the registers.

**Description**

This warning message comes when some registers selected for safety inference based on spfm_loss doesn't have a target rule, so they are ignored for safey inference. So not to ignore such registers, user can provide common rule for all the registers using command 'set_safety_register_rule -target ...'.

## SFT-1013

(info) Current design SPFM '%f' is more than the target design SPFM '%f' set by command 'set_spfm_target', so safety register inference is skipped.

**Description**

This information message comes to let user know that safety register inferece is going to be skipped as current design SPFM is already more than the target design SPFM specified by command 'set_spfm_target'.

## SFT-1025

(error) Deleting safety register group %s as %s.

**Description**

This message says why safety register group is getting deleted.

# SG

## SG-001

(Error) Could not find '%s' contributing to any given reference library

**Description**

You passed a contributing library file name to the *define_scaling_lib_group* command, and this library file name could not be found in any reference library. By default, all reference libraries are searched. A specific set of reference libraries to be searched using the *-reflib* option.

Feedback

### What Next

Make sure the specification of library file names to define_scaling_lib_group is correct. You can use report_lib to look at the contributing files to any open reference library.

---

## SG-002

(Error) All panes must come from the same reference library

### Description

Your arguments to the *define_scaling_lib_group* command yielded panes from multiple reference libraries. A scaling group cannot span more than one reference library. It's possible that a DB was "split" so that some of its cells are in one reference library and others are in another.

### What Next

Make sure the specification of library file names to *define_scaling_lib_group* is correct. You can use *report_lib* to look at the contributing files to any open reference library. By default, all reference libraries are searched. A specific set of reference libraries to be searched using the *-reflib* option. In the case of a "split" DB, you will need to use *-reflib* to be specific.

---

## SG-003

(Warning) Invalid library '%s' ignored: %s

### Description

A library that was explicitly passed to the *define_scaling_lib_group* command using the *-reflib* option is not valid. Either it's a reference library with no timing content, or it's a design library. It was ignored. By default, all reference libraries are searched. A specific set of libraries to be searched using the *-reflib* option.

### What Next

This is just a warning. You should not be passing physical-only reference libraries or design libraries to this command with the *-reflib* option. That option is designed to steer a scaling group into a single reference library that has timing content.

---

## SG-004

(Error) No reference libraries to search

Feedback

### Description

Your arguments to the *define_scaling_lib_group* command yielded no matching reference libraries. Either no reference libraries are loaded, or all reference libraries passed to the command contain only physical information, or an empty library was found, or you tried to use a design library. In the latter cases, you should have seen some SG-003 warnings.

### What Next

Make sure the specification of library file names to *define_scaling_lib_group* is correct. You can use *report_lib* to look at the contributing files to any open reference library. By default, all reference libraries are searched. A specific set of reference libraries to be searched using the *-reflib* option.

---

## SG-005

(Error) Must have at least two panes for a scaling group

### Description

Your arguments to the command *define_scaling_lib_group* could not form a scaling group. Either only one pane matching was found for the file names contributing to the reference libraries searched, or more than one was found, but all of the panes have the same PVT. A scaling group requires at least two panes with different PVTs.

### What Next

Make sure the specification of library file names to *define_scaling_lib_group* is correct. You can force a specific set of reference libraries to be searched using the *-reflib* option. Without this option, all reference libraries in memory are searched. All panes must come from the same reference library. You can use *report_lib* to look at the contributing files to any open reference library.

---

## SG-006

(Error) %s thresholds do not match

### Description

When creating a scaling group with the *define_scaling_lib_group* command, some thresholds did not match. Each pane in the group must have the same thresholds. As well, for each library cell, each pin must have the same thresholds in each pane. The threshold values are input_threshold_pct_fall, input_threshold_pct_rise, output_threshold_pct_fall, output_threshold_pct_rise, slew_lower_threshold_pct_fall, slew_lower_threshold_pct_rise, slew_upper_threshold_pct_fall, slew_upper_threshold_pct_rise, and slew_derate_from_library, as defined in Liberty. Thresholds can be viewed using *report_lib*

Feedback

### What Next

You can go back to the Library Manager and adjust the thresholds there. Or, pick a different set of contributing libraries to form a different group.

---

## SG-007

(Error) All panes in a scaling group must have the same process information

### Description

While creating a scaling group with the *define_scaling_lib_group* command, the command found more than one process label or number between the panes. All panes must share the same process label and process number. Process information can be found using *report_lib*.

### What Next

You can go back to the Library Manager and adjust the process information there. Or, pick a different set of contributing libraries to form a different group.

---

## SG-008

(Error) Cannot use pre-scaled panes from '%s' in a scaling group

### Description

Your arguments to the *define_scaling_lib_group* command matched a pane from a reference library which was created by an older version of the Library Manager. This pane is the result of a Library Manager generated scaling group and cannot be used in this context.

### What Next

The application can still perform scaling using the reference library, but it cannot be combined into a new scaling group.

---

## SG-009

(Warning) Scaling group already exists %s

### Description

You tried to create a scaling group but have already done so with the same name, or same set of panes, and so on. The message will give you the specific reason.

Feedback

**What Next**

Check for a typo in your command.

---

## SG-010

(information) Removed scaling library group '%s' in reference lib '%s'

**Description**

The scaling library group was removed by the *remove_scaling_lib_group* command.

**What Next**

No action required.

---

## SG-019

(Error) No scaling library group named '%s'.

**Description**

The scaling library group specified has not been defined by the *define_scaling_lib_group* command.

**What Next**

Define the scaling library group with the *define_scaling_lib_group* command.

---

## SG-023

(Error) %s characterization point:%s

**Description**

While creating a scaling group, one of two things is wrong. In one case, there is a missing characterization point. If the panes you selected share two temperatures and three voltages for a single rail, you would need six panes to cover those points.

In the other case, more than one pane has the same characterization point. Each point can occur only once.

**What Next**

Try using a different set of panes for your scaling group.

Feedback

### SG-024

(Information) For %s, found values:%s

**Description**

While creating a scaling group, the application found missing or duplicate characterization points (PVTs). This message tells you which temperatures and rail voltages were found. To get complete coverage, each combination of these values needs to match one pane to allow a valid scaling group.

This informational message is paired with other messages, which will indicate if this is allowed or if it is an error which will prevent creation of the scaling group.

## SIG

### SIG-6006

(Information) Unable to load design attachment. Inconsistency in congestion estimation database.

**Description**

This message informs that reading the congestion estimation database from NDM attachment has failed. The database is used to estimate the metrices related to congestion.

### SIG-6086

(Information) Parsing error for string '%s', could not convert into a number.

**Description**

This message informs that parsed string could not be converted to a number while deserializing the signature DB attachment.

## SIGEM

### SIGEM-001

(error) unsupported syntax '%s', line %d.

Feedback

**Description**

A unsupported syntax is found in the EM seciton While reading TLU plus file.

**What Next**

Check the TLU plus file and reload it.

---

## SIGEM-002

(error) syntax error in line %d.

**Description**

A syntax error is found in the EM seciton while reading TLU plus file.

**What Next**

Check the TLU plus file and reload it.

---

## SIGEM-003

(error) line should be start with token '#', line %d .

**Description**

The EM syntax should be started with token '#'.

**What Next**

Check the TLU plus file and reload it.

---

## SIGEM-004

(error) '%s' has no corresponding token '%s' line %d.

**Description**

This message will be reported when a corresponding token is missing in the EM file. For example, token '{' must have a corresponding '}'.

**What Next**

Check the TLU plus file and reload it.

---

## SIGEM-005

(error) The current library has no technology information.

Feedback

**Description**

The current library must have technology information to correctly read the EM file.

**What Next**

Load the technology file to current library, then reload the EM file.

---

## SIGEM-006

(warning) Layer %s is not defined in technology file, the constraints in the layer ignored.

**Description**

The Layer is not defined in technology file, correcsponding EM data will be ignored.

**What Next**

Check the layer definitation in technology file, then reload the technology file and EM file.

---

## SIGEM-008

(error) unsupported syntax in line %d: %s

**Description**

An unsupported syntax is found While reading ALF file for signal EM.

**What Next**

Check the file syntax and reload it.

---

## SIGEM-021

(warning) The net '%s' is ignored in signal EM analysis because its toggle rate is zero.

**Description**

If the toggle rate of the net is zero, it is ignored in signal EM analysis.

**What Next**

Check the toggle rate setting.

---

## SIGEM-022

(warning) The net '%s' is ignored in signal EM analysis because it is constant net or case analysis is set.

Feedback

**Description**

If the net is constant (logic 0 or logic 1) or case analysis is set, there is no AC signal propagating through the net. It is ignored in signal EM analysis.

**What Next**

Check the net setting.

---

## SIGEM-023

(warning) Net '%s' has unfixable signal EM violation because the violation is inside block.

**Description**

Signal EM analysis is performed on the nets in top level and also on the nets in the block abstract. However, the violation inside the block cannot be fixed at the top level.

**What Next**

Analyze and fix the signal EM violations inside block at the block level.

---

## SIGEM-024

(warning) Net '%s' has unfixable signal EM violation because the shape with violation is locked.

**Description**

The net with EM violation or the shape with EM violation is locked, so the violation cannot be fixed.

**What Next**

To fix signal EM violation, please update the status on the net/shape.

---

## SIGEM-025

(error) Cannot analyze signal EM for scenario '%s' because '%s'.

**Description**

The signal EM analysis use both early and late data to compute the worst situation. For setup only scenario, please use option -max to perform signal EM analysis with only late data. For hold only scenario, please use option -min to perform signal EM analysis with only early data.

Feedback

**What Next**

To perform signal EM analysis correctly, please set the scenario status accordingly.

---

## SIGEM-026

(warning) '%s' is ignored in signal EM analysis, line %d.

**Description**

This syntax is only supported in Prime Rail for power EM analysis. It is ignored in signal EM analysis.

**What Next**

Check the Signal EM ITF file.

---

## SIGEM-027

(error) Cannot analyze signal EM for scenario '%s' because parasitic data is not available.

**Description**

The signal EM analysis is not performed when the parasitic data is not available because the parasitic data are used for computing current.

**What Next**

Please check the parasitic tech setup for the constraint corner of current scenario.

---

## SIGEM-028

(warning) The net '%s' is ignored in signal EM analysis because some parasitic data don't have layer information.

**Description**

Signal EM analysis use parasitics, timing and routing data to compute current and check violations. Layer information is needed to identify the shape based violation.

**What Next**

Please check routing and parasitics. Please recreate abstract if the net is located inside block.

---

## SIGEM-029

(warning) There are no signal electormigration rules defined.

Feedback

### Description

If there are no signal electromigration rules defined, signal electromigration analysis calculates the current, but does not check for violations.

### What Next

Use the *read_signal_em_constraints* command to load the signal electromigration rules.

### See Also

- fix_signal_em
- read_signal_em_constraints
- report_signal_em
- set_scenario_status

---

## SIGEM-030

(warning) The net '%s' is ignored in signal EM analysis because parasitic data is not available.

### Description

Signal EM analysis calculates current using parasitc data.

### What Next

Please check extraction and timing for the net.

---

## SIGEM-031

(warning) The net '%s' is ignored in signal EM analysis because it is not routed.

### Description

Signal EM analysis calculates current and limit for every routing shape. It is not performed if the net is unrouted.

### What Next

Please check routing for the net.

---

## SIGEM-032

(warning) The net '%s' is ignored in signal EM analysis because parasitic data don't have node locations.

Feedback

**Description**

Signal EM analysis use parasitics, timing and routing data to compute current and check violations. Parasitic node locations are needed for the shape based analysis.

**What Next**

Please check extraction data for the net.

---

## SIGEM-033

(warning) The net '%s' is ignored in signal EM analysis because it has no driver.

**Description**

Signal EM analysis uses timing data to compute current. The current cannot be calculated for nets without driver data.

**What Next**

Please check driver for the net.

---

## SIGEM-034

(warning) The net '%s' is ignored in signal EM analysis because the driver '%s' doesn't actually drive anything.

**Description**

Signal EM analysis uses timing data to compute current. The current cannot be calculated for nets without driver data.

**What Next**

Please check driver for the net.

---

## SIGEM-035

(warning) The net '%s' is ignored in signal EM analysis because it is unconnected.

**Description**

Signal EM analysis works on routed nets only.

**What Next**

Please check the connection for the net.

Feedback

## SIGEM-036

(error) syntax error in line %d. Invalid value for "EM_PEAK_EQUATION_SOURCE_TECHFILE".

**Description**

A syntax error is found while reading ITF file. The value of "EM_PEAK_EQUATION_SOURCE_TECHFILE" should be either 0 or 1.

**What Next**

Correct the value of "EM_PEAK_EQUATION_SOURCE_TECHFILE" and reload it.

## SIGEM-037

(warning) The net '%s' is ignored in signal EM analysis because it is inside block abstract and it is not fully retained.

**Description**

Signal EM analysis use parasitics, timing and routing data to compute current and check violations. This net is not fully retained in abstract.

**What Next**

Please check the net in block level.

## SIGEM-038

(error) syntax error in line %d. '%s' is incomplete.

**Description**

A syntax error is found while reading ITF or ALF file. Either the index table or the value table is missing.

**What Next**

Check input file and complete the table.

## SIGEM-039

(error) syntax error in line %d. Data count in '%s' is inconsistent (index %d vs. value %d).

Feedback

**Description**

A syntax error is found while reading ITF or ALF file. The total number in index table is inconsistent with the number in value table.

For factor value table, the number should be same as the number of index table.

For current value table, the number should be multiplication of the numbers of all index tables. For example, if it has two indexes, one with 2 values and the other one with 4 values, we are expecting there are 2*4=8 values in the current value table.

**What Next**

Check input file and correct the table.

---

## SIGEM-040

(warning) The %s transition time for net '%s' is missing and '%f' ns from setting is used.

**Description**

Signal EM analysis uses transition time from timer to compute current. When it is not available, the transition set by app option sigem.min_trans is used. The default value 10ps is used if the app option is not set.

**What Next**

Please check transition time on the net.

---

## SIGEM-041

(warning) Layer %s is not defined in technology file, its FIT parameters are ignored.

**Description**

The Layer is not defined in technology file, correcsponding FIT parameters will be ignored.

**What Next**

Check the layer definitation in technology file and FIT parameters file.

---

## SIGEM-042

(warning) The pin %s is missing in timing graph.

**Description**

Signal EM analysis uses data from timer to compute current. This pin is dangling.

Feedback

**What Next**

Please check the connectivity of the pin.

---

## SIGEM-043

(warning) Net '%s' has unfixable signal EM violation because the violation is inside pin.

**Description**

Signal EM analysis is checking routing shapes inside pin. However, the violation inside the pin cannot be fixed.

**What Next**

Check the driver and loads, reduce the driver current to resolve the signal EM violation.

---

# SIT

---

## SIT-001

(error) No physical information present for %s%s.

**Description**

Physical/Block information is not present. Operations involving physical data can't be performed.

**What Next**

Include physical data into the current design.

---

## SIT-002

(error) %s%s%s.

**Description**

Associated data provided for site array creation is incorrect.

**What Next**

Rectify the input data as per details in error message.

---

## SIT-003

(Information) %s%s.

Feedback

**Description**

General information about site arrays, like default site array creation, etc.

# SKW

## SKW-001

(Warning) '%s' cell is not a macro. So it is ignored.

**Description**

The cell specified is not a macro, and hence ignored.

**What Next**

Correct the specification

## SKW-002

(Warning) Some previous set_skew_macros command specified '%s' macro in a different bank. Removing the macro from the previous bank and adding in the current one.

**Description**

The macro specified in command is already present in another bank. Remove the macro from the previous bank and add into the current bank.

**What Next**

Remove the macro from one of the banks.

## SKW-003

(Warning) Some previous set_skew_macros command had specified bank '%s'. Removing the previous specification and overriding with the current one.

**Description**

The data in the bank due to previous specification is overridden with current one.

**What Next**

Do the necessary correction

Feedback

## SKW-004

(Info) Skew macro specification for bank '%s' is removed.

**Description**

Removed set_skew_macros specification on the bank

**What Next**

Ensure that removal is intended

## SKW-005

(Warning) Bank '%s' is not found.

**Description**

The specified bank is not found in the database

**What Next**

Correct as necessary

## SKW-006

(Warning) Some previous set_skew_registers command had specified bank '%s'. Removing the previous specification and overriding with the current one.

**Description**

The data in the bank due to previous specification is overridden with current one.

**What Next**

Do the necessary correction

## SKW-007

(Warning) '%s' cell is a macro. So it is ignored.

**Description**

The cell specified is a macro, and hence ignored.

**What Next**

Correct the specification

Feedback

## SKW-008

(Warning) Some previous set_skew_registers command specified '%s' register in a different bank. Removing the register from the previous bank and adding in the current one.

**Description**

The register specified in command is already present in another bank. Remove the register from the previous bank and add into the current bank.

**What Next**

Remove the register from one of the banks.

## SKW-009

(Info) Skew register specification for bank '%s' is removed.

**Description**

Removed set_skew_registers specification on the bank

**What Next**

Ensure that removal is intended

## SKW-010

(Warning) Bank '%s' is not found.

**Description**

The specified bank is not found in the database

**What Next**

Correct as necessary

## SKW-101

(Info) BankName = %s, ImproveSide = %s, Critical = {%s}, BankInputSlack = %f, BankOutputSlack = %f, SkewVal = {%s}.

**Description**

Message gives information about skew macro results

Feedback

### SKW-102

(Info) BankName = %s, ImproveSide = %s, BankInputSlack = %f, BankOutputSlack = %f, SkewVal = %f.

**Description**

Message gives information about skew registers results

## SLG

### SLG-001

(fatal) This is NOT a code crash but an intentional bailout to get your attention. There are scaling-related setting errors or library data issues that are considered fatal to your analysis, and need to be resolved as soon as possible. See the SLG-001 man page.

**Description**

The *define_scaling_lib_group* command specifies a group of libraries for scaling and exact-match usages. These features have these major requirements:

- Library data consistency requirements.
- Scaling formation requirements which are needed by scaling only.
- Cell instance operating condition setting cannot go beyond the acceptable range defined by the scaling group.

If these requirements are not met, PrimeTime issues SLG and DEL errors and warnings. You got the SLG-001 message and bailout because the script has one or more of these major errors that seriously affect QoR or even cause calculation failure and fatal errors. PrimeTime stops before *update_timing* so you can get a list of related errors.

If you choose to accept the error and want timing analysis to continue, you can set the *timing_continue_on_scaling_error* variable to true to bypass SLG-001 errors. In that case, you will still see SLG-418 warnings with a list of offending error IDs. However, by bypassing SLG-001 errors, calculation failure or QoR effects can occur.

**What Next**

Use the *check_timing -include operating_conditions* command and fix the following check failures:

```
SLG-202(n)
SLG-209(n)
SLG-211(n)
SLG-217(n)
```

Feedback

```
SLG-303(n)
SLG-310(n)
SLG-311(n)
SLG-316(n)
SLG-317(n)
SLG-320(n)
SLG-321(n)
SLG-330(n)
```

### See Also

- define_scaling_lib_group

---

## SLG-201

(error) A library can be in only one scaling library group.

### Description

The *define_scaling_lib_group* command specifies a group of libraries to interpolate between for voltage and temperature scaling. The command must be issued after the *link_design* commands occur (either explicitly or implicitly).

If you try to use a library in more than one group, the SLG-201 error message occurs.

After design linking occurs, the remaining libraries in scaling library groups will be loaded to "complete" the groups. During the completion step, data for the same arcs and pins across libraries are associated with each other; if this association fails, the scaling library groups cannot be completed, and the SLG-202 error message is issued.

### What Next

Ensure that the set of arcs and pins is identical across the members of the scaling library groups. The positions of the arcs and pins within the libraries must be the same as well.

### See Also

- define_scaling_lib_group
- SLG-201
- SLG-202
- SLG-203
- SLG-205

Feedback

## SLG-202

(error) Completion of scaling library groups failed %s.

### Description

The *define_scaling_lib_group* command specifies a group of libraries to interpolate between for voltage and temperature scaling. The command must be issued after the design is read.

If you try to use a library in more than one group, the SLG-201 error message occurs.

After design linking occurs, the remaining libraries in scaling library groups will be loaded to "complete" the groups. During the completion step, data for the same arcs and pins across libraries are associated with each other; if this association fails, the scaling library groups cannot be completed and the SLG-202 error message is issued.

### What Next

Ensure that the set of arcs and pins is present across the members of the scaling library groups. This message is issued when some of the arcs or cells are missing across the members of the scaling library groups.

If you are using Base-curve Technology CCS Timing libraries, also verify if the base curve information in the libraries have been discretized and normalized in identical manner; that is, check if curve_x has identical values across the scaling libraries in a library group.

### See Also

- define_scaling_lib_group
- SLG-201
- SLG-202
- SLG-203

## SLG-203

(information) The members of a scaling library group have significantly different output-capacitance indexes; this can adversely affect scaling accuracy.

### Description

The *define_scaling_lib_group* command specifies a group of libraries to interpolate between for voltage and temperature scaling. The command must be issued after the *read_db* and *link_design* commands occur (either explicitly or implicitly).

If you try to use a library in more than one group, the SLG-201 message occurs.

Feedback

After design linking occurs, the remaining libraries in scaling library groups are loaded to complete the groups. During the completion step, data for the same arcs and pins across libraries are associated with each other. If this association fails, the scaling library groups cannot be completed, and the SLG-202 error message is issued.

**What Next**

Ensure that the set of arcs and pins is identical across the members of the scaling library groups. The positions of the arcs and pins within the libraries must be the same as well.

The output-capacitance indexes used for Composite Current Source (CCS) driver data is better to have identical values across the libraries. If this is not satisfied, the SLG-203 message is issued.

**See Also**

- define_scaling_lib_group
- SLG-201
- SLG-202
- SLG-203

## SLG-204

(warning) The timing arcs %s cannot be uniquely determined across the scaling libraries, it is assumed that the timing arcs are defined in the same order across all the libraries.

**Description**

The *define_scaling_lib_group* command specifies a group of libraries to interpolate between for voltage and temperature scaling. The command must be issued after the design is linked.

If you try to use a library in more than one group, the SLG-201 message occurs.

After design linking occurs, the remaining libraries in scaling library groups are loaded to complete the groups. During the completion step, data for the same arcs and pins across libraries are associated with each other. If the timing arcs cannot be associated across the scaling libraries because the timing arcs cannot be identified uniquely within a library, the SLG-204 warning message is issued. Since the timing arcs cannot be uniquely identified, the arcs are assumed to be in the same order across the libraries in a library group.

Note that this is only an informational warning; the scaling relationship for the scaling library group is still created.

Feedback

### What Next

Ensure that the set of arcs that cannot be uniquely identified are present in the same order across the members of the scaling library groups. Run LC scaling check to detect and correct all major violations.

### See Also

- define_scaling_lib_group
- SLG-201
- SLG-202
- SLG-203

---

## SLG-205

(warning) CCS Noise information is inconsistent for %s across the libraries in the scaling library group. Scaling for CCS noise will not be supported.

### Description

The *define_scaling_lib_group* command specifies a group of libraries to interpolate between for voltage and temperature scaling. The command must be issued after the design is read.

If you try to use a library in more than one group, the SLG-201 message occurs.

After design linking, the remaining libraries in scaling library groups are loaded to complete the groups. During the completion step, data for the same arcs and pins across libraries are associated with each other. If this association fails, the scaling library groups cannot be completed, and SLG-202 error message is issued. If the association fails only due to missing or inconsistent CCS noise data, the SLG-205 warning message is issued and the inconsistent CCS noise data is not used in calculation.

### What Next

Ensure that the CCS noise data for the set of arcs and pins is present across the members of the scaling library groups. This message is issued when some of the arcs or pins have at least one of the following conditions:

- Missing CCS noise model information across the members of the scaling library groups
- CCS noise model information with the input voltage and output voltage indexes not in the same percent vdd in all libraries of a scaling group

Feedback

- Inconsistent or missing conditions for CCS noise models across the members of a scaling library group
- Inconsistent ordering of conditional CCS noise models across the members of a scaling library group

**See Also**

- define_scaling_lib_group
- SLG-201
- SLG-202
- SLG-203

---

## SLG-206

(warning) Completion of scaling library group failed for power analysis: %s.

**Description**

This message is issued when some of the power arcs are missing across the members of the scaling library group.

The *define_scaling_lib_group* command specifies a group of libraries to interpolate between for voltage and temperature scaling. The command must be issued after the design is read.

After design linking, the remaining libraries in scaling library group are loaded to complete the group. During the completion step, data for the same arcs and pins across libraries are associated with each other. If this association fails for associating power arcs across the libraries, the SLG-206 warning message is issued, and scaling is not enabled for power analysis.

Note that this warning is for power analysis, the scaling relationship for the scaling library group is still created. The scaling for timing analysis is not affected. The SLG-202 message is issued when some of the timing arcs or cells are missing across the members of the scaling library group.

**What Next**

Ensure that the set of power arcs is present across the members of the scaling library group.

Feedback

### See Also

- define_scaling_lib_group
- SLG-202

---

## SLG-207

(warning) Some of the power arcs cannot be uniquely determined across the scaling libraries, it is assumed that the power arcs are defined in the same order across all the libraries.

### Description

The *define_scaling_lib_group* command specifies a group of libraries to interpolate between for voltage and temperature scaling. The command must be issued after the design is linked.

After design linking, the remaining libraries in scaling library group are loaded to complete the group. During the completion step, data for the same arcs and pins across libraries are associated with each other. If the power arcs cannot be associated across the scaling libraries because the power arcs cannot be identified uniquely within a library, the SLG-207 warning message is issued. Since the power arcs cannot be uniquely identified, the arcs are assumed to be in the same order across the libraries in a library group.

Note that this is only an informational warning, the scaling relationship for the scaling library group is still created.

### What Next

Ensure that the set of arcs that cannot be uniquely identified are present in the same order across the members of the scaling library group.

### See Also

- define_scaling_lib_group
- SLG-204
- SLG-206

---

## SLG-208

(error) The command define_scaling_lib_group needs to be defined before the command set_variation_library.

Feedback

### Description

You must use the *define_scaling_lib_group* command before using the *set_variation_library* command.

### What Next

Use the *define_scaling_lib_group* command before using the *set_variation_library* command.

### See Also

- define_scaling_lib_group
- SLG-201
- SLG-202
- SLG-203

---

## SLG-209

(error) Invalid set of libraries passed to define_scaling_lib_group.

### Description

A valid set of libraries should be passed to *define_scaling_lib_group*. Check the characterization operating conditions of the libraries in the group.

For example, if you want to use operating conditions between 0.8 and 0.9 V and temperatures between -40 and 125 degrees, libraries characterized at all four voltage and temperature combinations should be specified:

```
pt_shell> define_scaling_lib_group { \\
          lib_max_125_0.8V_cccs.db \\
          lib_max_125_0.9V_cccs.db \\
          lib_max_-40_0.8V_cccs_v2.db\\
          lib_max_-40_0.9V_cccs_v2.db }
```

If any of the four libraries are missing, the scaling library group is rejected. When specifying multiple temperatures or voltages, all libraries on the resulting voltage-temperature grid must be provided.

### What Next

Check the characterization operating conditions of the libraries provided to *define_scaling_lib_group*, and ensure that the grid is fully populated.

Feedback

**See Also**

- define_scaling_lib_group
- report_lib

---

## SLG-210

(error) Zero or more than one scaling library is in the link_path or link_path_per_instance. Creation of scaling group failed.

**Description**

Only one library in each scaling library group should be in the link_path, or be the min library of a library in the linked path. If there is none or more than one scaling library in the link_path, the tool does not create this library group.

**What Next**

Fix the scaling library group, and rerun your script.

**See Also**

- define_scaling_lib_group

---

## SLG-211

(warning) Loading of CCS data for multi-rail scaling support has failed , the reason is: %s.

**Description**

The reasons that might cause the loading to fail include: if the libraries are non-PG-pin libraries; if the libraries do not have any CCS data; if more than one or none of the specified libraries is a link library or a min library of a link library, and finally a series of data inconsistencies such as if a specific arc has driver data but no receiver data.

When this warning is issued, multirail scaling is not possible for that scaling group, and the scaling calculations using the scaling library group revert to regular single-rail scaling.

**What Next**

Ensure that libraries specified are CCS PG-pin libraries, and exactly one of them is a link library or min library of a linked library. Also, ensure that corresponding receiver and driver data is available for every arc in each of the libraries.

**See Also**

- define_scaling_lib_group

Feedback

## SLG-212

(information) The variable %s cannot be turned on because %s.

### Description

If you set the *variation_enable_analysis* variable after you use the *define_scaling_lib_group* command, the variable setting change does not take effect.

### What Next

Set the *variation_enable_analysis* variable before using the *define_scaling_lib_group* command.

## SLG-213

(information) The number of dimensions in dimension_names and dimension_values list are not consistent.

### Description

The size of the list in *dimension_names* and *dimension_values* should match.

When this information is issued, the command does not take effect.

### What Next

Reissue the command with correct set of *dimension_names* and *dimension_values*.

### See Also

- define_scaling_lib_group

## SLG-214

(warning) Scaling will not be done for cell %s, since it does not have consistent scaling information across the scaling libraries.

### Description

Scaling cannot be done for specific arcs or pins of the specified cell because of inconsistent or missing scaling data.

### What Next

Check the scaling libraries using the Library Compiler utility to correct the inconsistencies. This might be due to missing arc or pin data in one of the libraries. Reissue the command with a correct set of scaling libraries.

Feedback

**See Also**

- define_scaling_lib_group
- report_lib
- SLG-209

---

## SLG-215

(warning) Scaling creation for the %s has failed due to missing or mismatching data.

**Description**

If there is an arc or pin with missing or mismatching data, the scaling relation for that arc or pin is not created. The other arcs or pins in the scaling library group are not affected. The calculation of that arc or pin uses linked library data instead.

**See Also**

- define_scaling_lib_group
- SLG-202

---

## SLG-216

(error) An unexpected exact-match has been detected for the cell timing %s in %s calculation

**Description**

If you use the *define_scaling_lib_groups* command with the *-exact_match_only* option, only exact matches are considered acceptable. If the cell operating condition requires scaling, or none of the library operating condition matches the desired one, it is considered an exact-match failure, and this message is issued. Then the linked library is used.

**What Next**

In the case or *-exact_match_only*, change the operating condition of the cell instance, or add a library in the scaling group with the specified operating condition. Operating conidtions include voltage, rail voltage, temperature, and process. Alternatively, consider using scaling.

You can use "check_timing -override_default operating_conditions -verbose" to determine detail information about the PVT mismatch between design and library. And use "report_lib_group -scaling -show { process voltage temperature} " can determine each library PVT, "report_power_pin_info" can display the cell instance voltage setting. Compare both reports, you can see the PVT difference.

Please do not ignore this message. This error must be fixed or it will impact QoR.

## SLG-217

(warning) Scaling will not be done for cell %s, since the associated scaling library group does not meet the scaling formation requirement.

### Description

Scaling cannot be mathematically performed if the data is insufficient for performing scaling, such as trying to do two-dimensional scaling with only two libraries. Scaling formation is done by the points representing the operating conditions of the libraries in a multidimensional space.

PrimeTime supports on-the-grid, n+1, and 2n+1 scaling formations. This message is issued when the scaling operating condition formation of an arc or pin data of a cell does not meet any of the three formations.

### What Next

Check the operating conditions of the libraries, including voltage, rail voltages, temperature, and so on. Provide libraries that meet the scaling formation requirements, and reissue the command.

### See Also

- define_scaling_lib_group
- report_lib
- SLG-209

## SLG-218

(information) Find %s, check your scaling libraries.

### Description

The *define_scaling_lib_group* command specifies a group of libraries to interpolate between for voltage and temperature scaling. This requires data consistency otherwise scaling might encounter issues. This message detects inconsistency issues, such as missing voltage_maps or mismatch voltage_map sequence across the scaling libs.

### What Next

This informational message indicates an issue that the tool attempts to fix internally. However, you should fix the library to avoid any potential issues and run the Library

Feedback

Compiler scaling check. All libraries used in scaling should pass the Library Compiler scaling check.

**See Also**

- define_scaling_lib_group

## SLG-220

(warning) Completion of scaling library group failed for rail analysis: %s.

**Description**

The *define_scaling_lib_group* command specifies a group of libraries to interpolate between for voltage and temperature scaling. The command must be issued after the design is read.

After design linking, the remaining libraries in scaling library group are loaded to complete the group. During the completion step, data for the same arcs and pins across libraries are associated with each other. If this association fails for associating power arcs across the libraries, the SLG-220 warning message is issued, and scaling is not enabled for analysis.

**What Next**

Ensure that the set of power arcs is present across the members of the scaling library group. This message is issued when some of the power arcs are missing across the members of the scaling library group.

**See Also**

- define_scaling_lib_group

## SLG-221

(warning) Some of the power arcs cannot be uniquely determined across the scaling libraries for rail analysis, it is assumed that the power arcs are defined in the same order across all the libraries.

**Description**

The *define_scaling_lib_group* command specifies a group of libraries to interpolate between for voltage and temperature scaling. The command must be issued after the design is linked.

After design linking, the remaining libraries in scaling library group are loaded to complete the group. During the completion step, data for the same arcs and pins across libraries are associated with each other. If the power arcs cannot be associated across the

Feedback

scaling libraries because the power arcs cannot be identified uniquely within a library, the SLG-221 warning message is issued. Since the power arcs cannot be uniquely identified, the arcs are assumed to be in the same order across the libraries in a library group.

Note that this is only an informational warning; the scaling relationship for the scaling library group is still created.

**What Next**

Ensure that the set of arcs that cannot be uniquely identified are present in the same order across the members of the scaling library group.

**See Also**

- define_scaling_lib_group

---

## SLG-222

(warning) Library scaling group: Mismatch detected in the number of POCV tables for %s in library %s.

**Description**

This warning occurs when defining a library scaling group, if a discrepancy is detected in the number of POCV tables across the different libraries.

**See Also**

- define_scaling_lib_group

---

## SLG-223

(warning) Library scaling group: Mismatch detected in the index sizes of POCV tables for %s in library %s.

**Description**

This warning occurs when defining a library scaling group, if a discrepancy is detected in the index sizes of POCV tables across the different libraries.

**See Also**

- define_scaling_lib_group

---

## SLG-224

(warning) Replacing library for '%s' with '%s'

Feedback

### Description

This message warns you that you have the same cell instance listed multiple times in the setting for the *define_cell_alternative_lib_mapping* variable. In the following example, the *uls/utlb_ctl/U112* instance is listed twice.

```
define_cell_alternative_lib_mapping liberty_ss_0p600v_m25c_mis.db -cell
 "uls/utlb_ctl/U112 uls/utlb_ctl/U112"
```

### What Next

Examine the setting for the *define_cell_alternative_lib_mapping* variable, and ensure that only one valid library exists per cell-instance.

---

## SLG-225

(error) The *define_cell_alternative_lib_mapping* command failed due to missing library group information for library '%s'

### Description

You need to create a scaling group for the library used in the *define_cell_alternative_lib_mapping* command.

### What Next

Specify the library group using the *define_scaling_lib_group* command as shown in the following example.

```
set link_path "* my_linked_lib.db"
read_verilog my_design.v
link_design my_design
define_scaling_lib_group -exact_match_only {my_linked_lib.db
 my_alt_lib.db}
define_cell_alternative_lib_mapping my_alt_lib.db -cells {U1 U2}
update_timing
```

### See Also

- define_scaling_lib_group

---

## SLG-226

(warning) Scaling creation for the %s has failed due to unsupported 3-D NLDM data.

Feedback

### Description

If there is an arc using 3-D NLDM data, the scaling relation for that arc will not be created. The other arcs or pins in the scaling library group are not affected. The calculation of that arc or pin uses linked library data instead.

### See Also

- define_scaling_lib_group
- SLG-202

---

## SLG-227

(error) An unexpected exact-match failure due to missing arc/pin has been detected for the cell timing %s in %s calculation

### Description

If you use the *define_scaling_lib_groups* command with the *-exact_match_only* option, only exact matches are considered acceptable. You get this message when the exact_match library does not have the matching arc/pin as the linked library.

### What Next

Please run Library Compiler scaling check to find the mismatching arc/pin and fix accordingly.

---

## SLG-228

(error) An unexpected scaling failure due to extrapolation has been detected for the cell timing %s in %s calculation

### Description

This is a special error on extrapolation. You receive this error because your operating conditon of the cell is too close to a missing scaling library cell or a missing scaling library. Therefore accuracy is impacted and it is considered extrapolation.

### What Next

Please change cell operating condition to stay away from the missing library or the missing library cell. Check report_power_pin_info/report_design and report_lib_group -scaling -show {process voltage temperature} to check why the cell PVT is not in the range of the scaling group. Also check whether there is any lib cell/pin/arc is missing message SLG-305, SLG-307 and SLG-308 in the previous define_scaling_lib_group processing log file.

Feedback

## SLG-229

(error) An unexpected scaling calculation failure has been detected for the cell timing %s in %s calculation

### Description

This is an error to let you know that CCS or NLDM data scaling has failed. This is a very rare failure, often due to bad library data causing scaling to go wrong, that triggers internal detection to issue this error.

### What Next

Please check for bad library data, in particular if any data that is way too large or too small than it should be. Also, whether multiple libraries with same operating conditons are used. This could happen when most of the rails of the libraries are excluded from scaling calculation, then scaling uses the left over rails to decide scaling connection. Scaling choose to use the first library among the libraries with same set of remaining rails. If the first library happends to have dramatically different data than other libraries used in scaling, you may get this error on bad data triggering scaling failure.

## SLG-231

(error) Detected a scaling failure because specified operating conditions is not on proper grid of the scaling group for the cell timing %s calculation.

### Description

You receive this error because your operating conditon of the cell is far to the desired range of the scaling library group. This error message is to attention customer that it may have bad scaling calculation results due to the inproper operating condition setting or scaling lib definition.

### What Next

Please change the cell operating condidtion or add more full grid library corner and make sure the specified cell operating condition is within desired scaling range.

If it is voltage-only scaling, use 'report_power_pin_info' to get the related cell voltage setting, and use 'report_lib_group -scaling -show {voltage}' to find out the scaling range of the group.

## SLG-300

(error) Scaling libraries cannot be defined after timing analysis.

Feedback

### Description

You tried to use the *define_scaling_lib_group* command after a timing update.

### What Next

Close the current block and reopen the block. Be sure to run the *define_scaling_lib_group* before the first timing update.

### See Also

- define_scaling_lib_group

---

## SLG-301

(warning) Scaling libraries are defined after timing analysis.

### Description

*define_scaling_lib_group* command is recommended to be used before any timing update.

### What Next

To modify your running script and make all *define_scaling_lib_group* commands are used before any timing update. If user doesn't change the script and wants to skip this warning message, user needs to run update_timing -full to honor the newly added scaling libraries.

### See Also

- define_scaling_lib_group

---

## SLG-302

(error) Scaling libraries can be defined only after the design is linked.

### Description

You tried to use the *define_scaling_lib_group* command before the *link_design* command.

### What Next

Read in and link the design with the *link_design* command. Then use the *define_scaling_lib_group* command. Do this before the first timing update.

### See Also

- define_scaling_lib_group

Feedback

## SLG-303

(error) Multiple scaling libraries '%s' are in the link path specfied by the link_path and link_path_per_instance variables. Creation of the scaling group failed.

### Description

The *define_scaling_lib_group* command requires exactly one library of a scaling library group to be in the link path specified by the *link_path* and *link_path_per_instance* variables. The remaining scaling libraries are automatically read in to create the group. This error message indicates that more than one scaling library was found in the link path.

### What Next

Change either the link path or the *define_scaling_lib_group* command so that exactly one library in the scaling library group exists in the link path.

### See Also

- define_scaling_lib_group

## SLG-305

(error) Cell '%s' is not present in scaling library '%s'. Scaling will not be performed for this cell.

### Description

The *define_scaling_lib_group* command requires every library cell in a link library to be also present in every scaling library. This error message reports a library cell in a link library that does not exist in a scaling library.

### What Next

Make sure that all the libraries listed in *define_scaling_lib_group* command contain all of the library cells in the linked libraries.

### See Also

- define_scaling_lib_group

## SLG-306

(error) define_scaling_lib_group '%s', cannot find scaling Library '%s', scaling will not be performed.

Feedback

### Description

*define_scaling_lib_groups* command requires all libraries in the scaling lib group are present. This message will be issued if there is one scaling library cannot be found.

### What Next

Correct the library in the define_scaling_lib_group and make sure all scaling libraries are present.

---

## SLG-307

(error) Pin '%s' of cell '%s' is present in link library '%s' but not in scaling library '%s'.

### Description

The *define_scaling_lib_group* command requires every pin of every library cell in a link library to be also present in every scaling library. This error message reports a pin of a library cell in a link library that does not exist in a scaling library. define_scaling_lib_group command will error out when detect this error.

### What Next

Make sure that all the libraries listed in *define_scaling_lib_group* command contain all of the library pins in the linked libraries. Please modify script remove or change the library that contain different pin.

### See Also

- define_scaling_lib_group

---

## SLG-308

(error) Arc '%s' of cell '%s' is present in link library '%s' but not in scaling library '%s'. Scaling might not be performed for this arc.

### Description

The *define_scaling_lib_group* command requires every timing arc of every library cell in a link library to be also present in every scaling library. This error message reports an arc of a library cell in a link library that does not exist in a scaling library. If the related scaling lib group is exact_match_only, calculation will fallback to link library arc lib data.

### What Next

Make sure that all the libraries listed in *define_scaling_lib_group* command contain all of the library arcs in the linked libraries.

Feedback

## SLG-309

(error) The voltage rail of scaling library '%s' does not match that of the link library '%s'. The scaling group cannot be formed.

### Description

*define_scaling_lib_groups* command requires each scaling library voltage rail is same with the link library among the library group. If any scaling library voltage rail is different with the link library, define_scaling_lib_group command will error out, and output this message.

### What Next

Replace the offending scaling library with library that has same library voltage rail as link library.

## SLG-310

(error) The pin thresholds of scaling library '%s' do not match those of the link library '%s'.

### Description

The *define_scaling_lib_group* command requires each scaling library threshold to be the same as that of the the link library in the library group. This message indicates that the *define_scaling_lib_group* command found a threshold difference.

### What Next

Replace the offending scaling library with a library that has the same library threshold definitions as the link library.

### See Also

- define_scaling_lib_group

## SLG-311

(error) The CCS driver data of scaling library '%s' does not match that of link library '%s'. The scaling group cannot be formed.

### Description

The *define_scaling_lib_group* command requires library CCS data to be consistent between each scaling library and the link library. This error message indicates that the command found inconsistent CCS library data.

Feedback

### What Next

Make sure that the *define_scaling_lib_group* command specifies a list of libraries with CCS driver data consistent with the link library.

### See Also

- define_scaling_lib_group

---

## SLG-312

(error) The user data of scaling library '%s' does not match that of the link library '%s'. The scaling group cannot be formed.

### Description

The *define_scaling_lib_group* command requires library user data to be consistent between each scaling library and the link library. This error message indicates that the command found inconsistent user data.

### What Next

Make sure that the *define_scaling_lib_group* command specifies a list of libraries with user data consistent with the link library.

### See Also

- define_scaling_lib_group

---

## SLG-313

(error) Arc %s in library %s arc group index is different from the index of the library group. The scaling group cannot be formed.

### Description

The *define_scaling_lib_group* command requires each library arc index in the arc group to match its library index in the library group. This error message indicates that the command found inconsistent arc group index data.

### What Next

Make sure that each library arc is present in all libraries in the scaling library group defined by the *define_scaling_lib_group* command.

### See Also

- define_scaling_lib_group

Feedback

## SLG-314

(error) Pin '%s' in library '%s' pin group index is different from the index of the library group. The scaling group cannot be formed.

### Description

The *define_scaling_lib_group* command requires each library pin index in the pin group to match its library index in the library group. This error message indicates that the command found inconsistent pin group index data.

### What Next

Make sure that each library pin is present in all libraries in the scaling library group defined by the *define_scaling_lib_group* command.

### See Also

- define_scaling_lib_group

## SLG-315

(error) Invalid set of libraries '%s' passed to define_scaling_lib_group. Multi-dimension scaling requires the number of libraries to be at least the number of dimensions plus one.

### Description

The *define_scaling_lib_group* command requires the number of libraries in a group to be at least the number of scaling dimensions plus one. This error message indicates that too few libraries were listed.

### What Next

Check numbers of process, temperature, and voltage rail values. Make sure that the number of scaling libraries is at least the number of scaling dimensions plus one.

### See Also

- define_scaling_lib_group

## SLG-316

(error) Library '%s' already belongs to a scaling library group.

### Description

A library listed in the *define_scaling_lib_group* command already belongs to a scaling library group. Each library can belong to no more than one scaling library group.

Feedback

### What Next

Report the existing library groups using the *report_lib_groups* command. You can create multiple scaling library groups to cover different parts of the design, but each library can belong to no more than one group.

### See Also

- define_scaling_lib_group
- SLG-201
- SLG-202
- SLG-203
- SLG-205

## SLG-317

(error) No library in the scaling library group was found in the link path specified by the link_path and link_path_per_instance variables. Creation of the scaling group failed.

### Description

The *define_scaling_lib_group* command requires exactly one library of a scaling library group to be in the link path specified by the *link_path* and *link_path_per_instance* variables. The remaining scaling libraries are automatically read in to create the group. This error message indicates that no scaling library was found in the link path.

### What Next

Change either the link path or the *define_scaling_lib_group* command so that exactly one library in the scaling library group exists in the link path.

### See Also

- define_scaling_lib_group

## SLG-318

(error) Parallel lib arc '%s' of cell '%s' is present in link library '%s' but not in scaling library '%s'. Scaling might not be performed for this arc.

### Description

The *define_scaling_lib_group* command requires every timing arc of every library cell in a link library to be also present in every scaling library. This error message reports a parallel

Feedback

arc of a library cell in a link library that does not exist in a scaling library. If the related scaling lib group is exact_match_only, calculation will fallback to link library arc lib data.

**What Next**

Make sure that all the libraries listed in *define_scaling_lib_group* command contain all of the library arcs in the linked libraries. Please check whether the same from lib pin and to lib pin contain different number of parallel arcs between link library and scaling library.

---

## SLG-320

(error) Cell '%s' hits scaling extrapolation problem, '%s' value '%f' is less than scaling range low value '%f'. Scaling calculation cannot be performed on this cell.

**Description**

Scaling calculation can only be done if one cell pvt is within the range of the scaling lib group.

**What Next**

Check problem cell operating condition and voltage setting, correct the design setting problem.

**See Also**

- define_scaling_lib_group

---

## SLG-321

(error) Cell '%s' hits scaling extrapolation problem, '%s' value '%f' is larger than scaling range high value '%f'. Scaling calculation cannot be performed on this cell.

**Description**

Scaling calculation can only be done if one cell pvt is within the range of the scaling lib group.

**What Next**

Check problem cell operating condition and voltage setting, correct the design setting problem.

**See Also**

- define_scaling_lib_group

Feedback

## SLG-328

(warning) Arc '%s' of cell '%s' is tagged as invalid in scaling library '%s'. This may affect scaling of this arc.

### Description

The *define_scaling_lib_group* command expects that every timing arc of every library cell in a link library will also be present in every scaling library. This warning message reports an arc of a library cell in a scaling library that has been explicitly tagged as invalid. Scaling calculations will be done for the arc as if it did not exist in that scaling library. One-dimensional scaling may have degraded accuracy or may result in an exprapolation error and multi-dimensional scaling may fail completely, with calculation falling back to link library arc data. If the related scaling lib group is exact_match_only and the invalid arc is an exact match, calculation will fall back to link library arc lib data.

### What Next

Confirm that this is the expected behavior, and that the scaling group structure and design operating conditions are compatible with the presence of the invalid arc data.

## SLG-330

(error) in the define_scaling_lib_group command, there are at least two libraries share the same name '%s', and they have same process, temperature and voltage definition. Creation of scaling group failed.

### Description

The libraries in one define_scaling_lib_group command should not have exactly same process, temperature and voltage definition, when detect their names are also same, define_scaling_lib_group command disallows such duplicate library usage.

### What Next

Fix the scaling library group, and rerun your script.

### See Also

- define_scaling_lib_group

## SLG-331

(warning) In the define_scaling_lib_group command, library '%s' and '%s' have same process, temperature and voltage definition.

Feedback

**Description**

In order to do scaling calculation, any two libraries in one define_scaling_lib_group command should not have exactly same process, temperature and voltage.

**What Next**

User should check and correct the wrong library in the define_scaling_lib_group command.

**See Also**

- define_scaling_lib_group

---

## SLG-401

(error) Pin '%s' of cell '%s' library '%s' data is present in link library '%s', but it is not present in scaling library '%s'. This type library data scaling will not be performed for this pin.

**Description**

Scaling requires library data is consistent across all libraires in the group. If one pin lib data exists in link library but doesn't exist in scaling library, this message will be displayed.

**What Next**

Check and correct the missing library data in the scaling library.

**See Also**

- define_scaling_lib_group

---

## SLG-402

(error) Arc '%s' of cell '%s' library '%s' data is present in link library '%s', but it is not present in scaling library '%s'. This type library data scaling will not be performed for this arc.

**Description**

Scaling requires library data is consistent across all libraires in the group. If one arc lib data exists in link library but doesn't exist in scaling library, this message will be displayed.

**What Next**

Check and correct the missing library data in the scaling library.

Feedback

**See Also**

- define_scaling_lib_group

---

## SLG-403

(error) Pin '%s' of cell '%s' library '%s' data isn't consistent between link library '%s' and scaling library '%s', this type library data scaling will not be performed for this pin.

**Description**

Scaling requires library data is consistent across all libraires in the group. If one pin lib data are not consistent between link library and scaling library, this message will be displayed.

**What Next**

Check and correct the missing library data in the scaling library. If the error is related to CCS Noise model consistency but library group has corresponding CCSN models, please enable rc_ccsn_enable_library_error_reporting to check if any CCSN model is invalidated by CCS Noise model checking.

**See Also**

- define_scaling_lib_group

---

## SLG-404

(error) Arc '%s' of cell '%s' library '%s' data isn't consistent between link library '%s' and scaling library '%s', this type library data scaling will not be performed for this arc.

**Description**

Scaling requires library data is consistent across all libraires in the group. If one arc lib data are not consistent between link library and scaling library, this message will be displayed.

**What Next**

Check and correct the missing library data in the scaling library.

**See Also**

- define_scaling_lib_group

Feedback

## SLG-405

(warning) Pin '%s' of cell '%s' library '%s' data isn't consistent between link library '%s' and scaling library '%s', partial scaling will be performed on those consistent data.

### Description

Scaling requires library data is consistent across all libraries in the group. If there are some partial lib data are consistent across the lib group, scaling will only be performed on those consistent lib data and this message will be displayed.

### What Next

Check and correct the missing library data in the scaling library.

### See Also

- define_scaling_lib_group

## SLG-417

(information) The arc '%s' in lib cell '%s' in scaling library '%s' has significantly different output-capacitance indexes with link library '%s'; this can adversely affect scaling accuracy.

### Description

The *define_scaling_lib_group* requires ccs receiver cap indexes are closed between scaling library and link library, big difference could result in accuracy problem.

### What Next

Ensure that the set of arcs and pins is identical across the members of the scaling library groups. The positions of the arcs and pins within the libraries must be the same as well.

The output-capacitance indexes used for Composite Current Source (CCS) driver data is better to have identical values across the libraries. If this is not satisfied, the SLG-417 message is issued.

### See Also

- define_scaling_lib_group

## SLG-418

(warning) Timing analysis accuracy will be degraded because of problems with scaling group or library data.

Feedback

**Description**

The *define_scaling_lib_group* requires library data is consistent across all scaling libraries in the scaling group. If there is severe library data missing or inconsistency, it will bail out with SLG-001. If user enable variable timing_continue_on_slg001 to ignore the problem, this message will be displayed with a list of offending scaling check errors.

**What Next**

Correct all the library data issues for the offending check errors.

**See Also**

- define_scaling_lib_group

## SLG-419

(error) DCALM failed on cell %s as its libcell %s is not found in the specified dcalm library.

**Description**

The *define_cell_alternative_lib_mapping* command requires the libcell of the specified cell to exist in the dcalm library specified. If not dcalm will not be applied on the cell.

**What Next**

Please use the library which has the libcell for the cell used in the DCALM.

# SLM

## SLM-1005

(Error) "The parameter list is parsed as key-value pair of parameter name and parameter value, please provide even number of arguments in the parameter list."

**Description**

This error message is sent when during set_slm_element command odd number of arguments passed and it is expecting arguments as key-value pair. In this case set_slm_element will not set any PMM configuration and put this error. For example,

Error: "The parameter list is parsed as key-value pair of parameter name and parameter value, please provide even number of arguments in the parameter list." (SLM-1005)

**What Next**

Please check for missing argument and values to pass in set_slm_element command.

Feedback

## SLM-1006

(Error) "The parameter name is invalid. Supported parameters are: %s"

**Description**

This error message is sent when during set_slm_element command wrong value for '-parameter' option is passed and it is not expected. In this case set_slm_element will not set any PMM configuration and put this error. For example,

Error: "The parameter name is invalid. Supported parameters are: "PMM_NUM_PMU, PMM_PATH, PMM_DELAY, PMM_PIPE_PMC, PMM_PIPE_INTER,..."." (SLM-1006)

**What Next**

Please check for typos and suppported '-parameter' option and value to pass in set_slm_element command.

## SLM-1007

(Error) "The type

**Description**

This error message is sent when during set_slm_element command wrong value for '-type' option is passed and it is not expected. In this case set_slm_element will not set any PMM configuration and put this error. For example,

Error: "The type "pimm_test" is invalid. Supported Values are "pmm_test | hsio_pcie | hsio_usb"." (SLM-1007)

**What Next**

Please check for typos and valid '-type' option and value to pass in set_slm_element command.

## SLM-1008

(Error) "The type

**Description**

This error message is sent when during set_slm_element command not supported for '-type' option and it is being used. In this case set_slm_element will not set any PMM configuration and put this error. For example,

Error: "The type "pmm_field" is currently unsupported. Supported Values are "pmm_test | hsio_pcie | hsio_usb | pmu_field | pmc_field | cdm"." (SLM-1008)

Feedback

**What Next**

Please check for suppported '-type' option and correct value to pass in set_slm_element command.

---

## SLM-1009

(Error) "The provided parameter width

**Description**

This error message is sent when during set_slm_element command wrong or invalid width for any PMM parameters is used. In this case set_slm_element will not set any PMM configuration and put this error. For example,

Error: "The provided parameter width "45" for "PMM_PATH" is invalid. Width for "PMM_PATH" can range from "2" to "32" and be incremented only in steps of 1." (SLM-1009)

**What Next**

Please check for typos and suppported width for '-parameter' options to pass in set_slm_element command.

---

## SLM-1010

(Error) "The interface list is parsed as key-value pair of interface name and interface value, please provide even number of arguments in the interface list."

**Description**

This error message is sent when during set_slm_element command odd number of arguments is passed for '-interface' option and it is expected as key-value pair. In this case set_slm_element will not set any PMM configuration and put this error. For example,

Error: "The interface list is parsed as key-value pair of interface name and interface value, please provide even number of arguments in the interface list.." (SLM-1010)

**What Next**

Please check for the correct number of arguments to pass in '-interface' option during set_slm_element command.

---

## SLM-1011

(Error) "The parameter value is invalid. %s"

Feedback

**Description**

This error message is sent when during set_slm_element command wrong or invalid option value is used for '-paramter' option. In this case set_slm_element will not set any PMM configuration and put this error. For example,

Error: "The parameter value is invalid. The PMM_DELAY values should be power of 2." (SLM-1011) Error: "The parameter value is invalid. The PMM_PATH values should be power of 2." (SLM-1011)

**What Next**

Please check for typos and suppported '-paramter' options/value to pass in set_slm_element command.

---

## SLM-1013

(Error) "The interface name %s is invalid."

**Description**

This error message is sent when during set_slm_element command interface keyword used is not correct or valid. In this case set_slm_element will not set any PMM configuration and put this error. For example,

Error: "The interface name "PMM_PMMU_CLOCK" is invalid." (SLM-1013)

**What Next**

Please check for typos and valid '-interface' keyword options to pass in set_slm_element command.

---

## SLM-1014

(Error) "The port %s does not exist provided in interface %s ."

**Description**

This error message is sent when during set_slm_element command wrong or non existing port name used for any PMM interface. In this case set_slm_element will not set any PMM configuration and put this error. For example,

Error: "The port "my_pmu_clock0" does not exist provided in interface "PMM_PMU_CLOCK"." (SLM-1014)

**What Next**

Please check for typos and provide correct/existing port for PMM interface to pass in set_slm_element command.

Feedback

## SLM-1015

(Error) "PMU instance %s not found in any set_slm_element specification."

### Description

This error message is sent when during set_slm_element command wrong or non existing PMU instance name is used and it is not specified during set_slm_element command. In this case set_slm_element will not set any PMM configuration and put this error. For example,

Error: "PMU instance "PMU1_u1" not found in any set_slm_element specification." (SLM-1015)

### What Next

Please check for typos and existing or correct PMU instance name to pass in set_slm_loop command.

## SLM-1101

(Error) "The interface list is parsed as key-value pair of interface name and portname <interfaceName, portName>, please provide even number of arguments in the interface list. List of mandatory interface keywords %s"

### Description

This error message is sent when during set_slm_element command wrong/less argument of interface and port name pair passed, error is basically for invalid Interface List Count. In this case set_slm_element will not set any HSIO configuration and put this error. For example,

See - o_SCAN_CLK -interface [list sub_inst/i_SO1 sub_inst/i_SO2 i_SE sub_inst/o_SE i_SCAN_CLK sub_inst/o_SCAN_CLK o_SI sub_inst/i_SI i_SO sub_inst/o_SO o_SE sub_inst/i_SE o_SCAN_CLK] Error: "The interface list is parsed as key-value pair of interface name and portname, please provide even number of arguments in the interface list. List of mandatory interface keywords is "aclk, aresetn, awaddr, awvalid, awready, wid,...."." (SLM-1101)

### What Next

Please check number arguments passed in '-interface' options during set_slm_element command.

Feedback

## SLM-1102

(Error) "The %s list is parsed as key-value pair of %s, please provide correct number of arguments."

### Description

This error message is sent when during set_slm_element command wrong number of arguments for parameter or configuration option is passed. List should be even and expected as key-value pair. In this case set_slm_element will not set any configuration and put this error for invalid List Count. For example,

Error: "The configuration list is parsed as key-value pair of option name and value, please provide correct number of arguments." (SLM-1102) Error: "The parameter list is parsed as key-value pair of parameter name and width, please provide correct number of arguments." (SLM-1102)

### What Next

Please provide correct number of argument as key-value pair to set_slm_element comman.

## SLM-1103

(Error) "The provided interface type

### Description

This error message is sent when during set_slm_element command wrong value/spelling of interface keyword is used. In this case set_slm_element will not set any configuration and put this error. For example,

Error: "The provided interface type "o_rvaalid" is incorrect. List of supported interface types: supported interface list"" (SLM-1103)

### What Next

Please remove invalid Interface Keyword Provided during set_slm_element and pass the correct one.

## SLM-1104

(Error) "The provided %s type

Feedback

**Description**

This error message is sent when during set_slm_element command wrong value of parameter and configuration option is used . In this case set_slm_element will not set any configuration and put this error. For example,

Error: "The provided configuartion type "jtagg" is incorrect. List of supported configuration types: "jtag, debug, security,..."." (SLM-1104)

**What Next**

Please check typos or supported option values to pass in set_slm_element command.

---

## SLM-1105

(Error) "The argument

**Description**

This error message is sent when during set_slm_element command wrong value option for any configuration is passed and it is not as expected. In this case set_slm_element will not set any configuration and put this error. For example,

Error: "The argument "dissable" provided with configuration "debug" is incorrect. Supported values are enable|disable" (SLM-1105)

**What Next**

Please check for valid and supported option values for set_slm_element command.

---

## SLM-1106

(Error) "The provided port name

**Description**

This error message is sent when during set_slm_element command wrong value/name of interface port is used. In this case set_slm_element will not set any configuration and put this error. For example,

Error: "The provided port name "sub_inst/i_rvalid" is incorrect. Please provide a valid port name." (SLM-1106)

**What Next**

Please check typos and correct values to pass as arguments during set_slm_element command.

Feedback

## SLM-1107

(Error) "The provided parameter width

**Description**

This error message is sent when during set_slm_element command wrong value of width used for any HSIO parameter. In this case set_slm_element will not set any configuration and put this error. For example,

Error: "The provided parameter width "280" for "AXI_DATA_WIDTH" is invalid. Width for "AXI_DATA_WIDTH" can range from "32" to "256" and be incremented only in steps of 1." (SLM-1107)

**What Next**

Please provide correct and supported HSIO option name/value to set_slm_element command.

## SLM-1108

(Error) "Please use set_slm_configuration -hsio enable option."

**Description**

This error message is sent when command set_slm_element -type hsio_pcie/hsio_usb is being used without enabling hsio. In this case set_slm_element will not set any configuration and put this error. For example,

Error: "Please use set_slm_configuration -hsio enable option." (SLM-1108)

**What Next**

Please check whether HSIO is enable using set_slm_configuration.

## SLM-1109

(Error) "The provided parameter list does not contain the mandatory parameters %s. List of mandatory parameters: %s"

**Description**

This error message is sent when during set_slm_element command mandatory parameter option keywords are not passed. In this case set_slm_element will not set any configuration and will ask to add those parameters by putting this error. For example,

Error: "The provided parameter list does not contain the mandatory parameters "[list AXI_ADDR_WIDTH 32 AXI_DATA_WIDTH 64 NUM_SCAN_DATA 64

Feedback

NUM_SCAN_CTRL_DYNAMIC_DATA 8 NUM_SCAN_CTRL_STATIC_DATA 8 NUM_SCAN_CLK 32 TAP_CTRL_WIDTH 1 AXI_WR_ID_WIDTH 4 AXI_RD_ID_WIDTH 4]". List of mandatory parameters:." (SLM-1109)

**What Next**

Please check missing Required Parameters to pass in set_slm_element command.

---

## SLM-1110

(Error) "The provided interface list does not contain the mandatory interfaces %s. List of mandatory interfaces: %s"

**Description**

This error message is sent when during set_slm_element command mandatory interface keywords are missing. In this case set_slm_element will not set any configuration and put this error. For example,

Error: "The provided interface list does not contain the mandatory interfaces [list aresetn i_aresetn awid i_awid awaddr i_awaddr awlen i_awlen awsize i_awsize awburst i_awburst]. List of mandatory interfaces: "aclk, aresetn, awaddr, awvalid, awready, wid,...."." (SLM-1110)

**What Next**

Please check for missing Required Interfaces options to pass in set_slm_element command.

---

## SLM-1111

(Error) "%s

**Description**

This error message is sent when during set_slm_element command when existing port connection used again or duplicately for any another connection. In this case set_slm_element will not set second connection and put this error. For example,

-interfcae [list aresetn_invalid i_aresetn awid i_awid_invalid awlen i_awlen awlen i_awlen awsize i_awsize] Error: ""awlen" "i_awlen" driving same "awlen" ports." (SLM-1111)

**What Next**

Please remove duplicate and redundant enteries of arguments in set_slm_element command.

Feedback

## SLM-1112

(Error) "Module definition of instance

**Description**

This error message is sent when during set_slm_element command module instance not found for the pin specified. In this case set_slm_element will not set any HSIO configuration and put this error. For example,

Error: "Module definition of instance "sub_inst1" not found." (SLM-1112)

**What Next**

Please provide correct instance/sub_instanace name to set_slm_element command.

## SLM-1113

(Error) "Required argument '-interface' was not found."

**Description**

This error message is sent when during set_slm_element command '-interface' option is not being used. In this case set_slm_element will not set any HSIO configuration and put this error. For example,

set_slm_element -configuration [jtag enable] -parameters [list AXI_DATA_WIDTH 48]
Error: "Required argument '-interface' was not found." (SLM-1113)

**What Next**

Please provide 'interface' option to set_slm_element command.

## SLM-1114

(Error) "Wrong bus width specified for the following interfaces %s."

**Description**

This error message is sent when during set_slm_element command wrong value of width used for any HSIO interface port. In this case set_slm_element will not set any configuration and put this error. For example,

Error: "Wrong bus width specified for the following interfaces "i_SO"." (SLM-1114)

**What Next**

Please provide correct and supported bus width for port/pin to set_slm_element command.

Feedback

## SLM-1115

(Error) "Expected width of

**Description**

This error message is sent when during set_slm_element command wrong value of width used for any HSIO parameter. In this case set_slm_element will not set any configuration and put this error. For example,

Error: "Expected width of "sub_inst/o_SI" being connected to sub_inst/o_SO is [10:2], but found it [20:0] in design." (SLM-1115)

**What Next**

Please provide correct and supported bus width to set_slm_element command.

## SLM-1116

(Error) "Expected direction of

**Description**

This error message is sent when during set_slm_element command when port or pin with wrong direction is used as part of -interface option. In this case set_slm_element will not set any configuration and put this error. For example,

Error: "Expected direction of "sub_inst/o_TRSTN" being connected to "sub_inst/i_TRSTN" is "in", but found it "out" in design" (SLM-1116)

**What Next**

Please check and provide valid Port Direction to set_slm_element command.

## SLM-1117

(Error) "Please enable ebc_mode when ebc_sram is enabled."

**Description**

This error message is sent when in command set_slm_element we enable ebc_sram configuration without enabling ebc_mode Configuration. For example, set_slm_element \ -type hsio_usb \ -parameter [list FIFO_PTR_WIDTH 9 AXI_ADDR_WIDTH 32 AXI_DATA_WIDTH 64 NUM_SCAN_DATA 64 NUM_SCAN_CTRL_DYNAMIC_DATA 8 NUM_SCAN_CTR L_STATIC_DATA 8 NUM_SCAN_CLK 32 TAP_CTRL_WIDTH 1 AXI_WR_ID_WIDTH 4 AXI_RD_ID_WIDTH 4] \ -configuration [list ebc_mode disable ebc_sram enable]\

Feedback

Error: "Please enable ebc_mode when ebc_sram is enabled." (SLM-1117)

**What Next**

Please check configuration in set_slm_element command

---

## SLM-2000

(Error) "%s

**Description**

This error message is sent when during set_slm_element command duplicate or multiple input enteries found for some option or its value. In this case set_slm_element will not set any PMM configuration and put this error. For example,

-parameter {PMM_NUM_PMU 5 PMM_PATH 4 PMM_PATH 8 PMM_DELAY 4 PMM_PIPE_PMC 2 PMM_PIPE_INTER 1 PMM_PIPE_PMU 3} Error: ""PMM_PATH 4" already specified. Ignoring duplicate "PMM_PATH 8"." (SLM-2000)

**What Next**

Please check and remove redundant and duplicate inputs passed during set_slm_element command.

---

## SLM-2001

(Error) "Number of PMU clocks are not equal as number of PMUs"

**Description**

This error message is sent when there is mismatch between number PMU clocks and number of PMUs. In this case set_slm_element will not set any PMM configuration and put this error. For example,

Error: "Number of PMU clocks are not equal as number of PMUs" (SLM-2001)

**What Next**

Please check for total number of PMU clocks and PMU's set during set_slm_element command.

---

## SLM-2100

(Warning) "Expected width of "%s" being connected to "%s" is "[%d:%d]", but found it "[%d:%d]" in design. Considering width of "[%d:%d]"."

Feedback

**Description**

This warning message is sent when bus width is not as expected as specified in command set_slm_element . In this case set_slm_element will not set any configuration and put this warning. For example,

warning: "Expected width of sub_inst/o_SI being connected to sub_inst/o_SO is [10:2], but found it [20:0] in design. Considering width of [10:2]" (SLM-2100)

**What Next**

Either provied correct bus width during set_slm_element or continue with considered bus width.

---

## SLM-2101

(Warning) "No port connection specifed for %s interface "%s", tying to constant "%s""

**Description**

This warning message is sent when port connection is not as expected as specified in command set_slm_element . In this case set_slm_element will set specified port with constant and put this warning. For example,

warning: "No port connection specifed for "o_rvalid" interface "rvalid", tying to constant "sub_inst/i_rvalid"." (SLM-2101)

**What Next**

Either provied correct connection for interface during set_slm_element or continue with considered connection.

---

## SLM-2102

(Error) "Please enable jtag when scan_over_tap is enabled."

**Description**

This Error message is sent when scan_over_tap is enable with jtag disable using configuartion option in command set_slm_element.

Error: "Please enable jtag when scan_over_tap is enabled." (SLM-2102)

**What Next**

Please enable jtag during set_slm_element.

Feedback

## SLM-2103

(Warning) "%s"%s" driven by %s "%s" ports"

**Description**

This warning message is sent when ports are not driven correctly, as specified in command set_slm_element . For example,

warning: ""i_TAP_CTRL sub_inst/o_TAP_CTR" driven by "i_TAP_CTRL sub_inst_1/o_TAP_CTR" ports." (SLM-2103)

**What Next**

Either provied correct port connection or continue with considered connection.

## SLM-2104

(Warning) "%s "%s:%s" already specified. Ignoring duplicate "%s:%s""

**Description**

This warning message is sent when port Duplicate Inputs are given, using interface option in command set_slm_element. In this case set_slm_element will set the value which is provided in the beginning rest of them will be ignored and put this warning. For example,

warning: "sub_inst/o_SI[12:8] already specified. Ignoring duplicate sub_inst/o_SI[14:8]." (SLM-2104)

**What Next**

Remove redundant duplicate enteries in interface option during set_slm_element.

## SLM-2105

(Warning) "%s should be enabled when %s interface keyword is specified."

**Description**

This warning message is sent when some interface port is specified without enabling correct configuration option. In this case set_slm_element will put this warning to enable particular configuration. For example,

warning: ""debug" should be enabled when "debugCtrlEn" interface keyword is specified." (SLM-2105)

Feedback

**What Next**

Either provied correct confuguration for interface option during set_slm_element or continue with considered connection.

---

## SLM-2106

(Warning) "%s option is not available when "%s" -type is specified."

**Description**

This warning message is sent when some configuration option is used when it is not supported with specific hsio type. In this case set_slm_element will ignore the configuration and put this warning. For example,

warning: "'usb_ebc_mode' option is not available when "hsio_pcie" -type is specified." (SLM-2106)

**What Next**

Either set correct type or avoid usage of configuration option during set_slm_element.

---

## SLM-2107

(Error) "PDL file passed as the target doesn't exist."

**Description**

This error message is sent when during write_test_protocol command

For example,

Error: "PDL file passed as the target doesn't exist." (SLM-2107)

**What Next**

Please check for correct PDL file name to be passed with target under write_test_protocol command.

---

# SML

---

## SML-001

(Warning) %s: Low memory availability: system memory usage is %s for the last %s minutes.

Feedback

**Description**

Host machine is running low on available memory. System memory usage percentage mentioned in message is percentage of RAM, swap and cache memories put together.

**What Next**

Run on a host with sufficient RAM or on a host where it is not competing with other processes for RAM.

---

## SML-002

(Warning) %s: High CPU usage: current process(%s); other processes(%s) for the last %s minutes.

**Description**

The current process is using less percent of total cpu but the host is running with very high load (greater than 95%).

**What Next**

Run on a host with sufficient cores to satisfy the number of cores the tool is specified to use.

---

## SML-003

(Warning) %s: Poor network. Packet retransmission rate is high(%s) for the last %s minutes.

**Description**

The current process is running on a host where more than 5 percent of all packets are being retransmitted due to loss or damage. It indicates the network is heavily congested.

**What Next**

Contact your IT department to check and address the poor network connectivity problem.

---

## SML-004

(Warning) %s: Poor network. Packet loss is high(%s) for the last %s minutes.

**Description**

The current process is running on a host where more than 5 percent of all packets it is sending are not being received by the recipient. It indicates poor network connectivity.

Feedback

**What Next**

Contact your IT department to check and address the poor network connectivity problem.

---

## SML-005

(Warning) %s: High network latency to these hosts: %s.

**Description**

The current process is running on a host that has high network latency (>300ms) to the hosts specified in the message. These hosts may include disk filers.

**What Next**

Contact your IT department to check and address the high network latency to specified hosts and/or disk filers.

---

## SML-006

(Warning) Failed to initialize SML.

**Description**

SML was not initialized due to internal errors in SML library.

**What Next**

Contact your AC.

---

# SPCNL

---

## SPCNL-001

(warning) No cells named '%s' found in top module %s.

**Description**

No cells with the given name is found on the top module.

**What Next**

Ensure the existence of cell name is correct.

---

## SPCNL-002

(error) The cell named '%s' is present in both -die1 and -die2 lists.

Feedback

**Description**

The given cell name is common in -die1 and -die2 command options. These lists are expected to be mutually exclusive.

**What Next**

Ensure that no cells are common in -die1 and -die2 command options.

# SQM

## SQM-0001

(error) Cannot map sequential element '%s' in module '%s'. Library has no flip-flops of the required type. Check that usable flip-flops with the required type of asynchronous set/reset signals are available in the library.

**Description**

This message is issued when there is no logically equivalent flip-flops available in library during initial mapping or when there are only flops with opposite clock edge which is not allowed by user setting.

**What Next**

Check that usable flip-flops with the required type of asynchronous set/reset signals are available in the library. For example, the design requires both asynchronous set/reset signals while library has reset-only cells. The cell can't be mapped in this case; Or check if library only contains flop with opposite clock and any of following app options is set to true: *seqmap.count_added_inverters_for_flops* => This app option deprecates in T release and is aliased to *seqmap.honor_clock_sense_for_registers compile.seqmap.preserve_rtl_clock_phase_strictly*

## SQM-0002

(error) Cannot map sequential elements of module '%s'. Library has no flip-flops.

**Description**

Cannot map sequential element. Library has no flip-flops.

**What Next**

Need to provide flip-flops in library. ~

Feedback

## SQM-0003

(error) Cannot map retention register '%s' in module '%s'. No replacement set available.

**Description**

This error message is issued when no replacement set available for the retention cell. For example, libcell is not specified in retention strategy.

**What Next**

Check library and retention strategy to make sure that repalcement set available for retention cell ~

## SQM-0004

(error) Cannot map sequential elements of module '%s'. Library has no latches.

**Description**

Cannot map sequential element. Library has no latches.

**What Next**

Need to provide latches in library. ~

## SQM-0005

(error) Cannot map sequential element '%s' in module '%s'. Library has flip-flops of the required type, but they are not usable in the module context. Check purpose, target library subset and voltage domain compatibility of library elements.

**Description**

This message is issued when there are logically equivalent flip-flops available in library but the library cell can't be used in the module context.

**What Next**

- Use *analyze_mv_feasibility* to report mapping failure for retention cells. This command generates a report indicating the failure reason for each mapping cell. A more detailed version of this report (in HTML format) can be created with *analyze_mv_feasibility -format html*. It's possible to set *-format html* as default option assigning the application option *mv.upf.enable_amvf_html_report* to *true* with the command *set_app_options -as_user_default -name mv.upf.enable_amvf_html_report -value true*.
- For retention mapping failure, check if a suitable lib cell is specified in *map_retention_cells* setting, and those lib cells are usable.

Feedback

- Check purpose, target library subset and voltage domain compatibility of library elements.
- Check if the qualified cells have dont_use attribute or do not match pvt parameter making them not usable.

**See Also**

- analyze_mv_feasibility
- set_app_options
- mv.upf.enable_amvf_html_report

---

## SQM-0011

(error) Cannot map sequential element '%s' in module '%s'. Library has no latch of the required type.

**Description**

This message is issued when there is no logically equivalent latches available in library during initial mapping.

**What Next**

Library has no latch of the required type. ~

---

## SQM-0015

(error) Cannot map sequential element '%s' in module '%s'. Library has latches of the required type, but they are not usable in the module context. Check purpose, target library subset and voltage domain compatibility of library elements.

**Description**

This message is issued when there is logically equivalent latches available in library but the libcell can't be used in the module context.

**What Next**

o Use *analyze_mv_feasibility* to report mapping failure for retention cells. This command generates a report indicating the failure reason for each mapping cell. A more detailed version of this report (in HTML format) can be created with *analyze_mv_feasibility -format html*. It's possible to set *-format html* as default option assigning the application option *mv.upf.enable_amvf_html_report* to *true* with the command *set_app_options -as_user_default -name mv.upf.enable_amvf_html_report -value true*.

Feedback

o For retention mapping failure, check if a suitable lib cell is specified in *map_retention_cells* setting, and those lib cells are usable.

o Check purpose, target library subset and voltage domain compatibility of library elements.

o Check if the qualified cells have dont_use attribute or do not match pvt parameter making them not usable.

**See Also**

- analyze_mv_feasibility
- set_app_options
- mv.upf.enable_amvf_html_report

---

## SQM-1005

(warning) In module '%s' : No negative-edge flip-flops available in library. Resorting to positive-edge flip-flops.

**Description**

Negative-edge flip-flops are prefered but no such flip-flops available in library. Resorting to positive-edge flip-flops for mapping.

**What Next**

If negative-edge flip-flops must be used, please check that usable negative-edge flip-flops are available in the library. ~

---

## SQM-1006

(warning) In module '%s' : No positive-edge flip-flops available in library. Resorting to negative-edge flip-flops. Adding inversions on the clock net.

**Description**

Positive-edge flip-flops are prefered but no such flip-flops available in library. Resorting to negative-edge flip-flops for mapping.

**What Next**

If positive-edge flip-flops must be used, please check that usable positive-edge flip-flops are available in the library. ~

Feedback

## SQM-1007

(warning) Unable to swap out libcell '%s' for sequential element '%s'. Libcell/instance either has dont_use attribute or must change scan type but couldn't find a match.

**Description**

Swap for dont_use/opposite scan preference libcell is attemped but not successful. No match is found.

**What Next**

Provide matching cell in the library for the target dont_use/opposite scan preference libcell. ~

## SQM-1008

(warning) For sequential element '%s' of module '%s' : No flip-flops available in library with requested dedicated scan output type. Instance '%s' is mapped to different scan output type flip-flop '%s'

**Description**

No flip-flops available in library with requested dedicated scan output type. Instance is mapped to different scan output type flip-flop. For example, flip-flop with dedicated scan output pin is requested but library only has flip-flop with no dedicated scan output pin.

**What Next**

If certain dedicated scan output type libcell must be used, provide such cell in the library. ~

## SQM-1009

(warning) For sequential element '%s' of module '%s' : No flip-flops available in library with requested scan type. Instance '%s' is mapped to non-scan flip-flop '%s'

**Description**

No flip-flops available in library with scan type. Instance is mapped to non-scan flip-flop.

**What Next**

Check that usable flip-flops with scan type are available in the library if necessary. ~

## SQM-1010

(warning) For sequential element '%s' of module '%s' : No non-scan flip-flops available in library. Instance '%s' is mapped to scan flip-flop '%s'

Feedback

**Description**

No non-scan flip-flops available in library. Instance is mapped to scan type flip-flop.

**What Next**

Check that usable non-scan type flip-flops are available in the library if necessary. ~

---

## SQM-1011

(warning) Sequential element '%s' of module '%s' was mapped to a a flip-flop without dedicated scan out pin. Constraints placed on dedicated scan out pin will be dropped

**Description**

Sequential element was mapped to a flip-flop without dedicated scan out pin. Constraints placed on dedicated scan out pin will be dropped.

**What Next**

Check that usable flip-flops with dedicated scan out pin are available in the library if necessary. ~

---

## SQM-1012

(warning) For sequential element '%s' of module '%s' : No non-scan flip-flops available in library. Trying to map to scan flip-flop.

**Description**

No non-scan flip-flops available in library. Instance is mapped to scan type flip-flop. This happens during swapping out cells that is marked as don't use.

**What Next**

Check that usable non-scan type flip-flops are available in the library if necessary. ~

---

## SQM-1015

(warning) Sequential element '%s' of module '%s' could not be mapped without state inversion. For this element, allowing mapping with state inversion

**Description**

This warning message is issued when no solution is found to map the sequential element without output inversion. We are trying to map the element with output inverted. The solution is not guaranteed.

Feedback

**What Next**

- Check library and provided cells with desired Q/QN pins if the cell has to be mapped without output inversion.
- If you want to leave cell unmapped and don't want output inversion, you can use the following app option: set_app_options -as_user_default -list {compile.seqmap.allow_output_inversion_if_unmapped false}

---

## SQM-1016

(warning) Sequential element '%s' of module '%s' could not be mapped without state inversion. For this element, changing asynchronous set and reset priority

**Description**

This warning message is issued when no solution is found via regular mapping flow and changing asynchronous set/reset priority is allowed for mapping. However, solution is not guranteed.

**What Next**

If you want to leave cell unmapped and don't want to change asynchronous set/reset priority, you can use the following app option: set_app_options -as_user_default -list {compile.seqmap.allow_async_set_reset_priority_inversion_if_unmapped false} ~

---

## SQM-1019

(warning) For sequential element '%s' of module '%s' : No latches available in library with requested scan type. Instance '%s' is mapped to non-scan latch '%s'

**Description**

No latches available in library with scan type. Instance is mapped to non-scan latch.

**What Next**

Check that usable latches with scan type are available in the library if necessary. ~

---

## SQM-1020

(warning) Multibit lib cell '%s' is ignored because of incompatible attributes.

**Description**

Multibit libcell is ignored because one or some of its library attributes is/are not valid. For example, no module is found; no timing design is found; multibit width is less or equal to 1, etc.

Feedback

**What Next**

Check the attribute of the reported libcell. Fix if possible. ~

---

## SQM-1021

(warning) Multibit lib cell '%s' is ignored because of no valid functional description and no single-bit representative.

**Description**

The warning message is issued when multibit cell has no valid functional description and no single-bit representative.

**What Next**

Check the library of the reported lib cell. Provide functional description and single-bit representative as needed. ~

---

## SQM-1022

(warning) Library cell %s has inconsistent port directions: port %s (of slice %u), direction %s vs. port %s (of slice 0), direction %s.

**Description**

This warning is issued when multibit lib cell has inconsistent port directions among different slices. Cell will be marked as invalid.

**What Next**

Check the library of the reported lib cell. Make sure multibit cell ports are consistent. ~

---

## SQM-1023

(warning) Library cell %s, port %s (id %u) has %zu corresponding single-bit pins.

**Description**

This warning is issued when a certain multibit cell port corresponds to more than two single-bit pins, which is not allowed. Cell will be marked as invalid.

**What Next**

Check the library of the reported lib cell. Fix pin relation as needed. ~

Feedback

## SQM-1024

(warning) Library cell %s, inconsistent scan enable pins: %s (id %u) vs. %s (id %u).

**Description**

This warning is issued when multibit cell scan enable pin id does not match its single-bit cell scan enable pin id.

**What Next**

Check the library of the reported lib cell. Fix pin id as needed. ~

## SQM-1025

(warning) Library cell %s, inconsistent scan input pins: %s (id %u) vs. %s (id %u).

**Description**

This warning is issued when multibit cell scan input pin id does not match its single-bit cell scan input pin id.

**What Next**

Check the library of the reported lib cell. Fix pin id as needed. ~

## SQM-1026

(warning) Library cell %s has no scan enable pin.

**Description**

This warning is issued when multibit cell does not have scan enable pin.

**What Next**

Check the library of the reported lib cell. Use a different cell if needed. ~

## SQM-1027

(warning) Library cell %s has no scan input pin.

**Description**

This warning is issued when multibit cell does not have scan input pin.

**What Next**

Check the library of the reported lib cell. Use a different cell if needed. ~

Feedback

## SQM-1028

(warning) Library cell %s has no scan output pin.

**Description**

This warning is issued when multibit cell does not have scan output pin.

**What Next**

Check the library of the reported lib cell. Use a different cell if needed. ~

## SQM-1029

(warning) Library cell %s, inconsistent serial scan chain: slice %u, scan out pin %s (id %u) vs. slice %u, scan in pin %s (id %u) are different -- they should be the same.

**Description**

This warning is issued when multibit cell scan out pin doesn't match scan in pin of the next slice when it's in serial scan mode.

**What Next**

Check multibit cell scan chain connection. ~

## SQM-1030

(warning) Library cell %s, inconsistent parallel scan chain: slice %u, scan out pin %s (id %u) and slice %u, scan in pin %s (id %u) are the same -- they should be different.

**Description**

This warning is issued when multibit cell scan out pin matches scan in pin of the next slice when it's in parallel scan mode, which should not happen.

**What Next**

Check multibit cell scan chain connection. ~

## SQM-1031

(warning) Library cell %s, slice %u, scan out pin %s (id %u) corresponds to single-bit degenerate cell %s, non-scan out pin %s (id %u).

Feedback

**Description**

This warning is issued when multibit cell dedicated scan out pin corresponds to single bit degenerate non-dedicated scan out pin. Cell is marked as invalid.

**What Next**

Check library for the reported cell for more information. ~

---

## SQM-1032

(warning) Library cell %s, slice %u, non-scan out pin %s (id %u) corresponds to single-bit degenerate cell %s, scan out pin %s (id %u).

**Description**

This warning is issued when multibit cell non-dedicated scan out pin corresponds to single bit degenerate dedicated scan out pin. Cell is marked as invalid.

**What Next**

Check library for the reported cell for more information. ~

---

## SQM-1033

(warning) Cell bus '%s' (register '%s') cannot be mapped to multibit cell.

**Description**

This warning message is issued when instance can't be mapped to multibit cell due to its own reason. Refer to SQM-1036 for exact reason.

**What Next**

Check SQM-1036 for exact reason. These two message should come together. ~

---

## SQM-1034

(warning) No cell bus will be mapped to multibit because multibit mapping has been disabled by user.

**Description**

This warning message will be issued when (1) seqmap.enable_multibit is false && compile.seqmap.enable_multibit is false (or) (2) compile.seqmap.enable_rtl_multibit_banking is false This basically mean multibit mapping is disabled by user

Feedback

### What Next

Check script for these settings to see if they are expected. ~

---

## SQM-1035

(warning) Cell '%s' cannot be mapped to multibit because constraint(s) on its shared pins (such as clock, enable, set/reset, scan enable, etc) is(are) incompatible to rest of the cell bus.

### Description

This warning message is issued when multibit mapping is enabled but cell can't be mapped to multibit due to constraints on its shared pins.

### What Next

Check constraints set on cell shared pins to see if they can be removed so that cell can be mapped to multibit. ~

---

## SQM-1036

(warning) %s

### Description

This warning message is issued when instance can't be mapped to multibit cell due to its own reason. This message contains three possible reasons: 1. Cell is excluded from mapping to multibit by 'set_multibit_options -exclude' directive. 2. Cell cannot be mapped to multibit because it is a DFT register. 3. Cell cannot be mapped to multibit because of non-timing size_only restriction on it.

### What Next

~

---

## SQM-1037

(warning) Sequential element '%s' could not be mapped, because there is no matching library cells available for it.

### Description

This warning message is issued when sequential element is left unmapped after all possible trial.

Feedback

**What Next**

Check library to provide matching library cells for the unmapped instance. ~

---

## SQM-1040

(warning) No cell bus will be mapped to multibit because there is no multibit cells in the cell libraries.

**Description**

This warning message is issued when bus instance is to be mapped but library doesn't have multibit cells.

**What Next**

Check library to provide multibit cells for bus mapping. ~

---

## SQM-1041

(warning) Could not map retention cell '%s' to multibit due to lack of compatible multibit library cells.

**Description**

This warning message is issued when instance is retention cell and retention repalcement set is available but there is no multibit cell in repalcement set.

**What Next**

Check library and retention strategy to make sure retention multibit cell is added to repalcement set. ~

---

## SQM-1046

(warning) Sequential element '%s' of module '%s' is mapped to library cell '%s' or its equivalents, which don't have nextstate:data defined on input port. This may result in synchronous input nets swapped.

**Description**

This warning message is issued when instance is mapped to a libcell where there is no nextstate_type attribute defined on its ports. In this case, functional input pairs like D/SET, D/CLEAR can be swapped during net connection as they are logically equivalent. Tool can't tell which port is the "true" data port.

Feedback

### What Next

User can ignore this warning if they don't care about pin connection. Otherwise choose another libcell which is properly defined with nextstate_type. For example: get_attribute [get_lib_pins -of_objects [get_lib_cells */DFPCRQX1HXP ] ] nextstate_type ~

---

## SQM-1047

(warning) Sequential element '%s' of module '%s' is mapped to library cell '%s' or its equivalents, which have multiple input ports defined as nextstate:data. This may result in synchronous input nets swapped.

### Description

This warning message is issued when instance is mapped to a libcell where there are multiple ports defined as nextstate:data. In this case, functional input pairs like D/SET, D/CLEAR can be swapped during net connection as they are logically equivalent. Tool can't tell which port is the "true" data port.

### What Next

User can ignore this warning if they don't care about pin connection. Otherwise choose another libcell which is properly defined with nextstate_type. Follow the example to query nextstate type value on libcell pins: get_attribute [get_lib_pins -of_objects [get_lib_cells */DFPCRQX1HXP ] ] nextstate_type ~

---

## SQM-1051

(warning) Sequential element '%s' of module '%s' cannot be mapped single-bit, allowing it to be mapped to multibit lib cells.

### Description

This warning message is issued when the sequential mapper fails to map the sequential cell to a single-bit libcell. It then tries to find a multibit mapping for the sequential cell in order to avoid leaving it unmapped.

### What Next

Check the attributes or restrictions set on the single-bit library cells that prevents them from being used for sequential mapping.

---

## SQM-1054

(warning) User provided DW synchronous cell list '%s' doesn't have valid single stage libcell. Required cell should be valid in library and have

Feedback

**Description**

This warning is issued when user didn't provide valid single stage cell for DW synchronizers via "set_syncff_libcells".

**What Next**

Check the cell list carefully and make sure there is single stage cell; Make sure the single stage cell has "ff_chain" attribute defined; Check that the single stage cell can be used(no dont_use restriction) ~

---

## SQM-1055

(warning) Could not swap out user size_only sequential element '%s' without state inversion. Allowing mapping with state inversion.

**Description**

This warning is issued when a sequential instance is set with user size_only restriction but can't find a solution where we can swap out the cell without phase inversion.

**What Next**

No other action needed. For user size_only cells, it's required to try remapping them without input/output inversion first.

~

---

## SQM-1056

(warning) User provided DW synchronous multistage cell '%s' is not compatible with single stage cell '%s' in terms of matching keys. Cell is discarded

**Description**

This warning is issued when any user provided multi-stage cell for DW synchronizers is not compatible with single stage cell via "set_syncff_libcells" in terms of functional keys.

**What Next**

Locate the single stage cell in the cell list and get the string value for its ff_chain attribute. Make sure the multi-stage reported is in that string; If it's in the string, check their library details. It could be a mistake to mix two types of libcells in script setting.

~

Feedback

## SQM-1057

(warning) User provided DW synchronous multistage cell '%s' is not compatible with current design PVT setting. Cell is discarded.

**Description**

This warning is issued when any user provided multi-stage cell for DW synchronizers is not compatible with current design PVT setting.

**What Next**

Check the PVT settings for single stage cell library and multi-stage cell library as well as the script setting, There could be a mismatch. For example, multi-stage cell has only one voltage pane 0.8, but the power root is set to 0.9.

~

## SQM-1058

(warning) Libcell '%s' has a dedicated scanout pin but does not have the requisite attributes for sequential mapping.

**Description**

This warning message is issued when the libcell has a dedicated scanout pin but it does not have the attributes on the pin that are required for use by sequential mapping.

**What Next**

Check the attributes set on the dedicated scanout pin of this libcell.

## SQM-1062

(warning) Gating asynchronous set/reset for scan cells on net %s can't be done because it is driven by port

**Description**

This warning is issued when scan register is qualified for gating but its asynchronous nets are driven by port directly.

**What Next**

No further action needed. Gating for this register will be skipped.

~

Feedback

## SQM-1063

(warning) Gating asynchronous set/reset for scan cells can't be done because user didn't provide valid control signal

**Description**

This warning is issued when user didn't provide gating signal in script or tool couldn't find the signal on top level hier.

**What Next**

Check the script and make sure gating signal is properly set with command "set_register_async_gating_pin"

~

## SQM-1065

(warning) User has explicitly set 'compile.seqmap.scan' to false, thus mapping to scan flop is disabled.

**Description**

This warning is issued when user set "compile.seqmap.scan" to false. This means cells will be mapped to non-scan type by default.

**What Next**

Check your script to see if this app option is set as desired.

~

## SQM-1066

(warning) Sequential cell '%s' has noScan attribute, thus it won't be mapped to scan cell.

**Description**

User has explicitly set 'set_scan_element false' on the cell or the module containing this cell. Thus the cell has noScan attribute and will be mapped to non-scan cell.

**What Next**

Check your script to see if this command is set as desired.

~

Feedback

## SQM-1067

(warning) Mapping to scan cell '%s' in library is not possible because cell has 'dont_use' attribute.

**Description**

This is to warn user that certain scan cell in library can't be used because they are marked as dont_use.

**What Next**

Check library to make sure there is required scan cell.

~

## SQM-1068

(warning) Prefer scan over multibit banking ratio as user set 'seqmap.prefer_scan_over_mb_banking' to true

**Description**

This is to warn user that when multibit scan solution is not found, tool will prefer single bit scan solution instead of multibit non-scan solution.

**What Next**

Check script and make sure the app option is set as expected.

~

## SQM-1069

(warning) Prefer multibit banking ratio over scan as user set 'seqmap.prefer_scan_over_mb_banking' to false

**Description**

This is to warn user that when multibit scan solution is not found, tool will prefer multibit non-scan solution instead of single bit scan solution.

**What Next**

Check script and make sure the app option is set as expected.

~

Feedback

## SQM-1072

(warning) Latch cell '%s' couldn't honor user setting 'compile.seqmap.preserve_rtl_clock_phase_strictly' because either 'compile.seqmap.allow_tieoffs_for_latches' is false or library doesn't have appropriate cell

### Description

This warning is issued when user explicitly set 'compile.seqmap.preserve_rtl_clock_phase_strictly' to true but latch cell mapping couldn't honor it. There are two possible reasons: 1. 'compile.seqmap.allow_tieoffs_for_latches' is set to false. It will deactivate above app option so that tool will not spend the effort; 2. 'compile.seqmap.allow_tieoffs_for_latches' is set to true but tool couldn't find appropriate libcell.

### What Next

Check user setting for 'compile.seqmap.allow_tieoffs_for_latches'. Correct the value to true if it's set to false(if possible).

~

## SQM-1073

(warning) Sequential mapper only sees LSSD type scan cells in library but LSSD mapping is not enabled. No muxed scan cell is available.

### Description

This warning is issued because LSSD mapping is not enabled but sequential mapper couldn't see any mux based scan cell in library. Instead, all scan cells in library are marked with LSSD attribute. It's possible that these libcells are marked with wrong scan type.

### What Next

Check to see if library scan cells are marked with the correct scan type. Or turn on LSSD mapping support.

~

## SQM-1074

(warning) Shift register Inferencing is enabled so only head flop will be replaced with scan cell and not all registers in shift register chains are scan replaced.

Feedback

**Description**

This warning is to tell user that shift register inferring is enabled and only head register will be mapped to scan type.

**What Next**

NA

~

---

## SQM-1075

(warning) Gating asynchronous set/reset for scan cells can't be done because design hasn't been scan stitched.

**Description**

This warning is issued because the design is not scan stitched. If the design is not scan stitched, then there's no need for cells to be gated.

**What Next**

NA

~

---

## SQM-1076

(warning) Gating asynchronous set/reset for scan cells can't be done because user provided control port '%s' is driving other loads.

**Description**

This warning is issued when control port is driving other loads. The port is set by user via "set_register_async_gating_pin -port xxx". It must be top level that doesn't drive any other loads.

**What Next**

Create a new port on top level and put its name as control port for gating

~

---

## SQM-1080

(warning) Complex sequential cell %s is not swapped to complex retention cell. App option mv.cells.infer_complex_retention_cells needs to be set to enable this feature.

Feedback

### Description

This warning is issued when user set a complex instance(doesn't have valid function id, valid functionality info or more than one stage) in retention strategy but app option mv.cells.infer_complex_retention_cells is set to false.

### What Next

Set mv.cells.infer_complex_retention_cells to true to enable retention complex cell mapping.

~

---

## SQM-1081

(warning) Logic has been pushed out of latch '%s' for mapping purpose. Set *compile.seqmap.disable_async_logic_push_out_for_latches* to disable it.

### Description

This warning message is issued when logic is pushed out on SEQGEN latch instance. If user don't want logic be pushed out, setting app option *compile.seqmap.disable_async_logic_push_out_for_latches* would help preserve the logic. This, however, will potentially result in latch unmapped.

### What Next

Check the app option value for compile.seqmap.disable_async_logic_push_out_for_latches.

---

## SQM-1082

(warning) Output port '%s' on parallel scan libcell is not marked with signal type: test_scan_out. May need to check library definition.

### Description

This warning message is issued when the tool identified a multibit sequential libcell as parallel scan but found no signal type of test_scan_out on a register bit. For robustness, libcell should mark all scan output pins with signal type of test_scan_out.

### What Next

Check the definition of this libcell in user library.

Feedback

## SQM-1083

(warning) Unable to maintain scan connection of original cell '%s' on its non-scan replacement '%s'.

**Description**

This warning message is issued when the tool trys to rebind scan chain connection but cell is being remapped from scan to non-scan type. This could happen in flow that forces a remapping/swapping solution. For example, swap out dont_use libcells or debanking by remapping.

**What Next**

Check the reason of forced remapping from scan to non-scan cells.

## SQM-1084

(warning) Skipping scan type swap for sequential cell '%s', as retention type cannot be maintained with provided replacement set.

**Description**

This warning message is issued when the tool trys to swap out register for a different scan type but skipped due to unavailablity of remap candidates in the replacement set to maintain the current retention type.

**What Next**

Check the library and UPF setup.

## SQM-2004

(warning) No single-bit representative was found for multibit lib cell '%s'.

**What Next**

Check library and provide single-bit degenerate for the reported multibit cell. ~

## SQM-2005

(warning) Inferred a shift register of length %ld with %s as head in module %s\n

**Description**

This message is issued when shift register is created.

Feedback

**What Next**

~

---

## SQM-2006

(warning) The sequential cells in module '%s' are to be retimed, they will not be mapped to multibit.\n

**Description**

This message is issued when the module is to be retimed and user instructs that no multibit mapping for retimed module.

**What Next**

~

---

## SQM-2007

(warning) Sequential cell '%s' of module '%s' is to be retimed, it will not be mapped to multibit.\n

**Description**

This message is issued when a single instance in bus will not be mapped to multibit cell because the module is to be retimed.

**What Next**

~

---

## SQM-2010

(info) Multibit sequential cell '%s' does not have bus info nor single_bit_list attribute (common for pre-instantiated multibit cells). Multibit naming style support will be limited.\n

**Description**

This warning is issued when tool cannot name a multibit sequential cell properly since both bus info and single_bit_list attribute are missing. Instead, the cell will be named by annotating bit indices in the original cell name.

**What Next**

Check if the multibit sequential cell has valid bus info or single_bit_list attribute for naming.

~

Feedback

## SQM-2011

(info) Multibit sequential cell '%s' does not support setting of 'multibit.naming.expanded_name_style = 0' due to missing bus info.\n

**Description**

This warning is issued when tool cannot name a multibit sequential cell without expansion due to missing bus info. Instead, the cell will be named as expansion of single-bit names.

**What Next**

Check if the multibit sequential cell has valid bus info for naming.

~

## SQM-2012

(info) Library has LSSD cell but mapping to them is not possible because user setting 'set_scan_configuration -style lssd' is missing.

**Description**

This warning is issued when tool sees LSSD cell in library but user didn't set "set_scan_configuration -style lssd" in script. This setting is needed to enable LSSD mapping.

**What Next**

Set "set_scan_configuration -style lssd" in script if LSSD mapping is desired. Or ignore the message.

~

## SQM-2030

(info) The design contains some dont_touch registers, through which constant propagation is not allowed. Refer man page for more details.

**Description**

This info message is issued when the tool finds some constant registers marked dont_touch. By default, the tool does not propagate constants through dont_touch registers.

**What Next**

In order to allow constant propagation through dont_touch registers, set the following app option set_app_options -name

Feedback

compile.flow.propagate_constants_through_dont_touch_cells -value true. Note that, this setting propagates constants through dont_touch registers, but does not remove the register. ~

---

## SQM-2031

(info) The design contains some size_only registers, through which constant propagation is not allowed. Refer man page for more details.

**Description**

This info message is issued when the tool finds some constant registers marked size_only. By default, the tool does not propagate constants through size_only registers.

**What Next**

In order to allow constant propagation through size_only registers, set the following app option set_app_options -name compile.flow.propagate_constants_through_size_only_cells -value true. Note that, this setting propagates constants through size_only registers, but does not remove the register. ~

---

## SQM-3001

(warning) Failed to analyze the equivalent pins of lib cell '%s'.

**Description**

Debugging message for internal use.

**What Next**

~

---

## SQM-3002

(warning) No flip-flops in the library.

**Description**

Debugging message for internal use.

**What Next**

~

Feedback

## SQM-3003

(warning) No latchs in the library.

**Description**

Debugging message for internal use.

**What Next**

~

## SQM-3004

(warning) No usable multibit sequential cells in the library.

**Description**

Debugging message for internal use.

**What Next**

~

## SQM-3005

(warning) Shift register body or tail '%s' is mapped to a scan cell '%s'.

**Description**

Debugging message for internal use.

**What Next**

~

## SQM-3006

(warning) Shift register head '%s' is mapped to a non-scan cell '%s'.

**Description**

Debugging message for internal use.

**What Next**

~

Feedback

## SQM-3011

(warning) Mapping of flip-flop '%s' to library cell '%s' required tying off some pins

**Description**

Debugging message for internal use.

**What Next**

~

## SQM-3012

(warning) Mapping of latch '%s' to library cell '%s' required tying off some pins

**Description**

Debugging message for internal use.

**What Next**

~

## SQM-3013

(warning) Library cell %s: skip internal port %s (its id %u > max %u).

**Description**

Debugging message for internal use.

**What Next**

~

## SQM-3014

(warning) Logic has been pushed out of register '%s' for mapping purpose.

**Description**

Debugging message for internal use.

**What Next**

~

Feedback

## SQM-3015

(warning) Library cell %s has no dedicated scan output pin.

**Description**

Debugging message for internal use.

**What Next**

~

## SQM-3100

(Information): '%d' registers are removed as constant. Use report_transformed_registers for a list.

**Description**

This information message is printed as summary of constant registers removed by tool. At the end of each pass constant register removal by tool during initial_map/logic_opto/ initial_opto, this message prints total number of constant registers removed.

**What Next**

This is an information-only message. No action is required.

## SQM-3101

(information) %d registers are removed as unloaded. Use report_transformed_registers for a list

**Description**

This information message is printed as summary of unloaded registers removed by tool. At the end of each pass of unloaded register removal by tool during initial_map/logic_opto/ initial_opto, this message prints total number of unloaded registers removed.

**What Next**

This is an information-only message. No action is required.

## SQM-4022

(warning) set_register_replication: No replicable path found %s cell '%s' %s register '%s'.

Feedback

### Description

This message is printed when Fusion Compile cannot determine replicable path from a register to an included fan out cell or from fan in cell to a register. Fan in and fan out cells are specified as -include_fanout_logic and -include_fanin_logic command line options of set_register_replication.

Replicable path may not exists under following cases:

(1) it contains a cell marked dont touch (2) constant propagation is disabled across a net on a path (3) path crosses power domain boundary. (3) path contains coarse grain switch cell.

### What Next

Trace the path and inspect if any of the conditions above are met. Revise constraints accordingly.

---

## SQM-4034

(warning) set_register_replication: Register '%s' not replicated as it's fanout is less than max fanout constraint.

### Description

This message is printed when register replication engine finds number of load pins on register's output net less than the max fanout constraint specified using -max_fanout switch. When number of load pins is less than the max fanout constraint, engine considers constraints as satisfied and it will not replicate register.

Here is how replication engine calculates number of load pins:

The engine deletes the buffer tree on the net of register's output pin. The number of load pins on the flat net, connected to input pins of logical cells or output ports is considered as a fanout count. When a hierarchy in the fanout of a register is ECO hierarchy, engine considers hierarchical pin of ECO hierarchy as a load pin. That is, it does not descend in ECO hierarchy to count pins of the logical cells.

The number of pins thus collected is compared to the -max_fanout constraint.

Please note that engine does not consider timing end point as a load pin of a register.

### What Next

Please disable replication of register inside ECO hierarchy.

Feedback

## SQM-4037

(warning) set_register_replication: Cell '%s' not replicated as it does not have enough loads.

**Description**

This message is printed when Fusion Compile cannot determine load pins of a cell whose replication is requested. Minimum of 2 loads are required for a successful replication. The number of load pins are found at the time of replication in the compile flow. The load pins to go down to less than 2 should be rare. But if that happens, it means one of the following:

(1) some cell such as ISO cell has been added at Q pin of a register after set_register_replication command is issued. Addition of a cell can reduce number of loads.

(2) The net driven by Q pin drives ECO hierarchy. The load pins inside ECO hierarchy are counted as one as port-punching of ECO hierarchy is not allowed.

If un-replicated cell is an end point of a path, the replication of the entire path will be aborted. Path replication is specified by setting either -include_fanin_logic or -include_fanout_logic switch of a set_register_replication command.

**What Next**

Analyze design on which replication is done and see if any of the above mentioned conditions are present.

## SQM-4044

(information) Un-merging previously merged register '%s' into '%s'.

**Description**

Register that was merged previously was found to be on critical path during optimization. This register is split again.

**What Next**

~

## SQM-4045

(warning) Register '%s' does not have asynchronous control pins. It will not be replicated.

Feedback

**Description**

Registers without asynchronous pins will not be required as compile.seqmap.replicate_registers_with_asynchronous_logic option is set to false.

**What Next**

If you want to allow replication of such registers, then remove compile.seqmap.replicate_registers_with_asynchronous_logic from the script and re-run compile. ~

---

## SQM-4046

(warning) Register '%s' is self gated. It will not be replicated.

**Description**

Registers that are self gated are not replicated as it provides challenges during verification.

**What Next**

If you want to allow replication of such registers, disable self gating on replicated registers. ~

---

## SQM-4047

(warning) Register '%s' is size only. It will not be replicated.

**Description**

Replication of size only register is disabled using compile.seqmap.replicate_size_only_registers app-option.

**What Next**

If you want to allow replication size-only registers, set compile.seqmap.replicate_size_only_registers to false or remove it from the script.

---

## SQM-4048

(warning) There is no enough space to insert requested replicated copies of register %s in scan chains. Reducing number of copies from %d to %d to accomodate them in scan chains.

Feedback

### Description

This message is issued when there is no enough space in the scan chains to insert all newly created registers in the scan chains. Number of requited copies is reduced so that newly replicated registers can be added to scan chains.

### What Next

This message indicate an error in the estimation of scan chain capacity. Please contact Synopsys customer support.

---

## SQM-4049

(warning) Register %s can't be replicated because DFT requirements are not met: %s

### Description

A register that is part of a scan chain can't be replicated because certain DFT requirements that need to be met in order to allow the copied register to be also included in the correct scan chain are not met. This might be not be possible, for example, if the register's copy can't be accommodated on another already existing DFT scan chain.

This situation can arise because of several reasons that prevent updating the scan chains, DFT connections or similar DFT considerations in a safely way, i.e. allowing the replication can harm or break the scan chain connectivity. A message will provide a reason for the rejection.

A (non-exhaustive) list of usual rejection reasons is:

dft feature disabled

Replication feature for scan stitched cells is disabled. Check for the application option (*opt.dft.utils_allow_register_replication*) in the test case setup.

target cell is not in SCANDEF

Original register is scan chain stitched but it is not part of a SCANDEF file's stub chain, therefore its scan chain connection cannot be updated safely.

target cell is SCANDEF ordered

Original register is part of a SCANDEF stub chain's ordered component, like shift registers and alike that are 'read only' for scan chain purposes, therefore a copy register cannot be allocated in that fixed structure.

unsupported anchor cell found

Original register is driven by a cell that is not supported for modifying its scan connection safely for allocating a register copy.

Feedback

no replication slots available

There are no extra slots available for replication in the scan chain, i.e. scan chain max length would be violated if replication had been allowed.

invalid ScanOutput of previous cell

Original register is driven by an element that cannot be disconnected to insert a valid copy without disrupting the scan chain.

**What Next**

- Please verify how the register that cannot be replicated relates to the scan chains it belongs to and other related DFT logic.
- Consider to enable and perform register replication over the target registers before DFT scan chain insertion (*insert_dft*).

---

## SQM-4050

(warning) Register %s can't be replicated as it is TMR register.

**Description**

TMR registers can't be replicated.

---

## SQM-4051

(warning) duplicate_registers command is aborted as it can't replicate some register. Please refer to messages and re-run duplicate_registers command.

**Description**

Some registers specified for duplication by duplicate_registers command can't be replicated. Please refer to log to find messages explaining rejection of replication.

**What Next**

Please analyze error messages, fix problem and rerun duplicate_registers command

---

## SQM-4052

(warning) Register %s can't be replicated due to potential simulation mismatch at critical end point %s."

**Description**

This register can't be replicated due to simulation mismatch at critical end point.

Feedback

**What Next**

Please analyze error messages, fix problem and rerun register replication command.

---

## SQM-4053

(warning) Replication of register in ECO region is not allowed. Register: '%s'

**Description**

Replication of register inside ECO region is not allowed. ECO region is any hierarchy on which port punching is disallowed using set_freeze_ports command. Name of register being replicated is given.

**What Next**

Please disable replication of register inside ECO hierarchy.

---

## SQM-4054

(warning) Replication of register or fanout logic cell is rejected. Register/cell: '%s' Reason: '%s'"

**Description**

Replication of a register or a fanout logic cell is rejected. The reason for rejection is given.

**What Next**

Please review reason and update set_register_replication directive accordingly.

---

## SQM-4055

(warning) "Register '%s' is a scanned register but its copies will not be added to scan chains during replication."

**Description**

This message is issued when a register replication is requested on a register on a scan chain of partially scanned design. The replication engine will create copies however, those copies will not be inserted in scan chains. If this is not acceptable, please remove set_register_replication directive from the register and re-issue compile_fusion

**What Next**

Please review set_register_replication directive and follow recommended test insertion strategy.

Feedback

## SQM-4100

(information) The register '%s' is removed as constant '%d'.

### Description

This information message is printed because when constant propagation is enabled and constant optimization removes any detected constant register unless the register has user restrictions like user size_only.

The removal of constant registers is controlled by below app options.

```
compile.flow.propagate_constants_through_registers
compile.flow.enable_high_effort_constant_propagation
```

Both of which are true by default.

### What Next

This is an information-only message. No action is required.

## SQM-4101

(information) The register %s is removed because it is unloaded.

### Description

This information message is printed because when a register is removed as unloaded because of optimization during compile_fusion. The removal of unloaded registers is controlled by app option *compile.seqmap.remove_unloaded_registers* which are true by default.

### What Next

This is an information-only message. No action is required.

## SQM-4102

(information) The register %s is removed because it is merged to register %s.

### Description

When register merging is enabled, compile will detect the set of equivalent registers and merge them with this info message.

To disable register merging on specific registers, use *set_register_merging* command.

Feedback

## SQM-4104

(information) Unloaded register %s is preserved in module %s due to restriction: %s.

### Description

This information message is printed when an unloaded register is not removed due to user restriction like user size_only or don’t_touch. The removal of unloaded registers is controlled by app option which is true by default. *compile.seqmap.remove_unloaded_registers*

### What Next

Please check user restrictions using report_size_only/report_dont_touch.

## SQM-5001

(error) set and reset cannot be applied at the same time for command set_constant_register_removal.

### Description

set_constant_register_removal can be used as "set_constant_register_removal f1 true" or "set_constant_register_removal f1 -reset". However, it cannot be used as "set_constant_register_removal f1 true -reset". See man page of set_constant_register_removal for more information.

### What Next

Please contact Synopsys customer support and request enhancement.

## SQM-5002

(warning) Cannot apply command set_constant_register_removal on a combinational cell %s.

### Description

set_constant_register_removal does not support and cannot be applied on any combinational cell. See man page of set_constant_register_removal for more information.

### What Next

Please contact Synopsys customer support and request enhancement.

Feedback

## SQM-5003

(error) set and reset cannot be applied at the same time for command set_unloaded_register_removal.

### Description

set_unloaded_register_removal can be used as "set_unloaded_register_removal f1 true" or "set_unloaded_register_removal f1 -reset". However, it cannot be used as "set_unloaded_register_removal f1 true -reset". See man page of set_unloaded_register_removal for more information.

### What Next

Please contact Synopsys customer support and request enhancement.

## SQM-5004

(warning) Cannot apply command set_unloaded_register_removal on a combinational cell %s.

### Description

set_unloaded_register_removal does not support and cannot be applied on any combinational cell. See man page of set_unloaded_register_removal for more information.

### What Next

Please contact Synopsys customer support and request enhancement.

## SQM-5009

(warning) Fusion Compiler will not replicate registers that do not have asynchrnous control pins as app option compile.seqmap.replicate_registers_without_asynchronous_logic is set to false.

### Description

Fusion Compiler has been instructed not to replicate registers which do not have asynchronous set or reset pins. Replication of such registers will not be done. Only registers with asynchronous set or reset pins will be replicated.

### What Next

If you wish to enable replication of such registers, please set compile.seqmap.replicate_registers_without_asynchronous_logic to true or remove compile.seqmap.replicate_registers_without_asynchronous_logic from the script.

Feedback

# SRM

## SRM-000

(info) Deleted redundant routing blockage %s (blockage_group_id %d, layer %s).

**Description**

This routing blockage has been associated to a repelling group bound core via its blockage_group_id, and is deemed redundant.

## SRM-001

(info) Created routing blockage %s (blockage_group_id %d, layer %s).

**Description**

This routing blockage has been created for the core of a repelling group bound.

## SRM-002

(error) Unable to create routing blockage for block %s.

**Description**

As part of create_repelling_group_bound_shapes, a routing blockage should be created and associated to the core of a repelling group bound, but it could not be created.

## SRM-003

(error) Could not assign routing blockage %s to the appropriate layer %s.

**Description**

As part of create_repelling_group_bound_shapes, the routing blockage has been created for a core of a repelling group bound, but it could not be assigned to the appropriate layer.

## SRM-004

(error) Could not set routing blockage %s's blockage_group_id to %d.

Feedback

**Description**

As part of create_repelling_group_bound_shapes, the routing blockage has been created for a core of a repelling group bound, but it could not be assigned the appropriate blockage_group_id.

---

## SRM-005

(error) Could not set routing blockage %s to be zero spacing.

**Description**

As part of create_repelling_group_bound_shapes, the routing blockage has been created for a core of a repelling group bound, but it could not be defined as zero spacing.

---

## SRM-006

(error) Unable to extract routing blockage for core %s in repelling group bound %s.

**Description**

The cells of this core are placed in such a way that no valid shape could be extracted.

**What Next**

The cell placement of the core may be overlapping with other cores causing the shape extraction to fail. Run create_placement and legalize_placement to place the cores with sufficient spacing, and then rerun create_repelling_group_bound_shapes.

---

## SRM-007

(error) Could not change the name of routing blockage %s to %s.

**Description**

As part of create_repelling_group_bound_shapes, the routing blockage has been created for a core of a repelling group bound, but its name could not be changed to include the specified name prefix.

---

## SRM-008

(info) Renaming routing blockage %s to %s (blockage_group_id %d, layer %s).

**Description**

This routing blockage has been created earlier for the core of a repelling group bound but has been renamed following the given name_prefix.

Feedback

## SRM-009

(info) Block %s contains FuSa intent (%s).

**Description**

This message informs about detected functional safety intent of the specified type.

## SRM-010

(error) No proper lib_cell defined for creating a guide %s.

**Description**

Provide the proper lib_cell(s) to insert_redundant_trees via -buffer_lib_cell/-inverter_lib_cell.

## SRM-011

(error) Unable to create guide cell in hierarchy %s using lib_cell %s.

**Description**

The guide buffer/inverter cell could not be created in this hierarchy.

## SRM-012

(error) Unable to connect %s with %s.

**Description**

The connection between the given pin/net could not be created.

## SRM-013

(info) Working on %d split pin(s), driven by %d unique driver(s). The split pins form %d group(s) which require splitting. The design contains %d safety register group(s) and %d safety core group(s).

**Description**

Information about objects that insert_redundant_trees works on.

Feedback

## SRM-014

(info) Added %d guide buffer(s) and %d guide inverter(s). The design contains %d guide buffer(s) and %d guide inverter(s) in total.

**Description**

Shows the result of running insert_redundant_trees.

## SRM-015

(info) Selected %u split pin(s) for %s %s.

**Description**

Shows the selected split pin count for this redundancy group. Selection is done via split pin (types) following the rule or group, or if specified via the -pin_types or -pins option to insert_redundant_trees.

## SRM-016

(warning) Physical checks are off by default in this shell.

**Description**

In RTL-Architect, the design is typically not legalized, and therefore report_safety_status disables physical checks when run in this shell. If your design has been legalized, you can enable the physical checks by setting the following app-option before running report_safety_status. set_app_options -name report.safety.enable_physical_checks_in_rtla -value true

## SRM-017

(info) This type of summary formatting is deprecated.

**Description**

The summary formatting has changed to employ tables for improved legibility. The use of "set_app_options -name report.safety.use_tables_in_summary -value false" is deprecated.

## SRM-018

(error) Unable to select %s %s.

Feedback

**Description**

A group object was provided which could not be selected.

---

## SRM-019

(error) No valid reference found for %ld registers shared by groups {%s } %s.

**Description**

Found registers requiring isolation for which no valid tap cell reference could be determined. The rules of the groups could be conflicting; make sure that their optional tap_mapping list have at least one reference in common. When you are running create_safety_tap_cells with option -tap_lib_cells, ensure that it has a reference in common with the rules. A third option is that the libraries do not contain any lib_cells of type well_tap.

---

## SRM-020

(error) None of the %ld registers requiring isolation were selected.

**Description**

There are still registers that require isolation for functional safety, but none are selected to work in.

---

## SRM-021

(info) Selected %ld out of %ld registers requiring isolation for functional safety; %ld registers still require work.

**Description**

This message is purely informational.

---

## SRM-022

(info) Created %ld new tap cells.

**Description**

This message is purely informational.

---

## SRM-023

(warning) The advanced legalizer is not enabled while Functional Safety intent is specified.

Feedback

**Description**

Your design contains functional safety intent which requires the advanced legalizer. Please enable the advanced legalizer with: set_app_options -name place.legalize.enable_advanced_legalizer -value true

---

## SRM-024

(warning) Isolation checks are turned off and violations will not be reported.

**Description**

Certain Functional Safety related rules may require isolation of specific registers. This will be implemented during chip finishing by inserting adjacent tap cells through the create_safety_tap_cells command. So, during the earlier stages of the flow, groups related to rules stating an isolation requirement will fail because of the missing tap cells.

It is helpful to see whether such groups fulfill all other requirements and temporarily skip the isolation requirements. This is exactly what report_safety_status -ignore_isolation does. Groups will be marked passed if they fulfill all requirements other than the isolation requirements.

After tap cells have been inserted, run a regular report_safety_status to confirm that the isolation requirements are abided to.

---

## SRM-025

(info) Deleted routing blockage %s (blockage_group_id %d, layer %s).

**Description**

This routing blockage has been associated via its blockage_group_id to a core of a safety core group (or repelling group bound).

---

## SRM-026

(info) Removed %ld obsolete tap cells.

**Description**

This message is purely informational.

---

## SRM-027

(info) Added tap cell %s.

Feedback

**Description**

This message is purely informational.

---

## SRM-028

(error) Cannot define any restrictions.

**Description**

Due to an unknown reason, restrictions cannot be created.

---

## SRM-029

(info) Port %s will not be buffered because its net has multiple drivers.

**Description**

This input port drives a net that exceeds the maximum fanout limit, but that net will not be buffered because it is connected to another driver.

---

## SRM-030

(info) Port %s will not be buffered because its net is protected by dont_touch.

**Description**

This input port drives a net that exceeds the maximum fanout limit, but because it is protected by dont_touch, it cannot be buffered.

---

## SRM-031

(error) Could not create a buffer for port %s.

**Description**

This input port drives a net that exceeds the maximum fanout limit, but there was a problem in creating a new cell for buffering the net.

---

## SRM-032

(info) Created %s to buffer port %s.

**Description**

This input port drives a net that exceeds the maximum fanout limit. To buffer that net, a new cell has been created.

Feedback

### SRM-033

(error) Failed to insert %s to buffer port %s.

**Description**

This input port drives a net that exceeds the maximum fanout limit, but there was a problem in insert the cell between the port and the net.

## SRT

### SRT-001

(Error) No valid pins to be routed.

**Description**

Among all the input pins, there is no any valid PG pins that can be routed.

**What Next**

Check the pin list to ensure there is at least one PG pin.

### SRT-002

(Error) stripe layers are not specified or valid.

**Description**

The stripe layer is either not specified or not routable.

**What Next**

Check the stripe layer setting.

### SRT-003

(Error) No valid routing layer combinations.

**Description**

The routing layer combination is not routable given the stripe layer.

Feedback

**What Next**

Check the routing layer setting, and make sure that the routing direction of stripe layer, main-trunk layer, sub-trunk layer is either horizontal, vertical, horizontal or vertical, horizontal, vertical.

---

## SRT-004

(Warning) The top layer of pin %s (%s) should be lower than the layer of sub-trunk (%s), skipping it.

**Description**

Please check if routing layers setting correctly.

---

## SRT-005

(Warning) Pin %s does not belong to a PG Net, skipping it.

**Description**

Please check if pin setting correctly.

---

## SRT-006

(Error) association can't generte in read mode.

**Description**

Read mode design can't set association on lib pins.

**What Next**

Please turn off the read mode and try it again.

---

## SRT-007

(Error) The structured route rule: %s has been created in previous flow.

**Description**

Please use remove_structured_route_rule to remove current existing rule.

---

## SRT-008

(Warning) Pin %s is not secondary pg pin, skipping it.

Feedback

**Description**

Please check if lib pin is secondary pg.

---

## SRT-009

(Error) The value of option -number_of_stripes_by_layer is wrong

**Description**

There are a few possible wrong directions here, please confirm. 1. The layer you define is not defined in NDM. 2. The layer cannot be defined by multiple number of stripes. 3. The number of stripes should be 1 or 2. 4. The format of -number_of_stripes_by_layer should be {{layer1 count1} {layer 2 count2} {layer3 count3} .....}

---

## SRT-010

(Error) The option of -stripe_layer, -main_trunk_layer, or sub_trunk_layer is wrong.

**Description**

Please check whether the layer is already defined in ndm. In addition,-stripe_layer(-main_trunk_layer/-sub_trnuk_layer) needs to be the same layer.

---

## SRT-011

(Error) The structured route rule: %s is not existing.

**Description**

Please use create_structured_route_rule to create this rule first.

---

## SRT-012

(Error) Create via rules and constraints failed

**Description**

Create via rule and constraint failed when running autogen.

**What Next**

Please check whether autogen can run on secpg pins or set user_defined_via_ladder to use user-defined via rule.

Feedback

### SRT-013

(Error) The number of rules has exceeded the upper limit of %d, please remove the unused rules.

**Description**

Please use remove_structured_route_rule to remove outdated rules.

### SRT-014

(Error) One of the options %s and %s listed below is required.

**Description**

You must choose one of the options listed below to specify.

### SRT-015

(Warning) The secpg pin: %s already has rule. Will overwrite pin's rule.\n

**Description**

Please note set_structured_route_rule will overwrite pins rule setting.

### SRT-016

(Warning) The secpg lib_pin: %s already has rule. Will overwrite lib_pin's rule.\n

**Description**

Please note set_structured_route_rule will overwrite lib_pins rule setting.

## SSF

### SSF-001

(error) Invalid scope specified for SSF - %s.

**Description**

The value provided for the scope option of the command is not valid. Only a single cell must be specified and it must be a logical hierarchy only.

Feedback

**What Next**

Provide a single valid logical hierarchy cell.

# STIL

## STIL-101

(error) %s

**Description**

This error is generated when you try to read a STIL file with syntax error. The syntax error happens at the next symbol.

**What Next**

Fix the syntax error and read in the file again.

## STIL-102

(error) %s block is not a top level STIL block.

**Description**

This error is generated when you try to read a STIL file without any top level STIL blocks. At least one top level STIL block (except STIL & Header) is required for a valid STIL file.

**What Next**

modify the STIL file to make it valid.

## STIL-103

(error) Can't read version %s of STIL.

**Description**

This error is generated when the STIL parser can not parse the STIL version mentioned. It is usually caused by reading in a ctldb file that was generated by early version of dft compiler.

**What Next**

User have to regenerate the ctldb file with current version of dft compiler.

Feedback

## STIL-104

(information) Using CTL parser version %s on a version %s model.

**Description**

The model's CTL version is no longer supported. This message informs you that the tool will attempt to read the model using a parser designed for a later version of CTL. Most constructs in newer versions are backwards compatible. In all likelihood, the read will be successful.

**What Next**

Pay attention to the next messages. They will inform you whether the read was successful or not.

## STIL-301

(error) Illegal STIL string: %s.

**Description**

This error is generated when the mentioned string in is illegal according to STIL syntax.

**What Next**

Check with STIL syntax, modify the string to make it legal.

## STIL-302

(error) Illegal STIL identifier: %s.

**Description**

This error is generated when the mentioned identifier is illegal according to STIL syntax.

**What Next**

Check with STIL syntax and modify it to legal identifier.

## STIL-303

(error) Cannot find PatternExec with name %s

**Description**

This error is generated when the PatternExec with the specified name is not found in the STIL file.

Feedback

**What Next**

Specify another PatternExec name or modify the existing PatternExec name in STIL file.

---

## STIL-304

(error) Signal %s already defined.

**Description**

This error is generated when the concerned signal is previously defined in the STIL file. Redefinition of signal is not allowed in STIL file.

**What Next**

Remove the one of the dual signal definitions.

---

## STIL-305

(error) Variable %s redefined.

**Description**

This error is generated when the concerned variable is previously defined in the same domain. Redefinition of variable is not allowed in the same domain.

**What Next**

Remove one of the dual variable definitions.

---

## STIL-306

(error) Signals block redefined.

**Description**

This error is generated when the Signals block was redefined. Only one and anonymous Signals block is allowed in STIL syntax.

**What Next**

Remove the duplicated definition of Signals block.

---

## STIL-307

(error) Anonymous SignalGroups block redefined.

Feedback

**Description**

This error is generated when the anonymous SignalGroups block was redefined. Only one anonymous SignalGroups block is allowed in STIL syntax.

**What Next**

Remove the duplicated definition of the anonymous SignalGroups block.

---

## STIL-308

(error) Named %s domain not allowed in this context.

**Description**

This error is generated when it is not allowed to defined a named domain in the current context. e.g. when you try to define a named SignalGroups domain in the CoreTypes block. Only an anonymous SignalGroups domain is allowed to be defined there.

**What Next**

Modify the named domain to an anonymous(unnamed) domain.

---

## STIL-309

(error) Cann't define SignalVariables in %s block.

**Description**

This error is generated when user try to define SignalVariables in the block mentioned. This definition is not supported currently.

**What Next**

Remove the SignalVariable definition from the concerned block.

---

## STIL-310

(error) Cann't define SignalVariableEnums in %s block.

**Description**

This error is generated when user try to define SignalVariableEnums in the block mentioned. This definition is not supported currently.

**What Next**

Remove the SignalVariableEnum definition from the concerned block.

Feedback

## STIL-311

(error) ScanChain Length and number of scancells don't match in ScanChain %s.

**Description**

This error is generated when the specified scan chain length doesn't match the number of scan cells in the ScanChain mentioned.

**What Next**

Check to make them consistent.

## STIL-312

(error) Time expression format are supported.

**Description**

This error is generated when the format of time expression user specified is not supported now. Currently only numerical time expression format is supported.

**What Next**

Modify the concerned time expression to numerical only format.

## STIL-313

(error) The ScanInversion must be 0 or 1 in ScanChain %s.

**Description**

This error is generated when the user specify the ScanInversion value as neither 0 nor 1. The ScanInversion value means the overall inversion from before the first scan cell to after the last scan cell.

**What Next**

Correct the ScanInversion value to either 0 or 1.

## STIL-314

(error) Anonymous ScanStructures block redefined.

**Description**

This error is generated when the anonymous ScanStructures block was redefined. Only one anonymous ScanStructures block is allowed in STIL syntax.

Feedback

**What Next**

Remove the duplicated definition of the anonymous ScanStructures block.

---

## STIL-315

(error) %s Domain %s redefined.

**Description**

This error is generated when the Domain mentioned above is redefined. The Domain can be SignalGroups, ScanStructures, Timing, Procedures or MacroDefs. Each named domain can only be defined once.

**What Next**

Remove the duplicated domain definition.

---

## STIL-316

(error) Anonymous Variables block redefined.

**Description**

This error is generated when the anonymous Variables block was redefined. Only one anonymous Variables block is allowed in STIL syntax.

**What Next**

Remove the duplicated definition of the anonymous Variables block.

---

## STIL-317

(error) Duplicated signal %s in the signal expression.

**Description**

This warning is generated when there exists duplicate signals in a signal expresion.

**What Next**

Check each element in the signal expression, find out and remove the duplicates.

---

## STIL-318

(error) Non-cyclized data is not supported.

Feedback

**Description**

This error is generated when user includes non-cyclized data in a vector statement. Currently the STIL reader only support single cyclized data in a vector statement. For exact meaning of cyclized and non-cyclized data, please refer to STIL syntax.

**What Next**

You may need to remove non-cyclized data and find alternative ways to express that.

---

## STIL-319

(warning) Only constant expression is allowed in %s.

**Description**

This error is generated under given context, integer expression or sigvar expression can only contains constant variables or values, while non-constant variables were used in those expressions.

**What Next**

Replace the non-constant variables with constant variables or values.

---

## STIL-320

(error) %s domain %s does not exist.

**Description**

This error is generated when the specified STIL domain was not defined before. The STIL domain can be SignalGroups, ScanStructures, Timing, Procedures or MacroDefs.

**What Next**

Please check if the domain is referred or defined correctly.

---

## STIL-324

(error) Can't create PatternBurst %s.

**Description**

This error is generated when PatternBurst mentioned above can not be created during model construction. The most probable reason is that the PatternBurst with the same name has already been created before.

Feedback

**What Next**

Check if the duplicated PatternBurst has already been created.

---

## STIL-326

(error) PatternExec block has multiple Timing statements.

**Description**

This error is generated when the PatternExec block has multiple Timing statements. Only one Timing statement is allowed according to STIL syntax.

**What Next**

Remove the duplicated Timing statements in PatternExec block.

---

## STIL-327

(error) PatternBurst %s does not exist.

**Description**

This error is genenrated when the PatternBurst mentioned in the PatternExec block was not defined before. It is required by the STIL syntax to define PatternBurst before referencing.

**What Next**

Check if the PatternBurst was referred or defined correctly.

---

## STIL-328

(error) PatternExec block has multiple PatternBurst statements.

**Description**

This error is generated when multiple PatternBurst statements is included in the PatternExec block. Currently only PatternBurst statement is allowed in PatternExec block according to STIL syntax.

**What Next**

Define a new PatternBurst block to include the multiple PatternBurst statements, and put the new defined PatternBurst in the PatternExec block.

Feedback

## STIL-329

(error) Can't create PatternExec %s.

**Description**

This error is generated when PatternExec mentioned above can not be created during model construction. The most probable reason is that the PatternExec with the same name has already been created before.

**What Next**

Check if the duplicated PatternExec has already been created.

## STIL-330

(error) Too many events in event list.

**Description**

This error is generated when you define a waveform character list, the number of events in the event list is more than the number of wfcs in the wfc list.

**What Next**

Modify your wfc defintion to make them consistent.

## STIL-331

(error) Waveform inheritance not supported.

**Description**

This error is generated when the user specify an InheritWaveform statement in WaveformTable definition. Currently, the waveform inheritance is not supported.

**What Next**

Try alternative ways to reach the goal.

## STIL-332

(warning) Unsupported construct in waveform definition.

Feedback

**Description**

This warning is generated when the construct in the waveform definition is legal but not supported now. e.g. in the definition list of wfcs, currently we only support time-value pair definition. Other constructs are not recognized.

**What Next**

Try alternative ways to reach the goal.

---

## STIL-333

(error) Incomplete time-value pair in waveform definition.

**Description**

This error is generated when there is an imcomplete time-value pair in waveform definition. Currently we only support time-value pair as the definition of wfcs.

**What Next**

Correct it or try alternative ways to reach the goal.

---

## STIL-334

(error) Waveform inheritance not supported.

---

## STIL-335

(error) WaveformTable inheritance not supported.

**Description**

This error is generated when the user specify an InheritWaveformTable statement in WaveformTable definition. Currently, the waveform table inheritance is not supported.

**What Next**

Try alternative ways to reach the goal.

---

## STIL-336

(error) SubWaveforms not supported.

**Description**

This error is generated when the user specify an SubWaveforms statement in WaveformTable definition. Currently, the SubWaveforms is not supported.

Feedback

**What Next**

Try alternative ways to reach the goal.

---

## STIL-337

(error) WaveformTable %s redefined.

**Description**

This error is generated when user redefined a WaveformTable in the same Timing domain. The WaveformTable definition should be unique within one Timing domain.

**What Next**

Remove the duplicated definition of WaveformTable.

---

## STIL-339

(error) Single signal expected for %s.

**Description**

This error is generated when a signal bus is passed in when a single signal is expected. e.g. in the scan chain definition, the scan in and scan out are required to be a single signal.

**What Next**

Modified the signal bus to a single signal.

---

## STIL-340

(error) Signal or SignalGroup %s not defined.

**Description**

This error is generated when user tries to refer to an undefined signal or signal group. Signals and SignalGroups must be defined before being referenced.

**What Next**

Define the signal or signal group before using it.

---

## STIL-341

(error) ScanCell %s not defined.

Feedback

**Description**

This error is generated when user tries to refer to an undefined scancell. ScanCells must be defined before being referenced. They are defined in the ScanStructures blocks.

**What Next**

Define the scancell before using it.

---

## STIL-342

(error) Anonymous BistStructures block redefined.

**Description**

This error is generated when the anonymous BistStructures block was redefined. Only one anonymous BistStructures block is allowed in STIL syntax.

**What Next**

Remove the duplicated definition of the anonymous BistStructures block.

---

## STIL-345

(error) Ambigous parameter matching '%s' in call to '%s'.

**Description**

This error is issued when a parameter is multiple defined in a macro or procedure call.

**What Next**

Modify the macro or procedure call and remove the multiple defined parameter.

---

## STIL-401

(warning) Domain name '%s' already exists in the test model. Overwriting the old one.

**Description**

This warning is issued when a domain name in a STIL file read by read_test_protocol command already exists in the existing test model.

The old domain in the test model will be replaced by the new one.

**What Next**

If you do not want to overwrite the old one, remove the '-overwrite' argument from the read_test_protocol command. This will ignore the new one and keep the old one.

Feedback

## STIL-402

(warning) Domain name '%s' already exists in the test model. Ignoring the new one.

**Description**

This warning is issued when a domain name in a STIL file read by read_test_protocol command already exists in the existing test model.

The old domain in the test model will be preserved, and the domain in the protocol file will be ignored.

**What Next**

If you want to overwrite the old one, specify the '-overwrite' argument in the read_test_protocol command. This will replace the old one by the new one.

## STIL-403

(warning) Overwriting the existing waveform table with new timing.

**Description**

This warning is issued when the existing waveform talbe is overwritten by a new waveform table defined in the test protocol read by read_test_protocol command.

**What Next**

Check if the new waveform table is compatible with the timing in other test modes.

## STIL-404

(error) The timing of signal '%s' is different from the existing one.

**Description**

This error is issued when the waveform of the signal in the test protocol is different from the existing one.

The sequence of events in a waveform should be in canonical order, and the waveforms in the test protocol read and in the existing protocol (or model) should be equal.

DFTCompiler allows only one timing block and waveform table. You cannot read a test protocol that has different timing block from the one in other test protocols in different test modes.

You can replace the old timing with the new timing only if you have no protocol, or you have only one test protocol and replace it in the same test mode.

Feedback

**What Next**

Write out the existing test protocol, and compare the timing block against the timing block of the test protocol you want to read.

## STIL-406

(Error) Signal or SignalGroup %s used in test setup procedure is not defined.

**Description**

This error is generated when user tries to refer to an undefined signal or signal group in test setup. Signals and SignalGroups must be defined before being referenced. Make sure all referenced signals are within double quotes.

**What Next**

Define the signal or signal group before using it.

# STREAM

## STREAM-001

(warning) Incorrect syntax '%s' specified for cell-rename entry. The entry will be ignored.

**Description**

The syntax for the cell-rename entry is not a valid one. Refer the man-page of write_gds for the appropriate syntax.

## STREAM-002

(error) Failed to open the cell-rename map file '%s'.

**Description**

The cell-rename map file could not be opened. Verify the file-permissions and retry.

## STREAM-003

(error) Failed to start the stream-merger.

**Description**

A non-recoverable error occured while merging the stream-files.

Feedback

**What Next**

This is an internal error.

---

## STREAM-004

(error) Failed to merge the %s files.

**Description**

A non-recoverable error occured while merging the stream-files.

**What Next**

This is an internal error.

---

## STREAM-005

(error) Reading directory is not supported. File %s is a directory.

**Description**

The input file is a directory.

---

## STREAM-006

(error) Input file '%s' is neither a GDSII nor an OASIS file.

**Description**

The input file can be either a GDSII file or an OASIS file.

---

## STREAM-007

(error) Invalid suffix '%s' specified for conflict cell renaming.

**Description**

The suffix string should be an alpha-numeric string of maximum size of 10 characters.

---

# STROKE

---

## STROKE-001

(Error) Specified stroke sequence string, %s, is not valid.

Feedback

**Description**

The specified stroke sequence string is not valid. The sequence is comprised of an optional sequence of key modifiers and separated from the path sequence which is a sequence of one or more grid numbers in the range 1-9. See the man page for the set_gui_stroke_binding command for detailed documentation on the format for stroke sequences.

---

## STROKE-002

(Error) Failed to set the stroke binding because the built-in stroke command %s does not exist

**Description**

The specified built-in stroke command name is not valid. Use the report_gui_stroke_builtins command to get a report listing the available built-in commands.

---

## STROKE-003

(Error) Failed to set the gui stroke preference %s because the value was not valid for that preference.

**Description**

The specified stroke preference value is not valid. Please refer to the man page for the set_gui_stroke_preferences command for more details on the preferences.

---

## STROKE-004

(Error) Failed to set the gui stroke preference %s because the preference key was not valid.

**Description**

The specified stroke preference key does not exist. Please refer to the man page for the set_gui_stroke_preferences command for more details on the preferences.

---

## STROKE-005

(Error) Stroke dictionary %s does not exist.

Feedback

**Description**

The specified stroke dictionary does not exist. Use the report_gui_stroke_bindings commands to view the dictionaries that are defined as well as the bindings that they specify.

# SVF

## SVF-1104

(error) Cannot open file %s.

**Description**

The file cannot be opened.

**What Next**

Check to see if the specified file exists and that you have the proper access permissions.

## SVF-1107

(error) "%s" is not an SVF file.

**Description**

The given file does not appear to be an SVF file.

## SVF-1108

(error) The SVF file format is not compatible with this executable.

**Description**

The SVF format of the provided file appears to be newer than the version supported by this executable.

**What Next**

A newer version of this product is needed to run this command on this SVF file.

## SVF-1109

(error) Cannot extract the contents of the currently opened SVF file.

Feedback

**Description**

The extract_svf command is being run on the currently opened SVF file. This file must be closed first using set_svf -off.

**What Next**

Close the SVF file first using "set_svf -off" command before re-running extract_svf command.

---

## SVF-1110

(error) Cannot append to file "%s". File does not exist.

**Description**

Unable to append to the given file because the file appears to be missing.

**What Next**

Check to see if the path to the file is correct.

---

## SVF-3207

(info) Environment variable CDPL_IPV6 enabled on IPV4 host. Unsetting CDPL_IPV6 environment variable.

**Description**

Setting environment variable CDPL_IPV6 on a IPV4 host will result in an error. This environment variable is unset internally in Descartes.

---

## SVF-3210

(error) The app option formality.svf.compact_mode may not be changed at this time.

**Description**

The formality.svf.compact_mode app option may not be changed while there is already an opened SVF.

**What Next**

In order to change the value of this app option, you must first close the currently opened SVF using set_svf -off.

Feedback

# SVR

## SVR-1

(error) Unable to open file '%s'.

**Description**

The Verilog file you tried to open does not exist or has incorrect permissions.

**What Next**

Verify the file name and permissions.

## SVR-2

(information) Verilog read failed.

**Description**

This is a summary informational message indicating that the Verilog file could not be read. Previous error messages will point you at the actual problem source. There are two types of syntax errors: a real violation of Verilog syntax, or reading a Verilog file that has non-structural or other unsupported constructs with this reader.

**What Next**

Review previous error messages. Then, either correct the actual Verilog syntax errors in the file and reread the file, or use a Verilog reader that supports the constructs that are flagged as syntax errors.

## SVR-3

(error) Unsupported construct '%s'\n \tat line %d in '%s'.

**Description**

The reader detected an unsupported construct. Not all unsupported constructs are trapped in this way. Many show up as more simple syntax errors.

**What Next**

Either correct the error by removing the construct or use a Verilog reader that supports the construct.

Feedback

## SVR-4

(error) Expected %s but found '%s'\n \tat line %d in '%s'.

**Description**

This message covers a wide variety of syntax errors. A token was expected, but something else was found. For example, after a module formal list, a close paren is expected, but if it is omitted and a semicolon is found, this message appears.

This can be an indication of a real syntax error. It is also possible for this message to be used when Verilog constructs, which are not supported by this reader, are encountered.

**What Next**

Review the line number reported in the message and determine if a real syntax error needs to be corrected or if the file contains unsupported constructs.

## SVR-5

(error) Expected identifier after %s\n \tat line %d in '%s'.

**Description**

An identifier is expected after the token shown and something else was found.

**What Next**

Review the line number reported in the message and determine if a real syntax error needs to be corrected or if the file contains unsupported constructs.

## SVR-6

(error) Premature end-of-file reading '%s'.

**Description**

The reader detected end-of-file before it was expected. For example, an end-of-file within a module construct before the endmodule would cause this message to be raised.

**What Next**

Correct the error and reread the file.

## SVR-8

(error) Port '%s' is not defined in module terminal list\n \tbut is defined in an %s statement \n \tat line %d in '%s'.

Feedback

**Description**

While processing an input, output, or inout statement, a port was found which did not exist anywhere in the module terminal list.

**What Next**

Correct the module terminal list or remove the port wire from the appropriate input/output/inout declaration.

---

## SVR-9

(error) Duplicate wire/tri/wand/wor declaration for '%s'\n \tat line %d in '%s'.

**Description**

The named wire was found in a wire, tri, wand, or wor statement. However, the wire was already declared in a similar statement.

**What Next**

Ensure that each wire is only declared once.

---

## SVR-10

(error) Illegal assignment %s\n \tat line %d in %s.

**Description**

An assign statement (or wire statement with inline assignment) has one of several errors:

- Nothing is on the left side of the '='. • The left side of the assignment contains an undeclared net or unexpected characters. • The right side of the assignment contains an undeclared net or unexpected characters.

Note that the following syntax for *assign* is not supported:

assign z = ( a );

Use of parentheses is considered non-structural.

**What Next**

Correct the syntax error and reread the file.

---

## SVR-11

(error) Cannot assign to constant%s\n \tat line %d in %s.

Feedback

### Description

There is a constant on the left side of an assignment, which is not allowed.

### What Next

Correct the error, and reread the file.

---

## SVR-13

(error) Number of ports of instance is inconsistent with other instances\n \tat line %d in %s.

### Description

All ordered (non-name based) instances of a design must have the same number of ports. The first such instance sets the number. The following example would raise this error.

```
ND2 i1 (a, b, c);
ND2 i2 (d, e);
```

The first instance has 3 ports, whereas the second has only 2. It's possible that the second instance is missing a comma. For example,

```
ND2 i1 (a, b, c);
ND2 i2 (d, , e);
```

### What Next

Examine the instances to determine which is correct. One might be missing a comma.

---

## SVR-14

(error) Indexing into non-array '%s' is not allowed\n \tat line %d in %s.

### Description

A connection has a wire or port with bus notation, and the wire or port was not declared as a bus. For example, the following would raise this error.

```
input A;

ND2 i1 (A[0], b, c);
```

### What Next

Determine if the wire or port declaration or usage is correct.

---

## SVR-15

(error) Width of port %s (%d) is inconsistent with other instances (%d)\n \tat line %d in %s.

Feedback

### Description

The first instance of a design's port sets its width. The width is inferred from the connections to that port. This error indicates, and a subsequent instance has, a different width for a particular port. The following example would raise this error.

```
D1 i1 (A[15:0], b, c);
D1 i2 (A[17:16], e, f);
```

### What Next

Examine the various instances of the design to see which one is correct.

---

## SVR-16

(error) Constant width overflow\n \tat line %d in %s.

### Description

The constant is too large to be represented. Constants cannot exceed 32K bits.

### What Next

Correct the error and reread the file.

---

## SVR-17

(error) %s constant requires %d bits\n \twhich is too large for %s width %d\n \tat line %d in %s.

### Description

The constant specified is too wide for the given width. The default width for a decimal constant is 32 bits. In all other cases, the width would be specified as in 64'b0. An example error would be specifying a decimal number 2535353678882999122 without a width.

### What Next

Correct the error and reread the file.

---

## SVR-18

(error) Slice direction [%d:%d] does not match array direction\n \tat line %d in %s.

### Description

An array (bus) was declared either ascending (like A[0:3]) or descending (like A[3:0]). The reference to this array is inconsistent with the way it was declared. For example, if the declaration is A[3:0], a reference like A[0:1] would raise this error.

Feedback

**What Next**

Correct the error and reread the file.

---

## SVR-19

(error) Index range [%d:%d] is not within bounds [%d:%d]\n \tat line %d in %s.

**Description**

Some or all of the array (bus) indecies of the reference are out of range of the declaration. For example, if the bus was declared A[15:0], A[31:0] would raise this error, because indecies 31 down to 16 are out of bounds.

**What Next**

Correct the error and reread the file.

---

## SVR-20

(error) Non-terminated comment starting at line %d\n \tof '%s'

**Description**

The reader detected end-of-file in the middle of a C-style comment.

**What Next**

Go to the line number mentioned, and find where the comment should be terminated.

---

## SVR-21

(information) %s converted to a wire with no special attributes\n \tat line %d in %s

**Description**

The verilog constructs wand, wor, and tri are converted to a simple wire with no special attributes. This is just an informational message.

**What Next**

No action.

---

## SVR-22

(error) Duplicate instantiation of '%s' (first instance at line %d)\n \tat line %d in %s

Feedback

**Description**

A duplicate instance name was found. The line number of both the duplicate and the original are shown.

**What Next**

Rename one of the duplicates.

---

## SVR-23

(error) Duplicate module '%s' (first occurrence at line %d)\n \tat line %d in %s

**Description**

A module is defined more than once in the file. The line number of both the duplicate and the original are shown.

**What Next**

Rename one of the duplicates.

---

## SVR-24

(warning) Ignored '%s' construct\n \tat line %d in '%s'.

**Description**

This message covers a variety of constructs which are ignored by the native verilog reader, including *specify*, *parameter*, and so on.

This can be an indication of a real problem, especially in the case of *parameter*. The native verilog reader does not support parameters, so if the parameter is being used, other syntax errors will follow. If the parameter is just defined but not used, then ignoring it will be sufficient.

**What Next**

In almost all cases, no action is necessary. If you have real parameters in your netlist, you must use a different verilog reader which understands parameters.

---

## SVR-25

(error) Global reference to '%s' not valid in this context\n \tat line %d in '%s'.

**Description**

Global references are inter-module references, for example, a connection to a wire in a different module like 'global.gnd'. Although this is part of the verilog language, the native

Feedback

verilog reader has very limited support for global references. They can only be used in connections - they cannot be used in assign statements, tran statements, and so on.

This error is generated when a global reference is used in an unsupported context.

**What Next**

For any usage of global references other than in a connection, you need to use a different verilog reader.

---

## SVR-26

(error) Port '%s' has no defined direction (input/output/inout)\n \tin module %s at line %d in file %s.

**Description**

You received this message because the named port was defined in the module terminal list, but did not appear in an input, output, or inout statement. The line number reported is the line number in the module statement where the port is referenced.

**What Next**

Add an appropriate input, output, or inout statement for the named port.

---

## SVR-27

(error) Duplicate declaration of port '%s' in instance '%s'\n \tat line %d in '%s'.

**Description**

You received this message because the named port was defined twice in the terminal list of the given instance, as in the following example:

```
AN2 i0 (.A(a), .A(b), .Z(z));
```

**What Next**

Remove or replace the duplicate port.

---

## SVR-28

(warning) Number %s is too big. It will be truncated to 32 bits\n \tat line %d in '%s'.

Feedback

### Description

When reading a bus (array) declaration or instance, the msb or lsb was bigger than 32 bits, and was truncated. For example:

```
BOX u1 (.A(a[36'hF00000001:0]), .Z(z));
```

The range of a will be truncated to 1:0.

### What Next

Verify that the netlist is correct. This could be a typo.

---

## SVR-31

(error) Expected %s '%s' to be declared as a range\n \tat line %d in '%s'.

### Description

A port has been declared in an input/output/inout statement as a scalar, but was referenced in the module terminal list using a range. In this example:

```
module test (a, b[0], z);
input b;
```

port 'b' is either declared incorrectly in the input statement, or it is referenced incorrectly in the module statement.

### What Next

Correct one of the errors.

---

## SVR-32

(error) Range of %s port '%s' (%d:%d) does not cover\n \tthe range required by the module statement (%d:%d)\n \tat line %d in '%s'.

### Description

A port has been declared in an input/output/inout statement as a range, but the range was inadequate to cover the range required by references to it from the module statement. In this example:

```
module test ({a, b[2:1]}, b[0], z);
input [1:0] b;
```

the module statement requires a range of 2:0, but the input statement declares 1:0. Either the module statement or the input statement is incorrect.

Feedback

### What Next

Correct one of the errors.

---

## SVR-33

(warning) Port '%s' (%s) has a different direction than\n \tother ports in module formal %d (declared line %d); %s statement\n \tat line %d in '%s'.

### Description

A port has been declared in an input/output/inout statement, but its usage in the module statement creates a bus with mixed directions. For example:

```
module test ({a, b}, z);
output b, z;
input [1:0] a;
```

Port number 1 of the module statement has both inputs and outputs. The first *output* statement sets the direction as output. When the input statement is processed, this warning will be issued, since 'a' and 'b' are concatenated in the first module port, and they are different directions. The first port in the concatenation defines the direction for the entire bus (in this case, an input).

### What Next

Verify that the ports are declared correctly, and consider that the netlist created may not have the port directions that you expect.

---

## SVR-34

(error) Slice direction for '%s' [%d:%d] in module port %d ('%s')\n \tdoes not match declared array direction\n \tat line %d in '%s'.

### Description

A port has been declared in an input/output/inout statement as an array with a particular order, like n:0 or 0:n. Its usage in the given port in the module statement is inconsistent with that order. For example:

```
module test ({a[0:1], b}, z);
input [1:0] a;
input b;
output z;
```

Port number 1 of the module statement references a[0:1], but 'a' is declared 1:0. Either the module statement or the input statement is incorrect.

Feedback

### What Next

Correct one of the errors.

---

## SVR-35

(error) Port wire '%s' was never declared in an input/output/inout\n \tstatement in module %s in file %s.

### Description

A port in the module statement references a wire which was never declared in an input/output/inout statement. For example:

```
module test ({a[0:1], b}, z);
input [1:0] a;
output z;
```

Port number 1 of the module statement references 'b', but 'b' was never declared. Either the module statement or the input statement is incorrect.

### What Next

Correct one of the errors.

---

## SVR-37

(warning) Port ‘%s’ at Port number %d of module '%s' was renamed to '%s'\n\tat line %d in '%s'.

### Description

A port has been found in the module terminal list which needs to be renamed. Ports will be renamed if they are explictly repeated, or if they are a single bit of a bus. For example, in this module statement:

```
module test (a, b[0], a, {a,b});
```

The first port will be named 'a'. The second port will be renamed Port2. The third port will be renamed Port3 because 'a' is already in use. Similarly, the fourth port will be renamed Port4, again because 'a' is already in use. In case one of these generated names is already in use, the Verilog reader will continue to append the port number until a free port name is found.

### What Next

No action necessary.

Feedback

## SVR-38

(error) You are declaring the direction of a port '%s'\n \twhose direction is already specified. (File: %s Line: %d)

### Description

Formality reports this error if a port direction is declared more than once. It is an error if the port re-declared with direction whose direction is already specified.

Example:

```
module test (a, b);
input a;
output b;
output a;  //Error
output b;  //Error
```

In above example port 'a' is declared as 'input' port and later re-declared with 'output' direction. Also port 'b' is declared as a 'output' port and re-declared again as 'output' port. Formality will issue an error in each of these cases.

### What Next

Verify that the ports are declared correctly and correct the netlist.

## SVR-40

(error) Could not open temp file in '%s'

### Description

While reading Verilog, it was necessary to create a temporary file, and the file could not be opened. This could be due to file permissions on the directory, incorrect directory, and so on.

### What Next

Verify that you have access to the given directory. If necessary, change the directory as directed by the application.

## SVR-41

(error) while writing to temp directory '%s':\n \t%s

### Description

An error occured while writing to a temporary file in the given directory. This could be due to insufficient space on the disk, or other disk errors.

Feedback

**What Next**

Use a different temporary directory.

---

## SVR-42

(warning) %s constant '%s' requires %d bits\n \twhich is too large for width %d. Truncated. \n \tAt line %d in %s.

**Description**

The constant specified is too wide for the given width. A simple example would be specifying 2'h1e which is truncated from 00011110 to 10. Another example would be specifying a decimal number 2535353678882999122 without a width. The default width for decimal is 32 so this would be truncated from the MSB.

---

## SVR-50

(error) %s

**Description**

A syntax error occurred while parsing the Verilog file during the preprocessor phase. The message specifies where the error appeared and why it was issued.

**What Next**

Action based on the text of the message.

---

## SVR-51

(error) End of file seen before %s block begun at line %d was ended

**Description**

During preprocessing, a comment or directive was found which crossed the end of file boundry, indicating it wasn't terminated appropriately. This problem likely involves an unbalanced "/*", which will need a terminating "*/", or an ifdef without a terminating endif.

**What Next**

For run-on comments, examine the lines immediately after the one specified in the error. Mark the boundry between code and comment with a terminating "*/".

For (preprocessor) directives, its very likely an `ifdef or `else was not completed with an `endif. Determine what code should fit within the `ifdef, and end it with an `endif.

Feedback

## SVR-52

(error) 'include' directive requires a filename enclosed in double quotes\n \t%s

**Description**

The Verilog preprocessor found a missing or misformed include filename. Here is an example of the correct usage of `include:

`include "myfile.v"

**What Next**

Provide the filename, if missing.

Preface the filename with a double-quote ("), and place another double-quote after the last character in the filename.

## SVR-53

(error) Recursive file inclusion detected for file "%s"

**Description**

While preprocessing the Verilog file, a circular dependency was found in the use of an "`include" directive. This error is issued to prevent a file from including itself in an endless loop. Breaking the need for an include loop will avoid triggering this error.

A file that directly includes itself most likely meant to include another file with a similiar name.

**What Next**

Examine the need for the nested "`include". Most likely, several files all depend on each other. The dependencies should be split up so that one file does not depend on any of the others. This can be achieved by moving needed code or `defines to one central file. When one file no longer depends on the others, the corresponding `include(s) can be removed from that file. This should break the `include loop and avoid this error.

## SVR-54

(error) Unable to open file '%s' included by the 'include'\n \tdirective %s

**Description**

The Verilog preprocessor was unable to open a file specified in an "`include" directive. This is most likely caused by the file being outside of the search path. This error is also triggered if the file permissions prohibit read access.

Feedback

**What Next**

If the file exists in the current directory, ensure that "." is present in the search path.

For files in other directories, add the directory to your search path.

Permissions problems can be ruled out by trying to view the file in an editor. Errors of this variety are not specific to Verilog.

---

## SVR-55

(error) Mismatched directive %s

**Description**

The Verilog preprocessor found a compiler directive used with incorrect coupling. For example, an "`endif" is used without a corresponding "`ifdef" before it.

**What Next**

Delete the mismatched directive or add the missed one.

---

## SVR-57

(warning) Macro %s is redefined %s

**Description**

The Verilog preprocess detected a macro redefinition. The same macro name is provided two different definitions, leading to a single macro name having different meanings in separate parts of the file. Although not an error, this is a potentially hard to find and confusing problem.

**What Next**

If the duplicate macro definition was accidental, provide a different name for one of the usages.

If the duplicate usage is intentional, this warning can be avoided by performing an `undef of the macro name before each duplicate definition.

---

## SVR-58

(error) You are declaring a variable '%s' whose name is already found in the same scope\n \tat line %d in %s

Feedback

### Description

This fatal error occurs when the Verilog Netlist reader encounters a variable name that is already used in that scope. For example:

```
module top(clk, out);
  input clk;
  output out;
  wire a, b;
  GTECH_BUF aaa ( .A(clk), .Z(aaa) );
endmodule
```

### What Next

Correct the source so that the port, net, instance or variable is not redeclared.

---

## SVR-59

(error) Instantiation '%s' has mixed ordered and named port connections\n \tat line %d in %s

### Description

This fatal error occurs when the Verilog Netlist reader encounters a Instantiation with mixed ordered and named port connections. For example:

```
module top (a,b,c,out,out1,out2,out3);
input [1:0] a,b,c;
output out,out1,out2,out3;
and_gate and_instance2 ( {b[0],b[1]},.out(out2));
endmodule
```

### What Next

Correct the source so that the Instantiation does not have mixed port connection.

---

## SVR-60

(error) Token beginning with '%s'\n \texceeds maximum allowable length (%d)\n \tat line %d in '%s'.

### Description

A token was found which exceeds the maximum allowable length.

### What Next

Look for a syntax error, such as missing punctutation, near the line referenced.

Feedback

## SVR-61

(warning) Empty port at Port number %d of module '%s' was renamed to '%s'\n\tat line %d in '%s'.

### Description

You receive this warning message when the port list for a module has empty port, with no local name by which to form connections between its body and the instantiation site.

### Examples

The following legal Verilog statement indicates a module with four ports, the last of which is unconnected. ... module m( a,b,c,); /* Legal, but perhaps not intended */ ...

### What Next

This is a warning message only. No action is required on your part.

However, if the unconnected port is not your intention, remove a comma and then run the command again.

## SVR-62

(error) Mixed Port declaration style in module '%s'

### Description

Verilog module can have port declarations in 'ANSI' and 'non-ANSI' formats. It is illegal to mix both the formats.

### Examples

Following example shows illegal use of non-ANSI style port declarations inside module 'myDesign'. ... module myDesign (input a,b, output c); /* ANSI style */ input a, b; /* non-ANSI style: illegal to have port declarations here*/ ... endmodule

### What Next

Edit source code to follow one port declaration format.

## SVR-63

(error) Type mismatch in assignment to supply net at line %d\n \tin %s.

### Description

There is a type mismatch b/w supply net in LHS & RHS, which is not supported.

Feedback

### What Next

Correct the error, and re-read the file.

---

## SVR-64

(warning) Size mismatch between port & wire declaration for the port \n \t '%s' at line %d in '%s'. \n

### Description

SVR throws this warning when a port is defined as a vector and its corresponding wire declaration is a scalar. For example :

```
module test(out);
output [15:0] out;
wire out = 16'b101;     //'out' was declared initially with range [15:0]
endmodule
```

### What Next

This is a warning message only. No action is required on your part. However, if the range mismatch is not your intention, modify the RTL & re-read the file.

---

## SVR-65

(warning) Continuous assignment statement present in netlist design '%s'\n \tat line %d in %s.

### Description

It is a common situation where assign statements result in a technology-mapped netlist sometimes due to feedthrough paths or two outputs being generated from the same logic where a buffer has not been inserted. Continuous assign statements present in technology-mapped netlist can cause problems in downstream Place and Route tools.

For example :

```
module top(in1,in2,out1,out2,out3,out4);
    input in1, in2;
    output out1,out2,out3,out4;
    wire w1,w2,w3;
    low inst1 (in1,in2,w1,w2);
    assign out1 = w1;        //Warning
    assign out2 = w1;        //Warning
    low inst2 (in1,in2,out1,out2);
    low inst3 (in1,in2,out2,out3);
    low inst4 (in1,in2,out3,out4);
endmodule
```

Feedback

```
module low (in1,in2,o1,o2);
    input in1, in2;
    output o1,o2;

    assign o1 = in1;          //Warning
    assign o2 = in2;          //Warning
endmodule
```

**What Next**

Check if the assign statement is valid.

---

## SVR-66

(warning) Statement at Line %d in '%s'\n \texceeds maximum allowable length (%d).\n \tIncomplete statement will be returned.

**Description**

A statement was found which exceeds the maximum allowable length.

---

## SVR-67

(warning) The undeclared symbol '%s' is assumed to have default\n \tnet type 'wire' at line %d in %s.

**Description**

The RHS of an assignment contains an undeclared net. It will be implicitly declared as wire for Verilog 2001 onwards.

Ex: module x (a, z); input a; output z; assign z = q; //q will be implicitly declared as 'wire' endmodule

**What Next**

Check your design to make sure that you intended an implicit declara- tion for this symbol. If not, add an explicit declaration.

---

# SVW

---

## SVW-001

(warning) Writing SystemVerilog from unlinked design '%s'.

Feedback

**Description**

The current design is not linked and could not be automatically linked. The generated SystemVerilog file will contain only modules from within the current design. Modules from other designs and libraries will not be included. The modules will not be hierarchically ordered.

**What Next**

Link the current design and rerun write_sverilog, if you want a complete or hierarchically ordered SystemVerilog file.

---

## SVW-002

(warning) Inapplicable option '%s' ignored.

**Description**

The specified option is not applicable for unlinked designs. The current design is unlinked.

**What Next**

Link the current design and rerun write_sverilog, if you want a hierarchically ordered SystemVerilog file.

---

## SVW-003

(warning) Non-unique module name '%s' (from library '%s' design '%s') renamed to '%s'.

**Description**

Multiple distinct modules exist with the specified name. Duplicate modules are renamed to a unique name.

**What Next**

The user must make sure that any external scripts are modified to match the now-unique SystemVerilog file.

---

## SVW-004

(error) A %s within the design %s does not have a valid name.

**Description**

The specified object within the design does not have a valid name. The object will be omitted from the SystemVerilog output file.

Feedback

**What Next**

The user should check the design and verify that it is valid.

---

## SVW-005

(warning) Skipping module '%s' from design %s.%s ...

**Description**

The module cannot be written because it is from a non-design design view and would not create valid SystemVerilog. The module is probably from an Abstract or an Outline design view. These design views are abstractions of the full designs. They do not contain the complete design information. Writing such a module would result in incomplete SystemVerilog.

**What Next**

Switch to the design view for this design, then write your SystemVerilog modules from that design.

---

## SVW-006

(error) Cannot write SystemVerilog for design %s.%s.

**Description**

The design cannot be written because it is from a non-design design view and would not create valid SystemVerilog. The module is probably from an Abstract or an Outline design view. These design views are abstractions of the full designs. They do not contain the complete design information. Writing such a module would result in incomplete SystemVerilog.

**What Next**

Switch to the design view for this design, then write your SystemVerilog modules from that design.

---

## SVW-007

(warning) Net name '%s' is omitted from the output file, because the net name does not match a connected port name. The net is implicitly renamed to match one of its connected ports.

Feedback

### Description

The indicated net name is omitted from the output SystemVerilog file, because the net name does not match a connected port name. The ports and pins of the net are connected together by a implicit net in the output file. This net is implicitly named to match one of its ports.

When this SystemVerilog file is read back into the tool, the net will have the same name as one of its ports. Hence the original net name is lost.

---

## SVW-008

(Error) Port bus member name '%s' in port bus '%s' of module '%s' is invalid. Similar errors for each port bus might be reported only once.

### Description

The indicated port bus member is invalid. If it is a legal name in SystemVerilog, ensure that the port is declared in ANSI-C style in the original RTL and read in as SystemVerilog by the tool.

### What Next

Correct the issue and run the *write_sverilog* command again.

---

## SVW-009

(Error) Module '%s' contains unmapped synthetic cell '%s' of '%s'.

### Description

The *write_sverilog* command does not write out a module that contains unmapped synthetic cells because these cells cannot be read back into the tool as RTL.

---

## SVW-010

(Error) An interface port is detected in module '%s'.

### Description

The indicated module has an interface port, which is not supported by the *write_sverilog* command. For example, the following ANSI-C style module declaration contains an interface port a:

```
interface simple_bus;     // Define the interface
   logic req, gnt;
   logic [7:0] addr, data;
   logic [1:0] mode;
```

Feedback

```
   logic start, rdy;
endinterface

module memMod(
   simple_bus a,          // Access the simple_bus interface
   input logic clk);
    ...
endmodule
```

## SVW-011

(Error) A type parameter is detected in module '%s'.

### Description

The indicated module has a type parameter, which is not supported by the *write_sverilog* command.For example, the following ANSI-C style module declaration contains a type parameter T:

```
module mid#(parameter type T = logic)
   (input var T in, output var T out);
 ...
endmodule
```

## SVW-012

(Error) An explicitly named port is detected in module '%s'.

### Description

The indicated module has an explicitly named port, which is not supported by the *write_sverilog* command. The following example shows ANSI-C style port declarations with explicitly named ports P1, P2, and Y:

```
module mymod (
   output .P1(r[3:0]),
   output .P2(r[7:4]),
   ref .Y(x),
   input R );

   logic [7:0] r;
   int x;
endmodule
```

## SVW-013

(Information) No module has been affected by the group_cells, ungroup_cells, or uniquify command, so no module has been written out.

Feedback

**Description**

No module is written out because the *write_sverilog* command writes out only modules that are affected by the *group_cells*, *ungroup_cells*, or *uniquify* command.

## SVW-014

(Information) Module '%s' has been affected by '%s' command.

**Description**

The indicated module has been affected by the *group_cells*, *ungroup_cells*, or *uniquify* command, so the affected module is to be written out by the *write_sverilog* command. In case the module cannot be written out, an appropriate error message will be issued.

## SVW-015

(Error) Missing module header information for module '%s'.

**Description**

The module parameter and port declarations for the mentioned module are missing. It can be caused by one of the following reasons:

- The module parameter and port declarations are not in ANSI-C style.
- The module is not read in SystemVerilog format, but in Verilog or VHDL format.
- The module parameter and port declarations contain unsupported elements, such as type parameter and interface.

# SX

## SX-0001

(Fatal) -custom does not support command %s

**Description**

The -custom option in StarRC is intended for transistor level extraction only. The reported command is not supported.

**What Next**

Refer to reference manual for the set of features supported by the custom license.

Feedback

## SX-0002

(Warning) Cannot initialize error data base %s.

**Description**

StarRC could not initialize data base of error messages.

**What Next**

Check to make sure that you are running a installation StarRC installation.

## SX-0003

(Warning) REMOVE_FLOATING_NETS and REMOVE_FLOATING_PORTS are turned on when XREF_REMOVE_FLOATING_NETS is set to YES.

**Description**

REMOVE_FLOATING_NETS and REMOVE_FLOATING_PORTS are turned on when XREF_REMOVE_FLOATING_NETS is set to YES.

**What Next**

Information only; no action is needed.

## SX-0004

(Warning) Star-CCP flow is the standard flow for transistor level extraction for this process technology, please correct the command setup to achieve accurate extraction results.

**Description**

Star-CCP flow is the standard flow for transistor level extraction for this process technology. But CCP engine is not provided and could result to inaccurate result.

**What Next**

Check setting of CCP_EXTRACTION_FILE and rerun StarRC if necessary.

## SX-0010

(Fatal) The NXTGRD file is no longer supported. Regenerate with grdgenxo version 2010.12 or later.

Feedback

**Description**

Using 2010.12 or later grdgenxo version for the NXTGRD file leads to a better accuracy for StarRC.

**What Next**

Regenerate the NXTGRD file with grdgenxo version 2010.12 or later.

---

## SX-0170

(Warning) Using the NETS command with the * wildcard overrides all other NETS commands.

**Description**

You can use multiple NETS files within a StarRC command file or in a file specified with the NETS_FILE command. All of the commands are honored unless the NETS command is used with the asterisk (*) wildcard. The NETS:* command overrides all other NETS commands. The effect is the same whether the NETS command appears in the command file or in a file specified with the NETS_FILE command.

**What Next**

If you intend to select all nets, you can ignore this message. Otherwise, remove the NETS:* command from the command file.

---

## SX-0171

(Fatal) '%s':HIGH cannot be used with a LEFDEF or Milkyway cell-level database.

**Description**

REDUCTION:HIGH and POWER_REDUCTION:HIGH are not supported in the LEFDEF or Milkyway cell-level flow.

**What Next**

Modify the REDUCTION or POWER_REDUCTION command setting.

---

## SX-0172

(Fatal) Invalid '%s' value specified: '%s'.

**Description**

Not available.

Feedback

### What Next

Contact the Synopsys Support Center.

---

## SX-0173

(Warning) -cleanD mode is chosen, you can NOT run -cleanX, -cleanN or incremental afterwards.

### Description

Because cleanD mode removes internal files as many as possible, certain cleanX|N and incremental run cannot be run on top of a cleanD run.

### What Next

This is a warning message for information only.

---

## SX-0174

(Fatal) Previous run was incremental, but ECO_MODE: YES is not set in your command file. If you want to perform a full extraction from the scratch, please specify '-clean' instead.

### Description

In order to use certain '-clean' options in the Eco extraction flow, the ECO_MODE option should remain unchanged in order to ensure consistent functionality.

### What Next

Please refer to section on ECO extraction in the StarRC user guide.

---

## SX-0175

(Fatal) FS flow(EXTRACTION:FSCOMPARE or FS_EXTRACT_NETS) does not support unequal number of corners and unique nxtgrd files in SIMULTANEOUS_MULTI_CORNER flow.

### Description

The number of corners is unequal to the number of unique nxtgrd files. FS flow(EXTRACTION: FSCOMPARE or FS_EXTRACT_NETS) does not support this.

### What Next

set SIMULTANEOUS_MULTI_CORNER as NO to run FS flow or make the number of corners the same as the number of unique nxtgrd files.

Feedback

## SX-0176

(Fatal) FS_QTF_FILE does not support the simultaneous multicorner flow.

**Description**

The FS_QTF_FILE command enables StarRC to run the Field Solver flow with a QTF file. However, the FS_QTF_FILE command does not support the simultaneous multicorner flow, which is activated by the CORNER_TCAD_GRD_FILES command.

**What Next**

Remove the CORNER_TCAD_GRD_FILES command from the command file and run StarRC (with FS_QTF_FILE) with only one corner.

## SX-0177

(Warning) Cannot open CALIBRE_PDBA_FILE: '%s'. Device properties might be lost.

**Description**

The specified CALIBRE_PDBA_FILE cannot be opened for reading.

**What Next**

Please check if the file exists and you have read permission.

## SX-0178

(Information) MOS_GATE_CAPACITANCE is set to zero because this is a DHE flow.

**Description**

MOS_GATE_CAPACITANCE is set to zero because this is an dual-hiararchy extraction flow (DHE) flow. Device properties are obtained from the CALIBRE_PDBA_FILE or ICV_ANNOTATION_FILE.

**What Next**

Information only; no action is needed.

## SX-0179

(Warning) Device "%s" seems to match the SKIP_PCELLS setting, however exists outside of a pcell. Please double check the input db to see if the device got flattened unintentionally.

Feedback

### Description

The device name is matching with the wildcard settings in SKIP_PCELLS command, but it exists outside of a pcell in the input database. Mismatch between SKIP_PCELL matched in database and SKIP_PCELL recognized may cause accuracy problems, please check that LVS is not run in Flat Mode. Typically, the pcells and the device inside the pcell share a common name, except that the pcells have a suffix after this name.

### What Next

This could happen with flattened input database in which case this message can be ignored; however, if this warning occurs in an extraction run with hierarchical input database, please check if the flattening of the pcell in the input database is intended and/or if star cmd settings need to be changed appropriately to handle the intra-device parasitics in this pcell.

---

## SX-0183

(Fatal) No block name provided.

### Description

No block name provided.

### What Next

BLOCK is a required command to run StarRC. Please set it in StarRC command file.

---

## SX-0186

(Warning) METAL_FILL_POLYGON_HANDLING is set to AUTOMATIC because TRANSLATE_FILL_AS_NET is set.

### Description

METAL_FILL_POLYGON_HANDLING is set to AUTOMATIC because TRANSLATE_FILL_AS_NET is set.

### What Next

You can ignore this message if you want to use TRANSLATE_FILL_AS_NET. Otherwise, to use other METAL_FILL_POLYGON_HANDLING settings, remove the TRANSLATE_FILL_AS_NET command.

Feedback

## SX-0187

(Warning) TRANSLATE_FLOATING_AS_FILL is set to NO because TRANSLATE_FILL_AS_NET is set.

### Description

TRANSLATE_FLOATING_AS_FILL is set to NO because TRANSLATE_FILL_AS_NET is set.

### What Next

You can ignore this message if you want to use TRANSLATE_FILL_AS_NET. Otherwise, to use TRANSLATE_FLOATING_AS_FILL: YES, remove the TRANSLATE_FILL_AS_NET command.

## SX-0188

(Fatal) MILKYWAY_EXPAND_HIERARCHICAL_CELLS is only valid with P&R Milkyway databases.

### Description

MILKYWAY_EXPAND_HIERARCHICAL_CELLS only works with MILKYWAY gate-level flow.

### What Next

Make sure you specify MILKYWAY_LIBRARY in StarRC command file. Make sure MILKYWAY_EXTRAC_VIEW is NO in StarRC command file.

## SX-0189

(Fatal) MILKYWAY_EXPAND_HIERARCHICAL_CELLS and SKIP_CELLS may not be used together.

### Description

MILKYWAY_EXPAND_HIERARCHICAL_CELLS automatically configures the SKIP_CELLS list. So don't specify SKIP_CELLS when MILKYWAY_EXPAND_HIERARCHICAL_CELLS is used.

### What Next

Remove SKIP_CELLS setting in StarRC command file.

Feedback

## SX-0190

(Warning) Disable default port handling for DDB flow because the option BOX_PORT is used.

**Description**

Disable default port handling for DDB flow because the option BOX_PORT is used.

**What Next**

Information only; no action is needed.

## SX-0191

(Fatal) BOX_PORT should be only used in specific flow.

**Description**

StarRC command BOX_PORT should be only used in specific flow.

**What Next**

Remove the option from StarRC command file.

## SX-0192

(Warning) Disable SPICE_SUBCKT_FILE because OA_PORT_ANNOTATION_VIEW is set.

**Description**

Both StarRC commands control the port ordering information. Cannot specify those two commands at the same time.

**What Next**

Remove improper StarRC command.

## SX-0193

(Warning) POWER_EXTRACT:DRV_SHORT should only works with MODEL_BASED_SHORT_HANDLING.

**Description**

POWER_EXTRACT:DRV_SHORT should only works with MODEL_BASED_SHORT_HANDLING. Reset to POWER_EXTRACT:NO.

Feedback

**What Next**

Modify one of the two StarRC command file option, POWER_EXTRACT or MODEL_BASED_SHORT_HANDLING.

---

## SX-0194

(Fatal) MODEL_BASED_SHORT_HANDLING needs connectivity information for short detection. It cannot work with SKIP_CELLS_COUPLE_TO_NET.

**Description**

StarRC command MODEL_BASED_SHORT_HANDLING cannot work with StarRC command SKIP_CELLS_COUPLE_TO_NET.

**What Next**

Remove improper StarRC command.

---

## SX-0195

(Information) SHORT_EQUIV_NODES option is used.

**Description**

StarRC command option SHORT_EQUIV_NODES is found.

**What Next**

You need to do nothing about this message.

---

## SX-0197

(Warning) DPT_SHIFT_BEFORE_PROCESS_EFFECT: YES is ignored because the DPT:SHIFT option is not used.

**Description**

The DPT_SHIFT_BEFORE_PROCESS_EFFECT option configures the sequence of events when both pattern shifts and process effects are present. This option does not have any effect unless DPT:SHIFT is also set.

**What Next**

Remove DPT_SHIFT_BEFORE_PROCESS_EFFECT from the StarRC command file or add DPT:SHIFT.

Feedback

## SX-0199

(Fatal) StarXtract cannot initialize the main library.

**Description**

StarXtract encountered an error while trying to initialize the main library.

**What Next**

Check the Milkyway database. If needed, regenerate the Milkyway data.

## SX-0200

(Warning) Could not open library '%s'

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

## SX-0201

(Fatal) Could not get cell boundary for '%s'.

**Description**

StarXtract is getting this cell boundary information from Milkyway database, but this operation is failed.

**What Next**

Confirm the Milkyway data is not broken or re-generate the Milkyway data.

## SX-0202

(Fatal) Found GDS record larger than 65535 bytes in %s.

**Description**

The length of a GDS record is represented by a 2-byte unsigned integer which can only represents 0-65535 range. The length should be in the range of [4, 65535] bytes.

**What Next**

Please check your GDS file.

Feedback

## SX-0203

(Warning) NUM_CORES is larger than the required partitions, and is reduced from '%d' to '%d'.

**Description**

The optimal number of partitions determined by StarRC is less than the value of NUM_CORES specified in the StarRC command file. The value of NUM_CORES is internally changed to match the number of partitions required to conserve compute resources.

**What Next**

Not further action is necessary.

## SX-0204

(Fatal) Command '%s' is not specified in CCP_EXTRACTION_FILE.

**Description**

The following commands are mandatory in the CCP command file specified by the CCP_EXTRACTION_FILE command:

STARRCXT_CCP_EXECUTABLE STARRCXT_CCP_VERSION
STARRCXT_INTERFACE_VIA_LAYERS
STARRCXT_INTERFACE_CONDUCTOR_LAYERS

**What Next**

Add the missing commands to the CCP command file and rerun the CCP flow.

## SX-0205

(Fatal) Found GDS record 0x%x with wrong data type (required: 0-6, actual: %d) in %s

**Description**

The data type of a GDS record must be in the range of [0, 6]. This record is ignored.

**What Next**

Please check your GDS file.

## SX-0206

(Warning) Found GDS record 0x%x with wrong data type (actual: %d, required: %d) in %s

Feedback

**Description**

The data type for the GDS record doesn't follow the GDS specification. Thie record is ignored.

**What Next**

Please reference the GDS specification and check the GDS file.

---

## SX-0208

(Fatal) Database layers are not initialized.

**Description**

Database layers are not correctly initialized; the flow cannot continue.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0209

(Fatal) Illegal interface via layer: '%s'.

**Description**

The layer specified by STARRCXT_INTERFACE_VIA_LAYERS is illegal. Possible causes are the layer is not found in the ITF layers, or the layer is not a via layer.

**What Next**

Correct the layer name and restart the run.

---

## SX-0210

(Fatal) Illegal connection for interface via layer: '%s'.

**Description**

The layer specified in STARRCXT_INTERFACE_VIA_LAYERS has an illegal connection. Possible causes are that its lower metal layer or upper metal layer is not found.

**What Next**

Correct the corresponding layer mapping and restart the run.

Feedback

## SX-0211

(Fatal) Illegal interface conductor layer: '%s'.

**Description**

The layer specified in STARRCXT_INTERFACE_CONDUCTOR_LAYERS is illegal. Possible causes are the layer is not found in the ITF layers, or the layer is not a conductor layer.

**What Next**

Correct the layer name and restart the run.

## SX-0212

(Fatal) Interface conductor layer should be above V0 layer(s).

**Description**

The layers specified in STARRCXT_INTERFACE_CONDUCTOR_LAYERS should be above the layers specified in STARRCXT_INTERFACE_VIA_LAYERS

**What Next**

Correct the layer setup in CCP command file and restart.

## SX-0215

(Fatal) Command line option '%s' previously set.

**Description**

A command-line option cannot be used more than once when running the StarXtract executable.

**What Next**

Check your command-line arguments for multiple occurrences of the same option and use each command-line option no more than once.

## SX-0216

(Fatal) Invalid value '%s' for command line option '%s'

**Description**

The value for the specified command-line option is invalid.

Feedback

**What Next**

Rerun StarRC by specifying the correct command-line option. Refer to the StarRC User Guide for more information.

---

## SX-0217

(Fatal) Missing value for command line option '%s'

**Description**

A command-line option to StarXtract expects a value but the value was not provided.

**What Next**

Rerun StarXtract by providing the required value to the command-line option. Refer to the StarRC User Guide for more information.

---

## SX-0218

(Fatal) Unrecognized command-line option '%s'

**Description**

The specified command-line option to StarXtract is not recognized.

**What Next**

Check the command-line options used. Refer to the StarRC User Guide for more information.

---

## SX-0219

(Fatal) No corners specified with SELECTED_CORNERS command

**Description**

You receive this message because there are no corner(s) specified to extract. You must select at least one corner to extract with SELECTED_CORNERS command.

**What Next**

Specify at least one of the corners defined in the corner file to extract.

---

## SX-0220

(Fatal) Cannot open user defined corners file '%s'

Feedback

**Description**

You receive this message because the user defined corners file specified by CORNERS_FILE command cannot be opened.

**What Next**

Make sure the path to the user defined corners file is correct and the file is not corrupted.

---

## SX-0221

(Fatal) Unexpected keyword '%s' found on line '%d' in CORNERS_FILE '%s'.

**Description**

A corner definition must consist of a corner name, an NXTGRD file and an operating temperature.

**What Next**

Clean the corner definition error before conducting any extraction.

---

## SX-0222

(Warning) Disable SKIP_INSTANCES option because the option BOX_PORT is used.

**Description**

SKIP_INSTANCES is ignored when BOX_PORT is used.

**What Next**

No further action is required.

---

## SX-0224

(Fatal) Unexpected ":" found on line '%d' in CORNERS_FILE '%s'

**Description**

You receive this message because a syntax error is detected during parsing the user defined corners file.

**What Next**

You must clean the syntax errors before conducting any extraction.

Feedback

## SX-0225

(Fatal) Corner '%s' cannot contain ":" on line '%d' in CORNERS_FILE '%s'

**Description**

You receive this message because a syntax error is detected during parsing the user defined corners file.

**What Next**

You must clean the syntax errors before conducting any extraction.

## SX-0226

(Fatal) Corner '%s' cannot contain a backslash on line '%d' in CORNERS_FILE '%s'

**Description**

A syntax error is detected in the corners file specified by the CORNERS_FILE command. The backslash character (\) is not allowed within a corners file.

**What Next**

Correct the syntax errors in the corners file.

## SX-0227

(Fatal) Corner '%s' cannot contain "/" on line '%d' in CORNERS_FILE '%s'

**Description**

A syntax error is detected in the corners file specified by the CORNERS_FILE command. The forward slash character (/) is not allowed within a corners file.

**What Next**

Correct the syntax errors in the corners file.

## SX-0228

(Fatal) Duplicate corner name '%s' found on line '%d' in CORNERS_FILE '%s'

**Description**

StarRC has detected duplicate corner names in the corners file specified by the CORNERS_FILE command. All corner names must be unique.

Feedback

**What Next**

Remove the duplicate corner definition or rename one of the corners.

---

## SX-0229

(Fatal) Cannot open TCAD_GRD_FILE '%s' found on line '%d' in CORNERS_FILE '%s'

**Description**

StarRC failed to open a TCAD_GRD_FILE defined in the CORNERS_FILE.

**What Next**

Make sure that the file specified by the TCAD_GRD_FILE command is not corrupted or missing and that StarRC has permission to read it.

---

## SX-0230

(Fatal) Invalid operating temperature of '%s' found on line '%d' in CORNERS_FILE '%s'

**Description**

You receive this message because a syntax error is detected during parsing an operating temperature defined in CORNERS_FILE.

**What Next**

An operating temperature must be defined in numeric format. Make sure the operating temperature is defined in correct format.

---

## SX-0231

(Fatal) OPERATING_TEMPERATURE is out of range. Use a value between '%f' and '%f' in corner '%s'

**Description**

The operating temperature in the corners file is out of range. The error message specifies the allowable temperature range, in Celsius.

**What Next**

Make sure that OPERATING_TEMPERATURE is defined in Celsius and falls within the supported temperature range.

Feedback

## SX-0232

(Fatal) Unexpected value in CORNER_TYPE is specified in '%s'.

**Description**

An invalid value specified by CORNER_TYPE is detected during parsing the corner definitions. The valid value of CORNER_TYPE must be one of the following values: NOMINAL | OTHER

**What Next**

Change the value of CORNER_TYPE.

## SX-0233

(Fatal) Unexpected text '%s' found on line '%d' in CORNERS_FILE '%s'

**Description**

StarRC has detected a syntax error in the corners file specified by the CORNERS_FILE command.

**What Next**

Correct the syntax errors in the corners file.

## SX-0234

(Fatal) Last corner defined in CORNERS_FILE '%s' is incomplete.

**Description**

A corner definition must consist of a corner name, an NXTGRD file and an operating temperature.

**What Next**

Clean the corner definition error.

## SX-0235

(Fatal) More than one corner is defined as CORNER_TYPE: NOMINAL

**Description**

At most one corner can be defined as the NOMINAL corner using the CORNER_TYPE command in the corners file specified by the CORNERS_FILE command.

Feedback

### What Next

Make sure that no more than one corner is defined as the NOMINAL corner.

---

## SX-0236

(Fatal) All selected corners must have OPERATING_TEMPERATURE: GLOBAL_TEMP to enable temperature coefficient output

### Description

You receive this message because to enable temperature coefficient output in a SIMULTANEOUS_MULTI_CORNER(SMC) flow, StarRC requires all selected corners to have OPERATING_TEMPERATURE: GLOBAL_TEMP. This message indicates that the corners you selected in the StarRC command file failed to meet the requirement.

### What Next

Make sure all selected corners meet the requirement by either removing the inapplicable corners from SELECTED_CORNERS or changing the definitions of the inapplicable corners.

---

## SX-0237

(Fatal) Corner '%s' specified in SELECTED_CORNERS command not found in CORNERS_FILE '%s'

### Description

The definition of a corner specified in the SELECTED_CORNERS command cannot be found in the corners file specified by the CORNERS_FILE command.

### What Next

Make sure the corners names in the SELECTED_CORNERS command are correct and that all of the selected corners are defined in the corners file specified by the CORNERS_FILE command.

---

## SX-0238

(Fatal) Corner '%s' specified in SELECTED_CORNERS has a duplicated TCAD_GRD_FILE, DENSITY_OUTSIDE_BLOCK and OPERATING_TEMPERATURE with corner '%s' specified in SELECTED_CORNERS

Feedback

**Description**

The corners that have same TCAD_GRD_FILE, DENSITY_OUTSIDE_BLOCK and OPERATING_TEMPERATURE are duplicated corners in essence.

**What Next**

Remove or modify the duplicated corners to avoid this error.

---

## SX-0239

(Warning) Corner '%s' will be ignored since it has the same NXTGRD file and DENSITY_OUTSIDE_BLOCK settings as corner '%s' and resistance is not being extracted.

**Description**

Temperature multi-corners, in other words, the extraction corners that have same NXTGRD file and different operating temperatures, are only effective in resistance extraction when running simultaneous multi-corner (SMC) flow.

**What Next**

You can ignore this message or remove the ignored corners from SELECTED_CORNERS to avoid this warning.

---

## SX-0240

(Fatal) Corner '%s' specified in SELECTED_CORNERS has a duplicated TCAD_GRD_FILE, DENSITY_OUTSIDE_BLOCK and OPERATING_TEMPERATURE or duplicated TCAD_GRD_FILE and different mapping file with corner '%s' specified in SELECTED_CORNERS

**Description**

You receive this message because multiple corners in SELECTED_CORNERS have same TCAD_GRD_FILE, DENSITY_OUTSIDE_BLOCK and OPERATING_TEMPERATURE or have same TCAD_GRD_FILE but different mapping file. Corners that have same TCAD_GRD_FILE and OPERATING_TEMPERATURE are essentially duplicated. Extracting duplicated corners should be avoided. Corners that have same TCAD_GRD_FILE should have same mapping file.

**What Next**

Make sure no duplicated corners are selected to extract and make sure the corners with same TCAD_GRD_FILE have same mapping file.

Feedback

## SX-0241

(Fatal) Macro %s doesn't define valid SIZE/PIN in LEF or DIEAREA/PINS in DEF for StarRC to calculate its placement at (%d, %d, %s) in the design %s!

**Description**

If a macro is rotated in the design, it may require valid macro size for StarRC to place it correctly in the design. StarRC uses SIZE/PIN in the MACRO for LEF, or DIEAREA/PINS for DEF.

**What Next**

Check the macro definition in the LEF or DEF files.

## SX-0242

(Warning) Macro %s doesn't define valid SIZE/PIN in LEF or DIEAREA/PINS in DEF for StarRC. Such macro placement may introduce problems in extraction!

**Description**

StarRC suggests to define valid SIZE/PIN in LEF or DIEAREA/PINS in DEF to avoid possible problems in extraction.

**What Next**

Please check the definition of the macro in LEF/DEF files.

## SX-0243

(Fatal) Invalid layer '%s' in the METAL_SHEET_OVER_AREA command.

**Description**

This error message might be caused by several conditions:

(1) The layer specified in the METAL_SHEET_OVER_AREA command is not valid. You must specify a database metal layer name from the mapping file.

(2) The specified layer might cause shorts.

**What Next**

Correct the layer name in the METAL_SHEET_OVER_AREA command in the StarRC command file.

Feedback

## SX-0244

(Fatal) Incorrect lower-left and upper-right coordinates: '%s' '%s' '%s' '%s' in the METAL_SHEET_OVER_AREA command.

**Description**

The geometry of the specified metal sheet is not correct. You must specify legal bounding box coordinates.

**What Next**

Correct the METAL_SHEET_OVER_AREA arguments in the StarRC command file.

## SX-0245

(Warning) With UNIMPORTANT_NETS command, both TRANSLATE_FLOATING_AS_FILL and REMOVE_FLOATING_NETS commands are missing.

**Description**

When using UNIMPORTANT_NETS command, you must set TRANSLATE_FLOATING_AS_FILL or REMOVE_FLOATING_NETS.

**What Next**

See the StarRC User Guide.

## SX-0246

(Information) Ignoring REMOVE_FLOATING_NETS:YES becuse TRANSLATE_FLOATING_AS_FILL:YES is specifed.

**Description**

The commands TRANSLATE_FLOATING_AS_FILL:YES and REMOVE_FLOATING_NETS:YES are incompatible. Ignoring the latter command.

**What Next**

See the StarRC User Guide.

## SX-0247

(Information) Since TRANSLATE_FLOATING_AS_FILL is set to YES, floating nets are translated as floating metal fill. Metal fill polygon handling applies only to floating metal fills.

Feedback

**Description**

This message gives more details about the handling of floating nets and metal fill. It is for informational purposes only.

**What Next**

Consult StarRC User Guide for more information.

---

## SX-0248

(Warning) DEF macro %s has no DIEAREA. Use (%d %d %d %d) calculated from PINS placement instead.

**Description**

If a DEF macro has no DIEAREA, StarRC only uses PIN placements to calculate the bounding box of the DEF macro for performance consideration. However, bounding box calculated in such way may not be accurate. If the DEF macro is rotated in the design, it may require the accurate bounding box to calculate the transformation and place the macro correctly.

StarRC suggests that DEF macros always define DIEAREA to avoid potential issues.

**What Next**

Please make sure the reported bounding box is accurate.

---

## SX-0249

(Fatal) Zone metal net names conflict with names in the input database.

**Description**

Net names referenced by the ZONE_METAL_GDSII_FILE conflict with names in the input database.

**What Next**

Check the net names used in the file specified by the ZONE_METAL_GDSII_FILE command.

---

## SX-0250

(Fatal) Failed to run CCP executable: '%s'

**Description**

Failed to run CCP executable.

Feedback

**What Next**

Check to make sure CCP executable is installed correctly. Contact the Synopsys Support Center if necessary.

---

## SX-0251

(Fatal) Zone metal net name conflicts with the SKIP_CELLS_COUPLE_TO_NET: %s command.

**Description**

A net name referenced by the ZONE_METAL_GDSII_FILE conflicts with the net specified in the SKIP_CELLS_COUPLE_TO_NET command.

**What Next**

Change the net name in either the ZONE_METAL_GDSII_FILE or the SKIP_CELLS_COUPLE_TO_NET command.

---

## SX-0252

(Fatal) Zone metal net name conflicts with the ZONE_COUPLE_TO_NET: %s command.

**Description**

A net name referenced by the ZONE_METAL_GDSII_FILE conflicts with the net specified in the ZONE_COUPLE_TO_NET command.

**What Next**

Change the net name in either the ZONE_METAL_GDSII_FILE or the ZONE_COUPLE_TO_NET command.

---

## SX-0253

(Fatal) Zone metal net name conflicts with the SHEET_COUPLE_TO_NET: %s command.

**Description**

A net name referenced by the ZONE_METAL_GDSII_FILE conflicts with the net specified in the SHEET_COUPLE_TO_NET command.

**What Next**

Change the net name in either the ZONE_METAL_GDSII_FILE or the SHEET_COUPLE_TO_NET command.

Feedback

## SX-0254

(Fatal) TSV Layer '%s' is not defined in the ITF file.

**Description**

Through-silicon via (TSV) extraction is enabled with the 3D_IC_TSV_COUPLING_EXTRACTION command, but the TSV layer is not defined in the ITF file. TSV extraction cannot be performed.

**What Next**

Check if the ITF file is correct. Alternatively, disable TSV extraction by setting the 3D_IC_TSV_COUPLING_EXTRACTION command to NONE or by removing it from the StarRC command file.

## SX-0255

(Severe) StarRC task %s failed. For more details, see task summary file '%s'.

**Description**

A StarRC task has failed to complete successfully.

**What Next**

For more details on the cause of the failure, check the summary file for this task located in ./star/summary/.xxxx.sum. Contact the Synopsys Support Center if you need further help.

## SX-0256

(Warning) RSUB_VS_SPACING table is not defined in ITF file.

**Description**

If you use the 3D_IC_TSV_COUPLING_EXTRACTION:RCSUB command in the StarRC command file, you must provide a RSUB_VS_SPACING table within the TSV block in the ITF file.

**What Next**

Check if the ITF file is correct or modify the 3D_IC_TSV_COUPLING_EXTRACTION:RCSUB command.

## SX-0257

(Warning) incompatible Rapid3d version. Rapid3d will check out its separate licenses.

Feedback

**Description**

Rapid3d version is incompatible with StarRC version.

**What Next**

Check the User Guide for more information on the Field Solver flow. Contact the Synopsys Support Center for further details.

---

## SX-0258

(Warning) FS option -perc_self is set to a value larger than 0.5. FSCOMPARE reports may not be accurate enough to validate StarRC extraction results.

**Description**

Larger accuracy tolerance for the field solver can result in faster runtime. However the FS results may not have sufficient accuracy to validate extraction results.

**What Next**

For more information on accuracy setting in the field solver flow, consult the User Guide.

---

## SX-0259

(Warning) CSUB_VS_SPACING table is not defined in ITF file.

**Description**

If you use the 3D_IC_TSV_COUPLING_EXTRACTION:RCSUB command in the StarRC command file, you must provide a CSUB_VS_SPACING table within the TSV block in the ITF file.

**What Next**

Check if the ITF file is correct or modify the 3D_IC_TSV_COUPLING_EXTRACTION:RCSUB command.

---

## SX-0260

(Fatal) FS flow does not support CCP_EXTRACTION_FILE together with SIMULTANEOUS_MULTI_CORNER.

**Description**

FS CCP flow does not support SIMULTANEOUS_MULTI_CORNER yet.

Feedback

**What Next**

Run FS CCP flow without SIMULTANEOUS_MULTI_CORNER command.

---

## SX-0261

(Fatal) Failed to run field solver using the command: '%s'

**Description**

Field solver run failed.

**What Next**

Make sure you are running from a proper StarRC installation. If you need further help, call the Synopsys Support Center.

---

## SX-0262

(Warning) LEF/NDM macro %s merged with FOREIGN (%d %d %s) and GDS/OASIS content is 5%% larger than its defined SIZE.

**Description**

If the GDS/OASIS merged LEF/NDM cell is larger than its SIZE in LEF/NDM, its FOREIGN in LEF/NDM or GDS content might be problematic and will result in possible extraction problems, s.t. false shorts.

**What Next**

Please check the FOREIGN in the LEF/NDM macro or GDS/OASIS content.

---

## SX-0267

(Fatal) INSTANCE_PORT: PORT_DIR is not supported!

**Description**

INSTANCE_PORT: PORT_DIR currently doesn't support port directions defined in SKIP_CELL_PORT_PROP_FILE when XREF != NO.

**What Next**

Please review your StarRC command file.

---

## SX-0268

(Fatal) INSTANCE_PORT: '%s' uses invalid type.

Feedback

**Description**

Invalid type for INSTANCE_PORT command. Currently supported types are: < NOT_CONDUCTIVE | CONDUCTIVE | EXPLODE | SUPERCONDUCTIVE | MULTIPLE | CAPACITIVE | SUFFIXED >.

**What Next**

Change type to the one of the supported.

---

## SX-0269

(Fatal) Parse error in INSTANCE_PORT command: %s .

**Description**

Incorrect syntax for the command INSTANCE_PORT command. Command syntax is: INSTANCE_PORT: <type> [<type2>] [[CELL <wildcard>] [INST <wildcard>] [PORT <wildcard>] [PORT_DIR <IN|OUT|INOUT>]].

**What Next**

Correct syntax of the command.

---

## SX-0271

(Fatal) Unknown command (option) name in file %s Line: %s.

**Description**

StarRC does not recognize the name of a command (also known as an option) in the StarRC command file.

For example, the following line in a command file results in a SX-0271 error because the command name is misspelled (it should be INSTANCE_PORT).

INSTACE_PORT: CONDUCTIVE

**What Next**

Verify that the intended command name is correctly spelled. Consult the StarRC User Guide for more information about accepted commands.

---

## SX-0272

(Fatal) FS_DB Failed (see file '%s')

Feedback

**Description**

FS_DB stage failed unexpectedly. See the log file mentioned in message above to find the cause.

**What Next**

Correct the run settings based on the log file mentioned in message above. If the issue still exists, contact the Synopsys Support Center.

---

## SX-0273

(Fatal) Unknown keyword in file %s Line: %s .

**Description**

StarRC does not recognize a keyword used in the StarRC command file. A keyword is a word used to specify the setting of a StarRC command.

For example, the following line in a command file results in a SX-0273 error because the keyword name is misspelled (it should be CONDUCTIVE).

INSTANCE_PORT: CONDUCIVE

**What Next**

Verify that the intended keyword is correctly spelled. Consult the StarRC User Guide for more information about accepted keywords for specific commands.

---

## SX-0274

(Fatal) INTERNAL Error: unable to parse technology file.

**Description**

A consistency check failed during the parsing of a StarRC command file or an internal file produced by StarRC.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0275

(Fatal) Netlist VI PostProcess command Failed.

**Description**

VI post process command failed.

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## SX-0276

(Fatal) Internal error: report file '%s' does not exist

**Description**

StarRC cannot find the internal file mentioned in message above.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0277

(Warning) Failed to create symbolic link file to '%s'

**Description**

A system call to create a symbolic link failed.

**What Next**

Check to make sure that disk permissions are correct and that disk response is normal. For further details, contact the Synopsys Support Center.

---

## SX-0280

(Fatal) %s '%s' does not exist.

**Description**

An expected input file does not exist. The missing file might be an input file specified in the StarRC command file, or it might be an internally generated intermediate file.

**What Next**

If a user-specified input file is missing, verify that the file exists and has the requisite permissions for reading.

If an internally generated intermediate file is missing, contact the Synopsys Customer Support Center.

Feedback

## SX-0281

(Fatal) Invalid DP submission command %s : '%s'.

**Description**

The DP submission command specified in the error message is set to an invalid value.

**What Next**

Modify the value of the DP submission command. The correct choice depends on your computing environment. Refer to the StarRC User Guide for more information.

## SX-0282

(Fatal) Command %s is not supported anymore. Please remove it from the command file.

**Description**

User is attempting to run an obsolete command.

**What Next**

See StarRC user guide.

## SX-0284

(Fatal) The NUM_PARTS command is not supported.

**Description**

The NUM_PARTS command is not supported. For more information about distributed processing in this version of the StarRC tool, see the StarRC User Guide.

**What Next**

Remove the NUM_PARTS command from the StarRC command file. Use the NUM_CORES command to specify distributed processing.

## SX-0285

(Fatal) Working directory must be specified with an absolute path.

**Description**

-chdir option is intended for use with remote batch mode execution. Consequently it must specify an absolute path.

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## SX-0286

(Fatal) Unable to change to specified working directory prior to execution.

**Description**

You must specify global absolute network path for the working directory, and it must be visible from the target host.

**What Next**

See StarRC user guide.

---

## SX-0289

(Fatal) MAGNIFICATION_FACTOR cannot be used in HALF NODE flow where the scaling factor is specified in the nxtgrd file.

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0290

(Warning) HALF_NODE_SCALE_FACTOR detected in nxtgrd file, setting NETLIST_UNSCALED_COORDINATES:YES for unshrunk coordinates.

**Description**

StarRC automatically sets NETLIST_UNSCALED_COORDINATES to YES when HALF_NODE_SCALE_FACTOR is detected in the nxtgrd file. This ensures that the netlist coordinates are unscaled, in other words, they are full node coordinates.

**What Next**

If your design or process contains a half-node scale factor, no action is required on your part; this message is for informational purpose only. Refer to the sections on NETLIST_UNSCALED_COORDINATES and HALF_NODE_SCALE_FACTOR in the StarRC user guide for additional details. If you see this warning message and your design does not contain a half-node scale factor, contact the Synopsys Support Center.

Feedback

## SX-0291

(Warning) HALF_NODE_SCALE_FACTOR detected in nxtgrd file, setting NETLIST_UNSCALED_RES_PROP:YES for unshrunk coordinates.

**Description**

StarRC automatically sets NETLIST_UNSCALED_RES_PROP to YES when HALF_NODE_SCALE_FACTOR is detected in the nxtgrd file. This ensures that the resistor properties are unscaled, in other words, they are full node values.

**What Next**

If your design or process contains a half-node scale factor, no action is required on your part; this message is for informational purpose only. Refer to the sections on NETLIST_UNSCALED_RES_PROP and HALF_NODE_SCALE_FACTOR in the StarRC user guide for additional details. If you see this warning message and your design does not contain a half-node scale factor, contact the Synopsys Support Center.

## SX-0292

(Warning) HALF_NODE_SCALE_FACTOR detected in nxtgrd file, setting MAGNIFY_DEVICE_PARAMS:NO for unshrunk device properties.

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

## SX-0293

(Fatal) MAGNIFICATION_FACTOR have to be greater than 0

**Description**

MAGNIFICATION_FACTOR can only accept positive number.

**What Next**

Assign positive number for MAGNIFICATION_FACTOR

## SX-0294

(Fatal) METAL_FILL_GDS_MAG/METAL_FILL_OASIS_MAG must to be larger than 0

Feedback

**Description**

METAL_FILL_GDS_MAG/METAL_FILL_OASIS_MAG can only accept positive number.

**What Next**

Assign positive number for METAL_FILL_GDS_MAG/METAL_FILL_OASIS_MAG.

---

## SX-0295

(Fatal) No block name provided

**Description**

StarRC requires BLOCK in StarRC command file be specified.

**What Next**

Provide BLOCK setting in StarRC command file.

---

## SX-0296

(Fatal) StarRC cannot detect the input database

**Description**

StarRC can determine whether it's MILKYWAY, LEF/DEF, CALIBRE, AGDS, or NDM flows.

**What Next**

Provide corresponding input database source setting in StarRC command file.

---

## SX-0297

(Fatal) BLOCK and DB top block are not the same BLOCK = '%s' DB_BLOCK = '%s'

**Description**

The setting in BLOCK command has to be the same as the top block name stored in the input database.

**What Next**

Provide correct BLOCK setting in StarRC command file.

---

## SX-0298

(Warning) Unable to open PARA view.

Feedback

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0299

(Warning) SHORT_PINS_IN_CELLS option is not supported for the selected database since it does not contain structres for EEQ pins. This option will be ignored.

**Description**

SHORT_PINS_IN_CELLS supports MILKYWAY, LEF/DEF, and NDM flow only. Other flows don't have EEQ information so this option will be ignored.

**What Next**

If this is expected, no action is needed.

---

## SX-0300

(Severe) Map file '%s' does not exist

**Description**

MAPPING_FILE is required by StarRC.

**What Next**

Provide MAPPING_FILE in StarRC command file.

---

## SX-0303

(Fatal) TCAD_GRD_FILE contains unrecognized syntax.

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

Feedback

## SX-0304

(Fatal) TCAD_GRD_FILE contains SEARCH_DISTANCE for one or more layers. This feature is not supported for Milkyway or Lef/Def databases. Please supply a TCAD_GRD_FILE which does not contain SEARCH_DISTANCE.

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

## SX-0305

(Fatal) Not enough layers specified in UNIMPORTANT_NETS_LAYER_CONNECT: '%s'

**Description**

StarRC needs at least 2 different layers set in option for reconnect.

**What Next**

Set correct layers for option and rerun StarRC.

## SX-0306

(Fatal) Invalid layer name '%s' specified in UNIMPORTANT_NETS_LAYER_CONNECT.

**Description**

Invalid layer is specified in UNIMPORTANT_NETS_LAYER_CONNECT.

**What Next**

Set correct layer and rerun StarRC.

## SX-0308

(Warning) Global wildcard '%s' is not supported in SKIP_INSTANCES option. This entry will be ignored.

**Description**

SKIP_INSTANCES doesn't support global wildcard '*', '!*'.

Feedback

**What Next**

Specify the instance names directly or provide more specific wildcard settings like "inst*" in StarRC command file.

---

## SX-0309

(Warning) Device '%s' set in 3D_IC_FILTER_DEVICE is not found.

**Description**

Device set in 3D_IC_FILTER_DEVICE is not found in input data.

**What Next**

Information only; no action is needed.

---

## SX-0310

(Fatal) Instance '%s' port number is not 2.

**Description**

Device set in 3D_IC_FILTER_DEVICE must have 2 ports to connect signal on front side and back side.

**What Next**

Check the devices set in 3D_IC_FILTER_DEVICE and rerun StarRC.

---

## SX-0311

(Warning) Net %s not extracted.

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0314

(Severe) SkipCell '%s' is not found in design

**Description**

StarRC cannot find this skip cell in the input database.

Feedback

**What Next**

Provide correct skip cell settings in StarRC command file.

---

## SX-0315

(Fatal) Found missing skip cells

**Description**

StarRC detects there are some skip cell names listed in SKIP_CELLS cannot be found in the input database.

**What Next**

Provide correct skip cell settings in StarRC command file.

---

## SX-0316

(Warning) Reducing the number of partitions from %d to %d.

**Description**

The NUM_PARTS is larger than this design need, StarRC automatically adjust the number of partitions.

**What Next**

Information only. No action is needed.

---

## SX-0317

(Fatal) Could not open file dist_cuts for writing.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0320

(Fatal) No compare directory provided in XREF flow.

**Description**

No compare directory provided in XREF flow.

Feedback

**What Next**

Please provide compare directory with COMPARE_DIRECTORY to run XREF flow. Or please use XREF: NO in StarRC command file.

---

## SX-0321

(Fatal) Schematic block '%s' is not a valid equiv point.

**Description**

In XREF:YES or XREF:COMPLETE, it is requested for schematic block to be a valid equiv point.

**What Next**

Please check LVS run result. Or you can use XREF:NO for StarRC run.

---

## SX-0322

(Fatal) Cannot use MODEL_TYPE:SCHEMATIC with XREF:NO.

**Description**

The MODEL_TYPE:SCHEMATIC command specifies that the reference model comes from the schematic. However, the XREF:NO command specifies to use the layout net names. These two commands cannot be used together.

If the MODEL_TYPE command is not present in the command file, its setting defaults to LAYOUT.

If the XREF command is not present in the command file, its setting defaults to NO.

**What Next**

Remove the MODEL_TYPE:SCHEMATIC command or use the XREF:YES command in the StarRC command file.

---

## SX-0323

(Fatal) Cannot use MODEL_TYPE:SCHEMATIC with XREF_USE_LAYOUT_DEVICE_NAME:YES.

**Description**

The MODEL_TYPE:SCHEMATIC command cannot be used with the XREF_USE_LAYOUT_DEVICE_NAME:YES command.

Feedback

If the MODEL_TYPE command is not present in the command file, its setting defaults to LAYOUT.

If the XREF_USE_LAYOUT_DEVICE_NAME command is not present in the command file, its setting defaults to NO.

**What Next**

Remove the MODEL_TYPE:SCHEMATIC command or use the XREF_USE_LAYOUT_DEVICE_NAME:NO command in the StarRC command file.

---

## SX-0324

(Fatal) Invalid MODEL_TYPE value : '%s'

**Description**

An invalid keyword is specified for the MODEL_TYPE command. Valid keywords are SCHEMATIC and LAYOUT. If the MODEL_TYPE command is not present in the command file, the setting defaults to LAYOUT.

**What Next**

Correct the MODEL_TYPE command to use a valid keyword or remove the command from the command file.

---

## SX-0325

(Fatal) NETLIST_TYPE: unknown type '%s'. Possible types are: [no_couple] [R | Cg | Cc | RCg | RCc].

**Description**

An invalid setup is detected in NETLIST_TYPE command.

**What Next**

Correct the command setting and restart.

---

## SX-0326

(Warning) %s is not supported when EXTRACT_RES_BODY_COUPLING:YES and will be ignored.

**Description**

StarRC cannot run with both EXTRACT_RES_BODY_COUPLING:YES and the specified option, therefore the specified option is ignored.

Feedback

**What Next**

If this is acceptable, no action is needed otherwise contact the Synopsys Support Center.

---

## SX-0329

(Warning) CAP_DISTRIBUTION_SEGMENT_LENGTH is automatically set equal to NET_SEGMENT_CUT_LENGTH.

**Description**

CAP_DISTRIBUTION_SEGMENT_LENGTH can be set as minimum threshold for deciding the capacitance distribution during extraction. NET_SEGMENT_CUT_LENGTH is the maximum value for polygon segmentation. The value set on NET_SEGMENT_CUT_LENGTH is smaller than CAP_DISTRIBUTION_SEGMENT_LENGTH. To maintain the effects of CAP_DISTRIBUTION_SEGMENT_LENGTH, its value is automatically set to NET_SEGMENT_CUT_LENGTH.

**What Next**

Remove CAP_DISTRIBUTION_SEGMENT_LENGTH or set it to smaller than NET_SEGMENT_CUT_LENGTH.

---

## SX-0330

(Warning) CAP_DISTRIBUTION_SEGMENT_LENGTH is larger than default NET_SEGMENT_CUT_LENGTH value, it does not take effect.

**Description**

CAP_DISTRIBUTION_SEGMENT_LENGTH can be set as minimum threshold for deciding the capacitance distribution during extraction. The value set on this option is larger than NET_SEGMENT_CUT_LENGTH which is the largest segmentation. So the value of CAP_DISTRIBUTION_SEGMENT_LENGTH can never be achieved during extraction, hence it is ignored.

**What Next**

Modify the value of CAP_DISTRIBUTION_SEGMENT_LENGTH equal to or less than NET_SEGMENT_CUT_LENGTH.

---

## SX-0334

(Fatal) CCP_NUM_PARTS is only supported in CCP flow.

Feedback

**Description**

CCP_NUM_PARTS is only supported in CCP flow (when CCP_EXTRACTION_FILE is set).

**What Next**

Change the command setting and restart.

---

## SX-0335

(Fatal) CCP_ANNOTATION_FILE can't work when CCP_EXTRACTION_FILE is not set.

**Description**

CCP_ANNOTATION_FILE can't work when CCP_EXTRACTION_FILE is not set.

**What Next**

Set CCP_EXTRACTION_FILE in StarRC command file.

---

## SX-0337

(Fatal) A MAPPING_FILE command must be specified in the StarRC command file.

**Description**

The MAPPING_FILE command specifies a file containing physical layer mapping information between the design database and the nxtrgrd file. The StarRC tool requires at least one mapping file to perform extraction.

**What Next**

Add the MAPPING_FILE command to the StarRC command file.

---

## SX-0338

(Warning) EXTRACT_VIA_CAPS: FS is set, but FS_EXTRACT_NETS is not set. Setting EXTRACT_VIA_CAPS to NO.

**Description**

EXTRACT_VIA_CAPS: FS must be used with FS_EXTRACT_NETS. Since FS_EXTRACT_NETS is not set, StarRC sets EXTRACT_VIA_CAPS to NO.

**What Next**

If this behavior is acceptable, no action is required. Otherwise, if you want to use EXTRACT_VIA_CAPS: FS, you must also specify FS_EXTRACT_NETS.

Feedback

## SX-0339

(Fatal) METAL_FILL_POLYGON_HANDLING must be IGNORE GROUNDED FLOATING or AUTOMATIC.

**Description**

METAL_FILL_POLYGON_HANDLING must be IGNORE GROUNDED FLOATING or AUTOMATIC.

**What Next**

Use valid settins for METAL_FILL_POLYGON_HANDLING in StarRC command file.

## SX-0340

(Fatal) SUBSTRATE_SEGMENT_CUT_LENGTH must to be larger than or equal to 1 micron.

**Description**

In the StarRC command file, the value of SUBSTRATE_SEGMENT_CUT_LENGTH is less than 1 micron.

**What Next**

In the StarRC command file, set SUBSTRATE_SEGMENT_CUT_LENGTH to be larger than or equal to 1 micron.

## SX-0341

(Fatal) METAL_SLOT_LONG_DIMENSION_LENGTH must to be larger than or equal to 1 micron.

**Description**

In the StarRC command file, the value of METAL_SLOT_LONG_DIMENSION_LENGTH is less than 1 micron.

**What Next**

In the StarRC command file, set METAL_SLOT_LONG_DIMENSION_LENGTH larger than or equal to 1 micron.

## SX-0342

(Fatal) Invalid MILKYWAY_REF_LIB_MODE, '%s'

Feedback

**Description**

Invalid MILKYWAY_REF_LIB_MODE, valid settings are: NONE, HIER, FILE.

**What Next**

Set MILKYWAY_REF_LIB_MODE with correct setting.

---

## SX-0344

(Fatal) No schematic netlist is provided for a run that specifies XREF: COMPLETE.

**Description**

The XREF: COMPLETE command requires a schematic netlist.

**What Next**

For Hercules flows, use the EVACCESS_DIRECTORY command in the StarRC command file.

For IC Validator flows, use the ICV_SCHEMATIC_NETLIST or ICV_RUNSET_REPORT_FILE command in the StarRC command file.

---

## SX-0345

(Fatal) No evaccess directory provided

**Description**

No evaccess directory provided in StarRC command file or ICV_RUNSET_REPORT_FILE.

**What Next**

Please make sure EVACCESS_DIRECTORY is set in StarRC command file or ICV_RUNSET_REPORT_FILE.

---

## SX-0346

(Fatal) Option CALIBRE_IXF_FILE is required for Calibre input.

**Description**

You are running cross reference flow (XREF:YES) with Calibre CCI database. Option CALIBRE_IXF_FILE is required for input.

Feedback

**What Next**

Set CALIBRE_IXF_FILE with correct Calibre IXF file.

---

## SX-0347

(Fatal) Option CALIBRE_NXF_FILE is required for Calibre input.

**Description**

CALIBRE_NXF_FILE is not specified in star command file for Calibre input.

**What Next**

Set CALIBRE_NXF_FILE in star command file to the Calibre NXF file.

---

## SX-0348

(Fatal) Option XREF_USE_LAYOUT_DEVICE_NAME is only supported for XREF: YES

**Description**

XREF_USE_LAYOUT_DEVICE_NAME specifies whether to use layout device names with XREF: YES, it is only supported for XREF: YES.

**What Next**

Set XREF:YES in command file if you really want to use layout device names with XREF:YES, or you can remove XREF_USE_LAYOUT_DEVICE_NAME from command file.

---

## SX-0349

(Fatal) Option MERGE_INSTANCE_PORTS is only supported for XREF: COMPLETE

**Description**

MERGE_INSTANCE_PORTS is only valid when XREF: COMPLETE is also used.

**What Next**

Set XREF: COMPLETE in the StarRC command file.

---

## SX-0351

(Fatal) NETLIST_INCREMENTAL: YES only supports NETLIST_FORMAT: SPEF.

Feedback

**Description**

NETLIST_INCREMENTAL: YES is only supported for NETLIST_FORMAT: SPEF.

**What Next**

Change the command setting and restart.

---

## SX-0352

(Fatal) NETLIST_NODENAME_NETNAME can only be used with NETLIST_FORMAT: NETNAME|SPF|STAR|OA.

**Description**

NETLIST_NODENAME_NETNAME can only be used with NETLIST_FORMAT: NETNAME|SPF|STAR|OA.

**What Next**

Use supported NETLIST_FORMAT (NETNAME|SPF|STAR|OA) for NETLIST_NODENAME_NETNAME. Or remove NETLIST_NODENAME_NETNAME in StarRC command file.

---

## SX-0353

(Fatal) Cannot set netlist_format to NONE for EXTRACTION: FSCOMPARE.

**Description**

NETLIST_FORMAT:NONE makes StarRC skip netlist stage. It cannot be used with EXTRACTION:FSCOMPARE.

**What Next**

Set NETLIST_FORMAT with other option and run again.

---

## SX-0354

(Fatal) Cannot set netlist_format to NONE for incremental extraction.

**Description**

Netlist format NONE is not a valid incremental extraction netlist option. The most common netlist format for incremental extraction is SPEF.

**What Next**

Please refer to section on ECO extraction in the StarRC user guide.

Feedback

## SX-0355

(Fatal) RSPEF/RSPF format is no longer supported. Currently supported NETLIST_FORMAT are: SPF | STAR | SPEF | MW | CONLY | NETNAME | NONE.

**Description**

RSPEF/RSPF format is no longer supported.

**What Next**

Set NETLIST_FORMAT to either one of supported netlist formats: SPF, STAR, SPEF, MW, CONLY, NETNAME and NONE.

## SX-0356

(Warning) DPT_COLOR_GDS_FILE does not contain any cell definition for BLOCK %s . Color objects must reside in same name cell as BLOCK.

**Description**

DPT_COLOR_GDS_FILE doesn't contain any cell that matches BLOCK. The DPT_COLOR_GDS_FILE should be associated with the original design by having the same top cell.

**What Next**

Provide DPT_COLOR_GDS_FILE that contains a cell with name equals to BLOCK.

## SX-0358

(Fatal) Unknown NETLIST_FORMAT: '%s'

**Description**

Unknown NETLIST_FORMAT: '%s'

**What Next**

Please use correct setting for NETLIST_FORMAT: SPF | STAR | SPEF | CONLY | NETNAME | NONE | OA

## SX-0359

(Fatal) Number of values specified in OPERATING_TEMPERATURE is not equal to number of corners nor 1.

Feedback

**Description**

You receive this message because you are using OPERATING_TEMPERATURE, and the values is not equal to TCAD_GRD_FILE number nor 1.

**What Next**

Please change OPERATING_TEMPERATURE in StarRC command file to match the number with TCAD_GRD_FILE or only provide one value.

---

## SX-0360

(Fatal) OPERATING_TEMPERATURE is specified in CELSIUS. Please use temperature between -200 and +300

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0361

(Fatal) Duplicated values in TCAD_GRD_FILE are allowed only when OPERATING_TEMPERATURE has multiple values

**Description**

You receive this message because you set duplicated values in TCAD_GRD_FILE, but OPERATING_TEMPERATURE does not have multiple values.

**What Next**

Please remove duplicated values in TCAD_GRD_FILE if you do not use temperature sensitivity Or set OPERATING_TEMPERATURE with multiple values to use temperature sensitivity flow.

---

## SX-0362

(Warning) Ignoring operating_temperature in the StarRC command file, will use operating_temperatures in corner file.

**Description**

When TEMPERATURE_SENSITIVITY and NETLIST_CORNER_NAMES are both set in the StarRC command file, the operating temperatures take precedence.

Feedback

**What Next**

This is only a warning message. No action is required.

---

## SX-0363

(Warning) Ignoring operating_temperature in StarRC command file, will output CRT sensitivities.

**Description**

In TEMPERATURE_SENSITIVITY flow the TC1/TC2 is outputted based on the global temperature definition. So specified OPERATING_TEMPERATURE is not used and it is ignored.

**What Next**

Remove OPERATING_TEMPERATURE from StarRC command file

---

## SX-0367

(Fatal) ECO_MODE: YES does not support POWER_EXTRACT:'%s'

**Description**

Incremental extrction does not support extraction of power nets.

**What Next**

Please refer to section on ECO extraction in the StarRC user guide.

---

## SX-0368

(Warning) No NETLIST_POWER_FILE specified. Power extraction not performed.

**Description**

No NETLIST_POWER_FILE specified. Power extraction not performed.

**What Next**

Please add NETLIST_POWER_FILE if you want to performance power extraction. Or you can ignore this message.

---

## SX-0369

(Fatal) Only one nxtgrd file may be specified for TEMPERATURE_SENSITIVITY:YES

Feedback

**Description**

Only one nxtgrd file may be specified for TEMPERATURE_SENSITIVITY:YES

**What Next**

Please only keep on nxtgrd file in TCAD_GRD_FILE in StarRC command file.

---

## SX-0370

(Fatal) No tcad file provided

**Description**

No tcad file provided.

**What Next**

Set TCAD_GRD_FILE or MULTI_CORNER_TCAD_GRD_FILES in StarRC command file.

---

## SX-0371

(Warning) TCAD_GRD_FILE will be ignored because MULTI_CORNER_TCAD_GRD_FILES is set. Stop the job now if this is not what you intended.

**Description**

MULTI_CORNER_TCAD_GRD_FILES is intended to list multiple corners including nominal corner in correct order, TCAD_GRD_FILE is intended to provide nominal corner. When both are provided, TCAD_GRD_FILE is ignored. Stop the job now if this is not what you intended!

**What Next**

This is a warning message for information only.

---

## SX-0372

(Fatal) TCAD_GRD_FILE must specify exactly one file if it is used with TCAD_MIN_GRD_FILE/TCAD_MAX_GRD_FILE.

**Description**

You receive this message because you set multiple values for TCAD_MIN_GRD_FILE/ TCAD_MAX_GRD_FILE. Only one value is supported.

Feedback

**What Next**

Change TCAD_MIN_GRD_FILE/TCAD_MAX_GRD_FILE in StarRC command file to only provide one value.

---

## SX-0373

(Warning) TCAD_MIN_GRD_FILE and TCAD_MAX_GRD_FILE must both be specified. Not extracting triplets.

**Description**

This warning informs you that only one of TCAD_MIN_GRD_FILE and TCAD_MAX_GRD_FILE is specified in StarRC command file.

**What Next**

If you want to extract both min and max corner, you should specify both TCAD_MIN_GRD_FILE and TCAD_MAX_GRD_FILE. If you only want to extract one corner, you don't need to take any action.

---

## SX-0378

(Fatal) Inductance extraction is not supported in this release

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0379

(Fatal) Attempting to run invalid extraction mode.

**Description**

Extraction type must be RC, C, R, or FSCOMPARE.

**What Next**

See StarRC user guide.

Feedback

## SX-0380

(Fatal) Command EXTRACTION: C is not supported in CCP flow. Please use EXTRACTION: RC and NETLIST_TYPE: Cc * to get C only netlist.

**Description**

CCP flow does not support EXTRACTION: C. Please use EXTRACTION: RC and NETLIST_TYPE: Cc * to get C only netlist.

**What Next**

Revise the command file and restart.

## SX-0381

(Fatal) SENSITIVITY: YES and TEMPERATURE_SENSITIVITY:YES are not supported for EXTRACTION: %s

**Description**

SENSITIVITY: YES and TEMPERATURE_SENSITIVITY:YES can be used only for EXTRACTIONset as RC, C, FSCOMPARE and R.

**What Next**

Set EXTRACTION as RC, C, FSCOMPARE or R.

## SX-0384

(Fatal) SENSITIVITY:YES and TEMPERATURE_SENSITIVITY:YES only support specification of one file in TCAD_GRD_FILE command

**Description**

SENSITIVITY:YES and TEMPERATURE_SENSITIVITY:YES only supports with one file in TCAD_GRD_FILE command.

**What Next**

Set TCAD_GRD_FILE to one file, or change one of the two commands SENSITIVITY and TEMPERATURE_SENSITIVITY as other options other than YES.

## SX-0386

(Fatal) TEMPERATURE_SENSITIVITY:YES is supported for SPF, STAR, NETNAME, SPEF netlist formats and OA parasitic view only.

Feedback

**Description**

TEMPERATURE_SENSITIVITY:YES can only be used when the output netlist format is SPF, STAR, NETNAME, SPEF, or OA parasitic view.

**What Next**

You should use supported netlist format in NETLIST_FORMAT command in StarRC command file.

---

## SX-0392

(Fatal) ECO_MODE: YES can not be used with EXTRACTION: %s

**Description**

Incremental extraction limits the types of extraction options due to scope and typical usage. Please use RC extraction type.

**What Next**

Please refer to section on ECO extraction in the StarRC user guide.

---

## SX-0405

(Fatal) The IGNORE_CAPACITANCE value must be ALL [RETAIN_GATE_DIFFUSION_COUPLING], DIFF, or NONE.

**Description**

StarRC does not recognize the keyword used for the IGNORE_CAPACITANCE command.

**What Next**

Modify the IGNORE_CAPACITANCE command to use one of the supported values:

```
ALL
ALL RETAIN_GATE_DIFFUSION_COUPLING
DIFF
NONE
```

---

## SX-0406

(Warning) NETLIST_COMPRESS_COMMAND is not supported in non-ASCII NETLIST_FORMAT

**Description**

NETLIST_COMPRESS_COMMAND is not supported in non-ASCII NETLIST_FORMAT

Feedback

**What Next**

NETLIST_COMPRESS_COMMAND is ignored.

---

## SX-0407

(Warning) NETLIST_POSTPROCESS_COMMAND is not supported in non-ASCII NETLIST_FORMAT

**Description**

NETLIST_POSTPROCESS_COMMAND is not supported in non-ASCII NETLIST_FORMAT

**What Next**

NETLIST_POSTPROCESS_COMMAND is ignored

---

## SX-0408

(Warning) EXTRACT_RES_BODY_COUPLING is not supported in the Field Solver flow.

**Description**

EXTRACT_RES_BODY_COUPLING is not supported in the Field Solver flow. Set it to NO or remove the command from the command file (it defaults to NO).

**What Next**

Set the EXTRACT_RES_BODY_COUPLING command to NO or remove it from the command file (the default is NO).

---

## SX-0409

(Fatal) Current setting of NETLIST_FORMAT is only valid with EXTRACTION: RC

**Description**

Current setting of NETLIST_FORMAT and EXTRACTION is conflict.

**What Next**

Revise either above option to avoid the conflict.

---

## SX-0411

(Warning) FSCompare mode only produces comparison output... NETLIST_FORMAT is ignored.

Feedback

**Description**

FSCompare mode only produces comparison output... NETLIST_FORMAT is ignored.

**What Next**

You can ignore this message. StarRC will not output NETLIST_FORMAT.

---

## SX-0412

(Fatal) FS_EXTRACT_NETS is not supported for EXTRACTION: FSCOMPARE.

**Description**

FS_EXTRACT_NETS and EXTRACTION: FSCOMPARE can not be both set in StarRC command file.

**What Next**

If you want to use FS_EXTRACT_NETS, set EXTRACTION: RC instead of EXTRACTION: FSCOMPARE. If you want to set EXTRACTION: FSCOMPARE, you should not set FS_EXTRACT_NETS.

---

## SX-0416

(Fatal) TRANSLATE_RETAIN_BULK_LAYERS:YES must be set for substrate extraction with SUBSTRATE_EXTRACTION:YES.

**Description**

TRANSLATE_RETAIN_BULK_LAYERS:YES must be set for substrate extraction with SUBSTRATE_EXTRACTION:YES.

**What Next**

Set TRANSLATE_RETAIN_BULK_LAYERS: YES if you want to do substrate extraction. Or use SUBSTRATE_EXTRACTION: NO.

---

## SX-0418

(Fatal) EXTRACTION: C and REDUCTION: NO_REDUCTION are incompatible options.

**Description**

You can not use EXTRACTION: C and REDUCTION: NO_REDUCTION together.

**What Next**

Please user other settings for EXTRACTION and REDUCTION in StarRC command file.

Feedback

## SX-0419

(Fatal) NETLIST_REMOVE_LOOPS must be YES or NO.

**Description**

NETLIST_REMOVE_LOOPS must be YES or NO.

**What Next**

Please set NETLIST_REMOVE_LOOPS to YES or NO in StarRC command file.

## SX-0420

(Fatal) NETLIST_TAIL_COMMENTS is not allowed with POWER_REDUCTION: YES or POWER_REDUCTION: HIGH.

**Description**

NETLIST_TAIL_COMMENTS is not allowed with POWER_REDUCTION: YES or POWER_REDUCTION: HIGH.

**What Next**

If you want to see NETLIST_TAIL_COMMENTS, please set POWER_REDUCTION to other settings. Or remove NETLIST_TAIL_COMMENTS in StarRC command file.

## SX-0421

(Fatal) NETLIST_TAIL_COMMENTS: YES is not allowed with REDUCTION: YES or REDUCTION: HIGH or REDUCTION: NO_EXTRA_LOOPS.

**Description**

When reduction takes place across layers, the tail comments are not correct. Therefore you cannot set the NETLIST_TAIL_COMMENTS command to YES if the REDUCTION command is set to YES, HIGH, or NO_EXTRA_LOOPS.

**What Next**

Modify the StarRC command file to use an allowable combination of commands.

## SX-0423

(Warning) VIA_COVERAGE must be used with NETLIST_TAIL_COMMENTS.

Feedback

**Description**

NETLIST_TAIL_COMMENTS: YES|ADVANCED is required for VIA_COVERAGE.

**What Next**

Set NETLIST_TAIL_COMMENTS: YES|ADVANCED.

---

## SX-0424

(Warning) VIA_COVERAGE is only available with SPEF output format

**Description**

VIA_COVERAGE is only available with SPEF output format

**What Next**

Set NETLIST_FORMAT: SPEF or remove VIA_COVERAGE in StarRC command file

---

## SX-0425

(Fatal) SKIP_CELLS_COUPLE_TO_NET option only valid with COUPLE_TO_GROUND: NO

**Description**

SKIP_CELLS_COUPLE_TO_NET option only valid with COUPLE_TO_GROUND: NO.

**What Next**

Remove SKIP_CELLS_COUPLE_TO_NET or set COUPLE_TO_GROUND: NO in StarRC command file.

---

## SX-0427

(Fatal) ZONE_COUPLE_TO_NET option only valid with COUPLE_TO_GROUND: NO

**Description**

ZONE_COUPLE_TO_NET option only valid with COUPLE_TO_GROUND: NO

**What Next**

Remove ZONE_COUPLE_TO_NET or set COUPLE_TO_GROUND: NO in StarRC command file

Feedback

## SX-0429

(Fatal) SHEET_COUPLE_TO_NET option only valid with COUPLE_TO_GROUND: NO

**Description**

SHEET_COUPLE_TO_NET option only valid with COUPLE_TO_GROUND: NO

**What Next**

Remove SHEET_COUPLE_TO_NET or set COUPLE_TO_GROUND: NO in StarRC command file.

## SX-0431

(Fatal) COUPLE_NONCRITICAL_NETS option only valid with COUPLE_TO_GROUND: NO

**Description**

COUPLE_NONCRITICAL_NETS option only valid with COUPLE_TO_GROUND: NO

**What Next**

Remove COUPLE_NONCRITICAL_NETS or set COUPLE_TO_GROUND: NO in StarRC command file.

## SX-0433

(Fatal) COUPLE_NONCRITICAL_NETS is not supported for XREF:COMPLETE flow.

**Description**

COUPLE_NONCRITICAL_NETS is not supported for XREF:COMPLETE flow.

**What Next**

Remove COUPLE_NONCRITICAL_NETS or use XREF:YES in StarRC command file

## SX-0434

(Fatal) COUPLE_NONCRITICAL_NETS is not supported in FS flow.

**Description**

COUPLE_NONCRITICAL_NETS is not supported in FS flow.

Feedback

**What Next**

Remove COUPLE_NONCRITICAL_NETS or turn off FS flow in StarRC command file

---

## SX-0435

(Warning) IGNORE_CAP_FOR_ALL_DIFF_LAYERS only supports IGNORE_CAPACITANCE: ALL or DIFF. Ignored.

**Description**

IGNORE_CAP_FOR_ALL_DIFF_LAYERS only supports IGNORE_CAPACITANCE: ALL or DIFF. Ignored.

**What Next**

If you want to use IGNORE_CAP_FOR_ALL_DIFF_LAYERS, please set IGNORE_CAPACITANCE: ALL or DIFF. Or you can ignore this message.

---

## SX-0436

(Fatal) The THIRD_PARTY_FIELD_SOLVER option has been discontinued in StarRC-2008.12 beta and forward

**Description**

THIRD_PARTY_FIELD_SOLVER option has been obsolated.

**What Next**

Remove the option from StarRC command file.

---

## SX-0437

(Fatal) FSCOMPARE_OPTIONS: option '%s' has no value

**Description**

Field Solver does NOT accept options specified in FSCOMPARE_OPTIONS.

**What Next**

Modify options in FSCOMPARE_OPTIONS.

---

## SX-0438

(Fatal) FSCOMPARE_OPTIONS: option '%s' has no value or negative value

Feedback

**Description**

Not acceptable input for FSCOMPARE_OPTIONS.

**What Next**

Modify StarRC command file.

---

## SX-0439

(Fatal) FSCOMPARE_OPTIONS: option '%s' should be followed by a floating number value

**Description**

Not acceptable input for FSCOMPARE_OPTIONS.

**What Next**

Modify StarRC command file.

---

## SX-0440

(Fatal) FSCOMPARE_OPTIONS: option '%s' should be set through FS_BOUNDARY_BOX command

**Description**

Use FS_BOUNDARY_BOX instead of "-bb" in FSCOMPARE_OPTIONS to set the default boundary box for the Field Solver.

**What Next**

```
FSCOMPARE_OPTIONS
FS_BOUNDARY_BOX
```

---

## SX-0441

(Fatal) RANXT_BOUNDARY_CONDITION is obsolete in 2011.06. Use FS_BOUNDARY_CONDITION instead

**Description**

RANXT_BOUNDARY_CONDITION is obsolete in 2011.06. Use FS_BOUNDARY_CONDITION instead FS_BOUNDARY_CONDITION and FS_BOUNDARY_BOX must be used together to apply Neuman or Periodic boundary conditions for a design (or part of a design)

Feedback

**What Next**

Use FS_BOUNDARY_CONDITION

---

## SX-0442

(Fatal) bad FS_BOUNDARY_CONDITION; please check: '%s'

**Description**

Not acceptable input for FS_BOUNDARY_CONDITION option.

**What Next**

Modify StarRC command file.

---

## SX-0443

(Fatal) Both FS_BOUNDARY_CONDITION and FS_BOUNDARY_BOX must be set

**Description**

FS_BOUNDARY_CONDITION and FS_BOUNDARY_BOX must be used together to apply Neuman or Periodic boundary conditions for a design (or part of a design)

**What Next**

Set FS_BOUNDARY_CONDITION and FS_BOUNDARY_BOX both in StarRC command file.

---

## SX-0444

(Fatal) "-replicate" has been obsolated.

**Description**

"-replicate" has been obsolated in FS_BOUNDARY_BOX.

**What Next**

Remove "-replicate" in FS_BOUNDARY_BOX.

---

## SX-0445

(Fatal) RANXT_BOUNDARY_BOX is obsolete in 2011.06. Use FS_BOUNDARY_BOX instead

Feedback

**Description**

RANXT_BOUNDARY_BOX has been obsolete in 2011.06. Use FS_BOUNDARY_BOX instead. FS_BOUNDARY_CONDITION and FS_BOUNDARY_BOX must be used together to apply Neuman or Periodic boundary conditions for a design (or part of a design).

**What Next**

Use FS_BOUNDARY_BOX instead

---

## SX-0447

(Fatal) Not support FS_BOUNDARY_BOX parameters: %s

**Description**

FS_BOUNDARY_BOX requires 6 boundary box corrdinetes as input.

**What Next**

Modify StarRC command file.

---

## SX-0448

(Fatal) FS_BOUNDARY_BOX options has a non-number '%s' specified

**Description**

Not acceptable input for FS_BOUNDARY_BOX option.

**What Next**

Modify StarRC command file.

---

## SX-0454

(Fatal) Process effect(s) '%s' is (are) not supported for '%s'.

**Description**

StarRC with third-party Field Solver does not support certain process effect.

**What Next**

Use Rapid3D instead of thrid-party Field Solver.

Feedback

## SX-0455

(Fatal) TARGET_PWRA: Cannot limit EXTRACTION to C since POWER_EXTRACT is set to RONLY.

**Description**

EXTRACTION: C is conflicted with TARGET_PWRA because POWER_EXTRACT will be RONLY in TARGET_PWRA flow.

**What Next**

Correct the command setting and restart.

## SX-0456

(Warning) TARGET_PWRA: Changing POWER_EXTRACT to RONLY

**Description**

The TARGET_PWRA command automatically generates and changes a set of StarRC commands for RC extraction in a power reliability analysis flow. Refer to StarRC User Guide for more information.

**What Next**

This is only a warning message. No action is required.

## SX-0457

(Warning) TARGET_PWRA: The allowed POWER_REDUCTION settings are NO or LAYER or LAYER_NO_EXTRA_LOOPS, changing POWER_REDUCTION to LAYER_NO_EXTRA_LOOPS

**Description**

The TARGET_PWRA command automatically generates and changes a set of StarRC commands for RC extraction in a power reliability analysis flow. Refer to StarRC User Guide for more information.

**What Next**

This is only a warning message. No action is required.

## SX-0458

(Warning) TARGET_PWRA: Changing NETLIST_CONNECT_SECTION to YES

Feedback

**Description**

The TARGET_PWRA command automatically generates and changes a set of StarRC commands for RC extraction in a power reliability analysis flow. Refer to StarRC User Guide for more information.

**What Next**

This is only a warning message. No action is required.

---

## SX-0459

(Warning) TARGET_PWRA: Changing NETLIST_NODE_SECTION to YES

**Description**

The TARGET_PWRA command automatically generates and changes a set of StarRC commands for RC extraction in a power reliability analysis flow. Refer to StarRC User Guide for more information.

**What Next**

This is only a warning message. No action is required.

---

## SX-0460

(Warning) TARGET_PWRA: Changing NETLIST_TAIL_COMMENTS to YES

**Description**

The TARGET_PWRA command automatically generates StarRC commands for RC extraction in a power reliability analysis flow. NETLIST_TAIL_COMMENTS is one of them. Refer to StarRC User Guide for more information.

**What Next**

This is only a warning message. No action is required.

---

## SX-0461

(Warning) TARGET_PWRA: Changing SHORT_PINS to NO

**Description**

The TARGET_PWRA command automatically generates and changes a set of StarRC commands for RC extraction in a power reliability analysis flow. Refer to StarRC User Guide for more information.

Feedback

**What Next**

This is only a warning message. No action is required.

---

## SX-0462

(Warning) EXTRA_GEOMETRY_INFO will be set to NODE for TARGET_PWRA:YES

**Description**

The TARGET_PWRA command automatically generates StarRC commands for RC extraction in a power reliability analysis flow. EXTRA_GEOMETRY_INFO is one of them. Refer to StarRC User Guide for more information.

**What Next**

This is only a warning message. No action is required.

---

## SX-0463

(Fatal) %s is not supported in 3D_IC flow.

**Description**

The highlighted commands are not supported in 3D_IC extraction flow.

**What Next**

Remove the unsupported command and restart.

---

## SX-0464

(Warning) The GROUND_NET will be '%s' for 3D_IC_FLOATING_SUBSTRATE:YES, If a different name is desired, please use NETLIST_GROUND_NODE_NAME to specify it.

**Description**

When 3D_IC_FLOATING_SUBSTRATE:YES is required, the item *GROUND_NET in netlist file can be customized by NETLIST_GROUND_NODE_NAME.

**What Next**

This is a warning message for information purpose only. No action is required unless a different GROUND_NET is desired.

Feedback

## SX-0465

(Warning) NETLIST_GROUND_NODE_NAME: '%s' will override the default ground name '%s' for 3D_IC_FLOATING_SUBSTRATE:YES.

**Description**

This indicates the default ground name is revised by NETLIST_GROUNG_NODE_NAME setting.

**What Next**

This message is only a remind and no further action is required.

## SX-0466

(Fatal) SIMULTANEOUS_MULTI_CORNER: YES is not supported for Direct ITF flow.

**Description**

SIMULTANEOUS_MULTI_CORNER is not supported for Direct ITF flow.

**What Next**

Disable SIMULTANEOUS_MULTI_CORNER or Direct ITF setting.

## SX-0468

(Fatal) TEMPERATURE_SENSITIVITY should not be set in SIMULTANEOUS_MULTI_CORNER(SMC) flow. If you need temperature coefficients output for in transistor-level SMC flow, please just set OPERATING_TEMPERATURE: GLOBAL_TEMP for the corners in the SMC CORNERS_FILE.

**Description**

You receive this message because StarRC found TEMPERATURE_SENSITIVITY has been set in the command file when running a SIMULTANEOUS_MULTI_CORNER(SMC) flow.

**What Next**

If you expect to output temperature coefficients for a transistor-level SMC flow, make sure setting OPERATING_TEMPERATURE: GLOBAL_TEMP in the SMC CORNERS_FILE for the selected corners.

Feedback

## SX-0470

(Warning) PRINT_SILICON_INFO may not reflect all corners in SIMULTANEOUS_MULTI_CORNER flow.

**Description**

In SIMULTANEOUS_MULTI_CORNER flow, if PRINT_SILICON_INFO is set, the $si_w can only reflect the first corner being extracted. The silicon information for other corners is omitted.

**What Next**

Contact the Synopsys Support Center.

## SX-0472

(Fatal) CONVERT_DIODE_TO_PARASITIC_CAP is not supported in SIMULTANEOUS_MULTI_CORNER flow.

**Description**

CONVERT_DIODE_TO_PARASITIC_CAP needs special handling during extraction, this is not yet supported in SIMULTANEOUS_MULTI_CORNER flow.

**What Next**

Remove CONVERT_DIODE_TO_PARASITIC_CAP from StarRC command file and re-extract using SIMULTANEOUS_MULTI_CORNER flow.

## SX-0477

(Fatal) SIMULTANEOUS_MULTI_CORNER: YES is not supported with ENHANCED_PROCESS_VARIATION_HANDLING: LEVEL3.

**Description**

SIMULTANEOUS_MULTI_CORNER: YES is not supported with ENHANCED_PROCESS_VARIATION_HANDLING: LEVEL3.

**What Next**

Please change the value for the option ENHANCED_PROCESS_VARIATION_HANDLING or just remove this option when SIMULTANEOUS_MULTI_CORNER is set to YES.

Feedback

## SX-0478

(Fatal) CORNERS_FILE command option must be given if SIMULTANEOUS_MULTI_CORNER: YES is used.

**Description**

CORNERS_FILE command option must be given if SIMULTANEOUS_MULTI_CORNER: YES is used.

**What Next**

Please specify the value for the option CORNERS_FILE if SIMULTANEOUS_MULTI_CORNER is set to YES.

## SX-0479

(Fatal) SELECTED_CORNERS command option must be given if SIMULTANEOUS_MULTI_CORNER: YES is used.

**Description**

SELECTED_CORNERS command option must be given if SIMULTANEOUS_MULTI_CORNER: YES is used.

**What Next**

Please specify the selected corners for the option SELECTED_CORNERS.

## SX-0480

(Warning) Detected equal / larger value in ER_VS_SI_SPACING table than nominal ER value in TCAD_GRD_FILE from CORNER_TYPE:MIN

**Description**

It's found that ER value in ER_VS_SI_SPACING table is equal to or larger than nominal ER value in TCAD_GRD_FILE from CORNER_TYPE:MIN. You are recommended to use smaller ER in ER_VS_SI_SPACING table than nominal ER value in TCAD_GRD_FILE from CORNER_TYPE:MIN. Otherwise, there will be potential accuracy problem.

**What Next**

You should double check each nxtgrd for ER values in ER_VS_SI_SPACING table to see if they are correctly defined.

Feedback

## SX-0481

(Warning) Detected equal / smaller value in ER_VS_SI_SPACING table than nominal ER value in TCAD_GRD_FILE from CORNER_TYPE:MAX

**Description**

It's found that ER value in ER_VS_SI_SPACING table is equal to or smaller than nominal ER value in TCAD_GRD_FILE from CORNER_TYPE:MIN. You are recommended to use larger ER in ER_VS_SI_SPACING table than nominal ER value in TCAD_GRD_FILE from CORNER_TYPE:MIN. Otherwise, there will be potential accuracy problem.

**What Next**

You should double check each nxtgrd for ER values in ER_VS_SI_SPACING table to see if they are correctly defined.

## SX-0482

(Fatal) SELECTED_CORNERS must contain at least 1 unique NXTGRD files. There are '%d' unique NXTGRD files specified.

**Description**

SELECTED_CORNERS must contain at least 1 unique NXTGRD files.

**What Next**

Please specify at least 1 unique NXTGRD files for SELECTED_CORNERS.

## SX-0483

(Fatal) SELECTED_CORNERS must contain 60 or less process corners and DOB (density out of boundary) corners. There are in total '%d' process corners and DOB corners specified.

**Description**

SELECTED_CORNERS must contain 60 or less process corners and DOB corners.

**What Next**

Please only specify 60 or less process corners and DOB corners for the option SELECTED_CORNERS.

Feedback

## SX-0484

(Fatal) Grouping of corners with colons in SELECTED_CORNERS is not supported with NETLIST_FORMAT: OA. Please rerun by either removing the colons (:) in between corner names in the SELECTED_CORNERS statement or use one of the allowed NETLIST_FORMAT values.

**Description**

In SIMULTANEOUS_MULTI_CORNER flow, depending on the format of SELECTED_CORNERS output netlist can be grouped or separated. In grouped format, the RC parasitic is listed as Rval_corner1:...:Rval_cornern or Cval_corner1:...:Cval_cornern where n is the corner number. This grouping format is supported in all other NETLIST_FORMAT except OA.

**What Next**

Modify SELECTED_CORNERS to disable grouping corners in NETLIST_FORMAT:OA. Or change NETLIST_FORMAT to other format than OA if grouping output is required.

## SX-0485

(Fatal) Specification of empty corners in SELECTED_CORNERS is only supported with NETLIST_FORMAT: SPEF.

**Description**

An empty corner in the the SELECTED_CORNERS option can be used with NETLIST_FORMAT: SPEF.

**What Next**

Correct the SELECTED_CORNERS specification by removing the empty corner name, or use NETLIST_FORMAT: SPEF. Refer to the StarRC User Guide for more information.

## SX-0486

(Warning) If SIMULTANEOUS_MULTI_CORNER is set to YES, %s in the StarRC command file is ignored since the it is defined for each corner in the corner definition file.

**Description**

If SIMULTANEOUS_MULTI_CORNER is set to YES, a command in the StarRC command file is ignored since the it is defined for each corner in the corner definition file.

**What Next**

Please remove the option in the StarRC command file.

Feedback

## SX-0487

(Warning) If SIMULTANEOUS_MULTI_CORNER is set to YES and CORNER_INDICES_TO_OUTPUT is set in the command file, the corner names specified in SIMULTANEOUS_MULTI_CORNER will not be appended to the filenames.

**Description**

If SIMULTANEOUS_MULTI_CORNER is set to YES and CORNER_INDICES_TO_OUTPUT is set in the command file, the corner names specified in SIMULTANEOUS_MULTI_CORNER will not be appended to the filenames.

**What Next**

No action is required.

## SX-0488

(Fatal) If CORNER_INDICES_TO_OUTPUT is set or USE_SENSITIVITY_MULTI_CORNER_OUTPUT: NO is set in the command file, All SELECTED_CORNERS must have the same operating temperature.

**Description**

If CORNER_INDICES_TO_OUTPUT is set or USE_SENSITIVITY_MULTI_CORNER_OUTPUT: NO is set in the command file, All SELECTED_CORNERS must have the same operating temperature.

**What Next**

Please check the operating temperature for all SELECTED_CORNERS. They must have the same operating temperature. If all SELECTED_CORNERS have the same operating temperature but still get this error message, please contact the Synopsys Support Center.

## SX-0489

(Fatal) CORNER_INDICES_TO_OUTPUT must be specified with CORNER_TCAD_GRD_FILES.

**Description**

CORNER_INDICES_TO_OUTPUT can only be specified when CORNER_TCAD_GRD_FILES is already specified.

**What Next**

Please specify CORNER_TCAD_GRD_FILES or remove CORNER_INDICES_TO_OUTPUT.

Feedback

## SX-0490

(Fatal) USE_SENSITIVITY_MULTI_CORNER_OUTPUT must be specified with CORNER_TCAD_GRD_FILES.

**Description**

USE_SENSITIVITY_MULTI_CORNER_OUTPUT can only be specified when CORNER_TCAD_GRD_FILES is already specified.

**What Next**

Please specify CORNER_TCAD_GRD_FILES or remove USE_SENSITIVITY_MULTI_CORNER_OUTPUT.

## SX-0491

(Fatal) TCAD_GRD_FILE command options must have exactly 1 NXTGRD file specified to use simultaneous multi-corner flow.

**Description**

In order to use simultaneous multi-corner flow, there should be one and only one NXTGRD file specified for the option TCAD_GRD_FILE.

**What Next**

Please check the number of NXTGRD files specified for the option TCAD_GRD_FILE.

## SX-0492

(Fatal) No values given for CORNER_INDICES_TO_OUTPUT.

**Description**

No values given for CORNER_INDICES_TO_OUTPUT.

**What Next**

Please specify a correct value for CORNER_INDICES_TO_OUTPUT.

## SX-0493

(Fatal) Only one value of CORNER_INDICES_TO_OUTPUT can be specified if USE_SENSITIVITY_MULTI_CORNER_OUTPUT is set to NO.

Feedback

**Description**

If USE_SENSITIVITY_MULTI_CORNER_OUTPUT is set to NO, only one value can be specified for CORNER_INDICES_TO_OUTPUT. Don't specify more than one value for CORNER_INDICES_TO_OUTPUT if USE_SENSITIVITY_MULTI_CORNER_OUTPUT is set to NO.

**What Next**

Please check and only specify one value for CORNER_INDICES_TO_OUTPUT.

---

## SX-0494

(Fatal) Negative or zero value of CORNER_INDICES_TO_OUTPUT cannot be specified.

**Description**

Negative or zero value of CORNER_INDICES_TO_OUTPUT cannot be specified.

**What Next**

Please specify a value that is larger than zero for the option CORNER_INDICES_TO_OUTPUT.

---

## SX-0495

(Fatal) Specified CORNER_INDICES_TO_OUTPUT value is greater than the number of corner NXTGRD files provided in TCAD_GRD_FILE and CORNER_TCAD_GRD_FILES.

**Description**

The value of CORNER_INDICES_TO_OUTPUT is greater than the total number of corner NXTGRD files provided in TCAD_GRD_FILE and CORNER_TCAD_GRD_FILES.

**What Next**

Please input a smaller value for CORNER_INDICES_TO_OUTPUT and make sure that this value is not greater than the total number of corners provided in TCAD_GRD_FILE and CORNER_TCAD_GRD_FILES.

---

## SX-0497

(Fatal) Invalid option FAST_EXTRACTION_MODE

**Description**

FAST_EXTRACTION_MODE cannot be set as NO.

Feedback

**What Next**

Remove this option in star_cmd.

---

## SX-0498

(Fatal) Invalid option specified for FAST_EXTRACTION_MODE. Possible settings are NO, YES, and AUTO.

**Description**

The value of FAST_EXTRACTION_MODE is not correct and expected. Extraction cannot proceed without valid setting of this option. Note that the option of FAST_EXTRACTION_MODE is not intended for external setting.

**What Next**

Remove the option from setting or contact Synopsys for support if there is still error after removing the option.

---

## SX-0499

(Warning) MODE is no longer required for this type of extraction. Ignoring MODE in StarRC command file.

**Description**

StarRC will automatically select the appropriate accuracy level. MODE is no longer required in the StarRC command file.

**What Next**

Remove the MODE command from the StarRC command file.

---

## SX-0500

(Fatal) QTF file is missing.

**Description**

QTF file is required if the case is to simulate TSMC N16 and lower technology node design with Field Solver extraction.

**What Next**

use FS_QTF_FILE option to specify QTF file.

Feedback

## SX-0502

(Warning) CLEAN_XIN_IN_XTRACTOR: NO is not supported for fast extraction mode.

**Description**

CLEAN_XIN_IN_XTRACTOR: NO is not supported for fast extraction mode.

**What Next**

This message can be ignored. StarRC will set CLEAN_XIN_IN_XTRACTOR to YES.

## SX-0503

(Fatal) invalid option DP_TILE_WIDTH

**Description**

invalid option DP_TILE_WIDTH

**What Next**

DP_TILE_WIDTH is for internal use only

## SX-0505

(Fatal) invalid option DP_TILES

**Description**

invalid option DP_TILES

**What Next**

DP_TILES is for internal use only

## SX-0507

(Warning) Ignoring DP_MODE:USER setting, continue with DP_MODE:AUTO!

**Description**

This message informs you that the DP_MODE:USER setting is replaced by DP_MODE:AUTO.

**What Next**

You don't need to take any action.

Feedback

## SX-0516

(Fatal) Please fix the errors before proceeding

**Description**

The is error reported before this message. They may be: SX-0724, SX-0512, SX-0513, SX-0514, SX-0515.

**What Next**

Fix the error reported before this message.

## SX-0524

(Fatal) Cannot use %s in %s flow.

**Description**

This error message is used to prohibit the use of some options in the StarRC command file in specific flows or use-models.

**What Next**

Remove the option from the StarRC command file. Refer to the StarRC User Guide for more information.

## SX-0528

(Fatal) Invalid selection for EXTRACT_VIA_CAPS option.

**Description**

EXTRACT_VIA_CAPS must be set with "NO | YES [IGNORE_GATE_CONTACT_COUPLING]".

**What Next**

Correct EXTRACT_VIA_CAPS with correct option in StarRC command file.

## SX-0529

(Warning) CONLY_NETS are only valid for COUPLE_TO_GROUND: NO extractions.

**Description**

The CONLY_NETS command reports cross-capacitance to noncritical net neighbors. If the COUPLE_TO_GROUND: YES command is used, CONLY_NETS has no effect. If

Feedback

the COUPLE_TO_GROUND: NO command is used, the NETS command controls the behavior.

**What Next**

You can ignore this message and CONLY_NETS will have no effect. Otherwise, if you want to use CONLY_NETS, set COUPLE_TO_GROUND:YES in command file.

---

## SX-0530

(Fatal) INTERNAL Error: INTERNAL error in calling register_FS_xtract_jobs

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0531

(Fatal) CORNER_TCAD_GRD_FILES cannot be specified if TCAD_GRD_FILE is not specified.

**Description**

CORNER_TCAD_GRD_FILES cannot be specified if TCAD_GRD_FILE is not specified.

**What Next**

Please remove the option CORNER_TCAD_GRD_FILES or specify the option TCAD_GRD_FILE.

---

## SX-0532

(Fatal) CORNER_TCAD_GRD_FILES can be specified only if just one NXTGRD file is provided in TCAD_GRD_FILE.

**Description**

CORNER_TCAD_GRD_FILES can be specified only if just one NXTGRD file is provided in TCAD_GRD_FILE.

**What Next**

Please make sure that there is only one NXTGRD file provided in TCAD_GRD_FILE. Or remove the option CORNER_TCAD_GRD_FILES.

Feedback

## SX-0533

(Fatal) RES_UPDATE_FILE cannot be used with SIMULTANEOUS_MULTI_CORNER.

**Description**

RES_UPDATE_FILE cannot be used with SIMULTANEOUS_MULTI_CORNER.

**What Next**

Please remove RES_UPDATE_FILE or SIMULTANEOUS_MULTI_CORNER in StarRC command file.

## SX-0535

(Information) Density based thickness variation is calculated based on FIXED_DENSITY_VALUE setting instead of layout density.

**Description**

Density used for thickness variation is not calculated from layout but spicified by FIXED_DENSITY_VALUE command.

**What Next**

Information only. No action is needed.

## SX-0536

(Fatal) FIXED_DENSITY_VALUE option value must be between 0 and 1

**Description**

A non-physical value is specified for FIXED_DENSITY_VALUE in the StarRC command file. By definition, density must be between 0 (empty) and 1 (fully dense).

**What Next**

Set the value of FIXED_DENSITY_VALUE to be between 0 and 1 in the StarRC command file.

## SX-0537

(Fatal) DENSITY_OUTSIDE_BLOCK option value must be between 0 and 1

Feedback

**Description**

A non-physical value is specified for DENSITY_OUTSIDE_BLOCK in the StarRC command file. By definition, density must be between 0 (empty) and 1 (fully dense).

**What Next**

Set the value of DENSITY_OUTSIDE_BLOCK to be between 0 and 1 in the StarRC command file.

---

## SX-0538

(Warning) DENSITY_EMULATED_FILL_CONTRIBUTION must have 4 floats as arguments, skipping

**Description**

DENSITY_EMULATED_FILL_CONTRIBUTION must have 4 floats as arguments.

**What Next**

This is a Waring message and option will be ignored in this case, if this option needs to be turn on, need to revise the option with 4 floats.

---

## SX-0540

(Fatal) MOS_GATE_NON_NEGATIVE_DELTA_RESISTANCE: YES and MOS_GATE_DELTA_RESISTANCE: YES can not be used together! Only one of mos gate resistance modeling methods can be used!

**Description**

When MOS_GATE_DELTA_RESISTANCE: YES is used, StarRC will use delta network model for MOS gate resistance and will introduce negative resistor. When MOS_GATE_NON_NEGATIVE_DELTA_RESISTANCE: YES is used, StarRC will use a network model which doesn't include negative resistor for MOS gate resistance. The 2 commands can't be used together.

**What Next**

You should remove one of these 2 commands based on your purpose (Whether or not to have a negative resistor).

---

## SX-0541

(Fatal) %s is only supported in an ICV or a Calibre flow.

Feedback

**Description**

Reported command is only supported in an ICV or a Calibre flow.

**What Next**

Remove the reported command from StarRC command file.

---

## SX-0542

(Fatal) EXTRACT_VIA_CAPS must be set to YES in order to use %s.

**Description**

EXTRACT_VIA_CAPS must be set to YES in order
to use RETAIN_GATE_CONTACT_COUPLING or
RETAIN_GATE_CONTACT_AND_DIFFUSION_COUPLING.

**What Next**

Set EXTRACT_VIA_CAPS to YES in the StarRC command file.

---

## SX-0543

(Fatal) GROUND_CROSS_COUPLING may only be set to YES, NO, or AUTO.

**Description**

You are using unsupported value for command GROUND_CROSS_COUPLING.

**What Next**

Please use correct value (YES, NO, AUTO) in StarRC command file.

---

## SX-0544

(Warning) MULTIGATE_MODELS: YES is ignored, no SCT multigate models into nxtgrd file.

**Description**

The message informs that command MULTIGATE_MODELS: YES will be ignored since the nxtgrd file does not contain 3D models.

**What Next**

If you know that 3D models are not present in the nxtgrd file, remove the command from the StarRC command file. Otherwise, make sure that the nxtgrd file you are using is correct.

Feedback

## SX-0545

(Warning) MULTIGATE_MODELS: NO is set, but SCT multigate models detected into nxtgrd file.

### Description

The warning informs that 3D models present in the nxtgrd file will be ignored since MULTIGATE_MODELS: NO is set in the StarRC command file.

### What Next

If this setting is intentional, no action is needed from your side. Otherwise, review the StarRC command file and consider using MULTIGATE_MODELS: YES to enable 3D models.

## SX-0546

(Warning) DEVICE_MODEL_LEVEL: 2 is ignored, no SCT planar models into nxtgrd file.

### Description

The warning informs the user that command DEVICE_MODEL_LEVEL: 2 will be ignored since no 3D models have been detected in the nxtgrd file.

### What Next

If you know that 3D models are not present in the nxtgrd file, remove the command from the StarRC command file. Otherwise, make sure that the nxtgrd file you are using is correct.

## SX-0547

(Warning) DEVICE_MODEL_LEVEL: 1 is set, but SCT planar models detected into nxtgrd file.

### Description

The warning informs that 3D models present in the nxtgrd file will be ignored since DEVICE_MODEL_LEVEL: 1 is set in the StarRC command file.

### What Next

If this setting is intentional, no action is needed from your side. Otherwise, review the StarRC command file and consider using DEVICE_MODEL_LEVEL: 2 to enable 3D models.

Feedback

## SX-0548

(Warning) MODE is not set: setting to 400 by default for DEVICE_MODEL_LEVEL/ MULTIGATE_MODELS.

**Description**

The warning indicates that the StarRC command file does not contain the command MODE to control accuracy. StarXtract will force MODE: 400 in presence of DEVICE_MODEL_LEVEL or MULTIGATE_MODELS commands.

**What Next**

If this setting is suitable for you, no action is needed from your part. Otherwise, review the StarRC command file and consider adding command MODE.

## SX-0549

(Fatal) SENS_REDU_ERROR is only applicable with SENSITIVITY:yes or TEMPERATURE_SENSITIVITY:yes.

**Description**

SENS_REDU_ERROR is to control sensitivity accuracy during reduction. It is on applicable with SENSITIVITY:yes or TEMPERATURE_SENSITIVITY:yes.

**What Next**

Remove SENS_REDU_ERROR if you are not using SENSITIVITY:yes or TEMPERATURE_SENSITIVITY:yes. Or set SENSITIVITY:yes or TEMPERATURE_SENSITIVITY:yes to use SENS_REDU_ERROR.

## SX-0550

(INTERNAL Warning) VIAPAD_MIN_AREA will be ignored since it is not compatible for EXTRACTION: C or FSCOMPARE

**Description**

Internal StarRC command VIAPAD_MIN_AREA will be ignored for EXTRACTION:C or EXTRACTION:FSCOMPARE.

**What Next**

Nothing needs to be done.

Feedback

## SX-0551

(Warning) VIAPAD_MIN_LENGTH will be ignored since it is not compatible for EXTRACTION: C or FSCOMPARE

**Description**

For setting EXTRACTION: C or FSCOMPARE, VIAPAD_MIN_LENGTH will be ignored.

**What Next**

No further action is required.

## SX-0552

(Fatal) MESH_RESISTANCE_EXTRACTION_FOR_DIFFUSION supported in transitor level flow only.

**Description**

MESH_RESISTANCE_EXTRACTION_FOR_DIFFUSION is only supported for transistor level flow extraction.

**What Next**

Remove command MESH_RESISTANCE_EXTRACTION_FOR_DIFFUSION and restart.

## SX-0553

(Warning) DETECT_FUSE can be set only for REDUCTION:NO

**Description**

DETECT_FUSE is only supprted for REDUCTION:NO. DETECT_FUSE is ignored.

**What Next**

If you want to use DETECT_FUSE, please set EXTRACTION:NO. Or you can ignore this message.

## SX-0554

(Warning) DETECT_FUSE can be set only for POWER_REDUCTION:NO

**Description**

DETECT_FUSE is only suported for POWER_EXTRACTION:NO. DETECT_FUSE is ignored.

Feedback

**What Next**

If you want to use DETECT_FUSE, please set POWER_EXTRACTION:NO. Or you can ignore this message.

---

## SX-0555

(Fatal) PRINT_RV_FUSE must be used with NETLIST_TAIL_COMMENTS and DETECT_FUSE.

**Description**

PRINT_RV_FUSE is only supported with NETLIST_TAIL_COMMENTS:YES and DETECT_FUSE:YES.

**What Next**

Please set NETLIST_TAIL_COMMENTS:YES and DETECT_FUSE:YES if you want to use PRINT_RV_FUSE:YES

---

## SX-0558

(Warning) DIFFUSION_RES_GATE_TO_CONT_THRESHOLD is used in StarRC command file. It will override nxtgrd DIFFUSION_RES_GATE_TO_CONT_THRESHOLD.

**Description**

DIFFUSION_RES_GATE_TO_CONT_THRESHOLD can be specified in both StarRC command file and ITF file. The command specified in StarRC command file take precedence.

**What Next**

This is only a warning message. No action is required.

---

## SX-0559

(Warning) DIFFUSION_RES_BEND_THRESHOLD is used in StarRC command file. It will override nxtgrd DIFFUSION_RES_BEND_THRESHOLD command.

**Description**

DIFFUSION_RES_BEND_THRESHOLD can be specified in both StarRC command file and ITF file. The command specified in StarRC command file take precedence.

**What Next**

This is only a warning message. No action is required.

Feedback

## SX-0560

(Warning) USER_DEFINED_DIFFUSION_RES needs NETLIST_POSTPROCESS_COMMAND to plug-in user defined equation.

**Description**

StarRC writes information into the tail comments of diffusion resistors in the netlist. The extra information can then be used to calculate resistance according to a script specified by the NETLIST_POSTPROCESS_COMMAND in the StarRC command file.

**What Next**

Specify NETLIST_POSTPROCESS_COMMAND in the StarRC command file or use a script to postprocess the netlist to calculate the resistance.

## SX-0564

(Fatal) EXTRA_GEOMETRY_INFO:RES must be used with REDUCTION:NO/ POWER_REDUCTION:NO.

**Description**

EXTRA_GEOMETRY_INFO:RES enables StarRC to print bounding box for metal and via resistors. Reduction may cause incorrect geometry information.

**What Next**

Set both REDUCTION and POWER_REDUCTION to NO.

## SX-0565

(Fatal) EXTRA_GEOMETRY_INFO:RES must be used with NETLIST_TAIL_COMMENTS.

**Description**

EXTRA_GEOMETRY_INFO:RES enables StarRC to print bounding box for metal and via resistors in the tail comment. It must be used with NETLIST_TAIL_COMMENTS: YES| ADVANCED.

**What Next**

Set NETLIST_TAIL_COMMENTS to YES or ADVANCED.

## SX-0566

(Fatal) NETLIST_PORT_LEVEL_COMMENTS must be used with EXTRA_GEOMETRY_INFO: NODE.

Feedback

**Description**

In order to print level information required by NETLIST_PORT_LEVEL_COMMENTS, EXTRA_GEOMETRY_INFO: NODE must be enabled.

**What Next**

Enable EXTRA_GEOMETRY_INFO: NODE and restart.

---

## SX-0567

(Fatal) DP_TILES_STITCHING cannot be less than 0 or greater than 3.

**Description**

DP_TILES_STITCHING is an internal command.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0568

(Fatal) INTERNAL error: invalid option IAPI

**Description**

You are using invalid optin IAPI.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0569

(Fatal) IAPI:YES cannot be combined with EXTRACTION:FSCOMPARE

**Description**

IAPI:YES cannot be combined with EXTRACTION:FSCOMPARE

**What Next**

Please use other EXTRACTION settings in StarRC command file

---

## SX-0570

(Warning) Improper value %f for StarRC command IAPI_CRICIAL_AREA_WIDTH.

Feedback

**Description**

IAPI_CRICIAL_AREA_WIDTH value cannot be zero, negative or larger than 100um. A default value of 4um will be used instead.

**What Next**

Set proper value for the StarRC command.

---

## SX-0572

(Warning) Improper value %f for StarRC command IAPI_OVERLAP_BAND_WIDTH.

**Description**

IAPI_OVERLAP_BAND_WIDTH value cannot be zero, negative or larger than 100um. A default value of 4um will be used instead.

**What Next**

Set proper value for the StarRC command.

---

## SX-0574

(Warning) MODEL_BASED_SHORT_HANDLING does not work with EXTRACTION: '%s'. Disable MODEL_BASED_SHORT_HANDLING.

**Description**

MODEL_BASED_SHORT_HANDLING does not work with EXTRACTION: R / C / RK / RKC / FSCOMPARE

**What Next**

If this option needs to be turn on, need revise the EXTRACTION option to solve the conflict.

---

## SX-0575

(Warning) REFERENCE_DIRECTION: VERTICAL setting in star_cmd will be used and REFERENCE_DIRECTION=HORIZONTAL in ITF will be ignored.

**Description**

This message informs you that you are setting REFERENCE_DIRECTION: VERTICAL in StarRC command file but also setting REFERENCE_DIRECTION=HORIZONTAL in ITF. The setting in StarRC command file takes precedence so the setting in ITF will be ignored.

Feedback

### What Next

If you want to set REFERENCE_DIRECTION as VERTICAL, you don't need to take any action; if you want to set REFERENCE_DIRECTION as HORIZONTAL, you should also set REFERENCE_DIRECTION: HORIZONTAL in StarRC command file.

---

## SX-0576

(Warning) REFERENCE_DIRECTION: HORIZONTAL setting in StarRC command file will be used and REFERENCE_DIRECTION=VERTICAL in ITF will be ignored.

### Description

This message informs you that you are setting REFERENCE_DIRECTION: HORIZONTAL in StarRC command file but also setting REFERENCE_DIRECTION=VERTICAL in ITF. The setting in StarRC command file takes precedence so the setting in ITF will be ignored.

### What Next

If you want to set REFERENCE_DIRECTION as HORIZONTAL, you don't need to take any action; if you want to set REFERENCE_DIRECTION as VERTICAL, you should also set REFERENCE_DIRECTION: VERTICAL in StarRC command file.

---

## SX-0577

(Fatal) REFERENCE_DIRECTION type must be VERTICAL or HORIZONTAL or NONE or GATE

### Description

REFERENCE_DIRECTION is option to configurate the reference direction during process variation. The valid options of this command include VERTICAL, HORIZONTAL, NONE, GATE. The value of option in this run is not equivalent to any of valid values.

### What Next

Refer to StarRC manual to correct the setting of REFERENCE_DIRECTION.

---

## SX-0578

(Warning) LARGE_EXTRACTION_WINDOW: YES will be ignored because it is not defined in the nxtgrd file.

### Description

LARGE_EXTRACTION_WINDOW:YES is only applicable when the nxtgrd file defines it as well.

Feedback

**What Next**

This is a warning message for information only. The extraction will proceed.

---

## SX-0579

(Fatal) Unknown ANALOG_SYMMETRIC_NETS command choice! '%s'

**Description**

An invalid setting was provided for the ANALOG_SYMMETRIC_NETS option in the StarRC command file.

**What Next**

Provide a valid setting for the ANALOG_SYMMETRIC_NETS option in the StarRC command file.

---

## SX-0580

(Fatal) Unknown PARALLEL_TRANSISTOR choice: '%s'

**Description**

Specified setting for PARALLEL_TRANSISTOR is not valid.

**What Next**

Use proper setting for PARALLEL_TRANSISTOR in your StarRC command file. Valid settings are: YES, ETCH, AUTO, NONE

---

## SX-0581

(Fatal) Unknown choice '%s' for command NANOMETER_DESIGN

**Description**

You are using unsuported settings for command NANOMETER_DESIGN

**What Next**

Please use corret setting (YES, NONE, AUTO)

---

## SX-0583

(Fatal) Unknown REMOVE_OVERLAPS choice: '%s'

Feedback

**Description**

The specified setting for REMOVE_OVERLAPS is not valid.

**What Next**

Use valid setting for REMOVE_OVERLAPS in StarRC command file. valid settings are: YES, AUTO, NONE.

---

## SX-0584

(Fatal) Unknown DEVICELAYER_MODE choice: '%s'

**Description**

Unknown DEVICELAYER_MODE choice: '%s'

**What Next**

Please use correct setting for DEVICELAYER_MODE, YES, NORMAL, HIGH, NONE.

---

## SX-0588

(Fatal) Unknown ENHANCED_PROCESS_VARIATION_HANDLING parameter specified: '%s'

**Description**

Unknown parameter is specified for ENHANCED_PROCESS_VARIATION_HANDLING. Expected values are: DEFAULT, NO, LEVEL1, LEVEL2, LEVEL3, LEVEL4

**What Next**

Change your StarRC command file to provide valid parameter.

---

## SX-0596

(Warning) NONCRITICAL_POLYGON_SCAN: YES does not have any effect unless fast extraction mode is enabled.

**Description**

The NONCRITICAL_POLYGON_SCAN option was added in conjunction with FAST_EXTRACTION_MODE capability. It does not affect FAST_EXTRACTION_MODE:NO functionality.

**What Next**

NONCRITICAL_POLYGON_SCAN will be ignored and you don't need to take any action.

Feedback

## SX-0601

(INTERNAL Error) ADAPTIVE_SEARCH_DISTANCE_LEVEL must be >= 0

**Description**

StarRC internal command ADAPTIVE_SEARCH_DISTANCE_LEVEL must be >=0.

**What Next**

Contact the Synopsys Support Center.

## SX-0605

(Warning) KEEP_SKIPCELL_PIN_CG:YES has no effect!
KEEP_SKIPCELL_PIN_CG:YES only has effect when
COUPLE_TO_SKIPCELL_PINS:YES is set!

**Description**

KEEP_SKIPCELL_PIN_CG:YES can only be used when
COUPLE_TO_SKIPCELL_PINS:YES is set.

**What Next**

You should use both COUPLE_TO_SKIPCELL_PINS:YES and
KEEP_SKIPCELL_PIN_CG:YES in StarRC command file.

## SX-0607

(Fatal) invalid RV_VIA_EXTRACT_PROPERTY: '%s'

**Description**

The value of RV_VIA_EXTRACT_PROPERTY can only be SAVL, SAVC or both.

**What Next**

Set RV_VIA_EXTRACT_PROPERTY to SAVL, SAVC or both.

## SX-0611

(Warning) NETLIST_PRINT_CC_TWICE: YES is not supported for NETLIST_FORMAT:
SPF. It will be ignored.

Feedback

**Description**

NETLIST_PRINT_CC_TWICE: YES is only supported in NETLIST_FORMAT:STAR and NETNAME

**What Next**

You can ingore this message if you want to use NETLIST_FORMAT: SPF. Otherwise, change NETLIST_FORMAT to STAR or NETNAME

---

## SX-0612

(Warning) NETLIST_INTRANET_CAPS will obsolete, use INTRANET_CAPS: YES/NO instead.

**Description**

NETLIST_INTRANET_CAPS will obsolete, use INTRANET_CAPS: YES/NO instead.

**What Next**

Please use INTRANET_CAPS in your StarRC command file.

---

## SX-0614

(Warning) NETLIST_INSTANCE_SECTION only supports NO/YES/SELECTED, '%s' is ignored.

**Description**

Option other than NO|YES|SELECTED is supplied to NETLIST_INSTANCE_SECTION. The option is ignored. No instance section will be printed in output netlist.

**What Next**

Modify NETLIST_INSTANCE_SECTION with correct option in StarRC command file to remove this warning message.

---

## SX-0615

(Fatal) COUPLING_REPORT_FILE has the same name as the netlist file.

**Description**

The value for the COUPLING_REPORT_FILE option in the StarRC command file is identical to the value of the NETLIST_FILE option.

Feedback

**What Next**

Modify the StarRC command file to use different values for the COUPLING_REPORT_FILE and NETLIST_FILE options.

---

## SX-0616

(Fatal) COUPLING_REPORT_NUMBER must be greater than zero.

**Description**

This error message informs you that COUPLING_REPORT_NUMBER is set to a number less than zero.

**What Next**

Reset COUPLING_REPORT_NUMBER to an integer greater than zero.

---

## SX-0618

(Fatal) NETLIST_MINCAP_THRESHOLD is supported in transitor level flow only.

**Description**

NETLIST_MINCAP_THRESHOLD is only supported for transistor level flow.

**What Next**

Remove NETLIST_MINCAP_THRESHOLD and restart.

---

## SX-0619

(Fatal) NETLIST_MINCAP_THRESHOLD must be greater than zero.

**Description**

NETLIST_MINCAP_THRESHOLD must be greater than zero.

**What Next**

Correct the command setting and restart.

---

## SX-0620

(Fatal) NETLIST_TOTALCAP_THRESHOLD supported in transitor level flow only.

**Description**

NETLIST_TOTALCAP_THRESHOLD is only supported in transistor level flow extraction.

Feedback

**What Next**

Remove the command and restart.

---

## SX-0621

(Fatal) NETLIST_TOTALCAP_THRESHOLD must be greater than zero.

**Description**

NETLIST_TOTALCAP_THRESHOLD must be greater than zero. Otherwise the tool aborts.

**What Next**

Correct the command setting and restart.

---

## SX-0622

(Warning) INTERNAL Warning: NETLIST_MAX_MEMORY option is no longer supported.

**Description**

NETLIST_MAX_MEMORY option is no longer supported.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0623

(Warning) NETLIST_COUPLE_UNSELECTED_NETS is only valid with NETLIST_SELECT_NETS.

**Description**

NETLIST_COUPLE_UNSELECTED_NETS only works when NETLIST_SELECT_NETS is set.

**What Next**

This is a warning message, NETLIST_COUPLE_UNSELECTED_NETS will be ignored and the run will continue.

---

## SX-0624

(Warning) NETLIST_COUPLE_UNSELECTED_NETS is set to an obsolete option 'YES'. StarRC will use IDEAL instead.

Feedback

**Description**

The supported options for NETLIST_COUPLE_UNSELECTED_NETS are NO, IDEAL and COMPLETE. YES is obsolete and StarRC automatically replaces it with IDEAL.

**What Next**

This is a warning message for information only.

---

## SX-0625

(Fatal) '%s' is not a known option for NETLIST_COUPLE_UNSELECTED_NETS; please specify NO, IDEAL, COMPLETE or ADVANCED.

**Description**

An invaid option is detected in NETLIST_COUPLE_UNSELECTED_NETS command, the valid options are NO, IDEAL, COMPLETE and ADVANCED.

**What Next**

Correct the command NETLIST_COUPLE_UNSELECTED_NETS and restart.

---

## SX-0626

(Fatal) NETLIST_MERGE_SHORTED_PORTS is only supported for NETLIST_FORMAT: STAR|NETNAME|SPF|OA.

**Description**

NETLIST_MERGE_SHORTED_PORTS is dependent on NETLIST_FORMAT and is only supported for STAR/NETNAME/SPF/OA format.

**What Next**

Change the command setting and restart.

---

## SX-0627

(Fatal) NETLIST_PIN_SUBNODE_RES_VALUE can not be negative.

**Description**

NETLIST_PIN_SUBNODE_RES_VALUE is set as a negative value, which is invalid.

**What Next**

Set NETLIST_PIN_SUBNODE_RES_VALUE to a zero or positive value.

Feedback

## SX-0628

(Fatal) RV_PRINT_RESISTOR_TYPE must be used with NETLIST_TAIL_COMMENTS.

**Description**

NETLIST_TAIL_COMMENTS needs to be YES or ADVANCED to print the resistor type as a comment in the netlist.

**What Next**

Set NETLIST_TAIL_COMMNETS to YES or ADVANCED in StarRC command file.

## SX-0629

(Fatal) RV_PRINT_RESISTOR_TYPE must be used with EXTRA_GEOMETRY_INFO:RES.

**Description**

In the StarRC command file, RV_PRINT_RESISTOR_TYPE is used, but EXTRA_GEOMETRY_INFO is not set to RES.

**What Next**

Set EXTRA_GEOMETRY_INFO: RES in the StarRC command file.

## SX-0631

(Fatal) EXTRA_GEOMETRY_INFO must be used with NETLIST_TAIL_COMMENTS.

**Description**

When EXTRA_GEOMETRY_INFO: RES is specified, current direction is printed as comment in the output netlist. Following options must be enabled: NETLIST_TAIL_COMMENTS:YES|ADVANCED, REDUCTION: NO, KEEP_VIA_NODES:YES

**What Next**

Set NETLIST_TAIL_COMMENTS to YES or ADVANCED.

## SX-0632

(Fatal) PRINT_THICKNESS_INFO must be used with NETLIST_TAIL_COMMENTS.

Feedback

**Description**

When PRINT_THICKNESS_INFO: YES is specified, the top thickness variation and bottom thickness variation are printed in tail comment of resistors in the netlist. It must be used with NETLIST_TAIL_COMMENTS: YES|ADVANCED.

**What Next**

Set NETLIST_TAIL_COMMENTS to YES or ADVANCED.

---

## SX-0633

(Fatal) PRINT_SILICON_INFO must be used with NETLIST_TAIL_COMMENTS.

**Description**

NETLIST_TAIL_COMMENTS: YES|ADVANCED is required for PRINT_SILICON_INFO.

**What Next**

Set NETLIST_TAIL_COMMENTS: YES|ADVANCED.

---

## SX-0634

(Warning) ADD_ZERO_GROUND_CAP is only available with SPEF output format.

**Description**

ADD_ZERO_GROUND_CAP is only available with SPEF output format. It is ignored.

**What Next**

Use other NETLIST_FORMAT if you want to use ADD_ZERO_GROUND_CAP. Otherwise, you can ignorethis message.

---

## SX-0635

(Fatal) Options NETLIST_CORNER_FILE and NETLIST_CORNER_NAMES are valid only in Statistical netlisting.

**Description**

Options NETLIST_CORNER_FILE and NETLIST_CORNER_NAMES are valid only in Statistical netlisting.

Feedback

**What Next**

Please remove options NETLIST_CORNER_FILE and NETLIST_CORNER_NAMES if none of options SENSITIVITY, TEMPERATURE_SENSITIVITY, USE_SENSITIVITY_MULTI_CORNER_OUTPUT is set.

---

## SX-0636

(Warning) Option NETLIST_CORNER_FILE is ignored because option NETLIST_CORNER_NAMES isn't set.

**Description**

Option NETLIST_CORNER_FILE is ignored because option NETLIST_CORNER_NAMES isn't set.

**What Next**

You can ignore this message if you do not use multi-corner extraction. Or set NETLIST_CORNER_NAMES in StarRC command file.

---

## SX-0637

(Fatal) NETLIST_REMOVE_DANGLING_BRANCHES is not supported with NETLIST_FORMAT:SPEF.

**Description**

NETLIST_REMOVE_DANGLING_BRANCHES does not work when NETLIST_FORMAT is set to SPEF.

**What Next**

Change the command setup and restart.

---

## SX-0638

(Fatal) NETLIST_REMOVE_DANGLING_BRANCHES is supported in transitor level flow only.

**Description**

NETLIST_REMOVE_DANGLING_BRANCHES is an option only valid for transistor level flow.

**What Next**

Remove the command NETLIST_REMOVE_DANGLING_BRANCHES and restart.

Feedback

## SX-0639

(Fatal) NETLIST_REMOVE_MINRES_THRESHOLD supported in transitor level flow only.

**Description**

NETLIST_REMOVE_MINRES_THRESHOLD supported in transitor level flow only.

**What Next**

Please remove NETLIST_REMOVE_MINRES_THRESHOLD in StarRC command file since you are running gate level fow.

## SX-0640

(Fatal) Option NETLIST_MINRES_THRESHOLD value should be positive or zero

**Description**

Option NETLIST_MINRES_THRESHOLD value should be positive or zero

**What Next**

Please use positive value or zero for NETLIST_MINRES_THRESHOLD in StarRC command file

## SX-0641

(Warning) NETLIST_MINRES_HANDLING: MERGE (also default) and NETLIST_MINRES_THRESHOLD > 0 combination is not allowed with NETLIST_TAIL_COMMENTS: YES. Merging is turned off

**Description**

When NETLIST_MINRES_HANDLING: MERGE, StarRC merges the resistor with a neighboring resistor if it is in series and is smaller than the threshold. It need NETLIST_TAIL_COMMENTS to be NO.

**What Next**

Set NETLIST_TAIL_COMMENTS to NO.

## SX-0642

(Fatal) File specified in option MINRES_THRESHOLD_GRD_FILE doesn't match any file in TCAD_GRD_FILE.

Feedback

**Description**

MINRES_THRESHOLD_GRD_FILE does not match any file in TCAD_GRD_FILE.

**What Next**

Change MINRES_THRESHOLD_GRD_FILE in StarRC command file to use one of the files in TCAD_GRD_FILE.

---

## SX-0643

(Warning) TCAD_GRD_FILE will be used in applying NETLIST_MINRES_THRESHOLD.

**Description**

Since MINRES_THRESHOLD_GRD_FILE is not set, StarRC will use TCAD_GRD_FILE in applying NETLIST_MINRES_THRESHOLD.

**What Next**

If you want to use a different nxtgrd file for NETLIST_MINRES_THRESHOLD, set MINRES_THRESHOLD_GRD_FILE in StarRC command file.

---

## SX-0644

(Fatal) NETLIST_PARASITIC_RESISTOR_MODEL is supported for SPF, STAR, NETNAME netlist formats only.

**Description**

NETLIST_PARASITIC_RESISTOR_MODEL is only supported for NETLIST_FORMAT: SPF|STAR|NETNAME.

**What Next**

Correct the command setting and restart.

---

## SX-0645

(Fatal) NETLIST_MMC_FORMULA is supported for SPF, STAR, NETNAME netlist formats only.

**Description**

NETLIST_MMC_FORMULA is supported for SPF, STAR, NETNAME netlist formats only.

**What Next**

Disable NETLIST_MMC_FORMULA or change NETLIST_FORMAT to a supported format.

Feedback

## SX-0646

(Fatal) NETLIST_MMC_FORMULA is supported with MERGE_MULTI_CORNER:YES only.

**Description**

NETLIST_MMC_FORMULA is only supported when MERGE_MULTI_CORNER: YES is set.

**What Next**

Change the command setup and restart.

## SX-0647

(Fatal) NETLIST_SMC_FORMULA is supported for SPF, STAR, NETNAME, OA netlist formats only.

**Description**

NETLIST_SMC_FORMULA: YES can only be used when the output netlist is SPF, STAR, NETNAME, or OA. If you use other format in NETLIST_FORMAT command, StarRC will give this error.

**What Next**

You should use supported netlist format in StarRC command NETLIST_FORMAT. Otherwise, you should remove command NETLIST_SMC_FORMULA: YES.

## SX-0648

(Fatal) NETLIST_SMC_FORMULA is supported with SIMULTANEOUS_MULTI_CORNER:YES only.

**Description**

In SIMULTANEOUS_MULTI_CORNER flow, a formula format output can be achieved by setting NETLIST_SMC_FORMULA:YES. The formula format can be achieved only in SIMULTANEOUS_MULTI_CORNER flow. The run has NETLIST_SMC_FORMULA but not in SIMULTANEOUS_MULTI_CORNER flow, so it errors.

**What Next**

Remove NETLIST_SMC_FORMULA from StarRC command file

Feedback

## SX-0650

(Fatal) SELECTED CORNERS HAD BEEN CHANGED WHEN USING cleanN

**Description**

SELECTED CORNERS ARE MISMATHCHED WITH PREVIOUS RUNNING

**What Next**

THE SELECTED CORNERS SHOULD BE THE SAME AS THE RUNNING TO GENERATING SPF

## SX-0651

(Warning) BUS_BIT and OA_BUS_BIT should be set first if OA_INSTANCE_BIT is used! command OA_INSTANCE_BIT will be ignored!

**Description**

OA_INSTANCE_BIT is used to change bit delimiters for instances in OA view. For example, if the original database use () as bit delimiters for instances but you want to use <> in the output OA view, you can set: BUS_BIT: () OA_BUS_BIT: <> OA_INSTANCE_BIT: [] OA_INSTANCE_BIT should be always used together with BUS_BIT and OA_BUS_BIT command. If BUS_BIT or OA_BUS_BIT is missing but OA_INSTANCE_BIT is used, StarRC will give this warning and the bit delimiters for instances in OA view will not be changed.

**What Next**

You should add BUS_BIT and OA_BUS_BIT command to specify the bit delimiters for instances used in original database to StarRC command file.

## SX-0653

(Warning) NETLIST_DEVICE_LOCATION_ORIENTATION only supports NO/YES/COMMENT, '%s' is ignored.

**Description**

NETLIST_DEVICE_LOCATION_ORIENTATION only supports NO/YES/COMMENT options. The option specified is invalid and the default "NO" option will be used.

**What Next**

Remove NETLIST_DEVICE_LOCATION_ORIENTATION, or set it to either YES or COMMENT options.

Feedback

## SX-0655

(Warning) NETLIST_FORMAT: OA does not support HIERARCHICAL_SEPARATOR: / or HIERARCHICAL_SEPARATOR: ., changing it to |.

**Description**

HIERARCHICAL_SEPARATOR: '/' cannot be used with NETLIST_FORMAT: OA, StarRC will change it to '|' internally.

**What Next**

Nothing needs to be done.

## SX-0656

(Warning) BUS_BIT should be set first if OA_BUS_BIT is used! command OA_BUS_BIT will be ignored!

**Description**

OA_BUS_BIT is used to change but bit delimiters in OA view. For example, if the original database use () as bus bit delimiters but you want to use <> in the output OA view, you can set: BUS_BIT: () BUS_BIT: <> OA_BUS_BIT should be always used together with BUS_BIT command which specify the bus bit delimiters used in original database. If BUS_BIT is missing but OA_BUS_BIT is used, StarRC will give this warning and the bus bit delimiters in OA view will not be changed.

**What Next**

You should add BUS_BIT command to specify the but bit delimiters used in original database to StarRC command file.

## SX-0657

(Fatal) OA_LIB_NAME is mandatory in OA flow.

**Description**

OA_LIB_NAME must be set for NETLIST_FORMAT: OA

**What Next**

Set a correct OA_LIB_NAME command and restart.

## SX-0658

(Fatal) OA_DEVICE_MAPPING_FILE is mandatory in OA flow.

Feedback

**Description**

OA_DEVICE_MAPPING_FILE must be set for NETLIST_FORMAT: OA.

**What Next**

Correct the command setting and restart.

---

## SX-0659

(Warning) OA_LAYER_MAPPING_FILE was not specified. The parasitic view will not contain physical data.

**Description**

When using NETLIST_FORMAT: OA, in order to include physical data in the parasitic view, OA_LAYER_MAPPING_FILE must be specified.

**What Next**

This is a warning message for information only. If physical data is required, add OA_LAYER_MAPPING_FILE and restart.

---

## SX-0660

(Warning) OA_SKIPCELL_MAPPING_FILE will be ignored because SKIP_CELLS is not specified.

**Description**

OA_SKIPCELL_MAPPING_FILE only takes effect when SKIP_CELLS is specified.

**What Next**

This is a warning message for information only. The extraction will proceed.

---

## SX-0661

(Warning) Schematic property annotation is not supported for XREF:NO; OA_PROPERTY_ANNOTATION_VIEW will be ignored.

**Description**

Schematic property annotation is only applicable when you use XREF feature.

**What Next**

This is a warning message for information only. The extraction will proceed.

Feedback

## SX-0662

(Fatal) OA_PROPMAP_CASE_SENSITIVE may only be set to YES, NO, or MIXED.

**Description**

OA_PROPMAP_CASE_SENSITIVE may only be set to YES, NO, or MIXED.

**What Next**

Set OA_PROPMAP_CASE_SENSITIVE with correct setting.

## SX-0663

(Warning) COUPLE_NONCRITICAL_NETS_SUBNODE_SUFFIX will be ignored unless you specify NETLIST_FORMAT:SPEF.

**Description**

COUPLE_NONCRITICAL_NETS_SUBNODE_SUFFIX is only supported for NETLIST_FORMAT: SPEF.

**What Next**

This is a warning message for information only.

## SX-0664

(Fatal) NONCRITICAL_COUPLING_REPORT_FILE has the same name as %s.

**Description**

Each output file should have a unique name.

**What Next**

Change the file name and restart.

## SX-0665

(Fatal) NETLIST_FORMAT: OA is mandatory in OA_MULTI_OUTPUT flow.

**Description**

Because OA_MULTI_OUTPUT flow will run OA flow, you need to make sure NETLIST_FORMAT: OA is set and all OA libraries and setting files are ready.

Feedback

**What Next**

Set NETLIST_FORMAT: OA in StarRC command file.

---

## SX-0666

(Warning) NONCRITICAL_COUPLING_REPORT_FILE will not be generated unless you specify NETLIST_FORMAT:SPEF.

**Description**

NONCRITICAL_COUPLING_REPORT_FILE is only supported with NETLIST_FORMAT: SPEF. The file will not be generated in other netlist format.

**What Next**

This is a warning message for information only. But if NONCRITICAL_COUPLING_REPORT_FILE is a required output, set NETLIST_FORMAT to SPEF and rerun.

---

## SX-0667

(Fatal) Both LPE_PARAM and LPE_DEVICES must be defined.

**Description**

LPE_PARAM and LPE_DEVICES must be both set. StarRC aborts if only one of them is set.

**What Next**

Correct the command setting and restart.

---

## SX-0668

(Fatal) LPE_FLAGS_SETTING can only be set to either NETLIST or EXTRACTION.

**Description**

LPE_FLAGS_SETTING can only be set to NETLIST or EXTRACTION. Other settings are invalid and StarRC will abort.

**What Next**

Correct the command setting and restart.

Feedback

## SX-0669

(Fatal) RV_VIA_ARRAY_GDS_FILE need to be set when RV_VIA_ARRAY_REPORT_FILE mode is activated.

**Description**

RV_VIA_ARRAY_REPORT_FILE need to be used together with RV_VIA_ARRAY_GDS_FILE.

**What Next**

Please set RV_VIA_ARRAY_GDS_FILE in StarRC command file.

## SX-0670

(Fatal) RV_VIA_ARRAY_GDS_LAYER_MAP_FILE need to be set when RV_VIA_ARRAY_REPORT_FILE mode is activated.

**Description**

RV_VIA_ARRAY_REPORT_FILE need to be used together with RV_VIA_ARRAY_GDS_LAYER_MAP_FILE

**What Next**

Please set RV_VIA_ARRAY_GDS_LAYER_MAP_FILE in StarRC command file.

## SX-0671

(Fatal) NETLIST_ISOLATED_PORTS: YES only works with MULTIPLE option in INSTANCE_PORT setting.

**Description**

NETLIST_ISOLATED_PORTS: YES only works for ports with "MULTIPLE" option enabled in INSTANCE_PORT command.

**What Next**

Add MULTIPLE option in INSTANCE_PORT settings, or disable NETLIST_ISOLATED_PORTS command.

## SX-0672

(Warning) NETLIST_ISOLATED_PORTS: YES will override command NETLIST_REMOVE_ISOLATED_PORTS: '%s'

Feedback

**Description**

When NETLIST_ISOLATED_PORTS: YES conflicts with NETLIST_REMOVE_ISOLATED_PORTS setup, NETLIST_ISOLATED_PORTS will take the precedence. The only compatible setting of NETLIST_REMOVE_ISOLATED_PORTS is NONE (or not set at all) when NETLIST_ISOLATED_PORTS: YES is set.

**What Next**

This is a warning message for information only.

---

## SX-0673

(Warning) Invalid setting for NETLIST_REMOVE_ISOLATED_PORTS: '%s'

**Description**

NETLIST_REMOVE_ISOLATED_PORTS only support the following three possible input <{AUTO}|YES|NO>.

**What Next**

Revise the star_cmd to one of above three allowed options.

---

## SX-0674

(Warning) Instance section port names may not match net port names because of INSTANCE_PORT

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0675

(Fatal) STARDIR_REDUCE_SIZE:NO cannot be used in -cleanD mode, please switch to -clean mode and restart.

**Description**

STARDIR_REDUCE_SIZE:NO cannot be used in -cleanD mode because cleanD is intended to remove as many files as possible.

Feedback

**What Next**

Change the setting and restart.

---

## SX-0679

(Fatal) NETLIST_PARASITIC_RESISTOR_MODEL is allowed with REDUCTION: LAYER_NO_EXTRA_LOOPS, TOPOLOGICAL, NO or LAYER only.

**Description**

NETLIST_PARASITIC_RESISTOR_MODEL is allowed with REDUCTION: LAYER_NO_EXTRA_LOOPS, TOPOLOGICAL, NO or LAYER only.

**What Next**

Set REDUCTION command to the supported options for NETLIST_PARASITIC_RESISTOR_MODEL.

---

## SX-0680

(Fatal) invalid extra_geometry_info: %s

**Description**

EXTRA_GEOMETRY_INFO is not set correctly.

**What Next**

Refer to user mannual and set EXTRA_GEOMETRY_INFO correctly.

---

## SX-0681

(Warning) ENHANCED_CAP_DISTRIBUTION is ignored as it is only valid under 28nm process.

**Description**

The process node in your run is equal to or larger than 28nm, but ENHANCED_CAP_DISTRIBUTION is ignored as it is only valid under 28nm process.

**What Next**

Remove ENHANCED_CAP_DISTRIBUTION command to suppress this warning, or contact the Synopsys support center if you have some conerns related to this command.

Feedback

## SX-0682

(Fatal) EXTRA_GEOMETRY_INFO:RES must be used with REDUCTION:NO/ POWER_REDUCTION:NO.

**Description**

EXTRA_GEOMETRY_INFO:RES enables StarRC to print bounding box for metal and via resistors. Reduction may cause incorrect geometry information.

**What Next**

Set both REDUCTION and POWER_REDUCTION to NO.

## SX-0683

(Fatal) NETLIST_COMMENTS_FILE '%s' does not exist.

**Description**

NETLIST_COMMENTS_FILE '%s' does not exist.

**What Next**

Please check if specified file exists.

## SX-0685

(Warning) DENSITY_BASED_THICKNESS: YES is ignored since there is no thickness variation modeling.

**Description**

DENSITY_BASED_THICKNESS: YES is specified StarRC command file, but there is no thickness variation modeling in the nxtgrd file, such as THICKNESS_VS_DENSITY, POLYNOMIAL_BASED_THICKNESS_VARIATION. DENSITY_BASED_THICKNESS: YES is ignored.

**What Next**

You can ignore this message if it is expected to have no thickness vaiation modeling in nxtgrd file. Or use a nxtgrd file with thickness variation modeling.

## SX-0686

(Fatal) EXTRACT_VIA_CAPS may not be NO in order to use GATE_TO_DIFFUSION_CAP table defined in nxtgrd file.

Feedback

**Description**

EXTRACT_VIA_CAPS is set to NO, but there is GATE_TO_DIFFUSION_CAP table defined in nxtgrd file.

**What Next**

Please set EXTRACT_VIA_CAPS to YES in StarRC command file to use GATE_TO_DIFFUSION_CAP in nxtgrd file.

---

## SX-0687

(Warning) IGNORE_CAPACITANCE:ALL RETAIN_GATE_DIFFUSION_COUPLING specified but ITF file does not have a GATE_TO_DIFFUSION_CAP or GATE_TO_DIFFUSION_CHANNEL_CAP table defined.

**Description**

StarRC need GATE_TO_DIFFUSION_CAP or GATE_TO_DIFFUSION_CHANNEL_CAP table for extracting the gate-to-diffusion component when "IGNORE_CAPACITANCE:ALL RETAIN_GATE_DIFFUSION_COUPLING" is specified.

**What Next**

If gate-to-diffusion capacitance is not required, ignore this warning.

---

## SX-0691

(Fatal) nxtgrd file includes PARALLEL_TO_REFERENCE/ PERPENDICULAR_TO_REFERENCE, but neither nxtgrd file nor StarRC command file defines REFERENCE_DIRECTION

**Description**

nxtgrd file includes PARALLEL_TO_REFERENCE/PERPENDICULAR_TO_REFERENCE, but neither nxtgrd file nor StarRC command file defines REFERENCE_DIRECTION

**What Next**

Please add PARALLEL_TO_REFERENCE/PERPENDICULAR_TO_REFERENCE in nxtgrd file or StarRC command file

---

## SX-0692

(Warning) StarRC command file has REFERENCE_DIRECTION, but TCAD_GRD_FILE do not have PARALLEL_TO_REFERENCE/PERPENDICULAR_TO_REFERENCE keywords. REFERENCE_DIRECTION will be ignored for table lookup.

Feedback

**Description**

StarRC command file has REFERENCE_DIRECTION, but TCAD_GRD_FILE do not have PARALLEL_TO_REFERENCE/PERPENDICULAR_TO_REFERENCE keywords. REFERENCE_DIRECTION will be ignored for table lookup.

**What Next**

You can ignore this message if you do not want to use REFERENCE_DIRECTION. Or please add PARALLEL_TO_REFERENCE/PERPENDICULAR_TO_REFERENCE in TCAD_GRD_FILE.

---

## SX-0693

(Warning) nxtgrd file has REFERENCE_DIRECTION, but do not have PARALLEL_TO_REFERENCE/PERPENDICULAR_TO_REFERENCE keywords. REFERENCE_DIRECTION will be ignored for table lookup.

**Description**

nxtgrd file has REFERENCE_DIRECTION, but do not have PARALLEL_TO_REFERENCE/PERPENDICULAR_TO_REFERENCE keywords. REFERENCE_DIRECTION will be ignored for table lookup.

**What Next**

You can ignore this message if you do not want to use REFERENCE_DIRECTION. Or please add PARALLEL_TO_REFERENCE/PERPENDICULAR_TO_REFERENCE in nxtgrd file.

---

## SX-0694

(Fatal) IGNORE_ETCH is not supported with RPSQ_VS_WIDTH_SPACING or RHO_VS_WIDTH_SPACING in nxtgrd file

**Description**

You can not use IGNORE_ETCH in StarRC command file and RPSQ_VS_WIDTH_SPACING or RHO_VS_WIDTH_SPACING in nxtgrd file

**What Next**

Please either remove IGNORE_ETCH in StarRC command file

Feedback

## SX-0695

(Warning) PRINT_SILICON_INFO in the StarRC command file: silicon width information in netlist tail comments may be inaccurate due to RPSQ_VS_WIDTH_AND_SPACING or RHO_VS_WIDTH_AND_SPACING in the nxtgrd file.

### Description

The silicon width might be inaccurate when ITF commands RPSQ_VS_WIDTH_AND_SPACING or RHO_VS_WIDTH_AND_SPACING are specified.

### What Next

Replace RPSQ_VS_WIDTH_AND_SPACING and RHO_VS_WIDTH_AND_SPACING with RPSQ_VS_SI_WIDTH and ETCH_VS_WIDTH_AND_SPACING.

## SX-0696

(Fatal) The starRC command USER_DEFINED_DIFFUSION_RES is set but the nxtgrd file has no USER_DEFINED_DIFFUSION_RESISTANCE model defined.

### Description

The nxtgrd file was generated without a USER_DEFINED_DIFFUSION_RESISTANCE statement in the ITF file so the user defined diffusion resistance model parameters are not defined.

### What Next

Either remove USER_DEFINED_DIFFUSION_RES in the starRC command file to disable the user defined diffusion resistance feature or update the nxtgrd file to enable it. Afterwards, run StarXtract again. To update the nxtgrd file: add a USER_DEFINED_DIFFUSION_RESISTANCE statement in the ITF file and run grdgenxo with -res_update.

## SX-0697

(Warning) NETLIST_CONNECT_OPENS is not set for all nets. This may cause failure to connect short equivalent nodes.

### Description

When you use SHORT_EQUIV_NODES: YES, StarRC will try to merge short equivalent nodes into a single net and connect them with short resistors. If NETLIST_CONNECT_OPENS is not set for all nets, StarRC may fail to add short resistors to connect the short equivalent nodes.

Feedback

**What Next**

You should use NETLIST_CONNECT_OPENS: * in StarRC command file when you use SHORT_EQUIV_NODES: YES.

---

## SX-0698

(Fatal) INTERNAL Error: SCHEMATIC_SKIP_CELLS list cannot be used with SKIP_CELLS list, if CELL_TYPE is set to LAYOUT.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0699

(Fatal) BLOCK "%s" did not compare.

**Description**

Specified BLOCK did not compare. This is required to run StarRC.

**What Next**

Please check LVS run to make sure the TOP block compared to run StarRC.

---

## SX-0700

(Fatal) There is no valid equiv cell in the SKIP_CELLS list.

**Description**

SKIP_CELLS is set but there is no valid equiv cell in the SKIP_CELLS list.

**What Next**

Please check your SKIP_CELLS list or check LVS database to make sure SKIP_CELLS are valid equiv cells.

Feedback

## SX-0701

(Fatal) HIERARCHICAL_SEPARATOR must be / | . or :

**Description**

HIERARCHICAL_SEPARATOR must be / | . or :

**What Next**

Set HIERARCHICAL_SEPARATOR to proper value in your StarRC command file.

## SX-0702

(Fatal) BUS_BIT prefix must be [ < ( { . or : and suffix must be ] > ) or }

**Description**

BUS_BIT prefix must be [ < ( { . or : and suffix must be ] > ) or }

**What Next**

Set BUS_BIT in correct format.

## SX-0703

(Fatal) OA_BUS_BIT prefix must be [ < ( { and suffix must be ] > ) or }

**Description**

OA_BUS_BIT is used to change but bit delimiters in OA view. For example, if the original database use () as bus bit delimiters but you want to use <> in the output OA view, you can set: BUS_BIT: () BUS_BIT: <> OA_BUS_BIT only supports 4 pairs of but bit delimiters: [], <>, (), {}. If other character is used in this command, StarRC will give this error.

**What Next**

You should remove unsupported character from OA_BUS_BIT command.

## SX-0704

(Fatal) NETLIST_DELIMITER must be : . / | or #

**Description**

Unsupported NETLIST_DELIMITER setting is specified.

Feedback

**What Next**

Please use supported values: . / | #

---

## SX-0705

(Fatal) OA_INSTANCE_BIT prefix must be [ < ( { and suffix must be ] > ) or }

**Description**

OA_INSTANCE_BIT is used to change bit delimiters for instances in OA view. For example, if the original database use () as bit delimiters for instances but you want to use <> in the output OA view, you can set: BUS_BIT: () OA_INSTANCE_BIT: <> OA_INSTANCE_BIT only supports 4 pairs of bit delimiters: [], <>, (), {}. If other character is used in this command, StarRC will give this error.

**What Next**

You should remove unsupported character from OA_INSTANCE_BIT command.

---

## SX-0706

(Fatal) Errors while writing file '%s', disk may be full.

**Description**

An internal file could not be successfully written to disk.

**What Next**

Check if the disk has space or contact your system administrator.

---

## SX-0708

(Fatal) INTERNAL Error: Syntax error while parsing command file at line '%d'

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0709

(Warning) INTERNAL Warning: Layer '%s' is not in layers file

Feedback

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0710

(Fatal) Can not open file '%s' for reading

**Description**

StarRC cannot open a file for reading and parsing.

**What Next**

Check to make sure that you have read permission on the file system and that the disk performance is good.

---

## SX-0711

(Information) The starRC command TEMPERATURE_SENSITIVITY is set and the nxtgrd file has USER_DEFINED_DIFFUSION_RESISTANCE model defined. The two features are not compatible.

**Description**

This is an information message. TEMPERATURE_SENSITIVITY and USER_DEFINED_DIFFUSION_RESISTANCE features are not compatible. The TEMPERATURE_SENSITIVITY is not applied to User Defined Diffusion Resistances (UDDR).

**What Next**

No action is needed.

---

## SX-0712

(Information) The starRC command SIMULTANEOUS_MULTI_CORNER is set and USER_DEFINED_DIFFUSION_RESISTANCE is set to VERBOSE. The two features are not compatible, USER_DEFINED_DIFFUSION_RESISTANCE is set to YES.

Feedback

**Description**

This is an information message. The USER_DEFINED_DIFFUSION_RESISTANCE:VERBOSE mode is only available for single corner run.

**What Next**

In order to run StarRC in USER_DEFINED_DIFFUSION_RESISTANCE VERBOSE mode, setup individual starRC command file and run StarXtract for each corner.

---

## SX-0716

(Fatal) Tclsam_init result: '%s'

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0717

(Fatal) Tksam_init result: '%s'

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0718

(Fatal) Tixsam_init result: '%s'

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

Feedback

## SX-0719

(Warning) Could not find any net for wildcard set '%s'

**Description**

StarRC cannot find any net with name matches this wildcard set.

**What Next**

Information only. No action is needed.

## SX-0720

(Warning) Could not find net "%s"

**Description**

StarRC cannot find this net in input database.

**What Next**

Information only. No action is needed.

## SX-0722

(Warning) Negation wildcard is not supported for power nets selection. Token '%s' will be ignored.

**Description**

POWER_NETS and POWER_NETS_SOLVE don't support negation wildcard '!'. This wildcard will be ignored.

**What Next**

Information only. No action is needed.

## SX-0725

(Warning) Could not find instance %s .

**Description**

StarRC cannot find this instance specified in SKIP_INSTANCES in the input database.

**What Next**

If this is expected, no action is needed.

Feedback

## SX-0726

(Warning) DANGLING_NET '%s' is removed from netlist.

**Description**

Dangling net is removed from netlist.

**What Next**

Information only. No action is needed.

## SX-0727

(Warning) Removed %d %s_NETS from the netlist (see cells.sum/translate.sum).

**Description**

StarRC reports how many nets are removed.

**What Next**

Information only. No action is needed.

## SX-0728

(Warning) Floating nets will be translated as fill.

**Description**

TRANSLATE_FLOATING_AS_FILL is YES, floating nets will be translated as fill.

**What Next**

Information only. No action is needed.

## SX-0730

(Warning) Environment variable STARRCXT_OA_LIBS is going to be replaced by STARRC_OA_LIBS in 2010.06

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

Feedback

## SX-0735

(Warning) METAL_FILL_POLYGON_HANDLING is set to IGNORE so METAL_FILL_GDS_FILE will be ignored.

**Description**

METAL_FILL_POLYGON_HANDLING is set to IGNORE so METAL_FILL_GDS_FILE is not used and will be ignored.

**What Next**

If this is expected, no action is needed.

## SX-0737

(Warning) METAL_FILL_POLYGON_HANDLING is set to FLOATING so METAL_FILL_GDS_FILE_NET_NAME/METAL_FILL_OASIS_FILE_NET_NAME option will be ignored.

**Description**

METAL_FILL_POLYGON_HANDLING is set to FLOATING so METAL_FILL_GDS_FILE_NET_NAME/METAL_FILL_OASIS_FILE_NET_NAME option will be ignored.

**What Next**

If this is expected, no action is needed.

## SX-0738

(Warning) METAL_FILL_GDS_FILE_NET_NAME/ METAL_FILL_OASIS_FILE_NET_NAME: '%s' is not a valid layout netname. Grounded metal fill objects will be assigned to net 0.

**Description**

StarRC cannot find this net name in the input database so the grounded metal fill objects will be assigned to net 0 instead of METAL_FILL_GDS_FILE_NET_NAME/ METAL_FILL_OASIS_FILE_NET_NAME.

**What Next**

If this is expected, no action is needed.

Feedback

## SX-0739

(Fatal) METAL_FILL_GDS_OFFSET/METAL_FILL_OASIS_OFFSET must have 2 float elements

**Description**

The syntax of METAL_FILL_GDS_OFFSET/METAL_FILL_OASIS_OFFSET is: METAL_FILL_GDS_OFFSET: offsetX offsetY METAL_FILL_OASIS_OFFSET: offsetX offsetY

**What Next**

Provide correct METAL_FILL_GDS_OFFSET/METAL_FILL_OASIS_OFFSET setting in StarRC command file.

## SX-0741

(Fatal) Invalid syntax in METAL_SHEET_OVER_AREA. Usage is: METAL_SHEET_OVER_AREA: <metal_layer_name> <left> <bottom> <right> <top> .

**Description**

Format of METAL_SHEET_OVER_AREA is not correct.

**What Next**

Reset METAL_SHEET_OVER_AREA in StarRC command file.

## SX-0742

(Warning) Net %s becomes %s due to instance port(s) explosion and is removed.

**Description**

The instance port(s) is exploded and removed, and then this net name changed.

**What Next**

Confirm this is expected data.

## SX-0743

(Fatal) Metal fill block '%s' is not found in the METAL_FILL_GDS_FILE/ METAL_FILL_OASIS_FILE.

Feedback

**Description**

The import of fills into the main design database often takes significant time and inconsistent fill design hierarchy imported will be difficult to debug and cause numerous problems in later extraction stage. To avoid possible problems as early as possible, StarRC errors out when it is unable to find consistent fill design hierarchy.

**What Next**

METAL_FILL_GDS_FILE/METAL_FILL_OASIS_FILE must contain top block with same name as main design block given by BLOCK in the command file. You should correct top block in the METAL_FILL_GDS_FILE/METAL_FILL_OASIS_FILE or use METAL_FILL_BLOCK_NAME to continue.

---

## SX-0752

(Fatal) The use of '%s' for option '%s' is not yet supported.

**Description**

The specified value for the option mentioned the error message is not yet supported by StarRC.

**What Next**

Change the value for the option being used. Refer to the StarRC User Guide for more information.

---

## SX-0754

(Fatal) ASTRO_TIMER is no longer supported.

**Description**

Setting of ASTRO_TIMER for the TARGET_ANALYSIS option in the StarRC command file is no longer supported.

**What Next**

Change the value for the TARGET_ANALYSIS option in the StarRC command file. Refer to the StarRC User Guide to get the list of supported values.

---

## SX-0755

(Warning) EXTRA_STAR_CMD_FILE '%s' does not exist.

Feedback

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0756

(Fatal) One or more prohibited option combinations or settings were found in star_cmd file. Please correct the issues listed above.

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0757

(Warning) '%s'. Setting '%s'...

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0758

(Severe) '%s'.

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0759

(Fatal) SIMULTANEOUS_MULTI_CORNER extraction condition violation at : %s

Feedback

**Description**

This message indicates that the nxtgrd file in the selected corner does not support SIMULTANEOUS_MULTI_CORNER extraction.

**What Next**

Use single corner extraction.

---

## SX-0760

(Fatal) ITF_FILE command only supports EXTRA_GEOMETRY_INFO:NONE

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0761

(Warning) Could not open library '%s'

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0762

(Warning) Could not open '%s' view.

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0763

(Warning) Failed to get dbUnit from library '%s' cell '%s' : '%s'

Feedback

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0764

(Warning) Can not find MW object: cellId='%s' objId='%s' typeId='%s'

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0765

(Warning) Layer name not found for layer number '%d'

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0766

(Warning) Lib: '%s' cell: '%s' view: '%s' not open!

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0767

(Warning) PARA net '%s' not found in cell '%s' in library '%s'

Feedback

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0768

(Warning) PARA view needs to have extra geometry info for node searching tasks.

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0770

(Warning) Can not iterate nets in PARA view!

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0771

(Fatal) Option %s should not be mixed with HERCULES/CALIBRE options %s

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0772

(Fatal) Option %s should not be mixed with HERCULES/LEFDEF options %s

Feedback

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0773

(Fatal) Option %s should not be mixed with HERCULES/MILKYWAY options %s

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0774

(Fatal) Option %s should not be mixed with LEFDEF/CALIBRE option %s

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0775

(Fatal) Internal error while processing PATTERN commands

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0776

(Fatal) Please specify input file using -i

Feedback

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0777

(Fatal) Please specify output file using -out

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0779

(Information) PIN_CUT_THRESHOLD for layer: '%s' set to value :'%d'.

**Description**

StarXtract reports this PIN_CUT_THRESHOLD is specified for this layer name.

**What Next**

If this is expected data, it is information only. No action is needed.

---

## SX-0780

(Warning) Layer '%s' specified in PIN_CUT_THRESHOLD cannot be found in layer mapping file.

**Description**

This layer name is specified for PIN_CUT_THRESHOLD, but StarXtract cannot find this name in layer mapping file.

**What Next**

Confirm this specified layer name is correct.

Feedback

## SX-0781

(Warning) Ignored '%d' entries in PROBE_TEXT_FILE due to bad syntax or invalid layer specification (see translate.sum).

**Description**

Bad syntax exists in PROBE_TEXT_FILE.

PROBE_TEXT_FILE syntax is as follow:

CELL <cell_name> TEXTSTRING LOCAL_X(um) LOCAL_Y(um) LAYER_NAME TEXTSTRING LOCAL_X(um) LOCAL_Y(um) LAYER_NAME CELL <cell_name>

**What Next**

Correct the PROBE_TEXT_FILE syntax.

## SX-0782

(Fatal) option %s should not be mixed with LEFDEF/MILKYWAY option %s

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

## SX-0783

(Warning) Could not open cell "'%s' : '%s'".

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

## SX-0784

(Warning) Could not find node '%s'

**Description**

Not available.

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## SX-0785

(Warning) Could not find node neighboring location ('%f', '%f')

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0786

(Fatal) Cannot find net id in hnfile.hn for '%s'

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0787

(Fatal) Cannot find pin id in hnfile.hn for '%s'

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0788

(Fatal) Need ASCII_DESIGN_FILE in quick_mode

**Description**

Not available.

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## SX-0789

(Fatal) Need TCAD_GRD_FILE in quick_mode

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0790

(Warning) Logical pin port '%s' in BLOCK is not placed in the layout. This port will be netlisted in the *PORTS section of SPEF and/or the .SUBCKT statement for SPF/STAR/NETNAME formats but will not appear in the detail parasitic section (since it has no physical characteristic).

**Description**

StarRC detects logical only port. There is no physical shapes associated with it.

**What Next**

If this is expected, no action is needed.

---

## SX-0791

(Warning) Invalid boundary: '%d' , '%d' : '%d' , '%d'

**Description**

Invalid boundary setting of RING_AROUND_THE_BLOCK_TOP_METAL.

**What Next**

Reset RING_AROUND_THE_BLOCK_TOP_METAL in StarRC command file.

---

## SX-0792

(Fatal) Incorrect %s metal name: '%s' in %s command. You must specify a database metal layer name from the mapping file.

Feedback

**Description**

Layer set in RING_AROUND_THE_BLOCK_TOP_METAL/ RING_AROUND_THE_BLOCK_BOTTOM_METAL is invalid.

**What Next**

Reset RING_AROUND_THE_BLOCK_TOP_METAL/ RING_AROUND_THE_BLOCK_BOTTOM_METAL in StarRC command file.

---

## SX-0793

(Warning) Polygons for marker can not be non-manhattan, check the gds file

**Description**

Marker polygon of ViaArray has non-manhattan geometry. It's not available and the polygon is ignored.

**What Next**

Check GDSII file set in RV_VIA_ARRAY_GDS_FILE.

---

## SX-0794

(Fatal) option %s should not be mixed with CALIBRE/MILKYWAY option %s

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0796

(Fatal) This function %s should not be mixed with HERCULES option

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

Feedback

## SX-0798

(Fatal) INTERNAL Error: cut polygon horizontal distance is less than 10nm: '%f'

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## SX-0799

(Warning) INTERNAL Warning: overlap distance is rounded to '%d' due to smaller dbUnit.

**Description**

A consistency check may have failed or StarRC may be operating in a mode that is not considered normal. This condition is not considered severe enough to abort StarRC's execution but a warning message is given to alert the StarRC support team in case of unexpected program behavior.

**What Next**

Contact the Synopsys Support Center if StarRC terminates abnormally or if other issues are found.

## SX-0800

(Fatal) INTERNAL Error: cut thickness can not be larger than overlap distance

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## SX-0801

(Fatal) INTERNAL Error: File '%s' did not contain sufficient data to generate the dist_report file.

**Description**

An internal consistency check has failed.

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## SX-0802

(Fatal) INTERNAL Error: File '%s' did not contain sufficient data to generate the dist_report file ('%s' : '%d')

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0803

(Fatal) Cannot open file '%s' for reading.

**Description**

The specified file cannot be opened.

**What Next**

Make sure the specified file exists and you have the permission to read.

---

## SX-0804

(Fatal) This function %s should not be mixed with CALIBRE option

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0805

(Fatal) This function %s should not be mixed with MILKYWAY option

**Description**

Not available.

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## SX-0806

(Fatal) This function %s should not be mixed with LEFDEF option

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0807

(Warning) Incremental extraction doesn't support change of option: %s between a pre-ECO run and post-ECO run."

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0811

(Fatal) Both -custom and -ultra license options are specified.

**Description**

-custom and -ultra are mutually exclusive!

**What Next**

See StarRC user guide.

---

## SX-0812

(Fatal) -custom only supports setting %s

**Description**

The -custom option in StarRC is intended for transistor level extraction only. Only the setting in this message is suported for this command.

Feedback

**What Next**

Refer to reference manual for the set of features supported by the custom license.

---

## SX-0813

(Warning) Duplicate placement of '%s' found in an %dx%d OASIS array reference record placed at (%d, %d).

**Description**

Duplicate placements in the OASIS array reference create unexpected polygons in extraction and may introduce significant performance/accuracy problems.

StarRC will ignores such duplicate placements and leave only first one in the OASIS array reference record.

**What Next**

Check the array reference element in the OASIS file.

---

## SX-0814

(Warning) System error: Failed to set file descriptor limit to '%d'

**Description**

StarRC has determined that it may need to have a larger number of files open. A system call to increase the the limit on file descriptors has failed.

**What Next**

Check with local system administrator about possible system level fix. Contact the Synopsys Support Center if necessary.

---

## SX-0818

(Fatal) Unsupported option IAPI is not available.

**Description**

The IAPI debug option is not supported, and is not available.

**What Next**

Contact the Synopsys Support Center.

Feedback

## SX-0819

(Fatal) Invalid syntax. Usage: StarXtract -sort_xin tech_file -i <xin_file_name>

**Description**

StarXtract -sort_xin command requires an input xin file.

**What Next**

Contact the Synopsys Support Center.

## SX-0820

(Fatal) Invalid syntax. Usage: StarXtract -sort_xin_by_net tech_file -i <xin_file_name>

**Description**

StarXtract -sort_xin_by_net command requires input xin file.

**What Next**

Contact the Synopsys Support Center.

## SX-0821

(Fatal) Invalid syntax. StarXtract -pcell_layers tech_file -i <xin_file_name>

**Description**

StarXtract -pcell_layers command requires an input xin file.

**What Next**

Contact the Synopsys Support Center.

## SX-0822

(Fatal) Quick mode processing failed. Check star/.xTract.messages for details.

**Description**

StarXtract Quick mode processing failed. Check star/.xTract.messages for details.

**What Next**

Contact the Synopsys Support Center.

Feedback

## SX-0823

(Fatal) Invalid syntax. Usage: StarXtract -partition tech_file -i <xin file name>.

**Description**

StarXtract -partition option requires an input xin file name.

**What Next**

Contact the Synopsys Support Center.

## SX-0824

(Fatal) Invalid syntax. Usage: StarXtract -post_partition tech_file -i <xin_file_name>

**Description**

StarXtract -post_partition command requires an input xin file name.

**What Next**

Contact the Synopsys Support Center.

## SX-0830

(Fatal) Licenses available allow a distributed run with a maximum of %d cores.

**Description**

The number of StarRC licenses currently available allow the use of only a limited number of cores for extraction.

**What Next**

Contact the Synopsys Support Center.

## SX-0835

(Fatal) '%s' should be declared as an absolute path, because STAR_DIRECTORY is specified relative to the parent of current directory.

**Description**

When STAR_DIRECTORY is specified relative to the parent of the current directory, this input path must be specified as an absolute path.

Feedback

**What Next**

Consult the StarRC user guide.

---

## SX-0837

(Fatal) Invalid specification for NUM_CORES command: '%d'

**Description**

NUM_CORES must be must be specified as an integer greater than 1.

**What Next**

Consult the StarRC user guide.

---

## SX-0838

(Warning) Specified value %d of NUM_CORES is larger than the number of partitions. Reducing to %d.

**Description**

NUM_CORES cannot be larger than the number of tiles into which StarRC has partitioned the input design.

**What Next**

Consult the StarRC user guide. Contact the Synopsys Support Center.

---

## SX-0839

(Warning) STARRC_DP_STRING is ignored because a -clean option was not specified in the command line.

**Description**

If StarXtract is run using command-line options, and if STARRC_DP_STRING is specified, the -clean option or one of its variants must be specified in the command line.

**What Next**

Consult the StarRC user guide.

---

## SX-0840

(Warning) There are %d slots specified in %s. However, %s is specified as %d. The run will use only the first %d slots from %s.

Feedback

**Description**

NUM_CORES specification overrides the number of DP processes specified in the DP string command if the latter is greater.

**What Next**

Consult the StarRC user guide.

---

## SX-0841

(Warning) There are only %d cores specified in %s. However, NUM_CORES is specified as %d. The value of NUM_CORES is reduced to %d.

**Description**

The number of cores specified in the DP string command overrides specified value of NUM_CORES when the latter is greater.

**What Next**

Consult the StarRC user guide.

---

## SX-0842

(Fatal) Invalid job submission command in STARRC_DP_STRING.

**Description**

The job submission specification in STARRC_DP_STRING is not supported.

**What Next**

Consult the StarRC user guide for currently supported job submission protocols and their syntax.

---

## SX-0844

(Fatal) Neither of the environment variables SNPSLMD_LICENSE_FILE or LM_LICENSE_FILE, which specify the paths of license files has been set.

**Description**

The path to the license servers or license files must be specified in one of these environment variables.

**What Next**

Consult the StarRC user guide.

Feedback

## SX-0845

(Fatal) Environment variable PATH is not set.

**Description**

The PATH variable must be defined in DP mode.

**What Next**

Consult StarRC user guide.

## SX-0846

(Information) Launching jobs with xterms running gdb.

**Description**

This message is for informational purposes only.

**What Next**

Contact the Synopsys Support Center.

## SX-0847

(Information) Submitting job %d via system call to localhost.

**Description**

This message is for informational purposes only.

**What Next**

Contact the Synopsys Support Center.

## SX-0848

(Information) Submitting job '%d' by %s to '%s'

**Description**

A job is submitted to a remote host using the protocol mentioned in the message.

**What Next**

No further action is necessary.

Feedback

## SX-0851

(Fatal) The tech_file file from the previous StarRC run does not exist.

**Description**

A file named tech_file in the STAR_DIRECTORY from the previous run that needed to run StarXtract with the -cleanN option is not found.

**What Next**

StarXtract cannot be run with the -cleanN option. Rerun StarRC with the -clean option to generate netlist files in the desired format.

## SX-0852

(Fatal) In this release, the "-cleanN" command does not allow the user to change NETLIST_FORMAT from '%s' to '%s'.

**Description**

The following is supported: SPF to SPEF

**What Next**

Consult the StarRC user guide.

## SX-0854

(Warning) Running 32-bit binary with -64 flag.

**Description**

StarXtract was started with -64 flag, but we are running a 32-bit binary. The -64 flag is ignored.

**What Next**

Make sure you are running from a properly installed StarRC configuration.

## SX-0857

(Fatal) No technology (command) file provided.

**Description**

StarXtract command line must contain least one tech_file (command file).

Feedback

**What Next**

Consult the StarRC user guide.

---

## SX-0859

(Warning) '%s' will be interpreted as SKIP_CELLS: !*

**Description**

In SKIP_CELLS setting, it only contains "!" wildcard. In that case, StarRC interprets this setting as not skipping every cell (include the cells not specified in this command).

**What Next**

If this is expected, you don't need to do anything. But if you expect to skip every cell except the entries specified in SKIP_CELLS command with "!", change the setting to SKIP_CELLS: * !<cell name 1> !<cell name 2> ...

---

## SX-0860

(Information) Command line option -b '%s' supersedes block name '%s' specified in the command file.

**Description**

The name of the block specified using the -b option in the command line takes precedence over the one specified in the command file.

**What Next**

For more information, consult the StarRC user guide.

---

## SX-0861

(Fatal) No block name set for input design.

**Description**

The top level block (cell) name must be specified.

**What Next**

Consult the StarRC user guide.

---

## SX-0862

(Warning) Could not write to the summary file '%s'.

Feedback

**Description**

StarRC failed to open the summary file for writing.

**What Next**

Check to make sure that the file system has available space and that the disk performance is good.

---

## SX-0863

(Information) Preparing to submit %d jobs with command from STARRC_DP_STRING: '%s' for extraction: '%s'

**Description**

StarRC is spawning DP jobs using the command in STARRC_DP_STRING.

**What Next**

For more details, consult the StarRC user guide.

---

## SX-0864

(Fatal) NUM_CORES is set to 0. It must be at least 1.

**Description**

NUM_CORES must have a value of 1 or greater.

**What Next**

Consult the StarRC user guide.

---

## SX-0868

(Fatal) StarRC ran to completion but writing of the .starrc_complete file failed: '%s'

**Description**

A file opening problem was encountered when writing the .starrc_complete file.

**What Next**

Check to see if file system has sufficent space that disk performance is good.

Feedback

## SX-0869

(Fatal) Syntax is : StarXtract -virtual_metal_fill_dump tech_file -i <xin_file_name> -out <gds_file_name>

**Description**

Syntax is : StarXtract -virtual_metal_fill_dump tech_file -i <xin_file_name> -out <gds_file_name>

**What Next**

Review the syntax

## SX-0955

(Warning) Found shorted nets (see file shorts_all.sum). The capacitances of these nets may be unreliable due to the shorts.

**Description**

Not available.

**What Next**

See star/shorts_all.sum and verify reported shorts in layout. If there is no short in layout, run -Display debug function to find the short. If the root cause is still not clear, please contact the Synopsys Support Center.

## SX-0957

(Warning) Found metal fill shorts (see file shorts_all.sum).

**Description**

Not available.

**What Next**

See star/shorts_all.sum and verify reported shorts in layout. If there is no short in the layout, run the -Display debug function to find the short. If the root cause is still not clear, please contact the Synopsys Support Center.

## SX-0959

(Warning) Found vias with one or no connection (see file vias.sum)

Feedback

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-0961

(Warning) Found less than minimum spacing (see file smin.sum).

**Description**

Spacing values between neighbor polygons are found to be much smaller than minimum spacing. This message lists spacing value and the minimum spacing on this layer.

**What Next**

Please check if the small spacing values in the layout are indeed desired.

---

## SX-0962

(Warning) File specified for option '%s': '%s' has zero-bytes file size.

**Description**

The file name used to specify an input file as the value of an option in the StarRC command file has 0-bytes file size.

**What Next**

Verify the path to the file name mentioned in the error message. Check for spelling errors, access permissions and file size.

---

## SX-0963

(Information) WARNING: top-level equivalent nets found in design. Please refer to shorted_nets file for more information.

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

Feedback

## SX-0964

(Fatal) sview tech_file [<tech_file> ... ]

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

## SX-0965

(Fatal) Query RC Viewer does not support XREF: COMPLETE. Please set XREF: YES or XREF: NO and re-run StarXtract.

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

## SX-0966

(Fatal) File name specified for option '%s': '%s' does not exist.

**Description**

The file name used to specify an input file as the value of an option in the StarRC command file does not exist.

**What Next**

Verify the path to the file name mentioned in the error message. Check for spelling errors and access permissions.

## SX-0967

(Warning) File specified for option '%s': '%s' can't be read.

**Description**

The file name used to specify an input file as the value of an option in the StarRC command file does not exist or can't be read due to permission.

Feedback

### What Next

Verify the path to the file name mentioned in the error message. Check for spelling errors and access permissions.

---

## SX-0968

(Fatal) '%s' is not a valid setting for the '%s' option. Valid choices are: '%s'

### Description

The value for an option in the StarRC command file that allows only a limited set of values has been incorrectly specified.

### What Next

Fix the incorrect value setting by using one of the valid choices mentioned in the error message. Refer to the StarRC Use Guide for more information.

---

## SX-0969

(Fatal) Input files specified for options can't be read or have zero-bytes file size.

### Description

Some input files specified for options can't be read or have zero-bytes file size.

### What Next

Verify the path to the file name mentioned in the error messages. Check for spelling errors and access permissions. Files with zero-bytes file size should be removed from your StarRC command file.

---

## SX-0973

(Warning) Option '%s' will become obsolete from the next release of StarRC. Remove this option from the StarRC command file.

### Description

An option specified in the StarRC command file will become obsolete from the next release of StarRC. The feature enabled by this option may be supported by another option, or it may no longer be necessary for the user to enable this feature explicitly.

### What Next

Consult the StarRC User Guide or contact the Synopsys Customer Support Center for more information.

Feedback

## SX-0974

(Fatal) Option '%s' used in the StarRC command file is obsolete. Remove this option from the command file.

### Description

An option used in the StarRC command file is obsolete and can no longer be used. Users need to remove this option from the command file in order to run StarRC.

An option is made obsolete if the feature enabled by it is supported by another option, or it becomes unnecessary for users to enable the supported feature explicitly.

### What Next

Consult the StarRC User Guide or contact the Synopsys Customer Support Center for more information.

## SX-0981

(Fatal) 3D_IC can not support view only layers. Remove viewonly_layers section in mapping file.

### Description

If you run 3D_IC flow with StarRC, you can't add view only layers in viewonly_layers section in StarRC mapping file. Otherwise, StarRC will give this error.

### What Next

You should remove or comment out the whole viewonly_layers section in StarRC mapping file.

## SX-0984

(Warning) Could not open via_conns file for read.

### Description

An internal consistency check has failed.

### What Next

Contact the Synopsys Support Center.

## SX-0985

(Information) \t\tTrap ('%d') +'%d' -'%d' del:'%d'

Feedback

**Description**

Internal debug message.

**What Next**

Information only; no action is needed.

---

## SX-0986

(Information) \t\tElem ('%d') +'%d' -'%d'

**Description**

Internal debug message.

**What Next**

Information only; no action is needed.

---

## SX-0987

(Warning) No unimportant net found. Skipping reconnect...

**Description**

StarRC does not find any net in input data base match the syntax defined in command UNIMPORTANT_NETS so reconnection work is skipped.

**What Next**

Check the syntax defined in command UNIMPORTANT_NETS.

---

## SX-0988

(Fatal) Corner name '%s' is reserved for internal use and cannot be used in a SELECTED_CORNERS statement.

**Description**

The corner name used in the CORNERS_FILE specified in the StarRC command file is reserved for internal use.

**What Next**

Replace all occurrences of the forbidden corner name in the CORNERS_FILE to something else other than the forbidden name.

Feedback

## SX-0989

(Fatal) Via array merge error 1

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## SX-0995

(Fatal) VIA_ARRAY_GDS_LAYER_MAP_FILE is required for RV_VIA_ARRAY_GDS_FILE.

**Description**

GDS layer mapping file is not provided for translation.

**What Next**

Set VIA_ARRAY_GDS_LAYER_MAP_FILE in StarRC command file.

## SX-0997

(Warning) RV_VIA_ARRAY_GDS_FILE(s) do not contain any cell definition for top BLOCK. Via array objects must reside in same name cell as BLOCK.

**Description**

No cell in RV_VIA_ARRAY_GDS_FILE has same name as option BLOCK in StarRC command file.

**What Next**

Check RV_VIA_ARRAY_GDS_FILE in StarRC command file.

## SX-0998

(Information) Group bbox ('%d' , '%d')~( '%d' , '%d') is covered by more than 1 array bbox, please check the design

**Description**

Via group box is covered by more than 1 via array box.

Feedback

**What Next**

Check the mark polygons in RV_VIA_ARRAY_GDS_FILE.

---

## SX-0999

(Information) Group bbox ('%d' , '%d')~( '%d' , '%d') has invalid size of array bbox, please check the design

**Description**

Via group box of bridge via is not covered corretly by via array box.

**What Next**

Check the mark polygons in RV_VIA_ARRAY_GDS_FILE.

---

## SX-1001

(Fatal) Error in parsing VIA_COVERAGE_OPTION_FILE '%s'

**Description**

Invalid syntax is in VIA_COVERAGE_OPTION_FILE.

**What Next**

Check data in VIA_COVERAGE_OPTION_FILE.

---

## SX-1002

(Fatal) Could not open VIA_COVERAGE_OPTION_FILE '%s'

**Description**

StarRC could not open VIA_COVERAGE_OPTION_FILE.

**What Next**

Reset VIA_COVERAGE_OPTION_FILE in StarRC command file.

---

## SX-1003

(Warning) Layer specified in VIA_COVERAGE_OPTION_FILE is not a valid via layer: '%s'

**Description**

Layer specified in VIA_COVERAGE_OPTION_FILE is not a valid via layer.

Feedback

**What Next**

Reset VIA_COVERAGE_OPTION_FILE in StarRC command file.

---

## SX-1004

(Warning) Cell '%s' contains zero-width polygon (%d, %d) (%d %d) on layer %s.

**Description**

StarRC cannot handle zero-width polygon.

**What Next**

Information only; no action needed.

---

## SX-1005

(Fatal) Syntax error in VIA_COVERAGE_OPTION_FILE: '%s'

**Description**

Syntax error in VIA_COVERAGE_OPTION_FILE.

**What Next**

Reset VIA_COVERAGE_OPTION_FILE in StarRC command file.

---

## SX-1006

(Fatal) '%s' : '%s'

**Description**

StarRC meets error when parsing VIA_COVERAGE_OPTION_FILE.

**What Next**

Check context in VIA_COVERAGE_OPTION_FILE.

---

## SX-1007

(Fatal) Not support BOUNDARY_BOX parameters: %s

**Description**

BOUNDARY_BOX requires 4 boundary box corrdinetes as input.

Feedback

**What Next**

Modify StarRC command file.

---

## SX-1008

(Fatal) BOUNDARY_BOX options has a non-number '%s' specified

**Description**

Not acceptable input for BOUNDARY_BOX option.

**What Next**

Modify StarRC command file.

---

## SX-1019

(Fatal) An internal density parameter must be positive: %f

**Description**

StarRC Internal Error.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1020

(Fatal) Hash table initialized with invalid max density: %f

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1022

(Fatal) corrupted file

**Description**

StarRC is not able to read a file because it has been corrupted.

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## SX-1023

(Fatal) Can not create/open file '%s', please check filesystem for space and permissions.

**Description**

StarRC was not able to create or open a file.

**What Next**

Check file system for space and read/write permissions in the directory.

---

## SX-1024

(Fatal) Unable to lock file '%s', error status '%s', host '%s', working directory '%s'

**Description**

StarRC is unable to lock a file because of issues in the runtime environment.

**What Next**

Try running StarRC again or contact your system administrator.

---

## SX-1027

(Fatal) Failed to open '%s' for writing, last system error : '%s'

**Description**

Unable to open an output file.

**What Next**

Check to make sure the file system has sufficient space, and write permissions.

---

## SX-1028

(Warning) Netlist exceeded the file size limit %d; it has been splited to %d files.

**Description**

Netlist file exceeded maximum file size. As a result, it has been split into multiple files.

Feedback

**What Next**

This message is for informational purposes only.

---

## SX-1030

(Warning) Resetting max file size to %d

**Description**

This is a message for Synopsys internal purposes.

**What Next**

Contact Synopsys Support Center.

---

## SX-1031

(Fatal) Internal Error when initializing data of type '%s'

**Description**

Internal data initialization error.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1032

(Fatal) Version mismatch found on '%s'. (found: '%d', require: '%d')

**Description**

The version of an input file format is not supported.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1033

(Warning) Unable to open %s '%s' for %s, error status '%s'

**Description**

StarRC was unable to open a file because of conditions in the runtime environment.

Feedback

**What Next**

Take remedial actions based on the error status reported by the operating system. Contact your system administrator if necessary.

---

## SX-1034

(Warning) Unable to close file '%s' that was opened for reading: '%s'

**Description**

StarRC was able to open a file for file for reading, but could not close it.

**What Next**

Check to make sure disk I/O speeds are acceptable.

---

## SX-1036

(Fatal) Unable to read %d bytes from file %s, error status: '%s'

**Description**

StarRC cannot complete a read operation on the specified file because of conditions in the runtime environment. The error message includes the error status returned by the operating system.

**What Next**

Carry out remedial actions based on the operating system error status. Contact your system administrator if necessary.

---

## SX-1037

(Fatal) Unable to write %d bytes to file %s, error status: %s

**Description**

StarRC cannot complete a read operation on the specified file because of conditions in the runtime environment. The error message includes the error status returned by the operating system.

**What Next**

Carry out remedial actions based on the operating system error status. Contact your system administrator if necessary.

Feedback

## SX-1038

(Warning) Internal error: while copying '%s' to '%s', bytes read = '%d', but bytes written = '%d'

**Description**

StarRC was not able to write to an output file during a file copy operation.

**What Next**

Check to make sure that the output file system has sufficient disk space.

## SX-1039

(Warning) Unable to open file '%s' with mode '%s', error status: '%s' (re-try %d times)

**Description**

StarRC is unable to open a file because of conditions in the run-time environment.

**What Next**

Carry out remedial actions based on the error status indicated by the operating system. Contact your system administrator if necessary.

## SX-1040

(Warning) Unable to close file '%s' that was opened for writing: '%s'

**Description**

A file that was opened for output could not be closed.

**What Next**

Check to make sure that the file system has sufficient space.

## SX-1041

(Fatal) Unable to open file '%s' for %s, error status: '%s'

**Description**

StarRC is unable to open a file because of conditions in the runtime environment.

Feedback

**What Next**

Carry out remedial actions based on the error status indicated by the operating system. Contact your system administrator if necessary.

---

## SX-1042

(Fatal) Unable to close file '%s', error status '%s'

**Description**

StarRC is unable to close a file because of conditions in the runtime environment.

---

## SX-1043

(Fatal) Call to fclose() failed, error status '%s'

**Description**

StarRC is unable to close a file because of conditions in the runtime environment.

---

## SX-1044

(Fatal) File system error : '%s'

**Description**

StarRC detected an error during a file access operation.

**What Next**

Check to make sure the file system is working properly.

---

## SX-1045

(Fatal) Call to fseeko failed: '%s'

**Description**

StarRC detected an error during a file access operation.

**What Next**

Check to make sure the file system is working properly.

Feedback

## SX-1046

(Fatal) Failed to write '%d' bytes to disk: '%s'

**Description**

StarRC failed to write to a file that it opened.

**What Next**

Check to make sure that file system is not full.

## SX-1047

(Fatal) Failed to read '%d' bytes from disk: '%s'

**Description**

StarRC failed to read from an open disk file.

**What Next**

Check to make sure that the file system is working correctly.

## SX-1048

(Fatal) Unable to remove file '%s' : '%s'

**Description**

StarRC failed to remove a file.

**What Next**

Check to make sure that the file system is working properly.

## SX-1049

(Warning) failed to rename '%s' to '%s': '%s'

**Description**

Failed to rename a disk file.

**What Next**

Check to make sure that the file system is working properly.

Feedback

## SX-1050

(Fatal) Unable to save data to disk: '%s'

**Description**

StarRC failed to write binary data to an open file.

**What Next**

Check to make sure that file system has sufficient space, and that I/O operations are working properly.

## SX-1051

(Fatal) Unable to load binary data from disk: '%s'

**Description**

StarRC failed to read binary data from an open file.

**What Next**

Check to make sure file system I/O operations are working properly.

## SX-1052

(Fatal) Unable to save data to disk: '%s'

**Description**

StarRC failed to write binary data to an open file.

**What Next**

Check to make sure that file system has sufficient space and that I/O operations are working properly.

## SX-1053

(Fatal) Unable to load binary data from disk: '%s'

**Description**

StarRC failed to load binary data from disk.

**What Next**

Check to make sure that file system I/O is working properly.

Feedback

## SX-1054

(Fatal) Failed to rename '%s' to '%s': '%s'

**Description**

Failed to rename a disk file.

**What Next**

Check to make sure that the file system is working properly.

## SX-1076

(Fatal) Unable to create directory %s, error status %s

**Description**

StarRC is unable to create a directory because of issues in the runtime environment.

**What Next**

Carry out remedial actions based on the operating system error status. Contact your system administrator if necessary.

## SX-1077

(Fatal) Unable to change directory from %s to %s, error status %s.

**Description**

StarRC is unable to complete a directory change operation because of issues in the runtime environment.

**What Next**

Carry out remedial actions based on the operating system error status. Contact your system administrator if necessary.

## SX-1078

(Fatal) Environment variable %s is not defined.

**Description**

An environment variable required to run StarRC is not defined.

Feedback

**What Next**

Define the environment variable using

```
setenv VARIABLE_NAME value
```

in csh and tcsh or

```
VARIABLE_NAME=value
    export VARIABLE_NAME
```

in bash.

---

## SX-1079

(Fatal) Could not determine executable location '%s'

**Description**

The location of one of the executables required to run StarRC cannot be determined.

**What Next**

Check your StarRC installation. Verify that your PATH environment variable includes a path to the location of StarRC executables you intend to use.

---

## SX-1080

(Fatal) Could not determine path to current executable '%s', check your StarRC installation.

**Description**

The location of one of the executables required to run StarRC cannot be determined.

**What Next**

Check your StarRC installation. Verify that your PATH environment variable includes a path to the location of StarRC executables you intend to use.

---

## SX-1086

(Fatal) Unable to stat('%s'), error status %s'

**Description**

StarRC is unable to obtain the status of a file from the operating system.

Feedback

**What Next**

Probable cause is due to unsupported OS. Please make sure to use OS that are supported by StarRC. If the OS is supported and issue is not clear, please contact the Synopsys Support Center.

---

## SX-1091

(Warning) unable to stat path '%s', error status '%s'

**Description**

StarRC is unable to obtain the status of a file from the operating system.

**What Next**

No action is necessary if StarRC runs to completion. This message may indicate the source of a problem is StarRC does not run to completion. In this case, rerun StarRC after carrying out remedial actions based on the error status reported by the operating system. Contact your system administrator if necessary.

---

## SX-1093

(Warning) Failed to rename '%s' to '%s': %s, will attempt to copy file.

**Description**

StarRC is unable to rename a file and will attempt to copy the file to the new location and remove the original file.

**What Next**

No action is needed if StarRC runs to completion. Failure to rename a file might indicate a problem with file system configuration.

---

## SX-1094

(Warning) Failed to copy '%s' to '%s'

**Description**

StarRC is unable to make a copy of a file.

**What Next**

Another message before this one may give reasons why copying of the file failed. Carry out remedial actions based on the reasons given in the other warning message. Contact your system administrator if necessary.

Feedback

## SX-1095

(Fatal) Unable to open handle for file '%s', error status %s.

**Description**

A file cannot be opened or created because of issues in the runtime environment.

**What Next**

Carry out remedial actions based on the error status reported by the operating system. Contact your system administrator if necessary.

## SX-1097

(Fatal) Failed to allocate memory.

**Description**

A StarRC process ran out of memory.

**What Next**

Rerun StarRC on a host with more memory available.

## SX-1098

(Fatal) Unable to determine the current working directory, error status %s

**Description**

StarRC is unable to determine the current working directory because of issues in the runtime environment.

**What Next**

Carry out remedial actions based on the error status reported by the operating system. Contact your system administrator if necessary.

## SX-1099

(Fatal) Could not obtain information from the local password database for user %s, error status %s.

**Description**

StarRC is unable to obtain information for the specified user from the local password database file (typically /etc/passwd).

Feedback

**What Next**

Carry out remedial actions based on the error status reported by the operating system. Contact your system administrator if necessary.

---

## SX-1101

(Fatal) segmentation fault

**Description**

StarRC errors out with a memory segmentation violation.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1102

(Fatal) A call to system function %s failed, error status %s.

**Description**

A call to a function provided by the operating system interface failed.

**What Next**

Check if any remedial action can be taken based on the error status indicated by the operating system. If no action can be taken, try running StarRC once again, or contact your system administrator.

---

## SX-1103

(Fatal) Could not close the file descriptor associated with stream %s, error status %s.

**Description**

The file descriptor associated with the specified stream could not be closed because of issues in the runtime environment.

**What Next**

Take remedial actions based on the status reported by the operating system or contact your system administrator if necessary.

---

## SX-1104

(Fatal) Could not obtain a duplicate file descriptor for stream %s, error status %s.

Feedback

**Description**

A duplicate file descriptor for the specified stream could not be obtained because of issues in the runtime environment.

**What Next**

Take remedial action based on the operating system error status or contact your system administrator.

---

## SX-1105

(Fatal) Could not close stdout output: %s.

**Description**

This is an error in system which makes StarRC can't close stdout output.

**What Next**

Check system running status and rerun StarRC. If this error still occurs, contact Synopsys for more information.

---

## SX-1107

(Information) NOT RUNNING %s !! Using previous results. Previous %s

**Description**

When StarRC is run with options which reuse results from an earlier run (such as -cleanN), steps in StarRC's flow that are not necessary to produce the desired output for the current run are not re-run. This message indicates which steps do not get re-run.

**What Next**

This message is for information only, no further action is needed.

---

## SX-1108

(Warning) INTERNAL Warning: The intermediate DP stage summary file may be corrupted, StarRC cannot calculate total wall time information.

**Description**

An internal file produced by StarRC to tabulate runtime information is corrupt. The final runtime summary appearing at the end of StarRC's summary file may be missing or incorrect.

Feedback

### What Next

This message has no bearing on the results of parasitic extraction produced by StarRC, so no action is recommended. Contact the Synopsys Support Center if this message persists or is seen for many designs.

---

## SX-1109

(Warning) There are %d unfinished jobs.

### Description

An internal consistency check revealed that some jobs in StarRC's distributed processing schedule did not complete.

### What Next

Check if the results produced by StarRC are complete, such as by checking the number of nets in the netlist file. Try running StarRC again and contact the Synopsys Support Center if this warning persists.

---

## SX-1111

(Fatal) INTERNAL Error: Unknown file type

### Description

An internal consistency check has failed.

### What Next

Contact the Synopsys Support Center.

---

## SX-1113

(Fatal) INTERNAL Error: unknown LineModified return status.

### Description

An internal consistency check has failed.

### What Next

Contact the Synopsys Support Center.

Feedback

## SX-1116

(Information) Slave process %d is terminated since the number of partitions has been reduced to %d.

**Description**

To save resources, StarRC terminates unneeded processes during a distributed processing run if the number of partitions into which the layout is decomposed is lower than the number of CPU cores used.

**What Next**

No further action is necessary.

## SX-1120

(Warning) Number of jobs already reached the NUM_CORES value %d, terminating current process.

**Description**

Users cannot launch more StarXtract jobs than the value of NUM_CORES specified in the StarRC command file.

**What Next**

Do not launch more StarXtract jobs than the NUM_CORES value or increase the NUM_CORES specification in the StarRC command file.

## SX-1121

(Fatal) Attempting to use more CPU cores than the available number of licenses.

**Description**

The number of StarRC processes that can be launched concurrently is limited by the number of licenses available. Attempting to launch more processes than this limit will result in StarRC jobs terminating with this error message.

**What Next**

The following actions could be taken:

1

Use fewer number of processes to run StarRC by reducing the value of NUM_CORES in the StarRC command file.

Feedback

2

If StarRC is being run on more than one design concurrently, wait for other jobs to complete.

3

Acquire more StarRC licenses.

---

## SX-1132

(Fatal) Command '%s' not recognized on line '%d' in file '%s'.

**Description**

The Eco file is used to provide ECO change information to StarRC and reduces runtime overhead. This file has been corrupted and contains invalid information.

**What Next**

Check the available disk space and rerun as needed. This file should not be modified during StarRC extraction.

---

## SX-1133

(Fatal) Wrong number of arguments for command '%s' on line '%d' in file '%s'.

**Description**

The Eco file is used to provide ECO change information to StarRC and reduces runtime overhead. This file has been corrupted and contains invalid information.

**What Next**

Check the available disk space and rerun as needed. This file should not be modified during StarRC extraction.

---

## SX-1134

(Fatal) Unknown option '%s' on line '%d' in file '%s'.

**Description**

The Eco file is used to provide ECO change information to StarRC and reduces runtime overhead. This file has been corrupted and contains invalid information.

Feedback

**What Next**

Check the available disk space and rerun as needed. This file should not be modified during StarRC extraction.

---

## SX-1135

(Fatal) Database type '#D' not set.

**Description**

The Eco file is used to provide ECO change information to StarRC and reduces runtime overhead. This file has been corrupted and contains invalid information.

**What Next**

Check the available disk space and rerun as needed. This file should not be modified during StarRC extraction.

---

## SX-1201

(Fatal) Found .SUBCKT mismatches between nl and PDBA file near "%s".

**Description**

Found .SUBCKT mismatches between nl and PDBA file near "%s".

**What Next**

You should revise your nl or PDBA to make them consistancy.

---

## SX-1202

(Warning) Unrecognized syntax in CALIBRE_AGF_NAME_MAP file "%s" at line number %d.

**Description**

StarRC can not recongize the syntax in CALIBRE_AGF_NAME_MAP file "%s" at line %d.

**What Next**

You should check the syntax in CALIBRE_AGF_NAME_MAP. If there have new syntax which StarRC can not support, you should contact Synopsys customer support.

---

## SX-1203

(Fatal) Cell Extents file %s does not exist.

Feedback

**Description**

Calibre Cell Extents file does not exist.

**What Next**

You should check that the Calibre Cell Extents file defined in calibre query file or calibre runset does exist or not.

---

## SX-1204

(Fatal) Top block only format of CALIBRE_CELLS_PORTS file is unsupported. Please use PORT TABLE CELLS WRITE command instead.

**Description**

Top block is not acceptable in CALIBRE_CELLS_PORTS.

**What Next**

You should revise your CALIBRE_CELLS_PORTS to remove the top block, or use the default CALIBRE_CELL_PORTS defined in calibre query command file.

---

## SX-1206

(Warning) CELLS_PORTS file references unknown cell "%s" on line number %d. SKIP_CELLS must be defined in the AGF_NETLIST. This entry will be ignored.

**Description**

The cell "%s" in CELLS_PORTS file is not recognized by StarRC. This entry will be ignored.

**What Next**

SKIP_CELLS must be defined in the AGF_NETLIST. You should check whether the cell is defined in CALIBRE_AGF_NETLIST (CCI flow) or ICV_ANNOTATED_LAYOUT_NETLIST (ICV AGDS flow) or not. If not, you should fix the mismatch between the CELLS_PORTS file and AGF_NETLIST.

---

## SX-1207

(Warning) Non-existent or removed layer name "%s" found in CELLS_PORTS file on line %d. This entry will be ignored.

Feedback

**Description**

The layer name "%s" in CELLS_PORTS file on line %d is not acceptable, and this entry will be ignored.

**What Next**

You should look into the MAPPING_FILE to check this layer is not exist or defined in the remove_layers section.

---

## SX-1208

(Warning) Encountered port "%s" in CELLS_PORTS_FILE at line %d which does not exist in the AGF_NETLIST. This entry will be ignored.

**Description**

The port name "%s" in CELLS_PORTS file on line %d is not exist in AGF_NETLIST, and this entry will be ignored.

**What Next**

You should check the CALIBRE_CELLS_PORTS file (CCI flow) or ICV_ANNOTATED_CELL_PORTS file (ICV AGDS flow) is compatible with the CALIBRE_AGF_NETLIST file (CCI flow) or ICV_ANNOTATED_LAYOUT_NETLIST file (ICV AGDS flow) or not.

---

## SX-1209

(Fatal) Unable to parse CALIBRE_AGF_CELL_EXTENTS command; expected format is: CALIBRE_AGF_CELL_EXTENTS: <llx> <lly> <urx> <ury> Coordinates are inmicrons.

**Description**

Unable to parse CALIBRE_AGF_CELL_EXTENTS command. Because StarRC can not support the input format.

**What Next**

You should revise the format of CALIBRE_AGF_CELL_EXTENTS. The expected format is: CALIBRE_AGF_CELL_EXTENTS: <llx> <lly> <urx> <ury> Coordinates are inmicrons.

---

## SX-1212

(Warning) CALIBRE_AGF_CELL_EXTENTS bounding box is incorrect. Please specify coordinates which result in a positive chip area; <llx> <lly> <urx> <ury>

Feedback

**Description**

CALIBRE_AGF_CELL_EXTENTS bounding box is incorrect because the bounding box has negative area(urx less than llx or ury less than lly).

**What Next**

You should specify coordinates which result in a positive chip area; <llx> <lly> <urx> <ury>

---

## SX-1213

(Warning) CALIBRE_AGF_CELL_EXTENTS are too large; '%d'umx'%d'umy.

**Description**

CALIBRE_AGF_CELL_EXTENTS are too large. If width or height bounding box is above 1e6 um, StarRC will report this warning.

**What Next**

You should check the CALIBRE_AGF_CELL_EXTENTS is reasonable or not.

---

## SX-1214

(Warning)INTERNAL Error: %s option is not set.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1215

(Warning) LVS_EXTRACTION_REPORT_FILE or CALIBRE_RUNSET option is not set.

**Description**

LVS_EXTRACTION_REPORT_FILE or CALIBRE_RUNSET option is not set.

**What Next**

CALIBRE_RUNSET option is set by user. LVS_EXTRACTION_REPORT_FILE option comes from calibre query command file or comes from user. You should check your calibre command file or StarRC command file is correct or not.

Feedback

## SX-1216

(Fatal) Please set the above missing options.

**Description**

There have some missed calibre runset files in this case.

**What Next**

You should set option in "SX-1214" to make the CCI runset complete. Or you should check you calibre input file(CALIBRE_RUNSET or CALIBRE_QUERY_FILE).

## SX-1217

(Warning) CASE_SENSITIVE: NO is required for NET_TYPE: LAYOUT selection when LVS_EXTRACTION_REPORT_FILE or CALIBRE_RUNSET sets LAYOUT CASE NO; setting CASE_SENSITIVE: NO...

**Description**

CASE_SENSITIVE: NO is required for NET_TYPE: LAYOUT selection when LVS_EXTRACTION_REPORT_FILE or CALIBRE_RUNSET sets LAYOUT CASE NO. But in this case, the CASE_SENSITIVE:YES, it has a conflict between StarRC option and calibre runset. Here set the CASE_SENSITIVE:NO to make compatible with calibre runset.

**What Next**

You should check why the CASE_SENSITIVE used in starRC command file have conflict with calibre runset.

## SX-1218

(Fatal) BLOCK: %s is not an equivalence point.

**Description**

You receive this message if the string value of BLOCK in StarRC command file cannot in .nxf file outputted by Calibre.

**What Next**

You should check the BLOCK name according to the CELL_TYPE. If the CELL_TYPE: LAYOUT, you should fill in LAYOUT block name. If the CELL_TYPE: SCHEMATIC, you should fill in SCHEMATIC name. If the format of .nxf file has changed, you should contact Synopsys Support Center for enhancement.

Feedback

## SX-1219

(Warning) No equivalence points are found in this design. BLOCK: %s will be extracted at device level (SKIP_CELLS: !*).

### Description

You receive this message if no cell name or only BLOCK name found in the .nxf file output by Calibre. And StarRC will extract this case without any skip_cell(SKIP_CELL:!*)

### What Next

You should check the .nxf is expected or not.

## SX-1220

(Warning) SKIP_CELLS: * is specified but no lower level equivalence points are found. BLOCK: %s will be extracted at device level (SKIP_CELLS: !*).

### Description

SKIP_CELLS: * is specified but no lower level equivalence points are found. BLOCK: %s will be extracted at device level (SKIP_CELLS: !*).

### What Next

You set SKIP_CELLS:* but no lower level equivalencepoint below top block. You should set SKIP_CELLS:!* in this case to avoid this warning.

## SX-1221

(Warning) SKIP_CELLS list member "%s" does not match any equivalence points and will be ignored.

### Description

SKIP_CELLS list member "%s" does not match any equivalence points and will be ignored.

### What Next

If you want to use SKIP_CELLS flow in this case , you should revise your SKIP_CELLS starRC command to match the cell name in this case. Or you should set SKIP_CELLS: !*.

## SX-1222

(Warning) SKIP_CELLS list member "%s" is not an equivalence point and will be ignored.

Feedback

**Description**

SKIP_CELLS list member "%s" is not an equivalence point and will be ignored.

**What Next**

If you want to use SKIP_CELLS flow in this case , you should revise your SKIP_CELLS starRC command to match the cell name in this case. Or you should set SKIP_CELLS: !*.

---

## SX-1223

(Warning) %s BLOCK: %s will be extracted at device level (SKIP_CELLS: !*).

**Description**

SKIP_CELLS is set but not match any cells in this case. StarRC set SKIP_CELLS: !* and top block %s will be extracted at device level.

**What Next**

You should check the SKIP_CELLS is set correctly or not. If you want to use SKIP_CELLS flow, you should set SKIP_CELLS and compatible with the case to avoid this warning message. If not, you should set SKIP_CELLS: !*.

---

## SX-1224

(Warning) %s LVS BOX cells will be added to SKIP_CELLS list automatically while the remaining hierarchies will be flattened.

**Description**

the value of SKIP_CELLS StarRC option do not match any cell in this case, but LVS BOX cell added to SKIP_CELLS list automatically. It means other hierarchies will be flattened and LVS BOX cells are in SKIP_CELLS list.

**What Next**

You should check the SKIP_CELLS value is correct or not. If you want add some cells into SKIP_CELLS, you should correct the SKIP_CELLS option. Or you should use SKIP_CELLS: !*.

---

## SX-1225

(Warning) Equivalence point "%s(L)::%s(S)" is an LVS BOX cell and must be a SKIP_CELL; adding this cell to the SKIP_CELLS list...

Feedback

**Description**

LVS BOX cell must be a SKIP_CELL. This case SKIP_CELLS not contains LVS BOX cell "%s(Layout)::%s(Schematic)". StarRC add this cell to the SKIP_CELLS list automatically.

**What Next**

You should add the LVS BOX cell into SKIP_CELLS list to avoid this warning message.

---

## SX-1226

(Fatal) CALIBRE_AGF_CELL_EXTENTS_FILE was imported but contains no entry for specified BLOCK "BLOCK". Please provide cell extents information manually for selected BLOCK with CALIBRE_AGF_CELL_EXTENTS or complete the content of CALIBRE_AGF_CELL_EXTENTS_FILE.

**Description**

ALIBRE_AGF_CELL_EXTENTS_FILE was imported but contains no entry for specified top block defined in BLOCK starRC command.

**What Next**

You should provide cell extents information manually for selected BLOCK with CALIBRE_AGF_CELL_EXTENTS or complete the content of CALIBRE_AGF_CELL_EXTENTS_FILE.

---

## SX-1227

(Fatal) Please provide cell extents information for BLOCK with either CALIBRE_AGF_CELL_EXTENTS or CALIBRE_AGF_CELL_EXTENTS_FILE.

**Description**

StarRC do not find any cell extents information for BLOCK.

**What Next**

You can fix this issue in three ways: 1. set CALIBRE_AGF_CELL_EXTENTS directly. 2. set CALIBRE_AGF_CELL_EXTENTS_FILE directly. 3. check the CALIBRE_AGF_CELL_EXTENTS in your calibre query command file is exsit or not.

---

## SX-1229

(Fatal) Expected integer %s instead of %s

Feedback

### Description

You receive this message if the precision defined in the DEVICE TABLE(defined in CALIBRE_QUERY_FILE) are not integer. You can find the precision defined in the 2nd column of the first line in the DEVICE TABLE which is outputted by Calibre.

### What Next

Calibre output DEVICE TABLE, and you should check there have issue in Calibre flow or not. If DEVICE TABLE format updated with Calibre in new version, please contact Synopsys Customer support for enhancement. If old Calibre output the precision not integer, please contact Calibre side.

---

## SX-1230

(Fatal) Invalid setting for %s; %d. Must be greater than zero.

### Description

You receive this message if the precision defined in the DEVICE TABLE(defined in CALIBRE_QUERY_FILE) less than zero. You can find the precision defined in the 2nd column of the first line in the DEVICE TABLE which is outputted by Calibre.

### What Next

Calibre output DEVICE TABLE, and you should check there have issue in Calibre flow or not. If DEVICE TABLE format updated with Calibre in new version, please contact Synopsys Customer support for enhancement. If old Calibre output the precision less than zero, please contact Calibre side.

---

## SX-1231

(Warning) Suspicious value for %s; %d. Expected value divisible by 10.

### Description

You receive this message if the precision defined in the DEVICE TABLE (defined in CALIBRE_QUERY_FILE) cannot be divided by 10. You can find the precision defined in the 2nd column of the first line in the DEVICE TABLE.

### What Next

Calibre output DEVICE TABLE, and you should check there have issue in Calibre flow or not. If DEVICE TABLE format updated with new Calibre new version, please contact Synopsys Customer support for enhancement. If old Calibre output the precision cannot be divided by 10, please contact Calibre side.

Feedback

## SX-1232

(Fatal) The number of terminals of device "%s" is inconsistent between runset and pin file.

**Description**

In calibre DEVICE_TABLE, the number of terminals in device "%s" is different with CALIBRE_OPTIONAL_DEVICE_PIN_FILE.

**What Next**

You should check compatibility between calibre runset and CALIBRE_OPTIONAL_DEVICE_PIN_FILE.

## SX-1233

(Fatal) Pin %d of device "%s" has inconsistent layer between runset and pin file.

**Description**

In calibre DEVICE_TABLE, the layer of terminals in device "%s" is different with CALIBRE_OPTIONAL_DEVICE_PIN_FILE.

**What Next**

You should check compatibility between calibre runset and CALIBRE_OPTIONAL_DEVICE_PIN_FILE.

## SX-1234

(Severe) Parse error on line %d in CALIBRE_DEVTAB.

**Description**

StarRC not support the syntax in the line %d in calibre DEVICE TABLE.

**What Next**

You should check the Calibre DEVICE TABLE which is defined in CALIBRE_QUERY_FILE is correct or not. If there have new syntax defined in Calibre DEVICE TABLE, you should contact Synopsys Support Center.

## SX-1236

(Warning) CALIBRE_RUNSET does not contain LVS POWER/GROUND NAME setting and POWER_NETS option is empty

Feedback

**Description**

CALIBRE_RUNSET does not contain LVS POWER/GROUND NAME setting and POWER_NETS option is empty.

**What Next**

You should check your runset and put the power nets into your runset and write into POWER_NETS if there have power net in this case.

---

## SX-1237

(Warning) CALIBRE_RUNSET contains invalid setting for UNIT LENGTH parameter: %s. Supported values are numeric or u mil mm cm inch or m. Default value of u will be applied.

**Description**

StarRC not reconginze the UNIT LENGTH parameter in CALIBRE_RUNSET, so StarRC set the default value u in this case.

**What Next**

You should revise UNIT LENGTH in the CALIBRE_RUNSET using the value which StarRC can support. Now StarRC only support numeric or u mil mm cm inch or m.

---

## SX-1238

(Warning) CALIBRE_RUNSET INCLUDE file "%s" does not exist. This statement will be ignored.

**Description**

In CALIBRE_RUNSET, INCLUDE file does not exist. StarRC ignore this statement.

**What Next**

You should check the INCLUDE file does exist or not. If you input the wrong file location, you should correct it. If you do not want this INCLUDE file, you should remove this line from CALIBRE_RUNSET.

---

## SX-1239

(Warning) CALIBRE_RUNSET pre-processor directives which refer to shell or ENV variables are not supported. Conditional expressions which refer to external variables will evaluate as though the variable is undefined. Please see layers.sum for a full list of affected directives.

Feedback

**Description**

StarRC can not support the shell or ENV variables in CALIBRE_RUNSET pre-processor directives. So if you use '$' in your pre-processor directives, StarRC will not recognize this variables.

**What Next**

You should remove '$' in your pre-processor directives and use the defined variables in the CALIBRE_RUNSET to use directives.

---

## SX-1240

(Information) CALIBRE_RUNSET environment variable interpreted as undefined; '%s'@'%d': '%s'

**Description**

StarRC can not support the shell or ENV variables in CALIBRE_RUNSET pre-processor directives at line '%d'. So if you use '$' in your pre-processor directives, StarRC will not recognize this variables.

**What Next**

You should remove '$' in your pre-processor directives and use the defined variables in the CALIBRE_RUNSET to use directives.

---

## SX-1241

(Warning) CALIBRE_RUNSET contains unsupported pre-processor directive "%s". Supported directives are DEFINE UNDEFINE IFDEF IFNDEF ELSE and ENDIF. The following directive will be ignored; '%s'@'%d': '%s'

**Description**

the pre-processor directives only support DEFINE UNDEFINE IFDEF IFNDEF ELSE and ENDIF. If directives do not include these keywords, StarRC will ignore these directives.

**What Next**

You should check the directives with this message is necessary or not. You should remove these directives in CALIBRE_RUNSET if they are not necessary. Or you should revise the CALIBRE_RUNSET.

Feedback

## SX-1242

(Warning) Composite STAMP BY operations are not supported in CALIBRE_RUNSET_FILE (line:%d). Derived connectivity for runset layer "%s" will be ignored;this layer will not be allowed in MAPPING_FILE.

### Description

Composite STAMP BY operations are not supported in CALIBRE_RUNSET_FILE. StarRC ingore this operation and the layer defined in this operation is not allowed in MAPPING_FILE.

### What Next

StarRC do not support composite operations in CALIBRE_RUNSET_FILE. You should revise your calibre runset file and simplify the operations.

## SX-1243

(Warning) Composite COPY operations not supported in CALIBRE_RUNSET_FILE (line:%d). Derived connectivity for runset layer "%s" will be ignored; this layer will not be allowed in MAPPING_FILE.

### Description

Composite COPY operations are not supported in CALIBRE_RUNSET_FILE. StarRC ingore this operation and the layer defined in this operation is not allowed in MAPPING_FILE.

### What Next

StarRC do not support composite operations in CALIBRE_RUNSET_FILE. You should revise your calibre runset file and simplify the operations.

## SX-1244

(Warning) Skip encrypted lines from line @%d in CALIBRE_RUNSET.

### Description

If the line in CALIBRE_RUNSET is begin with #DECRYPT, StarRC treats this line and below as encrypted lines until the line begin with #ENDCRYPT. And StarRC print this warning message. StarRC ignore all the encrypted lines in CALIBRE_RUNSET.

### What Next

You should check this line is encrypted as expected or not. If you want StarRC read the encrypted lines, you should decrypted these lines and remove #DECRYPT.

Feedback

## SX-1245

(Warning) Resume to parse runset after line @'%d' in CALIBRE_RUNSET

**Description**

If the line in CALIBRE_RUNSET begin with #ENDCRYPT, StarRC will resume to parse CALIBRE_RUNSET and believe the encrypt section has ended.

**What Next**

You should check this line is encrypted as expected or not. If you want StarRC read the encrypted lines, you should decrypted these lines and remove #DECRYPT and #ENDCRYPT.

## SX-1246

(Fatal) Invalid selection for MARKER_GENERATION. Must be AUTOMATIC or USER.

**Description**

MARKER_GENERATION must be set as AUTOMATIC or USER. Other settings is not acceptable.

**What Next**

You should use AUTOMATIC or USER in MARKER_GENERATION.

## SX-1247

(Fatal) INTERNAL Error:Cell %s doesn't exist in AGF.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## SX-1249

(Fatal) BLOCK_BOUNDARY must have an even number of entries [%d].

**Description**

BLOCK_BOUNDARY must have an even number of entries.

Feedback

**What Next**

You should revise your BLOCK_BOUNDARY as an even number to describe the block boundary.

---

## SX-1250

(Fatal) BLOCK_BOUNDARY must have at least 4 points.

**Description**

BLOCK_BOUNDARY must have at least 4 points to describe the block boundary.

**What Next**

You should revise your BLOCK_BOUNDARY to have at least 4 points to describe the block boundary.

---

## SX-1254

(Information) BLOCK Boundary = {%s }

**Description**

StarRC recognize the BLOCK boundary as '%s'.

**What Next**

You can check the block boundary is expected or not.

---

## SX-1256

(Warning)Internal: In cell %s, cannot find device "%s" in HN database.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1257

(Warning) LVS_EXTRACTION_REPORT_FILE does not contain LVS POWER/GROUND NAME setting and POWER_NETS option is empty

Feedback

**Description**

LVS_EXTRACTION_REPORT_FILE does not contain LVS POWER/GROUND NAME setting and POWER_NETS option is empty.

**What Next**

You should check you LVS POWER/GROUND NAME setting in you LVS_EXTRACTION_REPORT_FILE>

---

## SX-1258

(Fatal) LVS_EXTRACTION_REPORT_FILE contains invalid setting for UNIT_LENGTH parameter: %s. Supported values are numeric or u mil mm cm inch or m.Default value of u will be applied.

**Description**

LVS_EXTRACTION_REPORT_FILE contains invalid setting for UNIT_LENGTH parameter: %s. Supported values are numeric or u mil mm cm inch or m.Default value of u will be applied.

**What Next**

You should revise the UNIT_LENGTH parameter.

---

## SX-1259

(Fatal) LVS_EXTRACTION_REPORT_FILE contains invalid setting for PRECISION parameter: %d.

**Description**

LVS_EXTRACTION_REPORT_FILE contains invalid setting for PRECISION parameter. You receive this message because the PRECISION parameter is empty or is negative number.

**What Next**

You should revise the PRECISION parameter in LVS_EXTRACTION_REPORT_FILE.

---

## SX-1260

(Warning) Without NETLIST_DEVICE_LOCATION_ORIENTATION, PLACEMENT_INFO_FILE will always report angle=0 for devices.

Feedback

**Description**

This case do not use NETLIST_DEVICE_LOCATION_ORIENTATION:YES, and without NETLIST_DEVICE_LOCATION_ORIENTATION, PLACEMENT_INFO_FILE will always report angle=0 for devices.

**What Next**

You should check the NETLIST_DEVICE_LOCATION_ORIENTATION and PLACEMENT_INFO_FILE setting is expected or not. If you want StarRC output the device angle, you should set NETLIST_DEVICE_LOCATION_ORIENTATION:YES.

---

## SX-1261

(Warning)Internal: Some devices are missing terminal coordinate information. (%s, etc.)

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1262

(Warning) %s.

**Description**

This is a multiple purpose message. In save_eco command, `target_file_name exists. Remove it.' indicates that target_file_name is already existed and will be removed. `STAR_DIRECTORY does not exist.' and `ECO_DIRECTORY does not exist.' are warning messages, which warn the absence of directories and save_eco cannot be done. `target_file_name is a directory.' is another warning message indicating that save_eco cannot be done because target_file_name exists and is a directory.

**What Next**

`target_file_name exists. Remove it.' is information only; no action is needed. For `STAR_DIRECTORY does not exist.' and `ECO_DIRECTORY does not exist.,' make sure incremental extraction is done successfully. For `target_file_name is a directory.,' make sure target_file_name is a valid file name.

Feedback

## SX-1263

(Warning) One or more *.DEVTMPLT entries referenced in the CALIBRE_AGF_NETLIST body could not be associated with a corresponding CALIBRE_DEVTAB Device Entry. %s

### Description

One or more *.DEVTMPLT entries referenced in the CALIBRE_AGF_NETLIST body could not be associated with a corresponding CALIBRE_DEVTAB Device Entry.

### What Next

There have some mismatch between *.DEVTMPLT and CALIBRE_DEVTAB. You should ensure that the CALIBRE_AGF_NETLIST is generated in accordance with CCI usage guidelines and that the CALIBRE_DEVTAB is consistent with the CALIBRE_AGF_NETLIST for best results.

## SX-1264

(Warning) One or more element devices referenced in the CALIBRE_AGF_NETLIST body could not be associated with either a *.DEVTMPLT or CALIBRE_DEVTAB Device Entry. %s

### Description

One or more element devices referenced in the CALIBRE_AGF_NETLIST body could not be associated with either a *.DEVTMPLT or CALIBRE_DEVTAB Device Entry.

### What Next

There have some mismatch between *.DEVTMPLT and CALIBRE_DEVTAB. You should ensure that the CALIBRE_AGF_NETLIST is generated in accordance with CCI usage guidelines and that the CALIBRE_DEVTAB is consistent with the CALIBRE_AGF_NETLIST for best results.

## SX-1266

(Fatal) Cell "%s" is both SKIP_CELL and SKIP_PCELL. Please remove all cells specified in SKIP_PCELLS from SKIP_CELLS or set SKIP_CELLS:!*.

### Description

Cell "%s" is both SKIP_CELL and SKIP_PCELL.

### What Next

You should remove all cells specified in SKIP_PCELLS from SKIP_CELLS or set SKIP_CELLS:!*.

Feedback

## SX-1267

(Fatal) Calibre runset syntax error! Improper specification of conditional pair(#IFDEF #ENDIF).

**Description**

#IFDEF and #ENDIF must be a pair in the CALIBRE_RUNSET file. But in this CALIBRE_RUNSET file, only have #ENDIF no #IFDEF.

**What Next**

You should add #IFDEF in the CALIBRE_RUNSET to make the conditional pair completely.

## SX-1308

(Fatal) Layer can not be negative at line %d

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

## SX-1309

(Fatal) Layer %d is larger than max_layer_no (%d) at line %d

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

## SX-1310

(Fatal) Syntax error while parsing command file at line %d

**Description**

Not available.

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## SX-1311

(Fatal) MAX_LAYER_NO must be positive

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1312

(Fatal) Search distance is set to an invalid value: %d

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1313

(Fatal) Unknown operand: %s

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1314

(Fatal) Invalid option %s in StarRC command file.

**Description**

The option specified in the StarRC command file is invalid.

Feedback

**What Next**

Remove this option from the StarRC command file and run again.

---

## SX-1317

(Fatal) Unsupported FS command

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1318

(Warning) Preprocessor encountered a nonrectangular shape, \it will be passed to output without any processing

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1319

(Fatal) Must enter either MAX_LAYER_NO or LAYER_FILE

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1322

(Fatal) Please specify input file using -i

**Description**

Not available.

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## SX-1323

(Fatal) Please specify output file using -out

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1326

(Fatal) Usage: xincompare2 <xin1> <xin2> <max_layer_no> <out>

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1327

(Fatal) %s

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1328

(Fatal) Please provide OA Layer Mapping File

**Description**

Not available.

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## SX-1333

(Fatal) Command file %s doesn't exist.

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1334

(Fatal) Please specify input file using -i

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1335

(Fatal) Please specify output file using -out

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1336

(Warning) sidvwd table incorrect; return input density

**Description**

Not available.

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## SX-1337

(Warning) Layers in density computation have no grd layers

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1338

(Warning) No thickness vs density secified in itf for layer %s. Why are we in density computation for it?

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1339

(Warning) Maximum size of all density box widths is zero; assuming default box width of %d microns

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1340

(Warning) Problem in density matrix, num_alive: %d num_resident %d

**Description**

Not available.

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## SX-1341

(Warning) %s edges remained in edge_scheduler

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1342

(Warning) GrdInfoNIT needs grd and layers info

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1343

(Warning) Layer %s is not in %s

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1346

(Fatal) Maximum number of cascaded AND operations (%d) is exceeded.

**Description**

Not available.

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## SX-1347

(Fatal) Usage: test <control_file> <input xin> <output xin>

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1348

(Fatal) nxtgrd file is missing for ThicknessIT

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1349

(Warning) FSpreprocess ATTENTION! More than 50% variation in metal thickness is detected in applying POLYNOMIAL_BASED_THICKNESS_VARIATION table.

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1350

(Information) More than 50% thickness variation in conductor level=%d at around (%d, %d)

**Description**

Not available.

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## SX-1351

(Fatal) conductor thickness is almost zero

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1352

(Fatal) StarXtract -cdnlicsvr unable to open output files %s

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1353

(Fatal) xin is not sorted- last left %d violating trap left %d

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1355

(Fatal) STARRC_AGF_NETLIST is missing.

**Description**

Direct DB flow needs STARRC_AGF_NETLIST as input for connectivity.

Feedback

**What Next**

Add STARRC_AGF_NETLIST in StarRC command file.

---

## SX-1356

(Fatal) File %s listed in STARRC_AGF_LIST doesn't exist.

**Description**

Files specified in STARRC_AGF_LIST must exist.

**What Next**

Provide correct filenames for STARRC_AGF_LIST.

---

## SX-1357

(Fatal) STARRC_AGF_LIST format: <file name> <mapping file name> <top-level macro name>

**Description**

The STARRC_AGF_LIST file format is: <file name> <mapping file name> <top-level macro name>

**What Next**

Follow the STARRC_AGF_LIST file format.

---

## SX-1358

(Warning) Device "%s" is not defined in STARRC_AGF_CONFIGURATION_FILE.

**Description**

StarRC use device definition in STARRC_AGF_CONFIGURATION_FILE
to determine the device terminal layers. If the device isn't defined in
STARRC_AGF_CONFIGURATION_FILE, StarRC cannot detect device terminal locations
properly.

**What Next**

Define this device in STARRC_AGF_CONFIGURATION_FILE.

---

## SX-1359

(Fatal) Error opening file '%s': %s

Feedback

**Description**

StarRC cannot open this file.

**What Next**

Provide correct filename.

---

## SX-1364

(Warning) Unknown GDS record '0x%x'

**Description**

StarRC parsed an unknown GDS record. This record is ignored.

**What Next**

Make sure this is a correct GDS file.

---

## SX-1384

(Warning) Polygon with property 30=CELL_BOUNDARY_POLYGON is missing. StarRC calculates the boundary box by iterating through all polygons in the top cell.

**Description**

StarRC uses the polygon with property 30=CELL_BOUNDARY_POLYON in the GDSII file as the boundary box for the top cell. If this information doesn't exist, StarRC iterates through the whole design to calculate the correct boundary box, requiring additional runtime.

**What Next**

Provide a polygon with property 30 in the GDSII file to specify the correct boundary box and reduce StarRC runtime.

---

## SX-1386

(Fatal) Could not find liboaReader-O.so in LD_LIBRARY_PATH.

**Description**

StarRC uses dynamic library liboaReader-O.so to read OpenAccess database.

**What Next**

Check whether your StarRC installation is correct.

Feedback

## SX-1387

(Fatal) A dynamic linking error occurred: %s

**Description**

StarRC is unable to link the dynamic library liboaReader-O.so

**What Next**

Check whether your StarRC installation is correct.

## SX-1389

(Fatal) Could not load symbols from OA.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## SX-1398

(Fatal) Specified models file %s is not a valid binary models database

**Description**

This is an internal fatal message

**What Next**

Contact the Synopsys Support Center.

## SX-1399

(Fatal) Specified models file %s is not compatible with this executable release.

**Description**

This is an internal fatal message

**What Next**

Contact the Synopsys Support Center.

Feedback

## SX-1400

(Fatal) Unknown marker %d encountered. Internal models database is corrupted.

**Description**

This is an internal fatal message

**What Next**

Contact the Synopsys Support Center.

## SX-1402

(Fatal) Could not open file "%s".

**Description**

StarRC cannot open one of the files specified in the SKIP_CELL_PORT_PROP_FILE command.

**What Next**

Ensure that the reported file exists and that you have read permission.

## SX-1405

(Warning) Skipping invalid line %d in PIO_FILE.

**Description**

There is invalid syntax in PIO_FILE. For pin direction, only "IN", "OUT", "BIDIR" are accepted.

**What Next**

Please correct the syntax in PIO_FILE.

## SX-1409

(Warning) Element device "%s" is not defined.

**Description**

A device definition is missing. Standard SPICE terminal order and names are used for instances of these devices in the ideal netlist.

Since the terminal layers of these devices cannot be identified, there are several possible consequences for parasitic extraction or netlisting: a) IGNORE_CAPACITANCE for these

Feedback

devices will not be correct. b) Shorts or opens might be reported for nets connected to instances of these devices. c) Parasitic structures for instances of these devices (such as *|I or instance calls in the instance section) might not be netlisted.

**What Next**

For best results, include all device definitions in the input.

---

## SX-1410

(Warning) Parse error at line %d in "%s".

**Description**

There is parse error in reported spice file.

**What Next**

Please check the syntax and correct it.

---

## SX-1412

(Warning) Missing .SUBCKT for referenced model "%s".

**Description**

The reported model does not has .SUBCKT defination in spice file.

**What Next**

Please check spice syntax and correct it.

---

## SX-1413

(Warning) Cannot determine parameter identity '%s'.

**Description**

The reported parameter in spice file does not have expected format.

**What Next**

Check spice syntax and correct it.

---

## SX-1414

(Warning) Bad parameter found in line: '%s'.

Feedback

**Description**

The reported parameter in spice file does not have expected format.

**What Next**

Check spice syntax and correct it.

---

## SX-1415

(Warning) Cell Name is not found for instance '%s'

**Description**

The cell name is not found for reported instance. The connection may have issue.

**What Next**

Check spice syntax and correct it.

---

## SX-1417

(Warning) Wrong Format in SKIP_CELL_PORT_PROP_FILE: %s

**Description**

You receive this message because there is more that 3 entries provided.

**What Next**

Please correct SKIP_CELL_PORT_PROP_FILE syntax. Expected syntax:

CELL cell1name pin1name pin1io pin_cap pin2name pin2io pin_cap

---

## SX-1418

(Warning) Running multiple netlist output flow, there might be mismatch between OA and %s output due to some flow-dependent commands.

**Description**

You receive this message because you are running multiple netlist output flow with OA_MULTI_OUTPUT. There are some star commands may be applicable to OA flow only and not to SPF/SPEF flow and also vice versa, these commands may cause mismatch between OA and SPF/SPEF outputs.

**What Next**

This is only a warning message. No action is required.

Feedback

## SX-1429

(Information) DEF macro %s does not have correspondent macro LEF. Use pin info from DEF instead.

**Description**

For those DEF macros which have correspondent LEF macro definitions, StarRC combine ports from both sources to form a complete port list for a cell. But, if a DEF macro has no correspondent LEF macro definition, StarRC will only use port info from DEF.

**What Next**

Information only; no action needed.

## SX-1430

(Fatal) Could not rename file %s to %s

**Description**

An internal file operation has failed.

**What Next**

Check if the working directory is full. If not, contact the Synopsys Support Center.

## SX-1433

(Warning) NETLIST_IDEAL_SPICE_TYPE is set to SCHEMATIC, but XREF is set to NO. NETLIST_IDEAL_SPICE_FILE will not be created.

**Description**

XREF is set to NO. Set NETLIST_IDEAL_SPICE_TYPE to SCHEMATIC is invalid. NETLIST_IDEAL_SPICE_FILE will not be created.

**What Next**

Correct the StarRC command file settings.

## SX-1434

(Warning) Can only attach DHE information to layout ideal spice file in XREF:NO run. Ignoring DHE...

Feedback

**Description**

NETLIST_IDEAL_SPICE_TYPE is set to LAYOUT but XREF is not set to NO. StarRC cannot attach dual-hiararchy extraction flow (DHE) in NETLIST_IDEAL_SPICE_FILE.

**What Next**

Check StarRC command file settings.

---

## SX-1435

(Warning) NETLIST_IDEAL_SPICE_HIER: YES is not allowed with CALIBRE_PDBA_FILE. Ignoring CALIBRE_PDBA_FILE setting...

**Description**

NETLIST_IDEAL_SPICE_HIER:YES doesn't work with CALIBRE_PDBA_FILE. CALIBRE_PDBA_FILE will be ignored.

**What Next**

Check StarRC command file settings.

---

## SX-1436

(Warning) NETLIST_IDEAL_SPICE_HIER: YES is not allowed with ICV_ANNOTATION_FILE . Ignoring ICV_ANNOTATION_FILE setting...

**Description**

NETLIST_IDEAL_SPICE_HIER:YES does not work with ICV_ANNOTATION_FILE. Information from ICV_ANNOTATION_FILE will be ignored.

**What Next**

Check StarRC command file settings.

---

## SX-1437

(Warning) %s

**Description**

The conditions that NETLIST_TYPE can specify must be covered by EXTRACTION and COUPLE_TO_GROUND commands.

**What Next**

Check StarRC command file settings.

Feedback

## SX-1438

(Warning) L or K must be used together with R.

**Description**

You must use R in NETLIST_TYPE with L and K.

**What Next**

Check StarRC command file settings.

## SX-1439

(Information) NETLIST_TYPE: Cc means netlisting both coupling and capacitance to ground.

**Description**

You specify both Cc and Cg in NETLIST_TYPE. It means netlisting both coupling and capacitance to ground.

**What Next**

Information only; no action is needed.

## SX-1440

(Warning) NETLIST_TYPE: no_couple option only goes with R|Cg|RCg|RL|RLCg.

**Description**

Cannot specify NO_COUPLE with CC.

**What Next**

Check StarRC command file settings.

## SX-1442

(Fatal) NETLIST_INDUCTANCE: YES only goes with NETLIST_TYPE: RLICc|RLICg|RLI|RLmCc|RLmCg|RLm.

**Description**

Cannot specify NETLIST_TYPE: K with NETLIST_INDUCTANCE:YES.

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## SX-1443

(Fatal) NETLIST_INDUCTANCE: NO only goes with NETLIST_TYPE: RKlCc|RKlCg|RKl|RKmCc|RKmCg|RKm.

**Description**

NETLIST_INDUCTANCE:NO cannot work with NETLIST_TYPE: Ll or NETLIST_TYPE: Lm

**What Next**

Check StarRC command file settings.

---

## SX-1444

(Fatal) NETLIST_FORMAT: SPEF does not support mutual inductance.

**Description**

NETLIST_FORMAT: SPEF does not work with NETLIST_TYPE: Lm

**What Next**

Contact the Synopsys Support Center.

---

## SX-1445

(Fatal) NETLIST_TYPE: RL|RLCg|RLCc is no longer supported. Try RLm or RLl.

**Description**

NETLIST_TYPE: RL|RLCg|RLCc is no longer supported. Try RLm or RLl.

**What Next**

Check StarRC command file settings.

---

## SX-1447

(Warning) NET "%s" is not set to be extracted in NETS and will not be netlisted.

Feedback

**Description**

NETLIST_SELECT_NETS and LOOP_OK_NETS can only contain nets that are specified in NETS.

**What Next**

Check StarRC command file settings. If this is expected, no action is needed.

---

## SX-1448

(Fatal) File : "%s" doesn't exist.

**Description**

Internal file does not exist.

**What Next**

Check the file system to see whether disk is full or connection is stable. Contact the Synopsys Support Center if receive this message again.

---

## SX-1449

(Fatal) Error(s) found while parsing commands

**Description**

StarRC encountered internal fatal error.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1451

(Fatal) Unknown command : %s

**Description**

StarRC encountered internal fatal error.

**What Next**

Retry with "StarXtract -clean" command. If receive this message again, contact the Synopsys Support Center.

Feedback

## SX-1452

(Fatal) Invalid command file : "%s" command line : "%s"

**Description**

Missing internal file.

**What Next**

Check the file system to see whether disk is full or connection is stable.

## SX-1454

(Warning) OVERLAP_FILL_SPACING set in mapping file is less than SMIN for layer "%s", will use SMIN.

**Description**

The keyword OVERLAP_FILL_SPACING is defined in conducting_layers section in StarRC mapping file. This keyword is used by metal fill reuse flow in an ECO loop. After ECO, some of the old metal fill structures might short to the new design changes. StarRC resolves these shorts by moving metal fill polygons away from signal nets by the value of OVERLAP_FILL_SPACING. However, the value you set for OVERLAP_FILL_SPACING is less than SMIN of that layer, which is not allowed in StarRC, so SMIN of that layer will be used instead of OVERLAP_FILL_SPACING to move metal fill polygons away from signal nets

**What Next**

To avoid this warning, you can set OVERLAP_FILL_SPACING to a value not less than the minimum spacing of that conductor layer. Or you can ignore this warning if minimum spacing of that conductor layer is also acceptable for OVERLAP_FILL_SPACING.

## SX-1455

(Warning) INTERNAL Warning: Invalid command no source: %s

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

Feedback

## SX-1456

(Warning) INTERNAL Warning: Invalid (number) resbody recognition layer: %s

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## SX-1457

(Warning) INTERNAL Warning: Unrecognizable Boolean Operation : %s

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## SX-1458

(Warning) INTERNAL Warning: Invalid command : %s

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## SX-1465

(Fatal) Could not write "%s"

**Description**

StarRC is not able to create a file.

**What Next**

Check the file system to see whether disk is full or connection is stable.

Feedback

## SX-1466

(Fatal) INTERNAL Error: Error found while processing edges.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## SX-1467

(Warning) INTERNAL Warning: Input file : %s is empty.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## SX-1469

(Warning) INTERNAL Warning: Invalid src layer: %s

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## SX-1470

(Warning) INTERNAL Warning: Unrecognizable Boolean Operation : %s

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

Feedback

## SX-1476

(Fatal) INTERNAL Error: unrecognizable line in fs.des file: %s

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## SX-1477

(Fatal) Please specify input file useing -i

**Description**

StarRC encountered fatal internal error.

**What Next**

Retry with "StarXtract -clean". If receive this message again, contact the Synopsys Support Center.

## SX-1478

(Fatal) Please specify output file using -o

**Description**

StarRC encountered fatal internla error.

**What Next**

Retry with "StarXtract -clean". If receive this message again, contact the Synopsys Support Center.

## SX-1479

(Fatal) Unknown boolean value : %s

**Description**

StarRC encountered fatal internal error.

**What Next**

Retry with "StarXtract -clean" option.

Feedback

## SX-1480

(Fatal) INTERNAL Error: Failed to create interal bin scheme for query.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## SX-1481

(Fatal) INTERNAL Error: Failed to allocate bin link memory.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## SX-1482

(Fatal) INTERNAL Error: Found Unknown code while processing MULTITABLE command

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## SX-1483

(Fatal) INTERNAL Error: Error found(invaild orientation) while processing Cfdata

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

Feedback

## SX-1484

(Warning) INTERNAL Warning: Invalid command no source: %s

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## SX-1485

(Warning) INTERNAL Warning: Unrecognizable Boolean Operation : %s

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## SX-1486

(Warning) INTERNAL Warning: Invalid command : %s

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## SX-1490

(Information) No overlap polygond found.

**Description**

StarRC didn't find overlap polygon.

**What Next**

You need to do nothing about this message.

Feedback

## SX-1491

(Warning) %d overlap polygons found and cleaned up. (See file overlap.sum)

**Description**

StarRC has cleaned up overlap polygons in design.

**What Next**

You need to do nothing about this message.

## SX-1492

(Warning) Overlap found around %s

**Description**

StarRC found overlap. This only report 1 location for your reference.

**What Next**

This is just information for your reference. If it is expected to have metal fill and signal overlap, you can ignore this message. Otherwise, please check your design in reported location.

## SX-1493

(Information) INTERNAL Warning: Layer %s is not in layers file

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## SX-1494

(Fatal) INTERNAL Error Unknown flag %d found

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

Feedback

## SX-1495

(Fatal) Error in setting FS_BOUNDARY_BOX

**Description**

FS_BOUNDARY_BOX only accept number inputs.

**What Next**

Only use number inputs for FS_BOUNDARY_BOX.

## SX-1496

(Warning) Some resistor body have too many terminals.

**Description**

Some resistor body have too many terminals. This may cause accuracy issue.

**What Next**

Check design database.

## SX-1497

(Fatal) Internal Error: Source Layer should exist

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## SX-1498

(Fatal) INTERNAL Error: Output/source Layer should exist

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

Feedback

## SX-1499

(Warning) Layer %s is not in mapping file.

**Description**

Layer not found in mapping file. StarRC will ignore it.

**What Next**

Set proper mapping information for all layers in design.

## SX-1500

(Warning) SKIP_PCELLS is not set along with SKIP_PCELL_LAYERS_FILE

**Description**

StarRC require to specify the PCELL or blocking layer, as well as the layers that are exploded, in a file specified by the SKIP_PCELL_LAYERS_FILE command.

**What Next**

Specify the PCELL or blocking layer with SKIP_PCELL_LAYERS_FILE.

## SX-1501

(Warning) PCELL %s is not set in SKIP_PCELLS (only print the first one), ignored

**Description**

PCELL is not set in SKIP_PCELLS. StarRC won't treat the PCELL as a fully characterized gray-box cell unit.

**What Next**

Check SKIP_PCELLS and SKIP_PCELL_LAYERS_FILE setting

## SX-1502

(Fatal) Parsing error in SKIP_PCELL_LAYERS_FILE at line %s

**Description**

Syntax error found in SKIP_PCELL_LAYERS_FILE.

**What Next**

Modify SKIP_PCELL_LAYERS_FILE.

Feedback

## SX-1503

(Fatal) Unrecognized input in SKIP_PCELL_LAYERS_FILE at %s

**Description**

Unrecognized input found in SKIP_PCELL_LAYERS_FILE.

**What Next**

Check SKIP_PCELL_LAYERS_FILE.

## SX-1504

(Fatal) Bad scale value in SKIP_PCELL_LAYERS_FILE at %s

**Description**

Scale parameter in SKIP_PCELL_LAYERS_FILE is not acceptable.

**What Next**

Modify scale parameter in SKIP_PCELL_LAYERS_FILE.

## SX-1505

(Warning) Layer "%s" is not found in database

**Description**

Layer specified in SKIP_PCELL_LAYERS_FILE is not found in database.

**What Next**

Check SKIP_PCELL_LAYERS_FILE and design database.

## SX-1506

(Warning) INTERNAL Warning: ITF layer "%s" is not found in nxtgrd file

**Description**

An internal consistency check has failed.

**What Next**

Check the input files MAPPING_FILE or SKIP_PCELL_LAYERS_FILE. Contact the Synopsys Support Center if StarRC terminates abnormally or other issues are found.

Feedback

## SX-1507

(Fatal) INTERNAL Error: Cannot find ws_mkrs hdr file

**Description**

Internal file does not exist.

**What Next**

Check the file system to see whether disk is full or connection is stable. Contact the Synopsys Support Center if receive this message again.

## SX-1508

(Fatal) ERROR: width=%d spacing=%d exceeding etch table size

**Description**

Etch table defined in ETCH_VS_WIDTH_AND_SPACING section is incorrect.

**What Next**

Check ETCH_VS_WIDTH_AND_SPACING section in ITF file.

## SX-1509

(Fatal) Some geometries missing.

**Description**

Some geometries missing in Field Solver stage.

**What Next**

Retry with "StarXtract -clean" option. If issue still exist, Contact the Synopsys Support Center.

## SX-1510

(Warning) More than 50% %s thickness variation(%d) in conductor level=%d at around (%d, %d) and (%d, %d)

**Description**

Thickness variation of certain polygon is more than 50% of it's thickness.

Feedback

**What Next**

Make sure the thickness variation setting in ITF file is correct.

---

## SX-1511

(Information) More polygons with more than 50% thickness variation

**Description**

More than 100 polygons with >50% thickness variation. StarRC will suppress this warning.

**What Next**

Make sure the thickness variation setting in ITF file is correct.

---

## SX-1512

(Warning) INTERNAL Error: Invalid command no source: %s

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1513

(Fatal) INTERNAL Error: Failed to find WS table for grd level " type(EtchVsWS)

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1514

(Fatal) INTERNAL Error: Bad bintable %s.

**Description**

An internal consistency check has failed.

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## SX-1516

(Warning) INTERNAL Warning Invalid command : %s

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1517

(Fatal) Conductor %s cannot placed on non-planner dielectric layer : %s

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1518

(Fatal) Can not find MEASURED_FROM(%s) for %s

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1519

(Fatal) Can not find FROM LAYER %s for VIA %s

**Description**

Not available.

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## SX-1520

(Fatal) Can not find TO LAYER %s for VIA %s

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1521

(Fatal) Can't find dielectric %s when converting itf to quickcap

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1522

(Fatal) No physical dielectric stack found

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1523

(Fatal) No conductor stack found

**Description**

Not available.

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## SX-1524

(Fatal) Can't find grd_layer of %s

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1525

(Fatal) Can't find %s in Itf2QuickCap

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1526

(Fatal) Can't find to_layer of %s

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1527

(Warning) Zmax of VIA %s is inconsistent with ITF file

**Description**

Not available.

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## SX-1528

(Fatal) Can't find %s in Itf2QuickCap

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1529

(Fatal) Can't find CONDUCTOR %s

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1530

(Fatal) Unsupported capacitance unit %s in QuickCap spice

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1531

(Fatal) Multiple totals for same net in QuickCap spice

**Description**

Not available.

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## SX-1532

(Fatal) Total-cap QuickCap spice can only contain '*cp' lines

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1533

(Fatal) unsupported capacitance format in QuickCap spice

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1534

(Information) Creating field solver input data

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1535

(Information) Done

**Description**

Not available.

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## SX-1536

(Information) Running Field Solver

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1537

(Fatal) Failed to run FS %s

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1539

(Warning) Errors found while process input database.

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1540

(Fatal) Invalid option for FS_EXTRA_COUPLING

**Description**

Not available.

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## SX-1541

(Fatal) Invalid option for FS_DEVICE_HANDLING

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1542

(Fatal) must specify FS_DEVICE_LAYERS for FS_DEVICE_HANDLING: FULL

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1543

(Fatal) Invalid option for FS_FLOW

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1544

(Fatal) FS_FLOW - NONE cannot be mixed with other options

**Description**

Not available.

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## SX-1545

(Fatal) FS_FLOW - must include RUN_FS with this set of options

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1546

(Fatal) QuickCap options do not include -spice. Star-RCFS analysis will not work.

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1547

(Fatal) QuickCap options do not include -p. Background execution is required for Star-RCFS.

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1548

(Fatal) FS_XREF_EXECUTABLE is required for XREF: YES or COMPLETE

**Description**

Not available.

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## SX-1549

(Fatal) Invalid value for FS_SCALE, must be > 0.0

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1550

(Warning) FS_CAP_USE_QC_EXTRACT is only supported with FS_EXTRACT_NETS.

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1551

(Warning) FS_RETAIN_BULK_LAYERS is only supported with FS_EXTRACT_NETS.

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1552

(Information) Running field solver with options:

**Description**

Not available.

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## SX-1553

(Information) %s

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1554

(Information) For progress see summary file 'fs/%s.cap.summary'

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1555

(Fatal) Field solver execution incomplete

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1556

(Information) Done

**Description**

Not available.

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## SX-1557

(Fatal) Could not write cap comparison file

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1558

(Fatal) Could not write quick cap file %s

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1559

(Fatal) invalid capacitance line found

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1560

(Fatal) INTERNAL Error: Failed to create interal bin scheme for query.

**Description**

An internal consistency check has failed.

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## SX-1561

(Fatal) INTERNAL Error: Failed to allocate internal bin link memory.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1562

(Fatal) INTERNAL Error: Corrupted hn file... could not find netname of %d.

**Description**

The net name corresponding to the net id is not found in HN db.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1563

(Fatal) Nothing selected in FS_EXTRACT_NETS.

**Description**

You receive this error message because there are no nets provided in FS_EXTRACT_NETS option of StarRC command file.

**What Next**

Provide names of the nets which need to be extracted by Field Solver in FS_EXTRACT_NETS option of StarRC command file.

---

## SX-1564

(Fatal) INTERNAL Error: Net index %d lookup failed... bad HN database.

**Description**

The net name corresponding to the net index is not found in HN database.

Feedback

### What Next

Contact the Synopsys Support Center.

---

## SX-1567

(Fatal) Could not write comparison output.

### Description

You receive this message because StarRC can't open the file for writing when reporting comparison results.

### What Next

Check whether the disk is full and rerun StarRC. If this message still occurs, contact Synopsys for more infomation.

---

## SX-1570

(Fatal) Could not write gnuplot cmd file.

### Description

You receive this message because StarRC can't open the file for writing when writing gnuplot cmd file.

### What Next

Check whether the disk is full and rerun StarRC. If this message still occurs, contact Synopsys for more infomation.

---

## SX-1572

(Fatal) FSCORRELATION is set to RC3, but could not find -rc3_cc in FSCOMPARE_OPTIONS.

### Description

You receive this message because the option FSCORRELATION is set as RC3 in StarRC command file, but -rc3_cc is not set in option FSCOMPARE_OPTIONS. When FSCORRELATION is set as RC3 in StarRC command file, -rc3_cc should be set in option FSCOMPARE_OPTION, the names of the nets which are intended to be extracted by RC3 should also be listed after -rc3_cc.

Feedback

**What Next**

Add -rc3_cc in FSCOMPARE_OPTIONS and list the names of the nets to be extracted by RC3 after -rc3_cc.

---

## SX-1573

(Fatal) FSCORRELATION is set to RC2, but could not find -rc2_cc in FSCOMPARE_OPTIONS.

**Description**

You receive this message because the option FSCORRELATION is set as RC2 in StarRC command file, but -rc2_cc is not set in option FSCOMPARE_OPTIONS. When FSCORRELATION is set as RC2 in StarRC command file, -rc2_cc should be set in option FSCOMPARE_OPTION, the names of the nets which are intended to be extracted by RC2 should also be listed after -rc2_cc.

**What Next**

Add -rc2_cc in FSCOMPARE_OPTIONS and list the names of the nets to be extracted by RC2 after -rc2_cc.

---

## SX-1574

(Fatal) Could not write correlation output.

**Description**

You receive this message because StarRC can't open the file for writing when reporting correlation results.

**What Next**

Check whether the disk is full and rerun StarRC. If this message still occurs, contact Synopsys for more infomation.

---

## SX-1575

(Fatal) Could not write comparison output.

**Description**

You receive this message because StarRC can't open the file for writing when reporting comparison results.

Feedback

### What Next

Check whether the disk is full and rerun StarRC. If this message still occurs, contact Synopsys for more infomation.

---

## SX-1578

(Fatal) Invalid value for FS_SCALE, must be > 0.

### Description

You receive this message because FS_SCALE option is StarRC command file is set to a value less than 0. FS_SCALE should be set to a positive value.

### What Next

Reset FS_SCALE to a value larger than 0.

---

## SX-1579

(Warning) FS_RETAIN_BULK_LAYERS layer "%s" is not a valid database layer or does not have a valid GRD Layer. Please verify MAPPING_FILE includes this layer in the CONDUCTING_LAYERS section and that the layer is not excluded from extraction.

### Description

You receive this message because the layer set in option FS_RETAIN_BULK_LAYERS is not a valid database layer or the database layer does not have a valid GRD Layer.

### What Next

Check whether the layers set in FS_RETAIN_BULK_LAYERS are database layers in the design; also check MAPPING_FILE to verify that the database layers set in FS_RETAIN_BULK_LAYERS are correctly mapped to GRD layers and are not excluded from extraction.

---

## SX-1580

(Warning) FS_RETAIN_BULK_LAYERS layer "%s" is not mapped to SUBSTRATE and will be ignored.

### Description

You receive this message because the layer set in FS_RETAIN_BULK_LAYERS is not mapped to SUBSTRATE.

Feedback

**What Next**

Check whether the layer is a bulk layer. If it's a bulk layer, map the layer to SUBSTRATE in mapping file.

---

## SX-1581

(Warning) FS_RETAIN_BULK_LAYERS is only supported for COUPLE_TO_GROUND: NO.

**Description**

You receive this message because while FS_RETAIN_BULK_LAYERS is set in StarRC command file, COUPLE_TO_GROUND is set to YES at the same time.

**What Next**

If FS_RETAIN_BULK_LAYERS is set, reset COUPLE_TO_GROUND option to NO.

---

## SX-1582

(Fatal) Failed to write "%s".

**Description**

You receive this message because StarRC can't write to the file.

**What Next**

Check whether the disk is full and rerun StarRC. If this message still occurs, contact Synopsys for more infomation.

---

## SX-1585

(Fatal) Could not write gnuplot data file.

**Description**

You receive this message because StarRC can't open the file for writing when writing gnuplot data file.

**What Next**

Check whether the disk is full and rerun StarRC. If this message still occurs, contact Synopsys for more infomation.

Feedback

## SX-1586

(Information) CapCompare: (Coupling) Avg = %f%% ; StdDev = %f%%

**Description**

This is an information message. It reports the summary of FSCOMPARE coupling results, including average error and standard deviation.

**What Next**

You need to do nothing about this message.

## SX-1587

(Fatal) Could not write %s.

**Description**

You receive this message because StarRC can not write to the file in FS_EXTRACT_NETS flow.

**What Next**

Check whether the disk is full and rerun StarRC. If the message still occurs, contact Synopsys for more information.

## SX-1589

(Warning) FS_RECONNECT_MODE: %d is illegal; reset to default.

**Description**

The FS_RECONNECT_MODE option can only be set to 0 or 1. When FS_RECONNECT_MODE is set to 1, the Field Solver preprocesser reconnect stage is enabled; when FS_RECONNECT_MODE is set to 0, the Field Solver preprocesser reconnect stage is disabled.

**What Next**

If you want to enable Field Solver preprocesser reconnect stage, set the option to 1; otherwise set it to 0.

## SX-1593

(Fatal) Could not write internal tech file.

Feedback

**Description**

StarRC can't open the file for writing when writing internal tech file.

**What Next**

Check whether the disk is full and rerun StarRC. If this message still occurs, contact Synopsys for more infomation.

---

## SX-1595

(Information) Post processing Field Solver net caps for multi-corner: %s.

**Description**

This information informs you that StarRC is post processing Field Solver net caps for multi-corner.

**What Next**

You don't need to take any action.

---

## SX-1596

(Information) Primary corner %s idx %d.

**Description**

This information informs you that the nxtgrd file for the primary corner and its corner number.

**What Next**

You don't need to take any action.

---

## SX-1597

(Information) Corner nxtgrd file %s idx %d.

**Description**

This information informs you that the nxtgrd file for the corner being handled and its corner number.

**What Next**

You don't need to take any action.

Feedback

## SX-1598

(Fatal) INTERNAL Error: Could not find %s in string list of CORNER_TCAD_GRD_FILES.

**Description**

This nxtgrd file is tended to be handled but it's not listed in CORNER_TCAD_GRD_FILES.

**What Next**

Contact the Synopsys Support Center.

## SX-1599

(Fatal) Could not write %s.

**Description**

StarRC can't open file to write when reporting Field Solver net caps.

**What Next**

Check whether the disk is full and rerun StarRC. If this message still occurs, contact Synopsys for more infomation.

## SX-1600

(Information) Post processing %s.

**Description**

This information informs you that StarRC is post processing Field Solver net caps for multi-corner.

**What Next**

You don't need to take any action.

## SX-1602

(Fatal) INTERNAL Error: Found error while processing field solver data on net: %d.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

Feedback

## SX-1603

(Fatal) INTERNAL Error: Field Solver data is corrupted.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## SX-1604

(Warning) INTERNAL Warning: Field Solver ignores coupling cap multiplication factor '%s' in '%s'.

**Description**

When there are more than 2 layers in IGNORE_CAP, Field Solver will ignore the other layers except first two layers.

**What Next**

If coupling cap between more than two layers needs to be ignored, instead of writing IGNORE_CAP layerA layerB layerC, should write: IGNORE_CAP layerA layerB IGNORE_CAP layerA layerC IGNORE_CAP layerB layerC

## SX-1605

(Fatal) INTERNAL Error: Parsing fs_ignore_cap_layer file: found invalid layer name %s.

**Description**

The layer shown in the message is not a valid database layer.

**What Next**

Contact Synopsys Customer Support. Check the layer name and confirm that a valid database layer is used.

## SX-1607

(Fatal) INTERNAL Error: Parsing fs_ignore_cap_layer file: found invalid layer names %s and %s.

Feedback

**Description**

The layers shown in the message are not valid database layers.

**What Next**

Contact Synopsys Customer Support. Check the layer names and confirm valid database layers are used.

---

## SX-1608

(Fatal) Failed to create pipe when getting Field Solver version.

**Description**

This is an error in Linux system which causes StarRC can't get Field Solver version.

**What Next**

Check Linux running status and rerun StarRC. If this error still occurs, contact Synopsys for more information.

---

## SX-1609

(Fatal) Failed to fork when getting Field Solver version.

**Description**

This is an error in system which causes StarRC can't get Field Solver version.

**What Next**

Check system running status and rerun StarRC. If this error still occurs, contact Synopsys for more information.

---

## SX-1610

(Fatal) INTERNAL Error: Failed to execute Field Solver: '%s'.

**Description**

You receive this message because Field Solver terminated abnormally.

**What Next**

Contact the Synopsys Support Center.

Feedback

## SX-1611

(Fatal) INTERNAL Error: Should find Field Solver's version, but failed.

**Description**

You receive this message because StarRC can't find Field Solver's version information.

**What Next**

Contact the Synopsys Support Center.

## SX-1612

(Fatal) INTERNAL Error: Failed to find Field Solver's version.

**Description**

You receive this message because StarRC can't find Field Solver's version information.

**What Next**

Contact the Synopsys Support Center.

## SX-1618

(Fatal) Failed to find Field Solver in search path.

**Description**

You receive this message because StarRC can't find Field Solver executable file in search path.

**What Next**

In order to run Field Solver, you should set correct Field Solver executable file path.

## SX-1619

(Fatal) Failed to create pipe when getting field solver version and built time.

**Description**

This is an error in system which causes StarRC can't get Field Solver version and built time.

Feedback

**What Next**

Check system running status and rerun StarRC. If this error still occurs, contact Synopsys for more information.

---

## SX-1620

(Fatal) Failed to fork when getting field solver version and built time.

**Description**

This is an error in system which causes StarRC can't get Field Solver version.

**What Next**

Check system running status and rerun StarRC. If this error still occurs, contact Synopsys for more information.

---

## SX-1621

(Fatal) INTERNAL Error: Failed to execute field solver: %s.

**Description**

Field Solver terminated abnormally.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1622

(Fatal) INTERNAL Error: Failed to find Field Solver's version.

**Description**

StarRC can't find Field Solver's version information.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1623

(Warning) Design '%s' is edit-locked by other process/person.

Feedback

**Description**

The opening design is edit-locked by other process/person. Although StarRC can open it in read-only mode, the design may be outdated and hence the translation result.

**What Next**

Information only.

---

## SX-1624

(Fatal) Could not duplicate stderr output.

**Description**

This is an error in system which makes StarRC can't duplicate stderr output.

**What Next**

Check system running status and rerun StarRC. If this error still occurs, contact Synopsys for more information.

---

## SX-1626

(Fatal) Could not duplicate stdout output.

**Description**

This is an error in system which makes StarRC can't duplicate stdout output.

**What Next**

Check system running status and rerun StarRC. If this error still occurs, contact Synopsys for more information.

---

## SX-1627

(Fatal) Failed to read fs.log file.

**Description**

StarRC can not read fs.log file.

**What Next**

Check whether fs.log exists. If fs.log exists, check read permission of the file. If fs.log doesn't exist, rerun StarRC. If this message still occurs after rerun, contact Synopsys for more information.

Feedback

## SX-1628

(Warning) Specified ranxt config file doesn't exist. Using default ranxt config file

**Description**

The ranxt config file specified in RANXT_CONFIG_FILE option doesn't exist. StarRC will use default ranxt config file instead.

**What Next**

Make sure that the ranxt config file specified in RANXT_CONFIG_FILE option exists.

## SX-1629

(Fatal) RANXT_RESOLUTION is obsolete in 2011.06. Please use FS_RESOLUTION instead.

**Description**

RANXT_RESOLUTION is an old StarRC command and has been replaced by FS_RESOLUTION from 2011.06.

**What Next**

Replace the command RANXT_RESOLUTION with FS_RESOLUTION.

## SX-1630

(Fatal) Invalid value for FS_RESOLUTION, must be greater than 0.

**Description**

You receive this message because the value set in FS_RESOLUTION is negative, it should be greater than 0.

**What Next**

Reset FS_RESOLUTION value to a number greater than 0.

## SX-1631

(Warning) FS_RESOLUTION value is too small, reset it to %f.

Feedback

**Description**

FS_RESOLUTION is used to set the box resolution during cutting non-Manhattan geometry to Manhattan boxes. By default, it's set as 0.1um. If its value is two small, it will be reset to a new value.

**What Next**

To avoid this warning, you need to set FS_RESOLUTION to a larger value than the reported number. Or you can ignore the warning if the reset number is acceptable for you.

---

## SX-1632

(Fatal) Could not write Field Solver technology file.

**Description**

You receive this message because StarRC can't write Field Solver technology file.

**What Next**

Check whether disk is full and rerun StarRC. If this message still occurs, contact Synopsys for more information.

---

## SX-1633

(Fatal) INTERNAL Error: Could not encrypt technology file.

**Description**

You receive this error message because StarRC can't encrypt Field Solver technology file.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1634

(Fatal) INTERNAL Error: Must read design to know if there is critical_substrate_polygon.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

Feedback

## SX-1635

(Warning) %s table is found, but QTF file is not set in FS flow.

**Description**

In order to run FS flow with %s defined in ITF, QTF should be specified to run StarQTF flow to get field solver results.

**What Next**

Turn off FS flows (do not use EXTRACTION: FSCOMPARE, or FS_EXTRACT_NETS) if you do not have a valid QTF file, or you could add the correct QTF file by using FS_QTF_FILE command.

## SX-1636

(Fatal) RANXT_BOUNDARY_BOX is obsolete in 2011.06. Please use FS_BOUNDARY_BOX instead

**Description**

RANXT_BOUNDARY_BOX is an old StarRC command and has been replaced by FS_BOUNDARY_BOX from 2011.06.

**What Next**

Replace the command RANXT_BOUNDARY_BOX with FS_BOUNDARY_BOX.

## SX-1637

(Fatal) Error in setting FS_BOUNDARY_BOX.

**Description**

You receive this error message because the setting of FS_BOUNDARY_BOX command is wrong.

**What Next**

The syntax of the command is: "FS_BOUNDARY_BOX: x_min y_min z_min x_max y_max z_max". You can refer to StarRC command reference for more information about setting FS_BOUNDARY_BOX.

## SX-1639

(Fatal) Failed to write out FS_BOUNDARY_BOX option.

Feedback

**Description**

You receive this error message because StarRC can't write out FS_BOUNDARY_BOX option to file.

**What Next**

Check whether the disk is full and rerun StarRC. If the message still occurs, contact Synopsys for more information.

---

## SX-1640

(Fatal) RANXT_BOUNDARY_CONDITION is obsolete in 2011.06. Use FS_BOUNDARY_CONDITION instead.

**Description**

RANXT_BOUNDARY_CONDITION is an old StarRC command and has been replaced by FS_BOUNDARY_CONDITION from 2011.06.

**What Next**

Replace the command RANXT_BOUNDARY_CONDITION with FS_BOUNDARY_CONDITION.

---

## SX-1641

(Fatal) Bad FS_BOUNDARY_CONDITION; please check: %s.

**Description**

You receive this message because you make a wrong setting for FS_BOUNDARY_CONDITION.

**What Next**

The syntax of FS_BOUNDARY_CONDITION is: "FS_BOUNDARY_CONDITION: [-bc_xn N | P] [-bc_xp N | P] [-bc_yn N | P] [-bc_yp N | P]". You can refer to StarRC command reference for more information.

---

## SX-1642

(Fatal) Failed to write out Field Solver boundary condition.

**Description**

You receive this error message because StarRC can't write out Field Solver boundary condition to file.

Feedback

**What Next**

Check whether the disk is full and rerun StarRC. If the message still occurs, contact Synopsys for more information.

---

## SX-1643

(Fatal) Obsoleted setting RANXT_DP_STRING.

**Description**

RANXT_DP_STRING has been obsoleted in 2011.06.

**What Next**

Use FS_DP_STRING instead. Refer to "Distributed Processing" section in StarRC manual for more information.

---

## SX-1644

(Information) Creating Field Solver input data.

**Description**

This information informs you that StarRC is writing Field Solver input data.

**What Next**

Information only, no action is needed.

---

## SX-1645

(Information) Done.

**Description**

This information informs you that the writing of Field Solver input data has finished.

**What Next**

Information only, no action is needed.

---

## SX-1646

(Information) Running Field Solver(%d).

**Description**

This information informs you that Field Solver has been invoked and is running.

Feedback

**What Next**

Information only, no action is needed.

---

## SX-1647

(Information) Invoking Field Solver with command:%s.

**Description**

This information informs you the command Field Solver is invoked with.

**What Next**

You can check the information to see whether the options in the command are set correctly. If any of the options are set incorrectly, you can reset them in FSCOMPARE_OPTIONS and rerun StarRC. You can also refer to StarRC command reference for the Field Solver options.

---

## SX-1649

(Fatal) Failed to run Field Solver.

**Description**

StarRC failed to run Field Solver due to unexpected reason.

**What Next**

Check <star directory>/fs.log file for detailed failure information.

---

## SX-1650

(Information) Done.

**Description**

This information informs you that Field Solver has finished running.

**What Next**

Information only, no action is needed.

---

## SX-1651

(Warning) INTERNAL Warning: Could not open fs_drop_layer file for reading.

Feedback

**Description**

StarRC can't open specified file for reading.

**What Next**

Contact the Synopsys Customer Support.

---

## SX-1652

(Fatal) Could not write Field Solver file %s.

**Description**

StarRC can't write specified Field Solver file.

**What Next**

Check whether the disk is full and rerun StarRC. If this message still occurs, contact Synopsys for more infomation.

---

## SX-1653

(Fatal) INTERNAL Error: Net index %d lookup failed... bad HN database.

**Description**

The net name corresponding to the net index is not found in HN database.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1654

(Fatal) INTERNAL error: net index '%d' lookup failed... bad HN database

**Description**

This is an internal fatal error in reading internal database

**What Next**

Contact the Synopsys Support Center.

---

## SX-1657

(Fatal) INTERNAL Error: Line %d unsupported Field Solver report format.

Feedback

**Description**

An internal consistency check has failed. StarRC can't recognize the Field Solver report on that line.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1658

(Fatal) INTERNAL Error: Unsupported capacitance format in Field Solver report.

**Description**

An internal consistency check has failed. StarRC can't recognize the capacitance format in Field Solver report.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1661

(Fatal) Multi-equivs are not yet supported for Calibre input. Layout cell "%s" is already assigned to "%s" and cannot be re-assigned to "%d".

**Description**

It is not supported that one layout cell equives to multiple schematic cell.

**What Next**

Check XREF data in Calibre input.

---

## SX-1662

(Warning) Schematic name "%s" is duplicated in XREF name table within equivalence point "%s(L):%s(S)", Original: %s(L) == %s(S) Duplicate: %s(L) == %s(S) %s will be netlisted with layout name.

**Description**

Schematic net/instance name is duplicated in XREF name table within equivalence point.

**What Next**

Information only; no action is needed.

Feedback

## SX-1663

(Fatal) Multiple schematic equivalencies for a single layout cell is not supported. Layout cell "%s" is already assigned to "%s".

**Description**

One layout cell could only has one equivalent schematic cell. Multiple schematic equivalencies for a single layout cell is not supported.

**What Next**

Check equivalent points data in input XREF data.

## SX-1664

(Information) CELL (L):%s (S):%s

**Description**

Layout and schematic equivalent point.

**What Next**

Information only; no action is needed.

## SX-1665

(Information) .alias %s %s

**Description**

Alias pair.

**What Next**

Information only; no action is needed.

## SX-1666

(Warning) Some schematic %s will not be netlisted because they belong to merged groups that contain more schematic elements than corresponding layout elements. Please see xrefhn.sum for list of excluded %s by schematic cell; .alias statements provide alternative element name from the same composite group that can be used for probing. The first field is the excluded element, the second field is the alternate element.

Feedback

**Description**

Some schematic entry will not be netlisted because they belong to merged groups that contain more schematic elements than corresponding layout elements.

**What Next**

Information only; no action is needed.

---

## SX-1667

(Information) Unused XfEntry %d(l),%d(s) in Cell %s(l);%s(s)

**Description**

An unused XREF data in cell.

**What Next**

Information only; no action is needed.

---

## SX-1668

(Fatal) Parse error on line number %d of file "%s". Line: "%s"

**Description**

An syntax error when StarRC parse input file.

**What Next**

Check content in input file.

---

## SX-1669

(Fatal) CALIBRE_IXF_FILE is required for XREF.

**Description**

StarRC fail to open file defined in CALIBRE_IXF_FILE.

**What Next**

Check setting of CALIBRE_IXF_FILE in StarRC command file.

---

## SX-1670

(Fatal) CALIBRE_NXF_FILE is required for XREF.

Feedback

**Description**

StarRC fail to open file defined in CALIBRE_NXF_FILE.

**What Next**

Check setting of CALIBRE_NXF_FILE in StarRC command file.

---

## SX-1671

(Fatal) Specified MACRO instance is not an equivalence point.

**Description**

Layout top block is not an equivalence point.

**What Next**

Check input XREF data.

---

## SX-1672

(Warning) Can not find schematic model of HN_NETLIST_MODEL_NAME:%s

**Description**

Schematic model set in HN_NETLIST_MODEL_NAME is not found in input XREF data.

**What Next**

Information only; no action is needed.

---

## SX-1673

(Warning) Can not find matched schematic model for HN_NETLIST_MODEL_NAME:%s

**Description**

Can not find matched schematic model for HN_NETLIST_MODEL_NAME.

**What Next**

Information only; no action is needed.

---

## SX-1674

(Warning) Global wildcard %s is not supported in SKIP_INSTANCES option. This entry will be ignored.

Feedback

**Description**

Global wildcard is not supported in SKIP_INSTANCES option. This entry will be ignored.

**What Next**

Set clear instance name in SKIP_INSTANCES.

---

## SX-1675

(Warning) Could not find instance "%s"

**Description**

Instance name set in SKIP_INSTANCES is not found in input data.

**What Next**

Information only; no action is needed.

---

## SX-1676

(Severe) Layout cell "%s" is identified as an EQUIV point but is not present in the instance cross-reference table.

**Description**

Layout cell is identified as an equivalence point but is not present in the instance cross-reference table.

**What Next**

Check input XREF data.

---

## SX-1677

(Severe) Layout cell "%s" is identified as an EQUIV point but is not present in the net cross-reference table.

**Description**

Layout cell is identified as an equivalence point but is not present in the net cross-reference table.

**What Next**

Check input XREF data.

Feedback

## SX-1678

(Warning) Layout element device "%s" does not have schematic EQUATE in runset or device table.

**Description**

Layout device does not have schematic equate device in runset or device table.

**What Next**

Information only; no action is needed.

## SX-1679

(Severe) Net and instance cross-reference tables contain inconsistent EQUIV points for layout cell "%s". Net table matches "%s" while instance table matches "%s".

**Description**

Net and instance cross-reference tables contain inconsistent equivalence points for layout cell.

**What Next**

Check layout cell input XREF data.

## SX-1680

(Fatal) There are composite matching MOS with RDC/RSC properties, please use XREF: YES/NO.

**Description**

Composite MOS device with RDC/RSC property is not allowed in XREF:COMPLETE mode.

**What Next**

Set XREF:YES/NO in StarRC command file.

## SX-1681

(Warning) Unknown property:%s

**Description**

Unknown property is found when StarRC parse XREF:COMPLETE data.

Feedback

**What Next**

Check device property in input file.

---

## SX-1682

(Fatal) Could not open ICV schematic netlist in %s

**Description**

Could not open ICV schematic netlist in the path.

**What Next**

Check ICV schematic netlist setting in StarRC command file.

---

## SX-1683

(Fatal) Could not open EvAccess library %s in %s

**Description**

Could not open EvAccess library in the path.

**What Next**

Check evaccess library setting in StarRC command file.

---

## SX-1684

(Fatal) Could not loop schematic netlist(Bgn).

**Description**

Could not loop schematic netlist.

**What Next**

Check schematic netlist in evaccess data.

---

## SX-1685

(Fatal) Could not open EvAccess view %s.

**Description**

Could not open EvAccess view.

Feedback

**What Next**

Check view data in evaccess directory.

---

## SX-1686

(Fatal) Could not open EvAccess cell %s in view %s.

**Description**

ould not open EvAccess cell in teh view.

**What Next**

Check cell data in evaccess directory.

---

## SX-1687

(Fatal) Schematic instance "%s%d%s" is connected to ground "0". To prevent incorrect result, please rename net "0" to another name.

**Description**

Schematic instance port is connected to ground net "0". That could result to incorrect result.

**What Next**

Rename net "0" to another name.

---

## SX-1688

(Fatal) Could not open cell %s.

**Description**

Could not open cell.

**What Next**

Check cell data in evaccess directory.

---

## SX-1689

(Fatal) Could not loop cell port(Bgn).

**Description**

Could not loop cell port data.

Feedback

**What Next**

Check cell port data in evaccess directory.

---

## SX-1690

(Fatal) Could not loop instance of cell %s.

**Description**

Could not loop instance of cell.

**What Next**

Check cell instance data in evaccess directory.

---

## SX-1691

(Fatal) Could not loop pin of instance %s.

**Description**

Could not loop pin of instance.

**What Next**

Check pin data of the instance in evaccess directory.

---

## SX-1693

(Fatal) Could not loop property of instance %s.

**Description**

Could not loop property of instance.

**What Next**

Check instance property data in evaccess directory.

---

## SX-1694

(Fatal) Could not close cell %s.

**Description**

Could not close cell.

Feedback

**What Next**

Check cell data in evaccess directory.

---

## SX-1695

(Warning) Unknown device type %d.

**Description**

Unknown device type.

**What Next**

Check device type in evaccess data.

---

## SX-1696

(Fatal) CDB database "%s" does not exist.

**Description**

StarRC could not CDB data for XREF.

**What Next**

Check CDB data generated by ICV or check path set in StarRC command file.

---

## SX-1698

(Warning) Binary Cdb file does not exist, using ASCII CDB file instead of it.

**Description**

Binary Cdb file does not exist, using ASCII CDB file instead of it.

**What Next**

Information only; no action is needed.

---

## SX-1699

(Fatal) Fail to open Binary CDB library.

**Description**

StarRC fail to open Binary CDB library from ICV.

Feedback

**What Next**

Check the binary CDB library generated by ICV. Generate CDB library again and rerun StarRC.

---

## SX-1700

(Fatal) Fail to open Binary CDB cell pool.

**Description**

StarRc fail to open Binary CDB cell pool from ICV.

**What Next**

Check the binary CDB library generated by ICV. Generate CDB library and rerun StarRC.

---

## SX-1701

(Warning) More than %ld consecutively duplicate elements found in the GDS files, it might incur extraction problems.

**Description**

Consecutively duplicate GDS elements are found in the GDS files. Most of the time, it does no harms to the extraction. However, in some extreme cases due to bugs in the GDS generation, the duplicate elements will increase number of polygons exponentially and cause serious problems.

**What Next**

Check the GDS files in IC WorkBench and make sure the duplicate elements are expected.

---

## SX-1702

(Warning) Fail to open Binary CDB cell for iterate cell %s.

**Description**

StarRc fail to open Binary CDB cell for iteration from ICV.

**What Next**

Check the binary CDB library generated by ICV. Generate CDB library and rerun StarRC.

Feedback

## SX-1703

(Warning) Found consecutively duplicate elements (%s )

**Description**

Consecutively duplicate elements are found in the GDS cell. Usually, they does no harms to the extraction if the duplicate ones are POLY or PATH elements. But, if the duplicate ones are SREF or AREF elements, they may increase the number of polygons exponentially and incur serious performance/accuracy problems in the extraction.

**What Next**

Check the GDS files in IC WorkBench and make sure the duplicate elements are expected.

## SX-1704

(Warning) Fail to open Binary CDB Instance in Cell%s

**Description**

StarRc fail to open Binary CDB instance from ICV.

**What Next**

Check the binary CDB library generated by ICV. Generate CDB library and rerun StarRC.

## SX-1705

(Warning) Fail to close Binary CDB Instance in Cell %s.

**Description**

StarRc fail to close Binary CDB instance from ICV.

**What Next**

Check the binary CDB library generated by ICV. Generate CDB library and rerun StarRC.

## SX-1706

(Warning) Fail to open Binary CDB net in Cell %s.

**Description**

StarRc fail to open Binary CDB net from ICV.

Feedback

**What Next**

Check the binary CDB library generated by ICV. Generate CDB library and rerun StarRC.

---

## SX-1707

(Warning) Fail to close Binary CDB net in Cell %s.

**Description**

StarRc fail to close Binary CDB net in cell.

**What Next**

Check the binary CDB library generated by ICV. Generate CDB library and rerun StarRC.

---

## SX-1708

(Warning) Fail to open Binary CDB Swapped port.

**Description**

StarRc fail to open Binary CDB swapped port from ICV.

**What Next**

Check the binary CDB library generated by ICV. Generate CDB library and rerun StarRC.

---

## SX-1709

(Warning) Fail to close Binary CDB Swapped port.

**Description**

StarRc fail to close Binary CDB swapped port from ICV.

**What Next**

Information only; no action is needed.

---

## SX-1710

(Warning) Failed to read some CDB cells (see xrefhn.sum)

**Description**

StarRc fail to read some CDB cells from ICV.

Feedback

**What Next**

Check the binary CDB library generated by ICV. Generate CDB library and rerun StarRC.

---

## SX-1713

(Warning) Fail to iterate Binary CDB cell pool.

**Description**

StarRc fail to iterate Binary CDB cell pool from ICV.

**What Next**

Check the binary CDB library generated by ICV. Generate CDB library and rerun StarRC.

---

## SX-1719

(Information) This cell %d is not Match

**Description**

StarRC finds a cross-referenced instance has unmatched cell when iterating Binary CDB library from ICV.

**What Next**

Information only; no action is needed.

---

## SX-1720

(Warning) Fail to close Binary CDB library.

**Description**

StarRc fail to close Binary CDB library pool from ICV.

**What Next**

Information only; no action is needed.

---

## SX-1724

(Fatal) CDB (Hercules Compare Database) version is "%s" which is not compatible with this release of StarXtract; CDB version "ver2.6" is required. Please re-run LVS with a compatible version of Hercules as indicated by StarRC release notes.

Feedback

**Description**

ICV CDB version is not compatible with this release of StarRC.

**What Next**

Rerun LVS with a compatible version of ICV as indicated by StarRC release notes.

---

## SX-1725

(Fatal) Encountered matched composite instance with permuted ports; not supported. Permuted ports may only be applied to non-device elements which cannot be composites.

**Description**

Unsupported data is found in CDB library from ICV.

**What Next**

Check ASCII CDB library generated by ICV. Generate CDB library without unsupported data and rerun StarRC.

---

## SX-1727

(Fatal) Swapped ports section has less than 2 ports in CDB database.

**Description**

Invalid number of ports is listed at swapped ports section in CDB database from ICV..

**What Next**

Check the binary CDB library generated by ICV. Generate CDB library and rerun StarRC.

---

## SX-1728

(Warning) The ICV/Hercules option, PUSH_DOWN_DEVICES, is not supported. Using this option may cause incorrect result.

**Description**

The ICV/Hercules option, PUSH_DOWN_DEVICES, is not supported. Using this option may cause incorrect result.

**What Next**

Information only; no action is needed.

Feedback

## SX-1729

(Warning) Failed in opening Composite Instance in binary CDB library.

**Description**

StarRC failed to open composite instance data in binary CDB library from ICV.

**What Next**

Check the binary CDB library generated by ICV. Generate CDB library and rerun StarRC.

## SX-1731

(Warning) Multiple schematic nets are apped to fewer layout nets, layout net %s will be netlisted ideally.

**Description**

Multiple schematic nets are apped to fewer layout nets. Root layout net will be netlisted ideally.

**What Next**

Information only; no action is needed.

## SX-1732

(Warning) Multiple schematic nets are mapped to zero layout net.

**Description**

Multiple schematic nets are mapped to zero layout net.

**What Next**

Information only; no action is needed.

## SX-1734

(Warning) Multiple schematic instances are mapped to fewer layout instance, layout instance %s will be netlisted ideally.

**Description**

Multiple schematic instances are mapped to fewer layout instance, root layout instance will be netlisted ideally.

Feedback

**What Next**

Information only; no action is needed.

---

## SX-1735

(Warning) Multiple schematic instances are mapped to zero layout instance.

**Description**

Multiple schematic instances are mapped to zero layout instance.

**What Next**

Information only; no action is needed.

---

## SX-1736

(Warning) Multiple schematic instances are mapped to one layout instances, layout instance %s will be netlisted ideally.

**Description**

Multiple schematic instances are mapped to one layout instances, root layout instance will be netlisted ideally..

**What Next**

Information only; no action is needed.

---

## SX-1737

(Warning) First ICV/Hercules XREF table name assignment in composite group within equivalence point %s(L):%s(S) involves a previously referenced schematic name. All members of this composite group will be netlisted with layout names.

**Description**

Duplicated schematic names are found in composite groups in CDB library from ICV/ Hercules.

**What Next**

Check the binary CDB library generated by ICV. Generate CDB library and rerun StarRC.

Feedback

## SX-1739

(Warning) First ICV/Hercules XREF table name assignment in composite group %s(L): %s(S) within equivalence point %s(L):%s(S) involves a previously referenced (duplicated) schematic name. All members of this composite group will be netlisted with layout names.

**Description**

Duplicated schematic names are found in composite group within the equivalence point in CDB library from ICV/Hercules.

**What Next**

Check the binary CDB library generated by ICV. Generate CDB library and rerun StarRC.

## SX-1740

(Warning) Fail in opening Composite Net.

**Description**

StarRC failed to open the composite net data in CDB library from ICV.

**What Next**

Check the binary CDB library generated by ICV. Generate CDB library and rerun StarRC.

## SX-1741

(Warning) Failed to open Binary cdb swap port.

**Description**

StarRC failed to open swap port data in CDB library from ICV.

**What Next**

Check the binary CDB library generated by ICV. Generate CDB library and rerun StarRC.

## SX-1742

(Warning) Encountered %s in permuted port group for matched instance "%s::%s" in EQUIV cell "%s::%s". Permutations involving %s are not supported, default port match for this group will be applied.

Feedback

**Description**

StarRC find unsupported port type for matched instance when reading swap ports section from CDB library.

**What Next**

Information only; no action is needed.

---

## SX-1743

(Information) Could not open file "%s".

**Description**

StarRC could not open the input file.

**What Next**

Information only; no action is needed.

---

## SX-1744

(Warning) Binary CDB file is not found

**Description**

StarRC does not find binary CDB file for XREF.

**What Next**

Check ICV result or data path set in StarRC command file.

---

## SX-1745

(Warning) TOP BLOCK "%s::%s" does not compare. LVS is not clean.

**Description**

Top block was not compared. This is not LVS-clean design.

**What Next**

Information only; no action is needed.

---

## SX-1746

(Warning) Multi-Sch equivalence found %s==%s && %s==%s

Feedback

**Description**

Schematic cell is equivalent to more than one layout cell.

**What Next**

information only; no action is needed.

---

## SX-1747

(Fatal) COMPARE_DIRECTORY is required for Hercules XREF: YES.

**Description**

ICV-StarRC flow needs compary directory when XREF:YES.

**What Next**

Provide correct setting for option COMPARE_DIRECTORY in StarRC command file.

---

## SX-1748

(Fatal) Schematic top block name not specified.

**Description**

StarRC could not find correct schematic top block from input data.

**What Next**

Set option SCHEMATIC_BLOCK in StarRC command file or check XREF data.

---

## SX-1749

(Fatal) INTERNAL Error: failed to load internal models database.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1751

(Fatal) Layout top block name is not specified.

Feedback

**Description**

StarRC could not find cell set in option BLOCK.

**What Next**

Reset option BLOCK in StarRC command file.

---

## SX-1752

(Warning) LVS_TOP_BLOCK is different from BLOCK in the command file. Parsing evaccess directory for %s.

**Description**

Cell names set in LVS_TOP_BLOCK and BLOCK are different.

**What Next**

Information only; no action is needed.

---

## SX-1755

(Warning) Both ICV_SCHEMATIC_NETLIST and EVACCESS_DIRECTORY are set. Using ICV_SCHEMATIC_NETLIST.

**Description**

Both ICV_SCHEMATIC_NETLIST and EVACCESS_DIRECTORY are set. Using ICV_SCHEMATIC_NETLIST.

**What Next**

Information only; no action is needed.

---

## SX-1756

(Warning) Ignoring syntax error in option XREF_SWAP_MOS_SD_PROPERTY: %s

**Description**

An invalid syntax was provided for option XREF_SWAP_MOS_SD_PROPERY.

**What Next**

Check StarRC command file.

Feedback

## SX-1757

(Fatal) For Model:%s, %s set more than once in XREF_SWAP_MOS_SD_PROPERTY.

**Description**

Property is set more than once for the model in XREF_SWAP_MOS_SD_PROPERTY.

**What Next**

Reset properties for the model in XREF_SWAP_MOS_SD_PROPERTY.

## SX-1758

(INTERNAL Warning) %s is default swapping property. Ignore setting in XREF_SWAP_MOS_SD_PROPERTY.

**Description**

Property is already considered for swapping. Ignore user setting in StarRC command file.

**What Next**

Information only; no action is needed.

## SX-1759

(Fatal) %s set more than once in XREF_SWAP_MOS_SD_PROPERTY.

**Description**

Property is set more than once in XREF_SWAP_MOS_SD_PROPERTY.

**What Next**

Check and correct setting in StarRC command file.

## SX-1760

(Fatal) Could not write internal tech file

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

Feedback

## SX-1761

(Warning) %s is the schematic top block and cannot be skipped.

**Description**

Schematic top block cannot be set as skip cell.

**What Next**

Information only; no action is needed.

## SX-1762

(Warning) %s is the layout top block and cannot be skipped

**Description**

Layout top block cannot be set as skip cell.

**What Next**

Information only; no action is needed.

## SX-1763

(Warning) %s skip cell %s is not a valid equiv. Verify cell type/equiv list.

**Description**

Layout/Schematic skip cell is not a valid equivalent point.

**What Next**

Verify cell type or equivalent point list from ICV/Calibre.

## SX-1764

(Warning) %s skip cell %s is not a valid equiv. Skip cell %s was exploded during LVS - %s

**Description**

This skip cell is not a valid equivalent point because it was exploded during LVS.

XREF_KEEP_IMBALANCED_EQUIV_SKIP_CELLS: YES can still keep them as SKIP_CELLS or SCHEMATIC_SKIP_CELLS, but extraction might not be expected.

Feedback

**What Next**

Information only; no action is needed.

---

## SX-1765

(Warning) %s was exploded during LVS; It will not be a SKIP_CELL anymore

**Description**

This cell was exploded during LVS so it will not be set as skip cell.

**What Next**

Information only; no action is needed.

---

## SX-1768

(Warning) OBSERVATION_POINT cell pattern "%s" does not match any non SKIP_CELL EQUIV points and will be ignored.

**Description**

Cell name pattern defined in OBSERVATION_POINT does not match any non-skip cell and will be ignored.

**What Next**

Information only; no action is needed.

---

## SX-1769

(Warning) OBSERVATION_POINT cell "%s" is not a non SKIP_CELL EQUIV point and will be ignored.

**Description**

Cell defined in OBSERVATION_POINT is not a non-skip cell and will be ignored.

**What Next**

Information only; no action is needed.

---

## SX-1770

(Warning) An asymmetric permutation was encountered in Hercules cross-reference table; two swapped nets do not have the same number of instance/device connections. This is typically caused by improper texting in the layout in conjunction with Hercules COMPARE

Feedback

options DETECT_PERMUTABLE_PORTS=TRUE and EQUATE_BY_NAME=TRUE. Please correctly text the layout or disable EQUATE_BY_NAME and re-run Hercules COMPARE.

**Description**

An asymmetric permutation was encountered in Hercules cross-reference table; two swapped nets do not have the same number of instance/device connections.

**What Next**

Check input XREF data.

---

## SX-1771

(Fatal) INTERNAL Error: Could not read layer information file

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1772

(Warning) INTERNAL Warning: Specified via coverage option for a layer which does not exist: %s

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1774

(Fatal) INTERNAL Error: Layer information file corrupted

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

Feedback

## SX-1776

(Fatal) INTERNAL Error: Must provide POSITIVE via_coverage either for all via layers or none: %s

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## SX-1777

(Fatal) INTERNAL Error: Could not write layer information file

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## SX-1778

(Fatal) INTERNAL Error: Could not write fs_drop_layer file

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## SX-1779

(Fatal) INTERNAL Error: Could not write layer density box file

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

Feedback

## SX-1780

(Warning) LAYER_INFO_ADDITIONAL_FILE will be appended following automated connection commands. No checking until xTractor.

### Description

Each time when you use LAYER_INFO_ADDITIONAL_FILE, the message will appear. The syntax of LAYER_INFO_ADDITIONAL_FILE will only be checked in xTract stage.

### What Next

If xTract stage shows no error, nothing needs to be done. Otherwise, you should check syntax of LAYER_INFO_ADDITIONAL_FILE based on messages shown in xTract stage.

## SX-1781

(Warning) LAYER_INFO_ADDITIONAL_FILE is directly appended to layer_info file. AUTO_RUNSET: YES will not change anything in LAYER_INFO_ADDITIONAL_FILE.

### Description

When LAYER_INFO_ADDITIONAL_FILE is used along with command AUTO_RUNSET: YES, the message will appear. AUTO_RUNSET: YES will generate separate layers for device terminals, which don't exist in original input database. As a result, the ignore cap settings defined in LAYER_INFO_ADDITIONAL_FILE may not work as you expect.

### What Next

You should avoid using AUTO_RUNSET: YES along with LAYER_INFO_ADDITIONAL_FILE.

## SX-1782

(Fatal) MAX_VIRTUAL_VIA_SEGMENTATION_NUM must be used with TRENCH_CONTACT_VIRTUAL_VIA_SEGMENTATION_RATIO.

### Description

Command MAX_VIRTUAL_VIA_SEGMENTATION_NUM can't be used alone. It must be used along with command TRENCH_CONTACT_VIRTUAL_VIA_SEGMENTATION_RATIO. StarRC finds that you use MAX_VIRTUAL_VIA_SEGMENTATION_NUM without TRENCH_CONTACT_VIRTUAL_VIA_SEGMENTATION_RATIO and gives this error.

Feedback

**What Next**

You should also add command TRENCH_CONTACT_VIRTUAL_VIA_SEGMENTATION_RATIO.

---

## SX-1783

(Fatal) Invalid value %s for WIDE_DEVICE_TERM_RESISTANCE.

**Description**

The valid value for command WIDE_DEVICE_TERM_RESISTANCE is RES or NONE. Any other value is invalid. If invalid value is detected, StarRC will give this error. Users can add device names after RES to apply the command to specified devices, which must be of type RESISTOR.

**What Next**

You should use either RES or NONE as value of command WIDE_DEVICE_TERM_RESISTANCE.

---

## SX-1784

(Warning) Contact layer %s or tap layer %s in REDUCE_SUBCONT_IN_TAP is not mapped

**Description**

The contact layer and tap layer in command REDUCE_SUBCONT_IN_TAP must be defined and mapped in StarRC mapping file. StarRC finds one or two of the shown layers are not defined in StarRC mapping file but used in command REDUCE_SUBCONT_IN_TAP and give this warning.

**What Next**

You should define and map the shown layers in StarRC mapping file.

---

## SX-1785

(Warning) No valid tap and contact layers specified and tap contact reduction will not run

**Description**

The warning means StarRC doesn't find valid tap and contact layers specified in command REDUCE_SUBCONT_IN_TAP. The reason may be all the contact layers and tap layers used in command REDUCE_SUBCONT_IN_TAP are not defined or mapped in StarRC mapping file. StarRC will ignore REDUCE_SUBCONT_IN_TAP command.

Feedback

### What Next

You should guarantee that all contact layers and tap layers used in command REDUCE_SUBCONT_IN_TAP are defined in StarRC mapping file. You can check layer names used in command REDUCE_SUBCONT_IN_TAP to see whether there is any typo.

---

## SX-1786

(Fatal) There is no keyword "DEVICE_LAYER" specified in the mapping file when POWER_EXTRACT:DEVICE_LAYERS

### Description

When POWER_EXTRACT:DEVICE_LAYERS is set, StarRC will extract resistance from power nets based on layers defined with keyword "DEVICE_LAYER" in StarRC mapping file. So in order to use POWER_EXTRACT:DEVICE_LAYERS, You should first define device layers with keyword "DEVICE_LAYER" in StarRC mapping file. If StarRC finds no layers are defined as device layer with keyword "DEVICE_LAYER" in StarRC mapping file but POWER_EXTRACT:DEVICE_LAYERS is set, it will give error.

### What Next

You should define device layers with keyword "DEVICE_LAYER" in StarRC mapping file and then use POWER_EXTRACT:DEVICE_LAYERS.

---

## SX-1787

(Fatal) POWER_EXTRACT:DEVICE_LAYERS doesn't work with LEFDEF flow.

### Description

POWER_EXTRACT:DEVICE_LAYERS doesn't work in LEFDEF flow.
StarRC gives this error once it finds the input database if LEFDEF but POWER_EXTRACT:DEVICE_LAYERS is used.

### What Next

You should use other options for command POWER_EXTRACT in LEFDEF flow.

---

## SX-1788

(Fatal) POWER_EXTRACT:DEVICE_LAYERS doesn't work with Milkyway flow.

Feedback

**Description**

POWER_EXTRACT:DEVICE_LAYERS doesn't work in Milkyway flow. StarRC gives this error once it finds the input database if Milkyway but POWER_EXTRACT:DEVICE_LAYERS is used.

**What Next**

You should use other options for command POWER_EXTRACT in Milkyway flow.

---

## SX-1790

(Warning) The precedence is not set for layers mapped to substrate in the mapping file when TRANSLATE_RETAIN_BULK_LAYERS:YES or CONLY is specified in your command file. Accuracy can be affected. Check the order of substrate layer and set precedence for them in your mapping file whenever the TRANSLATE_RETAIN_BULK_LAYERS:YES or CONLY command is used.

**Description**

Once TRANSLATE_RETAIN_BULK_LAYERS:YES or TRANSLATE_RETAIN_BULK_LAYERS:CONLY is used, you should set precedence for all layers mapped to SUBSTRATE in StarRC mapping file. StarRC give this warning if it finds precedence is not set for any layer mapped to SUBSTRATE in StarRC mapping file.

**What Next**

If you want to use TRANSLATE_RETAIN_BULK_LAYERS:YES or TRANSLATE_RETAIN_BULK_LAYERS:CONLY, you should check the order of all layers mapped to SUBSTRATE in StarRC mapping file and set correct precedence. If you only want to use TRANSLATE_RETAIN_BULK_LAYERS:NO, just change the value of this command to NO.

---

## SX-1792

(Warning) The precedence is not set for all the layers mapped to SUBSTRATE in the mapping file. For those layers not set with precedence, their precedence is set to zero automatically.

**Description**

Once TRANSLATE_RETAIN_BULK_LAYERS:YES or TRANSLATE_RETAIN_BULK_LAYERS:CONLY is used, precedence for all layers mapped to SUBSTRATE should be set in StarRC mapping file. StarRC give this warning if it finds precedence is not set for a few layers mapped to SUBSTRATE in StarRC mapping file. Precedence for those layers will be set to zero by StarRC automatically.

Feedback

**What Next**

You should check the order of all layers mapped to SUBSTRATE in StarRC mapping file and set correct precedence.

---

## SX-1794

(Warning) The precedence is not set for ignored cap layer: %s

**Description**

The precedence is not set for ignored cap layer: %s. The precendence is defined in SUBSTRATE layers. When two polygons ovelapped in two layers which are mapping to SUBSTRATE without precendence but have different ignore cap layer, StarRC can not decide which layer should be used with ignore cap layer.

**What Next**

You should revise you mapping file to define precedence on SUBSTRATE layers.

---

## SX-1795

(Warning) Layer: %s is connected to layers with different precedence! Fail to set its precedence according to connections!

**Description**

Layer: %s is connected to layers with different precedence! Fail to set its precedence according to connections. StarRC set the prcedence of the layer: %s as -1.

**What Next**

You should connect all the layer with same precedence or defined precedence directly in mapping file.

---

## SX-1796

(Fatal) To model DPT effect with ER_VS_SI_SPACING in nxtgrd file, please set DPT: YES. For non-DPT flow, please provide nxtgrd file without ER_VS_SI_SPACING

**Description**

If you want to model DPT effect with ER_VS_SI_SPACING in NXTGRD file, you should set DPT: YES in command file and use a nxtgrd file with ER_VS_SI_SPACING. If you don't want to model DPT effect with ER_VS_SI_SPACING in NXTGRD file, you should set DPT: NO in command file and use a nxtgrd file without ER_VS_SI_SPACING. StarRC

Feedback

gives this error when the nxtgrd file has ER_VS_SI_SPACING but DPT: YES is not set in command file.

**What Next**

If you want to model DPT effect with ER_VS_SI_SPACING in NXTGRD file, you should set DPT: YES in command file. If you don't want to model DPT effect with ER_VS_SI_SPACING in NXTGRD file, you should use a nxtgrd file without ER_VS_SI_SPACING.

---

## SX-1800

(Warning) DPT shift on layer %s is larger than allowed DPT_MAX_SHIFT ", the shift value on this layer is ignored.

**Description**

StarRC gives this warning when it finds the x or y shift defined in color_layers section in StarRC mapping file for the shown layer is larger than DPT_MAX_SHIFT defined in ITF file or nxtgrd file for this layer.

**What Next**

You should define a smaller shift value for this layer in color_layers seciton in StarRC mapping file which is not larger than DPT_MAX_SHIFT in ITF file.

---

## SX-1801

(Fatal) Only 1 color layer for grd layer %s

**Description**

StarRC gives this error when it finds there is only one color layer defined in color_layers section in StarRC mapping file for the shown grd layer. In color-aware DPT flow, you should define at least 2 color layers in StarRC mapping file for a physical layer.

**What Next**

You should define at least 2 color layers in StarRC mapping file for the shown physical layer.

---

## SX-1802

(Fatal) INTERNAL Error: Could not write layer pairs file

**Description**

An internal consistency check has failed.

Feedback

### What Next

Contact the Synopsys Support Center.

---

## SX-1803

(Fatal) VIA_COVERAGE must be <via_layer_name> <lower_full_coverage_parameter> <lower_quarter_coverage_parameter> [<lower_semi_coverage_parameter>] <upper_full_coverage_parameter> <upper_quarter_coverage_parameter> [<upper_semi_coverage_parameter>]. check your via_coverage for %s

### Description

StarRC gives this error when it finds the number of values set in command VIA_COVERAGE is less than 5. The syntax of command VIA_COVERAGE is: VIA_COVERAGE: <via_layer_name> <lower_full_coverage_parameter> <lower_quarter_coverage_parameter> [<lower_semi_coverage_parameter>] <upper_full_coverage_parameter> <upper_quarter_coverage_parameter> [<upper_semi_coverage_parameter>]

### What Next

You should check and correct settings in command VIA_COVERAGE.

---

## SX-1804

(Fatal) VIA_COVERAGE requires full_coverage_parameter >= quarter_coverage_parameter. VIA_COVERAGE requires quarter_coverage_parameter >= 0. Check your VIA_COVERAGE for %s

### Description

The syntax of command VIA_COVERAGE is: VIA_COVERAGE: <via_layer_name> <lower_full_coverage_parameter> <lower_quarter_coverage_parameter> [<lower_semi_coverage_parameter>] <upper_full_coverage_parameter> <upper_quarter_coverage_parameter> [<upper_semi_coverage_parameter>] Besides, the following rules need to be satisfied: <lower_full_coverage_parameter> >= <lower_quarter_coverage_parameter> >= 0 <upper_full_coverage_parameter> >= <upper_quarter_coverage_parameter> >= 0 StarRC gives this error when it finds the above rules are not satisfied in VIA_COVERAGE command.

### What Next

You should check VIA_COVERAGE command and correct the values for the shown layer.

Feedback

## SX-1805

(Fatal) VIA_COVERAGE requires quarter_coverage_parameter >= semi_coverage_parameter. VIA_COVERAGE requires semi_coverage_parameter >= 0. Check your VIA_COVERAGE for %s

### Description

The syntax of command VIA_COVERAGE is: VIA_COVERAGE: <via_layer_name> <lower_full_coverage_parameter> <lower_quarter_coverage_parameter> [<lower_semi_coverage_parameter>] <upper_full_coverage_parameter> <upper_quarter_coverage_parameter> [<upper_semi_coverage_parameter>] Besides, the following rules need to be satisfied: <lower_quarter_coverage_parameter> >= <lower_semi_coverage_paramete >= 0 <upper_quarter_coverage_parameter> >= <upper_semi_coverage_parameter> >= 0 StarRC gives this error when it finds the above rules are not satisfied in VIA_COVERAGE command.

### What Next

You should check VIA_COVERAGE command and correct the values for the shown layer.

## SX-1806

(Warning) specified via coverage option for a via layer which does not exist or is not a via layer: %s

### Description

StarRC gives this warning when it finds the layer name in one VIA_COVERAGE command is not defined in StarRC mapping file, or is not defined as a VIA. This VIA_COVERAGE command will be ignored by StarRC.

### What Next

If you don't need via coverage settings for the shown layer, nothing needs to be done. Otherwise, you should check and define the via layer in via_layers section in StarRC mapping file.

## SX-1807

(Fatal) you must provide POSITIVE VIA_COVERAGE either for all via layers or none: %s

### Description

StarRC gives this error when it finds only a few via layers has VIA_COVERAGE settings.

Feedback

**What Next**

You should define via coverage for either all via layers or none of the via layers.

---

## SX-1808

(Warning) Frequency dependent R disabled for layer %s because its RPSQ overwritten in mapping file.

**Description**

If you use StarRC command OPERATING_FREQUENCY to include substrate effects during high-frequency extractions for a few layers, you should not specify RPSQ value in StarRC mapping file for those layers. StarRC will not include substrate effects during high-frequency extractions for those layers with RPSQ set in StarRC mapping file and give this warning

**What Next**

If you don't need the shown layer to include substrate effects during high-frequency extractions, nothing needs to be done. Otherwise, you should remove RPSQ parameter for the shown layer in StarRC mapping file.

---

## SX-1809

(Fatal) INTERNAL Error: Could not write layer info file

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1810

(Fatal) The diffusion_res_model %s defined for the db layer %s in the mapping file does not match any of the USER_DEFINED_DIFFUSION_RESISTANCE model in the nxtgrd file.

**Description**

The nxtgrd file does not contain a user defined diffusion resistance model of the same name than the dblayer.

Feedback

**What Next**

Verify that the names of the USER_DEFINED_DIFFUSION_RESISTANCE model defined in the ITF file and the diffusion_res_model in the mapping file match. Correct the mapping file and run StarXtract again.

---

## SX-1811

(Fatal) Could not open LAYER_INFO_ADDITIONAL_FILE for read.

**Description**

StarRC gives this error when it can't open the file specified in StarRC command LAYER_INFO_ADDITIONAL_FILE. This may due to incorrect file path or you don't have read access permission for this file.

**What Next**

You should check whether the file path in LAYER_INFO_ADDITIONAL_FILE is correct or not and check whether you have read access permission for this file or not.

---

## SX-1812

(Fatal) VIA_TAG need to be specified in layer mapping file in RV_VIA_ARRAY_REPORT_FILE mode.

**Description**

When you use StarRC command RV_VIA_ARRAY_REPORT_FILE, you should also add via tags (VIA_D/VIA_T/VIA_H/VIA_GROUP_BBOX/VIA_ARRAY_BBOX) to at least one conducting layer in StarRC mapping file. Once StarRC finds command RV_VIA_ARRAY_REPORT_FILE is used but no conducting layers in StarRC mapping file have via tags, StarRC will give this error.

**What Next**

You should add via tags(VIA_D/VIA_T/VIA_H/VIA_GROUP_BBOX/VIA_ARRAY_BBOX) to StarRC mapping file. If you don't want to use command RV_VIA_ARRAY_REPORT_FILE, you should remove this command from StarRC command file.

---

## SX-1813

(Warning) MAPPING_FILE layer "%s" is not a valid database layer and will be removed.

Feedback

### Description

A valid database layer means a database layer which has connections. StarRC will check the connections of database layers except a few special layers such as view_only_layers. If the layer defined in mapping file doesn't have connection, StarRC will remove this layer and give this warning.

### What Next

If you don't need this layer, nothing needs to be done. Otherwise, you should check why there is no connection defined for this layer in database.

---

## SX-1814

(Warning) Runset gate layer "%s" is now mapped to GRD IS_GATE layer "%s" since the IS_GATE tag is used.

### Description

Runset gate layer is not mapped to gate layer in mapping file. "%s" is now mapped to GRD IS_GATE layer "%s" since the IS_GATE tag is used.

### What Next

You should map the runset gate layer to GRD gate layer in mapping file to avoid this warning message.

---

## SX-1815

(Warning) Database diffusion layer "%s" is not included in MAPPING_FILE. Mapping to "%s".

### Description

Once StarRC finds a database diffusion layer (usually based on device definition) is not defined in StarRC mapping file, StarRC will try to assign a correct physical layer (usually based on LAYER_TYPE definition in ITF file or nxtgrd file) to the diffusion layer and give this warning.

### What Next

If the automatic mapping is correct, nothing needs to be done. Otherwise, you should define correct mapping for the diffusion layer in StarRC mapping file.

---

## SX-1816

(Warning) Database gate layer "%s" is not included in MAPPING_FILE. Mapping to "%s".

Feedback

**Description**

Once StarRC finds a database gate layer (usually based on device definition) is not defined in StarRC mapping file, StarRC will try to assign a correct physical layer (usually based on LAYER_TYPE definition in ITF file or nxtgrd file) to the gate layer and give this warning..

**What Next**

If the automatic mapping is correct, nothing needs to be done. Otherwise, you should define correct mapping for the database layer in StarRC mapping file.

---

## SX-1817

(Warning) Database bulk layer "%s" is not included in MAPPING_FILE. Mapping to "%s".

**Description**

Once StarRC finds a database bulk layer (usually based on device definition) is not defined in StarRC mapping file, StarRC will try to assign a correct physical layer to the bulk layer and give this warning.

**What Next**

If the automatic mapping is correct, nothing needs to be done. Otherwise, you should define correct mapping for the bulk layer in StarRC mapping file.

---

## SX-1818

(Warning) Database layer "%s" is not included in MAPPING_FILE. Mapping to "%s".

**Description**

Once StarRC finds a device substrate layer (usually based on device definition) is not defined in StarRC mapping file, StarRC will try to assign a correct physical layer to the device substrate layer and give this warning..

**What Next**

If the automatic mapping is correct, nothing needs to be done. Otherwise, you should define correct mapping for the layer in StarRC mapping file.

---

## SX-1819

(Severe) Connected database layer "%s" does not have a valid ITF layer.

Feedback

**Description**

StarRC gives this error once it finds the shown layer is defined in database and has connections, but it is not defined in StarRC mapping file.

**What Next**

You should define the shown layer in StarRC mapping file.

---

## SX-1820

(Severe) INTERNAL Error: VIA layer "%s" connects two %d CONDUCTORS at different picture levels. %s[%s] and %s[%s]

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1821

(Warning) VIA layer "%s" connects MARKER layer "%s". MARKER layers can only connect CONDUCTOR layers. This connection will be ignored.

**Description**

StarRC gives this warning once it finds the VIA layer is connected to a MARKER layer (based on connection definitions or layer operations in database), which is not valid. This connection will be ignored.

**What Next**

If the connection can be ignored, nothings need to be done. Otherwise, you should check the database and correct the invalid connection definition.

---

## SX-1822

(Warning) The via layer %s defined from DB (runset) connects down to %s, which is mapped to ITF layer %s. The mapped ITF layer is different from the via's ITF from-layer %s. Check the mapping file.

**Description**

For a runset (DB) via layer, the down layer connected to the via layer must be mapped to from-layer of the via's ITF layer. For example, if via layer A is mapped to ITF layer VIA1

Feedback

(from-layer of VIA1 is M1 in ITF file), and layer A connects down to layer B, layer B must be mapped to M1 in StarRC mapping file. Otherwise, StarRC will give this warning and the connection will be ignored.

**What Next**

You should either correct the connection definition in database, correct the definition in the mapping file, or correct the from-layer definition in ITF file.

---

## SX-1823

(Severe) Marker layer "%s" must be connected to a conductor.

**Description**

StarRC gives this error once it finds a marker layer has on connections.

**What Next**

You should connect the marker layer to a conductor in database.

---

## SX-1825

(Warning) IGNORE_CAP_LAYERS refers to layer "%s" which is removed; all IGNORE_CAP pairs involving this layer will be ignored.

**Description**

StarRC gives this warning once it finds the layer used in ignore_cap_layers section in StarRC mapping file is removed. The causes can be: 1. the layer is defined in remove_layers section; 2. the layer is not defined in StarRC mapping file. You should define the layer in StarRC mapping file as a conductor or via. 3. the layer has no connections in database;

**What Next**

If you don't need the layer, nothing needs to be done. Otherwise, you should check the reason why the layer is removed. The possible causes are: 1. the layer is defined in remove_layers section; 2. the layer is not defined in StarRC mapping file. You should define the layer in StarRC mapping file as a conductor or via. 3. the layer has no connections in database;

---

## SX-1826

(Severe)INTERNAL Error: Layer %s is mapped to grd via layers with more than one common cap level (a)

Feedback

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1827

(Severe) INTERNAL Error: Layer %s must be mapped to grd via layers with one common cap level

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1828

(Severe)INTERNAL Error: Layer %s is mapped to grd layers with more than one common cap level (c)

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1829

(Severe) Multi layer contacts(one database via layer mapped to several ITF layers) detected in the mapping file. Must set AUTO_RUNSET: YES for successful run

**Description**

Multi layer contacts(one database via layer mapped to several ITF layers)detected in the mapping file. StarRC needs to use AUTO_RUNSET:YES to seperate the polygon on the database via layer into different ITF layer.

**What Next**

You should set AUTO_RUNSET:YES in this case.

Feedback

## SX-1830

(Fatal) AUTO_RUNSET_FILE %s can not be opened

**Description**

StarRC gives this error once it can't open file specified in command AUTO_RUNSET_FILE.

**What Next**

You should check whether the file used in AUTO_RUNSET_FILE exists or not, and whether you have read permission for that file.

## SX-1832

(Fatal) No nxtgrd file defined

**Description**

StarRC gives this error once it finds nxtgrd file is not specified.

**What Next**

You should specify nxtgrd file through StarRC command TCAD_GRD_FILE.

## SX-1834

(Fatal) INTERNAL Error: Could not writ FS_LAYER_INFO_ADDITIONAL_FILE.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## SX-1835

(Fatal) Could not open LAYER_INFO_ADDITIONAL_FILE for read.

**Description**

StarRC gives this error once it can't open the file specified in StarRC command LAYER_INFO_ADDITIONAL_FILE.

Feedback

**What Next**

You should check whether the file specified in StarRC command LAYER_INFO_ADDITIONAL_FILE exists or not, and whether you have permission to read it or not.

---

## SX-1836

(Warning) Drop factor is not defined in nxtgrd file %s. No need to generate drop factor layers

**Description**

If you use command DROP_LAYERS_STAGE: YES, you need to define drop factor in nxtgrd file. Otherwise, StarRC will give this warning and not generate drop factor layers.

**What Next**

If you don't need to generate drop factor layers, nothing needs to be done. Otherwise, you should define drop factor in ITF file or nxtgrd file.

---

## SX-1837

(Severe) Layer "%s" is not mapped to an ITF VIA layer.

**Description**

StarRC gives this error once it finds the shown database VIA layer is not mapped to a VIA ITF layer defined in ITF file.

**What Next**

In StarRC mapping file, You should map this database layer to a VIA defined in ITF file.

---

## SX-1838

(Warning) DB layer %s does not have parasitic resistor model name

**Description**

Once you use StarRC command NETLIST_PARASITIC_RESISTOR_MODEL: YES, you need to specify model name through MODEL=<model_name> parameter for conducting layers and via layers in StarRC mapping file. StarRC gives this warning once it finds the shown layer has no model name defined in StarRC mapping file.

**What Next**

You should add MODEL=<model_name> for this shown layer in StarRC mapping file.

Feedback

## SX-1839

(Severe) VIA layer "%s" must connect layer mapped to an ITF CONDUCTOR instead of "%s".

### Description

A VIA should connect conductors. So all layers connected to this VIA should be mapped to ITF conductor layers in StarRC mapping file. StarRC will give this error once it finds a layer (shown at the end of this message) connected to the shown VIA is not mapped to an ITF conductor layer.

### What Next

You should map the layer connected to the VIA to an ITF conductor in StarRC mapping file.

## SX-1841

(Severe) Connections of via layer %s do not correspond to connections of the mapped ITF layer. ITF layer connects %s to %s. Runset layer connects %s[%s] to %s[%s].

### Description

The connections for runset via layer defined in database should correspond connections of ITF via layer defined in ITF file or nxtgrd file. For example, in database, if via A connects conductor B (from layer) and C (to layer), and via A is mapped to VIA1. In ITF, the definition of VIA1 is: VIA VIA1 { FROM = M1 TO = M2 ... } That defines M1/M2 as from/to layer of VIA1. So runset layer B and C should be mapped to ITF layer M1 and M2 in StarRC mapping file. Otherwise, StarRC will give this error.

### What Next

You should either modify mappings in StarRC mapping file, or modify from/to layer definition in ITF file to make the connections consistent.

## SX-1842

(Warning) The distance over which the interlayer capacitance should be ignored can be specified for only MOS transistors. Ignore distance setting on "%s" and "%s".

### Description

In ignore_cap_layers section of StarRC mapping file, if you use L=<length> parameter to specify the distance over which the interlayer capacitance should be ignored, you can only set L parameter between gate and diffusion, or between diffusion and SUBSTRATE of MOS transistor. For ignore-cap layer pair which doesn't contain SUBSTRATE, the 2

Feedback

layers should be either MOS gate or MOS diffusion layer. Otherwise, StarRC will give this warning and ignore the distance setting.

**What Next**

You should check whether the shown layer pair is the one you expected. If it is not, you need to correct corresponding statement in ignore_cap_layers section in StarRC mapping file.

---

## SX-1843

(Warning) The distance over which the interlayer capacitance should be ignored can be specified for only MOS transistors. Ignore distance setting on "%s" and "SUBSTRATE".

**Description**

In ignore_cap_layers section of StarRC mapping file, if you use L=<length> parameter to specify the distance over which the interlayer capacitance should be ignored, you can only set L parameter between gate and diffusion, or between diffusion and SUBSTRATE of MOS transistor. For ignore-cap layer pair which contains SUBSTRATE, the layer other than SUBSTRATE should be MOS diffusion layer. Otherwise, StarRC will give this warning and ignore the distance setting.

**What Next**

You should check whether the shown layer pair is the one you expected. If it is not, you need to correct corresponding statement in ignore_cap_layers section in StarRC mapping file.

---

## SX-1844

(Warning) The distance over which the interlayer capacitance should be ignored is not supported in FSCOMPARE/FS_EXTRACT_NETS flow. The final Field Solver report will not consider the distance.

**Description**

Field Solver flow (FSCOMPARE/FS_EXTRACT_NETS) doesn't support L=<length> parameter to specify the distance over which the interlayer capacitance should be ignored. If you use L=<length> parameter in StarRC mapping file and use Field Solver flow, L parameter will have no impact on the Field Solver result.

**What Next**

If you don't need L parameters, nothing needs to be done. Otherwise, you shouldn't use Field Solver flow and should remove FSCOMPARE/FS_EXTRACT_NETS StarRC command from command file.

Feedback

## SX-1845

(Severe) GATE layer "%s" must be mapped to ITF CONDUCTOR.

### Description

If a database layer is used as gate terminal layer of a device (The definition can be found in device table file in CCI database, or ICV_RUNSET_REPORT_FILE in ICV database), the layer must be mapped to an ITF CONDUCTOR in ITF file or nxtgrd file.

### What Next

You should map this layer to an ITF conductor layer in StarRC mapping file.

## SX-1846

(Severe) DIFF layer "%s" must be mapped to SUBSTRATE or ITF CONDUCTOR.

### Description

If a database layer is used as source/drain terminal layer of a device (The definition can be found in device table file in CCI database, or ICV_RUNSET_REPORT_FILE in ICV database), the layer must be mapped to an ITF CONDUCTOR in ITF file or nxtgrd file, or mapped to SUBSTRATE.

### What Next

You should map this layer to an ITF conductor layer or SUBSTRATE in StarRC mapping file.

## SX-1847

(Warning) Runset GATE layer and runset DIFFUSION layers are not vertically adjacent; For example, there is a CONDUCTOR between GATE and DIFFUSION in the process cross-section.

### Description

The gate layer and diffusion layers are usually vertically adjacent. If there is any other conductor between them in the process, StarRC will give this warning.

### What Next

If your process really contains this special scenario, nothing needs to be done. Otherwise, you should check ITF file to see whether there is any conductor defined between gate and diffusion, and also check StarRC mapping file to see whether the mappings for device gate and diffusion layers are correct.

Feedback

## SX-1848

(Warning) MOS gate and diffusion layers are co-vertical to each other or to a third layer like field poly. Check your process cross-section and see if the nxtgrd file is correct.

### Description

The warning messages appears when StarRC detects gate layers and diffusion layers are co-vertical to each other, or are co-vertical to same third layer based on nxtgrd file.

### What Next

If the process does have gate layers and diffusion layers which are co-vertical to each other, no action is needed. Otherwise, you should check nxtgrd file or ITF file to see whether the 3D geomerty parameters of gate or diffusion layers is incorrect or not, and correct wrong parameters.

## SX-1849

(Severe) CONDUCTOR layer "%s" must be mapped to an ITF CONDUCTOR or SUBSTRATE.

### Description

Database conducting layers defined in conducting_layers section in StarRC mapping file must be mapped to an ITF conductor or SUBSTRATE.

### What Next

You should map the shown layer to an ITF conductor or SUBSTRATE in StarRC mapping file.

## SX-1850

(Warning) The diffusion layer being mapped to a substrate layer can cause capacitance extraction inaccuracy.

### Description

The diffusion layer is mapped to a substrate layer and not to a conductor in the ITF. As a consequence, StarRC results may be inaccurate.

### What Next

The diffusion layer has to be mapped to a conductor layer in the ITF and LAYER_TYPE=DIFFUSION can be specified in the conductor statement in order to improve the capacitance extraction accuracy.

Feedback

## SX-1851

(Warning) Layer "%s" has no connections.

### Description

Layer '%s' has no connections. If the layer is a device terminal layer and there have the polygon on this layer, it may cause opens.

### What Next

If this layer is designed to have no connection(e.g., dummy or mask layer), please ignore this message. If this is a device layer, you should add layer connection on this terminal layer to a conductor layer which can touch the terminal layer polygon.

## SX-1852

(Warning) Found %d WMIN/SMIN violations when comparing database's technology file to TCAD_GRD_FILE (see layers.sum for details). Any violations of TCAD_GRD_FILE minimum width or spacing parameters for these layers in the layout may cause severe inaccuracies in parasitic extraction.

### Description

StarRC will compare WMIN/SMIN settings in database's technology file (in Milkyway database or LEF file) to those settings in nxtgrd file or ITF file. It there is any difference between the settings in database's technology file and nxtgrd file, StarRC will give this warning and output detailed information to layers.sum file.

### What Next

You should check detailed difference in layers.sum, and check which settings are correct. Use the correct settings in both StarRC ITF file and database technology file.

## SX-1853

(Information) Detail WMIN/SMIN violation report. Please note that reported database technology file parameters include application of MAGNIFICATION_FACTOR.

### Description

StarRC will output detailed WMIN/SMIN difference between database technology file and StarRC nxtgrd file in layers.sum. This is a message to inform that the MAGNIFICATION_FACTOR is already applied on values from database technology file in detailed report.

Feedback

**What Next**

You should be aware that the MAGNIFICATION_FACTOR is already applied on values from database technology file in detailed report.

---

## SX-1854

(Information) WMIN Violations (database routing minWidth is less than ITF CONDUCTOR WMIN)

**Description**

StarRC will output detailed WMIN/SMIN difference between database technology file and StarRC nxtgrd file in layers.sum. This is the beginning of detailed WMIN difference.

**What Next**

You should check detailed WMIN difference following this message.

---

## SX-1856

(Information) SMIN Violations (database routing minSpace is less than ITF CONDUCTOR SMIN)

**Description**

StarRC will output detailed WMIN/SMIN difference between database technology file and StarRC nxtgrd file in layers.sum. This is the beginning of detailed SMIN difference.

**What Next**

You should check detailed SMIN difference following this message.

---

## SX-1860

(Severe) Attempting to CONNECT two layers at two ITF layer %s %s. If you are sure the new connections should be done, you can try AUTO_RUNSET:YES

**Description**

StarRC find two layers mapped to different ITF layer need to connect together.

**What Next**

You should using AUTO_RUNSET:YES if you are sure the connections should be done.

Feedback

## SX-1861

(Fatal) TABLE_NAME %s specified on %s layer in the layer mapping file does not match any table name in %s table in the ITF file.

### Description

The table name specified in the layer mapping file does not match any table name in the table in the ITF file. StarRC does not know which sub table to look up.

### What Next

Specify correct TABLE_NAME for the shown layer in StarRC mapping file.

## SX-1862

(Fatal) Missing TABLE_NAME on layer %s in the layer mapping file. Corresponing ITF layer has multiple %s tables defined in the ITF file.

### Description

The table shown in this message has multiple sub tables in the ITF file. Each sub table has a table name. A TABLE_NAME is needed on the layer shown in this message in the layer mapping file to indicate which sub table to look up for the database layer.

### What Next

Specify TABLE_NAME for the shown layer in StarRC mapping file.

## SX-1863

(Warning) No TABLE_NAME is given for %s layer in Mapping file, while there exist%s tables in ITF.

### Description

There are a few tables which contains multiple sub-tables in ITF file, such as GATE_TO_DIFFUSION_CAP and GATE_TO_DIFFUSION_CHANNEL_CAP. If in StarRC mapping file, a database layer is mapped to an ITF layer which contains such kind of table, you should specify which sub-table should be used through parameter TABLE_NAME. For a gate layer, the possible tables which contains multiple sub-tables are: ETCH_VS_CONTACT_AND_GATE_SPACINGS GATE_TO_DIFFUSION_CAP GATE_TO_DIFFUSION_CHANNEL_CAP ER_TABLE RAISED_DIFFUSION_TO_GATE_SMIN_TABLE SIDE_TANGENT_VS_SI_WIDTH_AND_CCO_SPACING If StarRC finds a gate layer is mapped to an ITF layer which contains at least one of above tables (the table names will

Feedback

be shown in this message), but in StarRC mapping file, the layer has no TABLE_NAME parameter, StarRC will give this warning.

**What Next**

You should specify TABLE_NAME for the shown layer in StarRC mapping file.

---

## SX-1864

(Warning) TABLE_NAME %s is provided for %s layer in Mapping file, but is not found in ITF file. TABLE_NAME for this layer will be ignored.

**Description**

There are a few tables which contains multiple sub-tables in ITF file, such as GATE_TO_DIFFUSION_CAP and GATE_TO_DIFFUSION_CHANNEL_CAP. If in StarRC mapping file, a database layer is mapped to an ITF layer which contains such kind of table, you should specify which sub-table should be used through parameter TABLE_NAME. The possible tables which contain multiple sub-tables are: ETCH_VS_CONTACT_AND_GATE_SPACINGS GATE_TO_DIFFUSION_CAP GATE_TO_DIFFUSION_CHANNEL_CAP ER_TABLE RAISED_DIFFUSION_TO_GATE_SMIN_TABLE SIDE_TANGENT_VS_SI_WIDTH_AND_CCO_SPACING UPPERDIFF_TO_CONTACT_CAP RAISED_DIFFUSION_TO_GATE_SMIN_TABLE RAISED_DIFFUSION_ETCH_TABLE If TABLE_NAME is defined in StarRC mapping file for a gate or diffusion layer, but StarRC finds the corresponding sub-table are not defined in ITF file or nxtgrd file, StarRC will give this warning.

**What Next**

You should check whether the table name is correct or not, and use table name which exists in the ITF file or nxtgrd file.

---

## SX-1865

(Warning) No TABLE_NAME is given for %s layer in Mapping file, while there exist UPPERDIFF_TO_CONTACT_CAP tables in ITF.

**Description**

There are a few tables which contains multiple sub-tables in ITF file, such as UPPERDIFF_TO_CONTACT_CAP and GATE_TO_DIFFUSION_CAP. If in StarRC mapping file, a database layer is mapped to an ITF layer which contains such kind of table, you should specify which sub-table should be used through parameter TABLE_NAME. For a UPPERDIFF layer, the possible table which contain multiple sub-tables is: UPPERDIFF_TO_CONTACT_CAP If StarRC finds a UPPERDIFF layer is

Feedback

mapped to an ITF layer which contains UPPERDIFF_TO_CONTACT_CAP, but in StarRC mapping file, the layer has no TABLE_NAME parameter, StarRC will give this warning.

**What Next**

You should specify correct TABLE_NAME for the shown layer in StarRC mapping file.

---

## SX-1866

(Warning) TABLE_NAME %s is provided for %s layer in Mapping file, but is not found in RPSQ_VS_SI_WIDTH definition in ITF file. TABLE_NAME in the context of RPSQ_VS_SI_WIDTH for this layer will be ignored.

**Description**

If TABLE_NAME is defined in StarRC mapping file for a layer, but StarRC finds the corresponding sub-table are not defined in RPSQ_VS_SI_WIDTH in ITF file or nxtgrd file, StarRC gives this warning.

**What Next**

You should check whether the table name is correct or not, and use table name which exists in the ITF file or nxtgrd file.

---

## SX-1868

(Warning) TABLE_NAME %s is specified for non-gate-level or non-diff-level or non-upperdiff-level layer %s in the mapping file. TABLE_NAME for this layer will be ignored.

**Description**

In StarRC mapping file, TABLE_NAME parameter can only be used for gate/diffusion layers. If StarRC finds TABLE_NAME parameter is used on a layer other than gate/diffusion layers, StarRC will give this warning and ignore the TABLE_NAME setting on this layer.

**What Next**

If TABLE_NAME is not needed for this layer, nothing needs to be done. Otherwise, you should check why this layer is not mapped to a gate/diffusion layer.

---

## SX-1869

(Warning) None of the VIA_ARRAY_BBOX layer is assigned, please check the mapping file

Feedback

### Description

None of the VIA_ARRAY_BBOX layer is assigned. This warning message only appears with RV_VIA_ARRAY_REPORT_FILE:YES.

### What Next

You should check you mapping file and include the VIA_ARRAY_BBOX layer in the mapping file.

---

## SX-1871

(Warning) No SUBSTRATE layers are to be extracted because none of SUBSTRATE layers has rpsq value in the layer mapping file.

### Description

Once you use SUBSTRATE_EXTRACTION: YES to perform substrate extraction on layers specified in the mapping file as "substrate" layers, you should specify each substrate layer with RPSQ value in StarRC mapping file. If RPSQ is not set for a substrate layer, the layer will be treated as ideal and no resistance extraction will be applied on this layer. StarRC gives this warning once it finds none of the substrate layers has RPSQ settings in StarRC mapping file. As a result, all substrate layers will be treated as ideal and SUBSTRATE_EXTRACTION: YES will have no effect.

### What Next

If you don't want to extract resistance from substrate layers, nothing needs to be done. Otherwise, you should add RPSQ value for each substrate layers in StarRC mapping file.

---

## SX-1872

(Warning) Set RPV*AREA value to 1e-6 due to RPV_VS_AREA and RPV_VS_WIDTH_AND_LENGTH for %s

### Description

StarRC gives this warning when RPV value for the shown layer is set to 0 in StarRC mapping file and RPV_VS_AREA or RPV_VS_WIDTH_AND_LENGTH is the mapped ITF layer used in ITF file. The setting in StarRC mapping file will take precedence and StarRC will set RPV*AREA value to a small value 1e-6 for 0 RPV value setting in StarRC mapping file.

### What Next

If you want to use RPV_VS_AREA or RPV_VS_WIDTH_AND_LENGTH defined in ITF file, you should remove RPV setting for this layer in StarRC mapping file. Otherwise, nothing needs to be done.

Feedback

## SX-1873

(Warning) Setting RPV*AREA value of 1e-6 for %s

**Description**

If you set RPV value as 0 in either StarRC mapping file or ITF file, or no RPV value or RPV table is defined in StarRC mapping file and ITF file, StarRC will set RPV*AREA to a small value 1e-6 and gives this warning.

**What Next**

If small RPV setting is what you expected, nothing needs to be done. Otherwise, you should define RPV value in StarRC mapping file or ITF file for the shown layer and use positive value.

## SX-1874

(Fatal) Invalid resistance information for layer %s

**Description**

StarRC gives this warning once it finds resistance information for the shown via layer is invalid. You can define resistance information for via through RPV parameter in StarRC mapping file or RPV/RHO/RPV_VS_AREA in ITF file. The value should be 0 or positive. If StarRC finds a negative resistance value for the shown via layer, it will give this error.

**What Next**

You should check the resistance settings for the shown layer in StarRC mapping file and ITF file and correct the incorrect settings.

## SX-1875

(Fatal) INTERNAL Error: Could not write fs_ignore_cap_layer information file

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## SX-1878

(Warning) Invalid IGNORE_CAP_LAYERS command. VIA layers must be listed alone with IGNORE_CAP_LAYERS. Layer pair "%s" and "%s" will be ignored.

Feedback

**Description**

This message appears when StarRC finds one layer of this ignore cap layer pair defined in ignore_cap_layers section of mapping file is a VIA. For a VIA, you should list it alone in the ignore_cap_layers section. For example: ignore_cap_layers: VIA1 VIA shouldn't be in a ignore cap layer pair, such as: ignore_cap_layers: VIA1 psub The invalid ignore cap layer pair will be ignored by StarRC

**What Next**

If you want to ignore capacitance between this VIA and all other layers, you should just list this VIA alone as a single line in ignore_cap_layers section. If you don't want to ignore capacitance of this VIA, no action is needed.

---

## SX-1879

(Fatal) Incorrect ignore cap description. "%s" is not via layer and cannot be listed alone with IGNORE_CAP_LAYERS.

**Description**

Incorrect ignore cap description."%s" is not via layer and cannot be listed alone with IGNORE_CAP_LAYERS in mapping file.

**What Next**

You should revise your mapping file to avoid list only one conductor layer in IGNORE_CAP_LAYERS section.

---

## SX-1882

(Warning) VIA layer

**Description**

VIA layer has no connections.(no FROM database layer and TO database layer defined in mapping file).

**What Next**

You should revise your mapping file to add FROM and TO layer on layer %s.

---

## SX-1883

(Fatal) INTERNAL Error: Unsupported layer flag %d found in layers DB.

Feedback

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1884

(Severe) SEARCH_DISTANCE for CONDUCTOR "%s" in TCAD_GRD_FILE is too large. SEARCH_DISTANCE must be no greater than 10um and no greater than 25 * (WMIN + SMIN).

**Description**

SEARCH_DISTANCE for specified CONDUCTOR in TCAD_GRD_FILE is too large. SEARCH_DISTANCE must be no greater than 10um and no greater than 25 * (WMIN + SMIN).

**What Next**

Set SEARCH_DISTANCE in ITF file to a value that no greater than 10um and no greater than 25 * (WMIN + SMIN), then re-generate TCAD_GRD_FILE.

---

## SX-1885

(Severe) SEARCH_DISTANCE for CONDUCTOR "%s" in TCAD_GRD_FILE must be positive.

**Description**

SEARCH_DISTANCE for specified CONDUCTOR in TCAD_GRD_FILE must be positive.

**What Next**

Set SEARCH_DISTANCE in ITF file to a positive value, then re-generate TCAD_GRD_FILE.

---

## SX-1886

(Severe) In layer %s, width of a density box entry cannot exceed 500.0

**Description**

Width of a density box entry for the specified layer cannot exceed 500.0.

Feedback

**What Next**

Set correct value in ITF file and re-generate TCAD_GRD_FILE.

---

## SX-1887

(Severe) Invalid search distance specification for one or more layers in TCAD_GRD_FILE. Please run concurrent release of grdgenxo on ITF file and regenerate the TCAD_GRD_FILE.

**Description**

Invalid search distance specification for one or more layers in TCAD_GRD_FILE. Please run concurrent release of grdgenxo on ITF file and regenerate the TCAD_GRD_FILE.

**What Next**

Set SEARCH_DISTANCE in ITF file to a positive value and no greater than 10um and no greater than 25 * (WMIN + SMIN), then re-generate TCAD_GRD_FILE.

---

## SX-1888

(Severe) Number of CONDUCTORs in nxtgrd file is limited to 7, if SEARCH_DISTANCE is specified.

**Description**

The message appears when StarRC finds number of conductors is large than 7 and at the same time, SEARCH_DISTANCE is specified in the nxtgrd file. StarRC only supports SEARCH_DISTANCE for processes which don't have more than 7 conductors.

**What Next**

You should remove SEARCH_DISTANCE defintion in ITF file since this process has more than 7 conductors.

---

## SX-1889

(Fatal) Could not read layer information file %s

**Description**

Could not read layer information file.

**What Next**

Make sure you have correctly set the layer infomation file and you have permission to read.

Feedback

## SX-1890

(Fatal) Layer information file corrupted - %s

**Description**

Layer information file corrupted. There's something wrong when parsing layer information file.

**What Next**

Make sure the layer information file is written in correct format.

## SX-1892

(Severe) TCAD_GRD_FILE is corrupted.

**Description**

TCAD_GRD_FILE is corrupted.

**What Next**

Make sure you have TCAD_GRD_FILE and you have permission to read.

## SX-1893

(Fatal) GRD File contains multiple diffusion layers. Layers %s and %s

**Description**

GRD File contains multiple diffusion layers.

**What Next**

Make sure you have only one diffusion layer defined in ITF file and re-generate TCAD_GRD_FILE.

## SX-1894

(Fatal) GRD File contains multiple gate layers. Layers %s and %s

**Description**

GRD File contains multiple gate layers.

Feedback

**What Next**

Make sure you have only one gate layer defined in ITF file and re-generate TCAD_GRD_FILE.

---

## SX-1899

(Information) Reading Process file %s

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1900

(Fatal) Conductor layer %s defined in grd file but not in process file

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1901

(Information) Done

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1902

(Fatal) While parsing process file %s at line %d: "%s" Conformal dielectrics not supported

**Description**

Not available.

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## SX-1903

(Fatal) Conductor layer %s found in process file but not in grd file

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1904

(Fatal) Physical layer %s is multiply defined in process file

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1905

(Fatal) Syntax error in process file %s at line %d: "%s"

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1906

(Warning) Incorrect MAP_LAYER %s

**Description**

You receivce this messsages because StarRC find invalid MAP_LAYER command. Valid MAP_LAYER command needs to have at least 2 strings: <layer_name> <layer_type>. All invalid MAP_LAYER commands will be ignored by StarRC.

Feedback

**What Next**

You should check the MAP_LAYER command shown in this messages and correct the command.

---

## SX-1907

(Warning) Section has not been described Line: %s

**Description**

You receive this message because StarRC can't find definition for a section before the line shown in this message when paring StarRC mapping file. The line shown in this messages will be ignored by StarRC. In StarRC mapping file, you should first define a section name such as "conducting_layers", "via_layers" and so on, and then add mapping content to each section.

**What Next**

You should add correct section name in StarRC mapping file before the line shown in this message.

---

## SX-1908

(Warning) Token %s should be an integer Line: %s

**Description**

Some parameters in StarRC mapping file needs to have an integer value, such as precedence. You receive this message because StarRC find the value for that parameter is not an integer.

**What Next**

You should correct the wrong value of the parameter with an integer.

---

## SX-1909

(Warning) Parsing mapping file, Wrong syntax in line %s

**Description**

You receive this message because StarRC finds there is no value for certain parameter in the shown line in StarRC mapping file.

**What Next**

You should add correct value for all parameters in the shown line.

Feedback

## SX-1911

(Warning) Parsing mapping file, There should be "=" after token %s in line %s

**Description**

You receive this message because StarRC find no "=" after a parameter in the shown line in StarRC mapping file. In StarRC mapping file, you should always use <parameter_name>=<value>

**What Next**

You should correct the parameter definition in the shown line.

## SX-1912

(Fatal) Layer %s multiply defined Line: %s

**Description**

You receive this message because StarRC finds the shown layer is defined more than once in StarRC mapping file. You should only define a conducting/via/marker/remove/view_only layer once.

**What Next**

You should remove all other definitions of the shown layer and keep only one.

## SX-1913

(Warning) No physical layer described for conducting layer Line: %s

**Description**

You receive this message because StarRC find no mapped physical layer for a database layer defined in color_layers section in StarRC mapping file. The syntax for a line in color_layers section should be: color_layer_name physical_layer_name [x=x_shift y=y_shift]

**What Next**

You should add mapped physical layer for the shown line.

## SX-1914

(Warning) Physical layer %s does not exist in process Line: %s

Feedback

**Description**

You receive this message because StarRC find the mapped physical layer for a database layer defined in color_layers section in StarRC mapping file is not defined in ITF or nxtgrd file. The syntax for a line in color_layers section should be: color_layer_name physical_layer_name [x=x_shift y=y_shift]. The physical_layer_name is a layer defined in ITF or nxtgrd file.

**What Next**

You should map the color layer to a layer defined in ITF or nxtgrd file.

---

## SX-1915

(Warning) Unknown color layer specification %s Line: %s

**Description**

You receive this message because StarRC can't recognize certain syntax defined in color_layers section in StarRC mapping file. The syntax for a line in color_layers section should be: color_layer_name physical_layer_name [x=x_shift y=y_shift]. Any other syntax will trigger this error.

**What Next**

You should remove the un-recognized syntax in the shown line.

---

## SX-1916

(Warning) the db layer:%s do not have the mapping in the map_qtf_layers section

**Description**

You receive this message because StarRC finds no mapped qtf layer for the database layer in the shown line in StarRC mapping file. The syntax for a line in map_qtf_layers section should be: <database_layer_name> <mapped_qtf_layer_name>

**What Next**

You should add mapped qtf layers in this line.

---

## SX-1917

(Warning) the qtf layer :%s in the qtf_layers section is not existed in the map_qtf_layers section

Feedback

**Description**

You receive this message because StarRC finds a qtf layer is defined in qtf_layers section but no database layer is mapped to this qtf layer in StarRC mapping file.

**What Next**

You should check whether there are any mappings missed in map_qtf_layer. If not, you can either remove this qtf layer from qtf_layers section or ignore this warning.

---

## SX-1918

(Warning) No physical layer described for conducting layer Line: %s

**Description**

You receive this message because StarRC find no mapped physical layer for a layer in conducting_layers section in StarRC mapping file. Each line in conducting_layers section needs at least a database layer name and a mapped physical layer name.

**What Next**

You should add the mapped physical layer to this line.

---

## SX-1919

(Warning) Physical layer %s does not exist in process Line: %s

**Description**

You receive this message because StarRC finds the mapped physical layer %s in StarRC mapping file is not defined in ITF file or nxtgrd file.

**What Next**

You should map the database layer to a physical layer defined in ITF file or nxtgrd file.

---

## SX-1920

(Warning) PRECEDENCE can be used only on SUBSTRATE Line: %s

**Description**

You receive this message because StarRC find there is precedence definition on non-substrate layer in StarRC mapping file. Precedence parameter can be defined only on substrate layer.

Feedback

**What Next**

You should remove the precedence definition from the non-substrate file in StarRC mapping file.

---

## SX-1921

(Warning) Invalid negative precedence value. Precedence value can be -1, 0, or any positive integer. Line: %s

**Description**

You receive this message because StarRC find the value for precedence parameter a negative value other than -1 in StarRC mapping file. The valid value for precedence is -1, 0, or any positive integer.

**What Next**

You should correct the precedence value with a valid integer.

---

## SX-1922

(Warning) Unknown floating as fill type: %s. Only "FLOATING" and "GROUND" are available! Line: %s

**Description**

You receive this message because StarRC find the value of floating_as_fill_type is neither "floating" nor "ground" in StarRC mapping file. Only "floating" and "ground" are valid value for floating_as_fill_type parameter.

**What Next**

You should use either "floating" or "ground" as the value of floating_as_file_type.

---

## SX-1923

(Warning) Unknown conducting layer specification %s Line: %s

**Description**

You receive this message because StarRC find invalid parameter on a conducting layer in StarRC mapping file. The syntax of a conducting layer is: db_LyrName itf_LyrName|SUBSTRATE [device_layer|capacitor_layer] [rpsq=r_value] [model=model_name] [mask=mask_num] [diffusion_res_model=diff_res_model_name] [net_segment_cut_length=cut_length] [overlap_fill_spacing=new_spacing] [table_name=cap_table]

Feedback

**What Next**

You should remove the invalid parameter on the conducting layer in StarRC mapping file.

---

## SX-1924

(Error) Database layer %s cannot map to SUBSTRATE with NXTGRD generated with NO_GROUND_PLANE=YES

**Description**

You receive this message because StarRC finds the database layer %s in StarRC mapping file maps to SUBSTRATE or SECOND_SUBSTRATE with nxtgrd generated with NO_GROUND_PLANE=YES

**What Next**

You should not use SUBSTARTE related layer with nxtgrd generated with NO_GROUND_PLANE=YES

---

## SX-1925

(Warning) AREA cannot be used with CRES or AREA cannot be set more than once

**Description**

You receive this message because StarRC find the AREA or CRES has been defined before for a via layer in StarRC mapping file. AREA parameter can be defined only once, and can't be used together with CRES.

**What Next**

You should remove the extra definition of CRES or AREA.

---

## SX-1926

(Warning) MAX_VIA_ARRAY_SPACING is specified for via layer %s connected to a diffusion layer in the mapping file and USER_DEFINED_DIFFUSION_RESISTANCE models are set for at least one gate layer. The user defined diffusion resistances having a merged via as the contact are merged as one resistor.

**Description**

Setting MAX_VIA_ARRAY_LENGTH or MAX_VIA_ARRAY_SPACING reduce the vias of the corresponding layer, if the conditions are right, to one via and eventually one resistor. Therefore, the user defined diffusion resistances having a merged via as the contact are merged as one resistor.

Feedback

**What Next**

If having merged resistors is the expected behavior, no action is needed. Otherwise, remove MAX_VIA_ARRAY_LENGTH or MAX_VIA_ARRAY_SPACING for the contact layers connected to a diffusion layer in the mapping file.

---

## SX-1928

(Warning) Unknown via layer specification %s Line: %s

**Description**

You receive this message because StarRC find invalid parameter used for a via layer in StarRC mapping file. The syntax of a via layer should be: db_layerName grd_layerName [device_layer] [rpv=rpv_value] [area=via_area] [MODEL= model_name] [MAX_VIA_ARRAY_LENGTH = length] [MAX_VIA_ARRAY_SPAING = spacing]

**What Next**

You should remove invalid parameters on the via layer.

---

## SX-1929

(Warning) Both the via resistance and area must be specified Line: %s

**Description**

You receive this message because StarRC find only AREA or only RPV is defined for a via layer in StarRC mapping file. You should either use both of the 2 parameters or use neither of the 2 parameters.

**What Next**

If yout want to use AREA or RPV, you should use both of them. If you don't want to use, you should remove the parameter.

---

## SX-1930

(Warning) FROM and TO must be used together for a VIA layer for line %s

**Description**

You receive this message because StarRC finds you use only "from" or only "to" parameter for a via layer. You should either use both of the 2 parameters or use neither of the 2 parameters.

Feedback

**What Next**

If yout want to specify 'from' layer or 'to' layer of the via layer, you should specify both. Otherwise, you should remove the parameter.

---

## SX-1931

(Warning) The value of MAX_VIA_ARRAY_SPACING or MAX_VIA_ARRAY_LENGTH could not be specified to a negative value %s

**Description**

You receive this message because StarRC find value of MAX_VIA_ARRAY_SPACING or MAX_VIA_ARRAY_LENGTH is negative, which is invalid in StarRC mapping file.

**What Next**

You should use positive value for parameter MAX_VIA_ARRAY_SPACING or MAX_VIA_ARRAY_LENGTH.

---

## SX-1932

(Fatal) MAX_VIA_ARRAY_SPACING must be specified to use via merge Line: %s

**Description**

If you set MAX_VIA_ARRAY_LENGTH or MAX_VIA_ARRAY_SPACING in StarRC mapping file, you should either set both of these 2 parameters or set MAX_VIA_ARRAY_SPACING alone. However, you can't set MAX_VIA_ARRAY_LENGTH along without setting MAX_VIA_ARRAY_SPACING. StarRC will give this error once it finds this invalid situation.

**What Next**

You should also set MAX_VIA_ARRAY_SPACING in this line.

---

## SX-1933

(Fatal) Layer %s multiply defined Line: %s

**Description**

You receive this message because StarRC finds the shown layer is defined more than once in StarRC mapping file. You should only define a conducting/via/marker/remove/view_only layer once..

Feedback

**What Next**

You should remove all other definitions of the shown layer and keep only one.

---

## SX-1934

(Fatal) FROM layer %s does not exist in process Line: %s

**Description**

You receive this message because StarRC finds the from layer %s of a via layer is not defined in StarRC mapping file.

**What Next**

You should use a layer already defined in StarRC mapping file as the from layer of a via layer.

---

## SX-1935

(Fatal) TO layer %s does not exist in process Line: %s

**Description**

You receive this message because StarRC finds the to layer %s of a via layer is not defined in StarRC mapping file.

**What Next**

You should use a layer already defined in StarRC mapping file as the to layer of a via layer.

---

## SX-1936

(Warning) Physical layer %s does not exist in process Line: %s

**Description**

This means the via defined in this line is mapped to a physical layer which is not defined in ITF or nxtgrd file.

**What Next**

You should correct the physical layer and map the via to a physical layer already defined in ITF or nxtgrd file.

Feedback

## SX-1937

(Warning) Unknown marker type %s Line: %s

**Description**

In marker_layers section, you should either only list the marker layer name or map the marker to MARKER. This message means the layer in this line is mapped to a layer other than MARKER

**What Next**

You should only list the marker layer name and remove any other tokens.

## SX-1938

(Warning) Incorrect remove description Line: %s

**Description**

Each line in remove_layers section in StarRC mapping file should only contain the removed layer name. Any other description is invalid and will trigger this warning.

**What Next**

You should only list removed layer name and remove any other tokens.

## SX-1939

(Warning) Incorrect graphic layers description Line: %s

**Description**

Each line in viewonly_layers section in StarRC mapping file should only contain the view only layer name. The message means this line is invalid.

**What Next**

You should only list view only layer name in this line and remove any other tokens.

## SX-1940

(Warning) The distance over which the interlayer capacitance should be ignored must be used together with exactly two layers in a single line. Line: %s

Feedback

**Description**

In ignore_cap_layers section of StarRC mapping file, if you use L=<length> parameter to specify the distance over which the interlayer capacitance should be ignored, you should only add 2 layers to ignore cap in this line. For example, you can only use: A B L=value, but you can't use: A B C L=value. StarRC will give this warning once it finds the invalid statement.

**What Next**

You should correct the ignore-cap statement in this line.

---

## SX-1941

(Warning) The distance over which the interlayer capacitance should be ignored must be positive. Line: %s

**Description**

In ignore_cap_layers section of StarRC mapping file, if you use L=<length> parameter to specify the distance over which the interlayer capacitance should be ignored, you should only use positive value for L parameter. Otherwise, StarRC will give this warning.

**What Next**

You should use a positive value for L parameter in this line.

---

## SX-1942

(Warning) Incorrect ignore cap description, must contain at least two layers Line: %s

**Description**

The description in ignore_cap_layers section in StarRC mapping file is incorrect. In ignore_cap_layers section, if you want to ignore cap for a via layer, you should only list the layer name of the via. If you want to ignore cap between 2 or more conducting layers, you should list layer names of the conducting layers (at least 2 layers).

**What Next**

Correct the description of ignore cap layers in this line.

---

## SX-1943

(Warning) The distance over which the interlayer capacitance should be ignored between "%s" and "%s" has been set twice. Line: %s

Feedback

**Description**

In ignore_cap_layers section of StarRC mapping file, if you use L=<length> parameter to specify the distance over which the interlayer capacitance should be ignored, you only need to specify L parameter once for same layer pair. Otherwise, StarRC will give this warning.

**What Next**

You should check whether the duplicate definition is due to typo mistake of layer name in StarRC mapping file. If not, you should remove this statement in ignore_cap_layers section in StarRC mapping file.

---

## SX-1944

(Information) License %s no longer valid, requesting again

**Description**

StarRC failed to renew a license. StarRC will repeatedly attempt to for a licenseuntil a timeout period is reached.

**What Next**

If this message appears repeatedly, check to make sure that license server is running, and that there are enough licenses available.

---

## SX-1945

(Information) ... Waiting for license (%d attempts).

**Description**

StarRC is making repeated attempts to check out a license.

**What Next**

If license checkout fails or is unduly delayed, please make sure sufficient StarRC licenses are available.

---

## SX-1946

(Information) Got license after %d attempts in %d min, continuing.

**Description**

StarRC successfully acquired a license after the reported number of attempts, and time waited.

Feedback

**What Next**

If StarRC requires unexpectedly large number of attempts before checking out a license, make sure that sufficient number of StarRC licenses are available.

---

## SX-1947

(Warning) %s

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1948

(Severe) Failed to obtain a license for %s %d with license key %d

**Description**

Failed to check out a license.

**What Next**

Check to make sure sufficient licenses are available.

---

## SX-1949

(Information) License %d is being re-validated.

**Description**

StarRC is re-validating a license that was previously checked out.

**What Next**

This message is for informational purposes only. No further action is necessary.

---

## SX-1959

(Fatal) This program has been built for Synopsys internal use only, and cannot be run outside Synopsys offcies.

Feedback

**Description**

This version of StarRC is a Synopsys internal build. It cannot be used outside Synopsys offices.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1960

(Warning) This executable is for DEMO purposes only.

**Description**

This build for StarRC is for demonstration purposes only, and should not be used for production runs.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1961

(Warning) This executable is an ALPHA version and is for testing purposes only.

**Description**

This build is an ALPHA version of StarRC. It should not be used for production runs.

**What Next**

Contact the Synopsys Support Center.

---

## SX-1962

(Warning) This version of StarRC is for DEMO purposes only.

**Description**

This is a special build of StarRC, intended for demo purposes only. It should not be used for production purposes.

**What Next**

Contact the Synopsys Support Center.

Feedback

## SX-1963

(Warning) Failed to read a license file. Checking out an Ultra license

**Description**

StarRC was not able to inherit a license by reading an encrypted license file. As a result, it will proceed to check out an Ultra license or equivalent.

**What Next**

Contact the Synopsys Support Center.

## SX-1964

(Fatal) Failed to stream-read NDM design %s for %s.

**Description**

StarRC was not able to stream-read NDM design. It may be caused by un-supported NDM objects.

**What Next**

Contact the Synopsys Support Center.

## SX-1965

(Fatal) failed to check out StarRC license

**Description**

StarRC failed to checkout a license.

**What Next**

Check to make sure you have sufficient StarRC licenses, and that the license servers are running properly.

## SX-1966

(Fatal) failed to check out DP manager license

**Description**

StarRC failed to check out a valid DP license.

Feedback

**What Next**

Check to make sure you have sufficient StarRC licenses, and that the license servers are running.

---

## SX-1967

(Warning) Fast field solver could not obtain required number of STARRC licenses. Trying again.

**Description**

The required number of licenses for running the field solver are not available.

**What Next**

If this message is repeated over several hours, check to make sure you have sufficient number of StarRC licenses and that the license server is running.

---

## SX-1971

(Warning) Fast field solver could not obtain required number of STARRC licenses. Will continue with %d processors

**Description**

StarRC was not able to obtain the required number of StarRC licenses to run the field solver using the desired number of processors. As a result, the field solver is being run with a reduced number of processors.

**What Next**

If the turn-around time for the field solver job is too slow, contact the Synopsys Customer Support Center.

---

## SX-1973

(Fatal) %s

**Description**

StarRC terminates execution because of an error.

**What Next**

Contact the Synopsys Support Center.

Feedback

## SX-1974

(Fatal) Invalid RECT geometry.

**Description**

RECT in DEF must have 2 points.

**What Next**

Correct the RECT in DEF.

## SX-1975

(Fatal) LAYER '%s' has no definition

**Description**

A referenced layer in DEF should have a valid defintion in LEFs. Physical geometries bound to it will be ignored during DEF translation. This missing LAYER may cause false opens in the later extraction stage.

**What Next**

Check your input LEF files and provide missing LAYER definition.

## SX-1976

(Fatal) Illegal PATTERN '%s' of VIARULE '%s'.

**Description**

Illegal PATTERN.

**What Next**

Check PATTERN in LEF/DEF specification for valid cut pattern.

## SX-1977

(Fatal) LEF VIA '%s' doesn't specify LAYER in its RECT statement.

**Description**

RECT statements for a LEF VIA must be defined on certain layer.

**What Next**

Check the definition of the VIA and provide LAYER.

Feedback

## SX-1978

(Fatal) INTERNAL Error: the x, y number of via array should be positive integer

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## SX-1981

(Warning) The BUSBITSCHARS is empty

**Description**

BUSBITCHARS in LEF/DEF is empty.

**What Next**

Check the BUSBITCHARS statement in your LEF/DEF files and refer to LEF/DEF specification about how to specify correct BUSBITCHARS.

## SX-1982

(Warning) Rounding error may happen when using floats to represent points in DEF

**Description**

Integer is recommended to be used in regular wiring. You are using floats, the decimals will be rounded off.

**What Next**

Check input DEF file and use integers instead.

## SX-1984

(Warning) Parse only for generated via in VIA section

**Description**

VIARULE sections in LEF are parse-only statements. Any vias created from VIARULE GENERATE rule that appear in the DEF NETS or SPECIALNETS sections must also appear in the DEF VIA section.

Feedback

**What Next**

Information only; no action required.

---

## SX-2020

(Fatal) DISCONNECT_PORT_AND_NET_IN_SKIP_CELLS only works for NDM flow or LEF/DEF flow

**Description**

DISCONNECT_PORT_AND_NET_IN_SKIP_CELLS command only works for NDM flow or LEF/DEF flow.

**What Next**

Remove DISCONNECT_PORT_AND_NET_IN_SKIP_CELLS in command file and run StarRC again.

---

## SX-2021

(Information) Duplicate cell definition for '%s' is found and ignored in '%s'.

**Description**

Multiple GDS/OASIS cell definitions are found for a skip cell. StarRC uses the first cell definition to synthesize the skip cell and ignores the duplicate cell definitions.

**What Next**

Information only; no action needed.

---

## SX-2022

(Fatal) lib '%s' is not defined in '%s'

**Description**

OA lib is not defined in OA_LIB_DEF file.

**What Next**

Make sure that the lib is defined in OA_LIB_DEF file.

---

## SX-2023

(Fatal) Cannot open design '%s', '%s', '%s'

Feedback

**Description**

Cannot open design lib/cell/view.

**What Next**

Make sure that the lib/cell/view exists and you have the permission to read.

---

## SX-2024

(Fatal) Cannot open design '%s', '%s', '%s'

**Description**

Cannot open design lib/cell/view.

**What Next**

Make sure that the lib/cell/view exists and you have the permission to read.

---

## SX-2025

(Warning) '%s'

**Description**

This message is from OA library, when StarRC tries to open OA_LIB_DEF file.

**What Next**

Contact the Synopsys Support Center.

---

## SX-2026

(Warning) syntax error in '%s': '%s'

**Description**

This message is from OA library, when StarRC tries to open OA_LIB_DEF file.

**What Next**

Contact the Synopsys Support Center.

---

## SX-2027

(Warning) '%s' load warning: '%s'

Feedback

**Description**

This message is from OA library, when StarRC tries to open OA_LIB_DEF file.

**What Next**

Contact the Synopsys Support Center.

---

## SX-2028

(Fatal) Cannot open design for '%s', '%s', '%s'

**Description**

This message is from OA library, when StarRC tries to open design.

**What Next**

Contact the Synopsys Support Center.

---

## SX-2029

(Fatal) lib '%s' is not defined in '%s'

**Description**

In OA flow, an error occurs when trying to open design.

**What Next**

Make sure that the lib is defined in OA_LIB_DEF.

---

## SX-2030

(Severe) GDS NETPROP NUMBER %d is not supported.

**Description**

StarRC do not support the "GDS NETPROP NUMBER %d" in Calibre query server commands.

**What Next**

You should revise your Calibre query server command file. StarRC only support "GDS NETPROP NUMBER 5".

---

## SX-2031

(Severe) GDS PLACEPROP NUMBER %d is not supported.

Feedback

**Description**

StarRC do not support the "GDS PLACEPROP NUMBER %d" in Calibre query server commands.

**What Next**

You should revise your Calibre query server command file. StarRC only support "GDS PLACEPROP NUMBER 6".

---

## SX-2032

(Severe) GDS DEVPROP NUMBER %d is not supported.

**Description**

StarRC do not support the "GDS DEVPROP NUMBER %d" in Calibre query server commands.

**What Next**

You should revise your Calibre query server command file. Now StarRC only support "GDS DEVPROP NUMBER 7".

---

## SX-2033

(Warning) NDM design database '%s' (schema: %s) doesn't support fast streaming read

**Description**

Old NDM database with schema version < 1.223 uses data model scheme that streaming read can't support. StarRC will switch to slower data reading for such design database.

**What Next**

No actions required.

---

## SX-2034

(Warning) Wildcards in SKIP_CELL_PORT_PROP_FILE are supported only in ICV flow.

**Description**

Wildcards are supported only in ICV flow. For other flows, wildcard settings are ignored and have no effects to the extraction.

**What Next**

No actions required.

Feedback

## SX-2035

(Warning) METAL_FILL_BLOCK_MAPPING_FILE is recommended for hierarchical metal fill extraction.

**Description**

METAL_FILL_BLOCK_MAPPING_FILE is recommended for designs with multiple MACRO_DEF_FILE(s) and METAL_FILL_GDS_FILE/METAL_FILL_OASIS_FILE.

**What Next**

No actions required.

## SX-2036

(Warning) More than 80%% (%d/%d) of the instances in the design '%s' are placed in its %s.

**Description**

If most of the instances (80%) are placed at corners of the design, it might introduce significant performance/memory issue during extraction of these corner partitions.

This is usually seen in the early-stage design which most of its instances are unplaced.

**What Next**

Check the design and see if most of the instances are unplaced.

## SX-2037

(Warning) Suspicious instance placement '%s' in desing '%s' - %s

**Description**

If an instance is placed outside die area, it may introduce accuracy problems, e.g., false open/short.

**What Next**

Check the instance in the design.

## SX-2038

(Warning) PIN '%s' is placed outside DIEAREA

Feedback

**Description**

If a pin is placed outside die area, it may introduce accuracy problems, e.g., false open/short.

**What Next**

Check the pin in the design.

---

## SX-2039

(Warning) Any NDM layer shape on layerPurpose %s (%d:%d) will %s ignored

**Description**

A Layer in technology file can declare itself as physical layer or user-purpose layer by nonMask in its associated LayerDataType struct. By default, StarRC honors this setting and automatically ignore shapes drawn on user-purpose layers (nonMask=1).

NDM_FORCE_IGNORE_USER_PURPOSE_SHAPE will bypass this checking and ignores shapes if their layerPurpose are specified in this option.

**What Next**

Review the technology database and check LayerDataType(s).

---

## SX-2040

(Fatal) Input Calibre query command file error.

**Description**

StarRC not support the value of GDS NETPROP NUMBER, GDS PLACEPROP NUMBER or GDS DEVPROP NUMBER defined in the Calibre query command file.

**What Next**

You should revise your Calibre query server commands file. Now StarRC only support "GDS NETPROP NUMBER 5", "GDS PLACEPROP NUMBER 6" and "GDS DEVPROP NUMBER 7".

---

## SX-2041

(Warning) GDS SEED PROPERTY is not set in query file.

**Description**

GDS SEED PROPERTY is not set in the query file.

Feedback

**What Next**

You should add GDS SEED PROPERTY in the calibre query file. Now StarRC only support GDS SEED PROPERTY ORIGINAL or GDS SEED PROPERTY DEVICE ORIGINAL.

---

## SX-2042

(Warning) GDS SEED PROPERTY ORIGINAL affects CCI AGF layer mapping; this command should occur before the GDS MAP command in Calibre query file. GDS MAP file could be invalid.

**Description**

GDS SEED PROPERTY ORIGINAL, when specified, must come before the GDS MAP command in the Calibre query command file.

**What Next**

You should revise your Calibre query server command file.

---

## SX-2043

(Warning) Non-RECT %s with SPACING %d on LAYER %s will not be shrunk

**Description**

Only RECT spacing OBS/BLOCKAGE are shrunk by SPACING and SMIN.

**What Next**

Check the POLYGON/PATH BLOCKAGE in the DEF. POLYGON/PATH OBS in the LEF.

---

## SX-2044

(Warning) Duplicated property: %d %s

**Description**

StarRC detects duplicated properties in GDSII elements. The first property is honored and the later one is ignored.

**What Next**

If the duplicated property is expected, no action is required.

Feedback

## SX-2045

(Warning) Found %s(s) with SPACING. %s shapes can be shrunk by their SPACING and layer SMIN in ITF if %s:SHRINK.

### Description

StarRC found OBS/BLOCKAGE(s) with SPACING. StarRC recommends to use

```
LEF_USE_OBS: SHRINK
TRANSLATE_DEF_BLOCKAGE: SHRINK
to shrink shape automatically by the SPACING and layer SMIN to improve
accuracy.

The shape is shrunk by the following rules but only RECT is shrunk and if
 the shrunk shape is NOT valid, the shape will be dropped.
SPACING == 0, shrinks the blockage by SMIN in ITF
SPACING > 0 and SMIN > SPACING, shrinks the blockage by (SMIN - SPACING)
```

### What Next

Information only.

## SX-2046

(Warning) %s ( %d %d ) ( %d %d ) with SPACING %d on LAYER %s (SMIN=%d) can not be shrunk and will be dropped!

### Description

StarRC shrinks RECT spacing blockages by their SPACING and layer SMIN. But, if the resulted shape is NOT valid, StarRC will still translate un-shrunk shape.

SMIN in the message is reported in the technology unit.

### What Next

Check the spacing blockages in the DEF.

## SX-2047

(Fatal) LAYOUT NETLIST SEPARATED PROPERTIES must be set as NO for PDBA/PDSP flow in query file.

### Description

LAYOUT NETLIST SEPARATED PROPERTIES must be set as NO for PDBA/PDSP flow in query file.

Feedback

**What Next**

You should revise your Calibre query server command file(set the LAYOUT NETLIST SEPARATED PROPERTIES as NO).

---

## SX-2048

(Fatal) LAYOUT NETLIST PIN LOCATIONS must be set as YES for non-PDBA flow in query file.

**Description**

LAYOUT NETLIST PIN LOCATIONS should be YES in non-PDBA flow(not specify CALIBRE_PDBA_FILE in StarRC command file).

**What Next**

You should revise your Calibre query server commands file(set LAYOUT NETLIST PIN LOCATIONS as YES).

---

## SX-2049

(Fatal) LAYOUT NETLIST HIERARCHY must be set as AGF in query file.

**Description**

StarRC only support LAYOUT NETLIST HIERARCHY AGF

**What Next**

You should revise your Calibre query server commands file(set LAYOUT NETLIST HIERARCHY as AGF).

---

## SX-2050

(Warning) LAYOUT NETLIST PIN LOCATIONS NO and LAYOUT NETLIST DEVICE TEMPLATES YES is recommended combination for PDBA/PDSP flow in query file.

**Description**

StarRC recommends to use LAYOUT NETLIST PIN LOCATIONS NO and LAYOUT NETLIST DEVICE TEMPLATES YES together.

**What Next**

You should revise your Calibre query server commands file.

Feedback

## SX-2051

(Warning) LAYOUT NETLIST TRIVIAL PINS YES is recommended in query file.

**Description**

LAYOUT NETLIST TRIVIAL PINS YES is recommended in query file.

**What Next**

You should revise your Calibre query server commands file(set LAYOUT NETLIST TRIVIAL PINS as YES).

## SX-2052

(Fatal) Active-fill METAL_FILL_RECONNECT flow requires METAL_FILL_POLYGON_HANDLING: FLOATING.

**Description**

Active-fill METAL_FILL_RECONNECT flow only supports METAL_FILL_POLYGON_HANDLING: FLOATING.

**What Next**

Set METAL_FILL_POLYGON_HANDLING: FLOATING in active fill METAL_FILL_RECONNECT flow.

## SX-2053

(Warning) Suggests to replace %s with LEF_USE_OBS:SHRINK.

**Description**

LEF_SPACING_OBS:YES and LEF_ZERO_SPACING_OBS:YES translate spacing OBS without considering the SMIN in ITF. This might introduce accuracy problem during extraction. StarRC suggests to replace LEF_SPACING_OBS:YES and LEF_ZERO_SPACING_OBS:YES with LEF_USE_OBS:SHRINK in the StarRC command file.

By default, spacing OBS and zero-spacing OBS are NOT translated (LEF_SPACING_OBS:NO and LEF_ZERO_SPACING_OBS:NO). LEF_USE_OBS:SHRINK overrides these two options and shrinks spacing OBS automatically.

**What Next**

Information only

Feedback

## SX-2054

(Warning) Found %s OBS being dropped. Suggest to use LEF_USE_OBS:SHRINK.

**Description**

LEF_SPACING_OBS:NO or LEF_ZERO_SPACING_OBS:NO is set in the StarRC command file.

**What Next**

Information only

## SX-2055

(Warning) BLOCKAGE(s) are being ignored due to TRANSLATE_DEF_BLOCKAGE_TYPE.

**Description**

Some BLOCKAGE(s) are being ignored due to TRANSLATE_DEF_BLOCKAGE_TYPE.

**What Next**

Information only

## SX-2056

(Warning) Option value %s in TRANSLATE_DEF_BLOCKAGE_TYPE is not supported.

**Description**

TRANSLATE_DEF_BLOCKAGE_TYPE supports

- *ALL (default) - Translate all blockages.*
- *FILLS - Translate FILLS blockage only.*
- *SLOT - Translate SLOT blockage only.*
- *PUSHDOWN - Translate PUSHDOWN blockage only.*
- *LAYER - Translate layer (EXCEPTPG, DESIGNRULEWIDTH and SPACING) blockages only.*

If TRANSLATE_DEF_BLOCKAGE_TYPE specifies no valid values, default value (ALL) is used.

Feedback

**What Next**

Information only

---

## SX-2057

(Warning) NDM blockage ( %d %d ) ( %d %d ) with SPACING %d on LAYER %s (SMIN= %d) can not be shrunk!

**Description**

StarRC shrinks rectilinear zero-spacing NDM blockages by their layer SMIN. But, if the resulted shape is NOT valid, StarRC will still translate un-shrunk shape.

SMIN in the message is reported in the technology unit.

---

## SX-2058

(Warning) NDM layer shapes on user layer purpose (%d:%d) are not ignored.

**Description**

A Layer in technology file can declare itself as physical layer or user-purpose layer by nonMask in its associated LayerDataType struct. By default, StarRC honors this setting and automatically ignore shapes drawn on user-purpose layers (nonMask=1).

NDM_FORCE_IGNORE_USER_PURPOSE_SHAPE will bypass this checking and ignores shapes if their layerPurpose are specified in this option.

**What Next**

Review the technology database and check LayerDataType(s).

---

## SX-2060

(Severe) The mapping file of OA syntax check has failed because: %s

**Description**

An exception has occurred while StarRC attempted to check the syntax of the mapping file related to OA that will be used to generate the OA view.

**What Next**

Ensure that the syntax of OA_DEVICE_MAPPING_FILE, OA_SKIPCELL_MAPPING_FILE, and OA_LAYER_MAPPING_FILE is correct.

Feedback

## SX-2061

(Fatal) No LEF_FILEs provided

**Description**

LEF_FILE in the command file must be set for LEF/DEF flow.

**What Next**

Provide valid values for LEF_FILE.

## SX-2063

(Fatal) No TOP_DEF_FILE provided

**Description**

TOP_DEF_FILE in the command file must be set for LEF/DEF flow.

**What Next**

Provide valid values for TOP_DEF_FILE.

## SX-2064

(Warning) VIA '%s' undefined.

**Description**

VIA is not defined in any LEF(s) and current parsing DEF. This via will be ignored during translation and may cause false open.

**What Next**

Check the input LEF(s) and DEF to make sure this VIA is defined.

## SX-2065

(Warning) Invalid path segment found for a NET %s at %d, %d (nm) on layer %s

**Description**

A path segment with invalid wire width or undefined layer will be ignored during translation.

**What Next**

Check this path segment in the DEF. It may be caused by undefined layer or wrong wire width from LEF and NONDEFAULTRULE.

Feedback

## SX-2073

(Fatal) Unknown selection '%s' for METAL_FILL_POLYGON_HANDLING.

**Description**

METAL_FILL_POLYGON_HANDLING command only accept AUTOMATIC, FLOATING, GROUNDED, or IGNORE.

**What Next**

Provide the correct setting for METAL_FILL_POLYGON_HANDLING in StarRC comman dfile.

## SX-2074

(Warning) No LayerDataType structs for any layer in technology file. All shapes such as fill shapes will be included during translation.

**Description**

There is no LayerDataType struct for any layer in technology file. nonMask is set to 0 by default for all LayerDataType. As a result, all shapes such as fill shapes will be included during translation.

**What Next**

Review the technology database and check LayerDataType(s).

## SX-2076

(Warning) LAYER '%s' has no VIA connection

**Description**

A LEF LAYER that has no VIA(s) reference to it is not useful layer in the design and it will be removed at Layers stage. Any geometries on the layer will also be ignored during translation.

**What Next**

Check the LAYER definition in the LEF(s). If it is needed in the design, please define a VIA that references it. If it is not needed in the design, it's safe to ignore this message.

## SX-2077

(Warning) DPT_COLOR_GDS_FILE is set, pre-color information will be ignored

Feedback

**Description**

When DPT_COLOR_GDS_FILE is set, all pre-color information defined in LEF(s)/DEF(s) will be ignored during translation.

**What Next**

Information only; no action needed.

---

## SX-2079

(Fatal) TYPE '%s' for LAYER '%s' is not supported

**Description**

The TYPE of a LAYER must be one of CUT, IMPLANT, MASTERSLIE, OVERLAP and ROUTING.

**What Next**

Refer to LEF specification for detailed description of these types.

---

## SX-2082

(Warning) VIARULE '%s' has wrong via_array_ratio='%f'

**Description**

The via array ratio of a VIARULE is derived from its attributes and it must be within (0.0, 1.0].

**What Next**

Check the definition of the VIARULE.

---

## SX-2084

(Fatal) VIA '%s' has wrong via_array_ratio='%f'

**Description**

The via array ratio of a VIA is derived from its attributes and it must be within (0.0, 1.0].

**What Next**

Check the definition of this VIA.

Feedback

## SX-2085

(Warning) NONDEFAULTRULE '%s' has previously declared LAYER '%s'

**Description**

If the previously-defined NONDEFAULTRULE has LAYER statement, the LAYER statement of newly-defined rule will be ignored, instead of merged.

**What Next**

Check all LEF files to find previously defined rule and avoid redefintion if not wanted.

## SX-2086

(Warning) LEF LAYER '%s' is not defined in mapping file

**Description**

LEF LAYER is not defined in the mapping file.

**What Next**

Information only; no action needed.

## SX-2087

(Warning) NONDEFAULTRULE '%s' has previously declared via '%s'

**Description**

If the previously-defined NONDEFAULTRULE has VIA statement, the VIA statement of newly-defined rule will be ignored, instead of merged.

**What Next**

Check all LEF files to find previously defined rule and avoid redefintion if not wanted.

## SX-2088

(Warning) Cell port '%s' is only defined in DEF macro '%s'.

**Description**

Cell ports defined in DEF macro only will be merged with ports defined in LEF macro to reduce design inconsistency.

Feedback

**What Next**

Information only; no action needed

---

## SX-2089

(Warning) Some VIA(s)/NONDEFAULTRULE(s) had previously defined (see lef_conflict.sum for details)

**Description**

Some VIA(s)/NONDEFAULTRULES(s) are multiply defined. Previous defintions will be overridden.

**What Next**

Please check your LEF/DEF to remove duplicate definitions if former definition is needed.

---

## SX-2090

(Warning) Duplicate VIA definition (%s) in '%s'

**Description**

Duplicate VIA definition is found and ignored.

**What Next**

Information only; no action needed.

---

## SX-2092

(Warning) NONDEFAULTRULE '%s' in '%s' had previously defined

**Description**

The NONDEFAULTRULE is redefined and its data will be merged during translation. For example, if previously-defined rule has no LAYER statement, StarRC will merge newly-defined LAYER statement into it.

**What Next**

Check all LEF files to find previously defined rule and avoid redefintion if not wanted.

---

## SX-2093

(Fatal) MACRO statement before any LAYER statement.

Feedback

**Description**

MACRO can't be defined before any LAYER statement as it requires LAYER information to construct.

**What Next**

Correct your LEF and move MACRO after LAYER definitions.

---

## SX-2094

(Fatal) MUSTJOIN pin '%s' is undefined.

**Description**

MUSTJOIN pin must be defined before reference.

**What Next**

Check the PIN defintion in the LEF.

---

## SX-2095

(Warning) LEF PIN '%s' in MACRO '%s' does not define any geometry.

**Description**

PIN definition in a MACRO does not include any geometry to extract.

**What Next**

Information only; no action needed.

---

## SX-2100

(Information) Wrote lef_conflict.sum.

**Description**

List of duplicate VIA and NONDEFAULTRULE definitions are written into lef_conflict.sum for information.

**What Next**

Information only; no action needed.

Feedback

## SX-2101

(Information) ECO_MODE is enabled. Currently we only support STAR DIRECTORY, GPD DIRECTORY and STAR COMMAND should in the same directory. Please make sure the setting in your star command.

**Description**

Current ECO flow setting.

**What Next**

No action required.

## SX-2102

(Warning) LEF macro '%s' has both FIXEDMASK & LAYERMASKSHIFT set

**Description**

LEF macro has both FIXEDMASK & LAYERMASKSHIFT which are mutually exclusive attributes.

**What Next**

No action required.

## SX-2103

(Information) BUSBITCHARS:%s is not consistent, use DEF's BUSBITCHARS:%s.

**Description**

The parsing DEF's BUSBITCHARS is not consistent with the one defined in the technology LEF. StarRC will use the one in DEF to translate.

**What Next**

Information only; no action needed.

## SX-2104

(Fatal) DEF component (%s %s) with MASKSHIFT '%s' - '%s' (in %s).

**Description**

DEF component's MASKSHIFT should not be specified due to the reason above.

Feedback

**What Next**

No action required.

---

## SX-2105

(Warning) DEF_MASKSHIFT_CONSISTENCY_CHECK may incur performance penalty in preXtraction.

**Description**

DEF_MASKSHIFT_CONSISTENCY_CHECK will run thorough consistency checks against each MASKSHIFT keyword in DEF files. It may incur performance penalty during HN stage if there are many mask-shifted components in the DEF(s).

If consistency checks for mask-shifting are not needed, it can be turned off for performance.

**What Next**

No action required.

---

## SX-2106

(Warning) Unsupported mask-shifting syntax in LEF - '%s'

**Description**

Technology LEF or macro LEF has unsupported syntax.

**What Next**

No action required.

---

## SX-2107

(Fatal) BLOCK_BOUNDARY required for RING_AROUND_THE_BLOCK in LEF/DEF flow.

**Description**

In LEF/DEF flow, BLOCK_BOUNDARY is required by RING_AROUND_THE_BLOCK to get bounding box of the BLOCK.

**What Next**

Review your star command file and add BLOCK_BOUNDARY for RING_AROUND_THE_BLOCK run.

Feedback

## SX-2108

(Warning) UNITS statements MUST precede the LAYER statements

### Description

The UNITS statement is optional and, when used, must precede any LAYER statements in LEF files.

### What Next

Check all LEF files and move UNITS statements before any LAYER statements.

## SX-2109

(Warning) Donut for GDS/OASIS cell %s is created successfully

### Description

GDS_DONUT is switched on to create donut ring for each GDS/OASIS cell in the run. GDS/OASIS cells that meet the criteria below can be donut'd.

- It has more than GDS_DONUT_NUM_LAYERS metal layers.
- A squared ring can be formed with GDS_DONUT_WIDTH as its ring width.

  If a GDS/OASIS cell can't be donut'd, StarRC will leave them as is.

### What Next

Information only. No action required.

```
GDS_DONUT: NO|YES
GDS_DONUT_WIDTH: 20 (microns)
GDS_DONUT_NUM_LAYERS: 2
```

## SX-2110

(Warning) BUSBITCHARS '%s' in DEF file '%s' is not consistent with '%s' in the top DEF file.

### Description

BUSBITCHARS is inconsistency between parsing DEF and top DEF file. StarRC will use BUSBITCHARS in top DEF file to translate current DEF.

### What Next

Contact the Synopsys Support Center.

Feedback

## SX-2111

(Warning) Metal fills on (%d:%d) in GDS_FILE/OASIS_FILE(s) are being ignored.

**Description**

Metal fills from GDS_FILE/OASIS_FILE(s) can be translated as FLOATING/GROUNDED/IGNORE based on the settings.

Metal fills are ignored if

- METAL_FILL_POLYGON_HANDLING:IGNORE or IGNORE being specified for IP_FILL mapping in GDS/OASIS mapping file.
- With METAL_FILL_POLYGON_HANDLING:AUTOMATIC, mappings without polygon handling mode in GDS/OASIS mapping file are ignored.

For performance consideration, only first metal fill object is reported.

**What Next**

Information only.

## SX-2112

(Warning) Invalid colored mapping for '%s' at line %d of GDS/OASIS layer map file '%s'

**Description**

The colored design database layer mapping doesn't exist. It might be caused by the missing coloring information in the NXTGRD file.

In this situation, StarRC will translate such GDS/OASIS metals to uncolored layers.

**What Next**

Please check if the colored design database layer is correctly generated at Layers stage.

## SX-2113

(Warning) Invalid database layer '%s' at line %d of GDS/OASIS layer map file '%s'

**Description**

A database layer named in either in the GDS/OASIS layer map file is not a valid layer name. The layer might be spelled incorrectly, or it might be missing from the mapping file specified by the MAPPING_FILE command.

Feedback

**What Next**

Check the layer name in the GDS/OASIS layer map file and also in the MAPPING_FILE.

---

## SX-2114

(Fatal) Syntax error '%s' at line %d of GDS/OASIS layer map file '%s'

**Description**

A line in the GDS/OASIS layer map file has invalid syntax.

**What Next**

Correct the line in the reported file.

---

## SX-2115

(Fatal) AGF file '%s' specified in SKIP_CELL_AGF_FILE for '%s' does not exist.

**Description**

The AGF file specified in SKIP_CELL_AGF_FILE for cell does not exist.

**What Next**

Check <agf file> column specifed in SKIP_CELL_AGF_FILE in StarRC command file.

---

## SX-2116

(Fatal) SPICE file '%s' specified in SKIP_CELL_AGF_FILE for '%s' does not exist.

**Description**

The SPICE netlist file specified in SKIP_CELL_AGF_FILE for cell does not exist.

**What Next**

Check <ideal spice netlist file> column specifed in SKIP_CELL_AGF_FILE in StarRC command file.

---

## SX-2117

(Warning) Metal fills on (%d:%d) in GDS_FILE/OASIS_FILE(s) are being translated as %s.

**Description**

Metal fills from GDS_FILE/OASIS_FILE(s) can be translated as FLOATING/GROUNDED/IGNORE based on the settings.

Feedback

For performance consideration, only first metal fill object is reported.

**What Next**

Information only.

---

## SX-2118

(Warning) DENSITY_BOX_WEIGHTING_FACTOR not specified in ITF. Use default value %f.

**Description**

The maximum value of DENSITY_BOX_WEIGHTING_FACTOR(s) is used to create GDS/OASIS donut. If it is not specified in ITF, StarRC uses default value.

**What Next**

Information only.

---

## SX-2119

(Warning) Donut for GDS/OASIS cell %s is not created - %s

**Description**

GDS_DONUT is switched on to create donut ring for each GDS/OASIS cell in the run. GDS/OASIS cells that meet the criteria below can be donut'd.

- It has more than GDS_DONUT_NUM_LAYERS metal layers.
- A squared ring can be formed with GDS_DONUT_WIDTH as its ring width.

  If a GDS/OASIS cell can't be donut'd, StarRC will leave them as is.

**What Next**

Information only. No action required.

```
GDS_DONUT: NO|YES
GDS_DONUT_WIDTH: 20 (microns)
GDS_DONUT_NUM_LAYERS: 2
```

---

## SX-2129

(Warning) Pin '%s' in the DEF macro is not placed and geometries are ignored.

Feedback

**Description**

Pin in the DEF macro is not placed. This pin will be nelisted in the *PORTS section of SPEF and/or the .SUBCKT statement fo r SPF/STAR/NETNAME formats but will not appear in the detail parasitic section (since it has no physical characteristic).

**What Next**

Information only; no action needed.

---

## SX-2131

(Warning) Port '%s' in subckt '%s' is duplicated.

**Description**

StarRC cannot accept duplicated ports, only one will be kept.

**What Next**

Check to make sure port definition in SUBCKT is correct in CALIBRE_AGF_NETLIST file.

---

## SX-2136

(Warning) Skip cell '%s' is not defined in either LEF or DEF files.

**Description**

The definition of the skip cell cannot be found in LEF_FILE or MACRO_DEF_FILE.

**What Next**

Check if the name of the cell is specified correctly in SKIP_CELLS or SKIP_CELLS_FILE.

---

## SX-2139

(Warning) Detected %d unconnected hierarchy pins which are not connected to parent cell for DEF '%s'

**Description**

Unconnected hierarchy pins are detected for DEF macro.

**What Next**

Information only; no action needed.

Feedback

## SX-2140

(Information) GDS Layer %s is removed by GDS_REMOVE_LAYER.

**Description**

GDS_REMOVE_LAYER will remove layers defined in GDS_LAYER_MAP_FILE.

**What Next**

Information only; no action is needed.

## SX-2141

(Fatal) Missing DIEAREA and no placements.

**Description**

There's no DIEAREA in def file and no placements in this design. StarRC can't translate this DEF file.

**What Next**

Correct this DEF file.

## SX-2142

(Information) Top Level Extents '%d' '%d' '%d' '%d'

**Description**

The extents of top-level block.

**What Next**

Information only; no action needed.

## SX-2143

(Information) Instance '%s' is multiply-defined.

**Description**

An instance is mutiply-defined in COMPONENTS section and ignored.

**What Next**

Information only; no action needed.

Feedback

## SX-2144

(Severe) The definition of cell '%s' instantiated as '%s' in DEF file '%s' is not found.

**Description**

The corresponding cell definition for the instance cannot be found in any of LEF_FILE or MACRO_DEF_FILE.

**What Next**

Check if LEF_FILE or MACRO_DEF_FILE includes the cell definition.

## SX-2145

(Warning) MHP: '%s' is not used in DEF macro '%s'

**Description**

Mismatched hierarchy pin (MHP) is detected.

For unconnected hierarchy pin (UHP) that has definition from both LEF and DEF, StarRC reports MHP for clarification.

**What Next**

Information only; no action needed.

## SX-2146

(Severe) Net '%s' connects to an undefined pin '%s' of LEF Macro '%s'.

**Description**

The pin to which the net is connected to is not defined in the LEF Macro.

**What Next**

If duplicate LEF Macro definitions are present, the first definition takes higher priority and if the first definition does not have the pin definition, StarRC reports this ERROR. Check the definition of the Macro in LEF_FILE.

## SX-2147

(Warning) NET '%s' is connected to an undefined PIN '%s'

Feedback

**Description**

In the DEF, a NET connection specified by ( PIN pinName ) must be connected to a pin that is previously defined in PINS section.

**What Next**

Check the DEF file and define missing PIN in PINS section.

---

## SX-2148

(Warning) UHP: %s macro(%s) %s

**Description**

Unconnected hierarchy pin (UHP) for DEF macro is detected.

A design that has a DEF macro in it and the DEF macro is flattened using

SKIP_CELLS: * !DEF_MACRO.

If there are nets in the DEF macro that do not connect to a top level net, they will be missing from the SPEF file and cause simulation problem. DEF pins on these nets are called unconnected hierarchy pin (UHP).

UHP will be netlisted with names of the format <instance>/<pin> since they are not pre-assigned a top level name in DEF. Such generated nets are eligible for re-assignment to ground if REMOVE_FLOATING_NETS and/or REMOVE_DANGLING_NETS are enabled and they meet the connection count criteria. If the unconnected DEF PINS have USE POWER or USE GROUND tags, the generated nets will be automatically recognized as POWER_NETS by StarRC unless the star_cmd file already sets POWER_NETS to user-defined value. If POWER_NETS is set by user and unconnected POWER or GROUND PINS exist in the design, they must be appended in the star_cmd file as 'POWER_NETS: */<pin> */*/<pin> ...' in order to be recognized by StarRC for special handling.

**What Next**

Information only; no action needed.

---

## SX-2149

(Warning) Illegal DEF dbuPerMicron

**Description**

The DEF dbuPerMicron must be less than or equal to the LEF dbuPerMicron, otherwise it may cause round-off errors and loss of accuracy during translation.

Feedback

**What Next**

See UNITS in LEF/DEF specification for valid dbuPerMicron pairings.

---

## SX-2150

(Warning) Invalid pairings of the LEF dbuPerMicron '%d' and the corresponding DEF dbuPerMircon '%d'

**Description**

StarRC requires dbuPerMicron pairings between technology LEF and parsing DEF to follow LEF/DEF specification to avoid possible round-off erros and loss of accuracy during translation.

**What Next**

Please refer to LEF/DEF specification and correct the pairings by the parings table.

---

## SX-2151

(Warning) Missing layer mapping for layernum:datatype(%d:%d) in layer map file '%s'.

**Description**

GDS/OASIS files require layer mapping specified in layer map files so as for StarRC to translate GDS/OASIS structs into appropriate layers in main design database.

**What Next**

Make sure that the layer mapping for the specified layer is correctly defined in GDS/OASIS layer map files.

---

## SX-2152

(Fatal) Layer '%s' can not be found for VIARULE '%s'

**Description**

A layer in the LAYERS statement in VIARULE defintion is not defined.

**What Next**

Correct the layer in the VIARULE defintion.

---

## SX-2153

(Warning) Duplicate layer:datetype (%d:%d) found.

Feedback

**Description**

Duplicate layer:datatype entry found in the GDS_LAYER_MAP_FILE/ OASIS_LAYER_MAP_FILE. StarRC uses the first mapping and ignore duplicate ones.

**What Next**

Remove the duplicate entries.

---

## SX-2154

(Fatal) LEF macro definition is missing for SKIP_CELL_AGF_FILE description " %s ".

**Description**

The reported AGF cell in SKIP_CELL_AGF_FILE has no correspondent LEF macro defined in LEF files.

**What Next**

Check input LEF files and provide LEF macro for reported AGF cell.

---

## SX-2155

(Warning) The pre-defined Via '%s' in LEF is covered by latest definition of DEF.

**Description**

Via defintion in LEF overridened by the one defined in DEF. StarRC will use the defintion from DEF to translate.

**What Next**

Information only; no action needed.

---

## SX-2156

(Warning) Missing layer mapping for ITF layer '%s'.

**Description**

None of any database layer maps to a specific ITF layer defined in TCAD_GRD_FILE. This ITF layer takes no effect for parasitic extraction.

**What Next**

Make sure that the physical layer mapping is correctly defined in MAPPING_FILE.

Feedback

## SX-2157

(Warning) STYLE #%d is not defined.

**Description**

A referenced STYLE is not defined in the DEF. The styling of geometries will be ignored during translation.

**What Next**

Correct the DEF with missing STYLE.

## SX-2158

(Information) Read unit %d from %s file %s. Using Units:%d to translate %s.

**Description**

StarXtract read unit from tech LEF or DEF file.

**What Next**

It is information only. No action is needed.

## SX-2165

(Information) Total %d metal fill cells are imported.

**Description**

Total number of fill cells StarRC imports.

**What Next**

Information only; no action required.

## SX-2166

(Warning) LAYER '%s' is not defined. Subsequent data for it will be ignored.

**Description**

The layer after RECT in a VIA defintion is not defined and geometries on this undefined layer will be ignored by StarRC during translation. If this via is referenced by any net, this may cause opens and other problems.

Feedback

**What Next**

Please check the layer definition in the LEFs.

---

## SX-2167

(Warning) VIA '%s' is not defined. Subsequent data for it will be ignored.

**Description**

VIA referenced by PIN statement is not defined.

**What Next**

Please check the via defintion in the LEFs and DEFs.

---

## SX-2168

(Warning) Geometries on the LAYER '%s' are found and ignored during translation.

**Description**

The LAYER is a valid LEF layer but may be marked as remove_layer in MAPPING_FILE. Geometries on the layer are ignored during translation.

**What Next**

Please check your MAPPING_FILE.

---

## SX-2169

(Warning) Duplicate cell definition is found and ignored in GDS/OASIS or fill GDS/OASIS files.

**Description**

Multiple GDS/OASIS cell defintions are found for a skip cell. When this happenes, StarRC uses the first cell defintion to synthesizethe skip cell and ignore the rest of duplicate cell defintions.

**What Next**

See stage summary file for the list of duplicate cells.

---

## SX-2170

(Information) GDS/OASIS cell '%s' applied to %s MACRO '%s' from '%s'

Feedback

**Description**

The GDS/OASIS cell definition from the file is applied and merged with macro to form a skip cell.

**What Next**

Information only; no action needed.

---

## SX-2172

(Fatal) Input in flex scanner failed. '%s'

**Description**

StarRC cannot read this GZIP properly.

**What Next**

Make sure the file is a correct GZIP file.

---

## SX-2175

(Fatal) Error parsing file '%s'

**Description**

LEF/DEF file parsing is not successful due to syntax errors.

**What Next**

Contact the Synopsys Support Center.

---

## SX-2182

(Fatal) File doesn not exist! ICV_ANNOTATION_FILE: %s

**Description**

StarXtract is trying to open this file, but this operation is failed.

**What Next**

Correct the setting of this file.

---

## SX-2183

(Fatal) Failed to open gzip file '%s'

Feedback

**Description**

StarXtract is trying to open this file, but this operation is failed.

**What Next**

Correct the setting of this file.

---

## SX-2184

(Fatal) ICV runset report file syntax error near line '%d', '%s' '%s'.

**Description**

There is unexpected syntax in the ICV runset report file.

**What Next**

Re-run the ICV or contact the Synopsys Support Center.

---

## SX-2185

(Fatal) ICV flow cannnot support MILKYWAY_EXTRACT_VIEW:NO.

**Description**

ICV flow only supports "MILKYWAY_EXTRACT_VIEW: YES".

**What Next**

Set "MILKYWAY_EXTRACT_VIEW: YES" or contact the Synopsys Support Center.

---

## SX-2186

(Fatal) NET_TYPE:SCHEMATIC is not available with XREF:NO.

**Description**

XREF: NO does not support NET_TYPE: SCHEMATIC.

**What Next**

Set either "XREF: YES" or "NET_TYPE: LAYOUT", or contact the Synopsys Support Center.

---

## SX-2189

(Warning) Ignore unknown option '%s' in line '%d' of ICV runset report file.

Feedback

**Description**

This is unknown option of the ICV runset report file.

**What Next**

Re-run the ICV or contact the Synopsys Support Center.

---

## SX-2190

(Fatal) Missing tagname_required in ICV runset report file.

**Description**

tagname_required information is required for the ICV runset report file.

**What Next**

Re-run the ICV or contact the Synopsys Support Center.

---

## SX-2191

(Fatal) ICV runset has SIMULATION_MODEL_NAME, which is only supported for "MODEL_TYPE: LAYOUT".

**Description**

The SIMULATION_MODEL_NAME of runset is only supported for MODEL_TYPE: LAYOUT.

**What Next**

Set "MODEL_TYPE: LAYOUT" in StarXtract or, contact the Synopsys Support Center.

---

## SX-2192

(Warning) ICV COMPARE option MEMORY_ARRAY_COMPARISON is not supported for XREF extractions.

**Description**

StarXtract finds invalid equiv points of the format SRAMblock<##>x<##>#<##>.

**What Next**

Set MEMORY_ARRAY_COMPARISON to FALSE and re-run ICV LVS, or contact the Synopsys Support Center.

Feedback

## SX-2193

(Fatal) The number of terminals of device '%s' is inconsistent between runset and pin file.

**Description**

There are terminal information discrepancy of the device between the runset and the pin file.

**What Next**

Correct either the runset data or the pin file data, or contact the Synopsys Support Center.

## SX-2194

(Fatal) Pin '%d' of device '%s' has inconsistent layer between runset and pin file.

**Description**

There is a layer information discrepancy of the device's pin between the runset and the pin file.

**What Next**

Correct either the runset data or the pin file data, or contact the Synopsys Support Center.

## SX-2196

(Severe) Could not iterate instances in cell '%s'.

**Description**

StarXtract is iterating the instances of this cell, but this operation is failed.

**What Next**

Confirm the Milkyway data is not broken or re-generate the Milkyway data.

## SX-2197

(Severe) Could not get instance data for '%s'.

**Description**

StarXtract is getting the instance data from Milkyway database, but this operation is failed.

**What Next**

Confirm the Milkyway data is not broken or re-generate the Milkyway data.

Feedback

## SX-2208

(Fatal) Requested view '%s' does not exist in main library or reference libraries.

**Description**

StarXtract is opening the requested view that depends on option settings, but it does not exist.

**What Next**

Check wheather there is any incorrect option setting about Milkyway view or there is any incorrect Milkyway input data.

## SX-2210

(Information) FORCE_REF_LIB_MODE must be used together with MILKYWAY_REF_LIB_MODE: HIER!

**Description**

FORCE_REF_LIB_MODE must be used together with MILKYWAY_REF_LIB_MODE: HIER!

**What Next**

If this is expected behavior, it is information only. No action is needed.

## SX-2211

(Fatal) MILKYWAY_REF_LIB_MODE cannot be used.

**Description**

MILKYWAY_REF_LIB_MODE cannot be used when dbSetLibRefCtrlFileMode flag is already set in the main library. dbSetLibRefCtrlFileMode controls the reference library tree source.

**What Next**

Remove MILKYWAY_REF_LIB_MODE from star_cmd file; the value of dbSetLibRefCtrlFileMode will be used.

## SX-2212

(Fatal) Failed to build FILE mode DCM table '%s'.

Feedback

**Description**

StarXtract is building this FILE mode DCM table, but this operation is failed.

**What Next**

Check whether there is any incorrect data.

---

## SX-2214

(Fatal) Failed to open reference libraries for '%s' in '%s' mode.

**Description**

StarXtract is opening this reference library, but this operation is failed.

**What Next**

Confirm the Milkyway data is not broken or re-generate the Milkyway data.

---

## SX-2215

(Fatal) Syntax error in SKIP_CELL_AGF_FILE: '%s'

**Description**

Wrong SKIP_CELL_AGF_FILE syntax in StarRC command file. The syntax is,

SKIP_CELL_AGF_FILE: <cell name> <agf file> <ideal spice netlist for agf>.

**What Next**

Check SKIP_CELL_AGF_FILE in your StarRC command file.

---

## SX-2216

(Information) Requested view '%s' does not exist in main library or reference libraries.

**Description**

StarXtract is opening the requested view that depends on option settings, but it does not exist.

**What Next**

If this is expected data, it is information only. No action is needed.

Feedback

## SX-2217

(Fatal) Failed to open reference libraries by control file for '%s'.

**Description**

StarXtract is opening this main reference library, but this operation is failed.

**What Next**

Confirm the Milkyway data is not broken or re-generate the Milkyway data.

## SX-2218

(Fatal) RefLib Control File inconsistent

**Description**

StarXtract finds RefLib control file inconsistecy.

**What Next**

Confirm the Milkyway data is not broken or re-generate the Milkyway data.

## SX-2219

(Warning) Could not locate cell '%s' in library '%s'.

**Description**

StarXtract is locating this cell, but this operation is failed.

**What Next**

Confirm this is expected data, or contact the Synopsys Support Center.

## SX-2221

(Severe) Could not resolve libId for reference library '%s'.

**Description**

This libId is not resolved for this reference library.

**What Next**

Confirm the Milkyway data is not broken or re-generate the Milkyway data.

Feedback

## SX-2222

(Severe) Failed to locate cell '%s' in specfied library.

**Description**

StarXtract is locating this cell, but this operation is failed.

**What Next**

Confirm the Milkyway data is not broken or re-generate the Milkyway data.

## SX-2223

(Warning) Failed to locate library EVRunset attach file.

**Description**

StarXtract is attempting to read cell EVRunset attach file.

**What Next**

Verify that all Hercules netlisting stages completed.

## SX-2224

(Warning) Attach file '%s';'%s' does not exist.

**Description**

StarXtract is accessing this attach file and it does not exist.

**What Next**

Confirm this is expected data, or contact the Synopsys Support Center.

## SX-2225

(Fatal) Could not determine '%s' attach file location. Milkyway '%s' is incomplete.

**Description**

StarXtract is accessing this attach file and it does not exist.

**What Next**

Confirm the Milkyway data is not broken or re-generate the Milkyway data.

Feedback

## SX-2226

(Fatal) Attach file '%s';'%s' does not exist.

**Description**

StarXtract is accessing this attach file and it does not exist.

**What Next**

Confirm the Milkyway data is not broken or re-generate the Milkyway data.

## SX-2227

(Warning) Detected the net '%s' connected to the P/G port '%s' of instance '%s' (cell '%s') would take them as power net

**Description**

The reported nets have no '+ USE POWER' attributes defined but they are connected to P/G pins of LEF macros. They will be treated as as power nets during translation.

**What Next**

Information only; no action needed.

## SX-2228

(Warning) BUSBITCHARS is only allowed 2 characters

**Description**

The BUSBITCHARS in LEF/DEF should only have 2 characters. If more than 2 characters are specified, the first two characters are taken as BUSBITCHARS.

**What Next**

Remove the redundant characters from BUSBITCHARS in your LEF/DEF files.

## SX-2230

(Information) Initializing milkyway library '%s'...

**Description**

StarXtract reports this information.

Feedback

**What Next**

If this is expected data, it is information only. No action is needed.

---

## SX-2231

(Severe) Could not open library '%s'.

**Description**

StarXtract is trying to open this library, but this operation is failed.

**What Next**

Confirm the procedures and settings to generate this data is correct, or re-generate the Milkyway data.

---

## SX-2232

(Information) Failed to read '%s' unit from database.

**Description**

StarXtract reports this information.

**What Next**

If this is expected data, it is information only. No action is needed.

---

## SX-2233

(Information) Invalid database unit '%d' for '%s'.

**Description**

StarXtract reports this information.

**What Next**

If this is expected data, it is information only. No action is needed.

---

## SX-2234

(Information) Invalid scale factor '%d' for '%s'.

**Description**

StarXtract reports this information.

Feedback

**What Next**

If this is expected data, it is information only. No action is needed.

---

## SX-2235

(Information) DBUperMicron = '%d'

**Description**

StarXtract reports this information.

**What Next**

If this is expected data, it is information only. No action is needed.

---

## SX-2236

(Information) CapacitanceScale = '%d'

**Description**

StarXtract reports this information.

**What Next**

If this is expected data, it is information only. No action is needed.

---

## SX-2237

(Information) ResistanceScale = '%d'

**Description**

StarXtract reports this information.

**What Next**

If this is expected data, it is information only. No action is needed.

---

## SX-2238

(Information) BusBitChars = '%s'

**Description**

StarXtract reports this information.

Feedback

**What Next**

If this is expected data, it is information only. No action is needed.

---

## SX-2239

(Information) HierchicalSep = '%c'

**Description**

StarXtract reports this information.

**What Next**

If this is expected data, it is information only. No action is needed.

---

## SX-2241

(Information) Milkyway Layer '%d' datatype# '%d' has nonMask set to 1, polygons from that dataType will be ignored.

**Description**

StarXtract ignores all polygons that the datatype has nonMask set to 1.

**What Next**

If this is expected data, it is information only. No action is needed.

---

## SX-2242

(Severe) MW TechFile layer '%s' is a CONNECTED layer. Either it or its mask layer '%s' must be specified in the MAPPING_FILE.

**Description**

StarXtract finds this CONNECTED layer, but cannot get its setting or its mask layer setting from MAPPING_FILE.

**What Next**

Set the correct layer mapping into MAPPING_FILE.

---

## SX-2244

(Severe) Layer '%s' is used by MWContactCode and must be in MAPPING_FILE.

Feedback

**Description**

StarXtract finds MWContactCode has this layer, but this layer is not in MAPPING_FILE. All layers mapping must be in MAPPING_FILE.

**What Next**

Set this layer mapping into MAPPING_FILE.

---

## SX-2245

(Fatal) Could not write file '%s'

**Description**

StarXtract is trying to write this file, but this operation is failed.

**What Next**

Correct the setting of this file.

---

## SX-2246

(Fatal) Could not read file '%s'

**Description**

StarXtract is trying to open this file, but this operation is failed.

**What Next**

Correct the setting of this file.

---

## SX-2247

(Fatal) Device DB is corrupted.

**Description**

StarXtract finds this incompatible data.

**What Next**

Run -clean to get updated device DB.

---

## SX-2248

(Fatal) Device DB is not compatible with this executable.

Feedback

**Description**

StarXtract finds this incompatible data.

**What Next**

Run -clean to get updated device DB.

---

## SX-2249

(Warning) Device recognition layer cannot be set to remove_layer in MAPPING_FILE when AUTO_RUNSET: YES is set.

**Description**

This in unexpected data of device recognition layer.

**What Next**

Confirm this is expected data. If it is not, correct the setting or the data.

---

## SX-2250

(Warning) Device terminal layer '%s' used by "'%s':'%s'" is removed in MAPPING_FILE or is the BULK terminal of another device.

**Description**

Terminals of this type will be netlisted ideally.

**What Next**

Confirm this is expected data. If it is not, confirm the procedures and settings to generate this data is correct, or re-generate the Milkyway data.

---

## SX-2251

(Severe) Invalid CONVERT_DIODE_TO_PARASITIC_CAP usage; wrong number of arguments in '%s'. '%s'

**Description**

StarXtract finds this invalid usage.

**What Next**

Correct the usage of CONVERT_DIODE_TO_PARASITIC_CAP.

Feedback

## SX-2252

(Severe) Invalid CONVERT_DIODE_TO_PARASITIC_CAP usage; area coefficient could not be converted to a positive number in '%s'. '%s'.

**Description**

StarXtract finds this invalid usage.

**What Next**

Correct the usage of CONVERT_DIODE_TO_PARASITIC_CAP.

## SX-2253

(Severe) Invalid CONVERT_DIODE_TO_PARASITIC_CAP usage; perimeter coefficient could not be converted to a positive number in '%s'. '%s'.

**Description**

StarXtract finds this invalid usage.

**What Next**

Correct the usage of CONVERT_DIODE_TO_PARASITIC_CAP.

## SX-2254

(Severe) Invalid CONVERT_DIODE_TO_PARASITIC_CAP usage; specified device '%s' is not a diode.

**Description**

StarXtract finds this invalid usage.

**What Next**

Correct the data of this device.

## SX-2255

(Severe) Invalid RETAIN_CAPACITANCE_CAP_MODELS usage; specified device '%s' is not a capacitor.

**Description**

StarXtract finds this device is not a capacitor.

Feedback

**What Next**

Confirm the Milkyway data is not broken. If it is not, confirm the procedures and settings to generate this data is correct, or re-generate the Milkyway data.

---

## SX-2256

(Warning) %s device "%s" specified in %s option does not exist in the runset.

**Description**

StarXtract finds this option does not exist in the runset.

**What Next**

Confirm this is expected data. If it is not, confirm the procedures and settings to generate this data is correct, or re-generate the Milkyway data.

---

## SX-2257

(Information) '%s' Option setting will be applied to all matching runset definitions.

**Description**

StarXtract is applying this option to all matching runset definitions.

**What Next**

If this is expected behavior, it is information only. No action is needed.

---

## SX-2258

(Fatal) Cannot open file '%s'.

**Description**

StarXtract is trying to open this file, but this operation is failed.

**What Next**

Confirm the setting for this file is correct.

---

## SX-2259

(Fatal) Element devices not suppported in Milkyway XTR view.

**Description**

StarXtract does not support element devices.

Feedback

**What Next**

Confirm the procedures and settings to generate this data is correct, and then regenerate this data.

---

## SX-2260

(Warning) Parse error in runset at line '%d' near token '%s'.

**Description**

StarXtract is having this parsing error for the runset.

**What Next**

Confirm the procedures and settings to generate this data is correct, and then regenerate this data.

---

## SX-2261

(Fatal) Must specify (P)DIODE_TYPE in Hercules '%s' (P)DIODE definition.

**Description**

StarXtract finds this invalid data. The (P)DIODE_TYPE must be specified in Hercules '%s' (P)DIODE definition.

**What Next**

Confirm the procedures and settings to generate this data is correct, and then regenerate this data.

---

## SX-2262

(Fatal) Invalid (P)DIODE_TYPE '%s' in Hercules '%s' (P)DIODE defintion. Must be PN or NP.

**Description**

StarXtract finds this invalid data. It must be PN or NP.

**What Next**

Confirm the procedures and settings to generate this data is correct, and then regenerate this data.

Feedback

## SX-2263

(Severe) Invalid RES_DIODE_TYPE for '%s'. Must be PN_DIODE or NP_DIODE.

**Description**

StarXtract finds this incorrect data. This RES_DIODE_TYPE must be PN_DIODE or NP_DIODE.

**What Next**

Confirm the procedures and settings to generate this data is correct, and then regenerate this data.

## SX-2264

(Severe) RES device '%s' with PDIODE through RES_DIODE_NAME/RES_DIODE_TYPE must have 3 terminals.

**Description**

StarXtract finds this incorrect data. This device through RES_DIODE_NAME/ RES_DIODE_TYPE must have 3 terminals, includes BULK.

**What Next**

Confirm the procedures and settings to generate this data is correct, and then regenerate this data.

## SX-2265

(Severe) RES device '%s' with PDIODE through RES_DIODE_NAME/RES_DIODE_TYPE must use same layer for terminal A and terminal B.

**Description**

StarXtract finds this incorrect data. This device through RES_DIODE_NAME/ RES_DIODE_TYPE must use same layer for terminal A and terminal B.

**What Next**

Confirm the procedures and settings to generate this data is correct, and then regenerate this data.

## SX-2266

(Warning) STARRCXT_WITH_2006_12_HERCULES is going to be replaced by STARRC_WITH_2006_12_HERCULES.

Feedback

**Description**

StarXtract replaces STARRCXT_WITH_2006_12_HERCULES with STARRCXT_WITH_2006_12_HERCULES.

**What Next**

Confirm this is expected data. If it is not, confirm the procedures and settings to generate this data is correct, or re-generate this data.

---

## SX-2267

(Warning) DEVICE_PREFIX for '%s' contains more than one character; the first character will be retained and the remainder ignored.

**Description**

StarXtract only retains the first character of DEVICE_PREFIX.

**What Next**

Confirm this is expected data. If it is not, confirm the procedures and settings to generate this data is correct, or re-generate this data.

---

## SX-2268

(Warning) INCLUDE_TECHNOLOGY_DEFAULTS=FALSE specified in Hercules runset.

**Description**

This option overrides the -rcxt command line option and may unnecessarily increase StarRC runtime.

**What Next**

Regenerate the XTR view database without this option for optimum performance.

---

## SX-2269

(Warning) MARKER_GENERATION: HERCULES is required when using PIN_LAYER_LIST command in Hercules runset.

**Description**

To run MARKER_GENERATION: HERCULES needs PIN_LAYER_LIST in the Hercules runset.

Feedback

**What Next**

Set MARKER_GENERATION: HERCULES.

---

## SX-2270

(Warning) MARKER_GENERATION: HERCULES requires PIN_LAYER_LIST in the Hercules runset.

**Description**

To run MARKER_GENERATION: HERCULES needs PIN_LAYER_LIST in the Hercules runset.

**What Next**

Set MARKER_GENERATION: AUTOMATIC.

---

## SX-2271

(Fatal) Not an LVS runset, cannot use XREF: YES/COMPLETE.

**Description**

XREF: YES/COMPLETE only supports LVS runset.

**What Next**

Make sure that WRITE_EXTRACT_VIEW and COMPARE are both run in the same call to Hercules.

---

## SX-2272

(Fatal) LVS_TOP_BLOCK: '%s' does not match runset BLOCK '%s'.

**Description**

StarXtract finds this inconsistent setting.

**What Next**

Correct one setting of them.

---

## SX-2273

(Warning) Hercules COMPARE option MERGE_PATHS=TRUE may result in duplicated schematic name assignments for individual layout elements during Hercules COMPARE.

Feedback

The first assignment encountered will be applied and the remaining conflicting assignments will be netlisted with layout element names.

**Description**

SX-5150 and/or SX-5151 will be issued at the XrefHN stage for any occurances of duplicated assignments.

**What Next**

Confirm this is expected data. If it is not, confirm the procedures and settings to generate this data is correct, or re-generate this data.

---

## SX-2274

(Warning) Hercules COMPARE option MEMORY_ARRAY_COMPARISON is not supported for XREF extractions.

**Description**

StarXtract does not support Hercules COMPARE option MEMORY_ARRAY_COMPARISON for XREF extractions.

**What Next**

Set MEMORY_ARRAY_COMPARISON to FALSE and re-run Hercules LVS.

---

## SX-2275

(Fatal) Hercules EVACCESS_OPTIONS option CREATE_VIEWS=TRUE is required for XREF.

**Description**

The setting of CREATE_VIEWS must be TRUE for XREF.

**What Next**

Correct this setting of Hercules runset.

---

## SX-2276

(Fatal) CELL_TYPE: SCHEMATIC is only available with XREF: YES or COMPLETE.

**Description**

StarXtract finds invalid setting of CELL_TYPE.

Feedback

**What Next**

Correct this setting in the StarXtract command file.

---

## SX-2277

(Fatal) NET_TYPE: SCHEMATIC is only available with XREF: YES or COMPLETE.

**Description**

StarXtract finds invalid setting of NET_TYPE.

**What Next**

Correct this setting in the StarXtract command file.

---

## SX-2278

(Fatal) NET_TYPE must be LAYOUT or SCHEMATIC.

**Description**

StarXtract finds there is the invalid setting of NET_TYPE.

**What Next**

Correct this setting in the StarXtract command file.

---

## SX-2279

(Fatal) Must use CASE_SENSITIVE: NO when IGNORE_CASE=TRUE in the Hercules runset.

**Description**

Hercules runset has IGNORE_CASE=TRUE. The setting of CASE_SENSITIVE in the StarXtract command file must be NO.

**What Next**

Correct the setting of StarXtract command file.

---

## SX-2280

(Warning) Specified '%s' does not exist.

**Description**

StarXtract is attempting to read this location from Hercules runset, but it does not exist.

Feedback

**What Next**

Correct this location setting.

---

## SX-2281

(Warning) '%s'::'%s'='%s' in the Hercules runset is not a valid path.

**Description**

StarXtract finds this is not a valid path.

**What Next**

Correct this setting of the path.

---

## SX-2282

(Warning) Could not determine '%s' location for Hercules runset.

**Description**

StarXtract could not determine this location for Hercules runset.

**What Next**

Correct this location setting for Hercules runset.

---

## SX-2283

(Warning) '%s' '%s' in the Hercules runset does not match '%s' in the StarXtract command file Hercules Runset: '%s' StarRC Cmd: '%s'.

**Description**

StarXtract expects they should be consistent.

**What Next**

Confirm this is expected data. If it is not, confirm the procedures and settings to generate this data is correct, or re-generate this data.

---

## SX-2284

(Warning) Schematic device '%s' equates to undefined layout device '%s'.

**Description**

StarXtract finds this schematic device equates to undefined layout device.

Feedback

**What Next**

Confirm this is expected data. If it is not, confirm the procedures and settings to generate this data is correct, or re-generate this data.

---

## SX-2285

(Severe) Hercules runset does not contain any valid EQUATE statement.

**Description**

There are some devices, but StarXtract does not find any valid EQUATE statement in the Hercules runset.

**What Next**

Insert lvsflow equates into user runset.

---

## SX-2286

(Warning) The '%s' database schema version is not compatible with StarXtract.

**Description**

This is incompatible Milkyway database schema version, the result of StarXtract may be incorrect.

**What Next**

Confirm this is expected data, or contact the Synopsys Support Center.

---

## SX-2287

(Fatal) Missing '%d' cells in database.

**Description**

StarXtract finds there are missing cells in database, could not continue. The detail information is in summary/hn.sum, SX-2325.

**What Next**

Confirm the Milkyway data is not broken. If it is not, confirm the procedures and settings to generate this data is correct, or re-generate the Milkyway data.

---

## SX-2288

(Warning) Missing FRAM views for '%d' SKIP_CELLs.

Feedback

**Description**

StarXtract reports there are these missing FRAM views. hn.sum file has the detail. The detail information is in summary/hn.sum, SX-2306.

**What Next**

Confirm this is expected data. If it is not, confirm the procedures and settings to generate this data is correct, or re-generate the Milkyway data.

---

## SX-2289

(Warning) Found '%d' SKIP_CELLS without POWER or GROUND ports.

**Description**

StarXtract reports there are these SKIP_CELLS without POWER or GROUND ports. hn.sum file has the detail. The detail information is in summary/hn.sum, SX-2293.

**What Next**

Confirm this is expected data. If it is not, confirm the procedures and settings to generate this data is correct, or re-generate the Milkyway data.

---

## SX-2290

(Warning) Total '%d' empty instance name errors will cause connecting to the ports fail.

**Description**

StarXtract reports there are these empty instance name errors. The detail information is in summary/hn.sum, SX-2320 and SX-2321.

**What Next**

Confirm this is expected data. If it is not, confirm the procedures and settings to generate this data is correct, or re-generate the Milkyway data.

---

## SX-2291

(Warning) Expanded '%d' hierarchical cells.

**Description**

StarXtract finds these hierarchical cells expanded. hn.sum file has the detail. The detail information is in summary/hn.sum, SX-2324.

Feedback

**What Next**

Confirm this is expected data. If it is not, confirm the procedures and settings to generate this data is correct, or re-generate the Milkyway data.

---

## SX-2292

(Warning) EEQ pin name '%s' in cell '%s' is already used by a primary port.

**Description**

StarXtract finds this pin name is already used by a primary port. This pin will not be netlisted uniquely.

**What Next**

Confirm this is expected data, or contact the Synopsys Support Center.

---

## SX-2293

(Information) SKIP_CELL '%s' does not contain POWER or GROUND ports.

**Description**

StarXtract finds this skipped cell does not contain POWER or GROUND ports.

**What Next**

If this is expected data, it is information only. No action is needed.

---

## SX-2294

(Information) physical only port found in '%s'

**Description**

StarXtract finds physical only port.

**What Next**

If this is expected data, it is information only. No action is needed.

---

## SX-2295

(Information) Too many physical only port found. Only print the first '%d' ports.

**Description**

StarXtract finds physical only port.

Feedback

**What Next**

If this is expected data, it is information only. No action is needed.

---

## SX-2296

(Severe) Could not iterate port instances from instance '%s'.

**Description**

StarXtract is iterating the port instances of this instance, but this operation is failed.

**What Next**

Confirm the Milkyway data is not broken or re-generate the Milkyway data.

---

## SX-2299

(Severe) Could not iterate extra port instances from instance '%s'.

**Description**

StarXtract is iterating the extra port instances of this cell, but this operation is failed.

**What Next**

Confirm the Milkyway data is not broken or re-generate the Milkyway data.

---

## SX-2300

(Severe) Could not determine extra port name of instance '%s' in cell '%s'.

**Description**

StarXtract could not determine this extra port name.

**What Next**

Confirm the Milkyway data is not broken or re-generate the Milkyway data.

---

## SX-2301

(Fatal) Layout instance "%s%c%s" is connected to ground "0".

**Description**

This kind of data will have incorrect result.

Feedback

**What Next**

Rename net "0" to another name.

---

## SX-2302

(Warning) Cell placement of '%s' has nonexistent port '%s'.

**Description**

StarXtract finds this nonexistent port.

**What Next**

Confirm this is expected data, or contact the Synopsys Support Center.

---

## SX-2303

(Fatal) Failed to compute cell extents for design.

**Description**

StarXtract is getting the cell extent information from Milkyway database, but this operation is failed.

**What Next**

Confirm the Milkyway data is not broken or re-generate the Milkyway data.

---

## SX-2304

(Warning) DIRTY_LAYOUT:YES is set in StarRC command file. This command is not for production purpose and the NETLIST_FILE with this command should not be considered as sign-off quality.

**Description**

DIRTY_LAYOUT:YES is set in StarRC command file. This command is not for production purpose and the NETLIST_FILE with this command should not be considered as sign-off quality.

**What Next**

Remove DIRTY_LAYOUT in StarRC command file for sign-off quality extraction.

---

## SX-2305

(Fatal) Cell '%s' is both SKIP_CELL and SKIP_PCELL.

Feedback

**Description**

StarXtract finds this skipped pcell is skipped cell.

**What Next**

Remove all cells specified in SKIP_PCELLS from SKIP_CELLS or set SKIP_CELLS:!*.

---

## SX-2306

(Information) Missing FRAM view for SKIP_CELL '%s'.

**Description**

StarXtract is getting the FRAM view data for this skipped cell, but this data does not exist.

**What Next**

If this is expected data, it is information only. No action is needed.

---

## SX-2308

(Severe) Could not get instance data for '%s' in cell '%s'.

**Description**

StarXtract is getting this instance data from Milkyway database, but this operation is failed.

**What Next**

Confirm the Milkyway data is not broken or re-generate the Milkyway data.

---

## SX-2309

(Warning) Instance '%s' of COUPLE_NONCRITICAL_NETS cell '%s' is not a SKIP_CELL and will be flattened.

**Description**

StarXtract will flatten this instance.

**What Next**

Confirm this is the expected behavior that depends on option settings.

---

## SX-2314

(Severe) Could not iterate port instances in cell '%s'.

Feedback

**Description**

StarXtract is iterating the port instances of this cell, but this operation is failed.

**What Next**

Confirm the Milkyway data is not broken or re-generate the Milkyway data.

---

## SX-2315

(Warning) The '%s' ('%s', '%s') has a port, portInstId = '%d', with netId = 0.

**Description**

StarXtract finds this net ID zero data.

**What Next**

Confirm this is expected data, or contact the Synopsys Support Center.

---

## SX-2316

(Warning) The '%s' has the port '%s', portId = '%d', with netId = 0.

**Description**

StarXtract finds this net ID zero data.

**What Next**

Confirm this is expected data, or contact the Synopsys Support Center.

---

## SX-2317

(Severe) Could not iterate nets in cell '%s'.

**Description**

StarXtract is iterating the nets of this cell, but this operation is failed.

**What Next**

Confirm the Milkyway data is not broken or re-generate the Milkyway data.

---

## SX-2320

(Warning) No instance name defined for placement of cell '%s'.

Feedback

**Description**

StarXtract finds there is no instance name defined for placement of this cell.

**What Next**

Confirm this Milkyway data is expected or contact the Synopsys Support Center.

---

## SX-2321

(Information) Empty instance name for placement '%s' in '%s' @ ['%d','%d'].

**Description**

StarXtract finds this empty instance name.

**What Next**

It is information only. No action is needed.

---

## SX-2322

(Severe) Unable to load cell '%s'.

**Description**

StarXtract is loading this cell, but this operation is failed.

**What Next**

Confirm the Milkyway data is not broken or re-generate the Milkyway data.

---

## SX-2324

(Information) Expanded hierarchical cell '%s' from library '%s'.

**Description**

StarXtract shows expanded hierarchical cell information.

**What Next**

It is information only. No action is needed.

---

## SX-2325

(Severe) Could not open cell "'%s'.'%s'".

Feedback

**Description**

StarXtract is opening this cell, but this operation is failed.

**What Next**

Confirm the Milkyway data is not broken or re-generate the Milkyway data.

---

## SX-2326

(Warning) Failed to iterate all '%s' in cell '%s' Total='%d' Iterated='%d'.

**Description**

StarXtract is iterating the Milkyway objects of this cell, but this operation is failed.

**What Next**

Confirm the Milkyway data is not broken or re-generate the Milkyway data.

---

## SX-2327

(Fatal) Failed to reset cell '%s'.

**Description**

StarXtract is resetting this cell, but this operation is failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-2329

(Severe) Could not get instance data for '%s' in cell '%s'.

**Description**

StarXtract is trying to get this instance data from Milkyway database, but this operation is failed.

**What Next**

Confirm the Milkyway data is not broken or re-generate the Milkyway data.

---

## SX-2332

(Severe) Unable to load cell '%s'.

Feedback

**Description**

StarXtract is trying to load the data of this cell from Milkyway database, but this operation is failed.

**What Next**

Confirm the Milkyway data is not broken or re-generate the Milkyway data.

---

## SX-2334

(Fatal) Cell '%s' is both SKIP_CELL and SKIP_PCELL.

**Description**

StarXtract reports this conflict data.

**What Next**

Remove all cells specified in SKIP_PCELLS from SKIP_CELLS or set SKIP_CELLS:!*.

---

## SX-2335

(Fatal) Only single nxtgrd file can be specified.

**Description**

StarRC detects that you have specified multiple nxtgrd files.

**What Next**

Make sure that only one nxtgrd file can be specified.

---

## SX-2389

(Fatal) Incorrect ContactCode '%d'.

**Description**

StarXtract reports this incorrect data.

**What Next**

Check tech file for this ContactCode.

---

## SX-2393

(Fatal) MARKER_GENERATION must be USER, AUTOMATIC or HERCULES.

Feedback

**Description**

The valid setting of MARKER_GENERATION are USER, AUTOMATIC or HERCULES.

**What Next**

Correct the setting in the StarXtract command file.

---

## SX-2395

(Fatal) Could not determine P&R cell boundary for '%s'.

**Description**

StarXtract is getting the cell boundary information from Milkyway database, but this operation is failed.

**What Next**

Confirm the Milkyway data is not broken or re-generate the Milkyway data.

---

## SX-2397

(Severe) Could not iterate instances array in cell '%s'.

**Description**

StarXtract is iterating the instances array of this cell, but this operation is failed.

**What Next**

Confirm the Milkyway data is not broken or re-generate the Milkyway data.

---

## SX-2398

(Warning) Virtual pin geometries are created on '%s'

**Description**

Pin in the macro is not placed in the layout. StarRC traces down the hierarchy for the first logically-connected instance and derives pin shapes from it for this unplaced pin (virtual PIO geometry). However, placement for this pin in parasitic netlist might be incorrect.

Due to the limitation in NDM streaming API, these virtual pin geometries are currently created on first metal layer.

**What Next**

Information only; no action needed.

Feedback

## SX-2399

(Warning) Pin '%s' does not have top-level routing shapes. Pin geometries are derived from the overlapping instance port shapes.

**Description**

Pin in the macro is not placed in the layout. StarRC traces down the hierarchy for the first logically-connected instance and derives pin shapes from it for this unplaced pin (virtual PIO geometry). However, placement for this pin in parasitic netlist might be incorrect.

**What Next**

Information only; no action needed.

## SX-2401

(Warning) Physical shape of logical pin port '%s' in BLOCK '%s' is not placed correctly and will be ignored.

**Description**

Physical shape of logical pin port must be placed on a physical layer.

**What Next**

Check the logical pin port that placed on a non-physical layer is expected. If not, contact the Synopsys Support Center.

## SX-2402

(Warning) Logical pin port '%s' in BLOCK is not placed in the layout.

**Description**

StarXtract is creating a virtual PIO geometry. Placement for this pin in parasitic netlist might be incorrect.

**What Next**

Confirm this is the expected data and behavior. If it is not the expected data, check Milkyway data is not broken or re-generate the Milkyway data.

## SX-2404

(Warning) Logical pin port '%s' in BLOCK is not placed in the layout.

Feedback

**Description**

This port will be nelisted in the *PORTS section of SPEF and/or the .SUBCKT statement for SPF/STAR/NETNAME formats but will not appear in the detail parasitic section (since it has no physical characteristic).

**What Next**

Confirm this is the expected data and behavior. If it is not the expected data, check Milkyway data is not broken or re-generate the Milkyway data.

---

## SX-2407

(Severe) Could not iterate rectangles in cell '%s'.

**Description**

StarXtract is iterating the rectangles of this cell, but this operation is failed.

**What Next**

Confirm the Milkyway data is not broken or re-generate the Milkyway data.

---

## SX-2411

(Warning) Embedded color information will not be used since DPT_COLOR_GDS_FILE is used.

**Description**

StarXtract will not use embedded color information while DPT_COLOR_GDS_FILE is used.

**What Next**

Confirm this is the expected behavior.

---

## SX-2412

(Warning) Unsupported OASIS record

**Description**

Circle and X-Geometry records are currently not supported. They are ignored during import.

**What Next**

Contact the Synopsys Support Center.

Feedback

## SX-2414

(Warning) Unrecognized OASIS record

**Description**

Records that are not documented in the OASIS specfication are unrecognized records. StarRC will skip any unrecognized record during import.

**What Next**

Contact the Synopsys Support Center.

## SX-2415

(Information) Failed to build FILE mode DCM table '%s'.

**Description**

StarXtract is building this FILE mode DCM table, but this operation is failed.

**What Next**

If this is expected behavior, it is information only. No action is needed.

## SX-2418

(Fatal) Running StarXtract with -cleanN and NETLIST_FORMAT: '%s' following a run with NETLIST_FORMAT: 'NONE' is not supported. Rerun with -clean to generate netlists with NETLIST_FORMAT '%s'.

**Description**

The only format that is supported with '-cleanN' after 'NETLIST_FORMAT: NONE' is NETLIST_FORMAT: 'SPEF'.

**What Next**

Consult the StarRC user guide.

## SX-2426

(Fatal) Found invalid option changes between previous run and this incremental run.

**Description**

There are inconsistent option commands between previous run and this incremental run. Please check command files.

Feedback

**What Next**

Please refer to section on ECO extraction in the StarRC user guide.

---

## SX-2434

(Fatal) Cannot open BLOCK '%s' in input database.

**Description**

Cannot open top BLOCK cell in input database.

**What Next**

Check whether the BLOCK setting is StarRC command file is correct.

---

## SX-2457

(Fatal) the executable is called from wrong location

**Description**

To invoke gui, please call StarXtract from correct location.

**What Next**

Check you StarXtract path or installation.

---

## SX-2458

(Fatal) Could not create tcl gui

**Description**

This is internal error message to indicate TCL interpreter cannot be created

**What Next**

Contact the Synopsys Support Center.

---

## SX-2465

(Information) Error: Could not connect to AServer.

**Description**

This is an internal error for -gui.

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## SX-2466

(Information) Error: Could not connect to Executable.

**Description**

This is an internal error for -gui.

**What Next**

Contact the Synopsys Support Center.

---

## SX-2467

(Information) Erro: No sesstion info from AServer.

**Description**

This is an internal error for -gui

**What Next**

Contact the Synopsys Support Center.

---

## SX-2468

(Information) ERROR: Lost connection to AServer.

**Description**

This is an internal error for -gui

**What Next**

Contact the Synopsys Support Center.

---

## SX-2469

(Information) ERROR: Unable to find '%s' session.

**Description**

This is an internal error for -gui

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## SX-2470

(Information) ERROR: Incorrect user id for '%s' session.

**Description**

This is an internal error for -gui

**What Next**

Contact the Synopsys Support Center.

---

## SX-2471

(Information) Connected to '%s' sId='%d' pId='%d'

**Description**

This is an internal error for -gui

**What Next**

Contact the Synopsys Support Center.

---

## SX-2472

(Information) Shutting down AServer connection...

**Description**

This is an internal error for -gui

**What Next**

Contact the Synopsys Support Center.

---

## SX-2473

(Information) ERROR: Unable to close connection with AServer.

**Description**

This is an internal error for -gui

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## SX-2474

(Information) Error: couldn't find hostname

**Description**

This is an internal error for -gui

**What Next**

Contact the Synopsys Support Center.

---

## SX-2475

(Information) ERROR: RPC connection not initialized yet.

**Description**

This is an internal error for -gui

**What Next**

Contact the Synopsys Support Center.

---

## SX-2476

(Information) ERROR: Time out waiting for application to respond.

**Description**

This is an internal error for -gui

**What Next**

Contact the Synopsys Support Center.

---

## SX-2477

(Information) ERROR: Sending command before connect.

**Description**

This is an internal error for -gui

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## SX-2478

(Information) ERROR: Connection lost while sending.

**Description**

This is an internal error for -gui

**What Next**

Contact the Synopsys Support Center.

---

## SX-2479

(Information) ERROR: Failed to send.

**Description**

This is an internal error for -gui

**What Next**

Contact the Synopsys Support Center.

---

## SX-2480

(Information) ERROR: Could not open library '%s'

**Description**

This is an internal error for -gui

**What Next**

Contact the Synopsys Support Center.

---

## SX-2481

(Information) Error: Could not read object details

**Description**

This is an internal error for -gui

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## SX-2482

(Fatal) Invalid violation file format

**Description**

This is an internal error for -gui

**What Next**

Contact the Synopsys Support Center.

---

## SX-2483

(Information) ERROR: No library opened. Cannot open cell '%s'

**Description**

This is an internal error for -gui

**What Next**

Contact the Synopsys Support Center.

---

## SX-2484

(Information) '%s'

**Description**

This is the message to pass all Error/Warning/Information message from Viewer->Prepare Viewer Data

**What Next**

If it is just a Information/Warning, please ignore it. If this is a Error, please contact Synopsys Suport Center.

---

## SX-2486

(Severe) '%s' Failed (see '%s' )

**Description**

If you get this message, that means Viewer->Prepare Viewer Data failed.

Feedback

**What Next**

Please check Error message before this message.

---

## SX-2489

(Fatal) LPE_PARAM: '%s' does not have a value defined for NORC.

**Description**

Wrong LPE_PARAM settings

**What Next**

Verify LPE_PARAM settings in StarRC command file.

---

## SX-2490

(Warning) LPE parameter '%s' was used for some devices but was not defined by LPE_PARAM statement.

**Description**

Wrong LPE_PARAM settings

**What Next**

Verify LPE_PARAM settings in StarRC command file.

---

## SX-2497

(Fatal) Could not write file '%s'.

**Description**

The file cannot be open or written.

**What Next**

Check if the disk is full.

---

## SX-2499

(Fatal) File offset is too large for 32-b application. Please run with 64-b.

**Description**

The file generated by StarRC is too large and the file offset is larger than 2^31.

Feedback

**What Next**

Run StarRC on 64-bits machine to overcome this issue.

---

## SX-2505

(Severe) Instance '%s' has no definition.

**Description**

Instances used in NETS section in DEF file aren't defined in COMPONENTS section.

**What Next**

Check the DEF file correctness.

---

## SX-2506

(Fatal) '%s' line '%d'. LPE_PARAM rule: expecting parameter name before parameter values.

**Description**

Wrong LPE_PARAM settings

**What Next**

Verify LPE_PARAM settings in StarRC command file.

---

## SX-2508

(Fatal) '%s' line '%d'. LPE_PARAM rule: unknown extraction mode '%s'.

**Description**

Wrong LPE_PARAM settings

**What Next**

Verify LPE_PARAM settings in StarRC command file.

---

## SX-2509

(Fatal) '%s' line '%d'. LPE_PARAM rule: the parameter values are not defined correctly.

**Description**

Wrong LPE_PARAM settings

Feedback

**What Next**

Verify LPE_PARAM settings in StarRC command file.

---

## SX-2510

(Fatal) '%s' line '%d'. PINEXCEPT rule: expecting list of pins (indices).

**Description**

Wrong LPE_PARAM settings

**What Next**

Verify LPE_PARAM settings in StarRC command file.

---

## SX-2511

(Fatal) '%s' line '%d'. PINOPERATION rule: expecting AND or OR operation.

**Description**

Wrong LPE_PARAM settings

**What Next**

Verify LPE_PARAM settings in StarRC command file.

---

## SX-2512

(Fatal) '%s' line '%d'. LPE_DEVICES rule: unexpected integer.

**Description**

Wrong LPE_DEVICES settings

**What Next**

Verify LPE_DEVICES settings in StarRC command file.

---

## SX-2513

(Fatal) INSTANCE_PORT_REPORT_FILE can only be set if INSTANCE_PORT is set.

**Description**

INSTANCE_PORT_REPORT_FILE is used to report instance port settings
by INSTANCE_PORT command. If INSTANCE_PORT is not set, StarRC will

Feedback

use default (CONDUCTIVE) for all instance ports. There is no need to use INSTANCE_PORT_REPORT_FILE.

**What Next**

Add INSTANCE_PORT in StarRC command file.

---

## SX-2514

(Fatal) Parse error at line '%d' of file: '%s' near unexpected token '%s'.

**Description**

Wrong LPE_PARAM settings

**What Next**

Verify LPE_PARAM settings in StarRC command file.

---

## SX-2515

(Fatal) LPE: 1 LPE_PARAM statement is missing.

**Description**

Wrong LPE_PARAM settings

**What Next**

Verify LPE_PARAM settings in StarRC command file.

---

## SX-2516

(Fatal) LPE: '%d' LPE_PARAM statements are missing.

**Description**

Wrong LPE_PARAM settings

**What Next**

Verify LPE_PARAM settings in StarRC command file.

---

## SX-2519

(Warning) Duplicate device name '%s' in CALIBRE_LVS_DEVICE_TYPE option.

Feedback

**Description**

Duplicate device name is defined in CALIBRE_LVS_DEVICE_TYPE option.

**What Next**

Make sure each device name is unique in CALIBRE_LVS_DEVICE_TYPE.

---

## SX-2520

(Warning) Duplicate device name '%s' in ICV_LVS_DEVICE_TYPE option.

**Description**

Duplicate device name is defined in ICV_LVS_DEVICE_TYPE option.

**What Next**

Make sure each device name is unique in ICV_LVS_DEVICE_TYPE.

---

## SX-2521

(Warning) Detected TSMC/3DIC design modules and exploded them automatically.

**Description**

TSMC 3DIC Design Module must be exploded (non-skipped cell) for StarRC to make them transparent in the design. Because of this, StarRC explodes them automatically if they are not marked non-skipped cells.

**What Next**

Information only; no actions required.

---

## SX-2522

(Warning) Cannot open CALIBRE_OPTIONAL_DEVICE_PIN_FILE:'%s'

**Description**

Cannot open CALIBRE_OPTIONAL_DEVICE_PIN_FILE.

**What Next**

Make sure the CALIBRE_OPTIONAL_DEVICE_PIN_FILE is correctly defined and you have permission to read.

Feedback

## SX-2523

(Warning) Cannot open ICV_OPTIONAL_DEVICE_PIN_FILE:'%s'

**Description**

Cannot open ICV_OPTIONAL_DEVICE_PIN_FILE.

**What Next**

Make sure the ICV_OPTIONAL_DEVICE_PIN_FILE is correctly defined and you have permission to read.

## SX-2526

(Warning) CALIBRE_OPTIONAL_DEVICE_PIN_FILE: Unsupport device type '%s'.

**Description**

CALIBRE_OPTIONAL_DEVICE_PIN_FILE: Unsupport device type. Only "M", "C" and "R" are supported device type.

**What Next**

Define correct device type in CALIBRE_OPTIONAL_DEVICE_PIN_FILE.

## SX-2527

(Warning) ICV_OPTIONAL_DEVICE_PIN_FILE: Unsupport device type '%s'.

**Description**

ICV_OPTIONAL_DEVICE_PIN_FILE: Unsupport device type. Only "M", "C" and "R" are supported device type.

**What Next**

Define correct device type in ICV_OPTIONAL_DEVICE_PIN_FILE.

## SX-2528

(Warning) CALIBRE_OPTIONAL_DEVICE_PIN_FILE: Device '%s' is not specified in CALIBRE_LVS_DEVICE_TYPE option.

**Description**

CALIBRE_OPTIONAL_DEVICE_PIN_FILE: Device is not specified in CALIBRE_LVS_DEVICE_TYPE option.

Feedback

**What Next**

Make sure that the devices defined in CALIBRE_OPTIONAL_DEVICE_PIN_FILE is specified in CALIBRE_LVS_DEVICE_TYPE option.

---

## SX-2529

(Warning) ICV_OPTIONAL_DEVICE_PIN_FILE: Device '%s' is not specified in ICV_LVS_DEVICE_TYPE option.

**Description**

ICV_OPTIONAL_DEVICE_PIN_FILE: Device is not specified in ICV_LVS_DEVICE_TYPE option.

**What Next**

Make sure that the devices defined in ICV_OPTIONAL_DEVICE_PIN_FILE is specified in ICV_LVS_DEVICE_TYPE option.

---

## SX-2530

(Warning) CALIBRE_OPTIONAL_DEVICE_PIN_FILE: Device type of '%s' is inconsistent to CALIBRE_LVS_DEVICE_TYPE specification.

**Description**

CALIBRE_OPTIONAL_DEVICE_PIN_FILE: Device type of the specified device is inconsistent to CALIBRE_LVS_DEVICE_TYPE specification. CALIBRE_LVS_DEVICE_TYPE represents for CALIBRE_LVS_DEVICE_TYPE_CAP, CALIBRE_LVS_DEVICE_TYPE_RES and CALIBRE_LVS_DEVICE_TYPE_MOS. For example: If command 'CALIBRE_LVS_DEVICE_TYPE_CAP: cap123' and 'CALIBRE_OPTIONAL_DEVICE_PIN_FILE: file_abc' are specified, and file_abc contains a line like this: R cap123 ..... Then, Warning message SX-2530 will be given to inform you the inconsistency.

**What Next**

Make sure that the device type is consistent in CALIBRE_OPTIONAL_DEVICE_PIN_FILE and CALIBRE_LVS_DEVICE_TYPE.

---

## SX-2531

(Warning) ICV_OPTIONAL_DEVICE_PIN_FILE: Device type of '%s' is inconsistent to ICV_LVS_DEVICE_TYPE specification.

Feedback

**Description**

ICV_OPTIONAL_DEVICE_PIN_FILE: Device type of the specified device is inconsistent to ICV_LVS_DEVICE_TYPE specification. ICV_LVS_DEVICE_TYPE represents for ICV_LVS_DEVICE_TYPE_CAP, ICV_LVS_DEVICE_TYPE_RES and ICV_LVS_DEVICE_TYPE_MOS. For example: If command 'ICV_LVS_DEVICE_TYPE_CAP: cap123' and 'ICV_OPTIONAL_DEVICE_PIN_FILE: file_abc' are specified, and file_abc contains a line like this: R cap123 ..... Then, Warning message SX-2530 will be given to inform you the inconsistency.

**What Next**

Make sure that the device type is consistent in ICV_OPTIONAL_DEVICE_PIN_FILE and ICV_LVS_DEVICE_TYPE.

---

## SX-2532

(Fatal) CALIBRE_OPTIONAL_DEVICE_PIN_FILE: In device '%s', terminal assignment format should be layer_name(terminal_name).

**Description**

Terminal assignment format should be layer_name(terminal_name) in CALIBRE_OPTIONAL_DEVICE_PIN_FILE.

**What Next**

Set correct format for device terminal in CALIBRE_OPTIONAL_DEVICE_PIN_FILE.

---

## SX-2533

(Fatal) ICV_OPTIONAL_DEVICE_PIN_FILE: In device '%s', terminal assignment format should be layer_name(terminal_name).

**Description**

Terminal assignment format should be layer_name(terminal_name) in ICV_OPTIONAL_DEVICE_PIN_FILE.

**What Next**

Set correct format for device terminal in ICV_OPTIONAL_DEVICE_PIN_FILE.

---

## SX-2534

(Warning) Model doesn't exist!! HN_NETLIST_MODEL_NAME: %s.

Feedback

**Description**

StarXtract finds this model name does not exist.

**What Next**

Confirm this is expected data. If it is not, correct this model name.

---

## SX-2536

(Fatal) Cannot open internal file '%s'

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-2537

(Fatal) Corrupted internal DHE file '%s'

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-2541

(Warning) Simple array via or base via in a via matrix '%s' with via definition '%s' is found with wrong array ratio '%.4f' in the database

**Description**

Multi-cut via with regular cut pattern can be abstracted with a via array ratio for StarRC to extract faster. If the array ratio is larger than 1.0, StarRC will ignore the cut abstraction and translate each cut instead.

It's most likely caused by a wrong multi-cut via definition which has wrong rows/columns properties.

**What Next**

Correct the via definition of the reported via instance.

Feedback

## SX-2542

(Fatal) Invalid HN_NETLIST_MODEL_NAME specification.

**Description**

The correct usage is HN_NETLIST_MODEL_NAME: <device model name> <netlist model name>.

**What Next**

Correct the usage of HN_NETLIST_MODEL_NAME.

## SX-2543

(Warning) NDM design '%s' references a non-existent or unreadble FILL design in disk (%s).

**Description**

The disk file for the FILL design might be deleted accidentally or chmod without proper permission. This leave its references broken in the design. The FILL design will be ignored in the translation.

**What Next**

Check if the FILL design is removed intentionally and fix it if it is not intentional.

## SX-2544

(Information) DONE

**Description**

This is information to show Viewer->Prepare Viewer Data finish successfully.

**What Next**

No action is needed.

## SX-2545

(Warning) A big but empty design cell '%s' scheduled for parallel SkipCell creation is found

Feedback

**Description**

A design cell scheduled for parallel SkipCell creation is found empty after DB import. If it is instantiated several times in the design, it may causes false opens or other issues at the extraction considering its significant size in the DB.

When all metal routings in a design cell are not mapped correctly in the MAPPING_FILE, metal routings are considered unimportant to the extraction and ignored. In this case, the design cell will be empty after DB import and its skip cell will be excluded from SkipCell creation.

**What Next**

Please check the design cell in the input DB or MAPPING_FILE.

---

## SX-2546

(Fatal) The character '/' is not allowed in COUPLE_NONCRITICAL_NETS_PREFIX.

**Description**

'/' is not allowed in the name of COUPLE_NONCRITICAL_NETS_PREFIX.

**What Next**

Change the COUPLE_NONCRITICAL_NETS_PREFIX name.

---

## SX-2548

(Fatal) Failed to compute cell extents for '%s'.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-2549

(Warning) PIN_CUT_THRESHOLD is not supported for non-manhattan pin shapes. Additional *P will not be generated for the following shape: '%s'

**Description**

StarRC detects non-manhattan shape pins. PIN_CUT_THRESHOLD only works with manhattan shape pins. So StarRC will not generate multiple *P for this pin.

Feedback

**What Next**

This is an informational message only. If this non-manhattan shape pin is expected, no action is required on your part.

---

## SX-2550

(Warning) Undefined GDS/OASIS cell '%s'

**Description**

A cell instance in the GDS_FILE/OASIS_FILE doesn't have cell definition. StarRC will ignore these instances during translation.

**What Next**

Information only; no action needed.

---

## SX-2552

(Fatal) SkipCellDB index file is corrupted or not compatible with this release. Please run -clean.

**Description**

An internal consistency check has failed.

**What Next**

This error message may due to version change in internal files. If you run StarRC with -clean command line option and you still see this error message, contact the Synopsys Support Center.

---

## SX-2553

(Fatal) SKIP_CELLS database version is not compatible with this release. Please run -clean.

**Description**

StarRC internal file version is changed. StarRC detects old version in STAR directory.

**What Next**

Run StarRC with -clean command line option. If this error still exists, contact the Synopsys Support Center.

Feedback

## SX-2555

(Fatal) %s_LAYER_MAP_FILE is missing.

**Description**

GDS_FILE, METAL_FILL_GDS_FILE, CONTEXT_GDS_FILE and SKIP_CELL_AGF_FILE require layer mappings specified GDS_LAYER_MAP_FILE so as for StarRC to translate GDS structs into appropriate layers in the main design database.

OASIS_FILE, METAL_FILL_OASIS_FILE and CONTEXT_OASIS_FILE require layer mappings specified in OASIS_LAYER_MAP_FILE so as for StarRC to translate OASIS structs into appropriate layers in main design database.

**What Next**

Specify GDS_LAYER_MAP_FILE/OASIS_LAYER_MAP_FILE in command file and run StarRC again.

## SX-2558

(Fatal) binary xin file is corrupted! suspect version mismatch.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## SX-2563

(Fatal) error while reading xin.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## SX-2566

(Warning) via_models file corrupted!

Feedback

**Description**

Invalid data is detected in via_models file.

**What Next**

An internal consistency check has failed.

---

## SX-2567

(Fatal) via_models file corrupted in via model section.

**Description**

Invalid data is detected in via_models file in via model section.

**What Next**

An internal consistency check has failed.

---

## SX-2569

(Fatal) via_models file corrupted in sav section.

**Description**

Invalid data is detected in via_models file in sav section.

**What Next**

An internal consistency check has failed.

---

## SX-2573

(Fatal) via_models file corrupted: cannot read corner RPV due to bad format.

**Description**

Format of via_models file is not correct.

**What Next**

An internal consistency check has failed.

---

## SX-2574

(Fatal) via_models file corrupted: cannot add corner RPV due to discrepancy from primary RPV.

Feedback

**Description**

Invalid RPV value is detected in via_models file.

**What Next**

An internal consistency check has failed.

---

## SX-2576

(Warning) Total number of via and via array models exceed limit of 32767. Via coverage codes may be incorrect!

**Description**

Total number of via models exceed the limit. The result of VIA_COVERAGE could be imfluenced.

**What Next**

Check if all the models are necessary.

---

## SX-2577

(Fatal) via_array_models file corrupted.

**Description**

Invalid data is detected in via_array_models file.

**What Next**

An internal consistency check has failed.

---

## SX-2578

(Warning) -compute_via_coverage is ON, but no via models provided, please re-run translation.

**Description**

via_models file is not found for via coverage computation.

**What Next**

Rerun StarRC.

Feedback

## SX-2579

(Fatal) could not write coverage_codes information file: '%s'

**Description**

Could not write coverage_codes information file.

**What Next**

An internal consistency check has failed.

## SX-2580

(Fatal) Some, but not all via models have semi coverage.

**Description**

Not every via model has Semi coverage.

**What Next**

An internal consistency check has failed.

## SX-2581

(Fatal) Internal, sav-dependent parameter table is empty.

**Description**

Sav-dependent table is empty.

**What Next**

An internal consistency check has failed.

## SX-2582

(Warning) Can't find coverage option for via on layer '%d'

**Description**

Can't find coverage option for this via layer.

**What Next**

Information only; no action is needed.

Feedback

## SX-2583

(Warning) Via model is empty in via_model file "via_models".

**Description**

Empty data is detected in via_models file.

**What Next**

Information only; no action is needed.

## SX-2584

(Warning) Can't find coverage option for via with ContactCode '%d' on layer '%d' with width '%d' length '%d'

**Description**

Can't find coverage option for via.

**What Next**

Information only; no action is needed.

## SX-2585

(Warning) check VIA_COVERAGE_OPTION_FILE and VIA_COVERAGE options in StarRC command file. Your VIA_COVERAGE_OPTION_FILE contains: '%s'

**Description**

Invalid data is detected in input data for via coverage.

**What Next**

Check VIA_COVERAGE_OPTION_FILE and VIA_COVERAGE options in StarRC command file.

## SX-2586

(Warning) Can't find coverage option for via width '%d' length '%d' on layer '%d'. Please check VIA_COVERAGE_OPTION_FILE and VIA_COVERAGE options in StarRC command file.

**Description**

Can't find coverage option for via on the layer.

Feedback

**What Next**

Check VIA_COVERAGE_OPTION_FILE and VIA_COVERAGE options in StarRC command file.

---

## SX-2588

(Warning) Can't find bridge via model for the via with width='%d', length='%d' level='%d', a default model for non-bridge via will be used.

**Description**

Can't find bridge via model for the via, a default model for non-bridge via will be used.

**What Next**

Check the models defined for bridge via.

---

## SX-2590

(Fatal) Could not write data to '%s'.

**Description**

StarRC is unable to write data to the specified file because of issues in the runtime environment.

**What Next**

Try running StarRC on a disk with ample space or contact your system administrator.

---

## SX-2592

(Fatal) Could not read file '%s' to completion.

**Description**

The specified file could not be read to completion.

**What Next**

Try rerunning StarRC. Contact the Synopsys Support Center if the error message persists.

---

## SX-2606

(Fatal) INTERNAL Error: attempt to write '%s' more than once in the flow.

Feedback

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-2607

(Fatal) INTERNAL Error: file '%s' does not exist.

**Description**

An internal consistency check has failed.

**What Next**

Please look for other errors/warnings in the sum file. Please try rerunning, if the original issue persists, please contact the Synopsys Support Center.

---

## SX-2609

(Information) Failed to open reference libraries for '%s' in '%s' mode.

**Description**

StarXtract is opening this reference library, but this operation is failed.

**What Next**

If this is expected data, it is information only. No action is needed.

---

## SX-2627

(Warning) INTERNAL Warning: Invalid NUM_PARTS specification : '%d', assuming 1.

**Description**

This warning is for information only.

**What Next**

No actions is necessary.

---

## SX-2632

(Fatal) The value of NUM_PARTS (%d) leads to invalid cut coordinates.

Feedback

**Description**

The value of NUM_PARTS in the StarRC command file leads to an invalid partitioning of the layout.

**What Next**

Rerun StarRC with a reduced value of NUM_PARTS.

---

## SX-2636

(Warning) INTERNAL Warning: design BBOX is not initialized.

**Description**

An internal consistency check has failed.

**What Next**

Check the input file to make sure bounding box is initialiezed. Contact the Synopsys Support Center if StarRC terminates abnormally or other issues are found.

---

## SX-2637

(Fatal) INTERNAL Error: Cannot specify both cut file name and NUM_PARTS.

**Description**

This error message applies to an internal debugging feature only.

**What Next**

No action is necessary.

---

## SX-2640

(Fatal) INTERNAL Error: the layout partitioning scheme results in too many interface nodes.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

Feedback

## SX-2643

(Fatal) INTERNAL Error: The number of pins in the design exceeds current limits in StarRC.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## SX-2646

(Fatal) INTERNAL Error: Could not find mapping for '%d'

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## SX-2650

(Fatal) Via coverage parameters must meet Full >= Quarter : '%s'

**Description**

In via coverage checking, parameters of Full pattern must >= Quarter pattern.

**What Next**

Check the parameters defined in every pattern and rerun StarRC.

## SX-2652

(Fatal) Via coverage parameters must meet Full >= Quarter and Full >= Semi : '%s'

**Description**

In via coverage checking, parameters of Full pattern must >= Quarter pattern and Semi pattern.

**What Next**

Check the parameters defined in every pattern and rerun StarRC.

Feedback

## SX-2654

(Fatal) incorrect via size range: '%d'

**Description**

Invalid via size is detected in via coverage definition.

**What Next**

Check the via size and rerun StarRC.

## SX-2655

(Fatal) Find overlapping range for via coverage '%s'

**Description**

Overlapping range is set in via coverage definition and will cause error.

**What Next**

Correct the range definition and rerun StarRC.

## SX-2666

(Fatal) Invalid INSTANCE_PORT, NOT_CONDUCTIVE|CONDUCTIVE|EXPLODE can only be used alone.

**Description**

INSTANCE_PORT command can only be set to NOT_CONDUCTIVE, CONDUCTIVE, or EXPLODE. It cannot be set to more than one value.

**What Next**

Correct INSTANCE_PORT setting in StarRC command file.

## SX-2667

(Fatal) Invalid type combination in INSTANCE_PORT, Only CONDUCTIVE|SUPERCONDUCTIVE can be used together with MULTIPLE|CAPACITIVE|SUFFIXED.

**Description**

Only CONDUCTIVE|SUPERCONDUCTIVE can be used together with MULTIPLE|CAPACITIVE|SUFFIXED in INSTANCE_PORT command.

Feedback

**What Next**

Check StarRC command file setting.

---

## SX-2668

(Severe) BitMap Overflow CELL_EXTENTS=['%d','%d':'%d','%d']

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-2673

(Fatal) Invalid xin bucket file name '%s'

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-2677

(Fatal) The layer_cut_info file is missing or corrupted!

**Description**

An internal consistency check has failed.

**What Next**

Turn off STARDIR_REDUCE_SIZE in your star command file and re-run. If the error still occurs, contact the Synopsys Support Center.

---

## SX-2678

(Fatal) Can't open xin file: %s

**Description**

An internal consistency check has failed.

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## SX-2679

(Fatal) MAX_TRAP_SORT can not be less than MIN_TRAP_SORT.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-2680

(Fatal) TRAP_SORT_STEP must be greater than 0.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-2683

(Fatal) The input design exceeds internal limitations of StarRC. Please rerun with TEXT_ID_COMPRESSION: yes in the StarRC command file.

**Description**

This design is too big and exceeds StarRC internal limitations.

**What Next**

Add TEXT_ID_COMPRESSION:YES in StarRC command file to overcome this problem.

---

## SX-2684

(Fatal) Cannot open output file '%s'. '%s'

**Description**

An internal consistency check has failed.

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## SX-2695

(Fatal) Could not open file '%s'

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-2700

(Fatal) Encountered polygon with negative width or height: ('%d' '%d'), ('%d' '%d')

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-2701

(Fatal) '%s'

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-2702

(Fatal) Encountered polygon with large coordinates: ('%d' '%d'), ('%d' '%d'). After scaling, it will be out of the integer range.

**Description**

An internal consistency check has failed.

Feedback

**What Next**

Check whether the input database contains incorrect polygon coordinates. If not, contact the Synopsys Support Center.

---

## SX-2703

(Warning) Encountered polygon with large coordinates: ('%d' '%d'), ('%d' '%d'). After scaling, it will be out of the integer range. Polygon is ignored.

**Description**

An internal consistency check has failed.

**What Next**

Check whether the input database contains incorrect polygon coordinates. If not, contact the Synopsys Support Center.

---

## SX-2706

(Fatal) Could not convert '%s' to a signed long long integer: '%s'

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-2708

(Fatal) Error reading '%s'

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-2709

(Fatal) Flush() must be explicitly called in this version of TrapMemSorter.

Feedback

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-2710

(Fatal) Could not write file '%s'

**Description**

An internal consistency check has failed.

**What Next**

Check whether the working directory is full. If not, contact the Synopsys Support Center.

---

## SX-2712

(Fatal) Failed to allocate memory for '%s' .

**Description**

An internal consistency check has failed.

**What Next**

Might be caused by different reasons. Try freeing up the memory on the current system and rerunning again. If the issue persists, please contact the Synopsys Support Center.

---

## SX-2714

(Fatal) ProbeTextTable version incompatible with this release.

**Description**

An internal consistency check has failed.

**What Next**

Rerun this case with -clean command line option. If this error still occurs, contact the Synopsys Support Center.

---

## SX-2717

(Fatal) Cannot open input file '%s'

Feedback

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-2718

(Fatal) An error has occurred during disk access.

**Description**

The operating system has detected an error related to disk access. No further information is available.

**What Next**

Check if you have run out of disk space. If not, contact your system administrator or the Synopsys Customer Support Center.

---

## SX-2734

(Fatal) Unexpected event during xin creation.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-2736

(Fatal) flush() must be explicitly called in this version of BucketTrapezoidWriter, please report this case to StarRC R&D.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

Feedback

## SX-2739

(Information) Missing GDS/OASIS cell description for %s MACRO '%s'.

**Description**

Skip cell has definition from LEF/NDM cell only in GDS_FILE/OASIS_FILE flow.

**What Next**

Information only; no action needed

## SX-2740

(Fatal) %s: '%s' is a directory, not a file.

**Description**

A directory was specified in the command file, but a file name is required.

**What Next**

Correct the entry in the command file to be a file name, not a directory.

## SX-2741

(Warning) NONDEFAULTRULE '%s' is undefined (used in NET: '%s')

**Description**

The NONDEFAULTRULE used in NET is not defined in current DEF or LEF files. The default rule will be used to translate this NET. This won't cause errors during translation but it is possible to cause false opens/shorts if different wire width should be defined and adopted for the NET.

**What Next**

Check the NONDEFAULTRULE setting and if it's not required, you can ignore this message.

## SX-2742

(Information) Failed to open reference libraries by control file for '%s'.

**Description**

StarXtract is opening this main reference library, but this operation is failed.

Feedback

**What Next**

If this is expected data, it is information only. No action is needed.

---

## SX-2744

(Information) Could not determine '%s' attach file location. Milkyway '%s' is incomplete.

**Description**

StarXtract is accessing this attach file and it does not exist.

**What Next**

If this is expected data, it is information only. No action is needed.

---

## SX-2748

(Information) Must specify (P)DIODE_TYPE in Hercules '%s' (P)DIODE definition.

**Description**

StarXtract finds this invalid data. The (P)DIODE_TYPE must be specified in Hercules '%s' (P)DIODE definition.

**What Next**

If this is expected data, it is information only. No action is needed.

---

## SX-2749

(Information) Invalid (P)DIODE_TYPE '%s' in Hercules '%s' (P)DIODE defintion. Must be PN or NP.

**Description**

StarXtract finds this invalid data. It must be PN or NP.

**What Next**

If this is expected data, it is information only. No action is needed.

---

## SX-2750

(Information) Not an LVS runset, cannot use XREF: YES/COMPLETE.

**Description**

XREF: YES/COMPLETE only supports LVS runset.

Feedback

**What Next**

Make sure that WRITE_EXTRACT_VIEW and COMPARE are both run in the same call to Hercules.

---

## SX-2751

(Information) CELL_TYPE: SCHEMATIC is only available with XREF: YES or COMPLETE.

**Description**

StarXtract finds invalid setting of CELL_TYPE.

**What Next**

If this is expected data, it is information only. No action is needed.

---

## SX-2752

(Information) NET_TYPE: SCHEMATIC is only available with XREF: YES or COMPLETE.

**Description**

StarXtract finds invalid setting of NET_TYPE.

**What Next**

If this is expected data, it is information only. No action is needed.

---

## SX-2753

(Information) NET_TYPE must be LAYOUT or SCHEMATIC.

**Description**

StarXtract finds invalid setting of NET_TYPE.

**What Next**

If this is expected data, it is information only. No action is needed.

---

## SX-2754

(Information) Must use CASE_SENSITIVE: NO when IGNORE_CASE=TRUE in the Hercules runset.

Feedback

**Description**

Hercules runset has IGNORE_CASE=TRUE. The setting of CASE_SENSITIVE in the StarXtract command file must be NO.

**What Next**

If this is expected data, it is information only. No action is needed.

---

## SX-2756

(Warning) Undefined LAYER '%s' in FILLS.

**Description**

FILLS on undefined LAYER will be ignored during translation.

**What Next**

Check the technology LEF for undefined LEF.

---

## SX-2757

(Information) RefLib Control File inconsistent.

**Description**

StarXtract finds RefLib control file inconsistecy.

**What Next**

If this is expected data, it is information only. No action is needed.

---

## SX-2758

(Information) Element devices are not suppported in Milkyway XTR view.

**Description**

StarXtract does not support element devices.

**What Next**

If this is expected data, it is information only. No action is needed.

---

## SX-2763

(Fatal) %s

Feedback

**Description**

Required parameters are missing. Correct usage: StarXtract -save_eco target_file_name or StarXtract -restore_eco source_file_name.

**What Next**

Issue the command with correct parameters.

---

## SX-2764

(Warning) Missing GDS/OASIS cell descriptions for some LEF MACROs.

**Description**

In GDS_FILE/OASIS_FILE flow, StarRC merges skip cell contents from both LEF and GDS/OASIS. If a skip cell only has defintion from LEF, StarRC will only translate its LEF and print out this warning. Normally, this won't cause translation error but it may incur design inconsistency for some cell definitions are not merged with proper GDS/OASIS cells.

**What Next**

Contact the Synopsys Support Center.

---

## SX-2766

(Warning) PARSING Warning: %s near line %d near token %s in file %s

**Description**

A non-critical parsing check has failed. This warning message is printed for use by the Synopsys Support Center in case a program error occurs during StarRC's execution.

**What Next**

Contact the Synopsys Support Center if StarRC terminates abnormally or other issues are found.

---

## SX-2767

(Fatal) PARSING Error: %s near line %d near token %s in file %s

**Description**

A critical parsing check has failed.

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## SX-2768

(Warning) dbuPerMicron in incremental LEF violates LEF specification.

**Description**

dbuPerMicron setting for the following MACRO LEFs is larger than technology LEF. This is a violation of LEF/DEF specification and may cause loss of accuracy during translation.

**What Next**

Correct the dbuPerMicron in the following MACRO LEFs.

---

## SX-2773

(Fatal) Invalid HN_NETLIST_SPICE_TYPE specification.

**Description**

The correct usage is; HN_NETLIST_SPICE_TYPE: <device model name> <netlist spice type>.

**What Next**

Correct the usage of HN_NETLIST_SPICE_TYPE.

---

## SX-2774

(Warning) Following option "JOIN" detected in Hercules/ICV runset is not supported in StarXtract.

**Description**

StarXtract finds this unsupported option in Hercules/ICV runset.

**What Next**

Confirm this is expected data. If it is not, confirm the procedures and settings to generate this data is correct, or re-generate this data.

---

## SX-2775

(Warning) Following option "DETECT_PERMUTABLE_PORTS" detected in Hercules/ICV runset is not supported in StarXtract.

Feedback

**Description**

StarXtract finds this unsupported option in Hercules/ICV runset.

**What Next**

Confirm this is expected data. If it is not, confirm the procedures and settings to generate this data is correct, or re-generate this data.

---

## SX-2776

(Warning) Following option "PUSH_DOWN_PINS" detected in Hercules/ICV runset is not supported in StarXtract.

**Description**

StarXtract finds this unsupported option in Hercules/ICV runset.

**What Next**

Confirm this is expected data. If it is not, confirm the procedures and settings to generate this data is correct, or re-generate this data.

---

## SX-2777

(Warning) Following option "PUSH_DOWN_DEVICES" detected in Hercules/ICV runset is not supported in StarXtract.

**Description**

StarXtract finds this unsupported option in Hercules/ICV runset.

**What Next**

Confirm this is expected data. If it is not, confirm the procedures and settings to generate this data is correct, or re-generate this data.

---

## SX-2778

(Warning) Hercules RUNSET does not contain LAYOUT_POWER/ SCHEMATIC_POWER,...,etc layout/schematic power/ground nets setting and POWER_NETS option is empty

**Description**

StarXtract does not find any valid setting about layout/schematic power/ground nets.

Feedback

**What Next**

Confirm this is expected data. If it is not, confirm the procedures and settings to generate this data is correct, or re-generate this data.

---

## SX-2780

(Error) Failed to locate required cells.

**Description**

StarXtract is locating this cell, but this operation is failed.

**What Next**

Check the reference library tree for completeness. Note that dbSetLibRefCtrlFileMode flag must be set in Milkyway control window when write permission for the library is available to enable use of control file.

---

## SX-2781

(Information) DCM lookup for " %s " from library " %s " failed.

**Description**

StarXtract is doing the DCM lookup, but this operation is failed.

**What Next**

If this is expected data, it is information only. No action is needed.

---

## SX-2782

(Severe) %s

**Description**

StarXtract could not locate this cell in this library.

**What Next**

Confirm the Milkyway data is not broken or re-generate the Milkyway data.

---

## SX-2783

(Fatal) %s

Feedback

**Description**

StarXtract could not locate this cell in this library.

**What Next**

Confirm the Milkyway data is not broken or re-generate the Milkyway data.

---

## SX-2784

(Warning) %s

**Description**

StarXtract could not locate this cell in this library.

**What Next**

Confirm this is expected data, or contact the Synopsys Support Center.

---

## SX-2785

(Information) %s

**Description**

StarXtract could not locate this cell in this library.

**What Next**

If this is expected data, it is information only. No action is needed.

---

## SX-2786

(Warning) Set DPT:YES but none of LAYER(s) in %s have MASK statements in their definitions.

**Description**

In DPT:YES flow, LAYER(s) in technology LEF and TCAD_GRD_FILE should have mask definition or the run may suffer from missing MASK(s).

**What Next**

Review technology LEF and TCAD_GRD_FILE and define missing MASK for LAYER(s) in question.

Feedback

## SX-2787

(Warning) The NDM design %s has critical data for physical sign-off tool behavior.

**Description**

The NDM design being opened has data that is critical for Physical Sign-off that StarRC does not understand. This happens when

- The schema of the opened NDM database is NEWER than the compiled-in version of NDM, AND
- There is new schema content, which will ignored by NDM as part of Forward-Compatibility support, is deemed critical for correct signoff-tool behavior, AND there is actually data in the ndmDesign representing such objects.

**What Next**

Contact the Synopsys Support Center.

## SX-2788

(Fatal) INTERNAL Error: %s

**Description**

An internal database access has failed.

**What Next**

Contact the Synopsys Support Center.

## SX-2789

(Warning) INTERNAL Warning: %s

**Description**

An internal database consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## SX-2790

(Information) INFORMATION: SMC automatic chosen default corner: %s

Feedback

**Description**

SMC worst corner selection

**What Next**

Please refer to StarRC manual.

---

## SX-2902

(Fatal) 3D_IC_TSV_COUPLING_EXTRACTION:CEFF should be used with OPERATING_FREQUENCY.

**Description**

3D_IC_TSV_COUPLING_EXTRACTION:CEFF should be used with OPERATING_FREQUENCY.

**What Next**

Set OPERATING_FREQUENCY in StarRC command file.

---

## SX-2903

(Fatal) 3D_IC_TSV_COUPLING_EXTRACTION:RCSUB should not be used with NETLIST_FORMAT:SPEF

**Description**

3D_IC_TSV_COUPLING_EXTRACTION:RCSUB could generate RC network between 2 different nets. That is not allowed in SPEF.

**What Next**

Reset NETLIST_FORMAT in StarRC command file.

---

## SX-2905

(Warning) CEFF_VS_FREQUENCY_AND_SPACING table is not defined in ITF file

**Description**

StarRC failed to find CEFF_VS_FREQUENCY_AND_SPACING table for 3D_IC_TSV_COUPLING_EXTRACTION:CEFF.

**What Next**

Check if the ITF file is correct or turn off 3D_IC_TSV_COUPLING_EXTRACTION:CEFF.

Feedback

## SX-2906

(Fatal) RLC model is not supported for NETLIST_FORMAT: SPEF

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

## SX-2907

(Warning) The command METAL_FILL_GDS_BLOCK would be removed in next release, use METAL_FILL_BLOCK_NAME instead

**Description**

The command METAL_FILL_GDS_BLOCK would be removed in next release, use METAL_FILL_BLOCK_NAME instead.

**What Next**

Use METAL_FILL_BLOCK_NAME instead of METAL_FILL_GDS_BLOCK.

## SX-2908

(Warning) The command METAL_FILL_GDS_MAG & METAL_FILL_OASIS_MAG would be removed in next release, please use METAL_FILL_MAG instead

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

## SX-2909

(Warning) The command METAL_FILL_GDS_OFFSET & METAL_FILL_OASIS_OFFSET would be removed in next release, please use METAL_FILL_OFFSET instead

**Description**

Not available.

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## SX-2910

(Warning) The command METAL_FILL_GDS_FILE_NET_NAME & METAL_FILL_OASIS_FILE_NET_NAME would be removed in next release, please use METAL_FILL_FILE_NET_NAME instead

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-2911

(Fatal) The starQTF flow does not support FS_VISUALIZATION :YES for the TCAD_GRD_FILE/FS_QTF_FILE

**Description**

The starQTF flow does not support FS_VISUALIZATION :YES for the TCAD_GRD_FILE/FS_QTF_FILE.

**What Next**

See user guide. For further details, contact the Synopsys Support Center to query what kind of TCAD_GRD_FILE/FS_QTF_FILE can be supported

---

## SX-2912

(Sereve) %s Please check %s: %s.

**Description**

The syntax of OA_Device_Mapping_file and Oa_Skip_Mapping_File need to check.

**What Next**

See user guide. For further details, contact the Synopsys Support Center to query the syntax of OA_Device_Mapping_file and Oa_Skip_Mapping_File.

Feedback

## SX-2913

(Warning) %s Please check %s: %s.

**Description**

The content of Oa_Layyer_Mapping_file need to check to make sure it is correct on token length.

**What Next**

See user guide. For further details, contact the Synopsys Support Center to query the syntax of Oa_Layyer_Mapping_file.

## SX-2973

(Information) Failed to open requested view '%s'

**Description**

StarXtract is opening the requested view that depends on option settings, but it does not exist.

**What Next**

If this is expected data, it is information only. No action is needed.

## SX-2981

(Fatal) file '%s' is corrupted!

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## SX-2999

(Warning) COUPLE_NONCRITICAL_NETS is only supported in AGF and Milkyway flow.

**Description**

COUPLE_NONCRITICAL_NETS is only supported in AGF and Milkyway flow.

Feedback

**What Next**

Remove COUPLE_NONCRITICAL_NETS from StarRC command file.

---

## SX-3001

(Warning) Translating automated fill macro '%s' at (%d, %d, %s).

**Description**

Translating a fill macro specified for design automation at the location in the current cell.

**What Next**

Information only; no action required.

---

## SX-3002

(Warning) Automated fill macro '%s' is instantiated.

**Description**

Automated fill macro are instantiated in the METAL_FILL_GDS_FILE/ METAL_FILL_OASIS_FILE. Cells. Their instantiations will be re-constructed at Translate stage based on the design automation.

**What Next**

Information only; no action required.

---

## SX-3003

(Information) [%s <-> %s] => [%d <-> %d]

**Description**

A design/fill macro association is specified and its cell id association is constructed.

**What Next**

Information only; no action required.

---

## SX-3004

(Warning) %s macro '%s' doesn't exist in %s database. Ignore its automation.

Feedback

**Description**

The design and fill macros specified for design automation in the METAL_FILL_BLOCK_MAPPING_FILE doesn't exist in the design or metal fill database. Their automation will be ignored.

**What Next**

Information only; no action required.

---

## SX-3005

(Fatal) Syntax error at line %d of METAL_FILL_BLOCK_MAPPING_FILE.

**Description**

The METAL_FILL_BLOCK_MAPPING should have line syntax as follows,

<design_macro> <fill_macro>

to specify a mapping between design and fill for design automation.

**What Next**

Correct the offending line.

---

## SX-3006

(Fatal) Automated hierarchical fill macro '%s' is instantiated in another automated hierarchical fill macro '%s'.

**Description**

If a fill macro is targeted for hierarchical design flow, it can't be referenced in another hierarchical fill macro.

**What Next**

Remove the offending instantiation in the hierarchical fill macro.

---

## SX-3007

(Fatal) The metal fill design automation METAL_FILL_BLOCK_MAPPING_FILE is not supported in this flow.

**Description**

The metal fill design automation METAL_FILL_BLOCK_MAPPING_FILE is only supported in LEF/DEF flow with METAL_FILL_GDS_FILE/METAL_FILL_OASIS_FILE.

Feedback

### What Next

Correct the StarRC command file.

---

## SX-3008

(Fatal) IGNORE_CAPACITANCE must be set to NONE, DIFF or ALL RETAIN_GATE_DIFFUSION_COUPLING when COUPLE_TO_GROUND: YES RETAIN_GATE_CONTACT_AND_DIFFUSION_COUPLING.

### Description

If the StarRC command file option COUPLE_TO_GROUND is set to YES RETAIN_GATE_CONTACT_AND_DIFFUSION_COUPLING, StarRC expects to extract the gate to contact and the gate to diffusion coupling capacitances. Therefore, the IGNORE_CAPACITANCE must be set to NONE, DIFF or ALL RETAIN_GATE_DIFFUSION_COUPLING to do so. If the gate to contact and the gate to diffusion coupling capacitances are ignored, the StarRC command file option COUPLE_TO_GROUND must be set to YES or NO.

### What Next

Modify one of the two StarRC command file option, IGNORE_CAPACITANCE or COUPLE_TO_GROUND.

---

## SX-3009

(Warning) RETAIN_GATE_CONTACT_COUPLING and RETAIN_GATE_CONTACT_AND_DIFFUSION_COUPLING are both set in FS flow. RETAIN_GATE_CONTACT_AND_DIFFUSION_COUPLING is enabled instead of RETAIN_GATE_CONTACT_COUPLING.

### Description

When RETAIN_GATE_CONTACT_COUPLING: YES is specified and COUPLE_TO_GROUND: YES RETAIN_GATE_CONTACT_AND_DIFFUSION_COUPLING is also specified, StarRC will retain the coupling from gate to contact and diffusion, not the coupling between gate and contact.

### What Next

If you want to retain coupling from gate to contact and diffusion, you don't have to take any action; if you just want to retain the coupling between gate and contact but not diffusion, you need to set COUPLE_TO_GROUND: YES RETAIN_GATE_CONTACT_COUPLING.

Feedback

## SX-3010

(Fatal) map_qtf_layers section is missing in StarRC mapping file when running StarQTF flow.

### Description

StarQTF flow (when FS_QTF_FILE is used) needs 2 additional sections in StarRC mapping file: map_qtf_layers: this section defines mappings between input database layers to QTF layers defined in QTF file. qtf_layers: this section defines QTF layer names for device layers (layers below metal1).

If you use FS_QTF_FILE command but don't define map_qtf_layers section in StarRC mapping file, StarRC will give this error.

### What Next

You should use a StarRC mapping file containing map_qtf_layers section.

## SX-3011

(Fatal) qtf_layers section is missing in StarRC mapping file when running StarQTF flow.

### Description

StarQTF flow (when FS_QTF_FILE is used) needs 2 additional sections in StarRC mapping file: map_qtf_layers: this section defines mappings between input database layers to QTF layers defined in QTF file. qtf_layers: this section defines QTF layer names for device layers (layers below metal1).

If you use FS_QTF_FILE command but don't define qtf_layers section in StarRC mapping file, StarRC will give this error.

### What Next

You should use a StarRC mapping file containing qtf_layers section.

## SX-3012

(Fatal) TCAD_GRD_FILE in CORNERS_FILE must use absolute path when CORNERS_FILE is specified out of current working directory.

### Description

You receive this message because TCAD_GRD_FILE defined in CORNERS_FILE does not use an absolute path to specify the Tcad file when CORNERS_FILE is specified out of current working directory.

Feedback

**What Next**

Move CORNERS_FILE into current working directory or use an absolute path to specify TCAD_GRD_FILE.

---

## SX-3013

(Warning) Library %s has no reference libraries.

**Description**

NDM_DATABASE doesn't specify any reference libraries. The cells in those reference libraries cannot be found.

**What Next**

Check your reference library setting in NDM database to see whether you've used command 'set_ref_libs' in ICC2 when you created the NDM library.

---

## SX-3014

(Warning) Implicit connected database layer "%s" does not have a valid ITF layer. Will remove this layer.

**Description**

StarRC gives this warning once it finds the shown layer is defined in database and has implicit connections (connections based on layer operations), but it is not defined in StarRC mapping file.

**What Next**

If you don't need this layer, nothing needs to be done. Otherwise, you should define mapping for this layer in StarRC mapping file.

---

## SX-3015

(Warning) Cannot find design

**Description**

NDM_DATABASE doesn't contain this design.

**What Next**

Check your NDM database setting to make sure the designs exist in main NDM library or reference libraries.

Feedback

## SX-3016

(Warning) Maskable user-defined layer '%s' is not defined in MAPPING_FILE and removed.

**Description**

A maskable user-defined layer must be defined in the MAPPING_FILE with a valid ITF layer.

**What Next**

Define the user-defined layer in the MAPPING_FILE and run again.

## SX-3017

(Fatal) database layer '%s' can not be a remove layer, because the database layer is mapped to a QTF layer, and this QTF layer is in the qtf_layers section. Remove this database layer from remove_layers section.

**Description**

If a database layer is mapped to a QTF layer in map_qtf_layers section and the QTF layer is included in qtf_layers section, StarRC needs this database layer. You can't put that database layer in remove_layers section.

**What Next**

You should remove the shown layer in remove_layers section.

## SX-3018

(Fatal) the qtf_layers in starRC mapping file are not consecutive, please add qtf layer: '%s' into the qtf_layers section.

**Description**

StarRC+QTF flow need to use all the layers stack from one ITF layer to the bottom. So StarRC need all the ITF layer which mapped to the DB layers defined in qtf_layers section are consecutive.

**What Next**

You should add the DB layer name in this message to qtf_layers section.

## SX-3019

(Warning) Attempt to sleep for '%lf' sec. failed: '%s'

Feedback

**Description**

An internal operating system call failed.

**What Next**

Please look for other errors/warning in the sum file. Please try rerunning, if the original issue persists, please contact the Synopsys Support Center.

---

## SX-3020

(Warning) Unable to test lock on file '%s', error status '%s'

**Description**

StarRC is unable to determine whether or not a file is locked because of issues in the runtime environment.

**What Next**

No action is necessary if StarRC successfully ran to completion. This warning may indicate the source of a problem if StarRC terminated abnormally. Contact your system administrator if necessary.

---

## SX-3021

(Fatal) View only layer '%s' can not be in the qtf_layers section

**Description**

If a database layer is mapped to a QTF layer in map_qtf_layers section and the QTF layer is included in qtf_layers section, you can't put that database layer in viewonly_layers section.

**What Next**

You should remove this layer from viewonly_layers section.

---

## SX-3022

(Warning) NDM design '%s' has references bound to unsupported views.

**Description**

StarRC supports frame and design views. If a design is bound to an unsupported view, StarRC will try to force open frame/design view based on the SKIP_CELLS/ NDM_DESIGN_VIEW options in the Star command file.

Feedback

However, in hierarchical design, macros may be bound to unsuported abstract views. These macros will be extracted with design/frame views instead of abstract views.

However, if these unresolved design views can't be inferred based on the options , their instantiations are ignored and may introduce accuracy issues. In such situation, SX-3015 will be issued and NDM_CELL_REPORT_FILE will report these designs.

**What Next**

Check NDM_CELL_REPORT_FILE and SX-3015 for unresolved designs. If unresolved designs found, please fix these errors to avoid possible accuracy problems. If not, this warning can be safely ignored.

---

## SX-3023

(Warning) NDM design '%s' without bounding box cached in its library will degrade Setup performance

**Description**

A NDM library with cached bouning box can speed up StarRC's Setup time. StarRC doesn't need to open top block for bounding box.

**What Next**

Update the library with bounding box by latest IC Compiler II.

---

## SX-3024

(Warning) NDM design '%s' with SHORT_PINS:NO will degrade Setup performance.

**Description**

SHORT_PINS:NO will force StarRC to read into Blk-aspect of NDM database which degrades its Setup performance.

**What Next**

If SHORT_PINS:NO is not required in the run, setting it YES will improve Setup performance.

---

## SX-3025

(Fatal) NDM streaming cache '%s' is not generated.

**Description**

Streaming cache must be generated at Layers_DP stage.

Feedback

### What Next

Contact Synopsys support.

---

## SX-3026

(Warning) A single huge METAL_FILL_GDS_FILE/METAL_FILL_OASIS_FILE is degrading metal fill import performance.

### Description

A single large METAL_FILL_GDS_FILE or METAL_FILL_OASIS_FILE with several performance-critical cells is downgrading StarRC's performance for metal fill import. To get better import performance, separate these performance-critical cells into different files and list them in the METAL_FILL_GDS_FILE/METAL_FILL_OASIS_FILE.

### What Next

Check the associated STAR/summary/.<stage>.messages and move reported big cells into different files.

---

## SX-3027

(Warning) NDM fills will be ignored because NDM_MULTIPLE_FILL_SOURCES is set to NO.

### Description

When NDM_MULTIPLE_FILL_SOURCES is set to NO, StarRC will *NOT* translate NDM fills in the presence of METAL_FILL_GDS_FILE/METAL_FILL_OASIS_FILE.

- For StarRC N-2017.12, this option is set to *NO* by default. If the metal fills from NDM and GDS/OASIS files should be included in the translation, this option must be set to YES.
- For StarRC O-2018.06 and above, this option is *NOT* available. Metal fills from NDM and GDS/OASIS files are automatically translated if they are specified in the StarRC command file.

### What Next

Please check your StarRC command file and make sure the NDM fills should be dropped in your run.

Feedback

## SX-3028

(Warning) UPDATE_XYTRANSFORM_WITH_FOREIGN doesn't support FOREIGN with <orient> in LEF MACRO '%s'

**Description**

UPDATE_XYTRANSFORM_WITH_FOREIGN doesn't support LEF FOREIGN with orient. The XY offset are still applied but orient will be ignored in the xform messages.

**What Next**

Information only

## SX-3029

(Warning) Option %s is ignored for it is supported only for XREF: COMPLETE.

**Description**

Option displayed in the warning message can only work under XREF:COMPLETE.

**What Next**

Disable the option in the message or change XREF:COMPLETE to other settings.

## SX-3031

(Fatal) INTERNAL Error: Error found (invalid orientation) while processing Cfidata.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## SX-3032

(Warning) More than one boundary element defined in the cell %s

**Description**

During parsing input databases to generate PLACEMENT_XYV_FILE, StarRC may take GDS cell boundary element defined at each cell in the input database as cell boundary.

If there are more than one boundary element defined for a cell, StarRC will only take the first one and ignore the rest.

Feedback

**What Next**

Contact the Synopsys Support Center.

---

## SX-3033

(Information) Cell definition for '%s' is read-in from '%s'.

**Description**

StarRC takes the cell definition from file.

**What Next**

Information only; no action needed.

---

## SX-3034

(Warning) Missing reference library '%s'

**Description**

A reference library is still missing after NDM_DATABASE and library paths in NDM_SEARCH_PATH being searched. Missing libraries can introduce polygon drops and lead to problems in extraction.

**What Next**

If it is intentional, the warning can be ignored. If not, please update the reference library paths in NDM_DATABASE or provide NDM_SEARCH_PATH.

---

## SX-3035

(Warning) Unable to remove file '%s': %s

**Description**

StarRC was not able to remove a file during execution.

**What Next**

Check to make sure you have disk permissions and that disk performance is adequate.

---

## SX-3036

(Fatal) Invalid option '%s' of TRANSLATE_RETAIN_BULK_LAYERS. Valid Values are: [NO] | YES | CONLY.

Feedback

**Description**

Valid values of command TRANSLATE_RETAIN_BULK_LAYERS are: [NO] | YES | CONLY. The value you used in invalid

**What Next**

You should use NO or YES or CONLY as value of command TRANSLATE_RETAIN_BULK_LAYERS.

---

## SX-3037

(Fatal) No %s defined for BLOCK '%s' in %s

**Description**

StarRC needs connectivity and physical information for BLOCK in order to extract successfully.

**What Next**

Review the BLOCK in StarRC command file or input files.

---

## SX-3038

(Warning) Encountered port "%s" in PORT TABLE CELLS WRITE which does not exist in the LAYOUT NETLIST WRITE in query_cmd. This entry will be ignored.

**Description**

A consistency check has failed.

**What Next**

Check the input files PORT TABLE CELLS WRITE and LAYOUT NETLIST WRITE in query_cmd. Contact the Synopsys Support Center if StarRC terminates abnormally or other issues are found.

---

## SX-3039

(Warning) The value of the %s option cannot be a negative number. A value of zero will be used internally.

**Description**

The value of the given option in the StarRC command file is a negative number. The option accepts positive numbers only.

Feedback

**What Next**

Change the value of the specified option to a positive number in the StarRC command file.

---

## SX-3043

(Fatal) SIMULTANEOUS_MULTI_CORNER only support TEMPERATURE_COEFFICIENT output in NETLIST_FORMAT: SPF | NETNAME | STAR | OA.

**Description**

In SIMULTANEOUS_MULTI_CORNER flow, TEMPERATURE_COEFFICIENT can be specified in intended corners for TC1/TC2 temperature coefficient output. The feature can only support limited NETLIST_FORMAT of SPF, NETNAME, STAR, OA.

**What Next**

Change NETLIST_FORMAT to SPF|NETNAME|STAR|OA if TEMPERATURE_COEFFICIENT is required.

---

## SX-3044

(Fatal) In SIMULTANEOUS_MULTI_CORNER merged-output format, NOMINAL corner type is needed for TEMPERATURE_COEFFICIENT output

**Description**

The corner type must be NOMINAL if the corner is selected for TEMPERATURE_COEFFICIENT output.

**What Next**

Modify corner type or select a NOMINAL corner.

---

## SX-3046

(Fatal) Duplicated TCAD_GRD files are used for TEMPERATURE_COEFFICIENT output. Please remove duplicated corners.

**Description**

The selected corners have same TCAD_GRD file, which is not allowed for TEMPERATURE_COEFFICIENT output.

**What Next**

Remove the duplicated corners.

Feedback

## SX-3047

(Fatal) Only REDUCTION:NO|LAYER is supported to TEMPERATURE_COEFFICIENT output in SIMULTANEOUS_MULTI_CORNER.

**Description**

In SIMULTANEOUS_MULTI_CORNER flow, TEMPERATURE_COEFFICIENT can be specified in intended corners for TC1/TC2 temperature coefficient output. The feature needs to set REDUCTION to NO or LAYER to ensure the correctness of TC1/TC2 values. The REDUCTION setting of the run is neither NO or LAYER, it caused the error.

**What Next**

Modify REDUCTION setting to either NO or LAYER for using TEMPERATURE_COEFFICIENT output in SIMULTANEOUS_MULTI_CORNER.

## SX-3048

(Warning) No device property match command PARAMETERIZED_DEVICE_PROPERTY.

**Description**

No property match settings in command PARAMETERIZED_DEVICE_PROPERTY. Please check settings and database for correcting.

**What Next**

Check setttings of command PARAMETERIZED_DEVICE_PROPERTY in StarRCXT cmd file.

## SX-3050

(Warning) File '%s' does not exist, sleeping for %d sec. before checking again.

**Description**

An intermediate file produced by StarRC on one host may not be accessible on another host because of issues with the file system. An attempt will be made to access the file again later to allow for latencies related to disk access.

**What Next**

No action is necessary if StarRC completes successfully. This message may indicate the source of a problem if StarRC does not run to completion. In this case, try re-running StarRC on a lightly loaded disk or network.

Feedback

## SX-3051

(Warning) Could not find file '%s' after %d attempts.

**Description**

StarRC is unable to access an intermediate file on one host that was written on another host because of issues with the file system.

**What Next**

No action is necessary if StarRC completes successfully. This message may indicate the source of a problem if StarRC does not run to completion. In this case, try re-running StarRC on a lightly loaded disk or network.

## SX-3052

(Warning) %s '%s' does not exist.

**Description**

An expected input file does not exist. The file could be an input file specified by the user in the StarRC command file or an internally generated intermediate file.

## SX-3053

(Fatal) Wrong mask ID is specified in the layer mapping file. Please specify a mask ID in interval (0, NUMBER_OF_MASKS in ITF file]. Line: %s

**Description**

Once you use mask=value parameter for a conductor in StarRC mapping file, the value should be a positive integer not larger than NUMBER_OF_MASKS value defined in ETCH_VS_WIDTH_AND_SPACING table in ITF file.

**What Next**

You should use an integer from 1 to the NUMBER_OF_MASKS value defined in ETCH_VS_WIDTH_AND_SPACING table in ITF file as value of mask parameter in this line.

## SX-3055

(Fatal) NETLIST_SMC_FORMULA does not work with TEMPERATURE_COEFFICIENT output in SIMULTANEOUS_MULTI_CORNER.

Feedback

**Description**

In SIMULTANEOUS_MULTI_CORNER flow, some corners can define GLOBAL_TEMP to enable TEMPERATURE_COEFFICIENT output. This feature cannot work with NETLIST_SMC_FORMULA because the formula cannot handle TC1/TC2 in a formula way.

**What Next**

Either disable NETLIST_SMC_FORMULA or disable TEMPERATURE_COEFFICIENT in selected corners.

---

## SX-3057

(Warning) The pre-defined NONDEFAULTRULE '%s' in LEF is covered by definition of DEF.

**Description**

NONDEFAULTRULE from LEF is redefined in DEF. StarRC will use the version in DEF to translate.

**What Next**

Contact the Synopsys Support Center.

---

## SX-3058

(Fatal) NETLIST_MINRES_HANDLING:MERGE with non-zero NETLIST_MINRES_THRESHOLD is not supported for TEMPERATURE_COEFFICIENT output in SIMULTANEOUS_MULTI_CORNER.

**Description**

In SIMULTANEOUS_MULTI_CORNER flow, TEMPERATURE_COEFFICIENT can be specified in intended corners for TC1/TC2 temperature coefficient output. The feature does not support NETLIST_MINRES_HANDLING:MERGE and NETLIST_MINRES_THRESHOLD:val when val is set as positive value. The merging behavior may perturb the correct value of TC1/TC2 for each corner.

**What Next**

Turn off NETLIST_MINRES_HANDLING:MERGE or NETLIST_MINRES_THRESHOLD to enable TEMPERATURE_COEFFICIENT output in SIMULTANEOUS_MULTI_CORNER.

Feedback

## SX-3059

(Fatal) MAPPING_FILE in CORNERS_FILE must use absolute path when CORNERS_FILE is specified out of current working directory.

**Description**

You receive this message because MAPPING_FILE defined in CORNERS_FILE does not use an absolute path to specify the mapping file when CORNERS_FILE is specified out of current working directory.

**What Next**

Move CORNERS_FILE into current working directory or use an absolute path to specify MAPPING_FILE.

## SX-3060

(Fatal) Cannot open MAPPING_FILE '%s' found on line '%d' in CORNERS_FILE '%s'

**Description**

You receive this message because StarRC failed to open the mapping file specified by MAPPING_FILE.

**What Next**

Make sure that the file specified by MAPPING_FILE isn't corrupted or missing and StarRC has permission to access it.

## SX-3061

(Fatal) Mapping file %s and mapping file %s are inconsistent.

**Description**

The error message informs you that the two mapping files reported in the message are inconsistent. The mapping files for different corners should be the same except the value of RPSQ and RPV.

**What Next**

Check the two mapping files reported in the message and make sure they are consistent.

## SX-3062

(Warning) Number of shorts exceeds %s: %d

Feedback

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-3063

(Warning) REPORT_METAL_FILL_STATISTICS incurs performance degradation

**Description**

REPORT_METAL_FILL_STATISTICS collects and reports fill information in the summary file and it will incur performance degradation. StarRC suggests to use this option only for debugging purpose.

**What Next**

Information only; no action is needed.

---

## SX-3064

(Warning) NDM_DB_READ_MEMORY_REDUCTION will incur performance degradation (NUM_CORES:%d reduced to %d for Cells).

**Description**

NDM_DB_READ_MEMORY_REDUCTION uses less cores for memory intensive NDM input DB read stages and sacrifice performance as a result. The memory reduction will be effective only when NUM_CORES >= 4.

The possible value for this option,

- NO: The default value.
- YES: 2X less cores are used at Cells_DP.
- HIGH: 4X less cores are used at Cells_DP if NUM_CORES >= 16. If NUM_CORES < 16, this mode is downgraded to YES which uses 2X less cores.
- AGGRESSIVE: Force only 1 core to process top cell at Cells_DP and leave other data types, s.t., macro cells and standard cells, unchanged.

**What Next**

Information only; no action is needed.

Feedback

## SX-3065

(Warning) FAST_RONLY_POWER_EXTRACTION is only supported with POWER_EXTRACT:RONLY and EXTRACTION:R. Normal power extraction is performed.

**Description**

You receive this message because you are using FAST_RONLY_POWER_EXTRACTION:YES, but POWER_EXTRACT:RONLY and EXTRACTION:R are not used. StarRC will perform normal power extraction flow. FAST_RONLY_POWER_EXTRACTION:YES is ignored.

**What Next**

If you want to use FAST_RONLY_POWER_EXTRACTION:YES, please set POWER_EXTRACT:RONLY and EXTRACTION:R. Otherwise you can ignore this message. It is only affect runtime.

## SX-3066

(Fatal) ECO_FILE option must be used with ECO_MODE: YES.

**Description**

Incremental extraction is required if an Eco file is input as a StarRC command option.

**What Next**

Please refer to section on ECO extraction in the StarRC user guide.

## SX-3067

(Fatal) Wrong mask ID is specified in the layer mapping file. Please specify a mask ID in interval (0, NUMBER_OF_MASKS in ITF file]. MASK=%d is specified on %s on %s while NUMBER_OF_MASKS is %d in ITF file.

**Description**

Once you use mask=value parameter for a conductor in StarRC mapping file, the value should be a positive integer not larger than NUMBER_OF_MASKS value defined in GATE_TO_DIFFUSION_ADJUSTMENT_CAP table in ITF file.

**What Next**

You should use an integer from 1 to the NUMBER_OF_MASKS value defined in GATE_TO_DIFFUSION_ADJUSTMENT_CAP in ITF file as value of mask parameter in this line.

Feedback

## SX-3068

(Fatal) No Cfi silicon marker layer is specified for %s with ITF=%s of MASK=%d in the layer mapping file.

**Description**

All non-zero MASK values of a Cfi table must be covered by silicon marker layers in the layer mapping file.

**What Next**

Please check correctness of the layer mapping file.

## SX-3069

(Fatal) No Cfi silicon marker layer for %s is found in the design and the layer is not covered by any silicon markers.

**Description**

No associated silicon marker layer for the specific gate layer, which has non-zero NUMBER_OF_MASKS Cfi tables, is found in the input database.

**What Next**

Please check correctness of the input database.

## SX-3070

(Warning) LVS Black Box cell '%s' is added to %s list automatically.

**Description**

The LVS black box cell is added to skip cell list.

**What Next**

Information only; no action is needed.

## SX-3071

(Warning) REPORT_SMIN_VIOLATION:YES may not report accurate SMIN violations. Refer to pre-flow DesignRuleCheck for accurate reporting.

Feedback

**Description**

REPORT_SMIN_VIOLATION:YES is used to report SMIN violation during extraction. However, the reporting may not be extremely accurate in StarRC parasitic extraction tool during RC extraction. It is suggested to use intended DesignRuleCheck (DRC) tool to accurately report the violation.

**What Next**

Use the SMIN violation report file only as estimated reference and refer to DRC tool for accurate violation.

---

## SX-3072

(Severe) Cannot map the marker layer "%s" to a valid GRD layer.

**Description**

Marker layer not mapped to a GRD layer and do not have layer connection with other GRD layer. So StarRC can not map the marker layer "%s" to a validate GRD layer.

**What Next**

You should revise your mapping file to add connection on marker layer or map to marker layer to a grd layer.

---

## SX-3073

(Warning) LVS_EXTRACTION_REPORT_FILE contains invliad ALIAS statement, ALIAS statement needs at least 2 layers. Invliad ALIAS statement will be ignored.

**Description**

LVS_EXTRACTION_REPORT_FILE contains invliad ALIAS statement, ALIAS statement needs at least 2 layers. Invliad ALIAS statement will be ignored.

**What Next**

You should check the ALIAS statement in LVS_EXTRACTION_REPORT_FILE and make ALIAS statement contains at least 2 layers.

---

## SX-3074

(Severe) Layer '%s' is in multiple ALIAS statements in LVS_EXTRACTION_REPORT_FILE.

Feedback

**Description**

Layer '%s' is in multiple ALIAS statements in LVS_EXTRACTION_REPORT_FILE.

**What Next**

You should check the layer name mentioned in this message in ALIAS statements in LVS_EXTRACTION_REPORT_FILE and revise it only in one ALIAS statement.

---

## SX-3076

(Information) Creating EcoFile for partition '%d'.

**Description**

During IncrementalDP stage, eco_file is generated by partition.

**What Next**

This message is for information only, no further action is needed.

---

## SX-3078

(Warning) EXTRACT_MESH_RES is only effective in transistor-level extraction. This option is ignored.

**Description**

EXTRACT_MESH_RES is a feature to enable mesh-Res extraction. This feature is limited at transistor-level only. The run is at gate-level, so the option is ignored.

**What Next**

Remove EXTRACT_MESH_RES from StarRC command file.

---

## SX-3079

(Fatal) VIA_COVERAGE_OPTION_FILE is mandatory for all corners if any corner has it specified in CORNERS_FILE.

**Description**

This error message is issued when some corners have VIA_COVERAGE_OPTION_FILE but the other corners don't have this field in CORNERS_FILE.
VIA_COVERAGE_OPTION_FILE should be specified for ALL or NONE of the corners for correct extraction handling.

Feedback

**What Next**

Correct the content CORNERS_FILE and restart. VIA_COVERAGE_OPTION_FILE should be specified for ALL or NONE of the corners in CORNERS_FILE. If the same VIA_COVERAGE_OPTION_FILE is applied to all corners, you can also choose to only specify it in the star command file..

---

## SX-3080

(Fatal) Different VIA_COVERAGE_OPTION_FILE '%s' and '%s' specified for the same nxtgrd file '%s' on its temperature corners;

**Description**

Same VIA_COVERAGE_OPTION_FILE value must be specified for the different temperature corners that use the same nxtgrd file.

**What Next**

Please check and make sure that the same VIA_COVERAGE_OPTION_FILE value is specified for the different temperature corners that use the same nxtgrd file.

---

## SX-3081

(Fatal) VIA_COVERAGE_OPTION_FILE in CORNERS_FILE must use absolute path when CORNERS_FILE is specified out of current working directory.

**Description**

You receive this message because VIA_COVERAGE_OPTION_FILE defined in CORNERS_FILE does not use an absolute path to specify the via coverage option file when CORNERS_FILE is specified out of current working directory.

**What Next**

Move CORNERS_FILE into current working directory or use an absolute path to specify VIA_COVERAGE_OPTION_FILE.

---

## SX-3082

(Fatal) Cannot open VIA_COVERAGE_OPTION_FILE '%s' found on line '%d' in CORNERS_FILE '%s'

**Description**

You receive this message because StarRC failed to open the via coverage option file specified by VIA_COVERAGE_OPTION_FILE.

Feedback

### What Next

Make sure that the file specified by VIA_COVERAGE_OPTION_FILE isn't corrupted or missing and StarRC has permission to access it.

---

## SX-3083

(Fatal) SMC via_coverage_option_file corrupted: cannot load corner via_coverage codes in sav section.

### Description

Failed to load sav dependent via_coverage code in SMC flow.

### What Next

Check and correct the VIA_COVERAGE_OPTION_FILE for each corner, and restart. If error persists, contact Synopsys Support Center.

---

## SX-3084

(Fatal) SMC via_coverage_option_file corrupted on layer '%d': cannot find corresponding codes in sav section.

### Description

Corner-dependent VIA_COVERAGE_OPTION_FILE in SMC flow should have consistent entries cross corners. VIA_COVERAGE_OPTION_FILE has discrepancy on the highlighted item from corner to corner, StarRC cannot process the data.

### What Next

Correct the input files for VIA_COVERAGE_OPTION_FILE and restart.

---

## SX-3085

(Fatal) SMC via_coverage_option_file corrupted on layer '%d' and sav key '%d': RPV discrepancy detected.

### Description

Corner-dependent VIA_COVERAGE_OPTION_FILE in SMC flow should have consistent entries cross corners. RPV has discrepancy on the highlighted item from corner to corner, StarRC cannot process the data.

### What Next

Correct the input files for VIA_COVERAGE_OPTION_FILE and restart.

Feedback

## SX-3086

(Fatal) SMC via_coverage_option_file corrupted on layer '%d': RPV table size is different cross corners.

### Description

Corner-dependent VIA_COVERAGE_OPTION_FILE in SMC flow should have consistent entries cross corners. The RPV table on the highlighted layer has different size from corner to corner.

### What Next

Correct the input files for VIA_COVERAGE_OPTION_FILE and restart.

## SX-3087

(Fatal) SMC via_coverage_option_file corrupted on layer '%d': RPV value '%f' and '%f' discrepancy cross corners.

### Description

Corner-dependent VIA_COVERAGE_OPTION_FILE in SMC flow should have consistent entries cross corners. Highlighted discrepancy is detected, StarRC cannot process the data.

### What Next

Correct the input files for VIA_COVERAGE_OPTION_FILE and restart.

## SX-3088

(Fatal) SMC via_coverage_option_file corrupted: some RPV values are missing from secondary corners.

### Description

Corner-dependent VIA_COVERAGE_OPTION_FILE in SMC flow should have consistent entries cross corners. Missing RPV values are detected on secondary corners while they present in the primary corner, StarRC cannot process the data.

### What Next

Correct the input files for VIA_COVERAGE_OPTION_FILE and restart.

## SX-3089

(Fatal) SMC via_coverage_option_file corrupted: RPV table size is different cross corners.

Feedback

**Description**

When using corner-dependent VIA_COVERAGE_OPTION_FILE in SMC flow, each RPV table should have consistent number of entries across corners.

**What Next**

Correct the input files for VIA_COVERAGE_OPTION_FILE and restart.

---

## SX-3090

(Fatal) SMC via_coverage_option_file corrupted.

**Description**

Failed to load corner-dependent VIA_COVERAGE_OPTION_FILE in SMC flow.

**What Next**

Make sure that each corner in CORNERS_FILE has correct VIA_COVERAGE_OPTION_FILE and restart. If error persists, contact Synopsys Support Center.

---

## SX-3091

(Fatal) SMC via_coverage_option_file corrupted: cannot load corner via_coverage codes.

**Description**

Failed to load via_coverage_option_file for one corner in SMC flow. It is usually caused by discrepancy on the input files.

**What Next**

Check and correct the VIA_COVERAGE_OPTION_FILE, and restart. If error persists, contact Synopsys Support Center.

---

## SX-3092

(Fatal) SMC via_coverage_option_file corrupted: cannot load corner via_coverage entry.

**Description**

Failed to load via_coverage entry from one of the corner-dependent VIA_COVERAGE_OPTION_FILE files. It is usually caused by discrepancy on input files.

Feedback

**What Next**

Check and correct VIA_COVERAGE_OPTION_FILE, and restart. If error persists, contact Synopsys Support Center.

---

## SX-3093

(Fatal) SMC via_coverage_option_file corrupted: different via_coverage dbu is detected.

**Description**

Different unit is used in various VIA_COVERAGE_OPTION_FILE in SMC flow.

**What Next**

Set same unit in each VIA_COVERAGE_OPTION_FILE.

---

## SX-3094

(Warning) ICV GDS flow does not need MILKYWAY_DATABASE.

**Description**

StarXtract finds this is ICV GDS flow. The setting of MILKYWAY_DATABASE is not needed for it.

**What Next**

Remove the setting of MILKYWAY_DATABASE.

---

## SX-3095

(Warning) Too old version of TCAD_GRD_FILE version is being used, it can result in accuracy loss. Please regenerate nxtgrd file.

**Description**

StarRC has general one-year recommendation on TCAD_GRD_FILE compatibility with StarRC executables. The version of TCAD_GRD_FILE is detected much older than the recommendation. Since TCAD_GRD_FILE provides models for extraction, this old TCAD_GRD_FILE can have impact final accuracy.

**What Next**

Recommend to regenerate TCAD_GRD_FILE as StarRC TCAD_GRD_FILE compatibility recommendation.

Feedback

## SX-3096

(Fatal) INTERNAL Error: inconsistency in DP schedule generation.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## SX-3097

(Fatal) Explicit setting of TEMPERATURE_SENSITIVITY:NO should not be set in SIMULTANEOUS_MULTI_CORNER(SMC) flow for temperature coefficient output

**Description**

You receive this message because the command TEMPERATURE_SENSITIVITY:NO is set in a SIMULTANEOUS_MULTI_CORNER(SMC) flow for temperature coefficient output

**What Next**

Remove the command "TEMPERATURE_SENSITIVITY:NO" from StarRC command file and run the flow again.

## SX-3098

(Warning) Reset the FS convergence goal of %s from %f to %f because only the license key StarRC2-MGR-INTEROP is available.

**Description**

Running with Interoperability license does not allow user to specify FS convergence goal.

**What Next**

For further details, contact the Synopsys Support Center.

## SX-3099

(Warning) FS option %s will be ignored because only the license key StarRC2-MGR-INTEROP is availabe.

Feedback

**Description**

When running with the StarRC2-MGR-INTEROP licnese, some field solver options cannot be specified or overridden.

**What Next**

Contact the Synopsys Support Center.

---

## SX-3100

(Warning) FS convergence goal is set to -perc_self 10 because only the license key StarRC2-MGR-INTEROP is availabe.

**Description**

When using the StarRC2-MGR-INTEROP license, FS convergence goal cannot be set lower than 10 percent.

**What Next**

Contact the Synopsys Support Center.

---

## SX-3101

(Fatal) StarRC only supports design view for top block. 'BLOCK: %s' in the command file is not supported.

**Description**

StarRC only supports design view for BLOCK and it can be specified explicitly or implicitly s.t.,

BLOCK: <your_block_name>.design OR BLOCK: <your_block_name>

Any other view name for top BLOCK is not supported and StarRC will error out.

**What Next**

Correct the BLOCK setting in the StarRC command file.

---

## SX-3102

(Warning) Suggest to adjust LEF_USE_OBS/TRANSLATE_DEF_BLOCKAGE for TSMC <= N7 process.

Feedback

**Description**

At TSMC advanced nodes (N7/N6/N5/N5P/N4/N3),

- OBS(s) in a LEF macro will be translated if considered as metal or dropped if considered as routing blockage automatically.
- BLOCKAGE(s) in a DEF macro should be dropped.

Suggests to use

```
LEF_USE_OBS: SHRINK
TRANSLATE_DEF_BLOCKAGE: NO
```

at these advanced nodes to avoid accuracy problem.

**What Next**

Information only

---

## SX-3103

(Fatal) Attempting to run StarRC feature unrelated to field solver. Only field solver features are available with the StarRC2-MGR-INTEROP license.

**Description**

Only FS flows can be launched with the license key StarRC2-MGR-INTEROP.

**What Next**

Contact the Synopsys Support Center.

---

## SX-3104

(Fatal) Attempting to run FS with selected option. This option is not available with the StarRC2-MGR-INTEROP license.

**Description**

Only FS all nets extraction is allowed for "FS_EXTRACT_NETS" with the license key StarRC2-MGR-INTEROP.

**What Next**

Contact the Synopsys Support Center.

Feedback

## SX-3105

(Information) Successfully copied file %s to %s.

**Description**

StarRC is successfully able to copy a file that could not be renamed earlier.

**What Next**

No action is necessary.

## SX-3106

INTENAL:(Warning) Manually set the CALIBRE_AGF_CELL_EXTENTS larger or smaller than the calibre runset may cause bad performance in StarRC

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

## SX-3107

(Information) No logical or physical changes detected. Skipping extraction and netlisting.

**Description**

If there are no logical or physical changes when comparing against a baseline layout, there is no need to perform incremental extraction. StarRC will exit without producing a netlist.

**What Next**

Information only; no action is needed.

## SX-3108

(Warning) XIN_ABUTT: YES is not supported for fast extraction mode.

**Description**

XIN_ABUTT: YES is not supported for fast extraction mode.

Feedback

**What Next**

You can ignore this message and StarRC will set XIN_ABUTT to NO.

---

## SX-3109

(Information) [STAR_MAIN_SUMMARY]Found '%s' percent ECO net change. Performing full extraction.

**Description**

When number of net changes is large, a full extraction, instead of incremental extraction, is performed.

**What Next**

This message is for information only, no further action is needed.

---

## SX-3111

(Fatal) PRINT_THICKNESS_INFO is not supported for this process.

**Description**

PRINT_THICKNESS_INFO can be used to output thickness variation as netlist tail comments. In the process under running, this feature is prohibited.

**What Next**

Remove PRINT_THICKNESS_INFO from StarRC command file. Or if PRINT_THICKNESS_INFO is needed, contact Synopsys or Foundry for more information.

---

## SX-3112

(Fatal) Invalid OPERATING_TEMPERATURE is specified. Floating number (-200~300 in unit of CELSIUS) is expected.

**Description**

OPERATING_TEMPERATURE has predefined range from -200 to 300 (unit:Celsius). In this run the temperature specified is out of the range, hence the extraction aborts.

**What Next**

Change the OPERATING_TEMPERATURE to in range of -200 to 300 (unit:Celsius).

Feedback

## SX-3113

(Fatal) Invalid OPERATING_TEMPERATURE is specified in corner: %s . Floating number (-200~300 in unit of CELSIUS) is expected.

**Description**

OPERATING_TEMPERATURE has predefined range from -200 to 300 (unit:Celsius). In some corner run, the temperature specified is out of the range, hence the extraction aborts.

**What Next**

Change the OPERATING_TEMPERATURE to in range of -200 to 300 (unit:Celsius).

## SX-3114

(Fatal) Specified NETLIST_CORNER_FILE does not exist!

**Description**

In TEMPERATURE_SENSITIVITY flow, NETLIST_CORNER_FILE is a necessary input. The input of this file cannot be accessed.

**What Next**

Correct the input file of NETLIST_CORNER_FILE and ensure it is readable.

## SX-3115

(Fatal) Cannot determine device gate direction.

**Description**

Some of the device gates in this design are in X direction while some of them are in Y direction. StarRC QTF flow only support one device gate direction in a design.

**What Next**

Make sure all device gates are in the same direction.

## SX-3116

(Warning) The DIEAREA syntax in TOP_DEF_FILE is not specified!

Feedback

**Description**

StarRC relies on DIEAREA syntax in DEF file to estimate the size of the design. If it's not available, it may affect StarRC performance.

**What Next**

Check whether DEF file is correct. Usually the front-end tools will generate this syntax in DEF file. If this DEF file is manually modified by users, please provide DIEAREA syntax if possible.

---

## SX-3117

(Warning) Instance '%s:%s' is UNPLACED in the design '%s'.

**Description**

An instance is UNPLACED in the design. UNPLACED and connected instances may cause peformacnce issue as they may congest in small area and introduce huge amount of polygons in that area at later extration stages.

For performance consideration, StarRC only reports few unplaced instances.

**What Next**

Information only; no action needed.

---

## SX-3118

(Information) Performing full extraction with ECO_MODE because %s.

**Description**

Perform full extraction for ECO_MODE for the given reason.

**What Next**

Information only; no action is needed.

---

## SX-3119

(Information) Backing up files required by ECO extraction.

**Description**

StarRC ECO extraction requires backing up extraction critical files following successful extraction. These files are required for subsequent iterations of ECO extraction.

Feedback

**What Next**

No action is necessary.

---

## SX-3120

(Information) Performing full extraction in ECO_MODE.

**Description**

Running full extraction because ECO_MODE is set to "RESET".

**What Next**

See StarRC user guide.

---

## SX-3121

(Fatal) ICC signature mismatch. Previous iteration signature '%s' does not match current iteration last signature '%s'

**Description**

StarRC Galaxy ECO flow requires consistency of the ICC2 signature between ECO iterations. There must be a match between the 'last' signature of the current iteration vs. the signature of the previous iteration.

**What Next**

Contact the Synopsys Support Center.

---

## SX-3122

(Warning) StarRC is unable to map LEE_BLK layer %s with maskName=%s to its real metal layer - more than one real metal layer found!

**Description**

ICC2 implements LEE_BLK mark shapes on the layers defined in technology database as below

Layer "LEE_BLK_7" { layerNumber = 2307 maskName = "metal8RCBlockage" blockLineEndExtensionMarkerWidth = 0.0038 ... }

To translate LEE_BLK_7 marker shapes on real metal layer, StarRC will need to obtain real metal layer from LEE_BLK_7's maskName.

Feedback

If more than one tech layers are given with same maskName (metal8 in this example), StarRC will only use the first one to translate LEE_BLK_7 and it might leads to wrong translation.

**What Next**

Contact the Synopsys Support Center.

---

## SX-3123

(Fatal) Unsupported CDB equivalence type: '%c'.

**Description**

Invalid equivalence type is found in equiv file.

**What Next**

Check data generated by ICV.

---

## SX-3124

(Information) [STAR_MAIN_SUMMARY]Performing incremental extraction on '%s' percent ECO change.

**Description**

Incremental extraction re-extracts those modified or added nets. This information shows the percentage of re-extracted nets.

**What Next**

Information only; no action is needed.

---

## SX-3125

(Fatal) INTERNAL Error: Invalid XOUT_NN_STITCH specification '%s', valid choices are any combination of %s.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

Feedback

## SX-3126

(Warning) Missing HN port for SPICE port '%s'

**Description**

Ports in input database might not be defined in SPICE_SUBCKT_FILE.

**What Next**

Contact the Synopsys Support Center.

## SX-3127

(Information) [STAR_MAIN_SUMMARY]Using -clean option in ECO mode because StarRC command option(s) have changed.

**Description**

StarRC is running with the -clean option in ECO mode because some StarRC command options have changed since the previous run.

**What Next**

See StarRC user guide.

## SX-3128

(Warning) LEE (Line End Extension) blockage from GDS_FILE/OASIS_FILE is not supported in LEF/DEF flow.

**Description**

Not supported.

**What Next**

Contact the Synopsys Support Center.

## SX-3130

(Fatal) FILL_COUPLING_THRESHOLD is recommended in transistor-level flow only.

**Description**

FILL_COUPLING_THRESHOLD is used for configurate the metal fill extraction accuracy. This option is intended for transistor-level flow only. As the run is gate run, it errors out to avoid any unexpected issue.

Feedback

**What Next**

Remove this option for the run.

---

## SX-3131

(Fatal) The process %s is not supported for extraction in 2014.06 or newer versions of StarRC.

**Description**

StarRC does not support the specified process since the 2014.06 version.

**What Next**

Use a StarRC version 2013.12 or older.

---

## SX-3135

(Warning) ECO_MODE_ACCURACY level not set to either NORMAL or HIGH. Setting value to NORMAL.

**Description**

Incremental (ECO) extraction accuracy vs. runtime tradeoff can be controlled by the ECO_MODE_ACCURACY level. The default value is HIGH, which provides the highest deegree of accuracy with minimal runtime impact. The NORMAL setting provides additional speedup in cases where the highest degree of accuracy is not required.

For more information on ECO extraction, see the StarRC user guide.

**What Next**

If the NORMAL setting is acceptable, no action is needed.

---

## SX-3136

(Fatal) ECO_MODE requires NETLIST_FORMAT: SPEF.

**Description**

Incremental (ECO) extraction only supports SPEF as the output parasitic file format.

For more information on ECO extraction, see the StarRC user guide.

**What Next**

Modify the NETLIST_FORMAT command to specify a SPEF netlist.

Feedback

## SX-3138

(Fatal) ECO_MODE does not support the CCP flow.

**Description**

Incremental (ECO) extraction does not support the CCP flow due to additional requirements on interface layer parasitics.

For more information on ECO extraction, see the StarRC user guide.

**What Next**

Modify the command file to run either the CCP flow or ECO extraction, but not both.

## SX-3139

(Warning) ECO_MODE requires COUPLE_TO_GROUND: NO. Switching to ECO_MODE:NO full extraction.

**Description**

Incremental (ECO) extraction requires that the COUPLE_TO_GROUND command be set to NO. Switching to ECO_MODE:NO full extraction.

For more information on ECO extraction, see the StarRC user guide.

**What Next**

Modify the command file to include the COUPLE_TO_GROUND: NO command.

## SX-3140

(Warning) ECO_MODE does not support the FS_EXTRACT_NETS flow. Switching to ECO_MODE:NO full extraction.

**Description**

Incremental (ECO) extraction does not support Field Solver flows. Switching to ECO_MODE:NO full extraction.

For more information on ECO extraction, see the StarRC user guide.

**What Next**

Modify the command file to remove all Field Solver commands.

Feedback

## SX-3141

(Warning) ECO_MODE does not support EXTRACTION: FSCOMPARE. Switching to ECO_MODE:NO full extraction.

**Description**

Incremental (ECO) extraction does not support Field Solver flows. Switching to ECO_MODE:NO full extraction.

For more information on ECO extraction, see the StarRC user guide.

**What Next**

Modify the command file to remove all Field Solver commands.

## SX-3142

(Warning) ECO_MODE does not support the POWER_EXTRACT flow. Switching to ECO_MODE:NO full extraction.

**Description**

Incremental (ECO) extraction does not support power extraction. Switching to ECO_MODE:NO full extraction.

For more information on ECO extraction, see the StarRC user guide.

**What Next**

Modify the command file to remove the POWER_EXTRACT command (it defaults to NO) or include an explicit POWER_EXTRACT: NO command.

## SX-3143

(Fatal) ECO_MODE does not support the 3D IC flow.

**Description**

Incremental (ECO) extraction does not support 3D IC extraction.

For more information on ECO extraction, see the StarRC user guide.

**What Next**

Modify the command flow to remove the 3D_IC command (it defaults to NO) or include an explicit 3D_IC: NO command.

Feedback

## SX-3144

(Fatal) ECO_MODE requires EXTRACTION: RC. Switching to ECO_MODE:NO full extraction.

**Description**

Incremental (ECO) extraction extracts both parasitic resistances and capacitances and produces a consolidated RC network, which describes the layout.

For more information on ECO extraction, see the StarRC user guide.

**What Next**

Modify the command file to remove the EXTRACTION command (it defaults to RC) or include an explicit EXTRACTION: RC command.

## SX-3145

(Fatal) ECO_MODE detected an unsupported clean option.

**Description**

StarRC has detected a clean option that is not compatible with incremental (ECO) extraction. Only the -clean and -cleanN options are supported.

For more information on ECO extraction, see the StarRC user guide.

**What Next**

Repeat the StarRC run with one of the supported options.

## SX-3146

(Fatal) ECO_MODE does not support transistor-level extraction.

**Description**

Incremental (ECO) extraction does not support transistor-level extraction.

For more information on ECO extraction, see the StarRC user guide.

**What Next**

Modify the command file to either perform a gate-level extraction or to disable ECO extraction.

Feedback

## SX-3147

(Fatal) Mapping file in CDLOut run directory does not exists.

**Description**

You receive this message because of the absence of the mapping files in CDLOut run directory while you specifed OA_CDLOUT_RUNDIR to use ihnl or cnl dir to map back to schematic net names in the OA extracted view.

**What Next**

Make sure that the mapping files in sub_directories of CDLOut run directory exist.

## SX-3148

(Fatal) Wrong format of net definition in CDLOut run directory.

**Description**

You receive this message because of the absence of the mapping files in CDLOut run directory while you specifed OA_CDLOUT_RUNDIR to use ihnl dir to map back to schematic net names in the OA extracted view.

**What Next**

Make sure that the format of net definition in both the file "globalmap" in CDLOut run directory and the files "map" in sub_directories of CDLOut run directory are correct.

The correct formats are:

>>In "globalmap": >>>>>>>>>>>>>>>>>>> $net net_name net_name_1 $endnet

>>In "map" at sub_directoires: >>>>>>>> (net (net_name net_name_1) //or: ("net_name" "net_name_1") )

## SX-3149

(Warning) Galaxy Parasitic Data (GPD) does not support the direct ITF flow with command ITF_FILE.

**Description**

Galaxy Parasitic Data (GPD) does not support the direct ITF flow. StarRC will run the non-GPD flow.

Feedback

**What Next**

Remove the ITF_FILE command to run StarRC with GPD netlisting or the non-GPD flow will run.

---

## SX-3150

(Warning) Galaxy Parasitic Data (GPD) is turned off for the StarQTF flow. Extraction is reverted to XOUT mode.

**Description**

Galaxy Parasitic Data (GPD) is turned off for StarQTF flow. StarRC will run the non-GPD flow.

**What Next**

To enable GPD in StarQTF flow, add option XTOR_GPD: YES or STARQTF_GPD:1 in star_cmd.

---

## SX-3151

(Warning) Mapping file in CDLOut run directory does not exists.

**Description**

You receive this message because of the absence of the mapping files in CDLOut run directory while you specifed OA_CDLOUT_RUNDIR to use ihnl or cnl dir to map back to schematic net names in the OA extracted view.

**What Next**

Make sure that the mapping files is in CDLOut run directory.

---

## SX-3192

(Fatal) EXTRACTION: RC must be specified with CLOCK_NETS option"

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

Feedback

## SX-3197

(Warning) Some blocking regions are not blocked any port, device terminal, or probe_text.

**Description**

At least one port, device terminal or probe_text need to be located in blocking regions with blocking layer methodology(defined in SKIP_PCELL_LAYERS_FILE). If not, the accuracy of capacitance which across the blocking region is not guaranteed.

**What Next**

Include a port, device terminal, or probe_text in the blocking boundary. The detailed information is in summary/pcelllayers.sum which have the bounding box of each blocking region do not have any port, terminal, or probe_text.

## SX-3198

(Fatal) ASCII_DESIGN_FILE is empty. There is no physical shape.

**Description**

ASCII_DESIGN_FILE is empty. There is no physical shape.

**What Next**

Confirm ASCII_DESIGN_FILE is not broken.

## SX-3199

(Fatal) The BBOX of "%s" is invalid. BBOX(%fum, %fum) (%fum, %fum).

**Description**

The cell boundary is not valid.

**What Next**

Confirm the Milkyway data is not broken or re-generate the Milkyway data.

## SX-3200

(Warning) No physical layer described for non-conductor layer Line: %s

**Description**

In ccp_reserved_layers section, you need to map the non-conductor database layer to a physical layer. If you only list the database layer name, StarRC will give this error.

Feedback

**What Next**

You should map the non-conductor database layer to a physical layer.

---

## SX-3201

(Fatal) INTERNAL Error: Cannot use coordinate-based stitcher because %s.

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-3202

(Fatal) CCP SMC: Cannot get skew name. Please define PROCESS_CORNER_NAME in ITF files and re-generate nxtgrd files.

**Description**

In order to use SIMULTANEOUS_MULTI_CORNER:YES in CCP flow, all the nextgrd files must have PROCESS_CORNER_NAME field set. This is required by CCP engine to get the mapping between nxtgrd files and the UP corners.

**What Next**

Modify the ITF files and regenerate nxtgrd files for all the related corners.

---

## SX-3203

(Fatal) In the CCP simultaneous multicorner flow, OPERATING_TEMPERATURE is only set for some of the corners in the CORNERS_FILE.

**Description**

When SIMULTANEOUS_MULTI_CORNER:YES is used in the CCP flow, if any corner listed in SELECTED_CORNERS sets OPERATING_TEMPERATURE in the CORNERS_FILE, all of the corners listed in SELECTED_CORNERS must set OPERATING_TEMPERATURE in the CORNERS_FILE as well.

If OPERATING_TEMPERATURE is varied across corners, each and every corner must have OPERATING_TEMPERATURE set in the CORNERS_FILE. If all corners are using the same OPERATING_TEMPERATURE, the temperature can be set either in the CORNERS_FILE or in the command file.

Feedback

**What Next**

Modify either the command file or the CORNERS_FILE to specify the operating temperature (or temperatures) correctly.

---

## SX-3204

(Fatal) In the CCP simultaneous multicorner flow, OPERATING_TEMPERATURE is not set in either the command file or the CORNERS_FILE.

**Description**

When SIMULTANEOUS_MULTI_CORNER:YES is used in the CCP flow and OPERATING_TEMPERATURE is not defined in the CORNERS_FILE, OPERATING_TEMPERATURE must be specified in the command file.

If OPERATING_TEMPERATURE is varied across corners, each and every corner must have OPERATING_TEMPERATURE set in the CORNERS_FILE. If all corners are using the same OPERATING_TEMPERATURE, the temperature can be set either in the CORNERS_FILE or in the command file.

**What Next**

Modify either the command file or the CORNERS_FILE to specify the operating temperature (or temperatures) correctly.

---

## SX-3205

(Warning) Blocking layer '%s' in SKIP_PCELL_LAYERS_FILE is not defined in either GDS_MAP or devtab.

**Description**

Block layer name '%s' in SKIP_PCELL_LAYERS_FILE is not match any layer name in either GDS_MAP or devtab in CCI flow.

**What Next**

There have some mismatch between layer name in SKIP_PCELL_LAYERS_FILE and calibre runset. You should check you SKIP_PCELL_LAYERS_FILE is compatible with the calibre runset.

---

## SX-3206

(Warning) Command CHECK_SKIP_PCELL_LAYER_NAMES can not be used in non-CCI flow, will be ignored.

Feedback

**Description**

Command CHECK_SKIP_PCELL_LAYER_NAMES can not be used in non-CCI flow, will be ignored.

**What Next**

CHECK_SKIP_PCELL_LAYER_NAMES is only for CCI flow. If you are going to use this command, use it in CCI flow, or it will be ignored.

---

## SX-3207

(Fatal) INTERNAL Error: Cannot use DP_STITCHING: yes without specifying XOUT_NN_STITCH.

**Description**

An internal consitency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-3208

(Warning) DIELECTRIC_FILL_GDS_FILE does not contain any cell definition for BLOCK %s . HPB objects must reside in same name cell as BLOCK.

**Description**

DIELECTRIC_FILL_GDS_FILE doesn't contain any cell that matches BLOCK. The DIELECTRIC_FILL_GDS_FILE should be associated with the original design by having the same top cell.

**What Next**

Provide DIELECTRIC_FILL_GDS_FILE that contains a cell with name equals to BLOCK.

---

## SX-3209

(Information) Applied GDS/OASIS file to cell '%s'.

**Description**

Add contents from GDS/OASIS file into the cell.

**What Next**

This is an informational message only. No action is required on your part.

Feedback

## SX-3210

(Fatal) The net name '%s' has hierarchy separator '%c' at first char.

**Description**

This kind of net name is not acceptable.

**What Next**

Correct this net name.

## SX-3211

(Warning) %s OBS on %s layer %s in LEF macro %s - '%s'

**Description**

At TSMC advanced nodes (N7/N6/N5/N5P/N4/N3), OBS(s) in a LEF macro will be translated if considered as metal or dropped if considered as routing blockage automatically.

When an OBS is defined on a DB layer mapped to a color ITF layer,

- If OBS is colored, the one with SPACING=0 is considered routing blockage:
  - Translated as metal if OBS has no SPACING.
  - Dropped as routing blockage if OBS has SPACING=0.
  - Translated as metal if OBS has SPACING > 0
    - If OBS is not colored, it is considered routing blockage:
      - This is not allowed and error out by default. When DEBUG_NON_COLOR_POLYGON:YES, this error is lifted and become a warning.
      - Dropped as routing blockage if OBS has SPACING >= 0.

        When an OBS is defined on a DB layer mapped to a non-color ITF layer,

        - Translated as metal if OBS has no SPACING.
        - Dropped as routing blockage if OBS has SPACING>=0.

          For performance consideration, this warning will be issued only once for each polygon dropping case above.

Feedback

**What Next**

Information only

---

## SX-3212

(Warning) Missing GDS/OASIS cell description for some cells.

**Description**

StarXtract reports the data information.

**What Next**

Confirm this is the expected data. If it is not, check the input data about it in the StarXtract command file.

---

## SX-3213

(Fatal) Memory allocation limit exceeded.

**Description**

StarRC was not able allocate sufficient memory to continue the job.

**What Next**

Contact your system administrator to see how much memory is available. If StarRC is using excessive amount of memory, contact Synopsys Customer Support.

---

## SX-3214

(Warning) Number of masks, %d, specified in LEF layer "%s" does not equal number of masks, %d, defined in GRD layer "%s".

**Description**

The number of masks defined in LEF_FILE (MASK) does not match the number of masks in TCAD_GRD_FILE (NUMBER_OF_MASK).

**What Next**

This is process-dependent (<=7nm). GRD contains all mask information for the layers. Layout (LEF) may use fewer masks than those in GRD. Please check if the number of masks mismatched is reasonable. If not clear, please contact the Synopsys Support Center.

Feedback

## SX-3215

(Warning) Redefinition of LAYER %s attribute %s

### Description

The LAYER is defined in multiple LEF(s) and the attribute is given differnent value than previous definition. The multiple defintions of a LAYER won't cause any error in DEF translation, but it may cause unexepected wire widths in the final database and thus false opens. It also introduces extra maintenance difficulty in the design flow.

### What Next

To keep the integrity of design input sources, StarRC suggests to define LAYER(s) only in your technology LEF.

## SX-3216

(Warning) MACRO_DEF_FILE "%s" contains the same DESIGN name as "%s".

### Description

The macro DEF defines same DESIGN with previouly parsed DEF and parsing of it may corrupt the main design data. It will be ignored during translation.

### What Next

Check the macro DEF. If it contains same design data with others, you can ignore this message or remove it from MACRO_DEF_FILE in the command file. If it should be parsed, you should correct the DESIGN and reference it corectly in the TOP_DEF_FILE or other macro DEF(s).

## SX-3217

(Fatal) MACRO_DEF_FILE "%s" contains the same DESIGN name "%s" as in TOP_DEF_FILE

### Description

The macro DEF defines same DESIGN name as in TOP_DEF_FILE.

### What Next

Check the macro DEF and TOP_DEF_FILE. Correct the DESIGN in the macro DEF file and its references in the TOP_DEF_FILE or other macro DEF(s).

Feedback

## SX-3218

(Warning) StarRC ignores the definition of cell '%s' in '%s' as there is no metal routing defined in it.

**Description**

The cell definition in the GDS_FILE/OASIS_FILE has no metal routing in it that can influence accuracy so that StarRC doesn't translate it.

This can be caused by empty GDS/OASIS cell or the GDS/OASIS cell doesn't define metals on the layers mapped in GDS_LAYER_MAP_FILE/OASIS_LAYER_MAP_FILE.

**What Next**

Information only; no action needed.

## SX-3219

(Warning) PROBE_TEXT_FILE is ignored in EXTRACTION: C.

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

## SX-3220

(Fatal) NETLIST_INDUCTANCE_UNIT is not supported with NETLIST_FORMAT:SPF

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

## SX-3221

(Warning) There is long active fill %d, %d, %d, %d in the design

**Description**

The new METAL_FILL_RECONNECT does not support active fill whose length is larger than 2 um.

Feedback

**What Next**

Check if that's expected to have such long active fill in input data.

---

## SX-3222

(Warning) Old metal fill reconnection (UNIMPORTANT_NETS) is invoked with via fills.

**Description**

Old metal fill reconnection (UNIMPORTANT_NETS) is invoked with via fills.

**What Next**

Information only; no action is needed.

---

## SX-3223

(Warning) New MFR is not supported in FS flow, old reconnect is used for metal fill reconnect.

**Description**

New MFR is not supported in FS flow, old reconnect is used for metal fill reconnect.

**What Next**

Information only; no action is needed.

---

## SX-3224

(Warning) New metal fill reconnection is invoked with active via fill. The new architecture does not handle via fill and can lead to accuracy issue. Please clean up the input data or use the old reconnectino flow.

**Description**

New metal fill reconnection is invoked with active via fill. The new architecture does not handle via fill and can lead to accuracy issue. Please clean up the input data or use the old reconnection flow.

**What Next**

Check the design to see if it's untended. Then clean up the input data and rerun StarRC or use the old reconnection flow.

Feedback

## SX-3225

(Warning) Fast MFR is not recommended for general usage when layout fills do not conform to the feature limitations, may impact extraction accuracy. Please refer to StarRC reference manual.

**Description**

Fast metal fill reconnect is enabled. In this mode, metal fill is considered only if it comes from a GDSII or OASIS file and is marked as active fill in the mapping file. This mode has a set of limitations including active fill polygon size, the sequence of active fill interaction with signal nets, and it does not consider via fills regardless of the setting of the TRANSLATE_VIA_FILLS command. Please refer to StarRC reference manual for more details.

**What Next**

Study StarRC reference manual to understand the limitations. Rerun StarRC using UNIMPORTANT_NETS for MFR if necessary.

## SX-3226

(Fatal) NETLIST format %s is not supported with CLOCK_NETS option; must be SPF or DSPF

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

## SX-3227

(Warning) ANALOG_SYMMETRIC_NETS is not needed for this extraction. It is ignored.

**Description**

ANALOG_SYMMETRIC_NETS can be set in certain process to improve the extraction accuracy. However at the process of this run, it is automatically detected that ANALOG_SYMMETRIC_NETS is no longer needed for accuracy purpose. The option is ignored.

**What Next**

Remove ANALOG_SYMMETRIC_NETS from the StarRC command file.

Feedback

## SX-3228

(Warning) All nets specified as clock nets for clock net inductance extraction.

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

## SX-3229

(Warning) Could not find any cell for wildcard set of MILKYWAY_CELL_VIEW.

**Description**

StarXtract cannot find any cell name for the wildcard set of MILKYWAY_CELL_VIEW.

**What Next**

Confirm the setting for MILKYWAY_CELL_VIEW is correct or expected.

## SX-3230

(Warning) DENSITY polygons are %s in the %s macro '%s'

**Description**

Real metal shapes always have higher precedence over DENSITY for sign-off extraction. If OBS or blockage polygons are defined in a LEF/NDM cell, StarRC will ignore its DENSITY polygons in the extraction.

DENSITY polygons can only be used in the extraction which real cell content is not given (OBS is not defined).

**What Next**

Information only.

## SX-3231

(Fatal) CALIBRE_REMOVE_ANALOG_DEVICES only supports CCI database and SPEF output format

Feedback

**Description**

CALIBRE_REMOVE_ANALOG_DEVICES:YES can only be used with CCI database as input database and SPEF format as output format.

**What Next**

You should use CCI database as input database and set NETLIST_FORMAT: SPEF. Otherwise, you can't use CALIBRE_REMOVE_ANALOG_DEVICES:YES in StarRC command file.

---

## SX-3232

(Fatal) INTERNAL Error: Can't generate file "ana_device_remove_info", CALIBRE_REMOVE_ANALOG_DEVICES fail to remove analog devices

**Description**

An internal consistency check has failed.

**What Next**

Contact the Synopsys Support Center.

---

## SX-3233

(Warning) %s

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-3234

(Fatal) TEMPERATURE_SENSITIVITY with NETLIST_CORNER_NAMES and NETLIST_CORNER_FILE is discontinued. Use SIMULTANEOUS_MULTI_CORNERS flow.

**Description**

Using TEMPERATURE_SENSITIVITY to generate multiple temperature netlist files is obsolete. SIMULTANEOUS_MULTI_CORNERS flow is recommended to generate multiple temperature netlist files.

Feedback

**What Next**

see option SIMULTANEOUS_MULTI_CORNERS for more information.

---

## SX-3235

(Information) CapCompare: (Total) Avg %s ; StdDev %s ; Range %s to %s.

**Description**

This is an information message. It reports the summary of FSCOMPARE total results, including average error, standard deviation and the range of the error.

**What Next**

You need to do nothing about this message.

---

## SX-3236

(Fatal) Cannot use FS_EXTRACT_NETS:<sch_net_name> when NET_TYPE:SCHEMATIC and XREF:NO are specified.

**Description**

You receive this message because NET_TYPE is set to SCHEMATIC and XREF is set to NO. This will cause a conflict in the net name type.

**What Next**

Specify NET_TYPE: SCHEMATIC and XREF: YES, then in FS_EXTRACT_NETS use schematic net name; or specify NET_TYPE: LAYOUT and XREF: NO, in FS_EXTRACT_NETS use layout net name.

---

## SX-3237

(Fatal) Version mismatch for internal file '%s': '%s' vs. '%s'.

**Description**

The current version of StarRC cannot read the specified file that was produced by an earlier version because of a change in the format of the file between two versions.

**What Next**

Rerun StarRC from an empty run directory.

Feedback

## SX-3238

(Fatal) Specified value for option '%s': '%s' is invalid.

**Description**

The executable name specified as the value of the given option in the StarRC command file was not found in the runtime environment.

**What Next**

Ensure that the executable is correctly specified by checking its spelling and verifying that it can be found from the PATH environment variable. If the value is specified using an absolute path, verify the path to the file.

Also verify that the user has the required privileges to execute the command.

## SX-3239

(Fatal) Specified layer for MARKER_LAYER: '%s' is invalid.

**Description**

The database layer specified in MARKER_LAYER must exist in the database.

**What Next**

Make sure the layer exists in the database

## SX-3240

(Fatal) Unable to open %s '%s' for %s.

**Description**

StarRC is unable to open a file because of conditions in the runtime environment.

**What Next**

Check your file system for errors and problem conditions such as running out of disk space. Contact your system administrator if necessary.

## SX-3241

(Fatal) Unable to open %s '%s' for %s, error status: '%s'

**Description**

StarRC is unable to open a file because of conditions in the runtime environment.

Feedback

**What Next**

Carry out remedial actions based on the error status indicated by the operating system. Contact your system administrator if necessary.

---

## SX-3242

(Warning) CALIBRE_REMOVE_ANALOG_DEVICES is not supported in Field Solver flow. This command will be ignored.

**Description**

Command CALIBRE_REMOVE_ANALOG_DEVICES is not supported in Field Solver flow. StarRC will set CALIBRE_REMOVE_ANALOG_DEVICES to NO automatically once it finds this is Field Solver flow and give this warning.

**What Next**

If you don't need CALIBRE_REMOVE_ANALOG_DEVICES: YES, nothing need to be done. Otherwise, you should remove any Field Solver flow related commands such as FS_EXTRACT_NETS or EXTRACTION: FSCOMPARE.

---

## SX-3243

(Fatal) Command %s is not supported in CCP flow.

**Description**

The highlighted command setup is not supported in CCP flow.

**What Next**

Change the StarRC command file and restart.

---

## SX-3244

(Warning) Command %s will be ignored in CCP flow.

**Description**

The highlighted command setup is related with device layers which is controlled by CCP extraction engine.

**What Next**

This is a warning message for information purpose. The program will not abort.

Feedback

## SX-3245

(Warning) Support of command %s in CCP flow is restricted to StarRC extracted interconnect layers.

**Description**

The highlighted command setup can only be applied on interconnect layers extracted by StarRC engine. It cannot impact the devicer layers which are extracted by CCP engine.

**What Next**

This is a warning message for information purpose. The program will not abort.

## SX-3246

(Error) MERGE_VIA_NODES: NO must be used with REDUCTION: NO.

**Description**

MERGE_VIA_NODES: NO keep all via nodes and the resistance between them. It requires REDUCTION: NO.

**What Next**

set REDUCTION: NO in the StarRC command file.

## SX-3247

(Warning) Rename conflicting pin %s to %s in cell %s

**Description**

In multi-pin flow, if a pin has name conflicted with other ports in the cell. StarRC will rename this pin to <port_name>_<pin_name>_<pin id>.

**What Next**

Information only.

## SX-3250

(Fatal) Specified value of command %s must be >= %d

**Description**

StarRC command value must be an integer above a threshold.

Feedback

**What Next**

Fix StarRC command value to conform to the permissible range.

---

## SX-3251

(Fatal) Invalid entry in command %s: %s

**Description**

Invalid entry for StarRC command

**What Next**

Fix StarRC command value to valid entry. Please consult the StarRC user guide or contact Synopsys Support Center.

---

## SX-3252

(Warning) Specified error suppression limit of %d for %s exceeds the maximum permitted value of %d. Resetting to the maximum recommended value.

**Description**

Large error suppression limit can significantly increase StarRC runtime and use more disk space.

**What Next**

See StarRC user guide description of the MESSAGE_SUPPRESSION command.

---

## SX-3253

(Warning) Specified message suppression limit of %d is larger than the recommended maximum of %d. As a result, StarRC runtime may increase significantly.

**Description**

Repeated output of messages can increase StarRC runtime significantly.

**What Next**

If StarRC runtime is too long, reduce the message suppression limit. Consult user guide for more details on the MESSAGE_SUPPRESSION_LIMIT command.

---

## SX-3254

(INTERNAL Warning) fs_assisted flow is not turned on for RANDOM ACCESS XOUT

Feedback

**Description**

fs_assisted flow is not turned on for RANDOM ACCESS XOUT

**What Next**

Nothing needs to be done. StarRC extracts all nets instead of activating fs_assisted flow.

---

## SX-3255

(Warning) KEEP_VIA_NODES is not supported with reduction HIGH and has been disabled.

**Description**

Using KEEP_VIA_NODES:YES results in so many nodes being preserved that REDUCTION:HIGH cannot function, so KEEP_VIA_NODES had been disabled.

**What Next**

If preservation of via nodes is required use REDUCTION:YES, NO or NO_EXTRA_LOOPS.

---

## SX-3258

(Warning) Device

**Description**

Once you use command CALIBRE/ICV_LVS_DEVICE_TYPE_MOS to re-cast a user-defined device as MOS, StarRC needs to map terminals of the user-defined device to standard terminals of a MOS device. For example, if a user-defined device "UD" has 4 terminals: A, B, C, D, and the device "UD" is re-casted as a MOS device through command CALIBRE_LVS_DEVICE_TYPE_MOS, StarRC needs to know how the terimal A, B, C, D is mapped to standard terminal (DRAIN, GATE, SOURCE, BULK) of a MOS device. StarRC has 2 mechanisms to do the mappings: <1 if the terminal names of the user-defined device are exactly D, G, S, B or DRAIN, GATE, SOURCE, BULK, StarRC will do mappings based on the terminal names; <2 You can define terminal mappings for this device in a optional pin file (CALIBRE/ICV_OPTIONAL_DEVICE_PIN_FILE), and StarRC will map terminals based on the mappings defined in that file.

If the user-defined device has no standard terminal names, and CALIBRE/ICV_OPTIONAL_DEVICE_PIN_FILE doesn't contain mapping information for this device, StarRC won't know the correct mappings of the device terminals. If the mappings are not correct, the result may be incorrect. So StarRC will not re-cast that device and give this warning.

Feedback

**What Next**

You should check whether you use a correct optional pin file through StarRC command CALIBRE/ICV_OPTIONAL_DEVICE_PIN_FILE. All devices in command CALIBRE/ICV_LVS_DEVICE_TYPE_MOS but have no standard terminal names should be included in the optional pin file.

---

## SX-3261

(Warning) TRANSLATE_VIA_FILLS is only effective in transistor-level extraction. This option is ignored.

**Description**

TRANSLATE_VIA_FILLS is a feature to enable translation of floating fill vias and accounting for them during floating fill reduction. This feature is limited at transistor-level only. The run is at gate-level, so the option is ignored.

**What Next**

Remove TRANSLATE_VIA_FILLS from StarRC command file.

---

## SX-3262

(Warning) A via fill %d, %d, %d, %d touches net id %d. This via is ignored.

**Description**

TRANSLATE_VIA_FILLS option does not support via fills touching signal nets. This via is ignored and will cause no effect.

**What Next**

Check the input data for errors.

---

## SX-3263

(Warning) A fill polygon %d, %d, %d, %d touches net id %d. This polygon is ignored

**Description**

Touching of fill polygons to signal nets is ignored. This fill polygon will cause no effect.

**What Next**

Check the input data for errors.

Feedback

## SX-3264

(Warning) TRANSLATE_RETAIN_BULK_LAYERS_CONLY_FORCE will be obsolete in next release

**Description**

TRANSLATE_RETAIN_BULK_LAYERS_CONLY_FORCE will be obsolete in next release

**What Next**

Remove TRANSLATE_RETAIN_BULK_LAYERS_CONLY_FORCE in StarRC command file

## SX-3265

(Fatal) Option %s should not be mixed with NDM/LEFDEF option %s

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

## SX-3266

(Fatal) Option %s should not be mixed with NDM/MILKYWAY option %s

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

## SX-3267

(Fatal) Option %s should not be mixed with NDM/HERCULES option %s

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

Feedback

## SX-3268

(Fatal) Option %s should not be mixed with NDM/CALIBRE option %s

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

## SX-3269

(Fatal) This function %s should not be mixed with NDM option

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

## SX-3280

(Warning) Gate layer %s and diff layer %s are in ignore_cap_layers, the coupling between them will be ignored.

**Description**

In StarRC command file, IGNORE_CAPACITANCE: ALL RETAIN_GATE_DIFFUSION_COUPLING is set. But the gate layer and diff layer reported in the message is also under ignore_cap_layers in mapping file. StarRC will ignore the coupling between the reported gate layer and diff layer.

**What Next**

If you want to ignore the coupling between the gate and diff layers, you don't need to do anything. If you want to extract the coupling between the gate and diff layers, remove them from ignore_cap_layers in mapping file.

## SX-3281

(Warning) NET_SEGMENT_CUT_LENGTH option will be obsolete for this flow in StarRC 2016.12 release. Please change your command file to reflect this change.

Feedback

**Description**

The option will be obsolete for gate-level flows in 2016.12 release.

**What Next**

Change your command file to avoid the warning.

---

## SX-3283

(Fatal) Macro '%s' specified in command 'DEF_ATTRIBUTE_FROM_LEF' is not found.

**Description**

The macro specified in the option is missing.

**What Next**

Checck if the macro is specified correctly.

---

## SX-3284

(Fatal) File '%s' specified in command 'DEF_ATTRIBUTE_FROM_LEF' is not found.

**Description**

The file specified in the option is missing.

**What Next**

Check if the file is specified correctly.

---

## SX-3285

(Fatal) Attribute '%s' specified in 'DEF_ATTRIBUTE_FROM_LEF' is not supported.

**Description**

The attribute specified in the option is not supported. Supported attributes are,

- *WIDTH*
- *WRONGDIRECTION*

**What Next**

Check if the attribute is specified correctly.

Feedback

## SX-3286

(Warning) Layer based NET_SEGMENT_CUT_LENGTH is no longer accepted. Please change your mapping file to reflect this change.

**Description**

Layer based NET_SEGMENT_CUT_LENGTH in the mapping file is obsolete in 2016.06 release.

**What Next**

Change your mapping file to avoid the warning.

## SX-3287

(Fatal) DEF cell '%s' specified in DEF_ATTRIBUTE_FROM_LEF must have only one LEF file.

**Description**

DEF_ATTRIBUTE_FROM_LEF supports the syntax as follows,

- DEF_ATTRIBUTE_FROM_LEF: WIDTH/WRONGDIRECTION <DEF cell> <LEF file>
- DEF_ATTRIBUTE_FROM_LEF: WIDTH <DEF cell 1> <LEF file> WRONGDIRECTION <DEF cell 2> <LEF file>

However, for a DEF cell, it is not allowed to have multiple LEF sources. For instance, the following specification is *not* allowed. DEF_ATTRIBUTE_FROM_LEF: WIDTH <DEF cell X> <LEF file> WRONGDIRECTION <DEF cell X> <LEF file>

Even though the LEF files are actually the same for DEF cell X.

**What Next**

Check if the attributes are specified correctly.

## SX-3288

(Warning) Gate layer %s and diff layer %s are in ignore_cap_layers, the coupling between them will be ignored.

**Description**

There is GATE_TO_LAYER_CAPS defined for layers mentioned above in ITF. But the gate layer and diff layer reported in the message is also under ignore_cap_layers in mapping file. StarRC will ignore the coupling between the reported gate layer and diff layer.

Feedback

**What Next**

If you want to ignore the coupling between the gate and diff layers, you don't need to do anything. If you want to extract the coupling between the gate and diff layers, remove them from ignore_cap_layers in mapping file.

---

## SX-3542

(Fatal) Incorrect Probe Text Coordinates: line %d: %s.

**Description**

StarXtract finds this incorrect probe text coordinates. The coordinates are too big or out of layout range.

**What Next**

fix the probe text coordinates.

---

## SX-3543

(Error) StarRC is not complete due to task error(s).

**Description**

StarRC has failed to complete successfully due to task error(s).

**What Next**

Check the errors shown in the summary file. Contact Synopsys Support Center if you need further help.

---

## SX-3544

(Fatal) Cannot use CCP_EXTRACTION_FILE and FS_QTF_FILE in the same case.

**Description**

StarRC command CCP_EXTRACTION_FILE and FS_QTF_FILE cannot be used at the same time.

**What Next**

Remove redundant syntax in StarRC command file

---

## SX-3545

(Fatal) The SDP feature is not supported during GPD conversion.

Feedback

**Description**

Use of the Synopsys Diagnostic Platform (SDP) is not supported during the conversion of a Galaxy Parasitic Database (GPD) to an ASCII netlist format.

**What Next**

Unset the SDP_ENABLED environment variable if set and run StarRC again.

---

## SX-3546

(Fatal) GPD converter usage error: %s

**Description**

Options specified to the Galaxy Parasitic Database (GPD) converter utility were incorrectly specified as indicated in the error message.

**What Next**

Specify the correct set of options for GPD conversion. Refer to the StarRC User Guide for more information.

---

## SX-3547

(Warning) No tech_file found inside GPD directory '%s', using default conversion options.

**Description**

The directory containing the Galaxy Parasitic Database (GPD) is expected to contain a file called tech_file, which is used to obtain options for converting the GPD to an ASCII netlist format.

The conversion proceeds with the default conversion options in a tech_file is not found inside the GPD directory.

**What Next**

Contact the Synopsys Support Center.

---

## SX-3548

(Warning) SNAP_RESISTOR_WIDTH is set to NO for NETLIST_TAIL_COMMENTS: NO.

**Description**

SNAP_RESISTOR_WIDTH:YES is to snap width to WMIN in the netlist tail comments, thus NETLIST_TAIL_COMMENTS needs to be set to YES.

Feedback

**What Next**

Set NETLIST_TAIL_COMMENTS to YES.

---

## SX-3550

(Fatal) When setting FSCOMPARE_TOTAL_COUPLING_THRESHOLD, COUPLING_ABS_THRESHOLD and COUPLING_REL_THRESHOLD should be set to 0.

**Description**

If using FSCOMPARE_TOTAL_COUPLING_THRESHOLD, COUPLING_ABS_THRESHOLD and COUPLING_REL_THRESHOLD should both be set to 0.

**What Next**

Set COUPLING_ABS_THRESHOLD and COUPLING_REL_THRESHOLD to 0 if you want to use FSCOMPARE_TOTAL_COUPLING_THRESHOLD.

---

## SX-3551

(Information) CapCompare: (Total Coupling) Avg %s ; StdDev %s ; Range %s to %s.

**Description**

This is an information message. It reports the summary of FSCOMPARE total coupling results, including average error, standard deviation and the range of the error.

**What Next**

You need to do nothing about this message.

---

## SX-3552

(Warning) Running StarRC in single corner mode will result in degraded runtime for design flows where multiple corners are used. For optimal runtime, add the necessary commands to configure the Simultaneous Multicorner flow (CORNERS_FILE and SELECTED_CORNERS).

---

## SX-3553

(Warning) Running StarRC in single corner mode will result in degraded runtime for design flows where multiple corners are used. For optimal runtime, add the necessary commands to configure the Simultaneous Multicorner flow (CORNERS_FILE and SELECTED_CORNERS) and remove the SIMULTANEOUS_MULTI_CORNER command.

Feedback

## SX-3554

(Fatal) When setting FSCOMPARE_TOTAL_COUPLING_THRESHOLD, POWER_EXTRACT should be set to NO.

**Description**

If using FSCOMPARE_TOTAL_COUPLING_THRESHOLD, POWER_EXTRACT should be set to NO.

**What Next**

Set POWER_EXTRACT to NO if you want to use FSCOMPARE_TOTAL_COUPLING_THRESHOLD.

## SX-3555

(Warning) When setting FSCOMPARE_TOTAL_COUPLING_THRESHOLD, if NETS is not set to *, fscomptotcoup report will not be generated.

**Description**

If using FSCOMPARE_TOTAL_COUPLING_THRESHOLD, NETS should also be set to *. If NETS is not set to *, fscomptotcoup report will not be generated, but fscomptot and fscompcoup reports will be generated as usual.

**What Next**

Set NETS to * if you want to generate fscomptotcoup report. For large design, you can select nets with FS_EXTRACT_NETS option.

## SX-3556

(Warning) Running StarRC with metal fill GDS files will result in degraded runtime. For optimal runtime, use the METAL_FILL_OASIS_FILE to include metal fills during extraction.

## SX-3560

(Fatal) When setting FSCOMPARE_TOTAL_COUPLING_THRESHOLD, COUPLE_TO_GROUND should be set to NO.

**Description**

If using FSCOMPARE_TOTAL_COUPLING_THRESHOLD, COUPLE_TO_GROUND should be set to NO.

Feedback

**What Next**

Set COUPLE_TO_GROUND to NO if you want to use FSCOMPARE_TOTAL_COUPLING_THRESHOLD.

---

## SX-3561

(Fatal) Transistor level design doesn't support FSCOMPARE_TOTAL_COUPLING_THRESHOLD.

**Description**

Only gate level design supports FSCOMPARE_TOTAL_COUPLING_THRESHOLD now.

**What Next**

If you are running a transistor level design, remove FSCOMPARE_TOTAL_COUPLING_THRESHOLD from StarRC command file.

---

## SX-3562

(Warning) Could not find any net for wildcard set '%s' in RETAIN_FLOATING_NETS.

**Description**

Could not find any net for wildcard set %s in RETAIN_FLOATING_NETS. The nets in RETAIN_FLOATING_NETS should be a floating net.

**What Next**

You should double check your case and revise the wildcard in RETAIN_FLOATING_NETS.

---

## SX-3563

(Warning) Could not find net '%s' in RETAIN_FLOATING_NETS.

**Description**

Could not find net %s in RETAIN_FLOATING_NETS. The nets in RETAIN_FLOATING_NETS should be a floating net.

**What Next**

You should double check your case and revise net name in RETAIN_FLOATING_NETS.

Feedback

## SX-3564

(Warning) StarRC's runtime could have been affected by adverse conditions in the runtime environment, see sml.sum in the STAR directory for more information.

### Description

StarRC's runtime can be affected by adverse conditions in the runtime environment such as high CPU loads, lack of sufficient memory on a host, network degradation and so on.

Diagnostic information about adverse conditions detected during a run is collected in the sml.sum file in the STAR directory.

### What Next

Users can refer to the sml.sum file and take remedial actions as necessary, such as running on a host with less load or hosts with higher memory.

Refer to the man pages for messages in the sml.sum file for additional information.

## SX-3565

(Warning) Galaxy Parasitic Data (GPD) does not support the CCP flow.

### Description

Galaxy Parasitic Data (GPD) does not support the CCP flow. StarRC will run the non-GPD flow.

### What Next

Remove the commands for the CCP flow to run StarRC with GPD netlisting or the non-GPD flow will run.

## SX-3567

(Warning) Galaxy Parasitic Data (GPD) only supports the RC, C or FSCOMPARE extraction modes for transistor level extraction.

### Description

Galaxy Parasitic Data (GPD) only supports the RC, C or FSCOMPARE extraction modes for transistor level extraction. StarRC will run the non-GPD flow.

### What Next

Set EXTRACTION: RC, C or FSCOMPARE to run StarRC with GPD netlisting or the non-GPD flow will run.

Feedback

## SX-3568

(Warning) Galaxy Parasitic Data (GPD) only supports POWER_EXTRACT: NO/YES

**Description**

Galaxy Parasitic Data (GPD) only supports POWER_EXTRACT: NO/YES

**What Next**

Set POWER_EXTRACT to YES/NO to run StarRC with GPD netlisting or the non-GPD flow will run.

## SX-3569

(Warning) Galaxy Parasitic Data (GPD) does not support the 3D IC flow.

**Description**

Galaxy Parasitic Data (GPD) does not support the 3D IC flow. StarRC will run the non-GPD flow.

**What Next**

Remove the 3D_IC command to run StarRC with GPD netlisting or the non-GPD flow will run.

## SX-3572

(Warning) Galaxy Parasitic Data (GPD) does not support NEW_IP_HANDLING: YES

**Description**

Galaxy Parasitic Data (GPD) does not support NEW_IP_HANDLING: YES

**What Next**

Remove the NEW_IP_HANDLING: YES command to run StarRC with GPD netlisting or the non-GPD flow will run.

## SX-3573

(Warning) Galaxy Parasitic Data (GPD) does not support the %s command

**Description**

Galaxy Parasitic Data (GPD) does not support the highlighted commmand

Feedback

**What Next**

Remove/change the highlighted command to run the GPD flow or the non-GPD flow will run.

---

## SX-3574

(Warning) Galaxy Parasitic Data (GPD) only supports SPEF/SPF formats.

**Description**

Galaxy Parasitic Data (GPD) only supports SPEF or SPF formats.

**What Next**

Set NETLIST_FORMAT: SPEF or SPF if you intend to run StarRC with GPD enabled.

---

## SX-3575

(Warning) Galaxy Parasitic Data (GPD) does not support CLOCK_NET_INDUCTANCE command

**Description**

Galaxy Parasitic Data (GPD) does not support CLOCK_NET_INDUCTANCE command.

**What Next**

Remove CLOCK_NET_INDUCTANCE if you intend to run StarRC with GPD netlisting.

---

## SX-3576

(Warning) Galaxy Parasitic Data (GPD) INSTANCE_PORT does not suuport option %s

**Description**

Galaxy Parasitic Data (GPD) INSTANCE_PORT does not support options EXPLODE, NOT_CONDUCTIVE, SUFFIXED, or MULTIPLE.

**What Next**

Remove unsupported option if you intend to run StarRC with GPD netlisting.

---

## SX-3577

(Warning) Galaxy Parasitic Data (GPD) does not support the transistor level extraction flow.

Feedback

**Description**

Galaxy Parasitic Data (GPD) only does not support the transistor level extraction flow.

**What Next**

None at your side.

---

## SX-3578

(Warning) Galaxy Parasitic Data (GPD) does not support the TEMPERATURE_SENSITIVITY extraction flow.

**Description**

Galaxy Parasitic Data (GPD) only does not support the TEMPERATURE_SENSITIVITY extraction flow.

**What Next**

None at your side.

---

## SX-3579

(Warning) Galaxy Parasitic Data (GPD) does not support one or more StarRC command options.

**Description**

Galaxy Parasitic Data (GPD) does not support one or more StarRC command options.

**What Next**

None at your side.

---

## SX-3580

(Fatal) NETLIST_FORMAT: GPD is not valid.

**Description**

NETLIST_FORMAT: GPD is not valid.

**What Next**

Set NETLIST_FORMAT to a different option and try to run StarRC again.

Feedback

## SX-3581

(Fatal) Commands -cleanT, -cleanXREF, -cleanX, -cleanFS, -cleanTFS, and -cleanXFS are obsoleted starting 2016.06 version.

**Description**

The highlighted commands are obsoleted starting 2016.06 version.

**What Next**

Please remove the obsoleted command and try again.

## SX-3582

(Warning) Command -clean is obsoleted for GATE level extraction and turned on by default.

**Description**

The -clean command is obsoleted for GATE level extraction and turned on by default.

**What Next**

This is a warning message for information purpose. The program will not abort.

## SX-3583

(Fatal) Command -cleanD is obsoleted starting 2016.06 version.

**Description**

The -cleanD command is obsoleted starting 2016.06 version.

**What Next**

Please remove the obsoleted command and try again.

## SX-3584

(Fatal) Command -cleanN is obsoleted for the GATE level extraction starting 2016.06 version.

**Description**

The -cleanN command for the GATE level extraction is obsoleted starting 2016.06 version.

Feedback

**What Next**

Please remove the obsoleted command and try again.

---

## SX-3585

(Warning) Netlist format %s is deprecated and will be obsoleted in the 2016.06 version.

**Description**

The highlighted netlist format is deprecated and will be obsoleted in the 2016.06 version.

**What Next**

This is a warning message for information purpose. The program will not abort.

---

## SX-3586

(Fatal) Netlist format %s is obsoleted.

**Description**

The highlighted netlist format is obsoleted.

**What Next**

Change the NETLIST_FORMAT setting and try again.

---

## SX-3587

(Fatal) Command CAPACITOR_TAIL_COMMENTS:YES can only be used in Calibre-StarRC or ICV-StarRC flow

**Description**

When command CAPACITOR_TAIL_COMMENTS: YES is used, you must use CCI or ICV output as StarRC input. Otherwise, StarRC will give this error message.

**What Next**

If you are not running a Calibre-StarRC or ICV-StarRC flow, remove this command from StarRC command file.

---

## SX-3588

(Fatal) Command CAPACITOR_TAIL_COMMENTS:YES can only be used when output netlist format is SPF/STAR/NETNAME

Feedback

### Description

Command CAPACITOR_TAIL_COMMENTS: YES only supports SPF/STAR/NETNAME formats. If other netlist format is specified, such as SPEF, StarRC will give this error message.

### What Next

You should change NETLIST_FORMAT to one of the supported formats, or remove this command from StarRC command file.

---

## SX-3589

(Fatal) Command CAPACITOR_TAIL_COMMENTS:YES can not be used in Field Solver flow except for FS_EXTRACT_NETS.

### Description

Command CAPACITOR_TAIL_COMMENTS:YES doesn't support Field Solver flow except for FS_EXTRACT_NETS. StarRC will give this error message when you run Field Solver flow with this command.

### What Next

If you want to run Field Solver flow, remove this command from StarRC command file.

---

## SX-3590

(Fatal) Only REDUCTION:NO|LAYER|LAYER_NO_EXTRA_LOOPS can be set when CAPACITOR_TAIL_COMMENTS:YES is used in a RC extraction run

### Description

When command CAPACITOR_TAIL_COMMENTS: YES is used in a RC extraction run (EXTRACTION: RC), you can only use REDUCTION:NO|LAYER| LAYER_NO_EXTRA_LOOPS as reduction setting. If other option is used, StarRC will give this error message.

### What Next

If you want to use command CAPACITOR_TAIL_COMMENTS: YES, change reduction setting to one of the supported options. If you don't need that command, remove it from command file.

Feedback

## SX-3591

(Information) [STAR_MAIN_SUMMARY] Found '%s' percent ECO net change for this iteration and totally '%s' percent ECO net change accumulated since last full chip extraction. Performing incremental extraction.

**Description**

When number of net changes is not large, incremental extraction is performed.

**What Next**

This message is for information only, no further action is needed.

## SX-3592

(Information) [STAR_MAIN_SUMMARY] Found '%s' percent ECO net change for this iteration and totally '%s' percent ECO net change accumulated since last full chip extraction. Performing full extraction.

**Description**

When number of net changes is large, a full extraction, instead of incremental extraction, is performed.

**What Next**

This message is for information only, no further action is needed.

## SX-3593

(Fatal) %s name '%s' is longer than %d characters.

**Description**

The length of the name string exceeds the limit.

**What Next**

Reduce the length of the string.

## SX-3595

(Information) [STAR_MAIN_SUMMARY] TCAD_GRD_FILE (nxtgrd file) generated by >4 years older version than the StarRC version is used for this extraction.

Feedback

**Description**

When the TCAD_GRD_FILE was generated by >4 years older version than StarRC version currently you run. This may or may not impact RC extraction accuracy depending on process being used.

**What Next**

Update TCAD_GRD_FILE to within 4 year of the StarRC version you currently run, or contact foundry for information

---

## SX-3596

(Fatal) Simultaneous multi-corner TC1/TC2 output only supports transistor level flow.

**Description**

Simultaneous multi-corner TC1/TC2 output only supports transistor level flow.

---

## SX-3597

(Warning) TEMPERATURE_SENSITIVITY will not be supported at gate-level flow. Switch to SIMULTANEOUS_MULTI_CORNER flow. TEMPERATURE_SENSITIVITY will be ignored in future release.

**Description**

TEMPERATURE_SENSITIVITY for outputting multiple netlist can be supported through SIMULTANEOUS_MULTI_CORNER flow. Single output with temperature coefficient is not supported at gate-level is not supported at gate-level is not supported at gate-level is not supported at gate-level.

---

## SX-3598

(Fatal) The setting of NETLIST_TAIL_COMMENTS:ADVANCED is obsolete since 2016.06.

**Description**

Since 2015.12, NETLIST_TAIL_COMMENTS: ADVANCED is deprecated. $m is printed in the tail comments on colored layer when NETLIST_TAIL_COMMENTS: YES. ITF layers are always printed in the LAYER_MAP section of the netlist file. It is obsolete since 2016.06

**What Next**

Set NETLIST_TAIL_COMMENTS: YES

Feedback

## SX-3599

(Fatal) The %s run is GPD which does not support %s command.

**Description**

GPD does not support -clean[N|X|T] command.

**What Next**

Remove -clean[N|X|T] from the command and try again.

## SX-3601

(Warning) Length dependent LATERAL_CAP_SCALING_VS_SI_SPACING will not be supported in 2016.06 and be replaced by DIELECTRIC_FILL_EMULATION_VS_SI_SPACING.

**Description**

Length dependent LATERAL_CAP_SCALING_VS_SI_SPACING is present in ITF. It will be be supported in 2016.06.

**What Next**

Consider using DIELECTRIC_FILL_EMULATION_VS_SI_SPACING.

## SX-3602

(Warning) Option 'NO' for TRANSLATE_RETAIN_BULK_LAYERS will become obsolete from the next release. Recommend to use 'CONLY'.

**Description**

Option 'NO' for TRANSLATE_RETAIN_BULK_LAYERS will become obsolete from the next release. Recommend to use 'CONLY'.

**What Next**

Consider using TRANSLATE_RETAIN_BULK_LAYERS: CONLY.

## SX-3603

(Fatal) The ITF_FILE option is supported in transistor-level and wafer-on-wafer extraction flows only.

Feedback

**Description**

The direct ITF flow with the ITF_FILE option is supported for transistor-level and wafer-on-wafer extraction flows only. It is not supported in general gate-level flows.

**What Next**

Use StarRC version 2015.06 or earlier to run with the ITF_FILE option in the gate-level flow.

---

## SX-3604

(Warning) Database resolution (dbu = %d) finer than 0.1 nm. Converting resolution to 0.1 nm (dbu = 10000) for extraction.

**Description**

The maximum resolution for extraction is 0.1 nm.

**What Next**

Use a database unit less than or equal to 10000.

---

## SX-3605

(Warning) Galaxy Parasitic Data (GPD) does not support NETLIST_COMPRESS_COMMAND and NETLIST_POSTPROCESS_COMMAND to be used together.

**Description**

Galaxy Parasitic Data (GPD) does not support the highlighted 2 commmands to be used together.

**What Next**

Remove/change one of the highlighted commands to run the GPD flow or the non-GPD flow will run.

---

## SX-3606

(Warning) Could not load netlist options from GPD (Galaxy Parasitic Data) directory "%s".

**Description**

When standalone GPD (Galaxy Parasitic Data) converter can't load netlist options stored in GPD directory, StarRC will give this warning.

Feedback

**What Next**

Check whether the specified GPD directory is correct and complete.

---

## SX-3607

(Warning) EM_DISABLE_REFINED_TRENCH_CONTACT_SEGMENTATION is only supported under REDUCTION:NO.

**Description**

EM_DISABLE_REFINED_TRENCH_CONTACT_SEGMENTATION is used to turn off the additional segmentation at trench contact ending segement. It is only supported under REDUCTION:NO for EM analysis purpose. This option is disabled if REDUCTION is set other than NO.

**What Next**

Change reduction to NO if EM_DISABLE_REFINED_TRENCH_CONTACT_SEGMENTATION is intended to be used for EM flow

---

## SX-3608

(Fatal) The bounding box defined in the cell_extents_file in this design is too large(over 100mm by 100mm). StarRC can not extract this kind of large design.

**Description**

The bounding box defined in the cell_extents_file is too large, and over 100mm by 100mm. It is not a resonable design bounding box.

**What Next**

You should check the cell_extents_file in the ICV_RUNSET_REPORT_FILE, and changed to a reasonable bounding box.

---

## SX-3609

(Fatal) HIERARCHICAL_COUPLING_REPORT_FILE has the same name as the netlist file.

**Description**

The value for the HIERARCHICAL_COUPLING_REPORT_FILE option in the StarRC command file is identical to the value of the NETLIST_FILE option.

Feedback

**What Next**

Modify the StarRC command file to use different values for the HIERARCHICAL_COUPLING_REPORT_FILE and NETLIST_FILE options.

---

## SX-3610

(Fatal) HIERARCHICAL_COUPLING_REPORT_FILE has the same name as COUPLING_REPORT_FILE.

**Description**

The value for the HIERARCHICAL_COUPLING_REPORT_FILE option in the StarRC command file is identical to the value of the COUPLING_REPORT_FILE option.

**What Next**

Modify the StarRC command file to use different values for the HIERARCHICAL_COUPLING_REPORT_FILE and COUPLING_REPORT_FILE options.

---

## SX-3611

(Fatal) HIERARCHICAL_COUPLING_REPORT_NUMBER must be greater than zero.

**Description**

This error message informs you that HIERARCHICAL_COUPLING_REPORT_NUMBER is set to a number less than zero.

**What Next**

Reset HIERARCHICAL_COUPLING_REPORT_NUMBER to an integer greater than zero.

---

## SX-3612

(Warning) HIERARCHICAL_COUPLING_REPORT_FILE command is designed only for gate level cases, and thus, is voided for transistor level cases.

**Description**

HIERARCHICAL_COUPLING_REPORT_FILE command is only designed for gate level cases. For transistor level cases, it will be voided and will not be effective.

**What Next**

If it is needed to produce skip cell coupling report files for transistor level cases, please contact StarRC team.

Feedback

## SX-3613

(Fatal) Invalid value for NFS_COHERENCY_DELAY %d, expected a value greater than zero and less than 200.

### Description

The NFS_COHERENCY_DELAY option is used to add delays in StarRC's distributed processing schedule to account for lag times in NFS coherency. The expected value (in seconds) must be greater than zero and less than 200.

### What Next

Change the value of this option to be in the specified range and rerun StarRC.

## SX-3614

(Warning) TRANSLATE_RETAIN_BULK_LAYERS will be ignored since it is not compatible with RVTV flow

### Description

StarRC command TRANSLATE_RETAIN_BULK_LAYERS will be ignored for processes using RVTV conductors, because such processes does not have BULK.

### What Next

Remove TRANSLATE_RETAIN_BULK_LAYERS from the StarRC command file

## SX-3615

(Warning) Use of the NFS_COHERENCY_DELAY option can lead to excessively long run times. Use this option only when coherency issues are observed in the network file system (NFS).

### Description

The NFS_COHERENCY_DELAY option is used to add delays in StarRC's distributed processing schedule to account for lag times in NFS coherency. Setting this option can lead to increases in run time, therefore this option is to be used only if file coherency delays in NFS are suspected.

One manifestation of a file coherency delay is when StarRC errors out with an error message that a file cannot be opened for reading when the file is actually found to exist.

Feedback

**What Next**

Remove this option from the StarRC command file if you are not experiencing coherency delays with NFS. Contact your system administrator for more information.

---

## SX-3617

(Fatal) POWER_EXTRACT:YES NONBULK_LAYERS doesn't work with LEFDEF flow.

**Description**

POWER_EXTRACT:YES NONBULK_LAYERS doesn't work in LEFDEF flow. StarRC gives this error once it finds the input database if LEFDEF but POWER_EXTRACT:YES NONBULK_LAYERS is used.

**What Next**

You should use other options for command POWER_EXTRACT in LEFDEF flow.

---

## SX-3618

(Fatal) POWER_EXTRACT:YES NONBULK_LAYERS doesn't work with Milkyway flow.

**Description**

POWER_EXTRACT:YES NONBULK_LAYERS doesn't work in Milkyway flow. StarRC gives this error once it finds the input database if Milkyway but POWER_EXTRACT:YES NONBULK_LAYERS is used.

**What Next**

You should use other options for command POWER_EXTRACT in Milkyway flow.

---

## SX-3619

(Fatal) Value of the INTRANET_CAPS option can be set to YES for transistor-level extractions only.

**Description**

Intranet capacitance is only applicable for high frequency simulation in analog / RF applications. It is not needed (or appropriate) for downstream analyses performed for gate-level designs such as timing or rail analysis.

**What Next**

Set the value of INTRANET_CAPS to NO in the StarRC command file.

Feedback

## SX-3620

(Warning) MILKYWAY_SHOW_CELL_INFO_DETAIL is a DEBUG ONLY command, subject to user interface and functionality change in future. Please avoid using this command in production flow.

### Description

MILKYWAY_SHOW_CELL_INFO_DETAIL is a DEBUG ONLY command, subject to user interface and functionality change in future. Please avoid using this command in production flow.

### What Next

Information only; no action is needed.

## SX-3621

(Fatal) MOS_GATE_NON_NEGATIVE_DELTA_RESISTANCE: YES and MOS_GATE_DELTA_RESISTANCE_LAYERS can not be used together. Only one of Rg modeling methods can be used.

### Description

MOS_GATE_DELTA_RESISTANCE/MOS_GATE_DELTA_RESISTANCE_LAYERS are used to extract gate resistance in delta topology (with negative resistors), and MOS_GATE_NON_NEGATIVE_DELTA_RESISTANCE is used to extract gate resistance without negative resistors but equal to delta topology. You should only use one setting for gate resistance extraction. They can't be used togther.

### What Next

Remove one of these 2 commands from StarRC command file.

## SX-3622

(Warning) Layer %s listed in MOS_GATE_DELTA_RESISTANCE_LAYERS is not mapped to an ITF layer with LAYER_TYPE=GATE. No delta resistance modifiaction will be applied on this layer.

### Description

Delta resistance modification is only allowed on gate layers. When you use MOS_GATE_DELTA_RESISTANCE_LAYERS to list layers to apply delta resistance modification, all the listed layers should be used by gate terminal and mapped to an ITF layer with LAYER_TYPE=GATE. If StarRC finds some layers in this command are NOT

Feedback

mapped to an ITF layer with LAYER_TYPE=GATE, StarRC will give this warning and doesn't do delta resistance modification on this layer.

**What Next**

You should map the reported layer to an ITF layer with LAYER_TYPE=GATE in StarRC mapping file. Or remove it from command MOS_GATE_DELTA_RESISTANCE_LAYERS if you don't want to have delta resistance modification on this layer.

---

## SX-3623

(Fatal) INCR_MAX_TCAP_ERROR_PCT: <value>, value must locate inside [0, 50].

**Description**

INCR_MAX_TCAP_ERROR_PCT is used in incremental extraction. It defines the threshold of relative tcap error in percentage for identifying eco affected nets. A ratio is calculated by summation of changed coupling caps for one net before and after eco divided by the pre-eco total cap of the net. If the ratio is less than INCR_MAX_TCAP_ERROR_PCT, the net is skipped as an eco affected net.

**What Next**

Change the value inside [0, 50]

---

## SX-3624

(Warning) HIERARCHICAL_COUPLING_REPORT_FILE is not compatible with COUPLE_TO_GROUND = YES, and thus, is voided.

**Description**

HIERARCHICAL_COUPLING_REPORT_FILE command is only compatible with COUPLE_TO_GROUND = NO. For COUPLE_TO_GROUND = YES, HIERARCHICAL_COUPLING_REPORT_FILE will be voided and will not be effective.

**What Next**

If it is needed to produce skip cell coupling report files for COUPLE_TO_GROUND = YES, please contact StarRC team.

---

## SX-3625

(Warning) The incremental NDM is not found. StarRC will detect ECO changes automatically.

Feedback

**Description**

There is no incremental NDM design as known as scratch view design in NDM library directory. StarRC performs XOR based incremental extraction flow to detect ECO changes by itself.

**What Next**

See StarRC user guide.

---

## SX-3626

(Information) The incremental NDM is found. StarRC will read incremental NDM directly.

**Description**

There is incremental NDM design as known as scratch view design in NDM library directory. StarRC performs fast incremental extraction flow by NDM ECO information.

**What Next**

See StarRC user guide.

---

## SX-3627

(Warning) Unable to unlock file '%s', error status '%s', host '%s', working directory '%s'

**Description**

StarRC is unable to unlock a file because of issues in the runtime environment.

**What Next**

This warning is benign unless a StarRC run with distributed processing fails in which case this message may indicate the source of the problem.

---

## SX-3628

(Warning) HIERARCHICAL_COUPLING_REPORT_FILE does not yet support NDM input, and thus, is voided.

**Description**

HIERARCHICAL_COUPLING_REPORT_FILE command is not compatible with NDM input yet. For NDM cases, HIERARCHICAL_COUPLING_REPORT_FILE will be voided and will not be effective.

Feedback

**What Next**

NDM support is already planned for HIERARCHICAL_COUPLING_REPORT_FILE.

---

## SX-3629

(Error) Invalid line '%s', line number = %d in input file '%s'.

**Description**

The line in input file is not valid based on file format.

**What Next**

Fix the input file based on the StarRC user guide and run again.

---

## SX-3630

(Warning) Incorrect silicon marker description Line: %s

**Description**

Each line in silicon_marker_layers section in StarRC mapping file should contain the silicon marker layer name and the layer it marks for. Any other description is invalid and will trigger this warning.

**What Next**

You should list the silicon marker layer and the layer it marks for.

---

## SX-3631

(Error) Option '%s' is not supported in FS flow (FSCOMPARE or FS_EXTRACT_NETS)

**Description**

This option is not supported in FS flow.

**What Next**

Remove the option and run again.

---

## SX-3633

(Fatal) TRANSLATE_RETAIN_BULK_LAYERS:CONLY is not supported for gate level flow

**Description**

TRANSLATE_RETAIN_BULK_LAYERS:CONLY is not supported for gate level flow

Feedback

**What Next**

Please remove command TRANSLATE_RETAIN_BULK_LAYERS

---

## SX-3634

(Fatal) Layer '%s' in silicon_marker_layers section in MAPPING_FILE does not have RPV_VS_AREA, RPV_VS_WIDTH_AND_LENGTH, RPV_VS_SI_WIDTH_AND_LENGTH or RPV_VS_SI_WIDTH_AND_LENGTH_AND_EXTENSION table.

**Description**

This layer apeared in silicon_marker_layers section in MAPPING FILE. The silicon marker layer can only be mapped to a fake via layer with RPV_VS_AREA, RPV_VS_WIDTH_AND_LENGTH, RPV_VS_SI_WIDTH_AND_LENGTH or RPV_VS_SI_WIDTH_AND_LENGTH_AND_EXTENSION table specified.

**What Next**

Specify RPV_VS_AREA, RPV_VS_WIDTH_AND_LENGTH, RPV_VS_SI_WIDTH_AND_LENGTH or RPV_VS_SI_WIDTH_AND_LENGTH_AND_EXTENSION table on this layer in the ITF file.

---

## SX-3635

(Fatal) GPD comand must be set for ECO_MODE YES or RESET

**Description**

GPD comand must be set for ECO_MODE YES or RESET

**What Next**

Please add GPD to StarRC command file

---

## SX-3636

(Information) StarXtract process for core %d terminated early because the number of cores used is reverted to %d.

**Description**

When the optimal number of partitions determined by StarRC is less than the value of NUM_CORES specified in the StarRC command file, the value of NUM_CORES is internally changed to match the number of partitions required. When this happens unneeded processes are terminated early to conserve compute resources.

Feedback

For example, if a user launches a job with NUM_CORES: 8 and the actual number of cores used is reverted to 3, processes for cores 4 through 8 will be terminated early.

**What Next**

This message is for information only. No action is necessary, but users can change the NUM_CORES value in the StarRC command file to the number of cores used as reported by this message to prevent StarRC from producing this message in subsequent runs.

---

## SX-3637

(Fatal) DIRECT_GPD cannot be set to 0 for ECO_MODE YES or RESET

**Description**

DIRECT_GPD cannot be set to 0 for ECO_MODE YES or RESET

**What Next**

Please set DIRECT_GPD to 1 or don't set DIRECT_GPD in StarRC command file

---

## SX-3638

(Warning) Galaxy Parasitic Data (GPD) does not support one or more StarRC command options. Extraction is reverted to Xout mode.

**Description**

Galaxy Parasitic Data (GPD) does not support one or more StarRC command options. Extraction is reverted to Xout mode.

**What Next**

None at your side.

---

## SX-3639

(Warning) ECO mode requires Galaxy Parasitic Data (GPD). Switching to ECO_MODE:NO full extraction.

**Description**

Incremental (ECO) extraction requires Galaxy Parasitic Data (GPD). Switching to ECO_MODE:NO full extraction.

**What Next**

None at your side.

Feedback

## SX-3640

(Warning) Galaxy Parasitic Data (GPD) does not support pipeline NETLIST_COMPRESS_COMMAND.

**Description**

Galaxy Parasitic Data (GPD) does not support pipeline NETLIST_COMPRESS_COMMAND.

**What Next**

Change NETLIST_COMPRESS_COMMAND to run the GPD flow or the non-GPD flow will run.

## SX-3641

(Warning) ENHANCED_SHORT_REPORTING is not supported for transistor level flow.

**Description**

ENHANCED_SHORT_REPORTING which is used to report shorts between signal and blockage, skip cell, unselectable, non-selected net is currently not supported for transistor level flow.

**What Next**

This is a warning message, nothing needs to be done from the user side. We will enhance the option to be supported for transistor level flow in future StarRC release.

## SX-3642

(Warning) StarXtract process for core %d terminated early because its parent process died.

**Description**

This message is produced when a worker process in StarRC's automatic distributed processing feature (enabled with STARRC_DP_STRING) terminates without killing all of its descendant processes. This usually indicates issues with the host on which the process was running.

Under these circumstances, the child process of the worker terminates itself and its descendants to prevent that StarRC process from continuing under abnormal environmental conditions.

Feedback

**What Next**

This message is for information only. A StarRC run may complete successfully even if this warning appears in the summary file because of in-built fault tolerance mechanisms.

---

## SX-3643

(Warning) SNAP_RESISTOR_WIDTH is set to NO because DETECT_FUSE: YES.

**Description**

SNAP_RESISTOR_WIDTH and DETECT_FUSE can not be used together.

**What Next**

Turn off SNAP_RESISTOR_WIDTH.

---

## SX-3644

(Fatal) The StarRC command ENHANCED_SHORT_REPORTING set to COMPLETE is not supported with the StarRC command %s.

**Description**

The StarRC command ENHANCED_SHORT_REPORTING set to COMPLETE is not supported with the StarRC command SKIP_CELLS_COUPLE_TO_NET or COUPLE_NONCRITICAL_NETS.

**What Next**

Remove the StarRC command SKIP_CELLS_COUPLE_TO_NET/ COUPLE_NONCRITICAL_NETS or set the StarRC command ENHANCED_SHORT_REPORTING to YES

---

## SX-3645

(fatal) DPT:SHIFT is discontinued. Please remove it from the command file and restart extraction.

**Description**

Stand alone DPT:SHIFT is discontinued from 2016.12 release. DPT:SHIFT will function only with FS_EXTRACT_NETS.

**What Next**

Either (1) remove the DPT:SHIFT command or (2) add FS_EXTRACT_NETS.

Feedback

## SX-3646

(Warning) DPT_MAX_SHIFT in ITF is not supported.

**Description**

DPT_MAX_SHIFT is not supported. The keyword DPT_MAX_SHIFT in the ITF (nxtgrd) file will have no effect on the extraction results.

**What Next**

This message is for information only. A StarRC run may complete successfully even if this warning appears in the summary file because of in-built fault tolerance mechanisms.

## SX-3647

(Fatal) Could not find executable for command '%s' used in %s

**Description**

The executable for the command specified to launch multiple StarRC jobs for distributed processing could not be found. The command should either be an absolute path to an existing executable or an executable with the given name should be found in the directories listed in the user's PATH environment variable.

**What Next**

Specify the correct path to an existing executable or modify the PATH environment variable so that the command is found in one of its component directories. Consult your system administrator for more information.

## SX-3648

(Fatal) Layer '%s' in silicon_marker_layers section in MAPPING_FILE does not map to a trench contact fake via layer.

**Description**

This layer apeared in silicon_marker_layers section in MAPPING FILE. The silicon marker layer can only be mapped to a trench contact fake via layer.

**What Next**

Modify the mapping file.

## SX-3649

(Warning) Unable to open res-excluded net file %s - ignore res-excluded nets

Feedback

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-3650

(Warning) Unable to open cap-excluded net file %s - ignore cap-excluded nets

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-3651

(Fatal) Fail to open file %s.

**Description**

StarRC is unable to open an input file.

**What Next**

Check the input file and try again

---

## SX-3652

(Fatal) Invalid line ' %s '.

**Description**

Invalid line in netlist file

**What Next**

Check the netlist file and try again

---

## SX-3653

(Warning) Huge resistor (%EOhm) found in net %s. P2Ps of this net might be incorrect.

Feedback

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-3654

(Warning) Unable to open rcNet file %s - ignore selected RC nets

**Description**

Not available.

**What Next**

Contact the Synopsys Support Center.

---

## SX-3655

(Fatal) CCP flow does not support SIMULTANEOUS_MULTI_CORNER(SMC) with Temperature Sensitivity.

**Description**

CCP flow does not support SIMULTANEOUS_MULTI_CORNER(SMC) with Temperature Sensitivity.

**What Next**

Use single-corner CCP flow to output Temperature Sensitivity.

---

## SX-3656

(Fatal) NETLIST_SMC_FORMULA cannot be used with Temperature Sensitivity.

**Description**

Temperature Sensitivity flow does not support NETLIST_SMC_FORMULA.

---

## SX-3657

(Warning) Connectivity of VIA layer "%s" is incorrectly assigned. Check Technology (e.g. LEF) or Mapping file.

Feedback

**Description**

The layer DB in the current configuration has led to wrong connectivity in the corresponding VIA layer. Check and see if the Tech file (e.g., LEF) or the corresponding mapping file is directing to the correct layers.

**What Next**

You should check your technology files (e.g., LEF) and mapping files and see if those files are designed correctly.

---

## SX-3658

(Warning) ECO_MODE does not support NETLIST_FORMAT: SPF. Switching to ECO_MODE:NO full extraction.

**Description**

Incremental (ECO) extraction does not support SPF format output. Switching to ECO_MODE:NO full extraction.

For more information on ECO extraction, see the StarRC user guide.

**What Next**

Modify the command file to use NETLIST_FORMAT: SPEF or NETLIST_FORMAT: GPD.

---

## SX-3659

(Fatal) The layer mapped to silicon marker layer '%s' is not defined in the conductor layer section in MAPPING_FILE

**Description**

The silicon_marker_layers should be mapped to the real database layer. This real database layer should be defined in the mapping file and input database.

**What Next**

1.You should add this database layer conductor_layers section. 2.You should make sure this database layer defined in the input database.

---

## SX-3660

(Fatal) The ITF layer '%s' defined in the silicon marker layer '%s' is not validate in the TCAD_GRD_FILE.

Feedback

**Description**

The ITF layer defined in the silicon_marker_layers "ITF=" section should be a valid ITF layer. This ITF layer should be defined in the TCAD_GRD_FILE.

**What Next**

You should make sure the ITF layer name is correct and defined in the TCAD_GRD_FILE.

---

## SX-3661

(Warning) Galaxy Parasitic Data (GPD) does not support StarRC command '%s'. Extraction is reverted to Xout mode.

**Description**

Galaxy Parasitic Data (GPD) does not support one or more StarRC command options. Extraction is reverted to Xout mode.

**What Next**

Remove the command to run StarRC with GPD netlisting or Xout flow will run.

---

## SX-3662

(Warning) Internally setting the 3D_IC option to NO since nxtgrd file '%s' does not have a TSV specification.

**Description**

The 3D_IC extraction feature of StarRC can be used only when the nxtgrd file contains a TSV statement. The 3D_IC feature will be automatically disabled internally if the user's nxtgrd file does not contain a TSV statement.

**What Next**

If 3D_IC extraction is desired, use a nxtgrd file that contains a TSV statement. Otherwise, remove the 3D_IC: yes specification from the StarRC command file in order to prevent this warning from being reported.

Refer to the StarRC User Guide for more information.

---

## SX-3663

(Warning) Layer %d contains a ranged VIA model whose minimum size is not zero, or multiple VIA models including one or many ranged VIA models. ENHANCED_VIA_MODEL_ASSIGNMENT_FOR_COVERAGE is thus turned off, which might affect VIA coverage accuracy.

Feedback

**Description**

A ranged VIA model is a VIA model where Xrange and Yrange are presented by a range [min_size, max_size]. ENHANCED_VIA_MODEL_FOR_VIA_COVERAGE supports only one ranged VIA model in which min_size must be 0. Furthermore, if a layer contains a ranged VIA model, it cannot contain other VIA models.

**What Next**

Contact the Synopsys Support Center.

---

## SX-3664

(Fatal) The syntax '%s' is not supported in silicon_marker_layers

**Description**

The silicon_marker_layers only accept "ITF=", "MASK=". Any other field is not acceptable for silicon_marker_layers.

**What Next**

You should remove all the fields which are not acceptable for the silicon_marker_layers.

---

## SX-3665

(Warning) ECO_MODE does not support NETLIST_INSTANCE_SECTION: YES. Switching to ECO_MODE:NO full extraction.

**Description**

Incremental (ECO) extraction does not support NETLIST_INSTANCE_SECTION: YES (Output instance section information to GPD directory) Switching to ECO_MODE:NO full extraction.

For more information on ECO extraction, see the StarRC user guide.

**What Next**

Remove NETLIST_INSTANCE_SECTION: YES from StarRC command file if you want to run incremental extraction.

---

## SX-3666

(Warning) NETLIST_INSTANCE_SECTION:YES doesn't support command: %s.

Feedback

**Description**

NETLIST_INSTANCE_SECTION:YES (Output instance section information to GPD directory) doesn't support the command specified in this message.

**What Next**

If you only need to generate SPEF (no instance section) from the GPD, please remove NETLIST_INSTANCE_SECTION: YES from StarRC command file. If you need to generate SPF with instance section from the GPD, please remove the specified command.

---

## SX-3667

(Warning) This is %s flow. NDM_CELL_REPORT_FILE command will be ignored.

**Description**

The command NDM_CELL_REPORT_FILE is only used for NDM flow.

**What Next**

Since this command is only used to generate cell log in NDM flow. When the flow is not NDM, the StarXtract will ignore this command.

---

## SX-3668

(Warning) There are no nets defined in the NETS and SPECAILNETS sections of cell '%s'.

**Description**

No nets defined in the cell. The final netlist may be empty.

**What Next**

Check the design to see whether missing nets in NETS and SPECAILNETS is expected.

---

## SX-3669

(Warning) Setting SMART_OPENS_RCG_LIMIT to a value > 1000 can lead to very long runtimes.

**Description**

StarRC runtime during extraction and post-extraction can be very long if the SMART_OPENS_RCG_LIMIT option is set to a large value.

Feedback

**What Next**

Remove or comment out this option from the StarRC command file or set it to a value less than 1000.

---

## SX-3670

(WARNING) MOS_GATE_NON_NEGATIVE_DELTA_RESISTANCE_TAIL_COMMENT: YES can only be use with MOS_GATE_NON_NEGATIVE_DELTA_RESISTANCE: YES and NETLIST_TAIL_COMMENTS: YES, command reset to NO.

**Description**

MOS_GATE_NON_NEGATIVE_DELTA_RESISTANCE_TAIL_COMMENT is used to add "$gate_drv" in resistor tail comment to distinguish single ended model or double ended model, and can not be set alone. You should set MOS_GATE_NON_NEGATIVE_DELTA_RESISTANCE: YES and NETLIST_TAIL_COMMENT: YES.

**What Next**

Set two commands in StarRC command file.

---

## SX-3671

(Warning) The option EM_PARAM_MAPPING_FILE is not supported in gate level flows.

**Description**

Currently EM_PARAM_MAPPING_FILE is only supported in transistor level flows, not gate level flows.

**What Next**

If this is acceptable, no action is needed otherwise contact the Synopsys Support Center.

---

## SX-3673

(Warning) MERGE_VIAS_IN_ARRAY is forced to NO when via coverage is used.

**Description**

When via coverage is used, StarRC does not merge vias in via arrays.

**What Next**

If this is acceptable, no action is needed otherwise contact the Synopsys Support Center.

Feedback

## SX-3674

(Fatal) STARRC_COUPLES_TO_BASELAYERS option is supported only in FS-CCP flow with FS_EXTRACT_NETS:*

**Description**

To use STARRC_COUPLES_TO_BASELAYERS option, both CCP_EXTRACTION_FILE and FS_EXTRACT_NETS: * must be specified.

**What Next**

Specify CCP_EXTRACTION_FILE and FS_EXTRACT_NETS:*

## SX-3675

(Information) WIDE_DEVICE_TERM_RESISTANCE:RES requires RES device to separate terminal layers from regular routing layers for correct current direction detection and resistance extraction. Not separating RES device terminal layers can lead to inaccuracy of resistance extraction.

**Description**

WIDE_DEVICE_TERM_RESISTANCE:RES requires RES device to separate terminal layers from regular routing layers for correct current direction detection and resistance extraction. If not separated from routing layer, the RES device terminal may not have proper current direction detection, this can impact overall resistance extraction accuracy on such layer.resistance

**What Next**

Confirm the RES device definition in LVS rule file has separated terminal layers from regular routing layers. If not please separate it in LVS. Contact StarRC suppor team for confirming this operation.

## SX-3676

(Fatal) the layer operation line "%s" is not supported in ADDITIONAL_LAYER_GENERATION_FILE.

**Description**

The layer operation is not complete and the syntax is not correct in ADDITIONAL_LAYER_GENERATION_FILE.

**What Next**

You should revise the layer operation.

Feedback

## SX-3677

(Fatal) The "%s" option is not suppored in SIZE operation in ADDITIONAL_LAYER_GENERATION_FILE.

**Description**

SIZE operation in ADDITIONAL_LAYER_GENERATION_FILE only support "-by" "-direction".

**What Next**

You should revise the layer operation.

## SX-3678

(Fatal) The Layer "%s" in ADDITIONAL_LAYER_GENERATION_FILE is not a validate layer which should be mapped to a ITF layer.

**Description**

The operating layer should be a phyiscal layer and mapping to a ITF layer.

**What Next**

You should revise the layer operation.

## SX-3679

(Fatal) Inconsistent format of nxtgrd files specified with CORNERS_FILE and SELECTED_CORNERS commands. Please regenerate nxtgrd files for all the related corners with the same grdgenxo version.

**Description**

In SMC flow, the nxtgrd files for all the selected corners must be consistent and have the same format.

**What Next**

Regenerate nxtgrd files for all the related corners with the same grdgenxo version.

## SX-3681

(Warning) The power net names in the short summary when StarRC commands POWER_EXTRACT is set to NO and ENHANCED_SHORT_REPORTING is set to YES or COMPLETE, is not supported with the StarRC command '%s'.

Feedback

**Description**

The short summary shorts_all.sum reports the shorts with power nets as shorts with unselectable.

**What Next**

Nothing to do.

---

## SX-3682

(Fatal) SUBSTRATE_ISLAND_SQUARE_WIDTH can only be used with TRANSLATE_RETAIN_BULK_LAYERS:CONLY [ALL_TO_CLOSEST_CONTACT].

**Description**

SUBSTRATE_ISLAND_SQUARE_WIDTH can be only used with TRANSLATE_RETAIN_BULK_LAYERS:CONLY [ALL_TO_CLOSEST_CONTACT].

**What Next**

Use TRANSLATE_RETAIN_BULK_LAYERS:CONLY [ALL_TO_CLOSEST_CONTACT] to enable SUBSTRATE_ISLAND_SQUARE_WIDTH behavior.

---

## SX-3683

(Warning) The TSV mapping is undefined in tsvFrontBackMap, use first tsv layer id instead.

**Description**

To map TSV to TSV layer, we read information from tsvFrontBackMap. However, this mapping seems not exist

**What Next**

Check tsvFrontBackMap and tsv mapping

---

## SX-3684

(Fatal) Host %s supports both IPv4 and IPv6 internet protocol addressing. StarRC requires the user to resolve this ambiguity by setting the %s option. Set %s: NO to use IPv4 addressing and %s: YES to use IPv6 addressing. Consult your system administrator on which IP address mode to use.

Feedback

**Description**

Addresses of hosts on the Internet can be specified using either 32 bits (IPv4) or 128 bits (IPv6). Usually a host supports only one addressing mode. For hosts that support both addressing modes, StarRC needs to know which addressing mode to use for interprocess communication during distributed processing. Set the option given in the error message to YES to use IPv6 and NO to use IPv4.

**What Next**

Contact your IT department to get guidance on which IP addressing protocol should be used.

---

## SX-3685

(Fatal) Hosts specified in %s support different internet protocol addressing modes.

**Description**

Addresses of hosts on the Internet can be specified using either 32 bits (IPv4) or 128 bits (IPv6). All hosts specified in the DP string option should support the same type of addressing otherwise interprocess communication will fail.

**What Next**

Change the DP string option to specify hosts that have the same addressing mode. Contact your IT department to get information on the addressing modes supported by hosts in your compute environment.

---

## SX-3686

(Fatal) Wrong usage. Using single GLOBAL_TEMP corner with TEMPERATURE_SENSITIVITY:YES is not supported.

**Description**

Using single GLOBAL_TEMP corner with TEMPERATURE_SENSITIVITY:YES is not supported.

**What Next**

Use either single corner temperature sensitivity flow or replace the keyword GLOBAL_TEMP with a specific temperature in the corner definition to output TC1/TC2.

---

## SX-3687

(Fatal) Invalid host name '%s' specified in %s

Feedback

**Description**

One of the host names specified in the a DP string option is incorrect.

**What Next**

Verify that the host name is correct and rerun StarRC after updating the DP string specification in the StarRC command file.

---

## SX-3688

(Fatal) Cannot use %s: %s for hosts that support the %s addressing mode only.

**Description**

Addresses of hosts on the Internet can be specified using either 32 bits (IPv4) or 128 bits (IPv6). StarRC automatically detects the addressing mode on the submit host when the STARRC_DP_STRING option is used and selects the appropriate addressing mode internally. The ENABLE_IPV6 option should be used only to specify the desired addressing mode when a host supports both IPv4 and IPv6. Specifying a value for ENABLE_IPV6 that is in conflict with the addressing mode of the submit host will result in this error message.

**What Next**

Change the value of the ENABLE_IPV6 option based on information provided by the error message. If the submit host supports IPV6, set ENABLE_IPV6 to YES, otherwise set it to NO.

Unless the host supports both IPv4 and IPv6, the ENABLE_IPV6 option need not be specified. Consult your IT department for more information about the IP address modes supported in your environment.

---

## SX-3689

(Fatal) ACTIVE_FILL_FROM_NDM_DESIGN_VIEW_ONLY option is supported only with '%s'.

**Description**

To use ACTIVE_FILL_FROM_NDM_DESIGN_VIEW_ONLY option, required option must be specified.

**What Next**

Specify the required option.

Feedback

## SX-3690

(Fatal) ACTIVE_FILL_FROM_NDM_DESIGN_VIEW_ONLY option is not supported with '%s'.

**Description**

To use ACTIVE_FILL_FROM_NDM_DESIGN_VIEW_ONLY option, unsupported option must be removed.

**What Next**

Remove unsupported option.

## SX-3691

(Severe) INDESIGN_VIRTUAL_SHIELDING option is supported only with '%s'.

**Description**

To use INDESIGN_VIRTUAL_SHIELDING option, required option must be specified.

**What Next**

Specify the required option.

## SX-3692

(Warning) A %s operation on file %s on host %s, working directory %s failed with system error '%s', attempting operation again.

**Description**

A file locking operation failed as indicated in the warning. This usually happens because of intermittent issues with file locking on NFS. StarRC attempts the file locking operation multiple times until it can be completed successfully based on internal criteria. In some cases, StarRC may terminate with an error message if all these attempts fail.

**What Next**

Check with your system administrator if there are any issues with NFS for the host given in the warning message and the disk from which files are being accessed.

## SX-3693

(Information) IC Compiler II InDesign StarRC's Virtual shielding feature is invoked. Virtual shielding is not meant for final signoff flow because virtual shielding may not represent the actual final shielding in a design.

Feedback

**Description**

The command INDESIGN_VIRTUAL_SHIELDING is used internally by IC Compiler II. Later in the flow, and in particular for final signoff, real shielding must be used.

**What Next**

Refer to IC Compiler II documentation to see how to create real shielding shapes.

---

## SX-3694

(Fatal) The NETLIST_FORMAT: PARAMETERIZED_SPICE option can be used only in the GPD flow.

**Description**

The NETLIST_FORMAT: PARAMETERIZED_SPICE option is intended to be used only for flows in which StarRC produces a Galaxy Parasitic Database (GPD). StarRC internally enables the GPD flow based on options in the command file. Users cannot enable or disable the GPD flow directly.

**What Next**

If the NETLIST_FORMAT: PARAMETERIZED_SPICE feature is not needed, remove it from the command-file or set its value to NO. If this feature is needed, contact StarRC customer support for information about the NETLIST_FORMAT: PARAMETERIZED_SPICE option and conditions under which it can be used.

---

## SX-3695

(Fatal) Use of the NETLIST_FORMAT: PARAMETERIZED_SPICE option requires %s (atleast two corners)

**Description**

The NETLIST_FORMAT: PARAMETERIZED_SPICE option can be used only when certain other options are set in the StarRC command file. This error message indicates the option that needs to be set in order to use NETLIST_FORMAT: PARAMETERIZED_SPICE.

**What Next**

Add the option mentioned in the error message to the StarRC command file. Contact StarRC customer support for more information about the NETLIST_FORMAT: PARAMETERIZED_SPICE option and conditions under which it can be used.

Feedback

## SX-3696

(Warning) fill shape {{%d %d} {%d %d}} on db layer '%s' intersects with shape {{%d %d} {%d %d}} on db layer '%s'.

### Description

This message indicates that fill shape intersects with another shape.

### What Next

This is a warining message, and not action is needed.

## SX-3697

(Fatal) Use of the NETLIST_FORMAT: PARAMETERIZED_SPICE option does not allow the use of %s

### Description

Some options in the StarRC command file cannot be used with the NETLIST_FORMAT: PARAMETERIZED_SPICE option. This error message indicates the option setting that is incompatible with the NETLIST_FORMAT: PARAMETERIZED_SPICE option.

### What Next

Remove or modify the option mentioned in the error message in the StarRC command file. Contact StarRC customer support for more information about the NETLIST_FORMAT: PARAMETERIZED_SPICE option and conditions under which it can be used.

## SX-3698

(Warning) Internally set option %s to %s for use with option %s

### Description

Use of some options in the StarRC command file results in other options being set internally, resetting their user-specified values if the option has been specified in the command file. This warning states the option whose value has been set or reset internally and the option setting which caused this change to occur.

### What Next

This is a warning message only, no action is necessary.

Feedback

## SX-3699

(Fatal) EEQ_CELL_PORT_FILE is not supported in gate level flow. This option is only used with ICV/CCI flow.

**Description**

EEQ_CELL_PORT_FILE can only be used with transistor level flow. Gate level flow can not be used with this option.

**What Next**

Remove the EEQ_CELL_PORT_FILE in the StarRC command file. Contact StarRC customer support for more information about the EEQ_CELL_PORT_FILE option and conditions under which it can be used.

## SX-3700

(Fatal) EEQ_CELL_PORT_FILE can not be used with PIN_CUT_THRESHOLD,PIN_CUT_LOCATION_CENTER or SPICE_SUBCKT_FILE.

**Description**

PIN_CUT_THRESHOLD,PIN_CUT_LOCATION_CENTER or SPICE_SUBCKT_FILE is not supported with EEQ_CELL_PORT_FILE option.

**What Next**

Remove the PIN_CUT_THRESHOLD,PIN_CUT_LOCATION_CENTER and SPICE_SUBCKT_FILE in the StarRC command file. Contact StarRC customer support for more information about the EEQ_CELL_PORT_FILE option and conditions under which it can be used.

## SX-3701

(Fatal) The syntax of EEQ_CELL_PORT_FILE is not correct at line '%s'.

**Description**

The syntax of EEQ_CELL_PORT_FILE is not correct.

**What Next**

The syntax of EEQ_CELL_PORT_FILE is not correct. Follow the syntax description in StarRC user guide.

Feedback

## SX-3702

(Warning) Gate-level Mutli-corner Temperature Sensitivity flow does not support Galaxy Parasitic Data (GPD) output. GPD output has been turned off.

**Description**

GPD output is not supported by Gate-level Multi-corner Temperature Sensitivity flow. GPD is automatcically turned off.

## SX-3703

(Fatal) EEQ_CELL_PORT_FILE can only be used with NETLIST_FORMAT:SPEF.

**Description**

EEQ_CELL_PORT_FILE can only be used with NETLIST_FORMAT:SPEF

**What Next**

Revise NETLIST_FORMAT to SPEF in StarRC command file. Contact StarRC customer support for more information about the EEQ_CELL_PORT_FILE option and conditions under which it can be used.

## SX-3704

(Fatal) INDESIGN_OPEN_NETS option is supported only with '%s'.

**Description**

To use INDESIGN_OPEN_NETS option, required option must be specified.

**What Next**

Specify the required option.

## SX-3705

(Fatal) RC scaling coefficients are supported only with '%s'.

**Description**

To use RC scaling feature, required option must be specified.

**What Next**

Specify the required option.

Feedback

## SX-3706

(Fatal) Scaling coefficient specified in the option '%s' of the corners file has to be positive.

**Description**

Scaling of RC parasitic values according to the specified coefficients.

**What Next**

Modify coefficient accordingly.

## SX-3707

(Warning) The GPD_IN_NDM command will be ignored because NDM_DATABASE is not defined.

**Description**

NDM_DATABASE does not be set when GPD_IN_NDM command is set as YES. GPD_IN_NDM is ignored.

## SX-3708

(Warning) The GPD command will be ignored because GPD_IN_NDM command is enabled.

**Description**

Ignoring the GPD command setting when GPD_IN_NDM is set as YES. The GPD directory name is design.<block_name>.gpd located in NDM attachement directory.

## SX-3709

(Fatal) MULTI_PHYSICAL_PINS_PREFIX option is supported only with '%s'.

**Description**

To use MULTI_PHYSICAL_PINS_PREFIX option, required option must be specified.

**What Next**

Specify the required option.

## SX-3710

(Warning) Incorrect %s pin position of transistor %s.

Feedback

**Description**

StarRC uses source/drain/gate-pin positions to detect gate direction (vertical or horizontal). Incorrect pin position may lead to incorrect gate direction.

**What Next**

Check the pin positions of input database.

---

## SX-3711

(Fatal) License error: %s

**Description**

StarRC encountered an error which checking out a license.

**What Next**

Check if (a) the LM_LICENSE_FILE environment variable is set correctly, (b) your license server is setup and working correctly and (c) You have enough available licenses to run StarRC using options specified in the command file. The use of some features of StarRC may have additional licensing requirements.

Contact Synopsys technical support for more information.

---

## SX-3712

(Fatal) POWER_EXTRACT:CONLY can only work with Transistor level Galaxy Parasitic Data(GPD).

**Description**

POWER_EXTRACT:CONLY can only work with Transistor level Galaxy Parasitic Data(GPD)

**What Next**

Specify the required option.

---

## SX-3713

(Fatal) There do not have any cell name in EEQ_CELL_PORT_FILE match current BLOCK '%s'.

**Description**

There do not have any cell name in EEQ_CELL_PORT_FILE match current BLOCK.

Feedback

**What Next**

Use the correct EEQ_CELL_PORT_FILE.

---

## SX-3714

(Fatal) SHORT_PINS:NO is not supported for NORC extraction.

**Description**

SHORT_PINS:NO is not supported for NORC extraction.

**What Next**

Specify the required option.

---

## SX-3721

(Warning) Net specified with netlist_connect_open command doesn't exist in the design.

**Description**

Net specified with the netlist_connect_open command doesn't exist in the design. This net will be ignored from further cosideration.

**What Next**

By default, we connect opens of nets in the design. NCO command is used to further narrow down this mechanism and should be used with real nets. This is a warning message telling that the design doesn't contain this net. Please be aware of this.

---

## SX-3722

(Warning) Distributed metal fill reconnect is invoked.

**Description**

Distributed metal fill reconnect is invoked instead of fast metal fill reconnect.

**What Next**

Information only. No action is needed.

---

## SX-3724

(Warning) NETLIST_INSTANCE_SECTION: %s is not supported with Virtual Block input, using NETLIST_INSTANCE_SECTION: NO.

Feedback

**Description**

The NETLIST_INSTANCE_SECTION feature is not supported for StarRC runs with Virtual Block input. StarRC will internally set the value of this option to NO to allow the run to continue.

**What Next**

Set NETLIST_INSTANCE_SECTION: no in the StarRC command file to prevent this warning from showing up.

---

## SX-3725

(Warning) PIN '%s' in DEF contains zero width/hieght RECT (%d, %d) (%d, %d). The polygon will be ignored.

**Description**

StarRC detects invalid zero width/height polygons.

**What Next**

See if the zero width/height polygon is reasonable.

---

## SX-3726

(Warning) File '%s' does not exist. It is ignored.

**Description**

StarRC detects a file that does not exist. The file is ignored.

---

## SX-3727

(Error) Cannot find INTER_DIE_BLOCK in NDM library

**Description**

StarRC cannot find INTER_DIE_BLOCK in NDM library

---

## SX-3728

(Warnning) StarRC read an empty TSV cell, will drop this cell.

**Description**

StarRC read an empty TSV cell, will drop this cell.

Feedback

**What Next**

Check the NDM database about the TSV design

---

## SX-3729

(Fatal) The syntax in VIA_ARRAY_COUNT_FILE: %s is invalid.

**Description**

VIA_ARRAY_COUNT_FILE has syntax error.

**What Next**

Modify the file according to the StarRC user's guide.

---

## SX-3730

(Warning) The layer %s in the VIA_ARRAY_COUNT_FILE: %s is not a via layer, ignored.

**Description**

The first column in the VIA_ARRAY_COUNT_FILE should be a database via layer.

**What Next**

Modify the VIA_ARRAY_COUNT_FILE file.

---

## SX-3731

(Fatal) The second column of "%s" in the VIA_ARRAY_COUNT_FILE: %s is invalid.

**Description**

The MAX_VIA_ARRAY_NUMBER must be a positive number.

**What Next**

Modify the VIA_ARRAY_COUNT_FILE file.

---

## SX-3732

(Warning) The MAX_VIA_ARRAY_SPACING and MAX_VIA_ARRAY_LENGTH are ignored on layer "%s" since MAX_VIA_ARRAY_NUMBER is specified in the VIA_ARRAY_COUNT_FILE.

Feedback

**Description**

The MAX_VIA_ARRAY_NUMBER in the VIA_ARRAY_COUNT_FILE has higher priority than MAX_VIA_ARRAY_SPACING and MAX_VIA_ARRAY_LENGTH in StarRC mapping file.

**What Next**

Remove MAX_VIA_ARRAY_SPACING and MAX_VIA_ARRAY_LENGTH from the StarRC mapping file.

---

## SX-3733

(Warning) %s is ignored by VIA_COVERAGE_OPTION_FILE or VIA_COVERAGE.

**Description**

%s cannot be specified together with VIA_COVERAGE_OPTION_FILE or VIA_COVERAGE.

**What Next**

Remove %s from StarRC command file.

---

## SX-3734

(Warning) %s is ignored in gate level flow.

**Description**

%s cannot be specified in gate level flow.

**What Next**

Remove %s from StarRC command file.

---

## SX-3735

(Warning) Value of the 3D_IC_FILTER_DEVICE option will be replaced with the value of the port_connector_name option from the ICV runset report file.

**Description**

If a port_connector_name option is specified in the ICV runset report file, its value will be used to replace the value of the 3D_IC_FILTER_DEVICE option from the StarRC command-file.

Feedback

### What Next

Remove the 3D_IC_FILTER_DEVICE option from the StarRC command-file to prevent this warning from appearing again.

---

## SX-3736

(Fatal) Target layer for connect inheritance %s has multiple implicit connects to source layer.

### Description

The CCI LVS extraction report file or ICV runset report file contains target layers for connect inheritance. In CCI LVS rextraction report file, target layers have layer type LVS_DB_CONNECTIVITY_LAYER. One target layer should be present once and only once in IMPLICIT_CONNECTIVITY section like this: IMPLICIT_CONNECT <target_layer> <source_layer> In ICV runset report file, target layers should be present in implicit_connect statement like this: implicit_connect <source_layer> <target_layer1> <target_layer2> ... StarRC will pass connectivity of source layer to target layer. One target layer should be present in only one impclit_connect. If a target layer presents in more than one implicit connects, StarRC will give this error message.

### What Next

Check Calibre/ICV runset to see whether the target layer is derived from more than one source layers or not.

---

## SX-3737

(Fatal) Target layer for connect inheritance %s has implicit connect to layers other than source layers.

### Description

The CCI LVS extraction report file contains target layers for connect inheritance (with layer type LVS_DB_CONNECTIVITY_LAYER). A target layer should be present once and only once in IMPLICIT_CONNECTIVITY section like this: IMPLICIT_CONNECT <target_layer> <source_layer> so that StarRC is able to find the source layer of the target layer and pass connectivity of source layer to target layer. Besides the implicit connect to source layer, target layer should not have any other connects or implicit connects. If target layer has implicit connect to layer other than source layer, StarRC will give this error message.

### What Next

Check Calibre runset to see whether there are any layers derived from this target layer or not.

Feedback

## SX-3738

(Fatal) Target layer for connect inheritance %s already has explicit connects.

### Description

The CCI LVS extraction report file or ICV runset report file contains target layers for connect inheritance. In CCI LVS rextraction report file, target layers have layer type LVS_DB_CONNECTIVITY_LAYER. One target layer should be present once and only once in IMPLICIT_CONNECTIVITY section like this: IMPLICIT_CONNECT <target_layer> <source_layer> In ICV runset report file, target layers should be present in implicit_connect statement like this: implicit_connect <source_layer> <target_layer1> <target_layer2> ... StarRC will pass connectivity of source layer to target layer. Besides the implicit connect to source layer, target layer should not have any other connects or implicit connects. If target layer has explicit connects defined in LVS extraction report file or ICV runset report file, StarRC will give this error message.

### What Next

Calibre/ICV runset should have no connects defined for a target layer. Check Calibre/ICV runset to see whether there are any connects defined for this target layer or not.

## SX-3739

(Fatal) Target layer for connect inheritance %s is not defined as a conductor or via or remove layer in StarRC mapping file.

### Description

The CCI LVS extraction report file or ICV runset report file contains target layers for connect inheritance. In CCI LVS rextraction report file, target layers have layer type LVS_DB_CONNECTIVITY_LAYER. One target layer should be present once and only once in IMPLICIT_CONNECTIVITY section like this: IMPLICIT_CONNECT <target_layer> <source_layer> In ICV runset report file, target layers should be present in implicit_connect statement like this: implicit_connect <source_layer> <target_layer1> <target_layer2> ... StarRC will pass connectivity of source layer to target layer. A target layer should be defined in conducting_layers or via_layers or remove_layers section in StarRC mapping file. Otherwise, StarRC will give this error message.

### What Next

Define this target layer in conducting_layers or via_layers or remove_layers section in StarRC mapping file.

Feedback

## SX-3740

(Fatal) Source layer for connect inheritance %s is not CONNECT type.

### Description

The CCI LVS extraction report file contains source layers for connect inheritance. A source layer should be present in IMPLICIT_CONNECTIVITY section like this: IMPLICIT_CONNECT <target_layer> <source_layer> where the target layer is a layer with layer type LVS_DB_CONNECTIVITY_LAYER. StarRC will pass connectivity of source layer to target layer. Source layer should have layer type CONNECT. Otherwise, StarRC will give this error message.

### What Next

Calibre runset should have explicit connect defined for source layer. Check Calibre runset to see whether there are connect defined for this source layer.

## SX-3741

(Fatal) Source layer for connect inheritance %s is also a target layer.

### Description

The CCI LVS extraction report file or ICV runset report file contains source layers for connect inheritance. In CCI LVS rextraction report file, a source layer should be present in IMPLICIT_CONNECTIVITY section like this: IMPLICIT_CONNECT <target_layer> <source_layer> where the target layer is a layer with layer type LVS_DB_CONNECTIVITY_LAYER. In ICV runset report file, a source layer should be present in implicit_connect statement like this: implicit_connect <source_layer> <target_layer1> <target_layer2> ...

StarRC will pass connectivity of source layer to target layer. A source layer should not be a target layer (with layer type LVS_DB_CONNECTIVITY_LAYER in CCI LVS extraction report file, or in other implicit_connect statement as target layer in ICV runset report file) derived from other source layer. Otherwise, StarRC will give this error message.

### What Next

Check Calibre/ICV runset to see whether this source layer is also derived from other source layer.

## SX-3742

(Fatal) Source layer for connect inheritance %s has implicit connect to layer without LVS_DB_CONNECTIVITY_LAYER type.

Feedback

### Description

The CCI LVS extraction report file contains source layers for connect inheritance. A source layer should be present in IMPLICIT_CONNECTIVITY section like this: IMPLICIT_CONNECT <target_layer> <source_layer> where the target layer is a layer with layer type LVS_DB_CONNECTIVITY_LAYER. StarRC will pass connectivity of source layer to target layer. Source layer should only have implicit connects to target layers (layer with LVS_DB_CONNECTIVITY_LAYER type). Otherwise, StarRC will give this error message.

### What Next

Check Calibre runset to see whether there are any layers not defined as LVS DB CONNECTIVITY LAYER are also derived from this source layer.

---

## SX-3743

(Fatal) Source layer for connect inheritance %s has no explicit connect.

### Description

The CCI LVS extraction report file or ICV runset report file contains source layers for connect inheritance. In CCI LVS rextraction report file, a source layer should be present in IMPLICIT_CONNECTIVITY section like this: IMPLICIT_CONNECT <target_layer> <source_layer> where the target layer is a layer with layer type LVS_DB_CONNECTIVITY_LAYER. In ICV runset report file, a source layer should be present in implicit_connect statement like this: implicit_connect <source_layer> <target_layer1> <target_layer2> ...

StarRC will pass connectivity of source layer to target layer. Source layer should have explicit connect. Otherwise, StarRC will give this error message.

### What Next

Calibre/ICV runset should have explicit connect defined for source layer. Check Calibre/ICV runset to see whether there are connect defined for this source layer.

---

## SX-3744

(Fatal) Source layer for connect inheritance %s is not in remove_layers section in StarRC mapping file.

### Description

The CCI LVS extraction report file or ICV runset report file contains source layers for connect inheritance. In CCI LVS rextraction report file, a source layer should be present in IMPLICIT_CONNECTIVITY section like this: IMPLICIT_CONNECT

Feedback

<target_layer> <source_layer> where the target layer is a layer with layer type LVS_DB_CONNECTIVITY_LAYER. In ICV runset report file, a source layer should be present in implicit_connect statement like this: implicit_connect <source_layer> <target_layer1> <target_layer2> ...

StarRC will pass connectivity of source layer to target layer. Source layer should not participate in parasitic extraction and should be defined as remove layer in StarRC mapping file. Otherwise, StarRC will give this error message.

**What Next**

Add this source layer to remove_layers section in StarRC mapping file.

---

## SX-3745

(Fatal) Target layer for connect inheritance %s has no implicit connect to source layer.

**Description**

The CCI LVS extraction report file contains target layers for connect inheritance (with layer type LVS_DB_CONNECTIVITY_LAYER). A target layer should be present once and only once in IMPLICIT_CONNECTIVITY section like this: IMPLICIT_CONNECT <target_layer> <source_layer> so that StarRC is able to find the source layer of the target layer and pass connectivity of source layer to target layer. If there is no implicit connect from this target layer to source layer, StarRC will give this error message.

**What Next**

Check Calibre runset to see whether the target layer is derived from a source layer or not.

---

## SX-3746

(Warning) The RDL block name %s, in %s, does not match TOP block name %s. Using the %s cell as the context/environment for block %s.

**Description**

For RDL pushdown, we assume the RDL block name should be same with TOP block name.

**What Next**

Check the RDL design name and TOP design name.

---

## SX-3747

(Fatal) SMC via_coverage from ITF corrupted: cannot load corner via_coverage entry.

Feedback

**Description**

Failed to load via_coverage entry from one of the corner-dependent via_coverage from ITF. It is usually caused by discrepancy on input ITF.

**What Next**

Check and correct via_coverage in ITF, and restart. If error persists, contact Synopsys Support Center.

---

## SX-3748

(Fatal) failed to check out licenses

**Description**

NETLIST_FORMAT: PARAMETERIZED_SPICE requires STAR-RC2-AG-TECHNOLOGY and STAR-RC_ULTRA_MANAGER license.

**What Next**

Check to make sure you have sufficient StarRC licenses, and that the license servers are running.

---

## SX-3749

(Fatal) COUPLE_TO_SPECIFIED_PCELL_PINS: '%s' uses invalid type.

**Description**

Invalid type for COUPLE_TO_SPECIFIED_PCELL_PINS command. Currently supported types are: < NO | YES | YES KEEP_CG | YES IGNORE_CG | YES AUTOMATIC_CG_HANDLING>.

**What Next**

Change type to the one of the supported.

---

## SX-3750

(Fatal) TCAD_GRD_FILE defined in CORNERS_FILE should be consistent in same format property.

**Description**

TCAD_GRD_FILE defined in CORNERS_FILE should be all non-compressed NXTGRD or all compressed NXTGRD. Mixed usage is not supported.

Feedback

**What Next**

Please make sure all TCAD_GRD_FILE defined in CORNERS_FILE are all non-compressed NXTGRD or all compressed NXTGRD.

---

## SX-3751

(Warning) DP_RECONNECT is not supported for %s. Metal fill reconnect is reverted to DP_RECONNECT: NO

**Description**

DP_RECONENECT is not supported. Metal fill reconnect is reverted to DP_RECONNECT: NO

**What Next**

This is a warning message only, no action is necessary.

---

## SX-3752

(Warning) TRANSLATE_VIA_FILLS is set to NO for transistor level designs in TSMC N5 process. This is not recommended due to accuracy concerns.

**Description**

TRANSLATE_VIA_FILLS is explicitly set to NO in the StarRC command file. There might be accuracy concerns. If not set, this options is automatically set to YES by default for transistor level designs in TSMC N5 process.

**What Next**

This is a warning message only, no action is necessary.

---

## SX-3753

(Fatal) Layer '%s' in silicon_marker_layers section in MAPPING_FILE must be the trench contact virtual via.

**Description**

This layer apeared in silicon_marker_layers section in MAPPING FILE. This type of silicon marker layer can only be mapped to a trench contact virtual via.

**What Next**

Revise silicon_marker_layers section in mapping file.

Feedback

## SX-3754

(Warning) Layer '%s' in silicon_marker_layers section in MAPPING_FILE have the same ITF layer with ITF= '%s'. StarRC removes this silicon marker layer.

**Description**

This layer apeared in silicon_marker_layers section in MAPPING FILE. This type of silicon marker layer mapped to the ITF layer is the same as "ITF=" in later section. So this type of silicon marker layer is not necessary.

**What Next**

Revise silicon_marker_layers section in mapping file.

## SX-3755

(Warning) Use of the '%s' option can result in a large increase in run time and disk usage of StarRC.

**Description**

Using the option mentioned in the message can result in a large increase in run time and disk usage of StarRC. This option is intended for use by Synopsys support staff only for troubleshooting purposes. It should not be used in production runs of the tool.

**What Next**

Remove the option for the StarRC command file to prevent this warning from showing up in future runs of the tool.

## SX-3756

(Warning) NETLIST_CONNECT_OPENS specified a net: %s that is not existent in the design. The non-existent net is ignored.

**Description**

If NETLIST_CONNECT_OPENS specified a net that is not existent in the design, it may have undesirable consequences. This may cause unpredicatble behavior including failure to connect short equivalent nodes.

**What Next**

Please correct the net name(s) in NETLIST_CONNECT_OPENS to avoid this warning and possible future issues in the design.

Feedback

## SX-3757

(Warning) The %s option should not be used in signoff production runs of StarRC.

**Description**

The option mentioned in the warning message is intended for use in experimental runs of StarRC or as to work around specific issues with the tool. It should not be used in signoff production runs.

## SX-3758

(Fatal) The %s option is supported in the transistor-level extraction flow only.

**Description**

The option listed in the error message can be used in the transistor-level extraction flow only. It should not be used in the gate-level extraction flow.

**What Next**

If the stated option is specified in the StarRC command file, remove it from the command file. If the option is a command-line option, do not specify this option on the command-line.

## SX-3759

(Fatal) The %s option cannot be used because the %s file does not exist inside the STAR directory.

**Description**

The option listed in the error message cannot be used because the file specified in the error message does not exist in the STAR directory.

**What Next**

Check the contents of the STAR directory. Regenerate the STAR directory by running StarRC without the option specified in the error message.

## SX-3760

(Fatal) XREF input database database has changed. Cannot reuse XREF result.

**Description**

The input database has changed. StarRC cannot reuse the XREF result.

Feedback

**What Next**

Remove -reuse_xref form the command options.

---

## SX-3761

(Warning) Ignoring multiply used StarRC command file '%s'

**Description**

If multiple files are used to specify options to StarRC, only the first occurrence of each file is parsed and subsequent uses of the file are ignored.

**What Next**

Examine the command-line options to StarRC and the INCLUDE_FILE options specified in each file. This warning will appear when a file name appears more than once in the combined set of files from command-line and INCLUDE_FILE options.

---

## SX-3762

(Warning) The layer %s in %s is not a via layer, ignored.

**Description**

The layer in the command should be a database via layer.

**What Next**

Modify the cmd.

---

## SX-3763

(Warning) The StarRC command ADD_GATE_ADJUSTMENT_RESISTANCE is turned off since there is not GATE_RESISTANCE_ADJUSTMENT_FACTOR defined in the nxtgrd file.

**Description**

The StarRC command ADD_GATE_ADJUSTMENT_RESISTANCE is turned off since there is not GATE_RESISTANCE_ADJUSTMENT_FACTOR defined in the nxtgrd file.

**What Next**

Remove ADD_GATE_ADJUSTMENT_RESISTANCE from StarRC command file.

Feedback

## SX-3764

(Fatal) The number "%s" in %s is invalid.

**Description**

The number in the command must be a positive number.

**What Next**

Modify the cmd.

## SX-3765

(Warning) MARKER_GENERATION: USER can not be used in ICV AGDS flow with cell_port_file. Change the setting to MARKER_GENERATION: AUTOMATIC

**Description**

For ICV AGDS flow with cell_port_file, all marker layers are removed and the ports are directly put on conductors in cell_port_file. Since all marker layers are removed, MARKER_GENERATION: USER can not be used.

**What Next**

No action is needed. StarRC will automatically generate the pin shapes based on connectivity.

## SX-3766

(Warning) XREF_REMOVE_ICV_MPROP_SUFFIX:YES is ignored when XREF_ICV_COMPOSITE_HANLDING:FLATTEN is used.

**Description**

XREF_REMOVE_ICV_MPROP_SUFFIX:YES cannot be used with XREF_ICV_COMPOSITE_HANLDING:FLATTEN.

**What Next**

Check if XREF_ICV_COMPOSITE_HANLDING:LEVEL_BY_LEVEL should be used.

## SX-3767

(Fatal) Option '%s' is not supported with enabled ENHANCED_CCP_FLOW command

Feedback

**Description**

This option is not supported in new CCP flow.

**What Next**

Remove the option and run again.

---

## SX-3768

(Fatal) ADJUST_RPV_BY_VIA_COUNT can only be set to YES in transistor flow.

**Description**

When ADJUST_RPV_BY_VIA_COUNT is set to YES, the flow must be a transistor-level flow.

**What Next**

If you are running gate-level flow, please remove this command.

---

## SX-3769

(Warning) The StarRC command ADJUST_RPV_BY_VIA_COUNT is turned off since there is no RPV_ADJUSTMENT_FACTOR_VS_VIA_COUNT defined in the nxtgrd file.

**Description**

The StarRC command ADJUST_RPV_BY_VIA_COUNT is turned off since there is no RPV_ADJUSTMENT_FACTOR_VS_VIA_COUNT defined in the nxtgrd file.

**What Next**

Remove ADJUST_RPV_BY_VIA_COUNT command in StarRC command file.

---

## SX-3770

(Warning) PREXTRACT_NONMANHATTAN_HANDLING is applied to '%s' layer.

**Description**

N/A

**What Next**

N/A

Feedback

## SX-3771

(Warning) Layer %s listed in MESH_RESISTANCE_EXTRACTION_LAYERS does not exist.

**Description**

Only db layer can be set in MESH_RESISTANCE_EXTRACTION_LAYERS command in StarRC command file.

**What Next**

Modify MESH_RESISTANCE_EXTRACTION_LAYERS setting in StarRC command file.

## SX-3772

(Fatal) Option '%s' is not supported with ENHANCED_VIA_EXTRACTION command

**Description**

This option is not supported with ENHANCED_VIA_EXTRACTION command.

**What Next**

Remove this option or turn off ENHANCED_VIA_EXTRACTION and run again.

## SX-3773

(Fatal) ENHANCED_GPD_POWER_REDUCTION:YES can only work with Transistor level Galaxy Parasitic Data(GPD).

**Description**

ENHANCED_GPD_POWER_REDUCTION:YES can only work with Transistor level Galaxy Parasitic Data(GPD).

**What Next**

Specify the required option.

## SX-3774

(Warning) IC Compiler II InDesign StarRC NDM_ZERO_SPACING_BLOCKAGE_RATIO option is invoked. NDM_ZERO_SPACING_BLOCKAGE_RATIO is not meant for final signoff flow because aggregate ratio is reflective of the reality of each macro gray material situation.

Feedback

**Description**

NDM_ZERO_SPACING_BLOCKAGE_RATIO is not meant for final signoff flow because aggregate ratio is reflective of the reality of each macro gray material situation.

**What Next**

Just take care

---

## SX-3775

(Warning) NDM_ZERO_SPACING_BLOCKAGE_RATIO %.6g should be smaller than or equal to 1, automaticall adjust the value to 1

**Description**

NDM_ZERO_SPACING_BLOCKAGE_RATIO should larger than 0 and smaller or equal to 1

**What Next**

adjust the ratio value

---

## SX-3776

(Warning) NDM_ZERO_SPACING_BLOCKAGE_RATIO %.6g should be larger than 0, so the ratio value is dropped.

**Description**

NDM_ZERO_SPACING_BLOCKAGE_RATIO should larger than 0 and smaller or equal to 1

**What Next**

adjust the ratio value

---

## SX-3777

(Error) NDM_ZERO_SPACING_BLOCKAGE_RATIO can only be used for NDM flow

**Description**

NDM_ZERO_SPACING_BLOCKAGE_RATIO cannot use in other flow.

**What Next**

remove the command or change flow.

Feedback

## SX-3778

(Warining) cell '%s' is not skip cell. It is ignored from NETLIST_LOCATION_TRANSFORMS_ADDITIONAL_CELLS.

**Description**

Cells must be skip cells to output netlist location transformation factors

**What Next**

Specify skip cells only for NETLIST_LOCATION_TRANSFORMS_ADDITIONAL_CELLS.

## SX-3779

(Fatal) Invalid negative cap scale value. CAP_SCALE value can be 0, or any positive value. Line: %s

**Description**

You receive this message because StarRC find the value for cap_scale parameter a negative value in StarRC mapping file. The valid value for cap_scale is 0, or any positive value.

**What Next**

You should correct the cap_scale value with a valid value.

## SX-3780

(Fatal) CAP_SCALE value must be defined with PRECEDENCE, Line: %s

**Description**

You receive this message because StarRC find cap_scale parameter not defined with PRECEDENCE value in StarRC mapping file. The CAP_SCALE parameter must be defined with PRECEDENCE.

**What Next**

You should add PRECEDENCE for this layer in StarRC mapping file

## SX-3781

(Fatal) CAP_SCALE parameter must be defined on the SUBSTRATE layers.Line: %s

Feedback

**Description**

You receive this message because StarRC find cap_scale parameter defined in non-SUBSTRATE layers in StarRC mapping file. The CAP_SCALE parameter must be defined on SUBSTRATE layers .

**What Next**

The CAP_SCALE parameter must be defined on SUBSTRATE layers .

---

## SX-3782

(Fatal) Directory '%s' is not a recognized GPD format

**Description**

StarXtract is unable to recognize directory '%s' as containing GPD formatted data.

**What Next**

Please check for the existence and contents of the README file contained in the GPD directory. If the README file is missing, the directory is either of some other type or the GPD directory is corrupt.

---

## SX-3783

(Fatal) Incorrect corner configuration. %s.

**Description**

A parasitic corner configuration error has been detected. Please check the detailed message for more information.

**What Next**

Please correct the problem(s) as described in the error message detail.

---

## SX-3784

(Information) %s.

**Description**

General information message generated during parasitic data processing.

**What Next**

No action required.

Feedback

## SX-3785

(Fatal) File operation error, %s for '%s'.

**Description**

A general file operation error occured during StarRC extraction.

**What Next**

Please correct the problem(s) as described in the error message detail.

## SX-3786

(Fatal) The syntax on POWER_EXTRACT is not correct .Line: %s

**Description**

You receive this message because StarRC find the syntax on POWER_EXTRACT option is not correct. The BOUNDRY parameter must have validate ITF layer .

**What Next**

The BOUNDRY parameter in POWER_EXTRACT must have validate ITF layer .

## SX-3787

(Fatal) The boundary ITF layer %s in POWER_EXTRACT option is not validate.

**Description**

You receive this message because StarRC find the boundary ITF layer in POWER_EXTRACT option is not correct. The BOUNDRY parameter must have validate ITF layer .

**What Next**

The BOUNDRY parameter in POWER_EXTRACT must have validate ITF layer .

## SX-3788

(Warning) Copying netlist file %s to %s across two disks. Cross-disk copying of large files may result in extraordinarily long run times for StarRC.

Feedback

**Description**

When the output file path specified for the NETLIST_FILE option is on a disk that is different from the disk used to run StarRC, StarRC has to copy netlist files between disks which can take a very long time depending on the disk and network configuration.

Using the same disk to run StarRC and store output files is recommended.

**What Next**

Modify the value of the NETLIST_FILE file option to use the same disk as the one used to run StarRC.

---

## SX-3789

(Fatal) Temperature sensitivity range requires valid lower and upper temperature bounds.

**Description**

You receive this message because the TEMPERATURE_SENSITIVITY_RANGE is not correctly set.

**What Next**

Check the setting of TEMPERATURE_SENSITIVITY_RANGE to be valid. Make sure lower and upper temperature bounds are valid numerical values.

---

## SX-3791

(Severe) input value to %s writer is '%s'.

**Description**

The input value to GPD/XOUT writer is not a number or is infinite.

**What Next**

Contact the Synopsys Support Center.

---

## SX-3792

(Warning) MESH_RESISTANCE_EXTRACTION_LAYERS is ignored for gate level flow.

**Description**

MESH_RESISTANCE_EXTRACTION_LAYERS is designed only for transistor level flow.

Feedback

**What Next**

remove MESH_RESISTANCE_EXTRACTION_LAYERS from StarRC command file.

---

## SX-3793

(Fatal) Could not find executable %s under directory %s : error code %s.

**Description**

The executable specified in the error message was not found in the specified directory. This usually indicates a problem with the StarRC installation.

**What Next**

Check your StarRC installation directory.

---

## SX-3794

(Fatal) Invalid %s specification '%s', need to use ssh when running under a container environment.

**Description**

The ssh protocol must be used When distributing jobs across multiple hosts using the "STARRC_DP_STRING: list ..." syntax under a container environment (i.e. either the SNPS_CONTAINER environment variable is set or StarXtract is launched with the -container option).

**What Next**

If no protocol is specified, the default protocol is rsh, so ssh must be specified explicitly. That is, replace

STARRC_DP_STRING: list hostA:1 hostB:2

with

STARRC_DP_STRING: list ssh hostA:1 hostB:2

If a protocol other than ssh is used, replace that with ssh. That is, replace

STARRC_DP_STRING: list rsh hostA:1 hostB:2

with

STARRC_DP_STRING: list ssh hostA:1 hostB:2

The above description applies to both STARRC_DP_STRING and GRDGENXO_DP_STRING.

Feedback

## SX-3795

(Warning) Only front stack of the design is extracted when TSV mapping/nxtgrd files are deteced without 3D_IC:YES.

### Description

When 3D_IC:NO is used together with TSV-mapping/nxtgrd files, only front stack of the design is extracted.

### What Next

This is a warning message only, no action is necessary.

## SX-3796

(Warning) Layer %s has no WMIN and SMIN specified in ITF or star cmd, use default %f um as max spacing to merge vias on this layer.

### Description

This is used for via merge in EM flows. If a via layer doesn't have WMIN and SMIN specified, then use StarRC's default max spacing as in timing flow.

### What Next

Specify WMIN and SMIN for the layer if want to use 1.25*(WMIN+SMIN) as max via merge spacing.

## SX-3799

(Fatal) Density outside block specified in the option '%s' of the corners file has to be between 0 and 1.

### Description

Density outside block specified in the corner file has to be between 0 and 1.

### What Next

Modify DENSITY_OUTSIDE_BLOCK in the corner file accordingly.

## SX-3800

(Warning) Galaxy Parasitic Data (GPD) does not support EXTRACTION: RFSCOMPARE flow.

Feedback

**Description**

Galaxy Parasitic Data (GPD) does not support resistance field solver (RFS) flow. StarRC will run the non-GPD flow.

**What Next**

Remove EXTRACTION: RFSCOMPARE to run StarRC with GPD netlisting or the non-GPD flow will run.

---

## SX-3801

(Fatal) Invalid syntax in PARTIAL_SECOND_GROUND_AREAS: %s

**Description**

There is a synatx error in the command.

**What Next**

Check if the command is written correctly.

---

## SX-3802

(Error) The command %s must have the same number of arguments than the number of bins %d specified by BINS_TCAP_THRESHOLD. Follow the guideline: %s.

**Description**

The syntax of the command is not right.

**What Next**

Follow the syntax of the command.

---

## SX-3803

(Fatal) The command ESTIMATED_CONNECT_OPENS must have three arguments (x-layer y-layer small-gap-threshold).

**Description**

The syntax of the command is not right.

**What Next**

Follow the syntax of the command.

Feedback

## SX-3804

(Warning) INSTANCE_PORT_LOCATION_CLOSER_TO_DRIVER is not supported for transistor level flow.

### Description

INSTANCE_PORT_LOCATION_CLOSER_TO_DRIVER which is used to assign instance port location closer to the driver is currently not supported for transistor level flow.

### What Next

This is a warning message, nothing needs to be done from the user side. We will enhance the option to be supported for transistor level flow in future StarRC release.

## SX-3805

(Fatal) Parasitic explorer (PE) analysis is on and XREF is not set to YES (PE only supports XREF: YES)

### Description

If XREF is not set to YEs, extracted names from layout may have format/syntactical issues which result in incorrect behavior. Therefore, only XREF: YES is supported

### What Next

Change setup for the extraction with proper settings with XREF:YES. Either disable Parasitic explorer analysis or change set up for XREF:YES

## SX-3806

(Warning) The TSV_TEMPERATURE_RESISTANCE_SCALING option is ignored when TEMPERATURE_SENSITIVITY: yes is used.

### Description

The TSV_TEMPERATURE_RESISTANCE_SCALING option is used to scale resistors in the prescribed TSV models by CRT1/CRT2 from the nxtgrd file. This option is ignored if the TEMPERATURE_SENSITIVITY option is set to YES since model resistors are handled differently to output TC1/TC2 parameters to the netlist in this case.

### What Next

Remove one of TSV_TEMPERATURE_RESISTANCE_SCALING or TEMPERATURE_SENSITIVITY: yes in the StarRC option file to make this warning go away.

Feedback

## SX-3807

(Warning) The TSV_TEMPERATURE_RESISTANCE_SCALING option is ignored if OPERATING_TEMPERATURE is not set.

### Description

The TSV_TEMPERATURE_RESISTANCE_SCALING option is used to scale resistors in the prescribed TSV models by CRT1/CRT2 from the nxtgrd file. The resitance scaling calculation requires the OPERATING_TEMPERATURE to be specified and hence the TSV_TEMPERATURE_RESISTANCE_SCALING option is ignored if OPERATING_TEMPERATURE is not set.

### What Next

Remove TSV_TEMPERATURE_RESISTANCE_SCALING from the StarRC command file or set the OPERATING_TEMPERATURE option to make this warning go away.

## SX-3808

(Warning) VMF parameterization/option file is specified but VIRTUAL_METAL_FILL_POLYGON_HANDLING is not set or set to IGNORE. VMF flow will be disabled.

### Description

VMF parameterization/option file is specified but VIRTUAL_METAL_FILL_POLYGON_HANDLING is not set or set to IGNORE. VMF flow will be disabled.

### What Next

Revise the syntax of the star command and specify VIRTUAL_METAL_FILL_POLYGON_HANDLING: ATTACKER/FLOATING/GROUNDED if you want to enable VMF engine.

## SX-3809

(Warning) Layer %s for which VMF parameterization is defined is not present in the database.

### Description

Message informs the user that Virtual Metal Fill flow has been disabled since layer %s for which VMF parameterization is defined is not present in the database.

Feedback

### What Next

If this is expected, there is nothing to be done. If unexpected, revise MF settings in star command file.

---

## SX-3810

(Warning) VMF flow will be disabled since real MF polygons are present in the design.

### Description

Virtual Metal Fill flow is supported only when running StarRC on designs where real Metal Fill polygons are not specified.

### What Next

VMF flow will be disabled and real MF polygons will be extracted. In order to run the case with VMF, remove MF polygons from the database.

---

## SX-3811

(Fatal) VMF parameterization file '%s' not specified, not found or in invalid format. Please specify a valid VMF parameterization file using VIRTUAL_METAL_FILL_PARAMETER_FILE option, when using VIRTUAL_METAL_FILL_POLYGON_HANDLING:FLOATING/GROUNDED.

### Description

Virtual Metal Fill parameter file %s specified by option VIRTUAL_METAL_FILL_PARAMETER_FILE is not accessible or in invalid format.

### What Next

Revise the syntax or the path of the file in VIRTUAL_METAL_FILL_PARAMETER_FILE or disable VMF flow by setting VIRTUAL_METAL_FILL_POLYGON_HANDLING: IGNORE.

---

## SX-3812

(Warning) Virtual Metal Fill flow is enabled with mode '%s'.

### Description

Message warns the user that Virtual Metal Fill flow is enabled.

### What Next

Nothing to be done.

Feedback

## SX-3813

(Warning) Virtual Metal Fill flow for layer '%s' will be disabled as VMF properties are not declared for such a layer.

**Description**

Message informs the user that Virtual Metal Fill flow is disabled for layer '%s' since there is no VMF property defined for the such a layer.

**What Next**

Revise properties in VMF file if the warning is unexpected.

## SX-3814

(Fatal) Unknown selection

**Description**

Error message informs the user that VIRTUAL_METAL_FILL_POLYGON_HANDLING command is badly set. Allowed modes for VMF are IGNORE (default), FLOATING or GROUNDED.

**What Next**

User should correctly set VIRTUAL_METAL_FILL_POLYGON_HANDLING command with IGNORE (default), FLOATING or GROUNDED modes.

## SX-3815

(Fatal) VMF parameterization file has invalid spacings and widths for layer '%s'.

**Description**

Message informs the user that VMF parameterization file has invalid spacings and widths for layer '%s'.

**What Next**

Revise properties in VMF file for the indicated layer, making sure that all parameters are respecting criteria indicated in the user guide (>WMIN and >SMIN).

## SX-3816

(Warning) Virtual Metal Fill flow has been enabled on a design with real MF.

Feedback

### Description

Message warns the user that Virtual Metal Fill flow has been enabled on a design with real MF.

### What Next

If this is expected, there is nothing to be done. If not intentional, revise MF settings in star command file (disable RF with METAL_FILL_POLYGON_HANDLING:IGNORE or disable VMF with VIRTUAL_METAL_FILL_POLYGON_HANDLING:IGNORE).

---

## SX-3817

(Fatal) NETLIST_NODENAME_NETNAME: YES must be set when using NODENAME_NETNAME_ON_DANGLING_PORTS: YES.

### Description

NODENAME_NETNAME_ON_DANGLING_PORTS: YES must be used along with NETLIST_NODENAME_NETNAME: YES. If NETLIST_NODENAME_NETNAME: YES is not set when NODENAME_NETNAME_ON_DANGLING_PORTS: YES is used, StarRC will give this error.

### What Next

You should either set NETLIST_NODENAME_NETNAME: YES in the command file or set NODENAME_NETNAME_ON_DANGLING_PORTS to NO.

---

## SX-3818

(Warning) NODENAME_NETNAME_ON_DANGLING_PORTS: YES is not supported in non-GPD flow. Set this command to NO.

### Description

NODENAME_NETNAME_ON_DANGLING_PORTS: YES is only supported in GPD flow. If non-GPD flow is run, StarRC will give this message and set this command to NO.

### What Next

If you use XTOR_GPD: NO to force StarRC to run non-GPD flow, you can remove XTOR_GPD: NO. Otherwise, you can check messages in StarRC summary file to find out which command make StarRC to revert to non-GPD flow and remove that command.

Feedback

## SX-3819

(Warning) No second ground is found in %s. Settings of IGNORE_ITF_LAYER_CAPS_TO_SECOND_SUBSTRATE are ignored by StarRC.

**Description**

IGNORE_ITF_LAYER_CAPS_TO_SECOND_SUBSTRATE have no effect in the processes without second ground.

**What Next**

Check if the command is written correctly.

## SX-3820

(Warning) Layer "%s" specified in IGNORE_ITF_LAYER_CAPS_TO_SECOND_SUBSTRATE cannot be found in ITF.

**Description**

A layer specified in IGNORE_ITF_LAYER_CAPS_TO_SECOND_SUBSTRATE cannot be found in ITF.

**What Next**

Check if the command is written correctly.

## SX-3821

(Warning) Layer "%s" found in IGNORE_ITF_LAYER_CAPS_TO_SECOND_SUBSTRATE is not supported and will be skipped.

**Description**

IGNORE_ITF_LAYER_CAPS_TO_SECOND_SUBSTRATE only supports conducting layers in ITF.

**What Next**

Check if the command is written correctly.

## SX-3822

(Warning) PARASITIC_EXPLORER_ENABLE_ANALYSIS is set to YES. However GPD flow is hindered, possibly due to unsupported GPD command.

Feedback

### Description

PE analysis requires GPD to be generated. However, GPD flow is hindered possibly due to the presence of an unssported command in command file.

### What Next

To complete extraction, you can set PARASITIC_EXPLORER_ENABLE_ANALYSIS: NO. If you would like to run PE after extraction, you need to remove the command not supported by GPD (specified at the end of the summary file).

---

## SX-3823

(Fatal) The command ESTIMATED_CONNECT_OPEN_NET can only be used if ESTIMATED_CONNECT_OPENS is also used.

### Description

The command ESTIMATED_CONNECT_OPEN_NET can only be used if ESTIMATED_CONNECT_OPENS is also used.

### What Next

Add ESTIMATED_CONNECT_OPENS to StarRC command file.

---

## SX-3824

(Fatal) TABLE_NAME %s is provided for %s layer in Mapping file, but is not found in ITF file.

### Description

There are a few tables which contains multiple sub-tables in ITF file, such as GATE_TO_DIFFUSION_CAP and GATE_TO_DIFFUSION_CHANNEL_CAP. If in StarRC mapping file, a database layer is mapped to an ITF layer which contains such kind of table, you should specify which sub-table should be used through parameter TABLE_NAME. The possible tables which contain multiple sub-tables are: ETCH_VS_CONTACT_AND_GATE_SPACINGS GATE_TO_DIFFUSION_CAP GATE_TO_DIFFUSION_CHANNEL_CAP ER_TABLE RAISED_DIFFUSION_TO_GATE_SMIN_TABLE SIDE_TANGENT_VS_SI_WIDTH_AND_CCO_SPACING UPPERDIFF_TO_CONTACT_CAP RAISED_DIFFUSION_TO_GATE_SMIN_TABLE RAISED_DIFFUSION_ETCH_TABLE If TABLE_NAME is defined in StarRC mapping file for a gate or diffusion layer, but StarRC finds the corresponding sub-table are not defined in ITF file or nxtgrd file, StarRC will give this warning.

Feedback

**What Next**

You should check whether the table name is correct or not, and use table name which exists in the ITF file or nxtgrd file.

---

## SX-3825

(Warning) Galaxy Parasitic Data (GPD) does not support Hercules database as input.

**Description**

Galaxy Parasitic Data (GPD) only does not support the Hercules flow.

**What Next**

None at your side.

---

## SX-3826

(Fatal) Error: %s '%s' specified in silicon_marker_layer '%s' in MAPPING_FILE is not found.

**Description**

The specified DAMAGE_THICKNESS_TABLE or RPSQ_VS_SI_WIDTH table is not found in the NXTGRD.

**What Next**

Check if the MAPPING_FILE is written correctly and consistent with the ITF.

---

## SX-3827

(Fatal) Invalid %s specification '%s': '%s'

**Description**

The value specified for the min cores option is incorrect as described in the error message.

**What Next**

Rectify the option value based on the hints provided in the error message. Refer to the StarRC User Guide for more information.

---

## SX-3828

(Warning) Setting of the %s option is ignored because %s

Feedback

### Description

Setting of the option to specify the minimum number of cores needed to begin a DP run is ignored because of the reason given in the message.

### What Next

Take action based on the warning message.

---

## SX-3829

(Fatal) Value for the %s option "%s" is incorrect, expected timeout in minutes followed by an optional job submission timeout value in minutes. All prescribed values should be non-negative integers.

### Description

The value of the DP timeout option is the number of minutes after which a StarRC run should abort if the number of live workers is less than the value prescribed in the corresponding min cores option. If not prescribed, StarRC will wait indefinitely for at least one worker to go live.

Optionally, the timeout value can be followed by another value prescribing the number of minutes StarRC should wait for job submission commands to complete. If not prescribed, StarRC will terminate execution if job submission commands do not complete within 1 minute.

### What Next

Set the timeout option values as described above and run StarRC again.

---

## SX-3830

(Fatal) Both %s and %s must be set to enable the DP timeout feature.

### Description

Both the min cores and timeout options must be set in order to use the DP timeout features.

### What Next

Set both the options in the StarRC command file to use the DP timeout feature, or remove both of them to disable it.

Feedback

## SX-3831

(Warning) Found non-color polygon on multi-patterning layer '%s'. BBox=(%g,%g),(%g, %g) %s

### Description

There is a non-color polygon found on the layer.

At TSMC advanced nodes, different behavior will be applied for non-color polygons automatically based on the process node.

- For TSMC N7/N6: StarRC drops non-color polygons (NON_COLOR_POLYGON_HANDLING:DROP).
- For TSMC N5/N5P/N4/N3: StarRC errors out when a non-color polygon is found (NON_COLOR_POLYGON_HANDLING:ERROR).

You can manually control the behavior by NON_COLOR_POLYGON_HANDLING in StarRC command file. Valid values,

- NONE - Disable the feature. Issue a SX-3931 warning if non-color polygon detected.
- DROP - Drop each non-color polygon. Issue a SX-3931 warning if non-color polygon detected.
- WARN - Issue this warning for each non-color polygon.
- DROP_AND_WARN - Drop and issue this warning for each non-color polygon.
- ERROR - Error out when a non-color polygon is found.

  In addition, StarRC provides a hidden option, DEBUG_NON_COLOR_POLYGON: {NO|YES} to ease the debugging of non-color polygons for NON_COLOR_POLYGON_HANDLING: ERROR. With the option set, StarRC will report non-color polygons in the summary file until Cells stage exit. To report all non-color polygons, please also increase message suppression limit for SX-3831 by MESSAGE_SUPPRESSION.

### SELL ALSO

```
SX-3214
SX-3831
SX-3931
NON_COLOR_POLYGON_HANDLING
DEBUG_NON_COLOR_POLYGON: YES
```

### What Next

See if the non-color polygon is expected.

Feedback

## SX-3832

(Warning) There are two nets with different net id has the same netname net1: %s net_id: %ld, net2: %s net_id: %ld. if they are different actually, please set CASE_SENSITIVE: YES.

**Description**

if case_sensitive:no, we think net1 and NET2 are the same.

**What Next**

Set CASE_SENSITIVE:YES

## SX-3833

(Warning) License information: %s

**Description**

StarRC has a status update for checking out a license.

**What Next**

Check if (a) the LM_LICENSE_FILE environment variable is set correctly, (b) your license server is setup and working correctly and (c) You have enough available licenses to run StarRC using options specified in the command file. The use of some features of StarRC may have additional licensing requirements.

Contact Synopsys technical support for more information.

## SX-3834

(Fatal) Incorrect formatting of file passed with option NETLIST_TERMINAL_MAPPING_FILE: %s.

**Description**

The formatting of the file passed with option NETLIST_TERMINAL_MAPPING_FILE is incorrect.

**What Next**

Follow the correct formatting of the file passed with option NETLIST_TERMINAL_MAPPING_FILE.

Feedback

## SX-3835

(Fatal) More than one terminal name mapping for model '%s' in NETLIST_TERMINAL_MAPPING_FILE: %s.

**Description**

Only one mapping is allowed for each device terminal name.

**What Next**

Ensure each terminal name has only one mapping.

## SX-3836

(Fatal) NETLIST_TERMINAL_MAPPING_FILE can only be used with XREF:YES.

**Description**

The NETLIST_TERMINAL_MAPPING_FILE command can only be used with XREF:YES command.

**What Next**

Use XREF:YES in order to use NETLIST_TERMINAL_MAPPING_FILE in the StarRC command file.

## SX-3837

(Fatal) NETLIST_TERMINAL_MAPPING_FILE cannot be used with XREF_USE_LAYOUT_DEVICE_NAME:YES.

**Description**

The NETLIST_TERMINAL_MAPPING_FILE command cannot be used with XREF_USE_LAYOUT_DEVICE_NAME:YES command.

**What Next**

Remove XREF_USE_LAYOUT_DEVICE_NAME:YES in order to use NETLIST_TERMINAL_MAPPING_FILE in the StarRC command file.

## SX-3838

(Fatal) NETLIST_TERMINAL_MAPPING_FILE cannot be used with XREF_USE_LAYOUT_TERMINAL_NAME:YES.

Feedback

**Description**

The NETLIST_TERMINAL_MAPPING_FILE command cannot be used with XREF_USE_LAYOUT_TERMINAL_NAME:YES command.

**What Next**

Remove XREF_USE_LAYOUT_TERMINAL_NAME:YES in order to use NETLIST_TERMINAL_MAPPING_FILE in the StarRC command file.

---

## SX-3839

(Fatal) Found non-color polygons in the design.

**Description**

There are non-color polygons found in the design.

**SEEL ALSO**

SX-3831

---

## SX-3840

(Fatal) Get a non-expected negative width value for net width from DEF file, value %d

**Description**

The width of a net should be non-negative value.

**What Next**

Update the DEF database

---

## SX-3841

(Error) Invalid setting for NETLIST_DEVICE_LOCATION_ORIENTATION! Please set it as 'NO' or re-run StarRC with YES/COMMENT.

**Description**

-clean_converter do not support NETLIST_DEVICE_LOCATION_ORIENTATION:YES|COMMENT if pre run is NO

**What Next**

set it as NO or rerun without -clean_converter option

Feedback

## SX-3842

(Information) ... Waiting for %s license (%d attempts).

**Description**

StarRC is making repeated attempts to check out a license.

**What Next**

If license checkout fails or is unduly delayed, please make sure sufficient StarRC licenses are available.

## SX-3843

(Warning) EXTRACT_RES_BODY_RESISTANCE is turned off because EXTRACT_RES_BODY_COUPLING is off.

**Description**

EXTRACT_RES_BODY_RESISTANCE must be set together with EXTRACT_RES_BODY_COUPLING.

**What Next**

Remove EXTRACT_RES_BODY_RESISTANCE from or add EXTRACT_RES_BODY_COUPLING: YES to StarRC command file.

## SX-3844

(Warning) RC_SCALING_FILE is ignored for field solver flow.

**Description**

RC_SCALING_FILE is not supported in field solver flow.

**What Next**

remove RC_SCALING_FILE from StarRC command file.

## SX-3845

(Warning) EXCLUDE_SADP_SHORTS_FROM_VIOLATION_REPORT is YES which suppresses violation reports of any shorts to SADP fill.

**Description**

Not available.

Feedback

**What Next**

EXCLUDE_SADP_SHORTS_FROM_VIOLATION_REPORT: NO if this violation filter is not intended.

---

## SX-3846

(Warning) clean_converter do not support %s : %s if original value is %s.

**Description**

-clean_converter

**What Next**

rerun without -clean_converter option

---

## SX-3847

(Error) Wrong syntax for CONTEXT_DEF/OASIS/GDS_FILE command usage.

**Description**

The usage of CONTEXT_DEF/OASIS/GDS_FILE should be follow with block or placement.

**What Next**

update command or remove it

---

## SX-3848

(Fatal) -clean_converter do not support %s.

**Description**

rerun full flow

**What Next**

set it as NO or rerun without -clean_converter option

---

## SX-3849

(Warning) RC_SCALING_FILE is set in StarRC command file. It is not meant for final signoff flow.

Feedback

**Description**

RC_SCALING_FILE is set in StarRC command file. RC values in the final netlist are scaled based on the scaling factor set in that file.

**What Next**

Make sure this option is not used during sign-off.

---

## SX-3850

(Fatal) Unable to perform a %s operation on file '%s', error status '%s', host '%s', working directory '%s'

**Description**

StarRC is unable to lock a file because of issues in the runtime environment.

**What Next**

Try running StarRC again or contact your system administrator.

---

## SX-3851

(Information) A %s operation on file '%s' completed after %d attempts

**Description**

This message is output when StarRC is able to complete locking operations on a file after multiple attempts if the first attempt had failed.

**What Next**

This message is informational, no further actions are required.

---

## SX-3852

(Warning) Unable to close file '%s', error status '%s'

**Description**

StarRC is unable to close a file because of conditions in the runtime environment.

---

## SX-3853

(Information) Failed to get ultraplus type licenses. Will release ultraplus type licenses and try again.

Feedback

**Description**

Not all the Ultraplus type licenses needed were checked out. All the Ultraplus_type licenses that have been checked out will be released and StarRC will try again to get the required licenses. If this happens frequently, more Ultraplus type licenses maybe needed on the server to meet demand.

---

## SX-3854

(Warning) INSTANCE_PORT_LOCATION_CLOSER_TO_DRIVER in transistor level flow needs SKIP_CELL_PORT_PROP_FILE

**Description**

INSTANCE_PORT_LOCATION_CLOSER_TO_DRIVER which is used to assign instance port location closer to the driver needs instance port direction information from SKIP_CELL_PORT_PROP_FILE.

**What Next**

This is a warning message, nothing needs to be done from the user side. We will enhance the option to be supported for transistor level flow in future StarRC release.

---

## SX-3860

(Warning) INPUT_NAMES_ESCAPE_REMOVAL is enabled, please make sure that the net select commands remove the escape at the same time.

**Description**

INPUT_NAMES_ESCAPE_REMOVAL command open by default.

---

## SX-3861

(Warning) GROUND_CAP_DISTRIBUTION_MODE is supported in transistor-level flows only.

**Description**

GROUND_CAP_DISTRIBUTION_MODE allows to control the distribution of the ground cap to power nets. This feature is limited at transistor-level only. The run is at gate-level, so the option is ignored.

**What Next**

Remove GROUND_CAP_DISTRIBUTION_MODE from StarRC command file.

Feedback

## SX-3862

(Warning) GROUND_CAP_DISTRIBUTION_MODE has no effect when TRANSLATE_RETAIN_BULK_LAYERS is set to NO.

**Description**

GROUND_CAP_DISTRIBUTION_MODE allows to control the distribution of the ground cap to power nets. This feature has an impact only when the substrate layers are translated. TRANSLATE_RETAIN_BULK_LAYERS is set to NO so the option is ignored.

**What Next**

Remove GROUND_CAP_DISTRIBUTION_MODE from StarRC command file or set TRANSLATE_RETAIN_BULK_LAYERS to appropriate value.

## SX-3863

(Fatal) Invalid option '%s' of GROUND_CAP_DISTRIBUTION_MODE. Valid values are: [NETLIST_GROUND_NODE_NAME], CLOSEST, FURTHEST or ALL

**Description**

GROUND_CAP_DISTRIBUTION_MODE allows to control the distribution of the ground cap to power nets. Valid values are: [NETLIST_GROUND_NODE_NAME], CLOSEST, FURTHEST or ALL

**What Next**

Remove GROUND_CAP_DISTRIBUTION_MODE from StarRC command file or set to valid value.

## SX-3864

(Warning) Field solver flows are not supported with non-default setting of GROUND_CAP_DISTRIBUTION_MODE.

**Description**

GROUND_CAP_DISTRIBUTION_MODE allows to control the distribution of the ground cap to power nets. By redistributing intentionally some ground caps to specific substrate shapes, power net accuracy can be impacted and correlation with field solver results is not ensured.

**What Next**

Remove GROUND_CAP_DISTRIBUTION_MODE from StarRC command file or disable field solver flows (EXTRACTION:FSCOMPARE, FS_EXTRACT_NETS).

Feedback

## SX-3865

(Fatal) Options %s are mutually exclusive.

**Description**

The options are listed in the error message cannot be used at the same time.

**What Next**

Use only one of the options.

## SX-3866

(Fatal) Valid value for '%s' are '%s'.

**Description**

Only listed values are valid for this command.

**What Next**

Use only valid values.

## SX-3867

(Fatal) %s is missing in %s file.

**Description**

Necessary setting is missing in the file.

**What Next**

specify all necessary settings in the file.

## SX-3868

(Fatal) layout does not meet restricted mode's feture restrictions.

**Description**

The layout would require features that are outside of the features enabled by STAR-RC2_ULTRA_MANAGER, STAR-RC2-AGP and/or STAR-RC2_AG3, and restricted mode only enables features under these 3 licenses.

**What Next**

Restart interactive mode in unrestricted mode or modify layout to meet restrictions.

Feedback

## SX-3869

(Fatal) layout does not meet restricted mode's restrictions.

**Description**

The layout has more than 10 nets in the xin file for extraction and restricted mode allows a maximum of 10 nets in the xin file.

**What Next**

Restart interactive mode without restrictions or modify layout to meet restrictions.

## SX-3870

(Fatal) File %s specified in TCNR_SIMULATION_SETUP_FILE is not executable.

**Description**

File specified for HSPICE_EXECUTABLE needs to be an executable file.

**What Next**

specify an executable file for HSPICE_EXECUTABLE in TCNR_SIMULATION_SETUP_FILE.

## SX-3871

(Fatal) %s specified in TCNR_SIMULATION_SETUP_FILE is not a directory.

**Description**

CMI_PATH needs to be a directory.

**What Next**

specify a directory for CMI_PATH in TCNR_SIMULATION_SETUP_FILE.

## SX-3872

(Fatal) entry name is missing in HSP_FILE specified in TCNR_SIMULATION_SETUP_FILE.

**Description**

entry name is missing in HSP_FILE specified in TCNR_SIMULATION_SETUP_FILE.

Feedback

**What Next**

specify hsp file name and entry name after HSP_FILE in TCNR_SIMULATION_SETUP_FILE.

---

## SX-3873

(Fatal) wrong syntax %s in TCNR_SIMULATION_SETUP_FILE.

**Description**

The syntax is not supported in TCNR_SIMULATION_SETUP_FILE.

**What Next**

correct the syntax in TCNR_SIMULATION_SETUP_FILE.

---

## SX-3874

(Fatal) TCNR simulation failed.

**Description**

TCNR_SIMULATION_SETUP_FILE is specified in StarRC command file. StarRC calls HSPICE to simulate TCNR but the simulation failed.

**What Next**

Check if TCNR_SIMULATION_SETUP_FILE has correct version of HSPICE executable and libraries. Check tcnr_spice.hspice.log under the star directory for more info on HSPICE simulation.

---

## SX-3875

(Fatal) The command ESTIMATED_CONNECT_OPENS_VIAR_SCALE must have one argument (scale).

**Description**

The syntax of the command is not right.

**What Next**

Follow the syntax of the command.

Feedback

## SX-3876

(Fatal) The command ESTIMATED_CONNECT_OPEN_VIAR_SCALE_NET can only be used if ESTIMATED_CONNECT_OPENS_VIAR_SCALE is also used.

**Description**

The command ESTIMATED_CONNECT_OPEN_VIAR_SCALE_NET can only be used if ESTIMATED_CONNECT_OPENS_VIAR_SCALE is also used.

**What Next**

Add ESTIMATED_CONNECT_OPENS_VIAR_SCALE to StarRC command file.

## SX-3877

(Fatal) The StarRC command %s should be set to either %s.

**Description**

The StarRC command is set to an unavailable choice.

**What Next**

Set the StarRC command to one of the available choices or unset it.

## SX-3878

(Fatal) SPICE_FILE: %s specified in TCNR_SIMULATION_SETUP_FILE does not have any device.

**Description**

SPICE_FILE specified in TCNR_SIMULATION_SETUP_FILE is used for TCNR simulation by HSPICE. HSPICE cannot simulate if there is no content.

**What Next**

modify or remove SPICE_FILE in TCNR_SIMULATION_SETUP_FILE.

## SX-3879

(Information) TCNR simulation is skipped since there is no transistor in the layout.

**Description**

Spice file is not generated if there is no transistor in the layout. TCNR simulation is not needed.

Feedback

**What Next**

Information only.

---

## SX-3880

(Warning) The %s violations are not checked and the %s not reported. A non-existing % summary file is not ensuring there is no violation.

**Description**

SKIP_VIOLATION_REPORT is set to OPENS, SHORTS or BOTH, or one of the options of the StarRC command file is setting SKIP_VIOLATION_REPORT to OPENS, SHORTS or BOTH.

**What Next**

If it's not intentional, remove SKIP_VIOLATION_REPORT in the StarRC command file or add SKIP_VIOLATION_REPORT set to NO to the StarRC command file.

---

## SX-3881

(Warning) EXTRACT_RES_BODY_RESISTANCE can not work for res bodys device %s on superconductive substrate.

**Description**

EXTRACT_RES_BODY_RESISTANCE must be set for res body when available R value.

**What Next**

Remove reported devices from EXTRACT_RES_BODY_RESISTANCE.

---

## SX-3883

(Warning) Nets have duplicate name '%s' are combined. INPUT_NAMES_ESCAPE_REMOVAL command can cause duplicate net names.

**Description**

Nets have duplicate name are combined. INPUT_NAMES_ESCAPE_REMOVAL command can cause duplicate net names.

**What Next**

Check net names and INPUT_NAMES_ESCAPE_REMOVAL command

Feedback

## SX-3888

(Fatal) Unsupported option for TRBL_GENDEV command: only one option - MOSLIKE or ALL - allowed.

**Description**

Unsupported option for TRBL_GENDEV command: only one option - MOSLIKE or ALL - allowed.

**What Next**

Please refer to user manual for supported options for TRBL_GENDEV command.

## SX-3889

(Fatal) TRBL_GENDEV command can only be used with TRANSLATE_RETAIN_BULK_LAYERS:CONLY.

**Description**

TRBL_GENDEV command can only be used with TRANSLATE_RETAIN_BULK_LAYERS:CONLY.

**What Next**

Please refer to user manual for supported options for TRBL_GENDEV command.

## SX-3890

(Fatal) %s parse error: %s

**Description**

There was a problem parsing the command.

**What Next**

Check the command syntax and fix the error.

## SX-3891

(Fatal) failed to submit %s %s

**Description**

A job could not be submitted to the farm. The tool may have produced more information that could help diagnose the reason for failure.

Feedback

**What Next**

This can happen because of errors in the GPD_DP_STRING option used for GPD conversion or STARRC_DP_STRING for regular StarRC runs. Check those options for errors.

---

## SX-3892

(Information) %s is ignored when TMIRC is set to NO.

**Description**

TMIRC related setting in StarRC command file is ignored since TMIRC is set to NO.

**What Next**

Information only.

---

## SX-3893

(Fatal) %s is missing when TMIRC is on.

**Description**

TMIRC related setting in StarRC command file is necessary when TMIRC is on.

**What Next**

Set necessary TMIRC related settings in StarRC command file.

---

## SX-3894

(Fatal) %s and %s must be set together.

**Description**

Two commands must be set together.

**What Next**

modify StarRC command file.

---

## SX-3895

(Warning) RCAPI is turned off for missing RCAPI_LIBRARY_NAME in the NXTGRD file.

Feedback

**Description**

RCAPI is specified in StarRC command file, but there is no RCAPI library specified in the NXTGRD file.

**What Next**

remove RCAPI in the StarRC command file or use an NXTGRD file with RCAPI library specified.

---

## SX-3896

(Fatal) Gate direction needs to be determined for etching trench contact layers with TC_ETCH_VS_WIDTH_AND_LENGTH table, this info is missing.

**Description**

Gate direction was not determined.

**What Next**

Gate direction needs to be determined.

---

## SX-3897

(Fatal) Invalid value %s for option %s, expected %s.

**Description**

The value specified for an option in the StarRC command file is invalid.

**What Next**

Change option value based on the error message.

---

## SX-3898

(Fatal) A %s operation with %s on host %s, directory %s, pid %d has timed out after %d seconds at %s. This indicates an issue with with the filesystem. Refer this to your IT department.

**Description**

This error message is generated whenever StarRC is run with DP_LOCK_TIME_OUT set and the run encounters a system issue while performing file locking operations. This indicates a problem with the filesystem and the error message is intended to provide maximum diagnostics in order for IT to investigate the problem.

Feedback

**What Next**

Refer the error message to your IT department.

---

## SX-3899

(Fatal) INTERNAL Error: CORNER_TYPE has been discontinued from 2020.06 release.

**Description**

CORNER_TYPE (one of CORNER options) has been discontinued from 2020.06 release.

**What Next**

Please don't use CORNER_TYPE, it has been discontinued.

---

## SX-3900

(Information) Failed to checkout %s licenses. Will release all licenses and try again in 5 min.

**Description**

Not all the licenses needed were checked out. All the ultra_manager and addon licenses will be released and StarRC will try again to get the required licenses. If this happens frequently, more licenses of the named type might be needed on the server to meet demand.

---

## SX-3901

(Information) %i attempts.

**Description**

Licenses have been attempted some number of times.

---

## SX-3902

(Fatal) RAPHAEL_NETS can only be used together with unencrypted itf in non SMC FSCOMPARE flow.

**Description**

RAPHAEL_NETS can only be used together with unencrypted itf in non SMC FSCOMPARE flow.

Feedback

**What Next**

RAPHAEL_NETS can only be used together with unencrypted itf in non SMC FSCOMPARE flow.

---

## SX-3903

(Warning) NETLIST_SKIP_RESSTAR_MODEL is only supported when TEMPERATURE_SENSITIVITY is set.

**Description**

NETLIST_SKIP_RESSTAR_MODEL is intended to remove resStar model in netlist when TEMPERATURE_SENSITIVITY is set.

**What Next**

Please confirm if TEMPERATURE_SENSITIVITY option is set correctly. Otherwise, you can ignorethis message.

---

## SX-3904

(Warning) very high CPU utilization %.1f%% on host %s at %s

**Description**

The host on which StarRC is running is experiencing heavy CPU load. This may lead to increased runtime.

**What Next**

This message is informational only. If runtime is critical, StarRC processes should be run on lightly loaded hosts. Consult your IT department for more information.

---

## SX-3905

(Warning) only %.2f GB %s available (%.1f%% of total) on host %s at %s

**Description**

The host on which StarRC is running has very little RAM or swap space available as indicated in the message. This may have a detrimental impact on StarRC's performance.

**What Next**

This message is informational only. If runtime is critical, StarRC processes should be run on lightly loaded hosts. Consult your IT department for more information.

Feedback

## SX-3906

(Warning) file system %s mounted on %s is full at %s

### Description

The file system on which StarRC's working directory resides is full. This could cause StarRC runs to terminate abnormally or produce unexpected results.

### What Next

This message is informational only. Move the StarRC run to another file system to prevent unexpected program behavior. Consult your IT department for more information.

## SX-3907

(Warning) only %.2f GB of space (%.1f%% of total) is available on file system %s mounted on %s at %s

### Description

This on which StarRC is running has very little space available, which can cause runs to terminate abnormally or produce unexpected results.

### What Next

This message is informational only. Move the StarRC run to another file system to prevent unexpected program behavior. Consult your IT department for more information.

## SX-3908

(Error) found polygon with net ID %d which exceeds StarRC's maximum net ID %d.

### Description

The number of nets in the design exceeds StarRC's maximum net ID limit or StarRC has internally assigned a net ID larger than the maximum net ID.

### What Next

Contact Synopsys support.

## SX-3909

(Warning) NETLIST_REMOVE_DANGLING_BRANCHES is reset to NO, when CAPACITOR_TAIL_COMMENTS: YES.

Feedback

**Description**

If NETLIST_REMOVE_DANGLING_BRANCHES: YES is
set along with CAPACITOR_TAIL_COMMENTS: YES, reset
NETLIST_REMOVE_DANGLING_BRANCHES to NO to avoid dangling branches being pruned on the net connected to the GATE, therefore coupling caps wouldn't get moved to the gate.

**What Next**

If you want to use command CAPACITOR_TAIL_COMMENTS: YES, please change NETLIST_REMOVE_DANGLING_BRANCHES to NO. If you don't need that command, remove it from command file.

---

## SX-3910

(Information) map_qtf_layers and qtf_layers sections will be generated from QTF_MAPPING_FILE.

**Description**

When QTF_MAPPING_FILE is specified in star_cmd for StarQTF low, the above two sections will be generated from QTF_MAPPING_FILE and then used for StarQTF flow.

**What Next**

Information only.

---

## SX-3911

(Warning) map_qtf_layers and qtf_layers sections in MAPPING_FILE will be overwritten.

**Description**

When QTF_MAPPING_FILE is specified in star_cmd, map_qtf_layers and qtf_layers will be generated from QTF_MAPPING_FILE even if they exist in the input MAPPING_FILE.

**What Next**

If you want to use the map_qtf_layers and qtf_layers sections defined in the input MAPPING_FILE for StarQTF flow, please remove QTF_MAPPING_FILE option from star_cmd.

---

## SX-3912

(Warning) failed to submit job using '%s', check your %s option for possible errors.

Feedback

**Description**

StarRC is unable to submit a job to the compute farm using a command based on a DP option provided by the user. This message indicates a possible error in the job submission command provided by the user or an issue with the user's compute farm management system.

**What Next**

Check the job submission command or consult your IT department.

---

## SX-3913

(Fatal) failed to launch GPD converter jobs.

**Description**

StarRC could not launch GPD converter jobs to resources specified using GPD_DP_STRING. This message should have been preceeded with other warnings to indicate what the problem may be.

**What Next**

Take actions based on warnings produced by StarRC before this error message.

---

## SX-3914

(Warning) command '%s' failed to complete in %d seconds.

**Description**

An external program launched by StarRC failed to complete in a stipulated interval of time.

StarRC sets a timeout value when it expects the programs that it launches to complete very quickly, ideally in fractions of a second. This error message indicates system overload or a problem with the program being executed.

**What Next**

Check for system overload or problems with the command being executed. Contact your IT department for more information if necessary.

---

## SX-3915

(Fatal) invalid %s value %s, expected %s.

**Description**

The value of specified for an option is incorrect as stated in the error message.

Feedback

**What Next**

Correct the option value based on recommendations in the error message.

---

## SX-3916

(Fatal) FSCOMPARE_OPTIONS: -fs_relax_co value cannot be smaller than 1.

**Description**

fs_relax_co value cannot be smaller than 1.

**What Next**

Define fs_relax_co larger than 1 or use the default value of 10.

---

## SX-3919

(Warning) NETLIST_TYPE is reset to RCC, when CAPACITOR_TAIL_COMMENTS: YES and EXTRACTION is C.

**Description**

If NETLIST_TYPE is set along with CAPACITOR_TAIL_COMMENTS: YES and EXTRACTION: C, reset it to RCC.

**What Next**

If you want to use command CAPACITOR_TAIL_COMMENTS: YES, please change NETLIST_TYPE to RCC. If you don't need that command, remove it from command file.

---

## SX-3920

(Warning) NETLIST_TYPE is reset to RCC, when CAPACITOR_TAIL_COMMENTS: YES and EXTRACTION is RC.

**Description**

If NETLIST_TYPE is set along with CAPACITOR_TAIL_COMMENTS: YES and EXTRACTION: RC, reset it to RCC.

**What Next**

If you want to use command CAPACITOR_TAIL_COMMENTS: YES, please change NETLIST_TYPE to RCC. If you don't need that command, remove it from command file.

Feedback

## SX-3931

(Warning) Polygon without MASK info detected on a layer with MASK info in ITF; Please check StarRC User Guide and run with NON_COLOR_POLYGON_HANDLING: %s to get list of polygons without MASK information.

### Description

Non-color polygon (polygon without MASK info) found on a color layer. Please check StarRC User Guide and run with NON_COLOR_POLYGON_HANDLING: WARN or DROP_AND_WARN to get list of polygons without MASK information.

At TSMC advanced nodes, different behavior will be applied for non-color polygons automatically based on the process node.

- For TSMC N7/N6: StarRC drops non-color polygons (NON_COLOR_POLYGON_HANDLING:DROP).
- For TSMC N5/N5P/N4/N3: StarRC errors out when a non-color polygon is found (NON_COLOR_POLYGON_HANDLING:ERROR).

You can manually control the behavior by NON_COLOR_POLYGON_HANDLING in StarRC command file. Valid values,

- NONE - Disable the feature. Issue this warning if non-color polygon detected.
- DROP - Drop each non-color polygon. Issue this warning if non-color polygon detected.
- WARN - Issue SX-3831 warning for each non-color polygon.
- DROP_AND_WARN - Drop and issue SX-3831 warning for each non-color polygon.
- ERROR - Error out when a non-color polygon is found.

  In addition, StarRC provides a hidden option, DEBUG_NON_COLOR_POLYGON: {NO|YES} to ease the debugging of non-color polygons for NON_COLOR_POLYGON_HANDLING: ERROR. With the option set, StarRC will report non-color polygons in the summary file until Cells stage exit. To report all non-color polygons, please also increase message suppression limit for SX-3831 by MESSAGE_SUPPRESSION.

### SELL ALSO

```
SX-3214
SX-3831
SX-3931
NON_COLOR_POLYGON_HANDLING
DEBUG_NON_COLOR_POLYGON: YES
```

Feedback

**What Next**

See if the non-color polygon is expected.

---

## SX-3932

(Fatal) OA_SEPARATE_PARASITICS cannot be set with OA_SEPARATE_INSTANCES.

**Description**

OA_SEPARATE_PARASITICS and OA_SEPARATE_INSTANCES can not be set simultaneously in StarRC command file.

**What Next**

Set either OA_SEPARATE_PARASITICS or OA_SEPARATE_INSTANCES in StarRC command file.

---

## SX-3933

(Fatal) invalid %s value '%s' : %s.

**Description**

The value of specified for an option is incorrect as stated in the error message.

**What Next**

Correct the option value based on recommendations in the error message.

---

## SX-3934

(Fatal) cannot set value for option %s without setting value for option %s.

**Description**

Some of the options in a StarRC configuration file can be used only when another option is used. This error message indicates an attempt to set the value of an option without setting the value of another option that is required to be set.

**What Next**

Correct the option values based on recommendations in the error message.

---

## SX-3935

(Fatal) invalid value %s for option %s, expected %s.

Feedback

**Description**

The value specified for an option in a StarRC configuration file is invalid.

**What Next**

Correct the option value based on recommendations in the error message.

---

## SX-3937

(Fatal) Recursive cell reference (cell name: '%s') in NDM library is detected. Duplicate cell names can cause this error.

**Description**

Recursive cell reference is illegal for NDM. Duplicate cell names in NDM library can also cause this error. StrRC cannot handle duplicate cell names and will lead to unexpected problems.

**What Next**

Check NDM library and cell names. If there is cell name duplicated, rename the duplicte cell name.

---

## SX-3938

(Warning) Galaxy Parasitic Data (GPD) does not support the %s command with %s

**Description**

Galaxy Parasitic Data (GPD) does not support the highlighted command when other options are used.

**What Next**

Remove/change the highlighted commands to run the GPD flow or the non-GPD flow will run.

---

## SX-3939

(Fatal) Unable to ID from CLL, caught exception: %s

**Description**

An error occurred while obtaining an ID from the CLL system.

**What Next**

Contact Synopsys support for more information.

Feedback

## SX-3940

(Fatal) The socket-based DP flow is not supported on the linuxppc64 platform.

**Description**

The socket-based DP feature is not supported on the linuxppc64 platform.

**What Next**

Remove SOCKET_BASED_DP: yes from the StarRC command file to get past this error.

## SX-3941

(Warning) call to system function %s failed: %s [%s], retrying.

**Description**

The call to a function provided by the operating system interface failed. This function call will be retried.

**What Next**

This warning message may indicate an issue with the runtime environment.

## SX-3942

(Fatal) call to system function %s failed: %s [%s] after multiple attempts.

**Description**

The call to a function provided by the operating system interface failed after multiple attempts.

**What Next**

This error message may indicate an issue with the runtime environment, consult your IT team.

## SX-3944

(Fatal) a %s operation on file %s has failed, see %s for more information.

**Description**

An operation performed by StarRC has failed, see the file mentioned in the error message for more information.

Feedback

**What Next**

Such issues usually occur intermittently because of instabilities in the runtime environment. Contact Synopsys support if this issue can be reproduced consistently.

---

## SX-3946

(Warning) FS_BOUNDARY_CONDITION_ON_RESISTANCE is turned off since there is no FS_BOUNDARY_BOX or FS_BOUNDARY_CONDITION specified.

**Description**

FS_BOUNDARY_CONDITION_ON_RESISTANCE only works when FS_BOUNDARY_BOX and FS_BOUNDARY_CONDITION are specified.

**What Next**

Remove FS_BOUNDARY_CONDITION_ON_RESISTANCE from StarRC command file.

---

## SX-3947

(Warning) FS_BOUNDARY_CONDITION_ON_RESISTANCE is turned off for non-FS flow.

**Description**

FS_BOUNDARY_CONDITION_ON_RESISTANCE only works for FS flow.

**What Next**

Remove FS_BOUNDARY_CONDITION_ON_RESISTANCE from StarRC command file.

---

## SX-3948

(Warning) FS_BOUNDARY_CONDITION is not applied in resistance extraction on big design.

**Description**

FS_BOUNDARY_CONDITION can only be applied in resistance extraction on small layout.

**What Next**

N/A

---

## SX-3949

(Warning) GDS SEED PROPERTY DEVICE ORIGINAL is not set in query file.

Feedback

**Description**

GDS SEED PROPERTY DEVICE ORIGINAL is not set in the query file, while EM_PARAM_MAPPING_FILE is used. Device region cannot be extracted without this query command.

**What Next**

You should add "GDS SEED PROPERTY DEVICE ORIGINAL" in the calibre query file.

---

## SX-3950

(Warning) The StarRC version is older than the nxtgrd file (grdgenxo version %s). It may cause unexpected extraction result.

**Description**

The nxtgrd file is generated by a newer version of grdgenxo than the StarXtract used for extraction, which may cause unexpected extraction result.

**What Next**

Use the StarRC version newer than the version of grdgenxo which generated the nxtgrd file.

---

## SX-3951

(Fatal) Via layer "%s" is not defined in ITF file.

**Description**

StarRC gives this error once it finds a via layer is not defined in ITF file.

**What Next**

You should modify the mapping file.

---

## SX-3952

(Warning) %s when TMIRC is set to YES.

**Description**

The StarRC command setting is not compatible with TMIRC. It is changed accordingly when TMIRC is on.

**What Next**

Change the StarRC command accordingly in the StarRC command file.

Feedback

## SX-3953

(Fatal) OA_USE_LAYOUT_DEVICE_NAME can only be used with XREF:YES.

**Description**

The OA_USE_LAYOUT_DEVICE_NAME command can only be used with XREF:YES command.

**What Next**

Use XREF:YES in order to use OA_USE_LAYOUT_DEVICE_NAME in the StarRC command file.

## SX-3954

(Fatal) OA_USE_LAYOUT_DEVICE_NAME cannot be used with XREF_USE_LAYOUT_DEVICE_NAME:YES.

**Description**

The OA_USE_LAYOUT_DEVICE_NAME command cannot be used with XREF_USE_LAYOUT_DEVICE_NAME:YES command.

**What Next**

Remove XREF_USE_LAYOUT_DEVICE_NAME:YES in order to use OA_USE_LAYOUT_DEVICE_NAME in the StarRC command file.

## SX-3956

(Fatal) THERMAL_PARAMETER_FILE is mandatory for all corners if any corner has it specified in CORNERS_FILE.

**Description**

This error message is issued when some corners have THERMAL_PARAMETER_FILE but the other corners don't have this field in CORNERS_FILE.
THERMAL_PARAMETER_FILE should be specified for ALL or NONE of the corners for correct extraction handling.

**What Next**

Correct the content CORNERS_FILE and restart. THERMAL_PARAMETER_FILE should be specified for ALL or NONE of the corners in CORNERS_FILE. If the same THERMAL_PARAMETER_FILE is applied to all corners, you can also choose to only specify it in the star command file.

Feedback

## SX-3957

(Fatal) Cannot open THERMAL_PARAMETER_FILE '%s' found on line '%d' in CORNERS_FILE '%s'

**Description**

You receive this message because StarRC failed to open the thermal parameter file specified by THEMRAL_PARAMETER_FILE.

**What Next**

Make sure that the file specified by THERMAL_PARAMETER_FILE isn't corrupted or missing and StarRC has permission to access it.

## SX-3958

(Fatal) THERMAL_PARAMETER_FILE in CORNERS_FILE must use absolute path when CORNERS_FILE is specified out of current working directory.

**Description**

You receive this message because THERMAL_PARAMETER_FILE defined in CORNERS_FILE does not use an absolute path to specify the via coverage option file when CORNERS_FILE is specified out of current working directory.

**What Next**

Move CORNERS_FILE into current working directory or use an absolute path to specify THERMAL_PARAMETER_FILE.

## SX-3960

(Error) NETLIST_COUPLE_UNSELECTED_NETS: ADVANCED must be used with NETLIST_FORMAT: SPEF.

**Description**

NETLIST_COUPLE_UNSELECTED_NETS: ADVANCED only works when NETLIST_FORMAT is SPEF.

**What Next**

set NETLIST_FORMAT: SPEF in the StarRC command file, or set NETLIST_COUPLE_UNSELECTED_NETS to NO, IDEAL or COMPLETE.

Feedback

## SX-3961

(WARNING) MOS_GATE_NON_NEGATIVE_DELTA_RESISTANCE_NETS can only be used with MOS_GATE_NON_NEGATIVE_DELTA_RESISTANCE: YES/NOPG, command is ignored.

**Description**

MOS_GATE_NON_NEGATIVE_DELTA_RESISTANCE_NETS can only be used when MOS_GATE_NON_NEGATIVE_DELTA_RESISTANCE is set to YES or NOPG.

**What Next**

Set MOS_GATE_NON_NEGATIVE_DELTA_RESISTANCE_NETS to YES/NOPG in StarRC command file.

## SX-3962

(Warning) REMOVE_SCHEDULER_FILES will be ignored when NUM_CORES is greater than 1 and STARRC_DP_STRING is not used.

**Description**

The REMOVE_SCHEDULER_FILES option removes used by StarRC's DP scheduler at the end of a run. This cannot be reliably done if the value of NUM_CORES is greater than 1 and STARRC_DP_STRING is not set as that implies running StarRC DP by launching multiple jobs on the command-line. Scheduler files cannot be reliably removed in this use model.

**What Next**

Use STARRC_DP_STRING to run StarRC DP. If this is not possible, remove or set REMOVE_SCHEDULER_FILES to in the StarRC command file.

## SX-3963

(Information) TMIRC_LIBRARY_VERSION=%s

**Description**

Information of TMIRC library version.

**What Next**

Information only.

Feedback

## SX-3964

(Warning) fake via segmentation is turned off for INTEL native flow in P1278 and later process.

**Description**

fake via has special model in INTEL native flow which require it is not segmented.

**What Next**

Remove TRENCH_CONTACT_VIRTUAL_VIA_SEGMENTATION_RATIO, TRENCH_CONTACT_VIRTUAL_VIA_SEGMENTATION_RATIO_FOR_POWER, and/or MAX_VIRTUAL_VIA_SEGMENTATION_NUMBER setting from StarRC command file.

## SX-3965

(Warning) Cell '%s' is not processed.

**Description**

Cell from metal fill file is not processed. It is not referenced in the design.

**What Next**

Information only; no action needed.

## SX-3966

(Information) Reading VMF file %s.

**Description**

StarRC is reading the option file in input.

**What Next**

Information only; no action needed.

## SX-3967

(Warning) Cell '%s' from '%s' may take lots of time to run GDS Donut flow in StarRC.

**Description**

It is not recommended to run GDS Donut flow in StarRC.

Feedback

**What Next**

Consider to use other tools to pre-run GDS Donut first.

---

## SX-3968

(Information) Using disk-based mechanisms for DP synchronization.

**Description**

StarRC will internally switch to using disk-based mechanisms for DP synchronization if the maximum number of open file descriptors per process supported on the host running the coordinator process is insufficient to support a large number of workers.

More information will be available in the dp_logs/star_logs/master.log file.

**What Next**

Set the NUM_CORES value to be less than the value prescribed in the dp_logs/star_logs/master.log file.

---

## SX-3969

(Warning) There is no fill shapes on layer '%s' and mask '%d' reading from input GDS(OASIS) files.

**Description**

Fill shapes on layer are empty. Layer may be undefined or wrong layer number in LAYER_MAP_FILE.

**What Next**

Verify the layer number in LAYER_MAP_FILE.

---

## SX-3970

(Severe) NETLIST_INSTANCE_SECTION:NO/SELECTED/not_set can only be used when at least one net is selected by NETS command.

**Description**

NETLIST_INSTANCE_SECTION:NO/SELECTED/not_set can only be used when at least one net is selected by NETS command.

**What Next**

Please check your NETS command, or set NETLSIT_INSTANCE_SECTION to YES.

Feedback

## SX-3971

(Warning) FS_EXTRACT_TALL_VIA_COUPLING is set, but FS_EXTRACT_NETS or EXTRACTION: FSCOMPARE is not set. FS_EXTRACT_TALL_VIA_COUPLING will do nothing when only StarRC is used.

**Description**

FS_EXTRACT_TALL_VIA_COUPLING is set, but FS_EXTRACT_NETS or EXTRACTION: FSCOMPARE is not set. FS_EXTRACT_TALL_VIA_COUPLING will do nothing when only StarRC is used.

**What Next**

No action is required.

## SX-3972

(Fatal) Inductance EM mode not recognized.

**Description**

Inductance_mode in star_cmd file does not have an acceptable value.

**What Next**

Check Inductance_mode setting.

## SX-3973

(Warning) GDS creation has pins that are points

**Description**

layout has pins that are points

**What Next**

no action needed, they will be given an area based on the containing polygon.

## SX-3974

(Fatal) Issue in inductance GDS creation.

**Description**

umbrella error to report issue in GDS creation

Feedback

**What Next**

report error to StarRC.

---

## SX-3976

(Fatal) INDUCTANCE_S_PARAM_FREQUENCY is not set.

**Description**

The EM flow for inductance only supports the INDUCTANCE_S_PARAM_FREQUENCY setting.

**What Next**

Check INDUCTANCE_S_PARAM_FREQUENCY is set.

---

## SX-3977

(Fatal) INDUCTANCE_S_PARAM_FREQUENCY is not set correctly.

**Description**

The EM flow for inductance only supports select modes of the INDUCTANCE_S_PARAM_FREQUENCY setting. Currently just LIN and DEC.

**What Next**

Check that INDUCTANCE_S_PARAM_FREQUENCY is set correctly with modes LIN or DEC.

---

## SX-3978

(Fatal) Absolute path '%s' is not permitted for star_cmd option SUMMARY_FILE.

**Description**

Absolute path is not permitted for star_cmd option SUMMARY_FILE, and only relative path is accepted.

**What Next**

Use relative path for star_cmd option SUMMARY_FILE.

---

## SX-3979

(Fatal) Cannot support the format of '%s %s'.

Feedback

**Description**

Only specific format is permitted for star_cmd option FSCOMPARE_OPTIONS -perc_coup and FSCOMPARE_OPTIONS -perc_slef.

**What Next**

Use correct format for star_cmd option FSCOMPARE_OPTIONS -perc_coup and FSCOMPARE_OPTIONS -perc_slef.

---

## SX-3980

(Fatal) No pins found.

**Description**

The EM flow for inductance requires pins. Please make sure there are pins in the nets to be extracted.

**What Next**

Check that the layout contains pins. Check that there are pins in the nets selected by INDUCTANCE_SELECT_NET.

---

## SX-3981

(Fatal) EM_Solver phase had errors.

**Description**

The Engine used by the EM_Solver phase did not compleate successfuly.

**What Next**

Check the inductance engine's logs for reported errors.

---

## SX-3982

(Fatal) 3D_IC_SUBCKT_FILE in CORNERS_FILE must use absolute path when CORNERS_FILE is specified out of current working directory.

**Description**

You receive this message because 3D_IC_SUBCKT_FILE defined in CORNERS_FILE does not use an absolute path to specify the via coverage option file when CORNERS_FILE is specified out of current working directory.

Feedback

**What Next**

Move CORNERS_FILE into current working directory or use an absolute path to specify 3D_IC_SUBCKT_FILE.

---

## SX-3983

(Fatal) StarRC gpd ran to completion but writing of the .gpd_complete file failed: '%s'

**Description**

A file opening problem was encountered when writing the .gpd_complete file.

**What Next**

Check to see if file system has sufficent space that disk performance is good.

---

## SX-3984

(Fatal) Found different TSV model '%s' from 3D_IC_SUBCKT_FILE '%s' in CORNERS_FILE '%s'

**Description**

You receive this message because StarRC found different TSV model settings in different 3D_IC_SUBCKT_FILE

**What Next**

Make sure that the TSV specified by 3D_IC_SUBCKT_FILE isn't corrupted or missing

---

## SX-3985

(Fatal) 3D_IC_SUBCKT_FILE should specify the same number of res and cap. \nStarRC only supports 1Res, or 2Res, or 2Res+1Cg models.

**Description**

You receive this message because StarRC found the different TSV settings specified by different 3D_IC_SUBCKT_FILE.

**What Next**

Make sure that the TSVs specified by 3D_IC_SUBCKT_FILE in different corners are set as the same res/cap number.

Feedback

## SX-3986

(Warning) Negative %s is specified for layer %s, which will be ignored.

**Description**

The specified value for the layer in layer mapping file should be non-negative. It will be ignored.

**What Next**

Check the value setting in layer mapping file.

## SX-3987

(Warning) %s of layer %s is %s than %s, it may introduce violations.

**Description**

The specified value in layer mapping file is larger than min_spacing or less than min_width of the layer. Though final line end extension may not trigger real violations, users still need to pay attention to the setting.

## SX-3988

(Fatal) Cannot open 3D_IC_SUBCKT_FILE '%s' found on line '%d' in CORNERS_FILE '%s'

**Description**

You receive this message because StarRC failed to open 3D_IC_SUBCKT_FILE specified in corner file.

**What Next**

Make sure that the file specified by 3D_IC_SUBCKT_FILE isn't corrupted or missing and StarRC has permission to access it.

## SX-3989

(Fatal) FS_QTF_FILE must be included for FS_ASSISTED_ESD

**Description**

FS_QTF_FILE is needed for FS_ASSISTED_ESD flow.

Feedback

**What Next**

Please include FS_QTF_FILE in star_cmd file.

---

## SX-3990

(Warning) Pin %s was dropped.

**Description**

This pin did not have a parent trapazoid/polygon.

**What Next**

check location of pin and make sure it is in or on the edge of a polygon.

---

## SX-3991

(Warning) GPD checker usage warning: Invalid check option "%s" found in check_options file, option skipped

**Description**

Undefined checker options found in user_defined check options file

**What Next**

Undefined checker options will be ignored

---

## SX-3992

(Fatal) GPD Checker usage error: %s

**Description**

Options specified to the Galaxy Parasitic Database (GPD) checker utility were incorrectly specified as indicated in the error message.

**What Next**

Specify the correct set of options for GPD checker. Refer to the StarRC User Guide for more information.

---

## SX-3993

(Warning) PRINT_VIA_VARIATION_MODEL is set to NO for NETLIST_TAIL_COMMENTS: NO.

Feedback

**Description**

PRINT_VIA_VARIATION_MODEL:YES print $vc_model in the netlist tail comments. NETLIST_TAIL_COMMENTS needs to be set to YES.

**What Next**

Set NETLIST_TAIL_COMMENTS to YES if $vc_model is needed.

---

## SX-3994

(Warning) Pin %s and Pin %s are to close on the same pic layer, pin %s was dropped.

**Description**

While creating area for pins, this pin was detected to have overlap with another pin that was kept or was not inside a parent polygon.

**What Next**

check location of pin and its neighbors on the same pic level.

---

## SX-3995

(Warning) Pin %s is on a via layer and was dropped.

**Description**

The pin was on a via layer and the selected inductance engine does not support via pins. This pin was dropped to enable continued running but should be addressed.

**What Next**

check location of pin place it on a non via layer.

---

## SX-3996

(Warning) Expected touchstone with %d pins but found one with %d pins.

**Description**

A pin was dropped by iris that was not anticipated.

**What Next**

investigate missing pin.

Feedback

## SX-3997

(Fatal) could not find valid touchstone file.

**Description**

something happened in EM solver that prevented creation of touchstone file.

**What Next**

investigate EM Solver logs.

## SX-3998

(Information) PRINT_VIA_VARIATION_MODEL is set to NO when the CCP_EXTRACTION_FILE is set in the StarRC command file.

**Description**

PRINT_VIA_VARIATION_MODEL is only supported in non-CCP flows.

**What Next**

Information only

## SX-3999

(Fatal) Unexpected error durring EM Flow stitching.

**Description**

umbrella error to report issue in touchstone spf stitching.

**What Next**

report error to StarRC

## SX-4000

(Warning) ITF layer "%s" specified in RESISTANCE_SCALING_VARIABLE_BY_LAYER is not found and will be skipped.

**Description**

Layer specified by RESISTANCE_SCALING_VARIABLE_BY_LAYER is not found.

**What Next**

Check if the command is written correctly.

Feedback

## SX-4001

(Warning) ITF layer "%s" has no scaling variable specified in RESISTANCE_SCALING_VARIABLE_BY_LAYER and will be skipped.

**Description**

Layer has no scaling variable specified by RESISTANCE_SCALING_VARIABLE_BY_LAYER.

**What Next**

Check if the command is written correctly. ITF layer and scaling variable should be in pair.

## SX-4005

(Fatal) ITF layer names and variable names are not in pairs in RESISTANCE_SCALING_VARIABLE_BY_LAYER

**Description**

ITF layer names and variable names must be defined by pairs in RESISTANCE_SCALING_VARIABLE_BY_LAYER

**What Next**

Change RESISTANCE_SCALING_VARIABLE_BY_LAYER setting in the StarRC command file.

## SX-4006

(Fatal) "%s" in RESISTANCE_SCALING_VARIABLE_BY_LAYER is not an ITF via layer name

**Description**

ITF via layer names and variable names must be defined by pairs in RESISTANCE_SCALING_VARIABLE_BY_LAYER

**What Next**

Change RESISTANCE_SCALING_VARIABLE_BY_LAYER setting in the StarRC command file.

## SX-4007

(Fatal) "%s" in RESISTANCE_SCALING_VARIABLE_BY_LAYER is not a fake via layer connecting diffusion and trench contact.

Feedback

**Description**

RESISTANCE_SCALING_VARIABLE_BY_LAYER only be specified on the fake via layer between diffusion and trench contact.

**What Next**

Change RESISTANCE_SCALING_VARIABLE_BY_LAYER setting in the StarRC command file.

---

## SX-4008

(Warning) NETLIST_NODE_SECTION: NO is not supported with OA_MULTI_OUTPUT in GPD flow, changed to NETLIST_NODE_SECTION :YES.

**Description**

In GPD flow, the NETLIST_NODE_SECTION will be changed to YES always with OA format even if user set NETLIST_NODE_SECTION: NO.

---

## SX-4009

(Warning) %s for TSMC N2 process.

**Description**

The StarRC command setting is not compatible with TSMC N2 process. It is changed accordingly.

**What Next**

Change the StarRC command accordingly in the StarRC command file.

---

## SX-4010

(Warning) NETLIST_LOWERCASE and CASE_SENSITIVE are used together.

**Description**

NETLIST_LOWERCASE is set, netlist file will be in lower case format, regardless the setting of CASE_SENSITIVE.

**What Next**

Check if case_sensitive can be ignored in netlist file.

Feedback

## SX-4011

(Warning) The option FS_ENHANCED_SRAM_ACCURACY is not supported in gate level flows.

**Description**

Currently FS_ENHANCED_SRAM_ACCURACY is only supported in transistor level flows, not gate level flows.

**What Next**

If this is acceptable, no action is needed otherwise contact the Synopsys Support Center.

## SX-4012

(Warning) The star_cmd option FS_QTF_FILE is not set in FS run. This might lead to cap inaccuracy for transistor-level case with TSMC advanced process.

**Description**

For transistor-level cases with TSMC advanced process, FS_QTF_FILE is needed for FS run to get accurate cap results.

**What Next**

Add FS_QTF_FILE in star_cmd file.

## SX-4013

(Warning) The star_cmd option NETLIST_MAX_LINE is ignored in GPD flow. This might have no action.

**Description**

For tx level cases with GPD flow, NETLIST_MAX_LINE has been ignored.

**What Next**

Delete NETLIST_MAX_LINE in star_cmd file.

## SX-4014

(Warning) The star_cmd option NETLIST_SWAP_TERMINAL is ignored in GPD flow. This might be useless.

Feedback

**Description**

Currently NETLIST_SWAP_TERMINAL has been ignored with tx level cases in GPD flow. For future supporting, it would be valid.

**What Next**

Delete NETLIST_SWAP_TERMINAL in star_cmd.

---

## SX-4015

(Warning) The star_cmd option %s is ignored when %s is set.

**Description**

Two options are confilct.

**What Next**

Take action based on the warning message.

---

## SX-4016

(Fatal) The REFERENCE_DIRECTION is not set when PARALLEL_TO_REFERENCE_ITF_LAYERS or PERPENDICULAR_TO_REFERENCE_ITF_LAYERS is set.

**Description**

The REFERENCE_DIRECTION is needed when PARALLEL_TO_REFERENCE_ITF_LAYERS or PERPENDICULAR_TO_REFERENCE_ITF_LAYERS is set.

**What Next**

set REFERENCE_DIRECTION in StarRC command file or ITF file.

---

## SX-4017

(Fatal) Detect 'R', 'RC', or 'RLC' in 3D_IC_SUBCKT_FILE. \nPlease setup 3D_IC_SUBCKT_FILE_TYPE for StarRC to select the correct subcircuit.

**Description**

You receive this message because StarRC found that users didnot setup 3D_IC_SUBCKT_FILE_TYPE.

Feedback

**What Next**

Make sure that users correctly setup 3D_IC_SUBCKT_FILE_TYPE.

---

## SX-4018

(Fatal) Specified layer for IRREGULAR_SHAPE_INTERFACE_LAYER: '%s' is invalid.

**Description**

The database layer specified in IRREGULAR_SHAPE_INTERFACE_LAYER must exist in the database.

**What Next**

Make sure the layer exists in the database

---

## SX-4019

(Fatal) IRREGULAR_SHAPE_INTERFACE_LAYER cannot work with MODEL_BASED_SHORT_HANDLING.

**Description**

StarRC command IRREGULAR_SHAPE_INTERFACE_LAYER cannot work with StarRC command MODEL_BASED_SHORT_HANDLING.

**What Next**

Remove improper StarRC command.

---

## SX-4021

(Fatal) StarRC does not allow using different kinds of ubump with size definitions, which are defined in 3D_IC_SUBCKT_FILE, in different corners.

**Description**

You receive this message because StarRC failed to find the same kind of ubump with size subcircuits in other corners.

**What Next**

Make sure that the file specified by 3D_IC_SUBCKT_FILE for each corner is correct.

---

## SX-4022

(Warning) virtual via segmentation is turned off in gate level.

Feedback

**Description**

In gate level flows, a single via resistor is extracted on a virtual via.

**What Next**

Remove TRENCH_CONTACT_VIRTUAL_VIA_SEGMENTATION_RATIO, TRENCH_CONTACT_VIRTUAL_VIA_SEGMENTATION_RATIO_FOR_POWER, and/or MAX_VIRTUAL_VIA_SEGMENTATION_NUMBER setting from StarRC command file.

---

## SX-4023

(Fatal) The total number of %s exceeds the upper limit, and may subsequently lead to crashes or inaccuracy at a later stage.

**Description**

StarRC temporarily only supports a design or process with total number of nets not exceeding 2 billion, any violations may cause crashes during xtract or stitch stages.

**What Next**

This message is a reminder of the cause when meeting unexpected internal errors due to limit exceeding.

---

## SX-4024

(Information) LAYER_TYPE=STACKED_CONDUCTOR in the nxtgrd file is ignored in gate level flows.

**Description**

LAYER_TYPE=STACKED_CONDUCTOR in the nxtgrd file is not supported in gate level flows.

**What Next**

Information only

---

## SX-4025

(Fatal) REDUCTION_NETS usage: %s

**Description**

There was an issue when parsing a REDUCTION_NETS command. See error message for details.

Feedback

**What Next**

Reexamine REDUCTION_NETS commands for issues.

---

## SX-4026

(Warning) SHORT_PINS was used in REDUCTION_NETS, use the SHORT_PINS command instead.

**Description**

SHORT_PINS is not supported on a net by net basis in the REDUCTION_NETS command. Use the global SHORT_PINS command instead.

**What Next**

Remove the SHORT_PINS option from all REDUCTION_NET commnds and add the SHORT_PINS command to your star_cmd file.

---

## SX-4027

(FATAL) IDEAL was used in REDUCTION_NETS, use the NETLIST_TYPE or EXTRACTION command instead.

**Description**

IDEAL is not supported on a net by net basis in the REDUCTION_NETS command. Use the NETLIST_TYPE command instead for net-by-net support or EXTRACTION for global support..

**What Next**

Remove the IDEAL setting from all REDUCTION_NET commnds and add the NETLIST_TYPE or EXTRACTION command to your star_cmd file.

---

## SX-4028

(FATAL) RONLY was used in REDUCTION_NETS, use the NETLIST_TYPE or EXTRACTION command instead.

**Description**

RONLY is not supported on a net by net basis in the REDUCTION_NETS command. Use the NETLIST_TYPE command instead for net-by-net support or EXTRACTION for global support.

Feedback

**What Next**

Remove the RONLY setting from all REDUCTION_NET commnds and add the NETLIST_TYPE or EXTRACTION commands to your star_cmd file.

---

## SX-4029

(Warning) Using NETLIST_NODENAME_NETNAME together with SPEF format might result in netlist reading issue.

**Description**

Using NETLIST_NODENAME_NETNAME together with SPEF format might result in netlist reading issue.

**What Next**

Use supported NETLIST_FORMAT (NETNAME|SPF|STAR|OA) for NETLIST_NODENAME_NETNAME or set NETLIST_NODENAME_NETNAME to NO

---

## SX-4030

(Fatal) The StarRC command %s must be set for %s.

**Description**

The StarRC command must be set for a specific scenario or flow.

**What Next**

Set the StarRC command with correct argument.

---

## SX-4031

(Warning) Internally setting the MODEL_BASED_SHORT_HANDLING option to NO for replacing backside bump layer's subcircuit.

**Description**

Disable MODEL_BASED_SHORT_HANDLING to resolve crash in 3dic flow for replacing backside bump.

**What Next**

Plan to fix this workaround afterwards.

Feedback

## SX-4032

(Information) All memory statistics reported in the run refers to peak memory used by one core during respective stage.

**Description**

Description of memory statistics.

**What Next**

Information only

## SX-4033

(Severe) Invalid value for option DISABLE_ASPECT_RATIO, %s'%s'

**Description**

Must specify correct orientation for option DISABLE_ASPECT_RATIO

**What Next**

Please check DISABLE_ASPECT_RATIO in command file

## SX-4034

(Fatal) NUM_CORES_FOR_HIGH_MEM_TASKS and STARRC_DP_STRING_FOR_HIGH_MEM_TASKS must be specified at the same time.

**Description**

NUM_CORES_FOR_HIGH_MEM_TASKS and STARRC_DP_STRING_FOR_HIGH_MEM_TASKS must be specified at the same time.

**What Next**

Specify both of NUM_CORES_FOR_HIGH_MEM_TASKS and STARRC_DP_STRING_FOR_HIGH_MEM_TASKS, or none of them.

## SX-4035

(Fatal) NUM_CORES_FOR_HIGH_MEM_TASKS/ STARRC_DP_STRING_FOR_HIGH_MEM_TASKS is only supported with socket based DP flow.

Feedback

**Description**

NUM_CORES_FOR_HIGH_MEM_TASKS/ STARRC_DP_STRING_FOR_HIGH_MEM_TASKS can only be specified with socket based DP flow.

**What Next**

Specify NUM_CORES_FOR_HIGH_MEM_TASKS/ STARRC_DP_STRING_FOR_HIGH_MEM_TASKS only with socket based DP flow.

---

## SX-4036

(Warning) Using SKIP_FULL_DEBUG: YES would leads to REPORT_METAL_FILL_STATISTICS, SKIP_VIOLATION_REPORT and PARASITIC_EXPLORER_ENABLE_ANALYSIS inconsistent with orignal setting in tech file.

**Description**

SKIP_FULL_DEBUG: YES should be equivalant with PARASITIC_EXPLORER_ENABLE_ANALYSIS: NO, SKIP_VIOLATION_REPORT: BOTH and REPORT_METAL_FILL_STATISTICS: NO, please monitor their consistency.

**What Next**

Check if the debug report is expected, else please remove SKIP_FULL_DEBUG from tech file.

---

## SX-4037

(Error) Cannot found skip cell '%s' in ICV_ANNOTATED_CELL_PORTS '%s'.

**Description**

All cells listed in SKIP_CELLS should be specified in ICV_ANNOTATED_CELL_PORTS when ICV_ENHANCED_CELL_PORT_FILE is set to YES.

**What Next**

check the ICV_ANNOTATED_CELL_PORTS.

---

## SX-4038

(Warning) Change ISOLATE_PINS_IN_CELLS to YES because ICV_ENHANCED_CELL_PORT_FILE is set.

Feedback

**Description**

ISOLATE_PINS_IN_CELLS has to be YES to support outputing multiple pins when ICV_ENHANCED_CELL_PORT_FILE is YES.

**What Next**

Delete ISOLATE_PINS_IN_CELLS in star_cmd file.

---

## SX-4039

(Warning) The StarRC command INCLUDE_SHORT_CATEGORIES is set and the StarRC command CLASSIFIED_SHORT_REPORTING is not set. Setting the StarRC command CLASSIFIED_SHORT_REPORTING to YES.

**Description**

The StarRC command CLASSIFIED_SHORT_REPORTING is set to YES for the StarRC command INCLUDE_SHORT_CATEGORIES to work.

**What Next**

This is info. If you don't want the classified short reporting please set the StarRC command CLASSIFIED_SHORT_REPORTING to NO.

---

## SX-4040

(Warning) The StarRC command INCLUDE_SHORT_CATEGORIES is set and the StarRC command CLASSIFIED_SHORT_REPORTING is set to NO. The StarRC command INCLUDE_SHORT_CATEGORIES is ignored.

**Description**

The StarRC command INCLUDE_SHORT_CATEGORIES is set and the StarRC command CLASSIFIED_SHORT_REPORTING is set to NO. The StarRC command INCLUDE_SHORT_CATEGORIES is ignored.

**What Next**

This is info.

---

## SX-4041

(Fatal) The argument %s provided to the StarRC command INCLUDE_SHORT_CATEGORIES %s. This command accepts only integer values within 0 and %s.

Feedback

**Description**

The argument provided to the StarRC command INCLUDE_SHORT_CATEGORIES is not an integer or not within the allowed range.

**What Next**

Please change the arguments of the StarRC command INCLUDE_SHORT_CATEGORIES or remove it from the starRC command file.

---

## SX-4042

(Fatal) the StarRC command CLASSIFIED_SHORT_REPORTING is not compatible with the StarRC command %s.

**Description**

The two StarRC commands are not supported together.

**What Next**

Please remove or unset one of the two StarRC command. Contact Synopsys for supporting both options together.

---

## SX-4043

(Fatal) HIGH_MEM_TASKS works only when NUM_CORES_FOR_HIGH_MEM_TASKS/ STARRC_DP_STRING_FOR_HIGH_MEM_TASKS are specified.

**Description**

HIGH_MEM_TASKS works only when NUM_CORES_FOR_HIGH_MEM_TASKS/ STARRC_DP_STRING_FOR_HIGH_MEM_TASKS are specified.

**What Next**

Specify NUM_CORES_FOR_HIGH_MEM_TASKS/ STARRC_DP_STRING_FOR_HIGH_MEM_TASKS.

---

## SX-4044

(Warning) The argument %s provided to the StarRC command INCLUDE_SHORT_CATEGORY is not a valid category of short when the StarRC command ENHANCED_SHORT_REPORTING is set to %s. Please set the StarRC command ENHANCED_SHORT_REPORTING to %s to report this category of shorts.

Feedback

**Description**

Please check with the Table of classified shorts and set the StarRC command ENHANCED_SHORT_REPORTING and the StarRC command INCLUDE_SHORT_CATEGORY accordingly.

**What Next**

Please remove the argument from the StarRC command INCLUDE_SHORT_CATEGORY or change the setting of the StarRC command ENHANCED_SHORT_REPORTING.

---

## SX-4045

(Fatal) Memory specified by STARRC_DP_STRING_FOR_HIGH_MEM_TASKS('%d') must be larger than memory specified by STARRC_DP_STRING('%d').

**Description**

Memory specified by STARRC_DP_STRING_FOR_HIGH_MEM_TASKS must be larger than memory specified by STARRC_DP_STRING.

**What Next**

Specify larger memory with STARRC_DP_STRING_FOR_HIGH_MEM_TASKS.

---

## SX-4046

(Warning) Add entry fail!! %s

**Description**

IXF entry not used in xref table

**What Next**

Check if IXF file is wirtten correctly

---

## SX-4047

(FATAL) RONLY_KEEPC was used in REDUCTION_NETS, use the NETLIST_TYPE or EXTRACTION commands instead.

**Description**

RONLY_KEEPC is not supported on a net by net basis in the REDUCTION_NETS command. Use the NETLIST_TYPE command instead for net-by-net support or EXTRACTION for global support.

Feedback

### What Next

Remove the RONLY_KEEPC setting from all REDUCTION_NET commnds and add the NETLIST_TYPE or EXTRACTION commands to your star_cmd file.

---

## SX-4093

(Warning) The short could be on the db layer or other db layers with the same height in star/short_all.sum.

### Description

For some nets, they don't have any polygon on the layer which reported short, it maybe caused by other same height db layers.

### What Next

This is just information for your reference. Please check other layers with same height.

---

## SX-4989

(Warning) GATE_TO_DIFFUSION_CAP_ABUTMENT_ONE_SIDE disabled for layer %s because it is not device gate layer.

### Description

If you use StarRC command GATE_TO_DIFFUSION_CAP_ABUTMENT_ONE_SIDE to enable one side cf table for a few layers, you should make sure that these layers are device gate layer. StarRC will not apply one side cf table for those layers without IS_GATE in StarRC layer_info and give this warning.

### What Next

If you need to apply one side cftable, you can use CALIBRE_LVS_DEVICE_TYPE_MOS and CAALIBRE_OPTIONAL_DEVICE_PIN_FILE to define device gate layers.

---

## SX-4990

(Warning) ABUTMENT_ONE_SIDE is specified for non-gate_to_diffusion_cap_table layer %s in the mapping file. ABUTMENT_ONE_SIDE for this layer will be ignored.

### Description

In StarRC mapping file, ABUTMENT_ONE_SIDE parameter can only be used for gate layers with GATE_TO_DIFFUSION_CAP TABLE. If StarRC finds ABUTMENT_ONE_SIDE parameter is used on a layer other than gate layers with GATE_TO_DIFFUSION_CAP, StarRC will give this warning and ignore the ABUTMENT_ONE_SIDE setting on this layer.

Feedback

**What Next**

If ABUTMENT_ONE_SIDE is not needed for this layer, nothing needs to be done. Otherwise, you should check why this layer is not mapped to a gate layer with GATE_TO_DIFFUSION_CAP TABLE.

---

## SX-4991

(Information) Force single partition for TMIRC in TSMC N2 process.

**Description**

This is to enable arbitrary TMIRC Rg network before it is supported in multiple partition flow.

**What Next**

Information only

---

## SX-4992

(Warning) the qtf_layers in starRC mapping file are not consistent with qtf file, please check qtf layer: '%s' into the qtf_layers section.

**Description**

StarRC+QTF flow: qtf_layers section are consistent with qtf layeralias.

**What Next**

You should check the qtf_layers or qtf file layer Alias name

---

## SX-4995

(Fatal) NETLIST_NPHY_COMMENTS: YES is only allowed with %s.

**Description**

NETLIST_NPHY_COMMENTS:YES is only allowed with the following commands combination: NETLIST_TAIL_COMMENS:YES REDUCTION:NO EXTRA_GEOMETRY_INFO:RES or NODE RES

**What Next**

Modify the StarRC command file to use an allowable combination of commands.

Feedback

## SX-4996

(Fatal) Syntax error at line %d of GDS_CELL_MAPPING_FILE.

**Description**

The GDS_CELL_MAPPING should have line syntax as follows:

1. starrc style lef_cell_name gds_cell_name
2. tsmc style <MASKSHIFTLAYER> m1 m0 </MASKSHIFTLAYER> <MASKSHIFTPATTERN> 00 10 01 11 </MASKSHIFTPATTERN> <CELLMAP> lef_cell_name None gds_cell_name None </CELLMAP>

**What Next**

Correct the offending line.

## SX-4997

(Warning) SCALE_RC_ON_ISO_IMPACTED_POLYGONS_ONLY is not supported in transistor level flow. RC_SCALING_FILE is applied to all polygons.

**Description**

SCALE_RC_ON_ISO_IMPACTED_POLYGONS_ONLY is only supported in gate level flow.

**What Next**

Remove the SCALE_RC_ON_ISO_IMPACTED_POLYGONS_ONLY command in the StarRC command file.

## SX-4998

(Fatal) RC_SCALING_FILE is not specified when SCALE_RC_ON_ISO_IMPACTED_POLYGONS_ONLY:YES

**Description**

SCALE_RC_ON_ISO_IMPACTED_POLYGONS_ONLY must be specified togehter with RC_SCALING_FACTOR.

**What Next**

Modify the StarRC command file to use an allowable combination of commands.

Feedback

## SX-9996

(Warning) SYSTEM Error: %s

**Description**

A system error has occured, the operation may be attempted later.

**What Next**

Contact your System Administrator.

## SX-9997

(Fatal) SYSTEM Error: %s

**Description**

A system error has occured.

**What Next**

Contact your System Administrator.

## SX-9998

(Warning) INTERNAL Warning: %s

**Description**

A non-critical internal consistency check has failed. This warning message is printed for use by the Synopsys Support Center in case a program error occurs during StarRC's execution.

**What Next**

Contact the Synopsys Support Center if StarRC terminates abnormally or other issues are found.

## SX-9999

(Fatal) INTERNAL Error: %s

**Description**

A internal consistency check has failed.

Feedback

**What Next**

Contact the Synopsys Support Center.

# TCK

## TCK-001

(Warning) The reported endpoint '%s' is unconstrained. Reason: %s.

**Description**

The error message occurs when there are unconstrained endpoints in the design. The design is not fully constrained. There are several reasons for the unconstrained endpoints: false path, no check, unclocked, disabeld check, case constant, or no arrival path.

**What Next**

The violation message identifies the unconstrained endpoint and the reason. You could check whether it is a expected behavior.

## TCK-002

(Warning) The register clock pin %s has no fanin clocks.

**Description**

The error message occurs when the register clock pin does not have a clock signal propagating to it.

**What Next**

The violation message identifies the clock pin that does not have a clock reaching it.There are several possible causes for this violation: the clock definition is missing, a generated clock is not expanded or the pin is blocked by a set_case_analysis, set_disable_timing, or some other constraint.

## TCK-003

(Info) The register clock pin '%s' has '%d' fanin clocks '%s'.

**Description**

The reported register clock pin has multiple clocks propating to it. There is a risk that the timing analysis will run slower and take more memory for a set of constraints with many clocks propagating to the same registers. There is also a risk that the design is not

Feedback

constrained for the functional modes as expected. The large number of clocks could also be due to an error in the case analysis settings for clock definitions in the design

**What Next**

The message provides the names of the clocks at the register clock pin. You coulc check the clock network to the register in the violation.

---

## TCK-004

(Warning) The generated clock '%s' has no path from master clock '%s'.

**Description**

This violation occurs whenever the clock network traversal can not find a path from the master clock to the generated clock. The master clock did exists at the -source pin of the generated clock, but there is no path to the generated clock.

**What Next**

Check the path between the master to the generated clock. Is there case values disabling the path? Is there unresolved library cell references blocking the path? Either fix the path or remove the generated clock.

---

## TCK-005

(Warning) The generated clocks '%s' form a loop.

**Description**

This violation occurs whenever the clock network traversal finds generated clocks in the design form a loop. For example, when the source pin of G_CLK1 is the definition point of G_CLK2, meanwhile the source pin of G_CLK2 is the definition point of G_CLK1, the tow generated clocks form a loop.

**What Next**

Check the definition of the violated generated clocks.

---

## TCK-006

(Warning) The '%s' of generated clock '%s' with source pin '%s' is not satisfiable

**Description**

This is an error message whenever the clock network traversal can not find a path which satisfies the sense relationship defined by create_generated_clock command.

Feedback

### What Next

Check for the generated clock definition to see if the generated clock is correctly defined. One example is that a divided_by 2 generated clock is driven by a inverter only. In this case, generated clock should be redefined with -invert with -divided_by 1. Another example is a divided_by 2 generated clock with preinverting. If master clock source pin is used as generated clock source pin, the warning message will be issued. In this case, generated clock source pin should be redefined to clock pin of divider.

---

## TCK-007

(Warning) The generated clock '%s' has not been expanded.

### Description

A generated clock will not expand if the master clock from which it is generated has not been created or activated. Also if the master clock was given with a -master_clock but does not reach the pin given with -source, this message will be given.

### What Next

Please create or activate the master of the generated clock or change the -source pin given. Use the *report_clock* command to see if the master clock is created or inactive.

---

## TCK-008

(Warning) The generated clock '%s' has no period specified.

### Description

This violation occurs whenever the generated clock has no clock period.

### What Next

Check the definition of the generated clock. Please create or activate the master of the generated clock or change the -source pin given. Use the *report_clock* command to see if the master clock is created or inactive.

---

## TCK-009

(Warning) The data check reference pin '%d' is driven by '%d' clocked signals '%s'

### Description

This warning message occurs when check_timing command finds no clocked signal or multiple clocked signal reach a data check register reference pin.

Feedback

### What Next

Check the clock propagating to the reported data check reference pin.

---

## TCK-010

(Warning) The register launch clock '%s' and capture clock '%s' are from different domains

### Description

This Warning message occurs whenever check_timing command finds clock interactions. If a clock launches one or more paths, which are captured by other clocks, it will be reported. If the two clocks are exclusive or asynchronous clocks(defined as exclusive/asyn clocks by set_clock_group), the path is marked with an asterisk (*). If the two clocks are set false path (set_false_path from clk1 to clk2), the path is marked with an asterisk (*).

### What Next

Check the propagating of reported clock to identify the clock interactions

---

## TCK-011

(Warning) A loop '%s' is detected, disable the arc '%s' to break the loop.

### Description

There is a combinational feedback loop in the design that was broken by the tool. THe message will give the pins that make up the combinational feedback loop and the disabled arc.

### What Next

Look at the combinational loop and determine the best location for it to be broken for this scenario. The *report_disable_timing* command can be used to see the disabled timing arcs in the design. Items in the report with the attribute l are disabled due to loop breaking.

---

## TCK-012

(Warning) The input port '%s' has no clock_relative delay specified.

### Description

The reported input port is not a clock source. The port does not have an input delay value, and it also has paths from it which are not disabled or false paths. In order to enable proper constraining of paths from the port, a realistic input delay needs to be provided for all data input ports.

Feedback

### What Next

The violation message will give you the input port which has no input delay specified. To resolve this issue, consider specifying input delay for each non-clock source input port. This input delay should also be relative to a clock in the design.

---

## TCK-013

(Warning) The clocks '%s' at reported gated clock point '%s' don't reach any register clock pins nor any output ports.

### Description

The error message occurs when the gated clock doesn't reach any register clock pins nor any output ports. This rule only check for ICG cells.

### What Next

Check the fanout of the reported clock point.

---

# TECH

---

## TECH-001

(warning) %s line %d, unsupported technology file section '%s'. It will be ignored.

### Description

The technology file section is ignored because it is not supported. In the case of incremental tech file update, only the following sections are supported in the tech file to be merged: Layer, ContactCode, DesignRule, ViaRule, DensityRule and LayerDataType. Other sections will be ignored, i.e. not merged.

### What Next

Check the technology file for possible errors.

---

## TECH-002

(warning) %s line %d, unsupported syntax '%s' in '%s' section. It will be ignored.

### Description

The syntax is ignored because it is not supported

Feedback

**What Next**

Check the technology file for possible errors.

---

## TECH-002e

(error) %s line %d, unsupported syntax '%s' in '%s' section.

**Description**

The syntax is not supported, and will error out.

**What Next**

Fix the syntactical errors within the technology file.

---

## TECH-003

(warning) %s line %d, layer '%s' referenced by '%s' is undefined. This rule will be ignored.

**Description**

The referenced layer hasn't been defined. The rule will be ignored.

**What Next**

Add the missing layer or remove the reference.

---

## TECH-003e

(error) %s line %d, layer '%s' referenced by '%s' is undefined. This rule will be ignored.

**Description**

The referenced layer hasn't been defined. The rule will be ignored.

**What Next**

Add the missing layer or remove the reference.

---

## TECH-004

(error) No technology information.

**Description**

There is no technology information.

Feedback

## TECH-005

(error) %s line %d, '%s' is not a valid value for property '%s'. It must be '%s'. This property will be ignored.

**Description**

The input value is not valid for the property. The property will be ignored.

**What Next**

Input a valid value.

## TECH-005w

(warning) %s line %d, '%s' is not a valid value for property '%s'. It must be '%s'. This property will be ignored.

**Description**

The input value is not valid for the property. The property will be ignored.

**What Next**

Input a valid value.

## TECH-006

(error) %s line %d, syntax error. This rule will be ignored.

**Description**

There is a syntax error. This rule will be ignored.

**What Next**

Fix the syntax error.

## TECH-007

(error) %s line %d, due to property '%s' not yet defined, property '%s' has unknown table size and will be ignored.

**Description**

The property is a table but its table size is not yet defined. The property will be ignored.

Feedback

**What Next**

Add related table size property before this property.

---

## TECH-008

(error) %s line %d, the table size defined by property '%s' is '%d'. However, the actual table size of property '%s' is '%d'. This property will be ignored.

**Description**

The property table size differs from the value of the property that defines this table size. The property will be ignored.

**What Next**

Correct the table size.

---

## TECH-008w

(warning) %s line %d, the table size defined by property '%s' is '%d'. However, the actual table size of property '%s' is '%d'. This property will be ignored.

**Description**

The property table size differs from the value of the property that defines this table size. The property will be ignored.

**What Next**

Correct the table size.

---

## TECH-009

(warning) %s line %d, '%s' is not a valid value for this property type. The value must be between '%d' and '%d'. This property will be ignored.

**Description**

The input value is not within legal range for the property. The property will be ignored.

**What Next**

Please make sure that the distance value is within the limit.

---

## TECH-010

(error) %s line %d, '%s' is not a valid value for property '%s'. It must be '%s'.

Feedback

**Description**

The input value is not valid for the property.

**What Next**

Input a valid value.

---

## TECH-011

(error) %s line %d, '%s' is not a valid value for this property type. The value must be between '%d' and '%d'.

**Description**

The input value is not within legal range for the property.

**What Next**

Please make sure that the input value is within the limit.

---

## TECH-012

(warning) %s line %d, syntax '%s' in '%s' section is obsolete and is replaced by '%s'. It will be ignored.

**Description**

The syntax is ignored because it is obsolete.

**What Next**

Use the new syntax.

---

## TECH-013

(warning) ContactCode not found for %s on Layer '%s'

**Description**

This warning message occurs when a valid contactCode is not found in the tech file for the width threshold range from enclosureTbl or when a valid contactCode is not found that is equivalent to or larger than the enclosures in enclosureTbl.

**What Next**

Please add a contactCode according to the enclosureTbl values or adjust enclosure width/height values in enclosureTbl. After making your changes, run the command again.

Feedback

## TECH-014

(warning) %sduplicate definition '%s = %s', line %d.

**Description**

This warning message occurs when a duplicate purpose name or purpose number definition is found in the technology file.

**What Next**

Check and correct the purpose definition, then reload the file.

## TECH-015

(warning) %smaskName = '%s' has been defined in layer '%s', line %d ignored.

**Description**

This warning message occurs when a duplicate maskName definition is found in the technology file.

## TECH-016

(warning) contactCodeNumber = '%d' has been defined in ContactCode '%s', ignored this ContactCode.

**Description**

This warning message occurs when a duplicate contactCodeNumber definition is found in the technology file.

## TECH-016e

(error) contactCodeNumber = '%d' has been defined in ContactCode '%s', ignored this ContactCode.

**Description**

This warning message occurs when a duplicate contactCodeNumber definition is found in the technology file. This can be in an incremental tech file.

## TECH-017

(warning) Skipped duplicate layer purpose number '%d' definition for layerNumber = %d.

Feedback

### Description

This warning message occurs when a duplicate layer purpose number definition is found in same layer.

### What Next

Check and correct the layer purpose number definition, then reload the file.

---

## TECH-018

(error) Line contains nested comments. (%s line %d)

### Description

This error message occurs when the technology file contains a nested comment starting on the specified line. Nested comments are not allowed in the technology file.

The following is an example of a nested comment and the resulting error message.

```
Line 10:
Line 11: ContactCode "via1Fat" {
Line 12:     contactCodeNumber = 12
Line 13: /*  isDefaultContact = 1 /* this is a nested comment */ */
Line 14:     isFatContact =1
Line 15: }
Line 16:

Error : Line contains nested comments. (line 13) (TECH-018)
```

### What Next

Remove the nested comment on the specified line and then rerun the command.

---

## TECH-019

(warning) Skipped conflict LayerDataType name definition %s.

### Description

This warning message occurs when a LayerDataType name has been defined for other LayerDataType or layer.

### What Next

Check and correct the name definition ,then reload the file.

Feedback

## TECH-020

(warning) %s '%s' '%s' and '%s' coexist.

### Description

This warning message occurs when two rules or attributes coexist in the technology file.

If both the minSpacingCornerKeepoutWidth attribute and the minSpacingXParallelLengthThresholdTbl and minSpacingYParallelLengthThresholdTbl attributes are defined for the same cut layer, the router honors the latter and ignores the former.

For RC blockage marker generation, clipTbl and endToZWireSpace rules can only be defined once in either LEE blockage marker layer (e.g. metal5RCBlockage) or its associated Metal Layer (metal5).

The following is an example of the warning message:

```
Warning: Layer 'V2' 'minSpacingCornerKeepoutWidth' and
'minSpacingXParallelLengthThresholdTbl' coexist. (TECH-020)
```

### What Next

You should remove the first rule, which is ignored from the technology file, and reload it.

## TECH-021

(warning) the size of '%s' is '%d' while the expected size is '%d' according to '%s', in Layer %s.

### Description

This error message occurs when the size of a table is conflict with its size definition.

### What Next

Check and correct the table size and reload it.

## TECH-022

(error) The specified technology database is in a lib_cell library. It is not editable with '%s'.

### Description

Lib-cell library data, including technology data, can only be edited in the Library Manager.

### What Next

If you want to edit the data in this technology database, use the Library Manager.

Feedback

## TECH-023

(error) %s line %d, Unsupported syntax '%s' in '%s' section.

**Description**

The tool cannot read the technology file because it contains unsupported syntax.

**What Next**

Remove the specified attribute from the technology file and then read the file again.

## TECH-024

(error) The specified library does not have an associated tech database.

**Description**

The library does not have a tech database or a tech reference. Tech editing operations cannot be performed on this library.

**What Next**

Use create_tech to create a tech database in the library or set its tech_lib to point to a library with an associated tech database.

## TECH-025

(warning) %s '%s' '%s' and '%s' coexist.

**Description**

This message occurs when two conflicting rules/attributes coexist.

For example, if uShapeMinSpacing attribute and uShapeExcludedLengthRange coexist, router does not know which rule to apply.

The following is an example of the warning message:

```
Warning: Layer 'M1' 'uShapeExcludedLengthRange' and 'uShapeMinSpacing'
 coexist. (TECH-025)
```

**What Next**

It is recommended that you choose the rule/attribute that you want and remove the other.

## TECH-026

(warning) %s '%s' is missing the attribute '%s'. (%s line %d)

Feedback

### Description

This warning message occurs when the specified object of the specified section is missing an important optional attribute, or partial attributes of a rule are specified. If the attribute is not specified in the technology file, the attribute has a default value of 0 or an empty string.

The following is an example of the warning message:

```
Warning : Layer 'M1' is missing the attribute 'uShapeMinLength'. (line
 758) (TECH-026)
```

### What Next

This is only a warning message. No action is required.

However, if you do not want to the use the default for the attribute, add the attribute with the desired value to the specified section of the technology file and then rerun the command.

---

## TECH-026e

(error) %s '%s' is missing the attribute '%s'. (%s line %d)

### Description

This error message occurs when the specified object of the specified section is missing an important attribute, or partial attributes of a rule are specified. If the attribute is not specified in the technology file, the attribute has a default value of 0 or an empty string.

The following is an example of the error message:

```
Error : Layer 'M1' is missing the attribute 'uShapeMinLength'. (line 758)
 (TECH-026e)
```

### What Next

Add the required attribute to the specified section and rerun the command.

---

## TECH-026i

(information) %s '%s' is missing the attribute '%s'. (%s line %d)

### Description

This information message occurs when the specified object of the specified section is missing an optional attribute, or partial attributes of a rule are specified. If the attribute is not specified in the technology file, the attribute has a default value of 0 or an empty string.

Feedback

The following is an example of the message:

```
Information: Layer 'M1' is missing the attribute 'uShapeMinLength'. (line
 758) (TECH-026i)
```

### What Next

This is only an information message. No action is required.

However, if you do not want to the use the default for the attribute, add the attribute with the desired value to the specified sec tion of the technology file and then rerun the command.

---

## TECH-027

(information) %s Non-metal layer '%s' has the metal layer attribute '%s', line %d.

### Description

This information message occurs when the specified Layer section is not a metal or poly layer, but it contains a metal layer attribute that is ignored.

The following are metal and poly layer attributes:

```
sameNetMinSpacing
stubThreshold
stubSpacing
maxLength
maxNumMinEdge
minEdgeLength
minArea
minEnclosedArea
maxWidth
fatContactThreshold
fatWireThreshold
fatThinMinSpacing
fatFatMinSpacing
fatWireExtensionRange
fatTblSpacing
fatTblParallelLength
fatTblMinEnclosedArea
```

The following is an example of the information message:

```
Information : Non-metal layer 'VIA12' has the metal layer
attribute 'fatTblMinEnclosedArea'. (line 740) (TECH-027)
```

### What Next

This is only an informational message. No action is required.

Feedback

However, if the result is not what you intended, you can either remove the specified attribute or change the mask name of the layer to a metal mask name, such as poly, metal1, metal2, or metal3.

---

## TECH-028

(warning) The rule attribute '%s' has a value of %s. This property will be ignored. (%s line %d)

### Description

s warning message occurs when the value of the specified rule attribute may not be optimal.

The value of the *minEnclosedArea* attribute for default metal layers should be greater than 0. Other values can affect the quality of the application results and can cause excessive design rule constraint violations during physical verification.

For a set of rule, completeness of rule parameters should be maintained:

For U-shape rule, if you specify uShapeMinLength or uShapeMinSpacing, you should also specify uShapeDepthThreshold and minSpacing. And uShapeMinSpacing > minSpacing.

For enclosed via minEdgeLength rule, if you specify enclosedViaMinEdgeLength, you must also specify a value for minEdgeLength that is within (0, enclosedViaMinEdgeLength) and a positive value for enclosedViaAdjacentEdgeLength.

For nonpreferred direction rule, the following should be met: nonPreferredWidth > defaultWidth, and nonPreferredMaxLength > nonPreferredWidth-defaultWidth

For multiple minArea rule, you should specify minArea if you specify specialMinArea, and specialMinArea > minArea.

For stubThreshold associated rules, you should specify stubThreshold if you specify stubMinLength, stubToStubSpacing, stubSpacing or endOfLineSideKeepoutLength.

For wire jog corner to corner width rule, the following should be met: alignedJogWidth >= minWidth and unalignedJogDiagWidth >= minWidth

For legal width rule, defaultWidth and nonPreferredWidth should be specifies and their values are in LegalWidthTbl.

defaultWidth >= minWidth for all layers.

The following is an example of the warning message:

```
Warning : Layer 'metal3' attribute 'minEnclosedArea' has a
value of 0. (TECH-028)
```

Feedback

### What Next

This is only a warning message. No action is required.

However, it is best practice to check the attribute value of the specified layer, and then change the value to a value greater than 0, if necessary. After making your changes, run the command again.

---

## TECH-029

(information) Non-via layer '%s' has the via layer attribute '%s'. (%s line %d)

### Description

This information message indicates that the specified Layer section does not define a via or poly contact layer, but it contains a via layer attribute, which is ignored.

The following are via and poly contact layer attributes:

```
cornerMinSpacing
maxStackLevel
fatTblExtensionContactNumber
fatTblExtensionMinCuts
fatTblFatContactNumber
fatTblFatContactMinCuts
maxNumAdjacentCut
adjacentCutRange
enclosedCutNumNeighbor
enclosedCutNeighborRange
enclosedCutMinSpacing
enclosedCutToNeighborMinSpacing
```

The following is an example of the information message:

```
Information : Non-via layer 'METAL2' has the via layer attribute
\'enclosedCutToNeighborMinSpacing'. (line 294) (TECH-029)
```

### What Next

This is only an informational message. No action is required.

However, if the result is not what you intended, you can either remove the specified attribute, or change the mask name of the layer to a via mask name, such as polyCont, via1, via2, or via3.

---

## TECH-030

(warning) Skipped LayerDataType '%s' since its layerNumber '%d' is invalid.

Feedback

### Description

This warning message occurs when a layerNumber of a LayerDataType is not defined.

### What Next

Define the layerNumber in the corresponding Layer section.

---

## TECH-031

(warning) %s '%s' attribute '%s' is assigned a list with an incorrect number of elements. (%s line %d)

### Description

This warning message occurs when the array attribute of the specified section does not have the correct number of elements, or the values are not separated by comma. minSpacingUpperEncCheckTbl table size should be cut1TblSize* cut2TblSize if used in cut to cut spacing rule and cutTblSize if used in cut to metal spacing rule.

The following is an example of the warning message:

```
Warning : Layer 'VIA12' has an incorrect number of elements listed
for array attribute 'fatTblExtensionMinCuts'. (line 753) (TECH-031)
```

For layer sections, the following array attributes must contain the indicated number of elements, based on the *fatTblDimension* attribute:

```
Attribute:                    Valid Number Of Elements:
    ------------------------------------------------------------
    fatTblThreshold              fatTblDimension
    fatTblSpacing                fatTblDimension x fatTblDimension
    fatTblParallelLength         fatTblDimension
    fatTblExtensionRange         fatTblDimension
    fatTblMinEnclosedArea        fatTblDimension
    fatTblFatContactNumber       fatTblDimension
    fatTblFatContactMinCuts      fatTblDimension
    fatTblExtensionContactNumber fatTblDimension
    fatTblExtensionMinCuts       fatTblDimension
```

For capacitance table sections, the array attribute must have the following number of elements, based on the wire width and spacing size:

```
Attribute      Valid Number Of Elements
    ---------------------------------------------
    wireWidth      wireWidthSize
    wireSpacing    wireSpacingSize
    capValue       wireWidthSize x wireSpacingSize
```

Feedback

For resistance models, the array attribute must have the following number of elements shown below, based on the size:

```
Attribute      Valid Number Of Elements
    -------------------------------------------
    wireWidth      size
    tempCoeff      size
    minRes         size
    nomRes         size
    maxRes         size
```

**What Next**

This is only a warning message. No action is required.

However, if the result is not what you intended, you can change the table dimension(s) or add or remove elements from the specified array attribute.

---

## TECH-032

(warning) %s '%s' has invalid table values '%s'. (%s line %d)

**Description**

This warning message occurs when the specified section contains an invalid table value list, e.g. thresholds. The value list must meet the following requirements:

- The number of the values must match the table dimension.
- All values must be nonnegative numbers and listed in ascending order.
- The first threshold value must be between 0 and minWidth inclusively.
- The second threshold value must be greater than the layer minWidth. •
  The first spacing value must not be smaller than minSpacing, and the first maxSpacingCheckRange value must not be smaller than minSpacing * 2 + minWidth.
- In 2-D spacing table between the same layer, the values must be symmetrical.

  In minCutsTbl, if area is specified, extension should be specified. In enclosureTbl, if no lower/upper width is specified or an equivalent ContactCode to the specified enclosures is not found, this message will occur.

In generalized cut spacing rule with parallel length, if minSpacingDiagonalCheckTbl[i][j] = 1, at least one of minSpacingXParallelLengthThresholdTbl[i][j] and minSpacingYParallelLengthThresholdTbl[i][j] should be < 0; otherwise, TECH-032 will be issued to ignore minSpacingDiagonalCheckTbl[i][j]. If minSpacingXParallelLengthThresholdTbl or minSpacingYParallelLengthThresholdTbl is defined, orthoSpacingExcludeCornerTbl[i][j] should be 1; otherwise, TECH-032 will be issued to warn users that the corners will be checked by x/y spacing.

Feedback

For the table of fatTblExtensionRange, if fatTblExtensionRangeDimension defined, the value list must meet the following requirements:

- The number of the values must match the dimension.
- All values must be larger than 0 and listed in ascending order.

  For fatWireViaEnc rule, fatWireViaEncMinWidthThresholdTbl is recommended to be in ascending order. However, the tool accepts this table even it is not in ascending order as long as the min and max thresholds are paired in fatWireViaEncMinWidthThresholdTbl and fatWireViaEncMaxWidthThresholdTbl. Furthermore, if missing a width range between fatWireViaEncMinWidthThresholdTbl and fatWireViaEncMaxWidthThresholdTbl, this message will be issued.

  For maxStackLevel rule, the value in maxStackLevelCutAsViaArrayTbl for the smallest squared cut should be 0.

  For separatedCut spacing rule, if defined between the same cut layer, the following should hold: 1) separatedCut1NameTbl=separatedCut2NameTbl 2) separatedCut1ToUpperMetalExcludedSpacingTbl=separatedCut2ToUpperMetalExcludedSpacingTbl 3) the 2D spacing tables separatedCutXMinSpacingTbl and separatedCutYMinSpacingTbl should be symmetrical.

  x|yLegalDimensionTbl table values should be in ascending order. voltageAreaThresholdTbl values should be in ascending order.

  For fatTblSpacing Layer property, each of its values must be greater than or equal to minSpacing value.

  The following is an example of the warning message:

```
Warning : Layer 'METAL1' has invalid table values '(0, 0,
1.505, 3.005, 4.505, 7.505)'. (line 225) (TECH-032)
```

**What Next**

Please change the value list of the specified section to meet the above requirements. After making your changes, run the command again.

---

## TECH-032e

(error) %s '%s' has invalid table values '%s'. (%s line %d)

Feedback

### Description

This error message occurs when the specified section contains an invalid table value list, e.g. thresholds. The value list must meet the following requirements:

- The number of the values must match the table dimension.
- All values must be nonnegative numbers and listed in ascending order.
- The first threshold value must be between 0 and minWidth inclusively.
- The second threshold value must be greater than the layer minWidth. • The first spacing value must not be smaller than minSpacing, and the first maxSpacingCheckRange value must not be smaller than minSpacing * 2 + minWidth.
- In 2-D spacing table between the same layer, the values must be symmetrical.

  In minCutsTbl, if area is specified, extension should be specified. In enclosureTbl, if no lower/upper width is specified or an equivalent ContactCode to the specified enclosures is not found, this message will occur.

In generalized cut spacing rule with parallel length, if minSpacingDiagonalCheckTbl[i][j] = 1, at least one of minSpacingXParallelLengthThresholdTbl[i][j] and minSpacingYParallelLengthThresholdTbl[i][j] should be < 0; otherwise, TECH-032e will be issued to ignore minSpacingDiagonalCheckTbl[i][j]. If minSpacingXParallelLengthThresholdTbl or minSpacingYParallelLengthThresholdTbl is defined, orthoSpacingExcludeCornerTbl[i][j] should be 1; otherwise, TECH-032e will be issued to warn users that the corners will be checked by x/y spacing.

For the table of fatTblExtensionRange, if fatTblExtensionRangeDimension defined, the value list must meet the following requirements:

- The number of the values must match the dimension.
- All values must be larger than 0 and listed in ascending order.

  For fatWireViaEnc rule, fatWireViaEncMinWidthThresholdTbl is recommended to be in ascending order. However, the tool accepts this table even it is not in ascending order as long as the min and max thresholds are paired in fatWireViaEncMinWidthThresholdTbl and fatWireViaEncMaxWidthThresholdTbl. Furthermore, if missing a width range between fatWireViaEncMinWidthThresholdTbl and fatWireViaEncMaxWidthThresholdTbl, this message will be issued.

  For maxStackLevel rule, the value in maxStackLevelCutAsViaArrayTbl for the smallest squared cut should be 0.

  For separatedCut spacing rule, if defined between the same cut layer, the following should hold: 1) separatedCut1NameTbl=separatedCut2NameTbl 2) separatedCut1ToUpperMetalExcludedSpacingTbl=separatedCut2ToUpperMetalExcludedSpacingTbl

Feedback

3) the 2D spacing tables separatedCutXMinSpacingTbl and separatedCutYMinSpacingTbl should be symmetrical.

x|yLegalDimensionTbl table values should be in ascending order. voltageAreaThresholdTbl values should be in ascending order.

For fatTblSpacing Layer property, each of its values must be greater than or equal to minSpacing value.

The following is an example of the error message:

```
Error : Layer 'METAL1' has invalid table values '(0, 0,
1.505, 3.005, 4.505, 7.505)'. (line 225) (TECH-032e)
```

**What Next**

Please change the value list of the specified section to meet the above requirements. After making your changes, run the command again.

---

## TECH-033

(error) Layer '%s' has invalid fat table contact number %s of table value %s.

**Description**

This error message occurs when the fat metal via rule or fat metal via extension rule in the specified Layer section contains an invalid contact code number. This could be a ContactCodeNumber for different cut layer or a negative number. Please note number 0 is valid as it denotes any valid ContactCodeNumber.

A valid contact Code number must correspond to a contact code defined in the technology file, and the contact code must be defined for the specified cut layer.

And due to router limitation, if an asymmetrical contact code, defined by geometry tables or cut array, is referenced in fatTblFatContactNumber or fatTblExtensionContactNumber table, the error message will also be issued.

The following is an example of the error message:

```
Error : Layer 'VIA12' has invalid fat table contact number 20 of table
value"1 6 20 28 29". (TECH-033)
```

**What Next**

Change the table contact code number or define the contact code for the specified Layer section and then run the command again.

Feedback

## TECH-033w

(warning) Layer '%s' has invalid fat table contact number %s of table value %s.

### Description

This warning message occurs when the fat metal via rule or fat metal via extension rule in the specified Layer section contains an invalid contact code number. This warning occurs under end user mode only.

A valid contact Code number must correspond to a contact code defined in the technology file, and the contact code must be defined for the specified cut layer.

And due to router limitation (unable to instantiate such a via), if an asymmetrical contact code, defined by geometry tables or cut array, is referenced in fatTblFatContactNumber or fatTblExtensionContactNumber table, this message will also be issued.

The following is an example of the warning message:

```
Warning : Layer 'VIA12' has invalid fat table contact number "20 29" of
table value"1 6 20 28 29 ". (TECH-033w)
```

### What Next

Change the table contact code number or define the contact code for the specified Layer section and then run the command again.

## TECH-034

(warning) DesignRule is defined with non-consecutive layers '%s' and '%s'. (%s line %d)

### Description

This warning message occurs when the specified DesignRule section contains two metal layers or two via layers that do not have consecutive mask names. For example, the mask names via1 and via2 are valid consecutive mask names. The mask names via2 and via4 are not consecutive and cause the warning message to occur and the design rule to be ignored. For cut to metal spacing rule with minSpacingUpperMetalWidthThresholdTbl, the rule should be defined in DesignRule section between cut and lower metal layers (via_x and metal_x). For forbiddenCut and separatedCut spacing rule, the attributes should be defined in DesignRule section between same cut layer; otherwise, TECH-034 will be issued.

The following is an example of the warning message:

```
Warning : DesignRule is defined with non-consecutive layers 'VIA2'
and 'VIA4'. (line 1954) (TECH-034)
```

Feedback

### What Next

This is only a warning message. No action is required.

However, to avoid this warning message, change the layers of the DesignRule section so that the mask names of the layers are consecutive. After making your changes, run the command again.

---

## TECH-035

(warning) DesignRule is defined with invalid metal-to-via enclosure layers '%s' and '%s'. (%s line %d).

### Description

This warning message occurs when the DesignRule section specifies either a minimum or an end-of-line enclosure rule, for which one layer is not a via layer and/or the other layer is not a metal layer. The application ignores the design rule. Enclosure rules are only valid between a metal and via layer.

The following is an example of the warning message:

```
Warning : DesignRule is defined with invalid metal-to-via
enclosure layers 'via2' and 'via1'. (line 436). (TECH-035)
```

### What Next

This is only a warning message. No action is required.

However, if the result is not what you intended, change the layers of the DesignRule section so that one is a via layer and the other is a metal layer.

---

## TECH-036

(error) %s line %d, %s '%s' attribute '%s' has a row with an incorrect number of elements '%d'. It must be %s.

### Description

The specified two dimensional table at least has one row with incorrect number of elements.

### What Next

Verify that the number of elements for the specified table is correct.

Feedback

## TECH-037

(error) %s line %d, missing attribute '%s' in %s section '%s', the %s will be ignored.

**Description**

There is a missing attribute in the specified section. Either the attribute or the whole section will be ignored.

**What Next**

Verify that necessary attributes are specified for the section.

## TECH-038

(error) Layers '%s' and '%s' are not adjacent. (%s line %d)

**Description**

The error message indicates that layer1 and layer2 of a DesignRule section that can only be defined between two adjacent layers are not adjacent.

**What Next**

Correct the erroneous layer specifications in the tech file and run the command again.

## TECH-039

(error) Valid contact code is missing for via layer '%s'.

**Description**

This error message occurs when a ContactCode section meets one of following criteria:

- The ContactCode section is not defined for the specified via layer.
- The ContactCode section is defined for the specified via layer, but it is invalid.

A ContactCode section is defined for a particular via layer by assigning the *cutLayer* or *cutLayerNumber* attribute to the corresponding via layer name or number. At least one valid contact code must be defined for each via layer.

The following is an example of the error message:

```
Error : Valid contact code is missing for the via layer 'via4'.
(TECH-039)
```

Feedback

### What Next

Define or correct the invalid ContactCode section for the specified via layer and then run the command again.

---

## TECH-039w

(warning) Valid contact code is missing for via layer '%s'.

### Description

This warning message occurs when a ContactCode section meets one of following criteria:

- The ContactCode section is not defined for the specified via layer.
- The ContactCode section is defined for the specified via layer, but it is invalid.

A ContactCode section is defined for a particular via layer by assigning the *cutLayer* or *cutLayerNumber* attribute to the corresponding via layer name or number. At least one valid contact code must be defined for each via layer.

The following is an example of the error message:

```
Warning : Valid contact code is missing for the via layer 'via4'.
(TECH-039w)
```

### What Next

Define or correct the invalid ContactCode section for the specified via layer and then run the command again.

---

## TECH-040

(warning) Valid SAV%s is missing in viaRule section for via layer '%s' of DesignRule '%s'.

### Description

This warning message occurs when a SAV/SAV cluster is missing in viaRule section meets one of following criteria:

- DesignRule defined attribute 'diffNetNonSavCenterMinSpacingTbl' or 'nonSelfAlignedViaCenterMinSpacingTbl', valid SAV via on layer1 should be defined in ViaRule section.
- DesignRule defined attribute 'selfAlignedViaClusterCenterMinSpacingTbl', valid SAV cluster via on layer1 should be defined in ViaRule section.

Feedback

Reference example:

```
ViaRule SAV1 {
 cutLayerNameTblSize = 1
 cutLayerNameTbl = (V1) # via layer name
 cutNameTbl = (Vsm) # cut name
 selfAlignedViaUpperLayerEncTblSize = 2 # N
 selfAlignedViaUpperLayerEncTbl = (0.00, 0.01)
}
DesignRule {
  layer1 = "V1"
  layer2 = "V1"
  cut1TblSize = 1
  cut2TblSize = 1
  cut1NameTbl = (Vsm)
  cut2NameTbl = (Vsm)
  nonSelfAlignedViaCenterMinSpacingTbl = (S)
}
```

If SAV via 'SAV1' is not defined in viaRule section, then the warning message TECH-040 will be issued for designRule 'V1-V1'.

The following is an example of the warning message:

```
Warning: Valid SAV is missing in viaRule section for via layer 'V1' of
DesignRule 'V1-V1'. (TECH-040)
```

**What Next**

Define or correct the invalid SAV/SAV cluster in viaRule section for the specified via layer of the DesignRule and then run the command again.

---

## TECH-041

(error) Layer is missing with mask name '%s'.

**Description**

This error message occurs when a Layer section with the specified mask name is missing from the technology file. All metal and via layers up to the highest defined physical layer must be defined in the technology file.

For example, if only metal1, via1, metal2, via2, metal3, and metal4 are defined, then via3 is missing and the error message appears.

The following is an example of the error message:

```
Error : Layer is missing with mask name 'via3'. (TECH-041)
```

**What Next**

Define a Layer section with the specified mask name and then run the command again.

Feedback

## TECH-042

(Error) Attribute '%s' of %s %s has an invalid value. (%s line %d)

### Description

The message indicates that the value of the specified attribute very likely violates some of the following conditions:

For ContactCode attribute excludedForSignalRoute, excludedForSignalRoute and isDefaultContact of the same ContactCode section cannot both be set to true.

For ContactCode attributes minNumRows and minNumCol, the valid value range is 1 to 16.

For ContactCode attributes upperLayerEncWidth, upperLayerEncHeight, lowerLayerEncWidth, and lowerLayerEncHeight, the following conditions must be satisfied:

```
(upperLayerEncWidth * 2) + cutWidth >= 0
(upperLayerEncHeight * 2) + cutHeight >= 0
(lowerLayerEncWidth * 2) + cutWidth >= 0
(lowerLayerEncHeight * 2) + cutHeight >= 0
```

The following is an example of the error message:

```
Error: Attribute 'minNumRows' of ContactCode VIA12 has an invalid
value. (in.tf line 1653) (TECH-042)
```

### What Next

Correct the erroneous value. After making the change, run the command again.

## TECH-042w

(warning) Attribute '%s' of %s %s has an invalid value. (%s line %d)

### Description

The message indicates that the value of the specified attribute very likely violates some of the following conditions:

For ContactCode attribute excludedForSignalRoute, excludedForSignalRoute and isDefaultContact of the same ContactCode section cannot both be set to true.

For ContactCode attributes minNumRows and minNumCol, the valid value range is 1 to 16.

Feedback

For ContactCode attributes upperLayerEncWidth, upperLayerEncHeight, lowerLayerEncWidth, and lowerLayerEncHeight, the following conditions must be satisfied:

```
(upperLayerEncWidth * 2) + cutWidth >= 0
(upperLayerEncHeight * 2) + cutHeight >= 0
(lowerLayerEncWidth * 2) + cutWidth >= 0
(lowerLayerEncHeight * 2) + cutHeight >= 0
```

The following is an example of the warning message:

```
Warning: Attribute 'minNumRows' of ContactCode VIA12 has an invalid
value. (in.tf line 1653) (TECH-042w)
```

**What Next**

Correct the erroneous value. After making the change, run the command again.

---

## TECH-043

(warning) Attributes '%s' and '%s' of %s '%s' cannot both be set to %s. (%s line %d)

**Description**

The message indicates that there is a conflict between the settings of the specified properties.

**What Next**

Correct the erroneous setting. After making the change, run the command again.

---

## TECH-044

(warning) The %s entry of attribute %s should be specified as -1, The specified value will be ignored. (%s line %d)

**Description**

The specified entry of the specified attribute should be specified as -1 because there is no spacing requirement on the cut layer corresponding to the entry.

**What Next**

Correct the erroneous table entry. After making the change, run the command again.

---

## TECH-045

(warning) The cut size of ContactCode %s does not have a match in Layer section %s. (%s line %d)

Feedback

**Description**

The cut size (cutWidth x cutHeight) of a ContactCode must have a match in the cut size table of the cut layer that is specified by the cutLayer attribute of the ContactCode.

**What Next**

Correct the cut size of the specifed ContactCode and then run the command again.

---

## TECH-045e

(error) The cut size of ContactCode %s does not have a match in Layer section %s. (%s line %d)

**Description**

The cut size (cutWidth x cutHeight) of a ContactCode must have a match in the cut size table of the cut layer that is specified by the cutLayer attribute of the ContactCode.

**What Next**

Correct the cut size of the specifed ContactCode and then run the command again.

---

## TECH-046

(warning) Cut name %s in DesignRule cut name table %s is not in the cut name table of layer %s. (%s line %d)

**Description**

Cut names specified in the cut1NameTbl attribute of a DesignRule must be defined in cutNameTbl of the cut layer that is specified by the layer1 attribute of the DesignRule and similarly cut names specified in cut2NameTbl of the DesignRule must be in cutNameTbl of the cut layer specified by layer2.

**What Next**

Correct the problematic cut name in the tech file and then run the command again.

---

## TECH-047

(information) CutMetal layer attribute %s is specified in layer %s which is not a cutMetal layer. (%s line %d)

**Description**

This informational message indicates that a cutMetal layer attribute is specified in a non-cutMetal layer section. The attribute specified in the message is ignored.

Feedback

### What Next

This is only an informational message. However, if the result is not what you intended, you can either remove the attribute or change the mask name of the layer to a cutMetal mask name such as cutMetal1, cutMetal2, cutMetal3, etc.

---

## TECH-048

(warning) Attribute %s is missing from the %s ContactCode section. The %s attribute will be ignored. (%s line %d)

### Description

In a ContactCode section if the lowerLayerMaskPattern attribte is specified, the lowerLayerRectTbl attribute of the same section must also be specified and if the upperLayerMaskPattern attribute is specified, the upperLayerRectTbl attribute must also be specified.

### What Next

Redefine the problematic ContactCode section. After making the change, run the command again.

---

## TECH-049

(warning) DesignRule is defined with different %s layers '%s' and '%s'. %s (%s line %d)

### Description

This warning message occurs when the specified DesignRule section must contain the same via layers or the same metal layers.

The following is an example of the warning message:

```
Warning : DesignRule is defined with different via layers 'VIA2'
and 'VIA4'. (line 1954) (TECH-049)
```

### What Next

If the result is not what you intended, Change the layers of the DesignRule section to make them the same.

---

## TECH-050

(warning) The value of the %s attribute of %s %s is invalid. Attribute %s %s attribute %s. (%s line %d)

Feedback

### Description

The message indicates that the value of the specified attribute violates some constraint as stated in the displayed message

The following is an example of the warning message:

```
Warning: The value of the maskEndTblSize attribute of Layer M2 is
 invalid.
Attribute maskEndTblSize should not be greater than attribute numMasks.
(line 225) (TECH-050)
```

### What Next

Correct the erroneous value. After making the change, run the command again.

---

## TECH-051

(warning) Value %s in attribute %s of %s %s is not one of the values specified by attribute %s. (%s line %d)

### Description

Sometimes the values specified by an attribute can only be values contained in a table attribute. The message indicates that such a rule is viaolated.

The following is an example of the warning message:

```
Warning: Value 0.05 in attribute sameColorSpanTblXExcludedSpacingRange of
 Layer
M1 is not one of the values specified by attribute
 sameColorSpanTblXThreshold.
(line 188) (TECH-051)
```

### What Next

Correct the erroneous value. After making the change, run the command again.

---

## TECH-052

(error) The table size of attribute %s %s, but its actual size is %d. This attribute is ignored. (%s line %d)

### Description

The tool issues this error message when the technology file contains an invalid number of entries for the specified attribute. The command ignores attributes with this error.

Feedback

**What Next**

Update the technology file to correct the number of entries for the specified attribute. After updating the technology file, rerun the command.

---

## TECH-053

(warning) Cut name %s in attribute %s of layer %s is not defined in cutNameTbl. (%s line %d)

**Description**

Any cut name specified in the attribute which is named by this message must be a cut name defined in the cutNameTbl attribute of the same layer. This message indicates that the above requirement is not met.

**What Next**

Correct the problematic cut name in the tech file and then run the command again.

---

## TECH-054

(warning) Cut %s in attribute %s of layer %s is a square cut. (%s line %d)

**Description**

Cuts specified in the attribute which is named by this message must be rectangular cuts.

**What Next**

Correct the error in the tech file and then run the command again.

---

## TECH-055

(error) Found invalid layer/layer stack for ContactCode '%s'. (line %d)

**Description**

The error message indicates that the multi-via-layer contact code has invalid layer/ layer stack. The multi-via-layer contact code syntax is as following: ContactCode "ccName" { contactCodeNumber = 312 layersRectTbl = (("MET1", "-0.100, ...", "0.035, ....") ("CUT12", "-0.100, ...", "0.035, ....") ("MET2", "-0.100, ...", "0.035, ....") ("CUT23", "-0.100, ...", "0.035, ....") ("MET3", "-0.100, ...", "0.035, ....")) } (1)The layers' name in property layersRectTbl must be valid layer name in layer section. (2)The layers‘ mask number order should be valid, it means the layer stack should be valid. For example: metal1,via1,metal2,via2,metal3 is valid, metal2,via1,metal3,via2,metal5 is not valid.

The contactcode doesn't meet the above criteria caused the error.

Here is an example of such contactcode: ContactCode "mcvTest" { contactCodeNumber = 3336 layersRectTbl = (("M1", "-0.01, -0.01, 0.01, 0.01", "0.1, 0.2, 0.3, 0.4") ("VIA1", "-0.0250, -0.0100, 0.0250, 0.0100") ("M2", "-0.0180, -0.0250, 0.0180, 0.0850,") ("VIA2", "-0.01, -0.01, 0.01, 0.01") ("M3", "-0.0250, -0.0100, 0.0250, 0.0100")) }

**What Next**

Correct the erroneous layer specifications of the ContactCode section and run the command again.

---

## TECH-055w

(warning) Found invalid layer/layer stack for ContactCode '%s'. (line %d)

**Description**

The warning message indicates that the multi-via-layer contact code has invalid layer/layer stack. The multi-via-layer contact code syntax is as following: ContactCode "ccName" { contactCodeNumber = 312 layersRectTbl = (("MET1", "-0.100, ...", "0.035, ....") ("CUT12", "-0.100, ...", "0.035, ....") ("MET2", "-0.100, ...", "0.035, ....") ("CUT23", "-0.100, ...", "0.035, ....") ("MET3", "-0.100, ...", "0.035, ....")) } (1)The layers' name in property layersRectTbl must be valid layer name in layer section. (2)The layers‘ mask number order should be valid, it means the layer stack should be valid. For example: metal1,via1,metal2,via2,metal3 is valid, metal2,via1,metal3,via2,metal5 is not valid.

The contactcode doesn't meet the above criteria caused the warning.

Here is an example of such contactcode: ContactCode "mcvTest" { contactCodeNumber = 3336 layersRectTbl = (("M1", "-0.01, -0.01, 0.01, 0.01", "0.1, 0.2, 0.3, 0.4") ("VIA1", "-0.0250, -0.0100, 0.0250, 0.0100") ("M2", "-0.0180, -0.0250, 0.0180, 0.0850,") ("VIA2", "-0.01, -0.01, 0.01, 0.01") ("M3", "-0.0250, -0.0100, 0.0250, 0.0100")) }

LONG TSV If cutLayer = "tsv" and lowerLayer is any metal including back metal, and upper layer is metal layer, even if the layers are not consecutive, this will not yield a TECH-055 since this is a "Long TSV". In this case, TECH-280 will be printed.

**What Next**

Correct the erroneous layer specifications of the ContactCode section and run the command again.

---

## TECH-056

(error) Ignore invalid layer shape %s of ContactCode '%s'.

Feedback

**Description**

The error message indicates that the multi-via-layer contact code has invalid layer shape. The multi-via-layer contact code syntax is as following: ContactCode "ccName" { contactCodeNumber = 312 layersRectTbl = (("MET1", "-0.100, ...", "0.035, ....") ("CUT12", "-0.100, ...", "0.035, ....") ("MET2", "-0.100, ...", "0.035, ....") ("CUT23", "-0.100, ...", "0.035, ....") ("MET3", "-0.100, ...", "0.035, ....")) } (1)One Rect shape value in layersRectTbl must be 4 numbers "d1,d2,d3,d4". So for the shapes in one layer, you can specify it as ("layer1", "d1, d2, d3, d4", "d5, d6, d7, d8", …) (2)One pair of parenthesis means the Rects of the layer.

The contactcode doesn't meet the above criteria caused the error.

Here is an example of such contactcode: ContactCode "mcvTest" { contactCodeNumber = 3336 layersRectTbl = (("M1", "-0.01, -0.01, 0.01, 0.01", "0.1, 0.2, 0.3, 0.4") ("VIA1", "-0.0250, -0.0100, 0.0250, 0.0100") ("M2", "-0.0180, -0.0250, 0.0180, 0.0850,") ("VIA2", "-0.01, -0.01, 0.01, 0.01") ("M3", "-0.0250, -0.0100, 0.0250, 0.0100")) }

**What Next**

Correct the erroneous layer shape specifications of the ContactCode section and run the command again.

---

## TECH-057

(error) Failed to create multiple via layers ContactCode '%s', the ContactCode will be ignored.

**Description**

The error message indicates that fail to create the multi-via-layer contact code.

**What Next**

Correct the TECH-055 and TECH-056 of the ContactCode section and run the command again.

---

## TECH-058

(error) Attribute %s is specified in a wrong Layer section. (%s line %d)

**Description**

Attributes maxLengthTblSize, maxLengthDataTypeTbl, and maxLengthTbl can be specified only in a Layer section whose mask name is either "metal0" or "poly". The message indicates that the attribute named in the message is specified in a Layer section that does not satisfy the mask name requirement.

Feedback

**What Next**

Correct the error in the tech file and then run the command again.

---

## TECH-059

(error) Layer2 of the DesignRule is not lower than Layer1. (%s line %d)

**Description**

This message indicates that the layer specified by layer2 of the DesignRule section that generated this message must be lower than the layer specified by layer1. However, the definition of this DesignRule does not satisfy this condition.

**What Next**

Correct the error in the tech file and then run the command again.

---

## TECH-060

(warning) The DesignRule is not between two consecutiv cut layer, nor is it between a cut layer and its adjacent metal layer. (%s line %d)

**Description**

This message indicates that this DesugnRule can be define only for two consecutive cut layers or a cut layer and its adjacent metal layer. However, the definition of this DesignRule does not satisfy this condition.

**What Next**

Correct the error in the tech file and then run the command again.

---

## TECH-061

(error) Attribute %s is not specified in a DesignRule section whose metal layer is a "hasRectangleOnly" layer. (%s line %d)

**Description**

This message indicates that the attribute named in the message can only be specified in metal-to-cut DesignRule sections and the "hasRectangleOnly" attribute of the metal layer must be 1. However, the definition of the DesignRule does not satisfy the condition.

**What Next**

Correct the error in the tech file and then run the command again.

Feedback

## TECH-062

(error) Attribute %s is specified in a DesignRule section that is not defined for two consecutive via layers. (%s line %d)

### Description

This message indicates that the attribute named in the message can only be specified in DesignRule sections that are defined for two consecutive via layers. However, the definition of the DesignRule that contains the attribute does not satisfy the condition.

### What Next

Correct the error in the tech file and then run the command again.

## TECH-063

(error) Attribute %s cannot be specified in ContactCode %s because the mask name of cut layer %s is not "tsv". The attribute will be ignored. (%s line %d)

### Description

The tsvKeepoutSpacing attribute can only be specified in ContactCode sections in which the cutLayer attribute specifies a layer whose mask name is "tsv". The display of this message indicates that the named ContactCode section, in which the tsvKeepoutSpacing attribute is specified, does not satisfy the requirement.

### What Next

Correct the error in the tech file and then run the command again.

## TECH-064

(error) Attribute %s cannot be specified in a tsv ContactCode. The attribute will be ignored. (%s line %d)

### Description

A ContactCode is a tsv ContactCode if the mask name of the layer specified by the cutLayer attribute of the ContactCode is "tsv". Attributes cutLayerRectTbl, lowerLayerRectTbl, upperLayerRectTbl, and layersRectTbl are not allowed to be specified in a tsv ContactCode.

### What Next

Correct the error in the tech file and then run the command again.

Feedback

## TECH-065

(error) The value specified by the tsvKeepoutSpacing attribute is invalid. ContactCode %s is ignored. (%s line %d)

### Description

The value of tsvKeepoutSpacing cannot be less than the value of upperLayerEncWidth, upperLayerEncHeight, lowerLayerEncWidth, or lowerLayerEncHeight.

### What Next

Correct the error in the tech file and then run the command again.

## TECH-066

(error) The technology file contains both noCutMetal and nonStdCellRegion attributes.

### Description

The set of noCutMetal* attributes and the set of nonStdCellRegion* attributes are mutually exclusive, hence cannot both be specified in the same technology file.

### What Next

Correct the error in the tech file and then run the command again.

## TECH-067

(information) Attribute '%s' is obsolete and is %s. (%s line %d)

### Description

Attributes constrainedCutPrefEncTblSize and constrainedCutPref*Tbl are obsolete.

If constrainedCutPrefEncTblSize and constrainedCutPrefEncNameTb are specified, they will be ignored.

If the constrainedCutPrefEncMaxWidthThresholdTbl attribute is specified, it will be replaced with the constrainedCutPrefEncMaxWidthThreshold attribute and the first entry of constrainedCutPrefEncMaxWidthThresholdTbl will be the value of constrainedCutPrefEncMaxWidthThreshold.

If the constrainedCutPrefEncConstrainingRangeTbl attribute is specified, it will be replaced with the constrainedCutPrefEncConstrainingRange attribute and the first entry of constrainedCutPrefEncConstrainingRangeTbl will be the value of constrainedCutPrefEncConstrainingRange.

Feedback

If the constrainedCutPrefEncConstrainingNumCutsTbl attribute is specified, it will be replaced with the constrainedCutPrefEncConstrainingNumCuts attribute and the first entry of constrainedCutPrefEncConstrainingNumCutstbl will be the value of constrainedCutPrefEncConstrainingNumCuts.

If the constrainedCutPrefEncMinEnclosureTbl attribute is specified, it will be replaced with the constrainedCutPrefEncMinEnclosure attribute and the first entry of constrainedCutPrefEncMinEnclosureTbl will be the value of constrainedCutPrefEncMinEnclosure.

**What Next**

No action is required.

---

## TECH-068

(information) Tech file attribute '%s' is obsolete and is %s.

**Description**

Tech file attributes constrainedCutPrefEncTblSize and constrainedCutPref*Tbl are obsolete.

If constrainedCutPrefEncTblSize and constrainedCutPrefEncNameTb are specified in the library, they will be ignored.

If the constrainedCutPrefEncMaxWidthThresholdTbl attribute is specified, it will be replaced with the constrainedCutPrefEncMaxWidthThreshold attribute and the first entry of constrainedCutPrefEncMaxWidthThresholdTbl will be the value of constrainedCutPrefEncMaxWidthThreshold.

If the constrainedCutPrefEncConstrainingRangeTbl attribute is specified, it will be replaced with the constrainedCutPrefEncConstrainingRange attribute and the first entry of constrainedCutPrefEncConstrainingRangeTbl will be the value of constrainedCutPrefEncConstrainingRange.

If the constrainedCutPrefEncConstrainingNumCutsTbl attribute is specified, it will be replaced with the constrainedCutPrefEncConstrainingNumCuts attribute and the first entry of constrainedCutPrefEncConstrainingNumCutstbl will be the value of constrainedCutPrefEncConstrainingNumCuts.

If the constrainedCutPrefEncMinEnclosureTbl attribute is specified, it will be replaced with the constrainedCutPrefEncMinEnclosure attribute and the first entry of constrainedCutPrefEncMinEnclosureTbl will be the value of constrainedCutPrefEncMinEnclosure.

**What Next**

No action is required.

Feedback

## TECH-069

(error) Attribute '%s' is specified in Layer '%s' whose mask name is not '%s'. (%s line %d)

**Description**

This message indicates that one of the following layer mask name requirements is not satisfied.

Layer section attributes "implantSubType", "vtGroupName", "vtGroupVerticalKeepoutParallelLength", and "vtGroupVerticalKeepoutSpacing" can be specified only in Layer sections whose mask name is "implant".

Layer section attributes "concaveConcaveMaxEdgeLength", "concaveConcaveMinEdgeLength", "convexConvexMaxEdgeLength", "convexConvexMinEdgeLength", "concaveConvexMaxEdgeLength", and "concaveConvexMinEdgeLength" can be specified only in Layer sections whose mask name is "diffusion".

Layer section attributes "jogSameLineContinuousMaxSpacing", "jogCrossLineContinuousMinParallelLength", "jogShapeEdgeLengthTblSize", "jogShapeWidthThresholdTbl", "jogTShapeMinEdgeLengthTbl", "jogCornerTShapeMinEdgeLengthTbl", "jogLShapeMinEdgeLengthTbl", "jogUShapeMinEdgeLengthTbl", and "jogUShapeMaxEdgeLengthTbl" can be specified only in Layer sections whose mask name is "diffusion".

Layer section attributes "widthTblSize", "widthThresholdTbl", "widthMaxLengthTbl", and "widthMaxRatioTbl" can be specified only in Layer sections whose mask name is "diffusion".

**What Next**

Correct the error in the tech file and then run the command again.

## TECH-070

(error) Values of attribute '%s' in Layer '%s' and '%s' are different.

**Description**

This message indicates that one of the following requirements is not satisfied.

All the "vtGroupName" attributes specified in the same tech file must have the the same value.

All the "vtGroupVerticalKeepoutParallelLength" attributes specified in the same tech file must have the same value.

Feedback

All the "vtGroupVerticalKeepoutSpacing" attributes specified in the same tech file must have the same value.

**What Next**

Correct the error in the tech file and then run the command again.

---

## TECH-071

(warning) Cannot find any valid mask specific ContactCode to pair with ContactCode %s. (%s line %d)

**Description**

If there is a ContactCode section in which the upperLayerMaskConstraint attribute is specified as either "mask1" or "mask2" and the isDefaultContact attribute is set to 1, then the ContactCode must be paired with another ContacCode for the same cut layer. The pair of ContactCode sections must have the same isDefaultContact attribute specification and the upperLayerMaskConstraint attribute is specified in both ContactCode sections with one being set to "mask1" and the other being set to "mask2".

**What Next**

Correct the error in the tech file and then run the command again.

---

## TECH-072

(warning) Cut name '%s' specified in DesignRule '%s' not found in the cut name table of the corresponding SAV ViaRule.

**Description**

This message is printed when a DesignRule contains self-aligned via (SAV) rule specification, and a cutName in this DesignRule was not found in any corresponding SAV ViaRule section.

**What Next**

Add the information of the cutName in the related SAV ViaRule section and/or correct the cutName specification in the DesignRule section.

---

## TECH-073

(warning) DesignRule missing cornerMinSpacing specification for entry %d of orthoSpacingExcludeCornerTbl. (%s line %d)

Feedback

### Description

This message is printed when a DesignRule contains orthoSpacingExcludeCornerTbl and for at least one of its non-zero entries, no valid cornerMinSpacing value was specified. This means that neither cornerMinSpacing property was specified with a non-zero value, nor was a non-zero value found at the corresponding position in any of the relevant *CornerMinSpacingTbl tables. These tables include the following : diffSegCenterMinSpacingTbl, sameSegCenterMinSpacingTbl, diffNetCornerMinSpacingTbl, diffNetCenterMinSpacingTbl, diffSegCornerMinSpacingTbl, sameSegCornerMinSpacingTbl, sameNetCenterMinSpacingTbl and sameNetCornerMinSpacingTbl.

### What Next

Either specify a valid non-zero value for cornerMinSpacing property, or a valid non-zero value at the corresponding position in the relevant *CornerMinSpacingTbl table property.

---

## TECH-074

(error) Attribute %s contains an incorrect number of entries. It should be a %s table. The attribute is ignored. (%s line %d)

### Description

The tool issues this error message when the technology file contains an invalid number of entries for the specified attribute. The command ignores attributes with this error.

### What Next

Update the technology file to correct the number of entries for the specified attribute. After updating the technology file, rerun the command.

---

## TECH-075

(warning) At lease one of the %s and %s attributes must be specified in the %s Layer section. (%sline %d)

### Description

The display of this message indicates that none of the following requirements is satisfied.

If the nonPrefLegalDimDiscreteLastValue attribute is specified in a Layer section and its value is 1, then at least one of the xLegalDimTbl and yLegalDimTbl attributes of the Layer section must be specified.

If the nonStdCellRegionNonPrefLegalDimDiscreteLastValue attribute is specified in a Layer section and its value is 1, then at least one of the nonStdCellRegionXLegalDimTbl and nonStdCellRegionYLegalDimTbl attributes of the Layer section must be specified.

Feedback

If the jogSameLineContinuousMaxSpacing and jogCrossLineContinuousMinParallelLength attributes are specified in a Layer section, at least one of the concaveConcaveMaxEdgeLength, convexConvexMaxEdgeLength, and concaveConvexMaxEdgeLength attributes must also be specified in the same Layer section.

**What Next**

Verify that necessary attributes are specified for the specified section.

---

## TECH-076

(error) Both viaShiftXDistance and viaShiftYDistance attributes of ContactCode %s are positive. Both attributes are ignored. (%s line %d)

**Description**

The viaShiftXDistance and viaShiftYDistance attributes of a ContactCode section cannot both be positive.

**What Next**

Correct the error in the tech file and then rerun the command.

---

## TECH-077

(error) %s is not a valid operator for the layerOperator attribute. (%s line %d)

**Description**

AND, OR, XOR, and NOT are the only valid operators that can be specified for the layerOperator attribute.

**What Next**

Correct the error in the tech file and then rerun the command.

---

## TECH-078

(error) %s is not a valid entry for the layerOperandTbl attribute. (%s line %d)

**Description**

Every entry in the layerOperandTbl table must be a layer name and the mask name of the layer must be "implant".

**What Next**

Correct the error in the tech file and then rerun the command.

Feedback

## TECH-079

(error) Layer %s is a virtual layer and its layerNumber attribute should not be specified. (%s line %d)

### Description

A virtual layers, such as an operated layer, cannot have a valid layer number. The display of this messge indicates that the layerNaumber attribute is specified in a virtual layer section.

### What Next

Correct the error in the tech file and then rerun the command.

## TECH-080

(warning) The stackViaCountUpperLayerOnly attribute is %s. (%s line %d)

### Description

The stackLevelForbiddenForMultipleViaTbl attribute can be specified in a Layer section only if the stackViaCountUpperLayerOnly attribute is specified in the Technology section and its value is 1.

### What Next

Correct the error in the tech file and then rerun the command.

## TECH-081

(warning) DesignRule is defined with non-distinct layers '%s' and '%s'. (line %d)

### Description

This warning message occurs when the layers of the specified DesignRule section are the same layer. The application ignores the design rule.

The following is an example of the warning message:

```
Warning : DesignRule is defined with non-distinct layers 'VIA3'
and 'VIA3'. (line 1960) (TECH-081)
```

### What Next

This is only a warning message. No action is required.

However, if the result is not what you intended, you can change one of the layers of the DesignRule section and then run the command again.

Feedback

## TECH-082w

(warning) ContactCode '%s' has an invalid cut width %g that is less than the cut layer minWidth %g. (line %d)

### Description

This message indicates that the specified ContactCode section's cutWidth (or cutHeight) or corresponding cut layer's minWidth is invalid. Each contact code's cutWidth (cutHeight) should be greater than or equal to its cut layer's minimum width.

### What Next

Change either the specified ContactCode section's cutWidth and/or cutHeight or corresponding cut layer's minWidth to meet this requirement.

### Examples

Warning : ContactCode 'VIA78fF' has an invalid cut width 0.2 that is less than the cut layer minWidth 0.4. (line 1599) (TECH-082w)

## TECH-083w

(warning) Cut layer '%s' has a non-cross primary default ContactCode '%s'. (line %d)

### Description

This message occurs when the cut layer has multiple default ContactCodes and the primary default ContactCode is a non-cross one. For a given cut layer, the primary ContactCode is defined to be the first one that appears in the technology file, and the primary default ContactCode is the first default ContactCode.

A non-cross (minus '-' shaped) ContactCode is one that has the same longitudinal direction of enclosures in both upper and lower layers. And a cross (plus '+' shaped) ContactCode is one that has different longitudinal directions of enclosures in both upper and lower layers. A cross ContactCode is determined by upperLayerEncWidth >= upperLayerEncHeight and lowerLayerEncWidth <= lowerLayerEncHeight or upperLayerEncWidth <= upperLayerEncHeight and lowerLayerEncWidth >= lowerLayerEncHeight and a non-cross contactCode is one that does not meet either of the above conditions. i.e. it is determined by upperLayerEncWidth > upperLayerEncHeight and lowerLayerEncWidth > lowerLayerEncHeight or upperLayerEncWidth < upperLayerEncHeight and lowerLayerEncWidth < lowerLayerEncHeight

The following is an example of the warning message:

```
Warning:  Cut layer 'VIA1' has a non-cross primary default ContactCode
 'V10'.
(line 254) (TECH-083w)
```

Feedback

### What Next

When non-cross default ContactCode is in the first position, please choose from other default ContactCodes for this cut layer, if any, to get optimal routing performance.

---

## TECH-084

(information) ContactCode '%s' has a minimum cut spacing %g that is less than the cut layer minimum spacing %g. (line %d)

### Description

This message indicates that the specified ContactCode section's minimum cut spacing is less than the corresponding cut layer's minimum spacing. Each contact code's minimum cut spacing should be greater or equal to its cut layer's minimum spacing.

If it is via cell master in the contactCode, and cutArrayXSpacing/cutArrayYSpacing is smaller than cut layer minSpacing rule, this warning message will also be issued.

### What Next

This is only an informational message. No action is required.

However, if the result is not what you intended, you can change either the specified ContactCode section's minimum cut spacing or corresponding cut layer's minimum spacing to meet your requirement.

### Examples

Information : ContactCode 'VIA78fF' has a minimum cut spacing 0 that is less than the cut layer minimum spacing 0.4. (line 1599) (TECH-084)

---

## TECH-085e

(error) ContactCode '%s' on cut layer '%s' has unsupported type '%s' of %s defined. (line %d)

### Description

This error message occurs when the specified section has an unsupported value assigned to the specified attribute.

For cutArrayMaskPattern, currently zRouter cannot support pattern other than 'uniform' on cut layers with more than two masks.

Feedback

The following is an example of the error message:

```
Error : ContactCode 'VIA1_VH' on cut layer 'CUT12' has unsupported type
 'alternate'
of cutArrayMaskPattern defined. (line 661) (TECH-085e)
```

### What Next

Change the attribute value of the specified section to a valid value. After changing the value, run the command again.

---

## TECH-086w

(warning) Layer '%s' has insignificant differences between fat table thresholds '%s'. (%s line %d)

### Description

This warning message occurs when the threshold list of the specified Layer section contains insignificant differences between the threshold values. It is best practice to have thresholds differ by more than half of the minimum width and more than ten percent of the greater threshold. If they do not differ by these amounts, then the fat table can contain redundant data, be unnecessarily large, and increase router run-time dramatically.

The following is an example of the warning message:

```
Warning : Layer 'VIA4' has insignificant differences between
fat table thresholds '(0, 0.5, 0.55, 1.2, 2)'. (line 724) (TECH-086w)
```

### What Next

This is only a warning message. No action is required.

However, you can check the fat tables for redundant data. If you find redundant data, then reduce the table dimension and the number of threshold values. After making your changes, rerun the command.

---

## TECH-087w

(warning) Tile '%s' has a width %g that is not a multiple of the metal layer pitch %g or %g.

### Description

This warning message occurs when the width of the specified tile is inconsistent with pitch of the first two metal layers.

The following is an example of the warning message:

```
Warning : Tile 'unit' has a width 0.32 that is not a multiple
of the metal layer pitch 0.2 or 0.28. (TECH-087w)
```

Feedback

### What Next

This is a warning message only. No action is required.

However, it is best practice to either change the width attribute of the corresponding tile section to the suggested pitch, or change the pitch of the first vertical wire track layer. After making your changes, run the command again.

---

## TECH-088w

(warning) Layer '%s' attribute '%s' has an invalid value of %g.

### Description

This warning message occurs when the value of the specified attribute is invalid or inconsistent with related attributes.

The following is checked: Each value in cutWidthTbl/cutHeightTbl should be compared against the minWdith on the layer. The cutWidth and cutHeight values should not be less than the minWidth on the layer.

The following is an example of the warning message:

```
Warning : Layer 'V2' attribute 'cutWidthTbl' has an invalid value of
 0.05. (TECH-088w)
```

### What Next

It is best practice to check the attribute value of the specified section, and then change the value to a valid value that is consistent with other related attributes in the same set of rule. After making your changes, run the command again.

---

## TECH-089

(warning) The %s attribute is missing from the proper DesignRule section. (%s line %d)

### Description

If the stackLevelForbiddenForMultipleViaTbl attribute is specified in a cut Layer section, the stackViaXEdgeSpacingThreshold, stackViaYEdgeSpacingThreshold, and stackViaCornerSpacingThreshold attributes must also be specified in the DesignRule section that is for the layer on which stackLevelForbiddenForMultipleViaTbl is specified vs. the layer's upper adjacent cut layer.

### What Next

Correct the error in the tech file and then rerun the command.

Feedback

## TECH-090

(error) '%s' is not a valid string for the 'endOfLineType' attribute. (%s line %d)

### Description

Strings 'cutEndOnly' and 'cutAndNaturalEnd' are the only two valid string values that the 'endOfLineType' attribute can take. The display of the message indicates that an invalid string was specified in the tech file for the 'endOfLineType' attribute.

### What Next

Correct the error in the tech file and then rerun the command.

## TECH-091

(warning) The corresponding cut metal layer of metal layer '%s' is missing from the tech file. (%s line %d)

### Description

If the 'endOfLineType' attribute is specified in a metal Layer section, the corresponding cut metal Layer section of the metal Layer must also be specified in the same tech file. Assuming that the mask name of the metal layer is 'metalx', where x is an integer, then the corresponding cut metal layer of the metal layer is the layer whose mask name is 'cutMetalx'.

### What Next

Correct the error in the tech file and then rerun the command.

## TECH-092

(error) String '%s' is not a valid entry for the '%s' attribute. (%s line %d)

### Description

The display of the message indicates that there is an invalid entry in the attribute that is specified in the displayed message. The valid entries for the metalCutMetalMapMetalMaskTbl and metalCutMetalMapCutMetalMaskTbl tables are 'noMask', 'mask1', 'mask2', 'mask3', and 'mask4'. The valid entries for the sameColorCutMetalCheckMetalMaskTbl table are 'mask1', 'mask2', 'mask3', 'mask4', and 'all'.

### What Next

Correct the error in the tech file and then rerun the command.

Feedback

## TECH-093

(error) The sameColorCutMetalCheckMetalMaskTbl table contains string "all" and other entries. (%s line %d)

**Description**

If 'all' is specified in the sameColorCutMetalCheckMetalMaskTbl table, the table cannot have any other entry.

**What Next**

Correct the error in the tech file and then rerun the command.

## TECH-094

(error) String '%s' in sameColorCutMetalCheckMetalMaskTbl is not in the metalCutMetalMapMetalMaskTbl table. (%s line %d)

**Description**

Any of the 'mask1', 'mask2', 'mask3', and 'mask4' strings can be specified in the sameColorCutMetalCheckMetalMaskTbl table only if it is an entry of the metalCutMetalMapMetalMaskTbl table of the same Layer section.

**What Next**

Correct the error in the tech file and then rerun the command.

## TECH-095

(error) Table metalCutMetalMapCutMetalMaskTbl contains duplicate entries of '%s'. (%s line %d)

**Description**

All entries of the metalCutMetalMapCutMetalMaskTbl table must be unique.

**What Next**

Correct the error in the tech file and then rerun the command.

## TECH-098w

(warning) Layer '%s' has incomplete '%s' defined. (line %d)

Feedback

**Description**

This warning message occurs when some tables or values are not defined completely or properly. It includes these types of checks:

In Layer, 1) For rectMinAreaTbl rule, rectMinAreaWidthThresholdTbl should start from minWidth to cover all width ranges;

**What Next**

User should complete/fix the tables/attributes or related sections and run the command again.

---

## TECH-101e

(error) Color attribute '%s' is assigned an out-of-range color intensity %s. (%s line %d)

**Description**

This error message occurs when the specified Color section contains an invalid color intensity value assigned to the specified attribute. Color intensity values must be between 0 and 255 inclusive.

**What Next**

Change the attribute value to a valid color intensity number from 0 through 255 and then run the command again.

---

## TECH-101w

(warning) Color attribute '%s' is assigned an out-of-range color intensity %s. (%s line %d)

**Description**

This warning message occurs when the specified Color section contains an invalid color intensity value assigned to the specified attribute. Color intensity values must be between 0 and 255 inclusive.

**What Next**

Change the attribute value to a valid color intensity number from 0 through 255 and then run the command again.

---

## TECH-102e

(error) Multiple default layers exists with mask name '%s'.

Feedback

### Description

This error message occurs when there are multiple layer sections with the specified mask name that are marked as default. Each physical layer can have only one default layer.

The following is an example of the error message:

```
Error : Multiple default layers exists with mask name 'via9'.
(TECH-102e)
```

### What Next

Check all layers with the specified mask name and mark only one as the default layer. After making your changes, run the command again.

---

## TECH-102w

(warning) Multiple default layers exists with mask name '%s'.

### Description

This warning message occurs when there are multiple layer sections with the specified mask name that are marked as default. Each physical layer can have only one default layer.

The following is an example of the warning message:

```
Warning : Multiple default layers exists with mask name 'via9'.
(TECH-102w)
```

### What Next

Check all layers with the specified mask name and mark only one as the default layer. After making your changes, run the command again.

---

## TECH-200

(warning) %s line %d, comment is not terminated.

### Description

This message is displayed when there is an comment which is not terminated properly.

### What Next

Correct the tech-file by terminating the comment and re-run the command.

Feedback

## TECH-204

(warning) %s line %d, %s section does not need a title. Title '%s' will be ignored.

**What Next**

If you do not want to see this warning messagei, remove the title provided to section.

## TECH-205

(warning) %s line %d, %s section expects a title of '%s' type. Title '%s' is invalid. Ignoring the section.

**What Next**

If you do not want to see this warning message provide title of expected type for section.

## TECH-216

(info) %s line %d, Property '%s' in section '%s' is obsolete. Replacing it with property '%s'.

**Description**

Property provided in technology file is obsolete. It is replaced with a more up to date property.

**What Next**

If you do not want to see this message, please replacve the obsolete property in tecnology file with the up to date property.

## TECH-219

(warning) Property '%s' is defined on incorrect layer '%s'. (%s line:%d)

**Description**

This message is displayed when a property in a section is defined on incorrect layer section. For example, Property 'prefDirForViaGen' should be defined on metal layers but instead defined on cut layer VIA0. In this case, the property will not be used in the fownstream flows. Property 'isSacrificialLayer' can only be defined as 1 for the frontside most or backside most metal and via layers. 1) frontside most metal layer is highest metalN 2) frontside most via layer is highest viaN 3) backside most metal layer is lowest bmetalN 4) backside most via layer is lowest bviaN.

**What Next**

Move the property to a correct layer section.

Feedback

## TECH-220

(warning) Property %s is re-defined in section %s. Ignoring previous definition.(%s line: %d)

### Description

This message is displayed when a property in a section is re-defined. Previous value for the property is ignored and new value is considered.

### What Next

Correct the tech-file by replacing multiple definitions of property with a single definition

## TECH-221

(error) Property '%s' is re-defined in section '%s'. (%s line:%d)

### Description

This message is displayed when a key property (layerNumber or maskName) in the section is re-defined in an add-on tech file with a different value from the one in the library or previous tech file. In this case, the add-on tech file data cannot be merged to the previous one.

### What Next

Correct the conflict in the property in either the add-on tech-file or the previous one.

## TECH-221w

(warning) Property '%s' is re-defined in section '%s'. Using previous definition. (%s line: %d)

### Description

This message is displayed when a property in a section is re-defined in an add-on tech file with a different value from the one in the library or previous tech file. The previous value for the property is used and the new value is ignored.

### What Next

Correct the conflict in the property in either the add-on tech-file or the previous one.

## TECH-223

(warning) Pitch %f for layer %s is less than sum(%f) of minSpacing(%f) and minWidth(%f). (%s line:%d)

Feedback

**Description**

Pitch value for a layer should be greater than or equal to sum of minSpacing and minWidth properties for that layer.

**What Next**

Correct the error in the tech-file and then run the command again.

---

## TECH-224

(warning) Mismatch in column widths for property table. Ignoring it. (%s line:%d)

**Description**

There are columns of different size for the property table which might be a typo.

**What Next**

Correct the error in the tech-file and then run the command again.

---

## TECH-225

(warning) Invalid unit name provided for %s in Technology section. (%s line:%d)

**Description**

Unit name provided for property in Technology section is wrong. Below are valid unit names and the corresponding Technology properties. Valid unit names for property 1) unitTimeName are "fs", "ps", "ns", "us", "ms" and "s" 2) unitLengthName are "micron" and "mil" 3) unitVoltageName are "fV", "pV", "nV", "uV", "mV" and "V" 4) unitCurrentName are "fA", "pA", "nA", "uA", "mA" and "A" 5) unitPowerName are "fw", "pw", "nw", "uw", "mw" and "w" 6) unitResistanceName are "mohm", "ohm", "kohm" and "Mohm" 7) unitCapacitanceName are "ff", "pf", "nf", "uf", "mf" and "f" 8) unitInductanceName are "fh", "ph", "nh", "uh", "mh" and "h"

**What Next**

Correct the error in the tech-file and then run the command again.

---

## TECH-226

(error) Technology section is missing. (%s line:%d)

**Description**

Technology section is mandatory. If it is missing then its an error and loading of tech-file fails.

Feedback

**What Next**

Correct the error in the tech-file and then run the command again.

---

## TECH-227

(error) %s in section %s has invalid value. (%s line:%d)

**Description**

Value provided for the property in the given section is not valid.

**What Next**

Correct the error in the tech-file and then run the command again.

---

## TECH-228

(error) Syntax error in section header. Ignoring it. (%s:%d)

**Description**

The beginning of section has a syntactical error. It could be either invalid section name, or invalid section value or a stray character etc

**What Next**

Correct the error in the tech-file and then run the command again.

---

## TECH-229

(error) Syntax error in property belonging to section %s. Ignoring it. (%s:%d)

**Description**

The property belonging to the section has a syntactical error. It could either be invalid property name, or invalid property value or a stray character etc

**What Next**

Correct the error in the tech-file and then run the command again.

---

## TECH-230

(warning) Missing value for property. Defaulting to empty string as property value. (%s:%d)

Feedback

**Description**

Value for the property is not specified. Only property name and "=" fields are specified. Property value is assumed to be empty string("").

**What Next**

Correct the error in the tech-file and then run the command again.

---

## TECH-231

(error) Missing comma in property table. (%s:%d)

**Description**

Values in property table are not separated by comma. These values are skipped.

**What Next**

Correct the error in the tech-file and then run the command again.

---

## TECH-232

(error) Invalid element %s in property table. Ignoring the property table. (%s:%d)

**Description**

Property table element is invalid. It should be one of bool, integer, float or string.

**What Next**

Correct the error in the tech-file and then run the command again.

---

## TECH-233

(error) Closing parentheses is missing for property table. (%s:%d)

**Description**

Property table has opening parentheses specified. But closing parentheses is not specified.

**What Next**

Correct the error in the tech-file and then run the command again.

---

## TECH-234

(error) Invalid hexadecimal value for property. Ignoring it. (%s:%d)

Feedback

**Description**

Hexadecimal value provided for property is not valid.

**What Next**

Correct the error in the tech-file and then run the command again.

---

## TECH-235

(error) Invalid octal value for property. Ignoring it. (%s:%d)

**Description**

Octal value provided for property is not valid.

**What Next**

Correct the error in the tech-file and then run the command again.

---

## TECH-236

(error) Nested comment found. Ignoring it. (%s:%d)

**Description**

Comments are nested one inside the other which is invalid. Ignoring the nested comment characters.

**What Next**

Correct the error in the tech-file and then run the command again.

---

## TECH-237

(error) Technology filename provided is empty.

**What Next**

Provide a valid technology file name.

---

## TECH-238

(error) Technology filename provided contains only white-spaces.

**What Next**

Provide a valid technology file name.

Feedback

## TECH-239

(error) Pre-processing of technology file failed.

**What Next**

Check the pre-processor directives in the technology file and correct them if wrong. Also check whether the pre-processor executable is available/accessible.

## TECH-240

(error) String has an un-matched quotation mark. (%s:%d)

**What Next**

Correct the syntax error in the tech-file and run the command again.

## TECH-241

(warning) %s line %d, property %s needed by %s in %s %s is missing.

**What Next**

Correct the syntax error in the tech-file and run the command again.

## TECH-242

(error) %s line %d, property %s in %s %s has invalid cut name %s.

**What Next**

Correct the syntax error in the tech-file and run the command again.

## TECH-243

(warning) %s line %d, Section by name %s already exists.

**What Next**

Remove the duplicate section by deleting or merging their contents into a single section.

## TECH-244

(warning) %s line %d, Floating-point property %s is assigned %s which is too precise and will be truncated.

Feedback

**What Next**

Provide a correct value within the limits of precision.

---

## TECH-245

(warning) %s line %d, in Layer %s, defaultWidth(%f) is less than minWidth(%f).

**What Next**

Provide a correct value either for defaultWidth or minWidth and re-run the command.

---

## TECH-246

(warning) %s line %d, in Stipple %s, pattern has a non-boolean value.

**Description**

Tabular property 'pattern' in Stipple section is expected to contain boolean values only. When a non-zero, non-boolean value is provided, as in this case, it will be converted to boolean value 1 and used.

**What Next**

If the boolean value 1, used in place of non-boolean value provided, is incorrect, then provide the correct value. Otherwise ignore this message.

---

## TECH-246e

(error) %s line %d, in Stipple %s, pattern has a non-boolean value.

**Description**

Tabular property 'pattern' in Stipple section is expected to contain boolean values only. When a non-zero, non-boolean value is provided, as in this case, it will be converted to boolean value 1 and used.

**What Next**

If the boolean value 1, used in place of non-boolean value provided, is incorrect, then provide the correct value. Otherwise ignore this message.

---

## TECH-247

(error) %s line %d, routing layer %s cannot have a zero value for '%s'.

Feedback

**Description**

Property 'minWidth' in Layer section should have a value greater than 0, if the layer is a routing layer(i.e. layer is either a metal layer or a via layer).

**What Next**

Provide a valid non-zero value for the property 'minWidth' in the routing layer in technology file and re-run the command.

---

## TECH-248

(error) %s line %d, metal layer %s cannot have a zero value for '%s'.

**Description**

Property 'pitch', 'defaultWidth' and 'minSpacing' in 'Layer' section should have a value greater than 0, if the layer is a metal layer.

**What Next**

Provide a valid non-zero value for the property 'pitch', 'defaultWidth' and 'minSpacing' in metal layer in technology file and re-run the command.

---

## TECH-249

(error) Extra comma or missing element or missing comma in property table. (%s:%d)

**Description**

There is either an stray comma in the property table or element in property table is missing or a comma is missing.

**What Next**

Remove the extra comma or add the missing property element or comma and then run the command again.

---

## TECH-250

(warning) %s line %d, ContactCode '%s' has undefined or zero-enclosures.

**Description**

This warning message occurs when the specified ContactCode section contains undefined or zero metal enclosure dimensions. If the zero metal enclosure values are set intentionally, this warning can be ignored. Otherwise, the values may cause the router to

Feedback

create enclosures that violate design rule constraints, such as completely enclosed spaces or donuts.

The following is an example of the warning message:

```
Warning: test.tf line 1551, ContactCode 'VIA67f' has undefined or zero
 metal enclosures. (TECH-250)
```

**What Next**

This is only a warning message. No action is required. However, if the zero metal enclosure values are not intentional, you can set the upper and lower enclosures of the ContactCode section to avoid potential design rule violations.

---

## TECH-251

(error) %s line %d, '%s' in '%s' section has zero value.

**Description**

This error message is displayed when any one of properties 'finFetXPitch' and 'finFetYPitch' in Technology section has zero value.

The following is an example of the warning message:

```
Error: test.tf line 101, 'finFetXPitch' in 'Technology' section has zero
 value. (TECH-251)
```

**What Next**

You need to set the correct non-zero value for the erroneous property in technology file and re-run the command again.

---

## TECH-252

(warning) %s line %d, 'cutNameTbl' on Layer '%s' has cut name '%s' with no matching ContactCode found.

**Description**

This warning message is displayed when cut name defined in 'cutNameTbl' does not have a ContactCode section defined in technology file, which matches the 'cutHeight' and 'cutWidth' of the cut.

The following is an example of the warning message:

```
Warning: test.tf line 1101, 'cutNameTbl' on Layer 'VIA2' has cut name
 'Alg'
with no matching ContactCode found. (TECH-252)
```

Feedback

### What Next

You need to define a ContactCode section in technology file which is compatible with the height and width for the cut which is causing the issue.

---

## TECH-253

(warning) %s line %d, Layer '%s' has duplicate cut sizes defined for cuts '%s' and '%s'.

### Description

This warning message is displayed when cut-width and cut-height of 2 cuts in the same layer are same. This might be a redundancy or might be a typo and needs user attention. While comparing the cut-sizes for 2 cuts cut-width and cut-height can used interchangeably.

The following is an example of the warning message:

```
Warning: test.tf line 1101, Layer 'VIA2' has duplicate cut sizes defined
for cuts 'Vsm' and 'Vsm1'. (TECH-253)
```

### What Next

You need to remove one of the duplicate cuts or modify the cut-sizes of the cut so that they are unique.

---

## TECH-254

(warning) %s line %d, '%s' ContactCode '%s' on Layer '%s' has cut size, for which no matching cut name is found.

### Description

This warning message is displayed when there is a ContactCode section with some cut-size(cutWidth and cutHeight) defined. But there is no cut-name in the cut-layer corresponding to the ContactCode which has same cut-size(part of cutWidthTbl or cutHeightTbl).

The following are examples of the warning messages:

```
Warning: test.tf line 1101, Default ContactCode 'VIA01' on Layer 'CUT12'
has cut size, for which no matching cut name is found. (TECH-254)
Warning: test.tf line 1201, Non-default ContactCode 'VIA01a' on Layer
'CUT12' has cut size, for which no matching cut name is found. (TECH-254)
```

### What Next

You need to add a cut-name in the cut-layer with right cut-size for the ContactCode or remove the ContactCode section.

Feedback

## TECH-255

(warning) %s line %d, ContactCode has lowerLayer '%s' which is above upperLayer '%s'.

### Description

This warning message occurs when the maskName associated with lowerLayer is above the maskName associated with upperLayer. For example, the mask name associated with lowerLayer 'M3' can be 'metal3' and maskName associated with upperLayer 'M2' can be 'metal2'. This is incorrect because metal3 comes above metal2.

The following is an example of the warning message:

```
Warning: test.tf line 1954, ContactCode has lowerLayer 'M3' which is
 above
upperLayer 'M2'. (TECH-255)
```

### What Next

To avoid this warning message, swap the values of upperlayer and lowerLayer of the ContactCode section so that the mask names of the layers are in right order. After making this change, run the command again.

## TECH-256

(warning) %s line %d, Layer '%s' has invalid maskName '%s'.

### Description

This warning message occurs when the maskName associated with Layer is invalid. It has to be one pre-defined values like metal0, metal1, via0, via1, bmetal1, bvia1, poly etc

The following is an example of the warning message:

```
Warning: test.tf line 1954, Layer 'MB' has invalid maskName 'bmetal'.
 (TECH-256)
```

### What Next

To avoid this warning message, provide a valid value for maskName.

## TECH-257

(warning) Technology file preprocessor '%s' does not exist in your installation.

### Description

The preprocessor executable used to handle directives in the technology file could not be found. The preprocessor direvtives in the technology file will be ignored.

Feedback

### What Next

Check the installation directory for the preprocessor executable.

---

## TECH-258w

(warning) ContactCode has an invalid maskName. Layer '%s' is not a pre-defined maskName. (line %d)

### Description

This warning message occurs when the maskName associated with lowerLayer, upperLayer, or cutLayer is not in the pre-defined list of maskNames.

The following is an example of the warning message:

```
Warning: ContactCode has an invalid maskName. Layer 'polycon' is not a
pre-defined maskName. (line 1691) (TECH-258w)
```

### What Next

To avoid this warning message, fix the maskNames to a pre-defined maskName. After making this change, run the command again.

---

## TECH-259e

(error) Metal Layer '%s' has keepout region with invalid values. (line %d)

### Description

This warning message occurs when the 2 keepout regions formed by L1o/L2o/So and L1s/L2s/Ss have invalid values. In techfile, these values are default positive but both L1o/L2o can be specified as negative values. If negative, their absolute value must be smaller than L2o/L1o. (same as in L1s/L2s). For reference: endToEndSpaceSameDirKeepoutWidth = Ss endToEndSpaceSameDirKeepoutLength = L2s endToEndSpaceSameDirSideKeepoutLength = L1s endToEndSpaceOppositeDirKeepoutWidth = So endToEndSpaceOppositeDirKeepoutLength = L2o endToEndSpaceOppositeDirSideKeepoutLength = L1o

The following is an example of the warning message:

```
Error: Metal Layer 'M1' has keepout region with invalid values. (line
 881) (TECH-259e)
```

### What Next

To avoid this warning message, fix the values. After making this change, run the command again.

Feedback

## TECH-259w

(warning) Metal Layer '%s' has keepout region with invalid values. (line %d)

### Description

This warning message occurs when the 2 keepout regions formed by L1o/L2o/So and L1s/L2s/Ss have invalid values. In techfile, these values are default positive but both L1o/L2o can be specified as negative values. If negative, their absolute value must be smaller than L2o/L1o. (same as in L1s/L2s). For reference: endToEndSpaceSameDirKeepoutWidth = Ss endToEndSpaceSameDirKeepoutLength = L2s endToEndSpaceSameDirSideKeepoutLength = L1s endToEndSpaceOppositeDirKeepoutWidth = So endToEndSpaceOppositeDirKeepoutLength = L2o endToEndSpaceOppositeDirSideKeepoutLength = L1o

The following is an example of the warning message:

```
Warning: Metal Layer 'M1' has keepout region with invalid values. (line
 881) (TECH-259w)
```

### What Next

To avoid this warning message, fix the values. After making this change, run the command again.

## TECH-260w

(warning) Missing layer section with maskName 'mimtop' or 'mimbottom'.

### Description

This warning message occurs when 'mimmiddle1' maskName appears, but 'mimtop' and/or 'mimbottom' do not appear in the tech file.

The following is an example of the warning message:

```
Warning: Missing layer section with maskName 'mimtop' or 'mimbottom'.
 (TECH-260w)
```

### What Next

To avoid this warning message, add a layer section with proper maskName. After making this change, run the command again.

Feedback

## TECH-261w

(warning) Layer '%s' attributes 'isDefaultLayer' and 'isParasiticOnlyLayer' cannot both be specified to value of '1'. (line %d)

### Description

This warning message occurs when 'isDefaultLayer' and 'isParasiticOnlyLayer' on Layer section are both specified to value of 1.

The following is an example of the warning message:

```
Warning: Layer 'V1' attributes 'isDefaultLayer' and
 'isParasiticOnlyLayer'
cannot both be specified to value of '1'. (line 453) (TECH-261w)
```

### What Next

To avoid this warning message, fix the values of the properties. After making this change, run the command again.

## TECH-262w

(warning) ViaRule '%s' has incorrect attribute 'contactNumberTbl'. %s

### Description

This warning message occurs when there is an issue with contactNumberTbl.

The following is an example of the warning message:

```
Warning: ViaRule 'CompTSV_3x3_cuts' has incorrect attribute
 'contactNumberTbl'.
The table size has to be 2 or 3. (TECH-262w)
```

### What Next

To avoid this warning message, fix the values of the property. After making this change, run the command again.

## TECH-263w

(warning) '%s' is not a valid maskName.

### Description

This warning message occurs when the maskName is not in the pre-defined list of maskNames.

Feedback

The following is an example of the warning message:

```
Warning: 'active' is not a valid maskName. (TECH-263w)
```

**What Next**

To avoid this warning message, fix the maskNames to a pre-defined maskName. After making this change, run the command again.

---

## TECH-264w

(warning) Layer '%s' attribute 'minCutsFieldNameTbl' has values in the wrong order. The order must be {numCuts, cutName, lowerWidth, upperWidth, cutSpacing, extRange}. (line %d)

**Description**

This warning message occurs when the minCutsFieldNameTbl is not in the proper order.

The following is an example of the warning message:

```
Warning: Layer 'V1' attribute 'minCutsFieldNameTbl' has values in the
 wrong
order. The order must be {numCuts, cutName, lowerWidth, upperWidth,
 cutSpacing,
extRange}. (line 244) (TECH-264w)
```

**What Next**

To avoid this warning message, fix the maskNames to a pre-defined maskName. After making this change, run the command again.

---

## TECH-266w

(warning) At least one of the %s and %s attributes must be specified in the DesignRule section. (line %d)

**Description**

The display of this message indicates that none of the following requirements is satisfied.

If the sameSegOneLayerOnlyTbl attribute is specified, at least one of the sameSegCornerMinSpacingTbl, sameSegXMinSpacingTbl, or sameSegYMinSpacingTbl properties must be defined in the same DesignRule section.

**What Next**

Verify that necessary attributes are specified for the specified section.

Feedback

## TECH-267e

(error) Layer '%s' has incorrect value for 'isDefaultLayer'. For CompoundTSV, pseudo layers must specify 'isDefaultLayer = 0' while parallel routing layers must specify 'isDefaultLayer = 1'.

### Description

The display of this message indicates that none of the following requirements is satisfied.

In a compound TSV, a pseudo-layer is created with a maskName that is the same as its parallel routing layer. The pseudo-layer is marked with isDefaultLayer = 0 so that it will not be used for routing. The parallel routing layer then must have isDefault = 1, otherwise the compoundTSV cannot be created.

### What Next

Fix the necessary attributes for the specified section.

## TECH-267w

(warning) Layer '%s' has incorrect value for 'isDefaultLayer'. For CompoundTSV, pseudo layers must specify 'isDefaultLayer = 0' while parallel routing layers must specify 'isDefaultLayer = 1'.

### Description

The display of this message indicates that none of the following requirements is satisfied.

In a compound TSV, a pseudo-layer is created with a maskName that is the same as its parallel routing layer. The pseudo-layer is marked with isDefaultLayer = 0 so that it will not be used for routing. The parallel routing layer then must have isDefault = 1, otherwise the compoundTSV cannot be created.

### What Next

Fix the necessary attributes for the specified section.

## TECH-268e

(error) ViaRule '%s' specifies cut name '%s' which is not found in the cutNameTbl of any via layer.

### Description

This message is printed when a ViaRule contains a cutName in its cutNameTbl which does not exist in the cutNameTbl of any via layer in the tech file.

Feedback

### What Next

Check the cutName in the ViaRule sections and/or correct the cutName specification in the via layer sections.

---

## TECH-268w

(warning) ViaRule '%s' specifies cut name '%s' which is not found in the cutNameTbl of any via layer.

### Description

This message is printed when a ViaRule contains a cutName in its cutNameTbl which does not exist in the cutNameTbl of any via layer in the tech file.

### What Next

Check the cutName in the ViaRule sections and/or correct the cutName specification in the via layer sections.

---

## TECH-269w

(warning) Layer %s has incorrect table values in fatTblSpacingExcludedPGMaxParallelLength.

### Description

The display of this message indicates that none of the following requirements is satisfied.

For any value of the fatTblSpacingExcludedPGMaxParallelLength table which is non-zero, the corresponding value in the fatTblSpacingExcludedPG table must be 1.

### What Next

Verify that the conditions are met in the specified section.

---

## TECH-270e

(error) Metal Layer '%s' property %s has invalid values. %s (line %d)

### Description

This error message occurs when a metal layer property has invalid values.

The following is an example of the error message:

```
Error: Metal Layer 'M1' property sameNetRDLMetalToPadMinSpacing has
 invalid values. It must be greater than or equal to minSpacing. (line
 881) (TECH-270e)
```

Feedback

### What Next

To avoid this error message, fix the values. After making this change, run the command again.

---

## TECH-270w

(warning) Metal Layer '%s' property %s has invalid values. %s (line %d)

### Description

This warning message occurs when a metal layer property has invalid values.

Specifications for RDL routing: sameNetRDLMetalToPadMinSpacing >= minSpacing diffNetRDLMetalToPadMinSpacing >= minSpacing jogRDLKeepoutRange >= minSpacing

Specifications for layer: cutoutMinWidth <= cutoutMaxWidth < maxWidth

The following is an example of the warning message:

```
Warning: Metal Layer 'M1' property sameNetRDLMetalToPadMinSpacing has
 invalid values. It must be greater than or equal to minSpacing. (line
 881) (TECH-270w)
```

### What Next

To avoid this warning message, fix the values. After making this change, run the command again.

---

## TECH-271w

(warning) Via Layer '%s' has different definitions for cutWidth/cutHeight in cutNameTbl and nonStdCellRegionCutNameTbl. (line %d)

### Description

This warning message occurs when for a given string(say, Vs) the cutWidth & cutHeight definition are different in cutNameTbl and nonStdCellRegionCutNameTbl.

The following is an example of the warning message:

```
Warning: Via Layer 'V3' has different definitions for cutWidth/cutHeight
 in cutNameTbl and nonStdCellRegionCutNameTbl. (line 176) (TECH-271w)
```

### What Next

To avoid this warning message, fix the values. After making this change, run the command again.

Feedback

## TECH-273w

(warning) DesignRule property viaProjectedForbiddenCheckFullProjectionTbl has invalid values. If viaProjectedForbiddenFullProjectionNonPrefParallelLength is defined, viaProjectedForbiddenCheckFullProjectionTbl must have at least a '1'. (line %d)

### Description

The following is an example of the warning message:

```
Warning: DesignRule property viaProjectedForbiddenCheckFullProjectionTbl
 has invalid values. If
 viaProjectedForbiddenFullProjectionNonPrefParallelLength is defined,
 viaProjectedForbiddenCheckFullProjectionTbl must have at least a '1'.
 (line 881) (TECH-273w)
```

### What Next

To avoid this warning message, fix the values. After making this change, run the command again.

## TECH-274w

(warning) Property 'forbidCutMetal' cannot be specified on both metal layers and cut-metal layers in the same tech file.

### Description

The following is an example of the warning message:

```
Warning: Property 'forbidCutMetal' cannot be specified on both metal
 layers and cut-metal layers in the same tech file. (TECH-274w)
```

### What Next

To avoid this warning message, fix the values. After making this change, run the command again.

## TECH-275w

(warning) Design Rule layers '%s' and '%s' are incorrect. If stackViaKeepout rule is defined, the two layers must be the top/bottom most via layer and the top most via layer's upper adjacent via layer. (line %d)

### Description

If existing stackViaKeepout* rule is specified in a DesignRule section, layer1 and layer2 cannot be adjacent via layers. layer1 and layer2 must be one of the following: - Top most

Feedback

via layer in the stack via and its upper adjacent via layer - Bottom most via layer in the stack via and top most via layer's upper adjacent via layer

The following is an example of the warning message:

```
Warning: Design Rule layers 'MD' and 'GD' are incorrect. If
 stackViaKeepout rule is defined, the two layers must be the top/bottom
 most via layer and the top most via layer's upper adjacent via layer.
 (line 176) (TECH-275w)
```

**What Next**

To avoid this warning message, fix the values. After making this change, run the command again.

---

## TECH-276w

(warning) DesignRule layers '%s' and '%s' are incorrect. %s (line %d)

The following is an example of the warning message:

```
Warning: DesignRule layers 'MD' and 'GD' are incorrect. If
 cutout2MinEnclosure is defined, the two layers must be the MCtop and
 MCmid layers. (line 176) (TECH-276w)
```

**What Next**

To avoid this warning message, fix the values. After making this change, run the command again.

---

## TECH-277w

(warning) Layer '%s' has the attribute '%s', which can only be defined on a poly layer. (line %d)

**Description**

This information message occurs when the specified Layer section is not a cpode poly layer, but it contains a cpode poly layer attribute that is ignored.

The following is an example of the information message:

```
Warning : Layer 'M2' has the attribute 'yMaxLength', which can only be
 defined on a poly layer. (line %d) (TECH-277w)
```

**What Next**

This is only an informational message. No action is required.

Feedback

## TECH-279

(information) Non-mim layer '%s' has the mim layer attribute '%s'. (line %d)

### Description

This information message indicates that the specified Layer section does not define a mim layer, but it contains a mim layer attribute, which is ignored.

The following is an example of the information message:

```
Information: Non-mim layer 'METAL2' has the mim layer attribute
\'cutoutMinWidth'. (line 294) (TECH-279)
```

### What Next

This is only an informational message. No action is required.

## TECH-280

(information) Long TSV '%s' found. LowerLayer is '%s' and UpperLayer is '%s'.

### Description

This information message indicates that a long tsv was found, only when cut layer has mask name of "tsv"

The following is an example of the information message:

```
Information: Long TSV 'TSV_MB1S' found. LowerLayer is 'MB1S' and
 UpperLayer is 'M17'. (TECH-280)
```

### What Next

This is only an informational message. No action is required.

## TECH-281w

(warning) Prune Tech- %s.

### Description

This message occurs warnings/errors during prune_tech_file command.

The following is an example of the information message:

```
Warning: Prune Tech- Layer m13 which was specified in 'layers.txt' does
 not have a routingDirection set. It will be ignored. (TECH-281w)
```

Feedback

### What Next

This is only an informational message. No action is required.

---

## TECH-282w

(warning) Poly Layer '%s' property %s has invalid values. %s (line %d)

### Description

This warning message occurs when a poly layer property has invalid values.

If both yMaxLength and yForbiddenParallelLengthThreshold are present in the same section, yForbiddenParallelLengthThreshold value must be <= yMaxLength.

The following is an example of the warning message:

```
Warning: Poly Layer 'M1' property yForbiddenParallelLengthThreshold has
 invalid values. It must be less than or equal to 'yMaxLength'. (line
 881) (TECH-282w)
```

### What Next

To avoid this warning message, fix the values. After making this change, run the command again.

---

## TECH-283w

(warning) Design Rule property %s has invalid values. %s (line %d)

### Description

This warning message occurs when a design rule property has invalid values.

The following is an example of the warning message:

```
Warning: Design Rule property
 stdCellRegionMaskEndToEndCutEncMaskNumTblSize has invalid values.
 'stdCellRegionMaskEndToEndCutEncMaskNumTblSize' cannot be greater than
 numMasks defined in the metal layer section. (line 881) (TECH-283w)
```

### What Next

To avoid this warning message, fix the values. After making this change, run the command again.

Feedback

## TECH-284w

(warning) Attribute '%s' cannot be specified in ContactCode %s because the mask name of cut layer %s is not 'tsv'. The attribute will be ignored. (%s line %d)

### Description

The following attributes can only be specified in ContactCode sections in which the cutLayer attribute specifies a layer whose mask name is "tsv". The display of this message indicates that the named ContactCode section, in which the attribute is specified, does not satisfy the requirement. 1) tsvExcludeTopRoutingBlockage 2) tsvExcludeBottomRoutingBlockage 3) tsvConnectWithinPin 4) tsvBulkCapKeepout 5) tsvToDiodeKeepoutSpacing

### What Next

Correct the warning in the tech file and then rerun the command

## TECH-285w

(warning) Layer '%s' property '%s' has an incorrect value. %s (line %d)

### Description

This warning message occurs for: 1) isSpecialMiddleEndOfLine is specified on layer whose mask name is not 'cutMetal0', 'metal0', or 'via0'.

The following is an example of the warning message:

```
Warning: Layer 'V0' property 'isSpecialMiddleEndOfLine' has an incorrect
 value. isSpecialMiddleEndOfLine can be specified only in layer
 sections whose maskName is 'metal0', 'via0' or 'cutMetal0'. (line 453)
 (TECH-285w)
```

### What Next

To avoid this warning message, fix the values of the properties. After making this change, run the command again.

## TECH-287e

(error) Core region name '%s' specified by the app_option does not exist in the techfile.

### Description

This error message occurs when the region name specified by the following app_options does not exist in the techfile: 1) file.tech.std_cell_region_name 2) file.tech.output_std_cell_region_name When a valid region name is specified and the

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original tech file has region dependent sections, the tech data for the tools will be specific to that region. If no region name is specified, i.e. an empty string, region dependent sections will be ignored and baselines will be used.

The following is an example of the error message:

```
Error: Core region name 'FB_D' specified by the app_option does not exist
 in the techfile. (TECH-287e)
```

**What Next**

To avoid this error message, fix the values of the app option. After making this change, run the read_tech_file or related command again.

---

## TECH-287w

(warning) Core region name '%s' specified by the app_option does not exist in the techfile.

**Description**

This warning message occurs when the region name specified by the following app_options does not exist in the techfile: 1) file.tech.std_cell_region_name 2) file.tech.output_std_cell_region_name When a valid region name is specified and the original tech file has region dependent sections, the tech data for the tools will be specific to that region. If no region name is specified, i.e. an empty string, region dependent sections will be ignored and baselines will be used. When the region name specified by file.tech.output_std_cell_region_name is not consistent with the one specified by std_cell_region_name, you will also see this message, and the region name specified by std_cell_region_name that exists in the tech file will be used. In the case of non-existent region name, the baseline tech will be kept.

The following is an example of the warning message:

```
Warning: Core region name 'FB_D' specified by the app_option does not
 exist in the techfile. (TECH-287w)
```

**What Next**

To avoid this warning message, fix the values of the app option. After making this change, run the read_tech_file or related command again.

---

## TECH-288

(error) Baseline %s '%s' section for region '%s' not defined.

**Description**

The baseline section is required when specifiying multi regions. This applies to Layer, DesignRule and ContactCode sections.

Feedback

The following is an example of the error message:

```
Error: Baseline Layer 'M1' section for region 'FB_A' not defined.
 (TECH-288)
```

**What Next**

To avoid this error message, fix the values of the properties. After making this change, run the command again.

---

## TECH-289

(warning) Property '%s' not defined in baseline %s '%s'.

**Description**

The superset of all rules/properties must be defined in baseline section when the property is defined in the multi region sections. This applies to Layer, DesignRule and ContactCode sections

The following is an example of the warning message:

```
Warning: Property 'minWidth' not defined in baseline Layer
 'M1' (TECH-289)
```

**What Next**

To avoid this warning message, fix the values of the properties. After making this change, run the command again.

---

## TECH-290

(error) %s '%s' is a region-dependent section and its %s attribute should not be specified. (line %d)

**Description**

Virtual layers, such as an operated layer and region dependent layers, cannot have a valid layerNumber or maskName specified. Region dependent ContactCode sections cannot have contactCodeNumber specified. The association of region dependent sections with their baseline sections is made by the section name.

**What Next**

Correct the error in the tech file and then rerun the command.

Feedback

## TECH-290w

(warning) %s '%s' is a region-dependent section and its %s attribute should not be specified. (line %d)

### Description

Virtual layers, such as an operated layer and region dependent layers, cannot have a valid layerNumber or maskName specified. Region dependent ContactCode sections cannot have contactCodeNumber specified. The association of region dependent sections with their baseline sections is made by the section name.

### What Next

Correct the warning in the tech file and then rerun the command.

## TECH-291w

(warning) ViaRule '%s' property %s has invalid values. %s (line %d)

### Description

This warning message occurs when a via rule property has invalid values.

### What Next

To avoid this warning message, fix the values. After making this change, run the command again.

## TECH-292

(warning) %s line %d, the size for entry %d of property %s is '%d'. However, the size for entry %d of property %s is '%d'. This property will be ignored.

### Description

The tool issues this warning message when the technology file contains an invalid size for the specified entry (entry number starts from 0) of property. The command ignores attributes with this warning.

### Examples

The following is an exmaple tech file syntax. The size for entry 3 of property is 2, but the size for entry 2 of property is 1.

```
fatTblFatContactNumber   = (4, 104, 1004,
                           "104, 1004", 1004, 1004)
fatTblFatContactMinCuts  = (1, 1, 1,
                           1, 1, 1)
```

Feedback

### Examples

Warning: fatContact.tf line 478, the size for entry 3 of property fatTblFatContactNumber is '2'. However, the size for entry 3 of property fatTblFatContactMinCuts is '1'. This property will be ignored. (TECH-292)

### What Next

Update the technology file to correct size for the specified entry of the property. After updating the technology file, rerun the command.

---

## TECH-293

(warning) Failed to set upper/lower/cut layer for TSV '%s'.

### Description

The warning message indicates that 1) the upper layer or lower layer is invalid. 2) the upper layer and lower layer in wrong order, which means maskName associated with lower layer is above the maskName associated with upper layer. For example, the mask name associated with lowerlayer 'M3' can be 'metal3' and maskName associated with upperLayer 'M2' can be 'metal2'. This is incorrect because metal3 comes above metal2. 3) the upper layer or lower layer is not default layer. For example, Layer "BTPM_MED" { layerNumber = 1230 ; maskName = "bmetal1" isDefaultLayer = 0 isParasiticOnlyLayer = 1 } Layer "BTSV_MED" { layerNumber = 1225 ; maskName = "tsv" } Layer "M0" { layerNumber = 180 ; maskName = "metal1" } ContactCode "VIABTSV_MED" { contactCodeNumber = 1503 cutLayer = "BTSV_MED" lowerLayer = "BTPM_MED" upperLayer = "M0" } Since layer "BTPM_MED" is marked with "isDefaultLayer = 0", tool issue the warning and customer must remove "isDefaultLayer = 0" or correct it as "isDefaultLayer = 1" to fix the issue.

### Examples

Warning: Failed to set upper/lower/cut layer for TSV '%s'.(TECH-293)

### What Next

Update the technology file to correct the ContactCode section or its corresponding layer section. After updating the technology file, rerun the command.

---

## TECH-293e

(error) Failed to set upper/lower/cut layer for TSV '%s'.

### Description

The error message indicates that 1) the upper layer or lower layer is invalid. 2) the upper layer and lower layer in wrong order, which means maskName associated with

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lower layer is above the maskName associated with upper layer. For example, the mask name associated with lowerlayer 'M3' can be 'metal3' and maskName associated with upperLayer 'M2' can be 'metal2'. This is incorrect because metal3 comes above metal2. 3) the upper layer or lower layer is not default layer. For example, Layer "BTPM_MED" { layerNumber = 1230 ; maskName = "bmetal1" isDefaultLayer = 0 isParasiticOnlyLayer = 1 } Layer "BTSV_MED" { layerNumber = 1225 ; maskName = "tsv" } Layer "M0" { layerNumber = 180 ; maskName = "metal1" } ContactCode "VIABTSV_MED" { contactCodeNumber = 1503 cutLayer = "BTSV_MED" lowerLayer = "BTPM_MED" upperLayer = "M0" } Since layer "BTPM_MED" is marked with "isDefaultLayer = 0", tool issue the warning and customer must remove "isDefaultLayer = 0" or correct it as "isDefaultLayer = 1" to fix the issue.

**Examples**

Error: Failed to set upper/lower/cut layer for TSV '%s'.(TECH-293e)

**What Next**

Update the technology file to correct the ContactCode section or its corresponding layer section. After updating the technology file, rerun the command.

---

## TECH-294

(error) %s line %d, END_PHYSICAL_RULE_SECTION is missing for PRF rule section.

**Description**

PRF rule section has opening token START_PHYSICAL_RULE_SECTION specified. But closing token END_PHYSICAL_RULE_SECTION is not specified.

**What Next**

Correct the error in the tech-file and then run the command again.

---

# TFCHK

---

## TFCHK-001

(error) Failed to read from file %s.

**Description**

This error message occurs when the specified file cannot be opened for read, or the file contain syntax error.

Feedback

The following is an example of this error message:

```
Error : Failed to read from file design.tf. (TFCHK-001)
```

**What Next**

Check that the specified file exists and is readable, and correct syntax errors and then run the command again.

---

## TFCHK-002

(error) Failed to write to file %s.

**Description**

This error message occurs when the specified file cannot be opened for writing.

The following is an example of this error message:

```
Error : Failed to write to file design.tf_backup. (TFCHK-002)
```

**What Next**

Check that the specified file exists and is writable, and then run the command again.

---

## TFCHK-003

(error) Technology file %s has a syntax error. (line %d)

**Description**

This error message occurs when the technology file contains a syntax error on or near the specified line. The following cases will cause syntax error: 1) illegal attribute name or section name 2) no space between = and value, such as minSpacing =0.1 3) missing or misplaced curly brackets {}, parentheses (), equal signs (=), and commas (,). 4) missing starting ; or # in comment that follows attribute value 5) missing comma between values in a table.

The following is an example of the error message:

```
Error : Technology file new.tf has a syntax error. (line 342) (TFCHK-003)
```

**What Next**

Check the specified line for syntax errors, such as missing or misplaced curly brackets {}, parentheses (), equal signs (=), and commas (,).

## TFCHK-003w

(warning) Technology file %s has a syntax error. (line %d)

### Description

This warning message occurs when the technology file contains a syntax error on or near the specified line. The following cases will cause this warning: 1) missing comma at the end of line in a table value list.

The following is an example of the error message:

```
Warning : Technology file new.tf has a syntax error. (line 342)
 (TFCHK-003w)
```

### What Next

Check the specified line for syntax errors, such as missing commas, and correct it.

## TFCHK-004

(error) Line exceeds %d characters. (line %d)

### Description

This error message occurs when the specified line in the technology file contains more than the maximum number of characters allowed.

The following is an example of the error message:

```
Error : Line exceeds 1024 characters. (line 342) (TFCHK-004)
```

### What Next

Reduce the number of characters on the specified line and then rerun the command.

## TFCHK-005

(error) Line contains nested comments. (line %d)

### Description

This error message occurs when the technology file contains a nested comment starting on the specified line. Nested comments are not allowed in the technology file.

The following is an example of a nested comment and the resulting error message.

```
Line 10:
Line 11: ContactCode "via1Fat" {
Line 12:     contactCodeNumber = 12
```

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```
Line 13: /*  isDefaultContact = 1 /* this is a nested comment */ */
Line 14:     isFatContact =1
Line 15: }
Line 16:

Error : Line contains nested comments. (line 13) (TFCHK-005)
```

### What Next

Remove the nested comment on the specified line and then rerun the command.

---

## TFCHK-006

(error) Line contains an open-ended comment. (line %d)

### Description

This error message occurs when the technology file contains an open-ended comment. If a comment begins with the /* string, it must end with the */ string. The open-ended comment is assumed to end on the specified line.

The example below shows an open-ended comment and the resulting error message.

```
Line 10:
Line 11: ContactCode "via1Fat" {
Line 12:     contactCodeNumber = 12
Line 13:     isDefaultContact = 1 /* this is an open-ended comment
Line 14:     isFatContact =1
Line 15: }
Line 16:

Error : Line contains an open-ended comment. (line 16) (TFCHK-006)
```

### What Next

Add the */ string to the end of the comment and then rerun the command.

---

## TFCHK-007

(warning) String has an unmatched quotation mark. (line %d)

### Description

This warning message occurs when the technology file contains unmatched quotation marks on the specified line. All quoted strings must begin and end with quotation marks ("). If unmatched quotation marks exist, the characters between the quotation marks and the next white space are interpreted as a string.

The following shows an example of unmatched quotation marks and the resulting warning message.

Feedback

```
Line 10:
Line 11: ContactCode "via1Fat {
Line 12:     contactCodeNumber = 12
Line 13:     isFatContact =1
Line 14: }
Line 15:

Warning : String has an unmatched quotation mark. (line 11) (TFCHK-007)
```

**What Next**

This is a only a warning message. No action is required.

However, if the result is not what you intended, you can add the closing quotation marks (") to the end of the string on the specified line and then rerun the command.

---

## TFCHK-008

(error) Section name '%s' is not a recognized section. (line %d)

**Description**

This error message occurs when the technology file contains an invalid section name on the specified line.

The following is an example of the error message:

```
Error : Section name 'CantactCade' is not a recognized section.
(line 312) (TFCHK-008)
```

**What Next**

Check the spelling of the section name, and either correct or remove the invalid section. After making your changes, rerun the command.

---

## TFCHK-009

(error) %s '%s' contains the unrecognized attribute '%s'. (line %d)

**Description**

This error message occurs when the specified section contains an unrecognized attribute name on the specified line of the technology file. For generalized metal to cut spacing rules in the DesignRule section, these three spacingTbl attributes are legal: diffNetXMinSpacingTbl, diffNetYMinSpacingTbl, and diffNetCornerMinSpacingTbl. Others such as same*MinSpacingTbl, diffSeg*MinSpacingTbl, and diffNetCenterMinSpacingTbl are illegal.

Feedback

The following is an example of the error message:

```
Error : ContactCode 'via1' contains the unrecognized attribute
 'isDefCnt'. (line 323) (TFCHK-009)
```

### What Next

Check the spelling of the attribute and either correct the spelling or remove the attribute from the specified section.

---

## TFCHK-009w

(warning) %s '%s' attribute '%s' is unrecognized and ignored. (line %d)

### Description

This error message occurs when the specified section contains an unrecognized attribute name on the specified line of the technology file. These attributes are not supported and so ignored in ICC/Milkyway but may be supported in other databases. These include: routingDirection routingOffset

The following is an example of the error message:

```
Warning: Layer 'M1' attribute 'routingDirection' is unrecognized and
ignored. (line 113) (TFCHK-009w)
```

### What Next

These unrecognized attributes are ignored just like they are not in the tech file. So you don't need to do anything.

---

## TFCHK-010

(informantion) %s '%s' contains the obsolete attribute '%s'. (line %d)

### Description

This information message occurs when the specified section contains an obsolete attribute. Obsolete attributes are ignored.

The *preferredDirection* attribute in the Layer section is obsolete and not valid in Astro. To specify preferred directions, use the *axgDefineWireTracks* scheme command instead of specifying the directions in the technology file.

The following is an example of the information message:

```
Information : Layer 'VIA12' contains the obsolete attribute
 'preferredDirection'.
(line 721) (TFCHK-010)
```

Feedback

### What Next

This is only a informational message. No action is required.

However, it is best practice to remove the obsolete attribute from the specified section and then run the command again.

---

## TFCHK-011

(error) %s is missing the attribute '%s'. (line %d)

### Description

This error message occurs when the specified section is missing a required attribute.

The following is an example of the error message:

```
Error : ContactCode is missing the attribute 'name'. (line 688)
 (TFCHK-011)
```

### What Next

Add the required attribute to the specified section and rerun the command.

---

## TFCHK-012

(warning) %s is missing the attribute '%s'. (line %d)

### Description

This warning message occurs when the specified section is missing an important attribute. If the attribute is not specified in the technology file, the default value is 0 for number attributes, or an empty string for string attributes.

Note: If the "minArea" attribute of a routing layer is missing, it will cause minimum area design rule violations in Astro.

For enclosed via minEdgeLength rule, if you specify enclosedViaMinEdgeLength, you must also specify a value for minEdgeLength that is within (0, enclosedViaMinEdgeLength) and a positive value for enclosedViaAdjacentEdgeLength.

For direction based minSpacing rule, Both xMinSpacing and yMinSpacing should be specified and the value should not be smaller than minSpacing.

In DesignRule section, minSpacing, minEnclosure and stackable should be specified. Otherwise, if none is specified, the whole section will not be dumped into tech file.

In DesignRule section, if 2-cut based rule is defined and orthoSpacingExcludeCornerTbl is specified, cornerMinSpacing should be specified.

Feedback

The following is an example of the warning message:

```
Warning : Technology section is missing the attribute 'currentPrecision'.
(line 19) (TFCHK-012)
```

**What Next**

If you do not want to use the default value for the attribute, add the attribute with the desired value to the specified section in the technology file and rerun the command.

---

## TFCHK-013

(error) %s '%s' is missing the attribute '%s'. (line %d)

**Description**

This error message occurs when the specified object of the specified section is missing a required attribute.

The following is an example of the error message:

```
Error : Default metal layer 'METAL5' is missing the attribute
 'minSpacing'.
(line 426) (TFCHK-013)
```

**What Next**

Add the required attribute to the specified section and then run the command again.

---

## TFCHK-014

(warning) %s '%s' is missing the attribute '%s'. (line %d)

**Description**

This warning message occurs when the specified object of the specified section is missing an important optional attribute, or partial attributes of a rule are specified. If the attribute is not specified in the technology file, the attribute has a default value of 0 or an empty string. In the case of a table attribute, if the associated table size attribute is not defined before the table attribute, this message may also occur.

In generalized cut spacing rule with parallel length, if minSpacingXParallelLengthThresholdTbl or minSpacingYParallelLengthThresholdTbl is defined, orthoSpacingExcludeCornerTbl[i][j] should be defined as 1. If missing, the corner will be checked by x/y spacing.

For fat metal via enclosure rule, 1) fatWireViaEnclosureTbl should be defined if fatWireViaEncCutNameTbl is defined. 2)

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fatWireViaEnclosureTbl/fatWireViaEnclosureOrthoTbl should be defined if fatWireViaEncSparseMinSpacingThresholdTbl is defined.

The following is an example of the warning message:

```
Warning : ContactCode 'RM6_M5_ND' is missing the attribute
\'unitMaxResistance'. (line 758) (TFCHK-014)
```

**What Next**

This is only a warning message. No action is required.

However, if you do not want to the use the default for the attribute, add the attribute with the desired value to the specified section of the technology file and then rerun the command.

---

## TFCHK-015

(error) %s '%s' attribute '%s' is assigned index %d, overriding %s '%s' previously assigned to the same index. (line %d)

**Description**

This error message occurs when the specified object of the specified section does not have the required unique index. The index is the same index as a previous section of the same type.

The Color, Layer, LayerDataType, ContactCode, and FringeCap sections must have unique index numbers. The index numbers must be unique among all sections of the same type. Sections of different types can have the same index.

The following example uses an index number more than once, and shows the resulting error message:

```
Line 30: ContactCode "via1" {
 Line 31:     contactCodeNumber = 1
 ...
 Line 50: ContactCode "via2" {
 Line 51:     contactCodeNumber = 1
 Line 52:
 ...

 Error : ContactCode 'via2' attribute 'contactCodeNumber' is assigned
 index 1, overriding ContactCode 'via1' previously assigned to the same
 index. (line 51) (TFCHK-015)
```

**What Next**

Change the index number on the specified section to a unique index number and then run the command again.

Feedback

## TFCHK-016

(error) %s attribute '%s' is assigned index %d, overriding the %s previously assigned to the same index. (line %d)

### Description

This error message occurs when the specified section does not have the required unique index. The index is the same index as a previous section of the same type.

The Color, Layer, LayerDataType, ContactCode, and FringeCap sections must have unique index numbers. The index numbers must be unique among all sections of the same type. Sections of different types can have the same index.

The following example uses an index number more than once, and shows the resulting error message.

```
Line 130: FringeCap 4 {
 Line 131:     layer1 = "M2"
 Line 132:     layer2 = "M2"
 ...
 Line 150: FringeCap 4 {
 Line 151:     layer1 = "M2"
 Line 152:     layer2 = "M3"
 ...

 Error : FringeCap attribute 'number' is assigned index 4, overriding the
 FringeCap previously assigned to the same index. (line 150) (TFCHK-016)
```

### What Next

Change the index number on the specified section to a unique index number and then run the command again.

## TFCHK-017

(error) %s '%s' attribute '%s' is assigned an out-of-range index %s. (line %d)

### Description

This error message occurs when the specified object of the specified section has an index that is either too small or too large.

The following table lists the valid ranges by section type:

```
Section Type            Valid Range
-------------------------------------
Color                   0 through 63
Layer                   1 through 255
LayerDataType           0 through 255
```

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```
ContactCode             1 through 255
FringeCap               1 or larger
```

The following is an example of an index that is too large and the error message that is generated.

```
Line 400: ContactCode "via2" {
 Line 401:     contactCodeNumber = 300
 ...

 Error : ContactCode 'via2' attribute 'contactCodeNumber' is assigned
 an out-of-range index 300. (line 401) (TFCHK-017)
```

### What Next

Change the index of the specified section index to a valid integer within the range specified in the table above, and then run the command again.

---

## TFCHK-017w

(warning) %s '%s' attribute '%s' is assigned an out-of-range index %s. (line %d)

### Description

This warning message occurs when the specified object of the specified section has an index that is either too small or too large. In ContactCode section, contactCodeNumber should be <256 in pre-2010.12 releases. In the Tech file, such contactCode's will be ignored when the variable suppress_TFCHK-017 is set to true. The following table lists the valid ranges by section type:

```
Section Type            Valid Range
-------------------------------------
Color                   0 through 63
Layer                   1 through 255
LayerDataType           0 through 255
ContactCode             1 through 255
FringeCap               1 or larger
```

The following is an example of an index that is too large and the message that is generated.

```
Line 400: ContactCode "via2" {
 Line 401:     contactCodeNumber = 300
 ...

 Warning : ContactCode 'via2' attribute 'contactCodeNumber' is assigned
 an out-of-range index 300. (line 401) (TFCHK-017w)
```

Feedback

### What Next

Change the index of the specified section index to a valid integer within the range specified in the table above, and then run the command again. Or simply ignore such a contactCode.

---

## TFCHK-018

(error) %s attribute '%s' is assigned an out-of-range index %s. (line %d)

### Description

This error message occurs when the specified section has an index that is either too small or too large.

The following table lists the valid ranges by section type:

```
Section Type            Valid Range
-------------------------------------
Color                   0 through 63
Layer                   1 through 255
LayerDataType           0 through 255
ContactCode             1 through 255
FringeCap               1 or larger
```

The following is an example of an index that is too small and the resulting error message.

```
Line 500: FringeCap -1 {
 Line 501:     layer1 = "M2"
 Line 502:     layer2 = "M3"
 ...

 Error : FringeCap attribute 'number' an out-of-range index -1. (line
 500)
 (TFCHK-018)
```

### What Next

Change the index of the specified section to a valid integer within the range specified above in the table above, and then run the command again.

---

## TFCHK-019

(warning) %s '%s' index attribute is reassigned from %d to %d. (line %d)

### Description

This warning message occurs when the index for the specified object of the specified section is reassigned to another index number. The reassignment occurs when the

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section's index is specified multiple times with different index numbers, or when another section has the same name but a different index number.

The following example shows the same index specified with two different index numbers and the resulting warning message.

```
Line 301: ContactCode "via12" {
 Line 202:     contactCodeNumber = 2
 ...
 Line 322:     contactCodeNumber = 3
 Line 323: }

 Warning : ContactCode 'via12' index attribute is reassigned from 2 to 3.
 (line 222) (TFCHK-019)
```

**What Next**

This is only a warning message. No action is required.

However, it is best practice to determine the cause of the index number reassignment. If the index is specified multiple times, you can remove the extra specifications. If another section exists with the same name but with a different index number, you can decide which section is correct and remove the incorrect section. After making the changes, run the command again.

---

## TFCHK-020

(warning) %s index attribute is reassigned from %d to %d. (line %d)

**Description**

This warning message occurs when the index for the specified section is reassigned to another index number. Reassignment occurs when the section's index is specified multiple times with different index numbers, or when another section has the same name but a different index number.

The following example shows an index reassignment and the resulting warning message:

```
Line 401: FringeCap 2 {
 Line 402:     number = 3
 ...
 Line 420: }

 Warning : FringeCap index attribute is reassigned from 2 to 3.
 (line 402) (TFCHK-020)
```

**What Next**

This is only a warning message. No action is required.

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However, it is best practice to determine the cause of the index number reassignment. If the index is specified multiple times, you can remove the extra specifications. If another section exists with the same name but with different index number, you can decide which section is correct and remove the incorrect section. After making the changes, run the command again.

## TFCHK-021

(error) %s '%s' attribute '%s' is assigned an invalid integer %s. (line %d)

### Description

This error message occurs when the specified section has a non-integer value assigned to the specified attribute, or an out-of-range integer value assigned to the specified attribute.

The range of the valid layer number for different objects are: DesignRule : The layers can be user layer that layer number is in range of [1, 187], or system layers. ContactCode: The layers can be user layer that layer number is in range of [1, 187] FringeCap : The layers can be user layer that layer number is in range of [1, 187] SlotRule : The layers can be user layer that layer number is in range of [1, 187] DensityRule: The layers can be user layer that layer number is in range of [1, 187]

The following is an example of the error message:

```
Error : Layer 'METAL1' attribute 'layerNumber' is assigned an invalid
integer 2.4. (line 46) (TFCHK-021)
```

The following are other examples of the error message, 1. when contactCode upperLayer is assigned an out-of-range layer that layer number is larger than 187 2. when onGrid is assigned an out-of-range value 2

```
Error: ContactCode 'VIA34' attribute 'upperLayer' is assigned an invalid
integer 334. (line 1338) (TFCHK-021)

Error: Layer 'M1L' attribute 'onGrid' is assigned an invalid
integer 2. (line 208) (TFCHK-021)
```

### What Next

Change the attribute value of the specified section to an integer value.The value cannot contain a decimal point, such as 16.0. Or Change the value to an integer value within the valid range. Such as 'onGrid' attribute has valid value 0 or 1 but not 2. After changing the value, run the command again.

## TFCHK-022

(error) %s attribute '%s' is assigned an invalid integer %s. (line %d)

Feedback

### Description

This error message occurs when the specified section has a non-integer value or an integer that is larger than MAX_INT (2147483647) assigned to the specified attribute.

The following is an example of this error message:

```
Error : Technology attribute 'inductancePrecision' is assigned an invalid
integer 100.01. (line 6) (TFCHK-022)
```

### What Next

Change the attribute value of the specified section to an integer value. The value cannot contain a decimal point, such as 16.0. After changing the value, run the command again.

---

## TFCHK-023

(error) %s '%s' attribute '%s' is assigned an invalid boolean %s. (line %d)

### Description

This error message occurs when the specified section has a non-Boolean value assigned to the specified attribute. All boolean attributes must have a 0 or 1 value.

The following is an example of this error message:

```
Error : ContactCode 'via2' attribute 'isDefaultContact' is assigned an
 invalid
boolean 2.5. (line 368) (TFCHK-023)
```

### What Next

Change the attribute value of the specified section to a 0 or 1 and run the command again.

---

## TFCHK-024

(warning) %s '%s' attribute '%s' is assigned an invalid boolean %s. (line %d)

### Description

This warning message occurs when the specified section has a non-Boolean value assigned to the specified attribute. All Boolean attributes must have a 0 or 1 value, or else the nonzero values default to 1.

The following is an example of this warning message:

```
Warning : ContactCode 'via2' attribute 'isDefaultContact' is assigned
an invalid boolean 2.5. (line 368) (TFCHK-024)
```

Feedback

### What Next

This is only a warning message. No action is required.

However, if the result is not what you intended, change the attribute value of the specified section to a 0 or a 1 and then run the command again.

---

## TFCHK-025

(error) %s '%s' references layer '%s' which does not exist. (line %d)

### Description

This error message occurs when the specified section references another section that does not exist. The reference occurs on the specified line.

The following is an example of this error message:

```
Error : ContactCode 'M3_M2WEST_ND' references layer 'METAL2' which does
not exist. (line 680) (TFCHK-025)
```

### What Next

Define the missing section, or redirect the reference to an existing section, or remove the reference to the missing section. After making your changes, run the command again.

---

## TFCHK-026

(warning) %s '%s' references layer '%s' which does not exist. (line %d)

### Description

This message indicates that the specified section references another section that does not exist. The reference occurs on the specified line. The section will be ignored.

### What Next

Define the missing section or remove the reference.

### Examples

Warning : ContactCode 'M3_M2WEST_ND' references layer 'METAL2' which does not exist. (line 680) (TFCHK-026)

---

## TFCHK-027

(error) %s '%s' has a non-boolean pattern. (line %d)

Feedback

### Description

This error message occurs when the the pattern of the specified section is not a list of Boolean values. All patterns must be a list of Boolean values.

The following is an example of this error message:

```
Error : LineStyle 'line3' has a non-Boolean pattern. (line 94)
 (TFCHK-027)
```

### What Next

Change the pattern of the specified section to a list of Boolean values and then run the command again.

---

## TFCHK-028

(warning) %s '%s' has a non-boolean pattern. (line %d)

### Description

This warning message occurs when the pattern of the specified section is not a list of Boolean values. All patterns must be a list of Boolean values. If any value is not a Boolean, it is automatically changed to 1.

The following is an example of this warning message:

```
Warning : LineStyle 'line3' has a non-Boolean pattern. (line 94)
 (TFCHK-28)
```

### What Next

This is only a warning message. No action is required.

However, if the result is not what you intended, you can change the pattern of the specified section to a list of Boolean values and then run the command again.

---

## TFCHK-029

(error) %s '%s' has a width and height that is not consistent with the pattern length. (line %d)

### Description

This error message occurs when the pattern length of the specified section does not match the size specified by the pattern width and height. The number of elements in the pattern list must equal the product of the pattern width and height.

Feedback

The following is an example of this error message:

```
Error : LineStyle 'line2' has a width and height that is not
consistent with pattern length. (line 89) (TFCHK-029)
```

### What Next

Change the width and height attribute values or add or remove Boolean values to the pattern list. After making your changes, run the command again.

---

## TFCHK-030

(warning) %s '%s' has a width and height that is not consistent with the pattern length. (line %d)

### Description

This warning message occurs when the pattern length of the specified section does not match the size specified by the pattern width and height. The number of elements in the pattern list must equal the product of the pattern width and height. The unexpected value of width or height are ignored, althought they are stored in the database.

The following is an example of this warning message:

```
Warning : LineStyle 'line2' has a width and height that is not consistent
with pattern length. (line 89) (TFCHK-030)
```

### What Next

This is a warning message only. No action is required.

However, if the result is not what you intended, change the width and height attribute values, or change the pattern list by adding or removing Boolean values, and then rerun the command.

---

## TFCHK-031

(error) %s '%s' attribute '%s' is assigned an out-of-range size %s. (line %d)

### Description

This error message occurs when the specified section has a invalid table size. The resistance model and capacitance table size must be between 1 and 16 inclusive.

The following is an example of this error message:

```
Error : CapTable 'poly_C_TOP_GP' has an out-of-range size 17
assigned to the attribute 'wireWidthSize'. (line 3231) (TFCHK-031)
```

Feedback

### What Next

Change the table size of the specified section to a value between 1 and 16 inclusive and run the command again.

---

## TFCHK-032

(warning) %s '%s' attribute '%s' is assigned an out-of-range size %s. (line %d)

### Description

This warning message occurs when the specified section has an invalid table size. The resistance model and capacitance table size must be between 1 and 16 inclusive. And some table sizes or dimensions have maximum 8 or 16. During open_mw_lib, any values in such tables that are beyond the max sizes will be cut out. For example, in 2010.03 releases, cutTblSize is expanded to 16 from 8; therefore, if you use an earlier version to load Milkyway library or design that are created using 2010.03 releases or later, the values in the related cut tables beyond 8th indexes wil be garbage.

The unexpected value of the table size is replaced by 1 if the value is less than 1, or by 16 if the value is more than 16. When enclosureTbl/minCutsTbl are defined, the converted fatTblThreshold should not have >32 values due to limitation on applications. In this case, only the first 32 values with fatTblDimension = 32 are used.

The following is an example of the warning message:

```
Warning : CapTable 'poly_C_TOP_GP' has an out-of-range size 17
assigned to the attribute 'wireWidthSize'. (line 3231) (TFCHK-032)
```

### What Next

This is only a warning message. No action is required.

However, if the result is not what you intended, change the table size of the specified section to a value between 1 and 16 inclusive and run the command again.

---

## TFCHK-033

(warning) %s '%s' attribute '%s' is assigned a list with an incorrect number of elements. (line %d)

### Description

This warning message occurs when the array attribute of the specified section does not have the correct number of elements, or the values are not separated by comma. minSpacingUpperEncCheckTbl table size should be cut1TblSize* cut2TblSize if used in cut to cut spacing rule and cutTblSize if used in cut to metal spacing rule.

Feedback

The following is an example of the warning message:

```
Warning : Layer 'VIA12' has an incorrect number of elements listed
for array attribute 'fatTblExtensionMinCuts'. (line 753) (TFCHK-033)
```

For layer sections, the following array attributes must contain the indicated number of elements, based on the *fatTblDimension* attribute:

```
Attribute:                    Valid Number Of Elements:
    ------------------------------------------------------------
    fatTblThreshold              fatTblDimension
    fatTblSpacing                fatTblDimension x fatTblDimension
    fatTblParallelLength         fatTblDimension
    fatTblExtensionRange         fatTblDimension
    fatTblMinEnclosedArea        fatTblDimension
    fatTblFatContactNumber       fatTblDimension
    fatTblFatContactMinCuts      fatTblDimension
    fatTblExtensionContactNumber fatTblDimension
    fatTblExtensionMinCuts       fatTblDimension
```

For capacitance table sections, the array attribute must have the following number of elements, based on the wire width and spacing size:

```
Attribute      Valid Number Of Elements
    ---------------------------------------------
    wireWidth      wireWidthSize
    wireSpacing    wireSpacingSize
    capValue       wireWidthSize x wireSpacingSize
```

For resistance models, the array attribute must have the following number of elements shown below, based on the size:

```
Attribute      Valid Number Of Elements
    ---------------------------------------------
    wireWidth      size
    tempCoeff      size
    minRes         size
    nomRes         size
    maxRes         size
```

**What Next**

This is only a warning message. No action is required.

However, if the result is not what you intended, you can change the table dimension(s) or add or remove elements from the specified array attribute.

---

## TFCHK-034

(error) Floating-point attribute '%s' is assigned %s which is too large to store in the database. (line %d)

Feedback

### Description

This error message occurs when the specified floating-point value is too large to store in the database. When stored in the database, the floating-point number is scaled by multiplying the floating-point number by the corresponding unit precision. If an overflow results, then this error message is generated.

The following is an example of the error message:

```
Error : Floating-point attribute 'unitMinResistance' is assigned
8e06 which is too large to store in the database. (line 78) (TFCHK-034)
```

### What Next

Decrease the corresponding unit precision or reduce the size of the floating-point number and then run the command again.

---

## TFCHK-035

(warning) Floating-point attribute '%s' is assigned %s which is too precise and will be truncated to %g. (line %d)

### Description

This warning message occurs when the specified floating-point number cannot be completely stored in the database and must be rounded off.

When storing floating-point numbers in the database, the number is scaled by multiplying the floating-point number by the corresponding unit precision. Digits to the right of the decimal point are dropped. When nonzero digits are dropped, this message appears.

The following is an example of the warning message:

```
Warning : Floating-point attribute 'lowerLayerEncHeight' is
assigned 0.35 which is too precise and will be truncated to 0.3.
(line 346) (TFCHK-035)
```

### What Next

This is a warning message only. No action is required.

However, if the result is not what you intended, you can increase the corresponding unit precision or reduce the number of digits in the floating-point number, and then run the command again.

If floating point rounding is insignificant and does not affect the results, it is not necessary to modify the precision.

Feedback

If rounding is severe and affects the results, modify the precision in the main and ALL reference libraries. This may involve reloading the Cell Library Format (CLF) timing information in the reference libraries.

## TFCHK-036

(information) Technology attribute '%s' is assigned %d which causes several %s values to be truncated. The recommended %s precision is %d. (line %d)

### Description

This information message occurs when the specified unit precision is not precise enough and causes some floating-point numbers to be truncated. The recommended precision is the minimum required precision to completely store the floating-point numbers to the database.

The following is an example of the information message:

```
Information : Technology attribute 'lengthPrecision' is
assigned 10 which causes several length values to be truncated.
The recommended length precision is 100. (line 12) (TFCHK-036)
```

### What Next

This is only an informational message. No action is required.

However, it is considered best practice to increase the unit precision to the specified recommended precision.

If floating-point rounding is insignificant and does not affect the results, it is not necessary to modify the precision. If rounding is severe and does affect the results, modify the precision modified in the main and ALL reference libraries. This may require reloading the CLF timing information in the reference libraries.

## TFCHK-037

(error) Technology attribute '%s' is assigned an invalid unit name '%s'. (line %d)

### Description

This error message occurs when the Technology section contains an invalid unit name.

The following table lists the valid unit names for each attribute. The unit names are not case-sensitive.

```
Attribute Name          Valid Unit Names
    ------------------------------------------
    unitLengthName          micron, mil
    unitTimeName            fs, ps, ns, us, ms, s
```

Feedback

```
    unitCapacitanceName     ff, pf, nf, uf, mf, f
    unitResistanceName      mohm, ohm, kohm, Mohm
    unitInductanceName      fh, ph, nh, uh, mh, h
    unitPowerName           fw, pw, nw, uw, mw, w
    unitVoltageName         fV, pV, nV, uV, mV, V
    unitCurrentName         fA, pA, nA, uA, mA, A
```

The following is an example of the error message:

```
Error : Technology attribute 'unitResistanceName' is assigned
an invalid unit name 'pohm'. (line 13) (TFCHK-037)
```

### What Next

Change the attribute value to a valid unit name and then run the command again.

---

## TFCHK-038

(warning) Technology attribute '%s' is assigned an invalid unit name '%s'. (line %d)

### Description

This warning message occurs when the Technology section contains an invalid unit name. The invalid unit names are replaced with application-specific default unit names.

The following table lists the valid unit names for each attribute. The unit names are case-sensitive.

```
Attribute Name          Valid Unit Names
    ------------------------------------------
    unitLengthName          micron, mil
    unitTimeName            fs, ps, ns, us, ms, s
    unitCapacitanceName     ff, pf, nf, uf, mf, f
    unitResistanceName      mohm, ohm, kohm, Mohm
    unitInductanceName      fh, ph, nh, uh, mh, h
    unitPowerName           fw, pw, nw, uw, mw, w
    unitVoltageName         fV, pV, nV, uV, mV, V
    unitCurrentName         fA, pA, nA, uA, mA, A
```

The following is an example of the warning message:

```
Warning : Technology attribute 'unitPowerName' is assigned an
invalid unit name 'V'. (line 14) (TFCHK-038)
```

### What Next

This is only a warning message. No action is required.

However, if the result is not what you intended, you can change the attribute value to a valid unit name and then run the command again.

Feedback

## TFCHK-039

(warning) Technology attribute '%s' is assigned an invalid mode %s. (line %d)

### Description

This warning message occurs when the Technology section contains an invalid mode. If an invalid mode is less than 0, the default mode of 0 is used. Application-specific defaults are used for all other invalid modes.

The following tables lists the valid modes by attribute name.

```
Attribute Name          Valid Modes
    ------------------------------------------
    fatTblSpacingMode       0, 1
    fatWireExtensionMode    0, 1, 2, 3
    minEdgeMode             0, 1, 2
```

The following is an example of the warning message:

```
Warning : Technology attribute 'minEdgeMode' is assigned an
invalid mode -1. (line 26) (TFCHK-039)
```

### What Next

This is only a warning message. No action is required.

However, if the result is not what you intended, you can change the attribute value to a valid mode and run the command again.

## TFCHK-040

(error) Technology section is missing entirely. (line %d)

### Description

This error message occurs when the Technology section is missing from the technology file.

The following is an example of the error message:

```
Error : Technology section is missing entirely. (line 1) (TFCHK-040)
```

### What Next

Add the Technology section to the technology file and run the command again.

## TFCHK-041

(error) PrimaryColor attribute '%s' is assigned an out-of-range color intensity %s. (line %d)

Feedback

### Description

This error message occurs when the PrimaryColor section contains an invalid color intensity value assigned to the specified attribute. Color intensity values must be between 0 and 255 inclusive.

The following shows an example of the error message:

```
Error : PrimaryColor attribute 'mediumGreen' is assigned an
out-of-range color intensity 256. (line 35) (TFCHK-041)
```

### What Next

Change the attribute value to a valid color intensity number from 0 through 255, and then run the command again.

---

## TFCHK-042

(warning) PrimaryColor attribute '%s' is assigned an out-of-range color intensity %s. (line %d)

### Description

This warning message occurs when the PrimaryColor section has contains an invalid color intensity value assigned to the specified attribute. Color intensity values must be between 0 and 255 inclusive. Invalid values that are less than zero default to 0. Invalid values that are more than 255 default to 255.

The following is an example of this warning message:

```
Warning : PrimaryColor attribute 'mediumGreen' is assigned
an out-of-range color intensity 256. (line 35) (TFCHK-042)
```

### What Next

Change the attribute value to a valid color intensity number from 0 to 255 and then run the command again.

---

## TFCHK-043

(error) Color '%s' attribute '%s' is assigned an out-of-range color intensity %s. (line %d)

### Description

This error message occurs when the specified Color section contains an invalid color intensity value assigned to the specified attribute. Color intensity values must be between 0 and 255 inclusive.

Feedback

The following is an example of this error message:

```
Error : Color 'oblue' attribute 'blueIntensity' is assigned
an out-of-range color intensity 256. (line 52) (TFCHK-043)
```

### What Next

Change the attribute value to a valid color intensity number from 0 through 255 and then run the command again.

---

## TFCHK-044

(warning) Color '%s' attribute '%s' is assigned an out-of-range color intensity %s. (line %d)

### Description

This warning message occurs when the specified Color section contains an invalid color intensity value assigned to the specified attribute. Color intensity values must be between 0 and 255 inclusive. Invalid values less than 0 use the default value of 0. Invalid values more than 255 use the default value of 255.

The following is an example of the warning message:

```
Warning : Color 'oblue' attribute 'blueIntensity' is assigned
an out-of-range color intensity 256. (line 52) (TFCHK-044)
```

### What Next

Change the attribute value to a valid color intensity number from 0 through 255 and then run the command again.

---

## TFCHK-045

(information) Non-via layer '%s' has the via layer attribute '%s'. (line %d)

### Description

This information message indicates that the specified Layer section does not define a via or poly contact layer, but it contains a via layer attribute, which is ignored.

The following are via and poly contact layer attributes:

```
cornerMinSpacing
maxStackLevel
fatTblExtensionContactNumber
fatTblExtensionMinCuts
fatTblFatContactNumber
fatTblFatContactMinCuts
maxNumAdjacentCut
adjacentCutRange
```

Feedback

```
enclosedCutNumNeighbor
enclosedCutNeighborRange
enclosedCutMinSpacing
enclosedCutToNeighborMinSpacing
```

The following is an example of the information message:

```
Information : Non-via layer 'METAL2' has the via layer attribute
\'enclosedCutToNeighborMinSpacing'. (line 294) (TFCHK-045)
```

**What Next**

This is only an informational message. No action is required.

However, if the result is not what you intended, you can either remove the specified attribute, or change the mask name of the layer to a via mask name, such as polyCont, via1, via2, or via3.

---

## TFCHK-046

(information) Non-metal layer '%s' has the metal layer attribute '%s'. (line %d)

**Description**

This information message occurs when the specified Layer section is not a metal or poly layer, but it contains a metal layer attribute that is ignored.

The following are metal and poly layer attributes:

```
sameNetMinSpacing
stubThreshold
stubSpacing
maxLength
maxNumMinEdge
minEdgeLength
minArea
minEnclosedArea
maxWidth
fatContactThreshold
fatWireThreshold
fatThinMinSpacing
fatFatMinSpacing
fatWireExtensionRange
fatTblSpacing
fatTblParallelLength
fatTblMinEnclosedArea
```

The following is an example of the information message:

```
Information : Non-metal lyera 'VIA12' has the metal layer attribute
\'fatTblMinEnclosedArea'. (line 740) (TFCHK-046)
```

Feedback

### What Next

This is only an informational message. No action is required.

However, if the result is not what you intended, you can either remove the specified attribute or change the make name of the layer to a metal mask name, such as poly, metal1, metal2, or metal3.

---

## TFCHK-047

(warning) Nondefault metal layer '%s' has the default-only attribute '%s'.

### Description

This warning message occurs when the specified Layer section is not a default metal layer, but it contains a default metal layer attribute.

Multiple layers can be defined for each metal layer, but only one should be set as the default layer. The default layer contains the following attributes:

```
pitch
defaultWidth
minWidth
minSpacing
```

The following is an example of the error message:

```
Warning : Nondefault metal layer 'METAL1Extra' has the default-only
attribute 'minSpacing'. (TFCHK-047)
```

### What Next

This is only a warning message. No action is required.

However, it is best practice to remove the specified attribute.

---

## TFCHK-048

(warning) Layer '%s' has a pitch %g that is less than the minimum spacing and width sum %g. (line %d)

### Description

This warning message occurs when the specified Layer section contains an invalid pitch, minimum spacing or minimum width. The pitch must be greater than or equal to the sum of the layer's minimum spacing and minimum width.

Feedback

The following is an example of the error message:

```
Warning : Layer 'METAL1' has a pitch 0.2 that is less than the
minimum spacing and width sum 0.24. (line 239) (TFCHK-048)
```

### What Next

This is a warning message only. No action is required.

However, if the result is not what you intended, change the pitch, minimum spacing, or minimum width to meet the requirement that the pitch be greater than or equal to the sum of the layer's minimum spacing and minimum width.

---

## TFCHK-049

(warning) Layer '%s' has a pitch %g that does not match the recommended wire-to-via pitch %s.

### Description

This warning message occurs when the pitch of the specified layer does not match the recommended pitch. The wire-to-via pitch is the recommended pitch and provides the best wire track resources for routing.

The wire-to-wire pitch is more aggressive, the via-to-via is more conservative and neither are recommended. The metal layer pitch must match the wire-to-via pitch in the known routing direction, which is determined by the wire track directions of the unit tile.

The horizontal wire-to-via pitch is calculated as follows:

```
(min spacing) + (min width)/2 + (via metal width)/2

The vertical wire-to-via pitch is calculated as follows:

    (min spacing) + (min width)/2 + (via metal height)/2
```

The via metal width and height are based on the larger of the lower and upper default contact codes.

If the layer routing directions are unknown, then two pitches may be recommended. One pitch is recommended for the horizontal wire-to-via pitch, and one pitch for the vertical wire-to-via pitch. You can chose either pitch according to your design requirements.

The following is an example of the warning message:

```
Warning : Layer 'metal3' has a pitch 1.9 that does not match
the recommended wire-to-via pitch 1.3. (TFCHK-049)
```

### What Next

This is only a warning message. No action is required.

Feedback

You can leave the pitch of the specified layer at the current setting, or you can change the pitch of the layer to the recommended pitch.

---

## TFCHK-050

(warning) Layer '%s' has a pitch %g that does not match the doubled pitch %g or tripled pitch %g.

### Description

This warning message occurs when the pitch of the specified layer is not double or triple the pitch of the lower metal layer with the same routing direction. Doubling or tripling the pitch value, maximizes routing resources and minimizes overlap with other metal layers.

If you do not know the layer routing directions, then double or triple the pitch of the second lower metal layer's pitch. This assumes that the routing directions alternate between layers.

The following is an example of the warning message:

```
Warning : Layer 'metal3' has a pitch 1.9 that does not match
the doubled pitch 3.8 or tripled pitch 5.7. (TFCHK-050)
```

### What Next

This is only a warning message. No action is required.

You can either leave the pitch of the specified layer unchanged, or change the pitch to the doubled or tripled pitch and run the command again.

---

## TFCHK-051

(error) Layer is missing with mask name '%s'.

### Description

This error message occurs when a Layer section with the specified mask name is missing from the technology file. All metal and via layers up to the highest defined physical layer must be defined in the technology file.

For example, if only metal1, via1, metal2, via2, metal3, and metal4 are defined, then via3 is missing and the error message appears.

The following is an example of the error message:

```
Error : Layer is missing with mask name 'metal9'. (TFCHK-051)
```

### What Next

Define a Layer section with the specified mask name and then run the command again.

Feedback

## TFCHK-052

(error) Layer '%s' is missing the fat table attribute '%s'. (line %d)

### Description

This error message occurs when the specified Layer section contains a fat table that is missing the specified attribute.

The following is an example of the error message:

```
Error : Layer 'METAL2' is missing the fat table attribute
\'fatTblThreshold'. (line 197) (TFCHK-052)
```

### What Next

Add the missing attribute or remove the fat table and then run the command again.

## TFCHK-053

(error) %s '%s' attribute '%s' is assigned an out-of-range dimension %s. (line %d)

### Description

This error message occurs when the specified Layer section contains a fat table dimension that is too small or too large. The fat table dimension for fatTblDimension, cutTblSize, cut1TblSize and cut2TblSize must be between 1 and 16 inclusive. For all others, the fat table dimension must be between 1 and 8 inclusive.

The following is an example of the error message:

```
Error : Layer 'METAL3' attribute 'fatTblDimension' is assigned
an out-of-range dimension 17. (line 324) (TFCHK-053)
```

### What Next

Change the fat table dimension for the specified Layer section to a valid integer and then run the command again.

## TFCHK-054

(error) Layer '%s' has invalid fat table thresholds '%s'. (line %d)

### Description

This error message occurs when the specified Layer section contains an invalid fat table threshold list. The threshold list must meet the following requirements:

- The number of threshold values must match the table dimension.

Feedback

- All threshold values must be non-negative numbers and listed in ascending order.
- The first threshold value must between 0 and the default width inclusively.
- The second threshold value must be greater than the layer default width.

The following is an example of the error message:

```
Error : Layer 'METAL1' has invalid fat table thresholds
\'(0, 0, 1.505, 3.005, 4.505, 7.505)'. (line 225) (TFCHK-054)
```

**What Next**

Change the threshold list for the specified Layer section to meet the above requirements and then run the command again.

---

## TFCHK-055

(warning) Layer '%s' has rounded fat table thresholds '%s'. (line %d)

**Description**

This warning message occurs when the threshold list of the specified Layer section contains rounded values. It is best practice to avoid ending threshold values with a 0 digit in the most precise position. This practice minimizes the confusion between tools that interpret the thresholds as exclusive, rather than inclusive, lower bounds.

The following is an example of the warning message:

```
Warning : Layer 'METAL6' has rounded fat table thresholds
\'(0, 0.215, 1.5, 3.005, 4.51, 7.505)'. (line 464) (TFCHK-055)
```

**What Next**

This is only a warning message. No action is required.

However, you can either increase the rounded threshold value by one grid resolution value and run the command again, or you can leave the threshold value at the current setting.

You need also check the other layers for the same case.

---

## TFCHK-056

(warning) Layer '%s' has insignificant differences between fat table thresholds '%s'. (line %d)

Feedback

### Description

This warning message occurs when the threshold list of the specified Layer section contains insignificant differences between the threshold values. It is best practice to have thresholds differ by more than half of the minimum width and more than ten percent of the greater threshold. If they do not differ by these amounts, then the fat table can contain redundant data, be unnecessarily large, and increase router run-time dramatically.

The following is an example of the warning message:

```
Warning : Layer 'VIA4' has insignificant differences between
fat table thresholds '(0, 0.5, 0.55, 1.2, 2)'. (line 724) (TFCHK-056)
```

### What Next

This is only a warning message. No action is required.

However, you can check the fat tables for redundant data. If you find redundant data, then reduce the table dimension and the number of threshold values. After making your changes, rerun the command.

---

## TFCHK-057

(error) Layer '%s' has an asymmetric fat table spacing array. (line %d)

### Description

This error message occurs when the fat table spacing array of the specified Layer section is not symmetric. Spacing arrays are two-dimensional and must be symmetric about the diagonal that connects the upper-left and lower-right corners.

The following is an example of the error message:

```
Error : Layer 'VIA12' has an asymmetric fat table spacing array.
(line 738) (TFCHK-057)
```

### What Next

Check that the fat table dimension of the specified layer is correct and then modify the spacing table to make it symmetric. After making your changes, run the command again.

---

## TFCHK-058

(error) Layer '%s' has an invalid fat table contact number %d.

### Description

This error message occurs when the fat metal via rule or fat metal via extension rule in the specified Layer section contains an invalid contact code number.

Feedback

A valid contact Code number must correspond to a contact code defined in the technology file, and the contact code must be defined for the specified cut layer.

And due to router limitation, if an asymmetrical contact code, defined by geometry tables or cut array, is referenced in fatTblFatContactNumber or fatTblExtensionContactNumber table, the error message will also be issued.

The following is an example of the error message:

```
Error : Layer 'VIA12' has an invalid fat table contact number 20.
 (TFCHK-058)
```

**What Next**

Change the table contact code number or define the contact code for the specified Layer section and then run the command again.

**SEE AISO**

```
cmCheckLibrary(2)
cmCreateLib(2)
cmReplaceTech(2)
dbCreateLib(2)
dbReplaceTechFile(2)
```

---

## TFCHK-058w

(warning) Layer '%s' has an invalid fat table contact number %d.

**Description**

This warning message occurs when the fat metal via rule or fat metal via extension rule in the specified Layer section contains an invalid contact code number. This warning occurs under end user mode only.

A valid contact Code number must correspond to a contact code defined in the technology file, and the contact code must be defined for the specified cut layer.

And due to router limitation (unable to instantiate such a via), if an asymmetrical contact code, defined by geometry tables or cut array, is referenced in fatTblFatContactNumber or fatTblExtensionContactNumber table, this message will also be issued.

The following is an example of the warning message:

```
Warning : Layer 'VIA12' has an invalid fat table contact number 20.
 (TFCHK-058w)
```

**What Next**

Change the table contact code number or define the contact code for the specified Layer section and then run the command again.

Feedback

### SEE AISO

```
cmCheckLibrary(2)
cmCreateLib(2)
cmReplaceTech(2)
dbCreateLib(2)
dbReplaceTechFile(2)
```

---

## TFCHK-059

(error) Multiple default layers exists with mask name '%s'.

### Description

This error message occurs when there are multiple layer sections with the specified mask name that are marked as default. Each physical layer can have only one default layer.

The following is an example of the error message:

```
Error : Multiple default layers exists with mask name 'via9'.
(TFCHK-059)
```

### What Next

Check all layers with the specified mask name and mark only one as the default layer. After making your changes, run the command again.

---

## TFCHK-060

(warning) Multiple default layers exists with mask name '%s'.

### Description

This warning message occurs when there are multiple layer sections with the specified mask name that are marked as default. Each physical layer is allowed only one default layer. When multiple default layers exist, the default layer with the lowest layer number becomes the actual default layer.

The following is an example of the warning message:

```
Warning : Multiple default layers exists with mask name 'via9'.
(TFCHK-060)
```

### What Next

This is only a warning message. No action is required.

However, it is best practice to check all layers with the specified mask name, and mark only one as the default layer.

Feedback

## TFCHK-061

(error) Layer '%s' has a default width %g that is less than its minimum width %g. (line %d)

### Description

This error message occurs when the default width or minimum width of the specified Layer section is invalid. The default width of each layer must be greater than or equal to its minimum width.

The following is an example of the error message:

```
Error : Layer 'METAL6' has a default width 0.12 that is less
than its minimum width 0.14. (line 473) (TFCHK-061)
```

### What Next

Change the default or minimum width of the specified Layer section so that the default width is greater than or equal to the minimum width. After making your changes, run the command again.

## TFCHK-062

(information) Layer '%s' has a same-net spacing %g that is greater than its minimum spacing %g. (line %d)

### Description

This information message occurs when the same-net spacing is greater than minimum spacing of the specified Layer. Usually, the same-net minimum spacing of each layer should be less than or equal to its minimum spacing. When sameNetSpacing greater than minSpcaing, U-shaped metal spacing rule is triggered. It requires bigger spacing between a net and a neighbour net with small notch. this is a new design rule introduced in 65nm technology to prevent or eliminate small notches.

The following is an example of the information message:

```
Information : Layer 'METAL6' has a same-net spacing 0.16 that is
greater than its minimum spacing 0.14. (line 473) (TFCHK-062)
```

### What Next

This is only an informational message. No action is required.

However, it is best practice to ensure that the correct values are specified for the same-net spacing and the minimum spacing of the specified Layer section.

Feedback

## TFCHK-063

(error) Layer '%s' attribute '%s' has an invalid value %g.

### Description

This error message occurs when the value of the specified layer attribute is invalid.

Default metal layers must have nonzero, positive values for the following attributes:

```
pitch
defaultWidth
minWidth
minSpacing
```

For enclosed via minEdgeLength rule, if you specify enclosedViaMinEdgeLength, you must also specify a value for minEdgeLength that is within (0, enclosedViaMinEdgeLength).

The following is an example of the error message:

```
Error : Layer 'METAL5' attribute 'minSpacing' has an invalid value 0.
(TFCHK-063)
```

### What Next

Change the attribute value of the specified layer to a nonzero, positive value and then run the command again.

## TFCHK-064

(warning) Layer '%s' attribute '%s' has a value of %g.

### Description

This warning message occurs when the value of the specified layer attribute may not be optimal.

The value of the *minEnclosedArea* attribute for default metal layers should be greater than 0. Other values can affect the quality of the application results and can cause excessive design rule constraint violations during physical verification.

For a set of rule, completeness of rule parameters should be maintained:

For U-shape rule, if you specify uShapeMinLength or uShapeMinSpacing, you should also specify uShapeDepthThreshold and minSpacing. And uShapeMinSpacing > minSpacing.

For enclosed via minEdgeLength rule, if you specify enclosedViaMinEdgeLength, you must also specify a value for minEdgeLength that is within (0, enclosedViaMinEdgeLength) and a positive value for enclosedViaAdjacentEdgeLength.

Feedback

For nonpreferred direction rule, the following should be met: nonPreferredWidth > defaultWidth, and nonPreferredMaxLength > nonPreferredWidth-defaultWidth

For multiple minArea rule, you should specify minArea if you specify specialMinArea, and specialMinArea > minArea.

For stubThreshold associated rules, you should specify stubThreshold if you specify stubMinLength, stubToStubSpacing, stubSpacing or endOfLineSideKeepoutLength.

For wire jog corner to corner width rule, the following should be met: alignedJogWidth >= minWidth and unalignedJogDiagWidth >= minWidth

For legal width rule, defaultWidth and nonPreferredWidth should be specifies and their values are in LegalWidthTbl.

defaultWidth >= minWidth for all layers. For Metal layer, refer to TFCHK-061.

The following is an example of the warning message:

```
Warning : Layer 'metal3' attribute 'minEnclosedArea' has a
value of 0. (TFCHK-064)
```

**What Next**

This is only a warning message. No action is required.

However, it is best practice to check the attribute value of the specified layer, and then change the value to a value greater than 0, if necessary. After making your changes, run the command again.

---

## TFCHK-065

(warning) Tile '%s' has dimensions %g x %g that does not match the '%s' cell's dimensions %g x %g. (line %d)

**Description**

This warning message occurs when the dimensions of the specified Tile section are inconsistent with the PR (place and route) boundary dimensions of the corresponding tile cell.

The following is an example of the warning message:

```
Warning : Tile 'unit' has dimensions 1.9 x 11.4 that does not match
the 'unitTile' cell's dimensions 1.9 x 22.8. (line 59) (TFCHK-065)
```

**What Next**

This is only a warning message. No action is required.

Feedback

However, it is best practice to either change the dimension attributes of the specified Tile section or regenerate the tile cell, and then run the command again.

## TFCHK-066

(warning) Tile '%s' has a width %g that is not a multiple of the metal layer pitch %s.

### Description

This warning message occurs when the width of the specified tile is inconsistent with pitch of the metal layer. The width of each tile must match the pitch of the first vertical wire track layer in the tile cell.

If the tile cell does not exist or its wire tracks have not been created, then two suggested pitch values appear in the warning message. The suggested pitch values are derived from the metal1 and metal2 layer pitches.

The following is an example of the warning message:

```
Warning : Tile 'unit' has a width 0.32 that is not a multiple
of the metal layer pitch 0.2 or 0.28. (TFCHK-066)
```

### What Next

This is a warning message only. No action is required.

However, it is best practice to either change the width attribute of the corresponding tile section to the suggested pitch, or change the pitch of the first vertical wire track layer. After making your changes, run the command again.

## TFCHK-067

(warning) FringeCap %d attribute '%s' is assigned a non-metal, non-poly layer '%s'. (line %d)

### Description

This warning message occurs when the specified FringeCap section does is not assigned a metal or poly layer. Unless corrected, this inaccurate information is stored in the database.

The following is an example of the warning message:

```
Warning : FringeCap 3 attribute 'layer2' is assigned a non-metal,
non-poly layer 'VIA2'. (line 1626) (TFCHK-067)
```

### What Next

This is only a warning message. No action is required.

Feedback

However, it is best practice to change the specified attribute to a metal or poly layer and then run the command again.

## TFCHK-068

(error) ContactCode '%s' has non-distinct lower and upper layer names '%s' and '%s'. (line %d)

### Description

This error message occurs when the lower and upper layers of the specified ContactCode section are the same layer. The ContactCode section is invalid and the command does not complete.

The following is an example of the error message:

```
Error : ContactCode 'VIA12A' has non-distinct lower and upper
layer names 'METAL2' and 'METAL2'. (line 1258) (TFCHK-068)
```

### What Next

Change one of the layers to another layer and then run the command again.

### Examples

```
cmCheckLibrary(2)
cmCreateLib(2)
cmReplaceTech(2)
dbCreateLib(2)
dbReplaceTechFile(2)
```

## TFCHK-069

(warning) ContactCode '%s' has non-distinct lower and upper layer names '%s' and '%s'. (line %d)

### Description

This warning message occurs when the lower and upper layers of the specified ContactCode section are the same layer. This ContactCode section is invalid and is ignored.

The following is an example of the warning message:

```
Warning : ContactCode 'VIA12A' has non-distinct lower and upper
layer names 'METAL2' and 'METAL2'. (line 1258) (TFCHK-069)
```

### What Next

This is only a warning message. No action is required.

Feedback

However, if the result is not what you intended, you can change one of the layers to another layer and then run the command again.

## TFCHK-070

(error) ContactCode '%s' has an invalid cut layer '%s' that does not correspond to a via layer between the lower and upper metal layers '%s' and '%s'. (line %d)

### Description

This error message occurs when the cut layer of the specified ContactCode section does not correspond to the via layer between the specified lower and upper metal layers. The ContactCode section is invalid and the command does not complete.

The following is an example of the error message:

```
Error : ContactCode 'via8_fat' has an invalid cut layer 'VIA89'
that does not correspond to a via layer between the lower and upper
metal layers 'METAL8' and 'METAL9'. (line 1442) (TFCHK-070)
```

### What Next

Change either the cut layer, or the lower or upper metal layer so that the cut layer corresponds to the specified via layer. After making your changes, run the command again.

## TFCHK-071

(warning) ContactCode '%s' has an invalid cut layer '%s' that does not correspond to a via layer between the lower and upper metal layers '%s' and '%s'. (line %d)

### Description

This warning message occurs when the cut layer of the specified ContactCode section does not correspond to the via layer between the specified lower and upper metal layers. This ContactCode section is invalid and is ignored.

The following is an example of the warning message:

```
Warning : ContactCode 'via8_fat' has an invalid cut layer 'VIA89'
that does not correspond to a via layer between the lower and
upper metal layers 'METAL8' and 'METAL9'. (line 1442) (TFCHK-071)
```

### What Next

This is only a warning message. No action is required.

However, if the result is not what you intended, you can change either the cut, lower, or upper metal layer so that the cut layer corresponds to the the specified via layer.

Feedback

## TFCHK-072

(information) ContactCode '%s' has a minimum cut spacing %g that is less than the cut layer minimum spacing %g. (line %d)

### Description

This message indicates that the specified ContactCode section's minimum cut spacing is less than the corresponding cut layer's minimum spacing. Each contact code's minimum cut spacing should be greater or equal to its cut layer's minimum spacing.

If it is via cell master in the contactCode, and cutArrayXSpacing/cutArrayYSpacing is smaller than cut layer minSpacing rule, this warning message will also be issued.

### What Next

This is only an informational message. No action is required.

However, if the result is not what you intended, you can change either the specified ContactCode section's minimum cut spacing or corresponding cut layer's minimum spacing to meet your requirement.

### Examples

Information : ContactCode 'VIA78fF' has a minimum cut spacing 0 that is less than the cut layer minimum spacing 0.4. (line 1599) (TFCHK-072)

## TFCHK-073

(warning) ContactCode '%s' has undefined or zero enclosures. (line %d).

### Description

This warning message occurs when the specified ContactCode section contains undefined or zero metal enclosure dimensions. If the zero metal enclosure values are set intentionally, this warning can be ignored. Otherwise, the values may cause the router to create enclosures that violate design rule constraints, such as completely enclosed spaces or donuts.

The following is an example of the warning message:

```
Warning : ContactCode 'VIA67f' has undefined or zero metal enclosures.
(line 1551). (TFCHK-073)
```

### What Next

This is only a warning message. No action is required.

However, if the zero metal enclosure values are not intentional, you can set the upper and lower enclosures of the ContactCode section to avoid potential design rule violations.

Feedback

## TFCHK-074

(error) ContactCode '%s' is defined as a fat and default contact code. (line %d)

### Description

This error message occurs when the specified ContactCode section is set as a fat and default contact code, which is not permitted. The contact code has both of the following attribute values:

**isDefaultContact** = 1

**isFatContact** = 1

The contact code cannot be marked simultaneously as a fat and default contact code.

The following is an example of the error message:

```
Error : ContactCode 'VIA23f' is defined as a fat and default
contact code. (line 1444) (TFCHK-074)
```

### What Next

Remove or set to 0 either the *isDefaultContact* or *isFatContact* attribute, and then run the command again.

## TFCHK-075

(warning) ContactCode '%s' is defined as a fake contact code. (line %d)

### Description

This warning message occurs when the specified ContactCode section contains the *_fake* string appended to the end of its name. This string is used to mark special contact codes to support special double vias for use during routing. However, this is a workaround and may not be supported in the future.

The following is an example of the warning message:

```
Warning : ContactCode 'via2_fake' is defined as a fake contact code.
(line 392) (TFCHK-075)
```

### What Next

This is only a warning message. No action is required.

However, if the workaround is no longer necessary, you can remove the specified ContactCode section.

Feedback

## TFCHK-076

(error) Default contact code %s defined for non-default via layer '%s'.

### Description

This error message occurs when the default ContactCode is defined for a non-default via layer. Default contactCode is allowed to be defined only for default via layer. Default via layer is identified by isDefaultLayer attribute, via layer without this attribute but is unique with specific mask name is also identified as default via layer.

The following is an example of the error message:

```
Error : Default contact code VIA12_X defined for non-default
via layer 'V12_X'. (TFCHK-076)
```

### What Next

Check the contact codes definition, either set the via layer used in contactCode as default layer, change via layer to another default via layer between the metal layers, or remove the isDefaultContact attribute in the contactCode can fix this error. After making your changes, run the command again.

## TFCHK-077

(warning) Multiple default contact codes exists for via layer '%s'.

### Description

This warning message occurs when there are multiple ContactCode sections with the specified via layer that are marked as the default. Each via layer is allowed only one default contact code with a *cutLayer* or *cutLayerNumber* attribute that corresponds to the specified via layer. If multiple default contact codes exist, the default contact code with the lowest contact code number is the used as the default.

The following is an example of the warning message:

### What Next

Warning : Multiple default contact codes exists for via layer 'VIA56'. (TFCHK-077)

### What Next

This is only a warning message. No action is required.

However, if the result is not what you intended, check all contact codes with the specified cut layer, and mark only one as the default contact code.

Feedback

## TFCHK-078

(error) Valid contact code is missing for via layer '%s'.

### Description

This error message occurs when a ContactCode section meets one of following criteria:

- The ContactCode section is not defined for the specified via layer.
- The ContactCode section is defined for the specified via layer, but it is invalid.

A ContactCode section is defined for a particular via layer by assigning the *cutLayer* or *cutLayerNumber* attribute to the corresponding via layer name or number. At least one valid contact code must be defined for each via layer.

The following is an example of the error message:

```
Error : Valid contact code is missing for the via layer 'via4'.
(TFCHK-078)
```

### What Next

Define or correct the invalid ContactCode section for the specified via layer and then run the command again.

## TFCHK-079

(warning) DesignRule attribute '%s' is assigned a non-physical layer '%s'. (line %d)

### Description

This warning message occurs when the specified DesignRule section does not have a physical layer, such as poly, poly contact, metal, or via layer, assigned to the specified attribute. The application ignores the design rule.

The following is an example of the warning message:

```
Warning : DesignRule attribute 'layer1' is assigned a non-physical
layer 'VIA12'. (line 790) (TFCHK-079)
```

### What Next

This is only a warning message. No action is required.

However, if you do not want the design rule ignored, change the attribute for specified DesignRule section to a valid physical layer, such as a layer with a valid mask name.

Feedback

## TFCHK-080

(warning) DesignRule is defined with non-consecutive layers '%s' and '%s'. (line %d)

### Description

This warning message occurs when the specified DesignRule section contains two metal layers or two via layers that do not have consecutive mask names. For example, the mask names via1 and via2 are valid consecutive mask names. The mask names via2 and via4 are not consecutive and cause the warning message to occur and the design rule to be ignored. For cut to metal spacing rule with minSpacingUpperMetalWidthThresholdTbl, the rule should be defined in DesignRule section between cut and lower metal layers (via_x and metal_x). For forbiddenCut and separatedCut spacing rule, the attributes should be defined in DesignRule section between same cut layer; otherwise, TFCHK-080 will be issued.

The following is an example of the warning message:

```
Warning : DesignRule is defined with non-consecutive layers 'VIA2'
and 'VIA4'. (line 1954) (TFCHK-080)
```

### What Next

This is only a warning message. No action is required.

However, to avoid this warning message, change the layers of the DesignRule section so that the mask names of the layers are consecutive. After making your changes, run the command again.

## TFCHK-081

(warning) DesignRule is defined with non-distinct layers '%s' and '%s'. (line %d)

### Description

This warning message occurs when the layers of the specified DesignRule section are the same layer. The application ignores the design rule.

The following is an example of the warning message:

```
Warning : DesignRule is defined with non-distinct layers 'VIA3'
and 'VIA3'. (line 1960) (TFCHK-081)
```

### What Next

This is only a warning message. No action is required.

However, if the result is not what you intended, you can change one of the layers of the DesignRule section and then run the command again.

Feedback

## TFCHK-082

(warning) DesignRule is defined with invalid metal-to-via enclosure layers '%s' and '%s'. (line %d).

### Description

This warning message occurs when the DesignRule section specifies either a minimum or an end-of-line enclosure rule, for which one layer is not a via layer and/or the other layer is not a metal layer. The application ignores the design rule. Enclosure rules are only valid between a metal and via layer.

The following is an example of the warning message:

```
Warning : DesignRule is defined with invalid metal-to-via
enclosure layers 'via2' and 'via1'. (line 436). (TFCHK-082)
```

### What Next

This is only a warning message. No action is required.

However, if the result is not what you intended, change the layers of the DesignRule section so that one is a via layer and the other is a metal layer.

## TFCHK-083

(warning) Valid capacitance model is missing for reference layer '%s', lower ground plane '%s', and upper ground plane '%s'.

### Description

This warning message appears when one of the following occurs in the CapModel section:

- A CapModel section is not defined for the specified layers.
- A CapModel section is defined for the specified layers, but it is invalid.

If one capacitance model is defined in the technology file, then define all of the models. Every poly and metal layers requires a model. Each of these layers needs one capacitance model for each pair of lower and upper ground planes. The number of capacitance models for the layer is equal to the number of lower ground planes multiplied by the the number of upper ground planes. The lower ground planes include the reference layer and all of the poly or metal layers below the reference layer. The upper ground planes include the reference layer and all of the metal layers above the reference layer.

Feedback

The following is an example of the warning message:

```
Warning : Valid capacitance model is missing for the reference
layer 'CP', lower ground plane '', and upper ground plane 'METAL2'.
(TFCHK-083)
```

### What Next

This is only a warning message. No action is required.

However, if the result is not what you intended, you can define or correct the invalid CapModel section for the specified layers and then run the command again.

---

## TFCHK-084

(warning) CapModel sections are missing. Capacitance models should be loaded with a TLU+ file later.

### Description

This warning message occurs when the technology file does not contain capacitance models. Load the TLU+ file later in the library preparation flow.

The following is an example of the warning message:

```
Warning : CapModel sections are missing. Capacitance models
should be loaded with a TLU+ file later. (TFCHK-084)
```

### What Next

This is only a warning message. No action is required.

However, it is best to practice to load the TLU+ file later in the library preparation flow.

---

## TFCHK-085

(error) CapModel '%s' references the non-existent table '%s'.

### Description

This error message occurs when the specified CapModel section references a CapTable section that does not exist.

The following is an example of the error message:

```
Error : CapModel 'polyConfig5' references the nonexistent
table 'polypoly_C_BOTTOM_GP_15MAX'. (TFCHK-085)
```

Feedback

### What Next

Define the missing CapTable section or remove the reference from the CapModel section and then run the command again.

---

## TFCHK-086

(warning) Layer '%s' has invalid fat table thresholds '%s'. (line %d)

### Description

This warning message occurs when the specified Layer section contains an invalid fat table threshold list. The threshold list must meet the following requirements:

- The number of threshold values must match the table dimension.
- All threshold values must be nonnegative numbers and listed in ascending order.
- The first threshold value must be between 0 and the minWidth inclusively.
- The second threshold value must be greater than the layer minWidth.

The following is an example of the warning message:

```
Warning : Layer 'METAL1' has invalid fat table thresholds '(0, 0,
1.505, 3.005, 4.505, 7.505)'. (line 225) (TFCHK-086)
```

### What Next

Please correct the tech file to meet the requirements described above. 1) All threshold values must be nonnegative numbers and listed in ascending order 2) The first threshold value must be between 0 and the minWidth inclusively 3) The second threshold value must be greater than minWidth on the layer.

After making your changes, run the command again.

---

## TFCHK-087

(error) Layer '%s' has invalid fatTblSpacing '%s'. (line %d)

### Description

This error message occurs when the specified Layer section contains an invalid fatTblSpacing list. The fatTblSpacing values must be equal to or greater than the layer minSpacing value. And fatTblSpacing[0] = minSpacing.

Feedback

### Examples

The following is an example of the error message:

```
Error : Layer 'METAL1' has invalid fatTblSpacing
 '(0.12,0.11,0.5,0.9,1.5,2.5,
0.11,0.17,0.5,0.9,1.5,2.5,
0.5,0.10,0.5,0.9,1.5,2.5,
0.9,0.9,0.9,0.9,1.5,2.5,
1.5,1.5,1.5,1.5,1.5,2.5,
2.5,2.5,2.5,2.5,2.5,2.5)'. (line 225) (TFCHK-087)
```

### What Next

Change the fatTblSpacing list of the specified Layer section so the values in the list are equal to or greater than the layer minSpacing value. After making your changes, run the command again.

---

## TFCHK-088

(warning) Layer '%s' has invalid fatTblSpacing '%s'. (line %d)

### Description

This warning message occurs when the specified Layer section contains an invalid fatTblSpacing list. The fatTblSpacing values must be equal to or greater than the layer minSpacing value. And fatTblSpacing[0] = minSpacing.

The following is an example of the warning message:

```
Warning : Layer 'METAL1' has invalid fatTblSpacing
 '(0.12,0.11,0.5,0.9,1.5,2.5,
0.11,0.17,0.5,0.9,1.5,2.5,
0.5,0.10,0.5,0.9,1.5,2.5,
0.9,0.9,0.9,0.9,1.5,2.5,
1.5,1.5,1.5,1.5,1.5,2.5,
2.5,2.5,2.5,2.5,2.5,2.5)'. (line 225) (TFCHK-088)
```

### What Next

This is only a warning message. No action is required.

However, it is best practice to change the fatTblSpacing list of the specified Layer section so the values are equal to or greater than the layer minSpacing value. After making your changes, run the command again.

---

## TFCHK-089

(warning) %s '%s' has invalid table values '%s'. (line %d)

Feedback

### Description

This warning message occurs when the specified section contains an invalid table value list, e.g. thresholds. The value list must meet the following requirements:

- The number of the values must match the table dimension.
- All values must be nonnegative numbers and listed in ascending order.
- The first threshold value must be between 0 and minWidth inclusively.
- The second threshold value must be greater than the layer minWidth. • The first spacing value must not be smaller than minSpacing, and the first maxSpacingCheckRange value must not be smaller than minSpacing * 2 + minWidth.
- In 2-D spacing table between the same layer, the values must be symmetrical.

  In minCutsTbl, if area is specified, extension should be specified. In enclosureTbl, if no lower/upper width is specified or an equivalent ContactCode to the specified enclosures is not found, this message will occur.

In generalized cut spacing rule with parallel length, if minSpacingDiagonalCheckTbl[i][j] = 1, at least one of minSpacingXParallelLengthThresholdTbl[i][j] and minSpacingYParallelLengthThresholdTbl[i][j] should be < 0; otherwise, TFCHK-089 will be issued to ignore minSpacingDiagonalCheckTbl[i][j]. If minSpacingXParallelLengthThresholdTbl or minSpacingYParallelLengthThresholdTbl is defined, orthoSpacingExcludeCornerTbl[i][j] should be 1; otherwise, TFCHK-089 will be issued to warn users that the corners will be checked by x/y spacing. Please note that the cutName in cutNameTbl should start with a letter, such as Vsm. Otherwise, you will see this message. For the table of fatTblExtensionRange, if fatTblExtensionRangeDimension defined, the value list must meet the following requirements:

- The number of the values must match the dimension.
- All values must be larger than 0 and listed in ascending order.

  For fatWireViaEnc rule, fatWireViaEncMinWidthThresholdTbl is recommended to be in ascending order. However, the tool accepts this table even it is not in ascending order as long as the min and max thresholds are paired in fatWireViaEncMinWidthThresholdTbl and fatWireViaEncMaxWidthThresholdTbl. Furthermore, if missing a width range between fatWireViaEncMinWidthThresholdTbl and fatWireViaEncMaxWidthThresholdTbl, this message will be issued.

  For maxStackLevel rule, the value in maxStackLevelCutAsViaArrayTbl for the smallest squared cut should be 0.

  For separatedCut spacing rule, if defined between the same cut layer, the following should hold: 1) separatedCut1NameTbl=separatedCut2NameTbl 2) separatedCut1ToUpperMetalExcludedSpacingTbl=separatedCut2ToUpperMetalExcludedSpacingTbl

Feedback

3) the 2D spacing tables separatedCutXMinSpacingTbl and separatedCutYMinSpacingTbl should be symmetrical.

x|yLegalDimensionTbl table values should be in ascending order.
voltageAreaThresholdTbl values should be in ascending order.

For minCutsTbl, if both lower and upper layer widths are specified with different values for extension rule, it also issues this message and takes both widths as thresolds. For via extension rule, only lower width needs to be specified.

The following is an example of the warning message:

```
Warning : Layer 'METAL1' has invalid table values '(0, 0,
1.505, 3.005, 4.505, 7.505)'. (line 225) (TFCHK-089)
```

**What Next**

Please change the value list of the specified section to meet the above requirements. After making your changes, run the command again.

---

## TFCHK-089e

(error) %s '%s' has invalid table values '%s'. (line %d)

**Description**

This error message occurs when the specified section contains an invalid table value list, e.g. thresholds. The value list must has all values be non-negative numbers and listed in ascending order.

The following is an example of the warning message:

```
Error : Layer 'VIA1' has invalid table values '(0, 0,
1.505, 3.005, 4.505, 7.505)'. (line 225) (TFCHK-089e)
```

**What Next**

Please change the value list of the specified section to meet the above requirements. After making your changes, run the command again.

---

## TFCHK-090

(warning) %s '%s' section occurred multiple times. (line %d)

**Description**

This warning message occurs when the same name section appeared in technology file more than once. Alternatively, when you redefine a predefined layer, it will issue the same

Feedback

message, for example, if you define Layer routeGuide, which is a predefined layer. In this case, always the values of the attributes in the last section will be used.

For example, if you load a tech file with Layer "V1" { layerNumber = 123 maskName = "via1" visible = 1 defaultWidth = 0.11 minWidth = 0.03 minSpacing = 0.1 } Layer "V1" { layerNumber = 24 maskName = "via1" defaultWidth = 0.12 minWidth = 0.04 } then the following will be used Layer "V1" { layerNumber = 24 maskName = "via1" visible = 1 defaultWidth = 0.12 minWidth = 0.04 minSpacing = 0.1 }

The following is an example of the warning message:

```
Warning : Layer 'M1' appeared multiple times. (line 758) (TFCHK-090)
```

**What Next**

This is only a warning message. No action is required.

However, it is recommended that you merge the data of the sections with the same section name into one section.

---

## TFCHK-091

(warning) ContactCode '%s' has an invalid cut width %g that is less than the cut layer minWidth %g. (line %d)

**Description**

This message indicates that the specified ContactCode section's cutWidth (or cutHeight) or corresponding cut layer's minWidth is invalid. Each contact code's cutWidth (cutHeight) should be greater than or equal to its cut layer's minimum width.

**What Next**

Change either the specified ContactCode section's cutWidth and/or cutHeight or corresponding cut layer's minWidth to meet this requirement.

**Examples**

Warning : ContactCode 'VIA78fF' has an invalid cut width 0.2 that is less than the cut layer minWidth 0.4. (line 1599) (TFCHK-091)

---

## TFCHK-092

(warning) Cut layer '%s' has a non-cross primary default ContactCode '%s'. (line %d)

**Description**

This message occurs when the cut layer has multiple default ContactCodes and the primary default ContactCode is a non-cross one. For a given cut layer, the primary

Feedback

ContactCode is defined to be the first one that appears in the technology file, and the primary default ContactCode is the first default ContactCode.

A non-cross (minus '-' shaped) ContactCode is one that has the same longitudinal direction of enclosures in both upper and lower layers. And a cross (plus '+' shaped) ContactCode is one that has different longitudinal directions of enclosures in both upper and lower layers. A cross ContactCode is determined by upperLayerEncWidth >= upperLayerEncHeight and lowerLayerEncWidth <= lowerLayerEncHeight or upperLayerEncWidth <= upperLayerEncHeight and lowerLayerEncWidth >= lowerLayerEncHeight and a non-cross contactCode is one that does not meet either of the above conditions. i.e. it is determined by upperLayerEncWidth > upperLayerEncHeight and lowerLayerEncWidth > lowerLayerEncHeight or upperLayerEncWidth < upperLayerEncHeight and lowerLayerEncWidth < lowerLayerEncHeight

The following is an example of the warning message:

```
Warning:  Cut layer 'VIA1' has a non-cross primary default ContactCode
 'V10'.
   (line 254) (TFCHK-092)
```

### What Next

When non-cross default ContactCode is in the first position, please choose from other default ContactCodes for this cut layer, if any, to get optimal routing performance.

---

## TFCHK-093

(warning) %s '%s' attribute '%s' has an invalid value of %g.

### Description

This warning message occurs when the value of the specified attribute is invalid or inconsistent with related attributes.

The value of the *minEnclosedArea* attribute for default metal layers should be greater than 0. Other values can affect the quality of the application results and can cause excessive design rule constraint violations during physical verification.

For a set of rule, completeness of rule parameters should be maintained:

For endOfLineViaWire spacing rule, the following should be met: (a) endOfLineViaWireMinWidth >= cutWidth + 2* minEnclosure (w or h) and endOfLineViaWireMinWidth > minWidth (b) endOfLineViaWireMinSpacing >= minSpacing + minEnclosure (c) endOfLineViaOrthoWireMaxThreshold >= minWidth

For direction based minSpacing rule, Both xMinSpacing and yMinSpacing should be specified and the value should not be smaller than minSpacing.

Feedback

For T-Junction rule, the following should be met: tJunctionStubToStubMinSpacing >= minSpacing *2 + minWidth tJunctionStubWireMaxThreshold >= minWidth tJunctionOrthoWireMaxThreshold >= minWidth

For cut geometry based cut rule, if orthoSpacingExcludeCornerTbl value = 1, cornerMinSpacing should be defined.

For asymmetric via to metal spacing rule, diffNetKeepoutMinWidth diffNetKeepoutMinLength diffNetKeepoutMinRadius should be defined on via and metal layer pair.

For fatWireViaEnc rule, the values in fatWireViaEncMinWidthThresholdTbl should be < fatWireViaEncMaxWidthThresholdTbl, respectively. For iso-dense rule only one entry in fatWireViaEnc[Min|Max]WidthThresholdTbl is required.

The following is an example of the warning message:

```
Warning : DesignRule '' attribute 'endOfLineViaWireMinWidth' has an
 invalid
value of 0.1. (TFCHK-093)
```

**What Next**

It is best practice to check the attribute value of the specified section, and then change the value to a valid value that is consistent with other related attributes in the same set of rule. After making your changes, run the command again.

---

## TFCHK-094

(warning) %s '%s' '%s' and '%s' coexist.

**Description**

This warning message occurs when the two rules/attributes coexist. If fatTblSpacing rule and consolidated fatTblXMinSpacing/fatTblYMinSpacing rule coexist, router will honor the latter and ignore the former. If fatTblParallelLengthDimension and fatTblSpacingMode coexist, router will ignore the latter since fatTblParallelLength is an independent threshold along with fatTblThreshold. If enclosedViaMetalMinLength rule is specified, minLengthMode in Technology section will be ignored. When minCutsTbl and fatTblFatContactMinCuts/fatTblExtensionMinCuts coexist, this message also will occur, and the router will honor the latter and ignore the former. For enclosedCut spacing rule, if both the table rule and the scalar rule are defined, the scalar rule will be ignored. If isDefaultContact = 1 is defined in ContactCode section, notAllowedForSignalRoute will be ignored. For fat metal via rule, the same contactNumber should not appear in both fatTblTotalEncContactNumber and fatTblExtraEncContactNumber. Otherwise, the one in fatTblExtraEncContactNumber will be ignored. onWireTrack should not coexist with onGrid attribute on the same layer section; otherwise, TFCHK-094 will be issued to ignore onGrid.

Feedback

The following is an example of the warning message:

```
Warning : Layer 'M1' 'fatTblSpacing' and 'fatTblXMinSpacing' coexist.
 (TFCHK-094)
```

### What Next

It is recommended that you remove the first rule that will be ignored from the technology file and reload it.

---

## TFCHK-095

(error) %s '%s' '%s' and '%s' coexist.

### Description

This message occurs when two conflicting rules/attributes coexist. For example, if minAreaFillMinLengthTbl/minAreaFillMinWidthTbl attributes and minAreaRectMinLengthTbl/minAreaRectMinWidthTbl coexist, router does not know which rule to apply. For fatWireViaEnc rule, fatWireViaEncMinWidthThresholdTbl / fatWireViaEncMaxWidthThresholdTbl and fatWireViaEncWidthThresholdTbl should not coexist. convexConvexMinEdgeLength scalar and table rules must not coexist.

The following is an example of the message:

```
Error: Layer 'M1' 'minAreaFillMinLengthTbl' and 'minAreaRectMinLengthTbl'
 coexist. (TFCHK-095)
```

### What Next

It is recommended that you choose the rule/attribute that you want and remove the other.

---

## TFCHK-096

(information) Unrecognized attributes in technology file %s were ignored.

### Description

This message indicates that the errors (TFCHK-009) for unrecognized attributes in the technology file were ignored because the following variable has been set to TRUE.

```
set ignore_tf_error true
```

This allows the tool to support a technology file with new attributes even if the new attributes are not yet recognized by the current tool version.

The following is an example of the message:

```
Information: Unrecognized attributes in technology file tech.tf were
 ignored. (TFCHK-096)
```

Feedback

### What Next

Verify that all of the unrecognized attributes in the technology file are safe to ignore. If the tool should not ignore the technology file errors reported by TFCHK-009, then the ignore_tf_error variable setting should be restored to the default value of FALSE.

---

## TFCHK-097

(warning) ContactCode '%s' with contactCodeNumber '%d' (>%d) ignored in %s.

### Description

This warning message occurs when contactCodeNumber in ContactCode section is out of range. There are two cases: 1) Using pre-2010.12 ICC/MW to open library created by 2010.12 ICC/MW or newer with contactCodeNumber>255. In this case, this ContactCode cannot be used in design and will be ignored. 2) Using 2010.12 ICC/MW or newer to create or open library that have contactCodeNumber>65535. In this case, library can still be generated, but design may not be able to use this ContactCode.

The following table lists a possible reason of this warning message:

```
Section Type            Attribute Value
-------------------------------------
ContactCode             contactCodeNumber > 255   (2010.03 and before)
ContactCode             contactCodeNumber > 65535 (2010.12 and later)
```

The following is an example of an index that is too large and the warning message that is generated.

```
Line 400: ContactCode "via2" {
 Line 401:     contactCodeNumber = 100000
 ...

 Warning : ContactCode 'via2' with contactCodeNumber 100000 (>65536)
 ignored in design. (TFCHK-097)
```

### What Next

Change the index of the specified section in tech file to a valid integer within the range specified in the table above, and then run the command again. ot if you see this message during open_mw_lib, you can ignore it since this contactCode cannot be used in design.

---

## TFCHK-098

(error) %s '%s' has incomplete '%s' defined. (line %d)

Feedback

**Description**

This error message occurs when some tables or values are not defined completely. It including these types of check:

In ContactCode 1) At least one valid Rect/PolygonTbl should be defined for each layer, or none of them should be defined under one contactCode section. 2) CutArrayXSpacing and cutArrayYSpacing must be defined together and their values must be larger than 0.

**What Next**

User should complete the tables/attributes and run the command again.

---

## TFCHK-098w

(warning) %s '%s' has incomplete '%s' defined. (line %d)

**Description**

This error message occurs when some tables or values are not defined completely or related section is missing. It including these types of check:

In ContactCode 1) At least one valid Rect/PolygonTbl should be defined for each layer, or none of them should be defined under one contactCode section. 2) CutArrayXSpacing and cutArrayYSpacing must be defined together and their values must be larger than 0. In DesignRule, if self-aligned via (SAV) related rules are defined, but SAV is not defined in ViaRule section.

In Layer, 1) For rectMinAreaTbl rule, rectMinAreaWidthThresholdTbl should start from minWidth to cover all width ranges;

**What Next**

User should complete the tables/attributes or related sections and run the command again.

---

## TFCHK-099

(warning) Unused attribute '%s' in %s '%s' is ignored and set to zero. (line %d)

**Description**

This warning message occurs when the Rect/PolygonTbl defined for ContactCode with via cell master correctly, cutWidth/cutHeight/EncWidth/EncHeight/minCutSpacing will not be used by router, and it will be set to 0.

Feedback

**What Next**

Remove these values from contactCode with via cell master, and the warning message will not be issued.

---

## TFCHK-100

(error) ContactCode '%s' has combined geometry table and cut array defined. (line %d)

**Description**

In contactCode with via cell master, either geometry tables (Rect/PolygonTbl) or cut array should be defined, but not both. This error message issued when such combined syntax definition detected.

**What Next**

User should check the contactCode and keep only the correct tables or cut array. After making your changes, run the command again.

---

## TFCHK-101

(error) ContactCode '%s' has no layer definition for geometry table %s. (line %d)

**Description**

In contactCode with via cell master, if geometry tables (Rect/PolygonTbl) defined, layer number must be specified prior to corresponding Rect/PolygonTbl.

**What Next**

Define layer number before the geometry table in contactCode. After making your changes, run the command again.

---

## TFCHK-102

(information) '%s' in ContactCode '%s' is not defined, using value of '%s [%g]' instead. (line %d)

**Description**

This information indicates that: When cut array used in contactCode via cell master, if cutArrayXSpacing and cutArrayYSpacing are not defined, value of minCutSpacing can be used to represent the array x/y spacing

**What Next**

This is only an informational message. No action is required.

Feedback

However, if user does not want to use minCutSpacing to represent cut array x/y spacing, please specify actual cutArrayXSpacing and cutArrayYSpacing.

---

## TFCHK-103

(error) Ambiguous '%s' and '%s' are defined in ContactCode '%s'. (line %d)

**Description**

This error message occurs if minCutSpacing, cutArrayXSpacing and cutArrayYSpacing are all defined in via cell master in contactCode, and their values are larger than 0, but not be equal.

**What Next**

User can remove minCutSpacing if want to set different array spacing on x/y direction. User can correct the value of these attributes and run the command again.

---

## TFCHK-104

(warning) 1x1 cut array defined in ContactCode '%s'. (line %d)

**Description**

This warning message occurs when cut array is used in via cell master, but cutArrayNumRows and cutArrayNumCols are both 1.

**What Next**

Assign correct value to these two attributes. User should not define 1x1 cut array, please use contactCode without via cell master instead.

---

## TFCHK-105

(warning) Default contact code '%s' defined as asymmetrical via, the contact code will be ignored. (line %d)

**Description**

This warning message occurs when the contact code defines an asymmetrical via with geometry tables or a cut array but is also marked as a default contact code. The router will not use this contact code.

**What Next**

Remove the isDefaultContact statement or remove the via definition from this contact code, and then rerun the command .

Feedback

### See Also

- read_tech_file

---

## TFCHK-106

(error) Valid ContactCode not found for %s on Layer '%s'.

### Description

This message occurs when a valid ContactCode is not found in the tech file for the width threshold range from enclosureTbl or when a valid ContactCode is not found that is equivalent to or larger than the enclosures in enclosureTbl. Alternatively, when ContactCodes are found from enclosureTbl but a different set of ContactCodes are found from minCutsTbl, this message also occurs. The definition of the tables are enclosureTbl = (cutName, lowEncWidth, lowEncHeight, upEncWidth, upEncHeight, lowMetalWidth, upMetalWidth) minCutsTbl = (numCuts, cutName, lowMetalWidth, upMetalWidth, area, extension) Please note that the cutName should start with a letter, such as Vsm. Otherwise, you may see this message.

The following is an example of the message:

```
Error : Valid ContactCode not found for threshold [0.1,0.16) on Layer
 'V2'.
```

### What Next

Please add a ContactCode according to the enclosureTbl values or adjust enclosure and/or width values in enclosureTbl. After making your changes, run the command again.

---

## TFCHK-106w

(warning) Valid ContactCode not found for %s on Layer '%s'.

### Description

This warning message occurs when a valid ContactCode is not found in the tech file for the width threshold range from enclosureTbl or when a valid ContactCode is not found that is equivalent to or larger than the enclosures in enclosureTbl. The definition of the tables are

enclosureTbl = (cutName, lowEncWidth, lowEncHeight, upEncWidth, upEncHeight, lowMetalWidth, upMetalWidth)

minCutsTbl = (numCuts, cutName, lowMetalWidth, upMetalWidth, area, extension)

Feedback

When either lower or upper threshold < default or min width and valid ContactCode is not found for the EoU syntax, this message will also occur as Warning. Please note that the cutName should start with a letter, such as Vsm. Otherwise, you may see this message.

The following is an example of the warning message:

```
Warning: Valid ContactCode not found for threshold [0.1,0.16) on Layer
 'V2'.
   (TFCHK-106w)
Warning: Valid ContactCode not found for enclosureTbl =
   (Vsm,0.025,0.025,0.250,0.250,...) on Layer 'V2'. (TFCHK-106w)
```

**What Next**

Add a ContactCode according to the enclosureTbl values or adjust the enclosure and width values in enclosureTbl. After making your changes, run the command again.

---

## TFCHK-107

(error) ContactCode '%s' has unsupported polygon defined, post-flow cannot use this type of contactCode. (line %d)

**Description**

In contactCode with via cell master, non-rectilinear polygon cannot be recognized and used by post-flow correctly. This type of contactCode will not work as intended.

**What Next**

Redefine the contactCode with rectangle or rectilinear polygon and run the command again.

---

## TFCHK-108

(warning) Object '%s' name '%s' exceeds %d characters. (line %d)

**Description**

This warning message occurs when the specified object name in the technology file contains more than the maximum number of characters allowed.

The length limitation of objects name are:

---------- Type Limits ----------
Layer 31 Color 31 Stipple 31 LineStyle 31 Tile 31 CapModel 31 ResModel 31 ContactCode 255 ----------

Feedback

The following is an example of the warning message:

```
Warning : Object 'Layer' name 'thisisatestingcaseforlayernamemorethan32'
exceeds 31 characters. (line 342) (TFCHK-108)
```

**What Next**

The layer name will be truncated to 31 characters and saved into library. All occurrences of the object name in the same tech file will be updated.

Also please review the number of characters of the specified object to make sure it does not exceed the allowed maximum length and then rerun the command.

---

## TFCHK-109

(error) Default ContactCode '%s' on layer '%s' has cut size with no matching cut name found.

**Description**

The via layer has cutNameTbl defined for generalized cut spacing rule, but the specific default contactCode has cutWidth/cutHeight that are not of the values defined in the cutWidthTbl/cutHeightTbl.

This may lead incorrect router behaviour lately.

The following is an example of the error message:

```
Error: Default ContactCode 'via12' on layer 'via1' has cut size with no
matching cut name found. (TFCHK-109)
```

**What Next**

Check the generalized cut spacing rule and the specific default contactCode, correct the technology file, and run the command again.

---

## TFCHK-109w

(warning) Default ContactCode '%s' on layer '%s' has cut size with no matching cut name found.

**Description**

The via layer has cutNameTbl defined for generalized cut spacing rule, but the specific default contactCode has cutWidth/cutHeight that are not of the values defined in the cutWidthTbl/cutHeightTbl.

This may cause router cannot check DRC for the via lately.

Feedback

The following is an example of the warning message:

```
Warning: Default ContactCode 'via12' on layer 'via1' has cut size with no
matching cut name found. (TFCHK-109w)
```

### What Next

Check the generalized cut spacing rule and the specific default contactCode, correct the file and run the command again.

---

## TFCHK-110

(warning) Non-default ContactCode '%s' on layer '%s' has cut size with no matching cut name found.

### Description

The via layer has cutNameTbl defined for generalized cut spacing rule, but the specific non-default contactCode has cutWidth/cutHeight that are not of the values defined in cutWidthTbl/cutHeightTbl.

If the contactCode is a via cell master specified with geometry tables, such as cutLayerRectTbl, the cut size of such a contactCode is not counted in the matching since router does not currently use it

The following is an example of the error message:

```
Warning: Non-Default ContactCode 'via1Ap' on layer 'via1' has cut size
 with
no matching cut name found. (TFCHK-110)
```

### What Next

Check the generalized cut spacing rule and the specific contactCode. If needed, correct cutWidthTbl/cutHeightTbl and/or contactCodes size (cutWidth/cutHeight).

---

## TFCHK-111

(warning) cutNameTbl on layer '%s' has cut name '%s' with no matching ContactCode found.

### Description

The specific layer has generalized cut size defined, but the specific cut width and cut height has no corresponding contactCode found. This warning message can be ignored if user defined the cut size for later use.

Feedback

**What Next**

Add corresponding contactCode with specific cut size and run the command again.

---

## TFCHK-112

(warning) Layer '%s' has duplicate cut sizes defined, '%s' and '%s'. (line %d)

**Description**

There're same cut width and cut height defined in the generalized cut spacing rule in the specific via layer with different cut name. Cut size for different cut name should be unique.

**What Next**

Redefine the cut width/cut height in the generalized cut spacing rule and run the command again.

---

## TFCHK-113

(error) Layer '%s' with mask name '%s' has layer number '%d' exceeding upper limit %d. (line %d)

**Description**

This error is issued since the specified layer has pre-defined mask name, and it is expected to be defined on layers below the upper limit.

These pre-defined mask names are: Metal Layer: poly, metal1 - metal15 Via Layer: polyCont, via1-via14 Back Metal Layer: bmetal1-bmetal3 Back Via Layer: tsv, bvia1-bvia2

Example: Error: Layer 'V13' with mask name 'via13' has layer number '212' exceeding upper limit 187. (line 101).

**What Next**

Redefine the layerNumber attribute, to use number below specified upper limit.

---

## TFCHK-114

(warning) Layer '%s' with layer number '%d' is overriding pre-defined system layer. (line %d)

**Description**

This error is issued since the specified layer was defined with layer number used by pre-defined system layer. The system layer is used by ICC, and overriding may cause issues in post-flow applications, like router.

Feedback

The range for the pre-defined system layer numbers is: 188 ~ 255

Example: Warning: Layer 'TEXT10' with layer number 192 is overriding pre-defined system layer. (line 1024)

**What Next**

Redefine the layerNumber attribute, to use number below specified range.

---

## TFCHK-115

(error) Multiple layers exists with mask name '%s'.

**Description**

This error message occurs when there are multiple layer sections with the specified mask name. This type of mask name can only defined on one physical layer.

The following is an example of the error message:

```
Error : Multiple layers exists with mask name 'mimtop'.
(TFCHK-059)
```

**What Next**

Check all layers with the specified mask name and remove redundant layers. After making your changes, run the command again.

---

## TFCHK-116

(error) Layer '%s' attribute '%s' has an invalid value '%c'.

**Description**

This error message occurs when the value of the specified layer attribute is invalid.

The value of the *routingDirection* attribute for default metal layers shold be 'V' and 'H'

The following is an example of the error message:

```
Error: Layer 'metal3' attribute 'routingDirection' has a
value of 'S'. (TFCHK-116)
```

**What Next**

Change the attribute value of the specified layer to correct value and then run the command again.

Feedback

## TFCHK-117

(error) Layer '%s' attribute '%s' has an invalid value '%s'.

### Description

This error message occurs when the value of the specified layer attribute is invalid.

The value of the *trackMaskOrder* attribute for default metal layers shold be one of values in following list.

```
mask1
mask2
mask1_mask2
mask2_mask1
any
```

The following is an example of the error message:

```
Error: Layer 'metal3' attribute 'trackMaskOrder' has a
value of 'mark'. (TFCHK-117)
```

### What Next

Change the attribute value of the specified layer to correct value and then run the command again.

## TFCHK-118

(error) Attribute '%s' can only be specified in %s Layer (line %d)

### Description

This error message occurs when the attribute is not specified in asked layer.

Following attributes should be defined in Metal layer.

```
trackMaskOrder
routingDirection
trackOffset
```

The following is an example of the error message:

```
Error: Attribute 'trackMaskOrder' can only be specified in
Metal Layer (line 120) (TFCHK-118)
```

### What Next

delete the attribute from the specified layer and then run the command again.

Feedback

## TFCHK-119

(warning) Layer '%s' attribute '%s' is specifid without attribute '%s'!

### Description

This warning message occurs when an attribute is specified alone. it should be set toghether with other attributes. Applications will ignore the attribute. For example, *trackOffSet* must be specified together with *routingDirection*

The following is an example of the error message:

```
Warning: Layer 'M1' attribute 'trackOffset' is specifid without
attribute 'routingDirection'! (TFCHK-119)
```

### What Next

remove the attribute or add the other attibute and then run the command again.

## TFCHK-120

(error) %s '%s' attribute '%s' is assigned an invalid string '%s'. (line %d)

### Description

This error message occurs when the specified section has an unrecognized string value assigned to the specified attribute.

The valid string of attribute cutMaskPattern in ContactCode is : uniform, alternate.

The following is an example of the error message:

```
Error : ContactCode 'Via1' attribute 'cutMaskPattern' is assigned an
 invalid
string 'pattern'. (line 46) (TFCHK-120)
```

### What Next

Change the attribute value of the specified section to a valid value. After changing the value, run the command again.

## TFCHK-121

(error) ContactCode '%s' on cut layer '%s' has unsupported type '%s' of %s defined. (line %d)

### Description

This error message occurs when the specified section has an unsupported value assigned to the specified attribute.

Feedback

For cutArrayMaskPattern, currently zRouter cannot support pattern other than 'uniform' on cut layers with more than two masks.

The following is an example of the error message:

```
Error : ContactCode 'VIA1_VH' on cut layer 'CUT12' has unsupported type
 'alternate'
of cutArrayMaskPattern defined. (line 661) (TFCHK-121)
```

**What Next**

Change the attribute value of the specified section to a valid value. After changing the value, run the command again.

---

## TFCHK-122

(information) >>>Reading technology file %s <<<

**Description**

This information indicates that: Another technology file is included by user input technology file, and it is being loaded into library now. The line number reported after this line is the actual line number in the including file.

**What Next**

This is only an informational message. No action is required.

---

## TFCHK-123

(information) >>>Finish reading file %s <<<

**Description**

This information indicates that: The included technology file has been loaded into library. After this line, all line number reported is the actual line number in user input technology file.

**What Next**

This is only an informational message. No action is required.

---

## TFCHK-124w

(warning) This techfile version contains syntax that can affect correlation to IC Compiler II. It is recommended to use Design Compiler NXT embedded links to IC Compiler II for correlation with this techfile release version.

Feedback

**Description**

This information indicates that: The included technology file has an attribute which is part of the ignoredWithWarning list

**What Next**

No action is required.

# TIM

## TIM-001

(Warning) Case propagation conflict at pin '%s'. User value of '%s' will be used.

**Description**

A propagated case value is different from the value given to set_case_analysis at this pin. The propagated value will be ignored and the value from the constraint will be used.

**What Next**

Check that your constraint files are meant to be used for this mode.

## TIM-002

(Warning) Case propagation conflict at inout pin '%s'. The %s value of '%s' will be used.

**Description**

A propagated case value to the driver side of a bidirectional pin conflicts with a propagated case value from the load side of a bidirectional pin. The message includes which side and value won.

**What Next**

Check that your constraint files are meant to be used for this mode.

## TIM-003

(Warning) Case propagation conflict from multiple drivers of pin: '%s'. A value of zero being used

**Description**

A load pin of a multi-driven net, has conflicting case values from the net drivers. The value of zero will be used.

Feedback

**What Next**

Check that your constraint files are meant to be used for this mode.

---

## TIM-004

(Warning) '%d' updated moded-ETM instances do not have set_cell_mode values in design mode %s. The first %d instances are: %s

**Description**

Moded ETM library cells are created with the ETM for of ICC2 Library Manager. Each instance of a moded ETM must have a set_cell_mode set on it in order for the timing to be correct. The initialial timing update checks all instances of moded-ETM instance to check if they have set_cell_mode. Incremental updates will only report instances that have been changed by case propagation or set_cell_mode or reset_cell_mode commands.

Use the command report_cell_modes on to see what the available values are for an instance.

**What Next**

use the command set_cell_mode to specify the cell mode that should be used for instances of moded ETM cells.

---

## TIM-005

(Warning) Case value conflict at pin '%s'. The new value of '%c' will be used the moded-ETM value has been over written.

**Description**

A set_case_analysis statement has been given on a moded-ETM cells and the value given is different than the case value for the current mode.

**What Next**

If you want to restore the moded-ETM value you must remove_case_analysis on the pin and then set_cell_mode on the cell.

---

## TIM-006

(Warning) removing moded-ETM case value at pin '%s'.

**Description**

A remove_case_analysis statement has been given on a moded-ETM cell output pin that had a case value in the current_cell_mode from the library.

Feedback

### What Next

If you want to restore the moded-ETM value, you must use reset_cell_mode and then set_cell_mode again on the cell

---

## TIM-007

(Warning) The value '%s' for option '%s' is not supported and will be ignored.

### Description

The given option value is valid but not supported yet. The option value will be ignored.

### What Next

Use valid supported values, please check man pages of related commands.

---

## TIM-010

(Warning) From_pin '%s' is not a timing startpoint and will be ignored.

### Description

The pin was given with the *-from*, *-rise_from*, or *-fall_from* options, but it is not a proper timing startpoint. It will be ignored for reporting. This can happen if a register's D or Q pin is given instead of its clock pin, which is the actual timing startpoint. It can also happen for a port if the port has no *set_input_delay* or other constraint.

### What Next

Choose the correct startpoint pin, or add a constraint to this pin.

### See Also

- report_timing
- get_timing_paths
- set_input_delay

---

## TIM-011

(Warning) To_pin '%s' is not a timing endpoint and will be ignored.

### Description

The pin was given with the *-to*, *-rise_to*, or *-fall_to* options, but it is not a proper timing endpoint. It will be ignored for reporting. This can happen if a register's clock or Q pin is

Feedback

given instead of its D pin, which is the actual timing endpoint. It can also happen for a port if the port has no *set_output_delay* or other constraint.

**What Next**

Choose the correct endpoint pin, or add a constraint to this pin.

**See Also**

- report_timing
- get_timing_paths
- set_output_delay

---

## TIM-012

(Warning) Auto-inferred clock-gating checks for setup and hold violations will be disabled. Note that it's not consistent with PrimeTime's default behavior.

**Description**

Auto-inferred clock-gating checks on objects that have not also had a set_clock_gating_check command run against them will be disabled.

Note that it's not consistent with *PrimeTime*'s default behavior. But it has better correlation with *IC Compiler*.

**See Also**

- time.clock_gating_user_setting_only

---

## TIM-013

(Information) Checking for disabled timing arcs because of auto loop breaking for mode

**Description**

Timer automatically detects and breaks the loops in the timing paths. The loops are broken through disabling the timing arcs. If any such disabled arcs are found for a mode, these are reported with the recommendation to set the user constraint set_disable_timing before doing the timing update.

**What Next**

Set the user constraint set_disable_timing to disable these arcs to have consistent timing across the flow and between sessions.

Feedback

## TIM-014

(Warning) Use set_disable_timing [ get_timing_arcs -from { %s } -to { %s } ] to avoid loop in the timing path.

### Description

Timer automatically detects and breaks the loops in the timing paths. The loops are broken through disabling the timing arcs. If any such disabled arcs are found for a mode, these are reported with the recommendation to set the user constraint set_disable_timing before doing the timing update.

### What Next

Set the user constraint set_disable_timing to disable these arcs to have consistent timing across the flow and between sessions.

SH "SEE ALSO"

```
TIM-013
TIM-015
```

## TIM-015

(Information) %u arcs found disabled for mode

### Description

Timer automatically detects and breaks the loops in the timing paths. The loops are broken through disabling the timing arcs. If any such disabled arcs are found for a mode, these are reported with the recommendation to set the user constraint set_disable_timing before doing the timing update.

### What Next

Set the user constraint set_disable_timing to disable these arcs to have consistent timing across the flow and between sessions.

SH "SEE ALSO"

```
TIM-013
TIM-014
```

## TIM-016

(Information) The parasitics of virtually routed nets are automatically saved to RC Cache.

Feedback

### Description

By default the parasitics of virtual routed nets (nets which are unrouted) are not saved to RC Cache. However if majority of nets in the design are routed, then the parasitics of unrouted nets as determined based on virtual routing are also saved to RC Cache.

---

## TIM-017

(Information) The parasitics of virtually routed nets will be automatically saved to RC Cache during next timing update.

### Description

By default the parasitics of virtual routed nets (nets which are unrouted) are not saved to RC Cache. However the optimization clients of timer have forced the option to save the parasitics of virtually routed nets to RC Cache during next timing update.

---

## TIM-018

(Information) Auto loop-breaking disables the arc from '%s' to '%s' in mode '%s' which originally disabled by case-analysis.

### Description

Timer automatically detects and breaks the loops in the timing paths. Sometimes, to avoid aggressive optimization, auto loop-breaking still breaks the arc in the loops which originally has been disabled by case-analysis.

### What Next

No action needed.

SH "SEE ALSO"

```
TIM-013
TIM-014
```

---

## TIM-019

(Warning) The ignore threshold of composite aggressor is larger than per-aggressor filtering threshold. This may lead to over-filtering.

### Description

In composite aggressor mode, the per-aggressor threshold is the largest of *time.si_filter_per_aggr_noise_peak_ratio*, *time.si_xtalk_composite_aggr_noise_peak_ratio*, and *set_scenario_si_filtering*. The current ignore threshold determined by

Feedback

*time.si_xtalk_composite_aggr_ignore_noise_peak_ratio*, is larger than the per-aggressor threshold. It may lead to over-filtering so it's bounded by the per-aggressor threshold.

**What Next**

Check if the thresholds are expected, set per-aggressor filtering threshold and ignore threshold properly.

SH "SEE ALSO"

```
time.si_filter_per_aggr_noise_peak_ratio(3)
time.si_xtalk_composite_aggr_noise_peak_ratio(3)
time.si_xtalk_composite_aggr_ignore_noise_peak_ratio(3)
time.si_xtalk_composite_aggr_mode(3)
set_scenario_si_filtering(2)
```

---

## TIM-020

(warning) The option '%s' is ignored when specified with option '%s'.

**Description**

The listed command options are exclusive. If both are given, the first one will be ignored.

**What Next**

Look at the manpage for this command for more information on command options.

---

## TIM-021

(Warning) No timing paths pass through -through pin set %d.

**Description**

One of the -through pin sets does not contain any timed pins, so no timing paths pass through it. This will prevent commands such as report_timing or get_timing_paths from finding any paths.

**What Next**

Edit the command's -through options so that every -through set contains at least one timed pin, or remove the problematic -through set.

**See Also**

- report_timing
- get_timing_paths

Feedback

## TIM-022

(Information) MIS is not enabled in scenario %s because all clocks are ideal.

### Description

When MIS (multi-input-switching) feature is turned on, the hold timing could be significantly impacted. Before clock trees are synthesized and propagated, doing MIS analysis is not desired because the arrival window is inaccurate without accurate clock latency/CRPR, etc.

### What Next

You don't need to do anything, MIS will be automatically turned on when clocks are made propagated in later stage of the flow.

### See Also

- time.enable_multi_input_switching_analysis

## TIM-023

(error) Can't move cell %s, because %s

### Description

This message occurs during reparent_cells when moving the cell violates the following TIM constraints: (1) moving a dontTouch cell; (2) moving a cell with not-allowed sizeOnly reason The detailed reason is listed in this message.

### See Also

- reparent_cells

## TIM-030

(warning) The exclusivity set with '%s' option on the output pin '%s' is ignored, due to the violation of condition %d.

### Description

This warning message is issued when the exclusivity relationship defined on an output pin is not exercised due to one or more of the following conditions:

1. No clock reaches the output pin of the exclusivity from its inputs.
2. There is a clock reaching the select lines of the MUX exclusivity.

Feedback

3. All the select lines of the MUX exclusivity are defined as constants.
4. Only one input pin of the MUX exclusivity has clock(s).
5. The output pin of the MUX exclusivity is a clock source.
6. Maximum number of MUX levels exceeded.

For 6, set time.max_exclusive_mux_levels (default is 3) to desired value. Larger value can increase memory required. A value greater than 8 is not recommended.

WHAT NEXT Verify if the warning message for the pin is correct and expected.

---

## TIM-031

(Warning) The option '%s' is not supported and will be ignored.

**Description**

The given option is valid but not supported yet. The option will be ignored.

**What Next**

Use valid supported options, please check man pages of related commands.

---

## TIM-032

(Warning) The '%s' command is not supported and will be ignored.

**Description**

The given command is valid but not supported yet. The command will be ignored.

---

## TIM-034

(Information) Muxed clock exclusivity enabled for mode %s with %d exclusive pins.

**Description**

Muxed clock exclusivity enabled by either set_clock_exclusivity command or time.enable_auto_mux_clock_exclusivity app option.

**What Next**

Check reported information is as expected. If number of exclusive pins is too high then runtime will be adversely impacted. Reduce exclusive pins to improve runtime. If number of exclusive pins is zero then no valid exclusive clock muxes were found.

Feedback

## TIM-035

(Information) The set_clock_balance_points command, when using "-consider_for_balancing true", becomes scenario-specific and therefore expects either "-mode" or "-corner" or "-scenario" to be used. If none of these is specified then the balance point will be applied to the current_scenario.

### Description

Balance-delays are completely scenario specific and expects either "-mode" or "-corner" or "-scenario" options. In none of them is supplied, tool applies this command to current_scneario. In absence of current_scenario, it will issue an error message.

### What Next

Look at the manpage for this command for more information on command options.

## TIM-050

(Information) Timer using '%s'.

### Description

This message shows some timing related features that used in current IC Compiler-II. These features include Zero-interconnect delay calculation, PrimeTime delaly calculation, SI(Signal Integrity), Timing Window Analysis, AWP(Advanced Waveform Propagation), AOCV(Advanced On-Chip Variation), POCV (Parametric On-Chip Variation), CRPR(Clock Reconvergence Pessimism Removal), LLE (Local Layout Effect),

For LLE feature, LLE on means at least one cell has the following annotations: 1) physical parameters annotation 2) elelctrial parameters annotation 3) the corresponding pane of the its lib cell has LLE related table data.

## TIM-051

(warning) The option '%s' will be ignored because %s.

### Description

This warning denotes that the command option will be ignored because one or more prerequisites are not satisfied.

### What Next

The option will be ignored because it doesn't take effect in the current session settings. If this is not intended, please check if the values of other related options or app_options are appropriate.

Feedback

## TIM-052

(warning) Interpolation/Extrapolation resulted in a negative sigma. Sigma will be clipped at zero. ('%s')

### Description

The process of interpolation or extrapolation resulted in a negative sigma. By definition, the sigma must be greater than or equal to zero.

### What Next

Check the sigma tables.

### See Also

- time.pocvm_enable_analysis

## TIM-053

(Information) %s: the %s command will override the %s app option.

### Description

The command and the app option can affect same feature in ICC2 and the command has higher priority than the app option. This message shows the command will override the app option when the comamnd is set after the app option.

## TIM-054

(Information) %s: the %s command will take precedence over the %s app option.

### Description

The command and the app option can affect same feature in ICC2 and the command has higher priority than the app option. This message shows the command will take precedence over the app option when the app option is set after the command.

## TIM-055

(Error) %s not turned on for the scenario '%s'.

### Description

Both setup and hold analyses are required for getting timing window information in *write_rh_file*.

Feedback

### What Next

Check scenario status.

### See Also

- set_scenario_status

---

## TIM-056

(Error) rail analysis failed, cannot do timing update based on IR voltage drop. Will do regular timing update.

### Description

This error will only appear during timing update when time.enable_dvd_analysis is true but some settings for analyze_rail is not configured correctly.

### What Next

If you want to do timing update considering ir drop, check the error or warning message printed by analyze_rail before timing update and fix the related problems. If you want to do timing update without considering ir drop, just disable the time.enable_dvd_analysis app option.

### See Also

- analyze_rail

---

## TIM-101

(Warning) No physical information exists for design '%s'. Zero interconnect delay is used in timing analysis.

### Description

Either the physical information for the design is not loaded or it is incorrectly initialized. The net parasitics can not be calculated and the timer will use the zero interconnect delay for net estimation.

### What Next

Check the set up for the design. Check if the design was placed correctly and has valid core box defined.

Feedback

## TIM-102

(Warning) The extractor can not be initialized for design '%s'. Zero interconnect delay is used in timing analysis.

**Description**

Either the physical information for the design is not loaded or it is incorrectly initialized. The net parasitics can not be calculated and the timer will use the zero interconnect delay for net estimation.

**What Next**

Check the set up for the design. Check constraint corner's parasitic tech setup in run script. Run report_parasitic_parameters to make sure all corners are correctly set.

## TIM-103

(Warning) The extractor can not be initialized for block '%s'.

**Description**

Either the physical information for the block is not loaded or it is incorrectly initialized. The extractor for the top block will be used to estimate the nets within the block.

**What Next**

Check the set up for the block. Check if the design was placed correctly and has valid core box defined.

## TIM-104

(Information) Coupling caps annotated for %u nets in block '%s' (Mem = %.4f MB).

**Description**

Information about the coupling caps annotated from an abstract block for the nets retained in the abstract netlist. The annotated coupling caps are used in signal integrity (SI) analysis during the timing update. The message also prints the total memory requirement for the annotated coupling caps.

## TIM-105

(Warning) No physical information exists for design '%s'. Extractor is not initialized for this block.

Feedback

**Description**

Either the physical information for the design is not loaded or it is incorrectly initialized. If the top block has physical data, the extractor from the top block is used to estimate the nets in the sub-block. This may result in inaccurate timing for the nets inside the sub-block. If crosstalk analysis has been enabled for timing, the nets crossing the physical hierarchy into this block will have their coupling caps disabled and may encounter issues during parasitics stitching across the physical hierarchy.

**What Next**

Check the set up for the design. Check if the design was placed correctly and has valid core box defined.

---

## TIM-106

(Warning) The parasitic stitching across the physical hierachy is skipped for net '%s'.

**Description**

The tool failed to stitch the parasitics across the physical hierarchy for the net. This may either be related to the failure to calculate the parasitics for a sub-block net or dangling net connections. The timer will revert to flat net based estimation for the net and ignore the physical hierarchy. If crosstalk analysis is enabled, it is strongly advised to fix these warnings before proceeding. Otherwise the coupling on the net may be ignored.

**What Next**

Check the set up for the design. Check if the sub-blocks were placed correctly and have valid core box defined. Check if the net connections are defined properly.

---

## TIM-107

(Warning) The aggressor block net '%s' is not valid and is skipped from crosstalk analysis of victim net '%s'

**Description**

The victim net is coupled to an aggressor block net which is not present. This will happen because of dangling connections at the physical hierarchy.

**What Next**

Check the set up for the design. Check if the design has physical hierarchy and the net connections are defined properly.

Feedback

## TIM-108

(Warning) The aggressor '%s [flatNetId=%u]' has delay annotated and is skipped from crosstalk analysis of victim net '%s'.

### Description

The design has physical hierarchy such that net delays have been annotated on the aggressor net. If the delays are annotated on the net, then the base parastics are not stitched for the net. As such we cant stitch the coupling caps for this net in the coupled network.

### What Next

Check the set up for the design. If we created and linked an abstract for the block containing the aggressor net, check if there is need of annotating the delays on the aggressor net.

## TIM-109

(Warning) The aggressor '%s [flatNetId=%u]' is ideal and is skipped from crosstalk analysis of victim net '%s'.

### Description

The design has physical hierarchy and the aggressor net is defined as an ideal net. If the net is defined as ideal, then the base parastics are not stitched for the net. As such we can't stitch the coupling caps for this net in the coupled network.

### What Next

Check the set up for the design.

## TIM-110

(Warning) The aggressor '%s [flatNetId=%u]' is HFN and is skipped from crosstalk analysis of victim net '%s'.

### Description

The design has physical hierarchy and the aggressor net is a high fanout net (HFN). If the net is HFN, then the base parastics are not stitched for the net. As such we can't stitch the coupling caps for this net in the coupled network.

### What Next

Check the set up for the design. Take care of processing/synthesizing the HFN nets before turning on the crosstalk analysis.

Feedback

## TIM-111

(Information) Update timing completed net estimation for all the timing graph nets

**Description**

Full update timing was invoked on the design which led the tool to estimate all the nets in the timing graph. If any timing graph net is across the physical hierachy, the parasitics from the constituent sub-blocks nets are stitched before estimating the net.

## TIM-112

(Information) Net estimation statistics: timing graph nets = %u, routed nets = %u, across physical hierarchy nets = %u, parasitics cached nets = %u, delay annotated nets = %u, parasitics annotated nets = %u, multi-voltage nets = %u.

**Description**

Show the full net estimation statistics with respect to timing graph net count.

## TIM-113

(Warning) Incremental crosstalk analysis is limited to post route optimization for designs with physical hierarchy.

**Description**

When you perform crosstalk analysis on designs with physical hierarchy, the tool stitches together the coupling capacitances on nets across the physical hierarchy during a full timing update. Postroute optimization on these cross-physical-hierarchy nets also updates the coupling capacitances for various on-route optimization techniques such as buffer insertion and buffer deletion. Postroute optimization should not perform off-route optimization on these cross-physical-hierarchy nets and do the incremental timing update.

**What Next**

Check if crosstalk analysis has been accidently enabled in the flow before running postroute optimization. The *time.si_enable_analysis* application option controls crosstalk analysis.

## TIM-114

(Information) Crosstalk analysis has been disabled.

Feedback

**Description**

Crosstalk anslysis has been disabled because of one of the following reasons: 1. Coupling caps does not exist for the design after full extraction. 2. Top design is using zero interconnect delay mode.

**What Next**

Check if the design is actually routed. Check if crosstalk analysis has been turned ON during pre-route flow. Check if zero interconnect mode has been specified. Check the design set up to make sure that extractor can be initialized for top design. If user intention is not to do crosstalk analysis then disable the crosstalk analysis through app option time.si_enable_analysis.

---

## TIM-115

(Warning) Stale parasitics found in RC Cache for net connected to driver %s.

**Description**

The net connections in the timing graph and the stored parasitics are not consistent for the net. The timer will discard the stored parastics and do the virtual route based extraction for the net.

---

## TIM-116

(Warning) The net's leaf connection in the timing graph is missing in the net's parasitics.

**Description**

The net connections in the timing graph and the parasitics are not consistent for the net. The delay calculation is not performed for the net arc with missing driver or load and zero net delays are used.

---

## TIM-117

(Warning) Coupling caps are already annotated for %d nets in block '%s'.

**Description**

The coupling caps have already been annotated for the block. It is expected that earlier annotated coupling caps are removed before trying to reannotate the coupling caps for the same block.

**What Next**

See message TIM-104.n in the log file.

Feedback

## TIM-118

(Information) In-design Star RC extraction was done.

**Description**

Users have specified to use in-design Star RC extraction. Star RC extraction was invoked and completed successfully. The timing for the design will be based on parasitics calculated from Star RC extraction.

## TIM-119

(Information) The netlist change observers are disabled for incremental timing updates.

**Description**

Timer logs the incremental netlist changes if the netlist observers are enabled for incremental timing updates. By default these observers are always enabled. If netlist change observers are disabled, the optimization engine updates the timing graph explicitly corresponding to netlist changes before invoking the incremental timing update.

**What Next**

See also TIM-120, TIM-121

## TIM-120

(Information) The netlist change observers are enabled for incremental timing updates.

**Description**

Timer logs the incremental netlist changes after the netlist observers are enabled for incremental timing updates. By default these observers are always enabled. If netlist change observers are disabled, the optimization engine updates the timing graph explicitly corresponding to netlist changes before invoking the incremental timing update.

**What Next**

See also TIM-119, TIM-121

## TIM-121

(Information) The timing has been updated before disabling netlist and extraction change observers.

Feedback

**Description**

Timer and extractor logs the incremental netlist changes after the netlist and extraction observers are enabled for further incremental timing updates. By default these observers are always enabled. If netlist and extraction change observers are disabled, the optimization engine updates the timing graph and parasitics explicitly corresponding to netlist changes before invoking the incremental timing update.

**What Next**

See also TIM-119, TIM-120

---

## TIM-122

(Information) %u nets have annotated parasitics in block %s.

**Description**

The parasitics for the block nets are annotated from linked abstract block. If any block net in the linked abstract block does not have annotated parasitics, it will be estimated based on virtual routing.

---

## TIM-123

(Information) The net parasitics of block %s are cleared.

**Description**

The net parasitics for the block are stored in RC cache repository. The extraction engine stores the extracted parasitics for the routed nets in RC cache for the block. The unrouted nets are estimated based on virtual routing and are also stored in RC cache. If the timer clients forces a full timing update or the design state changes, both the timing delay data and the stored net parasitics are cleared. The net parasitics will be re-calculated (extracted) during next timing update.

---

## TIM-124

(Information) RDE mode is turned %s.

**Description**

Route driven estimate mode is changed in the flow. The design will undergo full extraction and the current timing delay data will be cleared.

Feedback

## TIM-125

(Information) The stitching and editing of coupling caps is turned %s for design '%s'.

**Description**

The extractor extract the coupling caps per physical block in the design. If SI analysis is turned on, coupling caps on the nets crossing the physical hierarchy are stitched on the parasitic network. The coupling caps are also edited during post route optimization flow corresponding to on-route optimization changes.

## TIM-126

(Information) The netlist change observers are disabled for incremental extraction.

**Description**

Extractor logs the incremental netlist changes if the netlist observers are enabled for incremental extraction. By default these observers are always enabled. If netlist change observers are disabled, the optimization engine updates the parasitics explicitly corresponding to netlist changes before invoking the incremental timing update.

**What Next**

See also TIM-119, TIM-120, TIM-121, TIM-127

## TIM-127

(Information) The netlist change observers are enabled for incremental extraction.

**Description**

Extractor logs the incremental netlist changes after the netlist observers are enabled for incremental extraction and timing updates. By default these observers are always enabled. If netlist change observers are disabled, the optimization engine updates the parasitics explicitly corresponding to netlist changes before invoking the incremental timing update.

**What Next**

See also TIM-119, TIM-121

## TIM-128

(Warning) The pin '%s' has overflow coordinates.

**Description**

The pin location is close to min/max integer. The virtual router will not run.

Feedback

**What Next**

Check the set up for the block. Check if the design was placed correctly.

---

## TIM-129

(Warning) For computing a common base period for a number of clocks ICC2 timer limits the waveform expansion of the smallest period to be no more than 1000 times and the waveform expansion of the largest period to be no more than 101 times. Since the largest period is too large compared to the smallest period, no common base period is possible satisfying these limits, and ICC2 timer has taken the largest period as the common base period but still has not expanded the smallest period beyond its limit. In certain situations, this can cause paths between these clocks to be unconstrained.

**Description**

In certain situations, this can cause paths between these clocks to be unconstrained.

---

## TIM-130

(Warning) There is no net conneted to term %s, skip it. This driver will be excluded from analyze_subcircuit analysis.

**Description**

Specify the term that has no nets connected to it, skip it

**What Next**

No action needed.

---

## TIM-131

(Warning) The pin %s is an internal pin, ignore it, and no measurement statement will be generated for simulation.

**Description**

Skip internal pins when tracing clock tree and data tree, not include them in SPICE simulation and analyze_subcircuit

**What Next**

No action needed.

Feedback

## TIM-132

(Warning) Moded graph is not initialized for constraint mode %s.

**Description**

Specify the constraint mode for which moded graph is not initialized

**What Next**

No action needed.

## TIM-200

(Warning) Update timing will revert to use native delay calculator.

**Description**

Even though the application option 'time.use_pt_delay' to use primetime delay has been turned on, timer can not use the primetime delay.

**What Next**

Check if the library DBs are available for the design with check_dbs command. Check if the design is routed. Check if the search path is corectly specified to find the library DB files.

## TIM-201

(Information) Update timing is using PrimeTime delay calculation.

**Description**

The application option 'time.use_pt_delay' to use primetime delay is a no-op. This app option will be deprecated soon.

## TIM-202

(Warning) DB file '%s' does not match NDM.

**Description**

The DB file being loaded is not same as the one used to generate the NDM. This inconsistency could result in issues during timing update with primetime delay calculation.

Feedback

**What Next**

Check if the DB file being loaded is correct. Regenerate NDM to make it consistent wth the available DBs.

---

## TIM-203

(Warning) The DB library object '%s' could not be found in NDM library.

**Description**

The DB libcell or libpin could not be found in the NDM library. This could be due to inconsistency between NDM and DB.

**What Next**

Check if the DB file being loaded is correct. Regenerate NDM to make it consistent wth the DBs.

---

## TIM-204

(Warning) Advanced receiver model has not been enabled for detailed routed design.

**Description**

Advanced reciever model has not been enabled for detailed routed design. For improving accuracy of delay calculation and to correlate with PrimeTime, it is recommended to set the application option 'time.enable_ccs_rcv_cap' to true.

**What Next**

Enable time.enable_ccs_rcv_cap for signoff accurate delay calculation.

---

## TIM-205

(Warning) Invalid user port delay or slew threshold setting, use default setting.

**Description**

The delay and slew threshold setting must meet some basic conditions: low slew threshold < 0.5, high slew threshold > 0.5 low slew threshold = 1 - high slew threshold low slew threshold < input delay threshold high slew threshold > input delay threshold Otherwise, this warning message will be issued and timing calculation will use default delay and slew threshold.

The delay and slew threshold settings are specified by following app options: time.port_input_threshold_fall time.port_input_threshold_rise

Feedback

time.port_slew_lower_threshold_rise time.port_slew_upper_threshold_rise time.port_slew_lower_threshold_fall time.port_slew_upper_threshold_fall

**What Next**

Check delay and delay threshold settings with report_app_options

---

## TIM-206

(Warning) A new advanced waveform propagation engine is available for better timing correlation. Set time.awp_compatibility_mode to false to enable the new engine.

**Description**

Starting from N-2017.09, a new AWP engine is available in ICC2. The new engine is more accurate for advanced nodes and ultra low voltage designs. It requires libraries built by M-2016.12-SP2 or later lm_shell. To switch to the new engine, use set_app_options -as_user_default -list {time.awp_compatibility_mode false}

**See Also**

- time.awp_compatibility_mode

---

## TIM-207

(Warning) Libraries do not have CCS noise model for accurate waveform analysis in advanced waveform propagation mode.

**Description**

When advanced waveform propagation analysis is enabled, it is required that the stage driver cell has CCS noise model(s) for its timing arcs in terms of either arc-based or pin-based CCS noise model(s). When such CCS noise model(s) are not present in the library, analysis with desired accuracy cannot be performed.

**What Next**

Fix the library to ensure that all library cell timing arcs have either arc-based or pin-based CCS noise models.

**See Also**

- time.delay_calc_waveform_analysis_mode
- time.awp_compatibility_mode

Feedback

## TIM-208

(Warning) Unexpected large derate value ecountered for arc delays. Timing results may not be accurate.

### Description

Derate value of equal to or larger than 2 has been specified. Delays calculated for hold paths may not be accurate.

### What Next

Check and fix the specified derate values.

## TIM-210

(Information) Update timing is using PrimeTime SI.

### Description

The application option 'time.use_pt_si' to use primetime crosstalk analysis has been turned on and timer is using the PrimeTime crosstalk analysis engine.

## TIM-214

(Information) Delta delay estimation has been turned on.

### Description

Timer estimates delta delays for global-routed designs when the signal integrity analysis has been enabled by using the *time.si_enable_analysis* app option. The delta delays are estimated and added to the net delays. Delta delay estimation is not done for the clock network.

### What Next

Please set *time.si_enable_analysis* to false to turn off the estimation of delta delays for global routed designs.

### See Also

- time.si_enable_analysis

## TIM-220

(Warning) High timing complexity detected at endpoint '%s' defined upon mode '%s'!

Feedback

**Description**

This message indicates design timing complexity is very high, at the problem endpoint, the mode total internal timing data number is huge and it exceeds one big threshold. This could result in very slow runtime.

**What Next**

Check the problem mode constraint definition at the problem endpoint, reduce number of constraints.

---

## TIM-260

(Warning) Library %s is pre-18.06 version %u.%u, Correlation with Primetime can be low.

**Description**

ICC2 Primetime delay calculation can work with reference library built by version later than 2018.06 only. If reference library is built before 2018.06, delay calculation can have poor corrleation with Primetime.

**What Next**

Rebuild the reference library with library manager version later than 2018.06, or turn on library configuration app option lib.configuration.update_timing_view_for_pt_delay

---

## TIM-261

(Warning) Library %s is pre-17.09 scaling library, Please switch to using define_scaling_lib_group with 18.06 libraries.\n"

**Description**

ICC2 Primetime delay scaling calculation can't work with the library manager scaled pane which is built by pre-17.09 library manager.

**What Next**

rebuild the reference library with 2018.06 or later version, use define_scaling_lib_group in ICC2 define the scaling library group.

---

## TIM-262

(Warning) Library %s has ETM mode: '%s' pane %d, library is not supported for pt_delay mode.

Feedback

**Description**

ICC2 Primetime delay scaling calculation can't work reference library created by pre-2018.06 library manager. It will revert to native delay calculation.

**What Next**

rebuild the reference library with 2018.06 or later version.

---

## TIM-263

(Error) Library %s pane %d is source pane for two scaling pane %d and %d

**Description**

One corner pane can be source pane of only one scaling pane. if user puts one db in two define_scaling_lib_group command, it will issue this message.

**What Next**

Check the given pane id in report_lib command, and correct user script define_scaling_lib_group command.

---

## TIM-351

(Warning) The hierarchical cell named '%s' that is being ungrouped has derate. This derate will be set on the lower level leaf cells if the leaf cell does not have a derate itself. This may cause timing inconsistency before and after ungrouping.

**Description**

By default, during ungroup, timing derate defined upon original hierarchical-cell will be copied to low-level leaf-cells.

If low-level leaf-cells already have some pre-defined derate settings, ungroup will not overwrite them.

**What Next**

No action needed.

---

## TIM-352

(Error) No replacement pins could be found to inherit constraints from constrained-pin '%s'!

Feedback

**Description**

During ungroup, timing-constraints defined upon hierarchical-pins/hierarchical-cell will be copied to corresponding replacement leaf-pins. However, for some reason, we can't find appropriate leaf-pins to receive the original timing-constraints.

**What Next**

Remove timing-constraints or enable ungroup buffer-adding by changing variable time.ungroup_allow_new_buffer as true.

---

## TIM-353

(Error) Can't create new-buffer to inherit constraints from constrained-pin '%s'!

**Description**

During ungroup, timing-constraints defined upon hierarchical-pins/hierarchical-cell will be copied to corresponding replacement leaf-pins. However, for some reason, we can't find appropriate leaf-pins to receive the original timing-constraints.

This error happens when tool tries to create some new-buffer which will be used to inherit timing-constraints. The error could be due to below reasons, 1) Tool can't find appropriate buffer-cell from library. 2) Tool hits internal fatal error when adding this new-buffer.

**What Next**

Check timing-constraints and make sure they have been set correctly. Check library files and make sure appropriate buffer-cells provided.

---

## TIM-354

(Error) Reject ungroup due to ('%s') defined upon '%s'!

**Description**

There are some timing-constraints which block the hierarchical-cell being ungrouped.

Some timing-constraints defined upon this hierarchical-cell could be lost if we continue to do ungroup.

**What Next**

Remove all timing-constraints which block the ungroup if you hope to continue.

---

## TIM-355

(Warning) Detect conflict constraints '%s'%sdefined upon '%s'%s.

Feedback

**Description**

There are conflict constraints defined upon some hierarchical cell/pins.

Tool will automatically resolve the conflict without the penalty of Qor-change.

**What Next**

No action needed.

---

## TIM-356

(Error) Constraints '%s' upon '%s' conflicts with constraints '%s' upon '%s'.

**Description**

There are conflict constraints defined upon two hierarchical cell/pins.

To avoid Qor change before/after ungroup, ungroup will stop once conflict constraints have been detected between original hierarchical-pin and target replacement leaf-pins.

**What Next**

To resolve the conflict, remove conflict constraints or enable 'time.ungroup_allow_new_buffer'.

---

## TIM-357

(Error) Cannot preserve timing constraints on IO pin '%s' during ungroup.

**Description**

Timing can't preserve the constraints defined upon IO hierarchical pin.

**What Next**

To resolve the error, you can remove the constraints upon IO pin.

---

## TIM-358

(Error) Reject ungroup due to unsupported constraints '%s' defined upon '%s'!

**Description**

Timing can't preserve the constraints defined upon specified objects due to unsupported constraints.

**What Next**

To resolve the error, you can remove those unsupported constraints before ungroup.

Feedback

## TIM-400

%sTIM-400%s (Warning) Timer query %s made when %s timer is out-of-date.

### Description

Timing analysis is outdated or non-deterministic because netlist changes are not consumed by the timer yet.

### What Next

Report this problem with error code ID and query information to Synopsys.

## TIM-666

(Warning) CCS pin capacitance data is not present in the '%s' reference library.

### Description

The tool issues this warning message when the timer is configured to use CCS pin capacitance data (the *time.enable_ccs_rcv_cap* application option is *true*) and the specified reference library contains incomplete CCS pin capacitance data. Incomplete CCS pin capacitance data occurs when the library source files contain CCS pin capacitance data but the reference library is created with a version of lm_shell prior to version K-2015.06-SP3.

### What Next

Recompile the specified reference library with version K-2015.06-SP3 or later of lm_shell.

### See Also

- time.enable_ccs_rcv_cap

## TIM-800

(Warning) Ignoring time.pba_optimization_mode setting 'none' as optimization already done with pba_mode '%s'.

### Description

Post route-optimization has been done with either path or exhaustive mode.
Default timing reports or subsequent optimization cannot be changed to GBA mode (time.pba_optimization_mode none). GBA timing does not provide any value once PBA optimization has been done and can lead to confusion and non-convergent optimization.

Feedback

**What Next**

Change the time.pba_optimization mode to either path or exhasitive.

---

## TIM-995

(Error) Timer unexpected state: '%s'

**Description**

Timer has some basic assuptions, if any assumption is violated, tool will issue this error message.

**What Next**

Report this problem with state information to Synopsys.

---

## TIM-998

(Error) Timer unexpected state: error code '%d', state information: '%s'

**Description**

Timer has some basic assuptions, if any assumption is violated, tool will issue this error message.

**What Next**

Report this problem with error code ID and state information to Synopsys.

---

## TIM-999

(Warning) Timer unexpected state: error code '%d', state information: '%s'

**Description**

Timer has some basic assuptions, if any assumption is violated, tool will issue this error message.

**What Next**

Report this problem with error code ID and state information to Synopsys.

Feedback

# TL

## TL-100

(error) Different instances of block reference %s are bound to different views. Skipping further checks on the Block.

### Description

Multiple Instantiated Block instances are bound to different views. For the TopLevel flow, all the Multiple Instantiated Block instances should be bound to one view of the block.

### What Next

Please change the view of the block instances using change_abstract/change_view so that they are bound to one view only.

### See Also

- change_abstract
- change_view

## TL-101

(error) Missing %s view for block reference %s in library %s.

### Description

The reported view for the design/abstract does not exist in the directory from which the design/abstract is referenced. The frame view of a block is expected to reside in the same directory as the view being referenced. The design view of a block is expected to reside in the same directory as the abstract view being referenced when the block constraints are missing as it may lead to issues during promote_clock_data.

### What Next

If the reported view exists in some other directory, copy it to the directory from where the design/abstract view is referenced.

For missing design view, you can optionally choose to tolerate the error by setting tolerate policy for *hier.block.abstract_missing_design_view* check using *set_early_data_check_policy* command. For missing frame view, you can optionally choose to tolerate or repair the error by setting tolerate or repair policy for *hier.block.missing_frame_view* check using *set_early_data_check_policy* command.

Feedback

- If tolerate policy is set, the severity of error will be reduced, converted into warning and flow will be allowed to proceed. Be aware of the possible flow implications due to the warning.
- If repair policy is set, tool will attempt to repair the error by creating frame view when *check_hier_design* command is invoked.

**See Also**

- check_hier_design
- promote_clock_data
- set_early_data_check_policy
- report_early_data_checks
- get_early_data_check_records

---

## TL-103

(error) Internal Consistency failure. All views for the block %s are not referenced from the same library %s.

**Description**

All views of a block should be present in the same directory.

**What Next**

Please copy all the views of the block to one directory so that the views may be consistent over the flow.

---

## TL-104

(warning) Frame view of block reference %s created using non-recommended options. Use 'create_frame -block_all [true | auto | used_layers]' to avoid flow runtime/memory penalty.

**Description**

The frame view for the block has been created without using the -block_all option or using the option with value other than the recommended values. This may lead to runtime/memory impact during the flow.

**What Next**

Please recreate frame view for the block using create_frame -block_all [true | auto | used_layers].

Feedback

**See Also**

- create_frame

---

## TL-105

(error) Order of port(s) %s is/are different in %s and %s view of design %s.

---

## TL-106

(error) Missing port(s) %s in %s.%s compared to %s.%s.

**Description**

There is a mismatch in the ports among the reported views of the same block.

**What Next**

The mismatch between abstract and design views may be resolved using merge_abstract. You may choose to recreate the abstract or frame views using create_abstract or create_frame commands respectively.

For port mismatch between abstract/design view and frame view, you can optionally choose to tolerate or repair the error by setting tolerate or repair policy for *hier.block.port_mismatch_between_views* check using *set_early_data_check_policy* command.

- If tolerate policy is set, the severity of error will be reduced, converted into warning and flow will be allowed to proceed. Be aware of the possible flow implications due to the warning.
- If repair policy is set, tool will attempt to repair the error by re-creating frame view from the bound view when *check_hier_design* command is invoked.

**See Also**

- check_hier_design
- merge_abstract
- create_abstract
- create_frame
- set_early_data_check_policy
- report_early_data_checks
- get_early_data_check_records

Feedback

## TL-107

(error) Abstract view of block reference %s created using non-recommended options. Can not be used for top-level timing analysis or optimization.

**Description**

The abstract linked in the design is not a timing abstract.

**What Next**

If you intend to do any timing analysis or optimization in the design, create the timing abstract using create_abstract.

**See Also**

- create_abstract

## TL-108

(error) Abstract view of block reference %s has not been created with implementation as target use. Cannot be used for top-level timing analysis or optimization.

**Description**

The abstract linked in the design is not targeted to be used for implementation.

**What Next**

If you intend to do any timing analysis or optimization in the design, create abstract with target use as implementation.

**See Also**

- create_abstract

## TL-109

(error) Missing port in block %s corresponding to block instance pin %s.

**Description**

The block is either inconsistent or corrupted.

**What Next**

Please merge or recreate the required view.

Feedback

You can optionally choose to tolerate or repair the error by setting tolerate or repair policy for *hier.block.reference_missing_port* check using *set_early_data_check_policy* command.

- If tolerate policy is set, the severity of error will be reduced, converted into warning and flow will be allowed to proceed. Be aware of the possible flow implications due to the warning.
- If repair policy is set, tool will attempt to repair the error by creating missing ports on the block reference when *check_hier_design* command is invoked.

**See Also**

- check_hier_design
- merge_abstract
- create_abstract
- set_early_data_check_policy
- report_early_data_checks
- get_early_data_check_records

---

## TL-110

(error) Missing pin on block instance %s (Reference: %s) corresponding to port %s.

**Description**

The block is either inconsistent or corrupted.

**What Next**

Please merge or recreate the required view.

**See Also**

- merge_abstract
- create_abstract

---

## TL-111

(error) Missing core area in block %s. Can not run top-level timing analysis and optimization.

Feedback

### Description

The block is missing core area. This may lead to incorrect timing analysis.

### What Next

Open the block.design and use the get_core_area command to confirm that core area is correctly defined. If floorplanning was done in IC Compiler II, review the floorplanning steps to understand the reasons for missing core area. Re-do floorplanning after resolving the same. If floorplan was input using ASCII like DEF, review the DEF to confirm that the core are definition is correct.

### See Also

- get_core_area
- create_abstract

---

## TL-112

(error) Block instance %s (Reference: %s) is bound to Frame view. Cannot be used for top-level optimization.

### Description

The block is linked to the frame view which excludes cell-internal information about the block.

### What Next

Please change the view of the block using change_view command.

You can optionally choose to tolerate the error by setting tolerate policy for *hier.block.instance_bound_to_frame* check using *set_early_data_check_policy* command. If tolerate policy is set, the severity of error will be reduced, converted into warning and flow will be allowed to proceed. Be aware of the possible flow implications due to the warning.

### See Also

- check_hier_design
- change_view
- set_early_data_check_policy
- report_early_data_checks

Feedback

## TL-113

(error) Power domain %s does not have associated voltage area.

**Description**

Every power domain is associated with atleast one Flat VA. The reported power domain does not have any flat VA associated with it.

**What Next**

Associate a voltage domain with the power domain.

## TL-114

(error) Internal Consistency Failure. Internal voltage area %s does not have a voltage area.

**Description**

Every hierBlk VA is associated with a flat VA. The reported hierBlk VA does not have an associated flat VA.

**What Next**

Please check the relevant voltage area in top-level and block-level by using the report_voltage_area command. Also, check for missing power domain (PD) definition in top-level VA. If found missing, recreate the voltage area with the correct PD definition. If the issue still persists, report this issue to Synopsys.

## TL-115

(error) Internal Consistency Failure. Voltage area %s does not have associated power domain.

**Description**

Every Flat VA should have atleast one PD associated with it. The reported flat VA does not have any Power Domain associated with it.

**What Next**

This is a severe error. Report this issue to Synopsys.

## TL-116

(error) Internal Consistency Failure. Internal voltage area %s does not have associated power domain.

Feedback

**Description**

Every hierBlk VA should have power domain associated with it. The reported hierBlk VA does not have a power domain associated with it.

**What Next**

This is a severe error. Report this issue to Synopsys.

---

## TL-117

(error) Internal Consistency Failure. Internal voltage area %s does not have associated voltage area region.

**Description**

Every hierBlk VA should have an associated hierBlk VA region associated with it. The reported hierBlk VA does not have an associated hierBlk VA region.

**What Next**

This is a severe error. Report this issue to Synopsys.

---

## TL-118

(error) Internal Consistency Failure. Voltage area %s does not have any internal voltage area.

**Description**

Every flat VA must have atleast one hierBlk VA associated with it.

**What Next**

This is a severe error. Report this issue to Synopsys.

---

## TL-119

(error) Top-level has estimated_corner. Can not be used for top-level optimization.

**Description**

The top design has an estimated corner. This will result in inaccurate timing data.

**What Next**

In case top has an estimated corner, please remove the estimated corner from top design.

Feedback

You can optionally choose to tolerate the error by setting tolerate policy for *hier.top.estimated_corner* check using *set_early_data_check_policy* command. If tolerate policy is set, the severity of error will be reduced, converted into warning and flow will be allowed to proceed. Be aware of the possible flow implications due to the warning.

**See Also**

- check_hier_design
- set_early_data_check_policy
- report_early_data_checks

---

## TL-120

(error) Abstract view of block reference %s created using estimated timing. Can not be used for top-level optimization.

**Description**

The abstract view of the block has an estimated corner. This will result in inaccurate timing data.

**What Next**

Please remove the estimated corner from the design view of the block and recreate the abstract using create_abstract.

**See Also**

- remove_abstract
- create_abstract

---

## TL-121

(warning) Internal Consistency Failure. Hierarchical connection %s (dir: %s) %s. %s

**Description**

The reported hierarchical pin has no driver and/or load.

**What Next**

This is a severe error. Report this issue to Synopsys.

Feedback

## TL-122

(warning) The lib %s is set as reference lib from different paths. Such duplicate settings may lead to possible runtime/memory penalty in the flow. %s

### Description

The lib is set as a reference library in the designs from different paths.

### What Next

Remove the conflicting settings for the reference libraries.

### See Also

- set_ref_libs
- report_ref_libs

## TL-123

(error) The lib %s is set as reference lib from different paths. %s

### Description

The lib is set as a reference library in the designs from different paths.

### What Next

Remove the conflicting settings for the reference libraries.

### See Also

- set_ref_libs
- report_ref_libs

## TL-124

(Warning) Voltage area %s does not have an effective voltage area region.

### Description

The reported VA does not have an effective voltage area region. One possibility is that there may be other VA regions higher in the stacking order which may be reducing the effective region to zero.

Feedback

### What Next

Check the stacking order of the VA regions.

---

## TL-126

(error) Missing physical location for %s %s of block reference %s.

### Description

The pins of the block are not placed.

### What Next

Place the pins of the block by running place_pins command on the block. Use command run_block_compile_pg to place the power and ground pins of the block.

You can optionally choose to tolerate or repair the error by setting tolerate or repair policy for *hier.block.reference_missing_port_location* check using *set_early_data_check_policy* command.

- If tolerate policy is set, the severity of error will be reduced, converted into warning and flow will be allowed to proceed. Be aware of the possible flow implications due to the warning.
- If repair policy is set, tool will attempt to repair the error by placing the signal ports on the block boundary when *check_hier_design* command is invoked.

### See Also

- place_pins
- run_block_compile_pg
- set_early_data_check_policy
- report_early_data_checks
- get_early_data_check_records

---

## TL-127

(error) %s %s of block reference %s is outside the block boundary.

### Description

The location of the ports of the block are outside the block boundary.

Feedback

### What Next

Use place_opt or create_placement command to place all the ports inside the block boundary.

You can optionally choose to tolerate or repair the error by setting tolerate or repair policy for *hier.block.reference_port_outside_boundary* check using *set_early_data_check_policy* command.

- If tolerate policy is set, the severity of error will be reduced, converted into warning and flow will be allowed to proceed. Be aware of the possible flow implications due to the warning.
- If repair policy is set, tool will attempt to repair the error by re-assigning the location for signal ports on the block boundary when *check_hier_design* command is invoked.

### See Also

- check_hier_design
- create_placement
- place_opt
- set_early_data_check_policy
- report_early_data_checks
- get_early_data_check_records

---

## TL-128

(error) Missing location of leaf cell %s inside block reference %s.

### Description

The leaf cells inside the block are not placed.

### What Next

Please run placement on the block using the create_placement command.

You can optionally choose to tolerate or repair the error by setting tolerate or repair policy for *hier.block.missing_leaf_cell_location* check using *set_early_data_check_policy* command.

Feedback

- If tolerate policy is set, the severity of error will be reduced, converted into warning and flow will be allowed to proceed. Be aware of the possible flow implications due to the warning.
- If repair policy is set, tool will attempt to repair the error by assigning a location for the leaf cell inside the block boundary when *check_hier_design* command is invoked.

**See Also**

- check_hier_design
- create_placement
- set_early_data_check_policy
- report_early_data_checks
- get_early_data_check_records

---

## TL-129

(error) Leaf cell %s inside block reference %s is outside block boundary.

**Description**

The leaf cells inside the block are placed outside the boundary of the block.

**What Next**

Use place_opt or create_placement command to place all the cells inside the block boundary.

You can optionally choose to tolerate or repair the error by setting tolerate or repair policy for *hier.block.leaf_cell_outside_boundary* check using *set_early_data_check_policy* command.

- If tolerate policy is set, the severity of error will be reduced, converted into warning and flow will be allowed to proceed. Be aware of the possible flow implications due to the warning.
- If repair policy is set, tool will attempt to repair the error by re-assigning leaf cell location to the nearest location inside block boundary when *check_hier_design* command is invoked.

**See Also**

- check_hier_design
- create_placement

Feedback

- place_opt
- set_early_data_check_policy
- report_early_data_checks
- get_early_data_check_records

---

## TL-130

(error) Missing location of block instance %s (Reference: %s) in parent block reference %s.

### Description

The block instance in the given parent block reference is not placed.

### What Next

Please take care of block placement at parent level in Design Planning step ( shape_block ).

For missing child block instance location in sub-blocks, you can optionally choose to tolerate or repair the error by setting tolerate or repair policy for *hier.block.missing_child_block_instance_location* check using *set_early_data_check_policy* command.

- If tolerate policy is set, the severity of error will be reduced, converted into warning and flow will be allowed to proceed. Be aware of the possible flow implications due to the warning.
- If repair policy is set, tool will attempt to repair the error by assigning a location for child block instance based on connectivity inside block boundary when *check_hier_design* command is invoked.

Please note that location missing for direct child block instances of top shall not be tolerated or repaired. You need to manually handle it.

### See Also

- check_hier_design
- create_placement
- set_early_data_check_policy
- report_early_data_checks
- get_early_data_check_records

Feedback

## TL-131

(error) Block instance %s (Reference: %s) in parent block reference %s is outside the parent block boundary.

### Description

The block instances are outside the boundary of their parent block.

### What Next

If only small portion of block boundary is outside the top boundary , user can manually pull it inside the top boundary , else take care of block placement inside the parent core boundary in Design Planning Step ( shape_block ).

For location errors of child block instances in sub-blocks, you can optionally choose to tolerate or repair the error by setting tolerate or repair policy for *hier.block.child_block_instance_outside_boundary* check using *set_early_data_check_policy* command.

- If tolerate policy is set, the severity of error will be reduced, converted into warning and flow will be allowed to proceed. Be aware of the possible flow implications due to the warning.
- If repair policy is set, tool will attempt to repair the error by re-assigning the child block instance location to nearest location inside parent block boundary when *check_hier_design* command is invoked.

Please note that location errors for direct child block instances of top shall not be tolerated or repaired. You need to manually handle it.

### See Also

- check_hier_design
- create_placement
- place_opt
- set_early_data_check_policy
- report_early_data_checks
- get_early_data_check_records

## TL-132

(Warning) Missing physical location for %s %s of block reference %s.

Feedback

**Description**

The PG pins of the block are not placed.

**What Next**

Use command run_block_compile_pg to place the power and ground pins of the block.

**See Also**

- run_block_compile_pg

---

## TL-133

(Warning) %s %s of block reference %s is outside the block boundary.

**Description**

The location of the PG ports of the block are outside the block boundary.

**What Next**

Use command run_block_compile_pg to place the power and ground pins of the block.

**See Also**

- run_block_compile_pg

---

## TL-134

(error) Block instance %s (Reference: %s) has design type macro. Cannot be used for top-level optimization.

**Description**

Blocks created in IC Compiler II usually have 'module' design type. The block has design type macro, which excludes cell-internal information about the block.

**What Next**

(1) Since all blocks created in IC Compiler II have design type as 'module', check how this block was created (2) To continue the flow, set the design type of the block to 'module' or 'blackbox' as required.

You can optionally choose to tolerate the error by setting tolerate policy for *hier.block.instance_with_design_type_macro* check using *set_early_data_check_policy* command. If tolerate policy is set, the severity of error will be reduced, converted into warning and flow will be allowed to proceed. Be aware of the possible flow implications due to the warning.

Feedback

**See Also**

- check_hier_design
- set_attribute
- create_abstract
- set_early_data_check_policy
- report_early_data_checks

---

## TL-135

(error) Top design has a name conflict with the sub-block %s. This usage is not suppoted.

**Description**

The Top design has the same name as the sub-blocks. Current naming style is not supported.

**What Next**

Resolve the name conflict. If you are using the same name to indicate the stage of the design (for eg. place_opt), use labels for that purpose.

---

## TL-136

(error) Block instance %s (Reference: %s) is not linked to Top.

**Description**

Blocks instance is not linked to the Top design. This may happen if change_view is called for the sub-block. In check_hier_design command, automatic repair of this issue happens so that all further checks in the block shall be successfully performed. Tool does not automatically repair this issue from inside mega commands since doing so can potentially lead to runtime data issues.

**What Next**

Run link_block -force to link the block to the Top.

**See Also**

- link_block

Feedback

## TL-140

(Error) Logical loop detected. Output pin '%s' is driving input pin '%s' of the same physical hierarchy instance '%s'.

### Description

There is a logical loop as detailed in the message. These type of logical loops are not supported.

### What Next

Modify the design to remove such logical loops.

## TL-151

(information) In mode %s, balance points are updated for physical hierarchy pin %s:%s.

### Description

The mentioned balance point(s) are newly created or updated on the given boundary clock pins of physical hierarchy based on the latency inside the block. This will help in building clock tree in top-level to the boundary clock pins of CCD done blocks.

### See Also

- promote_clock_data
- report_clock_balance_points

## TL-152

(warning) In mode %s, no new balance point setting is made on pin %s. %s

### Description

No new balance point setting is made on the given boundary clock pin of the block instance. This may be because block did not have the port latency data for the given pin. Latency inside the block may not be used in top-level clock tree building.

If the given block boundary pin is connected to a block feedthrough net, no port latecy data will be promoted from the block since it can otherwise lead to inaccuracies and also ignoring clock tree on the output side of the feedthrough in top-level.

Feedback

### What Next

Check for the clock feedthrough nets in the block level or whether there are any sinks present from the fanout of the clock port. If required, proceed by manually setting the balance point on given boundary clock pin of the block.

### See Also

- promote_clock_data
- report_clock_balance_points

---

## TL-153

(warning) In mode %s, no new balance point setting is made on any of the boundary clock pins of block instance %s.

### Description

No new balance point setting is made on any of the boundary clock pins of the given block instance. This may be because block did not have the port latency data for the given block in the given mode. Either *compute_clock_latency* command was not run in the block-level on this mode or all clock ports of the block are connected to feedthrough nets.

This may also occur if port latencies are already promoted. The promoted data can be checked by using *report_clock_balance_points* command.

### What Next

Check if port latencies are already promoted. Check for clock feedthrough nets. Otherwise, activate required modes in the block , run *compute_clock_latency* command in the block and save the block. This will help create the necessary port latency data in the block. Then call *promote_clock_data -port_latency* in the top-level.

### See Also

- compute_clock_latency
- promote_clock_data
- report_clock_balance_points

---

## TL-154

(warning) Mode specific port latencies are being promoted from sub-block instance %s, the corner data can possibly be inconsistent.

Feedback

### Description

Port latencies being promoted are mode specific which are generated in older versions (before T-2022.03-SP3). The promoted data may have inconsistencies with respect to the corners especially when there are mismatches between the top-level and block-level corners.

### What Next

Skip saving or remove these promoted clock balance points at the top-level. At the block-level, re-link the design with link_block and save with save_block in T-2022.03-SP3 or later release. Then promote the port latencies afresh at top-level. This will ensure the promoted port latencies are scenario specific and have no inconsistencies with respect to the corners.

### See Also

- compute_clock_latency
- promote_clock_data
- report_clock_balance_points

---

## TL-156

(error) Cannot promote data for instance %s since it is neither design view nor abstract view.

### Description

Requested data cannot be promoted for the given physical hierarchy instance since the reference of the instance is neither design view nor abstract view. The *promote_block_data* command can be used to promote data only from design view or abstract view.

### What Next

Link to appropriate view - design or abstract and proceed with the promotion of data.

### See Also

- report_abstracts

---

## TL-157

(error) Cannot promote data for instance %s since the abstract view is not optimizable.

Feedback

**Description**

Requested data cannot be promoted for the given physical hierarchy instance since the abstract view is not optimizable. The *promote_block_data* command command needs to be run after *init_interface_optimization* command in top linked with abstracts flow.

**What Next**

Rerun *promote_block_data* command after *init_interface_optimization* command.

**See Also**

- report_abstracts

---

## TL-158

(information) Promoting dont_touch/size_only for instance '%s'.

**Description**

Dont_touch and size_only constraints are being promoted from the given lower-level instance to the top-level design.

---

## TL-159

(Error) Promoting dont_touch/size_only for instance '%s' failed.

**Description**

An error occured while promoting dont_touch/size_only constraints from the lower-level instance to the top-level design. The data has not been promoted or is only partially promoted.

**What Next**

Check the error messages that preceded this one for more details.

---

## TL-160

(Error) Invalid config name %s specified.

**Description**

Config name specified is invalid. The pre-defined configs available are default and auto_repair.

Feedback

**What Next**

Re-run command by specifying the correct config name.

---

## TL-161

(Error) Invalid check name %s specified.

**Description**

Check name specified is invalid. Command supports only pre-defined list of checks. Refer command manpage or *report_hier_check_policy* command output for list of supported checks.

**What Next**

Re-run command by specifying the correct check name.

---

## TL-162

(Error) Unsupported policy specified for check %s.

**Description**

Policy specified for the given check is not supported. Refer command manpage or *report_hier_check_policy* command output for list of supported checks and policy.

**What Next**

Re-run command by specifying appropriate policy name.

---

## TL-163

(Error) Skipping invalid reference name %s for %s check(s).

**Description**

Reference name specified is invalid. Reference name must be valid physical hierarchy name or top design name as relevant to the check. Command will fail for the invalid reference specified.

**What Next**

Re-run command by specifying the correct reference name.

---

## TL-164

(Error) Repair policy currently not supported.

Feedback

**Description**

Repair policy is not supported in the current release.

**What Next**

Re-run command by specifying supported policy only or fix issue manually.

---

## TL-165

(Error) Skipping lower most sub-block reference %s for %s check.

**Description**

Checks top_level.missing_sub_block_instance_location and top_level.sub_block_instance_location_outside_boundary can only be performed on block references which have lower-level sub-block instances.

**What Next**

Re-run command by specifying the appropriate reference name.

---

## TL-170

(Error) Repair for %s will not be attempted on sub-block %s because %s.

**Description**

The given block cannot be repaired for given check. There can be several reasons for this. First, editability might not be enabled on the block on which repair had to be performed. Second, library containing the block might not be opened for edit. Third, library containing the block might not be writable. Lastly, the bound view might either be an editable abstract or a read-only design. This restriction on the bound view type is taken into consideration to avoid issues/conflicts in the forthcoming merge.

**What Next**

Re-run check_hier_design command after performing the suggested actions listed below depending on the type of situation encountered.

1. Enable the editability on the blocks on which repair has to be performed using set_editability command. 2. Use -edit or -ref_libs_for_edit of open_lib or open_block to open the library in edit mode. 3. Check file access permissions and make the library containing the blocks writable. 4. Change the bound view to either a read-only abstract or an editable design.

Feedback

### See Also

- set_editability
- check_hier_design
- open_lib
- open_block
- create_abstract
- change_abstract

---

## TL-171

(Error) Repair related operations on parent block %s required due to repair of %s check in child block %s cannot be performed because %s.

### Description

Some of the checks might require parent repair when the child block is repaired. But the parent block could not be repaired. There can be several reasons for this. First, editability might not be enabled on the parent block on which repair had to be performed. Second, library containing the parent block might not be opened for edit. Third, library containing the parent block might not be writable. Lastly, the bound view of parent block might either be an editable abstract or a read-only design. This restriction on the bound view type is taken into consideration to avoid issues/conflicts in the forthcoming merge.

### What Next

Re-run check_hier_design command after performing the suggested actions listed below depending on the type of situation encountered.

1. Enable the editability on the parent block on which repair has to be performed using set_editability command. 2. Use -edit or -ref_libs_for_edit of open_lib or open_block to open the library in edit mode. 3. Check file access permissions and make the library containing the parent block writable. 4. Change the bound view to either a read-only abstract or an editable design.

### See Also

- set_editability
- check_hier_design
- open_lib
- open_block

Feedback

- create_abstract
- change_abstract

---

## TL-180

(Information) Current config is set to %s.

**Description**

Indicates the current working hierarchical check congfig.

---

## TL-181

(Information) The policy settings of the config include user modified values. Use ‘set_hier_check_policy -reset’ if the pre-defined policy settings of the config are to be restored.

**Description**

Policy settings of the current working config have values different (due to user modifications) from that of the predefined policy settings for the config.

**What Next**

If this is not intended, use -reset option of set_hier_check_policy command to revert back the policy values to the predefined settings.

---

## TL-182

(Warning) Unsupported policy specified for check %s.

**Description**

Policy specified for the given check is not supported. Policy setting remains unchanged for the check. Refer command manpage or *report_hier_check_policy* command output for list of supported checks and policy.

**What Next**

Re-run command by specifying appropriate policy name.

---

## TL-200

(error) Command has modified (%s) the readonly block '%s'. Report this issue to Synopsys with below details.

Feedback

### Description

This error message is issued when a readonly block is being modified.

### What Next

This is a severe error. Report this issue to Synopsys.

---

## TL-220

(Warning) RDE data missing for sub-blocks %s. In Top level RDE flow, it is recommended to have RDE for all blocks once TOP has RDE.

### Description

This warning message is printed when TOP has RDE data but one or more sub blocks do not have RDE data available. This might be because of an oversight. If this message is issued, it is recommended to enable RDE during place_opt -from final_opto at block level, save the block and then use this block in TOP. Block's RDE data will be used in TOP only if TOP RDE is available/generated.

If block RDE is not available, then tool will use VR for blocks during net estimation. If user is fine with this, then no action is needed.

### What Next

If block RDE is needed, run place_opt -from final_opto at block level with RDE enabled.

---

## TL-221

(Warning) RDE data missing for blocks %s. In Top level RDE flow, it is recommended to have RDE for all blocks including TOP.

### Description

This warning message is printed when some blocks/TOP have RDE data and some blocks/TOP do not have RDE data available. For blocks/TOP with missing RDE, either they are fully abutted or it might be because of an oversight. If this message is issued and block is not fully abutted, it is recommended to enable RDE during place_opt -from final_opto at block level, save the block and then use this block in TOP.

If block RDE is not available, then tool will use VR for blocks during net estimation. If user is fine with this, then no action is needed.

### What Next

If block RDE is needed, run place_opt -from final_opto at block level with RDE enabled.

Feedback

## TL-300

(error) The abstract of block %s has constraints information that is out of date.

**Description**

This error message is issued when a timing abstract does not contain timing constraints information or contains timing constraint information that is out of date (possibly because it was created using an earlier version of the tool).

**What Next**

Create the abstract again using the latest executable.

## TL-301

(error) Cannot find information about block constraints for block %s

**Description**

This error message is issued when the command check_hier_design is used to check the consistency of constraints between top and block and there is no information about the block constraints associated with the abstract of a block.

**What Next**

Use the command create_abstract to create a new abstract for the blocks.

## TL-302

(error) Cannot find mapping of top level mode %s in block %s

**Description**

This error message is issued when a mode that is defined at the top-level does not have a corresponding mode in the block.

**What Next**

Use the command set_block_to_top_map to associate the top level mode to a mode defined in the block.

## TL-303

(error) Cannot find mapping of top level corner %s in block %s

Feedback

**Description**

This error message is issued when a corner that is defined at the top-level does not have a corresponding corner in the block.

**What Next**

Use the command set_block_to_top_map to associate the top level corner to a corner defined in the block.

---

## TL-304

(error) Cannot find mapping of top level scenario %s in block %s

**Description**

This error message is issued when a scenario that is defined at the top-level does not have a corresponding scenario in the block.

**What Next**

Use the command set_block_to_top_map to associate the top level corner/mode to a corner/mode defined in the block.

---

## TL-305

(error) The leaf cell %s in the block reference %s is unmapped.

**Description**

The leaf cell in the block is unmapped.

**What Next**

Please go through the appropriate synthesis flow to avoid any unmapped netlist.

---

## TL-310

(Error) Skipping invalid reference name: %s.

**Description**

Reference name specified is invalid. Reference name must be valid physical hierarchy name or top design name. check_hier_design command will fail for the invalid reference specified.

**What Next**

Re-run check_hier_design command by specifying the correct reference name.

Feedback

**See Also**

- check_hier_design

---

## TL-401

(warning) Missing %s view for block reference %s in library %s.

**Description**

The reported view for the design/abstract does not exist in the directory from which the design/abstract is referenced. It is expected that all the views of one block reside in the same directory.

**What Next**

If the reported view exists in some other directory, copy it to the directory from where the design/abstract view is referenced.

---

## TL-406

(warning) Missing port(s) %s in %s.%s compared to %s.%s.

**Description**

There is a mismatch in the ports among the reported views of the same block.

**What Next**

The mismatch between abstract and design views may be resolved using merge_abstract. You may choose to recreate the abstract or frame views using create_abstract or create_frame commands respectively.

**See Also**

- merge_abstract
- create_abstract
- create_frame

---

## TL-409

(warning) Missing port in block %s corresponding to block instance pin %s.

**Description**

The block is either inconsistent or corrupted.

Feedback

**What Next**

Please merge or recreate the required view.

---

## TL-410

(warning) Missing pin on block instance %s (Reference: %s) corresponding to port %s.

**Description**

The block is either inconsistent or corrupted.

**What Next**

Please merge or recreate the required view.

---

## TL-412

(warning) Block instance %s (Reference: %s) is bound to Frame view. Cannot be used for top-level optimization.

**Description**

The block is linked to the frame view which excludes cell-internal information about the block.

**What Next**

Please change the view of the block using change_view command.

**See Also**

- change_view

---

## TL-419

(warning) Top-level has estimated_corner. Can not be used for top-level optimization.

**Description**

The top design has an estimated corner. This will result in inaccurate timing data.

**What Next**

In case top has an estimated corner, please remove the estimated corner from top design.

Feedback

## TL-420

(warning) Abstract view of block reference %s created using estimated timing. Can not be used for top-level optimization.

**Description**

The abstract view of the block has an estimated corner. This will result in inaccurate timing data.

**What Next**

Please remove the estimated corner from the design view of the block and recreate the abstract using create_abstract.

**See Also**

- remove_abstract
- create_abstract

## TL-426

(warning) Missing physical location for %s %s of block reference %s.

**Description**

The pins of the block are not placed.

**What Next**

Place the pins of the block by running place_pins command on the block. Use command run_block_compile_pg to place the power and ground pins of the block.

**See Also**

- place_pins
- run_block_compile_pg

## TL-427

(warning) %s %s of block reference %s is outside the block boundary.

**Description**

The location of the ports of the block are outside the block boundary.

Feedback

**What Next**

Use place_opt or create_placement command to place all the ports inside the block boundary.

**See Also**

- create_placement
- place_opt

---

## TL-428

(warning) Missing location of leaf cell %s inside block reference %s.

**Description**

The leaf cells inside the block are not placed.

**What Next**

Please run placement on the block using the create_placement command.

---

## TL-429

(warning) Leaf cell %s inside block reference %s is outside block boundary.

**Description**

The leaf cells inside the block are placed outside the boundary of the block.

**What Next**

Use place_opt or create_placement command to place all the cells inside the block boundary.

---

## TL-430

(warning) Missing location of block instance %s (Reference: %s) in parent block reference %s.

**Description**

The block instance in the given parent block reference is not placed.

**What Next**

Please take care of block placement at parent level in Design Planning step ( shape_block ).

Feedback

## TL-431

(warning) Block instance %s (Reference: %s) in parent block reference %s is outside the parent block boundary.

### Description

The block instances are outside the boundary of their parent block.

### What Next

If only small portion of block boundary is outside the top boundary , user can manually pull it inside the top boundary , else take care of block placement inside the parent core boundary in Design Planning Step ( shape_block ).

### See Also

- create_placement
- place_opt

## TL-434

(warning) Block instance %s (Reference: %s) has design type macro. Cannot be used for top-level optimization.

### Description

Blocks created in IC Compiler II usually have 'module' design type. The block has design type macro, which excludes cell-internal information about the block.

### What Next

(1) Since all blocks created in IC Compiler II have design type as 'module', check how this block was created (2) To continue the flow, set the design type of the block to 'module' or 'blackbox' as required.

### See Also

- set_attribute
- create_abstract

## TL-451

(warning) Inconsistent value(s) of app option in block abstract(s) and top design '%s'. Details are in the following table.

Feedback

**Description**

There is an inconsistency in app option value(s) in block abstract(s) and top design.

**What Next**

The inconsistency could be avoided by setting the correct app option value in both the views.

---

## TL-452

(warning) '%s' app option name is disabled/invalid in block '%s'. Will not be checked for inconsistency.

**Description**

App option is disabled/invalid. Will be skipped for the inconsistency check.

**What Next**

The app option name in the file provided needs to be corrected.

---

## TL-501

(information) Frame view for block reference %s in library %s was found missing. Repaired and created new frame for the block.

**Description**

Frame view for the design/abstract did not exist in the directory from which the design/abstract is referenced. New frame has been created.

---

## TL-506

(information) Port(s) %s were found missing in %s.%s compared to %s.%s. Repaired and created new frame for the block.

**Description**

There was a mismatch in the ports among the reported views of the same block. Repaired and created new frame for the block.

Feedback

**See Also**

- merge_abstract
- create_abstract
- create_frame

## TL-509

(Information) New Port %s created on block %s corresponding to block instance pin(s) %s.

**Description**

Port was found missing in physical hierarchy reference corresponding to the pins on the physical hierarchy instance(s). New port was created on the physical hierarchy reference.

## TL-526

(information) Location of port %s in block reference %s was found missing. Repaired and updated the location of the port to %d , %d.

**Description**

The port location has been updated with respect to the block.

**What Next**

Please run check_hier_design command to make sure that location is updated correctly.

## TL-527

(information) Location of port %s of block reference %s was found outside the block boundary. Repaired and updated the location of the port to %d , %d.

**Description**

The port location has been updated with respect to the block.

**What Next**

Please run check_hier_design command to make sure that location is updated correctly.

## TL-528

(information) Location of leaf instance %s inside block reference %s was found missing. Repaired and updated the location coordinates of the instance to X: %d Y: %d.

Feedback

**Description**

The instance location has been updated with respect to the block.

**What Next**

Please run check_hier_design command to make sure that location is updated correctly.

---

## TL-529

(information) Location of leaf cell %s inside block reference %s was found outside the boundary. Repaired and updated the location coordinates of the instance to X: %d Y: %d.

**Description**

The instance location has been updated with respect to the block.

**What Next**

Please run check_hier_design command to make sure that location is updated correctly.

---

## TL-530

(information) Location of block instance %s in parent block reference %s was found missing. Repaired and updated the location coordinates of the instance to X: %d Y: %d.

**Description**

The instance location has been updated with respect to the block.

**What Next**

Please run check_hier_design command to make sure that location is updated correctly.

---

## TL-531

(information) Location of block instance %s in parent block reference %s was found outside the parent boundary. Repaired and updated the location coordinates of the instance to X: %d Y: %d

**Description**

The instance location has been updated with respect to the block.

**What Next**

Please run check_hier_design command to make sure that location is updated correctly.

Feedback

## TL-551

(information) New frame created for block reference %s to honor the %s repair performed on the block.

### Description

Frame created for the block that was repaired either for location related error or missing port error.

### See Also

- create_frame
- TL-526
- TL-527
- TL-530
- TL-531
- TL-509

## TL-552

(information) A new frame was created. User will need to save the library %s.

### Description

New frame created for the block. Original version of the frame, if any existed, has been cleared. You will need to save the library for this change to be saved.

### See Also

- create_frame
- TL-526
- TL-527
- TL-530
- TL-531
- TL-509

Feedback

## TL-555

(information) Since '-report_repair_feasibility' option is given, only checking for issues and repair feasibility reporting will happen. Tolerate/repair action will not be attempted.

### Description

'-report_repair_feasibility' option of check_hier_design command only prints information on feasibility of repair to be performed on each of the sub-block reference with relevant reasons. Tolerate or repair action will not be performed.

## TL-601

(error) Missing frame view for block reference %s in library %s. Attempt to repair failed.

### Description

The reported view for the design/abstract does not exist in the directory from which the design/abstract is referenced. It is expected that all the views of one block reside in the same directory. Repair was tried to create a new frame, but it didnot succeed.

### What Next

If the reported view exists in some other directory, copy it to the directory from where the design/abstract view is referenced.

## TL-606

(error) Missing port(s) %s in %s.%s compared to %s.%s. Attempt to repair failed.

### Description

There is a mismatch in the ports among the reported views of the same block. Repair was tried to create a new frame, but it didnot succeed.

### What Next

The mismatch between abstract and design views may be resolved using merge_abstract. You may choose to recreate the abstract or frame views using create_abstract or create_frame commands respectively.

### See Also

- merge_abstract
- create_abstract
- create_frame

Feedback

## TL-609

(error) Port %s was found missing in block %s corresponding to block instance pin(s) %s. Attempt to create port failed.

### Description

Port was found missing in physical hierarchy reference corresponding to the pins on the physical hierarchy instance(s). Attempt to create port on the physical hierarchy reference failed.

## TL-626

(error) Missing physical location for port %s of block reference %s. Attempt to repair location failed.

### Description

The port location could not be updated with respect to the block.

### What Next

Place the pins of the block by running place_pins command on the block. Use command run_block_compile_pg to place the power and ground pins of the block.

### See Also

- place_pins
- run_block_compile_pg

## TL-627

(error) Location of port %s of block reference %s was found outside the block boundary. Attempt to repair location failed.

### Description

Port location could not be updated with respect to the block.

### What Next

Use place_opt or create_placement command to place all the ports inside the block boundary.

Feedback

**See Also**

- create_placement
- place_opt

---

## TL-628

(error) Missing location of leaf cell %s inside block reference %s. Attempt to repair location failed.

**Description**

The instance location could not be updated with respect to the block.

**What Next**

(1) Please run placement on the block using the create_placement command. (2) Make sure the block reference has enough placed cells as to meet the threshold requirement.

**See Also**

- create_placement
- TL-650

---

## TL-629

(error) Location of leaf cell %s inside block reference %s is outside the boundary. Attempt to repair location failed.

**Description**

The instance location could not be updated with respect to the block.

**What Next**

Use place_opt or create_placement command to place all the cells inside the block boundary.

**See Also**

- create_placement
- place_opt

Feedback

## TL-630

(error) Missing location of block instance %s in parent block reference %s. Attempt to repair location failed.

### Description

The instance location could not be updated with respect to the block.

### What Next

(1) Please take care of block placement at parent level in Design Planning step ( shape_block ). (2) Make sure the block reference has enough placed cells as to meet the threshold requirement.

### See Also

- create_placement
- TL-650

## TL-631

(error) Block instance %s in parent block reference %s is outside the parent block boundary. Attempt to repair location failed.

### Description

The instance location could not be updated with respect to the block.

### What Next

If only small portion of block boundary is outside the top boundary , user can manually pull it inside the top boundary , else take care of block placement inside the parent core boundary in Design Planning Step ( shape_block ).

### See Also

- create_placement
- place_opt

## TL-650

(error) Block reference %s does not have enough placed instances to meet the threshold for repair. Attempt to repair location skipped.

Feedback

### Description

Block needs to have atleast 90% of its instances placed to invoke repair.

### What Next

Please run placement on the block using the create_placement command.

### See Also

- create_placement
- TL-628
- TL-630

---

## TL-651

(error) Failed to create new frame view for block reference %s that honors the %s repair performed on the block.

### Description

Creation of frame for the block that was repaired either for location related error or missing port error failed.

### What Next

You may choose to recreate the frame view using create_frame command.

### See Also

- create_frame
- TL-526
- TL-527
- TL-530
- TL-531

---

## TL-801

(warning) The editable abstract/read-only design view '%s' has been repaired by c heck_hier_design. Merge operation to the design view will not be allowed.

Feedback

**Description**

The editable abstract/read-only design view '%s' has been repaired by check_hier_design. Merge operation to the design view will not be allowed.

**See Also**

- check_hier_design

---

## TL-802

(warning) The design view has been modified due to repair in check_hier_design. The abstract view cannot be merged back.

**Description**

The design view has been modified due to repair in check_hier_design. The abstract view cannot be merged back.

**See Also**

- check_hier_design

---

## TL-803

(warning) The abstract view has been modified due to repair in check_hier_design. It cannot be merged to the design view.

**Description**

The abstract view has been modified due to repair in check_hier_design. It cannot be merged to the design view.

**See Also**

- check_hier_design

---

# TLNL

---

## TLNL-001

(information) Wrote top-level Verilog file '%s'.

Feedback

**Description**

The top-level netlist was successfully generated and the corresponding Verilog file was written out. This means that the command ran successfully.

**What Next**

The generated Verilog file can be used for further analysis.

---

## TLNL-002

(information) Created top-level design '%s'.

**Description**

The top-level netlist was successfully generated and the corresponding design was created. This means that the command ran successfully.

**What Next**

Run "list_blocks" to view the newly generated design. You can save the design on disk or write out Verilog using "write_verilog" or do further analysis on the design.

---

## TLNL-003

(error) Could not write top-level Verilog file.

**Description**

There was an issue while writing out the Verilog file for the generated top-level netlist. Please check if some Error or Warning messages have been printed before this message and correct your inputs accordingly.

**What Next**

Provide the correct inputs and rerun this command

---

## TLNL-004

(error) No connections file specified.

**Description**

There were no valid text files provided with the "-connections" option. Top-level netlist cannot be generated.

**What Next**

Provide valid and readable input files to "-connections" option.

Feedback

## TLNL-005

(error) Unsupported text format '%s'.

**Description**

The format specified with "-text_format" option is not supported currently.

**What Next**

Provide a valid value, e.g. "-text_format csv"

## TLNL-006

(warning) Ignoring '-output' as it cannot be specified with '-format design'.

**Description**

The option '-output' is used for specifying an output file name if the output format is 'verilog'. It does not mean anything if it is specified with 'design' output format as there is no text file to write out. The command will ignore '-output' option and continue.

**What Next**

Do not specify '-output' if '-format design' is specified.

## TLNL-007

(error) Unsupported output format '%s'.

**Description**

Output format provided with '-format' option is unsupported.

**What Next**

Provide a valid output format like 'verilog' or 'design'

## TLNL-008

(error) Design '%s' already exists in the library '%s'.

**Description**

The design with the specified name already exists in the current library. If not specified, it default to "TOP". Another design cannot be created with the same name.

Feedback

**What Next**

Provide a different name using "-design_name" option, or delete the existing design.

---

## TLNL-009

(error) Unsupported input file format '%s'.

**Description**

The input file format specified with "-text_format" is not supported. Hence the command cannot parse the text file.

**What Next**

Provide a valid value to "-text_format" option, e.g. "csv".

---

## TLNL-010

(error) Cannot specify '%s' as an ignored block.

**Description**

This error occurs when an invalid value is provided to the "-ignore_blocks" option. e.g. You cannot specify "_TOP_" as an ignored block, because that keyword refers to the top-level design.

**What Next**

Remove the invalid instance name from "-ignore_blocks" option and rerun the command.

---

## TLNL-011

(error) Could not parse map file '%s'.

**Description**

There was some error while parsing the column name mapping file specified using "-map_file" option. The file should be a CSV file which maps custom column names in the connections file to standard Synopsys column names in the form <custom_name>,<standard_name> per line.

**What Next**

Correct the map file and rerun the command.

Feedback

## TLNL-012

(error) Could not parse connection file '%s'.

**Description**

There was a major syntax error in the specified connections file. It should be a single text file containing information of all the connections in the system. The different text fields should be separated by comma. Please refer to messages printed before this message

**What Next**

Please correct the input text file and rerun the command

## TLNL-013

(error) Could not generate top-level design.

**Description**

There was an issue while converting the input connection information into a design. The command was unsuccessful. Refer to Error or Warning messages printed before this message.

**What Next**

Correct the input data and rerun the command.

## TLNL-014

(warning) Ignoring entry as number of columns in map file should be exactly 2. (%s line %u)

**Description**

The map file does not contain data according to the expected format. The specific issue is on the line number mentioned in the message. It maps custom column names in the connections file to standard Synopsys column names in the form <custom_name>,<standard_name> per line.

**What Next**

Please correct the map file and rerun the command.

## TLNL-015

(error) Syntax error in map file. Column name '%s' contains invalid characters. (%s line %u)

Feedback

**Description**

The custom column name in the map file contains some invalid characters like '?' or '*'. This cannot be used as a column name. The command cannot continue successfully.

**What Next**

Correct the column names in the map file.

---

## TLNL-016

(error) Syntax error in map file. Standard column name '%s' specified in the first column. (%s line %u)

**Description**

The map file maps custom column names in the connections file to standard Synopsys column names in the form <custom_name>,<standard_name> per line. An existing standard_name cannot be specified as a custom_name.

**What Next**

Correct the entry at the specified line of the map file.

---

## TLNL-017

(information) Setting '%s' as a custom name for column name '%s'.

**Description**

The column name mapping specified the map file was successfully parsed for that particular column name.

**What Next**

There was no issue, the command will proceed further.

---

## TLNL-018

(warning) Unrecognized column name '%s' in map file, ignoring it. (%s line %u)

Feedback

**Description**

The second column at the specified line number of the map file contains a column name which is not a standard column name. The command is unable to recognize it. Currently, following standard column names are supported:

Module Instance Module Reference Module Port Module Port Direction Connected Module Instance Connected Module Reference Connected Module Port Connected Module Port Direction

**What Next**

Use one of the standard column names in the second entries of the map file rows.

---

## TLNL-019

(error) Incorrect column names or missing columns in file %s. The whole file will be ignored.

**Description**

The connections file could not be parsed correctly because of incorrect or missing data. The possible reasons could be: (1) At least one of the mandatory columns is missing. Currently, following columns are mandatory:

Module Instance Module Reference Module Port Connected Module Instance Connected Module Reference Connected Module Port

(2) One of the columns was repeated.

(3) The column names were blank or have an incorrect spelling.

**What Next**

Correct the entries of the first header row of the connections file and rerun the command.

---

## TLNL-020

(warning) Syntax error or missing data on line %u. This row will be ignored.

**Description**

The specified row of the connections file had a syntax issue, or some of the mandatory entries were missing or incorrectly specified. That particular row will be ignored, and hence that connection will not be present in the top-level design.

**What Next**

Fix the syntax issue and provide valid data in that row.

Feedback

## TLNL-021

(warning) Column name '%s' is invalid. Ignoring the column.

**Description**

The specified column in the connection file is not a valid column name. The name should not contain special characters like '*' or '?'.

**What Next**

Provide valid column names and rerun the command.

## TLNL-022

(warning) Unrecognized column name '%s'. Ignoring the column.

**Description**

The specified column name is not a standard column name, and hence the command cannot recognize it. The whole column will be ignored. Currently, following standard column names are supported:

Module Instance Module Reference Module Port Module Port Direction Connected Module Instance Connected Module Reference Connected Module Port Connected Module Port Direction

If '-map_file' option was used, the column name was unrecognized even after applying the mapping.

**What Next**

Please use the a standard column name or provide a mapping using -map_file option

## TLNL-023

(error) Column '%s' is repeated.

**Description**

The specified column occurs more than once, which is not a valid syntax. The command cannot continue further. This could also be due to an errorneous entry in the map file if provided.

**What Next**

Provide only one column with the given name and rerun the command.

Feedback

## TLNL-024

(error) Column '%s' is missing.

**Description**

One of the mandatory columns is missing from the connections file. Currently, following columns are mandatory:

Module Instance Module Reference Module Port Connected Module Instance Connected Module Reference Connected Module Port

**What Next**

Please add the missing column data in the connections file and rerun the command

## TLNL-025

(warning) Number of columns in this row is different than that of row 1. (%s line %u)

**Description**

The number of columns in all the rows must be equal. If there is a mismatch in the number of columns in some row with respect to the first row, this message will be issued. The resultant connection in the top-level netlist might be incorrect or missing altogether.

**What Next**

Ensure that all rows have uniform number of entries.

## TLNL-026

(error) No valid connections found.

**Description**

There were no valid connections found in the connections file. This might be due to empty file or syntax errors in each of the line. The top-level netlist cannot be generated.

**What Next**

Provide valid connections data and rerun the command.

## TLNL-027

(error) Could not create module '%s'.

Feedback

**Description**

There was an issue while creating one of the modules which is to be instantiated at the top-level. This could be due to incorrect or invalid entries or some other issue.

**What Next**

Ensure that the input data is correct.

---

## TLNL-028

(warning) Top-level port is connected to itself. (%s line %u)

**Description**

This message is printed when both the sides of a connection in the connections file are top-level ports and they have the same name, which means it is a connection of a port to itself. This is an invalid connection specification and will be ignored.

**What Next**

Remove that particular row from the input file and rerun the command.

---

## TLNL-029

(warning) Could not create top-level '%s'. (%s line %u)

**Description**

This message is printed when there was an error while trying to create a top-level port or a pin in the design. This could be due to incorrect or invalid entries or some other issue.

**What Next**

Check whether the input data is correct and consistent.

---

## TLNL-030

(error) Cannot instantiate a module within itself. (%s line %u)

**Description**

On the specified row of the connections file, the module reference name was specified to be _TOP_ whereas the instance name was not _TOP_. This tries to suggest that top-level module is instantiated within itself, which is invalid. Hence this row will be ignored.

**What Next**

Correct the entry in the connections file and rerun the command.

Feedback

## TLNL-031

(warning) Module port direction is different from the previous value '%s'. Maintaining the original value. (%s line %u)

### Description

This message is printed when a port or pin appears more than once, and the directions specified in each of these occurences is not the same. e.g. It was "in" in some row, and "out" in another row.

### What Next

Please ensure that the port direction entries are consistent across the input files.

## TLNL-032

(warning) Reference module name '%s' of instance '%s' is different from '%s' which was was previously specified.

### Description

This message occurs when the same instance name is used with two or more different reference names. This is invalid as the instance names at the top-level must be unique.

### What Next

Ensure that the same instance name is not repeated across different modules.

## TLNL-033

(warning) Invalid %s name found. Ignoring this connection (%s line %d)

### Description

This message occurs when the instance, reference, or port name is an invalid string. e.g. if it contains special characters like $, ^, ? or *. There will be no connection for that line.

### What Next

Remove the special characters from the name and run the command again.

Feedback

# TLUP

## TLUP-001

(Warning) Ignore data with unknown Keyword '%s' in section '%s'.

**Description**

This is a keyword that is not known by tool.

**What Next**

Check whether the keyword is valid. If it is valid, try the newest tool to read it. Otherwise, re-generate the TLU+ file correctly.

## TLUP-002

(Error) Cannot parse TLUPlus file because of syntax error.

**Description**

The TLUPlus file has syntax error.

**What Next**

To perform TLUPlus file parse without error, please make sure the TLUPlus file is generated correctly or try the newest tool.

## TLUP-003

(Error) TLUPlus file syntax error. Unsupported value type for unknown '%s' table in section '%s'.

**Description**

The TLUPlus parser will skip unknown self-contained table. Supported table element type are int, double and char. This error means TLU+ parser encounters unknown type, so it cannot skip the unknown table.

**What Next**

To skip unknown tables, please check ITF file and then re-generate TLUPlus file again.

## TLUP-004

(Error) Mask pair (%s, %s) is missing in TLUPlus file.

Feedback

### Description

Fully populated mask pairs are expected when we read a TLUPlus file. This error message is shown by *read_parasitic_tech* when a mask pair is missing.

### What Next

Please check ITF file and then re-generate TLUPlus file again.

### See Also

- read_parasitic_tech

---

## TLUP-005

(Error) TLUPlus file content error. Values in "%s" table in %s section in %s layer "%s" are not in ascending order.

### Description

This table should be in ascending order.

### What Next

Sort values in the table, then regenerate the TLUPlus file. Turn on command option -diable_sanity_check of read_parasitic_tech to surpress this error, but letting read_parasitic_tech move forward silently may cause problem in subsequent operations. The result is uncertain.

---

## TLUP-006

(Error) TLUPlus file content error. Values in "%s" table in %s section in %s layer "%s" are not in descending order.

### Description

This table should be in descending order.

### What Next

Sort values in the table, then regenerate the TLUPlus file. Turn on command option -diable_sanity_check of read_parasitic_tech to surpress this error, but letting read_parasitic_tech move forward silently may cause problem in subsequent operations. The result is uncertain.

Feedback

## TLUP-009

(Error) Layer mapping file limitation error. Number of mapping pairs defined in layer mapping file exceeds limitation.

**Description**

Max number of conducting layers supported in layer mapping file is 32. Max number of via layers supported in layer mapping file is 31.

**What Next**

Remove unused layers from layer mapping file, then reload the TLUPlus file again. Turn on command option -diable_sanity_check of read_parasitic_tech to surpress this error, but letting read_parasitic_tech move forward silently may cause problem in subsequent operations. The result is uncertain.

## TLUP-010

(Error) Layer mapping file error. Tech layer '%s' mapped in layer mapping file cannot be found in technology section.

**Description**

All tech layers mapped in layer mapping file should be defined in technology section.

**What Next**

If layer '%s' is not a valid tech layer, remove it from layer mapping file, then reload TLUPlus file again. If layer '%s' is a valid tech layer, define it in technology file, then reload technology file and TLUPlus file again. Turn on command option -diable_sanity_check of read_parasitic_tech to surpress this error, but letting read_parasitic_tech move forward silently may cause problem in subsequent operations. The result is uncertain.

## TLUP-011

(Warning) Layer mapping file warning. Tech layer '%s' mapped in layer mapping file is mask name of '%s'.

**Description**

Layer name is more preferred than mask name.

**What Next**

This is only a warning message. If you don't want this message to appear, please change mask name mapped in layer mapping file to its corresponding layer name.

Feedback

## TLUP-012

(Warning) Layer mapping file warning. Tech layer '%s' is not a metal/via/mimtop/mimbottom/poly layer defined in technology section.

### Description

Each tech layer mapped in layer mapping file should be metal/via/mimtop/mimbottom/poly defined in technology section.

### What Next

If layer '%s' is not a valid tech layer, remove it from layer mapping file, then reload TLUPlus file again. If layer '%s' is a valid metal/via/mimtop/mimbottom/poly layer, define it in technology file, then reload technology file and TLUPlus file again.

## TLUP-013

(Error) Layer mapping file error. ITF layer '%s' mapped in layer mapping file cannot be found in TLUPlus file.

### Description

All ITF layers mapped in layer mapping file should appear in TLUPlus file.

### What Next

If layer '%s' is not a ITF layer, remove it from layer mapping file, then reload TLUPlus file again. If layer '%s' is a ITF layer, define it in ITF file, then regenerate TLUPlus file from ITF file and reload the TLUPlus file again. Turn on command option -diable_sanity_check of read_parasitic_tech to surpress this error, but letting read_parasitic_tech move forward silently may cause problem in subsequent operations. The result is uncertain.

## TLUP-015

(Error) Layer mapping file error. Tech layer '%s' is missing in layer mapping file.

### Description

Layer mapping should be continues on tech layer's mask name. If technology section has mimcap layers, they should be in layer mapping file.

### What Next

Please add layer mapping for the missing tech layer to layer mapping file, then reload TLUPlus file again. Turn on command option -diable_sanity_check of read_parasitic_tech to surpress this error, but letting read_parasitic_tech move forward silently may cause problem in subsequent operations. The result is uncertain.

Feedback

## TLUP-016

(Information) Layer mapping file information: There are extra tech layer(s) %s.

**Description**

Both technology section and TLUPlus file have layer stacking. Metal layers that appear in tech layer stacking but are not defined in ITF layer stacking are extra tech metal layers. Extra layers are always on top of tech layer stacking.

## TLUP-017

(Error) Layer mapping file error. Layer mapping (%s, %s) is missing.

**Description**

All tech layers in tech layer stacking should be mapped. Missing few top layers in layer mapping file is not allowed.

**What Next**

Please add the missing layer mapping to layer mapping file, then reload TLUPlus file again. Turn on command option -diable_sanity_check of read_parasitic_tech to surpress this error, but letting read_parasitic_tech move forward silently may cause problem in subsequent operations. The result is uncertain.

## TLUP-018

(Error) Layer mapping file error. Via layer '%s' is missing in layer mapping file.

**Description**

If there are metal layers in layer mapping file, all via layers in between should also be in layer mapping file. This warning means layer mapping for the via layer is missing.

**What Next**

Please add layer mapping for the missing via layer to layer mapping file, then reload TLUPlus file again. Turn on command option -diable_sanity_check of read_parasitic_tech to surpress this error, but letting read_parasitic_tech move forward silently may cause problem in subsequent operations. The result is uncertain.

## TLUP-019

(Warning) Layer mapping file warning. Layer mapping (%s, %s) is corrected for (%s, %s).

Feedback

**Description**

This message indicates that one of ITF layer names that appear in layer mapping file cannot be found in TLUPlus file, but we think this could be caused by mistake. We infer the correct layer mapping for you from TLUPlus context.

**What Next**

Please check the corrected layer mapping.

---

## TLUP-020

(Error) Layer mapping file error. ITF layer '%s' is missing in layer mapping file.

**Description**

ITF side layer mapping must be continuous in the order of ITF layer that appears in the TLUPlus file.

**What Next**

Add layer mapping for the missing ITF layer to the layer mapping file, then reload the TLUPlus file again. Turn on the -disable_sanity_check option of the read_parasitic_tech command to suppress this error. But letting the read_parasitic_tech command proceed silently might cause problems in the subsequent operations. The result is uncertain.

---

## TLUP-021

(Information) Layer mapping file information: There are extra ITF layer(s) %s.

**Description**

Both technology section and TLUPlus file have layer stacking. ITF layers that appear in ITF layer stacking but are not defined in tech layer stacking are extra ITF layers. Extra layers are always on top of ITF layer stacking.

---

## TLUP-024

(Error) Layer mapping file error. ITF layer '%s' and '%s' are not continues in TLUPlus file.

**Description**

ITF side layer mapping should be continues on the order of ITF layer appear in TLUPlus file.

Feedback

### What Next

Please add layer mappiing for the missing ITF layer(s) to layer mapping file, then reload TLUPlus file again. Turn on command option -diable_sanity_check of read_parasitic_tech to surpress this error, but letting read_parasitic_tech move forward silently may cause problem in subsequent operations. The result is uncertain.

---

## TLUP-030

(Warning) %s file %s isn't preserved because it has been changed since the last time it was loaded.

### Description

This message means TLUPlus file or layer map file that specified by read_parasitic_tech command has been changed since the last time it was loaded. It's dangerous to preserve the changed file arbitarily because its content isn't match corresponding TLUPlus model.

### What Next

Change original TLUPlus file or layer map file back or use command read_parasitic_tech to reload the changed file.

---

## TLUP-031

(Warning) Preserved %s file %s has been changed since the last time it was saved.

### Description

This message means preserved TLUPlus file or layer map file has been changed since the last time it was saved. This is likely because the preserved file is changed by user manually. Content of changed file isn't match corresponding TLUPlus model anymore.

### What Next

Change TLUPlus file or layer map file back, otherwise don't use it as original file of corresponding TLUPlus model.

---

## TLUP-032

(Warning) Preserved %s file %s doesn't belong to any TLUPlus model.

### Description

Preserved TLUPlus file name should contain information of the TLUPlus model it belongs to. This warning means the preserved file name is not correct. The tool cannot deduce the

Feedback

TLUPlus model name from the file name. This is likely because the preserved file name is changed by user manually or user add his own file to TLUPlus file folder.

**What Next**

If this file is really a TLUPlus file or layer map file for a certain TLUPlus model, please change the file name back. If this file is added by user, please ignore this message.

---

## TLUP-033

(Warning) %s file %s isn't preserved because it has been deleted since the last time it was loaded.

**Description**

This warning means TLUPlus file or layer map file that specified by read_parasitic_tech command has been deleted since the last time it was loaded. So the file cannot be preserved.

**What Next**

Restore original TLUPlus file or layer map file back.

---

## TLUP-034

(Warning) Preserved %s file %s has been deleted since the last time it was saved.

**Description**

This message means preserved TLUPlus file or layer map file has been deleted since the last time it was saved. This is likely because the preserved file is deleted by user manually.

**What Next**

Restore the preserved TLUPlus file or layer map file back if the file was accidentally deleted.

---

## TLUP-035

(Warning) TLUPlus file (%.2fMB) will not be preserved because it exceeds file size limit (%dMB).

**Description**

In general, the tool saves the original TLUPlus file as an attachment in the design library so you can use it for extraction validation and debugging. However, if the TLUPlus file exceeds the size limit, the tool issues this warning message and does not save it in the design library to conserve disk space. The default size limit is 20MB.

Feedback

### What Next

Without the original TLUPlus file, it can be hard to identify the root cause when a problem occurs in extraction. To keep the TLUPlus file, set the *file.tlup.max_preservation_size* application option to a value greater than or equal to the TLUPlus file size and read the TLUPlus file again.

### See Also

- read_parasitic_tech
- file.tlup.max_preservation_size

---

## TLUP-040

(Error) Old version TLUPlus file is not supported.

### Description

This TLUPlus file is too old to support.

### What Next

Regenerate TLUPlus file by the newest tool.

---

## TLUP-041

(Error) TLUPlus file syntax error. Unknown lead marker '%s'.

### What Next

To parse TLUPlus file without error, please make sure the TLUPlus file is generated correctly or try the newest tool.

---

## TLUP-042

(Error) TLUPlus file syntax error. Unknown table keyword '%s' in section '%s'.

### Description

This keyword isn't known by tool.

### What Next

Check whether the keyword is valid. If it is valid, try the newest tool to read it. Otherwise, regenerate the TLU+ file correctly.

Feedback

## TLUP-043

(Error) TLUPlus file content error. Parasitic table %s size is out of boundary.

**What Next**

Edit ITF file, then regenerate the TLUPlus file.

## TLUP-044

(Error) TLUPlus file content error. Mask-aware '%s' table misses DR2SIDEN content for some mask pairs.

**Description**

DR2SIDEN is categorized under MASK pairs and must be repeated for each mask pair. This error indicates that mask is defined but DR2SIDEN is NOT under all MASK pairs.

**What Next**

Please regenerate the TLUPlus file to have DR2SIDEN populated for all mask pairs.

## TLUP-045

(Warning) minSpacing value of layer '%s' does not match : ITF (%f) vs technology file (%f).

**Description**

The minimum spacing value is differs between the ITF file and the technology file. The values printed in the warning message for both the ITF file and the technology file include any scale factor defined by the HALF_NODE_SCALE_FACTOR entry in the ITF file.

**What Next**

This is only a warning message. To prevent this message, check the minimum spacing value in the ITF file and its corresponding layer in the technology file.

## TLUP-046

(Warning) minWidth value of layer '%s' does not match : ITF (%f) vs technology file (%f).

**Description**

The minimum width value is differs between the ITF file and the technology file. The values printed in the warning message for both the ITF file and the technology file include any scale factor defined by the HALF_NODE_SCALE_FACTOR entry in the ITF file.

Feedback

**What Next**

This is only a warning message. To prevent this message, check the minimum width value in the ITF file and its corresponding layer in the technology file.

# TMGR

## TMGR-001

(Error) Unexpected state. Information: queueNode node %d, level %d, processPass %d isn't greater than lot->_processPass %d".

**Description**

This error message indicates tool task manager problem.

**What Next**

Report this problem with the message information to Synopsys.

# TSMAP

## TSM-0006

(information) Design has %d hiers, %d insts, %d tristates (%d generic tristates).

**Description**

Generates the summary report of tristate activity inside initial_map. It lists the hier count, inst count, tristate count and generic tristate count.It runs in the LBO thread environment.

# UI

## UI-001

(error) %s

**Description**

A script or complex command failed, but the error was suppressed because the value of the app option shell.common.noTclErrors is true.

Feedback

**What Next**

Identify the error based on the error message, and fix the error if necessary

# UIAT

## UIAT-5

(warning) Cannot get attribute for more than one object.

**Description**

While getting the value of an attribute, you specified more than one object. You can only get the attribute for a single object.

**What Next**

Specify a single object.

# UIC

## UIC-001

(error) %s '%s' is not in mode '%s'.

**Description**

Constraints in one mode cannot refer to objects in other modes.

**What Next**

Specify objects that are within the current mode.

## UIC-002

(error) Unable to set %s on '%s'; %s.

**Description**

The command failed to set the information on the named object due to the reason specified.

**What Next**

Examine the reason in the error message.

Feedback

## UIC-003

(error) Clock source net '%s' is multi-driven.\n \tOnly driver pin/port '%s' will be used as the source.

### Description

You should generally use ports or pins as clock or generated clock sources. If a net is specified, a driver leaf pin or port is chosen as the source. This error indicates that a clock source net has multiple drivers, and only one will be used as the clock source.

### What Next

Identify a valid set of sources and reapply the *create_clock* or *create_generated_clock* command.

### See Also

- create_clock
- create_generated_clock

## UIC-004

(error) Clock source net '%s' has no leaf driver pins or ports.

### Description

You should generally use ports or pins as clock or generated clock sources. If a net is specified, a driver leaf pin or port is chosen as the source. This error indicates that a clock source net has no drivers, and the clock cannot be created.

### What Next

Identify a valid set of sources and reapply the *create_clock* or *create_generated_clock* command.

### See Also

- create_clock
- create_generated_clock

## UIC-005

(Warning) Unable to set %s delay with -reference_pin '%s'. Cannot specify '-reference_pin' with '-network_latency_included or -source_latency_included'.

Feedback

**DESCRIPTIONi**

The -network_latency_included and -source_latency_included options cannot be used at the same time as the -reference_pin option. If you use this option and the propagated clocking is being used, the delay value is related to the arrival time at the specified reference pin, which is clock source latency plus its network latency from the clock source to this reference pin. For ideal clock network, only source latency is applied.

**What Next**

Apply the constraint without using options '-network_latency_included or -source_latency_included'.

---

## UIC-006

(Warning) Unable to set %s delay with -reference_pin '%s'. Reference pin should be leaf pin or port in a clock network.

**DESCRIPTIONi**

The pin specified with the -reference_pin option should be a leaf pin or port in a clock network, and in the direct or transitive fanout of a clock source specified with the -clock option.

**What Next**

Determine correct reference_pin to specify or use it without the -reference_pin option.

---

## UIC-007

(warning) Reference pin '%s' specified with external delay is not in clock network and hence ignored.

**Description**

The pin specified with the -reference_pin option should be a leaf pin or port in a clock network, and in the direct or transitive fanout of a clock source specified with the -clock option.

**What Next**

Specify valid reference pin from clock network.

---

## UIC-008

(Error) Unable to set clock sense on %s.

Feedback

### Description

The object specified for the *set_clock_sense* command is invalid.

Valid objects include ports, leaf pins and combinational propagation cell arcs. Net arcs, timing check arcs (such as setup or hold), or sequential edge propagation arcs are not supported. For hierarchical pin, it is valid only if it is in clock network and the pin is the clock source of specified clock.

### What Next

Use the *set_clock_sense* command on valid ports, pins or cell timing arcs.

---

## UIC-009

(warning) Object '%s' is not a valid %s. The %s command will not match this object.

### Description

The specified object is neither a valid timing startpoint nor endpoint. Commands such as *set_false_path*, *set_multicycle_path*, and *group_path* require the *-from* option *from_list* objects to be valid timing startpoints and the *-to* option *to_list* objects to be valid timing endpoints.

One important limitation to note is that the call to *update_timing* command may cause the creation of path endpoints at combinational pins. One major example is clock gating checks if the pin connects to the signal gating the clock signal. In that case, entering an exception before an *update_timing* would emit this message, whereas doing so after an *update_timing* would not.

### What Next

Use input ports or register clock pins for the *from_list* objects. Use output ports or register data pins for the *to_list* objects.

### See Also

- group_path
- set_false_path
- set_multicycle_path
- update_timing

---

## UIC-010

(warning) Invalid direction for port '%s' in a %s specifier.

Feedback

### Description

The entered port direction does not match the specifier used. For example, an input port is used with a -to specifier.

### What Next

Re-enter the command with valid port directions.

---

## UIC-011

(warning) %s is forcing pin '%s' to be a timing %s.

### Description

The specified pin is neither a valid timing startpoint nor endpoint. The *set_max_delay* and *set_min_delay* commands are point-to-point timing exception commands. In this case, these commands override the default single-cycle timing relationship for affected timing paths, so the matched component of each path has a new startpoint (if you specify the *-from* option) and a new endpoint (if you specify the *-to* option). The remaining portions of the path are left unconstrained at the specified pin.

*set_input_delay* will force startpoints. and *set_output_delay* will force endpoints. Forcing a startpoint or endpoint breaks all paths that would have been seen through those pins.

### What Next

The tool assumes that the behavior described above is intended. If not, use input ports or register clock pins for the *from from_list* objects, and output ports or register data pins for the *-to to_list* objects.

### See Also

- set_max_delay
- set_min_delay

---

## UIC-012

(Warning) Clock edge values are not monotonically increasing, so waveform has been adjusted.

### Description

The clock waveform must be an even number of monotonically increasing values. In the special case where two edges are specified and the second value is smaller than the first by less than one period, one clock period is added to the second value. For example, if the

Feedback

period is 10 and the waveform is {5.0 3.0}, the adjusted waveform will be {5.0 13.0}. This means that the clock rises at 5 and falls at 13.

**What Next**

You can check the *waveform* attribute on the clock to ensure that the adjusted value is correct.

---

## UIC-013

(warning) Creating a generated clock on input or inout port '%s'.

**Description**

When creating a generated clock on an input or inout port, please note that the generated clock would only derive its waveform characteristics from the master clock. Delay, on the other hand, is derived from delay information at the port itself. Additionally, creating a generated clock to drive an inout port can cause unpredictable circuit behavior. If the port is functioning as an output while the clock is driving, bus contention occurs. If the clock is three-stated, any registers in the fanout of the clock can lose state.

**What Next**

Verify that the generated clock is on the intended object.

---

## UIC-014

(Error) The -edges spec of generated clock '%s' has edge number\n \tless than 1. The edge number should be 1 or greater.

**Description**

The edge_list in create_generated_clock command cannot use edge number of 0 or less.

**What Next**

Change your edge_list and re-enter the create_generated_clock command.

---

## UIC-015

(Error) In the edge list of create_generated_clock\n \t'%s', the edge numbers must be in increasing order.

**Description**

In the -edges specification of a create_generated_clock command, the edge numbers specified cannot be decreasing. Note that two subsequent edges can be equal for a pulse clock, such as *-edges {1 1 3}*.

Feedback

**What Next**

Check the edge list value in the create_generated_clock command.

---

## UIC-016

(warning) Creating a clock source on inout port '%s'.

**Description**

A circuit where a clock signal drives an inout port can be unpredictable. If the port is functioning as an output while the clock is driving, bus contention occurs. If the clock is three-stated, any registers in the fanout of the clock can lose state.

**What Next**

Verify that the circuit is safe according to your design rules. Ensure that any inout port with a clock source will not have contention or three-state when the clock is being used.

---

## UIC-017

(error) The 'create_clock' command cannot be used on output port '%s'.

**Description**

Clock sources must be input ports or internal pins. Inout ports can be used, but they are not recommended because of bus contention issues.

**What Next**

Identify a valid set of sources and reapply the *create_clock* command.

---

## UIC-018

(warning) Creating '%s' on a hierarchical pin '%s'.

**Description**

Defining a constraint at a hierarchical pin causes the timing arcs (net arcs) from leaf pins driving this pin to leaf pins driven by this pin to be broken. This gives rise to two distinct limitations. First, the hierarchical boundary is a virtual designation and, generally, does not map to a specific physical location. Hence, timing analysis cannot make any assumptions as to distributing the interconnect delay to the left and right of the bidirectional boundary. Second, breaking the original timing arcs may introduce a loss of timing information. For example, if the hierarchical pin were driven by three leaf pins and drives three other leaf pins, then breaking at the hierarchical boundary would reduce nine timing arcs to six, or worse, three considering the first limitation.

Feedback

### What Next

If possible, you may want to move the constraint to a leaf driver or load pin on the same net.

### See Also

- create_clock
- set_clock_sense
- set_input_delay
- set_output_delay

---

## UIC-019

(warning) Creating a clock on internal pin '%s'.

### Description

Clock sources should normally be input ports. Internal pins are supported but not recommended because they segment timing paths and prevent slew propagation. If the clock is derived from an incoming clock signal, use *create_generated_clock* instead of *create_clock*.

### What Next

Identify a valid set of sources and reapply the *create_clock* command.

### See Also

- create_clock

---

## UIC-020

(error) Invalid waveform. Edges must be an even number of\n \tmonotonically increasing values less than one period in duration.

### Description

The specified clock waveform is not valid.

### What Next

Specify a valid waveform for *create_clock*.

Feedback

## UIC-021

(warning) Creating virtual clock named '%s' with no sources.

**Description**

This warning occurs when a virtual clock is created. A virtual clock has a name but no sources. This means it is not applied to any ports or pins in the design. A virtual clock can be used to specify input or output delay.

**What Next**

The command *create_clock -period 10 -name CLK* does not apply the clock to any sources. If you want to apply the clock to a pin or port, you must specify the pin or port as in *create_clock -period 10 CLK* or *create_clock -period 10 -name CLK ff1/CP*.

## UIC-022

(error) The number of edges specified '%d' is not an odd number larger than or equal to 3.

**Description**

The number of edges to make one period of the generated clock waveform has to be an odd number larger than or equal to 3.

**What Next**

Carefully specify edges and ensure that you specify one full clock cycle using the edges.

## UIC-023

(error) The number of edge_shifts specified '%d'\n \tusing '-edge_shift' option is not equal to the number of edges specifed '%d'\n \tusing '-edges' option.

**Description**

The number of edge_shifts specified using the '-edge_shifts' option must be equal to the number of edges specified using the '-edges' option.

**What Next**

Specify an edge shift for each edge in the edge list.

## UIC-024

(error) You cannot specify an output port '%s' to be a generated clock master source.

Feedback

**Description**

A generated clock master can be an input or inout port or a pin.

**What Next**

Determine an appropriate master source for the generated clock.

---

## UIC-025

(warning) Virtual clock '%s' cannot be made propagated.

**Description**

A virtual clock cannot be made propagated as it has no source and does not affect any register in the design.

**What Next**

Remove the virtual clock from the clock list.

**See Also**

- remove_clock

---

## UIC-026

(warning) Converting propagated %s '%s' to ideal.

**Description**

The direct setting of clock network latency on a propagated clock converts it to a ideal clock.

**What Next**

Verify that this is the intended behavior.

**See Also**

- remove_clock_latency
- set_clock_latency

---

## UIC-027

(warning) Converting %s object '%s' from ideal to propagated.

Feedback

### Description

The direct setting of a propagated_clock attribute on a clock, pin, or port can convert all latches in the transitive fanout to propagated. This occurs if they were already marked ideal and had some network latencies set. The user-specified network latencies are removed and can not be recovered. This message is generated if you use the *set_propagated_clock* command on the objects (pin, port or clock) after setting network latencies by using the *set_clock_latency* command on the same objects.

### What Next

Verify that this is the intended behavior.

### See Also

- remove_clock_latency
- set_clock_latency
- set_propagated_clock

---

## UIC-028

(warning) Negative clock latency of %g specified for %s '%s'.

### Description

A negative value to *set_clock_latency* is legal, but may be unintended.

### What Next

Ensure that a negative value is appropriate for the latency.

---

## UIC-029

(error) Clock '%s' exists in more than one group.

### Description

The *set_clock_groups* command allows each clock can be defined in only one clock group.

### What Next

To define multiple groups related to the same clock, use multiple *set_clock_groups* commands.

Feedback

## UIC-030

(Warning) Clock groups with same clocks are already set in '%s'.

### Description

You receive this warning message because the clock groups you specified are already set by previous command.

### What Next

Use the *report_clock* with *-groups* option to check what clock groups have been set. To remove the existing clock groups, use the *remove_clock_groups* command.

### See Also

- report_clock

## UIC-031

(warning) Clock group '%s' exists. Overwriting it with the new description.

### Description

You receive this warning message because a clock group with the same name has already been specified. The old group will be removed and will be overwritten with the new command.

### What Next

Use the *report_clock* with *-groups* option to check what clock groups have been set. To remove the existing clock groups, use the *remove_clock_groups* command.

### See Also

- set_clock_groups
- remove_clock_groups
- report_clock

## UIC-032

(error) Clock group '%s' does not exist.

### Description

All names must be predefined by the set_clock_groups.

Feedback

### What Next

Use the set_clock_groups command to define clock groups. Use the report_clock command with -groups option to see existing clock groups.

### See Also

- remove_clock_groups
- set_clock_groups
- report_clock

---

## UIC-033

(information) '%s' has been renamed to '%s'.

### Description

You receive this message because the -exclusive option has been renamed to -logically_exclusive since -physically_exclusive is added.

### What Next

Check the set_clock_group documentation for better understanding of the new options and use the one which is more suited.

### See Also

- set_clock_groups

---

## UIC-034

(warning) Redefining clock '%s'. %s

### Description

This message is printed when the *create_clock* command creates a clock with the same name of an existing clock. The clock will be redefined with the period, waveform and sources specified in the most recent command.

### What Next

Determine if the redundant *create_clock* command is desired and modify constraint scripts as needed.

### See Also

- create_clock

Feedback

## UIC-035

(warning) Removing clock '%s'\n \tAll of its sources are now assigned to a new clock.%s

### Description

This message is printed when the *create_clock* command is issued on a source that already has a clock of a different name, and the *-add* option is not used. The existing clock(s) will be removed from these sources, and will be removed entirely if they have no remaining sources.

### What Next

If existing clocks should be preserved, use *-add* in the later *create_clock* command.

If only the later clock is desired, change the constraint script to eliminate previous *create_clock* commands on those sources.

### See Also

- create_clock

## UIC-036

(warning) Modifying clock '%s'; some of its sources are now assigned to a new clock.%s

### Description

This message is printed when the *create_clock* command is issued on a source that already has a clock of a different name, and the *-add* option is not used. This message indicates that a clock had some of its sources in common with the new clock. Those sources are removed from the previous clock because *-add* was not used. The previous clock remains but now has fewer sources.

### What Next

If existing clocks should not be modified, use *-add* in the later *create_clock* command. Otherwise ensure that the *create_clock* commands do not reference the same sources unless *-add* is used.

### See Also

- create_clock

## UIC-037

(warning) Ignoring option '%s' because '%s' is not specified.

Feedback

**Description**

The option specified is valid only if specified along with the other option.

**What Next**

This option will be ignored, so you can remove this option from the command.

---

## UIC-038

(warning) Negative clock uncertainty specified: %g

**Description**

A negative value to *set_clock_uncertainty* is legal, but may be unintended.

**What Next**

Ensure that a negative value is appropriate for the uncertainty.

---

## UIC-039

(warning) '%s' command option '%s' is not supported.

**Description**

The specified command option is not supported in this application. It will be ignored.

**What Next**

There may be another command or set of commands that performs a similar action. If possible, ensure that the script uses commands and options which are supported in all of the tools where the script will be used.

---

## UIC-040

(warning) Adjusting %s %s on %s '%s' so that %s.

**Description**

The tool automatically adjusts delay values so that either min is never greater than max or neither of the values are left uninitialized. This message occurs during either set_input_delay or set_clock_tree_balance_point constraint command. For input delays, this message indicates that one input delay value was changed to prevent min values from being larger than the max values only; unitialized values are left untouched. For clock tree constraints, this message can indicate both the situations.

Feedback

### What Next

Check your script to determine what value was intended for each of min_rise, max_rise, min_fall and max_fall.

---

## UIC-041

(warning) Clock '%s' is specified multiple times in the same group of a clock group.

### Description

The *set_clock_groups* command allows each clock to be defined only once in a group.

### What Next

Check if another clock should have been defined instead of the duplication.

---

## UIC-042

(warning) Unable to apply '%s' on non-current design '%s'.

### Description

The constraints can be applied only to the current_design.

### What Next

Check if the current_design setting is as expected or the object passed to the command is correct. You may want to use the instance name instead if the intention is to set it on a hierarchical sub-design.

### See Also

- current_design

---

## UIC-043

(warning) Object '%s' is not on clock network for '%s' mode's '%s' clock. The %s command cannot be applied to this object.

### Description

The constraints can only be applied to any pin/port on a clock network.

Feedback

## UIC-044

(warning) Object '%s' had a previously specified constraint set on it for clock '%s', which will be overwritten.

**Description**

A set_clock_tree_balance_point constraint has already been applied to this object-clock pair. The new constraint will overwrite this one.

## UIC-045

(info) %s of %s objects specified were not clock tree balance points.

**Description**

No clock tree balance constraints were applied to the specified objects previously. You should check if the object names provided were correct.

## UIC-046

(error) No pin specified in the given '%s' list belong to specified '%s' list.

**Description**

The report command has specifed both a pin list and a clock list and none of the clocks in the clock list pass through the pins in the pin list.

## UIC-047

(warning) Pins specified by the given '%s' list that do not belong to the specified '%s' list have been dropped from the current report.

**Description**

The report command has specifed both a pin list and a clock list and some of the pins in the pin list do not belong to any clock network specified in the clock list. These pins will not be considered during report generation.

## UIC-048

(error) None of the pins in the '%s' list is present in the clock networks of the given mode(s).

Feedback

**Description**

The pins in the pin list specified in the report command are not present on any of the clocks of the given/implicit modes.

---

## UIC-049

(warning) Pins that are not in a clock network are dropped from the '%s' list.

**Description**

Some of the pins in the pin list specified in the report command are not present on any of the clocks of the given/implicit modes.

---

## UIC-050

(error) 'from_pin' list and 'to_pin' list contain no pins that are clocked by the same clock; therefore, the skew cannot be reported.

**Description**

For the skew report the from pins and the to pins given in the command should have at least one pin which are part of the same clock network for local skew to be reported.

---

## UIC-051

(error) '%s' is a %s clock and cannot be specified for the '%s' constraint.

**Description**

This error happens when the clock is either a generated or a virtual clock. Since this is a clock tree synthesis constraint it cannot be applied to a virtual clock. Also, clock tree synthesis balances the generated clock network as part of the master clock network. Hence clock tree constraints can only be applied with respect to a master clock.

---

## UIC-052

(error) %s name '%s' already exists.

**Description**

The constraint with the given name already exists.

**What Next**

Specify another name which is unique.

Feedback

## UIC-053

(error) '%s' already has a skew group '%s' defined for it with respect to the clock '%s'.

**Description**

The constraint for the given object already exists with respect to the specified clock.

## UIC-054

(error) Unable to apply %s on design '%s' because it is not properly linked.

**Description**

The specified design must be successfully opened and linked.

**What Next**

Check the design object passed to the command.

## UIC-056

(warning) The '%s' command is not supported by read_sdc.%s

**Description**

The command is not supported within the *read_sdc* command and will be ignored.

**What Next**

The message text will include additional information if there is an alternate way to apply the constraint.

## UIC-057

(error) None of the specified scenarios are active for %s analysis.

**Description**

None of the specified scenarios have been activated for setup or hold analysis. Because of Thus, no setup or hold timing paths can be found.

**What Next**

Use the *-modes* or *-corners* options to specify currently-active scenarios, or use the *set_scenario_status* command to activate some scenarios for setup or hold analysis.

Feedback

## UIC-058

(warning) Scenario %s is not configured for %s analysis: skipping.

**Description**

The specified scenario has not been activated for setup or hold analysis. Because of this, no setup or hold timing paths can be found for it.

**What Next**

If setup or hold analysis is deliberately being skipped for this scenario, this warning can be ignored. Otherwise, use the set_scenario_status command to activate this scenario for setup or hold analysis.

## UIC-059

(warning) The option '%s' has been renamed to '%s'.

**Description**

The specified option has been replaced by a new option with the same functionality. The old command option still works but will stop working soon.

**What Next**

Update the scripts to use the new option.

## UIC-060

(warning) Mode '%s' does not exist on cell '%s'.

**Description**

The requested cell mode could not be found on the specified instance. This cell mode cannot be activated.

**What Next**

Update the scripts to use a valid set mode.

## UIC-061

(warning) Implicitly setting the '%s' option for the set_timing_derate command.

Feedback

**Description**

There were no option specified on the set_timing_derate command. The tool will apply derates assuming the -cell_delay option.

**What Next**

If this was not your intent, update your script by using at least one of the following options -cell_delay, -check_delay or -net_delay

---

## UIC-062

(error) Cannot activate modes '%s' on cell '%s' because they belong to the same mode group. No mode is activated.

**Description**

You are trying to activate two or more cell modes belonging to the same mode group. Only one mode in each mode group can be activated at a time. The cell mode status for that particular cell will be unchanged.

**What Next**

Update the scripts to activate only one mode per mode group.

---

## UIC-063

(error) CRPR is currently switched off.

**Description**

The command works only when CRPR is enabled but it was called when CRPR was disabled.

**What Next**

Update the script to enable CRPR, if desired. To enable CRPR set timer app option 'remove_clock_reconvergence_pessimism' as follows: set_app_options -global {time.remove_clock_reconvergence_pessimism true}

---

## UIC-064

(error) Pin '%s' is not an input clock pin of a sequential device.

Feedback

### Description

You receive this message if you execute report_crpr with an input pin (either from_pin or to_pin) that is not a sequential clock pin. Both, the from_pin and the to_pin, must be sequential clock pins.

### What Next

Run report_crpr command again and ensure that the pins specified for -from pin and -to pin are correct. For designs with multiple modes check if the correct mode is specified.

### See Also

- set_app_options
- report_crpr

---

## UIC-065

(error) Could not find clock '%s' for %s '%s'.

### Description

You receive this error message because the clock you specified does not correspond to the input clock pin or the port(either from_pin or to_pin) specified with the report_crpr command.

### What Next

Run report_crpr again and ensure that clocks specified can reach the from_pin and to_pin sequential clock pins or the ports are constrained by the specified clock. For designs with multiple modes, make sure the clocks are from the correct mode.

### See Also

- set_app_options
- report_crpr

---

## UIC-066

(error) Port '%s' is not an %s port%s.

### Description

You receive this message if you execute report_crpr with a port (either -from or -to) that is not correct input/output type or is not constrained by specified clock. Option -from expects an input port and -to expects an output port.

Feedback

### What Next

Rerun the command with correct port for -from and -to options with correct clock name. For designs with multiple modes, check if the clocks correspond to the mode specified or current mode if a mode was not specified.

### See Also

- report_crpr
- set_app_options

---

## UIC-067

(error) %s '%s' not in clock network.

### Description

You receive this error message because the -from pin you specified with report_crpr command is not in any clock network. Report_crpr allows data pins of sequential devices as -from pins when timer app option crpr_remove_clock_to_data_crp is set to true.

### What Next

Ensure that pins specified are either clock pins or if clock to data crp report is desired then the app option crpr_remove_clock_to_data_crp is set to true and the specified -from pin is a data pin of a sequential device driven by clock network. Remove_clock_to_data_crp mode can be set as follows: set_app_options -global {time.crpr_remove_clock_to_data_crp true}

### See Also

- set_app_options
- report_crpr

---

## UIC-068

(error) pulse clock sense merging at pin: '%s' for clock: '%s' The clock will not propagate forward from this pin.

### Description

A pulse clock is combined through combinational logic with another sense of the same clock. The tool does not resolve this conflict. Any registers down stream from this point will NOT have this clock assigned to them.

Feedback

### What Next

Resolve the conflict by specifying the sense of the clock by using the 'set_clock_sense' or 'set_sense -type clock' commands.

### See Also

- set_clock_sense
- set_sense

---

## UIC-069

(warning) Object '%s' is not a valid timing endpoint for mode '%s'. The %s command cannot be applied to this object.

### Description

The constraints can only be applied to a valid timing endpoint like register clock pin, output port, etc.

---

## UIC-070

(error) No scenarios were specified.

### Description

The modes and corners specified by the *-modes* and *-corners* options did not specify any scenarios.

### What Next

Use values for the *-modes* or *-corners* options that specify existing scenarios, or use the *-scenarios* option to explicitly specify the desired scenarios.

---

## UIC-071

(error) Scenarios from more than one mode were specified.

### Description

The specified scenarios belong to more than one mode. However, the command can only process scenarios that all belong to the same mode.

### What Next

You can specify a single mode with the *-modes* option to ensure that only appropriate scenarios are specified.

Feedback

## UIC-072

(error) Scenarios from more than one corner were specified.

**Description**

The specified scenarios belong to more than one corner. However, the command can only process scenarios that all belong to the same corner.

**What Next**

You can specify a single corner with the *-corners* option to ensure that only appropriate scenarios are specified.

## UIC-073

(error) More than one scenario was specified.

**Description**

Multiple scenarios were specified. However, the command can only process a single scenario.

**What Next**

You can specify a single scenario with the *-scenarios* option.

## UIC-074

(error) No active scenarios were specified.

**Description**

The *-modes*, *-corners*, *-scenarios* options did not specify any active scenarios.

**What Next**

Use values for the *-modes*, *-corners*, or *-scenarios* options that specify existing scenarios that are active for setup or hold analysis.

## UIC-075

(error) Current scenario does not exist or is inactive.

**Description**

By default, the command operates on the current scenario, but the current scenario does not exist, or is not active for setup or hold analysis.

Feedback

**What Next**

Use values for the *-modes*, *-corners*, or *-scenarios* options that specify existing active scenarios.

Set the current scenario to an existing active scenario.

Use the *set_scenario_status* command to make the current scenario active.

---

## UIC-076

(error) Current scenario does not exist.

**Description**

By default, the command operates on the current scenario, but the current scenario does not exist.

**What Next**

Set the current scenario to an existing scenario.

Use values for the *-modes*, *-corners*, or *-scenarios* options that specify existing scenarios.

---

## UIC-077

(warning) Data check from '%s' to '%s' is ignored since %s is on clock network

**Description**

This warning happens when either the from pin or the to pin of data check is on clock network. In this case the data check is ignored.

---

## UIC-078

(warning) Ignoring data check %s hierarchical pin '%s'

**Description**

This warning happens when either the from pin or the to pin of data check is a hierarchical pin. The data check on hierarchical pin is not supported.

---

## UIC-079

(Error) Cannot specify from/through/to group with special (async, clock_gating, in2reg, reg2out or in2out) path groups.

Feedback

**Description**

Inbuilt group path are special group path for async, clock_gating or boundary paths and no object can be added or removed from these path group using group_path command or remove_path_group command. These group paths can be enabled or disabled using timer app option.

---

## UIC-080

(warning) Ignoring data check %s pin '%s'

**Description**

This warning happens when either the from pin or the to pin of data check is not found on the graph and hence the data check is ignored.

---

## UIC-081

(error) report_ocvm currently supports only AOCVM.

**Description**

This error happens when either the type option is not provided or it is something other than aocvm.

---

## UIC-082

(warning) report_ocvm does not support check arc.

**Description**

This warning happens when a check arc is provided as an option to the report_ocvm command.

---

## UIC-083

(Error) Cannot create a group_path using reserved name used for special group paths like: default, async, clock_gating, in2reg, reg2out or in2out path groups.

**Description**

Inbuilt group path are special group path for default, async, clock_gating or boundary paths and each one of them have reserved names.

Feedback

## UIC-084

(warning) CTS Exception cannot be applied with respect to generated clock: %s and would be applied to its primary master clock.

**Description**

CTS Exceptions (balance and Ignore) cannot be applied on generated clocks. If specified with repect to generated clock, tool will transfer it to primary master clock. It is highly recommended to specify the constraint on a master clock for clean and unambiguous application.

## UIC-085

(Error) The priority value specified for group_path is less than 0.\n \tThe priority value should be greater than 0. Constraint is ignored.

**Description**

A negative value for priority option for *group_path* is not allowed.

**What Next**

Reapply with priority value larger than 0.

## UIC-086

(error) Specified corner does not exist for mode: %s.

**Description**

The corners specified by the *-corners* options does not exist for current mode.

**What Next**

Specify corners which exist in current mode.

## UIC-087

(info) PrimeTime does not support I/O path groups; for consistency with PrimeTime, disable this app-option.

**Description**

PrimeTime does not support I/O path groups and hence enabling them will make group paths different for both the tools.

Feedback

**See Also**

- get_app_option_value
- set_app_options
- report_app_options
- report_path_groups
- time.enable_io_path_groups
- time.deprioritize_io_path_groups

---

## UIC-088

(Warning) Timing constraint set on unmapped DesignWare operator '%s' may be ignored.

**Description**

The timing constraint is set on cells or pin of unmapped DesignWare operator. The timing constraint implied size_only may be ignored by the datapath optimization step. The constraint on the DesignWare operator may be lost.

**What Next**

Please try not to set timing constraint on unmapped DesignWare operator.

---

## UIC-090

(warning) There is no clock named '%s' in mode '%s'.

**Description**

Clock-related constraints in one mode cannot refer to clocks in other modes. If a clock from a different mode is provided, the tool will try to substitute a clock with the same name in the correct mode. This warning will be issued if such a clock cannot be found.

**What Next**

Specify clocks that are within the current mode.

---

## UIC-091

(warning) Unrealistically large(absolute) derate value specified: %s.

Feedback

**Description**

This warning message is issued when set_timing_derate_command is called with a value that makes total timing derate value greater than 2.0 or less than 0. The expected range for total derate value is [0 to 2].

Total derate value is the sum of base and incremental derates. A total derate value of more than 2 may result into unrealistic timing. And when the calculated total derate value is less than zero it is set to zero for timing.

**What Next**

It is likely that the specified derate value is incorrect, because of a user input error. Please specify a more realistic derate factor within the range [0 to 2].

---

## UIC-092

(warning) Detected '%d' empty groups, setting up specified relationship among the remaining '%d' clock groups.

**Description**

An invalid/empty/undefined clock or clocks are detected in set_clock_groups.

**What Next**

Carefully check the clocks used in set_clock_groups. ICC2 continues the run with the remaining valid clocks.

---

## UIC-093

(warning) Clock '%s' is not a generated clock.

**Description**

The *remove_generated_clock* command removes only generated clocks, and the given clock is a regular clock, not a generated clock.

**What Next**

Use *remove_clock* instead of *remove_generated_clock* to remove this non-generated clock.

---

## UIC-094

(warning) Clock '%s' is not a valid option.

Feedback

### Description

Clock option can be used only if app-option time.gen_clock_skew_group is set to true. Also, only generated clock can be specified as part of this app-option.

### What Next

Check the app-option and/or the skew group specification.

---

## UIC-095

(error) '%s' is not an input pin of a cell.

### Description

The command expects an input pin of a MUX or other standard cell. Any other type of pin will generate this error.

### What Next

Please specify valid input pin of a cell.

### See Also

- set_clock_exclusivity

---

## UIC-096

(error) Input pins not specified for user_defined type.

### Description

The command expects at least two input pins for user_defined type. Without input pins, clock exclusivity cannot be determined.

### What Next

Please specify valid input pins for clock exclusivity.

### See Also

- set_clock_exclusivity

---

## UIC-097

(error) Minimum two input pins are required for clock exclusivity.

Feedback

### Description

The command expects at least two input pins. The clocks on these specified input pins will be treated as exclusive beyond the output pin.

### What Next

Please specify minimum two valid input pins for clock exclusivity.

### See Also

- set_clock_exclusivity

---

## UIC-098

(warning) Non-output pins in the list are ignored.

### Description

The command expects only output pin but other pins were specified along with output pin(s). Such non-output pin(s) are ignored by the command.

### What Next

Please check the object list and make sure all objects are specified are correct. If objects are correctly specified then this message can be ignored. Otherwise, fix the list of objects.

---

## UIC-099

(error) '%s' is not an output pin of a cell.

### Description

The command expects an output pin of a MUX or other standard cell. Any other type of pin will generate this error.

### What Next

Please specify output pin of a cell.

### See Also

- set_clock_exclusivity
- remove_clock_exclusivity

Feedback

## UIC-100

Information: DRC disabled for '%s' nets.

**Description**

DRC is disabled for specfied nets through *set_auto_disable_drc_nets* command.

**What Next**

Use the *set_auto_disable_drc_nets -none* command to re-enable DRC for specfied nets.

## UIC-101

(info) CTS Exception cannot be applied in logical hierarchal pin %s.

**Description**

CTS Exceptions (balance and Ignore) can not be applied on logical hierarchal pins. If specified then it will ignored.

## UIC-102

(Warning) Delay value of balance point will be copied to all corners.

**Description**

set_clock_balance_point command is issued without corner option. Even though corners are not equivalent(different Process, Voltage or temperature) same delay value will be copied to all corners.

## UIC-103

(Warning) set_sense cannot be applied on hierarchical pin %s beacuse the clocks %s are not defined on the pins.

**Description**

The object specified for the *set_sense* command is invalid. For hierarchical pin, it is valid only if it is in clock network and the pin is the clock source of specified clock.

**What Next**

Use the *set_sense* command on valid ports, pins or cell timing arcs.

Feedback

## UIC-104

(Warning) Cannot find any clocks defined on the hierarchical pins %s.

**Description**

The object specified for the *set_sense* command is invalid. For hierarchical pin, it is valid only if it is in clock network and the pin is the clock source of specified clock.

**What Next**

Use the *set_sense* command on valid ports, pins or cell timing arcs.

## UIC-105

(warning) Offset latency can only be specified for clock-sink pins or clock-pin of ICG cell. Object '%s' is neither a clock sink pin nor a clock-pin of ICG and hence offset latency is dropped for analysis purpose.

**Description**

Offset latency can only be applied to any clock sink pin or a clock-pin of ICG.

## UIC-106

(error) The clock %s is not a multiply_by type of generated clock.

**Description**

A generated clock which is not multiply_by type cannot be used in 'set_clock_jitter' and 'remove_clock_jitter' commands.

**What Next**

To execute the command, use a *master clock* or multiple_by generated clock.

## UIC-107

(warning) CST-Exceptions can only be specified on clock network pins. Exception specified on object: '%s' will be dropped from analysis.

**Description**

CST-Exceptions can only be specified on clock network pins.

Feedback

## UIC-108

(warning) Cannot perform "%s" operation on pin %s. The new setting conflicts with the previous operation on the same pin: "%s"

### Description

The command cannot perform operation because of a conflict and ignores the operation with this warning.

### What Next

Please check the object list and make sure all objects specified are correct. User can either remove conflicting setting on the pin or ignore this message.

## UIC-109

(error) Cannot specify '-type mux' with '-inputs'.

### Description

The command doesn't allow specifying input pins with MUX type. Exclusivity set on mux output pin with this type takes ALL valid inputs automatically. Please use -type user_defined, if fewer mux inputs are to be considered for exclusivity or if the cell is not a mux.

### What Next

Please remove -input option or use -type user_defined.

### See Also

- set_clock_exclusivity

## UIC-110

(error) The exclusivity set with '-type mux' option on the output pin '%s' is dropped.

### Description

This error message is issued when there is a MUX-exclusivity relationship defined on an output pin and the cell is not of a MUX type.

### What Next

Make sure to specify MUX exclusivity only on MUX cells.

Feedback

### See Also

- set_clock_exclusivity

---

## UIC-111

(warning) Corner independent clock-tree-balance-point is not supported. Balance delay on pin '%s' is dropped.

### Description

This warning message is issued when there is corner independent clock-tree-balance-point is specified on a clock network pin. This scenario can exists only in older blocks saved using older builds where tool used to support corner-independent balance-delays.

### What Next

Make sure to specify balance delays with valid mode, corner or scenarios.

### See Also

- set_clock_balance_points

---

## UIC-112

Information: Max transition sigma is used for library constraints only. Some violations may be shown with nominal/mean transition values.

### Description

When app option, \Btime.pocvm_max_transition_sigma_for_lib_only\P is set to true, costing applies max transition sigma only against library constraints. User set constraints on design, clock, port and pin, using set_max_transition command, use nominal/mean transition value. Related app option, time.pocvm_max_transition_sigma, specifies nsigma to be used.

Max transition slack is calculated as follows:

MaxTransitionCost = Max (LibrearyCost, UserCost)

1) LibraryCost = (TransMean + MaxTrans_Nsigma * TransSigma) - (LibraryConstraint)

2) UserCost = (TransMean) - (UserSetConstraint)

Slack = - MaxTransitionCost

Feedback

**What Next**

If this mode is needed, expect nominal transitions with user set constraints in slack calculation.

## UIC-113

(Error) Limit exceeded: Cannot define multicycle path with a path_multiplier of value %d.

**Description**

The max threshold value of multi-cycle path is 100000.

**What Next**

Rerun the set_multicycle_path command with a smaller value for path_multiplier.

**See Also**

- set_multicycle_path

# UID

## UID-2

(error) There are no designs to be linked.

## UID-3

(warning) Can't read link_library file '%s'.

**Description**

This warning message will be printed out if the given link library file is protected or doesn't exist. In the absense of the given library file, there will be some unresolved references during linking.

**What Next**

Check the link_library and the search_path variables and set their value accordingly.

## UID-4

(error) Current design is not defined.

Feedback

**Description**

The tool issues this error when the current design is not defined.

**What Next**

If no designs are currently loaded into memory, read a design into the tool. To set the current design, run the *current_design* command.

---

## UID-5

(error) Current design '%s' has no schematic.

**Description**

The current design does not have a schematic.

**What Next**

You can create a schematic with the *create_schematic* command. Check the man page by typing *help create_schematic*.

---

## UID-6

(error) Technology library has not been specified.

---

## UID-7

(error) Couldn't read technology library '%s'.

---

## UID-8

(error) Technology library '%s' is not a library.

---

## UID-9

(error) Can't read '%s' files '%s'.

---

## UID-10

(error) Cannot specify output port %s as a path startpoint.

Feedback

**Description**

This error message occurs when the specified output ports are not valid as path startpoints. Output ports cannot be the startpoint for a timing path.

**What Next**

Examine the design to determine the correct startpoint for the path. After correcting the startpoint, run the command again.

---

## UID-11

(error) Cannot specify input port %s as a path endpoint.

**Description**

This error message occurs when the an input port is specified as a path endpoint. Input ports are not valid as timing path endpoints.

**What Next**

Examine the design to determine the correct endpoint for the path. After correcting the endpoint, run the command again.

---

## UID-12

(error) Cannot specify hierarchical cell %s as a path named %s.

**Description**

This error message occurs because hierarchical cell names are not valid as path startpoints or endpoints.

**What Next**

Use a clock, port, leaf or hierarchical pin, or leaf cell as the path startpoint or endpoint.

---

## UID-13

(error) '%s' doesn't specify a unique design\n \tPlease use complete specification: full_file_name:design_name

---

## UID-14

(error) Design '%s' is not in the system.

Feedback

**Description**

This error indicates that the current command cannot find the design in the system.

**What Next**

Provide the design in the system or load in a design.

---

## UID-15

(error) '%s' doesn't specify a unique library\n \tPlease use complete specification: full_file_name:library_name

---

## UID-16

(warning) No input ports on design '%s'.

---

## UID-17

(warning) No output ports on design '%s'.

---

## UID-18

(error) Invalid pin list.

---

## UID-19

(error) '%s' doesn't specify a unique instance\n

**Description**

The instance name given matches more than one instance.

**What Next**

Specify a unique name for the instance you want.

---

## UID-20

(error) Write command is not available.

---

## UID-21

(error) Invalid file or design list.

Feedback

## UID-22

(error) No files or designs were specified.

**Description**

This error message is displayed when no files or design objects have been supplied to the *write* command, or when the specified files or design objects are not in memory.

**What Next**

Re-issue the *write* command, correctly specifying the files or designs to be written.

## UID-23

(error) Only designs can be specified with File '%s' is ignored.

## UID-24

(error) No valid designs or design files were specified.

## UID-25

(error) Write command failed.

**Description**

This error is generated when the *write* command failed. Errors include

- an invalid format (see EXPT-3), - writing to a write permission denied directory. (see UID-30) - no valid designs or design files. (see UID-24) - other causes during writing.

**What Next**

Identify and correct the cause of the write command failed and reinvoke the *write* command.

## UID-26

(warning) Can't find in memory specified design or design file '%s'.

**Description**

This error is generated when there is no specified design or design file in memory, but invoke the *write* command.

Feedback

### What Next

Use *list_designs* commands to see whether given design in the memory or use *read* command to read a design in. Then invoke *current_design* command to set the desired current design and reinvode *write* command.

---

## UID-27

(error) '%s' doesn't specify a unique design file\n \tPlease use complete path name to specify this file.

### Description

The specified file is present in more than on directories listed in the search path.

### What Next

One possible solution is to use the complete path name to specify the file. Alternately, the search path can be modified so that only one of the directories in the path has the specified file.

---

## UID-28

(error) Cannot write the '%s' design to '%s'.

### Description

The tool issues this error when it fails to open or write to the specified file. There can be several reasons for this. The most common reasons are: you do not have write permissions in the directory, the disk is full, or you already have a file with the same name that cannot be deleted.

### What Next

Try to manually create a file with the same name. Fix any problems that you encounter, and then delete the file and run the command again.

---

## UID-29

(error) Cannot write the '%s' file.

### Description

The tool issues this error when it fails to open or write to the specified file. There can be several reasons for this. The most common reasons are: you do not have write permissions in the directory, the disk is full, or you already have a file with the same name that cannot be deleted.

Feedback

**What Next**

Try to manually create a file with the same name. Fix any problems that you encounter, and then delete the file and run the command again.

---

## UID-30

(error) Cannot write the specified designs to '%s'.

**Description**

The tool issues this error when it fails to open or write to the specified file. There can be several reasons for this. The most common reasons are: you do not have write permissions in the directory, the disk is full, or you already have a file with the same name that cannot be deleted.

**What Next**

Try to manually create a file with the same name. Fix any problems that you encounter, and then delete the file and run the command again.

---

## UID-31

(error) Multiple designs per file are not permitted by '%s' format.

---

## UID-32

(error) Format '%s' is not a valid write format.

**Description**

This error is generated when *write* command encounters an invalid write format.

**What Next**

Invoke *write* -help command for valid write formats. Identify and correct format, then reinvoke the *write* command.

---

## UID-33

(error) LSI Netlist format is not enabled.

**Description**

The LSI Netlist format is not enabled.

Feedback

**What Next**

The LSI Netlist format is not enabled. Check your permissions.

---

## UID-34

(error) TDL Netlist format is not enabled.

**Description**

The TDL Netlist format is not enabled.

**What Next**

The TDL Netlist format is not enabled. Check your permissions.

---

## UID-35

(warning) Option -add when timing_enable_multiple_clocks_per_reg is not set to true, creates the clock but analysis is not done with multiple clocks

**What Next**

Set variable timing_enable_multiple_clocks_per_reg to be true

---

## UID-36

(error) Verilog netlist format is not enabled.

**Description**

The Verilog netlist format is not enabled.

**What Next**

The Verilog netlist format is not enabled. Check your permissions.

---

## UID-37

(error) Mentor format is not enabled.

**Description**

The Mentor format is not enabled.

**What Next**

The Mentor format is not enabled. Check your permissions.

Feedback

## UID-38

(error) EDIF format is not enabled.

**Description**

The EDIF format is not enabled.

**What Next**

The EDIF format is not enabled. Check your permissions.

## UID-39

(error) Design '%s' isn't mapped.

**Description**

You are trying to write the current design to other format other than synopsys db format, but the design isn't mapped.

**What Next**

Invoke compile -map_effort for mapping the current design, then reinvoke write command.

## UID-40

(error) Wire load '%s' not found.

## UID-41

(error) No files are specified.

## UID-42

(error) Invalid file list.

## UID-43

(warning) File '%s' was not found in the search_path.

## UID-44

(error) Either -all or cell_list must be specified.

Feedback

## UID-45

(error) Invalid cell list.

## UID-46

(error) No valid cells are specified for ungrouping.

**Description**

The ungroup command was called without any valid cells being specified for ungrouping.

**What Next**

Call the ungroup command with a set of valid cells.

## UID-47

(warning) Can't %s unmapped logic cell '%s'.

**Description**

Only hierarchical cells can be ungrouped. The cell mentioned in the error message is not hierarchical and so will be ignored by the ungroup command.

**What Next**

No further action is required, this is merely a warning message.

## UID-48

(warning) Can't %s logic one cell '%s'.

**Description**

Only hierarchical cells can be ungrouped. The cell mentioned in the error message is not hierarchical and so will be ignored by the ungroup command.

**What Next**

No further action is required, this is merely a warning message.

## UID-49

(warning) Can't %s logic zero cell '%s'.

Feedback

**Description**

Only hierarchical cells can be ungrouped. The cell mentioned in the error message is not hierarchical and so will be ignored by the ungroup command.

**What Next**

No further action is required, this is merely a warning message.

---

## UID-50

(warning) Can't %s cell '%s'; its reference is unresolved.

**Description**

This error message is generated when you try to invoke a command on a cell that doesn't have a reference.

**What Next**

Use the *link* command, then reinvoke the current command.

---

## UID-51

(warning) Can't %s leaf cell '%s'.

**Description**

Only hierarchical cells can be ungrouped. The cell mentioned in the error message is not hierarchical and so will be ignored by the ungroup command.

**What Next**

No further action is required, this is merely a warning message.

---

## UID-052

(warning) A non-unate path in clock network for clock '%s'\n from pin '%s' is detected.

**Description**

The clock tree for the specified clock contains non-unate paths. Clock networks normally do not contain cells such as exclusive-OR gates which have non-unate behavior. This implies that both inverting and noninverting clock waveforms reach the specified pin. The sense that is propagated from this pin is non-deterministic.

Feedback

### What Next

Since there is ambiguity on which sense to choose to propagate, the tool picks one arbitrarily to propagate. If this is not acceptable, create a clock using the output of the non-unate gate as a source.

---

## UID-52

(warning) -return_null_values is not supported for attribute '%s' and will be ignored.

### Description

get_attribute returns a collection instead of a list for some attributes. For these attributes, -return_null_values will have no effect.

### What Next

No further action is required, this is merely a warning message.

---

## UID-53

(error) Cannot create_cell of type '%s' because the physical library cell has a non-power/ground pin '%s'. The cell type must be in the logical library for types with non-power/ground pins.

### Description

This error occurs because you are trying to create a logical netlist cell with the wrong notation or you are using the *create_cell* command on a cell type that is only in the physical library, but has a non-power/ground pin. (In this tool, *create_cell* requires that any cell type with a non-power/ground pin be in the logical library.)

### What Next

If you are trying to create a logical netlist cell, you might have to specify a library by using the slash (/) notation; for example:

```
create_cell my_cell library_name/AND4.
```

If you are using the *create_cell* command on a cell type that is only in the physical library, but has a non-power/ground pin , add a logical library cell type to the logical library and rerun.

### See Also

- create_cell

Feedback

## UID-56

(error) Read format '%s' is not supported.

**Description**

The specified read format is not supported.

**What Next**

Please enter a valid read format.

## UID-57

(error) No design file specified to read.

## UID-58

(error) Cannot read file '%s'.

**Description**

This error is generated when you try to read a file which is protected or does not exist.

**What Next**

Check and correct search_path, then reinvoke the read command.

## UID-59

(error) Can't read '%s' file '%s'.

**Description**

This error is generated when reading a given format file unsuccessfully.

This error is generated when a file cannot be read successfully using the specified format.

**What Next**

Check to make sure the correct file format is being specified and reinvoke the *read* command. The problem persists, check the file contents.

## UID-60

(error) Verilog HDL Compiler is not enabled.

Feedback

**Description**

A license is not available for Verilog HDL Compiler.

**What Next**

Check too see if your licenses have run out.

---

## UID-61

(error) Invalid list of operating conditions.

---

## UID-62

(error) No operating conditions were found.

**Description**

The named operating condition was not found in specified library or in any of the link libraries.

**What Next**

Use the 'report_lib' command to see names of operating conditions available in your libraries.

---

## UID-63

(warning) Operating conditions '%s' not found.

**Description**

The named operating condition was not found in specified library or in any of the link libraries.

**What Next**

Use the 'report_lib' command to see names of operating conditions available in your libraries.

---

## UID-64

(error) Only one of: cell list, -pla, -leaf, -logic, -fsm, -hdl_all_blocks \n\tor -hdl_bussed can be specified. -hdl_block can only be used separately \n\tor with -hdl_all_blocks or -hdl_bussed.

Feedback

## UID-65

(error) Either cell list or -pla or -leaf or -logic must be specified.

## UID-66

(error) No valid cells are specified for grouping.

## UID-67

(error) A design with name '%s' already exists in the same\n \tdesign file as the given design: '%s'.

**Description**

The design name you specified in the -design_name option of the 'group' command is already being used. The 'group' command needs to create a new design to represent the new level of hierarchy, so the name you specify must not conflict with existing designs in the system.

**What Next**

Specify a unique design name.

## UID-68

(error) A cell %s already exists in design '%s'.

**Description**

The cell name you specified in the -cell_name option of the 'group' command is already being used for another cell in the design. The 'group' command needs to create a new cell in the design to represent the new level of hierarchy, so the name you specify must not conflict with any existing cells in the design.

**What Next**

Specify a unique cell name.

## UID-69

(error) Except set option cannot be used with -hdl_block, \n-hdl_all, or -hdl_bussed option.

Feedback

## UID-70

(error) Can't find HDL block %s.

**Description**

The block name specified in the -hdl_block option of the 'group' command could not be found in the current_design. The named block needs to have been specified in the HDL description using block labels. Block labels are separated by a backslash (/), for example, my_process/my_function.

## UID-71

(warning) No designs or libraries are available to free.

## UID-72

(error) Only one type of object allowed.

## UID-73

(error) Invalid %s list.

## UID-74

(error) No valid %ss specified.

## UID-75

(error) Unknown port '%s'.

## UID-76

(error) The specified POCVM corner sigma %5.2f is negative. Variable setting is not changed.

**Description**

The POCVM corner sigma value must be greater than or equal to zero.

**What Next**

Use a positive POCVM corner sigma value.

Feedback

## UID-79

(warning) Timing options specified without -timing.

## UID-80

(error) Illegal -path option of '%s'.

**Description**

Valid options for -path are "short", "full", "only", "end". Type help 'report_timing' to see a description of the options.

**What Next**

Re-type command with correct options.

## UID-81

(error) Illegal -delay option of '%s'.

**Description**

Valid option strings for the -delay option are "min", "min_rise", "min_fall", "max", "max_rise", "max_fall". Type help 'report_timing' to see a description of the options.

**What Next**

Re-type command with correct options.

## UID-82

(error) -max_paths cannot be negative.

## UID-83

(information) Updating graph...

**Description**

This information message confirms that the timing graph update is in progress.

**What Next**

This is only an information message. No action is required.

Feedback

## UID-84

(warning) Hierarchy options specified without -hierarchy.

## UID-85

(information) Updating design information...

**Description**

This information message confirms that the design information update is in progress.

**What Next**

This is only an information message. No action is required.

## UID-86

(error) Could not update the design.

**Description**

Could not complete timing analysis because of an error.

**What Next**

There should be another error message describing what went wrong.

## UID-87

(error) Cannot read this library, it is old.

## UID-88

(error) Invalid port list.

## UID-89

(error) 'set_unconnected' cannot be set on an input port '%s'.

## UID-90

(error) 'set_unconnected' cannot be set on an unknown port '%s'.

Feedback

## UID-91

(error) '%s' cannot be set on an output port '%s'.

## UID-92

(error) '%s' cannot be set on an unknown port '%s'.

## UID-93

(error) Invalid object list.

## UID-94

(error) '%s' is not a valid attribute type.

## UID-95

(warning) Can't find %s '%s' in design '%s'.

**Description**

This error message is generated when you try to find a nonexistent object of a given type in the specified design or the design is hidden and so the object cannot be shown.

**What Next**

Check the object by invoke report_<object>, then reinvoke the current command.

## UID-96

(information) Creating new attribute '%s' on %s '%s'.

## UID-97

(error) '%s' doesn't have an attribute '%s' type '%s'.

## UID-98

(error) Invalid attribute type.

Feedback

## UID-99

(error) Attribute '%s' on %s '%s' is type boolean;\n \tcannot be set to '%s'.

## UID-100

(warning) '%s' is not a valid object type.

## UID-101

(warning) Attribute '%s' does not exist on %s '%s'.

**Description**

This error message is issued by commands related to attributes of objects in a design like *get_attribute*, *remove_attribute*, or *report_attribute*. When the command does not find the given attribute, the tool prints out this warning message.

**What Next**

Provide a correct attribute name for this object. If you are not sure about attribute names, do not specify this attribute.

## UID-102

(warning) Ignoring %s '%s' type.

**Description**

This error message is issued when the type of object is not supported by the command you are running.

**What Next**

Run the command with the *-help* option or check the man page of the current command for supported types of objects.

## UID-103

(fatal) Synopsys database corrupted.

## UID-104

(error) Design '%s' not found in target library.

Feedback

**Description**

The design specified was not found in the target library which is specified by the variable target_library.

**What Next**

Verify that the specified design is the correct one, set the correct target_library variable.

---

## UID-105

(error) Example design '%s' is not a flip-flop.

---

## UID-106

(error) Cannot determine flip-flop type of design '%s'.

---

## UID-107

(warning) Multiple operating conditions specified;\n \tonly '%s' accepted.

**Description**

This error message is issued by the *set_operationg_conditions* command. It prints out this message when it can not process the command with multiple operating conditions. The command only accepts one of them.

**What Next**

None.

---

## UID-108

(error) Can't find timing range '%s'.

**Description**

The timing range specified could not be found in the specified library or in the local_link_library or link_library.

**What Next**

Check that the correct timing range is specified and then check the library to make sure the timing range is present.

Feedback

## UID-109

(error) Can't find %s '%s'.

**Description**

The object of the specified type could not be found.

**What Next**

Check the man page for the current command.

## UID-110

(error) VHDL HDL Compiler is not enabled.

## UID-111

(error) Can't check security for format '%s'.

## UID-112

(error) '%s' is an invalid %s.

## UID-113

(error) No report section specified.

**Description**

This error is issued by report_test command because there is an incorrect or missing option.

**What Next**

Check the man page for report_test command.

## UID-114

(warning) %s exists on port '%s'. Cannot %s.

**Description**

This warning message is displayed when a conflict is found between a logic value that has already been set on a port and a command such as *set_logic_one*, *set_logic_zero*, or *set_logic_dc* attempting to set a different logical attribute value. Each of these commands

Feedback

checks for pre-existing logic attributes, and issues this warning message when such attributes are found.

**What Next**

Remove the conflicting logic attribute from the port via the *remove_attribute* or *reset_design* commands.

---

## UID-115

(warning) '%s' is not a library cell.

**Description**

This error message is issued by the *set_prefer* command. The message is printed out when the command cannot find this cell in the target library.

**What Next**

Use the *find* command to check for library cells in the target library, then reenter the *set_prefer* command with the correct library cell.

---

## UID-116

(error) Can't find %s '%s' in design '%s'.

**Description**

This error message is generated when you try to find a nonexistent object of a given type in the specified design or the design is hidden and so the object cannot be shown.

**What Next**

Check the object by invoke report_<object>, then reinvoke the current command.

---

## UID-117

(warning) Can't find %s '%s' in library '%s'.

**Description**

This message is issued when the current command cannot find this object in the specified library. Therefore, it does not perform the current command on this object.

**What Next**

Provide the correct object in the specified library. You can use the *find* command to find objects in the library.

Feedback

## UID-118

(error) Port '%s' is in design '%s', but port '%s' is in design '%s'.

### Description

This error message is displayed when ports in different designs are supplied as arguments to the *set_equal* or *set_opposite* command. These commands require their port arguments to be in the same design.

### What Next

Modify the arguments to only include ports on the same design and re-issue the command.

## UID-119

(warning) %s '%s' is of the wrong type.

### Description

This message is issued when you specify an incorrect object type as the command argument.

For example, the *all_connected* command accepts net, pin, or port objects. If you specify a cell object as the command argument, the tool issues the following message:

```
prompt> all_connected [get_cells U0]
Warning: cell 'U0' is of the wrong type. (UID-119)
{}
```

### What Next

Check the command man page to determine the valid object types and then rerun the command.

## UID-120

(error) The clock period value can't be negative.

### Description

The clock period value specified in your command is negative. The command doesn't accept a negative clock period value.

### What Next

Provide the valid value.

Feedback

## UID-121

(error) Invalid timing range list; specify one or two timing range names.

**Description**

This error message is issued by the *set_timing_ranges* command. The comman accepts only one or two timing range names.

**What Next**

Provide the valid timing range list.

## UID-122

(error) The '-exact' option must be used when the example design is a latch.

**Description**

This error message is displayed when an example latch is given to the *set_register_type* command without using the command line switch '-exact'. The *set_register_type* command requires that the '-exact' switch is used when specifying an example latch.

**What Next**

Re-issue the *set_register_type* command using the '-exact' command line switch.

## UID-123

(error) Can't set flip-flop type on non sequential cell '%s'.

**Description**

This error message is displayed because the *set_register_type* command requires that the components in the *cell_or_design_list* be valid sequential cells (latches or flip flops). Non-sequential cells are not valid arguments for the command.

**What Next**

Modify the argument list so that it only includes valid sequential components.

## UID-124

(error) Design '%s' is of "unknown" flip-flop type.

Feedback

**Description**

This error message is displayed when an example flip flop given to the *set_register_type* command is not of the correct sequential type.

**What Next**

Re-issue the command with a different example device.

---

## UID-125

(error) Attribute '%s' can't be %s on this %s with this command.

---

## UID-126

(warning) Design '%s' is in state table format. Writing only port information.

**Description**

This error message is issued by the *write* command. When the command finds the design is in state table format, it writes only port information.

**What Next**

None.

---

## UID-127

(error) Can't find library pin '%s'.

---

## UID-128

(error) No drive specified on library pin '%s'.

**Description**

This message is issued by the *drive_of* command. The command prints out the message when it finds no drive on the specified pin.

**What Next**

Provide the drive for the specified library.

---

## UID-129

(error) Either wire load model or mode has to be specified.

Feedback

**Description**

This error message is issued by the *set_wire_load* command. The command prints out this message when the wire load model or mode is missing in your current command input.

**What Next**

Provide either wire load model or mode for the command.

---

## UID-130

(error) Mode named %s is not a legal mode.

**Description**

This error message occurs if the argument to the *-mode* option of the *set_wire_load* command is not one of the recognized modes. The recognized modes are top, enclosed, or segmented.

**What Next**

Check for a spelling mistake or missing argument for the *-mode* option of the *set_wire_load* command.

---

## UID-131

(error) Can't find the specified library '%s' in memory.

**Description**

This error indicates that the current command cannot find the specified library in the current memory database of the program.

**What Next**

Check the search_path to include the path to the library. Otherwise, read in the library and rerun the command if you needs the information in the specified library to complete your work.

---

## UID-132

(error) The write command cannot be used: \n\ \t\t-- with the "-format %s" option \n\ %s\t\t-- and with a design_file_list of more than one file. \n\ \tUse the "-hierarchy" option.

**Description**

For the *edif* and *vhdl* formats, there are restrictions when the design list must have only a single design. See the *write* command for details.

Feedback

**What Next**

Review the details in the help page for the *write* command for the format you have selected, and reissue the command either with the *-hierarchy* option or with a design list of only one design.

---

## UID-133

(error) Attribute flag value must be either true or false.

**Description**

This error indicates the command you ran required an attribute flag value of either true or false.

**What Next**

Provide the valid attribute flag value.

---

## UID-134

(error) Clock '%s' is undefined for design '%s'.

**Description**

This error indicates that the command cannot find the clock in the design.

**What Next**

Provide the clock that is defined in the design.

---

## UID-135

(error) No clock specified for remove_clock().

**Description**

This error is issued by the *remove_clock* command when a clock is missing in the command input.

**What Next**

Provide the clock name for the command.

---

## UID-136

(error) The filter expression is not a Boolean expression.

Feedback

**Description**

The second argument to the filter command must be a Boolean expression that returns 1 for objects that pass the filter.

**What Next**

See the EXAMPLES section of the man page for the *filter* command for a description of Boolean expressions (also called conditional or logical expressions in some places).

---

## UID-137

(error) Error in expression. '%c' must be followed by an attribute name.

**Description**

This error is issued by the *filter* command when the attribute is missing after the indicator in the expression.

**What Next**

Provide the attribute name after the filter indicator.

---

## UID-138

(warning) Attribute '%s' does not exist for %s '%s'.

**Description**

This error message is issued by commands related to attributes of objects in a design like *get_attribute*, *remove_attribute*, or *report_attribute*. When the command does not find the given attribute, it prints out this warning message.

**What Next**

Provide a correct attribute name for this object. If you are not sure about attribute names, do not specify this attribute.

---

## UID-139

(error) Invalid inferred bus naming style specification.

**Description**

The setting of the environment variable "bus_inference_style" is invalid, so the bussing of ports could not be inferred on the design which was read.

Feedback

**What Next**

Correct the setting of the "bus_inference_style" variable.

---

## UID-140

(error) You cannot set the '%s' variable.

**Description**

The variable indicated is a derived variable. Its value is derived from some other information, so you cannot change its value directly.

**What Next**

None.

---

## UID-141

(error) Can not set -min_block_size if mode is not "enclosed".

**Description**

Cannot use the *-min_block_size* option to *set_wire_load* without specifying *-mode enclosed*. The *-min_block_size* option makes sense only when used on a hierarchical design whose top level *-mode* is *enclosed*.

**What Next**

Use *-mode enclosed* along with the *-min_block_size* option.

---

## UID-142

(warning) Could not convert '%s' into a piece_index for library\n \tpin '%s. Using '%s' for index value instead.

**Description**

This error message is issued by the *drive_of* command when it cannot convert the piece index string for the library pin. The command uses the index value instead.

**What Next**

None.

Feedback

## UID-143

(error) Cannot set a wire_load mode on a port.\n \tOnly a wire_load model can be set on a port.

**Description**

This error message is issued by the *set_wire_load* command when it only accepts a wire_load model on a port.

**What Next**

Provide the correct model to be set on a port.

## UID-144

(error) %s not allowed in object_list.

**Description**

Objects of the specified type are not allowed in the command's *object_list*.

**What Next**

Remove objects of the specified type from the *object_list* and try the command again. Please refer to the command's man-page for more information on the command's usage.

## UID-145

(information) Truncating parameter value '%f' to '%d'.

**Description**

Some commands (and command options) only accept integer arguments. This information message informs the user that an integer is expected, but a floating-point value was supplied. As a result, the floating-point value will be truncated to the integer portion only.

**What Next**

The parameter has been automatically truncated.

## UID-146

(error) Invalid combination of options enabled.

Feedback

**Description**

Not all options may be utilized simultaneously for certain commands. The specific combination of options that were enabled for this command was invalid.

**What Next**

Please refer to the command's man-page for more information on the command's usage and valid option combinations.

---

## UID-147

(error) Can not set a wire_load on a port\n and a cluster at the same time.

**Description**

Tried to use both the port and the *-cluster* options to *set_wire_load_model* command simultaneously. The *-port* option can only be used when the wire_load is being set on a design; clusters do not have explicit ports.

**What Next**

Specify either port or *-cluster*, as desired.

---

## UID-148

(error) Cannot set a wire_load mode on a cluster.

**Description**

It does not make sense to set a *-mode* on a cluster, as mode makes sense only for a top-level design. On a cluster, it is meaningless.

**What Next**

Remove the *-mode* switch from the command and run it again.

---

## UID-149

(error) The design '%s' does not have a cluster named '%s'.

**Description**

Could not find the cluster specified in the design indicated.

**What Next**

Run the *report_clusters* command to find out what clusters exist in this design, then re-execute the command which failed.

Feedback

## UID-150

(warning) Duplicate signals ignored.

## UID-151

(error) Undeclared optional signals.

## UID-152

(error) You cannot set a selection group on a port.\n \tOnly a wire_load model can be set on a port.

**Description**

The *set_wire_load* command was used on a port and the *-selection_group* option was used. This combination is not valid.

**What Next**

Either remove the *-selection_group* option or specify a design or cluster instead of a port.

## UID-153

(error) You cannot set a selection group if the mode is not "enclosed".

**Description**

An attempt was made to use the *set_wire_load* command with the *-selection_group* option when the wire load mode was not "enclosed". This is not valid.

**What Next**

Either change the wire load mode, or else do not use the *-selection_group* option.

## UID-154

(error) Specified selection group '%s' not found.

**Description**

The wire load selection group name specified was not found in the user specified library or in the first library in the link library path. Therefore, it was not set on the design/cluster.

Feedback

### What Next

Verify that the specified name is the correct one, specify on the command line the library which contains the selection group, or update the link library path such that the named group can be found.

---

## UID-155

(error) Specified path group %s not found.

### Description

This error message occurs when the specified path group name is not found. The path group must first be created with the *group_path* command.

### What Next

Verify that the specified name is correct. Use the *report_path_group* command to determine if the path group was created correctly. You can recreate the group using the *group_path* command.

### See Also

- group_path
- report_path_group

---

## UID-156

(error) Either -path_group or object list must be specified.

### Description

This error message occurs when the *set_noise_slack_range* command requires that either a path group or a object list be specified.

### What Next

Execute the *set_noise_slack_range* command again and specify a path group with the *-path_group* option, an object list with the *object_list* option, or both a path group and an object list.

---

## UID-157

(error) '-to' option must be specified for bidirectional \ pin.

Feedback

**Description**

The *set_annotated_delay* command requires that when a delay is annotated on a net from a bidirectional pin, that the load pin or pins also must be specified with the '-to' option.

**What Next**

Use the '-to' option to specify the loads on the net that should be annotated.

---

## UID-158

(warning) %s is a library. Use the find %s command.

**Description**

This warning message occurs because the *find* command does not accept an object type of cell when trying to find a library cell nor does it accept a an object type of pin when trying to find a library pin.

**What Next**

This is a only a warning message. No action is required.

However, if this is not the result you expected, run the *find* command again using either *find lib_cell* or *find lib_pin*.

---

## UID-159

(warning) Predefined methodology selected, other options ignored.

---

## UID-160

(error) Pin %s is not a scan-out pin.

---

## UID-161

(error) Cell %s does not have a scan-out pin.

---

## UID-162

(error) Duplicate cell entries found in test routing order.

**Description**

The pin list to the *set_test_routing_order* contains some duplicates, which means the scan chain order is not well defined.

Feedback

**What Next**

Revise the pin list so that each pin occurs at most once. See the help page for *set_test_routing_order*.

---

## UID-163

(error) Unrecognized test methodology %s.

---

## UID-164

(error) Test report options selected without -test.

---

## UID-165

(warning) Unknown test fault class, all classes will be displayed.

---

## UID-166

(error) Fault report options selected without -fault.

**Description**

The *report_test* options "-class", "-incremental", and "-inst" only apply when selecting the "-fault" reporting option.

**What Next**

Review the help page for *report_test*, and correct the options to the command. The most likely change is to add the "-fault" option.

---

## UID-167

(error) Scan-chain index must be positive.

---

## UID-168

(error) Non-sequential cell/reference %s can not be in a scan-chain.

**Description**

The reported cell or cell reference was in the argument list to a *set_scan_chain* command but it is not sequential. See the help page for *set_scan_chain*.

Feedback

### What Next

One possibility is that the name of the cell is incorrect, such as forgetting some levels of hiearchy or a typing error, and a nonsequential cell has been selected by mistake. Either correct the name or remove it from the cell list.

---

## UID-169

(error) Non-sequential cell/reference/design %s can not be made scannable.

### Description

The design object list for the *set_scan* and *set_scan_element* commands must consist of sequential cells only. The indicated object is not sequential. See the help page for the command for details.

### What Next

One possibility is that the name of the cell is incorrect, such as forgetting some levels of hiearchy or a typing error, and a nonsequential cell has been selected by mistake. Either correct the name or remove it from the cell list.

---

## UID-170

(warning) '%s' cannot be interpreted as a list of local link library files.

---

## UID-171

(warning) Design '%s' contains parameterized ECL components.\n Writing this design to the '%s' netlist format will\n cause a loss of parameterized information.

---

## UID-172

(information) Writing synthetic library implementations for design '%s'.\n \tUse "write -no_implicit" to get just the design.

---

## UID-173

(warning) Can't %s an uncompiled synthetic cell '%s'.

---

## UID-174

(error) Syntax error in line '%s'.

Feedback

## UID-175

(warning) Design '%s' contains unmapped components.\n \tThe description might not simulate and might not be read back into Design Compiler.

## UID-176

(Error) -name is required. Please specify snapshot name.

**Description**

This warning message occurs when the require option of the command create_qor_snapshot is not specified.

**What Next**

Specify the -name option.

## UID-177

(error) File '%s' is a directory name.

## UID-178

(error) The '%s' attribute of a %s cannot be set with this command.

## UID-179

(error) %s format is not enabled.

**What Next**

Invoke the *read* -help command for valid read formats. Identify and correct format, then reinvoke the *read* command.

## UID-181

(warning) Can't write sequential_type for %s '%s'.\n \tCannot find example design in target_library.

## UID-182

(warning) %s '%s' is of the wrong type. It seems to work but might not work in the future releases. Please fix the script.

Feedback

### Description

This message is issued when you specify an incorrect object type as the command argument and the tool ignores the object type and searches for objects using just the object name.

For example, the *all_connected* command accepts net, pin, or port objects. If you specify a cell object as the command argument, the tool issues the following message:

```
prompt> get_clocks [get_ports A]
Warning: port 'A' is of the wrong type. It seems to work but might
not work in the future releases. Please fix the script. (UID-182)
{}
```

### What Next

Check the command man page to determine the valid object types and then rerun the command.

---

## UID-183

(error) Unrecognized scan style '%s'.

### Description

The specified scan style is not recognized by the *set_test_methodology* command.

### What Next

Review the help page for *set_test_methodology* for a description of the recognized styles, and correct the command.

---

## UID-184

(error) Minimum fault coverage must be at least 80 percent.

---

## UID-185

(error) Minimum fault coverage cannot exceed 100 percent.

---

## UID-186

(information) Attribute '%s' is set on %d objects.

Feedback

## UID-187

(Error) Directory (%s) is not accessible. Please check the setting on icc_snapshot_storage_location variable.

**Description**

This warning message occurs when the snapshot directory specified by the variable is not available.

**What Next**

Check the setting of the snapshot directory.

## UID-188

(error) Data type '%s' does not match attribute type '%s'.

**Description**

This error message is displayed if The data type for an attribute does not matech predefined attribute type.

**What Next**

Re-issue the command with correct data type.

## UID-190

(error) The -design_name and -cell_name options are not used with the \n -hdl_all_blocks or -hdl_bussed options.

**Description**

The *group* command has been used with conflicting options.

**What Next**

Review the help page for the *group* command if necessary and change the options being used. The most likely change is to remove the use of the "-design_name" and "-cell_name" options.

## UID-191

(error) The -design_name option is required unless you are using the \n -hdl_all_blocks, -hdl_bussed, or -hdl_block options.

Feedback

**Description**

The "-design_name" option is required for the *group* command unless you have specified one of the following options: "-hdl_all_blocks", "-hdl_bussed", or "-hdl_block".

**What Next**

Review the help page for the group *command if necessary, and then correct the options to your group command.*

---

## UID-192

(error) -all_instances option must be used with instance cells when the parent instance is not unique.

**Description**

The *set_size_only* or *change_link* command has been used with conflicting options.

**What Next**

Use -all_instances option together with cell instances. Either add -all_instances option to set_size_only command or uniquify the parent design instance of the cell before calling set_size_only.

---

## UID-193

(information) %s %s for all instances of cell '%s' in subdesign '%s'.

**Description**

The *set_size_only* or *change_link* command has been applied to all instances of the cell in a design.

---

## UID-194

(error) -all_instances option must be used with instance cells.

**Description**

The *set_size_only* command has been used with conflicting options.

**What Next**

Use -all_instances option together with cell instances. The most likely change is to add -all_instances option.

Feedback

## UID-195

(error) Cannot change current_design from '%s' to '%s' after Milkyway CEL has been written by IC Compiler.

### Description

You receive this message if you are trying to change the current_design after a Milkyway CEL has already been written by IC Compiler. IC Compiler writes a Milkyway CEL automatically during *link*, *write_milkyway*, and some other commands.

Once a MW CEL is written the current_design cannot be changed to a sub-design.

### What Next

Utilize *current_design* to apply constraints, write reports, etc. without linking or writing to a Milkyway CEL. Then set the current_design to be the top-level design for physical synthesis. Perform *link* to write the Milkyway CEL and then proceed with physical synthesis commands.

You may also use *close_mw_cel* and *remove_design -all* to start again and establish a new current_design.

## UID-196

(error) size_cell can not perform when the parent instance is not unique.

### Description

The *size_cell* command can not perform when the parent instance is not unique.

### What Next

uniquify the parent design instance of the cell before calling size_cell.

## UID-200

(warning) Signal index values can not be negative.

## UID-201

(warning) Cannot rename nets in buffer tree.

### Description

When the variable *icc_track_net_names* is TRUE, IC Compiler renames the nets during *create_buffer_tree* and other optimizations so that the resulting nets have names with the

Feedback

same prefix as the original driving net. Note that nets touching (hierarchical) ports must have names the same as the (hierarchical) port name, so those nets cannot be renamed.

*report_net_changes* reports the groups of nets that are associated by being the result of the same optimization.

**What Next**

Report the problem to Synopsys if it is important to have the net names in the buffer tree share the same prefix.

---

## UID-202

(error) Design contains only combinational cells but scan style is not set to "combinational"; test patterns cannot be saved.

**Description**

You receive this message from *set_scan_configuration* if your design contains only combinational cells but the scan style is set to some value other than *combinational*. Although this condition does not prevent the *check_test* or *create_test_patterns* commands from running, *create_test_patterns* cannot save test patterns.

By default, DFT Compiler uses the scan style defined by the variable *test_default_scan_style*; the default value of this variable is *multiplexed_flip_flop*. You use *set_scan_configuration -style* to override the value of the *test_default_scan_style* variable for the current design. You have attempted to set a non-combinational scan style on a design with no sequential cell. The specification is rejected and the scan style remains unchanged.

You may specify the scan style for a design using the *set_scan_configuration -style* design. If you do not specify the scan style directly then DFT Compiler uses the default value defined in the variable *test_default_scan_style*.

**What Next**

To avoid receiving this message, and to allow *create_test_patterns* to save the test patterns, explicitly set the scan style of the current design to *combinational* using the *set_scan_configuration -style combinational* command.

If you want only to run *check_test*, or if you want only to run *create_test_patterns* to analyze your design without saving the test patterns, you can ignore this message and continue.

---

## UID-207

(error) The %s command is only supported in xg mode.

Feedback

**Description**

This error message occurs when the specified command is not supported in the dc_shell mode.

**What Next**

Start the shell in xg mode and run your scripts again.

---

## UID-208

(error) The cell %s is combinational. It cannot be the startpoint or endpoint of an ignore-path for retiming.

**Description**

This error message occurs when a combinational cell is specified as the startpoint or endpoint of an ignore-path for retiming. Only ports, sequential cells, pins of sequential cells, and clocks can be startpoints and endpoints of ignore-paths for retiming.

**What Next**

Use a suitable startpoint or endpoint for the path and reissue the command.

---

## UID-209

(error) A %s license must be authorized for the set_retiming_ignore_path command.

**Description**

This error message occurs because the *set_retiming_ignore_path* command requires the authorization for the specified license.

**What Next**

Check your license key file.

---

## UID-210

(error) Proprietary design '%s' can't become the current design.

---

## UID-211

(warning) Can't write proprietary design '%s'.

Feedback

### Description

Proprietary designs cannot be written in the selected format, so the named design was skipped. This might occur for example if you try to do a hierarchical write on a design which instantiates a proprietary design.

### What Next

Normally there is nothing to do, but if the warning is unexpected, you can review the link step in case there is an problem, e.g., with the search path, which is causing a proprietary design to be linked when you were expecting a non-proprietary design.

---

## UID-220

(error) Unrecognized test assert value '%s', (use one of "0","1").

---

## UID-221

(error) Can't set assume value as direction of hierarchical pin %s is inout.

### Description

This error message is displayed when a pin in the pin list supplied to the *set_test_assume* command is a hierarchical bidirectional pin. The *set_test_assume* command requires that all hierarchical pin arguments have directions that are either 'in' or 'out'.

### What Next

Re-issue the *set_test_assume* command using hierarchical pins that have direction either 'in' or 'out'.

---

## UID-222

(error) Can't set assume value as pin %s from a combinational cell.

### Description

This error message is displayed when a pin in the pin list supplied to the *set_test_assume* command belongs to a combinational component. The *set_test_assume* command requires that none of its pin arguments are part of combinational components in the design.

### What Next

Re-issue the *set_test_assume* command without a pin list containing combinational pins.

Feedback

## UID-223

(error) Can't set assume value on pin %s as not test isolated.

### Description

This error message is displayed when a pin in the pin list supplied to the *set_test_assume* command belongs to a sequential component which has not been marked as being test isolated via the *set_test_isolate* command. The *set_test_assume* command may only be used on pins of a sequential component if those pins (or their cells) have been marked as logically isolated via *set_test_isolate*.

### What Next

Remove the named pin from the argument list and then re-issue the *set_test_assume* command. Or, if it is correct to do so, use the *set_test_isolate* command on the pin or its cell, indicating that it is logically isolated and should be considered untestable during test design rule checking

## UID-224

(error) Can't set hold value as port %s is not an input.

### Description

This error message is displayed when a port in the port list supplied to the *set_test_hold* command is not an input port. The *set_test_hold* command requires that all port arguments have direction 'in'.

### What Next

Re-issue the *set_test_hold* command using only ports that have direction 'in'.

## UID-225

(error) Can't set require value as direction of pin %s not in/out.

### Description

This error message is displayed when a pin in the pin list supplied to the *set_test_require* command is neither an output pin nor an input pin. The *set_test_require* command requires that all pin arguments have directions that are either 'out' or 'in'.

### What Next

Re-issue the *set_test_require* command, using only input or output pins in the list of pin arguments.

Feedback

## UID-226

(error) Can't set require value as pin %s is test isolated.

### Description

This error message is displayed when a pin in the pin list supplied to the *set_test_require* command belongs to a sequential component which has been marked as being test isolated via the *set_test_isolate* command. The *set_test_require* command may only be used on pins of a sequential component if those pins (or their cells) have not been marked as logically isolated via *set_test_isolate*.

### What Next

Remove the named pin from the argument list and then re-issue the *set_test_require* command. Or, if it is correct to do so, use the *remove_attribute* command on the pin or its cell, removing the test_isolate attribute.

## UID-227

(error) Can't set require value as pin %s from an unknown cell.

### Description

This error message is displayed when a pin in the pin list supplied to the *set_test_require* command belongs to an unknown component. The *set_test_require* command may only be used on pins of a component which can be recognized by Design Compiler.

### What Next

Remove the named pin from the argument list and then re-issue the *set_test_require* command.

## UID-228

(warning) Design has no hierarchy. No cells can be ungrouped.

### Description

This warning message is displayed when the *ungroup* command is issued without a valid cell list, or in a design that contains no hierarchy. It indicates that without at least one valid hierarchical cell to operate on, the *ungroup* command will do nothing.

### What Next

Verify that the arguments supplied are correct, and that the design does indeed have hierarchy. Then, re-issue the *ungroup* command.

Feedback

## UID-229

(warning) All hierarchical cells are don't touched. No cells can be ungrouped.

## UID-230

(warning) All specified hierarchical cells are don't touched. No cells can be ungrouped.

## UID-231

(warning) All designs specified are un-changed and not written.

## UID-232

(information) Updating design information for flat timing report...

## UID-233

(warning) Can't resolve reference '%s' for cell '%s'

**Description**

This error is generated when there is no design reference for the given cell.

**What Next**

Check and correct the link_library and search_path. Invoke *link* command, then reinvoke the current command.

## UID-234

(error) Invalid port direction '%s'.

**Description**

This error message is displayed when the *create_port* command is issued with an invalid port direction. Valid port directions for the *create_port* command are 'in', 'out', or 'inout'.

**What Next**

Re-issue the *create_port* command using a valid direction argument.

## UID-235

(error) Port '%s' already exists.

Feedback

## UID-236

(error) Cannot specify port_list with -unused.

**Description**

It is illegal to specify a port_list when using the "-unused" argument.

**What Next**

Reissue the command either without the port list or without the "-unused" argument.

## UID-237

(error) Unable to remove port '%s'.

**Description**

This error message is displayed when a port cannot be removed from a design. Ports which are bussed may not be removed individually from a design.

**What Next**

Reissue the command without the named port.

## UID-238

(error) The '-associated_clock' switch can only be used with 'clocked_on_also' signal type.

**Description**

This error message is displayed when the *set_signal_type* command has been issued with the '-associated_clock' switch and the signal type being set is not 'clocked_on_also'. This is because the '-associated_clock' switch makes sense only in the context of defining Master and Slave clock ports with the 'clocked_on_also' signal type.

**What Next**

Re-issue the *set_signal_type* command either removing the '-associated_clock' switch, or changing the signal type to be 'clocked_on_also'.

## UID-239

(error) Cannot run this command without the -mpc option.

Feedback

### Description

This error message is displayed when the *write_physical_script* command is called without the *-mpc* option.

The *write_physical_script* command will be enhanced in the future to print physical information other than mpc options. Always use the *-mpc* option to this command to avoid breaking the script later.

### What Next

Run the command again with the *-mpc* option.

---

## UID-240

(error) Either the '-flip_flop' or the '-latch' argument must be used.

---

## UID-241

(error) Unable to find example %s '%s'.

### Description

This error message is displayed when an example gate given to the *set_register_type* command cannot be found in the current target library. The *set_register_type* command requires that the example gate given with the '-latch' or '-flip_flop' switches exist in the target library.

### What Next

Change the target library, or change the argument given as an example gate to *set_register_type* and reissue the command.

---

## UID-242

(error) Example design '%s' is not a %s.

### Description

This error message is displayed when an example gate given to the *set_register_type* command is not a valid example component. The *set_register_type* command requires that the example gate given with the '-latch' or '-flip_flop' switches actually be a valid latch, or flip flop, respectively.

### What Next

Change the argument given as an example gate to *set_register_type* and reissue the command.

Feedback

## UID-243

(error) The %s design '%s' is of "unknown" sequential type.

**Description**

This error message is displayed if the latch or flip-flop specified to the *set_register_type* command cannot be recognized by *Design Compiler*.

**What Next**

Re-issue the *set_register_type* command with a recognizable sequential device.

## UID-244

(error) Cell '%s' is a latch. \n The latch type was NOT specified on the command line.

## UID-245

(error) Cell '%s' is a flip-flop. \n The flip-flop type was NOT specified on the command line.

## UID-246

(error) Cell '%s' is neither a latch nor a flip flop.

## UID-247

(error) Non-sequential cell/reference/design %s can not be made transparent.

**Description**

Only level-sensitive sequential elements can be made transparent for ATPG.

**What Next**

Verify that the list of devices given to the *set_scan_transparent* are all level-sensitive. The *all_registers* command can be used to get a list of such registers in the design.

## UID-248

(error) The %s design '%s' is of "unknown" combinational type.

**Description**

The replacement gate in the set_combinational_type command does not have a recognized combinational function which can be used for exact replacement.

Feedback

**What Next**

The set_combinational_type command requires the replacement gate to have a recognized combinational function that can be used for exact replacement. Specify another gate as the replacement_gate for set_combinational_type.

---

## UID-249

(error) Cell '%s' is not a combinational gate.

**Description**

One of the cells specified in the cell_list for set_combinational_type is not a combinational gate. All of these cells must be combinational gates. Set_combinational_type will only work on those cells in the cell list which are combinational.

**What Next**

Only specify combinational cells in the cell list for set_combinational_type.

---

## UID-250

(error) Cannot find %s '%s'.

**Description**

This error message occurs if the object of the given type and name could not be found.

**What Next**

Correct the type or name of the object passed to the issued command and re-issue it.

---

## UID-251

(error) Cannot specify both %s and %s.

**Description**

This error message is issued if a command is invoked with two incompatible options.

**What Next**

Issue the command with only one of the two options displayed in the error message.

---

## UID-252

(error) '%s' is not an instance of design '%s'.

Feedback

**Description**

This error message is issued if a path to an instance does not represent a valid instance of the design.

**What Next**

Correct the path to the instance and re-issue the command.

---

## UID-253

(error) Cannot specify %s without %s.

**Description**

This error message is displayed if only one of two options, which need to be specified together, is specified.

**What Next**

Specify both the options displayed in the error message and re-issue the command.

---

## UID-254

(error) Invalid delay direction for port '%s'.

**Description**

This error message is displayed if a port argument to a command has a direction which is not valid for the command.

**What Next**

Re-issue the command with ports of the required directions only.

---

## UID-255

(error) Value for %s cannot be negative.

---

## UID-256

(error) Either %s or %s must be specified.

**Description**

This error message is issued if a command is issued without one of the two mandatory options being specified.

Feedback

### What Next

Re-issue the command with one of the two options in the error message.

---

## UID-257

(error) Invalid waveform. Edges must be an even number of monotonically\n increasing values less than one period in duration.

### Description

This error message is displayed if the waveform specified for a clock is invalid. There must be an even number of numbers specifying the waveform of the clock. In addition, these numbers must be in a monotonically increasing sequence. This error message is issued if either of these rules is violated.

### What Next

Re-issue the *create_clock* command with a valid specification for the waveform.

---

## UID-258

(error) Group weight must be between 0.0 and 100.0.

---

## UID-259

(error) Must specify one of %s, %s, or %s.

### Description

This error message is displayed if none of the three mandatory options for a command are specified.

### What Next

Re-issue the command by specifying one of the three options in the error message.

---

## UID-260

(error) The '%s' command is obsolete. It is ignored.

---

## UID-261

(error) Cannot set current instance to object '%s'.

Feedback

**Description**

This error message is displayed if the *current_design* does not own the instance which is being set as the *current_instance*.

**What Next**

Re-issue the *current_instance* command using an instance that is owned by the *current_design*.

---

## UID-262

(error) '%s' doesn't specify a unique object.

**Description**

This error message is displayed if multiple objects are passed as an argument to a command that requires a single object only.

**What Next**

Re-issue the command by specifying only a single object as argument.

---

## UID-263

(error) Cannot specify multiple objects for command %s.

**Description**

This error message is displayed if multiple objects are specified as an argument to a command that requires only as single object.

**What Next**

Re-issue the command with a single object as argument.

---

## UID-264

(error) Cannot specify multiple objects.

**Description**

The error message is displayed if multiple objects are specified as an argument to a command that requires a single object.

**What Next**

Re-issue the command with a single object as argument.

Feedback

## UID-265

(error) Cannot specify library object '%s' for command %s.

**Description**

This error message is displayed if library objects (library, library design, or library pin) are passed as an argument to a command that does not accept them.

**What Next**

Re-issue the command without library objects in the argument.

## UID-266

(error) Cannot specify library object '%s'.

## UID-267

(error) Command %s cannot be performed on instance %s within a hierarchy.\n Please re-issue the command in the design referenced by the current instance.

## UID-268

(error) Cannot perform this command on instance %s within a hierarchy.\n Please re-issue the command in the design referenced by the current instance.

## UID-269

(error) Can't find design corresponding to instance '%s'.

## UID-270

(error) Cannot write the '%s' script file.

**Description**

The tool issues this error when it fails to open a script file in write mode. There can be several reasons for this. The most common reasons are: you do not have write permissions in the directory, the disk is full, or you already have a file with the same name that cannot be deleted.

Feedback

**What Next**

Try to manually create a file with the same name. Fix any problems that you encounter, and then delete the file and run the command again.

---

## UID-271

(error) Current instance is not defined.

---

## UID-272

(error) Cannot set current design to library design '%s'.

**Description**

The current design can only be set to a non-library designs.

**What Next**

check if the specified design is correct. Try specifying the complete path to the design as follows:

current_design = <db_file_name>:<design_name>

---

## UID-273

(error) Command '%s' cannot be performed within a hierarchy.\n Please re-issue the command in the design referenced by the current\n instance or the top-level of the current design.

**Description**

The specified command can only be invoked from the top-level of the hierarchy. If the current instance is set to a subdesign, the command will fail.

**What Next**

If you want to execute the command in the design reffered by the current instance, set the current design to point to the design current instance is pointing to. Now invoke the command. Otherwise, set current instance to the top-level design. See the man pages for commands current_design and current_instance for further details.

---

## UID-274

(error) This command cannot be performed within a hierarchy. Please re-issue it\n in the design referenced by the current instance or the top-level of\n the current design.

Feedback

### Description

The specified command can only be invoked from the top-level of the hierarchy. If the current instance is set to a subdesign, the command will fail.

### What Next

If you want to execute the command in the design reffered by the current instance, set the current design to point to the design current instance is pointing to. Now invoke the command. Otherwise, set current instance to the top-level design. See the man pages for commands current_design and current_instance for further details.

---

## UID-275

(warning) No %s to list.

---

## UID-276

(error) Can't set current instance to leaf cell '%s'.

### Description

The *current_instance* command does not work on leaf cells. If you attempt to run *current_instance* on a leaf cell, you get this error.

### What Next

Make sure when you use *current_instance* the cell you specify is not a leaf cell.

---

## UID-277

(warning) No library contain the wire load model '%s' set on design '%s'.

### Description

The indicated design has a wire-load model that could not be found in the libraries in the search path. This situation can occur if the wire-load model was created with *create_wire_load*, which creates wire-load models and saves them on designs.

### What Next

To report on the specified wire load model, use *report_wire_load -name*. Here is an example of a report for a wire-load model:

```
****************************************
Report : wire loads
Design : counter
Version: v3.1-development
Date   : Mon Jan 25 10:47:24 1993
```

Feedback

```
****************************************

Wire load model:   counter_wl
Location       :   counter
Resistance     :   2
Capacitance    :   1
Area           :   0
Slope          :   1.333
                                 Average  Standard  % Standard
Fanout   Length   Points            Cap  Deviation  Deviation
------------------------------------------------------------
     1     0.58     1565           0.58       0.01       0.99
     2     1.46      168           1.46       0.04       2.78
     3     2.40       64           2.40       0.12       4.80
     4     3.33       56           3.33       0.15       4.59
     5     4.03       30           4.03       0.24       5.88
------------------------------------------------------------
Weighted Average Standard Deviation:                    2.08
```

Write a technology file describing the wire-load model, and update this file into a library on your search path with *update_lib*. Here is an example of a technology file with a wire-load model described in the previous report:

```
wire_Load("counter_wl") {
        resistance : 2;
        capacitance : 1;
        area : 0;
        slope : 1.333;
        fanout_length(1,0.58);
        fanout_length(2,1.46);
        fanout_length(3,2.4);
        fanout_length(4,3.33);
        fanout_length(5,4.03);
}
```

If the design still has the annotated capacitances that were used to create the wire-load model, use *create_wire_load -output* to write the wire-load model directly to a technology file.

With *report_wire_load -name*, you can verify that the technology file describes the same model as the one described.

Then use *update_lib* to update the technology file into a library on your search path. The *set_wire_load_model* line in the script written can then be updated with the *-library* option to identify the library where the wire-load model can be found.

---

## UID-278

(error) No annotated timing check information was removed.

Feedback

### Description

The timing checks defined in *remove_annotated_check* did not exist, and *remove_annotated_check* could not succeed.

### What Next

Use *report_annotated_check* to verify what timing checks are annotated.

---

## UID-279

(error) Cannot have current_instance set for this command.

### Description

This error message is displayed if the *current_instance* is defined. *Write_script* and *write_sdc* cannot recognize the current_instance because these commands write out information relative to the current_design.

### What Next

Re-issue the *current_instance* with no instance specified.

---

## UID-280

(error) Period value conflicts with waveform on clock '%s'.

---

## UID-281

(warning) Path group '%s' has no paths; it will not affect optimization.

### Description

The *group_path* command can be used to create or modify path groups. You can use *group_path -name clock -weight 5* to change the weight of an existing group. This warning is to let you know when you have created a path group that has no paths. This path group will not affect optimization unless paths are added to it by additional *group_path* commands.

### What Next

You should use *group_path -name clock -to clock* if you wish to include all paths to object "clock" in the path group. Use *report_path_group* to see the path groups that are currently defined.

Feedback

## UID-282

(information) Annotated '%s' delays are assumed to include load delay.

**Description**

Load delay (also known as extra source gate delay) is the portion of cell delay which is due to the capacitive load of the net being driven. Some SDF interfaces include load delay as part of the net delay - others include it as part of the cell delay. By default, Design Compiler assumes that load delay is included in cell delays.

**What Next**

Make sure that you understand how both your SDF reader and writer will treat load delay. Use the -load_delay option of the read_timing, write_timing, write_constraints, or set_annotated_delay commands to specify whether Design Compiler should include load delay in cell or net delays.

## UID-283

(error) Load delay location '%s' should be 'cell' or 'net'.

**Description**

Load delay (also known as extra source gate delay) is the portion of cell delay which is due to the capacitive load of the net being driven. Some SDF interfaces include load delay as part of the net delay - others include it as part of the cell delay. By default, Design Compiler assumes that load delay is included in cell delays.

**What Next**

The allowable values for the -load_delay option are "cell" or "net".

## UID-284

(warning) In ICC, use remove_buffer_tree instead of clean_buffer_tree.

**Description**

ICC does not support the *clean_buffer_tree* command.

Please use *remove_buffer_tree* instead.

## UID-285

(warning) '%f' value is negative.

Feedback

### Description

You receive this error message when a negative value has been specified with the *set_annotated_delay* command. Annotated delay values are usually positive, but negative delay values will be accepted.

### What Next

If you do not want the value to be negative, re-execute the *set_annotated_delay* command and specify a positive value.

---

## UID-286

(error) Rise/fall exceptions not supported.

### Description

You receive this error message when a *-rise_to*, *-fall_to*, *-rise_from*, *-fall_from*, *-rise_through*, or *-fall_through* option has been specified with any of the exception commands. Although these options are supported in PrimeTime, they are not supported in Design Compiler.

### What Next

Reexecute the desired command and specify the *-rise*, *-fall*, and *-through* options as necessary.

---

## UID-290

(error) Can't set test assertion on hierarchical object %s.

---

## UID-295

(error) The command set_clock_skew is obsolete.\n \tIt has been replaced by set_clock_latency, set_clock_uncertainty \n \tand set_propagated_clock commands.

### Description

The *set_clock_skew* command has been obsoleted. The functionality has been replaced by a set of commands, which have capability beyond the set_clock_skew command had implemented.

The *set_clock_skew* command has been replaced by the following commands:

*set_propagated_clock* replaces the -propagated option of *set_clock_skew*

*set_clock_latency* replaces the -delay option of *set_clock_skew*

Feedback

*set_clock_uncertainty* replaces the -uncertainty options of *set_clock_skew*

Example:

*set_clock_skew -propagated -delay 1.0 -plus_uncertainty 2.1 clk* \n

should be replaced as:

*set_propagated_clock clk*\n *set_clock_latency 1.0 clk*\n *set_clock_uncertainty -hold 2.1 clk*\n

**What Next**

You should substitute the *set_clock_skew* command with the equivalent commands as suggested above. Please consult the man pages for *set_clock_latency* *set_clock_uncertainty* and *set_propagated_clock*.

**See Also**

- set_propagated_clock
- set_clock_latency
- set_clock_uncertainty

---

## UID-296

(error) The option -cell is obsolete. Please use -lib_cell instead.

**Description**

The option *-cell* for the command *set_driving_cell* has been obsolete and will not be supported in XG mode. Please use the *-lib_cell* option instead.

**What Next**

You should substitute the *-cell* option with *-lib_cell* option for all *set_driving_cell* commands. Plese refer to the *set_driving_cell* man page for detailed information on obsolete options for this command.

**See Also**

- set_driving_cell

---

## UID-297

(error) The command set_wire_load is obsolete in XG mode.\n \tIt has been replaced by set_wire_load_model, set_wire_load_mode, \n \tset_wire_load_selection_group, and set_wire_load_min_block_size \n \t commands.

Feedback

**Description**

The *set_wire_load* command has been obsoleted in XG mode. The functionality has been replaced by a set of commands, which have capability beyond the set_wire_load command had implemented.

The *set_wire_load* command has been replaced by the following commands:

*set_wire_load_model*

*set_wire_load_mode*

*set_wire_load_selection_group*

*set_wire_load_min_block_size*

**What Next**

You should substitute the *set_wire_load* command with the equivalent commands as suggested above. Please consult the man pages for *set_wire_load_model*, *set_wire_load_mode*, *set_wire_load_selection_group* and *set_wire_load_min_block_size*.

---

## UID-300

(error) Cannot use hdl options with the soft option.

---

## UID-301

(error) 'get_attribute' can't get value for unknown attribute type.

---

## UID-302

(warning) Invalid value for link_force_case variable: default value check_reference will be used.

---

## UID-303

(error) Cannot set initial value on pin %s from a non-scan cell.

---

## UID-304

(error) Cannot set initial value on pin %s from an unknown cell.

Feedback

## UID-305

(error) Period value unspecified.

## UID-306

(error) Waveform set unspecified.

## UID-307

(error) Waveform set must consist of two floating point values.

## UID-308

(error) Period value cannot be zero.

## UID-309

(error) Value for period is different from the default value specified by test_default_period/ custom test protocol.

**Description**

Period is the length of time required for each test vector cycle on a tester. Typically, only a single period is defined for a test set. Therefore, all the test clocks you created for a particular ATPG run should have same value for the period as the one specified by the test_default_period variable or in a custom test protocol.

**What Next**

Omit the period specification, or specify the same value as the test_default_period variable.

## UID-310

(error) Port '%s' is not an input port.

**Description**

The definition of the timing of a clock can only be applied to input ports.

**What Next**

Use the *report_port* command to find the complete list of input ports. Select the correct port for applying the clock constraints. Reissue the command with the new port.

Feedback

## UID-311

(error) Cannot set testsim_input_delay on port '%s'.\n It is not an input or bidirectional port. Command ignored.

## UID-312

(error) Cannot set testsim_output_strobe on port '%s'.\n It is not an output or bidirectional port. Command ignored.

## UID-313

(error) The specified input delay %5.2f is more than the period %5.2f. Command ignored.

**Description**

The input delay must be less than the clock period.

**What Next**

The currently specified input delay has been ignored. Specify the input delay value again and ensure that the new value is less than the clock period.

## UID-314

(error) The specified output strobe %5.2f is more than the period %5.2f. Command ignored.

**Description**

The output strobe must be less than the clock period.

**What Next**

The currently specified output strobe value has been ignored. Specify the output strobe value again and ensure that the new value is less than the clock period.

## UID-315

(error) The specified input delay %5.2f is negative. Command ignored.

**Description**

The input delay must be greater than or equal to zero.

Feedback

## UID-316

(error) The specified output strobe %5.2f is negative. Command ignored.

**Description**

The output strobe must be greater than or equal to zero.

**What Next**

The currently specified output strobe value has been ignored. Specify the output strobe value again and ensure that the new value is a non-negative real number.

## UID-317

(error) Attribute '%s' has not been defined for %s.

**Description**

The *set_user_attribute* command requires that the attribute first be defined using either the *define_user_attribute* command or the *set_attribute* command.

**What Next**

Check that the attribute name is correct and define the attribute using *define_user_attribute*, or else just use *set_attribute*.

## UID-318

(error) Object class '%s' is not supported.

**Description**

The *-class* argument to the command requires a recognized object class such as design, cell, or net.

**What Next**

Use the *-help* option of the command or see the command man page to find out what object classes are supported by the command.

## UID-319

(error) Attribute '%s' is already defined as type %s for object class %s.

Feedback

**Description**

The attribute data type for a *define_user_attribute* or *set_attribute* command does not match the data type already defined for this attribute.

**What Next**

Check the attribute name and data type against any previous attribute definitions, and correct the command.

---

## UID-324

(warning) '%s' has '%d' unresolved design%s or reference%s.

**Description**

Elaboration of the indicated Verilog/SystemVerilog configuration did not completely link. Either a top design template, or a lower level design reference was not resolved within the specified design libraries.

**What Next**

Check the defined paths of design libraries using report_design_lib, or verify that the necessary link_libraries have loaded.

---

## UID-325

(warning) Option '%s' is ignored because '%s' is not specified.

**Description**

You receive this warning message because you have specified only one option with the *balance_buffer* command. You must specify both options or specify no options. The specified option is ignored.

**What Next**

Reissue the *balance_buffer* command and specify both options or do not specify either option.

---

## UID-326

(error) Can't find pin '%s' on cell '%s'.

---

## UID-327

(warning) Arcs dont exist or have been disabled from pin '%s' to pin '%s' on cell '%s'.

Feedback

### Description

It is invalid to disable timing arcs between two pins where there are no arcs between the pins. Either the pins dont exist on the library cell or they have been disabled for some reason.

---

## UID-330

(error) '%s' in command get_design_parameter is not a valid parameter name for the current design. Valid parameter names are '%s'.

### Description

The specified parameter name in the *get_design_parameter* command is not one of the parameters for the current design.

### What Next

Check your spelling of the parameter name, or check the current design.

---

## UID-331

(error) Parameter '%s' does not have a value.

### Description

The specified parameter for the current design has not been assigned a value, and has no default value.

### What Next

Specify a default value for this parameter in the library, or *elaborate* the cell with a value for this parameter.

---

## UID-332

(error) The -library option is only valid with the db format.

### Description

The *-library* option is used to write db designs into a design library.

### What Next

To store vhdl or verilog designs in a design library, use the *analyze* command. (Currently, only vhdl, verilog, and db designs can be stored in a design library.)

Feedback

## UID-333

(error) The -out flag cannot be used with the -library flag.

**Description**

When you write a db design into a library, the filename and path are determined from the db design name and the library name respectively. As a result, it is illegal to use the *-out* flag when the *-library* flag is specified.

**What Next**

Re-issue the command, using only one of these flags.

## UID-334

(error) Tried to write the root object '%s' into a design library.

**Description**

It is illegal to write root objects into a design library.

**What Next**

On the command line, specify the individual designs that you would like to write, instead of the root object.

## UID-335

(warning) The assignment operator "=" appears in the filter string.

**Description**

The assignment operator is illegal in a filter string.

**What Next**

You may have typed the assignment operator "=" in place of the comparison operator "==" (is equal to). Try the filter command again, but be sure to use "==" for comparison.

## UID-336

(error) You cannot perform the command '%s' on design '%s'\n \tbecause it is a limited design.

Feedback

### Description

Access to a design can be limited for either of two reasons: (1) The design is fully licensed, but it has not yet been mapped. (2) The only licence that could be checked out for the design was a limited license. As a result, the internals of the design are hidden. You cannot write the design out, view a schematic, or execute any commands that would allow you to see the internals. There are also certain commands (such as group) that do not apply to limited designs.

### What Next

If the design is simply not mapped, then compiling the design will allow you access to it. If the design is limited because of licenses, then the only way to get access to the design is to check out a license that gives full access.

---

## UID-337

(error) You cannot perform the command '%s' on design '%s'\n \tbecause it contains the limited design '%s'.

### Description

Access to a design can be limited for either of two reasons: (1) The design is fully licensed, but it has not yet been mapped. (2) The only licence that could be checked out for the design was a limited license. As a result, the internals of the design are hidden. You cannot write the design out, view a schematic, or execute any commands that would allow you to see the internals. There are also certain commands (such as group) that do not apply to limited designs.

### What Next

If the design is simply not mapped, then compiling the design will allow you access to it. If the design is limited because of licenses, then the only way to get access to the design is to check out a license that gives full access.

---

## UID-338

(error) Cannot access design parameters on object '%s'.\n \tDesign parameter values are only valid on designs and references.

### Description

By convention an attribute name beginning with a dollar sign ($) indicates a design parameter (i.e. VHDL generic or Verilog parameter). Design parameters only exist on design or reference objects.

Feedback

**What Next**

Either specify a valid design or reference, or don't use ``$" in your attribute name. You may need to specify objects using the command *find(design, my_des)* or *find(reference, my_ref)* to insure that you are working on a design or reference.

---

## UID-339

(warning) Can't remove attribute '%s' from object '%s'.

**Description**

You cannot remove a static or a derived attribute from this object.

**What Next**

Don't try to remove a static or derived attribute.

---

## UID-340

(error) The connection class attribute can be set only on design ports.

**Description**

The *set_connection_class* command works only on ports that are connected to a design. You can't use this command to change the connection class attribute for ports in a library.

**What Next**

To change the connection class for library ports, you need to change the text file description of the library (usually in a '.lib' file) and run Library Compiler on that library. If you intended to set the connection class for ports in a design, check that you have specified the port(s) correctly.

---

## UID-341

(warning) Design '%s' has '%d' unresolved references. For more detailed information, use the "link" command.

---

## UID-342

(error) The cell '%s' is not a synthetic library module, so set_implementation was not applied to it.

Feedback

**Description**

Implementations can only be set on synthetic library modules. You cannot use set_implementation on non-synthetic library parts. You also cannot use set_implementation on synthetic library operators. It works for modules, not for operators. Use report_cell to see which cells are synthetic library modules.

This error can sometimes arise if the synthetic library in which this implementation resides has not been referred to in the link_library variable.

**What Next**

Check the setting of the link_library variable to make sure it lists the synthetic library in which the module to which the cell refers resides. If not, update link_library and re-issue the set_implementation command.

If you want to set the implementation of a synthetic library operator, you can include directives in your HDL source.

---

## UID-343

(error) Invalid read_timing context. Valid contexts are vhdl, verilog.

---

## UID-344

(warning) Can't %s synthetic operator '%s'.

---

## UID-345

(error) The value for the 'max_levels' argument should be\n \tbetween 1 and %d for design '%s'.

---

## UID-346

(warning) The -max_levels argument is ignored when -hierarchy\n \toption is not given.

---

## UID-347

(warning) Can't link reference '%s' to design '%s'.

---

## UID-348

(warning) Creating virtual clock named '%s' with no sources.

Feedback

**Description**

This warning occurs when a virtual clock is created. A virtual clock has a name but no sources. This means it is not applied to any ports or pins in the design. A virtual clock may be used to specify input or output delay.

**What Next**

The command *create_clock -period 10 -name CLK* does not apply the clock to any sources. If you want to apply the clock to a pin or port, you must specify the pin or port as in *create_clock -period 10 CLK* or *create_clock -period 10 -name CLK ff1/CP*.

---

## UID-349

(warning) Could not read any link_library files. The link_library setting %s may be invalid.

---

## UID-350

(warning) Could not infer bus %s for design %s, because its index range does not match its width.

---

## UID-351

(warning) The variable '%s' is \n\ \tobsolete. Use %s.

**Description**

This variable has been replaced and no longer has any effect.

**What Next**

In your script, replace the setting of this variable with the new setting.

---

## UID-352

(error) At least one argument must be specified with this command.

**Description**

This command requires at least one argument. Please refer to the man page for the command for more information.

---

## UID-353

(error) Arguments '%s' and '%s' are mutually exclusive.

Feedback

**Description**

Some of the arguments specified in the command are mutually exclusive. Please refer to the man page for the command for more information.

---

## UID-354

(error) '%s' is a bad option. Please specify '%s'.

**Description**

The argument is not what is expected by the command. Please refer to the man page for the command for more information.

---

## UID-360

(error) Options -transparent and -user only valid when setting scan FALSE.

---

## UID-361

(error) Options -transparent and -user can not both be specified.

---

## UID-362

(error) Option -transparent can not be specified with non-latch cell/design %s.

---

## UID-363

(warning) Port information will be listed for the design of the current\n\tinstance. This information is not used for analysis or optimization\n\tunless current_design is set to '%s'.

**Description**

Attributes on subdesign ports are not considered for timing or optimization. Only those attributes on ports at the top level (as specified by the *current_design* command) are used. The *report_port* command shows the attributes for the design of the *current_instance*, but these attributes will not have an effect unless *current_design* is used to change the top-level design.

**What Next**

If you want information on the ports of the current design, use the *current_instance* command, with no arguments, before issuing the *report_port* command. If you want the information about ports for the design of the current instance, you can ignore this warning message.

Feedback

## UID-364

(error) The current_instance is not a unique instantiation \n\t of design %s. Please rerun command %s after running uniquify.

## UID-365

(error) Cell %s is not owned by the current_instance. \n\tOnly cells in the current_instance can be specified.

## UID-366

(error) Cell %s is not owned by the current_design. \n\tOnly cells in the current_design can be specified.

## UID-367

(error) Design %s is not referenced by any cell in current_design. \n\tOnly designs referenced in current_design can be specified.

## UID-368

(error) \n\tOnly one design can be specified for top-down copying of attributes.

## UID-369

(warning) \n\t The -all option to the link command is now obsolete.\n\t Please add "*" at the head of your link_library variable to get this functionality.

### Description

Make sure the astericks is in quotes when using the "*" option. A valid link_library setting is link_library = {your_library.db, "*"}

### What Next

Use the "*" in your link_library variable instead of the -all option.

## UID-370

(error) The design '%s' has clusters and cannot be modified by\n\ the 'group' command. Use the remove_clusters command to remove the\n\ clusters from the design first.

Feedback

**Description**

The *group* command cannot be executed on a design with physical hierarchy because this would result in the hierarchical names of the design's cells changing. This situation would make it impossible for the tool that created the physical hierarchy to recognize the names.

**What Next**

If appropriate, run the *remove_clusters* command to remove clusters from the design. After the *group* command creates a new hierarchy, you may have to update your PDEF file so that the clusters in it correspond to the new logical hierarchy.

---

## UID-371

(error) The '.' implementation can only be used with mapped synthetic modules.

**Description**

Using '.' as the implementation name locks a module to keep its current implementation. Since unmapped modules do not yet have their implementation chosen, it is impossible to know what implementation choice to lock.

**What Next**

Use *compile* or *replace_synthetic* first to pick module implementations.

---

## UID-372

(warning) \n\t No cells found to copy attributes from.

---

## UID-373

(error) \n\t The specified range is different from the number of specified objects.

---

## UID-374

(error) \n\t No cells in design '%s' reference any of the specified designs.

---

## UID-375

(error) Cannot use '%s' clock skew option on object '%s' \nbecause the '%s' attribute is already set on that object.

Feedback

### Description

The *rise_delay* and *fall_delay* clock skew attributes are incompatible with the *propagated_clock* attribute. In other words, the network delay to register clock pins is either determined by the delay attributes (ideal), or by evaluating the delays along the clock network (propagated). This error occurs in two situations. If there are already *rise_delay* or *fall_delay* attributes on an object, and *set_clock_skew -propagated* is used, it is an error. Also, it is an error if the *propagated_clock* attribute is present on an object when *set_clock_skew* is used with one of the following options: *-delay, -rise_delay, -fall_delay*.

### What Next

The *remove_attribute* command should be used to remove the conflicting attribute before applying the new information.

---

## UID-376

(warning) It is dangerous to create a clock source on inout port '%s'.

### Description

A circuit in which a clock signal drives an inout port may be unpredictable. If the port is functioning as an output while the clock is driving, bus contention will occur. If the clock is three-stated, any registers in the clock's fanout may lose state.

### What Next

Verify that the circuit is safe according to your design rules. DesignTime will assume that the clock is valid

---

## UID-377

(error) Clock '%s' is not ideal. Cannot set clock transition.

### Description

The indicated clock has a propagated clock skew. Clock transitions can only be specified for ideal clocks.

### What Next

To indicate ideal clocks, use set_clock_skew -ideal.

---

## UID-378

(error) '%s' is not a valid format.

Feedback

**Description**

The format option for the write_script command must be either dctcl or dcsh.

**What Next**

Either specify a valid -format option for write_script or don't specify that option and the result will be in the same format that the shell is currently being run in.

---

## UID-379

(error) Cannot perform %s on the instance %s because\n\tparent is not a unique instantiation of design %s. Please\n\trerun the command after uniquifying the instance.

**Description**

You have issued the command when the parent design is not unique. This command expects the parent design to be unique to make sure that the reference of the design is consistent.

**What Next**

If the instance was passed by mistake please rerun the command after fixing it. Otherwise call uniquify command with -cell option to uniquify the parent instance and rerun the command.

---

## UID-380

(error) Invalid minimum porosity constraint, must be between 0 and 90.

**Description**

The *set_min_porosity* command sets the minimum porosity constraint on the design. The minimum porosity constraint value must be a number between 0 and 90. If set to 0, no porosity optimization will be done. The maximum porosity constraint is 90. For example, if set to 20, Design Compiler optimizes the design so that, on average, 20% of the total cell area is available for over-the-cell routing.

**What Next**

Use *set_min_porosity* with a number between 0 and 90.

---

## UID-381

(error) You are not licensed to run routability optimization.

Feedback

### Description

The *set_min_porosity* command sets the minimum porosity constraint on the design. That is, it constrains the design for a minimum over-the-cell routing area. The routing layers use the over-the-cell routing area to route the design. Design Compiler can improve the routability of designs by favoring cells with over-the-cell routing.

### What Next

To improve the routability of your design, contact your sales representative to acquire a license to run optimization.

---

## UID-382

(warning) Setting a minimum porosity constraint to less than the default %g in link libraries.

### Description

The *set_min_porosity* command sets the minimum porosity constraint on the design. That is, it constrains the design for a minimum over-the-cell routing area. The routing layers use the over-the-cell routing area to route the design. The porosity constraint given is less than the default minimum porosity set in the technologies libraries in the link path. *report_constraint* uses the default minimum porosity constraint specified in the link libraries.

### What Next

With *report_lib*, verify the *default_min_porosity* values in the link libraries. If necessary, set the porosity constraint to a value greater than the minimum default in the link libraries.

---

## UID-383

(error) The porosity constraint value is less than the default %g in target libraries.

### Description

The *set_min_porosity* command sets the minimum porosity constraint on the design. That is, it constrains the design for a minimum over-the-cell routing area. The routing layers use the over-the-cell routing area to route the design. The porosity constraint given is less than the default minimum porosity set in the technologies libraries in the target path. *compile -routability* and *reoptimize_design -routability* use the default porosity constraint from the target libraries by default.

### What Next

Verify with *report_lib* the *default_min_porosity* values in the target libraries. Set the porosity constraint to a value greater than the minimum default in the target libraries if necessary.

Feedback

## UID-384

(error) \n\t Invalid value specified for bus_naming_style : '%s'.

## UID-385

(error) Invalid value specified for bus_range_separator_style %s.

**Description**

This error message occurs when you specify an invalid value for the *bus_range_separator_style* variable.

Invalid values are character strings that contain the following an odd number of % (percent symbols) followed by either the letter d or the letter s. All of the following are invalid values:

%d

abc%%%d43

xv%ssx

%%%s

**What Next**

Provide a valid value for the *bus_range_separator_style* variable and reexecute the command.

## UID-386

(warning) The variable '%s' is now\n\ \tobsolete. The following, equivalent variable is being set:\n\ \t\t%s = "%s"\t

**Description**

This variable no longer has any effect. It has been replaced.

**What Next**

Replace the setting of this variable in your script with the new setting.

## UID-387

(warning) The variable '%s' is now\n\ \tobsolete. The following, equivalent variables are being set:\n\ \t\t%s = "%s", and,\n\ \t\t%s = "%s"\t

Feedback

**Description**

These variables no longer have any effect. They have been replaced.

**What Next**

Replace the settings of these variables in your script with the new settings.

---

## UID-388

(error) Attribute '%s' does not exist on %s '%s'.

**Description**

The *set_attribute* command changes the value of existing attributes. If the attribute does not exist, then that attribute's value cannot be changed.

**What Next**

The *set_attribute* command can be used to create a new attribute and set that attribute's value when *set_attribute* is used with the *-type* option. The *-type* option is used to specify the data type of the new attribute to be created. See the *set_attribute* manual page for more information on *set_attribute* and the *-type* option.

---

## UID-389

(warning) Current operating conditions specify best_case_tree\n \tas the interconnect model. Annotated resistance will be ignored in\n \tthis case.

**Description**

The best_case_tree model for interconnect assumes 0 interconnect delay. In this case, all annotated resistance values will be ignored.

**What Next**

If best_case_tree is the desired interconnect model, then this message can be ignored. If the resistance values need to be accounted for during timing analysis, then choose a set of operating conditions which do not use the best_case_tree model to model the interconnect delay.

---

## UID-390

(warning) All data in the names file '%s' is invalid.

Feedback

## UID-391

(error) The 'create_clock' command cannot be used on output port '%s'.

**Description**

Clock sources should be input ports or internal pins. In/out ports can be used, but they are not recommended because of bus contention issues.

**What Next**

Identify a valid set of sources and reapply the *create_clock* command.

## UID-392

(error) The object '%s' is a hierarchical object. You cannot set an\n IDDQ-invalid condition on a hierarchical object.

**Description**

Invalid IDDQ conditions must be specified in terms of logic values at ports of the design and at cell pins.

## UID-393

(error) No valid nodes specified for the set_iddq_invalid_state command.

**Description**

The 'set_iddq_invalid_state' was used without specifying a pin/port list for either the -logic_zero or -logic_one conditions. In this case the command does nothing.

**What Next**

Specify a list of ports and/or leaf pins for either the -logic_one or -logic_zero conditions.

## UID-394

(error) Error in command input : set_iddq_invalid_state command ignored.

**Description**

All of the pins specified for the -logic_zero or -logic_one conditions of the 'set_iddq_invalid_state' command were on hierarchical cells. Only leaf pins or ports are allowed for this command.

Feedback

### What Next

Specify only leaf pins or ports for the -logic_zero and -logic_one conditions of the 'set_iddq_invalid_state' command.

---

## UID-395

(warning) Could not link design '%s'.\n \tTherefore, the results of the filter command may be incorrect.

### Description

The *filter* command automatically attempts to link the design before executing the filtering process. Therefore, this warning message is generated only if there were errors during the link.

If a design is not linked, some derived attributes will revert to their default values instead of having the values expected if the design were linked. Therefore, if the design cannot be linked, the *filter* command might not produce correct results.

For example, for a design that uses some sequential library cells that are not properly linked, the *is_sequential* attribute reverts to its default value of *false*, because there is no way to determine whether or not the design is sequential. This situation could occur if the design has not yet been linked, or if errors during the linking process prevent the sequential library cells from being linked properly.

Once the design is linked properly, however, then the *is_sequential* attribute maintains the correct value, and the *filter* command executes correctly.

### What Next

Resolve the link errors and then re-execute *filter*.

---

## UID-396

(error) The cell '%s' does not reference a synthetic library operator DW03_mult_n_stage, so set_pipeline_stages was not applied to it.

### Description

Cannot apply the 'set_pipeline_stages' command to the cell because it does not reference a synthetic library operator, DW03_mult_n_stage.

### What Next

Use report_cell to see which cells reference synthetic library operators DW03_mult_n_stage. Check the man page for the 'set_pipeline_stages' command.

Feedback

## UID-397

(error) You must chose one of {fixed min auto}.

## UID-398

(warning) Pin '%s' does not belong to a cell in design '%s'.

**Description**

This warning occurs when calling cell_of command on an instance pin. The given pin belongs to a hierarchical cell of the current design.

**What Next**

Set the current design to the design of that pin, then issue the cell_of command on the pin.

For example, if the message is

Warning: Pin 'U1/U2/U3/A' does not belong to a cell in design 'TOP'

cell_of can accept pins that belong to cells in the design TOP only. You are calling cell_of on an instance pin 'U1/U2/U3/A'. With this example, set current design to the design of instance 'U1/U2', then issue cell_of("U3/A");

## UID-399

(error) Cannot perform %s on the instance %s because\n\tit is not unique. Please\n\trerun the command after uniquifying the instance.

**Description**

You have issued the command when the instance is not unique. This command expects the instance to be unique to make sure that the reference of the design is consistent.

**What Next**

If the instance was passed by mistake please rerun the command after fixing it. Otherwise call uniquify command with -cell option to uniquify the parent instance and rerun the command.

## UID-400

(error) Design is already scan inserted, scan replaced or test ready.

Feedback

### Description

This error is generated when a design have been scan_replaced , is a test_ready or is scan_inserted. Since in all these cases the design have already been modified to have scan cells with an existing scan style, Hence it is not possible to change the scan style now.

### What Next

You can re read the design and set the new scan style and repeat the old flow to take into account the new scan_style.

---

## UID-401

(warning) Design rule attributes from the driving cell will be\n \tset on the port.

### Description

Design rule attributes from the driving cell will be set on the port when the set_driving_cell command is used. This is in addition to timing related parameters that are set on the port.

### What Next

Use the -no_design_rule option for set_driving_cell to avoid having design rules inferred.

---

## UID-402

(warning) Specifying an input delay without an associated clock\n \tcan cause the port's timing to be analyzed incorrectly.

### Description

If a port input delay is specified without reference to a clock, a default clocking waveform is inferred for the port's drivers, based on the flip-flops and latches in the port's transitive fanout. The input data will be assumed to be coming from an edge-triggered flip-flop clocked by the inferred waveform. The program attempts to figure out what the appropriate clock waveform should be, but there is no guarantee that the inferred clock will be appropriate for all cases. Therefore, it is safer to explicitly specify the clock whenever you set an input delay.

### What Next

To avoid having default clocks inferred, use the -clock option with set_input_delay. In some cases, the -level_sensitive and -clock_fall options might also be needed.

Feedback

## UID-403

(error) Can't read in the DB file '%s'.\n\tFile '%s' is created using program version '%s'\n \twhich is newer than '%s'.

### Description

This error indicates that your db file is created using Synopsys program version which is newer than the current program.

### What Next

Check your db file, make sure it is created with Synopsys product whose version is not newer than the current program you are using.

## UID-404

(error) No setup or hold margin was specified.

### Description

Because no setup or hold margin was specified, no clock gating checks will be created.

### What Next

To specify margins for the clock gating checks, use the *-setup* or *-hold* options.

## UID-405

(error) The set_auto_ideal_nets command is obsolete. Use set_auto_disable_drc_nets instead.

### Description

Ideal nets are now defined as having ideal timing (zero latency and transition by default), no DRC checking, and *dont_touch* attributes set on them. The now-obsolete *set_auto_ideal_nets* command was not consistent with this definition of ideal, and has therefore been renamed to *set_auto_disable_drc_nets*. This command works in exactly the same way as *set_auto_ideal_nets*; it accepts the same arguments and by default disables DRC on clock and constant nets.

### What Next

Replace this command with the *set_auto_disable_drc_nets* command. Alternatively, if you want to set nets to be ideal, use *set_ideal_net* or *set_ideal_network* instead.

Feedback

### See Also

- set_auto_disable_drc_nets
- set_ideal_network

---

## UID-406

(information) set_ideal_net disables DRC, sets dont_touch on the net, and makes its timing ideal.

### Description

This message informs you about the current behavior of the *set_ideal_net* command. Since release 2001.08, ideal nets are defined as having ideal timing (zero latency and transition by default), no DRC checking, and *dont_touch* attributes set on them. The previous behavior of the *set_ideal_net* command was only to switch off DRC checking for the net. To revert to the previous behavior, set the variable *use_old_ideal_net* to true.

### What Next

This is an informational message only. No action on your part is required.

### See Also

- set_ideal_network

---

## UID-408

(warning) Current design named %s is hidden.

### Description

This warning message occurs only in Incremental Mode.

You receive this warning message if the current design is hidden, either by a limited license or by a don't show reference.

Access to a design can be limited for either of the following reasons:

- The design is fully licensed, but it has not yet been mapped.
- The only license that could be checked out for the design was a limited license.

The result of limited access is that the internals of the design are hidden. You are not permitted to write the design out, view a schematic, or execute any commands that would allow you to see the internals. Certain commands, such as the *group* command, do not apply to limited designs.

Feedback

**What Next**

This is a warning message only. No action is required.

However if the result is not what you intended, you can do one of the following:

- If the design is not mapped, compile the design.
- If the design is limited because of licenses, check out a license that allows full access.

After making your changes, you can reexecute the command that issued this warning message.

---

## UID-409

(error) Cannot group the cell %s because its parent is not %s.\n\tOnly cells whose parents are same can be grouped.

**Description**

You have issued the command when the cells passed to group command have different parents. This command expects all the cells to have the same parent.

**What Next**

Please rerun the command after fixing the error.

---

## UID-410

(error) Unable to find replacement gate '%s'.

**Description**

The replacement gate specified in the set_combinational_type command could not be found in the target library.

**What Next**

Either change the target_library variable to include the library which contains the replacement gate or change the replacement gate.

---

## UID-411

(error) Replacement design '%s' is not a combinational gate.

**Description**

The replacement_gate in the set_combinational_type command is not a combinational gate.

Feedback

**What Next**

Set_combinational_type must have a combinational gate as its replacement gate. Specify a new replacement gate.

---

## UID-412

(warning) Overriding compile directives for cell %s.

**Description**

The compile directives set by the *set_compile_directives* command are ignored for this cell. The *set_combinational_type* command must override the compile directives to ensure that this cell is only translated into the given replacement cell.

**What Next**

Do not use *set_combinational_type* and *set_compile_directives* on the same cell.

---

## UID-413

(warning) Overriding previous replacement cell for cell %s.

**Description**

There was a previous *set_combinational_type* command issued for this cell. The old command is ignored.

**What Next**

The *set_combinational_type* overrides any previous *set_combinational_type* commands issued for this cell.

---

## UID-414

(error) Cannot set_compile_directives on cell %s because it already has set_combinational_type

**Description**

There was a previous *set_combinational_type* command issued for this cell. You cannot *set_compile_directives* for the same cell because the *set_combinational_type* command needs to ensure that this cell is only translated into the particular replacement cell.

**What Next**

Remove the *set_combinational_type* information by doing a remove_attribute <cell_name> combinational_type_exact.

Feedback

## UID-415

(error) Cell list contains generic cell %s.

**Description**

The *set_compile_directives* command has been called on a list of cells containing generic cells .

**What Next**

Use *set_compile_directives* only on list of technology specific cells.

## UID-416

(information) The compile directive given by option %s had already been %s on all specified cells.

**Description**

A *set_compile_directives* command option has no effect because the corresponding directive had already been set for the specified cells.

**What Next**

Check if the options of *set_compile_directives* are set correctly. Note that the default is TRUE.

## UID-417

(error) Cell list contains synthetic operator cell %s.

**Description**

The *set_compile_directives* command has been called on a list of cells containing synthetic operators.

**What Next**

Use *set_compile_directives* only on list of technology specific cells.

## UID-418

(error) Cell list contains cell %s which is control logic of SELECT_OPs.

Feedback

**Description**

The *set_compile_directives* command has been called on a list of cells containing control logic of SELECT_OPs.

**What Next**

Use *set_compile_directives* only on list of technology specific cells.

---

## UID-420

(error) %s cannot be used on port with clock specified.

---

## UID-421

(error) bc_dont_ungroup cannot be applied after the current design is scheduled.

**Description**

You receive this message if you attempt to use *bc_dont_ungroup* after the current design is scheduled. Currently, you must use *bc_dont_ungroup* before scheduling the current design.

**What Next**

Apply *bc_dont_ungroup* before scheduling your design.

---

## UID-422

(error) get_timing_paths requires a mapped design.

**Description**

You receive this error message if you attempted to use *get_timing_paths* with an unmapped design. You cannot generate a timing report for an unmapped design because the correct timing arcs are not available.

**What Next**

Compile your design using the *compile* command before executing *get_timing_paths*.

**See Also**

- get_timing_paths

Feedback

## UID-423

(error) Invalid path group name %s supplied with -group option.

### Description

This error message occurs if you specify a non-existent path group name when using the *-group* option with the *report_timing* command.

### What Next

Specify a valid (existing) path group name when using the *-group* option and then run the *report_timing* command again.

Path groups are created implicitly according to the clock of the destination register of a path. You can also define a path group with via the *group_path* command. To see a list of all valid path groups, issue the *report_path_group* command. Any of the path group names listed can be supplied with the *-group* option in the *report_timing* command.

### See Also

- group_path
- report_path_group
- report_timing

## UID-424

(warning) The min ideal latency is greater than the max ideal latency\n \t for the object '%s'. Using the latest value for both min and max.

### Description

You receive this message because you set the minimum ideal latency higher than the maximum ideal latency for the specified object. The max ideal latency must be greater than or equal to the min ideal latency for the same object. The message informs you that the latest value entered is being used for both minimum and maximum ideal latency.

### What Next

If it is acceptable to you for the latest value to be used for both minimum and maximum ideal latency, no action is required on your part. However, if you want different values for the minimum and maximum ideal latency, reset these values using the *set_ideal_latency* command, and ensure that the maximum value is greater than or equal to the minimum value.

Feedback

### See Also

- set_ideal_latency

---

## UID-425

(warning) You cannot set ideal latency on object %s, \n because it is not a pin of a leaf cell or a port of the current design.

### Description

You receive this message if you attempt to set an ideal latency on an object that is not a pin of a leaf cell (a cell that does not contain other cells), or a port of the current design. You can set an ideal latency only on leaf cell pins or top-level ports.

### What Next

Reexecute the *set_ideal_latency* command with an *object_list* that contains only leaf cell pins and/or top-level ports.

### See Also

- set_ideal_latency

---

## UID-426

(warning) You cannot set ideal transition time on object %s, \n \t because it is not a pin of a leaf cell or a port of the current design.

### Description

You receive this message if you attempt to set an ideal transition time on an object that is not a pin of a leaf cell (a cell that does not contain other cells) or a port of the current design. You can set an ideal transition time only on leaf cell pins or top-level ports.

### What Next

Reexecute the *set_ideal_transition* command with an *object_list* that contains only leaf cell pins and/or top-level ports.

### See Also

- set_ideal_transition

Feedback

## UID-427

(warning) The clock source '%s' has ideal timing specified, \n \t but the clock was just created as an ideal clock. \n \t Ignoring previously-specified ideal timing for this clock source.

### Description

You receive this message from the *create_clock* command to warn you that the specified clock source has ideal timing previously specified. Clocks created by the *create_clock* command are ideal clocks, which ignore ideal timing. This message warns you that the previously-specified ideal timing is being ignored for the specified clock source.

An ideal clock uses latency and transition specified by *set_clock_latency* and *set_clock_transition* and ignores an ideal timing. Therefore, you cannot use the *set_ideal_latency* and *set_ideal_transition* commands on this ideal clock.

### What Next

If it is acceptable to you that the previously-specified ideal timing is ignored for this clock, no action is required on your part. Otherwise, verify that the specified clock source was the one you intended to use for creating an ideal clock. If so, and if you want to set latency and transition time on the ideal clock, use the *set_clock_latency* and *set_clock_transition* commands.

### See Also

- create_clock
- set_clock_latency
- set_clock_transition
- set_ideal_latency
- set_ideal_transition

## UID-428

(warning) You cannot set an ideal latency on the clock source\n \t %s, because it is an ideal clock. Ignoring the specified ideal latency unless the clock source is marked as ideal.

### Description

You receive this message if you attempt to set an ideal latency on a clock source created using the *create_clock* command. Clocks created using *create_clock* are ideal clocks, and you cannot set an ideal latency on an ideal clock. Instead, to set its latency, use the *set_clock_latency* command.

Feedback

### What Next

If it is acceptable to you that the ideal latency is ignored for the specified ideal clock, no action on your part is required. Otherwise, if you want to set latency on the clock, do so using the *set_clock_latency* command.

### See Also

- create_clock
- set_clock_latency
- set_ideal_latency

---

## UID-429

(warning) You cannot set an ideal transition time on the clock source \n \t %s, because it is an ideal clock. Ignoring the specified ideal transition time unless the clock source is marked as ideal.

### Description

You receive this message if you attempt to set an ideal transition time on a clock source created using the *create_clock* command. Clocks created using *create_clock* are ideal clocks, and you cannot set an ideal transition time on an ideal clock. Instead, to set its transition time, use the *set_clock_transition* command.

### What Next

If it is acceptable to you that the ideal transition time is ignored for the specified ideal clock, no action on your part is required. Otherwise, if you want to set a transition time on the clock, do so using the *set_clock_transition* command.

### See Also

- create_clock
- set_clock_transition
- set_ideal_transition

---

## UID-430

(error) No object was specified for the set_clock_uncertainty comand.

Feedback

### Description

The clock uncertainty can be specified on a clock a clock or a clock pin. You can also specify inter-clock uncertainty by using the *-from* and *-to* to specify from which clock to which clock to apply this uncertainy. The *rise_delay* and *fall_delay* clock skew The source clock latency is the time it takes for a clock signal to propagate from its actual ideal waveform origin point to the clock definition point in the design. This attribute can be set on a clock source, or on the clock itself. In case it is set on the clock, it is applied to all the clock sources.

### What Next

Reapply the command with the relevant object(s).

---

## UID-431

(error) -to clock_name is missing.

### Description

This error message occurs when a required clock name is not specified. To set interclock uncertainty, specify a *-from from_clock* and *-to to_clock* pair. Use the *-from* and *-to* options only for interclock uncertainty.

To set uncertainty on a clock or a clock pin, specify the object name at the end of the command line.

### What Next

Run the command again using the correct options.

### See Also

- set_clock_uncertainty

---

## UID-432

(error) -from clock_name is missing.

### Description

This error message occurs when the required clock name is missing. To set interclock uncertainty specify both the *-from from_clock* and *-to to_clock*>. The *-from* and *-to* option are used only for interclock uncertainty.

To set uncertainty on a clock or a clock pin, specify the object name at the end of the command line.

Feedback

### What Next

Rerun the command using the correct options.

### See Also

- set_clock_uncertainty

---

## UID-433

(error) Too many objects for inter clock uncertainty.

### Description

To set inter-clock uncertainty use *-from* <from_clock> and *-to* <to_clock> option. To set uncertainty on a clock or a clock pin, just specify the object name at the end of the command line. These 2 different types of clock uncertainty values cannot be set in one command line.

### What Next

Reapply the command(s) with the correct options.

---

## UID-434

(error) '%s' is not a name of a clock.

### Description

To specify inter-clock skew the *-from* and *-to* options should be followed by a clock name. To specify a skew to a register, the object should be specified without the *-from* and *-to* option.

### What Next

Reapply the command with the correct syntax.

---

## UID-435

(error) Cannot set clock source latency on '%s' .This attribute can be set on a clock or a clock source pin.

### Description

The source clock latency is the time it takes for a clock signal to propagate from its actual ideal waveform origin point to the clock definition point in the design. If you receive this message, you have attempted to set clock source latency on an object other than a clock

Feedback

or a clock source pin. This attribute can be set on a clock source, or on the clock itself. If it is set on the clock, the attribute is applied to all clock sources.

**What Next**

If you want to set clock source latency, you need to identify the relevant clock or clock source where you want to set it. If you want to set actual latency on a clock pin, use the *set_clock_latency* command without the *-source* option.

---

## UID-436

(error) '-rise' and '-fall' options are valid only for inter-clock uncertainty

**Description**

You receive this message if you attempt to specify *-rise* and *-fall* options for an endpoint uncertainty; you can specify only one value for the setup check and one value for the hold check. The value is applied to both the rise and the fall transition. The *-rise* and *-fall* options are valid only for inter-clock uncertainty.

**What Next**

Reapply the command with the correct syntax.

---

## UID-437

(warning) '%s' option is obsoleted. Although it will still function normally now, please use '%s' command to replace this option in the future.

**Description**

The specified option is obsolete, although it is still functional in the current software version. In the future, please use the replacement command.

**What Next**

Replace the obsolete option with the specified command.

---

## UID-438

(warning) Attribute '%s' is removed from from object '%s'.

**Description**

The command executed caused the specified attribute to be removed from the specified object, either because the command must proceed without this attribute, or because the value of the attribute is invalid for this command.

Feedback

### What Next

This message is for your information. No action is required on your part.

---

## UID-439

(warning) '%s' command is obsoleted. Although it will still function normally now, please use '%s' command to replace this command in the future.

### Description

This command is obsolete. However, it is still functional in this version. In the future, please use the replacement command.

### What Next

Remove the obsolete commands, and replace it with the new command.

---

## UID-440

(error) Cannot write the '%s' constraints file.

### Description

The tool issues this error when it fails to open or write to the specified file. There can be several reasons for this. The most common reasons are: you do not have write permissions in the directory, the disk is full, or you already have a file with the same name that cannot be deleted.

### What Next

Try to manually create a file with the same name. Fix any problems that you encounter, and then delete the file and run the command again.

---

## UID-441

(warning) You cannot specify an input transition time for the %s '%s'\n \t using set_input_transition. Ignoring the transition time for this port.

### Description

You receive this message if you specify an input transition for the specified ideal network port or ideal clock source, using *set_input_transition*. You cannot use *set_input_transition* for these objects; you must use *set_ideal_transition* for ideal network ports, and *set_clock_transition* for ideal clock sources. This message informs you that the input transition specified by *set_input_transition* is being ignored for the specified object.

Feedback

### What Next

If it is acceptable to you that the input transition time is ignored, no action is required on your part. Otherwise, if you want to set an input transition time, use *set_ideal_transition* for an ideal network port and *set_clock_transition* for a clock source.

### See Also

- set_clock_transition
- set_ideal_transition
- set_input_transition

---

## UID-444

(error) '%s' is a PrimeTime option that is not supported by this tool.

### Description

The tool issues this error when you run a command with an option that is supported by PrimeTime, but is not supported by this tool. To see the options supported by this tool, run the command with the *-help* option.

### What Next

Remove the unsupported option and rerun the command.

---

## UID-445

(warning) Ignoring %s objects in collection '%s' because they are not of type %s.

### Description

You receive this error message when you specify a collection as an argument to a command, and some objects in the collection are not of a type accepted by the command. For example, you receive this error message if you execute the *report_net* command and not all of the objects in the specified collection are nets.

### What Next

Use the command *query_objects -verbose* to see the details of the objects in the collection and their types.

---

## UID-446

(Error) For pattern '%s', %s.

Feedback

### Description

A problem has occured while trying to compile the regular expression pattern shown.

### What Next

Review the description in the man page for regexp to identify and correct the problem. Make sure that you intended the pattern to be a regular expression instead of a simple string match. A common mistake is to mix up the use of the wildcard '*' in string match with the corresponding regular expression '.*'. Also, remember that in regular expressions, the square brackets '[', ']' by default enclose a range of characters, and must be escaped with a backslash to be treated as regular characters.

---

## UID-447

(warning) Your SDC output may contain ambiguous names because you have\n \tset the variable sdc_write_unambiguous_names to FALSE.

### Description

Beginning with version 1.2, the Synopsys Design Constraints (SDC) format has features which ensure that the cell, net, pin, lib_cell, and lib_pin names written to the file are unambiguous. Some third party applications do no understand these features, and to suppress them, you set the variable *sdc_write_unambiguous_names* to FALSE. This warning reminds you that the setting of this variable may cause your SDC output to be ambiguous.

### What Next

Verify that you really want to disable the writing of unambiguous names. Review the man pages for *write_sdc* and *sdc_write_unambiguous_names*.

---

## UID-448

(warning) The sdc version specified is not valid. Valid options are 1.2, 1.3, 1.4, 1.5, and latest.

### Description

You receive this message if you specify an invalid version with the *-version* option of *write_sdc*. Valid values for *-version* are 1.2, 1.3, 1.4, 1.5, and latest (the default). Notice that in this case, specifying latest or 1.5 are equivalent, as is issuing *write_sdc* without the *-version* option.

### What Next

Reexecute *write_sdc* and either specify a valid version with the *-version* option, or omit the *-version* option and get the latest version.

Feedback

**See Also**

- write_sdc

---

## UID-449

(warning) The sdc version specified is not valid for db mode. Valid options are 1.2, 1.3, 1.4, and latest.

**Description**

You receive this message if you specify an invalid version with the *-version* option of *write_sdc*. When in DB mode the valid values for *-version* are 1.2, 1.3, or 1.4. Notice that in this case, specifying 1.4 is equivalent to issuing *write_sdc* without the *-version* option.

**What Next**

Reexecute *write_sdc* and either specify a valid version with the *-version* option, or omit the *-version* option and get the latest supported version.

**See Also**

- write_sdc

---

## UID-450

(Warnning) Value for %s is negative.

**Description**

A negative value was given in the command line; the value was accepted.

**What Next**

It is the user's responsibility to verify that the negative value is a valid value to describe this property.

---

## UID-451

(Warnning) '%s' option is ignored when '%s' argument is present.

**Description**

The given option is ignored to the current command when the specified argument is present. The command performs without the given option.

Feedback

## UID-452

(Warning) Empty object list.

### Description

You receive this message if you execute the *report_net_fanout* command with an empty pair of braces ({}) for the *net_list* argument, or with all members of *net_list* invalid. This condition could be caused by spelling errors or typos, or by listing nets for the wrong instance.

### What Next

If you want *report_net_fanout* to generate the report for all nets in the current instance, execute the command without the optional *net_list* argument. Otherwise, verify that the net list is correct for the current instance and that the names are spelled correctly. Then reexecute the *report_net_fanout* command.

### See Also

- report_net_fanout

## UID-453

(error) The argument of the -bound option must be \n \t larger than the argument of the -threshold option.

### Description

You receive this message if you execute the *report_net_fanout* command and specify a value for the *-bound* option that is smaller than or equal to the value for the *-threshold* option. The *-bound* option represents the upper level of fanout for displaying nets, while the *-threshold* option represents the lower level. Therefore, the value of *-bound* must be larger than the value of *-threshold*.

### What Next

Reexecute the *report_net_fanout* command and specify a value for the *-bound* option that is larger than the value for the *-threshold* option.

### See Also

- report_net_fanout

## UID-455

(error) The value %s is not a legal option for the\n \tset_transform_for_retiming command.

Feedback

**Description**

The *set_transform_for_retiming* command can only have the values *-multiclass*, *-decompose*, or *-dont_retime*.

**What Next**

Check the spelling and choose either the *-multiclass*, *-decompose* or *-dont_retime* option for the *set_transform_for_retiming* command.

---

## UID-456

(error) The value %s is not a legal option for the\n \tset_state_for_retiming command.

**Description**

The comand *set_state_for_retiming* can only have the values *-preserve* or *-dont_care*.

**What Next**

Check the spelling and specify the *-preserve* or *-dont_care* option for the *set_state_for_retiming* command.

---

## UID-457

(error) Cannot set size_only on cell '%s'.

**Description**

The *set_size_only* command works only on leaf cells.

**What Next**

When you use *set_size_only*, make sure the cell you specify is a leaf cell.

---

## UID-458

(warning) Cannot perform '%s' on object '%s' because it is not a port.

**Description**

The specified object is not a port. The command cannot be applied.

---

## UID-459

(warning) Cannot perform '%s' on object '%s' because it is not a port.

Feedback

**Description**

The specified object is not a port. The command can not be applied.

**What Next**

Varify that the specified object is not a port.

---

## UID-460

(error) Option '%s' is only available for dc_shell-t.

**Description**

The indicated option is only available in Tcl mode (dcshell-t).

**What Next**

Start dc_shell as dc_shell-t or dc_shell -tcl_mode.

---

## UID-461

(error) The -xg_mode option is not available.

**Description**

This error message occurs when the *-xg_mode* option is not available for the current release, or when there is a problem with accessing proper license features.

**What Next**

Start dc_shell without using the *-xg_mode* option.

---

## UID-462

(error) Invalid implementation '%s' for cell '%s'.

**Description**

The specified implementation for the synthetic cell does not exist in the specified synthetic libraries.

**What Next**

Verify the existence of the specified implementation in the synthetic libraries. You can use the report_synlib command to list the contents of the synthetic library.

Feedback

## UID-463

(error) %s name is required to set implementation on operator '%s'.

**Description**

Either module name or implementation name is missing when setting implementation on operator.

**What Next**

Reissue the *set_implementation* command with the implementation string that contains the module name and implementation name in the "<module_name>/<impl_name>" format.

## UID-464

(error) Invalid module '%s' or implementation '%s' for cell '%s'.

**Description**

The specified module name or implementation name on operator cell does not exist in the synthetic libraries.

**What Next**

Verify the existence of the specified implementation in the synthetic libraries. You can use the report_synlib command to list the contents of the synthetic library.

## UID-465

(error) Incorrect arguments to the set_retiming_bound command. Choose either the -forward or -backward option to specify the direction of the retiming_bound.

**Description**

This error message occurs because the *set_retiming_bound* command can only have the value of *-forward* or *-backward*.

**What Next**

Check the spelling and rerun the *set_retiming_bound* command with either *-forward* or the *-backward*.

## UID-466

(error) Missing arguments to command set_retiming_bound. Specify the direction of the retiming_bound using the -forward or -backward option.

Feedback

### Description

This error message occurs because the *set_retiming_bound* command requires either the *-forward* or *-backward* option to specify the direction that the registers move.

### What Next

Rerun the *set_retiming_bound* command with either *-forward* or *-backward*.

---

## UID-467

(error) The '%s' option is not supported in Design Compiler.

### Description

This error message occurs when the specified option is not supported by Design Compiler.

### What Next

Rerun the command without the option.

### See Also

- report_qor

---

## UID-468

(warning) The '%s' option is not supported in Design Compiler and will be ignored.

### Description

This warning message occurs when the specified option is not supported by Design Compiler.

### What Next

Rerun the command without the option.

---

## UID-470

(warning) It is dangerous to create a generated clock on an inout port '%s'.

### Description

A circuit in which a generated clock drives an inout port may be unpredictable. If the port is functioning as an output while the generated clock is driving, bus contention will occur. If the generated clock is three-stated, any registers in the generated clock's fanout may lose state.

Feedback

**What Next**

Verify that the circuit is safe according to your design rules. DesignTime will assume that the generated clock is valid.

---

## UID-471

(error) The 'create_generated_clock' command cannot be used on output port '%s'.

**Description**

Generated clocks should be input ports or internal pins. In/out ports can be used, but they are not recommended because of bus contention issues.

**What Next**

Identify a valid set of sources and reapply the *create_generated_clock* command.

---

## UID-472

(error) You can specify only a single object for master clock source.

**Description**

The -source option takes only a single object as an argument.

**What Next**

You can have a generated clock derived from a single clock. You cannot generate a clock from more than one master clock.

---

## UID-473

(error) The number of edges specified '%d' is not an odd number > 3.

**Description**

The number of edges to make one period of the generated clock waveform has to be an odd number > 3.

**What Next**

Carefully specify edges and ensure that you specify one full clock cycle using the edges.

---

## UID-474

(Error) In the -edge specification of create_generated_clock\n \t'%s', the edge numbers must be greater than zero and in increasing order.

Feedback

### Description

In the -edge specification of a create_generated_clock command, the edge numbers specified must be in increasing order. The edge numbers should also be greater than zero, since it refers to the edge-number of the master clock.

### What Next

Check the -edge spec in create_generated_clock command and edge numbers increasing.

---

## UID-475

(Error) The master source must be specified.

### Description

For *create_generated_clock*, the -source option must be specified. This option specifies the source of the generated clock.

### What Next

Specify -source in the create_generated_clock command.

---

## UID-476

(warning) Converting propagated clock at '%s' to ideal clock.

### Description

You receive this message if you use *set_clock_latency* to set network latency on a propagated clock. Clocks processed by *set_clock_latency* are automatically converted to ideal clocks. The *set_propagated_clock* command is normally used to set propagated clock network latency. This message warns you that the specified propagated clock is being converted to an ideal clock.

### What Next

This is a warning message only. If it is acceptable to you that the specified propagated clock is converted to an ideal clock, no action is required on your part. Otherwise, if you want the clock to be propagated, do not use *set_clock_latency* on the clock; use *set_propagated_clock* to specify a propagated clock latency.

### See Also

- set_clock_latency
- set_propagated_clock

Feedback

## UID-477

(warning) Converting ideal clock at '%s' to propagated clock.

### Description

You receive this message if you use *set_propagated clock* to set the *propagated_clock* attribute on a clock on which you previously set latency or transition. Clocks for which latency or transition have been set are automatically considered to be ideal clocks. Setting the *propagated_clock* attribute on an ideal clock converts it to a propagated clock. This message warns you that the specified ideal clock is being converted to a propagated clock, and the previously-specified latency or transition will not be used.

### What Next

This is a warning message only. If it is acceptable to you that the specified ideal clock is converted to a propagated clock, no action is required on your part. Otherwise, if you want the clock to remain ideal, remove the *propagated_clock* attribute using the *remove_propagated_clock* command, and the previously-specified latency and transition values will be used.

### See Also

- set_clock_latency
- set_propagated_clock

## UID-478

(Error) Virtual clock `%s` cannot be made propagated.

### Description

You receive this message if you attempt to set a *propagated_clock* attribute on a virtual clock. You cannot specify a virtual clock as a propagated clock, because virtual clocks do not have any source and cannot be made propagated.

### What Next

Review your design to verify your intended clock scheme.

### See Also

- set_propagated_clock

Feedback

## UID-479

(Error) Clock transition cannot be specified for virtual clock `%s` .

### Description

You receive this message if you attempt to use the *set_clock_transition* command on a virtual clock. You cannot specify clock transitions on virtual clocks, because they do not have any source.

### What Next

Review your design to verify your intended clock scheme.

### See Also

- set_clock_transition

## UID-480

(Warning) Clock transition specified for propagated clock `%s` will be ignored.

### Description

You receive this message if you attempt to specify a clock transition on a propagated clock using *set_clock_transition*. You can specify clock transitions only for ideal clocks. This message warns you that the clock transition you specified is being ignored for the clock.

### What Next

If it is acceptable to you that the clock transition is ignored for this clock, no action is required on your part. Otherwise, if you want to specify a clock transition for the clock, remove the *propagated_clock* attribute from it using *remove_propagated_clock*. Then reexecute *set_clock_transition*.

### See Also

- remove_propagated_clock
- set_clock_transition

## UID-481

(error) White spaces are not allowed in the string '%s'.

Feedback

**Description**

This error message occurs when white spaces appear in the input. This command does not allow any white space, such as tabs or spaces, in the string argument.

**What Next**

Rerun the command without any white space in the string argument.

---

## UID-482

(error) An empty string argument was found.

**Description**

This error message occurs when an empty string is entered as an argument. This command does not allow empty strings as arguments.

**What Next**

Rerun the command with a non-empty string argument.

---

## UID-483

(error) Not allowed to remove default power (or ground) net '%s'.

**Description**

This error occurs because the tool attempted to remove from the design the default power net stored in the *mw_logic1_net* variable (or the *mw_logic0_net* variable), which is not allowed. You are allowed to remove other power nets.

**What Next**

Eliminate any command that attempts to remove the default power net.

---

## UID-485

(error) Cannot specify %s without -source.

**Description**

This error message occurs when a specified command option requires another option, which is omitted.

**What Next**

Refer to the manual page for this command for detailed information on valid options and rerun the command.

Feedback

## UID-486

(warning) The output of \ the propagate_constraints command is \ in DCSH mode. Use the UNIX utility dc-transcript to \ convert DCSH script to DC-Tcl script.

### Description

You receive this message because you executed the *propagate_constraints* command from dc_shell-t; that is, in DC-Tcl mode. This message warns you that whether you execute the *propagate_constraints* command in DCSH-mode or in DC-Tcl mode, the output is always in DCSH mode. Thus, the script you generate is in DCSH mode.

### What Next

To convert the script from DCSH format to DC-Tcl format, use the UNIX dc-transcript utility.

## UID-487

(warning) The '%s' option is obsolete. It is ignored.

## UID-488

(error) Cannot specify both "-high" and "-low" as the noncontrolling value for '%s'.

### Description

You receive this error message if you attempt to set the noncontrolling value for a particular object to both high and low through *set_clock_gating_check* command.

### What Next

Only use the *-high* or *-low* option when you are certain about the noncontrolling value for a particular cell. Only one of the two options can be specified.

### See Also

- set_clock_gating_check

## UID-489

(error) Cannot specify %s option for '%s'. Can only specify on cells or pins.

### Description

You receive this message if you execute the *set_clock_gating_check* command and specify both the *-hi* and *-low* options on objects that are neither cells nor pins.

Feedback

### What Next

Use the *-high* or *-low* options only on cells or pins.

### See Also

- set_clock_gating_check

---

## UID-490

(error) Cannot specify both high and low for the same object.

### Description

You receive this message if you executed the *set_clock_gating_check* command and specified both the *-high* and *-low* options for the same object. This message informs you that you can set the noncontrolling value of the clock for this particular clock gating check to be the high or the low interval of the clock, but not both.

### What Next

Reexecute the *set_clock_gating_check* command and specify either the *-high* or *-low* option. Use these options only when you are very certain about the controlling value of the clock for a particular cell.

### See Also

- set_clock_gating_check

---

## UID-491

(error) Only one clock can be specified.

### Description

You receive this message if you executed the *set_input_delay*, *set_output_delay*, *remove_input_delay*, or *remove_output_delay* and specified multiple clocks in the -clocks option. You can only specify one clock for this option.

### What Next

Reexecute the command and only choose one clock.

### See Also

- set_input_delay
- set_output_delay

Feedback

- remove_input_delay
- remove_output_delay

## UID-492

(error) Object specified in -clock option must be a clock type.

**Description**

You receive this message if you executed the *set_input_delay*, *set_output_delay*, *remove_input_delay*, or *remove_output_delay* and specified a non-clock object in the -clocks option. You can only specify one clock object for this option.

**What Next**

Reexecute the command and only choose one clock object.

**See Also**

- set_input_delay
- set_output_delay
- remove_input_delay
- remove_output_delay

## UID-493

(error) Timing derate value should be bigger than 0 and smaller than 2.0.

**Description**

This error message occurs when the timing value is not within the acceptable range. The timing derate value must be larger than 0 and smaller than 2.0.

**What Next**

Specify a value within the range of 0 to 2.0 and run the command again.

**See Also**

- set_timing_derate

## UID-494

(warning) No -early or -late option is specified.

Feedback

### Description

This warning message occurs when running the *set_timing_derate* command without specifying either the *-early* or the *-late* option. When neither option is specified, the default is that both are set to *true*, and the derate value is applied to both.

### What Next

This is only a warning message. No action is required on your part.

However, if you do not want the derate value to apply to both options, run the command again and specify either the *-early* or the *-late* option.

### See Also

- set_timing_derate

---

## UID-495

(error) Cannot specify %s option with object list.

### Description

This error message occurs if you execute the *set_timing_derate* command and specify the *-net_delay* option on the object list.

### What Next

Execute the command again with the *-net_delay* option but without the object list.

### See Also

- set_timing_derate

---

## UID-496

(warning) SDC does not allow -min and -max deratings

### Description

This warning occurs if you execute the *write_sdc* command and there is a difference between the -min and -max settings of a*set_timing_derate* . These switches are not found in SDC and as a result this information is lost when producing the SDC output. To be safe, the *write_sdc* command will produce the lesser of the two numbers for the early derating, and the greater of these two numbers for the late derating, although this may give overly pessimistic results further downstream.

Feedback

### What Next

Examin the instances of *set_timing_derate* in the SDC output to determine whether the values are the most appropriate for tools that will make use of the SDC constraints.

### See Also

- set_timing_derate
- write_sdc

---

## UID-497

(information) Drop incorrect(obsolete) 'high/low' set_clock_gating_checks.

### Description

This warning occurs if you execute the *write_sdc* command and there was some incorrect set_clock_gating_checks stored in old designs.

We will ignore those incorrect set_clock_gating_checks during design loading.

### What Next

Ignore this message.

### See Also

- set_clock_gating_check
- write_sdc
- write_script

---

## UID-499

(warning) The clock %s is not a generated clock. This command removes only generated clocks.

### Description

One or more clock(s) specified in your command arguments is not a generated clock. The *remove_generated_clock* command can remove only generated clocks. This is not an error though.

Feedback

### What Next

You can use remove_clock command to remove non-generated (or Master) clocks. To know more about this, see help for *create_clock, create_generated_clock, remove_clock, remove_generated_clcok*.

---

## UID-500

(warning) Pin '%s' is hierarchical and not valid for case analysis.

### Description

You receive this warning message because you specified hierarchical pins as objects for *set_case_analysis* and *remove_case_analysis*. Therefore, case analysis is not being set or removed on the specified pin. If there are multiple pins specified in the object_list, case analysis values will still be set or removed on nonhierarchical pins or ports.

### What Next

Do not specify hierarchical pins for *set_case_analysis* and *remove_case_analysis*.

### See Also

- remove_case_analysis
- set_case_analysis

---

## UID-501

(error) There are no valid objects specified for case analysis.

### Description

You receive this error message because you did not provide a list of nonhierarchical pins or ports when you executed the *set_case_analysis* command or the *remove_case_analysis* command.

### What Next

Specify a valid set of non-hierarchical pins or ports.

### See Also

- remove_case_analysis
- set_case_analysis

Feedback

## UID-502

(error) Value '%s' is not a valid value.

### Description

You receive this error message because you did not specify a valid value with the *set_case_analysis* command. Valid values are 0, 1, zero, and one.

### What Next

Specify a valid value for the *set_case_analysis* command.

### See Also

- set_case_analysis

## UID-503

(error) Rising and falling values are not currently supported.

### Description

You receive this error message because you did not specify valid values with the *set_case_analysis* command. Valid values are 0, 1, zero, and one. Although PrimeTime supports rise, fall, rising, and falling, these options are not currently supported in DesignTime.

### What Next

Specify valid values for the *set_case_analysis* command.

### See Also

- set_case_analysis

## UID-504

(error) Must specify either -all or object_list.

### Description

You receive this error message because you have not specified a value with the *remove_case_analysis* command. You must specify either an object list or the *-all* option with this command.

Feedback

### What Next

Reexecute the *remove_case_analysis* command and specify either an object list or the *-all* option.

### See Also

- remove_case_analysis

---

## UID-505

(error) Cannot specify both -all and object_list.

### Description

You receive this error message because you have executed the *remove_case_analysis* command and have specified the *-all* option and an object list. You can only specify one of these arguments with the *remove_case_analysis* command.

### What Next

Reexecute the *remove_case_analysis* command and specify either the *-all* option or an object list.

### See Also

- remove_case_analysis

---

## UID-506

(error) Cannot change link for instance cell '%s'.

### Description

You receive this error message because you have executed the *change_link* command for an instance cell.

### What Next

Reexecute the *change_link* command without instance cells.

---

## UID-507

(error) The syntax of the pin mapping specified by user is not valid.

Feedback

**Description**

You receive this error message because you have executed the *change_link* command and specified an incorrect pin mapping. This error occurs if the syntax of the pin map entries are incorrect.

**What Next**

Reexecute the *change_link* command with a valid pin map.

---

## UID-508

(error) The replacement cell specified has lesser number of pins that the original cell.

**Description**

You receive this error message because you have executed the *change_link* command and specified a replacement cell which has lesser number of pins than the original cell.

**What Next**

Reexecute the *change_link* command with a valid replacement cell.

---

## UID-509

(error) Pin mapping from %s to %s is not valid.

**Description**

You receive this error message because you have executed the *change_link* command and specified an incorrect pin mapping.

This error may be because of the following reason. The pins specified in the mapping may not be compatible, ie,. they may have mismatched pin number or pin direction.

**What Next**

Reexecute the *change_link* command with a compatible replacement cell.

---

## UID-510

(error) In Dcl mode, you can specify only one mode with the set_mode command.

**Description**

You receive this error message because you have executed the *set_mode* command and specified more than one mode. This is invalid and, as a result, no modes are set by this command.

Feedback

### What Next

Reexecute the *set_mode* command and specify only one mode.

---

## UID-511

(error) Specified instance %s not valid. Must be a cell.

### Description

You receive this error message because you have issued the *set_mode* command on an instance that is not a cell. This is invalid and, as a result, no modes are set by this command.

### What Next

Reexecute the *set_mode* command and specify a mode on a cell or a cell instance.

---

## UID-512

(error) Pin mapping for pin %s is not available and no pin %s is in the replacement cell.

### Description

You receive this error message because you have executed the *change_link* command and specified an incorrect pin mapping.

This error may be because of the following reason.There may not not exist a mapping corresponding to an original pin and no pin of the original name may exist in the specified replacement object.

### What Next

Reexecute the *change_link* command with a valid replacement cell.

---

## UID-513

(error) Pin mapping from %s to %s is not valid.

### Description

You receive this error message because you have executed the *change_link* command and specified an incorrect pin mapping.

This error may be because of the following reason. A mapping was corresponding to an original pin was specified, however, there does not exists a pin by that name in the specified replacement cell.

Feedback

### What Next

Reexecute the *change_link* command with a valid pin map or a valid replacement cell.

---

## UID-520

(error) Ungroup -start_level does not work when the variable disable_2001_ungroup is true.

### Description

This message is issued when the variable disable_2001_ungroup is true and ungroup -start_level option is used. The -start_level option is not supported in the old version of ungroup and so this error message is issued. The same error message will be generated when trying to use ungroup -start_level for Behavioral compiler designs. A Behavioral Compiler design will have the attribute is_post_scheduled_design.

### What Next

Set the variable disable_2001_ungroup to false. If the error message still occurs, verify that the design is not a Behavioral Compilerdesign. If the design is a BC design, the -start_level option is not supported. To ungroup different levels of the design, use a combination of current_instance and ungroup. See the man page of ungroup for more details.

---

## UID-521

(error) Ungroup -start_level cannot be used in conjunction with -small option.

### Description

This message is issued when the -small and -start_level options are used together with the ungroup command. Both of these two options represent completly different strategies.

### What Next

Use either the -small option or the -start_level option.

---

## UID-523

(Warning) Some valid instances of design '%s' are not ungrouped due to dont_touches or other MV-related constraints in other instances.

### Description

This message is issued when ungroup -flatten, ungroup -small or ungroup -start_level is used where one or more instances of the design's multiply-instantiated sub-blocks

Feedback

has an instance-specific dont_touch attribute placed on one of its child cells, or has instance specific operating condition that is different from the owner block. Ungroup cannot be performed where it may change certain instances such that all instances of a sub-design can no longer point to the same reference. This problem does not occur if the dont_touches or Mv-related constraint are not instance specific.

**What Next**

If it is not OK, remove the instance specific dont_touch or constraints and run ungroup again. Or use uniquify command to uniquify the design.

---

## UID-524

(Warning) Some valid instances of design '%s' are not ungrouped because of some instances outside the start_level"

**Description**

This message is issued when ungroup -start_level option is used in the case where one or more instances of the design's multiply-instantiated sub-block has a child cell which does not fall into the range of specified levels to be ungrouped. Ungroup cannot be performed where it may change certain instances such that all instances of a sub-design can no longer point to the same reference.

**What Next**

Use ungroup -flatten option to flatten the design. Or, simply use the uniquify command to uniquify the design.

**See Also**

- uniquify

---

## UID-526

(error) No valid cells to ungroup within the level specified.

**Description**

This message is issued when there are simply no valid cells to be ungrouped from the specified -start_level downwards. In this case, nothing will be ungrouped.

**What Next**

None.

Feedback

## UID-527

(error) Writing designs in DB format is no longer supported. Option xg_force_db is ignored.

**Description**

This message is issued when you are trying to write the design in XG mode in DB Format. Due to capacity reasons, it is recomended to write the netlist using DDC format instead of DB format.

**What Next**

Please use a different format such as ddc to write out the netlist.

## UID-528

(warning) Writing designs in DB format is no longer supported.

**Description**

This message is issued when you are trying to write the design in XG mode in DB Format. Due to capacity reasons, it is recommended to write the netlist using DDC format instead of DB format.

**What Next**

Modify your scripts to write DDC format or ignore this error message if the usage was intentional.

## UID-529

(warning) A previous '%s' attribute on net '%s' will be overwritten.

**Description**

The *set_attribute* command can be used to specify the wired logic type of a net. This warning is given when a previous wire logic type exists and is different from the one being set.

**What Next**

To avoid this warning message, use *remove_attribute* command to remove the existing wired logic attribute before setting the new wired logic attribute on the net.

## UID-530

(Error) Writing designs in DB format is no longer supported.

Feedback

### Description

This message is issued when you are trying to write the design in XG mode in DB Format. Due to capacity reasons, it is recommended to write the netlist using DDC format instead of DB format.

### What Next

Modify your scripts to write DDC format or ignore this error message if the usage was intentional.

---

## UID-531

(Warning) Some valid instances of design '%s' passed as an argument to ungroup -all_instances option are not ungrouped due to dont_touches in other instances or other MV-related constraints in this instance or other instances.

### Description

This message is issued when ungroup -all_instances is used where one or more instances of the design's multiply-instantiated sub-blocks has an instance-specific dont_touch attribute or datapath_optimization attribute placed on one of its child cells, or has instance specific operating condition/ power domains that is different from the owner block. Ungroup cannot be performed where it may change certain instances such that all instances of a sub-design can no longer point to the same reference. This problem does not occur if the dont_touches or Mv-related constraint are not instance specific.

### What Next

If it is not OK, remove the instance specific dont_touch or constraints and run ungroup again. Or use uniquify command to uniquify the design.

---

## UID-540

(error) -scenarios and -active_scenarios may only be used with ddc format

### Description

The *-scenarios* and *-active_scenarios* options to the *read_file* command may only be used if *-format ddc* is specified.

### What Next

Verify that the correct format option has been supplied to the *read_file* command.

Feedback

## UID-541

(error) The -active_scenarios option cannot be used without the -scenarios option.

### Description

This error message occurs because the *read_file* command with the *-active_scenarios* option can be used only when the *-scenarios* option is also specified and the *-scenarios* list is not empty.

### What Next

See the *read_file* command man page for information about controlling which scenarios' constraints are read from the .ddc file.

### See Also

- all_scenarios

## UID-542

(error) All scenario names specified in -active_scenarios\n must also be specified in -scenarios

### Description

The *read_file* command's *-active_scenarios* option may only contain scenario names that are also listed in the *-scenarios* option. The *-scenarios* option, however, may contain names that are not included in *-active_scenarios*.

### What Next

See *read_file*(2) for more information on controlling which scenarios' constraints are read from the ddc file.

### See Also

- all_scenarios

## UID-550

(error) Unable to read the library '%s' from '%s'.

### Description

This error message occurs when the library has been pruned to provide optimal performance of the tool. The library will be re-read from the disk on demand to restore the necessary information to memory.

Feedback

The library has been removed or moved to a different location. This message occurs because the tool does not expect the library files to be removed within a session.

**What Next**

Copy the library file to the appropriate location and rerun the command.

---

## UID-551

(error) The library '%s' from '%s' has been modified after reading.

**Description**

This error message occurs when the library has been pruned to provide optimal performance of the tool. The library will be re-read from the disk on demand to restore the necessary information to memory.

The library has been modified after the tool has read the library in memory. This message occurs because the tool does not expect the library files to modified while the tool is still running. A combination of timestamp and checksum are used to detect if the library has been modified after it has been read in by the tool.

**What Next**

Start a new session and run the same set of commands and make sure that the library is not updated after it has been read in by the tool.

---

## UID-560

(warning) this command will be ignored since option %s is not supported.

**Description**

This warning message occurs if you execute the *set_timing_derate* command and specify the *-dynamic* option.

**What Next**

Execute the command again without the *-dynamic* option

**See Also**

- set_timing_derate

---

## UID-561

(error) The option %s has to be specified with option %s.

Feedback

### Description

This error message occurs if you execute the *set_timing_derate* command and specify the *-static* option without the *-net_delay* option.

### What Next

Execute the command again with the *-net_delay* option

### See Also

- set_timing_derate

---

## UID-562

(warning) the option %s will be ignored, this command will be executed as if this option is not specified.

### Description

This warning message occurs if you execute the *set_timing_derate* command and specify the *-static* option.

### See Also

- set_timing_derate

---

## UID-600

(error) Illegal type '%s' specified with the -type option. The value must be either "buffer" or "inverter".

### Description

The tool issues this error if you run the *set_isolate_ports* command and use an invalid argument for the *-type* option. Valid arguments for *-type* are *buffer* and *inverter*. To see all valid options and arguments for the *set_isolate_ports* command, see the man page or the command help (*set_isolate_ports -help*).

### What Next

Rerun the *set_isolate_ports* command with a valid argument for the *-type* option.

---

## UID-601

(error) Cannot isolate port '%s' because it is an '%s' port; only input or output ports can be isolated.

Feedback

### Description

You receive this message if you execute *set_isolate_ports* and specify a port that is an inout port. You can isolate only input or output ports.

### What Next

Reexecute the *set_output_ports* command and ensure that the list of ports contains only input or output ports.

---

## UID-602

(error) Can't specify <power_value> or <power_unit> when -delete_all is specified

### Description

You receive this message if you specify <power_value> or <power_unit> along with '-delete_all' option to *set_cell_internal_power* command.

### What Next

Reexecute the *set_cell_internal_power* command and ensure that you specify either '-delete_all' option or <power_value> and/or <power_unit>.

---

## UID-603

(error) There are no valid objects specified for set_cell_internal_power command.

### Description

You receive this message if you do not specify <pin_objects> or '-delete_all' option to *set_cell_internal_power* command.

### What Next

Re-execute the *set_cell_internal_power* command and ensure that you specify either '-delete_all' option or list of pins where you want to annotate the power number.

---

## UID-604

(information) It is recommended to use set_ideal_network command instead of set_ideal_net.

### Description

This information message occurs because the *set_ideal_net* and *set_ideal_network* commands are being combined. The *set_ideal_network* command is replacing *set_ideal_net*.

Feedback

In the 2004.12 release, the *set_ideal_net* command is replaced by the following command, which is applied to the same objects:

```
set_ideal_network -no_propagate
```

### What Next

This is only an informational message. No action is required.

### See Also

- set_ideal_network

---

## UID-605

(warning) Replacing set_ideal_net command with set_ideal_network -no_propagate command.

### Description

This warning message occurs because the *set_ideal_net* and *set_ideal_network* commands are being combined. The *set_ideal_network* command is replacing *set_ideal_net*.

The attributes are applied on the global driver pins of the specified nets, instead of applying the *ideal_net* attribute on the specified nets.

In the 2004.12 release, the *set_ideal_net* command is replaced by the following command, which is applied to the same objects:

```
set_ideal_network -no_propagate
```

### What Next

This is only an informational message. No action is required.

### See Also

- set_ideal_network

---

## UID-606

(warning) Transferring ideal_net attribute onto the driver pin %s of net %s.

Feedback

### Description

This warning message occurs because the *set_ideal_net* and *set_ideal_network* commands are being combined. The *set_ideal_network* command is replacing *set_ideal_net*.

In the 2004.12 release, the *set_ideal_net* command is replaced by the following command:

```
set_ideal_network -no_propagate
```

The attribute for *ideal_net* is transferred from the specified nets to the global driver pins of those nets, and is translated to the corresponding ideal network attributes.

### What Next

This is only a warning message. No action is required.

### See Also

- set_ideal_network

---

## UID-607

(error) The option -instance is obsolete in XG mode. Please use the complete path to the net instance with the %s command instead.

### Description

You have used the option -instance which is obsolete in XG mode. In XG mode one can use the complete hierarcical net name to perform remove_net and create_net operations. Please check the man page of these commands for details and examples.

### What Next

Please re-issue the command with the new format.

---

## UID-608

(information) Performing automatic 'set_ideal_net' on net '%s'.

### Description

This message occurs because *set_ideal_net* is being automatically performed at the named net. This behavior can be controlled using the following variables:

*icc_auto_ideal_nets* (default is false) *icc_auto_ideal_fanout_threshold* (default is 500) *icc_auto_ideal_multi_driver_threshold* (default is 100)

Feedback

This feature is meant to provide better default behavior during optimization for high-fanout nets that cannot be decomposed by buffering (for example, because they are dont_touch).

**What Next**

No action is required.

**See Also**

- set_ideal_network

---

## UID-610

(error) '%s' is not a supported value for the '%s' option.

**Description**

The tool issues this error when you specify an invalid or unsupported value for a command option.

To get a list of the supported options and arguments for the command, run the command with the *-help* option. For detailed information about the supported options and arguments, see the man page.

**What Next**

Change the option value to one that is supported and rerun the command.

---

## UID-611

(warning) Analysis type '%s' is not supported. Using '%s' instead.

**Description**

The tool issues this warning when you specify an unsupported analysis type. The tool uses the closest available analysis type, which should be suitable in most cases. You can check which options are supported by running the command with the *-help* option or by viewing the man page.

**What Next**

Verify that the substituted analysis type is acceptable for your purposes. Otherwise, change the analysis type to a supported value and rerun the command.

---

## UID-620

(error) Number of pins on the cell being changed is not same as the number of pins on the new reference cell.

Feedback

**Description**

You receive this error message because the number of pins on the cell or reference being changed with change_link/size_cell, does not match the number of pins on the new reference.

**What Next**

Please provide a compatible reference library cell for change_link to work correctly

---

## UID-621

(error) write_milkyway is not permitted, because the design contains a PG netlist obtained from the original RTL. Please use write_file -pg -f verilog.

**Description**

write_milkyway is not permitted in DC if the design contains a PG netlist obtained from the original RTL.

The PG netlist would have been obtained from the source RTL while dc_allow_rtl_pg was true. In this case, the only permitted output format is verilog. It is also possible to output to ddc, but only for reading back into DC.

**What Next**

Use write_file -pg -f verilog to communicate the design to a downstream tool.

---

## UID-622

(error) PG net '%s' connects to signal pin '%s'.\n This connection will be lost.

**Description**

DC does not currently allow a PG network described in source RTL to connect to signal pins. Such connections will be lost.

The PG netlist would have been obtained from the source RTL while dc_allow_rtl_pg was true. In this case, the PG network allowed by DC is limited, and does not permit PG connections to signal pins.

---

## UID-623

(error) PG net '%s' is driven by constant value '%s'.\n DC does not support constant values on RTL PG nets. This connection will be lost.

Feedback

### Description

DC does not currently allow a PG network described in source RTL to be driven by constant values. Such connections will be lost.

The reason for this limitation is that the DC RTL readers may merge constant nets together, thus losing the intended connectivity of individual PG nets. Instead, PG nets should be driven by top-level ports.

Examples of illegal constant PG connections are wire vdd; assign vdd = 1'b1; AND2 U1(.VDD(vdd), ...); supply1 vdd; AND2 U1(.VDD(vdd), ...); AND2 U1(.VDD(1'b1), ...);

The PG netlist would have been obtained from the source RTL while dc_allow_rtl_pg was true. In this case, the PG network allowed by DC is limited, and does not permit PG connections to constant values.

---

## UID-624

(error) Some requested RTL PG connections were dropped earlier. \n Please review errors LINK-31, LINK-32, UID-622, UID-623 in the log. \n You need to revise your RTL to avoid undesired logic.

### Description

At some point earlier in the DC session, RTL PG connection errors occurred, and some requested connections were dropped.

Check your log for errors UID-622, UID-623, LINK-31, and LINK-32. Those messages will give details about why the connections were dropped.

This error is issued at the start of compile as a reminder, and compile will not proceed. If the RTL is not corrected to avoid unsupported connections, then the resulting logic may not operate as desired.

---

## UID-625

(error) Can only set one default mode on library.

### Description

This error message occurs when users want to set multiple default modes on library.

### What Next

This is error message.

Feedback

## UID-632

(warning) PG net '%s' connects to signal pin '%s'. This connection will be lost.

**Description**

Currently DC does not allow a PG network described in source RTL to connect to signal pins. Such connections will be lost.

## UID-633

(warning) PG net '%s' is driven by constant value '%s'. DC does not support constant values on PG nets. This connection will be lost.

**Description**

Currently DC does not allow a PG network described in source RTL to be driven by constant values. Such connections will be lost.

The reason for this limitation is that the DC RTL readers may merge constant nets together, thus losing the intended connectivity of individual PG nets. Instead, PG nets should be driven by top-level ports.

Examples of illegal constant PG connections are wire vdd; assign vdd = 1'b1; AND2 U1(.VDD(vdd), ...); supply1 vdd; AND2 U1(.VDD(vdd), ...); AND2 U1(.VDD(1'b1), ...);

## UID-660

(error) Option%s is mutually exclusive to option%s.

**Description**

You receive this error message if you specify two or more options on one command line that have adverse effects and are therefore mutually exclusive.

**What Next**

Refer to the man page for the command that you invoked for information about the options. Do not use options that cannot be specified together on one command line.

## UID-661

(warning) Data missing for physically aware net delay estimation.

**Description**

This warning message occurs when you start to use the physically aware net model without all of the essential data.

Feedback

Any of the following causes the warning message:

- The physical library is not specified.
- The library is specified, but it could not be read in successfully.
- The library did not provide all of the necessary data, including the site's row_height, average wire_width, capacitance, and resistance per unit length.

**What Next**

This is only a warning message. No action is required.

However, if the result is not what you intended, you can do either of the following:

- Continue using the conventional wire load mode
- Make sure that you have specified a valid physical library.

## UID-662

(warning) %s data missing in physical library.

**Description**

This warning message occurs when the physical library currently set does not define all of the values necessary to correctly activate the physically aware wire load estimation method.

**What Next**

This is only a warning message. No action is required.

However, if the result is not what you intended, check that you are using the correct physical library.

## UID-663

(warning) Physical library not specified.

**Description**

You receive this warning message when the *physical_library* variable is not set or the specified libraries cannot be loaded.

**What Next**

This is only a warning message. No action is required.

However, you can check that you are using the correct physical library.

Feedback

## UID-665

(error) P/G net %s used in command %s.

**Description**

P/G nets can not be manipulated using connect_net or disconnect_net commands. connect_net and disconnect_net can only be used for signal nets.

**What Next**

Use one of *connect_pg_nets* , *connect_logic_zero* or *connect_logic_one* commands to do the necessary netlist changes for P/G nets.

## UID-666

(error) No Milkyway library is open.

**Description**

This error is generated when an ascii design is being read in without opening a Milkyway library first.

**What Next**

Use *open_mw_lib* command to open an existing Milkyway library Use *create_mw_lib* command to create a Milkyway library if one does not exist. Run the read command again.

## UID-667

(error) Some clocks have been specified in multiple groups.

**Description**

The messages prior to this show the clock that has been specified multiple times.

**What Next**

A clock should be specified only once.

## UID-668

(error) Can not specify -allow_paths with -logically_exclusive or -physically_exclusive.

**Description**

The allow_paths option is invalid for logically_exclusive or physically_exclusive option of set_clock_groups.

Feedback

**What Next**

Logically exclsuive relationship has no SI impact and only sets the false paths between the specified clocks. Hence -allow_paths does not make sense with -logically_exclusive option.

Physically exclusive clocks imply that clocks are not present at the same time and paths between the clocks can not exist.

---

## UID-670

(error) Design CEL is already read-in and created. Please close current CEL before reading in another design.

**Description**

The design CEL design is already read and created, so can't create another design.

**What Next**

The command read_ddc will create the design CEL, so if you want to load more than one ddc files, make sure read them in at once.

---

## UID-700

(error) There is unplaced eco cell in the given list when the option -legalize_only is used with -cells.

**Description**

When the option -legalize_only is used with -cells, it expects all the eco cells have been already placed. The option allows only doing legalization.

**What Next**

Use place_eco_cells -unplace to place the eco cells.

---

## UID-701

(error) This option -legalize_only is only used with -cells or -eco_changed_cells option. It cannot be used with -unplaced_cells.

**Description**

The option -legalize_only must be used with -cells or -eco_changed_cells.

Feedback

**What Next**

Use place_eco_cells -cells/-eco_changed_cells when the option -legalize_only is specified.

---

## UID-702

(warning) Skip to legalize the cell (%d) since it is not placed.

**Description**

When the option -legalize_only is used with -eco_changed_cells, the command will skip the cells that is not placed (attribute is_placed == false). The command should not legalize the unplaced cells.

**What Next**

Use place_eco_cells -unplaced(-cells) to place and legalize the cell.

---

## UID-820

(error) Floorplan block '%s' is not defined.

**Description**

You have specified a command which works on an existing fp_block. The block you specified could not be found.

**What Next**

Double check the names of defined blocks with write_fp_script.

---

## UID-821

(error) Floorplan atom '%s' is not defined.

**Description**

You have specified a command which works on an existing fp_atom. The atom you specified could not be found.

**What Next**

Double check the names of defined atoms with write_fp_script.

---

## UID-822

(error) Atom '%s' is already defined.

Feedback

**Description**

You are trying to redefine the binding of a floorplan atom which is already bound to another cell.

**What Next**

Use a different atom name, or first undefine the atom name in question using "define_fp_atom -name atom_name -undef".

---

## UID-823

(error) Atom '%s' is already defined for cell '%s'.

**Description**

You are attempting to redefine the the atom for a cell of the netlist.

**What Next**

Define the new atom for a different cell, or first undefine the current atom using "define_fp_atom -name old_atom_name -undef".

---

## UID-824

(error) Design '%s' (instance '%s') is multiply instantiated.\n\tFloorplan atoms are only permitted within uniquified hierarchies.

**Description**

You have attempted to define an atom in a part of the hierarchy which is instantiated multiple times. The designs for atoms may only be instantiated once in your design.

**What Next**

Use the "uniquify" command to force the multiple copies of the same design to be changed into multiple designs.

---

## UID-825

(error) You must specify an atom name or cell name.

**Description**

You have used define_fp_atom without specifying -name flag or a cell name. This is not supported.

Feedback

**What Next**

Specify a -name flag.

---

## UID-826

(error) The floorplan abstracts have not been created or are out of date.\n\tPlease (re)run create_fp_abstracts.

**Description**

This command requires create_fp_abstracts to be run before it can execute.

**What Next**

Please re(run) create_fp_abstracts.

---

## UID-827

(error) The floorplan abstract placement has not been created or is out of date.\n\tPlease (re)run place_fp_abstracts.

**Description**

This command requires place_fp_abstracts to be run before it can execute.

**What Next**

Please re(run) place_fp_abstracts.

---

## UID-828

(error) You can only select one of -select, -flyline, -expand, and -unexpand

**Description**

Only one of the given flags may be specified in a single command call.

**What Next**

Specify only one of these flags per call.

---

## UID-829

(error) You must specify a cell for -expand or -unexpand

**Description**

The -expand and -unexpand options require you to specify a cell to work on.

Feedback

**What Next**

Specify a cell name.

---

## UID-830

(error) You must specify an abstract name for -select abstract

**Description**

The -select abstract option requires you to also specify the name of the abstract that you want to select.

**What Next**

Specify the name of an abstract.

---

## UID-831

(warning) Cell '%s' has been assigned an abstract area of zero

**Description**

You have either omitted the -area flag or explitly specified zero to the -area flag. This will cause the given cell to be ignored for area analysis of the floorplan.

**What Next**

If you have assigned a zero area in error, you can either redefine the area by running "define_fp_abstract" again, or you can remove the abstract by using "define_fp_abstract -remove".

---

## UID-832

(warning) You need to define floorplan atoms or trial atoms before running this command

**Description**

This command analyzes design statistics based on the currently defined floorplan atoms and trial atoms. Without these, there is nothing to report.

**What Next**

Create floorplan atoms with define_fp_atom. Create trial atoms with define_trial_atoms.

---

## UID-833

(error) Floorplan block '%s' has no bound logic

Feedback

### Description

The specified floorplan block is currently empty. So there is nothing to show.

### What Next

Select a different floorplan block as a connection source, or do not specify any connection source.

---

## UID-834

(error) Cannot use '%s' as the clock reference. The clock reference used must be a clock object.

### Description

The clock network latency is the time it takes a clock signal to propagate from the clock definition point to a register clock pin. In the case of multiple clocks pass through the same pin or port object, clock network latency can be specified on this object with respect to a specific clock. In this way, different values can be applied as clock network latency for different clocks. With the *-clock* option, a clock object must be provided to specify which clock the clock network latency is referring to. Note, the clock network latency is only applied on ideal clock network.

### What Next

Make sure to provide a clock object when using *-clock* option. Use *report_clock* to report all clocks in the current deign.

---

## UID-835

(warning) Cannot use clock object '%s' in the object list when -clock option is provided. The -clock option will be ignored for '%s'.

### Description

The clock network latency is the time it takes a clock signal to propagate from the clock definition point to a register clock pin. In the case of multiple clocks pass through the same pin or port object, clock network latency can be specified on this object with respect to a specific clock. In this way, different values can be applied as clock network latency for different clocks. With the *-clock* option, only non-clock object can be provided in the object list. Otherwise, the *-clock* option will be ignored for these objects.

### What Next

Make sure to provide non-clock object in object list when using *-clock* option. Use *report_clock* to report all clocks in the current design.

Feedback

## UID-836

(error) Cannot use -clock option and any of the options -source, -early and -late at the same time.

### Description

See *set_clock_latency* man page for the differernce between clock network latency and clock source latency.

### What Next

Only set clock network latency or clock soruce latency at one time.

## UID-841

(error) Fail to rebuild timing design from database

### Description

Failed to load the timing design from the database. This is usually caused by database corruption.

### What Next

Quit and restart the session. Save the database to a new name to avoid corrupting the previously valid database. If you are using CEL as input, check the CEL by opening in the new session.

## UID-842

(warning) Command is not trace-mode compliant, and is ignored

### Description

You are running a trace-mode session, and the command executed is not trace-mode compliant. The command will be ignored.

### What Next

If you want to execute this command, please end the trace-mode session.

## UID-853

(warning) The extract_rc command is needed to report accurate AWE delays.

Feedback

**Description**

This warning message occurs when the Asymptotic Waveform Evaluation (AWE) delay method is set in the preroute stage but extraction has not yet been done.

**What Next**

The extract_rc command is needed to report accurate AWE delays.

---

## UID-900

(error) The -type argument must provide at least one cell.

**Description**

You receive this error message if you issue *set_scan_register_type -type* without an argument for the required *-type* switch. The *set_scan_register_type* command requires that at least one example cell be given with *-type*.

**What Next**

Reissue *set_scan_register_type -type* with at least one name of a valid example cell from the target library, as the argument to *-type*.

---

## UID-901

(error) The %s cell already has the set_register_type attribute set.

**Description**

You can only use *set_scan_register_type* after removing the *set_register_type* attribute.

**What Next**

Remove the *set_register_type* attribute and reissue the command.

---

## UID-902

(error) The cell %s issued as an argument to set_scan_register_type -type or set_scan_replacement options -lssd, -muxd, or -clocked_scan is a non-scan cell.

**Description**

You receive this error message when you issue the name of a non-scan cell as an argument to the *set_scan_register_type -type* command or the *set_scan_replacement* command with the *-lssd*, *-muxd*, or *-clocked_scan* option.

Feedback

The *set_scan_register_type* command *-type* switch requires a scan cell as an argument. The *-type* switch must specify a list of scan cells for the *insert_scan* or the *compile -scan* command to use for scan replacement of non-scan sequential cells or designs.

The *set_scan_replacement* command with the *-lssd*, *-muxd*, or *-clocked_scan* switches requires a scan cell as an argument. These switches specify a scan replacement for each scan style for the set of non-scan cells specified through the *-nonscan* option.

**What Next**

Reexecute the *set_scan_register_type* command and specify a scan cell as an argument.

---

## UID-903

(error) The %s cell is not valid for the methodology specified.

**Description**

You receive this error message when an example cell issued as an argument to the *set_scan_register_type -type* command or the *set_scan_replacement* command with the *-lssd*, *-muxd*, or *-clocked_scan* option is not valid for the methodology defined.

The methodology for the *set_scan_register_type* command is the one for the current design. The methodology for the *set_scan_replacement* command is specified by the option.

**What Next**

Verify the defined methodology using the *report_test -methodology* command.

You can reexecute the *set_scan_register_type* command with the *-type* argument, specifying only cells that are valid for that methodology for the *-type* argument.

Find the valid cell for the methodology and then you can reissue the *set_scan_replacement* command.

---

## UID-904

(error) Unable to find example %s '%s'.

**Description**

The tool issues this error if an example cell that you specify as an argument to the *set_scan_register_type -type* option cannot not be found in the target library. All example cells specified as arguments to the *-type* option must exist in the target library.

Possible causes for this error condition might be a spelling error in the example cell name or an incorrect library specified in the *target_library* variable.

Feedback

**What Next**

To correct this error,

1. Verify that the library that contains the example cell is specified in the *target_library* variable by running the *printvar target_library* command.
2. If necessary, reset the *target_library* variable so that it contains the appropriate target library.
3. Verify the correct capitalization and spelling of the example cell names and rerun the *set_scan_register_type* command.

---

## UID-905

(error) The %s design named %s is of "unknown" sequential type.

**Description**

You receive this error message if the type specified to the *set_scan_register_type* command or the *set_scan_replacement* command options cannot be recognized by the tool.

**What Next**

Reissue the *set_scan_register_type* or the *set_scan_replacement* command with a recognizable sequential device.

---

## UID-906

(error) There are no valid cells specified on which to assert set_scan_register_type.

**Description**

You receive this error message if you issue *set_scan_register_type* with the *cell_or_design_list* argument, but none of the cells on the list are valid. There is nothing for the command to operate on.

**What Next**

Examine the list of cells you entered, verify the spelling and capitalization of the names, or otherwise determine the reason why the cells were invalid. Then reissue the *set_scan_register_type* command with a list of valid cells for *cell_or_design_list*.

---

## UID-907

(warning) Cell '%s' is not a sequential cell.

Feedback

### Description

You receive this message if you issue *set_scan_register_type* with a nonsequential cell in the *cell_or_design_list*. *set_scan_register_type* must specify only sequential cells for *insert_scan* or *compile -scan* to replace with their scan equivalents.

### What Next

Reissue *set_scan_register_type* using only sequential cells in the *cell_or_design_list*.

---

## UID-908

(warning) Cell '%s' has the dont touch attribute set on it; cannot scan-replace.

### Description

You receive this message if *insert_scan* or *compile -scan* encounters a sequential cell that has the *dont_touch* attribute set to *true* on it. *insert_scan* and *compile -scan* attempt to perform scan replacement on your design by replacing each sequential cell with its scan equivalent. The *dont_touch* attribute is designed to protect cells from being modified or optimized, so cells that have this attribute cannot be scan replaced. This message informs you that the specified cell was not scan replaced.

### What Next

If you intended to protect the specified cell from scan replacement, no action is required on your part. However, if you want the specified cell to be scan replaced, first remove the *dont_touch* attribute, either with *remove_attribute* or by executing *set_dont_touch false*. Then reissue *insert_scan* or *compile -scan*.

### See Also

- set_dont_touch

---

## UID-909

(error) %s cannot be set on an input port %s.

### Description

This error message occurs when you attempt to set a *model_load* attribute on an input port. This attribute can only be set on output or inout ports.

### What Next

Reexecute the *set_model_load* command and ensure that the *port_list* contains only output and inout ports.

Feedback

## UID-910

(error) The argument of the -bound option must be larger than the argument of the -threshold option.

### Description

You receive this error message when you execute the *compare_rc* or *estimate_rc* command and specify a value for the *-bound* option that is smaller than or equal to the value for the *-threshold* option. The *-bound* option represents the upper level of back-annotated capacitance, and the *-threshold* option represents the lower level. For this reason, the value of the *-bound* must be larger than the value of the *-threshold* option.

### What Next

Reexecute the *compare_rc* or the *estimate_rc* command and specify a value for the *-bound* option that is larger than the value for the *-threshold* option.

## UID-911

(error) The -nonscan option requires at least one cell when mapping is specified.

### Description

You receive this error message if you do not specify at least one nonscan cell for the *-nonscan* option in the *set_scan_replacement* command for mapping.

### What Next

Reissue the *set_scan_replacement* command and specify a nonscan cell as an argument for the *-nonscan* option. The *-nonscan* argument is not needed when you use the *-remove* option alone.

## UID-912

(error) The replacement cell specified for the methodology %s is not a valid scan cell for the methodology.

### Description

You receive this error message for the *set_scan_Replacement* command when the scannable cell you specified for a given scan style is not a valid scan cell. The scan cell might not be valid for the scan style (*-lssd*, *-muxd*, or *clocked_scan*) that you specified.

### What Next

Reissue the *set_scan_replacement* command and specify a valid scan cell as an argument for the scan style option.

Feedback

## UID-913

(error) One of the -lssd, -muxd, and -clocked_scan options should have at least one cell when mapping is specified.

**Description**

You receive this error message when you do not specify at least one scannable cell for any of the *-lssd*, *-muxd*, or *clocked_scan* mapping options in the *set_scan_replacement* command.

**What Next**

Reissue the *set_scan_replacement* command and specify a scannable cell as an argument for the *-lssd*, *-muxd*, or *clocked_scan* option. These arguments are not needed when you use the *-remove* option alone.

## UID-914

(error) Cannot find target_library.

**Description**

This error message occurs when you do not specify a target library.

**What Next**

Set the *target_library* variable to the appropriate library and run the command again.

## UID-915

(error) Cannot find %s gate in the target_libray

**Description**

You receive this message if you do not specifiy a flop that exists in the target library in any of the options of set_scan_replacement.

**What Next**

Reissue *set_scan_replacement* , and specify a valid scan cell as an argument for the options. These arguments are not needed for -remove alone.

## UID-916

(error) Non-scan cell %s has more functional pins than the scan cell %s.

Feedback

### Description

You receive this message if a non-scan flop specified for -nonscan option of set_scan_replacement has more no of pins than a scan flop specified in one of the options . For example the non-scan flop might have async pins while scan flop is a simple scan flop without async pins.This is an error, since if scan replacement tries to use this to swap in a scan flop, it will result in bad logic.

### What Next

Re-issue *set_scan_replacement* , and specify a proper scan cell as an argument for the options.

---

## UID-917

(warning) Non-scan cell %s has less functional pins than the non-scan cell %s.

### Description

You receive this message if a scan flop specified for any of the options of set_scan_replacement has more no of pins than the non-scan flop. For example the scan flop might have async pins while non-scan flop is a plain D flip-flop.

### What Next

Re-issue *set_scan_replacement* , and specify a proper scan cell as an argument for the options. If not scan replacement will tie these pins to their default values.

---

## UID-918

(information) The timing reported below is an estimate. Please run vendor P&R tool to find out the actual design performance.

### Description

During timing analysis, DC FPGA uses cell delays and routing delays to calculate the path delays. Cell delays are provided by FPGA vendor and are thus accurate. The routing delays are estimates because DC FPGA does not have information on the cell placement and routing. Therefore, the timing reported by DC FPGA is an estimate. It is recommended that you always run vendor Place and Route tool to determine the actual design perfomance. Based on P&R timing result, you then decide whether and how you take the next step to improve timing.

### What Next

This is an informational message only. No action is required on your part.

Feedback

## UID-919

*UID-919* (Error) The -exact option has a list of cells to use for mapping.

### Description

This is shown when the -exact option in set_scan_register_type has a list of cells provided for mapping. Only a single cell newm should be provided.

### What Next

Usea single cell for -exact option and re-execute the command.

## UID-920

(error) The %s command is not allowed on %s, which is inside the noneditable block %s.

### Description

This error occurs because the tool does not support netlist editing within a noneditable block. The restricted commands are the following:

```
create_cell
create_net
create_port
connect_net
define_bus
disconnect_net
connect_pin
connect_logic_zero
connect_logic_one
remove_port
remove_cell
remove_net
insert_buffer
remove_buffer
size_cell
undefine_bus
```

### What Next

Restrict the collection of objects that you are editing to exclude any objects inside noneditable blocks and reapply the netlist editing commands.

### See Also

- create_cell
- create_net
- create_port

Feedback

- connect_net
- disconnect_net
- remove_cell
- remove_net
- insert_buffer
- remove_buffer
- size_cell

---

## UID-921

(warning) No shift-registers found matching given input pattern

### Description

This warning occurs because the tool executed the command get_shift_register_chains receiving a pattern, but no shift-register chains were found where at least one of the shift-register name's matches the pattern.

### What Next

Check that the pattern you are providing is correct, or that the objects you arexi providing as a collection contains valid shift-register cells. Also, check that you are using is a compile -scan flow in a test-ready compile, and the variable *compile_seqmap_identify_shift_registers* is set to TRUE.

---

## UID-922

(warning) Skipped shift-register head cells not connected to any shift-register flops

### Description

This warning occurs because the tool executed the command *get_shift_register_chains* and found a shift-register chains that contain a single SR cell in the chain, so it is actually not a chain, and it is not returned to the user.

### What Next

This warning message may arise if the tool optimized out the remaining of the SR chain, or this SR head cell was part of a Synchronous SR chain, where the next SR element was not directly connected to the head (or connected by inverters/buffers) but throught combinational logic. Synchronous SR chains are not supported by the command and are not expected to be reported.

Feedback

If the user wants to get this orphan SR cells reported as a valid SR chain, it must set the variable *get_shift_register_chains_filter_orphans* to FALSE.

---

## UID-950

(error) Unsupported pin name '%s' or its synonym '%s'.

**Description**

This error message occurs when the pin name or its synonym has unsupported cg=haracters. Wildcards and regular expressions are not supported. Forward slash '/' is also not supported for simple pin name synonyms.

**What Next**

Correct pin name or synonym string.

---

## UID-951

(error) Pin name synonym '%s' exists already.

**Description**

This error message occurs when there is a conflict in pin name synonym. Multiple synonym definitions for a pin name synonym is not supported.

**What Next**

Use '-force' switch to over-write the existing synonym. Use command report_pin_name_synonym to check the existing synonyms.

---

## UID-952

(information) Synonym '%s' not found.

**Description**

This error message occurs when trying to delete a pin name synonym that does not exist.

**What Next**

Use command report_pin_name_synonym to check the existing pin name synonyms.

---

## UID-955

(warning) Cannot find %s '%s' in design '%s'. This may be because the object has been modified or optimized away during group, ungroup, or optimization activity.

Feedback

**Description**

This warning message occurs when the specified object cannot be found in the specified design. One reason might be that the object was modified or optimized away during group, ungroup, or optimization activity. Another reason is that the design is hidden, so the object cannot be shown.

**What Next**

This is only a warning message. No action is required.

---

## UID-956

(error) Cannot specify '%s' with '%s'.

**Description**

This error message occurs when conflicts setting have been used together.

**What Next**

This is error message. Don't use conflict setting.

---

## UID-957

(warning) verification_priority attribute can not be set on %s, because it has dont_touch attribute.

**Description**

This error message is generated when you try to invoke command "set_verification_priority" on an object which has dont_touch attribute on it.

**What Next**

Check the object by invoke get_attribute, then reinvoke the current command.

---

## UID-958

(warning) verification_priority attribute can not be set on %s, because it is not a hierarchical cell.

**Description**

This error message is generated when you try to invoke command "set_verification_priority" on an object which is not a hierarchical cell.

Feedback

### What Next

Check the object by invoke get_attribute, then reinvoke the current command.

---

## UID-960

(error) connect_logic_net command failed to perform the connection as some of the specified ports do not exist in the design '%s'.

### Description

This error message occurs when one or more invalid objects are specified in the *-ports* option of *connect_logic_net* command.

The *connect_logic_net* progresses if all the specified objects exist. Else the command errors out and no connection is established.

### What Next

Specify the correct port(s) and rerun the command.

### See Also

- connect_logic_net

---

## UID-990

(error) Command '%s' is obsolete. Please use command '%s' instead.

### Description

The command is obsolete. Please try the alternative command.

### What Next

Try the alternative command.

---

## UID-991

(error) The driving cell %s has multiple inputs, and -from_pin is required when -input_transition_rise or -input_transition_fall is specified.

### Description

This error message is displayed if only one of two options, which need to be specified together in case of multiple input cells, is specified.

Feedback

**What Next**

This error message is displayed if -input_transition_rise or -input_transition_fall is specified, without specifying -from_pin for driving cells with multiple input pins.

---

## UID-992

(warning) Cannot find the top design for object %s.

**Description**

The top design for the given object could not be found. This may be caused either by a naming mismatch or an inconsistency in the hierarchy. As a result, the *setup* and *hold* attributes will not be propagated for this cell.

**What Next**

Report this problem to Synopsys.

---

## UID-993

(error) Cannot find the specified driving cell in memory.

**What Next**

Please use a driving cell from link library or from target library.

---

## UID-995

(error) Cannot find the target library.

**Description**

The specified target library could not be found. This may be caused by incomplete search path. As a result, the *set_driving_cell* will not be performed.

**What Next**

Please make sure the specified target library exists through search_path.

---

## UID-996

(error) Cannot find the output pin '%s'.

Feedback

### Description

This error message is issued by the *set_driving_cell* command. You receive this message if the specified driving pin is not found or its type is not output. This could be caused by a spelling error or typo.

### What Next

Verify the correct name of the specified output pin, then reexecute *set_driving_cell* with the correct name.

### See Also

- set_driving_cell

---

## UID-997

(warning) The set_dont_touch_network command is used for clock %s, for which no sources are specified.

### Description

This warning message occurs when the *set_dont_touch_network* command is used for a clock and the *dont_touch* attribute is assigned to the transitive fanout of the ports of the clock. The clock is created without specifying a list of ports, and the *set_dont_touch_network* command has no effect for this clock.

### What Next

This is only a warning message. No action is required.

Refer to the man pages for the *create_clock* command for more information.

### See Also

- create_clock
- set_dont_touch_network

---

## UID-998

(warning) Buffer tree at driver pin %s has cells or nets with dont_touch attribute.

### Description

This warning message occurs when the *clean_buffer_tree* command is applied to a buffer tree that has buffers, inverters, or nets with the *dont_touch* attribute. The specified cells and the cells that follow will not be removed.

Feedback

**What Next**

This is only a warning message. No action is required.

Refer to the man page for the *clean_buffer_tree* command for more information.

**See Also**

- set_dont_touch
- set_dont_touch_network

---

## UID-999

(error) Must specify either %s or %s.

**Description**

This error message occurs when neither of the two mandatory options for a command are specified.

**What Next**

Reissue the command and specify one of the two options in the error message.

---

## UID-1000

(error) Invalid scenario name.

**Description**

This error is generated when providing and invalid scenario name.

**What Next**

Use *all_scenarios* to see the names of the currently defined scenarios.

---

## UID-1001

(error) Current scenario is not defined.

**Description**

This error is generated when current scenario is not defined.

**What Next**

Use *all_scenarios* to see the names of the currently defined scenarios. Then invoke *current_scenario* to set the desired current_scenario.

Feedback

## UID-1002

(information) Cell %s is a special cell with always on pins and it is preserved with size_only attribute.

### Description

The tool issues this message to inform that cells with always on pins are marked with size_only attribute. Preserving these cells during optimization ensures power management logic is synthesized correctly. Only cells that are in power down regions are considered. Also, cells that are marked dont_touch and cells that are already marked size_only are not considered. TCL variable mark_always_on_logic_size_only can be used to turn off size_only marking of such cells. The default value of this variable is true.

### What Next

If you want to turn off automatic marking of size_only attribute on such cells, set TCL variable mark_always_on_logic_size_only to false.

## UID-1003

(warning) Driving cell was specified without the -library option. Using library_cell %s from library %s:%s.

### Description

This warning message is displayed if the set_driving_cell command is issued without the -library option.

### What Next

Specify the library for the driving cell using the -library option. This is very important for multi voltage designs where library cell of a given name may be present in multiple libraries. It is important for user to check if the correct library cell was picked.

## UID-1004

(error) Cannot use the '%s' option of save_mw_cel after ILM creation

### Description

This error is seen when, after creating an ILM, you run *save_mw_cel* with an option that is not supported. Currently, this includes only the '-overwrite' options.

### What Next

The '-overwrite' option is not currently supported after *create_ilm* is run. A fresh version of the ILM view must be created.

Feedback

## UID-1005

(error) Format '%s' is not a valid format.

**Description**

This error is generated when *write* command encounters an invalid write format. Please note that -power option works only for "verilog" format.

**What Next**

Identify and correct format, then reinvoke the *write* command.

## UID-1006

(information) Existing back annotation will be deleted.

**Description**

This message informs the user that the existing back annotation will be deleted.

**What Next**

This is only an information message. No action is required.

## UID-1007

(error) Static noise calculation requires ' set_si_options -static_noise true ' setting. Please set ' set_si_options -static_noise true ' and then run report_noise and report_noise_calculation.

**Description**

This error message occurs when the report_noise or report_noise_calculation is called without setting -static_noise option to true.

**What Next**

Use 'set_si_options -static_noise true ' and then run report_noise and report_noise_calculation command.

## UID-1008

(warning) Discarding all scenario specific information previously defined in this session.

Feedback

### Description

When issue the create_scenario command to create the first scenario, this warning will appear to remind you that any previously created scenario specific information will be discarded, and will not carry over into the context of the first scenario.

### What Next

No action is required, though you may want to review your script to ensure that scenario specific constraints are not erroneously created prior to the first create_scenario command.

### See Also

- create_scenario

---

## UID-1009

(warning) Current scenario changed to '%s'.

### Description

This warning is generated when current scenario changes without explicit use of the *current_scenario* command.

### What Next

Confirm that the new current scenario is the correct scenario for your purposes.

---

## UID-1010

(error) Invalid scenario '%s'.

### Description

This error is generated when a non-existent scenario is specified.

### What Next

Use *all_scenarios* to see the names of the currently defined scenarios.

---

## UID-1011

(error) Scenario is not active.

Feedback

### Description

This error is generated when a non active scenario is specified for current scenario. Only active scenarios can be set as the current scenario.

### What Next

Use *all_active_scenarios* to see the names of the currently active scenarios.

---

## UID-1012

(warning) %s scenario '%s' %s %s.

### Description

This warning message is issued because a command needs to temporarily activate inactive scenarios to avoid constraint or attribute losses during design changes. These scenarios are restored to their original states before the command returns.

### What Next

This is only a warning message and usually there is no action to be taken by user. However, active scenarios could use more memory.

### See Also

- all_scenarios

---

## UID-1015

(warning) Resetting 'min_delta_delay' to false when 'delta_delay' is set to false.

### Description

This warning message occurs when the *-delta_delay* option is being set to false while 'min_delta_delay' option is set to true.

### What Next

This is only a warning message. No action is required.

---

## UID-1016

(error) Cannot set 'min_delta_delay' to true without having 'delta_delay' option set to true.

Feedback

### Description

This error message is displayed if the option '-min_delta_delay' is specified to be true without having 'delta_delay' set to true.

### What Next

Set both '-min_delta_delay' and '-delat_delay' options to true, and re-run the command.

---

## UID-1017

(warning) Using the command-line option -nocase in rule-based matching could cause a high runtime.

### Description

This warning message occurs when using the *set_query_rules* command with the *-nocase* option.

During rule-based matching, switching the case-sensitivity mode using the command-line option *-nocase* forces rule-based matching to reconfigure each time the mode is changed, which incurs a high runtime.

### What Next

This is only a warning message. No action is required.

However, to avoid incurring a high runtime do one of the following:

- Set the *find_ignore_case* variable to true so that the rule-based matching always works in case-insensitive mode.
- Set the *find_ignore_case* variable to false and do not use the *-nocase* option with any command, so that rule-based matching always works in case-sensitive mode.

### See Also

- set_query_rules

---

## UID-1018

(error) Cannot set 'timing_window' to true without having 'delta_delay' or 'static_noise' option set to true.

### Description

This error message is displayed if the option '-timing_window' is specified to be true without having 'delta_delay' or 'static_noise' option set to true.

Feedback

### What Next

Set '-static_noise' or '-delat_delay' option to true, and re-run the command.

---

## UID-1019

(warning) Resetting 'timing_window' to false when 'delta_delay' and 'static_noise' are both set to false.

### Description

This warning message occurs when the *-delta_delay* option and *-static_noise* are being set to false while 'timing_window' option is set to true.

### What Next

This is only a warning message. No action is required.

---

## UID-1020

(error) the -all parameter can't be used with other options.

### Description

This error is generated when using the *set_active_scenarios* command, and both a list of scenarios and the *-all* parameter are specified.

### What Next

Either use *-all* or specify a list of scenario names.

---

## UID-1021

(error) You must specify at least one active scenario.

### Description

This error message occurs when using the *set_active_scenarios* command without specifying which scenarios to set active.

### What Next

Run the command again either using the *-all* option or specifying a list of scenario names.

---

## UID-1022

(Information) Scenario %s is no longer active.

Feedback

**Description**

This messages informs that a given scenario is now in the inactive state, and is not considered for optimization.

---

## UID-1023

(Information) Scenario %s is now active.

**Description**

This messages informs that a given scenario is now in the active state, and is considered for optimization.

---

## UID-1024

(information) Removed scenario %s from memory.

**Description**

This message indicates that the named scenario has been removed from memory. Any constraints associated with the scenario have been deleted.

**What Next**

This is an information-only message. No action is required.

---

## UID-1025

(Information) Mark scenario %s as CTS scenario.

**Description**

This messages informs that a given scenario was marked as CTS scenario, compile_clock_tree and optimize_clock_tree will work on CTS scenario if CTS scenario is marked.

---

## UID-1026

(Information) Scenario %s is no longer a CTS scenario.

**Description**

This messages informs that a given scenario was no longer CTS scenario, it happened when another scenario is marked and CTS scenario or the CTS scenario is removed.

Feedback

## UID-1027

(information) CTS scenario %s is already the only active scenario.

**Description**

This messages issued during compile_clock_tree and optimize_clock_tree to informs that the CTS scenario is the only active scenario inthe flow.

## UID-1028

(information) The current scenario has been switched to the CTS scenario %s.

**Description**

This messages is issued prior to clock tree optimization to inform you that the current scenario has been switched to the CTS scenario. Depending on the command, other scenarios may also be activated or deactivated. These are listed in the log file.

After this command is complete, the current and active scenarios will be restored.

## UID-1029

(error) Can't switch to CTS scenario %s, current scenario is %s

**Description**

This error is generated when compile_clock_tree and optimize_clock_tree trying to switch to CTS scenario but failed.

## UID-1030

(error) Cannot restore current scenario to previous value %s, current value is %s

**Description**

This error is generated when compile_clock_tree and optimize_clock_tree trying to restore current scenario from CTS scenario but failed.

## UID-1031

(information) Restore active scenarios, set current scenario to previous value %s

**Description**

This message informs you that compile_clock_tree and optimize_clock_tree have successfully restored the active scenarios and current scenario.

Feedback

## UID-1032

(error) Can't set both leakage-only and hold-only on current scenario!

### Description

This error is issued by the *set_scenario_options* command when trying to set both leakage-only and hold-only options on current scenario. The command will return fail and the current setting will be unchanged.

### What Next

Use *report_scenarios* to check the current setting and fix the problem in script.

### See Also

- report_scenarios

## UID-1033

(information) Using default format ddc because -format was not specified.

### Description

This information message occurs when the *-format* option of the *read_file* command is not specified. The file type cannot be inferred from the file name(s), so the default format of ddc is being used. It is good practice to always specify the format on the command line by using the *-format* option.

### What Next

This is an information-only message. No action is required.

### See Also

- UID-1034
- UID-1039

## UID-1034

(information) Inferring file format %s based on file name extension(s).

### Description

This information message occurs when the *-format* option of the *read_file* command is not specified. The command inferred the format based on the file name extension. It is good practice to always specify the format on the command line by using the *-format* option.

Feedback

### What Next

This is an information-only message. No action is required.

### See Also

- UID-1033
- UID-1039

---

## UID-1035

(error) Can't set leakage-only or hold-only on scenario %s.

### Description

This error is issued by the *set_scenario_options* command when trying to set leakage-only or hold-only options on an inactive scenario. The command will return fail and the current setting will be unchanged.

### What Next

Use *report_scenario_options* to check the current setting and fix the problem in script.

### See Also

- report_scenarios

---

## UID-1036

(warning) Can't find scenario %s.

### Description

This warning is issued by the *set_scenario_options* command when an invalid scenario name is passed to the -scenarios option. The command will skip the entry and continue with other scenarios in the list.

### What Next

Use *all_scenarios* to check all valid scenario names and fix the problem in script.

### See Also

- all_scenarios

Feedback

## UID-1037

(warning) Setting dynamic_power to false for scenario %s.

### Description

This warning is issued by the *set_scenario_options* command when trying to set dynamic_power to true without setting setup to true or when setup is set to false with dynamic_power left to true. Dynamic power computation needs scenario options setup and dynamic_power both set to true, to work properly. The command will finish successfully but will force dynamic_power to false. Set dynamic_power and setup to true to avoid this warning and enable dynamic power.

### What Next

Use *report_scenario_options* to report the current settings and fix the problem in script.

## UID-1038

(Warning) setting -cts_mode to true while -setup is false for scenario %s .

### Description

This warning is issued when the scenario option -cts_mode is set to true while the scenario option -setup is set to false. This will lead to multi-mode compile_clock_tree and optimize_clock_tree to abort.

### What Next

Use *report_scenario_options* to check the current settings and fix the problem in script.

## UID-1039

(error) Attempted to infer the file format based on file name extensions\n\t but there appear to be multiple file formats in the list. All\n\t the files in the list must be of the same format and it is\n\t recommended to always use the -format option.

### Description

This error message occurs when the *read_file* command determines that the file list may contain files of different formats. If the *-format* option of the *read_file* command is not specified, the tool attempts to infer the format from commonly-used file name extensions. The file list may contain only files of a single format, and if the file name extensions imply that multiple file formats were specified, this error occurs.

Feedback

### What Next

Use the *-format* option to force the command to use a particular format, and check that all of the files in the list are of the same format.

### See Also

- UID-1033

---

## UID-1040

(information) Removing all scenarios because all designs have been removed.

### Description

This information message occurs when the *remove_design* command has been used to remove all designs from memory, which triggers an automatic removal of any existing scenarios. If necessary, the *create_scenario* command can be used to re-create scenarios.

### What Next

This is an information-only message. No action is required.

### See Also

- create_scenario
- UID-1024

---

## UID-1041

(warning) Scenario '%s' does not exist on block %s.

### Description

This error message occurs when a nonexistent scenario is specified with the *select_block_scenario* command. The mapping has no effect on this block.

### What Next

This is only a warning message. No action is required.

However, if the result is not what you intended, check the scenarios on reference blocks or fix the script for *select_block_scenario*.

---

## UID-1042

(information) Scenario %s not exists. %s will skip it.

Feedback

## UID-1050

(error) Only %d character string is accepted as %s.

### Description

The *-hierarchical_separators* option only accepts a list of single character strings; The *-bus_name_notations* option only accepts a list of two character strings.

### What Next

Fix the value of option *-hierarchical_separators* or *-bus_name_notations* and re-issue the command *set_query_rules*.

## UID-1051

(error) '%s' is not supported as %s.

### Description

The single character *-hierarchical_separators* option does not accept the following characters: 0-9, *, ?, \\, +, (, ), [, ], ^. In addition, '/' is not allowed in the multi-character hierarchy separator. The *-bus_name_notations* option does not accept the following characters 0-9, *, ?, \\, +, ^. The *-bus_name_notations* option does not accept unpaired brackets such as "][", "(_".

The *-hierarchical_separators* option only accepts at most one alphabetical character a-Z or one multi-character hierarchy separator.

The *-bus_name_notations* option only accepts at most one matching pair of alphabetical characters a-Z.

The *-suffix* option only accepts a list of elements "_reg" and "_tri".

### What Next

Fix the value of option *-hierarchical_separators*, *-bus_name_notations* or *-suffix*, and re-issue the command *set_query_rules*.

## UID-1052

(warning) %s is missing as %s. This may cause incorrect result and long runtime.

### Description

'/' is often used as hierarchical separator for in-memory netlist; "[]" is often used as bus name notation for in-memory netlist. Not including them in the rule-based matching options may cause the program fail to find the object.

Feedback

**What Next**

Consider adding '/' in the *-hierarchical_separators* list, and "[]" in the *-bus_name_notations* list.

---

## UID-1054

(error) Only cell, pin, port and net are supported.

**What Next**

Fix the value of option *-class* and re-issue the command *set_query_rules*.

---

## UID-1055

(information) %d %s objects are matched according to rule-based query options.

**Description**

This message is issued in verbose mode in case query had matched some object according to the options set by command *set_query_rules*.

---

## UID-1056

(information) Rule-based matching enabled. Please turn it off once rule-based matching is no longer needed.

**Description**

Rule-based matching could slow down the program considerably, and use more memory. To avoid runtime and memory overhead, turn it off once it is no longer needed.

---

## UID-1057

(error) '-reset' option can not be used together with other options except for '-show'.

---

## UID-1058

(error) There must be at least 2 elements in %s list.

---

## UID-1059

(information) Scenario %s either not exists or is inactive. %s will skip it.

Feedback

## UID-1060

(error) There must be at least 3 elements in %s list.

### Description

This error is generated when there are not enough elements for -regsub option for command 'set_query_rules'.

### What Next

Check out the man page for Tcl Built-In regsub then reset the options for this.

## UID-1061

(warning) '-regsub' option can not be used together with '-exhaustive' option, '-regsub' option will be ignored.

### Description

When '-exhaustive' is specified, '-regsub' will not be supported for run time concern.

### What Next

Set one of the two options only and rerun the command.

## UID-1062

(information) Object %s is found using matching pattern %s.

### Description

The tool issues this message to inform that object is found with which pattern during rule-based mathcing. This information can be used to double check the correctness of the SDC file sourcing.

### What Next

If you want to disable this message, turn off verbose mode by command *set_query_rules*.

### See Also

- set_query_rules

## UID-1063

(warning) Design %s doesn't have scenario %s

Feedback

### Description

This warning is generated by *report_scenario* when a scenario exists in the session but the top level design doesn't have that scenario. if the design is hierarchical, the scenario may exists in some hierarchical block. Or the scenario is create on other designs which has been closed. For hierarchical design, you may need to create the scenario for top-level design explicitly using *create_scenario*. Or, you may want to remove the scenario using *remove_scenario* to reduce memory.

---

## UID-1064

(error) Save operation not allowed for ILM designs.

### Description

This error occurred because the design being saved is an ILM design. ILM designs are not allowed to be saved with save_mw_cel.

### What Next

Use create_ilm to generate a new ILM view.

---

## UID-1065

(information) Overwrite is already the default. You do not need to specify this option.

### Description

As of 2007.12, by default, save_mw_cel will overwrite a cell, so this option is no longer needed.

### What Next

You do not need to specify this option.

---

## UID-1066

(information) Global options '-regsub' and '-regsub_cumulative' are obsolete. They are converted to port, pin and net regsub rules.

### Description

The global regsub query rules are obsolete. For backward compatibility purposes, they are converted to the corresponding regsub rules for port, pin and net.

### What Next

Update your script to use object class specific regsub query rules.

Feedback

## UID-1067

(error) At most %d alphabetical or multi-character %s is allowed.

### Description

The *-hierarchical_separators* option accepts a list of single character strings; The *-hierarchical_separators* option accepts a list of strings as the hierarchy separators. If the list contains an alphabetical element or an string of more than one characters, the total number of such elements can not be greater than 1.

### What Next

Fix the value of option *-hierarchical_separators* and re-issue the command *set_query_rules*.

## UID-1068

(warning) String pattern '%s' is ambiguous for hierarchy separator '%s', that could lead to longer runtime and possible wrong object matches.

### Description

When the *-hierarchical_separators* option defines a hierarchy separator of multiple characters for the rule-based queries, unclean design name scheme could lead to ambiguity in object names when the multiple-character hierarchy separator string overlaps in an object name. This could cause longer runtime and possible wrong object matches.

For example, if the hierarchy separator rule is defined as {/ _H_}, and if a cell is named "a_H_H_b" after running *ungroup* and *change_names* commands, it is ambiguous because the original cell name could be either "a/H_b" or "a_H/b", in addition to the possibility of being plain "a_H_H_b".

### What Next

Avoid using a string of multiple characters in change_name rules as the replacement for hierarchy separator, if the string itself could be part of the object name in the design. It is even better to not to use string of multiple characters at all as the replacement for hierarchy separator.

## UID-1072

(information) Object %s is found using matching pattern %s.

Feedback

### Description

The tool issues this message to inform that object is found with which pattern during rule-based mathcing. This information can be used to double check the correctness of the SDC file sourcing.

### What Next

If you want to disable this message, turn off verbose mode by command *set_query_rules*.

### See Also

- set_query_rules

---

## UID-1080

(error) In the embedded scripts of the design %s, the '.' implementation is used in set_implementation command.

### Description

Using '.' as the implementation name is not allowed when allow_hier_refs_in_embedded_scripts is set to false.

### What Next

Use the real implementation name in embedded scripts or set the variable allow_hier_refs_in_embedded_scripts to true.

---

## UID-1081

(warning) In the embedded scripts of the design %s, the -check_impl option is used with set_implementation command.

### Description

Using -check_impl in embedded scripts is not supported when allow_hier_refs_in_embedded_scripts is set to false. The option will be ignored.

### What Next

Ignore the warning or set the variable allow_hier_refs_in_embedded_scripts to true.

---

## UID-1087

(warning) Ignoring %s because the net %s is multiply driven or has inout pins.

Feedback

**Description**

You receive this message when you use -source_of option and there are muliply driven nets or nets with inout pins in the fanin of the pin.

**What Next**

If the argument passed was incorrect, please fix it and rerun the command.

---

## UID-1088

(error) Must specify one of %s, %s, %s, or %s.

**Description**

This error message is displayed if none of the four mandatory options for a command are specified.

**What Next**

Re-issue the command by specifying one of the four options in the error message.

---

## UID-1089

(warning) Ignoring %s because the %s is %s.

**Description**

You receive this message when you give an inout, output or hierachical pin or an inout or input port, as an argument to -source_of option to clean_buffer_tree command.

**What Next**

If the argument passed was incorrect, please fix it and rerun the command.

---

## UID-1098

(error) Need to specify one and only one port type.

**Description**

This error is generated when *set_output_clock_port_type* command without specifying correct port type.

**What Next**

Please specify port type using one and only one option -clock or -data.

Feedback

## UID-1099

(error) Need to specify port_list or use -all option.

**Description**

This error is generated when *set_output_clock_port_type* command without correctly specifying port_list.

**What Next**

Please specify port_list, or use -all option to apply the port type to all output ports.

## UID-1100

(warning) Don't apply this command on input port. Ignore it.

**Description**

This error is generated when user apply *set_output_clock_port_type* command on input port. Ignore this setting.

## UID-1101

(error) The design '%s' has physical/floorplan data and cannot be modified by\n\ the 'group' command. Use the merge_fp_hierarchy command instead.

**Description**

The *group* command cannot be executed on a design with physical hierarchy.

**What Next**

Please use merge_fp_hierarchy command instead.

## UID-1102

(error) This older version of snapshot utility is no longer supported.

**Description**

This error message occurs when using the older version of snapshot utility is used.

**What Next**

Switch to using the newer version of snapshot utility by setting the variable enable_concise_qor_snapshot to TRUE (default).

Feedback

### See Also

- create_qor_snapshot
- report_qor_snapshot
- remove_qor_snapshot

---

## UID-1103

(information) This newer version of snapshot utility does not support these options: %s. Ignored.

### Description

This error message occurs when running the qor snapshot command with the above options

### What Next

User can ignore the message or re-run without the options.

### See Also

- create_qor_snapshot
- report_qor_snapshot
- remove_qor_snapshot

---

## UID-1104

(error) Not able to perform command due to read/write permission issue.

### Description

Read/Write permission problem. There can be several reasons for this. The most common reasons are: the user does not have read/write permissions in the directory, the disk is full or the user already has a file with the same name which can't be deleted.

### What Next

Try to create a file with the same name from the ICC shell. Fix any problem that you might encounter. Delete the file and invoke the icc_shell command again.

---

## UID-1105

(error) -lib_cell must specify a valid library cell in the form <library>/<cell>.

Feedback

## UID-1106

(error) -from_pin and -to_pin must specify different pins.

## UID-1107

(error) -from_pin must specify an input pin.

## UID-1108

(error) -to_pin must specify an output pin.

## UID-1200

(error) cannot recognize the given value for option '%s' in this SI command.

**Description**

check the command and give a valid value for the option.

## UID-1201

(warning) Overriding result from previous set_ideal_network command on object %s.

**Description**

The previous setting of the ideal network on an object is being over-written.

**What Next**

This is only a warning message. No action is required.

## UID-1202

(warning) expecting %s objects of type %s.

## UID-1203

(Error) This newer version of report_signal_em does not support these options: '%s'.

**Description**

This error message occurs when running electromigration command with the above options.

Feedback

### What Next

User could use command set_em_options to define the options.

### See Also

- report_signal_em
- fix_signal_em

---

## UID-1220

(error) Cannot find operating condition %s in library %s.

### Description

This error message is displayed if the specified operating conditions can't be found in the library from -max_library or -min_library option of set_operating_conditions command.

### What Next

Specify correct operating condition which can be found in the library.

---

## UID-1221

(information) %s command do not work on physical informaiton in ICC.

### Description

Please be aware that group/ungroup commands in ICC work only on the netlist view of a design and do not handle physical information of the design. The above mentioned commands are planned to be EoL once a replacement is available.

### What Next

This is only an informational message. Check manpages of group/ungroup for more informaion.

---

## UID-1301

(error) The field '%s' has already been specified for the current '%s'

### Description

This error occurs when a field is specified more than once for a given table. The error can also occur if the user does not include all fields during a table specification.

Feedback

**What Next**

Correct the AOCVM syntax by ensuring that each field is only specified once for a given table.

---

## UID-1302

(error) The current table is missing one or more fields

**Description**

This error occurs when a table is found at the end of the aocvm file without the required number of fields.

**What Next**

Correct the AOCVM syntax by adding the missing fields to the file.

---

## UID-1303

(error) Expected to find a valid field name but found '%s'.

**Description**

This error occurs when the parser cannot find a valid field name.

**What Next**

Correct the AOCVM syntax by ensuring that each field is
specified in the format name: data\n where name is one of
version,table,distance,depth,object_spec,derate_type,delay_type,rf_type,object_type.

---

## UID-1304

(error) Cannot specify '%s' for field '%s'.

**Description**

The data found in the AOCVM file is not valid for the given field.

**What Next**

Check the AOCVM spec to see allowed values for the field and update the AOCVM file appropriately.

---

## UID-1305

(warning) Cannot associate table with leaf cell '%s'.

Feedback

**Description**

AOCVM does not support the association of AOCVM data with leaf cells.

**What Next**

Associate AOCVM tables with hierarchical cells, library cells and designs only.

---

## UID-1306

(warning) Cannot find any objects of type '%s' for specification '%s'.

**Description**

Could not find any objects for the given specification. The table is not annotated on any objects but parsing and annotation will not stop due to this warning.

**What Next**

Check that the specification is correct.

---

## UID-1307

(error) Expected table of size %d but found table of size %d.

**Description**

The size of the 2D AOCVM table must equal the number of depth indices by the number of distance indices

**What Next**

Correct the size of the table

---

## UID-1308

(error) Cannot finish AOCVM file on a line continuation

**Description**

The AOCVM file ends with a line continuation. The parser expects more data but there is none.

**What Next**

Remove the line continuation character

Feedback

## UID-1309

(error) Could not find the specified AOCVM file

**Description**

The AOCVM file could not be found

**What Next**

Correct the name and path of the AOCVM file

## UID-1310

(error) The field '%s' must be specified before '%s'

**Description**

While parsing an AOCVM 2D table a field cannot be set because another field has not been specified

**What Next**

Change the order of the field specifications.

## UID-1311

(error) The version '%s' is not valid. Allowable versions are '%s'

**Description**

An invalid version number has been specified

## UID-1312

(error) An aocvm file version must be specified.

**Description**

The version number is used for backward compatibility purposes if the AOCVM file format changes in the future. It does affect the results of the analysis.

The version of the aocvm file must be specified on the first line of the AOCVM file. e.g version:1.0

**What Next**

Add the version information to the start of the AOCVM file.

Feedback

## UID-1313

(error) The %s array does not increase or decrease sequentially

**Description**

Sequential elements in the specified array must sequentially increase or decrease.

**What Next**

Fix the array so that it sequentially increases or decreases.

## UID-1314

(error) Negative values are not allowed

**Description**

Negative values are not allowed for the distance,depth arrays or in the 2D tables.

**What Next**

Remove negative values from the AOCVM file.

## UID-1315

(error) Cannot specify delay type %s with object type %s

**Description**

This combination of delay type and object types is not allowed as it makes no sense.

**What Next**

Correct the AOCVM file to remove the invalid combination of delay type and object type.

## UID-1316

(error) Cannot specify a non-monotonic derate table.

**Description**

This message is issued because the derate table specified in the message is incorrectly defined. The table has not been annotated.

The timing derates specified in each row and each column of the derate table must increase or decrease monotonically.

Feedback

### What Next

Fix the specified table so that sequential elements in each row and each column of the table sequentially increase or decrease.

---

## UID-1317

(error) Derate table variability is not decreasing with increasing depth.

### Description

This message is issued because the derate table specified in the message is incorrectly defined. The table has not been annotated.

For a late derate table, the timing derates specified in each row of the derate table must decrease monotonically as depth increases. For an early derate table, the timing derates specified in each row of the derate table must increase monotonically as depth increases.

No further restriction is placed on the table contents; however, typically, derate variability should be proportional to 1/sqrt(depth).

### What Next

Fix the specified table so that sequential elements in each row of the early (or late) table sequentially increase (or decrease) as depth increases.

---

## UID-1318

(error) Derate table variability is not increasing with increasing distance.

### Description

This message is issued because the derate table specified in the message is incorrectly defined. The table has not been annotated.

For a late derate table, the timing derates specified in each column of the derate table must increase monotonically as distance increases. For an early derate table, the timing derates specified in each row of the derate table must decrease monotonically as distance increases.

No further restriction is placed on the table contents; however, typically, derate variability should be proportional to distance.

### What Next

Fix the specified table so that sequential elements in each column of the early (or late) table sequentially decrease (or increase) as distance increases.

Feedback

## UID-1319

(information) Excluding block abstractions, ILMs, and physical hierarchy \n blocks from the list of designs to be written.

### Description

This message occurs when the *write_file* command detects one or more block abstractions, interface logic models (ILMs), or physical hierarchy blocks in the requested set of designs to be written out. Such designs are normally excluded by the *write_file* command unless the design is the top design in the hierarchy that is being written out.

### What Next

This is an information-only message. No action is required.

## UID-1320

(error) No Milkyway cell is open.

### Description

This error indicates that the current command needs the Milkyway cell to be opened first.

### What Next

Use *open_mw_cel* command to open a Milkyway cell. Run the current command again.

## UID-1321

(information) Excluding block abstractions from the list of designs to be written.

### Description

This message occurs when the *write_file* command detects one or more block abstractions in the requested set of designs to be written. Block abstractions are excluded from .ddc files unless the only thing being written is the block abstraction itself.

In this case, "block abstraction" refers to either the full design that has been processed by the *create_block_abstraction* command or the block abstraction (interface logic only) as specified by the *set_top_implementation_options* command.

### What Next

This is an information-only message. No action is required.

Feedback

## UID-1322

(error) Library cell '%s' belong to the other family '%s'.

**Description**

A library cell can only belong to one libcell subset. So in order to create a libcell subset, all libcells in the -libcell_list should not belong to any other libcell subset.

**What Next**

Remove the families which the specified libcells belong to using 'remove_libcell_subset -family_name' and re-run define_libcell_subset.

## UID-1323

(error) Could not find library cells '%s'.

**Description**

All library cells in -libcell_list should be able to be found in the link libraries.

**What Next**

Check the libcell list and the link libraries.

## UID-1324

(error) Library cell '%s' is not in the same function class as others.

**Description**

All library cells in the libcell list should be in the same function class.

**What Next**

Check the (user) function class of the library cells in the libcell list.

## UID-1325

(error) Cell '%s' is niether a sequential cell nor a lib-cell. The commands define_libcell_subset and set_libcell_subset supports sequential cells or instantiated lib-cells only.

**Description**

All (library) cells given to define_libcell_subset and set_libcell_subset should be sequential cells or instantiated lib-cells.

Feedback

### What Next

Specify seqential cells and seqential library cells in set_libcell_subset and define_libcell_subset, respectively.

---

## UID-1326

(error) Could not find the family. The family needs to be defined first.

### Description

Prior to execute set_libcell_subset, a family of library cells should be defined first using define_libcell_subset.

### What Next

Execute define_libcell_subset to create a family of library cells, and use the family name in set_libcell_subset.

---

## UID-1327

(error) Library cell '%s' of '%s' is not defined in family '%s'.

### Description

The *set_libcell_subset* command cannot set specifed cells in family if their library cells are not defined in the same family in advance.

### What Next

Please use remove_libcell_subset to remove the family, then use define_libcell_subset to redefine the family.

---

## UID-1328

(error) Reading user designs from db files is no longer supported.

### Description

This message occurs when you attempt to read a .db file that contains user-created designs instead of libraries. Reading user designs from .db files is no longer supported by the tool.

### What Next

Read the designs from RTL, or use an older version of the tool to convert the designs from a .db file to a .ddc file.

Feedback

## UID-1329

(warning) overriding result from previous dont_touch_network command on object %s.

**Description**

This warning is issued when dont_touch_network is applied to an object more than once.

**What Next**

This is only a warning message. No script fix is mandated.

## UID-1330

(warning) ignore net %s because -no_propagate is not specified.

**Description**

When set_ideal_network is applied to net object, -no_propagate option has to be used. Note also that it is no longer recommended to set a new object as ideal.

**What Next**

Consider applying set_ideal_network to the driver of the net, or using -no_propagate.

**See Also**

- set_ideal_network

## UID-1331

(information) %s %s for all instances of cell '%s' in subdesign '%s'.

**Description**

The *ssize_only* or *change_link* has been applied to all instances of the cell in a design.

## UID-1332

(Error) Path from %s to %s cannot be found.

**Description**

The set_register_replication command cannot find the path from specified register to the cell specified in -include option.

Feedback

## UID-1333

(Error) Sequential cell %s specified in -include option.

**Description**

The set_register_replication command should have only one register specified as the argument for the command itself.

## UID-1334

(Error) Path specified has boundary optimization set to false. Split cannot be done.

**Description**

If the path specified in set_register_replication command has loads across hieracchy and the hireachy has boundary optimization set to false, this warning is shown. The compile command will not split for loads across this hierarchy.

## UID-1335

(Error) cell %s has loads that cross a hirearchy with no bounray optimization.

**Description**

If the path specified in set_register_replication command has loads across hieracchy and the hireachy has boundary optimization set to false, this warning is shown. The compile command will not split for loads across this hierarchy.

## UID-1336

(Warning) Cell %s has dont_touch.

**Description**

Register replication will stop at cells that have dont_touch.

## UID-1337

(Error) Path (%s to %s) crosses power domains.

**Description**

Register (or path) replication will not be done if path crosses power domains.

Feedback

## UID-1338

(Error) Path (%s to %s) crosses more than %d hierarchies.

**Description**

Register (path) replication will not be done if path crosses more than the mentioned hierarchies.

## UID-1339

(Warning) Cell %s is not valid for split

**Description**

Register replication will not be perfnormed if the cell is dont_touch, retimed or is multibit.

## UID-1340

(Error) Sequential cell %s has dont_touch.

**Description**

Register replication will not be perfnormed if the sequential cell has dont_touch.

## UID-1341

(error) Field '%s' cannot be specified when ocvm_type is specified as '%s'.

**Description**

Invalid field specified.

**What Next**

Refer to the table format in documentation.

## UID-1342

(error) Field '%s' cannot be specified with field '%s'.

**Description**

Invalid fields specified.

**What Next**

Refer to the table format in documentation.

Feedback

## UID-1343

(warning) AOCVM analysis are not yet supported in POCVM. Turning POCVM analysis off.

### Description

POCVM analysis is incompatible with AOCVM analysis. If AOCVM is enabled concurrently with POCVM, POCVM analysis is not performed.

### What Next

This is a warning message only. The analysis will proceed disabling POCVM analysis internally. To ensure POCVM is used, disable AOCVM analysis.

### See Also

- time.aocvm_enable_analysis
- time.pocvm_enable_analysis

## UID-1344

(warning) '%s' analysis is only supported in IC Compiler.

### Description

AOCV and POCV analysis are only supported in IC Compiler, and they cannot be enabled in the current tool.

### What Next

This is a warning message only. The tool will proceed without POCV or AOCV analysis enabled.

## UID-1345

(Error) %s is not supported in Design Compiler.

### Description

AOCV analysis is not supported in Design Compiler.

## UID-1346

(Error) %s is supported only in DC Graphical and IC Compiler.

Feedback

## UID-1347

(warning) '%s' analysis is supported only in DC Graphical and IC Compiler.

### Description

POCV analysis is supported only in DC Graphical and IC Compiler, and they cannot be enabled in the current tool.

### What Next

This is a warning message only. The tool will proceed without POCV analysis enabled.

## UID-1400

(information) This command 'all_registers' now returns a collection containing integrated clock-gating cells since the variable 'all_registers_include_icg' is set to TRUE.

### Description

When the variable *all_registers_include_icg* is set to TRUE, command *all_registers* returns a collection containing integrated clock-gating cells.

### What Next

This is an informational message only. No action is required on your part.

### See Also

- all_registers

# UIED

## UIED-037

(Warning) Location {%s} on route shape %s is outside the core area. Buffer/inverter cannot be added at this location.

### Description

This warning message occurs when the location for buffer/inverter is outside the core area. The command could not place the buffer on this kind of location, so it is neglected.

### What Next

This is only a warning message.

Feedback

**See Also**

- add_buffer

---

## UIED-50

(error) The %s '%s' is of direction inout. Ports/pins of direction inout is not supported.

**Description**

This command does not support a port/pin which has a direction inout.

---

## UIED-060

(error) The inverter number from the driver to load pin/port '%s' are non-even.

**Description**

This error occurs when add_buffer_on_route detects that it will insert non-even inverter for the path from the driver to the load pin or top port.

**What Next**

Please re-estimate the "-repeater_distance" or "-location" to ensure that it could insert even buffer from the driver to all its loaders.

---

## UIED-066

(error) Internal error - %s.

**Description**

Internal error.

**What Next**

Report the problem to Synopsys

---

## UIED-114

(error) No valid net specified.

**Description**

This command requires a valid net to work on.

Feedback

**What Next**

Please provide correct inputs to the command

---

## UIED-123

(error) Cannot find the specified reference %s.

**Description**

The specified reference can not be found in any reference library path

**What Next**

Please provide correct inputs to the command

---

## UIED-156

(warning) No locations for the buffer can be found for net (%s)

**Description**

add_buffer_on_route doesn't find any buffer location for the specified net.

It may be caused by: 1. Route is too short for buffering comparing with repeater_distance/first_distance. Note: Buffers must be a little away from load pins, so route that is just as long as repeater_distance/first_distance may not be buffered as well. This feature could be controlled by eco.add_buffer_on_route.min_distance_buffer_to_load . 2. No gas station (whose is_primary is false) on route, while user requires add buffer only inside gas stations. 3. Cannot find bufferable location along route. Unbufferable situation include: i. Route over blockage/macro ii. Route outside core area iii. Route topology mismatches hierarchical netlist connection. (Detail will be reported by UIED-331.) iv. Route over/cross voltage areas that mismatch the power domains its net is in/cross.

**What Next**

Please re-specify suitable option value or modify the route if you still want to buffer the net by this command. You can also try -location.

---

## UIED-157

(error) Route segments of the net (%s) are not fully interconnected.

**Description**

The connection of route segments is not complete.

Feedback

**What Next**

Please verify the net's route segments are fully interconnected before using "add_buffer_on_route". For users do not expecting incomplete route feature of abor, they could use check_routes(2) to make sure the net to buffer is not open. For users expecting incomplete route feature of abor, they could open GUI and do mannually checking.

---

## UIED-158

(warning) delete wire '%d' for it is covered by wire '%d'.

**Description**

One wire is covered by another, so it is removed from database.

---

## UIED-159

(error) The net '%s' has dangling route.

**Description**

The net has dangling route. It will caused the incorrect buffer insertion behavior.

**What Next**

Please make sure the net has no dangling route.

---

## UIED-164

(warning) No locations for the buffer can be found for net (%s): New inserted buffers are out of core area.

**Description**

add_buffer_on_route doesn't find any buffer location for the specified net. New inserted buffers are out of core area.

---

## UIED-165

(error) incorrect input format, should be {{layer_name scaler} {layer_name scaler} ...}.

**What Next**

Please correct the input format.

Feedback

**See Also**

- add_buffer_on_route

---

## UIED-166

(error) cannot find layer with the name '%s'.

**Description**

the specified layer name doesn't exist.

**What Next**

Please correct the layer name.

**See Also**

- add_buffer_on_route

---

## UIED-167

(error) The value '%s' must be less than the value '%s'.

---

## UIED-170

(Error) Multiple driver net '%s' is not supported.

**Description**

This error message occurs when the operating net has more than one driver pin

**What Next**

*add_buffer_on_route* cannot operate on nets having multiple driver pins.

**See Also**

- add_buffer_on_route
- add_buffer
- remove_buffers

---

## UIED-171

(Error) library cell '%s' is not a buffer or an inverter.

Feedback

### Description

You recieve this error message, if the library cell matching the specified name is not a buffer or inverter library cell from the reference library.

### What Next

Choose the correct reference name of buffer/inverter.

---

## UIED-172

(Error) Invalid argument '%s'. '%s'.

### Description

You recieved this error message because an option in the command has an incorrect value. Either the tool could not find the expected value for the argument or the number of arguments is invalid.

### What Next

Clarify the proper use of the option, see the man page of the command.

---

## UIED-173

(Warning) Specified location (%.4f, %.4f) for buffer/inverter is outside the core area.

### Description

This warning message occurs when the coordinates provided for *-location* option of *add_buffer_on_route* command is not inside the core area of the design.

### What Next

This is only a warning message.

If you don't want this message to appear provide locations for the buffers which are inside the core area.

### See Also

- add_buffer_on_route

---

## UIED-174

(Warning) Library cell '%s' is an inverter; assuming -inverter_pair option.

Feedback

### Description

You recieve this message, if you have specified an inverter library cell for *buffer_lib_cell* option without specifying the *-inverter_pair* option.

### What Next

This is just a warning message, if result is not what you expected.

a) specify a library buffer cell.

b) specify -inverter_pair option along with inverter library cell.

### See Also

- add_buffer_on_route
- report_lib

---

## UIED-176

(error) Failed to find valid locations to add buffers

### Description

This error happens when *add_buffer_on_route* command fails to find valid locations along the net route, to place buffers on.

### What Next

Make sure the net is routed. If *"-location"* option was used, please make sure that the locations used are actually on the route.

---

## UIED-178

(error) Number of locations on net %s must be even when library cell is an inverter

### Description

The number of locations provided as input to command *add_buffer_on_route*, must be even when the library cell is an inverter. This is because inverters must be added as pairs.

### What Next

Provide an even number of locations.

Feedback

## UIED-179

(error) Failed to legalize new buffers on net %s. please try -max_distance_for_legalized_location to increase the search area for legalization.

**Description**

The command failed to find legal locations for the new buffers to be added along the route. No buffers are added to any nets.

**What Next**

Run the command with different parameters

## UIED-180

(error) Failed to add buffers

**Description**

The command failed to add any buffers.

**What Next**

Review errors from the command, before this one to proceed.

## UIED-183

(error) Failed to reroute new nets

**Description**

The command failed to reroute new nets created when buffers were added along the route of original net.

**What Next**

Run the command with different parameters

## UIED-184

(error) Location not specified correctly.

**Description**

Each of the locations provided to add_buffer_on_route, need to be a triplet containing the x coordinate, y coordinate and the routing layer name. The routing layer name is used to restrict the chosen route segment when there are multiple candidates in different layers

Feedback

for the specifed x and y coordinates. If the layer of route segment does not matter, the option "-detect_layer" can be used and the routing layer name is no need. When option "-detect_layer" is specified, the first route segment encountered is selected.

**What Next**

Specify the location with the format {x y layer_name} or {x y} -detect_layer.

**See Also**

- add_buffer_on_route

---

## UIED-185

(error) The PG net or the tie net is not allowed for this command.

**Description**

The PG net or the tie net is not allowed for this command.

**What Next**

Run the command with correct data

---

## UIED-186

(error) Failed to add buffer on net %s, due to mismatch between physical and logical hierarchy

**Description**

The physical location of the load pins of the net results in buffer locations which does not match the logical hierarchy. The logical nets end up driving load pins in different logical hierarchies.

**What Next**

Change the repeater_distance so that the number of buffers changes.

---

## UIED-189

(information) The section: '%s' starts.

**Description**

The section is running.

Feedback

## UIED-197

(error) duplicated layer name '%s' in the layer scaling list

**Description**

the specified layer name was used more than one time.

**What Next**

Please check the layer scaling list.

**See Also**

- add_buffer_on_route

## UIED-199

(error) Failed to snap new buffers to site rows.

**Description**

The command add_buffer_on_route failed snap new buffers to site rows. The snapping failed due to the site infomation of the specified library cells are not mathced to site rows in the design.

**What Next**

specify other library cells whose site infomation matched site rows in the desin. Or create new site rows.

## UIED-200

(error) %s.

**Description**

Input data error.

**What Next**

Please input correct data.

## UIED-201

(error) The location (%.4f %.4f) for buffer %s is out of search area (%.4f %.4f) (%.4f %.4f).

Feedback

**Description**

Although legal location has been returned but it is out within the search area requested by the add_buffer_on_route command. The search area might be out of with voltage area assigned to this buffer.

**What Next**

report_voltage_areas

---

## UIED-255

(error) Value for %s cannot be negative.

---

## UIED-300

(error) Classical router is not supported

**Description**

The command failed due to classical router not supported.

---

## UIED-303

(error) physical connectivity tracing error

**Description**

The command failed due to the physical connectivity tracing error.

---

## UIED-304

(error) Can't find place to insert buffer by given location (%.4f, %.4f).

**Description**

This error occurs when add_buffer_on_route can't find a route object to insert buffer on, within a region around the user given location. The region is specified by -max_distance_to_route.

**What Next**

You can re-assign the location (e.g. to be more close to the route) and try again.

**See Also**

- add_buffer_on_route

Feedback

## UIED-306

(error) inconsistent VA errror

### Description

The command failed due to inconsistent VA.

## UIED-307

(warning) Diode pin/port (%s): Routes that only drive this pin/port will be treated as dangling. No buffer is allowed to only drive a diode pin/port.

### Description

add_buffer_on_route find a diode pin/port as a load. It will be treated as dangling. If "-repeater_distance" is used, a buffer that only drives a diode pin/port will not be added. If "-location" is used, a buffer that only drives a diode pin/port will cause an error(UIED-159).

## UIED-308

(error) Bad location : Buffer location (%.4f, %.4f) is inside an unbufferable region %s.

### Description

The location to add buffer is inside an invalid region, e.g. placement blockage, macro, physical-logical inconsistent route segment.

Physical-logical inconsistent route segment means the route topology does not match with logical hierarchy. For example, in layout, a buffer (to be added on such route segment) drives module_A/module_B/cell1/I and module_A/module_C/cell2/I, but does not drive module_A/module_B/cell3/I. However, logically, if inserting such buffer in module_A (to be able to drive both cell1/I and cell2/I), cell3/I will be driven as well (which should not be driven).

### What Next

If -location is specified, please modify the location. In physical-logical inconsistent situation, you could also try to find out what is the inconsistency: 1. start_gui and identify the reported buffer location. Then find out which loads are physically driven by the buffer to add on layout. 2. Get the reported hierarchical net that tool derived to insert such buffer. 3. Find hierarchical ports and pins driven by the such reported hierarchical net. 4. For each one of these hierarchical ports and pins, you should check whether part loads of such hierarchical pin/port are physically driven by such buffer while the others are not.

Feedback

## UIED-309

(Error) Add buffer on route fail to get or process route of net %s

**Description**

Could not get or process route of a net

**What Next**

Check whether the net is valid

## UIED-310

(Error) Add buffer on route fail to check voltage area of net %s

**Description**

Meet unsupported voltage area case.

**What Next**

Check voltage areas.

## UIED-311

(Error) The add_buffer_on_route command does not support the route shape which has more than 2 points.

**Description**

*add_buffer_on_route* supports only horizontal and vertical route shapes.

## UIED-312

(error) Failed to add buffer on net %s, due to region %s mismatch between physical and logical hierarchy. %s

**Description**

The physical location of route crossing region does not match the netlist crossing logical hierarchy.

## UIED-313

(Warning) Failed to add buffer on net %s

Feedback

**Description**

Could not add buffer with current parameters on the net

**What Next**

Run the command with different parameters

---

## UIED-314

(Warning) The add_buffer_on_route_min_distance_buffer_to_load variable is ignored when you use the -inverter_pair option.

**Description**

The tool issues this warning when you run the *add_buffer_on_route* command with the *-inverter_pair* option and you have previously set the *add_buffer_on_route_min_distance_buffer_to_load* variable. When both are specified, the tool ignores the *add_buffer_on_route_min_distance_buffer_to_load* variable.

**What Next**

If this is not the desired behavior, rerun the *add_buffer_on_route* command without the *-inverter_pair* option.

**See Also**

- add_buffer_on_route

---

## UIED-318

(Error) Failed to split routes of net %s.

**Description**

The command failed when trying to split routes of a net for buffer insertion.

**What Next**

Please check database first.

---

## UIED-319

(Warning) This option is for a customer-specific flow. When you use the -load_pin_hierarchy option, buffers are added into the specified hierarchy.

Feedback

**Description**

When you use the -load_pin_hierarchy option, the tool adds buffers into the specified hierarchy.

---

## UIED-328

(Warning) Repeater distance length ratio >= 1

**Description**

Repeater distance length ratio of a value >=1 should only be used when user want to only add buffer(s) specified by -first_distance or -first_distance_length_ratio.

**What Next**

Check command argument value, unless it is just as expectation.

**See Also**

- add_buffer_on_route

---

## UIED-329

(error) net (%s) is dangling.

**Description**

The net is dangling. add_buffer_on_route does not support it.

**What Next**

Please check netlist before using "add_buffer_on_route".

---

## UIED-330

(Error) %s is an unsupported object.

**Description**

It is an unsupported object for add_buffer_on_route.

**What Next**

Please use get_nets to get nets for add_buffer_on_route (or use valid net name). If you have to use $flat_net which is returned by get_flat_net, please use [get_nets -of [get_pins -of $flat_net -filter "direction==out"]].

Feedback

## UIED-331

(Warning) No buffer will be added on route of net %s after location (%f, %f)%s, due to mismatch between route topology and hierarchical netlist connection topology.

### Description

After such location, route topology does not match with hierarchical netlist connection topology. add_buffer_on_route will not add buffers after such location, or the netlist connection and route of added buffer will drive different loads.

### What Next

Use other methods. (e.g. add "anchor buffer", then re-route) or use add_buffer_on_route -punch_port.

## UIED-332

(Error) Net %s has multi highest hier nets. It is not supported.

### Description

If request adding buffers in highest hierarchy, there should NOT be multi highest hier nets.

### What Next

Change netlist to meet command request.

## UIED-333

(error) Can not find unique layer to insert buffer near given location (%0.4f, %0.4f).

### Description

This error occurs when -location find multi possible layer to split for adding buffer, within a region around the user given location. The region radius is specified by -max_distance_for_legalized_location.

### What Next

If you use -detect_layer to let command auto choosing layer, you can assign a particular layer in format -location {x y layer_name} instead.

### See Also

- add_buffer_on_route

Feedback

## UIED-334

(error) Track assignment or detail route data is not supported by global routed net buffering.

### Description

This error message occurs if add_buffer_on_route -only_global_routed_nets is on and input net is track assigned or detail routed.

### What Next

Turn off -only_global_routed_nets or remove the route data then do only global route for the input net.

### See Also

- add_buffer_on_route
- route_global

## UIED-336

(error) user_specified_buffer not specified correctly.

### Description

Each of the "user_specified_buffer" provided, need to be a fiveplet containing the name, library cell, x coordinate, y coordinate and the routing layer name. If option "-detect_layer" is specified, then the routing layer name is no need and the tool detects the layer of location coordinates automatically.

### What Next

Run the command with correct data

## UIED-337

(error) Global route "%s" are not supported when bufferring track assigned or detail routed net.

### Description

This error message occurs if add_buffer_on_route finds global routes on the net when option -only_global_routed_nets is off.

Feedback

### What Next

If net is supposed to be track assigned or detail routed, check the route segment and re-route the net before run add_buffer_on_route. If net is global routed, turn on -only_global_routed_nets and try add_buffer_on_route again.

---

## UIED-338

(warning) New repeater will be named as "%s" (User specified cell name "%s" is already used by an existed cell).

### Description

User specified cell name is already used by an existed cell, tool will rename the new repeater.

---

## UIED-339

(error) Unsupport type %s of object %s .

### Description

This error message occurs if add_buffer_on_route finds unsupported type of objects on the net. The supported type of objects are "port", "leaf pin", "vertical and horizontal path", "global route" and "via".

### What Next

Check the type of reported objects. If it is path which contains non-vertical or non-horizontal wire, re-route the net and try add_buffer_on_route again. If it is other types, report bug to Synopsys.

---

## UIED-340

(Info) Cannot add repeaters from point (%0.4f, %0.4f) %s of net %s, due to inconsistent connection topology between logical hierarchy and physical route.

### Description

On some route segments, the physical location of the load pins of the net does not match logical hierarchy to insert buffer. Tool will not add buffers on such route segments, or their logical nets will drive load pins in different logical hierarchies. But tool still will add buffers on other route segments of such net.

Feedback

**What Next**

Please check logical hierarchy connection topology and physical route topology after the reported point.

---

## UIED-341

(error) Failed to find route segments in max search distance for opened pin %s.

**Description**

add_buffer_on_route failed to find route within search area for incomplete route.

**What Next**

Specify proper value of -max_distance_for_incomplete_route and try again.

**See Also**

- add_buffer_on_route

---

## UIED-342

(error) Can't find place to insert buffer by given location (%0.4f, %0.4f).

**Description**

This error occurs when add_buffer_on_route -location can't find a route object to insert buffer on, within a region around the user given location. The region radius is specified by -max_distance_to_route.

**What Next**

You can re-assign the location (e.g. to be more close to the route) or the search region radius and try again.

---

## UIED-343

(error) Unable to insert repeaters on net %s. Tool meets a route tracing error when crossing region border at point (%0.4f, %0.4f) %s.

**Description**

Tool meets an internal error when processing region (plangroup, voltage area, macro or blockage). E.g. there is a route overlapping with a blockage border somewhere before the reported point.

Feedback

**What Next**

Please check the route topology. If there is a wire overlapping with a blockage border, you can try to move the wire away from border and rerun this command. Please report this issue to Synopsys.

---

## UIED-344

(Info) Resuming buffering from point (%0.4f, %0.4f) %s of net %s, after route segments with inconsistent connection topology between logical hierarchy and physical route.

**Description**

On some route segments, the physical location of the load pins of the net does not match logical hierarchy to insert buffer. Tool will not add buffers on such route segments, or their logical nets will drive load pins in different logical hierarchies. But tool still will add buffers on other route segments of such net.

---

## UIED-345

(info) Port(s) %s are punched for connecting repeater %s and its load(s) %s on specified net %s due to inconsistent connection topology between logical hierarchy and physical route.

**Description**

On some route segments, the physical location of the load pins of the net does not match logical hierarchy to insert buffer. Tool will punch ports to connect such loads.

---

## UIED-346

(Error) %s is not a valid voltage area.

**Description**

User must specify a valid voltage area name or collection with a single VA, or DEFAULT_VA for default voltage area.

**What Next**

Check whether the voltage area is right.

---

## UIED-347

(error) %s not specified correctly.

Feedback

**Description**

This option ask user provide pairs which contains a valid voltage_area name and a valid library cell.

**What Next**

Run the command with correct option value.

---

## UIED-348

(info) In option %s, %s %s has already been specified with same %s.

**Description**

Duplicated items are specified in this option.

**What Next**

If it is a typo, run the command with correct option value.

---

## UIED-349

(error) In option %s, %s %s has already been specified with different %s.

**Description**

Conflict items are specified in this option.

**What Next**

Run the command with correct option value.

---

## UIED-354

(Warning) After buffering net %s, hierarchical port %s will be driven by an inverted signal.

**Description**

There will be odd number inverters from original driver till this hierarchical port. So the hierarchical port will be driven by an inverted signal.

**What Next**

A user could make inverters even from original driver till this hierarchical port by adjusting inverter locations.

Feedback

## UIED-355

(Error) Cannot find terminal of %s.

**Description**

This command cannot support a pin or port without terminal.

**What Next**

Make user each related pin or port has its terminal.

## UIED-370

(Error) No driver net '%s' is not supported.

**Description**

This error message occurs when the operating net has no valid driver pin.

**What Next**

*add_buffer_on_route* cannot operate on nets having no driver pin.

**See Also**

- add_buffer_on_route
- add_buffer
- remove_buffers

## UIED-371

(Error) Not support %s '%s' which is in a sub block.

**Description**

This error message occurs when the operating object is in a sub block.

**What Next**

Cannot operate on objects in sub blocks.

## UIED-372

(error) Cannot specify '%s' with '%s' when '%s' is used.

Feedback

**Description**

The listed command options are exclusive if the specified option is used. Only one of them can be specified.

**What Next**

Look at the manpage for this command for more information on command options.

---

## UIED-373

(error) Failed to find the legal location to add buffer due to max fanout violation (real fanout:%d > max fanout: %d) around {(%0.4f, %0.4f) (%0.4f, %0.4f)} on the net:%s.

**Description**

The max fanout violation has been detected, but there is no any legal location to add the buffer. The reason can be either placement blockage, macro, or macro keepout margin.

**What Next**

Adjust the location or shape for the placement blockage, macro or macro keepout margin.

---

## UIED-374

(error) Failed to find the legal location to add buffer due to max fanout violation (real fanout:%d > max fanout: %d) after (%0.4f, %0.4f) on the net:%s.

**Description**

The max fanout violation has been detected, but there is no any legal location to add the buffer. The reason can be either placement blockage, macro, or macro keepout margin.

**What Next**

Adjust the location or shape for the placement blockage, macro or macro keepout margin.

---

## UIED-375

(error) Based on the route topology, max-fanout:%d can not be satisfied near to (%0.4f, %0.4f) on the net:%s.

**Description**

According to the route topology, the buffer can not be able to add near the reported location in order to meet the input specified max_fanout number. It could be due to the route topology, placement blockage, macro or macro keepout margin.

Feedback

**What Next**

Adjust the route topology or route shape, placement blockage, mcro keepout margin or macro location, or use higher max_fanout number.

---

## UIED-376

(error) Based on the route topology, the buffer can not be added to drive the specified loads near to (%0.4f, %0.4f) on the net:%s.

**Description**

According to the route topology, the buffer can not be able to add near the reported location in order to drive the specified loads. It could be due to the route topology, placement blockage, macro or macro keepout margin.

**What Next**

Adjust the route topology or route shape, placement blockage, mcro keepout margin or macro location, or different input loads.

---

## UIED-378

(error) It is not supported to specify gas station %s (whose is_primary is false) in %s.

**Description**

This feature is for primary voltage area only. This command does not support physical only feed through gas stations.

**What Next**

Create supported gas stations. This command only support gas station whose power domain is same with the primary voltage area surrounding it. And it cannot be physical only crossed. Or you can create disjoint regions of primary voltage areas.

---

## UIED-379

(warning) Buffer %s is added inside voltage area %s. However, power domain of its hierarchy is %s, which does not belong to the voltage area.

**Description**

This buffer is inserted in a hierarchy whose power domain does not match the voltage area this buffer is inside in layout. It usually happens when you allow add buffer in a physical only feed through voltage area. A second possibility is the voltage area is a not supported gas station whose power domain is different with the primary voltage area surrounding it.

Feedback

**What Next**

If the voltage area is primary, you could use set_eco_power_intention to update the buffer's power domain. If the voltage area is not primary, you could replace it with a disjoint primary voltage area. You could also manually set the power domain of the buffer.

---

## UIED-380

(error) Library '%s' is missing technology information.

**Description**

The technology information is missing in the current library. This information is necessary in order to read the input layer mapping file.

**What Next**

Load a technology file into the library using the *read_tech_file* command.

---

## UIED-381

(error) Neither conducting_layers nor via_layers is found in the layer map file.

**Description**

Both conducting_layers and via_layers is required in the layer map file

**What Next**

Specified the conducting_layers or via_layers if either of them is missing.

---

## UIED-382

(error) No any %s layers mapping are specified

**Description**

There is no any conducting layers mapping or via layers mapping are found and stored.

**What Next**

At least one layer mapping is required to specified.

---

## UIED-383

(error) The layer '%s' cannot be found.

Feedback

**Description**

The layer with the given name cannot be found in the technology database.

---

## UIED-384

(error) '%s' is not a single rail buffer. It cannot be added inside gas-station %s.

**Description**

User should add single rail buffer inside gas stations (whose is_primary is false).

**What Next**

Specify a single rail buffer.

---

## UIED-385

(Error) The port punching on port will not be supported for -respect_voltage_areas.

**Description**

The port punching on port will not be operated for -respect_voltage_area.

**What Next**

remove -respect_voltage_areas.

---

## UIED-386

(error) user_specified_bus_buffer not specified correctly.

**Description**

Each of the "user_specified_bus_buffer" provided, need to be a 4-tuple containing eco bus buffer pattern, library cell, x coordinate and y coordinate.

**What Next**

Run the command with correct data

---

## UIED-387

(error) Fail to apply pattern %s on route at %s %s.

**Description**

Pattern can only be added at route segment which is fully parallel and meet pattern style.

Feedback

**What Next**

Select other patterns or change location to add buffers.

---

## UIED-388

(error) bus_cell_prefix not specified correctly.

**Description**

Each of the "bus_cell_prefix" provided, need to be a pair containing eco bus buffer pattern and cell prefix string.

**What Next**

Run the command with correct data

---

## UIED-389

(error) The specified minimum porosity should between 0 and 1.

**What Next**

Please correct the given porosity value.

**See Also**

- add_group_repeaters

---

## UIED-390

(error) Can not insert buffer in overlapping area at location (%0.4f, %0.4f).

**Description**

This error occurs when the specified location is in overlapping area of different wires.

**What Next**

Please specify location outside the overlapping area.

---

## UIED-391

(error) Can not find unique wire near location (%0.4f, %0.4f).

Feedback

**Description**

This error occurs when on the same layer, there are multiple possible wires that have the same distance to the specified location.

**What Next**

Please specify location that is closer to the wire that you want to insert buffer.

# UIG

## UIG-001

(warning) -from and -to objects must be on the same library cell.\n \tIgnoring invalid combination from '%s' to '%s'.

**Description**

This command expects that from and to library pins will be on the same library cell. It is ok to specify multiple library pins on different library cells in -from if -to is not specified, or in -to if -from is not specified. If -from objects are on a different library cell than -to objects, those combinations are ignored.

**What Next**

Specify from and to objects that are on a single library cell.

# UIL

## UIL-0

(information) Note that the ".synopsys_lc.setup" init file of\n \tlc_shell is a subset of the dc_shell ".synopsys_dc.setup" init file.\n \tIf a command other than an lc_shell command is encountered in the init file,\n \tan error message is issued and the processing might terminate.\n \tRefer to the Library Compiler manuals for supported commands.

**Description**

During the initialization phase, the Library Compiler command line tool *lc_shell and the graphical tool library_compiler read in a setup file .synopsys_lc.setup. Both tools perform a subset of the Design Compiler dc_shell commands. If a dc_shell command is encountered such as read_design command), Library Compiler tools issue an error. Depending on the semantics of the command read, it might terminate.*

Feedback

### What Next

Refer to the Library Compiler User Guide and Reference Manuals for supported commands.

---

## UIL-1

(warning) Overwriting an old symbol library '%s' file with a new one.

### Description

The symbol library, specified in your *read_lib* command, already exists in the database. The old library is deleted from the database and replaced by the new one. This message is generated when

```
- Reading a symbol library directly or
        - Reading an EDIF file with the "-symbol option" set to the
          same symbol filename
```

### What Next

Use *list -lib* to find all available libraries. No action is required if you decide to overwrite a library name reported by *list -lib*.

### Examples

Reading a symbol library explicitly:

```
lc_shell> read_lib uil1_symbol.slib
             Reading '/tmp/uil1_symbol.slib' ...
             Symbol library 'uil1' read successfully
   lc_shell> read_lib uil1_symbol.slib
             Reading '/tmp/uil1_symbol.slib' ...
             Warning: Overwriting an old symbol library '/tmp/uil1.sdb'
 file with a  new one. (UIL-1)
```

Reading a symbol library though the EDIF file:

```
lc_shell> read_lib -format edif uil1.edif -symbol uil1_symbol.slib
             Reading '/tmp/uil1_symbol.slib' ...
             Symbol library 'uil1' read successfully
   lc_shell> read_lib -format edif uil1.edif -symbol uil1_symbol.slib
             Reading '/tmp/uil1_symbol.slib' ...
             Warning: Overwriting an old symbol library '/tmp/uil1.sdb'
 file with a new one. (UIL-1)
```

### Examples

```
Warning: Overwriting an old symbol library '/tmp/uil1.sdb' file with a
 new one. (UIL-1)
```

Feedback

## UIL-2

(warning) Overwriting an old technology library '%s' file with a new one.

### Description

The technology library, specified in the *read_lib* command, already exists in the database. The old library is deleted from the database and is replaced by the new one.

### What Next

Use *list -lib* to find all available libraries. No action is required if you decide to overwrite a library name reported by *list -lib*.

### Examples

```
lc_shell> read_lib uil2.lib
                  Reading '/tmp/uil2.lib' ...
                  Technology library 'uil2' read successfully
        lc_shell> read_lib uil2.lib
                  Reading '/tmp/uil2.lib' ...
                  Warning: Overwriting an old technology library
 '/tmp/uil2.db' file with a new one. (UIL-2)
```

### Examples

```
Warning: Overwriting an old technology library '/tmp/uil2.db' file with a
 new one. (UIL-2)
```

## UIL-3

(warning) The '%s' library has not been read in yet.

### Description

The command issued tries to access a library that does not exist in the database. A typo in the library name can also result in this problem. Any of the following commands might trigger the generation of this warning: *add_module, compare_lib, write_lib, report_lib and update_lib.*

### What Next

Use *list -lib* to find all available libraries. Make sure the library name is among the ones being reported by *list -lib*. Otherwise, load the library into the database using *read* for a db file, or *read_lib* for a library source file, then reissue the command.

### Examples

```
Warning: The 'uil3' library has not been read in yet. (UIL-3)
```

Feedback

## UIL-4

(error) The '%s' library source file is not found.

### Description

The library source file is not found either because

```
- The file is not in the search_path, assuming that the full
          path is not specified or,
        - There is a typo in the filename.
```

### What Next

Issue the command with the correct file name and the full path. Alternatively, you can add the path to the current *dc_shell* or *lc_shell search_path*, and issue the command with the correct file name.

### Examples

```
Error: The 'uil4.lib' library source file is not found. (UIL-4)
```

## UIL-5

(error) The read_lib command does not support '%s' as the\n \t'-format' switch value. The only format allowed is 'edif'.

### Description

Currently, *read_lib* accepts only *edif* as *-format* switch value for reading in symbol library in EDIF format. It is reported as an error if you specify any other *-format* switch value.

### What Next

If you want to read in an EDIF symbol library, make sure the *read_lib* switch *-format* has the value *edif*. Otherwise, do not specify the *-format* switch with the *read_lib* command. Reinvoke the *read_lib* with the correct switch value. For quick reference, invoking *read_lib* with -help switch, causes *read_lib* to print out its legal command syntax.

### Examples

```
dc_shell> read_lib -help
        Usage: read_lib
             -format      (EDIF symbol format; default is Synopsys
 format)
             -symbol      (with EDIF, name of Synopsys library file to
 create)
             <file_name>  (technology or symbol library file)
             -no_warnings (disable warning messages)
             -names_file  <file_list>  (one or more names files)
```

Feedback

## UIL-6

(warning) Library Compiler is not enabled. The cell\n \tfunctionality is ignored.

### Description

This site is not licensed for Library Compiler. The attempt to compile a technology library is restricted to timing information only. All cell functions are removed, so they become black-boxes.

### What Next

If you are compiling a technology library for timing analysis use only, ignore this warning message. However, if you want to have full Library Compiler capability, contact your Synopsys system administrator to purchase Library Compiler.

### Examples

```
Warning: Library Compiler is not enabled. The cell
        functionality will be ignored. (UIL-6)
```

## UIL-7

(error) Cannot write the library to the '%s' file.

### Description

An attempt to write out a database (db or sdb) file w with *write_lib* failed. The reasons you get this error could be because either

```
- You do not have write permission to the target directory or,
        - There is not enough disk space for the file.
```

### What Next

Check disk space and permission of the target directory. Make sure you have enough disk space and write permission before using *write_lib*.

## UIL-8

(error) The '%s' pin of the '%s' cell in the '%s' technology library\n \tis not found in the same cell of the '%s' symbol library.

### Description

The Library Compiler detects unmatched pin names between a technology library and a symbol library using the *compare_lib* command.

Feedback

### What Next

Correct the mismatch either in the technology library or the symbol library, and recompile the modified library.

### Examples

```
Error: The 'D' pin of the 'uil8' cell in the 'techlib' technology library
        is not found in the same cell of the 'symblib' symbol library.
 (UIL-8)
```

---

## UIL-9

(error) The '%s' cell does not exist in the '%s' symbol library.

### Description

A correspondent symbol cell cannot be found in the symbol library. This error is caused by a typo in a cell name in either the technology library file or the symbol library file, or a missing symbol cell. This error results in the use of a generic symbol library symbol or a black box symbol in a schematic containing the cell.

### What Next

If the error is due to a typo, correct the cell name typo. If the error is due to a missing symbol, add it to the symbol library. Recompile the modified library, and run compare_lib command.

### Examples

```
Error: The 'uil9' cell does not exist in the 'symblib' symbol library.
 (UIL-9)
```

---

## UIL-10

(error) Trying to compare a '%s' library with a '%s' library.\n \tcompare_lib compares a technology library and a symbol library.

### Description

The two libraries submitted to *compare_lib* are not a legitimate combination. *compare_lib* compares a technology library and a symbol library.

### What Next

Use *report_lib* to obtain the types of libraries. Make sure that you compare a symbol library and a technology library.

Feedback

### Examples

```
Error: Trying to compare a 'Technology' library with a 'Technology'
 library.
        compare_lib compares a technology library and a symbol library.
 (UIL-10)
```

---

## UIL-12

(error) Trying to compare a library with itself. compare_lib\n \tcompares a technology library and a symbol library.

### Description

The two libraries submitted to *compare_lib* are identical. The *compare_lib* command compares a technology library and a symbol library.

### What Next

Compare a symbol library and a technology library. If the technology library and symbol library share the same library name, specify the symbol library with its full name in *format file_path:library_name*. For example, the libraries are both named *uil12*. Issue the commands

```
dc_shell> list -library
              Library           File                    Path
              -------           ----                    ----
              uil12               uil12.db
  /synopsys/libraries
              uil12               uil12.sdb
  /synopsys/libraries
      1

      dc_shell> compare_lib uil12.db:uil12 uil12.sdb:uil12
```

### Examples

```
dc_shell> compare_lib uil12 uil12
```

### Examples

```
Error: Trying to compare a library with itself. compare_lib
        compares a technology library and a symbol library. (UIL-12)
```

---

## UIL-13

(error) The specified '%s' library cannot be found in memory.

Feedback

### Description

You try to access a library that has not been read in yet. This error could result from typos in the library name.

### What Next

Use *list -lib* to find all available libraries. Make sure that you type in the correct library name with your command. If the library is not reported by the *list -lib* command, load the library with *read_lib* for a library source file, or *read* for a db file, before reissuing the command.

---

## UIL-15

(warning) No libraries in the '%s' library file.

### Description

A file is read in, but no library is found in the file. It is most likely that the file is empty.

### What Next

Read in the correct file with a library.

---

## UIL-16

(error) Cannot find the '%s' library or library file in memory.

### Description

This error is obsoleted.

---

## UIL-17

(error) Could not open the '%s' file for reading.

### Description

You do not have read permission of the specified EDIF file.

### What Next

Check the permissions for the file. Make sure you have read permissions for the file.

### Examples

```
Error: Could not open the 'uil17.edif' file for reading. (UIL-17)
```

Feedback

## UIL-18

(error) Could not open the '%s' file for writing.

### Description

An attempt to open an output file has failed. This problem might be caused by

```
1. Trying to write to a directory for which you do not have write
 permission.
       2. Trying to write to an existing file for which you do not have
 write permission.
       3. The output file name specifies a non-existing path.
       4. The file system is full.
```

### What Next

Check the correctness of file name and path, and make sure you have write permission for the file. If the file system is full, provide more disk space.

## UIL-19

(error) The '%s' cell does not exist in the '%s' technology library.

### Description

This error message is issued by either *compare_lib* or *report_lib*. In the *compare_lib* case, if it cannot find a matched technology library cell for a symbol component, it reports this error with the message. In the *report_lib* case, if you specify a nonexistent cell in the cell list, this error message is displayed. This error usually is caused by a typo.

### What Next

Provide a correct name with each cell in a cell list. If you are not sure about cell names, do not specify a cell list.

## UIL-20

(error) The '%s' symbol in the '%s' symbol library has an extra '%s' pin.

### Description

This error message is issued by the *compare_lib* command. For each cell of a symbol library, Library Compiler checks for pin names in the same cell of the technology library and issues an error if the pin does not exist.

Feedback

### What Next

Correct the symbol library by removing the pin name or adding it to the technology library. Use Library Compiler to recompile the modified library, and run the compare_lib command.

---

## UIL-21

(error) The '%s' synthetic module does not exist in the '%s' library.

### Description

This error message is issued by the *report_lib* command. You might have specified a nonexistent synthetic module in the module list. This error is usually caused by a typo.

### What Next

Provide a correct name with each module in a module list. If you are not sure about module names, do not specify a module list.

---

## UIL-23

(error) Invalid output format '%s'.

### Description

Currently, for dc_shell *write_lib* accepts only *db, edif*, and *vhdl* as *-format* switch value for writing libraries in the specifBed format. For lc_shell, only *db* and *vhdl* switch values are accepted. It is reported as an error if you specify any other *-format* switch value.

### What Next

If your want to write a db library, do not specify the *-format* switch with the *write_lib* command. Otherwise, make sure that the *write_lib* switch *-format* has the value edif or vhdl.

Reinvoke the *write_lib* with the correct switch value. For quick reference, invoking *write_lib* with the -help switch, causes *write_lib* to print out its correct command syntax.

### Examples

```
dc_shell> write_lib -help
             Usage: write_lib
             <library_name> (a library already loaded into dc_shell)
             -format        db|edif|vhdl (write library in the specified
 format;
                            default is db)
             -output        <file_name>  (write library to the specified
 file)
             -names_file    <file_list>  (one or more names files)
```

Feedback

## UIL-24

(error) Unacceptable VHDL library generation.\n \tYou must read in the `%s` library from a technology library source file.

### Description

This error is issued when the *write_lib -format vhdl* uses a library read as a db file using the *read* command.

### What Next

Read in the library from a technology library source file. Reissue the write_lib -format vhdl command.

### Examples

```
Error: Unacceptable VHDL library generation.
        You must read in the `uil24` library  from a technology library
 source file. (UIL-24)
```

## UIL-25

(error) The '-symbol' option can only be used with the '-format EDIF' option.

### Description

Currently, *read_lib* accepts the *-symbol* option only with *edif* as a *-format* switch value for reading in symbol library in EDIF format. It is reported as an error if you specify -symbol without the edif format.

### What Next

If your want to read in an EDIF library and select the Synopsys file to create a symbol library, make sure the *read_lib* switch *-format* has the value edif followed by the *-symbol filename* option. Otherwise, do not specify the *-symbol* switch with the *read_lib* command. Reinvoke the *read_lib* with the correct switch value. For quick reference, invoking *read_lib* with the -help switch, causes *read_lib* to print out its correct command syntax.

### Examples

```
dc_shell> read_lib -help
        Usage: read_lib
             -format      (EDIF symbol format; default is Synopsys
 format)
             -symbol      (with EDIF, name of Synopsys library file to
 create)
             <file_name>  (technology or symbol library file)
             -no_warnings (disable warning messages)
             -names_file  <file_list>  (one or more names files)
```

Feedback

## UIL-26

(error) Unacceptable list of cells for the '%s' library.

### Description

This error message is issued by the *report_lib* command. You might have specified an empty or a corrupted cell list. This error usually is caused by a typo.

### What Next

Provide a correct name with each cell in a cell list. If you are not sure about cell name, do not specify a cell list.

## UIL-27

(error) Invalid technology library timing report.\n \tYou must read in the `%s` library from a technology library source file.

### Description

This error message is issued by *report_lib -timing_arcs* or *-timing* command. The timing options are used only on a library that has been read from a ".lib" technology library source file by the *read_lib* command.

### What Next

To use the report_lib with the -timing option, read in the technology source library instead of the db file.

## UIL-28

(warning) The '%s' vendor specific delay model is used in this library.\n \tThe library timing report might be incomplete.

### Description

This message is issued during a *report_lib -timing* command. The *delay_model* attribute in the technology source file is set to a non-generic delay model, such as *generic_cmos* or *table_lookup*. In the case of a *cmos2*, *report_lib* does not display any information about the slope.

### What Next

You can choose to ignore the warning or to modify the delay model attribute in the technology source file if you want more timing information. Reread the library, and reissue the report_lib command.

Feedback

### Examples

```
Warning: The 'cmos2' vendor specific delay model is used in this library.
        The library timing report might be incomplete. (UIL-28)
```

---

## UIL-29

(error) The '%s' library is a synthetic library.\n \tUse the report_synlib command.

### Description

This error is issued when the *report_lib* command is used for reporting on a synthetic library. A dc_shell command *report_synlib* is used to report on synthetic libraries.

### What Next

Use *report_synlib* command to report on synthetic libraries.

---

## UIL-30

(error) The '%s' library is not a synthetic library.\n \tUse report_lib command.

### Description

This error is issued when the *report_synlib* command is used on reporting on a technology library. A Library Compiler command *report_lib* is used to report on technology libraries.

### What Next

Use *report_lib* command to report on technology libraries.

---

## UIL-32

(warning) The '%s' model library is different from the current link library.

### Description

The *model* command might generate this message whenever the model library is different from the current link library. This situation should not occur frequently, but whenever a *model* is used with a 'for' loop construct to get the model library, it is possible that the link library will be different from the model library.

### What Next

Redefine *link_library* to be the same as the model library.

Feedback

## UIL-33

(error) Invalid technology library state table report.\n \tYou must read in library `%s` from a technology library source file.

**Description**

The report_lib options '-table' and '-full_table' require that the technology library source file is read in by read_lib. These options print out the state table information in the technology library.

**What Next**

Read the technology library source file, then reissue the *report_lib* command with the *-table* or *-full_table* options.

## UIL-34

(information) Updating technology library (please save) ...

**Description**

Every db technology library written out by a version of Library Compiler older than 4.0, is processed again every time that it is read in. To cut out this very significant amount of overhead, after reading in the db technology library, write the library back out and use the updated library.

The variables 'read_db_lib_warnings' and 'read_translate_msff' control options for this processing step. Please refer to their man pages for more details.

**What Next**

Write the technology library back out by using the *write_lib* command.

## UIL-35

(warning) operating conditions '%s' not overwritable in lib '%s'

**Description**

The command issued tries to overwrite an operating condition group that has been declared permanent before. create this operating condition with a different name.

**What Next**

Use *report_lib* to find all available operating conditions for that lib. Choose a name which is not there and create a new one.

Feedback

### Examples

```
Warning: operating conditions 'WCCOM'  not overwritable in lib
 'my_lib' (UIL-35)
```

## UIL-36

(warning) operating condition already defined in lib '%s'

### Description

The command issued tries to overwrite an existing group with the same name.

### What Next

Use -overwrite option. If it is still not overwritable use *report_lib* to find all available operating conditions for that lib. Choose a name which is not there and create a new one.

### Examples

```
operating condition already defined in lib 'mylib' (UIL-36)
```

## UIL-37

(information) The DCM vendor specific delay model is used in this library.\n \tThe library report will not contain timing-related information.

### Description

This message is issued during a *report_lib* command. The *delay_model* attribute in the technology source file is set to "dcm" delay model. The *report_lib* command does not display any timing information.

### Examples

```
Information: The DCM vendor specific delay model is used in this library.
        The library report will not contain timing-related information.
 (UIL-37)
```

## UIL-39

(error) The psuedo techonlogy file %s not found.\n

### Description

This message is issued during a *read_lib -format gdsii* command. The given pusedo library file doesn't exist in the directory.

Feedback

**What Next**

1. Set the search_path variable correctly to point to the pusedo library directory. 2. Check to see if the file exists and has read permissions.

---

## UIL-40

(warning) The library '%s' is not a physical library. Ignore -macro_only option.

**Description**

You receive this warning message because you have executed the *write_lib* command with the *-macro_only* option and specified a library that is not a physical library. The *-macro_only* option with the *write_lib* command applies only to the physical library.

**What Next**

Do not use the *-macro_only* option when writing a non-physical library. Or, ensure that the specified library is in the physical library before executing the *writ_lib -macro_only* argument.

---

## UIL-41

(error) Invalid technology library noise report.\n \tYou must read in the `%s` library from a technology library source file.

**Description**

This error message is issued by *report_lib -noise command. The timing options are used only on a library that has been read from a ".lib" technology library source file by the read_lib command.*

**What Next**

To use the report_lib with the -noise option, read in the technology source library instead of the db file.

---

## UIL-42

(error) %s '%s' is duplicated.

**Description**

This error message is issued by *merge_lib* command.

Feedback

## UIL-43

(error) Library '%s' is not a NLDM library and can't be merged.

**Description**

This error message is issued by the *merge_lib* command.

**What Next**

Only native nldm libraries can be merged.

## UIL-44

(error) The '%s' of object '%s' and of object '%s' are different.

**Description**

This error message is issued by the *merge_lib* command.

**What Next**

Some key properties of the two objects to merge should be the same.

## UIL-45

(error) Can not open the Milkyway reference library '%s'.

**Description**

You receive this warning message because you have executed the *write_lib* command with the *-mw_ref_lib* option and specified a Milkyway reference library. However the Milkyway reference library can not be opened. The generated .db/.pdb file is not written into the Milkyway reference library

**What Next**

Check if the Milkyway reference library specified exists and is not locked by other applications.

## UIL-46

(warning) The operating condition '%s' is not valid. Use the default value of 'min'.

Feedback

### Description

You receive this warning message because you have executed the *write_lib* command with the *-mw_oc_type* option and specified an operating condition that is not valid. The value of -fB-mw_oc_type *can only be 'max', 'min', 'typical' or 'other'.*

---

## UIL-47

(error) The option '-mw_ref_lib' is not specified while using the '-mw_oc_type' option.

### Description

You receive this warning message because you have executed the *write_lib* command with the *-mw_oc_type* option and specified an operating condition without specifying a Milkyway reference library.

### What Next

Specify the '-mw_ref_lib' option along with the option '-mw_oc_type'

---

## UIL-48

(error) the output format is not in db format while trying to write the output into a Milkyway reference library.

### Description

You receive this warning message because you have executed the *write_lib* command with the *-mw_ref_lib* option and specified an output format which is not in db format. Only db format file can be written into an Milkyway reference library by *write_lib*

### What Next

Do not use the *-mw_ref_lib* option when writing a non-db format file.

---

## UIL-49

(error) Invalid technology library user-defined data report.\n \tThe 'report_user_data' library_feature is not specified in the '%s' library.\n \tYou must read in the `%s` library from a technology library source file.

### Description

This error message is issued by *report_lib -user_defined_data command. The user-defined_data options are used only on a library that has been read from a ".lib" technology library source file by the read_lib command.*

Feedback

**What Next**

To use the report_lib with the -user_defined_data option, read in the technology source library instead of the db file.

---

## UIL-50

(error) The '%s' cell does not exist in the '%s' physical library.

**Description**

This error message is issued by *report_lib*. In the *report_lib* case, if you specify a nonexistent cell in the cell list, this error message is displayed. This error usually is caused by a typo.

**What Next**

Provide a correct name with each cell in a cell list. If you are not sure about cell names, do not specify a cell list.

---

## UIL-51

(error) The '%s' library is a symbol library.\n \tIt is not supported by the get_lib_attribute command.

**Description**

This error is issued when the *get_lib_attribute* command is used for a synthetic library. The gerht_lib_attribute command can only be used for technology libraries.

**What Next**

Use *get_attribute* command for synthetic libraries.

---

## UIL-52

(error) The '%s' library is a physical library.\n \tIt is not supported by the get_lib_attribute command.

**Description**

This error is issued when the *get_lib_attribute* command is used for a physical library. The gerht_lib_attribute command can only be used for technology libraries.

**What Next**

Use *get_attribute* command for physical libraries.

Feedback

## UIL-53

(error) Unacceptable list of library objects for the '%s' library.

**Description**

This error message is issued by the *get_lib_attribute* command. You might have specified an empty or a corrupted library object list. This error usually is caused by a typo.

**What Next**

Provide a correct name with each object in a library object list. If you are not sure about library object name, do not specify a library object list.

## UIL-54

(error) The '%s' object does not exist in the '%s' technology library.

**Description**

This error message is issued by *get_lib_attribute*. If you specify a nonexistent object in the library object list, this error message is displayed. This error usually is caused by a typo.

**What Next**

Provide a correct name with each object in a libray object list. If you are not sure about object names, do not specify a object list.

## UIL-55

(error) The '%s' library_feature is not specified in the '%s' library.

**Description**

This error message is issued by *set_lib_attribute*. If you use set_lib_attribute command for a library which does not have 'allow_update_attribute' library feature, this error message is displayed.

**What Next**

Set the 'allow_update_attribute' library_feature in the source library as follows: library(sample) { ... library_features(allow_update_attribute); ... }

## UIL-56

(error) The value '%s' is not valid for the attribute '%s', or the attribute '%s' is not modifiable.

Feedback

**Description**

This error message is issued by *set_lib_attribute*.

**What Next**

Correct the attribute value.

---

## UIL-57

(error) Invalid technology library power report.\n \tYou must read in the `%s` library from a technology library source file.

**Description**

This error message is issued by the *report_lib -power* command when the library is read from a db file that is not enabled for reporting power information. In such cases, the *-power* option is enabled only if that library is read from a ".lib" technology library source file by the *read_lib* command.

A library db file can be enabled for reporting power information by compiling it from a ".lib" file that enables the library feature *report_power_calculation*.

**What Next**

To use *report_lib* with the *-power* option, read in the technology source library instead of the db file.

---

## UIL-58

(error) The format of the -%s option is incorrect.

**Description**

Command "merge_lib" requires that the -baseline option define a pair of baseline library name and calc_mode. Command "merge_lib" requires that the -library option define pairs of library name and calc_mode.

---

## UIL-59

(error) Calc_mode '%s' is not valid.

**Description**

This error message is issued by the *merge_lib* command. 1.If the baseline defines the default_operating_conditions, then the name of that default operating condition can't be used as a calc_mode for other libraries to be merged. 2.If the baseline does not defines the default_operating_conditions, then the name of the baseline library can't be used as

Feedback

a calc_mode for other libraries to be merged. 3. The name of the baseline library plus "_CALC_MODE" can't be used as a calc_mode for other libraries to be merged.

---

## UIL-60

(error) Cannot attach test model to library cell '%s'.

**Description**

This error is generated because a test model in CTL format is successfully read in but cannot be associated with the named cell because it does not have a CTL interface or it does not exist.

**What Next**

Make sure the named design exists and has the same CTL interface.

---

## UIL-61

(warning) Override the test model attached to the library cell '%s'.

**Description**

The library cell already has a test model attached to it but a newly generated test model is going to override the old one and attach to the library cell.

**What Next**

Make sure it is intended to override a previous attached test model to the library cell

---

## UIL-62

(warning) Operating condition '%s' is not found in library '%s'.

**Description**

The command issued tries to access an operating condition that does not exist in the specified library. A typo in the operating condition name can also result in this problem. Any of the following commands might trigger the generation of this warning: *report_operating_conditions, delete_operating_conditions*.

**What Next**

Use *report_operating_conditions -lib* to list all operating conditions. Make sure the operating condition name is among the ones being reported by *report_operating_conditions -lib*.

Feedback

### Examples

```
Warning: Operating condition 'OC1' is not found in library 'a_lib'.
 (UIL-62)
```

## UIL-63

(warning) Operating condition '%s' can't be deleted since it is not created by users using command 'create_operating_conditions'.

### Description

Only those operating conditions created by users using command *create_operating_conditions and not being used by any design can be deleted from a library.*

### Examples

```
Warning: Operating condition 'OC1' can't be deleted since it is not
 created by users using
command 'create_operating_conditions'. (UIL-63)
```

## UIL-64

(warning) Operating condition '%s' can't be deleted since it's still used by design : %s.

### Description

Only those operating conditions created by users using command *create_operating_conditions and not being used by any design can be deleted from a library.*

### Examples

```
Warning: Operating condition 'OC1' can't be deleted since it's still used
 by design : DESIGN1 DESIGN2. (UIL-64)
```

## UIL-65

(error) The '%s' library_feature is not specified in the '%s' library.

### Description

This error message is issued by *report_lib -yield*. If you use 'report_lib -yield' command for a library which does not have 'report_yield' library feature, this error message is displayed.

Feedback

### What Next

Set the 'report_yield' library_feature in the source library as follows: library(sample) { ... library_features(report_yield); ... }

---

## UIL-66

(error) Cannot write the library '%s'. The Composite Current Source information has been stripped while reading the library.

### Description

This library was read into memory when db_load_ccs_data was set to false. Hence the Composite Current Source driver and receiver model from the library has been stripped while reading. To be able to write the library, the variable db_load_ccs_data has to be set to true before reading the library.

### What Next

Set the variable db_load_ccs_data to true in a new dc_shell session, read the library and then perform write_lib.

---

## UIL-67

(error) Could not close the '%s' file.

### Description

The file cannot be closed.

### What Next

Check to see if there are any error messages before this file closing operation.

### Examples

```
Error: Could not close the 'a.lib' file. (UIL-67)
```

---

## UIL-70

(warning) Cannot load the Composite Current Source information since you are not authorized for the license Galaxy-Beta.

### Description

The license Galaxy-Beta is a requirement for using Composite Current Source (CCS) information. This message will be issued when the site is not authorized for Galaxy-Beta license and the user is trying to set the variable db_load_ccs_data to true. The same

Feedback

message will be repeated if he is not authorize for Galaxy-Beta license and he is reading a library with Composite Current Source driver and receiver model when the variable db_load_ccs_data is true.

---

## UIL-71

(error) Cannot write the library '%s'. Writing of the library which has Composite Current Source timing information is not supported inside '%s' in this release. Please use lc_shell.

**Description**

In this release, writing of the library which has CCS timing information is supported only in lc_shell. Hence issue the write_lib command inside lc_shell.

---

## UIL-72

(warning) The Composite Current Source information in the library '%s' is not reported. Reporting of CCS information is not supported inside '%s' in this release.

**Description**

In this release, report_lib -timing reports the CCS timing information only in lc_shell. If you want to view the CCS timing information, please use lc_shell.

---

## UIL-73

(error) Invalid technology library switch cell report.\n \tYou must read in the `%s` library from a technology library source file.

**Description**

This error message is issued by *report_lib -switch* command. The switch option is used only on a library that has been read from a ".lib" technology library source file by the *read_lib* command.

**What Next**

To use the report_lib with the -switch option, read in the technology source library instead of the db file.

---

## UIL-74

(error) Invalid technology library pg pin report.\n \tYou must read in the `%s` library from a technology library source file.

Feedback

### Description

This error message is issued by *report_lib -pg_pin* command. The pg_pin option is used only on a library that has been read from a ".lib" technology library source file by the *read_lib* command.

### What Next

To use the report_lib with the -pg_pin option, read in the technology source library instead of the db file.

---

## UIL-75

(warning) The directory '%s' exists and the files with same name under it will be updated.

### Description

This message is given if the directory exists. If so, the files with same name under it will be updated.

### What Next

If you want to keep the old files under this directory, just rename the output directory name by option *datasheet_output_dir* for Datasheet generator or *veriloglib_output_dir* for VERILOG generator.

### Examples

```
datasheet_output_dir=test_datasheet
write_lib -format datasheet  dbvh13

OR

veriloglib_output_dir=test_verilog
write_lib -format verilog dbvh13
```

### Examples

```
Warning: The directory already exists and the files with same name under
 it will be updated. (UIL-75)
```

---

## UIL-76

(error) No/Invalid option value specified for '%s'.

Feedback

### Description

In calculate_caa_based_yield2db command, some options have dependency and requirement:

(1) Operating target should be a valid library name (already loaded in Library Compiler), or a valid db file (that can be read into LC successfully). (2) If the operating target is a db file, not a library name, The -output option with a valid db file name should be specified. And the output db file name should be different from input db file. (3) Currently, for '-data_kit_type' option, only 'tsmc' and 'tsmc_encr' are valid values. (4) When there are more than one routing layers defined in the library, The '-layer_dsd_alias' option must be provided. If there is only one routing layer in library, use default {0 0 0 0 0} as its value. (5) "-layer_dsd_alias" option value must contains 5 integer values, for the value of alias layers: Mx, My, Mz, Mr, Mt by order.

### What Next

Check the options to this command, specify all necessary and valid options.

### Examples

```
calculate_caa_based_yield2db -data_kit_type tsmc \
 -particle_distr_func_file tsmc.ddk a.db -output a.db

OR
calculate_caa_based_yield2db -data_kit_type tsmc \
 -particle_distr_func_file tsmc.ddk -layer_dsd_alias {0 0} test

OR
calculate_caa_based_yield2db -data_kit_type tsmc \
 -particle_distr_func_file tsmc.ddk nonexist.db
```

### Examples

```
Error: No/Invalid option value specified for '-output'.(UIL-76)
Error: No/Invalid option value specified for '-layer_dsd_alias'.(UIL-76)
Error: No/Invalid option value specified for 'nonexist.db'.(UIL-76)
```

---

## UIL-77

(error) The directory '%s' doesn't exist.

### Description

This message is given if the directory doesn't exist. If so, the custom function file directory is wrong.

### What Next

If you want to define the directory for custom function files, define the correct directory name by option *veriloglib_custom_func_dir* for VERILOG generator.

Feedback

### Examples

```
veriloglib_custom_func_dir=test_dir
write_lib -format verilog dbvh13
```

### Examples

```
Error: The directory 'test_dir' doesn't exit. (UIL-77)
```

---

## UIL-78

(information) The mapping file "func.map" for custom udp flow doesn't exist.

### Description

This message is given if the custom function directory is defined, but "func.map" file doesn't exit in the directory.

In custom UDP flow, verilog generator will check for mapping file "func.map" in the directory first. If the file doesn't exit or there is no mapping information for the cell, verilog generator will search for file "[cell_name].func" as the single cell function part for the cell. If this file doesn't exit, verilog generator will search for file "[library_name].func", and search the cell's function part in this packet file. If this can't be found either, verilog generator will not use custom UDP flow and comes to the general flow.

### What Next

Users can ignore this information if they have provided [cell_name].func, or [library_name].func for custom UDP flow, or they haven't provided any kind of function files by don't want to use custom UDP flow. If users want to use mapping file approach in custom UDP flow, you must provide the "func.map" file as in the description part.

---

## UIL-79

(warning) Syntax error of mapping file "%s".

### Description

This message is given if the mapping file has some syntax error.

Mapping file must be defined by the following syntax. Each line contains the mapping information of a specific file to multiple cells. Cell name is separate by blank. The bracket is needed here. No comment is applied here.

<file name> [<cell name1> <cell name2>]

Feedback

### What Next

If user want to use mapping file approach in verilog and datasheet , user need to edit the mapping file and correct the syntax error.

### Examples

```
Foo2.func [FDQ1X1 FDQ1X2]
```

---

## UIL-80

(error) The %s format isn't supported by datasheet generator.

### Description

This error is given if the format is not "text" and "html".

### What Next

If user want to generate html format datasheet, set datasheet_enable to "html"; if user want to generate text format datasheet, set datasheet_enable to "text".

### Examples

```
html format:
set datasheet_enable html

text format:
set datasheet_enable text
```

---

## UIL-81

(error) The directory [%s] doesn't exist.

### Description

This error is given if the directory doesn't exist or there is a typo in the directory name.

### What Next

check whether the directory exist or type the correct directory name.

### Examples

```
Error: The directory [/slowfs/xxx] doesn't exist. (UIL-81)
```

---

## UIL-82

(error) The file [%s] can not be found

Feedback

### Description

This error is given if the file does not exist.

### What Next

check whether the file exist or type the correct file name.

### Examples

```
Error: The file [/slowfs/xxx/yunz.txt] can't be found. (UIL-82)
```

---

## UIL-83

(error) Can't merge power db '%s' to timing db '%s'.

### Description

This error is given because the two db don't pass merge availability checking.
The checking includes, (1) Library-level attributes "time_unit", "voltage_unit", "capacitive_load_units" must be equal individually in the two libraries power_db and timing_db. (2) The two libraries power_db and timing_db should have same-number and same-name designs. (3) Every two same-name cells from power_db and timing_db should have same-number and same-name pins. (4) Every two same-name cells from power_db and timing_db should have same-number and same-name pg_pins.

### What Next

Find another proper power.db to merge.

### Examples

```
Error: Can't merge power db 'power.db' to timing db 'timing.db'. (UIL-83)
```

---

## UIL-84

(warning) Attribute '%s' cannot be used in group %s, it will be ignored.

### Description

This warning indicates that the attribute name specified in the -by option of sort_objects command cannot be used to sort the supplied groups. It will be ignored.

### What Next

Remove the invalid attribute and run again.

### Examples

```
Warning: Attribute 'wire_load' cannot be used in group cell, it will be
 ignored. (UIL-84)
```

Feedback

## UIL-85

(warning) Sort objects failed, either no valid attribute name nor the object has no name.

### Description

This warning indicates that the sort_objects failed to sort the supplied objects, it could be: (1) If user supplied attribute(s) name with -by option, there's no valid attribute can be used to sort the group or (2) If user does not specify -by option, name of the objects will be used to sort, but the object has no name, e.g. timing group.

### What Next

Specify correct attribute to be used to sort the objects.

### Examples

```
Warning: Sort objects failed, either no valid attribute name nor the
 object has no name. (UIL-85)
```

## UIL-90

(warning) write_lib in %s could result in library data inaccuracy. Please use lc_shell to perform library operation tasks instead. Please consult with Synopsys support if write_lib flow is needed in %s.

### Description

It is recomended to use lc_shell to perform library operation.

### What Next

Use write_lib in lc_shell or consult Synopsys support if write_lib is needed in dc_shell or de_shell.

## UIL-91

(error) write_lib in %s is not enabled. Please start new session and do enable_write_lib_mode to enable it, or use lc_shell instead.

### Description

By default, write_lib is not enabled. Use enable_write_lib_mode at the beginning of tool session to enable it. It is recommended to use lc_shell to perform library operation. Please refer Solvnet article 037767 on restriction of write_lib usage in Design Compiler and IC Compiler. Also, Solvnet article 030453 about on-the-fly PG library update in ICC 2010.03 updated training.

Feedback

### What Next

Use enable_write_lib_mode or use write_lib in lc_shell or consult Synopsys support if write_lib is needed in icc_shell or dc_shell or de_shell.

---

## UIL-92

(error) write_lib mode cannot be enabled since a library is found to be already existing in memory.

### Description

Unable to enable write_lib mode for the tool. Since,a library is found to be already existing in memory. This message informs that enable_write_lib_mode command failed.

### What Next

To enable write_lib mode, make sure no library is loaded.

---

## UIL-93

(warning) This %s session is for library operations only. You should not perform optimization on the design. Please exit when you are finished with the library operation.

### Description

It is recommended to use lc_shell to perform library operation. Please refer Solvnet article 037767 on restriction of write_lib usage in Design Compiler and IC Compiler. Also, Solvnet article 030453 about on-the-fly PG library update in ICC 2010.03 updated training.

---

## UIL-94

(Error) Read the JCR file failed.

### Description

The message is given if read the JCR file failed. The reasons are list as below: 1) JCR file is not found 2) JCR information is not found in file 3) Invalid data in the file a.run status cannot be negative b.start time cannot be earlier than the end time c.number of files cannot be negative d.warning as error is a boolean value - it can be eitehr 0 or 1 e.number of threads cannot be negative or very large number f.number of cores cannot be negative

### What Next

If the error occurs, user need to update the JCR file.

Feedback

### Examples

```
Error: Read the JCR file failed. (UIL-94)
```

---

## UIL-95

(Error) Store the JCR failed.

### Description

The message is given if store the JCR file failed.

### What Next

If the error occurs, user need to update the JCR file.

### Examples

```
Error: Store the JCR file failed. (UIL-95)
```

---

## UIL-96

(Error) No jcr info can be reported.

### Description

The message is given if the option '-jcr' is attach to the command "report_lib", but no jcr info is stored in the report db.

### What Next

If the error occurs, user need to delete the option 'jcr' or re-generate a .db with jcr info.

### Examples

```
Error: No jcr info can be reported. (UIL-96)
```

---

## UIL-99

(information) Message '%s' will be changed from Warning to Error starting from '%s' release.

### Description

This message is issued at the end of a *read_lib* command. The reported "Warning" ID will be upgraded to an "Error" in the furture release. User should fix their library in this period of time, otherwise the library won't be complied.

Feedback

### What Next

Search this message ID in the screener log file, and fix all these "Warnings" in this library.

### Examples

```
Information: Message 'LBDB-993' will be changed from Warning to Error
 starting from '2016.12' release. (UIL-99)
```

---

## UIL-100

(error) The command is not enabled, it's support in advanced tcl mode only.

### Description

This command is supported in Library Compiler advanced tcl mode only.

### What Next

Check with Synopsys Library Compiler support team, to know how to enable the advanced tcl mode.

---

## UIL-101

(Error) %s objects from '%s' were of the %scorrect class.

### Description

Some commands do not allow heterogeneous collections as argument. The collection contained some objects that were of a class which is not accepted by the command. The message will indicate whether some objects or no objects were accepted by the command.

### What Next

Please make sure the command was applied to the objects which were expected.

---

## UIL-102

(warning) No libraries at all in memory.

### Description

The message is given if Library Compiler have not read in a library successfully at all.

### What Next

If the warning occurs, user need to read in a valid library.

Feedback

### Examples

```
Warning: No libraris at all in memory. (UIL-102)
```

### See Also

- read_lib
- read_db

---

## UIL-103

(error) Can't find %s '%s'.

### Description

The object of the specified type and name could not be found.

### What Next

Check if Librariy Compiler has read in the specified objects.

### Examples

```
Error: Can't find library 'z0'. (UIL-103)
```

### See Also

- read_lib
- read_db

---

## UIL-104

(error) Cannot get %s '%s' in library '%s' due to lack of required library_features value.

### Description

The object/attribute of the specified type and name could not be accessed, because the library_features in this library has no required value.

library_features lets other Synopsys products use the command features that you specify as attribute values. Related values of library_features are: report_delay_calculation, report_power_calculation, report_noise_calculation, report_user_data

Missing any of the value in library_features will make those related objects/attributes inaccessible.

It has no impact if the library is read from ASCII file.

Feedback

### What Next

Add required value to library_features in the original liberty file, or consult library provider.

### Examples

```
Error: Cannot get object 'cell_rise' in library test due to lack of
 required library_features value. (UIL-104)
```

### See Also

- read_lib
- read_db

---

## UIL-110

(error) Can't find %s '%s'.

### Description

The object of the specified type could not be found.

### What Next

Check the man page for the current command.

---

## UIL-111

(error) Attribute '%s' can't be %s on this %s with this command.

### Description

This error message is issued by *set_attribute*.

---

## UIL-112

(error) The '%s' attribute of a %s cannot be set with this command.

### Description

This error message is issued by *set_attribute*.

---

## UIL-113

(error) '%s' doesn't specify a unique library\n \tPlease use complete specification: full_file_name:library_name

Feedback

**Description**

This error is generated when there are more than one library who have the same library_name in memory.

---

## UIL-114

(error) Cannot read file '%s'.

**Description**

This error is generated when you try to read a file which is protected or does not exist.

**What Next**

Check and correct search_path, then reinvoke the read command.

---

## UIL-115

(Error) For pattern '%s', %s.

**Description**

A problem has occured while trying to compile the regular expression pattern shown.

**What Next**

Review the description in the man page for regexp to identify and correct the problem. Make sure that you intended the pattern to be a regular expression instead of a simple string match. A common mistake is to mix up the use of the wildcard '*' in string match with the corresponding regular expression '.*'. Also, remember that in regular expressions, the square brackets '[', ']' by default enclose a range of characters, and must be escaped with a backslash to be treated as regular characters.

---

## UIL-116

(error) In advanced tcl mode, the input library objects can only\n \tbe specified by collections. It doesn't accept library object name(string) as input.

**Description**

Only collection type data is allowed under advanced tcl mode only.

**What Next**

Check with Synopsys Library Compiler support team, to know how to enable the advanced tcl mode.

Feedback

## UIL-117

(Error) . The specified %s is not a user defined %s.

**Description**

The attribute name(object class type) is not user defined.

**What Next**

Remove the -user_defined_data option or change the attribute name(object class type).

## UIL-118

(warning) Process label '%s' from command line overwrites value '%s' from Liberty source

**Description**

In the library Compiler, you used *read_db* or *read_lib* to read a logical library source file. The source file already had a process label specified on the default operating conditions, and the value you specified on the command line differs from the source file.

**What Next**

Ensure that the process label you are using is correct.

## UIL-119

(error) Failed to update process_label for library %s!

**Description**

This error indicates that the set_process_label failed to set process_lable attribute on default operating_conditions group, it could be:

(1) The library is not read in successfully. or (2) There is no default operating_conditions group.

**What Next**

Please make sure library is loaded successfully and source file is correct

## UIL-200

(warning) The following cells do not have FRAM view:\n { %s}

Feedback

### Description

This message is for missing frame cells in physical library when write_physical_lib. In this case, if the library contains partial frames, for example: 10 cells have design view, but only 6 have frame, the library can still be written out.

### What Next

Use create_frame to create FRAM view for the cells which do not have FRAM view, and then rerun write_physical_lib.

### See Also

- create_frame

---

## UIL-201

(error) None of the cells have FRAM view!

### Description

This message is for missing all frame cells in physical library when write_physical_lib. In this case, write_physical_lib will error out and no disk file will be written.

### What Next

Use create_frame to create FRAM view for all the cells which do not have FRAM, and then rerun write_physical_lib.

### See Also

- create_frame

---

## UIL-210

(Error) The fusion_lib opened, please use %s to %s library.

### Description

This message indicates if there is fusion_lib library being opened in lc_shell. Before closing the currently opened fusion_lib library, user can only edit the library using fusion_lib comands.

### What Next

Please close the currently opened fusion_lib library, or use fusion_lib commands to edit the currently opened fusion_lib library.

Feedback

## UIL-211

(Warning) The %s set on command %s can only affect DB loaded after the command.

**Description**

If user uses this command set_fusion_lib_process_label when any fusion library is opened in memory, this warning message will be issued out that only the db loaded after this command will be affected.

**What Next**

Please use command set_fusion_lib_process_label before open_fusion_lib/ create_fusion_lib/compile_fusion_lib.

## UIL-212

(Error) Command '%s' can not be used in Physical Library mode.

**Description**

This message indicates if there is physical library being opened in lc_shell. Before closing the currently opened physical library, user can not use the command. for example, read_db cannot be used during create_physical_lib/open_physical_lib and close_physical_lib.

**What Next**

Please close the currently opened physical library.

## UIL-213

(Error) Command '%s' can not be used outside fusion library session!

**Description**

This message indicates if there is no fusion_lib library being opened in lc_shell. Customer can only use the command after open a fusion_lib library.

**What Next**

Please open a fusion_lib library.

## UIL-214

(Error) The option '%s' set on command '%s' can only work with app option '%s 'set to '%s'.

Feedback

### Description

This message indicates the command is used outside of fusion library virtual frame flow. Customer need to set the required app option to true to enable the feature.

### What Next

Please add "set_app_options -name fusion_lib.create_missing_macro_frame -value true" before the command.

---

## UIL-215

(Warning) There is logical library in memory, please use "remove_lib -all" to remove all.

### Description

This message indicates there is logical library opened when entering fusion lib session by using command like create_fusion_lib, compile_fusion_lib, open_fusion_lib. This will not be allowed in the future.

### What Next

Please use "remove_lib -all" to remove all logical libraries before create/open a fusion library.

---

## UIL-550

(information) The "%s" variable(%.5g) will be used in LBDB-550 check

### Description

This message indicating one of the 4 global float global variables was defined: - lc_lvf_mean_shift_threshold#if abs(mean_shift)<lc_lvf_mean_shift_threshold, then any mean_shift related checks will be skipped - lc_lvf_skewness_threshold# if abs(skewness)<lc_lvf_skewness_threshold, then any skewness related checks will be skipped - lc_lvf_sigma_threshold#if abs(sigma_early-sigma_late)<lc_lvf_sigma_threshold, then any sigma related checks will be skipped - lc_lvf_normalized_skewness_threshold# if abs(skewness/std_dev)^3)<lc_lvf_normalized_skewness_threshold, then any skewness related checks will be skipped

Effect on 3 rules: If either lc_lvf_mean_shift_threshold or lc_lvf_skewness_threshold or lc_lvf_normalized_skewness_threshold defined and satisfied, skip Rule 1 check (Rule 1) mean_shift * skewness < 0 If either lc_lvf_skewness_threshold or lc_lvf_normalized_skewness_threshold or lc_lvf_sigma_threshold defined and satisfied, skip Rule 2 check (Rule 2) skewness * (sigma_late – simga_early) < 0 If lc_lvf_sigma_threshold defined and satisfied, skip Rule 3 check (Rule 3) (std_dev – sigma_early) * (std_dev – sigma_late) > 0

Feedback

**What Next**

No action required.

---

## UIL-551

(Error) Option '%s' cannot be empty.

**Description**

Value specified for the option is empty. Non empty value is required.

**What Next**

Specify non-empty value.

---

## UIL-552

(Error) Options '%s' and '%s' cannot have same value.

**Description**

Values specified for the options are same. Different values are required.

**What Next**

Specify different values for the two options.

---

## UIL-553

(Error) Failed to rename rail: %s.

**Description**

Indicates rename rail failed with given reason.

---

# UIM

---

## UIM-001

(warning) Filter expression has %d characters, which exceeds the maximum of %d characters.

Feedback

**Description**

The command is not accepted because the filter expression exceeds the maximum length supported by the tool. Very long filter expressions can also cause long runtime for the command.

**What Next**

Break the command into multiple separate commands, using a smaller filter for the get_ command, and multiple *filter_collection* commands to obtain the final desired collection of objects.

---

## UIM-002

(error) Invalid format string '%s'.

**Description**

The format string does not match the available values. For example, using "%s" as the format for a floating-point value.

**What Next**

Specify a format string appropriate for this command.

---

## UIM-003

(Error) Found %d objects but expected %d.

**Description**

A collection command was entered with the *-expect* option. The number of matching objects is not equal to the expected number.

**What Next**

Use the collection command without *-expect* to determine which objects did match.

---

## UIM-004

(Error) Found %d objects but expected at least %d.

**Description**

A collection command was entered with the *-expect_at_least* option. The number of matching objects is less than the minimum expected number.

Feedback

**What Next**

Use the collection command without *-expect_at_least* to determine which objects did match.

---

## UIM-005

(warning) Application option '%s' is not supported

**Description**

This application option is not yet supported in this tool. Setting this option will have no effect.

**What Next**

Determine if there is another way to achieve the same effect.

---

## UIM-006

(error) Application option '%s' cannot be specified on a design

**Description**

This application option is only supported for global scope. Not all application options can be specified on a design.

**What Next**

Use "set_app_options" without "-design" option when modifying this option to refer to the global scope. Note that global app options will affect all designs and are not persistent.

---

## UIM-007

(error) Cannot modify read-only application option '%s'.

**Description**

This application option is read-only. You cannot change its value.

**What Next**

Some application options can be set only in the .synopsys_icc2.setup file, and become read-only after the file has been read. Determine if that is the case for this option.

---

## UIM-008

(error) Invalid column specification '%s':\n %s

Feedback

**Description**

The column specification for a tabular report is not valid. It must be one of the following: a valid column name such as "area", a valid column name and width such as "area:10", or a valid column name, width and number of significant digits such as "area:10.4".

The specified column name must be one of the valid choices for this report. The optional column width must be between 1 and a maximum value (currently 10000). The significant digits must be between 1 and 13, and must be less than the column width.

**What Next**

Enter a valid column specification.

---

## UIM-009

(error) Application option '%s' must be specified with a %s unit.

**Description**

This application option is a typed value and must be specified with a unit. It is not allowed to specify its value without a unit.

**What Next**

Specify a correct unit with the option value. For example, "20ps" for time values, "0.1um" for lengths, etc.

---

## UIM-010

(error) '%s' is not a valid report name for '%s'.\n %s

**Description**

The command is used to configure certain reports. The report name given is not one of the reports that is configurable by the specified command.

**What Next**

Enter a valid report name.

---

## UIM-011

(error) Command '%s' is not available in this shell.

**Description**

The command is not available in the shell being used. The command will be ignored.

Feedback

**What Next**

Check the command name and the shell you are using.

---

## UIM-012

(error) Command '%s' option '%s' is not available in this shell.

**Description**

The command option is not available in the shell being used. The command will be ignored.

**What Next**

Check the command option name and the shell you are using.

---

## UIM-013

(error) Container Technology doesnot support 'rsh'.

**Description**

The specified submit command(-submit_command) or submit protocol(-submit_protocol) used in the command is not a supported type for the container Technology. Currently 'rsh' is not supported in container technology.

**What Next**

Set a valid submit command or submit protocol for Container Technology.

---

## UIM-014

(warning) Container Technology doesnot support 'rsh', 'ssh' will be used intead.

**Description**

The specified submit command(-submit_command) or submit protocol(-submit_protocol) used in the command is not a supported type for the container Technology. Currently 'rsh' is not supported in container technology. Instead 'ssh' will be used.

**What Next**

Set a valid submit command or submit protocol for Container Technology.Otherwise, 'ssh' will be used.

Feedback

### UIM-015

(warning) Named host options are not available in this shell.

**Description**

The *-name* argument of the *set_host_options* command is not available in the shell being used. The argument will be ignored.

**What Next**

Check the shell you are using.

## UIT

### UIT-2

(warning) Unrecognized compaction mode ignored.

**Description**

You receive this message because you have entered an incorrect value for the *-compaction_effort* option. Valid values for this option are *low*, *medium*, or *high*. Or, you can use the *-no_compaction* option.

The *create_test_patterns* command will continue to run with a default value of *low*.

**What Next**

If you are willing to accept the results of low effort compaction, no action is required on your part. However, using low effort compaction typically results in a higher vector count than is a result of using medium or high effort compaction.

If you want to use *-compaction_effort medium* or *-compaction_effort high*, rerun the *create_test_patterns* command with the appropriate option.

### UIT-3

(error) Unrecognized test program format or missing license key.

**Description**

You receive this message if *write_test* does not recognize the format you specified with *write_test -format*, or if you do not have a required license key. *write_test* recognizes only formats that are contained in the variable *write_test_formats*; by default, *write_test_formats* contains the formats *synopsys*, *tds*, *wgl*, *verilog*, and *vhdl*. If you want

Feedback

to use a different format (for example, tdl91), you must enter it into *write_test_formats* so that *write_test* will recognize it. In addition, the tdl91 format requires a CTV key.

**What Next**

If you receive this message, do the following:

1. Within *dc_shell*, list the *write_test_formats* variable to see whether it contains the format you want to use.

```
dc_shell> list write_test_formats
    write_test_formats = {"synopsys", "tssi_ascii", "tds", "verilog",
 "vhdl", "wgl"}
```

1. If the required format (for example, tdl91) is not contained in the variable *write_test_formats*, add it, as follows.

```
dc_shell> write_test_formats = write_test_formats + tdl91
```

1. Ensure that you have the correct key (in this case, the CTV key).
2. Re-issue the *write_test* command and specify the required format.

```
dc_shell> write_test -format tdl91
```

---

## UIT-4

(warning) You have asked for scan replacement, but attributes show that the design already has scan circuitry. Your request is ignored.

**Description**

This message tells you that you have asked DFT Compiler to replace non-scan cells with scan cells. Design attributes tell DFT Compiler that this process has already occurred. Your request is ignored.

**What Next**

To eliminate the warning, change command options so that you do not ask scan insertion to do scan replacement.

---

## UIT-5

(warning) No existing test circuitry to route, scan will be inserted.

**Description**

You receive this message if you run *insert_dft* on a design with scan replacement disabled and scan routing enabled, but where attributes on the design show that the design is not

Feedback

scan replaced. For example, you receive this message if you enter the following sequence of commands:

```
read_file -format verilog rtl.v
compile
set_scan_configuration -replace false -route true
insert_dft
```

The message informs you that *insert_dft* is overriding your specification to not scan replace the design since *insert_dft* cannot route a design that is not scan replaced.

You receive this message if you read in a design from an ASCII format (Verilog or VHDL) that you have compiled with test-ready compile (*compile -scan*) and have run *insert_dft* with scan replacement disabled and scan routing enabled. Following is another example of a command sequence that will result in the return of this error message:

```
read rtl.v -format verilog
current_design = top
compile -scan
write -hierarchy -format verilog -out design_tr.v
remove_design -designs
read -format verilog design_tr.v
current_design top
set_scan_configuration -replace false -route true
insert_dft
```

After invoking *insert_dft*, you receive the following message:

```
Warning:  No existing test circuitry to route, scan will be inserted.
Loading desing 'top'
Using test design rule information from previous dft_drc run
Architecting Scan Chains
Inserting Scan Cells
Routing Global Signals
Mapping New Logic
Transferring design 'top' to database 'top.db'
```

This is an invalid flow. You should only save and read the test-ready designs from .ddc format. The scan configuration is correct for a test-ready design that you have saved in .ddc format because the .ddc format includes attributes to indicate to the *insert_dft* command that the design is scan replaced. However, these attributes are missing from the ASCII format.

**What Next**

Correct your script to avoid running the *insert_dft* command with replacement disabled but routing enabled on a design that does not contain the attributes to indicate that the design is scan replaced.

If you intended to perform scan replacement on the design, no action is required on your part and you can use the resulting design.

Feedback

If your design was already scan replaced but did not have the necessary attributes, the *insert_dft* command has built valid scan chains (verify this with *dft_drc*). However, this is not an optimum flow for your design and your design may not be optimum for area and timing. If your design does not meet your area and timing constraints, correct your compile scripts to save the test-ready design in .ddc format and rerun both compilation and scan insertion.

---

## UIT-6

(error) Cannot load test vector database file.

### Description

The test vector database file cannot be loaded, because of one or more of these errors:

```
wrong search path for the file
        nonexistent test vector database file
        misspelled test vector database filename
        the file is write-protected
        the file has wrong permissions set
        the file contains no vectors
```

### What Next

Make sure that the file exists, has the correct search path, correct spelling, appropriate permissions, and contains test vectors.

---

## UIT-8

(warning) Design has no scan path. Generated vectors will not be saved.

### Description

You receive this message when attributes indicate that a design has no scan chain and the scan style is not set to combinational.

Under these circumstances, DFT Compiler performs a "what if" analysis to predict the coverage if you were to insert scan chains. Without scan chains, the vectors that are generated are meaningless and are not saved.

### What Next

If you wish only to check the coverage that DFT Compiler can achieve and do not wish to save the vectors, no action is required on your part. However, you can check what the achievable coverage is more quickly by using the *-dft* option of *create_test_patterns*.

If your design has an existing scan path, establish attributes that indicate to DFT Compiler that the scan chain exists. To do this, annotate test ports using *set_dft_signal*. Then declare the existing scan using the command *set_scan_configuration -existing_scan true*

Feedback

and run *dft_drc*. For example, for a circuit with an existing multiplexed flip-flop style scan chain and test ports TEST_SI_PORT, TEST_SE_PORT, and TEST_SO_PORT, do the following:

1. Issue the command *set_dft_signal -type ScanDataIn -port TEST_SI_PORT*.
2. Issue the command *set_dft_signal -type ScanEnable -port TEST_SE_PORT*.
3. Issue the command *set_dft_signal -type ScanDataOut -port TEST_SO_PORT*.
4. Issue the command *set_scan_configuration -style multiplexed_flip_flop*.
5. Issue the command *set_scan_onfiguration -existing_scan true*.
6. Run *dft_drc*.
7. Run *create_test_patterns*.

If you have a combinational design and wish to save the ATPG vectors, you must set the combinational scan style using the *set_scan_configuration -style combinational* command.

---

## UIT-9

(error) Test clock period of %.2f on port '%s' is inconsistent with period of %.2f specified in the test protocol.

**Description**

The test clock period on the port is inconsistent with the period you have specified in the test protocol.

**What Next**

Check that your test clock period specification is consistent with your test protocol.

---

## UIT-10

(error) internal failure, write_test vector formatter (stran) failed.

**Description**

Attempt to format vector failed. This may be caused by improperly installed formatter executable, bin_path which points to the formatter is improperly specified, corrupted formatter executable, formatter executable has been tampered with, not enough disc space to write vectors, formatter run time execution error etc.

**What Next**

Make sure that the formatter executable is properly installed, that your bin_path is properly set, and that there is sufficient disc space to generate vectors.

Feedback

## UIT-11

(error) internal failure, write_test vector formatter (stran) failed. (%d).

**Description**

Vector formatting process was abruptly terminated. Possible causes are: 1. Not enough disc space to continue writing vectors. 2. Formatter executable was accidentally deleted by another process. 3. Does not have write permission to either write to the directory, or to overwrite an existing vector file. 4. Temporary network problems.

**What Next**

Make sure that sufficient disc space is available when generating vectors, all the right permissions are set, then reinvoke write_test.

## UIT-13

(Information) The variable atpg_test_asynchronous_pins is obsolete. It has no effect.

**Description**

In release 1999.10, the variable atpg_test_asynchronous_pins was made obsolete. The create_test_pattern command now always generates tests for faults on asynchronous pins.

**What Next**

No action required.

## UIT-14

(error) internal error, key error for write_test formatter (stran).

**Description**

Intermediate (SIF) file used as input to the formatter executable contained illegal key. The intermediate file may be corrupted or have been manually tampered with.

**What Next**

Reinvoke write_test command

## UIT-15

(error) internal error, write_test formatter (stran) can not generate '%s' format.

Feedback

**Description**

Unsupported test vector format indicated in the intermediate (SIF) input file for the write_test formatter executable. The file may be corrupted or may have been manually tampered with.

**What Next**

Reinvoke write_test command

---

## UIT-16

(error) No scan chains extracted for design - patterns cannot be formatted.

**Description**

Formatting scan test patterns into a set of manufacturing test vectors requires information about the scan chains within the design. This information includes the cells in each chain, the order in which they are connected, and scan chain inversions.

*DFT Compiler* automatically extracts scan chain information from a design when the *dft_drc* or *create_test_patterns* commands are executed. Execution of these two commands stores a copy of the design in *db* format in memory. If the design is saved in *db* format, scan chain information is saved with the design.

If the design is not saved in *db* format, the scan chain information is not available and the *write_test* command in unable to format the patterns. Typically this happens when a design is read in as a netlist and an existing pattern database file is used as input to the *write_test* command.

**What Next**

Run the *dft_drc* command to automatically extract the details of the design scan chains; use the *report_test -scan* command to verify that this information has been extracted. The extracted scan chain information is used by the *write_test* command to format test patterns.

---

## UIT-19

(error) No test protocol exists for design '%s'.

**Description**

A test protocol is required for the successful completion of the current command. But no test protocol was found for the specified design.

**What Next**

Install a test protocol.

Feedback

Test protocols can be user-specified or automatically inferred. To specify a test protocol, use the commands read_test_protocol or read_init_protocol. To make DFT Compiler infer a test protocol, use the command dft_drc.

---

## UIT-20

(error) Cannot open output test protocol file '%s'.

**Description**

You receive this message if you execute the *write_test_protocol* command and did not succeed.

**What Next**

Check if the destination directory has write permission.

---

## UIT-21

(warning) Unrecognized backtrack effort '%s' ignored.

**Description**

You receive this message when you have entered an incorrect value for the *-backtrack_effort* option of *create_test_patterns*. Valid values for this option are *low*, *medium*, and *high*.

The *create_test_patterns* command will continue to run with a default value of *low*.

**What Next**

If you are willing to accept the results of low backtrack effort, no action is required on your part. However, some circuits running *create_test_patterns* with a low backtrack effort may encounter aborted faults when *create_test_patterns* is unable to generate patterns without exceeding the backtrack limit.

If the results of *create_test_patterns* do not meet your fault coverage requirements and there are many aborted faults, re-execute *create_test_patterns* with a backtrack effort of *medium* or *high*.

---

## UIT-22

(warning) Only %d%% of faults were considered in ATPG run.

**Description**

You receive this message if you use *restore_test* or *write_test* on a vdb file generated by *create_test_patterns -sample*.

Feedback

This message reminds you that the vectors were generated only for a sample of faults in the design.

**What Next**

If you wish to generate vectors for all faults, you need to either regenerate the vectors from scratch using *create_test_patterns*, or use the same vdb file again as a starting point for ATPG by using the *-input* option of *create_test_patterns*.

---

## UIT-23

(error) No test protocol defined for design - patterns cannot be formatted.

**Description**

Formatting scan test patterns into a set of manufacturing test vectors is controlled by the test protocol defined for a design. The protocol specifies the steps of the scan test sequence (scan-in, parallel measure/capture, and scan-out).

The *dft_drc* and *create_test_patterns* commands automatically infers a default protocol for a design. If the design is stored in *db* format, this protocol is saved with the design. Custom protocols may be necessary for designs that require preconditioning vectors or designs that employ nonstandard scan test sequences. A custom protocol can be defined for a design and specified using the *read_test_protocol* command.

In this case, this information is not available, and the *write_test* command in unable to format the patterns. Typically this happens when a design is read in as a netlist and an existing pattern database file is used as input to the *write_test* command.

**What Next**

Run the *dft_drc* command to automatically infer a default protocol for the design. If a custom protocol is read in using the *read_test_protocol* command, execute *dft_drc* to extract detail's of the design's scan chains.

*Note:* If you create a protocol for a design, make sure that it is the same protocol used when existing pattern database files were created. If the protocols are different, the *write_test* command fails because the scan test patterns are inconsistent with the protocol. An example of an inconsistency is the use of different scan cells. Ideally, the design and its protocol should always be saved in *db* format after execution of the *create_test_patterns* command.

---

## UIT-24

(warning) vhdlout_single_bit="%s" not supported; assuming vhdlout_single_bit="VECTOR".

Feedback

**Description**

The *write_test* command issues this warning if you are writing out your vectors in VHDL format and the *vhdlout_single_bit* variable is not set to *VECTOR* or *BIT*. The warning message indicates which value the*vhdlout_single_bit* variable is set to.

The *vhdlout_single_bit* variable defines whether bussed ports are written as individual single bits in the test pattern. The *write_test* command only supports VECTOR or BIT values for this variable. However, the default value for this variable is *USER*. If you use the value *USER*, the *write_test* commmand will issue this message and automatically change the *vhdlout_single_bit* variable to *VECTOR*. Your vectors will be written out with the ports bussed.

**What Next**

To avoid this violation, set the *vhdlout_single_bit* variable for *VECTOR* or *BIT* before issuing the *write_test* command.

If you have used the value *USER* and wish to write out your vectors in bit-blasted fashion, you will need to set the *vhdlout_single_bit* variable to *BIT* and re-execute the *write_test* command.

If you have used the value *USER* and wish to write out your vectors in bussed fashion, then no action is required on your part.

---

## UIT-26

(error) Strobe value is required to be less than period value.

**Description**

Your strobe value has exceeded the period value.

**What Next**

Change your strobe value.

---

## UIT-27

(error) %s is a serial-only vector format.

**Description**

Invalid vectors are generated if you invoke *write_test* with the *-parallel* option for serial-only vector formats. When such an error occurs, this error message displays.

**What Next**

for serial-only vector formats, invoke *write_test* without the *-parallel* option.

Feedback

## UIT-28

(error) Current test protocol cannot be used with a parallel load format.

### Description

Currently the only protocols inferred by dft_drc can be used with a parallel load format. Therefore any custom test protocols cannot use this format due to the fact that it is difficult to determine which bits correspond to which registers. This error message is displayed when such an error occurs. This error message will also be issued when it does not make sense to write out vectors in parallel load format, such as trying to write out parallel vectors for a combinational design or for a design which contains no scan element.

### What Next

Either let dft_drc infer a test protocol to use a parallel load format or do not use the parallel load format.

## UIT-29

(error) No nets have global tracing enabled in design '%s'.

### Description

*BACKGROUND INFORMATION*

Using the *dft_drc* command, DFT Compiler design rule checking (DRC) is performed by doing a symbolic simulation of the test protocol for a design.

If the data applied at the scan in ports is successfully and predictably loaded into the scan chain, the cells are scan controllable. If data can be captured into the scan cells during the parallel capture cycle, and successfully and predictably unloaded, the cells are scan observable. When sequential cells are neither fully scannable (both scan controllable and scan observable) nor valid nonscan cells (for partial scan designs), fault coverage suffers.

DFT Compiler provides a net tracing capability which allows you to view the results of the symbolic simulation to help in the identification and correction of test protocol problems. DFT Compiler displays the logic values of the nets as information messages. All output from the design rule checker, including information messages, is linked to the design schematic. You can debug your protocol by selecting nets with certain values, then viewing the corresponding section in the schematic through the Design Analyzer.

Net tracing is enabled in either of two fashions. If *trace_nets statements* are added to vector or stream groups within a custom test protocol supplied by the designer, they enable tracing within a *local* scope limited to that vector or stream group. Alternatively, by specifying nets using the *trace_nets command*, nets can be traced within the *global* scope of the entire current test protocol. The *untrace_nets* command only acts on nets specified by the *trace_nets* command.

Feedback

*MANIFESTATION*

This error message is generated when the *untrace_nets* command has been invoked when there are *no* nets in the current design which have *global* tracing enabled as a result of a previous *trace_nets* command.

**What Next**

This is a benign error message indicating that you have asked that tracing be disabled on nets for which *global* tracing is not enabled. If you are still seeing net tracing for any nets, it must be the result of *local trace_nets statements* in your custom protocol. Remove those statements from the custom protocol.

---

## UIT-30

(error) Global tracing not enabled for net '%s' in design '%s'.

**Description**

*BACKGROUND INFORMATION*

Using the *dft_drc* command, DFT Compiler design rule checking (DRC) is performed by doing a symbolic simulation of the test protocol for a design.

If the data applied at the scan in ports is successfully and predictably loaded into the scan chain, the cells are scan controllable. If data can be captured into the scan cells during the parallel capture cycle, and successfully and predictably unloaded, the cells are scan observable. When sequential cells are neither fully scannable (both scan controllable and scan observable) nor valid nonscan cells (for partial scan designs), fault coverage suffers.

DFT Compiler provides a net tracing capability which allows you to view the results of the symbolic simulation to help in the identification and correction of test protocol problems. DFT Compiler displays the logic values of the nets as information messages. All output from the design rule checker, including information messages, is linked to the design schematic. You can debug your protocol by selecting nets with certain values, then viewing the corresponding section in the schematic through the Design Analyzer.

Net tracing is enabled in either of two fashions. If *trace_nets statements* are added to vector or stream groups within a custom test protocol supplied by the designer, they enable tracing within a *local* scope limited to that vector or stream group. Alternatively, by specifying nets using the *trace_nets command*, nets can be traced within the *global* scope of the entire current test protocol. The *untrace_nets* command only acts on nets specified by the *trace_nets* command.

*MANIFESTATION*

This error message is generated when a net specified by the *untrace_nets* command does not currently have *global* tracing enabled as a result of a previous *trace_nets* command.

Feedback

If an asterisk (*) is encountered within a net name passed to *untrace_nets*, the set of all nets in the design containing the associated partial net name specified will be checked for tracing. If no net within the set was found to have tracing enabled, this error message is invoked for *all* the nets in the set.

**What Next**

This is a benign error message indicating that you have asked that tracing be disabled on a net for which *global* tracing is not enabled. If you are still seeing net tracing for this net, it must be the result of a *local trace_nets statements* in your custom protocol. Remove that statement from the custom protocol.

If this error message is being produced multiple times it probably indicates that you have made imprecise use of a wildcard asterisk (*) when specifying your net. Check your use of asterisks within your list of net names passed to the *untrace_nets* command.

---

## UIT-31

(error) Cannot find vector file '%s' written by TestSim.

**Description**

This error message will appear when you are trying to perform the *write_test* command on a test program created by a fault simulation run. During a fault simulation run, two *.vdb* files are created : *<test_program_name>.vdb* and *<test_program_name>.testsim.vdb*. The *write_test* command needs both of these files to write out the output vector file.

**What Next**

Confirm that you have not accidentally deleted this file. If you have deleted this file, you will have to re-run fault simulation.

---

## UIT-32

(error) Bad protocol format '%s'.

**Description**

You receive this message if you specify a test protocol format other than stil or tpf; these are the only formats allowed by *write_test_protocol*. By default, *write_test_protocol* uses tpf format.

**What Next**

Re-execute *write_test_protocol* and specify either stil or tpf as the protocol format. Alternatively, if you want to use the default format (tpf), you can execute *write_test_protocol* with no format specification.

Feedback

## UIT-33

(error) Bad test model format '%s'.

### Description

You receive this message if you specify a test model format other than ctl; this is the only format allowed by *write_test_model*.

### What Next

Re-execute *write_test_model* and specify ctl as the test model format, or do not specify any format.

## UIT-34

(warning) Redundant TCK period specification `%f' ignored.\n \tExisting TCK period ` %f' (frequency `%e') retained.

### Description

This warning message is generated when *insert_jtag -tck_period* is invoked on a design with an existing boundary scan Test Access Port (TAP) TCK port where a clock has been previously specified at this port. The period and resulting frequency of this clock is reported in the message. In this case, DFT Compiler ignores the new specification and retains the old specification.

### What Next

If the old specification (probably generated using the *create_clock* command) is unsatisfactory, either remove the clock attribute on the TAP TCK port and invoke *insert_jtag -tck_period <correct_period>*, or change the existing clock period specification and invoke *insert_jtag* without the *-tck_period* option.

## UIT-35

(error) Physical dft preview doesn't support the shadow logic client.

### Description

You receive this message if you've specifed the shadow logic DFT client to be executed during DFT logic preview with the command: set_dft_configuration -order {wrapper} To perform physical dft preview, you have used the following commands: preview_dft -physcial Currently, physical dft preview doesn't support that client. This will prevent the completion of DFT logic previewing .

Feedback

**What Next**

Reset your DFT configuration: remove_dft_configuration Only set the Autofix client if you wish to use it as well as to perform physical dft preview. set_dft_configuration -order {autofix}. Proceed to DFT logic previewing.

---

## UIT-36

(Error) Test protocol already exists for design '%s'.

**Description**

A valid test protocol already exists in memory. It was either created by using create_test_protocol command or read by using read_test_protocol command.

**What Next**

You have to use remove_test_protocol command to avoid this message.

---

## UIT-37

(Warning) Test protocol that already exists for design '%s' is invalid.

**Description**

An invalid test protocol exists in memory. It was either created by using create_test_protocol command or read by using read_test_protocol command. This invalid test protocol will be removed and a new one will be created.

**What Next**

You can use remove_test_protocol command to avoid this message and read in a new protocol or create it from scratch.

---

## UIT-39

(error) The test_require attribute cannot be set on the three-state driver '%s'.

**Description**

DFT Compiler will not be able to account for the specified *test_require* on the three-state driver. The specification is ignored.

**What Next**

There are multiple ways to specify required values around a three-state bus: up-stream from the driverds, after the bus, or on the enables. In order to obtain the effect of a require on a single three-state driver, do the following. Identify the driver's input and enable pins.

Feedback

Manually insert buffers driving these pins. Set the desired test_require value on the driver of the data pin. Set a test_require value of 1 on the driver of the enable pin.

---

## UIT-41

(error) No implementation for component "%s" supplied.

**Description**

You must specify an implementation for the passed component type.

The implementation must be available in the jtag.db library. It may be a default implementation supplied with DFT Compiler, or a user supplied implementation added to the jtag.db.

Refer to the DFT Compiler reference manual for details of how to add custom components to the jtag.db

**What Next**

Re-execute the command, specifying an implementation type from those available in the jtag.db.

---

## UIT-42

(error) '%s' is not a valid value for %s option of the dft_drc command.

**Description**

This message tells you that you have given the *dft_drc* command invalid *-check_contention* or *-check_float* option arguments.

**What Next**

Repeat the *dft_drc* command using valid option arguments.

---

## UIT-43

(information) %s won't force conditioned bidirectionals.

**Description**

The default protocol inferred by DFT Compiler assumes that all bidirectional ports are turned in the same direction during scan shift. They can be turned inwards or outwards.

This message tells you that scan insertion does not insert logic to reconfigure ports that are turned in the opposite direction during scan shift.

Feedback

### What Next

Set "test_force_conditioned_bidir_pads" to "true" to change this behavior.

---

## UIT-44

(information) Assuming %s for combinational methodology.

### Description

This message tells you that DFT Compiler does not insert disabling logic when the test methodology is combinational.

### What Next

If you want to insert disabling logic, change the test methodology. To make the warning go away, ask for no disabling logic.

---

## UIT-45

(information) %s won't find the best scan-out pins.

### Description

This message tells you that scan insertion does not look for the best scan-out pins when it routes scan chains.

### What Next

Set "test_disable_find_best_scan_out" to "false" to change this behavior.

---

## UIT-46

(information) %s will create dedicated subdesign scan-out ports.

### Description

You receive this message because DFT Compiler does not look for existing connections and creates dedicated subdesign scan-out ports if no port has a *test_scan_out signal_type* attribute.

When connecting scan chains from subdesigns to parent designs, DFT Compiler first looks for subdesign ports that have *test_scan_out signal_type* attributes. DFT Compiler then checks to see whether an existing connection can be used. If nothing is found, DFT Compiler creates a dedicated subdesign scan-out port.

Feedback

### What Next

Set the *test_dedicated_subdesign_scan_outs* environment variable to *false* to change this behavior.

---

## UIT-47

(warning) Dedicated subdesign scan-out ports will be created for sub-designs when hierarchical isolation is specified. \n\tIgnoring hierarchical isolation if there are exsting dedicated subdesign scan-out ports.

### Description

You receive this message because DFT Compiler does not create dedicated subdesign scan-out ports when you also requested hierarchical isolation logic. The request for hierarchical isolation logic takes precedence. If the sub-designs already have dedicated scan-out ports hierarchical isolation request will not be valid.

When connecting scan chains from subdesigns to parent designs, DFT Compiler first looks for subdesign ports that have *test_scan_out signal_type* attributes. DFT Compiler then checks to see if an existing connection can be used. If it finds none, it creates a dedicated subdesign scan-out port.

You can set the *test_dedicated_subdesign_scan_outs* environment variable to *true* to direct DFT Compiler to ignore existing connections and create dedicated subdesign scan out ports.

### What Next

To obtain dedicated subdesign scan-outs, do not ask for hierarchical isolation logic. Set the *test_dedicated_subdesign_scan_outs* environment variable to *false* to eliminate the warning message.

---

## UIT-48

(warning) Scan insertion will not add clock gating logic.

### Description

Sequential ATPG assumes that valid nonscan cells in designs using multiplexed flip-flop scan styles retain their states during scan shift. To insert clock gating logic into your design and falidate this assumption, use the *-clock_gating* option of *set_scan_configuration*.

This message tells you that you decided not to insert clock gating logic. *dft_drc* violates nonscan cells that do not retain their states during scan shift, and *create_test_patterns* cannot use them to control or observe test vectors.

Feedback

**What Next**

Confirm that you do not want scan insertion to add clock gating logic.

---

## UIT-49

(warning) Scan style '%s' does not implement scan clock gating. Ignoring the set_scan_configuration -clock_gating command.

**Description**

Only the multiplexed_flip_flop scan style implements scan clock gating.

This message tells you that you asked DFT Compiler to implement scan clock gating, but have specified a scan style that is not a multiplexed_flip_flop. DFT Compiler is telling you that it is ignoring your *-clock_gating* option.

**What Next**

To get rid of the warning, remove the *-clock_gating* from the *set_scan_configuration* option list, or set the scan style to multiplexed_flip_flop.

---

## UIT-50

(warning) Test methodology '%s' does not implement scan clock gating. Ignoring the set_scan_configuration -clock_gating command.

**Description**

Only the partial scan test methodology implements scan clock gating.

This message tells you that you asked DFT Compiler to implement scan clock gating, but specified a test methodology that is not partial_scan. DFT Compiler is telling you that it is ignoring your *-clock_gating* option.

**What Next**

To get rid of the warning, remove the *-clock_gating* from the *set_scan_configuration* option list, or set the test methodology to partial_scan.

---

## UIT-51

(warning) Area-critical partial scan is not enabled. Ignoring the set_scan_configuration -area_critical true command.

Feedback

**Description**

Ultimately, DFT Compiler covers the entire scan spectrum. By default, DFT Compiler selects scan cells to maximize fault coverage, subject to both area and timing constraints. Tell DFT Compiler to ignore area constraints by setting a *set_scan_configuration -area_critical* option *false*, and tell it to ignore timing constraints by setting a *set_scan_configuration -timing_critical* option *false*.

The user model sees timing-critical partial scan as the first step to scan spectrum coverage. Tell DFT Compiler to maximize fault coverage subject to timing constraints, and ignore area constraints. Give *set_scan_configuration* an *-area_critical* option, and ask for timing-critical partial scan by giving it a mandatory *false* argument.

**What Next**

To get rid of the warning, remove the *-area_critical true* from the *set_scan_configuration* option list.

---

## UIT-52

(warning) Test methodology '%s' cannot not scan cells on timing critical paths. Ignoring the set_scan_configuration -area_critical command.

**Description**

Only the partial scan test methodology cannot scan cells on timing critical paths.

This message tells you that you asked DFT Compiler to implement not scan cells on timing critical paths, but specified a test methodology that is not partial_scan. DFT Compiler is telling you that it is ignoring your *-area_critical* option.

**What Next**

To get rid of the warning, remove the *-area_critical* from the *set_scan_configuration* option list or set the test methodology to partial_scan.

---

## UIT-53

(information) Enabling area critical partial scan.

**Description**

DFT Compiler can make scan cells non-scan to meet area constraints. Use this capability when you need to back off scan in a small percentage of cells to meet critical area constraints.

This message tells you that the capability is enabled.

Feedback

### What Next

Confirm that you want to use the capability. To tell the DFT Compiler that you do not want to use the capability, use *set_scan_configuration -area_critical false*.

---

## UIT-54

(Error) The format tpf is no longer supported.

### Description

You receive this error message because you invoked one of the following commands: read_init_protocol or read_test_protocol or write_test_protocol with the -format option set to tpf. The tpf format has been obsoleted and the only valid format is the stil format.

### What Next

While writing the test protocol, replace the tpf format by the stil format and reissue the command. While reading an initialization protocol or a test protocol, ensure that the protocol has the stil format, replace the tpf format by the stil format and reissue the command.

---

## UIT-55

(Error) All the scan signals have to be routed in the compression flow.

### Description

You receive this error message because the scan_signals option of the set_scan_configuration command has to be set to all for the scan compression feature.

### What Next

Set to all the scan signals for the set_scan_configuration command.

---

## UIT-56

(Error) Scan methodology must be multipexed_flip_flop and all the scan signals have to be routed in the logic BIST flow.

### Description

You receive this error message related to the set_scan_configuration command because the methodology must be set to multiplexed_flip_flop, the scan_signals option must be set to all and the route option must be set to true for the logic BIST flow.

Feedback

### What Next

Set to multiplexed_flip_flop the methodology, set to all the scan signalsand set to true the route for the set_scan_configuration command.

---

## UIT-85

(error) The 'compile' and 'compile_ultra' commands are disabled in the boundary-scan RTL generation flow.

### Description

The *compile* and *compile_ultra* commands are disabled in the boundary-scan RTL generation flow.

The RTL generation flow is enabled with the *-rtl enable* option of the *set_bsd_configuration* command. In this flow, use the *write_bsd_rtl* command to write out the RTL for the generated boundary-scan logic inserted in the top-level design.

You get this message even if you disable the RTL generation flow and attempt to use the *compile* or *compile_ultra* commands on the in-memory boundary-scan RTL.

To use the *compile* or *compile_ultra* commands to compile the boundary-scan design in the RTL generation flow, you must write out the boundary-scan RTL, remove the design in memory, then reread it and compile. For example,

```
insert_dft
remove_design CORE
write_bsd_rtl -format verilog -all -output TOP_bsd_rtl.v

remove_design -designs

analyze TOP_bsd_rtl.v
elaborate TOP
# also read in core logic, etc.
current_design TOP
link
compile
```

---

## UIT-86

(error) Write command disabled for BSDC RTL insertion flow. Instead use write_bsd_rtl command.

### Description

*write* command is disabled for BSDC RTL insertion flow. To write RTL output file, one should use *write_bsd_rtl* command.

Feedback

BSDC RTL flow is enabled using *-rtl enable* option of *set_bsd_configuration* command. In this flow *write_bsd_rtl* command can be used to write out the BSDC inserted RTL file for top level design. One can use *write* command in Netlist flow, during which *-rtl disable* is set. Netlist flow is the default flow.

### What Next

In RTL flow use *write_bsd_rtl* command in place of *write* command.

---

## UIT-87

(warning) Assignment of %s-only BSR cells to port `%s`\n\ \tis not compliant with the IEEE 1149.1 JTAG standard.

### Description

Unless the user requests a user-defined Boundary Scan Register (BSR) cell implementation using the *set_jtag_implementation* command, DFT Compiler selects default BSR cells from among the following 10 defined types:

```
Type      Default Name      BSR Cell Function

        BSRINBOTH       JTAG_BSRINBOTH  set & observe data input cell
        BSRINSET        JTAG_BSRINSET   set-only data input cell
        BSRINOBS        JTAG_BSRINOBS   observe-only data input cell
        BSRINCLKBOTH    JTAG_BSRINCLKBOTH       set & observe clock input
 cell
        BSRINCLKSET     JTAG_BSRINCLKSET        set-only clock input cell
        BSRINCLKOBS     JTAG_BSRINCLKOBS        observe-only clock input
 cell
        BSROUTBOTH      JTAG_BSROUTBOTH set & observe data output cell
        BSROUTSET       JTAG_BSROUTSET  set-only data output cell
        BSROUTOBS       JTAG_BSROUTOBS  observe-only data output cell
        BSRCTL  JTAG_BSRCTL     control cell
```

IEEE 1149.1 Standard rule 10.1.1 specifies legal boundary scan register cell types. Several of the 10 types listed above are not compliant with the standard. Nonetheless, DFT Compiler supports them in order to allow designers the greatest possible flexibility when using Programmable JTAG.

```
10.1.1(b) "For a unidirectional system input pin (clock or nonclock), a
boundary-scan register cell shall be connected between the system pin and
 the
on-chip system logic to allow the state of the system pin to be observed
 and,
optionally, for the state of the system logic to be controlled."
```

Thus, legal data and clock input BSR cell types for unidirectional system input pins are observe-only and both control and observe. BSRINSET and BSRINCLKSET are non-compliant types.

Feedback

**10.1.1(c)** "For 2-state and open-collector system output pins, a boundary-scan register cell shall be provided to allow the state of the system pin to be controlled and the state of the on-chip system logic output to be observed."

Thus, the legal output BSR cell type for system 2-state and open-collector output pins is both control and observe. BSROUTSET and BSROUTOBS are non-compliant types.

**10.1.1(d)** "For 3-state system output pins, boundary-scan register cells shall be provided to allow both the data value and the output drive state (active or inactive) to be controlled and for the corresponding system logic outputs to be observed."

Thus, the legal output BSR cell type for system 3-state output pins is both control and observe. BSROUTSET and BSROUTOBS are non-compliant types.

**10.1.1(e)** "For a bidirectional system pin, boundary-scan register cells shall be provided to allow the value and direction of data at the system pin to be controlled *or* observed and, where appropriate, for the corresponding system logic inputs to be controlled."

This is somewhat ambiguous. Test Compiler interprets this rule to indicate that as applied to bidirectional signals BSRINSET, BSROUTSET and BSROUTOBS are non-compliant types.

The types `*BSR*xxx*BOTH*' (control and observe) `*BSR*xxx*SET*' (control-only) and `*BSR*xxx*OBS*' (observe-only) differ. They correspond respectively to the three modes `both', `control' and `observe' specified with the *set_jtag_port_mode* command. When this command is invoked for one or more ports of the design, BSR cells for those ports will have BSR cells of the specified type associated with them.

If the specified type will result in a non-compliant BSR cell being synthesized for a port, this warning message is issued.

**What Next**

Determine whether non-compliant BSR cells can be tolerated in your design. If not, modify the port mode settings for the offending ports.

---

## UIT-88

(error) Data register `%s` pin of type `%s` not found.

Feedback

### Description

You receive this message because the specified data register pin is not recognized as a port at the top level of the core design.

As a designer customizing the JTAG chip interface, you may want the JTAG Test Access Port (TAP) to access core test structures that DFT Compiler cannot automatically recognize. For example, suppose you want to activate a RAM test using an embedded test data register. You must do the following:

1. Initiate the test.
2. Capture the result of the test in the data register.
3. Shift the test result out through *TDO*.

To perform this test using the JTAG TAP, you require:

- An activation signal enabled by the associated JTAG instruction. This signal enters the data register through its *enable_pin*.
- Connection of the data register's serial input to *TDI*. This connection is optional, required only if data must be shifted *into* the data register to execute the test. This signal enters the data register through its *load_pin*.
- An input to the JTAG MUX (which selects between data registers) dedicated to this data register. This input is selected when the associated JTAG instruction is active. This signal leaves the data register through its *capture_pin*.

In order to retain the functional and structural identity of the core logic, the DFT Compiler's JTAG logic synthesis interacts with the design's core logic at the top level of the design's hierarchy. This ensures that simulation of the core logic, before and after JTAG synthesis, remains identical. Thus, for DFT Compiler to automatically connect to the JTAG TAP the *capture_pin*, *load_pin*, and *enable_pin* of a user-specified core data register, these pins must exist as pseudo-ports at the top level of the original circuit. That is, it is the responsibility of the designer to ensure that the signals controlling and observing a core test register traverse the circuit hierarchy until they are visible at the top level of the design.

### What Next

Determine the reason that the pin has not been found. If it is appropriately named, check to ensure that the pin appears as a pseudo-port of the top level of the design core.

---

## UIT-89

(error) An illegal implementation `%s`\n\ \thas been specified for JTAG component type `%s`.

Feedback

### Description

Any alternate implementation for one of the JTAG component types

```
- the Test Access Port (TAP) Controller: TAP,
- the Bypass Register: BR
- the device Identification Register: ID
- the JTAG instruction register: IR, or
- one of the Boundary Scan Register cells: BSRINBOTH, BSRINSET,
BSRINOBS, BSROUTBOTH, BSROUTSET, BSROUTOBS,
BSRCTL, BSRINCLKBOTH, BSRINCLKSET or BSRINCLKOBS
```

must have all the I/O ports that exist on the default implementation.

### What Next

Compare the I/O of the user-specified implementation with that of the default component of the specified component type.

---

## UIT-90

(error) set_jtag_implementation must be invoked with a legal\n\ \tcomponent type. `%s` is not a recognized component type.

### Description

*set_jtag_implementation* must be invoked with one of the JTAG component types

```
- the Test Access Port (TAP) Controller: TAP,
- the Bypass Register: BR
- the device Identification Register: ID
- the JTAG instruction register: IR, or
- one of the Boundary Scan Register cells: BSRINBOTH, BSRINSET,
BSRINOBS, BSROUTBOTH, BSROUTSET, BSROUTOBS,
BSRCTL, BSRINCLKBOTH, BSRINCLKSET or BSRINCLKOBS
```

These are the *only* legal JTAG component types.

### What Next

Invoke the dc_shell JTAG command with the *-help* option to review the correct command syntax and make sure a legal JTAG component type has been selected.

---

## UIT-91

(warning) Illegal JTAG ground_bounce value `%d` replaced by `%d`.

Feedback

### Description

The environmental variable *test_jtag_ground_bounce_limit* has been added to the .synopsys setup environment to provide the user with a mechanism to ensure that the `ground bounce' problem does not occur for designs with JTAG synthesized by DFT Compiler.

This problem arises when the pins of a chip that does not normally support simultaneous switching or enabling of all outputs are controlled from the Boundary Scan Register (BSR), rather than from the on-chip system logic. For example, this situation arises when the EXTEST instruction is selected. It is possible that the tests supplied through the boundary-scan path will cause all outputs to change state and/or be active simultaneously. If precautions are not taken, the sum of the switching currents of all pins may produce VCC and VSS glitches that could exceed the tolerance level of the IC, and the core logic as well as the BSR cells and TAP would then be subject to interference. The power consumption could be increased beyond the capacity of the power pins for a prolonged period, perhaps resulting in incorrect operation of the chip.

Delays that only effect the signal path when the chip is in JTAG test mode can be added to prevent simultaneous switching of outputs when pins are driven from the BSR. The added delays should be small in comparison with the minimum period of TCK, but should be sufficient to ensure that the power-current demand arising from the change of state at one pin or group of pins does not overlap with that from another group producing the excessive power consumption.

However, if the process to optimize the number of BSR cells results in a single BSR cell (for example a bus three-state enable signal) driving too many enable signals, there is no way to introduce delays between the signals controlled by that BSR cell and ground bounce may not be prevented.

The environmental variable test_jtag_ground_bounce_limit puts a cap on the number of ouputs that can be driven by a single BSR cell. A default value is establised for this variable. The user can establish another positive non-zero limit, by changing the value of the variable in the .synopsys_dc.setup file.

The value of the ground-bounce limit for a particular JTAG design can be determined by executing *report_test -jtag*.

A detailed discussion of the Ground-Bounce problems is presented in Chapter 15, `Providing Boundary-Scan on Chips with Power or Output-Switching limitations' of `The Test Access Port and Boundary-Scan Architecture' by Colin M. Maunder and Rodham E. Tulloss, IEEE Computer Society Press Tutorial, 1990.

### What Next

Reset the value of the *test_jtag_ground_bounce_limit* to a legal value.

Feedback

## UIT-92

(information) User defined instruction '%s' not associated with\n\ \tidentifiable data register. Dangling signals will be synthesized.

### Description

Each JTAG instruction activates a single target data register. By definition, the target register of any user defined instruction is not specified in the IEEE 1149.1 JTAG Standard; therefore, that target register cannot be inferred by DFT Compiler. The user must identify the target register by using either *-reg_name core_scan_chain_name* or *-reg_coordinates capture_pin* [*load_pin*] and *-reg-enable enable_pin*.

When invoking *set_jtag_instruction instruction_name* [*-code code*] if you do not specify *-reg_name core_scan_chain_name*, *-reg_coordinates capture_pin* [*load_pin*], or *-reg-enable enable_pin*, DFT Compiler will synthesize all the necessary instruction decode logic and create two new design ports: an input port called <*instruction_name*>*CAPTURE* and an output port called <*instruction_name*>*ENABLE*. The port <*instruction_name*>*ENABLE* is active *HIGH* when the instruction *instruction_name* is loaded into the IR. The designer can then manually edit the circuit to connect these two signals to the appropriate data register circuitry.

### What Next

Manually connect the two signals <*instruction_name*>*CAPTURE* and <*instruction_name*>*ENABLE* to the appropriate data register circuitry.

## UIT-93

(warning) One or more ports have been excluded from the JTAG\n\ \tBoundary Scan Register (BSR). JTAG logic synthesized based on this\n\ \tconfiguration will not be compliant with the IEEE 1149.1 JTAG standard.

### Description

The IEEE 1149.1 JTAG Standard specifies that *all* non-TAP I/O ports must be included in the BSR. Exclusion of one or more ports results in this warning message.

### What Next

If a non-compliant JTAG implementation is satisfactory, ignore this message. Otherwise, compliance with this condition may be met by executing *set_jtag_port -default*.

## UIT-94

(error) Erroneous JTAG data register '%s' associated with JTAG\n\ \tinstruction '%s'.

Feedback

### Description

The IEEE 1149.1 JTAG Standard specifies and requires three mandatory JTAG instructions: BYPASS, SAMPLE/PRELOAD and EXTEST. If a JTAG device Identification Register (IDreg) exists, then a fourth JTAG instruction, IDCODE, is mandatory. The standard also defines three other instructions: INTEST, RUNBIST and USERCODE. The target register selected by each of these instructions is designated by the standard as follows:

```
Instruction     Target Data Register

        BYPASS  JTAG_BYPASS_REG
        SAMPLE/PRELOAD  JTAG_BSR
        EXTEST  JTAG_BSR
        INTEST  JTAG_BSR
        IDCODE  JTAG_ID_REG
        USERCODE        JTAG_ID_REG
        RUNBIST JTAG_BIST_REG
```

Because only one data register is associated with each of these instructions, it need never be specified by the designer. If the target data register *is* specified, using *set_jtag_instruction instruction_name -reg_name name*, a check is made that the legal data register has been specified. This error message appears if an illegal register is specified.

Note, however, that if you specify a particular core BIST register with RUNBIST, you need to include register coordinate information.

### What Next

Since specifying the data register with any of these seven JTAG instructions is redundant, avoid this error by simply removing the erroneous specification.

---

## UIT-95

(error) JTAG logic already inserted into this design.\n\ \tinsert_dft not invoked.

### Description

Invoking JTAG synthesis on a design twice returns an error. The danger is that preexisting JTAG logic might be confused with core logic and result in duplicate JTAG logic synthesis.

When DFT Compiler successfully inserts JTAG into a design, it sets a design attribute flag. A call to *insert_dft* is not executed if a check for the presence of this design attribute indicates pre-existing JTAG logic.

Feedback

**What Next**

If you want to reinvoke JTAG synthesis, make sure you begin with a clean design without JTAG.

---

## UIT-96

(error) Port '%s' excluded from the JTAG Boundary Scan Register (BSR).\n\ \tNo BSR cells can be associated with it.

**Description**

BSR cells are only synthesized for ports that are included in the JTAG BSR. Thus it is an error to specify a routing order for BSR cells relative to ports that are excluded from the BSR.

**What Next**

Either remove reference to those BSR cells associated to the excluded port or specify inclusion of the port in the BSR by executing *set_jtag_port* TRUE *port_name*.

---

## UIT-97

(error) '%s' not an input or output port.\n\ \t(You must specify '/out', '/in' or '/ctl'.)

**Description**

For bidirectional ports, the *port_name* is insufficient to differentiate between the Boundary Scan Register (BSR) input, output, and control cells.

**What Next**

A trailing specifier */in*, */out* or */ctl* must be appended to *port_name* to identify each bidirectional port BSR cell.

---

## UIT-98

(error) '%s' not a port of the current design.

**Description**

JTAG Boundary Scan Register (BSR) cells can only be specified in *set_jtag_port_routing_order* relative to ports in the current design.

**What Next**

Check the ports of the current design to verify the set of legal *port_name* candidates.

Feedback

## UIT-99

(error) Unrecognized BSR cell specifier '%s'.\n\ \t(Must be '/out', '/in' or '/ctl'.)

### Description

The *ordered_list_of_annotated_BSR_port_names* contains members of four classes:

- *<input_or_output_port_name>* which, depending on the direction of the port, specifies a BSR input or output cell;
- *<bidirectional_port_name>***/in** which specifies the BSR input cell associated with this bidirectional port;
- *<bidirectional_port_name>***/out** which specifies the BSR output cell associated with this bidirectional port;
- *<tristate_or_bidirectional_port_name>***/ctl** which specifies the BSR three-state control cell associated with this output or bidirectional port.

### What Next

Only three legal specifiers (*/in*, */out* or */ctl*) may be appended to a *port_name* to identify a BSR cell.

## UIT-100

(error) Instruction Register (IR) too small.\n\ \tThis design requires an %d-bit register.

### Description

The IEEE 1149.1 JTAG Standard mandates that the JTAG Instruction Register (IR) be at least two bits wide. DFT Compiler supports IRs of between 2 and 32 bits inclusively. The size of the IR is set in three ways:

- specifying the size using **insert_jtag -ir_size** *bit_width*;
- forcing the IR size by specifying an instruction code **set_jtag_instruction -code** "*code*";
- DFT Compiler will automatically infer the IR size from the set of specified and mandatory instructions selected.

## UIT-101

(error) JTAG Test Access Port (TAP) I/O port '%s' may not be included\n\ \tin the boundary scan register (BSR).

### Description

The IEEE 1149.1 JTAG Standard specifies that the Boundary Scan Register (BSR) is a data register connected between TDI and TDO. The BSR may not include any of the JTAG

Feedback

Test Access Port (TAP) ports. An error will result if you include ports carrying the following signals:

```
- TDI: JTAG Test Data In
- TDO: JTAG Test Data Out
- TMS: JTAG Test Mode Select
- TCK: JTAG Test Clock
- TRST: JTAG Test Reset
```

---

## UIT-102

(error) Illegal JTAG boundary scan port mode '%s' specified.

**Description**

The JTAG circuitry associated with each port allows that port to be *controlled* only, *observed* only, or *both* controlled and observed. This is the JTAG operational mode of a port.

The default mode can be initially set in the .synopsys file using *jtag_bsr_port_mode* = `<mode_string>' where allowable port modes are *control*, *observe*, or *both*.

---

## UIT-103

(error) Illegal JTAG boundary scan port signal type '%s' specified.

**Description**

The JTAG circuitry associated with a port can be optimized depending on the signal type, *clock*, *enable* or *data*, of that port.

The JTAG circuitry associated with an *input* port can be optimized based on the signal type (*clock* or *data*) of that port. Ports that drive clock trees can be associated with specialized JTAG circuitry. An input port can be explicitly identified as carrying one of the "clock" or "data" categories.

When the three-state elements driving three-state and bidirectional *outputs* are found in the core logic of the IC (i.e. not embedded in the pads themselves) DFT Compiler requires that these three-state elements be located at the top hierarchy of the core of the design so that Boundary Scan Register (BSR) cells can be placed on the *inside* of the three-state elements in order to control and observe the data and enable lines. In these cases, only a single three-state line exits the core logic.

However, in the cases where the three-state controller for output signals will be embedded in the three-state output or bidirectional input/ouput pad associated with these ports of a design two signals exit the core logic; a *data* line and a *enable* line. Unless, TC knows the distinction between these two types of exit lines, it will always assume that they are data lines and not insert the correct type of BSR cell on the enable lines. By specifying "enable"

Feedback

or "data" on a port the designer instructs TC on which type of BSR cell to place on these output ports.

Where possible, DFT Compiler will extract this information from the circuit design. A port may be explicitly identified as carrying one of these catagories of data.

**What Next**

If you are defining the characteristics of input ports, choose between the "clock" and "data" options. If you are defining the characteristics of ouput ports, choose between the "enable" and "data" options.

---

## UIT-104

(error) Requested Instruction Register size '%d' is less than\n\ \tpreviously specified Instruction code size '%d'.

**Description**

The IEEE 1149.1 JTAG Standard mandates that the JTAG Instruction Register (IR) be at least two bits wide. DFT Compiler supports IRs of 2- to 32-bit widths inclusively. The size of the IR is set in three ways:

```
- specifying the size using insert_jtag -ir_size bit_width;
- forcing the IR size by specifying an instruction code
set_jtag_instruction -code "code";
- DFT Compiler will automatically infer the IR size from the set of
specified and mandatory instructions selected.
```

This error is generated if one or more JTAG instructions with specified codes of length *n* have been previously specified, but *insert_jtag -ir_size m* is invoked where *n* > *m*.

If *m* > *n* then DFT Compiler can pad the *m-n* high order bits of the previously specified instruction codes with 0s. But it cannot truncate previously specified codes when *m* > *n*.

**What Next**

Either invoke *insert_jtag -ir_size* with a size greater or equal to the length of all previously specified codes, or, if a smaller IR is required, specify codes for all instructions with previously specifed codes, so that their code length is less than or equal to the desired IR size.

---

## UIT-105

(error) Instruction '%s' has code fixed by the 1149.1 JTAG standard.\n\ \tYou cannot assign the code '%s' to this instruction.

Feedback

### Description

You receive this message because DFT Compiler generates it when an attempt is made to reset *BYPASS* and *EXTEST* JTAG code.

Each JTAG instruction requires the following triplet of information:

- Instruction name
- Target data register
- Instruction code

DFT Compiler can automatically generate some of this information.

The mandatory *BYPASS* and *EXTEST* JTAG instructions have mandatory codes. The *BYPASS* code is all *1*s (ones). The *EXTEST* code is all *0*s (zeros). All other mandatory and nonmandatory JTAG instructions support arbitrary codes.

Since the *BYPASS* and *EXTEST* instructions have mandatory codes, the designer never has to specify them. DFT Compiler automatically extracts all information relative to these two instructions.

---

## UIT-106

(warning) The 1149.1 JTAG standard instruction '%s' does not\n\ \tsupport register coordinate specification.\n\ \t(Specifications ignored.)

### Description

The IEEE 1149.1 JTAG Standard specifies and requires three mandatory JTAG instructions: BYPASS, SAMPLE/PRELOAD, and EXTEST. If a JTAG device Identification Register (IDreg) exists, then a fourth JTAG instruction, IDCODE, is mandatory. The standard also defines three other instructions: INTEST, RUNBIST, and USERCODE. The target register selected by each of these instructions is designated by the standard as follows:

```
Instruction     Target Data Register

        BYPASS  JTAG_BYPASS_REG
        SAMPLE/PRELOAD  JTAG_BSR
        EXTEST  JTAG_BSR
        INTEST  JTAG_BSR
        IDCODE  JTAG_ID_REG
        USERCODE        JTAG_ID_REG
        RUNBIST JTAG_BIST_REG
```

For the first six JTAG instructions above only one data register is associated with each of these instructions; the register does not need to be specified. In these six cases,

Feedback

if the target data register *is* specified, using *set_jtag_instruction instruction_name -reg_coordinates capture_pin* [*load_pin*] *-reg_enable enable_pin*, the register coordinate information is ignored and this warning message is generated.

If you specify a particular core BIST register with RUNBIST, you need to include register coordinate information.

**What Next**

Since specifying the data register with any of these seven JTAG instructions is redundant, avoid this error by simply removing the redundant coordinate specification.

---

## UIT-107

(error) Code length '%d' of instruction '%s' conflicts with a\n\ \tpreviously specified instruction code length '%d'.

**Description**

The IEEE 1149.1 JTAG Standard mandates that the JTAG Instruction Register (IR) be at least two bits wide. DFT Compiler supports IRs of 2- to 32- bit widths inclusively. The size of the IR is set in three ways:

```
- specifying the size using insert_jtag -ir_size bit_width;
- forcing the IR size by specifying an instruction code
set_jtag_instruction -code "code";
- DFT Compiler will automatically infer the IR size from the set of
specified and mandatory instructions selected.
```

If you specify codes to be assigned to specific instructions, the length of these codes must be consistent with any previously specified codes or the previously specified Instruction Register (IR) size.

**What Next**

Modify the length of the offending codes to be equal that of previously specified instruction codes and/or IR size.

---

## UIT-108

(error) Code '%s' for instruction '%s' conflicts with the\n\ \tcode for instruction '%s'.

**Description**

You receive this message because the code for an instruction conflicts with the code specified for another instruction.

Feedback

Each instruction must have a unique code that does not conflict with the code specified for another instruction. In the absence of a user-specified code for an instruction, DFT Compiler generates a unique code for that instruction.

Each JTAG instruction requires a triplet of information:

- Instruction name
- Target data register
- Instruction code

DFT Compiler can automatically generate some of this information. The mandatory *BYPASS* and *EXTEST* JTAG instructions have mandatory codes. The *BYPASS* code is all *1*s (ones). The *EXTEST* code is all *0*s (zeros). All other mandatory and nonmandatory JTAG instructions support arbitrary codes.

**What Next**

Eliminate the coding conflict by executing a legal change of code for one of the two conflicting instructions. (You cannot change the mandatory codes of the *BYPASS* and *EXTEST* instructions.)

Again execute the *set_jtag_instruction* command.

While changing the conflicting code, you must keep all other specifications relative to the instruction the same.

---

## UIT-109

(error) BSR cell '%s' has been assigned multiple routing positions.

**Description**

Each JTAG Boundary Scan Register (BSR) cell only appears once in the BSR. Therefore, when specifying its location in the BSR using the *set_jtag_port_routing_order* instruction, it is an error to specify more than one location for each BSR cell.

**What Next**

Remove multiple specifications of the BSR cell reported in the error message, leaving at most one reference to it in the *ordered_list_of_annotated_BSR_port_names* when executing *set_jtag_port_routing_order*.

---

## UIT-110

(warning) Compliance with the IEEE's 1149.1 JTAG Standard requires\n\ \ta BYPASS Register. This implementation is non-compliant.

Feedback

### Description

If you specify that a JTAG BYPASS register not be synthesized, the resulting JTAG architecture will *not* be IEEE 1149.1 compliant.

### What Next

If you want a compliant JTAG architecture, do not invoke *insert_jtag* with the *-no_bypass* option.

---

## UIT-111

(warning) JTAG component implementation '%s' not found.\n\ \tThe default implementation will be used instead.

### Description

The basic JTAG components (*BSRin*, *BSRout*, *BSRcontrol*, *TAP*, *BYPASS*, *ID*, and *IR*) must have functionality that conforms the the IEEE 1149.1 standard. Each can be implemented in a number of different architectures. There are several different functionality implementations (set-only, observe-only, or both set and observe) of the JTAG Boundary Scan Register (BSR) cells. DFT Compiler has a default implementation for each of the three types (input, output, and control) of BSR cells.

You can override the default implementation of all the JTAG components:

```
- one of the Boundary Scan Register (BSR) cells BSRin, BSRout, or
BSRcontrol
- the Test Access Port (TAP) Controller TAP
- the Bypass Register BYPASS
- the device Identification Register ID
- the JTAG instruction register IR
```

by executing

```
set_jtag_implementation component_type implementation
```

where *component_type* is *BSRin*, *BSRout*, *BSRcontrol*, *TAP*, *BYPASS*, *ID*, or *IR*.

You can override the implementation of one or more of the BSR cells of the specified type in the BSR by executing the *set_jtag_implementation* instruction with the correct arguements:

```
- set_jtag_implementation BSRin implementation [list_of_ports]
- set_jtag_implementation BSRout implementation [list_of_ports]
- set_jtag_implementation BSRcontrol implementation [list_of_ports]
```

The *implementation* is the name of the desired JTAG component in the JTAG component library. If DFT Compiler cannot find this component in the library, it issues this message and uses the default implementation for this type of BSR cell.

Feedback

### What Next

If you want to override the default implementation, make sure that the JTAG component named *implementation* is in the JTAG library.

---

## UIT-112

(error) Target core register '%s' not found.

### Description

If you want to specify an instruction and, optionally, the code for a specific internal scan chain, execute *set_jtag_instruction instruction_name -reg_name name* [*-code code*].

The *name* of each internal scan chain is the name of its parallel scan-in port; that is, the port which has the *test_scan_in* attribute assigned to it. This guarantees the uniqueness of names for these scan chains and provides DFT Compiler with a mechanism to identify each target scan chain. If DFT Compiler cannot find this unique parallel scan-in port, this message is generated.

### What Next

Verify the names of the parallel scan-in ports of the design using *report_test -scan*.

---

## UIT-113

(information) Creating user defined instruction '%s'.\n\ \t(not a 1149.1 JTAG standard instruction)

### Description

DFT Compiler recognizes the seven standard instructions defined by the IEEE 1149.1 JTAG Standard:

```
Instruction     Target Data Register

        BYPASS  JTAG_BYPASS_REG
        SAMPLE/PRELOAD  JTAG_BSR
        EXTEST  JTAG_BSR
        INTEST  JTAG_BSR
        IDCODE  JTAG_ID_REG
        USERCODE        JTAG_ID_REG
        RUNBIST JTAG_BIST_REG
```

DFT Compiler also accepts user-defined JTAG instructions. These instructions may access core scan chains or invoke other test circuitry in the design. When a user-defined instruction has been successfully parsed, this information message is issued.

Feedback

The existence of this message also provides a check for mispelled names of the seven standard JTAG instructions. It verifies that you in fact want to create a user-defined instruction.

## UIT-114

(information) Creating 1149.1 JTAG standard instruction '%s'.

### Description

DFT Compiler recognizes the seven standard instructions defined by the IEEE 1149.1 JTAG Standard:

```
Instruction     Target Data Register

        BYPASS  JTAG_BYPASS_REG
        SAMPLE/PRELOAD  JTAG_BSR
        EXTEST  JTAG_BSR
        INTEST  JTAG_BSR
        IDCODE  JTAG_ID_REG
        USERCODE        JTAG_ID_REG
        RUNBIST JTAG_BIST_REG
```

DFT Compiler also accepts user-defined JTAG instructions. This information message confirms that DFT Compiler has recognized and parsed one of the standard instructions.

The existence of this message also provides a check for user-defined instructions which have been mislabeled with standard instruction names.

## UIT-115

(error) Illegal joint specification of -no_internal scan and\n\ \t-daisy_chained internal scan. (Specify only one of these options.)

### Description

By default DFT Compiler will attach core test structures that it recognizes (for example, internal scan chains) to the JTAG Test Access Port (TAP). Internal scan chains will not be connected to the TAP if you execute *insert_jtag -no_internal_scan*. If you want all of the recognized core scan chains accessible as a single long register comprised of the individual scan chains serially daisy-chained together, execute *insert_jtag -daisy_chained* [*ordered_set_of_scan_chain_indexes*].

These two options are mutually exclusive. This error message is invoked if both options are specified at once.

Feedback

### What Next

Determine which core scan chain option is desired and invoke *insert_jtag* with the appropriate argument.

---

## UIT-116

(information) Instruction '%s' has been cancelled.

### Description

Any non-mandatory instruction may be removed from the set of instructions "understood" by the synthesized JTAG instruction decode logic. This message confirms removal of the specified instruction. As a result, decode logic will not be synthesized to recognize this instruction.

---

## UIT-117

(error) Mandatory instruction '%s' may not be cancelled.

### Description

The IEEE 1149.1 JTAG Standard requires the following mandatory instructions:

```
- BYPASS,
- SAMPLE/PRELOAD
- EXTEST.
```

If a Device Identification (ID) Register is part of the JTAG logic, then IDCODE is also mandatory.

The first three instructions will *always* be included in the set of instructions "understood" by the synthesized JTAG instruction decode logic. If you request an ID register, you must include the fourth instruction, IDCODE. An error message will appear if you attempt to cancel any one of these instructions.

---

## UIT-118

(error) JTAG insertion terminated abnormally.

### Description

Anytime DFT Compiler encounters a fatal designer JTAG specification error after *insert_dft* has been invoked, an error message identifying the problems appears and DFT Compiler executes a graceful abnormal termination of JTAG synthesis. Any changes to the design resulting from execution of *insert_dft* are eliminated.

Feedback

Fatal designer JTAG specification errors include encountering a three-state cell controlling a port which is not at the top level of the design (see *UIT-119*); and JTAG component library not found (see *TEST-234*).

**What Next**

Identify and correct the cause of the abnormal termination and reinvoke *insert_dft*.

---

## UIT-119

(error) Three-state cell `%s` controlling port(s)\n\ \tin the JTAG Boundary Scan Register not at the top level of the design.

**Description**

Up to three Boundary Scan Register (BSR) cells can be associated with each port of the design. A simple input port has a single BSR input cell associated with it. If it is a two state output, an output port will have one BSR cell associated with it. If it is a three-state output, it will have two BSR cells associated with it. A bidirectional port can have three BSR cells associated with it (input, output and control BSR cells). When DFT Compiler inserts boundary scan cells for three-state ports into the design, it determines their location by locating the data and control inputs to generic three-state devices.

Whether or not you want to group all system logic into a level of (core) hierarchy, you must ensure that the three-state devices controlling the behavior of any ports are visible at the top level of the core hierarchy. This is necessary for pad optimization since three-states can be absorbed into a pad.

The following script fragment demonstrates this:

```
dc_shell> read -f lsi AMPM.NET

/* get the top level design */
dc_shell> current_design  = DESIGN

/* need to separate any top level port three_state control cells from the
 core.
   In this case they are buried in sub-modules U6 and U7 */
dc_shell> ungroup  {U6 U7}

/* group the core logic with the exception of the 6 three-state cells */
dc_shell> group -design_name  CORE -except
{U6/UU2 U6/UU3 U6/UU4 U7/UU4 U7/UU5 U7/UU6} -cell_name  AMPM

/* need to set port_is_pad to do pad insertion */
dc_shell> set_port_is_pad find  (port,"*")

dc_shell> insert_dft
```

Feedback

This error message results when DFT Compiler encounters a three-state cell controlling a port which is not at the top level of the design. The complete hierarchical name of the offending cell is reported in the error message. DFT Compiler then gracefully aborts JTAG insertion.

**What Next**

Bring the offending three-state cell to the top level as shown in the script fragment above to eliminate this error.

---

## UIT-120

(warning) -no_asynchronous_reset option invoked\n\ \tbut port '%s' with jtag_trst attribute found.\n\ \tPort '%s' will NOT be included in the JTAG Boundary Scan\n\ \tRegister (BSR). \n

**Description**

The JTAG Test Access Port (TAP) provides the test interface in JTAG mode between the chip and the outside world. It includes the three mandatory input ports (test data in *TDI*, test mode select *TMS*, and test clock *TCK)* and one optional input (test reset *TRST**). It also includes a single output port (test data out *TDO*).

The active-LOW asynchronous reset signal, if present, provides asynchronous initialization of the TAP controller and other JTAG test logic in the circuit. In the absense of the *TRST** signal, the TAP controller will synchronously enter the *Test-Logic-Reset* controller state following five rising edges of *TCK*, provided *TMS* is held high.

If none of the TAP I/O ports exist, the mandatory ports will be automatically generated. By default, the JTAG logic will be synthesized with an asynchronous reset. The optional *TRST** port will not be generated if requested using *insert_jtag -no_asynchronous_reset*. If however, the *TRST** port already exists, and it has been identified using *set_dft_signal -type trst* with *insert_jtag -no_asynchronous_reset* invoked, this warning message is generated to flag the apparent discrepancy in designer intent.

**What Next**

Determine whether an asynchronous reset of the TAP controller is desired and act accordingly. If you want the port with the jtag_trst attribute included in the BSR, remove this attribute using *remove_attribute jtag_trst port_name*.

---

## UIT-121

(error) Erroneous JTAG Manufacturer ID Number '%d'.

Feedback

### Description

The JTAG Device Identification Register (IDreg) is permanently loaded with a 32-bit number comprised of the JEDEC chip ID number, the design version number, and the manufacturer ID number.

Since the JTAG Device Identification register has a fixed size with four fixed-size fields, legal values for these fields must fall within the specified ranges. If DFT Compiler encounters an illegal value for the manufacturer ID field, this error message is generated along with error message UIT-124 specifying the erroneous value and the range of legal values.

### What Next

Enter a legal *integer* for *integer_in_range_0_to_2047* where *0* <= *integer* <= *2407* when executing *set_jtag_manufacturer_id* (see UIT-124).

---

## UIT-122

(error) Erroneous JTAG Part Number '%d'.

### Description

The JTAG Device Identification Register (IDreg) is permanently loaded with a 32-bit number comprised of the JEDEC chip ID number, the design version number, and the manufacturer ID number.

Since the JTAG Device Identification register has a fixed size with four fixed-size fields, legal values for these fields must fall within the specified ranges. If DFT Compiler encounters an illegal value for the part number field, this error message is generated along with error message UIT-124 specifying the erroneous value and the range of legal values.

### What Next

Enter a legal *integer* for *integer_in_range_0_to_65535* where *0* <= *integer* <= *65535* when executing *set_jtag_part_number* (see UIT-124).

---

## UIT-123

(error) Erroneous JTAG Version Number '%d'.

### Description

The JTAG Device Identification Register (IDreg) is permanently loaded with a 32-bit number comprised of the JEDEC chip ID number, the design version number, and the manufacturer ID number.

Feedback

Since the JTAG Device Identification register has a fixed size with four fixed-size fields, legal values for these fields must fall within the specified ranges. If DFT Compiler encounters an illegal value for the design version field, this error message is generated along with error message UIT-124 specifying the erroneous value and the range of legal values.

**What Next**

Enter a legal *integer* for *integer_in_range_0_to_15* where *0* <= *integer* <= *15* when executing *set_jtag_version_number* (see UIT-124).

---

## UIT-124

(information) \tNumber out of legal range.\n\ \t(Must be between %d and %d inclusively.)

**Description**

Since the JTAG Device Identification register has a fixed size with four fixed-size fields, legal values for these fields must fall within the specified ranges. Legal values for these ranges are specified as non-negative integers within the inclusive range specified in the error message.

**What Next**

Enter a legal *integer* for *integer_in_range_X_to_Y* where *X* <= *integer* <= *Y* when executing *set_jtag_version_number*, *set_jtag_part_number* or *set_jtag_manufacturer_id*. The erroneous field is identified by the previous error message (see UIT-121, UIT-122 or UIT-123).

---

## UIT-125

(warning) -cancel mode should be invoked with instruction name only.\n\ \t(Other arguments will be ignored.)

**Description**

Any non-mandatory instruction may be removed from the set of instructions "understood" by the synthesized JTAG instruction decode logic by executing *set_jtag_instruction instruction_name* [*-cancel*].

It is sufficient to supply *instruction_name* to identify which instruction to cancel, so the existence of other arguements to *set_jtag_instruction* is ignored when [*-cancel*] is included.

**What Next**

To eliminate this warning, remove the superfluous arguments when executing [*-cancel*].

Feedback

## UIT-126

(error) `00001111111' (127) is an illegal Manufacturer ID when\n\ \tused in JTAG ID registers.

### Description

The JTAG Device Identification Register (IDreg) is permanently loaded with a 32-bit number comprising the JEDEC chip ID number, the design version number, and the manufacturer ID number.

With respect to the manufacturer number, Rule 11.2.1(b) of the IEEE 1149.1 standard specifies that "00001111111 shall not be used in components that are otherwise compatible with this standard."

DFT Compiler generates this error message if this number is illegally defined using *set_jtag_manufacturer_id integer_in_range_0_to2047* [*list_of_designs*].

### What Next

Enter another manufacturer number using *set_jtag_manufacturer_id integer_in_range_0_to2047* [*list_of_designs*]. If you require (integer) 127 as the manufacturer ID number, enter (integer) 0 and manually rewire the code into the IDreg.

## UIT-127

(error) Direction of pin '%s' is erroneous.

### Description

You may want to control and access core test structures that DFT Compiler is unable to recognize automatically. For example, you may want to activate a RAM test, capture a response from that test structure, and shift it out through TDO.

To support such a scenario, two or three "hooks" are required:

```
enable_pin        (input) - the existence of an activation signal enabled
 by
                           the associated JTAG instruction
        load_pin  (input) - optional connection to TDI of the serial
 input
                           of the core test structure data register
        capture_pin         (output)        - a dedicated MUX input for
 selecting this
                           data register as the source of TDO output.
```

You need to select this latter MUX input when the associated JTAG instruction is active. Specify the existence of these "hooks" with the command *set_jtag_instruction instruction_name* [*-code code*] *-reg_coordinates capture_pin* [*load_pin*] *reg_enable*

Feedback

*enable_pin*. If you specify *capture_pin*, *load_pin* and *enable_pin*, the inputs will be connected to the appropriate synthesized JTAG instruction control logic.

Note that if you specified *-reg_coordinates*, the *capture_pin* is mandatory and the *load_pin* is optional. Test compiler checks that the specified pins are of the correct direction, and issues this error message if a pin has an erroneous direction.

If you do not specify the *enable_pin* and *capture_pin*, the result is a synthesis of all the necessary JTAG instruction decode and control logic as well as the creation of two new (bogus) ports connected to the design; an input port named "<*instruction_name*>CAPTURE" and an output port named "<*instruction_name*>ENABLE". The port <*instruction_name*>ENABLE is active HIGH when *instruction_name* is loaded into the JTAG Instruction Register (IR). You can then manually edit the circuit to connect these two (bogus) ports to the appropriate input and output pins of the target core test structure (thus "removing" them as ports).

**What Next**

If the specified pin direction is erroneous, then this is not the correct pin for this signal. Verify the design to identify to correct pin.

---

## UIT-128

(error) Port '%s' must have pad attribute set.

**Description**

The JTAG Boundary Scan Register (BSR) associates a register with each pad of a device which can be used to either `capture' (observe) the value at the pad and/or `apply' (control) a value at the pad distinct from the value applied externally or internally. These registers are serially chained to form the BSR from which their values may be serially set and read.

Since the BSR only includes design ports which will eventually have pads associated with them (that is, *not* module ports internal to the final top level design), DFT Compiler only considers ports which have been designated as requiring pads as candidates for inclusion in the BSR. At least one port must be specified as requiring a pad, since the BSR must have at least one element.

DFT Compiler will generate this error if a port processed by *set_jtag_port* (or any other command processing ports to be included in the BSR) does not have its *port_is_pad* attribute set.

**What Next**

Set the *port_is_pad* attribute using the *set_port_is_pad* command on each port to be considered for inclusion in the BSR. Note that this does not automatically mean that the

Feedback

port *will* be included in the BSR. For example, if a JTAG TAP port has the *port_is_pad* attribute set TRUE, it will nonetheless be excluded from the BSR.

One easy way to specify ports as pads is to execute *set_port_is_pad find (port,"*")* on the current_design.

---

## UIT-129

(warning) -default should be invoked alone.\n\ \t(Other arguments will be ignored.)

### Description

Many of the dc_shell JTAG commands include a *-default* argument which resets the specified JTAG settings to their default values. The *-default* argument overrides all other arguments so it should be invoked alone. This error message results from specification of *-default* with other arguments.

### What Next

Remove the conflicting arguments and reexecute the command.

---

## UIT-130

(error) No options specified for the "%s" command.\n\ \tFor available options, execute "%s -help".

### Description

You must specify at least one optional argument with this command. This error message is generated when no options are specified with the command, and when the command is invoked with the *-help* option.

### What Next

Invoke the command with the *-help* option to review the correct command syntax. Then, re-execute the command, using one of the optional arguments on the command line.

---

## UIT-131

(error) JTAG Port '%s' is not an input or bidirectional port.

### Description

Only input or bidirectional ports can take the */in* specifier. There are no input BSR cells associated with an output port. Thus, <*output_port_name*>*/in* is an illegal BSR cell identifier.

Feedback

In *set_jtag_port_routing_order*, the *ordered_list_of_annotated_BSR_port_names* contains members of four classes:

```
- <input_or_output_port_name>, which specifies
a BSR input or output cell (depending on the
direction of the port),
- <bidirectional_port_name>/in, which
specifies the BSR input cell associated with
this bidirectional port,
- <bidirectional_port_name>/out, which
specifies the BSR output cell associated with
this bidirectional port, or
- <tristate_or_bidirectional_port_name>/ctl,
which specifies the BSR three-state control
cell associated with this output or
bidirectional port.
```

Thus there are only three legal specifiers (`*/in*', `*/out*' or `*/ctl*') that can be appended to a *port_name* to identify a boundary scan register (BSR) cell. Each of these specifiers can only be associated with a subset of the port types (input, two-state output, three-state output and bidirectional ports) that can be included in the BSR.

DFT Compiler generates this message if you associate a non-existent boundary scan register (BSR) cell type (`*/in*', `*/out*' or `*/ctl*') with a port. DFT Compiler also generates UIT-138, which identifies the nature of the problem.

**What Next**

An illegal specifier for *output_port_name* indicates that a port of that type does not have a BSR input cell associated with it. Therefore, either delete this reference to a non-existent BSR input cell or correct the reference to an existing BSR cell by correcting *output_port_name*.

(See UIT-138)

---

## UIT-132

(error) JTAG Port '%s' is not an output or bidirectional port.

**Description**

Only output or bidirectional ports can take the */out* specifier. There are no output BSR cells associated with an input port. Thus, *<input_port_name>/out* is an illegal BSR cell identifier.

In *set_jtag_port_routing_order*, the *ordered_list_of_annotated_BSR_port_names* contains members of four classes:

```
- <input_or_output_port_name>, which specifies
a BSR input or output cell (depending on the
direction of the port),
```

Feedback

```
- <bidirectional_port_name>/in, which
specifies the BSR input cell associated with
this bidirectional port,
- <bidirectional_port_name>/out, which
specifies the BSR output cell associated with
this bidirectional port, or
- <tristate_or_bidirectional_port_name>/ctl,
which specifies the BSR three-state control
cell associated with this output or
bidirectional port.
```

Thus there are only three legal specifiers (`*/in*', `*/out*' or `*/ctl*') that can be appended to a *port_name* to identify a boundary scan register (BSR) cell. Each of these specifiers can only be associated with a subset of the port types (input, two-state output, three-state output and bidirectional ports) that can be included in the BSR.

DFT Compiler generates this message if you associate a non-existent boundary scan register (BSR) cell type (`*/in*', `*/out*' or `*/ctl*') with a port. DFT Compiler also generates UIT-138, which identifies the nature of the problem.

**What Next**

An illegal specifier for *input_port_name* indicates that a port of that type does not have a BSR output cell associated with it. Therefore, either delete this reference to a non-existent BSR output cell or correct the reference to an existing BSR cell by correcting *input_port_name*.

(See UIT-138)

---

## UIT-133

(error) JTAG Port '%s' is not three-stated.

**Description**

Only three-state ports can take the */ctl* specifier. There are no control BSR cells associated with an input or a two-state output port. Thus, *<two_state_input_or_output_port_name>/ctl* is an illegal BSR cell identifier.

In *set_jtag_port_routing_order*, the *ordered_list_of_annotated_BSR_port_names* contains members of four classes:

```
- <input_or_output_port_name>, which specifies
a BSR input or output cell (depending on the
direction of the port),
- <bidirectional_port_name>/in, which
specifies the BSR input cell associated with
this bidirectional port,
- <bidirectional_port_name>/out, which
specifies the BSR output cell associated with
```

Feedback

```
this bidirectional port, or
- <tristate_or_bidirectional_port_name>/ctl,
which specifies the BSR three-state control
cell associated with this output or
bidirectional port.
```

Thus there are only three legal specifiers (`*/in*', `*/out*' or `*/ctl*') that can be appended to a *port_name* to identify a boundary scan register (BSR) cell. Each of these specifiers can only be associated with a subset of the port types (input, two-state output, three-state output and bidirectional ports) that can be included in the BSR.

NOTE that even when a port is in fact three-stated, if the three-state is not visible at the top level of the core hierarchy, this error message may be generated and must be resolved. DFT Compiler determines whether a port is three-stated by checking for three-states at the top level of hierarchy. If it does not find them there, it assumes that the port is not three-stated and issues this error message when an operation requiring a three-stated port is invoked.

**What Next**

An illegal specifier for *two_state_port_name* indicates that a port of that type does not have a BSR control cell associated with it. Therefore, either delete this reference to a non-existent BSR control cell or correct the reference to an existing BSR cell by attaching the */ctl* specifier to the correct *three-state_or_bidirectional_port_name*.

If the offending port is in fact three-stated, you must regroup your core logic so that the three-state controlling this port is visible at the top level.

(See UIT-138)

---

## UIT-134

(error) Insufficient information included with user-specified\n \tinstruction '%s'. You must include either a register name or both\n \treg_coordinate(s) and reg_enable specifications.

**Description**

Each JTAG instruction activates a single target data register. The target register of a user-defined instruction is (by definition) not specified in the IEEE 1149.1 JTAG Standard. Therefore that target register cannot be inferred by DFT Compiler, and you must identify it by using either *-reg_name core_scan_chain_name* or *-reg_coordinates capture_pin* [*load_pin*] and *-reg-enable enable_pin*.

This message is also generated if you only partially identify the data register by using only one of the two options *-reg_coordinates capture_pin* [*load_pin*] or *-reg-enable enable_pin*. For complete identification of the target data register, you must specify both.

Feedback

**What Next**

Determine what information is missing and supply it.

If you want to specify an instruction and (optionally) the code for a specific internal scan chain, execute *set_jtag_instruction instruction_name -reg_name name* [*-code code*].

The *name* of each internal scan chain is the name of its parallel scan-in port; that is, the port which has the *test_scan_in* attribute assigned to it. This guarantees the uniqueness of names for these scan chains and provides DFT Compiler with a mechanism to identify each target scan chain.

If you want to specify some other core test structure, identify it by using *-reg_coordinates capture_pin* [*load_pin*] and *-reg-enable enable_pin*.

[See UIT-92]

---

## UIT-135

(error) Instruction '%s' has illegal JTAG instruction\n \tcode string '%s'.\n \t(The code string must contain only '0's and '1's.)

**Description**

JTAG instruction codes are loaded `as is' into the JTAG Instruction register. Therefore, they must be fully specified binary codes. Code cubes (which include `dont care' X bits) are not allowed. The only characters permitted in these code strings are '0's and '1's.

**What Next**

Re-write the instruction code string using only '0's and '1's.

---

## UIT-136

(error) Incompatible arguments. Specify either a register name or\n\ \tboth register coordinate(s) and register enable.

**Description**

Each JTAG instruction activates a single target data register. The target register of a user-defined instruction is (by definition) not specified in the IEEE 1149.1 JTAG Standard. Therefore that target register cannot be inferred by DFT Compiler, and you must identify it by using either *-reg_name core_scan_chain_name* or *-reg_coordinates capture_pin* [*load_pin*] and *-reg-enable enable_pin*.

Specifying both *-reg_name core_scan_chain_name* and *-reg_coordinates capture_pin* [*load_pin*] or *-reg-enable enable_pin* is not supported and generates this error message.

Feedback

### What Next

If you want to specify an instruction and (optionally) the code for a specific internal scan chain, execute *set_jtag_instruction instruction_name -reg_name name* [*-code code*].

The *name* of each internal scan chain is the name of its parallel scan-in port; that is, the port which has the *test_scan_in* attribute assigned to it. This guarantees the uniqueness of names for these scan chains and provides DFT Compiler with a mechanism to identify each target scan chain.

If you want to specify some other core test structure, identify it by using *-reg_coordinates capture_pin* [*load_pin*] and *-reg-enable enable_pin*.

---

## UIT-137

(error) Instruction Register (IR) must be between 2 and 32 bits wide.

### Description

The IEEE 1149.1 JTAG Standard mandates that the JTAG Instruction Register (IR) be at least two bits wide. DFT Compiler supports IRs of between 2 and 32 bits inclusively. There are three ways to set the size of the IR:

```
- You specify the size using insert_jtag
-ir_size bit_width,
- You force the IR size by specifying an
instruction code set_jtag_instruction
-code "code", or
- DFT Compiler automatically infers the
IR size from the set of specified and
mandatory instructions selected.
```

### POSSIBLE CAUSES

This error occurs if you have *not* surrounded your specified code in quotes (" "). In the absence of quotes, the command line parser interprets the first character as an independent string and assumes that a 1-bit code has been specified, forcing a 1-bit IR.

### What Next

Specify an IR width between 2 and 32 bits. If you forgot the quotes in your *set_jtag_instruction* command, re-issue the command with quotes around the code.

---

## UIT-138

(error) Non-existent BSR '%s' cell with routing\n \tposition '%d' associated with port '%s'.

Feedback

### Description

In *set_jtag_port_routing_order*, the *ordered_list_of_annotated_BSR_port_names* contains members of four classes:

```
- <input_or_output_port_name>, which specifies
a BSR input or output cell (depending on the
direction of the port),
- <bidirectional_port_name>/in, which
specifies the BSR input cell associated with
this bidirectional port,
- <bidirectional_port_name>/out, which
specifies the BSR output cell associated with
this bidirectional port, or
- <tristate_or_bidirectional_port_name>/ctl,
which specifies the BSR three-state control
cell associated with this output or
bidirectional port.
```

Thus there are only three legal specifiers (`*/in*', `*/out*' or `*/ctl*') that can be appended to a *port_name* to identify a boundary scan register (BSR) cell. Each of these specifiers can only be associated with a subset of the port types (input, two-state output, three-state output and bidirectional ports) that can be included in the BSR.

DFT Compiler generates this message if you associate a non-existent boundary scan register (BSR) cell type (`*/in*', `*/out*' or `*/ctl*') with a port. DFT Compiler also generates a message (UIT-131, UIT-132 or UIT-133) that tells you why the specifier you used is illegal.

### What Next

An illegal specifier for a port indicates that that type of port does not have that type of BSR cell associated with it. Therefore, either delete this reference to a non-existent BSR cell or correct the reference to an existing BSR cell by correcting the associated *port_name*.

(See UIT-131, UIT-132 or UIT-133)

---

## UIT-139

(error) -no_internal_scan invoked but instruction '%s' specifies core scan chain '%s'.

### Description

If you want to specify an instruction and (optionally) the code for a specific internal scan chain, execute *set_jtag_instruction instruction_name -reg_name name* [*-code code*].

The *name* of each internal scan chain is the name of its parallel scan-in port; that is, the port which has the *test_scan_in* attribute assigned to it. This guarantees the uniqueness of names for these scan chains and provides DFT Compiler with a mechanism for identifying each target scan chain.

Feedback

By default DFT Compiler will attach core test structures that it recognizes (e.g. internal scan chains) to the JTAG Test Access Port (TAP). Internal scan chains will *not* be connected to the TAP if you execute *insert_jtag -no_internal_scan*.

DFT Compiler generates this error message if *-no_internal_scan* has been invoked, but you have explicitly specified a JTAG instruction for one of the internal scan chains using *set_jtag_instruction instruction_name -reg_name name* [*-code code*].

**What Next**

Resolve the conflict. If you want to cancel the scan chain instruction, execute *set_jtag_instruction instruction_name -cancel*. Otherwise invoke *insert_jtag* without *-no_internal_scan*.

---

## UIT-140

(warning) The DFT Compiler specific JTAG instruction '%s'\n\ \tdoes not support register name or coordinate specification.\n \tSpecifications ignored.

**Description**

It is meaningless to specify *-reg_name*, *-reg_coordinates*, or *-reg_enable* with the *DAISY_CORE_SCAN* instruction. Doing so results in this error message.

The *DAISY_CORE_SCAN* instruction selects a single scan chain made up of all internal scan chains daisy-chained together. To assign a specific code to this instruction, issue the command *set_jtag_instruction DAISY_CORE_SCAN -code code*. This is equivalent to executing *insert_jtag -daisy_chained*, so you do not need to invoke *-daisy_chained* explicitly, UNLESS you want to specify the routing order of the serially daisy-chained scan chains.

**What Next**

Re-invoke *set_jtag_instruction DAISY_CORE_SCAN -code code* without the *-reg_name*, *-reg_coordinates*, and/or *-reg_enable* specifications.

---

## UIT-141

(error) Bad model scale factor, must be positive.

**Description**

For the command, set_model_scale, the scale value must be positive. You have entered a zero or negative number.

**What Next**

Reenter the command with a positive scale factor.

Feedback

## UIT-142

(error) Bad model load value, should not be negative.

**Description**

For the command, set_model_load, the load value cannot be negative. You have entered a negative number.

**What Next**

Reenter the command with a non-negative load value.

## UIT-143

(error) Bad model drive value, should not be negative.

**Description**

For the command, set_model_drive, the drive value cannot be negative. You have entered a negative number.

**What Next**

Reenter the command with a non-negative drive value.

## UIT-144

(warning) Port '%s/%s' is not an input or inout port; model_drive not set.

**Description**

The *set_model_drive* command works only on input or inout ports.

**What Next**

Examine the design to determine the correct ports for the command.

## UIT-145

(error) Erroneous specification of a list of ports since\n\ \tset_jtag_implementation has not been invoked with a Boundary Scan\n\ \tRegister (BSR) component.

**Description**

*set_jtag_implementation* only supports a *list_of_ports* when invoked with Boundary Scan Register (BSR) components (*BSRin*, *BSRout* or *BSRcontrol*), associating a JTAG component implementation of a BSR cell with specific ports in the BSR. A port list is

Feedback

meaningless with respect to the JTAG TAP controller, Bypass Register, Instruction Register or device Identification Register.

**What Next**

Resolve the inconsistency either by removing the list_of_ports, or by specifying a BSR cell component.

---

## UIT-146

(error) Three-state cell `%s`\n\ \t(technology library cell `%s`) is dont_touch.

**Description**

For successful JTAG synthesis, DFT Compiler requires that all three-state devices associated with circuit I/O be at the top level of the design. When DFT Compiler inserts boundary scan cells for three-state ports into the design, it determines their location by locating the data and control inputs to the associated generic three-state devices. In order to translate a three-state device into its generic equivalent, the original three-state device must *not* have its *dont_touch* attribute set *TRUE*. If it is ` set *TRUE*, this error message is generated.

There are two ways that a device can have its *dont_touch* attribute set *TRUE*; the user can explicitly set the attribute, or the technology library cell with which the device is implemented may specify that the device should not be touched.

**What Next**

Remove the *dont_touch* attribute associated with the offending three-state device.

---

## UIT-147

(error) Invalid sample percentage value '%d'.

**Description**

The *-sample* option of the command *create_test_patterns* specifies the percentage of faults to be considered for test generation.

**What Next**

The value of this option should be in the 0-100 range.

---

## UIT-149

(Information) The set_jtag_part_number command is overriding the\n\ \tenvironmental variable "jtag_part_number" value `%d'\n\ \tby `%d'.

Feedback

### Description

It is possible to specify the value of the 16-bit binary part number field of the JTAG ID register in two ways: using the *set_jtag_part_number* command or using the environmental variable *jtag_part_number*. This message indicates that the value set by *set_jtag_part_number* is overriding the default value set by *jtag_part_number*.

### What Next

Using environmental variables for those JTAG synthesis constraints that rarely change in your synthesis environment reduces the number of explicit JTAG synthesis commands you must invoke.

---

## UIT-150

(Information) The set_jtag_manufacturer_id command is overriding the\n\ \tenvironmental variable "jtag_manufacturer_id_number" value `%d'\n\ \tby `%d'.

### Description

There are two ways to specify the value of the 11-bit binary manufacturer ID field of the JTAG ID register: by using the *set_jtag_manufacturer_id* command, or by using the environmental variable *jtag_manufacturer_id_number*. This message indicates that the default value set by the latter is being overridden by the value specified by the former.

### What Next

To reduce the number of explicit JTAG synthesis commands you must invoke, use environmental variables for those JTAG synthesis constraints that rarely change in your synthesis environment.

---

## UIT-151

(Information) The set_jtag_version_number command is overriding the\n\ \tenvironmental variable "jtag_version_number" value `%d'\n\ \tby `%d'.

### Description

There are two ways to define the value of the 4-bit binary version number field of the JTAG ID register: by using the *set_jtag_version_number* command, or by using the environmental variable *jtag_version_number*. This message indicates that the default value set by the latter is being overridden by the value defined by the former.

### What Next

To reduce the number of explicit JTAG synthesis commands you must invoke, use environmental variables for those JTAG synthesis constraints that rarely change in your synthesis environment.

Feedback

## UIT-152a

(warning) The '%s' command is ignored. Either a '%s' license\n\t or '%s' license is required for this command.

### Description

You receive this warning message as a result of the *write_test* command. To run this command, you must have either the Test-Compiler license or the Test-IEEE-Std-1149-1 license.

### What Next

Verify that your installation is correct and that you are authorized for this license feature. If you are not authorized for this feature, contact Synopsys.

## UIT-152

(warning) The '%s' command is ignored. A '%s' license\n\t is required for this command.

### Description

Some dc_shell commands require specific licenses. This error may occur when the required feature is not authorized.

### What Next

Make sure your installation is correct and that you are supposed to be authorized for this license feature.

## UIT-153

(error) The sum of strobe time (%10.2f) and strobe width value (%10.2f) is required to be less than period value (%10.2f).

### Description

You have specified a strobe time and strobe width value such that the end of the strobe period stretches beyond the period.

### What Next

Check your strobe time specifications.

## UIT-154

(error) Unable to fault-simulate vectors because %s.

Feedback

### Description

The current vectors could not be fault-simulated, either because the the test protocol is not parallel-loadable, or because the *vdb* file was not created by a v3.1 or later version of ATPG.

During fault simulation of *vdb* files, scan operations are performed in a parallel-load fashion. Thus, fault simulation of *vdb* files is restricted to parallel-loadable protocols only. Currently, custom test protocols are NOT parallel-loadable. Initialization protocols read with the *read_init_protocol* command are acceptable.

Only *vdb* files written by v3.1 (or later) ATPG can be directly fault- simulated by TestSim; *vdb* files written by TestSim itself cannot be directly fault-simulated by TestSim.

### What Next

If you have a *vdb* file with vectors that are not parallel-loadable, you can use the *write_test* command to write these vectors out into a *tds* or a *wgl* vector format and then fault-simulate that *tds* or *wgl* file. The *wgl* format is preferable because its vector file is usually smaller in size. Note, however, that if you simulate a scan vector set in a serial format such as *tds* or *wgl*, then fault simulation of scan operations will be serial and therefore considerably slower.

If you got this message because your *vdb* file was created by a previous release version of ATPG, then you will need to create a new *vdb* file or write out the vectors to *tds* or *wgl* as suggested above.

---

## UIT-155

(error) write_test could not find the intermediate (SIF) file to be formatted.

### Description

Write_test formatter executable could not find the required input file (SIF file) to be formatted. The file may be deleted accidentally from the directory where this program is being run or it may not have been created due to errors such as not enough disc space to write vector file, occuring during write_test.

### What Next

Re-run write_test after ensuring that you have enough disk space

---

## UIT-156

(error) write_test failed to open temporary file to be used during vector formatting.

Feedback

**Description**

Write_test could not open temporary file(s) needed during the vector formatting process. This may be caused by lack of write permission to the directory, or failure due to insufficient disc space to write the file(s).

**What Next**

Make sure that sure that there is sufficient disc space, and that all the right permissions are set.

---

## UIT-157

(error) Insufficient memory, write_test failed to format vectors.

**Description**

Write_test ran out of run-time memory during vector formatting.

**What Next**

Ensure that sufficient memory is available.

---

## UIT-158

(error) internal error, write_test formatter executable (stran) failed due to syntax error in input (SIF) file.

**Description**

The write_test formatter failed due to syntax error in the intermediate (SIF) file used as its input. The file may have been corrupted, incompletely generated (e.g. due to insufficient disc space) or have been manually tampered with

**What Next**

Re-invoke write_test and notice any error and warning messages issued. Report any relevant messages to Synopsys.

---

## UIT-159

(error) internal error, unexpected end-of-file found in input file to the write_test formatter executable (stran).

Feedback

**Description**

The write_test formatter failed due to unexpected end-of-file found in the intermediate (SIF) file used as its input. The file may have been corrupted, incompletely generated (e.g. due to insufficient disc space) or have been manually tampered with

**What Next**

Ensure that sufficient disc space is available, and that the write_test command does not issue any error messages.

---

## UIT-160

(error) Unsupported vector format '%s' specified in the -format option.

**Description**

The vector format specified in the *-format* option of the *fault_simulate* command is not supported. Use one of the supported formats. The formats currently supported for the *fault_simulate* command are "sif", "tds", "vdb", "wgl", and "wif".

For the *prepare_testsim_vectors* command, all except the "vdb" format are supported. If you are using the "vdb" format, input this directly to the *fault_simulate* command without going through the *prepare_testsim_vectors* command.

**What Next**

Use one of the supported vector formats.

---

## UIT-163

(error) JTAG Port '%s' is not an input port.

**Description**

Only input ports can take the *clock* specifier. The JTAG circuitry associated with an *input* port can be optimized based on the signal type (*clock* or *data*) of that port. Ports that drive clock trees can be associated with specialized JTAG circuitry. An input port can be explicitly identified as carrying one of the "clock" or "data" categories.

DFT Compiler generates this message if you attempt to assign an output or bidirectional port the *clock* specifier.

The only legal output port specifiers are *enable* or *data*. (See error message UIT-164.)

**What Next**

Remove all output ports from the list of ports receiving the *clock* specifier. If you are trying to specify the type of an output port, select between *enable* or *data*.

Feedback

## UIT-164

(error) JTAG Port '%s' is not a two-state output port.

**Description**

Only two-state output ports can take the *enable* specifier.

When the three-state elements driving three-state and bidirectional *outputs* are found in the core logic of the IC (i.e. not embedded in the pads themselves) DFT Compiler requires that these three-state elements be located at the top hierarchy of the core of the design so that Boundary Scan Register (BSR) cells can be placed on the *inside* of the three-state elements in order to control and observe the data and enable lines. In these cases, only a single three-state line exits the core logic.

However, in the cases where the three-state controller for output signals will be embedded in the three-state output or bidirectional input/ouput pad associated with these ports of a design a pair of two state signals exit the core logic; a *data* line and a *enable* line. (It does not make sense for one of these ports to be a bidirectional port.) Unless, TC knows the distinction between these two types of exit lines, it will always assume that they are data lines and not insert the correct type of BSR cell on the enable lines. By specifying "enable" or "data" on a port the designer instructs TC on which type of BSR cell to place on these output ports.

DFT Compiler generates this message if you attempt to assign an input or bidirectional port the *enable* specifier, or if you have assigned the *enable* specifier to a three-state output.

The only legal input port specifiers are *clock* or *data*. (See error message UIT-163.)

**What Next**

Remove all input ports from the list of ports receiving the *enable* specifier. If you are trying to specify the type of an input port, select between *clock* or *data*.

## UIT-165

(error) %s time (%10.2f) must be less than strobe time (%10.2f).

**Description**

Data application time/delay must happen before strobe time

**What Next**

Change strobe time, and/or delay or bidirectional delay time such that data application time happens before strobe time, and rerun create_test_protocol. To set the strobe time

Feedback

use the test_default_strobe variable, and to set the bidirectional delay time use the test_default_bidir_delay variable.

---

## UIT-167

(error) Test program '%s' does not match the design '%s'.\n.

**Description**

This message indicates that the test program does not match the design. The design has probably been changed since the test program was written.

**What Next**

If you need to use this test program, you must change the design back to what it was when the test program was written out. Otherwise, the test program cannot be used and a new one will have to be generated.

---

## UIT-168

(error) Test program '%s' does not match the design '%s'. Could not get value for port '%s' in test program. This port is present in the design.

**Description**

This message indicates that the test program does not match the design. The design has probably been changed since the test program was written.

**What Next**

If you need to use this test program, you must change the design back to what it was when the test program was written out. Otherwise, the test program cannot be used and a new one will have to be generated.

---

## UIT-169

(Error) No port has been specified.

**Description**

The set_dft_signal command allows design objects to specify ports. The port specicification is mandatory.

**What Next**

Specify a port with the -port option and re-issue the command

Feedback

## UIT-170

(error) Value '%s' is not supported for the clock_gating option.\n \tUse either 'entire_design' or 'leaf_cell' as values.

**Description**

The *clock_gating* option currently supports the\n options *entire_design* and *leaf_cell*.

The default behavior if the *clock_gating* option is not specified is to do entire design clock gating.

**What Next**

Specify either *entire_design* and *leaf_cell* as values for the *clock_gating* option.

## UIT-171

(error) The test program sequence contained in the test program '%s'\n \tdoes not match the expected test program sequence.

**Description**

The test program sequence information contained in the test program specified in this error message is incorrect. It does not match the test program sequence information that was expected. The expected test program sequence information was based on the information in the test program that was input to this 'write_test' command.

This will happen when the original vdb file for the test program specified in this error message was overwritten by a later fault processor run.

When you lose or corrupt a vdb file for a test program, all test programs which depended on this test program (i.e. all test programs which were after it in the same sequence) are also invalidated.

**What Next**

Recreate the test program which is in error. You may have to also regenerate all test programs in the sequence following this test program.

## UIT-172

(information) Formatting of test vectors was successful for test program '%s'.

Feedback

### Description

You receive this message when you use the *-cummulative* option in a multi-pass test (*multi_pass_test_generation* set to *true*) after *write_test* successfully formats each test program.

### What Next

No action is required on your part.

---

## UIT-173

(warning) Formatting of test vectors failed for test program '%s'.

### Description

This message indicates that a serious error occured while *write_test* was processing your patterns. *write_test* has failed to save your patterns in the format you requested. Earlier messages will give you details of the specific problems encountered by *write_test*.

### What Next

Review previous error messages in the transcript and take the appropriate action based on those messages. See the man pages for the previous messages if you need more details on what action to take.

---

## UIT-174

(warning) You cannot use the -output option with the -cumulative option.\n \t The -output option will be ignored.

### Description

You receive this message when you attempt to use both the *-cummulative* and *-output* options in a multi-pass test (*multi_pass_test_generation* set to *true*). You cannot set both options because when you use the *-cummulative* option, you are writing out more than one file. The *-output* option specifies only one filename.

The *write_test* command ignores the *-output* option and will write out more than one file.

### What Next

To avoid this violation, change your script so you do not use the *-cummulative* and *-output* options simultaneously.

If you wish to format your patterns into a single file, you must use single pass test generation.

Feedback

## UIT-177

(warning) You cannot use the -all option as well as specify the object list with the set_test_unmask_fault command.\n \t The -all option wins. All faults in the entire design will be unmasked.

**Description**

You receive this message if specify both an object list and the *-all* option for the *set_test_unmask_fault* command. For example:

dc_shell> *set_test_unmask_fault U1 -all*

Object list and the *-all* option are mutually exclusive. The message warns you that the *set_test_unmask_fault* command has honored the *-all* option and ignored object list.

**What Next**

If you intend to unmask all faults, no action is required on your part. However, you can prevent future occurences of this error message by removing the object list.

If you intended to unmask only those faults specified by the object list, you need to re-establish fault masking on the faults you wish to mask. Use the *set_test_mask_fault* command to do this. Or, if you are using a script, you can correct and rerun the script.

## UIT-178

(error) You must either use the -all option or specify the object list with the set_test_unmask_fault command.

**Description**

You receive this message if you do not specify either an object list or the *-all* option. The following command will result in this error message:

dc_shell> *set_test_unmask_fault -sa1*

You must specify either the *-all* option or an object list to identify which masked faults you wish to unmask.

**What Next**

Issue the *set_test_unmask_fault* command with either an object list or the *-all* option specified.

## UIT-184

(error) JTAG instruction '%s' not yet declared, so it cannot be cancelled.

Feedback

**Description**

An instruction may not be deleted from the specification of the Boundary Scan Instruction Register instruction set, if it has not previously been added to that set.

In other words, *set_jtag_instruction <instruction_name> -cancel* may not be invoked if *set_jtag_instruction <instruction_name> has not been previously invoked.*

**What Next**

Remove this call to cancel the instruction from your script as it is effects no useful operation.

---

## UIT-186

(error) Illegal IDDQ fault model '%s'. Choose between 'shorts' and 'toggle'.

**Description**

The IDDQ fault model specified in the *-fault_model_iddq* option of the *fault_simulate* command is not supported. The IDDQ fault models currently supported are *shorts*, which targets transistor shorts and models 6 shorts per transistor; and *toggle*, which targets node toggle coverage.

**What Next**

Reissue *fault_simulate* and specify either of the valid fault models *shorts* or *toggle* for the *-fault_model_iddq* option.

---

## UIT-189

(error) Can't write output vdb file '%s'.

**Description**

This message indicates that the output *vdb* file cannot be written, because a file of the same name already exists and is protected from being overwritten.

**What Next**

Use a different name for the output test program, or remove the write protection from the existing *vdb* file so that it can be overwritten. Then re-execute.

---

## UIT-191

(error) '%s' is not a known clock domain restriction.

Feedback

**Description**

The set_scan_configuration command accepts a clock_domains option. This tells DFT Compiler how to define clock domains. Option arguments must be known restrictions.

**What Next**

Replace the option argument with a known restriction, for example "no_mix", "mix_edges", or "mix_clocks".

---

## UIT-192

(error) '%s' is not a known clock gating technique.

**Description**

The set_scan_configuration command accepts a clock_gating option. When a partial scan methodology and a multiplexed flip flop scan style is chosen, this tells DFT Compiler what technique to use to gate the clocks connected to non-scan cells. Option arguments must be known techniques.

**What Next**

Replace the option argument with a known technique, for example "entire_design", "superbuffer", or "leaf_cell".

---

## UIT-193

(error) Can not insert '%s' scan chains.

**Description**

The set_scan_configuration command accepts a chain_count option. This tells DFT Compiler the number of scan chains to implement in the testable design. Option arguments must be positive integers or "default" strings.

**What Next**

Repeat the set_scan_configuration command using a positive integer or the string "default".

---

## UIT-194

(error) '%s' is not a known bidirectional mode.

Feedback

### Description

The set_scan_configuration command accepts a dedicated_scan_ports option. This tells DFT Compiler whether to turn bidirectional ports inwards or outwards during scan. Arguments specified must be known bidirectional modes.

### What Next

Replace the option argument with a known bidirectional mode, for example "input" or "output".

---

## UIT-195

(warning) Arguments override the default script file name, but do not ask for a script. None is generated.

### Description

You receive this message because you are overriding a default script file name, but have not asked for a script file.

When you use *preview_dft* command with the *-script* option, DFT Compiler generates a script that completely specifies the scan design. If you do not specify the *-script* option, no script is generated.

The script file has a default name that you can override by specifying a script file name.

### What Next

Specify the *preview_dft* command with the *-script* option, if you want a script file. Otherwise, ignore the warning, or do request the script file name.

---

## UIT-196

(warning) '%s' is a %s and can not be used to specify a scan signal port.

### Description

The set_dft_signal command allows design objects to specify ports. These must be ports. Other kinds of design objects cannot be used to specify scan signal ports, and are rejected.

### What Next

Remove the object from the argument list to get rid of the warning. Use hierarchical instance names, with respect to the top-level design, to locate design objects.

Feedback

## UIT-197

(warning) '%s' is a %s and can not be used to specify a scan signal pin.

**Description**

The set_dft_signal command allows design objects to specify pins. These must be pins or pin instances. Other kinds of design objects cannot be used to specify scan signal ports, and are rejected.

**What Next**

Remove the object from the argument list to get rid of the warning. Use hierarchical instance names, with respect to the top-level design, to locate design objects.

## UIT-198

(error) '%s' is not a scan signal type.

**Description**

The *set_dft_signal* command lets you associate scan signal semantics with design ports. Specified signal types must be scan signal types in test_scan_enable, test_scan_enable_inverted, test_scan_clock, test_scan_clock_a, test_scan_clock_b, test_clock, test_scan_in, and test_scan_out. Other signal types do not have scan signal semantics and are rejected.

**What Next**

Repeat the *set_dft_signal* command using a scan signal type.

## UIT-199

(error) '%s' is not a signal type for the '%s' scan style.

**Description**

The set_dft_signal command lets you associate scan signal semantics with design ports. Specified signal types must be consistent with the specified scan style. Other signal types do not have scan signal semantics in the specified scan style and are rejected.

**What Next**

Repeat the set_dft_signal command using a scan signal type that has meaning in the specified scan style.

Feedback

## UIT-200

(warning) %s port '%s' can not be a %s.

**Description**

You can use the *set_dft_signal* command to put associated scan semantics with design ports. This warning is generated when you try to set the signal type of a port to something that is inconsistent with its direction. For example, signal type test_scan_out is associated with scan chain outputs. It cannot be associated with an input port.

**What Next**

Repeat the command using the correct port.

## UIT-201

(warning) did not complete set_dft_signal. '%s' is a %s, not a design port.

**Description**

You can use the set_dft_signal command to put test-specific signal types on design ports. This warning is generated when you try to set the signal type of an object that is not a design port.

**What Next**

Repeat the command using the correct port name.

## UIT-202

(warning) did not complete set_dft_signal. %s port '%s' can not be a %s.

**Description**

You can use the set_dft_signal command to put test-specific signal types on design ports. This warning is generated when you try to set the signal type of a port to something that is inconsistent with its direction. For example, signal type test_scan_out is associated with scan chain outputs. It cannot be associated with an in port.

**What Next**

Repeat the command using the correct port.

## UIT-203

(warning) %s pin '%s' can not be a %s.

Feedback

**Description**

You can use the set_dft_signal command to specify hookup pins for design ports that have scan semantics. This warning is generated when you try to set the hookup pin for a port to something that is inconsistent with its direction. For example, ports with signal type test_scan_out should be associated with output pins. They cannot be associated with input pins.

**What Next**

Repeat the command using the correct hookup pin.

---

## UIT-204

(error) There is no valid scan port.

**Description**

You use the set_dft_signal command to associate scan signal semantics with design ports. You see this error when nothing you specify is a valid design port.

**What Next**

Study accompanying messages for the reasons why specified ports were not acceptable, and repeat the command using valid ports.

---

## UIT-205

(error) No specified hookup pin is valid.

**Description**

You can use the set_dft_signal command to associate hookup pins with scan signal ports. You see this error when nothing you specify is a valid design pin.

**What Next**

Study accompanying messages for the reasons why the specified access pins were not acceptable. Repeat the command using valid pin instances.

---

## UIT-206

(warning) %d scan ports have been specified. Only the first will be processed.

**Description**

You use the set_dft_signal command to associate scan signal semantics with design ports. You see this error when you try to specify the signal type of multiple design ports using

Feedback

the same command. To simplify commands and eliminate ambiguity, separate commands must be used.

**What Next**

Split outstanding scan port specifications among multiple commands.

---

## UIT-207

(warning) %d hookup pins have been specified. Only the first will be processed.

**Description**

You can use the set_dft_signal command to associate hookup pins with scan signal ports. You see this warning when you try to associate more than one hookup pin with a design port. Only one can be specified.

**What Next**

If you want another hookup pin to be used, repeat the command specifying only that pin.

---

## UIT-208

(warning) There is no scan chain called '%s'. Specification is ignored.

**Description**

You can use the set_dft_signal command to associate scan chains with scan signals. You see this warning when you try to associate a non-existent chain with a scan signal. Chains must already have been specified using set_scan_path command.

**What Next**

Correct the chain name, if you've made a mistake, or specify the chain using the set_scan_path command before repeating the specification.

---

## UIT-209

(warning) %d scan ports have been specified, but access pins or scan chains have been specified also. Only the first scan port will be processed.

**Description**

You use the set_dft_signal command to associate scan signal semantics with design ports. You see this error when you try to specify the signal type of multiple design ports, and associate access pins of scan chains with them, using the same command. To simplify commands and eliminate ambiguity, separate commands should be used.

Feedback

### What Next

Split outstanding scan port specifications among multiple commands.

---

## UIT-210

(error) '%s' is not a known scan link type.

### Description

Use the set_scan_link command to declare scan links for the current design. Known scan link types include "wire" and "scan_out_lockup". You see this error when you specify an unknown scan link type.

### What Next

Repeat the specification using a known scan link type.

---

## UIT-211

(warning) '%s' is not a scan signal type. Discarding specified access point.

### Description

The *set_scan_group* command lets you specify access points with scan signal semantics. Specified signal types must be scan signal types in testdb_scan_enable, testdb_scan_enable_inverted, testdb_scan_clock, testdb_scan_clock_a, testdb_scan_clock_b, testdb_clock, testdb_scan_in, and testdb_scan_out. Other signal types do not have scan signal semantics and are rejected.

### What Next

Repeat the *set_scan_group* command using a scan signal type.

---

## UIT-212

(warning) '%s' does not identify a design pin. Discarding specified access point.

### Description

The *set_scan_group* command lets you specify access points with scan signal semantics. Specified access points must identify design pins. Other kinds of design objects cannot be used to specify access points, and are rejected.

### What Next

Remove the object from the argument list to get rid of the warning. Use hierarchical instance names, with respect to the top-level design, to locate design objects.

Feedback

## UIT-213

(warning) Access point specification is not paired. Last specification is incomplete and is discarded.

**Description**

The *set_scan_group* command lets you specify access points with scan signal semantics. Specified access points must be pairs, where each pair includes a signal type and a design port or pin. Incomplete pairs cannot be used to specify access points, and are rejected.

**What Next**

Remove the last object from the argument list to get rid of the warning.

## UIT-214

(error) Link name '%s' identifies a design cell. Link specification creates ambiguity and is discarded.

**Description**

The set_scan_link command lets you define scan links. Specified link names cannot identify design cells. Otherwise future scan chain specifications would be ambiguous.

**What Next**

Rename the scan link.

## UIT-215

(error) Segment name '%s' identifies a design cell. Specification creates ambiguity and is discarded.

**Description**

The *set_scan_group* command lets you define scan segments. Specified segment names cannot identify design cells. Otherwise future scan chain specifications would be ambiguous.

**What Next**

Rename the scan segment.

## UIT-216

(warning) %s pin '%s' can not be a %s.

Feedback

### Description

You can use the *set_scan_group* command to specify access pins for scan segment using scan semantics. This warning is generated when you try to set the access point for a segment to something that is inconsistent with its direction. For example, ports with signal type test_scan_out should be associated with output pins. They cannot be associated with input pins.

### What Next

Repeat the command using the correct access pin or signal type.

---

## UIT-217

(error) '%s' is not a signal type for the '%s' scan style.

### Description

The *set_scan_group* command lets you associate scan signal semantics with design ports and pins. Specified signal types must be consistent with the specified scan style. Other signal types do not have scan signal semantics in the specified scan style and are rejected.

### What Next

Repeat the set_dft_signal command using a scan signal type that has meaning in the specified scan style.

---

## UIT-218

(warning) Segment access list has multiple hookup points for signal type '%s'. All but the first is discarded.

### Description

The *set_scan_group* command lets you associate scan signal semantics with design ports or pins. You can associate at most one hookup point with each signal type. This message tells you that you have tried to associate more than one hookup point with a signal type, and that DFT Compiler is discarding all but the first.

### What Next

Repeat the *set_scan_group* command if you want to specify a different hookup point.

Feedback

## UIT-219

(warning) Segment access list has multiple signal types for hookup point '%s'. All but the first is discarded.

### Description

The *set_scan_group* command lets you associate scan signal semantics with design ports or pins. You can associate at most one signal type with each access point. This message tells you that you have tried to associate more than one signal type with an access point, and that DFT Compiler is discarding all but the first.

### What Next

Repeat the *set_scan_group* command if you want to specify a different signal type.

## UIT-220

(warning) Scan chain '%s' has scan style '%s' that differs from design scan style '%s'. Specification is ignored.

### Description

You can use the set_dft_signal command to associate scan chains with scan signals. You see this warning when you try to associate a chain that you have specified under one scan style with a scan signal you specify under another. Chains must have been specified using the same scan style.

### What Next

Change the scan style and reapply the set_dft_signal command, or reapply your earlier set_scan_path command using the new scan style.

## UIT-221

(warning) '%s' identifies multiple pins. Discarding specified access point.

### Description

The *set_scan_group* command lets you specify access points with scan signal semantics. Specified access points must identify one design pin. Wildcards cannot be used to specify access points, and are rejected.

### What Next

Remove the object from the argument list to get rid of the warning. Use hierarchical instance names, with respect to the top-level design, to locate design objects.

Feedback

## UIT-222

(warning) Segment '%s' does not have scan style '%s' and is not being added to the scan chain.

### Description

The set_scan_path command accepts an ordered list of design objects. These must be cells, instances, links or segments. DFT Compiler does not support scan chains with mixed scan styles. Segments must have the same scan style as the scan chain. You cannot add segments with other scan styles to a scan chain.

### What Next

If you really do want to add the segment to the chain, simply re-specify it using the same scan style as the chain.

## UIT-223

(warning) '%s' is not a cell, instance, scan segment, or scan link, and is not being added to the scan chain.

### Description

The set_scan_path command accepts an ordered list of design objects. These must be cells, instances, links or segments. Other kinds of design objects cannot be added to scan chains, and are rejected.

### What Next

Remove the object from the argument list to get rid of the warning. Use hierarchical instance names, with respect to the top-level design, to locate design objects.

## UIT-224

(warning) write_test_max_scan_patterns value specified (%d) ignored.

### Description

Illegal value specified for this variable: the value can not be a negative or a 1. The value specified is ignored.

### What Next

Use a value of greater than 1 for this variable

Feedback

## UIT-225

(warning) write_test_max_cycles value specified (%d) ignored.

**Description**

Illegal value specified for this variable: the value can not be a negative. The value specified is ignored.

**What Next**

Use a positive value for this variable

## UIT-226

(warning) Scan chain element '%s' is a scan link\n\tand is not being added to the scan chain.

**Description**

The *set_scan_path* command accepts an ordered list of design objects. The first object in the list can not be a scan link because there is no preceding cell to associate it with.

**What Next**

Remove the object from the argument list to get rid of the warning.

## UIT-227

(warning) Scan style '%s' does not add lockup latches to the scan chain.\n \tIgnoring the set_scan_configuration -add_lockup command.

**Description**

Only the multiplexed_flip_flop scan style adds lockup latches to the scan chain.

This message tells you that you have asked DFT Compiler to add lockup latches to the scan chain, but have specified a scan style that is not multiplexed_flip_flop. DFT Compiler is telling you that it is ignoring your *-add_lockup* option.

**What Next**

To get rid of the warning, remove the *-add_lockup true* from the *set_scan_configuration* option list, or set the scan style to multiplexed_flip_flop.

## UIT-228

(warning) Ignoring preview_dft -show option argument '%s'.

Feedback

**Description**

The *preview_dft* command has a *-show* option, that gives users the flexibility they need to customize what they see.

The option takes a list argument, containing case-insensitive keywords in "cells", "scan", "segments", "scan_clocks", "scan_signals", and "all".

This message says that you have used a *-show* option argument that is not a keyword. DFT Compiler ignores the specification.

**What Next**

To get rid of the warning, remove the unknown keyword from the *preview_dft -show* option argument list.

---

## UIT-229

(warning) Scan style '%s' does not need scan clock domain constraints. Ignoring the set_scan_configuration -clock_mixing command.

**Description**

Only the multiplexed_flip_flop scan style needs scan clock domain constraints.

This message tells you that you have asked DFT Compiler to implement scan clock domain constraints, but have specified a scan style that is not multiplexed_flip_flop. DFT Compiler is telling you that it is ignoring your *-clock_mixing* option.

**What Next**

To get rid of the warning, remove the *-clock_mixing* from the *set_scan_configuration* option list, or set the scan style to multiplexed_flip_flop.

---

## UIT-230

(information) To support new scan insertion capabilities, this command is being replaced by the set_scan_path command, which is used with the insert_scan command. Please see the DFT Compiler Scan Synthesis User Guide for more information about set_scan_path and insert_scan.

**Description**

To support new scan insertion capabilities, the *insert_test* command is being replaced by *insert_scan*.

The *set_scan_chain* command is not recognized by *insert_scan*. The *set_scan_path* command is used instead.

Feedback

DFT Compiler continues to support *set_scan_chain* and *insert_test* for a transition period, so legacy *dc_shell* scripts continue to work.

We believe that *set_scan_path* and *insert_scan* supports all *set_scan_chain* and *insert_test* features, and ultimately intend obsoleting *set_scan_chain* and *insert_test*.

**What Next**

See the DFT Compiler Scan Synthesis User Guide for more information about *set_scan_path*, *insert_scan*, and the hierarchical scan user interface. Let Synopsys know about any *set_scan_chain* or *insert_test* feature that you rely on, and is not supported by *set_scan_path* and *insert_scan*.

---

## UIT-231

(Error) The options -chain and -cell cannot be specified at the same time.

**Description**

This message indicates that a report_scan_path command has been issued, but the two options -chain and -cell cannot be specified at the same time. You choose either to report the scan signal information for the scan chains with the -chain option or to report the cells of the scan chains with the -cell option.

**What Next**

Remove one of the options (-chain or -cell) and reissue the command.

---

## UIT-232

(warning) Coverage threshold option ignored. Coverage threshold must be greater than 0%%.

**Description**

The specified coverage threshold for fault_simulate was ignored because it was less than or equal to 0%.

**What Next**

The coverage threshold must be greater than 0% and less than or equal to 100%.

---

## UIT-233

(warning) Coverage threshold option ignored. Coverage threshold must be less than or equal to 100%%.

Feedback

### Description

The specified coverage threshold for fault_simulate was ignored because it was greater than 100%.

### What Next

The coverage threshold must be greater than 0% and less than or equal to 100%.

---

## UIT-235

(error) cannot generate TSTL2 format, too many timing types.

### Description

All the inputs with same delay are grouped into one input timing type. All the outputs with same delay (or strobe time) are grouped into one output timing type. TSTL2 format allows a maximum of seven (6 + 1 dedicated for DT waveform) types of input timing and two types of output timing. One TIMESET statement is generated for each timing type. Input to the write_test requires it to generate more than allowed TIMESET statements.

### What Next

To generate TSTL2 format, change the timing values on inputs and/or outputs to reduce the number of timing types. Read TIMESET statement section of TSTL2 specifications for detailed information.

---

## UIT-236

(warning) You have invoked preview_dft with -script and one or more of the following options: -show, -test_points, -test_wrapper, -core_integration, -mbist, -bsd, -test_mode, -verbose. These options are ignored.

### Description

The *preview_dft* command option *-script* is mutually exclusive to the following options *-show*, *-test_points*, *-test_wrapper*, *-core_integration*, *-mbist*, *-bsd*, *-test_mode*, and *-verbose*. This message says that you have asked for one or more of the options. DFT Compiler ignores the additional option(s) and generates a script.

### What Next

To get rid of the warning, invoke *preview_dft* with either the *-show* option and/or the following options: *-test_points*, *-test_wrapper*, *-core_integration*, *-mbist*, *-bsd*, *-test_mode*, *-verbose* OR just the *-script* option, but not both.

Feedback

## UIT-237

(information) You have invoked remove_scan_configuration with no options. The command is ignored.

**Description**

The *remove_scan_specification* command does nothing when it is invoked without options.

This, in itself, is not a problem. No scan specifications are removed. You might, however, have meant to use the "-all" option or one of the other more limited options.

**What Next**

To get rid of the information message, invoke *remove_scan_configuration* with options.

## UIT-238

(information) To support new scan insertion capabilities, this command is being replaced by the insert_scan command. Please see the DFT Compiler Scan Synthesis User Guide for more information about insert_scan.

**Description**

To support new scan insertion capabilities, the *insert_test* command is being replaced by *insert_scan*.

DFT Compiler continues to support both commands for a transition period, so legacy *dc_shell* scripts continue to work.

*insert_scan* supports all *insert_test* features and ultimately intend obsoleting *insert_test*.

**What Next**

See the DFT Compiler Scan Synthesis User Guide for more information about *insert_scan* and the hierarchical scan user interface. Let Synopsys know about any *insert_test* feature that you rely on, and is not supported by *insert_scan*.

## UIT-239

(information) To support new scan insertion capabilities, this command is being replaced by the set_scan_path command, which is used with the insert_scan command. Please see the DFT Compiler Scan Synthesis User Guide for more information about set_scan_path and insert_scan.

Feedback

**Description**

To support new scan insertion capabilities, the *insert_test* command is being replaced by *insert_scan*.

The *set_test_routing_order* command is not recognized by *insert_scan*. The *set_scan_path* command is used instead.

DFT Compiler continues to support *set_test_routing_order* and *insert_test* for a transition period, so legacy *dc_shell* scripts continue to work.

*set_scan_path* and *insert_scan* support all *set_test_routing_order* and *insert_test* features, and ultimately intend obsoleting *set_test_routing_order* and *insert_test*.

**What Next**

See the DFT Compiler Scan Synthesis User Guide for more information about *set_scan_path*, *insert_scan*, and the hierarchical scan user interface. Let Synopsys know about any *set_test_routing_order* or *insert_test* feature that you rely on, and is not supported by *set_scan_path* and *insert_scan*.

---

## UIT-240

(information) Some design changes may have occurred after "create_test_patterns" or "create_test_patterns" was run in a different dc_shell session.

**Description**

write_test prints this message if it detects any design changes after "create_test_patterns" or if "create_test_patterns" was run in a different dc_shell session. Make sure that any changes made to the design after the test vectors were generated do not invalidate the patterns. This can cause simulation mismatches.

**What Next**

If there are any design changes that invalidate the patterns, please rerun "create_test_patterns".

---

## UIT-241

(warning) Ignoring the -complete true option because scan chain '%s' is empty.

**Description**

Specify scan chains using the *set_scan_path* command. Tell DFT Compiler to not add elements to your scan chain by setting the *-complete* option to *true*.

This message tells you that DFT Compiler ignores a *-complete true* option because it is applied to an empty scan chain.

Feedback

### What Next

Remove the option from the command line or specify valid scan chain elements to get rid of the warning. If you did specify chain elements, study accompanying warning messages to see why they were discarded.

---

## UIT-242

(information) The %s attribute on the library cell %s now has priority over this attribute on an instance, reference or design.

### Description

You receive this message when you perform *set_scan_element* or *set_scan_transparent* on a library cell. The message indicates that the command has been executed (and the specified attribute is now present on the library cell), but warns you that this attribute now has precedence and so cannot be overwritten by performing *set_scan_element* or *set_scan_transparent* on an instance, reference or design.

### What Next

If it is acceptable to you that the library cell attribute takes precedence over the attribute on an instance, reference or design, no action on your part is required. If this is not acceptable, remove the attribute from the library cell using *set_scan_element -false*, *set_scan_transparent -false*, or *remove_attribute*.

---

## UIT-243

(warning) The design %s corresponding to the object %s has the %s\n attribute set to true. This specification will not have any effect.

### Description

The value of the scan element or scan transparent attribute on a design has precedence over the the value of the scan element or scan transparent attribute on an instance or a reference. If the scan element or the scan transparent attribute is set on an instance or a reference, and the design corresponding to this instance or reference has the scan element or scan transparent attribute set, the specification does not have any effect because of the precedence rules.

### What Next

Remove the attribute on the design if you want to see the effect of the scan element or scan transparent attribute on an instance or reference.

Feedback

## UIT-244

(warning) The design %s corresponding to the object %s has the %s\n attribute set to true. This specification will not have any effect for the level-\n sensitive elements inside %s.

### Description

The value of the scan element or scan transparent attribute on a design has precedence over the the value of the scan element or scan transparent attribute on an instance or a reference. If the scan element or the scan transparent attribute is set on an instance or a reference, and the design corresponding to this instance or reference has the scan element or scan transparent attribute set, the specification does not have any effect because of the precedence rules.

### What Next

Remove the attribute on the design if you want to see the effect of the scan element or scan transparent attribute on an instance or reference.

## UIT-245

(warning) Deleting the scan transparent attribute on object %s.

### Description

If the value of the *scan_element* attribute on a design object is true, it implies that all flip-flops and all latches inside that design object, that do not have violations, must be made a part of the scan chain. Therefore, setting the *scan_element* attribute value on a design object that already has the scan transparent value set to true causes the scan transparent attribute to be deleted.

### What Next

If you do not want the scan transparent attribute deleted, use the *set_scan_element command to set the scan_element attribute false. To reapply the scan transparent attribute, use the set_scan_transparent command.*

## UIT-246

(error) Compliance enable specification is not paired.

### Description

The *set_bsd_compliance* command lets you specify a compliance enable pattern for a boundary scan design. Each compliance enable pattern consists of a set of port_name value pairs. Incomplete pairs cannot be used to specify a compliance enable pattern and are rejected.

Feedback

**What Next**

Make sure that the compliance enable pattern has port_name value pairs.

---

## UIT-247

(error) '%s' is not a port of the design.

**Description**

The *set_bsd_compliance* command lets you specify a compliance enable pattern for a boundary scan design. Each compliance enable pattern consists of a set of port_name value pairs. One of the port names specified in the compliance enable pattern does not correspond to a design port.

**What Next**

Change the compliance pattern specification such that all port names appearing in a compliance pattern are ports of the design.

---

## UIT-248

(error) Multiple design objects found with name '%s'.

**Description**

The *set_bsd_compliance* command lets you specify a compliance enable pattern for a Boundary Scan Design. Each compliance enable pattern consists of a set of port_name-value pairs. One of the port names specified in the compliance enable pattern corresponds to multiple design objects.

**What Next**

Re-issue the *set_bsd_compliance* command making sure that each port name in a compliance enable pattern corresponds to a unique design object.

---

## UIT-249

(error) The value specified at a compliance enable port must be either a '0' or a '1'.

**Description**

The *set_bsd_compliance* command lets you specify a compliance enable pattern for a boundary scan design. Each compliance enable pattern consists of a set of port_name value pairs. The value at a compliance enable port must be either a logic zero or a logic one.

Feedback

**What Next**

Make sure that the value at each compliance enable port in a compliance enable pattern is either a logic zero or a logic one.

---

## UIT-250

(error) The following compliance enable port is not an input port: %s.

**Description**

The *set_bsd_compliance* command lets you specify a compliance enable pattern for a boundary scan design. Each compliance enable pattern consists of a set of port_name value pairs. Each port name in a compliance enable specification must correspond to an input port of the design.

**What Next**

Make sure that each port in the compliance enable pattern corresponds to an input port of the design.

---

## UIT-251

(error) The following compliance enable port is a TAP port: %s.

**Description**

The *set_bsd_compliance* command lets you specify a compliance enable pattern for a boundary scan design. Each compliance enable pattern consists of a set of port_name value pairs. No compliance enable pattern may include a TAP port.

**What Next**

Remove the TAP ports from the compliance pattern specification.

---

## UIT-252

(warning) No TAP ports were found for the design.

**Description**

The *set_bsd_compliance* command issues an error if any compliance enable port is a TAP port. You get this warning if this check is not performed because no IEEE 1149.1 Test Access Ports were found for the design.

**What Next**

No further action is necessary.

Feedback

## UIT-253

(error) The following port appears more than once in a\n \tcompliance enable pattern: '%s'.

**Description**

The *set_bsd_compliance* command lets you specify a compliance enable pattern for a boundary scan design. Each compliance enable pattern consists of a set of port_name value pairs. No port can appear more than once in a compliance enable specification.

**What Next**

Make sure that the compliance enable pattern has no duplicate ports.

## UIT-254

(information) Writing '%s' package description.

**Description**

The *read_pin_map* command lets you read port-to-pin map files for Boundary Scan Designs. The *read_pin_map* command produces this information message when the user reads in a port-to-pin map file.

**What Next**

No further action is necessary.

## UIT-255

(information) Overwriting '%s' package description.

**Description**

The *read_pin_map* command lets you read port-to-pin map files for Boundary Scan Designs. If a port-to-pin map file with some package name has been read in and the user tries to read in another port-to-pin map file with the same package name, the *read_pin_map* command produces this information message.

**What Next**

No further action is necessary.

## UIT-256

(information) Deleting '%s' package description.

Feedback

**Description**

The *remove_pin_map* command lets you delete a package description for a Boundary Scan Design. The *remove_pin_map* command produces this information message when the user tries to delete a package description.

**What Next**

No further action is necessary.

---

## UIT-257

(error) Could not find package '%s'.

**Description**

The *remove_pin_map* command lets you delete a package description for a Boundary Scan Design. The *remove_pin_map* command produces this information message when the package name specified by the user has not been read in.

**What Next**

Check the packages read in for the design using the *report_pin_map* command and reissue the *remove_pin_map* command with a valid package name.

---

## UIT-258

(error) Naming check must be one of 'VHDL', 'BSDL' or 'none'.

**Description**

The only valid values for the naming_check argument to *write_bsdl* are "VHDL", "BSDL" or "none". This error message indicates that the naming_check argument has some value other than these.

**What Next**

Make sure that the value of the naming check argument is valid.

---

## UIT-259

(error) Unable to open output file '%s' for writing.

**Description**

This error message occurs when the output file cannot be opened for writing.

Feedback

**What Next**

Make sure that the output file and the directory have write permissions and run the command again.

---

## UIT-260

(information) Opened BSDL file '%s' for writing.

**Description**

This information message indicates that the BSDL generation subsystem has successfully opened the output file for writing.

**What Next**

No further action is necessary.

---

## UIT-261

(error) Cannot read file '%s'.

**Description**

This error appears when you try to read a protected pin map file or the file does not exist.

**What Next**

Check and correct search_path, then reinvoke the *read_pin_map* command.

---

## UIT-262

(error) File '%s' is a directory name.

**Description**

You receive this message from the *read_pin_map* command if the argument you issued with that command is a directory name and not a filename. The argument of *read_pin_map* must be the full name of a file containing the port-to-pin mapping for your design.

**What Next**

Verify the full pathname of your port-to-pin mapping file, then reissue *read_pin_map* using the complete, correct filename as the argument.

Feedback

## UIT-263

(warning) %d ports have been specified. Only the first will be processed.

**Description**

You use the set_dft_signal command to associate test mode signal semantics with exactly one design port. You see this error when you try to specify the test mode of multiple design ports using the set_dft_signal command.

**What Next**

Ensure that you want to associate test mode semantics with the first design port. Remove additional specifications to silence the warning.

## UIT-264

(warning) %s port '%s' can not be a %s.

**Description**

You can use the set_dft_signal command to associate test mode signal semantics with exactly one design input port. This warning is generated when you apply the set_dft_signal command to a port that is not an input port.

**What Next**

Repeat the command using the correct port.

## UIT-265

(error) There is no valid test mode port.

**Description**

You use the set_dft_signal command to associate test mode signal semantics with exactly one design input port. You see this error when nothing you specify identifies a design input port.

**What Next**

Study accompanying messages for the reasons why the port you specified was not acceptable, and repeat the command using a valid input port.

## UIT-266

(error) '%s' is not a test mode signal type.

Feedback

**Description**

The set_dft_signal command lets you associate test mode signal semantics with exactly one design input port. Specified signal types must be test mode signal types in test_hold_logic_zero and test_hold_logic_one. Other signal types do not have test mode signal semantics and are rejected.

**What Next**

Repeat the set_dft_signal command using a test mode signal type.

---

## UIT-267

(warning) Port '%s' has a '%s' signal_type attribute. This may conflict with '%s' signal semantics.

**Description**

The set_dft_signal command lets you associate signal semantics with design ports. This message tells you that a port has a signal_type attribute that may conflict with these semantics.

**What Next**

Satisfy yourself that there is no conflict.

---

## UIT-268

(warning) Port '%s' has a test_hold '%s' attribute. This conflicts with '%s' signal semantics.

**Description**

The set_dft_signal command lets you associate signal semantics with exactly one design input port. This message tells you that the port has a test_hold attribute that conflicts with these semantics.

**What Next**

Satisfy yourself that there is no conflict.

---

## UIT-269

(error) The type of a Test Access Port must be one\n \tof %s, %s, %s, %s, or %s.

**Description**

You can use the *set_bsd_port* command to put test-specific signal types on design ports. The only valid signal types for 1149.1 TAP ports are tdi, tdo, tms, trst, or tck. This error

Feedback

message appears when you try set the signal type of a TAP port to a value other than tdi, tdo, tms, trst, or tck.

**What Next**

Repeat the command using the correct TAP port type.

---

## UIT-270

(error) More than one port specified with the set_1149.1_port command.

**Description**

You can use the *set_bsd_port* command to put test-specific signal types on a design port. This error occurs when you try to perform *set_bsd_port* on a set of ports.

**What Next**

Specify exactly one port when using the *set_bsd_port* command.

---

## UIT-271

(error) Did not complete set_bsd_port. '%s' is a %s, not a design port.

**Description**

You can use the *set_bsd_port* command to put test-specific signal types on design ports. This error is generated when you try to set the signal type of an object that is not a design port.

**What Next**

Repeat the command using the correct port name.

---

## UIT-272

(error) Did not complete set_bsd_port. %s port '%s' can not be a %s.

**Description**

You can use the *set_bsd_port* command to put test-specific signal types on design ports. This error is generated when you try to set the signal type of a port to something that is inconsistent with its direction. For example, signal type tdo is associated with boundary scan test data output. It cannot be associated with an input port.

**What Next**

Repeat the command using the correct port.

Feedback

## UIT-273

(error) A port, '%s', with a signal type of '%s' already exists in the design.

**Description**

You can use the *set_bsd_port* command to put test-specific signal types on a design port. This error message is produced if a port with the given signal type already exists in the design.

**What Next**

Remove the signal_type attribute from the existing port and reissue the command.

## UIT-274

(error) The '%s' port is a system clock.

**Description**

You can use the *set_bsd_port* command to put test-specific signal types on a design port. This error message is produced if you try to declare a system clock port as the TDI, TDO, TMS or TRST pin of the TAP.

**What Next**

Reissue the *set_bsd_port* command specifying a non-clock port as the TDI, TDO, TMS or TRST pin of the TAP.

## UIT-275

(error) The '%s' port is a test clock.

**Description**

You can use the *set_bsd_port* command to put test-specific signal types on a design port. This error message is produced if you try to declare a test clock port as the TDI, TDO, TMS or TRST pin of the TAP.

**What Next**

Reissue the *set_bsd_port* command specifying a port that is not a test clock as the TDI, TDO, TMS or TRST pin of the TAP.

## UIT-276

(Warning) Overwriting the signal type attribute for the port '%s'.

Feedback

**Description**

You can use the *set_bsd_port* command to put test-specific signal types on a design port. This warning message is produced if you try to perform *set_bsd_port* on a port that already has a signal type attribute.

**What Next**

No further action is necessary.

---

## UIT-277

(error) The '<signal_port_bit_value_pairs>' list has no members.

**Description**

The *set_bsd_compliance* command lets you specify a compliance enable pattern for a Boundary Scan design. The signal_port-bit_value list in a *set_bsd_compliance* cannot be empty.

**What Next**

Reissue the command with a nonempty list of signal_port-bit_value pairs.

---

## UIT-278

(information) You invoked remove_bsd_specification with no\n \toptions. The command is ignored.

**Description**

The *remove_bsd_specification* command does nothing when it is invoked without options.

This, in itself, is not a problem. No IEEE 1149.1 specifications are removed. However, you might have meant to use the *-all* option or one of the other more limited options.

**What Next**

To get rid of the information message, invoke *remove_bsd_specification* with options.

---

## UIT-279

(warning) The sense value '%s' is ignored.

**Description**

In command *set_scan_signal*, the option *-sense* is applicable only if a hookup pin has been specified. If there is no hookup pin, the sense value is then ignored.

Feedback

**What Next**

If you want to specify a hookup pin, use the option *-hookup*.

---

## UIT-280

(error) '%s' is not a correct hookup sense type.

**Description**

The *set_dft_signal* command lets you associate scan signal semantics with design ports. With respect to a specific design port, if a hookup pin also is specified, *insert_dft* connects wires to the hookup pin. The sense between the port and the hookup pin is specified by the option of *-sense*. The valid values are *non_inverted* and *inverted*, and the default value is *non_inverted*.

**What Next**

Repeat the command using a correct hookup sense value.

---

## UIT-281

(information) Setting the default package for the design '%s'.

**Description**

The *-default_package* switch of the *set_bsd_configuration* command lets you set the the default package for a Boundary Scan Design. You get this information message when you invoke the *set_bsd_configuration* command to set the default package for a Boundary Scan Design.

**What Next**

No further action is necessary.

---

## UIT-282

(information) Overwriting the default package for the design '%s'.

**Description**

The *-default_package* switch of the *set_bsd_configuration* command lets you set the the default package for a Boundary Scan Design. If the default package for a Boundary Scan Design has already been set, and you invoke the *set_bsd_configuration* command to set the default package for the Boundary Scan Design, you get this information message.

Feedback

### What Next

No further action is needed.

---

## UIT-283

(warning) Signature is not supported for '%s' instruction. Ignored the signature specification.

### Description

This error message occurs when you specify a signature for another instruction. The *set_bsd_instruction* command allows you to set the signature for the RUNBIST instruction.

### What Next

The signature specification is ignored for this instruction. To avoid this warning, remove the signature specification.

---

## UIT-284

(error) The signature for the RUNBIST instruction is invalid.

### Description

The *set_bsd_instruction* command allows you to set the parameters for the RUNBIST instructions. This error message indicates that you either ignored signature specification for RUNBIST or specified a RUNBIST signature that contains values other than a logic zero or a logic one.

### What Next

Make sure you specify signature if time or clock cycle is specified for RUNBIST instruction. Also make sure that the signature specified for the RUNBIST instruction is a stream of logic ones and zeros of a length equal to the length of the test data register selected by the RUNBIST instruction.

---

## UIT-285

(error) The wait specification for the RUNBIST/INTEST instruction is missing.

### Description

This error message occurs when you have not specified the duration required for the RUNBIST or INTEST instruction to complete. The *set_bsd_instruction* command allows you to set the timing parameters for the RUNBIST or INTEST instructions.

Feedback

### What Next

Specify the wait duration required for the RUNBIST or INTEST instructions to complete using either the *-time* or the *-clock_cycles* options.

---

## UIT-286

(error) The time specified for RUNBIST/INTEST\n \tto complete must be a positive real number.

### Description

This error message occurs when you specify a nonpositive duration for the RUNBIST or INTEST instructions to complete. The *set_bsd_instruction* command allow you to set the parameters for the RUNBIST or INTEST instructions

### What Next

Specify a positive wait duration for the RUNBIST or INTEST execution and run the the command again.

---

## UIT-287

(error) The clock_port num_cycles list in the wait specification\n \tfor the RUNBIST/INTEST instruction is empty.

### Description

This error message occurs when the the argument passed using the *-clock_cycles* option contains no members. The *set_bsd_instruction* command allows you to set the parameters for the RUNBIST and INTEST instructions. One way to specify the RUNBIST or INTEST duration is to provide a list of clock_port integer pairs using the *-clock_cycles* option for these instructions.

### What Next

Make sure that the argument passed using the *-clock_cycles* option has a positive, even number of members and run the command again.

---

## UIT-288

(error) The RUNBIST/INTEST execution time specification is not paired.

### Description

This error message occurs when the argument passed using the *-clock_cycles* option contains an odd number of members. The *set_bsd_instruction* command allows you to

Feedback

set the parameters for the RUNBIST and INTEST instructions. One way to specify the RUNBIST or INTEST duration is to provide a list of clock_port integer pairs using the *-clock_cycles* option for these instructions.

**What Next**

Make sure that the argument specified with the *-clock_cycles* option contains an even number of members.

---

## UIT-289

(error) The number of clock cycles in the wait specification must be a positive integer.

**Description**

This error message occurs when you have specified a non-integer or a non-positive integer with the *-clock_cycles* option of the *set_bsd_instruction* command. The *set_bsd_instruction* command allows you to set the parameters for the EXTEST_PULSE, EXTEST_TRAIN, RUNBIST, and INTEST instructions. The number of clock cycles specified using the *-clock_cycles* option for this command must be a positive integer.

**What Next**

Make sure that the number of clock cycles specified via the *-clock_cycles* option is a positive integer.

---

## UIT-290

(error) Could not find port '%s' in the design '%s'.

**Description**

This error message occurs when a port name specified using the *-clock_cycles* option of the previous commands is not a port of the design.

The *set_bsd_instruction* command allows you to set the parameters for the EXTEST_PULSE, EXTEST_TRAIN, RUNBIST, and INTEST instructions.

**What Next**

Make sure that the port name specified with the *-clock_cycles* option is a port of the design.

---

## UIT-291

(error) The '%s' port in the wait specification is not an input/inout port.

Feedback

**Description**

This error message occurs when a port specified using the *-clock_cycles* option of the prior commands is not an input or inout port.

The *set_bsd_instruction* command allows you to set the parameters for the RUNBIST and INTEST instructions.

**What Next**

Make sure that all ports in the wait specification are input or inout ports.

---

## UIT-292

(error) The '%s' port in the RUNBIST/INTEST duration specification is not a system or test clock.

**Description**

The *set_bsd_runbist* and the *set_bsd_intest* commands allow you to set the parameters for the RUNBIST and INTEST instructions respectively. This error message indicates that a port specified using the *-clock_cycles* switch of the above commands is not a system or test clock.

**What Next**

Make sure that all ports in the wait specification are either system clocks or test clocks.

---

## UIT-293

(error) The '%s' port appears more than once in the RUNBIST/INTEST wait specification.

**Description**

The *set_bsd_runbist* and the *set_bsd_intest* commands allow you to set the parameters for the RUNBIST and INTEST instructions respectively. This error message indicates that a clock port appears more than once in the wait specification for the above commands.

**What Next**

Make sure that no clock ports appear more than once in the RUNBIST/INTEST wait specification.

---

## UIT-294

(information) Writing the parameters for the INTEST instruction.

Feedback

**Description**

The *set_bsd_intest* command allows you to set the parameters for the INTEST instructions. This information message indicates that the parameters associated with the INTEST instruction are being written to the db design.

**What Next**

No further action is needed.

---

## UIT-295

(information) Overwriting the parameters for the INTEST instruction.

**Description**

The *set_bsd_intest* command allows you to set the parameters for the INTEST instructions. This command is not cumulative. This information message indicates that the existing parameters associated with the INTEST instruction are being overwritten by the new user specification for the INTEST parameters.

**What Next**

No further action is needed.

---

## UIT-296

(information) Writing the parameters for the RUNBIST instruction.

**Description**

The *set_bsd_runbist* command allows you to set the parameters for the RUNBIST instructions. This information message indicates that the parameters associated with the RUNBIST instruction are being written to the db design.

**What Next**

No further action is needed.

---

## UIT-297

(information) Overwriting the parameters for the RUNBIST instruction.

**Description**

The *set_bsd_runbist* command allows you to set the parameters for the RUNBIST instructions. This command is not cumulative. This information message indicates that the

Feedback

existing parameters associated with the RUNBIST instruction are being overwritten by the new user specification for the RUNBIST parameters.

**What Next**

No further action is needed.

---

## UIT-298

(error) The '%s' port is not an IEEE 1149.1 Test Access Port.

**Description**

The *remove_bsd_port* command allows you to remove the IEEE 1149.1 attributes from the test access ports. This error message indicates that the port you specified does not have the attributes associated with IEEE 1149.1 Test Access Ports.

**What Next**

Run *report_test -port* to identify the IEEE 1149.1 test access ports and reissue the command.

---

## UIT-299

(error) Could not remove the '%s' IEEE 1149.1 test access port.

**Description**

The *remove_bsd_port* command allows you to remove the IEEE 1149.1 attributes from the test access ports. This error message tells you that the *remove_bsd_port* cannot remove the IEEE 1149.1 test attibutes from the port.

**What Next**

Make sure that the port is indeed an IEEE 1149.1 test access port.

---

## UIT-300

(warning) The specified routing signals are ignored because the -route option is set to false.

**Description**

In command *set_scan_configuration*, the option *-route_signals* is not applicable when the option *-route* is set to false.

**What Next**

Set the option *-route* to true, or just take the default value.

Feedback

## UIT-301

(error) The specified routing signal type '%s' is incorrect.

**Description**

In command *set_scan_configuration", the correct routing scan signal types for the option -route_signals are {all, global, serial, scan_enables, clocks}.*

**What Next**

Repeat the command using the correct routing scan signal types.

## UIT-302

(warning) The specified routing signal type '%s' is not valid for the scan style '%s'.

**Description**

In command *set_scan_configuration", the correct routing scan signal types for the option -route_signals are {all, global, serial, scan_enables, clocks}.*

However, some route signal types do not belong to certain scan styles. For example, the routing signal type *clocks* is not valid for *multiplexed-flip-flop* scan style, and *scan_enables* is not valid for *lssd* scan style.

**What Next**

Check your set_scan_configuration -style and -route_signals specifications are consistent.

## UIT-303

(warning) The -internal_clocks option will be ignored. It only applies to multiplexed-Flip-Flop scan style.

**Description**

You receive this message if you are using a scan style other than *multiplexed_flip_flop* and specified the *-internal_clocks* option with either the *set_scan_configuration* or *set_dft_signal* command. You can use the *-internal_clocks* option only with the *multiplexed_flip_flop* scan style; the option is ignored for all other scan styles.

**What Next**

If you want to use the *-internal_clocks* option, set the scan style to *multiplexed_flip_flop* using *set_scan_configuration*, and re-apply the *-internal_clocks* option. If you want to keep your current scan style, or if it is acceptable for the *-internal_clocks* option to be ignored, you do not need to take any action.

Feedback

## UIT-304

(warning) Restoring Constrained value for port '%s'.

**Description**

In the STIL procedures, you can force a constrained port to a state other than the requested constrained value, but usually this is allowed only for a few tester cycles, and then the port must be returned to the constrained value. For example, you might want to hold a global reset port to an off state for general ATPG patterns but allow it to be asserted to initialize the design.

**What Next**

This message is purely an informational message; no action is required.

## UIT-305

(error) '%s' is not a DFT signal type.

**Description**

You receive this message if you execute the *set_dft_signal* command and use a signal type that is not a valid DFT signal type. Valid DFT signal types are test_point_normal_data_source, test_point_normal_data_sink, test_point_clock and test_mode. Other signal types do not have DFT signal semantics and are rejected.

**What Next**

Reexecute the *set_dft_signal* command using a valid DFT signal type.

## UIT-306

(warning) %d DFT ports have been specified, but access pins have been specified also. Only the first port will be processed.

**Description**

You receive this message if you execute the *set_dft_signal* command and specify both the *-hookup pin* and *-port port_list* options, where *port_list* contains more than one port. If you use the *-hookup* option, you can use only one design port for *port_list*.

**What Next**

Re-execute the *set_dft_signal* command, and specify only one port if you are also specifying access pins with the *-hookup* option. If you want to specify multiple DFT ports, use a single command for each port.

Feedback

## UIT-307

(warning) '%s' is a %s and cannot be used to specify a DFT signal port.

**Description**

The set_dft_signal command allows design objects to specify ports. These must be ports. Other kinds of design objects cannot be used to specify DFT signal ports, and are rejected.

**What Next**

Remove the object from the argument list to get rid of the warning. Use hierarchical instance names, with respect to the top-level design, to locate design objects.

## UIT-308

(warning) '%s' is a %s and cannot be used to specify a DFT signal pin.

**Description**

The set_dft_signal command allows design objects to specify pins. These must be pins or pin instances. Other kinds of design objects cannot be used to specify DFT signal ports, and are rejected.

**What Next**

Remove the object from the argument list to get rid of the warning. Use hierarchical instance names, with respect to the top-level design, to locate design objects.

## UIT-309

(warning) %s port '%s' cannot be a %s.

**Description**

You can use the *set_dft_signal* command to put associated DFT semantics with design ports. This warning is generated when you try to set the signal type of a port to something that is inconsistent with its direction. For example, signal type test_point_normal_data_source is associated with input ports. It cannot be associated with an output port.

**What Next**

Repeat the command using the correct port.

Feedback

## UIT-310

(error) There is no valid DFT port.

**Description**

You receive this error message if you execute the *set_dft_signal* command and do not specify a valid design port.

**What Next**

Study accompanying messages to determine why specified ports were not acceptable, and repeat the command using valid ports.

## UIT-311

(error) No specified hookup pin is valid.

**Description**

You receive this message if you execute the *set_dft_signal* command with the *-hookup* option, and its argument is not a valid design pin.

**What Next**

Study accompanying messages to determine why the specified access pins were not acceptable. Repeat the command using valid pin instances.

## UIT-312

(warning) %s pin '%s' cannot be a %s.

**Description**

You can use the set_dft_signal command to specify hookup pins for design ports that have DFT semantics. This warning is generated when you try to set the hookup pin for a port to something that is inconsistent with its direction. For example, ports with signal type test_point_normal_data_source should be associated with output pins. They cannot be associated with input pins.

**What Next**

Repeat the command using the correct hookup pin.

## UIT-313

(warning) %d DFT ports have been specified. Only the first will be processed.

Feedback

### Description

You use the *set_dft_signal* command to associate DFT signal semantics with design ports. You receive this error message when both of the following are true:

```
1. The set_dft_signal specification has multiple ports specified using
 the -port
switch.

2. The set_dft_signal specification has a hookup pin specified using
 the -hookup
switch.
```

### What Next

Execute the *set_dft_signal* command specifying a single port each time you wish to specify a hookup pin for the DFT signal.

---

## UIT-314

(warning) %d hookup pins have been specified. Only the first will be processed.

### Description

You receive this message if you execute *set_dft_signal -hookup* with more than one pin name as the argument to *-hookup*. You can specify only one pin name for the *-hookup* option. This message informs you that all pin names after the first one are ignored.

### What Next

Re-execute *set_dft_signal -hookup* with a single pin as the argument. If you want an additional hookup pin to be used, re-execute the command specifying that pin for the design port.

---

## UIT-315

(information) The list of ports from which the DFT signals are to be removed is empty.

### Description

You receive this message if you issue the *remove_dft_signal* command and the *object_list* argument is empty. The *object_list* argument is required; the command does nothing when it is invoked with an empty set of boundary scan ports to be removed.

### What Next

If you did not intend to remove DFT signals from any ports, no action is required on your part. Otherwise, re-issue the *remove_dft_signal* command with at least one port as argument.

Feedback

## UIT-316

(error) At present, the '%s' command only supports the full scan test methodology.

### Description

You receive this error message if you execute *insert_dft* or *preview_dft* and have previously specified a scan test methodology of *partial_scan* or *none*. Currently the *insert_dft* and *preview_dft* commands support only the *full_scan* test methodology.

### What Next

Specify the *full_scan* test methodology by executing *set_scan_configuration -methodology full_scan*. Then reexecute *insert_dft* or *preview_dft*.

## UIT-317

(error) At present, the '%s' command does not support the '%s' scan style.

### Description

You receive this error message if you execute *insert_dft* or *preview_dft* and have previously specified a scan style other than *multiplexed_flip_flop* or *lssd*. Currently the *insert_dft* and *preview_dft* commands support only the *multiplexed_flip_flop* scan style and the *lssd* scan style.

### What Next

Specify the *multiplexed_flip_flop* scan style or the *lssd* scan style by executing *set_scan_configuration -scan_style multiplexed_flip_flop* or *set_scan_configuration -scan_style lssd*. Then reexecute *insert_dft* or *preview_dft*.

## UIT-318

(information) The '%s' command will become obsolete in the 2000.04 release. Please use '%s' instead.

### Description

This message informs you that the *set_scan*, *set_scan_chain*, *set_test_routing_order* and *insert_dft* commands will become obsolete in the 2000.04 release of DFT Compiler. You are advised to use the specified replacement commands instead.

### What Next

Use the *set_scan_element*, *set_scan_path* or *insert_dft* commands instead.

Feedback

## UIT-319

(error) Cannot change the scan style for a '%s' design.

### Description

You receive this error message if you attempt to use *set_scan_configuration -style* to change the scan style of a design whose scan state is test ready, scan replaced, scan routed or existing scan. In all these cases, the design has already been modified to have scan cells according to the current scan style. Therefore, you cannot change the scan style for such designs.

### What Next

If you want to change the scan style for a design, you must start with a pre-scan design.

## UIT-320

(warning) You have executed insert_dft with both the -dont_fix_constraint_violations and -map_effort options. Ignoring -map_effort.

### Description

You receive this message if you invoke *insert_dft* with both the *-dont_fix_constraint_violations* and *-map_effort* options. If you use the *-dont_fix_constraint_violations* option, *insert_dft* does not fix constraint violations after scan synthesis. Therefore, the *-map_effort* option has no meaning and is ignored.

### What Next

If it is acceptable to you that the *-map_effort* option is ignored, no action is required on your part. Otherwise, reissue *insert_dft -map_effort* without the *-dont_fix_constraint_violations* option.

## UIT-321

(warning) The model command is obsolete as of release 2000.10. Use the PrimeTime STAMP functionality instead.

### Description

This message informs you that the *model* command is obsolete as of release 2000.10; it has been replaced by the PrimeTime STAMP functionality.

### What Next

If you have questions or issues, contact your local Synopsys representative.

Feedback

## UIT-322

(warning) Invalid argument for -chain_length: '%s'. Argument must be a postive integer or the string "default". If an invalid argument is provided the "default" value will be used.

**Description**

You receive this message if you execute the *set_scan_configuration* command and use an invalid argument for the *-chain_length* option. Valid arguments are positive integers or the string "default".

**What Next**

Reexecute the *set_scan_configuration* command and use a valid argument for the *-chain_length* option.

## UIT-324

(error) Library cell %s not found in any of the libraries currently available in dc_shell.

**Description**

You get this error message when the library cell specified by the *define_dft_design* command, is not found in any of the libraries available in dc_shell.

**What Next**

Insure that the library cell name is correct and the library it belongs to is available in the dc_shell. Check the *link_library* variable. Check the *search_path* variable.

## UIT-325

(error) Pin named "%s" not found on the library cell "%s".

**Description**

You get this error message when the library cell specified by the *define_dft_design* command does not have a pin as specified in the access list.

**What Next**

Insure that correct library cell name and pin names are specified.

## UIT-326

(information) The '%s' command will become obsolete in the '%s' release. Please use '%s' instead.

Feedback

**Description**

This message informs you that the command specified in the error message will be obsolete in the release specified in the error message. You are advised to use the specified replacement command instead.

**What Next**

Use the replacement command instead.

---

## UIT-327

(Warning) Disable result '%s' is not valid for pad type '%s'. Ignoring the specified disable result and using the default value.

**Description**

You get this warning message when the disable result specified by the *-disable_res* option of the *define_dft_design* command, is not valid for the pad type.

Following are valid disable results for different pad types:

*tristate_output - Z bidirectional - Z open_drain_output - WEAK0, PULL0 open_source_output - WEAK1, PULL1 open_drain_bidirectional - WEAK0, PULL0 open_source_bidirectional - WEAK1, PULL1*

**What Next**

Specify a valid disable result for the pad design and rerun.

---

## UIT-328

(error) Signal types port and/or port_inverted not specified for differential pad design.

**Description**

You get this error message when the option *-differential* of the command *define_dft_design* is set to true and the *access_list* does not specify both *port* and *port_inverted* signal types. For differential pad designs both *port* and *port_inverted* signal types must be specified in the access list.

**What Next**

Specify both *port* and *port_inverted* signal types for the pad design and rerun.

Feedback

## UIT-329

(Error) Instruction '%s' has illegal opcode "%s". (The opcode string must contain only binary values '0's and '1's.)

### Description

BSD instruction opcodes must be binary codes. Code cubes (which include `dont care' X bits) are not allowed. The only characters permitted in these code strings are '0's and '1's.

### What Next

Re-write the binary opcode for the instruction using only '0's and '1's.

## UIT-330

(information) TPF protocol format will no longer be the default format from 2003.03 release. Please switch to STIL format in future releases.

### Description

The tool shows this message to indicate that the default format of the test protocol will be changed in the future release. Please switch to the new test protocol format.

## UIT-331

(information) The '%s' command will become obsolete in the 2002.05 release. Please use TetraMAX ATPG tool instead.

### Description

This message informs you that the *create_test_patterns* command will become obsolete in the 2002.05 release of DFT Compiler. You are advised to refer to TetraMAX ATPG tool for test pattern generation.

### What Next

Refer to the TetraMAX ATPG User Guide for more information about using TetraMAX ATPG tool.

## UIT-332

(warning) Port '%s' has a driven_by_logic_%s attribute. This conflicts with '%s' signal semantics.

Feedback

**Description**

The set_dft_signal command lets you associate signal semantics with exactly one design input port. This message tells you that the port has a driven_by_logic attribute that conflicts with these semantics.

**What Next**

Satisfy yourself that there is no conflict.

---

## UIT-333

(warning) Port '%s' has a dft_signal_type '%s' attribute. The signal semantics conflicts with the test_hold '%s' attribute.

**Description**

The set_test_hold command sets the test_hold attribute to a logic value to be assumed on specified input ports during testing. This message tells you that a port has a dft_signal_type attribute and the signal semantics conflicts with the value of the test_hold attribute.

**What Next**

Satisfy yourself that there is no conflict.

---

## UIT-334

(error) Incorrect value for the -async_fix option argument. Allowed values are mux and MUX; or gate and GATE.

**Description**

This error message informs you that you used the *set_autofix_configuration* command with an invalid argument for the *-async_fix* option. Allowed values for the *-async_fix* option argument are mux and MUX; or gate, GATE.

**What Next**

Use the *set_autofix_configuration* command with the *-async_fix* option and specify one of the allowed values as the argument.

---

## UIT-335

(warning) You have specified the command '%s' for the port '%s' with a sense inout. Make sure you apply the correct test constraints for test design rule checking.

Feedback

**Description**

This warning message informs you that you used one of the following commands - *set_dft_signal*, *set_autofix_clock*, *set_autofix_async* - to specify a port with a direction inout.

**What Next**

Make sure you apply the correct test constraints for test design rule checking.

---

## UIT-336

(error) Boundary scan data register length should be a positive integer.

**Description**

You receive this message if you execute the *set_bsd_register* command and specify an invalid value for the *legth*.

**What Next**

Specify a positive integer value for the boundary scan data register length.

---

## UIT-337

(error) License feature '%s' required for the '%s' option of the '%s' command could not be found in the license file.

**Description**

Some dc_shell commands require specific licenses for some of their options. This error may occur when the required feature is not authorized.

**What Next**

Make sure your installation is correct and that you are supposed to be authorized for this license feature.

---

## UIT-338

(error) '%s'is not a valid value for the '%s' option of the '%s' command.

**Description**

You get this error when the value specified for the command option is invalid.

Feedback

**What Next**

Specifiy a valid value for the command option and rerun. See the command man page for a list of valid values for the option.

---

## UIT-339

(error) Specified pin '%s' is not found in the design '%s'.

**Description**

You receive this message if the named pin is not found in the current design.

**What Next**

Reexecute the command with the correct pin that exists in the current design.

---

## UIT-349

(error) Register not specified for the instruction %s. The -register argument is required for user-defined and RUNBIST instructions.

**Description**

A standard data register or a user-defined register previously declared with the *set_scan_path* command must be specified for user-defined and RUNBIST instruction by *-register* argument. For user-defined and RUNBIST instructions there is no default register to be connected for serial access between TDI and TDO when the instruction is active.

**What Next**

Specify a standard data register or a user-defined register for the *-register* argument of the instruction.

---

## UIT-350

(error) '%s' is not a standard instruction.

**Description**

The *set_bsd_instruction* command is used to specify a standard boundary scan instruction as defined by IEEE Std 1149.1. You have used this command to specify a nonstandard boundary scan instruction.

**What Next**

Issue the command with a standard boundary scan instruction.

Feedback

## UIT-351

(error) The '%s' instruction is not supported.

**Description**

The *set_bsd_instruction* command is used to specify a standard boundary-scan instruction as defined by IEEE Std 1149.1. At present, the *set_bsd_instruction* command only supports the following instructions: BYPASS, EXTEST, SAMPLE, HIGHZ, CLAMP and IDCODE. You have used this command to specify one of the following instructions: RUNBIST, INTEST or USERCODE.

**What Next**

Reissue *set_bsd_instruction* command specifying a set of boundary scan instructions supported by BSD Compiler.

## UIT-352

(information) Writing boundary-scan instruction '%s'.

**Description**

You receive this message from the *set_bsd_instruction* command to inform you that the specified instruction is being written to the database, to be implemented during the Boundary Scan logic insertion performed by *insert_dft*.

**What Next**

If you intend that the specified instruction be implemented, no action is required on your part. Otherwise, you can remove the instruction using *remove_bsd_instruction*.

## UIT-353

(warning) Overwriting boundary-scan instruction '%s'.

**Description**

You receive this message if you issue the *set_bsd_instruction* command and specify an instruction that is already on the list of instructions to be implemented by *insert_dft*. This message informs you that the specified instruction is being overwritten in the list by the *set_bsd_instruction* command you just issued.

**What Next**

This is an informational message only. No action is required on your part.

Feedback

## UIT-354

(warning) %d boundary scan TAP ports have been specified. Only\n \tthe first boundary scan port will be processed.

**Description**

Use the *set_bsd_signal* command to associate boundary scan signal semantics with design ports. This error occurs when you try to specify the signal type of multiple design ports.

**What Next**

Split outstanding boundary scan TAP port specifications among multiple commands.

## UIT-355

(error) There is no valid boundary scan TAP port.

**Description**

Use the *set_bsd_signal* command to associate boundary scan signal semantics with design ports. This error occurs when nothing you specified is a valid design port.

**What Next**

Study accompanying messages for the reasons why specified ports were not acceptable, and repeat the command using valid ports.

## UIT-356

(warning) %d boundary scan ports are specified. Only the\n \tfirst will be processed.

**Description**

Use the *set_bsd_signalcommand to associate boundary scan signal semantics with design ports. This error occurs when you try to specify the signal type of multiple design ports using the same command. To simplify commands and eliminate ambiguity, use separate commands.*

**What Next**

Split outstanding boundary scan port specifications among multiple commands.

## UIT-357

(error) Instruction encoding must be one of 'default' or 'one_hot'.

Feedback

**Description**

You can use the *set_bsd_configuration* command to specify the instruction encoding for the boundary scan instructions synthesized by the *create_bsd* command. The only valid values for instruction encoding are *one_hot* or *default*. You have specified an instruction encoding other that *default* or *one_hot*.

**What Next**

Specify the instruction encoding to be either *default* or *one_hot*.

---

## UIT-358

(error) You must specify at least one option.\n\ For available options, input "set_bsd_configuration -help".

**Description**

The *set_bsd_configuration* command does nothing when it is invoked without options.

**What Next**

To get rid of the information message, invoke *set_bsd_configuration* with at least one option.

---

## UIT-359

(information) The set of boundary scan instructions to be removed is empty.

**Description**

The *remove_bsd_instruction* command does nothing when it is invoked with an empty set of instructions to be removed.

**What Next**

To get rid of the information message, invoke *remove_bsd_instruction* with at least one instruction to be removed.

---

## UIT-360

(information) Removing boundary scan instruction '%s'.

**Description**

The *remove_bsd_instruction* command is used to remove boundary scan instructions previously specified with the *set_bsd_instruction* command.

Feedback

**What Next**

Empty.

---

## UIT-361

(warning) Could not remove boundary scan instruction '%s'.

**Description**

The *remove_bsd_instruction* command is used to remove boundary scan instructions previously specified with the *set_bsd_instruction* command. This warning message indicates that the boundary scan instruction specified by the user could not be removed.

**What Next**

Check if the instruction was previously specified using the *set_bsd_instruction* command.

---

## UIT-362

(warning) Could not find boundary scan instruction '%s'.

**Description**

The *remove_bsd_instruction* command is used to remove boundary scan instructions previously specified with the *set_bsd_instruction* command. This warning message indicates that the boundary scan instruction specified by the user could not be found.

**What Next**

Check if the instruction was previously specified using the *set_bsd_instruction* command.

---

## UIT-363

(information) The list of ports from which the boundary scan signals are to be removed is empty.

**Description**

The *remove_bsd_signal* command does nothing when it is invoked with an empty set of boundary scan ports to be removed.

**What Next**

To get rid of the information message, invoke *remove_bsd_signal* with at least one boundary scan port to be removed.

Feedback

## UIT-365

(warning) The direction switch may only be used with bidirectional ports.

**Description**

The *-direction* switch of the *set_bsr_cell_type* command should only be used for bidirectional ports. You have used the *-direction* switch with some input or output ports. The *set_bsr_cell_type* command issued will only operate on bidirectional ports in the list of ports specified.

**What Next**

None.

## UIT-366

(warning) Ignoring the command for the %s ports.

**Description**

This warning message occurs when you invoke the *set_bsr_cell_type* command with the *-direction* switch and there are some input or output ports in the port list. This warning message also occurs when you specify a boundary scan cell type of *observe_only* for an output port or the output side of a bidirectional port.

**What Next**

Correct your *set_bsr_cell_type* specification and re-issue the command.

## UIT-367

(warning) Observe-only is not a valid boundary scan cell type for %s port.

**Description**

This warning message occurs when you specify a boundary scan cell type of *observe_only* for an output port or the output side of a bidirectional port.

**What Next**

Correct your *set_bsr_cell_type* specification and re-issue the command.

## UIT-368

(error) Invalid boundary scan cell type '%s'. Boundary scan cell type must be one of 'none', 'observe_only' or 'control_observe'.

Feedback

**Description**

You have specified an invalid boundary scan cell type. Boundary scan cell type must be one of none, observe_only or control_observe.

**What Next**

Specify one of the valid boundary scan cell types and re-issue the *set_bsr_cell_type* command.

---

## UIT-369

(error) Invalid direction argument '%s'. The direction must be one of 'in' or 'out'.

**Description**

You have specified an invalid direction argument for a bidirectional port. The direction argument must be one of in or out, corresponding to the input and output sides of a bidirectional port.

**What Next**

Re-issue the *set_bsr_cell_type* command with the correct direction.

---

## UIT-370

(error) Scan specification on cell '%s' would make multibit '%s' in design '%s' non-homogeneous.

**Description**

Multibits have the property that each bit implements the same function. The scan specification is such that, if insert_dft implements it, the bits of this multibit no longer have the same function.

In order to avoid breaking the homogeneity property of multibits, DFT Compiler ignores this scan specification.

**What Next**

If you still want the scan specification on this cell, you have two options:

1) Remove the multibit with the remove_multibit command. For this operation, you will need to set the current design to the design containing the multibit.

2) Issue the scan specification for the whole multibit.

Feedback

## UIT-371

(Warning) Multibit '%s' in design '%s' is not a scan segment.

**Description**

You have used a multibit in a context where a scan segment is expected. This particular multibit is not a segment. Either the global default (set using the set_scan_configuration) calls for multibits not to be segments, or you have marked this multibit as not a segment.

The multibit and its constituant cells are ignored.

**What Next**

You can make a multibit into a scan segment using the *set_scan_group* command. You can also list the individual bits of the multibit.

## UIT-372

(Warning) The '%s' option of the set_scan_group command does not apply to synthesizable segments; the option is ignored.

**Description**

The command you have issued refers to a synthesizable segment. You can not specify access points or the contained cells of a synthesizable segment. These options are used for user segments.

The extra option is ignored.

**What Next**

See the description of the *set_scan_group* command for an explanation of the various options and their interrelations.

## UIT-373

(Warning) The '%s' option of the set_scan_group command does not apply to user segments; the option is ignored.

**Description**

The command you have issued refers to a user segment. In a user segment, the order of the cells are determined by the connectivity of the design. You cannot alter this through scan specifications.

The -reverse_order option has no effect, it is ignored.

Feedback

**What Next**

See the description of the *set_scan_group* command for an explanation of the various options and their interrelations.

---

## UIT-374

(warning) Multibit '%s' is already a synthesizable segment.

**Description**

The specification is redundant. It is ignored.

**What Next**

No action required if you don't mind the extra message.

---

## UIT-375

(warning) '%s' is already a segment.

**Description**

The specification is redundant. It is ignored.

**What Next**

No action required if you don't mind the extra message.

---

## UIT-376

(information) There is no explicit synthesizable segment specification for multibit '%s'; it remains%s a synthesizable segment, as specified in the scan configuration.

**Description**

You have attempted to remove a segment specification which you have not given explicitly using the *set_scan_group* command.

**What Next**

If you want to override the default provided by the scan configuration, use the *set_scan_group* command.

---

## UIT-377

(information) The explicit synthesizable segment specification for multibit '%s' is removed; now, it is%s a synthesizable segment, as specified in the scan configuration.

Feedback

### Description

You have removed a segment specification which you previously set using the *set_scan_group* command. This message informs you of the default specification this segment inherits from the scan configuration.

### What Next

If you are happy with the current state of this segment, no action is required. Otherwise, you can issue an explicit scan configuration using the *set_scan_group* command.

---

## UIT-378

(error) Instance '%s' is not in the system.

### Description

This error indicates that the current command cannot find the instance in the system.

### What Next

Use a different -instance option.

---

## UIT-379

(error) Incorrect value of -clock option argument. Allowed values are true, TRUE, on, and ON; or false, FALSE, off, and OFF.

### Description

You receive this message if you execute *set_autofix_configuration* or *set_autofix_element* with an incorrect argument for the *-clock* option. Allowed values of the *-clock* option argument are *true, TRUE, on*, and *ON*; or *false, FALSE, off*, and *OFF*.

### What Next

Re-issue the command with one of the allowed values for the *-clock* option argument.

---

## UIT-380

(error) Incorrect value of -async option argument. Allowed values are true, TRUE, on, and ON; or false, FALSE, off, and OFF.

### Description

You receive this message if you execute *set_autofix_configuration* or *set_autofix_element* with an incorrect argument for the *-async* option. Allowed values of the *-async* option argument are *true, TRUE, on*, and *ON*; or *false, FALSE, off*, and *OFF*.

Feedback

### What Next

Re-issue the command with one of the allowed values for the *-async* option argument.

---

## UIT-381

(error) Incorrect value of -fix_protocol option argument. Allowed values are true, TRUE, on, and ON; or false, FALSE, off, and OFF.

### Description

You receive this message if you execute *set_autofix_configuration* or *set_autofix_element* with an incorrect argument for the *-fix_protocol* option. Allowed values of the *-fix_protocol* argument are *true, TRUE, on*, and *ON*; or *false, FALSE, off*, and *OFF*.

### What Next

Re-issue the command with one of the allowed values for the *-fix_protocol* option argument.

---

## UIT-382

(error) Incorrect value of -string option argument. Allowed values are true, TRUE, false, or FALSE.

### Description

You receive this message if you execute *set_autofix_element* with an incorrect argument for the *-string* option. Allowed values of the *-string* option argument are *true*, *TRUE*, *false*, or *FALSE*.

### What Next

Re-issue the command with one of the allowed values for the *-string* option argument.

---

## UIT-383

(error) No port of name %s was found in design %s.

### Description

This is an error message. The port %s was not found in design %s.\n \tCheck the name and re-invoke the command.

### What Next

You may want to use find command to list all ports.

Feedback

## UIT-384

(error) Port %s is not a test clock.

### Description

The port %s is not a test clock. Use the *set_dft_signal* command to create a test clock associated with port %s.

### What Next

Use *report_test -port* to report all test clocks available on the current design.

## UIT-385

(error) Port %s is not of type input.

### Description

You receive this error message if the specified port, given with the *set_autofix_clock* command or the *set_autofix_async* command, is not an input port. This argument must be an input port.

### What Next

Re-issue the *set_autofix_clock* command or the *set_autofix_async* command and specify an input port.

## UIT-386

(warning) Cannot find cell %s in design %s.

### Description

You receive this message if the specified cell, which you gave as part of the *object_list* argument of the *set_autofix_clock* command or the *set_autofix_async* command, or the *cellNames* argument of the *set_dft_drc_rules* command, cannot be found. Often, typos or spelling errors cause messages of this type.

### What Next

Use the *find* command to list the cells in your design, and re-execute the command using the name of a valid cell.

## UIT-387

(warning) Cell %s is combinational; you must use a sequential cell or a hierarchical cell with sequential elements.

Feedback

**Description**

This message warns you that the specified cell is combinational. *set_autofix_clock*, *set_autofix_async* and *set_autofix_element* require sequential cells or hierarchical cells with sequential elements. You cannot use combinational cells.

**What Next**

Re-issue the command and make sure that *object_list* does not contain any combinational cells.

---

## UIT-388

(error) You cannot specify more than one clock.

**Description**

You receive this error message if you execute *set_autofix_clock* and specify more than one clock. You can specify only one clock.

**What Next**

Re-execute *set_autofix_clock* and specify only one clock.

---

## UIT-389

(error) You cannot specify more than one asynchronous port.

**Description**

This message informs you that you used the *set_autofix_async* command and specified more than one asynchronous port. You can specify only one asynchronous port with this command.

**What Next**

Rerun *set_autofix_async* and specify only one asynchronous port.

---

## UIT-390

(error) Clock port %s has a hold value of %s. You cannot\n \tspecify a clock port that has a hold value set on it.

**Description**

You receive this message from *set_autofix_clock* if the specified clock port has the specified hold value set on it. Clocks with hold values are not valid for *set_autofix_clock*. A hold value is set on a clock by using the *set_test_hold* command, which places

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the *test_hold* attribute on the clock. You can remove the *test_hold* attribute using the *remove_attribute* command; *reset_design* removes all attributes, including *test_hold*.

**What Next**

Remove the *test_hold* attribute from the clock port using the *remove_attribute* command. Then reissue the *set_autofix_clock* command and respecify the clock port.

**See Also**

- reset_design

---

## UIT-391

(information) Performing set_autofix_clock of\n \tclock "%s" on cell/instance "%s".

**Description**

This message informs you that *set_autofix_clock* is being performed on the specified clock on the specified cell or instance.

**What Next**

This is an informational message only; no action is required on your part.

---

## UIT-392

(warning) The variable disable_transition_degradation is now obsolete. Use disable_library_transition_degradation instead.

**Description**

This message informs you that the variable *disable_transition_degradation* is now obsolete, and has been replaced by *disable_library_transition_degradation*. However, *disable_transition_degradation* will continue to be supported for backward compatibility.

**What Next**

Use *disable_library_transition_degradation* instead of *disable_transition_degradation* in future designs.

---

## UIT-393

(error) '%s' is not a valid compliance enable pattern name.

Feedback

**Description**

The *remove_bsd_specification -compliance <all | pattern name>* command let you remove all compliance enable patterns, or one compliance enable pattern with given pattern name.

**What Next**

You mush specify a valid pattern name, or "all". To view all existing compliance patterns, use command *report_test -bsd*.

---

## UIT-394

(error) Port %s is not a test asynch or a test asynch inverted.

**Description**

The port %s is neither a test asynch nor a test asynch inverted. Use *set_dft_signal* to create a test asynch or test asynch inverted associated with the port %s.

**What Next**

Use *report_test -port* to report all test asynchs available on the current design.

---

## UIT-395

(error) Port %s is already a test asynch or a test asynch inverted and can not be a test clock.

**Description**

The port %s is a test asynch or test asynch inverted and can not be a test clock. Use *remove_attribute* if you do not want the port %s to be a test asynch (or a test asynch inverted) and repeat the *set_dft_signa* clock definition command.

**What Next**

Use *report_test -port* to report all test clocks and test asynchs available on the current design.

---

## UIT-396

(error) Port %s is already a test clock and can not be a test asynch or a test asynch inverted.

Feedback

**Description**

The port %s is a test clock and can not be a test asynch or a test asynch inverted. Use *remove_attribute* if you do not want the port %s to be a test clock and repeat *set_dft_signal*.

**What Next**

Use *report_test -port* to report all test clocks and test asynchs available on the current design.

---

## UIT-397

(error) You cannot specify more than two async ports.

**Description**

You receive this error message if you execute *set_autofix_async* and specify more than two asyncs. You can specify one or two asyncs.

**What Next**

Re-execute *set_autofix_async* and specify one or two asyncs.

---

## UIT-398

(error) Options '-function' and '-direction' cannot be specified together.

**Description**

You receive this error message if you execute *set_bsd_data_cell* command and specified both the '-function' and '-direction' options. Currently *set_bsd_data_cell* command supports two alternate options to specify the BSR data cell function at a port. The '-function' option provides a direct way of specifying the BSR data cell function at a port. In the alternate way of specification the BSR data cell function is inferred from the '-direction' option and the port_direction. You should use exactly one of the '-function' and '-direction' options in your specification.

**What Next**

Re-execute *set_bsd_data_cell* and specify exactly one of the '-function' and '-direction' option.

---

## UIT-399

(error) Invalid value %s specified for the '-function' option.

Feedback

**Description**

You receive this error message if you execute *set_bsd_data_cell* command and specified an incorrect value for the '-function' option. The '-function' option provides a direct way of specifying the BSR data cell function at a port. The valid values for this option are: INPUT INPUT_INVERTED OUTPUT OUTPUT_INVERTED BIDIR BIDIR_INVERTED OBSERVE

**What Next**

Re-execute *set_bsd_data_cell* and specify a valid value for the '-function' option.

---

## UIT-400

(error) internal error, write_test vector formatter (stran) is unable\n to write out status.

**Description**

The *write_test* command invokes the vector formatter *stran* in order to translate .sif files to different vector formats. Once *stran* is run, it communicates back its result status to *write_test* through a file named '.stran.status'.

The error message above is issued by the *write_test* command when the '.stran.status' file is not writable. Two reasons for that to happen: either '.stran.status' has WRITE permission or no disk space left.

**What Next**

Check for write permission on the .stran.status file or disk space.

---

## UIT-401

(error) Unknown -command option argument '%s' for preview_scan, assuming 'insert_scan' (default).

**Description**

If you receive this message, you have probably used an invalid argument for the *-command* option of the *preview_scan* command. The valid arguments are *insert_scan* (the default) or *reoptimize_design*; this message informs you that the default, *insert_scan*, is being used.

When *-command* is set to *insert_scan*, *preview_scan* works only in logical domain. When *-command* is set to *reoptimize_design*, *preview_scan* takes the physical design information into account.

Feedback

### What Next

If you do not want *preview_scan* to use the *insert_scan* argument to the *-command* option, reapply the *preview_scan* command with the *reoptimize_design* option. For details, see the *preview_scan* man page.

---

## UIT-402

(error) BSD configuration style must be either synchronous or asynchronous.

### Description

You receive this message if you issue *set_bsd_configuration -style* with an invalid argument for the *-style* option. Valid values are *synchronous* and *asynchronous*. The *set_bsd_configuration* command specifies the configuration style for the boundary scan design synthesized by the *insert_dft* command.

### What Next

Reissue *set_bsd_configuration -style* with a valid argument for the *-style* option.

---

## UIT-403

(error) BSD instruction width must be in the range of 2 to 31.

### Description

You receive this message if you issue the *set_bsd_configuration* command with an invalid value for the *-ir_width* option argument. Valid values are 2 through 31 only. The *insert_dft* command uses the *-ir_width* option value to specify the number of bits in the instruction register when synthesizing the boundary scan design.

### What Next

Reissue the *set_bsd_configuration* with a valid value for the *-ir_width* option argument.

---

## UIT-404

(error) Invalid boundary scan register type for a bidirectional port.

### Description

You receive this message if you issue the *set_bsd_data_cell* command and specify an invalid value for the boundary scan register cell type to be used for a bidirectional port. Valid values of boundary scan register cell types for input, output, and bidirectional ports are listed in the man page and the online help for the *set_bsd_data_cell* command.

Feedback

The *insert_dft* command uses the specified boundary scan register cell types when synthesizing the boundary scan design.

**What Next**

Refer to the man page or the online help for the *set_bsd_data_cell* command to determine the current valid boundary scan register cell types, and reissue the *set_bsd_data_cell* command with a valid type.

---

## UIT-405

(error) Invalid boundary scan register cell type for an input port.

**Description**

You receive this message if you issue the *set_bsd_data_cell* command and specify an invalid value for the boundary scan register cell type to be used for an input port. Valid values of boundary scan register cell types for input, output, and bidirectional ports are listed in the man page and the online help for the *set_bsd_data_cell* command. The *insert_dft* command uses the specified boundary scan register cell types when synthesizing the boundary scan design.

**What Next**

Refer to the man page or the online help for the *set_bsd_data_cell* command to determine the current valid boundary scan register cell types, and reissue the *set_bsd_data_cell* command with a valid type.

---

## UIT-406

(error) Invalid boundary scan register cell type for an output port.

**Description**

You receive this message if you issue the *set_bsd_data_cell* command and specify an invalid value for the boundary scan register cell type to be used for an output port. Valid values of boundary scan register cell types for input, output, and bidirectional ports are listed in the man page and the online help for the *set_bsd_data_cell* command. The *insert_dft* command uses the specified boundary scan register cell types when synthesizing the boundary scan design.

**What Next**

Refer to the man page or the online help for the *set_bsd_data_cell* command to determine the current valid boundary scan register cell types, and reissue the *set_bsd_data_cell* command with a valid type.

Feedback

## UIT-407

(error) Invalid port type '%s' specified for user-defined BSR, TAP, or BSD Register.

### Description

You receive this message if you issue the *set_bsd_bsr_element*, *set_bsd_tap_element*, or *set_bsd_register* command and specify an invalid port type. Valid port types are found in the man page or online help for each of the three commands. The *insert_bsd* command uses the user-defined registers when synthesizing the boundary-scan design.

### What Next

Refer to the man pages or the online help for the *set_bsd_bsr_element*, *set_bsd_tap_element*, or *set_bsd_register* command to determine the current valid BSR, TAP, and BSD register port types, respectively. Then reissue the appropriate command with a valid port type.

## UIT-408

(error) Invalid port or control BSR name '%s'.

### Description

You receive this message if you issue the *set_bsd_path* command and the *identifier_list* contains a name that is neither a valid design port nor a valid control BSR cell identifier; this could be caused by a typo or spelling error. The *identifier_list* must contain a list of names of valid ports and valid control BSR identifiers. The order of the names in the list determines the order that the *insert_dft* command uses when it implements the boundary scan design.

### What Next

Verify the correct names of the ports and control BSR cell identifiers, then reissue the *set_bsd_path* command using the correct names and identifiers.

## UIT-409

(error) Port direction must be either "in" or "out".

### Description

You receive this message if you execute *set_bsd_data_cell* or *set_bsd_path* and specify an invalid port direction. Valid port directions are "in" and "out". For *set_bsd_data_cell*, you specify the port direction using the *-direction* option. For *set_bsd_path*, you can specify the direction of a bidirectional port by attaching /in or /out to the port name in the *identifier_list*.

Feedback

The *insert_dft* command uses the port direction information when synthesizing the boundary scan design.

**What Next**

Choose a valid port direction for the port, and reissue the appropriate command using the valid port direction.

---

## UIT-410

(error) You cannot specify the direction of a nonbidirectional port.

**Description**

You receive this message if you execute the *set_bsd_path* command and specify the direction of a port that is not a bidirectional port. You can specify the direction of a bidirectional port in the *identifier_list* as *port_name*/in or *port_name*/out; however, it is an error to attach /in or /out to the name of a nonbidirectional port. *insert_dft* uses the *identifier_list* to determine the order in which the BSR cells are to be connected in the synthesized boundary scan design.

**What Next**

Verify the port types in the *identifier_list*, and reexecute the *set_bsd_path* command specifying the direction for only the bidirectional ports.

---

## UIT-411

(error) Cannot find the specified pin '%s'.

**Description**

You receive this message if you issue the *set_bsd_register* command and a pin specified in the *access_list* cannot be found on the user-specified boundary scan data register cell. This could be caused by a typo or spelling error. The *insert_dft* command uses the user-specified boundary scan data register cell in synthesizing the boundary scan design. You may also get this message if you intend to specify a scan-through tap register, and issue the *set_bsd_register* command with the reserved words *SR_SI* or *SR_SO* in the access list, however the register name is not the reserved word *STT_REG*.

**What Next**

Verify the correct name for the specified pin, and reissue the *set_bsd_register* command using the correct pin name. If you want to specify a scan-through-tap register use the reserved name *STT_REG* for the register and reissue the command.

Feedback

## UIT-412

(error) You cannot specify a control BSR cell for a non-tristate port '%s'.

### Description

You receive this message if you execute the *set_bsd_control_cell* command and the *port_list* contains a port that is not connected to a tristate pad. You specify a control BSR cell to control tristate port pads only. The *insert_dft* command uses the control BSR cell when synthesizing the boundary scan design.

### What Next

Reexecute the *set_bsd_control_cell* command with a *port_list* that contains only names of ports that are connected to tristate pads.

## UIT-413

(error) Cannot find the specified port '%s' in the design or instance.

### Description

You receive this message if you execute the *set_bsd_bsr_element*, *set_bsd_tap_element*, or *set_bsd_register* command and a port you specify cannot be found in the design or instance. This could be caused by a typo or spelling error.

### What Next

Verify the name of the port. Then reissue the appropriate command using the correct port name for the design or instance.

## UIT-414

(error) Cannot find cell name '%s' in the current design.

### Description

You receive this message if you issue the *set_bsd_register* command and the name you specify for the *hierarchical_cell* argument cannot be found in the current design. This could be caused by a typo, a spelling error, or an incorrect path.

The *insert_dft* command uses the user-specified boundary scan data register cell in synthesizing the boundary scan design.

### What Next

Verify the correct name and hierarchical path of the cell you want to specify as a boundary scan data register, and reissue the *set_bsd_register* command using the correct name.

Feedback

## UIT-415

(error) Invalid cell type for boundary scan register element.

### Description

You receive this message if you issue the *set_bsd_bsr_element* command and specify an invalid value for the boundary scan register cell type. Valid values of boundary scan register cell types are listed in the man page and the online help for the *set_bsd_bsr_element* command. The *insert_dft* command uses the specified boundary scan register cell type when synthesizing the boundary scan design.

### What Next

Refer to the man page or the online help for the *set_bsd_bsr_element* command to determine the current valid boundary scan register cell types, and reissue the *set_bsd_bsr_element* command with a valid cell type.

## UIT-416

*UIT-416* (error) Invalid cell type for boundary scan tristate control register.

### Description

You receive this message if you issue the *set_bsd_control_cell* command and specify an invalid cell type for the boundary scan tristate control register. Valid values of cell types for a boundary scan tristate control register are listed in the man page and the online help for the *set_bsd_control_cell* command. The *insert_dft* command uses the specified boundary scan control register cell type when synthesizing the boundary scan design.

### What Next

Refer to the man page or the online help for the *set_bsd_control_cell* command to determine the current valid cell types for the boundary scan tristate control register, and reissue the *set_bsd_control_cell* command with a valid cell type.

## UIT-417

(error) Invalid boundary scan register cell type for ports.

### Description

You receive this message if you issue the *set_bsd_data_cell* command and specify an invalid value for the boundary scan register cell type to be used for specified ports. Valid values of boundary scan register cell types for input, output, and bidirectional ports are listed in the man page and the online help for the *set_bsd_data_cell* command.

Feedback

The *insert_dft* command uses the specified boundary scan register cell types when synthesizing the boundary scan design.

**What Next**

Refer to the man page or the online help for the *set_bsd_data_cell* command to determine the current valid boundary scan register cell types, and reissue the *set_bsd_data_cell* command with a valid cell type.

---

## UIT-418

(error) Cannot find the specified design in memory.

**Description**

You receive this message if you execute the *set_bsd_bsr_element* or *set_bsd_tap_element* command and the name you specify for the *design_name* argument cannot be found in memory. This could be caused by a typo or spelling error. The *insert_dft* command uses the specified design as a boundary scan register or tap controller when synthesizing the boundary scan design.

**What Next**

Verify that the design you want to use does exist in memory, and verify its spelling. Then re-execute the appropriate command using the correct design name.

---

## UIT-419

(error) You cannot use a reserved word as a user-defined boundary scan data register name.

**Description**

You receive this message if you execute the *set_scan_path* command and specify a reserved word (for example, BOUNDARY, BYPASS, or DEVICE_ID) as the *register_identifier*. Reserved words are predefined names, reserved for boundary scan test data; you cannot use a reserved word to name a user-defined data register. Reserved words are listed in the man page for the *set_bsd_register* command. The *insert_dft* command uses the user-defined data register when synthesizing the boundary scan design.

**What Next**

Refer to the man page of the *set_bsd_register* command for the list of reserved words, and specify a *register_identifier* that is not a reserved word.

Feedback

## UIT-420

(error) Invalid bidirectional port is specified

**Description**

You can use the *set_bsd_path* command to specify the order in which the BSR cells need to hooked up in the boundary scan design synthesized by the *create_bsd* command. This command takes a list of design ports or control BSR names as input. For a bidirectional port the direction can be specified as <port_name>/in or <port_name>/out if the input BSR cell and output BSR cell for the port need to be routed in a particular order. Direction can't be specified for a non bidirectional port. You have specified direction for a non bidirectional port.

**What Next**

Don't specify direction for a non bidirctional port.

## UIT-421

(error) Invalid bidirectional port is specified

**Description**

You can use the *set_bsd_path* command to specify the order in which the BSR cells need to hooked up in the boundary scan design synthesized by the *create_bsd* command. This command takes a list of design ports or control BSR names as input. For a bidirectional port the direction can be specified as <port_name>/in or <port_name>/out if the input BSR cell and output BSR cell for the port need to be routed in a particular order. Direction can't be specified for a non bidirectional port. You have specified direction for a non bidirectional port.

**What Next**

Don't specify direction for a non bidirctional port.

## UIT-422

(error) Invalid bidirectional port is specified

**Description**

You can use the *set_bsd_path* command to specify the order in which the BSR cells need to hooked up in the boundary scan design synthesized by the *create_bsd* command. This command takes a list of design ports or control BSR names as input. For a bidirectional port the direction can be specified as <port_name>/in or <port_name>/out if the input BSR cell and output BSR cell for the port need to be routed in a particular order. Direction can't

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be specified for a non bidirectional port. You have specified direction for a non bidirectional port.

**What Next**

Don't specify direction for a non bidirctional port.

---

## UIT-423

(error) Invalid bidirectional port is specified

**Description**

You can use the *set_bsd_path* command to specify the order in which the BSR cells need to hooked up in the boundary scan design synthesized by the *create_bsd* command. This command takes a list of design ports or control BSR names as input. For a bidirectional port the direction can be specified as <port_name>/in or <port_name>/out if the input BSR cell and output BSR cell for the port need to be routed in a particular order. Direction can't be specified for a non bidirectional port. You have specified direction for a non bidirectional port.

**What Next**

Don't specify direction for a non bidirctional port.

---

## UIT-424

(error) Specified BSD pad design "%s" does not exist.

**Description**

You receive this error message because you have executed the *define_dft_design* command and specified a design that does not exist in memory. You can use *define_dft_design* to specify and characterize a design as a pad design. Such a design is treated like a pad cell by the boundary-scan insertion.

**What Next**

Use the *list_designs* command to see what designs are in memory, then use the *read* command to read in the design. Next, execute the *define_dft_design* command to set the desired design as a pad design.

---

## UIT-425

(error) Signal type "%s" specified in the access list of the pad design is invalid.

Feedback

### Description

You receive this error message because you have executed the *define_dft_design* command and have specified an invalid signal type as part of the access list for the pad design. You use the *define_dft_design* command to specify and characterize a design as a pad design. The access list for the pad design is specified as a list of pairs of signal types and pin names. Valid signal types are data_in, data_in_inverted, data_out, data_out_inverted, enable, and enable_inverted.

### What Next

Reexecute the *define_dft_design* command and specify a valid signal type in the access list for the *-access* option.

---

## UIT-426

(error) Specified pad design "%s" does not have a pin named "%s".

### Description

You receive this error message when you execute the *define_dft_design* command and specify a pin name in the access list that does not exist in the pad design. The *define_dft_design* command is used to specify and characterize a design as a pad design. The access list for the pad design specifies a list of pairs of signal types and pin names.

### What Next

Reexecute the *define_dft_design* command and specify a valid pin name in the access list for the *-access* argument.

---

## UIT-427

(error) "%s" is not a valid value for the "%s" option of the define_dft_design command.

### Description

You receive this error message because you have specified an invalid argument for the *-type* option, the *-disable_res* option, or the *-differential* option of the *define_dft_design* command. Use the *define_dft_design* command to specify and characterize a design as a pad design. Valid arguments for the *-type* option are input, output, tristate_output, and bidirectional. Valid arguments for the *-disable_res* option are WEAK0, WEAK1, PULL0, and PULL1. Valid arguments for the *-differential* option are true and false.

### What Next

Reexecute the *define_dft_design* command and specify a valid option argument.

Feedback

## UIT-428

(error) Insufficient BSD instruction width '%d' specified. Requires at least '%d'.

**Description**

This error message tells you that you used the *set_bsd_configuration* command and specified an invalid value for the *-ir_width* option. The specified width is not sufficient to hold all the instruction opcodes specified with the *set_bsd_instruction* command. The *insert_dft* command uses the *-ir_width* option value to specify the number of bits in the instruction register when synthesizing the boundary scan design.

**What Next**

Use the *set_bsd_configuration* command with the *-ir_width* option and specify a value sufficient to hold all the instruction opcodes.

## UIT-429

(error) BSD instruction opcode width must be in the range of 2 to 31.

**Description**

This error message indicates that you ran the *set_bsd_instruction* command with the *-code* option and specified an invalid opcode width. Valid opcode widths are 2 through 31. The *insert_dft* command uses the opcode width to specify the number of bits in the instruction register when synthesizing the boundary scan design.

**What Next**

Rerun the *set_bsd_instruction* command and specify a valid opcode width for the *-code* option argument.

## UIT-430

(error) Signal type '%s' specified twice or specified with its inversion.

**Description**

This message informs you that you ran the *define_dft_design* command with an access list that specifies a signal type twice, or that specifies a signal type and its inversion. For example, an access list specifying both enable and enable_inverted signal types, or specifying two data_in signal types results in this error message.

**What Next**

Rerun the *define_dft_design* command with a valid access list.

Feedback

## UIT-431

(error) Signal type '%s' specified twice.

### Description

You receive this message if you issue the *define_dft_design* command with an access list specifying a signal type or it's inverted type twice. For instance an access list specifying both enable and enable_inverted signal types, or an access list specifying two data_in signal types.

### What Next

Reissue the *define_dft_design* with a valid access list.

## UIT-432

(error) Incorrect %s field is specified for input/output ports.

### Description

You receive this message if you execute *set_bsd_data_cell* command and specify an incorrect direction/function for an input or output port. The direction field in *set_bsd_data_cell* command for input/output ports should be unspecified(in which case it is assumed to be the same as the port direction) or should be consistent with the port direction. The function field in *set_bsd_data_cell* command for input ports should be unspecified(in which case it is assumed to be INPUT) or should be either INPUT or INPUT_INVERTED or OBSERVE. The function field in *set_bsd_data_cell* command for output ports should be unspecified (in which case it is assumed to be OUTPUT) or should be either OUTPUT or OUTPUT_INVERTED. Please note that options direction and function are mutually exclusive.

### What Next

For input/output ports, leave the direction/function field blank or specifiy a direction/function consistent with the port direction. For bidirectional port, you should specify the direction/function, or leave it blank in case you specify the cell_type as "BC_7", or "none".

## UIT-433

(error) The output format '%s' was not supported by new translation engine.

### Description

You receive this message if you execute *write_test* command with *write_test_new_translation_engine* = true(default), but specify a test program format which is not currently supported by new translation engine.

Feedback

### What Next

For valid test program formats currently supported by new translation engine, execute "write_test -help" for more information.

---

## UIT-434

(error) The design does not have a BSD clock port.

### Description

You receive this message if you execute *write_test* command to write out the BSD test program in or through STIL format, but the design does not have a BSD clock port.

### What Next

Please check if the jtag clock of the design was defined correctly.

---

## UIT-435

(error) The unix environment variable LTRAN_ROOT is not defined.

### Description

You receive this message if you execute *write_test* command to write out test program with format tdl91, tstl2, or ftdl, with environment variable *write_test_new_translation_engine" set to true(default). In this case, the environment variable LTRAN_ROOT must be correctly set to invoke Ltran executable.*

### What Next

You should set it as,

setenv LTRAN_ROOT $SYNOPSYS/$TARGET_ARCH/syn/tran

---

## UIT-436

(warning) Assigning name "%s" to the bsr element.

### Description

You get this warning when the (optional) name parameter for the bsr element is not specified. When unspecified, a name identical to the bsr element type("BC_1", "BC_2", "BC_4" or "BC_7") is assigned to the bsr element. As BSD compiler recognizes the bsr elements named as "BC_1", "BC_2", "BC_4" and "BC_7" as the default bsr elements of types BC_1, BC_2, BC_4 and BC_7 respectively, this bsr element will become the default bsr element for the type. If a bsr element named so(explicitly or implicitly) is not defined, then Design Ware bsr element remains the default bsr element for the type. A default

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bsr element is the one chosen from different implementations for bsr cells not having a "set_bsd_data_cell" or "set_bsd_control_cell" specification.

**What Next**

Specify a valid name other than "BC_1", "BC_2", "BC_4" and "BC_7" for the bsr element if you do not intend to use this bsr element as the default for the type, and rerun the command.

---

## UIT-437

(error) Reserved name %s cannot be used as bsr element name for type %s.

**Description**

You get this warning when the name specified for the bsr element is one of the reserved names "BC_1", "BC_2", "BC_4" or "BC_7" and it does not match with the specified type of the bsr element. Reserved bsr elements names "BC_1", "BC_2", "BC_4" and "BC_7" are used to explictly define the default bsr elements of types BC_1, BC_2, BC_4 and BC_7 respectively (else Design Ware bsr elements are used as default), and must match with the type. A default bsr element is the one chosen from different implementations for bsr cells not having a "set_bsd_data_cell" or "set_bsd_control_cell" specification.

**What Next**

Correct the name or the type of the bsr element. Specify a unique name other than "BC_1", "BC_2", "BC_4" and "BC_7" for the bsr element if you do not intend to use this bsr element as the default for the type.

---

## UIT-438

(error) Reserved name "%s" cannot be used as bsr element name.

**Description**

You get this warning when the name specified for the bsr element is one of the reserved names "DW_BC_1", "DW_BC_2", "DW_BC_4", "DW_BC_7" or "NONE". These reserved bsr elements names are used to convey the following meaning when used as the bsr element idenetifier in the "set_bsd_data_cell" and "set_bsd_control_cell" commands:

DW_BC_1 - Insert a Design Ware BSR cell of type BC_1 DW_BC_2 - Insert a Design Ware BSR cell of type BC_2 DW_BC_4 - Insert a Design Ware BSR cell of type BC_4 DW_BC_7 - Insert a Design Ware BSR cell of type BC_7 NONE - Do not insert a BSR cell

**What Next**

Specify a valid name for the bsr element other than "DW_BC_1", "DW_BC_2", "DW_BC_4", "DW_BC_7" or "NONE".

Feedback

## UIT-439

(error) BSR element named %s has already been defined.

### Description

You get this warning when the name specified for the bsr element has already been used to define a bsr element by an earlier "set_bsd_bsr_element" command.

### What Next

Specify a unique and valid name for the bsr element, and rerun the command.

## UIT-440

(warning) This bsr cell becomes the default bsr element for type %s.

### Description

BSD compiler recognizes the bsr elements named as "BC_1", "BC_2", "BC_4" and "BC_7" as the default bsr elements of types BC_1, BC_2, BC_4 and BC_7 respectively. If a bsr element named so is not defined, then Design Ware bsr element remains the default bsr element for the type. Note that if the name is not specified for a bsr element then BSD compiler assigns a name identical to the bsr element type("BC_1", "BC_2", "BC_4" or "BC_7") for the bsr element. A default bsr element is the one chosen from different implementations for bsr cells not having a "set_bsd_data_cell" or "set_bsd_control_cell" specification.

### What Next

Specify a valid name other than "BC_1", "BC_2", "BC_4" and "BC_7" for the bsr element if you do not intend to use this bsr element as the default for the type, and rerun the command.

## UIT-441

(error) Ilegal width %d for port '%s'. Port width for signal type '%s' must be %d.

### Description

You get this error when you specify a custom tap design port signal type and the actual width of the port is not found valid. The bus port widths must be as follows for different access types:

instructions = 32-bit tap_state = 16-bit user_code_val = 32-bit ver = 4-bit part_num = 16-bit mnfr_id = 11-bit

Rest of the access types are scalar with expected port width of 1.

Feedback

### What Next

Provide a correct interface for the tap design.

---

## UIT-442

(error) Usercode value is not specified for the %s instruction.

### Description

You get this error when you specify the USERCODE instruction, but don't provide a usercode value. USERCODE instruction must have the usercode value specified using the "-user_code_val" option of the "set_bsd_instruction" command.

### What Next

Correct the specification by specifying the usercode value. The 32-bit usercode value can be specified as design pin(s) and/or sized binary constant and/or sized hex constant. Design pins must be specified as full hierarchical full pin names. Binary and hex constants must be specified using the verilog syntax.

Example: -------

Following is an incomplete and hence invalid USERCODE specification:
set_bsd_instruction USERCODE -code {0110}

The above USERCODE specification should be corrected by specifying the usercode value. For instance following is a valid USERCODE specification:

set_bsd_instruction USERCODE -code {0110} -user_code_val {version_reg/v3, version_reg/v2, version_reg/v1, version_reg/v0, 16'b1111111110000000, 12'h2ab}

---

## UIT-443

(error) Usercode value cannot be specified for %s instruction. Usercode value can only be specified for the USERCDOE instruction.

### Description

You get this error when you specify the usercode value for a non USERCODE instruction. An instruction is treated as the USERCODE instruction when the instruction name is spelt as u-s-e-r-c-o-d-e (case insensitive).

### What Next

Correct the specification. If you have USERCODE specification mixed with other instructions, you will get this error for other instructions in the list. If this is the case, separate the USERCODE specification.

Feedback

If you intend to implement the USERCODE instruction and get this message make sure that the instruction name is spelt correctly(it must be spelt as case insensitive u-s-e-r-c-o-d-e).

---

## UIT-444

(error) '%s' is a PrimeTime command that is not supported by Design Compiler.

### Description

Although this command is supported by PrimeTime, it is not currently supported by Design Compiler. You can check which commands are supported by Design Compiler by executing *list -commands* or *help* in dc_shell.

### What Next

Remove the command that is not supported.

---

## UIT-445

(warning) Invalid value '%s' for spdm_mode. The valid values are either false or true.Setting default value as false.

### What Next

Set it to one of the valid values.

### Examples

```
Error: Invalid value 'false' for spdm_mode. The valid values are either
 false or true. Setting default value as false.(UIT-445)
```

---

## UIT-446

(error) Invalid argument in the usercode value list. '%s' is not a valid design pin, sized binary or a sized hex constant.

### Description

You get this error when you specify an invalid value in the usercode value argument list for the USERCODE instruction. The 32-bit usercode value can be specified as any combination of design pin(s) and/or sized binary constant and/or sized hex constant.

Feedback

**What Next**

Check and correct the usercode value argument syntax. Design pins must be specified as full hierarchical full pin names. Binary and hex constants must be specified using the following verilog syntax:

<size>'[b|B|h|H]<binary or hexadecimal value>

Example: ------- Following is a valid USERCODE specification:

set_bsd_instruction USERCODE -code {0110} -user_code_val {version_reg/v3, version_reg/v2, version_reg/v1, version_reg/v0, 16'b1111111110000000, 12'h2ab}

---

## UIT-447

(error) Invalid size %d of the usercode value argument '%s'. Usercode value argument's size should be in the range of 1 to 32-bits.

**Description**

You get this error when you specify a binary or hexadecimal constant value in the usercode value argument list and the size of the argument is not in the valid range of 1 to 32-bits.

Binary and hex constants are specified using the following verilog syntax:

<size>'[b|B|h|H]<binary or hexadecimal value>

Example: ------- Following is a valid USERCODE specification:

set_bsd_instruction USERCODE -code {0110} -user_code_val {version_reg/v3, version_reg/v2, version_reg/v1, version_reg/v0, 16'b1111111110000000, 12'h2ab}

**What Next**

Check and correct the usercode value argument's size.

---

## UIT-448

(error) %s usercode value '%s' cannot be represented in the specified %d bits.

**Description**

You get this error when you specify a binary or hexadecimal constant value in the usercode value argument list and the specified size of the argument is not sufficient enough to represent the value you have specified.

Binary and hex constants are specified using the following verilog syntax:

<size>'[b|B|h|H]<binary or hexadecimal value>

Feedback

**What Next**

Check and correct the usercode value argument's size.

Example: ------- Following is a wrong USERCODE specification, leading to the above error as the hex value 5'h2ab cannot be represented in 5 bits:

set_bsd_instruction USERCODE -code {0110} -user_code_val {version_reg/v3, version_reg/v2, version_reg/v1, version_reg/v0, 16'b1111111110000000, 5'h2ab}

Following would be a corrected specification:

set_bsd_instruction USERCODE -code {0110} -user_code_val {version_reg/v3, version_reg/v2, version_reg/v1, version_reg/v0, 16'b1111111110000000, 12'h2ab}

---

## UIT-449

(error) A %s value is specified as the usercode value. Usercode value must be a 32-bits value.

**Description**

You get this error when the cumulative bit width of the arguments specified in the '-user_code_val' option of the set_bsd_instruction command is not 32-bits.The 32-bit usercode value can be specified as any combination of design pin(s) and/or sized binary constant and/or sized hex constant.

Binary and hex constants are specified using the following verilog syntax: <size>'[b|B|h|H]<binary or hexadecimal value>

**What Next**

Check and correct the usercode value argument's size.

Example: ------- Following is a wrong USERCODE specification, leading to the above error as the cumulative bit width of the usercode value here is 31:

set_bsd_instruction USERCODE -code {0110} -user_code_val {version_reg/v3, version_reg/v2, version_reg/v1, version_reg/v0, 16'b1111111110000000, 11'h2ab}

Following would be a corrected specification:

set_bsd_instruction USERCODE -code {0110} -user_code_val {version_reg/v3, version_reg/v2, version_reg/v1, version_reg/v0, 16'b1111111110000000, 12'h2ab}

---

## UIT-450

(error) Cannot read test model file '%s'.

Feedback

**Description**

This error occurs when you try to read a test model file that is protected or does not exist.

**What Next**

Check that the file exists, check the search_path, and check what protections are set on the file. Then, rerun the *read_test_model* command.

---

## UIT-451

(error) Cannot write test model file '%s'.

**Description**

This error is generated when you run the *write_test_model* command and attempt to write the test model file to a nonexistent or protected directory, or over a file that is protected.

**What Next**

Check the existence and the write protection of the directory.

Check the protection of the existing file with the same name. Make the necessary change to the file or directory, and rerun the *write_test_model* command.

---

## UIT-452

(error) Cannot attach test model to design '%s'.

**Description**

This error is generated because a test model in CTL format is successfully read in but cannot be associated with the named design because it does not have a CTL interface.

**What Next**

Make sure the named design exists and has a CTL interface. Then, rerun the *read_test_model* command.

---

## UIT-453

(error) Design '%s' has no test model.

**Description**

This error is generated when you request a test model operation on a design that has no test model.

Feedback

### What Next

Make sure you specified the design you intended. To see a list of designs that have test models, use the *list_test_models* command. You may try to read a test model with the *read_test_model\P command or infer one with the check_dft\P command.*

---

## UIT-454

(error) Format '%s' is not a valid test model format.

### Description

This error message indicates that you specified an invalid format with the *read_test_model* command. Valid formats are db (the default) and ctl.

### What Next

Rerun the *read_test_model* command and specify a valid format.

---

## UIT-455

(error) The CTL model in '%s' is not accepted.

### Description

This error occurs when you run the *read_test_model* command and specify a CTL file that has errors that prevent the correct interpretation of the file. The exact errors are reported as CTL-* messages immediately preceding this message.

### What Next

Fix the reported errors and reissue the command.

---

## UIT-456

(error) The CTL model on design '%s' is not accepted.

### Description

This error occurs when the test model stored on the specified design contains errors that prevent its correct interpretation. The errors are reported as CTL-* messages immediately preceding this message.

### What Next

This error is most likely due to a corrupted db file. Verify that the file is not corrupted by reading it using the *read* command and using the design in a context other than test models. If the file is not corrupted, report the problem to Synopsys immediately.

Feedback

## UIT-457

(information) The name of the design is included in the model file; the -design option is ignored.

### Description

This message informs you that the *-design* option does not need to be specified with the *read_test_model* command. For test models in .ddc format, the model file contains the design name.

### What Next

To avoid receiving this message, use the *read_test_model* command without the *-design* option.

## UIT-458

(error) Asynchronous port %s has a hold value of %s. You cannot\n \t specify an asynchronous port that has a hold value set on it.

### Description

This error message tells you that the asynchronous port given to the *set_autofix_async* command has the specified hold value set on it. Asynchronous ports with hold values are not valid for *set_autofix_async*. A hold value is set on an asynchronous port using the *set_test_hold* command, which places the *test_hold* attribute on the asynchronous port. You can remove the *test_hold* attribute by using the *remove_attribute* command; *reset_design* removes all attributes, including *test_hold*.

### What Next

Remove the *test_hold* attribute from the asynchronous port using the *remove_attribute* command. Then, run the *set_autofix_async* command and specify the asynchronous port.

### See Also

- reset_design

## UIT-459

(information) Running set_autofix_async on\n \t asynchronous port(s) "%s" to autofix the asynchronous signal(s) on the cell or instance "%s".

Feedback

**Description**

This message informs you that *set_autofix_async* is being run on the specified asynchronous port to autofix the asynchronous signal(s) of the specified cell or instance.

**What Next**

This is an informational message only. No action is required on your part.

---

## UIT-460

(error) Mode decoding style must be one of 'default' or 'one_hot'.

**Description**

This error message tells you that you have specified a decoding style other than default or one_hot for the *-mode_decoding_style* option to the *set_core_integration_configuration* command. You use *set_core_integration_configuration* to specify the decoding style for the test mode ports used during core integration by the *insert_dft* command.

**What Next**

Specify default or one_hot as the mode decoding style.

---

## UIT-461

(information) You must specify the -mode_decoding_style option and the default or one_hot switch.

**Description**

This informational message tells you that you did not specify any options with the *set_core_integration_configuration* command.

**What Next**

Use the *set_core_integration_configuration* command and specify an option.

---

## UIT-462

(information) You must specify at least one of the following switches: -all.

**Description**

This message informs you that you did not specify an option when you used the *remove_core_integration_configuration* command. The *remove_core_integration_configuration* command does nothing when it is used without options.

Feedback

**What Next**

Use the *remove_core_integration_configuration* command with at least one option.

---

## UIT-463

(error) Design '%s' already has a wrapper.

**Description**

This error message indicates that you attempted to insert a wrapper around a design using the *set_dft_configuration -wrapper enable* command, but the design is already wrapped.

**What Next**

If the design is already wrapped, remove the core wrapping test mode definition and configuration commands from your script, then rerun it.

---

## UIT-464

(warning) Resolution of tristate port '%s' failed.

**Description**

This warning message tells you that the core wrapper architect only supports simple tristate drivers; for example, single-cell drivers at the top-level. Complex drivers are not supported.

**What Next**

Use simple tristate drivers in the design. Alternatively, the port can be excluded from wrapping.

---

## UIT-465

(warning) Resolution of bidirectional port '%s' failed.

**Description**

This warning message tells you that the core wrapper architect only supports simple bidirectional drivers, which are single cell drivers at the top level. Complex drivers are not supported.

**What Next**

Use simple bidirectional drivers in the design. Alternately, exclude the port from wrapping.

Feedback

## UIT-466

(warning) Degenerate bidirectional port : %s.

**Description**

This warning messages means that the core wrapper architect excludes the specified port from wrapping because it is a degenerate bidirectional port.

**What Next**

Rerun the command and specify a port that is not a degenerate bidirectional port.

## UIT-467

(warning) Degenerate tristate port : %s.

**Description**

This warning message informs you that the core wrapper architect excludes the specified port from wrapping because it is a degenerate tristate port.

**What Next**

Rerun the command and specify a port that is not a degenerate tristate port.

## UIT-468

(error) You must specify at least one of the following switches: -default_wrapper_cell, -default_safe_state.

**Description**

This error occurs because you used the *set_core_wrapper_configuration* command but did not specify any options.

**What Next**

Run the *set_core_wrapper_configuration* with at least one option.

## UIT-469

(warning) Bad type for wrapper cell. \n

**Description**

This warning message informs you that the specified wrapper cell type is invalid. The default specification is used.

Feedback

**What Next**

Rerun the command and specify a valid wrapper cell type.

---

## UIT-470

(error) You must specify at least one of the following switches: -cell_type, -safe_value, or -port_list.

**Description**

This error occurs because you used the *set_core_wrapper_cell* command and did not specify and option.

**What Next**

Run the *set_core_wrapper_cell* command with at least one option.

---

## UIT-471

(warning) Bad port: '%s'.

**Description**

This warning message informs you that you specified a port that is not in the design.

**What Next**

Run the command and specify a port that is in the design.

---

## UIT-472

(error) Error when running Core Wrapper.

**Description**

This message informs you that the Core Wrapper client could not be run successfully. This indicates an error in the design or in the specifications for core wrapping. Causes might be one of the following:

```
1. Trying to directly wrap a CTL model
2. Trying to wrap a design without any connected ports
3. An error while instantiating wrapper cells due to incorrect libraries
4. Unsupported scan methodology
```

**What Next**

Check the design. Make sure you are not directly trying to wrap a test model. Check the specifications.

Feedback

## UIT-473

(error) '%s' identifies multiple pins.

### Description

This error occurs if you use the *set_bsd_register* command and specify a pin in the access list that contains a wildcard that identifies multiple pins on the register cell. Each signal type and pin name pair must identify only one pin. If you want to identify multiple pins of the same type, you must specify multiple signal types and pin name pairs.

### What Next

Reissue the *set_bsd_register* command and specify multiple pin name pairs, or use a more specific wildcard.

## UIT-474

(error) The test model for design '%s' is stored in version %s of CTL. It cannot be retrieved.

### Description

The version of CTL in which the test model is expressed is incompatible with the tool's capabilities.

CTL is an evolving standard (P1450.6) which is not under the control of the Synopsys developers. Since the Beta version, some incompatible changes were introduced into the language.

Note that the CTL version is not an official denomination of P1450.6. It is an arbitrary number assigned to Synopsys' implementation of a snapshot of the evolving standard. The first production release supports version 1.00.

### What Next

There are two options. If the model is on a design for which full implementation is available, then you can run dft_drc to regenerate the model in the current version of CTL. If the design implementation is not readily available, then the process by which the model was created, most likely insert_dft, needs to be repeated using the current version of the tool.

## UIT-475

(information) The test model on design '%s' is updated to the current CTL version. Writing out the design to disk will make this update permanent.

Feedback

### Description

The version of CTL in which the stored test model is expressed is not compliant with the current state of the proposed standard. The model was retrieved and automatically translated to the newer version of the CTL syntax.

CTL is an evolving standard (P1450.6) which is not under the control of the Synopsys developers. Since the previous version of DFT Compiler, some incompatible changes were introduced into the language.

Note that the CTL version is not an official denomination of P1450.6. It is an arbitrary number assigned to Synopsys' implementation of a snapshot of the evolving standard. DFT Compiler currently supports version 1.70.

### What Next

There are two options. 1- Do nothing, in which case the same message will be reissued each time this test model is used in this version of DFT Compiler. 2- Save the db file containing this test model back to disk.

---

## UIT-476

(error) The test model for design '%s' cannot be retrieved.

### Description

The version of CTL in which the test model is expressed is compatible with the tool's capabilities, but the information cannot be retrieved. This may be due to data corruption on the .db file, or incomplete compatibility between tool versions.

### What Next

There are multiple options.

If the .db file containing the model is very old, reading it into a previous version of DFT Compiler and writing it back to disk will automatically update the text of the model to the DFT Compiler version. The current version of DFT Compiler will most likely be able to read the newly updated model. Otherwise, the model can be regenerated from the current text, from the current netlist, or from the pre-DFT netlist.

To update the model from the current text, edit the ASCII version of the model contained in file <design_name>_obsolete_model.ctl, in the current directory. It can then be read back in and re-attached to the design using the *read_test_model* command.

If the model is on a design for which full implementation is available, then the model may be regenerated from the netlist: you can run DRC to regenerate the model in the current version of CTL. If the design implementation is not readily available, then the process by which the model was created, most likely by the *insert_dft* command, needs to be repeated using the current version of the tool.

Feedback

## UIT-477

(information) The '%s' command is obsoleted in the 2003.03 release. Please use '%s' instead.

### Description

This message informs you that the *set_test_signal* command is obsoleted in the 2003.03 release of DFT Compiler. You are advised to use the specified replacement commands instead.

### What Next

Use the *set_dft_signal* command instead.

## UIT-478

(Error) Valid index required for scan-in and scan-out ports

### Description

This message indicates that you are attempting to describe the scan-in and scan-out ports in the design without associated indices. In order to to correctly extract scan chains using descriptive specifications, the test design rule checker needs information associating scan access pins to their respective scan chains.

### What Next

Specify the appropriate indices.

## UIT-479

(information) The option -order '{%s}' is obsolete in the 2003.12 release. Please use the options -autofix and/or -shadow_wrapper.

### Description

This message informs you that the *-order* option of the *set_dft_configuration* command is obsolete in the 2003.12 release of DFT Compiler. You are advised to use the specified replacement options instead.

### What Next

Use the the *-autofix* and/or *-shadow_wrapper* options instead.

Feedback

## UIT-480

(error) The -order option cannot not be used with the -autofix and/or -shadow_wrapper options.

**Description**

The *-order* option will become obsolete in the 2003.12 release. You are advised to use the *-autofix* and/or the *-shadow_wrapper* options instead.

**What Next**

Modify the options accordingly to the recommendations and repeat the command.

## UIT-481

(warning) Resolution of tristate port '%s' connected to pin '%s' of cell '%s' failed. The port will be excluded from wrapping.

**Description**

This warning message tells you that the core wrapper architect didn't find any tristate driver for the specified port connected to a tristate pin of a cell with test model.

**What Next**

Use simple tristate drivers in the design for the speciied port. Alternatively, the port can be excluded from wrapping.

## UIT-482

(warning) Resolution of bidirectional port '%s' connected to pin '%s' of cell '%s' failed. The port will be excluded from wrapping.

**Description**

This warning message tells you that the core wrapper architect didn't find any bidirectional driver for the specified port connected to a bidir pin of a cell with test model.

**What Next**

Use simple bidirectional drivers in the design. Alternately, exclude the port from wrapping.

## UIT-483

(warning) Mapping Wrapper Cell Designs is not successfull, turning off wrapper insertion.

Feedback

### Description

This warning message tells you that the core wrapper insertion couldn't map wrapper cell designs successfully. Wrapper insertion can't proceed without mapped wrapper cells. Wrapper insertion will be turned off.

### What Next

Check if link library is setup properly to include the libraries used in mapping wrapper cells. Check if there is a license available to map wrapper cells.

---

## UIT-484

(error) Instance '%s' of design '%s' has no test model.

### Description

This error is generated when there is no test model for the design corresponding to the cell instance you specified in a *set_test_target* command.

### What Next

Make sure you specified the instance you intended. To see a list of designs that have test models, use the *list_test_models* command. You may need to use the read_test_model command to attach a test model to the design specified in the error message.

---

## UIT-485

(error) Test mode '%s' does not exist.

### Description

This error is generated when the *-test_mode* argument to the current command is invalid.

### What Next

Use the *list_test_modes* command to see the valid *-test_mode* arguments. If the desired test mode is not listed, you can create a new test mode with the *define_test_mode* command.

---

## UIT-486

(error) Cell '%s' is repeated.

Feedback

### Description

This error is generated when a cell is repeated in the cell list specified by any of the following options:

- the *-cores* option of the *set_test_target* command
- the *-target* option of the *define_test_mode* command
- the *-include* option of the *define_dft_schedule* command

### What Next

Ensure there are no repeated cells in the specification and repeat the command. You can use the *report_test_model* command to see the available test modes for a given design.

---

## UIT-487

(error) Cell '%s' can not be in wrapped mode '%s' when -purpose is top_internal_test.

### Description

This error is generated when you try to put a core into a wrapped internal test mode (i.e. wrp_if mode) when you have specified the purpose of the mode as a top-level scan test. These are incompatible, because the top-level scan test implies that you will apply a set of patterns to the top level and the wrapped internal test mode implies that you will apply the patterns that were generated for the core. There is no way to apply both patterns at once.

### What Next

You should use the *define_test_mode* command to create two test modes. Use *set_test_target -purpose top_internal_test* without cores for one mode, and *set_test_target -purpose core_internal_test* with the desired cores for the other mode.

---

## UIT-488

(error) Test mode '%s' does not exist or is not supported for instance '%s'. of design '%s'.

### Description

This error is generated when the *-cores* argument to the *set_test_target* command specifies an invalid mode or *-target* list includes a test mode other than internal or external test mode.

### What Next

Use the *report_test_model* command to see the valid test modes for the design indicated in the error message. Retry the *set_test_target* command using a valid mode.

Feedback

If using *define_test_mode* command with *-target* option, make sure that the targeted mode is an internal or external test mode. The cores that are not included in the *-target* list are put in mission mode if they are wrapped or are put in an internal scan mode if they are not wrapped.

---

## UIT-489

(error) No mode specified for instance %s.

**Description**

This error is generated when the *-cores* argument to the *set_test_target* command specifies an instance without a mode.

**What Next**

Make sure that the *-cores* argument contains complete instance/mode pairs when you retry the *set_test_target* command.

---

## UIT-490

(error) Bad -purpose '%s'.

**Description**

This error is generated when you you specify an unrecognized *-purpose* for the *set_test_target* command.

**What Next**

Retry the command with a valid *-purpose*.

---

## UIT-491

(error) You must specify -cores when -purpose is core_internal_test.

**Description**

This error is generated when you specify *-purpose core_internal_test* without the *-cores* argument for the *set_test_target* command.

**What Next**

Retry the command with a the *-cores* argument, or change *-purpose* to *top_internal_test*.

Feedback

## UIT-492

(error) Current test mode is not defined.

### Description

This error is generated when the *-test_mode* argument is omitted from the current command, and the current test mode has not been set.

### What Next

Use the *current_test_mode* command to define the current test mode, or specify a test mode directly with the *-test_mode* argument when you retry the command.

## UIT-493

(error) Test mode '%s' must be type InternalTest.

### Description

This error is generated when the test mode for the *set_test_target* command does not have type InternalTest. Since the *set_test_target* command only specifies internal tests (either at the top level or core level) it doesn't make sense for the test mode to be anything other than InternalTest.

### What Next

Use the *define_test_mode* command to create a test mode with *-type InternalTest*, and specify that test mode with either the *current_test_mode* command, or the *-test_mode* argument when you retry the *set_test_target* command.

## UIT-494

(error) Cannot write CTL model with exec name '%s' to file '%s'.

### Description

This error is generated when you run the *write_test* command and failed to write out the CTL model with the specified pattern exec name to the given file.

### What Next

Check the existence and the write protection of the directory.

Check the protection of the existing file with the same name.

Check the CTL model if the specified PatternExec exists. Be sure not to have quoted pattern exec name in the command line.

Feedback

## UIT-495

(error) There is no valid pattern info in file '%s'.

**Description**

This error occurs when you run the *read_pattern_info* command and there is no valid pattern information block in the input file although it is a valid CTL file.

**What Next**

Check the input file to add pattern info block.

## UIT-496

(error) The patinfo for Pattern '%s' already existed.

**Description**

This error occurs when you run the *read_pattern_info* command and the patinfo blocks in the input file already exist in the CTL model. Unless '-overwrite' switch is added, it is an error to re- define the existing patinfo.

**What Next**

Add switch '-overwrite' if you want to redefine the existing patinfo.

## UIT-497

(error) Cannot find pattern matching name '%s' in existing CTL model.

**Description**

This error occurs when you run the *read_pattern_info* command and the pattern name in the patinfo file does not have a match in the existing CTL model.

**What Next**

The user should change the pattern block name in the test program and patinfo file to match the pattern name in the CTL model.

## UIT-498

(Warning) The following ports of design '%s' have no corresponding Signals in its test model:%s.

Feedback

### Description

The error occurs when you run the *read_test_model* command, if some ports of the design don't have corresponding Signals in the read test model. This means that there is no test information for these ports. This will make the test model unusable.

### What Next

The user should check the validity of the test model and whether it realy is the test model of the design it is being attached to.

---

## UIT-499

(Warning) The following test model Signals have no corresponding ports in design '%s': %s.

### Description

The error occurs when you run the *read_test_model* command, if some Signals of the model don't have corresponding ports in the design it is being associated with. This means that the model provides information about non-existing port. This will make the test model unusable.

### What Next

The user should check the validity of the test model and whether it realy is the test model of the design it is being attached to.

---

## UIT-500

(Warning) The following ports of sub-design '%s' have no corresponding Signals in its test core:%s.

### Description

The error occurs when some ports of a sub-design don't have corresponding Signals in its test core referenced by the test model of current design. This will make the CoreType unusable.

### What Next

The user should check the validity of the test core defined in the test model of current design.

Feedback

## UIT-501

(Warning) The following test core Signals have no corresponding ports in sub-design '%s': %s.

### Description

The error occurs when some Signals of a core referenced by the model of current design don't have corresponding ports in the sub-design it is being associated with. This means that the CoreType provides information about non-existing port. This will make the CoreType unusable.

### What Next

The user should check the validity of the CoreType in the test model of current design.

## UIT-502

(error) Mode decoding style must be one of 'binary' or 'one_hot'.

### Description

This error message tells you that you have specified a decoding style other than binary or one_hot for the *-mode_decoding_style* option to the *set_dft_configuration* command. You use *set_dft_configuration* to specify the decoding style for the test mode ports used during multi-mode DFT insertion by the *insert_dft* command.

### What Next

Specify binary or one_hot as the mode decoding style.

## UIT-503

(Error) The signal '%s' must be predefined as an Oscillator signal.

### Description

This error tells you that one of the specified PLL clocks has not been predefined as an Oscillator signal and hence the clock controller specification cannot be accepted. The *set_dft_clock_controller* command specification is ignored due to the error.

### What Next

To correct this error, specify the PLL clocks signals as having the Oscillator signal type.

A PLL clock pll/out is defined using

dc_shell> *set_dft_signal -view existing_dft -type Oscillator -hookup_pin pll_out -active_state < 0 | 1 >*

Feedback

## UIT-504

(Warning) The port '%s' has not predefined as an Oscillator signal.

**Description**

This warning tells you that one of the specified ATE clocks has not been predefined as an Oscillator signal. If the ATE clock is a reference clock, it has to be predefined as an Oscillator signal; otherwise, this warning can be ignored. The *set_dft_clock_controller* command specification is not ignored due to this warning.

**What Next**

You can ignore this warning if the ATE clock is not a reference clock. If the ATE clock is a reference clock, specify the ATE clock signal as having the Oscillator signal type and reissue the set_dft_clock_controller command.

An ATE clock ate_clk is defined using

dc_shell> *set_dft_signal -view existing_dft -type Oscillator -port ate_clk -active_state < 0 | 1 >*

## UIT-505

(Error) An ATE clock must be a port of the design; '%s' is not a valid port name.

**Description**

This error tells you that one of the specified ATE clock names is not a valid port name. The *set_dft_clock_controller* command specification is discarded due to this error.

**What Next**

If you want to specify a hookup pin for an ATE clock, you should specify the hookup pin with the *set_dft_signal* command and reissue the *set_dft_clock_controller* command.

## UIT-506

(Warning) The port '%s' has not predefined as a clock signal.

**Description**

This error tells you that the specified port has not been predefined as a clock signal and that the Oscillator signal specification cannot be accepted. If the port is a reference clock it has to be first defined as a clock and then defined as a reference clock. The *set_dft_signal* command specification is ignored due to this error.

Feedback

**What Next**

If the port is a reference clock, specify the ATE clock signal as having the a clock signal type and reissue the *set_dft_signal* command.

---

## UIT-510

(Error) Test Model of core instance '%s' is not valid for integration.

**Description**

SoC BIST does not allow integration of unencapsulated cores during core integration. This error tells you that the specified core instance is not encapsulated and hence a test mode of the core instance can't be used with *set_test_target* command. The *set_test_tagret* command specification is ignored due to the error.

**What Next**

To avoid the error, specify only encapsulated core test modes with *set_test_target* command.

---

## UIT-511

(Error) More than one bist test modes are specified, bist test modes can't be tested in parallel.

**Description**

SoC BIST does not allow bisted cores to be tested in parallel currently. This error tells you that the *set_test_target* specification includes more than one bist test modes. The *set_test_target* command specification is ignored due to the error.

**What Next**

To avoid the error, specify only one encapsulated bist test mode with *set_test_target* command.

---

## UIT-512

(Error) Bist test modes are specified along with scan test modes, bist test modes can't be tested in parallel.

**Description**

SoC BIST does not allow bisted cores to be tested in parallel currently. This error tells you that the *set_test_target* specification includes one or more bist test modes and one or

Feedback

more scan test modes. The *set_test_target* command specification is ignored due to the error.

**What Next**

To avoid the error, don't mix bist test modes with scan test modes with *set_test_target* command.

---

## UIT-515

(Error) The -setup_instructions and -bsd_test_setup options are mutually exclusive.

**Description**

The test_setup procedure for the boundary-scan patterns can be generated in two ways:

- Using the *-setup_instructions* option of the *create_bsd_patterns* command

  This method creates a test_setup procedure by initializating the user-defined test data registers (UTDRs) associated with the specified instructions. The initialization data is specified with the *-bsd_init_data* option of the *set_bsd_instruction* command.

- Using the *-bsd_test_setup* option of the *create_bsd_patterns* command

  This method inserts the test_setup section from the provided STIL protocol file into the created patterns. The setup protocol must be read using the *read_test_protocol -section test_setup* command prior to pattern creation.

The *-setup_instructions* and *-bsd_test_setup* options are mutually exclusive.

**What Next**

Use only one of the two methods previously described to create test_setup vectors.

---

## UIT-517

(Error) Cannot specify -jtag_reset disable without -setup_instructions or -bsd_test_setup. f

**Description**

This message will appear when neither *-setup_instructions* nor *-bsd_test_setup* is specified along with *-jtag_reset disable* in create_bsd_patterns.

Allowed options are:

The -jtag_reset disable option is valid only when it is used either with -setup_instructions or with -bsd_test_setup.

By default, -jtag_reset option is *enable*

Feedback

## UIT-520

(error) Specified dft design "%s" does not exist.

### Description

You receive this error message because you have executed the *define_dft_design* command and specified a design that does not exist in memory. You can use *define_dft_design* to specify and characterize a design as a dft design. Such a design is treated like a user dft design of the specified type for dft insertion.

### What Next

Use the *list_designs* command to see what designs are in memory, then use the *read* command to read in the design. Next, execute the *define_dft_design* command to set the desired design as a pad design.

## UIT-521

(error) Specified dft design "%s" does not have a pin named "%s".

### Description

You receive this error message when you execute the *define_dft_design* command and specify a pin name in the interface list that does not exist in the specified dft design. The *define_dft_design* command is used to specify and characterize a design as a dft design of a type. The interface list for the dft design specifies a list of triplets of signal type, pin name and pin polarity.

### What Next

Reexecute the *define_dft_design* command and specify a valid pin name in the interface list for the *-interface* argument.

## UIT-522

(error) Syntax error with the list specified with -interface option, expects a list of triplets <signal_type> <pin_name> <pin_polarity> .

### Description

You receive this error message when you execute the *define_dft_design* command and specify an incorrect interface list. The *define_dft_design* command is used to specify and characterize a design as a dft design of a type. The interface list for the dft design specifies a list of triplets : signal type, design pin name and design pin polarity.

Feedback

### What Next

Reexecute the *define_dft_design* command and specify a valid list for the *-interface* argument.

---

## UIT-523

(error) Invalid design name '%s' - this design name is reserved.

### Description

You receive this error message when you execute the *define_dft_design* command and specify a design name that is reserved. The *define_dft_design* command is used to specify and characterize a design as a dft design of a type. The design name specified with this command can not be one of the following names as the name conflicts with our designware implementation of the dft component.

DW01_decode DW_control_force DW_Z_control_force DW_observ_dgen DW03_shftreg DW_bc_1 DW_bc_2 DW_bc_3 DW_bc_4 DW_bc_5 DW_bc_7 DW_bc_8 DW_bc_9 DW_bc_10 DW_wc_d1_s DW_wc_s1_s DW_INSTRREG DW_tap DW_tap_uc DFT_bistc DFT_codec DFT_xbistc DFT_xcodec DFT_clk_mux DFT_clk_chain DW_mbist_ctrl DW_mbist_wrapper

### What Next

Reexecute the *define_dft_design* command and specify a valid design name for the *-design_name* argument.

---

## UIT-524

(error) Illegal width %d for pin '%s' of signal type '%s', legal width must be %d.

### Description

You receive this error message when you execute the *define_dft_design* command and specify a design pin of invalid width. The *define_dft_design* command is used to specify and characterize a design as a dft design of a type. The dft design pins are checked for legal width according to the specified dft design type.

The legal widths of bus pins for different interface pin types as as follows.

DFT Design Type : TAP

instructions = 32-bit tap_state = 16-bit

DFT Design Type : TAP_UC

instructions = 32-bit tap_state = 16-bit user_code_val = 32-bit ver = 4-bit part_num = 16-bit mnfr_id = 11-bit

Feedback

**What Next**

Reexecute the *define_dft_design* command and specify a valid design with bus pins of legal width.

---

## UIT-525

(error) Incorrect syntax specified for list param, list param should be terminated with $end_list$

**Description**

You receive this error message when you execute the *define_dft_design* command and specify a list parameter with no list terminator. The syntax of a list paramter is as follws.

<parameter name> <parameter type> <param value1> <param value2> ... $end_list$

where <parameter type> is string .

**What Next**

Reexecute the *define_dft_design* command and specify the correct syntax for list parameters.

---

## UIT-530

(error) Specified pin '%s' is not found in the current design.

**Description**

You receive this error message when you execute the *set_bsd_instruction* command and specify a pin name that does not exist in the current design.

**What Next**

Reexecute the *set_bsd_instruction* command and specify a valid pin name.

---

## UIT-531

(error) Register '%s' can't be used with standard instruction(s) : %s .

**Description**

You receive this error message when you execute the *set_bsd_instruction* command and specify an incorrect register name for a standard instruction.

Here are the standard instructions and the corresponding register names.

Feedback

EXTEST BOUNDARY EXTEST_PULSE BOUNDARY EXTEST_TRAIN BOUNDARY SAMPLE BOUNDARY PRELOAD BOUNDARY INTEST BOUNDARY

BYPASS BYPASS HIGHZ BYPASS CLAMP BYPASS

IDCODE DEVICE_ID USERCODE DEVICE_ID

This error message is issued when a non standard register is specified for a standard instruction.

**What Next**

Reexecute the *set_bsd_instruction* command and specify thye correct register name.

---

## UIT-532

(error) Invalid Input Clock Conditioning '%s' specified.

**Description**

You receive this error message when you execute the *set_bsd_instruction* command and specify an incorrect input clock conditioning value.

Here are the allowed values for input clock conditioning.

PI

TCK

**What Next**

Reexecute the *set_bsd_instruction* command and specify thye correct input clock conditioning value with option -input_clock_condition.

---

## UIT-533

(error) Invalid Output Conditioning '%s' specified.

**Description**

You receive this error message when you execute the *set_bsd_instruction* command and specify an incorrect out conditioning value.

Here are the allowed values for output conditioning.

BSR

HIGHZ

NONE

Feedback

**What Next**

Reexecute the *set_bsd_instruction* command and specify thye correct output conditioning value with option -output_condition.

---

## UIT-534

(error) Input Clock Conditioning can't be specified for a short BSR chain instruction.

**Description**

You receive this error message when you execute the *set_bsd_instruction* command and specify input clock conditioning for an instruction that selects a short BSR chain.

**What Next**

Reexecute the *set_bsd_instruction* command with out input clock conditioning.

---

## UIT-535

(error) Output Conditioning can't be specified for a short BSR chain instruction.

**Description**

You receive this error message when you execute the *set_bsd_instruction* command and specify output conditioning for an instruction that selects a short BSR chain.

**What Next**

Reexecute the *set_bsd_instruction* command with out output conditioning.

---

## UIT-536

(error) Invalid Excluded BSR Conditioning '%s' specified.

**Description**

You receive this error message when you execute the *set_bsd_instruction* command and specify an incorrect excluded bsr cell onditioning value.

Here are the allowed conditioning values for BSR cells excluded from the current instruction.

CLAMP

NONE

Feedback

### What Next

Reexecute the *set_bsd_instruction* command and specify thye correct excluded bsr conditioning value with option -excluded_bsr_condition.

---

## UIT-537

(error) Excluded BSR Conditioning can't be specified for a non short BSR chain instruction.

### Description

You receive this error message when you execute the *set_bsd_instruction* command and specify excluded bsr conditioning for an instruction that doesn't select a short BSR chain.

### What Next

Reexecute the *set_bsd_instruction* command with out excluded bsr conditioning.

---

## UIT-538

(error) Invalid BSR chain name '%s'.

### Description

You receive this error message when you execute the *set_scan_path* command and specify a reserved name as the BSR chain (path) name.

Here are the reseved names that are not allowed for BSR chains:

BYPASS DEVICE_ID STT_REG

### What Next

Reexecute the command and specify a different name for the BSR chain name.

---

## UIT-539

(error) Conflicting BSD register name '%s'.

### Description

You receive this error message when you execute the *set_scan_path* command and specify a conflicting name as the BSD register name. The name is also used to describe a short BSR chain.

### What Next

Reexecute the command and specify a different name for BSD register name.

Feedback

## UIT-540

(error) Conflicting BSR chain name '%s'.

**Description**

You receive this error message when you execute the *set_scan_path* command and specify a conflicting name as the BSR chain name. The name is also used to describe a short BSD register.

**What Next**

Reexecute the command and specify a different name for the BSR chain name.

## UIT-541

(warning) Unable to find the bsr cell for '%s', ignoring the member.

**Description**

You receive this error message when you execute the *set_scan_path* command and specify a short BSR chain and a member of the short chain doesn't point to a valid BSR cell.

Please note that input/output/control BSR cells of a tristate/bidir ports can be included separately in a short BSR chain only if they are defined.

EX:-

set_bsd_data_cell -function input -port out set_bsd_data_cell -function output -port out set_bsd_control_cell CTRL1 -type BC_1 -port out

set_bsd_path -name SHORT_BSR1 { out/in out1 out/out out3 CTRL1 }

Individual BSR cells of BSR embedded pads can't be included in a short BSR chain, either alll of embedded BSR cells can be included in a short BSR chain or none.

**What Next**

Reexecute the command and specify a valid BSR cell name.

## UIT-542

(error) Invalid or missing %s Condition specified for standard instructions : %s .

Feedback

### Description

You receive this error message when you execute the *set_bsd_instruction* command and specify *-input_clock_condition* or output_condition *options. These options are not allowed for the following standard instructions.*

EXTEST EXTEST_PULSE EXTEST_TRAIN SAMPLE PRELOAD BYPASS HIGHZ CLAMP IDCODE USERCODE

The standard predefines condition if any required for these instuctions and user is not allowed to change that.

You also see this error message for INTEST or RUNBIST instructions if *-input_clock_condition* or output_condition *options are not specified or incorrect value specified for the option output_condition*.

The standard allows only BSR or HIGHZ output condition for these instructions.

There are no restrictions on output condition for user defined instructions.

### What Next

Reexecute the command with the correct options.

---

## UIT-543

(error) Invalid bit '%c' specified in capture value.

### Description

You receive this error message when you execute the *set_bsd_instruction* command and specify *-capture_value* option and specify an invalid capture value string.

The format of capture value string is as follows.

nn'bXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

where nn is between 1 and 32 and X can be 0 or 1, -

nn'hXXXXXXXX

where nn is betweeen 1 and 32 and X can be 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, a, A, b, B, c, C, d, D, e, E, f, F, -

### What Next

Reexecute the command with correct capture value.

Feedback

## UIT-544

(error) Invalid capture value length '%d'.

**Description**

You receive this error message when you execute the *set_bsd_instruction* command and specify *-capture_value* option and specify an invalid capture value string.

The format of capture value string is as follows.

nn'bXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

where nn is between 1 and 32 and X can be 0 or 1, -

nn'hXXXXXXXX

where nn is between 1 and 32 and X can be 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, a, A, b, B, c, C, d, D, e, E, f, F, -

The bit stream length in binary form can't exceed 32.

**What Next**

Reexecute the command with correct capture value.

## UIT-545

(error) Invalid LSB for IDCODE capture value.

**Description**

You receive this error message when you execute the *set_bsd_instruction* command and specify *-capture_value* option and specify an invalid capture value string. The LSB bit of the capture value string for IDCODE instruction should be '1'.

**What Next**

Reexecute the command with correct capture value.

## UIT-546

(error) BSD Test Data Register length is mandatory, specify the length with -exact_length option.

Feedback

**Description**

You receive this error message when you execute the *set_scan_path* command to specify a User-defined Test Data Register (UTDR) and don't specify the length of the Test Data Register.

**What Next**

Reexecute the command and specify the length of the TDR using -exact_length option.

---

## UIT-547

(error) One of the options -ordered_elements or -hookup_elements is mandatory for BSD.

**Description**

You receive this error message when you execute the *set_scan_path* command to specify a BSR chain or User-defined Test Data Register (UTDR) and don't specify either of the required options.

In view spec, the -ordered_elements option specifies a BSR chain.

In view spec/existing_dft, the -hookup_elements option specifies a TDR.

**What Next**

Reexecute the command and specify either of the options as required.

---

## UIT-548

(error) Options -ordered_elements, -hookup_elements are mutually exclusive for BSD.

**Description**

You receive this error message when you execute the *set_scan_path* command to specify a BSR chain or User-defined Test Data Register (UTDR) and specify both of the above options. These options are mutually exclusive, specify only one of the options.

In view spec, the -ordered_elements option specifies a BSR chain.

In view spec/existing_dft, the -hookup_elements option specifies a TDR.

**What Next**

Reexecute the command and specify only one of the above options as required.

---

## UIT-549

(error) BSR chains can not be specified in view existing_dft.

Feedback

### Description

You receive this error message when you execute the *set_scan_path* command in view existing_dft to specify a BSR chain. BSR chains can not be specified in view existing_dft.

BSR chains can be specified in view spec using the option -ordered_elements in conjunction with the option -class bsd.

### What Next

Fix the error and reexecute the command.

---

## UIT-550

(error) Invalid argument in the capture value list. '%s' is not a valid design pin, sized binary or a sized hexadecimal constant.

### Description

This error message occurs when an invalid value is specified in the capture value argument list for the USERCODE instruction. The 32-bit capture value can be specified as any combination of design pin(s), sized binary constant(s), and/or sized hexadecimal constant(s).

Specify binary and hexadecimal constants using the following Verilog syntax:

```
<size>'[b|B|h|H]<binary or hexadecimal value>
```

For example, the following is a valid USERCODE specification:

```
set_bsd_instruction USERCODE -code {0110}
   -capture_value {version_reg/v3, version_reg/v2, version_reg/v1, \\
   version_reg/v0, 16'b1111111110000000, 12'h2ab}
```

### What Next

Check and correct the capture value argument syntax. Specify design pins as full hierarchical pin names.

---

## UIT-551

(error) Invalid size %d of the capture value argument '%s'. Capture value argument's size should be in the range of 1 to 32 bits.

### Description

This error message occurs when you specify a binary or hexadecimal constant value in the capture value argument list, and the size of the argument is not in the valid range of 1 to 32 bits.

Feedback

Specify binary and hexadecimal constants using the following Verilog syntax:

```
<size>'[b|B|h|H]<binary or hexadecimal value>
```

For example, the following is a valid USERCODE specification:

```
set_bsd_instruction USERCODE -code {0110}
   -capture_value {version_reg/v3, version_reg/v2, version_reg/v1, \\
   version_reg/v0, 16'b1111111110000000, 12'h2ab}
```

**What Next**

Check and correct the size of the capture value argument, and run the command again.

---

## UIT-552

(error) %s capture value '%s' cannot be represented in the specified %d bits.

**Description**

This error message occurs when you specify a binary or hexadecimal constant value in the capture value argument list and the specified size of the argument is not sufficient to represent the specified value.

Specify binary and hexadecimal constants using the following Verilog syntax:

```
<size>'[b|B|h|H]<binary or hexadecimal value>
```

For example, the following is a valid USERCODE specification:

```
set_bsd_instruction USERCODE -code {0110}
   -capture_value {version_reg/v3, version_reg/v2, version_reg/v1, \\
   version_reg/v0, 16'b1111111110000000, 12'h2ab}
```

**What Next**

Check and correct the size of the capture value argument, and run the command again.

---

## UIT-553

(error) A %s value is specified as the capture value. Capture value must be a 32-bit value.

**Description**

This error message occurs when the cumulative bit width of the arguments specified in the *-capture_value* option of the *set_bsd_instruction* command is not 32 bits. The 32-bit capture value can be specified as any combination of design pin(s), sized binary constant(s), and/or sized hexadecimal constant(s).

Feedback

Specify binary and hexadecimal constants using the following Verilog syntax:

```
<size>'[b|B|h|H]<binary or hexadecimal value>
```

For example, the following is a valid USERCODE specification:

```
set_bsd_instruction USERCODE -code {0110}
   -capture_value {version_reg/v3, version_reg/v2, version_reg/v1, \\
   version_reg/v0, 16'b1111111110000000, 12'h2ab}
```

**What Next**

Check and correct the size of the capture value argument, and run the command again.

---

## UIT-554

(error) '%s' cell does not exist in the design.

**Description**

This error message occurs when the cell that was specified as part of of the *state_cells* argument of the *set_tap_elements* command, cannot be found.

Typographical or spelling errors often cause this type of error message.

**What Next**

Use the *find* command to list the cells in your design, and rerun the *set_tap_elements* command using the name of a valid cell.

---

## UIT-555

(error) Invalid data. Recheck the file format or type.

**Description**

This error message indicates that an invalid data is read with the *read_bsdl* command. This might happen when a BSDL file read is not in proper unix format. You might observe this for files transferred from different platforms or operating systems.

**What Next**

Check the file type using unix *file* command. Recheck the BSDL file data.

---

## UIT-580

(error) The test_modes value, %s, specified is not a valid value.

Feedback

**Description**

The -test_modes option requires a valid integer number of modes required in the design.

---

## UIT-581

(error) The client %s specified is not a valid dft client.

**Description**

You receive this message if you issue *set_dft_configuration -order* and specify an invalid value for the *-order* option argument. Valid values are *autofix*, to specify the Autofix utility as dft client; or *wrapper*, to specify the Shadow LogicDFT utility as dft client. For more information, see the man page for the *set_dft_configuration* command.

**What Next**

Reissue *set_dft_configuration -order* with a valid value for the *-order* option argument.

---

## UIT-582

(error) Core Integration of cores is mutually exclusive with options : -shadow_wrapper | -core_wrapper | -testability | -mbist | -scan_compression .

**Description**

You receive this message if you issue *set_dft_configuration -core_integration* and also specify one of the following options. *-shadow_wrapper -core_wrapper -testability -mbist -scan_compression* UI option *-core_integration* is mutually exclusive with the above options. For more information, see the man page for the *set_dft_configuration* command.

**What Next**

Reissue *set_dft_configuration* command with valid options.

---

## UIT-583

(error) Core Integration of MBIST cores doesn't support the options : -clock_controller | -bist .

**Description**

You receive this message if you issue *set_dft_configuration -core_integration* and also specify one of the following options. *-bist -clock_controller* UI option *-core_integration* is mutually exclusive with the above options when the design contains MBIST(Tech2) cores. UI option *-core_integration* is mutually exclusive with the option *-clock_controller* when

Feedback

the design contains MBIST(Tech1) cores. For more information, see the man page for the *set_dft_configuration* command.

**What Next**

Reissue *set_dft_configuration* command with valid options.

---

## UIT-584

(error) Object '%s' is not valid, it's either not found or not a cell of current design, '%s'.

**Description**

Object specified in the command has to be present as a cell under the current design.

**What Next**

Check whether the name of the object is correct and current design is specified correctly.

---

## UIT-587

(error) Cell '%s' doesn't have a test model.

**Description**

Cell specified in the command should have a test model. The specification can't be executed if there is no test model for the specified cell.

**What Next**

Check whether the name of the cell is correct and it has a test model. Correct the specification and reexecute the command. Note that test model can be attached to a cell using read_test_model command.

---

## UIT-588

(error) MBIST(Tech2) encapsulated core modes can't be used with set_test_target command.

**Description**

You receive this message if you issue *set_test_target* command and also specify an encapsulated MBIST(Tech2) core mode with *-cores* option. DFT architect creates a single top level mode to target all MBIST(Tech2) encapsulated core modes of the same type and user can't change this behavior with set_test_target command. The name of such a test mode can be controlled with *define_test_mode* and *set_dft_logic_usage* commands.

Feedback

**What Next**

Don't specify *set_test_target* command with MBIST(Tech2) core modes.

---

## UIT-590

(error) A configuration for port %s already exits.

**Description**

The *set_port_configuration* command has previously been specified for this port.

**What Next**

Check to see if the *set_port_configuration* command has previously been specified for this port.

---

## UIT-591

(error) Duplicate control specification for %s data port of %s.

**Description**

The *set_port_configuration* command has previously been specified for this port.

**What Next**

Verify that the *set_port_configuration* command has previously been specified for this port.

---

## UIT-592

(error) %s has a clock port specified for it that is different from the port specified in the command.

**Description**

A previous *set_port_configuration* command has specified a different clock port for this port.

**What Next**

Check if the *set_port_configuration* command has previously been used to specify a clock for this port.

---

## UIT-593

(error) %s specified as clock port in the command does not exist on the cell

Feedback

### Description

You receive this message if you issue the *set_port_configuration* command and the clock port you specified was not found on the specified cell. This could be caused by a spelling error or typo.

### What Next

Verify that the port exists on the cell, and verify the correct spelling. Then reexecute *set_port_configuration*.

---

## UIT-594

(error) Multiple ports named %s on Object %s

### Description

You receive this message if you execute *set_port_configuration* and specify a port name that occurs more than once on the specified object. There can be only one port associated with a given port name.

### What Next

Examine the library description of the cell to determine why there are two ports with the same name. Ensure that there is only one port associated with each port name. Then reexecute *set_port_configuration*.

---

## UIT-595

(error) The read control pattern is not paired (port value)

### Description

You receive this message if you issue the *set_port_configuration* command with the *-read* option, and a *signal_value_pin_pairs* argument to *-read* does not contain two values. Each *-read* argument must be a pair that includes both a signal value and the name of the input pin associated with that value. One of those values is missing.

### What Next

Examine your execution of *set_port_configuration* for an unpaired pin name or signal value. Reexecute *set_port_configuration -read* and ensure that all arguments to *-read* are signal_value/pin_name pairs.

---

## UIT-596

(error) port %s not found on %s

Feedback

**Description**

You receive this message if you issue *set_port_configuration* and the *port_name* cannot be found on the specified object. This could be caused by a spelling error or typo.

**What Next**

Verify that the port exists on the object, and verify the correct spelling. Then reexecute *set_port_configuration*.

---

## UIT-597

(error) Illegal value %s speicified in the read control for port %s

**Description**

You receive this message if you issue the *set_port_configuration* command with the *-read* option and specify an illegal value for the *signal_value* part of the *signal_value_pin_pairs* argument pair. *signal_value* can have values of only 0 or 1.

**What Next**

Reexecute *set_port_configuration -read* with a valid value for *signal_value*.

---

## UIT-598

(error) Illegal direction for port %s

**Description**

You receive this message if you issue the *set_port_configuration* command with the *-read* option and specify a non-input port as the control port of the *signal_value_pin_pairs* argument. You can specify only input ports.

**What Next**

Reexecute *set_port_configuration -read* using the name of an input port as the control port for the *signal_value_pin_pairs* argument.

---

## UIT-599

(error) port %s has already been specified in the read control pattern

**Description**

You receive this message if you issue the *set_port_configuration* command with the *-read* option and the specified port name appears more than once in the list of *signal_value_pin_pairs*. Each port name can be used only once.

Feedback

**What Next**

Reexecute *set_port_configuration -read* without duplicating any port names in the *signal_value_pin_pairs* arguments.

---

## UIT-600

(error) %s, %s specification conflicts with previous value specified on %s

**Description**

You receive this message if you issue the *set_port_configuration* command with the *-read* option and a signal value assigned to a pin as one of the *signal_value_pin_pairs* conflicts with the value already assigned to that pin through a previous execution of *set_port_configuration*. Each pin can have only one value assigned to it.

**What Next**

Remove the ealier specification using remove_port_configuration command if the previous specification was incorrect and then issue both correct set_port_configuration commands. If the current specification is wrong then reissue only the current command with correct specification.

---

## UIT-601

(error) cound not find object named %s

**Description**

You receive this message if you issue *set_port_configuration* or *remove_port_configuration* and the specified object cannot be found in the design. This could be caused by the wrong current design setting, or by a spelling error or typo.

**What Next**

Verify that the current design is set as intended. Verify that the object exists in the design, and verify the correct spelling. Then reexecute the command.

---

## UIT-602

(error) %s is an output port. It can't have an write control.

**Description**

You receive this message if you issue the *set_port_configuration* command with the *-write* option and specify an output port as a control port for the *signal_value_pin_pairs* argument. You can specify only input ports as control ports.

Feedback

### What Next

Reexecute *set_port_configuration -write* using the name of an input port as part of the *signal_value_pin_pairs* argument.

---

## UIT-603

(error) The read control pattern is missing or empty

### Description

You receive this message if you issue the *set_port_configuration* command with the *-tristate* option and do not specify a read control using the *-read* option. A tristate port must have a read control.

### What Next

Reexecute *set_port_configuration* and use the *-read* option to specify a read control for the tristate port.

---

## UIT-604

(error) The current design does not exist

### Description

You receive this message if you issue the *set_port_configuration* command and the specified current design cannot be found in memory. This could be caused by a spelling error or typo.

### What Next

Verify that the current design exists in memory, and verify the correct spelling. If necessary, load the design into memory or respecify using *current_design*. Then reexecute *set_port_configuration*.

### See Also

- current_design

---

## UIT-605

(error) The direction of port %s is missing

### Description

You receive this message if you issue the *set_wrapper_element* command and do not supply a direction for the specified port. Each port must have a specified direction.

Feedback

### What Next

Check the model for the cell, and find out the port direction. Then reexecute *set_wrapper_element* and specify the direction of the port.

---

## UIT-606

(error) The port configuration information is missing for %s

### Description

You receive this message if you issue *set_wrapper_element* and the specified port does not have port configuration information. Each port must either be modeled using Library Compiler's memory syntax, or the port interface must be described using *set_port_configuration* for each output or bidirectional port.

### What Next

Either modify the module description in the library or provide the interface information using the *set_port_configuration* command. Then reexecute *set_wrapper_element*.

---

## UIT-607

(error) No objects specified in the command

### Description

You receive this message if you issue *set_wrapper_element* and do not specify an object. You must specify at least one object.

### What Next

Reexecute *set_wrapper_element* and specify at least one object.

---

## UIT-608

Wrapper attribute not found on %s

### Description

You receive this message if you issue the *remove_wrapper_element* command and the *wrapper_element* attribute was not found on the specified object. You can remove the *wrapper_element* attribute from an object only if the attribute has been set on that object using the *set_wrapper_element* command.

You might not have been aware that the specified object did not have the attribute; or you might have inadvertently specified the wrong object when you executed the *remove_wrapper_element* command.

Feedback

**What Next**

Determine whether the object specified in this warning message is the object from which you intended to remove the *wrapper_element* attribute. If necessary, reexecute the *remove_wrapper_element* command and specify only objects on which the *wrapper_element* attribute has been set.

---

## UIT-609

(error) %s is an output port; can't have a write control

**Description**

You receive this message if you issue the *set_port_configuration* command with the *-write* option and specify an output port as part of the *signal_value_pin_list* argument. Only input ports can have write control.

**What Next**

Verify the correct spelling of the input ports you want to use as control ports. Then reexecute *set_port_configuration* and specify only input ports as part of the *-write signal_value_pin_list* argument.

---

## UIT-610

(error) Illegal object type. Need reference or design

**Description**

Only reference of design can be specified in the command. You receive this message if you issue *set_port_configuration* or *remove_port_configuration* with an illegal object type as part of the *cell_design_ref_list* argument. Objects can be only cells, designs, or references.

**What Next**

Reexecute the command and specify only cells, designs, or references for the *cell_design_ref_list* argument.

---

## UIT-611

(error) Black_boxes are not supported.

**Description**

Only modules with descriptions or empty boxes are supported by the command.

Feedback

**What Next**

create an empty box i.e. an interface description for the module in the design.

---

## UIT-612

(error) Port %s not found on %s

**Description**

Port specified in the command should be there on the cell specified for command to be successful.

**What Next**

check whether cell name and port name are correct.

---

## UIT-613

(error) Object %s not found in %s

**Description**

Object specified in the command has to be present under the current design.

**What Next**

Check whether the name of the object is correct and current design is specified correctly.

---

## UIT-614

(error) Wrapping a constant is not supported

**Description**

The module to be wrapped has to be a non-constant.

**What Next**

Check whether name of the object is correct.

---

## UIT-615

(warning) %s has already been specified for wrapping

**Description**

An Object can be specified for wrapping only once. You receive this message if you issue the *set_wrapper_element* command and the *wrapper_element* attribute is already set

Feedback

on the specified object. You can set the *wrapper_element* attribute only once on a given object.

You might not have been aware that the specified object already had the attribute, or you might have inadvertently specified the wrong object, when you executed the *set_wrapper_element* command.

**What Next**

Determine whether the object specified in this warning message is the object on which you intended to set the *wrapper_element* attribute. If necessary, reexecute the *set_wrapper_element* command and specify only objects on which the *wrapper_element* attribute has not already been set.

---

## UIT-616

(warning) %s has already been wrapped

**Description**

An object can be wrapped only once.

**What Next**

Check whether the object name is correct.

---

## UIT-617

(warning) %s has already been specified as clock port for %s port

**Description**

Duplicate specification is detected.

**What Next**

Check the names of clock port and data port.

---

## UIT-618

(error) '%d' is not a valid value for %s option of the dft_drc command.

**Description**

You receive this message when you have assigned an invalid value to the *-max_detail_lines* option for the *dft_drc* command. The value must be greater than or equal to 0.

This message is only issued in the RTL test DRC flow.

Feedback

**What Next**

Reexecute the *dft_drc* command using a valid value.

---

## UIT-619

(error) '%s' is not a valid disabling option for a tristate net.

**Description**

You receive this message if you issue *set_scan_configuration* or *set_scan_tristate* with an invalid value for the scan shift tristate disabling option for internal or external tristate nets. Valid values for the tristate disabling option in scan shift are *enable_one*, *disable_all*, or *no_disabling*.

**What Next**

Reissue the *set_scan_tristate* or *set_scan_configuration* command and specify a valid scan shift disabling option for tristate nets in the design.

---

## UIT-620

(information) Writing the disable option for the tristate net '%s'.

**Description**

You receive this message to inform you that the *set_scan_tristate* command is writing to the persistent database the scan shift disabling option you specified for this tristate net.

**What Next**

This is an informational message only. No action is required on your part.

---

## UIT-621

(warning) Overwriting the disable option for the tristate net '%s'.

**Description**

You receive this message to inform you that there was already a scan shift disabling option set for this tristate net, and that setting is being overwritten by the scan shift disabling option you just specified using the *set_scan_tristate* command.

**What Next**

No action is required on your part.

Feedback

## UIT-622

(error) The '%s' object is a '%s'.

### Description

You receive this message if you issue the *set_scan_tristate* command and the *net_list* argument includes the name of an object that is not a net. You can specify only net names with the *set_scan_tristate* command.

### What Next

Reissue the *set_scan_tristate* command and specify a *net_list* argument that lists only net names.

## UIT-623

(error) '%s' is not a bidirectional port.

### Description

You receive this message if you issue the *set_scan_bidi* command and the *port_list* argument includes the name of an object that is not a bidirectional port. You can specify only bidirectional ports with the *set_scan_bidi* command.

### What Next

Reissue the *set_scan_bidi* command and specify a *port_list* argument that lists only bidirectional port names.

## UIT-624

(information) Writing the mode in scan shift for the bidirectional port '%s'.

### Description

You receive this message to inform you that the *set_scan_bidi* command is writing to the persistent database the scan shift mode you specified for this bidirectional port.

### What Next

This is an informational message only. No action is required on your part.

## UIT-625

(warning) Overwriting the mode in scan shift for the bidirectional port '%s'.

Feedback

**Description**

You receive this message to inform you that there was already a scan shift mode set for this bidirectional port, and that setting is being overwritten by the scan shift mode you just specified using the *set_scan_bidi* command.

**What Next**

No action is required on your part.

---

## UIT-626

(error) '%s' is not a valid disabling option for internal tristate nets.

**Description**

You receive this message if you issue *set_scan_configuration* with an invalid value of the scan shift tristate disabling option for internal tristate nets. Valid values are *disable_all*, *enable_one* (the default) or *no_disabling*.

**What Next**

Reissue the *set_scan_configuration* command and specify a valid scan shift tristate disabling option for internal tristate nets in the design.

---

## UIT-627

(error) '%s' is not a valid value for the -prtool option argument.

**Description**

You receive this message if you execute *set_scan_configuration* with an incorrect argument for the *-prtool* option. Currently allowed values are *cadence* and *avant*.

**What Next**

Reissue the *set_scan_configuration* command with one of the allowed values for the *-prtool* option argument.

---

## UIT-628

(warning) The design '%s' is not scan completed and cannot be represented with a test model.

**Description**

Only a scan completed design may be represented using a test model. You have specified a design that is not scan completed as a scan core.

Feedback

**What Next**

Make sure that your design is scan completed and re-issue the *set_scan_core* command.

---

## UIT-629

(Error) The scan path specification has been rejected.

**Description**

The error displayed by the tool caused the current scan path specification to be ignored. Please correct the error and enter the command again. Common mistakes associated with this error are:

- Scan path name not specified
- No elements specified for the current scan path
- Incorrect test mode name specified

**What Next**

Make sure that your specification is correct and re-issue the command.

---

## UIT-630

(error) Pin '%s' is not sticky.

**Description**

You receive this message if you issue the *set_rtl_load* command and specify a pin that is not sticky. A sticky pin is a pin or port that is guaranteed to be persistent throughout the early stages of optimization, so that annotated capacitance and resistance can "stick" to those pins and not be lost. You can perform *set_rtl_load* only on sticky pins, or on nets connected to sticky pins.

For a complete explanation of sticky pins, see the man page for the *set_rtl_load* command.

**What Next**

If you want to set an RTL load value on the specified pin, you must make it sticky; you can do so by using *set_dont_touch* or *set_size_only* on the pin. Then reissue the *set_rtl_load* command.

If you do not need to set an RTL load on the specified pin, remove it from the *pin_net_list* and reissue the *set_rtl_load* command.

Feedback

### See Also

- set_dont_touch
- set_size_only

---

## UIT-631

(error) Net '%s' has no sticky pins.

### Description

You receive this message if you issue the *set_rtl_load* command and specify a net that has no sticky pins. A sticky pin is a pin or port that is guaranteed to be persistent throughout the early stages of optimization, so that annotated capacitance and resistance can "stick" to those pins and not be lost. You can perform *set_rtl_load* only on sticky pins, or on nets connected to sticky pins.

For a complete explanation of sticky pins, see the man page for the *set_rtl_load* command.

### What Next

If you want to set an RTL load value on the specified net, you must make one or more pins of the net sticky. You can make pins sticky by using *set_dont_touch* or *set_size_only*; alternatively, you can make the net sticky by using *set_dont_touch* on the net. Then reissue the *set_rtl_load* command.

If you do not need to set an RTL load on the specified net, remove it from the *pin_net_list* and reissue the *set_rtl_load* command.

### See Also

- set_dont_touch
- set_size_only

---

## UIT-632

(error) You must specify at least a capacitance or resistance value.

### Description

You receive this message if you issue the *set_rtl_load* command and do not specify a capacitance or a resistance value on the pin or net. You must specify at least one; otherwise, the command has no effect. For more information, see the man page for the *set_rtl_load* command.

Feedback

### What Next

Reissue the *set_rtl_load* command and specify a capacitance or resistance value, or both, using the *-cap cvalue* or *-res rvalue* arguments.

---

## UIT-633

(error) You must specify a net or pin object or -all.

### Description

You receive this message if you issue the *remove_rtl_load* command and do not specify either *-all* or *pin_net_list*. You must specify one or more target objects on which the *remove_rtl_load* command is to operate.

### What Next

Reissue the *remove_rtl_load* command and specify either *-all* or *pin_net_list*.

---

## UIT-634

(error) You must specify either or both of the -capacitance and -delay options.

### Description

You receive this message if you issue the *calculate_rtl_load* command without either the *-capacitance* or *-delay* option. At least one of these options is required, and you can specify both.

### What Next

Reissue the *calculate_rtl_load* command and specify at least one of the *-capacitance* and *-delay* options.

---

## UIT-635

(error) Net '%s' must have a set_load value.

### Description

You receive this message if you issue *calculate_rtl_load -capacitance* and the specified net does not have a *set_load* value on it. In order for the *-capacitance* option to be processed, you must have previously placed a *set_load* value on the specified net, or on the net that is connected to the specified pin.

Feedback

### What Next

Set a load on the specified net using *set_load*. Then reissue the *calculate_rtl_load* command.

### See Also

- set_load

---

## UIT-636

(Error) The scan path removal specification has been rejected.

### Description

The error displayed by the tool caused the current scan path removal specification to be ignored. Please correct the error and enter the command again. Common mistakes associated with this error are:

- Scan path name not specified
- No elements specified for the current scan path
- Incorrect test mode name specified

### What Next

Make sure that your specification is correct and re-issue the command.

---

## UIT-650

(warning) A set_test_point_element command has already been specified for the pin %s and for the type %s.\n \tThe previous set_test_point specification will be deleted and replaced with the current one.

### Description

You have used set_test_point_element to set a new test point on pin %s \n \tA user-defined test point specification already exists on that pin. This is a warning message \n \tThe previous user-defined test point specification will be deleted.

### What Next

If you do not want to remove a test point which already exists on the specified pin, do not specify this pin with the set_test_point_element command.

Feedback

## UIT-651

(Information) The previous user-defined test point specification in location %s of type %s \n \tis being deleted.

### Description

This is an information message. The previous user-defined test point specification in location %s of type %s \n \tis being deleted.

### What Next

You can specify another test point location with the set_test_point_element command.

## UIT-652

(warning) Cell %s has a dont_touch attribute. \n \tThe asynchronous violation associated with it will not be autofixed.

### Description

You receive this warning message because the asynchronous violation associated with cell %s will not be autofixed because of a *dont_touch* attribute on this cell.

### What Next

If you are satisfied with this result, no action is required on your part. If you want the asynchronous violation associated with the specified cell to be autofixed, remove the *dont_touch* attribute from the specified cell and rerun autofix.

## UIT-653

(warning) Cell %s has a dont_touch attribute. \n \tThe clock violation associated with it will not be autofixed.

### Description

You receive this warning message because the clock violation associated with cell %s will not be \n autofixed because of a *dont_touch* attribute on this cell.

### What Next

If you are satisfied with this result, no action is required on your part. If you want the clock violation associated with the specified cell to be autofixed, remove the *dont_touch* attribute from the specified cell.

Feedback

## UIT-654

(information) There are %d other cells with the same violation.

### Description

This message summarizes the total number of violations identified in UIT-652 and UIT-653.

### What Next

This is an informational message. No action is required on your part. For more information, see the man page for UIT-652 and UIT-653.

## UIT-655

(warning) Scan style '%s' does not add lockup latches to the scan chain.\n \t Ignoring the set_scan_configuration -insert_end_of_chain_lockup_latch command.

### Description

This warning message tells you that you have asked DFT Compiler to add lockup latches to the scan chain but have specified a scan style that is not multiplexed_flip_flop. The compiler is ignoring the *-insert_end_of_chain_lockup_latch* option. Only the multiplexed_flip_flop scan style adds lockup latches to the scan chain.

### What Next

To avoid receiving this warning message, remove *-insert_end_of_chain_lockup_latch true* from the *set_scan_configuration* option list, or set the scan style to multiplexed_flip_flop.

## UIT-656

(warning) port '%s' is the wrong direction to be a %s.

### Description

This warning message informs you that you tried to set the access point for a register port to something that is inconsistent with its direction. The *set_bsd_register* command lets you specify access points for a user-defined register using boundary scan signal semantics. Ports with signal type "tdo" must be associated with output ports. Ports with all other signal types should not be associated with output ports or tristate ports.

### What Next

Issue the *set_bsd_register* command and specify a valid access port or signal type.

Feedback

## UIT-657

(error) Register port list has multiple signal types for hookup point '%s'.

### Description

This error message informs that you tried to associate more than one signal type with a port. The *set_bsd_register* command lets you specify ports for user-defined registers using boundary-scan signal semantics. However, you can associate only one signal type with each port.

### What Next

Use the *set_bsd_register* command and specify only one signal type per port.

## UIT-658

(warning) Port specification is not paired. Last specification is incomplete and is discarded.

### Description

This warning message informs you that you specified an incomplete port with the *set_bsd_register* command. The *set_bsd_register* command lets you specify ports for user-defined registers using boundary-scan signal semantics. Specified ports must be pairs, and each pair must include a signal type and a design port or pin. Incomplete pairs cannot be used to specify ports and are discarded.

### What Next

To avoid receiving this warning message, remove the last object from the argument list.

## UIT-659

(error) Port type '%s' is repeated.

### Description

This error message informs you that multiple ports have been specified for the same signal type in the access list. The *set_bsd_register* command lets you specify ports for user-defined registers using boundary-scan signal semantics. Port signal types "tdi" and "tdo" can be specified multiple times only for register named as *STT_REG*. For all other registers signal types "tdi" and "tdo" can be specified only once. All other signal types can always be repeated.

Feedback

### What Next

Run the *set_bsd_register* command and specify only one "tdi" and one "tdo" port if the register is not *STT_REG*. If you want to specify a scan-through-tap register use the reserved name *STT_REG* for the register and reissue the command.

---

## UIT-660

(error) Access point specification is missing signal type '%s'.

### Description

The *set_bsd_register* command lets you specify access points for a user-defined register with boundary-scan signal semantics. This error is generated because the register cannot be wired correctly without a signal of the designated type.

### What Next

Rerun the *set_bsd_register* command and include ports for the required signal type using the *-access* option.

---

## UIT-661

(Error) Missing ports for user-defined BSR or TAP elements.

### Description

This error is generated because some of the required access ports for the BSR and TAP elements are missing from the specification. The *set_bsd_bsr_element* or *set_bsd_tap_element* command lets you specify user-defined BSR or TAP elements.

### What Next

Rerun one of the commands and specify all required access ports using the *-access* option. For a list of required access ports, see the man page for the *set_bsd_bsr_element* or *set_bsd_tap_element* commands.

If you are getting this error on set_bsd_tap_element, it might be due to the reason that you are using an old custom tap imlementation and signal type 'instructions' is missing in the access list. In this case you might want to set the variable 'bsd_use_old_tap' to true and rerun the command.

---

## UIT-662

(error) Mismatched signal types 'tdi' and 'tdo'.

Feedback

**Description**

The *set_bsd_register* command lets you specify access points 'tdi' and 'tdo' multiple times for the scan-through-tap register *STT_REG*. This error is generated either because access points 'tdi' and 'tdo' do not match in number or because they are not specified in alternate order.

**What Next**

Correct the access points 'tdi' and 'tdo' specified in the *-access* option and rerun the *set_bsd_register* command.

---

## UIT-663

(error) Multiple signal types cannot be specified for hookup port '%s'.

**Description**

This error message informs that you tried to associate more than one signal type with a hookup port. The *set_dft_signal* command lets you specify hookup ports for diffrent types. However, you can associate only one signal type with each port.

**What Next**

Use the *set_dft_signal* command and specify only one signal type per hookup port.

---

## UIT-664

(error) -bsd_style can only be specified with -view spec.

**Description**

You receive this error message when you execute the *set_scan_path* command with view existing_dft and -bsd_style options. The -bsd_style option can not be specified with view existing_dft.

**What Next**

Fix the error and reexecute the command.

---

## UIT-665

(error) -bsd_style can only be specified for user defined test data registers.

Feedback

### Description

You receive this error message when you execute the *set_scan_path* command with -bsd_style option for Boundary or Short BSR Chains. -bsd_style can be specified for User Defined Test Data Registers only.

### What Next

Use -bsd_style for User Defined Test Data Registers only.

---

## UIT-666

(warning) Scan-in and scan-out ports for compression mode %s should be specified only in the base mode.

### Description

This warning message occurs when scan-in and scan-out ports are specified in the compression mode.

The compression mode specifications are specifically used to guide the tool to build scan chains outside compressor/decompressor in the compression mode. These chains can be specified in the compression mode by using the *set_scan_path command* and assigning the flops to be routed between the external scan-in and scan-out ports.

If you do not intend to build any chains outside compressor/decompressor, specify scan-in and scan-out specifications only in the all_dft mode or the base scan mode.

For example, the following commands instruct the tool to build scan chains outside compressor/decompressor in compression mode:

```
set_dft_signal -type ScanDataIn -port my_si \\
  -view spec -test_mode ScanCompression_mode

set_dft_signal -type ScanDataOut -port my_so \\
  -view spec -test_mode ScanCompression_mode

set_scan_path my_ext_chain -view spec -scan_data_in my_si \\
  -scan_data_out my_so -include_elements {Flop1 Flop2 Flop3} \\
  -complete true -test_mode ScanCompression_mode
```

### What Next

This is only a warning message. No action is required.

However, if the result is not what you intended, use the commands shown above for correct specifications, or remove the scan-in and scan-out specifications from the compression mode and specify them in the all_dft or the base scan mode.

Feedback

## UIT-667

(error) Pin name '%s' specified more than once.

**Description**

This message informs you that you ran the *define_dft_design* command with an access list that specifies a pin name more than once.

**What Next**

Rerun the *define_dft_design* command with a valid access list.

## UIT-668

(error) Signal type '%s' specified more than once.

**Description**

This message informs you that you ran the *define_dft_design* command with an access list that specifies any of the following signal types more than once: data_in, data_out or enable.

**What Next**

Rerun the *define_dft_design* command after making sure that the signal types viz. data_in, data_out and enable are specified only once.

## UIT-669

(error) Polarity high(H) specified more than once for signal type '%s'.

**Description**

This message informs you that you ran the *define_dft_design* command with an access list that specifies polarity "high" more than once for any of the following signal types: data_in, data_out and port.

**What Next**

Rerun the *define_dft_design* command after making sure that the polarity "high" is specified exactly once if the signal type is data_in, data_out or port.

## UIT-670

(error) Polarity low(L) specified more than once for signal type '%s'.

Feedback

### Description

This message informs you that you ran the *define_dft_design* command with an access list that specifies polarity "low" more than once for any of the following signal types: data_in, data_out and port.

### What Next

Rerun the *define_dft_design* command after making sure that the polarity "low" is specified exactly once if the signal type is data_in, data_out or port.

---

## UIT-671

(warning) Signal type '%s' should not be specified for 'hybrid' PAD. Ignored the specification.

### Description

This message informs you that you ran the *define_dft_design* command for a hybrid PAD with an interface list that specifies any of the following signal types: port, data_in, data_out, or enable. These signal types are not used for a hybrid PAD. For a hybrid PAD this information is collected from the *$bsr_segment$* specification of the *-params* option of the *define_dft_design* command.

### What Next

This is only a warning message. No action is required.

However, if the result is not what you intended you can make sure that the port, data_in, data_out, and enable signal types are not specified for the hybrid PAD, and then run the *define_dft_design* command again.

---

## UIT-700

(warning) Invalid scan state.

### Description

This warning message informs you that you have specified an invalid scan state with the *set_scan_state* command.

### What Next

Rerun the *set_scan_state* command and specify a valid scan state.

Feedback

## UIT-701

(error) Incorrect value of -power_saving_on option argument. Allowed values are true, TRUE; or false, FALSE.

### Description

You receive this message if you execute *set_testability_configuration* with an incorrect argument for the *-power_saving_on* option. Allowed values of the *-power_saving_on* option are *true, TRUE; or false, FALSE.*

### What Next

Re-issue the command with one of the allowed values for the *-power_saving_on* option argument.

## UIT-702

(warning) '%s' is not an external port and cannot be used to specify a port.

### Description

This warning informs you that you used an object other than an external port to specify a port. The *set_test_hold* command allows only external ports to specify ports. Other kinds of objects cannot be used to specify ports and are rejected.

### What Next

Remove the offending object from the argument list. Use hierarchical instance names, with respect to the top-level design, to locate external ports. Then, rerun the command with a valid external port.

## UIT-703

(warning) The test point '%s' is not inserted, because the instance location '%s' does not exist due to insert_dft optimization.

### Description

You receive this warning message if you execute *insert_dft* after the *set_test_point_element* command. The set_test_point_element command allows design objects to specify pin instances on which test points need to be inserted. Some test points might be rejected if their corresponding pin instances do not exist after the optimization.

### What Next

If the test point is required then use the *set_dont_touch* command to prevent the cell from being modified or replaced during optimization. Another way consists

Feedback

in using *insert_dft* with optimization switched off. Use *-dont_fix_constraints* and *-ignore_compile_design_rules* options to switch off optimization.

**NAME**

*UIT-703* (warning) The test point '%s' is not inserted because the instance location '%s' does not exist due to insert_dft optimization.

**Description**

This messages informs you that the specified test point is rejected because corresponding pin instances do not exist after optimization. You receive this warning message if you use the *insert_dft* command after the *set_test_point_element* command. The *set_test_point_element* command allows design objects to specify pin instances on which test points must be inserted.

**What Next**

Use the *set_dont_touch* command to prevent the cell from being modified or replaced during optimization; or, use *insert_dft* with optimization switched off. Use the *-dont_fix_constraints* and *-ignore_compile_design_rules* options with the *insert_dft* command to switch off optimization.

**See Also**

- set_dont_touch

---

## UIT-704

(warning) Cell '%s' has a dont_touch attribute. \n \tThe violation(s) associated with it will not be autofixed.

**Description**

This message informs you that the *dont_touch* attribute exists on the specified cell and prevents violations associated with this cell from being autofixed. You receive this warning message when you use the *preview_dft* command. The full list of instances with the *dont_touch* attribute is reported using the *preview_dft -verbose* command. Without the *-verbose* switch, the *preview_dft* command reports the first violation and the total number occurrences of the *dont_touch* attribute.

**What Next**

If you are satisfied with this result, no action is required on your part. If you want the violation(s) associated with the specified cell to be autofixed, remove the *dont_touch* attribute from the specified cell and rerun AutoFix.

Feedback

## UIT-705

(information) There are %d other cells with the same violation.

### Description

This message summarizes the total number of violations identified in UIT-704. Specifically, this message reports the first violation and the total number of occurrences of the *dont_touch* attribute in the design.

### What Next

This is an informational message. No action is required on your part. For more information, see the man page for UIT-704.

### See Also

- UIT-704

## UIT-706

(information) The port %s is used as a test mode signal to AutoFix asynchronous sets and resets and as a test scan enable signal.

### Description

This message informs you that the port %s is shared between the test mode signal and the test scan enable signal to Autofix asynchronous sets and resets. You have used the *set_autofix_configuration* command with the *-fix_async_with_se true* option to use the test scan enable signal to Autofix asynchronous sets and resets.

### What Next

This is an informational message only. No action is required on your part.

## UIT-707

(error) Incorrect value for -fix_async_with_scan_en option argument. Allowed values are true, TRUE, on, and ON; or false, FALSE, off, and OFF.

### Description

This error message informs you that you used the *set_autofix_configuration* command with an invalid argument for the *-fix_async_with_scan_en* option. Allowed values for the *-fix_async_with_scan_en* option argument are true, TRUE, on, and ON; or false, FALSE, off, and OFF.

Feedback

### What Next

Use the *set_autofix_configuration* command with the *-fix_async_with_scan_en* option and specify one of the allowed values as the argument.

---

## UIT-708

(Warning) Overwriting the async port specification '%s' to autofix the cell/instance '%s'.

### Description

You can use the *set_autofix_async* command to put a specific async port to autofix a cell/instance. This warning message is produced if you try to perform *set_autofix_async* on a cell/instance that already has a specified async port.

### What Next

No further action is necessary.

---

## UIT-709

(information) One asynchronous pin of the cell %s has already been AutoFixed. As a consequence, another handler %s will be used to AutoFix the second asynchronous pin %s.

### Description

You receive this information message because one of the asynchronous signals of the cell has been AutoFixed. The same handler cannot be used to AutoFix the asynchronous pins of the same register. Therefore, two different handlers will be used to AutoFix the two asynchronous pins of the same register.

### What Next

Use the *set_dft_signal* command to specify an additional test data port to be available to AutoFix to fix asynchronous signals.

---

## UIT-710

(error) Invalid cell type for core wrapper insertion.

### Description

You receive this message if you issue the *set_core_wrapper_design* command and specify an invalid value for the wrapper cell type. Valid values of wrapper cell types are listed in the man page and the online help for the *set_core_wrapper_design* command. The

Feedback

*insert_dft* command uses the specified wrapper cell type when synthesizing wrapper into the design.

**What Next**

Refer to the man page or the online help for the *set_core_wrapper_design* command to determine the current valid wrapper cell types, and reissue the *set_core_wrapper_design* command with a valid wrapper cell type.

---

## UIT-711

(error) Cannot find the specified design in memory.

**Description**

You receive this message if you execute the *set_core_wrapper_design* command and the name you specify for the *design_name* argument cannot be found in memory. This could be caused by a typo or spelling error. The *insert_dft* command uses the specified design as a wrapper cell design when synthesizing wrapper into the design.

**What Next**

Verify that the design you want to use does exist in memory, and verify its spelling. Then re-execute the appropriate command using the correct design name.

---

## UIT-712

(error) Invalid port type '%s' specified for user-defined wrapper cell design.

**Description**

You receive this message if you issue the *set_core_wrapper_design* command and specify an invalid port type. Valid port types are found in the man page or online help for the above command. The *insert_dft* command uses the user-defined wrapper cell design when synthesizing wrapper into the design.

**What Next**

Refer to the man pages or the online help for the *set_core_wrapper_design*, command to determine the current valid wrapper cell design port types, respectively. Then reissue the appropriate command with a valid port type.

---

## UIT-713

(Error) Missing ports for user-defined wrapper cell design : %s.

Feedback

**Description**

This error is generated because some of the required access ports for the wrapper cell design are missing from the specification. The *set_core_wrapper_design* command lets you specify user-defined wrapper cell design.

**What Next**

Rerun one of the commands and specify all required access ports using the *-access* option. For a list of required access ports, see the man page for the *set_core_wrapper_design* command.

---

## UIT-714

(error) Cannot find the specified port '%s' in the design.

**Description**

You receive this message if you execute the *set_core_wrapper_design* command and a port you specify cannot be found in the design. This could be caused by a typo or spelling error. The *insert_dft* command uses the specified wrapper cell type when synthesizing wrapper into the design.

**What Next**

Verify the name of the port. Then reissue the appropriate command using the correct port name for the design or instance.

---

## UIT-715

(error) Scan chain '%s' exists in another mode.

**Description**

DFT Compiler does not allow same scan chains to be specified using multiple *set_scan_path* commands. This warning tells you that an earlier specification for the same scan chain exists for another mode. Scan chain names should be unique for the whole design.

**What Next**

Establish whether you want the specification to be overwritten. If not, combine the two specifications into one *set_scan_path* command for the same mode.

---

## UIT-717

(error) The option test_use_test_models is not supported by write_layout_scan.

Feedback

**Description**

You receive this message if you set the env variable *test_use_test_models* to TRUE before the command write_layout_scan. This flow is not supported for write_layout_scan.

**What Next**

Reset the env variable *test_use_tes_models* before the command *write_layout_scan*

---

## UIT-718

(error) The -power_saving_on option is only available when the -method option has the tdvr argument.

**Description**

You receive this message if you execute *set_testability_configuration* with the *-power_saving_on* option and without tdvr as an argument for the *-method* option.

**What Next**

Re-issue the command with tdvr for the *-method* option argument or re-issue the command without the fB-power_saving_on *option.*

---

## UIT-719

(Error) The type %s is not a valid type for the set_test_point_element command.

**Description**

This message indicates that a set_test_point_element command has been issued, but the type is not a valid type for the command.

**What Next**

Check the valid types in the manpage of the set_test_point_element, modify the type and re-issue the command.

---

## UIT-720

(Error) The parameter for the %s option has to be enable or disable.

**Description**

This message indicates that a set_test_point_element command has been issued, but the parameter for the option is not a valid one. The valid parameters are enable or disable.

Feedback

### What Next

Change the parameter for the option and re-issue the command.

---

## UIT-721

(Error) Port %s is not a test mode.

### Description

The port %s is not a test mode. Use *set_dft_signal* to create a test mode associated with port %s.

### What Next

After the dft signal specification, re-issue the command.

---

## UIT-722

(Warning) The -scan_test_point_enable option is ignored for the type %s.

### Description

This message indicates that a set_test_point_element command has been issued, but the -scan_test_point_enable option is not an option that can be specified for the type and the option is ignored. The -scan_test_point_enable option is a valid option for the control types.

### What Next

Remove the -scan_test_point_enable option.

---

## UIT-723

(Warning) The -test_points_per_test_point_enable option is ignored for the type %s.

### Description

This message indicates that a set_test_point_element command has been issued, but the -test_points_per_test_point_enable option is not an option that can be specified for the type and the option is ignored. The -test_points_per_test_point_enable option is a valid option for the control types.

### What Next

Remove the -test_points_per_test_point_enable option.

Feedback

## UIT-724

(Warning) The -test_point_enable option is ignored for the type %s.

**Description**

This message indicates that a set_test_point_element command has been issued, but the -test_point_enable option is not an option that can be specified for the type and the option is ignored. The -test_point_enable option is a valid option for the control types.

**What Next**

Remove the -test_point_enable option.

## UIT-725

(Error) The port or the pin %s does not have the correct direction.

**Description**

This message indicates that a set_test_point_element command has been issued, but the port or the pin does not have the correct direction according to the option in which the port or the pin has been specified.

**What Next**

Specify another port or another pin with the correct direction and re-issue the command.

## UIT-726

(Error) The pin %s does not have the correct direction for a test point location.

**Description**

This message indicates that a set_test_point_element command has been issued, but a pin from the list does not have the correct direction. A test point location has to be an output pin.

**What Next**

Specify another pin with the correct direction for the test point location and re-issue the command.

## UIT-727

(Error) The port %s cannot be specified for a test point location.

Feedback

**Description**

This message indicates that a set_test_point_element command has been issued, but a port cannot be the location of a test point. A test point location has to be an output internal pin.

**What Next**

Specify an internal output pin and re-issue the command.

---

## UIT-728

(warning) Cell '%s' has no function specification. \n \tThe asynchronous pin violation(s) associated with it will not be autofixed.

**Description**

This message informs you that the specified cell has no function specification and this prevents the asynchronous pin violations associated with this cell from being autofixed. You receive this warning message when you use the *preview_dft* and *insert_dft* commands.

**What Next**

If you are satisfied with this result, no action is required on your part. If you want the violation(s) associated with the specified cell to be autofixed, provide the library for the cell; the test_simulation_library does not provide enough information to autofix the asynchronous pins of the specified cell.

---

## UIT-730

(Error) Unsupported client '%s' for RTL Insertion.

**Description**

This message informs you that RTL insertion is incompatible with any other DFT clients.

**What Next**

Remove DFT clients from set_dft_configuaration and insert_dft UI commands.

---

## UIT-731

(Error) Unrecognized HDL format '%s'.

Feedback

### Description

This message informs you that the specified HDL format for RTL insertion is not valid. Only 'verilog' and 'VHDL' are valid values.

### What Next

Change the HDL format to 'verilog' or 'VHDL'.

---

## UIT-732

(error) '%s' is not a directory.

### Description

You have specified an invalid directory; or, the directory or one or more of its parents are read or search protected.

### What Next

Check your project file to ensure that variables specifying directories are correctly set.

Change your permissions to enable reading and searching of the appropriate directory and all of its parents. For example,

```
chmod a+rx <dir_name>
```

---

## UIT-733

(error) Cannot perform RTL insertion because the variable hdlin_enable_rtldrc_info is set to FALSE

### Description

The variable hdlin_enable_rtldrc_info must be set to TRUE to perform RTL insertion. FALSE is the hdlin_enable_rtldrc_info variable default value.

This message is only issued in the RTL test DRC flow.

### What Next

Set the variable hdlin_enable_rtldrc_info to TRUE For example,

```
set hdlin_enable_rtldrc_info TRUE
```

Feedback

## UIT-770

(Warning) The option -identify_shift_register will be ignored in 2007.03 SP4. Shift register identification will be moved fully to DC Ultra in Z2007.03 SP4.

### Description

This options is ignored in 2007.03 release. Added back in 2007.03 SP1. This option will again be ignored from 2007.03 SP4. The functionality is incorporated into *compile_ultra -scan*. The variable compile_seqmap_identify_shift_registers can be used to enable/disable the identification in compile_ultra -scan

### What Next

Execute *compile_ultra -scan*

## UIT-800

(warning) Bad value for the number of test points per observe flop option. Eight test points per observe flop will be considered during test point insertion.

### Description

In order to provide balanced xor tree, the only accepted values for *-test_points_per_observe_flop* option are 1, 2, 4 and 8. If a different value is set, then the default value (8) will be considered.

### What Next

Set number of test points per observe clock with a correct value.

## UIT-801

(warning) No observe test point will be inserted.

### Description

This warning message appears when *-max_observe_points* and/or *-max_observe_logic_area* are set to zero. In this case, no observe test point will be inserted in the design.

### What Next

Set *-observe_points_per_scan_cell* and/or *-max_observe_logic_area* options with the correct value.

Feedback

## UIT-802

(Error) Unable to read the TRC file %s given.

**Description**

This is issued from the command *read_trc_file* . The command is not able to read the file specified due to errors given before.

**What Next**

Check the file syntax as well is as per the standard supported.

## UIT-803

(information) reverting value of environment variable '%s' to its default value '%s'.

**Description**

DFT Compiler validates the values of environment variables it uses. You see this message when an environment variable value is invalid. DFT Compiler supplies a default.

**What Next**

Confirm that the environment variable is invalid. Modify the environment variable value to remove the warning.

## UIT-804

(Warning) '%s' is not a valid scan style.

**Description**

DFT Compiler validates the values of environment variables it uses. You see this message when an environment variable that needs to specify a scan style specifies an invalid one. DFT Compiler will use a default scan style.

**What Next**

Confirm that the environment variable is invalid. Modify the environment variable value to remove the warning.

## UIT-805

(Warning) '%s' is not 'true' or 'false'.

Feedback

**Description**

DFT Compiler validates the values of environment variables it uses. You see this message when an environment variable that needs to specify either 'true' or 'false' specifies something else. DFT Compiler will use a default value.

**What Next**

Confirm that the environment variable is invalid. Modify the environment variable value to remove the warning.

---

## UIT-806

(warning) Cannot associate a '%s' signal with chain '%s'.

**Description**

DFT Compiler uses one scan enable signal to enable scan mode. This message warns that you used the *set_dft_signal* command to associate a scan enable with a specific scan chain. Because the scan enable is associated with all scan chains, the specific association is ignored.

**What Next**

Remove the association to suppress the warning.

---

## UIT-807

(information) reverting value of environment variable '%d' to its default value '%d'.

**Description**

DFT Compiler validates the values of environment variables it uses. You see this message when an environment variable value is invalid. DFT Compiler supplies a default.

**What Next**

Confirm that the environment variable is invalid. Modify the environment variable value to remove the warning.

---

## UIT-808

(Warning) '%s' does not include the substring '%%s'.

Feedback

**Description**

DFT Compiler validates the values of environment variables it uses. You see this message when an environment variable that needs include the substring "%s" does not. DFT Compiler will use a default value.

**What Next**

Confirm that the environment variable is invalid. Modify the environment variable value to remove the warning.

---

## UIT-809

(warning) Hookup sense value '%s' conflicts with the old value '%s'.

**Description**

DFT Compiler only associates exactly one hookup sense between a design port and an internal hookup pin. This message warns that you have used the *set_dft_signal* command to associate a second hookup sense, and that the second specification has overwritten the first.

**What Next**

Check that you do want to overwrite the specification. Repeat the correct specification if the overwrite was mistaken.

---

## UIT-810

(error) Argument -memory_cells should be specified for a single controller.

**Description**

This error occurs because you used the *set_mbist_controller* command to assign a list of memory cells to two or more controllers.

**What Next**

Run the *set_mbist_controller* command spcifying one controller with -controller_names and a list of memory cells using -memory_cells.

---

## UIT-811

(error) Memory '%s' has already been assigned to a controller. Cannot change notouch value to TRUE.

Feedback

### Description

This error occurs because you either used the *set_mbist_controller* command to assign a memory cell, which has already been assigned to a controller, to a new controller or you used *set_mbist_element* command to set notouch value to TRUE for that memory cell.

### What Next

Run the *remove_mbist_configuration* command to remove the current MBIST configuration. Assign the memory cell to the new controller using *set_mbist_controller* command spcifying one controller with -controller_names and the memory cell using -memory_cells.

---

## UIT-812

(error) -notouch and -ip_parameters are mutually exclusive.

### Description

This error occurs because you used the *set_mbist_element* command to simultaneously set the -notouch value and assign the -ip_parameters for the memory cells.

### What Next

Run the *set_mbist_element* command with either -notouch or the -ip_parameter switch.

---

## UIT-813

(error) Odd number of items in the -ip_parameters list.

### Description

This error occurs because you used the *set_mbist_element* or *set_mbist_controller* command to pass odd number of items in the -ip_parameters list.

### What Next

Run the *set_mbist_element* or *set_mbist_controller* command with even number of items in the -ip_parameter list.

---

## UIT-814

(error) -notouch value can be either TRUE or FALSE.

### Description

This error occurs because you used the *set_mbist_element* command to set -notouch to a value other than TRUE or FALSE.

Feedback

**What Next**

Run the *set_mbist_element* command with -notouch value as TRUE or FALSE

---

## UIT-815

(error) Memory '%s' has notouch value set to TRUE and cannot be assigned ip parameters or to a controller .

**Description**

This error occurs because you either used the *set_mbist_controller* command to assign a memory cell, which has already been excluded from MBIST consideration, to controller or tried to assign ip parameters using /fBset_mbist_element/fP.

**What Next**

Run the *remove_mbist_configuration* command to remove the current MBIST configuration. Assign the memory cell to the new controller using *set_mbist_controller* command spcifying one controller with -controller_names and the memory cell using -memory_cells.

---

## UIT-816

(error) Invalid value for the -mbist parameter

**Description**

This error occurs because you specified an invalid value for the -mbist parameter.

**What Next**

See the list of available values

---

## UIT-817

(error) "%s" is not a valid command line option for set_mbist_run command.

**Description**

You receive this message because the parameter(s) name or value specified in the set_mbist_run command are not valid.

**What Next**

Please enter valid parameter name and values in the command line.

Feedback

## UIT-818

(error) Parameter "fail_limit" can be defined only for debug mode.

**Description**

You receive this message because the parameter "fail_limit" was specified for a test_mode other than debug mode.

**What Next**

Please change the test_mode to debug mode or remove the "fail_limit" parameter from the command line.

## UIT-819

(error) Memory instance %s does not exist in the top level model.

**Description**

You receive this message because either the memory instance is missing from the top model or the corresponding reference object is missing.

**What Next**

Please add the memory instance to the netlist. If the instance does not exist in the design, remove the instance name from the list of memories in the command line. If the memory instance is in the design, read the library cell or the design object corresponding to the reported memory instance in the dc_shell.

## UIT-820

(error) Odd number of items in the -parameters list.

**Description**

This error occurs because you used the *set_mbist_wrapper*, *set_mbist_controller* or *set_mbist_configuration* command to pass an odd number of items in the *-parameters* list.

**What Next**

Run the respective command with an even number of items in the -parameter list.

## UIT-821

(error) Controller '%s' has no memories assigned to it, and can not have -complete TRUE.

Feedback

**Description**

This error occurs because you used the *set_mbist_controller* command to set a controller as complete (no more memories can be assigned), but you never assigned any memories. This means you tried to specify a controller that controls nothing.

**What Next**

Try the same *set_mbist_controller* command again, while including some -memory_cells.

---

## UIT-822

(error) 'memory_cells' (alias 'target' in XG mode) parameter is required when view is set to SPEC

**Description**

This is issued from the command *set_mbist_element* (or *set_mbist_wrapper* in XG mode) because the 'memory_cells' (alias 'target' in XG mode) parameter is missing. This is required when view is set to SPEC.

**What Next**

Look at *set_mbist_element* (or *set_mbist_wrapper* in XG mode) man page

---

## UIT-823

(warning) '-mem_id' is ignored when 'view' option is set to EXISTING

**Description**

This is issued from the command *set_mbist_element* (or *set_mbist_wrapper* in XG mode) because the 'mem_id' option is specified in the case of 'view' option set to SPEC. The 'mem_id' option is required only when 'view' is set to EXISTING.

**What Next**

Look at *set_mbist_element* (or *set_mbist_wrapper* in XG mode) man page

---

## UIT-824

(warning) '-no_touch', '-memory_cells' (alias '-target' in XG mode) and '-ip_parameters' (alias '-parameters' in XG mode) options are ignored when 'view' option is set to EXISTING

Feedback

**Description**

This is issued from the command *set_mbist_element* (or *set_mbist_wrapper* in XG mode) because the '-no_touch', '-memory_cells' (alias '-target' in XG mode) and '-ip_parameters' (alias '-parameters' in XG mode) options are used when 'view' option is set to SPEC and are ignored when 'view' option is set to EXISTING.

**What Next**

Look at *set_mbist_element* (or *set_mbist_wrapper* in XG mode) man page

---

## UIT-825

(error) the test model has not been generated by MBISTC Tech 2

**Description**

This is issued from the command *set_mbist_element* (or *set_mbist_wrapper* in XG mode) because the test model of the current design has not been generated by MBISTC Tech 2 and so it can not be modified.

**What Next**

Read the correct test model and set the correct current design

---

## UIT-826

(error) 'content_file' option supports only one file name in the case of 'view' option set to EXISTING

**Description**

This is issued from the command *set_mbist_element* (or *set_mbist_wrapper* in XG mode) because the 'content_file' option value is a list of file names. The 'content_file' value can be a file name list in the case of 'view' option set to SPEC; only one file name is supported for 'content_file' option in the case of 'view' option set to EXISTING.

**What Next**

Look at the *set_mbist_element* (or *set_mbist_wrapper* in XG mode) man page

---

## UIT-827

(error) At least one of the trace_file parameters must be specified

Feedback

**Description**

This is issued from the command *report_mbist_trace* as no trace file was specified. You must specify at least one of the following parameters: *-trace_file*, *-repair_trace_file*

**What Next**

Specify a trace file parameter

---

## UIT-900

(warning) '%s' is not a cell, instance, scan segment, or scan link, and is not being added to the scan group.

**Description**

The set_scan_group command accepts an unordered list of design objects. These must be cells, instances, links or segments. Other kinds of design objects cannot be added to scan groups, and are rejected.

**What Next**

Remove the object from the argument list to get rid of the warning. Use hierarchical instance names, with respect to the top-level design, to locate design objects.

---

## UIT-901

(warning) Scan group element '%s' is a scan link\n\tand is not being added to the scan group.

**Description**

The *set_scan_group* command accepts an unordered list of design objects. The first object in the list can not be a scan link because there is no preceding cell to associate it with.

**What Next**

Remove the object from the argument list to get rid of the warning.

---

## UIT-902

(Error) Ignoring the scan group specification as the scan group '%s' is empty.

**Description**

Specify scan groups using the *set_scan_group* command.

Feedback

This message tells you that DFT Compiler ignores a *set_scan_group* command because the scan group is empty .

**What Next**

Specify valid scan group elements to get rid of the error. If you did specify group elements, study accompanying warning messages to see why they were discarded.

---

## UIT-903

(error) Scan %s '%s' element '%s' has internal and wrapper segments.

**Description**

This error message is generated when you try to include a core cell into a scan group or scan segment. A core cell is a cell with a test model and has both internal scan and wrapper segments. As internal scan segments and wrapper segments can't be mixed during core integration, you can't include such a cell with other elements of a scan group or a scan segment.

**What Next**

Check if you have specified core_integration flag of dft configuration. If you are not integrating cores, remove such a specification and reinvoke the current command.

---

## UIT-904

(error) Scan Group '%s' has %s segments.

**Description**

This error message is generated when you try to include a wrapper or an internal scan segment with other segments in a scan group during core integration. Wrapper or internal scan segments can not be mixed with other segments during core integration.

**What Next**

Check if you have specified core_integration flag of dft configuration. If you are not integrating cores, remove such a specification and reinvoke the current command.

---

## UIT-905

(warning) Scan group '%s' overwrites an earlier specification.

Feedback

**Description**

DFT Compiler does not allow scan groups to be specified using multiple *set_scan_group* commands. This warning tells you that an earlier specification of scan group is being overwritten.

**What Next**

Establish whether you want the specification to be overwritten. If not, combine the two specifications into one *set_scan_group* command.

---

## UIT-906

(error) Scan group '%s' exists in another mode.

**Description**

DFT Compiler does not allow same scan group to be specified using multiple *set_scan_group* commands. This warning tells you that an earlier specification for the same scan group exists for another mode. Scan group names should be unique for the whole design.

**What Next**

Establish whether you want the specification to be overwritten. If not, combine the two specifications into one *set_scan_group* command for the same mode.

---

## UIT-907

(error) Scan groups '%s' and '%s' have common elements.

**Description**

DFT Compiler does not let scan groups share scan elements (cells or design instances). This error tells you that some elements belong to more than one group.

**What Next**

If the common elements can be removed from the scan group you're specifying, remove them and reapply the *set_scan_group* command. Otherwise, apply the *remove_scan_group* command and start again specifying without common elements between groups.

---

## UIT-908

(error) Signal of data type '%s' is not valid for internal pins flow.

Feedback

**Description**

The signal data types that are valid for internal pins flow are MasterClock, ScanMasterClock, Reset, Constant, ScanDataIn, ScanDataOut, ScanEnable, and TestMode.

**What Next**

Please specify a valid data type using the *set_dft_signal* command and turn on the internal pins flow using the *set_dft_drc_configuration*.

---

## UIT-909

(error) Internal pin '%s' of type '%s' direction is not valid for internal pins flow.

**Description**

If the internal pin specified is not a hierarchical pin and is of data type MasterClock, ScanMasterClock, Reset, Constant, ScanDataIn, ScanEnable, and TestMode, the pin should be of direction Out and if the data type is ScanDataOut, the pin should be of direction In. For the hierarchical pins, pins with both In and Out directions are valid.

**What Next**

Please specify a pin with valid direction using the *set_dft_signal* command and turn on the internal pins flow using the *set_dft_drc_configuration*.

---

## UIT-910

(error) The hookup pin '%s' of type '%s' is not a valid hookup pin.

**Description**

Driver pins specified with the *-hookup_pin* option of the *set_dft_signal* command must be driven by potentially active combinational or sequential logic, or they must be output pins of black-box cells.

In this case, the specified hookup pin is not valid. Possible causes are:

- The pin is driven by constant 0 or constant 1 logic value.
- The pin is undriven.

**What Next**

Change the netlist to ensure that the specified hookup pin is properly driven or specify another hookup pin with a valid driver. If the pin is expected to be a black-box output pin, check that the cell is indeed a black box by using the *report_cell* command to check for the 'b' (black box) attribute.

Feedback

## UIT-911

(error) Numbers of pipeline head and tail registers do not match the respective numbers in previously specified scan paths

### Description

You have specified several scan paths with their corresponding head and tail registers. But the number of head pipeline registers specified for the current chain is different from the one specified in previous chains or the number of tail pipeline stages specified for the current chain is different from the one specified in previous chains. The number of head pipeline registers should be the same across all chains. The number of tail pipeline registers should be the same across all chains.

### What Next

Correct the list of register name for *set_scan_path -pipeline_head* or *-pipeline_tail* command

## UIT-912

(error) The pin '%s' is not a valid hookup pin.

### Description

When hookup pins are specified with the *-hookup* option of the *set_scan_path* command, they must be previously defined with the correct signals types using the *set_dft_signal* command.

In this case, the tool could not find any pin type information for the indicated hookup pin.

### What Next

Verify that the pins are defined with the correct signal types using the *set_dft_signal* command.

If you are using a multiple test mode flow, be sure that the *set_dft_signal* commands are defined on the *all_dft* mode. Do this either by defining the signals before defining the test modes, or by defining the signals explicitly with the *-test_mode all_dft* option.

## UIT-913

(warning) Same hookup pin specified more than once. Duplicate hookup pins will be removed.

Feedback

**Description**

The hookup pins specified for set_scan_path are expected to be unique. Multiple use of a hookup pin gives this warning. The duplicate hookup pin specification is ignored.

**What Next**

Use unique hookup pins with *-hookup* option of *set_scan_path* command. Use *report_scan_path* command to observe the scan path set.

---

## UIT-914

(warning) Specification of a signal of type Oscillator in spec view is ignored.

**Description**

You receive this message because you have specified a signal of type Oscillator with -view spec or without the -view option. In this case, the command will be ignored because a signal of type Oscillator must be specified in view existing_dft.

**What Next**

Please specify the signal in existing view using the *set_dft_signal* command.

---

## UIT-915

(error) Hookup pin is not allowed for the '%s' boundary scan port.

**Description**

This error message occurs when a hookup pin is specified while using the *set_dft_signal* command with the *-type TAP* ports option. A hookup pin cannot be specified for the TCK, TDI, TDO, TMS and TRST boundary scan TAP ports.

**What Next**

Remove the *-hookup_pin* option for BSD TAP ports. See the *set_dft_signal* man page for more information.

---

## UIT-916

(error) Only one port is allowed for the '%s' boundary scan signal type.

**Description**

This error message occurs when multiple ports for a BSD signal type are specified with the *-port* option of the *set_dft_signal* command. Only one port should be used as a boundary scan TAP port.

Feedback

### What Next

Use only one port with the *-port* option of the *set_dft_signal* command for boundary scan TAP ports.

---

## UIT-917

(warning) BSD Test Data Register hookup pins are going to be obsoleted in existing_dft view for synthesis, specify them in spec view.

### Description

Considering that BSD Test Data Registers can be useful only at spec view, it is decided to remove the -existing_dft option of *set_scan_path* and *set_dft_signal* commands for BSD Test Data Registers.

### What Next

Use -hookup_pins option with option -view spec for Test Data Registers in BSD, while using *set_dft_signal* or *set_scan_path* commands.

---

## UIT-918

(warning) BSD Test Data Register hookup pin of type reset is obsoleted, specify the pin type as bsd_reset, changing the pin type to bsd_reset.

### Description

The hookup pin type *reset* is obsoleted in BSDC, instead it is suggested to use the new option *bsd_reset* for the same functionality. You get this message if -type reset is used while specifying hookup pin for Test Data Register using *set_dft_signal* command.

### What Next

You can ignore this warning as hookup pin type *reset* is read as *bsd_reset* by BSDC. Instead hookup pin type *bsd_reset* can be used to not get this warning.

---

## UIT-919

(warning) Incorrect active state '%d' specified for TRST port, resetting the active state of the port to 0.

### Description

This warning message occurs when the *set_dft_signal -active_state* option is not set to 0 (zero) for the TRST port type. If the TRST port *-active_state* is not set to 0, BSDC forces the active state to 0.

Feedback

**What Next**

This is only a warning message. No action is required.

You can ignore this warning as BSDC sets the active state to 0 internally. To avoid this message, use 0 with the *-active_state* option for a TRST port while using *set_dft_signal* command.

---

## UIT-920

(error) The option -include_elements of set_scan_path command does not accept a collection as an input.

**Description**

You have specified a collection as an input parameter for the -include_elements option of the set_scan_path command. This option accepts only a list of elements.

**What Next**

Use a list of elements as an input parameter

---

## UIT-921

(warning) Incorrect active state '%d' specified for TAP port, resetting the active state of the port to 1.

**Description**

This warning message occurs when the *set_dft_signal -active_state* option is not set to 1 (one) for the TAP port type (TCK/ TMS/ TDI / TDO). If the TAP port *-active_state* is not set to 1, BSDC forces the active state to 1.

**What Next**

This is only a warning message. No action is required.

You can ignore this warning as BSDC sets the active state to 1 internally. To avoid this message, use 1 with the *-active_state* option for a TAP port while using *set_dft_signal* command.

---

## UIT-923

(error) Pipeline Scan Data is not enabled.

Feedback

**Description**

Pipeline scan data configuration specification is discarded because pipeline scan data is not enabled. You should enable the pipeline scan data flow using the set_dft_configuration -pipeline_scan_data enable command before specifying pipeline scan data configuration specifications.

**What Next**

Please enable pipeline scan data flow using *set_dft_configuration* command.

---

## UIT-924

(warning) Cannot find cell %s in design %s.

**Description**

You receive this message because you have specified a cell using the *-include_elements* option that cannot be found in the current design.

**What Next**

Use the *get_cells* command to verify the presence of the cell you want to specify.

You can ignore this message if it was issued for a CTL-modeled black-box cell.

---

## UIT-925

(warning) -test_mode option not specified; specification applies only to the indicated test mode.

**Description**

You receive this message because you have issued a DFT configuration command command without the *-test_mode* option after user-defined test modes have been defined. In this case, the specification is applied only to the current test mode, which is indicated by the subsequent specification acceptance message.

If the expectation is for the specification to be global because the *-test_mode* option was omitted, then the specification must be corrected.

**What Next**

If the specification should be global, add the *-test_mode all_dft* option to the command.

If the specification is intended for the indicated test mode, you can eliminate the warning by explicitly specifying the test mode with the *-test_mode* command.

Feedback

If you have many such occurrences of this warning and want to suppress them all, use the *suppress_message* command.

---

## UIT-930

(error) Signal of data type '%s' is not valid for -connect_to option.

**Description**

The indicated signal type is not supported for the *-connect_to* or *-associated_clock* option.

For the *-connect_to* option, the supported signal types are:

- *MasterClock*
- *ScanMasterClock*
- *ScanClock*
- *Oscillator*
- *ScanDataIn*
- *ScanDataOut*
- *ScanEnable* (with *-usage scan* or *-usage clock_gating*)
- *wrp_shift*
- *TestMode* (with *-usage clock_gating*)

In addition, each signal type supports only a particular set of connection object types.

For *-associated_clock*, the supported signal type is *ScanEnable*, and the domain-based scan-enable feature must also be enabled with the following command:

```
prompt> set_scan_configuration -domain_based_scan_enable true
```

**What Next**

See the *set_dft_signal* man page for a complete description of what signal type and object type combinations are supported by the *-connect_to* or *-associated_clock* option.

---

## UIT-931

(error) Pin '%s' direction is not valid for -connect_to option.

**Description**

You receive this message because you specified a wrong pin direction for *-connect_to* option. In this case, the pin should be of direction In.

Feedback

**What Next**

Please specify a pin with a valid direction using the *set_dft_signal* command.

---

## UIT-932

(Warning) In FIDO dft architecture the -outputs specification will be ignored. The tool will build the output configuration as needed.

**Description**

The -outputs specification of FIDO dft architecture is ignored

---

## UIT-935

(error) Invalid clock controller specification due to inconsistent pll clock sources.

**Description**

This error message occurs when a single clock controller is specififed for pllclocks sources that come from different hierarchical blocks.

**What Next**

Specify a clock controller for each group of pllclocks that belong to the same hierarchy using the *set_dft_clock_controller* command.

---

## UIT-936

(error) Retiming Flop Specification is incorrect.

**Description**

You must specify '/begin_only', '/end_only', '/none' or '/begin_and_end'.

---

## UIT-999

(Error) The scan state is invalid for running this command. The scan state should be scan existing.

**Description**

You receive this error message because the scan state should be either *scan_existing* (scan routed) for the *disconnect_scan_chains* command to run. If the current state of the design is not scan existing, this error message prevents disconnect_scan_chains command from proceeding.

Feedback

**What Next**

Set the scan state using the *set_scan_state* command before using the *disconnect_scan_chains* command.

---

## UIT-1000

(error) Invalid argument for -use_register_io: '%s'. Argument must be either true or false.

**Description**

You receive this message if you execute either *set_core_wrapper_configuration* or *set_core_wrapper_cell* command and use an invalid argument for the *-use_register_io* option. Valid arguments are true or false.

**What Next**

Reexecute the command and use a valid argument for the *-use_register_io* option.

---

## UIT-1002

(error) Can not insert '%s' scan chains.

**Description**

The set_core_wrapper_configuration command accepts a chain_count option. This tells DFT Compiler the number of scan chains to implement in the testable design. Option arguments must be positive integers or "default" strings.

**What Next**

Repeat the set_core_wrapper_configuration command using a positive integer or the string "default".

---

## UIT-1003

(error) Cannot find the specified port '%s' in the current design.

**Description**

You receive this message if you execute the *set_core_wrapper_path* command and a port you specify cannot be found in the current design. This could be caused by a typo or spelling error or an invalid name being used as a port name.

**What Next**

Verify the name of the port. Then reissue the command *set_core_wrapper_path* using the correct port name of the current design.

Feedback

## UIT-1004

(error) Wrapper Cell type must be either "in" , "out" or "en".

### Description

You receive this message if you execute *set_core_wrapper_path* and specify an invalid wrapper cell type. Valid cell types are "in", "out" or "en". For *set_core_wrapper_path*, you can specify the type of a wrapper cell by attaching /in , /out or /en to the port name in the *identifier_list*. The cell types /in, /out, /en can be used to identify wrapper cells attached to a bidirectional port. The cell types /out, /en can be used to identify wrapper cells attached to a tr-state port. No cell type can be used to identify wrapper cells attached to input ports.

### What Next

Choose a valid cell type for the Wrapper cell, and reissue the command *set_core_wrapper_path* using the valid wrapper cell type.

## UIT-1005

(warning) Ignoring the -complete true option because wrapper chain '%s' is empty.

### Description

The *set_core_wrapper_path* command accepts a list of ports and *-complete* option. This tells DFT Compiler the order of wrapper cells in a wrapper chain to implement in the testable design. This error occurred as user didn't specify any valid ports and *-complete* option is set to TRUE for the command.

### What Next

Repeat the *set_core_wrapper_path* command using either a set of valid ports or setting the *-complete* option to false.

## UIT-1006

(information) You have invoked remove_core_wrapper_specification with no options. The command is ignored.

### Description

The *remove_core_wrapper_specification* command does nothing when it is invoked without options.

This, in itself, is not a problem. No core wrapper specifications are removed. You might, however, have meant to use the "-all" option or one of the other more limited options.

Feedback

**What Next**

To get rid of the information message, invoke *remove_core_wrapper_specification* with options.

---

## UIT-1007

(error) Pipelining is only supported with the "no_mix" scan clock domain constraints.

**Description**

You get this message when you have selected something else than *no_mix* after the option *-clock_mixing* and you have enabled scan enable pipelining. Pipelining scan enable supports only the no_mix scan clock domain constraints.

**What Next**

Modify the scan clock domain constraint to *no_mix*, using the set_scan_configuration command

---

## UIT-1008

(error) Invalid argument for -register_io_implementation: '%s'. Argument must be either swap or in_place.

**Description**

You receive this message if you execute either *set_core_wrapper_configuration* or *set_core_wrapper_cell* command and use an invalid argument for the *-register_io_implementation* option. Valid arguments are swap or in_place.

**What Next**

Reexecute the command and use a valid argument for the *-register_io_implementation* option.

---

## UIT-1009

(error) Invalid argument for -one_wrapper_clock: '%s'. Argument must be either true or false.

**Description**

You receive this message if you execute either *set_core_wrapper_configuration* or *set_core_wrapper_cell* command and use an invalid argument for the *-one_wrapper_clock* option. Valid arguments are swap or in_place.

Feedback

### What Next

Reexecute the command and use a valid argument for the *-one_wrapper_clock* option.

---

## UIT-1010

(error) Scan %s '%s' element '%s' has internal and wrapper segments.

### Description

This error message is generated when you try to include a core cell into a scan path or scan segment. A core cell is a cell with a test model and has both internal scan and wrapper segments. As internal scan segments and wrapper segments can't be mixed during core integration, you can't include such a cell with other elements of a scan path or a scan segment.

### What Next

Check if you have specified core_integration flag of dft configuration. If you are not integrating cores, remove such a specification and reinvoke the current command.

---

## UIT-1011

(error) Scan Chain '%s' has %s segments.

### Description

This error message is generated when you try to include a wrapper or an internal scan segment with other segments in a scan path during core integration. Wrapper or internal scan segments can not be mixed with other segments during core integration.

### What Next

Check if you have specified core_integration flag of dft configuration. If you are not integrating cores, remove such a specification and reinvoke the current command.

---

## UIT-1012

(Error) Incorrect lockup element type specified.

### Description

This is issued from the command *set_scan_configuration* or *set_scan_path*, when the user intends to specify the type of lockup element to be used (-lockup_type). Values for the lockup type are restricted to either "latch" or "flip_flop". By default, the lockup type is set to "latch"

Feedback

**What Next**

Check the value entered after the option -lockup_type.

---

## UIT-1013

(Error) The period %f must evenly divide the default period %f.

**Description**

This error occurs when you use the -mbist_clock argument and the specified period does not evenly divide the default period. This relation is required to create the MBIST protocols.

**What Next**

Specify a different clock period with the *create_test_clock* command, or change the default period with the *test_default_period* command.

---

## UIT-1014

(Error) The parameter for the -write_ip_rtl option has to be enable, generate_only or disable.

**Description**

This message indicates that a set_dft_insertionj_configuration command has been issued, but the parameter for the option is not a valid one. The valid parameters are enable, generate_only or disable.

**What Next**

Change the parameter for the option and re-issue the command.

---

## UIT-1015

(Error) The parameter for the -ip_rtl_format option has to be verilog or vhdl.

**Description**

This message indicates that a set_dft_insertionj_configuration command has been issued, but the parameter for the option is not a valid one. The valid parameters are verilog or vhdl.

**What Next**

Change the parameter for the option and re-issue the command.

Feedback

## UIT-1016

(error) %s.

### Description

This error is generated when the *-design* or *-memory_ref* arguments of the *create_mbist_standalone* command specify a wrong design name.

### What Next

Make sure that the *-design* or *-memory_ref* arguments specify a valid design name then retry the *create_mbist_standalone* command.

## UIT-1020

(error) : Component signal type '%s' is not specified in user design interface signals.

### Description

This error message is generated when you try to define a dft design with command define_dft_design. The error message is issued when a pin of the specified signal type is not included in the interface of the dft design.

### What Next

Reexecute the *define_dft_design* command and specify a dft design pin of the signal type with the *-interface* argument.

## UIT-1021

(error) Unknown signal type '%s' is specified in user design interface signals.

### Description

This error message occurs when attempting to define a DFT design with the *define_dft_design* command. The error message indicates that an unknown signal type is specified with the interface of the DFT design.

### What Next

If the port of the specified signal type is used in DFT insertion, add the correct signal type for the port and rerun the *define_dft_design* command.

## UIT-1022

(error) : Unable to find the specified design type '%s', make sure that proper dw libs are included in link_library.

Feedback

### Description

This error message is generated when you try to define a dft design with command define_dft_design. The error message is issued when the specified design type is not a known design type. This can happen if all dw libraries are not specified with the link_library variable.

### What Next

Specify all dw libraries with link_library variable and rerun the define_dft_design command.

---

## UIT-1023

(error) : One of the options '-design_name' or '-type' or '-all' is mandatory.

### Description

This error message is generated when you try to remove a dft design with command remove_dft_design. The error message is issued when the command is run with insufficient options.

### What Next

Specify one of the options '-design_name' or '-type' or -all' with the remove_dft_design command.

---

## UIT-1024

(error) : Options '-design_name', '-type', '-all' are mutually exclusive, specify only one of them.

### Description

This error message is generated when you try to remove/report a dft design with command remove_dft_design/report_dft_design. The error message is issued when the command is run with invalid combination of options. Please note that options '-design_name', '-type', '-all' of the command remove_dft_design/report_dft_design are mutually exclusive.

### What Next

Specify only one of the options '-design_name' or '-type' or -all' with the remove_dft_design/ report_dft_design commands.

Feedback

## UIT-1025

(error) : Design '%s' is not specified with define_dft_design command.

### Description

This error message is generated when you try to remove/report a dft design with command remove_dft_design/report_dft_design. The error message is issued when the command is run with the '-design_name' option and the specified design is not defined using define_dft_design command.

### What Next

Check if the specified design is defined using define_dft_design command. You can use report_dft_design command to see all the user dft designs.

## UIT-1026

(information) Memory-driven specification mode is enabled. You can not specify any Tests with the set_mbist_controller command.

### Description

You just enabled the MBIST configuration, and memory-driven specification mode is enabled, because you set the test_mbc_memory_driven_spec environment variable to true.

### What Next

All memory models should contain complete Test specifications. You can not specify any Tests with the *set_mbist_controller* command. This means all Test parameters (ie Test0, Test1, ...) are invalid.

## UIT-1027

(warning) Memory-driven specification mode is enabled. The test you just defined will not be synthesized.

### Description

The memory-driven specification mode is enabled, because you set the test_mbc_memory_driven_spec environment variable to true. This means that all synthesized tests are specified in the memory models. Any tests you define with the *define_mbist_program* command can only be used at run-time by the *set_mbist_run* command.

Feedback

### What Next

Do not attempt to specify any Tests with the *set_mbist_controller* command. This means all Test parameters (ie Test0, Test1, ...) are invalid.

---

## UIT-1028

(warning) Invalid option(s) specified to the set_optimize_registers command.

### Description

You receive this message if you execute the *set_optimize_registers* command with options that are not supported in non-XG mode.

### What Next

Please see the man page of the *set_optimize_registers* command for a list of options that are supported in XG/non-XG modes.

---

## UIT-1030

(error) Option -time, -clock_cycles and -signature can not be specified for instruction '%s'.

### Description

You receive this error message when you execute the *set_bsd_instruction* command and specify time constant or signature for an invalid instruction. Time constants are supported only for the following instructions.

INTEST RUNBIST EXTEST_PULSE EXEST_TRAIN

Signature is supported only for RUNBIST instruction.

For EXTEST_PULSE instruction, options -time specifies minimum wait duration in real time and option -clock_cycles specifies minimum wait duration in TCK pulses. These two option are mutually exclusive for this instruction.

For EXTEST_TRAIN instruction minimum number of TCK pulses should be specified with -clock_cycles option in addition to maximum time with -time option.

### What Next

Reexecute the *set_bsd_instruction* command with out time constant and signature specifications.

Feedback

## UIT-1031

(error) Invalid IEEE1149 Std. version '%s', valid values are : ieee1149.1_2001 | ieee1149.1_1993, ieee1149.6_2003 .

### Description

You receive this error message when you execute the *set_bsd_configuration* command and specify an incorrect std. version. The following std. versions are leagl.

"ieee1149.1_1993" "ieee1149.1_2001" "ieee1149.1_2001 ieee1149.6_2003" "ieee1149.1_1993 ieee1149.6_2003" "ieee1149.6_2003"

Std. version "ieee1149.6_2003" is same as "ieee1149.1_2001 ieee1149.6_2003" .

The default valud for the Std. option is "ieee1149.1_2001" .

### What Next

Reexecute the *set_bsd_configuration* command with the correct std. version name(s).

## UIT-1032

(Warning) IEEE1149 Std. versions ieee1149.1_2001, ieee1149.1_1993 are mutually exclusive, ignoring std. version ieee1149.1_1993 .

### Description

You receive this error message when you execute the *set_bsd_configuration* command and specify an incorrect std. version. The following std. versions are leagl.

"ieee1149.1_1993" "ieee1149.1_2001" "ieee1149.1_2001 ieee1149.6_2003" "ieee1149.1_1993 ieee1149.6_2003" "ieee1149.6_2003"

Std. version "ieee1149.6_2003" is same as "ieee1149.1_2001 ieee1149.6_2003" .

The default valud for the Std. option is "ieee1149.1_2001" .

### What Next

Reexecute the *set_bsd_configuration* command with the correct std. version name(s).

## UIT-1033

(error) Incorrect port '%s' with direction '%s' specified, only ports with direction 'out' or 'inout' are allowed.

Feedback

**Description**

You receive this error message when you execute the *set_bsd_ac_port* command and specify an incorrect port. Only ports of type 'out' or 'inout' are allowed as ac ports.

**What Next**

Reexecute the *set_bsd_ac_port* command with the correct ports.

---

## UIT-1034

(error) Insufficient number of options, specify either of options : -port_list | -all

**Description**

You receive this error message when you execute the *remove_bsd_ac_port* command and doesn't specify the required options. Either of the options -all or -port_list are required for this command.

**What Next**

Reexecute the *remove_bsd_ac_port* command with the required options.

---

## UIT-1035

(error) Options -port_list and -all are mutually exclusive, specify only one of them.

**Description**

You receive this error message when you execute the *remove_bsd_ac_port* command and specified conflicting options. Options -all and -port_list are mutually exclusive.

**What Next**

Reexecute the *remove_bsd_ac_port* command with the either of -all or -port_list options.

---

## UIT-1036

(Warning) Port '%s' is not an ac port, not removing the port.

**Description**

You receive this error message when you execute the *remove_bsd_ac_port* command and specified a port which is not an AC port. You can only remove AC ports with this command.

**What Next**

Reexecute the *remove_bsd_ac_port* command with the AC ports.

Feedback

## UIT-1037

(error) No Ac Ports found on the current design.

**Description**

You receive this error message when you execute the *remove_bsd_ac_port* command and there are no AC ports to be removed. Check if the current design is correct.

**What Next**

If the current design is correct, the *remove_bsd_ac_port* command is not required as there are no AC ports to be removed.

## UIT-1038

(error) Capture value not specified for the IDCODE or USERCODE instructions.

**Description**

You receive this error message when you execute the *set_bsd_instruction* command and doesn't specify capture value. Capture value is mandatory for these instructions.

**What Next**

If the current design is correct, the *set_bsd_instruction* command with capture value.

## UIT-1039

(Warning) Instruction '%s' does not exist, could not remove the instruction.

**Description**

You receive this error message when you execute the *remove_bsd_instruction* command and specify an unexisting instruction.

**What Next**

Check for instruction name and reexecute the *remove_bsd_instruction* command with the correct instruction names.

## UIT-1040

(error) No instructions exist, cannot remove instructions.

Feedback

### Description

You receive this error message when you execute the *remove_bsd_instruction* command and there are no instructions specified for the current design.

### What Next

If the current design is correct, the *remove_bsd_instruction* command is unnecessary as there are instructions to be removed.

---

## UIT-1041

(error) Boundary Cell Types AC_1, AC_2, AC_7, AC_SELU, AC_SELX can't be used for ports which are not AC ports.

### Description

You receive this error message when you execute the *set_boundary_cell* command on specify AC BSR cells on a set of ports that are not AC ports. AC BSR cells can be used only on AC ports.

### What Next

Fix the problem and Reexecute the *set_boundary_cell* command with the correct specification.

---

## UIT-1042

(error) Invalid cell type '%s' specified with function '%s' for boundary scan.

### Description

You receive this error message when you execute the *set_boundary_cell* command on specify a BSR cell type that is not supported with the specified function.

Here is the list of cell types allowed for different functions.

FUNCTION BSR Cell Types ==========
input BC_1 | BC_2 | BC_4 | none input_inverted BC_1 | BC_2 | BC_4 | none
output BC_1 | BC_2 | AC_1 | AC_2 | none output_inverted BC_1 | BC_2 | AC_1 | AC_2 | none bidir BC_7 | AC_7 | none bidir_inverted BC_7 | AC_7 | none control BC_1 | BC_2 | none observe BC_4 | none receiver_p BC_4 | BC_1 | BC_2 | none receiver_n BC_4 | BC_1 | BC_2 | none ac_select AC_SELU | AC_SELX | none
==========

### What Next

Fix the problem and Reexecute the *set_boundary_cell* command with the correct specification.

Feedback

## UIT-1043

(error) Specified design '%s' is not a user dft design, allowed design types : %s

### Description

You receive this error message when you execute the *set_boundary_cell* command with a design name that is not of a standard DFT design type.

The list of allowed DFT design types is:

```
WC_D1   WC_D1_S   WC_S1   WC_S1_S
BC_1    BC_2      BC_4    BC_7
AC_1    AC_2      AC_7    AC_SELU   AC_SELX
```

### What Next

Fix the problem and reexecute the *set_boundary_cell* command with the correct specification.

## UIT-1044

(Warning) Option '-share' is applicable only for Control/Ac Select Boundary Cells, ignoring the option.

### Description

You receive this error message when you execute the *set_boundary_cell* command and specify option -share with an unspported function. This option can can be specified only with option -function values : control | ac_select .

### What Next

Fix the problem and Reexecute the *set_boundary_cell* command with the correct specification.

## UIT-1045

(error) Boundary Cell Name is mandatory for Control/Ac Select boundary cells.

### Description

You receive this error message when you execute the *set_boundary_cell* command and specify option -function with value control or ac_select and doesn't specify a name for the boundary cell. Option -name is mandatory for these function values.

Feedback

**What Next**

Fix the problem and Reexecute the *set_boundary_cell* command with the correct specification.

---

## UIT-1046

(error) Option '-ports' is mandatory with options '-class core_wrapper', '-class bsd'.

**Description**

You receive this error message when you execute the *set_boundary_cell* command and doesn't specify a list of ports with option -class values core_wrapper or bsd. Port list is mandatory for these class values.

**What Next**

Fix the problem and Reexecute the *set_boundary_cell* command with the correct specification.

---

## UIT-1047

(error) Option '-cells' is not allowed with options '-class core_wrapper', '-class bsd'.

**Description**

You receive this error message when you execute the *set_boundary_cell* command and specified a list of cells with option -class values core_wrapper or bsd. Cell list is not allowed for these class values.

**What Next**

Fix the problem and Reexecute the *set_boundary_cell* command with the correct specification.

---

## UIT-1048

(error) Options '-type' and 'design' are mutually exclusive.

**Description**

You receive this error message when you execute the *set_boundary_cell* command and specified options -type and -design. These options are mutually exclusive.

**What Next**

Fix the problem and Reexecute the *set_boundary_cell* command with the correct specification.

Feedback

## UIT-1049

(error) Invalid boundary cell type with '-class core_wrapper', allowed values : WC_D1 | WC_D1_S | WC_S1 | WC_S1_S .

**Description**

You receive this error message when you execute the *set_boundary_cell* command and specified a cell tyep not suitable for option -class value core_wrapper. Only the following cell types are allowed for this class value.

WC_D1 WC_D1_S WC_S1 WC_S1_S

**What Next**

Fix the problem and Reexecute the *set_boundary_cell* command with the correct specification.

## UIT-1050

(error) Invalid functin value with '-class core_wrapper', allowed values : input | output | control | bidir .

**Description**

You receive this error message when you execute the *set_boundary_cell* command and specified a function tyep not suitable for option -class value core_wrapper. Only the following function types are allowed for this class value.

input output control bidir

**What Next**

Fix the problem and Reexecute the *set_boundary_cell* command with the correct specification.

## UIT-1051

(error) Ports '%s' can't be specified with function '%s' .

**Description**

You receive this error message when you execute the *set_boundary_cell* command and specified a function tyep not suitable for the specifed ports.

Here is a list of functions and allowed port types.

Function Name Port Direction ========== input input | bidir input_inverted input | bidir output output | bidir output_inverted

Feedback

output | bidir bidir bidir bidir_inverted bidir control output | bidir observe input | output | bidir receiver_p input | bidir receiver_n input | bidir ac_select output | bidir
==========

**What Next**

Fix the problem and Reexecute the *set_boundary_cell* command with the correct specification.

---

## UIT-1052

(Warning) Boundary Cell '%s' overwrites an earlier specification.

**Description**

You receive this error message when you execute the *set_boundary_cell* command and the specification overwrites an earlier specification.

Here are situations where earlier specification is overwritten.

1. Option -name value is already specified for a previous Boundary Cell. Previous Boundary Cell is overwritten.
2. Option -function value is already specified for a previous Boundary Cell with common ports. Common ports are removed from the previous Boundary Cell.
3. A Boundary Cell with common ports and unspecified -function value is found. Common ports are removed from the previous Boundary Cell.
4. Option -function value is not specified and a Boundary Cell with common ports is found. Common ports are removed from the previous Boundary Cell.

**What Next**

No action is required if overwriting earlier specification is ok with you.

---

## UIT-1053

(error) %s delay time (%10.2f) is required to be less than rise time value (%10.2f) for port '%s'.

**Description**

Input/Bidirectional delay time is greater than specified rise time of the port.

**What Next**

Check your test_default_delay/test_default_bidir_delay and timing specification for the specified port.

Feedback

## UIT-1054

(error) Rise time (%10.2f) of port '%s' is required to be less than strobe time value (%10.2f).

**Description**

You have specified rise time of the specified port greater than the strobe time.

**What Next**

Check your test_default_strobe and timing specification for the specified port.

## UIT-1055

(warning) Bidir delay (%10.2f) must be less than strobe time (%10.2f), setting bidir delay to input delay (%10.2f) .

**Description**

You have specified bidirectional delay time greater than the strobe time. Bidurectional delay time is reset to input delay time.

**What Next**

Check your test_default_strobe and test_default_bidir_delay variables.

## UIT-1056

(warning) Bidir delay (%10.2f) must be less than period (%10.2f), setting bidir delay to input delay (%10.2f) .

**Description**

You have specified bidirectional delay time greater than the period. Bidurectional delay time is reset to input delay time.

**What Next**

Check your test_default_period and test_default_bidir_delay variables.

## UIT-1057

(warning) Bidir delay (%10.2f) must be less than TCK rise time (%10.2f), setting bidir delay to input delay (%10.2f) .

Feedback

**Description**

You have specified bidirectional delay time greater than the TCK rise time. Bidurectional delay time is reset to input delay time.

**What Next**

Check your test_default_bidir_delay variable and rise time of TCK.

---

## UIT-1058

(error) Invalid Paramter Type '%s' specified for parameter '%s' .

**Description**

You have specified an unsupported paramater type with *define_dft_design* command. Only the following parameter types are supported.

string

boolean

float

list string

**What Next**

Correct the specification and reissue the *define_dft_design* command.

---

## UIT-1059

(error) Invalid Paramter value '%s' specified for parameter '%s', expected value of type '%s' .

**Description**

You have specified an unsupported paramater value with *define_dft_design* command. Only the following parameter values are supported.

Parameter Type Parameter Value ========== string string

boolean true | false

float floating point number

list string list of strings ==========

**What Next**

Correct the specification and reissue the *define_dft_design* command.

Feedback

## UIT-1060

(error) Invalid List Paramter member type '%s' specified for parameter '%s' .

**Description**

You have specified an unsupported list paramater member type with *define_dft_design* command. Only the following parameter types are supported.

list string

**What Next**

Correct the specification and reissue the *define_dft_design* command.

## UIT-1061

(error) Incorrect Pin Polarity '%s' specified for pin '%s', expected polarity : high | low .

**Description**

You have specified an unsupported pin polarity value with *define_dft_design* command. Only the following polarity values are supported.

high | h

low | l

**What Next**

Correct the specification and reissue the *define_dft_design* command.

## UIT-1062

(error) Option '-params' is only allowed for dft designs of type 'PAD'.

**Description**

You have specified option '-params' for a dft design that is not of type 'PAD' with *define_dft_design* command. This option is supported only for dft designs of type 'PAD'.

**What Next**

Correct the specification and reissue the *define_dft_design* command.

## UIT-1063

(error) Invalid Pad Access Pin Type '%s' specified for the PAD design '%s'.

Feedback

### Description

You have specified an unsupported pad access pin type for a pad design with *define_dft_design* command. Only the following pad access pin types are supported.

```
Pad Type                              Allowed Access Pin Types
=============================================================================
=====

any pad                         capture_clk  capture_en  update_clk
 update_en
                                port                     low          high
                                ac_mode      shift_dr    si           so

input                           data_out     receiver_p  receiver_n
 ac_init_data_p
                                mode_in      ac_init_clk ac_init_data_n
 observe

output                          data_in      mode_out    ac_test

tristate_output                 data_in      mode_out    ac_test      enable
                                highz

bidirectional                   data_out     receiver_p  receiver_n
 ac_init_data_p
                                data_in      mode_out    ac_test      enable
                                highz        mode_in     ac_init_clk
 ac_init_data_n
                                mode1_inout  mode2_inout observe

open_drain_output               enable       mode_out    ac_test      highz

open_drain_bidirectional        enable       receiver_p  receiver_n
 ac_init_data_p
                                data_out     mode_out    ac_test
                                highz        mode_in     ac_init_clk
 ac_init_data_n
                                mode1_inout  mode2_inout observe

open_source_output              enable       mode_out    ac_test      highz

open_source_bidirectional       enable       receiver_p  receiver_n
 ac_init_data_p
                                data_out     mode_out    ac_test
                                highz        mode_in     ac_init_clk
 ac_init_data_n
                                mode1_inout  mode2_inout observe

=============================================================================
=====
```

Feedback

**What Next**

Correct the specification and reissue the *define_dft_design* command.

---

## UIT-1064a

(error) No pad type found for the PAD design '%s'.

**Description**

This error message occurs because the pad type was not specified with the *-param {$pad_type$}* option of the *define_dft_design* command.

**What Next**

Rerun the *define_dft_design* command with the *-param {$pad_type$}* option specifying a valid pad type.

---

## UIT-1064

(error) Invalid Pad type '%s' specified for the PAD design '%s'.

**Description**

You have specified an unsupported pad type for a pad design with *$pad_type$* parameter of *define_dft_design* command. Only the following pad types are supported.

input

output

tristate_output

bidirectional

open_drain_output

open_drain_bidirectional

open_source_output

open_source_bidirectional

Note that the pad types need to be specified in the following format.

define_dft_design ... -params { $pad_type$ string <pad_type> ... }

where <pad_type> is one of the pad types mentioned above.

Feedback

**What Next**

Correct the specification and reissue the *define_dft_design* command.

---

## UIT-1065

(error) Invalid Parameter '%s' of type '%s' specified, expected type '%s' .

**Description**

You have specified an unsupported parameter or unsupported parameter type for a pad design with *define_dft_design* command. Only the following parameters are supported.

Parameter Name Parameter Type ================================ $pad_type $ string $disable_res$ string | boolean $differential$ string | boolean $bsr_segment$ list string $lib_cell$ string | boolean $lp_time$ string $hp_time$ string $on_chip$ string | boolean ================================

**What Next**

Correct the specification and reissue the *define_dft_design* command.

---

## UIT-1066

(error) Invalid active_state value %s specified.

**Description**

You have specified an unsupported active_state value with *set_bsd_power_up_reset* command. Only the following active_state values are supported.

high low

**What Next**

Correct the specification and reissue the *set_bsd_power_up_reset* command.

---

## UIT-1067

(error) Incorrect number of ports/values specified with option '-pattern'.

**Description**

You have specified an insufficient number of port/value pairs with option *-pattern* of *set_bsd_compliance* command. The syntax of the compliance pattern is as follows.

{ [<port_name> <port_value>]+ }

Feedback

where <port_name> is the name of the port of the current design and <port_value> can be 0 or 1.

**What Next**

Correct the specification and reissue the *set_bsd_compliance* command.

---

## UIT-1068

(error) Specified IR width %d is not within 2-63 inclusive, ir_width specification discarded.

**Description**

You have specified an unsupported Instruction Register(IR) Width with *set_bsd_configuration* command. The IR width should be a positive number between 2 and 63.

**What Next**

Correct the specification and reissue the *set_bsd_configuration* command.

---

## UIT-1069

(error) Invalid value %s specified for option -asynchronous_reset.

**Description**

You have specified an invalid value for option *-asynchronous_reset* of *set_bsd_configuration* command. The allowed values for this option are : true | false

**What Next**

Correct the specification and reissue the *set_bsd_configuration* command.

---

## UIT-1070

(error) Invalid value %s specified for option -instruction_encoding.

**Description**

You have specified an invalid value for option *-instruction_encoding* of *set_bsd_configuration* command. The allowed values for this option are : binary | one_hot

**What Next**

Correct the specification and reissue the *set_bsd_configuration* command.

Feedback

## UIT-1071

(error) Invalid value %s specified for option -style.

**Description**

You have specified an invalid value for option *-style* of *set_bsd_configuration* command. The allowed values for this option are : synchronous | asynchronous

**What Next**

Correct the specification and reissue the *set_bsd_configuration* command.

## UIT-1072

(warning) UI option -ieee1149.1_1993 will be obsolete, please use option -std to specify the std. version.

**Description**

You have specified option -ieee1149.1_1993 with *set_bsd_configuration* command. This option is obsolete, use the new option *-std ieee1149.1_1993* to enable IEEE1149.1_1993 standard.

**What Next**

Correct the specification and reissue the *set_bsd_configuration* command.

## UIT-1073

(warning) Ignoring %s constant of pad design '%s'.

**Description**

You have specified LP_time or HP_time constants for a pad design and the value of this time constant is not valid.

The time is not valid if the time constant is less than 0.0

LP_time constant is also invalid if HP_time contant is not specified.

The time constant should be a positive real number in units of nano seconds.

**What Next**

Correct the specification and reissue the *define_dft_design* and *preview_dft*, *insert_dft* commands.

Feedback

## UIT-1074

(error) Std. Instruction '%s' is not allowed if support for ieee1149.6_2003 is not enabled.

### Description

You have specified EXTEST_PULSE or EXTEST_TRAIN instruction with *set_bsd_instruction* command, but didn't enable IEEE1149.6_2003 standard. These two instructions are supported only if IEEE1149.6_2003 standard support is enabled with option *-std ieee1149.6_2003* of *set_bsd_configuration* command.

### What Next

Correct the specification and reissue the *set_bsd_configuration* and *preview_dft*, *insert_dft* commands.

## UIT-1075

(warning) Port '%s' doesn't need an AC/DC Selector cell, not adding the selector cell for the port.

### Description

You have specified that AC/DC selector cell should be added for the specified port and the tool determined that the port doesn't need such a BSR cell. AC/DC selector cells are added to control the AC mode pin of AC BSR cells. If an AC BSR cell(AC_1, AC_2, AC_7) doesn't drive a port, no AC/DC slection BSR cell is needed for the port.

### What Next

Correct the specification and reissue the *set_boundary_cell* and *preview_dft*, *insert_dft* commands.

## UIT-1076

(error) Invalid cell type '%s' for boundary scan tristate control register of port '%s', adding default control BSR cell to the port.

### Description

You have specified an invalid BSR cell as a control BSR cell for a port with *set_oundary_cell* command. Valid control BSR cells are : BC_1 | BC_2 | none. A default BSR cell will be added as a control cell for the port. The default control BSR cell is BC_1 cell if INTEST instruction is specified, otherwise the default control cell is BC_2 BSR cell.

Feedback

### What Next

Correct the specification and reissue the *set_boundary_cell* and *preview_dft*, *insert_dft* commands.

---

## UIT-1077

(error) Invalid cell type '%s' for boundary scan ac/dc select register of port '%s', not adding any ac/dc selector BSR cell to the port.

### Description

You have specified an invalid BSR cell as a ac/dc selector BSR cell for a port with *set_oundary_cell* command. Valid ac/dc selector BSR cells are : AC_SELU | AC_SELX | none. No AC/DC selector BSR cell is added to the specified port.

### What Next

Correct the specification and reissue the *set_boundary_cell* and *preview_dft*, *insert_dft* commands.

---

## UIT-1078

(warning) BC_1 cell should be used for output control pins when INTEST is specified, using BC_1 cell for control pin of port '%s'.

### Description

You have specified BC_2 BSR cell as the control cell for a port and also specified INTEST instruction. When INTEST instruction is specified, BC_2 BSR cell can't be used as a control BSR cell, adding BC_1 BSR cell as the control cell for the specified port.

### What Next

Correct the specification and reissue the *set_boundary_cell* and *preview_dft*, *insert_dft* commands.

---

## UIT-1079

(error) Polarity '%s' is not allowed of pad access pins of type '%s'.

### Description

You have specified an incorrect polarity for a pad access pin. Here is a list of pad access pins and the allowed polaritiues for each of them.

Padd Acces Pin Type Allowed Polarites
========== port high low ac_mode high

Feedback

low shift_dr high low capture_clk high low capture_en high low update_clk high low update_en high low data_out high low ac_init_data_p high low ac_init_data_n high low ac_init_clk high low mode_in high low data_in high low mode_out high low ac_test high low mode1_inout high low mode2_inout high low enable high low highz high low

observe high low high high high si high so high receiver_p high receiver_n high
==========

**What Next**

Correct the specification and reissue the *define_dft_design* command.

---

## UIT-1080

(Warning) Port '%s' is not a linkage port, not removing the port.

**Description**

You receive this warning message when you execute the *remove_bsd_linkage_port* command and specified a port which is not a Linkage port. You can only remove Linkage ports with this command.

**What Next**

Reexecute the *remove_bsd_linkage_port* command with Linkage ports.

---

## UIT-1081

(Error) No Linkage Ports found on the current design.

**Description**

You receive this error message when you execute the *remove_bsd_linkage_port* command and no linkage ports found on the current design. You can only remove Linkage ports with this command.

**What Next**

Reexecute the *remove_bsd_linkage_port* command with Linkage ports.

---

## UIT-1082

(Error) No Power Up Reset Ports found on the current design.

Feedback

**Description**

You receive this error message when you execute the *remove_bsd_power_up_reset* command and no power up reset ports found on the current design. You can only remove Power Up Reset ports with this command.

**What Next**

Reexecute the *remove_bsd_power_up_reset* command with Power Up Reset ports.

---

## UIT-1083

(Warning) Pattern-name '%s' is not in bsd compliance, not removing pattern-name.

**Description**

You receive this warning message when you execute the *remove_bsd_compliance* command and specified a pattern-name which is not set using set_bsd_compliance. You can only remove compliance pins with this command.

**What Next**

Reexecute the *remove_bsd_compliance* command with compliance pins.

---

## UIT-1084

(Error) No Compliance Ports found on the current design.

**Description**

You receive this error message when you execute the *remove_bsd_compliance* command and no compliance pins found on the current design. You can only remove Compliance pins with this command.

**What Next**

Reexecute the *remove_bsd_compliance* command with Compliance pins.

---

## UIT-1085

(Warning) Cell '%s' is not a boundary cell IO, not removing the cell.

**Description**

You receive this warning message when you execute the *remove_boundary_cell_io* command and specified a cell which is not a Boundary Cell IO. You can only remove Boundary Cell IOs with this command.

Feedback

**What Next**

Reexecute the *remove_boundary_cell_io* command with Boundary cell IOs.

---

## UIT-1086

(Error) No Boundary Cell IOs found on the current design.

**Description**

You receive this error message when you execute the *remove_boundary_cell_io* command and no boundary cell IOs found on the current design. You can only remove Boundary IO cells with this command.

**What Next**

Reexecute the *remove_boundary_cell_io* command with Boundary cell IOs.

---

## UIT-1087

(Error) Insufficient number of options, specify either of options : -cell | -all.

**Description**

You receive this error message when you execute the *remove_boundary_cell_io* command and doesn't specify the required options. Either of the options -all or -cell are required for this command.

**What Next**

Reexecute the *remove_boundary_cell_io* command with the required options.

---

## UIT-1088

(Error) Options -cell and -all are mutually exclusive, specify only one of them.

**Description**

You receive this error message when you execute the *remove_boundary_cell_io* command and specified conflicting options. Options -all and -port_list are mutually exclusive.

**What Next**

Reexecute the *remove_boundary_cell_io* command with the either of -all or -port_list options.

Feedback

## UIT-1089

(error) Insufficient number of options, specify either of options : -name | -all

**Description**

You receive this error message when you execute the *remove_bsd_compliance* command and doesn't specify the required options. Either of the options -name or -all are required for this command.

**What Next**

Reexecute the *remove_bsd_compliance* command with the required options.

## UIT-1090

(Warning) The '-name' option is applicable only for control and ac_select boundary cells; ignoring the option.

**Description**

You receive this warning message when you execute the *set_boundary_cell* command and specify the *-name* option with an unsupported function type. The *-name* option can be specified only when the *-function* option is set to *control* or *ac_select*; as a result, it is ignored for this specification.

**What Next**

To resolve the warning, update the *set_boundary_cell* command with the correct specification.

## UIT-1091

(Warning) Option '-share' is applicable only for multiple Control/Ac Select Boundary Cells, ignoring the option.

**Description**

You receive this warning message when you execute the *set_boundary_cell* command and specify option -share with a single port in -ports option. The option -share is applicable only when used with multiple Control/Ac Select Boundary Cells.

**What Next**

Fix the problem and Re-execute the *set_boundary_cell* command with the correct specification.

Feedback

## UIT-1093

(error) Invalid value %s specified for the -rtl option.

**Description**

This error message occurs when an invalid value is specified for the *-rtl* option of the *set_bsd_configuration* command. The allowed values for this option are *enable* or *disable*.

**What Next**

Reissue the *set_bsd_configuration* command with either the *-rtl enable* or the *-rtl disable* option.

## UIT-1094

(warning) The -output option must be used with the -rtl enable option. Discarded the option.

**Description**

This warning message occurs when the *-output* option of the *set_bsd_configuration* command is specified, but the *-rtl* option is not set to *enable*. The *-output* option must be used with the *-rtl enable* option of the *set_bsd_configuration* command.

**What Next**

This is only a warning message. No action is required.

However, if the result is not what you intended, reissue the *set_bsd_configuration* command with *-rtl enable* and *-output* options.

## UIT-1095

(error) The -rtl enable option must be used with the -output option.

**Description**

This error message occurs when the the *-rtl enable* option of *set_bsd_configuration* command is specified without using the *-output* option. The *-rtl enable* option must be used with the *-output* option of the *set_bsd_configuration* command.

**What Next**

Reissue the *set_bsd_configuration* command using both the *-rtl enable* option and the *-output* option.

Feedback

## UIT-1096

(error) Test model does not exist for the design '%s'.

### Description

This error message occurs when there is no test model for the design corresponding to the design you specified in the *use_test_model* command.

### What Next

Make sure you specified the design you intended, and that the design has an attached CTL model. To see a list of designs that have test models, use the *list_test_models* command. You may need to use the *read_test_model* command to attach a test model to the design specified in the error message.

## UIT-1097

(error) Option '-test_mode' is not allowed with the other specified options.

### Description

This error message occurs when you run the *set_wrapper_configuration* command and specify the *-test_mode* option along with other options that are not allowed with this option.

The multimode wrapper configuration is specified with the *-test_mode* option.

The multimode wrapper configuration specification works only with the following options:

```
-max_length
-chain_count
-mix_cells
-input_shift_enable
-output_shift_enable
-shift_enable
```

### What Next

Rerun the *set_wrapper_configuration* command with the correct specification and options.

### See Also

- reset_wrapper_configuration

## UIT-1098

(warning) Pad Access Pin type 'ac_mode' unspecified, though pad contains '%s'.

Feedback

**Description**

You have specified one of these Pad Access Pin type: ac_init_clk, ac_test, ac_init_data_p or ac_init_data_n. But missed ac_mode. ac_mode Pad Access Pin type should be specified alongwith any of these pad access types.

**What Next**

This might result in wrong pattern generation. Correct the specification and reissue the *define_dft_design* command.

---

## UIT-1099

(warning) Streaming compression enabled will take precedence over scan compression.

**Description**

You have enabled scan compression and streaming compression simultaneously for DFT insertion. In this scenario the tool implements streaming compression in the netlist.

**What Next**

If you want to insert scan compression please disable streaming compression feature by issuing the command *set_dft_configuration -streaming_compression disable* command.

---

## UIT-1100

(Error) Invalid option value '%s'

**Description**

The value specified for the option is invalid

**What Next**

Check the value

---

## UIT-1101

(Error) Could not find valid link/reference for %s.

**Description**

You receive this error message in set_bsd_register command when the cell specified in the '-cell' option of the command does not have a valid reference. This might happen when the specified cell does not have a reference or when it has failed to link.

Feedback

**What Next**

Run the link command and fix the LINK-3 or LINK-5 errors reported on the cell.

---

## UIT-1102

(Error) Could not find valid link/reference for %s.

**Description**

You receive this error message in set_dft_signal command when the cell specified in the '-hookup_pin' option of the command does not have a valid reference. This might happen when the specified cell does not have a reference or when it has failed to link.

**What Next**

Run the link command and fix the LINK-3 or LINK-5 errors reported on the cell.

---

## UIT-1103

(Error) %s requires that %s also be enabled.

**Description**

You receive this message because the feature or option you have attempted to use requires that another feature or option also be enabled.

**What Next**

Enable the required feature.

---

## UIT-1110

(warning) DFT Hierarchy Location specification is not honored when %s is enabled.

**Description**

You receive this warning message when you execute the *set_dft_location* command and also enable the following DFT clients for DFT insertion.

Wrapper Boundary Scan Logic Bist

When these clients are specified, user specified DFT Hierarchy location is not used in DFT insertion.

**What Next**

Dft logic not added to user specified DFT Hierarchy during DFT insertion.

Feedback

## UIT-1111

(warning) DFT hierarchy location specification is ignored for test-point insertion.

**Description**

This warning message occurs when running the *set_dft_location* command and also enabling the test-point insertion during DFT insertion.

When test-point insertion is enabled, test-point DFT logic is not added to the user-specified DFT hierarchy location.

**What Next**

This is only a warning message. No action is required.

## UIT-1112

(warning) Specified hierarchy name '%s' doesn't exist in the current design '%s'; it will be created.

**Description**

You receive this warning message when you execute the *set_dft_location* command and specify a hierarchical instance name that doesn't exist in the current design. In this case, the tool creates the specified level of hierarchy for you during DFT insertion.

**What Next**

If you wanted the new hierarchy level to be created, no action is needed. If the hierarchy level was expected to already exist prior to DFT insertion, check the indicated hierarchy name specification against the design.

## UIT-1113

(error) Specified hierarchy name '%s' is not valid.

**Description**

You receive this warning message when you execute the *set_dft_location* command and specify an invalid instance name with the command.

**What Next**

Check if the specified instance name is valid and reissue the command with the correct instance name.

Feedback

## UIT-1114

(error) Specified name '%s' is not an instance of hierarchical design.

**Description**

You receive this warning message when you execute the *set_dft_location* command and specify a leaf level instance with the command.

The specified name should be an instance of a hierarchical design. Instances of black box modules, unresolved modules or library cells are not allowed as valid instance names for this command.

**What Next**

Reissue the command with a valid hierarchical instance name.

## UIT-1115

(error) Specified instance '%s' %s.

**Description**

This error message occurs when an invalid cell instance is specified with the *set_dft_location* command.

The cell specified with the *set_dft_location* command must be modifiable by DFT insertion.

The following cells are not modifiable by DFT insertion:

- CTL-modeled cells
- Interface logic model (ILM) cells
- Block abstraction cells

You cannot specify any of these cells with the *set_dft_location* command. In addition, you cannot specify hierarchical subblock cells within any of these cells.

**What Next**

Reissue the command with a valid cell instance name that is modifiable by DFT insertion.

## UIT-1130

(warning) Setting -control_cell_max_fanout to 0, since -cross_hierarchy is specified.

Feedback

### Description

This warning message occurs when running the *set_bsd_configuration* command with the *-cross_hierarchy* option, and the *-control_cell_max_fanout* option was specified earlier with a value other than 0. The tool needs to optimize the control cells when the *-cross_hierarchy* option is specified, so the command automatically sets the *-control_cell_max_fanout* value to 0.

### What Next

This is only a warning message. No action is required.

---

## UIT-1131

(error) Specified -cross_hierarchy name '%s' is not a valid hierarchical path name in the current design.

### Description

This error message occurs when running the *set_bsd_configuration* command with the *-cross_hierarchy* option value set to an invalid hierarchical path name. The command expects a valid hierarchical path name in the current design as the value for this option.

### What Next

Rerun the *set_bsd_configuration* command with the *-cross_hierarchy* option and specify a valid hierarchical path name.

---

## UIT-1132

(error) Invalid buffer direction '%s' specified for signal '%s'.

### Description

This error message occurs when attempting to run the *set_bsd_configuration* command with an invalid signal direction specified by the *-signals_to_buffer* option.

The options expects a list of signal strings where each signal string has the following format:

```
signal_name signal_direction
```

Allowed values *signal_name* are as follows:

- `clock_dr`
- `update_dr`
- `shift_dr`
- `capture_en`

Feedback

- update_en
- mode_in
- mode_out
- mode1_inout
- mode2_inout
- ac_mode
- ac_test
- ac_init_clk

The following values are allowed *signal_direction*:

```
tdi - Buffering starts from tdi
tdo = Buffering starts from tdo.
```

**What Next**

Rerun the *set_bsd_configuration* command with the correct value for the *-signals_to_buffer* option.

---

## UIT-1133

(error) Specified signal name '%s' is not a valid signal to be buffered.

**Description**

This error message occurs when attempting to run the *set_bsd_configuration* command with an invalid signal name specified by the *-signals_to_buffer* option.

The option expects a list of signal strings where each signal string has the following format:

```
signal_name signal_direction
```

Allowed values for *signal_name* are as follows:

- clock_dr
- update_dr
- shift_dr
- capture_en
- update_en
- mode_in
- mode_out
- mode1_inout
- mode2_inout
- ac_mode
- ac_test
- ac_init_clk

Feedback

The following values are allowed for *signal_direction*:

```
tdi - Buffering starts from tdi
tdo = Buffering starts from tdo
```

**What Next**

Rerun the *set_bsd_configuration* command with the correct value for the *-signals_to_buffer* option.

---

## UIT-1134

(error) Cannot specify -signals_to_buffer option without also specifying the -bsr_buffer_fanout option.

**Description**

This error message occurs when you run the *set_bsd_configuration* command and specify the *-signals_to_buffer* option without also specifying the *-bsr_buffer_fanout* option.

The *-signals_to_buffer* option adds buffers on the specified BSR control signals. The tool needs to know the number of BSR cells driven by each of the buffers. This buffer fanout is specified with the *-bsr_buffer_fanout* option.

**What Next**

Run the *set_bsd_configuration* command again and specify both *-signals_to_buffer* and *-bsr_buffer_fanout*.

---

## UIT-1135

(error) A "-port" specification is not supported for DFT signals of type '%s'; only "-hookup_pin" is supported.

**Description**

You receive this message because you attempted to define a DFT signal using the *-port* option of the *set_dft_signal* command, but that signal type only supports specifications using the *-hookup_pin* option (without the *-port* option).

**What Next**

Adjust the specification to specify the internal hookup pin using the *-hookup_pin* option.

---

## UIT-1200

(error) The -none option cannot not be used with the -all option.

Feedback

**Description**

You are advised to use the *-none* option to turn off optimizations during scan insertion and use *-all* option for performing all the optimizations during scan insertion.

**What Next**

Modify the options accordingly to the recommendations and repeat the command.

---

## UIT-1300

(Warning) Resetting wire load mode to '%s'.

**Description**

The wire load mode is set to a default value.

**What Next**

Use the *set_wire_load_mode* command to adjust.

---

## UIT-1310

(error) Core Integration of cores requires the option : -scan .

**Description**

You receive this message if you issue *set_dft_configuration -integration enable* and also specify the following option. *-scan disable* UI option *-integration* of command *set_dft_configuration* requires the option *-scan enable*. For more information, see the man page for the *set_dft_configuration* command.

**What Next**

Reissue *set_dft_configuration* command with valid options.

---

## UIT-1311

(error) Core Integration of cores is mutually exclusive with options : \n\t-fix_set | -fix_reset | -fix_bus | -fix_bidirectional | -fix_clock | -fix_xpropagation | \n\t-control_points | -observe_points | -logicbist | -mbist | -wrapper | -scan_compression .

**Description**

You receive this message if you issue *set_dft_configuration -integration enable* and also specify one of the following options. *-fix_set enable -fix_reset enable -fix_bus enable -fix_bidirectional enable -fix_clock enable -fix_xpropagation enable -control_points*

Feedback

*enable -observe_points enable -logicbist enable -mbist enable -wrapper enable -scan_compression enable*

UI option *-integration* of command *set_dft_configuration* is mutually exclusive with the above options. For more information, see the man page for the *set_dft_configuration* command.

**What Next**

Reissue *set_dft_configuration* command with valid options.

---

## UIT-1312

(error) Core Integration of cores doesn't support the options : -clock_controller .

**Description**

You receive this message if you issue *set_dft_configuration -integration enable* and also specify one of the following options. *-clock_controller enable* UI option *-integration* of command *set_dft_configuration* is mutually exclusive with the option *-clock_controller enable* when the design doesn't contain MBIST(Tech2) cores. For more information, see the man page for the *set_dft_configuration* command.

**What Next**

Reissue *set_dft_configuration* command with valid options.

---

## UIT-1313

(error) Boundary Scan Synthesis is mutually exclusive with options : \n\t-scan | -fix_set | -fix_reset | -fix_bus | -fix_bidirectional | -fix_clock | -fix_xpropagation | \n\t-control_points | -observe_points | -wrapper | -scan_compression .

**Description**

You receive this message if you issue *set_dft_configuration -bsd enable* and also specify one of the following options. *-scan enable -fix_set enable -fix_reset enable -fix_clock enable -fix_xpropagation enable -control_points enable -observe_points enable -wrapper enable -scan_compression enable*

UI option *-bsd* of command *set_dft_configuration* is mutually exclusive with the above options. For more information, see the man page for the *set_dft_configuration* command.

**What Next**

Reissue *set_dft_configuration* command with valid options.

Feedback

## UIT-1314

(error) Boundary Scan Synthesis doesn't support the options : -mbist | -logicbist | -clock_controller .

### Description

You receive this message if you issue *set_dft_configuration -bsd enable* and also specify one of the following options. *-logicbist enable -mbist enable -clock_controller enable* UI option *-bsd* of command *set_dft_configuration* is mutually exclusive with the option *-logicbist enable*. UI option *-bsd* of command *set_dft_configuration* is mutually exclusive with the option *-mbist enable* when not synthesizing MBIST(Tech1) logic. UI option *-bsd* of command *set_dft_configuration* is mutually exclusive with the option *-clock_controller enable* when not synthesizing MBIST(Tech2) logic. For more information, see the man page for the *set_dft_configuration* command.

### What Next

Reissue *set_dft_configuration* command with valid options.

## UIT-1315

(error) Boundary Scan Integration of IEEE 1500 logic requires the option : -scan .

### Description

You receive this message if you issue *set_dft_configuration -bsd enable* and also specify the following option. *-scan disable -ieee_1500 enable -integration enable*

IEEE 1500 core integration is enabled with the following UI options. *-bsd enable -scan enable -ieee_1500 enable -integration enable*

For more information, see the man page for the *set_dft_configuration* command.

### What Next

Reissue *set_dft_configuration* command with valid options.

## UIT-1316

(warning) The option -integration is not recommended and will be obsoleted in the 2007.03 release.

### Description

You receive this message if you issue *set_dft_configuration -integration enable* command in XG mode or *set_dft_configuration -core_integration* command in DB mode.

Feedback

It is not receommended to use Core Integration functionality, the functionality is going to be obsoleted in 2007.03 release.

In DB mode, core integration functionality is still supported only with *-bist* option.

**What Next**

Reissue *set_dft_configuration* command with valid options.

---

## UIT-1800

(error) '%s' is not defined as a TestMode port.

**Description**

The port defined in the *-encoding* option of the *define_test_mode* command is not defined as a *TestMode port*. The type of the port is defined by using the *set_dft_signal* command.

---

## UIT-1801

(error) The value '%s' for the TestMode port is not correct.

**Description**

The value defined for the port in the *-encoding* option must be either equal to 0 or 1.

---

## UIT-1802

(warning) The current mode already contains encoding.

**Description**

The current mode already contains encoding. The previous encoding specification will be overwritten.

---

## UIT-1803

(error) The following compliance enable port is not an input or inout port: %s.

**Description**

The *set_bsd_compliance* command lets you specify a compliance enable pattern for a boundary scan design. Each compliance enable pattern consists of a set of port_name value pairs. Each port name in a compliance enable specification must correspond to an input or inout port of the design.

Feedback

**What Next**

Make sure that each port in the compliance enable pattern corresponds to an input or inout port of the design.

---

## UIT-1804

(error) No patterns found to write native Verilog test bench file.

**Description**

This error is generated when the create_bsd_patterns does not generate any patterns that could be written to the native Verilog test bench.

**What Next**

Check what caused no test bench patterns to be generated.

---

## UIT-1805

(Error) Specification of hookup sense without port specification is invalid.

**Description**

You receive this message because you have specified a hookup sense for a hookup pin without specifying the port name corresponding to this hookup pin. In the particular case of an internal pin specification, you should drop both the hookup sense and the port name.

**What Next**

If you are in an internal pin flow, please drop the hookup sense specification. If you are not in an internal pin flow please specify the port name corresponding to your hookup pin. *set_dft_signal* command.

---

## UIT-1806

(error) '%s' is defined several times in the encoding definition .

**Description**

The port defined in the *-encoding* option must be defined once.

---

## UIT-1807

(warning) Schedule specification '%s' overwrites a previous specification.

Feedback

### Description

This warning is generated when a schedule name is defined with the *define_dft_schedule* command that was already defined by a previous command. The new specification overwrites the previous specification.

### What Next

If the redefinition was not intentional, choose a unique name for the both schedules and repeat the schedule definition commands.

---

## UIT-1808

(error) Instance '%s' of design '%s' has no test model.

### Description

When a core is specified for scheduling in the automatic or manual DFT scheduling flows, the core must have test model information that defines the available test modes. This error is issued when there is no test model for the design instantiated by the specified cell instance.

In the automatic DFT scheduling flow, cell instances are specified with the *-include* option of the *define_dft_schedule* command. In the manual DFT scheduling flow, cell instances are specified with the *-target* option of the *define_test_mode* command.

### What Next

Make sure you specified the intended cell instance. To see a list of designs that have test models, use the *list_test_models* command. You might need to use the *read_test_model* command to attach a test model to the design specified in the error message.

---

## UIT-1810

(warning) The '%s' option does not apply to the '%s' target.

### Description

You receive this warning message because you have specified a target/option combination with the *set_testability_configuration* command that is not supported. As a result, that option is ignored for that target.

All other valid option/target specifications are accepted.

### What Next

If you specified only a single target, then the option you specified is not supported by that target. Edit your script to remove the option.

Feedback

If you specified multiple targets and the option applies to some targets but not others, you can do any of the following:

- Ignore this warning.
- Split the *set_testability_configuration* specification into multiple commands, and then apply the option only to the targets that support it.
- Use the *suppress_message* command to suppress this warning.

# UITCL

## UITCL-100

(error) Can't get object name of multiple objects;\n \tOnly a single object collection accepted.

**Description**

This command only accepts a collection that contains only one object.

**What Next**

Try the command again with one object collection.

## UITCL-101

(error) Cannot split list elements of value '%s' for variable '%s'.

**Description**

The variable value is not a valid list. Therefore, its value cannot be processed correctly.

**What Next**

Check the value of the given variable that is printed out by the error message.

## UITCL-102

(warning) The update_script command is not supported in Tcl mode;\n \tUpdate_script is available in dcsh only.\n \tTo translate dcsh to tcl format, use the dc-transcript program.

**Description**

This command is not supported in dc_shell Tcl mode.

Feedback

# UIV

## UIV-001

(warning) Application variable '%s' is not supported

**Description**

This application variable is not supported in this tool. Setting this variable will have no effect.

**What Next**

Use *report_app_options* to determine if there is an application option providing similar control.

## UIV-002

(warning) The '%s' variable is obsolete. Please use '%s' instead.

**Description**

Application options are now used instead of variables to control tool operation.

**What Next**

Change your script to use the recommended command. If the recommendation is to use the *set_app_option* command, decide whether you want to specify the option globally (to apply to all designs in the session), or to a specific design (which allows the option setting to be saved with the design data).

# UNDO

## UNDO-001

(error) Undo is not enabled.

**Description**

This command requires that undo is enabled, however undo is not enabled.

**What Next**

Please enable undo to execute the command.

Feedback

## UNDO-002

(error) Marker '%s' not found.

**Description**

The specified undo marker either does not exist or is not visible from within the current command hierarchy.

**What Next**

Please specify the name of an existing and visible marker.

## UNDO-003

(error) Marker '%s' cannot be undone to.

**Description**

The specified undo marker corresponds to a later point in the command history, so it cannot be undone to.

**What Next**

Use the 'redo' command to restore the system state to this marker.

## UNDO-004

(error) Marker '%s' cannot be redone to.

**Description**

The specified undo marker corresponds to a previous point in the command history, so it cannot be redone to.

**What Next**

Use the 'undo' command to restore the system state to this marker.

## UNDO-005

(error) Cannot undo %i levels.

**Description**

Either there are less than the specified number of undoable commands in the undo history, or the specified number is an invalid value.

Feedback

**What Next**

Please specify a positive number of levels less than or equal to the number of undoable commands in the undo history.

---

## UNDO-006

(error) Cannot redo %i levels.

**Description**

Either there are less than the specified number of redoable commands in the latest undo history, or the specified number is an invalid value.

**What Next**

Please specify a positive number of levels less than or equal to the number of redoable commands in the latest undo history.

---

## UNDO-007

(warning) Nothing to redo.

**Description**

This command restores the system to the most recent state in the undo history, however the system is already at the most recent state. Therefore this command has no effect.

**What Next**

No action needs taken.

---

## UNDO-008

(error) Marker name cannot be empty.

**Description**

Undo marker names must not be empty.

**What Next**

Please specify a non-empty marker name.

---

## UNDO-009

(error) Marker name '%s' is in system format.

Feedback

**Description**

The specified undo marker name matches the format used internally for system-created markers, and is therefore disallowed for user-created markers.

**What Next**

Please specify a name with a different format.

---

## UNDO-010

(error) Marker '%s' already exists.

**Description**

An undo marker with the specified name already exists within the current command hierarchy.

**What Next**

Please specify a unique marker name.

---

## UNDO-011

(error) Atomic name cannot be empty.

**Description**

The atomic name of the command group must not be empty.

**What Next**

Please specify a non-empty atomic name.

---

## UNDO-012

(error) Command name cannot be empty.

**Description**

The specified command name argument value must not be empty.

**What Next**

Please specify a non-empty command name argument value.

---

## UNDO-013

(info) The command '%s' is independent of the undo system.

Feedback

**Description**

The execution of the specified command will not directly affect the undo command history.

---

## UNDO-014

(info) The command '%s' is undoable.

**Description**

The specified command can be undone.

---

## UNDO-015

(info) The command '%s' is not undoable.

**Description**

The specified command cannot be undone.

---

## UNDO-016

(info) The command '%s' cleared the undo history.

**Description**

The specified command is not undoable, therefore its execution has cleared the undo history.

---

## UNDO-017

(info) A command within an eval_with_undo command group cleared the undo history.

**Description**

One or more commands within an eval_with_undo command group is not undoable, therefore its execution has cleared the undo history.

---

# UNG

---

## UNG-1001

(information) The hierarchy %s is ungrouped due to auto_ungrouping.

Feedback

**Description**

The specified hierarchy is automatically ungrouped due to the enabling of automatic ungrouping during initial_map stage of compile_fusion.

**What Next**

Automatic ungrouping during initial_map can be controlled with the app option compile.flow.autoungroup

---

## UNG-1002

(information) The hierarchy %s is ungrouped due to set_ungroup directive.

**Description**

The specified hierarchy is ungrouped due to the enabling of set_ungroup command on a cell or a module.

---

## UNG-1003

(information) %d of %d hierarchies are ungrouped. Use report_ungroup for a list.

**Description**

The specified number of hierarchies were automatically ungrouped (or) ungrouped by user.

---

## UNG-1004

(information) The hierarchy %s is not ungrouped due to %s.

**Description**

The specified hierarchy has not ungrouped due to some restrictions on it.

---

## UNG-1005

(information) %d of %d hierarchies are not ungrouped because of restrictions. Use report_ungroup for a list.

**Description**

The specified number of hierarchies were not ungrouped due to user or internal restrictions.

Feedback

## UNG-1006

(information) %d of %d hierarchies are not met auto_ungroup criteria.

**Description**

The specified number of hierarchies were not ungrouped because they did not meet auto_ungroup threshold criteria.

## UNG-1007

(information) Ungroup Summary Report

**Description**

Reports the high-level details of hierarchies that are ungrouped due to auto_ungroup or due to set_ungroup command. Also lists the hierarchies that are not ungrouped due to different restrictions.

## UNG-1008

(warning) The auto-ungroup setting might impact QoR.

**Description**

Auto-ungroup options set by the command *set_autoungroup_options* will limit the auto-ungrouping, hence could potentially affect QoR.

# UPF

## UPF-001

(error) Leaf cell '%s' is not valid as a UPF scope.

**Description**

Within the UPF file, scope must be applied to a hierarchical cell, not leaf level cells.

**What Next**

Determine the correct cell to apply the scope to and change the UPF accordingly.

Feedback

### See Also

- get_cells
- report_cells

---

## UPF-002

(warning) Supply nets or ports '%s' and '%s' are both defined in the power state table '%s' even though they are connected together or electrically equivalent.

### Description

This warning message occurs when the specified supply nets or ports in the supply list is connected together or declared as electrically equivalent. The electrical-equivalent supplies in the supply list of power state table are redundant.

### What Next

Remove the redundant supply net(s) or port(s) from the supply list.

### See Also

- create_pst

---

## UPF-003

(error) Invalid name specified: '%s'

### Description

An invalid identifier was specified. The name violates UPF naming conventions or the requirements for a specific command.

UPF identifiers must start with a letter. All other characters of the identifier must be either an alphanumeric character, "A-Z a-z 0-9", or the underscore character, "_". UPF names are case sensitive.

### What Next

The error message will give the name causing the error, using this, check which create_command in the UPF file is responsible and change the associated name accordingly.

### See Also

- create_power_domain
- create_supply_set

Feedback

- create_supply_net
- create_supply_port
- create_pst

---

## UPF-004

(error) Invalid hierarchical name specified: '%s'

**Description**

The command option requires a simple name, with no hierarchy separator character, for example, '/'. UPF objects may have hierarchical names, but the simple name required is not permitted to contain the hierarchy separator.

**What Next**

Re-enter the command with a valid simple name specified. If you choose to deviate from IEEE Std 1801 recommendation of using simple names, you can try setting the app option mv.upf.input_enforce_simple_names to false. You will be able to continue using hierarchical names if the app option is honored by the command causing this error.

**See Also**

- get_cells
- get_nets
- mv.upf.input_enforce_simple_names

---

## UPF-005

(error) Unable to create %s '%s': %s.

**Description**

The command failed to create an object, detailed reason for the failure is given.

**What Next**

From the reason given, understand why the object was not created and determine if the command should be re-entered with different options. If an object by this name already exists, you may need to remove the old one first. If the object that failed to be created is a bias function of a supply set, make sure that the enable_bias design attribute is set to true/derived prior to supply set creation.

Feedback

### See Also

- query_objects

---

## UPF-006

(error) %s '%s' does not exist.

### Description

The specified object does not exist.

### What Next

Understand why the specified object does not exist, create if required and rerun the command again.

### See Also

- get_objects_by_location
- query_objects

---

## UPF-007

(error) Design element '%s' is not within the scope '%s'

### Description

A design element has been specified that is not contained within the specified power scope.

### What Next

All of the leaf and hierarchical instances used with the -elements or -exclude_elements option must be contained within the current power scope, or the -scope instance if specified. Use get_power_domains / report_power_domains to find the scope of the instance.

### See Also

- get_power_domains
- report_power_domains
- report_power_scopes

Feedback

## UPF-008

(error) Cell '%s' is included in power domains '%s' and '%s'.

### Description

A cell has been specified to include in two different power domains. A cell cannot belong to more than one power domain.

### What Next

Ensure the list of instances is valid for the relevant power domains by reviewing the create_power_domain statements in the UPF file. In addition you can use the get_cells command to determine the power domain associated with a cell(s).

### See Also

- create_power_domain
- get_cells

## UPF-009

(error) Incorrect number of arguments specified with the option '-function'.

### Description

This error message occurs when an invalid number of arguments is specified with the *-function* option of the *create_supply_set* command.

### What Next

Provide the correct number of arguments for the *-function* option in the following format:

```
-function {function_name supply_net_name}
```

### See Also

- create_power_domain
- create_supply_set

## UPF-010

(error) Invalid supply set handle '%s' specified.

Feedback

### Description

This error message indicates that one or more supply set handle names specified with the *-supply* option of the *create_power_domain* command are invalid.

### What Next

Provide valid supply set handle to associate the supply sets successfully.

### See Also

- create_supply_set

---

## UPF-011

(error) Incorrect number of arguments specified with '-supply' option.

### Description

This error message occurs when an invalid number of arguments are specified with the -supply option of the create_power_domain command.

### What Next

Provide the correct number of arguments for the *-supply* option in the following format:

**-supply** {*supply_set_handle supply_set_name*}

### See Also

- create_supply_set

---

## UPF-012

(error) Domain %s supply net for domain '%s' has already been specified.

### Description

This error message occurs when the domain supply nets for the power domain have already been specified. Specifying the domain supply net multiple times is not allowed.

### What Next

Check the existing domain supply nets.

Feedback

**See Also**

- create_power_domain
- create_supply_set
- report_supply_nets

---

## UPF-013

(error) Unable to update %s '%s': %s

**Description**

The UPF command with the option -update failed to update an object. The detailed reason for the failure is given.

**What Next**

Understand why the object was not updated and determine if the command should be re-entered with different option arguments.

**See Also**

- create_power_domain
- set_isolation
- set_level_shifter
- set_retention

---

## UPF-014

(error) No supply net functions have been specified with the -update option.

**Description**

The error message occurs because function names and supply net names are required when the *create_supply_set command is used with the -update option.*

**What Next**

Provide the function names to which you want to associate the function of the supply set along with the supply net names and run the command again.

**See Also**

- create_supply_set

Feedback

## UPF-015

(error) Supply set '%s' does not exist.

### Description

This error message occurs when the supply set referenced for update by the -update option to the create_supply_set command does not exist. The -update option can only be used on an already existing supply set.

### What Next

Create the supply set and then run the *-update* option.

### See Also

- create_supply_set

## UPF-016

(error) Function '%s' has been specified multiple times for this supply set.

### Description

This error message occurs when a valid *function_name* is specified multiple times with the *-function* option.

### What Next

Ensure that a valid *function_name* is specified only once with multiple *-function* arguments.

### See Also

- create_supply_set

## UPF-017

(error) Supply set handle '%s' has been specified multiple times for this power domain.

### Description

This error message occurs when the supply set handle has been specified multiple times for the specified power domain.

The *primary*, *default_retention*, and *default_isolation* handles can be specified only once with the *-supply* option of the *create_power_domain* command.

Feedback

### What Next

Correct the syntax of the command to make it consistent with the above rules.

### See Also

- create_power_domain
- create_supply_set

---

## UPF-018

(error) Cell '%s' already belongs to power domain '%s'.

### Description

The cell corresponding to the scope has previously been added to another power domain. A cell cannot be added to more than one power domain.

### What Next

From the error message you can determine which cell is causing this issue and the initial domain it is associated with. Check the UPF file to find the additional domain that is being created and causing this error and correct the UPF accordingly.

### See Also

- create_power_domain

---

## UPF-019

(error) An existing top-level domain '%s' already exists.

### Description

This error message occurs when you are trying to create a design-level power domain that covers the design top and one already exists.

### What Next

The error message will occur after the "second" top level power domain is attempted to be created and will contain the name of the original top level domain name. From this correct your UPF file such that only one top level power domain exists.

### See Also

- create_power_domain

Feedback

## UPF-020

(error) Save and restore signals '%s' and the active levels are identical for retention strategy '%s' of power domain '%s'.

### Description

This message occurs when the save and restore control signals and their active levels are identical. As a result, the save and restore events of the retention cells can be triggered at the same time.

### What Next

Check the specified save and control signals. Also check the their active levels.

### See Also

- set_retention
- set_retention_control

## UPF-021

(error) %s connection of supply port %s is already connected to supply net %s.

### Description

This error message occurs when the specified supply port already has LowConn(inside)/ HighConn(outside) connections to the specified supply net.

A supply port can be connected by only one supply net on each side (inside/outside).

### What Next

Use the *save_upf* command to check existing connections.

### See Also

- connect_supply_net

## UPF-022

(error) Invalid connection from supply net %s to supply port %s.

Feedback

### Description

This error message occurs when attempting to make a connection from a supply net that is below the scope of the supply port. A supply port can be connected only by a supply net that is at or above the scope of the supply port.

### What Next

Specify a supply net that is at or above the scope of the supply port.

### See Also

- create_supply_net
- create_supply_port

---

## UPF-023

(error) Supply net '%s' cannot be connected to pin '%s' since it is not a supply port or leaf pin.

### Description

All pins specified in option '-ports' of command *connect_supply_net* must be a supply port or leaf pg pin.

### What Next

Change to a supply port or leaf pg pin and try again. A common cause for this is specifying a pg pin of a hierarchical cell of the logical hierarchy.

### See Also

- connect_supply_net

---

## UPF-024

(Warning) Supply net '%s' cannot be connected to pin '%s' because it is already connected to '%s'.

### Description

The power and ground pin has already been connected to an existing supply net, and the specified supply net connection will be skipped.

### What Next

Check the supply net connection of the power and ground pin, contained in the message the existing connection will be given.

Feedback

**See Also**

- connect_supply_net

---

## UPF-025

(error) UPF command '%s' is not allowed after the power intent is finalized.

**Description**

The UPF command is not allowed after the UPF intent is finalized with the commit_upf command.

**What Next**

Reset the UPF constraints of the design, modify the power intent as needed and re-issue commit_upf.

**See Also**

- commit_upf
- reset_upf

---

## UPF-026

(error) Supply net '%s' cannot be connected to pin '%s' because it is not a power/ground pin.

**Description**

All pins specified in option '-ports' of command connect_supply_net must be a supply port or leaf power/ground pin.

**What Next**

Check if the supply port or leaf power/ground pin was specified correctly. If pin was correct, check if the port_type was specified correctly during library preparation.

**See Also**

- connect_supply_net

---

## UPF-027

(error) could not set related supply net(s) on pin '%s' since it is a hierarchical pin.

Feedback

**Description**

The specified pin must be a top-level signal port or leaf signal pin.

**What Next**

Ensure you only specify a top-level signal port or leaf signal pin and try again.

**See Also**

- set_related_supply_net

---

## UPF-028

(error) UPF command '%s' is not allowed outside load_upf.

**Description**

Individual UPF commands are not allowed outside load_upf.

**What Next**

Include UPF commands in a UPF file and invoke load_upf to enter UPF constraints.

**See Also**

- load_upf

---

## UPF-029

(warning) Option '%s' of UPF command '%s' is not supported and will be ignored.

**Description**

This error message occurs when an unsupported option is specified. Other specified options, if any, will be honored.

**What Next**

Check the option of the UPF command and ensure it is not necessary for power intent. Otherwise, consider specifying the intent in another manner.

**See Also**

- load_upf

Feedback

## UPF-030

(error) Inconsistent resolution type on supply net %s: previously is '%s'.

### Description

This error message occurs when the *create_supply_net* command with the *-reuse* option is specified and the *-resolve* option is inconsistent with the -resolve setting in a previous create_supply_net command for the same net.

### What Next

According to the IEEE 1801 (UPF) standard, the -resolve specification for the same net must be consistent for all supply net definitions.

### See Also

- create_supply_net

## UPF-031

(error) Supply net %s cannot be connected to the pin %s in domain %s.

### Description

A supply net can only be connected to power pins that lie within the extent (set of instances) of the power domain(s) in which the supply net is defined.

### What Next

Check if the power domain of the supply net contains the cell. Use report_supply_nets to see the power domain of the supply net. Then, check the power domain of the cell associated with the pin by using get_power_domain.

### See Also

- create_supply_net
- connect_supply_net
- get_power_domains
- get_related_supply_nets
- get_supply_nets
- report_supply_nets

Feedback

## UPF-032

(warning) Cannot set port attribute '%s' on invalid object '%s'%s.

### Description

This message occurs when port attribute are defined on invalid objects. The invalid object will be ignored.

### See Also

- set_port_attributes

## UPF-033

(warning) could not set related supply net(s) on pin '%s' since it is a power/ground pin.

### Description

The specified pin must be a signal port or leaf signal pin in order to specify the related supply net.

### What Next

Change to a signal port or leaf signal pin and try again.

### See Also

- set_related_supply_net

## UPF-034

(information) UPF constraints and voltage areas of the design have been removed.

### Description

This is an informational message indicating that the UPF intent of the design are reset. In addition, all the voltage areas of power domains are removed.

### What Next

You may need to read in the UPF constraints with load_upf and recreate the voltage areas of the design.

Feedback

### See Also

- reset_upf
- load_upf

---

## UPF-035

(error) Supply net %s cannot be connected to the pin %s which does not belong to any power domains.

### Description

A supply net can only be connected to power pins that lie within the extent of the power domain(s) in which the supply net is defined.

### What Next

Make sure the top power domain exists. Also, check the power domain of the supply net contains the cell.

### See Also

- create_supply_net
- connect_supply_net
- create_power_domain

---

## UPF-036

(error) Port state format is not acceptable: %s.

### Description

This error message occurs when the supply port state specified in the add_port_state command does not comply with the UPF standard:

add_port_state supply_port_name -state {state_name <nom | <min nom max> | off> }

### What Next

Use the correct syntax shown and rerun the command.

Correct the appropriate error:

* Do not capitalize any letters of "off" for the off state.

* Use correct voltage value for the state in correct floating point format.

Feedback

* Ensure only two parameters are supplied in the list argument for the -state option for the "off" state.

* Ensure the number of parameters supplied in the list argument for the -state option is either two or four depending on if you are specifying nominal, or minimum, nominal and maximum triplets.

* Check that the voltage triplets are specified in the order of minimum, nominal and maximum.

**See Also**

- add_port_state

---

## UPF-037

(error) Port state '%s' has been defined on supply '%s' with voltage(s) %s.

**Description**

This error message occurs when the specified port state name is already defined either on the same supply or an equivalent supply. The voltage(s) of the same or equivalent supply is(are) shown.

Examples:

UPF-037 (error) Port state 'VON' has been defined on supply 'vdd' with voltage(s) 0.8.

UPF-037 (error) Port state 'VHigh' has been defined on supply 'vdd' with voltage(s) {0.7, 0.8, 0.9}.

UPF-037 (error) Port state 'VSLEEP' has been defined on supply 'vddsw' with voltage(s) 'off'.

**What Next**

Specify a unique name for the port state and rerun the command. To redefine a port state, use reset_upf, followed by load_upf of the updated UPF file with the new add_port_state command.

**See Also**

- add_port_state

---

## UPF-038

(error) The definition of power states is incompatible with the current template.

Feedback

### Description

The definition of power states must be compatible with the template created using the *create_pst* command. For example:

```
prompt> create_pst my_pst -supplies {VDD VDDG VDDY}

prompt> add_pst_state overdrive -pst my_pst -state {HV HV}
```

In this example, the number of supplies listed in the template (3) does not match those listed in *add_pst_state* (2).

### What Next

Rerun the *add_pst_state* command listing the correct number of supplies.

### See Also

- add_pst_state
- create_pst

---

## UPF-039

(error) UPF intent has not been committed.

### Description

UPF intent has not been committed with commit_upf command. commit_upf must be used to indicate that all UPF has been loaded. MV operations cannot be performed until commit_upf runs successfully.

### What Next

Commit the UPF constraints of the design with commit_upf command and try again.
If there is an error during commit_upf, please refer to the individual man page of the associated error message.

### See Also

- commit_upf

---

## UPF-040

(warning) UPF command '%s' is not currently supported and will be ignored.

### Description

This warning message occurs when an unsupported command is specified.

Feedback

### What Next

Check the UPF command.

### See Also

- load_upf

---

## UPF-041

(error) Invalid function '%s' has been specified for supply set.

### Description

This error message occurs when an invalid function_name is specified with the -function option. Currently, only "power", "ground", "nwell" and "pwell" functions are supported.

### What Next

Ensure that a valid function_name is specified.

### See Also

- create_supply_set

---

## UPF-042

(error) Supply Set '%s' already exists.

### Description

This error message occurs when an already existing supply set is specified without the *-update* option for create_supply_set. The existing supply sets are only support with *create_supply_set* if *-update* is being specified. In other words, user is updating and existing supply set functions.

### What Next

Ensure that a new and valid *supply set name* is specified.

### See Also

- create_supply_set

---

## UPF-043

(error) %s '%s' cannot be invoked with default power domain constraints.

Feedback

### Description

This error message occurs when the specified command or option is invoked and the default power domain that is automatically created by the tool exists. That is, the currently loaded UPF is a mixture of tool-generated and user-provided intent which can lead to unintended consequences. The tool generates basic UPF for non-multivoltage designs and it is not meant to be expanded with user intent.

### What Next

Please generate a complete UPF so that no tool-generated UPF is required and the UPF remains consistent.

### See Also

- load_upf

---

## UPF-044

(error) '%s' is a tool-generated default %s that cannot be used in user-specified power intent.

### Description

This error message occurs when the specified UPF object is automatically created by the tool and the user attempts to expand the UPF with user-specified power intent. The loaded UPF cannot be a mixture of tool-generated and user-generated power intent as this can be error-prone.

### What Next

Please generate a complete UPF so that no tool-generated UPF is required and the UPF remains consistent.

### See Also

- load_upf

---

## UPF-045

(error) Supply Set '%s' is not predefined handle.

### Description

This error message occurs when the supply set specified with -handle option is not a predefined supply set handle and does not belong to a power domain or strategy. In other words, user-defined supply sets cannot be used for -handle option.

Feedback

### What Next

Ensure that a predefined and valid supply set name is specified. Check the man page on associate_supply_set for the set of acceptable predefined supply set handles (not user-defined).

### See Also

- associate_supply_set

---

## UPF-046

(error) Supply Set handle '%s' is already associated.

### Description

This error message occurs when the supply set specified with *-handle* option is already associated with another supply set. This command allows a supply set handle to be associated only once.

### What Next

Check the supply set associations. If this command was not in error, remove the previous associations by resetting the UPF.

### See Also

- associate_supply_set
- reset_upf
- load_upf

---

## UPF-047

(error) Command '%s' is not allowed to be a derived UPF constraint.

### Description

This error message occurs when a UPF command is specified as a 'derived' constraint in the UPF file. A derived constraint is one that is added by the tool to the UPF. Derived UPF commands are preceded in the UPF file with either the tcl variables "snps_upf_is_derived_constraint" or "derived_upf" set to true. These Tcl variables can be set and changed multiple times in the UPF file to indicate how the UPF commands following the variable will be treated. Derived constraints are considered as supplemental UPF.

Feedback

### What Next

Check the UPF for "snps_upf_is_derived_constraint" or "derived_upf" tcl variables. These variables are not allowed to be set to true before the specified command in the Error message. Examples of disallowed UPF commands are create_supply_set, create_power_domain, and create_power_switch. Examples of UPF commands that can be derived are set_design_attributes, set_scope and create_supply_net, which can also be applied at the command line.

### See Also

- load_upf

---

## UPF-048

(error) Supply net %s is not available in the power domain %s.

### Description

This error message occurs when the specified supply net is not defined for the specified domain. A supply net can be used only in the domains for which it has been defined.

Some examples where this error may occur are -When defining supplies for power domains, -When creating a power switch (both the input and output supplies need to be available in the power domain), -Isolation and Retention cell checks.

### What Next

First, ensure that the specified supply is the intended supply for use. If the intention is correct, use the *report_power_domains* command to see if the specified supply is available in the power domain. If the supply is not available, you must modify the UPF. Please see the man pages for the *create_supply_net* and *create_power_domain* commands for information about creating and associating supply nets to power domains.

### See Also

- report_power_domains
- create_power_domain
- create_supply_net

---

## UPF-049

(warning) Association of retention cells with retention strategies is not supported in current release.

Feedback

### Description

This message indicates that association of retention cells with UPF retention strategies is not currently supported. Instead, explicit supply net connections to the retention cells are required.

### What Next

Specify explicit supply net connections to the retention cells.

### See Also

- set_retention
- connect_supply_net

---

## UPF-050

(error) Option '%s' cannot be specified with object_type '%s'.

### Description

The specified command option cannot be used with the specified object type.

### What Next

Please see the command manpage for for more information about command option use.

### See Also

- find_objects

---

## UPF-051

(error) Supply set '%s' and supply set handle '%s' cannot be associated because they are not at the same scope.

### Description

This error message occurs during supply set association using associate_supply_set command. The supply set specified with -handle option is not at or below the scope of the specified supply set.

### What Next

Please check the UPF to ensure that the scope of the specified supply set includes the power domain of the supply set handle.

Feedback

### See Also

- associate_supply_set

---

## UPF-052

(error) Implicit supply set handle '%s' cannot be updated to supply set function '%s' using the command '%s'.

### Description

This error occurs when an implicit Supply Set handle is being updated with functions of other supply sets, without the use of associate_supply_set command. For example, using the -update option of create_supply_set to associate an implicit supply set with a new supply set function.

### What Next

The UPF needs to be changed to fix the error. To make an association between supply sets, use the associate_supply_set command.

### See Also

- create_supply_set
- associate_supply_set

---

## UPF-053

(error) Power state format is not acceptable: %s.

### Description

This error message occurs when the supply state specified in the *add_power_state* command does not comply with the UPF standard:

```
add_power_state supply_set
   -state <name> {-supply_expr <boolean function> }
prompt> add_power_state PDA.primary -state ON {-supply_expr {{power ==
 `{FULL_ON,0.8}}}}
```

### What Next

Use the correct syntax shown and rerun the command.

### See Also

- add_power_state

Feedback

## UPF-054

(error) Unable to process %s because of conflicting power state '%s' with %s.

### Description

This message occurs while attempting to update a supply net in a supply set. During the update, a power state was found with the same name as the updated supply set but is defined differently.

### What Next

Remove or rename the power state that is causing the conflict before updating or associating the supply set.

### See Also

- add_power_state
- create_supply_set
- associate_supply_set

## UPF-055

(error) Supply %s '%s' cannot be created in '%s'.

### Description

This error message occurs when a hierarchical name is specified as a supply port or supply net, but the logical instance path is invalid. The logical instance must already exist and cannot be a leaf cell.

### What Next

Use the *get_cells* command to verify that the logical instance exists. Use the *report_cell* command to ensure that the logical instance is not a leaf cell.

### See Also

- get_cells
- report_cell
- create_supply_port
- create_supply_net

Feedback

## UPF-056

(error) Conflicting power states '%s' are defined on '%s' and '%s'.

### Description

This message occurs when two conflicting power states with same name are defined.

### What Next

Remove or rename the power states that are causing the conflict.

### See Also

- add_power_state
- add_pst_state

## UPF-057

(error) %s Strategy %s has been already defined in the power domain %s.

### Description

This error occurs when a duplicated strategy name is used within the same power domain for the reported type of strategy. Name duplication is not allowed for strategies in same power domain.

### What Next

The UPF needs to be changed to fix the error. Identify the referenced duplicate strategy names associated with the referenced power domain and strategy type. If the strategy is a complete duplicate, remove one of the duplicated strategies. If the strategies are different, use a different strategy name for one of the duplicate strategies.

### See Also

- set_isolation
- set_retention
- set_level_shifter
- set_repeater

## UPF-058

(warning) Strategy element %s has direction inout which is not supported.

Feedback

### Description

This message occurs when the element's direction is inout, this is not supported, direction should be either input or output.

### What Next

Check the element name or the direction from the library. If not corrected, the element will be dropped.

### See Also

- set_level_shifter
- set_isolation
- set_repeater
- set_isolation_control

---

## UPF-059

(warning) Element %s of strategy %s is not a domain boundary port or pin of domain %s.

### Description

This warning occurs when the element object defined for the specified strategy is not a domain boundary port or pin.

### What Next

Elements specified in the message are by default dropped from the specified strategy. Check the specified element to ensure the pin or port exists and that it is on a domain boundary.

### See Also

- set_isolation
- set_level_shifter

---

## UPF-060

(warning) UPF won't be split because commit_upf has not been successfully done yet.

Feedback

### Description

This warning occurs when UPF has been loaded but commit_upf has not been run or successfully completed. IC Compiler II requires that commit_upf is run after all UPF information have been loaded to signal completion of UPF intent.

### What Next

Execute commit_upf and clean up any errors it reports before attempting to run split_constraints again.

### See Also

- commit_upf

---

## UPF-061

(warning) None of element specified in strategy '%s' in power domain '%s' is valid.

### Description

This warning message occurs when all elements specified with set_repeater/set_isolation/set_level_shifter command are illegal. As a result, the effective element list becomes empty and the strategy will not be considered for implementation. Such strategies will be deleted during commit_upf and hence will not be written out by save_upf.

### What Next

Revisit the power intent and specify valid elements for the strategy.

### See Also

- set_repeater
- set_isolation
- set_level_shifter

---

## UPF-062

(error) Strategy '%s' specified by command '%s' does not exist in power domain '%s'.

### Description

This message occurs when the strategy cannot be found in the power domain.

Feedback

### What Next

Check the domain name or the strategy name to make sure the strategy is created before this command is issued.

### See Also

- set_level_shifter
- set_isolation
- set_isolation_control
- set_retention_control
- map_retention_cell

---

## UPF-063

(warning) %s command specified library cell %s is not %s type for strategy %s.

### Description

This message occurs when the library cell type does not match the mapping command for the strategy, or is not a valid library cell view. The referenced library cell in the warning message will not be used for mapping.

### What Next

Ensure the strategy type matches with the type of cells used in the mapping list. For a level shifter strategy, the library cell should have the attribute is_level_shifter set to true. For isolation strategy, the library cell should have the attribute is_isolation set to true. For power switch cells, the library cell should have the attribute switch_cell_type set to coarse_grain. For retention cells, the library cell should have the attribute retention_cell set to certain retention cell type.

This warning can also occur when the matching library cell does not have a timing view. When the mapping list only specifies the library cell names without specifying a specific library, all libraries are searched for the library cell. In cases where a linked physical-only library contains cells with names found in the mapping list, a frame view will be retrieved. Since a frame view is not usable for optimization, this warning is issued. In such a case, the reported cell name will end with the view name "/frame".

### See Also

- map_level_shifter_cell
- map_isolation_cell

Feedback

- map_power_switch
- map_retention_cell
- map_retention_clamp_cell
- get_attribute

---

## UPF-064

(error) Power switch port '%s' is used more than once for power switch strategy %s.

**Description**

This error occurs when the same port name is specified multiple times using the -control_port or -ack_port or -input_supply_port or -output_supply_port options of create_power_switch command. Each port of a power switch instance can only be used once.

**What Next**

Ensure unique port name is used for each power switch port.

**See Also**

- create_power_switch

---

## UPF-065

(error) Power switch '%s' already exists.

**Description**

This error message occurs when an already existing power switch strategy is specified for create_power_switch command.

**What Next**

Ensure that a new and valid power switch name is specified.

**See Also**

- create_power_switch

---

## UPF-066

(error) Invalid Boolean expression '%s' in argument %s.

Feedback

### Description

The expected Boolean expression must comply with the SystemVerilog requirements.

### What Next

The Boolean expression must be a combination of instance control port names and logical or bitwise operators (!, |, ||, &, &&, ^ and ==). SystemVerilog keywords are not acceptable port names.

### See Also

- create_power_switch

---

## UPF-067

(error) Supply '%s' used in power state table '%s' is not defined.

### Description

This error occurred because the supply port or net listed in the create_pst command has not been defined in the UPF.

### What Next

Check the UPF to ensure the specified supply has been defined in the UPF. In a hierarchical block, if the supply is inside a macro or an abstract, make sure the supply is defined in the macro UPF and the macro is correctly linked in the block.

### See Also

- create_supply_net
- create_supply_port
- create_pst

---

## UPF-068

(warning) The %s name '%s' conflicts with existing signal %s name within the same scope.

### Description

This warning message occurs when the specified supply net or port name is the same as the name of a logical net or port within the same scope.

### What Next

Invoke resolve_pg_nets command to resolve the name conflict.

Feedback

This warning may occur when a PG-netlist is read before UPF is loaded, causing conflict between existing nets created by the PG-netlist and the create_supply_net or create_supply_port commands in the UPF. These warnings can be ignored when issued before commit_upf has been run.

**See Also**

- create_supply_net
- create_supply_port
- resolve_pg_nets
- commit_upf

---

## UPF-069

(error) pst_state %s of power state table %s is not fully included in lower-level power state table %s.

**Description**

This error message occurs when a pst_state defined at the higher-level power state table is not included in the power state table of a lower-level scope.

**What Next**

Remove this pst_state from higher-level power state table if it's wrongly defined or add the pst_state to the power state table in the lower-level scope.

**See Also**

- add_pst_state

---

## UPF-070

(warning) pst_state %s of power state table %s defined is ignored by implementation.

**Description**

This warning message occurs when a pst_state defined in a lower-level scope is not included or used by the higher-level power state table and therefore is ignored by implementation.

**What Next**

The referenced pst_state is ignored in the current block during optimization because the pst_state cannot occur in the current block. If the referenced pst_state is not needed in

Feedback

the current block, you can ignore this warning. If the pst_state should be used, investigate the power state table of the current block to ensure the power state tables are defined as intended.

**See Also**

- add_pst_state

---

## UPF-071

(error) Valid supply ports or PG leaf pins have not been specified.

**Description**

This error occurs when none of the object names specified with connect_supply_net are valid. Only existing leaf PG pin and supply ports can be connected to a supply net. A SEL-003 warning message will be echoed for every invalid pin and port name prior to this error.

**What Next**

Ensure the specified object exists, the object_class attribute of the object is a pin or port, and that the port_type attribute is power or ground.

**See Also**

- connect_supply_net
- get_attribute

---

## UPF-072

(information) UPF has been successfully committed.

**Description**

This message shows that the UPF has already been successfully committed.

**See Also**

- commit_upf

---

## UPF-073

(information) Total %d netlist change(s) and disconnections have been made to resolve conflicts between power intent and PG netlist.

Feedback

### Description

This message shows the number of netlist changes that have been made to resolve conflicts between power intent (based on UPF) and pg net connections (from DEF or PG netlist). UPF intent takes precedence over DEF or PG netlist connection during conflict resolution.

### What Next

Check details of changes to resolve conflicts between power intent and pg netlist by running 'resolve_pg_nets -verbose' or 'connect_pg_net -verbose'.

### See Also

- commit_upf
- resolve_pg_nets

---

## UPF-074

(Information) %s attribute '%s' is read and stored, but not used.

### Description

This message occurs when the attribute specified in set_design_attributes or set_port_attributes command is not used by the tool.

These attributes are read during load_upf, and stored in the database, and saved in the output UPF created by save_upf. But they have no impact on the tool's optimization or power intent behavior.

Typically, these attributes are either user defined, or are used by other tools. Just because ICC II does not support this attribute, it does not mean that ICC II lacks support for the feature enabled by this attribute in another tool. The implementation of that feature in ICC II may be different and not rely on an attribute setting, or may rely on a different attribute or app option setting.

### What Next

Usually no further action is needed. Check the supported attributes in man pages of set_design_attributes and set_port_attributes.

### See Also

- set_design_attributes
- set_port_attributes

Feedback

- load_upf
- save_upf

---

## UPF-075

(warning) UPF constraints and voltage areas of the design have been removed because hierarchical instance of domain root cell or power scope is deleted.

### Description

This is a warning message indicating that the UPF intent of the design are reset because of hierarchical instance of domain root cell or power scope is deleted. In addition, all the voltage areas of power domains are removed.

### What Next

You may need to read in the UPF constraints and recreate the voltage areas of the design.

### See Also

- reset_upf
- load_upf

---

## UPF-076

(information) Power intent from %sblock %s has been propagated.

### Description

This message shows that the power intent from the block has already been propagated.

---

## UPF-077

(error) '%s' is not a valid power and ground type of supply net.

### Description

This message occurs when the specified power and ground type of supply net is not valid.

### See Also

- set_design_attributes

Feedback

## UPF-078

(warning) Cannot find %s '%s' for %s attribute '%s'.

### Description

This message occurs when the object specified in the set_design_attributes and set_port_attributes command do not exist.

### What Next

Check the supported attributes in man pages of the related commands.

### See Also

- set_design_attributes
- set_port_attributes

## UPF-079

(warning) Design element %s is not on domain boundary.

### Description

This warning message occurs when attempting to specify a design element in the *-elements* option of the *set_isolation* or *set_level_shifter* or *set_repeater* command that is not a root cell of the power domain for which the strategy is being defined.

### What Next

Look at the design element in the error message, verify that it is not a root cell of the power domain, and change it to the root cell to which the strategy applies.

### See Also

- set_isolation
- set_level_shifter
- set_repeater

## UPF-080

(warning) The direction of pin/port '%s' isn't consistent with the value of '-applies_to' of strategy '%s' in power domain '%s'.

Feedback

### Description

This warning message occurs when you define an isolation strategy that combines the -applies_to and -element options, and at least one pin is listed in the -element list whose direction doesn't match with the direction implied in the -applies_to option.

### What Next

This pin will be ignored by the isolation strategy. If it is not what you want, please input another correct 'set_isolation' command line.

### See Also

- set_isolation
- set_level_shifter

---

## UPF-081

(warning) Element '%s' does not belong in power_domain '%s'.

### Description

This message occurs when the specified elements are not in the power domain where the strategy is defined.

At present, for isolation, level-shifter strategy and repeater, only the root cells of the power domain or the pins/ports on the root cells are acceptable in strategy setting.

### What Next

Use the proper elements, and run the command again.

### See Also

- set_isolation
- set_level_shifter
- set_retention
- set_repeater

---

## UPF-082

(error) Isolation or level shifter strategy '%s' will insert isolation or level shifter cells outside the scope of its power domain.

Feedback

### Description

This error message occurs when an isolation or level shifter strategy is specified with a location parent, which requires the tool to insert the power management cells outside the scope of power domain. Isolation or level shifter constraints should not refer to things outside the scope of domain to which they belong.

### What Next

Try changing the UPF such that the isolation cells are inserted within the scope of the power domain, or applying the isolation strategy after top-level integration.

---

## UPF-083

(warning) The strategy %s is conflicting with the strategy %s in power domain %s.

### Description

This error message occurs when same combination of filters (-source, -sink and/or applies_to and/or diff_supply_only and/or -applies_to_boundary) are being applied on the domain or specified elements twice. As a result, the strategy will not be considered for implementation. Such strategies will be deleted during commit_upf and hence will not be written out by save_upf.

The *set_isolation*, *set_level_shifter* and *set_repeater* commands allow you to define isolation / level shifter / repeater strategies for a power domain or its elements with different directions being specified by '-applies_to' or different source/sink supply set being specified by '-source / -sink', or different domain boundaries the strategy will apply to with '-applies_to_boundary'. But note same combination of those filters can't be redefined on the domain or same element.

For example, if you first define an isolation strategy that applies to input ports, and you secondly define an isolation strategy that applies to output ports, then you can't later define a default isolation strategy that applies to both kinds of ports, or just input or output ports.

### What Next

Define only one isolation/level-shifter/repeater strategy for a given port type or a given source/sink supply set on specified elements or domain.

### See Also

- set_isolation
- set_level_shifter
- set_repeater

Feedback

## UPF-084

(warning) For level shifter strategy %s, options %s are not permitted when option -no_shift is specified.

### Description

This warning message can occur whenever a level shifter strategy is specified with the command set_level_shifter. It signals that a level shifter strategy is specified with both the option -no_shift and one or more of the following options: -applies_to, -rule, -threshold, or -location. Such a combination is forbidden by the IEEE 1801-2009 standard, and it is handled as follows: If the options -rule, -threshold, or -location are specified, they are dropped as if the user didn't specified them. The option -applies_to is a special case as it can be useful despite its illegality. This combination of options will be supported as an extension to the standard.

### What Next

Drop one or more of the conflicting options from the source where the level shifter strategy is being specified.

### See Also

- set_level_shifter

## UPF-085

(error) %s is not a valid %s.

### Description

The specified object is not of the correct type.

### What Next

Specify the object with the correct type and rerun the command again.

## UPF-086

(warning) Isolation strategy %s in power domain %s is missing %s.

### Description

The isolation strategy is not complete, and this strategy will be skipped at insertion and association.

Feedback

### What Next

This isolation strategy will not be used in inserting and associating isolation cells. If it is not what you want, please input another correct 'set_isolation/set_isolation_control' command line.

### See Also

- set_isolation
- set_isolation_control

---

## UPF-087

(warning) Isolation strategy %s in power domain %s has force_isolation option, which is not supported right now. Option force_isolation will be ignored at insertion and association.

### Description

The isolation strategy option "force_isolation" is not supported yet.

### What Next

This isolation strategy option "force_isolation" will not be ignored at inserting isolation cells. If it is not what you want, please input another correct 'set_isolation/set_isolation_control' command line.

### See Also

- set_isolation
- set_isolation_control

---

## UPF-088

(error) Cannot associate cell %s with %s strategy, this cell will be skipped.

### Description

The isolation strategy can only be associated with isolation cell, level shifter strategy can only be associated with level shifter cells, same goes for retention cells and power switches.

### What Next

Specify correct strategy and rerun the command.

Feedback

### See Also

- get_power_strategies

---

## UPF-089

(error) Cannot find the real signal preservation of control signal '%s'.

### Description

This error message occurs when specified isolation_signal, save_signal, restore_signal or switch control signal isn't driven by a unique real driver. The real drivers include output pins of leaf level cells, input hierarchical pins and input top level design ports. In case of switch ack signal, this error occurs if ack signal is not connected to real load. The real loads include input pins of leaf level cells, output hierarchical pins and output top level design ports.

### What Next

Modify the command so that it specifies a control signal that is being driven by a real driver/load.

### See Also

- set_isolation
- set_isolation_control
- set_retention
- set_retention_control
- create_power_switch

---

## UPF-090

(warning) Clamp value of library cell %s doesn't match the clamp value of isolation strategy %s. Discarding this library cell.

### Description

This message occurs when the library cell type does not match the mapping command for the strategy.

### What Next

Clamp of library cell should match isolation strategy. If it's not corrected, the library cell will be dropped in the mapping cell list.

Feedback

### See Also

- map_isolation_cell

---

## UPF-091

(information) Related supply net, driver_supply or receiver_supply set on hierarchical pin %s is ignored.

### Description

This is an informational message indicating that the set_related_supply_net or set_port_attribute -driver_supply/-receiver_supply command on any hierarchical pins is ignored. According to the UPF language reference manual, -driver_supply and -receiver_supply specifications of the set_port_attribute statement is permitted only on top-level ports and macro leaf pins. Related supply net will be gotten from the real driver/receiver(s) when the connections are in the netlist. save_upf does not output these ignored constraints.

### See Also

- set_related_supply_net
- set_port_attributes

---

## UPF-092a

(warning) Power Domain %s strategy %s applied on pin %s applies to lower domain boundary with -location other, cannot be used for split_constraints on voltage violation path.

### Description

This error message occurs when running the split_constraints command. If there is a level-shifter violation on any pin of the block under consideration, -location other on lower domain boundary port cannot be used for split_constraints.

### What Next

Use the set_level_shifter command and specify a valid level_shifter strategy.

### See Also

- split_constraints
- set_level_shifter

Feedback

## UPF-092

(error) There is a voltage violation on the net connected to pin %s. Specify a level-shifter strategy with location parent/self or set 'no_shift'.

### Description

This error message occurs when running the split_constraints command. If there is a level-shifter violation on any pin of the block under consideration, a clear location must be specified for the level-shifter strategy with location parent/self, or set the -no_shift option for this pin.

### What Next

Use the set_level_shifter command and specify a level_shifter strategy.

### See Also

- split_constraints
- set_level_shifter

## UPF-093

(error) Combination of level shifter and isolation location on pin %s is not supported.

### Description

This error message occurs when running the split_constraints command. A combination of location parent for the level shifter and location self for isolation is not allowed for the input pin, and location self for the level shifter and location parent for isolation is not allowed for the output pin.

### What Next

Change the location combination for the level shifter and isolation strategy and run the command again.

### See Also

- split_constraints
- set_isolation
- set_level_shifter

Feedback

## UPF-094

(information) %s is set to %s.

### Description

This is a informational message indicate that Isolation or Lever Shifter cell name prefix or suffix default string is set by name_format command.

### See Also

- name_format

## UPF-095

(information) ignore %s command for creating %s %s.

### Description

This is a informational message indicating that the create_supply_port/net command in top only UPF for supply port/net in physical block is ignored. Such supply port/net should be created inside the block UPF.

### See Also

- create_supply_port
- create_supply_net

## UPF-096

(warning) supply port %s already exists, ignore create_supply_port command for creating this supply_port again.

### Description

This is a warning message indicating that the create_supply_port command is ignored because this supply port already exists.

### What Next

Remove the duplicate create_supply_port command.

### See Also

- create_supply_port

Feedback

## UPF-097

(error) Power cell %s is not associated with any power strategies or supply net connections.

### Description

This is an error message indicating that either isolation cells, level shifter cells, power switch cells, or retention cells in the design are not properly associated with any power strategies.

### What Next

There are several ways to fix this problem. First is to fix power strategies in UPF file to cover all power cells in the design. Second is to associate the cells manually with set_power_strategy command, or connect_supply_net. Last is to remove the existing power cells.

### See Also

- connect_supply_net

## UPF-098

(information) Isolation strategy '%s' specifies no elements with the -no_isolation option.

### Description

This warning message occurs because set_isolation command has no objects specified with the -no_isolation option. No isolation cells will be inserted for the strategy's power domain unless other isolation strategies explicitly specify elements for isolation.

### What Next

This is an information-only message. No action is required. However, you might want to check whether the elements should be specified.

### See Also

- set_isolation

## UPF-099

(warning) The related supply net of pin '%s' cannot be determined.

Feedback

### Description

This warning message occurs because the related supply net of the specified pin or port cannot be determined.

### What Next

Check the related PG pin of the specified pin and its supply net.

### See Also

- set_domain_supply_net
- connect_supply_net
- report_cell

---

## UPF-100

(warning) Enabled level shifter %s ignored in map_level_shifter command.

### Description

This message occurs when the library cell type is enabled level shifter instead of level shifter for map_level_shifter command.

### What Next

For level shifter strategy, we don't support enabled level shifters used as level shifter cells. Enabled level shifters can be used for isolation cells but not level shifter cells.

### See Also

- set_level_shifter
- set_isolation
- set_isolation_control

---

## UPF-101

(error) Value '%s' for %s signal sense of retention strategy '%s' of power domain '%s' is not valid. Specify one of: high, low.

### Description

This message occurs when the signal sense specified for retention strategy is not supported or invalid.

Feedback

### What Next

Check the signal sense and invoke the command again.

### See Also

- set_retention
- set_retention_control

---

## UPF-102

(error) %s has already been defined on the retention strategy '%s' of power domain '%s'.

### Description

This error message occurs when set_retention_control or map_retention_cell is issued but save/restore signals or another set_retention_control or map_retention_cell command have already been defined for this retention strategy.

### What Next

Check the *set_retention*, *set_retention_control* or *map_retention_cell* constraint.

### See Also

- set_retention
- set_retention_control
- map_retention_cell

---

## UPF-103

(error) Library cell '%s' specified by retention strategy '%s' of power domain '%s' is not a retention cell type.

### Description

This message occurs when the library cell type does not match the mapping command for the strategy.

### What Next

Check the library cell type.

### See Also

- map_retention_cell

Feedback

## UPF-104

(warning) Isolation or level shifter strategy '%s' can't insert isolation or level shifter cells inside leaf or macro cell '%s'.

### Description

This error message occurs when an isolation or level shifter strategy is specified with a location self on leaf cell or macro cell, which requires the tool to insert the power management cells inside leaf cell or macro cell.

### What Next

Try changing the location of the strategy. If not the element will be dropped.

### See Also

- set_isolation
- set_isolation_control
- set_level_shifter

## UPF-105

(warning) Strategy element %s is located on lower domain boundary of strategy '%s', which does not work with location parent or fanout.

### Description

This warning message occurs when you specify a value of *parent* or *fanout* for the *-location* option, and an element specified with the *-elements* option is on the lower-domain boundary.

### What Next

Try changing the *-location* setting. Otherwise, the command drops the element specified in the warning message.

### See Also

- set_isolation
- set_level_shifter

Feedback

## UPF-106

(error) Cannot apply UPF %s on retention strategy '%s' of power domain '%s' because -no_retention is specified.

### Description

This message occurs when set_retention_control or map_retention_cell is specified for the retention strategy with -no_retention.

### What Next

Check the specified retention strategy.

### See Also

- set_retention
- set_retention_control
- map_retention_cell

## UPF-107

(warning) Element '%s' is already an element of retention strategy '%s' of power domain '%s' and cannot be set as element of %s.

### Description

This message occurs when two different set_retention or map_retention_cell commands mention the element either explicitly in their elements list or by default (not specifying any elements and thereby defaulting to the power domain root cell).

### What Next

Check the correctness of the set_retention or map_retention_cell commands and their elements options used for the strategies, and assign the cell to one strategy only.

### See Also

- set_retention
- set_retention_control
- map_retention_cell

Feedback

## UPF-108

(error) None of library cell specified in map_%s_cell of %s strategy %s is legal.

### Description

This error message occurs when all library cells specified with map_isolation_cell or map_level_shifter_cell or map_retention_cell, or map_retention_clamp_cell are illegal.

### See Also

- map_isolation_cell
- map_level_shifter_cell
- map_retention_cell
- map_retention_clamp_cell

## UPF-109

(warning) Retention strategy %s in power domain %s is missing %s.

### Description

The retention strategy is not complete, and this strategy will be skipped for association and mapping of retention cells.

### What Next

This retention strategy will not be used in association and mapping of retention cells. If it is not what you want, please input another correct 'set_retention/set_retention_control' commands.

### See Also

- set_retention
- set_retention_control

## UPF-110

(error) Driver or receiver supply attribute of '%s' are already defined.

### Description

This error message occurs when the ports already have driver or receiver supply attribute using -driver_supply or -receiver_supply argument of set_port_attributes and

Feedback

set_related_supply_net is used on the same ports again. The driver and receiver supply attributes have higher precedence than the related_supply_net attribute.

**What Next**

Change the related supply net by using -driver_supply or -receiver_supply argument of set_port_attributes.

**See Also**

- set_port_attributes
- set_related_supply_net

---

## UPF-111

(error) Found duplicated supply name '%s' in power state table '%s'.

**Description**

This error message occurs when the duplicated supply net or port name is specified for the power state table.

**What Next**

Re-run the command without any duplicated supply net or port names.

**See Also**

- create_pst

---

## UPF-112

(warning) %s supply of %s port '%s' will be ignored.

**Description**

This message occurs when the driver or receiver supply of the port will be ignored due to the port direction.

**What Next**

Check the port direction and the driver and receiver supplies of the port.

**See Also**

- set_port_attributes

Feedback

## UPF-113

(error) Supply Set handle '%s' is already associated with supply net '%s'.

### Description

This error message occurs when the supply set specified with *-handle* option is already associated with supply net(s). This command allows a supply set handle to be associated only once. This could occur if supply set handle was updated to supply nets using -update option.

### What Next

Remove the previous associations and updates with resetting the UPF.

### See Also

- associate_supply_set
- create_supply_set

## UPF-114

(error) Associating '%s' with '%s' creates a loop of associations.

### Description

This error message occurs when the association of supply set specified by *-handle* option creates a cycle or loop of associations. This means supply sets have A->B->C->A type of associations which are not allowed.

### What Next

Remove the previous associations and updates with resetting the UPF.

### See Also

- associate_supply_set

## UPF-115

(error) Supply set reference '%s' is same as supply set handle '%s'.

### Description

This error message occurs when the supply set specified by *-handle* option is same as the supply set reference.

Feedback

### What Next

Specify a different supply set for associations.

### See Also

- associate_supply_set

---

## UPF-116

(warning) Please do 'set_app_options -global {upf.exec_logic_eco_command true}' before loading the UPFs from split_constraints, so control signal(s) can be connected to the block(s) correctly.

### Description

This message occurs when any control signal for block UPF strategy originally is not available in the block. In order to connect control signal to the block, 'create_logic_port', 'create_logic_net' and 'connect_logic_net' commands will be added to the UPFs from split_constraints. These commands are not formally supported yet. User has to set the option 'upf.exec_logic_eco_command' to true, so the commands will be executed.

### What Next

Do 'set_app_options -global {upf.exec_logic_eco_command true}' before loading UPFs generated from split_constraints.

### See Also

- set_app_options

---

## UPF-117

(error) Unable to set related supply net for %s as supply net %s is not available in the block. %s

### Description

This error message occurs when running the *split_constraints*. The supply net to be set as the related supply net for the specified pin is unavailable for use.

### What Next

Make the supply net available for use in the block and run the command again.

### See Also

- split_constraints

Feedback

## UPF-118

(warning) Unable to set related supply net of pin %s because the driver or load supply is not uniform.

### Description

This warning message occurs when running the *split_constraints* command if there are multiple drivers or loads for the pin and they are not all powered by the same supply.

### What Next

This is only a warning message. No action is required.

However, you can run the *set_related_supply_net* command manually on such pins, if needed.

### See Also

- split_constraints

## UPF-119

(error) Unable to split the UPF for block %s because domain %s is nested inside but its scope is outside the block.

### Description

This error message occurs when running the split_constraints command. Domain nested inside block has to be scoped at or inside the block.

### What Next

Change the scope of the nested domain to be at or inside the block.

### See Also

- split_constraints
- create_power_domain

## UPF-120

(Error) The power net and ground net specified in %s strategy %s come from different supply sets.

Feedback

### Description

Both power and ground implicit supply nets from the same supply set must be used simultaneously. It's an error if only the power or the ground is specified using an implicit supply net or if both nets belong to different supply sets.

### What Next

Specify the supply nets as above description.

### See Also

- set_isolation
- set_retention

---

## UPF-121

(error) %s strategy %s does not apply to cell %s, this cell will be skipped.

### Description

The isolation strategy can only be associated with isolation cell with matching location, iso clamp, sense and iso signal.

---

## UPF-122

(warning) Logic net '%s' does not have any flat driver pins.

### Description

This message occurs when the specified net does not have any flat driver pins after execute connect_logic_net command.

### What Next

Check the net connections.

### See Also

- connect_logic_net
- report_net

---

## UPF-123

(error) Due to the error(s) above, UPF can't be split. Please fix all the errors.

Feedback

### Description

This error message occurs when running the split_constraints command. No block and top only UPFs will be created if any serious error is found.

### What Next

Fix all the errors and run split_constraints again.

### See Also

- split_constraints

---

## UPF-124

(error) Cannot set related supply net for pin %s because at least two conflicting level-shifter strategies are applied to this pin: %s of domain %s and %s of domain %s.

### Description

This error message occurs when you run the *split_constraints* command and the tool cannot determine the correct related supply net for the block pin because there are at least two conflicting level-shifter strategies. For example, one strategy adds the level shifter inside the block but the other one add the level shifter outside the block.

### What Next

Fix the error and run the *split_constraints* command again.

### See Also

- split_constraints
- set_level_shifter

---

## UPF-125

(error) Power domain %s is nested inside block %s but its scope is outside the block.

### Description

This error occurs when the tool identifies a power domain that is nested inside a block, but its scope is outside the block. The scope of a power domain must be at or inside the block associated with the power domain.

### What Next

Change the scope of the nested power domain to be at or inside the block.

Feedback

### See Also

- create_power_domain

---

## UPF-126

(error) Cannot create power domain %s at the top level because it contains element %s, which is inside a block.

### Description

This error occurs when the tool detects that an element associated with a power domain is inside a block. A power domain that is nested inside block must be created in the block.

### What Next

Load the UPF that creates the power domain in the block design, rather than at the top level.

### See Also

- load_upf
- create_power_domain

---

## UPF-127

(error) Cannot accept %s '%s' for logic net connection.

### Description

This message indicates that the net is not logic or digital. connect_logic_net command should only connect a digital logic net to digital logic ports.

### What Next

Check the port name, net name and its type.

### See Also

- connect_logic_net
- UPF-374

---

## UPF-128

(error) Cannot create power domain %s at the top level.

Feedback

### Description

Block domain has to be created in the block design except scoped-parent domain for top of the block.

### What Next

Load the UPF that creates the power domain in the block design, rather than at the top level.

### See Also

- load_upf
- create_power_domain

---

## UPF-129

(error) Invalid attribute format for '%s' attribute '%s': %s.

### Description

This error message occurs when the attribute option specified in the *set_design_attributes* or *set_port_attributes* command does not follow the UPF standard.

### What Next

Use the following syntax and rerun the command:

```
set_design_attributes
   [-attribute {name value}]
set_port_attributes
   [-attribute {name value}]
```

### See Also

- set_design_attributes
- set_port_attributes

---

## UPF-130

(error) [MIM] %s UPF constraints are different in multi-instantiated block:

### Description

This error message occurs when the UPF constraints are different in multi-instantiated blocks. In such case, split_constraints won't be able to split UPF for the multi-instantiated block.

Feedback

### What Next

- Modify UPF constraints for the blocks so the blocks can share one UPF file. • You can force the creation of the multi-instantiated module (MIM) UPF by setting the app_option *mv.hierarchical.ignore_mib_checking_error*. It is even possible to select which instance is used to create the MIM UPF assigning the cell attribute *is_upf_mim_primary_instance* to the desired instance.

### See Also

- split_constraints
- mv.hierarchical.ignore_mib_checking_error
- cell_attributes
- set_attribute

---

## UPF-131

(error) Cannot find %s to re-associate with voltage area %s.

### Description

This error message occurs when the tool tries to re-associate voltage areas (retained after reset_upf) with power domain(s) and supply net(s) in commit_upf after new UPF is loaded, but failed to find the power domain(s) or supply net(s).

### What Next

Delete the voltage area if it is no longer needed or correct, otherwise modify UPF file to include all the objects needed for this voltage area.

### See Also

- load_upf
- reset_upf
- commit_upf
- create_voltage_area
- remove_voltage_areas

---

## UPF-132

(information) UPF constraints of the design have been removed.

Feedback

### Description

This is a informational message indicating that the UPF intent of the design are reset.

### What Next

You may need to read in the UPF constraints of the design.

### See Also

- reset_upf

---

## UPF-133

(warning) The retention cell type '%s' of the element '%s' does not match the library cell type '%s' of the retention strategy '%s'.

### Description

This message occurs when the library cell type of retention strategy does not match the library cell type of the specified element. Retention cell elements can still be associated to the retention strategy if other criteria for association passes.

### What Next

See the *map_retention_cell* command man page for information about the *-lib_cell_type* option.

### See Also

- map_retention_cell

---

## UPF-134

(warning) The library cell of the element '%s' does not match the library cells specified in the retention strategy '%s'.

### Description

This message occurs when library cell of the retention cell does not match the library cells of the strategy. Retention cell elements will be skipped during association of retention cells if their library cell does not match the library cells or library cell type specified in the retention strategy.

### What Next

See the *map_retention_cell* command man page for information about the *-lib_cells* option.

Feedback

**See Also**

- map_retention_cell

---

## UPF-135

(error) Library cell type '%s' of retention strategy '%s' is illegal.

**Description**

This error message occurs in one of the following situations.

- there are no retention registers in the library

- specified library cell type has blank space in it

- none of the retention registers in the library have the specified type

**What Next**

Check the library cell type.

**See Also**

- map_retention_cell
- report_mv_lib_cells

---

## UPF-136

(warning) Logic net '%s' has more than one flat driver pins ('%s' and '%s').

**Description**

This message occurs when the specified net has multiple flat driver pins after execute connect_logic_net command.

**What Next**

Check the net connections.

**See Also**

- connect_logic_net
- report_net

Feedback

## UPF-137

(error) Supply set %s which is associated with %s is not available in block %s.

### Description

This error message occurs when running the split_constraints command. Tool cannot split UPF constraints for the block because the supply set which is needed for associate_supply_set command is not available in the block.

### What Next

Modify UPF constraints and re-run split_constraints.

### See Also

- split_constraints
- associate_supply_set

## UPF-138

(warning) The strategy %s on element %s conflicts with the strategy %s in power domain %s.

### Description

This warning message occurs when same combination of filters (-source, -sink and/or applies_to and/or diff_supply_only and/or -applies_to_boundary) are being applied on the domain or specified elements twice.

The *set_isolation*, *set_level_shifter* and *set_repeater* commands allow you to define isolation / level-shifter/repeater strategies for a power domain or its elements with different directions being specified by '-applies_to' or different source/sink supply set being specified by '-source / -sink' or different domain boundaries the strategy will apply to with '-applies_to_boundary'. But note same combination of those filters can't be redefined on same element.

For example, if you first define an isolation strategy that applies to input ports, then you can't later define another default isolation strategy that applies to both kinds of ports or just input ports.

### What Next

Define only one isolation/level-shifter/repeater strategy for a given port type or a given source/sink supply set on specified elements or domain.

Feedback

**See Also**

- set_isolation
- set_level_shifter
- set_repeater

---

## UPF-139

(warning) Strategy %s on element %s will insert cells outside the scope of its power domain.

**Description**

This error message occurs when an isolation or level shifter strategy is specified with a location parent, which requires the tool to insert the power management cells outside the scope of power domain. Isolation or level shifter constraints should not refer to things outside the scope of domain to which they belong.

**What Next**

Try changing the UPF such that the isolation/level shifter cells are inserted within the scope of the power domain.

---

## UPF-140

(error) Unable to split the UPF for block %s because it is not root cell of its domain (%s).

**Description**

This error message occurs when running the split_constraints command. Currently the tool only supports splitting UPF for block which is domain root cell.

**What Next**

Modify the UPF and run split_constraints again.

**See Also**

- split_constraints
- create_power_domain

---

## UPF-141

(error) Logic net '%s' cannot be connected to the logic port '%s'. Connection across power-domain boundaries is not allowed as per UPF semantics. No connections done.

Feedback

### Description

This error is seen when there is at least one port in the port list that does not lie within the extent of the power domain in which the net is defined. A net can only be connected to ports that lie within the extent of the power domain(s) in which the net is defined. All ports listed in port list are ignored and no connections are performed.

### What Next

Remove the non-qualifying ports and rerun this command. If the indicated logic port and net must be connected, create the appropriate logic port(s) and net(s) in the intermediate power domain(s), and then connect the logic port to the corresponding net within the same power domain.

### See Also

- create_logic_net
- create_logic_port
- connect_logic_net

---

## UPF-142

(error) %s type net '%s' cannot %s %s type net '%s' through the hierarchy pin '%s'. No connections done.

### Description

This error is seen when there is at least one hierarchical pin in the port list has an existing net connected with other side of the hierarchical pin, and the net type is illegal with specified logic net.

Below illegal connection will be checked and reported: - signal net drives tie high/low net. - tie high/low net connected with tie low/high net.

All ports listed in port list are ignored and no connections are performed.

### What Next

Remove the non-qualifying ports and rerun this command.

### See Also

- connect_logic_net

Feedback

## UPF-143

(error) Logic port '%s' is already connected to net '%s'. No connections done.

### Description

This error is seen when there is at least one port in the port list has already been connected.

All ports listed in port list are ignored and no connections are performed.

### What Next

Remove the non-qualifying ports and rerun this command. If the indicated logic port and net must be connected, try to use '-reconnect' option.

### See Also

- connect_logic_net

## UPF-144

(error) Domain %s and %s cannot be merged because %s.

### Description

This message occurs when the tool tries to merge two domains, but these two domains have conflict UPF constraints. The basic rules for two domains to be merged are:

1. Have equivalent domain primary supplies.
2. No conflict in power_switch, retention, Level-Shifter and Isolation strategies.

Note: If a child hierarchy is a root cell of top Power Domain, Tool automatically merges sub-block Power Domain into top Power Domain.

### What Next

- Fix the problem in the UPFs described in the message. Reload and commit the UPF.
- Ensure merged Power Domains have the same primary supply. If merging is not expected for child and parent hierarchies, please remove the parent Power Domain from UPF file, so Tool can reuse child Power Domain.

### See Also

- commit_upf
- create_power_domain

Feedback

## UPF-145

(error) Cannot bring control signal for %s %s to block %s because pin %s doesn't connect to any net.

### Description

This message occurs when split_constraints cannot bring a control signal to the block for strategy or power state in the block.

### What Next

Connect the pin to a net.

### See Also

- split_constraints

## UPF-146

(error) the location for isolation strategy '%s' of domain '%s' is automatic which cannot be supported in split_constraints.

### Description

This message occurs when the location of isolation strategy for the top domain of the block is automatic.

### What Next

Specify location self, parent or fanout for this isolation strategy.

### See Also

- split_constraints
- set_isolation
- set_isolation_control

## UPF-147

(error) UPF for black_box design cannot be loaded from top.

### Description

This error occurs when the tool detects UPF for black_box design is being loaded from the top. UPF for black_box design must be loaded in the black_box design.

Feedback

### What Next

Open the black_box design, load the UPF for it and then save the design.

### See Also

- load_upf

---

## UPF-148

(error) Supply net %s cannot be connected to the pin %s that is in different physical hierarchy.

### Description

A supply net cannot be connected to power pins that are in different physical hierarchy.

### What Next

Make the supply net connection in the design of the physical block instead.

### See Also

- connect_supply_net

---

## UPF-149

(error) Adding new UPF constraint into block design %s in %s view is not yet supported.

### Description

The message occurs when tool needs to add new UPF constraint (because of adding new power_switch or new feedthrough port) into block in the view type indicated in the message which is not yet supported.

### What Next

Link the blocks with only supported view type.

### See Also

- change_abstract

---

## UPF-150

(error) MV constraints cannot be added for new port %s because neither its driver supply (%s) nor its receiver supply (%s) are available in block %s.

Feedback

### Description

The message occurs when the tool cannot add correct MV constraints to new port because the supplies needed are not available in the block.

### What Next

possible actions:

- modify the UPF to make either of the supplies reported available inside block (note that this might have an impact on pg implementation)
- If possible disable feedthrough creation on the reported block (or specific net) to prevent the feedthrough port from being created.
- The created feedthrough port will not have a related supply net annotated, therefore at the block level the domain primary supplies will be used. A set_dont_touch must be applied on the net connected to the specified port at the block level, if the UPF-150 message has been ignored.

### See Also

- create_supply_net
- set_related_supply_net
- set_isolation
- set_level_shifter
- place_pins
- push_down_objects

---

## UPF-151

(error) MV constraints cannot be added for new port %s because existing ports on same net %s have upf strategies on them: %s

### Description

The message occurs when the tool cannot add correct MV constraints to the new port because existing ports on the net have upf strategies on them, so the tool cannot automatically derive a strategy for the new port.

Feedback

### What Next

possible actions:

- Remove upf strategies from any existing port on the net so that the UPF-151 message is not issued.
- Manually add a strategy for the new port (and related supply net information).
- Disable feedthrough creation on the block, or specific net to prevent feedthroughs on the block boundary.
- The created feedthrough port will not have a related supply net annotated, and will not have any UPF strategies, therefore at the block level the domain primary supplies will be used. A set_dont_touch must be applied on the net connected to the specified port at the block level, if the UPF-151 message has been ignored.

### See Also

- set_related_supply_net
- set_isolation
- set_level_shifter
- place_pins
- push_down_objects

---

## UPF-152

(error) Isolation strategy '%s' of power domain '%s' uses a location of parent or fanout, but the ports identified for isolation go to both parent and fanout domains.

### Description

This error indicates a situation where an isolation strategy is defined to place the isolation cells in either a parent or fanout domain, but the ports identified for isolation go to both parent and fanout domains. As a result, the strategy cannot be fully honored due to the ambiguity in the constraint.

This situation typically happens due to 'set_design_attributes -elements {.} -attribute lower_domain_boundary' being true. When this attribute is true, the domain boundary is identified not just by the ports of the hierarchy elements where the domain is defined (upper domain boundary), but also by the ports of any nested domain hierarchies in that domain (lower domain boundary).

If set_isolation is used without -elements, then it will pick up ports belonging to both the upper and lower domain boundaries, if there are nested domains. The upper domain

Feedback

boundary goes to a parent placement location, and the lower domain boundary goes to a fanout placement location. This causes ambiguity in the strategy setting if -location parent or fanout is used, and so the error is issued.

For example, suppose there is a top domain PD_TOP, with a second domain PD_A nested inside it, and PD_A has another domain PD_B nested inside of it.

create_power_domain PD_TOP create_power_domain PD_A -elements {a} create_power_domain PD_B -elements {a/b}

Now you want to isolate the inputs of PD_A with the isolation buffers placed in PD_TOP. So you apply:

set_isolation ISO_STRATEGY -domain PD_A -applies_to inputs set_isolation_control ISO_STRATEGY -domain PD_A -location parent

If the lower_domain_boundary attribute is set true, this will cause the UPF-152 error during commit_upf. This is because the '-applies_to inputs' directive will pick up not just the ports going in to PD_A from PD_TOP, but will also pick up the ports going out of PD_B into PD_A. The '-location parent' constraint makes no sense on those ports between PD_A and PD_B, since PD_B is a fanout location of PD_A, not a parent. So the error is issued.

**What Next**

You can avoid the error in two ways. The least disruptive approach is to use 'set_isolation -elements' to more explicitly specify which ports should be isolated. In the example above, you can avoid the error by modifying the set_isolation constraint:

set_isolation ISO_STRATEGY -domain PD_A -applies_to inputs -elements {a}

This restricts the isolated ports to only those on the 'a' hierarchy, for which a parent location makes sense. The ports on the 'a/b' hierarchy which conflict with the parent location directive are now excluded. This constraint can now be committed by commit_upf.

An alternate solution to avoid the UPF-152 error is to change the lower_domain_boundary setting:

set_design_attributes -elements {.} -attribute lower_domain_boundary false

This changes domain boundary identification to not consider the lower boundary when nested domains are present. So, the ports of hierarchy 'a/b' are no longer considered boundaries of PD_A.

This solution would not work for '-location fanout' in the example given above. It also has possible side-effects on the interpretation of other UPF constraints, since it can impact the ports considered by all set_isolation and set_level_shifter '-applies_to' options. It can also impact objects returned by the find_objects command.

So, using 'set_isolation -elements' is typically the less disruptive solution to avoiding this error, rather than changing the lower_domain_boundary setting.

Feedback

### See Also

- set_isolation
- set_isolation_control
- set_design_attributes

---

## UPF-153

(error) Multi-level physical hierarchy UPF design is not supported yet. %s is a physical block in this design.

### Description

This error message occurs when executing create_abstract command with -preserve_block_instance on design with physical hierarchy inside.

### What Next

Make design only one-level physical hierarchy or remove -preserve_block_instance option.

### See Also

- create_abstract

---

## UPF-154

(error) UPF design with multi-level physical hierarchy is not supported yet. This design contains more than two-level physical hierarchy. For example: %s is a physical block inside physical block %s which inside current top design.

### Description

This error message occurs when applying UPF constraints to a design with more than two-level physical hierarchy.

### What Next

Make design no more than two-level physical hierarchy.

### See Also

- commit_upf

Feedback

## UPF-155

(error) UPF cannot be applied to an ETM with the load_upf command.

### Description

This error message occurs when the load_upf command is used to apply UPF to an ETM. load_upf can only be used to apply UPF for blocks, not hard macros.

### What Next

ETM UPF is generated from the original block by running save_upf -for_etm. This UPF should then be applied in a top-level closure flow using ETM abstractions for one or more blocks. load_upf is not used to apply the UPF in this case. Instead, the constraint mapping file is used.

Specify the constraint mapping file that points to the ETM UPFs with the set_constraint_mapping_file command. This file does not have to include all constraint mapping information, just that needed for the ETM UPFs. These constraint mappings will be appended to any existing constraint mappings stored on the design. Alternately, set_constraint_mapping_file -reset can be used to remove all constraint file mappings, and then they can all be reapplied using set_constraint_mapping_file again.

The format for the constraint mapping file is as follows:

<block_reference_name> ETM_UPF <UPF_file>

The mapped ETM UPF constraints will be loaded automatically when required. Only the mapping file is saved in the design. The ETM UPFs must remain in the locations specified in the mapping file for them to be successfully loaded.

For example, suppose a block instance u_block1 in the top design is linked to an ETM reference block1_etm:

set_constraint_mapping_file etm_mapping.txt

The mapping file might contain:

block1_etm ETM_UPF ./upf/block1_etm.upf

This UPF for the ETM will be read as needed from the path specified in the mapping file.

### See Also

- set_constraint_mapping_file
- load_upf
- save_upf

Feedback

## UPF-156

(error) Retention strategy %s will not be implemented for cell %s in power domain %s. This cell has no mapping constraint defined using the map_retention_cell command.

### Description

This error indicates that the specified cell targeted for retention synthesis does not have mapping directive specified using the *map_retention_cell* command. As a result, the strategy will not be implemented for the specified cell. If this cell is a hierarchical cell, sequential elements inside it might remain unmapped.

### What Next

Add the specified cell to *-elements* list of the *map_retention_cell* command.

### See Also

- set_retention
- map_retention_cell

## UPF-157

(error) The element %s specified in map_retention_cell is not included in the element list of retention strategy %s in power domain %s. The retention strategy will be skipped for association and insertion.

### Description

This error indicates that the specified cell which is included in *-elements* list of *map_retention_cell* is not included in the *-elements* list of *set_retention* strategy. This is an error as per the UPF standard.

As per the UPF standard, following are the rules.

- If *map_retention_cell* has *-elements* list specified, *set_retention* must have same elements in its *-elements* list. In other words, *set_retention* cannot be domain-level strategy.

- If *set_retention* has *-elements* list specified, *map_retention_cell* need not have *-elements* list.

These rules ensure that all the elements of the strategy will have *map_retention_cell* directive.

### What Next

Make sure one of the rules specified above are met.

Feedback

### See Also

- set_retention
- map_retention_cell

---

## UPF-158

(warning) Retention strategy %s for power domain %s has no mapping constraint defined using map_retention_cell command.

### Description

This warning indicates that mapping constraints have not been defined on the retention strategy for the specified power domain using the map_retention_cell command. As a result, no retention cells will be inserted for the specified strategy during Synthesis. However, tools will associate the existing retention cell(s) with this retention strategy.

### What Next

Define the mapping constraints for the retention strategy using the map_retention_cell command.

### See Also

- set_retention
- map_retention_cell
- report_power_domain

---

## UPF-159

(error) Conflicting supply states '%s' and '%s' are specified for the connected supply nets or ports '%s' and '%s'%s.

### Description

This error message occurs when two different supply states are specified in the power state table for the supply nets or ports that are connected together. For example:

```
prompt> add_port_state VDD0 -state {s0 off}
prompt> add_port_state VDD1 -state {s1 1.0}
prompt> connect_supply_net VDD -ports {VDD0 VDD1}
prompt> create_pst pt -supplies {VDD0 VDD1}
prompt> add_pst_state PS1 -pst pt -state {s0 s1}
Error: Conflicting supply states 's0' and 's1' are specified for the
   connected supply nets or ports 'VDD0' and 'VDD1'. (UPF-159)
```

Feedback

### What Next

Specify the same supply state for the connected supply nets or ports.

### See Also

- create_pst
- add_port_state
- add_power_state
- add_pst_state

---

## UPF-160

(error) element(s) of power domain %s has(have) no correct reference library.

### Description

This error message occurs when element(s) of a power domain has(have) no correct reference library.

### What Next

Check the ref_lib list for this library to see if the desired target block exists. Update the ref_lib list to include the missing library, then relink the block.

### See Also

- set_ref_libs
- link_block
- shape_blocks

---

## UPF-161

(error) UPF constraints for block '%s' cannot be found.

### Description

This error message occurs when tool tries to link a block for a MV design but cannot find the UPF constraints for the block.

### What Next

Open the block and load its UPF constraints, then do commit_upf and save the block. If it's abstract view, you have to load the UPF into design view and re-create abstract view.

Feedback

**See Also**

- load_upf
- commit_upf
- create_abstract

## UPF-162

(error) %s of cell %s doesn't match %s strategy %s, and will not be associated to it.

**Description**

This error is issued by the set_power_strategy_attribute command when it fails to associate the cell to the power strategy (either an isolation, power switch, retention, or level shifter strategy). Association fails if the clamp, control, isolation power, location, or power domain of the cell do not match the specified power strategy.

**What Next**

Check the pin connections and power domain of the reported cell, and ensure that they match the strategy with which the cell is being associated.

**See Also**

- set_power_strategy_attribute
- associate_mv_cells
- set_level_shifter
- set_isolation
- set_isolation_control
- set_retention
- set_retention_control
- create_power_switch

## UPF-163

(error) Can't find driver for differential level shifter pin %s.

Feedback

### Description

The differential level shifter input has no driver. A T/B macro cell or differential signal driven by buffers and/or inverters should be used to drive the differential inputs of the level shifter.

### What Next

Ensure that the differential inputs of the level shifter are driven by buffers, inverters, or a T/B macro cell.

### See Also

- set_level_shifter
- commit_upf
- associate_mv_cells
- create_mv_cells

---

## UPF-164

(error) Cell %s is not a buffer, inverter, or T/B macro cell, and can't be the driver of differential level shifter pin %s.

### Description

This error may be issued by commit_upf, associate_mv_cells, or create_mv_cells.

A differential level shifter cell has two inputs that must be driven by a pair of differential signals. Both differential inputs of the level shifter expect to have either buffers, inverters, or a T/B macro cell as the driver. This error occurs when these conditions are not met.

### What Next

Check the differential level shifter and ensure that its differential inputs are properly driven by buffers, inverters, or a T/B macro cell.

### See Also

- set_level_shifter
- commit_upf
- associate_mv_cells
- create_mv_cells

Feedback

## UPF-165

(error) UPF for block %s cannot be split because its %s hierarchy %s is in the block list already.

### Description

These two blocks have parent-child relationship but split_constraints command cannot support multi-level-physical-hierarchy.

### What Next

Do split_constraints for multi-level-physical-hierarchy UPF two-level at a time in top-down manner.

### See Also

- split_constraints
- set_budget_options

## UPF-166

(error) For a multilevel physical hierarchy MV design, this is not supported: %s

### Description

Certain UPF design styles are not supported in the current release for multilevel physical hierarchy (MPH). This message warns of those situations. It may be encountered during the commit_upf command or the create_abstract command.

For commit_upf, there are two unsupported cases:

(1) A physical hierarchy block in the design is not a power domain root element. In simple terms, this means that a power domain is not defined at the same hierarchy level as the top hierarchy of the block. For example, if a block is called U1, both 'create_power_domain PD1 -elements {U1}' and create_power_domain PD1 -scope U1' would create a power domain where U1 is the root element. In some cases it may seem like a hierarchy would be a power domain root element, but it is not. Suppose there is a block U1 and another block inside U1 called U1/U2. If a power domain is created with 'create_power_domain PD1 -elements {U1 U1/U2}', it may appear that U1/U2 is a root element for the power domain. But because U1/U2 is contained completely inside U1, only U1 is the root element, so U1/U2 would trigger a UPF-166 error.

(2) A self-scoped physical hierarchy block has an isolation or level-shifter strategy with -location parent. Self-scoped means that the power domain root element is also the scope of the power domain. For example, 'set_scope /; create_power_domain PD1 -elements {U1}' would create a power domain PD1 at U1 hierarchy but it is scoped to the

Feedback

top hierarchy. So it is not self scoped. But, 'set_scope U1; create_power_domain PD1' would instead create a power domain U1/PD1 which exists at the U1 scope. This is a self scoped power domain. For a self-scoped power domain, using parent location for isolation or level shifter cells puts those cells outside the scope of the power domain. For two-level physical hierarchy, this is supported, but not yet for MPH.

The UPF-166 message can also be issued by the create_abstract command, under three circumstances:

(1) A block inside the abstract being created is not a domain root element. See explanation (1) above for commit_upf for a description of domain root elements.

(2) The top domain of a block inside the abstract being created is not self-scoped. All blocks inside the abstract must be self-scoped. See explanation (2) above for commit_upf for a description of self-scoped power domains.

(3) The top domain of a block inside the abstract being created cannot have any isolation or level-shifter strategies with parent location.

**What Next**

Given the restriction above for multilevel physical hierarchy designs, update the UPF constraints to match what the tool can support.

**See Also**

- commit_upf
- create_abstract
- set_scope
- create_power_domain
- set_isolation
- set_isolation_control
- set_level_shifter

---

## UPF-167

(error) set_isolation %s -domain %s has -source/-sink/-diff_supply_only with it. This is not yet supported in UPF hierarchical flow.

**Description**

Isolation strategy with -source/-sink/-diff_supply_only is not yet supported in UPF hierarchical flow. with physical hierarchy inside.

Feedback

### See Also

- set_isolation
- split_constraints

---

## UPF-168

(error) cannot find %s '%s' for %s strategy %s.

### Description

This message indicates that a power domain, source supply set, or sink supply set name from a set_isolation or set_level_shifter constraint could not be found.

### What Next

Check the name of the reported power domain or supply set and ensure that it is correct and has previously been created using either the create_power_domain or create_supply_set command.

### See Also

- set_isolation
- set_level_shifter
- create_power_domain
- create_supply_set

---

## UPF-169

(warning) There is a voltage violation on the net connected to pin %s but no level-shifter strategy specified for this path.

### Description

This message occurs when running the split_constraints command. If there is a level-shifter violation on any pin of the block under consideration, a clear location should be specified for the level-shifter strategy, or set the -no_shift option for this pin.

### What Next

Use the set_level_shifter command and specify a level_shifter strategy.

Feedback

### See Also

- split_constraints
- set_level_shifter

---

## UPF-170

(error) Terminal boundary attribute is set on non-domain boundary "%s".

### Description

This message occurs when running commit_upf command. If design attribute "terminal_boundary" is set on a cell which is not a root element of a power domain, the error will be reported.

### What Next

Modify UPF and remove the non-domain root cell from "-element" list of the command "set_design_attribute".

### See Also

- create_power_domain

---

## UPF-171

(error) Port/power state '%s' for %s is used in pst '%s' but has not been defined.

### Description

UPF statements should only reference already defined objects. If a UPF statement references an object (such as a port or power state) that has not been defined, load_upf typically gives a UPF-006 error. In the case of hierarchical design, sometimes it is expected that an object may not exist yet. For example, a top level PST table created with create_pst may reference supply nets or ports from blocks whose UPF has not been read yet. In these situations, the object name checking is deferred to allow all UPF files to be read.

This error occurs during commit_upf or change_abstract when a port state (add_port_state) for a supply port, or power state (add_power_state) for a supply set, that is referenced by UPF from a higher level of hierarchy never gets defined in the block UPF files. For example:

## Top Level UPF create_pst ... top_pst -supplies {VDD VSS U1/VDD1 U1/VSS1}
add_pst_state ... -pst top_pst {on on on1 on1}

Feedback

## Block U1 UPF create_supply_port VDD1 create_supply_port VSS1 ... add_port_state VDD1 -state {on2 0.8}

The top level UPF references power state on1 for both of the U1 block supplies in the top_pst PST table. But in the U1 block UPF, only state on2 is defined, not on1. A UPF-171 error would occur in this situation.

**What Next**

Check the top UPF add_pst_state commands, and compare the state names to those from the block UPF add_port_state and add_power_state commands. Ensure that all of the state names associated with the block supplies from the top UPF are defined in block UPF.

**See Also**

- add_port_state
- add_power_state
- add_pst_state
- create_pst
- commit_upf
- change_abstract

---

## UPF-172

(error) Wildcard '*' for %s is used in pst '%s' but no port/power state has been defined for this supply and its equivalent supplies.

**Description**

This message occurs when wildcard is specified in the PST for a supply but there is no port/power state has been defined for this supply and its equivalent supplies.

**What Next**

Defined the port/power state in UPF.

**See Also**

- add_port_state
- add_power_state
- add_pst_state
- commit_upf

Feedback

## UPF-173

(warning) Value of app option mv.upf.max_supply_count_without_driver is out of range. The maximum supply count without driver is 1000.

### Description

This message occurs when a value greater than 1000 is specified for app option mv.upf.max_supply_count_without_driver. UPF-177 error will be reported when the supply count without driver is over 1000.

## UPF-174

(information) Design attribute 'correlated_supply_group' is ignored because port and power state triplets are considered as correlated by default.

### Description

This message occurs when the 'correlated_supply_group' design attribute is specified. The port and power state triplets are considered as correlated by default, and so the value of the design attribute will be ignored.

### What Next

Check the voltages of the supplies.

### See Also

- set_design_attributes

## UPF-175

(error) cannot find domain '%s' for set_level_shifter %s

### Description

The power domain specified in set_level_shifter cannot be found.

### See Also

- set_level_shifter
- create_power_domain

Feedback

## UPF-176

(warning) Supply %s %s has no driver%s. This could potentially indicate missing connect_supply_net commands in the UPF.

### Description

This message, issued by commit_upf, indicates a UPF supply net that has no valid driver. Missing drivers indicate incomplete power intent, and could cause problems later in the flow. For example, two equivalent power supplies may not be detected as equivalent due to missing connect_supply_net statements, which can affect buffering decisions. Missing drivers for supply nets can also lead to incorrect or overly large full-chip PSTs, which can impact runtime.

### What Next

Check your UPF constraints related to this supply net and update them to provide a valid driver. A missing driver could be caused by several different situations.

(1) A supply net missing a supply port (create_supply_port), or unconnected to its related supply port due to missing connect_supply_net statements. (2) An input power port on a lower-level power domain not connected to the related supply net in the higher-level power domain with connect_supply_net. (3) A switched supply net that hasn't been connected to a power switch due to a missing or incorrect create_power_switch statement.

### See Also

- create_supply_net
- create_supply_port
- connect_supply_net
- commit_upf
- change_abstract

## UPF-177

(error) Max count exceeded for supply nets without a valid driver. A UPF-176 message has been issued for individuation violations.

### Description

This error message indicates the maximum count for supply nets without a valid drivers has been exceeded. A valid supply driver includes: - a supply port - out and inout P/G pin type of an instance or macro - internal P/G pin in a macro (for example: output of switch cell inside a fine-grain macro)

Feedback

If a supply cannot be associated with a driver, a UPF-176 warning messages is issued. Too many supplies without valid drivers can lead to incorrectly derived power state table (PST) , which can impact runtime, memory and cause incorrect optimization. By default the limit for supplies without drivers is set to seven (7), but it is recommended that all UPF-176 messages are investigated and resolved by the user.

**What Next**

A common reason for a UPF-176 warning is missing connect_supply_net commands in the UPF . For example, top "VDD" connects to port "U1/VDD", and net "U1/VDD". To connect VDD and U1/VDD, supplies need to be part of the connect_supply_net command, connecting to U1/VDD port. If one or both connect_supply_net are missing, top "VDD" will become a "different" supply from U1/VDD. This will cause the derived supplies to increase, which leads to increased use of resources and may cause incorrect P/G pin connections during implementation.

Another cause can be missing power switch strategies in the UPF. If shutdown power domains exist with internal supplies, a switch cell is required to drive the internal supply. User should check the create_power_switch commands to ensure UPF strategies are complete. In case where power switch cells are manually inserted (without UPF strategies), manual PG connection of the switch pins will also resolve the UPF-176 warning messages since a driver will be established.

If you are aware of the reasons for the UPF-176 messages and chose to ignored them, you can use the app_option mv.upf.max_supply_count_without_driver to change the default limit of 7 to greater value. This is generally not recommended due to implications mentioned earlier.

**See Also**

- create_supply_net
- create_supply_port
- connect_supply_net
- commit_upf
- change_abstract

---

## UPF-178

(error) Full-chip derived PST for all the supplies cannot be built because there are too many derived pst_state.

Feedback

### Description

This usually caused by design has too many independent supplies which possibly caused by incomplete supply net/port connections (UPF-176) and/or incomplete power-state-state (PST) constraints for legal states between supplies.

### What Next

Add proper supply net/port connections and/or provide complete PST / pst_state.

### See Also

- create_pst
- add_pst_state
- commit_upf
- change_abstract

---

## UPF-179

(error) Location fanout isolation strategy '%s' of domain '%s' is not supported in UPF hierarchical: %s

### Description

For location fanout isolation strategy in UPF hierarchical flow, the scenario stated in the message is not yet supported:

### What Next

Update UPF constraints to match what tool can support.

### See Also

- set_isolation
- set_isolation_control
- create_power_domain
- split_constraints

---

## UPF-180

(error) The value of "-lib_model_name" of "map_retention_cell" is not a valid format for retention strategy '%s'.

Feedback

### Description

This message occurs when the value of "-lib_model_name" of command "map_retention_cell" is not a valid format. The correct format is "<model_name> {-port <port_name> <net_ref>}*".

### What Next

See the *map_retention_cell* command man page for information about the *-lib_model_name* option.

### See Also

- map_retention_cell

---

## UPF-182

(error) The net_ref value %s is not present or has no signal driving it for retention strategy %s. The map_retention_cell constraint is ignored.

### Description

This error message is reported when,

1. You specify the *net_ref* value that is not a logic net name or supply net name from the current scope, or a symbolic reference. The symbolic reference can be one of the following:

- retention_ref.function_name {a retention supply set function, where function_name refers to the supply net corresponding to the function it provides to the retention ret_supply_net}
- primary_ref.function_name {a primary supply set function, where function_name refers to the supply net corresponding to the function it provides to the primary supply set of the domain}
- save_signal {refers to the save signal specified in the corresponding retention strategy}
- restore_signal {refers to the restore signal specified in the corresponding retention strategy}
- UPF_GENERIC_CLOCK
- UPF_GENERIC_DATA
- UPF_GENERIC_ASYNC_LOAD
- UPF_GENERIC_OUTPUT

Feedback

1. The design has a logic net present in current scope with the name specified as *net_ref*, but the logic net has no signal driving it or is dangling net.

**What Next**

Specify the correct *net_ref* value for the port connection.

**See Also**

- map_retention_cell

---

## UPF-183

(information) List of supplies grouped by equivalence - %s.

**Description**

This message shows the list of equivalent supplies.

**What Next**

This message is informational. No action is required.

**See Also**

- check_supply_equivalence

---

## UPF-184

(error) The '%s' in "-element" of "create_power_domain" is not a valid type of element for power domain.

**Description**

This message occurs when the cell of "-element" of command "create_power_domain" is not a valid type.

**What Next**

After checking the types of cells in the power domain, correct the cells of invalid types or modify the list of elements.

**See Also**

- create_power_domain

Feedback

## UPF-185

(error) Cannot reuse the domain-independent supply net '%s'.

### Description

This message occurs when a domain-independent supply net is specified to be reused in the create_supply_net command. When '-reuse' option is specified, the supply net name must have been defined for other power domain(s) by another create_supply_net command.

### What Next

Check the supply net name and make sure the existing supply net is made available for use in another domain by the '-domain' option of create_supply_net command.

### See Also

- create_supply_net

## UPF-186

(warning) None of library cell specified in map_isolation_cell of strategy '%s' in power domain '%s' is bias insulated.

### Description

This message occurs when all library cells specified with map_isolation_cell has no bias insulated. Isolation strategy will be dropped.

### See Also

- map_isolation_cell

## UPF-187

(warning) For level shifter strategy %s, option %s is not permitted when option -force_shift is specified.

### Description

This warning message can occur whenever a level shifter strategy is specified with the command set_level_shifter. It signals that a level shifter strategy is specified with both the option -force_shift and the option -threshold. Such a combination is handled as follows: If the option -threshold is specified, it is dropped as if the user didn't specified it.

Feedback

### What Next

Drop the option -threshold from the source where the level shifter strategy is being specified.

### See Also

- set_level_shifter

---

## UPF-188

(error) UPF constraints have been loaded with incomplete_upf mode turned on, without reset_upf, incomplete_upf mode cannot be turned off.

### Description

This message occurs when UPF constraints have been loaded with incomplete_upf mode turned on, and user tries to set the mode to off.

### What Next

The UPF netlist and netlist used in incomplete_upf mode is a throw-away netlist/ constraints. To turn off incomplete_upf mode, user has to start over with clean netlist and UPF constraint.

### See Also

- set_app_options
- load_upf
- commit_upf
- reset_upf

---

## UPF-189

(error) Adding leaf cell %s to the element list of domain %s in linked physical block is not supported.

### Description

This message occurs when new buffer/inverter is adding to linked physical block and assigning it to be domain root element.

### See Also

- create_power_domain

Feedback

## UPF-190

(error) Power state '%s' has already been defined on the supply set with voltage(s) %s.

### Description

This error message occurs when the specified port state name is already defined on the same supply set. The voltage(s) is(are) shown.

Examples:

UPF-190 (error) Power state 'FULL_ON' has already been defined on the supply set with voltage(s) 0.8.

UPF-190 (error) Power state 'FULL_ON' has already been defined on the supply set with voltage(s) {0.7, 0.8, 0.9}.

UPF-190 (error) Power state 'OFF' has already been defined on the supply set with voltage(s) 'OFF'.

### What Next

Specify a unique name for the power state and rerun the command. To redefine a power state, use reset_upf, followed by load_upf of the updated UPF file with the new add_power_state command.

### See Also

- add_power_state

## UPF-191

(error) No %s ports for port attribute 'feedthrough' on instance '%s' for the specified model '%s'.

### Description

This message occurs when the ports specified in set_port_attributes -model -feedthrough command are all driver ports or load ports.

### See Also

- set_port_attributes

## UPF-192

(error) Cannot set port attribute '%s' on %s%s '%s' due to %s.

Feedback

### Description

This message occurs when the attribute specified in the set_port_attributes and set_port_attributes command cannot apply to the specified object.

### See Also

- set_port_attributes

---

## UPF-193

(error) This is an internal database file and cannot be loaded.

### Description

An internal database file in the design library cannot be loaded in load_upf.

### What Next

Avoid loading the internal database UPF file

### See Also

- load_upf

---

## UPF-194

(error) Domain-dependent supply net '%s' cannot be set as equivalent supply net.

### Description

This message occurs when specified the supply net in set_equivalent command is a domain-dependent supply net. Only domain-independent supply net can be specified with 'set_equivalent -nets'.

### What Next

Check the supply net is a domain-independent supply net or not.

### See Also

- set_equivalent

---

## UPF-195

(warning) Equivalent supply '%s' does not exist and will be ignored.

Feedback

### Description

This message occurs when specified the supply net, supply port or supply set in set_equivalent command does not exist.

### What Next

Check the supply name is valid.

### See Also

- set_equivalent

---

## UPF-196

(error) The supply sets '%s' and '%s' have bias mismatch.

### Description

This error message occurs when you are trying to declare the specified supply sets with bias mismatch as equivalent supplies. The bias mismatch occurs when supply sets have different bias supplies or one of supply sets belongs to non bias blocks.

### What Next

Ensure you specify the correct supply sets.

### See Also

- create_supply_set
- create_power_domain
- set_equivalent

---

## UPF-197

(error) Supply net %s and %s are NOT equivalent and both have to be created in block %s with same name.

### Description

This message occurs when tool tries to split full-chip UPF constraints and finds two supply nets have to be created in same block with same name. This will cause supply net name conflict.

Feedback

### What Next

Check if these two supply nets should be equivalent. If yes, connect_supply_net command(s) could be missing. If not, user has to change the supply net name(s).

### See Also

- create_supply_net
- connect_supply_net
- split_constraints

---

## UPF-198

(warning) Library cell %s is %s-style isolation cell. Can't be used with isolation strategy %s. Discarding this library cell.

### Description

This message occurs when the library cell type does not match the mapping command for the strategy.

### What Next

Clamp of library cell should match isolation strategy. If it's not corrected, the library cell will be dropped in the mapping cell list.

### See Also

- map_isolation_cell

---

## UPF-199

(error) Due to the error(s) above, UPF implementation fails. Please fix all the errors.

### Description

This error message occurs during UPF implementation. Due to inconsistencies in UPF specification across instances of (a) multiply-instantiated module(s), power management logic cannot be uniformly implemented.

### What Next

Fix the UPF commands so that they are consistent across all instances of a multiply-instantiated module.

Feedback

## UPF-200

(error) The %s command/option is not supported in a bias block.

### Description

This error message occurs when set_domain_supply_net is used inside a bias block (a UPF element with 'set_design_attributes -attribute enable_bias TRUE'). set_domain_supply_net can only be used to associate the primary power and ground supply nets of a domain, and so cannot be used to set the pwell and/or nwell bias supplies.

### What Next

Instead of set_domain_supply_net, use 'create_power_domain -supply' to define the bias supplies available in the domain, or use the implicit supply set handles automatically created during create_power_domain. For bias blocks, all implicit supply sets (primary, default_isolation, and default_ground), have power, ground, nwell, and pwell functions created automatically.

### See Also

- set_design_attributes
- create_power_domain
- create_supply_set

## UPF-201

(error) The %s supply set handle and %s supply set have bias mismatch.

### Description

This error message occurs when you are trying to associate supply set with bias mismatch. The bias mismatch occurs when supply sets have different bias supplies or one of supply sets belongs to non bias blocks.

### What Next

Ensure you specify the correct supply sets.

### See Also

- create_supply_set
- create_power_domain

Feedback

## UPF-202

(error) Full-chip UPF cannot be saved because %s.

### Description

This error message indicates a failure of save_upf -full_chip to write a UPF file. This can happen for the following reasons:

* The top block is a blackbox * Any sub-block in the design is a blackbox * The top block is not an MV design * There are no valid sub-blocks with UPF intent

For the first two reasons, the general requirement for saving full-chip UPF is that all blocks must be linked to either their design view or abstract view. Full-chip UPF cannot be exported when blackboxes are present.

For the third reason, a block is considered an MV design if load_upf was used to apply UPF constraints on it. As such, if no UPF constraints were ever applied on the block, then it is not possible to write out full-chip UPF.

### What Next

Ensure that all blocks are linked to their abstract or design views, and ensure that UPF has previously been applied before attempting save full-chip UPF. Also, ensure that the design is a hierarchical design and valid sub-blocks with UPF intent are correctly specified in the save_upf command.

### See Also

- save_upf
- set_reference
- change_abstract

## UPF-203

(error) Bias block "%s" defined under non-biased block.

### Description

This message occurs when running commit_upf command. If the design attribute enable_bias is TRUE on a block which is under a non-bias block, this error occurs.

### What Next

Allowable use models are to have the top and all sub-blocks as all biased or all non-biased. Additionally, a higher level of hierarchy can be biased with a sub-block being non-biased. But, a biased block cannot exist in a lower hierarchy than a non-biased block.

Feedback

Use 'set_design_attributes -attribute enable_bias' to modify the biasing attributes in the UPF to adhere to the requirements.

**See Also**

- set_design_attributes

---

## UPF-204

(warning) The scope of power domain %s will be changed to the top of its block in the UPF generated by split_constraints.

**Description**

This scope of the domain originally is outside the its block which
is not supported in UPF hierarchical flow. Because the app_option
'mv.hierarchical.convert_scoped_outside_nested_domain' is set to 'true' by user,
split_constraints will change the scope of this domain to the top of its block in the output UPF.

**See Also**

- split_constraints
- create_power_domain

---

## UPF-205

(warning) Strategy '%s' is defined on power domain on leaf cell '%s', which is not supported.

**Description**

This message occurs when an isolation or level shifter strategy is defined on power domain on leaf cell, and this is not supported. Only no_isolation strategy is supported on power domain on leaf cell at the moment.

**What Next**

The element will be dropped.

**See Also**

- set_isolation
- set_isolation_control
- set_level_shifter

Feedback

## UPF-206

(warning) Library cell %s %s.

### Description

This message, when issued by map_power_switch, warns if either the specified power switch library cell does not exist, or if the specified cell is not a power switch.

map_retention_cell also issues this warning message when no library cells are found for the name specified in -lib_cells option.

### What Next

If the specified library cell does not exist, check the name of the cell and the library and correct it. Check for existence of library cells using the get_lib_cells command.

If the specified library cell is not a power switch, this means that the .lib attribute is_power_switch is not true. Check this attribute using either report_attributes or get_lib_cells -filter "is_power_switch==true|false". Either the library cell is not a power switch cell, or the library files need to be corrected to indicate that it is a power switch cell.

### See Also

- map_power_switch
- map_retention_cell
- get_lib_cells
- report_attributes

## UPF-207

(error) Using set_design_attributes to set '%s' '%s' must be done before other UPF commands.

### Description

This error message occurs when the set_design_attributes command is used to set the 'enable_bias' attribute after power domains or supply sets have already been created. Substrate biasing can impact available supplies in power domains and supply sets, and so must be specified before executing UPF commands like create_power_domain and create_supply_set.

### What Next

Change the order of commands in your UPF file to specify set_design_attributes before running any other UPF commands.

Feedback

**See Also**

- set_design_attributes
- create_power_domain
- create_supply_set

---

## UPF-208

(error) Bias legacy check fails for domain '%s'.

**Description**

This error message occurs because domain scope is bias enabled while one of its root cells is bias disabled.

Legacy domains are domains whose scope and all root cells are bias disabled when top design is bias enabled.

**What Next**

Specify correct scope for domain creation.

**See Also**

- create_power_domain
- set_design_attributes

---

## UPF-209

(error) The %s option is not supported in a bias block.

**Description**

This error message occurs when a UPF command option is used that is incompatible inside a bias block (a UPF element with 'set_design_attributes -attribute enable_bias TRUE').

The follow UPF command options cannot be used inside bias blocks:

1. set_isolation -isolation_power_net 2. set_isolation -isolation_ground_net 3. set_retention -retention_power_net 4. set_retention -retention_ground_net

For retention cells, the error will occur if the domain of the retention strategy is a bias block. For isolation cells, it depends on the set_isolation_control -location setting. Whether self or parent, if the insertion location for isolation cells is a bias block, then these options cannot be used.

Feedback

### What Next

For all four cases, the problem is that the bias supply nets cannot be specified using these options. For a bias flow, use supply sets instead. On set_isolation, use the -isolation_supply_set option. On set_retention, use the -retention_supply_set option.

### See Also

- set_design_attributes
- set_isolation
- set_isolation_control
- set_retention

---

## UPF-210

(warning) %s not allowed when %s feature is %s.

### Description

This warning indicates some restrictions on the command option usage or its value when the mentioned feature are turned on. commit_upf will issue an error on such usage thus invalidating UPF command having such usage.

Here are some of the restrictions current being imposed.

- Location fanout or parent is not allowed for isolation and level-shifter strategy with lower_domain_boundary feature. For isolation strategy only location self is allowed. For level-shifter strategy, only location automatic or self is allowed. commit_upf will error out on such strategies, thus making them ineffective.

### What Next

Make sure the restrictions are met.

### See Also

- set_design_attributes
- set_isolation
- set_isolation_control
- set_level_shifter
- commit_upf

Feedback

## UPF-211

(error) %s strategy %s of power domain %s cannot have location %s.

### Description

This error indicates that mentioned isolation or level-shifter strategy has location parent or fanout, which is not an allowed value when lower_domain_boundary feature is on. This restriction is because of other alternate approaches being available to achieve the same result.

### What Next

Change the location to self for isolation strategy and self/automatic for level-shifter strategy.

### See Also

- set_design_attributes
- set_isolation
- set_isolation_control
- set_level_shifter
- commit_upf

## UPF-212

(error) Cannot apply repeater supply to the object '%s'.

### Description

This error message occurs when *set_port_attributes* command is used with *-repeater_supply* argument or *repeater_power_net* and *repeater_ground_net* attributes on an object that does not meet the following criteria.

- If *-ports* argument is specified, objects in the list - must be ports or pins on the power domain root elements - can be ports of the design with direction output only - can be pins of the leaf instance with direction input only - can be pins of the hierarchical instance with direction in or out

- If *-elements* argument is specified, objects in the list - must be the power domain root elements - can be current design with *-applies_to outputs* only - can be leaf instance with *-applies_to inputs* only - can be a hierarchical instance

Feedback

### What Next

Specify the correct *-ports*, *-elements* or *-applies_to* to make sure correct value is specified for the objects.

### See Also

- set_port_attributes

---

## UPF-213

(error) Specified repeater attribute format is not correct.

### Description

This error message occurs when *set_port_attributes* command is used with wrong repeater attribute format. Here are two correct formats:

- Specify attribute *repeater_power_net* and *repeater_ground_net* together as a pair. For example, set_port_attributes -ports {P1} -attribute repeater_power_net VDD \ -attribute repeater_ground_net VSS

- Specify option *-repeater_supply* without other attributes. For example, set_port_attributes -ports {P1} -repeater_supply SS

### What Next

Specify the attribute *repeater_power_net* and *repeater_ground_net* together.

### See Also

- set_port_attributes

---

## UPF-214

(error) Cannot create non-bias power domain '%s' as bias supplies are available in the scope where it is being created.

### Description

This error message occurs because you are creating a power domain in a non-bias block, and bias supply sets are available to the power domain.

### What Next

Restrict the available supply of the power domain to non-bias supply sets only by using the -supply option of the create_power_domain command.

Feedback

### See Also

- create_power_domain

---

## UPF-215

(error) Cannot generate UPF for create_abstract because current block has other block(s) inside (listed below).

### Description

Currently for multi-level physical hierarchy MV designs, middle-level can only be design view. create_abstract will fail if the block has other block(s) inside for MV designs.

### What Next

Use design view for all middle-level blocks.

### See Also

- create_abstract

---

## UPF-216

(error) UPF is not supported when the "preserve_multiply_instantiated_module" attribute is used.

### Description

This error message occurs during UPF implementation. You have specified the "preserve_multiply_instantiated_module" attribute on one or more design modules and loaded a UPF. UPF is not supported on designs with the "preserve_multiply_instantiated_module" attribute.

### What Next

Remove the "preserve_multiply_instantiated_module" attribute on design modules using the remove_attribute command, or remove the UPF using the reset_upf command.

---

## UPF-217

(error) power_state with the following settings is not fully included in lower_level scope %s

### Description

This error message occurs when a power_state defined at the higher-level scope is not included in power_states in lower-level scope.

Feedback

### What Next

Remove this power_state from higher-level power state table if it's wrongly defined or add the power_state to the lower-level scope.

### See Also

- add_power_state

---

## UPF-218

(warning) power_state with the following settings in scope %s is ignored by implementation.

### Description

This warning message occurs when a power_state defined in a lower-level scope is not included or used by the higher-level power state and therefore is ignored by implementation.

### What Next

The referenced power_state is ignored in the current block during implementation because the power_state cannot occur in the current block. If the referenced power_state is not needed in the current block, you can ignore this warning. If the power_state should be used, investigate the power states of the current block to ensure the power states are defined as intended.

### See Also

- add_power_state

---

## UPF-219

(error) could not set related supply net(s) on pin '%s' from top since it is inside different physical hierarchy.

### Description

Related_supply_net for a pin cannot be set across physical hierarchy.

### What Next

Open the block design view which contains the pin and set the related_supply_net. Re-create abstract if it's needed.

Feedback

**See Also**

- set_related_supply_net

---

## UPF-220

(error) Cannot resolve power_state_group '%s' state '%s'.

**Description**

This error message occurs when the state defined in power_state_group includes undefined supply set state or undefined group state, or another unresolvable group state.

**What Next**

Define the supply state or resolve the unresolvable group state referred to.

**See Also**

- add_power_state

---

## UPF-221

(error) add_power_state command supply set object cannot be found when -supply is specified.

**Description**

This error message occurs when -supply is specified in add_power_state command but supply set is not specified or object cannot be found(not created yet).

**What Next**

Define the supply set correctly.

**See Also**

- add_power_state
- create_supply_set

---

## UPF-222

(error) add_power_state command has conflict with both %s and %s options specified.

Feedback

### Description

This error message occurs because those options cannot be issued in the meantime for add_power_state command.

### What Next

Correct the options for the command.

### See Also

- add_power_state

---

## UPF-223

(error) add_power_state and create_pst commands' version has conflict.

### Description

This error message occurs because UPF3.0 version of add_power_state command is used in the meantime with UPF2.0 version. e.g create_pst command and add_power_state -group command cannot be used together. create_pst command cannot be supported with add_power_state -supply usage.

### What Next

ICC II cannot support conflict version of add_power_state command. Select to use one version for the entire flow.

### See Also

- add_power_state
- create_pst

---

## UPF-224

(error) Power state group %s in add_power_state -group command is not defined yet.

### Description

This error message occurs when the power state group is not defined yet for add_power_state -group option.

### What Next

Create the power state group in the right scope.

Feedback

### See Also

- create_power_state_group

---

## UPF-225

(error) Power state group '%s' has already been defined in the scope %s.

### Description

This error message occurs when the specified port state group is already defined in the scope.

### What Next

Specify a unique name for the power state group in the specified scope. And make sure only create once.

### See Also

- set_scope
- add_power_state

---

## UPF-226

(error) add_power_state command requires -update for new state definition of power state group.

### Description

This error message occurs when power state group already has a state defined, adding new state definition does not have '-update' option in the command line. For the first state definition, this is not required, but any new state definition in a separate add_power_state command requires -update option. For the command line with multiple -state definition, -update is also not required.

### What Next

Add -update in the command line.

### See Also

- add_power_state

---

## UPF-227

(information) Incomplete UPF support is enabled.

Feedback

### Description

This message shows the incomplete_UPF mode is enabled.

### What Next

This message is informational. No action is required.

### See Also

- set_app_options

---

## UPF-228

(information) Total %s modifications have been made for incomplete UPF support.

%s.

### Description

This message shows the total number of modifications have been made for incomplete UPF support.

### What Next

This message is informational. User can use report_incomplete_upf command to report details.

### See Also

- report_incomplete_upf

---

## UPF-229

(error) Unable to set related supply net for %s as this pin is driven by NOR/NAND style isolation, but the driver of the isolation signal for isolation strategy %s of domain %s cannot be found.

### Description

This error message occurs when running the *split_constraints*. The supply net to be set as the related supply net for the specified pin cannot be got from the control signal's driver of NOR/NAND style isolation strategy.

### What Next

Specify valid isolation signal with legal driver in the NOR/NAND style isolation strategy and run the command again.

Feedback

**See Also**

- split_constraints
- set_isolation
- set_isolation_control

---

## UPF-230

(error) Unable to declare the supplies '%s' and '%s' to be electrically equivalent: supply %s.

**Description**

This error message occurs when when the *set_equivalent* command is unable to establish electrical equivalence relationship on the specified supplies.

Electrical equivalence may only be defined in a limited set of circumstances:

- Each supply is driven only by a top-level supply port;
- Each supply is driven only by an output or internal PG pin on the same macro cell;
- Each supply is driven only by an output supply port on the same blackbox cell;
- Each supply is driven only by an output of a pad cell;
- Each supply is already physically connected to the others.

**What Next**

Consider using the *-function_only* argument to *set_equivalent*.

**See Also**

- set_equivalent

---

## UPF-231

(error) Supply net '%s' state '%s' voltage not defined.

**Description**

This error message occurs when when the supply state is defined but no voltage specified.

**What Next**

Specify voltage for the supply state by add_power_state -update

Feedback

**See Also**

- add_power_state

---

## UPF-232

(warning) setting related supply net for pin(s) not in current block, the constraint will be session only and will not be saved.

**Description**

When setting supply net for pin(s) not in current block but inside different physical hierarchy, the constraint is not part of current block, so it will be session only and won't be saved.

**What Next**

If the constraint needs to be saved into the block contains the pin(s), user has to open the block design view and set the related_supply_net. Re-create abstract if it's needed.

**See Also**

- set_related_supply_net

---

## UPF-233

(error) Pins specified are in different physical hierarchies.

**Description**

The error happens when some of the pins specified for set_related_supply_net command are in current block, some are in other physical hierarchy.

**What Next**

Use separated commands to set the related supply net for these pins.

**See Also**

- set_related_supply_net

---

## UPF-234

(error) MV strategies have not be associated to mv cells yet.

Feedback

### Description

The error happens when CTS doing splitting or merging operation while the MV strategies have not be associated to mv cells yet.

### What Next

Do associate_mv_cells or commit_upf before CTS.

### See Also

- associate_mv_cells
- commit_upf

---

## UPF-235

(Information) The app option mv.upf.allow_unresolved_supply is set to true but the resolved supply checks cannot be skipped since the netlist is mapped.

### Description

This information occurs when the app option mv.upf.allow_unresolved_supply is set to true for allowing unresolved supplies in the commands. But unresolved supplies are not allowed when the netlist is mapped, indicating that the command could have issued from backend flow.

### What Next

If you are sure that you are in frontend flow or you want to use unresolved supplies (deliberately knowing the consequences), you need to set the app option mv.upf.allow_unresolved_supply_in_mapped_netlist to true along with the above app option.

### See Also

- commit_upf

---

## UPF-236

(warning) Crosstool app option is set to true, UPF may not match Verilog or SPEF.

### Description

This Warning message occurs when the app option mv.upf.write_crosstool_wrappers is set to true. The app option mv.upf.write_crosstool_wrappers is needed for exporting UPF for tools that do not support the use of power domain on instance (PDOI) in UPF. With the app option set to true, instances that are PDOI are grouped under derived hierarchical

Feedback

modules to resolved compatibility issues with other tools. The derived hierarchies are only added during ASCII output and are not represented in the block. The ASCII outputs that are affected are Verilog and SPEC from write_verilog and write_parasitics commands. The app option does not impact the output of save_upf command. That means logical mismatches can exist between the UPF from save_upf command and SPEF or Verilog netlist.

To write a compatible UPF that includes the hierarchical wrappers, use the write_verilog command. When the app option is set to true, write_verilog will write a Verilog netlist and a compatible UPF at the same time. The UPF will have the same name as the Verilog file with the added extension ".SNPS.upf".

**What Next**

If you have a multi-voltage design and are allowing physical feedthrough buffers (opt.common.allow_physical_feedthroughset is true) to be added during optimization and CTS, then you should use the compatible UPF created by the write_verilog command for ASCII import to other tools.

If you are not allowing any physical feedthrough buffers during optimization and CTS, you can ignore this warning or set the app option mv.upf.write_crosstool_wrappers to false to avoid this message.

**See Also**

- write_verilog
- write_parasitics
- save_upf
- opt.common.allow_physical_feedthrough
- mv.upf.write_crosstool_wrappers

---

## UPF-237

(warning) Cannot specify -no_retention option: %s.

**Description**

This warning occurs when the retention strategy is defined with -no_retention option but a retention element list or an element of a retention element list is specified as an element of the strategy. The strategy will be discarded.

**What Next**

Check the -no_retention option is intentional.

Feedback

**See Also**

- set_retention
- set_retention_elements

---

## UPF-238

(error) Cell '%s' of retention element list '%s' is not retained when other element(s) in the element list are retained.

**Description**

This error occurs when the element of specified retention element list is not retained when other elements of the same retention element list are retained.

**What Next**

Make sure the retention strategies of the design include the specific cell.

**See Also**

- set_retention
- set_retention_elements

---

## UPF-239

(error) Cell '%s' of retention element list '%s' is not retained when other cell(s) in the same power domain '%s' are retained.

**Description**

This error occurs when the element of specified retention element list is not retained when other cells in the same power domain are retained.

**What Next**

Make sure the retention strategies of the design include the specific cell.

**See Also**

- set_retention
- set_retention_elements

Feedback

## UPF-240

(error) Element '%s' has a no_retention strategy applied to it and cannot be part of the retention element list '%s'.

### Description

This error occurs when the specified cell cannot be part of a retention element list because it is already defined as an element of a retention strategy with -no_retention option.

### What Next

Check the -no_retention option of the retention strategy is intentional.

### See Also

- set_retention
- set_retention_elements

## UPF-241

(warning) The cell '%s' is not a sequential element.

### Description

This message occurs when the cell specified in set_retention_elements command is not a sequential element and so it won't be mapped to a retention cell.

### What Next

Check whether the cell should be part of the retention element list.

### See Also

- set_retention_elements

## UPF-242

(Warning) NOR/NAND style isolation cell %s is not associated with any isolation strategies nor supply net connections.

### Description

This is a warning message indicating that either NOR or NAND style isolation cells in the design are not properly associated with any isolation strategies.

Feedback

### What Next

Although NOR/NAND isolation cell is not properly associated with any isolation strategy. But as NOR/NAND style isolation cell normally use domain primary as isolation power, this maybe not be a problem. It is user's responsibility to make sure it is used properly.

### See Also

- connect_supply_net

---

## UPF-243

(error) UPF constraints cannot be uncommitted because the block(s) is(are) not immediate child block of current top: %s

### Description

This error occurs when user is trying to uncommit block which is not immediate child of current top on MV design.

### What Next

User has to uncommit Multi-Level-Physical-Hierarchy (MPH) MV design level-by-level bottom-up.

### See Also

- uncommit_block

---

## UPF-244

(information) Skipped power state table '%s' because none of the specified supplies are defined in the table.

### Description

This message occurs when the supplies specified by users are not defined in the power state table and so the table is skipped in the report.

### What Next

Check whether the supply names are correct.

### See Also

- report_pst

Feedback

## UPF-245

(warning) set_isolation_control command specified different '%s' from set_isolation command for isolation strategy '%s'.

### Description

This message occurs when the set_isolation_control specified different isolation_signal/sense/location from set_isolation command.

### What Next

set_isolation_control command will be dropped.

### See Also

- set_isolation
- set_isolation_control

## UPF-246

(warning) Cannot add set_related_supply_net constraint for %s and all other feedthrough ports on the same path because this path is driven by a ground net!

### Description

This message occurs when tool tries to add set_related_supply_net constraints for new feedthrough ports, but the path is driven by a ground net. For such path, tool won't be able to add any set_related_supply_net for new ports on this path.

### See Also

- place_pins
- set_related_supply_net

## UPF-247

(error) Failed to write out strategy %s

### Description

This message occurs when tool tries to write out the strategy shown in the message.

### What Next

Check if the strategy is supported for the command.

Feedback

**See Also**

- split_constraints

---

## UPF-248

(error) Supply net '%s' is not visible in the power domain '%s'.

**Description**

This error message occurs when the specified supply net is not visible for the given domain. A supply net can only be visible at the scope it is defined or scopes below.

**What Next**

Please specify a supply net that is visible in the scope of given power domain.

**See Also**

- report_power_domains
- create_power_domain
- create_supply_net

---

## UPF-249

(warning) UPF constraint tracking has been disabled due to name changes.

**Description**

This is a message indicating that the UPF constraint tracking is disabled due to name changes. So, the UPF commands written out by the tool may be different from the original input UPF.

**See Also**

- save_upf

---

## UPF-250

(information) %s strategy '%s' is invalid, and will not be used to %s cell association and insertion.

**Description**

This information message appears when specified strategy is specified with incomplete information. Missing information could be power supply or control signal.

Feedback

### What Next

Fix the UPF and specify all the required information needed for a power strategy.

### See Also

- set_isolation
- set_isolation_control
- set_retention
- set_retention_control

---

## UPF-251a

(Warning) Disconnecting non-constant net '%s' and connecting net '%s' to the port/pin '%s'. This is not IEEE 1801 compliant.

### Description

The warning message is issued when a non-constant net is disconnected and a new net is connected using *connect_logic_net -reconnect*. The tool supports disconnection of non-constant net, but it does not comply with the IEEE 1801 (UPF) standard.

### What Next

This is only a warning message. No action is required.

### See Also

- connect_logic_net

---

## UPF-251b

(Error) Cannot reconnect port/pin '%s' as its connected to a non-constant net '%s'.

### Description

This error message is reported when reconnection is requested for a port/pin which is connected to a non-constant net. By default, reconnection is allowed for both constant and non-constant connections. But if upf_reconnect_non_constant_net is set to FALSE, then reconnecting a non-constant net is not allowed.

### What Next

Unset upf_reconnect_non_constant_net, to allow reconnection.

Feedback

### See Also

- connect_logic_net

---

## UPF-251

(information) Disconnecting constant net '%s' driven by constant value %d and connecting net '%s' to the port/pin '%s'.

### Description

The information messages is issued when an existing net that is a constant, is disconnected from the port/pin and a new net is connected to it using *connect_logic_net -reconnect*.

### What Next

This is just an information message. No action is required.

### See Also

- connect_logic_net

---

## UPF-252

(Information) Connecting net '%s' to dangling port/pin '%s'.

### Description

The information messages is issued when *connect_logic_net -reconnect* is used to connect a net to an undriven port/pin.

### See Also

- connect_logic_net

---

## UPF-253

(Information) Net '%s' is already connected to port/pin '%s'.

### Description

The information messages is issued when *connect_logic_net* command is used to connect the same net to the specified port/pin, that it is already connected to. It's a no-op scenario.

### What Next

This is just an information message. No action is required.

Feedback

**See Also**

- connect_logic_net

---

## UPF-254

(error) Only 2 parameters "netname" and "sense" are supported for option '%s'.

**Description**

This error message occurs when the number of parameters specified with the assert_mutex option is wrong.

**What Next**

Check the parameter and invoke the command again.

**See Also**

- set_retention
- set_retention_control

---

## UPF-255

(error) Supply set %s is not visible in the power domain %s.

**Description**

This error message occurs when the specified supply set is not visible for the specified domain. A supply set is visible only at the scope where it is defined or scopes below.

**What Next**

See the man pages for the *create_supply_set* and *create_power_domain* commands for information about creating and associating supply sets to power domains.

**See Also**

- report_power_domains
- create_power_domain
- create_supply_set

---

## UPF-256

(warning) '%s' '%s' already exists%s.

Feedback

### Description

A net/port/pin with the specified name already exists. Two nets/ports/pins cannot have the same name in the same cell.

### What Next

Specify a different name for the net/port/pin.

---

## UPF-257

(error) Value supplemental is not allowed when Golden UPF mode is off.

### Description

This error message indicates that *save_upf -format supplemental* can only be used when Golden UPF mode is active.

### What Next

Turn on Golden UPF mode by using the following setting. *set_app_options -list {mv.upf.enable_golden_upf true}*

### See Also

- mv.upf.enable_golden_upf

---

## UPF-258

(error) UPF constraint tracking has been disabled and '-format' option is not allowed.

### Description

This message occurs when '-format' option is specified and UPF constraint tracking is disabled due to name changes.

### What Next

Invoke the command without '-format' option.

### See Also

- save_upf
- UPF-249

Feedback

## UPF-259

(error) Only one element should be specified for the contains_switches design attribute

**Description**

This error message indicates that command set_design_attribute is specifying multiple elements for the contain_switches attribute. Only one element should be specified.

**What Next**

Modify the set_design_attribute command to specify only one element.

## UPF-260

(Warning) Location specified for power switch %s is not one of the root cells of its power domain. Creating this switch in its scope hierarchy.

**Description**

This warning message indicates that command set_design_attribute command has specified a location for the switch using the contains_switches attribute but the location is not one of the root cells of the power domain of the switch. The power domain of the switch is specified using the -domain argument to the create_power_switch command.

In this case, the switch cell will be created in the hierarchy representing the scope of the switch.

**What Next**

Modify the set_design_attribute command specifying the location of the switch in one of the root cells of switch's domain.

## UPF-262

(error) Resolution function '%s' defined on supply net '%s' is different from resolution function '%s' of supply net '%s'.

**Description**

This message occurs when attempting to connect the supply net to a supply port that is already connected to another supply net with different resolution function. It's an error that connected supply nets have different resolution functions.

**What Next**

Check the resolve type of the specified supply nets.

Feedback

**See Also**

- connect_supply_net
- create_supply_net

---

## UPF-263

(Information) Created place holder cell %s for upf power switch %s

**Description**

This message indicates that a placeholder cell has been created for upf power switch.

---

## UPF-264

(Information) Removed the switch place holder cell %s.

**Description**

This message indicates that a placeholder cell for power switch has been removed.

---

## UPF-265

(error) UPF constraints cannot be uncommitted because %s

**Description**

This error occurs when user is trying to uncommit block(s) but it cannot be done because of the reason listed.

**What Next**

Such kind of block cannot be uncommitted.

**See Also**

- uncommit_block

---

## UPF-266

(error) Only '-format supplemental' is allowed for save_upf in Golden UPF mode.

**Description**

This message occurs when other value is specified in '-format' option and Golden UPF mode is turned on. In Golden UPF mode, only supplemental UPF could be dumped by '-format supplemental' or by default without '-format' option specified.

Feedback

### What Next

Invoke the command with supplemental format only.

### See Also

- save_upf

---

## UPF-267

(Warning) Crosstool wrapper option is not compatible with golden UPF flow. Ignoring the golden UPF setting.

### Description

This warning occurs when the app options mv.upf.write_crosstool_wrappers and mv.upf.enable_golden_upf are both set to true. The crosstool app options modifies both the Verilog and UPF to remove power domain on instances (PDOI). The removal of PDOI may require modification of the golden UPF content, therefore the two app option are not compatible. When the crosstool app option is enabled, UPF created by write_verilog command will ignore the mv.upf.enable_golden_upf setting and will always write a UPF" to match the Verilog output.

### What Next

You will need to avoid one of these 2 conditions: 1. Use a UPF" flow instead a golden UPF flow (mv.upf.enable_golden_upf false) 2. Avoid using physical feedthroughs during optimization (opt.common.allow_physical_feedthrough false & mv.upf.write_crosstool_wrappers false)

### See Also

- mv.upf.enable_golden_upf
- opt.common.allow_physical_feedthrough
- mv.upf.write_crosstool_wrappers

---

## UPF-268

(Information) Command '%s' is disabled for Golden UPF reapplication and will be ignored.

### Description

This message occurs in Golden UPF reapplication when you load the golden upf and it has a command that is only supported on initial application, so it will not be applied again.

Feedback

### What Next

This is only an information message, no action is required.

### See Also

- mv.upf.enable_golden_upf
- create_logic_port
- create_logic_net
- connect_logic_net

---

## UPF-269

(error) Option '%s' of UPF command '%s' is not supported and will be ignored.

### Description

This error message occurs when an unsupported option is specified. Other specified options, if any, will be honored.

### What Next

Check the option of the UPF command and ensure it is not necessary for power intent. Otherwise, consider specifying the intent in another manner.

### See Also

- load_upf

---

## UPF-270

(error) The keyword 'extra_supplies' must be followed by "".

### Description

This error message occurs because it is required that the keyword *extra_supplies* be followed by *""* (two sets of double quotation marks).

### What Next

Ensure that *extra_supplies* is followed by *""* and rerun the script.

### See Also

- create_power_domain

Feedback

## UPF-271

(error) create_power_state_group is not supported when power state propagation is enabled

### Description

create_power_state_group command is disallowed when power state propagation is enabled.

### See Also

- add_power_state
- create_pst

## UPF-272

(warning) Dropping port attribute '%s' since %s.

### Description

This message occurs when a port attribute, specified through the set port attributes command, is being dropped for the given reason.

### See Also

- set_port_attributes

## UPF-273

(error) Design is not in incomplete_upf mode.

### Description

This message indicates that the design is not incomplete_upf mode. Command report_incomplete_upf will not report any information related to incomplete_upf.

### What Next

User need enable the app_option 'mv.incomplete_upf.enable" before executing the command report_incomplete_upf. And incomplete_upf mode will not be available if DCM mode is enabled with set_early_data_check_policy.

Feedback

### See Also

- report_incomplete_upf
- set_early_data_check_policy

---

## UPF-274

(error) UPF constraints have been loaded with incomplete_upf missing_va mode turned on, without reset_upf, incomplete_upf missing_va mode cannot be changed.

### Description

This message occurs when UPF constraints have been loaded with incomplete_upf missing_va mode turned on, and user tries to change the mode.

### What Next

To turn off incomplete_upf missing_va mode, user has to start over with clean netlist and UPF constraint.

### See Also

- set_app_options
- load_upf
- commit_upf
- reset_upf

---

## UPF-275

(error) The unpropagated power state %s can not be used in add_pst_state

### Description

This message occurs when the power states of the supply set are not propagated to its functions ( its nets) but they are then used in add_pst_state.

To propagate the power states of the supply state, the upf design attribute enable_state_propagation_in_add_power_state should be set to true at or above the scope of the supply set

### What Next

If the power state of a supply set is to be used in add_pst_state, set enable_state_propagation_in_add_power_state upf attribute to true at or above the scope of the supply set.

Feedback

Example:

**See Also**

- set_design_attributes
- create_supply_set
- add_pst_state

---

## UPF-276

(error) The power state %s can not have more than one supply state

**Description**

This message occurs when the power states of the supply set are propagated to its functions ( its nets) and the user tries to have more than one supply states in a power state of this supply set.

If the power states of the supply set are propagated, each power state of this supply state should not have more than one supply state.

**What Next**

If more than one supply states are desired in the power state of a supply set, then state propagation should not be enabled. Please disable it using upf design attribute enable_state_propagation_in_add_power_state

**See Also**

- set_design_attributes
- create_supply_set
- add_pst_state

---

## UPF-277

(error) Attribute value cannot be changed for %s, as this scope already has supply set states.

**Description**

This error message occurs when the design attribute enable_state_propagation_in_add_power_state cannot be set to a different value at a particular scope. This happens when the scope already contains supply set states

Feedback

or groups. Once supply set states or groups are defined at a particular scope, it is not allowed to change the value of the attribute.

**See Also**

- set_design_attributes
- add_pst_state
- create_supply_set

---

## UPF-278

(error) '%s' has an %s ('%s') with the attribute value as %s. This is not a supported flow

**Description**

This error message occurs when an invalid configuration of the design attribute enable_state_propagation_in_add_power_state is detected. The valid attribute configurations are the following - 1. Attribute is set to FALSE for the entire design. 2. Attribute is set to TRUE for the entire design. 3. Attribute is set to TRUE for the block, and set to FALSE for the top design.

**See Also**

- set_design_attributes
- add_pst_state
- create_supply_set

---

## UPF-279

(warning) UPF constraints have been loaded with incomplete_upf missing_va mode turned off, commit_upf needs to be done after incomplete_upf missing_va mode is enabled.

**Description**

This message occurs when UPF constraints have been loaded with incomplete_upf missing_va mode turned off, and user tries to change the mode.

**What Next**

Do commit_upf after enabling incomplete_upf missing_va mode.

Feedback

**See Also**

- set_app_options
- commit_upf

---

## UPF-280

(error) The %s voltage specification is not supported for %s supplies.

**Description**

This error message occurs when a negative voltage is specified for a power supply net or a non-zero value for a ground supply net.

**What Next**

Please specify a valid voltage value.

**See Also**

- add_power_state
- add_port_state

---

## UPF-281

(error) The supply set %s , with negative voltage in power or non-zero in ground, cannot be used as domain primary or in strategies.

**Description**

This error message occurs when negative voltage is specified for power or non-zero for ground function for a supply set that is being used for as a domain primary. Also, this supply set cannot be used in any of the strategies for the same reason.

**What Next**

Please specify a valid voltage value.

**See Also**

- add_power_state
- add_port_state

Feedback

## UPF-282

(error) The non-zero voltage specification is not supported for ground net %s.

**Description**

This error message occurs when non-zero voltage is specified for ground supply net.

**What Next**

Please specify a valid voltage value.

**See Also**

- add_power_state
- add_port_state

## UPF-283

(error) The %s supply net %s, with a negative voltage, is connected to the PG pin %s. And that PG pin is not one of the following: %s.

**Description**

This error message occurs when a supply net with a negative voltage is connected to a PG pin which does not meet the specified requirements.

**What Next**

Check the connections.

**See Also**

- add_power_state
- add_port_state
- connect_supply_net
- set_related_supply_net

## UPF-284

(error) The group power state %s is empty

Feedback

### Description

This message occurs when the UPF creates a group power states and does not specify any states for it.

### What Next

Specify states for it using add_power_state command. add_power_state -update can also be used if the add_power_state was already used for this group but that command did not specify any states.

### See Also

- add_power_state
- create_power_state_group

---

## UPF-285

(error) The state %s of group power state %s is empty

### Description

This message occurs when the UPF creates a group power states and then uses add_power_state to add states to it but leaves those states empty.

### What Next

Specify the logic expression of the state using the add_power_state -update command for this state.

### See Also

- add_power_state
- create_power_state_group

---

## UPF-286

(error) The state %s of supply set %s is empty

### Description

This message occurs when the UPF creates a supply set state using add_power_state but then leaves it empty.

Feedback

### What Next

Specify the supply expression of the state using the add_power_state -update command for this state.

### See Also

- add_power_state

---

## UPF-287

(error) mv.upf.upf_enable_state_propagation_in_add_power_state can not be set to true. Design already has supply set states or group states.

### Description

This message occurs when the user tries to set the app option mv.upf.upf_enable_state_propagation_in_add_power_state to true after the supply set states or group power states have already been created.

### What Next

change the flow to set this app option before any supply set states or group states are created.

### See Also

- add_power_state
- create_power_state_group

---

## UPF-288

(warning) Supply '%s' is already available in Power Domain '%s'. It will be ignored.

### Description

The supply was previously made available in the given power domain.

### What Next

No further action is required.

### See Also

- create_power_domain

Feedback

## UPF-289

(warning) The logic expression in the supply set state is ignored

### Description

This message occurs when the add_power_state is used to specify a state for a supply set and the state includes a logic_expr. The logic expression for supply sets states is ignored

### See Also

- add_power_state

## UPF-290

(Information) Usage of add_power_state -update will be made mandatory in a future release.

### Description

This message occurs when the add_power_state is used to define a supply set state for a supply set and that supply set already has other power states defined for it. -update should be used to define power states for supply set if it already has other power states

### What Next

-update option of add_power_state command should be used to define power states for supply set if it already has other power states

### See Also

- add_power_state

## UPF-291

(Error) Illegal power state %s cannot be used in a legal state.

### Description

This message occurs when an illegal power state is used in a legal state. A legal state is one that is defined either by using the add_pst_state command or by using the add_power_state command without the -illegal option.

### What Next

Rewrite the add_pst_state or add_power_state command without using states that are defined as illegal.

Feedback

### See Also

- add_power_state

---

## UPF-292

(Error) All state of group/domain %s are declared illegal.

### Description

This message occurs when the add_power_state is used to define a power state for a group/domain and all the states are defined with the option illegal. All states of group/domain can not be declared illegal.

### What Next

Rewrite the add_power_state command without declaring all state of a group/domain illegal.

### See Also

- add_power_state

---

## UPF-293

(Error) Power domain '%s' has already been defined in the scope %s

### Description

This message occurs when the UPF is trying to create a power state group and the name of the power state group is same as an existing power domain. Power state groups and power domains can not have same name.

### What Next

Rewrite the UPF so that power state groups and power domains do not have conflicting names.

### See Also

- create_power_state_group
- create_power_domain

---

## UPF-294

(Error) Ambiguous supply name %s. %s is both a power domain supply handle and a supply set

Feedback

### Description

This message occurs when the add_power_state command is used to define a power state for a power domain. If the logic expression uses a supply name that is the supply handle of the power domain and a declared supply set, then we can not be sure if the command is referring to the power domain supply handle or supply set name.

For Example, in the UPF below, we can not tell whether add_power_state command is referring to my_handle the supply_set or my_handle the supply handle of PD. In this case we will get this error.

```
create_supply_set my_handle
create_power_domain PD -supply {my_handle}
add_power_state -domain PD -state s1 {-logic_expr {my_handle == .....
```

### What Next

Rewrite the UPF so that power state groups and power domains do not have conflicting names.

### See Also

- create_power_state_group
- create_power_domain

---

## UPF-295

(Error) Found %d bias rail order violations.

### Description

This is a summary message stating the number of bias rail order violations found. A bias rail order violation occurs when the bias supplies of a design element are less always-on than its power/ground supplies.

### What Next

Fix the sources of any UPF-297 or UPF-298 errors.

### See Also

- UPF-297
- UPF-298

---

## UPF-296

(Error) simstate update for %s is not correct.

Feedback

### Description

This message occurs when the add_power_state command is used to update the simstate for a supply set power state. When add_power_state -update is used to specify a simstate, only following changes are supported.

1. If the simstate was previously not defined for a state, then -update may be used to specify any type of simstate. 1. If the simstate was previously defined as NOT_NORMAL, then -update may be used to specify and simstate other than NORMAL.

all other changes to the simstate of a state are not supported.

### What Next

Modify the UPF to follow the rules above.

### See Also

- add_power_state
- create_power_domain

---

## UPF-297

(Error) %s bias function of power domain %s is less-on than the power/ground supply of the %s %s strategy.

### Description

Tool requires that bias supplies of the power domain where the reported strategy would be implemented have to be ON when the power and ground supplies of the reported strategy are ON, unless library cells with insulated wells are provided.

This error message is issued when the bias supplies of a power domain do not meet the above condition and no library cells with insulated wells are found in the map_isolation_cell (or map_retention_cell) command instance associated to the reported isolation (or retention) strategy

### What Next

Provide insulated cells on the map_isolation_cells (or map_retention_cell) command associated to the reported strategy.

### See Also

- set_isolation
- map_isolation_cell

Feedback

- map_retention_cell
- set_retention
- create_supply_set
- add_power_state

## UPF-298

(Error) Supply set %s's bias %s function is less always-on than its power/ground function.

**Description**

This error message appears when a bias rail order violation is found between a supply set's functions. The nWell(pWell) function of a supply set is required to be ON when both power and ground are ON.

**What Next**

Review the UPF and remove these violations.

**See Also**

- create_supply_set
- add_power_state

## UPF-299

(Information) Created power switch cell %s for upf power switch %s

**Description**

This message indicates that a power switch cell has been created for upf power switch.

## UPF-300

(Information) -repeater_supply option is deprecated in the standard. Consider using set_repeater.

**Description**

This information message occurs when you run *set_port_attributes -repeater_supply* command. *-repeater_supply* is deprecated in the standard. But tool will still honor *-repeater_supply*. For better compliance with the standard, you may want to migrate all *set_port_attributes -repeater_supply* usages to *set_repeater*.

Feedback

### What Next

This is an information message. If you don't want to see this message, use *set_repeater* instead of *set_port_attributes -repeater_supply*.

### See Also

- set_port_attributes
- set_repeater

---

## UPF-301

(Error) Found domain boundary port/pin %s with both repeater strategy %s and repeater_supply port attribute setting.

### Description

This error message occurs when the port/pin has both repeater strategy and repeater_supply port attribute applicable to it. Application is checked irrespective of whether it is on HighConn or LowConn side and irrespective of the granularity of the strategy/attribute.

In the following example, commit_upf will show error because mid/in has both repeater_supply attribute and repeater strategy setting on it:

```
prompt> create_power_domain PD_MID \
                -elements {mid}
prompt> set_port_attributes -ports mid/in \
                -repeater_supply SS1
prompt> set_repeater RP -domain PD_MID \
                -repeater_supply SS1
```

In the following example, commit_upf will show error because mid/in has both repeater_supply attribute and repeater strategy setting on it:

```
prompt> set_design_attributes -element {.} \
               -attribute lower_domain_boundary TRUE
prompt> create_power_domain PD_TOP
prompt> create_power_domain PD_MID \
                -elements {mid}
prompt> set_port_attributes -ports mid/in \
                -repeater_supply SS1
prompt> set_repeater RP -domain PD_TOP \
                -elements {mid} \
                -repeater_supply SS1 \
                -applies_to output
```

Feedback

### What Next

Make sure port/pin has only one of the settings.

### See Also

- set_port_attributes
- set_repeater

---

## UPF-302

(Warning) The specified vct name '%s' already exists.

### Description

This warning message occurs when the specified vct name is predefined or has been used by other user vcts.

### What Next

Ensure that a new and valid *vct name* is specified.

### See Also

- create_upf2hdl_vct
- create_hdl2upf_vct

---

## UPF-303

(error) The specified vct '%s' doesn't exist.

### Description

This error message occurs when the specified vct is neither predefined nor user created vct.

### What Next

Ensure that a valid existing vct is specified.

### See Also

- create_upf2hdl_vct
- create_hdl2upf_vct
- connect_supply_net

Feedback

## UPF-304

(error) Incorrect -hdl_type specification for the commands *create_upf2hdl_vct* and *create_hdl2upf_vct*.

### Description

The error message issues when tool encounters an illegal specification for *-hdl_type*.

### What Next

Check the man page of *create_upf2hdl_vct* and *create_hdl2upf_vct*.

### See Also

- create_upf2hdl_vct
- create_hdl2upf_vct

## UPF-305

(warning) UPF for this ETM has been loaded and applied to all instantiations already. Ignore this UPF file!

### Description

This message occurs when ETM UPF has been loaded before for instance of same ETM cell, but user tried to load ETM UPF again.

### What Next

Skip loading this UPF.

### See Also

- load_upf

## UPF-306

(Warning) State %s of power state group or domain %s is not reachable due to conflicts.

### Description

This message indicates the upf has specified a power state using add_power_state -group/-domain that is not reachable. This can happen when reaching the that state requires a supply net or connected supply nets to be at different voltages or on/off states. Consider the example below

Feedback

```
add_power_state SS1 -state SS1_1 {-supply_expr {power == {FULL_ON 1.0} &&
 ground == {OFF}}}
add_power_state SS2 -state SS2_1 {-supply_expr {power == {FULL_ON 1.0} &&
 ground == {FULL_ON 0}}}

create_supply_set SS1 -function {ground SS2.ground} -update

add_power_state -domain TOP -state TOP_S1 { -logic_expr {SS1 == SS1_1 &&
 SS2 == SS2_1}}
```

The state TOP_S1 of the domain TOP is not reachable because it will require SS1.ground to be OFF and SS2.ground to be at 0 volts. This is not possible because SS2.ground and SS1.ground are same net because of the create_supply_set -update command given.

---

## UPF-307

(Warning) State %s of power state group or domain %s is not reachable due to conflicts with other power state groups and/or psts.

### Description

This message indicates the upf has specified a power state using add_power_state -group/-domain that is not reachable. This can happen when the UPF has other power state groups and or psts defined and another group or pst or their combination is incompatible with this state.

In the example below, the state TOP_S1 of TOP is in conflict with the group G. Group G requires that SS1 be at SS1_0 all the times because in all of its states, SS1 is set to SS1_0.

However states G_S1 and G_S2 are reachable because TOP has a state (TOP_S2), that allows SS1 to be in any state.

By similar logic, state G_S2 is in conflict with the domain state TOP.

```
add_power_state SS1 -state SS1_1 {-supply_expr {power == {FULL_ON 1.0} &&
 ground == {OFF}}}
add_power_state SS1 -state SS1_0 {-supply_expr {power == {FULL_ON 1.2} &&
 ground == {OFF}}}
add_power_state SS2 -state SS2_1 {-supply_expr {power == {FULL_ON 1.0} &&
 ground == {FULL_ON 0}}}
add_power_state SS2 -state SS2_2 {-supply_expr {power == {FULL_ON 1.5} &&
 ground == {FULL_ON 0}}}


add_power_state -domain TOP -state TOP_S1 { -logic_expr {SS1 == SS1_1 &&
 SS2 == SS2_1}} \
                             -state TOP_S2 {-logic_expr { SS2 == SS2_1}
create_power_state_group G
```

Feedback

```
add_power_state G -state G_S1 { -logic_expr {SS1 == SS1_0 && SS2 ==
 SS2_1}} \
                  -state G_S2 { -logic_expr {SS1 == SS1_0 && SS2 ==
 SS2_2}
```

The state TOP_S1 of the domain TOP is not reachable because it will require SS1.ground to be OFF and SS2.ground to be at 0 volts. This is not possible because SS2.ground and SS1.ground are same net because of the create_supply_set -update command given.

---

## UPF-308

(Warning) Illegal states are not supported and will not be honored during the creation of system PST.

### Description

This warning message is to inform user that tool supports only the syntax of -illegal option but not its semantics. i.e. Illegal states will not be explicitly discarded from system PST. However, the illegal states will be written out in UPF'.

### What Next

Make sure that the legal states are defined such that they never include states which are marked illegal.

### See Also

- add_power_state

---

## UPF-309

(Warning) State %s of pst %s is not reachable due to conflicts with other power state groups and/or psts or some of the power states were reconciled.

### Description

This message indicates the upf has specified a pst state using add_pst_state that is not reachable. This can happen when the UPF has other power state groups and or psts defined and another group or pst or their combination is incompatible with this state. If the design attributes upf_reconciliation_automatic or upf_reconcile_boundary were set, some power states may have been modified leaving this state unreachable.

---

## UPF-310

(warning) Strategy '%s' has same precedence as strategy '%s' of same power domain '%s', and can't be resolved on power domain boundary port '%s'.

Feedback

### Description

This error message occurs when a strategy defined has same precedence as another same type strategy from same power domain, and tool can't resolve which one to use.

### What Next

Change either strategy and make strategy with different precedence.

### See Also

- set_isolation
- set_isolation_control
- set_level_shifter
- set_repeater
- commit_upf
- associate_mv_cells
- create_mv_cells

---

## UPF-311

(warning) Isolation strategy '%s' can't be implemented on power domain boundary port '%s' with location '%s' to isolate '%s'.

### Description

This error message occurs when a strategy defined can't implemented due of location of the strategy.

### What Next

Change either strategy with different location, or add another strategy with different location for the load.

### See Also

- set_isolation
- set_isolation_control
- create_mv_cells

Feedback

## UPF-312

(information) Repeater strategy %s belonging to power domain %s is ignored for implementation.

### Description

This information message is issued when the repeater strategy doesn't have a valid element list or the repeater supply is not visible inside the domain. The strategy will not be used for repeater cell association and insertion.

### What Next

Fix the repeater strategy and run the command again.

### See Also

- set_repeater

## UPF-313

(information) Domain %s will be treated as an Always-On Block Synthesis Domain, since the main_rail supply is specified.

### Description

This message indicates that the given domain will be treated as an Always-On Block Synthesis Domain, meaning it will be synthesized with dual-rail cells with primary PG pin connected to the main_rail supply and backup PG pin connected to the primary supply of the domain.

### What Next

This is an information-only message. No action is required. If user intention was to only create a regular user supply set of name 'main_rail', please change the name to something else. "main_rail" keyword is reserved for AO Block Synthesis Domains.

### See Also

- create_power_domain

## UPF-314

(warning) Multiple drivers on unresolved supply net '%s'

Feedback

### Description

This message occurs when multiple power sources are connected to a net specified as unresolved with the create_supply_net command -resolve option.

### What Next

Supply nets specified as unresolved may receive only one driver. Either disconnect other sources or specify the supply net as parallel.

### See Also

- create_supply_net

---

## UPF-315

(error) Update of tool-derived strategy %s of domain %s is not allowed.

### Description

This error is issued when you attempt to update a tool derived isolation, retention or level-shifter strategy. Updating the UPF constraint that was generated by the tool is not allowed.

### What Next

Create a new strategy instead of updating the existing tool-derived strategy.

### See Also

- set_isolation
- set_retention
- set_level_shifter

---

## UPF-316

(warning) Repeater strategy '%s' cannot be applied to the object '%s'.

### Description

This warning message occurs when a repeater strategy is specified on macro cell, which requires the tool to insert the repeater cells inside macro cell.

### What Next

This is just a warning message. No action is required.

Feedback

**See Also**

- set_repeater

---

## UPF-317

(error) Detected circular references to the file %s while loading the UPF file %s.

**Description**

This error message occurs when you load a UPF file that references itself. This results in a circular reference, which is illegal. An example of a circular reference is as follows :-

File 1.upf: load_upf 2.upf

File 2.upf load_upf 2.upf

**What Next**

Remove the load_upf command that causes the circular reference.

**See Also**

- load_upf

---

## UPF-318

(error) UPF for this ETM has been loaded already and cannot be loaded again.

**Description**

This message occurs when UPF for this ETM has been loaded, but user tried to load UPF again.

**What Next**

User has to do reset_upf if different ETM UPF file has to be loaded.

**See Also**

- load_upf
- reset_upf

---

## UPF-319

(error) Supply state '%s' does not exist on supply '%s'. It has not been propagated from supply set.

Feedback

### Description

This error occurs when the user tries to use a non existent supply state in add_pst_state command. The state name is present on one of the connected supply sets but has not been propagated to the supply nets and ports. The propagation of the power states from the supply sets to connected supply nets and ports can be enabled using the design attribute enable_state_propagation_in_add_power_state on the scope of the supply set.

### What Next

Please enable the propagation of the power states from supply sets to connected supply nets and ports using the design attribute enable_state_propagation_in_add_power_state.

### See Also

- set_design_attributes

---

## UPF-321

(error) Associated supply sets are resolved to different supply nets %s(%s) and %s(%s) in same scope.

### Description

For associated supply sets in the same scope, they must be resolved to the same explicit supply net if resolved.

### What Next

Ensure associated supply sets in the same scope are resolved to the same explicit supply net.

### See Also

- create_supply_set
- create_power_domain

---

## UPF-322

(error) Supply sets cannot be associated because there is no supply set in common ancestor scope.

### Description

For associated supply sets, the supply set list must contain a supply set which is defined in their common ancester scope.

Feedback

### What Next

Ensure associated supply sets contains a supply set in their common ancestor scope.

### See Also

- create_supply_set

---

## UPF-323

(error) The state %s of supply set %s doesn't have a -supply_expr defined, when this supply state is used in group %s.

### Description

This message occurs when a supply set state is used in a group, and it doesn't have a -supply_expr defined. E.g., it just has -simstate or just has -logic_expr.

### What Next

Specify the supply expression of the state using the add_power_state -update command for this state.

### See Also

- add_power_state

---

## UPF-324

(warning) The app_option "mv.upf.enable_missing_voltage_area" will not take effect until incomplete_upf mode is turned on".

### Description

This message occurs when "mv.incomplete_upf.enable" is off and "mv.upf.enable_missing_voltage_area" is set.

### What Next

Turn on "mv.incomplete_upf.enable" to make it work

### See Also

- set_app_options
- mv.incomplete_upf.enable

Feedback

## UPF-325

(error) The value "auto" of app_option "mv.incomplete_upf.missing_va" cannot be set, when app_option "mv.upf.enable_missing_voltage_area" is enabled.

### Description

This message occurs when "mv.upf.enable_missing_voltage_area" is on, and the value "auto" is used to set the app_option "mv.incomplete_upf.missing_va".

### What Next

Turn off or don't set "mv.upf.enable_missing_voltage_area" and set "mv.incomplete_upf.missing_va" again

### See Also

- set_app_options
- mv.incomplete_upf.missing_va

## UPF-326

(error) The option "-force_bias_connections" is not supported in bias mode or designs.

### Description

This message occurs when running connect_pg_net command with option "-force_bias_connections" and the design attribute "enable_bias" is TRUE, The option "-force_bias_connections" is used to connect nwell and pwell pg pins to the domain primary power and ground respectively, it is not allowed in bias pin connection support design.

### What Next

Check if the design is supporting bias and there is bias supplies in the UPF.

Use 'set_design_attributes -attribute enable_bias' to modify the biasing attributes in the UPF to adhere to the requirements.

### See Also

- connect_pg_net
- set_design_attributes

Feedback

## UPF-327

(warning) pst_state %s of power state table %s defined is ignored by implementation because supply %s and %s have voltage conflict in this pst_state, but they become equivalent because both are connected to same supply %s outside the block.

### Description

This warning message occurs when a pst_state defined in PST from linked block has equivalent supplies with voltage conflict, but these supplies become equivalent because they are connected to the same supply outside the block.

### What Next

Make sure these supplies are made equivalent outside the block intentionally.

### See Also

- add_pst_state
- connect_supply_net

## UPF-328

(Error) Reference-only supply %s cannot be used in this option %s of the command %s.

### Description

This error message appears when a reference-only supply net or supply set is specified in the option that doesn't support for reference-only supplies.

Reference-only supply is marked by: set_design_attribute -element {.} -attribute reference_only {supply_net_or_supply_set}

### What Next

Rewrite the UPF to avoid using reference-only supply in the option.

## UPF-329

(information) UPF constraints in database contains non-standard hierarchical name(s), and app option "mv.upf.input_enforce_simple_names" is automatically set to false for this design.

### Description

Based on IEEE Std 1801, some UPF command options require a simple name with no hierarchy separator character. This is an informational message that

Feedback

some hierarchical names in the UPF constraints of the database deviate from IEEE Std 1801 recommendation. The tools have automatically set the app option "mv.upf.input_enforce_simple_names" to false for backward compatibility.

**What Next**

Review the UPF constraints of the design and consider to change the hierarchical names to simple names.

**See Also**

- mv.upf.input_enforce_simple_names

---

## UPF-330

(information) Ignore internal design attribute '%s': %s '%s' does not exist.

**Description**

This message occurs when the object name specified in the internal attribute does not exist.

**What Next**

This message is informational. No action is required.

---

## UPF-331

(warning) Supply set '%s' does not exist and will be ignored.

**Description**

This message occurs when specified supply set in associate_supply_set command does not exist.

**What Next**

Check the supply set name is valid.

**See Also**

- associate_supply_set

---

## UPF-335

(error) Retention cell insertion is not supported in AO-Block Synthesis domain %s.

Feedback

### Description

The domain for which this map_retention strategy is specified is an AO-Block Synthesis Domain and retention cell insertion is not supported in such domains.

### See Also

- create_power_domain

---

## UPF-336

(Warning) Name space conflict with %s type object '%s' while create %s object.

### Description

This warning message occurs when an UPF command creates an UPF object or a design object with the name of an existing UPF object or design object, for the current design.

The warning could happen for below cases and the object creation is skipped:

- A logic net with the same name as a logic net
- A logic port with the same name as a logic port

The warning also happens when opening an old block which contains name conflict according to the new rules, but for backward compatibility consideration, we report warning only.

### What Next

This is only a warning message. No action is required. For backward compatibility, you can turn off mv.upf.object_namespace_checks to skip this name space check. Nevertheless, is advised to avoid name space conflicts altogether by renaming affected UPF or design objects.

### See Also

- UPF-337
- mv.upf.object_namespace_checks

---

## UPF-337

(Error) Name space conflict with %s type object '%s' while create %s object.

### Description

This error message occurs when an UPF command creates an UPF object or a design object with the name of an existing UPF object or design object, for the current design.

Feedback

There are two allowed exceptions:

- A logic or supply net with the same name as a logic or supply port
- A logic or supply port with the same name as a logic or supply net

Normally this condition is an error and the object creation is aborted. By setting app option *mv.upf.object_namespace_checks* to false, the complete name space conflict check is turned off.

**What Next**

Ensure that objects specified in your UPF file are not identical to the existing objects in the given name space.

---

## UPF-338

(error) Invalid '%s' name specified: '%s'.

**Description**

There are some keywords that are predefined in VCS-NLP: UNDEFINED_STATE, INVALID, illegal_state, DEFAULT_NORMAL, DEFAULT_CORRUPT.

1. The port state of any port cannot have the name UNDEFINED_STATE. 2. Power switch cannot have a state with the name INVALID. 3. PST cannot have a state with the name illegal_state. 4. Supply set cannot have a power state with the name DEFAULT_NORMAL or DEFAULT_CORRUPT.

**What Next**

Change the state name to avoid these keywords.

**See Also**

- add_port_state
- create_power_switch
- add_pst_state
- add_power_state

---

## UPF-339

(error) Cannot find supply pin %s in cell %s.

Feedback

### Description

This error message occurs when the connect_supply_net attempts to connect a specific supply pin of a given library cell to a supply net. In this case, the pin must exist in the interface.

### What Next

Check the interface of the library cell and correct the pin name.

### See Also

- connect_supply_net

---

## UPF-340

(warning) Command '%s' executed on ETM or power model scope is skipped.

### Description

This message occurs when specified command is skipped when it is executed on ETM or power model scope.

### What Next

Check the command is valid on ETM or power model scope.

### See Also

- set_scope

---

## UPF-341

(warning) set_scope .. is not supported nested inside ETM or power model.

### Description

This message occurs when set_scope .. is executed when the current scope is nested under ETM or power model.

### What Next

Use full hierarchical name starting with "/" instead of ".."

### See Also

- set_scope

Feedback

## UPF-342

(error) The app_option "mv.upf.upf_add_power_state_21_syntax" can not be set if power states have already been created

### Description

The app option mv.upf.upf_add_power_state_21_syntax can not be set after power states are created. Power states are created using add_power_state commands.

### What Next

Please set mv.upf.upf_add_power_state_21_syntax before any power states are created.

### See Also

- set_app_options
- add_power_state

## UPF-343

(warning) That UPF command '%s' is inputted through interactive shell will not be preserved "as-is" by save_upf.

### Description

This warning message occurs when the UPF command is inputted through interactive shell, not by load_upf. Now the implementation tools allow all UPF commands to be loaded outside load_upf. However, when it's interactively inputted, save_upf won't preserve the command as what it is.

### What Next

It's just a warning message. If it's not what you want, reload the command by load_upf.

### See Also

- load_upf

## UPF-345

(error) Power switch state '%s' is used more than once for power switch strategy %s.

Feedback

### Description

This error occurs when the same state name is specified multiple times using the -on_state or -error_state or -off_state or -on_partial_state options of create_power_switch command. State name should be unique in this power switch strategy definition.

### What Next

Ensure unique state name is used in the same power switch.

### See Also

- create_power_switch

---

## UPF-346

(error) Incorrect "set_port_attributes -attribute function" setting because: %s

### Description

This error message is shown when there is a non sintatic issue with a usage instance of the "-attribute function" option of the "set_port_attributes" command.

### What Next

Please review any "set_port_attributes -attribute function" settings involving the objects referenced in the error message

### See Also

- set_port_attributes

---

## UPF-347

(information) Implementation tool will read and ignore this command.

### Description

This command is only applicable for simulation and does not have impact on implementation tool. It won't be preserved by save_upf as well.

### What Next

Just an information message.

Feedback

**See Also**

- sim_corruption_control
- sim_assertion_control
- sim_replay_control
- set_simstate_behavior

## UPF-348

(warning) Specified %s supply does not exist. Related %s supply will be determined from the related PG pin found in the reference library model.

**Description**

This message occurs when a non-exist supply is used by set_related_supply_net in ETM or power model scope. If the specified supply cannot be found, then the related supply will be determined from the related PG pin found in the reference library model.

## UPF-349

(error) The specified use_interface_cell strategy %s already exists in the power domain %s.

**Description**

This message occurs when the specified use_interface_cell strategy already exists. Please check the existing use_interface_cell command(s), or use another use_interface_cell strategy name.

**See Also**

- use_interface_cell

## UPF-350

(warning) Using both use_interface_cell and map_isolation_cell/map_level_shifter_cell commands are not allowed in a single power domain. The tool only uses the specification from use_interface_cell.

**Description**

This message occurs when both use_interface_cell and map_isolation_cell/ map_level_shifter_cell specified for a single power domain. Only the specification from use_interface_cell is used.

Feedback

### See Also

- use_interface_cell

---

## UPF-351

(Error) Find error in the option -strategy. %s

### Description

Only one isolation strategy, one level-shifter strategy, or a pair of isolation and level-shifter strategies can be used in the option -strategy. This message occurs when more than one isolations strategy, more than one level-shifter strategy, or none strategy is specified in the option -strategy.

### What Next

Please fix the -strategy option accordingly.

### See Also

- use_interface_cell

---

## UPF-352

(error) The syntax used for add_power_state -state is inconsistent with previous syntax used for object %s

### Description

This error occurs when the add_power_state was used to define a power state for the object but the syntax used for the -state option is not consistent with the previously define power_states for the object. For example, this error will occur if first add_power_state used 2.0 syntax to define -state for supply set SS1 and then the next add_power_state command used 2.1 syntax

```
#this command uses 2.0 syntax for -state
add_power_state -supply SS1 -state S1 {-supply_expr {power == {OFF}}}

#this command uses 2.1 syntax for -state and will cause UPF-352 error.
add_power_state -supply SS1 -state {S0 -supply_expr {power == {OFF}}}
```

### What Next

Please make sure all add_power_state commands for the object in question uses the consistent syntax for -state option ( either user 2.0 syntax everywhere or 2.1 syntax everywhere)

Feedback

### See Also

- add_power_state

---

## UPF-353

(error) The syntax used for -state must be consistent throughout the command. Mixing syntax in the same command is not allowed.

### Description

This error happens when the add_power_state command has multiple -state argument and the syntax used to specify -state is not consistent throughout the command. This error happens if some -state arguments are using UPF 2.0 syntax (i.e. -state has two value, one for state name and the other for state definition) and some -state arguments are using UPF 2.1 syntax (i.e. -state has only value combining state name and state definition in one string)

For example, in the following command, state S1 is specified using UPF 2.0 syntax and S2 is specified using UPF 2.1 syntax. This is not supported.

```
add_power_state SS1 -state S1 {-supply_expr {power == {FULL_ON, 1}}} \\
   -state {S2 -supply_expr {power == {FULL_ON, 0.9}}}
```

### What Next

Please modify the add_power_state command to make sure -state uses either UPF 2.1 or UPF 2.0 consistently throughout the command.

### See Also

- add_power_state

---

## UPF-354

(error) Supply net reconnection failed because existing supply net on pin already has power states declared

### Description

This error happens if the connect_supply_net command is trying to reconnect a new net to a pin. If the existing net connected to the pin already has power/port states declared on it ( or if something connected to it has power/port states declared on it), then the reconnection is not allowed

Feedback

### What Next

Please modify the UPF to put the connect_supply_net command before any power states or ports states are added to the existing net of the port or pin where reconnection is being attempted.

### See Also

- add_power_state

---

## UPF-355

(Information) The lower_domain_boundary design attribute will be made obsolete in future. Consider using -applies_to_boundary option of set_isolation/set_level_shifter/set_repeater strategy commands.

### Description

This information message is issued when user specifies lower_domain_boundary design attribute to make isolation/level-shifter strategies apply to the lower boundary of the power domain. This attribute will soon be made obsolete and it is suggested that the -applies_to_boundary option of set_level_shifter/set_isolation/set_repeater commands be used instead. If the -applies_to_boundary option is not specified, strategies will by default only apply to the upper domain boundary. For strategies to apply on the lower domain boundary or on both boundaries, -applies_to_boundary can be set to lower and both respectively.

### What Next

Please make use of the -applies_to_boundary option of strategy commands.

### See Also

- set_isolation
- set_level_shifter
- set_repeater

---

## UPF-356

(Error) State %s of power state group or domain %s is not reachable due to conflicts within it.

### Description

This message indicates the upf has specified a power state using add_power_state -group/-domain that is not reachable due to conflicts within this state. this can happen

Feedback

if the state can only be reached by putting same supply two different supplies that are connected together in two different states.

```
add_power_state SS1 -state SS1_1 {-supply_expr {power == {FULL_ON 1.0} &&
 ground == {OFF}}}
add_power_state SS2 -state SS2_1 {-supply_expr {power == {FULL_ON 1.0} &&
 ground == {FULL_ON 0}}}

create_supply_set SS1 -function {ground SS2.ground} -update

add_power_state -domain TOP -state TOP_S1 { -logic_expr {SS1 == SS1_1 &&
 SS2 == SS2_1}}
```

The state TOP_S1 of the domain TOP is not reachable because it will require SS1.ground to be OFF and SS2.ground to be at 0 volts. This is not possible because SS2.ground and SS1.ground are same net because of the create_supply_set -update command given.

---

## UPF-357

(Error) State %s of pst %s is not reachable due to conflicts within it

### Description

This message indicates that a pst state is not possible because it specifies conflicting states for a supply. This can happen if pst state supplies include connected nets or ports and it then specifies conflicting states for them.

for example imagine a pst supplies include supply port SP1 and SP2 and SP1 and SP2 are connected together. If the PST state specifies that SP1 should be at 0.7 volt and SP2 should be at 0.9 volt, then it is a conflict. This means that this pst state is not possible.

### What Next

Please see if the conflict is due to a mistake in UPF. If not then please remove the conflicting pst state.

---

## UPF-358

(error) The cell '%s' has no valid power domain. This cell is being excluded in the power domain: '%s'.

### Description

This error message occurs when the specified cell is on the *-exclude_elements* list of a power domain and is left with no valid power domain. A cell being excluded from a power domain will have the domain of the parent hierarchy, if the cell is being excluded from that domain too or there is no parent hierarchy domain, the cell will have no domain. The

Feedback

cell with no domain can be specified as a root cell of another domain which is not being excluded for it to have a domain.

### What Next

Please include the specified cell in a power domain with *-elements* option. Here is an example:

```
prompt> create_power_domain PD_TOP -elements {.} -exclude_elements {u1}
prompt> create_power_domain PD_MID -elements {u1}
```

### See Also

- create_power_domain

---

## UPF-360

(error) Power domain '%s' has no elements.

### Description

This error message occurs when you are trying to create a power domain which is not the top-level power domain of a scope and the power domain does not have any elements specified. Only the top-level power domain of a scope can have no elements specified.

### What Next

Please update the power domain with -elements and -update options. Here is an example:

```
prompt> create_power_domain PD_TOP -include_scope
prompt> create_power_domain PD_MID -elements { }
prompt> set_level_shifter LS_STG -domain PD_MID -elements {u1 u2}
prompt> create_power_domain PD_MID -elements {u1 u2} -update
```

### See Also

- create_power_domain

---

## UPF-361

(warning) %s '%s' does not exist.

### Description

The specified object does not exist.

Feedback

### What Next

Understand why the specified object does not exist, create if required and rerun the command again.

### See Also

- get_objects_by_location
- query_objects

---

## UPF-362

(error) None of library cell specified in %s of %s strategy '%s' is legal.

### Description

This error message occurs when all library cells specified for isolation or level-shifter strategy are illegal.

### See Also

- use_interface_cell
- map_isolation_cell
- map_level_shifter_cell

---

## UPF-363

(warning) %s is not on the %s boundary of domain %s, and will be dropped.

### Description

This warning message occurs when given element is not on the upper (lower) domain boundary of the specified domain, and when -applies_to_boundary lower (upper) is specified in the strategy.

### What Next

Check the applies_to_boundary option of the specified strategy.

### See Also

- set_level_shifter
- set_isolation
- set_repeater

Feedback

## UPF-364

(Information) Power State %s defined using %s syntax for -state.

### Description

This information message notifies the user about the syntax used for -state in the add_power_state command.

### See Also

- add_power_state

## UPF-365

(Warning) The %s library cell setting, by the previous use_interface_cell %s command, is replaced by the current use_interface_cell %s command.

### Description

The message occurs when the current use_interface_cell command is trying to set up library cells on an isolation or a level-shifter strategy, but this strategy already has library cell setting by a previous use_interface_cell command. The current use_interface_cell command is overwriting the previous one.

### What Next

This is only a warning message.

### See Also

- use_interface_cell

## UPF-366

(warning) The use of the '%c' to match the hierarchy separator is only allowed with glob type matching; it is not allowed with -regexp.

### Description

This warning message occurs when you are specifying a matching pattern with hierarchical separator in the find_objects command with -regexp option.

### What Next

Check whether the matching pattern is correct or not.

Feedback

**See Also**

- find_objects

---

## UPF-367

(error) UPF cannot be saved because this is not a MV design.

**Description**

This error occurs when user tries to save_upf with -for_empty_blackbox on a non-MV design.

**What Next**

Load and commit UPF for the design.

**See Also**

- load_upf
- commit_upf

---

## UPF-369

(warning) Element '%s' of isolation strategy '%s' in power domain '%s' is connected with strategy control signal '%s', and will be dropped.

**Description**

This warning message occurs when the element defined in an isolation strategy will also be used as control signal in the same strategy. The element will be dropped from the isolation strategy.

**What Next**

Review the issued strategy for the self dependency definition.

**See Also**

- set_isolation
- set_isolation_control

---

## UPF-370

(warning) Isolation supply of strategy %s in domain %s cannot be made available to power domain %s.

Feedback

### Description

This warning message occurs when the specified isolation supply net cannot be made available for the domain where isolation cell will be inserted. Isolation cells will not be inserted with this warning message.

### What Next

First, ensure that the supply specified is the intended supply for use. If the intention is correct, use the *report_power_domains* command to see if the specified supply is available in the power domain. If the supply is not available, you must modify the UPF. See the man pages for the *create_supply_net* and *create_power_domain* commands for information about creating and associating supply nets to power domains.

### See Also

- report_power_domains
- set_isolation
- set_isolation_control

---

## UPF-371

(information) default_pst (one pst_state with all possible port_state/power_state for all supplies) has been applied to the design since there is no PST/power_state_group created in top scope.

### Description

When there is no PST/power_state_group created in the top scope, tool will automatically apply default_pst (one pst_state with all possible port_state/power_state for all supplies) to the design.

### See Also

- create_pst
- commit_upf

---

## UPF-372

(warning) Design has no PST or power_state_group in the top scope, and default_pst generation is disabled. Optimization can possibly create MV violations that cannot be detected without a proper PST.

Feedback

### Description

When design has no PST or power_state_group in the top scope and mv.upf.apply_default_pst has been set to be false, tool won't apply default_pst (one pst_state with all possible port_state/power_state for all supplies) to the design and will assume there is no MV violation (voltage, on-off/off-on) between any pair of supplies regardless of whether any port/power state is defined or not, and regardless of any PST/power_state_group defined in sub-scope or not. This means check_mv_design will not be able to detect any MV violation, buffering engine will assume all paths have no MV violation, and all supplies will be 'pst-equivalent'.

### What Next

Create proper PST or power_state_group in the top scope, or set mv.upf.apply_default_pst to be true, so tool can apply default_pst to the design

### See Also

- create_pst
- commit_upf

---

## UPF-373

(Error) Failed to set main_rail supply on domain '%s' since its always-on strategy is single_power.

### Description

When the always-on strategy has been set on a power domain as single_power, that power domain can't be marked as AO-Block Domain.

### What Next

You can change the power domain's always-on strategy to dual_power before you update the power domain as AO-Block Domain.

### See Also

- create_power_domain
- set_power_domain_constraints

---

## UPF-374

(warning) Net '%s' is connected to analog pin '%s'. The net will be derived as analog net.

Feedback

### Description

This message occurs when a net is connected to an analog pin. The net will be derived as analog net after this connection.

### What Next

Check the name and type of net or connected pin.

### See Also

- connect_logic_net
- connect_power_switch

---

## UPF-375

(error) Can't %s%s in %s because it violates the power intent of power model '%s'.

### Description

User tries to run a UPF command in the specified scope to which a power model has already been applied. However, tool cannot execute the UPF command because it will modify or create UPF objects in the specified scope and therefore violate the integrity of the power model.

Example:

```
define_power_model MODEL {
  create_power_domain PD_MACRO
}
apply_power_model MODEL -elements {macro_0}
set_level_shifter LS_STG -domain macro_0/PD_MACRO -location self
```

Error: Can't create level shifter strategy LS_STG in macro_0 because it violates the power intent of power model 'MODEL'. (UPF-375)

### What Next

Remove or modify the specified UPF command in order to keep the integrity of the power model intact.

### See Also

- define_power_model
- apply_power_model
- set_level_shifter

Feedback

- set_isolation
- set_retention

---

## UPF-376

(warning) Skipped power model '%s' from %s because it has already been defined in %s.

### Description

User has defined a power model in a file specified in power model search path through app option *mv.upf.power_model_search_path* and power model library through app option *mv.upf.power_model_library*. However, an existing power model with the same name has already been created. Duplicated power model is not allowed.

### What Next

Use a different name for the duplicated power model.

### See Also

- define_power_model
- apply_power_model
- mv.upf.power_model_search_path
- mv.upf.power_model_library

---

## UPF-377

(error) Cannot apply power model '%s' to cell '%s' because UPF commands have already been run in cell's power scope.

### Description

User tries to apply the specified power model to the specified cell through the *apply_power_model* command. However, tool cannot apply the specified power model because UPF commands have already been run in cell's power scope through the *load_upf* command. Power model can be applied to a cell only if no *load_upf* command has been run in the cell's power scope.

Example:

```
define_power_model MODEL -for {lib_cell} {
  create_supply_net VDD
  create_supply_net VSS
  create_supply_set SS -function {power VDD} -function {ground VSS}
  create_power_domain PD_TOP -supply {primary SS} -include_scope
```

Feedback

```
}
load_upf -scope u1/macro macro.upf
apply_power_model MODEL -elements {u1/macro}

Error: Cannot apply power model 'MODEL' to cell 'u1/macro' because UPF
commands have already been run in cell's power scope. (UPF-377)
```

### What Next

Do not run the *load_upf* command in the specified cell's power scope but include commands from the UPF file in the specified power model.

### See Also

- define_power_model
- apply_power_model
- load_upf

---

## UPF-378

(warning) Cannot apply power model%s%s to cell '%s' because design attribute 'UPF_is_%s_macro' is %s.

### Description

User tries to apply the specified power model to the specified cell through the *apply_power_model* command. However, tool cannot apply the specified power model because the design attribute *UPF_is_hard_macro* or *UPF_is_soft_macro* is not set on the specified cell or it is set to false.

### What Next

Please assign design attribute *UPF_is_hard_macro* or *UPF_is_soft_macro* to the specified cell through the *set_design_attributes* command.

Example:

```
define_power_model model -for {lib_cell_1 lib_cell_2} {
  create_supply_net VDD
  create_supply_net VSS
  create_supply_set SS -function {power VDD} -function {ground VSS}
  create_power_domain PD_TOP -supply {primary SS} -include_scope
}
set_design_attributes -models {lib_cell_1 lib_cell_2}
                      -attribute UPF_is_hard_macro true
apply_power_model model -elements {u1/macro u2/macro}
```

Feedback

### See Also

- define_power_model
- apply_power_model

---

## UPF-379

(error) Cannot apply design attribute 'UPF_is_%s_macro' or -is_%s_macro option to library cell '%s' because it is not a %s macro.

### Description

User set design attribute *UPF_is_hard_macro* or *UPF_is_soft_macro* (or use *-is_hard_macro*/*-is_soft_macro* options) to the specified library cell through the *set_design_attributes* command, However, tool cannot apply the design attribute because the specified library cell is not a hard or soft macro.

### What Next

Correct the design attribute assignment so that it is applied to the correct hard or soft macro.

### See Also

- define_power_model
- apply_power_model

---

## UPF-380

(warning) Power model '%s' contains no UPF commands.

### Description

User created a power model through the *define_power_model* command but the command does not contain any UPF commands. The power model is therefore empty.

### What Next

Include UPF commands in curly braces using the *define_power_model* command.

Example:

```
      define_power_model model -for {lib_cell_1 lib_cell_2} {
create_supply_net VDD
create_supply_net VSS
create_supply_set SS -function {power VDD} -function {ground VSS}
create_power_domain PD_TOP -supply {primary SS} -include_scope
```

Feedback

```
}
```

### See Also

- define_power_model
- apply_power_model

---

## UPF-381

(warning) Duplicated %s '%s' found in %s option to %s command on power model '%s'.

### Description

User specified a library cell, supply net, or supply port multiple times in the specified option to a define_power_model or apply_power_model command on the specified power model. Only one instance of each object can be included in the specified option. Duplicated objects will be dropped from the command option.

Example 1:

```
define_power_model MODEL -for {lib_cell_1 lib_cell_1} {
  create_supply_net VDD
  create_supply_net VSS
  create_supply_set SS -function {power VDD} -function {ground VSS}
  create_power_domain PD_TOP -supply {primary SS} -include_scope
}
Warning: Duplicated library cell 'lib_cell_1' found in -for option to
define_power_model command on power model 'MODEL'. (UPF-381)
```

Example 2:

```
apply_power_model MODEL -elements {macro_u1} \\
   -supply_map {{VDD VDD1_TOP} {VDD VDD2_TOP}}
Warning: Duplicated supply net 'VDD' found in -supply_map option to
apply_power_model command on power model 'MODEL' (UPF-381)
```

### What Next

Remove duplicated object from the command option.

### See Also

- define_power_model
- apply_power_model

Feedback

## UPF-382

(error) Power model '%s' has already been defined in %s.

### Description

User tries to create a new power model through the *define_power_model* command in the specified scope. However, an existing power model with the same name has already been created. Duplicated power model is not allowed.

Example:

```
set_scope /
define_power_model MODEL -for {lib_cell_1} {
  create_supply_net VDD
  create_supply_net VSS
  create_supply_set SS -function {power VDD} -function {ground VSS}
  create_power_domain PD_TOP -supply {primary SS} -include_scope
}
define_power_model MODEL -for {lib_cell_2} {
  create_power_domain PD_TOP
}

Error: Power model 'MODEL' has already been defined in top scope.
 (UPF-382)
```

### What Next

Correct the name of the new power model.

### See Also

- define_power_model
- apply_power_model

## UPF-383

(error) Can't find power model '%s' specified in apply_power_model command.

### Description

User tries to apply a power model through the *apply_power_model* command. However, an existing power model with the specified name cannot be found.

Example:

```
define_power_model MODEL -for {lib_cell_1} {
  create_supply_net VDD
  create_supply_net VSS
```

Feedback

```
  create_supply_set SS -function {power VDD} -function {ground VSS}
  create_power_domain PD_TOP -supply {primary SS} -include_scope
}
apply_power_model MODEL_2 -elements {u1/macro}

Error: Can't find power model 'MODEL_2' specified in apply_power_model
 command.
```

### What Next

- Correct the power model name specified in apply_power_model command.
- Check that define_power_model command is run before apply_power_model command.
- Make sure that the power model file <model_name.upf> is specified in app option mv.upf.power_model_library. For example:

  ```
  set_app_options -as_user_default
                  -name mv.upf.power_model_library
                  -value "MODEL.upf"
  ```

- Make sure that the power model search path is specified correctly in app option mv.upf.power_model_search_path. For example:

  ```
  set_app_options -as_user_default
                  -name mv.upf.power_model_search_path
                  -value "/usr/library/power_models"
  ```

### See Also

- define_power_model
- apply_power_model
- set_app_options

---

## UPF-384

(error) Can't apply power model '%s' to '%s' because power model '%s' has already been applied.

### Description

User tries to apply a power model through the *apply_power_model* command to the specified cell. However, an existing power model has already been applied to the same cell through a previous *apply_power_model* command.

Example:

```
define_power_model MODEL_1 -for {lib_cell_1} {
  create_power_domain PD_TOP
}
define_power_model MODEL_2 -for {lib_cell_1} {
  create_power_domain PD_TOP
}
apply_power_model MODEL_1 -elements {u1/macro}
apply_power_model MODEL_2 -elements {u1/macro}
```

Error: Can't apply power model 'MODEL_2' to 'u1/macro' because power model 'MODEL_1' has already been applied.

### What Next

Change that only one power model is applied to the specified cell.

### See Also

- define_power_model
- apply_power_model

---

## UPF-385

(warning) Can't apply power model%s%s to any cells in %s.

### Description

User tries to apply a power model through the *apply_power_model* command without *-elements* option. However, tool cannot find any cell such that:

```
1. no power model has been applied to the cell; and
2. the cell uses a library cell matching -for option
 to define_power_model command
or if -for option is not specified, the name of the specified power
 model; and
3. the library cell does not have UPF_is_hard_macro attribute set to
 false.
```

As a result, tool cannot determine which cells should the power model be applied to.

Example:

```
set_scope /
define_power_model MODEL {
  create_supply_net VDD
  create_supply_net VSS
  create_supply_set SS -function {power VDD} -function {ground VSS}
  create_power_domain PD_TOP -supply {primary SS} -include_scope
```

Feedback

```
}
apply_power_model MODEL
```

Warning: Can't apply power model MODEL to any cells in top scope. (UPF-385)

### What Next

Specify *-elements* option to *apply_power_model* command or specify *-for* option to *define_power_model* command.

### See Also

- define_power_model
- apply_power_model

---

## UPF-386

(error) Can't apply power model '%s' because %s '%s' specified in %s cannot be found.

### Description

User tries to apply a power model through the *apply_power_model* command with *-supply_map* or *-port_map* option. However, tool cannot find the specified supply set, supply net, or logic net in the current scope. It can also be because tool cannot find the specified logic port in the power model as logic port cannot be generated in a hard-macro.

Example:

```
create_supply_set TOP_SS
define_power_model MODEL {
  create_supply_net VDD
  create_supply_net VSS
  create_supply_set SS -function {power VDD} -function {ground VSS}
  create_power_domain PD_TOP -supply {primary SS} -include_scope
}
apply_power_model MODEL -supply_map {{SS TOP_SS_2}}
```

Error: Can't apply power model 'MODEL' because supply set 'TOP_SS_2' specified in -supply_map cannot be found. (UPF-386)

### What Next

Correct the specified supply set, supply net, logic net, or logic port in *-supply_map* or *-port_map* option to *apply_power_model* command.

Feedback

### See Also

- define_power_model
- apply_power_model

---

## UPF-387

(error) Can't apply power model '%s' to cell '%s' because cell %s.

### Description

User tries to apply a power model through the *apply_power_model* command with a cell specified in *-elements* option. However, tool cannot apply the power model because the specified cell is not a hard macro. In this version of the tool, power model can only be applied to hard macro cells.

Example:

```
define_power_model MODEL {
  create_supply_net VDD
  create_supply_net VSS
  create_supply_set SS -function {power VDD} -function {ground VSS}
  create_power_domain PD_TOP -supply {primary SS} -include_scope
}
apply_power_model MODEL -elements {non_hard_macro_instance}
```

Error: Can't apply power model 'MODEL' to cell 'non_hard_macro_instance' because cell is not a hard macro. (UPF-387)

### What Next

Include a hard macro cell in *-elements* option to *apply_power_model* command.

### See Also

- define_power_model
- apply_power_model

---

## UPF-388

(error) Can't apply power model '%s' to cell '%s' because its reference library cell is not included in the power model.

### Description

User tries to apply a power model through the *apply_power_model* command with a cell specified in *-elements* option. However, tool cannot apply the power model because

Feedback

the specified cell is using a reference library cell which is not included in *-for* option of *define_power_model* command.

Example:

```
define_power_model MODEL -for {lib_cell_1} {
  create_supply_net VDD
  create_supply_net VSS
  create_supply_set SS -function {power VDD} -function {ground VSS}
  create_power_domain PD_TOP -supply {primary SS} -include_scope
}
apply_power_model MODEL -elements {an_instance_not_using_lib_cell_1}
```

Error: Can't apply power model 'MODEL' to cell 'an_instance_not_using_lib_cell_1' because its reference library cell is not included in the power model. (UPF-388)

### What Next

Include the reference library cell used by the specified cell in *-for* option to *define_power_model* command.

### See Also

- define_power_model
- apply_power_model

---

## UPF-389

(error) Can't create power model '%s' because it is defined inside another power model '%s'.

### Description

User tries to create a power model through the *define_power_model* command. However, tool cannot create the power model because it is defined inside another existing power model. Nested power models are not allowed.

Example:

```
define_power_model MODEL {
  create_power_domain PD_TOP
  define_power_model SUB_MODEL {
    create_power_domain PD_SUB
  }
  apply_power_model SUB_MODEL -elements {u1/macro}
}
apply_power_model MODEL -elements {u0/macro}
```

Feedback

Error: Can't create power model 'SUB_MODEL' because it is defined inside another power model 'MODEL'. (UPF-389)

**What Next**

Move the specified *define_power_model* command outside of the existing power model.

**See Also**

- define_power_model
- apply_power_model

---

## UPF-390

(error) Can't add parameter '%s' because add_parameter command is not executed inside a power model.

**Description**

User tries to add a parameter through the *add_parameter* command. However, tool cannot add the parameter because the *add_parameter* command can only be executed inside a power model (through the *define_power_model* command).

Example:

```
define_power_model MODEL {
  create_power_domain PD_TOP
}
add_parameter VAR -default_value 1 -description "a parameter"
```

Error: Can't add parameter 'VAR' because add_parameter command is not executed inside a power model. (UPF-390)

**What Next**

Move the specified *add_parameter* command inside an existing power model.

**See Also**

- define_power_model
- apply_power_model

---

## UPF-391

(warning) Found duplicated parameter '%s' defined in power model '%s'.

Feedback

### Description

User tries to add a parameter through the *add_parameter* command. However, user has already added another parameter with the same name in the power model. The duplicated add_parameter command will be ignored.

Example:

```
        define_power_model MODEL {
  add_parameter VAR -default_value 1 -description "a parameter"
  add_parameter VAR -default_value 2 -description "duplicated parameter"
}
```

Error: Found duplicated parameter 'VAR' defined in power model 'MODEL'. (UPF-391)

### What Next

Remove the duplicated *add_parameter* command inside the specified power model.

### See Also

- define_power_model
- apply_power_model

---

## UPF-392

(error) Parameter '%s' specified in -parameters of apply_power_model command is not defined in power model '%s'.

### Description

User specified a parameter through the *-parameters* option to *apply_power_model* command. However, the specified parameter is not defined through any *add_parameter* command in the specified power model. A parameter needs to be defined inside a power model before it can be used in *apply_power_model* command.

Example:

```
define_power_model MODEL {
  add_parameter VAR_1 -default_value 1 -description "a parameter"
}
apply_power_model MODEL -parameters {{VAR_1 1} {VAR_2 2}}
```

Error: Parameter 'VAR_2' specified in -parameters of apply_power_model command is not defined in power model 'MODEL'.

Feedback

### What Next

Add the missing *add_parameter* command inside the specified power model.

### See Also

- define_power_model
- apply_power_model

---

## UPF-393

(error) Can't set_scope into '%s' because a power model has already been applied to the scope.

### Description

User tries to change the current scope to the specified scope through the *set_scope* command. However, a power model has already been applied to the specified scope. It is illegal to set_scope into a scope which already has a power model applied.

Example:

```
define_power_model MODEL {
  create_power_domain PD_BLK
}
apply_power_model MODEL -elements {macro_0 macro_1}
set_scope macro_0
```

Error: Can't set_scope into 'macro_0' because a power model has already been applied to the scope. (UFP-393)

### What Next

Remove the illegal *set_scope* command.

### See Also

- define_power_model
- apply_power_model

---

## UPF-394

(warning) Can't implement -location parent(other) option for %s strategy '%s' because %s cells cannot be inserted outside of a power model.

Feedback

### Description

User specifies an isolation or level shifter strategy through the *set_isolation* or *set_level_shifter* command with *-location parent(other)* option inside a power model. However, the strategy applies to a cell or pin on the boundary of the power model. Because tool is not allowed to insert an isolation or level shifter cell outside of a power model, it cannot implement the specified isolation or level shifter strategy.

Example:

```
define_power_model MODEL {
  create_power_domain PD_MACRO_TOP
  set_level_shifter LS_STG -domain PD_MACRO_TOP -location parent
}
apply_power_model MODEL -elements {macro_0 macro_1}
```

Warning: Can't implement -location parent option for level shifter strategy 'LS_STG' because level shifter cells cannot be inserted outside of a power model. (UPF-394)

### What Next

Move the specified isolation or level shifter strategy outside of the power model.

Example:

```
define_power_model MODEL {
  create_power_domain PD_MACRO_TOP
}
apply_power_model MODEL -elements {macro_0 macro_1}
set_level_shifter LS_STG1 -domain macro_0/PD_MACRO_TOP -location parent
set_level_shifter LS_STG2 -domain macro_1/PD_MACRO_TOP -location parent
```

### See Also

- define_power_model
- apply_power_model
- set_level_shifter
- set_isolation
- set_isolation_control

---

## UPF-395

(information) Read power model file "%s" %s.

Feedback

### Description

user specified a power model search path through app option *mv.upf.power_model_search_path* and power model library through app option *mv.upf.power_model_library*. Tool reads the specified power model file successfully or with errors.

### What Next

No action needed if the specified power model file is read successfully.

### See Also

- define_power_model
- apply_power_model
- mv.upf.power_model_search_path
- mv.upf.power_model_library

---

## UPF-396

(Error) apply_power_model with -all_hard_macros option has already been run in %s.

### Description

User executed an *apply_power_model* command with the *-all_hard_macros* option in the specified power scope. However, the same command with the same option has already been run in the specified power scope. The *apply_power_model* command with the *-all_hard_macros* option can only be run once per power scope.

### What Next

Remove the extra *apply_power_model* command with the *-all_hard_macros* option.

### See Also

- define_power_model
- apply_power_model

---

## UPF-397

(error) Can't apply power model '%s': %s port '%s' cannot be connected to %s net '%s' according to -port_map because %s.

Feedback

### Description

User tries to apply a power model through the *apply_power_model* command with *-port_map* option. However, tool cannot connected the specified supply/logic port to the specified supply/logic net mentioned in the *-port_map* option.

### What Next

Correct the specified supply/logic net or supply/logic port in *-port_map* option to *apply_power_model* command.

### See Also

- define_power_model
- apply_power_model

---

## UPF-398

(information) Applied power model '%s' in %s to %d instance%s.

### Description

User applied a power model through the *apply_power_model* command in the specified scope. This message shows how many instances to which the specified power model has been applied.

Example:

```
set_scope /
define_power_model MODEL {
  create_power_domain PD_MACRO_TOP
}
apply_power_model MODEL -elements {macro_0 macro_1}
```

Information: Applied power model 'MODEL' in top scope to 2 instances. (UPF-398)

### What Next

No action is needed.

### See Also

- define_power_model
- apply_power_model

Feedback

## UPF-399

(error) Can't apply power model '%s' to cell '%s' because cell is not in the current physical block.

### Description

User tries to apply a power model through the *apply_power_model* command with a cell specified in *-elements* option. However, tool cannot apply the power model because the specified cell is from a physical block which is different from the current block.

### What Next

Do not include the specified cell in *-elements* option of the *apply_power_model* command.

### See Also

- define_power_model
- apply_power_model

## UPF-400

(warning) Element list '%s' in the strategy '%s' of power domain '%s' is redundant since all the instances of the element list are present in conflicting state with some other strategy(s).

### Description

This message occurs when all the individual elements of an element list in any strategy are in a conflicting state with some other strategy. This means that the elements of the element list are used as an instance in some other retention strategy(s).

### What Next

Check the correctness of set_retention and set_retention_elements commands and their elements used for the strategy, and assign the cell to one strategy only. If possible, remove the element list from the mentioned strategy.

### See Also

- set_retention
- set_retention_elements

Feedback

## UPF-401

(error) Signal level power domain '%s' is specified more than once with the same signal net '%s' inside the macro scope in power model.

### Description

This error occurs when user creates a signal level power domain inside a macro scope in a power model where the power domain with the same name and signal net is already present.

Example:

```
define_power_model MODEL {
   create_power_domain PD1 -elements {S1}
   create_power_domain PD1 -elements {S2 S1}
}
apply_power_model MODEL -elements {u0/macro}
```

### What Next

Please delete the second (and so) create_power_domain command with the domain name for which this error message is generated.

### See Also

- create_power_domain

## UPF-402

(error) Signal level power domain '%s' is already specified on the signal net '%s' inside the macro scope in power model and hence cannot be specified in the new domain '%s'.

### Description

This error occurs when user creates a signal level power domain inside a macro scope in a power model with a signal net but a power domain is already created on that signal net.

Example:

```
define_power_model MODEL {
   create_power_domain PD1 -elements {S1}
   create_power_domain PD2 -elements {S1}
}
apply_power_model MODEL -elements {u0/macro}
```

Feedback

### What Next

Please delete the second (and so) create_power_domain command with the signal net for which this error message is generated.

### See Also

- create_power_domain

---

## UPF-403

(error) The retention strategy '%s' with the element '%s' belonging to the power domain '%s' is errored out because a different power domain '%s' is defined on the same signal.

### Description

This error occurs when user gives a retention strategy on a signal in macro is defined in a power domain where the signal doesn't belong to. A different signal level power domain is created on the signal.

Example:

```
define_power_model MODEL {
   create_power_domain PD1 -elements {S1}
   create_power_domain PD2 -elements {S2}
   set_retention -domain PD1 -elements {S2} -retention_supply TOP.primary
}
apply_power_model MODEL -elements {u0/macro}
```

### What Next

Please make sure that the domain on which the signal is defined matches with the domain in which the retention strategy is specified.

### See Also

- create_power_domain
- set_retention

---

## UPF-404

(warning) The domain '%s' cannot be used for '%s' strategy '%s' because it is defined inside hard macro via a power model.

Feedback

### Description

The power domain specified in the strategy is not a valid power domain. It may be a power domain on signal level.

### What Next

Please remove the path based strategy defined on the power domain. Only retention strategies is applicable on these domains.

### See Also

- set_isolation
- set_repeater
- set_level_shifter
- create_power_domain

---

## UPF-405

(error) Can not accept mixed UPF 2.0/2.1 style %s attribute in one command.

### Description

This error message occurs when the attribute option specified in the *set_design_attributes* or *set_port_attributes* command has mixed style usage in one command.

In UPF 2.1 style, -attribute option will accept 1 value, the syntax looks like:

```
set_design_attributes
   [-attribute {name value}]
set_port_attributes
   [-attribute {name value}]
```

In UPF 2.0 style, -attribute option will accept 2 value, the syntax looks like:

```
set_design_attributes
   [-attribute name value]
set_port_attributes
   [-attribute name value]
```

### What Next

Use consistent syntax and rerun the command.

Feedback

### See Also

- set_design_attributes
- set_port_attributes

---

## UPF-406

(error) Error while reading the UPF file : %s. Use error_info command for more information.

### Description

This error message is generated by the load_upf command when a syntax error occurs while reading a UPF file. These kind of syntax errors are caught in the TCL interpreter. The text of the message shows the specific error.

Example:

```
        create_power_domain mid -elements {MID
Error: Error while reading the UPF file : missing close-brace. Use
 error_info command for more information (UPF-406)
```

### What Next

Use the error_info command to see the file name and the line number of the syntax error. See what error message is printed with this message ID.

### See Also

- load_upf

---

## UPF-407

(warning) The cells specified in the retention strategy %s for power domain %s will be mapped to a non retention library cell as no mapping constraint is defined for the strategy using map_retention_cell command.

### Description

This warning indicates that mapping constraints have not been defined on the retention strategy for the specified power domain using the map_retention_cell command. As a result, tool will go ahead and map the cells specified in set_retention with the best available non retention library cell.

Feedback

### What Next

Define the mapping constraints for the retention strategy using the map_retention_cell command.

### See Also

- set_retention
- map_retention_cell
- report_power_domain

---

## UPF-408

(information) Converting control signal net specified in the strategy '%s' to its driver pin '%s' as the net seems to be deleted.

### Description

This information message occurs when you specify a net or hierarchical pin as the control signal using the set_isolation_control or set_retention_control command. Tool then uses the driver pin of the net as the control signal.

### What Next

This is only an information message. No action is required.

### See Also

- set_retention_control
- set_isolation_control

---

## UPF-410

(information) Applying power model '%s' in %s to %s.

### Description

User applied a power model through the *apply_power_model* command in the specified scope. This message shows that the tool is applying the specified model to the specified instance. This message is printed when the app option *mv.upf.power_model_verbose* is set to TRUE.

Example:

```
set_app_options -as_user_default -list {mv.upf.power_model_verbose TRUE}
set_scope /
```

Feedback

```
define_power_model MODEL {
  create_power_domain PD_MACRO_TOP
  create_supply_net VDD
}
apply_power_model MODEL -elements {macro_0 macro_1}

Information: Applying power model 'MODEL' in top scope to macro_0.
 (UPF-410)
set_scope macro_0
create_power_model PD_MACRO_TOP
create_supply_net VDD
set_scope /
Information: Applying power model 'MODEL' in top scope to macro_1.
 (UPF-410)
set_scope macro_1
create_power_model PD_MACRO_TOP
create_supply_net VDD
set_scope /
Information: Applied power model 'MODEL' in top scope to 2 instances.
 (UPF-398)
```

### What Next

No action is needed.

### See Also

- define_power_model
- apply_power_model
- mv.upf.power_model_verbose

---

## UPF-411

(Error) Applied power model '%s' in %s to %s with user errors.

### Description

User applied a power model through the *apply_power_model* command in the specified scope. The power model is applied to the specified instance but errors are encountered during the application.

### What Next

Check the messages above UPF-411 to find out the reasons why the specified power model is applied with errors.

Please run *reset_upf* command before applying another power model to the same instance.

Feedback

### See Also

- define_power_model
- apply_power_model
- report_power_model
- mv.upf.power_model_verbose

---

## UPF-412

(warning) Disconnected %s '%s' from %s '%s' and connected it to %s '%s' because apply_power_model %s -port_map has higher priority.

### Description

User tries to apply a power model through the *apply_power_model* command with *-port_map* option. However, tool finds that the specified supply/logic port is already connected to another supply/logic net. Because *apply_power_model -port_map* has higher priority than existing connections, tool disconnected the specified supply/logic port and reconnected it to the specified supply/logic net according to *apply_power_model -port_map*.

### What Next

Correct the specified supply/logic net or supply/logic port in *-port_map* option to *apply_power_model* command if the supply/logic port connection is not the design intention.

### See Also

- define_power_model
- apply_power_model

---

## UPF-413

(error) Can't apply power model '%s' to '%s' because power model '%s' has already been applied to the same library cell '%s'.

### Description

User tries to apply a power model through the *apply_power_model* command to the specified cell. However, an existing power model has already been applied to the same reference library cell used by the specified cell through a previous *apply_power_model* command.

Feedback

For example, assume that u1/macro and u2/macro are instantiations of the same library cell lib_cell_1:

```
define_power_model MODEL_1 -for {lib_cell_1} {
  create_power_domain PD_TOP_1
}
define_power_model MODEL_2 -for {lib_cell_1} {
  create_power_domain PD_TOP_2
}
apply_power_model MODEL_1 -elements {u1/macro}
apply_power_model MODEL_2 -elements {u2/macro}
```

Error: Can't apply power model 'MODEL_2' to 'u2/macro' because power model 'MODEL_1' has already been applied to the same library cell 'lib_cell_1'. (UPF-413)

**What Next**

Change that only one power model is applied to the specified library cell.

**See Also**

- define_power_model
- apply_power_model

---

## UPF-420

(error) State '%s' referred in transition '%s' option '%s' is not defined on object '%s'.

**Description**

This message occurs when the power state specified in state transition mismatch with the specified object.

**What Next**

Specify power state match with the object specified in state transition.

**See Also**

- describe_state_transition
- add_state_transition

---

## UPF-421

(error) Transition '%s' specified as -legal, while final state '%s' is illegal.

Feedback

**Description**

This message occurs when -legal is specified for a transition where final state is illegal.

**What Next**

Specify a legal final state for the transition.

**See Also**

- describe_state_transition
- add_state_transition

---

## UPF-422

(error) State '%s' is specified in both the from state and the to state of transition '%s'.

**Description**

This message occurs when the same state is specified as both the from state and the to state.

**What Next**

Specify different state for from and to state for the transition.

**See Also**

- describe_state_transition
- add_state_transition

---

## UPF-423

(error) Invalid transition format for transition '%s': %s.

**Description**

This message occurs when syntax format issue detected in the specified transition.

**What Next**

Check the man page of the command to follow transition format.

**See Also**

- add_state_transition

Feedback

## UPF-424

(error) Supply net '%s' [%s] has lower voltage than supply net '%s' [%s], while the former used as power and the latter used as ground in supply set %s.

### Description

This message occurs when turn on relative PG support, and there is power ground pair voltage conflict in a supply set.

### What Next

Check the pst voltage defined for both supply net.

### See Also

- create_supply_set
- report_supply_nets
- report_pst

## UPF-425

(error) Power domain '%s' use relative supply as primary while it contains element '%s' which is not leaf cell support relative supply.

### Description

When turn on relative PG support, it is allow to use relative supply as domain primary. However, the domain should not contain instance that do not have relative PG pin.

### What Next

Check domain primary supply and elements.

### See Also

- create_power_domain
- report_power_domain

## UPF-426

(error) %s strategy '%s' use relative supply as %s.

Feedback

### Description

This message occurs when relative supply nets or supply sets are used by power strategies.

### What Next

Check the strategy definition.

### See Also

- set_isolation
- set_level_shifter
- set_repeater
- set_retention
- create_power_switch

---

## UPF-427

(warning) Dropping library cell %s from map_retention_cell command of retention strategy %s belonging to power domain %s as the library cell is not suitable when -use_retention_as_primary is %s.

### Description

The warning message is issued :- 1. The retention strategy is specified with -use_retention_as_primary option but the library cell mentioned above has its data output pin not related with the backup PG pin. 2. The retention strategy is not specified with -use_retention_as_primary option but the library cell mentioned above has its data output pin related with the backup PG pin

This library cell will be dropped from the map_retention_cell command and will not be used from mapping.

### What Next

Please specify proper library cells in map_retention_cell command which has its data output pin related with the backup PG pin if -use_retention_as_primary is specified. If not specified, normal retention cell i.e. whose data output pin related with the primary PG pin should be used.

### See Also

- set_retention
- map_retention_cell

Feedback

## UPF-429

(error) An existing %s strategy with the same name '%s' already exists in the same power domain %s inside power model scope.

### Description

This error occurs when a duplicated strategy name is used within the same power domain for the reported type of strategy inside power model scope Name duplication is not allowed for strategies in same power domain.

### What Next

The UPF needs to be changed to fix the error. Identify the referenced duplicate strategy names associated with the referenced power domain and strategy type. If the strategy is a complete duplicate, remove one of the duplicated strategies. If the strategies are different, use a different strategy name for one of the duplicate strategies.

### See Also

- define_power_model
- apply_power_model
- set_isolation
- set_retention
- set_level_shifter
- set_repeater

## UPF-430

(error) There are power domains with missing voltage areas in the design.

### Description

This message occurs when compile runs in the legal-handoff mode, and there are power domains in the design with missing voltage areas,

### What Next

Use the "create_voltage_area" command to assign voltage areas to power domains, or turn on the "mv.incomplete_upf.enable" and "mv.upf.enable_missing_voltage_area" app_options to enable incomplete UPF missing VA flow to make the tool automatically assign voltage areas to power domains.

Feedback

**See Also**

- create_voltage_area
- set_app_options
- mv.incomplete_upf.enable
- mv.upf.enable_missing_voltage_area

---

## UPF-431

(warning) UPF command '%s' is not supported in hierarchical flow, and will be dropped.

**Description**

This warning message occurs when an UPF command which is not supporting hierarchical flow is found by related commands. It will not be preserved in UPF saved during hierarchical flow.

**What Next**

check the UPF command.

**See Also**

- split_constraints
- save_upf
- uncommit_block
- add_state_transition
- describe_state_transition

---

## UPF-432

(Error) Failed to mark reference-only on supply %s because %s.

**Description**

This error message appears when set_design_attribute fails to mark a supply net or supply set as reference-only.

Reference-only supply is marked by: set_design_attribute -element {.} -attribute reference_only {supply_net_or_supply_set}

When a supply has been set as domain primary, isolation supply, repeater supply or retention supply, it can't be marked as reference-only.

Feedback

### What Next

Rewrite the UPF to avoid using the supplies as reference-only.

---

## UPF-433

(information) The grouped cell %s contains leaf cell %s. Will follow the regular split_constraints procedure.

### Description

This message occurs when the grouped cell contains at least one leaf cell. In this case, we will following the regular procedure to perform split_constraints.

### See Also

- split_constraints

---

## UPF-434

(error) Cannot change scope of domain %s into block %s because %s

### Description

This message occurs when try to split_constraints on blocks with outside-block nested power domains, and we cannot change the scope of nested power domain into the block due to the given reason. Split_constraints will thus fail.

### See Also

- split_constraints

---

## UPF-435

(information) The grouped cell %s contains non-domain-root cell %s.

### Description

This message occurs when the grouped cell contains at least one non-domain-root cell (i.e. the cell is not the root element of power domain). In this case, we will not limit the supplies in this block UPF. And the primary of the top domain of this block will be the same as the parent domain of this block before split_constraints.

### See Also

- split_constraints

Feedback

## UPF-436

(Warning) Strategy %s in power domain %s is applied to the lower boundary. Will follow the regular split_constraints procedure.

### Description

This message occurs when a strategy is applied to the lower boundary from the top domain. In this case, we will follow the regular procedure to perform split_constraints.

### See Also

- split_constraints

## UPF-437

(warning) The split hierarchy %s is the root element of a power domain. Will follow the regular split_constraints procedure.

### Description

This message occurs when the target split hierarchy is the root element of a power domain. This case is currently not supported by the new split_constraints feature. Therefore, we will follow the regular procedure to perform split_constraints.

### See Also

- split_constraints

## UPF-438

(warning) The primary supply of user-selected power domain %s is not available by the new power domain on the grouped cell %s.

### Description

This message occurs when the user selects a power domain and applies its primary supply to the new power domain on the grouped cell but the primary supply of the selected power domain is not available by the new power domain created on the grouped cell.

### See Also

- split_constraints

Feedback

## UPF-439

(information) The sub-domains under the grouped cell %s have different primary supplies.

### Description

The information message is issued when the sub-domains under the grouped cell don't have the same primary supply

### See Also

- split_constraints

## UPF-440

(information) Use the primary supply of power domain %s as the primary of the new power domain on the grouped cell %s.

### Description

The information message is issued when the tool determines the primary supply of the new power domain on the grouped cell during limit-supply flow of split_constraints.

### See Also

- split_constraints

## UPF-441

(warning) The primary supplies of the all power domains under the hierarchy of grouped cell %s are not available by the new power domain. Apply the primary supply of the top domain %s.

### Description

This message occurs when none of the primary supplies of the sub-domains under the hierarchy of the grouped cell is available by the new power domain. In this case, we will apply the primary supply of the top domain as the primary supply of the new domain.

### See Also

- split_constraints

Feedback

## UPF-442

(error) The name pair %s has syntax error. The legal expression is {group_cell_name sub_pd_name}.

### Description

During the limit-supply flow of split_constraints, when users define group_cell-power_domain pairs (i.e. applying the primary supply of the power_domain to the new power domain created on the group_cell), each pair should follow the format {group_cell_name sub_pd_name}. Otherwise an error will be issued and the user-defined pair list will be ignored.

### What Next

Check the new_pd_primary_name_list app option for any syntax error.

### See Also

- split_constraints

## UPF-443

(error) Unable to split the UPF for block %s because partial power domain list of shared_voltage_area attribute is specified and domain '%s' is not inside the block.

### Description

This error message occurs when running the split_constraints command. The block cannot contain only partial power domain list of shared_voltage_area attribute.

### What Next

Check the shared_voltage_area attribute.

### See Also

- split_constraints

## UPF-444

(warning) set_equivalent pattern %s matched supply net and supply port/pg pin %s. The supply port or pg pin will be ignored.

Feedback

### Description

The pattern given in set_equivalent matched a supply net and either a supply port or pg pin of the same name, and they are not directly connected. Only the supply net will be selected for the set_equivalent command.

### What Next

This is only a warning message. No action is required.

### See Also

- set_equivalent

---

## UPF-445

(warning) set_design_attribute -exclude_elements option is not supported by all tools in the SNPS flow.

### Description

This command is only applicable for simulation and does not have impact on implementation tool. Option -exclude_elements should go together with option -elements and attribute ret_mode. It will be preserved by save_upf as well.

---

## UPF-446

(error) set_design_attribute -exclude_elements option is not supported by all tools in the SNPS flow. The option is only for attribute ret_mode.

### Description

This command is only applicable for simulation attribute ret_mode, Option -exclude_elements should go together with option -elements and attribute ret_mode. The command will error out for other attribute except attribute ret_mode.

### See Also

- UPF-445

---

## UPF-447

(warning) %s attribute '%s' has been set more than once by one command.

Feedback

### Description

This warning message occurs when the same attribute specified in the *set_design_attributes* or *set_port_attributes* command more than once in one command.

The attribute value that is set later override the attribute value that is set first. In other words, the last value setting for the same attribute will take effect.

### What Next

Please check if the attribute value is expected. Remove the unnecessary setting.

### See Also

- set_design_attributes
- set_port_attributes

---

## UPF-448

(warning) Invalid hierarchical name specified: '%s'

### Description

The design attribute reference_only requires the supply object specified with a simple name.

UPF objects may have hierarchical names, but the simple name required is not permitted to contain the hierarchy separator.

### What Next

Re-enter the command with a valid simple name specified.

### See Also

- set_design_attributes

---

## UPF-449

(warning) The library cell %s specified in command %s is marked as dont_use.

### Description

This warning message occurs when a library cell is used as mapped cell for a strategy and that library cell is marked as dont_use or excluded for power.

Feedback

### What Next

Change library cell setting or just avoid using that library cell mapping for the specified strategy.

### See Also

- map_retention_clamp_cell

---

## UPF-450

(information) Explicit supply net connections to isolation and retention cells will be written out. Use mv.upf.save_upf_include_supply_exceptions_for_iso_retn to control this.

### Description

This information message informs that explicit supply net connections to PG pins of isolation and retention cells will be written out by save_upf. If this behavior is not desired, it can be turned off by setting the app option *mv.upf.save_upf_include_supply_exceptions_for_iso_retn* to false.

### What Next

No action is required.

### See Also

- save_upf
- mv.upf.save_upf_include_supply_exceptions_for_iso_retn

---

## UPF-451

(warning) Removed control signal %s from %s strategy %s. Deleting it will remove the corresponding %s control command.

### Description

This warning message occurs when objects that are used as a reference for a UPF control signal are removed. As a result, the related UPF constraint is also removed. Design objects such as ports and pins can be removed by direct commands such as the remove_design command.

### What Next

This is only a warning message. No action is required.

Feedback

However, to avoid this warning message, apply UPF constraints after each modification to the design. The design object can be found while loading the UPF constraints.

**See Also**

- set_retention
- set_retention_control
- set_isolation
- set_isolation_control
- create_power_switch

## UPF-453

(warning) Design attribute 'shared_voltage_area' is ignored.

**Description**

This message occurs when the 'shared_voltage_area' design attribute is specified and will be ignored. The 'shared_voltage_area' design attribute is applicable only when 'mv.upf.multi_pd_shared_voltage_area_flow' app option is set to 'new'.

**What Next**

Check the 'shared_voltage_area' design attribute and 'mv.upf.multi_pd_shared_voltage_area_flow' app option.

**See Also**

- set_design_attributes

## UPF-455

(error) Required argument '-golden_upf_files' was not found. Argument is required when 'mv.upf.enable_golden_upf' is true.

**Description**

This error message indicates that a required argument is missing and needs to be provided for successful execution of the command.

**What Next**

Provide the required argument.

Feedback

## UPF-456

(error) Unable to ungroup cell %s because %s.

### Description

This message occurs when the check for UPF restructuring during ungroup_cells. This message specifies the cell that cannot be ungrouped due to the UPF restructuring check failure. In order to continue ungrouping the cell, the user should fix the problem listed in this message.

### See Also

- ungroup_cells

## UPF-457

(error) Invalid boolean expression '%s' in logic expression for retention %s.

### Description

Invalid boolean expression in logic expression for retention strategy.

### What Next

The boolean expression must be a combination of UPF generic references, logic nets and ports, and logical or bitwise operators (!, |, ||, &, &&, ^, ~, ==, !=). SystemVerilog keywords are not acceptable ports.

### See Also

- set_retention

## UPF-458

(warning) Boolean logic expression '%s' of %s cannot resolve object reference, and will be dropped.

### Description

This warning message occurs when an object reference in a boolean logic expression cannot be updated because the object was removed.

### What Next

Check the object references in the boolean expression.

Feedback

**See Also**

- set_retention
- save_upf

---

## UPF-460

(warning) The '-traverse_macros' option would implicitly set '-transitive' as true.

**Description**

This warning message occurs when you are specifying '-traverse_macros' with '-transitive false'.

**What Next**

Check whether the specified options are correct.

**See Also**

- find_objects

---

## UPF-461

(error) Adding user-defined %s '%s' with tool-derived power domain '%s' is not allowed.

**Description**

This error is issued when you attempt to add a user-defined constraints with a tool-derived power domain.

**What Next**

Check whether the power domain and the specified constraint should be tool-derived or not.

**See Also**

- create_power_domain

---

## UPF-462

(error) The application option "mv.upf.preserve_logic_in_boolean_expr" cannot be set if constraints have already been loaded

Feedback

### Description

The application option mv.upf.preserve_logic_in_boolean_expr cannot be set after constraints are loaded.

### What Next

Set mv.upf.preserve_logic_in_boolean_expr before any constraints are loaded.

### See Also

- set_app_options

---

## UPF-465

(error) Cell '%s' is already included in power domain '%s'.

### Description

A cell has been specified that is already included in a power domain. A cell cannot belong to more than one power domain.

### What Next

Ensure the list of instances is valid for the relevant power domains by reviewing the create_power_domain statements in the UPF file. In addition you can use the get_cells command to determine the power domain associated with a cell(s).

### See Also

- create_power_domain
- get_cells

---

## UPF-466

(Error) Issue found with '%s' %s attribute setting on '%s': %s.

### Description

This message occurs when a conflict is found with an existing port or design attribute setting in the reported object.

### See Also

- set_port_attributes
- set_design_attributes

Feedback

## UPF-467

(information) Related supplies are not explicitly specified on %d port(s) and primary supplies (%s, %s) of top power domain will be assumed as the related supply.

### Description

This informational message is printed when primary supplies of top power domain are assumed as related supply of ports.

### What Next

You can query the ports using the Tcl attribute 'related_supply_default_primary' and check whether related supply or driver/receiver supply are supposed to be explicitly specified on them.

### See Also

- set_related_supply_net
- set_port_attributes
- get_related_supply_nets

## UPF-468

(information) Reference-only supply %s has been removed and related supply of %d ports are reset.

### Description

This informational message is to show the number of ports that their related supply net or driver/receiver supplies are reset by the command due to deletion of the specified reference-only supply.

### What Next

This message is informational. No action is required.

### See Also

- remove_reference_only_related_supply

## UPF-469

(Error) Reference-only supply %s cannot be removed : %s.

Feedback

### Description

This error message is printed when the reference-only supply cannot be removed.

### What Next

Please check the reason why the reference-only related supply cannot be removed.

### See Also

- remove_reference_only_related_supply

---

## UPF-470

(warning) UPF on the grouped cell %s can't split because it belongs to a MIM %s whose instances have different power constraints

### Description

This message is issued after group_cells, when the grouped cell is MIM and will fail split_constraints due to different power constraints on instances of this MIM. Note that this message is just the result of MIM split checking; no split_constraints is performed at this stage. If users want to split_constraints on this grouped cell later, they should modify the UPF to meet the following conditions:

- UPF constraints on all instances of the MIM must be identical. This includes the name and arguments of all declared power objects and power strategies;
- There must be a level shifter strategy at the MIM boundary if there is voltage violation;
- Boundary conditions must be identical. This includes that the driver and receiver supplies of isolation source/sink strategies, level shifter strategies and all feedthrough paths must be identical;
- Matching power ports on all instances must be connected to the same external supply;
- PVT of all MIM instances must be identical;

### See Also

- group_cells
- split_constraints

---

## UPF-471

(error) Can't move cell %s to %s, because %s

Feedback

### Description

This message occurs during reparent_cells when moving the cell violates the following UPF constraints: (1) moving a cell to another domain root cell; (2) moving a cell to another power scope. The detailed reason is listed in this message.

### See Also

- reparent_cells

---

## UPF-472

(warning) UPF intent of blackbox design '%s' will not be captured in the full-chip UPF.

### Description

This warning message indicates that the UPF intent of the design is not captured in the full-chip UPF because the design is a blackbox. The general requirement for saving full-chip UPF is that all blocks must be linked to either their design view or abstract view.

### What Next

Ensure that the block is linked to abstract or design view.

### See Also

- save_upf
- set_reference
- change_abstract

---

## UPF-473

(Error) Mismatch in the %s design attribute value between top design and block design.

### Description

This Error message indicates that there is a mismatch in the value of the design attribute between the top design and the block design.

### What Next

Ensure that the top design and block Design has same design attribute value.

---

## UPF-474

(Error)Cannot create the power domain %s because the legacy block criteria is violated.

Feedback

### Description

This error message occurs during create power domain when the *legacy_block* attribute specified using the *set_design_attribute* command to a cell does not satisfy the legacy block criteria. The cell must satisfy the following criteria to be considered a legacy block:

1. Domain-independent supply nets or supply sets are not defined or used.
2. Implicit supply handles of a power domain cannot be used.
3. All parts of the legacy block must belong to power domains scoped within the block.

   Example:

   ```
   prompt> create_power_domain PD1 -elements {mid1 mid2 mid3}
   prompt> set_design_attributes -elements {mid1 mid2} -attribute
    legacy_block true
   ```

   Error: cannot create the power domain PD1 because multiple elements are specified in the element list while creating a power domain corresponding to a legacy block.

   To correct this error, specify only one element in the -elements option of create_power_domain command while creating power domain corresponding to a legacy block.

   ```
   prompt> set_design_attribute -elements {mid1 mid2 mid3} -attribute
    legacy_block TRUE
   ```

### What Next

Ensure that the legacy block criteria is satisfied.

### See Also

- create_power_domain

---

## UPF-475

(error) Cannot specify a domain-independent supply net or supply set with UPF command %s for legacy block/domain.

### Description

This error message occurs when a UPF command, such as *create_supply_net*, *create_supply_set*, *associate_supply_set*, *set_domain_supply_net*,*connect_supply_net*, *set_isolation*, or *set_retention*, using a domain-independent supply net or supply set in its arguments is applied to a cell with a *legacy_block* design attribute, or a legacy power domain.

Feedback

### What Next

Please remove any domain-independent supply net or supply set from the argument list of the UPF command before applying it on a legacy_block, or an element inside a legacy block.

### See Also

- create_supply_net
- create_supply_set
- set_domain_supply_net
- connect_supply_net
- set_isolation
- set_retention

---

## UPF-476

(error) Cannot use external supply handles in a power domain in a legacy block.

### Description

This error message occurs when a *create_power_domain* command attempts to specify an external supply set for a power domain in a legacy block. All power domains in a legacy block must be fully-restricted.

### What Next

Please remove the use of external supply handles with the *create_power_domain* command in a legacy block.

### See Also

- create_power_domain

---

## UPF-477

(error) Cannot connect the domain-independent supply net to a pin or a port which is inside a legacy block.

### Description

This error message occurs when you use the *connect_supply_net* command to connect a domain-independent supply net to a port/pin which is inside a legacy block. The only exception is when the supply port is at the boundary of a legacy block.

Feedback

Example:

```
prompt> create_supply_net di_net
prompt> create_power_domain PD1 -elements {B1}
prompt> create_supply_net dd_net -domain PD1
prompt> connect_supply_net di_net PD1/dd_net
```

Error:Cannot connect the domain-independent supply net to a pin or a port which is inside a legacy block.(UPF-477)

The *connect_supply_net* command generates a UPF-477 error message, as it attempts to connect a domain-independent supply net (di_net) to a domain-dependent supply net ( PD1/dd_net).

**What Next**

Set the *legacy_block* design attribute on the cell connected to the supply port:

Example:

```
prompt> set_design_attributes -elements B1 -attribute legacy_block true
```

or Specify the supply nets that have consistent domain dependency.

**See Also**

- connect_supply_net
- create_supply_net

---

## UPF-478

(Error) %s strategy must contain -%s_power_net and -%s_ground_net options and should not contain -source/-sink options if it belongs to a legacy block power domain.

**Description**

This error message is printed because the %s strategy is specified in a power domain inside a legacy block, but the -%s_power_net and -%s_ground net options are not used and if -source or -sink options are used.

**What Next**

Ensure that the %s strategy has -%s_power_net and -%s_ground_net options specified and remove -source or -sink options if specified.

Feedback

**See Also**

- set_isolation
- set_retention

---

## UPF-479

(warning) Ignore user's name (%s) for the new domain. The domain name should be a simple name.

**Description**

This messages occurs when user specify the name of the new domain on the grouped cell during split_constraints with the app option mv.hierarchical.new_pd_name_on_group_cell. It is required that the provided name is simple. Otherwise, the provided name will be ignored and this warning message is issued.

**See Also**

- split_constraints

---

## UPF-480

(Error) Cannot set legacy block attribute on %s.

**Description**

This error message is printed due to the following reasons. 1) User tries to set legacy block attribute on a leaf cell. 2) The element specified in the set_design_attribute belongs to a power domain which has domain independent supplies defined in the current scope. 3) Legacy block attribute value is set to false on an element that belongs to a legacy block.

**What Next**

Ensure that legacy block attribute is set correctly.

---

## UPF-481

(error) The command '%s' is supported only when mv.upf.multi_pd_shared_voltage_area_flow is set to 'new'.

**Description**

This message occurs when the command is executed and the app option 'mv.upf.multi_pd_shared_voltage_area_flow' is not set to 'new'. It is supported only when the value of 'mv.upf.multi_pd_shared_voltage_area_flow' is set to 'new'.

Feedback

### What Next

Check the app option value of 'mv.upf.multi_pd_shared_voltage_area_flow'.

### See Also

- mv.upf.multi_pd_shared_voltage_area_flow

---

## UPF-482

(information) Power domain '%s' is skipped because %s.

### Description

This message occurs when the get_equivalent_power_domains or check_equivalent_power_domains command skip the power domain due to the specified reason.

### What Next

Check the specified power domain.

### See Also

- get_equivalent_power_domains
- check_equivalent_power_domains

---

## UPF-483

(information) List of equivalent power domains - %s.

### Description

This message shows the list of equivalent power domains that satisfy the criteria to share a primary voltage area.

### What Next

This message is informational. No action is required.

### See Also

- check_equivalent_power_domains

---

## UPF-484

(warning) cannot automatically add SPA/SRSN for the punched port %s, because %s

Feedback

### Description

This message occurs during split_constraints when the tool fails to add SPA/SRSN for the punched port automatically. Next step: check the reason of failure; then either adding the SPA/SRSN manually to the split UPF or modify the original UPF to avoid the condition.

### See Also

- split_constraints

---

## UPF-487

(error) UPF constraints have been loaded with DCM check policy set to 'tolerate' or 'repair'. Without reset_upf, the policy can not be changed to 'error'.

### Description

This message occurs when UPF constraints have been loaded with DCM check policy set to 'tolerate' or 'repair', and user tries change the policy to 'error'.

### What Next

The UPF netlist and netlist used in tolerate/repair mode is a throw-away netlist/constraints. To changed to tape_out mode, user has to start over with clean netlist and UPF constraint.

### See Also

- load_upf
- commit_upf
- reset_upf

---

## UPF-488

(Information) -location fanout option will be deprecated in future release.

### Description

This information message occurs when you run *set_isolation -location fanout* command. *-location fanout* will be deprecated in future release.

### What Next

This is an information message.

### See Also

- set_isolation

Feedback

## UPF-489

(warning) Current design is not a UPF design

### Description

Current design is not upf design. this command should be run on the UPF designs. Please load the UPF for the design.

### What Next

```
Please load the UPF for the design.
```

## UPF-490

(error) %s failed because the legacy block setting is violated.

### Description

This error message occurs in *%s* if the legacy block setting is violated due to one of the following reasons:

1. Element marked as legacy block does not have a self-scoped power domain defined on it.
2. The scope to which the legacy block's power domain belongs has a domain independent supply defined in it.

### What Next

Please define a self-scoped power domain on the legacy block or remove the domain independent supply within the scope of legacy block.

### See Also

- create_power_domain

## UPF-491

(Error) split_constraints was not successful because rule for splitting a legacy block was violated.

### Description

This error message is printed when user attempts split_constraints at the boundary of a legacy block or on a block which is inside a legacy block.

Feedback

### What Next

Ensure that elements specified in set_budget_options command adheres to legacy block rules.

### See Also

- set_budget_options
- split_constraints

---

## UPF-492

(Warning) Isolation strategies with -location fanout is not supported in a legacy block.

### Description

This warning message is printed when the user defines a isolation strategy with -location option as fanout in the presence of a legacy block in the design.

### What Next

Ensure that isolation strategies does not have fanout as their location when the design contains a legacy block.

### See Also

- set_isolation

---

## UPF-493

(Warning) Supply net '%s' is reconnected to pin '%s' that was previously connected to supply net '%s'.

### Description

The power and ground pin had existing explicit supply net connection, and it is reconnected to a different supply net specified by the command.

### What Next

Check the power and ground pin to ensure that its supply net connection is correct.

Feedback

**See Also**

- connect_supply_net
- get_supply_nets
- mv.upf.enable_pg_pin_reconnection

---

## UPF-494

(warning) The design element %s is not on the boundary of the power domain. The %s might not %s.

**Description**

This error message occurs when you specify a design element in the *-elements* or *-exclude_elements* option of *set_port_attributes* command that is not on the upper or lower boundary of the power domain.

**What Next**

If the design element is at the boundary of the power domain before running the *commit_upf*, *create_mv_cells*, or any other action commands, no action is required. Otherwise, specify the correct design element for the command.

**See Also**

- set_port_attributes

---

## UPF-495

(Warning) The design element %s is not on power domain boundary. The %s supply attribute has been dropped for pin %s.

**Description**

This warning message occurs when you specify with *-elements* or *-ports* option of the *set_port_attributes* command, a black-box cell or pin that is not on power domain boundary.

**What Next**

Verify that the design element mentioned in the error message is not on the power domain boundary. Specify black-box design elements that are on power domain boundary, for *set_port_attributes* command.

Feedback

**See Also**

- set_port_attributes

---

## UPF-496

(warning) The 'set_port_attributes %s %s' setting on macro pin %s is not functionally equivalent to the related supplies derived from its liberty model (%s - %s).

**Description**

This message occurs when a set_port_attributes -receiver_supply (or -driver_supply) defined at the scope of a hard macro input (or ouput) pin is found to not be functionally equivalent to the power supplies that can be derived from the related PG pin data on its liberty model.

The purposes of the message is to show an inconsistency between the UPF at the macro scope and its liberty model. For implementation purposes, the related supplies derived from the liberty model will be used

**What Next**

Review the UPF to make sure the setting is correct.

**See Also**

- set_port_attributes

---

## UPF-497

(error) The macro_as_domain_boundary attribute cannot be set on the %s %s.

**Description**

The message occurs when the set_design_attribute command is trying to set macro_as_domain_boundary attribute on an instance, which has neither of these two properties: 1. Liberty attribute is_macro_cell set to TRUE, and 2. Design attribute UPF_is_hard_macro set to TRUE.

**What Next**

This is an error message. Please revisit your UPF intent and make the necessary change.

---

## UPF-498

(error) Value '%s' for %s signal sense of isolation strategy '%s' of power domain '%s' is not valid. Specify one of: high, low.

Feedback

### Description

This message occurs when the signal sense specified for isolation strategy is not supported or invalid.

### What Next

Check the signal sense and invoke the command again.

### See Also

- set_isolation

---

## UPF-499

(warning) option %s is specified without %s in set_isolation, and it will be ignored.

### Description

This error message occurs when set_isolation options -async_set_reset/-async_clamp_value are not all specified.

### See Also

- set_isolation

---

## UPF-500

(Information) Usage of -model will be made mandatory with -is_analog in a future release.

### Description

This information message occurs when is_analog attribute is set on pins/ports without -model option. Use of set_port_attributes -is_analog without this option will cause downstream tools to fail.

### See Also

- set_port_attributes

---

## UPF-502

(Information) Propagating is_analog setting on port(s) %sof %s from block %s to top. At top level, all instances of the model will get this is_analog setting.

Feedback

### Description

This information message indicates that is_analog setting on library cell is being propagated from block to top. At top level, all instances of this model will have the same is_analog setting.

### See Also

- set_port_attributes

---

## UPF-503

(error) Equivalent supplies %s and %s in block %s have mismatched drivers.

### Description

This message occurs when tool tries to split full-chip UPF constraints and finds two supply nets in the block with mismatched drivers to be declared as equivalent.

### What Next

Check if these two supply nets should be equivalent.

### See Also

- split_constraints

---

## UPF-504

(error) Unable to split/merge the UPF for block, because the design attribute enable_state_propagation_in_add_power_state is set true above the block, and the same attribute is set false inside the block.

### Description

This error message occurs when running the split_constraints or link command.

### What Next

Check and revise the design attribute enable_state_propagation_in_add_power_state, and rerun the design.

### See Also

- split_constraints

Feedback

## UPF-505

(error) Supply net connections to both HighConn and LowConn side of supply port '%s' of direction 'internal' is not supported.

**Description**

Check the supply net connection(s) to the supply port.

**See Also**

- create_supply_port
- connect_supply_net

## UPF-506

(error) Specifying additional UPF constraints to the scope of hard macro '%s' after domain merging is not allowed.

**Description**

Check whether the domain of the scope should be merged.

**What Next**

Specify complete UPF constraints of the scope before commit_upf.

**See Also**

- load_upf
- create_power_domain
- commit_upf

## UPF-507

(warning) Domain merging for %s is not allowed and will be skipped.

**Description**

This message occurs when domain merging is not allowed for the specified instance.

**What Next**

Check whether domain merging is expected.

Feedback

**See Also**

- commit_upf
- create_power_domain

---

## UPF-508

(error) PST state %s of PST %s refers to state %s which has conflicting definitions on %s.

**Description**

This message occurs when a given supply state is not uniquely defined on a netgroup and has conflicting voltage values.

**What Next**

Check the supply states defined in the UPF to identify and fix the conflict.

**See Also**

- commit_upf
- add_pst_state
- add_port_state
- add_supply_state

---

## UPF-509

(warning) Retention strategy %s for power domain %s will not result in ZPR insertion, as its control singals are not from same source. map_retention_clamp_cell will not take effect until next design setup is fixed.

**Description**

This message occurs when retention strategy has different save/restore signal, and map_retention_clamp_cell specified.

**What Next**

Check whether ZPR and clamp cell insertion is expected.

**See Also**

- map_retention_clamp_cell

Feedback

## UPF-510

(warning) Retention strategy %s for power domain %s will not result in ZPR insertion, as its mapping constraint does not have zero-pin library cell. map_retention_clamp_cell will not take effect until next design setup is fixed.

### Description

This message occurs when retention strategy does not have zero-pin lib cell specified in map_retention_cell, but with map_retention_clamp_cell specified.

### What Next

Check whether ZPR and clamp cell insertion is expected.

### See Also

- map_retention_clamp_cell

## UPF-511

(error) Unable to set related supply net for %s based on the isolation strategy %s defined on it.

### Description

This error message occurs when running the *split_constraints* command. The supply net to be set as the related supply net for the specified pin could not be determined from the isolation strategy defined on the pin.

### What Next

Check the isolation strategies defined and re-run split_constraints.

### See Also

- split_constraints

## UPF-512

(warning) The related supply set of %s '%s' cannot be determined.

### Description

This warning message occurs because the related power and ground supply nets of the specified pin or port cannot be resolved to an available supply set of its parent domain.

Feedback

### What Next

Check the related PG pin of the specified pin and its supply net. Check the available supplies of the parent domain of the pin or port.

### See Also

- set_domain_supply_net
- report_power_domain

---

## UPF-513

(warning) Cannot determine the related supply for any of the referenced pins and ports

### Description

This warning message occurs because the related power and ground supply nets of the specified pins and ports cannot be resolved to an available supply set of its parent domain.

### What Next

Check the related PG pin of the specified pins and its supply nets Check the available supplies of the parent domain of the pins and ports.

### See Also

- set_domain_supply_net
- report_power_domain

---

## UPF-515

(error) Attribute value specified %s is invalid. Supported values for enable_bias attribute are : true, false, derived.

### Description

This error message indicates incorrect attribute value setting for enable_bias attribute.

### What Next

Check the enable_bias attribute setting.

### See Also

- set_design_attributes

Feedback

## UPF-516

(information) Supply net '%s' has connections to macro internal pg pin(s).

### Description

This informational message is printed when the specified supply net has connections to macro internal pg pins. Connection to macro internal pg pins can change the supply availability and whether the supply net and its connected/associated supply nets should be implemented in netlist.

### What Next

Please check the connections and supply availability of the specified supply net and its connected or associated supply net(s).

### See Also

- connect_supply_net

## UPF-517

(error) Tools does not support design attribute "hetero_fanout_isolation" with value false.

### Description

This message occurs when design attribute "hetero_fanout_isolation" is set to false. This attribute will be dropped.

### What Next

Use the following syntax and rerun the command:

```
set_design_attributes
   [-attribute {name value}]
```

### See Also

- set_design_attributes

## UPF-518

(Information) The design attribute "hetero_fanout_isolation" is not set. Tool will insert isolation cells on hetero-fanout path. It is recommended to set the attribute "hetero_fanout_isolation" to TRUE in the input UPF to ensure cross tool consistency for path based isolation cell insertion on heterogeneous fanouts.

Feedback

### Description

This information message is issued when design attribute "hetero_fanout_isolation" is not set in input UPF and UPF has isolation strategy with -sink/-diff_supply_only. Tool will insert isolation cells on hetero-fanout path. It's recommended to set the attribute "hetero_fanout_isolation" to TRUE in the input UPF to ensure cross tool consistency for path based isolation cell insertion on heterogeneous fanouts.

### See Also

- set_design_attributes

---

## UPF-519a

(information) DCM mode is enabled, app options "mv.incomplete_upf.iso_infer_csn_rule" will not be ignored. Use corresponding DCM check "mv.cells.iso_missing_csn" to set CSN infer rule.

### Description

This message indicates that the design is in DCM mode,and app options "mv.incomplete_upf.iso_infer_csn_rule" will not be effective.

### What Next

DCM mode is enabled with set_early_data_check_policy, set option with corresponding DCM checks.

### See Also

- set_early_data_check_policy

---

## UPF-519b

(information) DCM mode is enabled, app options "mv.incomplete_upf.ls_infer_csn_rule" will not be ignored. Use corresponding DCM check "mv.cells.ls_missing_csn" to set CSN infer rule.

### Description

This message indicates that the design is in DCM mode,and app options "mv.incomplete_upf.ls_infer_csn_rule" will not be effective.

### What Next

DCM mode is enabled with set_early_data_check_policy, set option with corresponding DCM checks.

Feedback

### See Also

- set_early_data_check_policy

---

## UPF-519c

(information) DCM mode is enabled, app options "mv.incomplete_upf.ret_infer_csn_rule" will not be ignored. Use corresponding DCM check "mv.cells.ret_missing_csn" to set CSN infer rule.

### Description

This message indicates that the design is in DCM mode,and app options "mv.incomplete_upf.ret_infer_csn_rule" will not be effective.

### What Next

DCM mode is enabled with set_early_data_check_policy, set option with corresponding DCM checks.

### See Also

- set_early_data_check_policy

---

## UPF-519

(information) DCM mode is enabled, app options "mv.incomplete_upf.enable" will be ignored.

### Description

This message indicates that the design is in DCM mode,and app options "mv.incomplete_upf.enable " will not be effective.

### What Next

DCM mode is enabled with set_early_data_check_policy, set option with corresponding DCM checks.

### See Also

- set_early_data_check_policy

Feedback

## UPF-519e

(information) DCM mode is enabled, app options "mv.incomplete_upf.ao_infer_csn_rule" will not be ignored. Use corresponding DCM check "mv.cells.ao_missing_csn" to set CSN infer rule.

### Description

This message indicates that the design is in DCM mode,and app options "mv.incomplete_upf.ao_infer_csn_rule" will not be effective.

### What Next

DCM mode is enabled with set_early_data_check_policy, set option with corresponding DCM checks.

### See Also

- set_early_data_check_policy

## UPF-520

(error) Supply Net '%s' does not have a voltage defined on corner '%s'.

### Description

This error is reported when the specified Supply Net has no voltage defined in the mentioned corner.

All the supply nets defined in the design must have an operating voltage defined for synthesis to proceed further.

### What Next

- If the specified corner is *default*, assign the operating voltage using *set_voltage* for the Supply Net. If not, use *set_voltage -corner* on the specified corner.

### See Also

- set_voltage

## UPF-521

(error) Supply Net '%s' has a voltage '%s' defined in corner '%s' which is out of the range of its valid states.

Feedback

### Description

This message is reported when the assigned Supply Net has an operating voltage higher or lower than the range defined in the Power State of the by the command *add_power_state*.

### What Next

- Set operating voltage in the valid range with command *set_voltage*. • Update the voltage range defined in the Power State.

### See Also

- add_power_state
- set_voltage

---

## UPF-522

(error) Supply voltage on supply net '%s' and its connected supply nets are not consistent in corner '%s'.

### Description

This error message occurs when you attempt to specify an operating voltage that differs from the operating voltages of other supply nets connected to it as a single metal(through supply ports and by association).

All supply nets connected together as a single metal must share common operating voltages. Supply nets having different operating voltages cannot be connected together.

### What Next

- If you have previously used the *set_voltage* command on several supply nets connected together, apply the *set_voltage* command on these supply nets simultaneously to change their operating voltages.
- To apply the *set_voltage* command, specify multiple supply nets with the *-object_list* option.

### See Also

- set_voltage

---

## UPF-523

(information) DCM mode is enabled, app options "mv.upf.enable_missing_voltage_area" will be ignored.

Feedback

### Description

This message indicates that the design is in DCM mode,and app options "mv.upf.enable_missing_voltage_area" will not be effective.

### What Next

DCM mode is enabled with set_early_data_check_policy, set option with corresponding DCM check "mv.va.missing_voltage_area".

### See Also

- set_early_data_check_policy

---

## UPF-525

(information) DCM mode is enabled, app options "mv.incomplete_upf.iso_infer_csn_rule" will be ignored. Use corresponding DCM check "mv.cells.iso_missing_csn" to set CSN infer rule.

### Description

This message indicates that the design is in DCM mode,and app options "mv.incomplete_upf.iso_infer_csn_rule" will not be effective.

### What Next

DCM mode is enabled with set_early_data_check_policy, set option with corresponding DCM checks.

### See Also

- set_early_data_check_policy

---

## UPF-526

(information) DCM mode is enabled, app options "mv.incomplete_upf.ls_infer_csn_rule" will be ignored. Use corresponding DCM check "mv.cells.ls_missing_csn" to set CSN infer rule.

### Description

This message indicates that the design is in DCM mode,and app options "mv.incomplete_upf.ls_infer_csn_rule" will not be effective.

Feedback

### What Next

DCM mode is enabled with set_early_data_check_policy, set option with corresponding DCM checks.

### See Also

- set_early_data_check_policy

---

## UPF-527

(information) DCM mode is enabled, app options "mv.incomplete_upf.ret_infer_csn_rule" will be ignored. Use corresponding DCM check "mv.cells.ret_missing_csn" to set CSN infer rule.

### Description

This message indicates that the design is in DCM mode,and app options "mv.incomplete_upf.ret_infer_csn_rule" will not be effective.

### What Next

DCM mode is enabled with set_early_data_check_policy, set option with corresponding DCM checks.

### See Also

- set_early_data_check_policy

---

## UPF-528

(information) DCM mode is enabled, app options "mv.incomplete_upf.psw_infer_csn_rule" will be ignored. Use corresponding DCM check "mv.cells.psw_missing_csn" to set CSN infer rule.

### Description

This message indicates that the design is in DCM mode,and app options "mv.incomplete_upf.psw_infer_csn_rule" will not be effective.

### What Next

DCM mode is enabled with set_early_data_check_policy, set option with corresponding DCM checks.

### See Also

- set_early_data_check_policy

Feedback

## UPF-529

(information) DCM mode is enabled, app options "mv.incomplete_upf.ao_infer_csn_rule" will be ignored. Use corresponding DCM check "mv.cells.ao_missing_csn" to set CSN infer rule.

### Description

This message indicates that the design is in DCM mode,and app options "mv.incomplete_upf.ao_infer_csn_rule" will not be effective.

### What Next

DCM mode is enabled with set_early_data_check_policy, set option with corresponding DCM checks.

### See Also

- set_early_data_check_policy

## UPF-530

(information) DCM mode is enabled, app options "mv.incomplete_upf.optimization_control" will be ignored. Use corresponding DCM check "mv.buf.mv_violation" to set buffering rule.

### Description

This message indicates that the design is in DCM mode,and app options "mv.incomplete_upf.optimization_control" will not be effective.

### What Next

DCM mode is enabled with set_early_data_check_policy, set option with corresponding DCM checks.

### See Also

- set_early_data_check_policy

## UPF-531

(warning) Block instance '%s' is skipped because it is not immediate child block of current design and higher-level block instance '%s' is not included in the full-chip UPF output.

### Description

This message indicates that the specified block instance is excluded from the full-chip upf output because higher-level block instance is not specified or included in the full-chip UPF.

Feedback

### What Next

Ensure that the all higher-level block instance(s) are specified to be included in the full-chip UPF.

### See Also

- save_upf

---

## UPF-532

(information) Summary of UPF Cells

### Description

This informational message is printed at the postlude to summarize the UPF information like number of power domain(s), supply net(s), voltage area info and MV cells summary.

### See Also

- connect_supply_net

---

## UPF-533

(information) Design has all mapped instances, create_mv_cells in mapped mode.

### Description

This informational message is printed if all the instances are mapped at the create_mv_cells step

### See Also

- create_mv_cells

---

## UPF-534

(information) Design has unmapped instance(s), create_mv_cells in %s mode.

### Description

This informational message is printed if there is any unmapped instance(s) in the design after running create_mv_cells.

### What Next

Please use check_mv_design to see more details.

Feedback

**See Also**

- create_mv_cells
- check_mv_design

---

## UPF-535

(information) Please run check_mv_design for more details.

**Description**

This informational message is printed if there is any MV violation flagged at postlude MV Check.

**What Next**

Please run check_mv_design to see more details of the violation.

**See Also**

- check_mv_design

---

## UPF-539

(error) Number of derived PST states is over limit %d. Derived PST reported is not complete.

**Description**

This error message occurs when the number of derived PST states is over the preset limit.

**What Next**

Please either increase the state limit through option '-pst_state_limit' or limit the number of supplies through option '-supplies'.

**See Also**

- report_pst

---

## UPF-540

(error) System power state table buidling time is over the limit.

Feedback

### Description

This error message occurs when the building time of system-level power state table is over the preset limit. The system power state table is incomplete and may result in false isolation/level shifter violations. This is usually due to system has large number of independent supply nets.

### What Next

Please check if design supply network is fully connected to reduce the number of independent supply nets or increase the time limit for building system power state table.

### See Also

- report_supply_net_groups
- mv.upf.system_pst_building_time_limit

---

## UPF-570

(error) The correlated supply group "%s" is invalid because %s.

### Description

This error message occurs when you run the *set_design_attributes* command with the *correlated_supply_group* attribute, but the group list is invalid.

Example: Invalid correlated supply groups

```
prompt> set_design_attributes -elements {.} -attribute
 correlated_supply_group ""
prompt> set_design_attributes -elements {.} -attribute
 correlated_supply_group "{ }"
prompt> set_design_attributes -elements {.} -attribute
 correlated_supply_group "VDD1 VDD2 VSS"
prompt> set_design_attributes -elements {.} -attribute
 correlated_supply_group "{VDD1 VDD2} VSS"
prompt> set_design_attributes -elements {.} -attribute
 correlated_supply_group "{VDD1 VDD2 VSS} { }"
```

### What Next

Specify a valid list of the supply groups with the *correlated_supply_group* attribute.

Example:

```
• set_design_attributes -elements {.} -attribute \
       correlated_supply_group "{VDD1 VDD2 VSS1} {VDD3 VDD4 VSS2}"
```

Feedback

```
• set_design_attributes -elements {.} -attribute \
      correlated_supply_group "*"
```

**See Also**

- set_design_attributes

---

## UPF-571

(error) The supply net %s is included multiple times in the correlated supply group {%s}.

**Description**

This error message occurs when you run the *set_design_attributes* command with the *correlated_supply_group* attribute and a supply net is included multiple times in the same correlated supply group.

In the following example, VDD1 is included twice in a correlated supply group:

```
prompt> set_design_attributes -elements {.} \
                  -attribute correlated_supply_group "{VDD1 VDD1 VSS}"
```

Error: The supply net VDD1 is included multiple times in the correlated supply group. (UPF-571)

**What Next**

Remove all duplicate entries of the supply net from the correlated supply group and rerun the command.

**See Also**

- set_design_attributes

---

## UPF-572

(warning) The operating voltage (%s=%.2f) for the supply net %s does not match any of its power states.

**Description**

For a specified power state of the supply net, the operating voltage must be one of the maximum, minimum, or nominal voltages. This warning message occurs when specify a

Feedback

operating voltage for the supply net that does not match any of its power states, as shown in the following example:

Example:

```
prompt> create_supply_net VDD
prompt> create_supply_port VDD
prompt> connect_supply_net VDD -ports { VDD }
prompt> add_port_state VDD -state { ON 0.85 0.90 0.95 }
prompt> set_voltage 0.70 -object_list { VDD }
prompt> check_mv_design
```

Warning: The operating voltage (max=0.70) for the supply net VDD does not match any of its power states.(UPF-572)

### What Next

1. Use the **set_voltage** command to assign an appropriate operating voltage to the supply net, or
2. Add a new port or power state that matches the operating voltage of the supply net.

### See Also

- set_voltage
- add_port_state
- add_power_state

---

## UPF-573

(Information) Correlated supply groups %s and %s are merged.

### Description

This information message occurs when you run the *set_design_attributes* command with the *correlated_supply_group* attribute and some supply nets in the group list belong to the same netgroup. Therefore, the correlated supply groups of these supply nets are merged.

Two supply nets belong to the same netgroup if any one of the following conditions are met:

- they are connected through a supply port by the **connect_supply_net** command.
- they belong to two supply sets that are associated because of **associate_supply_set** command.
- one belongs to a supply set whose function is updated to the other by **create_supply_set -update** command.

Feedback

• one belongs to a supply set and the other belongs to a supply set handle of
a power domain; and the supply set and supply handle are associated
by the **create_power_domain -supply** command.

### What Next

This is an information message. No action is required.

### See Also

- set_design_attributes
- connect_supply_net
- associate_supply_set
- create_supply_set
- create_power_domain

---

## UPF-574

(error) The correlated supply groups %s are set in an invalid scope '%s' because the scope does not have its own power domain.

### Description

This error message occurs when you specify the *correlated_supply_group* attribute with the *set_design_attributes* command in the specified scope. However, the scope does not have its own power domain. The *correlated_supply_group* attribute can only be set in the top scope of a power domain.

### What Next

Create a power domain in the specified scope.

### See Also

- set_design_attributes

---

## UPF-575

(error) Cannot propagate correlated supply group %s from scope '%s' because some supply nets are in different correlated supply groups in the top-level scope.

Feedback

### Description

The commit_upf command failed because it cannot propagate design attribute *correlated_supply_group* from the specified scope to the top-level scope. Two or more supply nets appear in the same correlated supply group in the specified scope but they appear in different correlated supply groups in the top-level scope.

### What Next

Correct the design attribute *correlated_supply_group* in the top-level scope so that the conflicting supply nets appear in the same correlated supply group.

---

## UPF-576

(error) Correlated supply group has already been set up in scope '%s'.

### Description

The set_design_attribute command failed because it tries to set up correlated supply group in the specified scope but another correlated supply group has already been set up in that scope.

Here is an example:

```
prompt> set_scope /U1
prompt> set_design_attribute -elements {.} \\
   -attribute correlated_supply_group "{VDD1 VDD2}"
prompt> set_scope /U1
prompt> set_design_attribute -elements {.} \\
   -attribute correlated_supply_group "{VDD3 VDD4}"
Error: Correlated supply group has already been set up in scope 'U1'.
 (UPF-576)
```

### What Next

Delete one of the correlated supply groups or combine them into one correlated supply group for the specified scope.

For the above example, we can combine the two correlated supply groups into one:

```
prompt> set_scope /U1
prompt> set_design_attribute -elements {.} \\
   -attribute correlated_supply_group "{VDD1 VDD2} {VDD3 VDD4}"
```

---

## UPF-578

(error) For Supply set handle %s, Invalid number of supply sets specified.

Feedback

### Description

This error message indicates that for one or more supply set handles, more than one supply sets are specified. The supply handles and their associated supply sets are specified using the *-supply* option of the *create_power_domain* command .

### What Next

Provide only one supply set for each supply handle

### See Also

- create_supply_set
- create_power_domain

---

## UPF-579

(error) %s Strategy with name %s has been already defined in the power domain %s.

### Description

This error occurs when a duplicated strategy name is used within the same power domain for strategy of isolation, level_shifter or retention. Name duplication is not allowed for strategies in same power domain.

### What Next

The UPF needs to be changed to fix the error. Identify the referenced duplicate strategy names associated with the referenced power domain, make sure to use a different strategy names.

### See Also

- set_isolation
- set_isolation_control
- set_level_shifter
- set_retention

---

## UPF-580

(error) The string specified in -elements is invalid for the %s attribute '%s'. %s

Feedback

### Description

This error message indicates that command set_design_attributes or set_port_attributes is specifying invalid elements for given attribute according to its rule.

### What Next

Modify the set_design_attributes/set_port_attributes command to specify correct element according to given attribute. For example, design attribute may require one '.' or hierarchical cell as element, port attribute may require power domain root cell as element.

### See Also

- set_design_attributes
- set_port_attributes

---

## UPF-581

(error) Resolution conflicts on net %s (%s) and net %s (%s).

### Description

This error message indicates that two nets in the same netgroup having conflicting net resolution. Those two nets could be connected through connect_supply_net, set_equivalent or associate_supply_set command.

### What Next

Use same resolution for nets in the same netgroup.

### See Also

- create_supply_net

---

## UPF-582

(Error) Retention strategy %s of domain %s is not applicable to any of the elements specified in the element list of the strategy.

### Description

This error message occur when the elements in the element list of set_retention command are either not part of the domain or another retention strategy applies to them.

### What Next

Make sure element list contain at least one valid element.

Feedback

### See Also

- set_retention

---

## UPF-583

(Warning) The cell(s) specified with map_retention_cell -elements for retention strategy %s will be ignored.

### Description

The tool ignores -elements option of map_retention_cell. Effectively, map_retention_cell will be applicable to all the cells of the corresponding retention strategy. Since tool ignores -elements option of map_retention_cell, save_upf will not write out this option.

### What Next

This is only a warning message. No action is required.

### See Also

- set_retention

---

## UPF-584

(error) Invalid boolean expression '%s' in supply set state %s logic expression.

### Description

Invalid boolean expression '%s' in logic expression for supply set state.

### What Next

```
The boolean expression must be a combination of instance control  ports
and  logical  or bitwise operators (!, |, ||, &, &&, ^ ). SystemVerilog
keywords are not acceptable ports.
```

---

## UPF-585

(information) add_power_state global -simstate outside of individual state curly braces will be obsolete in a near future release.

### Description

add_power_state -simstate outside of individual state curly braces will be obsolete in a near future release, customer should specify -simstate in individual state curly braces.

Feedback

### What Next

```
stop using -simstate globally for add_power_state
```

## UPF-586

(warning) Multiple set_variation %s. Keeping value %.2f, discarding value %.2f.

### Description

multiple set_variation command issued for same supply net or globally, the latest value is kept, the previous value is discarded

Example:

### What Next

```
make sure the right value is specified for supply voltage tolerance.
```

## UPF-587

(Error) %s value specified in set_variation command -tolerance option.

### Description

set_variation command -tolerance option accept one or two positive values. cannot specify negative value or more than two values.

Example:

### What Next

```
make sure the right value is specified for supply voltage tolerance.
```

### See Also

- set_variation

## UPF-588

(error) set_port_attributes terminates for the following reason: '%s%s'.

### Description

This message occurs when the command set_port_attribute cannot procced further because of the specified reason on the message.

Feedback

**See Also**

- set_port_attributes

---

## UPF-590

(error) The top power domain %s of soft macro instance %s is scoped above itself.

**Description**

This message occurs when a soft macro instance does not have its own top-level power domain.

**What Next**

Define a top-level power domain for the soft macro (e.g., *create_power_domain -elements {.}* .

**See Also**

- create_power_domain

---

## UPF-591

(warning) set_variation command only supported in top design UPF.

**Description**

This occurs when the tool detects set_variation command in block level UPF. supply variation for voltage reconciliation can be only set in top level UPF. The command in block level UPF will be ignored.

**What Next**

set_variation in top level UPF.

**See Also**

- load_upf

---

## UPF-592

(warning) set_scope command %s crosses soft macro boundary.

**Description**

This message occurs when *set_scope ..* is executed at the top scope of a soft macro, or when *set_scope /* is executed anywhere inside a soft macro.

Feedback

### What Next

No action required. This is a warning message.

### See Also

- set_scope

---

## UPF-593

(error) Conflict upf_reconcile_boundary setting for module : %s.

### Description

This message occurs when conflict upf_reconcile_boundary setting is done for the same module.

### See Also

- report_pst

---

## UPF-594

(error) Cannot implement %s strategy '%s' because it is defined inside a soft macro, but accesses or inserts objects outside of the soft macro.

### Description

User specifies an isolation, level shifter or retention strategy through a *set_isolation*, *set_level_shifter*, or *set_retention* command inside a soft macro, but accesses or inserts objects outside of the soft macro.

Examples: 1. use of *-location parent* or-*location fanout* on a boundary pin of the soft macro. 2. references a control signal outside of the soft macro.

### What Next

Move the specified isolation or level shifter strategy outside of the soft macro.

### See Also

- set_design_attributes
- set_level_shifter
- set_isolation

Feedback

- set_isolation_control
- set_retention

---

## UPF-595

(information) Voltage reconciliation tolerance setting from supply net %s, other setting(s) on the same netgroup will be ignored.

### Description

set_variation for supply net reconciliation voltage tolerance can be set from top level design or block level design. If set from top level supply net, all other settings in block level will be ignored. If no top level setting is done, first setting from block level will be taken, all other settings in other blocks will be ignored. The tolerance will be applied to entire netgroup.

### What Next

### See Also

- set_variation
- report_pst

---

## UPF-596

(error) Conflict voltage reconciliation tolerance setting from : %s.

### Description

set_variation for supply net reconciliation voltage tolerance can be set from top level design or block level design. you get this Error message because conflict tolerances are set for same level supply nets in the netgroup, or conflict tolerances are set for equivalent supplies, this is considered as conflict, the settings are dropped.

### What Next

```
Resolve the conflict by eliminating settings from different blocks, or
 set the desired tolerance from higher hierarchical level,
or set only from one block or top design.
```

Feedback

**See Also**

- set_variation
- report_pst
- set_equivalent

## UPF-597

(error) Cannot connect supply net %s to supply port %s because %s is inside a soft macro.

**Description**

This error message occurs when the specified supply port is inside a soft macro, and the specified supply net is not.

Supply net connections cannot cross soft macro boundaries.

**See Also**

- connect_supply_net

## UPF-598

(error) Cannot associate supply set %s to supply handle %s because the association crosses a soft macro boundary.

**Description**

This error message occurs when the specified supply handle is inside a soft macro, and the specified supply set is not, or vice versa.

Supply set association cannot cross soft macro boundaries.

**See Also**

- associate_supply_set

## UPF-599

(error) set_isolation_control -location command cannot be used after set_isolation -location -update.

Feedback

### Description

This message occurs when the set_isolation_control -location is used on a strategy that already has the location set with set_isolation -location -update.

### What Next

set_isolation_control command will be dropped.

### See Also

- set_isolation
- set_isolation_control

---

## UPF-600

(warning) The level-shifter strategy %s of power domain %s cannot be applied with location other on Upper domain boundary.

### Description

This message occurs when *set_level_shifter ..* is set for top domain upper boundary port with -location other, tool will not insert level shifter cell outside the scope of this power domain.

### What Next

This is a warning message. Check the set_level_shifter command.

### See Also

- set_level_shifter

---

## UPF-601

(warning) The level-shifter strategy %s of power domain %s cannot be applied with location other on hard macro lower domain boundary port %s.

### Description

This message occurs when *set_level_shifter ..* is set for lower boundary port with -location other, the lower domain element is hard macro, level shifter cell cannot be inserted.

### What Next

This is a warning message. Check the set_level_shifter command.

Feedback

### See Also

- set_level_shifter

---

## UPF-602

(warning) The pin '%s' has already been inserted in the strategy '%s' and will not be inserted in the strategy '%s' of type '%s'.

### Description

This warning happens when a pin has more than one set_dft_isolation setting and ends up with more than one valid set_isolation_strategy. The pin will be added to only one strategy to avoid resolutions conflict.

### What Next

Please check that your set_dft_isolations settings are correct.

### See Also

- check_isolation_coverage

---

## UPF-603

(error) The pin %s is not an analog macro pin and is connected to a negative ground supply.

### Description

This error message occurs when a pin with a negative ground supply is connected to a non-analog and/or non-macro pin.

### What Next

Check the connections.

### See Also

- add_power_state
- add_port_state
- connect_supply_net
- set_related_supply_net

Feedback

## UPF-604

(warning) The user defined attribute '%s' has the following issue: %s.

### Description

This warning message occursduring commit_upf when an user defined attribute that is being used in UPF does not meet the specified restriction.

### What Next

Check the reason of the issue described in the message.

### See Also

- define_user_attribute
- set_attribute

## UPF-605

(warning) Uniquification of instance '%s' of module '%s' was triggered because: %s.

### Description

This warning message occurs when a multiple instanciated module is uniquified during commit_upf or load_upf and that uniquification was not prevented because of the given reason.

### What Next

Check the reason of the issue described in the message.

### See Also

- commit_upf
- load_upf

## UPF-908

(information) All %s cells can be %s.

### Description

For current design and UPF, tool will be able to map all isolation/enable level shifter cells, or all level shifter cells can be inserted to fix voltage violation. If -element specified in the command, the report is specific to the element pin, port, cell or net.

Feedback

**See Also**

- analyze_mv_feasibility

---

## UPF-909a

(warning) Not all level shifter cells can be inserted to fix voltage violation, refer to the summary for general issue.

**Description**

For current design and UPF specified, tool will not be able to insert level shifter cell to fix voltage violation on some path. Please refer to the output of this command or do analyze_mv_feasiblity for more detail information.

**See Also**

- analyze_mv_feasibility

---

## UPF-909

(error) Not all the %s cells can be %s.

**Description**

For current design and UPF specified, tool will not be able to map all the isolation and/or enable level shifter elements to technology library cells, or level shifter cell canot be inserted to fix the voltage violation for some path. Please refer to the output of this command analyze_mv_feasiblity and/or the HTML report generated with -format html for details on which elements cannot be mapped or inserted and why.

**See Also**

- analyze_mv_feasibility

---

## UPF-910

(error) Element '%s' in the extent of atomic power domain '%s' has been defined as root cell of another power domain '%s'. Applied strategies will be re-evaluated once this error is addressed.

**Description**

This error occurs when a cell in the extent of an atomic power domain is not excluded from the power domain and the cells belongs to a different power domain than the atomic one.

Feedback

### What Next

Please exclude the specified cell of the atomic power domain with *-update* option. Here is an example:

```
prompt> create_power_domain PD_TOP -elements {m2 m1/b1} -include_scope
prompt> create_power_domain PD_MID -elements {m1} -atomic
prompt> create_power_domain PD_MID -exclude_elements {m1/b1}
```

### See Also

- create_power_domain

---

## UPF-911

(error) Unable to merge domain '%s' and '%s' because '%s' is atomic but '%s' is not atomic.

### Description

This warning occurs when a block domain tries to merge to a TOP domain but they do not share the same -atomic setting. Atomic power domains can only be merged to other atomic domains.

### What Next

Add the atomic option to the domain which is missing the setting to keep the domains as atomic.

### See Also

- create_power_domain

---

## UPF-912

(warning) Element '%s' in the extent of the atomic power domain '%s' but belongs to another power domain '%s'.

### Description

This warning occurs when a block domain is propagated from the block to TOP and elements of this domain are in the extent of an atomic domain.

### What Next

Exclude this instances from the atomic domain and run commit_upf to validate the change.

Feedback

**See Also**

- create_power_domain

---

## UPF-913

(error) Set_isolation strategy '%s' with multiple control signals failed because: %s.

**Description**

When defining a set_isolation strategy with multiple -isolation_signal, -isolation_sense, -clamp_value or -isolation_supply. The strategy can only be applying inside a hard macro and, if specified, all the mentioned options must have the same number of elements as the others or only one.

**See Also**

- set_isolation

---

## UPF-914

(error) The command "%s" can only be used before any MV object is created.

**Description**

The specified command needs to be called before any MV objects like supply ports, supply nets, supply sets and power domains are created.

**What Next**

Move the command before the creation of any MV objects.

**See Also**

- set_design_attributes

---

## UPF-915

(warning) Update to the option '%s' of implemented %s strategy '%s' found.

**Description**

Updates to elements list (e.g. -elements, -exclude_elements) or filtering options (-applies_to, -source, -sink, etc) of power management strategies can cause already implemented power management cells to change or lose their currently associated strategy. Thuis can lead to reimplementation or removal of existing power management cell.

Feedback

### What Next

If changes to power management strategy association is not desired please refrain from performing updates after implementation.

### See Also

- set_isolation
- set_retention
- set_repeater
- set_level_shifter

---

## UPF-5191

(information) DCM mode is enabled, app options "mv.incomplete_upf.iso_infer_csn_rule" will not be ignored. Use corresponding DCM check "mv.cells.iso_missing_csn" to set CSN infer rule.

### Description

This message indicates that the design is in DCM mode,and app options "mv.incomplete_upf.iso_infer_csn_rule" will not be effective.

### What Next

DCM mode is enabled with set_early_data_check_policy, set option with corresponding DCM checks.

### See Also

- set_early_data_check_policy

---

## UPF-5192

(information) DCM mode is enabled, app options "mv.incomplete_upf.ls_infer_csn_rule" will not be ignored. Use corresponding DCM check "mv.cells.ls_missing_csn" to set CSN infer rule.

### Description

This message indicates that the design is in DCM mode,and app options "mv.incomplete_upf.ls_infer_csn_rule" will not be effective.

Feedback

### What Next

DCM mode is enabled with set_early_data_check_policy, set option with corresponding DCM checks.

### See Also

- set_early_data_check_policy

---

## UPF-5193

(information) DCM mode is enabled, app options "mv.incomplete_upf.ret_infer_csn_rule" will not be ignored. Use corresponding DCM check "mv.cells.ret_missing_csn" to set CSN infer rule.

### Description

This message indicates that the design is in DCM mode,and app options "mv.incomplete_upf.ret_infer_csn_rule" will not be effective.

### What Next

DCM mode is enabled with set_early_data_check_policy, set option with corresponding DCM checks.

### See Also

- set_early_data_check_policy

---

## UPF-5194

(information) DCM mode is enabled, app options "mv.incomplete_upf.psw_infer_csn_rule" will not be ignored. Use corresponding DCM check "mv.cells.psw_missing_csn" to set CSN infer rule.

### Description

This message indicates that the design is in DCM mode,and app options "mv.incomplete_upf.psw_infer_csn_rule" will not be effective.

### What Next

DCM mode is enabled with set_early_data_check_policy, set option with corresponding DCM checks.

### See Also

- set_early_data_check_policy

Feedback

## UPF-5195

(information) DCM mode is enabled, app options "mv.incomplete_upf.ao_infer_csn_rule" will not be ignored. Use corresponding DCM check "mv.cells.ao_missing_csn" to set CSN infer rule.

**Description**

This message indicates that the design is in DCM mode,and app options "mv.incomplete_upf.ao_infer_csn_rule" will not be effective.

**What Next**

DCM mode is enabled with set_early_data_check_policy, set option with corresponding DCM checks.

**See Also**

- set_early_data_check_policy

## UPF-5196

(information) DCM mode is enabled, app options "mv.incomplete_upf.optimization_control" will not be ignored. Use corresponding DCM check "mv.buf.mv_violation" to set buffering rule.

**Description**

This message indicates that the design is in DCM mode,and app options "mv.incomplete_upf.optimization_control" will not be effective.

**What Next**

DCM mode is enabled with set_early_data_check_policy, set option with corresponding DCM checks.

**See Also**

- set_early_data_check_policy

# UR

## UR-001

(Error) No regions associated with the flat-voltage-area %s.

Feedback

**Description**

The flat-voltage-area does not have any regions defined. Utilization can not be calculated for the voltage-area with no regions.

---

## UR-002

(Warning) Capacity area is zero

**Description**

The accumulated area of all the exclusion regions covers the entire capacity region.

---

## UR-003

(Error) No regions associated with the move-bound %s.

**Description**

The move-bound has no regions defined. Utilization cannot be defined for the move-bound with no regions.

---

## UR-004

(Error) Move bound %s completely overlaps with other exclusive move bounds.

**Description**

All the regions associated with the given move bound overlap with other exclusive move bounds. Capacity of a move bound is the effective region in the move bound, after deducting the regions overlapping with other exclusive move bounds. In this case, the capacity of the given move bound is zero.

---

## UR-005

(Error) Effective shapes of site-array %s is not defined.

**Description**

The given site-array does not have the effective shapes defined. Utilization of this site-array cannot be reported.

---

## UR-006

(Error) Site-array not defined for block %s.

Feedback

**Description**

Site-array has not been defined for the block. Utilization report cannot be based on site-array.

---

## UR-007

(Error) Utilization configuration '%s' does not exist.

**Description**

The requested utilization configuration is not defined.

**What Next**

Please provide a valid configuration name

**See Also**

- create_utilization_configuration
- get_utilization_configurations

---

## UR-008

(Warning) A configuration with name '%s' already exists.

**Description**

A utilization configuration with the same name already exists. Use '-force' option to overwrite the values in the existing configuration.

**See Also**

- create_utilization_configuration

---

## UR-009

(Warning) Existing configuration '%s' has same values for capacity and inclusions/exclusions.

**Description**

An existing configuraiton has same capacity and inclusion/exclusion values. Use '-force' option to overwrite the configuration name.

Feedback

**See Also**

- create_utilization_configuration

---

## UR-010

(Error) Could not find the utilization configuration in the given scope.

**Description**

The requested utilization configuration was not found in the given scope. Use 'get_utilization_configurations -scope <scope>' to get the available configs in the given scope.

**See Also**

- get_utilization_configurations

---

## UR-011

(Error) Cannot store utilization configurations in %s '%s', as it is %s.

**Description**

The given library is read-only and cannot be used to store the utilization configurations. Use a writable library or choose another scope.

**See Also**

- create_utilization_configuration

---

## UR-012

(Error) Cannot delete utilization configuration(s) in %s '%s', as it is %s.

**Description**

The given utilization configuration is associated with a read-only library(or tech) and hence cannot be deleted.

**See Also**

- create_utilization_configuration

Feedback

## UR-013

(Error) Block '%s' does not have any site-rows.

**Description**

Site-rows have not been defined for the block. Utilization report cannot be based on site-row.

## UR-014

(Error) Cannot store utilization configurations in tech '%s', as it is associated with a read-only library '%s'.

**Description**

The given tech is read-only and cannot be used to store the utilization configurations.

**See Also**

- create_utilization_configuration

## UR-015

(Error) Cannot modify the utilization configuration '%s', as it is associated with a read-only %s

**Description**

The utilization-configuration belongs to a read-only library/tech and it cannot be modified.

**See Also**

- create_utilization_configuration

## UR-016

(error) Category '%s' cannot be specified in both '-include' and '-exclude' options.

**Description**

Inclusion and exclusion categories must be distinct.

**What Next**

Correct the option value to avoid repeatation of the same category.

Feedback

**See Also**

- create_utilization_configuration

---

## UR-017

(Information) Using user default configuration with name '%s' in scope %s for utilization reporting.

**Description**

Reports the user defined configuration implicitly used for utilization computation.

**See Also**

- create_utilization_configuration

---

## UR-018

(Error) The minimum grid size must be 3 times the height %0.4f of placement site_def '%s'.

**Description**

The specified value of "-grid_size" option is incorrect. Please specify a value at-least 3 times the height of the reported site_def of minimum height.

---

## UR-019

(Error) Effective shapes of site-array %s doesn't overlap with the user specified region.

**Description**

The user specified region is completely outside the capacity of the reported site-array. Utilization of the site-array cannot be reported for the specified region.

**What Next**

Please correct the user specified region to overlap with the default or bottom-most site-array for utilization of capacity_type site_array.

---

## UR-020

(Error) The specified grid size %0.4f results in %u sub-regions which is too high.

Feedback

**Description**

Utilization of these many sub-regions seems impractical as the reporting would be difficult to analyze and comprehend.

**What Next**

Please reduce the number of sub-regions by increasing the grid-size or use the -region option to speficy a smaller region of reasonable area.

---

## UR-021

(Error) %s '%s' and %s '%s' are not same.

**Description**

In hybrid mode, width of composite site-def and corresponding elementary site-defs should be same. Also, height of composite site-def and sum of elementary site-defs should be same.

**What Next**

Rectify the input data as per details in error message.

---

# VER

---

## VER-1

(warning) %s The value %s is too large for the numeric data type being used

**Description**

This is an overflow condition detected during constant evaluations of the RTL when non-zero bits carry beyond the numeric precision (width) of the expression's data type.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

---

## VER-2

(warning) %s Unknown debug option %s

**Description**

This is a syntax error.

Feedback

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

---

## VER-3

(error) %s always_comb do not support RP(Relative Placement) registerbank, please remove RP directive '%s'.

**Description**

This error message occurs when relative placement register bank directives are put inside *always_comb*.

**What Next**

Remove the relative placement directives that are inside *always_comb*.

---

## VER-4

(warning) %s Incompatible port connection, port %d of instance %s in module %s.

**Description**

This is a syntax error.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

---

## VER-5

(warning) %s Mismatch in number of ports for instance %s

**Description**

This is a syntax error.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

---

## VER-6

(warning) %s The unsized Verilog number %s was left-truncated from %d bits to 32 bits, which can be prevented with sized %d%s.

Feedback

**Description**

During analysis of the Verilog or SystemVerilog RTL a based literal constant without an explicit size constant before the apostrophe was encountered for which the requested digits would need more than 32 bits to represent. The leftmost bits were truncated to fit into the default size of 32 bits.

**What Next**

Insert an explicit size specifier before the apostrophe.

---

## VER-7

(information) %s %s.

**Description**

This informational message is issued when analyzing or reading Verilog or SystemVerilog RTL when the variable hdlin_analyze_verbose_mode is overridden to a positive value. For an explanation of how that variable works, please see its man page.

**What Next**

This is an informational message. If you would prefer that it not be issued, please examine the current setting of hdlin_analyze_verbose_mode.

---

## VER-8

(warning) %s Hierarchical component name %s is a macromodule and not expandable.

**Description**

This is a syntax error.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

---

## VER-9

(warning) %s Illegal synopsys label pragma, it will be ignored.

**Description**

When synopsys label pragma is put wrong place, this warnning message will be dumped. Presto ignores this label pragma.

Feedback

**What Next**

Put the label pragma into right place.

---

## VER-12

(error) %s Hierarchical component name %s is not a module, task, function, or block

**Description**

This is a syntax error.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

---

## VER-15

(error) %s The call to '%s' does not have enough arguments to satisfy all format specifiers.

**Description**

This error message occurs if you supply a format string when using the '$display' (or similar) system function, and there are not enough subsequent parameters to supply each token in the specifier.

In the following example, the format specifier "%s%s%d" contains three tokens and requires at least three values, which are a, b, and c.

```
$display ("%s%s%d", a, b, c);
```

**What Next**

Correct any occurrences of insufficient parameters for the specified format string in the Verilog code.

---

## VER-16

(error) %s First parameter of $dumpvars must be an integer

**Description**

This is a syntax error.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

Feedback

## VER-17

(error) %s Incompatible number of arguments, usage versus declaration.

**Description**

This is a syntax error.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

## VER-18

(error) %s Bounds %s for vector %s must be integer.

**Description**

This is a syntax error.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

## VER-19

(error) %s +liborder and +librescan conflict; only one can be specified.

**Description**

This is a syntax error.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

## VER-20

(error) %s Package %s does not export the name '%s'.

**Description**

This message issues when the System Verilog scope operator (::) tries to extract content from the specified package, but the name was either not found in the global scope of that package, or its package-level definition is marked as not exportable.

Feedback

### What Next

Check that the package name is correct and that the source code with the desired definition has been analyzed without errors more recently than any other package with the same name. If the supplying package is at fault, re-analyze it and also the present file. If the consuming reference is not correct, adjust the RTL and re-analyze. The setting of hdlin_sv_packages can also affect whether names imported into a package can be re-exported.

### See Also

- VER-26
- VER-934

---

## VER-21

(error) %s Multiple conflicting definitions of '%s' are available via wildcard imports.

### Description

This message issues when you use a name which you have not previously declared that has more than one distinct definition supplied through several wildcard import routes ( import pkg_A::*, pkg_B::*; ) into the same surrounding scope.

When all imported packages are analyzed with hdlin_sv_packages set to *dont_chain*, this error corresponds to the prohibition in section 19.2.1 of IEEE-1800-2005: a name may come from at most one wildcarded package. Packages analyzed with hdlin_sv_packages set to *chain* or *enable*, may also export names that they imported from elsewhere. Thus a single common definition for name might arrive via several paths, VER-21 issues only when multiple source definitions are import candidates, and no import is more local than the others at the point of use.

### What Next

There are many RTL source code adjustments of the import statements and/or the name reference that could clear this condition: You could place the wildcard imports at different levels in the scope stack so that the prefered supplier shadows the unwanted supplier. You could add an explicit "import desired_supplier::name;" you could prefix the package scope extraction operator "desired_supplier::" to the name reference to indicate the desired package. You might delete the import of an unused package that causes the conflict. You could correct a spelling to avoid the collision. You could supply a missing local declaration of the name in the overloaded scope to override the wildcard imports. You could explicitly declare the name with a local meaning.

If you believe the RTL as written is correct, check that the current search_path reaches .pvk files that were produced by successfully analyzing the current versions

Feedback

of these packages' source files. The setting of *hdlin_sv_packages at the time those packages were analyzed can also affect whether names imported into them will be re-exported, which might be either producing or exposing the observed name conflict.*

**See Also**

- ELAB-106

---

## VER-22

(error) %s Too many parameters found on task %s.

**Description**

This is a syntax error.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

---

## VER-23

(error) %s The %s parameter is not completely writable and it is being passed to an OUTPUT or INOUT formal parameter.

**Description**

This is a syntax error.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

---

## VER-25

(error) %s Module '%s' is redefined.

**Description**

You receive this message to inform you that during execution of the *analyze* or *read* command the compiler has encountered in your design a module declaration that has the same name as a previously defined module. Each module name must be unique.

**What Next**

Rename one of the modules so that all modules have unique names.

Feedback

## VER-26

(warning) %s The package %s has already been analyzed. It is being replaced.

### Description

You receive this warning message if a package with the same name was analyzed by a previous read or analyze command, possibly during a different DC shell session. This can occur in three possible ways:

- The file currently being analyzed has already been analyzed.
- A package with the same name is declared in another file analyzed in the current work directory.
- You have modified the package's source file and are reanalyzing it.

The previous package contents are removed, and possibly overwritten by the declaration currently being analyzed.

### What Next

This is only a warning message.

However, you can check the overwritten file. If the result is not what you intended, change the package name and run the command again.

All references to a revised package should be re-analyzed to insure that they are coherent. Failure to do this is one source of ELAB-106 warnings, VER-20, and various type-mismatch errors.

### See Also

- ELAB-106
- VER-20
- VER-930

## VER-27

(error) %s Package %s has already been used or declared during this command.

### Description

This error issues if a package is used before being (re)declared within the same analyze or read command. There are many ways this condition can arise. In addition to use-before-declaration or name collision within a single file: the file currently being analyzed, a copy of that file, or another file declaring or using a package with the same name may

Feedback

have appeared earlier in the file list of this command, been `include'd there, or been indirectly imported by earlier package reference(s).

**What Next**

Arrange your command line(s) so that every package is analyzed or read exactly once before its first use. Conditional compilations such as the following allow declarations to be parsed just once.

```
`ifndef PKG_X
`define PKG_X
package X;
   . . .
endpackage
`endif
```

Use a separate analyze command to update or change a package's definition. All references to a re-analyzed package should also be re-analyzed to insure that they are coherent. Failure to do this is one source of ELAB-106 ("homograph") warnings and type-mismatch errors.

---

## VER-28

(warning) %s The configuration %s has already been analyzed; it is being replaced.

**Description**

You receive this warning message if a SystemVerilog config with the same name as the one cited was analyzed by a previous read or analyze command, possibly during a different DC shell session. This can occur in three possible ways:

- The file currently being read or analyzed has already been read or analyzed.
- A config with the same name is declared in another file analyzed into the current work directory.
- You have modified the config's source file and are now reanalyzing it.

The previous configuration description is removed; if the current description is successfully analyzed, it will replace the defunct version. If it fails to analyze, no configuration of this name will exist come elaborate.

**What Next**

This is only a warning message.

However, you can check the overwritten file. If the result is not what you intended, change the one of the configuration names and re-analyze the config you destroyed.

Feedback

## VER-29

(error) %s Configuration %s has already been declared.

### Description

This error issues if more than one config with the cited name is declared within the same analyze or read command. There are many ways this condition can arise. In addition to a blatant name collision in the same source file: the file currently being analyzed, a copy of that file, or another file declaring or using a package with the same name may have appeared earlier in the file list of this command or been `include'd by some earlier file.

### What Next

Arrange your command line(s) and/or `include(s) so that configurations enter the input stream no more than once. Conditional compilation directives could help to prevent repetition of the same declarations.

Use a separate analyze command to update or change a configuration.

## VER-30

(error) %s Stack overflow. %s

### Description

This error message occurs when RTL compilation cannot complete because the internal stack overflowed. All applications and tools need an internal stack to execute. This stack can grow up to a limit set by the "limit" command on csh interpreters, or the "ulimit" command on sh interpreters.

The following conditions may cause the internal stack to overflow:

- Your design has an infinite or very long iterative loop and you set a high value for the *hdlin_while_loop_iterations* variable.
- Your design has a deeply nested construct, such as a deeply parenthesized expression. This occurs with code generated by other tools, but not with hand coded sources.

Note that this message may be shown for loops that finish in simulation, but are infinite in synthesis. One example of an infinite loop, and so is unsupported, is this design:

```
module sum (input [3:0] in, output reg [7:0] out);
reg [3:0] i;
always @* begin
  i = in;
  out = 0 ;
  while (i>0) begin
```

Feedback

```
    out = out + i;
    i = i - 1;
  end
end
endmodule
```

This example will not finish in synthesis because the terminating condition depends on a port and so is never a proper constant.

**What Next**

Review your code and ensure that the terminating condition of an iterative loop will eventually be constant true. It is always difficult to determine if an iterative loop will eventually finish, but this message is usually shown when it is infinite.

In very unusual cases, the VER-30 message is shown because the stack size limit is not big enough. Check the current stack size limit with the "limit stacksize" command on csh interpreters, or the "ulimit -s" command on sh interpreters. Increase the stack size significantly, for example with the "limit stacksize 100m" command on csh interpreters, or the "ulimit -S -s 104857600" command on sh interpreters. Restart Design Compiler. Analyze and elaborate your design again. If the message persists or the current stack size is unlimited, re-evaluate the possibility of an infinite loop.

Note that if your design has an infinite loop and the stack size limit is increased, Design Compiler will also take more time to show the stack overflow message.

If your design has been generated by another tool, this error message may be due to a deeply nested construct. In this case increase the stack size limit as described above, restart Design Compiler, and analyze and elaborate your design again.

**See Also**

- ELAB-900
- ELAB-901

---

## VER-35

(warning) %s Register (%s) is the target of both blocking and nonblocking assignments in the same process.

**Description**

This is a syntax error.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

Feedback

## VER-36

(error) %s Internal error: %s, in %s at line %d

### Description

This message issues when RTL compilation cannot complete because the compiler reached some detectably invalid internal state. Internal cross-checks are used extensively in the Presto RTL compilers to prevent software bugs from producing erroneous results - often called "bad logic". RTL code which conforms closely to the IEEE standards ( 1364, 1800, or 1076 ) and lies within the synthesizable subsets of those languages should, of course, not encounter internal compiler errors.

### What Next

Contact Synopsys support who can help you further diagnose this problem and can often help you find a coding workaround, or a compiler release where the bug has been fixed.

If you encounter an internal error when using the most recent release from Synopsys, submit a testcase that reproduces the problem to the Synopsys Support Center by using Enter A Call at http://solvnet.synopsys.com/EnterACall.

For information about the latest software releases, go to the Synopsys SolvNet Release Library at http://solvnet.synopsys.com/ReleaseLibrary.

For information about creating, packaging, and sending a testcase, go to http://www.synopsys.com/testcase.

### See Also

- ELAB-255
- VER-37

## VER-37

(error) %s Internal assertion failure.

### Description

This error message occurs if the HDL Compiler encounters an internal assertion failure and cannot proceed. The command shell is not affected and there is no loss of data.

The HDL Compiler contains thousands of internal checks on its operation designed to catch miscompilation problems as close as possible to their source and to prevent the production of incorrect results.

Feedback

**What Next**

If you encounter an internal assertion failure when using the most recent release from Synopsys, submit a testcase that reproduces the problem to the Synopsys Support Center by using Enter A Call at http://solvnet.synopsys.com/EnterACall.

For information about the latest software releases, go to the Synopsys SolvNet Release Library at http://solvnet.synopsys.com/ReleaseLibrary.

For information about creating, packaging, and sending a testcase, go to http://www.synopsys.com/testcase.

---

## VER-38

(error) %s Internal error: %s

**Description**

This is a syntax error.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

---

## VER-40

(error) %s Too many errors; can't continue.

**Description**

The Presto HDL Compiler issues this error from the read, analyze, or elaborate commands when it has encountered too many errors to continue the compilation.

**What Next**

Correct all of the errors reported before this one, and repeat the command.

---

## VER-41

(error) %s Unable to open file `%s': %s.

**Description**

This error message occurs when you issue a *read* or *analyze* command on a file that cannot be opened for reading or writing.

Feedback

**What Next**

To write to a file, make sure that the directory named by the *hdlin_output_dir* variable is writable, or change *hdlin_output_dir* to point to a writable directory. To read a file, make sure the file exists.

---

## VER-43

(error) %s Out of memory.

**Description**

This may indicate syntax errors.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

---

## VER-44

(warning) %s Little argument or return value checking implemented for system task or function %s.

**Description**

This is a syntax error.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

---

## VER-51

(error) %s internal error: %s

**Description**

This is a syntax error.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

---

## VER-55

(error) %s Illegal character in module, interface, or package name.

Feedback

**Description**

You receive this error message when the name you specified for a module, interface, or package cannot be directly used to name a file in the host file system. This is usually due to some reserved characters, such as '/' or ':'. Names in the standard *definitions name space* exist as file system objects and must conform to host or network naming conventions in addition to Verilog lexical rules.

For example, \a/b/c is an invalid module name.

**What Next**

Change the module name in the relevant code to conform to the rules for file names in the host file system and run the command again.

---

## VER-59

(error) %s Array index out of bounds %s.

**Description**

This is a syntax error.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

---

## VER-60

(error) %s Array index out of bounds %s.

**Description**

This is a syntax error.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

---

## VER-61

(warning) %s Statement unreachable %s.

**Description**

This warning is issued when statements inside of conditional branches can't be reached, because the condition is always false or a previous condition is always true.

Feedback

**What Next**

Check the conditions of conditional statements like: if, case, when, ?, ...

---

## VER-64

(warning) %s Overwriting existing design element '%s' with newer version.

**Description**

You receive this message to inform you that during execution of the *analyze* or *read* command the compiler has encountered in your design a module or interface declaration that has the same name as a previously defined module or interface. This is non-standard according to 3.13(a) of IEEE Std 1800-2017. An identifier defined within a namespace is unique to that namespace and cannot be redefined within that namespace.

**What Next**

Rename one of the modules so that all modules have unique names, or guard definitions with an `ifndef compiler directive.

---

## VER-65

(error) %s Option 'hdlin_decoder_min_use_percentage' must be in the range 0..100 (currently %d).

**Description**

This is a syntax error.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

---

## VER-66

(warning) %s Won't infer decoders with input width > %d; option '%s' set too large.

**Description**

This is a syntax error.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

Feedback

## VER-86

(warning) %s Comparison against 'x' values not inside a casex is always false.

**Description**

This is a syntax error.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

## VER-87

(warning) %s Begin instance %s... with a design identifier.

**Description**

This warning issues when the hierarchical path given for an instance clause does not begin with one of the design identifiers listed by the configuration. No topmost or hierarchical use of the config can apply this instance clause as written. That is probably not what you intend.

**What Next**

Change the indicated code to conform to IEEE Stds 1364-2005 or 1800-2005 or later.

## VER-97

(error) %s No input files specified.

**Description**

You receive this error message because no input files were specified with the *analyze* or *read* command for the Presto HDL Compiler. These commands require that one or more input files be specified.

**What Next**

Execute the *analyze* or *read* command and specify one or more input files.

## VER-98

(error) %s Relative placement group '%s' is defined multiple times.

**Description**

This error message occurs when a relative placement group is defined multiple times.

Feedback

**What Next**

Remove the duplicate relative placement group and run the command again.

---

## VER-99

(error) %s The column or row of relative placement group should be integer, not `%s'.

**Description**

This error message occurs when the column or row of the relative placement group is not an integer.

**What Next**

Specify the correct column or row for the relative placement group and run the command again.

---

## VER-100

(error) %s Cell %s was placed outside of the boundaries of rp_group %s at column %d row %d.

**Description**

This error message occurs when the user defines an explicit size of the rp_group (column and row) using the synopsys rp_group pragma, an later at least one cell was placed outside of the predefined boundaries.

**What Next**

There are three ways to solve this error: 1) Specify the correct column and row size of the group modifying the rp_group definition to a size big enough to allocate all RP placed cells. 2) Specify or modify the rp_fill pragma to ensure all cells are placed within the boundaries of the defined RP group. 3) Remove the rp_group size definition to allow RP to freely allocate the cells without a hardcoded limit that triggers the error.

---

## VER-102

(warning) %s Variable '%s' does not accept the value '%d'. The value has been reset to a compatible value.

**Description**

This warning message occurs when a variable uses a value that is not supported by the current flow. The variable is reset to a compatible value as documented in the man page of the variable.

Feedback

### What Next

Verify that the value for the variable is within the allowed range, and that the value matches your design requirement.

---

## VER-103

(warning) %s Pragma is obsolete or not implemented. It will be ignored.

### Description

A pragma is a compiler directive in the source code. The Presto HDL compiler does not implement certain pragmas that the Synopsys HDL Compiler does. This warning, which has occurred during *analyze* or *read* command activity, notifies you that the pragma in the design will not have the same effect it would have if you were to compile the design using the Synopsys HDL Compiler.

The Presto HDL compiler implements most of the commonly used compiler directives, but not some of the less frequently used ones.

### What Next

If your design requires the use of the pragma the message refers to, use the Synopsys HDL Compiler to analyze and elaborate your design.

---

## VER-104

(warning) %s The '%s' construct is not supported. It will be ignored.

### Description

This warning message occurs during an *analyze* or *read* command activity, and notifies you that the given construct in your design is not supported for synthesis.

### What Next

Check your design to verify that it still functions correctly, even though the construct the message refers to is ignored.

To avoid this warning, surround the construct with the following compiler directives:

```
`ifndef SYNTHESIS

     <construct>

`endif
```

When the Presto HDL Compiler sees these compiler directives, it does not interpret the construct.

Feedback

## VER-105

(warning) %s Call to function '%s' with no parameters is not standard Verilog.

### Description

You receive this message because, during a *read* or *analyze* command activity, the compiler encountered a call to a function or functions that used parentheses to indicate an empty parameter list.

IEEE standard Verilog requires that a function have at least one input argument. The Presto HDL compiler accepts functions with no arguments; however, many Verilog tools do not.

```
...
     x = f();  // non-standard call to zero-argument function
   ...
```

### What Next

Consider adding at least one input argument to the function in question to conform to IEEE standards.

### See Also

- VER-175

## VER-106

(error) %s Function declaration redeclares symbol '%s'.

### Description

You see this message because you have assigned a function declaration to a symbol that has already been defined.

In the following example, which demonstrates such a situation, the symbol *f* is declared twice: first as a register and then as a function:

```
reg f;          // initial declaration of 'f'

   function f;     // illegal redeclaration of 'f'
   ...
   endfunction
```

The example elicits the VER-106 message.

Feedback

### What Next

Rename either the function declaration or the other declaration and all corresponding uses of the symbol.

---

## VER-107

(error) %s Functions cannot contain blocking delays.

### Description

You are seeing this message because, during the execution of the *read* or *analyze* command, the compiler has encountered blocking delays in your design. The Verilog standard states that functions execute in one (simulation) time unit and cannot contain time-controlling statements, such as blocking delays.

The following example demonstrates the incorrect use of blocking delays in a function body:

```
...
  function f;
   input a;
   begin
    #10;       // illegal use of blocking delay inside a function
    f = a;
   end
  endfunction
  ...
```

### What Next

For synthesis, remove the the blocking delays because the compiler does not accept them and terminates with this error.

---

## VER-108

(error) %s Functions cannot contain event control statements.

### Description

You receive this error message because you have attempted to use an event control statement inside of a function. According to the Verilog standard, functions shall execute in one simulation time unit, and are not allowed to contain time-controlling statements. Therefore, event control statements are not allowed inside functions.

Feedback

The following example demonstrates the illegal use of an event control statement in a function body:

```
...
  function f;
   input a;
   begin
    @(posedge clk);   // illegal use of an event control statement in a
 function.
    f = a;
   end
  endfunction
  ...
```

**What Next**

Rewrite the source code so that the functions do not contain any time-controlling statements.

---

## VER-109

(error) %s Function arguments cannot be declared as 'inout'.

**Description**

You receive this error message because you have declared a function argument as 'inout'. According to the Verilog standard, functions can have arguments declared as 'input' only. Declaring a function argument as 'inout', as illustrated in the following example, results in this error message.

```
...
  function f;
   inout a;    // illegal declaration of a function argument as 'inout'.
   ...
  endfunction
  ...
```

**What Next**

Rewrite the code so that the function has 'input' arguments only. If you cannot avoid using 'inout' arguments, consider using a task instead of a function.

---

## VER-110

(error) %s Function '%s' not defined.

**Description**

You receive this error message because no function with the given name has been defined.

Feedback

### What Next

Check to see if the specified function name was misspelled. If not, add a function declaration to define the missing function.

---

## VER-111

(error) %s Function arguments cannot be declared as 'output'.

### Description

You receive this error message because, according to IEEE Std 1364, functions can only have arguments declared as 'input'. Declaring a function argument as 'output', as illustrated in the following example, results in this error message.

```
...
  function f;
   output a;    // illegal declaration of a function argument as
 'output'.
   ...
  endfunction
  ...
```

### What Next

Rewrite the code so that the function only has 'input' arguments. If you cannot avoid using 'output' arguments, consider using a task instead of a function.

---

## VER-112

(error) %s Illegal function declaration using empty parameter list.

### Description

You receive this error message because the function declaration was written with an empty parameter list.

The following example shows a function declaration that would generate this error.

```
function foo();
   ...
   endfunction
```

The correct function declaration:

```
function foo;
   ...
   endfunction
```

Feedback

### What Next

Rewrite the function declaration as shown in the example.

---

## VER-113

(error) %s Functions cannot enable tasks.

### Description

You receive this error message because you attempted to enable a task from a function. According to the IEEE Std 1364, tasks cannot be enabled from functions. The following is an example of this situation.

```
task t;
    ...
  endtask


  function f;
    ...
    t;          // error: tasks cannot be enabled from functions.
  endfunction
```

### What Next

Rewrite the source code so that tasks are enabled only from processes or other tasks.

---

## VER-114

(error) %s Functions cannot contain 'wait' statements.

### Description

You receive this error message when you attempt to use a 'wait' statement inside of a function. According to the IEEE Verilog Standard 1364, functions shall execute in one simulation time unit and are not allowed to contain time-controlling statements. Therefore, 'wait' statements are not allowed inside functions.

The following example demonstrates the illegal use of a 'wait' statement in a function body.

```
...
  function f;
    input a;
    begin
      wait (!a) f = a;   // 'wait' not allowed inside a function.
    end
  endfunction
  ...
```

Feedback

### What Next

Rewrite the Verilog source code so that functions do not contain any time-controlling statements.

---

## VER-115

(error) %s Redefinition of compiler directive '%s' not supported.

### Description

Redefining compiler directives using 'define' is not allowed. The following example is the illegal attempt to redefine the compiler directive 'define' itself:

```
`define define 2      // error: illegal redefinition of a
                         // compiler directive
```

### What Next

If a redefinition of the the original directive was not intended, just rename the symbol such that it does not collide with a compiler directive.

---

## VER-116

(error) %s Redeclaration of symbol '%s' as parameter.

### Description

You receive this message if a symbol in a parameter declaration has already been declared, either as a parameter or as something else. In the following example, "in" is first declared as an input, then erroneously redeclared as a parameter.

```
input in;
 ...
 parameter in = 1'b0;     // erroneous redeclaration of input 'in'
                          // as a parameter.
```

### What Next

Decide which of the declarations is the intended one and remove or change the name in the other declaration. If you change the name in a declaration, ensure that you change all occurrences of that name.

---

## VER-117

(error) %s Read-only expression '%s' passed to 'output' or 'inout' argument of a task or function.

Feedback

### Description

Task (or SV function) arguments declared with direction 'output' or 'inout' must receive an actual register, selection within a register, or other legal target of an assignment statement.

For example, the following two task calls are incorrect because neither an input nor a wire are registers, so they are not valid arguments to task output or inout parameters.

```
input a;
   wire b;

   task t;
     output x;
     begin
       x = 1'b0;
     end
   endtask

  always @(posedge clk)
    begin
      t(a);       // error: input is not a register
      t(b);       // error: wire is not a register
    end
```

### What Next

Check the argument name or its positional order. If the task or function definition does not assign to the formal argument, restrict its direction to 'input'. Otherwise change the calling RTL to receive the task's (or function's) assignment into a register variable.

### See Also

- VER-118

---

## VER-118

(error) %s Constant passed to an 'output' or 'inout' argument of a task or function.

### Description

Task (or SV function) arguments declared with direction 'output' or 'inout' must receive an actual register, selection within a register, or other legal target of an assignment statement.

The following task call is illegal because 1b0 is a constant value and not a valid argument to a task output or inout parameter.

```
task t;
    output x;
    ...
   endtask
```

Feedback

```
...

t(1'b0);     // error: passing a constant value to an
             // 'output' parameter
```

### What Next

Check the argument name or its positional order. If the task or function definition does not assign to the formal argument, restrict its direction to 'input'. Otherwise change the calling RTL to receive the task's (or function's) assignment into a register variable.

### See Also

- VER-117

---

## VER-119

(error) %s Wrong number of arguments in call to function or task '%s'.

### Description

You receive this error message because the actual number of arguments in the function or task call does not match the number of formal arguments expected by the corresponding function or task.

### What Next

Specify the correct number of arguments. Refer to the function or task declaration to determine the number and order of the expected arguments.

---

## VER-120

(error) %s Port '%s' not declared as 'input', 'output' or 'inout'.

### Description

The IEEE Verilog Standard 1364 requires that each port listed in a module port list be declared as input, output, or inout.

You receive this error because a port listed in the module port list is not declared subsequently as input, output, or inout.

The following example shows a correct port declaration:

```
module m (i, o);
     input i;       // omitting this input declaration is an error.
     output o;
     reg o;

     always ...
```

Feedback

```
    endmodule
```

If the input declaration of port *i* is not present, an error is issued.

**What Next**

Add the missing port declaration, as either input, output, or inout.

---

## VER-121

(error) %s Redeclaration of port '%s' is not allowed.

**Description**

You receive this message because, during *analyze* or *read* command activity, the Presto HDL Compiler has detected that the same port is declared twice inside the same function, task, or module. Following is an example that illustrates this type of error:

```
function f;
   input a;
   input a;      // redeclaration of already declared port
    ...
  endfunction;
```

**What Next**

Remove or rename as appropriate one of these ports.

---

## VER-122

(error) %s Redeclaration of symbol '%s' as a real number is not allowed.

**Description**

You receive this message because, during *analyze* or *read* command activity, the Presto HDL Compiler has detected a symbol declared twice, which is not valid.

In the following example, "r" is declared both as an integer and as a real number.

```
integer r;
   real r;   // error: invalid redeclaration of "r" as a real number
```

**What Next**

Remove or rename as appropriate one of the declarations.

Feedback

## VER-123

(error) %s Redeclaration of symbol '%s' not allowed.

**Description**

This is a syntax error.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

## VER-124

(error) %s Task or function argument must be specified.

**Description**

This message issues when the Presto HDL Compiler parses an incomplete parameter list, or one with empty comma-delimited positions. It is an error to omit a parameter (argument) in the list used to enable (invoke) a task or function (unless, in SystemVerilog, the corresponding formal argument was declared with a default value initializer). Default initializers are not yet supported by Presto HDL.

The following example demonstrates a correct and an incorrect task call:

```
task not;
   input a;
   output y;
   ...
  endtask

  ...
  not(,o);        // error: incomplete parameter list: first argument
 missing
  not(a,b);       // correct task call
  ...
```

**What Next**

Check the location of commas in this argument list. Specify any missing arguments.

## VER-125

(error) %s Called function (or task) '%s' not defined as a function (or task).

Feedback

### Description

This syntax error issues from *analyze* or *read* commands when no suitable definition of a named function or task can be found in the lexical scopes immediately enclosing the cited invocation of that function or task.

The function or task name is either not defined at all, defined as something other than a function (or task), defined as a task but being used as a function, or vice versa.

You may be seeing this error due to a language restriction (in the synthesis subset). Verilog and SystemVerilog simulators support a larger scope for invoking function and task definitions; they include all scopes in the instance hierarchy above the call site (i.e. the definition can reach references made in the same scope where it is defined, in any module instance in that scope, and recursively in any modules they may instantiate). But in synthesis by separate-compilation, that hierarchy may be edited, expanded, and/or relinked long after the function or task call has been elaborated. So-called "hierarchical" tasks and functions are not supported, except that they can be invoked lexically-before they are defined within a single scope, or its lexically-nested scopes.

In Verilog-2005 and SystemVerilog standards, functions (or tasks) defined within anonymously scoped generate blocks are not accessible outside that block. For example:

```
module m #(parameter P=32) (output o);
    generate
      if (P>16)
        function f (input i);
           f = 1'b0 ;
        endfunction
      else
        function f (input i);
           f = 1'b1 ;
        endfunction
    endgenerate

    assign o = f(1'b0);          /* Function 'f' is not visible  */
                                 /* outside the generate block */
  endmodule
```

### What Next

Check that the name of the function (or task) is spelled correctly, and that the name is a valid definition of the function (or task).

If the identifier refers to a function or task defined elsewhere in the design hierarchy, copy or move the definition to a location that can be made commmon to all call sites. In Verilog, that would be an `include file brought into each module scope that needs the definition. In SystemVerilog it could additionally be a package from which the definition is imported, an interface instantiated by (or ported into) the calling modules, or the $unit scope of the input stream preceding all the call sites. Only the interface option allows signals from the

Feedback

defining scope to be used in the function, and these may also need to be listed individually in modports of the interface which are used to import the function or task defininion.

---

## VER-126

(error) %s Called task '%s' not defined as a task.

**Description**

You receive this message because, during *analyze* or *read* command activity, the Presto HDL Compiler has detected a task that is not properly defined.

The task, which a task call refers to, is not defined at all or it is defined as something other than a task.

**What Next**

Check to verify that the name of the task is spelled correctly. If it is, then you need to supply a different (correct) definition of the task.

---

## VER-127

(error) %s Called task '%s' not defined as a task.

**Description**

You receive this message because, during *analyze* or *read* command activity, the Presto HDL Compiler has detected a task that is not properly defined.

The task, which a task call refers to, is not defined at all or it is defined as something other than a task.

**What Next**

Check to verify that the name of the task is spelled correctly. If it is, then you need to supply a different (correct) definition of the task.

---

## VER-128

(error) %s Parameterless task call to task '%s' should not have parentheses.

**Description**

Task calls that do not pass any parameters should not use parentheses to indicate an empty parameter list.

Feedback

The following example demonstrates the right and the wrong way to call a task without parameters:

```
task t;
   ...
  endtask

  ...
   t();          // error: incorrect syntax for empty parameter list
   t;            // correct syntax without parentheses
  ...
```

**What Next**

Rewrite the task call as shown in the example above.

---

## VER-129

(warning) %s Intraassignment delays for blocking assignments are ignored.

**Description**

You receive this message because, during *analyze* or *read* command activity, the Presto HDL Compiler has detected an intraassignment delay specified for a blocking assignment. Such delays have no meaning in synthesis, and the compiler ignores them.

Following is an example of an intraassignment delay.

```
y = #10 a;
```

**What Next**

Synthesis ignores intraassignment delays for blocking assignments, so it is acceptable to remove them in instances such as this.

---

## VER-130

(warning) %s Intraassignment delays for nonblocking assignments are ignored.

**Description**

You receive this message because, during *analyze* or *read* command activity, the Presto HDL Compiler has detected an intraassignment delay specified for a nonblocking assignment. Such delays have no meaning in synthesis, and the compiler ignores them.

Following is an example of an intraassignment delay.

```
y = #10 a;
```

Feedback

### What Next

Synthesis ignores intraassignment delays for nonblocking assignments, so it is acceptable to remove them in instances such as this.

---

## VER-131

(error) %s Intra-assignment events for blocking assignments are unsupported in synthesis.

### Description

This error message occurs because intra-assignment events in blocking assignments are not supported for synthesis. For example:

```
y = @(posedge clk) a;
```

### What Next

Rewrite the block so it does not use an intra-assignment event.

---

## VER-132

(error) %s Intra-assignment events for non-blocking assignments are not supported.

### Description

Intraassignment events for non-blocking assignments are not supported during synthesis. For example:

```
y <= @(posedge clk) a;
```

The correct synthesizable version is:

```
y <= a;
```

### What Next

Remove the intraassignment event as shown above.

---

## VER-133

(warning) %s Intra-assignment repeat-event controls for blocking assignments are ignored.

Feedback

### Description

Intra-assignment repeat-event for blocking assignments do not have a meaning for synthesis and are ignored. For example:

```
y = repeat(3) @(posedge clk) a;
```

The correct synthesizable version is:

```
y = a;
```

### What Next

Take out the repeat event specification as shown above.

---

## VER-134

(error) %s Variable '%s' is the target of both blocking and nonblocking assignments in the same always block.

### Description

Mixing of blocking and nonblocking assignments to the same variable in the same always block is not supported in synthesis. For example:

```
always @(...) begin
    y <= a;
    y = a;
  end
```

Verilog simulator performance is not improved by the mixing of blocking and nonblocking assignments in the same always block.

### What Next

If the always block represents combinational logic, rewrite the always block to use only blocking assignments. If the always block represents sequential logic, rewrite the always block to use only nonblocking assignments.

---

## VER-135

(warning) %s Intra-assignment repeat-event controls for non-blocking assignment are ignored.

### Description

Intra-assignment repeat-event for non-blocking assignments do not have a meaning for synthesis and are ignored. For example:

```
y <= repeat(3) @(posedge clk) a;
```

Feedback

The correct synthesizable version is:

```
y <= a;
```

**What Next**

Take out the repeat event specification as shown above.

---

## VER-136

(warning) %s Function '%s' with non-empty body is mapped to '%s'; body will be ignored.

**Description**

By mapping a function to a module, the module (instead of the function body) determines the implementation of the function. This warning is given to remind you the body of the function will be ignored.

The following example contains a function for which this warning message will be issued:

```
function f;     // synopsys map_to_module add
   input a;
   input b;
   begin
    f = a+b;      // function body will be ignored.
   end
  endfunction
```

**What Next**

Make sure that the body of the function is indeed intended to be ignored in favor of the module the function is mapped to.

---

## VER-137

(error) %s Return port name '%s' conflicts with names of function input parameters.

**Description**

The return port name specified by the 'return_port_name' pragma must be different from all formal parameters of the function the pragma is attached to. This error indicates that there is a conflict with one of the formal parameter names.

In the following example such a conflict is demonstrated: The return port name is specified to be "a", but a function parameter with the same name exists.

```
function f;
        // synopsys map_to_module down
        // synopsys return_port_name a
   input a;
```

Feedback

```
   ...
  endfunction
```

To resolve this error, the name of the function parameter is changed to "b" in the following example:

```
function f;
        // synopsys map_to_module down
        // synopsys return_port_name a
   input b;
   ...
  endfunction
```

**What Next**

Change the name of the conflicting function parameter or the port name of the module the function is mapped to so that they don't conflict.

---

## VER-138

(warning) %s Extraneous argument for system function or task '%s' ignored.

**Description**

The system function has been passed an unexpected additional argument. This argument will be ignored.

**What Next**

Remove the extra argument from the call. Refer to the function/task declaration to find out which arguments are actually required.

---

## VER-139

(warning) %s Parameterless call to function '%s' should not have parentheses.

**Description**

In Verilog, calls to functions without parameters should not use parentheses to indicate an empty parameter list; only the function identifier is used to call the function.

The following example indicates the correct and incorrect way to write a parameterless Verilog function call:

```
...
     x = f();    // incorrect syntax: using parentheses
     x = f;      // correct syntax: using the function identifier only
   ...
```

Feedback

### What Next

Remove the parentheses as shown in the example above.

---

## VER-140

(warning) %s Nonblocking assignments and blocking delays in the same process; potential simulation or synthesis mismatch.

### Description

This warning message occurs during *analyze* or *read* command activity, when the Presto HDL Compiler detects a nonblocking assignment and a blocking delay in the same process. This can cause simulation or synthesis mismatches.

The following is an example that causes the warning message:

```
always @(posedge clk) begin
     #10;              // blocking delay is ignored
     out <= in;        // nonblocking assignment
   end
```

### What Next

Review the code carefully to explore the potential for a simulation or synthesis mismatch.

---

## VER-141

(error) %s Blocking delays and nonblocking assignments are not allowed in the same process.

### Description

You receive this message because, during *analyze* or *read* command activity, the Presto HDL Compiler has detected a nonblocking assignment and a blocking delay in the same process, which is invalid for synthesis.

Even though synthesis ignores blocking delays, it is important to recognize them in the two forms in which they occur. Blocking delays can stand alone or as part of an intra-assignment delay. The following example shows an instance of each form:

```
always @(posedge clk) begin
     a <= x;         // nonblocking assignment
     #10;            // blocking delay, not allowed with nonblocking
 assignment
     b = #10 y;    // intraassignment delay, not allowed with nonblocking
 assignment
   end
```

Feedback

### What Next

Synthesis ignores blocking delays, so it is acceptable to remove them in instances such as this.

---

## VER-142

(error) %s Nonblocking assignments and blocking delays are not allowed in the same process.

### Description

You receive this message because, during *analyze* or *read* command activity, the Presto HDL Compiler has detected a nonblocking assignment and a blocking delay in the same process.

The following example demonstrates this invalid description:

```
always @(posedge clk) begin
     #10;            // blocking delay
     out <= in;      // nonblocking assignment, not allowed with blocking
 delay
   end
```

### What Next

Synthesis ignores blocking delays, so it is acceptable to remove them in instances such as this.

---

## VER-143

(error) %s Blocking delays and nonblocking assignments are not allowed in the same process.

### Description

You receive this message because, during *analyze* or *read* command activity, the Presto HDL Compiler has detected a nonblocking assignment and a blocking delay in the same process, which is invalid for synthesis.

Even though synthesis ignores blocking delays, it is important to recognize them in the two forms in which they occur. Blocking delays can stand alone or as part of an intra-assignment delay. The following example shows an instance of each form:

```
always @(posedge clk) begin
     a <= x;         // nonblocking assignment
     #10;            // blocking delay, not allowed with nonblocking
 assignment
```

Feedback

```
    b = #10 y;   // intraassignment delay, not allowed with nonblocking
assignment
  end
```

**What Next**

Synthesis ignores blocking delays, so it is acceptable to remove them in instances such as this.

---

## VER-144

(error) %s Incorrect naming of port found.

**Description**

You receive this message because the *read* or *analyze* command has detected an incorrect port name. A missing port name after '.' can cause the problem, as shown in the following example:

```
module test(a, . (b), c);
      input a, b;
      output c;

    endmodule
```

**What Next**

Make sure the port name is complete.

---

## VER-145

(error) %s Declarations are supported in named blocks only

**Description**

This error message occurs because declarations are allowed only in named blocks.

**What Next**

Correct the code by either removing the declarations inside the block, or giving the block a name.

---

## VER-146

(warning) %s Time precision is less precise than the time unit.

Feedback

**Description**

This warning message advises you that time precision is usually more precise than the time unit.

**What Next**

This is a warning message only. No action is required on your part.

However, it is best practice to check the code and change the time precision to be more precise than the time unit where necessary.

---

## VER-147

(error) %s Instantiation %s has mixed ordered and named port connections

**Description**

This error message occurs because instantiation is using both ordered port connections and named port connections. Mixed ordered and port connections are not supported.

**What Next**

Change the code to use only ordered or only port connections.

---

## VER-148

(error) %s Gate output %s must be a net

**Description**

This error message occurs when the variable type for gate output is not set to net.

**What Next**

Change the variable type to net in the code.

---

## VER-149

(error) %s Symbol %s not included in portlist

**Description**

This error message occurs when the symbol declared in input and output lists is not included in the port list.

**What Next**

Add the missing input and output lists in the port list to the code.

Feedback

## VER-150

(error) %s Net assignment %s found in net declaration list.

### Description

You receive this message because the *read* or *analyze* command has detected a net assignment in a net declaration, a construct that the Presto HDL Compiler does not support.

The following is one example that can cause the problem:

```
module test( a, b, c);
      input a, b;
      output c;
      wire d, e=a;

    endmodule
```

### What Next

Check your design for invalid constructs like this and correct them.

## VER-151

(warning) %s port %s have inconsistent declarations.

### Description

You receive this message because the *read_verilog* or *analyze* command has detected an inconsistency of the port type.

The following is one example that can cause the problem:

```
`define N 4

   module m1 (a,b) ;
   input [`N-1:0] a, b ;
   parameter P = `N ;
   wire [`N-1:0] a ;
   wire [P-1:0] b ;
   endmodule
```

This a mismatch for b, because the parameter P could be overridden, so it cannot be assumed to have its default value.

Another example that can cause the problem is:

```
module m2 (a,b);

   parameter A=2;
```

Feedback

```
   parameter B=A;

   input [A:0] a;
   input [B:0] b;

   wire [B:0] a;
   wire [A:0] b;

   endmodule
```

Even though B has the same default value as A, that value can be overridden.

### What Next

Use identical expressions in corresponding pairs of port and signal declarations.

### See Also

- read_verilog

---

## VER-152

(error) %s Vector or memory size mismatch on port %s using last declaration.

### Description

You receive this message because the *read_verilog* or *analyze* command has detected vector or memory size mismatches.

The following is one example that can cause the problem:

```
`define N 4

   module m1 (a,b) ;
   input [`N-1:0] a, b ;
   parameter P = `N+1 ;
   wire [`N-1:0] a ;
   wire [P-1:0] b ;
   endmodule
```

This a mismatch for b, because the parameter P's default value is not the same as `N. As a result, the width of array can not be determined.

Another example that can cause the problem is:

```
module m2 (a,b);

   parameter A=2;
   parameter B=A;
```

Feedback

```
input [A:0] a;
input [B:0] b;

wire [B:0] a;
wire [A:0] b;

endmodule

module top(a, b);
input [2:0] a;
output [3:0] b;

m2 #(.B(3)) inst2(a, b);
emdmodule
```

```
Even though B has the same default value as A, that value is overridden
in the module instantiation when design top is evaluated.
As a result, the port a and b in module m2
has mismatched sizes in their declarations.
```

### What Next

Use identical expressions in corresponding pairs of port and signal declarations.

### See Also

- read_verilog

---

## VER-153

(error) %s Port %s occurs more than once in instance %s of module %s.

### Description

You see this message because during *read* or *analyze* command activity, the compiler has detected that you have used the same port name more than once in an instantiation, which is not valid.

The following is one example that can cause the problem:

```
module test( a, b, c);
    input a, b;
    output c;

    my_test c1(.p1(a), .p1(b), .p2(c));

    endmodule
```

### What Next

Check your design for duplicate port names and correct them.

Feedback

## VER-154

(error) %s Gate instance with too few ports

**Description**

This error message occurs when there are not enough ports in the gate instance.

**What Next**

Change the code to use the correct number of ports for the gate instance.

## VER-155

(warning) %s non constant assignment to specparam %s.

**Description**

This warning message occurs when a non constant is assigned to the *specparam* variable and may not be accepted by other simulators.

**What Next**

This is a warning message only. No action is required on your part.

However, it is best practice to assign a constant expression to the *specparam* variable.

## VER-156

(error) %s UDP initialization value %s is not 0, 1, or X.

**Description**

This error message occurs when the user-defined primitive (UDP) initialization value is not assigned as 0, 1, or X.

**What Next**

Assign 0, 1, or X to the UDP initialization value and run the command again.

## VER-157

(error) %s UDP initialization variable %s must be output.

**Description**

This error message occurs when the user-defined primitive (UDP) initialization variable is not output.

Feedback

**What Next**

The UDP initialization statement specifies the value of the output port when simulation begins. Use output port as the UDP initialization variable and run the command again.

---

## VER-158

(error) %s too few input entries in table specification.

**Description**

This error message occurs when there are not enough input entries in the table specification. The number of input entries must be equal to the number of input variables.

**What Next**

Modify the table specification, ensuring that the number of input entries is equal to the number of input variables, and then run the command again.

---

## VER-159

(error) %s too many input entries in the table specification.

**Description**

This error message occurs when there are too many input entries in the table specification. The number of input entries must be equal to the number of input variables.

**What Next**

Modify the table specification, ensuring that the number of input entries is equal to the number of input variables, and then run the command again.

---

## VER-160

(error) %s vector declaration %s not supported in UDPs.

**Description**

This error message occurs because vector declaration is not supported in user-defined primitives (UDPs).

**What Next**

Do not use vector declaration in UDPs, or use the Synopsys compiler directives *translate_off* and *translate_on* to comment it out as shown in the following example.

Feedback

```
/* synopsys translate_off */
Vector declaration
/* synopsys translate_on */
```

After making corrections, run the command again.

## VER-161

(error) %s path element %s should be wire.

**Description**

This error message occurs if the path element is not wire.

**What Next**

Change the specified path element to wire and run the command again.

## VER-162

(error) %s module path source %s not port.

**Description**

This error message occurs if the module path source is not port.

**What Next**

Change the module path source to port and run the command again.

## VER-163

(error) %s module path destination %s not port.

**Description**

This error message occurs if the module path destination is not port.

**What Next**

Change the module path destination to port and run the command again.

## VER-164

(error) %s path pulse source %s not defined.

**Description**

This error message occurs if the path pulse source is not defined.

Feedback

**What Next**

Define the specified path pulse source and run the command again.

---

## VER-165

(error) %s path pulse destination %s is not defined.

**Description**

This error message occurs if the specified path pulse destination is not defined.

**What Next**

Define a valid path pulse destination and rerun the command.

---

## VER-166

(error) %s path pulse source %s is not a port.

**Description**

This error message occurs when the path pulse source should be a port.

**What Next**

Change the specified path pulse source to an input port and rerun the command.

---

## VER-167

(error) %s path pulse destination %s is not a port.

**Description**

This error message occurs when the path pulse destination should be a port.

**What Next**

Change the specified path pulse destination to an output port and rerun the command.

---

## VER-168

(error) %s timing notifier malformed.

**Description**

This error message occurs when an expression (instead of an identifier) is used as a timing modifier.

Feedback

**What Next**

Change the timing notifier to use an identifier and rerun the command.

---

## VER-169

(error) %s illegal reference event: expressions are not allowed.

**Description**

This error message occurs when an expression is used as a reference event. An expression cannot be used as a reference event.

**What Next**

Use a legal reference event and rerun the command.

---

## VER-170

(error) %s timing notifier %s should be a reg.

**Description**

This error message occurs because the timing notifier should be a *reg*.

**What Next**

Correct the timing notifier to use a *reg* and rerun the command.

---

## VER-171

(error) %s timing function %s has an incorrect number of parameters.

**Description**

This error message occurs when the timing function has an incorrect number of parameters.

**What Next**

Modify the timing function so that it has the correct number of parameters and rerun the command.

---

## VER-172

(error) %s wire (%s) is connected to both the input and output of a built-in gate.

Feedback

**Description**

This error message occurs when the same wire is connected to both the input and the output of a built-in gate, resulting in combinational feedback.

**What Next**

Modify the design so that the same wire is not connected to both the input and output of the built-in gate, and then run the command again.

---

## VER-173

(warning) %s delays for continuous assignment are ignored.

**Description**

This warning message advises that delays for continuous assignment are ignored in synthesis.

**What Next**

This is a warning message only. No action is required on your part.

However, it is best practice to ensure that the ignored delay is not integral to the functionality of the design.

---

## VER-174

(error) %s '%s' is listed more than once in port list.

**Description**

This error message occurs when the specified port name is used more than once in a port list.

**What Next**

Correct the code to use a port name only once in each port list and then rerun the command.

---

## VER-175

(warning) %s Function '%s' should have at least one input argument.

**Description**

You receive this message because, during *read* or *analyze* command activity, the compiler encountered a function that has no input arguments. The IEEE standard for Verilog (1364-1995) requires that a function have at least one input argument.

Feedback

For example, the following function would trigger this warning:

```
function [0:2] f;
       begin
         f = 3'b101;
       end
     endfunction
```

The Presto HDL compiler accepts functions with no arguments; however, many tools do not.

**What Next**

Consider adding at least one input argument to the function in question to conform to IEEE standards.

**See Also**

- VER-105

---

## VER-176

(warning) %s delay controls are ignored for synthesis.

**Description**

This warning message occurs because your RTL description contains a delay control. Delay controls are ignored for synthesis.

The following is an example of a delay control.

```
module foo;
  ...
  always
  #(5)   // delay control, will be ignored.
     b = a;
endmodule
```

**What Next**

This is a warning message only. No action is required on your part. You can leave the delay controls in your description, but be aware that the delay controls have no meaning for synthesis.

---

## VER-177

(error) %s real declarations are not supported by synthesis.

Feedback

### Description

This error message occurs because the real data type is not supported by synthesis.

The following example shows a real declaration.

```
module foo;
  real r;          // data type real not supported by synthesis
endmodule
```

### What Next

Rewrite your RTL code so that it does not contain real data types. Depending on your specific design, you may need to use integer types instead of real types. Rerun the command after making the changes.

---

## VER-178

(error) %s realtime declarations are not supported by synthesis.

### Description

This error message occurs because the realtime data type is not supported by synthesis.

The following example shows a realtime declaration.

```
module foo;
  realtime r;       // data type realtime not supported by synthesis
endmodule
```

### What Next

Realtime variables do not have a meaning in synthesis. Rewrite your RTL code so that it does not use realtime data types. Rerun the command after making the changes.

---

## VER-179

(error) %s triand declarations are not supported by synthesis.

### Description

This error message occurs because your RTL description contains triand declarations, which are not supported by synthesis.

The following example shows a triand declaration.

```
module foo;
  triand d;
endmodule
```

Feedback

### What Next

Rewrite your design so that it does not use triand declarations. Since triand nets imply an and logic function, one solution is to explicitly describe the logic in your RTL code. Rerun the command after making your changes.

---

## VER-180

(error) %s trior declarations are not supported by synthesis.

### Description

This error message occurs because your RTL description contains trior declarations, which are not supported by synthesis.

The following example shows a trior declaration.

```
module foo;
  trior d;
endmodule
```

### What Next

Rewrite your design so that it does not use trior declarations. Since trior nets imply an or logic function, one solution is to explicitly describe the logic in the RTL code. Rerun the command after making the changes.

---

## VER-181

(error) %s tri0 declarations are not supported by synthesis.

### Description

This error message occurs because your RTL description contains tri0 declarations, which are not supported by synthesis.

The following example shows a tri0 declaration:

```
module foo;
  tri0 d;
endmodule
```

### What Next

Rewrite your design such that it does not use tri0 declarations and then rerun the command.

Feedback

## VER-182

(error) %s tri1 declarations are not supported by synthesis.

### Description

This error message occurs because your RTL description contains tri1 declarations, which are not supported by synthesis.

The following example shows a tri1 declaration:

```
module foo;
  tri1 d;
endmodule
```

### What Next

Rewrite your design so that it does not use tri1 declarations and run the command again.

## VER-183

(error) %s trireg declarations are not supported by synthesis.

### Description

This error message occurs because your RTL description contains trireg declarations, which are not supported by synthesis.

The following example shows a trireg declaration.

```
module foo;
  trireg d;
endmodule
```

### What Next

Rewrite your design so that it does not use trireg declarations and run the command again.

## VER-184

(error) %s pulldown declarations are not supported by synthesis.

### Description

This error message occurs because your RTL description contains pulldown declarations, which are not supported by synthesis.

The following example shows a pulldown declaration.

Feedback

```
module foo (c);
  output c;
  ..
  pulldown c1(c);
endmodule
```

**What Next**

Rewrite your design so that it does not use pulldown declarations and then run the command again.

---

## VER-185

(error) %s pullup declarations are not supported by synthesis.

**Description**

This error message occurs because your RTL description contains pullup declarations, which are not supported by synthesis.

The following example shows a pullup declaration.

```
module foo (c);
  output c;
  ..
  pullup c1(c);
endmodule
```

**What Next**

Rewrite your design so that it does not use pullup declarations and then run the command again.

---

## VER-186

(error) %s user-defined primitives (UDPs) are not supported by synthesis.

**Description**

This error message occurs because user-defined primitives (UDPs) are not supported by synthesis.

**What Next**

Enclose the UDPs in the following statement and run the command again.

```
`ifndef SYNTHESIS
  ...
`endif
```

Feedback

## VER-187

(error) %s fork and join constructs are not supported by synthesis.

**Description**

This error message occurs because fork and join constructs are not supported by synthesis.

**What Next**

Enclose the fork-join construct as follows and rerun the command.

```
`ifndef SYNTHESIS
 ...
`endif
```

## VER-188

(error) %s wait statements are not supported by synthesis.

**Description**

This error message occurs because wait statements are not supported by synthesis.

**What Next**

Enclose the wait statements as follows and run the command again.

```
`ifndef SYNTHESIS
 ...
`endif
```

## VER-189

(error) %s case equality (===) is not supported by synthesis.

**Description**

This error message occurs because case equality (===) is not supported by synthesis.

**What Next**

If the statement containing === is for synthesis, try replacing === with ==, if it does not change the semantics.

If the statement containing === is not for synthesis, enclose the statement as follows.

```
`ifndef SYNTHESIS
 ...
`endif
```

Feedback

After making your changes, run the command again.

## VER-190

(error) %s case inequality (!==) is not supported by synthesis.

**Description**

This error message occurs because case inequality (!==) is not supported by synthesis.

**What Next**

If the statement containing !== is for synthesis, try replacing !== with !=, if it does not change the semantics.

If the statement containing !== is not for synthesis, enclose the statement with the following.

```
`ifndef SYNTHESIS
 ...
`endif
```

After making the changes, run the command again.

## VER-191

(error) %s time declarations are not supported by synthesis.

**Description**

This error message occurs because time declarations are not supported by synthesis.

**What Next**

Enclose the time declarations with the following and then run the command again.

```
`ifndef SYNTHESIS
 ...
`endif
```

## VER-192

(warning) %s In initial block, only $power and $retain are supported, other statements are ignored.

**Description**

This warning message advises you that in initial block, only $power and $retain calls are supported in synthesis. All other statements are not supported in synthesis.

Feedback

### What Next

This is a warning message only. No action is required on your part.

However, to avoid receiving this warning message, you can suppress the message, or enclose the initial block with the following.

```
`ifndef SYNTHESIS
 ...
`endif
```

## VER-193

(error) %s event triggers are not supported.

### Description

This error message occurs because event triggers are not supported.

### What Next

Enclose the event trigger with the following and then rerun the command.

```
`ifndef SYNTHESIS
 ...
`endif
```

## VER-194

(error) %s Base of subscript operator %s must be a vector.

### Description

You receive this message because during *read* or *analyze* command activity, the compiler has detected a subscript operator whose base is not a vector, which is invalid.

The following is one example that can cause the problem:

```
module test(a, b);
        input a;
        output b;
        reg b;

        always
          b = a[0];

        endmodule
```

### What Next

Check your design for invalid constructs like this and correct them.

Feedback

## VER-195

(error) %s Illegal digit %c in based number.

**Description**

You see this message because during *read* or *analyze* activity, the compiler has detected an invalid digit in a based number.

**What Next**

Check your design for invalid digits in based numbers and correct them.

## VER-196

(error) %s Unable to open %s for writing: %s

**Description**

This I/O error issues when the attempt to write a result file (usually into the current design library) fails. The file system's explanation for the error is given at the end of the message. The file name usually identifies the RTL module template, interface, package, etc. being written.

**What Next**

Check that the parent directories all exist and offer write permission to the current user. Address any other concerns raised by the I/O subsystem portion of this message.

## VER-197

(information) %s writing module %s to file %s.

**Description**

This information message advises you that the tool is writing the specified module to an analyzed file format (.pvl) file.

**What Next**

This is an informational message only. No action is required on your part.

## VER-198

(error) %s mixed direction ports %s must be nets.

Feedback

**Description**

This error message occurs because the code contains mixed direction ports that should be net-types.

**What Next**

Rewrite code so that the ports with mixed direction are net-types and then rerun the command.

---

## VER-199

(error) %s UDP table has invalid edge specification.

**Description**

This error message occurs when the UDP table has an invalid edge specification, such as (0 0), (1 1) or (X X). The two level symbols in the edge indicator should be different. So (0 0), (1 1) and (X X) are invalid, while (1 0) is valid.

**What Next**

Change the invalid edge indicator to a valid one and run the command again.

---

## VER-200

(error) %s port %s must be declared as input.

**Description**

This error message occurs because the port must be declared as an input port.

**What Next**

Correct the RTL code to declare the port as input port and then run the command again.

---

## VER-201

(error) %s port %s must be declared as output.

**Description**

This error message occurs when the port must be declared as an output port.

**What Next**

Correct the RTL code to declare the port as an output port and then run the command again.

Feedback

## VER-202

(error) %s path element %s should be reg or wire.

**Description**

This error message occurs because the path element should be either reg or wire.

**What Next**

Correct the code to declare the path element as either reg or wire and run the command again.

## VER-203

(error) %s pulse limit for path not defined.

**Description**

This error message occurs because the pulse limit for path is not defined. The valid *specparam* assignment format is one of the following.

```
specparam_identifier = constant_mintypmax_expression

specparam_identifier = pulse_control_specparam
```

The *pulse_control_specparam* format is as follows.

```
$PATHPULSE$... = (reject_limit_value, [, error_limit_value]).
```

You receive this error message when the required *reject_limit_value* is missing.

**What Next**

Add the *reject_limit_value* to the code and then run the command again.

## VER-204

(error) %s timing reference event %s not port.

**Description**

This error message occurs because the timing reference event specified is not port.

**What Next**

Change the timing reference event to port and run the command again.

Feedback

## VER-205

(error) %s timing reference event %s should have edge identifier.

**Description**

This error message occurs because the timing reference event requires an edge identifier.

**What Next**

Add an edge identifier to the timing reference event and run the command again.

## VER-206

(error) %s Too many parameters passed to system task '%s'.

**Description**

The given system task is passed more arguments than expected.

**What Next**

Find out how which arguments the system task expects. Remove the extraneous arguments from the task call.

## VER-207

(error) %s The %s systask parameter must be writable.

**Description**

Actual parameters passed to 'output' or 'inout' formal task parameters must be writable (because they are conceptually used as left-hand-side values of assignments). In the following example a constant is passed as an argument to the system function '$random', which indeed expects the argument to be writable:

```
...
   r = $random(3);      // error: constant "3" is not writable.
   ...
```

**What Next**

Make sure the actual argument in question is writable, eg. a 'register'.

## VER-208

(warning) %s Register '%s' is the target of both blocking and non-blocking assignments in the same process.

Feedback

**Description**

Presto does allow mixing blocking and non-blocking assignments in the same process. However, this coding style can lead to unexpected results and the code may be hard to comprehend.

This warning is informational only. It tells you that targets assigned by both kinds of assignments actually exist.

**What Next**

Make sure the mixture of blocking and non-blocking assignments is really what you intended.

---

## VER-209

(warning) %s Little argument or return value checking implemented for system task or function '%s'.

**Description**

Argument and return value checking for system tasks and functions is limited. This is a warning message indicating that some errors potentially remain uncaught because of incomplete checking.

**What Next**

The user should check system task or function all carefully to make sure the argument and return value are correct.

---

## VER-210

(error) %s timing reference event %s has wrong edge.

**Description**

This error message occurs when the wrong edge is specified in the timing reference event.

**What Next**

Specify the edge as either posedge or negedge and then run the command again.

---

## VER-211

(error) %s timing limit malformed.

Feedback

### Description

This error message occurs when the timing limit is malformed because both the event and the normal condition exist in the timing limit. Only one of these conditions is allowed.

### What Next

Remove either the event or the normal condition and then run the command again.

---

## VER-212

(error) %s timing limit should be positive.

### Description

This error message occurs when the timing check limit should be positive.

### What Next

Specify blocks are not supported in synthesis. Prevent the block from being read by enclosing it in the following compiler directives.

```
`ifndef SYNTHESIS
 ...
`endif
```

After making the changes, run the command again.

---

## VER-213

(error) %s timing event malformed.

### Description

This error message occurs when the timing check event is malformed.

### What Next

Specify blocks are not supported in synthesis. Prevent the specify block from being read by enclosing it in the following compiler directives.

```
`ifndef SYNTHESIS
 ...
`endif
```

After making the changes, run the command again.

---

## VER-214

(error) %s timing notifier %s malformed.

Feedback

### Description

This error message occurs when the optional notifier on the timing check is malformed.

### What Next

Specify blocks are not supported in synthesis. Prevent the specify block from being read by enclosing it in the following compiler directives.

```
`ifndef SYNTHESIS
 ...
`endif
```

After making the changes, run the command again.

---

## VER-215

(error) %s timing reference event %s is not input or inout port.

### Description

This error message occurs because the reference event for a timing check must be an input or inout port.

### What Next

Specify blocks are not supported in synthesis. Prevent the specify block from being read by enclosing the block in the following compiler directives.

```
`ifndef SYNTHESIS
 ...
`endif

After making the changes, run the command again.
```

---

## VER-216

(error) %s %s is not a parameter in module %s

### Description

You receive this error message because the name used in module instantiation (component instantiation in VHDL) or DEFPARAM is not a parameter in the specified module.

### What Next

Change the name in the instantiation to a supported parameter.

Feedback

## VER-217

(error) %s wrong number of ports in UDP instantiation, not the same as declaration.

**Description**

This error message occurs when the number of ports in the UDP instantiation is different than the number of ports in the UDP declaration.

**What Next**

UDPs are not supported in synthesis. Modify the code so it does not use UDPs and then rerun the command.

## VER-219

(error) %s Non-standard replication constant is not supported.

**Description**

This message is a follow-up to the ELAB-364 warning message, but is issued with "Error" severity, so that violations of the prevailing language standard halt compilation.

**What Next**

Change the relevant code to conform to the level of IEEE Std 1364 selected by hdlin_vrlg_std, or choose a different value for hdlin_repeat_count_0.

## VER-220

(error) %s Unresolved hierarchical name %s in module %s.

**Description**

This is a syntax error.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

## VER-221

(error) %s Module %s to be elaborated is not defined in analyzed source files.

**Description**

This is a syntax error.

Feedback

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

---

## VER-222

(warning) %s Assignment output %s should be a register.

**Description**

This is a syntax error.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

---

## VER-223

(warning) %s Task %s is not implemented; statement skipped.

**Description**

This is a syntax error.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

---

## VER-224

(error) %s Package '%s' has not been analyzed for import or content extraction.

**Description**

This syntax error issues when a reference to the given package is analyzed before its declaration has been analyzed. Successfully analyzed packages leave a <identifier>.pvk file in DC shell's working directory. This file must be found and read when the package is later referenced for import or to extract its contents. Note that packages may never refer to each other in a cycle of dependence.

**What Next**

Compare the spelling of the package identifier in the reference to its intended declaration. Remove cycles of package reference for which no analyze ordering is possible. Insure that the source file containing the package declaration is analyzed before the the source files that refer to it, or that the package declaration precedes all uses within the same source file. Check DC shell's search paths to insure that packages analyzed by other DC shell sessions can be found by this session. Inspect file and directory search permissions to

Feedback

be certain you can access the package. Check whether an intervening read or analyze command reported a syntax error within a package of the same name, thereby removing the good package's .pvk file.

## VER-225

(warning) %s Recommend parentheses when a reduction-or follows bitwise-or.

**Description**

A reduction-or operator follows a bitwise-or operator in the Verilog source. This is often a typographical error, when a space is inserted between the two characters in '||' (logical-or), resulting in '| |' (bitwise-or followed by reduction-or). Furthermore, some Verilog tools do not accept this syntax.

**What Next**

If you intended to use a reduction-or, surround it with parentheses to inhibit this warning.

## VER-226

(warning) %s Single element concatenation has unsized number.

**Description**

This is a syntax error.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

## VER-227

(error) %s Package '%s' directly or indirectly refers to the current package '%s'.

**Description**

This syntax error issues within a package if a cyclic reference is made (by scope-extraction operator) to a package whose definition appears to depend upon the package currently being analyzed. This error is reported aggressively, based upon any mention of the current package in the source of the target package or recursively in the packages to which it may refer, regardless of whether an actual cycle of definitional dependence is being formed. Aggressive enforcement prevents VER-224 errors when all the packages are re-analyzed in an empty work directory; i.e. it prevents incremental closure of package reference cycles which would make it impossible to analyze them in an acyclic order.

Feedback

### What Next

You might not be extracting from the right package. See the VER-224 help page for suggestions related to this contingency.

Perhaps you chose a name for your package that matches another package name deeper in the supply-chain of the package libraries you are using. Choose a unique package name.

If you are analyzing a older library package that used to work, perhaps the tree of packages upon which it depends has grown to include a package name which collides with this one. Since VER-227 is reported for any name collision in the transitive closure of scope extractions, it could take a while to track down by trial and error. Use 'strings *.pvk | fgrep my-package' or similar search tools to locate uses of the current package name in the *.pvk (or the package source files) of your project. Once located, the remedy may involve renaming one or the other package, or splitting either one so that the supply trees are disjoint (have no name collisions).

---

## VER-228

(error) %s Import symbol %s has already been defined.

### Description

This error message issues when you attempt to import a name that is already declared in the immediately surrounding scope.

### What Next

Edit the RTL source to remove or rename whatever has been declared or imported with the same name, or to enclose the import statement and all references to the name in a more local sequential block. Rerun the command.

---

## VER-229

(error) %s Value '%s' for enumerated type '%s' should be an integer or sized constant.

### Description

You receive this error message when a parameter declaration with an attached 'synopsys enum' pragma declares a parameter whose value is not an integer or sized constant. For example:

```
parameter [2:0] /* synopsys enum colors */ red = 3'b001,
                                           green = 3'b010,
                                           yellow = "YELLOW";
```

The third value in this example would result in this error message.

Feedback

**What Next**

Modify the design so that all parameters declared as members of an enumerated type are given an integer value or sized constant value.

---

## VER-230

(warning) %s Recommend parentheses when a reduction-and follows bitwise-and.

**Description**

A reduction-and operator follows a bitwise-and operator in the Verilog source. This is often a typographical error, when a space is inserted between the two ampersands in '&&' (logical-and), resulting in '& &' (bitwise-and followed by reduction-and). Furthermore, some Verilog tools do not accept this syntax.

**What Next**

If you intended to use a reduction-and, surround it with parentheses to inhibit this warning.

---

## VER-231

(warning) %s Range ignored for integer variables.

**Description**

This is a syntax error. The range will be ignored. In future releases this usage may be disallowed.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

---

## VER-232

(warning) %s Specifying strengths with the 'strength' keyword is archaic.

**Description**

This message notifies you that during *read* or *analyze* command activity, the compiler has detected use of 'strength' as a keyword. This word is archaic and may not be supported by some Verilog tools. Synthesis ignores all drive strengths, including one specified in this way.

**What Next**

Avoid the use of the 'strength' keyword.

Feedback

## VER-233

(error) %s Symbol '%s' cannot be used as an argument to the 'disable' statement because it is not a task or named block identifier.

### Description

You receive this message because, during *analyze* or *read* command activity, the compiler has detected an invalid argument to a disable statement. The invalid argument is a symbol that does not refer to a task or named block. In the following example, the symbol *x* does not refer to a named block and will trigger this error.

```
module test(in1);
       input in1;
       reg x;

       always begin : block
          if(in1)
             disable x;
       end
    endmodule
```

### What Next

Correct the *disable* statement to refer to a named block or task.

## VER-234

(error) %s Strengths highz0 and highz1 cannot be paired together in a strength specification.

### Description

You receive this error message because, during *read* or *analyze* command activity, the compiler has detected a strength specification that uses both highz0 and highz1 strengths, which is invalid. Furthermore, all strength specifications are ignored for synthesis.

### What Next

Correct the strength specification so that it does not use both highz1 and highz0.

## VER-235

(error) %s Strength %s for pullup/pulldown must be a strength0.

Feedback

### Description

You receive this message because, during *analyze* or *read* command activity, the Presto HDL Compiler has detected an invalid strength value in a position that requires a *strength0* value.

Synthesis ignores all strength specifications and does not support pullup and pulldown gates.

### What Next

Replace the invalid value with a *strength0* value. Acceptable are *supply0*, *strong0*, *pull0*, *weak0*, and *highz0*.

---

## VER-236

(error) %s Strength %s for pullup/pulldown must be a strength1.

### Description

You receive this message because the *read* or *analyze* command has detected an invalid strength value in a position that requires a strength1 value. Synthesis ignores all strength specifications, and does not support pullup and pulldown gates.

### What Next

Replace the strength value with a strength1 value. Acceptable values are *supply1*, *strong1*, *pull1*, *weak1*, and *highz1*.

---

## VER-237

(error) %s Strength specifications must have strength0 and strength1 components.

### Description

You receive this error message because you have specified a strength that does not have one strength0 and one strength1 component. With the exception of 'pullup' and 'pulldown' gates, a strength must have one strength0 and one strength1 component.

Note that all strength specifications are ignored for synthesis.

In the example below, both strengths are from the strength1 category, resulting in this error message.

```
wire  x, y, z;
   and (weak1,strong1) a1 (x, y, z);
```

### What Next

Specify a strength from the strength0 category and a strength from the strength1 category.

Feedback

## VER-238

(error) %s Strength declaration in incorrect place.

### Description

There is a syntax error in the Verilog source because a strength declaration appears at an incorrect place. Often, this indicates a second strength specification in a gate instantiation.

For example, in the following code segment, the second strength specification will trigger this error because it is not valid Verilog.

```
wire  a, b, c, x, y, z;
   and (weak1,weak0) a1(a, b, c), (weak1,weak0) a2(x, y, z);
```

### What Next

Examine the given file and line to determine which strength specification needs to be moved or deleted.

## VER-239

(error) %s A wire declaration cannot have a drive strength, at symbol '%s'.

### Description

A wire declaration contained a drive strength. A drive strength can only be specified in a net assignment.

### What Next

Remove the drive strength from the declaration.

## VER-240

(warning) %s Strength %s for pullup is redundant.

### Description

A 'pullup' gate has only a strength0 specified, and no strength1. This is likely an error, because strength0 specifications are ignored for 'pullup' gates.

Note, however, that 'pullup' gates are not supported for synthesis, and drive strengths are ignored for synthesis.

### What Next

Check if you intended to specify a strength1 for the pullup gate instead of a strength0. If not, ignore the warning.

Feedback

## VER-241

(warning) %s Strength %s for pulldown is redundant. 0-strength defaulting to strong0.

### Description

A 'pulldown' gate has only a strength1 specified, and no strength0. This is likely an error, because strength1 specifications are ignored for 'pulldown' gates.

Note, however, that 'pulldown' gates are not supported for synthesis, and drive strengths are ignored for synthesis.

### What Next

Check if you intended to specify a strength0 for the pulldown gate instead of a strength1. If not, ignore the warning.

## VER-242

(error) %s Enumerated type '%s' needs a size specification.

### Description

You receive this error message because a parameter declaration that is being used to declare an enumerated type using the 'synopsys enum' pragma is missing a size specification. The following declaration results in this error message.

```
parameter /* synopsys enum colors */ red = 3'b001,
                                       green = 3'b010,
                                       yellow = 3'b100;
```

### What Next

Add a size specification to the declaration. The above example can be corrected as follows:

```
parameter [0:2] /* synopsys enum colors */ red = 3'b001,
                                              green = 3'b010,
                                              yellow = 3'b100;
```

## VER-244

(error) %s Enumerated type '%s' has not been declared.

### Description

You receive this error message because the design has declared a symbol as a member of an enumerated type using the 'synopsys enum' pragma, but the enumerated type has not been declared.

Feedback

### What Next

Declare each enumerated type before using it. If you have declared the enumerated type, verify that no errors occurred while processing it.

---

## VER-245

(warning) %s Enumerated type '%s' is not compatible with this declaration.

### Description

The design declares a symbol as a member of an enumerated type using the 'synopsys enum' pragma, but the enumerated type has a different underlying type than the symbol. In the following example, the declaration of 'current_state' would trigger this error because its type is not '[0:2]'.

```
parameter [0:2] /* synopsys enum colors */ red = 3'b001,
                                               green = 3'b010,
                                               yellow = 3'b100;

    reg [0:3] /* synopsys enum colors */ y;
```

### What Next

If it is typo, correct the type of either the enumerated type declaration or the symbol declaration so that they match. A corrected declaration for 'y' in the above example would change its bounds to '[0:2]':

```
reg [0:2] /* synopsys enum colors */ y;
```

If user intends to let the symbol and the enumerated type have different type, the user have to make sure that the synthesis behavior is what the user wants.

---

## VER-246

(error) %s Port %s declared as being two different enums.

### Description

This error occurs when port has both a port declaration determining its direction, and a register declaration, and has a different enumerated type attached to the two declarations, using the 'synopsys enum' pragma. In the following example, the two declarations of 'y' would cause this error to be issued:

```
parameter [0:2] /* synopsys enum colors */ red = 3'b001,
                                               green = 3'b010,
                                               yellow = 3'b100;
    parameter [0:2] /* synopsys enum numbers */ one = 3'b001,
                                                   two = 3'b010;
```

Feedback

```
output [0:2] /* synopsys enum colors */ y;
reg [0:2] /* synopsys enum numbers */ y;
```

**What Next**

Choose one enumerated type for the port in question, and correct the declaration.

---

## VER-247

(error) %s Enumerated type '%s' has multiple declarations.

**Description**

You receive this error message because the same enumerated type is declared twice. The following example illustrates this situation.

```
parameter [0:2] /* synopsys enum colors */ red = 3'b001,
                                           green = 3'b010,
                                           yellow = 3'b100;

parameter [0:4] /* synopsys enum colors */ brown = 5'b001,
                                           black = 5'b010,
                                           gray = 5'b100;
```

**What Next**

Specify a new name for one of the enumerated types.

---

## VER-248

(error) %s Encoding '%s' for '%s' is the wrong size for this enumerated type.

**Description**

You receive this error message because a value given for an enumerated type has a different size than the enumerated type itself.

In the following example, the state values given are 8 bits, while the enumerated type is declared as 12 bits. This difference will result in this error message.

```
parameter [11:0] /* synopsys enum states */ state1 = 8'b00000001,
                                                state2 = 8'b00000010;
```

**What Next**

Modify the values or the type declaration so that they are the same size.

Feedback

## VER-249

(error) %s force output %s must be a register or a net.

### Description

This error message occurs because a force statement can only be used to assign to a variable or a net.

### What Next

Force statements are not supported in synthesis. Enclose the force statement in the following compiler directives to prevent the statement from being read.

```
`ifndef SYNTHESIS
 ...
`endif
```

## VER-251

(error) %s Expecting a pair of bounds [msb:lsb].

### Description

This error message occurs when an array dimension has not specified exactly two bounds. Verilog bit vectors, Verilog memories, and all packed dimensions of SystemVerilog array types must be declared using the subscript values for both the most and the least significant bits as shown in the following example:

```
reg [ most_significant_bit_index : least_significant_bit_index ]
 packed_vector;
```

### What Next

Adjust the bounds syntax. Or, in SystemVerilog, move the dimension brackets to the right of the array or type identifier to specify that this dimension should be treated as unpacked.

## VER-252

(error) %s illegal use of identifier %s.

### Description

This error message occurs when an identifier that is defined as a function is being used as a task.

Feedback

**What Next**

Modify the source code to change the call to a function call and then run the command again.

---

## VER-253

(error) %s Illegal reference to memory %s.

**Description**

This is a syntax error.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

---

## VER-254

(error) %s invalid symbol %s found in activation expression.

**Description**

This error message occurs when there is an invalid symbol in an activation expression. The symbol in an activation expression should be a net, variable or event.

**What Next**

Change the symbol in the activation expression to a net, variable or event, and then rerun the command.

---

## VER-255

(error) %s invalid symbol %s found in expression.

**Description**

This error message occurs when there is an invalid symbol in an expression. The symbol in an expression should be a net or variable.

**What Next**

Change the symbol in the activation expression to a net or variable and then rerun the command.

---

## VER-256

(error) %s illegal part selection has been found.

Feedback

**Description**

This error message occurs when an illegal part selection is found. The index of a vector in part selection should be constant valued, and its sign bit cannot be Z or X value.

**What Next**

Check the indexes of vectors in the part selection and correct the offending index. Run the command again.

---

## VER-257

(error) %s invalid symbol %s found as system task argument.

**Description**

This error message occurs when there is an invalid symbol in a system task argument list. The symbol used as a system task argument should be a net, variable, instance, module, event, or function.

**What Next**

Change the symbol in the system task argument list to a net, variable, instance, module, event, or function.

---

## VER-258

(error) %s no semicolon at end of module name.

**Description**

This error message occurs when the semicolon is missing from the end of a module name.

**What Next**

Add a semicolon to the end of the module name and run the command again.

---

## VER-259

(error) %s multiple delays found.

**Description**

This error message occurs when the tool finds multiple delays. Only one delay is allowed.

**What Next**

Correct the code to ensure that there is only one delay, and then run the command again.

Feedback

## VER-260

(error) %s symbol %s must be a constant or parameter.

### Description

This error message occurs because the symbol indicated in the error message must be a constant or a parameter, as required by the language.

In the example below, the expression in the generate case statement is required to evaluate to a constant value at elaboration time. The use of an input port in that expression violates this requirement.

```
module m (a);
    input a;

    generate
      case (a)         // error: not constant
      ...
    endgenerate
  endmodule
```

### What Next

Change the specified symbol to a constant or a parameter and then run the command again.

## VER-261

(error) %s continuous assignment output %s must be a net.

### Description

This error message occurs because the output of a continuous assignment must be a net.

### What Next

Change the output of the continuous assignment to net and then run the command again.

## VER-262

(error) %s net symbol %s has already been defined.

### Description

This error message occurs when the redeclaration of a net symbol is detected. A net can only be declared once.

Feedback

**What Next**

Remove or rename the nets declared with the same name and then run the command again.

---

## VER-263

(error) %s bit- or part-selected destination of a force %s must be a net.

**Description**

This error message occurs because the bit-selected or part-selected destination of a force statement must be a net and not a variable.

**What Next**

Change the destination of the force statement from a variable to a net and run the command again.

---

## VER-264

(error) %s bad hierarchical name (%s).

**Description**

For Verilog, hierarchical references are not supported for synthesis.

For SystemVerilog, the preceding expression does not evaluate to a struct or union.

**What Next**

For Verilog, exclude the code from synthesis by enclosing the hierarchical reference in the conditional compilation directives shown below.

```
`ifndef SYNTHESIS
 ...
`endif
```

For SystemVerilog, if the name is a struct or union member, make sure the preceding expression evaluates to a struct or union. If the name is not a struct or union, follow the instructions in the previous paragraph, and enclose the code in the conditional compilation directives.

---

## VER-265

(error) %s RCMOS switches are not supported.

Feedback

### Description

This error message occurs because RCMOS switches are not supported for synthesis.

### What Next

Exclude the code from synthesis by enclosing the code in the conditional compilation directives shown below.

```
`ifndef SYNTHESIS
 ...
`endif
```

## VER-266

(error) %s RNMOS switches are not supported.

### Description

This error message occurs because RNMOS switches are not supported for synthesis.

### What Next

Exclude the code from synthesis by enclosing the code in the conditional compilation directives shown below.

```
`ifndef SYNTHESIS
 ...
`endif
```

## VER-267

(error) %s RPMOS switches are not supported.

### Description

This error message occurs because RPMOS switches are not supported for synthesis.

### What Next

Exclude the code from synthesis bvy enclosing the code in the conditional compilation directives shown below.

```
`ifndef SYNTHESIS
 ...
`endif
```

## VER-268

(error) %s RTRAN switches are not supported.

Feedback

### Description

This error message occurs because RTRAN switches are not supported for synthesis.

### What Next

Exclude the code from synthesis by enclosing the code in the conditional compilation directives shown below.

```
`ifndef SYNTHESIS
 ...
`endif
```

## VER-269

(error) %s RTRANIF0 switches are not supported.

### Description

This error message occurs because RTRANIF0 switches are not supported for synthesis.

### What Next

Exclude the code from synthesis by enclosing the code in the conditional compilation directives shown below.

```
`ifndef SYNTHESIS
 ...
`endif
```

## VER-270

(error) %s RTRANIF1 switches are not supported.

### Description

This error message occurs because RTRANIF1 switches are not supported for synthesis.

### What Next

Exclude the code from synthesis by enclosing the code in the conditional compilation directives shown below.

```
`ifndef SYNTHESIS
 ...
`endif
```

## VER-271

(error) %s TRAN switches are not supported.

Feedback

### Description

This error message occurs because TRAN switches are not supported for synthesis.

### What Next

Exclude the code from synthesis by enclosing the code in the following conditional compilation directives.

```
`ifndef SYNTHESIS
 ...
`endif
```

---

## VER-272

(error) %s TRANIF0 switches are not supported.

### Description

This error message occurs because TRANIF0 switches are not supported for synthesis.

### What Next

Exclude the code from synthesis by enclosing the code in the following conditional compilation directives

```
`ifndef SYNTHESIS
 ...
`endif
```

---

## VER-273

(error) %s TRANIF1 switches are not supported.

### Description

This error message occurs because TRANIF1 switches are not supported for synthesis.

### What Next

Exclude the code from synthesis by enclosing the code in the following conditional compilation directives

```
`ifndef SYNTHESIS
 ...
`endif
```

---

## VER-274

(warning) %s Unsupported system task '%s' will be ignored for synthesis.

Feedback

### Description

Many system tasks are used only for simulation or testbench purposes; they are not supported for synthesis, and are ignored. This warning alerts you that your design contains an unsupported system task (or unsupported system function whose result is discarded).

### What Next

If your design functions correctly without the system task, then no action is needed.

---

## VER-275

(error) %s User-defined primitives (UDP's) are not supported.

### Description

User-defined primitives (UDP's) are not supported.

### What Next

Change your source file to remove UDPs or enclose them with translate on and translate off, or enclose them with `ifndef SYNTHESIS ... `endif.

---

## VER-276

(error) %s Specify blocks are not supported.

### Description

This error message occurs because specify blocks are not supported for synthesis.

### What Next

Exclude the specify blocks from the synthesis source code by enclosing the blocks in conditional compilation directives as shown below.

```
`ifndef SYNTHESIS
     specify
        ...      /* specparams that interfere with synthesis */
     endspecify
  `endif
```

---

## VER-277

(warning) %s Charge strengths are ignored.

Feedback

### Description

Charge strengths are ignored for synthesis.

### What Next

No further action required.

---

## VER-278

(error) %s Array bounds must be of integer type.

### Description

You receive this error message because you have not provided the correct value for array bounds. For synthesis, array bounds must be of integer type.

### Examples

The following erroneous example tries to declare an array with a float value as an array bound:

```
...
   reg [3.0:0] r;   /* ERROR: '3.0' is not an integer bound */
   ...
```

### What Next

Correct the values of the array bounds in your design so that they are expressed as integers.

---

## VER-279

(error) %s Syntax error in parameter value list at or near token '%s' (string position %d).

### Description

This error message occurs when there is a syntax error in a parameter value list given by the *elaborate* command *-parameters* argument.

### What Next

Correct the syntax error in the parameter value list and then run the *elaborate* command again.

See the man page for the *elaborate* command for information on the accepted parameter list syntax.

Feedback

## VER-280

(warning) %s In UPF shell mode, the RTL power construct $isolate, $power and $retain are ignored.

### Description

This warning message advises you that in UPF mode, the $isolate, $power and $retain statements are ignored.

### What Next

This is a only a warning message. No action is required.

However, to prevent this warning message, remove the $isolate, $power and $retain power constructs from the RTL code.

## VER-281

(warning) %s The statements in initial blocks are ignored.

### Description

This warning message advises you that initial blocks are not supported in synthesis.

### What Next

This is a warning message only. No action is required on your part.

However, to avoid receiving this warning message, you can enclose the initial block with the following:

```
`ifndef SYNTHESIS
 ...
`endif
```

## VER-282

(warning) All isolation_upf and retention_upf pragmas are ignored.

### Description

This warning message advises you that all isolation_upf and retention_upf pragmas are ignored.

### What Next

This is only a warning message. No action is required.

Feedback

However, to prevent this warning message, remove the isolation_upf and retention_upf pragmas from input Verilog.

---

## VER-286

(warning) %s 'event' declarations are not supported by synthesis.

**Description**

The Verilog 'event' declaration is not supported for synthesis.

**What Next**

Rewrite your design so that it does not use the 'event' declaration.

---

## VER-287

(error) %s In UPF shell mode, the RTL power construct $isolate, $power and $retain are not supported.

**Description**

This error message advises you that in UPF shell mode, the $isolate, $power and $retain statements are not supported.

**What Next**

Remove the specified line of code which contains the $isolate, $power or $retain constructs, and correct the design correspondingly. Or, you may set hdlin_upf_allow_legacy_construct to true to let the tool strip off the logic for you.

---

## VER-288

(error) %s Block name symbol %s already defined.

**Description**

This error message occurs when the name used for the block label has been previously used as an object name or a block label.

For example:

```
module test (input in, output reg out1, out2);
always @(in) begin:BLK
  out1 = in;
end
always @(in) begin:BLK
  out2 = in;
```

Feedback

```
end
endmodule
```

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

---

## VER-289

(error) %s Instance name symbol %s already defined.

**Description**

This is a syntax error.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

---

## VER-290

(error) %s Error in edge control specifier.

**Description**

This is a syntax error.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

---

## VER-291

(error) %s Event symbol %s already defined.

**Description**

This is a syntax error.

**What Next**

Change the relevant code to conform to the IEEE std 1364.

---

## VER-292

(error) %s Unable to open file `%s': %s.

Feedback

**Description**

This is a syntax error.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

---

## VER-293

(error) %s `%s' is not a regular file.

**Description**

This is a syntax error.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

---

## VER-294

(error) %s Syntax error at or near token '%s'%s.

**Description**

You receive this error message because your code contains a syntax error or there's a syntax error in a /* Synopsys */ comment.

**What Next**

Review your code and make appropriate corrections.

---

## VER-295

(error) %s CMOS switches are not supported.

**Description**

This is a syntax error.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

---

## VER-296

(error) %s NMOS switches are not supported.

Feedback

**Description**

This is a syntax error.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

---

## VER-297

(error) %s PMOS switches are not supported.

**Description**

This is a syntax error.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

---

## VER-298

(warning) %s Keyword vectored is ignored.

**Description**

This warning message advises you that *vectored* is an optional advisory keyword. Presto warns about IEEE-1364 conforming uses of this net accessibility specification, but it is otherwise ignored for synthesis.

The following example shows the declaration of a vectored net. Since Presto ignores the keyword *vectored*, the effect is the same as if the keyword had been omitted.

```
...
  wire vectored [1:0] x;
  ...
```

**What Next**

This is a warning message only. No action is required on your part.

---

## VER-299

(error) %s Illegal use of event '%s' as module connection.

Feedback

### Description

You receive this message because the *read_verilog* or *elaborate* command detects an event in a module connection list. Events cannot be module connections in a module instantiation statement.

The IEEE 1364 Standard defines the term "module connection" as an element connected to a port of the instantiated module. Expressions in a parenthesized list in a module instantiation statement are valid module connections.

The following example shows an invalid attempt to use an event as a module connection:

```
event e;
  ...
  SUBMODULE my_submodule(e);  /* error: event used as module connection
 */
```

### What Next

Determine the original intent of the module connection and replace the event with an element that correctly connects to a module.

### See Also

- read_verilog

---

## VER-300

(error) %s Illegal attempt to declare hierarchical reference '%s'.

### Description

Hierarchical names are illegal in contexts where they would introduce or declare a new name in the current scope.

For example it is illegal to do a cross module declaration, as shown in the following code fragment:

```
reg a.b; /* illegal to declare a register in another scope */
```

### What Next

Check if this is a typo, where a period should be a comma. Check that the names in the hierarchical reference are properly declared and accessible in the current scope. Reduce cross-module references by adding explicit ports to carry signals between modules.

---

## VER-301

(error) %s Specparam symbol '%s' already defined.

Feedback

### Description

You receive this error message if you attempt to redefine an already defined symbol as a specification parameter.

### Examples

The following example demonstrates a case that results in this error message.

```
reg x;                    /* declaration of 'x' as a register */
  ...
  specify
    specparam x = 1;      /* illegal attempt to redeclare 'x' */
    ...
  endspecify
```

### What Next

Rename the original symbol or the specification parameter and all corresponding uses so that there is no conflict.

---

## VER-302

(warning) %s Pragma '%s' for object '%s' will be ignored because the object could not be found.

### Description

A pragma is a compiler directive in the source code. This warning appears because, during *read* or *analyze* command activity, the symbol used in an *infer_multibit* pragma is not declared in the source code.

### What Next

Possibly you misspelled the name of the symbol or the symbol you used exists only in an old version of the source code and is no longer valid. Correct the symbol name or consider removing the pragma.

---

## VER-305

(warning) %s Drive strength specifications for gate instantiations are ignored.

### Description

You receive this warning message because a drive strength specification has been found in a gate instantiation statement. Drive strength specifications have no meaning for synthesis and are ignored.

Feedback

### Examples

The following example shows a gate instantiation statement containing a drive strength specification, which results in this warning message.

```
not (highz1, strong0) n1 (a,y);  /* highz1 and strong0 are drive
 strengths */
```

### What Next

The drive strength specification is ignored during synthesis. If the code is intended for synthesis purposes only, consider removing the drive strength specification.

---

## VER-306

(warning) %s Drive strength specification for tri-state gate instantiation is ignored.

### Description

You receive this warning message because a drive strength specification has been found in a tristate gate instantiation statement. Drive strength specifications have no meaning for synthesis and are ignored.

### Examples

The following example shows a gate instantiation statement containing a drive strength specification that results in this warning message.

```
bufif (highz1, strong0) b1 (y,a,c);  /* highz1 and strong0 are drive
 strengths */
```

### What Next

If the code is intended for synthesis purposes only, consider removing the drive strength specification.

---

## VER-307

(warning) %s Drive strength specification for pull gate instantiation is ignored.

### Description

You receive this warning message because a drive strength specification has been found in a pull gate instantiation statement. Be aware that drive strength specifications have no meaning for synthesis and are ignored.

### Examples

The following example shows a gate instantiation statement containing a drive strength specification that results in this warning message.

Feedback

```
pullup (strong1) p1 (y);    /* highz1 and strong0 are drive strengths */
```

**What Next**

If the code is intended for synthesis purposes only, consider removing the drive strength specification.

---

## VER-308

(warning) %s Drive strength specification for module instantiation is ignored.

**Description**

You receive this warning message because a drive strength specification has been found in a module instantiation statement. Drive strength specifications have no meaning for synthesis and are ignored.

**Examples**

The following example shows a module instantiation statement containing a drive strength specification that results in this warning message.

```
...
 down (strong1, strong0) d (a,y);  /* module instantiation with drive
 strengths */
 ...
```

**What Next**

If the code is intended for synthesis purposes only, consider removing the drive strength specification.

---

## VER-309

(warning) %s Drive strength specification for continuous assignment is ignored.

**Description**

You receive this warning message because a drive strength specification has been found in a continuous assignment statement. Be aware that the drive strength specifications have no meaning for synthesis and are ignored.

**Examples**

The following example shows a continuous assignment statement containing a drive strength specification that results in this warning message.

Feedback

```
...
 assign (strong1, strong0) y = a; /* containing assignment with drive
 strengths */
 ...
```

**What Next**

If the code is intended for synthesis purposes only, consider removing the drive strength specification.

---

## VER-310

(warning) %s Keyword scalared is ignored.

**Description**

This warning message advises you that *scalared* is an optional advisory keyword. Presto warns about IEEE-1364 conforming uses of this net accessibility specification, but it is otherwise ignored for synthesis.

The following example shows the declaration of a scalared net. Since Presto ignores the keyword *scalared*, the effect is the same as if the keyword had been omitted.

```
...
  wire scalared [1:0] x;
  ...
```

**What Next**

This is a warning message only. No action is required on your part.

---

## VER-311

(warning) %s Parameter range specification is new feature of Verilog 2001. Synthesis and non-Verilog 2001 compatible simulation may have different results.

**Description**

This warning appears during *read* or *analyze* command activity to let you know that some simulators might ignore the range specification in the parameter declaration of your design and expand the parameter to 32 bits. It is thus possible that the circuit synthesized by the Presto HDL Compiler and the non-Verilog 2001 compatible simulation results will not match.

In the following example, the value of *out* from the simulator might be 0, but Presto HDL Compiler gives the value 16.

*Verilog*

Feedback

```
output [4:0] out;
        parameter [3:0] a = 0;

        assign out = {1'b1, a};
```

In the following example, the value of *a* from the simulator might be 10, but the Presto HDL Compiler gives the value 2. The bit on the left is ignored.

*Verilog*

parameter [2:0] a = 4'b1010;

**What Next**

Check accordance between synthesis and simulation at the location in the source code that refers to the bit width of the parameter.

---

## VER-312

(error) %s Cannot write architecture file %s: %s.

**Description**

You receive this error when, during a *read* or *analyze* command, the Presto HDL compiler cannot open or write a file describing the source language (and for VHDL, the architecture) of the module or entity being analyzed.

**What Next**

Make sure that the output directory is writable.

---

## VER-313

(warning) %s Verilog 'signed' data types are present; Verilog 'integer' data types become signed as well.

**Description**

You receive this warning when you use the Verilog 2000 'signed' keyword to declare signed data types, using the Presto HDL Compiler in release 2000.11-1 and subsequent releases. Using signed types causes signed logic for 'integer' types to be generated, as well (that is, any arithmetic using the variable of type integer will generate signed GTECH components in your netlist.)

Ordinarily the interpretation of integers as signed or unsigned data types in generated logic is controlled by the variable hdlin_unsigned_integers. However, when any variables in your design are declared using the 'signed' keyword, the variable is ignored, and integers are always interpreted as signed quantities.

Feedback

This is a departure from releases of Presto prior to 2000.11-1, and from the default HDL Compiler in all previous releases. In earlier releases, the integer data type was not considered a hardware data type, but instead was intended only for compile-time calculations, such as counting loop iterations. As such, the integer data type generated unsigned logic.

This change reflects a reclassification of the integer data type under the Synopsys synthesis subset and conforms more closely to the IEEE Standard 1364-1995.

If you use integer data types to produce logic in your design, you might see a difference between designs compiled with earlier Presto HDL Compiler or HDL Compiler versions, as compared to Presto HDL Compiler version 2000.11-1 and subsequent versions. Make sure that expressions involving integer data types that produce logic are correctly computed when using signed arithmetic.

**What Next**

Ensure that your use of the integer data type is consistent with signed arithmetic.

---

## VER-314

(warning) %s Starting with the 2000.11-1 release, the Presto Verilog reader treats Verilog 'integer' types as signed; synthesized result may not match earlier versions of HDL Compiler.

**Description**

You receive this warning when you use the Verilog 'integer' data type, when compiling your design with the Presto HDL Compiler, in release 2000.11-1 and subsequent releases.

By default, when expressions involving integer variables produce logic in the netlist, signed logic for will be generated (that is, any arithmetic using the variable of type integer will generate signed GTECH components in your netlist.)

This is a departure from Presto HDL Compiler in releases before 2000.11-1 and from the default HDL Compiler in all releases before 2001.08. In those earlier releases, the integer data type was not considered a hardware data type, but instead was intended only for compile-time calculations, such as counting loop iterations. As such, the integer data type generated unsigned logic.

This change reflects a reclassification of the integer data type under the Synopsys synthesis subset and conforms more closely to the IEEE Standard 1364-2001.

If you use integer data types to produce logic in your design, make sure that expressions involving integer data types that produce logic are correctly computed when using signed arithmetic.

Feedback

The interpretation of integers as signed or unsigned data types in generated logic is controlled by the variable hdlin_unsigned_integers. However, when any variables in your design are declared using the 'signed' keyword, the variable is ignored, and integers are always interpreted as signed quantities.

**What Next**

Ensure that your use of the integer data type is consistent with signed arithmetic. If your design requires unsigned arithmetic generated by integer datatypes, and your design does not use Verilog 2000 'signed' types, consider setting hdlin_unsigned_integers to true.

---

## VER-315

(warning) %s Module '%s' was renamed to '%s' with architecture '%s'.

**Description**

You receive this warning because the named module contains the special separator string "__" and was thus renamed to include the module and architecture. This warning informs you of the situation and this man page instructs you what to do if you did *not* intend the renaming.

When a Verilog module name contains the special separator string "__" (two underscores), and the *hdlin_module_arch_name_splitting* variable is set to *true*, Presto HDL Compiler interprets the module name as specifying both a module and a specific architecture or implementation for that module. The portion of the original module name before the separator string becomes the new module name, and the portion after becomes the architecture.

For example, a module named "mod__impl" would be renamed to "mod" and marked as having architecture "impl".

If the variable *hdlin_module_arch_name_splitting* is set to *false*, this renaming never occurs and modules always retain their original names. If it is set to *true*, the "__" is used to divide the module name into module and architecture, as described above.

This capability exists primarily to allow the creation of synthetic libraries using Verilog, with multiple architectures per module.

**What Next**

If you did not intend to specify a module and architecture pair separated by "__", either set *hdlin_module_arch_name_splitting* to *false* or rename your module so that it doesn't contain the separator string.

If you intended to imply a specific architecture, no further action on your part is required.

Feedback

## VER-316

(error) %s The aggregated port expression in the declaration for port '%s' has a mixed direction.

### Description

This error indicates that the compiler has found a port declaration with an aggregated port expression that has a mixed direction.

Modules with mixed-direction port declarations are not supported in synthesis.

In Verilog, port directions are not declared directly, but are instead implied by the directions declared for port identifiers within port expressions.

When the port expression is a concatenation and the port identifiers within it are declared with different directions, the port has a mixed direction.

```
module mixed_direction (.p({a,e})) ;
   input a;
   output e;
   not(e,a);
   endmodule
```

### What Next

If you did not intend to specify a port with a mixed direction, change the expression in the port declaration or change the directions of some the identifiers it uses. If you intended to specify a port expression with mixed direction, split the port into smaller ports that do not have mixed directions.

## VER-318

(warning) %s %s to %s %s occurs.

### Description

You receive this warning message when the Presto HDL Compiler performs one of the following conversions or assignments:

- Implicitly converts an unsigned expression to a signed expression
- Implicitly converts a signed expression to an unsigned expression
- Assigns an unsigned right side to a signed left side
- Assigns a signed right side to an unsigned left side.

Feedback

The following is an example of implicit signed/unsigned conversion:

```
reg signed [3:0] a;
reg [7:0] c;

a = 4'sb1010;
c = a+7'b0101011;
```

According to Verilog standard 2001, the tool converts the signed operand *a* to an unsigned value in the second assignment. The *a* operand is not sign-extended and the Presto HDL Compiler issues the following warning message:

```
signed to unsigned conversion occurs
```

However, if the operand being converted is a constant with leading zeros (in this case, converting between signed and unsigned does not change logic), the Presto HDL Compiler does not issue a warning. For example:

```
c = 5'sb01101+7'b0101011;
```

The first operand on the right side is implicitly converted to an unsigned value, but no warning is issued since the msb of the operand is 0.

Explicit type casting might also disable warnings about conversions. In the example above, Presto HDL Compiler does not issue any warnings if *a* is coerced to be unsigned; for example, by assigning *c* as follows:

```
c = $unsigned(a)+7'b0101011;
```

The following is an example of a signed/unsigned assignment:

```
reg [7:0] a;

a = 4'sb1010;
```

The Presto HDL Compiler issues the following warning message:

```
signed to unsigned assignment occurs
```

Although the left side is unsigned, the right side is still sign-extended, so after the assignment, *a* equals 8'b11111010.

**What Next**

This is a warning message only. No action is required on your part if you want to have signed/unsigned conversions or assignments. However, you can avoid implicit conversion or signed/unsigned assignments by changing your Verilog code.

---

## VER-319

(error) %s defparam to local parameter '%s' inside generate block not supported.

Feedback

### Description

You receive this error message because parameter overwrites inside *generate* blocks to local parameters are not supported by synthesis.

```
module m ();
    parameter p = 1;
    generate
      defparam p = 2;      // not allowed
    endgenerate
  endmodule
```

### What Next

Rewrite your source file, removing the parameter overwrites inside the generate blocks to local parameters, and then run the command again.

### See Also

- read_verilog

---

## VER-320

(warning) %s right hand side of defparam statement should only contain constants or references to parameters.

### Description

You receive this warning message because according to the Verilog standard, the right side of a defparam statement should be a constant expression involving only numbers and references to parameters.

```
module m ();
    genvar i;

    generate
      for (i=0; i<10; i=i+1) begin : B1
        defparam p = i;   // warning: genvar is not allowed on the RHS
      end;
    endgenerate
  endmodule
```

### What Next

No action is required on your part.

However, to strictly comply with the language standard, you should rewrite your code to use only constants or references to parameters on the right side of defparam statements.

You can avoid the warning message by changing the source file so that the defparam statements conform to the Verilog standard.

Feedback

### See Also

- read_verilog

---

## VER-321

(error) %s genvar '%s' used outside generate loop.

### Description

You receive this error message because a genvar can only be used inside the generate loop that uses the genvar as a loop index variable. Using the genvar outside of this loop is not permitted.

```
module m ();
    genvar i;

    generate
      for (i=0; i<10; i=i+1) begin : B1
      ...
      end;
      assign y = i;   // use pf genvar outside generate loop.
    endgenerate
  endmodule
```

### What Next

Change your source file so that all genvars are only used inside the generate loop that uses the genvar as a loop index variable, and then run the command again.

### See Also

- read_verilog

---

## VER-322

(error) %s A packed object cannot contain unpacked data: %s.

### Description

This error message occurs because a structure or union declared as packed cannot contain an unpacked member; an array with packed dimensions cannot have elements whose type is unpacked; or a Verilog-style concatenation cannot include unpacked ingredients.

```
struct packed {
    struct {
    } a;        // error: unpacked structure inside packed structure
  } b;
```

Feedback

```
struct { byte car, cdr } cons;       // This unpacked type cannot be:
cons [12] register_file;             // on a packed dimension, or
register_file = {12{ cons'{0,0} }};  // in a packed concatenation.
```

### What Next

Change your source file so that either the element (and all of its subelements) are declared as packed, or else declare all outer structs and/or unions as unpacked, and write outer array dimensions to the right of their variable name (which connotes an unpacked data arrangement).

In concatenations, you can rewrite the code to treat unpacked data in a separate structure. If this is not possible, then cast the concatenation to the appropriate unpacked type. As a final option, you can cast the offending element to a packed type of the correct width.

---

## VER-323

(error) %s Trying to redeclare typename '%s'.

### Description

You receive this error message because an identifier declared as a typename can't be redeclared later.

```
typedef logic SPOCK;
  ...
  reg SPOCK     // error: 'SPOCK' already declared as a typename
```

### What Next

Change your source file by renaming either the typename or the identifier causing the name clash to a new name.

### See Also

- read_verilog

---

## VER-324

(error) %s Illegal slice name.

### Description

You receive this error message because you are applying more than one slice to an array or the slice is not the last element in the select chain.

The example below demonstrates legal and illegal situations.

Feedback

```
reg [0:3] [4:7] [8:11] a;

  ... = a[1:2];      // legal: only one slice at the end of the select
 chain
  ... = a[2][5:6];   // legal: only one slice at the end of the select
 chain

  ... = a[1:2][4];   // illegal: slice not last element of select chain
  ... = a[1:2]5:6];  // illegal: more than one slice in select chain
```

**What Next**

Change your source file to use select chains with only one slice per select chain and make sure the slice is always the last element in the select chain.

---

## VER-325

(error) %s This structure constructor is missing a value for member '%s'.

**Description**

This error message occurs because structure construction expressions must specify a value for every member of the structure. The cited member was not mentioned in the list of member:value specifications, and none of the type:value or default:value specifications could be applied successfully.

**What Next**

Check the nesting of curly braces around this source line. Check the declaration of the structure type(s) and typedef names involved. After making the necessary changes, run the command again.

---

## VER-326

(error) %s conditionally generated declaration of port '%s'.

**Description**

You receive this error message because the declaration of a port is conditionally generated. All 3 port declarations in the following example are illegal because they are either fully or partly contained in a conditional generate region.

```
module m (a, b, c);
 reg c;
 output b;

 generate
  if (...)
   output a;
   reg a;
```

Feedback

```
   reg b;
   output c;
 endgenerate
endmodule
```

**What Next**

Check the nesting of curly braces around this source line. Check the declaration of the structure type(s) and typedef names involved.

---

## VER-327

(error) %s Non-constant index expression on left hand side of continuous assignment '%s'.

**Description**

You receive this error message because the left hand side of a continuous assignment contains a non-constant index expression. This is not legal Verilog.

```
module m (a, b);
 reg [0:7] a;
 integer b;

 assign a[b] = 1'b0;    // error, 'b' is non-constant

endmodule
```

**What Next**

Rewrite the design such that all left hand sides of continuous assignments contain only constant index expressions. Depending on the context one way of achieving this is to turn the continuous assignment into a nonblocking assignment inside a combinational always block.

---

## VER-328

(error) %s Syntax error on apparent Verilog 2001 construct.

**Description**

You receive this error message because hdlin_vrlg_std has been set to 1995, but the Verilog code apparently contains a Verilog 2001 construct.

**What Next**

Either rewrite the code so that it does not use Verilog 2001 constructs or do not override the default value of hdlin_vrlg_std.

Feedback

## VER-329

(warning) %s Parameter keyword used in local parameter declaration.

### Description

This warning issues when the 'parameter' keyword is used to declare a strictly local parameter. Since this coding style often repeats, only one citation occurs for each locale.

According to the 2nd paragraph of 3.11.1 in the Verilog-2001 LRM, "If any param_assignments appear in a module_parameter_port_list, then any param_assignments that appear in the module become local parameters and shall not be overridden by any method." For example, because p2 in the following is a local parameter:

```
module bottom #(parameter p1 = 13) (output [31:0] o) ;
  parameter p2 = 17 ;
  assign o = p1 + p2 ;
  endmodule
```

an instantiation that attempted to override it would be disallowed:

```
module top (output [31:0] o) ;
  bottom #(.p2(29)) bot ( o ) ;
  endmodule
```

In SystemVerilog there are several additional scope contexts in which the parameter keyword is a synonym for the localparam keyword. According to 6.3.2 in the IEEE SystemVerilog LRM, "Unlike nonlocal parameters, local parameters can be declared in a generate block, in a package, or in a compilation-unit scope. In these contexts, the parameter keyword can be used as a synonym for the localparam keyword."

### What Next

If there is no need to override the parameters in a module or interface declaration either do nothing or use the 'localparam' keyword instead. If there is a need to override the parameter in this declaration, declare them in the list_of_parameter_declarations instead.

The parameters in a package, compilation-unit scope or generate block are always local parameters and cannot be overridden.

## VER-330

(warning) %s No operator for pragma label %s

### Description

In the statement, there is no operator to which the label could be applied.

Feedback

**Verilog**

```
a = b ; // synopsys label my_oper
```

A label can be associated only with an infix operator, such as +, or with a task or function call.

**What Next**

Modify the Verilog source code to remove the pragma or to introduce an operator for the label.

---

## VER-331

(warning) %s Redeclaration of port '%s'.

**Description**

You receive this message because, during *analyze* or *read* command activity, the Presto HDL Compiler has detected that a port previously declared using an ANSI-style port declaration (12.3.4 in the Verilog 2001 LRM) has been redeclared in the body of the module.

Following is an example that illustrates this type of problem:

```
module m ( input a, output b);
    wire a; // redeclaration of input port
    reg b; // redeclaration of ouput port
   endmodule
```

**What Next**

Remove the redeclaration and move any additional information it contains into the ANSI-style port declaration.

---

## VER-332

(warning) %s The '%s' pragma is ignored when applied to 'if' statements.

**Description**

This warning message occurs during *analyze* or *read* command activity when the compiler finds a full_case or parallel_case pragma applied to an "if" statement.

The full_case or parallel_case pragmas are only effective when applied to case statements.

The infer_mux pragma is meaningful when applied to "if" statements.

Feedback

### What Next

Remove any full_case or parallel_case pragmas from "if" statements in your design.

---

## VER-333

(error) %s Widths of ports on 'tran' instantiation are inconsistent.

### Description

This error message occurs when a Verilog "tran" instance has an input and output port that have different widths.

The following module produces this error message because the widths of "a" and "b" are not equal.

```
module E (a, b);

  input [2:0] a;
  output [3:0] b;

  tran (a, b);

endmodule
```

### What Next

Correct the widths of the ports so that they are consistent and then invoke the compiler again.

---

## VER-334

(error) %s Width of interface port is not consistent with width of expression.

### Description

This error message occurs when the design being compiled contains an interface instantiation in which the declared width of the interface's modport does not match the width of the connection in the interface instantiation.

### What Next

Correct the width of the ports so that they are consistent and then invoke the compiler again.

---

## VER-335

(error) %s 'defparam' is not supported inside generate blocks

Feedback

### Description

You receive this error message when running the *read* or the *elaborate* command if your design contains a *defparam* statement within a *generate* block.

Presto HDL Compiler does not support *defparam* parameter value override assignments within *generate* blocks.

The following example demonstrates a supported coding style for parameter value override assignments.

```
module sub (...);
   PARAMETER P1 = 10;
   ...
endmodule


module top (...);

  ...
  generate if(1) begin: blk
    sub u1 (...) ;           //error: defparams not supported
    defparam u1.P1 = 20;     //       inside generate blocks.
  end

  ...

  generate if(1) begin: blk2
    sub #(20) u1 (...) ;    //supported method to assign parameter
  end                       //value overrides

  ...
endmodule
```

### What Next

Rewrite the parameter value assignment as shown in the example above.

---

## VER-336

(warning) %s 'defparam' that accesses a hierarchical name may be ignored.

### Description

The *read* or the *analyze* command issues this message when your design contains a *defparam* statement that modifies a parameter through hierarchical reference.

Presto HDL Compiler ignores *defparam* overrides of parameters inside functions, tasks or named blocks. Since the actual target is not known at the point where the defparam statement is analyzed, this message is just a warning and may be a false negative indication.

Feedback

**What Next**

You may ignore this message if the defparam overrides a parameter which will later be declared in the top level of a module instantiated from within the current design. Otherwise, recode the design and try again.

---

## VER-337

(error) %s PARAMETER assignment statement has same LHS and RHS '%s'.

**Description**

This error message occurs when the PARAMETER assignment statement has the same LHS and RHS. For example: parameter P0 = P0.

**What Next**

Correct the PARAMETER assignment statement and make sure RHS is constant.

---

## VER-338

(warning) %s Net '%s' was renamed to '%s' in module '%s'.

**Description**

This warning message occurs when the Verilog Netlist Reader renames a net to prevent possible name conflicts with other objects.

**What Next**

This is only a warning message. No action is required.

However, if you did not want the Verilog file to be processed by the Netlist Reader, use the *read* command with the *-rtl* option to force it to use Presto HDL Compiler instead.

---

## VER-339

(error) %s Delay, event, or process control is prohibitted in always_comb

**Description**

Issues when a statement within an always_comb block tries to manage the sequence of simulation events. Examples include @*, fork, or disable. Explicit delays may also be flagged as errors when in always_comb, since do not describe combinational logic.

**What Next**

Perhaps you meant to use a traditional always block which is not guaranteed to produce combinational logic. contains error hence latches are inferred unexpectedly.

Feedback

## VER-400

(error) %s Bad format character in template_naming_style or template_parameter_style.

**Description**

Bad format character in template_naming_style or template_parameter_style.

**What Next**

Change the relevant code to conform to the IEEE Std.

## VER-401

(warning) %s Formality does not support %s .

**Description**

Formality does not support this feature now, please do not use it, otherwise, mismatch will happen.

## VER-402

(error) %s Improper usage of internally defined macros: %s

**Description**

Following are Presto Verilog Compiler specific macros, DC, PRESTO, VERILOG_1995, VERILOG_2000, VERILOG_2001 and SYSTEMVERILOG. SYNTHESIS is a reserved macro in IEEE Verilog synthesizable subset. All of the above have a pre-defined value of "". They should only be used with `ifdef, `ifndef and `elsif.

## VER-403

(warning) %s Improper usage of internally defined macros: %s

**Description**

Following are Presto Verilog Compiler specific macros, DC, PRESTO, VERILOG_1995, VERILOG_2000, VERILOG_2001, VERILOG_2005 and SYSTEMVERILOG. SYNTHESIS is a reserved macro in IEEE Verilog synthesizable subset. All of the above have a pre-defined value of "". They should only be used with `ifdef, `ifndef and `elsif.

## VER-404

(error) %s Incorrect value for hdlin.verilog.standard. Valid values are 1995, 2001, or 2005.

Feedback

**Description**

This error message occurs when a nonexistent version of the Verilog standard is requested.

**What Next**

Set the value of *hdlin.verilog.standard* to the intended version and run the command again.

---

## VER-405

(error) %s A non-locally static choice must be the single choice, file support call for workaround.

**Description**

A non-locally static choice must be the single choice, file support call for workaround.

---

## VER-406

(warning) %s A non-locally static choice has more than one choice may not be supported correctly, you need to make sure the generated design is correct.

**Description**

A non-locally static choice has more than one choice may not be supported correctly, you need to make sure the generated design is correct.

---

## VER-407

(error) %s Return port name of type function must be '%s'.

**Description**

You receive this error message because the return port name of the type function is not as required .

**What Next**

Correct the return port name for the type function by using -- pragma *return_port_name*.

---

## VER-408

(error) %s Bad clock expression: '%s'.

Feedback

**Description**

This error message occurs when the clock expression is unacceptable due to the specified issue.

When the specified problem is an unsupported clock expression, be aware that in VHDL only simple constants are supported; for example:

```
generic ( CLK_EDGE: std_logic := '1' );
 ...
 if (clk'event and clk = CLK_EDGE)
 ...
```

However, the following valid code is not simple enough to be supported:

```
generic( CLK_EDGE: std_logic := 1 );
 ...
 constant CLK_EDGE2 : std_logic := CLK_EDGE;
 if (clk'event and clk = CLK_EDGE2)  -- not simple enough
 ...
```

Constant clock expressions that are not simple will be supported in a future release.

**What Next**

Use only simple constants in clock expressions.

---

## VER-409

(error) %s Attributes with non-locally static values on integer/constants are not supported.

**Description**

Attributes with non-locally static values on integer/constants are not supported.

**What Next**

Change the relevant code to conform to the IEEE Std.

---

## VER-410

(warning) %s Expecting explicit event control immediately after always_ff.

**Description**

Expecting explicit event control immediately after always_ff.

---

## VER-411

(error) %s Interface instance is expected here : %s

Feedback

**Description**

Failed to get the expected interface instance.

The most likely cause of this error message is that an interface has been instantiated before declaration. In synthesis, interfaces, unlike modules, must be declared before they are instantiated.

**What Next**

Please check your RTL for typographical errors, or that you have declared the interface before instantiating it and try again.

---

## VER-412

(error) %s Illegal digit %c in unbased number.

**Description**

The digit should be one of the four basic values (0, 1, x, z).

**What Next**

Check your design for invalid digits in unbased numbers and correct them.

---

## VER-413

(error) %s Invalid argument for system task or function : %s

**Description**

This is a syntax error.

**What Next**

Change the relevant code to conform to the IEEE Std.

---

## VER-414

(warning) %s Unsupported hdlin_field_naming_style : %s.

**Description**

Unsupported hdlin_field_naming_style encountered when analyzing/reading SystemVerilog RTL with Presto HDL Compiler.

The hdlin_field_naming_stle variable defines the parts of the net names that Presto HDL Compiler generates corresponding to the fields in VHDL records and SystemVerilog structs.

Feedback

By default, the hdlin_field_naming_style is derived from the bus_naming_style and bus_dimension_separator_style. But when these are "%s[%d]" and "][" or are "%s<%d>" and "><", then it is possible to specify an independent hdlin_field_naming_style, such as "%s<%s>", "%s[%s]", or "%s.%s", in which the first %s stands for the name up to the field and the second %s stands for the field. The hdlin_field_naming_style must be of the form "%sX%s" or "%sX%sY", where X and Y are (possibly identical) non-whitespace characters.

**What Next**

Modify the bus_naming_style, bus_dimension_separator_style, and/or hdlin_field_naming_style.

---

## VER-415

(error) %s POLICY WARNING: missing constant expression in clk event.

**Description**

POLICY WARNING: missing constant expression in clk event.

---

## VER-416

(warning) %s The -preserve command-line option contains the %s hierarchical name.

**Description**

This warning message occurs when the *-preserve* option specifies a hierarchical name. Only simple names are allowed when the *-preserve* option is used with the *analyze*, *read_file*, *read_verilog*, or *read_vhdl* command.

**What Next**

This is only a warning message. No action is required.

However, if the result is not what you intended, change the hierarchical name to a simple name in the relevant script.

**See Also**

- read_verilog

---

## VER-418

(error) %s Empty string used as bus_dimension_separator_style.

Feedback

**Description**

An empty bus_dimension_separator_style string ("") was encountered when elaborating a multidimensional bus.

**What Next**

Do not override the bus_dimension_separator_style to the empty string ("").

---

## VER-419

(warning) %s Empty string used as bus_dimension_separator_style.

**Description**

An empty bus_dimension_separator_style string ("") was encountered when elaborating a multidimensional bus.

**What Next**

Do not override the bus_dimension_separator_style to the empty string ("").

---

## VER-441

(warning) %s Port number %d of '%s' was named '%s'.

**Description**

You receive this warning when a complex port such as the third port in

```
module m(clk, out, .P({a,b}));
```

is declared without an explicit port name, as in

```
module m(clk, out, {a,b});
```

**What Next**

If you want to control the naming of the port, add an explicit name to the port declaration.

---

## VER-444

(error) %s Incorrect value for hdlin_sverilog_std. Valid values are 2005, 2009, 2012, or 2017.

**Description**

You receive this error message when a nonexistent version of the SystemVerilog standard is requested.

Feedback

**What Next**

Set the value of *hdlin_sverilog_std* to the intended version.

---

## VER-445

(error) %s Incorrect value for hdlin_vhdl_std. Valid values are 1987, 1993 or 2008.

**Description**

This error message occurs when a nonexistent version of the VHDL standard is requested.

**What Next**

Set the value of *hdlin_vhdl_std* to *1987*, *1993* or *2008*.

---

## VER-446

(warning) %s The variable hdlin_vhdl_87 is obsolete and will not take effect.

**Description**

This warning message occurs when you use the *hdlin_vhdl_87* variable in your script, The variable is obsolete and will not take effect. Please update your script by replacing occurrences of *hdlin_vhdl_87* with the new *hdlin_vhdl_std* variable. The *hdlin_vhdl_std* default setting is *1993*. To use the 1987 VHDL standard, set *hdlin_vhdl_std* to *1987*.

**What Next**

To avoid this message, replace the *hdlin_vhdl_87* variable in your script with the new *hdlin_vhdl_std* variable.

---

## VER-500

(error) %s %s

**Description**

A syntax or internal error has surfaced during verilog traversering. The message specifies where the error appeared and why it was issued.

**What Next**

Internal errors indicate a problem which should be reported to your Synopsys representitive.

Feedback

## VER-501

(error) %s Redundant %s directives on a chain of if-else-if statements.

### Description

You receive this error message when running the *read* or the *elaborate* command if your SystemVerilog design contains a chain of if-else-if statements with redundant *unique* or *priority* or *unique0* keywords. For example:

```
unique if ( x == 0 )
    y = 1;
  else unique if ( x == 1 )
    y = 2;
```

The second *unique* keyword is not permitted. Only the top if statement should be labeled.

### What Next

Verify that you intend to use a unique or priority or unique0 chain of if statements, and remove the redundant directive.

## VER-502

(error) %s Conflicting directives on a chain of if-else-if statements; outer statements are labeled '%s', inner statement '%s'.

### Description

You receive this error message when running the *read* or the *elaborate* command when your SystemVerilog design contains a chain of if-else-if statements with conflicting *unique* or *priority* or *unique0* keywords. For example:

```
unique if ( x == 0 )
    y = 1;
  else if (x == 1)
    y = 2;
  else priority if ( z == 0 )
    y = 2;
  else if ( z == 0 )
    y = 3;
```

The inner if statement is labeled with a conflicting directive, which is not permitted. Use a directive only on the top if statement.

### What Next

If you want to start a new, separate chain of if-else-if statements inside the final else branch of the outer chain, use a block around the inner chain as shown below.

Feedback

```
unique if ( x == 0 )
    y = 1;
  else if (x == 1)
    y = 2;
  else
    begin
      priority if ( z == 1 )
        y = 2;
      else if ( z == 0 )
        y = 3;
    end
```

Otherwise, decide whether you intend to use a *unique* or *priority* or *unique0* chain of if statements, and label only the top level if statement, removing all other directives in the chain of if-else-if statements.

---

## VER-503

(warning) %s Case statement marked unique has overlapping conditions.

### Description

You receive this warning message when running the *read* or the *elaborate* command if your SystemVerilog design contains a *unique* case statement with overlapping branches. This means that more than one condition can be true at the same time, which contradicts the definition of unique case.

For example, the following case is not unique because the third branch matches any value of c, including the values matched by the first two branches. Therefore, the case requires priority encoding for correct execution.

```
unique casex ( c )
    2'bx1: y = 2;
    2'b1x: y = 1;
    2'bxx: y = 3;
  endcase
```

The synthesis tool may make optimizations based on the *unique* keyword so that the priority encoding logic is simplified. However, this may bring simulation/synthesis mismatch.

If you are unsure whether or not a case is *unique*, do not mark it *unique*.

### What Next

Check the case statement conditions. Make sure that the case labels are intended to overlap. Otherwise, either correct the conditions so that they do not overlap, or remove the *unique* keyword.

Feedback

## VER-504

(warning) %s Case statement marked %s does not cover all possible conditions.

### Description

You receive this warning message when running the *read* or the *elaborate* command if your SystemVerilog design contains a *unique* or *priority* case statement, where it is possible that no branch condition will evaluate to true. This means the case statement may execute, but no branch is taken, which contradicts the definition of *unique* and *priority* case statements.

For example, the following case is not a *unique* or *priority* case, because when c is 3'b011, 3'b010, or 3'b001, no branch is executed.

```
unique casex ( c )
    3'b111: y = 5;
    3'b11x: y = 4;
    3'b1xx: y = 3;
    3'b000: y = 2;
  endcase
```

However, based on your knowledge of the design, you may know that only the listed values will occur. In this case, you can ignore this warning.

The synthesis tool cannot check all *unique* and *priority* case statements for this condition. If a case is marked *unique* or *priority*, but the case statement may execute without having any branch taken, your design may produce unexpected results.

If you are unsure whether or not a case statement is a *unique* or *priority* case statement, do not mark it unique.

### What Next

Check the case statement conditions. Based on your knowledge of the design, if you know that any values that are omitted from the case conditions cannot occur, you may ignore the warning. Otherwise, either add branches to cover the missing values, add a default branch, or remove the *unique* or *priority* keyword.

## VER-505

(warning) %s If statement marked unique has overlapping conditions.

### Description

You receive this warning message when running the *read* or the *elaborate* command if your SystemVerilog design contains a *unique* if statement whose branches overlap. This means that more than one condition can be true at the same time, which contradicts the definition of a *unique* statement.

Feedback

For example, the following if statement is not *unique* because the third branch overlaps with the first. Therefore, the if statement requires *priority* encoding for correct execution.

```
unique if ( c == 2'b01 )
    y = 2;
  else if ( c == 2'b10 )
    y = 1;
  else if ( c == 2'b11 || c == 2'b01 )
     y = 3;
```

The synthesis tool may make optimizations based on the *unique* keyword so that the priority encoding logic is simplified. However, this may bring simulation/synthesis mismatch.

If you are unsure whether or not an if statement is *unique*, do not mark it *unique*.

**What Next**

Check the if statement conditions. Make sure that the case labels are intended to overlap. Otherwise, either correct the conditions so that they do not overlap, or remove the *unique* keyword.

---

## VER-506

(warning) %s If statement marked %s does not cover all possible conditions.

**Description**

You receive this warning message when running the *read* or the *elaborate* command if your SystemVerilog design contains a *unique* or *priority* if statement where it is possible that no branch condition will evaluate to true. This means that the if statement may execute, but no branch is taken, which contradicts the definition of *unique* and *priority* if statements.

For example, the following if chain is not a *unique* or *priority* if statement because only two of the eight possible values of c are checked.

```
priority if ( c == 3'b111 )
    y = 5;
  else if ( c == 3'b110 )
    y = 4;
```

However, based on your knowledge, you may know that only those two values will occur. In this case, you may ignore this warning.

The synthesis tool cannot check all *unique* and *priority* if statements for this error. If an if statement is marked *unique* or *priority*, but the statement may execute without having any branch taken, your design may produce unexpected results.

Feedback

If you are unsure whether or not an if statement is a *unique* or *priority* if statement, do not mark it as *unique* or *priority*.

**What Next**

Check the if statement conditions. Based on your knowledge of the design, if you know that any values that are omitted from the if conditions cannot occur, you may ignore the warning. Otherwise, either add branches to cover the missing values, add a final else branch, or remove the *unique* or *priority* keyword.

---

## VER-507

(error) %s Enumerated type member %s does not match size of type; (size %d, expecting %d).

**Description**

You receive this error message when running the *read* or *elaborate* command if your SystemVerilog design contains an enumerated type declaration, and a member of the enumerated type is the wrong size.

A value can be the wrong size because it will not fit into an explicitly declared data type, or because it does not match the size of a previous enum member that has a sized value.

```
enum short { red = 600000, blue = 2, green = 1} e; // red is too large to
 fit

enum { red = 5'd4, blue = 4'd3, green } e; // blue's size does not match
 red's
```

Consult the SystemVerilog language reference manual for more information on how values are assigned to members of enumerated types.

**What Next**

Examine the members of the enumerated type, and make the changes needed to ensure that each member has the same size.

---

## VER-508

(error) %s Enumerated type member %s duplicates a previously seen value %s.

**Description**

You receive this error message when running the *read* or *elaborate* command if your SystemVerilog design contains an enumerated type declaration, and two members of the enumerated type are assigned to the same value.

Feedback

The error is issued either for members that have an explicit value assigned to them, or for members whose value is determined automatically from the values of the preceding members.

Values of SystemVerilog enumerated type members are determined as follows.

- If an explicit value is given, then that value is used.
- If no explicit value is given, and the member is the first in the enumerated type, it is assigned to 0.
- If no explicit value is given, and the member is not the first in the enumerated type, then it is given a value 1 greater than that of the previous member.

In the following example red and green are explicitly assigned the same value.

```
enum { red = 1, blue = 2, green = 1} e;
```

In the next example, red is assigned explicitly to 2, while green is given the value 2 because the previous member is given the value 1.

```
enum { red = 2, blue = 1, green } e;
```

Consult the SystemVerilog language reference manual for more information on how values are assigned to members of enumerated types.

**What Next**

Examine the members of the enumerated type, and make the changes needed to ensure that each member has a unique value.

---

## VER-509

(error) %s Enumerated type member %s cannot be assigned a four-state value (%s).

**Description**

You receive this error message when running the *read* or *elaborate* command if your SystemVerilog design contains an enumerated type declaration, the data type of the enumerated type is a two-state type, and a member of the enumerated type is assigned to a four-state value.

In the first example below, the int type is a two-state type and cannot represent x or z values, so red cannot be assigned to 3'bxxx. If no data type is specified for the enumerated type, then it defaults to int, which is a two-state type.

```
enum int { red = 3'bxxx, blue = 2, green = 1} e;   // error

enum { red1 = 3'bxxx, blue1 = 2, green1 = 1} e;   // error since default
 is int
```

Feedback

To use four-state values in enumerated types, choose a data type as shown below.

```
enum integer { red = 3'bxxx, blue = 2, green = 1} e;   // OK with integer
```

Consult the SystemVerilog language reference manual for more information on how values are assigned to members of enumerated types.

**What Next**

Either use a four-state data type, or remove any four-state values from the list of enumerated type members.

---

## VER-510

(error) %s Value of enumerated type member %s cannot be determined since previous value was four-state.

**Description**

You receive this error message when running the *read* or *elaborate* command if your SystemVerilog design contains an enumerated type declaration, the specified member of the type is not explicitly assigned a value, and the previous member of the enumerated type is assigned to a four-state value.

Values of SystemVerilog enumerated type members are determined as follows.

- If an explicit value is given, then that value is used.
- If no explicit value is given, and the member is the first in the enumerated type, it is assigned to 0.
- If no explicit value is given, and the member is not the first in the enumerated type, then it is given a value 1 greater than that of the previous member.

When the value of a member of the enumerated type contains x or z, then it is an error not to specify an explicit value for the following member.

In the following example, red contains x, so blue must be explicitly assigned a value.

```
enum int { red = 3'bxxx, blue, green = 1} e;   // error
```

**What Next**

Supply an explicit value for any enumerated type member that follows a member with a four-state value.

---

## VER-511

(error) %s Subscript or range expression of the enum member %s is not a literal constant.

Feedback

### Description

You receive this error message when running the *read* or *elaborate* command if your SystemVerilog design contains an enumerated type declaration that uses a subscript or range expression to produce a series of values in the type, and the subscript or range expression does not consist solely of constants.

The Presto HDL Compiler supports only a literal constant in such subscript or range expressions. For example, the following design fragment generates an error, because a parameter is used in a subscript expression for an enumerated type member.

```
parameter n = 8;
  typedef enum { state[n] } states;
```

The following example showns an acceptable declaration.

```
typedef enum { state[8] } states;
```

### What Next

Supply a constant value for all subscript or range expressions in enumerated type declarations.

---

## VER-512

(error) %s Only integer types or one dimensional packed arrays are permitted as the underlying type of an enumeration.

### Description

You receive this error message when running the *read* or *elaborate* command if your SystemVerilog design contains an enumerated type declaration that specifies an underlying data type, and the underlying data type is not permitted.

Only integer types or one dimensional packed arrays are permitted as the underlying type of an enumeration. The following is an example of an unacceptable declaration.

```
/* Not acceptable */
  typedef enum logic signed [0:0][0:0] { state1, state2, state3, state4 }
 states;
```

This example shows an acceptable declaration.

```
typedef enum logic signed [1:0] { state1, state2, state3, state4 }
 states;
```

### What Next

Correct the design to use an integer type or one dimensional packed array as the underlying type of the enumeration.

Feedback

## VER-513

(error) %s redeclaration of symbol %s as constant.

### Description

You receive this error message if a symbol in a constant declaration has already been declared, either as a constant or as something else. In the following example, in is first declared as an input, then erroneously redeclared as a constant.

```
input in;
 ...
 const in = 1'b0;       // erroneous redeclaration of input 'in'
                        // as a constant
```

### What Next

Decide which of the declarations is the intended one and remove or change the name in the other declaration, and then run the command again.

## VER-514

(error) %s Return statement in non-void function %s requires an expression as return value.

### Description

You receive this error message when running the *read* or *elaborate* command if your SystemVerilog contains a function with a non-void return type, and a return statement occurs without an expression to use as the return type.

When a function is declared with a return type, all return statements must be accompanied by an expression.

### What Next

Supply a return value in the return statement and run the command again.

## VER-515

(warning) %s Discarding return value of function %s.

### Description

You receive this warning message when running the *read* or *elaborate* command if your SystemVerilog design contains a task-style call to a function where the function's return value is not used.

The warning is not issued if the return type of the function is void.

Feedback

### What Next

This is a warning message only. No action is required. If you intend to always ignore the return value of this function, change its return type to void. Otherwise, embedd this function call in an assignment statement or other expression that will demand its result value.

---

## VER-516

(error) %s the name %s after the end construct does not match the name %s of the construct.

### Description

You receive this error message when the name after the end construct (module, interface, block, function, or task) does not match the construct name.

The following is an example of mismatched names after endmodule.

```
module name1(input i, output o);
assign o = i;
endmodule :name2  //name2 is not the same as name1
```

The following shows another example of mismatched names for sequential block.

```
module name1(input i, output reg o);
always begin
  o = i;
end : name2  //the name of the sequential block is not specified.
endmodule : name1
```

### What Next

Correct the names so that they match and run the command again.

---

## VER-517

(error) %s A case statement is marked with both a unique or priority keyword, and a full_case or parallel_case directive.

### Description

You receive this error message from the *read* or *elaborate* commands when your SystemVerilog design contains a case, casex, or casez statement marked with the 'priority' or 'unique' keyword that is also marked with the '//synopsys parallel_case' or '//synopsys full_case' pragma.

In Verilog, pragmas were the only means of communicating information about a case statement's properties to the compiler. In SystemVerilog, the 'unique' and 'priority'

Feedback

keywords convey that information. The Presto HDL Compiler accepts either style, but does not accept a mixture of the two styles on the same case statement.

**What Next**

If knowledge of the design guarantees that :

```
1. At least one case condition will evaluate to true
BUT NOT:
  2. No more than one case condition will evaluate to true
```

then use the 'priority' keyword on the case.

If knowledge of the design guarantees that :

```
1. At least one case condition will evaluate to true
AND:
  2. No more than one case condition will evaluate to true
```

then use the 'unique' keyword on the case.

If knowledge of the design guarantees that :

```
1. No more than one case condition will evaluate to true
BUT NOT:
  2. At least one case condition will evaluate to true
AND:
```

then use the '//synopsys parallel_case' pragma for the case statement.

If neither of these conditions is known to be true, then use no keyword or pragma.

---

## VER-518

(error) %s Cannot recover from previous errors.

**Description**

This error message occurs when the errors reported so far have left the compiler in a state where further error reports would be inaccurate, misleading, or omitted. The *analyze* or *read* command stops even if some of the source text has not yet been processed. For example, if the $unit scope of a SystemVerilog compilation issues any errors, compilation of modules or interfaces with visibility into that $unit may not proceed.

**What Next**

Resolve the previously reported error message(s), and rerun the *analyze* or *read* command.

Feedback

## VER-519

(warning) %s Return value of function '%s' is not set.

### Description

This warning message occurs during a *read* or *elaborate* command, when your SystemVerilog design contains an automatic function with a non-void return value that does not set the return value inside the function body. The return value of the function is not specified by the language standard. Change the your code to clearly specify the result.

The warning is not issued if the return type of the function is *void*.

```
function automatic f;
   input a;
   reg r;
   r = 2 * a;
 endfunction

 always begin
  y = f (a);
 end
```

### What Next

If you intended to ignore the return value of the function you can disregard the warning message.

If the result is not what you intended, change the code to clearly specify the result.

## VER-520

(warning) %s Task enable in always_comb block.

### Description

This warning message occurs during a *read* or *analyze* command when your SystemVerilog design contains a task enable within an always_comb block. This is a possible simulation-synthesis mismatch, because the language standard does not require simulators to be sensitive to the contents of a task.

### What Next

Modify your code to use calls to void functions instead of task enables within always_comb blocks and then run the command again.

## VER-521

(error) %s Using nonexistent return value of void function '%s'.

Feedback

### Description

You receive this error message from the *read* or *analyze* commands when your SystemVerilog design uses a call to a void function as an expression instead of like a task enable.

```
function automatic void f ( a , output [1:0] b ) ;
 reg r;
    b = 2 * a;
 endfunction

 always begin
    y = f (a , b);
 end
```

### What Next

Modify the void function to one that returns a value or to pass back the result through an output port.

---

## VER-522

(error) %s The 'inout' port '%s' is not of net type.

### Description

The Presto HDL Compiler issues this error message during execution of the *analyze* or *read* command when it encounters an inout port declared with a non-net type, such as 'struct', 'bit', 'reg' or 'logic'. According to IEEE Std 1800-2005, section 19.12.1, a variable data type is not permitted on either side of an inout port.

### What Next

Rewrite the SystemVerilog code to use a net type on this port or to use two ports instead of just one. (In simulation, a 'ref' port could also be used, but 'ref' ports are not synthesizable.)

---

## VER-523

(error) %s Static %s '%s' is not synthesizable in %s, expecting 'automatic' keyword.

### Description

This error message occurs when a function or task that is not automatic is encountered in $unit (formerly known as $root), in a package declaration, or in an interface declaration. The synthesizable SystemVerilog subset requires that each task and function in $unit, in a package declaration, or in an interface declaration be declared *automatic*.

Feedback

**What Next**

Add the *automatic* keyword after the *task* or *function* keyword.

---

## VER-524

(error) %s Multiple implicit port connection tokens in instantiation.

**Description**

This syntax error message occurs when the .* implicit port connection token is used more than once in a single module or interface instantiation:

xtend xtend(.* , .dout(dataout[7:0]), .*, din(alu_out[7]));

**What Next**

Remove all but one of the .* tokens from the instantiation. Run the *analyze* or *read* command again.

---

## VER-525

(error) %s Invalid interface access %s.

**Description**

This error message occurs when the Presto HDL Compiler finds an error when accessing the interface instance or the interface port.

For example:

```
I inst();   // I is an interface definition, with modport mp
            // assign a = inst.mp.b;
```

**What Next**

Correct the access to the interface and then run the compiler again.

---

## VER-526

(error) %s Invalid name %s in the modport %s of the interface %s.

Feedback

### Description

This error message occurs when the Presto HDL Compiler finds an invalid name in the modport definition. This error may be caused by one of the following:

- There is no valid symbol declared in the interface.
- The symbols declared are inconsistent with the usage in the modport declaration. In the following example, the function *foo* is used as a port and signal *s* is used as a function or task:

```
interface I();
function foo ... endfunction
logic s;
modport MP (input foo, import function s());
endinterface
```

- The name has been used in the modport declaration. In the following example, there are two ports named *s*:

```
interface I();
function foo ... endfunction
logic s;
modport MP (input s, input s, import foo());
endinterface
```

### What Next

Correct the name in the modport definition and then run the compiler again.

---

## VER-527

(error) %s The input port %s is not writable.

### Description

This error message occurs when the input port expanded from modport is not writable.

In the following example, *a* is an input port expanded from modport port *i* in module *M*. The assignment to *i.a* causes the error message.

```
interface I ();
logic a;
modport MP(input a);
endinterface

module M(I.MP i);
assign i.a = 1'b1;
endmodule
```

Feedback

**What Next**

Correct any errors in the code and run the command again.

---

## VER-528

(warning) %s Invalid symbol name '%s' in the pragma signal name list. It is being ignored.

**Description**

This warning message occurs because the Presto HDL Compiler found an invalid symbol name in the pragma signal name list. This problem may be caused by one of the following:

- There is no valid symbol declared in the module.
- The usage of the symbol in the pragma signal name list is not consistent with the symbol declaration in the module.
- The type of the symbol (such as constant or function name) is not allowed to appear in the pragma signal name list.

**What Next**

This is only a warning message. You can eliminate this warning by following the instructions below.

If you do not want the symbol to be ignored from the pragma signal name list, correct the symbol name in the pragma signal name list and then run the compiler again.

---

## VER-530

(warning) %s Treating %s '%s' as automatic in %s; possible simulation-synthesis mismatch.

**Description**

This warning message occurs when a function or task that is not declared *automatic* is encountered in $unit (formerly known as $root), in a package declaration, or in an interface declaration. The synthesizable SystemVerilog subset requires that each task and function in $unit, in a package declaration, or in an interface declaration be declared *automatic*.

**What Next**

Add the *automatic* keyword after the *task* or *function* keyword and run the command again.

Feedback

## VER-531

(information) %s Variable '%s' is being read, but is not initialized. The subprogram default value will be used.

### Description

You receive this message from the Presto HDL Compiler when your design contains a variable which is read, but has not been initialized yet. The subprogram default value will be used.

The subprogram default value can be 0's (as default) or tick-LEFT, which is controlled by the switch:

hdlin_subprogram_default_values

### What Next

Verify that you intended to use the subprogram default value as the initial value of the uninitialized variable.

## VER-532

(warning) %s If statement with %s directive but no else clause.

### Description

You receive this warning message when running the *read* or the *elaborate* command if your SystemVerilog design contains a unique or priority if that has no has no else clause. For example:

```
unique if ( x == 0 )
    y = x;
```

According to the language standard, the unique or priotity keyword on an if with no else clause, is an assertion that the condition is always true. Hence, the example is equivalent to an assertion that x is always 0, followed by the assignment. It could be optimized to

```
y = 0;
```

### What Next

Verify that you intend to assert that the condition is always true. If so, consider using a formal assertion. If not, remove the unique or priority keyword.

## VER-533

(warning) %s Using default enum base size of 32.

Feedback

### Description

You receive this warning from the *read* or *analyze* commands when your SystemVerilog design contains an enum declaration without an explicit size declaration after the enum keyword. According to the language standard, the default size of an enum is 32. For example, in

```
typedef enum { RED, BLUE, GREEN, YELLOW } Color ;
  Color current, next ;
```

current and next will be 32 bits wide, not 2 bits wide, as in

```
typedef enum [1:0] { RED, BLUE, GREEN, YELLOW } Color ;
  Color current, next ;
```

### What Next

Modify your code to use an explict size declaration for the enum.

---

## VER-537

(error) %s Illegal %s direction for %s '%s' in modport.

### Description

This error message occurs when the Presto HDL Compiler finds a variable listed in a modport definition with the direction of *inout*, or a net listed with the direction of *ref*. These combinations are not possible for ordinary module ports, so they cannot be used to specify how to deliver the given interface content as part of an interface modport connection.

### What Next

Correct either the direction setting for this element of the modport list, or the variable versus net declaration of the interface content. Variables may pass as *ref*, *input*, or *output*. Nets may pass as *inout*, *input*, or *output*. Reanalyze the changed interface source.

---

## VER-538

(warning) %s Nonstandard %s direction for %s '%s' in modport.

### Description

This warning message occurs when the Presto HDL Compiler finds a variable listed in a modport definition with the direction of *inout*, or a net listed with the direction of *ref*. These combinations are illegal for ordinary module ports. The *hdlin_sv_allow_inout_modport_vars* switch setting has allowed Presto to form a wired connection to continuously transmit the value of this interface content.

Feedback

### What Next

This is only a warning message. No action is required.

However, to avoid this message you can correct either the direction setting for this element of the modport list, or the variable versus net declaration of the interface content. Variables may pass as *ref*, *input*, or *output*. Nets may pass as *inout*, *input*, or *output*. Reanalyze the changed interface source.

---

## VER-540

(warning) %s Redefining macro '%s' with value '%s' to '%s'.

### Description

A macro is being defined at least twice with different values in the RTL code. If this is not really intented by customer, it may produce unexpected syntesys results.

The macro redefinition may happend if it is defined several times in different files or when macros are defined in files included by other include files.

### What Next

If a redefinition of the macro was not intended, just rename the macro such that it does not collide with the other macro definition, or keep files using one set of macro definitions separate when reading in the designs, to avoid macro names clashes.

---

## VER-541

(warning) %s Parameters of macro '%s' have been redefined.

### Description

A macro is being defined at least twice with a different list of parameters in the RTL code. If this is not really intented by customer, it may produce unexpected synthesis results.

The macro redefinition may happen if it is defined several times in different files or when macros are defined in files included by other include files.

The redefinition checks for literal parameter definition. If in one file exists this definition:

`define SUM(x,y) x+y

and in another verilog file exists another definition of SUM:

`define SUM(a,b) a+b

and if both files are used while compiling the same unit scope, VER-541 warning message will be issued, even if the macros are semantically equal.

Feedback

When a macro is being redefined, usually the definition of macro will change too, so an additional VER-540 warning message may be issued.

**What Next**

If a redefinition of the macro was not intended, just rename the macro so that it does not collide with the other macro definition, or keep files using one set of macro definitions separate when reading in the designs, to avoid macro names clashes.

---

## VER-553

(error) %s Pack operation's result width is less than the total width of the source expressions.

**Description**

The Presto HDL Compiler issues this error message when the right-hand side of an assignment or the operand of a cast is a streaming operator ({<<{...}} or {>>{...}}) that is producing more bits than the left-hand side or the casting-type can utilize.

**What Next**

Increase the number of bits allocated in the target type, or shorten the data sizes of the source expressions.

---

## VER-554

(error) %s Unpack operation using streaming operator requires that width of source bit stream be greater than or equal to target.

**Description**

The Presto HDL Compiler issues this error message when the left-hand side of an assignment is a streaming operation that consumes more bits than the right-hand side can provide.

**What Next**

Correct any errors like this in your code.

---

## VER-555

(warning) %s Truncation discards bits.

**Description**

The Presto HDL Compiler issues this warning message when the left-hand side of an assignment discards some of the bits provided by the right-hand side.

Feedback

### What Next

Correct any errors like this in your code.

---

## VER-561

(error) %s Subscript expression of the enum member %s is not a positive literal constant.

### Description

You receive this error message when running the *read* or *elaborate* command if your SystemVerilog design contains an enumerated type declaration that uses [N] to produce a series of members in the type, but the N is not positive.

Only a positive literal constant is legal in that context.

For example, the following design fragment generates an error, because [0] is used to produces a series of members.

```
typedef enum { state[0] } states;
```

### What Next

Supply a positive literal value for expressions in enumerated type declarations.

---

## VER-563

(warning) %s Case statement marked unique0 has overlapping conditions.

### Description

You receive this warning message when running the *read* or the *elaborate* command if your SystemVerilog design contains a *unique0* case statement with overlapping branches. This means that more than one condition can be true at the same time, which contradicts the definition of unique0 case.

For example, the following case is not unique0 because the third branch matches any value of c, including the values matched by the first two branches. Therefore, the case requires priority encoding for correct execution.

```
unique0 casex ( c )
    2'bx1: y = 2;
    2'b1x: y = 1;
    2'bxx: y = 3;
  endcase
```

The synthesis tool may make optimizations based on the *unique0* keyword so that the priority encoding logic is simplified. However, this may bring simulation/synthesis mismatch.

Feedback

If you are unsure whether or not a case is *unique0*, do not mark it *unique0*.

**What Next**

Check the case statement conditions. Make sure that the case labels are intended to overlap. Otherwise, either correct the conditions so that they do not overlap, or remove the *unique0* keyword.

---

## VER-565

(warning) %s If statement marked unique0 has overlapping conditions.

**Description**

You receive this warning message when running the *read* or the *elaborate* command if your SystemVerilog design contains a *unique0* if statement whose branches overlap. This means that more than one condition can be true at the same time, which contradicts the definition of a *unique0* statement.

For example, the following if statement is not *unique0* because the third branch overlaps with the first. Therefore, the if statement requires *priority* encoding for correct execution.

```
unique0 if ( c == 2'b01 )
    y = 2;
  else if ( c == 2'b10 )
    y = 1;
  else if ( c == 2'b11 || c == 2'b01 )
     y = 3;
```

The synthesis tool may make optimizations based on the *unique0* keyword so that the priority encoding logic is simplified. However, this may bring simulation/synthesis mismatch.

If you are unsure whether or not an if statement is *unique0*, do not mark it *unique0*.

**What Next**

Check the if statement conditions. Make sure that the case labels are intended to overlap. Otherwise, either correct the conditions so that they do not overlap, or remove the *unique0* keyword.

---

## VER-700

(error) %s The construct '%s' is not supported in synthesis.

**Description**

This error message occurs when the syntactic construct cited is not supported for synthesis.

Feedback

### What Next

Exclude non-synthesizable constructs by enclosing the code with the conditional compilation directives shown below.

```
module mt( a );
  `ifndef SYNTHESIS
    localparam realtime bus_stop=42;
  `endif
```

## VER-701

(error) %s Variable '%s' index not supported.

### Description

You receive this error from the *read -f verilog -netlist* command, which invokes the Verilog netlist reader, because your code contains constructs the Verilog netlist reader does not support. To read a limited set of constructs more efficiently, the Verilog netlist reader supports only those Verilog constructs that commonly occur in netlists.

The Verilog netlist reader does not support the use of variables within subscripts or array declarations.

### What Next

If your code contains any of these constructs, use HDL Compiler to read your Verilog design. Invoke HDL Compiler by omitting the "-netlist" option from the *read* command, or by using the *analyze* and *elaborate* commands.

## VER-702

(error) %s Unknown wire type '%s'.

### Description

This error message occurs error message is generated by the *read -f verilog -netlist* command, which invokes the Verilog netlist reader, because the code contains constructs the Verilog netlist reader does not support. To read a limited set of constructs more efficiently, the Verilog netlist reader supports only those Verilog constructs that commonly occur in netlists, in order to read a limited set of constructs more efficiently.

The Verilog netlist reader does not support the use of some Verilog variable declarations, such as the *reg* data type.

Feedback

**What Next**

If your code contains these constructs, use HDL Compiler to read your Verilog design. Invoke HDL Compiler by omitting the *-netlist* option of the *read* command, or by using the *analyze* and *elaborate* commands.

---

## VER-703

(error) %s Unknown constant value '%d'

**Description**

You receive this message because The Verilog netlist reader was unable to read the design; the reader could not determine a constant value for the variable named in the message.

**What Next**

Use HDL Compiler to read your Verilog design. Invoke HDL Compiler by omitting the "-netlist" option from the *read command*, or by using the *analyze* and *elaborate* commands.

---

## VER-704

(error) %s Constant width not matched '%d' vs '%d'.

**Description**

This is a syntax error.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

---

## VER-705

(error) %s Constant signal '%s' cannot be at the left side.

**Description**

This is a syntax error.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

---

## VER-706

(error) %s Verilog netlist reader encountered trouble to build the named port in this design.

Feedback

### Description

Verilog netlist reader has limited support for named port handling. The following examples demonstrate the typical cases netlister reader now has trouble handling.

```
module inouta (.a(i),.b(i));
        inout i;
    endmodule

    module bit_select ( .a(i), .b1(j[1]), .b2(j[0]) );
        input        i;
        output [1:0] j;
    endmodule
```

### What Next

Please use read command without the -netlist option.

---

## VER-707

(error) %s Variable index not supported.

### Description

This error message occurs when running the *read -f verilog -netlist* command, which invokes the Verilog netlist reader, because your code contains constructs the Verilog netlist reader does not support. To read a limited set of constructs more efficiently, the Verilog netlist reader supports only those Verilog constructs that commonly occur in netlists.

The Verilog netlist reader does not support the use of variables or arithmetic operations, within subscripts or array declarations.

### What Next

If your code contains any of these constructs, use HDL Compiler to read your Verilog design. Invoke HDL Compiler by omitting the *-netlist* option from the *read* command, or by using the *analyze* and *elaborate* commands.

---

## VER-708

(warning) %s The construct '%s' is not supported in synthesis; it is ignored.

### Description

This warning message issues when the cited construct cannot be synthesized, but might be ignorable. This can lead to behavior which differs from simulation.

Feedback

### What Next

Exclude non-synthesizable constructs by enclosing the code with the conditional compilation directives shown below.

```
module mt( a );
     logic static_init
  `ifndef SYNTHESIS
                       = 1'b0
  `endif
                              ;
     always ...
        if( !static_init ) ...
```

## VER-709

(warning) %s The interconnect net will be treated as a wire net in synthesis.

### Description

This warning message issues when a SystemVerilog interconnect net declaration is encountered, because it will be treated as a wire net declaration in synthesis.

There are fewer restrictions on the usage of wire nets than there on interconnect nets, so it's possible that some illegal usages of interconnect nets will be accepted in synthesis but correctly rejected in simulation.

### What Next

Exclude illegal usages of interconnect nets by also reading the design into a simulator. If there are no illegal usages, no further action is needed.

## VER-710

(error) %s The construct '%s' is not supported when %s = %s.

### Description

This error message occurs because the syntactic construct cited is not supported by the current configuration of the HDL Compiler or optimizer cited in the message.

### What Next

Adjust the global option named in the message. Note that not all option combinations are possible. After making your changes, run the command again.

## VER-720

(error) %s The construct '%s' is not supported in this language.

Feedback

**Description**

This error message occurs when the syntactic construct cited is not supported for the language specified by the *-format*, *-rtl*, or *-netlist* option of your *read* or *analyze* command.

Verilog (*-format verilog*) reports this error for syntax where SystemVerilog (*-format sverilog*) has extended the language beyond IEEE Std 1364.

This error may also be reported in rare situations where SystemVerilog does not implement syntax that is only available in older editions of Verilog.

The *-netlist* reader reports this error message about features that are not supported in gate-level netlists.

**What Next**

Adjust your use of the *-format* option or Tcl commands with the *read_* prefix to parse the intended source language.

Switching from *-netlist* to *-rtl* may not be a practical solution for a large design. Use the *-rtl* reader on a small sample of the cited syntax, write the equivalent gate-level output, and then recode your *-netlist* input in the pattern suggested by the *-rtl* compiler.

---

## VER-721

(error) %s The construct '%s' is not supported.

**Description**

This error message occurs when the syntactic construct cited is not supported in this compiler release.

**What Next**

Check whether or not you are running a current version of the synthesis tool.

If you are interested in more information about this feature, contact the Synopsys Support Center by using Enter A Call at http://solvnet.synopsys.com/EnterACall.

For information about the latest software releases, go to the Synopsys SolvNet Release Library at http://solvnet.synopsys.com/ReleaseLibrary.

For information about creating, packaging, and sending a test case, go to http://www.synopsys.com/testcase.

---

## VER-722

(error) %s The relative placement directive '%s' has a syntax error.

Feedback

**Description**

This error message occurs when the relative placement directive has a syntax error.

For example, the orientation in may be *rp_orient* incorrect. The orientation should be N|W|S|E|FN|FW|FS|FE.

Another example is that the alignment default is not *sw* in *rp_align*.

**What Next**

Correct the relative placement syntax error.

---

## VER-730

(warning) %s The '%s' syntax is non-standard. %s

**Description**

This is a generic warning message which issues at every use of certain non-standard language features. The report includes a sentence to clarify what semantics are provided in lieu of the standard-prescribed error.

The setting of hdlin_vrlg_std at the time the (System)Verilog source is analyzed can influence which syntax gets this citation and what semantics are supplied.

**What Next**

Make sure you agree that the non-standard meaning is the intended behavior for this RTL. Check the setting of hdlin_vrlg_std if you believe that this construct is meant to invoke a standard definition. If you need more choices of behavior when migrating legacy codes to current standards, contact Synopsys.

---

## VER-731

(warning) %s The '%s' syntax is non-standard. %s\n\tOnly the first use of this construct will be cited.

**Description**

This is a generic warning message which issues on the first use of oft-repeated, but non-standard, language features. The report includes a sentence to clarify what semantics are provided in lieu of the standard-prescribed error. Other uses of the construct will be handled similarly, but silently.

The setting of hdlin_vrlg_std at the time the (System)Verilog source is analyzed can influence which syntax gets this citation and what semantics are supplied.

Feedback

### What Next

Make sure you agree that the non-standard meaning is the intended behavior for this RTL. Check all uses of the non-standard construct. Check the setting of hdlin_vrlg_std if you believe that this construct is meant to invoke a standard definition. If you need more choices of behavior when migrating legacy codes to current standards, contact Synopsys.

---

## VER-733

(warning) %s Forward references to '%s' are non-standard. %s

### Description

This message warns about violations of the SystemVerilog standard requirement that data variables be declared before use. Only the first reference to each name is cited. The references have been tolerated without this warning because the selection syntax you used could also be a legal form of hierarchical reference, but the explicit local declaration for this name became available before a hierarchical reference was located.

This is an unsound language extension, since forward references to variables cannot be resolved in all possible contexts. To aid the migration of legacy codes that may have relied upon forward reference, this message has warning level severity; it will rise to an error level severity in a future release of Presto SystemVerilog.

The setting of the *hdlin_sverilog_std* variable at the time the SystemVerilog source is analyzed can influence which syntax gets this citation and what semantics are supplied.

### What Next

Make sure you agree that the non-standard meaning is the intended behavior for this RTL. Check the setting of the *hdlin_sverilog_std* variable if you believe that this construct is meant to invoke a standard definition. If you need more choices of behavior when migrating legacy codes to current standards, contact Synopsys.

---

## VER-734

(warning) %s Nonstandard package access '%s::'; this name has a more local definition.

### Description

This message warns about your use of a nonstandard language extension which might not be available in other SystemVerilog implementations. The SystemVerilog standard prescribes that the identifier left operand to the scope operator (::) can resolve to a local data object, function, block label, or a wildcard import candidate for such an object, even though doing so inevitably leads to an error for misuse of the scope extraction syntax. Your access to a package with the same name as this more local definition will not be supported on a minimally conforming language processor. Notwithstanding this language

Feedback

extension, the scope operator always resolves the name of a local type even if this leads to an error, despite the existence of a package of the same name.

**What Next**

No action is required. The scope operator found a package with the given name. To make this use conform to the IEEE standard, rename either the package or its local homograph.

**See Also**

- suppress_message

---

## VER-735

(information) %s Variables crossing hierarchy: %s%s%s.

**Description**

This information message occurs for a few RTL coding styles involving variables where simulation and synthesis use different approaches.

In simulation, an RTL variable is allocated storage at its point of declaration. References to simulation variables fetch the most recent value written into that storage, regardless of any process or hierarchical boundaries between various writers and readers.

In synthesis, storage is inferred whenever a process uses a variable as a way to feed its output back onto its input. These well-known conventions are not entirely incompatible, in fact it is typical for several processes to read the stored output of one other process, and for the RTL to express this using a variable.

More aggressive uses, for example with multiple processes writing, or with a single process using explicit delays to write at distinct simulation times, have no synthesizable counterparts. These will lead to multiple driver errors, or to race conditions that the tool may only warn you about by displaying "ignoring delays".

Presto informs you when a design hierarchy is added that separates two processes sharing a simulation variable. These situations include the following:

Explicit uses of *ref* port direction

In simulation, *ref* ports are used to connect simulation variables across the design hierarchy. When synthesized, *ref* ports connect wires and are continuously driven. This could cause nets to be multiply-driven or cause value sequences, controlled by simulation time delays, to resolve differently outside the module.

Feedback

Implicit connection of *ref* ports

Variables within an interface that are not given *input* or *output* direction by a modport declaration, will be connected via *ref* ports as described above.

In each case, there may be another RTL coding style that will present simulation with a more precise picture of what is being synthesized.

**What Next**

This is an information-only message. No action is required.

Implicit communication of variables can sometimes be avoided by adding a modport declaration that excludes the variable, and using it to modify the interface instance when it is passed as a port. Also, consider presenting the data on a signal wire instead. If your simulator does not permit a data type to be declared for a wire, use size'() casting to extinguish the type information as it is assigned to or read from a net.

**See Also**

- suppress_message

---

## VER-736

(warning) %s Task '%s' with non-empty body is mapped to '%s'; body will be ignored.

**Description**

This warning message occurs when you map a task to a module because the module (instead of the task body) determines the implementation of the task. The body of the task is ignored.

The following example contains a task for which this warning message is issued:

```
task so_msff_t(output so, input si);  // synopsys map_to_module r00foobar
  begin
     if(si === 1'b1) // task body will be ignored
       so <= si;
  end
  endtask
```

**What Next**

This only a warning message.

Make sure that you intended the body of the task to be ignored in favor of the module that the task is mapped to.

Feedback

## VER-737

(warning) %s No module instance name.

**Description**

The Presto HDL Compiler issues this warning message when a module instantiation without an instance name is detected during the *analyze* or *read* command.

According to the Verilog language standard, module instantiations must include an explicit name. For backward compatibility reasons, the Presto HDL Compiler issues a warning instead of an error on such usage and invents an instance name.

**What Next**

Add an explicit instance name so that the instantiation conforms with the language standard.

## VER-738

(error) %s No structure of instances name.

**Description**

The Presto HDL Compiler issues this error message when a structure of instances without a name is detected during the *analyze* or *read* command. This syntactic restriction is consistent with the syntactic restrictions the Verilog langauage standard imposes on arrays of instances.

**What Next**

Add an explicit name for the structure of instances.

## VER-739

(error) %s No array of instances name.

**Description**

The Presto HDL Compiler issues this error message when an array of instances without a name is detected during the *analyze* or *read* command. According to the Verilog language standard, every array of instances, even of gate instances, must have an explicit name specified.

**What Next**

Add an explicit name for the array of instances.

Feedback

## VER-746

(warning) %s The declaration of module '%s' is being ignored, because the module name is in hdlin_sv_blackbox_modules.

### Description

This warning message issues when the declaration of a module with its name in hdlin_sv_blackbox_modules is encountered. All text until the next endmodule token is skipped.

### What Next

If this module declaration should not be ignored, remove the module name from hdlin_sv_blackbox_modules.

## VER-747

(warning) %s The body of module '%s' is being discarded, because the module name is in hdlin_sv_interface_only_modules.

### Description

This warning message issues when the declaration of a module with its name in hdlin_sv_interface_only_modules is encountered. The body will be discarded, except for port declarations.

### What Next

If the body of this module declaration should not be ignored, remove the module name from hdlin_sv_interface_only_modules.

## VER-791

(warning) %s An end_keywords directive without a matching begin_keywords directive.

### Description

The Presto HDL Compiler issues this warning when excess end_keywords directives are encountered. For example,

```
`begin_keywords "1364-1995"
      module test (in, generate);
      input in;
      output generate;
      assign generate = in;
      endmodule
   `end_keywords
   `end_keywords
```

Feedback

### What Next

Determine whether there is a missing begin_keywords directive or too many end_keywords directives.

---

## VER-792

(warning) %s A begin_keywords directive without a matching end_keywords directive.

### Description

The Presto HDL Compiler issues this warning when insufficient end_keywords directives are encountered. For example,

```
`begin_keywords "1364-1995"
   `begin_keywords "1364-1995"
      module test (in, generate);
      input in;
      output generate;
      assign generate = in;
      endmodule
   `end_keywords
```

### What Next

Determine whether there are too many begin_keywords directives or not enough end_keywords directives to match them.

---

## VER-793

(error) %s Unknown begin_keywords directive version specifier '%s'.

### Description

The Presto HDL Compiler issues this error when the begin_keywords directives is used with an unknown version specifier. For example,

```
`begin_keywords "1364-1999"
      module test (in, generate);
      input in;
      output generate;
      assign generate = in;
      endmodule
   `end_keywords
```

### What Next

Determine if there is typo in the version specifier, which must be one of "1364-1995", "1364-2001", "1364-2001-noconfig", "1364-2005", or "1800-2005".

Feedback

## VER-794

(error) %s Missing or unknown version specifier in begin_keywords directive.

### Description

The Presto HDL Compiler issues this error when the begin_keywords directives is used alone, instead of with a version specifier, or when the double quotes in the version specifier are omitted. For example,

```
`begin_keywords
      module test (in, generate);
      input in;
      output generate;
      assign generate = in;
      endmodule
   `end_keywords
```

or

```
`begin_keywords 1364-1995
      module test (in, generate);
      input in;
      output generate;
      assign generate = in;
      endmodule
    `end_keywords
```

### What Next

Determine if there is a version specifier, which must be one of "1364-1995", "1364-2001", "1364-2001-noconfig", "1364-2005", or "1800-2005". If so, then check if the double quotes have been forgot.

## VER-795

(error) %s The begin_keywords directive is only supported in SystemVerilog format.

### Description

The Presto HDL Compiler issues this error when the begin_keywords directives is used with the Verilog format instead of with the SystemVerilog format.

### What Next

Read the design as SystemVerilog.

Feedback

## VER-796

(error) %s A begin_keywords directive within a design element.

**Description**

The Presto HDL Compiler issues this error when

**What Next**

Do something wonderful.

## VER-797

(error) %s An end_keywords directive within a design element.

**Description**

The Presto HDL Compiler issues this error when

**What Next**

Do something wonderful.

## VER-900

(error) %s The 'defparam' construct is not supported.

**Description**

Presto HDL Compiler does not support the use of defparam statements. The following is an example of a defparam statement.

```
module top;
    defparam b.parm = 0;
    bottom b ();
  endmodule

  module bottom;
    parameter parm = 1;
    . . .
  endmodule
```

**What Next**

Replace defparam statements with module instance parameter overrides as shown in the following example.

Feedback

```
module top;
    bottom #(.parm(0)) b ();
  endmodule
```

## VER-901

(error) %s The '`define' directive requires that commas separate formal arguments.

### Description

The '`define' directive allows you to parameterize macros. An example is:

```
`define ADD(A,B) ((A)+(B))
```

This error message is generated whenever two formal arguments to the macro are not separated by a comma.

```
`define ADD(A B) ((A)+(B))
             ^  The missing comma will result in an error message.
```

### What Next

Delete whitespace or add a comma to form the identifier(s) needed by the macro's definition.

## VER-902

(error) %s The '`define' directive requires a non-empty argument list if parentheses are present.

### Description

The following macro definition is rejected because no identifiers were found inside the parentheses.

`define MACRO() 10

### What Next

When a macro does not need arguments, omit the parentheses.

`define MACRO 10

## VER-903

(error) %s the `default_nettype directive cannot occur inside a module, interface, or package.

Feedback

### Description

This error message occurs because when using the `default_nettype directive, it must be placed outside the scope of a module declaration as shown in the following example.

```
`default_nettype wand

   module top;
   endmodule
```

The following example shows an invalid use of the `default_nettype directive.

```
module top;
    `default_nettype wand
   endmodule
```

### What Next

Remove any `default_nettype directives from within module bodies. Ensure that the desired default net type is in effect from a directive outside the module body.

---

## VER-904

(error) %s The `ifdef directive must have an identifier name after it.

### Description

This error message occurs when the token following an `ifdef directive is not an identifier name. A common mistake is to use a literal constant after the `ifdef.

The following is a example of the valid use of an `ifdef directive.

```
`ifdef SYNTHESIS
`endif
```

This is an example of an invalid use of the `ifdef directive, which generates this error message.

```
`ifdef 5
`endif
```

### What Next

Correct your design so that the `ifdef directive is used with a valid identifier, and then run the command again.

---

## VER-905

(error) %s The `else directive was found without a corresponding `ifdef directive.

Feedback

### Description

This error message occurs when an `else directive is found without a corresponding `ifdef directive. Two common causes of this error message are that the corresponding `ifdef either does not exist, or that the `ifdef is not followed by an identifier name.

The following is an example of an `else directive without a corresponding `ifdef directive.

```
`else
`endif

// the `ifdef is missing
```

This is an example of an `ifdef directive without the following identifier name.

```
`ifdef
`else
`endif

// the `ifdef exists, but no identifier name follows it
```

This is an example of the correct use of an `else directive.

```
`ifdef SYNTHESIS
// ...

`else
// ...

`endif
```

### What Next

Correct the design so that the `ifdef, `else, and `endif directives are matched, and then run the command again.

---

## VER-906

(error) %s The `endif directive was found without a corresponding `ifdef directive.

### Description

This error message occurs when the `endif directive is found without a corresponding `ifdef directive. Two common causes of this error message are that the corresponding `ifdef either does not exist, or that the `ifdef is not followed by an identifier name.

The following is an example of an `endif directive without a corresponding `ifdef directive.

```
`endif

// the `ifdef is missing
```

Feedback

This is an example of an `ifdef directive without the following identifier name.

```
`ifdef
`endif

// the `ifdef exists, but no identifier name follows it
```

This is an example of the correct use of an `endif directive.

```
`ifdef SYNTHESIS
// ...

`endif
```

**What Next**

Correct the design so that the `ifdef and `endif directives match, and then run the command again.

---

## VER-907

(error) %s The `timescale directive was used incorrectly.

**Description**

This error message occurs when the syntax of the `timescale directive is incorrect.

The correct syntax of the `timescale directive is shown below.

```
`timescale time_unit / time_precision
```

The following is example of the correct use of the `timescale directive.

```
`timescale 1 ns / 1 ps
```

The `timescale directive has no effect on synthesis.

**What Next**

Correct the directive to use a valid timescale. Consult the IEEE 1364 standard for valid values. After making your changes, run the command again.

---

## VER-908

(error) %s The `unconnected_drive directive requires pull1 or pull0 to be used.

**Description**

This error message occurs when either the `unconnected_drive directive is used without either pull1 or pull0.

Feedback

The following example shows the correct use of the `unconnected_drive directive.

```
`unconnected_drive pull1
```

This directive has no effect for synthesis.

**What Next**

Correct the `unconnected_drive directive to use either use pull0 or pull1 and then run the command again.

---

## VER-909

(error) %s The `default_nettype directive requires a valid wire type or 'none'.

**Description**

This error message occur when using the `default_nettype directive, without specifying a valid wire type or 'none'. The following list shows the valid wire types.

```
wire      tri         tri0
wand      triand      tri1
wor       trior       trireg
```

This example shows a valid use of the `default_nettype directive with the wand wire type.

```
`default_nettype wand
```

When the `default_nettype is set to 'none', all nets must be explicitly declared.

**What Next**

Correct the `default_nettype directive to use a valid wire type or 'none' and then run the command again.

---

## VER-910

(error) %s macro calls are nested too deeply.

**Description**

This error message occurs when there are too many nested macro calls. There is a limit to the amount of nesting that can occur in macro calls. The following is an example of a nested macro call.

```
`a( `b( `c( ... ) ) )
```

The Verilog standard sets the nesting limit to 50 levels, although the compiler may support more levels.

Feedback

### What Next

Recode your design to reduce the nesting of macro calls and then run the command again.

---

## VER-911

(error) %s The '`define' directive must have an identifier name after it.

### Description

The Presto HDL Compiler issues this error message during execution of the *analyze* or *read* command when it finds that the token following an `define directive is not an identifier name. A common mistake is to use a literal constant, which the following example shows:

```
`define 0 5
`endif
```

The following example shows a valid use of `define.

```
`define ZERO 0
`endif
```

### What Next

Correct any errors like this in your code.

---

## VER-912

(error) %s The macro specified with the '`define' directive has a bad parameter list.

### Description

The Presto HDL Compiler issues this error message during execution of the *analyze* or *read* command when it encounters a `define directive that specifies a macro that in turn takes parameters, and the parameters are not a list of identifier names.

The following example generates this error message.

```
`define NOT(0) ~X
```

### What Next

Be sure that any macro you specify with a `define directive has a valid parameter list of identifier names.

---

## VER-913

(error) %s The macro '%s' has not been defined.

Feedback

### Description

This message occurs when an attempt is made to use a macro that has not been previously defined. An example of this error is:

```
module top (out);
        output out;

        assign out = `UNDEFINED_MACRO;
endmodule
```

### What Next

Either define the macro, correct the macro reference to refer to a defined macro, or remove the macro invocation. After making your changes, rerun the command.

---

## VER-914

(error) %s End-of-file found before `endif found.

### Description

This message occurs when an `ifdef or `ifndef compiler directive is not closed with a corresponding `endif compiler directive.

### What Next

Insert an `endif to close the `ifdef or `ifndef.

---

## VER-915

(error) %s Syntax error in system timing check at token '%s'.

### Description

Standard Verilog system timing checks can be grouped into those ($setup, $hold, $setuphold, $recovery, $removal, $recrem) that use a stability window and those ($skew, $timeskew, $fullskew, $period, $width, $nochange) that are for clock and control signals.

Although the syntax looks like system task enables, the Verilog standard makes clear that these timing checks are not actual system task enables.

This error is issued when an actual system task enable is encountered where a system timing check is expected.

### What Next

Modify the code to use standard system timing check syntax.

Feedback

## VER-916

(error) %s '%s' is a system function and not a system task.

### Description

You receive this error when a system function call is used as a statement instead of as an expression. Often this is an indication that the value of the system function call is being inadvertently thrown away.

```
module m ;
 always
   $random() ;
endmodule
```

### What Next

If you really do not wish to use the return value, you could assign it to a dummy variable that will not be used or wrap the system function call in a dummy if statement.

```
module m ;
 always
   if ($random()) ;
endmodule
```

## VER-917

(error) %s The 'inout' port '%s' is incompatibly declared as 'real'.

### Description

The Presto HDL Compiler issues this error message during execution of the *analyze* or *read* command when it encounters an inout port declared a real type, which is invalid.

### What Next

Correct any errors like this in your code.

## VER-918

(error) %s The 'inout' port '%s' is incompatibly declared as 'reg'.

### Description

The Presto HDL Compiler issues this error message during execution of the *analyze* or *read* command when it encounters an inout port declared a register type (reg), which is invalid.

Feedback

### What Next

Correct any errors like this in your code.

---

## VER-919

(error) %s The 'input' port '%s' is incompatibly declared as 'real'.

### Description

The Presto HDL Compiler issues this error message during execution of the *analyze* or *read* command when it encounters an input port declared a real type, which is invalid.

### What Next

Correct any errors like this in your code.

---

## VER-920

(error) %s The 'input' port '%s' is incompatibly declared as 'reg'.

### Description

The Presto HDL Compiler issues this error message during execution of the *analyze* or *read* command when it encounters an input port declared a register type (reg), which is invalid.

### What Next

Correct any errors like this in your code.

---

## VER-921

(warning) %s The macro '%s' you are attempting to undefine with the '`undef' directive is not defined.

### Description

When using the '`undef' directive to undefine a macro, it must be defined. For example, the following examples would result in error messages:

```
`define CAT
    `undef KAT

    `define MY_MACRO
    `undef MY_MACRO
    `undef MY_MACRO
```

Feedback

**What Next**

Examine the code to see if perhaps the name of the macro has been misspelled, the `undef occurs twice, or the macro never got defined.

---

## VER-923

(error) %s A valid identifier must follow the '`undef' directive.

**Description**

The Presto HDL Compiler issues this error message during execution of the *analyze* or *read* command when it finds that the `undef directive lacks the required identifier.

The following example shows an instance of an invalid statement (0 is not a valid identifier):

```
`undef 0
```

**What Next**

Correct any instances of this type by replacing an invalid identifier with a valid one.

---

## VER-924

(error) %s Newline in argument list of `define %s.

**Description**

This error message occurs when the `define directive's argument list contains a new line. All parameters of a macro and its definition must occur on the same line.

**What Next**

Remove the newline, or if the newline is part of a formal argument's default text, use a backslash character (\\) immediately before the newline as an escape.

---

## VER-925

(error) %s Input files nested too deep '%s'.

**Description**

The Presto HDL Compiler issues this error message during execution of the *analyze* or *read* command when it finds that input source files are nested too deeply, using the `include directive in Verilog.

When this happens, the message is generated.

Feedback

### What Next

Reduce the number of nested input files to a workable amount.

---

## VER-926

(error) %s The '`include' directive requires a filename enclosed in double quotes.

### Description

When using the `include directive to input another file, you must provide the filename enclosed in double quotation marks, as shown in the following example:

```
`include "my_include.h"
```

### What Next

Be sure to use double quotation marks when using the `include directive in this way.

---

## VER-927

(error) %s The number of parameters for the '%s' macro doesn't match its definition.

### Description

The Presto HDL Compiler issues this error message during execution of the *analyze* or *read* command when it finds that the number of parameters you pass to a macro does not match the number of parameters specified by the macro definition.

The following example shows such an error:

```
`define ADD(A,B) A+B

module top (IN, OUT);
        input IN;
        output OUT;

        assign OUT = `ADD(IN);
endmodule
```

### What Next

Correct any errors like this in your code.

---

## VER-929

(error) %s The input file %s specified by the '`include' directive cannot be recursive.

Feedback

### Description

This message occurs when the file specified by an '`include' directive contains an '`include' directive to the current file or to another file that has one or so on. The Verilog language standard expresses it as: "A file included in the source using the `include compiler directive may contain other `include compiler directives. The number of nesting levels for include files shall be finite."

### What Next

Remove one of more of the `include directives from the files, or wrap the contents of the included file in an `ifndef. For example,

```
`ifndef SEEN_HEADER_VH
  `define SEEN_HEADER_VH
     < the original contents of file >
  `endif
```

---

## VER-930

(error) %s System function %s has been defined.

### Description

This is a syntax error.

### What Next

Change the relevant code to conform to the IEEE Std 1364.

---

## VER-931

(error) %s System task %s has been defined.

### Description

This may be a syntax error.

### What Next

This message is no longer issued.

---

## VER-932

(error) %s '%s' is a system task and not a system function.

Feedback

### Description

This message is issued when a system task enable is used as an expression, that is, when it used as if were a system function call and had a return value. For example,

```
module m ( output reg o ) ;
         always o = $display("Hello, world") ;
      endmodule
```

### What Next

Modify code as appropriate.

---

## VER-933

(error) %s The '%s' system call should not have parentheses unless it has at least one parameter.

### Description

The Presto HDL Compiler issues this error message during execution of the *analyze* or *read* command when it encounters invalid parentheses. In Verilog, when calling a system task or system function, if the task or function does not take parameters, you cannot have empty parentheses in the code.

The following example illustrates an invalid use of parentheses:

```
assign Q = $time();
```

Because $time takes 0 parameters, you must omit the parentheses, as shown here:

```
assign Q = $time;
```

### What Next

Correct any invalid instances like this in your code.

---

## VER-934

(information) %s The imported name '%s' is %svisible to clients of this package.

### Description

This information message issues for every name imported into the most global scope of a package declaration. It indicates whether or not the prevailing SystemVerilog dialect makes this name visible to the package's clients.

Feedback

### What Next

Because early SV implementations differ on this aspect of the *import* statement, you may need to adjust the setting of *hdlin_sv_packages* to bring the set of exported names into agreement with other tools in the flow which use your package. The reported visibility is final; settings of hdlin_sv_packages at the time that clients refer to this package will have no effect on which of its names will be (in)visible.

### See Also

- VER-20

---

## VER-935

(error) %s Identifier name '%s' has not been declared.

### Description

This error is issued when the name of a variable is used before it has been declared.

### What Next

This error is no longer issued.

---

## VER-936

(warning) %s the undeclared symbol '%s' assumed to have the default net type, which is '%s'.

### Description

The Verilog standard says that if no explicit declaration is given for an identifier and the identifier is not in the scope of a '`default_nettype none' compiler directive, then it shall be assumed to be implicitly declared as a net of the default net type, which is 'wire' unless overridden with the `default_nettype compiler directive. Specifically, this is assumed for undeclared identifiers used in a port expression declaration, for undeclared identifiers used in the terminal list of a primitive instance or a module instance, and, in Verilog 2001, for undeclared identifiers that appear on the left-hand side of a continuous assignment statement.

### What Next

Check your design to make sure that you intended an implicit declaration for this symbol. If not, add an explicit declaration.

Feedback

## VER-937

(warning) %s Invalid argument for system function or task %s ignored.

**Description**

This is a syntax error.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

## VER-938

(warning) %s Missing expected argument for system function or task %s.

**Description**

This is a syntax error.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

## VER-939

(warning) %s The '%s' directive is not supported and will be ignored.

**Description**

This message is issued when a compiler directive is encountered that is not currently supported. The result is that the directive is just ignored. For example,

```
`delay_mode_unit
```

**What Next**

No action is necessary.

## VER-940

(error) %s Implicit wire declaration inside .%s port connection.

**Description**

The System Verilog standard says that a .port_identifier connection is semantically equivalent to the named port connection .port_identifier(port_identifier), with the exception that it creates an implicit wire declaration. This error message occurs because your code is attempting to create such a wire.

Feedback

### What Next

Modify your design, either by writing the full name of the port connection, or by declaring the wire before the instantiation.

---

## VER-941

(warning) %s Invalid escape sequence '\\%c' in call to '%s'.

### Description

When using '$display' (or similar) this message is issued when the format specifier contains an invalid escape sequence for printing special characters.

```
$display("Hello, \\zWorld.") ;
```

### What Next

Rewrite the format specifier to use a valid escape sequence.

---

## VER-942

(warning) %s Unescaped Verilog-2001 keyword '%s' used as identifier.

### Description

The Presto HDL Compiler issues this error message during execution of the *analyze* or *read* command when you use one of the identifiers 'design', 'instance' or 'cell', which are reserved keywords in Verilog-2001, without escaping it. In future releases, this restriction will be enforced as a syntax error.

### What Next

Either modify the code, so that it doesn't use a keyword without escaping it, or disable Verilog 2001 features by setting hdlin_vrlg_std to 1995.

---

## VER-943

(error) %s Implicit net declaration for '%s' prohibited by '`default_nettype none' compiler directive.

### Description

The Presto HDL Compiler issues this error message during execution of the *analyze* or *read* command when a net is implicitly declared within the scope of a '`default_nettype none' compiler directive.

Feedback

### What Next

Modify the code, so that either the net is explicitly declared or its implicit declaration is not in the scope of a '`default_nettype none' compiler directive.

---

## VER-944

(warning) %s A unnamed generate block with an LRM-defined name '%s' is detected, which is incompatible with Verilog standard 2001 or 1995.

### Description

This warning message is issued by the Presto HDL Compiler when running the *analyze* or *read* command and there is a unnamed generate block, which is not a scope in Verilog standard 2001 or 1995, but it is a scope in Verilog standard 2005 with an LRM-defined name attached.

For example:

```
17  generate
18    if (1) begin
19      sub u1 (a[1], b[1], clk, y[1]);
20    end
21  endgenerate
```

In Verilog standard 2005, the unnamed generate-if block at line 18 is a scope and will be assigned an LRM-defined name; for example 'genblk1'. The instance u1 within the generate block can be referred as 'genblk1.u1'; in Verilog standard 2001, the unnamed generate block is not a scope, and the u1 instance will be referred as 'u1'.

The Verilog standard 2005 is not backward compatible with Verilog standard 2001 or 1995. It is not recommended to read in designs with mixing Verilog standards.

### What Next

Make sure all of your designs are compiled under the same Verilog standard setting, for example 2005, and all the object names referred to in your scripts are consistent with the Verilog standard you are using.

It is considered best practice to attach a name to the unnamed generate block to prevent possible naming incompatibility issues caused by the different standard versions.

---

## VER-945

(warning) %s Potential simulation/synthesis mismatch on obsolete Verilog-2001 construct %s.

Feedback

### Description

This warning message occurs because you are using an obsolete construct that is now enforced as a SystemVerilog syntax error. Standalone generate blocks were allowed in the old Verilog-2001 standard (IEEE Std 1364-2001), but were later prohibited by the Verilog-2005 standard (IEEE Std 1364-2005). This prohibition was inherited by the SystemVerilog standards (IEEE Std 1800-2005 and IEEE Std 1800-2009).

For example:

```
17  generate
18    begin
19      sub u1 (a[1], b[1], clk, y[1]);
20    end
21  endgenerate
```

The standalone generate block on lines 18-20 was legal in Verilog-2001, but is illegal in Verilog-2005 and SystemVerilog.

### What Next

This is only a warning message. No action is required.

However, to avoid this message rewrite the code to remove the *begin* and *end* keywords.

---

## VER-946

(error) %s Syntax error on an obsolete Verilog 2001 construct %s.

### Description

This error message occurs when you use an obsolete Verilog 2001 construct standalone generate block. Standalone generate blocks were allowed in the old Verilog-2001 standard (IEEE Std 1364-2001), but were later prohibited by the Verilog-2005 standard (IEEE Std 1364-2005). This prohibition was inherited by the SystemVerilog standards (IEEE Std 1800-2005 and IEEE Std 1800-2009).

For example, an anonymous standalone block:

```
17  generate
18    begin
19      sub u1 (a[1], b[1], clk, y[1]);
20    end
21  endgenerate
```

For example, a named standalone block:

```
22  generate
23    begin:BLK
24      sub u1 (a[1], b[1], clk, y[1]);
```

Feedback

```
25    end
26  endgenerate
```

The standalone generate blocks on lines 18-20 and 23-25 were legal in Verilog-2001, but are illegal in Verilog-2005 and SystemVerilog.

For example, the following named generate block is legal, beacause it is part of a conditional generate construct:

```
27  generate
28    if (P==0) begin:BLK
29      sub u1 (a[1], b[1], clk, y[1]);
30    end
31  endgenerate
```

In the final example, it would have been legal to omit the generate block name, but in Verilog-2001 the instance could have been referred to from outside of the conditional generate, but in Verilog-2005 and SystemVerilog it could not. It is recommended to give names to all generate blocks.

**What Next**

Rewrite the code to remove the *begin* and *end* keywords.

---

## VER-951

(error) %s The destination of a procedural continuous assignment must not be an array reference, bit select, or part select.

**Description**

The Presto HDL Compiler issues this error message during execution of the *analyze* or *read* command when it encounters an invalid destination for a procedural continuous assignment. The destination must not be an array reference, bit select, or part select.

Eiher of the procedural continuous assignments in the following example causes this error.

```
module foo (A, B);
        output [1:0] A;
        input B;
        reg [1:0]A;
        reg C[1:0];

always
begin
        assign  A[1] = B;       // This is an error.
        assign  C[1] = B;       // This is an error.
end

endmodule
```

Feedback

### What Next

Correct any similar invalid instances in your code and then run the command again.

---

## VER-952

(error) %s Assigment to '%s' requires that it be a register.

### Description

The Presto HDL Compiler issues this error message during execution of the *analyze* or *read* command when it encounters assignments that are not registers but should be.

In the following example, *a* and *c* should be registers.

```
always
begin
        a = b;
        c = d;
end
```

### What Next

Redefine the assignments as registers.

---

## VER-953

(error) %s Deassigning argument '%s' requires that '%s' be a register.

### Description

The Presto HDL Compiler issues this error message during execution of the *analyze* or *read* command when it encounters a value for a deassignment that is not a register.

The code in the following example causes the error because *B* is not a register type:

```
module foo (A, B);
        input A;
        output B;

always
        deassign B;
endmodule
```

### What Next

Change the value of the deassignment to a register.

Feedback

## VER-954

(error) %s The register symbol '%s' has already been defined.

### Description

The Presto HDL Compiler issues this error message during execution of the *analyze* or *read* command when it encounters a register defined twice in the code. It is illegal to define a register twice.

The code in the following example generates this error:

```
module foo;
        reg C;
        reg C;
endmodule
```

### What Next

Correct the code so that it defines a register once only.

## VER-955

(error) %s Assigning to output '%s' requires that it be a register.

### Description

The Presto HDL Compiler issues this error message during execution of the *analyze* or *read* command when it encounters assignments that should be registers.

In the following example, *a* and *c* should be registers:

```
always
begin
        assign a = b;
        assign c = d;
end
```

### What Next

Redefine the assignments as registers.

## VER-956

(error) %s The symbol '%s' is not defined.

Feedback

### Description

The Presto HDL Compiler issues this error message during execution of the *analyze* or *read* command when it encounters the use of a symbol or variable that has not been defined.

For example, the code in the following example generates the error:

```
module foo (A);
        input A;

always
        undefined = A;
endmodule
```

### What Next

Define the symbol or variable.

---

## VER-957

(error) %s Too many delay values supplied in the delay list.

### Description

When specifying the delay of gates, you cannot supply more delay values than the gate supports.

For example, the following code attempts to instantiate a buffer:

```
module foo (A, B);
        input A;
        output B;

        buf # (0, 1, 2, 3, 4, 5) (B,A);
endmodule
```

The buf gate uses only 2 delay values, so providing 6 delay values is an error.

### What Next

To correct the above example, the code should read as follows:

```
module foo (A, B);
        input A;
        output B;

        buf # (0, 1) (B,A);
endmodule
```

Feedback

## VER-958

(error) %s The triggered named event '%s' has not been declared.

### Description

This error is issued whenever the named event to be triggered on has not been declared. (NOTE: Event triggers are not supported by synthesis.)

For example,

```
module m;
     always
       begin
                -> nonexistent ;
       end
   endmodule
```

generates an error because the event 'nonexistent' has not been defined.

### What Next

Add an 'event' declaration to declare the named event, and wrap the event trigger with "`ifndef SYNTHESIS ... `endif" to avoid a VER-193 error.

For example,

```
module m;
     event nonexistent ;
     always
       begin
       `ifndef SYNTHESIS
                -> nonexistent ;
       `endif
       end
   endmodule
```

## VER-959

(error) %s The triggered event '%s' not declared as an event.

### Description

This error is issued when the event name to be triggered was declared as some type other than 'event'. (NOTE: Event triggers are not supported by synthesis.)

For example,

```
module foo;
     reg nonevent;
     always
        begin
```

Feedback

```
                -> nonevent ;
        end
      endmodule
```

generates an error because the name 'nonevent' has been declared as a 'reg' instead of an event.

**What Next**

Modify the declaration be an 'event' declaration, and wrap the event trigger with "`ifndef SYNTHESIS ... `endif" to avoid a VER-193 error.

For example,

```
module m;
     event nonevent ;
     always
       begin
       `ifndef SYNTHESIS
                -> nonevent ;
       `endif
       end
   endmodule
```

## VER-960

(error) %s Lack of register declaration requires combinational primitive.

**Description**

This is a syntax error.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

## VER-961

(error) %s Register declaration '%s' supported only for output.

**Description**

This is a syntax error.

**What Next**

Change the relevant code to conform to the IEEE Std 1364.

Feedback

## VER-962

(error) %s Existence of register declaration requires sequential primitive.

### Description

This is a syntax error.

### What Next

Change the relevant code to conform to the IEEE Std 1364.

## VER-963

(error) %s Illegal part select of memory '%s'.

### Description

In Verilog, it is illegal to index a slice of a memory using a part select. The following example illustrates a part select of a memory that results in this error message.

```
module foo;
     reg [10:0] mem [10:0];
     reg [10:0] temp [3:0];

     always
        temp[3:0] = mem[3:0];
   endmodule
```

### What Next

Rewrite this part of your design with a looping construct. Or, if available, read the file as SystemVerilog, in which it is legal to index a slice of a memory with a part select.

## VER-964

(error) %s Port '%s' incompatibly declared as scalar and vector.

### Description

You get this message when the width declarations for a port are inconsistent. For example,

```
module m(dataout);
    output dataout;
    reg [2:0] dataout;
    endmodule
```

Feedback

### What Next

Modify the declarations to be consistent with each other. Usually, but not necessarily, this entails adding vector range information to the scalar declaration.

---

## VER-965

(error) %s Premature end-of-file encountered in a '%s' block begun at line %d.

### Description

This message, issued during execution of the *analyze* or *read* command, indicates that the Presto HDL Compiler has encountered a block of code that does not have its terminating token. For example, a *module* that has no *endmodule* or a comment that begins with '/*' but has no '*/' to terminate it.

### What Next

Review your code to locate such missing tokens and insert them.

---

## VER-966

(error) %s Procedural-continuous assignments are not supported by synthesis.

### Description

This message, issued during execution of the *analyze* or *read* command, indicates that the Presto HDL Compiler has encountered a procedural-continuous assignment. Synthesis ignores such assignments.

The following example illustrates a procedural-continuous assignment, assign B = A, that synthesis ignores:

```
module foo (A, B);
        input A;
        output B;
        reg B;

always
        assign B = A;      //  this is a procedural-continuous
 assignment.

endmodule
```

### What Next

Because synthesis ignores procedural-continuous assignments, consider removing them if the code is intended for synthesis only.

Feedback

## VER-967

(error) %s The 'force' construct is not supported by synthesis.

### Description

This message, issued during execution of the *analyze* or *read* command, indicates that the Presto HDL Compiler has encountered a force construct. Synthesis ignores such constructs.

The following example illustrates a force construct, force B, that synthesis ignores:

```
module foo (B);
        output B;
        reg B;

always
        force B = 1'b1;           // force is not supported

endmodule
```

### What Next

Because synthesis ignores force constructs, consider removing the force construct if the code is intended for synthesis only.

## VER-968

(error) %s The 'release' construct is not supported by synthesis.

### Description

This warning, issued during execution of the *analyze* or *read* command, indicates that the Presto HDL Compiler has encountered a release construct. Synthesis ignores such constructs.

The following example illustrates a release construct, release B, that synthesis ignores:

```
module foo (B);
   output B;
   reg B;

always
        release B;        // release is not supported

endmodule
```

### What Next

Because synthesis ignores release constructs, consider removing the release construct if the code is intended for synthesis only.

Feedback

## VER-969

(error) %s The 'deassign' construct is not supported by synthesis.

### Description

This warning, issued during execution of the *analyze* or *read* command, indicates that the Presto HDL Compiler has encountered a deassign construct. Synthesis ignores such constructs.

The following example illustrates a deassign construct, deassign B, that synthesis ignores:

```
module foo (B);
   output B;
   reg B;

always
   deassign B;          // deassign is not supported

endmodule
```

### What Next

Because synthesis ignores deassign constructs, consider removing the deassign construct if the code is intended for synthesis only.

## VER-970

(warning) %s The delay specification for gate instantiation is ignored.

### Description

This warning, issued during execution of the *analyze* or *read* command, indicates that the Presto HDL Compiler has encountered a delay information for gate instantiation. Synthesis ignores such a delay specification.

The following example illustrates delay information, #(1,2), that synthesis ignores:

```
module foo (A,B,C);
   input A,B;
   output C;
   and #(1,2) (C,A,B);
endmodule
```

### What Next

Because synthesis ignores delay information on gate instantiation, consider removing the delay information if the code is intended for synthesis only.

Feedback

## VER-971

(warning) %s The delay specification for tri-state gate instantiation is ignored.

### Description

This warning, issued during execution of the *analyze* or *read* command, indicates that the Presto HDL Compiler has encountered delay information for a tri-state gate instantiation. Synthesis ignores such delay information.

The following example illustrates delay information, #(1,2), that synthesis ignores:

```
module foo (A,B,C);
   input A,B;
   output C;
   bufif0 #(1,2) (C,A,B);
endmodule
```

### What Next

Because synthesis ignores delay information on tri-state gate instantiation, consider removing the delay information if the code is intended for synthesis only.

## VER-972

(warning) %s The delay specification for mos switch instantiation is ignored.

### Description

This warning, issued during execution of the *analyze* or *read* command, indicates that the Presto HDL Compiler has encountered delay information for a mos switch instantiation. Synthesis ignores such delay information.

The following example illustrates delay information, #(1,2), that synthesis ignores:

```
module foo (A,B,C);
   input A,B;
   output C;
   nmos #(1,2) (C,A,B);
endmodule
```

### What Next

Because synthesis ignores delay information for mos switch instantiation, consider removing the delay information if the code is intended for synthesis only.

## VER-973

(warning) %s The delay specification for cmos switch instantiation is ignored.

Feedback

### Description

This warning, issued during execution of the *analyze* or *read* command, indicates that the Presto compiler has encountered delay information for a cmos switch instantiation. Synthesis ignores such delay information.

The following examples illustrates delay information, #(1,2), that synthesis ignores:

```
module foo (A,B,C,D);
   input A,B,C;
   output D;
   cmos #(1,2) (D,A,B,C);
endmodule
```

### What Next

Because synthesis ignores delay information on a cmos switch instantiation, consider removing the delay information if the code is intended for synthesis only.

---

## VER-974

(warning) %s The delay specification for bi-directional switch instantiation is ignored.

### Description

This warning, issued during execution of the *analyze* or *read* command, indicates that the Presto compiler has found delay information for a bidirectional switch instantiation. Synthesis ignores such delay information.

The following example illustrates delay information, #(1,2), that synthesis ignores:

```
module foo (A,B,C);
   input C;
   inout A,B;
   tranif0 #(1,2) (A,B,C);
endmodule
```

### What Next

Because synthesis ignores delay information on bidirectional switch instantiation, consider removing the delay information if the code is intended for synthesis only.

---

## VER-975

(warning) %s The delay specification for pull gate instantiation is ignored.

### Description

This warning, issued during execution of the *analyze* or *read* command, indicates that the compiler has found a delay specification for a pull gate instantiation.

Feedback

The following example illustrates a delay specification, #(1), that synthesis ignores:

```
module foo (A);
        output A;
        pullup #(1) (A);
endmodule
```

**What Next**

Because synthesis ignores delay specifications on pull gate instantiations, consider removing the delay specification if the code is intended for synthesis only.

---

## VER-976

(warning) %s The delay specification for net declaration is ignored.

**Description**

This warning, issued during execution of the *analyze* or *read* command, indicates that the compiler has found a delay specification in a net declaration.

The following example illustrates a delay specification, #(1), that synthesis ignores:

```
module foo;
        wire #(1) w;
endmodule
```

**What Next**

Be aware that synthesis ignores delay specifications on nets. If the code is intended for synthesis only, consider removing the delay specification.

---

## VER-977

(warning) %s The strength specification for a net declaration is ignored by synthesis.

**Description**

This warning, issued during execution of the *analyze* or *read* command, indicates that the compiler has found a strength specification in a net declaration.

**What Next**

Be aware that synthesis ignores strength specifications on nets. If the code is intended for synthesis purposes only, consider removing the strength specification.

---

## VER-978

(error) %s Ambiguous genvar ('%s') used in generate loop statement.

Feedback

### Description

You receive this error message because the two genvar assignments used in a generate loop specification must have the same genvar on the left hand sides. For example:

```
generate
     for (i=0; i<5; j=i+1)      // 2 different genvars "i" and "j" used
     ....
   endgenerate
```

### What Next

Change the generate loop so that the same genvar is used on the left side of both assignments, and then run the command again.

### See Also

- read_verilog

---

## VER-979

(error) %s Two nested generate loops using the same genvar '%s'.

### Description

You receive this error message because the same genvar is used as loop index by two nested generate loops. According to the language standard, using the same genvar by two nested generate loops is not allowed. The situation is demonstrated by the following example:

```
generate
  for (i=0; i<5; i=i+1) // genvar 'i' as index
    for (i=1; i<4; i=i+1) //ERROR: nested loop using the same genvar 'i'
 as index
  ....
endgenerate
```

### What Next

Rewrite one of the generate loops to use a different genvar as index.

### See Also

- read_verilog

---

## VER-980

(error) %s Illegal assignment to genvar '%s'.

Feedback

### Description

You receive this error message because the design contains an assignment to a genvar outside of a generate loop header. The value of genvars can only be defined by assignments inside generate loop headers. For example:

```
generate
     for (i=0; i<5; i=i+1) // only valid assignments to genvar 'i'
     ...
     i = 5;  // ERROR: invalid assignment to genvar 'i'
     ...
   endgenerate
```

### What Next

Rewrite the RTL description so that genvars are only assigned values inside the generate loop headers.

### See Also

- read_verilog

---

## VER-981

(error) %s Macro definition is recursive '%s'.

### Description

The Presto HDL Compiler issues this error message when a recursively defined macro is called while executing the *analyze* or *read* command.

Macros are defined using the define directive in Verilog.

### What Next

Redefine the macro to eliminate all self-references, including indirect self-references occurring in the definitions of other macros.

---

## VER-982

(error) %s The '`elsif' directive was found without a corresponding '`ifdef' directive.

### Description

You receive this error message when the `elsif directive is found with no corresponding `ifdef. The `ifdef might not exist, or the `ifdef might not be followed by an identifier name.

Feedback

The following is an example of a missing `ifdef:

```
`elsif SYNOPSIS_SYNTHESIS
`else
`endif

// the `ifdef is missing
```

The following shows an example of an `ifdef without an identifier name:

```
`ifdef
`elsif SYNOPSYS_SYNTHESIS
`else
`endif

// the `ifdef exists, but no identifier name follows it
```

### What Next

Correct the `elseif directive so it includes the `ifdef directive as shown in the example:

```
`ifdef SOMETHING_ELSE
// ...

`elsif SYNOPSYS_SYNTHESIS
// ...

`else
// ...

`endif
```

### See Also

- read_verilog

---

## VER-983

(error) %s The "`ifndef' directive must have an identifier name after it.

### Description

The tool issues this error message when the token following an `ifndef directive is not an identifier name. A common mistake is using a literal constant.

The following example shows an invalid identifier name, resulting in this error message.

```
`ifndef 5
`endif
```

Feedback

### What Next

Correct the identifier name after the `ifndef directive, as shown in the following example:

```
`ifndef SYNOPSYS_SYNTHESIS
`endif
```

### See Also

- read_verilog

---

## VER-984

(error) %s The '`elsif' directive must have an identifier name after it.

### Description

The tool issues this error message because the token following an `elsif directive is not an identifier name. A common mistake is using a literal constant.

The following example shows an invalid identifier name, resulting in this error message.

```
`elsif 5
```

### What Next

Correct the identifier name after the `elsif directive, as shown in the example of a valid `elsif:

```
`elsif SYNOPSYS_SYNTHESIS
```

### See Also

- read_verilog

---

## VER-985

(warning) %s The '`line' directive requires a %s.

### Description

You receive this warning message because the Verilog `line compiler directive cannot find the expected arguments.

The expected arguments are a nonnegative decimal number, a double-quoted string, and a level of 0, 1, or 2.

### What Next

No action is required on your part.

Feedback

## VER-986

(warning) %s Empty port in %s declaration.

### Description

You receive this warning message when the port list for a module, interface, primitive, or checker is empty, with no local name by which to form connections between its body and the instantiation site.

### Examples

The following legal Verilog statement indicates a module with four ports, the last of which is unconnected.

```
...
   module m( a,b,c,);   /* Legal, but perhaps not intended */
   ...
```

### What Next

This is a warning message only. No action is required on your part.

However, if the unconnected port is not your intention, remove a comma and then run the command again.

## VER-987

(warning) %s Port %s is implicitly typed

### Description

The port warned has been declared previously with a different type.

**Verilog**

```
module test ( ..., dataout);
output dataout;
reg [2:0] dataout;
```

Here dataout has been previously declared as one-bit, and later on declared as [2:0]. This may cause simulation, synthesis and formal verification mismatch due to the different interpretation in different tools as what's type of dataout.

### What Next

Change your Verilog source code to remove the inconsistent declaration for the port.

Feedback

## VER-988

(information) %s List () of one, unnamed, port is ignored.

### Description

This informational message issues when the port list of a module or interface declaration is entirely empty. The IEEE standards have always said this indicates one unconnected port, but legacy implementations all dissented; like them, we produce zero ports. The resulting module is the same as that produced when the empty parentheses are omitted. It will match an unresolved instantiation in which the parentheses are provided (as required!) and the list of port connections is similarly empty.

### Examples

The following Verilog statement declares a module with zero ports, not one.

```
module m( );   /* Ambiguous intention */
   ...
   endmodule
```

### What Next

This is an informational message only. No action is required on your part, unless the module has trouble linking to instances which supply one port.

If your intention is to produce one port which cannot be used within the module, use the syntax for a named port but omit the port_expression.

```
module m( .Port1() );   /* The explicit name matches the port's implicit
 name */
   ...
   endmodule
```

## VER-999

(error) %s Only simple identifiers are allowed in pragma signal list.

### Description

You receive this error message when the pragma signal list contains expression list or non-simple identifiers.

### Examples

The following verilog code won't work as identifier associated with pragam is not an simple identifier.

```
// synopsys sync_set_reset "reset[0]"

  reg y ;
```

Feedback

```
always @ (posedge clk)
begin : synchronous_reset
if (reset[0])
y = 1'b0;
else
y = d1;
end
```

**What Next**

Use a wire and replace non-simple identifier.

---

## VER-1000

(warning) The hdlin_enable_presto variable will be ignored in a future release.

**Description**

The HDL Compiler (Presto Verilog) tool was introduced as the default Verilog reader in the 2001.08 release. In a future release, you will no longer be able to disable it using the *hdlin_enable_presto* variable.

**What Next**

Remove the *hdlin_enable_presto* variable from your scripts.

---

## VER-1003

(warning) %s attribute of interface port %s is ignored.

**Description**

The port is interface and has atttribute_instance.

```
module test (output reg out,
           (*eco_dont_tap*) clientIF.fifoMod   cIF [2], ...);
         sub inst (out, ...);
       endmodule
```

Attribute "eco_dont_tap" will be ignored, because "clientIF" is interface.

---

## VER-1004

(warning) %s Net %s connected to instance %s is declared as reg data type but is not driven by an always block.

Feedback

### Description

The net connected to the instance is not driven by an always block but is declared as reg data type.

```
module test (output reg out, ...);
         sub inst (out, ...);
       endmodule
```

You should use sequential statements to assign values to nets with reg data type. Using continuous assignments for nets with reg type is illegal. When HDL Compiler finds no explicit driver from sequential statements and the net is connected to instances, the net is likely to be driven by an instance. In Verilog, a connection to an output port of an instance is a continuous assignment, so the tool issues a warning because it is illegal in Verilog to connect a net of reg type.

### What Next

Remove the inconsistent declaration of the net in the source code, or re-analyze the design as SystemVerilog because a continuous assignment to a variable is legal in SystemVerilog.

---

## VER-1005

(warning) %s Port %s of type input is being assigned.

### Description

The port of type input is being assigned.

**Verilog**

```
module test ( ..., datain);
input datain;
assign datain = ...;
```

The port of type input has been assigned. This may cause simulation, synthesis and formal verification mismatch due to the different interpretation in different tools as what's type of dataout.

### What Next

Change your Verilog source code to remove the inconsistent declaration for the port.

---

## VER-1006

(error) %s Entity of architecture '%s' cannot be found.

Feedback

**Description**

This error message occurs during an *analyze* or *read* command activity if the entity of an architecture cannot be found.

**What Next**

Ensure the desired entity exists and has been specified correctly. Check if there are any unsupported embedded configurations inside the architecture.

---

## VER-1007

(error) %s Only one event control allowed on always_ff block.

**Description**

You receive this error message because SystemVerilog does not support multiple event statements on an always_ff block.

**What Next**

Change the source code to remove the use of multiple events in the always_ff block.

---

## VER-1008

(error) %s Incompatible DW encrypted file %s, the encrypted file is version %s, only version lower than or equal to %s can be read by the current DC.

**Description**

This error message occurs when the tool detects that there is an incompatible version of an encrypted DW file.

**What Next**

You need another version of DC to read the file.

---

## VER-1009

(error) %s No delays are allowed on an always_ff, always_comb or always_latch block.

**Description**

You receive this error message because SystemVerilog does not support delays on an always_ff, always_comb or always_latch block.

Feedback

### What Next

Change the source code to remove the use of delays in the always_ff, always_comb or always_latch block.

---

## VER-1010

(error) %s Variable input port '%s' is being assigned.

### Description

This error message occurs when SystemVerilog adds variable input ports. In Verilog, there were two kinds of output ports, net output ports and variable output ports, but only one kind of input port, net input ports.

SystemVerilog added variable input ports. Unlike the net input ports of Verilog, it is not legal to assign to SystemVerilog variable input ports.

```
module test (input logic in1, in2, output out);
  assign in1 = in2;
  assign out = in1;
endmodule
```

### What Next

Change the direction of the variable port or remove the assignment to that port. It is not recommended to assign to a net input port, even though it is legal.

---

## VER-1011

(error) %s Tasks cannot contain event control statements.

### Description

You receive this error message because you have attempted to use an event control statement inside of a task. This is not supported by Presto.

### What Next

Rewrite the source code so that tasks don't contain events.

---

## VER-1012

(error) %s Nonblocking assignments to TASK local variable '%s';

### Description

Nonblocking assignments to static variables declared inside a TASK are not supported in synthesis.

Feedback

```
module ShiftR (Out, In);
output Out;
input In;
reg Out;
always@(In)
     tShift(Out, In );
task tShift(output Q, input D);
      Q <= D ;
endtask
endmodule
```

### What Next

To avoid receiving this error message, do not set the TASK local variable using a nonblocking assignment. You can change the variable as a global variable. The above example can be rewritten as follows:

```
module ShiftR (Out, In);
output Out;
input In;
reg Out;
always@(In)
     tShift(In);
task tShift(input D);
      Out <= D ;
endtask
endmodule
```

---

## VER-1013

(error) Format in VCS command file '%s' is not consistent with the format option '%s' in analyze command.

### Description

Format mentioned in the VCS command file is not consistent with the format argument of analyze command.

### What Next

Remove the format option from the analyze command.

---

## VER-1014

(error) %s Loop index variable '%s' of generate loop not a genvar.

Feedback

### Description

You receive this error message because the loop index variable of a generate loop has not been declared as a genvar. For example:

```
integer i; // integer variable as loop index of a generate loop

   generate
     for (i=0; i<5; i=i+1)
     ...
   endgenerate
```

### What Next

Declare the variable using 'genvar' keyword.

```
genvar i;

   generate
     for (i=0; i<5; i=i+1)
     ...
   endgenerate
```

or

```
generate
     for (genvar i=0; i<5; i=i+1)
     ...
   endgenerate
```

### See Also

- read_verilog

---

## VER-1015

(information) %s %s in encrypted file

### Description

You receive this message because the tool detected an issue in the protected region of the file. You need to contact the IP Vendor to understand and resolve the issue.

---

## VER-1016

(error) %s Error in reading protected envelope - %s %s

Feedback

### Description

You receive this message because the tool was unable to read a protected envelope in the input file. You need to correct any syntax issues in the envelope or contact the IP Vendor to resolve the issue.

---

## VER-1017

(error) %s Option %s is only supported for VHDL.

### Description

You receive this message because the option specified by you is only supported for VHDL. You must un-set the option and rerun the command.

---

## VER-1023

(warning) %s Break statement in multidimensional foreach iterator over '%s' is ambiguous, and terminates the foreach completely.

### Description

You receive this warning message because there are multiple foreach loop index variables and an ambiguous break statement associated with them. For example:

```
logic [3:0] array [0:1];
  logic [3:0] out;

  always_comb begin
   out = '0;
   foreach( array[j,k] ) begin     // j 0->1 then k 3->0
     if( array[j][k] == 1'b1 ) begin
       out[k] = 1;
       break;                    // terminate j or k ??
     end
   end
  end
```

Here the intent is probably to terminate only the inner "k" loop and then continue with the outermost "j" loop, but the actual expected behavior is to terminate the outermost "j" loop and jump out of the foreach.

### What Next

If the intent doesn't match the expected behavior, add a nested foreach with only a single loop variable. For example:

```
logic [3:0] array [0:1];
  logic [3:0] out;
```

Feedback

```
always_comb begin
  out = '0;
  foreach( array[j,] ) begin            // j 0->1
    foreach( array[,k] ) begin          // k 3->0
      if( array[j][k] == 1'b1 ) begin
        out[k] = 1;
        break;                          // terminate inner loop (k)
      end
    end
  end
end
```

## VER-1024

(warning) %s Overriding the same parameter, %s, multiple times.

### Description

You receive this warning message when a parameter is overridden more than once.

### Examples

The following module instantiation will warn on the parameter P being assigned multiple times

```
...
   module top;
     bot#(.P(1), .P(2)) U1(); /* Will be warned on P*/
   endmodule

   ...
```

### What Next

This is a warning message only. No action is required on your part.

However, if the resulting parameter assignment is not what is desired, remove the ambiguity to ensure the right value is being assigned.

## VER-1025

(error) %s Invalid argument within '{}': %s

### Description

Invalid argument given within {}. For example: analyze -format vhdl { -filelist flist.f}

-filelist is an invalid argument.

Feedback

**What Next**

Check if it is a supported argument for the command or misspelled commands

---

## VER-1026

(error) %s Value not specified for the argument in '{}': %s

**Description**

Invalid argument given within {}. For example: analyze -format vhdl {-f}

value not specified for the argument -f in {}.

**What Next**

In case of a name-value pair argument, specify value for the argument.

---

## VER-1051

(warning) %s Cannot redefine higher precedence macro '%s' with value '%s' to '%s'.

**Description**

A higher precedence macro is being redefined with a different value in the RTL code. It might produce unexpected synthesys results if not intended.

**What Next**

If the redefinition was not intended, just rename the macro such that it does not collide with the higher precedence definition. If intended, the define prioritization can be turned off by setting *hdlin_analyze_prioritize_command_line_defines* to *false*.

---

# VHD

---

## VHD-1

(error) %s The library '%s' is not mapped to a directory.

**Description**

This message indicates that you attempted to access a library which does not have a valid directory path associated with it.

Feedback

**What Next**

For information on mapping between design libraries and paths, refer to the *define_design_lib* command manual page. Mapping can also be done via the .synopsys_vss.setup file.

---

## VHD-3

(error) %s You have declared more than one architectures with the same name '%s'.

**Description**

You receive this error message if you have declared more than one architectures with the same name.

The last declared implementation of the architecture will overwrite its previous declared implementations.

**What Next**

Verify that each of the architecture has been assigned a different name.

---

## VHD-4

(warning) %s The architecture %s has already been analyzed. It is being replaced.

**Description**

You receive this warning message if an architecture with the same name has already been analyzed, which occurs if either of the following are true:

- The current analyzed file has already been analyzed.
- An architecture with the same name is declared in another file and has already been analyzed.

The implementation of the architecture is overwritten by the latest declared implementation.

**What Next**

This is only a warning message.

However, you can check the overwritten file. If the result is not what you intended, change the architecture name and then run the command again.

Feedback

## VHD-5

(warning) %s the architecture %s has already been analyzed by a compiler other than Presto HDL. It is being replaced.

**Description**

You receive this warning message if an architecture with the same name has already been analyzed by a compiler other than Presto HDL.

The implementation of the architecture is overwritten by the latest declared implementation.

**What Next**

This is a warning message only. No action is required on your part.

However, you can check the overwritten file. If the result is not what you intended, change the architecture name and then run the command again.

## VHD-6

(warning) %s The entity '%s' has multiple architectures defined. The last defined architecture '%s' will be used to build the design by default.

**Description**

You receive this warning message if there are multiple architectures associated with the same entity. Just one architecture will be chosen to build the design. In PRESTO-VHDL, the last defined architecture will be chosen by default.

**What Next**

Verify if the last defined architecture is what you want to use to build the design.

## VHD-7

(warning) %s Initial values for signals/variables are not supported for synthesis by default. They are ignored.

**Description**

You receive this warning message if the signals/variables have initial values when they are declared. Presto VHDL ignores the initial values and uses the subprogram default value instead. This may cause simulation mismatch.

Feedback

If variables are declared in a function, you can enable the variable initial value support by setting the following switch to true (the default is false):

```
hdlin_support_subprogram_var_init
```

**What Next**

This is only a warning message.

If the subprogram default value is the same as the declared initial value or is acceptable, no action is required.

Otherwise, you can do either of the following:

- Use the *hdlin_support_subprogram_var_init* switch to solve the problem.
- Instead of the initial value, use an explicit assignment to set the initial value.

---

## VHD-8

(information) %s Variable '%s' is being read, but is not initialized. The subprogram default value will be used.

**Description**

You receive this information message from the Presto HDL Compiler when your design contains a variable which is read, but is not yet initialized. The subprogram default value is used.

The subprogram default value can be 0s (as default) or ` (grave accent), which is controlled by the following switch:

```
hdlin_subprogram_default_values
```

**What Next**

This is only an information message. No action is required.

However, it is best practice to verify that you intended to use the subprogram default value as the initial value of the uninitialized variable.

---

## VHD-9

(error) %s Illegal character in entity, architecture, package, or configuration name.

Feedback

**Description**

You receive this error message when the name you specified for an entity, architecture, package, or configuration cannot be directly used to name a file in the host file system. This is usually due to some reserved characters, such as '/' or ':'. Names in the standard *definitions name space* exist as file system objects and must conform to host or network naming conventions in addition to VHDL lexical rules.

For example, \ab/cd\ is an invalid entity, architecture, package, or configuration name.

**What Next**

Change the entity, architecture, package, or configuration name in the relevant code to conform to the rules for file names in the host file system and run the command again.

---

## VHD-100

(error) %s Elements of aggregate or string literal are out of bound, overlap or do not cover all cases.

**Description**

This error occurs when the values specified for an aggregate do not cover all the possible elements or if more than one value is specified for the same element or if the values specified are for elements that are not part of this aggregate.

**What Next**

Modify your HDL description to ensure that there is one to one correspondence between elements of the aggregate and the values specified for it.

---

## VHD-101

(error) %s Invalid return value for function.

**Description**

The output of the subprogram doesn't exist or is not valid

**What Next**

Look at the subprogram body and see if all return paths exist and are valid

---

## VHD-102

(error) %s The value type of attribute %s is not %s.

Feedback

### Description

The attribute type mismatch.

### What Next

Check the attribute declaration and see if it conflicts with synopsys attribute declaration.

---

## VHD-103

(error) %s The type of formal generic subprogram %s does not match the type of the actual subprogram %s.

### Description

The types of the formal generic subprogram and the actual subprogram do not match.

### What Next

Check at the instantiation if the actual subprogram has the compatiable types as the generic subprogram specification.

---

## VHD-104

(error) %s Only simple names are supported for configuration port mapping.

### Description

You must use only simple names to map the ports of a configuration. In particular, a simple name for a port is the identifier that appears in the port declaration of a configuration, package, procedure, or function. HDL Compiler does not support other port naming styles for configuration port mapping, such as selected names, indexed names, slice names, and attribute names. For example, the following port mapping uses indexed names, which is not supported:

```
for all : my_AND
          use entity work.AND2 (rtl)
          port map (A => D_in(0), B => D_in(1), Z => Z);
        end for;
```

### What Next

You should create a wrapper to match the ports. For example:

```
entity mux21_wrapper is
  port(
    D_in : in  std_logic_vector (1 downto 0);
    S0   : in  std_logic;
    Z    : out std_logic
  );
```

Feedback

```
end mux21_wrapper;

architecture rtl of mux21_wrapper is
component mux21 is
    port(D0 : in  std_logic;
         D1 : in  std_logic;
         S0 : in  std_logic;
         Z  : out std_logic);
end component;
begin
    inst: mux21 port map (D0 => D_in(0), D1 => D_in(1), S0 => S0, Z =>
 Z);
end rtl;

In the configuration:

   for all : mux21_noarc
      use entity work.mux21_wrapper(rtl)
      port map ( D_in => D_in , S0=> S, Z => Z) ;
   end for;
```

## VHD-200

(error) Unexpected %s\n\t %s. Analysis Failed.

### Description

This message indicates an internal problem in the VHDL analysis for synthesis.

### What Next

Contact your Synopsys application engineer or the Synopsys hotline staff. Document beforehand the test case that produced the error to assist Synopsys personnel in analyzing your problem. In the meantime, do not use the construct that caused the error.

# VHDL

## VHDL-1

(warning) Can't find types for all ports on design '%s'.\n \tIt will be written out with its ports bit-blasted.

### Description

This error is generated when vhdlout_single_bit is not set to BIT and if there are no bus information, then the design probably came from a format that doesn't put bus information on, and the user will expect the ports to be bit-blasted anyways.

Feedback

### What Next

Try to read in a different format of netlist which can represent buses. They can be EDIF,Verilog, VHDL, etc. Or change vhdlout_single_bit to BIT.

---

## VHDL-2

(warning) '%s' is both a %s and a %s.\n \tThe %s is being renamed to '%s'.

### Description

There was a name clash between these two objects. Please refer to the IEEE Standard VHDL Language Reference Manual (normally referred to as the VHDL LRM) for details on scoping rules for identifiers.

There is a limitation in the intelligence of VHDLout when issuing this warning. If there is a port with a name that clashes with a type-name, where the type is used as a sub-type of another type in a design being written out, and this sub-type is defined in STD.STANDARD (a predefined package that is part of the language definition), then this warning is issued even though no type declaration for the sub-type is written out. In other words this warning may be issued for a type name that is actually not written out. With the current version of the VHDL LRM, IEEE Std 1076-1987, this will only happen for the following example:

```
entity e is
  port ( bit: out bit_vector(1 downto 0));
end e;

architecture a of e is
begin
        ...
end a;
```

### What Next

You can change the port names so it will not be clashed with type-name.

---

## VHDL-3

(warning) Design '%s' has invalid vhdl types.\n \tIt will be written out with its ports bit-blasted.

### Description

This error is generated when vhdlout_single_bit is not set to BIT, and type information is not supported.

Feedback

**What Next**

Change the types to the supported types. one of the following: Bit,Array,Enumeration and Record

---

## VHDL-4

(error) Design '%s' is not linked. Use the link command on the design\n \tif you would like to write out vhdl for it.

**Description**

The design needs to be linked first in order for the VHDL writer to write out the design.

**What Next**

Run the link command in dc_shell.

---

## VHDL-5

(error) Could not process the types for the design.

---

## VHDL-6

(error) Could not process the types in the design. Try setting vhdlout_dont_write_types to TRUE.

---

## VHDL-7

(error) You must change 'vhdlout_bit_type' to a bit type that includes a\n \tthree-state value when writing out %s.

**Description**

*vhdlout_bit_type* determines what VHDL type to use when writing out ports, single-bit-wide ports, or ports that were buses in the original source but are written out bit-blasted.

This error can occur if, for instance, a generic three-state device is written out and *vhdlout_bit_type* is set to type "BIT," which does not contain a three-state value. Such a description cannnot simulate the high-impedance value, so VHDLout issues this error.

Note that VHDLout, apart from the type "BIT," does not recognize possible values of a type. So, if you use a type other than "BIT," but it does not contain a three-state value, this error will go undetected.

Feedback

### What Next

Set *vhdlout_bit_type* to a type that contains a three-state value. For example *std_logic*.

### Examples

This example will cause the error if read in and then written out with *vhdlout_bit_type* = *BIT*.

```
library IEEE;
use     IEEE.std_logic_1164.all;

entity TRI_VHDL is
  port ( cond : in boolean; a : in std_logic; r : out std_logic);
end TRI_VHDL;

architecture test of TRI_VHDL is
begin
  process ( cond, a) begin
    if( cond) then
      r <= a;
    else
      r <= 'Z';
    end if;
  end process;
end test;
```

### Examples

```
Error: You must change 'vhdlout_bit_type' to a bit type that includes a
        three-state value when writing out generic three-state devices.
 (VHDL-7)
```

---

## VHDL-8

(error) This pre-v2.0 db file contains inconsistent type information.\n

### Description

The current version of Design Compiler is not able to read the db generated before version 2.0.

### What Next

\tPlease re-read in the original VHDL file.

---

## VHDL-9

(error) You tried to write out two designs ('%s' and '%s') that contained\n \t\tdifferent types with the same name ('%s').\n

Feedback

**Description**

VHDL writer is not able to write out two designs (object) of the same names but of different types.

**What Next**

Please write out the designs separately.

---

## VHDL-10

(warning) The type '%s' in design '%s' was renamed to '%s'.

**Description**

This warning is generated when the type needs to be renamed because of the illegal VHDL character. It is generated internally for the returned types of subprograms.

**What Next**

Check the returned type defined for the subprograms; if it is an anonymous type, try to change it to be a real type.

---

## VHDL-11

(warning) The bits in port '%s' in design '%s' have inconsistent '%s' constraints.\n \tSome constraint information is being lost.

**Description**

This warning is generated when there is more than one constraint set for the same port. It takes the first constraint you set.

**What Next**

Make sure the constraints are setting once, or the first constraint is the one you want.

---

## VHDL-12

(warning) EQUAL and OPPOSITE attributes on multi-bit ports are being ignored.

**Description**

This warning is generated when EQUAL and OPPOSITE attributes exist on multi-bit ports.

**What Next**

Make sure the attributes are consistently set for multi-bit ports.

Feedback

## VHDL-13

(warning) The '%s' '%s' is not a legal vhdl identifier.\n \tIt has been renamed to '%s'.

**Description**

This warning is generated when the identifier contains illegal VHDL character.

**What Next**

Please refer to VHDL reference manual and change it to legal VHDL character if you do not it to be renamed.

## VHDL-14

(error) Record '%s' has duplicate field names '%s'

## VHDL-15

(error) The variable '%s' should be a list of lists with each\n \tsub-list having exactly three strings in it.

**Description**

The variable vhdlout_conversion_functions should contains list of list which contains exactly three strings in it. They are "from_type", "to_type" and "function_name".

**What Next**

Type "help vhdlout_conversion_functions" for more details.

## VHDL-16

(error) Unable to model inout ports '%s' and '%s', which are wired together.

**Description**

The *write -f vhdl* command is intended primarily for the creation of a simulation model of the contents of the *dc_shell* database. This error occurs if there is more than one bidirectional port (port mode "inout") in the same net. Note that a bidirectional port driving a bidirectional pin will not cause this error.

This connectivity cannot be correctly modeled for simulation in VHDL, because it would require one assign statement for port A driving port B and one for port B driving port A. This circular assignment does not work.

Feedback

### What Next

The preferred remedy is to change the HDL source so that it does not contain more than one bidirectional port per net.

The least preferred remedy is to use the *dc_shell disconnect_net* command to disconnect all but one of the bidirectional ports in the same net. Then re-write the design. If necessary, these ports can be reconnected manually in the resulting VHDL text file.

---

## VHDL-17

(error) Unable to write a conversion function for the inout %s port '%s'.

### Description

VHDLout encountered a bidirectional port when it was creating type conversion functions. Because information about the current direction of the signal is not available at any given time, it is not possible to create a working type conversion function for bidirectional signals. Hence, one does not know whether to convert for a source or a sink signal.

Beyond looking at the port's mode, VHDLout also analyzes the net to see if the port is used both as a sink and a source. If it is used only as a unidirectional signal, VHDLout treats it as such and creates a type conversion for it. Of course, this error does not occur in that case.

### What Next

Note that type conversion functions are created in two situations:

*vhdlout_single_bit* = *VECTOR* and *vhdlout_write_top_configuration* = *TRUE*.

To write a VHDL description for the design in this situation, change the setting of *vhdlout_single_bit* to "not equal VECTOR" or set *vhdlout_write_top_configuration* = *FALSE*.

*vhdlout_single_bit* = *USER* and *vhdlout_preserve_hierarchical_types* = *VECTOR* and

The suggested remedy for this situation is to set *vhdlout_preserve_hierarchical_types* = *USER*

To see the possible settings and their meanings, refer to the man page for the *vhdlout_single_bit* variable. With either of these two changes, no type conversions are created.

If the above setting does not work, then it could be this port connects to an component pin that is buffer.

Feedback

## VHDL-18

(warning) '%s' is not a legal simulator.\n \tVariable '%s' is being ignored.

**Description**

This warning is generated when vhdlout_target_simulator is set to an unknown simulater.

**What Next**

Reset vhdlout_traget_simulator to a valid simulator.

## VHDL-19

(warning) Target simulator '%s' is not supported when\n \t%s is set to "TRUE".

**Description**

The warning is generated when vhdlout_target_simulator is set to a "xp" or "xp100" simulator and vhdlout_equations is set to TRUE.

**What Next**

Change vhdlout_equations to FALSE or change vhdlout_target_simulator.

## VHDL-20

(warning) There are inout ports of type '%s' that are\n \tmultiply driven. The type '%s' must be a resolved type or\n \tthe vhdl will not simulate.

**Description**

VHDLout cannot determine whether the port is of a resolved type. This warning is issued whenever a multiply driven inout port is found.

**What Next**

Check that the type reported in the message is a resolved type. If it is, ignore this warning. If the type is not resolved, two remedies exist.

Modify the setting of *vhdlout_single_bit* or *vhdlout_preserve_hierarchical_types* as appropriate. Change the variable setting to "BIT" or "VECTOR." When necessary, VHDLout creates a resolved type. For more information on the use of these variables, see the appropriate man pages.

Alternately modify your VHDL source to use a resolved type for the inout ports, recreate the design, and write again. Note that the warning is issued only once for each type. In a hierarchical design, multiple entities with the same problem may exist.

Feedback

### Examples

Example of warning that is safe to ignore. "std_logic" is a resolved type.

```
library IEEE;
use     IEEE.std_logic_1164.all;

entity M_VHDL is
  port ( a, b : in std_logic; r : inout std_logic);
end M_VHDL;

architecture test of M_VHDL is
begin
  process ( a) begin
    r <= a;
  end process;

  process ( b) begin
    r <= b;
  end process;
end test;
```

Example where warning is correct. "BIT" is not a resolved type. This will cause problems in simulation.

```
entity M_VHDL is
  port ( a, b : in bit; r : inout bit);
end M_VHDL;

architecture test of M_VHDL is
begin
  process ( a) begin
    r <= a;
  end process;

  process ( b) begin
    r <= b;
  end process;
end test;
```

### Examples

```
Warning: There are inout ports of type 'std_logic' that are
        multiply driven. The type 'std_logic' must be a resolved type or
        the vhdl will not simulate. (VHDL-20)
```

---

## VHDL-21

(warning) STD.TEXTIO package used by design '%s' in file '%s' is not supported for synthesis.

Feedback

**Description**

STD.TEXTIO package is not supported for synthesis.

**What Next**

Remove the use of this package from the use clause

---

## VHDL-22

(error) Stop processing due to a corrupted db file (%s).

**Description**

Somehow the db file contains incomplete type information. Please find out which command caused this problem and report it to the Support Center or CAE.

**What Next**

Please regenerate the db file.

---

## VHDL-23

(error) Write command failed due to system error.

**Description**

It could be the file system is full.

---

## VHDL-24

(error) Can not write out the assignment statement for this inout port %s.

**Description**

This ocurrs probably due to the renaming on the inout port signals.

**What Next**

Check your design and the port assignment.

---

## VHDL-25

(error) Can not write out this design '%s' with memory operations in it using non-levelized writer.

**Description**

The design needs to be flatten before writting out.

Feedback

**What Next**

Please set vhdlout_levelize to TRUE.

---

## VHDL-100

(error) Parameter '%s' multiply defined %s

---

## VHDL-101

(error) Variable or signal '%s' defined more than once %s

---

## VHDL-102

(error) Port '%s' defined more than once %s

---

## VHDL-103

(error) Entity '%s' defined more than once %s

---

## VHDL-104

(warning) 'SIGNAL' declaration for subprogram input port ignored %s

---

## VHDL-105

(error) Unknown type '%s' used %s

---

## VHDL-106

(error) Undefined function, variable or type '%s' %s

---

## VHDL-107

(error) Undefined variable '%s' %s

---

## VHDL-108

(warning) 'Transport' construct is not supported. It is ignored %s

Feedback

## VHDL-109

(warning) Signal assignment delays are not supported. They are ignored %s

## VHDL-110

(error) Invalid based integer %s %s

## VHDL-111

(warning) GUARDED is not supported. It is ignored %s

## VHDL-112

(warning) TRANSPORT is not supported. It is ignored %s

## VHDL-113

(warning) GUARDED and TRANSPORT are not supported. They are ignored %s

## VHDL-114

(warning) Beginning designator '%s' does not match end designator '%s' \n for construct %s

## VHDL-115

(warning) Access types are not supported. They are ignored %s

## VHDL-116

(warning) File types are not supported. They are ignored %s

## VHDL-117

(warning) Record types are not supported. They are ignored %s

## VHDL-118

(warning) Physical types are not supported. They are ignored %s

Feedback

## VHDL-119

(warning) Library clauses are not supported. They're ignored %s

## VHDL-120

(warning) Bus-resolution function '%s' ignored %s

## VHDL-121

(error) Allocators are not supported %s

## VHDL-122

(error) Cannot index into scalar variable '%s' %s

## VHDL-123

(error) Cannot evaluate to expression or range %s

## VHDL-124

(error) Cannot evaluate to range %s

## VHDL-125

(error) Cannot cast expression to type '%s' %s

## VHDL-126

(error) Multiple 'others' clause in aggregate %s

## VHDL-127

(error) Cannot mix named and positional elements in same aggregate %s

## VHDL-128

(error) Expected named element in aggregate %s

Feedback

## VHDL-129

(error) Illegal attribute or cast for '%s' %s

## VHDL-130

(error) Cannot open file '%s'

## VHDL-131

(error) Syntax error %s

## VHDL-132

(error) Can only assign to signals using '<=' %s

## VHDL-133

(error) Multiple waveform elements are not supported %s

## VHDL-134

(error) Cannot read from output port %s

## VHDL-135

(error) Cannot assign to input port %s

## VHDL-136

(error) Function parameters can only be of mode 'in' %s

## VHDL-137

(error) Procedure parameters cannot be of mode 'buf' or 'linkage'

## VHDL-138

(error) Cannot assign to signals using ':=' %s

Feedback

## VHDL-139

(warning) Label declarations are not valid VHDL %s

## VHDL-140

(error) Unexpected unconstrained range specification %s

## VHDL-141

(error) Unexpected paranthesised list %s

## VHDL-142

(error) Undefined variable %s %s

## VHDL-143

(error) Unexpected range specification %s

## VHDL-144

(error) Cannot constrain type %s

## VHDL-145

(error) Syntax error: identifier expected

## VHDL-146

(error) Syntax error: Illegal constraint %s

## VHDL-147

(error) Syntax error: Empty paranthesised list %s

## VHDL-148

(error) Cannot evaluate to expression %s

Feedback

## VHDL-149

(error) Cannot use type '%s' as variable %s

## VHDL-150

(error) Port and generic names can only be simple identifiers %s

## VHDL-151

(error) Predefined attributes not supported for enumerated value '%s' %s

## VHDL-152

(error) Predefined attributes supported only for arrays and ranges %s

## VHDL-153

(error) Cannot evaluate expression to type %s

## VHDL-154

(error) Return statement attempts to return expression in a procedure

## VHDL-155

(error) Type '%s' multiply defined %s

## VHDL-156

(error) Package '%s' defined more than once %s

**Description**

More than one package body has been analyzed for a given package header. There are two ways this can happen: there are two bodies for a package in the file that was analyzed, or the package header and body are in separate files and the body was analyzed a second time.

**What Next**

A package can have only one body. If a package has more than one body, remove all but one of them.

Feedback

If the package header and body are in separate files and you need to reanalyze the package body (if it has changed, for example), you must reanalyze the package header before analyzing the new package body.

---

## VHDL-157

(error) Package '%s' not found %s

---

## VHDL-158

(error) Entity class must be 'type' for enumeration encoding %s

---

## VHDL-159

(error) Attributes not supported for operator symbols %s

---

## VHDL-160

(error) 'OTHERS' and 'ALL' not supported for attribute specification %s

---

## VHDL-161

(error) Encoding for multiple types not supported in same specification %s

---

## VHDL-162

(error) Type '%s' has already been assigned default encoding %s

---

## VHDL-163

(error) Number of enumerations specified is not equal to the number of \nenumerations specified in the type %s

---

## VHDL-164

(error) All encodings must be of equal length in enumeration %s

---

## VHDL-165

(error) 'while' statement not supported %s

Feedback

## VHDL-166

(error) 'NEXT' statement is not in loop %s

## VHDL-167

(error) 'EXIT' statement is not in loop %s

## VHDL-168

(error) Cannot use OTHERS along with other choices in a choice expression %s

## VHDL-169

(error) Process name '%s' is not unique %s

## VHDL-170

(error) Constant '%s' defined more than once %s

## VHDL-171

(error) Cannot assign to constants %s

## VHDL-172

(error) Component '%s' defined more than once %s

## VHDL-173

(error) Attribute '%s' not supported %s

## VHDL-174

(error) Incorrect form of 'wait' statement %s

## VHDL-175

(error) Wait statements not allowed in processes with sensitivity lists %s

Feedback

## VHDL-176

(error) Resource '%s' defined more than once %s

## VHDL-177

(error) Synthesis attribute '%s' must be scoped inside a design %s

## VHDL-178

(error) Undefined resource '%s' %s

## VHDL-179

(error) Iteration scheme required %s

## VHDL-180

(warning) Configurations are not supported. They are ignored %s

## VHDL-181

(warning) Disconnection specifications are not supported. They are ignored %s

## VHDL-182

(warning) Aliases are not supported. They are ignored %s

## VHDL-183

(warning) The REGISTER keyword is not supported. It is ignored %s

## VHDL-184

(warning) The BUS keyword is not supported. It is ignored %s

## VHDL-185

(warning) Files are not supported. They are ignored %s

Feedback

## VHDL-186

(warning) Incomplete type declarations are ignored %s

## VHDL-187

(warning) Directive '%s' must be scoped inside a subprogram %s

## VHDL-188

(error) Error in manual resource binding syntax %s

## VHDL-189

(warning) Attribute '%s' not supported %s

## VHDL-190

(warning) Guard expression not supported for blocks %s

## VHDL-191

(warning) Ports and generics not supported in block statements %s

## VHDL-192

(warning) Entity statement parts are not supported. They are ignored %s

## VHDL-193

(warning) Assertion statements are not supported. They are ignored %s

## VHDL-194

(warning) Generics are not supported. They are ignored %s

## VHDL-195

(warning) Timeout clause not supported in wait statement %s

Feedback

## VHDL-196

(warning) Sensitivity list not supported in wait statement %s

## VHDL-197

(warning) %s assumed to be of type 'integer' %s

## VHDL-198

(error) Entity class must be 'type' for data_class attribute %s

## VHDL-199

(error) Type '%s' has already been assigned a data class %s

## VHDL-200

(warning) Resource sharing attributes are effective only in processes and subprograms %s

## VHDL-201

(warning) Sensitivity list for process not supported. It is ignored %s

## VHDL-202

(error) Enumeration literal '%s' defined more than once in same enumeration type %s

## VHDL-203

(warning) Expected semicolon (;) at end of statement %s

## VHDL-204

(warning) Expected "%s" %s

## VHDL-205

(warning) 'THEN' expected after condition following 'IF' keyword %s

Feedback

## VHDL-206

(warning) 'SELECT' expected after expression following 'WITH' keyword %s

## VHDL-207

(warning) 'IS' expected after the expression following 'CASE' keyword %s

## VHDL-208

(warning) Initial values for signals not supported %s

## VHDL-210

(error) '%s' declaration is not legal in a %s %s

## VHDL-211

(error) An 'if' statement is not a concurrent statement %s

## VHDL-212

(error) A loop statement is not a concurrent statement %s

## VHDL-213

(warning) Dc script in %s '%s' ignored

## VHDL-214

(error) Signal '%s' illegally redefined %s

## VHDL-215

(error) End of file seen before %s block begun at line %d was ended

## VHDL-216

(error) Illegal reuse of '%s' %s

Feedback

## VHDL-217

(error) Multiple definition of '%s' %s

## VHDL-218

(error) Variable %s is not defined as a record %s

## VHDL-219

(error) Suffix of record is not a simple name %s

## VHDL-220

(warning) Port '%s' does not exist %s\n \t"multiple_port" attribute ignored.

## VHDL-221

(error) Architecture '%s' defined more than once for same entity %s

## VHDL-222

(error) Constant records are not supported %s

## VHDL-223

(error) %s is supported only when it is of type 'integer'\n \t %s

## VHDL-224

(error) Cannot find declaration for component %s

## VHDL-225

(error) %s is not a formal port of component %s

## VHDL-226

(error) 'others' is not legal as a formal name in a port map statement %s

Feedback

## VHDL-227

(error) In a port map statement, formals that are not simple names \n \tor simple subelements are not supported %s

## VHDL-228

(warning) Initial values are not supported for variables %s

## VHDL-229

(error) '%s' must be a procedure %s

## VHDL-230

(error) Compilation aborted. Too many errors.

## VHDL-231

(error) Identifier %s is reserved for internal use. Please use a different\n \tname for this entity %s

## VHDL-232

(error) Cannot find entity %s

**Description**

The entity for the current architecture could not be found in the library.

**What Next**

Provide the entity for the architecture.

## VHDL-233

(warning) Port '%s' of component '%s' includes a default value, which is not supported for synthesis.

**Description**

Default values for ports of a component are not supported for synthesis. For example:

```
entity e is
  port (a: in bit;
```

Feedback

```
        z: out bit);
end;

architecture a of e is
  component c
    port (p: in bit;
          q: in bit := '1';
          x: out bit);
  end component;

begin
  u1: c port map (p => a, x => z);
end;
```

**What Next**

Remove the default value for the component port and add an equivalent association in the port map of the component instantiation. For example:

```
architecture a of e is
  component c
    port (p: in bit;
          q: in bit;
          x: out bit);
  end component;

  constant c1: bit := '1';
begin
  u1: c port map (p => a, q => c1, x => z);
end;
```

## VHDL-240

(warning) Dc_shell variable 'bin_path' has incorrect value.

## VHDL-241

(warning) Cannot execute vhdl parser. Execution will continue with less error\n \t checking for the VHDL source.

## VHDL-250

(error) Cannot create work directory for VHDL parser

## VHDL-251

(error) Unable to initialize workspace for VHDL parser

Feedback

## VHDL-252

(error) Abnormal exit of child process

## VHDL-261

(error) Component name is not a simple name %s

## VHDL-262

(warning) Only simple configurations (specification of architecture for a top\n \tlevel entity) are supported. Nested block specifications and component\n \tconfigurations are ignored %s

## VHDL-263

(error) Expected simple name for architecture %s

## VHDL-264

(warning) Configuration specifications are not supported. They are ignored %s

## VHDL-265

(warning) The value '%s' of\n \t'vhdlout_preserve_hierarchical_types' is greater than\n \tthat of 'vhdlout_single_bit'.\n \t'vhdlout_preserve_hierarchical_types' is changed to be equal to\n \t'vhdlout_single_bit'.

**Description**

This error occurs when *vhdlout_preserve_hierarchical_types* is set to a higher value than *vhdlout_single_bit*.

The variables *vhdlout_single_bit* and *vhdlout_preserve_hierarchical_types* control how VHDLout handles types on ports and pins. See the manual pages on these variables for information on possible settings.

**What Next**

Change the value of *vhdlout_single_bit* and/or *vhdlout_preserve_hierarchical_types* and retry the *write* command.

Feedback

### Examples

Error: The value 'ENUM' of *vhdlout_preserve_hierarchical_types* is greater than that of *vhdlout_single_bit*. *vhdlout_preserve_hierarchical_types* is changed to be equal to *vhdlout_single_bit*.

---

## VHDL-266

(error) '%s' contains the invalid string '%s'.

### Description

This error occurs when *vhdlout_single_bit* or *vhdlout_preserve_hierarchical_types* is set to an invalid value.

The variables *vhdlout_single_bit* and *vhdlout_preserve_hierarchical_types* control how VHDLout handles types on ports and pins. See the manual pages on these variables for information on possible settings.

### What Next

Correct the setting of the variable and retry the *write* command.

### Examples

Error: 'vhdlout_preserve_hierarchical_types' contains the invalid string 'RNUM'.

---

## VHDL-267

(error) The current setting, '%s', of 'vhdlout_package_naming_style' does not produce a valid VHDL identifier.

### Description

VHDLout infers the name of the type-conversion package (if it is needed) from the variable *vhdlout_package_naming_style*. The setting of this variable must result in a legal VHDL identifier. See the manual page for *vhdlout_package_naming_style* for more details.

This error occurs when vhdlout_package_naming_style is set to a value that results in an illegal name for the type conversion package.

### What Next

Correct the setting of the variable and retry the *write* command.

### Examples

Error: The current setting, '%s', of *vhdlout_package_naming_style* does not produce a valid VHDL identifier.

Feedback

## VHDL-268

(error) '%s' is not valid for 'vhdlout_package_naming_style'.

### Description

VHDLout infers the name of the type-conversion package (if it is needed) from the variable *vhdlout_package_naming_style*. The setting of this variable must result in a legal VHDL identifier. See the manual page for *vhdlout_package_naming_style* for more details.

This error occurs when *vhdlout_package_naming_style* is set to a string that cannot be understood by VHDLout. The string (or substring) begins with a '%' followed by an unexpected character.

### What Next

Correct the setting of the variable and retry the *write* command.

### Examples

Error: '%' is not valid for *vhdlout_package_naming_style*.

## VHDL-269

(warning) The dc_script '%s' will be applied to all subsequent entities in this file.

### Description

This use of dc_scripts is discouraged because it does not have the same scope as *library* and *use* clauses.

### What Next

To apply a dc_script to a single design unit, the script must be declared in the declarative region of that design unit.

You can suppress this message by setting the *suppress_errors* variable.

## VHDL-270

(warning) Configurations are written out only when the\n \t'vhdlout_single_bit' mode is 'vector'.

### Description

Configurations between the test bench and your top-level design are written out only when the variable *vhdlout_single_bit* is set to type *vector*.

Feedback

The purpose of the configuration is to map between the original and new types. If *vhdlout_single_bit* is set to *user*, the original types are used, and a configuration is unnecessary. If *vhdlout_single_bit* is set to *bit*, the new ports do not map well to the original types. In this case, a configuration is impractical.

**What Next**

If you are using the original types (*vhdlout_single_bit* is set to *user*), you can ignore this message, or, to disable the writing of configurations, you can set *vhdlout_write_top_configuration* to "FALSE."

If you want to write out a configuration with your design, set *vhdlout_single_bit* to *vector*, and rewrite the design.

*NOTE:* Setting *vhdlout_single_bit* to *vector* does not affect the acceleration of your design.

---

## VHDL-271

(error) Design %s contains a type with name matching\n \tthe current setting of 'vhdlout_bit_type' (%s), that is more\n \tthan one bit wide. Cannot write a VHDL description including this design.

**Description**

The 'vhdlout_bit_type' must be a one-bit wide type for VHDLout to be able to produce a legal description.

**What Next**

Check your type definition and ensure it is one bit wide. If the type definition is an enumerated type with more than two possible values, it may be missing the ENUM_ENCODING attribute.

---

## VHDL-272

(warning) You must change 'vhdlout_bit_type' to a bit type that includes a three-state value when writing out three-state resolution functions. If left as is, this description may or may not simulate.

**Description**

Vhdlout found a possible clash between two connected objects. An object A, that may drive a three-state value. An object B, being driven by object A and that is of a type not including a three-state value.

For example: a signal of type 'bit' may drive a port in mode 'inout'. This is legal and works as long as that port is never driven by the outside world. This may also happen if the source description language was less rigorous on types than VHDL.

Feedback

Note that such a clash is cought by simulation during run-time.

**What Next**

The alternatives are:

A - locate the clash and correct the HDL description, re-analyze, re-synthesize and re-write vhdl. In the example above, one can either use a resolved type for the signal or change the port's mode to 'out'.

B - leave as is and simulate to find out if this was a true error.

---

## VHDL-273

(error) There is a latch in this un-compiled design. Unfortunately, this release cannot write VHDL for latches pre-compile.

**Description**

This will be fixed in a future release.

**What Next**

Work around, compile design before writing out VHDL.

---

## VHDL-274

(warning) Library cell %s has a pin of unsupported three-state driver type. It will be treated as a regular three-state driver type. The description may not simulate correctly.

**Description**

Driver type may be specified for three-state cell pins in Synopsys' Library format, which is used to describe technology libraries. This warning is issued when VHDLout is writing a behavioral description and don't recognize the driver type at hand. The resulting description may have problems simulating weak signal values properly.

Please inform Synopsys when this happens.

**What Next**

This warning is issued only when vhdlout_equations = TRUE. The warning can be avoided by setting the variable to FALSE in which case a structural description is written. For such a description, the driver type isn't relevant to VHDLout.

An alternate approach is to use the dont_use attribute on the library cell in question, prior to compiling the design. Please refer to man-page for the dc_shell command set_dont_use for details.

Feedback

## VHDL-275

(warning) Library cell %s have three-state pins of different driver types, VHDLout cannot handle more than one type on same cell. All pins on this cell will be treated as regular three-state driver type. The description may not simulate correctly.

### Description

Driver type may be specified for three-state cell pins in Synopsys' Library format, which is used to describe technology libraries. This warning is issued when VHDLout is writing a behavioral description and encounters different pin driver types on the same library cell. VHDLout is not capable of handling more than one driver type per cell.

The resulting description may have problems simulating weak signal values properly.

Please inform Synopsys when this happens.

### What Next

This warning is issued only when vhdlout_equations = TRUE. The warning can be avoided by setting the variable to FALSE in which case a structural description is written. For such a description, the driver type isn't relevant to VHDLout.

An alternate approach is to use the dont_use attribute on the library cell in question, prior to compiling the design. Please refer to man-page for the dc_shell command set_dont_use for details.

## VHDL-276

(warning) The acceleratable VHDL description that you requested could not be written, because your design, '%s', cannot be described using the XP-supported VHDL subset. The VHDL description is being written without setting the BACKPLANE attribute.

### Description

The XP accelerator accepts as input only a limited subset of VHDL constructs. A design that cannot be described using this subset is not acceleratable by the XP accelerator.

By setting the variable *vhdlout_target_simulator* to *XP*, you requested a VHDL output that is targeted at the XP accelerator. If *vhdlout_target_simulator* is set to *XP*, *write -f vhdl* attempts to describe the design using the XP-supported VHDL subset. If this is successful, then *write -f vhdl* adds the BACKPLANE attribute to the output VHDL file, indicating that the design is to be simulated using the XP Accelerator.

If, however, *vhdlout_target_simulator* is set to *XP* and the design cannot be described using the XP-supported VHDL subset, then *write -f vhdl* does not add the BACKPLANE attribute to the generated VHDL description, and issues this warning message.

Feedback

The XP-supported VHDL subset cannot describe type conversions or user-defined types. (For more information on allowed types, refer to the ZyCad XP manuals.) If you receive this warning message, your design might require type conversions and/or might contain user-defined types.

**What Next**

If your design contains user-defined types or requires type conversions, check the setting of the variable *vhdlout_single_bit* and, if it is set to USER, change it to VECTOR or BIT. Then re-execute *write -f vhdl*.

---

## VHDL-277

(error) The VHDL PROCESS on line %d must be labelled.

**Description**

The process statement contains an attributed object in its scope and, as a result, must be labelled.

**What Next**

Label the process statement.

---

## VHDL-278

(error) The port '%s' in design '%s' has unknown direction.

**Description**

The VHDL writer does not know the port direction, so it cannot write it out. This is probably caused by some modules that are missing when reading Verilog files.

**What Next**

Read in the complete design that has specified the port direction and try again.

---

## VHDL-279

(warning) STD.TEXTIO package is not supported for synthesis.

**Description**

STD.TEXTIO package is not supported for synthesis.

**What Next**

Remove the use of this package from the use clause

Feedback

## VHDL-280

(error) Only global constants are supported as the index constraint for aliasing.

### Description

Only global constants are supported as the index constraint for aliasing. Generics are also not supported in this position.

### What Next

Modify the alias declaration so that index constraint is a global constant.

## VHDL-281

(error) You cannot take an index or slice of a non-vector alias %s.

### Description

You received this message because you attempted to apply indexing or slicing to an alias that was not declared as a vector type.

Note that this error can occur even if the VHDL language permits indexing or slicing to be applied to the alias. For example, consider the following:

```
subtype T is bit_vector(0 to 2);
  alias b: T is a(1 to 3);
```

In the alias declaration, HDL Compiler does not recognize 'T' as a vector type. Thus, this error will occur if any attempt is made to apply slicing or indexing to 'b'; for example, "b(0)" or "b(0 to 1)".

### What Next

Modify the alias declaration so that the type of the alias is an array type; for example, std_logic_vector. The example shown above could be rewritten as follows:

```
alias b: bit_vector(0 to 2) is a(1 to 3);
```

## VHDL-283

(error) Aliased objects may only be a simple name, an index of a simple name, or a slice of a simple name.

Feedback

### Description

An attempt was made to declare an alias to a non-simple name. Aliases may be declared only for simple names, simple names with indexes, or simple names with slices. For example, the following aliases are acceptible:

```
alias a: bit is foo;
  alias a: bit is foo(0);
  alias a: bit_vector(0 to 1) is foo(2 to 3);
```

### What Next

Modify the alias declaration so that the aliased object is either a simple name, an index of a simple name, or a slice of a simple name.

---

## VHDL-284

(error) Arbitrary expression for bussed clock index is not supported.

### Description

Arbitrary expression for a bussed clock index is not supported. Only single variable index or integer expressions are supported.

### What Next

Use supported expressions for a bussed clock index.

---

## VHDL-285

(warning) "BAD CONNECTION - primary in/out port %s of %s is driven or read by an internal node. Using intermediate signal to generate a syntactically correct VHDL description. Consequently, this will violate the definition of structural VHDL description syntax."

### Description

VHDL syntax rule does not allow primary port of "in" mode be updated by an internal node, or a primary port of "out" mode be read by an internal node. VHDL writer is generating an intermediate signal of the form portName_port to handle this bad connection, so it can write out a syntatically correct VHDL file. The intermediate signal will be assigned a value in the body of the architecture, and this signal assignment will violate the rule for structural VHDL description.

### What Next

Fix this bad connection in your design.

Feedback

## VHDL-286

(warning) There is a data discrepancy between synopsys database and the output file.

**Description**

VHDL/verilog writer detects that the names of the netlist objects have changed to fit syntax and semantic requirement for VHDL/verilog. The change might cause a back-annotation problem in the flow. For example, if the VHDL/verilog output file is used to generate sdf or saif file. The netlist object names in the sdf or sail might not able to back-annotate to synopsys database.

**What Next**

Run the *change_names* command with the *-rule vhdl/verilog* and then rewrite the netlist.

**See Also**

- change_names

## VHDL-287

(error) "BAD CONNECTION - port %s of mode other than out is unconnected. Consequently, this will violate the definition of structural VHDL description syntax."

**Description**

VHDL syntax rule does not allow port of mode other than "out" be unconnected.

**What Next**

Run change_names to create a dummy net to connect the port.

## VHDL-288

(error) "BAD CONNECTION - bussed port %s is partially connected."

**Description**

VHDL syntax rule does not allow bussed port to be partially connected.

**What Next**

Run change_names to create a dummy net to connect each bit of the port.

## VHDL-289

(error) Bussed clock is not longer supported in VHDLC. Use Presto VHDL instead.

Feedback

**Description**

Bussed clock is no longer supported in VHDLC.

**What Next**

Please either recode the VHDL source code, or use Presto VHDL instead.

---

## VHDL-290

(warning) A dummy net '%s' is created to connect open pin '%s'.

**Description**

VHDL writer detects that a pin is not connected. The writer creates a dummy net connected to the pin to satisfy syntax requirement for VHDL. You may use variable vhdlout_unconnected_pin_prefix to control the prefix of the dummy nets. The dummy net appears in the output VHDL file only. It does not exist in the db file. To let db file contain dummy nets, change_names needs to be executed before VHDL writer.

**See Also**

- change_names

---

## VHDL-291

(warning) You are using an older version of the VHDL netlist reader/writer, which will be obsoleted in the X-2005.06 release.

**Description**

You are using an older version of the VHDL netlist reader/writer, which will be obsoleted in the X-2005.06 release. To use the current VHDL reader/writer, please make sure these variables are set as follows:

VHDL netlist reader: enable_2003.03_vhdl_reader true VHDL netlist writer: enable_2003.03_vhdl_writer true

---

## VHDL-292

(error) Verilog In Error, %d modules are possible top modules.

**Description**

Unsupport more then one top modules.

Feedback

**What Next**

Please check the design file, and make sure there is only one top module.

---

## VHDL-2001

(warning) Statements in an entity declaration are not supported for synthesis. They are ignored in entity %s %s

---

## VHDL-2021

(warning) 'BUS' and 'REGISTER' signal kinds are not supported for synthesis. They are ignored %s

---

## VHDL-2022

(warning) Initial values for signals are not supported for synthesis. They are ignored %s

---

## VHDL-2023

(warning) Type of the generic is assumed to be 'Integer' in synthesis %s

---

## VHDL-2024

(error) Only generics of type INTEGER are supported for synthesis. - %s\n

**Description**

The synthesis VHDL subset only includes generics of type integer.

---

## VHDL-2035

(warning) Type %s has already been assigned default encoding %s

---

## VHDL-2040

(warning) Attribute %s not supported for synthesis %s

---

## VHDL-2041

(warning) Alias declarations are not supported for synthesis. They are ignored %s

Feedback

## VHDL-2042

(warning) File declarations are not supported for synthesis. They are ignored %s

## VHDL-2043

(warning) Disconnection specifications are not supported for synthesis. They are ignored %s

## VHDL-2045

(warning) Guard conditions for blocks are not supported %s

## VHDL-2046

(warning) Declaration and use of generics and ports in a block header is not supported %s

## VHDL-2047

(error) The attribute '%s' defined on line %d is not\n \tcontained within an architecture. Attribute '%s' only\n \tapplies to processes, and it must be defined inside of the\n \tarchitecture that contains the process.

**Description**

Attributes that need to be applied to processes must occur inside of the architecture that contains them.

**What Next**

Move the attribute to inside of the architecture.

## VHDL-2048

(error) Could not find process '%s' inside of the architecture%s\n \tat line %d. The process is referred to by the attribute\n \t'%s' defined on line %d.

**Description**

The attribute '%s' is meant to apply to processes only, and the process defined could not be found.

**What Next**

Check the names of the processes and the definition of the attribute for errors.

Feedback

## VHDL-2049

(Warning) The attribute '%s' is only valid when applied to\n \tone of the following:\n \t\t%s. \n \tOn line %d, the attribute is applied to a(n) %s.\n \tThe attribute will be ignored.

### Description

The name attribute is only allowed to attribute certain classes.

### What Next

Check to make sure that the attribute is applied to the correct object.

## VHDL-2050

(warning) Timeout clause not supported for synthesis in wait statement %s

## VHDL-2060

(error) The translate_on/translate_off pragma has been incorrectly used here\n \tinside a construct %s.

### Description

Only use the translate_on and translate_off pragmas on complete constructs, not on parts of constructs. For example, do not use these pragmas to comment out one of more branches of a "case" statement or the "if" construct as shown in the following examples.

```
Example 1

-- pragma translate_off
if (Is_X(svec))  then
  assert (now < 20 ns)
    report "logic vector contains U, X, Z or W"
    severity error;
else
-- pragma translate_on
  if ( t > 20) then  -- synthesize this RTL
    . . .
-- pragma translate_off
end if;
-- pragma translate_on

Example 2

case (expr_1) is
  -- pragma translate_off
  when value_1 =>  -- incomplete case statement missing this branch
    . . .
```

Feedback

```
  -- pragma translate_on
  when value_2 =>
    . . .
  when . . .
    . . .
end case;
```

**What Next**

Modify the placement of translate_on and translate_off pragmas. When you use the translate_on and translate_off pragmas, use them around the entire "if"/"case" statement. When this isn't possible, then "translate_off" parts of a construct that aren't part of VHDL syntax. In the modified example below, the "if", "then" and "else" constructs are all "translated", the "else" clause is interpreted to synthesis as a "null".

```
Example 1

temp := (others => "X");
if (svec /= temp)  then
  if ( t > 20) then  -- synthesize this RTL
    . . .
else
-- pragma translate_off
  assert (now < 20 ns)
    report "logic vector contains U, X, Z or W"
    severity error;
-- pragma translate_on
end if;

Example 2

case (expr_1) is
  when value_1 =>
  -- pragma translate_off
    . . .
  -- pragma translate_on
  when value_2 =>
    . . .
  when . . .
    . . .
end case;
```

## VHDL-2090

(warning) Declarations in a configuration declaration statement are not supported for synthesis. They are ignored %s

## VHDL-2091

(warning) Configuration specifications are not supported for synthesis %s

Feedback

## VHDL-2092

(warning) Only simple configurations (specification of architecture for a top\n \tlevel entity) are supported for synthesis. Nested block specifications and component\n \tconfigurations are ignored %s

## VHDL-2093

(error) Access types are not supported for synthesis %s

## VHDL-2094

(warning) File types are not supported for synthesis. They are ignored %s

## VHDL-2095

(warning) Physical types are not supported for synthesis. They are ignored %s

## VHDL-2096

(warning) Incomplete type declarations are not supported for synthesis. They are ignored %s

## VHDL-2097

(warning) Signal assignment delays are not supported for synthesis. They are ignored %s

## VHDL-2098

(warning) 'Transport' construct is not supported for synthesis. It is ignored %s

## VHDL-2099

(warning) Assert and report statements are not supported for synthesis. They are ignored %s

**Description**

The VHDL "assert" and "report" statements have no meaning for synthesis. These statements are ignored by HDL Compiler.

Feedback

## VHDL-2100

(error) Allocators are not supported for synthesis %s

## VHDL-2103

(error) %s

**Description**

Slang functions cannot have a return data-type that is unconstrained. The return-type cannot be a record or multi-dimensional array.

The legal data types allowed are:

Scalar types: enumeration, integer, physical, floating-point Composite types: single dimensional array only Access Types File Types

Slang procedures and functions cannot have parameters whose data-type is a record or multi-dimensional array.

The legal data types allowed are:

Scalar types: enumeration, integer, physical, floating-point Composite types: single dimensional array only Access Types File Types

## VHDL-2104

(error) indexing of access-type line is not supported %s

## VHDL-2105

(error) The suffix all is not supported %s

## VHDL-2106

(error) generalized Wait statements are supported only in architectures that are attributed as TESTBENCH %s

## VHDL-2107

(error) generalized Wait statements are not supported in architectures whose TESTBENCH attribute is not set TRUE %s

Feedback

## VHDL-2108

(error) Wait statements are not supported in subprograms %s

## VHDL-2109

(error) Event and Stable attributes are not supported in subprograms %s

## VHDL-2110

(error) An expression of this kind is supported for synthesis, when only one variable is used. This expression has two variables %s and %s %s

## VHDL-2111

(error) Aggregate assignment by name is not supported for the field names of a record.

### Description

The fields of a record were assigned using named aggregate assignment, which is not supported for synthesis. The following is an example of this style of assignment:

```
type myrec_type is record
    vecpart : bit_vector(0 to 4);
    count   : integer range 0 to 4;
  end record;

  signal myrec : myrec_type;

    ...

  myrec <= (vecpart => input_vecpart, count => input_count);
```

In the above example, "vecpart" and "count" are fields of a record. For records, aggregate assignment using named notation is not supported.

### What Next

Assign each field of the record individually, or use positional notation. The following example shows both positional notation and assignment of each field individually.

```
type myrec_type is record
    vecpart : bit_vector(0 to 4);
    count   : integer range 0 to 4;
  end record;

  signal myrec_positional : myrec_type;
  signal myrec_eachfield  : myrec_type;
```

Feedback

```
  ...

myrec_eachfield.vecpart <= input_vecpart;
myrec_eachfield.count   <= input_count;

myrec_positional <= (input_vecpart, input_count);
```

## VHDL-2112

(error) %s

**Description**

File names in the file declaration should be constants, generics or constant string literals. Use of expression or argument of the subprogram parameter is not supported by Cyclone.

## VHDL-2120

(error) In a port map statement, formals that are not simple names or simple subelements are not supported for synthesis %s

## VHDL-2121

(error) Type conversion on formal associations is not supported for synthesis %s

## VHDL-2130

(error) Only simple configurations are supported for synthesis. This configuration declaration is not supported %s

## VHDL-2131

(error) Configurations are not supported for direct instantiation during synthesis %s.

**Description**

The following usage is not supported in VHDL Compiler.

```
LABEL:  configuration library_name.configuration_name
                generic map (...)
                port map (...);
```

Feedback

## VHDL-2132

(warning) Explicit architecture specification is not supported for direct instantiation during synthesis %s.

### Description

The following usage is not correct in VHDL Compiler.

```
LABEL:  library_name.entity_name(architecture_name)
                generic map (...)
                port map (...);
```

VHDL Compiler reads the architecture to be used from the .mra file. Therefore, only the most recently analyzed one is meaningful to synthesis.

### What Next

Remove the architecture specification from the statement and analyze the architecture you want to use first.

## VHDL-2140

(error) Multi-dimensional arrays are not supported for synthesis %s

## VHDL-2141

(error) Multiple waveform elements are not supported for synthesis %s

## VHDL-2142

(error) Attribute `RANGE or `REVERSE_RANGE can't have index with it %s.

### Description

Multiple-dimensional array is not supported for synthesis. So `RANGE(i) or `REVERSE_RANGE(i) is meaningless for synthesis.

## VHDL-2150

(error) This form of wait statement is not supported for synthesis %s

## VHDL-2151

(error) Attribute %s%s is not supported for synthesis %s

Feedback

## VHDL-2152

(error) Literal '%s' is not supported for synthesis %s

## VHDL-2153

(error) Physical types are not supported for synthesis %s

## VHDL-2154

(error) Cannot mix named and positional elements in same aggregate for synthesis %s

## VHDL-2155

(error) Deferred constants are not supported for synthesis %s

## VHDL-2156

(error) Floating point types are not supported for synthesis %s

**Description**

The VHDL predefined type "REAL" is an example of a floating point type.

**What Next**

Remove any floating point types from the VHDL source.

## VHDL-2157

(error) Type %s not found %s

**Description**

This error occurs if a type is used but not declared. It is likely to happen if the type declaration is enclosed in *translate_off* or *synthesis_off* comments.

**What Next**

Declare the type before you use it.

## VHDL-2158

(error) Record types (%s) are not supported in generic\n \tdeclarations for synthesis %s

Feedback

### Description

Record types are not support in generics, only in ports, signals variables, and constants.

### What Next

Use multiple smaller generics.

---

## VHDL-2159

(error) Empty string constants are not supported for synthesis %s.

### Description

A string with no characters or spaces is not supported by VHDL Compiler. For example, the following is not supported:

```
entity test is
  generic(str : string := "");  -- This is not supported
  port(data_in  : in std_logic;
       data2_in : in std_logic;
       data_out : out std_logic);
end test;

architecture one of test is
constant c1 : string := "";  -- This is not supported
begin
end test;
```

### What Next

Use a blank string (i.e. " ") instead of an empty string (i.e. ""). The above code should be modified to:

```
entity test is
  generic(str : string := " ");  -- Add a space.  This is allowed.
  port(data_in  : in std_logic;
       data2_in : in std_logic;
       data_out : out std_logic);
end test;

architecture one of test is
constant c1 : string := " ";  -- Add a space.  This is allowed.
begin
end test;
```

---

## VHDL-2160

(error) The 'event or 'stable attribute (%s) is supported only when the attribute is used in conformance with the style described in the Synopsys manual for the VHDL compiler.

Feedback

### Description

An example of the use of the *'event* attribute is:

```
if (clk'event and clk = '1') then ...

                or

        wait until clk'event and clk = '1';
```

An example for the use of the 'stable attribute is:

```
if (not clk'stable and clk = '1') then ...

                or

        wait until not clk'stable and clk = '1';
```

This error will occur in cases such as the following:

```
if(clock'event and clock = '1' and enable = '1') then ...

                or

        wait until clk'stable and clk = '1';
```

### What Next

Modify the statement containing the *'stable* or *'event* attribute so that it is in conformance with the suggested style.

---

## VHDL-2161

(warning) Synopsys attribute %s's value is not set to TRUE, the attribute is ignored %s

### Description

This is a Boolean attribute. To have any effect, it must be set to "TRUE."

### What Next

Check that the attribute specification assigns the value "TRUE." Case doesn't matter.

---

## VHDL-2162

(error) Usage of parameters or variables declared in parent subprogram is not supported %s

Feedback

### Description

VHDL Compiler doesn't allow parameters or variables declared in parent subprogram to be used in nested child subprograms. For Example, the following usage of "j" and "v" are prone to bad logic:

```
library ieee ;
use ieee.std_logic_1164.all ;

entity my_test is
port( d                 : in std_logic ;
       q                 : out std_logic
    );
end  my_test ;

architecture synthesize of my_test is
begin
    process(d)
        procedure outer_procedure(signal j : out std_logic) is
        variable v : std_logic;
            procedure inner_procedure is
            begin
                j <= v;
            end ;
        begin
            v := '1';
            inner_procedure(j);
        end ;
    begin
        outer_procedure(q);
    end process;
end  synthesize ;
```

### What Next

Change the usage of parameters or variables in your VHDL code.

---

## VHDL-2163

(error) The rising_edge or falling_edge function (%s) is supported only when used in conformance with the style described in the VHDL Compiler Reference Manual.

### Description

You receive this error message if you use the *rising_edge* or *falling_edge* function in a way that does not conform to the style described in the *VHDL Compiler Reference Manual*.

The following are examples of uses that would generate this error message.

```
if rising_edge( ck ) and enable = '1' then ...

                or
```

Feedback

```
        if falling_edge( ck ) and ck = '0' then ...
```

The following is an example of a supported use of the *rising_edge* function.

```
if rising_edge( ck ) then ...

                or

        wait until rising_edge( ck );
```

Similarly, the following is an example of a supported use of the *falling_edge* function.

```
if falling_edge( ck ) then ...

                or

        wait until falling_edge( ck );
```

**What Next**

Refer to the *VHDL Compiler Reference Manual* and modify the statement containing the *rising_edge* or *falling_edge* function so that it conforms to the suggested style. Then, re-execute the command.

---

## VHDL-2201

(error) An entity named '%s' already exists in this library. (It may be an internal Synopsys design). Please use a different name for your entity.

---

## VHDL-2202

(error) %s is the name of an internal Synopsys design. Please use a different name for your entity.

---

## VHDL-2203

(error) Cannot find entity %s %s

---

## VHDL-2204

(error) Could not find object %s %s. This error can occur if a\n \tpackage (%s) that exists in memory also exists in another library referred to\n \tin a Library statement. This error can also occur if the package uses\n \tsynthesis_off/synthesis_on directives around the object.

Feedback

### Description

This error occurs because the synthesis tool searches for packages in memory before it searches the libraries.

A package exists in memory if it has been analyzed in the current session of the tool. The error can also occur if two or more libraries have packages of the same name and if the packages are used in the same design.

If the object contains synopsys synthesis_off/on or translate_off/on directives around it, the object is not synthesized. As a result, the object cannot be found when it is referenced in the design. The problem can also occur if the directives are placed incorrectly around part of the object. For example, consider the following procedure declaration:

```
procedure my_proc (variable data_in: in std_logic;
                   variable clk: in std_logic;
                   variable data_out: out std_logic;
-- synopsys synthesis_off
                   variable delay: in time);
-- synopsys synthesis_on
```

In this case, the directives have not been inserted properly, so the procedure is not synthesized. To make the code usable for both synthesis and simulation, this procedure declaration should be:

```
procedure my_proc (variable data_in: in std_logic;
                   variable clk: in std_logic;
                   variable data_out: out std_logic
-- synopsys synthesis_off
                   ;
                   variable delay: in time
-- synopsys synthesis_on
                   );
```

### What Next

To get around this problem, if you have more than one package of the same name (even if they are in different libraries), rename one of the packages. If you do not have more than one package of the same name, analyze the package and the design that uses the package in separate sessions of *dc_shell* or *design_analyzer.*

If the problem has to do with the synthesis_off/on or translate_off/on directives, modify the code to use the directives appropriately. Please see the DESCRIPTION section above for additional information on this.

---

## VHDL-2205

(warning) Package '%s' defined more than once %s

Feedback

**Description**

This warning occurs if the same package is multiply declared in same HDL source file.

It also occurs if multiple HDL source files are read by the same *read* or *analyze* command *and* if the same package is defined in more than one of the files.

**What Next**

Be sure to read or analyze a package only once.

---

## VHDL-2206

(error) You have declared two architectures with the same name %s %s

**Description**

You are using two (or more) architectures with the same name, belonging to the same entity.

**What Next**

Rename the architecture(s) so that they have unique names.

---

## VHDL-2207

(error) You have declared a component inside a for generate loop.

**Description**

VHDL-compiler does not allow component declaration inside a for generate loop.

**What Next**

Move the component declaration outside the for loop.

---

## VHDL-2221

(error) Type %s is not defined %s

---

## VHDL-2230

(error) '%s' used more than once as the label for an instance.

**Description**

This error occurs if two or more instances in the same scope are labelled with the same name using the "label" pragma.

Feedback

**What Next**

Change the names of one instances to make them distinct. You can do this by changing the label pragma for each instance.

---

## VHDL-2231

(error) Undefined resource %s %s

---

## VHDL-2232

(error) Entity class must be 'type' for data_class attribute %s

---

## VHDL-2233

(error) Type %s has already been assigned a data class %s

---

## VHDL-2234

(error) Object %s not found %s

**Description**

An object was referenced but not declared. This can happen if a pragma *synthesis_off* hides the declaration.

**What Next**

Change the HDL description so that the declaration is visible for the HDL Compiler.

---

## VHDL-2235

(warning) Type %s not found %s

---

## VHDL-2236

(warning) Pragma "out_port_type" must have exactly two arguments %s

---

## VHDL-2237

(warning) The pragma '%s' on line %d is not a supported pragma\n \tIt will be ignored.

Feedback

**Description**

This error is given when the user has a comment of the form

-- pragma <pragma_name>

And the pragma_name is not one of the pragmas that is supported.

**What Next**

This is just a warning, so no action is required. However, the user may want to check the pragma_name to see if it has been mis-typed.

---

## VHDL-2241

(error) Cannot encode multiple types in same attribute specification %s

---

## VHDL-2242

(error) All encodings must be of equal length in enumeration %s

---

## VHDL-2243

(error) The value of Synopsys attribute %s is expected to be name of port, signal or variable %s

**Description**

The attribute value does not match the name of an object of the class of port, signal, or variable.

**What Next**

Change the attribute value.

---

## VHDL-2244

(error) Synopsys attribute %s's value is expected to be name of a process label %s

**Description**

The attribute value does not match name of a process label.

**What Next**

Change the attribute value.

Feedback

## VHDL-2245

(error) Label %s already has a Synopsys set_reset attribute %s

**Description**

The process label has more than one Synopsys *set_reset* attribute, which is illegal.

**What Next**

Change source code to put only one *set_reset* attribute on the process.

## VHDL-2246

(error) More than one Synopsys attribute master_process_is and/or slave_process_is specified for same process label %s

**Description**

A process label can have only one *master_process_is/slave_process_is* attribute.

**What Next**

Change source code to put only one attribute on the process.

## VHDL-2247

(error) Pragma translate_off inside an expression is not supported %s

**Description**

Inside an expression, the VHDL Analyzer cannot deal with pragma *translate_off* or *translate_on*.

**What Next**

In the case of an after-clause, like the example below, take out the pragmas. After-clauses are ignored for synthesis. A warning is issued in that case.

To suppress warnings, refer to the man page for the *suppress_errors* variable.

**Examples**

The following example will cause error VHDL-2247:

```
entity pragma is
        port ( iput : in bit; oput : out bit);
    end pragma;

    architecture test of pragma is
```

Feedback

```
    begin
        oput <= iput
        -- synopsys translate_off
        AFTER 7 ns
        -- synopsys translate_on
        ;

    end test;
```

## VHDL-2248

(error) Body of package %s seen before or without its declaration %s

### Description

The package declaration must be analyzed before the corresponding package body. There was probably a problem with the declaration part related to the synthesis subset, causing the analysis of the declaration to terminate. Since the analysis of the package declaration was terminated, the body cannot be analyzed.

### What Next

Correct the problem in the package declaration, and this error will go away.

### Examples

This example will cause error VHDL-2248:

```
package P is
          constant    C : bit; -- deferred constant, not supported by
 synthesis.
        end P;

        package body P is
          constant    C : bit := '0';
        end P;
```

## VHDL-2249

(warning) The attribute '%s' requires a list of more than one signal %s.

## VHDL-2250

(warning) Incorrect computation may result because range of the left hand side\n \tdoes not cover range of the right hand side for assignment %s.

Feedback

### Description

When the value of the right-hand side is outside the range of the left-hand side, a run-time error occurs in simulation. However, no logic is built by the VHDL Compiler to detect this kind of run-time error.

### What Next

Declare a proper range for the left-hand side. Or type-convert the right-hand side to ensure that the range of the left-hand side covers that of the right-hand side.

---

## VHDL-2251

(error) Enabling expression not permitted outside wait statements %s

### Description

This error occurs with statements of the form: if(clk'event and clock = '1' and enable = '1') then .. Such expressions are only permitted in: wait until clk'event and clock = '1' and enable = '1' ... at present

### What Next

Modify the if statement with the enabling condition to conform to suggested style.

---

## VHDL-2252

(information) Make sure there is an assertion statement in the design\n \tfor attribute '%s' %s.

### Description

Attribute 'one_hot' indicates at most one object in its group can have a *Logic1* value at any instance of time, all other objects must have a *Logic0* value. Attribute 'one_cold' indicates at most one object in its group can have a *Logic0* value at any instance of time, all other objects must have a *Logic1* value.

### What Next

For *"attribute one_hot of c1, c2, c3 : signal is "true"*, add a process in the design like the following exmaple.

```
-- synopsys synthesis_off
process (c1, c2, c3)
begin
  assert not ((c1='1' and c2='1') or
              (c1='1' and c3='1') or
              (c2='1' and c3='1'))
    report "One-hot violation"
```

Feedback

```
      severity Error;
end process
-- synopsys synthesis_on
```

For *"attribute one_cold of c1, c2, c3 : signal is "true"*, add a process in the design like the following exmaple.

```
-- synopsys synthesis_off
process (c1, c2, c3)
begin
  assert not ((c1='0' and c2='0') or
              (c1='0' and c3='0') or
              (c2='0' and c3='0'))
    report "One-cold violation"
      severity Error;
end process
-- synopsys synthesis_on
```

---

## VHDL-2253

(error) Unsupported type of value\n \t in attribute specification - %s.

### Description

Synthesis VHDL subset does not include this type of a value in a attribute specification. Value could be of concatenation type.

### What Next

Rewrite the attribute specification using the VHDL synthesis subset.

---

## VHDL-2254

(error) "Time" is an unsupported type - %s\n

### Description

Synthesis VHDL subset does not include the VHDL type TIME. Your HDL should not depend on simulation-specific delays for its proper behavior.

### What Next

You can prevent synthesis from using simulation-specific HDL by enclosing it within the synthesis_off / synthesis_on pragmas.

### Examples

The following is a sample usage of the TIME type used in a declaration:

GENERIC( DelayAmount : TIME );

Feedback

To prevent the TIME type from going through synthesis, change the above HDL to:

GENERIC( --synopsys synthesis_off DelayAmount : TIME --synopsys synthesis_on );

This will ensure the HDL is present for simulation, but is ignored during synthesis.

---

## VHDL-2255

(error) Generics of type string are not supported - %s\n

**Description**

Synthesis VHDL subset does not include generics of type string.

**What Next**

Rewrite the attribute specification using the VHDL synthesis subset.

---

## VHDL-2256

(error) Range of enum types is not supported %s.

**Description**

Ranges of enumerated types using the *TO* construct are not supported in VHDL Compiler. The following *CASE* statement using the *TO* construct is not supported.

```
type STATE_TYPE is (STATE0, STATE1, STATE2, STATE3);
  signal STATE, NEXT_STATE : STATE_TYPE;

  process
  begin
    case STATE is
    when STATE0 =>
         DOUT <= '0';
    when STATE1 to STATE3  =>
         DOUT <= '1';
    end case;
  end process;
```

**What Next**

The workaround is to use the "|" construct which performs an *OR* operation.

```
type STATE_TYPE is (STATE0, STATE1, STATE2, STATE3);
  signal STATE, NEXT_STATE : STATE_TYPE;

  process
  begin
    case STATE is
    when STATE0 =>
```

Feedback

```
        DOUT <= '0';
    when STATE1 | STATE2 | STATE3 =>
        DOUT <= '1';
    end case;
  end process;
```

## VHDL-2257

(error) unsupported function call %s

### Description

Some remote functions can not be called.

### What Next

Move to a local package.

## VHDL-2260

(error) Bit-slice is not supported in generic for '%s',\n \t%s.

### Description

Bit slices in the generic map of a component instantiation are not supported. For example:

```
U1: GATE1  generic map ( INIT(0) => '0', INIT(1 to 7 ) => init_val)
           port map ( DINPUT(0) => d0, DINPUT(1) => d1, DOUTPUT =>
 dout );
```

### What Next

Concatenate the values assigned to slices in the generic map or port map. Assign this concatenation to an intermediate signal or variable, and use this signal or variable in the component instantiation. For example:

```
aggr1 <= ('0' & init_val);
U1: GATE1  generic map ( INIT => aggr1 )
           port map ( DINPUT(0) => d0, DINPUT(1) => d1, DOUTPUT =>
 dout );
```

## VHDL-2262

(error) Enumeration values may not be used as for or for-generate loop bounds

Feedback

**Description**

Enumeration values may not be used to define the bounds of a for or for-generate loop in Synopsys' synthesizable subset of VHDL. These bounds must be integer values that can be computed at compile time.

---

## VHDL-2263

(error) Number of enumeration encoding values does not match the number of enumeration values, line %d.

**Description**

The number of enumeration encoding values defined via the ENUM_ENCODING attribute must match the number of enumeration values for the type.

**What Next**

Modify the number of enumeration or enumeration attribute values.

---

## VHDL-2264

(error) Incorrect way to use '%s attribute .

**What Next**

event and 'stable attributes are allowed to be used only when inferring a clock edge . For example

(clock'event and clock = '1')

and

((not clock'stable) and clock = '1')

following are not allowed

((not clock'event and clock = '1')

and

(clock'stable and clock = '1')

---

## VHDL-2270

(error) The alias declaration%sis invalid because object %s is an alias. Aliases to existing aliases are not supported for synthesis .

Feedback

### Description

HDL Compiler supports aliases to signals, variables and constants only. HDL Compiler does not support aliases that refer to existing aliases.

### What Next

Either remove the alias to an alias and replace references to the first alias with references to the second, or replace the second alias with an alias to the original object. For example, given:

signal B : std_logic_vector(0 to 31); alias MSB : std_logic_vector(0 to 7) of B(24 to 31); alias MSN : std_logic_vector(0 to 3) of MSB(4 to 7);

...

MSN <= "0000";

You could remove the definition of MSN and replace its use with:

MSB(4 to 7) <= "0000";

Or, you could transform the declaration of MSN and leave its use intact:

alias MSN : std_logic_vector(o to 3) of B(28 to 31);

...

MSN <= "0000";

---

## VHDL-2271

(error) Cannot use predefined attributes with alias %s%s, as the alias's range is not locally static.

### Description

HDL Compiler does not support the use of predefined attributes (such as 'LEFT, 'HIGH, 'RANGE, and so on) with aliases when the alias range is not locally static.

### What Next

Replace the predefined attribute with the corresponding components of the alias's range. For example, given

procedure foo(B : inout std_logic_vector) is begin constant b_left : integer := B'length-1; alias b_alias : std_logic_vector(b_left downto 0) of B;

...

for I in b_alias'RANGE loop ... end loop;

Feedback

... end foo;

replace the loop with

for I in 0 to b_left loop ... end loop;

---

## VHDL-2280

(error) 'IMAGE is used outside of an attribute statement in a for-generate loop.

**Description**

HDL Compiler supports the 'IMAGE attribute only as part of an attribute specification of the following type:

attribute RELATIVE_LOCATION of <object> : label is <string_literal> & INTEGER'IMAGE(<indexvar>) & <string_literal>

Where <indexvar> is the index variable of a for-generate loop.

**What Next**

The following is an example of the situation in which 'IMAGE might be used.

G : FOR d IN 0 TO 7 GENERATE B : BLOCK ATTRIBUTE RELATIVE_LOCATION OF flop : LABEL IS "R" & INTEGER'IMAGE(d) & "C0"; BEGIN flop : dff PORT MAP (D => INP(d), Q=>OUTP(d), CK=>CLK); END BLOCK; END GENERATE G;

In this case, you're instantiating a set of eight DFFs that will be used to latch the value of an eight-bit bus. Note that 'IMAGE can be used only

```
+ Within a block within a for-generate loop.
+ As part of an attribute statement, where the attribute name is
  "RELATIVE_LOCATION" and the value is a concatination of three strings:
  a prefix, the 'IMAGE expression, and a suffix. In the previous example,
 "R"
  is the prefix and "C0" is the suffix. The prefix and suffix must be
  literals.
+ With the index variable for the for-generate loop.
```

Note that choices for the following are left to you:

```
+ The label for the for-generate loop and the block.
+ The number and contents of the instantiation statements within the
 block.
```

You can also specify more than one entity name in the attribute statement. For example, you can use

ATTRIBUTE RELATIVE_LOCATION OF flop,rflop : LABEL IS "R" & INTEGER'IMAGE(d) & "C0";

Feedback

if you were instantiating two series of entities:

flop : dff PORT MAP (D => INP(d), Q=>OUTP(d), CK=>CLK); rflop : dff PORT MAP (D => INP(d), Q=>R_OUTP(8-d), CK=>CLK);

You can also have more than one such attribute statement if you want to specify a different prefix or suffix.

---

## VHDL-2281

(error) The attribute RELATIVE_LOCATION does not have a value of the form <string_literal> & INTEGER'IMAGE(<indexvar>) & <string_literal>

**Description**

HDL Compiler supports the use of the RELATIVE_LOCATION attribute only with a very specific value. The attribute statement must be of the form:

attribute RELATIVE_LOCATION of <object> : label is <string_literal> & INTEGER'IMAGE(<indexvar>) & <string_literal>

Where <indexvar> is the index variable of a for-generate loop.

**What Next**

The following is an example of the situation in which you may use the attribute RELATIVE_LOCATION:

G : FOR d IN 0 TO 7 GENERATE B : BLOCK ATTRIBUTE RELATIVE_LOCATION OF flop : LABEL IS "R" & INTEGER'IMAGE(d) & "C0"; BEGIN flop : dff PORT MAP (D => INP(d), Q=>OUTP(d), CK=>CLK); END BLOCK; END GENERATE G;

In this case, we're instantiating a set of eight DFFs that will be used to latch the value of an eight-bit bus. Note that 'IMAGE can be used only: + Within a block within a for-generate loop. + As part of an attribute statement, where the attribute name is "RELATIVE_LOCATION" and the value is a concatination of three strings: a prefix, the 'IMAGE expression, and a suffix. In the above example, "R" is the prefix and "C0" is the suffix. The prefix and suffix must be literals. + With the index variable for the for-generate loop.

Note that choices for the following are left to you: + The label for the for-generate loop and the block. + The number and contents of the instantiation statements within the block.

You may also specify more than one entity name in the attribute statement. For example, you could use

ATTRIBUTE RELATIVE_LOCATION OF flop,rflop : LABEL IS "R" & INTEGER'IMAGE(d) & "C0";

Feedback

if you were instantiating two series of entities:

flop : dff PORT MAP (D => INP(d), Q=>OUTP(d), CK=>CLK); rflop : dff PORT MAP (D => INP(d), Q=>R_OUTP(8-d), CK=>CLK);

You may also have more than one such attribute statement, if you want to specify a different prefix and/or suffix.

---

## VHDL-2282

(error) The argument to 'IMAGE is not a for-generate loop index variable.

**Description**

HDL Compiler supports the 'IMAGE attribute only as part of an attribute specification of the following type:

attribute RELATIVE_LOCATION of <object> : label is <string_literal> & INTEGER'IMAGE(<indexvar>) & <string_literal>

Where <indexvar> is the index variable of a for-generate loop.

**What Next**

The following is an example of the situation in which 'IMAGE might be used.

G : FOR d IN 0 TO 7 GENERATE B : BLOCK ATTRIBUTE RELATIVE_LOCATION OF flop : LABEL IS "R" & INTEGER'IMAGE(d) & "C0"; BEGIN flop : dff PORT MAP (D => INP(d), Q=>OUTP(d), CK=>CLK); END BLOCK; END GENERATE G;

In this case, you're instantiating a set of eight DFFs that will be used to latch the value of an eight-bit bus. Note that 'IMAGE can be used only

```
+ Within a block within a for-generate loop.
+ As part of an attribute statement, where the attribute name is
  "RELATIVE_LOCATION" and the value is a concatination of three strings:
  a prefix, the 'IMAGE expression, and a suffix. In the previous example,
 "R"
  is the prefix and "C0" is the suffix. The prefix and suffix must be
  literals.
+ With the index variable for the for-generate loop.
```

Note that choices for the following are left to you:

```
+ The label for the for-generate loop and the block.
+ The number and contents of the instantiation statements within the
 block.
```

You can also specify more than one entity name in the attribute statement. For example, you can use

Feedback

ATTRIBUTE RELATIVE_LOCATION OF flop,rflop : LABEL IS "R" & INTEGER'IMAGE(d) & "C0";

if you were instantiating two series of entities:

flop : dff PORT MAP (D => INP(d), Q=>OUTP(d), CK=>CLK); rflop : dff PORT MAP (D => INP(d), Q=>R_OUTP(8-d), CK=>CLK);

You can also have more than one such attribute statement if you want to specify a different prefix or suffix.

---

## VHDL-2283

(error) Only the type INTEGER is supported for the prefix of the attribute 'IMAGE %s

**Description**

Presently 'IMAGE attribute has a limited support in HDL compiler. Only Integer may appear as a prefix to it

**What Next**

Please rewrite the use of 'image attribute so that it has integer as its prefix

---

## VHDL-2284

(error) Declarative regions of generate statements is not supported %s

**Description**

This version of VHDL Compiler doesn't support declarative regions of generate statements.

**What Next**

Use block statement inside the generate statement.

Example:

You replace the following VHDL codes

---------- build_blks: FOR i IN 1 TO 4 GENERATE

SIGNAL int_connect: unsigned(3 DOWNTO 0); BEGIN blk1: blk_in PORT MAP(a_in(i - 1), int_connect(i - 1)); blk2: blk_out PORT MAP(int_connect(i - 1), b_out(i - 1)); END GENERATE build_blks; ----------

by

Feedback

---------- build_blks: FOR i IN 1 TO 4 GENERATE

THE_BLOCK : block SIGNAL int_connect: unsigned(3 DOWNTO 0); BEGIN blk1: blk_in PORT MAP(a_in(i - 1), int_connect(i - 1)); blk2: blk_out PORT MAP(int_connect(i - 1), b_out(i - 1)); end block;

END GENERATE build_blks; ----------

## VHDL-2285

(warning) VHDL-93 generates different concatenation results from VHDL-87 %s.

### Description

You receive this message if *read* or *elaborate* detects that the concatenation result generated by VHDL-93 for your design differs from the result that would have been generated by VHDL-87. This message is to warn you to expect different results from those you might have become accustomed to with VHDL-87.

By default, currently VHDL Compiler uses VHDL-93 concatenation, which generates a different left bound and/or direction from VHDL-87. (For details of the concatenation definition, see the IEEE *VHDL-93 Language Reference Manual*.) You can override the default behavior by setting the *hdlin_vhdl93_concat* environment variable to *false*.

### What Next

If you receive this message, proceed as follows:

1. Read the *VHDL-93 Language Reference Manual* to understand the concatenation results generated by VHDL-93.
2. If this is acceptable for your design, you need take no further action.
3. If you want to revert to VHDL-87 behavior, set *hdlin_vhdl93_concat* to *false*.

# VL

## VL-001

(warning) lib-pin "%s" doesn't have required EM type via ladder candidates.

### Description

The lib-pin has attribute "is_em_via_ladder_required" set to true, and therefore it needs to have at least one EM type via ladder in its via ladder candidate list.

Feedback

**What Next**

Use set_via_ladder_candidate to add at least one EM type via ladder to the lib-pin's via ladder candidate list.

---

## VL-002

(Error) Via ladder candidate "%s" of lib-pin "%s" is undefined.

**Description**

The via ladder candidate defined on the lib-pin does not exist.

**What Next**

There are four options to resolve this.

- Use *create_via_rule* command to create the via ladder.
- Check technology file to see if the via ladder is properly defined.
- Check if the technology file is correctly read in.
- Use *report_via_ladder_candidates* command to check if the via ladder candidates are correctly defined on the lib-pin. If the via ladder candidates are incorrectly defined, please use reset_via_ladder_candidates and *set_via_ladder_candidate* commands to correct the settings.

---

## VL-003

(Warning) Illegal via rule "%s" - Pin layers on via ladder ("%s") and on "%s" ("%s") are different, tool will ignore this rule.

**Description**

Via rule is illegal due to layer is mismatch with pin.

**What Next**

Please check tech file or via rules.

---

## VL-004

(Warning) Illegal via rule "%s" - Terminal counts on via ladder ("%d") and on "%s" ("%d") are different, tool will ignore this rule.

**Description**

Via rule is illegal due to temninal counts are mismatch.

Feedback

**What Next**

Please check tech file or via rules.

---

## VL-005

(Warning) Illegal via rule "%s" - Top layer of via ladder ("%s") is over max layer ("%s") of the net, tool will ignore this rule.

**Description**

Via rule is illegal due to top layer is over max layer.

**What Next**

Please check tech file or via rules.

---

# VR

---

## VR-001

(information) Ignoring reference to '%s' in module '%s'\n \tbecause it has no connections.

**Description**

This message may occur during *read_verilog*. If all instances of a given reference name are unconnected within a module, the reference will be ignored and no instances of that reference will be created. Typically such instances are for filler cells or other physical objects that have no meaning in a logical netlist.

**What Next**

If you want to create instances in such cases, set the application variable *svr_keep_unconnected_cells* to "true".

---

## VR-002

(information) Renamed scalar %s '%s' to '%s' in design '%s'.

**Description**

While reading a Verilog file, two net or port names were found to be in conflict, requiring one of the objects to be renamed. This can occur for one of two reasons: an ambiguous bus naming style or escaped names.

Feedback

Given a bus_naming_style of "%s[%d]" and the following verilog port declarations:

```
output \\z[1] ;
output [1:0] z;
```

the bus reference z[1] conflicts with the scalar port z[1], declared with the first output statement. The scalar is renamed to something like z[1]1, unless there is already a port z[1]1, in which case z[1]2 is tried, and so on.

**What Next**

No action is required. This is an informational message.

---

## VR-004

(error) Invalid global reference '%s' - %s\n \tin module ending at line %d in %s

**Description**

You receive this message if we find an invalid global reference while reading a Verilog file. A global reference is a connection to a wire in another module. There are several restrictions on the use of global references. For a reference of the form *module.wire*, all of the following must be true:

- The module must exist in the file being read.
- The module must precede the module that is making the reference.
- The wire must exist in the module.
- The wire must be a logic constant.

Any deviation from these rules generates this error message.

**What Next**

There is no other support for global signals. If you cannot conform to the rules, you must remove these signals from your netlist.

---

## VR-005

(warning) Failed to find the specified top module '%s'.

**Description**

The user-specified top module was not declared in any Verilog file. The Verilog reader will attempt to determine the top module and set the current design.

**What Next**

Check the current design using the *current_design* command. If it is not correct, set the correct current design using the *current_design* command, or rerun read_verilog with the correct module passed through the *-top* option.

Feedback

## VR-006

(warning) Failed to determine the top module.

**Description**

The Verilog reader attempted to determine the top module based upon the module instantiations. However it was unable to find any possible top modules. The current design is not set.

**What Next**

Set the current design using the *current_design* command.

## VR-007

(error) Multiple possible top modules exist.

**Description**

The Verilog reader attempted to determine the top module based upon the module instantiations. However it found multiple modules that could be the top module, and was unable to determine the correct one. No design has been read.

**What Next**

Rerun the command and specify the top module with the *-top* option. You may use the *-verbose* option to display a list of possible top modules.

## VR-008

(warning) Module '%s' already exists (duplicate found at line %d in %s).

**Description**

The module is already defined in the design. The duplicate module definition is ignored. If reading multiple Verilog files into the same design, an earlier file may have already defined this module.

**What Next**

Rename or remove the duplicate.

## VR-009

(error) Unable to rename %s '%s'.

Feedback

**Description**

The object name conflicts with an existing object name. The Verilog reader attempted to find a new name for the object, but was unable to find a unique name. The object will be unnamed.

**What Next**

Check the input Verilog file.

---

## VR-010

(error) Duplicate declaration of port '%s'.

**Description**

The port has already been declared. Each port should be declared once and only once.

**What Next**

Check the ports in the input Verilog file.

---

## VR-011

(error) Design '%s' in library '%s' is currently opened; reading Verilog into an opened design is not allowed.

**Description**

The Verilog file cannot be read into a design that is currently opened in memory.

**What Next**

Close the currently opened design and then run *read_verilog* again.

---

## VR-012

(information) Reading Verilog into new design '%s' in library '%s'.

**Description**

A new design has been created by *read_verilog*. The Verilog file is read into the new design.

**What Next**

If your intention was to read the Verilog file into a new design, there is nothing further to do. However if this is not your intention, review the options that were used when calling *read_verilog*.

Feedback

## VR-013

(information) Reading Verilog into existing design '%s' in library '%s'; the existing design contents has been truncated.

**Description**

The Verilog file is read into the specified existing design. Incremental Verilog updates are not allowed. So *read_verilog* opens the design in truncate mode, discards the existing contents, and then reads the new Verilog file into the design.

**What Next**

When saving truncated designs, use the *save_block -as* option to avoid overwriting your existing design on disk.

## VR-014

(error) Unable to rename top module from '%s' to '%s'.

**Description**

The design and top module names must match. This command attempted to rename the top module to the same name as the design, but another module with that name already exists. As a result, this command failed and no design is read.

**What Next**

Run the command again and specify a different design name with the *-design* option.

## VR-015

(information) Multiple possible top modules exist.

**Description**

The Verilog reader attempted to determine the top module based upon the module instantiations. However it found multiple modules that could be the top module, and the first read module was chosen as the top module of the block.

**What Next**

Continue the reading flow and complete the block by linking it while selecting the correct top module, with the set_top_module command.

## VR-016

(error) Current block is already linked.

Feedback

### Description

The Verilog reader attempted to write to the current block, but the block is already linked. The current block must be unlinked to allow multiple read_verilog commands.

### What Next

Either create a new block or use the -design option to specify a new block. If the intended target was indeed the current block, ensure that it is not linked.

---

## VR-017

(warning) Net name '%s' is reserved and has been renamed to '%s' in module '%s'.

### Description

The Verilog file declares a signal net with the name "*Logic0*" or "*Logic1*". However net names "*Logic0*" and "*Logic1*" are reserved by Synopsys for the tie-low and tie-high constant nets. They cannot be used as names of signal nets in the netlist and have been renamed.

### What Next

No action is required. The Verilog file may have been incorrectly generated. If the Verilog file was generated by a Synopsys tool, try running the *change_names* command to fix the net name prior to writing the Verilog file.

---

## VR-018

(warning) Found redefinition of module '%s' (%s found at line %d in %s).

### Description

The module is already defined in the design. The duplicate module definition is ignored or the original overwritten. If reading multiple Verilog files into the same design, an earlier file may have already defined this module. Preference is given to a module definition if the source file was specified as a full pathfile name.

### What Next

Resolve the collision if necessary.

---

## VR-019

(error) Decryption Failed. %s

Feedback

**Description**

This message occurs when failing to read an encrypted verilog file, currently, *read_verilog* only can decrypt IEEE 1735 standard Verilog file.

**What Next**

Specify appropriate verilog file and then run *read_verilog* again.

---

## VR-020

(error) Module %s cannot be created because same name design cannot be found in current library and its reference libraries.

**Description**

*read_3d_verilog* requires physical blocks to be created firstly (except the top), then read in a 3D-Verilog file. All modules in Verilog file except the top module should have same name physial blocks existed in its reference libraries.

**What Next**

Please use option -design to specify the top design name. If the error still exists, please check whether modules in source Verilog file can match with designs in current library's reference libraries.

---

# VW

---

## VW-001

(warning) Writing verilog from unlinked design '%s'.

**Description**

The current design is not linked and could not be automatically linked. The generated Verilog file will contain only modules from within the current design. Modules from other designs and libraries will not be included. The modules will not be hierarchically ordered.

**What Next**

Link the current design and rerun write_verilog, if you want a complete or hierarchically ordered Verilog file.

---

## VW-002

(warning) Inapplicable option '%s' ignored.

Feedback

**Description**

The specified option is not applicable for unlinked designs. The current design is unlinked.

**What Next**

Link the current design and rerun write_verilog, if you want a hierarchically ordered Verilog file.

---

## VW-003

(warning) Non-unique module name '%s' (from library '%s' design '%s') renamed to '%s'.

**Description**

Multiple distinct modules exist with the specified name. Duplicate modules are renamed to a unique name.

**What Next**

The user must make sure that any external scripts are modified to match the now-unique Verilog file.

---

## VW-004

(error) A %s within the design %s does not have a valid name.

**Description**

The specified object within the design does not have a valid name. The object will be omitted from the Verilog output file.

**What Next**

The user should check the design and verify that it is valid.

---

## VW-005

(warning) Skipping module '%s' from design %s.%s ...

**Description**

The module cannot be written because it is from a non-design design view and would not create valid Verilog. The module is probably from an Abstract or an Outline design view. These design views are abstractions of the full designs. They do not contain the complete design information. Writing such a module would result in incomplete Verilog.

Feedback

### What Next

Switch to the design view for this design, then write your Verilog modules from that design.

### See Also

- read_verilog_outline

---

## VW-006

(error) Cannot write Verilog for design %s.%s.

### Description

The design cannot be written because it is from a non-design design view and would not create valid Verilog. The module is probably from an Abstract or an Outline design view. These design views are abstractions of the full designs. They do not contain the complete design information. Writing such a module would result in incomplete Verilog.

### What Next

Switch to the design view for this design, then write your Verilog modules from that design.

### See Also

- read_verilog_outline

---

## VW-007

(warning) Net name '%s' is omitted from the output file, because the net name does not match a connected port name. The net is implicitly renamed to match one of its connected ports.

### Description

The indicated net name is omitted from the output Verilog file, because the net name does not match a connected port name. The ports and pins of the net are connected together by a implicit net in the output file. This net is implicitly named to match one of its ports.

When this Verilog file is read back into the tool, the net will have the same name as one of its ports. Hence the original net name is lost.

To retain the original net name in the output Verilog file, assign statements must be written out. Set application option, file.verilog.write_pg_net_assigns, to true to output assign statements for renamed PG nets. Set application option, file.verilog.write_non_pg_net_assigns, to true to output assign statements for renamed non-PG nets.

Feedback

## VW-008

(error) Bus port '%s' of module '%s' is written out as a scalar port because it is in an invalid state. %s

### Description

The bus port is in an invalid state as indicated and cannot be written out using the Verilog bus format. It is written out as a scalar port.

A bus port is valid only if the port's owning bus name is a non-empty string and the port's name is the concatenation of the owning bus name, opening bus delimiter, bus index, and closing bus delimiter (e.g., bus[0]).

### What Next

Check the bus port and correct the data so it is in a valid state.

## VW-009

(error) The bus port '%s' of module '%s' has a port element with an irregular name '%s'; similar errors for each bus port might be reported only once. Run change_names before write_verilog.

### Description

The indicated bus port element name is irregular and cannot be written out using the Verilog bus format.

A bus port element name is acceptable only if its owning bus name is a non-empty string and the element name is the bus name followed by only one non-negative subscript.

The following examples show the unsupported bus port element names:

```
a[2][3]
  a.b
  a.b[2].c
  a[-1]
  a[b]
```

The following example shows the bus port element name that can be accepted:

```
a[4]
```

### What Next

Correct the issue, such as renaming the bus elements cited as being in an unsupported format. The *change_names -hierarchy -rule verilog* command will "flatten_port_bus" in most such cases. Then run the *write_verilog* command again.

Feedback

**See Also**

- change_names
- define_name_rules

---

## VW-010

(warning) Net name '%s' is renamed to '%s' in module '%s' in output file.

**Description**

The indicated net name is renamed to avoid name confliction in the module, because the net and same name port is not connected together in the module.

Otherwise, when this Verilog file read cannot read back into the tool, or the connection will be wrong.

**WHAT'S NEXT**

Check the net and same name port connection, and correct the name confliction in the module.

---

## VW-011

(error) object name '%s' contains space.

**Description**

Spaces are allowed in database object names, however the Verilog syntax does not support spaces in object names.

**WHAT'S NEXT**

Run change_names before write_verilog

---

## VW-012

(warning) Incorrect syntax '%s' specified for cell-rename entry. The entry will be ignored.

**Description**

The syntax for the cell-rename entry is not a valid one. Refer the man-page of write_verilog for the appropriate syntax.

---

## VW-013

(error) Failed to open the cell-rename map file '%s'.

Feedback

**Description**

The cell-rename map file could not be opened. Verify the file-permissions and retry.

---

## VW-014

(error) Invalid value for option -mask_shift_layers.

**Description**

There can not be duplicated layers in the specified value.

---

## VW-016

(error) Mask shifted cell '%s' does not have its corresponding mapping in the map file.

**Description**

There is not a corresponding mapping for the mask shifted cell in the map file.

**What Next**

Please add the corresponding mapping to the map file and then try again.

---

## VW-017

(error) Mask shift property mismatch on cell '%s'.

**Description**

There is mask shift property mismatch between different map files.

**What Next**

Please correct the mask shift property and then try again.

---

## VW-018

(error) Fail to remove the file '%s'.

**Description**

The verilog file is removed if failure occurs.

**What Next**

Please ignore the file.

Feedback

## VW-019

(error) The port '%s' of module '%s' has a duplicate name with other port or port bus. Run change_names before write_verilog.

**Description**

The Verilog syntax does not support two module ports with the same name.

**WHAT'S NEXT**

Run "change_names -rules verilog" before write_verilog.

## VW-020

(information) Output mask-shifted cell name according to rename files.

**Description**

This message occurs when -rename_cell is specified to write_verilog and the rename files are for mask-shifted cells.

**What Next**

No action required.

## VW-021

(error) Encrypt Verilog file failed.

**Description**

This message occurs when use -encrypt option and encrytion failed.

**What Next**

If encryption fails, please don't use -encrypt option temporary, and contact support for help.

## VW-022

(warning) Command will ignore analysis block %s.

**Description**

A message indicating that this command cannot be applied to an analysis block. An analysis block is for analysis purposes and isn't suitable for other usages.

Feedback

**What Next**

If you want to write out a shell netlist from the analysis design, use the -shell_only option.

---

## VW-023

(information) Writing from view '%s' as specified by the -switch_view_list option.

**Description**

An alternate view was found for the block as specified by the *-switch_view_list* option. This alternate view will be written to the generated Verilog, instead of the bound view.

---

## VW-024

(information) Writing from bound view '%s' instead of switching views as specified by the -switch_view_list option, because no alternate view exists.

**Description**

No alternate view was found for the block as specified by the *-switch_view_list* option. The bound view will be written to the generated Verilog.

---

## VW-025

(warning) Current design is analysis design, -key_file option will be ignored.

**Description**

RTLA analysis design doesn't support using user-specified keys to encrypt Verilog file.

---

## VW-026

(warning) Executing write_verilog with %s may change the NDM database.

**Description**

This is a warning message to indicate that write_verilog will not behave as a read_only command under the reported circumstances. Although the database contents will be restored to the pre-write_verilog state, the collections and the variables populated before write_verilog might be affected.

**What Next**

Use the reported message to change the settings according to the requirement.

Feedback

## VW-027

(error) The port bus '%s' of module '%s' is a malformed port bus.

**Description**

The reported port bus is a malformed port bus. Please check the design.

**WHAT'S NEXT**

Please use report_port_buses command to investigate and rectify the port bus reported.

## VW-028

(warning) The ports of the port bus '%s' of module '%s' are written out as scalar ports because the port bus is malformed.

**Description**

The reported port bus is a malformed port bus. Please check the design.

**WHAT'S NEXT**

Please use report_port_buses command to investigate and rectify the port bus reported.

## VW-029

(warning) In module '%s', the instance name '%s' is the same as the net name and this might cause issues.

**Description**

This warning is generated when the instance name has the same name as the port net name. In Verilog, both port and module have their own namespace, which include instance names and net names. Implicit nets are used for those nets that are connected to ports. This means that the port net uses the same name as its port. Because instance names and net names share the same space, they can not use the same names.

**WHAT'S NEXT**

Please use change_names command to correct before executing the write_verilog command.

## VW-108

(error) Port bus '%s' of module '%s' is invalid; its member '%s' is not a valid one-dimensional bus port. Similar errors may only report once for each port bus.

Feedback

**Description**

The indicated port bus is invalid, because its member name is invalid in Verilog; only one-dimensional array is supported.

**What Next**

Correct the issue and try write_verilog again.

# WINSEL

## WINSEL-001

(Warning) Invalid object class %s. Supported objects are : %s

**Description**

The specified object class does not exist.

**What Next**

Verify that the object class exists and the spelling is correct.

## WINSEL-002

*WINSEL-002* (ERROR) Bad syntax for rectangle: %s

**Description**

The syntax specified for the rectangle was not acceptable. Correct syntax is, a rectangle must be of the form

{ <lower_left_point> <upper_right_point> }

and the point is

{ <x_dist > <y_dist> }

Note that distances are fixed point numbers, like 17.25 or 13000.1 or 2500.

**What Next**

Please fix the syntax and try again.

## WINSEL-003

(error) Nothing matched for search constraints.

Feedback

**Description**

No objects satisfied the constraints you specified.

**What Next**

Check the command arguments you gave, and see if the constraints can be changed or relaxed to make a match.

# ZRT

## ZRT-001

(error) Could not save the design view.

**Description**

This error message indicates that the tool could not save the design view into the design library. Possible reasons for this are

- You do not have write permission.
- No disk space is available.

**What Next**

Verify that you have write permission and that there is sufficient disk space available.

## ZRT-002

(error) Cannot load the design.

**Description**

This error message indicates that the router could not read in all the required information from the design database. Possible reasons for this are

- There is an inconsistency issue in the design.
- There is a problem in the technology file.
- Certain information that is required for the router is missing.

**What Next**

Please check the design and the technology file.

Feedback

## ZRT-003

(warn) Cannot create the ECO route attachment file.

**Description**

This message indicates that the router could not create or update an attachment file that describes the nets that have been changed by the ECO routing command. Without this file, the signoff flow might be broken. Possible reasons for this are

- You do not have write permission.
- No disk space is available.

**What Next**

Verify that you have write permission and that there is sufficient disk space available.

## ZRT-004

(error) Number of routing layers has to be greater than 1.

**Description**

The design only has one routing layer and the router can only route design with two or more routing layers.

**What Next**

Please add more routing layer to the design.

## ZRT-005

(error) Unable to support design with more than %d routing layers.

**Description**

The router currently only supports maximum of 31 routing layers.

**What Next**

Please reduce the number of routing layers.

## ZRT-006

(warn) Both adjacent cut rule and enclosed cut rule are specified, supporting only enclosed cut rule.

Feedback

**Description**

Enclosed cut rule is a super set of adjacent cut rule which is an older rule the router supports for backward compatibility. So if both rules are specified, only enclosed cut rule is honored.

**What Next**

Please remove adjacent cut rule and use enclosed cut rule to specify adjacent cut rule.

---

## ZRT-007

(warn) Ignore fat rule on layer %s.

**Description**

The reason for receiving this warning could be one of the following:

- Fat threshold table is not in ascending order.
- Fat spacing table is not symmetric.

**What Next**

Please check the technology file and make sure the fat rule is specified correctly.

---

## ZRT-008

(warn) Ignore protrusion rule on layer %s.

**Description**

The reason for receiving this warning could be one of the following:

- Protrusion threshold table is not in ascending order.

**What Next**

Please check the technology file and make sure the protrusion rule is specified correctly.

---

## ZRT-009

(warn) Ignore %s rule on layer %s.

**Description**

This warning is issued because the spacing for this rule is not specified in increasing order.

Feedback

**What Next**

Please check the technology file and make sure the rule is defined correctly.

---

## ZRT-010

(warning) Cannot find a fat contact for layer '%s'.

**Description**

This error message indicates that the tool cannot find a suitable fat contact for the specified layer.

This error restricts the router's ability to resolve fat contact violations.

**What Next**

This error is most likely due to a missing fat contact definition in the technology file. Check the technology file and, if necessary, add a fat contact definition.

---

## ZRT-011

(warning) Found more than one fat contact for layer '%s'.

**Description**

This warning message indicates that the tool found more than one fat contact for the specified layer. This is most likely due to multiple fat contact definitions in the technology file.

When the technology file contains multiple fat contact definitions, the router arbitrarily uses one of the fat contacts to fix fat contact violations.

**What Next**

Check the technology file and remove the extra fat contact definitions.

---

## ZRT-012

(error) Found illegal fat contact '%s' for layer '%s'.

**Description**

This error message indicates that the tool found an illegal fat contact definition in the technology file.

This error restricts the router's ability to fix fat contact violations.

Feedback

**What Next**

Check the technology file and fix the fat contact definition.

---

## ZRT-013

(error) Found illegal extension fat contact '%s' for layer '%s'.

**Description**

This error message indicates that the tool found an illegal extension fat contact definition in the technology file.

This error restricts the router's ability to fix extension fat contact violations.

**What Next**

Check the technology file and fix the extension fat contact definition.

---

## ZRT-014

(warn) Ignore stub spacing definition on layer %s.

**Description**

The reason for this warning message is most likely because the stub spacing value is less or equal to 0

**What Next**

Please check the technology file and make sure the stub spacing is defined correctly.

---

## ZRT-015

(warn) Ignore contact %s.

**Description**

The reason for receiving this warning could be one of the following:

- Lower, upper, or cut layer of the contact is not a valid routing layer.
- The contact has the same lower and upper layer.

**What Next**

Please check the technology file and make sure the contact is defined correctly.

Feedback

## ZRT-016

(warn) Inconsistent %s design rule found between %s and %s.

**Description**

This warning message indicates that inconsistent design rule values are found. This is most likely due to the design rule being defined multiple times with the same two layers but with different values. Only the first definition read by the router is honored.

**What Next**

Please check the technology file and make sure the design rule is only defined once for the same two layers.

## ZRT-017

(warn) No diode ratio set for %s, turning off antenna rule mode %d.

**Description**

When an antenna rule of diode mode 5 or greater is defined, diode ratio also needs to be specified for every routing layer. In this case, one or more layers are found without diode ratio and so the antenna rule is ignored.

**What Next**

Please use technology file antenna definitions to specify diode ratio for every routing layer.

## ZRT-018

(warn) No scale factor set for %s, turning off antenna rule mode %d.

**Description**

When an antenna rule of diode mode 11 or 12 is defined, scale factor also needs to be specified for every routing layer. In this case, one or more layers are found without scale factor and so the antenna rule is ignored.

**What Next**

Please use define_antenna_layer_rule to specify scale factor for every routing layer.

## ZRT-019

(warn) Illegal min/max layer constraint on net %s.

Feedback

**Description**

This warning occurs when the min layer constraint is on a higher layer than the max layer constraint. This constraint is ignored for the net.

**What Next**

Please use set_net_routing_layer_constraints to redefine the min/max layer constraint or use remove_net_routing_layer_constraints to remove the constraint on the net.

---

## ZRT-020

(warn) Illegal cell min/max layer constraint.

**Description**

This warning occurs when the min layer constraint is on a higher layer than the max layer constraint. This cell constraint is ignored.

**What Next**

Please redefine or reset this cell layer constraint.

---

## ZRT-021

(info) A partial via ladder for user debugging has been detected. Please run refresh_via_ladders to remove these structures so that routing may proceed.

**Description**

This design contains partial via ladder for user debugging. All the Zroute commands are forbidden to run on this block, except for check_routes, insert_via_ladders, refresh_via_ladders, and other full routing commands with via ladder stage.

**What Next**

Please run refresh_via_ladders to remove these structures so that routing may proceed.

---

## ZRT-022

(warn) Cannot find a default contact code for layer %s.

**Description**

This warning message indicates that a default contact code was not found for the specified layer. This is most likely due to a missing default contact code definition in the technology file.

Feedback

**What Next**

Add a default contact code definition to the technology file. If the specified layer is a nonrouting layer, such as V0, you can ignore this message.

---

## ZRT-023

(warn) create_frame on %d macros were done with "treat blockage as thin".

**Description**

"Treat blockage as thin" flag was on during the create_frame process and thus blockages inside these macros will be treated as thin by the router.

**What Next**

If treating the blockages as thin is not the desired behavior, please re-run create_frame.

---

## ZRT-024

(error) %s has no corresponding % layer.

**Description**

The given layer does not have a corresponding wire/via layer. This limits the router's ability to move from one layer to another.

**What Next**

Please define the corresponding wire/via layer in the technology file.

---

## ZRT-025

(warn) Layer %s does not have a preferred direction, assigning to %s.

**Description**

This message indicates that the layer does not have a preferred direction assigned to it. If the adjacent layer has a preferred direction, the opposite direction is set for the layer. Otherwise, metal 1 is assigned to horizontal, metal 2 to vertical, and so forth.

**What Next**

Please use the set_attribute command to assign the preferred direction as follows:
set_attribute -name routing_direction -value vertical [get_layer M9]

Feedback

## ZRT-026

(warn) Layer %s default pitch %.3f may be too small to handle up/down via. Note, that minimum requirements are "wire/via-down %.3f" & "wire/via-up %.3f".

### Description

The pitch defined for the layer may be too small due to a larger "up via" and/or "down via" requirement. The router will use the larger spacing when placing those via.

### What Next

This is only a warning message.

## ZRT-027

(warn) Ignore %d top cell ports with no pins.

### Description

This warning message indicates that the design's top cell has logical ports which have no physical shapes attached to them. The router will ignore such ports.

### What Next

Please check the top cell and create physical shapes for those ports if needed.

## ZRT-028

(error) Failed to open child cell %s.%s.

### Description

This warning message indicates that the router was not able to open the child cell for read. This could be because the reference library where the child cell resides does not have read permission or is missing.

### What Next

Please make sure the reference library is linked and readable to the current design.

## ZRT-029

(error) Cell %s has inconsistent number of logical ports in the library and in the top cell.

Feedback

**Description**

This warning message indicates that the counts for the logical ports are different between the child library cell's record and the top cell's record. This could happen if the library cell got modified after the top cell was created.

**What Next**

Please check and update the design hierarchy.

---

## ZRT-030

(warn) Ignore pin (cell %s, pin %s) on layer %s.

**Description**

This warning message occurs because physical pin shape on non-routing layer was found. The router will ignore such pin.

By default, routing commands print only 10 messages of this type. If user wishes to have more messages of this type printed to log file, set_message_info command is to be used.

Example is: set_message_info -id {ZRT-030} -limit 300

**What Next**

Please check to see if the pin shape is on the correct layer.

---

## ZRT-031

(warn) Invalid routing layer '%s' on wire

**Description**

This warning message indicates that a wire from the database is on an invalid routing later. The wire will be ignored by the router.

**What Next**

Please check the database and fix the wire to be on valid routing layer.

---

## ZRT-032

(warn) Invalid routing layer '%s' on path

**Description**

This warning message indicates that a path from the database is on an invalid routing later. The path will be ignored by the router.

Feedback

**What Next**

Please check the database and fix the path to be on valid routing layer.

---

## ZRT-033

(warn) Invalid contact array of via def %s.

**Description**

This warning message indicates that an invalid contact array is in the database. The router was unable to find a master via corresponding to the specified via def from database.

**What Next**

Please check the database and specify the correct via def for the contact array.

---

## ZRT-034

(warn) Invalid contact with via def %s.

**Description**

This warning message indicates that an invalid contact is in the database. The router was unable to find a master via corresponding to the specified via def from database.

**What Next**

Please check the database and specify the correct via def for the contact.

---

## ZRT-035

(warn) Invalid routing layer '%s' on global route

**Description**

This warning message indicates that a global route from database in on an invalid routing layer. The router will ignore the global route.

**What Next**

Please check the database and fix the global route to be on valid routing layer.

---

## ZRT-036

(warn) Ignore polygon pin (cell %s, port %s) on layer %d.

Feedback

**Description**

This warning message occurs because polygon pin shape was found. The router currently does not handle polygon pin and thus this pin will be ignored.

**What Next**

Please modify the polygon into rectangles.

---

## ZRT-037

(warn) Ignore polygon via-region in cell %s.

**Description**

This warning message occurs because polygon via-region was found. The router currently does not handle polygon via-region and thus this via region will be ignored.

**What Next**

Please re-generate the frame view.

---

## ZRT-038

(warn) Cannot find via def %s for via-region in cell %s.

**Description**

This warning message indicates that via def cannot be found for the the via-region. This via-region will be ignored by the router.

**What Next**

Please use a correct via def for the via-region and re-generate the frame view.

---

## ZRT-039

(warn) Contact %s for via-region in cell %s cannot be rotated.

**Description**

This warning message indicates that non-rotatable contact was found for a rotated via-region. The via-region is ignored in this case.

**What Next**

Please re-generate the frame view.

Feedback

## ZRT-040

(warn) Cell %s's boundary has exceeded the limit.

**Description**

This warning message occurs because the cell boundary size is larger than maximum 32-bit integer value. The cell boundary has been adjusted to fit the limit.

**What Next**

Please re-size the cell boundary to be smaller.

## ZRT-041

(error) Cannot find port %s in cell %s.

**Description**

This warning message occurs when master port cannot be found during the instance port processing. This could happen if the library cell got modified after the top cell was created and the hierarchical information is out of date. This error may also be issued in antenna flows where this port information is not available in frame view. In such cases, antenna analysis may be incorrect.

**What Next**

Please check data and update the design hierarchy.

## ZRT-042

(warn) Contact %s's %s layer enclosure %s does not satisfy end of line enclosure design rule.

**Description**

The router depends on correct construction of contacts to avoid the end of line enclosure rule. In this case, the contact's metal enclosure does not satisfy the requirement and there may be end of line enclosure design rule violation at the end of routing stage.

**What Next**

Please modify the contact to satisfy the end of line enclosure requirement or remove the design rule from the technology file.

## ZRT-043

(warn) Found must join pins on port %s without any net.

Feedback

**Description**

Must join pins without net information were found. The router won't be able to connect them unless net is assigned to them.

**What Next**

Please create net for those pins.

---

## ZRT-044

(warn) Standard cell pin %s/%s has no valid via regions.

**Description**

Via regions are useful in avoiding DRCs in accessing pins and in accessing pins off the routing grid. It is defined with the matching via specified by the contact code number.

**What Next**

Please check for DRCs in accessing this pin, and check ZRT-038 for error in contact code number, and generate the via regions if necessary.

---

## ZRT-045

(warn) Via regions for port %s/%s may be blocked completely.

**Description**

Via regions are useful in avoiding DRCs in accessing pins and in accessing pins off the routing grid.

**What Next**

Please check for DRCs in accessing this pin and generate the via regions if necessary.

---

## ZRT-046

(warn) Conflicting voltage-areas for instance %s.

**Description**

This warning message indicates that the instance belongs to two conflicting voltage-areas. In the case of a conflict, the voltage-area of the smallest enclosing instance will take precedence. The reasons for this could be one of the following.

- The leaf-level instance has been assigned a voltage-area that conflicts with the enclosing hierarchical voltage-area.

Feedback

**What Next**

Please make sure that each instance belongs to only one voltage-area.

---

## ZRT-047

(error) Ignoring voltage-area %s because maximum number of voltage-areas allowed is %d.

**Description**

This error message indicates that the number of voltage-areas in this design exceeds the maximum allowed. The reasons for this could be one of the following.

- Too many voltage-areas are being used in this design.

**What Next**

Please make sure that the number of voltage-areas is within the limit.

---

## ZRT-048

(error) Failed to write block %s.

**Description**

The router issues this error when it cannot write the block to the design library.

**What Next**

Check the preceding error messages for the specific failure reasons.

---

## ZRT-049

(error) Failed to load block %s.

**Description**

The router issues this error when it cannot read the block from the design library.

**What Next**

Check the preceding error messages for the specific failure reasons.

---

## ZRT-050

(warn) Invalid rotated polygon in master cell '%s'

Feedback

**Description**

This warning message indicates that a rotated polygon in the master cell has invalid dimensions. This rotated polygon will be ignored by the router.

**What Next**

Please check the database and fix the rotated polygon dimensions.

---

## ZRT-051

(warn) Illegal Jog wire end of line rule (%s, %s).

**Description**

This warning message indicates that end of line rule is specified between two inconsistent layers.

**What Next**

Please check the technology file and make sure the end of line rule is specified between a cut and metal layer.

---

## ZRT-052

(warn) Illegal Jog wire end of line rule (%s, %s), (%d %d).

**Description**

This warning message indicates that end of line rule is specified between two inconsistent layers.

**What Next**

Please check the technology file and make sure the end of line rule is specified between adjacent cut and metal layer.

---

## ZRT-053

(warn) Illegal Special end of line enclosure rule (%s, %s).

**Description**

This warning message indicates that end of line rule is specified between two inconsistent layers.

Feedback

**What Next**

Please check the technology file and make sure the end of line rule is specified between a cut and metal layer.

---

## ZRT-054

(warn) Illegal Special end of line enclosure rule (%s, %s), (%d %d).

**Description**

This warning message indicates that end of line rule is specified between two inconsistent layers.

**What Next**

Please check the technology file and make sure the end of line rule is specified between adjacent cut and metal layer.

---

## ZRT-055

(warn) Illegal T-shape end of line enclosure rule (%s, %s).

**Description**

This warning message indicates that end of line rule is specified between two inconsistent layers.

**What Next**

Please check the technology file and make sure the end of line rule is specified between a cut and metal layer.

---

## ZRT-056

(warn) Illegal T-shape end of line enclosure rule (%s, %s), (%d %d).

**Description**

This warning message indicates that end of line rule is specified between two inconsistent layers.

**What Next**

Please check the technology file and make sure the end of line rule is specified between adjacent cut and metal layer.

Feedback

## ZRT-057

(warn) Illegal Jog wire off limits rule (%s, %s).

**Description**

This warning message indicates that end of line rule is specified between two inconsistent layers.

**What Next**

Please check the technology file and make sure the end of line rule is specified between a cut and metal layer.

## ZRT-058

(warn) Illegal Jog wire off limits rule (%s, %s), (%d %d).

**Description**

This warning message indicates that end of line rule is specified between two inconsistent layers.

**What Next**

Please check the technology file and make sure the end of line rule is specified between adjacent cut and metal layer.

## ZRT-059

(warn) Illegal Fat wire off limits rule (%s, %s).

**Description**

This warning message indicates that end of line rule is specified between two inconsistent layers.

**What Next**

Please check the technology file and make sure the end of line rule is specified between a cut and metal layer.

## ZRT-060

(warn) Illegal Fat wire off limits rule (%s, %s), (%d %d).

Feedback

**Description**

This warning message indicates that end of line rule is specified between two inconsistent layers.

**What Next**

Please check the technology file and make sure the end of line rule is specified between adjacent cut and metal layer.

---

## ZRT-061

(warn) Port %s (cell %s) of net %s has no pin.

**Description**

This warning message indicates that the port has no physical shape attached to it. The router will ignore such port and the net will be left open at the end of the routing stage.

**What Next**

Please check the cell and create pin shape for the port.

---

## ZRT-062

(warn) Value %s is not one among the predefined values for option %s. Assigning the default value.

**Description**

This warning message indicates that the value specified for the string token option is not among the list of predefined values. This can happen if the set of predefined values change and an old value which no longer exists is read from a saved cell. In this case, the option is assigned the default value.

**What Next**

Please set the right value for the option, if the default value is not correct.

---

## ZRT-063

(warn) Cover cell '%s' contains a non PG port '%s'. Ignoring it.

**Description**

This warning message indicates that the cover cell contains a port which is connected to non PG net. This is not handled currently based on our methodology.

Feedback

**What Next**

Please make sure that the cover cell does not contain non PG port.

---

## ZRT-064

(error) Cell %s is not placed.

**Description**

Placement status of the cell indicates that it is not legally placed and will be ignored by the router.

**What Next**

Please make sure all the cells are placed before running the router.

---

## ZRT-065

(warn) Ignore net %s's connection to port %s (cell %s).

**Description**

This warning message indicates that the port will be ignored by the router, and the reason is most likely due to some data error with the cell.

**What Next**

Please check the port's cell and fix any problem associated with it.

---

## ZRT-066

(error) Net %s not found.

**Description**

Failed to find the net in the current design.

**What Next**

Check the spelling of the net name. Make sure net of that name exists.

---

## ZRT-067

(error) Net %s is not a power or ground net.

**Description**

The found net is not a power or ground net.

Feedback

**What Next**

Make sure the net is a power or a ground net.

---

## ZRT-068

(error) Could not find ground net.

**Description**

The tool could not find the ground net in the current block.

**What Next**

Add a ground net to the current block.

**See Also**

- connect_pg_net

---

## ZRT-069

(error) Too many ground nets, please specify one.

**Description**

This command found more than one ground net in the current design.

**What Next**

Specify a ground net as an argument to the command.

---

## ZRT-070

(error) Could not find any nets in the design that match the specification.

**Description**

The command expects the list of nets to be specified. Either the specification is incorrect, or there are no nets in the design that match the specification.

**What Next**

Please check to make sure that the syntax for specifying the nets is correct.

---

## ZRT-071

(warn) Skip physical_status locked shape(%s) on layer %s of net %s.

Feedback

**Description**

This warning message occurs because remove_routes find the physical status of the shape is locked.

**What Next**

Please unlock the shape and re-run remove_route if you want to remove it.

---

## ZRT-072

(error) Top cell not open.

**Description**

The top cell is not open.

**What Next**

Open the top cell to run the current command.

---

## ZRT-073

(error) The -from_file option specifies: %s but that file is not found .

**Description**

This warning message occurs because route_group cannot find a file by the name specified.

**What Next**

Please ensure that the file exits and has been generated correctly.

---

## ZRT-074

(error) The file: %s does not contain any valid net names.

**Description**

This warning message occurs because route_group cannot find any valid net names in the file specified by the -from_file option.

**What Next**

Please ensure that the file was generated correctly and contains some valid net names.

Feedback

## ZRT-075

(warn) The the file: %s specifies a net: %s which does not exist.

**Description**

This warning message occurs because the -from_file option for the route_group command has encountered a putative net name which does not specify a net.

**What Next**

Please ensure that the file was generated correctly and contains some valid net names.

## ZRT-076

(warn) Invalid contact array pitch (via master %s) (location %d %d).

**Description**

The contact array at the specified location has pitch smaller than the pitch required by the router. The router will continue, using the larger pitch. DRC's reported against the via arrays will be conservative.

**What Next**

Please check the database and verify that contact array pitch is even multiple of min grid.

## ZRT-077

(warn) %s has different horizontal and vertical net weight.

**Description**

The net has different horizontal weight and vertical weight specified. The router currently only supports one net criticality value per net, and the larger of the two values will be used. This message is only printed once for the first net found with the issue.

**What Next**

Please check the setup for the net and make sure it is correct.

## ZRT-078

(warn) Illegal enclosed via metal minLength rule (%s, %s).

Feedback

**Description**

This warning message indicates that enclosed via metal minLength rule is defined between inconsisten layers. It should be defined between a cut layer and the metal layer just above it.

**What Next**

Please check the technology file and make sure the enclosed via metal minLength rule is specified between a cut layer and metal layer.

---

## ZRT-079

(warn) Illegal enclosed via metal minLength rule (%s, %s), (%d %d).

**Description**

This warning message indicates that enclosed via metal minLength rule is defined between inconsisten layers. It should be defined between a cut layer and the metal layer just above it.

**What Next**

Please check the technology file and make sure the enclosed via metal minLength rule is specified between a cut layer and the metal layer jus above it.

---

## ZRT-080

(error) Layer %s has no routing tracks.

**Description**

The given layer does not have any routing tracks. This prevents the router from assigning wires and vias on the layer.

**What Next**

Please check if the routing tracks have been correctly defined on the layer. If the intent is to have the router automatically generate routing tracks then please make sure that the value of the router common option "-track_auto_fill" is set to true (the default value).

---

## ZRT-081

(warn) NDR width is smaller than default width (%s, %s).

Feedback

**Description**

This warning message indicates that the NDR width defined on one routing layer is smaller than the default width of the same layer.

**What Next**

Please check the NDR rule definition in tcl command to make sure that the NDR width is no less than the default width on each routing layer.

---

## ZRT-082

(warn) Turning off legal width checking on layer %s as the rule is not specified correctly.

**Description**

The number of legal widths specified cannot exceed 16. They must be in increasing order and must contain the default width on the layer.

**What Next**

Please check the technology file and make sure the legal width rules are correctly specified.

---

## ZRT-083

(warn) Contact %s's %s layer enclosure does not satisfy legal width requirement.

**Description**

The router depends on correct construction of contacts that satisfy the legal width rule. In this case, the contact's metal enclosure does not satisfy the legal width requirement and it may result in long routing run time.

**What Next**

Please modify the contact to satisfy the legal width requirement or remove the design rule from the technology file.

---

## ZRT-084

(warn) Turning off viaOnGrid rule checking on cut layer %s as the lower and the upper routing layers have the same preferred direction, and viaOnGrid rule is not well-defined.

Feedback

**Description**

If the upper and the lower routing layers of a cut layer have the same preferred direction, then the following conditions have to be satisfied in order to correctly check viaOnGrid rule on the cut layer.

- All routing tracks on both layers must be completely aligned.
- There must be a perpendicular direction layer, only one side of either below the lower layer or above the upper layer.

**What Next**

Please check the technology file and make sure the viaOnGrid rule and routing tracks on adjacent layers are correctly specified.

---

## ZRT-085

(error) Cannot run timing based detail route command with the timing driven option set to false.

**Description**

This error indicates that a timing based detail route command is being run without turning on the detail route timing driven option.

**What Next**

Please set the detail route timing driven option to true before running the command (set_app_options -global {route.detail.timing_driven true}).

---

## ZRT-086

(warn) Via ladder is not connected to all pin shapes on Cell %s Pin %s ( via rule %s ).

**Description**

The via-rule set to this pin is not having enough number of via cuts to connect all pin shapes.

**What Next**

Please check the via-rule setting flow or confirm the floating pin shape is fine.

---

## ZRT-087

(warn) The design contains a huge X-oriented obstruction.

Feedback

**Description**

This warning message indicates that the current design contains an unusually large X-oriented (45 or 135 degree) obstruction. Large X-oriented obstructions may cause inefficiency in the router and result in slow runtime.

**What Next**

Check the design to see if the large X-oriented obstruction is indeed intended. The user may need to break it into smaller ones if possible.

---

## ZRT-088

(error) NDR rule %s has illegal widths on layer %s.

**Description**

This error message indicates that either the wire width, shield width or the via width on the metal layer is illegal or larger than the signal route max width defined on that layer. Router will ignore this NDR rule on the associated nets in the design.

**What Next**

Please make sure that the NDR rule widths are legal and are less than or equal to the signal route max width defined in the tech file.

---

## ZRT-089

(warn) Value %s for layer %s is very big.

**Description**

This warning is issued because the value for a spacing is very big. Check if it is intentional.

**What Next**

If the large value was set by accident, correct it and rerun the routing command.

---

## ZRT-090

(warn) Master cell %s's place and route boundary does not contain all pin shapes.

**Description**

This warning message occurs when there is a pin shape does not overlap its master cell's place and route boundary.

Feedback

**What Next**

Please check master cell's place and route boundary.

---

## ZRT-091

(warn) Unsupported tech file value: %s: %s = %s. %s

**Description**

This warning message occurs when there is an unsupported tech file value.

**What Next**

Please check and correct the tech file.

---

## ZRT-092

(warning) Cell %s contains more than %d obstructions (%d). This can increase routing runtime; you might want to regenerate the frame view.

**Description**

A cell contains a large number of obstructions. This is atypical and might increase routing runtime.

**What Next**

If this is not expected, regenerate the frame view. To reduce the number of obstructions in the frame view, try using the *remove_non_pin_shapes* or *merge_metal_blockage* options.

**See Also**

- check_workspace
- create_frame
- lib.physical_model.remove_non_pin_shapes
- lib.physical_model.merge_metal_blockage

---

## ZRT-093

(error) Found global route in DR stage. %s

**Description**

This error message indicates that global route types are inputted in detail routing stage.

Feedback

**What Next**

Please check the database.

---

## ZRT-094

(error) Skipping a cell instance record from database as it has no name.

**Description**

There is a cell instance record in database with no name.

**What Next**

The database is corrupt. Please report this problem along with the commands run until this stage.

---

## ZRT-095

(warn) Ignore double pattern rule on layer %s.

**Description**

The reason for receiving this warning could be one of the following:

- The width threshold for an end is either not defined or is zero. • The side spacing is either not defined or is zero. • The corner spacing is either not defined or is zero.

**What Next**

Please check the technology file and make sure that all the double pattern rule values are specified correctly.

---

## ZRT-096

(error) Database contains global routing GCells that are too large.

**Description**

The database contains GCells that are too large. The number of track intersections in a GCell for any two adjacent perpendicular layers should not meet or exceed 1600. For example, if a square GCell contains 40 M2 tracks and 40 M3 tracks, this error will be triggered, since 40 x 40 >= 1600. The global routing data was likely not created with route_global.

**What Next**

Please run route_global before executing your flow.

Feedback

## ZRT-097

(warning) Pin %s in cell %s.frame does not contain the complete information required to honor x-shaped pins in routing. Please re-generate the frame view.

**Description**

The information required to process x-shapes in a pin is incomplete. The x-shapes may be ignored.

**What Next**

Please re-generate the frame view.

## ZRT-098

(warning) Contact code %s contains polygons and will be ignored by the router.

**Description**

The router does not support contact codes that have polygons. Such contact codes will be ignored by the router.

## ZRT-099

(error) Via ladder %s refers to an invalid cut name.

**Description**

A ViaRule specified in the tech file contains an invalid entry in cutNameTbl.

**What Next**

Please correct the tech file and try again.

## ZRT-100

(warning) Pin %s of net %s has no valid shapes to connect.

**Description**

This message indicates that pin has no shapes inside design boundary.

**What Next**

Please make sure that the pin has valid shapes to connect.

Feedback

## ZRT-101

(warning) Power net %s has no power preroutes, skip tie-off.

**Description**

Power net has no power preroutes. Tie-off will be turned off for this net.

**What Next**

Please add power preroutes.

## ZRT-102

(info) %d of the existing global routes are deleted.

**Description**

All existing global routes are deleted at the beginning of global routing stage, unless incremental option is set to true. In route group flow, global routes of nets not specified in the net group will be deleted even if incremental option is true.

**What Next**

No action necessary.

## ZRT-103

(info) Global Routing %s early.

**Description**

Global routing terminates early because either the congestion has been minimized or it does not seem like the congestion can be improved further.

**What Next**

No action necessary.

## ZRT-104

(warning) Routed net %s through blocked area.

**Description**

Global router routed the net through blocked area. Pin might be blocked or there is no routing solution.

Feedback

### What Next

Please check if the pins are blocked.

---

## ZRT-105

(warning) Macro pin (%s %s) does not have access edge.

### Description

The global router expects macro pins to have access edges to help the router to connect to the pins. The tool issues this warning when it finds a macro pin without an access edge. Possible causes are

- The pins are physically blocked.
- create_frame could not generate the access edges even though the pins are close to the edge of the macro.

This warning can significantly affect the runtime, memory, and QoR.

### What Next

Open the design view and verify that

- There is a clear path for the pin to exit the macro.
- It is possible for create_frame to generate the access edges.

  Try to rerun create_frame in the latest release to see if it generates more access edges.

### See Also

- check_routability

---

## ZRT-106

(warning) 'set_app_options route.global.comb_distance' option will be deprecated and replaced with 'set_app_options route.common.comb_distance' option.

### Description

The 'set_app_options route.global.comb_distance' option will be deprecated and replaced with 'set_app_options route.common.comb_distance' option.

### What Next

Please update scripts to use the new option.

Feedback

## ZRT-107

(warning) 'set_app_options route.global.clock_topology' option will be deprecated and replaced with 'set_app_options route.common.clock_topology' option.

**Description**

The 'set_app_options route.globalclock_topology' option will be deprecated and replaced with 'set_app_options route.commom.clock_topology' option.

**What Next**

Please update scripts to use the new option.

## ZRT-108

(warning) 'set_app_options route.global.comb_max_connections' option will be obsolete.

**Description**

The 'set_app_options route.global.comb_max_connections' option will be obsolete and not honored.

**What Next**

Please do not use the option in your scripts.

## ZRT-109

(error) Cannot map from non regular via %s.

**Description**

This error message indicates that the router does not support the via in the given via mapping.

## ZRT-110

(Warning) The global router sees the pin %s of net %s as blocked and might route the net through the blockages.

**Description**

The global router sees the pin as blocked in its modeling and might route the net through the blockages. The pin could be physically blocked due to blockages in the macro or a bad floorplan or due to conservative modeling of the global routing cell capacity. The pin could also be blocked due to incorrect min/max layer setting. If that's the case, the pin might be left unrouted.

Feedback

### What Next

View the pin in the GUI or run the *check_routability* command to see if the pin is physically blocked. If it is, fix the root cause and reroute the design. Also check if the pin layer is outside of min/max layer constraint setting.

### See Also

- check_routability

---

## ZRT-111

(error) The pin constraint on module %s pin %s is inconsistent with the constraint on module %s pin %s. Constraint will be ignored.

### Description

Two pin constraints are inconsistent. For example, a typical inconsistency would be when mod1 abuts to mod2 and abutting pins are placed at different locations or layers on the abutted edge. The program will discard the conflicting constraint and continue on.

### What Next

You should correct any inconsistent pin constraints; otherwise, you may get unexpected results.

---

## ZRT-112

(warning) Net %s could not be routed inside routing corridor %s.

### Description

The net could not be routed inside the specified routing corridor. This could be caused by the routing corridor not cover all pins or disconnected.

### What Next

Please run report_routing_corridors to see if the routing corridor is a legal one.

---

## ZRT-113

(warning) Net %s has unused floating user-enter shapes.

### Description

When the *connect_floating_shapes* option has a value of *false*, the net might have some unused user-enter shapes.

Feedback

**What Next**

If you want to connect the pins and all the user-enter shapes, use the *set_app_options* command to set the *route.common.connect_floating_shapes* common route option to *true* and rerun global routing.

---

## ZRT-114

(warning) The corridor setting of net %s will be ignored in incremental global routing.

**Description**

Corridors are used during initial global routing only. They should not be used during incremental global routing (*-reuse_existing_global_route true*) due to potential conflicts, such as when existing global routes are outside the corridor area. The global router ignores the corridor setting when the *-reuse_existing_global_route* option is enabled.

**What Next**

To prevent this message, remove the net routing corridor setting before running incremental global routing.

---

## ZRT-115

(Warning) The routing layer constraints of the %s net and the %s routing corridor conflict.

**Description**

The minimum or maximum routing layer setting of the routing corridor is not contained in the routing layer constraints of the net.

**What Next**

To prevent this message, adjust the minimum or maximum routing layer setting of the routing corridor or the net.

---

## ZRT-116

(Warning) The net %s is routed over blockages within the routing corridor %s.

**Description**

This warning occurs when the specified net is routed over blockages. The blockages can be macros, route guides, and so on.

**What Next**

To prevent this message, adjust the routing corridor to avoid blockages.

Feedback

## ZRT-117

(error) The location of the instance pin %s/%s is inconsistent with the location of the pin %s/%s on an abutted edge. The constraint is ignored.

**Description**

The tool issues this error when two block pin locations on an abutted edge are inconsistent. The tool discards the conflicting constraint and continues on.

**What Next**

You should correct any inconsistent pin placements; otherwise, you might get unexpected results.

## ZRT-118

(warning) 'set_app_options route.global.dpt_layer_utilization' option will be deprecated and replaced with 'set_app_options route.global.double_pattern_utilization_by_layer_name' option.

**Description**

The 'set_app_options route.global.dpt_layer_utilization' option will be deprecated and replaced with 'set_app_options route.global.double_pattern_utilization_by_layer_name' option.

**What Next**

Please update scripts to use the new option.

## ZRT-119

(warning) Setting double_pattern_utilization_by_layer_name %d on %s is invalid. Ignored.

**Description**

Setting double_pattern_utilization_by_layer_name on non-DPT layer is invalid and will be ignored.

**What Next**

Please check your tech file and scripts to make sure that you only set the DPT layer utilization on DPT layers.

## ZRT-120

(warn) Contact %s's %s layer enclosure does not satisfy legal dimension requirement.

Feedback

**Description**

The router depends on correct construction of contacts that satisfy the legal dimension rule. In this case, the contact's metal enclosure does not satisfy the legal dimension requirement and it may result in long routing run time.

**What Next**

Please modify the contact to satisfy the legal dimension requirement or remove the design rule from the technology file.

---

## ZRT-121

(warn) Not enough routing capacity for bus %s on side %d of module %s at the requested spacing of %d tracks between pins.

**Description**

There is not enough routing capacity for this bus on the selected edge with the requested pin spacing. This may be due to the size of the edge, or the number of other routes already directed through this edge. The router may have difficulty with this bus. The pin placer will attempt to place the bus on this side, but may not be able to maintain the requested spacing.

**What Next**

Set the "tracks between pins" to a smaller number. Another possible solution might be to guide the tool to place the bus on a longer or less congested edge.

---

## ZRT-122

(warn) There was an error processing physical bus %s.

**Description**

This physical bus may be empty or otherwise inconsistent.

**What Next**

Check the bus for consistency using "report_physical_bus". Use "remove_physcial_bus" to delete the bus if it is no longer needed.

---

## ZRT-123

(warn) Router ignores pin guide %s for intra module net %s.

Feedback

### Description

A pin guide was attached to the given net, but the net is entirely inside the module and therefore will not generate a pin on the module.

### What Next

Remove this net from the pin guide definition to eliminate this warning.

---

## ZRT-124

(warn) A pin guide for net %s is ignored as it implies a forbidden feedthrough on module %s.

### Description

A pin guide forces a feedthrough on the given module. However, feedthroughs are forbidden on this module, or on this net. Therefore this pin guide can not be implemented.

### What Next

Change the feedthrough settings to allow feedthroughs on this net or module. Altertavely, edit or delete the pin guide so that it does imply creation of this feedthrough.

---

## ZRT-125

(Warning) Router cannot access %s %s\n \ton net %s, forcing access.

### Description

The global router can NOT connect to this macro cell or IO pin using the pin access direction. This happens when the pin is totally blocked or when the pin is blocked in the assigned access direction. For instance, if a pin is located at the left edge of die area and has pin access direction from the left, the router will print this message. Or, in virtual flat routing mode, if a pin is located at edge of a block and has an access direction from outside the block, but needs to be connected to other pins inside the block, this message will be issued. In such cases, pin access direction cannot be honored and the router will connect to the pin without obeying the access direction.

### What Next

The pin may be blocked because it is on the edge of the die or block, because it is under an overlapping cell, or because it is blocked by some other obstruction. While exploration mode routing will still attempt to connect the pin, blocked macro pins are not expected and the floorplan should be adjusted to remove this condition.

Feedback

## ZRT-126

(Warning) Numerous standard cells have blocked pins. Assuming design is not legalized.

**Description**

The global router cannot access a number of standard cell pins because they are blocked, probably by overlapping standard cells. This is expected if the design has not been legalized.

**What Next**

If router is running in exploration mode and the placement is not legalized, such configurations are expected and you may ignore this message.

## ZRT-127

(Warning) The global router sees net %s having %d/%d blocked pins.

**Description**

The global router detects whether access to a pin is totally blocked. For example, during placement when the design is not legalized, a pin may be blocked by in-cell shapes and pin shapes of overlapping cells. Also, due to floorplan issues, sometimes a macro pin is completely embedded inside adjacent macros and cannot be accessed without going through routing blockages. The router will print a warning message for every net with blocked pins detected: how many pins are blocked among the total number of pins.

**What Next**

While the global router will still attempt to connect the pins, any floorplan issues causing blocked pins should be corrected to remove this condition.

## ZRT-128

(Warning) Total number of blocked macro pins according to global router modelling: %d. With blocked pins, global routing runtime can be higher.

**Description**

The global router detects whether access to a pin is totally blocked. For example, during placement when the design is not legalized, a pin may be blocked by in-cell shapes and pin shapes of overlapping cells. Also, due to floorplan issues, sometimes a macro pin is completely embedded inside adjacent macros and cannot be accessed without going through routing blockages. The router will print a warning message summarizing the total number of blocked macro pins in the design, if non-zero.

Feedback

### What Next

While the global router will still attempt to connect the pins, any floorplan issues causing blocked pins should be corrected to remove this condition.

---

## ZRT-129

(Warning) Total number of blocked standard cell pins according to global router modelling: %d. With blocked pins, global routing runtime can be higher.

### Description

The global router detects whether access to a pin is totally blocked. For example, during placement when the design is not legalized, a pin may be blocked by in-cell shapes and pin shapes of overlapping cells. Also, due to floorplan issues, sometimes a macro pin is completely embedded inside adjacent macros and cannot be accessed without going through routing blockages. The router will print a warning message summarizing the total number of blocked standard cell pins in the design, if non-zero.

### What Next

While the global router will still attempt to connect the pins, any floorplan issues causing blocked pins should be corrected to remove this condition.

---

## ZRT-130

(Warning) Port %s%s%s of net %s is blocked due to layer constraint settings. Routing will leave this net open.

### Description

The global router detects whether access to a port is totally blocked by layer constraint settings. For example, set the max_routing_layer to M7 in hard mode with number_of_vias_over_global_max_layer to 0, the M8 ports may be blocked by this layer constraint.

### What Next

Please check the layer constraint settings or set the net frozen if you intend to skip routing.

### See Also

- check_routability
- set_ignored_layers

Feedback

## ZRT-131

(Warning) Net %s is unroutable due to layer constraint settings. Routing will leave this net open.

### Description

The global router detects a net is totally unroutable due to layer constraint settings. For example, layer M3 is a frozen layer but a net has ports on M2 and M4, the net could be unroutable due to this layer constraint.

### What Next

Please check the layer constraint settings or set the net frozen if you intend to skip routing.

### See Also

- check_routability
- set_ignored_layers

## ZRT-132

(Error) Top port %s on net %s is at same location on the same layer as port %s on net %s inside multiple instantiated block (MIB) cells, shorted pins will be created

### Description

The global router detects top port terminal is inside a MIB instance while another top port terminal is at same location on other MIB instance, however, the top port direction is conflict (one is input and another one is output). Since the block pin will be placed at same location under the top level terminals, the pins will be shorted with each other on MIB reference block

### What Next

please check the top terminal location and layer.

## ZRT-133

(Warning) The net %s has port(s) outside the routing corridor %s.

### Description

This warning occurs when the specified net has port(s) outside the routing corridor.

### What Next

To prevent this message, please check if all the ports inside the routing corridor.

Feedback

**See Also**

- report_routing_corridors

---

## ZRT-134

(Error) Net %s is unroutable because pin %s/%s is not inside block %s. Routing will ignore this net.

**Description**

The global router detects that a net is unroutable because one or more of its pins is not in the intended block.

**What Next**

This condition can also be detected by report_placement -physical_hierarchy_violations, which should run clean before attempting virtual flat global routing. To fix the problem, either move the cell that is incorrectly placed, or edit the boundary of the block.

**See Also**

- report_placement

---

## ZRT-135

(Error) Failed to find reference block for instance %s

**Description**

This error message indicates that the router was not able to find the reference block for this cell. This may due to the linking error or reference design is missing. When router running in design planning hierarchinal routing mode, the routing results cannot be used to create feedthroughs or place_pins on the nets connected to this instance.

**What Next**

Please make sure the reference block is linked to the current design.

---

## ZRT-136

(Error) Coarse gCell capacity overflows. Global routing will proceed with this capacity bounded to maximum limit %u.

Feedback

**Description**

This Error message indicates that the capacity of this gCell overflows. This typically happens when the pitch size is too small comparing the minimum width and spacing requirement.

**What Next**

Please make sure the pitch size of design is within an appropriate range.

---

## ZRT-137

(Warning) If the placement is not legalized, the presence of blocked standard cell pins might be expected. If the placement is legalized, the number of blocked standard cell pins should be minimized by design entering routing to ensure routing quality. In addition, global routing reported blocked standard cell pins can be different from the set reported by check command because they use different mechanism.

**Description**

Warning message about the presence of standard cell blocked pins during placement. This is to clarify that blocked standard cell pins are expected in unlegalized designs.

**What Next**

While the global router will still attempt to connect the pins, any floorplan issues causing blocked pins should be corrected to remove this condition.

---

## ZRT-138

(info) %s layer %s will be routed with target utilization ratio of %d per cent.

**Description**

Information message about the target utilization ratio ZGR is going to use for a layer. Layers with double patterning rules will use a ratio of lower than 100 per cent unless specified by the user.

**What Next**

No action necessary.

---

## ZRT-139

(warning) custom width track(s) found while running route_global.

Feedback

### Description

Custom width track flow does not trigger in standalone global route; it only triggers with route auto or route group.

### What Next

Please delete custom width track(s) or re-run with either route_auto or route_group.

---

## ZRT-140

(error) Non-PG net %s has a %s shape with shape_use as shield_route. The shape's BBox is (%.3f, %.3f) (%.3f, %.3f). Changed the shape_use to detail_route.

### Description

A non-PG net's shape could not have shape_use attribute as shield_route. This illegal attribute might generate wrong data inside Zroute's infrastructure. It might corrupt Zroute's data and generate unexpected incorrect result of routing.

Zroute automatically changes the shape_use attribute from shield_route to detail_route and save it to database.

### What Next

User could run check_routability to get a report of shapes with illegal shape_use as shield_route.

---

## ZRT-141

(warn) The shield data checker detects: %s

### Description

An illegal or suspicious shield data is detected by the shield data checker. Please follow the description in the message. Not all issues detected by shield data checker is an error. Sometimes an issue is suspicious data rather than illegal data. The user has to be sure the data is correct before running any Zroute command.

### What Next

Please review and correct the issue specified in the description.

The user could run check_routability -check_shield true to get a report of shield data issues. The user could also use set_app_options -name route.common.check_shield -value true to enable shield data checker in all Zroute commands.

Feedback

## ZRT-142

(warn) PG net %s has a %s shape with shape_use detail_route and it is too long. Long part of the shape is located at (%.3f, %.3f) (%.3f, %.3f). The shape's BBox is (%.3f, %.3f) (%.3f, %.3f).

**Description**

In general, a PG wire with shape_use detail_route is a tie-off wire. However, a very long PG wire could be a part of PG mesh or rail. Please review and correct the shape_use of the shape reported. Please be noted, this check only detects suspicious ones, not all of them.

**What Next**

Review and correct the shape_use of the shape reported.

The user could run check_routability -check_shield true to get a report of shield data issues. The user could also use set_app_options -name route.common.check_shield -value true to enable shield data checker in all Zroute commands.

## ZRT-143

(warning) Secondary PG net %s has no power stripe, skip to route.

**Description**

Secondary PG net has no power stripe to connect.

**What Next**

Please add power stripe.

## ZRT-144

(warning) Option route.common.verbose_level is ON. This may increase the global router run time significantly.

**Description**

Option route.common.verbose_level may increase the router's run time significantly in commands where this warning is issued.

**What Next**

Disabling this option is recommended.

Feedback

## ZRT-145

(info) Shielding is up-to-date and re-shielding is not needed.

**Description**

Shielded nets are not modified during routing; therefore, there is no need to do re-shielding.

**What Next**

No action is required.

## ZRT-147

(warning) Failed to find reference port for pin %s

**Description**

This error message indicates that the router was not able to find the reference port for this pin. This maybe because cell instance is not linked correcty or the previously linked abstract view/design view/frame view is out-of-sync for this pin

**What Next**

Please check the flow to make sure that the linked views are in sync

## ZRT-148

(warning) Routing of %s could be slow

**Description**

Progress of this PG net seems to be slow. Possible reasons are:

1. Few or/and far away PG stripes to PG ports or shield shapes.
2. If it is a secondary PG net, maximum fanout limit could be too strict.
3. route.common.tie_off_mode is set to rail_only and rail for the net is not setup properly.

**What Next**

Please check PG structure setup for the net, or/and setting of number_of_secondary_pg_pin_connections or tie_off_mode.

## ZRT-149

(warn) macro cell %s is illegally placed inside block cell %s

Feedback

**Description**

This implies that there are macro cells that have been placed inside a block cell which doesn't contain those macro cells. Those cells will NOT being considered when calculate the channels during design planning global route

**What Next**

If this is NOT what you intend to do, please place the macro cells legally

---

## ZRT-150

(warn) top level port %s has no pin on net %s, feedthrough will not be created for this net

**Description**

Global route detects that some of the top level ports don't have pins or terminals. Router will skip those ports when routing connected nets and no feedthrough will be created on the connected nets

**What Next**

If this is NOT what you intend to do, please place the top level terminals first before running global route

---

## ZRT-151

(warning) clock comb/fishbone routing net %s has no mesh/stripe found, please create it or check layer constraint settings or check if mesh/stripe is blocked.

**Description**

comb/fishbone clock net has no mesh/stripe to connect.

**What Next**

please create clock mesh/stripe for the net or check layer constraint settings or check if mesh/stripe is blocked.

---

## ZRT-152

(Warning) Net %s has shape(s) %s that can't be connected due to layer constraint settings.

**Description**

The global router detects floating shapes that can't be routed to due to layer constraint settings. Global router will not connect to those route shapes.

Feedback

**What Next**

Please check the net's existing route shapes or/and layer constraint settings.

---

## ZRT-153

(warn) %s layer %s utilization ratio of %d might be too low.

**Description**

The utilization ratio specified by the user might be too conservative. Low utilization could lead to additional runtime spent on resolving congestion for the layer.

**What Next**

Check double_pattern_utilization_by_layer_name setting.

---

## ZRT-200

(error) Skipping track assignment because this design has no global routes.

**Description**

This message indicates that there are no global routes in this design, so the router cannot perform track assignment. This happens because either global routing is not correctly finished, or track assignment is already done.

**What Next**

First check whether track assignment was already completed. Then check whether global routing was completed before running track assignment.

---

## ZRT-201

(error) Skipping track assignment because this design has global route based via ladder.

**Description**

This message indicates that there are global route based via ladder in this design, so the router cannot perform track assignment. This happens because global routing was done with route.global.insert_gr_via_ladders set to true, and refresh_via_ladders hasn't been done.

**What Next**

Run refresh_via_ladders before track assignment.

Feedback

**See Also**

- route.global.insert_gr_via_ladders
- refresh_via_ladders

---

## ZRT-300

(warn) Antenna diodes cannot protect any pin.

**Description**

Under certain antenna calculation modes, diode cells do not offer any protection to violating pins.

**What Next**

Please check the antenna calculation mode. If diode cells are desired, set the antenna diode calculation mode appropriately.

---

## ZRT-301

(warn) Antenna diode insertion will be off.

**Description**

Antenna diode insertion is set on by the input options, but will be off in this run because diode insertion could not be initialized.

**What Next**

Please refer to the messages about problems in initializing diode insertion before this message. Fix those issues and rerun the routing command.

---

## ZRT-302

(warn) No antenna diodes found, or could not link them.

**Description**

Antenna diode insertion is set on by the input options, but no diode cells were found in the linked library.

**What Next**

Check the library to find diode cells.

Feedback

## ZRT-303

(warn) Failed to initialize legalizer for antenna diode insertion.

**Description**

Antenna diode insertion is set on by the input options, but the router could not initialize the legalizer for diode insertion.

**What Next**

Check the preceding messages from the legalizer for specific problems.

## ZRT-304

(info) Skip final DRC merging, too many DRCs.

**Description**

DRC merging reduces redundant DRCs by merging aligned and coincident DRCs. However, DRC merging is skipped if there are too many DRCs, as it might take too log.

**What Next**

No action necessary.

## ZRT-305

(info) Merged away %d aligned/redundant DRCs.

**Description**

DRCs can be merged if at least two opposing edges are aligned and they are of the same type, layer, and nets. For example, if two overlapping different-net-metal-spacing DRCs are between shapes of the same layer, from the same two nets, and have top and bottom edges aligned, then they will be merged to one DRC. Merging is done to give the user a more realistic number of DRCs, and to minimize the effect of the router's internal representation of the shapes on the DRC count.

**What Next**

No action necessary.

## ZRT-306

(info) Discarded %d drcs of internal-only type.

Feedback

**Description**

DRCs of internal-only type are not violations of technology rules and may be discarded.

**What Next**

No action necessary.

---

## ZRT-307

(info) Skip DR iteration %d for new DRC type to start.

**Description**

Some DRC analysis may be deferred until later global iterations. In that case, the router may skip iterations while the number of DRCs is zero, until all DRC analysis is on.

**What Next**

No action necessary.

---

## ZRT-308

(error) Design has global route wires. Please run track assignment before detailed route.

**Description**

Track assignment is required before detailed routing.

**What Next**

Run track assignment.

---

## ZRT-309

(warn) No antenna rules defined, Skip antenna analysis.

**Description**

Antenna rules must be defined in the library in order to do antenna analysis.

**What Next**

Define antenna rules in the library.

---

## ZRT-310

(error) Failed to store hierarchical antenna property.

Feedback

**Description**

The tool issues this message when an attempt to store hierarchical antenna property in top-level pins failed.

**What Next**

This is not expected behavior.

---

## ZRT-311

(warning) Skipping antenna analysis for net %s. The pin %s on cell %s does not have enough gate area information.

**Description**

One or more input pins of this net do not have the correct gate area values. So antenna analysis will not be done for this net.

Limit for number of ZRT-311 messages is set to 100 by default for readability of the log file. In order to print more or all the ZRT-311 messages, set this message limit to a very large number using set_message_info command.

**What Next**

Check and set correct gate area for all the input pins of the net on all the metal & cut layers. If gate area of some pins is not available, consider using app options route.detail.default_gate_size and route.detail.default_port_external_gate_size. Value set for these options will be used for pins without gate area information during antenna analysis.

---

## ZRT-312

(info) Detail Routing terminated early because DRCs were not converging: %s

**Description**

Detail Routing terminated early because DRCs were not converging.

**What Next**

No action necessary.

---

## ZRT-313

(warning) Ignoring the user defined partition as it is too big.

Feedback

**Description**

There is a limit on the size of the user defined partition as it can cause huge memory and run time overhead. As a rough guide, the area of the partition should be less than 600 pitches by 600 pitches on metal 1 (i.e., 600 tracks by 600 tracks).

**What Next**

Please use a smaller partition size.

---

## ZRT-314

(warning) Skipping diode insertion for port %s of cell instance %s as enough information is not available.

**Description**

The diode insertion values specified for this port are incorrect or a port and cell with that name does not exist.

**What Next**

Check the input specification for this port.

---

## ZRT-315

(info) Inserted/reused %d out of the specified %d diodes for port %s(cell %s).

**Description**

This message is for success rate of user-specified diode insertion for each port/cell. It is possible that diodes may not be inserted or reused for ports.

Diode insertion may fail for one of the following: - there are no legal locations in the region new diodes are attempted. or - there is no DRC clean solution to connect diode pin to the specified port. - specified diode(s) do not satisfy MV based constraints , or there is no voltage set on supply nets. ZRT-331 messages are issued for this condition.

**What Next**

If diodes are not inserted for a port, - Consider a modified diode distance, choice or number of diodes and rerun the command. - Check and reduce placement utilization in the area where diodes are to be inserted.

---

## ZRT-316

(warn) Zero equivalent gate area for net %s, port %s on layer %s for gate class %d and diode mode %d. The antenna analysis may not be correct.

Feedback

**Description**

For the case connected to diode, the equivalent gate area is a function of the diode ratio, diode protect and gate area. It is zero for this case. This may result in incorrect antenna analysis.

**What Next**

Check the antenna rules, gate area of the cells and the diodes in the design for non-zero diode protect.

---

## ZRT-317

(warn) Ignoring diode_preference setting as diode_insertion_mode is not set to new_and_spare.

**Description**

The diode_preference chooses preference between spare diode and new diodes. It is required that user allows new diodes and spare diodes for the diode preference to be effective. One of them is currently disabled with diode_insertion_mode.

**What Next**

Set the diode insertion mode to new_and_spare if diode preference should be honored.

---

## ZRT-318

(error) "%s" is not a supported route type.

**Description**

remove_shorts supports only the following route types:

user_enter, signal_route_detail, pg, clk, bus, shield_fixed, shield_dynamic

Other route types are not supported.

**What Next**

Specify only the supported route types.

---

## ZRT-319

(error) Only one region can be specified for short removal.

**Description**

Currently, only one region can be specified for remove_shorts to perform short removal.

Feedback

**What Next**

Specify only one region to perform short removal.

---

## ZRT-320

(error) Design still contains global route wires; cannot continue.

**Description**

remove_shorts can only be used after detail routing of the design. It cannot work when the design still contains global route wires.

**What Next**

Perform detail routing on the design before calling the remove_shorts command.

---

## ZRT-321

(info) Starting from the %s release, the default value of option %s of command %s will be %s.

**Description**

There is a change in default value.

**What Next**

No action necessary.

---

## ZRT-322

(warning) Ignoring diode mode 13 setting. Use diode mode 14 instead.

**Description**

Diode mode 13 is not supported as it is a subset of diode mode 14.

**What Next**

Modify antenna rule definition with diode mode 14 and re-run the routing commands.

---

## ZRT-323

(info) Filtered %d drcs of internal-only type.

Feedback

**Description**

DRCs of internal-only type are not violations of technology rules and are filtered for reporting purpose.

**What Next**

No action necessary.

---

## ZRT-324

(warn) remove_redundant_shapes cannot change net '%s' because it is open or has DRC violations.

**Description**

The tool issues this warning when remove_redundant_shapes is used to change open nets or nets with DRC violations.

**What Next**

Verify that the net is fully connected and free of DRC violations.

---

## ZRT-325

(warning) Skipping antenna analysis for net %s as it is open.

**Description**

Antenna analysis cannot be accurate when the net is open. So this net is skipped for antenna analysis.

**What Next**

Connect the net with incremental routing.

---

## ZRT-326

(error) partition bloating is too large to generate a valid metal sbin

**Description**

The metal bloat partion is huge, and the size of the sbin is larger than max int value

**What Next**

Please set "set_route_zrt_detail_options -continue_after_large_design_rule_value_error false", and run detail router to see the message.

Feedback

## ZRT-327

(Information) End Off Preferred Grid violations will be ignored since preferred grid extensions have been trimmed in this design.

**Description**

End Off Preferred Grid extensions on fixed shapes have been trimmed with the standalone command add_metal_extension_cut_metal -trim_end_off_preferred_grid_extension true.

**What Next**

Please add End Off Preferred Grid extensions on fixed shapes with add_metal_extension_cut_metal or route.detail.enable_end_off_preferred_grid_patching_on_fixed_shapes to allow for a complete DRC analysis.

## ZRT-328

(Warning) router fails to find preferred and bufferable voltage areas for net %s, global router will allow routing to go through none-default voltage areas and pin placement will allow pins on this net to be placed within none-default voltage areas

**Description**

Please check related supply net for the pins connnected to this net

## ZRT-329

(warning) Ignoring antenna mode %d as it is defined with diode mode %d.

**Description**

Floating metal based diode mode is to be defined with accumulative area type antenna modes only. They are antenna modes 3 or 6. If this diode mode is defined with any other antenna mode, that antenna mode based antenna rule will be ignored.

**What Next**

Please check and correct the antenna rule definitions.

## ZRT-330

(warn) Zero equivalent gate area for net %s, port %s on layer %s for gate class %d well type %d and diode mode %d. The antenna analysis may not be correct.

Feedback

**Description**

For the case connected to diode, the equivalent gate area is a function of the diode ratio, diode protect and gate area. It is zero for this case. This may result in incorrect antenna analysis.

**What Next**

Check the antenna rules, gate area of the cells and the diodes in the design for non-zero diode protection.

---

## ZRT-331

(warning) Skipping diode insertion for port %s of cell %s. There are no legal diodes available for the port that satisfy MV requirements, or there is no voltage set on supply nets.

**Description**

Diodes provided have the following issues: - they do not satisfy MV requirements for the specified port or - there is no voltage set on supply nets (please check OPT-006 messages). So diodes will not be used to fix antenna violations of the port.

**What Next**

Please check

1. antenna_diode_type library cell attribute of diodes 2. is_diode library pin attribute of diodes 3. The reference libraries have multiple diode cells with different voltage 4. set_voltage is to be set for each supply net

Selects the proper diode cell by using the following criteria: (a) If the load cell's supply voltage is lower than driver cell’s supply voltage, use a ground diode (b) If the load cell’s power supply is switchable and less always on than the driver cell‘s power supply, use a ground diode (c) If the load cell’s ground supply is switchable and less always on than the driver cell‘s ground supply, use a power diode (d) If there are conflicting conditions, do not insert a diode (e) For other cases, use any diode type (power, ground, or power-ground)

If there are more questions about this message, contact MV team for which requirements are not satisfied for the diode port.

---

## ZRT-332

(warning) Skipping antenna analysis for net %s. The pin %s on cell %s does not have enough nwell area information.

Feedback

**Description**

One or more pins of this net do not have the correct nwell area values. So antenna analysis will not be done for this net for diode mode 21.

Limit for number of ZRT-332 messages is set to 100 by default for readability of the log file. In order to print more or all the ZRT-332 messages, set this message limit to a very large number using set_message_info command.

**What Next**

Check and set correct nwell area for all the pins of the net on all the metal & cut layers.

---

## ZRT-333

(warning) Ignoring port protection diode for port %s as no diode with name %s found.

**Description**

Diode libcell name specified for port protection is not valid. Please correct the input and rerun routing command.

**What Next**

Check if libcell name specified is correct. Library name of the diode should not be specified in the input.

---

## ZRT-334

(warning) Ignoring port-protection diodes for port %s of net %s. This net has no no input ports.

**Description**

Net of the port(top-level pin) specified does not have any input ports. Port protection diodes are inserted only on nets with input ports.

**What Next**

Check and correct the port protection diode setting for the port.

---

## ZRT-335

(warning) Ignoring port-protection diodes for port %s of net %s. This net has no %s ports.

**Description**

Net of the port(top-level pin) specified does not have the needed other lib-cell ports to insert diodes. If port direction is input, diodes are inserted near the net's lib-cell pins with

Feedback

direction input. If port direction is output, diodes are inserted near the net's lib-cell pins with direction output.

**What Next**

Check and correct the port protection diode setting for the port.

---

## ZRT-336

(info) Deleted redundant diode %s that had pin connected to net %s.

**Description**

This diode does not help with any antenna violation for the connected net. So it is deleted.

**What Next**

Review the design if there are remaining antenna violations for the net.

---

## ZRT-337

(warning) Port protection diode %s does not connect within layer for top level port %s.

**Description**

Port protection diode cannot protect the port as it is not connected within the metal layer of the top level port.

**What Next**

Routing is to be modified so connection from the diode to top level port is within the layer of top level port. Set app option route.detail.shift_diode_locations_for_port_protection to TRUE and run detail route. With this app option setting, diode gets shifted and connected within layer of top level port.

---

## ZRT-338

(warning) Skipping antenna fixing for net %s due to large pin count of %d.

**Description**

The net has a large fanout. Antenna fixing is skipped to save runtime. Once the net is optimized later in the flow, antenna analysis and fixing will be done for the optimized net.

Power and ground nets are excluded for the high fanout check.

Feedback

**What Next**

If antenna fixing is needed for the high fanout-net nets, set app option route.detail.max_antenna_fixing_pin_count_threshold to -1 and rerun the routing commands.

---

## ZRT-339

(warn) Skipping diode insertion for port %s of cell %s based on user constraint allow_diode_insertion.

**Description**

User set constraint allow_diode_insertion to false on a top level port or lib-cell pin. If there are any antenna violations on this port, diode insertion will not be used for the fix. Instead layer hopping will be attempted.

**What Next**

Check if it is intended to skip didoe insertion for this port or lib-pin.

---

## ZRT-340

(warning) Skipping antenna analysis for pin %s of libCell %s.

**Description**

User specified lib cell pins to be skipped for antenna using app option route.detail.libcell_pin_names_to_skip_antenna. For all instances of the lib cell in top design, specified pin is skipped during antenna analysis and fixing by Zroute.

**What Next**

Check the setting of app option route.detail.libcell_pin_names_to_skip_antenna.

---

## ZRT-400

(error) Cannot open timing file '%s'.

**Description**

The reasons for this could be one of the following.

- The file does not exist.
- You do not have the required permissions to read it.

Feedback

**What Next**

Please make sure the timing file exists and that you have the required permissions to read it.

---

## ZRT-401

(error) Corrupted or incomplete timing file.

**Description**

The reasons for this could be one of the following.

- There was not enough disk space when the timing file was saved.
- The timing file was saved using a different executable from the one that is being used to read it.

**What Next**

Please re-run after making sure that there is enough disk space available.

---

## ZRT-402

(error) Cannot find the user specified timing file '%s'.

**Description**

The reasons for this could be one of the following.

- The file does not exist or the full path to the file was not specified.
- You do not have the required permissions to read it.

**What Next**

Please make sure the full path to the file is specified and that you have the required permissions to read it.

---

## ZRT-403

(warn) Hidden %s parameter %s of %s is obsolete in router.

**Description**

This debug/hidden parameter has been set or has a different value than the default value, but the parameter to option translator does not support translation of this parameter.

Feedback

**What Next**

If possible, please remove this parameter from the script or reset the value if it's coming from the cell parameter setting.

---

## ZRT-404

(warn) Unable to translate %s of %s because %s -%s does not support value %d.

**Description**

The parameter is set to a value that is not supported by the option and thus can not be translated.

**What Next**

Please do not use the unsupported value with the parameter.

---

## ZRT-405

(warn) Unable to translate %s -%s because %s of %s does not support mode %s.

**Description**

The option is set to an additional mode that is not supported by the parameter and thus can not be translated.

**What Next**

Please do not set the option to this mode if the translation of this option to parameter is required.

---

## ZRT-406

(warn) Unable to translate %s to %s because %s.

**Description**

The translator cannot perform the translation because there is no exact correspondence between the option and the parameter.

**What Next**

Please review the setting of the option or the parameter.

---

## ZRT-407

(error) Failed in separate process.

Feedback

**Description**

This error message indicates that the command failed in a separate process. The reasons for this could be one of the following:

- A crash occurred in the separate process flow.
- No disk space is available.

**What Next**

Verify that there is sufficient disk space available.

---

## ZRT-408

(error) Using -nets option with -antenna, -open_net, and -voltage_area set to false.

**Description**

This error message indicates that the command check_routes failed. The option "-nets" is specified with -antenna, -open_net, and -voltage_area options set to false.

**What Next**

1) Specify one or more of these three options in this Tcl command. or 2) Remove the "-nets" option and rerun the command.

---

## ZRT-409

(warn) %s parameter %s is not supported by router.

**Description**

The translator cannot perform the translation because the parameter is currently not supported by router.

**What Next**

Please review the setting of the parameter.

---

## ZRT-410

(info) %s parameter %s is obsolete in router.

**Description**

The translator cannot perform the translation because the parameter is obsolete in router. Most likely router is already handling the parameter automatically and there is no need for an user option, or the parameter is useless in router.

Feedback

## ZRT-411

(warn) %s parameter %s is supported only in the technology file.

**Description**

The tool issues this warning when you specify a routing option that is supported only in the technology file.

**What Next**

Define the design rule criteria in the technology file.

## ZRT-412

(warn) %s parameter %s cannot be translated exactly to corresponding router option.

**Description**

Router has a similar option, but the approach it takes is different and does not match the parameter with one to one correspondence.

**What Next**

Please set the option manually and refer to the application notes for more details.

## ZRT-413

(error) Invalid coordinates for area routing.

**Description**

The router cannot perform area routing because the specified coordinates are invalid.

**What Next**

Please use the correct format to specify the coordinates.

## ZRT-414

(error) Abnormal exit of child process

**Description**

Child process crashed.

**What Next**

This is not expected behavior.

Feedback

## ZRT-415

(warn) The number of threads for this run is reduced to %d.

**Description**

Based on the maximum number of threads set,an attempt is made to checkout the required number of license keys of each type.However all the required keys are not available.Based on available number of keys of each type,a reduced number of threads is calculated and used for this run.

**What Next**

Please check if license keys of each type are available for this run.

## ZRT-416

(warning) %d %s cell PG pins connected to net NULL and not PG.

**Description**

This warning message indicates that the filler cell removal command detects PG pins of some cells in the design connected to NULL net. Short violations between the PG pins connected to NULL net and PG pins connected to valid PG nets will be detected during the command, and filler cells with such violations will be removed.

**What Next**

Please run the connect_pg_net command to ensure valid PG connecttions to PG pins.

## ZRT-417

(warning) Deleted instance %s is not a standard filler cell

**Description**

This warning message indicates that the filler cell removal command detects a violation on a cell which name matches a given name pattern, but is not classified as a standard filler cell. Filler cells are standard cells without any connections to signal nets. Filler cells that were inserted with create_stdcell_fillers command have a prefix of "xofiller".

**What Next**

Please check the given name pattern if it is indeed intended for filler cell deletion. You may use a default name pattern if only filler cells with a prefix of "xofiller" were intended for deletion.

Feedback

## ZRT-418

(warning) Instance %s which is connected to signal nets is not deleted.

**Description**

This warning message indicates that the filler cell removal command detects a violation on a cell which name matches a given name pattern, but has connections to signal nets. Such cells can not be removed.

**What Next**

Please check the given name pattern if it is indeed intended for filler cell deletion.

## ZRT-419

(warning) Cell %s has internal violations

**Description**

This warning message indicates that the filler cell removal command detects a cell which has violation on objects inside the cell. Such violation is ignored for cell removal purpose.

**What Next**

Please check the filler library cell for violations.

## ZRT-420

(error) It is not recommended to use classic router commands after router commands.

**Description**

Routing was done with router on the current design. Now an attempt is made to run commands of classic router. This is not recommended because the DRC analysis and routing graph interpretations are different between the two routers.

**What Next**

Continue to run all the routing commands in router. If the intention to use the classic router for the flow, delete all the existing routes and start the flow with classic router.

## ZRT-421

(error) Cannot open file '%s' for '%s'.

Feedback

**Description**

The file cannot be opened.

**What Next**

Please make sure you have the required permissions on the directory.

---

## ZRT-422

(warn) %s option "%s" will be obsolete in %s release. %s.

**Description**

The option specified will be obsoleted in future release.

**What Next**

Please define the design rule in the technology file.

---

## ZRT-423

(warn) %s parameter %s is supported by router by command line option.

**Description**

The parameter is supported in router as command line option.

**What Next**

Please use command line option instead.

---

## ZRT-424

(warning) "-max_number_of_threads" option is deprecated

**Description**

The "-max_number_of_threads" option is being phased out, in favor of using the "-max_cores option" for the "set_host_options" command.

**What Next**

Please update scripts to use "set_host_options -max_cores" instead.

**See Also**

- set_host_options

Feedback

## ZRT-425

(warning) The "set_host_options -max_cores" option will be honored instead of the "set_app_options route.common.max_number_of_threads" option.

### Description

The tool issues this warning when you specify both the recommended *set_host_options -max_cores* command and the deprecated *set_app_options route.common.max_number_of_threads* command.

Because the *set_app_options route.common.max_number_of_threads* command is being phased out, the *set_host_options -max_cores* setting is used and the *set_app_options route.common.max_number_of_threads* setting is ignored.

### What Next

Update your scripts to use the *set_host_options -max_cores* command instead of the *set_app_options route.common.max_number_of_threads* command.

### See Also

- set_host_options

## ZRT-426

(error) This error message indicates that invalid route type is specified.

### Description

The valid values for the route type are fixed_route and nonfixed_route.

### What Next

Check the command syntax from the man page and use the correct value.

## ZRT-427

(warning) Net %s was modified, need to be reshielded.

### Description

The indicated shielded net was modified outside of router environment. It will be automatically unshielded by router. If option is set: set_app_options route.common.reshield_modified_nets reshield it will be automatically reshielded at the end of signal routing.

Feedback

### What Next

To leave shielding of modified nets intact, you should set: set_app_options route.common.reshield_modified_nets off To delete shielding of modified nets, you should set: set_app_options route.common.reshield_modified_nets unshield To reshield the modified nets, please set: set_app_options route.common.reshield_modified_nets reshield or run create_shield command.

---

## ZRT-428

(warning) Net %s is a reserved channel net, but feedthroughs are permitted on this net.

### Description

The net is designated to be routed in "reserved channels", but it is also permitted to create feedthroughs on blocks. Therefore the net may not be routed through the channels as expected. To guard against the possibility of very large log files, the number of these warnings is limited to fifty, so it is possible that not all problem nets are reported.

### What Next

If you want this net to be routed in channel areas as much as possible, you should set it to prohibit feedthroughs using the "set_individual_pin_constraints -nets <net_names>" command with the "-allow_feedthroughs false" argument.

---

## ZRT-429

(warning) One of avoid metal area (AMA) candidate set of lithography hotspot %s does not overlap with any changeable wire/via.

### Description

None of the avoid metal areas of the AMA candidate set overlap with any changeable wire or via. Since an AMA set will be valid for router only if at least one of its avoid metal areas overlaps with router changeable wire or via. This AMA candidate set will not have any effect in router and will not be output as an AMA set in the output file.

### What Next

You should check if the hotspot marker was generated correctly, since the AMA candidate set are generated based on the hotspot marker. For example, a bridge hotspot should have at least two polygons nearby, otherwise there is nothing to be "bridged". However, if one of the polygons is an unchangeable shape, like pin or obstacle shapes, while the other one is a changeable wire or via, then you can ignore this warning since router still have chance to use the other AMA set to fix the hotspot.

Feedback

## ZRT-430

(warning) All of avoid metal area (AMA) candidate set of lithography hotspot %s do not overlap with any changeable wire/via.

### Description

None of the avoid metal areas of all the AMA candidate sets overlaps with any changeable wire or via. An AMA set will be valid for router only if at least one of its avoid metal areas overlaps with router changeable wire or via. No AMA candidate set will be output for this hotspot in the output file. Thus router can not fix this hotspot by rip-up and reroute.

### What Next

You should check if the hotspot marker was generated correctly, since the AMA candidate set are generated based on the hotspot marker. You should also check if the settings in config_file_name are set to proper values, especially the DefaultSpacing field. A fake or false alarm hotspot marker often causes this warning.

## ZRT-431

(warning) Found conflict constraints between block %s and voltage area %s, voltage area constraints will be ignored.

### Description

This message is issued when you have a block boundary coincident with voltage area boundary and the feethrough constraints setting are conflict. When those constraints are conflict, block related pin constraints win and voltage area feedthrough constraints will be ignored.

### What Next

Please use report_block_pin_constraints and report_fp_voltage_area_constraints to check the constraint setting and examine the confliction.

## ZRT-432

(Warning) found conflict feedthrough constraint setting for voltage area and plangroup on net %s, voltage area based constraints will be ignored

### Description

This message is issued when you have feethrough constraints setting conflict from pin constraints and voltage area constraints on a net. When those constraints are conflict, plangroup related pin constraints win and voltage area feedthrough constraints will be ignored.

Feedback

### What Next

Please use report_fp_pin_constraints and report_fp_voltage_area_constraints to check the constraint setting and examine the confliction.

---

## ZRT-433

(Warning) Router cannot access %s %s\n \ton net %s, forcing access.

### Description

The global router can NOT connect to this pin using the pin access direction. This happens when the pin is totally blocked or when the pin is blocked in the assigned access direction. For instance, if a pin is located at the left edge of die area and has pin access direction from the left, the router will print this message. Or, in virtual flat routing mode, if a pin is located at edge of a block and has an access direction from outside the blcok, but needs to be connected to other pins inside the block, this message will be issued. In such cases, pin access direction cannot be honored and the router will connect to the pin without obeying the access direction.

### What Next

The pin may be blocked. If router is running in exploration mode and the placement is not legalized, such configurations are expected and you may ignore this message. Otherwise, please check to see if the pin is blocked and adjust the placement if necessary. If the pin is not blocked, please check if the pin can be connected using pin access direction.

---

## ZRT-434

(warning) reshield_rerouted_nets option is deprecated

### Description

The -reshield_rerouted_nets option is being phased out, in favor of using the 'set_app_options route.common.reshield_modified_nets'.

### What Next

Please update scripts to use 'set_app_options route.common.reshield_modified_nets' instead of using 'set_app_options route.detail.reshield_rerouted_nets'.

---

## ZRT-435

(warning) off_grid_routing_mode option is deprecated

Feedback

**Description**

The -off_grid_routing_mode option is being phased out, in favor of using layer onGrid parameter in technology file.

**What Next**

Please update scripts to remove 'set_app_options route.common.off_grid_routing_mode'. Instead, define onGrid parameter in the Layer section of technology file.

---

## ZRT-436

(error) Could not find plan group or soft macro for db Id %d.

**Description**

An object ID in the internal pdc constraint structure could not be located in the database. This is an internal error.

**What Next**

This is not expected behavior.

---

## ZRT-437

(warning) Could not find port named %s specified in the physical pin constraints.

**Description**

A port name specifed in the physical pin constraint file could not be matched in the current database. This constraint will be ignored.

**What Next**

Fix the port names in the physical pin constraint file, or delete this constraint.

---

## ZRT-438

(error) router is not allowed to run in XP mode.

**Description**

router is not licensed in XP mode.

**What Next**

Please check your license.

Feedback

## ZRT-439

(warning) Cannot route net %s (dbId %lu).

### Description

The global router has failed to route a net. This may happen when the router cannot find a legal routing solution for the net. Normally this is caused by inconsistent data or constraint settings. Here are potential reasons that can cause this message:

a). If feedthroughs are prohibited in a fully abutted designs, if a net cannot be routed without generating an illegal feedthrough this message will be issued.

b). This message may also due to a placement violation. For instance, if a cell that logically belongs to a block has been placed outside of its block or a top terminal is placed inside an unconnected block which has feedthrough disallowed for the net, this message will be issued because the router can not route to those pins/terminals.

c). In addition, if the net is clock net, by default, the router will not create feedthrough routing. User can control the clock net feedthrough routing by app option "plan.pins.exclude_clocks_from_feedthroughs" or use command "set_individual_pin_constraints".

d). There are fixed block pins which has wrong pin access direction.

e). The connected pins of nets are on different layers which are restricted due to the net's min/max layer connection settings (from app options route.common.global_min_layer_mode/route.common.global_max_layer_mode or command set_ignored_layers).

### What Next

Check to see if contraint settings are correct or if the placement has any violations. Try running report_placement -physical_hierarchy_violations. Or check if the net is clock net and feedthrough is disabled. Also check if there are fixed block pins with wrong access direction. Also check the routing app option settings (route.common.global_min_layer_mode/route.common.global_max_layer_mode) or layer settings from command set_ignored_layers.

### See Also

- route_global

## ZRT-440

(error) Illegal RDB file input or format.

Feedback

**Description**

RDB file is the input file to the RDB2RGF translator, which translate the route guidance from RDB format to RGF format that router takes. If the input file does not comply with the RDB format, the translator can not produce the correct RGF file and will abort the translation.

**What Next**

You should check if the format of the RDB file is correct. The possible error in the RDB file could be: empty or wrong (ther must be at least two different values for horizontal or vetical values) vertices of a hotspot; some necessary fields defined in the config file are missing, ..., etc. Please examine the RDB file and correct the incorrect place(s).

---

## ZRT-441

(warning) route design in floorplan mode, floorplan global routing results should NOT be used for track assignment or detail route

**Description**

Floorplan global routing is intended to be used in floorplan exploration and design planning, it is NOT supported to run track assignment or detail route on floorplan global route results

**What Next**

Please don't use floorplan global route results for track assignment or detail route

---

## ZRT-442

(error) exploration global routing results can not be used for track assignment or detail route, set -stop_after_global_route to true

**Description**

Exploration router global routing is mainly used for floorplan exploration and design planning, it is NOT supported to run track assignment or detail route on exploration global route results

**What Next**

Please use -stop_after_global_route true in exploration mode

Feedback

## ZRT-443

(error) It is not recommended to use the result from -exploration global routing. Please re-run global route.

**Description**

The -exploration mode global route is used for design planning purpose and the result should not be used for subsequent routing steps (track assignment, detail routing).

**What Next**

Please re-run global route without the -exploration option.

## ZRT-444

(info) Using %d threads for routing.

**Description**

This message reports the number of threads used in multi-threaded routing.

**What Next**

No action is required.

**See Also**

- set_host_options

## ZRT-445

(warning) timing driven mode is NOT supported in exploration routing mode, design will be routed without timing driven

**Description**

Exploration router global routing is mainly used for floorplan exploration and design planning, it is NOT supported to run global route in both timing driven and exploration mode

**What Next**

Please use either exploration or timing driven mode, but NOT both

Feedback

## ZRT-446

(warning) cross talk driven mode is NOT supported in exploration routing mode, design will be routed without cross talk driven

**Description**

Exploration router global routing is mainly used for floorplan exploration and design planning, it is NOT supported to run global route in both cross talk driven and exploration mode

**What Next**

Please use either exploration or cross talk driven mode, but NOT both

## ZRT-447

(warning) option -plan_group_aware_quality will be obsolete in 2010.12 release, please use -exploration option for route_global or route_group

**Description**

Option -plan_group_aware_quality will be obsolete in 2010.12 release, please use "-exploration true | false" to turn on/off exploration mode

**What Next**

Please switch to use -exploration option for route_global/route_group commands. For more information on -exploration option, please see the man page of route_global/route_group commands.

## ZRT-448

(warning) "place_fp_pins -block_level -effort high" does not support plan group aware routing. Setting "-plan_group_aware off".

**Description**

"place_fp_pins -block_level -effort high" uses the global router to determine blcok level pin positions. The router cannot observe plan group constraints and block level constraints concurrently.

**What Next**

Use the setting "set_app_options route.common.plan_group_aware off" when doing block level pin placement.

Feedback

## ZRT-449

(warning) Timing-driven mode is NOT supported in hierarchical design planning; design will be routed without timing-driven constraints.

### Description

Timing-driven routing is not supported in hierarchical design planning (pin placement) mode. Timing-driven routing constraints will be ignored.

### What Next

If you are performing global routing on a design that contains plan groups, you can resolve this warning by using the *set_app_options* command to set the *route.common.plan_group_aware* option to *off*.

## ZRT-450

(warning) Crosstalk-driven mode is NOT supported in hierarchical design planning; design will be routed without crosstalk constraints.

### Description

Crosstalk constraints are not supported in hierarchical design planning (pin placement) mode. Crosstalk constraints will be ignored.

### What Next

If you are performing global routing on a design that contains plan groups, you can resolve this warning by using the *set_app_options* command to set the *route.common.plan_group_aware* option to *off*.

## ZRT-451

(info) The application variable "max_parallel_computations" is too large. It will not override "set_host_options -max_cores".

### Description

The application variable "max_parallel_computations" is set to be larger than the option in "set_host_options -max_cores". Router will use "set_host_options -max_cores" option for routing.

"max_parallel_computations" should be set to a value between 0 and the "set_host_options -max_cores" option value in order to override the "set_host_options -max_cores" option. Otherwise, "set_host_options -max_cores" will continue to be used for multiple computations.

Feedback

### What Next

Reduce "max_parallel_computations" to a value no greater than the "set_host_options -max_cores" option value.

### See Also

- set_host_options

---

## ZRT-452

(info) The application variable "max_parallel_computations" overrides the "set_host_options -max_cores" option.

### Description

The application variable "max_parallel_computations" is set to override the option in "set_host_options -max_cores". Router will use "max_parallel_computations" value for multiple computations.

### What Next

No action is required.

### See Also

- set_host_options

---

## ZRT-453

(error) Value %s specified for option %s is not in correct format.

### Description

Setting for the option will be ignored due to incorrect format.

### What Next

Please see man page for correct specification.

---

## ZRT-454

(warn) Voltage area name %s specified for option -voltage_area_weight does not exist.

### Description

Weight setting for this voltage area will be ignored.

Feedback

**What Next**

Please check for correct voltage area name.

---

## ZRT-455

(error) Via ladder constraint %s is used on pin but is not defined in tech file.

**Description**

A via ladder ViaRule must be defined in the tech file before the constraint is set on a pin.

**What Next**

Create a corresponding ViaRule in the tech file or remove this constraint from the pin, then try again.

---

## ZRT-456

(error) %s. No ContactCode uses this cut.

**Description**

A via ladder has been specified to use this cut, but the tech file does not define a via that uses the cut. The via ladder is therefore invalid.

**What Next**

Modify the tech file so that any cut used in a via ladder is a cut that is used in at least one via, then try again.

---

## ZRT-457

(error) Too many via ladders listed in constraint for pin. No more than %d are allowed.

**Description**

The list of via ladder preferences for the pin contains too many entries.

**What Next**

Reduce the number of entries in the via ladder constraint and try again.

---

## ZRT-458

(error) %s. Cuts are required to be on increasing adjacent layers.

Feedback

**Description**

A via ladder has been specified with illegal non-adjacent cuts. Cuts in a via ladder must increase by 1 cut layer for each level of the via ladder.

**What Next**

Modify the tech file so that the via ladder uses adjacent layer-increasing cuts on each level and try again.

---

## ZRT-459

(error) Failed to read ViaRule %s. %s.

**Description**

A via ladder has been specified with incomplete or illegal settings. This can happen if a non-existent layer is given, or if the number of entries in each field do not match the specified table size.

**What Next**

Modify the tech file so that the ViaRule is legitimate, then try again.

---

## ZRT-460

(warn) Net %s has a pin guide AND a topological constraint. The toptological constraint takes precedence.

**Description**

In the case of multiple constraint types for one net, topological constraints take precedence over pin guides. You cannot mix constraint types on a single net.

**What Next**

Either delete the topological constraint, or delete the pin guide.

---

## ZRT-461

(error) The design contains plan group pin shapes, which are a floor planning feature. Track assignment and detailed routing are not supported for plan group pins.

**Description**

Plan group pin shapes are floorplanning objects and are not understood by track assignment and detailed routing. You have likely created them using "place_fp_pins" or

Feedback

"create_fp_pins". Normally, you should run "commit_fp_plan_groups" to convert your plan groups into soft macros with real physical pins before doing detailed route.

**What Next**

Run commit_fp_plan_groups before doing physical routing. It is also possible to rerun your command after turning off the plan_group_aware flag using "set_app_options -global {route.common.plan_group_aware off}".

---

## ZRT-462

(info) Coordinates are a guide for verification area, errors may be flagged outside of them.

**Description**

The rectangles specified by the -coordinates option in the check_routes command is only a guide for the area where the command performs the DRC check. The actual area checked by the command may exceed the specified rectangles. As a result, DRC errors maybe flagged outside the boundary of the specified rectangles.

**What Next**

No action is required.

**See Also**

- check_routes

---

## ZRT-463

(info) Certain net based DRC checks are not performed with area based DRC check.

**Description**

When the -coordinates option is specified in the check_routes command. The following net based DRCs are not checked:

1. Open net
2. Antenna
3. Tie to rail; Tie to rail directly 4. Secondary PG pin directly connection; Secondary PG pin maximum connections
4. Comb routing directly connection; Comb routing maximum connections; Comb routing daisy-chain distance

Feedback

### What Next

No action is required.

### See Also

- check_routes

---

## ZRT-464

(info) -coordinates option is specified with the command check_routes. -nets, -open_net, -report_all_open_nets, -voltage_area and -antenna options will be ignored.

### Description

When the -coordinates option is specified in the check_routes command. The net based DRCs are not checked. Net related options: -nets, -open_net, -report_all_open_nets, -voltage_area and -antenna are also ignored.

### What Next

No action is required.

---

## ZRT-465

(info) Complete DRC check is recommended after area based DRC checks are complete.

### Description

Area based DRC check does not check net based DRCs. To obtain a full DRC report, please run a complete DRC check (check_routes without -coordinate option).

### What Next

No action is required.

---

## ZRT-466

(error) -drc is turned off when -coordinates is specified.

### Description

Area based DRC check checks only DRC violations, so -drc should not be turned off when -coordinates is specified.

### What Next

Remove -drc false from the options when -coordinates is specified.

Feedback

## ZRT-467

(warn) A specified rectangular area extends outside the design boundary.

**Description**

Part of or the whole rectangle specified in the command is outside the boundary of the design.

**What Next**

Please check and see if there is any mistake in the input coordinates of the rectangle.

## ZRT-468

(error) Open the top cell to set this option.

**Description**

Some of the options require information from the design. Setting such options including this one require the top cell to be open.

**What Next**

Open the top cell and set this option.

## ZRT-469

(error) Cannot use comb routing topology with plan-group-aware routing. Using normal clock topology.

**Description**

The comb topology for routing clocks is not applicable to plan-group-aware routing. This error is not fatal; routing proceeds using normal topology for clocks.

**What Next**

To avoid this message, set the clock topology to normal before doing plan-group-aware routing.

## ZRT-470

(warning) Both exploration mode and effort are set. Exploration mode takes priority.

Feedback

**Description**

If you specify both exploration mode and effort for global routing, Router ignores the effort setting. This is just a warning; exploration routing proceeds as usual.

**What Next**

To avoid this message, do not set the global routing effort when using exploration mode.

---

## ZRT-471

(error) Invalid pin constraint specifies side %d on block %s that has only %d sides.

**Description**

This constraint is ignored because it specifies an invalid side number for the given module.

**What Next**

This is not a fatal error, but this constraint is dropped. Correct the side number (which should be 1 to n, where n is the number of sides in the block boundary) or delete the constraint. Note that sides are numbered clockwise from the lowest leftmost side.

---

## ZRT-472

(warning) The source-target pair on line %d of the feedthrough map file implies no feedthrough and overrides the previous feedthrough setting.

**Description**

If the feedthrough map input file has a source-target pair, such as "{Block A} {Block B}", it implies that the two must be connected directly, without feedthroughs. If there is a "{Feedthrough}" statement in the same net entry that permits feeds, that this directive is ignored.

**What Next**

This is just a warning. If you want to clean up the file, you can delete the "{Feedthrough}" statement.

---

## ZRT-473

(Warning) The virtual pin at (%.3f, %.3f) is outside of the %s boundary. It will be ignored.

**Description**

This warning occurs when the virtual pin location is outside of the corresponding plan group boundary.

Feedback

A virtual pin location must be on or inside the corresponding plan group boundary. If it is outside, it is ignored.

**What Next**

It is possible that the plan group boundaries were moved after the virtual pins were created. You should delete these virtual pins.

---

## ZRT-474

(Error) The '%s' value specified for the '%s' option is not of type %s.

**Description**

This error occurs when the value specified for an option has the incorrect type. The tool ignores this option setting.

**What Next**

See the command man page for information about the valid values for this option.

---

## ZRT-475

(Error) The '%s' value specified for the '%s' option is not in the range %s to %s.

**Description**

This error occurs when the value specified for an option is outside the acceptable range. The tool ignores this option setting.

**What Next**

See the command man page for information about the valid values for this option.

---

## ZRT-476

(Error) The '%s' value specified for the '%s' option is invalid.

**Description**

This error occurs when the value specified for an option is invalid. The tool ignores this option setting.

**What Next**

See the command man page for information about the valid values for this option.

Feedback

## ZRT-477

(Warning) Skip net: %s. Intended break point is too far away from legalized cell location. Current distance %d is more than %d.

**Description**

Intended break point is too far away from legalized cell location.

**What Next**

Please set a proper maximum distance in the split_net command option.

## ZRT-478

(Warning) Skipping net: %s. Breakpoint (%d, %d, %d) does not exist on layer %d.

**Description**

This warning message occurs when a breakpoint specified in the *-breakpoints* option of the *split_net* command does not exist on the specified layer.

**What Next**

Ensure that the layer specified in the breakpoint is valid for the specified coordinates.

## ZRT-479

(Warning) Skipping net: %d. Source net %d is invalid for the split_net command.

**Description**

This warning message occurs when the net specified in the *-from_nets* option of the *split_net* command is invalid.

**What Next**

Check the net specified in the *-from_nets* option in the *split_net* command.

## ZRT-480

(warn) Layer '%s' is not a frozen layer.

**Description**

This warning message occurs because the route.common.freeze_via_to_frozen_layer_by_layer_name option for the set_app_options command has encountered a metal layer which is not frozen.

Feedback

**What Next**

Please ensure that the layers specified in the list of the arguments are all frozen layers.

---

## ZRT-481

(warn) The similar option '%s' is specified earlier. The previous option is overwritten by the current option '%s'.

**Description**

This warning message occurs because the route.common.freeze_via_to_frozen_layer and the route.common.freeze_via_to_frozen_layer_by_layer_name options for the *set_app_options* command are used sequentially to freeze vias. The option specified later will overwrite the previous one.

**What Next**

It is recommended to only use the option route.common.freeze_via_to_frozen_layer_by_layer_name.

---

## ZRT-482

(error) The options route.common.freeze_via_to_frozen_layer and route.common.freeze_via_to_frozen_layer_by_layer_name cannot be specified at the same time.

**Description**

This error message occurs because both the route.common.freeze_via_to_frozen_layer and the route.common.freeze_via_to_frozen_layer_by_layer_name options for the *set_app_options* command are used at the same time.

**What Next**

It is recommended to only use the option route.common.freeze_via_to_frozen_layer_by_layer_name.

---

## ZRT-483

(warning) The pin guide that contains point (%.3f %.3f) has a non-rectilinear edge and will be ignored.

**Description**

A pin guide shape must be rectilinear. If any edges are not vertical or horizontal, the pin guide will be ignored.

Feedback

**What Next**

Pin guide coordinates that are manually entered might not be rectilinear. Please check the pin guide shape.

---

## ZRT-484

(warning) Cannot route net %s (dbId %lu) with pin guides or pin blockages or individual pin constraint(s).

**Description**

The global router has failed to route a net with pin guides, or with pin blockages, or with individual pins/nets constraint(s).

**What Next**

Check to see that the pin guide/pin blockages/individual pins/nets constraints make sense in terms of the routing targets in this net or pins of this net.

---

## ZRT-485

(warn) cell %s is placed outside of partition or block cell %s

**Description**

This implies that the cell has been placed outside of partition or block cell boundary. The cell placement is violating hierarchical placement constraints and will NOT be considered in design planning pin placement and global route.

**What Next**

Fix placement errors for those cells

---

## ZRT-486

(warn) cell %s is NOT placed or placed illegally

**Description**

This implies that there are cells have NOT been placed in the design or placed illegally (for instance, cell is placed outside of top block boundary). Those cells will NOT be considered in design planning pin placement and global route. This could be the case if child level design has NOT been implemented yet if the cell is inside child block.

Feedback

**What Next**

If this is NOT what you intend to do, please place the cells inside the partition/block so that they will be consideredd in design planning global route and pin placement

---

## ZRT-487

(warn) Ignoring the bus ordering for the %s bus because the %s net has plan group pins.

**Description**

The tool issues this message when you have enabled the *-keep_buses_together* option of the *set_fp_pin_constraints* command, but one or more bits in the specified bus already have plan group pins. The router must obey the plan group pins, which might be incompatible with bus pin ordering, so it ignores the *-keep_buses_together* setting.

**What Next**

Either delete the plan group pins or disable the *-keep_buses_together* option.

---

## ZRT-488

(error) Plan group pin shape %s/%s is not on a routing layer.

**Description**

The tool issues this error when the layer of the specified plan group pin is not within the minimum and maximum routing layers specified by the *set_ignored_layers* command. The router cannot connect to the plan group pin shape under these conditions.

**What Next**

Either delete the plan group pin or reset the routing layers. To reset the routing layers, use the *set_ignored_layers* command. For example, to set the maximum routing layer as METAL8, run the following command: *set_ignored_layers -max_routing_layer METAL8*.

---

## ZRT-489

(error) Shield cannot be created because the '%s' shielding net is marked as frozen.

**Description**

The tool issues this error during shield creation when the shielding net is marked as frozen. When the shielding net is marked as frozen, the tool cannot make any changes to the net.

Feedback

**What Next**

Check the attributes of the shielding net.

---

## ZRT-490

(error) Shield cannot be created because the router cannot find the shielding net.

**Description**

Shield cannot be created because the router cannot find the shielding net.

**What Next**

Verify that the shielding net exists in the design database.

---

## ZRT-491

(warn) Shield cannot be created for the '%s' net because the router cannot retrieve the shielding options from the design database during incremental shielding.

**Description**

The tool issues this warning when the net has inconsistent shielding options.

**What Next**

Rerun the shielding flow for the specified net.

---

## ZRT-492

(warn) The -pg_via_tie_effort_level and -shield_via_tie_effort_level options are both set to low.

**Description**

The tool issues this warning when you set both the *-pg_via_tie_effort_level* and *-shield_via_tie_effort_level* options of the *create_shield* command to low. This could result in a low shielding ratio if there is no structured power and ground mesh.

**What Next**

Provide a structured power and ground mesh to which to connect the shielding wires.

---

## ZRT-493

(warn) Skip net '%s' for shielding since it is not a valid P/G net.

Feedback

**Description**

The tool issues this warning when there is invalid P/G net specified in *route.common.shielding_nets* of the *set_app_options* command.

**What Next**

Provide P/G net that has ring/rail/stripe structure.

---

## ZRT-494

(error) No valid P/G net specified in route.common.shielding_nets.

**Description**

The tool issues this error when there is no valid P/G net specified in *route.common.shielding_nets* of the *set_app_option* command.

**What Next**

Please specify at least one P/G net that has rail/ring/stripe structure.

---

## ZRT-495

(info) No-trimming shielding mode is activated. All shielding shapes with DRC violations will be kept. No further operations are recommended.

**Description**

In no-trimming shielding mode, all shielding shapes with DRC violations are kept. Therefore, no further operations are recommended.

---

## ZRT-496

(warning) Cannot route net %s (dbId %lu) with topological constraint.

**Description**

The router has failed to route a net that has a toplogical constraints. This may be due to unsatisfiable constraints. For example, if the constraint specifies a direct connection from block A to C, but there is no direct path from A to C without creating a feedthrough. In some cases, this failure could also occur if the topological constraint specifications greatly overload an edge segment, particularly in an abutted design.

**What Next**

Check the topological constraints for this net. Correct them if not valid.

Feedback

## ZRT-497

(warning) Cannot find flat net for supply net %s on voltage area %s.

**Description**

The router could not find a flat (global) net corresponding to the given supply net in the voltage area. The router will not be able to check for feedthrough compatibility between the driver and this supply net in this voltage area, and may therefore avoid the voltage area when not strictly necessary.

**What Next**

Check your specification of the power net connectivity.

## ZRT-498

(error) Feedthrough map file error on net %s. %s on side %d of block %s is obstructed by abutted block %s.

**Description**

The feedthrough map file specifies a location through which to connect two blocks, but that location is obstructed by a different, abutted block. The router will proceed, but will ignore this offset.

**What Next**

Fix the specification in the feedthrough map file.

## ZRT-499

(error) Layer '%s' does not exist in design.

**Description**

You receive this message because you used a layer that does not exist in design.

**What Next**

Check the layers from the design and use the right routing layer name.

## ZRT-500

(Warn) Incorrect special minimum area rule defined on layer %s.

Feedback

**Description**

The rectangle-based special minimum area rule is obsolete in router. You cannot define the *specialMinArea* attribute without the *minAreaEdgeThreshold* attribute.

**What Next**

Define one of the valid minimum area rules as described in the routing rules documentation.

---

## ZRT-501

(Warn) Invalid short metal rule defined on layer %s.

**Description**

The length, width, and spacing thresholds must be greater than zero; otherwise, the rule is ignored for that layer.

**What Next**

Define a valid short metal rule as described in the routing rules documentation.

---

## ZRT-502

(warn) No antenna rule assigned for net %s.

**Description**

When multiple antenna rules are defined and rules assigned to nets, a rule must be assigned to every net in the design. Otherwise that net will be skipped for antenna analysis and fixing.

**What Next**

Assign antenna rules for all the nets.

---

## ZRT-503

(warn) NDR %s rule '%s' is rounded up to the next %s multiple of the min grid rule on layer %s.

**Description**

The NDR width/spacing rule defined is not an even/integer multiple of the min grid rule. It is automatically rounded up to the next even/integer multiple of the min grid rule to avoid the conflict between the two rules.

Feedback

**What Next**

Please define the NDR width/spacing rule as an even/integer multiple of the min grid rule.

---

## ZRT-504

(error) Mismatch in soft rule effort level and soft rule spacing weights for rule: %s, layer: %s, efforts: %s, weights: %s.

**Description**

This error message indicates that the router found a mismatch between the number of soft rule effort levels and soft rule spacing weights specified.

**What Next**

Please ensure that the number of effort levels specified matches the number of soft spacing weights. High effort is always required for the hard weight rule, which may not have been specified by the user.

---

## ZRT-505

(warning) Skip net %s for pin guide %s on plan group %s (too few intersections).

**Description**

The pin guide specified for this net has NO or has only one intersection with a parent plan group and there is NO net connection (logical port) on this plan group. Since the net has no port on this plan group, the guide is considered a feedthrough guide for this plan group. A feedthrough guide must have at least 2 intersections with the plan group boundary.

**What Next**

Please ensure that the number of intersections between pin guide and plan group larger than 1

---

## ZRT-506

(warn) %s setting will be ignored.

**Description**

The setting is being ignored because constraints defined by define_voltage_area_routing_rule are found, and router does not honor the settings together.

Feedback

**What Next**

Please make sure only one type of constraint is set.

---

## ZRT-507

(warn) skipping path geometry with octagonal or round encap on net %s.

**Description**

Path geometry with an octagonal or round endcap is not supported and will be ignored.

**What Next**

Please use either a zero or half-width rectangular endcap.

---

## ZRT-508

(warn) skipping path geometry with octagonal or round encap.

**Description**

Path geometry with an octagonal or round endcap is not supported and will be ignored.

**What Next**

Please use either a zero or half-width rectangular endcap.

---

## ZRT-509

(error) Layer %s has odd default width defined in tech file.

**Description**

Odd default width for routing layer is strongly discouraged since it leads to routing wires to be off the manufacturing grid. The wires that are off-manufacturing-grid will not be reported as design rule violations.

**What Next**

Please specify even default width for each routing layer.

---

## ZRT-510

(error) Failed to get design.

**Description**

The design could not be opened.

Feedback

**What Next**

Open the design and run the command.

---

## ZRT-511

(Warning) Cell contains tie connections which are not connected to real PG.

**Description**

Cell contains some tie nets (tie_low or tie_high) which are not connected to real PG. These nets will not contain route information unless they are realized with PG rail. To connect to PG rail, users need to call "connect_pg_net" command.

An example for query such a net is get_nets -hierarchical -filter net_type=~tie_low

**What Next**

Run "connect_pg_net".

---

## ZRT-512

(warning) The '%s' via enclosure width is less than NDR width constraint of '%s' layer.

**Description**

You receive this message because the via enclosure width is less than NDR width constraint of the layer. Router may report "Less than NDR width" violation for these via enclosure.

**What Next**

Please increase NDR via enclosure to meet metal layer NDR width.

---

## ZRT-513

(warn) Ignoring interlayer via metal spacing check for the %s and %s layers.

**Description**

The tool issues this warning when entries in the minSpacingUpperEncCheck table are set to 0.

To enable the interlayer via metal spacing check, entries in minSpacingUpperEncCheck table must be set to 1.

**What Next**

Review the interlayer via metal spacing rule definition in the technology file.

Feedback

## ZRT-514

(warn) The error view specified with the -error_view option uses an invalid file extension (%s); using the file extension "err" instead.

### Description

The tool issues this warning when the error view specified with the *-error_view* option has an invalid file extension.

The *check_routability* command requires a file extension of "err" for the error view. If you specify a different file extension, the tool keeps the error view name, but replaces the file extension with "err".

Note that you can also specify the error view name without an extension; in this case, the tool adds the "err" extension, but does not issue a warning.

### What Next

To avoid this warning message, either use the "err" file extension or no file extension when you specify the error view name.

### See Also

- check_routability

## ZRT-515

(error) The error view specified with the -error_view option is invalid; using the default error view name instead.

### Description

The tool issues this error when the error view specified with the *-error_view* option is invalid. For example, the name consists only of the file extension and does not contain an error view name.

When this error occurs, the *check_routability* command ignores the specified error view name and uses the default error view name of *top_cell_name*.err instead.

### What Next

The *-error_view* option is not required. If you specify this option. ensure that you specify a valid name for the error view.

### See Also

- check_routability

Feedback

## ZRT-516

(info) Creating error view %s.err.

### Description

The results of the *check_routability* command are stored in the created error view and can be viewed by the GUI error browser.

### See Also

- check_routability

## ZRT-517

(warn) The %s.err error view already exists; the current results overwrite the existing error view.

### Description

The tool issues this warning when it writes to an existing error view. The contents of the existing error view are replaced with the results of the current run. All previous data stored in the error view is lost.

### What Next

If you need to keep the existing error view, such as when you want to compare the results of two *check_routability* runs, use the *-error_view* option to specify a new error view name.

### See Also

- check_routability

## ZRT-518

(warn) The value specified for the -blocked_range option is too large; the tool will use a value of 40.

### Description

To avoid performance issues, the maximum value allowed for the *-blocked_range* option is 40. When you specify a value larger than 40, the tool issues this warning and sets the value to 40.

### What Next

Specify a value less than 40 for the *-blocked_range* option.

Feedback

### See Also

- check_routability

---

## ZRT-519

(warn) The value specified for the -blocked_range_via_side option is too large; the tool will use a value of 40.

### Description

To avoid performance issues, the maximum value allowed for the *-blocked_range_via_side* option is 40. When you specify a value larger than 40, the tool issues this warning and sets the value to 40.

### What Next

Specify a value less than 40 for the *-blocked_range_via_side* option.

### See Also

- check_routability

---

## ZRT-520

(warn) The check_routability command uses a single CPU process; ignoring the set_host_options -max_cores setting of %d.

### Description

The tool issues this warning when run the *check_routability* command and have specified multiple cores with the *set_host_options -max_cores* command.

The *check_routability* command works only with a single CPU. Requests for multicore processing are ignored only for this command; it does not affect any other router commands.

Please be noted, The global route congestion map overflow check supports multi-thread. This warning is issued only for subsequent checks after global route congestion map overflow check.

### See Also

- check_routability

Feedback

## ZRT-521

(info) PG preroute in signal net %s.

**Description**

The signal net contains routing objects of PG route type.

**What Next**

No action necessary.

## ZRT-522

(warn) The value specified for the -standard_cell_search_range option is too large; the tool will use a value of 10.

**Description**

To avoid performance issues, the maximum value allowed for the *-standard_cell_search_range* option is 10. When you specify a value larger than 10, the tool issues this warning and sets the value to 10.

**What Next**

Specify a value less than 10 for the *-standard_cell_search_range* option.

**See Also**

- check_routability

## ZRT-523

(warn) The via cut and enclosure sizes for %s might cause the via surrounds to be off the minimum grid.

**Description**

The tool issues this warning when the via cut and enclosure sizes might cause the via surrounds to be off the minimum grid.

**What Next**

Check the the via cut and enclosure sizes in the technology file and correct if necessary.

Feedback

## ZRT-524

(warn) The via cut, minimum cut spacing, and enclosure sizes for %s might cause the via array surrounds to be off the minimum grid. Turn off min grid checking in check_routability.

**Description**

The tool issues this warning when the via cut, minimum cut spacing, and enclosure sizes might cause the via array surrounds to be off the minimum grid. Min grid checking in check_routability is turned off since there will be too many off minimum grid violations and runtime will be long if the via array surrounds off minimum grid issue is not fixed.

**What Next**

Check the via cut, minimum cut spacing, and enclosure sizes in the technology file and correct if necessary.

## ZRT-525

(warn) router does not support the rotated unfixed custom via %s and will ignore it.

**Description**

The tool issues this warning when it reads in an unfixed custom via while reading the vias from the database.

**What Next**

Double-check the usage of this via.

## ZRT-526

(warn) Invalid fat wire extension spacing rule on the %s layer.

**Description**

The tool issues this warning when both the forbidden space rule and the fat wire extension spacing rule are defined on the specified layer. When both rules are defined, the router ignores the fat wire extension spacing rule.

**What Next**

Remove the fat wire extension spacing rule from the technology file.

Feedback

## ZRT-527

(warn) The '%s' rule is specified with the create_routing_rule -spacing_weight_levels option. The effort level defined in the route.common.routing_rule_effort_level option is ignored.

### Description

The tool issues this warning when you specify the soft rule weights in the *create_routing_rule* command by using the *-spacing_weight_levels* option and also specify the effort level for the rule by using the *route.common.routing_rule_effort_level* option. In this case, the behavior can be ambiguous. If the *-spacing_weight_levels* option is used to specify the spacing weights, the entry in the *route.common.routing_rule_effort_level* option is ignored and the value from the *route.common.soft_rule_weight_to_effort_level_map* option is used.

### What Next

Remove the entry for the rule from the *route.common.routing_rule_effort_level* option.

### See Also

- create_routing_rule

## ZRT-528

(warn) %d ports are not checked because they include diagonal shapes %s.

### Description

The *check_routability* command has limited support for diagonal shapes. Diagonal shapes are checked only for minimum grid violations. Ports with diagonal shapes are not checked for accesibility. The warning message lists only the first 100 encountered ports with diagonal shapes.

### See Also

- check_routability

## ZRT-529

(warn) %d ports are not checked because they might be blocked by diagonal shapes %s.

### Description

The *check_routability* command has limited support for diagonal shapes. Diagonal shapes are checked only for minimum grid violations. Ports that might be blocked by diagonal

Feedback

shapes are not checked for accesibility. The warning message lists only the first 100 encountered ports that might be blocked by diagonal shapes.

**See Also**

- check_routability

---

## ZRT-530

(error) A technology file attribute (%s) on layer %s contains a larger value than expected (%f), which is above the limit of %f.

**Description**

This error message indicates that a larger than expected value has been found in the technology that might be due to a technology file coding error. In general, the design rule values should be smaller than the threshold limit specified in the error message (20 times the width of the global routing cell).

**What Next**

If this is a technology file coding error, you should reduce the value of the specified attribute to be less than the given threshold. If you want to perform routing using the current value, use the *set_app_options* command to set the *route.detail.continue_after_large_design_rule_value_error* option to *true*. This might result in excessive runtime and memory overhead.

---

## ZRT-531

(error) Skip splitting net %s, gap_length is not even multiple of min grid.

**Description**

The input of -gap_length is not even multiple of min grid. This will cause the wires after splitting to be off min grid.

**What Next**

Check the input of -gap_length and correct it to be even multiple of min grid.

---

## ZRT-532

(warn) Min layer of net %s cannot be updated to via ladder top layer of %s since the net has an existing max layer setting of %s.

Feedback

**Description**

This net contains via ladders that will imply a minimum routing layer for the net. See option route.common.via_ladder_top_layer_overrides_net_min_layer . The top layer across all via ladders on this net is higher than the maximum routing layer of this net. Therefore the minimum routing layer of this net cannot be updated.

**What Next**

Please check the maximum routing layer setting for this net and confirm it is compatible with the via ladder constraints for the pins of this net.

---

## ZRT-533

(warn) Ignore the onGrid rule defined on layer %s.

**Description**

The design has both "onGrid" and "onWireTrack" rule defined. We will ignore the "onGrid" value for all layers.

**What Next**

Clean up the onGrid rule in the tech file and try to use onWireTrack.

---

## ZRT-534

(warn) Illegal tech file: %s.

**Description**

The tool issues this error when the tech file specified a non-recoverable error.

**What Next**

Correct the error in the technology file listed in the message.

---

## ZRT-535

(error) Skip splitting net %s, breakpoint (%d, %d, %d) is not on min grid.

**Description**

The breakpoint is not on min grid. This will cause the wires after splitting to be off min grid.

**What Next**

Check the input of -breakpoints and correct it to be on min grid.

Feedback

## ZRT-536

(warn) There is no constraint to limit maximum number of secondary power and ground (PG) pins in a cluster.

**Description**

The option value of route.common.number_of_secondary_pg_pin_connections in set_app_options is 0. The routing/drc-check behavior for secondary PG pin will be exactly the same as signal pin.

**What Next**

Please set option value of route.common.number_of_secondary_pg_pin_connections in set_app_options

## ZRT-537

(warn) minSpacing in layer %s is not integer multiple of min grid in tech file.

**Description**

minSpacing is not integer multiple of min grid (gridResolution/lengthPrecision in tech file) for routing layer is strongly discouraged since it may lead to routing wires to be off the manufacturing grid.

**What Next**

Please specify minSpacing as integer multiple of min grid for each routing layer.

## ZRT-538

(error) An instance of cell %s contains too many pins or too many must-join terminals. The total for any cell cannot be greater than %d.

**Description**

The cell instance contains a large number of pins or the cell has ports with a large number of must-join terminals. The sum of all pins and must-join terminals cannot exceed the maximum value for any single cell.

**What Next**

Please check the cell to see if the number of pins exceeds the maximum supported number. If not, check if any of the terminals have a must_join_class setting that is non-zero. Then, either reduce the number of pins in the cell or set the must_join_class to 0 on some of the terminals. Or remove the instance. Then try again.

Feedback

## ZRT-539

(warn) Incorrect input data, %s.

**Description**

Found input data have error(s) as described.

By default, routing commands print only 10 messages of this type. If user wishes to have more messages of this type printed to log file, set_message_info command is to be used.

Example is: set_message_info -id {ZRT-539} -limit 300

**What Next**

Please check on the source that the data was from, and try again.

## ZRT-540

(warn) Illegal %s rule (%s, %s).

**Description**

This warning message indicates that the rule is specified between two inconsistent layers.

**What Next**

Please check the technology file and make sure the rule is specified between consistent layers.

## ZRT-541

(warn) The %s rule refers to an invalid cut name %s.

**Description**

A cut name specified in the tech file for the rule contains an invalid entry.

**What Next**

Please correct the tech file and try again.

## ZRT-542

(warn) Cell %s is marked as filler but has diode pin.

Feedback

**Description**

Diodes may be inserted or reused during routing. Filler cell location may be used for diodes too. Having the cell with both these settings causes difficulty for routing.

**What Next**

Do the necessary data clean up to avoid both markings and rerun the routing commands.

---

## ZRT-543

(warn) Shape {%d %d %d %d} on layer %s is not on min manufacturing grid. Snap it to {%d %d %d %d}. The shape belongs to Net: %s.

**Description**

Each edge of a shape should always be on the min manufacture grid. If this is not satisfied, DRC checking can't be accurate. This rule should apply to all the features on the layout.

**What Next**

If the off minGrid shape is a fixed shape from the libary, try to fix them manully. If the shape is a routable shape, it should be automaticlly fixed after running detailed route.

---

## ZRT-544

(warn) Ignore polygon shape : Cell %s, Layer %s, BBox [%d,%d..%d,%d].

**Description**

This warning message occurs because polygon shape was found. The router currently does not handle polygon shape and thus this shape will be ignored.

**What Next**

Please modify the polygon into rectangles.

---

## ZRT-545

(warn) A library cell %s contains too many pins or too many must-join terminals. The total for any cell instance cannot be greater than %d. Any attempt to instantiate this cell in a design will result in an error.

Feedback

**Description**

A cell in the library contains a large number of pins or the cell has ports with a large number of must-join terminals. The sum of all pins and must-join terminals cannot exceed the maximum value for any single cell.

These cells can exist in the library, but any attempt to instantiate such cells will result in an error.

**What Next**

No action is necessary if the cell is not instantiated in the design. A separate error will be issued if the cell is instantiated.

---

## ZRT-546

(warn) Unrecognized pattern type %s for maximum number of pattern route guide located at {{%.3f %.3f} {%.3f %.3f}} in cell %s.

**Description**

The maximum number of pattern route guide contains pattern type that cannot be recognized. Only the following pattern types are recognized: 1. NonPrefDirEdge

**What Next**

Please correct the pattern type in the route guide.

---

## ZRT-547

(warn) Skipping the maximum number of pattern route guide located at {{%.3f %.3f} {%.3f %.3f}} in cell %s because it is too big for pattern type %s.

**Description**

The maximum number of pattern route guide has the following size limit. Route guides width width or height exceeding this limit are ignored. 1. NonPrefDirEdge: 6 routing pitch of the layer it is defined on.

**What Next**

Please reduce the size of the route guide.

---

## ZRT-548

(warn) NDR rule %s has '%s' or '%s' defined with layer based tapering. Taper distance will be igored.

Feedback

**Description**

The NDR tapper has the control on "taper_distance" and "driver_taper_distance". When the layer based tappering is defined, i.e. 'taper_to_number_layer_over_pins/ taper_to_number_layer_under_pins', we automatically ignore the taper distance.

**What Next**

Don't define taper distance if using layer based tapering.

---

## ZRT-549

(warn) Ignore NDR width (%.2f) smaller than minimum width (%.2f) (%s, %s).

**Description**

The NDR width rule defined is less than the minimum width of the layer.

**What Next**

Please check the NDR rule definition in tcl command to make sure that the NDR width is no less than the minimum width on each routing layer.

---

## ZRT-550

(warn) Ignore a DRC with unknown/undefined layer(s), with DB layer numbers %d and %d.

**Description**

Ignore a DRC with unknown/undefined layer(s).

**What Next**

Please check the DRC marks' layers of the design.

---

## ZRT-551

(warn) Contact array %s-pitch (via master %s) (location %.3f %.3f) is not a even multiple of min grid. Router will use %.3f instead.

**Description**

The contact array at the specified location has pitch that is not an even multiple of min grid. Such a pitch value will result in lower/center/upper coordinate of the via to be off min grid. The router will continue using the larger pitch.

Feedback

**What Next**

Please check the database and verify that contact array pitch is even multiple of min grid.

---

## ZRT-552

(warn) Illegal via mask_constraint (%d) found in cell %s object %d.

**Description**

A via is found with mask_constraint value outside of the legal range. The legal values for mask_constraint are: 0, 1, 2, 3.

**What Next**

Please check the via mask_constraint assigned to the specified object.

---

## ZRT-553

(error) A net %s connects too many ports and out of the limit of router. The total net port cannot be greater than %d.

**Description**

The net connects a large number of ports The number of ports cannot exceed the maximum value for each net.

**What Next**

Please check the design to see if the number of ports of the net exceeds the maximum supported number.

---

## ZRT-554

(error) Cannot open design %s.%s for %s.

**Description**

You received this message when one of the following occurs: 1. the specified cell is locked or read-only and you tried to write to it, 2. you do not have permission to access the cell or the library where it resides, 3. the path (e.g. in stream out, a reference library path) or the cell name is incorrect, 4. too many files or reference libraries are open.

**What Next**

Check the path names, especially of reference libraries, and cell names to make sure they are correct, and also their file permissions. Check file descriptors using "ulimit -a"

Feedback

command. In Sun SparcOS5, the limit is 256. And increase the allowable number to its maximum value.

---

## ZRT-555

(error) Must specify at least one shape use.

**Description**

You receive this message because no shape use was specified in the command.

**What Next**

Correct the option.

---

## ZRT-556

(warn) Net pattern did not match any nets.

**Description**

The net pattern specified did not match with any nets.

**What Next**

Verify that the net pattern specified is correct.

---

## ZRT-557

(error) %s

**Description**

Basic input error checking.

**What Next**

Please recheck the command inputs and correct any errors reported in the log file.

---

## ZRT-558

(error) Found %d master cells that contain more than %d obstructions each. The total number of obstructions %d exceeds the maximum allowed value of %d. Please refer to ZRT-092 warnings in the log for additional detail. These obstructions will cause a very long routing runtime without any benefit. Please re-generate the frame view.

Feedback

**Description**

Multiple cells contain a very large number of obstructions. This is atypical.

**What Next**

Please re-generate the frame view.

---

## ZRT-559

(info) Routes in non-preferred voltage areas = %d

**Description**

Voltage areas are a soft constraint, and routing outside of preferred voltage areas is occasionally necessary and not indicative of an error.

For example, routes may cross through non-preferred voltage areas to avoid unacceptably long detours. Short routes may also enter a nearby non-preferred voltage area along the border. These behaviors are normal.

**What Next**

No action necessary.

---

## ZRT-560

(error) Cell %s is placed overlapping with other cells at {0 0}.

**Description**

This error message indicates that the cell has not been legally placed and it will be ignored by the router.

**What Next**

Please make sure all the cells are placed legally before running the router.

---

## ZRT-561

(error) Non default routing rule (%s) layer (%s) has width (%g) with no matching track

**Description**

The design has width constraint for tracks defined on the specified layer and non default routing rule with width not matching any track width.

This not routable, since the non default width nets have no legal routing location.

Feedback

**What Next**

Please check the track system is defined for the non default routing rule width.

---

## ZRT-562

(warn) Layer %s does not have a preferred routing direction.

**Description**

This message indicates that the layer does not have a preferred routing direction. Router cannot route without preferred routing direction.

**What Next**

Please use the set_attribute command to assign the preferred routing direction as follows: set_attribute -name routing_direction -value vertical [get_layer M9]

---

## ZRT-563

(warn) Non default routing rule exceeds 510 limitation.

**Description**

This message indicates that non-default routing rule number exceeds 510 limitation. Router may crash or has long runtime issue.

**What Next**

Please use remove_routing_rules command to reduce the non-default routing_rule number.

---

## ZRT-564

(info) Min layer of net %s was set to %s due to via ladders on the net and will be treated as a soft constraint.

**Description**

This net contains via ladders that will imply a minimum routing layer for the net. The top layer across all via ladders on this net is the new minimum routing layer for the net. This minimum routing layer will be treated as a soft constraint during all stages of routing.

**What Next**

If this minimum routing layer behavior is unexpected, please check the setting of app option route.common.via_ladder_top_layer_overrides_net_min_layer .

Feedback

## ZRT-565

(warning) Cell %s port %s contains more than %d pins (Number of pins: %d). This may increase routing runtime.

**Description**

A port contains a large number of pins. This is atypical and may increase routing runtime.

**What Next**

If this is not expected, please reduce the number of pins for the port.

## ZRT-566

(warn) Fixed contact array %s-pitch (via master %s) (location %.3f %.3f) is not a even multiple of min grid. Suggested value is %.3f.

**Description**

The contact array at the specified location has pitch that is not an even multiple of min grid. Such a pitch value will result in lower/center/upper coordinate of the via to be off min grid. This can cause the DRC checking not being accurate. The router can't change the fixed via array.

**What Next**

Please check the database and verify that contact array pitch is even multiple of min grid.

## ZRT-567

(error) No ECO changed area found.

**Description**

This error message indicates that the router could not get the ECO changed area for region-based DRC checking and removal on filler cells.

**What Next**

First check whether command read_eco_changes was already applied, then check whether ECO operations (e.g. place_eco_cells, legalize_placement, route_eco, create_stdcell_fillers, connect_pg_net) were already applied before running command remove_stdcell_fillers_with_violation.

Feedback

## ZRT-568

(error) Failed to read a via definition because there are more than %d via definitions defined in the design.

**Description**

The design has more via definitions defined than Zroute can support.

**What Next**

Check the design to determine if the large number of via definitions was created due to a problem in the flow.

## ZRT-569

(error) A shape at metal layer %s location {%f %f %f %f} is missing mask_constraint in DB or has other DB related problem

**Description**

Metal shape don't have a mask_constraint or other DB related mask_constraint problem happens

**What Next**

Please check the mask_constraint of the shape at location specified

## ZRT-570

(Warning) Cannot run timing driven route command when setup timing information is not available.

**Description**

This warning message indicates that a timing-driven routing command is being run on a design that does not have any timing constraints or extraction setup. When the tool issues this message, it does not perform detail routing.

**What Next**

To perform timing-driven routing, set the timing constraints or extraction setup before running the routing command.

Feedback

To disable timing-driven routing, set the following application options to *false* before running the routing command:

- *route.detail.timing_driven*
- *route.global.timing_driven*
- *route.track.timing_driven*

**See Also**

- route_auto
- route_detail
- route_eco

---

## ZRT-571

(error) In rule %s, via enclosure of %s has illegal widths on layer %s.

**Description**

This error message indicates that NDR via enclosure on the metal layer is illegal or larger than the signal route max width defined on that layer. If such via is the only candidate defined for the NDR rule, it may cause inefficiency in the router and result in slow runtime. Router will use the default vias instead.

**What Next**

Please make sure that the NDR via enclosure is legal and is less than or equal to the signal route max width defined in the tech file.

---

## ZRT-572

(warn) Pin %s of cell %s has more than %d access edges.

**Description**

The global router expects macro pins to have access edges to help the router to connect to the pins. This may affect the runtime, memory, and QoR significantly. There are some pins with unreasonably large number of access edges. Zroute will skip reading many of these access edges.

**What Next**

Check if it is possible for the frame creation to generate fewer number of access edges. Try to re-generate the frame view in latest release to see if reasonable number of access edges are generated.

Feedback

## ZRT-573

(warn) Design has custom via def %s with %d cuts.

**Description**

The design has custom via def with more than thousand cuts. Instances of such a via def in the design can impact runtime of Zroute. Correct the data by creating those vias as via arrays in NDM.

**What Next**

Check the design and create those via defs as of via array type.

## ZRT-574

(Information) Freeing timing information from routing.

**Description**

This information indicates that timing update has happened after route_global or route_track command and all timing information used in route_global or route_track command is freed. To be consistent with route_auto ICC2 use the same timing information for route_global, route_track and route_detail steps when timing driven is on. But if the user calls update_timing explicitly between route_global/route_track/route_detail, the existing timing information will be freed and the next route_command will use up-to-date timing information.

## ZRT-575

(error) Unfixed X shape wires at (%.4g, %.4g) on layer %s will be treated as fixed by router.

**Description**

ZRoute does not support unfixed X shape wires. Any unfixed X shape wires will be treated as fixed by router.

**What Next**

Please mark the X shape wires as fixed.

## ZRT-576

(error) via cut routing blockage {%.3f %.3f} {%.3f %.3f} in layer %s is treated as real via.

Feedback

### Description

This via cut blockage is treated as real via because these six attribuites are all false:
1. icc_route_guide 2. icc_system_layer_blockage 3. is_design_rule_blockage 4. is_boundary_blockage 5. is_external_boundary_blockage 6. is_zero_spacing

And the size of via cut blockage doesn't fit any legal cut size (cutWidthTbl, cutHeightTbl, minWidth) in the layer specified by technology file.

Thus it is incorrect real metal via cut.

In general, the 1. to 5. are special attributes generated from tool flow. The user could set 6. as true to represent a routing blockage, or as false to represent a real metal.

The router applies DRC rule checking on them and the performance could be degraded. These via cut blockages might also generate unreasonable DRCs, which might lead to unexpected router behavior.

### What Next

Please review the via cut blockage attributes and the technology file.

---

## ZRT-577

(error) There should be at least one horizontal routing layer and at least one vertical routing layer.

### Description

This error message indicates that all the routing layers have the same preferred direction. This is illegal input to the router. Zroute will print an error message and quit.

### What Next

Please check the input and make sure that there is at least one horizontal routing layer and at least one vertical routing layer.

---

## ZRT-578

(warn) Ignore bad shield association attached to shield wire at (%.3f, %.3f, %.3f, %.3f) metalLay: %s

### Description

The shield association of the shield wire is wrong. Either the shielded net of the shield association is PG net or the shielded net of the shield association does not exist.

Feedback

**What Next**

Please use remove_shield_association command to remove the shield association.

---

## ZRT-579

(warn) Ignore bad shield association attached to shield via at (%.3f, %.3f) cutLay: %s

**Description**

The shield association of the shield via is wrong. Either the shielded net of the shield association is PG net or the shielded net of the shield association does not exist.

**What Next**

Please use remove_shield_association command to remove the shield association.

---

## ZRT-580

(error) There is no shielding target to connect

**Description**

There is no PG stripe and PG ring in the design. Besides, both -ignore_shielding_net_pins and -ignore_shielding_net_rails of create_shields are set as true. Therefore, There is no shielding target to connect.

**What Next**

Check the usage of -ignore_shielding_net_pins and -ignore_shielding_net_rails

---

## ZRT-581

(error) A track must have mask_constraint of mask1 or mask2. Unexpected track mask_constraint is found on layer %d.

**Description**

A track should be with a mask_constraint of either mask1 or mask2.

**What Next**

Please check the track mask_constraint on the layer reported.

---

## ZRT-582

(Warning) Starting from %s release, the '%s' command will no longer support the '%s' option

Feedback

**Description**

This command option will no longer be available in a future release.

**What Next**

Consult the man page of the command for suggestions regarding other commands or options to accomplish the same effect. Consult the release notes.

---

## ZRT-583

(Warning) Starting with the %s release, the %s command will no longer support the %s option. Use the %s option instead.

**Description**

This command option will no longer be available in a future release. Please use the recommended alternative instead.

**What Next**

See the release notes or command man page for suggestions regarding other commands or options to accomplish the same effect.

---

## ZRT-584

(warning) %s.frame does not exist.

**Description**

This message indicates that Zroute cannot find frame cell for a design in physical library.

**What Next**

If not having frame view for this design is intended, ignore this warning. If frame view is expected for this design, please check the designs in the library and man page of LNK-055 for more details.

---

## ZRT-585

(info) Skipping internal port %s of %s.%s as it is not physical.

**Description**

Zroute skips pins of ports with port_direction internal and port_is_physical false for routing and DRC checks.

Feedback

**What Next**

If these pins are intended to be routed, check the data preparation.

---

## ZRT-586

(Warning) No existing global route is available for reuse in incremental global routing.

**Description**

Option *-reuse_existing_global_route true* is set for global router. However, no existing global route can be found. As a result, no global route is reused in incremental global routing.

**What Next**

To prevent this message, create global route before running global router with option *-reuse_existing_global_route true*, or run global router with the default option (*-reuse_existing_global_route false*).

---

## ZRT-587

(warn) NDR rule (%s) wide track requirement conflicts with tapering.

**Description**

The NDR rule has a conflict between wide track requirement and tapering.

**What Next**

Disable tapering for the NDR rule.

---

## ZRT-588

(error) The design is not legalized. Skip blocked port checkings.

**Description**

If the placement is not legalized and standard cells are overlapped, the presence of blocked port are expected and routing quality will be very bad. check_routability skips checking blocked port because those checkings are meaningless and time consuming when design is not legalized.

**What Next**

Please do legalize_placement before doing check_routability.

Feedback

## ZRT-589

Error: Net fails consistency check. Skip net graph extraction. (%s)

**Description**

The net fails consistency check before RT graph extraction. The net will be skipped for graph extraction.

**What Next**

Turn on interactive routing verbose to find out why the net fails the consistency check and fix it.

## ZRT-590

Fail to set in-cell mask (%s)

**Description**

Fail to set in-cell mask. This will result in some objects in an instance cell missing DPT colors.

**What Next**

Stop at the error code to see why it is failed.

## ZRT-591

(error) Found clock net %s has topology issue

**Description**

Expect clock net topology to be a tree.

**What Next**

Double check on the net's route, debug and fix the topology if necessary.

## ZRT-592

(warn) Skip shielded net '%s' since it is an RDL net created in 45 degree.

**Description**

The tool issues this warning when there is a invalid shielded net.

Feedback

**What Next**

Please make sure shielding only supports RDL nets which are created in 90 degree for all layers. Please refer to *flip_chip.route.layer_routing_angles*

---

## ZRT-593

(Information) The design has %ld vias.

**Description**

The design has a very large number of vias. Please review if this is intended or a problem with data preparation.

**What Next**

Check the design and correct the data if necessary.

---

## ZRT-594

(warn) Found must join pins on port %s of instance %s without any net.

**Description**

Must join pins without net information were found. The router won't be able to connect them unless net is assigned to them.

**What Next**

Please create net for those pins.

---

## ZRT-595

Design has at least one Macro with no-color shapes when DPT color-based flow is enabled and/or with via-color rule.

**Description**

Design has at least one Macro with no-color shapes when DPT color-based flow is enabled and/or with via-color rule.

**What Next**

Check message reported during router dbIn in the log to see which macro cell has incorrect shape color and fix the coloring problem.

Feedback

## ZRT-596

(warn) Incorrect syntax found in layer %s table %s.

**Description**

Error was encountered parsing the table entry in the tech file. The router may not interpret the rule as the user intends.

**What Next**

Please double check the tech file entry.

## ZRT-597

(warn) Incorrect syntax found in layers %s,%s table %s.

**Description**

Error was encountered parsing the table entry in the tech file. The router may not interpret the rule as the user intends.

**What Next**

Please double check the tech file entry.

## ZRT-598

(info) Min layer of net %s was overridden from %s to %s due to via ladders on the net. The original min layer mode was %s and the new min layer mode will be soft.

**Description**

This net contains via ladders that will imply a minimum routing layer for the net. The top layer across all via ladders on this net is the new minimum routing layer for the net. Please note this minimum routing layer will be treated as a soft constraint during all stages of routing. It may be a relaxation of the original minimum routing layer mode.

**What Next**

If this minimum routing layer behavior is unexpected, please check the setting of app option route.common.via_ladder_top_layer_overrides_net_min_layer .

## ZRT-599

(warn) Spare cell %s is placed overlapping with %d other spare cells at {%.3f %.3f}.

Feedback

**Description**

This warning message indicates that there are over 100 other spare cells overlapping with this spare cell and these spare cells will be ignored by the router.

**What Next**

Please make sure all the spare cells are placed legally before running the router.

---

## ZRT-600

Ignore unsupported NDM mask %s

**Description**

Reading an unsupported NDM mask which will be ignored in router. This indicates some objects from DB may have unsupported colors.

**What Next**

Stop at the error code to see the detail of the NDM mask.

---

## ZRT-601

(warn) Eco route ignores -open_net_driven and -nets options when route.detail.enable_end_off_preferred_grid_patching_on_fixed_shapes is set to true or end extension for via enclosure rule is defined.

**Description**

When route.detail.enable_end_off_preferred_grid_patching_on_fixed_shapes is set to true or end extension for via enclosure rule is defined, fixed shape patches in the entire design need to be deleted and redone. So the options to restrict the eco like -open_net_driven and -nets do not make sense. Instead we run regular eco route as if those options were not specified.

**What Next**

Update script to run regular eco route when route.detail.enable_end_off_preferred_grid_patching_on_fixed_shapes is set to true or end extension for via enclosure rule is defined.

---

## ZRT-602

(warn) Ignore via array %s longer than the chip.

Feedback

### Description

The via array from DB is longer than the top design boundary. Router ignores the via array.

### What Next

Review the via array and correct its definition.

---

## ZRT-603

(warning) Net %s has locked shields, it's not recommended in reshield mode.

### Description

The indicated shielding net has physical_status locked shield shapes. It's not recommended in reshield mode. Any locked shield in reshield mode might cause degradation of run-time, shielding rates, or introducing new opens in shields.

### What Next

It is recommended to set physical_status of shield shapes to be minor_change (fixed shield) or unrestricted (dynamic shield) in reshield mode. Reshield mode is controlled by route.common.reshield_modified_nets option.

### See Also

- set_attribute
- report_app_options
- set_app_options
- route.common.reshield_modified_nets

---

## ZRT-604

(info) Shielded%s%d%% side-wall of (%s)

### Description

Reports the shielding rates of a net. Any shielding rates below 1% is rounded to 1%.

### What Next

While the total rates of all nets are high, it's OK when the user sees low shielding rates of a few nets. To increate the shielding rates, please review shield related rules, settings, or reserve more space for shields.

Feedback

## ZRT-605

(Error) Ignoring unfixed via matrix on layer %s at (%d, %d).

**Description**

Zroute does not support unfixed via matrix objects. Please correct the input.

**What Next**

Mark the via matrix objects as fixed and rerun the routing command.

## ZRT-606

(Warning) Layer %s does not have a preferred direction track.

**Description**

There is no preferred direction track defined for the layer.

**What Next**

Set the correct preferred direction for the layer or correct the direction for the track.

## ZRT-607

(info) Skipping global routing as it has already been run in the global_route_opt stage of clock_opt.

**Description**

Message indicates that the tool detected a situation where global routing is already done previously in the global_route_opt stage of clock_opt. Therefore, global routing will be skipped. This is applicable to route_global and route_auto commands. Application option route.global.force_rerun_after_global_route_opt can be used to override this behavior.

**What Next**

No action necessary.

## ZRT-608

(info) Forcing rerun of global router after global_route_opt due to app option route.global.force_rerun_after_global_route_opt being set.

Feedback

**Description**

Message indicates that the tool detected route.global.force_rerun_after_global_route_opt as set. In this case, global router will be forced to rerun after global_route_opt.

**What Next**

No action necessary.

---

## ZRT-609

(error) Failed in cell reader (%s)

**Description**

Failed in reading cell in router DbIn.

**What Next**

Stop at the error code to see why it is failed.

---

## ZRT-610

(error) Failed to start IMRD manager due to %s

**Description**

Failed to start IMRD manager due to a router dbIn error.

**What Next**

Stop at the error code to see why it is failed.

---

## ZRT-611

(Warn) Found %s

**Description**

Found a number of unplaced cells. (Some of the unplaced cells are printed using ZRT-064.)

**What Next**

Please make sure all the cells are placed before running the router.

Feedback

## ZRT-612

(warn) PG structure in the region (%.4f %.4f)_(%.4f %.4f) on layer %s might cause router to fail to route long wires on same layer.

**Description**

Message indicates that the PG structue might cause router to fail to route long straight wires without changing layers.

**What Next**

No action necessary unless the user thinks it is not allowed on the particular layer.

## ZRT-613

(info) RC layer preference is turned %s for this design.

**Description**

RC layer preference is turned off if the design does not have enough variation in RC value among the reasonable routing layers. A layer is reasonable if it is within min and max routing layers, and its pitch is within six times the average pitch of the lower five metal routing layers.

**What Next**

No action necessary.

## ZRT-614

(warn) The global router sees multiple (%d) terminals for the port %s on net %s, cannot perform feedthrough creation on this net.

**Description**

Message indicates that more than one terminals exist for the reported port. Due to abutment, it forces more than one feedthrough creations to connected with the port, which is an unsupported behavior by pin creation. Thereby, the feedthrough creation is canceled. Feedthrough creation for other connections of the net may also be affected.

**What Next**

Verify that the port have a single terminal.

## ZRT-615

(warn) %d tracks on layer %s will cause high memory consumption.

Feedback

**Description**

Message indicates that the number of tracks on the given layer is unusually high which will cause high memory consumption.

**What Next**

Verify if tracks are defined correctly and check if size of design is reasonable for routing command.

---

## ZRT-616

(warn) Average track distance %f is much smaller than pitch %f on layer %s.

**Description**

Message indicates that the the average track distance deviates considerably from the pitch on the given layer. This might point to unintentional or erroneous track definitions.

**What Next**

Verify if tracks are defined correctly.

---

## ZRT-617

(warn) The global router sees signal net %s connecting with %u power or ground %s.

**Description**

Message indicates that the top level signal net is connected with power or ground pins on stand cells or macro cells inside of child blocks. The global router will skip routing to most of these power and ground pins in hierarchical global routing to avoid potential runtime issues.

**What Next**

Verify the tie off results of this signal net.

---

## ZRT-618

(warn) %d redundant duplicated PG %s at {%.3f %.3f} {%.3f %.3f} in layer %s.

**Description**

Redundant duplicated PG shapes are overlapped at the same location, which might degrade the DRC results and performance.

Feedback

### What Next

Please review whether the redundant duplicated PG shapes should be removed or not.

---

## ZRT-619

(info) Via ladder engine would be activated for pattern must join connection in certain commands. Please refer to man-page for the command list.

### Description

This design contains pattern must join pins, which is required to be connected by pattern must join connection shapes. Pattern must join connection has similar structure to via ladder, and is also constructed by via ladder engine. It would be processed during via ladder related commands, and full routing commands. Here is the command list: insert_via_ladders, verify_via_ladders, refresh_via_ladders, route_group, route_auto, route_eco, route_opt. To separate route_auto into several commands, please run through: insert_via_ladders, route_global, route_track, route_detail.

### What Next

If this pattern must join connection is unexpected, please check the attribute of pin, pattern_must_join.

---

## ZRT-620

(warning) Core area is (partially) outside boundary area.

### Description

This message indicates that the core area is not completely inside the block boundary. This may create problems for routing.

### What Next

This likely means that the block core area is not set up correctly. It can be fixed by using initialize_floorplan on the block.

### See Also

- initialize_floorplan
- check_physical_constraints

Feedback

## ZRT-621

(warn) The library real metal blockage at (%.4f %.4f)-(%.4f %.4f) on layer %s is touching or overlapping a library pin.

**Description**

A real metal routing blockage touching or overlapping a library pin might cause routing time degradation and DRC non-convergence issues.

This blockage is treated as a piece of real metal because these six attribuites are all false: 1. icc_route_guide 2. icc_system_layer_blockage 3. is_design_rule_blockage 4. is_boundary_blockage 5. is_external_boundary_blockage 6. is_zero_spacing

Thus it is treated as a piece of real metal.

In general, the 1. to 5. are special attributes generated from tool flow. The user could set 6. as true to represent a routing blockage, or as false to represent a real metal.

The coordinates reported here are relative to the library block inside, not the top design block.

**What Next**

Please review the library routing blockage attributes and the flow of library generation. When a library routing blockage is not representing a piece of real metal, it should have the attribute "is_zero_spacing" to be true.

## ZRT-622

(warn) X-oriented shapes are not supported, %s.

**Description**

This warning message indicates that the current design contains X-oriented (non-manhattan direction) shapes.

**What Next**

X-oriented shapes should be pre-processed.

## ZRT-623

(warn) Option value "%s" of app option "%s" will be obsolete in "%s" release.

**Description**

The option value specified will be obsoleted in future release. Please refer to man page of that option for complete details.

Feedback

**What Next**

Please specify valid option value for this option as currently specified value would be obsolete.

---

## ZRT-624

(warn) Option %s is set to %s, reset %s to %s with this setting.

**Description**

Conflicting app options are detected. Reset internally. Please refer to man page of that option for complete details.

**What Next**

Please check if any option is set unexpectedly.

---

## ZRT-625

(warn) Master cell %s has duplicated redundant library pin shapes at {%.3f %.3f} {%.3f %.3f} on layer %s.

**Description**

Duplicated redundant library pin shapes are found at the give location. The coordinates are relative to the library inside. These incorrect data will impact the router performance and DRC convergence.

**What Next**

Please remove the redundant pin shapes. Please also review the tool flow, they might be generated from incorrect library preparation flow.

---

## ZRT-626

(warning) High fanout net %s can affect runtime for antenna fixing.

**Description**

The design has high fanout nets and antenna fixing is turned on. Runtime of antenna check and fixing is affected by high fanout nets.

**What Next**

Please check the design and address the high fanout net issue. If antenna check is not needed for these nets, consider setting app option route.detail.max_antenna_pin_count.

Feedback

Option "route.detail.max_antenna_pin_count" specifies the maximum number of pins on a net on which antenna checking is performed.

---

## ZRT-627

(warning) Skipping route_group command as global_route_opt has been run.

**Description**

Message indicates that the tool detected a situation where global_route_opt has already been run. Therefore, route_group command will be skipped. Application option route.global.force_rerun_after_global_route_opt can be used to override this behavior.

**What Next**

No action necessary.

---

## ZRT-628

(info) Forcing rerun of route_group command after global_route_opt due to application option route.global.force_rerun_after_global_route_opt being set.

**Description**

Message indicates that the tool detected application option route.global.force_rerun_after_global_route_opt as setafter global_route_opt. In this case, route_group command will be forced to rerun. This will erase global routing data and therefore user must ensure rerunning global routing by setting route.global.force_rerun_after_global_route_opt to true.

**What Next**

No action necessary.

---

## ZRT-629

(Warning) Shape at (%.4g %.4g) on layer %s in frozen net %s (type %s) has no color.

**Description**

Found a shape on a frozen (usually locked) net has no dpt color on a dptColor enabled layer when router reads in the design.

**What Next**

Please look for the shape at the specified location to study why it has no color and fix the problem. Please note all shapes on a dptColor enabled layer in a frozen (usually locked)

Feedback

net should be assigned a color properly. Router does not change shape color in frozen nets.

---

## ZRT-630

(warn) Abutted feedthrough pin may be created on cell %s to connect with top level port %s that has different related supply nets.

**Description**

Hierarchical global router detects that the routing needs to create abutted feedthrough to connect with the top level port. However, the abutted child block and the top level port have different related supply nets.

**What Next**

Please review the floorplan and power domain configuration.

---

## ZRT-631

(warn) Net %s may create feedthroughs in unbufferable voltage area.

**Description**

Hierarchical global router detects that the net may need to route through unbufferable voltage area in order to be fully routed. It could be due to the floorplan issue which a voltage area for the given net is completely isolated by unbufferable voltage areas, or the applied routing constraints do not allow connectivity through bufferable voltage areas for the given net.

**What Next**

Please review floorplan and constraints to allow net connectivity within the bufferable voltage areas.

---

## ZRT-632

(warn) Option route.detail.force_end_on_preferred_grid is set to true due to techfile missing %s on layer %s.

**Description**

Forcing end on preferred grid since techfile is out-dated. Please refer to man page of that option for complete details.

**What Next**

Please check if the techfile is up-to-date.

Feedback

## ZRT-633

(Error) Zroute commands do not support designs with 40-bit NDM object Ids.

**Description**

If NDM has 40-bit object Ids, Zroute commands can't support it.

**What Next**

Please check your design for 40 bit objects. Modify the design to use 32-bit NDM objects to run Zroute commands.

## ZRT-634

(warn) Skipping non-manhattan pin cut shape at location {%f %f} on layer %s in master cell %s.

**Description**

Zroute does not support non-manhattan pin cut shapes.

**What Next**

Please check the database and fix the not supported shape.

## ZRT-635

(warn) Metal extensions generated with option route.detail.force_end_on_preferred_grid set to true are found in the design.

**Description**

Metal extensions forcing on preferred grid are found in this design. It could lead to longer wirelength and less optimized result with option route.detail.force_end_on_preferred_grid set to false.

**What Next**

Check if these out-of-date metal extensions are desirable.

## ZRT-636

(warn) Ignore %s due to %s.

Feedback

**Description**

Incorrect or conflicting options are detected. Reset internally. Please refer to man page of that option for complete details.

**What Next**

Please check if any option is set unexpectedly.

---

## ZRT-637

(warn) Lib-pin '%s' does not have required 'pattern_must_join' attribute. There are multiple pin shapes on metal layer '%s'.

**Description**

If the top layer terminals of a lib-pin are with multiple pin shapes, then the 'pattern_must_join' attribute should be set as true on this lib-pin.

**What Next**

If you want to connect all multiple pin shapes of the lib-pin, please update the 'pattern_must_join' attribute on lib-pin or set 'must_join_group' attribute on terminals.

---

## ZRT-638

(warn) Instance cell pin '%s' does not have required high performance or electromigration via ladder. Current via ladder list: %s.

**Description**

The instance cell pin's lib-pin has attribute 'is_em_via_ladder_required' set to true, and therefore it needs to have at least one high performance or electromigration via ladder in its via ladder list.

**What Next**

Please try one of below methods to fix the problem. 1. change current via ladder type by setting 'for_high_performance' or 'for_electro_migration' attribute. 2. add at least one high performance or electromigration via ladder to the instance cell pin's via ladder list 3. reset the 'is_em_via_ladder_required' attribute on its lib-pin.

---

## ZRT-639

(warn) The %s application option is set to %s; suggested value for the via ladder flow is %s.

Feedback

**Description**

For better via ladder insertion result, please use suggested app-option value

**What Next**

Please reset the app-option value.

---

## ZRT-640

(error) Cell %s is placed overlapping with %d other cells at {%.3f %.3f}.

**Description**

This warning message indicates that there are over 1000 other cells overlapping with this cell.

**What Next**

Please make sure all the cells are placed legally before running the router and run check_legality before running check_routability.

---

## ZRT-641

(warn) Current design technology file is inconsistent with the technology file used to create frame views of reference library %s. Update via regions for the reference frame views (in memory only) with current design technology file.

**Description**

Current design technology file is inconsistent with the technology file used to create frame views of the reference library. The tool will re-generate via regions by current technology file, but only in memory. The original via regions saved in the reference frame views, including the auto generated via regions by frame creation process or the user created via regions, will not be overwritten in the reference library. But the tool will use the new in memory via regions for later process.

**What Next**

Please confirm that current design technology file is correct.

---

## ZRT-642

(warn) %s.

Feedback

**Description**

The router detected unreasonable routing track settings. Please review routing track settings and the layer pitches in technology file.

There are three types of warning: 1. The step of a set of track is less than the pitch of layer defined in technology file. 2. The set of track is defined in a non-routing layer. 3. The quantity of tracks in a rectilinear chip area is unreasonably high. The reasonable quantity of tracks is less than or equal to the width or height of area divided by the pitch of layer. Please also check for redundant routing tracks in this area.

Incorrect routing track settings could degrade router performance and unexpected router behavior.

**What Next**

Please correct any incorrect track settings and layer pitch values.

---

## ZRT-643

(warn) Ignoring pin access route guide at (%.4f %.4f) (%.4f %.4f) with missing macro information.

**Description**

DRC checks for pin access route guides in Zroute require information about associated macros. If this information is not available for a route guides of this type, they will be ignored for "Metal over pin access route guide" DRC check.

**What Next**

Create the pin access route guides using command "derive_pin_access_route_guides" to have them associated with macro instances in the design.

---

## ZRT-644

(warn) Cannot find contact code that realizes sufficient line-end trimming at location {%.4f %.4f} on layer %s, skipping contact code exchange for %s.

**Description**

The message indicates that for the via at given location no contact code can be found that realizes sufficient line-end trimming, according to spacing value defined in trim_shapes command. The contact code is not exchanged and the original line-end spacing is maintained.

Feedback

**What Next**

Verify if tech file contains all contact codes necessary for trimming line-ends according to spacing values defined in trim_shapes command.

---

## ZRT-645

(warn) Invalid attribute of pattern_must_join/via_ladder object %s.

**Description**

The "shape_use" attribute of pattern_must_join/via_ladder object is invalid.

**What Next**

Please remove the invalid object or update the "shape_use" attribute of object. The valid value is "user_route".

---

## ZRT-646

(warning) %s

**Description**

Specified command will be deprecated. Please follow the instruction to update your script accordingly.

**What Next**

Please update scripts to remove the command specified, and migrate to new one if any.

---

## ZRT-647

(Warning) Current dominant scenario %s for timing driven route %s is different from dominant scenario for the previous timing driven route which is %s.

**Description**

The message tells dominant scenario which drives timing driven route changed from the previous timing driven route. Timing driven route behavior will be different from the previous timing driven route. Especially, in RDE flow, if the dominant scenario for RDE capturing is different from initial route stage, it will impact the correlation.

**What Next**

Use route.common.focus_scenario to control dominant scenario for timing driven route

Feedback

## ZRT-648

(warning) %s

**Description**

A situation may need case-by-case analysis to ensure the correctness.

**What Next**

A warning that needs user's attention.

## ZRT-649

(error) Invalid cut layer %s specified for route.common.redundant_via_include_weight_group_by_layer_name or route.common.redundant_via_exclude_weight_group_by_layer_name.

**Description**

A string supplied to redundant_via_include_weight_group_by_layer_name or redundant_via_exclude_weight_group_by_layer_name does not name a valid cut layer.

**What Next**

Please review the layer names supplied to these options.

## ZRT-650

(error) Invalid weight group %d specified for route.common.redundant_via_include_weight_group_by_layer_name or route.common.redundant_via_exclude_weight_group_by_layer_name.

**Description**

A weight group supplied to redundant_via_include_weight_group_by_layer_name or redundant_via_exclude_weight_group_by_layer_name is invalid or out of range.

**What Next**

Please review the weight groups supplied to these options.

## ZRT-651

(warn) Ignore rule %s on layer %s.

Feedback

**Description**

This warning message occurs because unexpected value is set for the specified rule. Ignore it.

**What Next**

Please check to see if the rule is correctly defined on this layer.

---

## ZRT-652

Error: rtPsvMngr::splitGrh is called on non-splittable shape. (%s)

**Description**

rtPsvMngr::splitGrh is called on non-splittable shape which may cause unexpected result.

**What Next**

Make sure to call rtPsvMngr::isSplitable to check if a shape is valid before calling rtPsvMngr::splitGrh.

---

## ZRT-653

(Error) An obstrcution in cell has no valid color. (%s)

**Description**

An obstrcution in cell has no valid color.

**What Next**

If the cell is a macro, user should have assigned all obstructions in the macro cell before routing. Please fix the input issue and re-run the tool. If it is not a macro, please pass a test case to router team to look into why the color is missed.

---

## ZRT-654

(info) Ignoring patch clean-up in iteration %d due to option route.detail.min_number_iterations_after_last_patch_clean_up_iteration.

**Description**

Patch clean-up is skipped in some iterations due to conflict to route.detail.min_number_iterations_after_last_patch_clean_up_iteration.

Feedback

**What Next**

Please review settings for route.detail.patch_clean_up_iteration_frequency, route.detail.patch_clean_up_max_number_iterations, route.detail.patch_clean_up_iterations and route.detail.min_number_iterations_after_last_patch_clean_up_iteration.

---

## ZRT-655

(error) In rule %s, var via %s is an special via that we don't support.

**Description**

This error message indicates that the NDR via is an special via.

**What Next**

Please specify the NDR via using non special ones.

---

## ZRT-656

(warn) Ignoring via at (%.3f %.3f) with missing via definition.

**Description**

Vias without a via definition in NDM will be ignored by Zroute.

**What Next**

Please check and correct the input vias with missing via definition.

---

## ZRT-657

(Warning) Rule %s has unexpected setting on lay %s.

**Description**

This warning message indicates that the rule has unexpected setting on layer for the specified rule.

**What Next**

Please check the technology file and make sure the rule is correct.

---

## ZRT-658

(Warning) Top level routing blockage on layer %s at (%.3f %.3f) (%.3f %.3f) is not marked as zero spacing. This may impact runtime of routing.

Feedback

**Description**

This warning message indicates that there are routing blockages at top level that are not marked as zero spacing. This can impact the DRC convergence and runtime of detail route.

**What Next**

Please check the input and set the blockages as zero spacing.

---

## ZRT-659

(Information) Ignoring %s at (%.3f %.3f) as it is marked as %s.

**Description**

Vias with this marking are ignored by routing commands.

**What Next**

This message is for the user to be aware of the ignored vias when reviewing the design.

---

## ZRT-660

(error) Failed to read via definition because there are too many of them in the design.

**Description**

The design has more via definitions defined than Zroute can support.

**What Next**

Check the design to determine whether this large number of via definitions was created due to a problem in the flow.

---

## ZRT-661

(Warning) An invalid global route shape is found. %s.

**Description**

When a shape, via or G-Link has these attributes: 1. Shape_use is global_route. 2. Owner net is none.

It is considered as an invalid global route shape. It could negatively impact the router's performance and quality of result.

Such global route shapes are deleted if the routing command being run involves creating or modifying the routes.

Feedback

**What Next**

Please review the reason invalid global route shapes get created before the routing command.

---

## ZRT-662

(warn) Eco route ignores -open_net_driven option when -nets option is defined.

**Description**

When -nets option is used, the -open_net_driven option will be ignored.

**What Next**

Update script to remove -open_net_driven option when -nets option is used.

---

## ZRT-663

(warn) router does not support the rotated unfixed custom via with def %s at (%.3f %.3f) and will ignore it.

**Description**

The tool issues this warning when it reads in an unfixed custom via while reading the vias from the database.

**What Next**

Double-check the usage of this via.

---

## ZRT-664

(error) Found global route in frozen nets in the input to track assignment. %s

**Description**

This error message indicates that there are some global routes in frozen nets in the input to track assignment. As frozen nets are skipped by track assinment, these global routes will not be assigned correctly and cause detail router error.

**What Next**

Please check the database and fix the issue of global routes in frozen net.

Feedback

## ZRT-665

(Warning) The input to track assignment has out-of-chip-boundary shape or graph node. %s

### Description

This error message indicates that there are some out-of-chip-boundary shapes or graph nodes in the input to track assignment. This may result in out-of-chip-boundary shapes in the output of track assignment.

### What Next

Please fix the issue in the input to track assignment.

## ZRT-666

(warn) Change the value of %s because the current setting is not compatible for the given technology node.

### Description

The warning message indicates that the value of app options is not compatibile for the given technology node so tool helps to correct the value.

### What Next

Even tool helps to correct the value, Users may need to check their script and setting.

## ZRT-667

(warn) The value of %s is different from what mega switch wants. It might degrade the result in the given technology node.

### Description

The warning message indicates that the value of app options is different from what mega switch wants. Tool still use the current value but it might degrade the result

### What Next

Users may need to check their script or setting to see if the value of this app options is what they want.

## ZRT-668

(error) This design has %d gCells with overflow >= %d. Please open global route congestion map to review it.

Feedback

**Description**

This error message indicates that there are some high overflow gCells in global route congestion map. In general, high overflow gCell areas might be unable to get a good routing result. In some worst cases, detail route stage hang might happen due to severe overflow.

**What Next**

Please open global route congestion map in GUI to review this design.

Please be noted, this check runs a fast global route to get a coarse result. When the user needs an accurate congestion map, please run report_congestion command.

---

## ZRT-669

(warn) Instant secondary PG pin %s at (%.3f %.3f) is in the instant cell's primary PG net %s. The primary PG pin is %s.

**Description**

Should not use zroute to route primary PG net.

**What Next**

Please use power tools to route PG pins to the primary PG net.

---

## ZRT-670

(warn) Number of Chunks (num = %d) is less than number of Workers (num = %d). Some workers will be pending.

**Description**

The number of chunks is less than the number of workers in this case. some workers will be pending.

**What Next**

Please review if we could start less number of workers to avoid idle workers.

---

## ZRT-671

(warn) Cluster at {%.3f %.3f} {%.3f %.3f} on layer %s exceeds max cluster limit.

**Description**

The cluster size exceeds the sav_on_grid_max_cluster_size for SAV rule. It could result in incorrect DRC checking and bad routing convergence.

Feedback

**What Next**

Please increase the sav_on_grid_max_cluster_size limit.

---

## ZRT-672

(warn) Automatically switch the option value of "%s" to true in "%s" mode of option "%s". \n.

**Description**

The option clip_rule_patch_mode is enabled by default and works under the option value of treat_preferred_grid_on_clip_rule_layer_as_clip_grid as true. If users set the option value of treat_preferred_grid_on_clip_rule_layer_as_clip_grid as false, it will automatically modified to true.

**What Next**

Please set the option value of treat_preferred_grid_on_clip_rule_layer_as_clip_grid as true if users do not want to see this warning message.

---

## ZRT-673

(warn) (rtapi) (@GOE) cannot find the MifsNet for the ndmBlkNet %s of driver term %s

**Description**

Error found in rtapi module. Optimization passes a net to router to route while router sees it is an invalid net.

**What Next**

Please recheck if the net have physical pins.

---

## ZRT-674

(warn) Cannot find any MifsNet to route in routeNetGrp!

**Description**

Error found in rtapi module. Optimization passes nets to router to route while router does not see any physical nets.

**What Next**

Please recheck if the net have physical pins. There may not be QoR impact as they are not routable nets.

Feedback

## ZRT-675

(warn) Cannot find MifsNet for the ndmBlkNet (%s) passed into routeNetGrp!

**Description**

Error found in rtapi module. Optimization passes the net to router to route while router does not consider it as a physical net.

**What Next**

Please recheck if the net have physical pins. There may not be QoR impact as it is not a routable net.

## ZRT-676

(warn) (rtapi) (@GOE): GIMRD via-ladder info corrupted

**Description**

Warning found in rtapi module. Please fix the described error.

**What Next**

Correct the described error.

## ZRT-677

(warn) (rtapi) (@GOE): With GR via-ladders enabled, net has a DR via-ladder: %s

**Description**

Warning found in rtapi module. Please fix the described error.

**What Next**

Correct the described error.

## ZRT-680

(error) voltage precision is not defined.

**Description**

The voltage precision is not defined in the tech file. This can cause incorrect voltage spacing violations.

Feedback

**What Next**

Please define unitVoltageName and voltagePrecision in the tech file.

---

## ZRT-681

(error) The via %s is set as default via but did not specify upperLayerMaskConstraint or lowerLayerMaskConstraint, please check the tech file to correct specified via definition

**Description**

The default via must specify upperLayerMaskConstraint or lowerLayerMaskConstraint

**What Next**

Please check the tech file to correct specified via definition

---

## ZRT-682

(warn) Fall back to non-distributed routing for better runtime.

**Description**

Due to the sparse distribution of input DRCs, disable distributed routing to get better runtime.

**What Next**

Please check if the design is good for distributed routing.

---

## ZRT-683

(warning) A net %s connects more than %d ports (Number of ports: %d). This may increase routing runtime.

**Description**

A net contains a large number of ports. This is atypical and may increase routing runtime.

**What Next**

If this is not expected, please reduce the number of pins for the port.

---

## ZRT-684

(Error) Incorrect %s routing guide is detected in this design. %s.

Feedback

**Description**

Please check the problematic routing guides and correct issues to proceed.

**What Next**

Correct the routing guide and rerun routing.

---

## ZRT-685

(Warn) Incorrect setting %s is detected in app option %s. %s.

**Description**

Please check the problematic setting and correct it.

**What Next**

Correct the app option setting and rerun.

---

## ZRT-686

(warning) Routing of PG net %s could be slow due to strict fanout constraints and/or non-optimal topology

**Description**

The routing of this PG net may take lots of time. Reasons are:

1. Maximum fanout limit could be too strict. More than 10 separate components need to be generated.
2. The ports which are over 10 cells row height away from a preroute account for more than 50 percent of the total unrouted ports.

.SH WHAT NEXT Please check PG structure setup for the net, or/and setting of number_of_secondary_pg_pin_connections.

---

## ZRT-687

(warn) Insufficient upper routing layers to connect pattern_must_join pin '%s/%s' on layer '%s'. Ignoring pattern_must_join.

**Description**

A pin on an upper layer is set as pattern_must_join. A pattern must join connection requires at least one metal layer above the pin. The required number of layers do not exist, so the router is ignoring the pattern_must_join attribute.

Feedback

**What Next**

Remove the pattern_must_join attribute on the library pin.

---

## ZRT-688

(warn) Skipping antenna fixing for net %s.

**Description**

User set some nets to be skipped for antenna fixing using app option route.detail.skip_antenna_fixing_for_nets. Those nets are reported in this message.

**What Next**

Check if it is intended to skip antenna fixing for the net.

---

## ZRT-689

(warning) The value of app option "route.common.concurrent_redundant_via_mode" is not compatible for the given technology node. This may increase routing runtime and reduce routability.

**Description**

The "route.common.concurrent_redundant_via_mode" option is not recommended to set for the given technology node. This may increase routing runtime and reduce routability.

**What Next**

Correct the app option setting and rerun.

---

## ZRT-690

(error) Zroute does not support more than %d metal layers in 3DIC Compiler.

**Description**

In 3DIC shell, there is a limit of number of routing layers that Zroute commands support.

**What Next**

Please modify number of metal layers for routing in the design or contact Synopsys support.

Feedback

## ZRT-691

(warn) Distance of neighboring grids of congestion map is too large between %s %d and %d (%d gcells)

**Description**

A large distance between neighboring grids may cause slowness and error in routing.

**What Next**

Check the setup of the congestion map to ensure reasonable distance between neighboring grids.

## ZRT-692

(warning) A net %s connects more than %d ports (Number of ports: %d), and the value of app option %s is %s. This may increase routing runtime.

**Description**

A net contains a large number of ports, and the app option setting is not recommended to set for this kind of net. This may increase routing runtime.

**What Next**

If this is not expected, please reduce the number of ports for the net or correct the app option setting.

## ZRT-693

(Information) Make adjustment on tech file : %s.

**Description**

This information indicates that the setting of endToEndForbiddenSpaceRangeMin/ endToEndForbiddenSpaceRangeMax and lengthPrecision/gridResolutionin tech file are inconsistent. Tools will do the conversion automatically on endToEndForbiddenSpaceRangeMin/endToEndForbiddenSpaceRangeMax.

## ZRT-694

(Information) Non preferred direction routing is disabled by option on layer %s.

**Description**

Non preferred direction routing is disabled by application option route.common.disable_non_preferred_direction_routing_by_layer_name.

Feedback

## ZRT-695

(warning) secondary PG net %s has cluster size of %.1f around location (%.4f %.4f), and the value of app option %s is %.1f. This may increase routing runtime.

**Description**

A net has a way large cluster than expected. This makes detailed routing difficut to fix secondary PG violations and it would increase routing runtime.

**What Next**

Please check net PG structure to see if it is setup properly. You can also save global routing and track assignment results to see if the problem aleady appears at these stages.

## ZRT-696

(error) router does not support the rotated unfixed custom via with def %s at (%.3f %.3f) and will ignore it.

**Description**

The tool issues this error when it reads in an unfixed custom via while reading the vias from the database.

**What Next**

Check the usage of this via.

## ZRT-697

(warning) secondary PG net %s has cluster size of %d around location (%.4f %.4f), and the value of app option %s is %d. This may increase routing runtime.

**Description**

A net has a way large cluster than expected. This makes detailed routing difficut to fix secondary PG violations and it would increase routing runtime.

**What Next**

Please check net PG structure to see if it is setup properly. You can also save global routing and track assignment results to see if the problem aleady appears at these stages.

## ZRT-698

(warning) Routing of PG net %s could be slow due to power PG structure coverage over ports.

Feedback

**Description**

The routing of this PG net may take lots of time. Reasons are:

1. The bounding box of PG structure only covers less than 50% percent of bounding box of ports.

.SH WHAT NEXT Please check PG structure setup for the net, or/and setting of number_of_secondary_pg_pin_connections.

---

## ZRT-699

(error) Definition of ViaRule %s will exceed the maximum allowed number of definitions. No more than %d are allowed.

**Description**

The number of via ladder definitions exceeds the total number supported. Too many ViaRule via ladders defined in the tech file or using create_via_rule.

**What Next**

Reduce the number of via ladder definitions.

---

## ZRT-700

(warning) %s layer %s has no enough gcell capacity for promoted nets.

**Description**

Too many promoted nets with layer constraints and no enough capacity for routing.

**What Next**

Reduce the number of promoted nets.

---

## ZRT-701

(Error) Potential memory overflow. Detail: %s

**Description**

Tool finds a situation that may cause memory overflow. More detail is printed.

**What Next**

Tool finds a situation that may cause memory overflow. Please check if the chip floor plan size is reasonable. If the chip size has problem, please fix the issue. Otherwise, please pass a test case to router team.

Feedback

## ZRT-702

(warn) Skipping the rectangle_only route guide located at {{%.3f %.3f} {%.3f %.3f}} because it doesn't cover the design boundary in the layer's preferred direction.

**Description**

The rectangle_only route guide need to cover the design boundary in the layer's preferred direction. If not, we will skip this route guide.

**What Next**

Please adjust the size of the route guide.

## ZRT-703

(info) Option route.detail.force_end_on_preferred_grid will be ignored since none of the layers have preferred grid.

**Description**

Ignore option route.detail.force_end_on_preferred_grid since there are no preferred grids in this design.

## ZRT-704

(warn) The value of %s is different from default value or mega switch value. Please make sure the value is expected.

**Description**

The warning message indicates that the user value if different from default value or mega switch value. Please make sure the value is expected as this kind of app option can easily impact the result.

**What Next**

Users may need to check their script or setting to see if the value of this app options is what they want.

## ZRT-705

(Error) Tech file error: Layer %s. %s

**Description**

Error found in the tech file design rule definition. The rule will be skipped.

Feedback

**What Next**

Correct the tech file error.

---

## ZRT-706

(Information) When applicable layer based tapering will taper up to %.2f in distance from the pin.

**Description**

Distance control for layer based tapering is turned on.

---

## ZRT-707

(Information) When applicable Min-max layer allow_pin_connection mode will allow paths of length %.2f outside the layer range.

**Description**

Distance threshold for min-max layer allow pin conn mode.

---

## ZRT-708

(warn) net %s has opens.

**Description**

The warning message indicates that the net is open in the end of global route.

**What Next**

Users may need to check whether and why the net is open and to solve open net issue.

---

## ZRT-709

(info) Detail Router is running in early mode.

**Description**

In early mode, Detail Router will reduce rip-up and rerouting in some locations to reduce runtime. This may have an impact on the DRC counts.

**What Next**

See app option route.detail.mode for different modes

Feedback

## ZRT-710

(Error) Closed routing DB due to an exceptional situation. (%s)

**Description**

Routing data-base needs to be shut down due to an unexpected situation to prevent crash. User should look for root cause and fix it.

**What Next**

Please look for the script or flow issues and discuss with the flow owner to fix it.

## ZRT-711

(info) The value of the app option "%s" is ignored. New value "%s" will be used for DR early mode.

**Description**

In early mode, Detail Router will trun some settings on by default.

**What Next**

See the reported app option for detailed information.

## ZRT-712

(info) Using early mode ZGR.

**Description**

This message reports that early mode ZGR is activated.

**What Next**

No action is required.

**See Also**

- set_host_options

## ZRT-713

(warn) Unable to launch distributed routing as the library is opened in readonly mode.

Feedback

### Description

The distributed routing requires the library to be opened in non-readonly mode. It will fall back to non-distributed routing.

### What Next

Open the library in non-readonly mode.

---

## ZRT-714

(warn) Recovered from an unexpected state. More information may be provided in following lines.

### Description

When a segmentation fault or a floating-point exception is encountered in Zroute detail route or via ladder stage, if the crash can be managed and skipped, the tool can complete the rest jobs and continue. ZRT-714 informs the user a crash is managed and skipped. The stack trace is printed if available. The tool terminates with ZRT-715 when the amount of ZRT-714 in a command exceeds a configurable threshold. If a crash cannot be managed and skipped, the tool terminates.

### What Next

When a ZRT-714 is issued in detail route or via ladder, the user can continue the work. The DRC closure may not be optimal due to skipped jobs. The DRC report may be inaccurate due to skipped jobs. The NDM database remains valid after a ZRT-714 recovery. It is recommended to run check_routes after a ZRT-714. When check_routes has ZRT-714, the DRC report is inaccurate due to skipped jobs. It is recommended to report a ZRT-714 with logs and data.

---

## ZRT-715

(error) The occurrence count of unexpected state recovery hits the maximum.

### Description

The tool terminates with ZRT-715 when the amount of ZRT-714 in a Zroute command exceeds the threshold. With default threshold value, many ZRT-714 may indicate problems in input data or persistent bugs in the router. The maximal recovery threshold is controlled by app_option route.common.unexpected_state_recovery_threshold

### What Next

Please review the input data and logs. The user could change the value of app_option route.common.unexpected_state_recovery_threshold. It is recommended to report ZRT-714 with logs and data.

Feedback

## ZRT-716

(Error) (rtapi) %s

**Description**

Error found in rtapi module. Please fix the described error.

**What Next**

Correct the described error.

## ZRT-717

(warn) Instance pin %s has a via ladder constraint but also has a physical custom via ladder attached. Current via ladder constraint list: %s. The custom via ladder may be removed by refresh_via_ladders and related operations.

**Description**

Custom via ladders are usually not expected on pins that have a conventional via ladder constraint.

**What Next**

Please confirm that the via ladder constraints are expected on the pin, and be aware that the router may remove the custom via ladder. If the custom via ladder is not described by a ViaRule in the constraint, the custom via ladder will be removed.

## ZRT-718

(Information) When applicable Min-max layer allow_pin_connection mode will allow paths of length %.2f outside the layer range on clock nets.

**Description**

Distance threshold for min-max layer allow pin conn mode on clock nets.

## ZRT-719

(warn) Could not find corresponding default via contactCode on %s layer enclosure of cut layer to match the track width constraint %.7g on layer %s.

**Description**

Routing layer has track width constraint less than default width. Router needs to update this track width as default width. But the corresponding track width via contactCode is

Feedback

missing or is not a default contactCode, which will cause router to use the wrong via during routing.

**What Next**

Please update the tech file to add or set the via contactCode that match the track width constraint as default contactCode.

---

## ZRT-720

(Warning) Port %s at {{%.3f,%.3f} {%.3f,%.3f}} is out of boundary. Cannot be routed.

**Description**

The port is out of boundary.

By default, routing commands print only 10 messages of this type. If user wishes to have more messages of this type printed to log file, set_message_info command is to be used.

Example is: set_message_info -id {ZRT-720} -limit 300

**What Next**

Please check the design.

---

## ZRT-721

(warn) An invalid shape is ignored because %s

**Description**

An invalid shape is ignored as described reason.

**What Next**

Please review and correct the shape.

---

## ZRT-722

(info) Applying settings based on technology node %s.

**Description**

The reminder of the technology node user had specified.

Feedback

## ZRT-723

(warning) Invalid or missing min cut spacing value for contactCode (%s). New value %g is used instead.

### Description

The min cut spacing value of the contactCode is invalid or missing. Router will use the min spacing value of the corresponding cut layer to replace it. Default value 0 will be used if the min spacing value is invalid too.

### What Next

Please check the contactCode value in the tech file and correct it if needed.

## ZRT-724

(warn) Skipping antenna analysis for net %s.

### Description

User set some nets to be skipped for antenna analysis using app option route.detail.skip_antenna_analysis_for_nets. Those nets are reported in this message.

### What Next

Check if it is intended to skip antenna analysis for the net.

## ZRT-725

(warn) Pin shape (cell %s, pin %s) is not on a routing layer %s.

### Description

The tool issues this error when the layer of the specified pin is not within the minimum and maximum routing layers. Please check the setting of *set_ignored_layers* command or the routing layer setting for this net. The router cannot connect to the pin shape under these conditions.

### What Next

Please check the setting of *set_ignored_layers* command or the routing layer setting for this net.

### See Also

- set_ignored_layers

Feedback

## ZRT-726

(warning) PG net %s has rail in the region (%.4f %.4f) (%.4f %.4f) on layer %s that crosses even number of tracks.

**Description**

PG rail with DPT mask crosses even number of tracks. It might reduce the routing resource.

**What Next**

Re-build the track or shift the rail.

## ZRT-727

(warning) the routing tracks on DPT layer %s are without any mask information.

**Description**

When the option route.common.color_based_dpt_flow is truned on, it is necessary to specify mask information for the routing tracks on DPT layer.

**What Next**

Please check your tech file and scripts to make sure that you set the mask information for the tracks on DPT layers.

## ZRT-728

(warning) Layer %s has no routing tracks.

**Description**

The given layer does not have any routing tracks. This prevents the router from assigning wires and vias on the layer.

**What Next**

Please check if the routing tracks have been correctly defined on the layer.

## ZRT-729

(warning) Track at {%.3f %.3f} {%.3f %.3f} on layer %s is out of boundary.

**Description**

The track is out of boundary. It may impact the routing result.

Feedback

**What Next**

Please check if the routing tracks have been correctly defined on the layer.

---

## ZRT-730

(warning) The value of stackViaPairLayerCount on layer %s is invalid.

**Description**

The value of stackViaPairLayerCount is not valid (is 0 or larger than the max number of stack levels of that via layer) Zroute ignores this rule to prevent unexpected behavior.

**What Next**

Please check if the value of stackViaPairLayerCount is correctly defined on the via layer.

---

## ZRT-731

(error) Reset %s on layer %s since the option route.detail.force_end_on_preferred_grid is set to true.

**Description**

Reset natural end on preferred grid center rule since force end on prferred grid is enabled.

**What Next**

Please check if the techfile is up-to-date.

---

## ZRT-732

(warning) The pg_pin_weight was lowered for %d pins of the secondary PG nets. These pins are unroutable with the original pg_pin_weight values due to missing enough tap-points. Use check_routes to highlight their locations.

**Description**

The pg_pin_weight was lowered for pins of the secondary PG nets. These pins are unroutable with the original pg_pin_weight values due to missing enough tap-points. Use check_routes to highlight their locations.

**What Next**

Please check if the pg_pin_weight of pins is up-to-date or if tap-points is enough.

Feedback

## ZRT-733

(warning) checkMetalEndThreshold is ignored for cut Metal layer %s since the endToEndSpaceMaxWidthThreshold of metal layer %s is not specified.

**Description**

When the checkMetalEndThreshold of the cutMetal layer is specified as 1, the endToEndSpaceMaxWidthThreshold of the corresponding metal layer must also be specified.

**What Next**

Please check and correct the techfile.

## ZRT-734

(warning) The option "route.detail.limit_m0_routing_pin_extension_for_1cpp_cell_gap" is defined but metal layer %s does not have preferred grid.

**Description**

When the option "route.detail.limit_m0_routing_pin_extension_for_1cpp_cell_gap" is defined, the preferred grid on metal layer must also be specified.

**What Next**

Please check and correct the option or specify the preferred grid on metal layer.

## ZRT-735

(warn) Option %s is set to %s while %s is not set to %s.

**Description**

Conflicting app options are detected. Please refer to man page of that option for complete details.

**What Next**

Please check if any option is set unexpectedly.

## ZRT-736

(error) Cannot run this command for multiple die.

Feedback

**Description**

The design is multiple die design that contains base die and interposer die and all zroute commands should stop for this kind of design.

**What Next**

Please modify the design or contact Synopsys support.

---

## ZRT-737

(warn) %s command line option "%s" has no effect since route.auto_via_ladder.apply_app_options_to_insert_via_ladders_command is true (default).

**Description**

The specified command line option will be obsoleted in a future release. Please use the route.auto_via_ladder.* app option with the same name.

**What Next**

Please migrate scripts to use route.auto_via_ladder.* app options for all insert_via_ladders and verify_via_ladders settings.

If the command line options must be used, please set "route.auto_via_ladder.apply_app_options_to_insert_via_ladders_command false." However, this temporary support will disappear in a future release.

---

## ZRT-738

(warn) route.auto_via_ladder.apply_app_options_to_insert_via_ladders_command will become obsolete in %s and will be treated as true by default.

**Description**

route.auto_via_ladder.apply_app_options_to_insert_via_ladders_command is now true by default and does not need to be explicitly set. Users are encouraged to migrate all scripts to use route.auto_via_ladder.* app options, and avoid use of any redundant insert_via_ladders or verify_via_ladders command line options.

Legacy command line controls can still be used by setting "route.auto_via_ladder.apply_app_options_to_insert_via_ladders_command false" but support will be made obsolete in a future release.

**What Next**

Please migrate scripts to use route.auto_via_ladder.* app options for all insert_via_ladders and verify_via_ladders settings.

Feedback

If the command line options must be used, please set "route.auto_via_ladder.apply_app_options_to_insert_via_ladders_command false." However, this temporary support will disappear in a future release.

---

## ZRT-739

(warning) Cell %s port %s contains more than %d pin shapes (Number of pin shapes: %d). This may increase routing runtime.

**Description**

A port contains a large number of pin shapes. This is atypical and may increase routing runtime.

**What Next**

If this is not expected, please reduce the number of pin shapes for the port.

---

## ZRT-740

(warning) %s is unplaced. This may impact routing result.

**Description**

A port is unplaced. This is atypical and may impact routing result.

**What Next**

If this is not expected, please fix the issue.

---

## ZRT-741

(warn) Reset %s on layer %s since there is no preferred grid defined.

**Description**

Reset natural end on preferred grid center rule since there is no preferred grid defined.

**What Next**

Please check if the techfile is up-to-date.

---

## ZRT-742

(warning) %s track is not on center of pin %s. This may impact via_ladder insertion. Please enable app-option route.auto_via_ladder.generate_center_track_on_off_grid_pattern_must_join_pin_shapes.

Feedback

**Description**

This track is not on center of pin. This is atypical and may impact via_ladder insertion.

**What Next**

If this is expected, please enable app-option route.auto_via_ladder.generate_center_track_on_off_grid_pattern_must_join_pin_shapes.

---

## ZRT-743

(error) Track at {%.3f %.3f} {%.3f %.3f} on layer %s without preferred grid is found. Please make sure track is set up properly.

**Description**

This layer has enablePrefGrid but corresponding track does not have consistent preferred grid setting.

**What Next**

Please check tracks are created correctly.

---

## ZRT-744

(Error) Invalid LEE blockage metal layer found. Please make sure the -exclude_layers of create_line_end_extension_blockages command is set up properly.

**Description**

Error out due to invalid LEE blockage metal layer specified.

**What Next**

Please check if the -exclude_layers of create_line_end_extension_blockages command is set up correctly.

---

## ZRT-745

(Error) Remove cut metals failed since %s.

**Description**

Command remove_cut_metals failed. The cut metals cannot be deleted or the design has no cut metal.

**What Next**

Please check the cut metals in the design accroding to the reported message.

Feedback

## ZRT-746

(error) Incorrect table size for syntax %s on layer %s.

**Description**

Set the correct table size for the specified syntax on that layer.

**What Next**

Please check if the table size of the specified syntax is correct.

## ZRT-747

(warning) Cell %s contains more than %d “forbidden_preferred_grid_extension” routing guides (%d). This can increase routing runtime; you might want to regenerate the frame view. The internal engine will ignore more than threshold route guides.

**Description**

A cell contains a large number of “forbidden_preferred_grid_extension” routing guides. This is atypical and might increase routing runtime.

**What Next**

If this is not expected, regenerate the frame view.

**See Also**

- check_workspace
- create_frame

## ZRT-748

(warn) Incorrect value or syntax missed for %s as %s has been defined on layer %s.

**Description**

Set the correct value of the syntax or add the required syntax on that layer.

**What Next**

Please check if the value of specified syntax is correct or any required syntax missed.

## ZRT-749

(error) Invalid syntax value %s found in %s on layer %s.

Feedback

**Description**

Set the valid value for the specified syntax on that layer.

**What Next**

Please check if the value of the specified syntax is correct.

---

## ZRT-750

(warning) Net %s has a min_layer and max_layer that are identical (%s).

**Description**

A net has a min layer and max layer that are identical and may be overly restrictive.

**What Next**

Relax either the min_layer or the max_layer to avoid DRC convergence issues.

---

## ZRT-751

(warning) App option "route.common.prefer_via_cuts_of_default_vias_as_ndr_vias" is set to TRUE, but cannot found user defined ndr via of rule (%s) that has cut size matching the default vias on layer %s.

**Description**

If app option "route.common.prefer_via_cuts_of_default_vias_as_ndr_vias" is set, user defined vias not matching the cut size of default vias will be filterd out. This warning indicates that there is no such valid user defined via, and all user defined vias will still be used.

**What Next**

Please check the routing rule and define the valid vias.

---

## ZRT-752

(error) %s %s of top level net %s is out of specified routing bounding box.

**Description**

This error message indicates that the specified routing bounding box should contain top level terminal or cell instance pin of top level net.

**What Next**

Please specify appropriate routing bounding box.

Feedback

## ZRT-753

(warning) Post-route command "%s" is disabled since ZRoute is running in early mode.

**Description**

Post-route commands/operations are disabled in ZRoute early flow to prevent long runtime on early designs. The early fow was set by "set_qor_strategy -mode early_design".

**What Next**

Please update the early flow script to exclude post-route commands.

## ZRT-754

(error) %s, options %s and %s cannot be used together.

**Description**

User specified options are mutually exclusive.

**What Next**

Please change the options.

## ZRT-755

(warn) The value of %s is different from mega switch value. Reset it to %s on %s to accomodate both user setting and mega switch.

**Description**

Conflicting options are detected. Reset internally.

**What Next**

Please check if any option is set unexpectedly.

## ZRT-756

(info) Dirty region identified at {{%.4f %.4f} {%.4f %.4f}} due to %s violation explosion.

**Description**

A region with extremely high violation number is detected which could cause hang in detailed routing or DRC checking.

Feedback

**What Next**

Please review the region of interest and see if any unexpected input in there.

---

## ZRT-757

(warning) When running the Early GR flow the removal of global routes shapes is prevented.

**Description**

The functionality of commands which are removing global route shapes is restricted in the Early GR flow.

---

## ZRT-758

(warning) In rule %s, var %s set to 0 on layer %s.

**Description**

This warning message indicates that the taper distance and/or the driver taper distance set to 0.

**What Next**

Please specify the NDR taper (driver) distance using non zero ones.

---

## ZRT-759

(warning) Convert via cut routing blockage {%.3f %.3f} {%.3f %.3f} on layer %s to zero spacing based on EDC records.

**Description**

The via cut blockage was identified as invalid blockage (ZRT-576) in check_routability and recorded in early data check (EDC). The blockage will be converted to zero spacing during routing in EDC repair mode.

**What Next**

See ZRT-576, report_early_data_checks

---

## ZRT-760

(warning) Remove shape {%.3f %.3f} {%.3f %.3f} on layer %s based on EDC records because of no net information.

Feedback

**Description**

The shape is without any net information and is recorded in early data check (EDC). The shape will be removed during routing in EDC repair mode.

**What Next**

See report_early_data_checks

---

## ZRT-761

(warning) pin is off track in the region (%.4f %.4f) (%.4f %.4f) on layer %s.

**Description**

This pin is off track.

---

## ZRT-762

(warning) Please run the command "check_routability" before running Zroute commands.

**Description**

When the design or setting is changed, please run the command "check_routability" before running Zroute commands. The command can detect incorrect desin and setting. Please fix unexpected error to get better result of router.

**What Next**

Run the command "check_routability" before running Zroute commands.

---

## ZRT-763

(warning) Cell %s is placed overlapping with other cells at {{%.3f %.3f} {%.3f %.3f}}.

**Description**

This warning message indicates that there are other cells overlapping with this cell. This is a roughly check for overlapped cell. For detailed information, please run "check_legality".

**What Next**

Please make sure all the cells are placed legally before running the router and run check_legality before running check_routability.

---

## ZRT-764

(warning) PG net %s has opens.

Feedback

**Description**

This warning message indicates the open PG nets. This is a roughly check. For detailed information, please run "check_lvs".

**What Next**

Please make sure all PG nets are connected and run check_lvs to fix the error.

---

## ZRT-765

(warning) Net %s skips pin-to-pin routing because the distance of the psv pair is larger than %d gcells (psv pair %d: from node %d to node %d, distance = %d).

**Description**

The open of the given is too large for incremental routing.

---

## ZRT-766

(warning) Net %s skips pin-to-pin routing because the distance of the psv pair is larger than %d gcells.

**Description**

The open of the given is too large for incremental routing.

---

## ZRT-767

(warning) Net %s rips up all edges because the distance of the psv pair is larger than %d gcells.

**Description**

The open of the given is too large for incremental routing.

---

## ZRT-768

(warning) All NDM blockages in cell %s are skipped based on EDC record.

**Description**

There are too many routing blockages detected in this cell in check_routability and recorded in early data check (EDC). These routing blockages will be skipped in EDC repair mode.

Feedback

**What Next**

See report_early_data_checks

---

## ZRT-769

(warning) All “forbidden_preferred_grid_extension” routing guides in cell %s are skipped based on EDC record.

**Description**

There are too many “forbidden_preferred_grid_extension” routing guides detected in this cell in check_routability (ZRT-747) and recorded in early data check (EDC). These routing guides will be skipped in EDC repair mode.

**What Next**

See ZRT-747, report_early_data_checks

---

## ZRT-770

(warning) All NDM existing duplicate shapes in block %s are removed based on EDC records by running the command "check_duplicates -remove".

**Description**

Redundant duplicated PG shapes are identified in check_routability and recorded in early data check (EDC). The shapes will be removed in EDC repair mode.

**What Next**

See ZRT-618, report_early_data_checks

---

## ZRT-771

(warn) %s is high fanout net, going to set it to locked.

**Description**

This warning occurs when there is high fanout net, and it is going to be set to locked.

**What Next**

See report_early_data_checks

---

## ZRT-772

(warn) %s is the net connecting to blocked ports, going to set it to locked.

Feedback

**Description**

This warning occurs when there is a blocked port, and the net connecting to blocked ports is going to be set to locked.

**What Next**

See report_early_data_checks

---

## ZRT-773

(info) Detected forbidden_preroute_access_layers attribute on net %s with {%s}.

**Description**

Net's forbidden_preroute_access_layers is defined.

**What Next**

Please double check the purpose of forbidden_preroute_access_layers, or make sure the net's stripes are all DRC clean.

---

## ZRT-774

(warning) Incorrect via ladder %s with via ladder rule %s set on pin %s.

**Description**

Found via ladder rules set on pins have error(s) as described.

**What Next**

Please check via ladder rules which are set on pins and make sure that each rule is set correctly, and try again.

---

## ZRT-775

(warn) PG structure in the region (%.4f %.4f) (%.4f %.4f) on layer %s might cause router to fail to drop default vias.

**Description**

The reported PG structure might cause router to fail to drop default vias on the transition layer.

**What Next**

Check the reported PG structure.

Feedback

## ZRT-776

(warning) When running the Early GR flow the option -rerun_global_router automatically maps to option -recompute_map.

**Description**

Rerunning the global router is restricted in the Early GR flow. Only recomputation of the congestion map is allowed based on the existing global route shapes.

## ZRT-777

(Information) %s based on technology file setting.

**Description**

The technology file has the setting "enableExplicitCutMetalGeneration = 1".
But the app options "route.common.enable_explicit_cut_metal_generation" or "route.common.enable_cut_metal_merge_abut" is set to false. Will use the technology file setting instead.

**What Next**

Check the technology section in the tech file and the option settings

## ZRT-778

(info) Eco change area percentage: %.2f \%

**Description**

Percentage of change area in route_eco

**What Next**

Check the changed area

## ZRT-779

(warning) PMJ/VL pin %s (%s layer) has preroute via overlapped.

**Description**

When a PMJ/VL pin has preroute via in the same layer overlapped, check_routability -via_ladder will report it as a warning.

Feedback

## ZRT-780

(warning) The attribute route_connection of pin %s is set to skip.

**Description**

When the attribute route_connection of the pin is set to skip, check_routability -check_pins will report it as a warning.

## ZRT-781

(warning) Port%s%s %s might be blocked.

**Description**

When the port might be blocked, check_routability will report it as a warning.

## ZRT-782

(warn) The option "route.detail.enable_via_in_non_preferred_region_check" is set, but the tech file syntax "contactDefaultPreferredRegionMode" is not set to 1. The option setting will be ignored.

**Description**

Option value has conflict with tech file syntax.

**What Next**

Please check if the option is set unexpectedly or update the tech file.

## ZRT-783

(warning) Blockage data overflowed. The router will ignore all blockage data and continue with incomplete input.

**Description**

The router identified a bad input pattern where too many blockage data are being loaded. The router will ignore all blockage data in order to continue. The routing behavior might not be desired.

**What Next**

Clear bad input problem and rerun routing.

Feedback

## ZRT-784

(Warning) Ignore viaShiftDistance setting on contact %s for oneSideAlignedVia rule.

**Description**

The router will ignore the viaShiftDistance if the layer of via enclosure has oneSideAlignedVia rule.

**What Next**

Check techfile about viaShiftDistance and oneSideAlignedVia.

## ZRT-785

(info) Via ladders found in areas with DRCs = %d

**Description**

Via ladder verification will count DRCs in the vicinity of any via ladder if the app option route.auto_via_ladder.report_via_ladder_in_area_with_drc_threshold is set to a non-negative value. This message reports the number of via ladders in areas with a DRC count that exceeds that threshold.

**What Next**

No action necessary.

## ZRT-786

(Warning) pin %s width is larger than techfile's maxWidth on layer %s.

**Description**

The incorrect pin width causes the maxWidth violation, and DRC will never converge.

**What Next**

Please check the pin width to meet the constraint of techfile's maxWidth.

## ZRT-787

(Warning) Skip fixing fixed shapes DRCs not involving nets specified in route_group command.

Feedback

### Description

The warning message occurs when -fix_general_fixed_shapes_drcs_involving_routed_nets_only command option specified in route_group command. This indicates that the router will not fix those fixed shapes DRCs which do not involve nets in -nets command option. User who uses this command option has the responsibility to take care of the miscorrelation issue as a result.

### What Next

Users need to be aware of where those DRCs which will have to be resolved in later stages involving DR.

---

## ZRT-788

(warning) Option '%s' is ignored when running the Early GR flow.

### Description

The specified option has no effect in the Early GR flow and its intention is ignored.